

Exotic External SUGRA Bases — Quiver Diagrams (v2)

gravity_blocks.v2 catalog ($c_{\text{ext}} \leq 16$ for $T > 1$), T_{max} via f_{multi}

Contents

1	$\hat{1}$ External (87 combos, 16930 bases)	2
1.1	$\mathfrak{e}_6 + \hat{1} \rightarrow (-1)$	2
1.2	$\mathfrak{e}_7 + \hat{1} \rightarrow (-1)$	3
1.3	$\mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$	7
1.4	$\mathfrak{e}_8 + \hat{1} \rightarrow (-1)$	8
1.5	$\mathfrak{f}_4 + \hat{1} \rightarrow (-1)$	15
1.6	$\hat{1} \rightarrow (-1)$	15
1.7	$\hat{1} \rightarrow (-2)$	18
1.8	$\mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1299
1.9	$\mathfrak{so}_8 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1300
1.10	$\mathfrak{su}_2 + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$	1300
1.11	$\mathfrak{su}_2 + \mathfrak{e}_7 + \hat{1} \rightarrow (-1)$	1301
1.12	$\mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$	1303
1.13	$\mathfrak{su}_2 + \mathfrak{e}_8 + \hat{1} \rightarrow (-1)$	1304
1.14	$\mathfrak{su}_2 + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$	1305
1.15	$\mathfrak{su}_2 + \hat{1} \rightarrow (-1)$	1306
1.16	$\mathfrak{su}_2 + \hat{1} \rightarrow (-1) + \hat{1} \rightarrow (-1)$	1307
1.17	$\mathfrak{su}_2 + \hat{1} \rightarrow (-2)$	1307
1.18	$\mathfrak{su}_2 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1361
1.19	$\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1362
1.20	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$	1362
1.21	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_7 + \hat{1} \rightarrow (-1)$	1363
1.22	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$	1363
1.23	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$	1363
1.24	$\mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow (-1)$	1364
1.25	$\mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow (-1) + \hat{1} \rightarrow (-1)$	1366
1.26	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1366
1.27	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$	1367
1.28	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$	1368
1.29	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow (-1)$	1368
1.30	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1371
1.31	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow (-1)$	1373
1.32	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1382
1.33	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow (-1)$	1383
1.34	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1393
1.35	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow (-1)$	1393
1.36	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow (-1)$	1393
1.37	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow (-1)$	1393
1.38	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1393
1.39	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1399
1.40	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1399
1.41	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1406
1.42	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1406
1.43	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1408
1.44	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1408
1.45	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1412
1.46	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1413
1.47	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1415
1.48	$\mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow (-1)$	1415

1.49	$\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$	1415
1.50	$\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$	1415
1.51	$\mathfrak{su}_2 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1416
1.52	$\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1417
1.53	$\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1417
1.54	$\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1418
1.55	$\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1418
1.56	$\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1418
1.57	$\mathfrak{su}_2 + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$	1419
1.58	$\mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$	1419
1.59	$\mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$	1419
1.60	$\mathfrak{su}_2 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$	1419
1.61	$\mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1420
1.62	$\mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$	1420
1.63	$\mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_7 + \hat{1} \rightarrow (-1)$	1420
1.64	$\mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$	1420
1.65	$\mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_8 + \hat{1} \rightarrow (-1)$	1421
1.66	$\mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow (-1)$	1421
1.67	$\mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow (-2)$	1422
1.68	$\mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow (-1)$	1423
1.69	$\mathfrak{su}_3 + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$	1424
1.70	$\mathfrak{su}_3 + \mathfrak{e}_7 + \hat{1} \rightarrow (-1)$	1424
1.71	$\mathfrak{su}_3 + \mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$	1424
1.72	$\mathfrak{su}_3 + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$	1424
1.73	$\mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1425
1.74	$\mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1425
1.75	$\mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1426
1.76	$\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1427
1.77	$\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1427
1.78	$\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$	1427
1.79	$\mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$	1427
1.80	$\mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$	1428
1.81	$\mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1428
1.82	$(-3) \rightarrow (-2) + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$	1428
1.83	$(-3) \rightarrow (-2) + \mathfrak{e}_7 + \hat{1} \rightarrow (-1)$	1428
1.84	$(-3) \rightarrow (-2) + \mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$	1429
1.85	$(-3) \rightarrow (-2) + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$	1429
1.86	$(-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$	1429
1.87	$(-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1430
2	$\mathfrak{su}_8 \rightarrow \mathfrak{so}_{16}$ External (7 combos, 12 bases)	1431
2.1	$\mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$	1431
2.2	$\mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$	1431
2.3	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$	1431
2.4	$\mathfrak{su}_2 + \mathfrak{su}_8 \rightarrow \mathfrak{so}_{16}$	1431
2.5	$(-3) \rightarrow (-2) + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$	1431
2.6	$\mathfrak{su}_8 \rightarrow \mathfrak{so}_{16}$	1431
2.7	$\mathfrak{su}_8 \rightarrow \mathfrak{so}_{16} + (-3) \rightarrow (-2)$	1432
3	$\mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$ External (67 combos, 581 bases)	1433
3.1	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + (-3) \rightarrow (-2)$	1433
3.2	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + (-3) \rightarrow (-2)$	1433
3.3	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + (-3) \rightarrow (-2)$	1433
3.4	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + (-3) \rightarrow (-2)$	1434
3.5	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2$	1434
3.6	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + (-3) \rightarrow (-2)$	1434

3.7	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2)$	1434
3.8	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + (-3) \rightarrow (-2)$	1434
3.9	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + (-3) \rightarrow (-2) + \mathfrak{su}_2$	1435
3.10	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3)$	1435
3.11	$\text{NHC}(-2)(-3) + \mathfrak{e}_7 + (-3) \rightarrow (-2)$	1435
3.12	$\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + (-3) \rightarrow (-2)$	1435
3.13	$\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + (-3) \rightarrow (-2)$	1436
3.14	$\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2)$	1436
3.15	$\text{NHC}(-2)(-3) + \mathfrak{f}_4 + (-3) \rightarrow (-2)$	1436
3.16	$\text{NHC}(-2)(-3) + \mathfrak{f}_4 + (-3) \rightarrow (-2) + \mathfrak{su}_2$	1437
3.17	$\text{NHC}(-2)(-3) + \mathfrak{f}_4 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3)$	1437
3.18	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2)$	1437
3.19	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2)$	1439
3.20	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + (-3) \rightarrow (-2)$	1439
3.21	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + (-3) \rightarrow (-2)$	1439
3.22	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + (-3) \rightarrow (-2)$	1440
3.23	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_2$	1441
3.24	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3)$	1443
3.25	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \text{su2n3mix}$	1443
3.26	$\text{NHC}(-2)(-3) + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_3$	1443
3.27	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + (-3) \rightarrow (-2)$	1444
3.28	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + (-3) \rightarrow (-2)$	1444
3.29	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{so}_8 + (-3) \rightarrow (-2)$	1444
3.30	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + (-3) \rightarrow (-2)$	1446
3.31	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2)$	1448
3.32	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{e}_6$	1474
3.33	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{f}_4$	1474
3.34	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8$	1475
3.35	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2$	1475
3.36	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	1476
3.37	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3)$	1477
3.38	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	1478
3.39	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \text{su2n3mix}$	1479
3.40	$\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_3$	1479
3.41	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + (-3) \rightarrow (-2)$	1481
3.42	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2)$	1481
3.43	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2)$	1483
3.44	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + (-3) \rightarrow (-2)$	1484
3.45	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{so}_8$	1488
3.46	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2$	1489
3.47	$\text{NHC}(-2)(-3) + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_3$	1491
3.48	$\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8 + (-3) \rightarrow (-2)$	1492
3.49	$\mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$	1492
3.50	$\mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$	1492
3.51	$\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$	1492
3.52	$\mathfrak{su}_2 + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$	1492
3.53	$\mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$	1492
3.54	$\mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$	1493
3.55	$\mathfrak{su}_2 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$	1493
3.56	$\mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1493
3.57	$\mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$	1494
3.58	$\mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$	1494
3.59	$\mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1494
3.60	$(-3) \rightarrow (-2) + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$	1494
3.61	$(-3) \rightarrow (-2) + \mathfrak{e}_7 + \hat{1} \rightarrow (-1)$	1495

3.62	$(-3) \rightarrow (-2) + \mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$	1495
3.63	$(-3) \rightarrow (-2) + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$	1495
3.64	$(-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$	1496
3.65	$(-3) \rightarrow (-2) + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$	1496
3.66	$(-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$	1496
3.67	$\mathfrak{su}_8 \rightarrow \mathfrak{so}_{16} + (-3) \rightarrow (-2)$	1497
4	NHC $(-2)(-3)$ External (1227 combos, 49200 bases)	1498
4.1	NHC $(-2)(-3)$	1498
4.2	NHC $(-2)(-3) + \mathfrak{e}_7$	1537
4.3	NHC $(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2$	1538
4.4	NHC $(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	1538
4.5	NHC $(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	1539
4.6	NHC $(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	1539
4.7	NHC $(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	1540
4.8	NHC $(-2)(-3) + \mathfrak{so}_8$	1541
4.9	NHC $(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2$	1541
4.10	NHC $(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	1541
4.11	NHC $(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	1541
4.12	NHC $(-2)(-3) + \mathfrak{su}_2$	1542
4.13	NHC $(-2)(-3) + \mathfrak{su}_2 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$	1546
4.14	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2$	1547
4.15	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	1550
4.16	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	1551
4.17	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	1554
4.18	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$	1554
4.19	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix}$	1555
4.20	NHC $(-2)(-3) + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2$	1555
4.21	NHC $(-2)(-3) + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$	1556
4.22	NHC $(-2)(-3) + \mathfrak{su}_2 + \text{su2n3mix} + \text{su2n3mix}$	1556
4.23	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3$	1557
4.24	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	1557
4.25	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	1557
4.26	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	1557
4.27	NHC $(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	1557
4.28	NHC $(-2)(-3) + \mathfrak{su}_3$	1558
4.29	NHC $(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2$	1559
4.30	NHC $(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	1561
4.31	NHC $(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	1562
4.32	NHC $(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	1562
4.33	NHC $(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	1562
4.34	NHC $(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	1562
4.35	NHC $(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	1562
4.36	NHC $(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$	1562
4.37	NHC $(-2)(-3) + \mathfrak{su}_3 + \text{su2n3mix} + \mathfrak{su}_2$	1563
4.38	NHC $(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	1563
4.39	NHC $(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	1563
4.40	NHC $(-2)(-3)$	1563
4.41	NHC $(-2)(-3) + \mathfrak{e}_6$	1741
4.42	NHC $(-2)(-3) + \mathfrak{e}_6 + \mathfrak{e}_6$	1750
4.43	NHC $(-2)(-3) + \mathfrak{e}_6 + \mathfrak{e}_6 + \mathfrak{su}_2$	1751
4.44	NHC $(-2)(-3) + \mathfrak{e}_6 + \mathfrak{e}_7$	1751
4.45	NHC $(-2)(-3) + \mathfrak{e}_6 + \mathfrak{e}'_7$	1751
4.46	NHC $(-2)(-3) + \mathfrak{e}_6 + \mathfrak{f}_4$	1751
4.47	NHC $(-2)(-3) + \mathfrak{e}_6 + \mathfrak{f}_4 + \mathfrak{su}_2$	1753

4.48	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3)$	1754
4.49	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \text{NHC}(-2)(-3)$	1755
4.50	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{so}_8$	1755
4.51	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$	1756
4.52	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2$	1758
4.53	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3$	1758
4.54	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$	1758
4.55	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$	1759
4.56	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2$	1760
4.57	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3$	1760
4.58	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8$	1760
4.59	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8 + \text{NHC}(-2)(-3)$	1763
4.60	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$	1763
4.61	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$	1770
4.62	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	1771
4.63	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$	1771
4.64	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	1772
4.65	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3$	1773
4.66	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$	1774
4.67	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2$	1774
4.68	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{e}_6$	1831
4.69	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{f}_4$	1831
4.70	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$	1832
4.71	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$	1834
4.72	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$	1835
4.73	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8$	1835
4.74	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$	1842
4.75	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$	1843
4.76	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$	1843
4.77	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$	1844
4.78	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$	1889
4.79	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$	1890
4.80	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$	1891
4.81	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	1891
4.82	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$	1908
4.83	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$	1909
4.84	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	1911
4.85	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$	1911
4.86	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$	1924
4.87	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$	1927
4.88	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	1928
4.89	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	1937
4.90	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$	1937
4.91	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix}$	1938
4.92	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	1939
4.93	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{su2n3mix}$	1941
4.94	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$	1955
4.95	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_3$	1955
4.96	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$	1956
4.97	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$	1983
4.98	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8$	1983
4.99	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$	1983
4.100	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	1996
4.101	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$	1998
4.102	$\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	1999

4.158NHC(-2)(-3) + e ₆ + su ₃ + su ₃ + su ₂ + su ₂	2147
4.159NHC(-2)(-3) + e ₆ + su ₃ + su ₃ + su ₂ + su ₃	2148
4.160NHC(-2)(-3) + e ₆ + su ₃ + su ₃ + su ₂ →(-3)	2149
4.161NHC(-2)(-3) + e ₆ + su ₃ + su ₃ + su ₃	2149
4.162NHC(-2)(-3) + e ₆ + su ₃ + su ₃ + su ₃ + su ₂	2150
4.163NHC(-2)(-3) + e ₆ + (-3)→(-2)	2150
4.164NHC(-2)(-3) + e ₆ + (-3)→(-2) + su ₂	2151
4.165NHC(-2)(-3) + e ₆ + (-3)→(-2) + su ₂ →(-3)	2151
4.166NHC(-2)(-3) + e ₇	2151
4.167NHC(-2)(-3) + e ₇ + e ₆	2153
4.168NHC(-2)(-3) + e ₇ + f ₄	2153
4.169NHC(-2)(-3) + e ₇ + NHC(-2)(-3)	2153
4.170NHC(-2)(-3) + e ₇ + NHC(-2)(-3) + su ₂	2154
4.171NHC(-2)(-3) + e ₇ + NHC(-2)(-3) + su ₂ →(-3)	2156
4.172NHC(-2)(-3) + e ₇ + NHC(-2)(-3) + su ₃	2156
4.173NHC(-2)(-3) + e ₇ + so ₈	2157
4.174NHC(-2)(-3) + e ₇ + so ₈ + su ₂	2157
4.175NHC(-2)(-3) + e ₇ + so ₈ + su ₂ + su ₂	2159
4.176NHC(-2)(-3) + e ₇ + so ₈ + su ₂ →(-3)	2160
4.177NHC(-2)(-3) + e ₇ + so ₈ + su ₃	2160
4.178NHC(-2)(-3) + e ₇ + su ₂	2161
4.179NHC(-2)(-3) + e ₇ + su ₂ + NHC(-2)(-3)	2175
4.180NHC(-2)(-3) + e ₇ + su ₂ + so ₈	2176
4.181NHC(-2)(-3) + e ₇ + su ₂ + so ₈ + su ₂	2178
4.182NHC(-2)(-3) + e ₇ + su ₂ + su ₂	2179
4.183NHC(-2)(-3) + e ₇ + su ₂ + su ₂ + so ₈	2210
4.184NHC(-2)(-3) + e ₇ + su ₂ + su ₂ + su ₂	2211
4.185NHC(-2)(-3) + e ₇ + su ₂ + su ₂ + su ₂ + su ₂	2226
4.186NHC(-2)(-3) + e ₇ + su ₂ + su ₂ + su ₂ + su ₂ →(-3)	2227
4.187NHC(-2)(-3) + e ₇ + su ₂ + su ₂ + su ₂ + su ₃	2228
4.188NHC(-2)(-3) + e ₇ + su ₂ + su ₂ + su ₂ →(-3)	2229
4.189NHC(-2)(-3) + e ₇ + su ₂ + su ₂ + su ₂ →(-3) + su ₂	2232
4.190NHC(-2)(-3) + e ₇ + su ₂ + su ₂ + su ₂ →(-3) + su ₃	2233
4.191NHC(-2)(-3) + e ₇ + su ₂ + su ₂ + su2n3mix	2233
4.192NHC(-2)(-3) + e ₇ + su ₂ + su ₂ + su ₃	2235
4.193NHC(-2)(-3) + e ₇ + su ₂ + su ₂ + su ₃ + su ₂	2240
4.194NHC(-2)(-3) + e ₇ + su ₂ + su ₂ + su ₃ + su ₂ →(-3)	2241
4.195NHC(-2)(-3) + e ₇ + su ₂ + su ₂ + su ₃ + su ₃	2241
4.196NHC(-2)(-3) + e ₇ + su ₂ + su ₂ →(-3)	2241
4.197NHC(-2)(-3) + e ₇ + su ₂ + su ₂ →(-3) + su ₂	2246
4.198NHC(-2)(-3) + e ₇ + su ₂ + su ₂ →(-3) + su ₂ + su ₂	2249
4.199NHC(-2)(-3) + e ₇ + su ₂ + su ₂ →(-3) + su ₂ + su ₃	2249
4.200NHC(-2)(-3) + e ₇ + su ₂ + su ₂ →(-3) + su ₂ →(-3)	2250
4.201NHC(-2)(-3) + e ₇ + su ₂ + su ₂ →(-3) + su2n3mix	2250
4.202NHC(-2)(-3) + e ₇ + su ₂ + su ₂ →(-3) + su ₃	2250
4.203NHC(-2)(-3) + e ₇ + su ₂ + su ₂ →(-3) + su ₃ + su ₂	2252
4.204NHC(-2)(-3) + e ₇ + su ₂ + su2n3mix	2252
4.205NHC(-2)(-3) + e ₇ + su ₂ + su2n3mix + su ₂	2260
4.206NHC(-2)(-3) + e ₇ + su ₂ + su2n3mix + su ₂ →(-3)	2261
4.207NHC(-2)(-3) + e ₇ + su ₂ + su2n3mix + su ₃	2261
4.208NHC(-2)(-3) + e ₇ + su ₂ + su ₃	2262
4.209NHC(-2)(-3) + e ₇ + su ₂ + su ₃ + su ₂	2277
4.210NHC(-2)(-3) + e ₇ + su ₂ + su ₃ + su ₂ + su ₂	2282
4.211NHC(-2)(-3) + e ₇ + su ₂ + su ₃ + su ₂ + su ₂ →(-3)	2283
4.212NHC(-2)(-3) + e ₇ + su ₂ + su ₃ + su ₂ + su ₃	2283

4.213NHC(-2)(-3) + $e_7 + su_2 + su_3 + su_2 \rightarrow (-3)$	2284
4.214NHC(-2)(-3) + $e_7 + su_2 + su_3 + su_2 \rightarrow (-3) + su_2$	2285
4.215NHC(-2)(-3) + $e_7 + su_2 + su_3 + su_{2n3mix}$	2286
4.216NHC(-2)(-3) + $e_7 + su_2 + su_3 + su_3$	2286
4.217NHC(-2)(-3) + $e_7 + su_2 + su_3 + su_3 + su_2$	2288
4.218NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3)$	2288
4.219NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + NHC(-2)(-3)$	2289
4.220NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + so_8$	2290
4.221NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2$	2290
4.222NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 + su_2$	2294
4.223NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 + su_2 + su_2$	2297
4.224NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 + su_2 + su_3$	2298
4.225NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 + su_2 \rightarrow (-3)$	2298
4.226NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 + su_{2n3mix}$	2298
4.227NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 + su_3$	2298
4.228NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 + su_3 + su_2$	2300
4.229NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 \rightarrow (-3)$	2300
4.230NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 \rightarrow (-3) + su_2$	2301
4.231NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 \rightarrow (-3) + su_2 \rightarrow (-3)$	2301
4.232NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 \rightarrow (-3) + su_{2n3mix}$	2301
4.233NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_2 \rightarrow (-3) + su_3$	2301
4.234NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_{2n3mix}$	2302
4.235NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_{2n3mix} + su_2$	2302
4.236NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_{2n3mix} + su_2 \rightarrow (-3)$	2302
4.237NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_3$	2303
4.238NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_3 + su_2$	2303
4.239NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_3 + su_2 + su_2$	2305
4.240NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_3 + su_2 \rightarrow (-3)$	2305
4.241NHC(-2)(-3) + $e_7 + su_2 \rightarrow (-3) + su_3 + su_3$	2305
4.242NHC(-2)(-3) + $e_7 + su_{2n3mix}$	2305
4.243NHC(-2)(-3) + $e_7 + su_{2n3mix} + su_2$	2309
4.244NHC(-2)(-3) + $e_7 + su_{2n3mix} + su_2 + su_2$	2317
4.245NHC(-2)(-3) + $e_7 + su_{2n3mix} + su_2 + su_2 \rightarrow (-3)$	2319
4.246NHC(-2)(-3) + $e_7 + su_{2n3mix} + su_2 + su_3$	2319
4.247NHC(-2)(-3) + $e_7 + su_{2n3mix} + su_2 \rightarrow (-3)$	2319
4.248NHC(-2)(-3) + $e_7 + su_{2n3mix} + su_2 \rightarrow (-3) + su_2$	2320
4.249NHC(-2)(-3) + $e_7 + su_{2n3mix} + su_2 \rightarrow (-3) + su_2 \rightarrow (-3)$	2320
4.250NHC(-2)(-3) + $e_7 + su_{2n3mix} + su_{2n3mix}$	2320
4.251NHC(-2)(-3) + $e_7 + su_{2n3mix} + su_3$	2321
4.252NHC(-2)(-3) + $e_7 + su_{2n3mix} + su_3 + su_2$	2322
4.253NHC(-2)(-3) + $e_7 + su_3$	2322
4.254NHC(-2)(-3) + $e_7 + su_3 + NHC(-2)(-3)$	2325
4.255NHC(-2)(-3) + $e_7 + su_3 + so_8$	2326
4.256NHC(-2)(-3) + $e_7 + su_3 + su_2$	2326
4.257NHC(-2)(-3) + $e_7 + su_3 + su_2 + su_2$	2342
4.258NHC(-2)(-3) + $e_7 + su_3 + su_2 + su_2 + su_2$	2347
4.259NHC(-2)(-3) + $e_7 + su_3 + su_2 + su_2 + su_2 \rightarrow (-3)$	2348
4.260NHC(-2)(-3) + $e_7 + su_3 + su_2 + su_2 + su_3$	2348
4.261NHC(-2)(-3) + $e_7 + su_3 + su_2 + su_2 \rightarrow (-3)$	2349
4.262NHC(-2)(-3) + $e_7 + su_3 + su_2 + su_2 \rightarrow (-3) + su_2$	2350
4.263NHC(-2)(-3) + $e_7 + su_3 + su_2 + su_{2n3mix}$	2350
4.264NHC(-2)(-3) + $e_7 + su_3 + su_2 + su_3$	2350
4.265NHC(-2)(-3) + $e_7 + su_3 + su_2 + su_3 + su_2$	2353
4.266NHC(-2)(-3) + $e_7 + su_3 + su_2 \rightarrow (-3)$	2353
4.267NHC(-2)(-3) + $e_7 + su_3 + su_2 \rightarrow (-3) + su_2$	2354

4.268NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$	2355
4.269NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$	2356
4.270NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	2356
4.271NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_{2n3mix}$	2356
4.272NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_{2n3mix} + \mathfrak{su}_2$	2357
4.273NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3$	2357
4.274NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$	2359
4.275NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	2362
4.276NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	2362
4.277NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$	2362
4.278NHC(-2)(-3) + $\mathfrak{e}_7 + (-3) \rightarrow (-2)$	2363
4.279NHC(-2)(-3) + $\mathfrak{e}'_7$	2363
4.280NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{e}_6$	2364
4.281NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{f}_4$	2365
4.282NHC(-2)(-3) + $\mathfrak{e}'_7 + \text{NHC}(-2)(-3)$	2365
4.283NHC(-2)(-3) + $\mathfrak{e}'_7 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$	2366
4.284NHC(-2)(-3) + $\mathfrak{e}'_7 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$	2367
4.285NHC(-2)(-3) + $\mathfrak{e}'_7 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$	2367
4.286NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{so}_8$	2367
4.287NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{so}_8 + \mathfrak{su}_2$	2368
4.288NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$	2370
4.289NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	2370
4.290NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{so}_8 + \mathfrak{su}_3$	2370
4.291NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2$	2371
4.292NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$	2385
4.293NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{so}_8$	2386
4.294NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$	2388
4.295NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$	2388
4.296NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$	2419
4.297NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	2420
4.298NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$	2424
4.299NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	2424
4.300NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	2427
4.301NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_{2n3mix}$	2427
4.302NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$	2429
4.303NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$	2433
4.304NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	2433
4.305NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$	2433
4.306NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	2433
4.307NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	2437
4.308NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3$	2440
4.309NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$	2440
4.310NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_{2n3mix}$	2440
4.311NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	2440
4.312NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2$	2442
4.313NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_{2n3mix}$	2442
4.314NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_{2n3mix} + \mathfrak{su}_2$	2449
4.315NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_{2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$	2451
4.316NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_{2n3mix} + \mathfrak{su}_3$	2451
4.317NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3$	2452
4.318NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$	2466
4.319NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	2470
4.320NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	2471
4.321NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$	2471
4.322NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	2471

4.323NHC(-2)(-3) + $e'_7 + su_2 + su_3 + su_2 \rightarrow (-3) + su_2$	2473
4.324NHC(-2)(-3) + $e'_7 + su_2 + su_3 + su_2n3mix$	2473
4.325NHC(-2)(-3) + $e'_7 + su_2 + su_3 + su_3$	2473
4.326NHC(-2)(-3) + $e'_7 + su_2 + su_3 + su_3 + su_2$	2475
4.327NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3)$	2475
4.328NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + NHC(-2)(-3)$	2476
4.329NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + so_8$	2476
4.330NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2$	2476
4.331NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2 + su_2$	2480
4.332NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2 + su_2 + su_3$	2483
4.333NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2 + su_2 \rightarrow (-3)$	2483
4.334NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2 + su_2n3mix$	2483
4.335NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2 + su_3$	2483
4.336NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2 + su_3 + su_2$	2485
4.337NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2 \rightarrow (-3)$	2485
4.338NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2 \rightarrow (-3) + su_2$	2486
4.339NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2 \rightarrow (-3) + su_2 \rightarrow (-3)$	2486
4.340NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2 \rightarrow (-3) + su_2n3mix$	2486
4.341NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2 \rightarrow (-3) + su_3$	2486
4.342NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2n3mix$	2486
4.343NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2n3mix + su_2$	2487
4.344NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_2n3mix + su_2 \rightarrow (-3)$	2487
4.345NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_3$	2487
4.346NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_3 + su_2$	2488
4.347NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_3 + su_2 + su_2$	2490
4.348NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_3 + su_2 \rightarrow (-3)$	2490
4.349NHC(-2)(-3) + $e'_7 + su_2 \rightarrow (-3) + su_3 + su_3$	2490
4.350NHC(-2)(-3) + $e'_7 + su_2n3mix$	2490
4.351NHC(-2)(-3) + $e'_7 + su_2n3mix + su_2$	2494
4.352NHC(-2)(-3) + $e'_7 + su_2n3mix + su_2 + su_2$	2501
4.353NHC(-2)(-3) + $e'_7 + su_2n3mix + su_2 + su_2 \rightarrow (-3)$	2503
4.354NHC(-2)(-3) + $e'_7 + su_2n3mix + su_2 + su_3$	2503
4.355NHC(-2)(-3) + $e'_7 + su_2n3mix + su_2 \rightarrow (-3)$	2504
4.356NHC(-2)(-3) + $e'_7 + su_2n3mix + su_2 \rightarrow (-3) + su_2$	2504
4.357NHC(-2)(-3) + $e'_7 + su_2n3mix + su_2 \rightarrow (-3) + su_2 \rightarrow (-3)$	2504
4.358NHC(-2)(-3) + $e'_7 + su_2n3mix + su_2n3mix$	2505
4.359NHC(-2)(-3) + $e'_7 + su_2n3mix + su_3$	2505
4.360NHC(-2)(-3) + $e'_7 + su_2n3mix + su_3 + su_2$	2506
4.361NHC(-2)(-3) + $e'_7 + su_3$	2506
4.362NHC(-2)(-3) + $e'_7 + su_3 + NHC(-2)(-3)$	2510
4.363NHC(-2)(-3) + $e'_7 + su_3 + so_8$	2510
4.364NHC(-2)(-3) + $e'_7 + su_3 + su_2$	2511
4.365NHC(-2)(-3) + $e'_7 + su_3 + su_2 + su_2$	2525
4.366NHC(-2)(-3) + $e'_7 + su_3 + su_2 + su_2 + su_2$	2529
4.367NHC(-2)(-3) + $e'_7 + su_3 + su_2 + su_2 + su_2 \rightarrow (-3)$	2530
4.368NHC(-2)(-3) + $e'_7 + su_3 + su_2 + su_2 + su_3$	2530
4.369NHC(-2)(-3) + $e'_7 + su_3 + su_2 + su_2 \rightarrow (-3)$	2530
4.370NHC(-2)(-3) + $e'_7 + su_3 + su_2 + su_2 \rightarrow (-3) + su_2$	2532
4.371NHC(-2)(-3) + $e'_7 + su_3 + su_2 + su_2n3mix$	2532
4.372NHC(-2)(-3) + $e'_7 + su_3 + su_2 + su_3$	2532
4.373NHC(-2)(-3) + $e'_7 + su_3 + su_2 + su_3 + su_2$	2534
4.374NHC(-2)(-3) + $e'_7 + su_3 + su_2 \rightarrow (-3)$	2534
4.375NHC(-2)(-3) + $e'_7 + su_3 + su_2 \rightarrow (-3) + su_2$	2535
4.376NHC(-2)(-3) + $e'_7 + su_3 + su_2 \rightarrow (-3) + su_2 + su_2$	2536
4.377NHC(-2)(-3) + $e'_7 + su_3 + su_2 \rightarrow (-3) + su_2 \rightarrow (-3)$	2536

4.378NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$	2537
4.379NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_{2n3\text{mix}}$	2537
4.380NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{su}_2$	2538
4.381NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$	2538
4.382NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$	2540
4.383NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$	2542
4.384NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$	2543
4.385NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$	2543
4.386NHC(-2)(-3) + $\mathfrak{e}'_7$ + $(-3) \rightarrow (-2)$	2543
4.387NHC(-2)(-3) + $\mathfrak{e}_8$	2543
4.388NHC(-2)(-3) + $\mathfrak{e}_8$ + NHC(-2)(-3)	2543
4.389NHC(-2)(-3) + $\mathfrak{e}_8$ + NHC(-2)(-3) + $\mathfrak{su}_2$	2544
4.390NHC(-2)(-3) + $\mathfrak{e}_8$ + NHC(-2)(-3) + $\mathfrak{su}_3$	2544
4.391NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{so}_8$	2544
4.392NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2$	2544
4.393NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$	2544
4.394NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + NHC(-2)(-3)	2547
4.395NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$	2547
4.396NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$	2547
4.397NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$	2555
4.398NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$	2559
4.399NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$	2560
4.400NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$	2560
4.401NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$	2562
4.402NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$	2562
4.403NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$	2565
4.404NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$	2567
4.405NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$	2567
4.406NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_3$	2567
4.407NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_3$ + NHC(-2)(-3)	2568
4.408NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$	2568
4.409NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$	2571
4.410NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$	2573
4.411NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$	2573
4.412NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$	2573
4.413NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$	2574
4.414NHC(-2)(-3) + $\mathfrak{f}_4$	2574
4.415NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{e}_6$	2584
4.416NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{e}_6$ + $\mathfrak{su}_2$	2585
4.417NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{e}_7$	2586
4.418NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{e}'_7$	2586
4.419NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{f}_4$	2586
4.420NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$	2587
4.421NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$	2590
4.422NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$	2590
4.423NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \rightarrow (-3)$	2590
4.424NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$	2591
4.425NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{f}_4$ + $\mathfrak{su}_{2n3\text{mix}}$	2591
4.426NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{f}_4$ + $\mathfrak{su}_3$	2591
4.427NHC(-2)(-3) + $\mathfrak{f}_4$ + NHC(-2)(-3)	2591
4.428NHC(-2)(-3) + $\mathfrak{f}_4$ + NHC(-2)(-3) + $\mathfrak{so}_8$	2593
4.429NHC(-2)(-3) + $\mathfrak{f}_4$ + NHC(-2)(-3) + $\mathfrak{su}_2$	2593
4.430NHC(-2)(-3) + $\mathfrak{f}_4$ + NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$	2594
4.431NHC(-2)(-3) + $\mathfrak{f}_4$ + NHC(-2)(-3) + $\mathfrak{su}_2 \rightarrow (-3)$	2594
4.432NHC(-2)(-3) + $\mathfrak{f}_4$ + NHC(-2)(-3) + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$	2595

4.433NHC(-2)(-3) + $\mathfrak{f}_4$ + NHC(-2)(-3) + su2n3mix	2595
4.434NHC(-2)(-3) + $\mathfrak{f}_4$ + NHC(-2)(-3) + $\mathfrak{su}_3$	2595
4.435NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$	2596
4.436NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + NHC(-2)(-3)	2599
4.437NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2$	2599
4.438NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$	2609
4.439NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$	2610
4.440NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2$ + su2n3mix	2610
4.441NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$	2610
4.442NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$	2611
4.443NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$	2612
4.444NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$	2613
4.445NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + su2n3mix	2613
4.446NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{su}_2$	2614
4.447NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + $\mathfrak{su}_3$	2614
4.448NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$	2616
4.449NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$	2616
4.450NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$	2616
4.451NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{e}_6$	2674
4.452NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{f}_4$	2675
4.453NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$	2677
4.454NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \rightarrow (-3)$	2677
4.455NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + NHC(-2)(-3)	2677
4.456NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + NHC(-2)(-3) + $\mathfrak{su}_2 \rightarrow (-3)$	2679
4.457NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$	2679
4.458NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2$	2687
4.459NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$	2688
4.460NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$ + su2n3mix	2689
4.461NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$ + $\mathfrak{su}_3$	2689
4.462NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$	2689
4.463NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{f}_4$	2731
4.464NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$	2731
4.465NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$	2733
4.466NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$	2738
4.467NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + su2n3mix	2739
4.468NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$	2746
4.469NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$	2755
4.470NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$	2756
4.471NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{f}_4$	2769
4.472NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + NHC(-2)(-3)	2770
4.473NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{so}_8$	2770
4.474NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$	2770
4.475NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$	2776
4.476NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + su2n3mix	2777
4.477NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$	2777
4.478NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$	2781
4.479NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + su2n3mix	2782
4.480NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + su2n3mix + $\mathfrak{so}_8$	2812
4.481NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + su2n3mix + $\mathfrak{su}_2$	2812
4.482NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$	2819
4.483NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + su2n3mix + $\mathfrak{su}_3$	2820
4.484NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$	2822
4.485NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{so}_8$	2848
4.486NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$	2849
4.487NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$	2858

4.488NHC(-2)(-3) + f ₄ + su ₂ + su ₃ + su ₂ →(-3)	2859
4.489NHC(-2)(-3) + f ₄ + su ₂ + su ₃ + su ₂ →(-3) + su ₂	2862
4.490NHC(-2)(-3) + f ₄ + su ₂ + su ₃ + su2n3mix	2863
4.491NHC(-2)(-3) + f ₄ + su ₂ + su ₃ + su ₃	2865
4.492NHC(-2)(-3) + f ₄ + su ₂ + (-3)→(-2)	2868
4.493NHC(-2)(-3) + f ₄ + su ₂ →(-3)	2868
4.494NHC(-2)(-3) + f ₄ + su ₂ →(-3) + f ₄	2873
4.495NHC(-2)(-3) + f ₄ + su ₂ →(-3) + f ₄ + su ₂	2873
4.496NHC(-2)(-3) + f ₄ + su ₂ →(-3) + NHC(-2)(-3)	2873
4.497NHC(-2)(-3) + f ₄ + su ₂ →(-3) + NHC(-2)(-3) + su ₂	2873
4.498NHC(-2)(-3) + f ₄ + su ₂ →(-3) + so ₈	2874
4.499NHC(-2)(-3) + f ₄ + su ₂ →(-3) + so ₈ + su ₂	2875
4.500NHC(-2)(-3) + f ₄ + su ₂ →(-3) + so ₈ + su ₃	2875
4.501NHC(-2)(-3) + f ₄ + su ₂ →(-3) + su ₂	2875
4.502NHC(-2)(-3) + f ₄ + su ₂ →(-3) + su ₂ + f ₄	2889
4.503NHC(-2)(-3) + f ₄ + su ₂ →(-3) + su ₂ + NHC(-2)(-3)	2889
4.504NHC(-2)(-3) + f ₄ + su ₂ →(-3) + su ₂ + so ₈	2889
4.505NHC(-2)(-3) + f ₄ + su ₂ →(-3) + su ₂ + su ₂	2890
4.506NHC(-2)(-3) + f ₄ + su ₂ →(-3) + su ₂ + su ₂ + su ₃	2895
4.507NHC(-2)(-3) + f ₄ + su ₂ →(-3) + su ₂ + su2n3mix	2896
4.508NHC(-2)(-3) + f ₄ + su ₂ →(-3) + su ₂ + su ₃	2896
4.509NHC(-2)(-3) + f ₄ + su ₂ →(-3) + su ₂ + su ₃ + su ₂	2900
4.510NHC(-2)(-3) + f ₄ + su ₂ →(-3) + su ₂ →(-3)	2901
4.511NHC(-2)(-3) + f ₄ + su ₂ →(-3) + su2n3mix	2901
4.512NHC(-2)(-3) + f ₄ + su ₂ →(-3) + su2n3mix + su ₂	2903
4.513NHC(-2)(-3) + f ₄ + su ₂ →(-3) + su ₃	2903
4.514NHC(-2)(-3) + f ₄ + su ₂ →(-3) + su ₃ + so ₈	2906
4.515NHC(-2)(-3) + f ₄ + su ₂ →(-3) + su ₃ + su ₂	2906
4.516NHC(-2)(-3) + f ₄ + su ₂ →(-3) + su ₃ + su ₂ + su ₂	2910
4.517NHC(-2)(-3) + f ₄ + su ₂ →(-3) + su ₃ + su ₃	2911
4.518NHC(-2)(-3) + f ₄ + su ₂ →(-3) + (-3)→(-2)	2912
4.519NHC(-2)(-3) + f ₄ + su2n3mix	2912
4.520NHC(-2)(-3) + f ₄ + su2n3mix + f ₄	2938
4.521NHC(-2)(-3) + f ₄ + su2n3mix + NHC(-2)(-3)	2938
4.522NHC(-2)(-3) + f ₄ + su2n3mix + so ₈	2939
4.523NHC(-2)(-3) + f ₄ + su2n3mix + so ₈ + su ₂	2940
4.524NHC(-2)(-3) + f ₄ + su2n3mix + su ₂	2940
4.525NHC(-2)(-3) + f ₄ + su2n3mix + su ₂ + so ₈	2970
4.526NHC(-2)(-3) + f ₄ + su2n3mix + su ₂ + su ₂	2970
4.527NHC(-2)(-3) + f ₄ + su2n3mix + su ₂ + su ₂ →(-3)	2978
4.528NHC(-2)(-3) + f ₄ + su2n3mix + su ₂ + su ₃	2978
4.529NHC(-2)(-3) + f ₄ + su2n3mix + su ₂ →(-3)	2980
4.530NHC(-2)(-3) + f ₄ + su2n3mix + su ₂ →(-3) + su ₂	2982
4.531NHC(-2)(-3) + f ₄ + su2n3mix + su2n3mix	2982
4.532NHC(-2)(-3) + f ₄ + su2n3mix + su ₃	2983
4.533NHC(-2)(-3) + f ₄ + su2n3mix + su ₃ + su ₂	2990
4.534NHC(-2)(-3) + f ₄ + su2n3mix + su ₃ + su ₃	2992
4.535NHC(-2)(-3) + f ₄ + su ₃	2993
4.536NHC(-2)(-3) + f ₄ + su ₃ + f ₄	3006
4.537NHC(-2)(-3) + f ₄ + su ₃ + NHC(-2)(-3)	3006
4.538NHC(-2)(-3) + f ₄ + su ₃ + so ₈	3007
4.539NHC(-2)(-3) + f ₄ + su ₃ + so ₈ + su ₂	3009
4.540NHC(-2)(-3) + f ₄ + su ₃ + so ₈ + su ₂ →(-3)	3009
4.541NHC(-2)(-3) + f ₄ + su ₃ + su ₂	3010
4.542NHC(-2)(-3) + f ₄ + su ₃ + su ₂ + so ₈	3036

4.543NHC(-2)(-3) + f ₄ + su ₃ + su ₂ + su ₂	3037
4.544NHC(-2)(-3) + f ₄ + su ₃ + su ₂ + su ₂ + su ₂ →(-3)	3046
4.545NHC(-2)(-3) + f ₄ + su ₃ + su ₂ + su ₂ →(-3)	3047
4.546NHC(-2)(-3) + f ₄ + su ₃ + su ₂ + su ₂ →(-3) + su ₂	3050
4.547NHC(-2)(-3) + f ₄ + su ₃ + su ₂ + su2n3mix	3051
4.548NHC(-2)(-3) + f ₄ + su ₃ + su ₂ + su ₃	3053
4.549NHC(-2)(-3) + f ₄ + su ₃ + su ₂ →(-3)	3056
4.550NHC(-2)(-3) + f ₄ + su ₃ + su ₂ →(-3) + so ₈	3059
4.551NHC(-2)(-3) + f ₄ + su ₃ + su ₂ →(-3) + su ₂	3059
4.552NHC(-2)(-3) + f ₄ + su ₃ + su ₂ →(-3) + su ₂ + su ₂	3062
4.553NHC(-2)(-3) + f ₄ + su ₃ + su ₂ →(-3) + su ₃	3063
4.554NHC(-2)(-3) + f ₄ + su ₃ + su2n3mix	3064
4.555NHC(-2)(-3) + f ₄ + su ₃ + su2n3mix + su ₂	3072
4.556NHC(-2)(-3) + f ₄ + su ₃ + su2n3mix + su ₃	3074
4.557NHC(-2)(-3) + f ₄ + su ₃ + su ₃	3074
4.558NHC(-2)(-3) + f ₄ + su ₃ + su ₃ + su ₂	3077
4.559NHC(-2)(-3) + f ₄ + su ₃ + su ₃ + su ₂ →(-3)	3080
4.560NHC(-2)(-3) + f ₄ + su ₃ + su ₃ + su2n3mix	3081
4.561NHC(-2)(-3) + f ₄ + su ₃ + su ₃ + su ₃	3082
4.562NHC(-2)(-3) + f ₄ + (-3)→(-2)	3082
4.563NHC(-2)(-3) + f ₄ + (-3)→(-2) + su ₂	3082
4.564NHC(-2)(-3) + f ₄ + (-3)→(-2) + su ₂ →(-3)	3083
4.565NHC(-2)(-3) + NHC(-2)(-3)	3083
4.566NHC(-2)(-3) + NHC(-2)(-3) + e ₆	3087
4.567NHC(-2)(-3) + NHC(-2)(-3) + e ₆ + NHC(-2)(-3)	3087
4.568NHC(-2)(-3) + NHC(-2)(-3) + e ₆ + so ₈	3087
4.569NHC(-2)(-3) + NHC(-2)(-3) + e ₆ + su ₂	3087
4.570NHC(-2)(-3) + NHC(-2)(-3) + e ₆ + su ₂ + su ₂	3089
4.571NHC(-2)(-3) + NHC(-2)(-3) + e ₆ + su ₂ + su ₃	3089
4.572NHC(-2)(-3) + NHC(-2)(-3) + e ₆ + su ₃	3090
4.573NHC(-2)(-3) + NHC(-2)(-3) + e ₆ + su ₃ + su ₂	3090
4.574NHC(-2)(-3) + NHC(-2)(-3) + e ₆ + su ₃ + su ₃	3090
4.575NHC(-2)(-3) + NHC(-2)(-3) + f ₄	3091
4.576NHC(-2)(-3) + NHC(-2)(-3) + f ₄ + so ₈	3091
4.577NHC(-2)(-3) + NHC(-2)(-3) + f ₄ + su ₂	3091
4.578NHC(-2)(-3) + NHC(-2)(-3) + f ₄ + su ₂ + su ₂ →(-3)	3093
4.579NHC(-2)(-3) + NHC(-2)(-3) + f ₄ + su ₂ →(-3)	3093
4.580NHC(-2)(-3) + NHC(-2)(-3) + f ₄ + su ₂ →(-3) + su ₂	3093
4.581NHC(-2)(-3) + NHC(-2)(-3) + f ₄ + su2n3mix	3093
4.582NHC(-2)(-3) + NHC(-2)(-3) + f ₄ + su ₃	3094
4.583NHC(-2)(-3) + NHC(-2)(-3) + so ₈	3094
4.584NHC(-2)(-3) + NHC(-2)(-3) + so ₈ + e ₆	3096
4.585NHC(-2)(-3) + NHC(-2)(-3) + so ₈ + f ₄	3096
4.586NHC(-2)(-3) + NHC(-2)(-3) + so ₈ + so ₈	3096
4.587NHC(-2)(-3) + NHC(-2)(-3) + su ₂	3096
4.588NHC(-2)(-3) + NHC(-2)(-3) + su ₂ + su ₂	3099
4.589NHC(-2)(-3) + NHC(-2)(-3) + su ₂ + su ₂ →(-3)	3100
4.590NHC(-2)(-3) + NHC(-2)(-3) + su ₂ →(-3)	3100
4.591NHC(-2)(-3) + NHC(-2)(-3) + su ₂ →(-3) + su ₂	3101
4.592NHC(-2)(-3) + NHC(-2)(-3) + su2n3mix	3101
4.593NHC(-2)(-3) + NHC(-2)(-3) + su ₃	3101
4.594NHC(-2)(-3) + NHC(-2)(-3) + su ₃ + su ₂	3102
4.595NHC(-2)(-3) + so ₈	3103
4.596NHC(-2)(-3) + so ₈ + e ₆	3123
4.597NHC(-2)(-3) + so ₈ + e ₆ + NHC(-2)(-3)	3124

4.598NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2$	3124
4.599NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$	3128
4.600NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	3130
4.601NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$	3130
4.602NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3)$	3130
4.603NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_3$	3131
4.604NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$	3132
4.605NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_7$	3132
4.606NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_2$	3133
4.607NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$	3133
4.608NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3)$	3133
4.609NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_3$	3134
4.610NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}'_7$	3134
4.611NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_2$	3134
4.612NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$	3135
4.613NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3)$	3135
4.614NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_3$	3135
4.615NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4$	3135
4.616NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \text{NHC}(-2)(-3)$	3137
4.617NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2$	3138
4.618NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$	3144
4.619NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	3145
4.620NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 + \text{su2n3mix}$	3145
4.621NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$	3146
4.622NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$	3146
4.623NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	3147
4.624NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	3148
4.625NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \text{su2n3mix}$	3148
4.626NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_2$	3148
4.627NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3$	3149
4.628NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$	3150
4.629NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	3151
4.630NHC(-2)(-3) + $\mathfrak{so}_8 + \text{NHC}(-2)(-3)$	3151
4.631NHC(-2)(-3) + $\mathfrak{so}_8 + \text{NHC}(-2)(-3) + \mathfrak{e}_6$	3152
4.632NHC(-2)(-3) + $\mathfrak{so}_8 + \text{NHC}(-2)(-3) + \mathfrak{f}_4$	3153
4.633NHC(-2)(-3) + $\mathfrak{so}_8 + \text{NHC}(-2)(-3) + \mathfrak{so}_8$	3153
4.634NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8$	3153
4.635NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \text{NHC}(-2)(-3)$	3157
4.636NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{so}_8$	3157
4.637NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2$	3157
4.638NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$	3170
4.639NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	3173
4.640NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \text{su2n3mix}$	3174
4.641NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$	3175
4.642NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	3178
4.643NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	3179
4.644NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$	3180
4.645NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	3180
4.646NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \text{su2n3mix}$	3180
4.647NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{su}_2$	3182
4.648NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{su}_3$	3183
4.649NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3$	3183
4.650NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$	3186
4.651NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	3189
4.652NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 + \text{su2n3mix}$	3189

4.653NHC(-2)(-3) + s08 + s08 + su3 + su3	3189
4.654NHC(-2)(-3) + s08 + su2	3190
4.655NHC(-2)(-3) + s08 + su2 + e6	3293
4.656NHC(-2)(-3) + s08 + su2 + e6 + su2	3296
4.657NHC(-2)(-3) + s08 + su2 + e6 + su2 + su2	3297
4.658NHC(-2)(-3) + s08 + su2 + e6 + su3	3297
4.659NHC(-2)(-3) + s08 + su2 + f4	3298
4.660NHC(-2)(-3) + s08 + su2 + f4 + su2	3300
4.661NHC(-2)(-3) + s08 + su2 + f4 + su2→(-3)	3301
4.662NHC(-2)(-3) + s08 + su2 + f4 + su2n3mix	3301
4.663NHC(-2)(-3) + s08 + su2 + f4 + su3	3302
4.664NHC(-2)(-3) + s08 + su2 + s08	3302
4.665NHC(-2)(-3) + s08 + su2 + s08 + su2	3312
4.666NHC(-2)(-3) + s08 + su2 + s08 + su2→(-3)	3315
4.667NHC(-2)(-3) + s08 + su2 + s08 + su2n3mix	3316
4.668NHC(-2)(-3) + s08 + su2 + s08 + su3	3317
4.669NHC(-2)(-3) + s08 + su2 + su2	3320
4.670NHC(-2)(-3) + s08 + su2 + su2 + s08	3363
4.671NHC(-2)(-3) + s08 + su2 + su2 + su3	3364
4.672NHC(-2)(-3) + s08 + su2 + su2→(-3)	3365
4.673NHC(-2)(-3) + s08 + su2 + su2→(-3) + s08	3376
4.674NHC(-2)(-3) + s08 + su2 + su2→(-3) + su2→(-3)	3376
4.675NHC(-2)(-3) + s08 + su2 + su2→(-3) + su2n3mix	3378
4.676NHC(-2)(-3) + s08 + su2 + su2n3mix	3381
4.677NHC(-2)(-3) + s08 + su2 + su2n3mix + s08	3417
4.678NHC(-2)(-3) + s08 + su2 + su2n3mix + su2→(-3)	3417
4.679NHC(-2)(-3) + s08 + su2 + su2n3mix + su2n3mix	3421
4.680NHC(-2)(-3) + s08 + su2 + su3	3422
4.681NHC(-2)(-3) + s08 + su2 + su3 + e6	3462
4.682NHC(-2)(-3) + s08 + su2 + su3 + f4	3463
4.683NHC(-2)(-3) + s08 + su2 + su3 + s08	3463
4.684NHC(-2)(-3) + s08 + su2 + su3 + su2	3466
4.685NHC(-2)(-3) + s08 + su2 + (-3)→(-2)	3467
4.686NHC(-2)(-3) + s08 + su2→(-3)	3468
4.687NHC(-2)(-3) + s08 + su2→(-3) + e6	3475
4.688NHC(-2)(-3) + s08 + su2→(-3) + f4	3475
4.689NHC(-2)(-3) + s08 + su2→(-3) + f4 + su2	3476
4.690NHC(-2)(-3) + s08 + su2→(-3) + f4 + su3	3476
4.691NHC(-2)(-3) + s08 + su2→(-3) + s08	3477
4.692NHC(-2)(-3) + s08 + su2→(-3) + s08 + su2	3477
4.693NHC(-2)(-3) + s08 + su2→(-3) + s08 + su2→(-3)	3478
4.694NHC(-2)(-3) + s08 + su2→(-3) + s08 + su3	3478
4.695NHC(-2)(-3) + s08 + su2→(-3) + su2	3478
4.696NHC(-2)(-3) + s08 + su2→(-3) + su2 + s08	3493
4.697NHC(-2)(-3) + s08 + su2→(-3) + su2 + su2→(-3)	3493
4.698NHC(-2)(-3) + s08 + su2→(-3) + su2 + su2n3mix	3495
4.699NHC(-2)(-3) + s08 + su2→(-3) + su2→(-3)	3499
4.700NHC(-2)(-3) + s08 + su2→(-3) + su2→(-3) + s08	3500
4.701NHC(-2)(-3) + s08 + su2→(-3) + su2→(-3) + su2	3500
4.702NHC(-2)(-3) + s08 + su2→(-3) + su2n3mix	3502
4.703NHC(-2)(-3) + s08 + su2→(-3) + su2n3mix + su2	3506
4.704NHC(-2)(-3) + s08 + su2→(-3) + su3	3510
4.705NHC(-2)(-3) + s08 + su2→(-3) + su3 + f4	3512
4.706NHC(-2)(-3) + s08 + su2→(-3) + su3 + s08	3512
4.707NHC(-2)(-3) + s08 + su2→(-3) + su3 + su2	3512

4.708NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2)$	3512
4.709NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_{2n3mix}$	3513
4.710NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_{2n3mix} + \mathfrak{f}_4$	3569
4.711NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_{2n3mix} + \mathfrak{f}_4 + \mathfrak{su}_2$	3570
4.712NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_{2n3mix} + \mathfrak{so}_8$	3570
4.713NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_{2n3mix} + \mathfrak{so}_8 + \mathfrak{su}_2$	3571
4.714NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_{2n3mix} + \mathfrak{so}_8 + \mathfrak{su}_3$	3572
4.715NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_{2n3mix} + \mathfrak{su}_2$	3572
4.716NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_{2n3mix} + \mathfrak{su}_2 + \mathfrak{so}_8$	3611
4.717NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_{2n3mix} + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	3611
4.718NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_{2n3mix} + \mathfrak{su}_2 + \mathfrak{su}_{2n3mix}$	3615
4.719NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_{2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$	3616
4.720NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_{2n3mix} + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	3620
4.721NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_{2n3mix} + \mathfrak{su}_{2n3mix}$	3624
4.722NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_{2n3mix} + \mathfrak{su}_{2n3mix} + \mathfrak{su}_2$	3628
4.723NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_{2n3mix} + \mathfrak{su}_3$	3630
4.724NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_{2n3mix} + \mathfrak{su}_3 + \mathfrak{so}_8$	3638
4.725NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_{2n3mix} + (-3) \rightarrow (-2)$	3638
4.726NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3$	3638
4.727NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6$	3659
4.728NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$	3659
4.729NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_7$	3660
4.730NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}'_7$	3660
4.731NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4$	3660
4.732NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$	3661
4.733NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$	3662
4.734NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8$	3662
4.735NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$	3665
4.736NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	3668
4.737NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_{2n3mix}$	3668
4.738NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3$	3668
4.739NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$	3669
4.740NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6$	3709
4.741NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4$	3710
4.742NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8$	3710
4.743NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	3713
4.744NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	3714
4.745NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4$	3716
4.746NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$	3716
4.747NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	3716
4.748NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_{2n3mix}$	3716
4.749NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_{2n3mix} + \mathfrak{so}_8$	3724
4.750NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3$	3724
4.751NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8$	3730
4.752NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$	3731
4.753NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + (-3) \rightarrow (-2)$	3732
4.754NHC(-2)(-3) + $\mathfrak{so}_8 + (-3) \rightarrow (-2)$	3732
4.755NHC(-2)(-3) + $\mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_2$	3734
4.756NHC(-2)(-3) + $\mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3)$	3736
4.757NHC(-2)(-3) + $\mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_{2n3mix}$	3736
4.758NHC(-2)(-3) + $\mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_3$	3736
4.759NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_{3n2mix}$	3737
4.760NHC(-2)(-3) + $\mathfrak{su}_2$	3737
4.761NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6$	4112
4.762NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{e}_6$	4121

4.818NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$	4282
4.819NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	4290
4.820NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$	4291
4.821NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4$	4299
4.822NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \text{NHC}(-2)(-3)$	4299
4.823NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$	4300
4.824NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	4300
4.825NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3$	4305
4.826NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix}$	4306
4.827NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	4307
4.828NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2$	4310
4.829NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \text{su2n3mix}$	4311
4.830NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{so}_8$	4326
4.831NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_2$	4326
4.832NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$	4333
4.833NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_3$	4334
4.834NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$	4336
4.835NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8$	4348
4.836NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$	4348
4.837NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	4356
4.838NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	4357
4.839NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	4361
4.840NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \text{su2n3mix}$	4361
4.841NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3$	4363
4.842NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + (-3) \rightarrow (-2)$	4366
4.843NHC(-2)(-3) + $\mathfrak{su}_2 + \text{NHC}(-2)(-3)$	4366
4.844NHC(-2)(-3) + $\mathfrak{su}_2 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$	4369
4.845NHC(-2)(-3) + $\mathfrak{su}_2 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$	4370
4.846NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8$	4370
4.847NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6$	4425
4.848NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2$	4427
4.849NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$	4428
4.850NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_3$	4428
4.851NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4$	4429
4.852NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2$	4431
4.853NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$	4432
4.854NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \text{su2n3mix}$	4433
4.855NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3$	4433
4.856NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$	4433
4.857NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2$	4441
4.858NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	4445
4.859NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \text{su2n3mix}$	4446
4.860NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3$	4447
4.861NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$	4450
4.862NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8$	4493
4.863NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$	4494
4.864NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	4495
4.865NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$	4501
4.866NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$	4501
4.867NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix}$	4502
4.868NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \text{su2n3mix}$	4506
4.869NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{so}_8$	4527
4.870NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$	4527
4.871NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \text{su2n3mix} + \text{su2n3mix}$	4530
4.872NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$	4531

4.103 N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₂ + su ₂	5892
4.103 N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₂ + su ₂ → (-3)	5892
4.104 N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₂ + su ₃	5892
4.104 N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₂ → (-3)	5892
4.104 N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₂ → (-3) + su ₂	5893
4.104 N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su _{2n3mix}	5893
4.104 N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₃	5894
4.104 N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ + su ₃ + su ₂	5894
4.104 N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ → (-3)	5894
4.104 N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ → (-3) + su ₂	5895
4.104 N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ → (-3) + su ₂ + su ₂	5895
4.104 N NHC(-2)(-3) + su ₃ + e ₇ + su ₂ → (-3) + su ₃	5896
4.105 N NHC(-2)(-3) + su ₃ + e ₇ + su _{2n3mix}	5896
4.105 N NHC(-2)(-3) + su ₃ + e ₇ + su _{2n3mix} + su ₂	5897
4.105 N NHC(-2)(-3) + su ₃ + e ₇ + su ₃	5897
4.105 N NHC(-2)(-3) + su ₃ + e ₇ + su ₃ + su ₂	5897
4.105 N NHC(-2)(-3) + su ₃ + e ₇ + su ₃ + su ₂ + su ₂	5898
4.105 N NHC(-2)(-3) + su ₃ + e ₇ + su ₃ + su ₂ → (-3)	5898
4.105 N NHC(-2)(-3) + su ₃ + e ₇ + su ₃ + su ₃	5898
4.105 N NHC(-2)(-3) + su ₃ + e' ₇	5898
4.105 N NHC(-2)(-3) + su ₃ + e' ₇ + NHC(-2)(-3)	5899
4.105 N NHC(-2)(-3) + su ₃ + e' ₇ + so ₈	5899
4.106 N NHC(-2)(-3) + su ₃ + e' ₇ + su ₂	5899
4.106 N NHC(-2)(-3) + su ₃ + e' ₇ + su ₂ + su ₂	5902
4.106 N NHC(-2)(-3) + su ₃ + e' ₇ + su ₂ + su ₂ + su ₂	5904
4.106 N NHC(-2)(-3) + su ₃ + e' ₇ + su ₂ + su ₂ + su ₂ → (-3)	5904
4.106 N NHC(-2)(-3) + su ₃ + e' ₇ + su ₂ + su ₂ + su ₃	5904
4.106 N NHC(-2)(-3) + su ₃ + e' ₇ + su ₂ + su ₂ → (-3)	5904
4.106 N NHC(-2)(-3) + su ₃ + e' ₇ + su ₂ + su ₂ → (-3) + su ₂	5905
4.106 N NHC(-2)(-3) + su ₃ + e' ₇ + su ₂ + su _{2n3mix}	5905
4.106 N NHC(-2)(-3) + su ₃ + e' ₇ + su ₂ + su ₃	5905
4.106 N NHC(-2)(-3) + su ₃ + e' ₇ + su ₂ + su ₃ + su ₂	5906
4.107 N NHC(-2)(-3) + su ₃ + e' ₇ + su ₂ → (-3)	5906
4.107 N NHC(-2)(-3) + su ₃ + e' ₇ + su ₂ → (-3) + su ₂	5906
4.107 N NHC(-2)(-3) + su ₃ + e' ₇ + su ₂ → (-3) + su ₂ + su ₂	5907
4.107 N NHC(-2)(-3) + su ₃ + e' ₇ + su ₂ → (-3) + su ₃	5907
4.107 N NHC(-2)(-3) + su ₃ + e' ₇ + su _{2n3mix}	5908
4.107 N NHC(-2)(-3) + su ₃ + e' ₇ + su _{2n3mix} + su ₂	5908
4.107 N NHC(-2)(-3) + su ₃ + e' ₇ + su ₃	5909
4.107 N NHC(-2)(-3) + su ₃ + e' ₇ + su ₃ + su ₂	5909
4.107 N NHC(-2)(-3) + su ₃ + e' ₇ + su ₃ + su ₂ + su ₂	5910
4.107 N NHC(-2)(-3) + su ₃ + e' ₇ + su ₃ + su ₂ → (-3)	5910
4.108 N NHC(-2)(-3) + su ₃ + e' ₇ + su ₃ + su ₃	5910
4.108 N NHC(-2)(-3) + su ₃ + f ₄	5910
4.108 N NHC(-2)(-3) + su ₃ + f ₄ + f ₄	5916
4.108 N NHC(-2)(-3) + su ₃ + f ₄ + NHC(-2)(-3)	5917
4.108 N NHC(-2)(-3) + su ₃ + f ₄ + so ₈	5917
4.108 N NHC(-2)(-3) + su ₃ + f ₄ + so ₈ + su ₂	5919
4.108 N NHC(-2)(-3) + su ₃ + f ₄ + so ₈ + su ₂ → (-3)	5919
4.108 N NHC(-2)(-3) + su ₃ + f ₄ + su ₂	5919
4.108 N NHC(-2)(-3) + su ₃ + f ₄ + su ₂ + so ₈	5945
4.108 N NHC(-2)(-3) + su ₃ + f ₄ + su ₂ + su ₂	5945
4.109 N NHC(-2)(-3) + su ₃ + f ₄ + su ₂ + su ₂ + su ₂ → (-3)	5954
4.109 N NHC(-2)(-3) + su ₃ + f ₄ + su ₂ + su ₂ → (-3)	5955
4.109 N NHC(-2)(-3) + su ₃ + f ₄ + su ₂ + su ₂ → (-3) + su ₂	5959

4.109 N NHC(-2)(-3) + su ₃ + f ₄ + su ₂ + su2n3mix	5959
4.109NHC(-2)(-3) + su ₃ + f ₄ + su ₂ + su ₃	5961
4.109 N NHC(-2)(-3) + su ₃ + f ₄ + su ₂ →(-3)	5964
4.109NHC(-2)(-3) + su ₃ + f ₄ + su ₂ →(-3) + so ₈	5967
4.109 N NHC(-2)(-3) + su ₃ + f ₄ + su ₂ →(-3) + su ₂	5967
4.109 S NHC(-2)(-3) + su ₃ + f ₄ + su ₂ →(-3) + su ₂ + su ₂	5971
4.109 S NHC(-2)(-3) + su ₃ + f ₄ + su ₂ →(-3) + su ₃	5971
4.110 O NHC(-2)(-3) + su ₃ + f ₄ + su2n3mix	5972
4.110NHC(-2)(-3) + su ₃ + f ₄ + su2n3mix + su ₂	5980
4.110 N NHC(-2)(-3) + su ₃ + f ₄ + su2n3mix + su ₃	5982
4.110 S NHC(-2)(-3) + su ₃ + f ₄ + su ₃	5982
4.110 N NHC(-2)(-3) + su ₃ + f ₄ + su ₃ + su ₂	5985
4.110 S NHC(-2)(-3) + su ₃ + f ₄ + su ₃ + su ₂ →(-3)	5988
4.110 O NHC(-2)(-3) + su ₃ + f ₄ + su ₃ + su2n3mix	5989
4.110 N NHC(-2)(-3) + su ₃ + f ₄ + su ₃ + su ₃	5990
4.110 S NHC(-2)(-3) + su ₃ + NHC(-2)(-3)	5990
4.110 S NHC(-2)(-3) + su ₃ + NHC(-2)(-3) + su ₂	5991
4.111 O NHC(-2)(-3) + su ₃ + so ₈	5992
4.111NHC(-2)(-3) + su ₃ + so ₈ + e ₆	6011
4.111 N NHC(-2)(-3) + su ₃ + so ₈ + e ₆ + su ₂	6012
4.111 S NHC(-2)(-3) + su ₃ + so ₈ + e ₇	6012
4.111 N NHC(-2)(-3) + su ₃ + so ₈ + e ₇ '	6012
4.111 S NHC(-2)(-3) + su ₃ + so ₈ + f ₄	6013
4.111 O NHC(-2)(-3) + su ₃ + so ₈ + f ₄ + su ₂	6014
4.111 N NHC(-2)(-3) + su ₃ + so ₈ + f ₄ + su ₂ →(-3)	6014
4.111 S NHC(-2)(-3) + su ₃ + so ₈ + so ₈	6015
4.111 S NHC(-2)(-3) + su ₃ + so ₈ + so ₈ + su ₂	6017
4.112 O NHC(-2)(-3) + su ₃ + so ₈ + so ₈ + su ₂ →(-3)	6020
4.112NHC(-2)(-3) + su ₃ + so ₈ + so ₈ + su2n3mix	6020
4.112 N NHC(-2)(-3) + su ₃ + so ₈ + so ₈ + su ₃	6021
4.112 S NHC(-2)(-3) + su ₃ + so ₈ + su ₂	6021
4.112 N NHC(-2)(-3) + su ₃ + so ₈ + su ₂ + e ₆	6062
4.112 S NHC(-2)(-3) + su ₃ + so ₈ + su ₂ + f ₄	6063
4.112 O NHC(-2)(-3) + su ₃ + so ₈ + su ₂ + so ₈	6063
4.112 N NHC(-2)(-3) + su ₃ + so ₈ + su ₂ + su ₂	6066
4.112 S NHC(-2)(-3) + su ₃ + so ₈ + su ₂ →(-3)	6067
4.112 S NHC(-2)(-3) + su ₃ + so ₈ + su ₂ →(-3) + f ₄	6069
4.113 O NHC(-2)(-3) + su ₃ + so ₈ + su ₂ →(-3) + so ₈	6069
4.113NHC(-2)(-3) + su ₃ + so ₈ + su ₂ →(-3) + su ₂	6069
4.113 N NHC(-2)(-3) + su ₃ + so ₈ + su2n3mix	6069
4.113 S NHC(-2)(-3) + su ₃ + so ₈ + su2n3mix + so ₈	6077
4.113 N NHC(-2)(-3) + su ₃ + so ₈ + su ₃	6078
4.113 S NHC(-2)(-3) + su ₃ + so ₈ + su ₃ + so ₈	6083
4.113 O NHC(-2)(-3) + su ₃ + so ₈ + su ₃ + su ₃	6084
4.113 N NHC(-2)(-3) + su ₃ + so ₈ + (-3)→(-2)	6085
4.113 S NHC(-2)(-3) + su ₃ + su ₂	6086
4.113 S NHC(-2)(-3) + su ₃ + su ₂ + e ₆	6254
4.114 O NHC(-2)(-3) + su ₃ + su ₂ + e ₆ + NHC(-2)(-3)	6260
4.114NHC(-2)(-3) + su ₃ + su ₂ + e ₆ + so ₈	6260
4.114 N NHC(-2)(-3) + su ₃ + su ₂ + e ₆ + su ₂	6260
4.114 S NHC(-2)(-3) + su ₃ + su ₂ + e ₆ + su ₂ + su ₂	6270
4.114 N NHC(-2)(-3) + su ₃ + su ₂ + e ₆ + su ₂ + su ₃	6273
4.114 S NHC(-2)(-3) + su ₃ + su ₂ + e ₆ + su ₃	6274
4.114 O NHC(-2)(-3) + su ₃ + su ₂ + e ₆ + su ₃ + su ₂	6277
4.114 N NHC(-2)(-3) + su ₃ + su ₂ + e ₆ + su ₃ + su ₃	6278

4.120 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$	6545
4.120 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3$	6545
4.120 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$	6546
4.120 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4$	6546
4.120 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$	6547
4.120 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$	6550
4.120 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \text{su2n3mix}$	6550
4.121 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3$	6551
4.121 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8$	6551
4.121 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8$	6556
4.121 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3$	6557
4.121 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$	6558
4.121 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	6562
4.121 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	6563
4.121 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$	6564
4.121 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$	6564
4.121 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8$	6569
4.122 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2)$	6569
4.122 N NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \text{su3n2mix}$	6570
4.122 N NHC(-2)(-3) + $\mathfrak{su}_3 + (-3) \rightarrow (-2)$	6570
4.122 N NHC(-2)(-3) + $\mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{so}_8$	6575
4.122 N NHC(-2)(-3) + $\mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2$	6575
4.122 N NHC(-2)(-3) + $\mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_3$	6577
4.122 N NHC(-2)(-3) + $\mathfrak{su}_3 + \text{su3n2mix}$	6578
4.122 N NHC(-2)(-3) + $\mathfrak{su}_3 + \text{su3n2mix} + \mathfrak{su}_3$	6579

5 NHC (-2)(-3)(-2) External (163 combos, 2345 bases)	6580
5.1 NHC(-2)(-3)(-2)	6580
5.2 NHC(-2)(-3)(-2) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$	6581
5.3 NHC(-2)(-3)(-2) + $\mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$	6581
5.4 NHC(-2)(-3)(-2) + $\mathfrak{su}_2$	6581
5.5 NHC(-2)(-3)(-2) + $\mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	6582
5.6 NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$	6583
5.7 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \text{NHC}(-2)(-3)(-2)$	6583
5.8 NHC(-2)(-3) + $\mathfrak{e}_6 + \text{NHC}(-2)(-3)(-2)$	6583
5.9 NHC(-2)(-3) + $\mathfrak{e}_6 + \text{NHC}(-2)(-3)(-2) + \mathfrak{su}_2$	6583
5.10 NHC(-2)(-3) + $\mathfrak{e}_6 + \text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3$	6584
5.11 NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)(-2)$	6584
5.12 NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)(-2)$	6584
5.13 NHC(-2)(-3) + $\mathfrak{e}_7 + \text{NHC}(-2)(-3)(-2)$	6584
5.14 NHC(-2)(-3) + $\mathfrak{e}'_7 + \text{NHC}(-2)(-3)(-2)$	6585
5.15 NHC(-2)(-3) + $\mathfrak{e}_8 + \text{NHC}(-2)(-3)(-2)$	6585
5.16 NHC(-2)(-3) + $\mathfrak{f}_4 + \text{NHC}(-2)(-3)(-2)$	6585
5.17 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \text{NHC}(-2)(-3)(-2)$	6586
5.18 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \text{NHC}(-2)(-3)(-2)$	6586
5.19 NHC(-2)(-3)(-2)	6586
5.20 NHC(-2)(-3)(-2) + $\mathfrak{e}_6$	6603
5.21 NHC(-2)(-3)(-2) + $\mathfrak{e}_6 + \mathfrak{e}_6$	6605
5.22 NHC(-2)(-3)(-2) + $\mathfrak{e}_6 + \mathfrak{f}_4$	6605
5.23 NHC(-2)(-3)(-2) + $\mathfrak{e}_6 + \text{NHC}(-2)(-2)(-3)$	6605
5.24 NHC(-2)(-3)(-2) + $\mathfrak{e}_6 + \text{NHC}(-2)(-3)$	6605
5.25 NHC(-2)(-3)(-2) + $\mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$	6606
5.26 NHC(-2)(-3)(-2) + $\mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$	6606
5.27 NHC(-2)(-3)(-2) + $\mathfrak{e}_6 + \mathfrak{so}_8$	6606
5.28 NHC(-2)(-3)(-2) + $\mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$	6607

5.29	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3$	6607
5.30	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2$	6607
5.31	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$	6613
5.32	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8$	6613
5.33	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$	6614
5.34	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$	6622
5.35	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	6623
5.36	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$	6623
5.37	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$	6627
5.38	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$	6627
5.39	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3)$	6627
5.40	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	6628
5.41	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$	6628
5.42	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \text{su2n3mix}$	6628
5.43	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3$	6628
5.44	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$	6630
5.45	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8$	6630
5.46	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$	6630
5.47	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	6634
5.48	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$	6634
5.49	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$	6635
5.50	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$	6635
5.51	$\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$	6636
5.52	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4$	6636
5.53	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{e}_6$	6637
5.54	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{f}_4$	6638
5.55	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \text{NHC}(-2)(-3)$	6638
5.56	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{so}_8$	6638
5.57	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$	6639
5.58	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	6640
5.59	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3$	6640
5.60	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2$	6640
5.61	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8$	6647
5.62	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$	6648
5.63	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$	6654
5.64	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	6655
5.65	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2 + \text{su2n3mix}$	6656
5.66	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$	6659
5.67	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$	6662
5.68	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$	6662
5.69	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$	6663
5.70	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	6663
5.71	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	6665
5.72	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \text{su2n3mix}$	6665
5.73	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_2$	6667
5.74	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_3$	6670
5.75	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_3$	6671
5.76	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8$	6673
5.77	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$	6673
5.78	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	6676
5.79	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	6677
5.80	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_3 + \text{su2n3mix}$	6677
5.81	$\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3$	6677
5.82	$\text{NHC}(-2)(-3)(-2) + \text{NHC}(-2)(-3)(-2)$	6678
5.83	$\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8$	6678

5.84	NHC(-2)(-3)(-2) + s0 ₈ + e ₆	6683
5.85	NHC(-2)(-3)(-2) + s0 ₈ + e ₆ + su ₂	6683
5.86	NHC(-2)(-3)(-2) + s0 ₈ + e ₆ + su ₃	6684
5.87	NHC(-2)(-3)(-2) + s0 ₈ + f ₄	6684
5.88	NHC(-2)(-3)(-2) + s0 ₈ + f ₄ + su ₂	6684
5.89	NHC(-2)(-3)(-2) + s0 ₈ + f ₄ + su ₂ →(-3)	6685
5.90	NHC(-2)(-3)(-2) + s0 ₈ + f ₄ + su ₃	6685
5.91	NHC(-2)(-3)(-2) + s0 ₈ + s0 ₈	6685
5.92	NHC(-2)(-3)(-2) + s0 ₈ + s0 ₈ + su ₂	6686
5.93	NHC(-2)(-3)(-2) + s0 ₈ + s0 ₈ + su ₂ →(-3)	6687
5.94	NHC(-2)(-3)(-2) + s0 ₈ + s0 ₈ + su2n3mix	6688
5.95	NHC(-2)(-3)(-2) + s0 ₈ + s0 ₈ + su ₃	6688
5.96	NHC(-2)(-3)(-2) + s0 ₈ + su ₂	6689
5.97	NHC(-2)(-3)(-2) + s0 ₈ + su ₂ →(-3)	6696
5.98	NHC(-2)(-3)(-2) + s0 ₈ + su ₂ →(-3) + su ₂	6697
5.99	NHC(-2)(-3)(-2) + s0 ₈ + su ₂ →(-3) + su ₂ →(-3)	6697
5.100	NHC(-2)(-3)(-2) + s0 ₈ + su ₂ →(-3) + su ₂ →(-3) + su ₂	6697
5.101	NHC(-2)(-3)(-2) + s0 ₈ + su ₂ →(-3) + su ₂ →(-3) + su ₂ + su ₂	6698
5.102	NHC(-2)(-3)(-2) + s0 ₈ + su ₂ →(-3) + su2n3mix	6699
5.103	NHC(-2)(-3)(-2) + s0 ₈ + su ₂ →(-3) + su2n3mix + su ₂	6699
5.104	NHC(-2)(-3)(-2) + s0 ₈ + su2n3mix	6700
5.105	NHC(-2)(-3)(-2) + s0 ₈ + su2n3mix + su ₂	6703
5.106	NHC(-2)(-3)(-2) + s0 ₈ + su2n3mix + su ₂ + su ₂	6704
5.107	NHC(-2)(-3)(-2) + s0 ₈ + su2n3mix + su ₂ →(-3)	6705
5.108	NHC(-2)(-3)(-2) + s0 ₈ + su2n3mix + su ₂ →(-3) + su ₂	6705
5.109	NHC(-2)(-3)(-2) + s0 ₈ + su2n3mix + su2n3mix	6705
5.110	NHC(-2)(-3)(-2) + s0 ₈ + su ₃	6706
5.111	NHC(-2)(-3)(-2) + s0 ₈ + su ₃ + e ₆	6709
5.112	NHC(-2)(-3)(-2) + s0 ₈ + su ₃ + f ₄	6710
5.113	NHC(-2)(-3)(-2) + s0 ₈ + su ₃ + s0 ₈	6710
5.114	NHC(-2)(-3)(-2) + s0 ₈ + (-3)→(-2)	6710
5.115	NHC(-2)(-3)(-2) + su ₂	6711
5.116	NHC(-2)(-3)(-2) + su ₂ + s0 ₈	6729
5.117	NHC(-2)(-3)(-2) + su ₂ + su ₂	6729
5.118	NHC(-2)(-3)(-2) + su ₂ + su ₂ + su ₂ →(-3)	6734
5.119	NHC(-2)(-3)(-2) + su ₂ + su ₂ →(-3)	6734
5.120	NHC(-2)(-3)(-2) + su ₂ + su ₂ →(-3) + su ₂	6742
5.121	NHC(-2)(-3)(-2) + su ₂ + su ₂ →(-3) + su ₂ + su ₂	6752
5.122	NHC(-2)(-3)(-2) + su ₂ + su ₂ →(-3) + su ₂ →(-3)	6752
5.123	NHC(-2)(-3)(-2) + su ₂ + su ₂ →(-3) + su ₂ →(-3) + su ₂	6755
5.124	NHC(-2)(-3)(-2) + su ₂ + su ₂ →(-3) + su ₂ →(-3) + su ₂ + su ₂	6758
5.125	NHC(-2)(-3)(-2) + su ₂ + su ₂ →(-3) + su2n3mix	6758
5.126	NHC(-2)(-3)(-2) + su ₂ + su ₂ →(-3) + su2n3mix + su ₂	6764
5.127	NHC(-2)(-3)(-2) + su ₂ + su ₂ →(-3) + su2n3mix + su ₂ + su ₂	6767
5.128	NHC(-2)(-3)(-2) + su ₂ + su2n3mix	6767
5.129	NHC(-2)(-3)(-2) + su ₂ + su2n3mix + su ₂	6797
5.130	NHC(-2)(-3)(-2) + su ₂ + su2n3mix + su ₂ + su ₂	6802
5.131	NHC(-2)(-3)(-2) + su ₂ + su2n3mix + su ₂ →(-3)	6802
5.132	NHC(-2)(-3)(-2) + su ₂ + su2n3mix + su ₂ →(-3) + su ₂	6808
5.133	NHC(-2)(-3)(-2) + su ₂ + su2n3mix + su ₂ →(-3) + su ₂ + su ₂	6810
5.134	NHC(-2)(-3)(-2) + su ₂ + su2n3mix + su2n3mix	6810
5.135	NHC(-2)(-3)(-2) + su ₃	6812
5.136	NHC(-2)(-3)(-2) + su ₃ + e ₆	6819
5.137	NHC(-2)(-3)(-2) + su ₃ + e ₆ + NHC(-2)(-3)	6820
5.138	NHC(-2)(-3)(-2) + su ₃ + e ₆ + s0 ₈	6820

5.139	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$	6821
5.140	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$	6824
5.141	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$	6824
5.142	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3$	6824
5.143	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$	6825
5.144	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$	6825
5.145	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{f}_4$	6825
5.146	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8$	6826
5.147	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$	6826
5.148	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$	6829
5.149	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$	6829
5.150	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \text{su2n3mix}$	6830
5.151	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3$	6830
5.152	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{so}_8$	6831
5.153	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{e}_6$	6833
5.154	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4$	6833
5.155	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8$	6834
5.156	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{su}_2$	6834
5.157	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	6836
5.158	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	6836
5.159	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	6837
5.160	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	6837
5.161	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	6837
5.162	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$	6838
5.163	NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \text{su2n3mix}$	6838

6	NHC (-2)(-2)(-3) External (186 combos, 2947 bases)	6839
6.1	NHC(-2)(-2)(-3)	6839
6.2	NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2$	6840
6.3	NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	6840
6.4	NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \text{su2n3mix} + \mathfrak{su}_2$	6841
6.5	NHC(-2)(-2)(-3)	6841
6.6	NHC(-2)(-2)(-3) + $\mathfrak{e}_6$	6885
6.7	NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{e}_6$	6887
6.8	NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{f}_4$	6887
6.9	NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \text{NHC}(-2)(-2)(-3)$	6888
6.10	NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \text{NHC}(-2)(-3)$	6888
6.11	NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$	6889
6.12	NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$	6889
6.13	NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{so}_8$	6890
6.14	NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$	6891
6.15	NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$	6892
6.16	NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	6893
6.17	NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3$	6893
6.18	NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2$	6894
6.19	NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$	6903
6.20	NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8$	6903
6.21	NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$	6905
6.22	NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$	6905
6.23	NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$	6919
6.24	NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$	6919
6.25	NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	6922
6.26	NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \text{su2n3mix}$	6923
6.27	NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$	6923
6.28	NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$	6930

6.29	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$	6933
6.30	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3)$	6934
6.31	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$	6935
6.32	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	6935
6.33	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$	6935
6.34	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	6936
6.35	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \text{su2n3mix}$	6936
6.36	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \text{su2n3mix} + \mathfrak{su}_2$	6937
6.37	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \text{su2n3mix} + \mathfrak{su}_3$	6937
6.38	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3$	6937
6.39	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$	6941
6.40	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8$	6941
6.41	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$	6941
6.42	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	6949
6.43	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$	6952
6.44	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	6953
6.45	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \text{su2n3mix}$	6953
6.46	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$	6953
6.47	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$	6954
6.48	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$	6955
6.49	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + (-3) \rightarrow (-2)$	6956
6.50	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_7$	6956
6.51	$\text{NHC}(-2)(-2)(-3) + \mathfrak{e}'_7$	6956
6.52	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4$	6956
6.53	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{e}_6$	6958
6.54	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{f}_4$	6959
6.55	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_2$	6959
6.56	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$	6959
6.57	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \text{NHC}(-2)(-3)$	6959
6.58	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8$	6959
6.59	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$	6961
6.60	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	6962
6.61	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8 + \text{su2n3mix}$	6963
6.62	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3$	6963
6.63	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2$	6964
6.64	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4$	6973
6.65	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8$	6973
6.66	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	6975
6.67	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	6978
6.68	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \text{su2n3mix}$	6978
6.69	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_3$	6983
6.70	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$	6984
6.71	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	6987
6.72	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \text{su2n3mix}$	6987
6.73	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$	6988
6.74	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4$	6989
6.75	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$	6989
6.76	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	6989
6.77	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3$	6992
6.78	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$	6993
6.79	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2$	6994
6.80	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \text{su2n3mix}$	6995
6.81	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{so}_8$	7000
6.82	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_2$	7000
6.83	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_2 + \mathfrak{su}_3$	7005

6.84	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_3$	7005
6.85	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_3 + \mathfrak{su}_2$	7007
6.86	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3$	7007
6.87	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8$	7012
6.88	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$	7012
6.89	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	7015
6.90	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \text{su2n3mix}$	7016
6.91	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	7016
6.92	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	7017
6.93	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \text{su2n3mix}$	7018
6.94	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \text{su2n3mix} + \mathfrak{su}_2$	7019
6.95	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3$	7020
6.96	$\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + (-3) \rightarrow (-2)$	7021
6.97	$\text{NHC}(-2)(-2)(-3) + \text{NHC}(-2)(-3)$	7021
6.98	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8$	7021
6.99	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}_6$	7028
6.100	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2$	7028
6.101	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$	7030
6.102	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_3$	7030
6.103	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4$	7031
6.104	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2$	7031
6.105	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$	7033
6.106	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \text{su2n3mix}$	7033
6.107	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3$	7033
6.108	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8$	7034
6.109	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2$	7035
6.110	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	7038
6.111	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8 + \text{su2n3mix}$	7039
6.112	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3$	7039
6.113	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2$	7041
6.114	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8$	7054
6.115	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$	7055
6.116	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$	7057
6.117	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$	7057
6.118	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix}$	7057
6.119	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix}$	7058
6.120	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{so}_8$	7064
6.121	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$	7064
6.122	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + \text{su2n3mix}$	7065
6.123	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3$	7066
6.124	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6$	7073
6.125	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4$	7073
6.126	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8$	7073
6.127	$\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + (-3) \rightarrow (-2)$	7074
6.128	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2$	7075
6.129	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2 + \mathfrak{so}_8$	7092
6.130	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$	7094
6.131	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2$	7095
6.132	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2 + \text{su2n3mix}$	7095
6.133	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3$	7096
6.134	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$	7096
6.135	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3$	7096
6.136	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6$	7113
6.137	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \text{NHC}(-2)(-3)$	7114
6.138	$\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8$	7115

6.139NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$	7115
6.140NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$	7122
6.141NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$	7124
6.142NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3$	7125
6.143NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$	7126
6.144NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$	7127
6.145NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4$	7127
6.146NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8$	7129
6.147NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$	7129
6.148NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$	7132
6.149NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \text{su2n3mix}$	7133
6.150NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$	7133
6.151NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	7134
6.152NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \text{su2n3mix}$	7135
6.153NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_2$	7136
6.154NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3$	7136
6.155NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \text{NHC}(-2)(-3)$	7137
6.156NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8$	7138
6.157NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{e}_6$	7143
6.158NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4$	7143
6.159NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8$	7144
6.160NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2$	7145
6.161NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$	7147
6.162NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$	7147
6.163NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$	7147
6.164NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \text{su2n3mix}$	7151
6.165NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3$	7153
6.166NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + (-3) \rightarrow (-2)$	7153
6.167NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2$	7153
6.168NHC(-2)(-3) + $\mathfrak{e}_6 + \text{NHC}(-2)(-2)(-3)$	7154
6.169NHC(-2)(-3) + $\mathfrak{e}_6 + \text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2$	7155
6.170NHC(-2)(-3) + $\mathfrak{e}_6 + \text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3$	7156
6.171NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-2)(-3)$	7156
6.172NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_3 + \text{NHC}(-2)(-2)(-3)$	7156
6.173NHC(-2)(-3) + $\mathfrak{e}_7 + \text{NHC}(-2)(-2)(-3)$	7157
6.174NHC(-2)(-3) + $\mathfrak{e}_7 + \text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2$	7158
6.175NHC(-2)(-3) + $\mathfrak{e}_7 + \mathfrak{su}_2 + \text{NHC}(-2)(-2)(-3)$	7158
6.176NHC(-2)(-3) + $\mathfrak{e}'_7 + \text{NHC}(-2)(-2)(-3)$	7158
6.177NHC(-2)(-3) + $\mathfrak{e}'_7 + \text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2$	7159
6.178NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_2 + \text{NHC}(-2)(-2)(-3)$	7159
6.179NHC(-2)(-3) + $\mathfrak{e}_8 + \text{NHC}(-2)(-2)(-3)$	7159
6.180NHC(-2)(-3) + $\mathfrak{e}_8 + \text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2$	7160
6.181NHC(-2)(-3) + $\mathfrak{e}_8 + \mathfrak{su}_2 + \text{NHC}(-2)(-2)(-3)$	7160
6.182NHC(-2)(-3) + $\mathfrak{f}_4 + \text{NHC}(-2)(-2)(-3)$	7160
6.183NHC(-2)(-3) + $\text{NHC}(-2)(-2)(-3)$	7160
6.184NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \text{NHC}(-2)(-2)(-3)$	7160
6.185NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \text{NHC}(-2)(-2)(-3)$	7161
6.186NHC(-2)(-3) + $\mathfrak{su}_3 + \text{NHC}(-2)(-2)(-3)$	7161

7 NHC (-2)(-4) External (1 combos, 1 bases)

7162

7.1 nhc ₂₄	7162
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1 $\hat{1}$ External (87 combos, 16930 bases)

1.1 $\mathbf{e}_6 + \hat{1} \rightarrow (-1)$

$$T = 3 \quad [0] \quad 1^1 1^1 + \mathbf{e}_6 + \hat{1} \rightarrow 1 \quad T = 3, T_{\max} = 184, \Delta = -291, \det = -6, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 184, T \in [13, 184]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & -1 & -1 \\ 2 & -3 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 4 \quad [1] \quad 1^1 21^1 + \mathbf{e}_6 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 173, \Delta = -262, \det = 5, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 173, T \in [13, 173]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & -1 & -1 \\ 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [2] \quad 1^1 221^1 + \mathbf{e}_6 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 101, \Delta = -233, \det = -4, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13, T_{\max} = 101, T \in [13, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & -1 & -1 \\ 2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [3] \quad 1^1 2221^1 + \mathbf{e}_6 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 100, \Delta = -204, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [4] \quad 1^1 22221^1 + \mathbf{e}_6 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 100, \Delta = -175, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^1 23131^1 + \mathbf{e}_6 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 101, \Delta = -192, \det = -3, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & 3 & 3 \\ 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^1 3 \quad 21^1 + \mathbf{e}_6 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 101, \Delta = -186, \det = -3, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 1 & -1 \\ 1 & 1 & 2 & 2 \\ 2 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [8] \quad 1^1 231321^1 + \mathbf{e}_6 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 101, \Delta = -164, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^1 223131^1 + \mathbf{e}_6 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 101, \Delta = -163, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & 4 & 4 \\ 2 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [9] \quad 1^1 3141321^1 + \mathbf{e}_6 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 102, \Delta = -162, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [11] \quad 1^1 23215131^1 + \mathbf{e}_6 + \hat{1} \rightarrow 1 \quad T = 10, T_{\max} = 102, \Delta = -159, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 17, T_{\max} = 102, T \in [17, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & 4 & 4 \\ 2 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1 \overset{1}{5} \overset{1}{131} \overset{1}{+ \mathbf{e}_6 + \hat{1} \rightarrow 1} \quad T = 10, \ T_{\max} = 102, \ \Delta = -157, \ \det = 2, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 17, \ T_{\max} = 102, \ T \in [17, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 1 & -1 \\ 1 & 1 & 3 & 3 \\ 2 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [12] \ 1 \overset{1}{6} \overset{1}{131} \overset{1}{+ \mathbf{e}_6 + \hat{1} \rightarrow 1} \quad T = 11, \ T_{\max} = 102, \ \Delta = -156, \ \det = -2, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 18, \ T_{\max} = 102, \ T \in [18, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

1.2 $\mathbf{e}_7 + \hat{1} \rightarrow (-1)$

$$T = 3 \quad [0] \ 1 \overset{1}{1} \overset{1}{+ \mathbf{e}_7 + \hat{1} \rightarrow 1} \quad T = 3, \ T_{\max} = 190, \ \Delta = -346, \ \det = -8, \ (n_+, n_-, n_0) = (1, 3, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.50 \rightarrow 15, \ T_{\max} = 190, \ T \in [15, 190]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -7 & 0 & -5 \\ 1 & 0 & -1 & -1 \\ 2 & -5 & -1 & -5 \end{pmatrix}_{b_0}$$

$$T = 4 \quad [1] \ 1 \overset{1}{21} \overset{1}{+ \mathbf{e}_7 + \hat{1} \rightarrow 1} \quad T = 4, \ T_{\max} = 188, \ \Delta = -317, \ \det = 7, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.57 \rightarrow 15, \ T_{\max} = 188, \ T \in [15, 188]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & -1 & -1 \\ 2 & -4 & -1 & -4 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [2] \ 1 \overset{1}{221} \overset{1}{+ \mathbf{e}_7 + \hat{1} \rightarrow 1} \quad T = 5, \ T_{\max} = 186, \ \Delta = -288, \ \det = -6, \ (n_+, n_-, n_0) = (1, 5, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.67 \rightarrow 15, \ T_{\max} = 186, \ T \in [15, 186]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & -1 & -1 \\ 2 & -3 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [3] \ 1 \overset{1}{2221} \overset{1}{+ \mathbf{e}_7 + \hat{1} \rightarrow 1} \quad T = 6, \ T_{\max} = 170, \ \Delta = -259, \ \det = 5, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.80 \rightarrow 15, \ T_{\max} = 170, \ T \in [15, 170]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & -1 & -1 \\ 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [4] \ 1 \overset{1}{22221} \overset{1}{+ \mathbf{e}_7 + \hat{1} \rightarrow 1} \quad T = 7, \ T_{\max} = 102, \ \Delta = -230, \ \det = -4, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & -1 & -1 \\ 2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1 \overset{1}{3} \overset{1}{21} \overset{1}{+ \mathbf{e}_7 + \hat{1} \rightarrow 1} \quad T = 7, \ T_{\max} = 114, \ \Delta = -241, \ \det = -5, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.80 \rightarrow 16, \ T_{\max} = 114, \ T \in [16, 114]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 1 & -3 \\ 1 & 1 & 2 & 2 \\ 2 & -3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [6] \ 1 \overset{1}{222221} \overset{1}{+ \mathbf{e}_7 + \hat{1} \rightarrow 1} \quad T = 8, \ T_{\max} = 102, \ \Delta = -201, \ \det = 3, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.33 \rightarrow 16, \ T_{\max} = 102, \ T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1 \overset{1}{231321} \overset{1}{+ \mathbf{e}_7 + \hat{1} \rightarrow 1} \quad T = 8, \ T_{\max} = 103, \ \Delta = -219, \ \det = 4, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & -1 & -1 \\ 2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \ 1 \overset{1}{2222221} \overset{1}{+ \mathbf{e}_7 + \hat{1} \rightarrow 1} \quad T = 9, \ T_{\max} = 102, \ \Delta = -172, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 16, \ T_{\max} = 102, \ T \in [16, 102]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[9] $1^1 2231321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1}$ $T = 9$, $T_{\max} = 103$, $\Delta = -190$, $\det = -3$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 16.33 \rightarrow 17$, $T_{\max} = 103$, $T \in [17, 103]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[10] $1^1 2231321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1}$ $T = 9$, $T_{\max} = 103$, $\Delta = -190$, $\det = -3$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 16.33 \rightarrow 17$, $T_{\max} = 103$, $T \in [17, 103]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 5 & 5 \\ 2 & -6 & 5 & -1 \end{pmatrix}_{b_0}$$

$T = 10$ [11] $1^1 22313221^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1}$ $T = 10$, $T_{\max} = 102$, $\Delta = -161$, $\det = 2$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 17$, $T_{\max} = 102$, $T \in [17, 102]$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[12] $1^1 23141321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1}$ $T = 10$, $T_{\max} = 104$, $\Delta = -189$, $\det = 3$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 17.33 \rightarrow 18$, $T_{\max} = 104$, $T \in [18, 104]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 1 & 1 \\ 2 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ [13] $1^1 231413221^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1}$ $T = 11$, $T_{\max} = 103$, $\Delta = -160$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 18$, $T_{\max} = 103$, $T \in [18, 103]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 2 & 2 \\ 2 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

[14] $1^1 231413221^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1}$ $T = 11$, $T_{\max} = 103$, $\Delta = -160$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 18$, $T_{\max} = 103$, $T \in [18, 103]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & 3 & 3 \\ 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[15] $1^1 231512321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1}$ $T = 11$, $T_{\max} = 105$, $\Delta = -184$, $\det = -3$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 18.33 \rightarrow 19$, $T_{\max} = 105$, $T \in [19, 105]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 1 & -2 \\ 1 & 1 & 3 & 3 \\ 2 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

[17] $1^1 231512321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1}$ $T = 11$, $T_{\max} = 105$, $\Delta = -186$, $\det = -3$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 18.33 \rightarrow 19$, $T_{\max} = 105$, $T \in [19, 105]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 5 & 5 \\ 2 & -6 & 5 & -1 \end{pmatrix}_{b_0}$$

[16] $1^1 231512321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1}$ $T = 11$, $T_{\max} = 105$, $\Delta = -186$, $\det = -3$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 18.33 \rightarrow 19$, $T_{\max} = 105$, $T \in [19, 105]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$T = 12$ [22] $1^1 2231512321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1}$ $T = 12$, $T_{\max} = 104$, $\Delta = -157$, $\det = 2$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 19$, $T_{\max} = 104$, $T \in [19, 104]$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[21] $1^1 2314141321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1}$ $T = 12$, $T_{\max} = 104$, $\Delta = -159$, $\det = 2$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 19$, $T_{\max} = 104$, $T \in [19, 104]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

[20] $1^1 2231512321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow \mathbf{1}$ $T = 12$, $T_{\max} = 104$, $\Delta = -157$, $\det = 2$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 19$, $T_{\max} = 104$, $T \in [19, 104]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 6 & 6 \\ 2 & -6 & 6 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{1^{12321}} 1^5 1321^1 + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1$ $T = 12$, $T_{\max} = 103$, $\Delta = -155$, $\det = 2$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 19$, $T_{\max} = 103$, $T \in [19, 103]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 1 & -2 \\ 1 & 1 & 4 & 4 \\ 2 & -2 & 4 & 0 \end{pmatrix}_{b_0}$

[19] $1^{1^{12321}} 1^5 1321^1 + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1$ $T = 12$, $T_{\max} = 103$, $\Delta = -155$, $\det = 2$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 19$, $T_{\max} = 103$, $T \in [19, 103]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 1 & -1 \\ 1 & 1 & 3 & 3 \\ 2 & -1 & 3 & 0 \end{pmatrix}_{b_0}$

[23] $1^{1^{12321612321}} 1^1 + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1$ $T = 12$, $T_{\max} = 106$, $\Delta = -185$, $\det = 3$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 19.33 \rightarrow 20$, $T_{\max} = 106$, $T \in [20, 106]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 1 & 1 \\ 2 & -2 & 1 & -1 \end{pmatrix}_{b_0}$

$T = 13$ [0] $1^{1^{12321}} 1^7 12321^1 + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1$ $T = 13$, $T_{\max} = 107$, $\Delta = -183$, $\det = -3$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 20.33 \rightarrow 21$, $T_{\max} = 107$, $T \in [21, 107]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 1 & 1 \\ 2 & -2 & 1 & -1 \end{pmatrix}_{b_0}$

$T = 14$ [1] $1^{1^{1231513151321}} 1^1 + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1$ $T = 14$, $T_{\max} = 106$, $\Delta = -157$, $\det = 2$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 21$, $T_{\max} = 106$, $T \in [21, 106]$
 $\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$

[2] $1^{1^{12321}} 1^8 12321^1 + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1$ $T = 14$, $T_{\max} = 107$, $\Delta = -182$, $\det = 3$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 21.33 \rightarrow 22$, $T_{\max} = 107$, $T \in [22, 107]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 1 & -2 \\ 1 & 1 & 3 & 3 \\ 2 & -2 & 3 & -1 \end{pmatrix}_{b_0}$

[3] $21^{1^{12321}} 1^8 12321^1 + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1$ $T = 14$, $T_{\max} = 107$, $\Delta = -182$, $\det = 3$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 21.33 \rightarrow 22$, $T_{\max} = 107$, $T \in [22, 107]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 1 & 1 \\ 2 & -2 & 1 & -1 \end{pmatrix}_{b_0}$

$T = 15$ [5] $1^{1^{12321613151321}} 1^1 + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1$ $T = 15$, $T_{\max} = 106$, $\Delta = -156$, $\det = -2$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 22$, $T_{\max} = 106$, $T \in [22, 106]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & 3 & 3 \\ 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$

[4] $1^{1^{12321613151321}} 1^1 + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1$ $T = 15$, $T_{\max} = 106$, $\Delta = -156$, $\det = -2$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 22$, $T_{\max} = 106$, $T \in [22, 106]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 2 & 2 \\ 2 & -2 & 2 & 0 \end{pmatrix}_{b_0}$

$T = 16$ [7] $1^{1^{123216131612321}} 1^1 + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1$ $T = 16$, $T_{\max} = 107$, $\Delta = -155$, $\det = 2$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 23$, $T_{\max} = 107$, $T \in [23, 107]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$

[6] $1^{1^{123217123151321}} 1^1 + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1$ $T = 16$, $T_{\max} = 107$, $\Delta = -155$, $\det = 2$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 23$, $T_{\max} = 107$, $T \in [23, 107]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 2 & 2 \\ 2 & -2 & 2 & 0 \end{pmatrix}_{b_0}$

[8] $1^{1^{123217123151321}} 1^1 + \mathbf{e}_7 + \hat{\mathbf{1}} \rightarrow 1$ $T = 16$, $T_{\max} = 107$, $\Delta = -155$, $\det = 2$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 23$, $T_{\max} = 107$, $T \in [23, 107]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & 3 & 3 \\ 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$T = 17 \quad [9] \quad 1^{12321}81223151321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 17, \quad T_{\max} = 108, \quad \Delta = -154, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 17, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 24, \quad T_{\max} = 108, \quad T \in [24, 108]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & 3 & 3 \\ 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{123151321}8^{12321} + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 17, \quad T_{\max} = 108, \quad \Delta = -154, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 17, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 24, \quad T_{\max} = 108, \quad T \in [24, 108]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 2 & 2 \\ 2 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{12321}81223151321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 17, \quad T_{\max} = 108, \quad \Delta = -154, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 17, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 24, \quad T_{\max} = 108, \quad T \in [24, 108]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 2 & 2 \\ 2 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{123151321}8^{12321} + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 17, \quad T_{\max} = 108, \quad \Delta = -154, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 17, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 24, \quad T_{\max} = 108, \quad T \in [24, 108]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & 3 & 3 \\ 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$T = 18 \quad [13] \quad 1^{12321}712321712321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 18, \quad T_{\max} = 109, \quad \Delta = -154, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 25, \quad T_{\max} = 109, \quad T \in [25, 109]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$T = 19 \quad [14] \quad 1^{12321}8^{12321}712321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 19, \quad T_{\max} = 110, \quad \Delta = -153, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 26, \quad T_{\max} = 110, \quad T \in [26, 110]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & 4 & 4 \\ 2 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{12321}8^{12321}712321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 19, \quad T_{\max} = 110, \quad \Delta = -153, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 26, \quad T_{\max} = 110, \quad T \in [26, 110]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$T = 20 \quad [16] \quad 1^{12321}8^{12321}1^{12321}812321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 20, \quad T_{\max} = 111, \quad \Delta = -152, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 20, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 27, \quad T_{\max} = 111, \quad T \in [27, 111]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 1 & -1 \\ 1 & 1 & 3 & 3 \\ 2 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$T = 22 \quad [17] \quad 1^{12321}8123151321812321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 22, \quad T_{\max} = 113, \quad \Delta = -153, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 22, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 29, \quad T_{\max} = 113, \quad T \in [29, 113]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$T = 24 \quad [0] \quad 1^{12321}812321712321812321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 24, \quad T_{\max} = 115, \quad \Delta = -152, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 24, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 31, \quad T_{\max} = 115, \quad T \in [31, 115]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$T = 25 \quad [1] \quad 1^{1^{2321812321}} \quad 12321812321^1 + \mathbf{e}_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 25, \quad T_{\max} = 116, \quad \Delta = -151, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 32, \quad T_{\max} = 116, \quad T \in [32, 116]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 2 & 2 \\ 2 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

1.3 $\mathbf{e}'_7 + \hat{\mathbf{i}} \rightarrow (-1)$

$$T = 3 \quad [0] \quad 1^{1^1} + \mathbf{e}'_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 3, \quad T_{\max} = 187, \quad \Delta = -318, \quad \det = -7, \quad (n_+, n_-, n_0) = (1, 3, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.57 \rightarrow 14, \quad T_{\max} = 187, \quad T \in [14, 187]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & -1 & -1 \\ 2 & -4 & -1 & -4 \end{pmatrix}_{b_0}$$

$$T = 4 \quad [1] \quad 1^{1^2} + \mathbf{e}'_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 4, \quad T_{\max} = 185, \quad \Delta = -289, \quad \det = 6, \quad (n_+, n_-, n_0) = (1, 4, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.67 \rightarrow 14, \quad T_{\max} = 185, \quad T \in [14, 185]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & -1 & -1 \\ 2 & -3 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [2] \quad 1^{1^2} + \mathbf{e}'_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 5, \quad T_{\max} = 171, \quad \Delta = -260, \quad \det = -5, \quad (n_+, n_-, n_0) = (1, 5, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.80 \rightarrow 14, \quad T_{\max} = 171, \quad T \in [14, 171]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & -1 & -1 \\ 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [3] \quad 1^{1^2} + \mathbf{e}'_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 6, \quad T_{\max} = 101, \quad \Delta = -231, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & -1 & -1 \\ 2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [4] \quad 1^{1^2} + \mathbf{e}'_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 7, \quad T_{\max} = 101, \quad \Delta = -202, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.33 \rightarrow 15, \quad T_{\max} = 101, \quad T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1^2} \quad 1^3 \quad 21^1 + \mathbf{e}'_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 7, \quad T_{\max} = 102, \quad \Delta = -213, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 1 & -2 \\ 1 & 1 & 2 & 2 \\ 2 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [6] \quad 1^{1^2} + \mathbf{e}'_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -173, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, \quad T_{\max} = 101, \quad T \in [15, 101]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{1^2} \quad 1^3 \quad 1321^1 + \mathbf{e}'_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 8, \quad T_{\max} = 102, \quad \Delta = -191, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.33 \rightarrow 16, \quad T_{\max} = 102, \quad T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \quad 1^{1^2} \quad 1^3 \quad 1321^1 + \mathbf{e}'_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -162, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 16, \quad T_{\max} = 102, \quad T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -7 & 0 & -5 \\ 1 & 0 & 5 & 5 \\ 2 & -5 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{1^2} \quad 1^3 \quad 1321^1 + \mathbf{e}'_7 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -162, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [10] \quad 1^{123141321^1} + \mathbf{c}_7' + \hat{\mathbf{1}} \rightarrow \mathbf{1} \quad T = 10, T_{\max} = 102, \Delta = -161, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 17, T_{\max} = 102, T \in [17, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [13] \quad 1^{1231} \quad 1^5 \quad 1321^1 + \mathbf{c}_7' + \hat{\mathbf{1}} \rightarrow \mathbf{1} \quad T = 11, T_{\max} = 102, \Delta = -156, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 18, T_{\max} = 102, T \in [18, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 1 & -1 \\ 1 & 1 & 3 & 3 \\ 2 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{1231512321^1} + \mathbf{c}_7' + \hat{\mathbf{1}} \rightarrow \mathbf{1} \quad T = 11, T_{\max} = 103, \Delta = -158, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 18, T_{\max} = 103, T \in [18, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -7 & 0 & -5 \\ 1 & 0 & 5 & 5 \\ 2 & -5 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1231512321^1} + \mathbf{c}_7' + \hat{\mathbf{1}} \rightarrow \mathbf{1} \quad T = 11, T_{\max} = 103, \Delta = -158, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 18, T_{\max} = 103, T \in [18, 103]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [14] \quad 1^{12321612321^1} + \mathbf{c}_7' + \hat{\mathbf{1}} \rightarrow \mathbf{1} \quad T = 12, T_{\max} = 104, \Delta = -157, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 19, T_{\max} = 104, T \in [19, 104]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [0] \quad 1^{12321} \quad 1^7 \quad 12321^1 + \mathbf{c}_7' + \hat{\mathbf{1}} \rightarrow \mathbf{1} \quad T = 13, T_{\max} = 104, \Delta = -155, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 20, T_{\max} = 104, T \in [20, 104]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [2] \quad 1^{12321} \quad 1^8 \quad 12321^1 + \mathbf{c}_7' + \hat{\mathbf{1}} \rightarrow \mathbf{1} \quad T = 14, T_{\max} = 105, \Delta = -154, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 21, T_{\max} = 105, T \in [21, 105]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 1 & -1 \\ 1 & 1 & 3 & 3 \\ 2 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 21^{12321} \quad 1^8 \quad 12321^1 + \mathbf{c}_7' + \hat{\mathbf{1}} \rightarrow \mathbf{1} \quad T = 14, T_{\max} = 105, \Delta = -154, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 21, T_{\max} = 105, T \in [21, 105]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

1.4 $\mathbf{e}_8 + \hat{\mathbf{1}} \rightarrow (-1)$

$$T = 3 \quad [0] \quad 1^{11^1} + \mathbf{e}_8 + \hat{\mathbf{1}} \rightarrow \mathbf{1} \quad T = 3, T_{\max} = 193, \Delta = -461, \det = -12, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.33 \rightarrow 19, T_{\max} = 193, T \in [19, 193]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -11 & 0 & -9 \\ 1 & 0 & -1 & -1 \\ 2 & -9 & -1 & -9 \end{pmatrix}_{b_0}$$

$$T = 4 \quad [1] \quad 1^{121^1} + \mathbf{e}_8 + \hat{\mathbf{1}} \rightarrow \mathbf{1} \quad T = 4, T_{\max} = 193, \Delta = -432, \det = 11, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.36 \rightarrow 19, T_{\max} = 193, T \in [19, 193]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -10 & 0 & -8 \\ 1 & 0 & -1 & -1 \\ 2 & -8 & -1 & -8 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& T = 5 \quad [2] \quad 1^1 221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 5, T_{\max} = 193, \Delta = -403, \det = -10, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 18.40 \rightarrow 19, T_{\max} = 193, T \in [19, 193] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -9 & 0 & -7 \\ 1 & 0 & -1 & -1 \\ 2 & -7 & -1 & -7 \end{pmatrix}_{b_0} \\
& T = 6 \quad [3] \quad 1^1 2221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 6, T_{\max} = 193, \Delta = -374, \det = 9, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 18.44 \rightarrow 19, T_{\max} = 193, T \in [19, 193] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & -1 & -1 \\ 2 & -6 & -1 & -6 \end{pmatrix}_{b_0} \\
& T = 7 \quad [4] \quad 1^1 22221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 7, T_{\max} = 193, \Delta = -345, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 18.50 \rightarrow 19, T_{\max} = 193, T \in [19, 193] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -7 & 0 & -5 \\ 1 & 0 & -1 & -1 \\ 2 & -5 & -1 & -5 \end{pmatrix}_{b_0} \\
& T = 8 \quad [5] \quad 1^1 222221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 8, T_{\max} = 192, \Delta = -316, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 18.57 \rightarrow 19, T_{\max} = 192, T \in [19, 192] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & -1 & -1 \\ 2 & -4 & -1 & -4 \end{pmatrix}_{b_0} \\
& T = 9 \quad [6] \quad 1^1 2222221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 9, T_{\max} = 190, \Delta = -287, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 18.67 \rightarrow 19, T_{\max} = 190, T \in [19, 190] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & -1 & -1 \\ 2 & -3 & -1 & -3 \end{pmatrix}_{b_0} \\
& [7] \quad 1^1 2231321^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 9, T_{\max} = 192, \Delta = -305, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.57 \rightarrow 20, T_{\max} = 192, T \in [20, 192] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & -1 & -1 \\ 2 & -4 & -1 & -4 \end{pmatrix}_{b_0} \\
& T = 10 \quad [8] \quad 1^1 22222221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 10, T_{\max} = 174, \Delta = -258, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 18.80 \rightarrow 19, T_{\max} = 174, T \in [19, 174] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & -1 & -1 \\ 2 & -2 & -1 & -2 \end{pmatrix}_{b_0} \\
& [9] \quad 1^1 22313221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 10, T_{\max} = 190, \Delta = -276, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.67 \rightarrow 20, T_{\max} = 190, T \in [20, 190] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & -1 & -1 \\ 2 & -3 & -1 & -3 \end{pmatrix}_{b_0} \\
& T = 11 \quad [10] \quad 1^1 222222221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 11, T_{\max} = 106, \Delta = -229, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19, T_{\max} = 106, T \in [19, 106] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & -1 & -1 \\ 2 & -1 & -1 & -1 \end{pmatrix}_{b_0} \\
& [11] \quad 1^1 231413221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 11, T_{\max} = 191, \Delta = -275, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.67 \rightarrow 21, T_{\max} = 191, T \in [21, 191] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -9 & 0 & -7 \\ 1 & 0 & 3 & 3 \\ 2 & -7 & 3 & -3 \end{pmatrix}_{b_0} \\
& T = 12 \quad [12] \quad 1^1 2222222221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 12, T_{\max} = 106, \Delta = -200, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.33 \rightarrow 20, T_{\max} = 106, T \in [20, 106] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 \end{pmatrix}_{b_0} \\
& [13] \quad 1^1 2231413221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 12, T_{\max} = 152, \Delta = -246, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$T_{\min} = 20.80 \rightarrow 21, T_{\max} = 152, T \in [21, 152]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -7 & 0 & -5 \\ 1 & 0 & 2 & 2 \\ 2 & -5 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1^1 2231} 1321^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 12, T_{\max} = 187, \Delta = -270, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 21.67 \rightarrow 22, T_{\max} = 187, T \in [22, 187]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -7 & 1 & -5 \\ 1 & 1 & 3 & 3 \\ 2 & -5 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1^1 2231512321} 1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 12, T_{\max} = 192, \Delta = -272, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 21.67 \rightarrow 22, T_{\max} = 192, T \in [22, 192]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & -1 & -1 \\ 2 & -3 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [0] \ 1^{1^1 22222222221} 1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 13, T_{\max} = 106, \Delta = -171, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20, T_{\max} = 106, T \in [20, 106]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1^1 2231} 13221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 13, T_{\max} = 120, \Delta = -241, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 21.80 \rightarrow 22, T_{\max} = 120, T \in [22, 120]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -7 & 1 & -5 \\ 1 & 1 & 4 & 4 \\ 2 & -5 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1^1 23141413221} 1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 13, T_{\max} = 147, \Delta = -245, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 21.80 \rightarrow 22, T_{\max} = 147, T \in [22, 147]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -9 & 0 & -7 \\ 1 & 0 & 4 & 4 \\ 2 & -7 & 4 & -2 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [3] \ 1^{1^1 223141413221} 1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 14, T_{\max} = 109, \Delta = -216, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 22, T_{\max} = 109, T \in [22, 109]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & 2 & 2 \\ 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [4] \ 1^{1^1 2314141413221} 1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 15, T_{\max} = 110, \Delta = -215, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23, T_{\max} = 110, T \in [23, 110]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -9 & 0 & -7 \\ 1 & 0 & 5 & 5 \\ 2 & -7 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1^1 2231513151321} 1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 15, T_{\max} = 138, \Delta = -243, \det = -5, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.80 \rightarrow 24, T_{\max} = 138, T \in [24, 138]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & -1 & -1 \\ 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 16 \quad [6] \ 1^{1^1 22314141413221} 1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 16, T_{\max} = 110, \Delta = -186, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.33 \rightarrow 24, T_{\max} = 110, T \in [24, 110]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & 2 & 2 \\ 2 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1^1 22315131513221} 1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 16, T_{\max} = 111, \Delta = -214, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24, T_{\max} = 111, T \in [24, 111]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & -1 & -1 \\ 2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1^1 23216131513221} 1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 16, T_{\max} = 134, \Delta = -242, \det = 5, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.80 \rightarrow 25, T_{\max} = 134, T \in [25, 134]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 3 & 3 \\ 2 & -6 & 3 & -2 \end{pmatrix}_{b_0}$$

$$T = 17 \quad [9] \quad 1^1 231414141413221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 17, \quad T_{\max} = 111, \quad \Delta = -185, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 17, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.33 \rightarrow 25, \quad T_{\max} = 111, \quad T \in [25, 111]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -9 & 0 & -7 \\ 1 & 0 & 6 & 6 \\ 2 & -7 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^1 232171231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 17, \quad T_{\max} = 124, \quad \Delta = -241, \quad \det = -5, \quad (n_+, n_-, n_0) = (1, 17, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.80 \rightarrow 26, \quad T_{\max} = 124, \quad T \in [26, 124]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 3 & 3 \\ 2 & -6 & 3 & -2 \end{pmatrix}_{b_0}$$

$$T = 18 \quad [11] \quad 1^1 2231414141413221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 18, \quad T_{\max} = 109, \quad \Delta = -156, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 25, \quad T_{\max} = 109, \quad T \in [25, 109]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 2 & 2 \\ 2 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^1 2321812231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 18, \quad T_{\max} = 120, \quad \Delta = -240, \quad \det = 5, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.80 \rightarrow 27, \quad T_{\max} = 120, \quad T \in [27, 120]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 3 & 3 \\ 2 & -6 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^1 223151321 \quad 12321^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 18, \quad T_{\max} = 120, \quad \Delta = -240, \quad \det = 5, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.80 \rightarrow 27, \quad T_{\max} = 120, \quad T \in [27, 120]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 3 & 3 \\ 2 & -6 & 3 & -2 \end{pmatrix}_{b_0}$$

$$T = 19 \quad [14] \quad 1^1 23141414141413221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 19, \quad T_{\max} = 110, \quad \Delta = -155, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 26, \quad T_{\max} = 110, \quad T \in [26, 110]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -9 & 0 & -7 \\ 1 & 0 & 7 & 7 \\ 2 & -7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^1 23151316131513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 19, \quad T_{\max} = 114, \quad \Delta = -213, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 27, \quad T_{\max} = 114, \quad T \in [27, 114]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 4 & 4 \\ 2 & -6 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 20 \quad [16] \quad 1^1 223151316131513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 20, \quad T_{\max} = 114, \quad \Delta = -184, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 20, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.33 \rightarrow 28, \quad T_{\max} = 114, \quad T \in [28, 114]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & 3 & 3 \\ 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^1 232161316131513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 20, \quad T_{\max} = 115, \quad \Delta = -212, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 20, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, \quad T_{\max} = 115, \quad T \in [28, 115]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 4 & 4 \\ 2 & -6 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 21 \quad [18] \quad 1^1 2315132181231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 21, \quad T_{\max} = 116, \quad \Delta = -212, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 21, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 29, \quad T_{\max} = 116, \quad T \in [29, 116]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 4 & 4 \\ 2 & -6 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 22 \quad [19] \quad 1^1 22315132181231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 22, T_{\max} = 116, \Delta = -183, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 29.33 \rightarrow 30, T_{\max} = 116, T \in [30, 116]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & 3 & 3 \\ 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$T = 23 \quad [0] \quad 1^1 231513161316131513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 23, T_{\max} = 117, \Delta = -183, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.33 \rightarrow 31, T_{\max} = 117, T \in [31, 117]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 5 & 5 \\ 2 & -6 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 232171232181231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 23, T_{\max} = 118, \Delta = -211, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31, T_{\max} = 118, T \in [31, 118]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 4 & 4 \\ 2 & -6 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 24 \quad [2] \quad 1^1 2231513161316131513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 24, T_{\max} = 115, \Delta = -154, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31, T_{\max} = 115, T \in [31, 115]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & 3 & 3 \\ 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^1 2321613161316131513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 24, T_{\max} = 117, \Delta = -182, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.33 \rightarrow 32, T_{\max} = 117, T \in [32, 117]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 5 & 5 \\ 2 & -6 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^1 2321 \quad 8 \quad 1232181231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 24, T_{\max} = 118, \Delta = -210, \det = 4, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32, T_{\max} = 118, T \in [32, 118]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 4 & 4 \\ 2 & -6 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 25 \quad [5] \quad 21 \quad \begin{matrix} 1^1 2231513221 \\ (12) \end{matrix} \quad 1223151321^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 25, T_{\max} = 120, \Delta = -211, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33, T_{\max} = 120, T \in [33, 120]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 4 & 4 \\ 2 & -6 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1 \quad \begin{matrix} 1 \\ (12) \end{matrix} \quad 1223151321^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 25, T_{\max} = 120, \Delta = -211, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33, T_{\max} = 120, T \in [33, 120]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 1 & -4 \\ 1 & 1 & 4 & 4 \\ 2 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 26 \quad [7] \quad 1 \quad \begin{matrix} 1 \\ (12) \end{matrix} \quad 12231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 26, T_{\max} = 119, \Delta = -182, \det = 3, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33.33 \rightarrow 34, T_{\max} = 119, T \in [34, 119]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 1 & -4 \\ 1 & 1 & 5 & 5 \\ 2 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[8] \quad 21 \quad \begin{matrix} 1^1 2231513221 \\ (12) \end{matrix} \quad 12231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow 1 \quad T = 26, T_{\max} = 119, \Delta = -182, \det = 3, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33.33 \rightarrow 34, T_{\max} = 119, T \in [34, 119]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & 3 & 3 \\ 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$T = 27 \quad [9] \quad 1^1 2315131613161316131513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 27, T_{\max} = 118, \Delta = -153, \det = -2, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 34, T_{\max} = 118, T \in [34, 118]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 6 & 6 \\ 2 & -6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^1 2315132181232181231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 27, T_{\max} = 120, \Delta = -182, \det = -3, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 34.33 \rightarrow 35, T_{\max} = 120, T \in [35, 120]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 5 & 5 \\ 2 & -6 & 5 & -1 \end{pmatrix}_{b_0}$$

$$T = 28 \quad [12] \quad 1^1 22315132181232181231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 28, T_{\max} = 119, \Delta = -153, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35, T_{\max} = 119, T \in [35, 119]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & 3 & 3 \\ 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^1 23216131613161316131513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 28, T_{\max} = 119, \Delta = -152, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35, T_{\max} = 119, T \in [35, 119]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 6 & 6 \\ 2 & -6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$T = 29 \quad [13] \quad 1^1 232171232181232181231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 29, T_{\max} = 122, \Delta = -181, \det = -3, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.33 \rightarrow 37, T_{\max} = 122, T \in [37, 122]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 5 & 5 \\ 2 & -6 & 5 & -1 \end{pmatrix}_{b_0}$$

$$T = 30 \quad [14] \quad 1^1 2321 \quad 8 \quad 1232181232181231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 30, T_{\max} = 123, \Delta = -180, \det = 3, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.33 \rightarrow 38, T_{\max} = 123, T \in [38, 123]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 5 & 5 \\ 2 & -6 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^1 2231513221 \quad (12) \quad 1223151321812321^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 30, T_{\max} = 124, \Delta = -210, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38, T_{\max} = 124, T \in [38, 124]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & 1 & 1 \\ 2 & -3 & 1 & -1 \end{pmatrix}_{b_0}$$

$$T = 33 \quad [0] \quad 1^1 2315132181232181232181231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 33, T_{\max} = 124, \Delta = -152, \det = -2, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40, T_{\max} = 124, T \in [40, 124]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 6 & 6 \\ 2 & -6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$T = 35 \quad [1] \quad 1^1 232171232181232181232181231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 35, T_{\max} = 126, \Delta = -151, \det = -2, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42, T_{\max} = 126, T \in [42, 126]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 6 & 6 \\ 2 & -6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 232181232181231513221(12)12231513221^1 + \mathbf{e}_8 + \hat{\mathbf{i}} \rightarrow \mathbf{1} \quad T = 35, T_{\max} = 129, \Delta = -209, \det = -4, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 43, T_{\max} = 129, T \in [43, 129]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 4 & 4 \\ 2 & -6 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 36 \quad [3] \quad 1 \quad \overset{1}{8} \quad 1232181232181232181231513221^1 + \mathbf{e}_8 + \hat{1} \rightarrow 1 \quad T = 36, T_{\max} = 127, \Delta = -150, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 43, T_{\max} = 127, T \in [43, 127]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -8 & 0 & -6 \\ 1 & 0 & 6 & 6 \\ 2 & -6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$T = 37 \quad [4] \quad 1 \quad \overset{1}{(12)} \quad 12231513221(12)12231513221^1 + \mathbf{e}_8 + \hat{1} \rightarrow 1 \quad T = 37, T_{\max} = 130, \Delta = -181, \det = -3, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.33 \rightarrow 45, T_{\max} = 130, T \in [45, 130]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 1 & -4 \\ 1 & 1 & 5 & 5 \\ 2 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$T = 38 \quad [5] \quad 1 \quad \overset{1}{(12)} \quad 12231513221(12)12231513221^1 + \mathbf{e}_8 + \hat{1} \rightarrow 1 \quad T = 38, T_{\max} = 129, \Delta = -152, \det = 2, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45, T_{\max} = 129, T \in [45, 129]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 1 & -3 \\ 1 & 1 & 5 & 5 \\ 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$T = 42 \quad [6] \quad 1 \quad \overset{1}{(12)} \quad 12231513221(12)1223151321812321^1 + \mathbf{e}_8 + \hat{1} \rightarrow 1 \quad T = 42, T_{\max} = 135, \Delta = -180, \det = 3, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.33 \rightarrow 50, T_{\max} = 135, T \in [50, 135]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & 1 & 1 \\ 2 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

$$T = 48 \quad [0] \quad 1 \quad \overset{1}{(12)} \quad 12231513221(12)1223151321^1 + \mathbf{e}_8 + \hat{1} \rightarrow 1 \quad T = 48, T_{\max} = 141, \Delta = -180, \det = 3, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 55.33 \rightarrow 56, T_{\max} = 141, T \in [56, 141]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -7 & 0 & -5 \\ 1 & 0 & 4 & 4 \\ 2 & -5 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 49 \quad [1] \quad 1 \quad \overset{1}{(12)} \quad 12231513221(12)12231513221(12)12231513221^1 + \mathbf{e}_8 + \hat{1} \rightarrow 1 \quad T = 49, T_{\max} = 140, \Delta = -151, \det = -2, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56, T_{\max} = 140, T \in [56, 140]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 1 & -4 \\ 1 & 1 & 6 & 6 \\ 2 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1 \quad \overset{1}{(12)} \quad 12231513221(12)12231513221^1 + \mathbf{e}_8 + \hat{1} \rightarrow 1 \quad T = 49, T_{\max} = 140, \Delta = -151, \det = -2, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56, T_{\max} = 140, T \in [56, 140]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -6 & 0 & -4 \\ 1 & 0 & 4 & 4 \\ 2 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$T = 54 \quad [0] \quad 1 \quad \overset{1}{(12)} \quad 12231513221(12)12231513221(12)1223151321812321^1 + \mathbf{e}_8 + \hat{1} \rightarrow 1 \quad T = 54, T_{\max} = 145, \Delta = -150, \det = 2, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 61, T_{\max} = 145, T \in [61, 145]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & 1 & 1 \\ 2 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

1.5 $\mathbf{f}_4 + \hat{\mathbf{1}} \rightarrow (-1)$

$$T = 3 \quad [0] \quad 1^1 1^1 + \mathbf{f}_4 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 3, T_{\max} = 174, \Delta = -265, \det = -5, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 174, T \in [12, 174]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -4 & 0 & -2 \\ 1 & 0 & -1 & -1 \\ 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 4 \quad [1] \quad 1^1 21^1 + \mathbf{f}_4 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 4, T_{\max} = 106, \Delta = -236, \det = 4, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 106, T \in [12, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 0 & -1 \\ 1 & 0 & -1 & -1 \\ 2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [2] \quad 1^1 221^1 + \mathbf{f}_4 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 5, T_{\max} = 99, \Delta = -207, \det = -3, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 99, T \in [13, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [3] \quad 1^1 2221^1 + \mathbf{f}_4 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 6, T_{\max} = 99, \Delta = -178, \det = 2, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 99, T \in [13, 99]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^1 1321^1 + \mathbf{f}_4 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 6, T_{\max} = 100, \Delta = -187, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 1 & -1 \\ 1 & 1 & 2 & 2 \\ 2 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^1 2231^1 + \mathbf{f}_4 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 6, T_{\max} = 100, \Delta = -187, \det = 14, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -3 & 1 & -1 \\ 2 & 1 & 2 & 2 \\ 2 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [6] \quad 1^1 1^2 321^1 + \mathbf{f}_4 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 7, T_{\max} = 99, \Delta = -160, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 2 \\ 2 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^1 13221^1 + \mathbf{f}_4 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 7, T_{\max} = 99, \Delta = -158, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -3 & 1 & -1 \\ 1 & 1 & 3 & 3 \\ 2 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^1 23131^1 + \mathbf{f}_4 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 7, T_{\max} = 100, \Delta = -166, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -5 & 0 & -3 \\ 1 & 0 & 3 & 3 \\ 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

1.6 $\hat{\mathbf{1}} \rightarrow (-1)$

$$T = 2 \quad [0] \quad 1^1 1^1 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 2, T_{\max} = 120, \Delta = -242, \det = 4, (n_+, n_-, n_0) = (1, 2, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 120, T \in [10, 120]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$T = 3 \quad [2] \quad 1^1 21^1 + \hat{\mathbf{1}} \rightarrow 1 \quad T = 3, T_{\max} = 98, \Delta = -213, \det = -4, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 98, T \in [10, 98]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 21^1 2 + \hat{1} \rightarrow 1 \quad T = 3, T_{\max} = 98, \Delta = -213, \det = -4, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 98, T \in [10, 98]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$T = 4 \quad [4] \ 1^1 221^1 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -184, \det = 4, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 98, T \in [10, 98]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 222^1 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -184, \det = 4, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 98, T \in [10, 98]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [6] \ 1^1 2221^1 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -155, \det = -4, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 2221^2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -155, \det = -4, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [9] \ 2^3 21^1 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 96, \Delta = -145, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 96, T \in [11, 96]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 22221^2 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 85, \Delta = -126, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 85, T \in [10, 85]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[8] \ 1^1 22221^1 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 85, \Delta = -126, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 85, T \in [10, 85]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [11] \ 1^1 222221^1 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 13, \Delta = -97, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 13, T \in [10, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[10] \ 222221^2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 13, \Delta = -97, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 13, T \in [10, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[12] \ 2^3 1321^1 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 30, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 30, T \in [11, 30]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [14] \ 1^1 2222221^1 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 10, \Delta = -68, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 6 \\ 2 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[15] \ 2^3 13221^1 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[13] \quad 2^{2222221^1 + \hat{1} \rightarrow 1} \quad T = 8, T_{\max} = 10, \Delta = -68, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 6 \\ 2 & 6 & 6 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [16] \quad 2^{22222221^1 + \hat{1} \rightarrow 1} \quad T = 9, T_{\max} = 10, \Delta = -39, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 7 \\ 2 & 7 & 7 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{122222221^1 + \hat{1} \rightarrow 1} \quad T = 9, T_{\max} = 10, \Delta = -39, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 7 & 7 \\ 2 & 7 & 7 \end{pmatrix}_{b_0}$$

$$[18] \quad 31^{1^{231}} 13 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 29, T \in [13, 29]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[19] \quad 2^{1^{23141}} 3^{2 + \hat{1} \rightarrow 1} \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [22] \quad 31^{1^{2231}} 5^{13 + \hat{1} \rightarrow 1} \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[23] \quad 2^{1^{23141}} 3^{2 + \hat{1} \rightarrow 1} \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 6 \\ 2 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1222222221^1 + \hat{1} \rightarrow 1} \quad T = 10, T_{\max} = 10, \Delta = -10, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 8 \\ 2 & 8 & 8 \end{pmatrix}_{b_0}$$

$$[20] \quad 2^{222222221^1 + \hat{1} \rightarrow 1} \quad T = 10, T_{\max} = 10, \Delta = -10, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 8 & 8 \\ 2 & 8 & 8 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1] \quad 321^{321} 8^{12321^1 + \hat{1} \rightarrow 1} \quad T = 12, T_{\max} = 28, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 16, T_{\max} = 28, T \in [16, 28]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{1^{2321}} 7^{12321^1 + \hat{1} \rightarrow 1} \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [2] \quad 31^{1^{2315131}} 6^{13 + \hat{1} \rightarrow 1} \quad T = 13, T_{\max} = 16, \Delta = -86, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [3] \quad 316^{31} 131612321^1 + \hat{1} \rightarrow 1 \quad T = 14, T_{\max} = 17, \Delta = -85, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 19 \quad [4] \quad 3221 \overset{1}{2} 231513221 (12) \quad 1223 + \hat{1} \rightarrow 1 \quad T = 19, T_{\max} = 22, \Delta = -85, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 5 \\ 2 & 5 & 5 \end{pmatrix}_{b_0}$$

1.7 $\hat{1} \rightarrow (-2)$

$$T = 12 \quad [11] \quad 22 \overset{1}{1} 181223151 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19, T_{\max} = 24, T \in [19, 24]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 321 \overset{512321}{8} 12 \overset{1}{1} + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 71, \Delta = -121, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18, T_{\max} = 71, T \in [18, 71]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 321 \overset{151321}{8} 12 \overset{1}{1} + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18, T_{\max} = 49, T \in [18, 49]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 18123215132 \overset{1}{1} 2 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 71, \Delta = -121, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19, T_{\max} = 71, T \in [19, 71]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1812315132 \overset{1}{1} 2 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19, T_{\max} = 49, T \in [19, 49]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 22 \overset{1}{1} 1812315123 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19, T_{\max} = 49, T \in [19, 49]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 22 \overset{1}{1} 1812321513 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 60, \Delta = -120, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19, T_{\max} = 60, T \in [19, 60]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[9] \quad 22 \overset{1}{1} 1812231512 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19, T_{\max} = 24, T \in [19, 24]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1 \overset{151321}{8} 12 \overset{1}{1} 2 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19, T_{\max} = 24, T \in [19, 24]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1 \overset{1512321}{8} 12 \overset{1}{1} 2 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 26, \Delta = -112, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19, T_{\max} = 26, T \in [19, 26]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 22 \overset{1}{1} 2181231512 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19, T_{\max} = 24, T \in [19, 24]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 22 \overset{1}{1} 2181232151 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 26, \Delta = -112, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19, T_{\max} = 26, T \in [19, 26]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

- [14] $22^1 218123151 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [15] $8122315132^1 2 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 49, \Delta = -119, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19, T_{\max} = 49, T \in [19, 49]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [16] $31^1 6^{21} 131 6^1 1 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 18, \Delta = -96, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0}$
- [17] $31^1 6^{21} 1316131 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 18, \Delta = -96, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [18] $31^1 6^{21} 1316122^1 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 18, \Delta = -96, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0}$
- [19] $1612321712^1 2 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [20] $3141321812^1 31 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [21] $21612321712^1 2 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [22] $14141321812^1 2 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 28, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19, T_{\max} = 28, T \in [19, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [23] $81231414132^1 2 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 82, \Delta = -123, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19, T_{\max} = 82, T \in [19, 82]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [0] $1^{2151321} 8^{12} 12^1 2 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [10] $22^1 181231513 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 39, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19, T_{\max} = 39, T \in [19, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [64] $1612^1 3217^1 1 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [65] $161232^1 17^1 1 + \hat{1} \rightarrow 2$ $T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

- [28] $1 \begin{smallmatrix} 1 \\ 8 \\ 151321 \end{smallmatrix} 12^1 3 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [29] $1 \begin{smallmatrix} 21 \\ 215132^1 1 \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [30] $1 \begin{smallmatrix} 21 \\ 1512^1 321 \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 26, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 26, T \in [19, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [31] $1 \begin{smallmatrix} 21 \\ 151232^1 1 \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 26, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 26, T \in [19, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [32] $1 \begin{smallmatrix} 21 \\ 15132^1 1 \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [33] $1 \begin{smallmatrix} 32^1 1513221 \\ 8 \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [34] $1 \begin{smallmatrix} 512321 \\ 8 \end{smallmatrix} 122^1 3 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 71, \Delta = -121, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 71, T \in [19, 71]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [35] $1 \begin{smallmatrix} 151321 \\ 8 \end{smallmatrix} 122^1 3 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [36] $1 \begin{smallmatrix} 1513221 \\ 8 \end{smallmatrix} 12^1 3 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [37] $1 \begin{smallmatrix} 32^1 151321 \\ 8 \end{smallmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [38] $1 \begin{smallmatrix} 215132^1 21 \\ 8 \end{smallmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [39] $1 \begin{smallmatrix} 15132^1 21 \\ 8 \end{smallmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [40] $1 \begin{smallmatrix} 215132^1 1 \\ 8 \end{smallmatrix} 122 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

- [41] $1 \overset{151232^1}{8} 122 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 26, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 26, T \in [19, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [42] $1 \overset{1512^1 321}{8} 122 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 26, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 26, T \in [19, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [43] $1 \overset{15132^1}{8} 122 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [44] $21 \overset{512321}{8} 12^1 3 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 71, \Delta = -121, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 71, T \in [19, 71]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [45] $21 \overset{151321}{8} 12^1 3 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [46] $2321 \overset{51321}{8} 12^1 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 71, \Delta = -121, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.33 \rightarrow 18, T_{\max} = 71, T \in [18, 71]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [47] $21 \overset{215132^1}{8} 12 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [48] $21 \overset{1512^1 321}{8} 12 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 26, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 26, T \in [19, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [49] $21 \overset{151232^1}{8} 12 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 26, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 26, T \in [19, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [50] $21 \overset{15132^1}{8} 12 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [51] $32^1 1613151315 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 82, \Delta = -123, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 82, T \in [19, 82]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [52] $32^1 2181223151 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [53] $21812231512^1 3 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [54] $22181231512^1 3 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$

- [24] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
 $1 \xrightarrow{32^1 151321} 8 \xrightarrow{1} 1 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [69] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 $161232^1 17122 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
- [70] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
 $17 \xrightarrow{21} 12^1 321612 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
- [71] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 $17 \xrightarrow{21} 1232^1 1612 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
- [72] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 $14132^1 1812313 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -103, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
- [73] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 $141321812^1 313 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -103, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
- [74] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
 $16123217122^1 3 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
- [75] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
 $32^1 1612321712 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 49, T \in [19, 49]$
- [76] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 $2161232^1 17122 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
- [77] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
 $21612^1 3217122 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
- [78] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 $41232^1 1812313 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
- [79] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
 $412^1 321812313 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
- [80] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 $412321812^1 313 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$
- [81] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 $412321812^1 32 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 20, \Delta = -99, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$

- [82] $412^1 32181232 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 20$, $\Delta = -99$, $\det = 9$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19$, $T_{\max} = 20$, $T \in [19, 20]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [83] $41232^1 181232 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 20$, $\Delta = -99$, $\det = 9$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19$, $T_{\max} = 20$, $T \in [19, 20]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [84] $4141321^{32^1 1} 8^1 1 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 82$, $\Delta = -123$, $\det = 9$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19$, $T_{\max} = 82$, $T \in [19, 82]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [85] $41413218122^1 3 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 82$, $\Delta = -123$, $\det = 9$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19$, $T_{\max} = 82$, $T \in [19, 82]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [86] $41413221812^1 3 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 82$, $\Delta = -123$, $\det = 9$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19$, $T_{\max} = 82$, $T \in [19, 82]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [27] $1^{11} \frac{8}{512321} 12^1 3 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 9$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19$, $T_{\max} = 71$, $T \in [19, 71]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [88] $1^{11} \frac{8}{3151232^1 1} 1 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 60$, $\Delta = -120$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 60$, $T \in [18, 60]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [89] $1^{11} \frac{8}{2151322^1 1} 1 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 24$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 24$, $T \in [18, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [90] $1^{11} \frac{8}{315132^1 1} 1 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 39$, $\Delta = -118$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 39$, $T \in [18, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [91] $1^{11} \frac{8}{151322^1 1} 1 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 24$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 24$, $T \in [18, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [92] $1^{11} \frac{8}{512^1 321} 123 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [93] $1^{512^1 315131} \frac{6}{1} 1 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 29$, $\Delta = -115$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 29$, $T \in [18, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [94] $1^{3215132^1 21} \frac{8}{1} 1 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

- [95] $1 \overset{1}{8} \overset{315132^1 21}{1} + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 39, \Delta = -118, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 39, T \in [18, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [96] $1 \overset{1}{8} \overset{512^1 321}{1223} + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 71, \Delta = -121, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 71, T \in [18, 71]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [97] $1 \overset{1}{8} \overset{3215132^1 1}{12} + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 49, T \in [18, 49]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [98] $1 \overset{1}{8} \overset{3151232^1 1}{12} + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 60, \Delta = -120, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 60, T \in [18, 60]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [99] $1 \overset{1}{8} \overset{2151322^1 1}{12} + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 24, T \in [18, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [100] $1 \overset{1}{8} \overset{315132^1 1}{12} + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 39, \Delta = -118, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 39, T \in [18, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [101] $1 \overset{1}{8} \overset{151322^1 1}{12} + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 24, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 24, T \in [18, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [102] $21 \overset{1}{8} \overset{512^1 321}{123} + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 71, \Delta = -121, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 71, T \in [18, 71]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [103] $1612315132^1 15 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 32, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 32, T \in [18, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [104] $1612^1 31513215 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 32, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 32, T \in [18, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [105] $1612^1 31513151 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 29, \Delta = -115, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 29, T \in [18, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [106] $2161315132^1 15 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 29, \Delta = -115, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 29, T \in [18, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [107] $512^1 321613151 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 35, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 35, T \in [18, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [108] $218122^1 315123 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 49, \Delta = -119, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18, T_{\max} = 49, T \in [18, 49]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

- [109] $218122^1 31513 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 39$, $\Delta = -118$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 39$, $T \in [18, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [110] $221812^1 315123 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [111] $221812^1 321513 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 60$, $\Delta = -120$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 60$, $T \in [18, 60]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [112] $221812^1 231512 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 24$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 24$, $T \in [18, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [113] $221812^1 31513 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 39$, $\Delta = -118$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 39$, $T \in [18, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [114] $221812^1 23151 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 24$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 24$, $T \in [18, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [115] $6122315132^1 15 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 32$, $\Delta = -116$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 32$, $T \in [18, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [116] $6122^1 31513215 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 32$, $\Delta = -116$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 32$, $T \in [18, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [117] $6122^1 31513151 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 29$, $\Delta = -115$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 29$, $T \in [18, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [118] $161232^1 171 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 39$, $\Delta = -118$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 39$, $T \in [18, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [119] $316131612^1 2 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 18$, $\Delta = -96$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 16.25 \rightarrow 17$, $T_{\max} = 18$, $T \in [17, 18]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 13 & 7 \\ 2 & 7 & 3 \end{pmatrix}_{b_0}$
- [120] $1712^1 3216123 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [121] $171232^1 161 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [122] $161232^1 1712 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 39$, $\Delta = -118$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 39$, $T \in [18, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [123] $41315132^1 171 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 39$, $\Delta = -118$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 39$, $T \in [18, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [124] $314132^1 181231 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 24$, $\Delta = -103$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 24$, $T \in [18, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [125] $1612^1 32171223 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [126] $32161232^1 1712 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [127] $4132171232^1 15 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [128] $41315132^1 1712 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 39$, $\Delta = -118$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 39$, $T \in [18, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [129] $41315132^1 2171 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 39$, $\Delta = -118$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 39$, $T \in [18, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [130] $18 \overset{21}{12} 1^1 314141 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 28$, $\Delta = -114$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 28$, $T \in [18, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [131] $18 \overset{1}{12} 2^1 314141 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 28$, $\Delta = -114$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 28$, $T \in [18, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [132] $1414132^1 18122 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 28$, $\Delta = -114$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 28$, $T \in [18, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [133] $1414132^1 21812 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 28$, $\Delta = -114$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 28$, $T \in [18, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [134] $413161232^1 161 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [87] $1 \overset{1}{\underset{3215132^1 1}{8}} 1 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.75 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [188] $413151322^1 171 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 39$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 39$, $T \in [18, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [137] $1 \overset{1}{\underset{15132^1 1}{8}} 123 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$

- [151] $1 \overset{31512^1 321}{8} 12 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 60$, $\Delta = -120$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 60$, $T \in [18, 60]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [152] $21 \overset{51232^1 1}{8} 123 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [153] $21 \overset{15132^1 1}{8} 123 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [154] $21 \overset{51321}{8} 12^1 32 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [155] $51232^1 1613151 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 35$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 35$, $T \in [18, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [156] $15132^1 1613151 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 29$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 29$, $T \in [18, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [157] $18122^1 3151232 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [158] $1812231512^1 32 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [159] $18122^1 315132 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [160] $32218122^1 3151 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [161] $322^1 181223151 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [162] $21812^1 2315123 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [163] $2181231512^1 32 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [164] $21812^1 3151232 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [165] $21812^1 3215132 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[166] $21812^1 231513 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 39$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 39$, $T \in [18, 39]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[167] $21812^1 315132 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[168] $22181232^1 1513 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 60$, $\Delta = -120$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 60$, $T \in [18, 60]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[169] $612^1 231513215 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 32$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 32$, $T \in [18, 32]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[170] $612^1 231513151 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 29$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 29$, $T \in [18, 29]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[171] $612^1 315131513 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 82$, $\Delta = -123$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 82$, $T \in [18, 82]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[172] $1612^1 32171 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 39$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 39$, $T \in [18, 39]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[173] $171232^1 16123 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[174] $1712^1 32161 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[175] $1612^1 321712 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 39$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 39$, $T \in [18, 39]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[176] $612321^{232^1} 71 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[177] $612^1 32171 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[178] $1316131612^1 23 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 18$, $\Delta = -97$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 18$, $T \in [18, 18]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[179] $161232^1 171223 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

- [180] $1612321712^1 23 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [181] $232161232^1 171 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [182] $232^1 161232171 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [183] $321612^1 321712 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [184] $41321712^1 3215 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [185] $4132^1 17123215 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [186] $4132^1 17123151 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [187] $41321712^1 3151 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [135] $1 \begin{smallmatrix} 8 \\ 31512^1 321 \end{smallmatrix} + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 60$, $\Delta = -120$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 60$, $T \in [18, 60]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [189] $1812^1 3141413 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 76$, $\Delta = -122$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 76$, $T \in [18, 76]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [190] $1812^1 2314141 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 28$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 28$, $T \in [18, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [191] $4132^1 1613161 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 35$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 35$, $T \in [18, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [192] $31414132^1 1812 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 76$, $\Delta = -122$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 76$, $T \in [18, 76]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [193] $14141322^1 1812 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 28$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 28$, $T \in [18, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [194] $31414132^1 2181 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 76$, $\Delta = -122$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 76$, $T \in [18, 76]$

- [195] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ $4141321812^1 23 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 82$, $\Delta = -123$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 82$, $T \in [18, 82]$
- [196] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ $4132^1 16131612 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 35$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 35$, $T \in [18, 35]$
- [197] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$ $4131612^1 32161 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
- [198] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$ $4132^1 1812314 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 82$, $\Delta = -123$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 82$, $T \in [18, 82]$
- [199] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$ $4132^1 18122314 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 82$, $\Delta = -123$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 82$, $T \in [18, 82]$
- [200] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ $413218122^1 314 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 82$, $\Delta = -123$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 82$, $T \in [18, 82]$
- [201] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ $141414132^1 171 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 28$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 28$, $T \in [18, 28]$
- [202] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$ $7122^1 31414141 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 28$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 28$, $T \in [18, 28]$
- [136] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ $1 \xrightarrow{1} \begin{matrix} 8 \\ 5123211 \end{matrix} 123 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.44 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
- [204] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ $1812^1 32151322 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 36$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.22 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
- [205] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ $1812^1 2315132 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 36$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.22 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
- [206] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ $1812^1 3151322 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 36$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.22 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
- [207] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ $8122^1 3151322 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 36$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.22 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
- [208] $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$ $231414132^1 181 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 82$, $\Delta = -123$, $\det = 36$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.22 \rightarrow 18$, $T_{\max} = 82$, $T \in [18, 82]$

- [209] $314141322^1 181 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 76$, $\Delta = -122$, $\det = 36$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.22 \rightarrow 18$, $T_{\max} = 76$, $T \in [18, 76]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [210] $8122^1 31414132 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 82$, $\Delta = -123$, $\det = 36$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.22 \rightarrow 18$, $T_{\max} = 82$, $T \in [18, 82]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [211] $712^1 314141413 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 76$, $\Delta = -122$, $\det = 36$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.22 \rightarrow 18$, $T_{\max} = 76$, $T \in [18, 76]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [212] $712^1 231414141 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 28$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.22 \rightarrow 18$, $T_{\max} = 28$, $T \in [18, 28]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [203] $1812^1 23151232 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 36$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.22 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [218] $31612^1 3151315 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 82$, $\Delta = -123$, $\det = 49$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18$, $T_{\max} = 82$, $T \in [18, 82]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [215] $1^{151322^1} 8^{123} + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [216] $1^{5132^1 21} 8^{1232} + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [217] $21^{5132^1 1} 8^{1232} + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [214] $1^{231512^1 321} 8^{1232} + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [213] $1^{5132^1 1} 8^{1232} + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [220] $3221812^1 23151 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [221] $2181232^1 15132 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18$, $T_{\max} = 71$, $T \in [18, 71]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [222] $812^1 23151322 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 49$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18$, $T_{\max} = 49$, $T \in [18, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [223] $61232^1 1^{2321} 7^1 + \hat{1} \rightarrow 2$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 17.06 \rightarrow 18, T_{\max} = 71, T \in [18, 71]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [224] $2321612^1 32171 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 71, \Delta = -121, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18, T_{\max} = 71, T \in [18, 71]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [225] $414132^1 1^1 8^1 1 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 82, \Delta = -123, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18, T_{\max} = 82, T \in [18, 82]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [226] $414132^1 181223 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 82, \Delta = -123, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18, T_{\max} = 82, T \in [18, 82]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [227] $812^1 314141322 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 82, \Delta = -123, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18, T_{\max} = 82, T \in [18, 82]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [228] $414132^1 218123 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 82, \Delta = -123, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18, T_{\max} = 82, T \in [18, 82]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [229] $812^1 231414132 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 82, \Delta = -123, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18, T_{\max} = 82, T \in [18, 82]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [230] $41321812^1 2314 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 82, \Delta = -123, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18, T_{\max} = 82, T \in [18, 82]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [219] $5132^1 16131513 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 82, \Delta = -123, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.06 \rightarrow 18, T_{\max} = 82, T \in [18, 82]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [231] $315132^1 161315 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 82, \Delta = -123, \det = 64, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 16.94 \rightarrow 17, T_{\max} = 82, T \in [17, 82]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [233] $4141322^1 18123 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 82, \Delta = -123, \det = 81, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 16.84 \rightarrow 17, T_{\max} = 82, T \in [17, 82]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [232] $181232^1 151322 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 71, \Delta = -121, \det = 81, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 16.84 \rightarrow 17, T_{\max} = 71, T \in [17, 71]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [234] $1^{51322^1 1} 8^1 1232 + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 71, \Delta = -121, \det = 100, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 16.76 \rightarrow 17, T_{\max} = 71, T \in [17, 71]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $T = 13$ [267] $21^{22^1 22221} (12) 122 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 99, \Delta = -135, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20, T_{\max} = 99, T \in [20, 99]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [258] $1^{22^1 2222221} (12) 1 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 99, \Delta = -135, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20, T_{\max} = 99, T \in [20, 99]$

66

[illegible]

- [290] $(12)1222222222^1 2 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20$, $T_{\max} = 99$, $T \in [20, 99]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [291] $1612321 \overset{1}{8} 1 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 25$, $\Delta = -109$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20$, $T_{\max} = 25$, $T \in [20, 25]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [292] $16123218122^1 2 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 25$, $\Delta = -109$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20$, $T_{\max} = 25$, $T \in [20, 25]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [293] $321 \overset{2}{8} 1232161 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 40$, $\Delta = -118$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19$, $T_{\max} = 40$, $T \in [19, 40]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [294] $1 \overset{22}{8} 12321612 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 25$, $\Delta = -109$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20$, $T_{\max} = 25$, $T \in [20, 25]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [295] $1612321812^1 2 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 36$, $\Delta = -117$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20$, $T_{\max} = 36$, $T \in [20, 36]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [296] $1712321712^1 2 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 25$, $\Delta = -108$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20$, $T_{\max} = 25$, $T \in [20, 25]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [297] $321612321812^1 2 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 40$, $\Delta = -118$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20$, $T_{\max} = 40$, $T \in [20, 40]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [298] $216123218122^1 2 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 25$, $\Delta = -109$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20$, $T_{\max} = 25$, $T \in [20, 25]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [299] $22^1 17123217122 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 25$, $\Delta = -108$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20$, $T_{\max} = 25$, $T \in [20, 25]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [300] $413151321812^1 2 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 36$, $\Delta = -117$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20$, $T_{\max} = 36$, $T \in [20, 36]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [237] $1 \overset{22}{(12)} 1 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -4$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20$, $T_{\max} = 99$, $T \in [20, 99]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [343] $21(12)1222222^1 22 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -9$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20$, $T_{\max} = 99$, $T \in [20, 99]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [301] $1 \overset{1}{(12)} 1 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -9$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20$, $T_{\max} = 99$, $T \in [20, 99]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

- [329] $222^1 221 (12) 122 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 99, \Delta = -135, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 99, T \in [20, 99]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [330] $222^1 21 (12) 1222 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 99, \Delta = -135, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 99, T \in [20, 99]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [331] $222^1 221 (12) 12^1 22 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 99, \Delta = -135, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 99, T \in [20, 99]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [332] $222^1 1 (12) 1222 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 99, \Delta = -135, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 99, T \in [20, 99]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [333] $32^1 17123151315 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 77, \Delta = -122, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 77, T \in [20, 77]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [334] $132181231512^1 3 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 21, \Delta = -99, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 21, T \in [20, 21]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [335] $31321812^1 31512 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 20, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [336] $3132^1 181231512 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 20, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [337] $3132^1 181232151 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -100, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [338] $31321812^1 32151 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -100, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [339] $3132181232^1 151 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -100, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [340] $31321812^1 3151 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 20, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [341] $3132^1 18123151 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 20, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [342] $1(12)12222222^1 22 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 99, \Delta = -135, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 99, T \in [20, 99]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [302] $1 (12) 1 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 99, \Delta = -135, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $222^1 22221$

- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
 [358] $1 \overset{32^1 21}{8} 1232161 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 40, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 40, T \in [20, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $\overset{1}{16} 12^1 32181222 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $\overset{1}{16} 12321812^1 22 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\overset{1}{16} 1232^1 181222 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\overset{21}{18} 12321612^1 3 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 40, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 40, T \in [20, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $\overset{32^1 1}{21} 8 1232161 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 40, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 40, T \in [20, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $\overset{221}{1} 8 12^1 321612 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $\overset{221}{1} 8 1232^1 1612 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\overset{21}{21} 8 1232^1 1612 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\overset{21}{21} 8 12^1 321612 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $\overset{1}{17} 123217122^1 3 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 36, \Delta = -117, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 36, T \in [20, 36]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $\overset{32^1 1}{1} 7 12321712 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 36, \Delta = -117, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 36, T \in [20, 36]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $\overset{21}{17} 12^1 3217122 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -108, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $\overset{21}{17} 1232^1 17122 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -108, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[372] \quad 612321^{2321} 8 \ 12^1 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 61, \Delta = -120, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.33 \rightarrow 19, T_{\max} = 61, T \in [19, 61]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[373] \quad 32^1 16131612315 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 72, \Delta = -121, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 72, T \in [20, 72]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[374] \quad 161315131612^1 3 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 61, \Delta = -120, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 61, T \in [20, 61]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[375] \quad 32^1 16123218122 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 40, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 40, T \in [20, 40]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[376] \quad 21612321812^1 22 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[377] \quad 21612^1 32181222 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[378] \quad 2161232^1 181222 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[379] \quad 217123217122^1 3 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 36, \Delta = -117, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 36, T \in [20, 36]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[380] \quad 221712^1 3217122 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 25, \Delta = -108, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[381] \quad 612315131612^1 3 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 72, \Delta = -121, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 72, T \in [20, 72]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[382] \quad 412^1 3218123214 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 20, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[383] \quad 41232^1 18123214 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 20, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[384] \quad 414141321812^1 3 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 77, \Delta = -122, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 77, T \in [20, 77]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[303] \quad 1 \begin{pmatrix} 12 \\ 222^1 2221 \end{pmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 99, \Delta = -135, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.22 \rightarrow 20, T_{\max} = 99, T \in [20, 99]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[illegible]

- [412] $17122^1 31513215 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 29, \Delta = -114, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 29, T \in [19, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [413] $17122^1 31513151 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 28, \Delta = -113, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 28, T \in [19, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [414] $21712315132^1 15 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 29, \Delta = -114, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 29, T \in [19, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [415] $21712^1 31513215 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 29, \Delta = -114, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 29, T \in [19, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [416] $21712^1 31513151 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 28, \Delta = -113, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 28, T \in [19, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [417] $512^1 3217123215 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 40, \Delta = -118, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [418] $15132171232^1 15 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 33, \Delta = -116, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 33, T \in [19, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [419] $1321812^1 315123 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [420] $1321812^1 321513 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 25, \Delta = -100, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 25, T \in [19, 25]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [421] $1321812^1 31513 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 20, \Delta = -98, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [422] $1(12)1222222^1 222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [423] $21(12)122222^1 222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [424] $221(12)12222^1 222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [425] $2221(12)12222^1 222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [386] $1 \begin{pmatrix} 12 \\ 2222^1 2221 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$

- [427] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 22221(12)1222¹222 + $\hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 99$, $T \in [19, 99]$
- [428] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ 2222¹21(12)122222 + $\hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 99$, $T \in [19, 99]$
- [429] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ (12)12222222¹222 + $\hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 99$, $T \in [19, 99]$
- [430] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 41315132¹18¹1 + $\hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 36$, $\Delta = -117$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 36$, $T \in [19, 36]$
- [431] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 16³¹1232¹18²¹1 + $\hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 36$, $\Delta = -117$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 36$, $T \in [19, 36]$
- [432] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ 17¹12¹3217¹123 + $\hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 36$, $\Delta = -117$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 36$, $T \in [19, 36]$
- [433] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$ 16¹13161232¹15 + $\hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 29$, $\Delta = -114$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 29$, $T \in [19, 29]$
- [434] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 16¹1315132¹161 + $\hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 27$, $\Delta = -112$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 27$, $T \in [19, 27]$
- [435] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 1³²²¹8¹1232¹161 + $\hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 40$, $\Delta = -118$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 40$, $T \in [19, 40]$
- [436] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ 18²¹12¹3216123 + $\hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 40$, $\Delta = -118$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 40$, $T \in [19, 40]$
- [437] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 21³²¹8¹1232¹161 + $\hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 40$, $\Delta = -118$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 40$, $T \in [19, 40]$
- [438] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 16³¹1232¹18122 + $\hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 36$, $\Delta = -117$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 36$, $T \in [19, 36]$
- [439] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ 17¹12¹32171223 + $\hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 36$, $\Delta = -117$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 36$, $T \in [19, 36]$
- [440] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 1³²¹7¹1232¹1712 + $\hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 36$, $\Delta = -117$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 36$, $T \in [19, 36]$

- [441] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 $^{512}1321 \quad 8 \quad 1 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 61, \Delta = -120, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 61, T \in [19, 61]$
- [442] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 $^{21}1321 \quad 1 \quad 8 \quad 1 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 36, \Delta = -117, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$
- [443] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 $^11321 \quad 1 \quad 2 \quad 1 \quad 8 \quad 1 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 36, \Delta = -117, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$
- [444] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 $^131513161 \quad 1 \quad 6 \quad 1 \quad 2 \quad 2 \quad 1 \quad 3 \quad 1 \quad 5 \quad 1 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 27, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 27, T \in [19, 27]$
- [445] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
 $^115 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 29, \Delta = -114, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 29, T \in [19, 29]$
- [446] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 $^1321613151 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 29, \Delta = -114, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 29, T \in [19, 29]$
- [447] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 $^1315131612 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 27, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 27, T \in [19, 27]$
- [448] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 $^1315132161 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 28, \Delta = -113, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 28, T \in [19, 28]$
- [449] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 $^118122 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 40, \Delta = -118, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$
- [450] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 $^132171223 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 36, \Delta = -117, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$
- [451] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
 $^115 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 61, \Delta = -120, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 61, T \in [19, 61]$
- [452] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 $^118122 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 36, \Delta = -117, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$
- [453] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 $^121812 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 36, \Delta = -117, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$
- [454] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 $^1315131612 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 27, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19, T_{\max} = 27, T \in [19, 27]$
- [455] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
 $^1315132161 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 28, \Delta = -113, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 28$, $T \in [19, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [456] $6122315132^1 161 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 28$, $\Delta = -113$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 28$, $T \in [19, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [457] $6122^1 315132216 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 28$, $\Delta = -113$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 28$, $T \in [19, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [458] $413161232^1 171 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 36$, $\Delta = -117$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 36$, $T \in [19, 36]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [459] $413161232^1 1712 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 36$, $\Delta = -117$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 36$, $T \in [19, 36]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [460] $4132171232^1 161 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 40$, $\Delta = -118$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 40$, $T \in [19, 40]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [387] $1 \begin{pmatrix} 12 \\ 2222^1 221 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -16$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.75 \rightarrow 19$, $T_{\max} = 99$, $T \in [19, 99]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [468] $1 \begin{pmatrix} 12 \\ 22222^1 22221 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 99$, $T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [537] $612^1 2315132161 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 28$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 28$, $T \in [19, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [464] $1 \begin{pmatrix} 12 \\ 22222^1 1 \end{pmatrix} 1222 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 99$, $T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [465] $1 \begin{pmatrix} 12 \\ 22222^1 21 \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 99$, $T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [466] $1 \begin{pmatrix} 12 \\ 22222^1 1 \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 99$, $T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [467] $21 \begin{pmatrix} 12 \\ 22222^1 1 \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 99$, $T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [462] $1 \begin{pmatrix} 12 \\ 22222^1 221 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 99$, $T \in [19, 99]$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [483] $15132^1 17123151 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 29, \Delta = -114, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 29, T \in [19, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [484] $132181232^1 1513 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 25, \Delta = -100, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 25, T \in [19, 25]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [485] $1(12)122222^1 2222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [486] $21(12)12222^1 2222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [487] $221(12)1222^1 2222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [488] $2221(12)1222^1 2222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [489] $22221(12)122^1 2222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [490] $22222^1 1(12)122222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [491] $7122^1 315131513 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 72, \Delta = -121, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 72, T \in [19, 72]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [492] $(12)1222222^1 2222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [493] $16^1 12^1 321^1 81 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 36, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [494] $171232^1 17123 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 36, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [495] $16131612^1 3215 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 29, \Delta = -114, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 29, T \in [19, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [496] $16131612^1 3151 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 27, \Delta = -112, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 27, T \in [19, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [497] $1^1 8^1 12^1 32161 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 40, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[498] \quad 1 \overset{322^1}{8} 1232161 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 40, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[499] \quad 1 \overset{1}{1812321612^1} 32 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 61, \Delta = -120, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 61, T \in [19, 61]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[500] \quad 1 \overset{1}{1812^1} 32161232 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 61, \Delta = -120, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 61, T \in [19, 61]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[501] \quad 1 \overset{21}{181232^1} 16123 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 40, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[502] \quad 21 \overset{321}{8} 12^1 32161 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 40, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[503] \quad 1 \overset{31}{6} 12^1 3218122 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 36, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[504] \quad 1 \overset{1}{171232^1} 171223 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 36, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[505] \quad 1 \overset{1}{1712321712^1} 23 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 36, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[506] \quad 1 \overset{2321}{7} 1232^1 171 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 50, \Delta = -119, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 50, T \in [19, 50]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[507] \quad 1 \overset{232^1}{7} 1232171 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 50, \Delta = -119, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 50, T \in [19, 50]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[508] \quad 1 \overset{321}{7} 12^1 321712 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 36, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[509] \quad 4132^1 \overset{512321}{1} 8^1 1 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 61, \Delta = -120, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 61, T \in [19, 61]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[510] \quad 41321 \overset{51232^1}{8} 1 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 61, \Delta = -120, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 61, T \in [19, 61]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

- [511] $41321 \overset{15132^1}{8} 1 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 40, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [512] $41321 \overset{15132^1}{1} 1 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 40, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [513] $413151322 \overset{1}{181} + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 36, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [514] $612 \overset{31}{1} 31513161 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 61, \Delta = -120, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 61, T \in [19, 61]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [515] $612 \overset{1}{1} 231513161 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 27, \Delta = -112, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 27, T \in [19, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [516] $612321 \overset{232^1}{8} 12 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 61, \Delta = -120, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 61, T \in [19, 61]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [517] $612 \overset{232^1}{1} 321 \overset{1}{8} 12 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 61, \Delta = -120, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 61, T \in [19, 61]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [518] $216131612 \overset{1}{1} 3215 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 29, \Delta = -114, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 29, T \in [19, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [461] $1 \overset{1}{(12)} 1 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 99, \Delta = -135, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $\overset{22222^1}{22221}$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [536] $612 \overset{1}{1} 3151321613 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 72, \Delta = -121, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 72, T \in [19, 72]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [519] $216131612 \overset{1}{1} 3151 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 27, \Delta = -112, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 27, T \in [19, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [520] $161232 \overset{1}{1} 1613151 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 29, \Delta = -114, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 29, T \in [19, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [521] $321612 \overset{1}{1} 3218122 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 40, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [522] $232161232 \overset{1}{1} 1812 + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 61, \Delta = -120, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 61, T \in [19, 61]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

- [523] $232^1 1612321812 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 61$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 61$, $T \in [19, 61]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [524] $21712321712^1 23 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 36$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 36$, $T \in [19, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [525] $2171232^1 171223 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 36$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 36$, $T \in [19, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [526] $4132^1 181223151 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 40$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 40$, $T \in [19, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [527] $413218122^1 3151 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 40$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 40$, $T \in [19, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [528] $4132^1 218123215 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 61$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 61$, $T \in [19, 61]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [529] $413221812^1 3215 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 61$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 61$, $T \in [19, 61]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [530] $413221812^1 3151 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 40$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 40$, $T \in [19, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [531] $4132^1 218123151 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 40$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 40$, $T \in [19, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [532] $413151322^1 1812 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 36$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 36$, $T \in [19, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [533] $612^1 3216131513 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 77$, $\Delta = -122$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 77$, $T \in [19, 77]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [534] $612^1 3151316123 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 72$, $\Delta = -121$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 72$, $T \in [19, 72]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [535] $612^1 2315131612 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 27$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 27$, $T \in [19, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [548] $4141321812^1 314 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 77$, $\Delta = -122$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 77$, $T \in [19, 77]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [547] $141414132^1 2181 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 28$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19$, $T_{\max} = 28$, $T \in [19, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

- [546] $141414132^1 1812 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 28, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 28, T \in [19, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [545] $1812^1 31414141 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 28, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 28, T \in [19, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [544] $4132^1 171232161 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 40, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [543] $41321712^1 32161 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 40, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 40, T \in [19, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [542] $4131612^1 321712 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 36, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [541] $4132^1 161316131 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 19, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [540] $4131612^1 32171 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 36, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 36, T \in [19, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [539] $612^1 2315132216 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 28, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 28, T \in [19, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [538] $61315131612^1 32 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 77, \Delta = -122, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 77, T \in [19, 77]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [463] $1 \begin{pmatrix} 221 \\ 22222^1 21 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.44 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [567] $3141414132^1 181 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 72, \Delta = -121, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 72, T \in [19, 72]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [566] $(12)122222^1 22222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [550] $1 \begin{pmatrix} 21 \\ 22222^1 21 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [551] $1 \begin{pmatrix} 221 \\ 22222^1 1 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

- [568] $1414141322^1 181 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 28, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 28, T \in [19, 28]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [569] $8122^1 314141413 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 72, \Delta = -121, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 72, T \in [19, 72]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [570] $712^1 3141414141 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 28, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 28, T \in [19, 28]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [549] $1 \begin{pmatrix} 12 \\ 222222^1 221 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.22 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [572] $1 \begin{pmatrix} 12 \\ 222222^1 1 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [573] $1 \begin{pmatrix} 12 \\ 222222^1 221 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [574] $1 \begin{pmatrix} 12 \\ 222222^1 21 \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [575] $1 \begin{pmatrix} 12 \\ 222222^1 1 \end{pmatrix} 122 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [576] $21 \begin{pmatrix} 12 \\ 222222^1 1 \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [577] $321712^1 3151315 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 77, \Delta = -122, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 77, T \in [19, 77]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [578] $5132^1 171231513 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 77, \Delta = -122, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 77, T \in [19, 77]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [579] $1(12)12222^1 222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [580] $21(12)1222^1 222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [581] $221(12)122^1 222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 99, T \in [19, 99]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

- [582] $2221(12)12^1 222222 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 99$, $T \in [19, 99]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [583] $(12)122222^1 222222 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 99$, $T \in [19, 99]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [584] $16^1 131612^1 315 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 61$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 61$, $T \in [19, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [571] $1^1 (12) 1 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 99$, $\Delta = -135$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $222222^1 21$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 99$, $T \in [19, 99]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [586] $1^1 7^1 12^1 32171 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 50$, $\Delta = -119$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 50$, $T \in [19, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [587] $61232^1 1^1 8^1 12 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 61$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 61$, $T \in [19, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [588] $3216131612^1 315 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 72$, $\Delta = -121$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 72$, $T \in [19, 72]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [589] $31612^1 32161315 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 77$, $\Delta = -122$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 77$, $T \in [19, 77]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [590] $31612^1 31513161 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 61$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 61$, $T \in [19, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [591] $2321612^1 321812 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 61$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 61$, $T \in [19, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [592] $41321812^1 23151 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 40$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 40$, $T \in [19, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [593] $41322^1 18123215 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 61$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 61$, $T \in [19, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [594] $41322^1 18123151 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 40$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 40$, $T \in [19, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [595] $61232^1 16131513 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 77$, $\Delta = -122$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 77$, $T \in [19, 77]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [596] $612315132^1 1613 + \hat{1} \rightarrow 2$ $T = 13$, $T_{\max} = 72$, $\Delta = -121$, $\det = -49$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19$, $T_{\max} = 72$, $T \in [19, 72]$

- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [597] $812^1 3141414132 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 77, \Delta = -122, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 77, T \in [19, 77]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [598] $812^1 2314141413 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 72, \Delta = -121, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 72, T \in [19, 72]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [599] $414132^1 1812314 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 77, \Delta = -122, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 77, T \in [19, 77]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [585] $181232^1 161232 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 61, \Delta = -120, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.06 \rightarrow 19, T_{\max} = 61, T \in [19, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [600] $1 \begin{pmatrix} 12 \\ 22222222^1 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.94 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [601] $1 \begin{pmatrix} 12 \\ 22222222^1 \end{pmatrix} 21 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.94 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [602] $1 \begin{pmatrix} 12 \\ 22222222^1 \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.94 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [603] $51321712^1 31513 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 77, \Delta = -122, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.94 \rightarrow 18, T_{\max} = 77, T \in [18, 77]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [604] $1(12)1222^1 2222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.94 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [605] $21(12)122^1 2222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.94 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [606] $221(12)12^1 2222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.94 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [607] $(12)1222^1 2222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.94 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [608] $16131612^1 31513 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 61, \Delta = -120, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.94 \rightarrow 18, T_{\max} = 61, T \in [18, 61]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [609] $3161232^1 161315 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 77, \Delta = -122, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.94 \rightarrow 18, T_{\max} = 77, T \in [18, 77]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$

- [610] $1 \overset{22222222^1}{(12)} 1 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -81, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.84 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [611] $1(12)122^1 22222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -81, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.84 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [612] $21(12)12^1 22222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -81, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.84 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [613] $(12)1222^1 22222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -81, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.84 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [614] $41414132^1 18123 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 77, \Delta = -122, \det = -81, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.84 \rightarrow 18, T_{\max} = 77, T \in [18, 77]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [615] $1(12)12^1 22222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -100, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.76 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [616] $(12)122^1 22222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -100, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.76 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [617] $(12)12^1 22222222 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 99, \Delta = -135, \det = -121, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.69 \rightarrow 18, T_{\max} = 99, T \in [18, 99]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [618] $6131612^1 315132 + \hat{1} \rightarrow 2$ $T = 13, T_{\max} = 72, \Delta = -121, \det = -121, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.69 \rightarrow 18, T_{\max} = 72, T \in [18, 72]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- $T = 14$ [630] $3221 \overset{1}{(12)} 1222^1 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [632] $1 \overset{21}{(12)} 12 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 87, T \in [21, 87]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [622] $1 \overset{221}{(12)} 1 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 87, T \in [21, 87]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [623] $1 \overset{22^1}{(12)} 1 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 87, T \in [21, 87]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [624] $1 \overset{1}{(12)} 1222^1 2 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 84, \Delta = -123, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 84, T \in [21, 84]$

- $$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [652] $221 \overset{132^1 21}{(12)} 12222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 31, T \in [21, 31]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$
- [653] $22^1 21 \overset{13221}{(12)} 1222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 31, T \in [21, 31]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [654] $2221 \overset{132^1 21}{(12)} 1222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 31, T \in [21, 31]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$
- [655] $3221 \overset{3221}{(12)} 12222^1 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20, T_{\max} = 87, T \in [20, 87]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [656] $21 \overset{2313221}{(12)} 122^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 87, T \in [21, 87]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [657] $22^1 1 \overset{2313221}{(12)} 122 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 87, T \in [21, 87]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [658] $21 \overset{22^1 313221}{(12)} 122 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 87, T \in [21, 87]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$
- [659] $21 \overset{22313221}{(12)} 12^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 87, T \in [21, 87]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [660] $21 \overset{313221}{(12)} 1222^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 84, T \in [21, 84]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [661] $221 \overset{313221}{(12)} 122^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 84, T \in [21, 84]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [662] $22^1 1 \overset{313221}{(12)} 1222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 84, T \in [21, 84]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [663] $22^1 181231513215 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 29, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 29, T \in [21, 29]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [664] $22^1 181231513151 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 28, \Delta = -112, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 28, T \in [21, 28]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [665] $22222221(12)122^1 31 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 21, T_{\max} = 31, T \in [21, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [666] $22^1 222221(12)12231 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 31, T \in [21, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [667] $22^1 2221(12)1223132 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 87, T \in [21, 87]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [668] $22^1 221(12)12231322 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 87, T \in [21, 87]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [669] $22221(12)1223132^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 87, T \in [21, 87]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [670] $22^1 22221(12)122313 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 84, T \in [21, 84]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [671] $8123151315132^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 73, \Delta = -121, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 73, T \in [21, 73]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [672] $1712321 \overset{1}{8} \overset{21}{123} + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [673] $1712321 \overset{21}{8} \overset{221}{1} + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 26, T \in [21, 26]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [674] $1 \overset{321}{7} 12321812^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 34, T \in [21, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [675] $17123218122^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 26, T \in [21, 26]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [676] $1 \overset{221}{8} 12321712^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 26, T \in [21, 26]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [677] $1 \overset{221}{8} 123217122 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 26, T \in [21, 26]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [678] $321 \overset{21}{8} 12321712 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [679] $21 \overset{21}{8} 12321712^1 2 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 26, T \in [21, 26]$

- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
 [680] $321612321812^1 31 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 21$, $\Delta = -98$, $\det = 4$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21$, $T_{\max} = 21$, $T \in [21, 21]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [681] $3221712321812^1 2 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 4$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21$, $T_{\max} = 34$, $T \in [21, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [682] $2217123218122^1 2 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -107$, $\det = 4$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21$, $T_{\max} = 26$, $T \in [21, 26]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [683] $22^1 171232181222 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -107$, $\det = 4$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21$, $T_{\max} = 26$, $T \in [21, 26]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [684] $4131612321812^1 2 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 4$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21$, $T_{\max} = 34$, $T \in [21, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 1 \\ 2313221 \end{pmatrix}$
 [621] $1 (12) 122^1 2 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 4$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21$, $T_{\max} = 87$, $T \in [21, 87]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 1 \\ 132^1 21 \end{pmatrix}$
 [620] $1 (12) 122222 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 4$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 1 \\ 13221 \end{pmatrix}$
 [619] $1 (12) 12222^1 2 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 4$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 32^1 1 \\ 1 \end{pmatrix}$
 [739] $1 7 123218122 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 34$, $T \in [21, 34]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 1 \\ 3132^1 21 \end{pmatrix}$
 [689] $1 (12) 12222 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 84$, $\Delta = -123$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 84$, $T \in [21, 84]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 21 \\ 1322^1 1 \end{pmatrix}$
 [690] $1 (12) 12222 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 21 \\ 13221 \end{pmatrix}$
 [691] $1 (12) 122^1 22 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 221 \\ 13221 \end{pmatrix}$
 [692] $1 (12) 12^1 22 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

- [693] $1 \begin{smallmatrix} 221 \\ 1322^1 1 \end{smallmatrix} 1222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 31, T \in [21, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [694] $2221 \begin{smallmatrix} 1 \\ 3221 \end{smallmatrix} (12) 122^1 3 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 87, T \in [21, 87]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [695] $1 \begin{smallmatrix} 21 \\ 313221 \end{smallmatrix} (12) 12^1 22 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 84, T \in [21, 84]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [696] $1 \begin{smallmatrix} 21 \\ 3132^1 21 \end{smallmatrix} (12) 1222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 84, T \in [21, 84]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [697] $1 \begin{smallmatrix} 221 \\ 3132^1 21 \end{smallmatrix} (12) 122 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 84, T \in [21, 84]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [698] $21 \begin{smallmatrix} 21 \\ 1322^1 1 \end{smallmatrix} (12) 1222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 31, T \in [21, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [699] $21 \begin{smallmatrix} 21 \\ 13221 \end{smallmatrix} (12) 12^1 22 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 31, T \in [21, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [700] $21 \begin{smallmatrix} 221 \\ 1322^1 1 \end{smallmatrix} (12) 122 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 31, T \in [21, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [701] $21 \begin{smallmatrix} 221 \\ 3221 \end{smallmatrix} (12) 122^1 3 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 87, T \in [21, 87]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [702] $21 \begin{smallmatrix} 221 \\ 3132^1 21 \end{smallmatrix} (12) 12 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 84, T \in [21, 84]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [703] $1 \begin{smallmatrix} 513151321 \end{smallmatrix} 8 \begin{smallmatrix} 221 \\ 1322^1 1 \end{smallmatrix} 12^1 3 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 73, \Delta = -121, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 73, T \in [21, 73]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [704] $1 \begin{smallmatrix} 1322^1 1 \end{smallmatrix} (12) 1222222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 31, T \in [21, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [705] $1 \begin{smallmatrix} 13221 \end{smallmatrix} (12) 12222^1 22 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 20.22 \rightarrow 21, T_{\max} = 31, T \in [21, 31]$$

[706] $1^{2313221} (12) 122^1 22 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 87$, $T \in [21, 87]$

[707] $1^{22313221}(12)12^122+\hat{1}\rightarrow\hat{2}$ $T=14$, $T_{\max}=87$, $\Delta=-124$, $\det=9$, $(n_+, n_-, n_0)=(1, 14, 0)$, $c_{\text{ext}}=15.0$
 $T_{\min}=20.22\rightarrow 21$, $T_{\max}=87$, $T\in[21, 87]$

[708] $1^{313221}(12)1222^122 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 84$, $\Delta = -123$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 84$, $T \in [21, 84]$

[709] $1 \overset{3132^1 21}{(12)} 122222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 84, T \in [21, 84]$

[710] $21 \overset{1322^1}{(12)} 122222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 31, T \in [21, 31]$

$$\begin{array}{l} \text{[711]} \quad 21 \overset{13221}{(12)} 1222 \overset{1}{2}2 + \overset{1}{\hat{1}} \rightarrow \overset{2}{2} \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 20.22 \rightarrow 21, T_{\max} = 31, T \in [21, 31] \end{array}$$

[712] $^{1322^1 1}$ 221 (12) 12222 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$

[713] $221(12)122^122 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$

[714] $2221 \overset{1322^1 1}{(12)} 1222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 31, T \in [21, 31]$

[715] $22\bar{2}^{13221}1(12)1222 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$

[716] $22221(12)12\bar{2}^13 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 87, T \in [21, 87]$

[717] $21 \overset{2313221}{(12)} 12^1 22 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 87, T \in [21, 87]$

[718] $21 \overset{3132^1 21}{(12)} 12222 + \overset{\textcolor{red}{1}}{\textcolor{red}{1}} \rightarrow \textcolor{red}{2} \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 84, T \in [21, 84]$

- [733] $1712^1 1321^{221} 81 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 26$, $T \in [21, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [734] $171232^1 181 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 26$, $T \in [21, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [735] $171232^1 1812 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 26$, $T \in [21, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [736] $1712^1 1321^{221} 812 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 26$, $T \in [21, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [737] $18123217122^1 3 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 34$, $T \in [21, 34]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [687] $1(12)12^{122} + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 87$, $T \in [21, 87]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [686] $1(12)1222^{122} + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [685] $1(12)12222 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [741] $1712321812^{122} + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 26$, $T \in [21, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [742] $1712^1 132181222 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 26$, $T \in [21, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [743] $171232^1 181222 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 26$, $T \in [21, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [744] $1812^1 13217122 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 26$, $T \in [21, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [745] $181232^1 17122 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 26$, $T \in [21, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [746] $218^{321} 12321712 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 34, T \in [21, 34] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[747] & \begin{matrix} 21 \\ 21812^1 3217122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 26, T \in [21, 26] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[748] & \begin{matrix} 21 \\ 2181232^1 17122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 26, T \in [21, 26] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[749] & 5131612321712^1 3 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 73, \Delta = -121, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 73, T \in [21, 73] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[750] & 1613151321712^1 3 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 51, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 51, T \in [21, 51] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[751] & 32^1 161315132171 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 51, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 51, T \in [21, 51] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[752] & 32^1 161232181231 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 21, \Delta = -98, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[753] & 32^1 217123218122 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 34, T \in [21, 34] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[754] & 221712321812^1 22 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 26, T \in [21, 26] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[755] & 221712^1 32181222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 26, T \in [21, 26] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[756] & 22171232^1 181222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 26, T \in [21, 26] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[757] & 7122315131612^1 3 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 51, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 51, T \in [21, 51] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[758] & 6123151321712^1 3 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 62, \Delta = -120, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 62, T \in [21, 62] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[759] & 6123216131612^1 3 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 62, \Delta = -120, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 62, T \in [21, 62] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[760] & \begin{matrix} 1 \\ 161316131612^1 3 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 14, T_{\max} = 37, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.22 \rightarrow 21, T_{\max} = 37, T \in [21, 37] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [738] $1 \overset{32^1 21}{8} 12321712 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 34, T \in [21, 34]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [688] $1 \overset{1}{(12)} 122^1 22 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21, T_{\max} = 84, T \in [21, 84]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [740] $2321 \overset{2^1 1}{8} 1232171 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 41, \Delta = -118, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.33 \rightarrow 20, T_{\max} = 41, T \in [20, 41]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [761] $1 \overset{1}{512315132^1 1} 8 1 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 29, \Delta = -113, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 29, T \in [20, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [762] $1 \overset{1}{512^1 3151321} 8 1 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 29, \Delta = -113, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 29, T \in [20, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [763] $1 \overset{1}{151315132^1 1} 8 1 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 28, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 28, T \in [20, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [764] $1 \overset{1}{(12)} 122^1 222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 31, T \in [20, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [765] $1 \overset{1}{(12)} 1222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [766] $1 \overset{1}{(12)} 12^1 222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 84, T \in [20, 84]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [767] $1 \overset{1}{(12)} 12222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 84, T \in [20, 84]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [768] $1 \overset{21}{(12)} 12^1 222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 31, T \in [20, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [769] $22^1 21 \overset{1}{(12)} 1223 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.25 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [770] $1 \overset{221}{(12)} 12 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 19.75 \rightarrow 20, T_{\max} = 87, T \in [20, 87] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[774] & 21 \begin{pmatrix} 12 & 12 + \hat{1} \rightarrow 2 \\ 23132^{121} \end{pmatrix} T = 14, T_{\max} = 87, \Delta = -124, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.75 \rightarrow 20, T_{\max} = 87, T \in [20, 87] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[775] & 21 \begin{pmatrix} 12 & 12 + \hat{1} \rightarrow 2 \\ 31322^{11} \end{pmatrix} T = 14, T_{\max} = 84, \Delta = -123, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.75 \rightarrow 20, T_{\max} = 84, T \in [20, 84] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[776] & 1 \begin{pmatrix} 512315132^{121} \\ 8 \end{pmatrix} 1 + \hat{1} \rightarrow 2 T = 14, T_{\max} = 29, \Delta = -113, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.75 \rightarrow 20, T_{\max} = 29, T \in [20, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[777] & 1 \begin{pmatrix} 512^{131513221} \\ 8 \end{pmatrix} 1 + \hat{1} \rightarrow 2 T = 14, T_{\max} = 29, \Delta = -113, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.75 \rightarrow 20, T_{\max} = 29, T \in [20, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[778] & 1 \begin{pmatrix} 151315132^{121} \\ 8 \end{pmatrix} 1 + \hat{1} \rightarrow 2 T = 14, T_{\max} = 28, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.75 \rightarrow 20, T_{\max} = 28, T \in [20, 28] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[779] & 1 \begin{pmatrix} 512^{13151321} \\ 8 \end{pmatrix} 12 + \hat{1} \rightarrow 2 T = 14, T_{\max} = 29, \Delta = -113, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.75 \rightarrow 20, T_{\max} = 29, T \in [20, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[780] & 1 \begin{pmatrix} 512315132^{11} \\ 8 \end{pmatrix} 12 + \hat{1} \rightarrow 2 T = 14, T_{\max} = 29, \Delta = -113, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.75 \rightarrow 20, T_{\max} = 29, T \in [20, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[781] & 1 \begin{pmatrix} 151315132^{11} \\ 8 \end{pmatrix} 12 + \hat{1} \rightarrow 2 T = 14, T_{\max} = 28, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.75 \rightarrow 20, T_{\max} = 28, T \in [20, 28] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[782] & 1 \begin{pmatrix} 512321 \\ 8 \end{pmatrix} 1232^{115} + \hat{1} \rightarrow 2 T = 14, T_{\max} = 37, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.75 \rightarrow 20, T_{\max} = 37, T \in [20, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[783] & 1 \begin{pmatrix} 512^{1321} \\ 8 \end{pmatrix} 123151 + \hat{1} \rightarrow 2 T = 14, T_{\max} = 31, \Delta = -115, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.75 \rightarrow 20, T_{\max} = 31, T \in [20, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[784] & 1 \begin{pmatrix} 13221 \\ (12)1222^{1222} + \hat{1} \rightarrow 2 \end{pmatrix} T = 14, T_{\max} = 31, \Delta = -115, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.75 \rightarrow 20, T_{\max} = 31, T \in [20, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[785] & 1 \begin{pmatrix} 23132^{121} \\ (12)1222 + \hat{1} \rightarrow 2 \end{pmatrix} T = 14, T_{\max} = 87, \Delta = -124, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.75 \rightarrow 20, T_{\max} = 87, T \in [20, 87] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[786] & 1 \begin{pmatrix} 2313221 \\ (12)12^{1222} + \hat{1} \rightarrow 2 \end{pmatrix} T = 14, T_{\max} = 87, \Delta = -124, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.75 \rightarrow 20, T_{\max} = 87, T \in [20, 87]
\end{aligned}$$

- $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 29$, $T \in [20, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [801] $221812^1 31513215 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 29$, $\Delta = -113$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 29$, $T \in [20, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [802] $221812^1 31513151 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 28$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 28$, $T \in [20, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [803] $151322181232^1 15 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 31$, $T \in [20, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [804] $2222^1 2221(12)12231 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 31$, $T \in [20, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [805] $2222^1 21(12)1223132 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 87$, $T \in [20, 87]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [806] $222221(12)122^1 3132 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 87$, $T \in [20, 87]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [807] $2222^1 1(12)12231322 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 87$, $T \in [20, 87]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [808] $2222221(12)12^1 2313 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 84$, $\Delta = -123$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 84$, $T \in [20, 84]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [809] $2222^1 221(12)122313 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 84$, $\Delta = -123$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 84$, $T \in [20, 84]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [810] $\overset{1}{16}1315132^1 \overset{1}{17}1 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 26$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 26$, $T \in [20, 26]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [811] $\overset{1}{17}12^1 32181223 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 34$, $T \in [20, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [812] $\overset{1}{17}12^1 3218^{\overset{21}{12}}3 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 34$, $T \in [20, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [813] $\overset{321}{1}7^{\overset{21}{12}}32^1 181 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 34$, $T \in [20, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [814] $\overset{1}{17}12^1 321613151 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 28$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20$, $T_{\max} = 28$, $T \in [20, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

- [815] $161315132^1 1712 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 26, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 26, T \in [20, 26]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [816] $1712^1 315131612 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 26, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 26, T \in [20, 26]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [817] $161315132^1 2171 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 26, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 26, T \in [20, 26]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [818] $1712^1 315132161 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 27, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 27, T \in [20, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [819] $1712315132^1 161 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 27, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 27, T \in [20, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [820] $16^3 1232^1 181231 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 20, \Delta = -97, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [821] $18^2 12^1 32171223 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 34, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [822] $1^3 8^2 1232^1 1712 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 34, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [823] $1^3 7^2 1232^1 18122 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 34, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [824] $21^3 8^2 1232^1 1712 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 34, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [825] $6122^1 315132171 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 27, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 27, T \in [20, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [826] $6122315132^1 171 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 27, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 27, T \in [20, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [827] $161232171232^1 15 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 31, \Delta = -115, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 31, T \in [20, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [828] $1612^1 3217123215 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 31, \Delta = -115, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 31, T \in [20, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[illegible]

- [844] $1613161232^1 161 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 27, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 27, T \in [20, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [845] $413161232^{21} 181 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 34, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [846] $4132171232^1 171 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 34, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [847] $21613161232^1 161 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 27, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 27, T \in [20, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [848] $414132181232^1 15 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 51, \Delta = -119, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 51, T \in [20, 51]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [849] $413161232^1 18122 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 34, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [850] $41322181232^1 161 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 37, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 37, T \in [20, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [851] $4132171232^1 1712 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 34, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [852] $4132181232^1 161 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 37, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 37, T \in [20, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [773] $21^{22} (12) 1223 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.25 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [771] $1^{21} (12) 1222 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 84, \Delta = -123, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 84, T \in [20, 84]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [772] $1^{221} (12) 122 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 84, \Delta = -123, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.75 \rightarrow 20, T_{\max} = 84, T \in [20, 84]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [920] $21613151322^1 171 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 26, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 26, T \in [20, 26]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [884] $15132^1 218123151 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 29, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 29, T \in [20, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

- [858] $1 \begin{smallmatrix} 21 \\ 223132^1 21 \end{smallmatrix} (12) \ 12 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [859] $21 \begin{smallmatrix} 221 \\ 3221 \end{smallmatrix} (12) 12^1 23 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [860] $21 \begin{smallmatrix} 21 \\ 231322^1 1 \end{smallmatrix} (12) \ 12 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [861] $1 \begin{smallmatrix} 5123151322^1 1 \\ 8 \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 29, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 29, T \in [20, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [862] $1 \begin{smallmatrix} 3151315132^1 1 \\ 8 \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 62, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 62, T \in [20, 62]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [863] $1 \begin{smallmatrix} 1513151322^1 1 \\ 8 \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 28, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 28, T \in [20, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [864] $1 \begin{smallmatrix} 512321 \\ 8 \end{smallmatrix} 12^1 3215 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 37, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 37, T \in [20, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [865] $1 \begin{smallmatrix} 51232^1 1 \\ 8 \end{smallmatrix} 123151 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 31, T \in [20, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [866] $1 \begin{smallmatrix} 512321 \\ 8 \end{smallmatrix} 12^1 3151 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 31, T \in [20, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [867] $1 \begin{smallmatrix} 151321 \\ 8 \end{smallmatrix} 12^1 3151 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 29, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 29, T \in [20, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [868] $1 \begin{smallmatrix} 13221 \\ (12) 122^1 2222 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 31, T \in [20, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [869] $1 \begin{smallmatrix} 231322^1 1 \\ (12) 12222 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [870] $1 \begin{smallmatrix} 223132^1 21 \\ (12) 1222 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

- [871] $1^{313221}(12)12^12222 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 84$, $\Delta = -123$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 84$, $T \in [20, 84]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [872] $21^{13221}(12)12^12222 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 31$, $T \in [20, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [873] $22221^{3221}(12)12^123 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 87$, $T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [874] $21^{231322^11}(12)1222 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 87$, $T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [875] $221^{231322^11}(12)122 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 87$, $T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [876] $21^{223132^121}(12)122 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 87$, $T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [877] $18122^{1315131513} + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 62$, $\Delta = -120$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 62$, $T \in [20, 62]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [878] $322^{118123151315} + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [879] $21812^{1231513215} + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 29$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 29$, $T \in [20, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [880] $21812^{1315131513} + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 62$, $\Delta = -120$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 62$, $T \in [20, 62]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [881] $21812^{1231513151} + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 28$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 28$, $T \in [20, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [882] $15132^{1218123215} + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 31$, $T \in [20, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [883] $1513221812^{13215} + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 31$, $T \in [20, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [856] $2221^1(12)12^123 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 87$, $T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

- [855] $1 \overset{221}{(12)} 1 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [854] $1 \overset{1}{(12)} 1222 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [887] $22222 \overset{1}{1}(12)1223132 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [888] $222221(12)12^1 23132 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [889] $22221(12)122^1 31322 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 87, \Delta = -124, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [890] $22222 \overset{1}{21}(12)122313 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 84, \Delta = -123, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 84, T \in [20, 84]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [891] $1712321812 \overset{1}{23} + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [892] $171232 \overset{1}{18}1223 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [893] $1 \overset{2321}{7} 1232 \overset{1}{18}1 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 41, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 41, T \in [20, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [894] $1 \overset{2321}{7} 1232181 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 41, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 41, T \in [20, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [895] $171232 \overset{1}{18}123 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [896] $1 \overset{321}{7} 12^1 32181 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 34, T \in [20, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [897] $171232 \overset{1}{16}13151 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 28, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 28, T \in [20, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [898] $161315132 \overset{31}{17}1 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 41, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20, T_{\max} = 41, T \in [20, 41]$

$$\begin{aligned}
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 27, T \in [20, 27] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[913] & 1612321712^1 3215 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 31, T \in [20, 31] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[914] & 161232^1 17123215 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 31, T \in [20, 31] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[915] & 31513161232^1 171 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 62, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 62, T \in [20, 62] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[916] & 161232^1 17123151 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 29, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 29, T \in [20, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[917] & 1612321712^1 3151 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 29, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 29, T \in [20, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[918] & 15131612^1 321712 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 28, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 28, T \in [20, 28] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[919] & 32161315132^1 171 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 51, \Delta = -119, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 51, T \in [20, 51] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[853] & 1 \begin{matrix} 1 \\ (12) 12^1 2222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 14, T_{\max} = 31, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& \begin{matrix} 13221 \\ 13221 \end{matrix} \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 31, T \in [20, 31] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[921] & 31612315132^1 171 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 51, \Delta = -119, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 51, T \in [20, 51] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[922] & 16123151322^1 171 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 27, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 27, T \in [20, 27] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[923] & 321612^1 32181231 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 21, \Delta = -98, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 21, T \in [20, 21] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[924] & 322^1 17123218122 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 34, T \in [20, 34] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[925] & 3221712^1 3218122 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 34, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 34, T \in [20, 34] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[926] & 612^1 32171231513 + \hat{1} \rightarrow 2 \quad T = 14, T_{\max} = 73, \Delta = -121, \det = 25, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.44 \rightarrow 20, T_{\max} = 73, T \in [20, 73] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [927] $712315131612^1 32 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [928] $613151321712^1 32 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [929] $7122^1 3151316123 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 51$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [930] $612^1 31513217123 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 62$, $\Delta = -120$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 62$, $T \in [20, 62]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [931] $612^1 23151321712 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 27$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 27$, $T \in [20, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [932] $7122^1 3151321613 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 51$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [933] $612^1 23151322171 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 27$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 27$, $T \in [20, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [934] $61223151322^1 171 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 27$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 27$, $T \in [20, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [935] $16131612^1 32161 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 27$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 27$, $T \in [20, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [936] $4131612^1 321^1 81 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 34$, $T \in [20, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [937] $41321712^1 32171 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 34$, $T \in [20, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [938] $4132^1 171232171 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 34$, $T \in [20, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [939] $612^1 3216131^1 61 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 51$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [940] $216131612^1 32161 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 27$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 27$, $T \in [20, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [941] $4141321812^1 3215 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 51$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$

- [942] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ 4141321812¹3151 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 37$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 37$, $T \in [20, 37]$
- [943] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ 4131612¹3218122 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 34$, $T \in [20, 34]$
- [944] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$ 413221812¹32161 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 37$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 37$, $T \in [20, 37]$
- [945] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$ 4132¹2181232161 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 37$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 37$, $T \in [20, 37]$
- [946] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$ 41321712¹321712 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 34$, $T \in [20, 34]$
- [947] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$ 4132¹1712321712 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 34$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 34$, $T \in [20, 34]$
- [948] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$ 612¹32161316123 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 62$, $\Delta = -120$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 62$, $T \in [20, 62]$
- [949] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 41321812¹32161 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 37$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 37$, $T \in [20, 37]$
- [950] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$ 4132¹181232161 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 37$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 37$, $T \in [20, 37]$
- [951] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ 61316131612¹32 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 51$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$
- [952] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$ 413161315132¹16 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 51$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$
- [886] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$ 22222¹221(12)12231 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 31$, $T \in [20, 31]$
- [857] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 1 (12)²²¹_{231322¹1} + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 87$, $T \in [20, 87]$
- [885] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 1513221812¹3151 + $\hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 29$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.44 \rightarrow 20$, $T_{\max} = 29$, $T \in [20, 29]$

- [954] $222^1 1(12)1223 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.89 \rightarrow 19, T_{\max} = 87, T \in [19, 87]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$
- [957] $1^{2231322^1} (12) 1222 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [958] $222^1 21(12)1223 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.89 \rightarrow 19, T_{\max} = 87, T \in [19, 87]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$
- [959] $21^{2231322^1} (12) 122 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [960] $1812^1 2315131513 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 62, \Delta = -120, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 62, T \in [20, 62]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [961] $1812^1 3151315132 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 73, T \in [20, 73]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [962] $222222^1 21(12)12231 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 31, \Delta = -115, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 31, T \in [20, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [963] $22221(12)12^1 231322 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 87, \Delta = -124, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 87, T \in [20, 87]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [964] $222222^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 84, \Delta = -123, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 84, T \in [20, 84]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [965] $8122^1 3151315132 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 73, T \in [20, 73]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [966] $712^1 2315131^1 61 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 41, \Delta = -118, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 41, T \in [20, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [967] $712^1 32161315132 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 73, T \in [20, 73]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [968] $712^1 31513161232 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 73, T \in [20, 73]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [969] $712^1 23151316123 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 51, \Delta = -119, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20, T_{\max} = 51, T \in [20, 51]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [970] $712^1 23151321613 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 51, \Delta = -119, \det = 36, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 19.22 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [971] $14141414132^1 181 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 28$, $\Delta = -112$, $\det = 36$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20$, $T_{\max} = 28$, $T \in [20, 28]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [972] $8122^1 3141414141 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 28$, $\Delta = -112$, $\det = 36$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20$, $T_{\max} = 28$, $T \in [20, 28]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [955] $1 \begin{pmatrix} 12 \\ 2231322^1 1 \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 36$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20$, $T_{\max} = 87$, $T \in [20, 87]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [956] $1 \begin{pmatrix} 12 \\ 13221 \end{pmatrix} 12^1 22222 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 36$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20$, $T_{\max} = 31$, $T \in [20, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [953] $1 \begin{pmatrix} 12 \\ 2231322^1 1 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 36$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.22 \rightarrow 20$, $T_{\max} = 87$, $T \in [20, 87]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [973] $1 \begin{pmatrix} 8 \\ 51315132^1 1 \end{pmatrix} 123 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [974] $1 \begin{pmatrix} 8 \\ 3151321 \end{pmatrix} 12^1 315 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [975] $3221812^1 3151315 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [976] $3218122^1 3151315 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [977] $5132^1 2181231513 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [978] $31513221812^1 315 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [979] $151322^1 18123215 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 31$, $T \in [20, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [980] $151322^1 18123151 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 29$, $\Delta = -113$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 29$, $T \in [20, 29]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [981] $2222222^1 1(12)12231 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 31$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 31$, $T \in [20, 31]$

- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
 [982] $812^1 23151315132 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [983] $812^1 31513151322 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [984] $1^{\frac{2321}{7}} 12^1 32181 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 41$, $\Delta = -118$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 41$, $T \in [20, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
 [985] $1^{\frac{2321}{7}} 12^1 321812 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 41$, $\Delta = -118$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 41$, $T \in [20, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
 [986] $21^{\frac{2321}{8}} 12^1 32171 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 41$, $\Delta = -118$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 41$, $T \in [20, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
 [987] $315131612^1 32171 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 62$, $\Delta = -120$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 62$, $T \in [20, 62]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
 [989] $31612321712^1 315 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [990] $31612^1 321712315 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
 [991] $161315132^1 17123 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 51$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [992] $31612^1 315132171 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 51$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [993] $61232^1 171231513 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [994] $612315132^1 17123 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 62$, $\Delta = -120$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 62$, $T \in [20, 62]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [995] $7122315132^1 1613 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 51$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [996] $61232^1 1613161 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 51$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [997] $1613161232^1 1613 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 51$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [998] $414132^1 18123215 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 51$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [999] $414132^1 18123151 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 37$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 37$, $T \in [20, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1000] $41316131612^1 315 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 51$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 51$, $T \in [20, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1001] $41322^1 181232161 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 37$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 37$, $T \in [20, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1002] $61232^1 161316123 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 62$, $\Delta = -120$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 62$, $T \in [20, 62]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1003] $812^1 31414141413 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 62$, $\Delta = -120$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 62$, $T \in [20, 62]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1004] $812^1 23141414141 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 28$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 28$, $T \in [20, 28]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
- [988] $513161232^1 17123 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 49$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.06 \rightarrow 20$, $T_{\max} = 73$, $T \in [20, 73]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1005] $1^{315132^1 1} 8^{12315} + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 64$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.94 \rightarrow 19$, $T_{\max} = 73$, $T \in [19, 73]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1006] $2222^1 1^{3221} (12) 1223 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 87$, $\Delta = -124$, $\det = 64$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.69 \rightarrow 19$, $T_{\max} = 87$, $T \in [19, 87]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1007] $513221812^1 31513 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 64$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.94 \rightarrow 19$, $T_{\max} = 73$, $T \in [19, 73]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1008] $315132^1 21812315 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 64$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.94 \rightarrow 19$, $T_{\max} = 73$, $T \in [19, 73]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [1009] $5131612^1 3217123 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 64$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.94 \rightarrow 19$, $T_{\max} = 73$, $T \in [19, 73]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1010] $3161232^1 1712315 + \hat{1} \rightarrow 2$ $T = 14$, $T_{\max} = 73$, $\Delta = -121$, $\det = 64$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.94 \rightarrow 19$, $T_{\max} = 73$, $T \in [19, 73]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$

[1011] $612321712^1 31513 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 64, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.94 \rightarrow 19, T_{\max} = 73, T \in [19, 73]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1012] $16131612^1 321613 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 51, \Delta = -119, \det = 64, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.94 \rightarrow 19, T_{\max} = 51, T \in [19, 51]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1013] $321812^1 23151315 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 81, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.84 \rightarrow 19, T_{\max} = 73, T \in [19, 73]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1014] $71232^1 161315132 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 81, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.84 \rightarrow 19, T_{\max} = 73, T \in [19, 73]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

[1015] $51322^1 181231513 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 100, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.76 \rightarrow 19, T_{\max} = 73, T \in [19, 73]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1016] $61315132^1 171232 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 100, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.76 \rightarrow 19, T_{\max} = 73, T \in [19, 73]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1017] $3151322^1 1812315 + \hat{1} \rightarrow 2$ $T = 14, T_{\max} = 73, \Delta = -121, \det = 121, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.69 \rightarrow 19, T_{\max} = 73, T \in [19, 73]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

$T = 15$ [1037] $21(12)^{21} 122314132^1 2 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 85, \Delta = -123, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 85, T \in [22, 85]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[1018] $1(12)^1 122^1 31 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 27, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1023] $22^1 1(12)^{313221} 12231 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 27, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1024] $221(12)^{313221} 122^1 31 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 27, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1025] $22^1 313221(12)122313 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 29, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 29, T \in [22, 29]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1026] $1^1 8^{22} 12321^1 8^1 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 27, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[1027] $18^{21} 12321^1 8^1 123 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 32, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21, T_{\max} = 32, T \in [21, 32]$

- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
 [1028] $1 \overset{22^1 1}{8} \overset{21}{12321} 8 \overset{1}{12} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\overset{22^1 2221}{1} (12) 1223141 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 31, T \in [22, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\overset{22^1 221}{21} (12) 1223141 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 31, T \in [22, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\overset{2221}{22^1 1} (12) 1223141 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 31, T \in [22, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\overset{22^1 21}{221} (12) 1223141 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 31, T \in [22, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\overset{22^1 1}{1} (12) 1223141322 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 85, \Delta = -123, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 85, T \in [22, 85]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\overset{221}{1} (12) 122314132^1 2 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 85, \Delta = -123, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 85, T \in [22, 85]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- $\overset{22^1 21}{1} (12) 122314132 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 85, \Delta = -123, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 85, T \in [22, 85]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\overset{22313221}{1} (12) 122^1 31 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\overset{22^1 313221}{1} (12) 12231 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- $\overset{21}{1} (12) 122^1 31 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\overset{22^1 21}{21} (12) 12231413 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 79, \Delta = -122, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 79, T \in [22, 79]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\overset{221}{22^1 1} (12) 12231413 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 79, \Delta = -122, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 79, T \in [22, 79]$

- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
 [1055] $21613151321812^1 2 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -4$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22$, $T_{\max} = 27$, $T \in [22, 27]$
- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
 [1056] $16123151321812^1 2 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -4$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22$, $T_{\max} = 27$, $T \in [22, 27]$
- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
 [1057] $22^1 2181232181222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -106$, $\det = -4$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22$, $T_{\max} = 27$, $T \in [22, 27]$
- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 [1058] $81232161315132^1 2 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -4$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22$, $T_{\max} = 63$, $T \in [22, 63]$
- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
 [1059] $61223151321812^1 2 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -4$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22$, $T_{\max} = 27$, $T \in [22, 27]$
- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
 [1060] $41321712321812^1 2 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -4$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22$, $T_{\max} = 32$, $T \in [22, 32]$
- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
 [1022] $21 \begin{pmatrix} 2313221 \\ 12 \end{pmatrix} 122^1 31 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -104$, $\det = -4$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22$, $T_{\max} = 27$, $T \in [22, 27]$
- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [1038] $21 \begin{pmatrix} 22^1 1 \\ (12) \end{pmatrix} 122314132 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -4$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22$, $T_{\max} = 85$, $T \in [22, 85]$
- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [1036] $1 \begin{pmatrix} 22^1 221 \\ (12) \end{pmatrix} 12231413 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 79$, $\Delta = -122$, $\det = -4$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22$, $T_{\max} = 79$, $T \in [22, 79]$
- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [1120] $71223151321712^1 3 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -9$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 42$, $T \in [22, 42]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
 [1086] $221 \begin{pmatrix} 222^1 1 \\ (12) \end{pmatrix} 1223141 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -9$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 31$, $T \in [22, 31]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
 [1066] $1 \begin{pmatrix} 22313221 \\ (12) \end{pmatrix} 12^1 231 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -104$, $\det = -9$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 27$, $T \in [22, 27]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 [1067] $1 \begin{pmatrix} 3132^1 21 \\ (12) \end{pmatrix} 1223132 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 29$, $\Delta = -112$, $\det = -9$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 29$, $T \in [22, 29]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [1068] $21 \begin{pmatrix} 2313221 \\ (12) \end{pmatrix} 12^1 231 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -104$, $\det = -9$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 27$, $T \in [22, 27]$

- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 [1069] $^{313221} 221 (12) 12^1 231 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -103, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $^{3132^{121}} 221 (12) 12231 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -103, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $^{313221} 32^1 21 (12) 1223 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 29, \Delta = -112, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 29, T \in [22, 29]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $^{313221} 21 (12) 122^1 313 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 28, \Delta = -111, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $^{313221} 22313221(12)122^1 313 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 29, \Delta = -112, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 29, T \in [22, 29]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $^{21} 18 123218 122^1 3 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 32, T \in [22, 32]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $^{1} 18 123218 1232 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 38, \Delta = -117, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.33 \rightarrow 21, T_{\max} = 38, T \in [21, 38]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- $^{221} 18 12^1 3218 1 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $^{21} 18 123218 12^1 3 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 32, T \in [22, 32]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $^{221} 18 1232^1 18 12 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $^{221} 18 12^1 3218 12 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $^{21} 218 12^1 3218 12 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $^{222221} 1 (12) 122^1 3141 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 31, T \in [22, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $^{222^1 221} 1 (12) 1223141 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 31, T \in [22, 31] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 22221 \\ [1083] \end{matrix} 21(12)122^1 3141 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 31, T \in [22, 31] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 222^1 21 \\ [1084] \end{matrix} 21(12)1223141 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 31, T \in [22, 31] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 2221 \\ [1085] \end{matrix} 221(12)122^1 3141 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 31, T \in [22, 31] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 21 \\ [1064] \end{matrix} 1(12)12231 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -103, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1063] \end{matrix} 13221(12)122^1 3 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1062] \end{matrix} 1(12)122^1 313 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 28, \Delta = -111, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 28, T \in [22, 28] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 222^1 1 \\ [1089] \end{matrix} 21(12)12231413 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 79, \Delta = -122, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 79, T \in [22, 79] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 32^1 1 \\ [1090] \end{matrix} 181232161315 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 63, T \in [22, 63] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1091] \end{matrix} 1812315131612^1 3 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 42, T \in [22, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 32^1 1 \\ [1092] \end{matrix} 181231513161 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 42, T \in [22, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 32^1 21 \\ [1093] \end{matrix} 18123218122 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 32, T \in [22, 32] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 221 \\ [1094] \end{matrix} 181232^1 181222 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 221 \\ [1095] \end{matrix} 1812^1 32181222 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27]
\end{aligned}$$

- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
 [1096] $1 \ 8 \ 12321812^1 22 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
 [1097] $21 \ 8 \ 123218122 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 32, T \in [22, 32]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [1098] $2321 \ 8 \ 12321812 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 38, \Delta = -117, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.33 \rightarrow 21, T_{\max} = 38, T \in [21, 38]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [1099] $21 \ 8 \ 12^1 32181222 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
 [1100] $21 \ 8 \ 12321812^1 22 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [1101] $21 \ 8 \ 1232^1 181222 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 [1102] $413221(12)1222 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 85, \Delta = -123, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 85, T \in [22, 85]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
 [1103] $6123151321 \ 8 \ 1 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 52, \Delta = -119, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 52, T \in [22, 52]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
 [1104] $1413221(12)1222^1 22 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 31, T \in [22, 31]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [1105] $14132^1 21(12)1222222 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 31, T \in [22, 31]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 [1106] $2231413221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 85, \Delta = -123, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 85, T \in [22, 85]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [1107] $231413221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 85, \Delta = -123, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 85, T \in [22, 85]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [1108] $31413221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 79, \Delta = -122, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 79, T \in [22, 79]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [1109] $51316123218122^1 3 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 63, T \in [22, 63]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1110] & 16131513218122^1 3 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 42, T \in [22, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1111] & 32^1 1613151321812 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 42, T \in [22, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[1112] & 16131513221812^1 3 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 42, T \in [22, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1113] & 32^1 1613151322181 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 42, T \in [22, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[1114] & 32^1 1712321712315 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 63, T \in [22, 63] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1115] & 17123151321712^1 3 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 42, T \in [22, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1116] & 222^1 181232181222 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0} \\
[1117] & 2221812^1 32181222 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[1118] & 61231513218122^1 3 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 52, \Delta = -119, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 52, T \in [22, 52] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1119] & 61231513221812^1 3 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 52, \Delta = -119, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 52, T \in [22, 52] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1061] & \begin{matrix} 1 \\ 12 \end{matrix} 12^1 231 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& \begin{matrix} 2313221 \\ 2313221 \end{matrix} \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1121] & 16131612321712^1 3 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 42, T \in [22, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1122] & 32^1 1613161232171 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 42, T \in [22, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[1088] & \begin{matrix} 222^1 21 \\ 12 \end{matrix} 12231413 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 79, \Delta = -122, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.22 \rightarrow 22, T_{\max} = 79, T \in [22, 79] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1065] $\begin{smallmatrix} 21 \\ 313221 \end{smallmatrix} 1 (12) 12^1 231 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 27, \Delta = -103, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 27, T \in [22, 27]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1087] $\begin{smallmatrix} 222^1 1 \\ 313221 \end{smallmatrix} 1 (12) 122314132 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 85, \Delta = -123, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 85, T \in [22, 85]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1126] $\begin{smallmatrix} 21 \\ 31322^1 1 \end{smallmatrix} 1 (12) 12231 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 27, \Delta = -103, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1127] $\begin{smallmatrix} 313221 \\ 31322^1 1 \end{smallmatrix} 1 (12) 122^1 3132 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 29, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 29, T \in [21, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1128] $\begin{smallmatrix} 31322^1 1 \\ 313221 \end{smallmatrix} 1 (12) 1223132 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 29, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 29, T \in [21, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1129] $\begin{smallmatrix} 23132^1 21 \\ 31322^1 1 \end{smallmatrix} 21 (12) 12231 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 27, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1130] $\begin{smallmatrix} 31322^1 1 \\ 23132^1 21 \end{smallmatrix} 221 (12) 12231 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 27, \Delta = -103, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1131] $\begin{smallmatrix} 313221 \\ 23132^1 21 \end{smallmatrix} 21 (12) 12^1 2313 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 28, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 28, T \in [21, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1132] $\begin{smallmatrix} 31322^1 1 \\ 23132^1 21 \end{smallmatrix} 23132^1 21 (12) 1223132 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 30, \Delta = -113, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 30, T \in [21, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1133] $\begin{smallmatrix} 31322^1 1 \\ 23132^1 21 \end{smallmatrix} 22313221 (12) 12^1 2313 + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 29, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 29, T \in [21, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1134] $\begin{smallmatrix} 1 \\ 6122315132^1 181 \end{smallmatrix} + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 27, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1135] $\begin{smallmatrix} 1 \\ 6122^1 3151321 181 \end{smallmatrix} + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 27, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1136] $\begin{smallmatrix} 1 \\ 161315132^1 181 \end{smallmatrix} + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 27, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1137] $\begin{smallmatrix} 1 \\ 161315132^1 2181 \end{smallmatrix} + \hat{1} \rightarrow 2$ $T = 15, T_{\max} = 27, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1138] $1712^1 1315132171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1139] $1812^{21} 132181223 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1125] $132^{11} 21(12)1223 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -104$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.25 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1124] $1(12)12^{13} 2313 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 28$, $\Delta = -111$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 28$, $T \in [21, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1123] $1(12)12231 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -104$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1143] $21(12)1223141 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1144] $21(12)12^{12} 23141 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1145] $221(12)12^{12} 23141 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1146] $1(12)122^{12} 31413 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 79$, $\Delta = -122$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 79$, $T \in [21, 79]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1147] $1(12)12231413 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 79$, $\Delta = -122$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 79$, $T \in [21, 79]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1148] $21(12)122^{12} 31413 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 79$, $\Delta = -122$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 79$, $T \in [21, 79]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1149] $221(12)122^{12} 31413 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 79$, $\Delta = -122$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 79$, $T \in [21, 79]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1150] $1^1 8^{512321} 1232^{12} 161 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$

- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [1151] $1 \overset{512^1 321}{8} 1232161 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 31, T \in [21, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [1152] $1 \overset{151321}{8} 1232^1 161 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 29, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 29, T \in [21, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [1153] $1 \overset{21}{8} 12^1 321613151 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 28, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 28, T \in [21, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [1154] $1 \overset{1}{16} 1315132^1 18122 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [1155] $1 \overset{21}{8} 12^1 315131612 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [1156] $1 \overset{1}{16} 1315132^1 21812 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [1157] $1 \overset{1}{18} 122^1 315131612 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [1158] $1 \overset{21}{8} 12^1 315132161 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [1159] $1 \overset{21}{8} 12315132^1 161 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [1160] $1 \overset{1}{18} 122^1 315132161 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [1161] $1 \overset{1}{18} 122315132^1 161 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [1162] $1 \overset{1}{17} 1232171232^1 15 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 30, \Delta = -113, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 30, T \in [21, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
 [1163] $1 \overset{1}{17} 12^1 3217123215 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 30, \Delta = -113, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21, T_{\max} = 30, T \in [21, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [1164] $1 \overset{1}{17} 12^1 3217123151 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 28, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 28$, $T \in [21, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1165] $1712315132^1 1712 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1166] $1712^1 3151321712 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1167] $1712^1 3151322171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1168] $1712315132^1 2171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1169] $1 \overset{321}{7} 1232^1 181231 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 21$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 21$, $T \in [21, 21]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1170] $1 \overset{3221}{8} 1232^1 18122 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1171] $21 \overset{321}{8} 1232^1 18122 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1172] $413221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.25 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1173] $6122^1 3151321 \overset{21}{8} 1 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1174] $6122315132^1 1 \overset{21}{8} 1 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1175] $6122^1 31513221 \overset{1}{8} 1 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1176] $6122315132^1 21 \overset{1}{8} 1 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1177] $1413221(12)1222^1 222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1178] $141322^1 1(12)1222222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 31, T \in [21, 31]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1179] \quad 231413221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 85, \Delta = -123, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 85, T \in [21, 85]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1180] \quad 314132^1 21(12)122222 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 79, \Delta = -122, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 79, T \in [21, 79]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1181] \quad 31413221(12)122^1 222 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 79, \Delta = -122, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 79, T \in [21, 79]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1182] \quad 1612^1 32181223151 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 29, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 29, T \in [21, 29]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1183] \quad 1513161232^1 18122 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 28, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 28, T \in [21, 28]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1184] \quad 2161315132^1 18122 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1185] \quad 2161315132^1 21812 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1186] \quad 1612315132^1 18122 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1187] \quad 1612^1 31513218122 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1188] \quad 1612315132^1 21812 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1189] \quad 1612^1 31513221812 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1190] \quad 21712^1 3217123215 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 30, \Delta = -113, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 30, T \in [21, 30]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1191] \quad 2171232171232^1 15 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 30, \Delta = -113, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 30, T \in [21, 30]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1192] \quad 21712^1 3217123151 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 28, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 28, T \in [21, 28]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1193] \quad 21712^1 3151321712 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 20.75 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$T = [20, 25] \cup [31, 37] \quad T \subset [21, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}$$

1203] 1712¹3216131612 + $\hat{1} \rightarrow 2$ $T = 15$, T

$$T_{\min} = 20.75$$

$$[1204] \ 4132171232^{\overline{21}}181 + \hat{1} \rightarrow 2 \quad T = 15, \ T_{\text{max}} = 15$$

$$T_{\min} = 20.75$$

$\dots \rightarrow 0 \rightarrow H^0(\mathcal{O}_X) \xrightarrow{\alpha} H^0(\mathcal{O}_Y) \xrightarrow{\beta} H^0(\mathcal{O}_{\tilde{Y}}) \xrightarrow{\gamma} H^0(\mathcal{O}_{\tilde{Z}}) \xrightarrow{\delta} H^0(\mathcal{O}_Z) \rightarrow 0$

$$T_{\min} = 20.75$$

$$\begin{pmatrix} 2 & 7 & 2 \end{pmatrix}_{b_0} \quad \begin{matrix} 1 & 1 \end{matrix}$$

$$T_{\text{eff}} = 20.75$$

$$\begin{pmatrix} 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

1207] 161316

- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [1208] $21613161232^1 1712 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1209] $1612^1 32171232161 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 29$, $\Delta = -112$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 29$, $T \in [21, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1210] $413161232^1 181231 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 21$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 21$, $T \in [21, 21]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1211] $4132171232^1 18122 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1212] $41322181232^1 1712 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1213] $4132181232^1 1712 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1214] $414132181232^1 161 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 35$, $\Delta = -116$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 35$, $T \in [21, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1142] $1^{2222} (12) 1223141 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1140] $1^{21} 8 12^{21} 3218123 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1141] $1^{222221} (12) 12^1 23141 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -16$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.75 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1260] $21^{232} 8^1 12321812 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 38$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 38$, $T \in [21, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1217] $1^{223132} (12) 12231 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1326] $4132^1 21812321712 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1328] $7123216131612^1 32 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

- [1219] $21 \text{ }^{231322^1 1} (12) 12231 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1220] $322^1 1 (12) 1223 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 29, \Delta = -112, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 29, T \in [21, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1221] $231322^1 1 (12) 1223132 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 30, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 30, T \in [21, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1222] $223132^1 21 (12) 122313 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 29, \Delta = -112, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 29, T \in [21, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1223] $612^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1224] $161315132^1 181 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 38, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 38, T \in [21, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1225] $1613151322^1 181 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 27, T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1226] $181232^1 181223 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 32, T \in [21, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1258] $21 \text{ }^{321} 812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 32, T \in [21, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1227] $1812321812^1 23 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 32, T \in [21, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1228] $1812^1 32181232 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 38, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 38, T \in [21, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1229] $1812321812^1 32 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 38, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 38, T \in [21, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1230] $181232^1 18123 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 32, T \in [21, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1231] $1 \text{ }^{22222^1 1} (12) 1223141 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 31, \Delta = -114, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21, T_{\max} = 31, T \in [21, 31]$

$$\begin{aligned}
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1246] & \overset{1}{1812^1 2315131612} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[1247] & \overset{31}{161315132^1 2181} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 38, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 38, T \in [21, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1248] & \overset{1}{1812^1 3151321613} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 42, T \in [21, 42] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1249] & \overset{1}{1812^1 2315132161} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[1250] & \overset{1}{171232^1 17123215} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 30, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 30, T \in [21, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1251] & \overset{1}{1712321712^1 3215} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 30, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 30, T \in [21, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1252] & \overset{1}{171232^1 17123151} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 28, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 28, T \in [21, 28] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1253] & \overset{1}{1712321712^1 3151} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 28, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 28, T \in [21, 28] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1254] & \overset{1}{17123151322^1 171} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1255] & \overset{321}{1712^1 32181231} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 21, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1256] & \overset{3221}{1712^1 3218122} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 32, T \in [21, 32] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1257] & \overset{322^1}{1712^1 123218122} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 32, T \in [21, 32] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1259] & \overset{2321}{21712^1 1812} + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 38, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 38, T \in [21, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1261] $413221(12)1222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1262] $4132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1263] $612^1 321 \quad 8 \quad 1 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1264] $613151321 \quad 232^1 1 \quad 8 \quad 1 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1265] $612^1 3151321 \quad 8 \quad 1 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1266] $612^1 23151321 \quad 8 \quad 1 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1267] $612^1 231513221 \quad 8 \quad 1 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1268] $61223151322^1 181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1269] $1413221(12)122^1 2222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1270] $2314132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1271] $3141322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 79$, $\Delta = -122$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 79$, $T \in [21, 79]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1272] $31413221(12)12^1 2222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 79$, $\Delta = -122$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 79$, $T \in [21, 79]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1273] $161232^1 181223151 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 29$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 29$, $T \in [21, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1274] $16123218122^1 3151 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 29$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 29$, $T \in [21, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1275] $5131612321812^1 23 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1276] $31513161232^1 1812 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1277] $15131612^1 3218122 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 28$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 28$, $T \in [21, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1278] $1613151321812^1 23 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1279] $232^1 161315132181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1280] $32161315132^1 1812 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1281] $32161315132^1 2181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1282] $21613151322^1 1812 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1283] $31612315132^1 1812 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1284] $16123151322^1 1812 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1285] $31612315132^1 2181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1286] $1712^1 32171231513 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1287] $2171232^1 17123215 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 30$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 30$, $T \in [21, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1288] $21712321712^1 3215 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 30$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 30$, $T \in [21, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1289] $21712321712^1 3151 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 28$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 28$, $T \in [21, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1290] $2171232^1 17123151 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 28$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 28$, $T \in [21, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1291] $1712^1 31513217123 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1292] $1712^1 23151321712 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1293] $1712^1 23151322171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1294] $322^1 171232181231 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 21$, $\Delta = -96$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 21$, $T \in [21, 21]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1295] $3221712^1 32181231 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 21$, $\Delta = -96$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 21$, $T \in [21, 21]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1296] $4132^1 18123151315 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1297] $612^1 315131613151 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1298] $612^1 321812231513 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1299] $8122315131612^1 32 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1300] $6131513221812^1 32 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1301] $612^1 315132181223 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1302] $6123151321812^1 23 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1303] $612^1 315132218123 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1304] $612^1 231513218122 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1305] $61223151322^1 1812 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1306] $612^1 231513221812 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1307] $7123151321712^1 32 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1308] $7122^1 31513217123 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1309] $16131612^1 321712 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1310] $171232^1 16131612 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1311] $16^1 13161232^1 171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 38$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 38$, $T \in [21, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1312] $41321712^1 32181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1313] $4132^1 171232181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1314] $413221812^1 32171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1315] $4132^1 2181232171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1316] $4132^1 181232171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1317] $41321812^1 32171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1318] $16131612^1 32171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1319] $321613161232^1 171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 42, T \in [21, 42] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1320] & 216131612^1 321712 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 27, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1321] & 161232^1 171232161 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 29, \Delta = -112, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 29, T \in [21, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1322] & 413161315132^1 171 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 38, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 38, T \in [21, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1323] & 4131612^1 32181231 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 21, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1324] & 41321712^1 3218122 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 32, T \in [21, 32] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1325] & 4132^1 17123218122 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 32, T \in [21, 32] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1327] & 413221812^1 321712 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 32, T \in [21, 32] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1329] & 6131612321712^1 32 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 63, T \in [21, 63] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1330] & 612^1 321712321613 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 63, T \in [21, 63] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1331] & 4132^1 1812321712 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 32, T \in [21, 32] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1332] & 41321812^1 321712 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 32, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 32, T \in [21, 32] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1333] & 413161315132^1 217 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 38, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 38, T \in [21, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1334] & 4141321812^1 32161 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 35, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 35, T \in [21, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1335] & 4131613161232^1 16 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 20.44 \rightarrow 21, T_{\max} = 42, T \in [21, 42] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1218] $1 \overset{313221}{(12)} 12^1 23132 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 29$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 29$, $T \in [21, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1215] $1 \overset{1}{(12)} 12231 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1216] $13221 \overset{1}{(12)} 12^1 23 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.44 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1356] $712^1 231513217123 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1358] $4131613151322^1 17 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 38$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 38$, $T \in [21, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1357] $712^1 321613161232 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1336] $1322^1 1(12)1223 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.89 \rightarrow 20$, $T_{\max} = 27$, $T \in [20, 27]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$
- [1337] $1 \overset{2231322^1 1}{(12)} 12231 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 27$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 27$, $T \in [21, 27]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1338] $3221 \overset{3132^1 21}{(12)} 1223 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 29$, $\Delta = -112$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.89 \rightarrow 20$, $T_{\max} = 29$, $T \in [20, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$
- [1339] $2231322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 29$, $\Delta = -112$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 29$, $T \in [21, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1340] $1(12)122^1 3141322 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1341] $1(12)12^1 2314132 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1342] $21(12)122^1 3141322 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1343] $21(12)12^1 2314132 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1344] $1613151322^1 181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 38$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 38$, $T \in [21, 38]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1345] $413221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.89 \rightarrow 20$, $T_{\max} = 85$, $T \in [20, 85]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[1346] $1413221(12)12^1 22222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1347] $22314132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1348] $23141322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1349] $231513161232^1 181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1350] $232161315132^1 181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1351] $321613151322^1 181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1352] $316123151322^1 181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1353] $8122^1 31513161232 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1354] $712^1 321712315132 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1355] $712^1 315132171232 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -36$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.22 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1401] $414132^1 181232161 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 35$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 35$, $T \in [21, 35]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1400] $61232171232^1 1613 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1403] $812^1 314141414141 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 28$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 28$, $T \in [21, 28]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[1359] $1^{18}1232^{121}181232 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 38$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 38$, $T \in [21, 38]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1360] $1^{221}(12)12^{121}23141322 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1361] $21^{21}(12)12^{121}23141322 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1362] $1^{18}1232^{116}131513 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1363] $1^{321}8^{12}12^{132}161315 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1364] $1^{5132^{11}}8^{12}12321613 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1365] $1^{51321}8^{12}1232^{116}13 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1366] $1^{321}8^{12}12^{131}1513161 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1367] $1^{18}12315132^{116}13 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1368] $21^{2321}8^{12}12^{132}1812 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 38$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 38$, $T \in [21, 38]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[1369] $41322^{13221}(12)1222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1370] $61232^{13151321}1^{18}8^{12}1 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1371] $612315132^{1321}1^{18}8^{12}1 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1372] $223141322^{11}(12)1222 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 85$, $\Delta = -123$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 85$, $T \in [21, 85]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [1373] $5132^1 16131613151 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 31$, $\Delta = -114$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 31$, $T \in [21, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1374] $1612321812^1 23151 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 29$, $\Delta = -112$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 29$, $T \in [21, 29]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1375] $513161232^1 181223 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1376] $315131612^1 321812 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [1377] $316123218122^1 315 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1378] $31612^1 3218122315 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1379] $161315132^1 181223 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1380] $161315132^1 218123 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1381] $31612^1 3151321812 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1382] $31612^1 3151322181 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1383] $171232^1 171231513 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1384] $321712^1 321712315 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1385] $321712321712^1 315 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1386] $1712315132^1 17123 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1387] $41321812^1 3151315 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1388] $61232^1 1812231513 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1389] $812^1 321613151322 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1390] $812^1 231513161232 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1391] $612315132^1 181223 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1392] $612315132^1 218123 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 52$, $\Delta = -119$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 52$, $T \in [21, 52]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1393] $7122315132^1 17123 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1394] $16^1 131612^1 32171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 38$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 38$, $T \in [21, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [1395] $41322^1 181232171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1396] $1613161232^1 17123 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1397] $3216131612^1 32171 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [1398] $41322^1 1812321712 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 32$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 32$, $T \in [21, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1399] $61232^1 1712321613 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 63$, $T \in [21, 63]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1402] $41316131612^1 3216 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.06 \rightarrow 21$, $T_{\max} = 42$, $T \in [21, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1412] $6123218122^1 31513 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -64$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.94 \rightarrow 20$, $T_{\max} = 63$, $T \in [20, 63]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1413] $16131612^1 3217123 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 42$, $\Delta = -118$, $\det = -64$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 19.94 \rightarrow 20, T_{\max} = 42, T \in [20, 42] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1414] & 612321712^1 321613 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.94 \rightarrow 20, T_{\max} = 63, T \in [20, 63] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1404] & 3221^{31322^1 1} (12) 1223 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 29, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.69 \rightarrow 20, T_{\max} = 29, T \in [20, 29] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1405] & 1^3 8 1232^1 161315 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.94 \rightarrow 20, T_{\max} = 63, T \in [20, 63] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[1406] & 1^{51321} 8^{12^1} 321613 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.94 \rightarrow 20, T_{\max} = 63, T \in [20, 63] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1407] & 612321^{315132^1 1} 8^1 1 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.94 \rightarrow 20, T_{\max} = 63, T \in [20, 63] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1408] & 5131612^1 32181223 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.94 \rightarrow 20, T_{\max} = 63, T \in [20, 63] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1409] & 3161232^1 18122315 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.94 \rightarrow 20, T_{\max} = 63, T \in [20, 63] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[1410] & 1712321712^1 31513 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 52, \Delta = -119, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.94 \rightarrow 20, T_{\max} = 52, T \in [20, 52] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1411] & 32171232^1 1712315 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.94 \rightarrow 20, T_{\max} = 63, T \in [20, 63] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[1418] & 71232^1 1712315132 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -81, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.84 \rightarrow 20, T_{\max} = 63, T \in [20, 63] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1419] & 71232^1 1613161232 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -81, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.84 \rightarrow 20, T_{\max} = 63, T \in [20, 63] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1417] & 6123151322^1 18123 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 52, \Delta = -119, \det = -81, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.84 \rightarrow 20, T_{\max} = 52, T \in [20, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1415] & 2315131612^1 32181 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 63, \Delta = -120, \det = -81, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.84 \rightarrow 20, T_{\max} = 63, T \in [20, 63] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[1416] & 1613151322^1 18123 + \hat{1} \rightarrow 2 \quad T = 15, T_{\max} = 42, \Delta = -118, \det = -81, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 19.84 \rightarrow 20, T_{\max} = 42, T \in [20, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1422] $61315132^1 2181232 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -100$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.76 \rightarrow 20$, $T_{\max} = 63$, $T \in [20, 63]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1424] $613161232^1 171232 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -100$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.76 \rightarrow 20$, $T_{\max} = 63$, $T \in [20, 63]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1423] $712315132^1 171232 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -100$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.76 \rightarrow 20$, $T_{\max} = 63$, $T \in [20, 63]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1420] $61315132^1 1^{2321} 8 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -100$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.76 \rightarrow 20$, $T_{\max} = 63$, $T \in [20, 63]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1421] $31612321812^1 2315 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -100$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.76 \rightarrow 20$, $T_{\max} = 63$, $T \in [20, 63]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1427] $712321712^1 315132 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -121$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.69 \rightarrow 20$, $T_{\max} = 63$, $T \in [20, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[1428] $6131612^1 32171232 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -121$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.69 \rightarrow 20$, $T_{\max} = 63$, $T \in [20, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[1425] $612321812^1 231513 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -121$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.69 \rightarrow 20$, $T_{\max} = 63$, $T \in [20, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1426] $81232^1 1613151322 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -121$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.69 \rightarrow 20$, $T_{\max} = 63$, $T \in [20, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[1429] $613151322^1 181232 + \hat{1} \rightarrow 2$ $T = 15$, $T_{\max} = 63$, $\Delta = -120$, $\det = -169$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 19.59 \rightarrow 20$, $T_{\max} = 63$, $T \in [20, 63]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

$T = 16$ [1503] $413221812321812^1 2 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 32$, $T \in [23, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[1502] $216131612321812^1 2 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[1461] $21^{21513221} (12) 1222^1 2 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1430] $1^{21} (12) 1 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $22^1 31513221$
 $T_{\min} = 23$, $T_{\max} = 43$, $T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1431] $1 \begin{smallmatrix} 22^1 1 \\ (12) \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 2321513221
 $T_{\min} = 23, T_{\max} = 64, T \in [23, 64]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[1432] $1 \begin{smallmatrix} 22^1 1 \\ (12) \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 231513221
 $T_{\min} = 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[1433] $1 \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 122^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 321513221
 $T_{\min} = 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1434] $1 \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 122^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 31513221
 $T_{\min} = 23, T_{\max} = 39, T \in [23, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1435] $1 \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 1222^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 21513221
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1436] $1 \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 1222^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 1513221
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1437] $1 \begin{smallmatrix} 2^1 21 \\ (12) \end{smallmatrix} 1223 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 1513221
 $T_{\min} = 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$

[1438] $1 \begin{smallmatrix} 22^1 1 \\ (12) \end{smallmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 321513221
 $T_{\min} = 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[1439] $1 \begin{smallmatrix} 22^1 1 \\ (12) \end{smallmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 31513221
 $T_{\min} = 23, T_{\max} = 39, T \in [23, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[1440] $1 \begin{smallmatrix} 21 \\ (12) \end{smallmatrix} 122^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 21513221
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1441] $1 \begin{smallmatrix} 21 \\ (12) \end{smallmatrix} 122^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 1513221
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1442] $1 \begin{smallmatrix} 221 \\ (12) \end{smallmatrix} 12^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 21513221
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1443] & 1 \quad (12) \quad 12^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
& T_{\min} = 23, T_{\max} = 28, T \in [23, 28] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1444] & 3221 \quad (12) \quad 12^1 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22, T_{\max} = 43, T \in [22, 43] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[1445] & 21 \quad (12) \quad 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23, T_{\max} = 28, T \in [23, 28] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1446] & 21 \quad (12) \quad 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23, T_{\max} = 28, T \in [23, 28] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1447] & 1 \quad (12) \quad 122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23, T_{\max} = 43, T \in [23, 43] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1448] & 1 \quad (12) \quad 12^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23, T_{\max} = 43, T \in [23, 43] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1449] & 1 \quad (12) \quad 122^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23, T_{\max} = 64, T \in [23, 64] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1450] & 1 \quad (12) \quad 122^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23, T_{\max} = 43, T \in [23, 43] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1451] & 1 \quad (12) \quad 1222^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23, T_{\max} = 43, T \in [23, 43] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1452] & 1 \quad (12) \quad 1222^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23, T_{\max} = 39, T \in [23, 39] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1453] & 1 \quad (12) \quad 12222^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23, T_{\max} = 28, T \in [23, 28] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1454] & 1 \quad (12) \quad 12222^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23, T_{\max} = 28, T \in [23, 28] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1455] 1513221 3221 (12) $1222^1 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22$, $T_{\max} = 43$, $T \in [22, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [1456] $^{22^1 31513221}$ 21 (12) $12 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 43$, $T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1457] 2321513221 21 (12) $12^1 2 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 64$, $\Delta = -120$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 64$, $T \in [23, 64]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1458] 231513221 21 (12) $12^1 2 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 43$, $T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1459] 321513221 21 (12) $122^1 2 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 43$, $T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1460] 31513221 21 (12) $122^1 2 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 39$, $T \in [23, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1505] $^{2^1 21}$ 413221(12)122314 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22$, $T_{\max} = 80$, $T \in [22, 80]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [1462] 1513221 21 (12) $1222^1 2 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1504] 1 41321812321812 $^1 2 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 32$, $T \in [23, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1464] 31513221 22 $^1 1$ (12) $122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 39$, $T \in [23, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1465] 21513221 22 $^1 1$ (12) $1222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1466] 21513221 221 (12) $122^1 2 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1467] 1513221 22 $^1 1$ (12) $1222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

- [1468] $221 \overset{1513221}{(12)} 122^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1469] $22^1 21(12)1223151322 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1470] $2221(12)122315132^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1471] $22^1 221(12)1223151232 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 64, T \in [23, 64]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1472] $22^1 221(12)122315132 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1473] $22^1 2221(12)122315123 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1474] $22^1 2221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 39, T \in [23, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1475] $22^1 22221(12)12231512 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1476] $22^1 22221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1477] $1 \overset{132^1 21}{(12)} 122314132 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1478] $21 \overset{132^1 21}{(12)} 12231413 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1479] $17123151321812^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1480] $1413221(12)1223132^1 2 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1481] $22^1 31413221(12)12231 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

- [1482] $2231413221(12)122^1 31 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1483] $512321712321812^1 2 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 30, \Delta = -112, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 30, T \in [23, 30]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1484] $151321712321812^1 2 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1485] $217123151321812^1 2 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1486] $171223151321812^1 2 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1487] $812321712315132^1 2 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 53, T \in [23, 53]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1488] $1^{22^1 221}(12)122314141 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 31, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 31, T \in [23, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1489] $21^{22^1 21}(12)122314141 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 31, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 31, T \in [23, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1490] $1^{21}(12)12231414132^1 2 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 80, \Delta = -122, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 80, T \in [23, 80]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1491] $22^1 1^{221}(12)122314141 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 31, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 31, T \in [23, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1492] $1^{22^1 1}(12)12231414132 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 80, \Delta = -122, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 80, T \in [23, 80]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1493] $1^{22^1 21}(12)1223141413 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 75, \Delta = -121, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 75, T \in [23, 75]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1494] $21^{22^1 1}(12)1223141413 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 75, \Delta = -121, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 75, T \in [23, 75]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1495] $1^{16131612321812^1 2} + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

- [1496] $41413221^{3221}(12)122^1 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 80, \Delta = -122, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 80, T \in [22, 80]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [1497] $141413221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 31, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 31, T \in [23, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1498] $22^1 3141413221(12)122 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 80, \Delta = -122, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 80, T \in [23, 80]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1499] $223141413221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 80, \Delta = -122, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 80, T \in [23, 80]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1500] $23141413221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 80, \Delta = -122, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 80, T \in [23, 80]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1501] $3141413221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 75, \Delta = -121, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 75, T \in [23, 75]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1463] $22^1 1^{321513221}(12) 122 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 43, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1591] $32^1 16131612321812 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 39, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 39, T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [1590] $161316123218122^1 3 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 39, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 39, T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1593] $612321712321712^1 3 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 53, T \in [23, 53]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1506] $1^{321513221}(12) 12^1 22 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 43, \Delta = -118, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1507] $1^{32^1 1513221}(12) 1222 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 43, \Delta = -118, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [1508] $1^{31513221}(12) 12^1 22 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 39, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 39, T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1509] $1^{21513221}(12) 122^1 22 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.22 \rightarrow 23, T_{\max} = 43, T \in [23, 43]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_b$$

$$T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_b$$

$$15132^1 21$$

$$\min = 22.2$$

$$T = \langle 22, 22 \rangle \setminus \langle 22 \rangle \quad T = \langle 64 \rangle \quad T \in [22, 64]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}$$

$$T \vdash_{\text{ma}} 22 \ 22 \rightarrow 23 \quad T \vdash_{\text{ma}} 13 \quad T \in [23 \ 13]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}.$$

$$T_{\min} = 22.22 \rightarrow 23, T_{\max} = 43, T \in [23, 43]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.22 \rightarrow 23, \quad T_{\max} = 43, \quad T \in [23, 43]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_b$$

$$I_{\min} = 22.22 \rightarrow 23, \quad I_{\max} = 39, \quad I \in [23, 39]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_b$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 2 & 5 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix},$$

$$\begin{array}{r} 1 \\ 15132 \end{array}$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_L$$

$$\begin{array}{r} 1 \\ 15132 \end{array}$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_1}$$

$$x = 0.170$$

- [1534] $^{1513221} 2221 \ (12) \ 122^1 3 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1535] $^{32^1 1513221} 21 \ (12) \ 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1536] $^{321513221} 21 \ (12) \ 12^1 22 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1537] $^{31\bar{5}13221} 21 \ (12) \ 12^1 22 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 39, T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1538] $^{215132^1 21} 21 \ (12) \ 12222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1539] $^{21513221} 21 \ (12) \ 122^1 22 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1540] $^{15132^1 21} 21 \ (12) \ 12222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1541] $^{1513221} 21 \ (12) \ 122^1 22 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1542] $^{32^1 1513221} 221 \ (12) \ 122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 43, T \in [23, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1543] $^{215132^1 21} 221 \ (12) \ 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1544] $^{21513221} 221 \ (12) \ 12^1 22 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1545] $^{1513221} 221 \ (12) \ 12^1 22 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1546] $^{15132^1 21} 221 \ (12) \ 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

- [1547] $222^1 1(12)1223151322 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 43$, $T \in [23, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1548] $222^1 21(12)1223151232 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 64$, $\Delta = -120$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 64$, $T \in [23, 64]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1549] $222^1 21(12)122315132 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 43$, $T \in [23, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1550] $222221(12)12231512^1 3 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 43$, $T \in [23, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1551] $222^1 221(12)122315123 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 43$, $T \in [23, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1552] $222^1 221(12)12231513 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 39$, $T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1553] $2222221(12)122^1 31512 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1554] $222^1 2221(12)12231512 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1555] $2222221(12)122^1 3151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1556] $222^1 2221(12)1223151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1557] $1^{2313221} (12) 122^1 3141 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [1558] $1^{1322^1 1} (12) 122314132 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1559] $32^{3221} 21(12)1223141 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1560] $21^{313221} (12) 122^1 3141 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -102$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 28$, $T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$

- [1561] $^{3132^1 21} 21 (12) 1223141 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -102, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- [1562] $^{3132^1 21} 1 (12) 12231413 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1563] $^{1322^1 1} 21 (12) 12231413 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -102, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1564] $^{32^1 1} 1 \ 8 \ 12321712315 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 53, T \in [23, 53]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1565] $^{51321} 1 \ 8 \ 12321712^1 3 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 53, T \in [23, 53]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1566] $^1 18123151321712^1 3 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 39, T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1567] $^{32^1 1} 1 \ 8 \ 12315132171 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 39, T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [1568] $^{32^1 21} 413221(12)12231 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1569] $^{32^1 1} 71223151321 \ 8 \ 1 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 39, T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1570] $^{14132^1 21} 14132^1 21(12)12231322 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1571] $^{2231413221} 2231413221(12)12^1 231 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1572] $^{231413221} 231413221(12)122^1 313 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 29, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1573] $^{513217123218122^1 3} 513217123218122^1 3 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 53, T \in [23, 53]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1574] $^{32^1 1} 17123218122315 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 53, T \in [23, 53]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

- [1575] $171231513218122^1 3 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 39$, $T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1576] $32^1 17123151321812 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 39$, $T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [1577] $171231513221812^1 3 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 39$, $T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1578] $32^1 17123151322181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 39$, $T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [1579] $712231513218122^1 3 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 39$, $T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1580] $712231513221812^1 3 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 39$, $T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1581] $1^{222^1 21} (12) 122314141 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 31$, $\Delta = -113$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 31$, $T \in [23, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- [1582] $21^{222^1 1} (12) 122314141 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 31$, $\Delta = -113$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 31$, $T \in [23, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- [1583] $1^{222^1 1} (12) 1223141413 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 75$, $\Delta = -121$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 75$, $T \in [23, 75]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1584] $1^{18123216131612^1 3} + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 39$, $T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1585] $1^{32^1 1} 8 12321613161 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 39$, $T \in [23, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [1586] $41413221^{32^1 21} (12) 122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 80$, $T \in [23, 80]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1587] $141413221(12)1222^1 22 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 31$, $\Delta = -113$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 31$, $T \in [23, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1588] $23141413221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 80$, $T \in [23, 80]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1589] $3141413221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 75$, $\Delta = -121$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 75$, $T \in [23, 75]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1592] $613151316131612^1 3 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 9$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 53$, $T \in [23, 53]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $^{2151322^1 1}$
 [1623] $21 \quad (12) \quad 12222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- 1
 [1711] $4132181232^1 18122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- 3221
 [1674] $413221(12)122^1 31 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.25 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- 1
 [1594] $1 \quad (12) \quad 1222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $^{3215132^1 21}$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 1
 [1595] $1 \quad (12) \quad 1222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $^{315132^1 21}$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 1
 [1596] $1 \quad (12) \quad 12222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $^{2151322^1 1}$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 1
 [1597] $1 \quad (12) \quad 12^1 222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 21513221
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 1
 [1598] $1 \quad (12) \quad 12^1 222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 1513221
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 1
 [1599] $1 \quad (12) \quad 12222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $^{151322^1 1}$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $^{22^1 1}$
 [1600] $1 \quad (12) \quad 1223 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 1513221
 $T_{\min} = 21.25 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- 221
 [1601] $1 \quad (12) \quad 12 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $^{3215132^1 21}$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1627] $221 \overset{315132^1}{(12)} 122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1628] $221 \overset{2151322^1}{(12)} 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1629] $221 \overset{151322^1}{(12)} 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1630] $512^1 3218123151315 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1631] $2222^1 1(12)1223151232 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 64, T \in [22, 64]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1632] $2222^1 1(12)122315132 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1633] $222221(12)122^1 315123 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1634] $2222^1 21(12)122315123 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1635] $2222^1 21(12)12231513 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1636] $222221(12)122^1 31513 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1637] $2222^1 221(12)12231512 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1638] $2222221(12)12^1 231512 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1639] $2222^1 221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1640] $2222221(12)12^1 23151 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$

- [1654] $21^{13221}(12)122^1 31413 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -102$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1655] $1712^1 32181223151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1656] $18^{21}12^1 3217123215 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1657] $18^{21}1232171232^1 15 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [1658] $1^{512321}8^{1232^1}1712 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1659] $1^{512^1}8^{321}12321712 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1660] $18^{21}12^1 3217123151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1661] $1^{151321}8^{1232^1}1712 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1662] $17^{12315132^1}18122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1663] $17^{12^1}131513218122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1664] $18^{21}12^1 3151321712 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1665] $18^{21}12315132^1 1712 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1666] $17^{12315132^1}21812 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1667] $17^{12^1}131513221812 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$

- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [1668] $\overset{1}{18122315132^1 1712} + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $\overset{1}{18122^1 3151321 1712} + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\overset{21}{1812315132^1 2171} + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $\overset{21}{1812^1 31513221 171} + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\overset{1}{18122^1 31513221 171} + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\overset{1}{18122315132^1 2171} + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $[1714] 414132181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 32, \Delta = -114, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 32, T \in [22, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $[1675] 141322^1 1(12)12231322 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $[1713] 414132181232^1 171 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 32, \Delta = -114, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 32, T \in [22, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $[1677] 314132^1 21(12)1223132 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 29, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 29, T \in [22, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $[1678] 231413221(12)12^1 2313 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 29, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 29, T \in [22, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- $[1679] 21712^1 32181223151 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- $[1680] 512^1 3217123218122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 30, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 30, T \in [22, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $[1681] 51232171232^1 18122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 30, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22, T_{\max} = 30, T \in [22, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

- $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1697] $1712^1 32171232161 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1698] $171232171232^1 161 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1699] $41413221^{3221}(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.25 \rightarrow 22$, $T_{\max} = 80$, $T \in [22, 80]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1700] $41322181232^1 181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1701] $4132181232^1 181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1702] $1613161232^1 181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1703] $1414132^1 21(12)122222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 31$, $\Delta = -113$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 31$, $T \in [22, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1704] $141413221(12)122^1 222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 31$, $\Delta = -113$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 31$, $T \in [22, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1705] $3141413221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 75$, $\Delta = -121$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 75$, $T \in [22, 75]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1706] $21613161232^1 18122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1707] $1612^1 321712321712 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1708] $161232171232^1 1712 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1709] $41322181232^1 18122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1710] $1612^1 32181232161 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 29$, $\Delta = -111$, $\det = 16$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.75 \rightarrow 22$, $T_{\max} = 29$, $T \in [22, 29]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1712] & 413221^{22^1 1}(12)122314 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 80, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.25 \rightarrow 22, T_{\max} = 80, T \in [22, 80] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1752] & 22222^{1^1}(12)122315123 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1867] & 41321812315132^{1^1}16 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1811] & 1712^{1^1}231513218122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[1715] & 1 \quad \begin{pmatrix} 221 \\ 12 \end{pmatrix} \quad 1 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& \begin{pmatrix} 23215132^{1^1}21 \\ 221 \end{pmatrix} \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 64, T \in [22, 64] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1716] & 1 \quad \begin{pmatrix} 221 \\ 12 \end{pmatrix} \quad 1 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& \begin{pmatrix} 232^{1^1}1513221 \\ 221 \end{pmatrix} \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 64, T \in [22, 64] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1717] & 1 \quad \begin{pmatrix} 221 \\ 12 \end{pmatrix} \quad 1 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& \begin{pmatrix} 2315132^{1^1}21 \\ 221 \end{pmatrix} \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1718] & 1 \quad \begin{pmatrix} 1 \\ 12 \end{pmatrix} \quad 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& \begin{pmatrix} 32151322^{1^1}1 \\ 12 \end{pmatrix} \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1719] & 1 \quad \begin{pmatrix} 1 \\ 12 \end{pmatrix} \quad 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& \begin{pmatrix} 3151322^{1^1}1 \\ 12 \end{pmatrix} \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 39, T \in [22, 39] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1720] & 1 \quad \begin{pmatrix} 221 \\ 12 \end{pmatrix} \quad 12^{1^1}23 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& \begin{pmatrix} 1513221 \\ 221 \end{pmatrix} \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1721] & 1 \quad \begin{pmatrix} 221 \\ 12 \end{pmatrix} \quad 1223 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& \begin{pmatrix} 15132^{1^1}21 \\ 221 \end{pmatrix} \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1722] & 1 \quad \begin{pmatrix} 21 \\ 12 \end{pmatrix} \quad 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0 \\
& \begin{pmatrix} 23215132^{1^1}21 \\ 21 \end{pmatrix} \\
& T_{\min} = 21.44 \rightarrow 22, T_{\max} = 64, T \in [22, 64] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1723] $1 \begin{smallmatrix} 21 \\ (12) \end{smallmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $232^1 1513221$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 64, T \in [22, 64]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1724] $1 \begin{smallmatrix} 21 \\ (12) \end{smallmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $231^1 15132^1 21$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1725] $1 \begin{smallmatrix} 221 \\ (12) \end{smallmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $32151322^1 1$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1726] $1 \begin{smallmatrix} 221 \\ (12) \end{smallmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $3151322^1 1$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1727] $21 \begin{smallmatrix} 21 \\ (12) \end{smallmatrix} 12^1 23 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 1513221
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1728] $21 \begin{smallmatrix} 21 \\ (12) \end{smallmatrix} 1223 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $15132^1 21$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1729] $21 \begin{smallmatrix} 21 \\ (12) \end{smallmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $32151322^1 1$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1730] $21 \begin{smallmatrix} 21 \\ (12) \end{smallmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $3151322^1 1$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1731] $1 \begin{smallmatrix} 232^1 1513221 \\ (12) \end{smallmatrix} 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 64, T \in [22, 64]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1732] $1 \begin{smallmatrix} 23215132^1 21 \\ (12) \end{smallmatrix} 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 64, T \in [22, 64]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1733] $1 \begin{smallmatrix} 2315132^1 21 \\ (12) \end{smallmatrix} 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1734] $1 \begin{smallmatrix} 32151322^1 1 \\ (12) \end{smallmatrix} 12222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

- [1735] $1 \overset{3151322^1}{(12)} 12222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1736] $1 \overset{21513221}{(12)} 12^1 2222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1737] $1 \overset{1513221}{(12)} 12^1 2222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1738] $2221 \overset{1513221}{(12)} 12^1 23 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1739] $2221 \overset{15132^1 21}{(12)} 1223 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1740] $21 \overset{23215132^1 21}{(12)} 122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 64, T \in [22, 64]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1741] $21 \overset{232^1 1513221}{(12)} 122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 64, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 64, T \in [22, 64]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1742] $21 \overset{2315132^1 21}{(12)} 122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1743] $21 \overset{32151322^1 1}{(12)} 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1744] $21 \overset{3151322^1 1}{(12)} 1222 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1745] $221 \overset{32151322^1 1}{(12)} 122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1746] $221 \overset{3151322^1 1}{(12)} 122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1747] $51232^1 18123151315 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

- [1748] $15132^1 18123151315 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 33$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 33$, $T \in [22, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1749] $22221(12)12231512^1 32 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 64$, $\Delta = -120$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 64$, $T \in [22, 64]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1750] $22221(12)122^1 3151232 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 64$, $\Delta = -120$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 64$, $T \in [22, 64]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1751] $22221(12)122^1 315132 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1870] $4131613161232^1 171 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 36$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 36$, $T \in [22, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1753] $222221(12)12^1 2315123 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1754] $22222^1 1(12)12231513 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1755] $222221(12)12^1 231513 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1756] $22222^1 21(12)12231512 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1757] $22222^1 21(12)1223151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1758] $171232^1 18123215 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1759] $1712321812^1 3215 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1760] $1712321812^1 3151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1761] $171232^1 18123151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1762] $17123151322^1 181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ^{231322¹}
 [1763] $1 (12) 1223141 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 54 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- ¹³²²¹
 [1764] $1 (12) 122^1 314132 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ³²²¹
 [1765] $322^1 1 (12) 1223141 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- ³¹³²²¹
 [1766] $1 (12) 12^1 231413 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ¹³²²¹
 [1767] $21 (12) 12^1 231413 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -102$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ¹
 [1768] $1812^1 32171231513 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ³¹⁵¹³²¹
 [1769] $1^1 8^1 1232^1 171 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ¹
 [1770] $17123218122^1 3151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- ¹
 [1771] $171232^1 181223151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- ²¹
 [1772] $181232^1 17123215 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- ²¹
 [1773] $1812321712^1 3215 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- ⁵¹²³²¹
 [1774] $1^1 8^1 12^1 321712 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- ^{51232¹}
 [1775] $1^1 8^1 12321712 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

- [1776] $18^{21}12321712^13151 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1777] $18^{21}1232^117123151 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1778] $1^{15132^11}8^{12321712} + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1779] $1^{151321}8^{12^1321712} + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1780] $1812^131513217123 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 39, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1781] $1^{321}8^{12315132^1171} + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 39, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1782] $17123151322^11812 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1783] $1812^123151321712 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1784] $18^{21}123151322^1171 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1785] $1812^123151322171 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1786] $181223151322^1171 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1787] $413221^{322^11}(12)12231 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1788] $4132^{3221}21(12)12231 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1789] $7123151321^{232^11}8^1 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 53, T \in [22, 53]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[1868] $4141321812^1 32171 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1791] $1413221(12)122^1 31322 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1792] $31413221(12)12^1 23132 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 29$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 29$, $T \in [22, 29]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1793] $3141322^1 1(12)1223132 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 29$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 29$, $T \in [22, 29]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1794] $2314132^1 21(12)122313 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 29$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 29$, $T \in [22, 29]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1795] $1513161315132^1 171 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1796] $1712^1 321812231513 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1797] $51321712321812^1 23 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1798] $315132171232^1 1812 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1799] $217123218122^1 3151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1800] $2171232^1 181223151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[1801] $51232^1 17123218122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1802] $512321712^1 3218122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1803] $15132^1 17123218122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1804] $151321712^1 3218122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$

- [1805] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$ 1712¹315132181223 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
- [1806] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 17123151321812¹23 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
- [1807] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ 232¹1712315132181 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
- [1808] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 321712315132¹1812 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
- [1809] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 1712¹315132218123 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
- [1810] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 321712315132¹2181 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
- [1813] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 171223151322¹1812 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
- [1812] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 217123151322¹1812 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
- [1869] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$ 4141321812¹321712 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
- [1814] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$ 1712¹231513221812 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
- [1815] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 7122¹315131613151 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$
- [1816] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ 71223151321812¹23 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
- [1817] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 7122¹315132181223 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
- [1818] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$ 81223151321712¹32 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
- [1819] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$ 71231513221812¹32 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [1820] $7122^1 315132218123 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 39, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 22221
 [1821] $1(12)12^1 2314141 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 31, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 31, T \in [22, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- 2221
 [1822] $21(12)12^1 2314141 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 31, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 31, T \in [22, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- 221
 [1823] $221(12)12^1 2314141 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 31, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 31, T \in [22, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- 2221
 [1824] $1(12)122^1 3141413 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 75, \Delta = -121, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 75, T \in [22, 75]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 221
 [1825] $21(12)122^1 3141413 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 75, \Delta = -121, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 75, T \in [22, 75]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 1
 [1826] $1812^1 32161316123 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 39, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 1
 [1827] $16131612^1 3218122 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- 21
 [1828] $181232^1 16131612 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 31
 [1829] $1613161232^1 1812 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 36, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 36, T \in [22, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 1
 [1830] $1712321712^1 32161 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- 1
 [1831] $171232^1 171232161 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- $^{322^1 1}$
 [1832] $41413221(12)122 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 80, \Delta = -122, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22, T_{\max} = 80, T \in [22, 80]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- 1
 [1833] $413161315132^1 181 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 36, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 36$, $T \in [22, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1834] $4132^1 21812321^1 81 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1835] $413221812^1 321^1 81 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1836] $612^1 315131613161 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 29$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 29$, $T \in [22, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1837] $6131612321^{232^1 1} 81 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1838] $41321812^1 321^1 81 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1839] $4132^1 1812321^1 81 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1840] $16131612^1 321^1 81 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1841] $1613161232^1 181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 36$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 36$, $T \in [22, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1842] $141413221(12)12^1 2222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 31$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 31$, $T \in [22, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1843] $14141322^1 1(12)122222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 31$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 31$, $T \in [22, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1844] $31414132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 75$, $\Delta = -121$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 75$, $T \in [22, 75]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1845] $16131612321812^1 23 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1846] $232^1 1613161232181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1847] $321613161232^1 1812 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$

- [1848] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 216131612¹3218122 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
- [1849] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$ 1612321712¹321712 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
- [1850] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$ 161232¹1712321712 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
- [1851] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$ 3161232171232¹171 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
- [1852] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 413161315132¹1812 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 36$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 36$, $T \in [22, 36]$
- [1853] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$ 413161315132¹2181 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 36$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 36$, $T \in [22, 36]$
- [1854] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$ 4132¹181232161315 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
- [1855] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$ 4132¹181231513161 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
- [1856] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$ 413221812¹3218122 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
- [1857] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$ 4132¹218123218122 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
- [1858] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 612¹3216131613151 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 31$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 31$, $T \in [22, 31]$
- [1859] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$ 612¹3151316131612 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 29$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 29$, $T \in [22, 29]$
- [1860] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ 612¹3217123217123 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
- [1861] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$ 161232¹181232161 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 29$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 29$, $T \in [22, 29]$
- [1862] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$ 4132¹18123218122 + $\hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [1863] $41321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [1864] $612^1 321812321613 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [1865] $4132^1 21^{221} 122314 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 80$, $T \in [22, 80]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [1866] $4132^1 181231513216 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [1790] $7122^1 3151321^{321} 81 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.44 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [1914] $4131613151322^1 181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 36$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 36$, $T \in [22, 36]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [1911] $231414132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 80$, $T \in [22, 80]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [1912] $314141322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 75$, $\Delta = -121$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 75$, $T \in [22, 75]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [1871] $1^{21} (12)^{21} 1 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [1872] $1^{221} (12)^{221} 1 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 64$, $\Delta = -120$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 64$, $T \in [22, 64]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [1873] $1^{221} (12)^{221} 1 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [1874] $1^{21} (12)^{21} 12 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 64$, $\Delta = -120$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 64$, $T \in [22, 64]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [1875] $1^{21} (12)^{21} 12 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$

- [1889] $3221(12)122^1 3141 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.89 \rightarrow 21$, $T_{\max} = 28$, $T \in [21, 28]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$
- [1890] $413221(12)12^1 231 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.89 \rightarrow 21$, $T_{\max} = 28$, $T \in [21, 28]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$
- [1891] $712^1 321^1 8^1 1 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1892] $712^1 3151321^1 8^1 1 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1893] $712^1 23151321^1 8^1 1 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1894] $1413221(12)12^1 231322 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1895] $22314132^1 21(12)12231 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 28$, $T \in [22, 28]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1896] $23141322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 29$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 29$, $T \in [22, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1897] $2315132171232^1 181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1898] $2321712315132^1 181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1899] $3217123151322^1 181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1900] $712^1 3151316131513 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 64$, $\Delta = -120$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 64$, $T \in [22, 64]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1901] $712^1 2315131613151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 30$, $\Delta = -112$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 30$, $T \in [22, 30]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1902] $712^1 3218122315132 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1903] $712^1 2315132181223 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.22 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$

- [1918] $1 \overset{223151322^1}{(12)} 122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1919] $2221 \overset{151322^1}{(12)} 1223 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1920] $21 \overset{223151322^1}{(12)} 12 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1921] $512321812^1 3151315 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1922] $151321812^1 3151315 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 33, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 33, T \in [22, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1923] $2221(12)12^1 23151322 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1980] $41316131612^1 32171 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 36, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 36, T \in [22, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [1924] $181232^1 171231513 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1925] $1 \overset{3151321}{8} 12^1 32171 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [1926] $1712321812^1 23151 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 28, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1927] $1 \overset{321}{8} 12321712^1 315 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 53, T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1928] $1 \overset{321}{8} 12^1 321712315 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 53, T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1929] $1 \overset{51321}{8} 1232^1 17123 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 53, T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1930] $1 \overset{5132^1}{8} 123217123 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 53, T \in [22, 53]$

- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [1931] $1812315132^1 17123 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 39, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 321
 [1932] $1\ 8\ 12^1 315132171 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 39, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- 3221
 [1933] $41322^1 1(12)12231 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 321
 [1934] $7122315132^1 1\ 8\ 1 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 39, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1935] $223141322^1 1(12)12231 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [1936] $171232^1 1812231513 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 43, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1937] $5132^1 171232181223 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 53, T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1938] $5132171232^1 181223 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 53, T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1939] $321712^1 3218122315 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 53, T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1940] $3217123218122^1 315 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 53, T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1941] $3151321712^1 321812 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 43, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 43, T \in [22, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [1942] $21712321812^1 23151 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 28, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1943] $1712315132^1 181223 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 39, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1944] $321712^1 3151321812 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 39, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22, T_{\max} = 39, T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1945] $1712315132^1 218123 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 39, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1946] $321712^1 3151322181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1947] $812^1 3217123151322 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1948] $7122315132^1 181223 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1949] $812^1 2315132171232 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1950] $7122315132^1 218123 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 21
 [1951] $1(12)122^1 314141322 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 80$, $T \in [22, 80]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 221
 [1952] $1(12)12^1 231414132 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 80$, $T \in [22, 80]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 21
 [1953] $21(12)12^1 231414132 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 80$, $T \in [22, 80]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 1
 [1954] $161316131612^1 315 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 29$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 29$, $T \in [22, 29]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 1
 [1955] $181232^1 161316123 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- 321
 [1956] $1812^1 321613161 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- 31
 [1957] $16131612^1 321812 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 36$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 36$, $T \in [22, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- 3221
 [1958] $414132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 80$, $T \in [22, 80]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 21
 [1959] $41322^1 181232181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1960] $1\overset{31}{6}131612^132181 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 36$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 36$, $T \in [22, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [1961] $2231414132^121(12)122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 80$, $T \in [22, 80]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [1962] $2314141322^11(12)1222 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 80$, $T \in [22, 80]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [1963] $2161316131612^1315 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 29$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 29$, $T \in [22, 29]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1964] $1613161232^1181223 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1965] $3216131612^1321812 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [1966] $31612^132171232171 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1968] $41321812^132161315 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1970] $41322^118123218122 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1971] $61232^116131613151 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 31$, $\Delta = -113$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 31$, $T \in [22, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1972] $61232171232^117123 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1973] $61232^117123217123 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1974] $61232^11812321613 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1975] $61232181232^11613 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 53$, $T \in [22, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

- [1976] $41322^1(12)122314 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 80$, $T \in [22, 80]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1977] $41321812^131513216 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 43$, $T \in [22, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1978] $414132^1181232171 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 32$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 32$, $T \in [22, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1969] $41321812^131513161 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 39$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.06 \rightarrow 22$, $T_{\max} = 39$, $T \in [22, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1981] $3221(12)12^123141 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 64$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.69 \rightarrow 21$, $T_{\max} = 28$, $T \in [21, 28]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1997] $612321812^1321613 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 64$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21$, $T_{\max} = 53$, $T \in [21, 53]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1983] $1^18^11232171 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 64$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21$, $T_{\max} = 43$, $T \in [21, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1984] $1^18^11232^11712315 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 64$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21$, $T_{\max} = 53$, $T \in [21, 53]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [1985] $1^18^112^13217123 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 64$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21$, $T_{\max} = 53$, $T \in [21, 53]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1986] $17123218122^131513 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 64$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21$, $T_{\max} = 43$, $T \in [21, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1987] $51321712^132181223 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 64$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21$, $T_{\max} = 53$, $T \in [21, 53]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1988] $32171232^118122315 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 53$, $\Delta = -119$, $\det = 64$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21$, $T_{\max} = 53$, $T \in [21, 53]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [1989] $315132^11712321812 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 43$, $\Delta = -118$, $\det = 64$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21$, $T_{\max} = 43$, $T \in [21, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1990] $1(12)12^12314141322 + \hat{1} \rightarrow 2$ $T = 16$, $T_{\max} = 80$, $\Delta = -122$, $\det = 64$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21$, $T_{\max} = 80$, $T \in [21, 80]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

- [1991] $1 \overset{321}{8} 1232^1 1613161 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 64, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21, T_{\max} = 39, T \in [21, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1992] $22314141322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 80, \Delta = -122, \det = 64, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21, T_{\max} = 80, T \in [21, 80]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1993] $16131612^1 32181223 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 64, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21, T_{\max} = 39, T \in [21, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1994] $3161232^1 171232171 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 64, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21, T_{\max} = 43, T \in [21, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1995] $4132181232^1 161315 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 64, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [1996] $612321712^1 3217123 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 64, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1982] $1812321712^1 31513 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 43, \Delta = -118, \det = 64, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.94 \rightarrow 21, T_{\max} = 43, T \in [21, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1998] $71232^1 1 \overset{23151321}{8} 1 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 81, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.84 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [1999] $23151321712^1 32181 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 81, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.84 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [2000] $17123151322^1 18123 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 81, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.84 \rightarrow 21, T_{\max} = 39, T \in [21, 39]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2001] $71232^1 18122315132 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 81, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.84 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [2002] $71223151322^1 18123 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 39, \Delta = -117, \det = 81, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.84 \rightarrow 21, T_{\max} = 39, T \in [21, 39]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2003] $4141322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 80, \Delta = -122, \det = 81, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.84 \rightarrow 21, T_{\max} = 80, T \in [21, 80]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2004] $23216131612^1 32181 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 81, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.84 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [2010] $613161232^1 1 \overset{2321}{8} 1 + \hat{1} \rightarrow 2 \quad T = 16, T_{\max} = 53, \Delta = -119, \det = 100, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.76 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2005] $712315132^1 8^1 1 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 100, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.76 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2006] $321712321812^1 2315 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 100, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.76 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2007] $2321712^1 315132181 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 100, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.76 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2008] $8122315132^1 171232 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 100, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.76 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2009] $712315132^1 2181232 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 100, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.76 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2016] $6131612^1 321^1 8^1 1 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 121, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.69 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2011] $712321^1 8^1 1 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 121, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.69 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2012] $1712321812^1 231513 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 43, \Delta = -118, \det = 121, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.69 \rightarrow 21, T_{\max} = 43, T \in [21, 43]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2013] $2315132^1 171232181 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 121, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.69 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[2014] $7123218122^1 315132 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 121, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.69 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2015] $81232^1 17123151322 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 121, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.69 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2017] $7123151322^1 181232 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 169, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.59 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2018] $812321712^1 3151322 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 196, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.55 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[2019] $712321812^1 2315132 + \hat{1} \rightarrow 2$ $T = 16, T_{\max} = 53, \Delta = -119, \det = 225, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20.52 \rightarrow 21, T_{\max} = 53, T \in [21, 53]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

- $T = 17$ [2028] $22^1 313221(12)1223151 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2050] $4141413221(12)12^1 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 76, \Delta = -121, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 76, T \in [23, 76]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [2047] $1(12)1223141414132^1 2 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 76, \Delta = -121, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 76, T \in [24, 76]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2049] $1(12)122314141413 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 65, \Delta = -120, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 65, T \in [24, 65]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2051] $14141413221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 31, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2046] $1(12)12231414141 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 31, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2045] $1612321712321812^1 2 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2044] $23141413221(12)122^1 31 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -102, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2043] $1(12)1223141413 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2042] $8123218122315132^1 2 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 44, \Delta = -118, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 44, T \in [24, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2041] $4131513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 37, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 37, T \in [24, 37]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2040] $2181223151321812^1 2 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2039] $22^1 181231513218122 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2038] $22^1 181232181223151 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

- [2037] $812321^{22^1 3151321} 8^1 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 44, \Delta = -118, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 44, T \in [24, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2036] $4131513221^{221} (12) 12^1 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 37, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 37, T \in [24, 37]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2035] $4131513221^{22^1 21} (12) 12 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 37, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 37, T \in [24, 37]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2034] $4131513221^{22^1 221} (12) 1 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 37, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 37, T \in [24, 37]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2033] $413221^{1513221} (12) 122^1 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 40, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 40, T \in [23, 40]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [2032] $181223151321812^{12} + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2031] $18^{21} 123151321812^{12} + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2030] $1^{151321} 8^1 12321812^{12} + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2026] $13221^{12} 122315132^{12} + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2025] $132^{121} 21^{12} 1223151322 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2024] $3221^{2^1 1513221} (12) 1223 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22, T_{\max} = 24, T \in [22, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 3 & 3 \\ 2 & 3 & 3 \end{pmatrix}_{b_0}$
- [2023] $21^{321513221} (12) 122^1 31 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2022] $1^{231513221} (12) 122^1 31 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2021] $1^{2321513221} (12) 122^1 31 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -100, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2020] $1 \overset{1}{(12)} 122^1 31 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 321513221
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2029] $1 \overset{512321}{8} 12321812^1 2 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 30, \Delta = -111, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2027] $22^1 313221(12)12231512 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2052] $22^1 314141413221(12)12 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 76, \Delta = -121, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 76, T \in [24, 76]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2053] $2314141413221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 76, \Delta = -121, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 76, T \in [24, 76]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2054] $314141413221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 65, \Delta = -120, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 65, T \in [24, 65]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2055] $4141321812321812^1 2 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 32, \Delta = -113, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2056] $413221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 76, \Delta = -121, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23, T_{\max} = 76, T \in [23, 76]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [2048] $21(12)12231414141 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 31, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2076] $3132^1 21(12)122315132 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2075] $3132^1 21(12)1223151232 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -108, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 29, T \in [24, 29]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2074] $22313221(12)122^1 3151 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2073] $22313221(12)122^1 31512 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2072] $2313221(12)12231512^1 3 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -107, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 29, T \in [24, 29]$

- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
 [2071] $1322^1 1(12)1223151322 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [2070] $32^1 21^{1513221} (12) 1223 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [2069] $32^1 21^{21513221} (12) 1223 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [2077] $1^3 8^{321} 12321812315 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 44, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 44, T \in [24, 44]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [2067] $21^{321} 1513221 (12) 12231 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
 [2066] $13221^{1513221} (12) 122^1 3 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [2065] $1^{31513221} (12) 122^1 313 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 29, T \in [24, 29]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 [2064] $1^{321513221} (12) 122^1 313 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 29, T \in [24, 29]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 [2063] $1^{321} 1513221 (12) 122313 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 29, T \in [24, 29]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
 [2062] $1^{151321} (12) 1223132 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [2061] $1^{2151321} (12) 1223132 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
 [2060] $1^{231513221} (12) 12^1 231 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 [2059] $1^{2321513221} (12) 12^1 231 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 29, \Delta = -100, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 29, T \in [24, 29]$

- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
 [2058] $\overset{1}{32^1 1513221} (12) \ 12231 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $\overset{1}{321513221} (12) \ 12^1 231 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\overset{1}{18123151321812^1 3} + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 37, T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $\overset{51321}{1} \ 8 \ 123218122^1 3 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 44, T \in [24, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $\overset{32^1 1}{1} \ 8 \ 123218122315 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 44, T \in [24, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\overset{1}{181231513218122^1 3} + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 37, T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $\overset{32^1 1}{1} \ 8 \ 123151321812 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 37, T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $\overset{1}{181231513221812^1 3} + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 37, T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $\overset{32^1 1}{1} \ 8 \ 123151322181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 37, T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- $\overset{222^1 21}{4131513221} (12) \ 1 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 37, T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\overset{222^1 1}{4131513221} (12) \ 12 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 37, T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $\overset{1}{5131613151321812^1 3} + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 65, \Delta = -120, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 65, T \in [24, 65]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $\overset{1}{32^1 218123218122315} + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24, T_{\max} = 44, T \in [24, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

- [2089] $2181231513218122^1 3 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 37$, $T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2090] $1812231513218122^1 3 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 37$, $T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2091] $32^1 181223151321812 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 37$, $T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [2092] $1812231513221812^1 3 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 37$, $T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2093] $4131513221(12)1222^1 22 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 37$, $T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2094] $1 \begin{pmatrix} 3132^1 21 \\ 12 \end{pmatrix} 122314141 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -101$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 29$, $T \in [24, 29]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- [2095] $1 \begin{pmatrix} 1322^1 1 \\ 12 \end{pmatrix} 1223141413 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -101$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 29$, $T \in [24, 29]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2096] $612321712321 \begin{pmatrix} 32^1 1 \\ 8 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 44$, $T \in [24, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2097] $23141413221(12)12^1 231 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -102$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 29$, $T \in [24, 29]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2098] $3141413221(12)122^1 313 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 29$, $T \in [24, 29]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2099] $14132^1 21(12)122314132 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -102$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 29$, $T \in [24, 29]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2100] $1712321712321712^1 3 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 40$, $T \in [24, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2101] $7123151316131612^1 3 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 44$, $T \in [24, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2102] $6123217123218122^1 3 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 44$, $T \in [24, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2103] $14132^1 21 \begin{pmatrix} 1 \\ 12 \end{pmatrix} 12231413 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -101$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 29$, $T \in [24, 29]$

- [2104] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}^1$ 612321812321712¹3 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 44$, $T \in [24, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2105] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}^{222^1 1}$ 1(12)12231414141 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 31$, $\Delta = -112$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- [2106] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}^{32^1 21}$ 4141413221(12)12 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 76$, $T \in [24, 76]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2107] 14141413221(12)122¹22 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 31$, $\Delta = -112$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2108] 314141413221(12)12¹22 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 65$, $\Delta = -120$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 65$, $T \in [24, 65]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2109] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}^1$ 613161316131612¹3 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 37$, $T \in [24, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2068] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}^{321513221}$ 21 (12) 12¹231 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -9$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 24$, $T \in [24, 24]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2137] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}^{11}$ 31322¹1(12)1223151232 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -108$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2136] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}^1$ 22313221(12)12¹23151 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2197] 414132181232¹18122 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 32$, $\Delta = -113$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 32$, $T \in [23, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2138] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}^1$ 31322¹1(12)122315132 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2139] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}^{21}$ 1812¹3218123215 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 30$, $\Delta = -111$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 30$, $T \in [23, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2140] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}^{21}$ 181232181232¹15 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 30$, $\Delta = -111$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 30$, $T \in [23, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [2141] $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}^{21}$ 1812¹3218123151 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.75 \rightarrow 23, \quad T_{\max} = 29, \quad T \in [23, 29]$$

[2142] $\overset{21}{1}8\overset{21}{12}1^4315132181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$

[2143] $\overset{1}{18}122\overset{21}{1}315132181 + \overset{1}{1}\rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$

[2144] $18122315132^1 181 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$

[2145] $\overset{1}{18}122\overset{1}{3}1513221\overset{1}{8}1 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$

[2146] $1\bar{8}12^132181223151 + \hat{1}\bar{1}\rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$

[2147] $1 \overset{512321}{8} 1232^1 18122 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 30, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23, T_{\max} = 30, T \in [23, 30]$

[2148] $1^{512} 8^{1321} 123218122 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 30$, $\Delta = -111$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 30$, $T \in [23, 30]$

[2149] $1^{\overline{151321}} \quad 8 \quad 1232^{\overline{1}}18122 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$

[2150] $1\bar{8}12315132^118122 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$

[2151] $1\bar{8}12^131513218122 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$

[2152] $18122315132^1 18122 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$

[2153] $18122^1 31513218122 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$

[2154] $18\ 12315132^1 21812 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$

- [2155] $^{21}1812^131513221812 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2156] $^{11}18122315132^121812 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2157] $^{11}18122^131513221812 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2158] $^{1513221}413221(12)12^12 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.25 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2159] $^{22221}41315132^121(12)1 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2160] $^{22221}4131513221(12)1 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2161] $^{2221}41315132^121(12)12 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2162] $^{221}41315132^121(12)122 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2163] $^{1612}1612^13218123151315 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 33$, $\Delta = -114$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 33$, $T \in [23, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2164] $^{513161232181232}115 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [2165] $^{161315132181232}115 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 33$, $\Delta = -114$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 33$, $T \in [23, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [2166] $^{221812}132181223151 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2167] $^{221812}131513218122 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2168] $^{218122315132}118122 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2169] $^{218122}131513218122 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2170] & 218122^1 31513221812 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2171] & 4131513221(12)122^1 222 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 37, T \in [23, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2172] & 41315132^1 21(12)122222 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 37, T \in [23, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2173] & 612315132181232^1 15 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 34, \Delta = -115, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 34, T \in [23, 34] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 313221 \\ 1 \end{matrix} \\
[2174] & 1 (12) 122^1 314141 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -101, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 31322^1 1 \\ 1 \end{matrix} \\
[2175] & 1 (12) 122314141 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -101, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 21 \\ 1 \end{matrix} \\
[2176] & 181232171232^1 161 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 21 \\ 1 \end{matrix} \\
[2177] & 1812^1 32171232161 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ 1 \end{matrix} \\
[2178] & 171232171232^1 1712 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -108, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ 1 \end{matrix} \\
[2179] & 1712^1 321712321712 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -108, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ 1 \end{matrix} \\
[2180] & 171232181232^1 161 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ 1 \end{matrix} \\
[2181] & 1712^1 32181232161 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ 1 \end{matrix} \\
[2182] & 1712^1 32171232171 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -108, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2183] 141413221(12)122¹3132 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -102$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2184] 1414132¹21(12)1223132 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -102$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2185] 3141413221(12)12¹2313 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2186] 141322¹1(12)122314132 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -102$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2187] 314132¹21(12)12231413 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2188] 161232171232¹18122 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2189] 1612¹3217123218122 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2190] 21712¹321712321712 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -108$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2191] 1413221¹(12)122¹31413 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -101$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2192] 141322¹1(12)12231413 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -101$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2193] 161232181232¹1712 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [2194] 1612¹321812321712 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2195] 414132181232¹181 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 32$, $\Delta = -113$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.75 \rightarrow 23$, $T_{\max} = 32$, $T \in [23, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2110] 1¹(12) 12231 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
3215132¹21

- $$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$
- $$[2123] \quad 1 \quad (12) \quad 12^1 2313 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$
- $$[2124] \quad 132^1 21 \quad (12) \quad 1223 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 22.25 \rightarrow 23, T_{\max} = 24, T \in [23, 24]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$
- $$[2125] \quad 21 \quad (12) \quad 12231 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 22.75 \rightarrow 23, T_{\max} = 24, T \in [23, 24]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$
- $$[2126] \quad 21 \quad (12) \quad 12231 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 23, \Delta = -97, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 22.75 \rightarrow 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$
- $$[2127] \quad 21 \quad (12) \quad 12^1 2313 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 23, \Delta = -97, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 22.75 \rightarrow 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$
- $$[2128] \quad 21 \quad (12) \quad 122313 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 23, \Delta = -97, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 22.75 \rightarrow 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$
- $$[2129] \quad 21 \quad (12) \quad 122313 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 23, \Delta = -97, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 22.75 \rightarrow 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$
- $$[2130] \quad 21 \quad (12) \quad 12^1 2313 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 23, \Delta = -97, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 22.75 \rightarrow 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$
- $$[2131] \quad 2313221(12)122^1 315123 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$
- $$[2132] \quad 23132^1 21(12)122315123 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$
- $$[2133] \quad 2313221(12)122^1 31513 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$
- $$[2134] \quad 23132^1 21(12)12231513 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 22.75 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$
- $$[2135] \quad 22313221(12)12^1 231512 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 22.75 \rightarrow 23, T_{\max} = 24, T \in [23, 24]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2196] 14141413221(12)12¹222 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 31$, $\Delta = -112$, $\det = -16$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.75 \rightarrow 23, T_{\max} = 31, T \in [23, 31]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2232] 1⁵¹³²¹8^{12321812¹23} + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2231] 1^{1812¹321812231513} + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 40, T \in [23, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2292] 612¹321712321³²¹8¹1 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2291] 7122¹315131613161¹ + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[2290] 1^{1712321712¹321712} + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -108$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[2289] 1^{171232¹1712321712} + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -108$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2288] 1^{1812¹321712321613} + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 40, T \in [23, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2287] 1^{181232¹171232161} + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2286] 1^{1812321712¹32161} + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[2285] 1^{1613161315132¹171} + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2284] 1¹³²²¹(12)122¹3141413 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -101$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2283] 1³¹³²²¹(12)12¹2314141 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -101$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[2282] 812231513221812¹32 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 32, T \in [23, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 32, T \in [23, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 34, T \in [23, 34]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 34, T \in [23, 34]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

- [2267] $218123151321812^1 23 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2266] $2218123218122^1 3151 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2265] $22181232^1 181223151 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2264] $21812^1 321812231513 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2263] $322^1 18123218122315 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2262] $1513161315132^1 2181 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 30$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 30$, $T \in [23, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2261] $1513161315132^1 1812 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 30$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 30$, $T \in [23, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2260] $1613151321812^1 3151 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 31$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 31$, $T \in [23, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2259] $1613151321812^1 3215 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 33$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 33$, $T \in [23, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2258] $5131612321812^1 3151 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 33$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 33$, $T \in [23, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2257] $5131612321812^1 3215 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2256] $161232^1 18123151315 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 33$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 33$, $T \in [23, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2255] $81223151321^{232^1 1} 8^1 1 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2254] $413151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2253] $413151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2252] $413151322^1 (12) 1 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [2251] $4132^1 21 (12) 122 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 40, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 40, T \in [23, 40]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [2250] $413221^{15132^1 21} (12) 122 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 40, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 40, T \in [23, 40]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
 [2249] $1812^1 231513221812 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2248] $181223151322^1 1812 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2247] $18123151322^1 1812 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
 [2246] $1812^1 231513218122 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2245] $1812315132^1 2181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [2244] $1812^1 315132218123 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2243] $1812315132^1 1812 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [2242] $12315132181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [2241] $18123151321812^1 23 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [2240] $1812^1 315132181223 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [2239] $123218122 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2238] $1 \quad 8 \quad 12^{1321} 3218122 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2237] $1 \quad 8 \quad 12^{1321} 3218122 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 30$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 30$, $T \in [23, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2236] $1 \quad 8 \quad 12^{512321} 23218122 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 30$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 30$, $T \in [23, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2235] $1 \quad 8 \quad 12^{21} 3218122^{1321} 181223151 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2234] $1 \quad 8 \quad 12^{21} 3218122^{1321} 3151 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2198] $1 \quad (12) \quad 12231 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2199] $1 \quad (12) \quad 12231 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 23$, $\Delta = -97$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 23$, $T \in [23, 23]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2200] $1 \quad (12) \quad 12231 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 29$, $\Delta = -100$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2201] $1 \quad (12) \quad 12231 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 29$, $\Delta = -100$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2202] $1 \quad (12) \quad 12231 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2203] $1 \quad (12) \quad 12^{21513221} 23132 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2204] $1 \quad (12) \quad 12^{1513221} 23132 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2205] $1 \quad (12) \quad 12^{32151322} 2313 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 29$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $\overset{1}{3151322}1$
 [2206] $1 \quad (12) \quad 122313 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 29$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $\overset{15132}{1}21$
 [2207] $13221 \quad (12) \quad 1223 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- $\overset{1513221}{1}$
 [2208] $13221 \quad (12) \quad 12^1 23 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- $\overset{32151322}{1}1$
 [2209] $21 \quad (12) \quad 12231 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $\overset{1}{3151322}1$
 [2210] $21 \quad (12) \quad 12231 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 23$, $\Delta = -97$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 23$, $T \in [23, 23]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $\overset{21513221}{1}$
 [2211] $322^1 1 \quad (12) \quad 1223 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- $\overset{1}{1513221}$
 [2212] $322^1 1 \quad (12) \quad 1223 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2213] $231322^1 1(12)122315123 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 29$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2214] $2313221(12)12^1 2315123 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 29$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $\overset{1}{2313221}(12)12^1 231513 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 29$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $\overset{1}{231322^1 1}(12)12231513 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 29$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2217] $223132^1 21(12)12231512 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2218] $223132^1 21(12)122315^1 1 + \hat{1} \rightarrow 2 \quad T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 24, T \in [23, 24] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2219] & 313221(12)12231512^1 32 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2220] & 313221(12)122^1 3151232 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2221] & 313221(12)122^1 315132 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2222] & 1812^1 32181231513 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 40, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 40, T \in [23, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2223] & 1812321812^1 3215 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 30, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 30, T \in [23, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2224] & 181232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 30, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 30, T \in [23, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2225] & 181232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2226] & 1812321812^1 3151 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2227] & 1812^1 31513218123 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2228] & 1812^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[2229] & 1812^1 23151322181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[2294] & 171232^1 181232161 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2295] & 171232^1 171232171 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[2296] 14141322¹1(12)1223132 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2297] 141413221(12)12¹23132 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2298] 31414132¹21(12)122313 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2299] 1413221(12)122¹314132 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2300] 3141322¹1(12)12231413 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2301] 15131613161232¹171 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 30$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 30, T \in [23, 30]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2302] 21613161315132¹171 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2303] 161232¹17123218122 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2304] 1612321712¹3218122 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[2305] 3161232171232¹1812 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 40, T \in [23, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2306] 1712¹3217123217123 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 40, T \in [23, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2307] 2171232¹1712321712 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -108$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2308] 4132¹1812321712315 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[2309] 41321812315132¹171 + $\hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

- [2310] $4132^1 1812315132171 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2311] $7122^1 3151316131612 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [2312] $612^1 32171232181223 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2313] $612321712321812^1 23 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2314] $712321712321712^1 32 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 54$, $\Delta = -119$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 54$, $T \in [23, 54]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2315] $1413221(12)12^1 231413 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -101$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2316] $161232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2317] $1612321812^1 321712 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2318] $3161232181232^1 171 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2319] $612^1 3218123217123 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2320] $41321812315132^1 217 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [2321] $4132^1 1812315132217 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2322] $1(12)122^1 31414141 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 31$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 31$, $T \in [23, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [2323] $21(12)122^1 31414141 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 31$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.44 \rightarrow 23$, $T_{\max} = 31$, $T \in [23, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [2324] $4141413221(12)12 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -25$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 76, T \in [23, 76] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2325] & 4141321812^1 32181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 32, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 32, T \in [23, 32] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2326] & 4131613161232^1 181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 34, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 34, T \in [23, 34] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2327] & 612^1 3216131613161 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[2328] & 141414132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 31, \Delta = -112, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 31, T \in [23, 31] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2329] & 4141321812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 32, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 32, T \in [23, 32] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2330] & 4131613161232^1 1812 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 34, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 34, T \in [23, 34] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2331] & 4132^1 1812321613161 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 37, T \in [23, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2332] & 612^1 32161316131612 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[2333] & 4132^1 21(12)12231414 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 76, \Delta = -121, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 76, T \in [23, 76] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2293] & 1712321812^1 32161 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2230] & 1812^1 315131613151 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 30, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 30, T \in [23, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2233] & 1^1 8^1 1232^1 1812 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 40, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.44 \rightarrow 23, T_{\max} = 40, T \in [23, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2337] & 3221^1 (12)^1 1223 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 21.89 \rightarrow 22, T_{\max} = 24, T \in [22, 24] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2338] $3221 \overset{15132^1 21}{(12)} 1223 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.89 \rightarrow 22, T_{\max} = 24, T \in [22, 24]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$
- [2339] $13221(12)122 \overset{1}{1} 3151322 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 24, T \in [23, 24]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2340] $2231322^1(12)12231512 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 24, T \in [23, 24]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2341] $2231322^1(12)1223151 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 24, T \in [23, 24]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2342] $313221(12)12^1 23151232 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -108, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2343] $313221(12)12^1 2315132 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2344] $1 \overset{23151321}{8} 1232^1 181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2345] $1 \overset{2321}{8} 12315132^1 181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2346] $1 \overset{321}{8} 123151322^1 181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2347] $8122^1 3151321 \overset{2321}{8} 1 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2348] $31513161315132^1 181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 54, \Delta = -119, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 54, T \in [23, 54]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2349] $15131613151322^1 181 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 30, \Delta = -111, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 30, T \in [23, 30]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2350] $1812^1 3218122315132 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2351] $1812^1 2315132181223 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

- [2352] $23218122315132^1 181 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2353] $1812^1 2315132218123 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2354] $8122^1 3151316131513 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 54$, $\Delta = -119$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 54$, $T \in [23, 54]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2355] $8122^1 3151322181232 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2356] $1^{13221} (12) 12^1 23141413 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -101$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2357] $712^1 3151316131^{31} 61 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2358] $712^1 2315131613161^1 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2359] $231414132^1 21(12) 12231 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -102$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2360] $314141322^1 1(12) 122313 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2361] $1413221(12) 12^1 2314132 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -102$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2362] $712^1 32161316131513 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 54$, $\Delta = -119$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 54$, $T \in [23, 54]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2363] $712^1 31513161316123 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2364] $712^1 23151316131612 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -109$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2365] $712^1 32171232171232 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 54$, $\Delta = -119$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 54$, $T \in [23, 54]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

- [2366] $413218123151322^1 17 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2367] $1(12)12^1 231414141 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 31$, $\Delta = -112$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 31$, $T \in [23, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2368] $21(12)12^1 231414141 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 31$, $\Delta = -112$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 31$, $T \in [23, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2369] $1(12)12^1 314141413 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 65$, $\Delta = -120$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 65$, $T \in [23, 65]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2370] $21(12)12^1 314141413 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 65$, $\Delta = -120$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 65$, $T \in [23, 65]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2371] $1414141322^1 1(12)1222 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 31$, $\Delta = -112$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 31$, $T \in [23, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2372] $3141414132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 65$, $\Delta = -120$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 65$, $T \in [23, 65]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2336] $1322^1 1(12)1223 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.89 \rightarrow 22$, $T_{\max} = 24$, $T \in [22, 24]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$
- [2335] $1(12)12^1 12231 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2334] $1(12)12^1 12231 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -100$, $\det = -36$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.22 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2441] $414132^1 18123218122 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 32$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 32$, $T \in [23, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [2373] $13221(12)1223 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [2378] $1812315132^1 18123 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2379] $181232^1 1812231513 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 40, T \in [23, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2380] & \begin{matrix} 51321 \\ 1 & 8 & 1232^1 181223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 44, T \in [23, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2381] & \begin{matrix} 5132^1 1 \\ 1 & 8 & 1232181223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 44, T \in [23, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2382] & \begin{matrix} 321 \\ 1 & 8 & 123218122^1 315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 44, T \in [23, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2383] & \begin{matrix} 321 \\ 1 & 8 & 12^1 3218122315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 44, T \in [23, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2384] & \begin{matrix} 3151321 \\ 1 & 8 & 12^1 321812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 40, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 40, T \in [23, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[2385] & \begin{matrix} 21 \\ 1 & 8 & 12321812^1 23151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 29, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 29, T \in [23, 29] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2386] & \begin{matrix} 1 \\ 1 & 8 & 12315132^1 181223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 37, T \in [23, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2387] & \begin{matrix} 321 \\ 1 & 8 & 12^1 3151321812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 37, T \in [23, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2388] & \begin{matrix} 1 \\ 1 & 8 & 12315132^1 218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 37, T \in [23, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2389] & \begin{matrix} 321 \\ 1 & 8 & 12^1 3151322181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 37, T \in [23, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2390] & \begin{matrix} 1513221 \\ 41322^1 1 & (12) & 122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 40, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 40, T \in [23, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2391] & \begin{matrix} 151322^1 1 \\ 413221 & (12) & 122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 40, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 40, T \in [23, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2392] & \begin{matrix} 223151321 \\ 812^1 321 & 8 & 1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.06 \rightarrow 23, T_{\max} = 44, T \in [23, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2393] $812^1 23151321^{\overline{2321}} 8 \ 1 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2394] $1612321812^1 3151315 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 33, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 33, T \in [23, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [2395] $513161232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2396] $513161232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 33, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 33, T \in [23, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2397] $161315132^1 18123215 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 33, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 33, T \in [23, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2398] $161315132^1 18123151 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 31, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 31, T \in [23, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2399] $31613151321812^1 315 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 65, \Delta = -120, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 65, T \in [23, 65]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2400] $3221812^1 3218122315 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [2401] $32218123218122^1 315 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 44, T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2402] $2181232^1 1812231513 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 40, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 40, T \in [23, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2403] $221812321812^1 23151 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 29, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 29, T \in [23, 29]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [2404] $21812315132^1 181223 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2405] $18122315132^1 181223 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2406] $3218122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2407] $18122315132^1 218123 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23, T_{\max} = 37, T \in [23, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

- [2408] $812^1 31513161315132 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 65$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 65$, $T \in [23, 65]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2409] $812^1 23151316131513 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 54$, $\Delta = -119$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 54$, $T \in [23, 54]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2410] $612315132^1 18123215 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 34$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 34$, $T \in [23, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2411] $612315132^1 18123151 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 32$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 32$, $T \in [23, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2412] $812^1 32181223151322 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2413] $812^1 23151322181232 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2414] $181232171232^1 1613 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2415] $181232^1 1712321613 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2416] $61232^1 1712321^1 81 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2417] $61232171232^1 1^1 81 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2418] $2314141322^1 1(12)12231 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [2419] $151316131612^1 32171 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 30$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 30$, $T \in [23, 30]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [2420] $31612321712^1 321812 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [2421] $31612^1 321712321812 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2422] $171232^1 17123217123 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$

- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [2423] $41321812^1 321712315 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
 [2424] $41321812321712^1 315 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [2425] $41321812^1 315132171 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [2426] $61232171232^1 181223 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2427] $61232^1 171232181223 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [2428] $31612^1 32181232171 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2429] $31612321812^1 32171 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [2430] $61232181232^1 17123 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2431] $61232^1 18123217123 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [2432] $41321812^1 315132217 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
 [2433] $1(12)122^1 3141414132 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 76$, $T \in [23, 76]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [2434] $1(12)12^1 2314141413 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 65$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 65$, $T \in [23, 65]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [2435] $21(12)12^1 2314141413 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 65$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 65$, $T \in [23, 65]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [2436] $414132^1 181232181 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 32$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 32$, $T \in [23, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

- [2437] $41316131612^1 32181 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 34$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 34$, $T \in [23, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [2438] $61232^1 16131613161 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [2439] $23141414132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 76$, $T \in [23, 76]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [2440] $31414141322^1 1(12)1222 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 65$, $\Delta = -120$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 65$, $T \in [23, 65]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [2376] $1 \ 8 \ 12^1 321812315 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [2442] $41316131612^1 321812 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 34$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 34$, $T \in [23, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [2443] $41321812^1 321613161 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 37$, $T \in [23, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2444] $61232^1 161316131612 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 29$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 29$, $T \in [23, 29]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [2445] $41322^1 1(12)12231414 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 76$, $T \in [23, 76]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2446] $413221(12)122^1 31414 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 76$, $T \in [23, 76]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [2375] $181232^1 181231513 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 40$, $T \in [23, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2377] $1 \ 8 \ 12321812^1 315 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 44$, $T \in [23, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2374] $13221(12)12^1 23151322 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 24$, $\Delta = -98$, $\det = -49$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.06 \rightarrow 23$, $T_{\max} = 24$, $T \in [23, 24]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2461] $612321712^1 321 \ 8 \ 1 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$

- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [2451] $1 \overset{1}{18} 123218122^1 31513 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 40, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 40, T \in [22, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $\overset{51321}{1} \overset{8}{8} 12^1 32181223 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 44, T \in [22, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $\overset{321}{1} \overset{8}{8} 1232^1 18122315 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 44, T \in [22, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- $\overset{315132^1}{1} \overset{8}{8} 12321812 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 40, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 40, T \in [22, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- $[2455] \overset{5131612^1}{1} 3218123215 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 37, \Delta = -116, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 37, T \in [22, 37]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $[2456] \overset{5131612^1}{1} 3218123151 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 33, \Delta = -114, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 33, T \in [22, 33]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $[2457] \overset{322181232^1}{1} 18122315 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 44, T \in [22, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- $[2458] \overset{218123218122^1}{1} 31513 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 40, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 40, T \in [22, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $[2459] \overset{613151321812^1}{1} 31513 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 65, \Delta = -120, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 65, T \in [22, 65]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $[2449] \overset{1}{18} 12321812^1 31513 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 40, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 40, T \in [22, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $\overset{151322^1}{3221} (12) 1223 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.69 \rightarrow 22, T_{\max} = 24, T \in [22, 24]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $\overset{2151322^1}{3221} (12) 1223 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.69 \rightarrow 22, T_{\max} = 24, T \in [22, 24]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $[2463] \overset{1712321712^1}{1} 3217123 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 40, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 40, T \in [22, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $[2464] \overset{4132181232^1}{1} 1712315 + \hat{1} \rightarrow 2 \quad T = 17, T_{\max} = 44, \Delta = -118, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22, T_{\max} = 44, T \in [22, 44]$

- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
 [2465] $612321712^1 32181223 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [2466] $3161232^1 181232171 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 40$, $T \in [22, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [2467] $612321812^1 3217123 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [2468] $1(12)122^1 31414141322 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 76$, $T \in [22, 76]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [2469] $1(12)12^{21} 23141414132 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 76$, $T \in [22, 76]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [2470] $223141414132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 76$, $T \in [22, 76]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [2471] $231414141322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 76$, $T \in [22, 76]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [2472] $4132181232^1 1613161 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 37$, $T \in [22, 37]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 [2450] $1 \ 8 \ 1232^1 1812315 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
 [2462] $3161232^1 1712321812 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 40$, $T \in [22, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 [2460] $1812321712^1 321613 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -64$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.94 \rightarrow 22$, $T_{\max} = 40$, $T \in [22, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [2473] $1 \ 8 \ 12^{23151321} 32181 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
 [2474] $18123151322^1 18123 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 37$, $T \in [22, 37]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
 [2475] $513161315132^1 18123 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 65$, $\Delta = -120$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 65$, $T \in [22, 65]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2476] $181232^1 18122315132 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [2477] $321812^1 23151321812 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 37$, $T \in [22, 37]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2478] $181223151322^1 18123 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 37$, $\Delta = -116$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 37$, $T \in [22, 37]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2479] $71232^1 161316131513 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 54$, $\Delta = -119$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 54$, $T \in [22, 54]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [2480] $71232^1 171232171232 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 54$, $\Delta = -119$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 54$, $T \in [22, 54]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [2481] $1(12)12^{11} 231414141322 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 76$, $T \in [22, 76]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2482] $41414132^{3221} 21(12)12 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 76$, $T \in [22, 76]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2483] $2231414141322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 76$, $T \in [22, 76]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2484] $413221(12)12^{21} 231414 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -81$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.84 \rightarrow 22$, $T_{\max} = 76$, $T \in [22, 76]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2485] $1\ 8\ 12321812^1 2315 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -100$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.76 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2486] $1\ 8\ 12^{2321} 315132181 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -100$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.76 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2487] $8122315132^1 1\ 8\ 1 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -100$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.76 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2488] $3221812321812^1 2315 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -100$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.76 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2489] $23218122^1 315132181 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -100$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.76 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$

- $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [2490] $61315132^1 181231513 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 65$, $\Delta = -120$, $\det = -100$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.76 \rightarrow 22$, $T_{\max} = 65$, $T \in [22, 65]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [2491] $8122315132^1 2181232 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -100$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.76 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [2492] $71232171232^1 171232 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 54$, $\Delta = -119$, $\det = -100$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.76 \rightarrow 22$, $T_{\max} = 54$, $T \in [22, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [2493] $1812321812^1 231513 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.69 \rightarrow 22$, $T_{\max} = 40$, $T \in [22, 40]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
 [2494] $1^{2315132^1} 8^{1232181} + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -121$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.69 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
 [2495] $81232^1 1^{223151321} 8^{1232181} + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -121$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.69 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
 [2496] $3161315132^1 1812315 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 65$, $\Delta = -120$, $\det = -121$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.69 \rightarrow 22$, $T_{\max} = 65$, $T \in [22, 65]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [2497] $18123218122^1 315132 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -121$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.69 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 [2499] $81232^1 181223151322 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -121$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.69 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
 [2500] $712321712^1 32171232 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 54$, $\Delta = -119$, $\det = -121$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.69 \rightarrow 22$, $T_{\max} = 54$, $T \in [22, 54]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 [2501] $414141322^1 1^{3221} (12)12 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 76$, $\Delta = -121$, $\det = -121$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.69 \rightarrow 22$, $T_{\max} = 76$, $T \in [22, 76]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [2498] $21812321812^1 231513 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 40$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.69 \rightarrow 22$, $T_{\max} = 40$, $T \in [22, 40]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
 [2502] $2321812^1 2315132181 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -169$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.59 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
 [2503] $81223151322^1 181232 + \hat{1} \rightarrow 2$ $T = 17$, $T_{\max} = 44$, $\Delta = -118$, $\det = -169$, $(n_+, n_-, n_0) = (1, 17, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 21.59 \rightarrow 22$, $T_{\max} = 44$, $T \in [22, 44]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

- [2504] $812321^{22315132^1 1} 1 + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 44, \Delta = -118, \det = -196, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.55 \rightarrow 22, T_{\max} = 44, T \in [22, 44]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [2505] $8123218122^{13151322} + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 44, \Delta = -118, \det = -196, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.55 \rightarrow 22, T_{\max} = 44, T \in [22, 44]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [2506] $1812321812^{12315132} + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 44, \Delta = -118, \det = -225, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.52 \rightarrow 22, T_{\max} = 44, T \in [22, 44]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [2507] $812321812^{123151322} + \hat{1} \rightarrow 2$ $T = 17, T_{\max} = 44, \Delta = -118, \det = -361, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 21.41 \rightarrow 22, T_{\max} = 44, T \in [22, 44]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- $T = 18$ [2508] $1^{22^1 315131513221} (12) 1 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 66, T \in [25, 66]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2509] $1^{315131513221} (12) 1 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 55, \Delta = -119, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 55, T \in [25, 55]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2510] $1^{51231513221} (12) 122^1 2 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 32, \Delta = -112, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 32, T \in [25, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2511] $1^{15131513221} (12) 122^1 2 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 31, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2512] $1^{5131513221} (12) 1223 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 66, T \in [24, 66]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [2513] $1^{51231513221} (12) 12 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 32, \Delta = -112, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 32, T \in [25, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2514] $1^{15131513221} (12) 12 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 31, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2515] $1^{1513221} (12) 1223151 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 32, \Delta = -112, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [2516] $1^{22^1 315131513221} (12) 12 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 66, T \in [25, 66]$

- $T_{\min} = 25, T_{\max} = 66, T \in [25, 66]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2531] $22^1 221(12)122315131513 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 55, T \in [25, 55]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2532] $22^1 2221(12)12231513215 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 32, T \in [25, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2533] $22^1 2221(12)12231513151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2534] $1^1 12321712321812^1 2 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 30, T \in [25, 30]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2535] $41413221 \begin{smallmatrix} 1513221 \\ (12) \end{smallmatrix} 12^1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24, T_{\max} = 38, T \in [24, 38]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [2536] $21712321712321812^1 2 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 30, T \in [25, 30]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2537] $1612321812321812^1 2 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 30, T \in [25, 30]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2538] $314141413221(12)122^1 31 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 30, T \in [25, 30]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2539] $1 \begin{smallmatrix} 22^1 1 \\ (12) \end{smallmatrix} 1223141414141 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2540] $1414141413221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2541] $22^1 31414141413221(12)1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 66, T \in [25, 66]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2542] $31414141413221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 55, T \in [25, 55]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2577] $1414141413221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2576] $414141413221 \begin{smallmatrix} 32^1 21 \\ (12) \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 66, T \in [25, 66]$

- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
 [2548] $1^{51231513221} (12) 122^1 22 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $1^{15131513221} (12) 122^1 22 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $1^{5131513221} 221 (12) 122^1 3 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 66, T \in [25, 66]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $1^{51231513221} 21 (12) 12^1 22 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $1^{15131513221} 21 (12) 12^1 22 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $1^{31513221} 513221 (12) 122^1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.33 \rightarrow 24, T_{\max} = 66, T \in [24, 66]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- $1^{222^1} 1(12)1223151315132 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 66, T \in [25, 66]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $1^{222^1} 21(12)122315131513 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 55, T \in [25, 55]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $1^{222^1} 221(12)12231513215 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $1^{222^1} 221(12)12231513151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $1^{3132^1} 21 4131513221 (12) 1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -104, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 30, T \in [25, 30]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $1^{14132^1} 21(12)1223151232 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 26, \Delta = -99, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 26, T \in [25, 26]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $1^{31413221} (12)12231512^1 3 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 30, T \in [25, 30]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- $1^{1812321712321712^1} 3 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 38, T \in [25, 38]$

- [2546] $1 \begin{smallmatrix} 2^1 1 \\ 31513221 \end{smallmatrix} (12) \quad 122315 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.33 \rightarrow 24, T_{\max} = 66, T \in [24, 66]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [2545] $1 \begin{smallmatrix} 21 \\ 5131513221 \end{smallmatrix} (12) \quad 122^1 3 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 66, T \in [25, 66]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2544] $1 \begin{smallmatrix} 1 \\ 15131513221 \end{smallmatrix} (12) \quad 12^1 22 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2547] $1 \begin{smallmatrix} 315131513221 \\ 1 \end{smallmatrix} (12) \quad 12^1 22 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 55, T \in [25, 55]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2543] $1 \begin{smallmatrix} 1 \\ 51231513221 \end{smallmatrix} (12) \quad 12^1 22 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2581] $1 \begin{smallmatrix} 221 \\ 512315132^1 21 \end{smallmatrix} (12) \quad 12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2582] $1 \begin{smallmatrix} 221 \\ 512^1 31513221 \end{smallmatrix} (12) \quad 12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2583] $1 \begin{smallmatrix} 221 \\ 151315132^1 21 \end{smallmatrix} (12) \quad 12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2584] $21 \begin{smallmatrix} 21 \\ 512^1 31513221 \end{smallmatrix} (12) \quad 12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2585] $21 \begin{smallmatrix} 21 \\ 512315132^1 21 \end{smallmatrix} (12) \quad 12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2586] $21 \begin{smallmatrix} 21 \\ 151315132^1 21 \end{smallmatrix} (12) \quad 12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2587] $1 \begin{smallmatrix} 21 \\ 512315132^1 21 \end{smallmatrix} (12) \quad 12^1 222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2588] $1 \begin{smallmatrix} 512315132^1 21 \\ 1 \end{smallmatrix} (12) \quad 12222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2589] \quad 1 \quad (12) \quad 12222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[2590] \quad 1 \quad (12) \quad 12^1 222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2591] \quad 1 \quad (12) \quad 12222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2592] \quad 22^1 1 \quad (12) \quad 1223 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.25 \rightarrow 24, T_{\max} = 66, T \in [24, 66]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2593] \quad 21 \quad (12) \quad 1222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[2594] \quad 21 \quad (12) \quad 1222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2595] \quad 21 \quad (12) \quad 1222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2596] \quad 221 \quad (12) \quad 122 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[2597] \quad 221 \quad (12) \quad 122 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2598] \quad 221 \quad (12) \quad 122 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2599] \quad 22^1 1 \quad (12) \quad 1223151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.25 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2600] \quad 2222^1 1 (12) 122315131513 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 55, T \in [24, 55]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2601] \quad 2222^1 21 (12) 12231513215 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$$

- [2615] $41315132^{121} (12) 1 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -104, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2616] $4131513221^{11} (12) 1 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -104, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2617] $41315132^{121} (12) 12 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 24, \Delta = -96, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2618] $141322^1 (12) 1223151232 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 26, \Delta = -99, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 26, T \in [24, 26]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2619] $141322^1 (12) 122315132 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 24, \Delta = -97, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2620] $31413221 (12) 122^1 315123 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2621] $314132^1 21 (12) 122315123 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2622] $314132^1 21 (12) 12231513 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -104, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2623] $31413221 (12) 122^1 31513 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -104, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2624] $231413221 (12) 12^1 231512 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 24, \Delta = -97, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2625] $231413221 (12) 12^1 23151 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 24, \Delta = -97, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2626] $21712^1 3218123151315 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 32, \Delta = -112, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2627] $5132171232181232^1 15 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 35, \Delta = -115, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 35, T \in [24, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [2628] $1712315132181232^1 15 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 33, \Delta = -113, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24, T_{\max} = 33, T \in [24, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [2629] $4131513221 (12) 122^1 3132 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 33$, $T \in [24, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [26444] $1612^1 32181232161315 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 33$, $\Delta = -113$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 33$, $T \in [24, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [26445] $1612^1 32181231513161 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 31$, $\Delta = -111$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [26446] $2171232171232^1 18122 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [26447] $21712^1 3217123218122 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [26448] $612315132181232^1 161 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 32$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [26449] $161232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [2650] $1612^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2651] $21712^1 321812321712 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2652] $14141413221(12)12^1 2313 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -100$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2653] $141413221(12)122^1 31413 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -100$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2654] $1414132^1 21(12)12231413 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -100$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2655] $3141413221(12)12^1 23141 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -100$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2579] $1 \quad \begin{matrix} 1 \\ 512315132^1 21 \end{matrix} \quad 1222 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 32$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2578] $1 \quad \begin{matrix} 1 \\ 512^1 31513221 \end{matrix} \quad 1222 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 32$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2603] $222221(12)122315132^1 15 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 32$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.75 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[2666] $1 \begin{pmatrix} 3151315132^1 21 \\ (12) \end{pmatrix} 1222 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 55$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 55$, $T \in [24, 55]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2660] $1 \begin{pmatrix} 21 \\ (12) \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 55$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 55$, $T \in [24, 55]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2661] $1 \begin{pmatrix} 221 \\ (12) \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 32$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2662] $1 \begin{pmatrix} 221 \\ (12) \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 31$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2663] $1 \begin{pmatrix} 21 \\ (12) \end{pmatrix} 122^1 3151 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 32$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[2664] $21 \begin{pmatrix} 21 \\ (12) \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 32$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2659] $1 \begin{pmatrix} 21 \\ (12) \end{pmatrix} 12^1 23 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 66$, $T \in [24, 66]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[2658] $1 \begin{pmatrix} 1 \\ (12) \end{pmatrix} 1222 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 31$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2657] $1 \begin{pmatrix} 1 \\ (12) \end{pmatrix} 1222 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 32$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2668] $1 \begin{pmatrix} 1513151322^1 1 \\ (12) \end{pmatrix} 12222 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 31$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2669] $221 \begin{pmatrix} 5131513221 \\ (12) \end{pmatrix} 12^1 23 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 66$, $T \in [24, 66]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

- [2670] $21^{3151315132^1 21} (12) 122 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 55, T \in [24, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2671] $21^{5123151322^1 1} (12) 1222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2672] $21^{1513151322^1 1} (12) 1222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2673] $221^{5123151322^1 1} (12) 122 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2674] $221^{1513151322^1 1} (12) 122 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2675] $221^{1513221} (12) 122^1 3151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2676] $22221(12)122^1 315131513 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 55, T \in [24, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2677] $22222^1(12)12231513215 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2678] $222221(12)12^1 231513215 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2679] $22222^1(12)12231513151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2680] $222221(12)12^1 231513151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2681] $1^{32151322^1 1} (12) 1223141 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 24, \Delta = -97, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 54 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [2682] $1^{3151322^1 1} (12) 1223141 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 24, \Delta = -96, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 54 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [2683] $1^{21513221} (12) 12^1 231413 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 24, \Delta = -96, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
¹₁₅₁₃₂₂₁
 [2684] $1(12)12^1 231413 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 24, \Delta = -96, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ¹
 [2685] $171232^1 18123151315 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- ³¹³²²¹
 [2686] $413151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ¹³²²¹
 [2687] $413151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 24, \Delta = -96, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2688] $1413221(12)12231512^1 32 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 26, \Delta = -99, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 26, T \in [24, 26]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2689] $1413221(12)122^1 3151232 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 26, \Delta = -99, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 26, T \in [24, 26]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ¹
 [2690] $1413221(12)122^1 315132 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 24, \Delta = -97, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2691] $31413221(12)12^1 2315123 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2692] $3141322^1 1(12)122315123 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ¹
 [2693] $31413221(12)12^1 231513 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ¹
 [2694] $3141322^1 1(12)12231513 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2695] $2314132^1 21(12)12231512 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 24, \Delta = -97, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ¹
 [2696] $2314132^1 21(12)1223151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 24, \Delta = -97, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2697] $2171232^1 18123151315 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$

$$\begin{aligned}
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[2713] & \begin{matrix} 1 \\ 1613161315132^1 2181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 30, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2714] & \begin{matrix} 1 \\ 1812^1 3217123217123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 38, T \in [24, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2715] & \begin{matrix} 321 \\ 1 \ 8 \ 1232171232^1 171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 38, T \in [24, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2716] & \begin{matrix} 1 \\ 1712321712^1 3218122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 30, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2717] & \begin{matrix} 1 \\ 171232^1 17123218122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 30, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2718] & \begin{matrix} 21 \\ 1 \ 8 \ 1232^1 1712321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 30, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2719] & \begin{matrix} 21 \\ 1 \ 8 \ 12321712^1 321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 30, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2720] & \begin{matrix} 15132^1 21 \\ 41413221 \ (12) \ 12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 38, T \in [24, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2721] & \begin{matrix} 51321 \\ 4132^1 1812321 \ 8 \ 1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 41, T \in [24, 41] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2722] & \begin{matrix} 1 \\ 41321812315132^1 181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 35, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 35, T \in [24, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2723] & \begin{matrix} 1 \\ 4132^1 1812315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 35, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 35, T \in [24, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2724] & \begin{matrix} 232^1 1 \\ 712321712321 \ 8 \ 1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 45, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 45, T \in [24, 45] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2725] & \begin{matrix} 1 \\ 171232^1 181232171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 18, T_{\max} = 30, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2726] $18^{21}123218^{12}132161 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2727] $18^{21}123218^{12}1232161 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2728] $18^{12}13218^{12}12321613 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2729] $17^{12}123218^{12}1321712 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2730] $17^{12}123218^{12}12321712 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2731] $612^{12}13218^{12}123218^11 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 41, T \in [24, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2732] $16^{12}1316131513218^11 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2733] $17^{12}12321712^{12}13218^11 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2734] $17^{12}123217123218^11 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2735] $16^{12}1316123218^{12}13215 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 33, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 33, T \in [24, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2736] $16^{12}1316123218^{12}13151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2737] $15^{12}13161316123218^{12}12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2665] $21^{21} \begin{pmatrix} 12 \\ 1513151322^11 \end{pmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2739] $216^{12}1316131513212181 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

- [2740] $161232^1 181232161315 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 33$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 33$, $T \in [24, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2741] $161232^1 181231513161 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 31$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2742] $1712^1 32171232181223 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2743] $1712321712321812^1 23 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2744] $232^1 171232171232181 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 45$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 45$, $T \in [24, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2745] $32171232171232^1 1812 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2746] $2171232^1 17123218122 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2747] $21712321712^1 3218122 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2748] $4132^1 18123218122315 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 41$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 41$, $T \in [24, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2749] $41321812315132^1 1812 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 35$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 35$, $T \in [24, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2750] $4132^1 18123151321812 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 35$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 35$, $T \in [24, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2751] $41321812315132^1 2181 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 35$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 35$, $T \in [24, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2752] $4132^1 18123151322181 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 35$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 35$, $T \in [24, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2753] $8123151316131612^1 32 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 55$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 55$, $T \in [24, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2754] $6131613151321812^1 32 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 55$, $\Delta = -119$, $\det = 25$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24$, $T_{\max} = 55$, $T \in [24, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

- [2755] $612^1 321812315131613 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 55, \Delta = -119, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 55, T \in [24, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2756] $6123151321812^1 32161 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 32, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2757] $612^1 315132181232161 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 32, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [2758] $161232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2759] $1612321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2760] $3161232181232^1 1812 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 38, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2761] $1712^1 3218123217123 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 38, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2762] $2171232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2763] $612^1 32181232181223 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 41, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 41, T \in [24, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2764] $612321812321812^1 23 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 41, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 41, T \in [24, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2765] $712321812321712^1 32 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 45, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 45, T \in [24, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2766] $16131613161232^1 171 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2767] $141414132^1 21(12)122313 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -100, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2768] $141413221(12)12^1 231413 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -100, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2769] $14141322^1 1(12)12231413 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 30, \Delta = -100, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2770] & 31414132^1 21(12)1223141 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -100, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 54 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[2771] & 216131613161232^1 171 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2772] & 4132181232171232^1 16 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 41, T \in [24, 41] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[2773] & 4132^1 18123217123216 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 41, T \in [24, 41] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2774] & 414141413221^{322^1 1} (12)1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 66, T \in [24, 66] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2775] & 4132^1 21^{11} (12)1223141414 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 66, T \in [24, 66] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2656] & 1^{221} (12)^{3151315132^1 21} 1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 55, T \in [24, 55] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2667] & 1^{5123151322^1 1} (12)^{12222} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 32, T \in [24, 32] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2738] & 21613161315132^1 1812 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.44 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2779] & 1^{231513151322^1 21} (12)^{122} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 36, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.22 \rightarrow 24, T_{\max} = 66, T \in [24, 66] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2780] & 1^{31513151322^1 1} (12)^{1222} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 36, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.22 \rightarrow 24, T_{\max} = 55, T \in [24, 55] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2781] & 21^{231513151322^1 21} (12)^{12} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 36, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.22 \rightarrow 24, T_{\max} = 66, T \in [24, 66] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2782] & 21^{31513151322^1 1} (12)^{122} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 36, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.22 \rightarrow 24, T_{\max} = 55, T \in [24, 55] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2783] $22^1 1^{31513221} (12) 122315 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.78 \rightarrow 23$, $T_{\max} = 66$, $T \in [23, 66]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 25 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [2784] $2221(12)122^1 3151315132 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 66$, $T \in [24, 66]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2785] $22221(12)12^1 2315131513 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 55$, $\Delta = -119$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 55$, $T \in [24, 55]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2786] $1413221(12)12^1 23151232 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 26$, $\Delta = -99$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 26$, $T \in [24, 26]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2787] $1413221(12)12^1 2315132 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 24$, $\Delta = -97$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 24$, $T \in [24, 24]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2788] $23141322^1 1(12)12231512 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 24$, $\Delta = -97$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 24$, $T \in [24, 24]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2789] $23141322^1 1(12)1223151 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 24$, $\Delta = -97$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 24$, $T \in [24, 24]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2790] $712^1 315132181231513 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 55$, $\Delta = -119$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 55$, $T \in [24, 55]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2791] $712^1 231513218123215 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 33$, $\Delta = -113$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 33$, $T \in [24, 33]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2792] $712^1 231513218123151 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 31$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2793] $1613161315132^1 181 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2794] $16131613151322^1 181 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -108$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2795] $8122^1 31513161316^1 1 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2796] $712^1 321712321^1 81 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 45$, $\Delta = -118$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 45$, $T \in [24, 45]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2797] $315131613161232^1 181 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 45$, $\Delta = -118$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 45$, $T \in [24, 45]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2798] $321613161315132^1 181 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 41$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 41$, $T \in [24, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2799] $21613161315132^1 181 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -108$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2800] $232171232171232^1 181 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 45$, $\Delta = -118$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 45$, $T \in [24, 45]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2801] $41321812315132^1 181 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 35$, $\Delta = -115$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 35$, $T \in [24, 35]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2802] $8122^1 31513161316123 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 41$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 41$, $T \in [24, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2803] $712^1 32181232171232 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 45$, $\Delta = -118$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 45$, $T \in [24, 45]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2804] $712^1 3216131613161 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2805] $1414141322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -100$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2806] $3141414132^1 21(12)12231 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -100$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2807] $314141322^1 1(12)1223141 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -100$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 77 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [2808] $712^1 321613161316123 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 41$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 41$, $T \in [24, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2809] $4131613161315132^1 17 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2810] $1(12)122^1 3141414141 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 31$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2811] $21(12)122^1 3141414141 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 31$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

- [2812] $14141414132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 31$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2778] $1 \begin{pmatrix} 21 \\ 12 \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 55$, $\Delta = -119$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $31513151322^1 1$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 55$, $T \in [24, 55]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2777] $1 \begin{pmatrix} 221 \\ 12 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 55$, $\Delta = -119$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $31513151322^1 1$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 55$, $T \in [24, 55]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2776] $1 \begin{pmatrix} 21 \\ 12 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 36$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $23151315132^1 21$
 $T_{\min} = 23.22 \rightarrow 24$, $T_{\max} = 66$, $T \in [24, 66]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2839] $1 \begin{pmatrix} 321 \\ 8 \end{pmatrix} 12^1 32171232171 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2813] $1 \begin{pmatrix} 1 \\ 12 \end{pmatrix} 1 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $223151315132^1 21$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 66$, $T \in [24, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [2818] $1 \begin{pmatrix} 223151315132^1 21 \\ 12 \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 66$, $T \in [24, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2819] $1 \begin{pmatrix} 231513151322^1 1 \\ 12 \end{pmatrix} 122 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 66$, $T \in [24, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2820] $221 \begin{pmatrix} 51315132^1 21 \\ 12 \end{pmatrix} 1223 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 66$, $T \in [24, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [2821] $21 \begin{pmatrix} 231513151322^1 1 \\ 12 \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 66$, $T \in [24, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2822] $221 \begin{pmatrix} 31513221 \\ 12 \end{pmatrix} 122^1 315 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 66$, $T \in [24, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2823] $221 \begin{pmatrix} 1513221 \\ 12 \end{pmatrix} 12^1 23151 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 32$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2824] $221(12)122^1 31513151322 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 66$, $T \in [24, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2825] $2221(12)12^1 23151315132 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 66$, $\Delta = -120$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 66, T \in [24, 66] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2826] & \overset{1}{1712321812^1 3151315} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 32, T \in [24, 32] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2827] & 21712321812^1 3151315 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 32, T \in [24, 32] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2828] & 5132^1 17123218123215 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 35, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 35, T \in [24, 35] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2829] & 5132171232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 35, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 35, T \in [24, 35] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2830] & 5132171232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 33, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 33, T \in [24, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2831] & 5132^1 17123218123151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 33, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 33, T \in [24, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2832] & 1712315132^1 18123215 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 33, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 33, T \in [24, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2833] & 1712315132^1 18123151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 31, T \in [24, 31] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2834] & 7122315132^1 18123215 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 33, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 33, T \in [24, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2835] & 7122315132^1 18123151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 31, T \in [24, 31] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2836] & \overset{1}{181232^1 16131613151} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 30, T \in [24, 30] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2837] & \overset{1}{181232^1 17123217123} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2816] & \overset{21}{1 \quad (12) \quad 122^1 315} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& \quad \quad \quad 31513221 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 66, T \in [24, 66] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2815] & \overset{21}{1 \quad (12) \quad 1223} + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& \quad \quad \quad 51315132^1 21 \\
& T_{\min} = 23.06 \rightarrow 24, T_{\max} = 66, T \in [24, 66]
\end{aligned}$$

- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
 [2814] $1 \begin{pmatrix} 21 \\ 12 \end{pmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $231513151322^1 1$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 66, T \in [24, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- 1513221
 [2841] $414132^1 21 \begin{pmatrix} 12 \\ 12 \end{pmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $151322^1 1$
 [2842] $41413221 \begin{pmatrix} 12 \\ 12 \end{pmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- $5132^1 1$
 [2843] $41321812321 \begin{pmatrix} 8 \\ 8 \end{pmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 41, T \in [24, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 51321
 [2844] $41321812^1 321 \begin{pmatrix} 8 \\ 8 \end{pmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 41, T \in [24, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- 1
 [2845] $41321812^1 315132181 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 35, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 35, T \in [24, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- 31
 [2846] $812^1 2315131613161 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 1
 [2847] $181232181232^1 1613 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 1
 [2848] $181232^1 1812321613 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- 1
 [2849] $61232181232^1 1 \begin{pmatrix} 321 \\ 8 \end{pmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 41, T \in [24, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 1
 [2850] $61232^1 1812321 \begin{pmatrix} 321 \\ 8 \end{pmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 41, T \in [24, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2851] $1613161232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 33, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 33, T \in [24, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2852] $1613161232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2853] $151316131612^1 321812 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 30, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 30, T \in [24, 30]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[2854] \quad 316131612321812^1 315 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 55, T \in [24, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2855] \quad 1612321812^1 32161315 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 33, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 33, T \in [24, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[2856] \quad 1612321812^1 31513161 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[2857] \quad 171232^1 171232181223 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[2858] \quad 171232171232^1 181223 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2859] \quad 321712321712^1 321812 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[2860] \quad 321712^1 321712321812 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[2861] \quad 413218123218122^1 315 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 41, T \in [24, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2862] \quad 41321812^1 3218122315 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 41, T \in [24, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[2863] \quad 41321812^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 35, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 35, T \in [24, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[2864] \quad 41321812^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 35, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 35, T \in [24, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[2865] \quad 812^1 321613161315132 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 55, T \in [24, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2866] \quad 812^1 315131613161232 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 55, T \in [24, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2867] \quad 812^1 231513161316123 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.06 \rightarrow 24, T_{\max} = 41, T \in [24, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2868] \quad 61315132181232^1 1613 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$$

- $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 55$, $T \in [24, 55]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2869] $61232^1 1812315131613 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 55$, $\Delta = -119$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 55$, $T \in [24, 55]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2870] $612315132^1 181232161 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 32$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [2871] $31612321812^1 321812 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2872] $31612^1 321812321812 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2873] $171232181232^1 17123 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2874] $171232^1 18123217123 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 38$, $T \in [24, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2875] $61232^1 181232181223 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 41$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 41$, $T \in [24, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2876] $61232181232^1 181223 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 41$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 41$, $T \in [24, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2877] $161316131612^1 32171 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [2878] $31414141322^1 1(12)12231 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -100$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [2879] $2161316131612^1 32171 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 30$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 30$, $T \in [24, 30]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [2880] $41321812321712^1 3216 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 41$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 41$, $T \in [24, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2881] $41321812^1 3217123216 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 41$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 41$, $T \in [24, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [2882] $1(12)12^{221} 23141414141 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 31$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$

- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
 [2883] $21 \begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}^{21} 12^1 23141414141 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
 [2884] $1 \begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}^{21} 122^1 31414141413 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 55, T \in [24, 55]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [2885] $141414141322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 31, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
 [2886] $314141414132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 55, T \in [24, 55]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
 [2887] $41322^1 1(12)1223141414 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 66, T \in [24, 66]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2888] $414132^1 21(12)12231414 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 66, T \in [24, 66]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2817] $1 \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}^{21} 12^1 23151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 32, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2840] $1 \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}^{321} 8 \ 12321712^1 32171 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
 [2838] $1 \begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}^{1} 181232171232^1 17123 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 23.06 \rightarrow 24, T_{\max} = 38, T \in [24, 38]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [2895] $51321712^1 3218123151 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 33, \Delta = -113, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23, T_{\max} = 33, T \in [23, 33]$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [2894] $51321712^1 3218123215 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 35, \Delta = -115, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23, T_{\max} = 35, T \in [23, 35]$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [2892] $221 \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}^{315132^1 21} 12^1 22315 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23, T_{\max} = 66, T \in [23, 66]$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [2891] $1 \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}^{2231513151322^1 21} 12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23, T_{\max} = 66, T \in [23, 66]$
- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [2890] $1 \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}^{21} 12^1 22315 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23, T_{\max} = 66, T \in [23, 66]$

- $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 66$, $T \in [23, 66]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2897] $1812321712^1 3217123 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 38$, $T \in [23, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2898] $181232^1 171232171 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 38$, $T \in [23, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2899] $4132181232^1 1^1 8^1 1 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 41$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 41$, $T \in [23, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [2900] $1812321812^1 321613 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 38$, $T \in [23, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2901] $612321812^1 321^1 8^1 1 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 41$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 41$, $T \in [23, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2902] $16131612^1 3218123215 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 33$, $\Delta = -113$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 33$, $T \in [23, 33]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2903] $16131612^1 3218123151 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 31$, $\Delta = -111$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 31$, $T \in [23, 31]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2904] $161232181232^1 161315 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 33$, $\Delta = -113$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 33$, $T \in [23, 33]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [2905] $1712321712^1 32181223 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 38$, $T \in [23, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2906] $32171232^1 1712321812 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 38$, $T \in [23, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2907] $4132181232^1 18122315 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 41$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 41$, $T \in [23, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [2908] $6131612321812^1 31513 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 55$, $\Delta = -119$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 55$, $T \in [23, 55]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2909] $613151321812^1 321613 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 55$, $\Delta = -119$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 55$, $T \in [23, 55]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2910] $3161232^1 1812321812 + \hat{1} \rightarrow 2$ $T = 18$, $T_{\max} = 38$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 22.94 \rightarrow 23$, $T_{\max} = 38$, $T \in [23, 38]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[29111] & 1712321812^1 3217123 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.94 \rightarrow 23, T_{\max} = 38, T \in [23, 38] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2912] & 612321812^1 32181223 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.94 \rightarrow 23, T_{\max} = 41, T \in [23, 41] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2913] & 4132181232^1 17123216 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.94 \rightarrow 23, T_{\max} = 41, T \in [23, 41] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[2914] & 1(12)122^1 314141414132 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.94 \rightarrow 23, T_{\max} = 66, T \in [23, 66] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2915] & 1(12)12^{21} 231414141413 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.94 \rightarrow 23, T_{\max} = 55, T \in [23, 55] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2916] & 2314141414132^{21} 21(12)12 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.94 \rightarrow 23, T_{\max} = 66, T \in [23, 66] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2917] & 3141414141322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.94 \rightarrow 23, T_{\max} = 55, T \in [23, 55] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2896] & 7123151321812^1 31513 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.94 \rightarrow 23, T_{\max} = 55, T \in [23, 55] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2893] & 221(12)12^1 231513151322 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.94 \rightarrow 23, T_{\max} = 66, T \in [23, 66] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2889] & 1 \quad \begin{pmatrix} 12 \\ 2231513151322^1 1 \end{pmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 64, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.94 \rightarrow 23, T_{\max} = 66, T \in [23, 66] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2927] & 71232^1 16131613161 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 38, \Delta = -116, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.84 \rightarrow 23, T_{\max} = 38, T \in [23, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2922] & 3151316131612^1 32181 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 45, \Delta = -118, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.84 \rightarrow 23, T_{\max} = 45, T \in [23, 45] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[2923] & 5131613161232^1 18123 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.84 \rightarrow 23, T_{\max} = 55, T \in [23, 55] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2924] & 1613161315132^1 18123 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 41, \Delta = -117, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.84 \rightarrow 23, T_{\max} = 41, T \in [23, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2925] $2321712321712^1 32181 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 45, \Delta = -118, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 45, T \in [23, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [2926] $71232^1 181232171232 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 45, \Delta = -118, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 45, T \in [23, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [2920] $4141322^1 1^{1513221} (12) 12 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 38, \Delta = -116, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 38, T \in [23, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2928] $71232^1 1613161316123 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 41, \Delta = -117, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 41, T \in [23, 41]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [2929] $1^{12} (12) 12^1 2314141414132 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 66, T \in [23, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2930] $22314141414132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 66, T \in [23, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2931] $23141414141322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 66, T \in [23, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2932] $413221^{12} 122^1 3141414 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 66, T \in [23, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2933] $4141322^1 1^{12} 12231414 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 66, T \in [23, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2921] $71232^1 1712321^{2321} 8 \ 1 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 45, \Delta = -118, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 45, T \in [23, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [2919] $221^{513151322^1 1} (12) 1223 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 66, T \in [23, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2918] $1^{21} (12) 1223 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 81, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.84 \rightarrow 23, T_{\max} = 66, T \in [23, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2934] $1^{21} (12) 12^1 2315 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 100, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.76 \rightarrow 23, T_{\max} = 66, T \in [23, 66]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2935] $221^{31513221} (12) 12^1 2315 + \hat{1} \rightarrow 2$ $T = 18, T_{\max} = 66, \Delta = -120, \det = 100, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 22.76 \rightarrow 23, T_{\max} = 66, T \in [23, 66]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2950] & 81232^1 1613161315132 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 121, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.69 \rightarrow 23, T_{\max} = 55, T \in [23, 55] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[2951] & 612321812^1 315131613 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 121, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.69 \rightarrow 23, T_{\max} = 55, T \in [23, 55] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2952] & 712321812^1 32171232 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 45, \Delta = -118, \det = 121, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.69 \rightarrow 23, T_{\max} = 45, T \in [23, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2953] & 4141414132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 121, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.69 \rightarrow 23, T_{\max} = 66, T \in [23, 66] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2954] & 413221(12)12^1 23141414 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 121, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.69 \rightarrow 23, T_{\max} = 66, T \in [23, 66] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2955] & 51316131612^1 3218123 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 144, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.64 \rightarrow 23, T_{\max} = 55, T \in [23, 55] \\
& \begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2956] & 316131612^1 321812315 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 169, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.59 \rightarrow 23, T_{\max} = 55, T \in [23, 55] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 64 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2957] & 613161315132^1 181232 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 55, \Delta = -119, \det = 169, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.59 \rightarrow 23, T_{\max} = 55, T \in [23, 55] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2958] & 41414141322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 18, T_{\max} = 66, \Delta = -120, \det = 169, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 22.59 \rightarrow 23, T_{\max} = 66, T \in [23, 66] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
T = 19 & [2974] 22^1 1(12)122315131612 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26, T_{\max} = 31, T \in [26, 31] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[2959] & 1^{315131513221} (12) 122^1 31 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 27, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26, T_{\max} = 27, T \in [26, 27] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[2964] & 16^{31} 131513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 39, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26, T_{\max} = 39, T \in [26, 39] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[2965] & 16^{31} 131513221(12)12 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 39, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26, T_{\max} = 39, T \in [26, 39] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2966] $16131513221(12)12222^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2967] $3221^{21} 12231513161 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 42, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 42, T \in [25, 42]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [2968] $1^{221} 1223151316123 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 42, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 42, T \in [26, 42]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2969] $1^{221} 12231513161232 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 56, \Delta = -119, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 56, T \in [26, 56]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2970] $1^{221} 122315131612 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2971] $1^{31} 16131513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 39, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 39, T \in [26, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2972] $21^{221} 1223151316123 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 42, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 42, T \in [26, 42]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2962] $16131513221^{21} (12)122^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2961] $16131513221^1 (12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2960] $132^1 21(12)1223151315132 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -100, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2976] $21^{221} 122315132161 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2977] $22^1 1(12)122315132161 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2978] $1^{221} 1223151321613 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 42, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 42, T \in [26, 42]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

- [2979] $21^{22^1 1}(12)1223151321613 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 42, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 42, T \in [26, 42]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2980] $413221^{5131513221}(12)12^1 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 56, \Delta = -119, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 56, T \in [25, 56]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [2981] $61231513221^{3221}(12)122^1 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 46, \Delta = -118, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25, T_{\max} = 46, T \in [25, 46]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [2982] $612231513221^{22^1 221}(12)1 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2983] $612231513221^{22^1 21}(12)12 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2984] $612231513221^{221}(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2985] $3216131513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 42, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 42, T \in [26, 42]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2986] $23216131513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 56, \Delta = -119, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 56, T \in [26, 56]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2987] $216131513221(12)12222^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2988] $161231513221(12)12222^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2989] $3161231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 42, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 42, T \in [26, 42]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2990] $22^1 18123218123151315 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 32, \Delta = -111, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 32, T \in [26, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2991] $612231513221(12)12222^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

- [2992] $18^{21}12321712321812^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2993] $1712321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2994] $22^1 18123217123218122 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2995] $21712321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2996] $141414141413221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2997] $(12)12231414141414132^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 56, \Delta = -119, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 56, T \in [26, 56]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [2963] $16131513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2975] $1^{22^1 221}(12)122315132161 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2973] $21^{22^1 21}(12)122315131612 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3012] $1^{222^1 21}(12)122315131612 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -108, \det = -9, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- [2998] $1^{315131513221}(12)12^1 231 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 27, \Delta = -99, \det = -9, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26, T_{\max} = 27, T \in [26, 27]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3003] $32^1 1513221(12)122315123 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -101, \det = -9, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [3004] $31^{32^1 1513221(12)12231} 1 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -100, \det = -9, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26, T_{\max} = 31, T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3005] $16131513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -108, \det = -9, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- ¹
 [3006] $16131513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- ²¹
 [3007] $16131513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 39$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 39$, $T \in [26, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- ¹
 [3008] $16131513221(12)1222^1 22 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- ^{32^1 21}
 [3009] $221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 42$, $T \in [26, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- ^{222^1 1}
 [3010] $1(12)1223151316123 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 42$, $T \in [26, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- ¹
 [3001] $1322^1 1(12)1223151315132 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -100$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- ¹⁵¹³¹⁵¹³²²¹
 [3000] $1(12)122^1 313 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 27$, $\Delta = -99$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 27$, $T \in [26, 27]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- ⁵¹²³¹⁵¹³²²¹
 [2999] $1(12)122^1 313 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -100$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- ²²¹
 [3014] $21(12)122315131612^1 3 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 42$, $T \in [26, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- ^{222^1 1}
 [3015] $21(12)122315131612 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- ^{222^1 21}
 [3016] $1(12)122315132161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- ^{222^1 1}
 [3017] $21(12)122315132161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- ^{222^1 1}
 [3018] $1(12)1223151321613 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 42$, $T \in [26, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- ^{32¹21}
 [3019] 61231513221(12)122 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 46$, $T \in [26, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- ^{222¹21}
 [3020] 612231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- ^{222¹1}
 [3021] 612231513221(12)12 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- ^{32¹16131513221(12)12222} + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 42$, $T \in [26, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [3023] 3216131513221(12)122¹22 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 42$, $T \in [26, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3024] 23216131513221(12)12¹22 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 56$, $T \in [26, 56]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3025] 216131513221(12)1222¹22 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3026] 161231513221(12)1222¹22 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3027] 3161231513221(12)122¹22 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 42$, $T \in [26, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- ^{32¹18123151321812315} + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 56$, $T \in [26, 56]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3029] 612231513221(12)1222¹22 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- ¹
 [3030] 18123217123218122¹3 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

- [3031] $1 \overset{32^1}{8} 12321712321812 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [3032] $1812321812321712^1 3 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3033] $1 \overset{32^1}{8} 1232181232171 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3034] $1812321712321812^1 3 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3035] $218123217123218122^1 3 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3036] $613151321812321712^1 3 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 46$, $T \in [26, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3037] $17123218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3038] $32^1 1712321812321812 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3039] $161316131612321812^1 3 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 39$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 39$, $T \in [26, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3040] $32^1 16131613161232181 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 39$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 39$, $T \in [26, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [3041] $41414141413221(12)122^1 3 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 56$, $T \in [26, 56]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3002] $3132^1 21(12)122315131513 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -107$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 31$, $T \in [26, 31]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3013] $16131513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 39$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 39$, $T \in [26, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3011] $1(12)122315131612^1 3 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 42$, $T \in [26, 42]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[3066] & \begin{matrix} 31 \\ 16 \end{matrix} 131513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 39, \Delta = -116, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 39, T \in [25, 39] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3047] & 2313221(12)122^1 31513215 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -101, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3044] & \begin{matrix} 51231513221 \\ 1 \end{matrix} (12) 12^1 2313 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3043] & \begin{matrix} 512^1 31513221 \\ 1 \end{matrix} (12) 122313 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[3049] & 2313221(12)122315132^1 15 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -101, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[3050] & 2313221(12)122^1 31513151 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3051] & 23132^1 21(12)12231513151 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3052] & 31322^1 1(12)122315131513 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3053] & 3215132^1 21(12)122315123 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -101, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3054] & \begin{matrix} 321513221(12)122^1 31 \\ 31 \end{matrix} 5 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3055] & \begin{matrix} 3215132^1 21(12)12231 \\ 31 \end{matrix} 5 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3056] & \begin{matrix} 31513221(12)122^1 31 \\ 31 \end{matrix} 5 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 27, \Delta = -99, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 27, T \in [25, 27] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [3057] $\overset{1}{161315132^1 21(12)12222} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -108, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3058] $\overset{1}{16131513221(12)12^1 222} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -108, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3059] $\overset{1}{161315132^1 21(12)^{21} 1222} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -108, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3060] $\overset{1}{161315132^1 21(12)^{221} 122} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -108, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3061] $\overset{1}{16131513221(12)122^1 222} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -108, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3062] $\overset{1}{161315132^1 21(12)122222} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -108, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3063] $\overset{3221}{22^1 1(12)12231513161} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.25 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3064] $\overset{2222^1 1}{1(12)122315131612} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -108, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3065] $\overset{22221}{1(12)122^1 315131612} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -108, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3042] $\overset{512315132^1 21}{1(12)^{122313}} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3067] $\overset{2221}{21(12)122^1 315131612} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -108, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3068] $\overset{221}{221(12)122^1 315131612} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -108, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3069] $\overset{22221}{1(12)122^1 315132161} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$

- $$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$
- [3070] $1^{22221}(12)122315132^1 161 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$
- [3071] $1^{2222^1 1}(12)122315132161 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$
- [3072] $21^{2221}(12)122315132^1 161 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$
- [3073] $21^{2221}(12)122^1 315132161 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$
- [3074] $221^{221}(12)122315132^1 161 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$
- [3075] $221^{221}(12)122^1 315132161 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$
- [3076] $18^{21}12^1 3218123151315 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 32, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$
- $$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$
- [3077] $1^{51321}8^{1232181232^1 15} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 35, \Delta = -114, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 35, T \in [25, 35]$
- $$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$
- [3078] $18^{112315132181232^1 15} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$
- $$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$
- [3079] $61231513221^{3221}(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.25 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$
- $$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$
- [3080] $6122315132^1 21^{22221}(12)1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$
- [3081] $612231513221^{2222^1 1}(12)1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$
- [3082] $6122^1 31513221^{22221}(12)1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

- [3083] $6122315132^{\overline{1}21}(12)12 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3084] $6122^{\overline{1}31513221}(12)12 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3085] $6122^{\overline{1}31513221}(12)122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3086] $6122315132^{\overline{1}21}(12)122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3087] $3216131513221(12)12^{\overline{1}222} + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 42$, $T \in [25, 42]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3088] $216131513221(12)122^{\overline{1}222} + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3089] $2161315132^{\overline{1}21}(12)122222 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3090] $1612^{\overline{1}31513221}(12)122222 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3091] $161231513221(12)122^{\overline{1}222} + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3092] $1612315132^{\overline{1}21}(12)122222 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3093] $3161231513221(12)12^{\overline{1}222} + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 42$, $T \in [25, 42]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3094] $221812^{\overline{1}3218123151315} + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 32$, $\Delta = -111$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3095] $51322181232181232^{\overline{1}15} + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 35$, $\Delta = -114$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 35$, $T \in [25, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$

- [3096] $21812315132181232^1 15 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [3097] $18122315132181232^1 15 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [3098] $612231513221(12)122^1 222 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3099] $6122315132^1 21(12)122222 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3100] $6122^1 31513221(12)122222 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3101] $1712^1 32181232161315 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 32$, $\Delta = -111$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3102] $1712^1 32181231513161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3103] $181232171232^1 18122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3104] $1812^1 3217123218122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3105] $612315132181232^1 171 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3106] $1712^1 32181232181 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3107] $171232181232^1 181 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3108] $1712^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3109] $171232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[3110] & 181232181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[3111] & 1812^1 321812321712 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3112] & 1812^1 32171232181 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3113] & 1414132^1 21(12)122315123 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 25, \Delta = -96, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3114] & 141413221(12)122^1 315123 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 25, \Delta = -96, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3115] & 1612^1 321812321712315 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 33, T \in [25, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3116] & 513161232181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 32, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 32, T \in [25, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3117] & 161315132181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3118] & 1612^1 321812315132171 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3119] & 221812^1 3217123218122 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3120] & 4131513221(12)122^1 31413 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -103, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3121] & 41315132^1 21(12)12231413 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -103, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3122] & 61232171232181232^1 15 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 35, \Delta = -114, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 35, T \in [25, 35] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[3123] & 612315132181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.75 \rightarrow 25, T_{\max} = 31, T \in [25, 31]
\end{aligned}$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3124] $7122315132181232^1 161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3125] $2171232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[3126] $21712^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3127] $1613161232181232^1 161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3048] $23132^1 21(12)12231513215 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -101$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.75 \rightarrow 25$$
, $T_{\max} = 31$, $T \in [25, 31]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3046] $1^{15131513221}(12)12^1 2313 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 27$, $\Delta = -99$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 27$, $T \in [25, 27]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3045] $1^{151315132^1 21}(12)122313 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 27$, $\Delta = -99$, $\det = -16$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.75 \rightarrow 25$, $T_{\max} = 27$, $T \in [25, 27]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3177] $32161315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25$$
, $T_{\max} = 42$, $T \in [25, 42]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3218] $181232^1 1712321^1 81 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3220] $141413221(12)12^1 2315123 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 25$, $\Delta = -96$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25$$
, $T_{\max} = 25$, $T \in [25, 25]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3221] $161232^1 1812321712315 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[3222] $5131612321812^1 321712 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 32$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[3128] $1^{3151315132^1 21}(12)12231 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 27$, $\Delta = -99$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 27$, $T \in [25, 27]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

- [3129] $1^{5123151322^1 1}(12)122313 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3130] $1^{1513151322^1 1}(12)122313 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 27, \Delta = -99, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 27, T \in [25, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3131] $231322^1 1(12)12231513215 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -101, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3132] $2313221(12)12^1 231513215 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -101, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3133] $231322^1 1(12)12231513151 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3134] $2313221(12)12^1 231513151 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3135] $313221(12)122^1 315131513 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3136] $32151322^1 1(12)122315123 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -101, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3137] $31^{321513221(12)12^1 231} 5 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3138] $31^{32151322^1 1(12)12231} 5 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3139] $31^{1} 31513221(12)12^1 231 5 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 27, \Delta = -99, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 27, T \in [25, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3140] $1^{1} 1613151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3141] $1^{1} 1613151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3142] & \begin{matrix} 1 & & 221 \\ 1613151322^1 1(12)122 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 31, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[3143] & \begin{matrix} 31 & & 1 \\ 161315132^1 21(12)1222 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 39, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 39, T \in [25, 39] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[3144] & \begin{matrix} 31 & & 221 \\ 161315132^1 21(12)12 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 39, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 39, T \in [25, 39] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[3145] & \begin{matrix} 1 \\ 1613151322^1 1(12)122222 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 31, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[3146] & \begin{matrix} 1 \\ 16131513221(12)12^1 2222 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 31, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[3147] & \begin{matrix} 322^1 1 \\ 221(12)12231513161 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 42, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 42, T \in [25, 42] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[3148] & \begin{matrix} 2221 \\ 1(12)122^1 3151316123 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 42, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 42, T \in [25, 42] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[3149] & \begin{matrix} 221 \\ 1(12)122315131612^1 32 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 56, \Delta = -119, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 56, T \in [25, 56] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[3150] & \begin{matrix} 22221 \\ 1(12)12^1 2315131612 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 31, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[3151] & \begin{matrix} 31 \\ 161315132^1 21(12)12222 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 39, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 39, T \in [25, 39] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[3152] & \begin{matrix} 221 \\ 21(12)122^1 3151316123 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 42, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 42, T \in [25, 42] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \end{matrix} \\
[3153] & \begin{matrix} 21 \\ 21(12)122315131612^1 32 + \hat{1} \rightarrow 2 & T = 19, T_{\max} = 56, \Delta = -119, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 24.44 \rightarrow 25, T_{\max} = 56, T \in [25, 56] \\ \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \end{matrix}
\end{aligned}$$

- [3154] $21(12)12^{2221}2315131612 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [3155] $221(12)12^{221}2315131612 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -108$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [3156] $1(12)12^{22221}2315132161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [3157] $21(12)12^{2221}2315132161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [3158] $221(12)12^{221}2315132161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [3159] $1(12)122^{2221}3151321613 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 42$, $T \in [25, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3160] $21(12)122^{221}3151321613 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 42$, $T \in [25, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3161] $181232^{21}18123151315 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 32$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3162] $18^{51321}12321812^13215 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 35$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 35$, $T \in [25, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3163] $18^{51321}12321812^13151 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3164] $1812^1131513218123215 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3165] $18123151321812^13215 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3166] $1812^1131513218123151 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3167] $18123151321812^13151 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [3168] $4132^{\mathbf{1}21}{}^{5131513221}(12)12 + \hat{\mathbf{1}} \rightarrow \mathbf{2} \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [3169] $61231513221^{\mathbf{322}^{\mathbf{1}1}}(12)122 + \hat{\mathbf{1}} \rightarrow \mathbf{2} \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [3170] $612^{\mathbf{1}31513221}(12)122 + \hat{\mathbf{1}} \rightarrow \mathbf{2} \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3171] $61223151322^{\mathbf{1}1}(12)1 + \hat{\mathbf{1}} \rightarrow \mathbf{2} \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
 [3172] $612^{\mathbf{1}231513221}(12)1 + \hat{\mathbf{1}} \rightarrow \mathbf{2} \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3173] $61223151322^{\mathbf{1}1}(12)12 + \hat{\mathbf{1}} \rightarrow \mathbf{2} \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
 [3174] $612^{\mathbf{1}231513221}(12)12 + \hat{\mathbf{1}} \rightarrow \mathbf{2} \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
 [3176] $612^{\mathbf{1}231513221}(12)122 + \hat{\mathbf{1}} \rightarrow \mathbf{2} \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [3178] $232^{\mathbf{1}16131513221}(12)1222 + \hat{\mathbf{1}} \rightarrow \mathbf{2} \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [3179] $216131513221(12)12^{\mathbf{1}2222} + \hat{\mathbf{1}} \rightarrow \mathbf{2} \quad T = 19, T_{\max} = 31, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3180] $21613151322^{\mathbf{1}1}(12)122222 + \hat{\mathbf{1}} \rightarrow \mathbf{2} \quad T = 19, T_{\max} = 31, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3181] $16123151322^{\mathbf{1}1}(12)122222 + \hat{\mathbf{1}} \rightarrow \mathbf{2} \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3182] $161231513221(12)12^{\mathbf{1}2222} + \hat{\mathbf{1}} \rightarrow \mathbf{2} \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3183] 31612315132¹21(12)12222 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3184] 22181232¹18123151315 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 32$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[3185] 513221812321812¹3215 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 35$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 35, T \in [25, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3186] 15132¹18123218122315 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[3187] 218123151321812¹3215 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3188] 21812¹31513218123215 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3189] 218123151321812¹3151 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3190] 21812¹31513218123151 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3191] 181223151321812¹3215 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3192] 18122¹31513218123215 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3193] 181223151321812¹3151 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3194] 18122¹31513218123151 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3195] 61223151322¹1(12)122222 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3196] 612¹231513221(12)122222 + $\hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[3197] $612231513221(12)12^1 2222 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3198] $171232^1 181232161315 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 32$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[3199] $171232^1 181231513161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3200] $1812321712321812^1 23 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3201] $1812^1 32171232181223 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3202] $1^1 8^1 1232171232181 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[3203] $1^1 8^1 1232171232^1 1812 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3204] $1^1 8^1 12321712^1 3218122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[3205] $1^1 8^1 1232^1 17123218122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3206] $612^1 315132181232171 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[3216] $712321812321^1 8^1 1 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[3208] $1712321812^1 3218^1 1 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[3209] $171232^1 18123218^1 1 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3210] $1812^1 3218123217123 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [3211] $1 \ 8 \ 1232181232^1 171 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [3212] $171232^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [3213] $1712321812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [3214] $181232^1 1812321712 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [3215] $1812321812^1 321712 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [3223] $1613151321812^1 321712 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3224] $3161315132181232^1 171 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3225] $1612321812315132^1 171 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [3226] $161232^1 1812315132171 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [3227] $21812^1 32171232181223 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [3228] $21812321712321812^1 23 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [3229] $22181232^1 17123218122 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3230] $413151322^1 1(12)12231413 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -103, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3231] $4131513221(12)12^1 231413 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -103, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$

- [3232] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ $612321712321812^1 3215 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 35$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 35$, $T \in [25, 35]$
- [3233] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ $612^1 3217123218123215 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 35$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 35$, $T \in [25, 35]$
- [3234] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ $612^1 3217123218123151 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$
- [3235] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ $612321712321812^1 3151 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$
- [3236] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$ $612^1 3151321812321712 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
- [3237] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$ $6123151321812^1 321712 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
- [3238] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$ $7122^1 315132181232161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
- [3239] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$ $71223151321812^1 32161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
- [3240] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ $171232181^1 2321812^1 23 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
- [3241] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ $1712^1 32181^1 232181223 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
- [3242] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$ $232^1 171232181^1 232181 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 42$, $T \in [25, 42]$
- [3243] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$ $32171232181^1 232^1 1812 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
- [3244] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$ $2171232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
- [3245] $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$ $21712321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$

- [3246] $16131613161232^1 181 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3247] $16131613161232^1 1812 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3248] $1812^1 32161316131612 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [3249] $4141413221^{15132^1 21} (12) 1 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3250] $216131613161232^1 1812 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3251] $16131612321812^1 32161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3252] $4132181232171232^1 171 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3253] $4132^1 181232171232171 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3254] $81232161316131612^1 32 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 46$, $T \in [25, 46]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3255] $61316131612321812^1 32 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 46$, $T \in [25, 46]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3256] $612^1 3218123216131613 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 46$, $T \in [25, 46]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3257] $612321812321812^1 314 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 39$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 39$, $T \in [25, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3258] $612^1 321812321812314 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 39$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 39$, $T \in [25, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3259] $414141413221(12)12^1 23 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.44 \rightarrow 25$, $T_{\max} = 56$, $T \in [25, 56]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3260] $414141413221(12)122^1 314 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -25$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[3217] \quad 1812^1 13217123218123 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3175] \quad 61223151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3219] \quad 14141322^1 1(12)122315123 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 25, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3207] \quad 6123151321812^1 32171 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3265] \quad 1613151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 39, \Delta = -116, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 39, T \in [25, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3266] \quad 1(12)12^1 23151316123 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3267] \quad 1(12)122^1 31513161232 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3268] \quad 1613151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 39, \Delta = -116, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 39, T \in [25, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3269] \quad 21(12)12^1 23151316123 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3270] \quad 21(12)122^1 31513161232 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3271] \quad 1(12)12^1 23151321613 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3272] \quad 21(12)12^1 23151321613 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3273] \quad 321613151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

15.0

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3274] \quad 232161315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3275] \quad 316123151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3276] \quad 1812^1 315132181231513 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3277] \quad 1812^1 231513218123215 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3278] \quad 1812^1 231513218123151 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3279] \quad 8122^1 315132181231513 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3280] \quad 1 \overset{2321}{8} 1232171232^1 181 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3281] \quad 712^1 321812321 \overset{1}{8} \overset{2321}{1} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3282] \quad 712^1 3151316131613151 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 32, \Delta = -111, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3283] \quad 712^1 3151321812321613 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3284] \quad 712^1 2315132181232161 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[3285] \quad 232171232181232^1 181 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3286] \quad 1 \overset{31}{6} 131613161232^1 181 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3287] \quad 3216131613161232^1 181 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 39, \Delta = -116, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$T_{\min} = 24.22 \rightarrow 25, T_{\max} = 39, T \in [25, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3288] $4131613161315132^1 181 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3289] $4131613161315132^1 218 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[3290] $41316131613161232^1 17 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[3264] $16^{31} 13151322^1 1(12) 1222 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 39, \Delta = -116, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 39, T \in [25, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3263] $313221(12) 12^1 2315131513 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3262] $13221(12) 122^1 3151315132 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3261] $1^{31513151322^1} (12) 12231 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 27, \Delta = -99, \det = -36, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.22 \rightarrow 25, T_{\max} = 27, T \in [25, 27]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3295] $1^{2221} (12) 122315132^1 1613 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3296] $21^{221} (12) 122315132^1 1613 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3297] $18^{21} 12321812^1 3151315 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 32, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[3298] $1^{51321} 8^{1232^1} 18123215 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 35, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 35, T \in [25, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3299] $1^{5132^1} 8^{123218123215} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 35, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 35, T \in [25, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[3300] $1^{51321} 8^{1232^1} 18123151 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

- [3301] $1 \overset{5132^1}{8} 123218123151 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3302] $1812315132^1 18123215 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3303] $1812315132^1 18123151 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3304] $41322^1 \overset{5131513221}{1} (12) 12 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3305] $41322^1 \overset{51315132^1 21}{(12)} 12 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [3306] $612315132^1 \overset{3221}{21} (12) 122 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3307] $2321613151322^1 1(12) 1222 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [3308] $31612^1 31513221 (12) 12222 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [3309] $221812321812^1 3151315 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 32, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [3310] $51322181232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 35, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 35, T \in [25, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3371] $4141413221 (12) 122^1 31414 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [3294] $21 \overset{21}{(12)} 12^1 231513161232 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3293] $1 \overset{221}{(12)} 12^1 231513161232 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3311] $5132^1 218123218123215 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 35, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 35, T \in [25, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3312] \quad 151321812^1 3218122315 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[3313] \quad 1513218123218122^1 315 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3314] \quad 3218123151321812^1 315 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3315] \quad 21812315132^1 18123215 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3316] \quad 21812315132^1 18123151 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3317] \quad 18122315132^1 18123215 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3318] \quad 18122315132^1 18123151 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3319] \quad 812^1 3151321812315132 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 56, T \in [25, 56]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3320] \quad 812^1 2315132181231513 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3321] \quad 1712321812^1 32161315 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 32, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[3322] \quad 1712321812^1 31513161 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[3323] \quad 181232^1 171232181223 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3324] \quad 181232171232^1 181223 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3325] \quad 1^1 812^1 321712321812 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

- [3326] $1\ 8\ 12321712^1 321812 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 36, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [3327] $612315132^1 181232171 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [3328] $181232181232^1 17123 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 36, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3329] $181232^1 18123217123 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 36, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3330] $1\ 8\ 12^1 32181232171 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 36, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3331] $1\ 8\ 12321812^1 32171 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 36, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3332] $181232^1 17123218123 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 36, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3333] $181232171232^1 18123 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 36, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3334] $1612321812321712^1 315 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 33, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3335] $1612321812^1 321712315 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 33, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [3336] $513161232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 32, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [3337] $161315132^1 1812321712 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [3338] $31613151321812^1 32171 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 42, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 42, T \in [25, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [3339] $1612321812^1 315132171 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 31, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [3340] $2181232171232^1 181223 + \hat{1} \rightarrow 2$ $T = 19, T_{\max} = 36, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 35, T \in [25, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 35, T \in [25, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 24.06 \rightarrow 25, \quad T_{\max} = 36, \quad T \in [25, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_1}$$

$$T = [25 \ 26]$$

$$\min = 24.06$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \end{pmatrix}$$

3352] 321712321812¹321812 + $\hat{1} \rightarrow 2$ $T = 1$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

3354] 161316131612+32181 + 1→2 $T = 19$
 $T = 24$ 06 + 25 $T = 31$ $T = [25, 31]$

$$\min = 24.06$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \end{pmatrix}$$

- [3355] $\overset{1}{161316131612^1 321812} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [3356] $\overset{1}{181232^1 161316131612} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [3357] $\overset{151322^1 1}{4141413221} (12) \ 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [3358] $\overset{1}{2161316131612^1 321812} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [3359] $\overset{1}{1613161232^1 181232161} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [3360] $\overset{1}{41321812^1 32171232171} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [3361] $\overset{1}{41321812321712^1 32171} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [3362] $\overset{1}{812^1 3216131613161232} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3363] $\overset{1}{61232^1 18123216131613} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3364] $\overset{1}{613161232181232^1 1613} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 46, T \in [25, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3365] $\overset{1}{61232^1 1812321812314} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 39, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 39, T \in [25, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3366] $\overset{1}{61232181232^1 1812314} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 39, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 39, T \in [25, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3367] $\overset{1}{41316131613151322^1 18} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3368] $\overset{21}{1(12)122^1 314141414141} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25, T_{\max} = 31, T \in [25, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$

- [3292] $^{3221}221(12)122^131513161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25$, $T_{\max} = 42$, $T \in [25, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [3291] $13221(12)12^123151315132 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -100$, $\det = -49$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3369] $1414141414132^121(12)122 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25$, $T_{\max} = 31$, $T \in [25, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [3370] $414141413221(12)12^12314 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -49$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.06 \rightarrow 25$, $T_{\max} = 56$, $T \in [25, 56]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [3378] $^{1171232181232^1161315 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 32$, $\Delta = -111$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [3372] $^{51321}1^8 12^13218123215 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 35$, $\Delta = -114$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 35$, $T \in [24, 35]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3373] $^{51321}1^8 12^13218123151 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 33$, $T \in [24, 33]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3374] $513221812^13218123215 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 35$, $\Delta = -114$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 35$, $T \in [24, 35]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3375] $15132181232^118122315 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 33$, $T \in [24, 33]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [3376] $18123151321812^131513 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 46$, $T \in [24, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3379] $^{11812321712^132181223 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 36$, $T \in [24, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3380] $^{321}1^8 1232^11712321812 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 36$, $T \in [24, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3381] $^{11812321812^13217123 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 36$, $T \in [24, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3382] $^{321}1^8 1232^1181232171 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 36$, $T \in [24, 36]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3383] $1812321712^1 3218123 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 36$, $T \in [24, 36]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3384] $161232181232^1 1712315 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 33$, $T \in [24, 33]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[3385] $5131612^1 321812321712 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 32$, $\Delta = -111$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[3386] $21812321712^1 32181223 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 36$, $T \in [24, 36]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3387] $612321712^1 3218123215 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 35$, $\Delta = -114$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 35$, $T \in [24, 35]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3388] $612321712^1 3218123151 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 33$, $\Delta = -112$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 33$, $T \in [24, 33]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3389] $613151321812^1 3217123 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 46$, $T \in [24, 46]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3390] $7123151321812^1 321613 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 46$, $T \in [24, 46]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3391] $1712321812^1 32181223 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 36$, $T \in [24, 36]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3392] $32171232^1 1812321812 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 36$, $T \in [24, 36]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3393] $16131612^1 32181232161 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 31$, $T \in [24, 31]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[3394] $4132181232^1 171232171 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 36$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 36$, $T \in [24, 36]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3395] $6131612321812^1 321613 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 46$, $T \in [24, 46]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3396] $612321812^1 321812314 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 39$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.94 \rightarrow 24$, $T_{\max} = 39$, $T \in [24, 39]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3397] $1(12)^{21} 12^1 2314141414141 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 31$, $\Delta = -110$, $\det = -64$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 23.94 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[3398] \quad 1(12)122^1 3141414141413 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.94 \rightarrow 24, T_{\max} = 46, T \in [24, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[3399] \quad 14141414141322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 31, \Delta = -110, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.94 \rightarrow 24, T_{\max} = 31, T \in [24, 31]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3400] \quad 31414141414132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.94 \rightarrow 24, T_{\max} = 46, T \in [24, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3377] \quad 81223151321812^1 31513 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.94 \rightarrow 24, T_{\max} = 46, T \in [24, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3416] \quad 4141413221(12)12^1 231414 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.84 \rightarrow 24, T_{\max} = 56, T \in [24, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3411] \quad 32161316131612^1 32181 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 39, \Delta = -116, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.84 \rightarrow 24, T_{\max} = 39, T \in [24, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[3401] \quad 221(12)12^1 231513161 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.84 \rightarrow 24, T_{\max} = 42, T \in [24, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[3402] \quad 413221^{513151322^1} (12) 12 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.84 \rightarrow 24, T_{\max} = 56, T \in [24, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3403] \quad 6123151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.84 \rightarrow 24, T_{\max} = 46, T \in [24, 46]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[3404] \quad 321812^1 3151321812315 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.84 \rightarrow 24, T_{\max} = 56, T \in [24, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3405] \quad 1^1 8 12321712^1 32181 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.84 \rightarrow 24, T_{\max} = 42, T \in [24, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[3406] \quad 71232^1 1812321^{2321} 8 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.84 \rightarrow 24, T_{\max} = 42, T \in [24, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[3407] \quad 2321712321812^1 32181 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 23.84 \rightarrow 24, T_{\max} = 42, T \in [24, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[3408] & \overset{31}{161316131612^1 32181} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.84 \rightarrow 24, T_{\max} = 36, T \in [24, 36] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[3409] & \overset{1513221}{41414132^1 21(12)} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.84 \rightarrow 24, T_{\max} = 36, T \in [24, 36] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3410] & 16131613161232^1 18123 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 39, \Delta = -116, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.84 \rightarrow 24, T_{\max} = 39, T \in [24, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3415] & 314141414141322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.84 \rightarrow 24, T_{\max} = 46, T \in [24, 46] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3414] & 231414141414132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.84 \rightarrow 24, T_{\max} = 56, T \in [24, 56] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3413] & \overset{1}{1(12)12^1 23141414141413} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.84 \rightarrow 24, T_{\max} = 46, T \in [24, 46] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3412] & 413161316131612^1 3217 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.84 \rightarrow 24, T_{\max} = 36, T \in [24, 36] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[3417] & 41414132^1 21(12)12231414 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -81, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.84 \rightarrow 24, T_{\max} = 56, T \in [24, 56] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3426] & \overset{1}{2321712^1 32181232181} + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.76 \rightarrow 24, T_{\max} = 42, T \in [24, 42] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3427] & 613161232^1 1812321613 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.76 \rightarrow 24, T_{\max} = 46, T \in [24, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3418] & 51322^1 18123218123215 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 35, \Delta = -114, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.76 \rightarrow 24, T_{\max} = 35, T \in [24, 35] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3419] & 151321812321812^1 2315 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 33, \Delta = -112, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.76 \rightarrow 24, T_{\max} = 33, T \in [24, 33] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[3420] & 1812315132^1 181231513 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} =
\end{aligned}$$

15.0

$$T_{\min} = 23.76 \rightarrow 24, T_{\max} = 46, T \in [24, 46]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3421] \ 8122315132^1 181231513 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.76 \rightarrow 24, T_{\max} = 46, T \in [24, 46]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3422] \ 1^1 8 \ 12^{2321} 32171232181 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.76 \rightarrow 24, T_{\max} = 42, T \in [24, 42]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[3423] \ 71232181232^1 1^1 8 \ 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.76 \rightarrow 24, T_{\max} = 42, T \in [24, 42]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3424] \ 61315132^1 18123217123 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.76 \rightarrow 24, T_{\max} = 46, T \in [24, 46]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3425] \ 712315132^1 1812321613 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.76 \rightarrow 24, T_{\max} = 46, T \in [24, 46]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3429] \ (12)122^1 314141414141322 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.76 \rightarrow 24, T_{\max} = 56, T \in [24, 56]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[3428] \ 2314141414141322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -100, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.76 \rightarrow 24, T_{\max} = 56, T \in [24, 56]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[3436] \ 414141322^1 1^{1513221} (12) \ 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 36, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.69 \rightarrow 24, T_{\max} = 36, T \in [24, 36]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3431] \ 8123151321812^1 315132 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -121, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.69 \rightarrow 24, T_{\max} = 56, T \in [24, 56]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[3432] \ 1^1 8 \ 1232^1 171232181 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -121, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.69 \rightarrow 24, T_{\max} = 42, T \in [24, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[3433] \ 712321812^1 321^1 8 \ 1 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -121, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.69 \rightarrow 24, T_{\max} = 42, T \in [24, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[3434] \ 3161315132^1 181232171 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 42, \Delta = -117, \det = -121, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 23.69 \rightarrow 24, T_{\max} = 42, T \in [24, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

- [3430] $321812315132^1 1812315 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -121$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.69 \rightarrow 24$, $T_{\max} = 56$, $T \in [24, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [3437] $81232^1 16131613161232 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -121$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.69 \rightarrow 24$, $T_{\max} = 46$, $T \in [24, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [3438] $612321812^1 3216131613 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -121$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.69 \rightarrow 24$, $T_{\max} = 46$, $T \in [24, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [3439] $6131612^1 321812321613 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -121$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.69 \rightarrow 24$, $T_{\max} = 46$, $T \in [24, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3440] $(12)12^1 2314141414141322 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -121$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.69 \rightarrow 24$, $T_{\max} = 56$, $T \in [24, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [3441] $4141414132^1 21(12)122314 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -121$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.69 \rightarrow 24$, $T_{\max} = 56$, $T \in [24, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3442] $414141322^1 1(12)12231414 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -121$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.69 \rightarrow 24$, $T_{\max} = 56$, $T \in [24, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3435] $232171232^1 181232181 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 42$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.69 \rightarrow 24$, $T_{\max} = 42$, $T \in [24, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3443] $161316131612^1 3218123 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 39$, $\Delta = -116$, $\det = -144$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.64 \rightarrow 24$, $T_{\max} = 39$, $T \in [24, 39]$
 $\begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3444] $812315132^1 1812315132 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -169$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.59 \rightarrow 24$, $T_{\max} = 56$, $T \in [24, 56]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3445] $6131613161232^1 181232 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -169$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.59 \rightarrow 24$, $T_{\max} = 46$, $T \in [24, 46]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [3446] $61232181232^1 16131613 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 46$, $\Delta = -118$, $\det = -169$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 23.59 \rightarrow 24$, $T_{\max} = 46$, $T \in [24, 46]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [3447] $414141414132^1 21(12)1223 + \hat{1} \rightarrow 2$ $T = 19$, $T_{\max} = 56$, $\Delta = -119$, $\det = -169$, $(n_+, n_-, n_0) = (1, 19, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 23.59 \rightarrow 24, T_{\max} = 56, T \in [24, 56] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[3448] & 41414141322^1 1(12)122314 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -169, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.59 \rightarrow 24, T_{\max} = 56, T \in [24, 56] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[3449] & 4141414141322^1 1(12)1223 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 56, \Delta = -119, \det = -225, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.52 \rightarrow 24, T_{\max} = 56, T \in [24, 56] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3450] & 61316131612^1 32181232 + \hat{1} \rightarrow 2 \quad T = 19, T_{\max} = 46, \Delta = -118, \det = -289, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 23.46 \rightarrow 24, T_{\max} = 46, T \in [24, 46] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 203 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
& T = 20 \quad [3460] \quad 21(12)12231513217123 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27, T_{\max} = 40, T \in [27, 40] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3459] & 1(12)1223151321712 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27, T_{\max} = 32, T \in [27, 32] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3452] & 171231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27, T_{\max} = 32, T \in [27, 32] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3453] & 171231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27, T_{\max} = 32, T \in [27, 32] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3454] & 171231513221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27, T_{\max} = 32, T \in [27, 32] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3455] & 171231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27, T_{\max} = 32, T \in [27, 32] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[3456] & 3221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26, T_{\max} = 40, T \in [26, 40] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[3457] & 1(12)12231513217123 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27, T_{\max} = 40, T \in [27, 40] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3458] & 1(12)122315132171232 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 47, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27, T_{\max} = 47, T \in [27, 47] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [3461] $21^{22^1 21}(12)1223151321712 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3462] $22^1 1^{221}(12)1223151321712 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3463] $1^{22^1 221}(12)1223151322171 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3464] $21^{22^1 21}(12)1223151322171 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3465] $22^1 1^{221}(12)1223151322171 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3466] $712231513221^{3221}(12)122^1 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 40, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 40, T \in [26, 40]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [3467] $23216131513221(12)122^1 31 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 28, \Delta = -99, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 28, T \in [27, 28]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3468] $32171231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 40, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 40, T \in [27, 40]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3469] $232171231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 47, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 47, T \in [27, 47]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3470] $2171231513221(12)12222^1 2 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3471] $1712231513221(12)12222^1 2 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3472] $41316131513221^{22^1 21}(12)1 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 37, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 37, T \in [27, 37]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3473] $41316131513221^{22^1 1}(12)12 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 37, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 37, T \in [27, 37]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

- [3474] $18^{21}123218^112321812^12 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3475] $5131612321812321812^12 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3476] $1613151321812321812^12 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3477] $41316131513221(12)1222^12 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 37, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 37, T \in [27, 37]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3478] $6123151321812321812^12 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3479] $22^118123218123218122 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3480] $413221(12)^{21}12231513161 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 40, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 40, T \in [26, 40]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [3481] $413221(12)^{21}12231513216 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 43, \Delta = -117, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 43, T \in [26, 43]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [3451] $1513221^{5131513221}(12)12^1 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 36, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26, T_{\max} = 36, T \in [26, 36]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [3483] $171231513221(12)^{21}12^122 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3484] $16^{31}131513221(12)122^1313 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -104, \det = 9, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3482] $171231513221(12)^{21}12^122 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 32, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27, T_{\max} = 32, T \in [27, 32]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3486] $221(12)^{32121}122315132171 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 40, \Delta = -116, \det = 9, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27, T_{\max} = 40, T \in [27, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [3487] $1(12)^{2221}1223151321712^13 + \hat{1} \rightarrow 2$ $T = 20, T_{\max} = 40, \Delta = -116, \det = 9, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27, T_{\max} = 40, T \in [27, 40]$

- $$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [3488] $1(12)12231513217123 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 9, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27, T_{\max} = 40, T \in [27, 40]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$
- $$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$$
- [3489] $1(12)1223151321712 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27, T_{\max} = 32, T \in [27, 32]$
- $$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$$
- $$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [3490] $21(12)1223151321712^1 3 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 9, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27, T_{\max} = 40, T \in [27, 40]$
- $$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$
- $$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$$
- [3491] $21(12)1223151321712 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27, T_{\max} = 32, T \in [27, 32]$
- $$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$$
- $$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$$
- [3492] $1(12)1223151322171 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27, T_{\max} = 32, T \in [27, 32]$
- $$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$$
- $$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$$
- [3493] $21(12)1223151322171 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27, T_{\max} = 32, T \in [27, 32]$
- $$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$$
- $$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [3494] $712231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 9, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27, T_{\max} = 40, T \in [27, 40]$
- $$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [3495] $14132^1 21(12)122315131513 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 27, \Delta = -98, \det = 9, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$
- $$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$
- [3496] $23216131513221(12)12^1 231 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 28, \Delta = -99, \det = 9, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27, T_{\max} = 28, T \in [27, 28]$
- $$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [3497] $32^1 16131513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27, T_{\max} = 32, T \in [27, 32]$
- $$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$
- [3498] $3216131513221(12)122^1 313 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27, T_{\max} = 32, T \in [27, 32]$
- $$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [3499] $3161231513221(12)122^1 313 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27, T_{\max} = 32, T \in [27, 32]$
- $$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [3500] $32^1 171231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 9, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27, T_{\max} = 40, T \in [27, 40]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[3501] 32171231513221(12)122¹22 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 40, T \in [27, 40]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3502] 232171231513221(12)12¹22 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 47, T \in [27, 47]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3503] 2171231513221(12)1222¹22 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 32, T \in [27, 32]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3504] 1712231513221(12)1222¹22 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 32, T \in [27, 32]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3505] 4131513221(12)12231512¹3 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 28$, $\Delta = -99$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 28, T \in [27, 28]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[3506] 41316131513221(12)1 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 37$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 37, T \in [27, 37]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3507] 613151321812321³²1¹8 1 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 43, T \in [27, 43]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[3508] 1812321812321812¹3 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 36, T \in [27, 36]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[3509] 18123218123218122¹3 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 36, T \in [27, 36]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[3510] 1³²1¹8 12321812321812 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 36, T \in [27, 36]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[3511] 41316131513221(12)122¹22 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 37$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 37, T \in [27, 37]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3512] 6131513218123218122¹3 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 43, T \in [27, 43]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[3513] 6123218123151321812¹3 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 47, T \in [27, 47]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

- [3514] $7123151321812321712^1 3 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27$, $T_{\max} = 43$, $T \in [27, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3515] $218123218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27$, $T_{\max} = 36$, $T \in [27, 36]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3516] $6131612321812321712^1 3 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27$, $T_{\max} = 43$, $T \in [27, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3485] $171231513221(12)1222^1 22 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27$, $T_{\max} = 32$, $T \in [27, 32]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3534] $171231513221(12)122^1 222 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3535] $1712315132^1 21(12)122222 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3523] $1712315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3524] $1712315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3525] $1712^1 31513221(12)122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3526] $16131513221(12)122^1 3132 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -97$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3527] $161315132^1 21(12)1223132 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -97$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3528] $1(12)122315131612 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -96$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3529] $1(12)122^1 315131612 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -96$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3530] $16131513221(12)12^1 2313 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -104$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[3531] \quad 1 \quad (12) \quad 122315132161 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 26, \Delta = -97, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3532] \quad 1 \quad (12) \quad 122315132^1 161 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 26, \Delta = -97, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3533] \quad 1 \quad (12) \quad 122^1 315132161 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 26, \Delta = -97, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3518] \quad 161315132^1 21(12)122313 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 26, \Delta = -96, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3517] \quad 16131513221(12)12^1 2313 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 26, \Delta = -96, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3536] \quad 1712^1 31513221(12)122222 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3537] \quad 22^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.25 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3538] \quad 1(12)122^1 3151321712 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3539] \quad 1(12)122315132^1 1712 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3540] \quad 1(12)1223151321712 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3541] \quad 21(12)122^1 3151321712 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3542] \quad 21(12)122315132^1 1712 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3543] \quad 221(12)122^1 3151321712 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -107, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3557] & 314132^1 21(12)12231513215 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 28, \Delta = -99, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 28, T \in [26, 28] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3558] & 31413221(12)122315132^1 15 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 28, \Delta = -99, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 28, T \in [26, 28] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[3559] & 31413221(12)122^1 31513215 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 28, \Delta = -99, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 28, T \in [26, 28] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3560] & 314132^1 21(12)12231513151 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 27, \Delta = -98, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 27, T \in [26, 27] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3561] & 31413221(12)122^1 31513151 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 27, \Delta = -98, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 27, T \in [26, 27] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3562] & 216131513221(12)122^1 3132 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 26, \Delta = -97, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3563] & 2161315132^1 21(12)1223132 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 26, \Delta = -97, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3564] & 3216131513221(12)12^1 2313 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3565] & 1612^1 31513221(12)1223132 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 27, \Delta = -98, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 27, T \in [26, 27] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3566] & 161231513221(12)122^1 3132 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 27, \Delta = -98, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 27, T \in [26, 27] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3567] & 1612315132^1 21(12)1223132 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 27, \Delta = -98, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 27, T \in [26, 27] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3568] & 3161231513221(12)12^1 2313 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[3569] 32171231513221(12)12¹222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3570] 21712315132¹21(12)122222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3571] 2171231513221(12)122¹222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3572] 21712¹31513221(12)122222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3573] 17122¹31513221(12)122222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3574] 1712231513221(12)122¹222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3575] 17122315132¹21(12)122222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3576] 4131513221(12)122¹315123 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 28$, $\Delta = -99$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 28, T \in [26, 28]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3577] 41315132¹21(12)122315123 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 28$, $\Delta = -99$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 28, T \in [26, 28]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3578] 4131513221(12)122¹31513 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 27, T \in [26, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3579] 41315132¹21(12)12231513 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 27, T \in [26, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3580] 6122315132¹21(12)1223132 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.75 \rightarrow 26, T_{\max} = 27, T \in [26, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3581] 6122¹31513221(12)1223132 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 27, T \in [26, 27] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3582] & 612231513221(12)122^1 3132 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 27, \Delta = -98, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 27, T \in [26, 27] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3583] & \begin{matrix} 51321 \\ 1 \end{matrix} 8^{12} 1232181232^1 161 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 33, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3584] & \begin{matrix} 21 \\ 1 \end{matrix} 8^{12} 1^1 32181232161315 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3585] & \begin{matrix} 21 \\ 1 \end{matrix} 8^{12} 1^1 32181231513161 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3586] & \begin{matrix} 1 \\ 1 \end{matrix} 8^{12} 15132181232^1 161 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3587] & \begin{matrix} 1 \\ 1 \end{matrix} 712^1 1^1 321812321712315 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3588] & \begin{matrix} 1 \\ 1 \end{matrix} 712^1 1^1 321812315132171 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3589] & 612315132181232^1 \begin{matrix} 21 \\ 1 \end{matrix} 8^1 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3590] & 7122315132181232^1 \begin{matrix} 1 \\ 1 \end{matrix} 7^1 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3591] & \begin{matrix} 21 \\ 1 \end{matrix} 8^{12} 1^1 321812321 \begin{matrix} 1 \\ 1 \end{matrix} 8^1 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3592] & \begin{matrix} 21 \\ 1 \end{matrix} 8^{12} 1232181232^1 18122 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[3593] & \begin{matrix} 21 \\ 1 \end{matrix} 8^{12} 1^1 3218123218122 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 32, T \in [26, 32] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3594] & 1612^1 1^1 3218123218122315 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 33, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.75 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [3595] $513161232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3596] $161315132181232^1 18122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3597] $1612^1 3218123151321812 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3598] $1612^1 3218123151322181 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3599] $171232171232181232^1 15 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 34$, $\Delta = -112$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 34$, $T \in [26, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [3600] $21712^1 321812321712315 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3601] $21712^1 321812315132171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3602] $41316131513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 37$, $\Delta = -115$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 37$, $T \in [26, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3603] $612315132181232^1 18122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3604] $7122315132181232^1 1712 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3605] $221812^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -105$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3606] $61232181232181232^1 15 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 35$, $\Delta = -113$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 35$, $T \in [26, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [3607] $1712^1 321812321613161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3608] $1613161232181232^1 1712 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3609] $61232171232181232^1 161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 16$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.75 \rightarrow 26$, $T_{\max} = 33$, $T \in [26, 33]$

- $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ¹
 [3617] $16131513221(12)12^1 23132 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -97$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ¹
 [3618] $1613151322^1 1(12)1223132 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -97$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ¹³²²¹
 [3619] $1(12)122^1 3151316123 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -97$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ³¹³²²¹
 [3620] $1(12)12^1 2315131612 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -96$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- ³¹
 [3621] $161315132^1 21(12)122313 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -104$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ³¹³²²¹
 [3622] $1(12)12^1 2315132161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -97$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- ¹³²²¹
 [3623] $1(12)122^1 3151321613 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -97$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ¹
 [3624] $171231513221(12)12^1 2222 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ¹
 [3625] $17123151322^1 1(12)122222 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ³²²¹
 [3626] $221(12)122315132^1 171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 40$, $T \in [26, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- ^{322^1 1}
 [3627] $221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 40$, $T \in [26, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- ²²²¹
 [3628] $1(12)122^1 31513217123 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 40$, $T \in [26, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- ²²¹
 [3629] $1(12)1223151321712^1 32 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

- [3643] $7122^1 31513221^{3221}(12)122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 40$, $T \in [26, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3644] $712231513221^{322^1 1}(12)122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 40$, $T \in [26, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3645] $1413221(12)122^1 315131513 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 27$, $T \in [26, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3646] $3141322^1 1(12)12231513215 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 28$, $\Delta = -99$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 28$, $T \in [26, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3647] $31413221(12)12^1 231513215 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 28$, $\Delta = -99$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 28$, $T \in [26, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3702] $1712321712321812^1 3215 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 34$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 34$, $T \in [26, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3648] $3141322^1 1(12)12231513151 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 27$, $T \in [26, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3703] $1712^1 3217123218123215 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 34$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 34$, $T \in [26, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3649] $31413221(12)12^1 231513151 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 27$, $T \in [26, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3650] $232^1 16131513221(12)12231 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 28$, $\Delta = -99$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 28$, $T \in [26, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3651] $21613151322^1 1(12)1223132 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -97$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3652] $216131513221(12)12^1 23132 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -97$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3653] $32161315132^1 21(12)122313 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3654] 16123151322¹1(12)1223132 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 27, T \in [26, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3655] 161231513221(12)12¹23132 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 27, T \in [26, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3656] 31612315132¹21(12)122313 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3657] 321712315132¹21(12)12222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3658] 232¹171231513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 47, T \in [26, 47]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[3659] 217123151322¹1(12)122222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3660] 2171231513221(12)12¹2222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3661] 1712231513221(12)12¹2222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3662] 171223151322¹1(12)122222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3663] 1712¹231513221(12)122222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[3664] 413151322¹1(12)122315123 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 28$, $\Delta = -99$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 28, T \in [26, 28]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3665] 4131513221(12)12¹2315123 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 28$, $\Delta = -99$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 28, T \in [26, 28]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

- [3666] $413151322^1 1(12)12231513 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 27$, $T \in [26, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3667] $4131513221(12)12^1 231513 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 27$, $T \in [26, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3668] $61223151322^1 1(12)1223132 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 27$, $T \in [26, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3669] $612^1 231513221(12)1223132 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 27$, $T \in [26, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [3670] $612231513221(12)12^1 23132 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 27$, $T \in [26, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3671] $1^1 8^1 12321812^1 32161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 33$, $T \in [26, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3672] $18^1 1232^1 181232161315 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3673] $18^1 1232^1 181231513161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3674] $1812^1 321812315131613 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 40$, $T \in [26, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3675] $18123151321812^1 32161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3676] $1812^1 315132181232161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [3677] $171232^1 1812321712315 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3678] $1712321812315132^1 171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3679] $171232^1 1812315132171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3680] \quad 413161315132^{2221}21(12)1 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 37, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 37, T \in [26, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3681] \quad 413161315132^{221}21(12)12 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 37, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 37, T \in [26, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3682] \quad 612^13151321812321^2181 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[3683] \quad 6123151321812^1321^2181 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3684] \quad 7122^13151321812321^171 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[3685] \quad 71223151321812^1321^171 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3686] \quad 1812^1321^1812321^18123 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3687] \quad 181232^{21}1812321^181 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3688] \quad 1812321812321812^123 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[3689] \quad 1812^1321^181232181223 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3690] \quad 1^232^11^1232181232181 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[3691] \quad 1^321^1812321^1812 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3692] \quad 1812321812^13218122 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.44 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

- [3693] $181232^{\overline{21}}18123218122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3694] $161232^{\overline{1}}18123218122315 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 33$, $T \in [26, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3695] $5131612321812^{\overline{1}}3218122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3704] $1712^{\overline{1}}3217123218123151 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3705] $1712321712321812^{\overline{1}}3151 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3706] $2171232^{\overline{1}}1812321712315 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3707] $21712321812315132^{\overline{1}}171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3708] $2171232^{\overline{1}}1812315132171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3709] $413161315132^{\overline{1}}21(12)12222 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 37$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 37$, $T \in [26, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3710] $613151321812321812^{\overline{1}}23 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 43$, $T \in [26, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3711] $612^{\overline{1}}32181231513218123 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 47$, $T \in [26, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3712] $6123151321812^{\overline{1}}3218122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3713] $612^{\overline{1}}31513218123218122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [3714] $71223151321812^{\overline{1}}321712 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3715] $7122^{\overline{1}}3151321812321712 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[3716] $21812321812321812^1 23 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3717] $21812^1 181232181223 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3718] $22181232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[3719] $4132^1 21(12)12231513161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 40$, $T \in [26, 40]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3720] $612^1 3218123218123215 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 35$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 35$, $T \in [26, 35]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3721] $612321812321812^1 3215 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 35$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 35$, $T \in [26, 35]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3722] $612^1 3218123218123151 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 33$, $T \in [26, 33]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3723] $612321812321812^1 3151 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 33$, $T \in [26, 33]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3724] $413221(12)122315132^1 16 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 43$, $T \in [26, 43]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[3725] $4132^1 21(12)12231513216 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 43$, $T \in [26, 43]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[3726] $171232^1 1812321613161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3727] $4132^1 181232171232181 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3728] $4132181232171232^1 181 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3729] $16131612321812^1 321712 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3730] $31613161232181232^1 171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 40$, $T \in [26, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3731] $4132^1 1812321712321812 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3732] $4132181232171232^1 1812 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3733] $612^1 32171232181232161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 33$, $T \in [26, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [3698] $161232^1 18123151321812 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3735] $4132^1 181232181232171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3736] $4132181232181232^1 171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3737] $414141413221(12)122^1 3151 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3696] $1613151321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3697] $3161315132181232^1 1812 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 40$, $T \in [26, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3734] $612321712321812^1 32161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.44 \rightarrow 26$, $T_{\max} = 33$, $T \in [26, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3738] $16^{31} 13151322^1 1(12)12231 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -96$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3742] $1(12)12^{221} 231513217123 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 40$, $T \in [26, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3743] $1(12)122^{221} 315132171232 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 47$, $T \in [26, 47]$

- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3757] 15131613161315132¹181 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [3758] 31612321812315132¹181 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 43$, $T \in [26, 43]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [3759] 16123218123151322¹181 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [3760] 4131613151322¹1(12)12222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 37$, $\Delta = -115$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 37$, $T \in [26, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [3761] 8122¹3151316131613151 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3762] 8122¹3151321812321613 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 43$, $T \in [26, 43]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3763] 712¹32171232181231513 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 47$, $T \in [26, 47]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3764] 712¹31513218123217123 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 43$, $T \in [26, 43]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3765] 712¹23151321812321712 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [3766] 712¹3151316131613161¹ + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [3767] 712¹32161316131613151 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [3768] 712¹31513161316131612 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [3769] 41316131613161232¹181 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [3741] 1¹³²²¹(12)12¹23151321613 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 26$, $\Delta = -97$, $\det = 36$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26$, $T_{\max} = 26$, $T \in [26, 26]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3740] $1613151322^1 1(12)122313 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -104, \det = 36, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3739] $1(12)12^1 23151316123 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 26, \Delta = -97, \det = 36, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.22 \rightarrow 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3770] $1(12)12^1 23151 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3771] $1(12)12^1 23151 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[3772] $21(12)12^1 23151 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[3773] $21(12)12^1 23151 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3774] $1(12)122315132^1 1613 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 26, \Delta = -97, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3775] $221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[3776] $1(12)122315132^1 17123 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3777] $1(12)12^1 2315132171232 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 47, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 47, T \in [26, 47]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3778] $21(12)122315132^1 17123 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3779] $21(12)12^1 2315132171232 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 47, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 47, T \in [26, 47]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3780] $7122315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3781] 2321613151322¹1(12)12231 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 28$, $\Delta = -99$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 28, T \in [26, 28]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[3782] 31612¹31513221(12)122313 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3783] 321712¹31513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3784] 23217123151322¹1(12)1222 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 47, T \in [26, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[3785] 1⁵¹³²8¹1232181232161 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[3786] 1⁵¹³²¹8¹1232¹181232161 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[3787] 1²¹8¹12321812¹32161315 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[3788] 1²¹8¹12321812¹31513161 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3789] 1¹181232¹1812315131613 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[3790] 1¹1812315132¹181232161 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[3791] 1¹1712321812¹321712315 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[3792] 1¹1712321812321712¹315 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3793] 1¹1712321812¹315132171 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

- [3794] $414132^{121} \quad (12) \quad 1 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 47, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 47, T \in [26, 47]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3795] $41413221^{513151321} \quad (12) \quad 1 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 47, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 47, T \in [26, 47]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [3796] $61315132181232^{321} 1 \ 8 \ 1 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 43, T \in [26, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3798] $7122315132^1 181232171 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [3799] $1^1 81232^1 1^1 81232^1 18123 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3800] $1^1 81232^1 1^1 81223 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3801] $1^1 81232^1 1^1 81232181223 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3802] $1^1 8 \ 12^{321} 321812321812 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3803] $1^1 8 \ 12^{321} 321812 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3804] $16123218123218122^1 315 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 33, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3805] $1612321812^1 3218122315 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 33, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [3806] $513161232^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [3807] $161315132^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [3808] $31613151321812^1 321812 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [3809] $1612321812^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[3810] \quad 31612^1 321812315132181 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 43, T \in [26, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[3811] \quad 1612321812^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[3812] \quad 171232^1 17123218123215 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 34, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 34, T \in [26, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3813] \quad 171232171232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 34, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 34, T \in [26, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3814] \quad 171232^1 17123218123151 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3815] \quad 171232171232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3816] \quad 21712321812^1 321712315 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[3817] \quad 21712321812321712^1 315 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3818] \quad 21712321812^1 315132171 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[3819] \quad 812^1 31513161316131513 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 47, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 47, T \in [26, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3820] \quad 812^1 23151316131613151 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3821] \quad 61315132181232^1 181223 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 43, T \in [26, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3822] \quad 61232^1 181231513218123 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 47, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 47, T \in [26, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3823] \quad 612315132^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[3824] \quad 812^1 23151321812321613 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 43, T \in [26, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3825] \quad 8122315132181232^1 1613 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 43, T \in [26, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3826] \quad 712315132181232^1 17123 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 43, T \in [26, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3827] \quad 7122315132^1 1812321712 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 32, T \in [26, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[3828] \quad 2181232181232^1 181223 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3829] \quad 2181232^1 181232181223 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3830] \quad 413221(12)122^1 31513161 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3831] \quad 41322^1 1(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 40, T \in [26, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[3832] \quad 61232181232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 35, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 35, T \in [26, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3833] \quad 61232^1 18123218123215 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 35, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 35, T \in [26, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3834] \quad 61232181232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 33, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3835] \quad 61232^1 18123218123151 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 33, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3836] \quad 413221(12)122^1 31513216 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 43, T \in [26, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[3837] \quad 41322^1 1(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 25.06 \rightarrow 26, T_{\max} = 43, T \in [26, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3838] \quad 1712321812^1 321613161 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

- $T_{\min} = 25.06 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [3839] $41321812321712^1 32181 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [3840] $41321812^1 32171232181 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [3841] $1613161232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [3842] $316131612321812^1 32171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26$, $T_{\max} = 40$, $T \in [26, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [3843] $41321812^1 321712321812 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [3844] $41321812321712^1 321812 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [3845] $613161232181232^1 17123 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26$, $T_{\max} = 43$, $T \in [26, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3846] $61232^1 171232181232161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26$, $T_{\max} = 33$, $T \in [26, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [3847] $61232171232^1 181232161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26$, $T_{\max} = 33$, $T \in [26, 33]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [3848] $41321812^1 32181232171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3849] $41321812321812^1 32171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3850] $414141413221(12)12^1 23151 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26$, $T_{\max} = 36$, $T \in [26, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [3797] $612315132^1 181232181 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.06 \rightarrow 26$, $T_{\max} = 32$, $T \in [26, 32]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [3865] $2171232181232^1 1712315 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[3866] 613151321812¹32181223 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 43$, $T \in [25, 43]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3856] 613151321812¹321³²¹8 1 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 43$, $T \in [25, 43]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3857] 1812321812¹321¹8123 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3858] 1812321812¹32181223 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3859] 1³²¹8 1232¹1812321812 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3860] 161232181232¹18122315 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[3861] 5131612¹3218123218122 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[3862] 3161232¹1812315132181 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 43$, $T \in [25, 43]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3863] 1712321712¹3218123215 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 34$, $\Delta = -112$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 34$, $T \in [25, 34]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3854] 1²¹8 1232181232¹161315 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[3853] 1⁵¹³²¹8¹²¹32181232161 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[3852] 141414141414132¹21(12)12 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[3867] 81223151321812¹321613 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 43$, $T \in [25, 43]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3868] 712321712321812¹31513 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 47$, $T \in [25, 47]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

- [3869] $7123151321812^1 3217123 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 43$, $T \in [25, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3870] $21812321812^1 32181223 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3871] $612321812^1 3218123215 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 35$, $\Delta = -113$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 35$, $T \in [25, 35]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3872] $612321812^1 3218123151 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3873] $171232181232^1 1613161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3874] $4132181232^1 171232181 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3875] $16131612^1 321812321712 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [3876] $4132181232^1 1712321812 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3877] $6131612321812^1 3217123 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 43$, $T \in [25, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3878] $612321712^1 32181232161 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 33$, $T \in [25, 33]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [3879] $4132181232^1 181232171 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 36$, $T \in [25, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3855] $171232181232^1 1712315 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [3864] $1712321712^1 3218123151 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3851] $1(12)122^1 31414141414141 + \hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 64$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 24.94 \rightarrow 25$, $T_{\max} = 32$, $T \in [25, 32]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3881] 1414141414141322¹1(12)12 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -109$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3884] 21^{513151322¹1}(12) 1223151 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3885] 221³²²¹(12)12¹2315132171 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 40, T \in [25, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[3886] 71223151322¹1³²²¹(12)122 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 40, T \in [25, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[3887] 4141322¹1⁵¹³¹⁵¹³²²¹(12) 1 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 47, T \in [25, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[3888] 41413221^{513151322¹1}(12) 1 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 47, T \in [25, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3889] 1²³²¹8¹12321812¹32181 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 40, T \in [25, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[3890] 612321812315132¹18123 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 47, T \in [25, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[3891] 71232¹171232181231513 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 47, T \in [25, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[3892] 413221²¹(12)12¹231513161 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 40, T \in [25, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[3893] 413221²¹(12)12¹231513216 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 43, T \in [25, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3894] 71232¹161316131613151 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 32$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 32, T \in [25, 32]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[3895] 413161316131612¹32181 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[3882] 3141414141414132¹21(12)1 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 81$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.84 \rightarrow 25, T_{\max} = 43, T \in [25, 43]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3883] & 1 \quad \begin{matrix} 1 \\ 513151322^1 1 \end{matrix} \quad 1223151 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.84 \rightarrow 25, T_{\max} = 36, T \in [25, 36] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[3880] & 1 \quad \begin{matrix} 1 \\ 313141414141414141 \end{matrix} + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 32, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.84 \rightarrow 25, T_{\max} = 32, T \in [25, 32] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 152 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[3896] & 3141414141414141322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 100, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.76 \rightarrow 25, T_{\max} = 43, T \in [25, 43] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[3897] & (12)122^1 314141414141414132 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 47, \Delta = -118, \det = 100, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.76 \rightarrow 25, T_{\max} = 47, T \in [25, 47] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3898] & \begin{matrix} 221 \\ 1(12)122315132^1 171232 \end{matrix} + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 47, \Delta = -118, \det = 100, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.76 \rightarrow 25, T_{\max} = 47, T \in [25, 47] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3899] & \begin{matrix} 21 \\ 21(12)122315132^1 171232 \end{matrix} + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 47, \Delta = -118, \det = 100, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.76 \rightarrow 25, T_{\max} = 47, T \in [25, 47] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3900] & 2321712^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 47, \Delta = -118, \det = 100, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.76 \rightarrow 25, T_{\max} = 47, T \in [25, 47] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3901] & 61315132^1 1812321 \begin{matrix} 321 \\ 8 \ 1 \end{matrix} + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 100, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.76 \rightarrow 25, T_{\max} = 43, T \in [25, 43] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3902] & 1 \quad \begin{matrix} 2321 \\ 8 \ 12 \end{matrix} \begin{matrix} 1 \\ 32181232181 \end{matrix} + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.76 \rightarrow 25, T_{\max} = 40, T \in [25, 40] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3904] & 31612321812^1 315132181 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 100, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.76 \rightarrow 25, T_{\max} = 43, T \in [25, 43] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3905] & 61315132^1 181232181223 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 100, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.76 \rightarrow 25, T_{\max} = 43, T \in [25, 43] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3906] & 8122315132^1 1812321613 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 43, \Delta = -117, \det = 100, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 24.76 \rightarrow 25, T_{\max} = 43, T \in [25, 43] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[3907] 71232171232¹181231513 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 100$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.76 \rightarrow 25, T_{\max} = 47, T \in [25, 47]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[3908] 712315132¹18123217123 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 100$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.76 \rightarrow 25, T_{\max} = 43, T \in [25, 43]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[3909] 613161232¹18123217123 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 100$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.76 \rightarrow 25, T_{\max} = 43, T \in [25, 43]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[3903] 1612321812321812¹2315 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 33$, $\Delta = -111$, $\det = 100$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.76 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3910] (12)12¹23141414141414132 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 121$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.69 \rightarrow 25, T_{\max} = 47, T \in [25, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[3911] 1812321812¹315131613 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.69 \rightarrow 25, T_{\max} = 40, T \in [25, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[3912] 1²³²¹8 1232¹181232181 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.69 \rightarrow 25, T_{\max} = 40, T \in [25, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3913] 3161315132¹1812321812 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.69 \rightarrow 25, T_{\max} = 40, T \in [25, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[3914] 612321812¹31513218123 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 121$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.69 \rightarrow 25, T_{\max} = 47, T \in [25, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[3915] 712321712¹32181231513 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 47$, $\Delta = -118$, $\det = 121$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.69 \rightarrow 25, T_{\max} = 47, T \in [25, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3916] 31613161232¹181232171 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 40$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.69 \rightarrow 25, T_{\max} = 40, T \in [25, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[3917] 6131612¹3218123217123 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 43$, $\Delta = -117$, $\det = 121$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.69 \rightarrow 25, T_{\max} = 43, T \in [25, 43]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3918] 4141414132¹21(12)1223151 + $\hat{1} \rightarrow 2$ $T = 20$, $T_{\max} = 36$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 24.69 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3919] \quad 316131612^1 32181232171 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 40, \Delta = -116, \det = 169, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.59 \rightarrow 25, T_{\max} = 40, T \in [25, 40]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[3920] \quad 41414141322^1 1(12)1223151 + \hat{1} \rightarrow 2 \quad T = 20, T_{\max} = 36, \Delta = -114, \det = 169, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 24.59 \rightarrow 25, T_{\max} = 36, T \in [25, 36]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$T = 21 \quad [3956] \quad 22^1 1(12)12231513221812 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[3925] \quad 1^3 8^{321} 1231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[3926] \quad 1^3 8^{21} 1231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3927] \quad 1^3 8^{21} 1231513221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3928] \quad 1^3 18^1 12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3929] \quad 1^3 18^1 12231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3930] \quad 1^3 18^1 12231513221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3931] \quad 1^3 (12)12231513161315132^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 58, \Delta = -119, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 58, T \in [28, 58]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[3932] \quad 1513221^{21} (12)12231513161 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 34, \Delta = -111, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3933] \quad 1^{221} (12)1223151316131513 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

- $T_{\min} = 28, T_{\max} = 48, T \in [28, 48]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3934] $1(12)122315131613151 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3935] $3221(12)12231513161315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 58, \Delta = -119, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 58, T \in [27, 58]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [3936] $21(12)122315131613151 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3937] $1(12)122315132181223 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3938] $1 \ 8 \ 1231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3939] $1 \ 8 \ 1231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 44, T \in [28, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3940] $1 \ 8 \ 1231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3924] $1 \ 8 \ 1231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3923] $1 \ 8 \ 1231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 44, T \in [28, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [3922] $1 \ 8 \ 1231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3944] $3221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27, T_{\max} = 38, T \in [27, 38]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [3945] $21(12)1223151321812^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 28, T_{\max} = 33, T \in [28, 33] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[3946] & 21^{22^1 21}(12)12231513218122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28, T_{\max} = 33, T \in [28, 33] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[3947] & 221^{221}(12)1223151321812^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28, T_{\max} = 33, T \in [28, 33] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[3948] & 22^1 1^{221}(12)12231513218122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28, T_{\max} = 33, T \in [28, 33] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[3949] & 1^{18} 12231513221(12)12222^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28, T_{\max} = 33, T \in [28, 33] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[3950] & 3221^{2^1 21}(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[3951] & 1^{22^1 21}(12)122315132218123 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28, T_{\max} = 38, T \in [28, 38] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3952] & 1^{22^1 1}(12)1223151322181232 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28, T_{\max} = 44, T \in [28, 44] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3953] & 1^{22^1 221}(12)12231513221812 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28, T_{\max} = 33, T \in [28, 33] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[3954] & 21^{22^1 1}(12)122315132218123 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28, T_{\max} = 38, T \in [28, 38] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3955] & 21^{22^1 21}(12)12231513221812 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28, T_{\max} = 33, T \in [28, 33] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[3921] & 1^{18} 1231513221^{2^1 21}(12)1223 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[3957] & 61231513221^{1513221}(12)12^1 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 35, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27, T_{\max} = 35, T \in [27, 35] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[3958] $22^1 3151316131513221(12)12 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 58, \Delta = -119, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 58, T \in [28, 58]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3959] $23151316131513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 58, \Delta = -119, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 58, T \in [28, 58]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3960] $3151316131513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 48, \Delta = -118, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 48, T \in [28, 48]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3961] $151316131513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3962] $232171231513221(12)122^1 31 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 28, \Delta = -98, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3963] $322181231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3964] $22181231513221(12)12222^1 2 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3965] $22^1 181231513221(12)122222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[3966] $321812231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3967] $2321812231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 44, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 44, T \in [28, 44]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3968] $21812231513221(12)12222^1 2 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3969] $51321712321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3970] $17123151321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} =$

15.0

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[3971] \quad 71223151321812321812^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[3972] \quad 413221(12)^{21} 122315132171 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3973] \quad 413221(12)^{21} 122315132217 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3974] \quad 16131612321812321812^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[3941] \quad 1(12)^{22221} 1223151321812^1 2 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[3943] \quad 21(12)^{2211} 122315132181223 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3942] \quad 1(12)^{221221} 12231513218122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[3978] \quad 181231513221(12)^1 122^1 22 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[3979] \quad 181231513221(12)^{3211} 12 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3980] \quad 181231513221(12)^{21} 12^1 22 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[3981] \quad 1812231513221(12)^1 122^1 22 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[3982] \quad 1812231513221(12)^{21} 12^1 22 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} =$$

15.0

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[3983] \quad 1(12)122315131613151 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -110, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[3984] \quad 21(12)12231513161315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 58, \Delta = -119, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 58, T \in [28, 58]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3985] \quad 513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 58, \Delta = -119, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.33 \rightarrow 27, T_{\max} = 58, T \in [27, 58]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3986] \quad 1(12)122315132181223 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[3987] \quad 1(12)12231513218122^1 3 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3988] \quad 1 \ 8 \ 1231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3989] \quad 1 \ 8 \ 1231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3990] \quad 1 \ 8 \ 1231513221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 44, T \in [28, 44]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3991] \quad 1 \ 8 \ 1231513221(12)1222^1 22 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3992] \quad 1(12)12231513218122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[3993] \quad 21(12)12231513218122^1 3 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

- [3994] $221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [3995] $21(12)12231513218122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 33$, $T \in [28, 33]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- [3996] $1812231513221(12)1222^1 22 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 33$, $T \in [28, 33]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3997] $221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [3998] $1(12)12231513221812^1 3 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3999] $1(12)122315132218123 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [4000] $1(12)12231513221812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 33$, $T \in [28, 33]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- [4001] $21(12)12231513221812^1 3 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [4002] $21(12)12231513221812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 33$, $T \in [28, 33]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- [4003] $6131513221(12)12^1 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.33 \rightarrow 27$, $T_{\max} = 58$, $T \in [27, 58]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [4004] $3151316131513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 48$, $\Delta = -118$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 48$, $T \in [28, 48]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [4005] $151316131513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -110$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 33$, $T \in [28, 33]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [4006] $232171231513221(12)12^1 231 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 28$, $\Delta = -98$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 28$, $T \in [28, 28]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[4007] $32^1 171231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -104$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[4008] $32171231513221(12)122^1 313 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -104$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[4009] $322181231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[4010] $32^1 2181231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[4011] $22181231513221(12)1222^1 22 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[4012] $321812231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[4013] $32^1 1812231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[4014] $2321812231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 44, T \in [28, 44]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[4015] $21812231513221(12)1222^1 22 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[4016] $7123151321812321 \overset{32^1 1}{8} 1 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 41, T \in [28, 41]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[4017] $32^1 1812321712321812315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 48$, $\Delta = -118$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 48, T \in [28, 48]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[4018] $17123218123151321812^1 3 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 41, T \in [28, 41]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[4019] $32^1 1712321812315132181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 41, T \in [28, 41]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[4020] $41316131513221(12)122^1 313 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -103$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 33, T \in [28, 33]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[4021] $71231513218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 41, T \in [28, 41]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[4022] $81223151321812321712^1 3 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 41, T \in [28, 41]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[4023] $6131612321812321^1 8^1 1 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 41, T \in [28, 41]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[4024] $81231513161316131612^1 3 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 41, T \in [28, 41]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[4025] $61316123218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 41, T \in [28, 41]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[3976] $1^1 8^1 1231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[3975] $1^1 181231513221(12)122^1 3 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[3977] $1^1 8^1 1231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.22 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[4106] $6122315132^1 21(12)12231413 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4026] $1^1 181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.25 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4031] $1^1 8^1 12^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4032] $1812315132^1 21^{221}(12)122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4033] $1812^1 131513221^{221}(12)122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4034] $18122315132^1 21^{12}(12)1222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4035] $18122^1 131513221^{12}(12)1222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4036] $1812231513221^{12}(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4037] $18122315132^1 21^{21}(12)1222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4038] $18122^1 131513221^{21}(12)1222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4039] $18122315132^1 21^{221}(12)122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4040] $18122^1 131513221^{221}(12)122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4041] $1712^1 131513221(12)1223132 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 27, \Delta = -96, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4042] $1712315132^1 21(12)1223132 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 27, \Delta = -96, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4043] $171231513221(12)122^1 3132 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4044] $181231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4045] $181231513221(12)122^1 222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4046] $1812^1 31513221(12)122222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4047] $1812315132^1 21(12)122222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4048] $1(12)122315132^1 18122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4049] $1(12)122^1 31513218122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4050] $1(12)12231513218122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4051] $22^1 1(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.25 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4052] $21(12)122^1 31513218122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4053] $21(12)122315132^1 18122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4054] $221(12)122315132^1 18122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& [4055] \quad 221^{221}(12)122^1 31513218122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [4056] \quad 1812231513221(12)122^1 222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
& [4057] \quad 18122315132^1 21(12)122222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& [4058] \quad 18122^1 31513221(12)122222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [4059] \quad 22^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.25 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [4060] \quad 1^{22221}(12)122315132^1 21812 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& [4061] \quad 1^{22221}(12)122^1 31513221812 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [4062] \quad 1^{2222^1 1}(12)12231513221812 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [4063] \quad 21^{2221}(12)122^1 31513221812 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [4064] \quad 21^{2221}(12)122315132^1 21812 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& [4065] \quad 221^{221}(12)122^1 31513221812 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [4066] $221(12)122315132^1 21812 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4067] $151316131513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -110, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4068] $2171231513221(12)122^1 3132 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 27, \Delta = -96, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4069] $21712^1 31513221(12)1223132 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 27, \Delta = -96, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4070] $21712315132^1 21(12)1223132 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 27, \Delta = -96, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4071] $32171231513221(12)12^1 2313 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4072] $17122^1 31513221(12)1223132 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 27, \Delta = -96, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4073] $17122315132^1 21(12)1223132 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 27, \Delta = -96, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4074] $1712231513221(12)122^1 3132 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 27, \Delta = -96, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4075] $322181231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 38, \Delta = -115, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4076] $221812315132^1 21(12)122222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4077] $221812^1 31513221(12)122222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4078] $22181231513221(12)122^1 222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} =$

15.0

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4079] 321812231513221(12)12¹222+ $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 38, \Delta = -115, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4080] 218122¹31513221(12)122222+ $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4081] 21812231513221(12)122¹222+ $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4082] 218122315132¹21(12)122222+ $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4083] ¹1712¹3218123218122315 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4084] ¹181232171232181232¹15 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 34, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[4085] ²¹1812¹321812321712315 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4086] ⁵¹³²¹181232181232¹1712 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4087] ¹1712¹3218123151321812 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4088] ²¹1812¹321812315132171 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4089] ¹1812315132181232¹1712 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

- [4090] $1712^1 3218123151322181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4091] $7122315132181232^1 181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4092] $1712^1 321812321812315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4093] $1712^1 321812315132181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4094] $141322^1 1(12)1223151316123 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4095] $141322^1 1(12)1223151321613 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4096] $31413221(12)122315132^1 161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4097] $31413221(12)122^1 315132161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4098] $314132^1 21(12)122315132161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [4099] $21712^1 3218123218122315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4100] $2181232171232181232^1 15 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 34$, $\Delta = -111$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [4101] $5132171232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4102] $1712315132181232^1 18122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4103] $21712^1 3218123151321812 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4104] $21712^1 3218123151322181 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4029] $1812315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4028] $1812^1 31513221(12)12222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4027] $181231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4108] $612231513221(12)122^1 31413 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 27, \Delta = -96, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4109] $7122315132181232^1 18122 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4110] $171232181232181232^1 15 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 34, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[4111] $41315132^1 21(12)1223151314 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 27, \Delta = -97, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4112] $1812^1 321812321613161 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4113] $61232171232181232^1 171 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.75 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

- [4114] $1613161232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4115] $1612^1 32181232171232171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4116] $61232171232181232^1 1712 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4117] $61232181232181232^1 161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -110$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4030] $1^1 8 12315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4107] $6122^1 31513221(12)12231413 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4105] $41316131513221(12)12^1 2313 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -103$, $\det = -16$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.75 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4227] $612231513221(12)12^1 231413 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4122] $1^1 8 12315132^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4123] $1^1 8 12315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4124] $1^1 8^1 1231513221(12)12 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 44$, $T \in [27, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [4125] $1^1 8 12315132^1 1(12)1222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$

15.0

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4138] $1^{13221} 171231513221(12)12^1 23132 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 27, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4139] $1^{13221} 1(12)122^1 31513217123 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 27, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4140] $1^{2221} 1(12)1223151321812^1 23 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4141] $1^{2221} 1(12)122^1 315132181223 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4142] $1^{321} 1812315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4143] $1^{232^1 1} 181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4144] $1^{21} 181231513221(12)12^1 2222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4145] $1^{21} 18123151322^1 1(12)122222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4146] $1^{2221} 1(12)12^1 231513218122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[4147] $1^{221} 21(12)1223151321812^1 23 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4148] $1^{221} 21(12)122^1 315132181223 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[4149] & \begin{matrix} 3221 \\ 221(12)122315132^1 1812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4150] & \begin{matrix} 322^1 1 \\ 221(12)1223151321812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[4151] & \begin{matrix} 2221 \\ 21(12)12^1 231513218122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[4152] & \begin{matrix} 221 \\ 221(12)12^1 231513218122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[4153] & \begin{matrix} 1 \\ 1812^1 231513221(12)122222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[4154] & \begin{matrix} 1 \\ 1812231513221(12)12^1 2222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[4155] & \begin{matrix} 1 \\ 181223151322^1 1(12)122222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4156] & \begin{matrix} 322^1 1 \\ 221(12)1223151322181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[4157] & \begin{matrix} 3221 \\ 221(12)122315132^1 2181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4158] & \begin{matrix} 2221 \\ 1(12)122^1 315132218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[4159] & \begin{matrix} 221 \\ 1(12)12231513221812^1 32 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.44 \rightarrow 27, T_{\max} = 44, T \in [27, 44] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [4160] $1(12)12^{2221}12^1 231513221812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [4161] $1(12)1223151322^1 1812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4162] $21(12)122^1 315132218123 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4163] $21(12)12231513221812^1 32 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 44$, $T \in [27, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [4164] $21(12)12^{2221}12^1 231513221812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [4165] $21(12)1223151322^1 1812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4166] $221(12)12^{221}12^1 231513221812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [4167] $221(12)1223151322^1 1812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4168] $612^{1513221}131513221(12)12 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 35$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 35$, $T \in [27, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4169] $61231513221^{15132^1}131513221(12)12 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 35$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 35$, $T \in [27, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4170] $1513161315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4171] $232^1 171231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 28$, $\Delta = -98$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 28$, $T \in [27, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[4172] $217123151322^1 1(12)1223132 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 27, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4173] $2171231513221(12)12^1 23132 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 27, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4174] $321712315132^1 21(12)122313 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4175] $1712^1 231513221(12)1223132 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 27, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[4176] $171223151322^1 1(12)1223132 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 27, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4177] $1712231513221(12)12^1 23132 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 27, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4178] $3221812315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4179] $322^1 181231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4180] $2218123151322^1 1(12)122222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4181] $22181231513221(12)12^1 2222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4182] $3218122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4183] $232^1 1812231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 44, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4184] $2181223151322^1 1(12)122222 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} =$

15.0

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4185] 21812231513221(12)12¹2222+ $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4186] 21812¹231513221(12)122222+ $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[4187] 171232¹18123218122315 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[4188] 1812¹3217123218123215 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 34, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4189] 1812321712321812¹3215 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 34, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4190] 1812¹3217123218123151 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4191] 1812321712321812¹3151 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4192] 181232²¹1812321712315 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[4193] 1⁵¹³²¹8¹²³²¹12321812¹321712 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4194] 1712321812315132¹1812 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4195] 171232¹18123151321812 + $\hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

- [4196] $1812321812315132^1 171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4197] $181232^1 1812315132171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [4198] $18123151321812^1 321712 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [4199] $1812^1 3151321812321712 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [4200] $171232^1 18123151322181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [4201] $1712321812315132^1 2181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4202] $7122^1 315132181232181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [4203] $71223151321812^1 32181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [4204] $171232^1 1812321812315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [4205] $171232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [4206] $1712321812315132^1 181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4207] $1413221(12)122^1 3151316123 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[4208] 1413221(12)122¹3151321613 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4209] 3141322¹1(12)122315132161 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 54 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[4210] 31413221(12)12¹2315132161 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[4211] 2171232¹18123218122315 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[4212] 21812¹3217123218123215 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 34$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4213] 21812321712321812¹3215 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 34$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4214] 21812321712321812¹3151 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4215] 21812¹3217123218123151 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4216] 51321712321812¹3218122 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4226] 61223151322¹1(12)12231413 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4218] 1712¹31513218123218122 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[4219] 17123151321812¹3218122 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4220] 2171232¹18123151321812 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[4221] \quad 21712321812315132^1 1812 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[4222] \quad 21712321812315132^1 2181 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[4223] \quad 2171232^1 18123151322181 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[4224] \quad 413161315132^1 21(12)122313 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -103, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[4225] \quad 612^1 231513221(12)12231413 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 27, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[4121] \quad 1 \quad 8 \quad 232^1 1 \quad 221 \quad 1231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[4120] \quad 1 \quad 8 \quad 321 \quad 1 \quad 12315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[4119] \quad 1 \quad 181231513221(12)12^1 23 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4229] \quad 8123151321812321712^1 32 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 48, T \in [27, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4230] \quad 7123218123151321812^1 32 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 48, T \in [27, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4231] \quad 71223151321812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[4232] \quad 7122^1 31513218123218122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} =$$

15.0

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[4233] \quad 1712^1 13218123218123215 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 34, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[4234] \quad 1712321812321812^1 3215 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 34, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4235] \quad 1712^1 13218123218123151 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[4236] \quad 1712321812321812^1 3151 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4237] \quad 4132^1 21(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4238] \quad 413221(12)122315132^1 171 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[4239] \quad 413221(12)122315132^1 217 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4240] \quad 4132^1 21(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4241] \quad 413151322^1 1(12)1223151314 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 27, \Delta = -97, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[4242] \quad 181232^1 1812321613161 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[4243] \quad 1812^1 13218123216131613 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4244] $612^1 32171232181232171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[4245] $612321712321812^1 32171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4246] $4132181232181232^1 181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 36$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 36, T \in [27, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4247] $4132^1 181232181232181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 36$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 36, T \in [27, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4248] $16131612321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4249] $31613161232181232^1 1812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4250] $161232^1 181232171232171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4251] $161232181232171232^1 171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4252] $6131612321812321812^1 23 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4253] $612^1 321712321812321712 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[4254] $612321712321812^1 321712 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4255] $4132181232181232^1 1812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 36$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.44 \rightarrow 27, T_{\max} = 36, T \in [27, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

- [4256] $4132^1 1812321812321812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 36$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 36$, $T \in [27, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4257] $612321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [4258] $612^1 32181232181232161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [4259] $612^1 31513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 41$, $T \in [27, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4260] $4132^1 18123218123151316 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 41$, $T \in [27, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [4118] $1812^1 31513221(12)1223 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4228] $7123151321812321812^1 23 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 41$, $T \in [27, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [4217] $1712^1 32181231513218123 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.44 \rightarrow 27$, $T_{\max} = 41$, $T \in [27, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [4264] $1^1 8^1 12315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27$, $T_{\max} = 44$, $T \in [27, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [4265] $1(12)122^1 3151316131513 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 48$, $\Delta = -118$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27$, $T_{\max} = 48$, $T \in [27, 48]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4266] $1(12)12^1 2315131613151 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -110$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.22 \rightarrow 27$, $T_{\max} = 33$, $T \in [27, 33]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4267] $21(12)122^1 3151316131513 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 48$, $\Delta = -118$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 48, T \in [27, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4268] & \begin{matrix} 221 \\ 21(12)12^1 2315131613151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -110, \det = -36, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4269] & \begin{matrix} 13221 \\ 1(12)12^1 231513217123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 27, \Delta = -96, \det = -36, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4270] & \begin{matrix} 2221 \\ 1(12)12^1 2315132181223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4271] & \begin{matrix} 321 \\ 1 \ 8 \ 123151322^1 1(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[4272] & \begin{matrix} 2321 \\ 1 \ 8 \ 12315132^1 21(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 44, T \in [27, 44] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[4273] & \begin{matrix} 221 \\ 21(12)12^1 2315132181223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4274] & \begin{matrix} 3221 \\ 221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[4275] & \begin{matrix} 2221 \\ 1(12)12^1 2315132218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4276] & \begin{matrix} 221 \\ 1(12)122^1 3151322181232 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 44, T \in [27, 44] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4277] & \begin{matrix} 221 \\ 21(12)12^1 2315132218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4278] & \begin{matrix} 21 \\ 21(12)122^1 3151322181232 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.22 \rightarrow 27, T_{\max} = 44, T \in [27, 44]
\end{aligned}$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4279] 31513161315132¹21(12)1222+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 48$, $\Delta = -118$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 48, T \in [27, 48]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4280] 15131613151322¹1(12)1222+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -110$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4281] 2321712315132¹21(12)12231+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 28$, $\Delta = -98$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 28, T \in [27, 28]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4282] 3217123151322¹1(12)122313+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4283] 32218123151322¹1(12)1222+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4284] 32181223151322¹1(12)1222+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4285] 23218122315132¹21(12)1222+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4286] 17123218123151322¹181+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4287] 712¹3151321812321³²¹81+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4288] 712¹23151321812321²¹81+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4289] 1413221(12)12¹23151316123+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4290] 1413221(12)12¹23151321613+ $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4291] $1812^1 32171232181231513 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4292] $321712321812315132^1 181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4293] $217123218123151322^1 181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4294] $4131613151322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -103$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4295] $712^1 315132181232181223 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4296] $712^1 321812315132181232 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 48$, $\Delta = -118$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 48, T \in [27, 48]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4297] $8122^1 31513218123217123 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4298] $712^1 231513218123218122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4299] $712^1 32181232181231513 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4300] $413221(12)1223151322^1 17 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[4301] $16131613161315132^1 181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4302] $8122^1 3151316131613161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4303] $151316131613161232^1 181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4304] $216131613161315132^1 181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4305] $8122^1 31513161316131612 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4306] $712^1 321712321812321613 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4307] $712^1 32161316131613161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4308] $712^1 321613161316131612 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4263] $1^{\frac{321}{8}} 123151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4262] $1^{\frac{2321}{8}} 12315132^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4261] $1^{\frac{321}{8}} 123151322^1 1(12)1222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -36$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.22 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4360] $321812321712321812^1 315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 48$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 48, T \in [27, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4313] $1^{\frac{2321}{8}} 123151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[4311] $1^{\frac{2321}{8}} 123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

- [4310] $1 \overset{321}{8} 12^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [4316] $21 \overset{1513221}{(12)} 122^1 31513161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 34$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [4317] $1 \overset{221}{(12)} 12^1 23151316131513 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 48$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 48$, $T \in [27, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [4318] $21 \overset{21}{(12)} 12^1 23151316131513 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 48$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 48$, $T \in [27, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [4319] $21 \overset{5132^1 21}{(12)} 122315131613 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 58$, $T \in [27, 58]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [4320] $1 \overset{13221}{(12)} 122315132^1 17123 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 27$, $\Delta = -96$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4321] $1 \overset{2221}{(12)} 122315132^1 181223 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4322] $1 \overset{321}{8} 12^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [4323] $1 \overset{2321}{8} 123151322^1 1(12)1222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 44$, $T \in [27, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [4324] $21 \overset{221}{(12)} 122315132^1 181223 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [4325] $221 \overset{3221}{(12)} 122^1 3151321812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [4326] $221 \overset{3221}{(12)} 122^1 3151322181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.06 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[4327] & \begin{matrix} 2221 \\ 1(12)122315132^1 218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4328] & \begin{matrix} 221 \\ 1(12)12^1 23151322181232 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 44, T \in [27, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[4329] & \begin{matrix} 221 \\ 21(12)122315132^1 218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4330] & \begin{matrix} 21 \\ 21(12)12^1 23151322181232 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 44, T \in [27, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[4331] & \begin{matrix} 151322^1 1 \\ 61231513221 (12) 12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 35, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 35, T \in [27, 35] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[4332] & \begin{matrix} 1513221 \\ 612315132^1 21 (12) 12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 35, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 35, T \in [27, 35] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4333] & 231513161315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 58, \Delta = -119, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 58, T \in [27, 58] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[4334] & 315131613151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 48, T \in [27, 48] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[4335] & 23217123151322^1 1(12)12231 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 28, \Delta = -98, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 28, T \in [27, 28] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[4336] & 321712^1 31513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[4337] & 3221812^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[4338] & 3218122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4339] $232181223151322^1 1(12)1222 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[4340] $1712321812^1 3218122315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[4341] $17123218123218122^1 315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4342] $181232^1 17123218123215 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 34$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4343] $181232171232^1 18123215 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 34$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4344] $181232^1 17123218123151 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4345] $181232171232^1 18123151 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4346] $1812321812321712^1 315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4347] $1812321812^1 321712315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[4348] $1^1 8^1 12321812321712 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[4349] $1^1 8^1 1232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[4350] $1712321812^1 3151321812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[4351] & \overset{21}{1} 8 12321812^1 315132171 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[4352] & \overset{1}{1} 812315132^1 1812321712 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[4353] & \overset{1}{1} 712321812^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[4354] & 712315132181232^1 \overset{321}{1} 8 1 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 41, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4355] & 7122315132^1 1812321 \overset{21}{8} 1 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[4356] & \overset{1}{1} 712321812321812^1 315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[4357] & \overset{1}{1} 712321812^1 321812315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[4358] & \overset{1}{1} 712321812^1 315132181 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[4359] & 1413221(12)122315132^1 1613 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 27, \Delta = -96, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4309] & \overset{1}{1} 812315132^1 21 \overset{221}{(12)} 1223 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[4361] & 217123218123218122^1 315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[4362] & 21712321812^1 3218122315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[4363] $2181232^1 17123218123215 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 34$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4364] $2181232171232^1 18123215 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 34$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4365] $2181232171232^1 18123151 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4366] $2181232^1 17123218123151 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4367] $5132^1 17123218123218122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[4368] $5132171232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[4369] $171232^1 181231513218123 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4370] $321712^1 321812315132181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4371] $1712315132^1 18123218122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[4372] $21712321812^1 3151321812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4373] $21712321812^1 3151322181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4374] $812^1 321712321812315132 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 48$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 48, T \in [27, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4375] $712315132181232^1 181223 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 41, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4376] $812^1 315132181232171232 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 48, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 48, T \in [27, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4377] $812^1 231513218123217123 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 41, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4378] $8122315132181232^1 17123 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 41, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4379] $7122315132^1 18123218122 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[4380] $171232^1 18123218123215 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 34, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4381] $171232181232^1 18123215 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 34, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4382] $171232^1 18123218123151 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4383] $171232181232^1 18123151 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 33, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4384] $413221(12)^{21} 122^1 315132171 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 38, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4385] $41322^1 1(12)^{21} 122315132171 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 38, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[4386] $413221(12)^{21} 122^1 315132217 + \hat{1} \rightarrow 2$ $T = 21, T_{\max} = 38, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[4387] $41322^1 1(12)122315132217 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4388] $1812321812^1 321613161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4389] $181232^1 18123216131613 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4390] $812^1 3151316131613161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4391] $812^1 23151316131613161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[4392] $613161232181232^1 181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4393] $61232171232^1 181232171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[4394] $61232^1 171232181232171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[4395] $41321812321812^1 32181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 36$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 36, T \in [27, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4396] $41321812^1 32181232181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 36$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 36, T \in [27, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4397] $1613161232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[4398] $316131612321812^1 321812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[4399] $1612321812^1 32171232171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4400] $1612321812321712^1 32171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[4401] $4141413221(12)122^1 3151315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[4402] $812^1 321613161316131513 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4403] $812^1 315131613161316123 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4404] $812^1 231513161316131612 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[4405] $613161232181232^1 181223 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4406] $61232171232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[4407] $61232^1 1712321812321712 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[4408] $71232171232181232^1 1613 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 44, T \in [27, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4409] $414132^1 21(12)12231513161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[4410] $41413221(12)122^1 31513161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4411] $41321812^1 321812321812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 36$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 36, T \in [27, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4412] $41321812321812^1 321812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 36$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 36, T \in [27, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[4413] $61232181232^1 181232161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[4414] $61232^1 181232181232161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 33, T \in [27, 33]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4415] $61231513221(12)122^1 31414 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[4416] $612315132^1 21(12)12231414 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[4417] $41321812^1 3218123151316 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 41, T \in [27, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[4315] $21(12)12231513161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 34$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[4312] $1^1 8^1 12^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[4314] $1(12)122^1 31513161315132 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -49$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.06 \rightarrow 27, T_{\max} = 58, T \in [27, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[4419] $1(12)12^1 231513161315132 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 58, T \in [26, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[4422] $2315131613151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 58, T \in [26, 58] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[4423] & \begin{matrix} 1 \\ 171232181232^1 18122315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[4424] & \begin{matrix} 1 \\ 1812321712^1 3218123215 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 34, \Delta = -111, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 34, T \in [26, 34] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[4425] & \begin{matrix} 1 \\ 1812321712^1 3218123151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[4426] & \begin{matrix} 21 \\ 181232181232^1 1712315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[4427] & \begin{matrix} 51321 \\ 1 \ 8 \ 12^1 321812321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[4428] & \begin{matrix} 321 \\ 7123151321812^1 321 \ 8 \ 1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 41, \Delta = -116, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 41, T \in [26, 41] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[4429] & \begin{matrix} 1 \\ 171232181232^1 1812315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[4430] & \begin{matrix} 1 \\ 1812321712321812^1 31513 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[4431] & \begin{matrix} 1 \\ 2171232181232^1 18122315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[4432] & \begin{matrix} 1 \\ 21812321712^1 3218123215 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 34, \Delta = -111, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 34, T \in [26, 34] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[4433] & \begin{matrix} 1 \\ 21812321712^1 3218123151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[4434] & \begin{matrix} 1 \\ 51321712^1 3218123218122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33]
\end{aligned}$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[4435] $32171232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[4436] $7123151321812^1 32181223 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4437] $81223151321812^1 3217123 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4438] $1712321812^1 3218123215 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 34$, $\Delta = -111$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 34, T \in [26, 34]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4439] $1712321812^1 3218123151 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4440] $712321812321812^1 31513 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 44, T \in [26, 44]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4441] $181232181232^1 1613161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[4442] $6131612321812^1 321812321812^1 31513 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4443] $612321712^1 32181232171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[4444] $4132181232^1 1812321812^1 321812321812^1 31513 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 36$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4445] $16131612^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -107$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[4446] $161232181232^1 171232171 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[4447] $6131612321812^1 32181223 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4448] $612321712^1 321812321712 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[4449] $712321712321812^1 321613 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 44, T \in [26, 44]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4450] $4132181232^1 1812321812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 36$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 36, T \in [26, 36]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4451] $612321812^1 32181232161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -110$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[4452] $4132181232^1 18123151316 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[4421] $2231513161315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 58, T \in [26, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[4418] $1(12)122^1 315131613151322 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 58, T \in [26, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[4420] $6131513221^{315132^1 21} (12) 12 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -64$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.94 \rightarrow 26, T_{\max} = 58, T \in [26, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4456] $21^{1513221} (12) 12^1 231513161 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 34$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.84 \rightarrow 26, T_{\max} = 34, T \in [26, 34]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[4454] $18123151322^1 1^{221} (12)1223 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -81$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.84 \rightarrow 26, T_{\max} = 38, T \in [26, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[4458] $221^{3221} (12)12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 38$, $\Delta = -115$, $\det = -81$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.84 \rightarrow 26, T_{\max} = 38, T \in [26, 38]$$

- $$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$
- [4459] $221^{3221}(12)12^1 23151322181 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26, T_{\max} = 38, T \in [26, 38]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$
- [4460] $1^{2221}(12)1223151322^1 18123 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26, T_{\max} = 38, T \in [26, 38]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$
- [4461] $21^{221}(12)1223151322^1 18123 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26, T_{\max} = 38, T \in [26, 38]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$
- [4462] $6123151322^{1513221} 1(12)12 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 35, \Delta = -112, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26, T_{\max} = 35, T \in [26, 35]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$
- [4463] $22315131613151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 58, \Delta = -119, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26, T_{\max} = 58, T \in [26, 58]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$
- [4464] $321812^1 231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26, T_{\max} = 38, T \in [26, 38]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$
- [4465] $181232^1 171232181231513 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26, T_{\max} = 44, T \in [26, 44]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$
- [4466] $321812^1 321712321812315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26, T_{\max} = 48, T \in [26, 48]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$
- [4467] $1712321812315132^1 18123 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 41, \Delta = -116, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$
- [4468] $71232^1 1812315132181232 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26, T_{\max} = 48, T \in [26, 48]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$
- [4469] $413221^{21}(12)12^1 2315132171 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26, T_{\max} = 38, T \in [26, 38]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$
- [4470] $71232^1 181232181231513 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[4471] & 413221(12)^{21}12^1 2315132217 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 38, T \in [26, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[4472] & 1513161316131612^1 32181 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[4473] & 41414132^1 21(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[4474] & 4141413221(12)12^1 23151315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[4475] & 71232^1 1712321812321613 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[4476] & 41413221(12)^1 12^1 231513161 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 38, T \in [26, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[4477] & 4141322^1 1(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 38, T \in [26, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[4478] & 6123151322^1 1(12)12231414 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 41, \Delta = -116, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 41, T \in [26, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[4479] & 61231513221(12)^1 12^1 231414 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 41, \Delta = -116, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 41, T \in [26, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[4480] & 71232^1 161316131613161 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[4481] & 71232^1 1613161316131612 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.84 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [4457] $21(12)122^{3221}131513161315 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -81$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26$, $T_{\max} = 58$, $T \in [26, 58]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [4455] $1(12)12^{12315131613151322} + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -81$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26$, $T_{\max} = 58$, $T \in [26, 58]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [4453] $1414141414141414132^{121(12)1} + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -108$, $\det = -81$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.84 \rightarrow 26$, $T_{\max} = 33$, $T \in [26, 33]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [4485] $1^1 8^{2321} 12^{131513221(12)12} + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.76 \rightarrow 26$, $T_{\max} = 44$, $T \in [26, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [4486] $21^{51322^{11}}(12)122315131613 + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.76 \rightarrow 26$, $T_{\max} = 58$, $T \in [26, 58]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [4487] $1^1 8^{2321} 12^{131513221(12)1222} + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.76 \rightarrow 26$, $T_{\max} = 44$, $T \in [26, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [4488] $1(12)122315132^{12181232} + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.76 \rightarrow 26$, $T_{\max} = 44$, $T \in [26, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [4489] $21(12)122315132^{12181232} + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.76 \rightarrow 26$, $T_{\max} = 44$, $T \in [26, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [4490] $61315132^{121(12)12} + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 58$, $\Delta = -119$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.76 \rightarrow 26$, $T_{\max} = 58$, $T \in [26, 58]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [4491] $2321712^{131513221(12)12231} + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 28$, $\Delta = -98$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.76 \rightarrow 26$, $T_{\max} = 28$, $T \in [26, 28]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [4492] $23218122^{131513221(12)1222} + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.76 \rightarrow 26$, $T_{\max} = 44$, $T \in [26, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [4493] $1712321812321812^{12315} + \hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 25.76 \rightarrow 26$, $T_{\max} = 33$, $T \in [26, 33]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4494] 712315132¹1812321³²¹ 8 1 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[4495] 181232171232¹181231513 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 44, T \in [26, 44]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[4496] 21712321812321812¹2315 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 33$, $\Delta = -109$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 33, T \in [26, 33]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[4497] 321712321812¹315132181 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[4498] 712315132¹181232181223 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[4499] 812315132181232¹171232 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 48$, $\Delta = -118$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 48, T \in [26, 48]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[4500] 8122315132¹18123217123 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[4501] 71232181232¹181231513 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 44, T \in [26, 44]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[4502] 613161232¹1812321³²¹ 8 1 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[4503] 613161232¹181232181223 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[4504] 71232171232¹1812321613 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 44$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 44, T \in [26, 44]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[4505] 41321812321812¹3151316 + $\hat{1} \rightarrow 2$ $T = 21$, $T_{\max} = 41$, $\Delta = -116$, $\det = -100$, $(n_+, n_-, n_0) = (1, 21, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 25.76 \rightarrow 26, T_{\max} = 41, T \in [26, 41]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[4484] & 1 \ 8 \ 12^{2321} 131513221^{221} (12)1 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -100, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.76 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[4482] & 14141414141414141322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 33, \Delta = -108, \det = -100, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.76 \rightarrow 26, T_{\max} = 33, T \in [26, 33] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[4483] & (12)122^1 3141414141414141413 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 41, \Delta = -116, \det = -100, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.76 \rightarrow 26, T_{\max} = 41, T \in [26, 41] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[4510] & 1812321712^1 32181231513 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[4506] & (12)12^1 23141414141414141413 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 41, \Delta = -116, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 41, T \in [26, 41] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[4507] & 21(12)12^{3221} 231513161315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 58, \Delta = -119, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 58, T \in [26, 58] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[4512] & 1712321812^1 31513218123 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 41, \Delta = -116, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 41, T \in [26, 41] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[4513] & 81232^1 1712321812315132 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 48, T \in [26, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[4514] & 812321712321812^1 315132 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 48, T \in [26, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[4515] & 8123151321812^1 32171232 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 48, T \in [26, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[4516] & 712321812^1 32181231513 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[4517] & 1812321812^1 3216131613 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 38, T \in [26, 38] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[4518] & 6131612^1 321812321^{321} 8 \ 1 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 41, \Delta = -116, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 41, T \in [26, 41] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[4519] & 31613161232^1 1812321812 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 38, T \in [26, 38] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[4520] & 414141322^1 1(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[4521] & 81232^1 1613161316131513 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[4522] & 6131612^1 32181232181223 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 41, \Delta = -116, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 41, T \in [26, 41] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[4523] & 712321712^1 321812321613 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[4508] & 21 (12) 122^1 315131613 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 58, \Delta = -119, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 58, T \in [26, 58] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[4511] & 32181232171232^1 1812315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 48, T \in [26, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[4509] & 6131513221^{513221} (12) 12 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 58, \Delta = -119, \det = -121, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.69 \rightarrow 26, T_{\max} = 58, T \in [26, 58] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[4524] & 32181232^1 1712321812315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -144, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.64 \rightarrow 26, T_{\max} = 48, T \in [26, 48] \\
& \begin{pmatrix} 0 & 12 & 2 \\ 12 & 59 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[4528] & 2321812^1 231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -169, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.59 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[4525] & 1(12)1223151322^1 181232 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -169, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 25.59 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[4530] & 81232171232^1 1812315132 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -169, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.59 \rightarrow 26, T_{\max} = 48, T \in [26, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[4531] & 812315132^1 181232171232 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -169, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.59 \rightarrow 26, T_{\max} = 48, T \in [26, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[4532] & 712321812315132^1 181232 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -169, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.59 \rightarrow 26, T_{\max} = 48, T \in [26, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[4533] & \overset{1}{181232181232^1 16131613} + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.59 \rightarrow 26, T_{\max} = 38, T \in [26, 38] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[4534] & 316131612^1 321812321812 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 38, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.59 \rightarrow 26, T_{\max} = 38, T \in [26, 38] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[4527] & \overset{31513221}{613151322^1 1(12)} 12 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 58, \Delta = -119, \det = -169, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.59 \rightarrow 26, T_{\max} = 58, T \in [26, 58] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[4529] & 321812321712^1 321812315 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -169, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.59 \rightarrow 26, T_{\max} = 48, T \in [26, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 64 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[4526] & \overset{21}{21(12)} 1223151322^1 181232 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 44, \Delta = -117, \det = -169, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.59 \rightarrow 26, T_{\max} = 44, T \in [26, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[4535] & \overset{513221}{21(12)} 12^1 2315131613 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 58, \Delta = -119, \det = -196, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.55 \rightarrow 26, T_{\max} = 58, T \in [26, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[4536] & 812321712^1 321812315132 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -196, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.55 \rightarrow 26, T_{\max} = 48, T \in [26, 48] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[4537] & 712321812^1 315132181232 + \hat{1} \rightarrow 2 \quad T = 21, T_{\max} = 48, \Delta = -118, \det = -225, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 25.52 \rightarrow 26, T_{\max} = 48, T \in [26, 48] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- $T = 22$ [26] $612321712321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 34, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [25] $413221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 38, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [3] $161316131513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 34, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [4] $3221(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 42, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 42, T \in [28, 42]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [5] $1(12)12231513161316123 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 42, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 42, T \in [29, 42]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [6] $1(12)1223151316131612 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 34, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [7] $161316131513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 39, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 39, T \in [29, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [8] $21(12)1223151316131612 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 34, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [9] $1^1 8^1 12321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 34, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [10] $18123151321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 34, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [11] $4132181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 38, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 38, T \in [29, 38]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [12] $4132181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 38, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 38, T \in [29, 38]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [13] $413221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 38, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [1] $712231513221(12)12^1 + \hat{1} \rightarrow 2$ $T = 22, T_{\max} = 34, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [14] $\overset{1}{161316131513221}(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [15] $\overset{1}{161316131513221}(12)12 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [16] $\overset{31}{161316131513221}(12)1 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 39, T \in [29, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [17] $\overset{22^1}{32161316131513221}(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 42, T \in [29, 42]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [18] $\overset{22^1}{232161316131513221}(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 49, \Delta = -118, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 49, T \in [29, 49]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [19] $\overset{22^1}{2161316131513221}(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [20] $\overset{22^1}{22^1 18123218123218122315} + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [21] $\overset{22^1}{22^1 18123218123151321812} + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [22] $\overset{22^1}{22^1 18123218123151322181} + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 34, T \in [29, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [0] $\overset{2^1}{1513221}(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [23] $\overset{2^1}{4132181231513221}(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 38, T \in [29, 38]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [24] $\overset{2^1}{413221}(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [2] $\overset{2^1}{3151316131513221}(12)122^1 31 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 29, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$

- [31] $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$ 151316131513221(12)122¹313 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 29$, $\Delta = -98$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 29$, $T \in [29, 29]$
- [41] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$ 21(12)1223151316131612¹3 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 42$, $T \in [29, 42]$
- [28] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$ 1 8 1231513221(12)122¹313 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 34$, $T \in [29, 34]$
- [29] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$ 71231513221 (12) 12¹ + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 49$, $\Delta = -118$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.33 \rightarrow 28$, $T_{\max} = 49$, $T \in [28, 49]$
- [30] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$ 3151316131513221(12)12¹231 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 29$, $\Delta = -98$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 29$, $T \in [29, 29]$
- [32] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$ 322181231513221(12)122¹313 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 34$, $T \in [29, 34]$
- [33] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$ 32¹2181231513221(12)122313 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 34$, $T \in [29, 34]$
- [34] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$ 321812231513221(12)122¹313 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 34$, $T \in [29, 34]$
- [35] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$ 32¹1812231513221(12)122313 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 34$, $T \in [29, 34]$
- [36] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$ 161316131513221(12)122¹22 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 34$, $T \in [29, 34]$
- [37] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$ 21(12)122315131613161 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 42$, $T \in [29, 42]$
- [38] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$ 1(12)1223151316131612¹3 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 42$, $T \in [29, 42]$
- [39] $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$ 1(12)1223151316131612 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 34$, $T \in [29, 34]$

- [40] $161316131513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 39$, $T \in [29, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [42] $18123218123151321812^1 3 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 39$, $T \in [29, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [43] $1 \ 8 \ 12321812315132181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 39$, $T \in [29, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [44] $4132181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 38$, $T \in [29, 38]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [45] $81223151321812321 \ 8 \ 1 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 39$, $T \in [29, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [46] $161316131513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 34$, $T \in [29, 34]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [47] $32161316131513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 42$, $T \in [29, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [48] $32^1 161316131513221(12)1222 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 42$, $T \in [29, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [49] $2161316131513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 34$, $T \in [29, 34]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [50] $32^1 21812321812315132181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 39$, $T \in [29, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [51] $218123218123151321812^1 3 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 39$, $T \in [29, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [52] $4132181231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 38$, $T \in [29, 38]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [53] $812231513218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 39$, $T \in [29, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [54] $32^1 1812321812321812315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 45$, $T \in [29, 45]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [55] $612321812321712321812^1 3 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 45$, $T \in [29, 45]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [56] $712321712321812321712^1 3 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 42$, $T \in [29, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [57] $812321613161316131612^1 3 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 39$, $T \in [29, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [27] $1^{\frac{32}{1}} 8^1 1231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 9$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 34$, $T \in [29, 34]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [79] $1712^1 32181232171232171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [59] $151316131513221(12)12^1 2313 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 29$, $\Delta = -98$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 29$, $T \in [28, 29]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [78] $181232171232181232^1 161 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [77] $2181232181232181232^1 15 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [60] $322181231513221(12)12^1 2313 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [61] $321812231513221(12)12^1 2313 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [62] $161316131513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [63] $1812^1 3218123218122315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [64] $1^{\frac{51321}{1}} 8^1 1232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$

- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [65] $1812315132181232^1 18122 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 21
 [66] $1812^1 3218123151321812 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [74] $221812^1 3218123151322181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [75] $4132181231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [76] $61315132181232181232^1 15 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 37, \Delta = -113, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 37, T \in [28, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- 21
 [67] $1812^1 3218123151322181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 1
 [68] $181232181232181232^1 15 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- 21
 [69] $1812^1 321812321812315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- 21
 [70] $1812^1 321812315132181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [71] $2161316131513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -107, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [72] $221812^1 3218123218122315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [73] $221812^1 3218123151321812 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 21
 [80] $61232171232181232^1 181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- 1
 [81] $61232181232181232^1 171 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$

- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [82] $1612^1 321812321712321812 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [83] $171232171232181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -107, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [84] $61232171232181232^1 18122 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [85] $171232181232181232^1 161 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
 [86] $61232181232181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [58] $1^1 8^1 1231513221(12)12^1 2313 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -103, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.75 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [148] $413221(12)^{21} 122315132^1 1812 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [158] $612321812321812^1 32171 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [134] $22181232^1 18123151321812 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [132] $18123218123151321812^1 32 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 45, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 45, T \in [28, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [146] $21812^1 3218123218123151 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [87] $21^1 (12)^{1513221} 122315132^1 171 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [88] $21^1 (12)^{15132^1 21} 122315132171 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [89] $1^1 8^1 1231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 28, \Delta = -97, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 28, T \in [28, 28]$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [90] $1 \ 8 \ 12315132^1 21(12)122313 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 1513221
 [91] $7122^1 31513221 (12) \ 12 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $15132^1 21$
 [92] $712231513221 (12) \ 12 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [93] $1513161315132^1 21(12)122313 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 29, \Delta = -98, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 29, T \in [28, 29]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [94] $3221812315132^1 21(12)122313 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [95] $322^1 181231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [96] $232^1 1812231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 28, \Delta = -97, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [97] $3218122315132^1 21(12)122313 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 1
 [98] $1613161315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $322^1 1$
 [99] $21(12)122315131613161 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 42, T \in [28, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 21
 [100] $1(12)1223151316131612^1 32 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 49, \Delta = -118, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 49, T \in [28, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- 2221
 [101] $1(12)122^1 3151316131612 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$

- [102] $21^{221}(12)122^1 3151316131612 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [103] $18^1 1232^1 18123218122315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [104] $1^{51321} 8^1 12321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [105] $1812^1 32181231513218123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [106] $1812^1 31513218123218122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [107] $18123151321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [108] $18^1 1232^1 18123151321812 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [109] $18^1 12321812315132^1 1812 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [110] $18^1 1232^1 18123151322181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [111] $18^1 12321812315132^1 2181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [112] $4132^1 181231513221^{2221}(12)1 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [119] $1812^1 3218123218123151 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [120] $1812321812321812^1 3151 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [121] $413221^{21}(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [122] $4132^1 21(12)122315132181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [123] $1613161315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [124] $1613161315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [125] $181232^1 1812321812315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [126] $181232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [127] $1812321812315132^1 181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [128] $232^1 161316131513221(12)122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 49$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 49$, $T \in [28, 49]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [129] $21613161315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [130] $22181232^1 18123218122315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [131] $322^1 1812321812315132181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [116] $8123151321812321^{232^1 1} 81 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 45$, $T \in [28, 45]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [133] $21812^1 32181231513218123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [147] $4132^1 21(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 49$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 49$, $T \in [28, 49]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

- [135] 221812321812315132¹1812+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [136] 22181232¹18123151322181+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [137] 221812321812315132¹2181+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [138] 4132¹181231513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [139] 41321812315132¹21(12)12222 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [140] 613151321812321812¹3215+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 37$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 37$, $T \in [28, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [141] 613151321812321812¹3151+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 35$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 35$, $T \in [28, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [142] 81223151321812321812¹23+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [143] 21812¹3218123218123215 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [144] 21812321812321812¹3215 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [145] 21812321812321812¹3151 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [115] 4132¹181231513221²²¹(12)12+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [114] 41321812315132¹21²²¹(12)12+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [113] 41321812315132¹21²²²¹(12)1+ $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

- [149] $4132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [150] $413221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [151] $4132^1 21(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [152] $1812321712321812^1 32161 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [117] $1812^1 3218123218123215 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [153] $1812^1 32171232181232161 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [154] $171232^1 181232171232171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [155] $171232181232171232^1 171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [156] $612^1 321712321812321 81 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [157] $612321712321812^1 321 81 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [175] $4132^1 181232181232161316 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [174] $4141413221(12)122315132^1 16 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [173] $4132^1 181232181231513217 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

- [172] $7122^1 31513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [171] $612321812321812^1 321712 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [170] $612^1 321812321812321712 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [169] $41413221(12)122315132^1 171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [168] $1712321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [167] $1712^1 32181232181232161 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [166] $612^1 3217123218123218122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [165] $612321712321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [164] $612^1 3218123217123218123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 45$, $T \in [28, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [163] $1712^1 321712321812321712 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [162] $1712321712321812^1 321712 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [161] $161232^1 1812321712321812 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [160] $161232181232171232^1 1812 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [159] $612^1 32181232181232171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

- [118] $1812321812321812^1 3215 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.44 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [176] $1^{\frac{2321}{8}} 12315132^1 21(12)12231 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 28, \Delta = -97, \det = 36, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [177] $1^{\frac{321}{8}} 123151322^1 1(12)122313 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -103, \det = 36, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [178] $712^1 31513221^{\frac{31513221}{(12)}} 12 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 49, \Delta = -118, \det = 36, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28, T_{\max} = 49, T \in [28, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [179] $712^1 231513221^{\frac{1513221}{(12)}} 12 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -110, \det = 36, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [180] $31513161315132^1 21(12)12231 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 29, \Delta = -98, \det = 36, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28, T_{\max} = 29, T \in [28, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [182] $32218123151322^1 1(12)122313 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -103, \det = 36, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [183] $23218122315132^1 21(12)12231 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 28, \Delta = -97, \det = 36, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [184] $32181223151322^1 1(12)122313 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -103, \det = 36, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [185] $16131613151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -107, \det = 36, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [186] $1^{\frac{221}{(12)}} 122^1 31513161316123 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 36, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28, T_{\max} = 42, T \in [28, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [187] $1^{\frac{2221}{(12)}} 12^1 23151316131612 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -107, \det = 36, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [188] $16^{\frac{31}{13}} 13161315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 36, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$

- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [189] $21^{21}(12)122^1 31513161316123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 42$, $T \in [28, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 221
 [190] $21(12)12^1 23151316131612 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- 321
 [191] $1\ 8\ 12321812315132^1 181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 21
 [192] $1\ 8\ 123218123151322^1 181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 2221
 [193] $413218123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 221
 [194] $413218123151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 321
 [195] $8122^1 3151321812321\ 8\ 1 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 1
 [196] $16131613151322^1 1(12)1222 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 221
 [197] $16131613151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 31
 221
 [198] $1\ 6\ 13161315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- 31
 21
 [199] $1\ 6\ 13161315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [200] $321613161315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 42$, $T \in [28, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [201] $216131613151322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[202] $3221812321812315132^1 181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[203] $1812^1 321812315132181232 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 45$, $T \in [28, 45]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[204] $2218123218123151322^1 181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[205] $413218123151322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[206] $8122^1 315132181232181223 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[207] $1812^1 32181232181231513 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 42$, $T \in [28, 42]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[208] $413221(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[209] $3161232181232171232^1 181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 42$, $T \in [28, 42]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[210] $712^1 3217123218123217123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 42$, $T \in [28, 42]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[211] $712^1 321812321812321613 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 42$, $T \in [28, 42]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[212] $712^1 231513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[213] $41321812321812315132^1 17 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[214] $161316131613161232^1 181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[215] $2161316131613161232^1 181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$

- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [181] $15131613151322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 29$, $\Delta = -98$, $\det = 36$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.22 \rightarrow 28$, $T_{\max} = 29$, $T \in [28, 29]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [244] $812^1 23151321812321^1 81 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [245] $181232^1 18123218123215 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [272] $2181232181232^1 18123151 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [247] $181232^1 18123218123151 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [246] $181232181232^1 18123215 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [216] $1^{5131513221} (12) 122^1 3151315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 42$, $T \in [28, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [217] $21^{1513221} (12) 122^1 315132171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
 [218] $21^{151322^1 1} (12) 122315132171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
 [219] $1^1 8^{2321} 123151322^1 1(12)12231 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 28$, $\Delta = -97$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 28$, $T \in [28, 28]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [220] $1^1 8^{321} 12^1 31513221(12)122313 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [221] $7122315132^1 21^{1513221} (12) 12 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
 [222] $712231513221^{151322^1 1} (12) 12 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

- [223] $315131613151322^1 1(12)12231 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 29$, $\Delta = -98$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 29$, $T \in [28, 29]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [224] $3221812^1 31513221(12)122313 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [225] $232181223151322^1 1(12)12231 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 28$, $\Delta = -97$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 28$, $T \in [28, 28]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [226] $3218122^1 31513221(12)122313 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [227] $1(12)12^{221} 231513161316123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 42$, $T \in [28, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [228] $1(12)122^{21} 315131613161232 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 49$, $\Delta = -118$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 49$, $T \in [28, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [229] $16^{31} 131613151322^1 1(12)1222 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [230] $21^{21} (12)12^1 231513161316123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 42$, $T \in [28, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [231] $18^{21} 12321812^1 3218122315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [232] $18^{21} 123218123218122^1 315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [233] $1^{5132} 8^1 123218123218122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [234] $1^{51321} 8^1 1232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [235] $1^{1} 181232^1 181231513218123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$

- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
 [236] $1\ 8\ 12^1 321812315132181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
 [237] $1812315132^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [238] $1\ 8\ 12321812^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [239] $1\ 8\ 12321812^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [240] $41321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [241] $41321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [248] $181232181232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [249] $41322^1 1(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
 [250] $413221(12)122^1 315132181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [251] $1\ 6\ 131613151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
 [252] $1\ 6\ 131613151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
 [253] $1\ 8\ 12321812^1 321812315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
 [254] $1\ 8\ 12321812321812^1 315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$
- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [255] $1\ 8\ 12321812^1 315132181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[256] \ 3216131613151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 42, T \in [28, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[257] \ 2321613161315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 49, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 49, T \in [28, 49]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[258] \ 2218123218123218122^1 315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[259] \ 221812321812^1 3218122315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[260] \ 3221812^1 321812315132181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[261] \ 2181232^1 181231513218123 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[262] \ 221812321812^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[263] \ 221812321812^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[264] \ 41321812^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[265] \ 61315132181232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 37, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 37, T \in [28, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[266] \ 61315132181232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 35, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[267] \ 812^1 2315132181232181223 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[268] \ 8122315132181232^1 181223 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[269] \ 321812321812321812^1 315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 45, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 45, T \in [28, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[270] $2181232181232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[271] $2181232^1 18123218123215 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[273] $2181232^1 18123218123151 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[274] $41322^1 1(12)12231513161315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 49, \Delta = -118, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 49, T \in [28, 49]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[275] $41322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[276] $413221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[277] $41322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[278] $413221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[279] $812^1 321812321812315132 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 45, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 45, T \in [28, 45]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[280] $181232^1 171232181232161 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[281] $181232171232^1 181232161 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[282] $1712321812321712^1 32171 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

- [283] $1712321812^1 32171232171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [242] $812^1 3151321812321^{2321} 8^1 1 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 45$, $T \in [28, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [312] $41321812^1 32181232161316 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [311] $41316131613161315132^1 18 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [310] $4141413221(12)122^1 31513216 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [309] $812^1 3216131613161316123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [308] $4141413221(12)122^1 31513161 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [307] $812^1 32161316131613161^1 61 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [306] $41321812^1 32181231513217 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [305] $7122315132^1 21(12)12231414 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [304] $712231513221(12)122^1 31414 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [303] $71232181232181232^1 1613 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 42$, $T \in [28, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [302] $61232181232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [301] $61232^1 1812321812321712 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[300] \quad 41413221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[299] \quad 414132^1 21(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[298] \quad 171232181232^1 181232161 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[297] \quad 171232^1 181232181232161 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[296] \quad 71232171232181232^1 17123 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 42, T \in [28, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[295] \quad 61232171232^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[294] \quad 61232^1 17123218123218122 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[293] \quad 61232^1 18123217123218123 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 45, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 45, T \in [28, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[292] \quad 171232^1 1712321812321712 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[291] \quad 171232171232^1 1812321712 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[290] \quad 31612^1 32181232171232181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 42, T \in [28, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[289] \quad 1612321812^1 321712321812 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[288] \quad 1612321812321712^1 321812 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 27.06 \rightarrow 28, T_{\max} = 34, T \in [28, 34]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[287] \quad 61232^1 181232181232171 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

- $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [286] $612321^1 81232^1 1812321^1 71 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [285] $61232171232^1 1812321^1 81 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 16 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [284] $61232^1 1712321812321^1 81 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 34$, $T \in [28, 34]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [243] $8122315132181232^1 1^1 81 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.06 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [321] $1^1 81232181232^1 1812315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [330] $21812321^1 812^1 3218123151 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [313] $71231513221^1 (12)^1 12 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 49$, $\Delta = -118$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 49$, $T \in [27, 49]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [314] $1^1 (12)12^1 2315131613161232 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 49$, $\Delta = -118$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 49$, $T \in [27, 49]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [315] $1^1 81232181232^1 18122315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [316] $1^1 8^1 12^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [317] $1^1 8^1 1232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 39$, $T \in [27, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [318] $81223151321812^1 321^1 81 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 39$, $T \in [27, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [319] $1^1 1812321812^1 3218123215 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$

- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [320] $1812321812^1 3218123151 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [322] $23216131613151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 49$, $\Delta = -118$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 49$, $T \in [27, 49]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [323] $22181232181232^1 18122315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
 [324] $322181232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 39$, $T \in [27, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 [325] $613151321812^1 3218123215 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 37$, $\Delta = -113$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 37$, $T \in [27, 37]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [326] $613151321812^1 3218123151 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 35$, $\Delta = -111$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 35$, $T \in [27, 35]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [327] $81223151321812^1 32181223 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 39$, $T \in [27, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [328] $1812321812321812^1 31513 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 42$, $T \in [27, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [344] $4132181232^1 181232161316 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 39$, $T \in [27, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
 [343] $4132181232^1 181231513217 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 39$, $T \in [27, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
 [342] $712321812321812^1 321613 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 42$, $T \in [27, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [341] $612321812^1 321812321712 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
 [340] $1712321812^1 32181232161 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [339] $712321712321812^1 3217123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 42$, $T \in [27, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

- [338] $612321712^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [337] $1712321712^1 321812321712 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [336] $3161232^1 181232171232181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 42$, $T \in [27, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [335] $161232181232^1 1712321812 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [334] $612321812^1 32181232171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [333] $612321712^1 32181232181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [332] $171232181232^1 171232171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [331] $1812321712^1 32181232161 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [329] $21812321812^1 3218123215 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 64$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.94 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [360] $31612321812321712^1 32181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 42$, $T \in [27, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [361] $61232181232171232^1 18123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 45$, $T \in [27, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [346] $21 \begin{pmatrix} 1513221 \\ 12 \end{pmatrix} 12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [347] $1 \begin{pmatrix} 13221 \\ 12 \end{pmatrix} 1223151322^1 18123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 27$, $\Delta = -95$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 27$, $T \in [27, 27]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [348] $71223151322^1 1 \begin{pmatrix} 1513221 \\ 12 \end{pmatrix} 12 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [349] $321812^1 231513221 \begin{pmatrix} 12 \\ 12 \end{pmatrix} 2313 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -103$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[350] \quad 21(12)122^1 1315131613161 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 42, T \in [27, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[351] \quad 1812321812315132^1 18123 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 39, T \in [27, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[362] \quad 71232^1 17123218123217123 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 42, T \in [27, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[352] \quad 413221(12)12^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[353] \quad 181232^1 1812315132181232 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 45, \Delta = -117, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[354] \quad 21812321812315132^1 18123 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 39, T \in [27, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[355] \quad 181232^1 181232181231513 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 42, T \in [27, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[356] \quad 321812^1 321812321812315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 45, \Delta = -117, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[357] \quad 413221(12)122^1 31513161315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 49, \Delta = -118, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 49, T \in [27, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[358] \quad 413221(12)12^1 23151321812 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[359] \quad 413221(12)12^1 23151322181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[363] \quad 4141322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 38, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.84 \rightarrow 27, T_{\max} = 38, T \in [27, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

- [364] $41413221(12)12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [365] $71232^1 1812321812321613 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 42$, $T \in [27, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [366] $71223151322^1 1(12)12231414 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [367] $712231513221(12)12^1 231414 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [368] $1613161316131612^1 32181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [369] $21613161316131612^1 32181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [370] $4141413221(12)12^1 231513161 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [371] $41414132^1 21(12)12231513161 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [372] $4141413221(12)12^1 231513216 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 39$, $T \in [27, 39]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [373] $41414132^1 21(12)12231513216 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 39$, $T \in [27, 39]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [345] $1^{5131513221}(12)12^1 23151315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.84 \rightarrow 27$, $T_{\max} = 42$, $T \in [27, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [380] $812315132181232^1 1^8 1 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 100$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27$, $T_{\max} = 45$, $T \in [27, 45]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [382] $221812321812321812^1 2315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -108$, $\det = 100$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[381] $8122315132^1 1812321 \overset{321}{8} 1 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 39, T \in [27, 39]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[383] $3221812321812^1 315132181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 39, T \in [27, 39]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[384] $61315132^1 18123218123215 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 37, \Delta = -113, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 37, T \in [27, 37]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[378] $1812321812321812^1 2315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -108, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[377] $23218122^1 31513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 28, \Delta = -97, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 28, T \in [27, 28]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[376] $712315132^1 21 \overset{31513221}{(12)} 12 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 49, \Delta = -118, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 49, T \in [27, 49]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[375] $1 \overset{2321}{8} 12^1 31513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 28, \Delta = -97, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 28, T \in [27, 28]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[374] $(12)122^1 314141414141414141 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 34, \Delta = -107, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 34, T \in [27, 34]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 189 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[385] $61315132^1 18123218123151 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 35, \Delta = -111, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 35, T \in [27, 35]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[386] $8122315132^1 181232181223 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 39, T \in [27, 39]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[387] $181232181232^1 181231513 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 42, T \in [27, 42]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[388] $31612321812^1 32171232181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 26.76 \rightarrow 27, T_{\max} = 42, T \in [27, 42]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[389] $71232171232^1 18123217123 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} =$

15.0

$$T_{\min} = 26.76 \rightarrow 27, T_{\max} = 42, T \in [27, 42]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[390] \quad 1 \quad 71232181232^1 1812321613 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.76 \rightarrow 27, T_{\max} = 42, T \in [27, 42]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[391] \quad 41321812321812^1 31513217 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.76 \rightarrow 27, T_{\max} = 39, T \in [27, 39]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[392] \quad 41321812321812^1 32161316 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.76 \rightarrow 27, T_{\max} = 39, T \in [27, 39]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[379] \quad 1 \quad 8 \quad 12321812^1 315132181 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 100, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.76 \rightarrow 27, T_{\max} = 39, T \in [27, 39]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[412] \quad 4132181232181232^1 161316 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.69 \rightarrow 27, T_{\max} = 39, T \in [27, 39]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[395] \quad 21(12)12^1 2315131613161 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.69 \rightarrow 27, T_{\max} = 42, T \in [27, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[396] \quad 1 \quad 1812321812^1 31513218123 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.69 \rightarrow 27, T_{\max} = 39, T \in [27, 39]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[397] \quad 8123151321812^1 321 \quad 8 \quad 1 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 45, \Delta = -117, \det = 121, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.69 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[398] \quad 21812321812^1 31513218123 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 39, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.69 \rightarrow 27, T_{\max} = 39, T \in [27, 39]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[399] \quad 1 \quad 1812321812^1 32181231513 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 42, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.69 \rightarrow 27, T_{\max} = 42, T \in [27, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[400] \quad 32181232181232^1 1812315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 45, \Delta = -117, \det = 121, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.69 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[401] \quad 1 \quad 413221(12)12^1 231513161315 + \hat{1} \rightarrow 2 \quad T = 22, T_{\max} = 49, \Delta = -118, \det = 121, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 26.69 \rightarrow 27, T_{\max} = 49, T \in [27, 49]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

- [402] $81232^1 1812321812315132 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 45$, $T \in [27, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [403] $812321812321812^1 315132 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 45$, $T \in [27, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [405] $612321812^1 3217123218123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 45$, $T \in [27, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [407] $712321812^1 321812321613 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 42$, $T \in [27, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [408] $81232^1 16131613161316^1 1 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [409] $414141322^1 1(12)12231513161 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 38$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 38$, $T \in [27, 38]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [410] $81232^1 16131613161316123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 39$, $T \in [27, 39]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [411] $414141322^1 1(12)12231513216 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 39$, $\Delta = -115$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 39$, $T \in [27, 39]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [393] $(12)12^1 2314141414141414141 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 34$, $\Delta = -107$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 34$, $T \in [27, 34]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 230 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$
- [394] $71231513221^1 (12) 12 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 49$, $\Delta = -118$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 49$, $T \in [27, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [406] $712321712^1 3218123217123 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 42$, $T \in [27, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [404] $3161232181232^1 171232181 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 42$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.69 \rightarrow 27$, $T_{\max} = 42$, $T \in [27, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [413] $32181232^1 1812321812315 + \hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 144$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 26.64 \rightarrow 27$, $T_{\max} = 45$, $T \in [27, 45]$
 $\begin{pmatrix} 0 & 12 & 2 \\ 12 & 59 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[414] 612321812321712¹3218123 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 144$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.64 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[415] 7123151322¹1³¹⁵¹³²²¹(12) 12 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 49$, $\Delta = -118$, $\det = 169$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.59 \rightarrow 27, T_{\max} = 49, T \in [27, 49]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[416] 2321812¹231513221(12)12231 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 28$, $\Delta = -97$, $\det = 169$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.59 \rightarrow 27, T_{\max} = 28, T \in [27, 28]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[417] 812315132¹1812321²³²¹8 1 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 169$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.59 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[418] 1812321812315132¹181232 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 169$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.59 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[419] 321812321812¹321812315 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 169$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.59 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 64 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[420] 81232181232¹1812315132 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 169$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.59 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[421] 61232181232¹17123218123 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 169$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.59 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[422] 812321812¹321812315132 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 196$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.55 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[423] 1812321812¹315132181232 + $\hat{1} \rightarrow 2$ $T = 22$, $T_{\max} = 45$, $\Delta = -117$, $\det = 225$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 26.52 \rightarrow 27, T_{\max} = 45, T \in [27, 45]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$T = 23$ [424] 181231513221¹(12)1223151 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -4$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[425] 1(12)1223151321812315132¹2 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -4$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30, T_{\max} = 50, T \in [30, 50]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[426] 1(12)122315132181231513 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -4$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 30, T_{\max} = 46, T \in [30, 46]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[427] \quad 1^{22^1 21}(12)12231513218123215 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 36, \Delta = -111, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[428] \quad 1^{22^1 21}(12)12231513218123151 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30, T_{\max} = 35, T \in [30, 35]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[429] \quad 3221^{2^1 1}(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 29, T_{\max} = 50, T \in [29, 50]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[430] \quad 1513221^{2^1 1}(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[431] \quad 21^{22^1 1}(12)12231513218123215 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 36, \Delta = -111, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[432] \quad 21^{22^1 1}(12)12231513218123151 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30, T_{\max} = 35, T \in [30, 35]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[433] \quad 1513221^{2^1 1}(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[434] \quad 22^{2^1 1}315132181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30, T_{\max} = 50, T \in [30, 50]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[435] \quad 2315132181231513221(12)12^{2^1 1} + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30, T_{\max} = 50, T \in [30, 50]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[436] \quad 315132181231513221(12)122^{2^1 1} + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30, T_{\max} = 46, T \in [30, 46]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[437] \quad 51232181231513221(12)122^{2^1 1} + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 36, \Delta = -111, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[438] 15132181231513221(12)1222¹2+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -4$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30, T_{\max} = 35, T \in [30, 35]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[439] 413161316131513221²²1(12)1+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -4$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30, T_{\max} = 39, T \in [30, 39]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[440] 1712321712321812321812¹2+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -4$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30, T_{\max} = 35, T \in [30, 35]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[441] 413161316131513221(12)122¹2+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -4$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30, T_{\max} = 39, T \in [30, 39]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[442] 612321812321812321812¹2+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -4$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30, T_{\max} = 35, T \in [30, 35]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[449] 812231513221³¹⁵¹³²²¹(12)12¹+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.33 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[447] 513221²1(12)12231513218123+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.33 \rightarrow 29, T_{\max} = 50, T \in [29, 50]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[460] 712321812321812321712¹3+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[459] 612321812321812321812¹3+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 43$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 43, T \in [30, 43]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[458] 7123217123218123218122¹3+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[457] 413161316131513221(12)12¹22+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 39, T \in [30, 39]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[456] 32¹171232181232171232181+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[455] 1712321812321712321812¹3 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[454] 712321712321812321³²¹8 1 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[453] 4132181231513221(12)122¹313 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -102$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 35, T \in [30, 35]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[452] 15132181231513221(12)122¹22 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 35, T \in [30, 35]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[451] 51232181231513221(12)122¹22 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 36$, $\Delta = -111$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[450] 315132181231513221(12)12¹22 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 46, T \in [30, 46]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[448] 21⁵¹³²²¹(12) 1223151321812¹3 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 50, T \in [30, 50]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[446] 21³²¹21(12) 1223151321812315 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 50, T \in [30, 50]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[445] 1²²²¹(12) 12231513218123151 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 35, T \in [30, 35]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[444] 1²²²¹(12) 12231513218123215 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 36$, $\Delta = -111$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.22 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[443] 31513221²¹(12) 122315132181 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -9$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.33 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[465] 512¹32181231513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 36$, $\Delta = -111$, $\det = -16$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[466] 15132181231513221(12)12¹222 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[464] 51232181231513221(12)12¹222+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 36, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[463] 81231513221²³¹⁵¹³²²¹(12) 12¹+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 50, \Delta = -118, \det = -16, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 6 \\ 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[462] 21²²¹(12)122315132181232¹15 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 36, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[461] 1²²²¹(12)122315132181232¹15 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 36, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[469] 1712¹32181232171232181 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 35, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[471] 181232171232181232¹1712 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 35, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[467] 4132181231513221(12)12¹2313+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 35, \Delta = -102, \det = -16, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[468] 712315132181232181232¹15 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 37, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 37, T \in [29, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[470] 1712¹321812321712321812 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 35, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[472] 1812¹32181232171232171 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 35, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[473] 181232181232181232¹161 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 35, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.75 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

- [474] $1712^1 132181232181232171 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.75 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [475] $61232181232181232^1 181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.75 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [476] $171232171232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.75 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [477] $21712^1 321812321712321812 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.75 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [478] $613161232181232181232^1 15 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 37$, $\Delta = -112$, $\det = -16$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.75 \rightarrow 29$, $T_{\max} = 37$, $T \in [29, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [479] $61315132181232181232^1 161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -110$, $\det = -16$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.75 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [480] $2181232181232181232^1 161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.75 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [481] $171232181232181232^1 1712 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.75 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [482] $61232181232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.75 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [532] $712321712321812321812^1 23 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [499] $612^1 31513221^{5131513221} (12) 1 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 39$, $T \in [29, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [509] $171232^1 181232171232181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[533] $812321712321812321712^1 32 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[508] $171232181232171232^1 181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[507] $7123151321812321812^1 3151 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[506] $7123151321812321812^1 3215 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 37$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 37, T \in [29, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[505] $41321812315132^1 21(12)122313 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[504] $4132^1 181231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[503] $151321812315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[502] $15132^1 181231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[501] $512321812315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 36$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[500] $51232^1 181231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 36$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[511] $171232181232171232^1 1812 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[498] $21 \overset{1513221}{(12)} 122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

- [497] $^{15132^1 21}$ 21 (12) 1223151322181 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [496] 221 21(12)1223151321812¹3151 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [495] 221 21(12)122¹31513218123151 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [494] 221 21(12)122¹31513218123215 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 36$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [493] 221 21(12)1223151321812¹3215 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 36$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [492] $^{15132^1 21}$ 21 (12) 1223151321812 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [491] 1513221 21 (12) 122315132¹1812 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [490] $^{322^1 1}$ 21(12)1223151321812315 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 50$, $T \in [29, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [489] 221 1(12)122¹31513218123151 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [488] 221 1(12)1223151321812¹3151 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [487] 221 1(12)122¹31513218123215 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 36$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [486] 221 1(12)1223151321812¹3215 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 36$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 36, T \in [29, 36] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[485] & 1 \quad \begin{pmatrix} 15132 & 1 & 21 \\ 12 & 12231513161315 & +\hat{1} \rightarrow 2 \end{pmatrix} \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 38, T \in [29, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[484] & 1 \quad \begin{pmatrix} 1812 & 1 & 31513221 & 21 \\ 12 & 1223151 & +\hat{1} \rightarrow 2 \end{pmatrix} \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[483] & 1 \quad \begin{pmatrix} 181231513221 & 21 \\ 12 & 122 & 13151 & +\hat{1} \rightarrow 2 \end{pmatrix} \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[512] & 1 \quad \begin{pmatrix} 1812 & 1 & 321712321812321712 & +\hat{1} \rightarrow 2 \end{pmatrix} \quad T = 23, T_{\max} = 35, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[513] & 1 \quad \begin{pmatrix} 1812321712321812 & 1 & 321712 & +\hat{1} \rightarrow 2 \end{pmatrix} \quad T = 23, T_{\max} = 35, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[514] & 1 \quad \begin{pmatrix} 18 & 1232 & 1 & 181232171232171 & +\hat{1} \rightarrow 2 \end{pmatrix} \quad T = 23, T_{\max} = 35, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[515] & 1 \quad \begin{pmatrix} 18 & 1232181232171232 & 1 & 171 & +\hat{1} \rightarrow 2 \end{pmatrix} \quad T = 23, T_{\max} = 35, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[516] & 1 \quad \begin{pmatrix} 1812 & 1 & 32181232181232161 & +\hat{1} \rightarrow 2 \end{pmatrix} \quad T = 23, T_{\max} = 35, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[517] & 1 \quad \begin{pmatrix} 1812321812321812 & 1 & 32161 & +\hat{1} \rightarrow 2 \end{pmatrix} \quad T = 23, T_{\max} = 35, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[518] & 1 \quad \begin{pmatrix} 41413221 & 1 & 122315132 & 1 & 181 & +\hat{1} \rightarrow 2 \end{pmatrix} \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 38, T \in [29, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[519] & 1 \quad \begin{pmatrix} 171232181232181232 & 1 & 171 & +\hat{1} \rightarrow 2 \end{pmatrix} \quad T = 23, T_{\max} = 35, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [520] $171232^1 181232181232171 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [521] $612^1 32181232181232181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [522] $612321812321812^1 32181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [523] $1712^1 3218123217123218123 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [534] $712321812321712321812^1 32 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 46$, $T \in [29, 46]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [535] $21812^1 32181232181232161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [536] $21812321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [537] $1712321812321812^1 321712 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [524] $1712321712321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [538] $1712^1 321812321812321712 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [525] $1712^1 3217123218123218122 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [539] $41413221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 38$, $T \in [29, 38]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[540] $41413221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[541] $612^1 3218123218123218123 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 43$, $\Delta = -116$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 43, T \in [29, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[542] $4132^1 21(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[543] $612^1 3218123218123218122 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[544] $612321812321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[545] $4132^1 1812321812315132218 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[546] $4141413221(12)122315132^1 171 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[547] $4141413221(12)122315132^1 217 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[526] $2171232181232171232^1 1812 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[527] $2171232^1 1812321712321812 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[528] $4132^1 1812321812315132181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.44 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[529] $6131612321812321812^1 3215 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 37$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 37$, $T \in [29, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [530] $6131612321812321812^1 3151 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [531] $613151321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- 1
 [510] $171232^1 1812321712321812 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.44 \rightarrow 29$, $T_{\max} = 35$, $T \in [29, 35]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [575] $41321812321812315132^1 218 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -36$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 39$, $T \in [29, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- 221
 [549] $1(12)122^1 315132181231513 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 46$, $T \in [29, 46]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 1
 [574] $712^1 3218123218123217123 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -36$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [576] $4141413221(12)1223151322^1 17 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -36$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 38$, $T \in [29, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- 1
 [573] $41413221(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -36$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 38$, $T \in [29, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- 1
 [572] $1812^1 321812321812321613 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -36$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [571] $712^1 32181232171232181232 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 46$, $T \in [29, 46]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [570] $712^1 32171232181232181223 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -36$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.22 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [569] $41321812321812315132^1 181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -36$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[568] 4131613161315132¹21(12)1222+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 39, \Delta = -114, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[567] 32171232181232171232¹181+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 40, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[566] 712¹321712321812321³²¹8 1+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 40, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[565] 4131613161315132²¹12(12)12+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 39, \Delta = -114, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[564] 4131613161315132²²¹21(12)1+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 39, \Delta = -114, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[563] 712¹31513218123218123151+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 35, \Delta = -110, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[562] 712¹31513218123218123215+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 37, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 37, T \in [29, 37]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[561] 413218123151322¹1(12)122313+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 35, \Delta = -102, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[560] 321613161315132¹21(12)12231+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 29, \Delta = -96, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[559] 1513218123151322¹1(12)12222+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 35, \Delta = -109, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[558] 5123218123151322¹1(12)12222+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 36, \Delta = -111, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[557] 3151321812315132¹21(12)1222+ $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 46, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} =$

15.0

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[556] \quad 8122^{\overset{31513221}{1}}1513221 \quad (12) \quad 12 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[555] \quad 21 \quad (12) \quad 1223151322^{\overset{1513221}{1}}181 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[554] \quad 21 \quad (12) \quad 12^{\overset{221}{1}}231513218123151 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[553] \quad 21 \quad (12) \quad 12^{\overset{221}{1}}231513218123215 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 36, \Delta = -111, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[552] \quad 21 \quad (12) \quad 122^{\overset{21}{1}}315132181231513 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[551] \quad 1 \quad (12) \quad 12^{\overset{2221}{1}}231513218123151 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[550] \quad 1 \quad (12) \quad 12^{\overset{2221}{1}}231513218123215 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 36, \Delta = -111, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[548] \quad 21 \quad (12) \quad 122315132^{\overset{31513221}{1}}181 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -36, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28.22 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[579] \quad 1 \quad (12) \quad 12231513161 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[577] \quad 181231513221 \quad (12) \quad 12^{\overset{21}{1}}23151 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[633] \quad 41316131613151322^{\overset{1}{1}}1(12)1222 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 39, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[632] 414132¹21(12)12231513161315 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[631] 21712321812321712¹321812 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[630] 21712321812¹321712321812 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[629] 171232171232¹18123218122 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[628] 171232¹17123218123218122 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[627] 321712¹32181232171232181 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[626] 171232¹18123217123218123 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[625] 61232181232¹1812321²¹81 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[624] 61232¹1812321812321²¹81 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[578] 1812315132¹21²¹(12)1223151 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[623] 1712321812321812¹32171 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[622] 1712321812¹32181¹232171 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[621] $414132^1 21(12)122315132181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[620] $41413221(12)122^1 315132181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[619] $181232^1 181232181232161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[618] $181232181232^1 181232161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[617] $71232171232181232^1 1^3 81 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[616] $41316131613151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[615] $41316131613151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[614] $1812321812^1 32171232171 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[613] $1812321812321712^1 32171 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[612] $181232^1 1712321812321712 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[611] $181232171232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[610] $\overset{1}{1712321812^1 321712321812} + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[609] $\overset{1}{1712321812321712^1 321812} + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$

[608] $\overset{1}{1712321812^1 32171232181} + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[607] $\overset{1}{1712321812321712^1 32181} + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$

[606] $\overset{1}{712315132181232^1 18123151} + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[605] $\overset{1}{712315132181232^1 18123215} + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 37, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 37, T \in [29, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[604] $\overset{1}{41321812^1 31513221(12)122313} + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -102, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[603] $\overset{1}{3216131613151322^1 1(12)12231} + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 29, \Delta = -96, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$

[602] $\overset{1}{151321812^1 31513221(12)12222} + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[601] $\overset{1}{512321812^1 31513221(12)12222} + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 36, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[600] $\overset{1}{31513218123151322^1 1(12)1222} + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$

[599] $\overset{1}{23151321812315132^1 21(12)122} + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 50, T \in [29, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$

- [598] $812^{1231513221} (12) 12 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [597] $812^{131513221} (12) 12 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 50, T \in [29, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [596] $612315132^{121} (12) 1 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 39, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [595] $61231513221^{51315132^{121}} (12) 1 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 39, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [594] $21^{151322^{121}} (12) 1223151322181 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [593] $21^{1513221} (12) 122^{13151322181} + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [592] $21^{221} (12) 122315132^{18123151} + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [591] $21^{221} (12) 122315132^{18123215} + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 36, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [590] $21^{151322^{121}} (12) 1223151321812 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [589] $21^{1513221} (12) 122^{13151321812} + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [588] $21^{21} (12) 12^{12315132181231513} + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [587] $21^{5132^{121}} (12) 12231513218123 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 50, T \in [29, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

- [586] $^{3221} 21(12)1223151321812^1 315 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 50, T \in [29, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [585] $^{2221} 1(12)122315132^1 18123151 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [584] $^{2221} 1(12)122315132^1 18123215 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 36, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [635] $812^1 31513161316131613151 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [636] $613161232181232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 37, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 37, T \in [29, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [637] $613161232181232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [638] $61315132181232^1 181232161 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [639] $71232171232181232^1 181223 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 40, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [640] $812^1 32171232181232171232 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [641] $^{1} 2181232181232^1 181232161 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [642] $^{1} 2181232^1 181232181232161 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [643] $^{1} 181232181232181232^1 1613 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 40, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.06 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[644] $1712321^{181232^1}1812321712 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[645] $1712321^{1812321812321}1812321712 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[646] $414132^{121(12)1223151321812} + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[647] $41413221^{1(12)122^13151321812} + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[648] $41413221^{1(12)122^13151322181} + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[649] $414132^{121(12)1223151322181} + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[650] $61232^{18123218123218123} + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 43$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 43, T \in [29, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[651] $41322^{11(12)122315131613161} + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[652] $61232181232^{18123218122} + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[653] $61232^{18123218123218122} + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 35, T \in [29, 35]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[654] $71232181232181232^{117123} + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[655] 413218123218123151322¹18 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[656] 41321812¹321812315132218 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[657] 4141413221(12)122¹315132171 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[658] 4141413221(12)122¹315132217 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[581] 1¹⁵¹³²²(12) 12231513161315 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[659] 413161316131613161232¹18 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[580] 1⁵¹³¹⁵¹³²²¹(12) 122¹31513161 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[583] 1²²¹(12)12¹2315132181231513 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 46, T \in [29, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[582] 1²¹(12)122¹3151321812315132 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 50, T \in [29, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[634] 41321812¹321812315132181 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -49$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.06 \rightarrow 29, T_{\max} = 39, T \in [29, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[662] 1²¹(12)12¹23151321812315132 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[664] 21²¹(12)1223151321812¹31513 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

- [661] $21 \quad (12) \quad 122315132181 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [663] $1(12)1223151321812^1 31513 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [692] $4132181232^1 1812315132218 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 39, \Delta = -114, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [691] $71232181^1 2321812^1 3217123 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 40, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [690] $612321812^1 3218123218122 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [689] $1712321812^1 321812321712 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -106, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [688] $1812321812321812^1 321613 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 40, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [687] $21812321812^1 32181232161 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [686] $712321712321812^1 32181223 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 40, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [685] $613151321812^1 32181232161 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -110, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [684] $6131612321812^1 3218123151 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -110, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [683] $6131612321812^1 3218123215 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 37, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.94 \rightarrow 28, T_{\max} = 37, T \in [28, 37]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[682] $4132181232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[681] $2171232181232^1 1712321812 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[680] $1712321712^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[679] $32171232^1 181232171232181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[678] $612321812^1 321812321^1 81 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[677] $171232181232^1 181232171 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[676] $1812321812^1 32181232161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -107$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[675] $712321712321812^1 321^1 81 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[674] $181232181232^1 171232171 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[673] $1812321712^1 321812321712 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[672] $171232181232^1 1712321812 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[671] $171232181232^1 171232181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -106$, $\det = -64$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[670] 7123151321812¹3218123151 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 35, \Delta = -110, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[669] 7123151321812¹3218123215 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 37, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 37, T \in [28, 37]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[668] 315132¹181231513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 46, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[667] 231513218123151322¹1(12)122 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 50, \Delta = -118, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[666] 223151321812315132¹21(12)12 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 50, \Delta = -118, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[665] 812231513221^{315132¹21}(12) 12 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 46, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[660] 1¹(12)122¹31513218123151322 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 50, \Delta = -118, \det = -64, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.94 \rightarrow 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[713] 71232¹171232181232181223 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 40, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.84 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[701] 21¹⁵¹³²²¹(12) 12¹23151322181 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 35, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.84 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[718] 4141322¹1(12)1223151322181 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 38, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.84 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[719] 41413221¹(12)12¹23151322181 + $\hat{1} \rightarrow 2$ $T = 23, T_{\max} = 38, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 27.84 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[720] $61232181232181232^1 18123 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 43, \Delta = -116, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 43, T \in [28, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[721] $413221(12)122^1 315131613161 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 40, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[722] $71232^1 18123218123217123 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 40, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

[723] $41414132^1 21(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[724] $4141413221(12)12^1 2315132171 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[725] $4141413221(12)12^1 2315132217 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[726] $41414132^1 21(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[716] $41413221(12)12^1 23151321812 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[693] $18123151322^1 1(12)1223151 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 35, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[694] $1^{5131513221} (12) 12^1 231513161 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[695] $1^{513151322^1 1} (12) 12231513161 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[696] $1^{1513221} (12) 122^1 31513161315 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

- [697] $1(12)12^1 231513218123151322 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 50$, $T \in [28, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [698] $21(12)122^1 3151321812315 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 50$, $T \in [28, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [699] $21(12)122315132^1 18123 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 50$, $T \in [28, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [712] $41413221(12)122^1 31513161315 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 46$, $T \in [28, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [711] $4141322^1 1(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 46$, $T \in [28, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [710] $321712321812321712^1 32181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 40$, $T \in [28, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [709] $171232181232171232^1 18123 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 40$, $T \in [28, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [708] $4141322^1 1(12)122315132181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [707] $41413221(12)12^1 2315132181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [714] $71232^1 181232171232181232 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 46$, $T \in [28, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [706] $71232^1 1712321812321^1 81 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 40$, $T \in [28, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [717] $4141322^1 1(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.84 \rightarrow 28$, $T_{\max} = 38$, $T \in [28, 38]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[715] $181232^1 1812321812321613 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.84 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[705] $1413221(12)1223151322^1 18123 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 28$, $\Delta = -94$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.84 \rightarrow 28, T_{\max} = 28, T \in [28, 28]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[704] $2231513218123151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.84 \rightarrow 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[703] $61231513221^{513151322^1 1} (12) 1 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.84 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[702] $6123151322^1 1^{5131513221} (12) 1 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.84 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[700] $21^{1513221} (12) 12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -109$, $\det = -81$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.84 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[735] $71232171232^1 1812321^{321} 8 1 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.76 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[744] $71232181232^1 18123217123 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.76 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[745] $41321812321812^1 315132218 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.76 \rightarrow 28, T_{\max} = 39, T \in [28, 39]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[743] $181232181232^1 1812321613 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.76 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[741] $71232171232^1 181232181223 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.76 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[740] $61315132^1 181232181232161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -110$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.76 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

- $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
 [739] 613161232¹18123218123151 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -110$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.76 \rightarrow 28$, $T_{\max} = 35$, $T \in [28, 35]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [738] 613161232¹18123218123215 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 37$, $\Delta = -112$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.76 \rightarrow 28$, $T_{\max} = 37$, $T \in [28, 37]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [737] 41321812321812¹315132181 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 39$, $\Delta = -114$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.76 \rightarrow 28$, $T_{\max} = 39$, $T \in [28, 39]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [736] 321712321812¹32171232181 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.76 \rightarrow 28$, $T_{\max} = 40$, $T \in [28, 40]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [734] 712315132¹18123218123151 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -110$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.76 \rightarrow 28$, $T_{\max} = 35$, $T \in [28, 35]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [733] 712315132¹18123218123215 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 37$, $\Delta = -112$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.76 \rightarrow 28$, $T_{\max} = 37$, $T \in [28, 37]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [732] 3151321812¹31513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.76 \rightarrow 28$, $T_{\max} = 46$, $T \in [28, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [731] 8122315132¹21³¹⁵¹³²²¹(12)12 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.76 \rightarrow 28$, $T_{\max} = 46$, $T \in [28, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [730] 21²¹(12)122315132¹181231513 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.76 \rightarrow 28$, $T_{\max} = 46$, $T \in [28, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [729] 21^{51322¹1}(12)12231513218123 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.76 \rightarrow 28$, $T_{\max} = 50$, $T \in [28, 50]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [728] 1²²¹(12)122315132¹181231513 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 27.76 \rightarrow 28$, $T_{\max} = 46$, $T \in [28, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [727] 21³¹⁵¹³²²¹(12)122¹315132181 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 27.76 \rightarrow 28, T_{\max} = 46, T \in [28, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[742] & 81232171232181232^1 171232 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -100, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.76 \rightarrow 28, T_{\max} = 46, T \in [28, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[757] & 32171232181232^1 171232181 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 40, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.69 \rightarrow 28, T_{\max} = 40, T \in [28, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[770] & 4131613161316131612^1 3218 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 38, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.69 \rightarrow 28, T_{\max} = 38, T \in [28, 38] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 21 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[748] & 1(12)1223151321812^1 315132 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -121, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.69 \rightarrow 28, T_{\max} = 50, T \in [28, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[749] & 21(12)122315132^1 1812315 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -121, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.69 \rightarrow 28, T_{\max} = 50, T \in [28, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[750] & 21(12)12^1 23151321812315 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -121, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.69 \rightarrow 28, T_{\max} = 50, T \in [28, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[751] & 21(12)122^1 31513218123 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -121, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.69 \rightarrow 28, T_{\max} = 50, T \in [28, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[752] & 81231513221^{2315132^1 21} (12) 12 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -121, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.69 \rightarrow 28, T_{\max} = 50, T \in [28, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[753] & 812231513221^{3151322^1 1} (12) 12 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -121, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.69 \rightarrow 28, T_{\max} = 46, T \in [28, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[754] & 2315132^1 181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -121, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.69 \rightarrow 28, T_{\max} = 50, T \in [28, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[755] & 712321712^1 321812321^1 8 1 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 40, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.69 \rightarrow 28, T_{\max} = 40, T \in [28, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[756] $1712321812^1 3217123218123 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[747] $21^1 (12) 122315132181 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[746] $1^1 (12) 12^1 231513161315 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[760] $6131612^1 3218123218123151 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 35$, $\Delta = -110$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 35, T \in [28, 35]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[761] $712321712^1 32181232181223 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[762] $812321712321812^1 32171232 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[763] $81232^1 171232181232171232 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[764] $1812321812^1 321812321613 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[758] $41413221(12)12^1 231513161315 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[765] $612321812^1 3218123218123 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 43$, $\Delta = -116$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 43, T \in [28, 43]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[766] $413221(12)12^1 2315131613161 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[767] $712321812^1 3218123217123 + \hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[768] 414141322¹1(12)122315132171+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[769] 414141322¹1(12)122315132217+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 38$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 38, T \in [28, 38]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[759] 6131612¹3218123218123215+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 37$, $\Delta = -112$, $\det = -121$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.69 \rightarrow 28, T_{\max} = 37, T \in [28, 37]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[771] 1712321812321712¹3218123+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -144$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.64 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[772] 612321812321812¹3218123+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 43$, $\Delta = -116$, $\det = -144$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.64 \rightarrow 28, T_{\max} = 43, T \in [28, 43]$$

$$\begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[773] 21³¹⁵¹³²²¹(12)¹12¹2315132181+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -169$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.59 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[774] 1²¹(12)122315132¹1812315132+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -169$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.59 \rightarrow 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[775] 812315132¹21²³¹⁵¹³²²¹(12)¹12+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -169$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.59 \rightarrow 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[777] 23151321812¹31513221(12)122+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 50$, $\Delta = -118$, $\det = -169$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.59 \rightarrow 28, T_{\max} = 50, T \in [28, 50]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[778] 171232181232¹17123218123+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 40$, $\Delta = -115$, $\det = -169$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.59 \rightarrow 28, T_{\max} = 40, T \in [28, 40]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[779] 81232171232¹181232171232+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -169$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.59 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[780] 71232181232171232¹181232+ $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -169$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 27.59 \rightarrow 28, T_{\max} = 46, T \in [28, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[781] & 61232181232^1 18123218123 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 43, \Delta = -116, \det = -169, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.59 \rightarrow 28, T_{\max} = 43, T \in [28, 43] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[776] & 81223151322^1 1^{31513221} (12) 12 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -169, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.59 \rightarrow 28, T_{\max} = 46, T \in [28, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[784] & 22315132^1 181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -196, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.55 \rightarrow 28, T_{\max} = 50, T \in [28, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[785] & 812321712^1 32181232171232 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -196, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.55 \rightarrow 28, T_{\max} = 46, T \in [28, 46] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[782] & 1^{12} (12) 1223151321812^1 3151322 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -196, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.55 \rightarrow 28, T_{\max} = 50, T \in [28, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[783] & 21^{513221} (12) 12^1 231513218123 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -196, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.55 \rightarrow 28, T_{\max} = 50, T \in [28, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[786] & 81231513221^{23151322^1 1} (12) 12 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -225, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.52 \rightarrow 28, T_{\max} = 50, T \in [28, 50] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[787] & 712321812^1 32171232181232 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 46, \Delta = -117, \det = -225, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.52 \rightarrow 28, T_{\max} = 46, T \in [28, 46] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[788] & 1^{12} (12) 122315132^1 18123151322 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -256, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.48 \rightarrow 28, T_{\max} = 50, T \in [28, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[789] & 223151321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -256, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.48 \rightarrow 28, T_{\max} = 50, T \in [28, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[790] & 8123151322^1 1^{231513221} (12) 12 + \hat{1} \rightarrow 2 \quad T = 23, T_{\max} = 50, \Delta = -118, \det = -289, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 27.46 \rightarrow 28, T_{\max} = 50, T \in [28, 50] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[791] 712321812321712¹32181232 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -289$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.46 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 203 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[792] 71232181232¹171232181232 + $\hat{1} \rightarrow 2$ $T = 23$, $T_{\max} = 46$, $\Delta = -117$, $\det = -324$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 27.43 \rightarrow 28, T_{\max} = 46, T \in [28, 46]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$T = 24$ [797] 61232181231513221³²²¹(12)¹12 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 4$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30, T_{\max} = 47, T \in [30, 47]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[804] 21812321712321812321812¹2 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 4$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31, T_{\max} = 36, T \in [31, 36]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[793] 1²²¹(12)1223151316131613151 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 4$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31, T_{\max} = 36, T \in [31, 36]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[794] 1²²¹²¹(12)122315132181232161 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 4$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31, T_{\max} = 36, T \in [31, 36]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[795] 21²²¹¹(12)122315132181232161 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 4$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31, T_{\max} = 36, T \in [31, 36]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[796] 1²²¹¹(12)1223151321812321613 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 4$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31, T_{\max} = 44, T \in [31, 44]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[798] 22¹31513161316131513221(12)1 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 4$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31, T_{\max} = 51, T \in [31, 51]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[799] 31513161316131513221(12)12¹2 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 4$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31, T_{\max} = 47, T \in [31, 47]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[800] 1513161316131513221(12)122¹2 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 4$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31, T_{\max} = 36, T \in [31, 36]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[801] 161232181231513221(12)122¹2 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 4$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31, T_{\max} = 36, T \in [31, 36]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[802] 3161232181231513221(12)122¹2 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 4$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 31, T_{\max} = 44, T \in [31, 44]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[803] \overset{1}{1812321712321812321812} \overset{32^1}{12} + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31, T_{\max} = 36, T \in [31, 36]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[805] \overset{1}{1712321812321812321812} \overset{32^1}{12} + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31, T_{\max} = 36, T \in [31, 36]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[815] \overset{1}{712321812321812321} \overset{32^1}{8 \ 1} + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 40, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 40, T \in [31, 40]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[806] \overset{1}{51232181231513221(12)122^1} 313 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 32, \Delta = -99, \det = 9, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 32, T \in [31, 32]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[807] \overset{32^1}{1(12)122315131613161315} + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 51, \Delta = -118, \det = 9, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 51, T \in [31, 51]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[808] \overset{222^1}{1(12)122315132181232161} + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -108, \det = 9, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 36, T \in [31, 36]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[809] \overset{32^1}{61232181231513221(12)12} + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 47, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 47, T \in [31, 47]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[810] \overset{1}{1513161316131513221(12)12^1} 22 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 36, T \in [31, 36]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[811] \overset{1}{161232181231513221(12)122^1} 22 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -108, \det = 9, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 36, T \in [31, 36]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[817] \overset{32^1}{32^1 1812321712321812321812} + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 40, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 40, T \in [31, 40]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[812] \overset{1}{3161232181231513221(12)12^1} 22 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 44, \Delta = -116, \det = 9, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 44, T \in [31, 44]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[818] \overset{1}{1812321812321812321712^1} 3 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 40, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 40, T \in [31, 40]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

- [819] $32^1 181232181232181232171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 40$, $T \in [31, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [820] $7123218123218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 40$, $T \in [31, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [816] $18123217123218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 40$, $T \in [31, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [813] $1812321812321712321812^1 3 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 40$, $T \in [31, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [814] $1^3 8^1 1232181232171232181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 40$, $T \in [31, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [824] $1^{2221} (12) 122315132181232^1 161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.75 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [843] $613161232181232181232^1 161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 16$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.75 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [825] $21^{221} (12) 122315132181232^1 161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.75 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [821] $512^1 32181231513221 (12) 122313 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 32$, $\Delta = -99$, $\det = 16$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.75 \rightarrow 30$, $T_{\max} = 32$, $T \in [30, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [822] $51232181231513221 (12) 12^1 2313 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 32$, $\Delta = -99$, $\det = 16$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.75 \rightarrow 30$, $T_{\max} = 32$, $T \in [30, 32]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [823] $15132181231513221 (12) 12^1 2313 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 30$, $\Delta = -97$, $\det = 16$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.75 \rightarrow 30$, $T_{\max} = 30$, $T \in [30, 30]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [826] $161232181231513221 (12) 12^1 222 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.75 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [827] $1612^1 32181231513221 (12) 12222 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[828] & 512^1 321812321812315132181 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 37, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[829] & 8122315132181232181232^1 15 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 37, \Delta = -111, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[830] & 181232171232181232^1 181 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[831] & 181232171232181232^1 18122 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[832] & 1812^1 321812321712321812 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[833] & 61315132181232181232^1 171 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[834] & 181232181232181232^1 171 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[835] & 181232181232181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[836] & 1712^1 321812321812321812 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[837] & 1812^1 32181232181232171 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[838] & 2181232171232181232^1 18122 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[839] & 61315132181232181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[840] 712315132181232181232¹161 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 16$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[841] 171232181232181232¹18122 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 16$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[842] 2181232181232181232¹1712 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 16$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.75 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[853] 1²²²¹(12)122¹315132181232161 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[880] 171232181232181232¹1812 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[845] 7122¹31513221⁵¹³¹⁵¹³²²¹(12) 1 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 38, T \in [30, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[846] 51232¹181231513221(12)122313 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 32$, $\Delta = -99$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 32, T \in [30, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[847] 512321812315132¹21(12)122313 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 32$, $\Delta = -99$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 32, T \in [30, 32]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[848] 15132¹181231513221(12)122313 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 30$, $\Delta = -97$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[849] 151321812315132¹21(12)122313 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 30$, $\Delta = -97$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[850] 1^{15132¹21}(12) 122315131613161 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[851] 1^{322¹1}(12)122315131613161315 + $\hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 51, T \in [30, 51]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

- [854] $21^{221}(12)122^1 315132181232161 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [852] $1^{2221}(12)1223151321812^1 32161 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [855] $21^{221}(12)1223151321812^1 32161 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [903] $4132^1 18123218123217123217 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [902] $6131612321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [901] $4141413221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 38, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30, T_{\max} = 38, T \in [30, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [900] $4141413221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 38, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30, T_{\max} = 38, T \in [30, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [899] $4141413221(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 38, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30, T_{\max} = 38, T \in [30, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [856] $61232181231513221^{322^1 1}(12)12 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 47, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30, T_{\max} = 47, T \in [30, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [857] $612^1 32181231513221^{3221}(12)12 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 47, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30, T_{\max} = 47, T \in [30, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [858] $1612321812315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [859] $161232^1 181231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[860] $51232^1 1812321812315132181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 37$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[861] $15132^1 1812321812315132181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[862] $81223151321812321812^1 3215 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 37$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[863] $81223151321812321812^1 3151 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[864] $4132^1 21(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 47, T \in [30, 47]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[865] $612^1 31513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 37$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[866] $612^1 31513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 38, T \in [30, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[867] $1812321712321812^1 32181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[868] $1812^1 32171232181232181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[869] $1812^1 3218123217123218123 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[870] $1812321712321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[871] $1812^1 3217123218123218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[872] $181232181232171232^1 1812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[873] $181232^1 1812321712321812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[874] $613151321812321812^1 32171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[875] $812321712321812321^{232^1} 8^1 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[876] $1812321812321812^1 32171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[877] $1812^1 32181232181232171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[878] $1812^1 321812321812321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[879] $1812321812321812^1 321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[898] $812321812321812321712^1 32 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[897] $712321812321812321812^1 23 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[896] $712321812321812321812^1 32 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[895] $21812^1 321812321812321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[894] $21812321812321812^1 321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[893] $1712^1 3218123218123218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[892] $1712321812321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[891] $32181232181232181232^1 171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[890] $7123151321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[889] $613151321812321812^1 321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[888] $21812^1 3217123218123218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[887] $21812321712321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[886] $32181232171232181232^1 1812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[885] $232^1 181232171232181232181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[884] $1812321712321812321812^1 23 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[883] $1^1 81232181232181232^1 171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.44 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
 [882] $181232^1 181232181232171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [881] $171232^1 1812321812321812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [844] $1^{5131513221} (12) 122315132^1 171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.44 \rightarrow 30$, $T_{\max} = 38$, $T \in [30, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [904] $712^1 231513221^{5131513221} (12) 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 38$, $T \in [30, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [905] $3151321812315132^1 21(12)12231 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 30$, $\Delta = -97$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 30$, $T \in [30, 30]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [906] $5123218123151322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 32$, $\Delta = -99$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 32$, $T \in [30, 32]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [907] $1513218123151322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 30$, $\Delta = -97$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 30$, $T \in [30, 30]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [908] $1^{221} (12)122^1 3151316131613151 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [909] $21^{21} (12)122^1 3151316131613151 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [910] $1^{2221} (12)12^1 2315132181232161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [911] $21^{221} (12)12^1 2315132181232161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [912] $1^{221} (12)122^1 3151321812321613 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 44$, $T \in [30, 44]$

- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [913] $21^{21}(12)122^1 3151321812321613 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [914] $15131613161315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [915] $1612321812315132^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [916] $31612321812315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [917] $512321812321812315132^1 181 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 37, \Delta = -111, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [918] $151321812321812315132^1 181 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [919] $8122^1 31513218123218123215 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 37, \Delta = -111, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [920] $8122^1 31513218123218123151 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [921] $1^1 8^{321} 1232181232171232^1 181 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
 [922] $712^1 3218^1 12321812321^1 8^1 1 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [923] $1812^1 32171232181232181223 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [924] $232181232171232181232^1 181 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 36, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$

- [928] $712^1 32181232181232181223 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [929] $4141413221(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 38$, $T \in [30, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [930] $4132181232181232171232^1 17 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [925] $712^1 315132181232181232161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [927] $712^1 32181232181232181232 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 44$, $T \in [30, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [926] $1812^1 3218123218123217123 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.22 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1000] $812^1 32181232181232171232 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 44$, $T \in [30, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [998] $2181232^1 1812321812321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [932] $1^{5131513221} (12) 122^1 315132171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 38$, $T \in [30, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [933] $7122315132^1 21^{5131513221} (12) 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 38$, $T \in [30, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [934] $712231513221^{51315132^1 21} (12) 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 38$, $T \in [30, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [935] $31513218123151322^1 1(12)12231 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 30$, $\Delta = -97$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 30$, $T \in [30, 30]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[936] $512321812^1 31513221(12)122313 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 32$, $\Delta = -99$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 32, T \in [30, 32]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[937] $151321812^1 31513221(12)122313 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 30$, $\Delta = -97$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[938] $1^{151322^1} (12) 122315131613161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[939] $1^{21}(12)122^1 31513161316131513 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 47, T \in [30, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1001] $4141413221(12)122^1 315132181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 38, T \in [30, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1002] $812^1 31513161316131613161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[940] $1^{221}(12)12^1 23151316131613151 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[941] $1^{5132^1 21} (12) 1223151316131613 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 51, T \in [30, 51]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[942] $21^{21}(12)12^1 23151316131613151 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[943] $1^{2221}(12)122315132^1 181232161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[944] $21^{221}(12)122315132^1 181232161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[945] $1^{221}(12)12^1 23151321812321613 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[946] $1(12)122315132181232^1 1613 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 44, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[947] $21(12)122315132181232^1 1613 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 44, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[948] $21(12)12^1 23151321812321613 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 44, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[949] $61232^1 181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 47, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 47, T \in [30, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[950] $315131613161315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 47, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 47, T \in [30, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[951] $151316131613151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[952] $1612321812^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[953] $31612^1 32181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 44, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[954] $316123218123151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 44, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[955] $512321812^1 321812315132181 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 37, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[956] $151321812^1 321812315132181 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1006] $812^1 321613161316131613151 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1007] $812^1 315131613161316131612 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[1008] $613161232181232^1 181232161 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1009] $41321812^1 3218123217123217 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[957] $812^1 315132181232181231513 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 47, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 47, T \in [30, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[958] $812^1 231513218123218123215 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 37, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[959] $8122315132181232^1 18123215 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 37, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[960] $8122315132181232^1 18123151 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[961] $812^1 231513218123218123151 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[962] $161315132^1 21(12)12231513161 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[963] $413221(12)1223151321812^1 315 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 47, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 47, T \in [30, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[964] $41322^1 1(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 47, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 47, T \in [30, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

- [965] $61231513221(12)^1 122^1 31513161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 37$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 37$, $T \in [30, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [966] $612315132^1 21(12)12231513161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 37$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 37$, $T \in [30, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [967] $612315132^1 21(12)12231513216 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 38$, $T \in [30, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [968] $181232^1 1712321812321^1 81 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [969] $181232171232^1 1812321^1 81 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [970] $181232^1 18123217123218123 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [971] $1^1 812^1 32181232171232181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [972] $181232^1 17123218123218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [973] $181232171232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [974] $1^1 812321812^1 321712321812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [975] $1^1 812321812321712^1 321812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [976] $61315132181232^1 181232171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.06 \rightarrow 30$, $T_{\max} = 36$, $T \in [30, 36]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[977] $812^1 321712321812321^{2321} 8^1 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 44, T \in [30, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[999] $71232181232181232^1 181223 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[978] $181232181232^1 181232171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[979] $181232^1 181232181232171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[980] $181232181232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[981] $181232^1 1812321812321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[982] $1712321812^1 321812321812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[983] $1712321812321812^1 321812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[984] $1812321812^1 32181232171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[985] $1812321812321812^1 32171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[986] $71232181232181232^1 1^1 8^1 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[987] $181232171232181232^1 181223 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[988] $321812321712321812^1 321812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[989] $2181232^1 17123218123218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[990] $2181232171232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[991] $61315132181232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[992] $712315132181232^1 181232161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[993] $181232181232181232^1 17123 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[994] $321812321812321812^1 32171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1004] $4141413221(12)122^1 3151322181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 38, T \in [30, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[931] $1^{51315132^1 21} (12) 122315132171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 38, T \in [30, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1005] $414132^1 21(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[995] $171232181232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[996] $171232^1 18123218123218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[997] $2181232181232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 36, T \in [30, 36]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1003] $4141413221(12)122^1 3151321812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.06 \rightarrow 30, T_{\max} = 38, T \in [30, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1041] $712321812321812^1 32181223 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.94 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1027] $181232181232^1 1712321812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.94 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1028] $613151321812^1 32181232171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.94 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1029] $1812321812^1 32181232171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.94 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1030] $1812321812^1 321812321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.94 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1010] $315132^1 181231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 30$, $\Delta = -97$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.94 \rightarrow 29, T_{\max} = 30, T \in [29, 30]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1011] $1(12)122^1 315131613161315132 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.94 \rightarrow 29, T_{\max} = 51, T \in [29, 51]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1012] $1(12)12^1 231513161316131513 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.94 \rightarrow 29, T_{\max} = 47, T \in [29, 47]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1013] $1(12)1223151321812^1 321613 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.94 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1014] $21(12)1223151321812^1 321613 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.94 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

- [1015] $61316131513221^{315132^1 21} (12) 1 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 51, \Delta = -118, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 51, T \in [29, 51]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1016] $2315131613161315132^{1 21} (12) 12 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 51, \Delta = -118, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 51, T \in [29, 51]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1017] $3151316131613151322^1 1 (12) 122 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 47, \Delta = -117, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 47, T \in [29, 47]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1018] $3161232^1 181231513221 (12) 1222 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 44, \Delta = -116, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1019] $51232181232^1 1812315132181 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 37, \Delta = -111, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 37, T \in [29, 37]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1020] $15132181232^1 1812315132181 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -109, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1021] $8123151321812321812^1 31513 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 47, \Delta = -117, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 47, T \in [29, 47]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1022] $81223151321812^1 3218123215 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 37, \Delta = -111, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 37, T \in [29, 37]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1023] $81223151321812^1 3218123151 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -109, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1024] $1812321712^1 32181232181 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1031] $171232181232^1 1812321812 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [1032] $181232181232^1 181232171 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$

- [1025] $1 \overset{321}{8} 1232^1 181232171232181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1026] $1812321712^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1033] $712321812321812^1 321 \overset{321}{8} 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1034] $1812321712321812^1 32181223 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1035] $21812321712^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1036] $613151321812^1 321812321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1037] $7123151321812^1 32181232161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1038] $1812321812321812^1 3217123 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1039] $1712321812^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1040] $21812321812^1 321812321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -105$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1043] $4132181232^1 18123217123217 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [1042] $6131612321812^1 32181232161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 64$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.94 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1076] $41321812321812321712^1 3217 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1071] $4141413221(12)12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1045] $1^{5131513221}(12)12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1046] $71223151322^1 1^{5131513221}(12)1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1047] $71223151322^1 1^{5131513221}(12)1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1048] $1^1(12)12^1 2315131613161315132 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 51, T \in [29, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1049] $1^{1513221}(12)122^1 315131613161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1050] $612321812315132^1 21^{3221}(12)12 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 47, T \in [29, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1051] $22315131613161315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 51, T \in [29, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1052] $2315131613161315132^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 51, T \in [29, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1053] $1613151322^1 1^1(12)12231513161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1054] $413221^1(12)122^1 3151321812315 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 47, T \in [29, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1055] $6123151322^1 1(12)12231513161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 37$, $\Delta = -111$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 37, T \in [29, 37]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1056] $61231513221^1 (12)12^1 231513161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 37$, $\Delta = -111$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 37, T \in [29, 37]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1057] $6123151322^1 1(12)12231513216 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 38$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1058] $181232181232171232^1 18123 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1059] $1^3 8 12321812321712^1 32181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1060] $71232^1 1812321812321^3 8 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1061] $181232^1 171232181232181223 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1062] $2321812321712321812^1 32181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1063] $321812^1 321712321812321812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1064] $181232^1 18123218123217123 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1065] $321812^1 32181232181232171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1066] $71232^1 181232181232181232 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1067] $71232^1 181232181232181223 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1068] $41414132^1 21(12)122315132181 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 38, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1069] $4141413221(12)12^1 2315132181 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 38, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1073] $41414132^1 21(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 38, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1074] $41413221(12)122^1 315131613161 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1075] $4141322^1 1(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1072] $4141413221(12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 38, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1070] $41414132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 38, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1044] $1^{513151322^1 1} (12) 122315132171 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 38, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.84 \rightarrow 29, T_{\max} = 38, T \in [29, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1099] $613161232^1 181232181232161 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -109, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.76 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1090] $61315132^1 181232181232171 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -108, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.76 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

- [1097] $71232181232^1 181232181223 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1096] $181232181232^1 18123217123 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1095] $712315132^1 181232181232161 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -109$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1100] $41321812321812^1 3217123217 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1098] $81232181232181232^1 171232 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 44$, $T \in [29, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1094] $61315132^1 1812321812321712 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 36$, $\Delta = -108$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 36$, $T \in [29, 36]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1093] $181232171232^1 181232181223 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1092] $71232181232^1 1812321 \overset{321}{8} 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1091] $81232171232181232^1 1 \overset{2321}{8} 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 44$, $T \in [29, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1077] $3151321812^1 31513221(12)12231 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 30$, $\Delta = -97$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 30$, $T \in [29, 30]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1078] $1 \overset{51322^1}{(12)} 1223151316131613 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 51$, $T \in [29, 51]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1079] $1 \overset{221}{(12)} 122315132^1 1812321613 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 100$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29$, $T_{\max} = 44$, $T \in [29, 44]$

- $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [1080] $21(12)122315132^1 1812321613 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [1081] $223151316131613151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 51, \Delta = -118, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29, T_{\max} = 51, T \in [29, 51]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
 [1082] $31612321812^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [1083] $512321812321812^1 315132181 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 37, \Delta = -111, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29, T_{\max} = 37, T \in [29, 37]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [1084] $151321812321812^1 315132181 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -109, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [1085] $812315132181232^1 181231513 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 47, \Delta = -117, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29, T_{\max} = 47, T \in [29, 47]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [1086] $8122315132^1 18123218123215 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 37, \Delta = -111, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29, T_{\max} = 37, T \in [29, 37]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [1087] $8122315132^1 18123218123151 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 36, \Delta = -109, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29, T_{\max} = 36, T \in [29, 36]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [1088] $61315132^1 21(12)122315131613 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 51, \Delta = -118, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29, T_{\max} = 51, T \in [29, 51]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [1089] $1 \ 8 \ 12321812^1 32171232181 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 100, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.76 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [1118] $32181232181232^1 181232171 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
 [1117] $1812321812^1 3218123217123 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 40, T \in [29, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1121] & 812321812321812^1 32171232 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 44, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 44, T \in [29, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1122] & 414141322^1 1(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 38, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 38, T \in [29, 38] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[1123] & 414141322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 38, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 38, T \in [29, 38] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[1124] & 414141322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 38, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 38, T \in [29, 38] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[1119] & 712321812^1 32181232181223 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 40, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 40, T \in [29, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1125] & 41413221(12)12^1 2315131613161 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 40, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 40, T \in [29, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1126] & 81232^1 1613161316131613151 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 36, T \in [29, 36] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[1127] & 6131612^1 32181232181232161 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 36, T \in [29, 36] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[1128] & 4132181232181232^1 17123217 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 40, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 40, T \in [29, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1101] & 1^{1513221} (12) 12^1 2315131613161 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 36, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 36, T \in [29, 36] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1102] & 1^{3221} (12) 122^1 315131613161315 + \hat{1} \rightarrow 2 \quad T = 24, T_{\max} = 51, \Delta = -118, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 28.69 \rightarrow 29, T_{\max} = 51, T \in [29, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1103] $61316131513221^{3151322^1 1} (12) 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29$, $T_{\max} = 51$, $T \in [29, 51]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1104] $61232181231513221^{3221} (12) 12 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29$, $T_{\max} = 47$, $T \in [29, 47]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1105] $612321812^{3221} 31513221 (12) 12 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29$, $T_{\max} = 47$, $T \in [29, 47]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1106] $8123151321812^{32181231513} + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29$, $T_{\max} = 47$, $T \in [29, 47]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1107] $413221^{1} (12) 122315132^{1} 1812315 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29$, $T_{\max} = 47$, $T \in [29, 47]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [1108] $413221^{1} (12) 12^{1} 23151321812315 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 47$, $\Delta = -117$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29$, $T_{\max} = 47$, $T \in [29, 47]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [1109] $6131513221^{1} (12) 122^{1} 315131613 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29$, $T_{\max} = 51$, $T \in [29, 51]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1110] $1812321812^{1} 3217123218123 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1111] $1^3 8^{321} 1232181232^{1} 171232181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [1112] $81232^{1} 1712321812321^{2321} 8^1 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29$, $T_{\max} = 44$, $T \in [29, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [1113] $812321712321812^{1} 321^1 8^1 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.69 \rightarrow 29$, $T_{\max} = 44$, $T \in [29, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1114] $712321812^{1} 321812321^{321} 8^1 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 28.69 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1115] 1812321712¹32181232181223+ $\hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.69 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1116] 32181232171232¹1812321812+ $\hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.69 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[1120] 81232¹181232181232171232+ $\hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.69 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1131] 32181232¹181232181232171+ $\hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 144, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.64 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1129] 1812321812321712¹3218123+ $\hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 144, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.64 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1130] 32181232¹1712321812321812+ $\hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 144, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.64 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 12 & 2 \\ 12 & 203 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1133] 613161315132¹21³¹⁵¹³²²¹(12) 1+ $\hat{1} \rightarrow 2$ $T = 24, T_{\max} = 51, \Delta = -118, \det = 169, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.59 \rightarrow 29, T_{\max} = 51, T \in [29, 51]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1134] 812315132¹181232181231513+ $\hat{1} \rightarrow 2$ $T = 24, T_{\max} = 47, \Delta = -117, \det = 169, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.59 \rightarrow 29, T_{\max} = 47, T \in [29, 47]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1135] 613151322¹1(12)122315131613+ $\hat{1} \rightarrow 2$ $T = 24, T_{\max} = 51, \Delta = -118, \det = 169, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.59 \rightarrow 29, T_{\max} = 51, T \in [29, 51]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1136] 181232181232¹17123218123+ $\hat{1} \rightarrow 2$ $T = 24, T_{\max} = 40, \Delta = -114, \det = 169, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.59 \rightarrow 29, T_{\max} = 40, T \in [29, 40]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1137] 81232171232¹1812321²³²¹8 1+ $\hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 169, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.59 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$$

- $$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$
 [1138] $2321812^1 32171232181232181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 169$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.59 \rightarrow 29$, $T_{\max} = 44$, $T \in [29, 44]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$
 [1139] $321812321712^1 321812321812 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.59 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$
 [1140] $321812321812^1 32181232171 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 40$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.59 \rightarrow 29$, $T_{\max} = 40$, $T \in [29, 40]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$
 [1141] $71232181232181232^1 181232 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 169$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.59 \rightarrow 29$, $T_{\max} = 44$, $T \in [29, 44]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$
 [1142] $81232181232^1 181232171232 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 169$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.59 \rightarrow 29$, $T_{\max} = 44$, $T \in [29, 44]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$
 [1132] $1(12)^{3221} 12^1 2315131613161315 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 169$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.59 \rightarrow 29$, $T_{\max} = 51$, $T \in [29, 51]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$
 [1145] $812321712^1 321812321^{2321} 8^1 1 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 196$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.55 \rightarrow 29$, $T_{\max} = 44$, $T \in [29, 44]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$
 [1144] $6131513221(12)^1 12^1 2315131613 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 196$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.55 \rightarrow 29$, $T_{\max} = 51$, $T \in [29, 51]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$
 [1146] $812321812^1 32181232171232 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 196$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.55 \rightarrow 29$, $T_{\max} = 44$, $T \in [29, 44]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$
 [1143] $1(12)^{513221} 122^1 3151316131613 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 51$, $\Delta = -118$, $\det = 196$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.55 \rightarrow 29$, $T_{\max} = 51$, $T \in [29, 51]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$
 [1148] $712321812^1 32181232181232 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 225$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 28.52 \rightarrow 29$, $T_{\max} = 44$, $T \in [29, 44]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$
 [1147] $232181232171232^1 181232181 + \hat{1} \rightarrow 2$ $T = 24$, $T_{\max} = 44$, $\Delta = -116$, $\det = 225$, $(n_+, n_-, n_0) = (1, 24, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 28.52 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[1149] $61316131513221^{31513221}1(12)1 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 51, \Delta = -118, \det = 256, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.48 \rightarrow 29, T_{\max} = 51, T \in [29, 51]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1152] $712321812321812^{132181232}1 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 289, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.46 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 203 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1151] $232181232^{171232181232181}1 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 289, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.46 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 373 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$$

[1150] $1(12)12^{513221}23151316131613 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 51, \Delta = -118, \det = 289, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.46 \rightarrow 29, T_{\max} = 51, T \in [29, 51]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[1154] $71232181232^{181232181232}1 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 324, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.43 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[1153] $2321812321712^{132181232181}1 + \hat{1} \rightarrow 2$ $T = 24, T_{\max} = 44, \Delta = -116, \det = 324, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 28.43 \rightarrow 29, T_{\max} = 44, T \in [29, 44]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 413 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

$T = 25$ [1174] $221^{221}(12)12231513221(12)12222^12 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32, T_{\max} = 37, T \in [32, 37]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[1178] $3221^{211}(12)122315132181232171 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 42, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1195] $218123218^{12321812321812}2 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32, T_{\max} = 37, T \in [32, 37]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[1194] $613151321812321812321812^12 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32, T_{\max} = 37, T \in [32, 37]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[1193] $21613161316131513221(12)122^12 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32, T_{\max} = 37, T \in [32, 37]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1192] & \begin{smallmatrix} 1 \\ 321613161316131513221 \end{smallmatrix} (12) 12^1 2 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1191] & \begin{smallmatrix} 1 \\ 1812321812321812321812 \end{smallmatrix} 2 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1190] & \begin{smallmatrix} 31 \\ 1613161316131513221 \end{smallmatrix} (12) 12^1 2 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1189] & \begin{smallmatrix} 22^1 1 \\ 1(12)12231513161316131612 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1155] & \begin{smallmatrix} 22^1 221 \\ 1(12)12231513221 \end{smallmatrix} \begin{smallmatrix} 1 \\ (12)12222 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1156] & \begin{smallmatrix} 22^1 21 \\ 1(12)12231513221 \end{smallmatrix} \begin{smallmatrix} 221 \\ (12)1223 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1157] & \begin{smallmatrix} 2221 \\ 1(12)12231513221 \end{smallmatrix} \begin{smallmatrix} 2^1 21 \\ (12)1223 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[1158] & \begin{smallmatrix} 22221 \\ 1(12)12231513221 \end{smallmatrix} \begin{smallmatrix} 21 \\ (12)122^1 2 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1159] & \begin{smallmatrix} 22^1 221 \\ 1(12)12231513221 \end{smallmatrix} \begin{smallmatrix} 21 \\ (12)1222 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1160] & \begin{smallmatrix} 22^1 221 \\ 1(12)12231513221 \end{smallmatrix} \begin{smallmatrix} 221 \\ (12)122 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1161] & \begin{smallmatrix} 22221 \\ 1(12)12231513221 \end{smallmatrix} \begin{smallmatrix} 221 \\ (12)12^1 2 \end{smallmatrix} + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1162] & 21^{221}(12)12231513221^{2^1 21}(12)1223 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[1163] & 21^{22^1 1}(12)12231513221^{221}(12)1223 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1164] & 21^{22^1 21}(12)12231513221^{21}(12)1222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1165] & 21^{22^1 21}(12)12231513221^{221}(12)122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1166] & 21^{2221}(12)12231513221^{221}(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1167] & 22^1 1^{221}(12)12231513221^{221}(12)122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1168] & 1^{22^1 221}(12)12231513221(12)122222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1169] & 1^{22221}(12)12231513221(12)12222^1 2 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1173] & 21^{22^1 21}(12)12231513221(12)122222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1188] & 1^{1613161316131513221}(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1175] & 22^1 1^{221}(12)12231513221(12)122222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1170] & \begin{matrix} 3221 \\ 221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1171] & \begin{matrix} 2^1 21 \\ 3221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1172] & \begin{matrix} 2221 \\ 21(12)12231513221(12)12222^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1176] & \begin{matrix} 22^1 2221(12)12231513221(12)122222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1187] & \begin{matrix} 1 \\ 1613161316131513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1186] & \begin{matrix} 22^1 1 \\ 2171232181231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1185] & \begin{matrix} 22^1 1 \\ 232171232181231513221(12)12^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 48, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 48, T \in [32, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1184] & \begin{matrix} 22^1 1 \\ 32171232181231513221(12)122^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1183] & \begin{matrix} 1 \\ 171232181231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1182] & \begin{matrix} 1 \\ 171232181231513221(12)122^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1181] & \begin{matrix} 22^1 1 \\ 21(12)1223151321812321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

[1180] $1(12)1223151321812321712 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[1179] $1(12)12231513218123217123 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 42, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32, T_{\max} = 42, T \in [32, 42]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1177] $171232181231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[1197] $1(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 41, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[1196] $1(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$

[1209] $21(12)12231513221(12)12222^1 2 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$

[1208] $221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 41, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1207] $221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 41, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$

[1206] $1(12)12231513221(12)12222^1 2 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$

[1205] $1(12)12231513221(12)1222^1 22 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1204] $21(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$

[1203] $21^{222^1 1} (12) 12231513221^{21} (12) 1222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$

[1202] $21^{221} (12) 12231513221^{221} (12) 122^1 3 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[1201] $1^{222^1 21} (12) 12231513221^{21} (12) 122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$

[1200] $1^{222^1 21} (12) 12231513221^{21} (12) 1222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$

[1199] $1^{22221} (12) 12231513221^{21} (12) 12^1 22 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[1198] $1^{2221} (12) 12231513221^{221} (12) 122^1 3 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[1211] $221^{221} (12) 12231513221 (12) 1222^1 22 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1212] $222^1 221 (12) 12231513221 (12) 122222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$

[1213] $1^{171232181231513221} (12) 122^1 22 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1214] $21^{32^1 21} (12) 122315132181232171 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$

[1215] $1^{221} (12) 1223151321812321712^1 3 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[1216] $1(12)1223151321812321712 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$

[1217] $21(12)1223151321812321712^1 3 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 42, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[1218] $171232181231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[1219] $32171232181231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 42, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1223] $1613161316131513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1220] $32^1 171232181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 42, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$

[1221] $2171232181231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1222] $32^1 21(12)1223151316131613161 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 42, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$

[1230] $32^1 1812321812321812321812 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 40, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 40, T \in [32, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[1229] $18123218123218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 40, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 40, T \in [32, 40]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[1228] $21613161316131513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1227] $32^1 1613161316131513221(12)122 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 42, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[1226] & 1 \overset{32^1}{8} 1232181232181232181 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 40, T \in [32, 40] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1225] & 1812321812321812321812^1 3 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 40, T \in [32, 40] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1224] & 1 \overset{21}{(12)} 12231513161316131612^1 3 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1210] & 21 \overset{2221}{(12)} 12231513221(12)1222^1 22 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -9, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 37, T \in [32, 37] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1262] & 21 \overset{221}{(12)} 122315132181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1247] & 1 \overset{22221}{(12)} 122315132^1 21(12)122222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1276] & 8122315132181232181232^1 161 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -108, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1277] & 71232171232181232181232^1 15 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 38, \Delta = -111, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[1278] & 712315132181232181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1279] & 2181232181232181232^1 18122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[1280] & 613161232181232181232^1 171 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1281] & 613161232181232181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} =
\end{aligned}$$

15.0

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1274] $1612^1 321812321812315132181 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -108, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1273] $1812^{21} 321812321812321812 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1272] $181232181232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[1271] $181232181232181232^1 181 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[1270] $712315132181232181232^1 171 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1269] $61315132181232181232^1 181 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1268] $21712^1 32181231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1267] $2171232181231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1266] $1612^1 32181231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 31, \Delta = -96, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1265] $161232181231513221(12)12^1 2313 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 31, \Delta = -96, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1264] $1712^1 32181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1263] $\overset{1}{1712^1 32181231513221(12)1222 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1261] $\overset{2221}{1(12)122315132181232^1 1712 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1260] $\overset{1}{171232181231513221(12)12^1 222 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1259] $\overset{1}{1712^1 32181231513221(12)12222 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1258] $\overset{2221}{2222^1 21(12)12231513221(12)122222 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1257] $\overset{2221}{222221(12)122^1 31513221(12)122222 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1256] $\overset{221}{221(12)122^1 31513221(12)122222 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1255] $\overset{221}{221(12)12231513221(12)122^1 222 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1254] $\overset{221}{221(12)122315132^1 21(12)122222 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1253] $\overset{2221}{21(12)122^1 31513221(12)122222 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1252] $\overset{2221}{21(12)122315132^1 21(12)122222 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1251] $\overset{2221}{21(12)12231513221(12)122^1 222 + \hat{1} \rightarrow 2}$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}^{3221} \\
[1250] & 221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}^{3221} \\
[1249] & 22^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.25 \rightarrow 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}^{22221} \\
[1245] & 1(12)12231513221(12)122^1 222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}^{221} \\
[1244] & 221(12)122^1 31513221(12)122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}^{2221} \\
[1243] & 21(12)122315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}^{2221} \\
[1242] & 21(12)122^1 31513221(12)122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}^{2221} \\
[1241] & 21(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}^{221} \\
[1240] & 21(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.25 \rightarrow 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}^{2222^1 1} \\
[1239] & 1(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}^{22221} \\
[1238] & 1(12)122315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}^{22221} \\
[1237] & 1(12)122^1 31513221(12)122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1236] & \begin{matrix} 22221 & 21 \\ 1(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1235] & \begin{matrix} 22221 & 21 \\ 1(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1234] & \begin{matrix} 2222^1 1 & 21 \\ 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1233] & \begin{matrix} 2221 & 22^1 1 \\ 1(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.25 \rightarrow 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1232] & \begin{matrix} 22221 & 1 \\ 1(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1231] & \begin{matrix} 2222^1 1 & 1 \\ 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1248] & \begin{matrix} 22221 \\ 1(12)122^1 31513221(12)122222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1246] & \begin{matrix} 2222^1 1 \\ 1(12)12231513221(12)122222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 35 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1275] & \begin{matrix} 61315132181232181232^1 18122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.75 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1362] & \begin{matrix} 1 \\ 4132^1 21(12)12231513218123216 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 45, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 45, T \in [31, 45] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1320] & \begin{matrix} 221 \\ 21(12)1223151321812^1 321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1327] \quad & 232^1 171232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 48, \Delta = -117, \det = -25, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 48, T \in [31, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1328] \quad & 21712321812315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -25, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1329] \quad & 2171232^1 181231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -25, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1330] \quad & 16131513221(12)122315132^1 171 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -109, \det = -25, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1331] \quad & 712^1 31513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -109, \det = -25, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1332] \quad & 61231513221(12)122315132^1 171 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -110, \det = -25, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1333] \quad & 612^1 31513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -110, \det = -25, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1334] \quad & 61231513221(12)122315132^1 217 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -110, \det = -25, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1335] \quad & 612^1 31513221(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -110, \det = -25, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1336] \quad & 1(12)1223151316131613161 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1337] \quad & 1(12)12231513161316131612^1 32 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 48, \Delta = -117, \det = -25, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 48, T \in [31, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1338] $613151321812321812^1 32181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1339] $7123151321812321812^1 32171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1340] $1812321812321812^1 32181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1341] $1812^1 32181232181232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1342] $1812^1 3218123218123218122 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1343] $1812321812321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1344] $1812^1 3218123218123218123 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 40$, $T \in [31, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1345] $181232181232181232^1 1812 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [1346] $181232^1 1812321812321812 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1347] $812321812321812321^{232^1} 81 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 42$, $T \in [31, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1348] $232^1 1613161316131513221(12)12 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 48$, $T \in [31, 48]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1349] $161232^1 1812321812315132181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -108$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1350] & 4132^1 21(12)122315131613161315 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 48, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 48, T \in [31, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1351] & 613151321812321812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1352] & 81223151321812321812^1 32161 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1353] & 712321712321812321812^1 3215 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 38, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1354] & 712321712321812321812^1 3151 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1355] & 7123151321812321812^1 321712 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1356] & 1812321812321812321812^1 23 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 40, T \in [31, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1357] & 232^1 181232181232181232181 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1358] & 32181232181232181232^1 1812 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 40, T \in [31, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1359] & 21812^1 3218123218123218122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1360] & 21812321812321812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1325] & 161232^1 181231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 31, \Delta = -96, \det = -25, (n_+, n_-, n_0) =
\end{aligned}$$

$(1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 31$, $T \in [31, 31]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1324] $1^{12}171232^1181231513221(12)12+\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1323] $1^{12}1712321812315132^121(12)12+\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1364] $4132^1181232181232171232181+\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 40$, $T \in [31, 40]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1365] $6131612321812321812^1321712+\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1366] $712321812321812321812^1314+\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 40$, $T \in [31, 40]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1282] $1^{12}1812^131513221(12)1223151315+\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 38$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1283] $1^{22221}(12)12^1231513221(12)12222+\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[1284] $1^{2221}(12)12231513221(12)12^123+\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 41$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 41$, $T \in [31, 41]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1285] $1^{2221}(12)122^131513221(12)1223+\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 41$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 41$, $T \in [31, 41]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1286] $1^{22221}(12)12^1231513221(12)1222+\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[1287] $1^{22221}(12)1223151322^11(12)1222+\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.44 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1288] & \begin{matrix} 22221 \\ 1(12)1223151322^1 1(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1289] & \begin{matrix} 22221 \\ 1(12)12^1 231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[1290] & \begin{matrix} 221 \\ 21(12)122^1 31513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1291] & \begin{matrix} 221 \\ 21(12)12231513221(12)12^1 23 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1292] & \begin{matrix} 2221 \\ 21(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[1293] & \begin{matrix} 2221 \\ 21(12)12^1 231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[1361] & \begin{matrix} 1 \\ 413221(12)122315132181232^1 16 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 45, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 45, T \in [31, 45] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1322] & \begin{matrix} 1 \\ 171232^1 181231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1363] & \begin{matrix} 1 \\ 6131612321812321812^1 32171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1294] & \begin{matrix} 2221 \\ 21(12)1223151322^1 1(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1295] & \begin{matrix} 221 \\ 221(12)12^1 231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1296] $1^{5131513221}(12)122315132^11812 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 38, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1326] $1612321812315132^121(12)122313 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 31, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1297] $1^{15132^121}(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 39, \Delta = -112, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 39, T \in [31, 39]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1298] $1^{5131513221}(12)122315132^12181 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 38, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1299] $1^{22221}(12)12231513221(12)12^12222 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1300] $1^{22221}(12)1223151322^11(12)122222 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1301] $1^{22221}(12)12^1231513221(12)122222 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1302] $221^{322^11}(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 41, \Delta = -114, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 41, T \in [31, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1303] $221^{3221}(12)122315132^121(12)12222 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 41, \Delta = -114, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 41, T \in [31, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1304] $21^{2221}(12)12231513221(12)12^12222 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1305] $21^{2221}(12)1223151322^11(12)122222 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1306] $21^{2221}(12)12^1231513221(12)122222 + \hat{1} \rightarrow 2$ $T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

15.0

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[1307] \quad {}^{221} 221(12)12^1 231513221(12)122222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[1308] \quad {}^{221} 221(12)12231513221(12)12^1 2222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1309] \quad {}^{221} 221(12)1223151322^1 1(12)122222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1310] \quad 22222^1 1(12)12231513221(12)122222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 54 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[1311] \quad 222221(12)12^1 231513221(12)122222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[1312] \quad {}^1 171232^1 181231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1313] \quad {}^1 1712321812315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1314] \quad {}^{322^1 1} 21(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1315] \quad {}^{3221} 21(12)122315132181232^1 171 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1316] \quad {}^{21} 1(12)1223151321812321712^1 32 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 48, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 48, T \in [31, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1317] \quad {}^{2221} 1(12)122^1 3151321812321712 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[1318] & \begin{matrix} 2221 \\ 1(12)1223151321812^1 321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1319] & \begin{matrix} 221 \\ 21(12)122^1 3151321812321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[1321] & \begin{matrix} 1 \\ 1712321812315132^1 21(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.44 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1380] & \begin{matrix} 1 \\ 321712321812315132^1 21(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1400] & \begin{matrix} 1 \\ 4132181232181232171232^1 181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 40, T \in [31, 40] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1377] & \begin{matrix} 1 \\ 17123218123151322^1 1(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1376] & \begin{matrix} 1 \\ 17123218123151322^1 1(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1375] & \begin{matrix} 221 \\ 21(12)12^1 23151321812321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1374] & \begin{matrix} 21 \\ 21(12)122^1 31513218123217123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1373] & \begin{matrix} 2221 \\ 1(12)12^1 23151321812321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1372] & \begin{matrix} 221 \\ 1(12)122^1 31513218123217123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[1371] $17123218123151322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1370] $221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 41$, $\Delta = -114$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 41$, $T \in [31, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1369] $1^{5131513221}(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 38$, $\Delta = -111$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1378] $16123218123151322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 31$, $\Delta = -96$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 31$, $T \in [31, 31]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1382] $712^1 31513221(12)122315131613 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 48$, $T \in [31, 48]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1383] $712^1 231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -109$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1384] $61231513221(12)1223151322^1 17 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -110$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1368] $21(12)12^1 231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 41$, $\Delta = -114$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 41$, $T \in [31, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1367] $1(12)12^1 231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 41$, $\Delta = -114$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 41$, $T \in [31, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1381] $217123218123151322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1385] $16131613161315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.22 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1379] $31612321812315132^1 21(12)12231 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 31$, $\Delta = -96$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 31, T \in [31, 31] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1386] & \begin{matrix} 1 \\ 16131613161315132^1 21(12)12+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1387] & \begin{matrix} 1 \\ 16131613161315132^1 21(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1388] & \begin{matrix} 221 \\ 1(12)122^1 31513161316131612+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1389] & \begin{matrix} 21 \\ 21(12)122^1 31513161316131612+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1390] & \begin{matrix} 1 \\ 712^1 315132181232181232171+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1391] & \begin{matrix} 321 \\ 1 \ 8 \ 1232181232181232^1 181+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 40, T \in [31, 40] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1392] & \begin{matrix} 1 \\ 216131613161315132^1 21(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1393] & \begin{matrix} 1 \\ 1612321812321812315132^1 181+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -108, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1394] & \begin{matrix} 1 \\ 8122^1 315132181232181232161+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -108, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1395] & \begin{matrix} 1 \\ 712^1 3217123218123218123215+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 38, \Delta = -111, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1396] & \begin{matrix} 1 \\ 712^1 3217123218123218123151+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -109, \det = -36, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[1397] $712^1 3151321812321812321712 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1398] $1812^1 32181232181232181223 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 40, T \in [31, 40]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1399] $232181232181232181232^1 181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1401] $712^1 321812321812321812314 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -36$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.22 \rightarrow 31, T_{\max} = 40, T \in [31, 40]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1413] $51316131513221(12)122^1 3151315 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1425] $1612321812^1 31513221(12)122313 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 31$, $\Delta = -96$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1447] $21(12)12^1 231513161316131612 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[1448] $61315132181232^1 181232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1449] $712315132181232^1 181232171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1450] $181232181232^1 181232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1451] $181232^1 181232181232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1452] $181232181232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1453] & \begin{matrix} 1 \\ 181232^1 18123218123218122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1454] & \begin{matrix} 1 \\ 181232^1 18123218123218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 40, T \in [31, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1455] & \begin{matrix} 321 \\ 1 \ 8 \ 12^1 32181232181232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 40, T \in [31, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1456] & \begin{matrix} 21 \\ 1 \ 8 \ 12321812^1 321812321812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1457] & \begin{matrix} 21 \\ 1 \ 8 \ 12321812321812^1 321812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1458] & \begin{matrix} 1 \\ 812^1 321812321812321 \ 8 \ 1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1459] & \begin{matrix} 2321 \\ 3216131613161315132^1 21(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[1460] & \begin{matrix} 2321 \\ 2161316131613151322^1 1(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[1461] & \begin{matrix} 2321 \\ 1612321812^1 321812315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1462] & \begin{matrix} 21 \\ 414132^1 21(12)1223151321812315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 45, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 45, T \in [31, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1463] & \begin{matrix} 21 \\ 41413221(12)1223151321812^1 315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 45, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 45, T \in [31, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1464] & \begin{matrix} 21 \\ 41322^1 1(12)122315131613161315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 48, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

(1, 25, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 48$, $T \in [31, 48]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1465] 61315132181232¹18123218122+ $\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1466] 812315132181232181232¹1613+ $\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 45$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 45$, $T \in [31, 45]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1467] 812¹3151321812321812321613+ $\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 45$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 45$, $T \in [31, 45]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1468] 812¹2315132181232181232161+ $\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -108$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[1469] 8122315132181232¹181232161+ $\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -108$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1470] 71232171232181232¹18123215+ $\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 38$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1471] 71232171232181232¹18123151+ $\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1472] 712315132181232¹1812321712+ $\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1473] 181232181232181232¹181223+ $\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 40$, $T \in [31, 40]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1474] 321812321812321812¹321812+ $\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 40$, $T \in [31, 40]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1475] 2181232181232¹18123218122+ $\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

- [1476] $2181232^1 18123218123218122 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1477] $413221(12)1223151321812^1 3216 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 45$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 45$, $T \in [31, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1478] $41322^1 1(12)12231513218123216 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 45$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 45$, $T \in [31, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1479] $613161232181232^1 181232171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1480] $41321812^1 32181232171232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 40$, $T \in [31, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1481] $613161232181232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1482] $71232181232181232^1 1812314 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 40$, $T \in [31, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1483] $812^1 321613161316131613161 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -105$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
- [1484] $812^1 3216131613161316131612 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -105$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 37$, $T \in [31, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
- [1408] $1^{151322^1} (12) 1223151321812315 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 39$, $\Delta = -112$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 39$, $T \in [31, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1407] $1^{5131513221} (12) 122^1 3151321812 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 38$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1406] $1^{51315132^1 21} (12) 1223151321812 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 38$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[1405] & \begin{matrix} 221 & & 221 \\ 21(12)122315132^1 21(12)1223 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 41, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1404] & \begin{matrix} 2221 & & 221 \\ 1(12)122315132^1 21(12)1223 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 41, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1403] & \begin{matrix} 1 & & 1 \\ 181231513221(12)122^1 3151315 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 38, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1402] & \begin{matrix} 1 & & 1 \\ 1812315132^1 21(12)1223151315 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 38, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1445] & \begin{matrix} 221 \\ 1(12)12^1 231513161316131612 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 37, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0} \\
[1444] & \begin{matrix} 21 \\ 1(12)122^1 315131613161316123 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 42, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1443] & \begin{matrix} 1 \\ 161316131613151322^1 1(12)1222 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 37, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[1446] & \begin{matrix} 31 \\ 16131613161315132^1 21(12)122 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 41, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[1442] & \begin{matrix} 31 & & 21 \\ 16131613161315132^1 21(12)1 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 41, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 41, T \in [31, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[1441] & \begin{matrix} 1 & & 21 \\ 161316131613151322^1 1(12)12 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 37, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[1440] & \begin{matrix} 1 & & 221 \\ 161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 37, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[1439] $61231513221(12)122^1 315132217 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1438] $612315132^1 21(12)122315132217 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1437] $61231513221(12)122^1 315132171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1410] $1^{5131513221}(12)122^1 3151322181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 38$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1436] $612315132^1 21(12)122315132171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1435] $7122315132^1 21(12)12231513161 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1434] $712231513221(12)122^1 31513161 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1433] $16131513221(12)122^1 315132171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1432] $161315132^1 21(12)122315132171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1431] $21712321812^1 31513221(12)12222 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1430] $2321712321812315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.06 \rightarrow 31, T_{\max} = 48, T \in [31, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[1429] $3217123218123151322^1(12)1222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$

[1428] $321712^132181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[1427] $31612^132181231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 31, \Delta = -96, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31, T_{\max} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[1426] $316123218123151322^11(12)12231 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 31, \Delta = -96, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31, T_{\max} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$

[1411] $1^{51315132^121}(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 38, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

[1412] $221^{3221}(12)122^131513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31, T_{\max} = 41, T \in [31, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[1414] $1712321812^131513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[1415] $21^{3221}(12)1223151321812^132171 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$

[1416] $1^{221}(12)12^1231513218123217123 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1417] $1^{221}(12)122315132181232^117123 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1418] $1^{21}(12)122^1315132181232171232 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 48, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 30.06 \rightarrow 31, T_{\max} = 48, T \in [31, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1419] $1^{2221}(12)122315132^11812321712 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[1420] & 21(12)12^{21}231513218123217123+\hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1421] & 21(12)122315132181232^1 17123+\hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1422] & 21(12)122315132^1 1812321712+\hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[1423] & 1712321812^1 31513221(12)1222+\hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1424] & 1712321812^1 31513221(12)12+\hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 37, T \in [31, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1409] & 1^{1513221} (12) 1223151321812^1 315+\hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 39, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.06 \rightarrow 31, T_{\max} = 39, T \in [31, 39] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1515] & 712321812321812^1 321812314+\hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.94 \rightarrow 30, T_{\max} = 40, T \in [30, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1514] & 6131612321812^1 321812321712+\hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1513] & 4132181232^1 181232171232181+\hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.94 \rightarrow 30, T_{\max} = 40, T \in [30, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1512] & 6131612321812^1 32181232171+\hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1511] & 21812321812^1 3218123218122+\hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37]
\end{aligned}$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1510] $1812321812321812^1 32181223 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1509] $7123151321812^1 321812321712 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1508] $712321712321812^1 3218123151 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -109$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1507] $712321712321812^1 3218123215 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 38$, $\Delta = -111$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 38, T \in [30, 38]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1506] $81223151321812^1 32181232161 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -108$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1505] $8123151321812321812^1 321613 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 45$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 45, T \in [30, 45]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1504] $613151321812^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1503] $161232181232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -108$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1501] $32161316131613151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 42, T \in [30, 42]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1500] $181232181232^1 1812321812 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1499] $181232^1 181232181232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1498] $1812321812^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1497] & \begin{matrix} 1 & 1 & 21 \\ 1812321812^1 32181232181 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 37, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\ \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[1496] & \begin{matrix} 1 \\ 7123151321812^1 32181232171 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 37, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\ \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[1495] & \begin{matrix} 21 \\ 613151321812^1 32181232181 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 37, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 29.94 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\ \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[1494] & \begin{matrix} 31 \\ 161316131613151322^1 1(12)122 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 41, \Delta = -114, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 29.94 \rightarrow 30, T_{\max} = 41, T \in [30, 41] \\ \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[1493] & \begin{matrix} 21 \\ 1(12)12^1 2315131613161316123 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 42, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 29.94 \rightarrow 30, T_{\max} = 42, T \in [30, 42] \\ \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[1492] & \begin{matrix} 1 \\ 1(12)122^1 3151316131613161232 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 48, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 29.94 \rightarrow 30, T_{\max} = 48, T \in [30, 48] \\ \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[1491] & \begin{matrix} 31 \\ 161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 41, \Delta = -114, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 29.94 \rightarrow 30, T_{\max} = 41, T \in [30, 41] \\ \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[1490] & \begin{matrix} 1 \\ 23217123218123151322^1 1(12)122 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 48, \Delta = -117, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 29.94 \rightarrow 30, T_{\max} = 48, T \in [30, 48] \\ \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[1489] & \begin{matrix} 1 \\ 32171232^1 181231513221(12)1222 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 42, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 29.94 \rightarrow 30, T_{\max} = 42, T \in [30, 42] \\ \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[1488] & \begin{matrix} 1 \\ 3161232^1 181231513221(12)12231 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 31, \Delta = -96, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 29.94 \rightarrow 30, T_{\max} = 31, T \in [30, 31] \\ \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[1487] & \begin{matrix} 21 \\ 21(12)1223151321812^1 3217123 + \hat{1} \rightarrow 2 & T = 25, T_{\max} = 42, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 29.94 \rightarrow 30, T_{\max} = 42, T \in [30, 42] \\ \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \end{matrix}
\end{aligned}$$

- [1486] $1(12)^{21}12^1 2315132181232171232 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.94 \rightarrow 30$, $T_{\max} = 48$, $T \in [30, 48]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [1485] $1(12)^{221}1223151321812^1 3217123 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.94 \rightarrow 30$, $T_{\max} = 42$, $T \in [30, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1502] $23216131613161315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -64$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.94 \rightarrow 30$, $T_{\max} = 48$, $T \in [30, 48]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1547] $321812^1 321812321812321812 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1528] $21(12)^{3221}122^1 315132181232171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 42$, $T \in [30, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1531] $712231513221(12)^1 12^1 231513161 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -109$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 37$, $T \in [30, 37]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1529] $16131513221(12)^1 12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -109$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 37$, $T \in [30, 37]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1527] $513161315132^1 21(12)1223151315 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 42$, $T \in [30, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1526] $51316131513221(12)12^1 23151315 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 42$, $T \in [30, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1525] $221(12)^{3221}12^1 231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 41$, $\Delta = -114$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 41$, $T \in [30, 41]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1524] $1(12)^{513151322^1 1} 1223151322181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 38$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 38$, $T \in [30, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1523] $1(12)^{5131513221} 12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 38$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 38, T \in [30, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1513221 \\ [1522] \end{matrix} 1 \quad (12) \quad 122^1 3151321812315 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 39, \Delta = -112, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 39, T \in [30, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 513151322^1 1 \\ [1521] \end{matrix} 1 \quad (12) \quad 1223151321812 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 38, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 38, T \in [30, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 5131513221 \\ [1520] \end{matrix} 1 \quad (12) \quad 12^1 23151321812 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 38, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 38, T \in [30, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 221 \\ [1519] \end{matrix} 21(12)1223151322^1 1(12)1223 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 41, \Delta = -114, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 41, T \in [30, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1550] \end{matrix} 71232^1 1812321812321812314 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 40, T \in [30, 40] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
& [1549] 41321812321812321712^1 32181 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 40, T \in [30, 40] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1548] \end{matrix} 413221(12)122^1 31513218123216 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 45, \Delta = -116, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 45, T \in [30, 45] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1546] \end{matrix} 2321812321812321812^1 32181 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 42, T \in [30, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1545] \end{matrix} 181232^1 181232181232181223 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 40, T \in [30, 40] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
& [1544] 71232^1 17123218123218123151 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 37, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
& [1543] 71232^1 17123218123218123215 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 38, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 38, T \in [30, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1542] $4141322^1 1(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 45$, $\Delta = -116$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 45$, $T \in [30, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1541] $41413221(12)122^1 3151321812315 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 45$, $\Delta = -116$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 45$, $T \in [30, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1540] $232161316131613151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 48$, $T \in [30, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[1539] $1^{321} 8 12321812321812^1 32181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

[1538] $1^{18} 1232181232181232^1 18123 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1537] $1^{12} 12^1 23151316131613161232 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 48$, $T \in [30, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1536] $6123151322^1 1(12)122315132217 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -110$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 37$, $T \in [30, 37]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1535] $61231513221(12)12^1 2315132217 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -110$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 37$, $T \in [30, 37]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1534] $6123151322^1 1(12)122315132171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -110$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 37$, $T \in [30, 37]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1533] $61231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -110$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 37$, $T \in [30, 37]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1518] $1^{2221} (12)1223151322^1 1(12)1223 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 41$, $\Delta = -114$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.84 \rightarrow 30$, $T_{\max} = 41$, $T \in [30, 41]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1517] $1^{18} 123151322^1 1(12)1223151315 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 38$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 38, T \in [30, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1516] & \begin{matrix} 1 \\ 181231513221(12)12^1 23151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 38, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 38, T \in [30, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1530] & \begin{matrix} 1 \\ 1613151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1532] & \begin{matrix} 1 \\ 71223151322^1 1(12)12231513161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.84 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1569] & \begin{matrix} 1 \\ 181232181232^1 181232181223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.76 \rightarrow 30, T_{\max} = 40, T \in [30, 40] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1570] & \begin{matrix} 1 \\ 613161232^1 181232181232171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.76 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1571] & \begin{matrix} 1 \\ 41321812321812^1 32171232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.76 \rightarrow 30, T_{\max} = 40, T \in [30, 40] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1572] & \begin{matrix} 1 \\ 613161232^1 1812321812321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.76 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1573] & \begin{matrix} 1 \\ 71232181232^1 1812321812314 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.76 \rightarrow 30, T_{\max} = 40, T \in [30, 40] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1568] & \begin{matrix} 1 \\ 712315132^1 1812321812321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.76 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1567] & \begin{matrix} 1 \\ 71232171232^1 18123218123151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 37, \Delta = -109, \det = -100, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.76 \rightarrow 30, T_{\max} = 37, T \in [30, 37] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1566] & \begin{matrix} 1 \\ 71232171232^1 18123218123215 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 38, \Delta = -111, \det = -100, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.76 \rightarrow 30, T_{\max} = 38, T \in [30, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1565] $8122315132^1 181232181232161 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -108$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1564] $812315132181232^1 1812321613 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 45$, $\Delta = -116$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 45, T \in [30, 45]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1560] $1 \overset{321}{8} 12321812^1 32181232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1551] $1 \overset{221}{(12)} 122315132^1 18123217123 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 42, T \in [30, 42]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1552] $1 \overset{21}{(12)} 122315132181232^1 171232 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 48, T \in [30, 48]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1553] $21 \overset{21}{(12)} 122315132^1 18123217123 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 42, T \in [30, 42]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1554] $31612321812^1 31513221(12)12231 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 31$, $\Delta = -96$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 31, T \in [30, 31]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1555] $321712321812^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 42, T \in [30, 42]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1556] $2321712^1 32181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 48, T \in [30, 48]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1557] $712315132^1 21(12)122315131613 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 48, T \in [30, 48]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1558] $61315132^1 1812321812321 \overset{21}{8} 1 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1559] $712315132^1 1812321812321 \overset{1}{7} 1 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1562] $1612321812321812^1 315132181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -108$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1561] $81232181232181232^1 1^8 1 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 42, T \in [30, 42]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1563] $61315132^1 18123218123218122 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.76 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1579] $1(12)1223151321812^1 32171232 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 48, T \in [30, 48]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1575] $1^{1513221}(12)122315132^1 1812315 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 39$, $\Delta = -112$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 39, T \in [30, 39]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1594] $32181232181232^1 1812321812 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1595] $413221(12)122315132^1 18123216 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 45$, $\Delta = -116$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 45, T \in [30, 45]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1596] $413221(12)12^1 231513218123216 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 45$, $\Delta = -116$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 45, T \in [30, 45]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1597] $6131612^1 32181232181232171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1598] $4132181232181232^1 171232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 40, T \in [30, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[1599] $6131612^1 321812321812321712 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 37$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1600] $712321812^1 321812321812314 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} =$

$$T_{\min} = 29.69 \rightarrow 30, \quad T_{\max} = 40, \quad T \in [30, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1601] 81232¹1613161316131613161¹+¹~~1~~→2 $T = 25$, $T_{\max} = 37$, $\Delta = -105$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 219 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[1602] 81232¹16131613161316131612+¹~~1~~→² $T = 25$, $T_{\max} = 37$, $\Delta = -105$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 219 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[1574] 1 (12) 12¹⁵¹³²²¹23151321812315+**1→2** $T = 25$, $T_{\max} = 39$, $\Delta = -112$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 39, T \in [30, 39]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1592] 712321712¹3218123218123151+ **$\hat{1} \rightarrow 2$** $T = 25$, $T_{\max} = 37$, $\Delta = -109$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 37, T \in [30, 37]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1591] 712321712¹3218123218123215+¹**→2** $T = 25$, $T_{\max} = 38$, $\Delta = -111$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 38, T \in [30, 38]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1590] 8123151321812¹321812321613+**1**→**2** $T = 25$, $T_{\max} = 45$, $\Delta = -116$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 45, T \in [30, 45]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1589] 413221(12)122¹315131613161315 + $\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 48, T \in [30, 48]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1588] 41413221(12)12¹23151321812315 + $\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 45$, $\Delta = -116$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 45, T \in [30, 45]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1587] 41413221(12)122315132¹1812315 + $\hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 45$, $\Delta = -116$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 45, T \in [30, 45]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1586] $\overset{1}{8}12321\overset{2321}{18}12321\overset{1}{3}21\overset{8}{8}1+\overset{1}{1}\rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, \quad T_{\max} = 42, \quad T \in [30, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1585] 81232¹1812321812321²³²¹ 8 1+**1→2** $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.69 \rightarrow 30, T_{\max} = 42, T \in [30, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1584] $1 \overset{321}{8} 1232181232^1 181232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.69 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[1583] $1812321812^1 3218123218123 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.69 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1582] $1(12)122^1 3151316131613161 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.69 \rightarrow 30$, $T_{\max} = 42$, $T \in [30, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1581] $71231513221(12)122^1 315131613 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.69 \rightarrow 30$, $T_{\max} = 48$, $T \in [30, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1580] $232171232^1 181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.69 \rightarrow 30$, $T_{\max} = 48$, $T \in [30, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[1578] $21(12)122315132^1 181232171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.69 \rightarrow 30$, $T_{\max} = 42$, $T \in [30, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[1577] $21(12)12^1 2315132181232171 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.69 \rightarrow 30$, $T_{\max} = 42$, $T \in [30, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1576] $5131613151322^1 1(12)1223151315 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.69 \rightarrow 30$, $T_{\max} = 42$, $T \in [30, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1593] $1812321812^1 32181232181223 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.69 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[1603] $1812321812321812^1 3218123 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -144$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.64 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[1604] $32181232^1 1812321812321812 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 40$, $\Delta = -113$, $\det = -144$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 29.64 \rightarrow 30$, $T_{\max} = 40$, $T \in [30, 40]$
 $\begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[1605] $1(12)122315132^1 181232171232 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 48$, $\Delta = -117$, $\det = -169$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 29.59 \rightarrow 30, T_{\max} = 48, T \in [30, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1607] & \begin{matrix} 1 \\ 7123151322^1 1(12)122315131613 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 48, \Delta = -117, \det = -169, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.59 \rightarrow 30, T_{\max} = 48, T \in [30, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1608] & \begin{matrix} 3221 \\ 1(12)12^1 23151316131613161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 29.59 \rightarrow 30, T_{\max} = 42, T \in [30, 42] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[1609] & \begin{matrix} 1 \\ 181232181232^1 18123218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 29.59 \rightarrow 30, T_{\max} = 40, T \in [30, 40] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1610] & \begin{matrix} 1 \\ 81232181232^1 1812321 \end{matrix} \begin{matrix} 2321 \\ 8 \end{matrix} 1 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 29.59 \rightarrow 30, T_{\max} = 42, T \in [30, 42] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1611] & 413221(12)12^1 2315131613161315 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 48, \Delta = -117, \det = -169, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.59 \rightarrow 30, T_{\max} = 48, T \in [30, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1612] & 812315132^1 1812321812321613 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 45, \Delta = -116, \det = -169, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 29.59 \rightarrow 30, T_{\max} = 45, T \in [30, 45] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1613] & \begin{matrix} 1 \\ 2321812^1 32181232181232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 29.59 \rightarrow 30, T_{\max} = 42, T \in [30, 42] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1614] & \begin{matrix} 1 \\ 321812321812^1 321812321812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 40, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 29.59 \rightarrow 30, T_{\max} = 40, T \in [30, 40] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1606] & 2321712321812^1 31513221(12)122 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 48, \Delta = -117, \det = -169, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.59 \rightarrow 30, T_{\max} = 48, T \in [30, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1615] & \begin{matrix} 1 \\ 71231513221(12)12^1 2315131613 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 25, T_{\max} = 48, \Delta = -117, \det = -196, (n_+, n_-, n_0) = \\
& (1, 25, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 29.55 \rightarrow 30, T_{\max} = 48, T \in [30, 48] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1616] & \begin{matrix} 1 \\ 812321812^1 321812321 \end{matrix} \begin{matrix} 2321 \\ 8 \end{matrix} 1 + \hat{1} \rightarrow 2 \quad T = 25, T_{\max} = 42, \Delta = -115, \det = -196, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 29.55 \rightarrow 30, T_{\max} = 42, T \in [30, 42] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1617] $232181232181232^1 181232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -225$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.52 \rightarrow 30, T_{\max} = 42, T \in [30, 42]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[1618] $232181232^1 181232181232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -289$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.46 \rightarrow 30, T_{\max} = 42, T \in [30, 42]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 203 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1619] $2321812321812^1 32181232181 + \hat{1} \rightarrow 2$ $T = 25$, $T_{\max} = 42$, $\Delta = -115$, $\det = -324$, $(n_+, n_-, n_0) = (1, 25, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 29.43 \rightarrow 30, T_{\max} = 42, T \in [30, 42]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$T = 26$ [1643] $7123151321812321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 4$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[1630] $413221(12)^{2^1 1} 12231513221(12)^{2221} 1 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 4$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32, T_{\max} = 41, T \in [32, 41]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[1636] $181232181231513221(12)^{21} 122^1 2 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 4$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[1637] $181232181231513221(12)^{22^1 1} 12 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 4$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[1638] $322181232181231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 4$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[1639] $22181232181231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 4$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[1640] $22^1 181232181231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 4$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[1641] $413221(12)^{21} 12231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 4$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[1634] $181232181231513221(12)^{2^1 1} 1223 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 4$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[1633] $413221(12)^{2^1 1} 12231513221(12)^{221} 12 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 41, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32, T_{\max} = 41, T \in [32, 41]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[1632] $413221(12)^{21} 12231513221(12)^{22^1 1} 12 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 41, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[1645] $6131612321812321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[1620] $3221(12)^{2^1 21} 12231513221(12) 122313 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1642] $413221(12)^{2^1 1} 12231513221(12) 12222 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 41, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32, T_{\max} = 41, T \in [32, 41]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[1629] $21(12)^{221} 1223151321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[1628] $21(12)^{22^1 1} 12231513218123218122 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[1627] $3221(12)^{2^1 1} 1223151321812321812 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 42, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1626] $1(12)^{2221} 1223151321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[1625] $1(12)^{22^1 21} 12231513218123218122 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[1624] $1^1 81232181231513221(12) 1222^1 2 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} =$

15.0

$$T_{\min} = 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[1623] \overset{2321}{1} \ 8 \ 1232181231513221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 46, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33, T_{\max} = 46, T \in [33, 46]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[1622] \overset{321}{1} \ 8 \ 1232181231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[1621] \overset{22^1}{1}(12)122315132181232181223 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1644] 4131613161316131513221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 41, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[1635] \overset{321}{1} \ 8 \ 1232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[1631] \overset{21}{413221}(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 41, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1657] \overset{32^1}{1} \ 8 \ 1232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1661] 32^1 2181232181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1662] 22181232181231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 38, T \in [33, 38]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1663] \overset{21}{413221}(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 41, \Delta = -113, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[1664] \overset{1}{6131513221}(12)1223151321812^1 3 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 49, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 49, T \in [33, 49]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1665] & 8123151321812321812321712^1 3 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 43, \Delta = -115, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.22 \rightarrow 33, T_{\max} = 43, T \in [33, 43] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1660] & 322181232181231513221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.22 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1646] & 221(12)^{3221} 12231513221(12)122^1 313 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -102, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.22 \rightarrow 33, T_{\max} = 38, T \in [33, 38] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1647] & 221(12)^{32^1 21} 12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -102, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.22 \rightarrow 33, T_{\max} = 38, T \in [33, 38] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[1648] & 1(12)^{221} 12231513218123218122^1 3 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.22 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1649] & 1 \ 8 \ 1232181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.22 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[1650] & 1 \ 8 \ 1232181231513221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.22 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1651] & 1 \ 8 \ 1232181231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.22 \rightarrow 33, T_{\max} = 38, T \in [33, 38] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1652] & 1(12)^{222^1 1} 12231513218123218122 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.22 \rightarrow 33, T_{\max} = 38, T \in [33, 38] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0} \\
[1658] & 1 \ 8 \ 1232181231513221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.22 \rightarrow 33, T_{\max} = 38, T \in [33, 38] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1659] & 1 \ 8 \ 1232181231513221(12)^{21} 12 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.22 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1656] $181232181231513221(12)122^13 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 42$, $T \in [33, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1655] $413221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 9$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 41$, $T \in [33, 41]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1654] $21(12)1223151321812321812 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 42$, $T \in [33, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [1653] $21(12)12231513218123218122^13 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 42$, $T \in [33, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1673] $1812^132181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 16$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.75 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1672] $1812^132181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 16$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.75 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1674] $22181232181231513221(12)12^1222 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 16$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.75 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1675] $221812^132181231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 16$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.75 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1676] $413221(12)12231513221(12)12^1222 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 16$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.75 \rightarrow 32$, $T_{\max} = 41$, $T \in [32, 41]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1677] $1712^1321812321812315132181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 16$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.75 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1678] $712315132181232181232^1181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 16$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.75 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1679] $8122315132181232181232^1171 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 16$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.75 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

15.0

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1680] 181232171232181232181232¹15+ $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[1682] 712315132181232181232¹18122+ $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1683] 8122315132181232181232¹1712+ $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1684] 712321¹81232181232181232¹15+ $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[1685] 613161232181232181232¹1²¹81+ $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1686] 613161232181232181232¹18122+ $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1687] 71232171232181232181232¹161+ $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1671] 21²²¹(12)122315132181232¹18122+ $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1670] 1²²²¹(12)122315132181232¹18122+ $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1669] 1²¹81232181231513221(12)12¹222+ $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1668] 1²¹812¹32181231513221(12)12222+ $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.75 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

- [1667] $^{3221}221(12)12231513221(12)12^12313+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -102$, $\det = 16$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.75 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1666] $^{3221}22^11(12)12231513221(12)122313+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -102$, $\det = 16$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.25 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 25 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1681] $21712^1321812321812315132181+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 16$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.75 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1713] $^{21}1812321812315132^121(12)1222+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.44 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1700] $^{221}21(12)122^131513218123218122+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.44 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [1704] $4132^121(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.44 \rightarrow 32$, $T_{\max} = 41$, $T \in [32, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1705] $413221(12)122315132^121(12)1+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.44 \rightarrow 32$, $T_{\max} = 41$, $T \in [32, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1706] $4132^121(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.44 \rightarrow 32$, $T_{\max} = 41$, $T \in [32, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1707] $413221(12)122315132^121(12)12+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.44 \rightarrow 32$, $T_{\max} = 41$, $T \in [32, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1708] $61231513221(12)122315132^1181+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.44 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1709] $612^131513221(12)122315132181+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.44 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [1710] $^{21}181232181231513221(12)12^123+\hat{1}\rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1711] \quad 1812^1 32181231513221(12)^{21} 1223 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1712] \quad 1 \quad 8 \quad 12321^{232} 1 \quad 1232181231513221(12)^{21} 1 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 46, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 46, T \in [32, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[1702] \quad 612^1 32181231513221(12)^{1513221} 1 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 39, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1714] \quad 1 \quad 8 \quad 1232^1 181231513221(12)^1 1222 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1715] \quad 181232^1 181231513221(12)^{21} 12 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1716] \quad 1812321812315132^1 21(12)^{221} 12 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1717] \quad 513161316131513221(12)122^1 3151 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 40, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 40, T \in [32, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1718] \quad 322^1 181232181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1719] \quad 221812321812315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1720] \quad 22181232^1 181231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1721] \quad 612^1 31513221(12)12231513161315 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 41, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1722] $16131513221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1723] $16131513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1724] $1712^1 31513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1725] $4132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 41, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1726] $413221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 41, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1727] $61231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1728] $612^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1729] $61231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1730] $612^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1731] $712231513221(12)122315132^1 171 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1732] $7122^1 31513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1733] $7122^1 31513221(12)122315132217 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1734] $171232^1 1812321812315132181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1735] $7123151321812321812^{21} 32181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1736] $81223151321812321812^{1} 32171 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1737] $1812321712321812321812^{1} 3215 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1738] $1812321712321812321812^{1} 3151 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1739] $2171232^{1} 1812321812315132181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1740] $7123151321812321812^{1} 3218122 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1741] $81223151321812321812^{1} 321712 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1742] $413221(12)122315132181232^{1} 171 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1743] $4132^{1} 21(12)122315132181232171 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1744] $712321812321812321812^{1} 3215 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1745] \quad 712321\overset{1}{8}12321812321812^1 3151 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1746] \quad 6131612321812321812^1 321\overset{21}{8}1 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[1747] \quad 4132^1 21(12)1223151316131613161 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1748] \quad 6131612321812321812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[1749] \quad 712321712321812321812^1 32161 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[1750] \quad 4132^1 181232181232181\overset{1}{2}32181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 40, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 40, T \in [32, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1751] \quad 41413221(12)122315132181232^1 16 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 43, \Delta = -115, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 43, T \in [32, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1698] \quad 21(12)122315132181232^1 1812 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1697] \quad 21(12)1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1696] \quad 21(12)1223151321812321812^1 23 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1695] \quad 1(12)1223151321812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1694] & \begin{matrix} 2221 \\ 1(12)122^1 31513218123218122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[1693] & \begin{matrix} 21 \\ 181232^1 181231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1692] & \begin{matrix} 21 \\ 1812321812315132^1 21(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1691] & \begin{matrix} 232^1 1 \\ 181232181231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 46, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.44 \rightarrow 32, T_{\max} = 46, T \in [32, 46] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1690] & \begin{matrix} 221 \\ 1(12)1223151321812321812^1 23 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.44 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1689] & \begin{matrix} 322^1 1 \\ 221(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -102, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1688] & \begin{matrix} 3221 \\ 221(12)122315132^1 21(12)122313 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -102, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1701] & \begin{matrix} 15132^1 21 \\ 61232181231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 39, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.44 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1699] & \begin{matrix} 221 \\ 21(12)1223151321812^1 3218122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1703] & \begin{matrix} 1 \\ 1812^1 31513221(12)12231513161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.44 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1770] & \begin{matrix} 1 \\ 8122^1 31513221(12)122315131613 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 46, \Delta = -116, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 46, T \in [32, 46]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1761] & \begin{matrix} 321 \\ 1 \ 8 \ 12321812315132^1 21(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1768] & \begin{matrix} 1 \\ 16131513221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1769] & \begin{matrix} 21 \\ 413221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 41, \Delta = -113, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1766] & \begin{matrix} 21 \\ 2218123218123151322^1 1(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1771] & \begin{matrix} 1 \\ 61231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1772] & \begin{matrix} 1 \\ 712^1 231513221(12)122315132171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1765] & \begin{matrix} 21 \\ 3221812321812315132^1 21(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1764] & \begin{matrix} 21 \\ 18123218123151322^1 1(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1760] & \begin{matrix} 21 \\ 413221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 41, \Delta = -113, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1759] & \begin{matrix} 21 \\ 413221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 41, \Delta = -113, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1773] & \begin{matrix} 1 \\ 712^1 231513221(12)122315132217 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1774] $1712321812321812315132^1 181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1775] $712^1 3151321812321812321^1 81 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1776] $8122^1 3151321812321812321^1 71 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1777] $1812^1 3217123218123218123215 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -110$, $\det = 36$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1778] $1812^1 3217123218123218123151 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 36$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1779] $21712321812321812315132^1 181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1780] $712^1 31513218123218123218122 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1781] $8122^1 3151321812321812321712 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1782] $712^1 3218123218123218123215 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -110$, $\det = 36$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1783] $712^1 3218123218123218123151 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 36$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1784] $712^1 32171232181232181232161 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 36$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1785] $4132181232181232181232^1 181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 40$, $\Delta = -112$, $\det = 36$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.22 \rightarrow 32$, $T_{\max} = 40$, $T \in [32, 40]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1752] & \begin{matrix} 3221 \\ 221(12)1223151322^1 1(12)122313+\hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -102, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1753] & \begin{matrix} 221 \\ 1(12)122^1 315132181232181223+\hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1754] & \begin{matrix} 321 \\ 1 \ 8 \ 12321812315132^1 21(12)1222+\hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1755] & \begin{matrix} 21 \\ 1 \ 8 \ 123218123151322^1 1(12)12222+\hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1756] & \begin{matrix} 2221 \\ 1(12)12^1 231513218123218122+\hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1757] & \begin{matrix} 21 \\ 21(12)122^1 315132181232181223+\hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1758] & \begin{matrix} 221 \\ 21(12)12^1 231513218123218122+\hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1763] & \begin{matrix} 321 \\ 1 \ 8 \ 12321812315132^1 21(12)12+\hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1767] & \begin{matrix} 1 \\ 316131513221(12)122315132^1 181+\hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 46, \Delta = -116, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 46, T \in [32, 46] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1762] & \begin{matrix} 21 \\ 1 \ 8 \ 123218123151322^1 1(12)1222+\hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.22 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1796] & \begin{matrix} 3221 \\ 21(12)1223151321812^1 321812+\hat{1} \rightarrow 2 \end{matrix} \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1831] $61231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1798] $61232181231513221^{1513221} (12) 1 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 39, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1799] $61232^1 181231513221^{1513221} (12) 1 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 39, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1800] $1812315132^1 21(12)12231513161 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1801] $181231513221(12)122^1 31513161 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1802] $41322^1 1(12)12231513221^{2221} (12) 1 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 41, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1803] $413221^{21} (12)122^1 31513221^{2221} (12) 1 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 41, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1804] $41322^1 1(12)12231513221^{2221} (12) 12 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 41, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1805] $413221^{21} (12)122^1 31513221^{2221} (12) 12 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 41, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1806] $61231513221(12)122^1 315132181 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1807] $612315132^1 21(12)122315132181 + \hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

- [1808] $181232^1 181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [1809] $181232^1 181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1810] $181232181231513221^1(12)1 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [1811] $1812321812315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 46$, $T \in [32, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [1812] $1812321812^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1813] $181232181231513221^1(12)12 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [1814] $181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1815] $1812321812^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1816] $161316131513221(12)122^1 3151315 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 40$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 40$, $T \in [32, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1817] $513161316131513221(12)12^1 23151 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 40$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 40$, $T \in [32, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1818] $161315132^1 21(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 40$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 40$, $T \in [32, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1819] $3221812^1 32181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1820] & 32218123218123151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[1821] & 221812321812^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1822] & 612315132^1 21(12)12231513161315 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 41, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1823] & 16131513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1824] & 161315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1825] & 161315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1826] & 16131513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1827] & 1712315132^1 21(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1828] & 41322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 41, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1829] & 413221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 41, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1830] & 812^1 231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 46, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.06 \rightarrow 32, T_{\max} = 46, T \in [32, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1795] $21(12)^{21}12^1 2315132181232181223 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1794] $21(12)^{21}122315132181232^1 181223 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1793] $1(12)^{2221}122315132^1 18123218122 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1834] $61231513221(12)^1 122^1 3151322181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1835] $7122315132^1 21(12)^1 122315132171 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1836] $712231513221(12)^1 122^1 315132171 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1837] $7122315132^1 21(12)^1 122315132217 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1838] $1712321812^1 321812315132181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1839] $712315132181232^1 181232181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1840] $812^1 2315132181232181232171 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
- [1841] $8122315132181232^1 181232171 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1842] $181232171232181232^1 18123215 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1843] \ 181232171232181232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1844] \ 21712321812^1 321812315132181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[1845] \ 812^1 32171232181232181231513 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 46, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 46, T \in [32, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1846] \ 812315132181232181232^1 17123 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 43, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 43, T \in [32, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1847] \ 812^1 31513218123218123217123 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 43, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 43, T \in [32, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1848] \ 712315132181232^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[1849] \ 8122315132181232^1 1812321712 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[1850] \ 812^1 23151321812321812321712 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[1851] \ 413221(12)1223151321812^1 32171 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[1852] \ 41322^1 1(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[1853] \ 71232181232181232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

- [1854] $71232181232181232^1 18123151 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1855] $41316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 41$, $T \in [32, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [1856] $613161232181232^1 181232181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1857] $41316131613161315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 41$, $T \in [32, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [1858] $41322^1 1(12)1223151316131613161 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1859] $613161232181232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1860] $71232171232181232^1 181232161 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1861] $41321812^1 32181232181232181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 40$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 40$, $T \in [32, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1862] $414132^1 21(12)12231513218123216 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 43$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 43$, $T \in [32, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1863] $41413221(12)1223151321812^1 3216 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 43$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 43$, $T \in [32, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1792] $1812321812^1 31513221(12)12222 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 38$, $T \in [32, 38]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1791] $1^1 812321812315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.06 \rightarrow 32$, $T_{\max} = 46$, $T \in [32, 46]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[1790] $1 \ 8 \ 12^{321} 12^1 32181231513221(12)1222+\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1789] $1 \ 8 \ 123218123151322^1 1(12)1222+\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[1788] $1(12)122315132181232^1 181223+\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1787] $1(12)12^{221} 12^1 2315132181232181223+\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1786] $221(12)122^1 31513221(12)122313+\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -102$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1833] $612315132^1 21(12)1223151322181+\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1832] $612315132^1 21(12)1223151321812+\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1797] $21(12)122315132^1 18123218122+\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.06 \rightarrow 32, T_{\max} = 38, T \in [32, 38]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1866] $1 \ 8 \ 123218123151322^1 1(12)122+\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.94 \rightarrow 31, T_{\max} = 46, T \in [31, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1867] $21(12)1223151321812^1 32181223+\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.94 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1868] $1 \ 8 \ 1232^1 181231513221(12)1+\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.94 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

- [1869] $1 \ 8 \ 123218123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 46$, $T \in [31, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1870] $1 \ 8 \ 1232^1 181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 42$, $T \in [31, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1871] $32171232^1 181231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 31$, $\Delta = -95$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 31$, $T \in [31, 31]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1872] $322181232^1 181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 42$, $T \in [31, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1873] $171232181232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1874] $7123151321812^1 32181232181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1875] $81223151321812^1 32181232171 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1876] $1812321712321812^1 3218123215 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -110$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1877] $1812321712321812^1 3218123151 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1878] $2171232181232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1879] $812321712321812321812^1 31513 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 46$, $T \in [31, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1880] $8123151321812321812^1 3217123 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 43$, $\Delta = -115$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 43$, $T \in [31, 43]$

- $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [1881] 7123151321812¹3218123218122+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1882] 81223151321812¹321812321712+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- ¹
 [1883] 712321812321812¹3218123215+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -110$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- ¹
 [1884] 712321812321812¹3218123151+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- ²¹
 [1885] 413161316131613151322¹1(12)1+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 41$, $T \in [31, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- ²¹
 [1886] 6131612321812¹32181232181+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1887] 413161316131613151322¹1(12)122 + $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 41$, $\Delta = -113$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 41$, $T \in [31, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1888] 6131612321812¹3218123218122+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1889] 712321712321812¹32181232161+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- ¹
 [1890] 4132181232¹181232181232181+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 40$, $\Delta = -112$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 40$, $T \in [31, 40]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- ³²¹
 [1865] 1 8 1232¹181231513221(12)1222+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.94 \rightarrow 31$, $T_{\max} = 42$, $T \in [31, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- ²²¹
 [1864] 1(12)1223151321812¹32181223+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 30.94 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1902] \quad 161316131513221(12)12^1 23151315 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 40, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 40, T \in [31, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1927] \quad 71232^1 171232181232181232161 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[1900] \quad 61231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1930] \quad 41413221(12)122^1 31513218123216 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 43, \Delta = -115, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 43, T \in [31, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1929] \quad 4141322^1 1(12)12231513218123216 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 43, \Delta = -115, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 43, T \in [31, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1891] \quad 1^{13221}(12)1223151322^1 1^{221}(12)1223 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 31, \Delta = -94, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1892] \quad 221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -102, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[1893] \quad 21^{3221}(12)122^1 3151321812321812 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[1894] \quad 612321812315132^1 21^{1513221}(12)1 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 39, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 39, T \in [31, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1895] \quad 1^{181231513221}(12)12^1 231513161 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1896] \quad 1^{18123151322^1 1}(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

15.0

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1911] \quad 16131513221(12)12^1 23151322181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1912] \quad 17123151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1913] \quad 413221(12)^{21} 12^1 231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 41, \Delta = -113, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 41, T \in [31, 41]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1914] \quad 6131513221(12)122315132^1 18123 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 49, \Delta = -117, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 49, T \in [31, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1915] \quad 6123151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1916] \quad 61231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1917] \quad 6123151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1918] \quad 61231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1919] \quad 71223151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1920] \quad 712231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1921] \quad 71223151322^1 1(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1922] & 181232^1 17123218123218123215 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1923] & 181232^1 17123218123218123151 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1924] & 413221(12)122^1 315132181232171 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.84 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1925] & 71232^1 18123218123218123215 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1926] & 71232^1 18123218123218123151 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.84 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1928] & 41321812321812321812^1 32181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 40, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.84 \rightarrow 31, T_{\max} = 40, T \in [31, 40] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1945] & 8122315132^1 181232181232171 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1950] & 812315132181232^1 18123217123 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 43, \Delta = -115, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 43, T \in [31, 43] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1954] & 71232181232^1 18123218123151 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1931] & 1^{221}(12)122315132^1 181232181223 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1932] & 1^{321} 8 12321812^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1933] $1 \ 8 \ 12^{2321} 32181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.76 \rightarrow 31$, $T_{\max} = 46$, $T \in [31, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1934] $21(12)122315132^1 181232181223 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.76 \rightarrow 31$, $T_{\max} = 42$, $T \in [31, 42]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1935] $1 \ 8 \ 12321812^1 31513221^{221}(12)1 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.76 \rightarrow 31$, $T_{\max} = 42$, $T \in [31, 42]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1936] $1 \ 8 \ 12^{2321} 32181231513221^{21}(12)1 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.76 \rightarrow 31$, $T_{\max} = 46$, $T \in [31, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1937] $1 \ 8 \ 12321812^1 31513221^{21}(12)12 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.76 \rightarrow 31$, $T_{\max} = 42$, $T \in [31, 42]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1938] $321712321812^1 31513221(12)12231 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 31$, $\Delta = -95$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.76 \rightarrow 31$, $T_{\max} = 31$, $T \in [31, 31]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1939] $3221812321812^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.76 \rightarrow 31$, $T_{\max} = 42$, $T \in [31, 42]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1940] $316131513221^1(12)122^1 315132181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.76 \rightarrow 31$, $T_{\max} = 46$, $T \in [31, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1941] $61315132^1 21(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 49$, $\Delta = -117$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.76 \rightarrow 31$, $T_{\max} = 49$, $T \in [31, 49]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1942] $8122315132^1 21(12)122315131613 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.76 \rightarrow 31$, $T_{\max} = 46$, $T \in [31, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1956] $613161232^1 18123218123218122 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.76 \rightarrow 31$, $T_{\max} = 38$, $T \in [31, 38]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1957] $71232171232^1 181232181232161 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 100$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1958] & 41321812321812^1 32181232181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 40, \Delta = -112, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 40, T \in [31, 40] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1951] & 712315132^1 18123218123218122 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1943] & 1712321812321812^1 315132181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1944] & 712315132^1 1812321812321 \overset{21}{8} 1 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1953] & 71232181232^1 18123218123215 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -110, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1955] & 613161232^1 1812321812321 \overset{21}{8} 1 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1946] & 181232171232^1 18123218123215 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -110, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1947] & 181232171232^1 18123218123151 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -108, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1948] & 21712321812321812^1 315132181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1949] & 81232171232181232^1 181231513 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 46, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 46, T \in [31, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1952] & 8122315132^1 1812321812321712 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 38, \Delta = -106, \det = 100, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.76 \rightarrow 31, T_{\max} = 38, T \in [31, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1969] & 16131513221(12)12^1 231513161315 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 40, \Delta = -112, \det = 121, (n_+, n_-, n_0) =
\end{aligned}$$

(1, 26, 0), $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 40, T \in [31, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1970] 61231513221(12)12¹231513161315 + $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 41, \Delta = -113, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 41, T \in [31, 41]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1971] 3161315132¹21(12)122315132181 + $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 46, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 46, T \in [31, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1968] 5131613161315132¹21(12)1223151 + $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 40, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 40, T \in [31, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1967] 16131613151322¹1(12)1223151315 + $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 40, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 40, T \in [31, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1966] 1²³²¹8 1232¹181231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 46, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 46, T \in [31, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[1965] 18123218123151322¹1(12)1223 + $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 42, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1964] 1812321812¹31513221(12)1223 + $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 42, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1963] 612321812¹⁵¹³²²¹31513221 (12) 1 + $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 39, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 39, T \in [31, 39]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1973] 812231513221(12)122¹315131613 + $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 46, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 46, T \in [31, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1974] 1812321712¹3218123218123215 + $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1975] 1812321712¹3218123218123151 + $\hat{1} \rightarrow 2$ $T = 26, T_{\max} = 38, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1976] 81232¹171232181232181231513+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 46, T \in [31, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1977] 812321712321812¹32181231513+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 46, T \in [31, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1978] 8123151321812¹3218123217123+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 43$, $\Delta = -115$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 43, T \in [31, 43]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1979] 413221(12)¹122315132¹181232171 + $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1980] 413221(12)¹12¹2315132181232171 + $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1981] 712321812¹3218123218123215+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1982] 712321812¹3218123218123151+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1983] 6131612¹321812321812321²¹81+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1984] 413221(12)¹122¹3151316131613161 + $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 42, T \in [31, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1985] 6131612¹3218123218123218122+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1986] 712321712¹32181232181232161+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 38$, $\Delta = -108$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 30.69 \rightarrow 31, T_{\max} = 38, T \in [31, 38]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1987] 4132181232181232¹181232181+ $\hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 40$, $\Delta = -112$, $\det = 121$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 30.69 \rightarrow 31, T_{\max} = 40, T \in [31, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1988] & 41413221(12)12^1 231513218123216 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 43, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.69 \rightarrow 31, T_{\max} = 43, T \in [31, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1989] & 41413221(12)122315132^1 18123216 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 43, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.69 \rightarrow 31, T_{\max} = 43, T \in [31, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1962] & 6123218123151322^1 1^{1513221} (12) 1 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 39, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.69 \rightarrow 31, T_{\max} = 39, T \in [31, 39] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1961] & 21(12)12^1 23151321812321812 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.69 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1960] & 21(12)122315132^1 1812321812 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 42, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.69 \rightarrow 31, T_{\max} = 42, T \in [31, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[1959] & 1^1 8^{2321} 1232^1 181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 46, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.69 \rightarrow 31, T_{\max} = 46, T \in [31, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[1972] & 6131513221(12)122^1 31513218123 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 49, \Delta = -117, \det = 121, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.69 \rightarrow 31, T_{\max} = 49, T \in [31, 49] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[1990] & 1^1 8^{2321} 12321812^1 31513221(12)122 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 46, \Delta = -116, \det = 169, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.59 \rightarrow 31, T_{\max} = 46, T \in [31, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1991] & 1^1 8^{2321} 12321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 46, \Delta = -116, \det = 169, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.59 \rightarrow 31, T_{\max} = 46, T \in [31, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1992] & 51316131613151322^1 1(12)1223151 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 40, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.59 \rightarrow 31, T_{\max} = 40, T \in [31, 40] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[1993] & 316131513221(12)12^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 26, T_{\max} = 46, \Delta = -116, \det = 169, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 30.59 \rightarrow 31, T_{\max} = 46, T \in [31, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1994] $613151322^1 1(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 49$, $\Delta = -117$, $\det = 169$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.59 \rightarrow 31$, $T_{\max} = 49$, $T \in [31, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1995] $81223151322^1 1(12)122315131613 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 169$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.59 \rightarrow 31$, $T_{\max} = 46$, $T \in [31, 46]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1996] $81232171232^1 181232181231513 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 169$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.59 \rightarrow 31$, $T_{\max} = 46$, $T \in [31, 46]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1997] $812315132^1 18123218123217123 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 43$, $\Delta = -115$, $\det = 169$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.59 \rightarrow 31$, $T_{\max} = 43$, $T \in [31, 43]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1998] $413221(12)12^1 23151316131613161 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 42$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.59 \rightarrow 31$, $T_{\max} = 42$, $T \in [31, 42]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$
- [1999] $31613151322^1 1(12)122315132181 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 196$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.55 \rightarrow 31$, $T_{\max} = 46$, $T \in [31, 46]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2000] $6131513221^1 (12)12^1 231513218123 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 49$, $\Delta = -117$, $\det = 196$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.55 \rightarrow 31$, $T_{\max} = 49$, $T \in [31, 49]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2001] $812231513221(12)12^1 2315131613 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 196$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.55 \rightarrow 31$, $T_{\max} = 46$, $T \in [31, 46]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2002] $812321712^1 32181232181231513 + \hat{1} \rightarrow 2$ $T = 26$, $T_{\max} = 46$, $\Delta = -116$, $\det = 196$, $(n_+, n_-, n_0) = (1, 26, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 30.55 \rightarrow 31$, $T_{\max} = 46$, $T \in [31, 46]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- $T = 27$ [2005] $1^{22} 1^{21} (12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -108$, $\det = -4$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34$, $T_{\max} = 39$, $T \in [34, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2009] $21^{22} 1^1 (12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -108$, $\det = -4$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34$, $T_{\max} = 39$, $T \in [34, 39]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2010] $21^{22} 1^{31} 513221 (12)12 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -4$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} =$

15.0

$$T_{\min} = 34, T_{\max} = 50, T \in [34, 50]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[2011] 21 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$ $\begin{smallmatrix} 231513221 \\ (12) \end{smallmatrix}$ 12231513221(12)12¹2+ $\hat{1} \rightarrow 2$ $T = 27, T_{\max} = 50, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34, T_{\max} = 50, T \in [34, 50]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[2012] 21 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$ $\begin{smallmatrix} 31513221 \\ (12) \end{smallmatrix}$ 12231513221(12)122¹2+ $\hat{1} \rightarrow 2$ $T = 27, T_{\max} = 47, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34, T_{\max} = 47, T \in [34, 47]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[2013] 1513221²¹ $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$ 12231513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[2014] 21 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$ $\begin{smallmatrix} 1513221 \\ (12) \end{smallmatrix}$ 12231513221(12)1222¹2+ $\hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34, T_{\max} = 39, T \in [34, 39]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[2015] 413221²¹ $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$ 12231513221(12)122313 + $\hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[2016] 4132181232181231513221²²¹ $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$ (12)1+ $\hat{1} \rightarrow 2$ $T = 27, T_{\max} = 42, \Delta = -113, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34, T_{\max} = 42, T \in [34, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[2017] 41413221¹ $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$ $\begin{smallmatrix} 22^1 21 \\ (12) \end{smallmatrix}$ 12231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 27, T_{\max} = 41, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34, T_{\max} = 41, T \in [34, 41]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[2018] 41413221¹ $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$ $\begin{smallmatrix} 22^1 1 \\ (12) \end{smallmatrix}$ 12231513221(12)12+ $\hat{1} \rightarrow 2$ $T = 27, T_{\max} = 41, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34, T_{\max} = 41, T \in [34, 41]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[2019] 15131613161316131513221(12)12¹2 + $\hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34, T_{\max} = 39, T \in [34, 39]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[2020] 22¹1812321812321812315132181+ $\hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34, T_{\max} = 39, T \in [34, 39]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[2021] 4132181232181231513221(12)122¹2 + $\hat{1} \rightarrow 2$ $T = 27, T_{\max} = 42, \Delta = -113, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[2022] & (12)1223151316131613161315132^1 2 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34, T_{\max} = 50, T \in [34, 50] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[2023] & 81223151321812321812321812^1 2 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34, T_{\max} = 39, T \in [34, 39] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[2024] & 41413221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 41, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[2006] & 1(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[2004] & 1(12)12231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34, T_{\max} = 47, T \in [34, 47] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[2003] & 1(12)12231513221(12)122315132^1 2 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34, T_{\max} = 50, T \in [34, 50] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[2008] & 21(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[2007] & 3221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33, T_{\max} = 50, T \in [33, 50] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[2034] & 71231513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 47, T \in [34, 47] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[2031] & 21(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 47, T \in [34, 47] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[2032] & 21(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -108, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 39, T \in [34, 39]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[2033] & 413221(12)^{21}12231513221(12)122^1313 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -101, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 39, T \in [34, 39] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[2029] & 21(12)^{21}12231513221(12)^{2^1 1}12231513 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.33 \rightarrow 33, T_{\max} = 47, T \in [33, 47] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[2035] & 8123151321812321812321^{32^1 1}8^1 1 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 43, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 43, T \in [34, 43] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[2036] & 41413221(12)^112231513221(12)^{222^1 1}1 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 41, \Delta = -112, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[2037] & 5131613161316131513221(12)122^13 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 50, T \in [34, 50] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[2038] & 4132181232181231513221(12)12^122 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[2039] & 81231513218123218123218122^13 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 43, \Delta = -114, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 43, T \in [34, 43] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[2040] & 41413221(12)^112231513221(12)122^122 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 41, \Delta = -112, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[2028] & 21(12)^{32^1 21}12231513221(12)^{21}122315 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 50, T \in [34, 50] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[2030] & 31513221(12)^{2^1 1}12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.33 \rightarrow 33, T_{\max} = 47, T \in [33, 47] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2027] & 21(12)^{3221}12231513221(12)^{2^1 1}122315 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.33 \rightarrow 33, T_{\max} = 50, T \in [33, 50] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2026] $1(12)12231513221(12)1223151+\hat{1}\rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -108, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34, T_{\max} = 39, T \in [34, 39]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2025] $1(12)12231513221(12)12231513+\hat{1}\rightarrow 2$ $T = 27, T_{\max} = 47, \Delta = -116, \det = -9, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.33 \rightarrow 33, T_{\max} = 47, T \in [33, 47]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [2041] $1(12)12231513221(12)122315132 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 50, \Delta = -117, \det = -16, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32, T_{\max} = 50, T \in [32, 50]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 12 & 6 \\ 2 & 6 & 3 \end{pmatrix}_{b_0}$
- [2042] $231513221(12)12231513221(12)122 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 50, \Delta = -117, \det = -16, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32, T_{\max} = 50, T \in [32, 50]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$
- [2043] $21(12)12231513221(12)12^1222 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -108, \det = -16, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.75 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2045] $1812^1321812321812315132181+\hat{1}\rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.75 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2046] $8122315132181232181232^1181+\hat{1}\rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.75 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2047] $221812^1321812321812315132181 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.75 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2048] $8122315132181232181232^118122 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -105, \det = -16, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.75 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2049] $181232181232181232181232^115+\hat{1}\rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.75 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [2050] $41413221(12)12231513221(12)12^1222 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 41, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.75 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2051] $71232171232181232181232^1171+\hat{1}\rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} =$

15.0

$$T_{\min} = 32.75 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2052] \quad 1612^1 32181232181232171232181 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.75 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2053] \quad 71232171232181232181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.75 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2054] \quad 71232181232181232181232^1 161 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.75 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2044] \quad 413221^{21}(12)12231513221(12)12^1 2313 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -101, \det = -16, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.75 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2059] \quad 21^{221}(12)122^1 31513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2084] \quad 4132^1 181232181231513221^{21}(12)12 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2061] \quad 2231513221^{2^1 1}(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.80 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 30 & 10 \\ 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2062] \quad 21^{15132^1 21}(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2063] \quad 21^{1513221}(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2064] \quad 1^{1513221}(12)122315132181232^1 171 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2065] \quad 1^{15132^1 21}(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2066] & \begin{matrix} 1 \\ 1812^1 31513221(12)122315132171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2067] & \begin{matrix} 1 \\ 181231513221(12)122315132^1 171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[2068] & \begin{matrix} 1 \\ 712231513221(12)122315132^1 181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2069] & \begin{matrix} 1 \\ 7122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[2070] & \begin{matrix} 1 \\ 51316131513221(12)122315132^1 171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 40, \Delta = -111, \det = -25, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2058] & \begin{matrix} 322^1 1 \\ 21(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 50, T \in [33, 50] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2057] & \begin{matrix} 2221 \\ 1(12)12231513221(12)122^1 3151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2071] & \begin{matrix} 1 \\ 7122^1 31513221(12)12231513161315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 40, \Delta = -111, \det = -25, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2072] & \begin{matrix} 1 \\ 171231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[2073] & \begin{matrix} 1 \\ 1712^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2074] & \begin{matrix} 1 \\ 171231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]
\end{aligned}$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[2075] $1712^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2076] $413221^{21}(12)122315132^1 21(12)122313 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -101$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2077] $4132^1 21(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -101$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2078] $7122^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[2079] $712231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2080] $712231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2081] $7122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[2082] $181232^{21} 1812321812315132181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2083] $4132^1 181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2056] $1(12)122^1 31513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -108$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2055] $1(12)12231513221(12)1223151322 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.80 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 20 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[2106] $712321812321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[2105] $712321712321812321812^1 321712 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[2104] $41413221(12)122315132181232^1 171 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2103] $161232^1 181232181232171232181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2102] $712321712321812321812^1 32171 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[2101] $4132^1 21(12)1223151321812321812 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2100] $413221(12)122315132181232^1 1812 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2099] $41413221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 41$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2098] $1812321812321812321812^1 3151 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2097] $1812321812321812321812^1 3215 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2096] $81223151321812321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[2095] $8123151321812321812321812^1 23 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 43$, $\Delta = -114$, $\det = -25$, $(n_+, n_-, n_0) =$

(1, 27, 0), $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 43, T \in [33, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2094] $612^1 31513221(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[2093] $4132^1 181232181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 42, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2092] $22181232^1 1812321812315132181 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2091] $5131613161316131513221(12)12^1 23 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 50, \Delta = -117, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 50, T \in [33, 50]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2090] $1613161316131513221(12)122^1 3151 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2089] $4132^1 21(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 42, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2088] $413221(12)122315132181232^1 181 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 42, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2087] $41413221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 41, \Delta = -112, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2086] $41413221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 41, \Delta = -112, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2085] $81223151321812321812^1 32181 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[2060] $21(12)12231513221(12)122^1 3151 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.44 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

- [2109] $21^{21}(12)122^1 31513221^{21}(12)12231513 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 47$, $T \in [33, 47]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2123] $41321812321812315132^1 21^{221}(12)1 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 42$, $T \in [33, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2111] $21^{31513221}(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 47$, $T \in [33, 47]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2112] $21^{1513221}(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -108$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2113] $712^{31513221} 32181231513221^{121}(12)1 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 50$, $T \in [33, 50]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2114] $712^{1231513221}(12)122315132181^1 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2115] $712^{1231513221}(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 40$, $\Delta = -111$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2116] $171231513221^{121}(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2117] $413221^{21}(12)1223151322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -101$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2118] $712^{131513221}(12)12231513218123^1 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 47$, $T \in [33, 47]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2119] $712^{1231513221}(12)1223151321812^1 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.22 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2120] $712231513221^{121}(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) =$

$$(1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2107] \quad 1(12)122^{\overset{21}{1}}31513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -36, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 47, T \in [33, 47]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2124] \quad 41321812321812315132^{\overset{21}{1}}21(12)12 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2125] \quad 8122^{\overset{21}{1}}315132181232181232181 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[2126] \quad 41413221(12)1223151322^{\overset{2221}{1}}1(12)1 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 41, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2127] \quad 41413221(12)1223151322^{\overset{221}{1}}1(12)12 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 41, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2128] \quad 221812321812321812315132^{\overset{1}{1}}181 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2129] \quad 41321812321812315132^{\overset{1}{1}}21(12)1222 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2130] \quad 8122^{\overset{1}{1}}31513218123218123218122 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[2131] \quad 1812^{\overset{1}{1}}3218123218123218123215 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -109, \det = -36, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2132] \quad 1812^{\overset{1}{1}}3218123218123218123151 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2133] \quad 41413221(12)1223151322^{\overset{1}{1}}1(12)12222 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 41, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2134] $712^1 32171232181232181232171 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2135] $161232181232181232171232^1 181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2136] $712^1 321712321812321812321712 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2137] $712^1 32181232181232181232161 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2108] $1(12)^{2221} 12^1 231513221(12)^{21} 1223151 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -108$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2121] $712^1 231513221(12)^1 1223151322181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2110] $21(12)^{221} 12^1 231513221(12)^{21} 1223151 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -108$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2122] $18^{21} 12321812321812315132^1 181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -105$, $\det = -36$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.22 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2142] $21(12)^{3221} 12231513221(12)^{21} 122^1 315 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 50, T \in [33, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2143] $21(12)^{21} 12^1 231513221(12)^{21} 12231513 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 47, T \in [33, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2144] $21(12)^{221} 12231513221(12)^{21} 12^1 23151 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -108$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

- [2145] $21^{221}(12)122315132^1 21^{21}(12)1223151 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2146] $21^{231513221}(12)122315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 50, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 50, T \in [33, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [2147] $21^{31513221}(12)1223151322^1 1(12)1222 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 47, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 47, T \in [33, 47]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [2148] $21^{1513221}(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2149] $21^{151322^1 1}(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [2150] $5132^1 181231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 43, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 43, T \in [33, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2151] $5132181231513221(12)122^1 3151315 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 43, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 43, T \in [33, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [2152] $1^{1513221}(12)1223151321812^1 32171 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [2153] $1^{151322^1 1}(12)122315132181232171 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [2154] $1^{1812315132^1 21}(12)122315132171 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2155] $1^{181231513221}(12)122^1 315132171 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [2156] $712231513221(12)122^1 315132181 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -107, \det = -49, (n_+, n_-, n_0) =$

(1, 27, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2157] $7122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2158] $51316131513221(12)122^1 315132171 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 40$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2159] $7122315132^1 21(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 40$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2160] $171231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2161] $1712315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2162] $171231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2163] $1712315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2164] $413221(12)122^1 31513221(12)122313 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -101$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2165] $41322^1 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -101$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2166] $7122315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2167] $712231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2168] & \quad 7122315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& \quad T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2169] & \quad 712231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& \quad T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2170] & \quad 1(12)122^{21} 31513161316131613151 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& \quad T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2171] & \quad 1^1 812321812^1 321812315132181 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& \quad T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2172] & \quad 413218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& \quad T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[2173] & \quad 41321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& \quad T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2174] & \quad 41321812^1 32181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& \quad T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2175] & \quad 413218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& \quad T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[2176] & \quad 812315132181232181232^1 1^1 81 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 43, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& \quad T_{\min} = 32.06 \rightarrow 33, T_{\max} = 43, T \in [33, 43] \\
& \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2177] & \quad 812^1 3151321812321812321^1 81 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 43, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& \quad T_{\min} = 32.06 \rightarrow 33, T_{\max} = 43, T \in [33, 43] \\
& \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2178] & \quad 812^1 23151321812321812321^1 81 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& \quad T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \quad \begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[2141] $1(12)122315132^{\overline{2221}}21(12)1223151+\hat{1}\rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2140] $1(12)12231513221(12)12^{\overline{21}}23151+\hat{1}\rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[2139] $1(12)12^{\overline{221}}231513221(12)12231513 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 47, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 47, T \in [33, 47]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[2182] $414132^{\overline{1}}21(12)12231513221(12)12 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 41, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[2183] $41413221(12)122^{\overline{1}}31513221(12)12 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 41, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2184] $413221(12)1223151321812^{\overline{1}}32181 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 42, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[2185] $41322^{\overline{1}}1(12)122315132181232181 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 42, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[2186] $1613161316131513221(12)12^{\overline{1}}23151 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[2187] $151316131613161315132^{\overline{1}}21(12)122 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$

[2188] $31613161316131513221(12)122^{\overline{1}}315 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 50, \Delta = -117, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 50, T \in [33, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[2189] $161316131513221(12)122^{\overline{1}}31513161 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[2190] $221812321812^{\overline{1}}321812315132181 + \hat{1}\rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2191] 413218123218123151322¹1(12)1222 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[2192] 41321812¹32181231513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2193] 612315132¹21(12)122315131613161 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[2194] 812¹315132181232181232181223 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 43$, $\Delta = -114$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 43, T \in [33, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2195] 812315132181232181232¹181223 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 43$, $\Delta = -114$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 43, T \in [33, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2196] 812¹231513218123218123218122 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -105$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[2197] 8122315132181232¹18123218122 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -105$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[2198] 181232181232181232¹18123215 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2199] 181232181232181232¹18123151 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2200] 414132¹21(12)12231513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 41$, $\Delta = -112$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2201] 41413221(12)122¹31513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 41$, $\Delta = -112$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.06 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2202] 41322¹1(12)1223151321812321812 + $\hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2203] & \overset{1}{413221(12)1223151321812^1 321812 + \hat{1} \rightarrow 2} \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2204] & \overset{1}{812^1 32181232181232181231513 + \hat{1} \rightarrow 2} \quad T = 27, T_{\max} = 44, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 44, T \in [33, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2205] & \overset{1}{71232171232181232^1 181232171 + \hat{1} \rightarrow 2} \quad T = 27, T_{\max} = 39, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[2206] & \overset{1}{1612321812^1 32181232171232181 + \hat{1} \rightarrow 2} \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2207] & \overset{1}{41413221(12)1223151321812^1 32171 + \hat{1} \rightarrow 2} \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[2208] & \overset{1}{414132^1 21(12)122315132181232171 + \hat{1} \rightarrow 2} \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[2209] & \overset{1}{81232171232181232181232^1 1613 + \hat{1} \rightarrow 2} \quad T = 27, T_{\max} = 44, \Delta = -115, \det = -49, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 44, T \in [33, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2210] & \overset{1}{812^1 321712321812321812321613 + \hat{1} \rightarrow 2} \quad T = 27, T_{\max} = 44, \Delta = -115, \det = -49, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 44, T \in [33, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2211] & \overset{1}{71232171232181232^1 1812321712 + \hat{1} \rightarrow 2} \quad T = 27, T_{\max} = 39, \Delta = -106, \det = -49, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[2212] & \overset{1}{71232181232181232^1 181232161 + \hat{1} \rightarrow 2} \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 39, T \in [33, 39] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2138] & \overset{21}{1(12)122^1 31513221(12)122315132 + \hat{1} \rightarrow 2} \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -49, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.06 \rightarrow 33, T_{\max} = 50, T \in [33, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[2181] $414132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 41$, $\Delta = -112$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 41$, $T \in [33, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[2179] $8122315132181232^1 181232181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -105$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 39$, $T \in [33, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

[2180] $41413221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 41$, $\Delta = -112$, $\det = -49$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.06 \rightarrow 33$, $T_{\max} = 41$, $T \in [33, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2213] $1(12)122^1 31513221(12)1223151322 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

[2214] $1(12)12^1 231513221(12)122315132 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

[2215] $1(12)12231513221(12)122^1 31513 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 47$, $T \in [32, 47]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[2216] $21(12)12231513221(12)122^1 31513 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 47$, $T \in [32, 47]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[2217] $21(12)12231513221(12)122^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2218] $21(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2219] $21(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 47$, $T \in [32, 47]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2220] $181232^1 181231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 32$, $\Delta = -94$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 32$, $T \in [32, 32]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

- [2221] $71232181231513221^{315132^1 21} (12) 1 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2222] $322181232^1 181231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 32$, $\Delta = -94$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 32$, $T \in [32, 32]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2223] $1(12)122^1 315131613161316131513 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 47$, $T \in [32, 47]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [2224] $1(12)12^1 231513161316131613151 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -108$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [2225] $181232181232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -105$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2226] $4132181232^1 181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2227] $4132181232^1 181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2228] $8123151321812321812^1 321^1 81 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 43$, $\Delta = -114$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 43$, $T \in [32, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2229] $81223151321812^1 32181232181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -105$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [2230] $3151316131613161315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 47$, $T \in [32, 47]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2231] $1513161316131613151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -108$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.94 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2232] $22181232181232^1 1812315132181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -105$, $\det = -64$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2233] & 4132181232^1 181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -64, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2234] & 613161316131513221(12)122^1 31513 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -64, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2235] & 8123151321812321812^1 32181223 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 43, \Delta = -114, \det = -64, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 43, T \in [32, 43] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2236] & 81223151321812^1 3218123218122 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -105, \det = -64, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2237] & 1812321812321812^1 3218123215 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2238] & 1812321812321812^1 3218123151 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2239] & 812321812321812321812^1 31513 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 44, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 44, T \in [32, 44] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2240] & 712321712321812^1 32181232171 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -106, \det = -64, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2241] & 161232181232^1 181232171232181 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -64, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2242] & 812321712321812321812^1 321613 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 44, \Delta = -115, \det = -64, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 44, T \in [32, 44] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2243] & 712321712321812^1 321812321712 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -106, \det = -64, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.94 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2244] & 712321812321812^1 32181232161 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} =
\end{aligned}$$

15.0

$$T_{\min} = 31.94 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2296] \quad 71232^1 1712321812321812321712 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[2287] \quad 181232^1 18123218123218123215 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2297] \quad 71232^1 181232181232181232161 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2295] \quad 41413221(12)122^1 315132181232171 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2294] \quad 4141322^1 1(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2293] \quad 1612321812321812321712^1 32181 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[2292] \quad 71232^1 171232181232181232171 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[2291] \quad 413221(12)122^1 3151321812321812 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2290] \quad 4141322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 41, \Delta = -112, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 41, T \in [32, 41]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2245] \quad 1(12)12^1 231513221(12)1223151322 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2246] \quad 1(12)1223151322^1 1(12)1223151 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2247] & \begin{matrix} 3221 & & 21 \\ 21(12)122^1 31513221(12)122315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2248] & \begin{matrix} 221 & & 21 \\ 21(12)1223151322^1 1(12)1223151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2249] & \begin{matrix} 2231513221 \\ 21(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2250] & \begin{matrix} 1513221 \\ 21(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2251] & \begin{matrix} 5132181231513221(12)12^1 23151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 43, \Delta = -114, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 43, T \in [32, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2252] & \begin{matrix} 51321812315132^1 21(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 43, \Delta = -114, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 43, T \in [32, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2253] & \begin{matrix} 3221 \\ 221(12)12^1 231513221(12)1223141 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 32, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 197 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[2254] & \begin{matrix} 1513221 \\ 1(12)122^1 315132181232171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2255] & \begin{matrix} 31513221 \\ 71232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2256] & \begin{matrix} 1 \\ 18123151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2257] & \begin{matrix} 1 \\ 181231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2258] $413221^{21}(12)12^1 231513221^{13221}(12)1 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 32$, $\Delta = -93$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.84 \rightarrow 32$, $T_{\max} = 32$, $T \in [32, 32]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2259] $71223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.84 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2260] $71223151322^1 (12)12^1 2315132181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.84 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2289] $41413221^1 (12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 41$, $\Delta = -112$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.84 \rightarrow 32$, $T_{\max} = 41$, $T \in [32, 41]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2261] $513161315132^1 21(12)122315132171 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 40$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.84 \rightarrow 32$, $T_{\max} = 40$, $T \in [32, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2262] $51316131513221(12)12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 40$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.84 \rightarrow 32$, $T_{\max} = 40$, $T \in [32, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2263] $712231513221(12)122^1 31513161315 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 40$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.84 \rightarrow 32$, $T_{\max} = 40$, $T \in [32, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2264] $71223151322^1 1(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 40$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.84 \rightarrow 32$, $T_{\max} = 40$, $T \in [32, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2265] $17123151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.84 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2266] $17123151322^1 (12)12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.84 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2267] $17123151322^1 (12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.84 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2268] $17123151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2269] & 413221(12)12^{21}231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -101, \det = -81, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2270] & 71231513221(12)122315132^118123 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -81, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 47, T \in [32, 47] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2271] & 71223151322^11(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2272] & 712231513221(12)12^123151321812 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2273] & 71223151322^11(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2274] & 712231513221(12)12^123151322181 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2275] & 1(12)12^12315131613161316131513 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -81, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 47, T \in [32, 47] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2276] & 41413221(12)12^1231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 41, \Delta = -112, \det = -81, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2277] & 4141322^11(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 41, \Delta = -112, \det = -81, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2278] & 41413221(12)12^1231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 41, \Delta = -112, \det = -81, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 41, T \in [32, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2279] & 4141322^11(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 41, \Delta = -112, \det = -81, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.84 \rightarrow 32, T_{\max} = 41, T \in [32, 41]
\end{aligned}$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2280] $413221(12)122^1 315132181232181 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2281] $23151316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2282] $31513161316131613151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 47, T \in [32, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2283] $1613161315132^1 21(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2284] $161316131513221(12)12^1 231513161 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2288] $181232^1 18123218123218123151 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2285] $61231513221(12)122^1 315131613161 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2286] $6123151322^1 1(12)122315131613161 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.84 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2298] $1(12)122315132^1 21(12)12231513 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -100, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.76 \rightarrow 32, T_{\max} = 47, T \in [32, 47]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2306] $41321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.76 \rightarrow 32, T_{\max} = 42, T \in [32, 42]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2308] $812315132181232^1 1812321^1 81 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 43, \Delta = -114, \det = -100, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.76 \rightarrow 32, T_{\max} = 43, T \in [32, 43]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2307] $41321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 42, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2299] & \begin{matrix} 3221 & & 21 \\ 21(12)12231513221(12)12^1 2315 + \hat{1} \rightarrow 2 & T = 27, T_{\max} = 50, \Delta = -117, \det = -100, (n_+, n_-, n_0) = \\ (1, 27, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2300] & \begin{matrix} 21 & & 21 \\ 21(12)122315132^1 21(12)12231513 + \hat{1} \rightarrow 2 & T = 27, T_{\max} = 47, \Delta = -116, \det = -100, (n_+, n_-, n_0) = \\ (1, 27, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 47, T \in [32, 47] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2301] & \begin{matrix} 31513221 \\ 21(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 & T = 27, T_{\max} = 47, \Delta = -116, \det = -100, (n_+, n_-, n_0) = \\ (1, 27, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 47, T \in [32, 47] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2302] & \begin{matrix} 321 \\ 1812321812^1 31513221(12)12231 + \hat{1} \rightarrow 2 & T = 27, T_{\max} = 32, \Delta = -94, \det = -100, (n_+, n_-, n_0) = \\ (1, 27, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2303] & \begin{matrix} 3221812321812^1 31513221(12)12231 + \hat{1} \rightarrow 2 & T = 27, T_{\max} = 32, \Delta = -94, \det = -100, (n_+, n_-, n_0) = \\ (1, 27, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2310] & \begin{matrix} 321 \\ 231513161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2 & T = 27, T_{\max} = 50, \Delta = -117, \det = -100, (n_+, n_-, n_0) = \\ (1, 27, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2311] & \begin{matrix} 31613161316131513221(12)12^1 2315 + \hat{1} \rightarrow 2 & T = 27, T_{\max} = 50, \Delta = -117, \det = -100, (n_+, n_-, n_0) = \\ (1, 27, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2312] & \begin{matrix} 221812321812321812^1 315132181 + \hat{1} \rightarrow 2 & T = 27, T_{\max} = 39, \Delta = -105, \det = -100, (n_+, n_-, n_0) = \\ (1, 27, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2313] & \begin{matrix} 41321812321812^1 31513221(12)1222 + \hat{1} \rightarrow 2 & T = 27, T_{\max} = 42, \Delta = -113, \det = -100, (n_+, n_-, n_0) = \\ (1, 27, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 42, T \in [32, 42] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2314] & \begin{matrix} (12)122^1 31513161316131613151322 + \hat{1} \rightarrow 2 & T = 27, T_{\max} = 50, \Delta = -117, \det = -100, (n_+, n_-, n_0) = \\ (1, 27, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2315] & \begin{matrix} 61315132^1 21(12)1223151316131613 + \hat{1} \rightarrow 2 & T = 27, T_{\max} = 50, \Delta = -117, \det = -100, (n_+, n_-, n_0) = \\ (1, 27, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 31.76 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2316] & \begin{matrix} 812315132181232^1 181232181223 + \hat{1} \rightarrow 2 & T = 27, T_{\max} = 43, \Delta = -114, \det = -100, (n_+, n_-, n_0) = \end{matrix}
\end{aligned}$$

(1, 27, 0), $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.76 \rightarrow 32, T_{\max} = 43, T \in [32, 43]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2317] $8122315132^1 18123218123218122 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.76 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2318] $181232181232^1 18123218123215 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -109, \det = -100, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.76 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2319] $181232181232^1 18123218123151 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.76 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2320] $81232181232^1 181231513 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 44, \Delta = -115, \det = -100, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.76 \rightarrow 32, T_{\max} = 44, T \in [32, 44]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2321] $71232171232^1 181232181232171 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -106, \det = -100, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.76 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2322] $1612321812321812^1 32171232181 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.76 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2323] $81232171232181232^1 1812321613 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 44, \Delta = -115, \det = -100, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.76 \rightarrow 32, T_{\max} = 44, T \in [32, 44]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2324] $71232171232^1 1812321812321712 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -106, \det = -100, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.76 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2325] $71232181232^1 181232181232161 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.76 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2309] $8122315132^1 181232181232181 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 39, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.76 \rightarrow 32, T_{\max} = 39, T \in [32, 39]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2304] $712315132^1 21(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 27, T_{\max} = 47, \Delta = -116, \det = -100, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 31.76 \rightarrow 32, T_{\max} = 47, T \in [32, 47]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2305] & \begin{matrix} 21 \\ 1812321812321812^1 315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 39, \quad \Delta = -105, \quad \det = -100, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.76 \rightarrow 32, \quad T_{\max} = 39, \quad T \in [32, 39] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2328] & \begin{matrix} 3221 \\ 21(12)122315132^1 21(12)122315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2330] & \begin{matrix} 21 \\ 21(12)12231513221(12)12^1 231513 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 47, \quad \Delta = -116, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, \quad T_{\max} = 47, \quad T \in [32, 47] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2331] & \begin{matrix} 2315132^1 21 \\ 21(12)12231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2332] & \begin{matrix} 3151322^1 1 \\ 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 47, \quad \Delta = -116, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, \quad T_{\max} = 47, \quad T \in [32, 47] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2333] & 51321812^1 31513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 27, \quad T_{\max} = 43, \quad \Delta = -114, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, \quad T_{\max} = 43, \quad T \in [32, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2334] & 513218123151322^1 1(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 27, \quad T_{\max} = 43, \quad \Delta = -114, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, \quad T_{\max} = 43, \quad T \in [32, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2335] & \begin{matrix} 1513221 \\ 1(12)12^1 2315132181232171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 39, \quad \Delta = -109, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, \quad T_{\max} = 39, \quad T \in [32, 39] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[2336] & \begin{matrix} 1513221 \\ 1(12)122315132^1 181232171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 39, \quad \Delta = -109, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, \quad T_{\max} = 39, \quad T \in [32, 39] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2337] & \begin{matrix} 3151322^1 1 \\ 71232181231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2338] & 5131613151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 27, \quad T_{\max} = 40, \quad \Delta = -111, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, \quad T_{\max} = 40, \quad T \in [32, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2339] $712231513221(12)12^1 231513161315 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 40$, $\Delta = -111$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 40$, $T \in [32, 40]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [2340] $71231513221(12)122^1 31513218123 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 47$, $T \in [32, 47]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [2341] $8123151321812^1 321812321^1 81 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 43$, $\Delta = -114$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 43$, $T \in [32, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2342] $413221(12)12^1 23151321812321^1 81 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2343] $413221(12)122315132^1 1812321^1 81 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [2344] $16131613161315132^1 21(12)1223151 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -109$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2345] $16131613151322^1 1(12)12231513161 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -109$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [2346] $613161316131513221(12)12^1 231513 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [2347] $(12)12^1 231513161316131613151322 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [2348] $61316131513221(12)122^1 315131613 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [2349] $61231513221(12)12^1 2315131613161 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -110$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [2350] $8123151321812^1 32181232181223 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 43$, $\Delta = -114$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 43$, $T \in [32, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

- [2351] $1812321812^1 3218123218123215 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -109$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2352] $1812321812^1 3218123218123151 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2353] $413221(12)122315132^1 1812321812 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [2354] $413221(12)12^1 23151321812321812 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2355] $81232^1 181232181232181231513 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 44$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 44$, $T \in [32, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [2356] $812321812321812^1 32181231513 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 44$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 44$, $T \in [32, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2357] $712321712^1 32181232181232171 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -106$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [2358] $161232181232181232^1 171232181 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 39$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 39$, $T \in [32, 39]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [2359] $41413221(12)12^1 2315132181232171 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [2360] $41413221(12)122315132^1 181232171 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 42$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 42$, $T \in [32, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [2361] $81232^1 1712321812321812321613 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 44$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 44$, $T \in [32, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [2362] $812321712321812^1 321812321613 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 44$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.69 \rightarrow 32$, $T_{\max} = 44$, $T \in [32, 44]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2363] & 712321712^1 321812321812321712 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[2364] & 712321812^1 32181232181232161 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2329] & 21(12)^{3221} 12^1 231513221(12)^{21} 122315 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -121, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2326] & 1(12)^{21} 12231513221(12)^{21} 122^1 315132 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -121, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2327] & 1(12)^{221} 12231513221(12)^{21} 12^1 231513 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -121, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.69 \rightarrow 32, T_{\max} = 47, T \in [32, 47] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2370] & 712321812315132^1 21(12)^{31513221} 1 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -169, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.59 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2365] & 1(12)^{21} 122315132^1 21(12)^{21} 122315132 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 50, \Delta = -117, \det = -169, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.59 \rightarrow 32, T_{\max} = 50, T \in [32, 50] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2367] & 21(12)^{21} 1223151322^1 1(12)^{21} 12231513 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -169, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.59 \rightarrow 32, T_{\max} = 47, T \in [32, 47] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2366] & 1(12)^{221} 1223151322^1 1(12)^{21} 12231513 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 47, \Delta = -116, \det = -169, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.59 \rightarrow 32, T_{\max} = 47, T \in [32, 47] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2372] & 812315132^1 1812321812321^{321} 8 1 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 43, \Delta = -114, \det = -169, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.59 \rightarrow 32, T_{\max} = 43, T \in [32, 43] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2373] & 161316131613151322^1 1(12)^{21} 1223151 + \hat{1} \rightarrow 2 \quad T = 27, T_{\max} = 39, \Delta = -109, \det = -169, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.59 \rightarrow 32, T_{\max} = 39, T \in [32, 39] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2374] $51316131613161315132^1 21(12)1223 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -169$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.59 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[2375] $613151322^1 1(12)1223151316131613 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -169$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.59 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2376] $613161315132^1 21(12)122315131613 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -169$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.59 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2377] $812315132^1 181232181232181223 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 43$, $\Delta = -114$, $\det = -169$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.59 \rightarrow 32$, $T_{\max} = 43$, $T \in [32, 43]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2378] $81232181232^1 181232181231513 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 44$, $\Delta = -115$, $\det = -169$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.59 \rightarrow 32$, $T_{\max} = 44$, $T \in [32, 44]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2379] $81232171232^1 1812321812321613 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 44$, $\Delta = -115$, $\det = -169$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.59 \rightarrow 32$, $T_{\max} = 44$, $T \in [32, 44]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2369] $21^1 (12) 12^1 231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -169$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.59 \rightarrow 32$, $T_{\max} = 47$, $T \in [32, 47]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[2371] $7123151322^1 1(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -169$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.59 \rightarrow 32$, $T_{\max} = 47$, $T \in [32, 47]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2368] $21^1 (12) 122^1 31513221(12)122 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -169$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.59 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2383] $71231513221^1 (12)12^1 231513218123 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 47$, $\Delta = -116$, $\det = -196$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.55 \rightarrow 32$, $T_{\max} = 47$, $T \in [32, 47]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2384] $6131513221(12)122^1 3151316131613 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -196$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.55 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2385] $61316131513221(12)12^1 2315131613 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -196$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 31.55 \rightarrow 32$, $T_{\max} = 50$, $T \in [32, 50]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2386] & \begin{matrix} 1 \\ 812321812^1 32181232181231513 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 44, \quad \Delta = -115, \quad \det = -196, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.55 \rightarrow 32, \quad T_{\max} = 44, \quad T \in [32, 44] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2387] & \begin{matrix} 1 \\ 812321712^1 321812321812321613 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 44, \quad \Delta = -115, \quad \det = -196, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.55 \rightarrow 32, \quad T_{\max} = 44, \quad T \in [32, 44] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ 1(12)12231513221(12)122^1 3151322 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -196, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.55 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 3221 \\ 21(12)1223151322^1 1(12)122315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -196, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.55 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2382] & \begin{matrix} 22315132^1 21 \\ 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -196, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.55 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[2388] & \begin{matrix} 21 \\ 1(12)12231513221(12)12^1 2315132 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -225, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.52 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2389] & \begin{matrix} 23151322^1 1 \\ 21(12)12231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -225, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.52 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[2390] & \begin{matrix} 31513221 \\ 712321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -225, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.52 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2391] & \begin{matrix} 513161316131613151322^1 1(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -225, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.52 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2392] & \begin{matrix} 1 \\ 1(12)122315132^1 21(12)1223151322 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -256, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.48 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[2393] & \begin{matrix} 2231513221 \\ 21(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -256, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.48 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2394] $7123218123151322^{\overline{1}1} (12)^{\overline{1}1} 1 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -256$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.48 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2395] $6131613161315132^{\overline{1}21}(12)12231513 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -256$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.48 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2396] $6131613151322^{\overline{1}1}(12)122315131613 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -256$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.48 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2397] $1(12)1223151322^{\overline{1}1}(12)122315132 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -289$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.46 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[2398] $21^{\overline{1}1}(12)^{\overline{1}1} 231513221(12)122 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -289$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.46 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[2399] $316131613161315132^{\overline{1}21}(12)122315 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -289$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.46 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 101 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2400] $6131513221(12)12^{\overline{1}1} 23151316131613 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -289$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.46 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[2403] $61316131613151322^{\overline{1}1}(12)12231513 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -361$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.41 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$$

[2401] $1(12)12231513221(12)12^{\overline{1}1} 23151322 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -361$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.41 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[2402] $21^{\overline{1}1}(12)^{\overline{1}1} 2231513221(12)12 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -361$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.41 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

[2404] $3161316131613151322^{\overline{1}1}(12)122315 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -400$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.39 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{pmatrix} 0 & 20 & 2 \\ 20 & 139 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2406] $21^{\overline{1}1}(12)^{\overline{1}1} 12^{\overline{1}1} 231513221(12)12 + \hat{1} \rightarrow 2$ $T = 27$, $T_{\max} = 50$, $\Delta = -117$, $\det = -441$, $(n_+, n_-, n_0) = (1, 27, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 31.37 \rightarrow 32, T_{\max} = 50, T \in [32, 50]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0} \\
[2405] & \begin{matrix} 1 \\ 1(12)1223151322^1 1(12)1223151322 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 27, \quad T_{\max} = 50, \quad \Delta = -117, \quad \det = -441, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 31.37 \rightarrow 32, \quad T_{\max} = 50, \quad T \in [32, 50] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
T = 28 & \quad [2411] \quad 315132181232181231513221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 48, \quad \Delta = -116, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 35, \quad T_{\max} = 48, \quad T \in [35, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[2410] & \begin{matrix} 22^1 315132181232181231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 51, \quad \Delta = -117, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 35, \quad T_{\max} = 51, \quad T \in [35, 51] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[2412] & \begin{matrix} 51232181232181231513221(12)122^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -110, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 35, \quad T_{\max} = 40, \quad T \in [35, 40] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[2413] & \begin{matrix} 15132181232181231513221(12)122^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -108, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 35, \quad T_{\max} = 40, \quad T \in [35, 40] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[2414] & \begin{matrix} 16131613161316131513221(12)12^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -105, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 35, \quad T_{\max} = 40, \quad T \in [35, 40] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[2415] & \begin{matrix} 4141413221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 41, \quad \Delta = -111, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 35, \quad T_{\max} = 41, \quad T \in [35, 41] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[2416] & \begin{matrix} 4141413221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 41, \quad \Delta = -111, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 35, \quad T_{\max} = 41, \quad T \in [35, 41] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[2417] & \begin{matrix} 216131613161316131513221(12)12^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -105, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 35, \quad T_{\max} = 40, \quad T \in [35, 40] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[2418] & \begin{matrix} 4141413221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 41, \quad \Delta = -111, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 35, \quad T_{\max} = 41, \quad T \in [35, 41] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[2419] & \begin{matrix} 712321712321812321812321812^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -105, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 35, \quad T_{\max} = 40, \quad T \in [35, 40] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[2407] & \begin{matrix} 1513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 34, \quad \Delta = -96, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0
\end{aligned}$$

$(1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34$, $T_{\max} = 34$, $T \in [34, 34]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[2409] $1(12)12231513218123218123151 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 4$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 35$, $T_{\max} = 40$, $T \in [35, 40]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[2408] $1(12)12231513218123218123215 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 4$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 35$, $T_{\max} = 40$, $T \in [35, 40]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[2424] $51232181232181231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 9$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 40$, $T \in [35, 40]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[2423] $413221(12)12231513221(12)12^1 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 9$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.33 \rightarrow 34$, $T_{\max} = 48$, $T \in [34, 48]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[2425] $15132181232181231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 9$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 40$, $T \in [35, 40]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[2426] $181231513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 45$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 45$, $T \in [35, 45]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[2427] $812231513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 45$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 45$, $T \in [35, 45]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[2428] $41413221(12)12231513221(12)122^1 313 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -100$, $\det = 9$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 40$, $T \in [35, 40]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[2429] $1(12)122315131613161316131612^1 3 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 44$, $T \in [35, 44]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[2430] $4141413221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 9$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 41$, $T \in [35, 41]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[2431] $16131613161316131513221(12)122^1 3 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 44$, $T \in [35, 44]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[2432] & 32^1 16131613161316131513221(12)12 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 44, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.22 \rightarrow 35, T_{\max} = 44, T \in [35, 44] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[2433] & 4141413221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 41, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.22 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[2434] & 812321712321812321812321712^1 3 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 44, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.22 \rightarrow 35, T_{\max} = 44, T \in [35, 44] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
& \begin{matrix} 513221 \\ 1(12)1223151321812321812^1 3 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.22 \rightarrow 35, T_{\max} = 51, T \in [35, 51] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[2420] & 31513221(12)12231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 34, \Delta = -96, \det = 9, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.33 \rightarrow 34, T_{\max} = 34, T \in [34, 34] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 12 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2421] & 1(12)1223151321812321812315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 9, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.22 \rightarrow 35, T_{\max} = 51, T \in [35, 51] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[2441] & 71232171232181232181232^1 181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.75 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2440] & 1712^1 32181232181232171232181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.75 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2438] & 512^1 32181232181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.75 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2437] & 21(12)122315132181232181232^1 15 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.75 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[2436] & 1(12)122315132181232181232^1 15 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 16, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.75 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[2443] & 21712^1 32181232181232171232181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 16, (n_+, n_-, n_0) =
\end{aligned}$$

(1, 28, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 33.75 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2444] 4141413221(12)12231513221(12)12¹222 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 16$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.75 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2445] 71232171232181232181232¹18122 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 16$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.75 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2446] 1812321¹81232181232181232¹161 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 16$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.75 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[2447] 71232181232181232181232¹1712 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 16$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.75 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2439] 41413221¹(12)12231513221(12)12¹2313 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -100$, $\det = 16$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.75 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2442] 71232181232181232181232¹171 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 16$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.75 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2435] 21¹⁵¹³²²¹(12)12231513221(12)12¹2313 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 34$, $\Delta = -96$, $\det = 16$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.75 \rightarrow 34$, $T_{\max} = 34$, $T \in [34, 34]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2467] 51316131513221(12)122315132¹2181 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2452] 1¹⁵¹³²²¹(12)122315132181232¹1812 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2453] 1^{15132¹21}(12)1223151321812321812 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2454] 1²²¹(12)1223151321812321812¹3215 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2455] & \begin{matrix} 221 \\ 1(12)1223151321812321812^1 3151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2456] & \begin{matrix} 21 \\ 21(12)1223151321812321812^1 3215 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2457] & \begin{matrix} 21 \\ 21(12)1223151321812321812^1 3151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2458] & \begin{matrix} 1 \\ 1812^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[2459] & \begin{matrix} 1 \\ 181231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[2460] & \begin{matrix} 1 \\ 1812^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2461] & \begin{matrix} 1 \\ 1812^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2462] & \begin{matrix} 1 \\ 181231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[2463] & \begin{matrix} 1 \\ 4132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 48, T \in [34, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2464] & \begin{matrix} 1 \\ 181232181231513221(12)122^1 3151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2465] & \begin{matrix} 1 \\ 1812^1 32181231513221(12)1223151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2451] & 1^{322^1 1} (12) 1223151321812321812315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 51, T \in [34, 51] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2450] & 1^{1812^1 31513221} (12) 12231513161315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2449] & 21^{15132^1 21} (12) 12231513221(12) 122313 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 34, \Delta = -96, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 34, T \in [34, 34] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2469] & 15132^{181232181231513221} (12) 1222 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2470] & 612^{132181231513221} (12) 1223151315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 43, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 43, T \in [34, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2471] & 612^{131513221} (12) 1223151321812315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 42, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2472] & 21812^{131513221} (12) 1223151321812 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2473] & 21812^{131513221} (12) 1223151322181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2474] & 2181231513221(12) 122315132^{12181} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[2475] & 18122^{131513221} (12) 1223151322181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2476] & 81231513221(12) 1223151321812^{132} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 25, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.44 \rightarrow 34, T_{\max} = 51, T \in [34, 51] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2477] $161316131513221(12)122315132^1 171 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2478] $7122^1 31513221(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [2479] $41413221(12)122315132^1 21(12)122313 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -100$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2480] $171232^1 181232181232171232181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2481] $4141413221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2482] $4141413221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2483] $41413221(12)122315132181232^1 181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 42$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 42$, $T \in [34, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2484] $712321712321812321812^1 32181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2485] $712321812321812321812^1 32171 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2486] $16131613161316131513221(12)12^1 23 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 44$, $T \in [34, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2487] $232^1 16131613161316131513221(12)1 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 48$, $T \in [34, 48]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2488] $2171232^1 181232181232171232181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.44 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2489] 4141413221(12)122315132¹21(12)12222 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33.44 \rightarrow 34, T_{\max} = 41, T \in [34, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2490] 41413221(12)122315132181232¹1812 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 42$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33.44 \rightarrow 34, T_{\max} = 42, T \in [34, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2491] 712321712321812321812¹3218122 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[2492] 1812321812321812321812¹32161 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2493] 712321812321812321812¹321712 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[2494] 4132¹181232181232181231513218 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 43$, $\Delta = -113$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33.44 \rightarrow 34, T_{\max} = 43, T \in [34, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[2448] 21¹⁵¹³²²¹(12)122315132¹21(12)122313 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 34$, $\Delta = -96$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33.44 \rightarrow 34, T_{\max} = 34, T \in [34, 34]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2468] 51232¹181232181231513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2466] 51316131513221(12)122315132¹1812 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33.44 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2498] 1²²¹(12)122¹31513218123218123215 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33.22 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2499] 1²²¹(12)122¹31513218123218123151 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 33.22 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2500] 21²¹(12)122¹31513218123218123215 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2501] & \begin{matrix} 21 \\ 21(12)122^1 31513218123218123151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 36, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2502] & \begin{matrix} 1 \\ 181231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2503] & \begin{matrix} 1 \\ 51316131513221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 36, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2504] & \begin{matrix} 1 \\ 512321812321812315132^1 21(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 36, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2505] & \begin{matrix} 1 \\ 151321812321812315132^1 21(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 36, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2506] & \begin{matrix} 1 \\ 1812^1 31513221(12)12231513218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 45, \Delta = -115, \det = 36, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 45, T \in [34, 45] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2507] & \begin{matrix} 1 \\ 2181231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2508] & \begin{matrix} 1 \\ 1812^1 231513221(12)1223151322181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2509] & \begin{matrix} 1 \\ 8122^1 31513221(12)12231513218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 45, \Delta = -115, \det = 36, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 45, T \in [34, 45] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2510] & \begin{matrix} 1 \\ 712^1 31513221(12)1223151316131613 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 36, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 48, T \in [34, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2511] & \begin{matrix} 1 \\ 712^1 231513221(12)122315131613161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 36, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2512] $41413221(12)1223151322^11(12)122313 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -100$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2513] $171232181232181232171232^1181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2514] $4141413221(12)1223151322^11(12)1 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2515] $4141413221(12)1223151322^11(12)12 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2516] $712^132171232181232181232181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2517] $712^132181232181232181232171 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2518] $2171232181232181232171232^1181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2519] $4141413221(12)1223151322^11(12)12222 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2520] $712^13217123218123218123218122 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2521] $1812^132181232181232181232161 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2522] $712^1321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.22 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2497] $1(12)122315132181232^1181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 36$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 48, T \in [34, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [2496] \quad 21 \quad (12) \quad 1223151322^1 1(12)122313 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 34, \Delta = -96, \det = 36, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 34, T \in [34, 34] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [2495] \quad 21 \quad (12) \quad 122315132^1 21(12)12231 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 34, \Delta = -96, \det = 36, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.22 \rightarrow 34, T_{\max} = 34, T \in [34, 34] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [2542] \quad 1812315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [2537] \quad 21(12)12^1 231513218123218123215 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [2528] \quad 1 \quad (12) \quad 12231513218123218123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 51, T \in [34, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
& [2529] \quad 1(12)122^1 315132181232181231513 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 48, T \in [34, 48] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [2530] \quad 1 \quad (12) \quad 1223151321812^1 321812 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0} \\
& [2531] \quad 1 \quad (12) \quad 1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
& [2532] \quad 1(12)122315132181232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [2533] \quad 1(12)12^1 231513218123218123215 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [2534] \quad 1(12)12^1 231513218123218123151 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2526] & \begin{matrix} 1 \\ 1812315132^1 21(12)12231513161315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -110, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, \quad T_{\max} = 40, \quad T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2525] & \begin{matrix} 151322^1 1 \\ 21 \quad (12) \quad 12231513221(12)122313 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 34, \quad \Delta = -96, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, \quad T_{\max} = 34, \quad T \in [34, 34] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[2524] & \begin{matrix} 1513221 \\ 21 \quad (12) \quad 122^1 31513221(12)122313 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 34, \quad \Delta = -96, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, \quad T_{\max} = 34, \quad T \in [34, 34] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2538] & \begin{matrix} 21 \\ 21(12)12^1 231513218123218123151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -108, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, \quad T_{\max} = 40, \quad T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2539] & \begin{matrix} 21 \\ 21(12)122315132181232^1 18123151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -108, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, \quad T_{\max} = 40, \quad T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2540] & \begin{matrix} 231513221 \\ 812^1 32181231513221 \quad (12) \quad 1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 51, \quad \Delta = -117, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, \quad T_{\max} = 51, \quad T \in [34, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2541] & \begin{matrix} 1 \\ 1812315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -106, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, \quad T_{\max} = 40, \quad T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[2523] & \begin{matrix} 31513221 \\ 21 \quad (12) \quad 1223151322^1 1(12)12231 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 34, \quad \Delta = -96, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, \quad T_{\max} = 34, \quad T \in [34, 34] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[2543] & \begin{matrix} 1 \\ 181231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -106, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, \quad T_{\max} = 40, \quad T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2544] & \begin{matrix} 1 \\ 1812315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -106, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, \quad T_{\max} = 40, \quad T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2545] & \begin{matrix} 1 \\ 181231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -106, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, \quad T_{\max} = 40, \quad T \in [34, 40]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2546] $413221^1(12)12231513221^{5132^1 21}(12)12+\hat{1}\rightarrow 2$ $T = 28, T_{\max} = 48, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 48, T \in [34, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2547] $41322^1 1(12)12231513221^{5132^1 21}(12)12+\hat{1}\rightarrow 2$ $T = 28, T_{\max} = 48, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 48, T \in [34, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2548] $181232^1 181231513221^1(12)1223151 + \hat{1}\rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[2549] $181232181231513221^1(12)12^1 23151 + \hat{1}\rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2550] $161315132^1 21(12)1223151321812315 + \hat{1}\rightarrow 2$ $T = 28, T_{\max} = 41, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 41, T \in [34, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2551] $16131513221(12)1223151321812^1 315 + \hat{1}\rightarrow 2$ $T = 28, T_{\max} = 41, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 41, T \in [34, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2552] $51316131513221(12)122^1 3151321812 + \hat{1}\rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2553] $51316131513221(12)122^1 3151322181 + \hat{1}\rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2554] $3151321812321812315132^1 21(12)122 + \hat{1}\rightarrow 2$ $T = 28, T_{\max} = 48, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 48, T \in [34, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[2555] $512321812^1 32181231513221(12)1222 + \hat{1}\rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2556] $5123218123218123151322^1 1(12)1222 + \hat{1}\rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[2557] $151321812^1 32181231513221(12)1222 + \hat{1}\rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2558] & 1513218123218123151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[2559] & 61232^1 181231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 43, \Delta = -113, \det = 49, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 43, T \in [34, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2560] & 61232181231513221(12)122^1 3151315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 43, \Delta = -113, \det = 49, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 43, T \in [34, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2561] & 61231513221(12)1223151321812^1 315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 42, \Delta = -112, \det = 49, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2562] & 612315132^1 21(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 42, \Delta = -112, \det = 49, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2563] & 21812315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2564] & 21812315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2565] & 2181231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2566] & 18122315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2567] & 812^1 31513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 49, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 51, T \in [34, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2568] & 812^1 231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 45, \Delta = -115, \det = 49, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 45, T \in [34, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2569] $161316131513221(12)122^1 315132171 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.06 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2570] $7122315132^1 21(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.06 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [2571] $41413221(12)122^1 31513221(12)122313 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -100$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.06 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2572] $414132^1 21(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -100$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.06 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [2573] $161316131613161315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.06 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [2574] $1(12)122^1 315131613161316131612 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.06 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
- [2575] $1712321812^1 32181232171232181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.06 \rightarrow 34$, $T_{\max} = 40$, $T \in [34, 40]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2576] $4141413221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.06 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2577] $4141413221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.06 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2578] $41413221(12)1223151321812^1 32181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 42$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.06 \rightarrow 34$, $T_{\max} = 42$, $T \in [34, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [2579] $414132^1 21(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 42$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.06 \rightarrow 34$, $T_{\max} = 42$, $T \in [34, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [2580] $71232171232181232^1 181232181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[2581] & 71232181232181232^1 181232171 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2582] & 161316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[2583] & 2161316131613161315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[2584] & 21712321812^1 32181232171232181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2585] & 4141413221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 41, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2586] & 41413221(12)1223151321812^1 321812 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 42, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[2587] & 414132^1 21(12)1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 42, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[2588] & 81232171232181232181232^1 17123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 44, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2589] & 812^1 3217123218123218123217123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 44, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2590] & 71232171232181232^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[2591] & 181232181232181232^1 181232161 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2592] & 81232181232181232181232^1 1613 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 44, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

15.0

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 44, T \in [34, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2593] \text{ } 812^1 321812321812321812321613 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 44, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 44, T \in [34, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2594] \text{ } 71232181232181232^1 1812321712 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[2595] \text{ } 41321812^1 32181232181231513218 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 43, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 43, T \in [34, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[2596] \text{ } 41321812321812321812315132^1 18 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 43, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 43, T \in [34, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[2536] \text{ } 21^{21}(12)122315132181232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2535] \text{ } 1^{221}(12)122315132181232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 40, T \in [34, 40]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2527] \text{ } 1^{3221}(12)1223151321812321812^1 315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 49, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 33.06 \rightarrow 34, T_{\max} = 51, T \in [34, 51]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2600] \text{ } 1^{21}(12)12^1 2315132181232181231513 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.94 \rightarrow 33, T_{\max} = 48, T \in [33, 48]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2601] \text{ } 1^{21}(12)1223151321812321812^1 31513 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.94 \rightarrow 33, T_{\max} = 48, T \in [33, 48]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2602] \text{ } 1^{221}(12)1223151321812^1 3218123215 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.94 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2603] \text{ } 1^{221}(12)1223151321812^1 3218123151 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.94 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2604] & \begin{matrix} 21 \\ 21(12)1223151321812^1 3218123215 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -110, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, \quad T_{\max} = 40, \quad T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2605] & \begin{matrix} 21 \\ 21(12)1223151321812^1 3218123151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -108, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, \quad T_{\max} = 40, \quad T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2606] & 23151321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 51, \quad \Delta = -117, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, \quad T_{\max} = 51, \quad T \in [33, 51] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2607] & 315132^1 181232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 48, \quad \Delta = -116, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, \quad T_{\max} = 48, \quad T \in [33, 48] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2608] & 31513218123218123151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 48, \quad \Delta = -116, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, \quad T_{\max} = 48, \quad T \in [33, 48] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2609] & 51232181232^1 181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -110, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, \quad T_{\max} = 40, \quad T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2610] & 15132181232^1 181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -108, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, \quad T_{\max} = 40, \quad T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2611] & 4132181232^1 181231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 28, \quad T_{\max} = 33, \quad \Delta = -93, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, \quad T_{\max} = 33, \quad T \in [33, 33] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2612] & \begin{matrix} 1 \\ 1(12)122^1 3151316131613161316123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 44, \quad \Delta = -114, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, \quad T_{\max} = 44, \quad T \in [33, 44] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2613] & \begin{matrix} 1 \\ 1613161316131613151322^1 1(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -105, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, \quad T_{\max} = 40, \quad T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2614] & \begin{matrix} 21 \\ 1(12)12^1 2315131613161316131612 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -105, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, \quad T_{\max} = 40, \quad T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2615] & \begin{matrix} 31 \\ 161316131613161315132^1 21(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 43, \quad \Delta = -113, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 43, T \in [33, 43] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2616] & \overset{1}{171232181232^1 181232171232181} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2617] & 712321712321812^1 32181232181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2618] & \overset{1}{712321812321812^1 32181232171} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[2619] & \overset{1}{1613161316131613151322^1 1(12)1} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2620] & \overset{31}{161316131613161315132^1 21(12)1} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 43, \Delta = -113, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 43, T \in [33, 43] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2621] & 32161316131613161315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 44, \Delta = -114, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 44, T \in [33, 44] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2622] & 21613161316131613151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2623] & 2171232181232^1 181232171232181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2624] & 812321712321812321812^1 3217123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 44, \Delta = -114, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 44, T \in [33, 44] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2625] & 712321712321812^1 3218123218122 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2626] & \overset{1}{1812321812321812^1 32181232161} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 64, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.94 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2627] $812321812321812321812^1 321613 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.94 \rightarrow 33, T_{\max} = 44, T \in [33, 44]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2628] $712321812321812^1 321812321712 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 64$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.94 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[2629] $4132181232^1 181232181231513218 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 43$, $\Delta = -113$, $\det = 64$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.94 \rightarrow 33, T_{\max} = 43, T \in [33, 43]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[2599] $1(12)122^1 3151321812321812315132 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.94 \rightarrow 33, T_{\max} = 51, T \in [33, 51]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2598] $1^{315132^1 21} (12) 122315132181232181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.94 \rightarrow 33, T_{\max} = 48, T \in [33, 48]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2597] $21^{315132^1 21} (12) 12231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 34$, $\Delta = -96$, $\det = 64$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.94 \rightarrow 33, T_{\max} = 34, T \in [33, 34]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2657] $218123151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2658] $218123151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2635] $1^{513221} (12) 122315132181232^1 18123 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 51, T \in [33, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2636] $1^{1513221} (12) 122^1 3151321812321812 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2637] $18123151322^1 1(12)122315132181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2638] $181231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) =$

$$(1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2639] \quad 18123151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2640] \quad 18123151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2641] \quad 181231513221^1 (12)12^1 23151322181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2642] \quad 413221^1 (12)122^1 31513221^1 (12)12 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 48, T \in [33, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2643] \quad 1812321812315132^1 21(12)1223151 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2644] \quad 16131513221(12)122^1 3151321812315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 41, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2645] \quad 1613151322^1 1(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 41, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2646] \quad 51316131513221(12)12^1 23151321812 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2647] \quad 513161315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2648] \quad 513161315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2649] \quad 51316131513221(12)12^1 23151322181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

- [2650] $223151321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2651] $231513218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2652] $612321812315132^1 21(12)1223151315 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 43$, $\Delta = -113$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 43$, $T \in [33, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2653] $61232181231513221(12)12^1 23151315 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 43$, $\Delta = -113$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 43$, $T \in [33, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2654] $6123151322^1 1(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 42$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 42$, $T \in [33, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2655] $61231513221(12)122^1 3151321812315 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 42$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 42$, $T \in [33, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2633] $1^{31513221} (12) 1223151321812^1 32181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 48$, $T \in [33, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [2632] $1^{181231513221} (12)122^1 31513161315 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2631] $1^{18123151322^1} 1(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2659] $1^{2181231513221} (12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2660] $1^{181223151322^1} 1(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2661] $1^{812231513221} (12)122315132^1 18123 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 45$, $\Delta = -115$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 45$, $T \in [33, 45]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2662] 1613161315132¹21(12)122315132171 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2663] 161316131513221(12)12¹2315132171 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2664] 71223151322¹1(12)122315131613161 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2665] 712231513221(12)122¹315131613161 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2666] 41413221¹(12)12¹231513221(12)122313 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -100$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2667] 4141322¹1(12)12231513221(12)122313 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -100$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2668] 1¹(12)12¹23151316131613161316123 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 44, T \in [33, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2669] 16³¹13161316131613151322¹1(12)12 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 43$, $\Delta = -113$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 43, T \in [33, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2670] 1712321812321812321712¹32181 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2671] 4141413221(12)12¹231513221²²²¹(12)1 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2672] 41414132¹21(12)12231513221²²²¹(12)1 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2673] 4141413221(12)12¹231513221²²¹(12)12 + $\hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 41$, $\Delta = -111$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$(1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2674] \quad 41414132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 41, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2675] \quad 41413221(12)122^1 315132181232181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 42, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2676] \quad 4141322^1 1(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 42, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2677] \quad 71232^1 171232181232181232181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[2678] \quad 71232^1 181232181232181232171 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2679] \quad 16^1 13161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 43, \Delta = -113, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 43, T \in [33, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2680] \quad 232161316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 48, T \in [33, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2681] \quad 321613161316131613151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 44, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 44, T \in [33, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2682] \quad 21712321812321812321712^1 32181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[2683] \quad 41414132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 41, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2684] \quad 4141413221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 41, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 32.84 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

- [2685] $4141322^1 1(12)1223151321812321812 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 42$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 42$, $T \in [33, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2686] $41413221(12)122^1 3151321812321812 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 42$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 42$, $T \in [33, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2687] $71232^1 17123218123218123218122 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [2688] $181232^1 181232181232181232161 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [2689] $71232^1 1812321812321812321712 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [2634] $1(12)12^1 23151321812321812315132 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2630] $21(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 34$, $\Delta = -96$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 34$, $T \in [33, 34]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2656] $181231513221(12)122315132^1 18123 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 45$, $\Delta = -115$, $\det = 81$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.84 \rightarrow 33$, $T_{\max} = 45$, $T \in [33, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [2692] $1(12)122315132181232^1 181231513 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.76 \rightarrow 33$, $T_{\max} = 48$, $T \in [33, 48]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2694] $1(12)122315132^1 18123218123151 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.76 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2695] $21(12)122315132^1 18123218123215 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.76 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2696] $21(12)122315132^1 18123218123151 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -108$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [2697] \quad 413221(12)12231513221 \overset{1}{(12)} 12 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 48, T \in [33, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
& [2698] \quad 22315132181232181231513221 \overset{51322}{1} (12)1 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 100, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
& [2699] \quad 3151321812 \overset{1}{32181231513221} (12)122 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 48, T \in [33, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& [2700] \quad 512321812321812 \overset{1}{31513221} (12)1222 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -110, \det = 100, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& [2701] \quad 151321812321812 \overset{1}{31513221} (12)1222 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 100, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& [2702] \quad 1812315132 \overset{1}{21} (12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 45, \Delta = -115, \det = 100, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 45, T \in [33, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [2703] \quad 8122315132 \overset{1}{21} (12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 45, \Delta = -115, \det = 100, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 45, T \in [33, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [2704] \quad 41321812321812 \overset{1}{31513221} (12)12231 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 33, \Delta = -93, \det = 100, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 33, T \in [33, 33] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& [2705] \quad 712315132 \overset{1}{21} (12)1223151316131613 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 48, T \in [33, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [2706] \quad 1712321812321812 \overset{1}{32171232181} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 100, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& [2707] \quad 71232171232 \overset{21}{1812321812321} 81 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 100, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.76 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2708] $71232181232^1 181232181232171 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2709] $2321613161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 48, T \in [33, 48]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2710] $21712321812321812^1 32171232181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2711] $81232171232181232^1 18123217123 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 44, T \in [33, 44]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2712] $71232171232^1 18123218123218122 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2713] $181232181232^1 181232181232161 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2714] $81232181232181232^1 1812321613 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 44, T \in [33, 44]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2715] $71232181232^1 1812321812321712 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2716] $41321812321812^1 32181231513218 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 43$, $\Delta = -113$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 43, T \in [33, 43]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2693] $1(12)122315132^1 18123218123215 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2690] $21 \begin{pmatrix} 12 \\ 122^1 31513221 \end{pmatrix} 1(12)12231 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 34$, $\Delta = -96$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 34, T \in [33, 34]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2691] $1 \begin{pmatrix} 51322^1 1 \\ 12 \end{pmatrix} 12231513218123218123 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 100$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 32.76 \rightarrow 33, T_{\max} = 51, T \in [33, 51]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2763] & \overset{1}{712321812^1 321812321812321712} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2721] & \overset{3221}{1(12)122^1 3151321812321812315} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2722] & \overset{3221}{1(12)122315132181232^1 1812315} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2723] & \overset{21}{1(12)1223151321812^1 32181231513} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 48, T \in [33, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2724] & \overset{1513221}{1(12)122315132^1 1812321812} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2725] & \overset{1513221}{1(12)12^1 23151321812321812} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[2726] & \overset{2315132^1 21}{81232181231513221(12)1} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2727] & \overset{231513221}{81232^1 181231513221(12)1} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[2728] & \overset{1}{413221(12)12^1 231513221(12)12} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 48, T \in [33, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2729] & \overset{1}{413221(12)122315132^1 21(12)12} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 48, T \in [33, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2730] & \overset{1}{18123218123151322^1 1(12)1223151} + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 40, T \in [33, 40]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2731] & \begin{matrix} 1 \\ 1812321812^1 31513221(12)1223151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -108, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, \quad T_{\max} = 40, \quad T \in [33, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2732] & \begin{matrix} 1 \\ 16131513221(12)122315132^1 1812315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 41, \quad \Delta = -111, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, \quad T_{\max} = 41, \quad T \in [33, 41] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2720] & \begin{matrix} 1 \\ 1(12)1223151321812321812^1 315132 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 51, \quad \Delta = -117, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, \quad T_{\max} = 51, \quad T \in [33, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2719] & \begin{matrix} 3151322^1 1 \\ 1(12)122315132181232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 48, \quad \Delta = -116, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, \quad T_{\max} = 48, \quad T \in [33, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2718] & \begin{matrix} 1 \\ 181231513221(12)12^1 231513161315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 40, \quad \Delta = -110, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, \quad T_{\max} = 40, \quad T \in [33, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2736] & \begin{matrix} 1 \\ 2315132^1 181232181231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 51, \quad \Delta = -117, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, \quad T_{\max} = 51, \quad T \in [33, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2737] & \begin{matrix} 1 \\ 315132181232^1 181231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 48, \quad \Delta = -116, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, \quad T_{\max} = 48, \quad T \in [33, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2738] & \begin{matrix} 1 \\ 612321812^1 31513221(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 43, \quad \Delta = -113, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, \quad T_{\max} = 43, \quad T \in [33, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2739] & \begin{matrix} 1 \\ 6123218123151322^1 1(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 43, \quad \Delta = -113, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, \quad T_{\max} = 43, \quad T \in [33, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2740] & \begin{matrix} 1 \\ 61231513221(12)12^1 23151321812315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 42, \quad \Delta = -112, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, \quad T_{\max} = 42, \quad T \in [33, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2741] & \begin{matrix} 1 \\ 61231513221(12)122315132^1 1812315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 42, \quad \Delta = -112, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, \quad T_{\max} = 42, \quad T \in [33, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2742] & \begin{matrix} 1 \\ 181231513221(12)122^1 31513218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 28, \quad T_{\max} = 45, \quad \Delta = -115, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 15.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 45, T \in [33, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
& [2743] \quad 812231513221(12)122^1 31513218123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 45, \Delta = -115, \det = 121, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 45, T \in [33, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
& [2744] \quad 16131613151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 121, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
& [2745] \quad 712231513221(12)12^1 2315131613161 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -108, \det = 121, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
& [2746] \quad 171232181232181232^1 171232181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 121, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
& [2747] \quad 414141322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 41, \Delta = -111, \det = 121, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 41, T \in [33, 41] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
& [2748] \quad 414141322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 41, \Delta = -111, \det = 121, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 41, T \in [33, 41] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
& [2749] \quad 41413221(12)12^1 2315132181232181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 42, \Delta = -112, \det = 121, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
& [2750] \quad 41413221(12)122315132^1 181232181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 42, \Delta = -112, \det = 121, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 42, T \in [33, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
& [2751] \quad 712321712^1 32181232181232181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
& [2752] \quad 712321812^1 32181232181232171 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
& [2753] \quad 2171232181232181232^1 171232181 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 40, \Delta = -105, \det = 121, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.69 \rightarrow 33, T_{\max} = 40, T \in [33, 40] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2754] $414141322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 41, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[2755] $41413221(12)12^1 23151321812321812 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 42, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2756] $41413221(12)122315132^1 1812321812 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 42, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 42, T \in [33, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[2757] $812321712321812^1 3218123217123 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 44, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 44, T \in [33, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2758] $81232^1 17123218123218123217123 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 44, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 44, T \in [33, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2759] $712321712^1 3218123218123218122 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[2760] $1812321812^1 32181232181232161 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 40, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 40, T \in [33, 40]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2761] $81232^1 1812321812321812321613 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 44, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 44, T \in [33, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2762] $812321812321812^1 321812321613 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 44, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 44, T \in [33, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2717] $21 \begin{pmatrix} 3151322^1 1 \\ 12 \end{pmatrix} 12231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 34, \Delta = -96, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 34, T \in [33, 34]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[2764] $4132181232181232^1 181231513218 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 43, \Delta = -113, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 43, T \in [33, 43]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2733] $16131513221(12)12^1 23151321812315 + \hat{1} \rightarrow 2$ $T = 28, T_{\max} = 41, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 32.69 \rightarrow 33, T_{\max} = 41, T \in [33, 41]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2735] $5131613151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.69 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[2734] $5131613151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 40$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.69 \rightarrow 33$, $T_{\max} = 40$, $T \in [33, 40]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

513221
[2765] $1(12)1223151321812^1 3218123 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 144$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.64 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

1
[2777] $81223151322^1 1(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 45$, $\Delta = -115$, $\det = 169$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.59 \rightarrow 33$, $T_{\max} = 45$, $T \in [33, 45]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

31513221
[2766] $21(12)12^1 231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 34$, $\Delta = -96$, $\det = 169$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.59 \rightarrow 33$, $T_{\max} = 34$, $T \in [33, 34]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

21
[2771] $1(12)122315132^1 181232181231513 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 169$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.59 \rightarrow 33$, $T_{\max} = 48$, $T \in [33, 48]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2772] $23151321812^1 32181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 169$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.59 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2773] $3151321812321812^1 31513221(12)122 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 169$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.59 \rightarrow 33$, $T_{\max} = 48$, $T \in [33, 48]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

1
[2774] $18123151322^1 1(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 45$, $\Delta = -115$, $\det = 169$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.59 \rightarrow 33$, $T_{\max} = 45$, $T \in [33, 45]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

1
[2775] $81231513221(12)122315132^1 181232 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 169$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.59 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

1
[2776] $812315132^1 21(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 169$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.59 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

31513221
[2767] $1(12)122^1 315132181232181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 169$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 32.59 \rightarrow 33, T_{\max} = 48, T \in [33, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[2778] & 7123151322^1 1(12)1223151316131613 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 169, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.59 \rightarrow 33, T_{\max} = 48, T \in [33, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2779] & 161316131613161315132^1 21(12)1223 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 44, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.59 \rightarrow 33, T_{\max} = 44, T \in [33, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[2780] & 81232171232^1 18123218123217123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 44, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.59 \rightarrow 33, T_{\max} = 44, T \in [33, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2781] & 81232181232^1 1812321812321613 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 44, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.59 \rightarrow 33, T_{\max} = 44, T \in [33, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2782] & 41321812321812321812^1 31513218 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 43, \Delta = -113, \det = 169, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.59 \rightarrow 33, T_{\max} = 43, T \in [33, 43] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2770] & 1^{3221}(12)12^1 23151321812321812315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 169, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.59 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2768] & 1^{3221}(12)122315132181232^1 1812315132 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 169, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.59 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2769] & 1^{3221}(12)1223151321812^1 321812315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 169, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.59 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 64 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2786] & 22315132^1 181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 196, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.55 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[2785] & 413221^{513221}(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 196, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.55 \rightarrow 33, T_{\max} = 48, T \in [33, 48] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2788] & 181231513221^{513221}(12)12^1 231513218123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 45, \Delta = -115, \det = 196, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.55 \rightarrow 33, T_{\max} = 45, T \in [33, 45] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2789] $812231513221(12)12^1 231513218123 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 45$, $\Delta = -115$, $\det = 196$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.55 \rightarrow 33$, $T_{\max} = 45$, $T \in [33, 45]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2790] $71231513221(12)122^1 3151316131613 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 196$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.55 \rightarrow 33$, $T_{\max} = 48$, $T \in [33, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2791] $812321712^1 3218123218123217123 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 196$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.55 \rightarrow 33$, $T_{\max} = 44$, $T \in [33, 44]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [2792] $812321812^1 321812321812321613 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 196$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.55 \rightarrow 33$, $T_{\max} = 44$, $T \in [33, 44]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [2784] $1(12)122^1 31513218123218123 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 196$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.55 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2783] $1(12)1223151321812^1 321812315132 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 196$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.55 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [2787] $2315132181232^1 181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 196$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.55 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [2793] $1(12)122315132^1 181232181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 225$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.52 \rightarrow 33$, $T_{\max} = 48$, $T \in [33, 48]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [2794] $81232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 225$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.52 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [2795] $81231513221(12)122^1 315132181232 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 51$, $\Delta = -117$, $\det = 225$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.52 \rightarrow 33$, $T_{\max} = 51$, $T \in [33, 51]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [2796] $1613161316131613151322^1 1(12)1223 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 44$, $\Delta = -114$, $\det = 225$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 32.52 \rightarrow 33$, $T_{\max} = 44$, $T \in [33, 44]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2797] $1(12)12^1 2315132181232181 + \hat{1} \rightarrow 2$ $T = 28$, $T_{\max} = 48$, $\Delta = -116$, $\det = 256$, $(n_+, n_-, n_0) = (1, 28, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 32.48 \rightarrow 33, T_{\max} = 48, T \in [33, 48] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
& [2798] \overset{1}{1}(12)122315132^1 1812321812315132 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 256, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.48 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
& [2799] 223151321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 256, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.48 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
& [2800] 23151321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 256, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.48 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
& [2802] \overset{513221}{1}(12)12^1 231513218123218123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 289, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.46 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
& [2805] \overset{1}{8}123151322^1 1(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 289, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.46 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
& [2806] 71231513221(12)12^1 23151316131613 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 48, \Delta = -116, \det = 289, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.46 \rightarrow 33, T_{\max} = 48, T \in [33, 48] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
& [2803] 812321812315132^1 21 \overset{231513221}{(12)} 1 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 289, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.46 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
& [2804] 22315132181232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 289, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.46 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
& [2801] \overset{3221}{1}(12)122315132^1 1812321812315 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 289, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.46 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 118 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
& [2807] \overset{513221}{1}(12)122315132^1 18123218123 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 324, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 32.43 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
& [2809] 223151321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 361, (n_+, n_-, n_0) = \\
& (1, 28, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 32.41 \rightarrow 33, T_{\max} = 51, T \in [33, 51] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2808] $812321812^1 31513221^{231513221} (12) \ 1 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 361, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.41 \rightarrow 33, T_{\max} = 51, T \in [33, 51]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$

[2810] $81231513221(12)12^1 2315132181232 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 361, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.41 \rightarrow 33, T_{\max} = 51, T \in [33, 51]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

[2811] $81232181231513221^1 (12) \ 1 + \hat{1} \rightarrow 2 \quad T = 28, T_{\max} = 51, \Delta = -117, \det = 441, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 32.37 \rightarrow 33, T_{\max} = 51, T \in [33, 51]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

$T = 29$ [2819] $712321812321812321812321812^1 2 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36, T_{\max} = 41, T \in [36, 41]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[2816] $161232181232181231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36, T_{\max} = 41, T \in [36, 41]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[2817] $3161232181232181231513221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[2818] $1812321712321812321812321812^1 2 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36, T_{\max} = 41, T \in [36, 41]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[2815] $1^{22^1 1}(12)122315132181232181232161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36, T_{\max} = 41, T \in [36, 41]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[2814] $1^{5131513221} (12) \ 12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36, T_{\max} = 41, T \in [36, 41]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[2813] $1^{5131513221} (12) \ 12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36, T_{\max} = 41, T \in [36, 41]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[2812] $1^{22^1 21} (12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36, T_{\max} = 41, T \in [36, 41]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

- [2820] $1(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -9$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36$, $T_{\max} = 41$, $T \in [36, 41]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [2821] $1(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 42$, $\Delta = -111$, $\det = -9$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.33 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [2823] $41413221(12)12231513221(12)12^1 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.33 \rightarrow 35$, $T_{\max} = 46$, $T \in [35, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [2824] $61232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 49$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36$, $T_{\max} = 49$, $T \in [36, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2825] $161232181232181231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -107$, $\det = -9$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36$, $T_{\max} = 41$, $T \in [36, 41]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2826] $61316131513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 49$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36$, $T_{\max} = 49$, $T \in [36, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2827] $812321712321812321812321(8)1 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 44$, $\Delta = -113$, $\det = -9$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36$, $T_{\max} = 44$, $T \in [36, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2828] $4141413221(12)12231513221(12)122^1 313 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 37$, $\Delta = -99$, $\det = -9$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36$, $T_{\max} = 37$, $T \in [36, 37]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2829] $8123217123218123218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 44$, $\Delta = -113$, $\det = -9$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36$, $T_{\max} = 44$, $T \in [36, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2830] $812321812321812321812321712^1 3 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 44$, $\Delta = -113$, $\det = -9$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36$, $T_{\max} = 44$, $T \in [36, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [2822] $1(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -9$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36$, $T_{\max} = 41$, $T \in [36, 41]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2834] $1612^1 32181232181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -107$, $\det = -16$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2835] & 812315132181232181232181232^1 15 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -16, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[2836] & \begin{matrix} 21 \\ 1812 \end{matrix} 132181232181232171232181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -16, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2837] & \begin{matrix} 1 \\ 1712 \end{matrix} 132181232181232181232181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -16, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2838] & \begin{matrix} 1 \\ 71232181232181232181232 \end{matrix} 181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -16, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2839] & 181232171232181232181232^1 18122 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -16, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2840] & 4141413221(12)12231513221(12)12^1 2313 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 37, \Delta = -99, \det = -16, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 37, T \in [35, 37] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2841] & \begin{matrix} 1 \\ 181232181232181232181232 \end{matrix} 1712 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -16, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[2842] & \begin{matrix} 1 \\ 71232181232181232181232 \end{matrix} 18122 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -16, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2832] & \begin{matrix} 221 \\ 1(12)122315132181232181232 \end{matrix} 161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -16, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2833] & \begin{matrix} 21 \\ 21(12)122315132181232181232 \end{matrix} 161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -16, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2831] & \begin{matrix} 5131513221 \\ 1 \end{matrix} (12) 12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -16, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.75 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[2845] $1^{15132^1 21} (12) 12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 42, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

[2847] $1712^1 32181231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 42, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[2848] $1712^1 31513221(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 41, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[2849] $7122^1 31513221(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 41, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[2850] $1812^1 31513221(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 41, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$

[2851] $1^{221} (12)1223151321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 41, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[2852] $21^{21} (12)1223151321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 41, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[2853] $612^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 49, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 49, T \in [35, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[2854] $61232181232181231513221^{322^1 1} (12)1 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 49, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 49, T \in [35, 49]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[2855] $161316131513221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 41, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2856] $161316131513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 41, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2857] $161232^1 181232181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 29, T_{\max} = 41, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2858] & 4132^1 21(12)1223151321812321812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 49, \Delta = -116, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 49, T \in [35, 49] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2859] & 612^1 32181231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[2860] & 8123151321812321812321812^1 3215 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2861] & 8123151321812321812321812^1 3151 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -108, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[2862] & 61232181231513221(12)122315132^1 16 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[2863] & 612^1 32181231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2864] & 181232^1 181232181232171232181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2865] & 171232^1 181232181232181232181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2866] & 712321812321812321812^1 32181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2867] & 1812321712321812321812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2868] & 4141413221(12)122315132^1 21(12)122313 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 37, \Delta = -99, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 37, T \in [35, 37] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2869] & 812321712321812321812321812^1 23 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -25, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.44 \rightarrow 35, T_{\max} = 44, T \in [35, 44]
\end{aligned}$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2870] $1812321812321812321812^1 321712 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2871] $712321812321812321812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[2846] $1^{5131513221} (12) 122315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2843] $1^{2221} (12) 122^1 31513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2844] $1^{5131513221} (12) 122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.44 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2872] $1^{2221} (12) 12^1 231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -36, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.22 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2873] $1^{5131513221} (12) 1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -36, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.22 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2874] $1^{5131513221} (12) 1223151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -36, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.22 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2875] $712^1 231513221(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -36, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.22 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2876] $1^{221} (12) 122^1 315132181232181232161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.22 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2877] $21^{21} (12) 122^1 315132181232181232161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.22 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

- [2878] 3161316131513221(12)122315132¹181 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -36$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 46$, $T \in [35, 46]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2879] 161316131513221(12)122315132¹181 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 41$, $T \in [35, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2880] 1612321812321812315132¹21(12)1222 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 41$, $T \in [35, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2881] 8122¹31513221(12)1223151316131613 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -36$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 46$, $T \in [35, 46]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2882] ²¹181232181232181232171232¹181 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 41$, $T \in [35, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2883] ¹1712321812321812321¹81232¹181 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 41$, $T \in [35, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2884] ¹712¹321812321812321812321²¹81 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 41$, $T \in [35, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2885] 1812¹3217123218123218123218122 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 41$, $T \in [35, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [2886] 4141413221(12)1223151322¹1(12)122313 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 37$, $\Delta = -99$, $\det = -36$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 37$, $T \in [35, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2887] 1812¹321812321812321812321712 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 41$, $T \in [35, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2888] 712¹3218123218123218123218122 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.22 \rightarrow 35$, $T_{\max} = 41$, $T \in [35, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2934] ¹71232181232181232¹1812321²¹81 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.06 \rightarrow 35$, $T_{\max} = 41$, $T \in [35, 41]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2927] & 812315132181232181232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -108, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& 151322^1 1 \quad 21 \\
[2894] & 1 (12) 12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& 5131513221 \\
[2895] & 1 (12) 122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
& 51315132^1 21 \\
[2896] & 1 (12) 12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[2897] & 171232^1 181231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2898] & 171232181231513221(12)122^1 3151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2899] & 1712315132^1 21(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2900] & 171231513221(12)1223151321812^1 315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2901] & 712231513221(12)1223151321812^1 315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2902] & 7122315132^1 21(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& 1 \\
[2903] & 1812315132^1 21(12)122315131613161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
& 221 \\
[2904] & 1(12)122315132181232^1 181232161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[2905] & 1(12)12^1 2315132181232181232161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0} \\
[2906] & 21(12)122315132181232^1 181232161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[2907] & 21(12)12^1 2315132181232181232161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0} \\
[2908] & 1(12)122315132181232181232^1 1613 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2909] & 1(12)122^1 3151321812321812321613 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2910] & 41413221(12)12231513221^{5132^1 21} (12) 12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2911] & 414132^1 21(12)12231513221^{513221} (12) 12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2912] & 61232^1 181232181231513221^{3221} (12) 1 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 49, \Delta = -116, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 49, T \in [35, 49] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[2913] & 161316131513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2914] & 161316131513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2915] & 16123218123218123151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[2916] 1612321812¹32181231513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2917] 31612¹32181232181231513221(12)122 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 46, T \in [35, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2918] 31612321812321812315132¹21(12)122 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 46, T \in [35, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[2919] 41322¹1(12)1223151321812321812315 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 49$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 49, T \in [35, 49]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2920] 413221(12)1223151321812321812¹315 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 49$, $\Delta = -116$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 49, T \in [35, 49]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2921] 812¹231513221(12)1223151316131613 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 46, T \in [35, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2922] 61232181231513221(12)122¹31513161 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2923] 61232¹181231513221(12)12231513161 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[2924] 812¹31513218123218123218123215 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2925] 812315132181232181232¹18123215 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2926] 812¹31513218123218123218123151 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -108$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2892] 1^{51315132¹21}(12)²²¹12231513221(12)12 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[2891] & 1^{5131513221} (12) 122^1 31513221^{221} (12) 12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2890] & 1^{2221} (12) 12231513221^1 (12) 122^1 3151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2930] & 1^{21} 1812321812^1 32181232171232181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2931] & 812^1 321712321812321812321^{321} 81 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 44, T \in [35, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2932] & 81232171232181232181232^1 181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 44, T \in [35, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2933] & 1^{1712321812^1 32181232181232181} + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2889] & 1^{2221} (12) 122315132^1 21^{12} 1223151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2935] & 181232171232181232^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 41, T \in [35, 41] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[2936] & 4141413221(12) 122^1 31513221(12) 122313 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 37, \Delta = -99, \det = -49, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 37, T \in [35, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2937] & 812^1 32171232181232181232181223 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 44, T \in [35, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2938] & 81232171232181232181232^1 181223 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.06 \rightarrow 35, T_{\max} = 44, T \in [35, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2939] & 1^{181232181232181232^1 1812321712} + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -49, (n_+, n_-, n_0) =
\end{aligned}$$

(1, 29, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 34.06 \rightarrow 35$, $T_{\max} = 41$, $T \in [35, 41]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2940] $81232181232181232181232^1 17123 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 44$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34.06 \rightarrow 35$, $T_{\max} = 44$, $T \in [35, 44]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2941] $812^1 3218123218123218123217123 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 44$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34.06 \rightarrow 35$, $T_{\max} = 44$, $T \in [35, 44]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2942] $71232181232181232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34.06 \rightarrow 35$, $T_{\max} = 41$, $T \in [35, 41]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2929] $61232^1 181231513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 42$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34.06 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[2893] $1^{1513221}(12)12231513221(12)122^1 315 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 42$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34.06 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2928] $61232181231513221(12)122^1 31513216 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 42$, $\Delta = -111$, $\det = -49$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 34.06 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2957] $812321712321812321812^1 32181223 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 44$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 44$, $T \in [34, 44]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2947] $161232181232^1 181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -107$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2948] $316123218123218123151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 46$, $T \in [34, 46]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2949] $3161232^1 181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 46$, $T \in [34, 46]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2950] $8123151321812321812^1 3218123215 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

- [2951] $8123151321812321812^1 3218123151 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -108$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2952] $181232181232^1 181232171232181 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2953] $812321712321812321812^1 321^1 81 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 44$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 44$, $T \in [34, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2954] $171232181232^1 181232181232181 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [2955] $712321812321812^1 321812321^1 81 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [2945] $1(12)1223151321812321812^1 321613 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 46$, $T \in [34, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2944] $21(12)1223151321812^1 32181232161 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -107$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [2943] $1(12)1223151321812^1 32181232161 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -107$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [2959] $812321812321812321812^1 3217123 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 44$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 44$, $T \in [34, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2960] $712321812321812^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [2961] $413161316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 43$, $\Delta = -112$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.94 \rightarrow 34$, $T_{\max} = 43$, $T \in [34, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2962] $413161316131613161315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 43$, $\Delta = -112$, $\det = -64$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 33.94 \rightarrow 34, T_{\max} = 43, T \in [34, 43] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2946] & 1(12)12^{\overset{21}{1}}23151321812321812321613 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -64, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.94 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2958] & 1812321812321812^{\overset{1}{1}}321812321712 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -64, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.94 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2956] & 1812321712321812^{\overset{1}{1}}3218123218122 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -64, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.94 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2963] & 1(12)12231513221(12)12^{\overset{1}{1}}23151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2964] & 1(12)12231513221^{\overset{2221}{1}}1(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2965] & 1^{\overset{5131513221}{1}}(12)12^{\overset{221}{1}}231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2966] & 1^{\overset{5131513221}{1}}(12)12231513221^{\overset{221}{1}}(12)12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = \\
& 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2967] & 1^{\overset{1513221}{1}}(12)122^{\overset{21}{1}}31513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2968] & 1^{\overset{5131513221}{1}}(12)12^{\overset{1}{1}}231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2969] & 1^{\overset{5131513221}{1}}(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2970] & 1712321812315132^{\overset{1}{1}}21(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 42, T \in [34, 42]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2971] & 171232181231513221(12)12^1 23151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2972] & 17123151322^1 1(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2973] & 171231513221(12)122^1 3151321812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2974] & 712231513221(12)122^1 3151321812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2975] & 71223151322^1 1(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2976] & \begin{matrix} 1 \\ 181231513221(12)122^1 315131613161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2977] & \begin{matrix} 1 \\ 18123151322^1 1(12)122315131613161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2978] & \begin{matrix} 513221 \\ 41413221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2979] & \begin{matrix} 513221 \\ 4141322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2980] & 161316131513221(12)12^1 23151321812 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2981] & 1613161315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2982] & 161316131513221(12)12^1 23151322181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2983] & 1613161315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2985] & 61232181231513221(12)12^1 231513161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2986] & 612321812315132^1 21(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2987] & 612321812315132^1 21(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2988] & 61232181231513221(12)12^1 231513216 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2989] & 18^{21} 12321812321812321712^1 32181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[2990] & 1712321812321812321812^1 32181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[2991] & 71232^1 181232181232181232181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[2992] & 181232^1 17123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2993] & 4141413221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 37, \Delta = -99, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 37, T \in [34, 37] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2994] & 41414132^1 21(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 37, \Delta = -99, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 37, T \in [34, 37] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2995] & 181232^1 1812321812321812321712 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2996] & 71232^1 18123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[2997] & 4131613161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 43, \Delta = -112, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 43, T \in [34, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2998] & 4131613161316131613151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 43, \Delta = -112, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 43, T \in [34, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2984] & 61316131513221(12)122315132^1 18123 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 49, \Delta = -116, \det = -81, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.84 \rightarrow 34, T_{\max} = 49, T \in [34, 49] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2999] & 1^{1513221} (12) 12231513221^{21} (12)12^1 2315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -100, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.76 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[3000] & 1^{221} (12)122315132^1 181232181232161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -100, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.76 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3001] & 21^{21} (12)122315132^1 181232181232161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -100, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.76 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3002] & 1^{21} (12)122315132181232^1 1812321613 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -100, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.76 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3003] & 41413221(12)12231513221^{51322^1} (12) 12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -100, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.76 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[3004] & 3161316131513221(12)122^1 315132181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -100, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.76 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3005] & 1612321812321812^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -100, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.76 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [3006] $31612321812^1 32181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 46$, $T \in [34, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3007] $8122315132^1 21(12)1223151316131613 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 46$, $T \in [34, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3008] $812315132181232^1 18123218123215 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3009] $812315132181232^1 18123218123151 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -108$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3010] $1812321812321812^1 32171232181 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3011] $81232171232181232^1 1812321 \overset{321}{8} 1 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 44$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 44$, $T \in [34, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3012] $1712321812321812^1 32181232181 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3013] $71232181232^1 1812321812321 \overset{21}{8} 1 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3014] $181232171232^1 18123218123218122 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [3015] $81232171232181232^1 181232181223 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 44$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 44$, $T \in [34, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3016] $181232181232^1 1812321812321712 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3017] $81232181232181232^1 18123217123 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 44$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.76 \rightarrow 34$, $T_{\max} = 44$, $T \in [34, 44]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[3018] $71232181232^1 18123218123218122 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -100, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.76 \rightarrow 34, T_{\max} = 41, T \in [34, 41]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3027] $181231513221(12)12^1 2315131613161 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.69 \rightarrow 34, T_{\max} = 41, T \in [34, 41]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[3023] $171231513221(12)122315132^1 1812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.69 \rightarrow 34, T_{\max} = 41, T \in [34, 41]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[3024] $171231513221(12)12^1 23151321812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.69 \rightarrow 34, T_{\max} = 41, T \in [34, 41]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[3025] $712231513221(12)122315132^1 1812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.69 \rightarrow 34, T_{\max} = 41, T \in [34, 41]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[3022] $1712321812^1 31513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.69 \rightarrow 34, T_{\max} = 42, T \in [34, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[3021] $17123218123151322^1 1(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.69 \rightarrow 34, T_{\max} = 42, T \in [34, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[3020] $1^{1513221} (12) 12^1 231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.69 \rightarrow 34, T_{\max} = 42, T \in [34, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[3029] $41413221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.69 \rightarrow 34, T_{\max} = 46, T \in [34, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[3030] $41413221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.69 \rightarrow 34, T_{\max} = 46, T \in [34, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[3031] $612321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 49, \Delta = -116, \det = -121, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 33.69 \rightarrow 34, T_{\max} = 49, T \in [34, 49]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[3032] $612321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 49, \Delta = -116, \det = -121, (n_+, n_-, n_0) =$

(1, 29, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 49$, $T \in [34, 49]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[3033] 16131613151322¹1(12)1223151321812 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[3034] 16131613151322¹1(12)1223151322181 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[3035] 3161232181232¹181231513221(12)122 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 46$, $T \in [34, 46]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[3036] 413221(12)122¹3151321812321812315 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 49$, $\Delta = -116$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 49$, $T \in [34, 49]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[3037] 413221(12)122315132181232¹1812315 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 49$, $\Delta = -116$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 49$, $T \in [34, 49]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[3038] 61316131513221(12)122¹31513218123 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 49$, $\Delta = -116$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 49$, $T \in [34, 49]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[3039] 612321812¹31513221(12)12231513161 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[3040] 6123218123151322¹1(12)12231513161 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[3041] 8123151321812¹3218123218123215 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3042] 8123151321812¹3218123218123151 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -108$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3043] 612321812¹31513221(12)12231513216 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 42$, $\Delta = -111$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 42$, $T \in [34, 42]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[3044] 6123218123151322¹1(12)12231513216 + $\hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 42$, $\Delta = -111$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3045] & \overset{21}{181232181232181232^1 171232181} + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -121, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[3046] & 812321712321812^1 321812321 \overset{321}{81} + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -121, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3047] & 81232^1 1712321812321812321 \overset{321}{81} + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -121, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[3048] & \overset{1}{171232181232181232^1} \overset{1}{181232181} + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -121, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3049] & 712321812^1 321812321812321 \overset{21}{81} + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -121, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3050] & 1812321712^1 3218123218123218122 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -121, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[3051] & 414141322^1 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 37, \Delta = -99, \det = -121, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 37, T \in [34, 37] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[3052] & 812321712321812^1 32181232181223 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -121, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3053] & 81232^1 171232181232181232181223 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -121, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[3054] & \overset{1}{1812321812^1} \overset{1}{321812321812321712} + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -104, \det = -121, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3055] & 812321812321812^1 3218123217123 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -121, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.69 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [3056] $81232^1 18123218123218123217123 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 44$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 44$, $T \in [34, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [3057] $712321812^1 3218123218123218122 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -104$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [3019] $1^{1513221} (12)^{21} 122315132^1 21 (12)^{21} 122315 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 42$, $\Delta = -111$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 42$, $T \in [34, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [3026] $712231513221 (12)^{21} 23151321812315 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [3028] $1^{21} (12)^{21} 1223151321812^1 321812321613 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.69 \rightarrow 34$, $T_{\max} = 46$, $T \in [34, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3060] $6123218123218123151322^1 1 (12)^1 1 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 49$, $\Delta = -116$, $\det = -169$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.59 \rightarrow 34$, $T_{\max} = 49$, $T \in [34, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [3062] $31612321812321812^1 31513221 (12)^1 122 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -169$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.59 \rightarrow 34$, $T_{\max} = 46$, $T \in [34, 46]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [3063] $413221 (12)^1 12^1 23151321812321812315 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 49$, $\Delta = -116$, $\det = -169$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.59 \rightarrow 34$, $T_{\max} = 49$, $T \in [34, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3064] $413221 (12)^1 1223151321812^1 321812315 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 49$, $\Delta = -116$, $\det = -169$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.59 \rightarrow 34$, $T_{\max} = 49$, $T \in [34, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 64 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [3065] $613161315132^1 21 (12)^1 12231513218123 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 49$, $\Delta = -116$, $\det = -169$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.59 \rightarrow 34$, $T_{\max} = 49$, $T \in [34, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [3066] $81223151322^1 1 (12)^1 1223151316131613 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 46$, $\Delta = -115$, $\det = -169$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.59 \rightarrow 34$, $T_{\max} = 46$, $T \in [34, 46]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [3067] $812315132^1 18123218123218123215 + \hat{1} \rightarrow 2$ $T = 29$, $T_{\max} = 41$, $\Delta = -110$, $\det = -169$, $(n_+, n_-, n_0) = (1, 29, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 33.59 \rightarrow 34$, $T_{\max} = 41$, $T \in [34, 41]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3068] & 812315132^1 18123218123218123151 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 41, \Delta = -108, \det = -169, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.59 \rightarrow 34, T_{\max} = 41, T \in [34, 41] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3069] & 81232171232^1 1812321812321 \overset{321}{8} 1 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -169, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.59 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3070] & 81232171232^1 181232181232181223 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -169, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.59 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3071] & 81232181232^1 18123218123217123 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -169, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.59 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3058] & 1(12)122315132^1 1812321812321613 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -169, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.59 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3059] & 61232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 49, \Delta = -116, \det = -169, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.59 \rightarrow 34, T_{\max} = 49, T \in [34, 49] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[3061] & 3161316131513221(12)12^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -169, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.59 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[3076] & 812231513221(12)122^1 3151316131613 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -196, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.55 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3074] & 31613161315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -196, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.55 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[3077] & 812321712^1 321812321812321 \overset{321}{8} 1 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -196, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.55 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[3078] & 812321712^1 32181232181232181223 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -196, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.55 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[3079] & 812321812^1 3218123218123217123 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 44, \Delta = -113, \det = -196, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 33.55 \rightarrow 34, T_{\max} = 44, T \in [34, 44] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
& [3072] \quad 1 \quad (12) \quad 1223151322^1 (12) 122315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 42, \Delta = -111, \det = -196, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.55 \rightarrow 34, T_{\max} = 42, T \in [34, 42] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& [3073] \quad 41413221(12)1223151322^1 (12) 12 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -196, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.55 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& [3075] \quad 61316131513221(12)12^1 231513218123 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 49, \Delta = -116, \det = -196, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.55 \rightarrow 34, T_{\max} = 49, T \in [34, 49] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& [3080] \quad 6131613151322^1 (12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 49, \Delta = -116, \det = -256, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.48 \rightarrow 34, T_{\max} = 49, T \in [34, 49] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
& [3081] \quad 612321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 49, \Delta = -116, \det = -289, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.46 \rightarrow 34, T_{\max} = 49, T \in [34, 49] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
& [3082] \quad 316131613151322^1 (12)122315132181 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -289, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.46 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 390 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0} \\
& [3083] \quad 413221(12)122315132^1 1812321812315 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 49, \Delta = -116, \det = -289, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.46 \rightarrow 34, T_{\max} = 49, T \in [34, 49] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 118 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
& [3084] \quad 812231513221(12)12^1 23151316131613 + \hat{1} \rightarrow 2 \quad T = 29, T_{\max} = 46, \Delta = -115, \det = -289, (n_+, n_-, n_0) = \\
& (1, 29, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 33.46 \rightarrow 34, T_{\max} = 46, T \in [34, 46] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
& T = 30 \quad [3095] \quad 171232181232181231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 4, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37, T_{\max} = 42, T \in [37, 42] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
& [3089] \quad 61231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 4, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37, T_{\max} = 42, T \in [37, 42] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
& [3090] \quad 61231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 4, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37, T_{\max} = 42, T \in [37, 42]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3091] $16131513221(12)^1 12231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37, T_{\max} = 42, T \in [37, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3092] $316131513221(12)^1 12231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 47, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37, T_{\max} = 47, T \in [37, 47]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3093] $61231513221(12)^1 12231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37, T_{\max} = 42, T \in [37, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3094] $171232181232181231513221(12)^1 1 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37, T_{\max} = 42, T \in [37, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3088] $6131513221(12)^1 12231513221(12)^{3221} 12^1 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36, T_{\max} = 50, T \in [36, 50]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[3096] $1(12)^{22^1 1} 1223151321812321812321712 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37, T_{\max} = 42, T \in [37, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[3097] $32171232181232181231513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 46, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37, T_{\max} = 46, T \in [37, 46]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3098] $2171232181232181231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37, T_{\max} = 42, T \in [37, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3099] $1812321812321812321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37, T_{\max} = 42, T \in [37, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3085] $1(12)^{22^1 21} 12231513221(12)^1 12231513161 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37, T_{\max} = 42, T \in [37, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3086] $21(12)^{22^1 1} 12231513221(12)^1 12231513161 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37, T_{\max} = 42, T \in [37, 42]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

- [3087] $1(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 47, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37, T_{\max} = 47, T \in [37, 47]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3109] $1(12)1223151321812321812321712^1 3 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 46, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3104] $16131513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 42, T \in [37, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3105] $316131513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 47, \Delta = -115, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 47, T \in [37, 47]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3106] $61231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -108, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 42, T \in [37, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3107] $1(12)122315132181232181232171 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 46, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [3108] $171232181232181231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 42, T \in [37, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3103] $61231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -108, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 42, T \in [37, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [3110] $32^1 171232181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 46, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [3111] $2171232181232181231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 42, T \in [37, 42]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3112] $812321812321812321812321 \ 8 \ 1 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 44, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3113] $8123218123218123218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 44, \Delta = -112, \det = 9, (n_+, n_-, n_0) =$

$$(1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3100] \quad 1 \quad \overset{5131513221}{(12)} \quad 12231513221(12)122^1 313 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 37, \Delta = -98, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3101] \quad 1 \overset{222^1 1}{(12)} 12231513221 \overset{1}{(12)} 12231513161 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 42, T \in [37, 42]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[3102] \quad 6131513221 \overset{1}{(12)} 12231513221 \overset{32^1 21}{(12)} 12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 9, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 50, T \in [37, 50]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3117] \quad 1712^1 32181232181231513221 \overset{221}{(12)} 1 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3118] \quad 1712^1 32181232181231513221 \overset{21}{(12)} 12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3119] \quad 1712^1 32181232181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3120] \quad 1 \overset{221}{(12)} 122315132181232181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3121] \quad 21 \overset{21}{(12)} 122315132181232181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3122] \quad 21712^1 32181232181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3123] \quad 1812^1 32181232181232181232181 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -103, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[3124] \quad 812315132181232181232181232^1 161 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -107, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3125] $181232181232181232181232^1 18122 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -103, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[3115] $16131513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -107, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[3114] $1^{5131513221}(12)12231513221(12)12^1 2313 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 37, \Delta = -98, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 37, T \in [36, 37]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3116] $61231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.75 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3127] $1812^1 32181231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3130] $21^{221}(12)122^1 31513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3131] $6131513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 50, T \in [36, 50]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3132] $612^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[3133] $61231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3134] $612^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[3135] $61231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3136] $21812^1 32181231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3137] $5132181232181231513221(12)122^1 3151 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 43$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3138] $21812^1 31513221(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3139] $5132181231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3140] $16131513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[3141] $61231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3142] $612^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -108$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[3143] $171232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3144] $171232^1 181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3145] $1(12)122315132181232181232171 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 46$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3146] $1(12)122315132181232181232^1 171 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 46$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.44 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3147] $1(12)1223151321812321812321712^1 32 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 50, T \in [36, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

[3148] $171232^1 181232181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3149] $1(12)1223151321812321812^1 321712 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[3150] $21(12)1223151321812321812^1 321712 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[3151] $16131513221(12)122315132181232^1 171 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3152] $232^1 171232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 50, T \in [36, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[3153] $2171232^1 181232181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3154] $612^1 31513221(12)122315132181232171 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$

[3155] $61231513221(12)122315132181232^1 171 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3156] $612^1 32181231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$

[3157] $61232181231513221(12)122315132^1 171 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3158] $612^1 32181231513221(12)122315132217 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

- [3159] $61232181231513221(12)122315132^1 217 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36$, $T_{\max} = 42$, $T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [3160] $181232^1 181232181232181232181 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -103$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36$, $T_{\max} = 42$, $T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3161] $8123151321812321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36$, $T_{\max} = 42$, $T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [3162] $1812321812321812321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -103$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36$, $T_{\max} = 42$, $T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3163] $812321812321812321812321812^1 23 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 44$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36$, $T_{\max} = 44$, $T \in [36, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3164] $4132^1 21(12)12231513218123218123216 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 47$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36$, $T_{\max} = 47$, $T \in [36, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3165] $413221(12)122315132181232181232^1 16 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 47$, $\Delta = -115$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36$, $T_{\max} = 47$, $T \in [36, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [3166] $4132^1 18123218123218123217123218 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 44$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36$, $T_{\max} = 44$, $T \in [36, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [3129] $1(12)122^1 31513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36$, $T_{\max} = 42$, $T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3128] $1812^1 31513221(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36$, $T_{\max} = 42$, $T \in [36, 42]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3126] $1(12)122315132^1 21(12)122313 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 37$, $\Delta = -98$, $\det = 25$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.44 \rightarrow 36$, $T_{\max} = 37$, $T \in [36, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3170] $1(12)122^1 31513221(12)122315131613 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 47$, $\Delta = -115$, $\det = 36$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36$, $T_{\max} = 47$, $T \in [36, 47]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3171] & 21(12)122^1 131513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.22 \rightarrow 36, T_{\max} = 47, T \in [36, 47] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3172] & 61231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3173] & 61231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3174] & 5132181231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[3175] & 16131513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -107, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3176] & 316131513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.22 \rightarrow 36, T_{\max} = 47, T \in [36, 47] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3177] & 61231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3178] & 1712321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[3179] & 1712321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[3180] & 1712321812321812315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[3181] & 1(12)122^1 13151321812321812321712 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [3182] $21(12)122^1 3151321812321812321712 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [3183] $21712321812321812315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3184] $712^1 32181231513221(12)122315131613 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36, T_{\max} = 50, T \in [36, 50]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3185] $61232181231513221(12)1223151322^1 17 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3186] $181232181232181232181232^1 181 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -103, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3187] $1812^1 3218123218123218123218122 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -103, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3169] $21(12)12^1 231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -107, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3168] $1(12)12^1 231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -107, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3167] $1^1 (12) 1223151322^1 1(12)122313 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 37, \Delta = -98, \det = 36, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.22 \rightarrow 36, T_{\max} = 37, T \in [36, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3221] $1(12)1223151321812321812^1 32171 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 46, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [3220] $17123218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [3193] $1812315132^1 21(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[3205] 2181232181231513221(12)122¹3151315 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[3206] 5132¹181232181231513221(12)1223151 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 43$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[3207] 5132181232181231513221(12)12¹23151 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 43$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[3208] 2181231513221(12)1223151321812¹315 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3209] 21812315132¹21(12)1223151321812315 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3210] 5132¹181231513221(12)1223151322181 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[3211] 5132181231513221(12)122¹3151322181 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3212] 161315132¹21(12)12231513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -107$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3213] 16131513221(12)122¹31513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -107$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[3214] 316131513221(12)1223151322¹1(12)1222 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 47$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 47, T \in [36, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[3215] 61231513221(12)122¹31513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3216] 612315132¹21(12)12231513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[3217] & \begin{matrix} 1 \\ 1712321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[3218] & \begin{matrix} 1 \\ 17123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[3219] & \begin{matrix} 1 \\ 1712321812^1 32181231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[3191] & \begin{matrix} 1 \\ 181232181231513221(12)122^1 3151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 42, \Delta = -110, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[3190] & \begin{matrix} 1 \\ 181232^1 181231513221(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 42, \Delta = -110, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[3189] & \begin{matrix} 51315132^1 21 \\ 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 37, \Delta = -98, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 37, T \in [36, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[3223] & \begin{matrix} 1 \\ 17123218123218123151322^1 1(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[3224] & \begin{matrix} 21 \\ 1(12)122^1 31513218123218123217123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[3225] & \begin{matrix} 21 \\ 1(12)122315132181232181232^1 17123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3226] & \begin{matrix} 221 \\ 1(12)122315132181232^1 1812321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[3227] & \begin{matrix} 221 \\ 1(12)12^1 23151321812321812321712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[3228] $21(12)122315132181232^1 1812321712 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[3229] $21(12)12^1 23151321812321812321712 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[3230] $161315132^1 21(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[3231] $16131513221(12)1223151321812^1 32171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[3232] $321712321812321812315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[3233] $321712^1 32181232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3234] $21712321812^1 32181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3235] $217123218123218123151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[3236] $612315132^1 21(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[3237] $61231513221(12)1223151321812^1 32171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[3238] $61232^1 181231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[3239] $61232181231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$$

- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
 [3240] $61232181231513221(12)122^1 315132217 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [3241] $61232^1 181231513221(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- $\begin{matrix} 21 & & 1 \\ 18 & 12321812 & 132181232181 \end{matrix} + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- $\begin{matrix} & 1 & & 321 \\ 812 & 1321812321812321812321 & 8 & 1 \end{matrix} + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 44, T \in [36, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- $\begin{matrix} & 1 & & 321 \\ 812321812321812321812321 & 1 & 8 & 1 \end{matrix} + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 44, T \in [36, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3245] $812^1 315132181232181232181232161 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
- [3246] $812315132181232181232^1 181232161 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- $\begin{matrix} & 1 \\ 181232181232181232^1 & 18123218122 & + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 42, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\begin{matrix} & 1 \\ 812^1 & 132181232181232181232181223 & + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 44, T \in [36, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- $\begin{matrix} & 1 \\ 81232181232181232181232^1 & 181223 & + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 44, T \in [36, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3250] $413221(12)1223151321812321812^1 3216 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.06 \rightarrow 36, T_{\max} = 47, T \in [36, 47]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3251] $41322^1 1(12)12231513218123218123216 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 47, T \in [36, 47] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3252] & 4132181232181232181232171232^1 18 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 44, T \in [36, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[3253] & 41321812^1 3218123218123217123218 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 44, T \in [36, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[3222] & 1712321812^1 32181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[3188] & 1^{5131513221}(12)122^1 31513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 37, \Delta = -98, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 37, T \in [36, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[3192] & 1^{181231513221(12)1223151321812^1 315} + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 49, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.06 \rightarrow 36, T_{\max} = 42, T \in [36, 42] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3257] & 1^{171232181232^1 181231513221(12)1222} + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 64, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.94 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3258] & 1^{21}(12)12^1 231513218123218123217123 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 64, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.94 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[3259] & 1^{21}(12)1223151321812321812^1 3217123 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 64, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.94 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[3260] & 1^{221}(12)1223151321812^1 321812321712 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 64, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.94 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[3261] & 1^{21}(12)1223151321812^1 321812321712 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 64, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.94 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[3262] & 3217123218123218123151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 64, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.94 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [3263] $32171232^1 181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 46$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 46$, $T \in [35, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3264] $2321712321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 50$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 50$, $T \in [35, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3265] $2171232181232^1 181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -105$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3266] $1(12)122^1 315131613161316131613151 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -107$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [3267] $1^1 81232181232^1 181232181232181 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -103$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [3268] $812321812321812321812^1 321^1 81 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 44$, $\Delta = -112$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 44$, $T \in [35, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3269] $1513161316131613161315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -107$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3270] $8123151321812321812^1 32181232161 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -107$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [3271] $1812321812321812^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -103$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3272] $812321812321812321812^1 32181223 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 44$, $\Delta = -112$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 44$, $T \in [35, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3273] $4132181232^1 18123218123217123218 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 44$, $\Delta = -112$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 44$, $T \in [35, 44]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [3256] $1(12)122^1 315132181232181232171232 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 50$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.94 \rightarrow 35$, $T_{\max} = 50$, $T \in [35, 50]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[3255] & \begin{matrix} 1 \\ 171232181232^1 181231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -105, \quad \det = 64, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.94 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3254] & \begin{matrix} 1 \\ 171232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -105, \quad \det = 64, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.94 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3306] & \begin{matrix} 1 \\ 61232181231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -109, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[3274] & \begin{matrix} 513151322^1 1 \\ 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 37, \quad \Delta = -98, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 37, \quad T \in [35, 37] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3279] & \begin{matrix} 1 \\ 18123151322^1 1(12)1223151321812315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -109, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3280] & \begin{matrix} 2221 \\ 1(12)1223151322^1 1(12)12231513161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -107, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3281] & \begin{matrix} 2221 \\ 1(12)12231513221(12)12^1 231513161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -107, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[3282] & \begin{matrix} 221 \\ 21(12)1223151322^1 1(12)12231513161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -107, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3283] & \begin{matrix} 221 \\ 21(12)12231513221(12)12^1 231513161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -107, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[3284] & \begin{matrix} 1 \\ 6131513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 50, \quad \Delta = -116, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 50, \quad T \in [35, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3285] & \begin{matrix} 1 \\ 61231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -108, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [3286] $6123151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3287] $6123151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3288] $61231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [3289] $21812321812315132^1 21(12)1223151315 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [3290] $2181232181231513221(12)12^1 23151315 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3291] $2181231513221(12)122^1 3151321812315 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3292] $218123151322^1 1(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [3293] $5132181231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [3294] $51321812315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3295] $1613151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [3296] $16131513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3297] $6123151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 34.84 \rightarrow 35$, $T_{\max} = 42$, $T \in [35, 42]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3298] $1(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[3299] $1(12)12^1 2315132181232181232171232 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[3300] $16131513221(12)122^1 315132181232171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[3301] $1613151322^1 1(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[3302] $23217123218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3303] $71232^1 181231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[3304] $61231513221(12)122^1 315132181232171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[3305] $6123151322^1 1(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[3277] $1812321812315132^1 21(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[3307] $612321812315132^1 21(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[3308] $61232181231513221(12)12^1 2315132217 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3309] $612321812315132^1 21(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 34.84 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3310] & \begin{matrix} 1 \\ 1(12)12^1 2315131613161316131613151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -107, \quad \det = 81, \quad (n_+, n_-, n_0) = (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3311] & \begin{matrix} 21 \\ 1812321812321812321812^1 32181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -103, \quad \det = 81, \quad (n_+, n_-, n_0) = (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[3312] & \begin{matrix} 1 \\ 31513161316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 47, \quad \Delta = -115, \quad \det = 81, \quad (n_+, n_-, n_0) = (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 47, \quad T \in [35, 47] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3313] & \begin{matrix} 1 \\ 15131613161316131613151322^1 1(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -107, \quad \det = 81, \quad (n_+, n_-, n_0) = (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3314] & \begin{matrix} 1 \\ 181232^1 18123218123218123218122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -103, \quad \det = 81, \quad (n_+, n_-, n_0) = (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[3276] & \begin{matrix} 1 \\ 181232181231513221(12)12^1 23151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -110, \quad \det = 81, \quad (n_+, n_-, n_0) = (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[3278] & \begin{matrix} 1 \\ 181231513221(12)122^1 3151321812315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -109, \quad \det = 81, \quad (n_+, n_-, n_0) = (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[3275] & \begin{matrix} 5131513221 \\ 1 \quad (12) \quad 12^1 231513221(12)122313 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 37, \quad \Delta = -98, \quad \det = 81, \quad (n_+, n_-, n_0) = (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.84 \rightarrow 35, \quad T_{\max} = 37, \quad T \in [35, 37] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[3317] & \begin{matrix} 1 & & 3221 \\ 61315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 50, \quad \Delta = -116, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.76 \rightarrow 35, \quad T_{\max} = 50, \quad T \in [35, 50] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3319] & \begin{matrix} 1 & & 221 \\ 1712321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -105, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.76 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3320] & \begin{matrix} 1 & & 21 \\ 1712321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, \quad T_{\max} = 42, \quad \Delta = -105, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 30, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.76 \rightarrow 35, \quad T_{\max} = 42, \quad T \in [35, 42] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[3321] $1(12)122315132181232181232^1 171232 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[3322] $1712321812321812^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3323] $1(12)122315132181232^1 18123217123 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 46, T \in [35, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[3324] $1(12)122315132^1 1812321812321712 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[3325] $21(12)122315132^1 1812321812321712 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[3326] $321712321812^1 32181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 46, T \in [35, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3327] $2321712^1 32181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3328] $21712321812321812^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -105, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3329] $1812321812321812^1 32181232181 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -103, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[3330] $81232181232181232^1 1812321^1 81 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[3331] $31513161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 47, T \in [35, 47]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

[3332] $(12)122^1 31513161316131613161315132 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.76 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3333] & 812315132181232^1 181232181232161 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -107, \det = 100, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.76 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3334] & 181232181232^1 18123218123218122 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -103, \det = 100, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.76 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3335] & 81232181232181232^1 181232181223 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 100, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.76 \rightarrow 35, T_{\max} = 44, T \in [35, 44] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3336] & 41321812321812^1 3218123217123218 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 100, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.76 \rightarrow 35, T_{\max} = 44, T \in [35, 44] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[3318] & 316131513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 100, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.76 \rightarrow 35, T_{\max} = 47, T \in [35, 47] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3315] & 1(12)122315132^1 21(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 100, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.76 \rightarrow 35, T_{\max} = 47, T \in [35, 47] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3316] & 21(12)122315132^1 21(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 100, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.76 \rightarrow 35, T_{\max} = 47, T \in [35, 47] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3369] & 181232181232181232^1 181232181 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -103, \det = 121, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3350] & 2181231513221(12)12^1 23151321812315 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 121, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[3342] & 21(12)12231513221(12)122^1 315131613 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 121, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 47, T \in [35, 47] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[3343] & 6131513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 121, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 50, T \in [35, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [3344] $6131513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [3345] $21812321812^1 31513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [3346] $218123218123151322^1 1(12)1223151315 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3347] $51321812321812315132^1 21(12)1223151 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 43, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 43, T \in [35, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3340] $181231513221(12)12^1 23151321812315 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [3339] $181231513221(12)122315132^1 1812315 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [3338] $18123218123151322^1 1(12)1223151315 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3351] $51321812^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [3352] $513218123151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 42, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [3353] $3161315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 47, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 47, T \in [35, 47]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [3354] $1(12)122^1 315132181232181232171 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 46, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 46, T \in [35, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [3355] $1(12)122315132181232^1 181232171 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 46, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.69 \rightarrow 35, T_{\max} = 46, T \in [35, 46]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[3356] & 1(12)1223151321812321812^1 32171232 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 50, T \in [35, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3357] & 1(12)1223151321812^1 3218123217123 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3358] & 16131513221(12)12^1 2315132181232171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3359] & 16131513221(12)122315132^1 181232171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[3360] & 32171232181232^1 181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[3361] & 232171232^1 181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 50, T \in [35, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[3362] & 71232181231513221(12)122^1 315131613 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 50, T \in [35, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[3363] & 61231513221(12)12^1 2315132181232171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3364] & 61231513221(12)122315132^1 181232171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[3365] & 612321812^1 31513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[3366] & 6123218123151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[3367] & 6123218123151322^1 1(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]
\end{aligned}$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[3368] $612321812^1 31513221(12)122315132217 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -109$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[3337] $1812321812^1 31513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[3370] $812321812321812^1 321812321 \overset{321}{8} 1 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 44$, $\Delta = -112$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.69 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3371] $81232^1 1812321812321812321 \overset{321}{8} 1 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 44$, $\Delta = -112$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.69 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[3372] $(12)12^1 231513161316131613161315132 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 50$, $\Delta = -116$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.69 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[3373] $8123151321812^1 32181232181232161 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -107$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[3374] $1812321812^1 3218123218123218122 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 42$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3375] $812321812321812^1 32181232181223 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 44$, $\Delta = -112$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.69 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3376] $81232^1 181232181232181232181223 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 44$, $\Delta = -112$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.69 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[3377] $413221(12)122315132181232^1 18123216 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 47$, $\Delta = -115$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.69 \rightarrow 35, T_{\max} = 47, T \in [35, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[3378] $413221(12)122^1 31513218123218123216 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 47$, $\Delta = -115$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 34.69 \rightarrow 35, T_{\max} = 47, T \in [35, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[3379] $41321812321812321812321712^1 3218 + \hat{1} \rightarrow 2$ $T = 30$, $T_{\max} = 44$, $\Delta = -112$, $\det = 121$, $(n_+, n_-, n_0) = (1, 30, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 44, T \in [35, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 21 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[3380] & 4132181232181232^1 18123217123218 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 121, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 44, T \in [35, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[3349] & 2181231513221(12)122315132^1 1812315 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -109, \det = 121, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 42, T \in [35, 42] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[3348] & 51321812^1 32181231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 43, \Delta = -111, \det = 121, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 43, T \in [35, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[3341] & \begin{matrix} 221 & & 1 \\ 1(12)12231513221(12)122^1 315131613 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 121, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.69 \rightarrow 35, T_{\max} = 47, T \in [35, 47] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[3381] & \begin{matrix} 221 & & 1 \\ 1(12)1223151322^1 1(12)122315131613 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 169, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.59 \rightarrow 35, T_{\max} = 47, T \in [35, 47] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[3385] & 5132181232^1 181231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 43, \Delta = -111, \det = 169, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.59 \rightarrow 35, T_{\max} = 43, T \in [35, 43] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[3386] & \begin{matrix} & & 1 \\ 316131513221(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 169, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.59 \rightarrow 35, T_{\max} = 47, T \in [35, 47] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[3387] & \begin{matrix} 3221 \\ 1(12)1223151321812^1 32181232171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.59 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[3388] & \begin{matrix} 3221 \\ 1(12)12^1 2315132181232181232171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.59 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[3389] & \begin{matrix} & & 1 \\ 1(12)122315132181232^1 181232171232 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 169, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.59 \rightarrow 35, T_{\max} = 50, T \in [35, 50] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3390] & \begin{matrix} 21 \\ 1(12)122315132^1 18123218123217123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.59 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[3391] $321712321812321812^1 31513221(12)122 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 46, T \in [35, 46]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[3392] $2321712321812^1 32181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[3393] $712321812315132^1 21(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[3394] $81232181232^1 1812321812321 \overset{321}{8} 1 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3395] $812315132^1 181232181232181232161 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 42, \Delta = -107, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 42, T \in [35, 42]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$

[3396] $81232181232^1 181232181232181223 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3397] $413221(12)1223151321812^1 3218123216 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 47, T \in [35, 47]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 64 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[3398] $413221(12)12^1 231513218123218123216 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 47, T \in [35, 47]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[3399] $41321812321812321812^1 3217123218 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 44, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

[3384] $513218123218123151322^1 1(12)1223151 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 43, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 43, T \in [35, 43]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[3383] $613151322^1 1(12)12231513221 \overset{3221}{(12)12} + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[3382] $21 \overset{21}{(12)1223151322^1 1(12)122315131613} + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 169, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.59 \rightarrow 35, T_{\max} = 47, T \in [35, 47]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

- [3400] $1(12)12231513221(12)12^1 2315131613 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 47, \Delta = -115, \det = 196, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.55 \rightarrow 35, T_{\max} = 47, T \in [35, 47]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3401] $21(12)12231513221(12)12^1 2315131613 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 47, \Delta = -115, \det = 196, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.55 \rightarrow 35, T_{\max} = 47, T \in [35, 47]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3403] $31613151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 47, \Delta = -115, \det = 196, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.55 \rightarrow 35, T_{\max} = 47, T \in [35, 47]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3404] $1(12)1223151321812^1 32181232171232 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 196, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.55 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [3405] $232171232181232^1 181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 196, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.55 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [3406] $71232181231513221(12)12^1 2315131613 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 196, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.55 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3407] $812321812^1 321812321812321^1 81 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 44, \Delta = -112, \det = 196, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.55 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [3408] $812321812^1 32181232181232181223 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 44, \Delta = -112, \det = 196, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.55 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [3409] $4132181232181232181232^1 17123218 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 44, \Delta = -112, \det = 196, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.55 \rightarrow 35, T_{\max} = 44, T \in [35, 44]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 41 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3402] $6131513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 196, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.55 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3410] $712321812^1 31513221(12)122315131613 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 225, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 34.52 \rightarrow 35, T_{\max} = 50, T \in [35, 50]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [3411] $1(12)122315132^1 181232181232171232 + \hat{1} \rightarrow 2$ $T = 30, T_{\max} = 50, \Delta = -116, \det = 256, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 34.48 \rightarrow 35, T_{\max} = 50, T \in [35, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[3412] & 2321712321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 256, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.48 \rightarrow 35, T_{\max} = 50, T \in [35, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[3413] & 7123218123151322^1 1(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 50, \Delta = -116, \det = 256, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.48 \rightarrow 35, T_{\max} = 50, T \in [35, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3414] & 51321812321812^1 31513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 43, \Delta = -111, \det = 289, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.46 \rightarrow 35, T_{\max} = 43, T \in [35, 43] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3415] & 1^{3221}(12)122315132^1 181232181232171 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 46, \Delta = -114, \det = 289, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.46 \rightarrow 35, T_{\max} = 46, T \in [35, 46] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 407 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[3416] & 413221(12)122315132^1 18123218123216 + \hat{1} \rightarrow 2 \quad T = 30, T_{\max} = 47, \Delta = -115, \det = 289, (n_+, n_-, n_0) = \\
& (1, 30, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 34.46 \rightarrow 35, T_{\max} = 47, T \in [35, 47] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 118 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
\\
T = 31 & [3419] 71231513221^1(12)12231513221^{3221}(12)12^1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -4, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[3432] & 22^1 181232181232181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -4, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38, T_{\max} = 43, T \in [38, 43] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[3417] & 1^{22^1 21}(12)12231513221^1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -4, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38, T_{\max} = 43, T \in [38, 43] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3418] & 21^{22^1 1}(12)12231513221^1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -4, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38, T_{\max} = 43, T \in [38, 43] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3420] & 712231513221^1(12)12231513221^{22^1 21}(12)1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -4, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38, T_{\max} = 43, T \in [38, 43] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[3421] & 712231513221^1(12)12231513221^{22^1 1}(12)12 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -4, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38, T_{\max} = 43, T \in [38, 43]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3422] $171231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38, T_{\max} = 43, T \in [38, 43]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3423] $712231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38, T_{\max} = 43, T \in [38, 43]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[3424] $181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38, T_{\max} = 43, T \in [38, 43]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3425] $181232181232181231513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -4, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3426] $181232181232181231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38, T_{\max} = 43, T \in [38, 43]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3427] $1(12)1223151321812321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38, T_{\max} = 43, T \in [38, 43]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3428] $1(12)12231513218123218123218122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38, T_{\max} = 43, T \in [38, 43]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[3429] $21(12)1223151321812321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38, T_{\max} = 43, T \in [38, 43]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3430] $322181232181232181231513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -4, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3431] $22181232181232181231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38, T_{\max} = 43, T \in [38, 43]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3448] $32^1 16131613161316131513221(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[3433] $513221(12)^{2^1 1} 12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.33 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[3434] $1(12)^{222^1 1} 12231513221(12)^1 122315132171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38, T_{\max} = 43, T \in [38, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[3435] $71231513221(12)^1 12231513221(12)^{32^1 21} 12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[3436] $712231513221(12)^1 12231513221(12)^{222^1 1} 1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38, T_{\max} = 43, T \in [38, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[3437] $171231513221(12)^1 12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38, T_{\max} = 43, T \in [38, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[3438] $712231513221(12)^1 12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38, T_{\max} = 43, T \in [38, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[3439] $181232181232181231513221(12)^1 122^1 3 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[3440] $1^1 8^{32^1 1} 1232181232181231513221(12)^{21} 1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$

[3441] $1(12)^{21} 12231513218123218123218122^1 3 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[3442] $1(12)^{32^1 21} 1223151321812321812321812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$

[3443] $1^1 8^{32^1 1} 1232181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$

- [3444] $18^{21}1232181232181231513221(12)12^122 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -9$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 43$, $T \in [38, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3445] $32^12181232181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 46$, $\Delta = -113$, $\det = -9$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 46$, $T \in [38, 46]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [3446] $22181232181232181231513221(12)12^122 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -9$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 43$, $T \in [38, 43]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3447] $6131513221(12)1223151321812321812^13 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 51$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 51$, $T \in [38, 51]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [3458] $8123151321812321812321^1171 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -105$, $\det = -16$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.75 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3449] $171231513221(12)12231513221(12)12^1222 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.75 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [3450] $61231513221(12)12231513221(12)12^12313 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 37$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.75 \rightarrow 37$, $T_{\max} = 37$, $T \in [37, 37]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3459] $812321712321812321812321^115 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.75 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [3460] $812315132181232181232181232^11712 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -105$, $\det = -16$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.75 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3451] $712231513221(12)12231513221(12)12^1222 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.75 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [3452] $18^{21}12^132181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -16$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.75 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3453] $18^{21}12^132181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -16$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.75 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3454] \quad 1812^{21}32181232181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.75 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3455] \quad 1(12)122315132181232181232^{221}18122 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.75 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3456] \quad 21(12)122315132181232181232^{21}18122 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.75 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3457] \quad 221812^{21}32181232181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.75 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3502] \quad 61232181231513221(12)122315132^{21}1812 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3516] \quad (12)122315131613161316131612^{21}32 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.44 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[3503] \quad 612^{21}32181231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[3461] \quad 1(12)12231513221(12)122315132^{2221}171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3462] \quad 1(12)122^{2221}31513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3463] \quad 21(12)122^{221}31513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3464] \quad 21(12)12231513221(12)122315132^{221}171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[3465] $71231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[3466] $7122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[3467] $712231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[3468] $7122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[3469] $712231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[3470] $171231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[3471] $1712^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[3472] $61231513221(12)122315132^1 21(12)122313 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 37, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 37, T \in [37, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[3473] $612^1 31513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 37, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 37, T \in [37, 37]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[3474] $7122^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[3475] $712231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[3476] $181232181232181231513221(12)12^1 23 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 46$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37$, $T_{\max} = 46$, $T \in [37, 46]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[3477] $1812^1 32181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 46$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37$, $T_{\max} = 46$, $T \in [37, 46]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[3478] $1^{232} 8^1 1232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 48$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37$, $T_{\max} = 48$, $T \in [37, 48]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[3479] $181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3480] $181232^1 181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3481] $1812^1 32181231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$

[3482] $1(12)1223151321812321812321812^1 23 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 46$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37$, $T_{\max} = 46$, $T \in [37, 46]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[3483] $1(12)1223151321812321812321812 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 46$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37$, $T_{\max} = 46$, $T \in [37, 46]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3484] $1(12)122315132181232181232^1 1812 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 46$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37$, $T_{\max} = 46$, $T \in [37, 46]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3485] $1^{232} 8^1 1232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 48$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37$, $T_{\max} = 48$, $T \in [37, 48]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[3486] $181232^1 181232181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3487] $1(12)1223151321812321812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[3488] $21(12)1223151321812321812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[3489] $61231513221(12)122315132181232^1 181 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3490] $612^1 31513221(12)122315132181232^1 181 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$

[3491] $61232181231513221(12)122315132^1 181 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3492] $612^1 32181231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$

[3493] $16131513221(12)122315132181232^1 1812 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3494] $171232181231513221(12)122315132^1 171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3495] $1712^1 32181231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$

[3496] $322^1 181232181232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3497] $22181232^1 181232181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3498] $612^1 32181232181231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[3499] & 61232181232181231513221(12)122^1 3151 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -110, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3500] & 612^1 31513221(12)1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[3504] & 61232181231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3505] & 612^1 32181231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[3506] & 712231513221(12)122315132181232^1 171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3507] & 7122^1 31513221(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[3508] & 4132^1 21(12)12231513221(12)1223151316 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[3509] & 8123151321812321812321812^1 32171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -105, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[3510] & 4132^1 21(12)122315132181232181232171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[3511] & 413221(12)122315132181232181232^1 171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3512] & 812321712321812321812321812^1 3215 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -109, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[3513] & 812321712321812321812321812^1 3151 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]
\end{aligned}$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3514] $8123151321812321812321812^1 321712 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[3515] $812321812321812321812321812^1 314 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 44$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 36.44 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3501] $61231513221(12)122315132181232^1 1812 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -108$, $\det = -25$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 36.44 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3517] $1(12)12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3518] $21(12)12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3519] $712^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 48$, $\Delta = -115$, $\det = -36$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3520] $712^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3521] $712231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3522] $712^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[3523] $712231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3524] $171231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

- [3525] $61231513221(12)1223151322^11(12)122313 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 37, \Delta = -96, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 37, T \in [37, 37]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3526] $712^1231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [3527] $712231513221(12)1223151322^11(12)1222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3528] $18^112321812321812315132^121(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3529] $18^112321812321812315132^121(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3530] $18^112321812321812315132^121(12)1222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3531] $1(12)122^1131513218123218123218122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [3532] $21(12)122^1131513218123218123218122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [3533] $316131513221(12)122315132181232^1181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3534] $221812321812321812315132^121(12)1222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3535] $61232181231513221(12)1223151322^1181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [3536] $712^1231513221(12)122315132181232171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 36.22 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[3591] & 61232^1 181231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[3544] & 7122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[3537] & 21(12)12231513221(12)122^1 3151315 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[3538] & 21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[3539] & 1(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[3540] & 1(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3546] & 712231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3547] & 171231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[3548] & 1712315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3549] & 61231513221(12)122^1 31513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 37, \Delta = -96, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 37, T \in [37, 37] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3550] & 612315132^1 21(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 37, \Delta = -96, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 37, T \in [37, 37]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[3551] $7122315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[3552] $712231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[3553] $181232^1 181232181231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[3554] $1^3 8^{321} 12321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[3555] $1^3 8^{321} 12^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3556] $18^{21} 123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[3557] $18^{21} 12321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3558] $18^{21} 123218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[3559] $18^{21} 12321812^1 32181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3560] $181232^1 181231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[3561] $181232181231513221(12)122^1 31513161 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[3562] & \overset{21}{1(12)122^1 315132181232181232181223} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[3563] & \overset{21}{1(12)122315132181232181232^1 181223} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[3564] & \overset{3221}{1(12)1223151321812321812^1 321812} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[3565] & \overset{321}{1 \ 8 \ 12^1 32181232181231513221(12)122} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[3566] & \overset{321}{1 \ 8 \ 12321812321812315132^1 21(12)122} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[3567] & \overset{21}{1 \ 8 \ 123218123218123151322^1 1(12)1222} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[3568] & \overset{21}{1 \ 8 \ 12321812^1 32181231513221(12)1222} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[3569] & \overset{21}{1(12)122315132181232^1 18123218122} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[3570] & \overset{221}{1(12)12^1 231513218123218123218122} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0} \\
[3571] & \overset{21}{21(12)122315132181232^1 18123218122} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[3572] & \overset{21}{21(12)12^1 231513218123218123218122} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -49, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0} \\
[3573] & \overset{1}{612315132^1 21(12)122315132181232181} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -49, (n_+, n_-, n_0) =
\end{aligned}$$

(1, 31, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 36.06 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[3574] 61231513221(12)1223151321812¹32181 + $\hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -108$, $\det = -49$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 36.06 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[3575] 61232¹181231513221(12)122315132181 + $\hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -108$, $\det = -49$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 36.06 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[3576] 61232181231513221(12)122¹315132181 + $\hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -108$, $\det = -49$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 36.06 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3577] 16131513221(12)1223151321812¹321812 + $\hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 36.06 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[3578] 161315132¹21(12)1223151321812321812 + $\hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 36.06 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[3579] 171232181231513221(12)122¹315132171 + $\hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 36.06 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3580] 171232¹181231513221(12)122315132171 + $\hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 36.06 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[3581] 3221812¹32181232181231513221(12)122 + $\hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 46$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 36.06 \rightarrow 37$, $T_{\max} = 46$, $T \in [37, 46]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3582] 3221812321812321812315132¹21(12)122 + $\hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 46$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 36.06 \rightarrow 37$, $T_{\max} = 46$, $T \in [37, 46]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[3583] 221812321812¹32181231513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 36.06 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[3584] 2218123218123218123151322¹1(12)1222 + $\hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 36.06 \rightarrow 37$, $T_{\max} = 43$, $T \in [37, 43]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[3585] 61232181232181231513221(12)12¹23151 + $\hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) =$

$$(1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[3586] \ 61232^1 181232181231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -110, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3587] \ 612315132^1 21(12)1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[3588] \ 61231513221(12)1223151321812^1 321812 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[3589] \ 61232^1 181231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[3590] \ 61232181231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[3541] \ 21(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[3592] \ 61232181231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[3593] \ 712231513221(12)1223151321812^1 32171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[3594] \ 7122315132^1 21(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[3595] \ 41322^1 1(12)12231513221(12)1223151316 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3596] \ 812^1 315132181232181232171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[3597] $812315132181232181232^1 181232171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

[3598] $413221(12)1223151321812321812^1 32171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$

[3599] $41322^1 1(12)122315132181232181232171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

[3600] $81232171232181232181232^1 18123215 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3601] $812^1 3217123218123218123218123215 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3602] $81232171232181232181232^1 18123151 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3603] $812^1 3217123218123218123218123151 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3604] $812315132181232181232^1 1812321712 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

[3605] $812^1 3151321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$

[3606] $812^1 321812321812321812321812314 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 44, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[3607] $81232181232181232181232^1 1812314 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 44, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[3545] $7122315132^1 21(12)12231513221^{221}(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

- [3542] $21(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3543] $712231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.06 \rightarrow 37, T_{\max} = 43, T \in [37, 43]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [3614] $1(12)1223151321812321812^1 32181223 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3617] $1^8 12321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3618] $181232181232^1 181231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3612] $181232181232^1 181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3611] $181232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3610] $1^8 1232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3609] $1^8 123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3608] $1^8 12321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3619] $1(12)1223151321812^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [3620] $21(12)1223151321812^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -64, (n_+, n_-, n_0) =$

$$(1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[3621] \quad 32218123218123218123151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3622] \quad 322181232^1 181232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[3623] \quad 22181232181232^1 181231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[3624] \quad 8123151321812321812^1 32181232171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -105, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[3625] \quad 812321712321812321812^1 3218123215 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3626] \quad 812321712321812321812^1 3218123151 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3627] \quad 8123151321812321812^1 321812321712 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -105, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[3628] \quad 812321812321812321812^1 321812314 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 44, \Delta = -111, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 44, T \in [36, 44]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3629] \quad 1613161316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3630] \quad 1613161316131613161315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3631] \quad 1(12)122^1 3151316131613161316131612 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[3632] $21613161316131613161315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[3616] $1^3 8 1232^1 181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[3613] $1^3 1(12)12^1 2315132181232181232181223 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

[3615] $1^3 8 123218123218123151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.94 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[3633] $513221 21(12)122315132^1 21(12)1223151315 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[3634] $513221 21(12)12231513221(12)12^1 23151315 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[3635] $2221 1(12)12231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[3636] $2221 1(12)1223151322^1 1(12)122315132171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[3637] $221 21(12)12231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[3638] $221 21(12)1223151322^1 1(12)122315132171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[3639] $71231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[3640] $71223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -106, \det = -81, (n_+, n_-, n_0) =$

$$(1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3641] \quad 712231513221^1(12)12^1231513221^{221}(12)1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[3642] \quad 71223151322^11(12)12231513221^{221}(12)12 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3643] \quad 712231513221^1(12)12^1231513221^{221}(12)12 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[3644] \quad 1613151322^11(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 36, \Delta = -95, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.84 \rightarrow 36, T_{\max} = 36, T \in [36, 36]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[3645] \quad 16131513221^1(12)12^1231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 36, \Delta = -95, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.84 \rightarrow 36, T_{\max} = 36, T \in [36, 36]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3646] \quad 17123151322^11(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[3648] \quad 61231513221^1(12)12^1231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 37, \Delta = -96, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.84 \rightarrow 36, T_{\max} = 37, T \in [36, 37]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[3649] \quad 6123151322^11(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 37, \Delta = -96, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.84 \rightarrow 36, T_{\max} = 37, T \in [36, 37]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3650] \quad 71223151322^11(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3651] \quad 712231513221^1(12)12^1231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[3652] \quad 1^2321123218123218123151322^11(12)1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 48, T \in [36, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3653] & \overset{1}{1812321812315132^1 21(12)12231513161} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3654] & \overset{1}{181232181231513221(12)12^1 231513161} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[3655] & \overset{2321}{1 \ 8 \ 123218123218123151322^1 1(12)12} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -81, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 48, T \in [36, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3656] & \overset{1}{61231513221(12)122^1 315132181232181} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -81, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[3657] & \overset{1}{6123151322^1 1(12)122315132181232181} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -81, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3658] & \overset{1}{612321812315132^1 21(12)122315132181} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -81, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3659] & \overset{1}{61232181231513221(12)12^1 2315132181} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -108, \det = -81, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[3660] & \overset{1}{16131513221(12)122^1 3151321812321812} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[3661] & \overset{1}{1613151322^1 1(12)1223151321812321812} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3662] & \overset{1}{316131513221(12)1223151321812^1 32181} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -81, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 48, T \in [36, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[3663] & \overset{1}{171232181231513221(12)12^1 2315132171} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[3664] $1712321812315132^1 21(12)122315132171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3665] $6131513221(12)122315132181232^1 18123 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 51, \Delta = -116, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 51, T \in [36, 51]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[3666] $6123151322^1 1(12)1223151321812321812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3667] $61231513221(12)122^1 3151321812321812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[3668] $61232181231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[3669] $612321812315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3670] $61232181231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[3671] $612321812315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3672] $712231513221(12)122^1 315132181232171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[3673] $71223151322^1 1(12)122315132181232171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3674] $413221(12)122^1 31513221(12)1223151316 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[3675] $16131613161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

- [3676] $16131613161316131613151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -81$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36$, $T_{\max} = 43$, $T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3677] $1(12)12^1 23151316131613161316131612 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -81$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36$, $T_{\max} = 43$, $T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 152 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [3678] $1613161316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 45$, $\Delta = -112$, $\det = -81$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36$, $T_{\max} = 45$, $T \in [36, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3679] $321613161316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 46$, $\Delta = -113$, $\det = -81$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36$, $T_{\max} = 46$, $T \in [36, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3680] $216131613161316131613151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -81$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36$, $T_{\max} = 43$, $T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3647] $171231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -106$, $\det = -81$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.84 \rightarrow 36$, $T_{\max} = 43$, $T \in [36, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [3687] $1812321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36$, $T_{\max} = 43$, $T \in [36, 43]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3700] $81232171232181232^1 18123218123151 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 43$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36$, $T_{\max} = 43$, $T \in [36, 43]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3681] $21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 46$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36$, $T_{\max} = 46$, $T \in [36, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3682] $712315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 48$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36$, $T_{\max} = 48$, $T \in [36, 48]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3683] $316131513221(12)122^1 31513221(12)12231 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 36$, $\Delta = -95$, $\det = -100$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36$, $T_{\max} = 36$, $T \in [36, 36]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3684] $1812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31$, $T_{\max} = 48$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 31, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 48, T \in [36, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3702] & \overset{1}{8}1232181232181232^1 1812321812314 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 44, \Delta = -111, \det = -100, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 44, T \in [36, 44] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3703] & \overset{31}{1}613161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 45, \Delta = -112, \det = -100, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 45, T \in [36, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[3704] & 3216131613161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -100, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[3705] & (12)122^1 315131613161316131613161232 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -100, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 48, T \in [36, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3686] & \overset{21}{1}812321812321812^1 31513221 \overset{221}{(12)}1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -100, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3685] & \overset{321}{1}812321812^1 32181231513221 \overset{21}{(12)}1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -100, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3688] & \overset{21}{1}(12)122315132181232^1 181232181223 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -100, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3689] & \overset{321}{1}812321812^1 32181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -100, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3690] & \overset{2321}{1}812^1 32181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -100, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 48, T \in [36, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3691] & \overset{21}{1}812321812321812^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -100, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3692] & \overset{221}{1}(12)122315132^1 18123218123218122 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -104, \det = -100, (n_+, n_-, n_0) = \\
& (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.76 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[3693] $21^{21}(12)122315132^1 18123218123218122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -100, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[3694] $3221812321812^1 32181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3695] $221812321812321812^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -104, \det = -100, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3696] $61315132^1 21(12)12231513218123218123 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 51, \Delta = -116, \det = -100, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36, T_{\max} = 51, T \in [36, 51]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[3697] $413221(12)12231513221(12)122^1 3151316 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -100, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[3698] $812315132181232^1 181232181232171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[3699] $81232171232181232^1 18123218123215 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -109, \det = -100, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[3701] $812315132181232^1 1812321812321712 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.76 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[3739] $612321812^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[3750] $81232^1 17123218123218123218123151 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[3728] $3161315132^1 21(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[3725] $6123218123151322^1 1(12)122315132181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

- [3724] $612321812^1 31513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [3723] $61231513221(12)12^1 2315132181232181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [3722] $61231513221(12)122315132^1 181232181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [3721] $1^2 8 1232^1 181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [3720] $1^2 8 1232181232^1 181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [3719] $1(12)122^1 3151321812321812321812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [3718] $1(12)122315132181232^1 1812321812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [3706] $21(12)1223151322^1 1(12)1223151315 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3717] $1(12)1223151321812^1 32181232181223 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3716] $1812321812^1 31513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [3715] $18123218123151322^1 1(12)12231513161 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [3714] $1^2 8 1232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) =$

$$(1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[3713] \quad 1 \quad 8 \quad 1232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[3712] \quad 1812321812^1 32181231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[3711] \quad 1812321812321812315132^1 21(12)1223 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3710] \quad 3161315132^1 21(12)12231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 36, \Delta = -95, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 36, T \in [36, 36]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[3727] \quad 16131513221(12)122315132^1 1812321812 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[3709] \quad 71231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[3729] \quad 1712321812^1 31513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[3741] \quad 712231513221(12)12^1 2315132181232171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[3730] \quad 17123218123151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[3731] \quad 322181232181232^1 181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[3732] \quad 612321812321812315132^1 21(12)1223151 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

- [3733] $612321812^1 32181231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [3734] $61231513221(12)12^1 23151321812321812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [3735] $61231513221(12)122315132^1 1812321812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [3736] $6123218123151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [3737] $612321812^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [3738] $6123218123151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [3740] $712231513221(12)122315132^1 181232171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [3708] $71231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [3707] $21(12)122^1 31513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [3742] $413221(12)12^1 231513221(12)1223151316 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [3743] $413221(12)122315132^1 21(12)1223151316 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [3744] $8123151321812^1 32181232181232171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[3745] & 413221(12)122^1 315132181232181232171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3746] & 413221(12)122315132181232^1 181232171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.69 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[3747] & 812321712321812^1 3218123218123215 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -109, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3748] & 81232^1 17123218123218123218123215 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -109, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[3749] & 812321712321812^1 3218123218123151 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3754] & (12)12^1 231513161316131613161232 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.69 \rightarrow 36, T_{\max} = 48, T \in [36, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[3753] & 812321812321812^1 321812321812314 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 44, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.69 \rightarrow 36, T_{\max} = 44, T \in [36, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3752] & 81232^1 1812321812321812321812314 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 44, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.69 \rightarrow 36, T_{\max} = 44, T \in [36, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[3751] & 8123151321812^1 321812321812321712 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[3726] & 16131513221(12)12^1 23151321812321812 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.69 \rightarrow 36, T_{\max} = 43, T \in [36, 43] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3755] & 6131513221(12)1223151321812^1 3218123 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 51, \Delta = -116, \det = -144, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.64 \rightarrow 36, T_{\max} = 51, T \in [36, 51] \\
& \begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[3775] & 413221(12)12^1 2315132181232181232171 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 46, T \in [36, 46]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[3779] & \overset{1}{81232181232^1 1812321812321812314} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 44, \Delta = -111, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 44, T \in [36, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3758] & \overset{1}{181232181232^1 181231513221(12)1223} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[3759] & \overset{1}{18123218123218123151322^1 1(12)1223} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[3760] & \overset{2321}{1 \ 8 \ 12321812^1 32181231513221(12)1} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 48, T \in [36, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[3761] & \overset{321}{1 \ 8 \ 12321812321812^1 31513221(12)1} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[3762] & \overset{21}{1(12)122315132^1 181232181232181223} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[3763] & \overset{3221}{1(12)1223151321812^1 321812321812} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[3764] & \overset{3221}{1(12)12^1 23151321812321812321812} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[3765] & \overset{321}{1 \ 8 \ 12321812321812^1 31513221(12)122} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 46, T \in [36, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[3766] & \overset{2321}{1 \ 8 \ 12321812^1 32181231513221(12)12} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 48, T \in [36, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[3767] & \overset{2321}{316131513221(12)122^1 315132181232181} + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.59 \rightarrow 36, T_{\max} = 48, T \in [36, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[3768] $3221812321812321812^1 31513221(12)122 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[3769] $61232181232^1 181231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -110, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

[3770] $6123218123218123151322^1 1(12)1223151 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -110, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[3771] $613151322^1 1(12)12231513218123218123 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 51, \Delta = -116, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 51, T \in [36, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[3772] $413221(12)12231513221(12)12^1 23151316 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[3773] $812315132^1 181232181232181232171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$

[3774] $413221(12)1223151321812^1 32181232171 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 46, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$

[3756] $712315132^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 48, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[3776] $81232171232^1 18123218123218123215 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -109, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3777] $81232171232^1 18123218123218123151 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[3778] $812315132^1 1812321812321812321712 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 43, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$

[3757] $316131513221(12)12^1 231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 31, T_{\max} = 36, \Delta = -95, \det = -169, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.59 \rightarrow 36, T_{\max} = 36, T \in [36, 36]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

- [3780] $^{513221} 21 (12) 12^1 231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 46, \Delta = -113, \det = -196, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.55 \rightarrow 36, T_{\max} = 46, T \in [36, 46]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3781] $^{3221} 71231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -196, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.55 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3782] $^1 31613151322^1 1(12)12231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 36, \Delta = -95, \det = -196, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.55 \rightarrow 36, T_{\max} = 36, T \in [36, 36]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3783] $^{2321} 1 \ 8 \ 1232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -196, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.55 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [3784] $^{2321} 1 \ 8 \ 1232181232^1 181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -196, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.55 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [3785] $^{2321} 31613151322^1 1(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -196, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.55 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [3786] $6131513221(12)122^1 31513218123218123 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 51, \Delta = -116, \det = -196, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.55 \rightarrow 36, T_{\max} = 51, T \in [36, 51]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3787] $^1 413221(12)1223151322^1 1(12)1223151316 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -196, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.55 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3788] $812321712^1 3218123218123218123215 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -109, \det = -196, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.55 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [3789] $812321712^1 3218123218123218123151 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 43, \Delta = -107, \det = -196, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.55 \rightarrow 36, T_{\max} = 43, T \in [36, 43]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [3790] $^1 812321812^1 321812321812321812314 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 44, \Delta = -111, \det = -196, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.55 \rightarrow 36, T_{\max} = 44, T \in [36, 44]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [3791] $316131513221(12)122315132^1 181232181 + \hat{1} \rightarrow 2 \quad T = 31, T_{\max} = 48, \Delta = -115, \det = -225, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 35.52 \rightarrow 36, T_{\max} = 48, T \in [36, 48]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[3792] & \begin{matrix} 2321 \\ 1 \end{matrix} 8 \ 12321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 31, \ T_{\max} = 48, \ \Delta = -115, \ \det = -256, \ (n_+, n_-, n_0) = \\
& (1, 31, 0), \ c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.48 \rightarrow 36, \ T_{\max} = 48, \ T \in [36, 48] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[3793] & \begin{matrix} 2321 \\ 1 \end{matrix} 8 \ 12321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 31, \ T_{\max} = 48, \ \Delta = -115, \ \det = -256, \ (n_+, n_-, n_0) = \\
& (1, 31, 0), \ c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.48 \rightarrow 36, \ T_{\max} = 48, \ T \in [36, 48] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[3794] & 316131513221(12)12^1 2315132181232181 + \hat{1} \rightarrow 2 \quad T = 31, \ T_{\max} = 48, \ \Delta = -115, \ \det = -256, \ (n_+, n_-, n_0) = \\
& (1, 31, 0), \ c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.48 \rightarrow 36, \ T_{\max} = 48, \ T \in [36, 48] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[3798] & 6131513221(12)12^1 231513218123218123 + \hat{1} \rightarrow 2 \quad T = 31, \ T_{\max} = 51, \ \Delta = -116, \ \det = -289, \ (n_+, n_-, n_0) = \\
& (1, 31, 0), \ c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.46 \rightarrow 36, \ T_{\max} = 51, \ T \in [36, 51] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[3795] & \begin{matrix} 1 \\ 18 \end{matrix} 12321812321812^1 31513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 31, \ T_{\max} = 46, \ \Delta = -113, \ \det = -289, \ (n_+, n_-, n_0) = \\
& (1, 31, 0), \ c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.46 \rightarrow 36, \ T_{\max} = 46, \ T \in [36, 46] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3797] & 612321812321812^1 31513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 31, \ T_{\max} = 43, \ \Delta = -110, \ \det = -289, \ (n_+, n_-, n_0) = \\
& (1, 31, 0), \ c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.46 \rightarrow 36, \ T_{\max} = 43, \ T \in [36, 43] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[3796] & \begin{matrix} 3221 \\ 1 \end{matrix} (12)122315132^1 1812321812321812 + \hat{1} \rightarrow 2 \quad T = 31, \ T_{\max} = 46, \ \Delta = -113, \ \det = -289, \ (n_+, n_-, n_0) = \\
& (1, 31, 0), \ c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.46 \rightarrow 36, \ T_{\max} = 46, \ T \in [36, 46] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 407 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[3799] & 413221(12)122315132^1 181232181232171 + \hat{1} \rightarrow 2 \quad T = 31, \ T_{\max} = 46, \ \Delta = -113, \ \det = -289, \ (n_+, n_-, n_0) = \\
& (1, 31, 0), \ c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.46 \rightarrow 36, \ T_{\max} = 46, \ T \in [36, 46] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 407 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[3800] & 6131513221(12)122315132^1 18123218123 + \hat{1} \rightarrow 2 \quad T = 31, \ T_{\max} = 51, \ \Delta = -116, \ \det = -324, \ (n_+, n_-, n_0) = \\
& (1, 31, 0), \ c_{\text{ext}} = 15.0 \\
& T_{\min} = 35.43 \rightarrow 36, \ T_{\max} = 51, \ T \in [36, 51] \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
\\
T = 32 & [4] \begin{matrix} 1 \\ 18 \end{matrix} 1231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 32, \ T_{\max} = 44, \ \Delta = -105, \ \det = 4, \ (n_+, n_-, n_0) = \\
& (1, 32, 0), \ c_{\text{ext}} = 15.0 \\
& T_{\min} = 39, \ T_{\max} = 44, \ T \in [39, 44] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[0] & \begin{matrix} 22^1 21 \\ 1 \end{matrix} (12)12231513221(12)12231513161315 + \hat{1} \rightarrow 2 \quad T = 32, \ T_{\max} = 44, \ \Delta = -109, \ \det = 4, \ (n_+, n_-, n_0) = \\
& (1, 32, 0), \ c_{\text{ext}} = 15.0 \\
& T_{\min} = 39, \ T_{\max} = 44, \ T \in [39, 44] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1] $^{22^1 1} 21(12)12231513221(12)12231513161315 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [2] $^1 181231513221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [3] $^1 181231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [5] $^{2^1 1} 3221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [6] $^{22^1 1} 1(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 48, T \in [39, 48]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [7] $^{22^1 21} 1(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [8] $^{22^1 1} 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [9] $^{22^1 21} 1(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [10] $^{22^1 1} 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [11] $^1 812231513221(12)12231513221(12)12^1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [12] $^1 51316131513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

- [13] $32181231513221(12)12231513221(12)122^12 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 48, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 48, T \in [39, 48]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [14] $232181231513221(12)12231513221(12)12^12 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 52, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 52, T \in [39, 52]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [15] $2181231513221(12)12231513221(12)1222^12 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [16] $1812231513221(12)12231513221(12)1222^12 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [17] $4132181232181232181231513221(12)12^12 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 46, \Delta = -112, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [18] $8123151321812321812321812321812^12 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [19] $513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.33 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [31] $32181231513221(12)12231513221(12)12^122 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 48, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [21] $61231513221(12)12231513221(12)12^12 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 45, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.33 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [22] $181231513221(12)12231513221(12)12^122 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [23] $181231513221(12)12231513221(12)122^122 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [24] $21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 48, \Delta = -114, \det = 9, (n_+, n_-, n_0) =$

$(1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 48$, $T \in [39, 48]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[25] $1(12)12231513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 48$, $T \in [39, 48]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[26] $1(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 9$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 44$, $T \in [39, 44]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[27] $21(12)12231513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 48$, $T \in [39, 48]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[28] $1(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 9$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 44$, $T \in [39, 44]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[29] $812231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 48$, $T \in [39, 48]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[30] $51316131513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -109$, $\det = 9$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 44$, $T \in [39, 44]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[36] $71232181231513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 49$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 49$, $T \in [39, 49]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[35] $71231513221(12)1223151321812321812^1 3 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 49$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 49$, $T \in [39, 49]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[34] $1812231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 9$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 44$, $T \in [39, 44]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[33] $2181231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 9$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 44$, $T \in [39, 44]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[32] $32^1 181231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.22 \rightarrow 39$, $T_{\max} = 48$, $T \in [39, 48]$

- $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
 [20] $1^{222^1 1}(12)12231513221(12)12231513161315 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 44, T \in [39, 44]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
 [37] $1^{181}1231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.75 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
 [38] $51316131513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.75 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
 [39] $2181231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.75 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
 [40] $1812231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.75 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
 [41] $812315132181232181232181232^1 181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -104, \det = 16, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.75 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [42] $812315132181232181232181232^1 18122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -104, \det = 16, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.75 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [43] $81232181232181232181232181232^1 15 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.75 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
 [44] $81232171232181232181232181232^1 161 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.75 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
 [49] $1^{181}12^1 31513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
 [76] $7122^1 31513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$

[51] $1812^1 31513221(12)^1 12231513221(12)^{221} 12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$

[50] $181231513221(12)^1 122315132^1 21(12)^{221} 12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[53] $181231513221(12)^1 122315132^1 21(12) 12222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[54] $21(12)^{322^1} 12231513221(12)^1 122315132181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[55] $1(12)^{21} 12231513221(12)^1 1223151321812^1 32 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 52, T \in [38, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

[56] $1(12)^{2221} 12231513221(12)^1 122315132^1 1812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[57] $1(12)^{2221} 122^1 31513221(12)^1 1223151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[58] $21(12)^{221} 122^1 31513221(12)^1 1223151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[59] $21(12)^{221} 12231513221(12)^1 122315132^1 1812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[60] $1(12)^{2221} 122^1 31513221(12)^1 1223151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[61] $1(12)^{2221} 12231513221(12)^1 122315132^1 2181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

- [62] $21(12)122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [63] $21(12)12231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [64] $6131513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -110$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [65] $812231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38$, $T_{\max} = 48$, $T \in [38, 48]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [66] $51316131513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [48] $181231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [47] $612^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 45$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [46] $21(12)122^1 31513221(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [45] $1(12)122^1 31513221(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [67] $232^1 181231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 52$, $\Delta = -116$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38$, $T_{\max} = 52$, $T \in [38, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [68] $2181231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.44 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [69] $21812^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$(1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[70] \quad 18122^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[71] \quad 1812231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[72] \quad 181232181231513221(12)122315132^1 171 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[73] \quad 1812^1 32181231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[74] \quad 181231513221(12)122315132181232^1 171 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[75] \quad 1812^1 31513221(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[99] \quad 812321712321812321812321812^1 32161 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[98] \quad 812321812321812321812321812^1 3151 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[97] \quad 812321812321812321812321812^1 3215 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[96] \quad 8123151321812321812321812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[95] \quad 413221(12)122315132181232181232^1 1812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[94] $4132^1 21(12)1223151321812321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[93] $4132^1 1812321812321812321513221(12)122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[92] $8123151321812321812321812^1 32181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[91] $4132^1 21(12)122315132181232181232181^1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[90] $413221(12)122315132181232181232^1 181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[89] $4132^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[88] $4132^1 21(12)12231513221(12)12231513217 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[87] $712231513221(12)122315132181232^1 1812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[86] $7122^1 31513221(12)1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[85] $171232181231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[84] $1712^1 32181231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[83] $171231513221(12)122315132181232^1 1812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[82] & 1712^1 31513221(12)1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[81] & 1712^1 32181231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[80] & 171232181231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[79] & 171232181232181231513221(12)122^1 3151 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 25, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[78] & 1712^1 32181232181231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 25, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[77] & 712231513221(12)122315132181232^1 181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 25, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[52] & 1812^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 25, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.44 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[100] & 1(12)12^1 231513221(12)12231513161315 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -109, \det = 36, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.22 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[101] & 21(12)12^1 231513221(12)12231513161315 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -109, \det = 36, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.22 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[102] & 181231513221(12)1223151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 36, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.22 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[103] & 181231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 36, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.22 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [104] $181231513221(12)1223151322^11(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [105] $21(12)12231513221(12)122315132^1181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 48$, $T \in [38, 48]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [106] $1(12)122^131513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 48$, $T \in [38, 48]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [107] $1(12)12^1231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [108] $21(12)122^131513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 48$, $T \in [38, 48]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [109] $21(12)12^1231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [110] $1(12)12^1231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [111] $1(12)12231513221(12)1223151322^1181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [112] $21(12)12231513221(12)1223151322^1181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [113] $21(12)12^1231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [114] $8122^131513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 36$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.22 \rightarrow 38$, $T_{\max} = 48$, $T \in [38, 48]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [115] $51316131513221(12)1223151322^11(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -109$, $\det = 36$, $(n_+, n_-, n_0) =$

$$(1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[116] \quad 32181231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[117] \quad 2181231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[118] \quad 1812^1 231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[119] \quad 1812231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[120] \quad 712^1 231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 36, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[121] \quad 171232181231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 36, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[122] \quad 712^1 32181232181231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 36, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 52, T \in [38, 52]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[123] \quad 712^1 31513221(12)12231513218123218123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 49, \Delta = -115, \det = 36, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[124] \quad 712^1 32181231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 49, \Delta = -115, \det = 36, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[125] \quad 712^1 231513221(12)1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 36, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[126] \quad 413221(12)12231513221(12)122315132^1 17 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.22 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

- [170] $1712315132^1 21(12)1223151321812321812 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [140] $1(12)122^1 31513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 52$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 52$, $T \in [38, 52]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [145] $21(12)12231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [147] $1(12)12231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [128] $21(12)12231513221(12)122^1 31513161 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [129] $1(12)122315132^1 21(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [130] $21(12)122315132^1 21(12)12231513161315 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [131] $61231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 45$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [132] $612315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 45$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [133] $1812315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [134] $181231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [135] $1812315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) =$

$(1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[136] $181231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[137] $1812315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[138] $181231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[139] $1(12)12^1 231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 48$, $T \in [38, 48]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[194] $81232171232181232181232^1 181232161 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[193] $812^1 32171232181232181232181232161 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[192] $6131513221(12)12231513221(12)122^1 31414 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 48$, $T \in [38, 48]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[191] $812^1 3218123218123218123218123151 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[190] $81232181232181232181232^1 18123151 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[189] $812^1 3218123218123218123218123215 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[188] $81232181232181232181232^1 18123215 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 37.06 \rightarrow 38$, $T_{\max} = 44$, $T \in [38, 44]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[187] 812315132181232181232¹18123218122 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -104$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[186] 812¹31513218123218123218123218122 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -104$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[185] 41322¹1(12)1223151321812321812321812 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 46$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 37.06 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[184] 413221(12)1223151321812321812¹321812 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 46$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 37.06 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[183] 41321812321812321812315132¹21(12)122 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 46$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 37.06 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[182] 41321812¹32181232181231513221(12)122 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 46$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 37.06 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[181] 812315132181232181232¹18123218²¹1 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -104$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[180] 812¹31513218123218123218123218²¹1 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -104$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[179] 41322¹1(12)12231513218123218123218¹1 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 46$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 37.06 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[178] 413221(12)1223151321812321812¹3218¹1 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 46$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 37.06 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[177] 41321812321812321812315132²¹21(12)1 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 46$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 37.06 \rightarrow 38, T_{\max} = 46, T \in [38, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[176] 41321812¹32181232181231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 46$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 46, T \in [38, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[175] & 41322^1 1(12)12231513221(12)12231513217 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[174] & 7122315132^1 21(12)1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[173] & 712231513221(12)1223151321812^1 321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[172] & 171232181231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[171] & 171232^1 181231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[169] & 171231513221(12)1223151321812^1 321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[168] & 171232181231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[167] & 171232^1 181231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[166] & 171232181232181231513221(12)12^1 23151 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[165] & 171232^1 181232181231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[164] & 7122315132^1 21(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[163] & 712231513221(12)1223151321812^1 32181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[162] & \overset{1}{181231513221}(12)1223151321812^1 32171 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[161] & \overset{1}{1812315132^1 21}(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[160] & \overset{1}{181232181231513221}(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[159] & \overset{1}{181232^1 181231513221}(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[158] & \overset{1}{18122315132^1 21}(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[157] & \overset{1}{1812231513221}(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[156] & \overset{1}{21812315132^1 21}(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[155] & \overset{1}{2181231513221}(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[154] & \overset{1}{232181231513221}(12)122315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 52, T \in [38, 52] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[153] & \overset{1}{32181231513221}(12)1223151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[152] & \overset{1}{51316131513221}(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -109, \det = 49, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[151] & 812^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[150] & 6131513221(12)12231513221(12)151322^1 1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[149] & 21(12)122315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[148] & 21(12)12231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[146] & 1(12)122315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[144] & 21(12)122315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[143] & 21(12)12^1 231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[142] & 1(12)12231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[141] & 1(12)122315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[127] & 21(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.06 \rightarrow 38, T_{\max} = 44, T \in [38, 44] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[195] & 1(12)12^1 231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.94 \rightarrow 37, T_{\max} = 52, T \in [37, 52]
\end{aligned}$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[196] $232181231513221 \overset{1}{(12)}1223151322^11(12)122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.94 \rightarrow 37, T_{\max} = 52, T \in [37, 52]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[197] $71232181232181231513221(12)122^131513 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.94 \rightarrow 37, T_{\max} = 52, T \in [37, 52]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[198] $413218123218123218123151322^11(12)1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.94 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[199] $4132181232^1181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.94 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[200] $8123151321812321812^132181232181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -104, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.94 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[201] $4132181232^1181232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.94 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[202] $413218123218123218123151322^11(12)122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.94 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[203] $8123151321812321812^13218123218122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -104, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.94 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[204] $812321812321812321812^13218123215 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.94 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[205] $812321812321812321812^13218123151 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.94 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[206] $812321712321812321812^132181232161 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 64, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 36.94 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[212] $21(12)1223151322^11(12)12231513161315 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -109, \det = 81, (n_+, n_-, n_0) =$

$$(1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[228] \quad {}^{221}_1 21(12)12231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[208] \quad {}^{513221}_1 21(12)122315132^1 21(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[209] \quad {}^{513221}_1 21(12)12231513221(12)12^1 231513161 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[211] \quad {}^{2221}_1 1(12)1223151322^1 1(12)12231513161315 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[213] \quad {}^{221}_1 21(12)12231513221(12)122^1 31513161315 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[214] \quad {}^{513221}_1 61231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 45, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 45, T \in [37, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[215] \quad {}^{513221}_1 6123151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 45, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 45, T \in [37, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[216] \quad {}^1_{18} 1231513221(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[217] \quad {}^1_{18} 123151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[218] \quad {}^1_{18} 1231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[219] \quad {}^1_{18} 123151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[220] & \begin{matrix} 1 & 1 \\ 18123151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[221] & \begin{matrix} 1 & 1 \\ 181231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[222] & \begin{matrix} 3221 & 1 \\ 21(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 48, \Delta = -114, \det = 81, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[210] & \begin{matrix} 2221 & 1 \\ 1(12)12231513221(12)122^1 31513161315 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 44, \Delta = -109, \det = 81, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[223] & \begin{matrix} 221 & 1 \\ 1(12)12231513221(12)122315132^1 18123 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 48, \Delta = -114, \det = 81, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[224] & \begin{matrix} 2221 & 1 \\ 1(12)1223151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[225] & \begin{matrix} 2221 & 1 \\ 1(12)12231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[226] & \begin{matrix} 21 & 1 \\ 21(12)12231513221(12)122315132^1 18123 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 48, \Delta = -114, \det = 81, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[227] & \begin{matrix} 221 & 1 \\ 21(12)1223151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[266] & \begin{matrix} 1 \\ 6131513221(12)122315132^1 21(12)12231414 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 48, \Delta = -114, \det = 81, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[265] & \begin{matrix} 1 \\ 6131513221(12)12231513221(12)12^1 231414 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 48, \Delta = -114, \det = 81, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 48, T \in [37, 48]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[264] & 413221(12)122^1 31513221(12)12231513217 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[263] & 71223151322^1 1(12)1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[262] & 712231513221(12)122^1 3151321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[261] & 71232181231513221(12)122315132^1 18123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 49, \Delta = -115, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 49, T \in [37, 49] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[260] & 71232^1 181231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 49, \Delta = -115, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 49, T \in [37, 49] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[259] & 71231513221(12)122315132181232^1 18123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 49, \Delta = -115, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 49, T \in [37, 49] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[258] & 71232^1 181232181231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 52, T \in [37, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[257] & 171232181231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[256] & 1712321812315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[255] & 17123151322^1 1(12)1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[254] & 171231513221(12)122^1 3151321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[253] & 171232181231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]
\end{aligned}$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[252] $1712321812315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[251] $712231513221(12)122^1 315132181232181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[250] $71223151322^1 1(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[249] $18123151322^1 1(12)122315132181232171 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[248] $181231513221(12)122^1 315132181232171 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[247] $181232181231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[246] $1812321812315132^1 21(12)122315132171 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[245] $71223151322^1 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 37$, $\Delta = -94$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[244] $712231513221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 37$, $\Delta = -94$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 37, T \in [37, 37]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[243] $1812231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[242] $181223151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

- [241] $218123151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [240] $2181231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [239] $321812^1 31513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37$, $T_{\max} = 48$, $T \in [37, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [238] $171231513221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 37$, $\Delta = -94$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37$, $T_{\max} = 37$, $T \in [37, 37]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [237] $17123151322^1 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 37$, $\Delta = -94$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37$, $T_{\max} = 37$, $T \in [37, 37]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [236] $51316131513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -109$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [235] $513161315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -109$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [234] $812231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37$, $T_{\max} = 48$, $T \in [37, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [233] $6131513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [232] $21(12)1223151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [231] $21(12)12231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [230] $1(12)1223151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) =$

$(1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 2221 & 1 \\ 1(12)12231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 32, T_{\max} = 44, \Delta = -105, \det = 81, (n_+, n_-, n_0) =$
 $(1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 $[207] \quad 4131613161316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 45, \Delta = -111, \det = 81, (n_+, n_-, n_0) =$
 $(1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.84 \rightarrow 37, T_{\max} = 45, T \in [37, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 $[267] \quad 41316131613161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 45, \Delta = -111, \det = 100, (n_+, n_-, n_0) =$
 $(1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.76 \rightarrow 37, T_{\max} = 45, T \in [37, 45]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 51322^1 1 \\ 21(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 32, T_{\max} = 44, \Delta = -110, \det = 100, (n_+, n_-, n_0) =$
 $(1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.76 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 51322^1 1 \\ 61231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 32, T_{\max} = 45, \Delta = -111, \det = 100, (n_+, n_-, n_0) =$
 $(1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.76 \rightarrow 37, T_{\max} = 45, T \in [37, 45]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 3221 & 1 \\ 21(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 100, (n_+, n_-, n_0) =$
 $(1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.76 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 221 & 1 \\ 1(12)122315132^1 21(12)12231513218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 100, (n_+, n_-, n_0) =$
 $(1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.76 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 21 & 1 \\ 21(12)122315132^1 21(12)12231513218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 100, (n_+, n_-, n_0) =$
 $(1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.76 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 $[273] \quad 61315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -110, \det = 100, (n_+, n_-, n_0) =$
 $(1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.76 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 $[274] \quad 8122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 100, (n_+, n_-, n_0) =$
 $(1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.76 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 $[275] \quad 32181231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 100, (n_+, n_-, n_0) =$
 $(1, 32, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 36.76 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[276] & 712315132^1 21(12)12231513218123218123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 49, \Delta = -115, \det = 100, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.76 \rightarrow 37, T_{\max} = 49, T \in [37, 49] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[277] & 413221(12)12231513221(12)122^1 31513217 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 100, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.76 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[278] & 41321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 100, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.76 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[279] & 812315132181232^1 181232181232181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -104, \det = 100, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.76 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[280] & 41321812321812^1 32181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 100, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.76 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[281] & 812315132181232^1 18123218123218122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -104, \det = 100, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.76 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[282] & 81232181232181232^1 18123218123215 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 100, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.76 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[283] & 81232181232181232^1 18123218123151 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 100, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.76 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[284] & 61315132^1 21(12)12231513221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 100, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.76 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[285] & 81232171232181232^1 181232181232161 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 100, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.76 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[291] & 61231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 45, \Delta = -111, \det = 121, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 45, T \in [37, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[294] $1(12)12231513221(12)122^1 31513218123 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 48, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[336] $812321712321812^1 32181232181232161 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[337] $81232^1 171232181232181232181232161 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 219 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$

[288] $1(12)12231513221(12)12^1 231513161315 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

[287] $21(12)1223151322^1 1(12)12231513161 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[286] $21(12)122^1 31513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[332] $81232^1 18123218123218123218123151 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[331] $81232^1 18123218123218123218123215 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[330] $812321812321812^1 3218123218123215 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[329] $8123151321812^1 3218123218123218122 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[328] $413221(12)122315132181232^1 1812321812 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 46, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[327] $413221(12)122^1 3151321812321812321812 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 46, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.69 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[326] & 4132181232181232^1 181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[325] & 8123151321812^1 321812321812321^1 81 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[324] & 413221(12)122^1 3151321812321812321^1 81 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[323] & 413221(12)122315132181232^1 181232181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[322] & 4132181232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[321] & 413221(12)12^1 231513221(12)12231513217 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[320] & 413221(12)122315132^1 21(12)12231513217 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[319] & 712231513221(12)12^1 23151321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[318] & 712231513221(12)122315132^1 1812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[317] & 71232181231513221(12)122^1 31513218123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 49, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 49, T \in [37, 49] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[316] & 71232181232181231513221(12)12^1 231513 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 52, T \in [37, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[315] & 1712321812^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[314] \quad 17123218123151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[290] \quad 61231513221(12)12^{513221} 231513221 (12) 12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 45, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 45, T \in [37, 45]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[333] \quad 812321812321812^1 3218123218123151 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[289] \quad 21^{221}(12)12231513221(12)12^1 231513161315 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[313] \quad 171231513221(12)122315132^1 1812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[312] \quad 171231513221(12)12^1 23151321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[311] \quad 17123218123151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[310] \quad 1712321812^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[309] \quad 1712321812321812315132^1 21(12)1223151 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[308] \quad 1712321812^1 32181231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[307] \quad 712231513221(12)122315132^1 181232181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[306] \quad 712231513221(12)12^1 2315132181232181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) =$$

$$(1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[305] \quad \overset{1}{181231513221}(12)122315132^1 181232171 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[304] \quad \overset{1}{181231513221}(12)12^1 2315132181232171 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[303] \quad \overset{1}{18123218123151322^1} 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[302] \quad \overset{1}{1812321812^1} 31513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[301] \quad \overset{1}{321812315132^1} 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[300] \quad \overset{1}{5131613151322^1} 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[299] \quad \overset{1}{812231513221}(12)1223151322^1 \overset{3221}{1}(12)12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[298] \quad \overset{1}{812231513221}(12)122^1 31513221 \overset{3221}{(12)12} + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[297] \quad \overset{1}{6131513221}(12)1223151322^1 \overset{1513221}{1}(12)1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[296] \quad \overset{1}{6131513221}(12)122^1 31513221 \overset{1513221}{(12)1} + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 36.69 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[295] \quad \overset{21}{21}(12)12231513221(12)122^1 31513218123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[334] & 6131513221(12)122^1 31513221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 121, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[292] & 21(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 121, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[293] & 21(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 121, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[335] & 6131513221(12)1223151322^1 1(12)12231414 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 121, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.69 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[338] & 71231513221(12)1223151321812^1 3218123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 49, \Delta = -115, \det = 144, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.64 \rightarrow 37, T_{\max} = 49, T \in [37, 49] \\
& \begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[361] & 812315132^1 18123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -104, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[349] & 17123218123218123151322^1 1(12)1223151 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[359] & 413221(12)1223151321812^1 321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[358] & 41321812321812321812^1 31513221(12)122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[357] & 812315132^1 1812321812321812321 81 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -104, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[356] & 413221(12)12^1 2315132181232181232181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [355] $413221(12)1223151321812^1 32181232181 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 46$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.59 \rightarrow 37$, $T_{\max} = 46$, $T \in [37, 46]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [354] $41321812321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 46$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.59 \rightarrow 37$, $T_{\max} = 46$, $T \in [37, 46]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [353] $413221(12)12231513221(12)12^1 231513217 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.59 \rightarrow 37$, $T_{\max} = 48$, $T \in [37, 48]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [363] $81232181232^1 18123218123218123151 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.59 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [352] $712321812315132^1 21(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 49$, $\Delta = -115$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.59 \rightarrow 37$, $T_{\max} = 49$, $T \in [37, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [351] $7123151322^1 1(12)12231513218123218123 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 49$, $\Delta = -115$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.59 \rightarrow 37$, $T_{\max} = 49$, $T \in [37, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [350] $171232181232^1 181231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -108$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.59 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [348] $2321812^1 31513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 52$, $\Delta = -116$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.59 \rightarrow 37$, $T_{\max} = 52$, $T \in [37, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [365] $81232171232^1 181232181232181232161 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 44$, $\Delta = -106$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.59 \rightarrow 37$, $T_{\max} = 44$, $T \in [37, 44]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$
- [364] $613151322^1 1(12)12231513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.59 \rightarrow 37$, $T_{\max} = 48$, $T \in [37, 48]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [346] $32181231513221(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 36.59 \rightarrow 37$, $T_{\max} = 48$, $T \in [37, 48]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [345] $81223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 32$, $T_{\max} = 48$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 32, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[344] & 613151322^1 1(12)12231513221^{1513221} (12) 1 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -110, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[343] & 21^{21} (12)1223151322^1 1(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[342] & 1(12)12231513221^{21} (12)122315132^1 181232 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 52, T \in [37, 52] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[341] & 1(12)122315132^{21} 21(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 52, T \in [37, 52] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[340] & 1(12)1223151322^{221} 1(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[339] & 21^{3221} (12)12231513221^{12} 12^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 48, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 48, T \in [37, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[362] & 81232181232^1 18123218123218123215 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -108, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[347] & 232181231513221^{12} 122^1 31513221(12)122 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 52, \Delta = -116, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 52, T \in [37, 52] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[360] & 413221(12)12^1 23151321812321812321812 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 46, \Delta = -112, \det = 169, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.59 \rightarrow 37, T_{\max} = 46, T \in [37, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[366] & 21^{513221} (12)12^1 231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 44, \Delta = -110, \det = 196, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.55 \rightarrow 37, T_{\max} = 44, T \in [37, 44] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[367] & 61231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 32, T_{\max} = 45, \Delta = -111, \det = 196, (n_+, n_-, n_0) = \\
& (1, 32, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 36.55 \rightarrow 37, T_{\max} = 45, T \in [37, 45]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[368] & \begin{matrix} 3221 & 1 \\ 21(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 48, \Delta = -114, \det = 196, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.55 \rightarrow 37, T_{\max} = 48, T \in [37, 48] & \end{matrix} \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[369] & \begin{matrix} 221 & 1 \\ 1(12)12231513221(12)12^1 231513218123 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 48, \Delta = -114, \det = 196, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.55 \rightarrow 37, T_{\max} = 48, T \in [37, 48] & \end{matrix} \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[370] & \begin{matrix} 21 & 1 \\ 21(12)12231513221(12)12^1 231513218123 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 48, \Delta = -114, \det = 196, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.55 \rightarrow 37, T_{\max} = 48, T \in [37, 48] & \end{matrix} \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[371] & \begin{matrix} 1 & 1513221 \\ 6131513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 44, \Delta = -110, \det = 196, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.55 \rightarrow 37, T_{\max} = 44, T \in [37, 44] & \end{matrix} \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[372] & \begin{matrix} 1 & 3221 \\ 812231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 48, \Delta = -114, \det = 196, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.55 \rightarrow 37, T_{\max} = 48, T \in [37, 48] & \end{matrix} \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[373] & \begin{matrix} 1 \\ 3218123151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 48, \Delta = -114, \det = 196, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.55 \rightarrow 37, T_{\max} = 48, T \in [37, 48] & \end{matrix} \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[374] & \begin{matrix} 1 \\ 71231513221(12)122^1 31513218123218123 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 49, \Delta = -115, \det = 196, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.55 \rightarrow 37, T_{\max} = 49, T \in [37, 49] & \end{matrix} \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[375] & \begin{matrix} 1 \\ 71232181231513221(12)12^1 231513218123 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 49, \Delta = -115, \det = 196, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.55 \rightarrow 37, T_{\max} = 49, T \in [37, 49] & \end{matrix} \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[376] & \begin{matrix} 1 & 1 \\ 413221(12)1223151322^1 1(12)12231513217 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 48, \Delta = -114, \det = 196, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.55 \rightarrow 37, T_{\max} = 48, T \in [37, 48] & \end{matrix} \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[377] & \begin{matrix} 1 \\ 812321812^1 3218123218123218123215 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 44, \Delta = -108, \det = 196, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.55 \rightarrow 37, T_{\max} = 44, T \in [37, 44] & \end{matrix} \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[378] & \begin{matrix} 1 \\ 812321812^1 3218123218123218123151 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 44, \Delta = -106, \det = 196, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.55 \rightarrow 37, T_{\max} = 44, T \in [37, 44] & \end{matrix} \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [379] $6131513221(12)12^1 231513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 48, \Delta = -114, \det = 196, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.55 \rightarrow 37, T_{\max} = 48, T \in [37, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [380] $812321712^1 32181232181232181232161 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -106, \det = 196, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.55 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$
- [383] $712321812^1 32181231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 52, \Delta = -116, \det = 225, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.52 \rightarrow 37, T_{\max} = 52, T \in [37, 52]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [381] $1(12)12231513221(12)122^1 315132181232 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 52, \Delta = -116, \det = 225, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.52 \rightarrow 37, T_{\max} = 52, T \in [37, 52]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [384] $712321812^1 31513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 49, \Delta = -115, \det = 225, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.52 \rightarrow 37, T_{\max} = 49, T \in [37, 49]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [382] $2321812315132^1 21(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 52, \Delta = -116, \det = 225, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.52 \rightarrow 37, T_{\max} = 52, T \in [37, 52]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [386] $7123218123151322^1 1(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 49, \Delta = -115, \det = 256, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.48 \rightarrow 37, T_{\max} = 49, T \in [37, 49]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [385] $712321812321812315132^1 21(12)12231513 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 52, \Delta = -116, \det = 256, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.48 \rightarrow 37, T_{\max} = 52, T \in [37, 52]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [389] $1712321812321812^1 31513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 44, \Delta = -108, \det = 289, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.46 \rightarrow 37, T_{\max} = 44, T \in [37, 44]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [391] $413221(12)122315132^1 181232181232181 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 46, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.46 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 407 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [392] $413221(12)122315132^1 1812321812321812 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 46, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.46 \rightarrow 37, T_{\max} = 46, T \in [37, 46]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 407 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [388] $232181231513221(12)12^1 231513221(12)122 + \hat{1} \rightarrow 2$ $T = 32, T_{\max} = 52, \Delta = -116, \det = 289, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 36.46 \rightarrow 37, T_{\max} = 52, T \in [37, 52]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[387] & \begin{matrix} 21 & 1 \\ 1(12)1223151322^1 1(12)122315132181232 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 52, \Delta = -116, \det = 289, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.46 \rightarrow 37, T_{\max} = 52, T \in [37, 52] & \end{matrix} \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[390] & \begin{matrix} 71231513221(12)12^1 231513218123218123 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 49, \Delta = -115, \det = 289, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.46 \rightarrow 37, T_{\max} = 49, T \in [37, 49] & \end{matrix} \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[394] & \begin{matrix} 71231513221(12)122315132^1 18123218123 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 49, \Delta = -115, \det = 324, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.43 \rightarrow 37, T_{\max} = 49, T \in [37, 49] & \end{matrix} \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[393] & \begin{matrix} 71232181232^1 181231513221(12)12231513 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 52, \Delta = -116, \det = 324, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.43 \rightarrow 37, T_{\max} = 52, T \in [37, 52] & \end{matrix} \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[397] & \begin{matrix} 7123218123218123151322^1 1(12)12231513 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 52, \Delta = -116, \det = 361, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.41 \rightarrow 37, T_{\max} = 52, T \in [37, 52] & \end{matrix} \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[395] & \begin{matrix} 21 & 1 \\ 1(12)12231513221(12)12^1 2315132181232 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 52, \Delta = -116, \det = 361, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.41 \rightarrow 37, T_{\max} = 52, T \in [37, 52] & \end{matrix} \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[396] & \begin{matrix} 23218123151322^1 1(12)12231513221(12)122 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 52, \Delta = -116, \det = 361, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.41 \rightarrow 37, T_{\max} = 52, T \in [37, 52] & \end{matrix} \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[398] & \begin{matrix} 712321812321812^1 31513221(12)12231513 + \hat{1} \rightarrow 2 & T = 32, T_{\max} = 52, \Delta = -116, \det = 576, (n_+, n_-, n_0) = \\ (1, 32, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 36.33 \rightarrow 37, T_{\max} = 52, T \in [37, 52] & \end{matrix} \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 407 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
T = 33 & [399] \begin{matrix} 22^1 21 \\ 1(12)12231513221(12)122315131613161 + \hat{1} \rightarrow 2 & T = 33, T_{\max} = 45, \Delta = -106, \det = -4, (n_+, n_-, n_0) = \\ (1, 33, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 40, T_{\max} = 45, T \in [40, 45] & \end{matrix} \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[400] & \begin{matrix} 22^1 1 \\ 21(12)12231513221(12)122315131613161 + \hat{1} \rightarrow 2 & T = 33, T_{\max} = 45, \Delta = -106, \det = -4, (n_+, n_-, n_0) = \\ (1, 33, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 40, T_{\max} = 45, T \in [40, 45] & \end{matrix} \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[401] & \begin{matrix} 22^1 1 \\ 1(12)12231513221(12)1223151316131613 + \hat{1} \rightarrow 2 & T = 33, T_{\max} = 49, \Delta = -114, \det = -4, (n_+, n_-, n_0) = \\ (1, 33, 0), c_{\text{ext}} = 15.0 & \\ T_{\min} = 40, T_{\max} = 49, T \in [40, 49] & \end{matrix} \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [402] $61316131513221(12)12231513221(12)12^1 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 50$, $\Delta = -115$, $\det = -4$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 39$, $T_{\max} = 50$, $T \in [39, 50]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [404] $3161316131513221(12)12231513221(12)122^1 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -4$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40$, $T_{\max} = 49$, $T \in [40, 49]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [405] $51232181232181232181231513221(12)12^1 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -4$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40$, $T_{\max} = 45$, $T \in [40, 45]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [406] $15132181232181232181231513221(12)12^1 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -107$, $\det = -4$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40$, $T_{\max} = 45$, $T \in [40, 45]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [407] $(12)1223151321812321812321812315132^1 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -4$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40$, $T_{\max} = 53$, $T \in [40, 53]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [403] $161316131513221(12)12231513221(12)1222^1 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -106$, $\det = -4$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40$, $T_{\max} = 45$, $T \in [40, 45]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [410] $1(12)12231513221(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -106$, $\det = -9$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 39.22 \rightarrow 40$, $T_{\max} = 45$, $T \in [40, 45]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- [418] $812231513221(12)1223151321812321812^1 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -9$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 39.22 \rightarrow 40$, $T_{\max} = 49$, $T \in [40, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [417] $32^1 181232181231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -9$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 39.22 \rightarrow 40$, $T_{\max} = 49$, $T \in [40, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [416] $181232181231513221(12)1223151321812^1 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -9$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 39.22 \rightarrow 40$, $T_{\max} = 49$, $T \in [40, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [415] $32^1 181232181232181231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 39.22 \rightarrow 40$, $T_{\max} = 53$, $T \in [40, 53]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [414] $5132181232181232181231513221(12)122^1 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -9$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 39.22 \rightarrow 40$, $T_{\max} = 53$, $T \in [40, 53]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

- [413] 3161316131513221(12)12231513221(12)12¹22 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -9$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 39.22 \rightarrow 40$, $T_{\max} = 49$, $T \in [40, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [412] 161316131513221(12)12231513221(12)12¹22 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -106$, $\det = -9$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 39.22 \rightarrow 40$, $T_{\max} = 45$, $T \in [40, 45]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [408] 513221²¹(12)12231513221(12)122315132171 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -9$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.33 \rightarrow 39$, $T_{\max} = 45$, $T \in [39, 45]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [409] 712231513221(12)12231513221⁵¹³²²¹(12)12¹ + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -9$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.33 \rightarrow 39$, $T_{\max} = 45$, $T \in [39, 45]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [411] 61316131513221(12)12231513221³²¹(12)12 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 50$, $\Delta = -115$, $\det = -9$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 39.22 \rightarrow 40$, $T_{\max} = 50$, $T \in [40, 50]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [420] 1²¹(12)122315132181232181232181232¹15 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.75 \rightarrow 39$, $T_{\max} = 45$, $T \in [39, 45]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [421] 161316131513221(12)12231513221(12)12¹222 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.75 \rightarrow 39$, $T_{\max} = 45$, $T \in [39, 45]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [422] 512¹32181232181232181231513221(12)122 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.75 \rightarrow 39$, $T_{\max} = 45$, $T \in [39, 45]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [423] 81232181232181232181232181232¹161 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -105$, $\det = -16$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.75 \rightarrow 39$, $T_{\max} = 45$, $T \in [39, 45]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [424] 81232171232181232181232181232¹171 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -104$, $\det = -16$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.75 \rightarrow 39$, $T_{\max} = 45$, $T \in [39, 45]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [425] 81232171232181232181232181232¹1712 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -104$, $\det = -16$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.75 \rightarrow 39$, $T_{\max} = 45$, $T \in [39, 45]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [419] 51316131513221(12)12231513221(12)12¹2313 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 39$, $\Delta = -97$, $\det = -16$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.75 \rightarrow 39$, $T_{\max} = 39$, $T \in [39, 39]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[427] $7122^1 31513221(12)12231513221^{513221}(12)12 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[428] $71231513221^1(12)12231513221^{15132^1 21}(12)^1 1 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[429] $51316131513221(12)122315132^1 21(12)122313 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 39, \Delta = -97, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 39, T \in [39, 39]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[430] $181232181231513221(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[431] $1812^1 32181231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[432] $1(12)122^1 31513221(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[433] $21(12)122^1 31513221(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[434] $181232181232181231513221(12)122^1 3151 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[435] $1812^1 32181232181231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[436] $1(12)1223151321812321812321812^1 3215 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[437] $1(12)1223151321812321812321812^1 3151 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

- [438] $1812^1 132181231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [439] $181232181231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [440] $181231513221(12)122315132181232^1 1812 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [441] $1812^1 131513221(12)1223151321812321812 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [442] $1812^1 32181231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [443] $181232181231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [444] $61316131513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 50, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 50, T \in [39, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [445] $161316131513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [446] $5132181232181232181231513221(12)12^1 23 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 53, \Delta = -116, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 53, T \in [39, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [447] $2181232181232181231513221(12)122^1 3151 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [448] $21812^1 32181232181231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [449] $51232^1 181232181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[450] & 15132^1 181232181232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& 513221 \\
[426] & 21(12)12231513221(12)122315132^1 171 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[452] & 21812^1 32181231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[453] & 21812^1 32181231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[454] & 2181232181231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[455] & 81231513221(12)1223151321812321812^1 32 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 53, T \in [39, 53] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[456] & 81232181231513221(12)1223151321812^1 32 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 53, T \in [39, 53] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[457] & 4132^1 21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 48, T \in [39, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[458] & 4132^1 21(12)12231513221(12)122315132218 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 48, T \in [39, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[459] & 812321812321812321812321812^1 32161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[460] & 812321712321812321812321812^1 32171 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -104, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[461] & 812321712321812321812321812^1 321712 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -104, \det = -25, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[451] \quad 2181232181231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.44 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[462] \quad 712^1 231513221(12)12231513221 \overset{513221}{(12)} 12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.22 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[463] \quad 712^1 31513221(12)12231513221 \overset{1}{(12)} 1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.22 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[464] \quad 51316131513221(12)1223151322^1 1(12)122313 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 39, \Delta = -97, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.22 \rightarrow 39, T_{\max} = 39, T \in [39, 39]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[465] \quad 1(12)12^1 231513221(12)122315131613161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.22 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[466] \quad 21(12)12^1 231513221(12)122315131613161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.22 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[467] \quad 1(12)122^1 31513221(12)1223151316131613 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.22 \rightarrow 39, T_{\max} = 49, T \in [39, 49]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[468] \quad 21(12)122^1 31513221(12)1223151316131613 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.22 \rightarrow 39, T_{\max} = 49, T \in [39, 49]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[469] \quad 181232181231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.22 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[470] \quad 161316131513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -106, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.22 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[471] \quad 3161316131513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.22 \rightarrow 39, T_{\max} = 49, T \in [39, 49]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

- [472] $1812^1 32181232181231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 50, \Delta = -115, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 50, T \in [39, 50]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [473] $1812^1 32181231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 49, \Delta = -114, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 49, T \in [39, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [474] $32181232181231513221(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 49, \Delta = -114, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 49, T \in [39, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [475] $2181232181231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [476] $8122^1 31513221(12)12231513218123218123 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 49, \Delta = -114, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 49, T \in [39, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [477] $413221^1(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [478] $413221^1(12)12231513221(12)122315132^1 218 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [479] $712^1 31513221(12)12231513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.22 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [485] $51316131513221(12)122^1 31513221(12)122313 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 39, \Delta = -97, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 39, T \in [39, 39]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [516] $2181232^1 181231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [483] $7122315132^1 21(12)12231513221^1(12)12 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [484] $71231513221^1(12)12231513221^1(12)1 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[481] & \begin{matrix} 513221 \\ 21(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[486] & \begin{matrix} 1 \\ 181232^1 181231513221(12)122315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[487] & \begin{matrix} 1 \\ 181232181231513221(12)122^1 315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[488] & \begin{matrix} 2221 \\ 1(12)122315132^1 21(12)122315131613161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 45, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[489] & \begin{matrix} 221 \\ 21(12)122315132^1 21(12)122315131613161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 45, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[490] & \begin{matrix} 221 \\ 1(12)12^1 231513221(12)1223151316131613 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.06 \rightarrow 39, T_{\max} = 49, T \in [39, 49] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[491] & \begin{matrix} 21 \\ 21(12)12^1 231513221(12)1223151316131613 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.06 \rightarrow 39, T_{\max} = 49, T \in [39, 49] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[492] & \begin{matrix} 1 \\ 181232181232181231513221(12)12^1 23151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[493] & \begin{matrix} 1 \\ 181232^1 181232181231513221(12)1223151 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[494] & \begin{matrix} 21 \\ 1(12)122315132181232181232^1 18123215 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[495] & \begin{matrix} 21 \\ 1(12)122^1 31513218123218123218123215 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[496] $1(12)122^1 31513218123218123218123151 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[497] $1(12)122315132181232181232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[498] $181232^1 181231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$

[499] $181232181231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[500] $1812315132^1 21(12)1223151321812321812 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

[502] $181232181231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[503] $181232^1 181231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$

[504] $161316131513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[505] $3161316131513221(12)1223151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 49, T \in [39, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$

[506] $5132^1 181232181232181231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 53, T \in [39, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[507] $32181232181232181231513221(12)122^1 315 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 53, T \in [39, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[508] $2181232181232181231513221(12)12^1 23151 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[509] 2181232¹181232181231513221(12)1223151 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[480] 21^{5132¹21}(12)12231513221(12)122315132171 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[510] 512321812¹32181232181231513221(12)122 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[511] 512321812321812321812315132¹21(12)122 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[512] 151321812321812321812315132¹21(12)122 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[513] 151321812¹32181232181231513221(12)122 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[514] 2181232181231513221(12)122¹3151321812 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -105$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[482] 712231513221(12)12231513221^{5132¹21}(12)12 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[515] 2181232¹181231513221(12)1223151321812 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -105$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[531] 812¹321712321812321812321812321712 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[530] 81232171232181232181232¹1812321712 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[529] 81232171232181232181232¹181232171¹ + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[528] $812^1 32171232181232181232181232171 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[527] $81232181232181232181232^1 181232161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[526] $812^1 32181232181232181232181232161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[525] $71231513221(12)12231513221(12)122^1 31414 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[524] $41322^1 1(12)12231513221(12)122315132218 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[523] $413221(12)12231513221(12)1223151322^1 18 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[522] $41322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[521] $812^1 231513221(12)12231513218123218123 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 49, T \in [39, 49]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[520] $812^1 32181231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 53, T \in [39, 53]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[519] $812^1 31513221(12)122315132181232181232 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 53, T \in [39, 53]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[518] $812^1 32181232181231513221(12)122315132 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 53, T \in [39, 53]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[517] $2181232181231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) =$

$$(1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[501] \quad 1^{12}181231513221(12)1223151321812^1 321812 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.06 \rightarrow 39, T_{\max} = 45, T \in [39, 45]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[532] \quad 1^{12}(12)122^1 315132181232181232181231513 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.94 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[534] \quad 1^{21}(12)1223151321812321812^1 3218123215 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.94 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[535] \quad 1^{21}(12)12^1 231513218123218123218123215 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.94 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[536] \quad 1^{21}(12)1223151321812321812^1 3218123151 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.94 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[537] \quad 1^{21}(12)12^1 231513218123218123218123151 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.94 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[538] \quad 181232181232181231513221(12)122^1 31513 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.94 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[539] \quad 3151321812321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.94 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[540] \quad 315132^1 181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.94 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[541] \quad 5123218123218123218123151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.94 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[542] \quad 51232181232^1 181232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.94 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

- [543] $1513218123218123218123151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.94 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [544] $15132181232^1 181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -107, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.94 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [545] $812321812321812321812^1 32181232161 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.94 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [546] $812321712321812321812^1 32181232171 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.94 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [547] $812321712321812321812^1 321812321712 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.94 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [533] $1(12)1223151321812321812321812^1 31513 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 50, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.94 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [584] $321812^1 32181231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 49, \Delta = -114, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [582] $181232^1 181231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 49, \Delta = -114, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [581] $31513218123218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 50, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [585] $2181232181231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [586] $21812321812315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [587] $2181232181231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

- [588] $21812321812315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [589] $812231513221(12)122315132181232^1 18123 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 49, \Delta = -114, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [590] $413221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [591] $413221(12)122^1 31513221(12)122315132218 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [592] $71231513221(12)12231513221(12)12^1 231414 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [593] $71231513221(12)122315132^1 21(12)12231414 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [594] $15131613161316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [575] $61316131513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 50, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [576] $1613161315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [577] $161316131513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [583] $181232181231513221(12)122315132^1 18123 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 49, \Delta = -114, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [578] $181232^1 181232181231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 50, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[579] $23151321812321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 53, \Delta = -116, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.84 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[580] $321812^1 32181232181231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 53, \Delta = -116, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.84 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[573] $181232181231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[572] $181231513221(12)122^1 3151321812321812 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[571] $18123151322^1 1(12)1223151321812321812 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[570] $181232181231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[569] $1812321812315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[568] $1(12)12^1 2315132181232181232181231513 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 50, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.84 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[567] $21(12)1223151322^1 1(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[566] $21(12)12231513221(12)122^1 315131613161 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[565] $1(12)1223151322^1 1(12)122315131613161 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[564] $1(12)12231513221(12)122^1 315131613161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[563] $181232181231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[562] $1812321812315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[561] $181223151322^1 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 38, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[560] $1812231513221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 38, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[559] $218123151322^1 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 38, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[558] $2181231513221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 38, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[557] $321812^1 31513221(12)12231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 38, \Delta = -94, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[556] $51316131513221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 39, \Delta = -97, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 39, T \in [38, 39]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[555] $513161315132^1 21(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 39, \Delta = -97, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 39, T \in [38, 39]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[554] $71231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[574] $1812321812315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[552] & \overset{1}{181231513221}(12)12^{\overset{1}{1}231513221(12)122313} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 38, \Delta = -93, \det = -81, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.84 \rightarrow 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[551] & 712231513221(12)122^{\overset{513221}{1}31513221}(12)12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -81, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[550] & 71223151322^{\overset{513221}{1}1}(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -81, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[549] & \overset{513221}{21}(12)122315132^{\overset{1}{1}21}(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -81, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[548] & \overset{513221}{21}(12)12231513221(12)12^{\overset{1}{1}2315132171} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -81, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.84 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[553] & \overset{1}{18123151322^{\overset{1}{1}1}}(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 38, \Delta = -93, \det = -81, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.84 \rightarrow 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[620] & 81232171232181232^{\overset{1}{1}1812321812321712} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -104, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[596] & 712231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[597] & 712315132^{\overset{1}{1}21}(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[598] & \overset{1}{32181231513221}(12)122^{\overset{1}{1}31513221}(12)12231 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 38, \Delta = -94, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[599] & \overset{221}{1}(12)122315132^{\overset{1}{1}21}(12)1223151316131613 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 49, T \in [38, 49]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[600] & 21^{21}(12)122315132^1 21(12)1223151316131613 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 49, T \in [38, 49] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[601] & 1^{12}(12)122315132181232181232^1 181231513 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 50, T \in [38, 50] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[595] & 51322^{11} 21^{12}(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[602] & 21^{12}(12)122315132181232^1 18123218123215 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[603] & 21^{12}(12)122315132181232^1 18123218123151 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[604] & 3161316131513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 49, T \in [38, 49] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[605] & 231513218123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 53, T \in [38, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[606] & 32181232181232181231513221(12)12^1 2315 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 53, T \in [38, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[607] & 3151321812^1 32181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 50, T \in [38, 50] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[608] & 512321812321812^1 32181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[609] & 151321812321812^1 32181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[610] & 32181232181231513221(12)122^1 315132181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -100, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 49, T \in [38, 49] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[611] & (12)122^1 31513218123218123218123151322 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 53, T \in [38, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[612] & 8122315132^1 21(12)12231513218123218123 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 49, T \in [38, 49] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[613] & 413221(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[614] & 413221(12)12231513221(12)122^1 315132218 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[615] & 712315132^1 21(12)12231513221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[616] & 81232181232181232^1 181232181232161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[617] & 81232171232181232^1 181232181232171 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -104, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[618] & 151316131613161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -106, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[619] & (12)122^1 31513161316131613161316131513 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -100, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.76 \rightarrow 38, T_{\max} = 49, T \in [38, 49] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[629] & 18123218123151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[632] & 21(12)12231513221(12)12^1 2315131613161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -106, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [657] $21812321812^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [658] $21812321812^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [659] $218123218123151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [660] $81232^1 181232181231513221(12)122315132 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 53, \Delta = -116, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [661] $81232181232181231513221(12)122^1 315132 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 53, \Delta = -116, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [628] $321812315132^1 21(12)12231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 38, \Delta = -94, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [662] $(12)12^1 231513218123218123218123151322 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 53, \Delta = -116, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [663] $81232^1 181231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 53, \Delta = -116, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [664] $413221(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [665] $413221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [666] $413221(12)12^1 231513221(12)122315132218 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [667] $413221(12)122315132^1 21(12)122315132218 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[668] & 71231513221(12)1223151322^1 1(12)12231414 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[669] & 71231513221(12)122^1 31513221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[670] & 81232^1 181232181232181232181232161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 21 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[671] & 812321812321812^1 32181232181232161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[672] & 812321712321812^1 32181232181232171 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[673] & 81232^1 171232181232181232181232171 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 219 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[674] & (12)12^1 23151316131613161316131513 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 49, T \in [38, 49] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[675] & 812321712321812^1 321812321812321712 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[676] & 81232^1 1712321812321812321812321712 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 219 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[634] & 1812321812321812315132^1 21(12)1223151 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[631] & 1(12)12231513221(12)12^1 2315131613161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[636] \quad & \overset{21}{1(12)1223151321812^1 3218123218123215} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[630] \quad & \overset{1}{1812321812^1 31513221(12)122315132181} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[633] \quad & \overset{1}{1(12)1223151321812321812^1 32181231513} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 50, T \in [38, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[637] \quad & \overset{21}{1(12)1223151321812^1 3218123218123151} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[638] \quad & \overset{1}{1812321812^1 31513221(12)1223151321812} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[639] \quad & \overset{1}{18123218123151322^1 1(12)1223151321812} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[640] \quad & \overset{1}{181231513221(12)12^1 23151321812321812} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[641] \quad & \overset{1}{181231513221(12)122315132^1 1812321812} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[642] \quad & \overset{1}{1812321812^1 31513221(12)1223151322181} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[643] \quad & \overset{1}{18123218123151322^1 1(12)1223151322181} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[644] \quad & \overset{3221}{61316131513221(12)1223151322^1 1(12)12} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -121, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.69 \rightarrow 38, T_{\max} = 50, T \in [38, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[645] \quad & \overset{3221}{61316131513221(12)122^1 31513221(12)12} + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -121, (n_+, n_-, n_0) =
\end{aligned}$$

- (1, 33, 0), $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 50$, $T \in [38, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [646] 16131613151322¹1(12)12231513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -106$, $\det = -121$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [647] 181232181232181231513221(12)12¹231513 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 50$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 50$, $T \in [38, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [648] 2315132¹181232181232181231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -121$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 53$, $T \in [38, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [649] 51321812¹32181232181231513221(12)1223 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -121$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 53$, $T \in [38, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [650] 315132181232¹181232181231513221(12)12 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 50$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 50$, $T \in [38, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [651] 21812321812321812315132¹21(12)1223151 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [652] 21812321812¹32181231513221(12)1223151 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [653] 51232181232181232¹181231513221(12)122 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -121$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [656] 218123218123151322¹1(12)1223151321812 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -105$, $\det = -121$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [654] 15132181232181232¹181231513221(12)122 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [655] 181232181231513221(12)122¹31513218123 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -121$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.69 \rightarrow 38$, $T_{\max} = 49$, $T \in [38, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [621] 21 (12) 122⁵¹³²²¹131513221(12)122315132171 + $\hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -121$, $(n_+, n_-, n_0) =$

$$(1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[622] \quad 21 \quad (12) \quad 1223151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[623] \quad 712231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[624] \quad 712231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[625] \quad 71231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[626] \quad 71231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[627] \quad 5131613151322^1 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 39, \Delta = -97, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.69 \rightarrow 38, T_{\max} = 39, T \in [38, 39]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[635] \quad 1812321812^1 32181231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.69 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[679] \quad 812231513221(12)1223151321812^1 3218123 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -144, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.64 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$$

$$\begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[677] \quad 32181232^1 181232181231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -144, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.64 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 12 & 2 \\ 12 & 59 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[678] \quad 32181232^1 181231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -144, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.64 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$$

$$\begin{pmatrix} 0 & 12 & 2 \\ 12 & 203 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[710] \quad 81232171232^1 1812321812321812321712 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.59 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$$

[699] $1812321812315132^1 21(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 49, \Delta = -114, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[705] $413221(12)12231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 48, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 48, T \in [38, 48]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[704] $81223151322^1 1(12)12231513218123218123 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 49, \Delta = -114, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[703] $81232181231513221(12)122315132^1 181232 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 53, \Delta = -116, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[702] $812315132^1 21(12)122315132181232181232 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 53, \Delta = -116, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[701] $81231513221(12)122315132181232^1 181232 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 53, \Delta = -116, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[700] $32181232181231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 49, \Delta = -114, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[698] $151321812321812321812^1 31513221(12)122 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[697] $512321812321812321812^1 31513221(12)122 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -109, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[696] $2181232181232^1 181231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

[695] $218123218123218123151322^1 1(12)1223151 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 45, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[694] $3151321812321812^1 32181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 33, T_{\max} = 50, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

- [693] $51321812321812321812315132^1 21(12)1223 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38$, $T_{\max} = 53$, $T \in [38, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [692] $5132181232^1 181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38$, $T_{\max} = 53$, $T \in [38, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [691] $23151321812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38$, $T_{\max} = 53$, $T \in [38, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [690] $3161316131513221(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38$, $T_{\max} = 49$, $T \in [38, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [689] $613161315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 50$, $\Delta = -115$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38$, $T_{\max} = 50$, $T \in [38, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [688] $1(12)122315132^1 18123218123218123151 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -107$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [687] $1(12)122315132^1 18123218123218123215 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [686] $181232181232^1 181231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -107$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [685] $18123218123218123151322^1 1(12)1223151 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -107$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [684] $1(12)122315132181232^1 181232181231513 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 50$, $\Delta = -115$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38$, $T_{\max} = 50$, $T \in [38, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [683] $21(12)1223151322^1 1(12)1223151316131613 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.59 \rightarrow 38$, $T_{\max} = 49$, $T \in [38, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [682] $1(12)1223151322^1 1(12)1223151316131613 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -169$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 37.59 \rightarrow 38, T_{\max} = 49, T \in [38, 49] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[681] & 32181231513221(12)12^1 231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 38, \Delta = -94, \det = -169, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.59 \rightarrow 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[680] & 7123151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -169, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.59 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[707] & 7123151322^1 1(12)12231513221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -169, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.59 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[708] & 81232181232^1 181232181232181232161 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -105, \det = -169, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.59 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[709] & 81232171232^1 181232181232181232171 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -104, \det = -169, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.59 \rightarrow 38, T_{\max} = 45, T \in [38, 45] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[706] & 413221(12)12231513221(12)12^1 2315132218 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -169, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.59 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[728] & 71231513221(12)12^1 231513221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 48, \Delta = -113, \det = -196, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.55 \rightarrow 38, T_{\max} = 48, T \in [38, 48] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[719] & 31613161315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -196, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.55 \rightarrow 38, T_{\max} = 49, T \in [38, 49] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[720] & 2315132181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -196, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.55 \rightarrow 38, T_{\max} = 53, T \in [38, 53] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[721] & 315132181232181232^1 181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -196, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.55 \rightarrow 38, T_{\max} = 50, T \in [38, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[722] & 181232181231513221(12)12^1 231513218123 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -196, (n_+, n_-, n_0) = \\
& (1, 33, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 37.55 \rightarrow 38, T_{\max} = 49, T \in [38, 49] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [723] $321812321812315132^1 21(12)122315132181 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -196$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.55 \rightarrow 38$, $T_{\max} = 49$, $T \in [38, 49]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [724] $(12)1223151321812321812321812^1 3151322 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -196$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.55 \rightarrow 38$, $T_{\max} = 53$, $T \in [38, 53]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [725] $812231513221(12)122^1 31513218123218123 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -196$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.55 \rightarrow 38$, $T_{\max} = 49$, $T \in [38, 49]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [726] $413221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 48$, $\Delta = -113$, $\det = -196$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.55 \rightarrow 38$, $T_{\max} = 48$, $T \in [38, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [727] $413221(12)1223151322^1 1(12)122315132218 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 48$, $\Delta = -113$, $\det = -196$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.55 \rightarrow 38$, $T_{\max} = 48$, $T \in [38, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [718] $61316131513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 50$, $\Delta = -115$, $\det = -196$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.55 \rightarrow 38$, $T_{\max} = 50$, $T \in [38, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [729] $812321812^1 32181232181232181232161 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -105$, $\det = -196$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.55 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 41 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [730] $812321712^1 32181232181232181232171 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -104$, $\det = -196$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.55 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$
- [731] $812321712^1 321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -104$, $\det = -196$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.55 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$
- [713] $71231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -196$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.55 \rightarrow 38$, $T_{\max} = 45$, $T \in [38, 45]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [715] $1(12)12231513221(12)122^1 3151316131613 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -196$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.55 \rightarrow 38$, $T_{\max} = 49$, $T \in [38, 49]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [711] $21(12)12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 45$, $\Delta = -109$, $\det = -196$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 37.55 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[712] \quad 712231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -109, \det = -196, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.55 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[716] \quad 21(12)12231513221(12)122^1 3151316131613 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -196, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.55 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[717] \quad 1(12)1223151321812^1 32181232181231513 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -196, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.55 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[714] \quad 3218123151322^1 1(12)12231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 38, \Delta = -94, \det = -196, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.55 \rightarrow 38, T_{\max} = 38, T \in [38, 38]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[732] \quad 1812321812^1 32181231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -225, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.52 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[736] \quad 81232181231513221(12)122^1 315132181232 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -225, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.52 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[735] \quad 81232181232181231513221(12)12^1 2315132 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -225, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.52 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[734] \quad 1812321812^1 31513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -225, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.52 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[733] \quad 513218123218123218123151322^1 1(12)1223 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -225, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.52 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[737] \quad 1(12)122315132^1 181232181232181231513 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -256, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.48 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[740] \quad 23151321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -256, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.48 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[739] \quad 1812321812321812315132^1 21(12)12231513 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -256, (n_+, n_-, n_0) =$$

$(1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.48 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

$[741] \ 3151321812321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -256, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.48 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

$[742] \ 18123218123151322^1 1(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -256, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.48 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

$[743] \ (12)122315132181232181232^1 18123151322 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -256, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.48 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

$[738] \ 6131613151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -256, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.48 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

$[744] \ 1(12)12231513221(12)12^1 23151316131613 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -289, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

$[746] \ 1812321812321812^1 31513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -289, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

$[747] \ 316131613151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -289, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38, T_{\max} = 49, T \in [38, 49]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 390 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0}$

$[748] \ 2315132181232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -289, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

$[749] \ 51321812321812^1 32181231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -289, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

$[750] \ 321812321812321812315132^1 21(12)122315 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -289, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 101 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[751] \ 21812321812321812^1 31513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 45, \Delta = -107, \det = -289, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38, T_{\max} = 45, T \in [38, 45]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

$[752] \ 3218123218123151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 49, \Delta = -114, \det = -289, (n_+, n_-, n_0) =$

$(1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38$, $T_{\max} = 49$, $T \in [38, 49]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 390 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0}$

[753] $(12)1223151321812321812^1 3218123151322 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -289$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38$, $T_{\max} = 53$, $T \in [38, 53]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$

[754] $8123151322^1 (12)122315132181232181232 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -289$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38$, $T_{\max} = 53$, $T \in [38, 53]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[755] $81231513221(12)1223151321812^1 32181232 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -289$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38$, $T_{\max} = 53$, $T \in [38, 53]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 203 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[756] $812321812315132^1 21(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -289$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38$, $T_{\max} = 53$, $T \in [38, 53]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[757] $812231513221(12)12^1 231513218123218123 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -289$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38$, $T_{\max} = 49$, $T \in [38, 49]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

21
[745] $21(12)12231513221(12)12^1 23151316131613 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -289$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.46 \rightarrow 38$, $T_{\max} = 49$, $T \in [38, 49]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[759] $321812321812^1 32181231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -324$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.43 \rightarrow 38$, $T_{\max} = 53$, $T \in [38, 53]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 125 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[761] $812231513221(12)122315132^1 18123218123 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -324$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.43 \rightarrow 38$, $T_{\max} = 49$, $T \in [38, 49]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[760] $321812321812^1 31513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 49$, $\Delta = -114$, $\det = -324$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.43 \rightarrow 38$, $T_{\max} = 49$, $T \in [38, 49]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 449 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[758] $181232181232^1 181231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 50$, $\Delta = -115$, $\det = -324$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.43 \rightarrow 38$, $T_{\max} = 50$, $T \in [38, 50]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 233 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$

[766] $(12)122315132181232^1 18123218123151322 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -361$, $(n_+, n_-, n_0) = (1, 33, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 37.41 \rightarrow 38$, $T_{\max} = 53$, $T \in [38, 53]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[767] $81231513221(12)122^1 315132181232181232 + \hat{1} \rightarrow 2$ $T = 33$, $T_{\max} = 53$, $\Delta = -116$, $\det = -361$, $(n_+, n_-, n_0) =$

$$(1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.41 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[768] \ 81232181231513221(12)12^1 2315132181232 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -361, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.41 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[769] \ 812321812^1 31513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -361, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.41 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[765] \ 812321812^1 32181231513221(12)122315132 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -361, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.41 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[764] \ 5132181232181232^1 181231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -361, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.41 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[763] \ 23151321812321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -361, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.41 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

$$[762] \ 18123218123218123151322^1 1(12)12231513 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -361, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.41 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[770] \ 3218123218123218123151322^1 1(12)122315 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -400, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.39 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 20 & 2 \\ 20 & 139 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[771] \ (12)1223151321812^1 3218123218123151322 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -400, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.39 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 20 & 2 \\ 20 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[773] \ 812321812321812315132^1 21(12)122315132 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -441, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.37 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[772] \ 32181232181232^1 181231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -441, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.37 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 167 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[774] \ 8123218123151322^1 1(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -441, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.37 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[775] \ (12)122315132^1 18123218123218123151322 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -484, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.36 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

$$[776] \ 51321812321812321812^1 31513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -529, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.34 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$$

$$[777] \ 81232181232^1 181231513221(12)122315132 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -529, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.34 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 413 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

$$[778] \ 81231513221(12)12^1 2315132181232181232 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -529, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.34 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 390 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0}$$

$$[779] \ 1812321812321812^1 31513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 50, \Delta = -115, \det = -576, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.33 \rightarrow 38, T_{\max} = 50, T \in [38, 50]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 407 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

$$[780] \ 8123218123218123151322^1 1(12)122315132 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -625, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.31 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 474 & 42 \\ 2 & 42 & 3 \end{pmatrix}_{b_0}$$

$$[781] \ 81231513221(12)122315132^1 181232181232 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -625, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.31 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 449 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$$[782] \ 321812321812321812^1 31513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -729, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.29 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 27 & 2 \\ 27 & 269 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[783] \ 812321812321812^1 31513221(12)122315132 + \hat{1} \rightarrow 2 \quad T = 33, T_{\max} = 53, \Delta = -116, \det = -961, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 37.25 \rightarrow 38, T_{\max} = 53, T \in [38, 53]$$

$$\begin{pmatrix} 0 & 31 & 2 \\ 31 & 743 & 53 \\ 2 & 53 & 3 \end{pmatrix}_{b_0}$$

$$T = 34 \quad [784] \ 1(12)12231513221(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41, T_{\max} = 46, T \in [41, 46]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[785] \ 21(12)12231513221(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41, T_{\max} = 46, T \in [41, 46]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[786] \ 5132181231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 4, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41, T_{\max} = 46, T \in [41, 46]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[787] \ 161232181232181232181231513221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -106, \det = 4, (n_+, n_-, n_0) =$$

(1, 34, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 41$, $T_{\max} = 46$, $T \in [41, 46]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[788] 812321712321812321812321812321812¹2 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -103$, $\det = 4$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41$, $T_{\max} = 46$, $T \in [41, 46]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[789] (12)12231513161316131613161316131612¹3 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 48$, $\Delta = -112$, $\det = 9$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 40.22 \rightarrow 41$, $T_{\max} = 48$, $T \in [41, 48]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[792] 513221²1¹(12)12231513221(12)1223151322181 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 9$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.33 \rightarrow 40$, $T_{\max} = 46$, $T \in [40, 46]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[793] 181231513221(12)12231513221²1¹(12)122315 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 9$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.33 \rightarrow 40$, $T_{\max} = 46$, $T \in [40, 46]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[794] 5132181231513221(12)12231513221(12)122¹22 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 9$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 40.22 \rightarrow 41$, $T_{\max} = 46$, $T \in [41, 46]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[795] 61232181232181232181231513221(12)122¹3 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 51$, $\Delta = -115$, $\det = 9$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 40.22 \rightarrow 41$, $T_{\max} = 51$, $T \in [41, 51]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[790] 1²²²1¹(12)12231513221(12)1223151321812315 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 9$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 40.22 \rightarrow 41$, $T_{\max} = 46$, $T \in [41, 46]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[791] 513221²1¹(12)12231513221(12)1223151321812 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 9$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.33 \rightarrow 40$, $T_{\max} = 46$, $T \in [40, 46]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[796] 5132181231513221(12)12231513221(12)12¹222 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 16$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.75 \rightarrow 40$, $T_{\max} = 46$, $T \in [40, 46]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[797] 1²¹(12)122315132181232181232181232¹161 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -106$, $\det = 16$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.75 \rightarrow 40$, $T_{\max} = 46$, $T \in [40, 46]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[798] 1612¹32181232181232181231513221(12)122 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -106$, $\det = 16$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.75 \rightarrow 40$, $T_{\max} = 46$, $T \in [40, 46]$

$$(1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[818] \quad 812321712321812321812321812^1 321 81 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[819] \quad 812321712321812321812321812^1 3218122 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[810] \quad 5132181231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[808] \quad 812231513221(12)12231513221 \quad (12) \quad 1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[807] \quad 1 \quad (12) \quad 12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[806] \quad 21 \quad (12) \quad 12231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[805] \quad 21(12)122^1 31513221(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[804] \quad 21 \quad (12) \quad 12231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[803] \quad 1(12)122^1 31513221(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.44 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[824] \quad 8122^1 31513221(12)12231513221 \quad (12) \quad 1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.22 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[820] \quad 1(12)12^1 231513221(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 39.22 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [821] \quad 21(12)12^1 231513221(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 36, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.22 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [827] \quad 8122^1 31513221(12)12231513221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 36, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.22 \rightarrow 40, T_{\max} = 48, T \in [40, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [823] \quad 1(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 36, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.22 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [825] \quad 5132181231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 36, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.22 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [826] \quad 41413221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 36, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.22 \rightarrow 40, T_{\max} = 48, T \in [40, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [822] \quad 21(12)12231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 36, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.22 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [835] \quad 21(12)12231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
& [836] \quad 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
& [837] \quad 812^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 54, \Delta = -116, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 54, T \in [40, 54] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [838] \quad 812231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
& [839] \quad 812^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[840] & \begin{matrix} 1 & & 21 \\ 181231513221(12)12231513221(12)122^1 315 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\ (1, 34, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[841] & \begin{matrix} 1 & & 21 \\ 1812315132^1 21(12)12231513221(12)122315 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\ (1, 34, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[842] & \begin{matrix} 5132181231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\ (1, 34, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[843] & \begin{matrix} 5132^1 181231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\ (1, 34, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[844] & \begin{matrix} 6131513221(12)12231513221(12)122^1 3151315 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 48, \Delta = -112, \det = 49, (n_+, n_-, n_0) = \\ (1, 34, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 48, T \in [40, 48] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[845] & \begin{matrix} 21 \\ 1(12)122315132181232181232^1 181232161 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 46, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\ (1, 34, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[846] & \begin{matrix} 21 \\ 1(12)122^1 315132181232181232181232161 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 46, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\ (1, 34, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0} \\
[847] & \begin{matrix} 1 \\ 1(12)122315132181232181232^1 1613 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 50, \Delta = -114, \det = 49, (n_+, n_-, n_0) = \\ (1, 34, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 50, T \in [40, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[848] & \begin{matrix} 1612321812^1 32181232181231513221(12)122 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 46, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\ (1, 34, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[834] & \begin{matrix} 5132^1 21 \\ 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\ (1, 34, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[850] & \begin{matrix} 31612^1 32181232181232181231513221(12)12 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 50, \Delta = -114, \det = 49, (n_+, n_-, n_0) = \\ (1, 34, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 50, T \in [40, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[851] & \begin{matrix} 61232^1 181232181232181231513221(12)1223 + \hat{1} \rightarrow 2 & T = 34, T_{\max} = 51, \Delta = -115, \det = 49, (n_+, n_-, n_0) = \end{matrix}
\end{aligned}$$

$$(1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 51, T \in [40, 51]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[852] \quad 414132^1 21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 48, T \in [40, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[853] \quad 812^1 231513221(12)12231513221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 48, T \in [40, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[854] \quad 812231513221(12)12231513221(12)122^1 31414 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 48, T \in [40, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[855] \quad 41322^1 1(12)12231513221(12)12231513161316 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 50, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[856] \quad 81232181232181232181232^1 1812321712 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[857] \quad 812^1 321812321812321812321812321712 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[858] \quad 81232181232181232181232^1 181232171 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[859] \quad 812^1 32181232181232181232181232171 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[860] \quad 812^1 32171232181232181232181232181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[861] \quad 81232171232181232181232^1 181232181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[862] \quad 812^1 3217123218123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0} \\
[863] & 81232171232181232181232^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
& 513221 \\
[830] & 21(12)12231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
& 2221 \\
[828] & 1(12)12231513221(12)1223151321812^1 315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& 221 \\
[833] & 21(12)122315132^1 21(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& 221 \\
[832] & 21(12)12231513221(12)1223151321812^1 315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& 5132^1 21 \\
[831] & 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0} \\
& 2221 \\
[829] & 1(12)122315132^1 21(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[849] & 1612321812321812321812315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.06 \rightarrow 40, T_{\max} = 46, T \in [40, 46] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[870] & 16123218123218123218123151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -106, \det = 64, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.94 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[869] & 161232181232^1 181232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -106, \det = 64, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.94 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& 1 \\
[868] & 1(12)122^1 3151321812321812321812321613 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 50, \Delta = -114, \det = 64, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.94 \rightarrow 39, T_{\max} = 50, T \in [39, 50] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [867] $1(12)1223151321812321812321812^1 321613 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 50, \Delta = -114, \det = 64, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.94 \rightarrow 39, T_{\max} = 50, T \in [39, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [866] $1(12)12^1 2315132181232181232181232161 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 46, \Delta = -106, \det = 64, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.94 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [865] $1(12)1223151321812321812^1 32181232161 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 46, \Delta = -106, \det = 64, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.94 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [864] $81231513221(12)12231513221 \begin{smallmatrix} 1 \\ 315132^1 21 \end{smallmatrix} (12) 1 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 54, \Delta = -116, \det = 64, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.94 \rightarrow 39, T_{\max} = 54, T \in [39, 54]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [876] $812321712321812321812^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 46, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.94 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [875] $812321712321812321812^1 321812321 \begin{smallmatrix} 21 \\ 81 \end{smallmatrix} + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 46, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.94 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [874] $812321812321812321812^1 321812321 \begin{smallmatrix} 1 \\ 171 \end{smallmatrix} + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 46, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.94 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [873] $812321812321812321812^1 321812321712 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 46, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.94 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [872] $3161232^1 181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 50, \Delta = -114, \det = 64, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.94 \rightarrow 39, T_{\max} = 50, T \in [39, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [871] $31612321812321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 50, \Delta = -114, \det = 64, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.94 \rightarrow 39, T_{\max} = 50, T \in [39, 50]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [885] $21(12)12231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 46, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [878] $21613161316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 46, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[891] & 51321812315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 81, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[893] & 6131513221(12)12231513221(12)12^1 23151315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 81, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 48, T \in [39, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[903] & 413221(12)122^1 31513221(12)12231513161316 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 50, \Delta = -114, \det = 81, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 50, T \in [39, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[902] & 812231513221(12)12231513221(12)12^1 231414 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 81, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 48, T \in [39, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[901] & 812231513221(12)122315132^1 21(12)12231414 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 81, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 48, T \in [39, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[900] & 4141322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 81, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 48, T \in [39, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[899] & 41413221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 81, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 48, T \in [39, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[898] & 316123218123218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 50, \Delta = -114, \det = 81, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 50, T \in [39, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[897] & 1(12)12^1 23151321812321812321812321613 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 50, \Delta = -114, \det = 81, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 50, T \in [39, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[896] & 1613161315132^1 21(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 39, \Delta = -94, \det = 81, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 39, T \in [39, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[895] & 161316131513221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 39, \Delta = -94, \det = 81, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.84 \rightarrow 39, T_{\max} = 39, T \in [39, 39] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [894] $6131513221(12)^1 122315132^1 21(12)1223151315 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 48$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39$, $T_{\max} = 48$, $T \in [39, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [892] $5132181231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [890] $181231513221(12)122^1 31513221(12)122315 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [889] $18123151322^1 1(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [888] $812231513221(12)122315132^1 21(12)^{1513221} 1 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [887] $1^{1513221} (12) 122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [886] $21^{513221} (12) 122315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [884] $21^{221} (12)1223151322^1 1(12)1223151321812315 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [883] $21^{221} (12)12231513221(12)122^1 3151321812315 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [882] $21^{513221} (12) 122315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [881] $21^{513221} (12) 12231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [880] $1^{2221} (12)12231513221(12)122^1 3151321812315 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) =$

$(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 2221 \\ [879] \end{matrix} 1(12)1223151322^1 1(12)1223151321812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 81, (n_+, n_-, n_0) =$
 $(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 1 \\ [877] \end{matrix} 16131613161316131613161315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 81, (n_+, n_-, n_0) =$
 $(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.84 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 $[907] (12)122^1 315131613161316131613161316123 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 100, (n_+, n_-, n_0) =$
 $(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.76 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 1 \\ [904] \end{matrix} 161316131613161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 100, (n_+, n_-, n_0) =$
 $(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.76 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
 $[906] 2161316131613161316131613151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 100, (n_+, n_-, n_0) =$
 $(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.76 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 51322^1 1 \\ [908] \end{matrix} 21 (12) 12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 100, (n_+, n_-, n_0) =$
 $(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.76 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 51322^1 1 \\ [909] \end{matrix} 21 (12) 12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 100, (n_+, n_-, n_0) =$
 $(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.76 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 25 & 3 \end{pmatrix}_{b_0}$
 $\begin{matrix} 1513221 \\ [910] \end{matrix} 1 (12) 12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 100, (n_+, n_-, n_0) =$
 $(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.76 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 $[911] 8122315132^1 21(12)12231513221 (12) 1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 100, (n_+, n_-, n_0) =$
 $(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.76 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 $[912] 181231513221(12)12231513221(12)12^1 2315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 100, (n_+, n_-, n_0) =$
 $(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.76 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 $[913] 61315132^1 21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 100, (n_+, n_-, n_0) =$
 $(1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.76 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$

[914] 3161316131513221(12)122¹31513221(12)12231 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 39$, $\Delta = -94$, $\det = 100$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.76 \rightarrow 39, T_{\max} = 39, T \in [39, 39]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[915] $1(12)122315132181232^1181232181232161 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.76 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[916] $1(12)122315132181232181232^11812321613 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 50, \Delta = -114, \det = 100, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.76 \rightarrow 39, T_{\max} = 50, T \in [39, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[917] 1612321812321812¹32181231513221(12)122 + **1**→**2** $T = 34$, $T_{\max} = 46$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.76 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[918] 31612321812¹32181232181231513221(12)12 + $\hat{\mathbf{i}} \rightarrow \mathbf{2}$ $T = 34$, $T_{\max} = 50$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.76 \rightarrow 39, T_{\max} = 50, T \in [39, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[919] $41413221(12)12231513221(12)122^1315132181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 100, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.76 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[920] $8122315132^1 21(12)12231513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 48$, $\Delta = -112$, $\det = 100$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.76 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[921] 81232181232181232¹1812321812321712 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -103$, $\det = 100$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.76 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[922] $\overset{1}{8}12321\overset{1}{8}1232181232^1181232181232171 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 100, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 38.76 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[923] 81232171232181232¹181232181232181 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -103$, $\det = 100$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.76 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[924] 81232171232181232¹18123218123218122 + **$\hat{\mathbf{i}} \rightarrow \mathbf{2}$** $T = 34$, $T_{\max} = 46$, $\Delta = -103$, $\det = 100$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 38.76 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[905] (12)122¹31513161316131613161316131613161 + $\hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 47$, $\Delta = -111$, $\det = 100$, $(n_+, n_-, n_0) =$

$$(1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.76 \rightarrow 39, T_{\max} = 47, T \in [39, 47]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[946] \ 16131613151322^1 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 39, \Delta = -94, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.69 \rightarrow 39, T_{\max} = 39, T \in [39, 39]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[945] \ 6131513221(12)1223151322^1 1(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.69 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[944] \ 6131513221(12)122^1 31513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.69 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[943] \ 51321812^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[942] \ 513218123151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[941] \ 181231513221(12)12^1 231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[940] \ 181231513221(12)122315132^1 21(12)122315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[939] \ 812231513221(12)122^1 31513221^{1513221} (12) \ 1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[938] \ 812231513221(12)1223151322^1 1(12) \ 1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[937] \ 81231513221(12)12231513221^{3151322^1 1} (12) \ 1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 54, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.69 \rightarrow 39, T_{\max} = 54, T \in [39, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[947] \ 1(12)1223151321812^1 32181232181232161 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[935] & 1(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
& 513221 \\
[934] & 21(12)1223151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
& 513221 \\
[933] & 21(12)122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
& 221 \\
[932] & 21(12)12231513221(12)122315132^1 1812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
& 221 \\
[931] & 21(12)12231513221(12)12^1 23151321812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
& 513221 \\
[930] & 21(12)1223151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
& 513221 \\
[929] & 21(12)122^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
& 2221 \\
[928] & 1(12)12231513221(12)12^1 23151321812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
& 2221 \\
[927] & 1(12)12231513221(12)122315132^1 1812315 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[926] & (12)12^1 231513161316131613161316123 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 48, T \in [39, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
& 31 \\
[925] & (12)12^1 23151316131613161316131613161 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 47, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 47, T \in [39, 47] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[948] $1(12)1223151321812321812^1 321812321613 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 50, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.69 \rightarrow 39, T_{\max} = 50, T \in [39, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[949] $161232181232181232^1 181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 46, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[950] $3161232181232^1 181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 50, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.69 \rightarrow 39, T_{\max} = 50, T \in [39, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[951] $612321812^1 32181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 51, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.69 \rightarrow 39, T_{\max} = 51, T \in [39, 51]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[952] $41413221(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 48, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.69 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[953] $41413221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 48, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.69 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[954] $812231513221(12)1223151322^1 1(12)12231414 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 48, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.69 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[955] $812231513221(12)122^1 31513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 48, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.69 \rightarrow 39, T_{\max} = 48, T \in [39, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[956] $413221(12)122315132^1 21(12)12231513161316 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 50, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.69 \rightarrow 39, T_{\max} = 50, T \in [39, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

[957] $413221(12)12^1 231513221(12)12231513161316 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 50, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.69 \rightarrow 39, T_{\max} = 50, T \in [39, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

[958] $812321812321812^1 321812321812321712 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 46, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[959] $81232^1 1812321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 34, T_{\max} = 46, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 21 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[960] & 81232^1 181232181232181232181232171 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 121, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 21 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[961] & 812321812321812^1 32181232181232171 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 121, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[962] & 81232^1 171232181232181232181232181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 121, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 219 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[963] & 812321712321812^1 32181232181232181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 121, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[964] & 812321712321812^1 3218123218123218122 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 121, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[965] & 81232^1 17123218123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 121, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 219 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[936] & 1^{1513221} (12) 12^1 231513221 (12) 122315132181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 121, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.69 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[980] & 413221^1 (12) 12231513221 (12) 122^1 31513161316 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 50, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.59 \rightarrow 39, T_{\max} = 50, T \in [39, 50] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[984] & 81232171232^1 18123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 169, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.59 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[977] & 612321812321812321812315132^1 21 (12) 1223 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 51, \Delta = -115, \det = 169, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.59 \rightarrow 39, T_{\max} = 51, T \in [39, 51] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[976] & 61232181232^1 181232181231513221 (12) 1223 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 51, \Delta = -115, \det = 169, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.59 \rightarrow 39, T_{\max} = 51, T \in [39, 51] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[975] & 31612321812321812^1 32181231513221 (12) 12 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 50, \Delta = -114, \det = 169, (n_+, n_-, n_0) =
\end{aligned}$$

(1, 34, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 50$, $T \in [39, 50]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[974] $1612321812321812321812^1 31513221(12)122 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -106$, $\det = 169$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[973] $1(12)122315132181232^1 1812321812321613 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 50$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 50$, $T \in [39, 50]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[972] $1(12)122315132^1 181232181232181232161 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -106$, $\det = 169$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$$

[971] $3161316131513221(12)12^1 231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 39$, $\Delta = -94$, $\det = 169$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 39$, $T \in [39, 39]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[970] $613151322^1 1(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 48$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 48$, $T \in [39, 48]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[969] $81223151322^1 1(12)12231513221^{1513221}(12)1 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -108$, $\det = 169$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[981] $81232181232^1 1812321812321812321712 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -103$, $\det = 169$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[968] $81231513221(12)122315132^1 21^{31513221}(12)1 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 54$, $\Delta = -116$, $\det = 169$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 54$, $T \in [39, 54]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[979] $81223151322^1 1(12)12231513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 48$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 48$, $T \in [39, 48]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[967] $812315132^1 21(12)12231513221^{31513221}(12)1 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 54$, $\Delta = -116$, $\det = 169$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 54$, $T \in [39, 54]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[978] $41413221(12)12231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 48$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 38.59 \rightarrow 39$, $T_{\max} = 48$, $T \in [39, 48]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[966] & \overset{1513221}{1} (12) 12231513221 \overset{1}{(12)12^1 2315132181} + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 169, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.59 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[983] & 81232171232^1 1812321812321812321 \overset{21}{81} + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.59 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[982] & \overset{1}{81232181232^1 181232181232181232171} + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.59 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[993] & 3161232181232181232^1 181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 50, \Delta = -114, \det = 196, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.55 \rightarrow 39, T_{\max} = 50, T \in [39, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[985] & \overset{513221}{21} (12) 12^1 231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 196, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.55 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[986] & \overset{513221}{21} (12) 12^1 231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 196, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.55 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[987] & \overset{1513221}{1} (12) 1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 196, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.55 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[988] & \overset{1}{812231513221(12)12^1 231513221} \overset{1513221}{(12)1} + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 196, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.55 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[989] & \overset{1}{181231513221(12)1223151322^1 1(12)122315} + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 46, \Delta = -108, \det = 196, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.55 \rightarrow 39, T_{\max} = 46, T \in [39, 46] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[990] & \overset{1}{6131513221(12)12^1 231513221(12)1223151315} + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 48, \Delta = -112, \det = 196, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.55 \rightarrow 39, T_{\max} = 48, T \in [39, 48] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[996] & \overset{1}{413221(12)1223151322^1 1(12)12231513161316} + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 50, \Delta = -114, \det = 196, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.55 \rightarrow 39, T_{\max} = 50, T \in [39, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [991] $31613161315132^1 21(12)12231513221(12)12231 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 39$, $\Delta = -94$, $\det = 196$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.55 \rightarrow 39$, $T_{\max} = 39$, $T \in [39, 39]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [992] $1(12)1223151321812^1 321812321812321613 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 50$, $\Delta = -114$, $\det = 196$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.55 \rightarrow 39$, $T_{\max} = 50$, $T \in [39, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [997] $812321812^1 321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -103$, $\det = 196$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.55 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 41 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [998] $812321812^1 32181232181232181232171 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -103$, $\det = 196$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.55 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 41 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [999] $812321712^1 32181232181232181232181 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -103$, $\det = 196$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.55 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$
- [1000] $812321712^1 3218123218123218123218122 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 46$, $\Delta = -103$, $\det = 196$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.55 \rightarrow 39$, $T_{\max} = 46$, $T \in [39, 46]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$
- [995] $812231513221(12)12^1 231513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 48$, $\Delta = -112$, $\det = 196$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.55 \rightarrow 39$, $T_{\max} = 48$, $T \in [39, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [994] $41413221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 48$, $\Delta = -112$, $\det = 196$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.55 \rightarrow 39$, $T_{\max} = 48$, $T \in [39, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1002] $6123218123218123218123151322^1 1(12)1223 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 51$, $\Delta = -115$, $\det = 225$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.52 \rightarrow 39$, $T_{\max} = 51$, $T \in [39, 51]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1001] $81231513221(12)122^1 31513221^{31513221} (12) 1 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 54$, $\Delta = -116$, $\det = 225$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.52 \rightarrow 39$, $T_{\max} = 54$, $T \in [39, 54]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [1003] $81231513221(12)1223151322^1 1^{31513221} (12) 1 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 54$, $\Delta = -116$, $\det = 256$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 38.48 \rightarrow 39$, $T_{\max} = 54$, $T \in [39, 54]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1004] $1(12)122315132^1 1812321812321812321613 + \hat{1} \rightarrow 2$ $T = 34$, $T_{\max} = 50$, $\Delta = -114$, $\det = 256$, $(n_+, n_-, n_0) = (1, 34, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 38.48 \rightarrow 39, T_{\max} = 50, T \in [39, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1005] & 31612321812321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 50, \Delta = -114, \det = 256, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.48 \rightarrow 39, T_{\max} = 50, T \in [39, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[1006] & 413221(12)12231513221(12)12^1 231513161316 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 50, \Delta = -114, \det = 256, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.48 \rightarrow 39, T_{\max} = 50, T \in [39, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1007] & 8123151322^1 1(12)12231513221^1 (12) 1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 54, \Delta = -116, \det = 289, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.46 \rightarrow 39, T_{\max} = 54, T \in [39, 54] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[1008] & 316131613151322^1 1(12)12231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 39, \Delta = -94, \det = 289, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.46 \rightarrow 39, T_{\max} = 39, T \in [39, 39] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 390 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0} \\
[1009] & 612321812321812^1 32181231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 51, \Delta = -115, \det = 289, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.46 \rightarrow 39, T_{\max} = 51, T \in [39, 51] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1010] & 81231513221(12)12^1 231513221^1 (12) 1 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 54, \Delta = -116, \det = 361, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.41 \rightarrow 39, T_{\max} = 54, T \in [39, 54] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[1011] & 61232181232181232^1 181231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 51, \Delta = -115, \det = 361, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.41 \rightarrow 39, T_{\max} = 51, T \in [39, 51] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[1012] & 612321812321812321812^1 31513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 34, T_{\max} = 51, \Delta = -115, \det = 529, (n_+, n_-, n_0) = \\
& (1, 34, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 38.34 \rightarrow 39, T_{\max} = 51, T \in [39, 51] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
T = 35 & [1015] 61232181231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -107, \det = \\
& -4, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42, T_{\max} = 47, T \in [42, 47] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1018] & 812321812321812321812321812321812^1 2 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -102, \det = -4, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42, T_{\max} = 47, T \in [42, 47] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1017] & 2171232181232181232181231513221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -104, \det = -4, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42, T_{\max} = 47, T \in [42, 47] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1016] $171232181232181232181231513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -4$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42$, $T_{\max} = 47$, $T \in [42, 47]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1013] $61232181231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -107$, $\det = -4$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42$, $T_{\max} = 47$, $T \in [42, 47]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1014] $61232181231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -107$, $\det = -4$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42$, $T_{\max} = 47$, $T \in [42, 47]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1023] $32^1 171232181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -9$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.22 \rightarrow 42$, $T_{\max} = 50$, $T \in [42, 50]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [1022] $171232181232181232181231513221(12)122^1 3 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -9$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.22 \rightarrow 42$, $T_{\max} = 50$, $T \in [42, 50]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1021] $1(12)1223151321812321812321812321712^1 3 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -9$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.22 \rightarrow 42$, $T_{\max} = 50$, $T \in [42, 50]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1020] $61232181231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -107$, $\det = -9$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.22 \rightarrow 42$, $T_{\max} = 47$, $T \in [42, 47]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1019] $61232181231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -107$, $\det = -9$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.22 \rightarrow 42$, $T_{\max} = 47$, $T \in [42, 47]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1031] $81232181232181232181232181232^1 181 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -16$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40.75 \rightarrow 41$, $T_{\max} = 47$, $T \in [41, 47]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1024] $5132181231513221(12)12231513221(12)12^1 2313 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 41$, $\Delta = -96$, $\det = -16$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40.75 \rightarrow 41$, $T_{\max} = 41$, $T \in [41, 41]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1030] $81232181232181232181232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -16$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40.75 \rightarrow 41$, $T_{\max} = 47$, $T \in [41, 47]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1029] $21712^1 32181232181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -16$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 40.75 \rightarrow 41$, $T_{\max} = 47$, $T \in [41, 47]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1028] & \begin{matrix} 1 \\ 1712^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, T_{\max} = 47, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.75 \rightarrow 41, T_{\max} = 47, T \in [41, 47] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1027] & \begin{matrix} 21 \\ 1(12)122315132181232181232181232^1 1712 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, T_{\max} = 47, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.75 \rightarrow 41, T_{\max} = 47, T \in [41, 47] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1026] & \begin{matrix} 1 \\ 1712^1 32181232181232181231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, T_{\max} = 47, \Delta = -104, \det = -16, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.75 \rightarrow 41, T_{\max} = 47, T \in [41, 47] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1025] & \begin{matrix} 1 \\ 61232181231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, T_{\max} = 47, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.75 \rightarrow 41, T_{\max} = 47, T \in [41, 47] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1040] & \begin{matrix} 1 \\ 6131513221(12)12231513221(12)122315132^1 16 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, T_{\max} = 47, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1033] & \begin{matrix} 15132^1 21 \\ 61316131513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, T_{\max} = 47, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1034] & \begin{matrix} 2221 \\ 61232181231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, T_{\max} = 47, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1035] & \begin{matrix} 2221 \\ 612^1 32181231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, T_{\max} = 47, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[1036] & \begin{matrix} 221 \\ 61232181231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, T_{\max} = 47, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1037] & \begin{matrix} 221 \\ 612^1 32181231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, T_{\max} = 47, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[1041] & \begin{matrix} 1 \\ 171232^1 181232181232181231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, T_{\max} = 47, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[1042] $1(12)1223151321812321812321812^1321712 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[1043] $171232^1181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1044] $171232181232181232181231513221(12)12^123 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 50, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.44 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1045] $1712^132181232181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 50, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.44 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1049] $812321812321812321812321812^132181 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -102, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

[1048] $812321812321812321812321812^13218122 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -102, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

[1047] $2171232^1181232181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -104, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1046] $232^1171232181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 52, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.44 \rightarrow 41, T_{\max} = 52, T \in [41, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[1038] $61232181231513221(12)122315132^121(12)12222 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1039] $612^132181231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.44 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$

[1032] $5132181231513221(12)122315132^121(12)122313 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 41, \Delta = -96, \det = -25, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.44 \rightarrow 41, T_{\max} = 41, T \in [41, 41]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1054] $61232181231513221(12)1223151322^11(12)12222 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.22 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

- [1053] $61232181231513221(12)1223151322^11(12)12 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.22 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1052] $61232181231513221(12)1223151322^11(12)1 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.22 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1051] $712^131513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 48, \Delta = -111, \det = -36, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.22 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1050] $5132181231513221(12)1223151322^11(12)122313 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 41, \Delta = -96, \det = -36, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.22 \rightarrow 41, T_{\max} = 41, T \in [41, 41]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1058] $61316131513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.06 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1075] $321712^132181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 50, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.06 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1057] $71231513221(12)12231513221(12)122^13151315 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 48, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.06 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1073] $1712321812321812321812315132^121(12)1 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.06 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$
- [1072] $1712321812^132181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.06 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1071] $1(12)122315132181232181232^11812321712 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.06 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1070] $1(12)122^13151321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.06 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
- [1074] $171232^1181232181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 50, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 50, \quad T \in [41, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1069] & \begin{matrix} 1 \\ 1712321812321812321812315132^1 21(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, \quad T_{\max} = 47, \quad \Delta = -104, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 35, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 47, \quad T \in [41, 47] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[1068] & \begin{matrix} 1 \\ 1712321812^1 32181232181231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, \quad T_{\max} = 47, \quad \Delta = -104, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 35, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 47, \quad T \in [41, 47] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1067] & \begin{matrix} 1 \\ 1(12)122315132181232181232181232^1 17123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, \quad T_{\max} = 50, \quad \Delta = -113, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 35, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 50, \quad T \in [41, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1066] & \begin{matrix} 1 \\ 6131513221(12)12231513221(12)122^1 31513216 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, \quad T_{\max} = 47, \quad \Delta = -110, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 35, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 47, \quad T \in [41, 47] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1065] & \begin{matrix} 1 \\ 6131513221(12)12231513221(12)122^1 31513161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, \quad T_{\max} = 47, \quad \Delta = -109, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 35, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 47, \quad T \in [41, 47] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1064] & \begin{matrix} 1 \\ 61232^1 181231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, \quad T_{\max} = 47, \quad \Delta = -107, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 35, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 47, \quad T \in [41, 47] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[1063] & \begin{matrix} 1 \\ 61232181231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, \quad T_{\max} = 47, \quad \Delta = -107, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 35, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 47, \quad T \in [41, 47] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1062] & \begin{matrix} 221 \\ 61232^1 181231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, \quad T_{\max} = 47, \quad \Delta = -107, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 35, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 47, \quad T \in [41, 47] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[1061] & \begin{matrix} 221 \\ 61232181231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, \quad T_{\max} = 47, \quad \Delta = -107, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 35, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 47, \quad T \in [41, 47] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1060] & \begin{matrix} 2221 \\ 61232181231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, \quad T_{\max} = 47, \quad \Delta = -107, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 35, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 47, \quad T \in [41, 47] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1059] & \begin{matrix} 2221 \\ 61232^1 181231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 35, \quad T_{\max} = 47, \quad \Delta = -107, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 35, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.06 \rightarrow 41, \quad T_{\max} = 47, \quad T \in [41, 47]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[1055] $5132^1 181231513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 41$, $\Delta = -96$, $\det = -49$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 40.06 \rightarrow 41, T_{\max} = 41, T \in [41, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[1076] $21712321812^1 32181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 40.06 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1077] $21712321812321812321812315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 40.06 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[1078] $812^1 3218123218123218123218123218122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 40.06 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[1079] $81232181232181232181232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 40.06 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1080] $81232181232181232181232^1 1812321^1 81 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 40.06 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1081] $812^1 321812321812321812321812321^1 81 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 40.06 \rightarrow 41, T_{\max} = 47, T \in [41, 47]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[1082] $414132^1 21(12)12231513221(12)12231513161316 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 40.06 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1056] $5132181231513221(12)122^1 31513221(12)122313 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 41$, $\Delta = -96$, $\det = -49$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 40.06 \rightarrow 41, T_{\max} = 41, T \in [41, 41]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1087] $1(12)12^1 23151321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.94 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1096] $812321812321812321812^1 321812321^1 81 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.94 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1084] $1(12)1223151321812321812321812^1 3217123 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) =$

(1, 35, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 50$, $T \in [40, 50]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1086] $17123218123218123218123151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1088] $1(12)1223151321812321812^1 321812321712 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1089] $17123218123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1090] $171232181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1091] $32171232^1 181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 50$, $T \in [40, 50]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1092] $321712321812321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 50$, $T \in [40, 50]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1093] $217123218123218123218123151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1094] $2171232181232^1 181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1095] $812321812321812321812^1 3218123218122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1085] $171232181232^1 181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1083] $1(12)122^1 31513218123218123218123217123 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.94 \rightarrow 40$, $T_{\max} = 50$, $T \in [40, 50]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1097] 5132181231513221(12)12¹231513221(12)122313+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 41$, $\Delta = -96$, $\det = -81$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.84 \rightarrow 40, T_{\max} = 41, T \in [40, 41]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1098] 51321812315132¹21(12)12231513221(12)122313+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 41$, $\Delta = -96$, $\det = -81$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.84 \rightarrow 40, T_{\max} = 41, T \in [40, 41]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1099] 71231513221(12)12231513221(12)12¹23151315+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 48$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.84 \rightarrow 40, T_{\max} = 48, T \in [40, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1100] 71231513221(12)122315132¹21(12)1223151315+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 48$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.84 \rightarrow 40, T_{\max} = 48, T \in [40, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1101] 61316131513221(12)122315132¹⁵¹³²²¹21 (12) 1 + $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -109$, $\det = -81$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.84 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1102] 61232181231513221(12)12¹231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.84 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1103] 612321812315132²²²¹21(12)12231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.84 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1104] 612321812315132²²¹21(12)12231513221(12)12 + $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.84 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1105] 61232181231513221(12)12²²¹231513221(12)12 + $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.84 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1107] 61232181231513221(12)12¹231513221(12)12222+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.84 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1108] 6131513221(12)12231513221(12)12¹231513161+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -109$, $\det = -81$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.84 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1110] $6131513221(12)122315132^1 21(12)12231513216+\hat{1}\rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.84 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1111] $6131513221(12)12231513221(12)12^1 231513216+\hat{1}\rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.84 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1112] $1(12)12^1 231513218123218123218123217123 + \hat{1}\rightarrow 2$ $T = 35, T_{\max} = 50, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.84 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1113] $2321712321812321812321812315132^1 21(12)1+\hat{1}\rightarrow 2$ $T = 35, T_{\max} = 52, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.84 \rightarrow 40, T_{\max} = 52, T \in [40, 52]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[1114] $3217123218123218123218123151322^1 1(12)12+\hat{1}\rightarrow 2$ $T = 35, T_{\max} = 50, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.84 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[1115] $41413221(12)122^1 31513221(12)12231513161316+\hat{1}\rightarrow 2$ $T = 35, T_{\max} = 50, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.84 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1116] $4141322^1 1(12)12231513221(12)12231513161316+\hat{1}\rightarrow 2$ $T = 35, T_{\max} = 50, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.84 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1106] $612321812315132^1 21(12)12231513221(12)12222+\hat{1}\rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.84 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1109] $6131513221(12)122315132^1 21(12)12231513161+\hat{1}\rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.84 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1117] $4131613161316131613161315132^1 21(12)+\hat{1}\rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -110, \det = -100, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.76 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 9 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$

[1118] $712315132^1 21(12)12231513221(12)1223151315+\hat{1}\rightarrow 2$ $T = 35, T_{\max} = 48, \Delta = -111, \det = -100, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.76 \rightarrow 40, T_{\max} = 48, T \in [40, 48]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1119] $61315132^1 21(12)12231513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -109, \det = -100, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 39.76 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1120] $61315132^1 21(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -110$, $\det = -100$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.76 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1121] $1(12)122315132181232181232^1 18123217123 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.76 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1122] $1712321812321812^1 32181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.76 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1123] $1(12)122315132181232^1 1812321812321712 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.76 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1124] $1712321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.76 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1125] $23217123218123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 52$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.76 \rightarrow 40, T_{\max} = 52, T \in [40, 52]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1126] $2321712^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 52$, $\Delta = -115$, $\det = -100$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.76 \rightarrow 40, T_{\max} = 52, T \in [40, 52]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1127] $321712321812^1 32181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.76 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1128] $21712321812321812^1 32181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.76 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1129] $81232181232181232^1 18123218123218122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -100$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.76 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1130] $81232181232181232^1 181232181232181 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -100$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.76 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1147] $6131513221(12)122^1 31513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -110$, $\det = -121$, $(n_+, n_-, n_0) =$

$$(1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1161] 41413221(12)122315132^1 21(12)12231513161316+\hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 50, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1131] 41316131613161316131613151322^1 1(12)+\hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 10 & 10 \\ 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1132] 51321812^1 31513221(12)12231513221(12)122313+\hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 41, \Delta = -96, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 41, T \in [40, 41]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1133] 513218123151322^1 1(12)12231513221(12)122313+\hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 41, \Delta = -96, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 41, T \in [40, 41]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[1134] 71231513221(12)1223151322^1 1(12)1223151315+\hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 48, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 48, T \in [40, 48]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1135] 71231513221(12)122^1 31513221(12)1223151315+\hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 48, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 48, T \in [40, 48]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1136] 61316131513221(12)122^1 31513221 \quad (12) \quad 1 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -109, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1137] 61316131513221(12)1223151322^1 1 \quad (12) \quad 1 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -109, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1138] 612321812^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1139] 6123218123151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[1143] 6123218123151322^1 1(12)12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[1144] $6131513221(12)1223151322^1 1(12)12231513161 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -109, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1145] $6131513221(12)122^1 31513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -109, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1146] $6131513221(12)1223151322^1 1(12)12231513216 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1140] $6123218123151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[1148] $1(12)1223151321812321812^1 3218123217123 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 50, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1149] $171232181232181232^1 181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[1150] $1(12)1223151321812^1 321812321812321712 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1151] $171232181232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[1152] $1712321812^1 32181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 50, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1153] $232171232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 52, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 52, T \in [40, 52]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[1154] $32171232181232^1 181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 50, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 39.69 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[1155] $2171232181232181232^1 181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 35, T_{\max} = 47, \Delta = -104, \det = -121, (n_+, n_-, n_0) =$

(1, 35, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 39.69 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[1156] $812321812321812^1 3218123218123218122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -121$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.69 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1157] $8123218123218123218123218123218122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -121$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.69 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 21 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1158] $812321812321812^1 32181232181232181 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -121$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.69 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1159] $81232181232181232181232181232181 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -102$, $\det = -121$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.69 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 21 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1160] $41413221(12)12^1 231513221(12)12231513161316 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.69 \rightarrow 40$, $T_{\max} = 50$, $T \in [40, 50]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1141] $612321812^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.69 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1142] $612321812^1 31513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.69 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1168] $1(12)122315132^1 1812321812321812321712 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -169$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.59 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$$

[1167] $1712321812321812321812^1 31513221(12)122 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -104$, $\det = -169$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.59 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1166] $1(12)122315132181232^1 18123218123217123 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -169$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.59 \rightarrow 40$, $T_{\max} = 50$, $T \in [40, 50]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1165] $613151322^1 1(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -110$, $\det = -169$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 39.59 \rightarrow 40$, $T_{\max} = 47$, $T \in [40, 47]$

$$\begin{aligned}
& T_{\min} = 39.59 \rightarrow 40, T_{\max} = 50, T \in [40, 50] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
& [1178] \text{ }^1 71231513221(12)12^1 231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 48, \Delta = -111, \det = -196, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.55 \rightarrow 40, T_{\max} = 48, T \in [40, 48] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& [1179] \text{ }^{1513221} 61316131513221(12)12^1 231513221(12) \text{ }^{1513221} 1 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -109, \det = -196, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.55 \rightarrow 40, T_{\max} = 47, T \in [40, 47] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& [1180] \text{ }^1 6131513221(12)12^1 231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -109, \det = -196, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.55 \rightarrow 40, T_{\max} = 47, T \in [40, 47] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& [1181] \text{ }^1 6131513221(12)12^1 231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -110, \det = -196, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.55 \rightarrow 40, T_{\max} = 47, T \in [40, 47] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& [1182] \text{ }^1 1(12)1223151321812^1 3218123218123217123 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 50, \Delta = -113, \det = -196, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.55 \rightarrow 40, T_{\max} = 50, T \in [40, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
& [1183] \text{ }^{181232181232} 232171232181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 52, \Delta = -115, \det = -196, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.55 \rightarrow 40, T_{\max} = 52, T \in [40, 52] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
& [1184] \text{ }^{181232181232} 32171232181232181232^1 181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 50, \Delta = -113, \det = -196, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.55 \rightarrow 40, T_{\max} = 50, T \in [40, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
& [1185] \text{ }^1 812321812^1 3218123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -102, \det = -196, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.55 \rightarrow 40, T_{\max} = 47, T \in [40, 47] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 41 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [1186] \text{ }^1 812321812^1 321812321812321812321 \text{ }^{21} 81 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 47, \Delta = -102, \det = -196, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.55 \rightarrow 40, T_{\max} = 47, T \in [40, 47] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 41 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [1187] \text{ }^{14143221} 41413221(12)1223151322^1 1(12)12231513161316 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 50, \Delta = -113, \det = -196, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.55 \rightarrow 40, T_{\max} = 50, T \in [40, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& [1188] \text{ }^{17123218123218123218123151322} 17123218123218123218123151322^1 1(12)1223 + \hat{1} \rightarrow 2 \quad T = 35, T_{\max} = 50, \Delta = -113, \det = -225, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 39.52 \rightarrow 40, T_{\max} = 50, T \in [40, 50] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1193] 41413221(12)12231513221(12)12¹231513161316+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -256$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.48 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1191] 2321712321812321812¹32181231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 52$, $\Delta = -115$, $\det = -256$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.48 \rightarrow 40, T_{\max} = 52, T \in [40, 52]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[1189] 6131613151322¹⁵¹³²²¹1(12)12231513221 (12) 1 + $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 47$, $\Delta = -109$, $\det = -256$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.48 \rightarrow 40, T_{\max} = 47, T \in [40, 47]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1190] 1(12)122315132¹18123218123218123217123+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -256$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.48 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1192] 321712321812321812321812¹31513221(12)12+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -256$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.48 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[1194] 1712321812321812¹32181231513221(12)1223+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -289$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.46 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1195] 232171232181232181232¹181231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 52$, $\Delta = -115$, $\det = -289$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.46 \rightarrow 40, T_{\max} = 52, T \in [40, 52]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[1196] 171232181232181232¹181231513221(12)1223+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -361$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.41 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1197] 2321712321812321812321812¹31513221(12)1+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 52$, $\Delta = -115$, $\det = -361$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.41 \rightarrow 40, T_{\max} = 52, T \in [40, 52]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[1198] 1712321812321812321812¹31513221(12)1223+ $\hat{1} \rightarrow 2$ $T = 35$, $T_{\max} = 50$, $\Delta = -113$, $\det = -529$, $(n_+, n_-, n_0) = (1, 35, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 39.34 \rightarrow 40, T_{\max} = 50, T \in [40, 50]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$$

$T = 36$ [1207] 1(12)12231513221(12)12231513221(12)1222²²²¹2 + $\hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -104$, $\det = 4$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43, T_{\max} = 48, T \in [43, 48]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[1208] 21(12)12231513221(12)12231513221(12)122³²²¹2 + $\hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 51$, $\Delta = -113$, $\det = 4$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43, T_{\max} = 51, T \in [43, 51]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1209] & \begin{matrix} 2^1 1 \\ 1 \end{matrix} 3221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 4, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42, T_{\max} = 51, T \in [42, 51] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[1210] & \begin{matrix} 221 \\ 1 \end{matrix} 21(12)12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 4, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1211] & \begin{matrix} 22^1 1 \\ 1 \end{matrix} 21(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 4, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1212] & \begin{matrix} 1 \\ 22^1 221 \end{matrix} (12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 4, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1213] & \begin{matrix} 22^1 21 \\ 1 \end{matrix} 1(12)12231513221(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 4, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1214] & \begin{matrix} 22^1 1 \\ 21 \end{matrix} 21(12)12231513221(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 4, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1215] & \begin{matrix} 3221 \\ 71232181231513221 \end{matrix} (12)12231513221(12)12^1 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 4, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[1216] & 171232181231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 4, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1217] & \begin{matrix} 21 \\ 181232181232181232181231513221 \end{matrix} (12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 4, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1199] & \begin{matrix} 22^1 21 \\ 1 \end{matrix} 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 4, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1200] & \begin{matrix} 221 \\ 1 \end{matrix} 1(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 4, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42, T_{\max} = 51, T \in [42, 51]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[1201] & 1(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1202] & 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1218] & 1(12)1223151321812321812321812321812^1 2 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1203] & 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1219] & 22^1 181232181232181232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1204] & 21(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 4, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42, T_{\max} = 51, T \in [42, 51] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[1205] & 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1206] & 1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1220] & 22181232181232181232181231513221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1229] & 21(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.22 \rightarrow 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1235] & 1(12)12231513218123218123218123218122^1 3 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.22 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

[1230] $222^1 21(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -104, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 48, T \in [43, 48]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[1236] $1 \ 8 \ 1232181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$

[1228] $21(12)12231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 51, \Delta = -113, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[1227] $21(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 51, \Delta = -113, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[1234] $181232181232181232181231513221(12)122^1 3 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[1239] $32^1 2181232181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$

[1226] $1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -104, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 48, T \in [43, 48]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[1225] $1(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -104, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 48, T \in [43, 48]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[1224] $21(12)12231513221(12)12231513221(12)122^1 3 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 51, \Delta = -113, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[1223] $1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -104, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 48, T \in [43, 48]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$

[1222] $1(12)12231513221(12)12231513221(12)122^1 3 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 51, \Delta = -113, \det = 9, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

$$\begin{aligned}
[1221] & 1^{222^1 1} (12) 12231513221 (12) 12231513221 (12) 1222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 9, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.22 \rightarrow 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0} \\
[1237] & 1^{32^1 1} 8 1232181232181232181231513221 (12) 1 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 9, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.22 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[1232] & 71232181231513221 (12) 12231513221 (12) 12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 9, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.22 \rightarrow 43, T_{\max} = 52, T \in [43, 52] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1233] & 171232181231513221 (12) 12231513221 (12) 122^1 22 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 9, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.22 \rightarrow 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1238] & 2181232181232181232181231513221 (12) 122^1 3 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 9, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.22 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[1231] & 1^{222^1 1} (12) 12231513221 (12) 122315132181232171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 9, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.22 \rightarrow 43, T_{\max} = 48, T \in [43, 48] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0} \\
[1246] & 1^{21} 8 12^1 32181232181232181231513221 (12) 1 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 16, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.75 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1245] & 1^{21} (12) 122315132181232181232181232^1 18122 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 16, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.75 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1247] & 221812^1 32181232181232181231513221 (12) 122 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 16, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.75 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1244] & 1^{21} 8 12^1 32181232181232181231513221 (12) 122 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 16, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.75 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1243] & 171232181231513221 (12) 12231513221 (12) 12^1 222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 16, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.75 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1242] & 2222^1 1 (12) 12231513221 (12) 12231513221 (12) 12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 16, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 41.75 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& [1240] \overset{2221}{1}(12)12231513221(\overset{1}{12})12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 16, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.75 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
& [1241] 21(\overset{221}{12})12231513221(\overset{1}{12})12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 16, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.75 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
& [1272] \overset{1}{1}(12)1223151321812321812321812321812^1 23 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 25, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.44 \rightarrow 42, T_{\max} = 50, T \in [42, 50] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
& [1253] \overset{221}{21}(\overset{1}{12})122^1 31513221(\overset{1}{12})12231513221(\overset{221}{12})12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 25, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
& [1262] \overset{221}{21}(\overset{1}{12})12231513221(12)122315132181232^1 171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 25, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [1263] \overset{221}{21}(\overset{1}{12})122^1 31513221(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 25, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
& [1264] \overset{322^1 1}{71}232181231513221(12)12231513221(\overset{1}{12})12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 25, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.44 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
& [1265] 1712^1 32181231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 25, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
& [1266] 171232181231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 25, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [1267] 6131513221(\overset{1}{12})12231513221(12)122315132^1 171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 25, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& [1268] 6131513221(\overset{1}{12})12231513221(12)122315132^1 217 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 25, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]
\end{aligned}$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1269] $71231513221(12)12231513221(12)122315132^1 16 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1270] $181232181232181232181231513221(12)12^1 23 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1271] $1812^1 32181232181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1273] $1^1 8^1 1232181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 52, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1248] $1(12)122^1 31513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[1274] $181232^1 181232181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1275] $1(12)1223151321812321812321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1276] $181232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1277] $21812^1 32181232181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1278] $2181232181232181232181231513221(12)12^1 23 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1279] $322^1 181232181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1280] $22181232^1 181232181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -103, \det = 25, (n_+, n_-, n_0) =$

$$(1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1252] \quad {}^{21}21(12)12231513221(12)12231513221(12)12^1 23 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1249] \quad {}^{221}1(12)12231513221(12)12231513221(12)12^1 23 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1250] \quad {}^{2221}1(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1251] \quad {}^{2221}1(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[1260] \quad {}^{2221}1(12)12231513221(12)122315132181232^1 171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1259] \quad {}^{22221}1(12)122^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1258] \quad {}^{221}21(12)122^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1257] \quad {}^{221}21(12)12231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1256] \quad {}^{322^1 1}21(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1255] \quad {}^{2221}1(12)122^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1254] \quad {}^{2221}1(12)12231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
& [1261] \quad 1(12)122^1 31513221(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 25, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.44 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0} \\
& [1286] \quad 21^{221}(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 36, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
& [1285] \quad 21(12)122^1 31513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 36, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 51, T \in [42, 51] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [1287] \quad 1(12)12231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 36, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
& [1288] \quad 1(12)12^1 231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 36, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [1289] \quad 21^{3221}(12)12231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 36, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 51, T \in [42, 51] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [1290] \quad 21^{221}(12)12231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 36, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
& [1291] \quad 21(12)12^1 231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 36, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [1292] \quad 1812^1 31513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 36, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [1293] \quad 22221(12)12^1 231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 36, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [1283] \quad 1(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 36, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1284] & 1(12)12^{\overset{2221}{1}}231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 36, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1294] & 8122^{\overset{1}{1}}31513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 36, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1295] & 1(12)12^{\overset{2221}{1}}231513221(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1296] & 21(12)12^{\overset{221}{1}}231513221(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1297] & 712^{\overset{3221}{1}}32181231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1298] & 171232181231513221(12)1223151322^{\overset{1}{1}}1(12)1222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1299] & 712^{\overset{1}{1}}31513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1300] & 6131513221(12)12231513221(12)1223151322^{\overset{1}{1}}17 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 36, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1301] & 712^{\overset{1}{1}}31513221(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1282] & 1(12)122^{\overset{221}{1}}31513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 36, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 51, T \in [42, 51] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1281] & 1(12)12^{\overset{2221}{1}}231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 36, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.22 \rightarrow 42, T_{\max} = 48, T \in [42, 48] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[1327] $1^{12}181232^1181232181232181231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[1328] $1^{12}1(12)122315132181232181232181232^1181223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1306] $21^{21}(12)12^1231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1307] $21^{221}(12)122315132^121(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

[1308] $1^{2221}(12)122315132^121(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1309] $1^{2221}(12)12231513221(12)122^131513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[1304] $1^{2221}(12)122315132^121(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

[1310] $21^{3221}(12)12231513221(12)1223151322^11(12)1222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1311] $21^{221}(12)122315132^121(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1312] $21^{221}(12)12231513221(12)122^131513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[1313] $1^{12}181231513221(12)12231513221(12)122^13151315 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

- [1314] $22221(12)122315132^1 21(12)12231513221(12)12222+\hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -104$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1315] $812^1 231513221(12)12231513221(12)1223151315+\hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1316] $812231513221(12)12231513221(12)122^1 3151315+\hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1317] $1(12)12231513221(12)1223151321812^1 32171 + \hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [1330] $1812321812^1 32181232181231513221(12)122 + \hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1325] $6131513221(12)12231513221(12)122^1 315132217+\hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1324] $6131513221(12)12231513221(12)122^1 315132171+\hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1323] $71231513221(12)12231513221(12)122^1 31513161+\hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1322] $171232181231513221(12)122^1 31513221(12)12222+\hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1321] $171232^1 181231513221(12)12231513221(12)12222+\hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [1320] $21(12)12231513221(12)1223151321812^1 32171 + \hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [1319] $21(12)122315132^1 21(12)122315132181232171 + \hat{1}\rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) =$

$(1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1318] $1(12)122315132^1 21(12)122315132181232171 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -105$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1331] $1812321812321812321812315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[1332] $1(12)122^1 31513218123218123218123218122 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[1333] $1(12)122315132181232181232^1 18123218122 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1334] $1812^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 50$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 50$, $T \in [42, 50]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1335] $1812321812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1336] $1812321812321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[1337] $2181232^1 181232181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 50$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 50$, $T \in [42, 50]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1338] $3221812^1 32181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 50$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 50$, $T \in [42, 50]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1339] $221812321812321812321812315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[1340] $221812321812^1 32181232181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 36$, $T_{\max} = 48$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 36, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.06 \rightarrow 42$, $T_{\max} = 48$, $T \in [42, 48]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1329] $1 \overset{321}{8} 12^1 32181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.06 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1303] $1(12)12^1 231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.06 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1302] $1(12)122315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.06 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1326] $71231513221(12)12231513221(12)122^1 31513216 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.06 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1305] $1(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 41.06 \rightarrow 42, T_{\max} = 48, T \in [42, 48]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1344] $1 \overset{321}{8} 12321812321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 40.94 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1345] $1 \overset{21}{8} 123218123218123218123151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 40.94 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1346] $1 \overset{21}{8} 1232181232^1 181232181231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 40.94 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1347] $1(12)12^1 231513218123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 40.94 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1348] $1(12)1223151321812321812^1 3218123218122 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 40.94 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1349] $1 \overset{321}{8} 12321812321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 40.94 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[1350] & \overset{321}{1\ 8\ 1232^1 181232181232181231513221(12)1} + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.94 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1343] & \overset{321}{1\ 8\ 1232^1 181232181232181231513221(12)12} + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.94 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1342] & \overset{1}{1(12)122^1 315132181232181232181232181223} + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.94 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1341] & \overset{1}{1(12)1223151321812321812321812^1 32181223} + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.94 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1351] & \overset{21}{1\ 8\ 123218123218123218123151322^1 1(12)1} + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.94 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1352] & \overset{21}{1\ 8\ 1232181232^1 181232181231513221(12)1} + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.94 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1353] & \overset{21}{3221812321812321812321812315132^1 21(12)12} + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.94 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1354] & \overset{21}{322181232^1 181232181232181231513221(12)12} + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.94 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1355] & \overset{21}{2218123218123218123218123151322^1 1(12)122} + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.94 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1356] & \overset{21}{22181232181232^1 181232181231513221(12)122} + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.94 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1375] & \overset{221}{21(12)1223151322^1 1(12)122315132181232171} + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1384] & \overset{1}{71231513221(12)122315132^1 21(12)12231513161} + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1377] \quad 71232181231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 52, T \in [41, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1374] \quad 1(12)12231513221(12)122^1 315132181232171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[1373] \quad 1(12)1223151322^1 1(12)122315132181232171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1372] \quad 812231513221(12)12231513221(12)12^1 23151315 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1371] \quad 812231513221(12)122315132^1 21(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1370] \quad 22221(12)1223151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1369] \quad 181231513221(12)12231513221(12)12^1 23151315 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1368] \quad 181231513221(12)122315132^1 21(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1367] \quad 21(12)1223151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1366] \quad 21(12)12231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1365] \quad 21(12)122^1 31513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 40.84 \rightarrow 41, T_{\max} = 51, T \in [41, 51]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1364] & \begin{matrix} 2221 & 1 \\ 1(12)1223151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1363] & \begin{matrix} 2221 & 1 \\ 1(12)12231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1362] & \begin{matrix} 221 & 1 & 221 \\ 21(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1361] & \begin{matrix} 21 & 1 & 21 \\ 21(12)12231513221(12)122315132^1 21(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1360] & \begin{matrix} 2221 & 1 & 221 \\ 1(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1359] & \begin{matrix} 2221 & 1 & 221 \\ 1(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1358] & \begin{matrix} 221 & 1 & 21 \\ 1(12)12231513221(12)122315132^1 21(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1357] & \begin{matrix} 2221 & 1 & 1 \\ 1(12)1223151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1378] & \begin{matrix} 3221 \\ 71232^1 181231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1379] & \begin{matrix} 171232181231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1380] & \begin{matrix} 1712321812315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1381] $61232181231513221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 41, \Delta = -95, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.84 \rightarrow 41, T_{\max} = 41, T \in [41, 41]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1382] $612321812315132^1 21(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 41, \Delta = -95, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.84 \rightarrow 41, T_{\max} = 41, T \in [41, 41]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1383] $71231513221(12)12231513221(12)12^1 231513161 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1387] $6131513221(12)122315132^1 21(12)122315132217 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1388] $6131513221(12)12231513221(12)12^1 2315132217 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -108, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1389] $71231513221(12)122315132^1 21(12)12231513216 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1390] $71231513221(12)12231513221(12)12^1 231513216 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1391] $1(12)12^1 2315132181232181232181232181223 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.84 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1392] $1^3 8^1 12321812321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 52, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.84 \rightarrow 41, T_{\max} = 52, T \in [41, 52]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1393] $1^3 8^1 123218123218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.84 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1394] $1^3 8^1 123218123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.84 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1395] $32218123218123218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1385] & 6131513221(12)12231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 81, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1386] & 6131513221(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 81, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1376] & 21(12)12231513221(12)122^1 315132181232171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 81, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.84 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1399] & 21(12)12231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1400] & 1812315132^1 21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1401] & 8122315132^1 21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1402] & 712315132^1 21(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1403] & 61315132^1 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1404] & 61315132^1 21(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1405] & 712315132^1 21(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -109, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1406] & 1(12)122315132181232181232^1 181232181223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 50, T \in [41, 50]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1407] & \begin{matrix} 2321 \\ 1 \ 8 \ 12^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1408] & \begin{matrix} 2321 \\ 1 \ 8 \ 123218123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1409] & \begin{matrix} 321 \\ 1 \ 8 \ 12321812^1 32181232181231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1410] & \begin{matrix} 21 \\ 1 \ 8 \ 12321812321812^1 32181231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1411] & \begin{matrix} 21 \\ 1(12)122315132181232^1 18123218123218122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1412] & \begin{matrix} 321 \\ 1 \ 8 \ 12321812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1413] & \begin{matrix} 21 \\ 1 \ 8 \ 12321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1414] & \begin{matrix} 321 \\ 3221812321812^1 32181232181231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1415] & \begin{matrix} 21 \\ 221812321812321812^1 32181231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1398] & \begin{matrix} 21 \\ 21(12)122315132^1 21(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1397] & \begin{matrix} 221 \\ 1(12)122315132^1 21(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 100, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.76 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1396] $(12)122^1 3151316131613161316131613151 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -105, \det = 100, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.76 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1453] $21812321812^1 32181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1445] $1812321812^1 32181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1455] $22181232181232181232^1 181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [1454] $322181232181232^1 181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [1448] $1^1 8 1232181232^1 181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [1452] $1^1 8 1232181232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [1451] $1^1 8 1232181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [1450] $1(12)1223151321812^1 3218123218123218122 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [1449] $1^1 8 1232181232181232^1 181231513221(12)122 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [1416] $(12)12^1 23151316131613161316131613151 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1417] $1(12)12231513221(12)122^1 31513221(12)1223 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 51, \Delta = -113, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 51, T \in [41, 51]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[1418] & \begin{matrix} 221 & 1 & 21 \\ 1(12)12231513221(12)1223151322^1 1(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1419] & \begin{matrix} 21 & 1 & 21 \\ 21(12)12231513221(12)1223151322^1 1(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1420] & \begin{matrix} 21 & 1 & 21 \\ 21(12)12231513221(12)122^1 31513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[1421] & \begin{matrix} 3221 & 1 \\ 21(12)122315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[1447] & \begin{matrix} 2321 \\ 1 \ 8 \ 1232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[1422] & \begin{matrix} 3221 & 1 \\ 21(12)12^1 231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1423] & \begin{matrix} 1 \\ 181231513221(12)122^1 31513221(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[1424] & \begin{matrix} 1 \\ 181231513221(12)1223151322^1 1(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1425] & \begin{matrix} 1 \\ 812231513221(12)122^1 31513221(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[1426] & \begin{matrix} 1 \\ 812231513221(12)1223151322^1 1(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1427] & \begin{matrix} 2221 \\ 1(12)12231513221(12)12^1 2315132181232171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1428] $1^{2221}(12)12231513221(12)122315132^1181232171 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[1429] $21^{221}(12)12231513221(12)12^12315132181232171 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1430] $21^{221}(12)12231513221(12)122315132^1181232171 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[1431] $71232181231513221(12)122^131513221(12)12 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 52, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 52, T \in [41, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1432] $71232181231513221(12)1223151322^11(12)12 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 52, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 52, T \in [41, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1433] $17123218123151322^11(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[1434] $1712321812^131513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1435] $612321812^131513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 41, \Delta = -95, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 41, T \in [41, 41]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1436] $6123218123151322^11(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 41, \Delta = -95, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 41, T \in [41, 41]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[1437] $71231513221(12)1223151322^11(12)12231513161 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

[1438] $71231513221(12)122^131513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

[1439] $6131513221(12)122^131513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1440] & \begin{matrix} 1 \\ 6131513221(12)1223151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 121, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1441] & \begin{matrix} 1 \\ 6131513221(12)122^1 31513221(12)122315132217 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 121, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[1442] & \begin{matrix} 1 \\ 6131513221(12)1223151322^1 1(12)122315132217 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 121, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1443] & \begin{matrix} 1 \\ 71231513221(12)1223151322^1 1(12)12231513216 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -109, \det = 121, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1444] & \begin{matrix} 1 \\ 71231513221(12)122^1 31513221(12)12231513216 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -109, \det = 121, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[1446] & \begin{matrix} 1 \\ 1(12)1223151321812321812^1 32181232181223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 121, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.69 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1468] & \begin{matrix} 1 \\ 1(12)122315132181232^1 181232181232181223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 169, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1478] & \begin{matrix} 1 \\ 221812321812321812321812^1 31513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 169, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1457] & \begin{matrix} 21 \\ 21(12)1223151322^1 1(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 169, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1458] & \begin{matrix} 3221 \\ 21(12)12231513221(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 51, \Delta = -113, \det = 169, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 51, T \in [41, 51] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1459] & \begin{matrix} 1 \\ 18123151322^1 1(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 169, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 48, T \in [41, 48]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1460] & 81223151322^1 1(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -110, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1461] & 712321812315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1462] & 7123151322^1 1(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1463] & 613151322^1 1(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1464] & 613151322^1 1(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1465] & 7123151322^1 1(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -109, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1466] & 181232181232^1 181232181231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1469] & 1^1 8 12321812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1470] & 1^1 8 12321812321812^1 32181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1471] & 1^1 8 12321812321812321812^1 31513221(12)122 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1472] & 1(12)122315132^1 18123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.59 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1473] $1 \overset{321}{8} 12321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.59 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1474] $1 \overset{21}{8} 12321812321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.59 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1475] $2181232181232^1 181232181231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.59 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

[1476] $21812321812321812321812315132^1 21(12)1223 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.59 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[1456] $1(12)1223151322^1 1(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 51, \Delta = -113, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.59 \rightarrow 41, T_{\max} = 51, T \in [41, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1477] $3221812321812321812^1 32181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.59 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1467] $1 \overset{1}{8} 12321812321812321812315132^1 21(12)1223 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 50, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.59 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[1479] $1(12)12231513221(12)12^1 231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 51, \Delta = -113, \det = 196, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.55 \rightarrow 41, T_{\max} = 51, T \in [41, 51]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1480] $21(12)12231513221(12)12^1 231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 51, \Delta = -113, \det = 196, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.55 \rightarrow 41, T_{\max} = 51, T \in [41, 51]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1481] $21(12)1223151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 51, \Delta = -113, \det = 196, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.55 \rightarrow 41, T_{\max} = 51, T \in [41, 51]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1482] $181231513221(12)12^1 231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -110, \det = 196, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.55 \rightarrow 41, T_{\max} = 48, T \in [41, 48]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1483] $812231513221(12)12^1 231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 36, T_{\max} = 48, \Delta = -110, \det = 196, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 40.55 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1484] & 71232181231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 196, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.55 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1485] & 71231513221(12)12^1 231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 196, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.55 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1486] & 6131513221(12)12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 196, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.55 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1487] & 6131513221(12)12^1 231513221(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -108, \det = 196, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.55 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1488] & 71231513221(12)12^1 231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 48, \Delta = -109, \det = 196, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.55 \rightarrow 41, T_{\max} = 48, T \in [41, 48] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1489] & 1(12)1223151321812^1 32181232181232181223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 196, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.55 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[1490] & 1 \overset{2321}{8} 1232181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 196, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.55 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[1491] & 1 \overset{321}{8} 1232181232181232^1 181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 196, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.55 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[1492] & 1 \overset{321}{8} 1232181232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 196, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.55 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[1493] & 322181232181232181232^1 181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 196, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.55 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[1496] & 218123218123218123218123151322^1 1(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 225, (n_+, n_-, n_0) = \\
& (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.52 \rightarrow 41, T_{\max} = 50, T \in [41, 50]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1495] & \overset{1}{18123218123218123218123151322^1}(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 225, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.52 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1494] & 712321812^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 225, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.52 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[1502] & 3221812321812321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.48 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[1501] & \overset{321}{1} 8 12321812321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.48 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[1500] & \overset{321}{1} 8 12321812321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.48 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[1499] & \overset{2321}{1} 8 12321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 256, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.48 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[1498] & \overset{1}{1}(12)122315132^1 181232181232181232181223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.48 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1497] & \overset{3221}{7123218123151322^1}(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 256, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.48 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1505] & 21812321812321812^1 32181231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.46 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1504] & \overset{2321}{1} 8 1232181232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 289, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.46 \rightarrow 41, T_{\max} = 52, T \in [41, 52] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1503] & \overset{1}{1812321812321812^1} 32181231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 40.46 \rightarrow 41, T_{\max} = 50, T \in [41, 50] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1506] $\overset{1}{181232181232181232^1 181231513221(12)1223} + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 361, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.41 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[1507] $\overset{2321}{1 \ 8 \ 12321812321812321812^1 31513221(12)1} + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 52, \Delta = -114, \det = 361, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.41 \rightarrow 41, T_{\max} = 52, T \in [41, 52]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$

[1508] $2181232181232181232^1 181231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 361, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.41 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[1509] $\overset{1}{1812321812321812321812^1 31513221(12)1223} + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 529, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.34 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$

[1510] $21812321812321812321812^1 31513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 36, T_{\max} = 50, \Delta = -112, \det = 529, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 40.34 \rightarrow 41, T_{\max} = 50, T \in [41, 50]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$

$T = 37$ [1521] $2181232181231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44, T_{\max} = 49, T \in [44, 49]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[1512] $\overset{1}{181232181231513221(12)12231513221} \overset{22^1 1}{(12)12} + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44, T_{\max} = 49, T \in [44, 49]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[1513] $\overset{1}{181232181231513221(12)12231513221(12)1222^1 2} + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44, T_{\max} = 49, T \in [44, 49]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[1514] $\overset{2^1 1}{3221(12)12231513221(12)122315132181232181} + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43, T_{\max} = 52, T \in [43, 52]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 6 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[1515] $\overset{22^1 1}{1(12)12231513221(12)12231513218123218123} + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44, T_{\max} = 52, T \in [44, 52]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1516] $\overset{22^1 21}{1(12)12231513221(12)1223151321812321812} + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44, T_{\max} = 49, T \in [44, 49]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

- [1517] $21^{22^1 1}(12)12231513221(12)1223151321812321812 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44, T_{\max} = 49, T \in [44, 49]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1518] $413221^1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 51, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1519] $32181232181231513221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44, T_{\max} = 52, T \in [44, 52]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1520] $232181232181231513221(12)12231513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 54, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44, T_{\max} = 54, T \in [44, 54]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [1511] $181232181231513221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44, T_{\max} = 49, T \in [44, 49]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1522] $413221^1(12)12231513221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 51, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1523] $181232181231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -9, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.22 \rightarrow 44, T_{\max} = 49, T \in [44, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1524] $181232181231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -9, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.22 \rightarrow 44, T_{\max} = 49, T \in [44, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1525] $21^{32^1 21}(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.22 \rightarrow 44, T_{\max} = 52, T \in [44, 52]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [1527] $1^{222^1 1}(12)12231513221(12)1223151321812321812 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -9, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.22 \rightarrow 44, T_{\max} = 49, T \in [44, 49]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$
- [1528] $21^{21}(12)12231513221(12)1223151321812321812^1 3 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.22 \rightarrow 44, T_{\max} = 52, T \in [44, 52]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1529] $32^1 181232181231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -9, (n_+, n_-, n_0) =$

$(1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 52$, $T \in [44, 52]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[1530] $32181232181231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 52$, $\Delta = -113$, $\det = -9$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 52$, $T \in [44, 52]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[1531] $2181232181231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -9$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 49$, $T \in [44, 49]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[1532] $413221(12)^1 12231513221(12)^1 12231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -9$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[1526] $1(12)^{221} 12231513221(12)1223151321812321812^1 3 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 52$, $\Delta = -113$, $\det = -9$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 52$, $T \in [44, 52]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[1533] $181232181231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -16$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.75 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1534] $2181232181231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -16$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.75 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1553] $4132^1 21(12)^1 12231513221(12)^1 12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.44 \rightarrow 43$, $T_{\max} = 51$, $T \in [43, 51]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1562] $4132^1 181232181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 50$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.44 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1536] $181232181231513221(12)122315132^1 21(12)^1 1222 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.44 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1537] $181232181231513221(12)122315132^1 21(12)^{221} 12 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.44 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1538] $1812^1 32181231513221(12)12231513221(12)^{221} 12 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.44 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[1539] $181232181231513221(12)122315132^1 21(12)12222+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1540] $1812^1 32181231513221(12)12231513221(12)12222+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[1541] $21(12)12231513221(12)122315132181232181+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 52$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1542] $1(12)12231513221(12)1223151321812321812^1 32+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 54$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 54, T \in [43, 54]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[1543] $1(12)122^1 31513221(12)1223151321812321812+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[1544] $1(12)12231513221(12)122315132181232^1 1812+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1545] $21(12)12231513221(12)122315132181232^1 1812+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1546] $21(12)122^1 31513221(12)1223151321812321812+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[1547] $4132^1 21(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1548] $4132^1 21(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1535] $1812^1 32181231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[1552] 2181232181231513221(12)122315132¹21(12)12222+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1549] 6131513221(12)12231513221(12)122315132¹181+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1550] 232¹181232181231513221(12)12231513221(12)122+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 54$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 54, T \in [43, 54]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1551] 21812¹32181231513221(12)12231513221(12)12222+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[1554] 612¹31513221(12)12231513221(12)122315132181+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -108$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1555] 6131513221(12)12231513221(12)122315132¹1812+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1556] 6131513221(12)12231513221(12)122315132¹2181+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1557] 71231513221(12)12231513221(12)122315132¹171+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1558] 812231513221(12)12231513221(12)122315132¹16+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -108$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1559] 71231513221(12)12231513221(12)122315132¹217+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1560] 4132¹21(12)12231513221(12)12231513218123217+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 52$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.44 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[1561] $4132^1 181232181232181232181231513221(12)1+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 50, \Delta = -111, \det = -25, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.44 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1584] $413221(12)12231513221(12)122315132181232^1 17+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1563] $181232181231513221(12)1223151322^1 1(12)1222+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1564] $181232181231513221(12)1223151322^1 1(12)12+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1565] $181232181231513221(12)1223151322^1 1(12)12222+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1566] $21(12)12231513221(12)122315132181232^1 181+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1567] $1(12)122^1 31513221(12)12231513218123218123+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1568] $1(12)12^1 231513221(12)1223151321812321812+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[1569] $21(12)122^1 31513221(12)12231513218123218123+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1570] $21(12)12^1 231513221(12)1223151321812321812+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[1571] $413221(12)12231513221(12)122315132^1 21(12)1+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 51, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1572] $413221(12)12231513221(12)122315132^1 21(12)12+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 51, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1573] $32181232181231513221(12)122315132^1 21(12)1222+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1574] $2181232181231513221(12)1223151322^1 1(12)12222+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1575] $413221(12)12231513221(12)122315132^1 21(12)1222+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 51, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1576] $1812^1 31513221(12)12231513221(12)12231513161+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1577] $61231513221(12)12231513221(12)122315132^1 181+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -108, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1578] $8122^1 31513221(12)12231513221(12)12231513161+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1579] $6131513221(12)12231513221(12)1223151322^1 181+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1580] $712^1 31513221(12)12231513221(12)122315132171+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1581] $8122^1 31513221(12)12231513221(12)12231513216+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -108, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1582] $712^1 31513221(12)12231513221(12)122315132217+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.22 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1583] $71231513221(12)12231513221(12)1223151322^1 17+ \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -36, (n_+, n_-, n_0) =$

$(1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.22 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1618] $812^1 231513221(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -108$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1619] $812231513221(12)12231513221(12)122^1 31513216 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -108$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1586] $181232181231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1587] $181232181231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1588] $181232^1 181231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[1589] $181232181231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1590] $181232^1 181231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[1591] $1(12)12^1 231513221(12)12231513218123218123 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 52$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 52$, $T \in [43, 52]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1592] $1(12)122^1 31513221(12)122315132181232181232 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 54$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 54$, $T \in [43, 54]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1593] $1(12)12231513221(12)1223151321812^1 321812 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.06 \rightarrow 43$, $T_{\max} = 49$, $T \in [43, 49]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[1594] $1(12)122315132^1 21(12)1223151321812321812 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 49, T \in [43, 49] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[1595] & 21(12)12^{21}231513221(12)12231513218123218123+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 52, T \in [43, 52] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1596] & 21(12)12231513221(12)1223151321812^{21}321812+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -49, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 49, T \in [43, 49] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 62 & 14 \\ 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[1597] & 21(12)122315132^{21}21(12)1223151321812321812+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -49, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 49, T \in [43, 49] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[1598] & 41322^1(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 51, \Delta = -112, \det = -49, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1599] & 413221(12)12231513221(12)1223151322^1(12)1+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 51, \Delta = -112, \det = -49, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1600] & 41322^1(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 51, \Delta = -112, \det = -49, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1601] & 413221(12)12231513221(12)1223151322^1(12)12+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 51, \Delta = -112, \det = -49, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1602] & 6131513221(12)12231513221(12)122^1315132181+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 49, T \in [43, 49] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1603] & 32181232181231513221(12)1223151322^1(12)1222+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -49, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 52, T \in [43, 52] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[1604] & 232181232181231513221(12)122315132^121(12)122+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -49, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 54, T \in [43, 54] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[1605] & 2181232181231513221(12)122^131513221(12)12222+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -49, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.06 \rightarrow 43, T_{\max} = 49, T \in [43, 49] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1606] $2181232^1 181231513221(12)12231513221(12)12222+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.06 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [1607] $61316131513221(12)12231513221(12)122^1 3151315+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 50, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.06 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1608] $41322^1 1(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 51, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.06 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1609] $413221(12)12231513221(12)1223151322^1 1(12)1222+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 51, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.06 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1610] $181231513221(12)12231513221(12)122^1 31513161+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.06 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1611] $612315132^1 21(12)12231513221(12)122315132181+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.06 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1612] $6131513221(12)12231513221(12)122^1 3151321812+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.06 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1613] $812^1 31513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 54, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.06 \rightarrow 43, T_{\max} = 54, T \in [43, 54]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1614] $812231513221(12)12231513221(12)122^1 31513161+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.06 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1615] $812^1 231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.06 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1616] $6131513221(12)12231513221(12)122^1 3151322181+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.06 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1620] $71231513221(12)12231513221(12)122^1 315132217+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -49, (n_+, n_-, n_0) =$

$$(1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.06 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[1621] \quad 41322^1 1(12)12231513221(12)12231513218123217 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.06 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1622] \quad 41321812^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.06 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[1623] \quad 41321812^1 32181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.06 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[1617] \quad 71231513221(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.06 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1585] \quad 181232^1 181231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.06 \rightarrow 43, T_{\max} = 49, T \in [43, 49]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[1626] \quad 4132181232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -64, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.94 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1627] \quad 41321812321812321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -64, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.94 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1628] \quad 4132181232^1 181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -64, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.94 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1629] \quad 41321812321812321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -64, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.94 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1625] \quad 232181232181231513221(12)1223151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.94 \rightarrow 42, T_{\max} = 54, T \in [42, 54]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1624] \quad 1(12)12^1 231513221(12)122315132181232181232 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.94 \rightarrow 42, T_{\max} = 54, T \in [42, 54]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1643] & 21(12)^{221}12231513221(12)122^13151321812321812+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1630] & 21(12)^{3221}122^131513221(12)^112231513221(12)12231+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 42, \Delta = -93, \det = -81, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.84 \rightarrow 42, T_{\max} = 42, T \in [42, 42] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1634] & 181232181231513221(12)12^1231513221(12)^{221}12+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1635] & 1812321812315132^121(12)12231513221(12)12222+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1636] & 181232181231513221(12)12^1231513221(12)12222+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1637] & 21(12)^{3221}12231513221(12)1223151321812^132181+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -81, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.84 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[1638] & 21(12)^{3221}122^131513221(12)122315132181232181+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -81, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.84 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1639] & 1(12)^{221}12231513221(12)122315132181232^118123+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -81, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.84 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1640] & 1(12)^{2221}12231513221(12)122^13151321812321812+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1641] & 1(12)^{221}1223151322^11(12)1223151321812321812+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1642] & 21(12)^{21}12231513221(12)122315132181232^118123+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -81, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.84 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1632] $181232181231513221(12)12^1 231513221(12)1222+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1644] $21(12)1223151322^1 1(12)1223151321812321812+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1645] $413221(12)122^1 31513221(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 51, \Delta = -112, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1646] $413221(12)122^1 31513221(12)12231513221(12)12+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 51, \Delta = -112, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1647] $6131513221(12)12231513221(12)12^1 2315132181+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1648] $6131513221(12)122315132^1 21(12)122315132181+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1649] $1712321812315132^1 21(12)12231513221(12)122313+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 42, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 42, T \in [42, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1650] $171232181231513221(12)12^1 231513221(12)122313+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 42, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 42, T \in [42, 42]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1651] $321812^1 32181231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1652] $21812321812315132^1 21(12)12231513221(12)12222+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1653] $2181232181231513221(12)12^1 231513221(12)12222+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1654] $61316131513221(12)12231513221(12)12^1 23151315+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 50, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1655] 61316131513221(12)122315132¹21(12)1223151315+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 50$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1656] 413221(12)122¹31513221(12)12231513221(12)1222+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -81$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1657] 181231513221(12)122315132¹21(12)12231513161+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1658] 181231513221(12)12231513221(12)12¹231513161+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1659] 6123151322¹1(12)12231513221(12)122315132181+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -108$, $\det = -81$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1660] 61231513221(12)122¹31513221(12)122315132181+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -108$, $\det = -81$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1661] 6131513221(12)122315132¹21(12)1223151321812+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1662] 6131513221(12)12231513221(12)12¹23151321812+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1663] 812231513221(12)122315132¹21(12)12231513161+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1664] 812231513221(12)12231513221(12)12¹231513161+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1665] 6131513221(12)122315132¹21(12)1223151322181+ $\hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

- [1666] $6131513221(12)12231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1667] $71231513221(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1668] $71231513221(12)12231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1669] $812231513221(12)122315132^1 21(12)12231513216 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1670] $812231513221(12)12231513221(12)12^1 231513216 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1671] $71231513221(12)12231513221(12)12^1 2315132217 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1672] $71231513221(12)122315132^1 21(12)122315132217 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1673] $413221(12)122^1 31513221(12)12231513218123217 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1674] $413221(12)12231513221(12)1223151321812^1 3217 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [1675] $413218123218123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 50, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1676] $413218123218123218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 50, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.84 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1631] $1812321812315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) =$

$$(1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1633] \text{ } 1812321812315132^1 21(12)12231513221^{221}(12)12+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.84 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1677] (12)122^1 31513161316131613161316131613161+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -102, \det = -100, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 189 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1678] (12)122^1 31513161316131613161316131612+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -102, \det = -100, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 189 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1679] 21(12)12231513221^1 (12)122^1 31513221(12)12231+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 42, \Delta = -93, \det = -100, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 42, T \in [42, 42]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1680] 1(12)122315132^1 21(12)12231513218123218123+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1681] 21(12)122315132^1 21(12)12231513218123218123+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1682] 413221^1 (12)12231513221^1 (12)122^1 31513221^{221}(12)1+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 51, \Delta = -112, \det = -100, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1683] 413221^1 (12)12231513221^1 (12)122^1 31513221^{21}(12)12+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 51, \Delta = -112, \det = -100, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1684] 61315132^1 21(12)12231513221(12)122315132181+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1685] 32181232181231513221(12)122^1 31513221(12)1222+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1686] 413221^1 (12)12231513221^1 (12)122^1 31513221(12)1222+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 51, \Delta = -112, \det = -100, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 51, T \in [42, 51]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1687] $1812315132^1 21(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1688] $61231513221(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -108$, $\det = -100$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1689] $61315132^1 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1690] $8122315132^1 21(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1691] $61315132^1 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1692] $712315132^1 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1693] $8122315132^1 21(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -108$, $\det = -100$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1694] $712315132^1 21(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1695] $41321812321812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 50$, $\Delta = -111$, $\det = -100$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1696] $41321812321812^1 32181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 50$, $\Delta = -111$, $\det = -100$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.76 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1732] $6131513221(12)122^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

- [1747] $4132181232181232^1 181232181231513221(12)12+\hat{1}\rightarrow 2$ $T = 37$, $T_{\max} = 50$, $\Delta = -111$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 50$, $T \in [42, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [1734] $812231513221(12)1223151322^1 1(12)12231513161+\hat{1}\rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [1733] $81231513221(12)12231513221(12)122^1 315131613+\hat{1}\rightarrow 2$ $T = 37$, $T_{\max} = 54$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 54$, $T \in [42, 54]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1736] $6131513221(12)122^1 31513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [1731] $6131513221(12)1223151322^1 1(12)1223151321812+\hat{1}\rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [1730] $61231513221(12)122315132^1 21(12)122315132181+\hat{1}\rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -108$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1729] $61231513221(12)12^1 231513221(12)122315132181+\hat{1}\rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -108$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1728] $181231513221(12)122^1 31513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1727] $181231513221(12)1223151322^1 1(12)12231513161+\hat{1}\rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1726] $413221(12)122315132^1 21(12)12231513221(12)1222+\hat{1}\rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 51$, $T \in [42, 51]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1725] $413221(12)12^1 231513221(12)12231513221(12)1222+\hat{1}\rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 51$, $T \in [42, 51]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1724] $61316131513221(12)1223151322^1 1(12)1223151315+\hat{1}\rightarrow 2$ $T = 37$, $T_{\max} = 50$, $\Delta = -111$, $\det = -121$, $(n_+, n_-, n_0) =$

$(1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 50$, $T \in [42, 50]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1723] $61316131513221(12)122^1 31513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 50$, $\Delta = -111$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 50$, $T \in [42, 50]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1722] $21812321812^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1721] $218123218123151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -104$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[1720] $1712321812^1 31513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 42$, $\Delta = -93$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 42$, $T \in [42, 42]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1719] $17123218123151322^1 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 42$, $\Delta = -93$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 42$, $T \in [42, 42]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[1718] $6131513221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1717] $6131513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1716] $413221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 51$, $T \in [42, 51]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1715] $413221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 51$, $T \in [42, 51]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1714] $413221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 51$, $T \in [42, 51]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1713] $413221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -121$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.69 \rightarrow 42$, $T_{\max} = 51$, $T \in [42, 51]$

[1701] $1812321812^1 131513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1700] $21(12)12^{3221} 12^1 231513221(12)12231513221(12)12231+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 42, \Delta = -93, \det = -121, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 42, T \in [42, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1699] $21(12)122315132^{3221} 21^1(12)12231513221(12)12231+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 42, \Delta = -93, \det = -121, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 42, T \in [42, 42]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1698] $(12)12^1 231513161316131613161316131612+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 230 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$

[1697] $(12)12^1 23151316131613161316131613161+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 230 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$

[1737] $6131513221(12)1223151322^1 1(12)1223151322181+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

[1738] $71231513221(12)122^1 31513221(12)122315132171+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

[1739] $71231513221(12)1223151322^1 1(12)122315132171+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

[1740] $812231513221(12)122^1 31513221(12)12231513216+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1741] $812231513221(12)1223151322^1 1(12)12231513216+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1742] $71231513221(12)122^1 31513221(12)122315132217+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1743] $71231513221(12)1223151322^1 1(12)122315132217+\hat{1} \rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -121, (n_+, n_-, n_0) =$

$(1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

$[1744] \overset{1}{413221}(12)12^1 231513221(12)12231513218123217 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -121, (n_+, n_-, n_0) =$
 $(1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

$[1745] \overset{1}{413221}(12)122315132^1 21(12)12231513218123217 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -121, (n_+, n_-, n_0) =$
 $(1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

$[1746] \overset{1}{4132181232181232^1 181232181231513221}(12)1 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -121, (n_+, n_-, n_0) =$
 $(1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

$[1735] \overset{1}{812231513221}(12)122^1 31513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -107, \det = -121, (n_+, n_-, n_0) =$
 $(1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.69 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

$[1748] \overset{221}{1}(12)12231513221(12)1223151321812^1 3218123 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -144, (n_+, n_-, n_0) =$
 $(1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.64 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

$[1749] \overset{21}{21}(12)12231513221(12)1223151321812^1 3218123 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -144, (n_+, n_-, n_0) =$
 $(1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.64 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 12 & 2 \\ 12 & 83 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

$[1750] \overset{1}{32181232^1 181231513221}(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -144, (n_+, n_-, n_0) =$
 $(1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.64 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 12 & 2 \\ 12 & 203 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

$[1759] \overset{1}{613151322^1 1}(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -107, \det = -169, (n_+, n_-, n_0) =$
 $(1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

$[1752] \overset{3221}{21}(12)12231513221(12)122^1 315132181232181 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -169, (n_+, n_-, n_0) =$
 $(1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

$[1753] \overset{221}{1}(12)1223151322^1 1(12)12231513218123218123 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -169, (n_+, n_-, n_0) =$
 $(1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

$[1754] \overset{21}{1}(12)12231513221(12)122315132181232^1 181232 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -169, (n_+, n_-, n_0) =$
 $(1, 37, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1755] & 1(12)122315132^1 21(12)122315132181232181232 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -169, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1758] & 413221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 51, \Delta = -112, \det = -169, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 51, T \in [42, 51] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1751] & 21(12)12231513221(12)12^1 231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 42, \Delta = -93, \det = -169, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 42, T \in [42, 42] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1760] & 32181232181231513221(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -169, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[1761] & 232181232181231513221(12)122^1 31513221(12)122 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -169, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1762] & 2321812^1 32181231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -169, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[1763] & 613161315132^1 21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -169, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 50, T \in [42, 50] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1764] & 413221(12)12231513221(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 51, \Delta = -112, \det = -169, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 51, T \in [42, 51] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1765] & 18123151322^1 1(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -107, \det = -169, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1766] & 61231513221(12)12231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -108, \det = -169, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1767] & 613151322^1 1(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -107, \det = -169, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.59 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1768] $81231513221^1(12)122315132^121(12)122315131613+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 54, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 54, T \in [42, 54]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1769] $812315132^121(12)12231513221(12)122315131613+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 54, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 54, T \in [42, 54]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1770] $81223151322^11(12)12231513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1771] $613151322^11(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1772] $7123151322^11(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1773] $81223151322^11(12)12231513221(12)12231513216+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -108, \det = -169, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1774] $7123151322^11(12)12231513221(12)122315132217+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 49, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 49, T \in [42, 49]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1775] $413221^1(12)12231513221(12)122^131513218123217+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1776] $41321812321812321812^132181231513221(12)1+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 50, \Delta = -111, \det = -169, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1777] $41321812321812321812^132181231513221(12)12+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 50, \Delta = -111, \det = -169, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 50, T \in [42, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1756] $21(12)1223151322^11(12)12231513218123218123+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.59 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1757] $413221^1(12)12231513221(12)12^1231513221(12)1+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 51, \Delta = -112, \det = -169, (n_+, n_-, n_0) =$

(1, 37, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 41.59 \rightarrow 42$, $T_{\max} = 51$, $T \in [42, 51]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1780] $1(12)12231513221(12)122^1 31513218123218123 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 52$, $\Delta = -113$, $\det = -196$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.55 \rightarrow 42$, $T_{\max} = 52$, $T \in [42, 52]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1781] $21(12)12231513221(12)122^1 31513218123218123 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 52$, $\Delta = -113$, $\det = -196$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.55 \rightarrow 42$, $T_{\max} = 52$, $T \in [42, 52]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1782] $413221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -196$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.55 \rightarrow 42$, $T_{\max} = 51$, $T \in [42, 51]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1783] $413221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -196$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.55 \rightarrow 42$, $T_{\max} = 51$, $T \in [42, 51]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1784] $6131513221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -196$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.55 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1785] $321812321812315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 52$, $\Delta = -113$, $\det = -196$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.55 \rightarrow 42$, $T_{\max} = 52$, $T \in [42, 52]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[1779] $21(12)1223151322^1 1(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 52$, $\Delta = -113$, $\det = -196$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.55 \rightarrow 42$, $T_{\max} = 52$, $T \in [42, 52]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[1787] $413221(12)1223151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 51$, $\Delta = -112$, $\det = -196$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.55 \rightarrow 42$, $T_{\max} = 51$, $T \in [42, 51]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1788] $181231513221(12)12^1 231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -196$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.55 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1789] $61231513221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -108$, $\det = -196$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 41.55 \rightarrow 42$, $T_{\max} = 49$, $T \in [42, 49]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1790] $6131513221(12)12^1 231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 37$, $T_{\max} = 49$, $\Delta = -107$, $\det = -196$, $(n_+, n_-, n_0) = (1, 37, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1791] & 81231513221(12)^1 12231513221(12)12^1 2315131613 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -196, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1792] & 812231513221(12)^1 12^1 231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -107, \det = -196, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1793] & 6131513221(12)^1 12^1 231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -107, \det = -196, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1794] & 71231513221(12)^1 12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -107, \det = -196, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1795] & 812231513221(12)^1 12^1 231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -108, \det = -196, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1796] & 71231513221(12)^1 12^1 231513221(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 49, \Delta = -107, \det = -196, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 49, T \in [42, 49] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1797] & 413221(12)^1 1223151322^1 1(12)12231513218123217 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -196, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1798] & 4132181232181232181232^1 181231513221(12)^1 1 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -196, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 50, T \in [42, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[1799] & 4132181232181232181232^1 181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -196, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 50, T \in [42, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[1778] & 21(12)^{3221} 1223151322^1 1(12)^1 12231513221(12)12231 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 42, \Delta = -93, \det = -196, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 42, T \in [42, 42] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1786] & 61316131513221(12)12^1 231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -196, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.55 \rightarrow 42, T_{\max} = 50, T \in [42, 50]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1800] & \begin{matrix} 3221 \\ 21(12)12231513221(12)122315132^1 181232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -225, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.52 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[1801] & \begin{matrix} 1 \\ 81231513221(12)122^1 31513221(12)122315131613 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -225, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.52 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[1802] & \begin{matrix} 1 \\ 413221(12)12231513221(12)122315132^1 18123217 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -225, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.52 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1803] & \begin{matrix} 3221 \\ 21(12)12231513221(12)12^1 2315132181232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -256, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.48 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1804] & \begin{matrix} 1 \\ 6131613151322^1 1(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -256, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.48 \rightarrow 42, T_{\max} = 50, T \in [42, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1805] & \begin{matrix} 1 \\ 81231513221(12)1223151322^1 1(12)122315131613 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -256, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.48 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1806] & \begin{matrix} 1 \\ 413221(12)12231513221(12)12^1 231513218123217 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -256, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.48 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1807] & \begin{matrix} 1 \\ 41321812321812321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -256, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.48 \rightarrow 42, T_{\max} = 50, T \in [42, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[1808] & \begin{matrix} 1 \\ 41321812321812321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 37, T_{\max} = 50, \Delta = -111, \det = -256, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.48 \rightarrow 42, T_{\max} = 50, T \in [42, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[1809] & \begin{matrix} 221 \\ 1(12)12231513221(12)12^1 231513218123218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 37, T_{\max} = 52, \Delta = -113, \det = -289, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.46 \rightarrow 42, T_{\max} = 52, T \in [42, 52] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[1810] & \begin{matrix} 21 \\ 1(12)12231513221(12)1223151321812^1 32181232 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -289, (n_+, n_-, n_0) = \\
& (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.46 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 203 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1811] $1(12)1223151322^1 1(12)122315132181232181232+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 54, \Delta = -115, \det = -289, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.46 \rightarrow 42, T_{\max} = 54, T \in [42, 54]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[1812] $21(12)12231513221(12)12^1 231513218123218123+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.46 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[1813] $3218123218123151322^1 1(12)12231513221(12)1222+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.46 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 390 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0}$

[1814] $232181232^1 181231513221(12)12231513221(12)122+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 54, \Delta = -115, \det = -289, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.46 \rightarrow 42, T_{\max} = 54, T \in [42, 54]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 373 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$

[1815] $232181232181231513221(12)12^1 231513221(12)122+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 54, \Delta = -115, \det = -289, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.46 \rightarrow 42, T_{\max} = 54, T \in [42, 54]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$

[1816] $8123151322^1 1(12)12231513221(12)122315131613+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 54, \Delta = -115, \det = -289, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.46 \rightarrow 42, T_{\max} = 54, T \in [42, 54]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[1817] $1(12)12231513221(12)122315132^1 18123218123+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -324, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.43 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1819] $321812321812^1 31513221(12)12231513221(12)1222+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -324, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.43 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 449 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[1818] $21(12)12231513221(12)122315132^1 18123218123+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 52, \Delta = -113, \det = -324, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.43 \rightarrow 42, T_{\max} = 52, T \in [42, 52]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1820] $1(12)12231513221(12)122^1 315132181232181232+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 54, \Delta = -115, \det = -361, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.41 \rightarrow 42, T_{\max} = 54, T \in [42, 54]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

[1821] $2321812321812315132^1 21(12)12231513221(12)122+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 54, \Delta = -115, \det = -361, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.41 \rightarrow 42, T_{\max} = 54, T \in [42, 54]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[1822] $81231513221(12)12^1 231513221(12)122315131613+\hat{1}\rightarrow 2$ $T = 37, T_{\max} = 54, \Delta = -115, \det = -361, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 41.41 \rightarrow 42, T_{\max} = 54, T \in [42, 54]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[1823] & 1^{21}(12)12231513221(12)12^{12315132181232181232+\hat{1} \rightarrow 2} \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -529, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.34 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 390 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0} \\
[1824] & 23218123218123151322^1 1(12)12231513221(12)122+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -529, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.34 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 666 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0} \\
[1826] & 2321812321812^1 31513221(12)12231513221(12)122+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -625, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.31 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 25 & 2 \\ 25 & 799 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0} \\
[1825] & 1^{21}(12)12231513221(12)122315132^1 181232181232+\hat{1} \rightarrow 2 \quad T = 37, T_{\max} = 54, \Delta = -115, \det = -625, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 41.31 \rightarrow 42, T_{\max} = 54, T \in [42, 54] \\
& \begin{pmatrix} 0 & 25 & 2 \\ 25 & 449 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
\\
T = 38 & [1833] 21^{22^1 1}(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45, T_{\max} = 50, T \in [45, 50] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1835] & 41413221(12)12231513221^1(12)12231513221(12)122^1 2+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1834] & 41413221(12)12231513221^1(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1832] & 1^{22^1 21}(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45, T_{\max} = 50, T \in [45, 50] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1831] & 1^{1513221}(12)12231513221^1(12)12231513221(12)122^1 2+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45, T_{\max} = 50, T \in [45, 50] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1830] & 1^{31513221}(12)12231513221^1(12)12^1 2+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 55, \Delta = -115, \det = 4, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45, T_{\max} = 55, T \in [45, 55] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1829] & 1^{22^1 31513221}(12)12231513221^1(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 4, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45, T_{\max} = 58, T \in [45, 58]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1828] & 1(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45, T_{\max} = 50, T \in [45, 50] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1827] & 21(12)12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45, T_{\max} = 50, T \in [45, 50] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1837] & 21(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.22 \rightarrow 45, T_{\max} = 50, T \in [45, 50] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1836] & 513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.33 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 15 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1838] & 1(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 9, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.22 \rightarrow 45, T_{\max} = 58, T \in [45, 58] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1839] & 1(12)12231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.22 \rightarrow 45, T_{\max} = 50, T \in [45, 50] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1840] & 1(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.22 \rightarrow 45, T_{\max} = 50, T \in [45, 50] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1841] & 1(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.33 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[1842] & 21(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.33 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[1843] & 41413221(12)12231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.22 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1845] & 1(12)122315132181232181232181232^1 15 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 16, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.75 \rightarrow 44, T_{\max} = 50, T \in [44, 50]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[1846] & 512^1 32181232181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 16, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.75 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& 513221 \\
[1844] & 21(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 16, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.75 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
& 15132^1 21 \\
[1854] & 71232181231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 25, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
& 1 \\
[1866] & 4132^1 21(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 52, \Delta = -112, \det = 25, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.44 \rightarrow 44, T_{\max} = 52, T \in [44, 52] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
& 1 \\
[1855] & 71231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 25, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& 1 \\
[1856] & 181231513221(12)12231513221(12)122315132^1 171 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 25, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1867] & 61316131513221(12)12231513221(12)122315132^1 16 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -109, \det = 25, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
& 221 \\
[1853] & 21(12)122^1 31513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 25, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& 221 \\
[1852] & 1(12)122^1 31513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 25, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& 15132^1 21 \\
[1851] & 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 25, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
& 15132^1 21 \\
[1850] & 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 25, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[1849] $1(12)12231513221(12)12231513221(12)122^1 3151 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1848] $1(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 58, \Delta = -116, \det = 25, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.44 \rightarrow 44, T_{\max} = 58, T \in [44, 58]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

[1847] $21(12)12231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1857] $7122^1 31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[1858] $71231513221(12)12231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1859] $812231513221(12)12231513221(12)122315132^1 171 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1860] $71231513221(12)12231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1861] $812231513221(12)12231513221(12)122315132^1 217 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[1862] $1(12)1223151321812321812321812321812^1 3215 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1863] $1(12)1223151321812321812321812321812^1 3151 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1864] $51232^1 181232181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -108, \det = 25, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.44 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1865] $15132^1 181232181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 25, (n_+, n_-, n_0) =$

(1, 38, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 43.44 \rightarrow 44$, $T_{\max} = 50$, $T \in [44, 50]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1886] $41413221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 51$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1889] $41413221(12)12231513221(12)122315132181232^1 17 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 52$, $\Delta = -112$, $\det = 36$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 52$, $T \in [44, 52]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1868] $21(12)12231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 50$, $T \in [44, 50]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1869] $1(12)122^1 31513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 50$, $T \in [44, 50]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1870] $1(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 50$, $T \in [44, 50]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1871] $1(12)12231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 50$, $T \in [44, 50]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1872] $1(12)12^1 231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 50$, $T \in [44, 50]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1873] $21(12)12^1 231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 50$, $T \in [44, 50]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1888] $413221(12)12231513221(12)122315132181232^1 181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 52$, $\Delta = -112$, $\det = 36$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 52$, $T \in [44, 52]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1874] $712^1 32181231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 36$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.22 \rightarrow 44$, $T_{\max} = 50$, $T \in [44, 50]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1875] $712^1 31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1876] & 1812^1 31513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1877] & 712^1 231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1887] & 41413221(12)12231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 36, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1878] & 712231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1879] & 712^1 31513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1880] & 8122^1 31513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1881] & 712^1 31513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1882] & 71231513221(12)12231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1883] & 8122^1 31513221(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1884] & 812231513221(12)12231513221(12)1223151322^1 17 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 36, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1885] & 41413221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 36, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.22 \rightarrow 44, T_{\max} = 51, T \in [44, 51]
\end{aligned}$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1919] $414132^1 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 51$, $\Delta = -111$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1904] $21(12)122315132^1 21(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1890] $21(12)12231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1891] $21(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 83 & 17 \\ 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[1892] $1(12)12231513221(12)12231513221(12)122^1 315 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 58$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 58, T \in [44, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1893] $1(12)122^1 31513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 55$, $\Delta = -115$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 55, T \in [44, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1928] $414132^1 21(12)12231513221(12)12231513218123217 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 52$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 52, T \in [44, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1894] $1(12)12231513221(12)12231513221(12)12^1 23151 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1895] $1(12)12^1 231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1896] $1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[1897] $1(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 55 & 13 \\ 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[1898] $1 \begin{smallmatrix} 31513221 \\ (12) \end{smallmatrix} 12231513221 \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 122315132^1 21(12)122+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 55, \Delta = -115, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.06 \rightarrow 44, T_{\max} = 55, T \in [44, 55]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1899] $1 \begin{smallmatrix} 151322^1 1 \\ (12) \end{smallmatrix} 12231513221 \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[1900] $1 \begin{smallmatrix} 1513221 \\ (12) \end{smallmatrix} 12231513221 \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 1223151322^1 1(12)1222+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1901] $1 \begin{smallmatrix} 2221 \\ (12) \end{smallmatrix} 12231513221(12)12231513221 \begin{smallmatrix} 21 \\ (12) \end{smallmatrix} 122^1 315+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1902] $1 \begin{smallmatrix} 2221 \\ (12) \end{smallmatrix} 122315132^1 21(12)12231513221 \begin{smallmatrix} 21 \\ (12) \end{smallmatrix} 122315+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1927] $61316131513221(12)12231513221(12)122^1 31513216+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[1926] $41322^1 1 \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 12231513221(12)122315132181232181+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 52, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.06 \rightarrow 44, T_{\max} = 52, T \in [44, 52]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[1925] $414132^1 21(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.06 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[1924] $41413221(12)12231513221 \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 1223151322^1 1(12)1222+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.06 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1923] $61316131513221(12)12231513221(12)122^1 31513161+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[1922] $151321812^1 32181232181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[1921] $512321812^1 32181232181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1920] & 41413221(12)12231513221(12)1223151322^1 1(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 49, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1918] & 414132^1 21(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 49, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1917] & 41413221(12)12231513221(12)1223151322^1 1(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 49, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1916] & 1(12)122315132181232181232181232^1 18123151+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1915] & 1(12)122315132181232181232181232^1 18123215+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 49, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1914] & 812^1 231513221(12)12231513221(12)122315132217+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1913] & 812231513221(12)12231513221(12)122^1 315132217+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1912] & 71231513221(12)12231513221(12)122^1 3151322181+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1911] & 812^1 231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1910] & 812231513221(12)12231513221(12)122^1 315132171+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1909] & 71231513221(12)12231513221(12)122^1 3151321812+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 49, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1908] $7122315132^1 21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1907] $181231513221(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1906] $71231513221(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1905] $71232181231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1903] $21(12)12231513221(12)12231513221(12)122^1 315 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 49$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.06 \rightarrow 44, T_{\max} = 50, T \in [44, 50]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1929] $1(12)122^1 31513221(12)12231513221(12)122315132 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 58$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.94 \rightarrow 43, T_{\max} = 58, T \in [43, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1943] $15132181232^1 181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 64$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.94 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1942] $151321812321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 64$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.94 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1941] $51232181232^1 181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.94 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1940] $512321812321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.94 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1939] $315132^1 181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 54$, $\Delta = -114$, $\det = 64$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.94 \rightarrow 43, T_{\max} = 54, T \in [43, 54]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1938] $1(12)1223151321812321812321812^1 3218123151 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 64$, $(n_+, n_-, n_0) =$

(1, 38, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 42.94 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

¹[1937] $1(12)122^1 31513218123218123218123218123151 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 64$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.94 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

¹[1936] $1(12)1223151321812321812321812^1 3218123215 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.94 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

¹[1935] $1(12)122^1 31513218123218123218123218123215 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 64$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.94 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

³¹⁵¹³²²¹[1933] $1(12)12231513221(12)1223151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 55$, $\Delta = -115$, $\det = 64$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.94 \rightarrow 43$, $T_{\max} = 55$, $T \in [43, 55]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

²³¹⁵¹³²²¹[1932] $1(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 58$, $\Delta = -116$, $\det = 64$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.94 \rightarrow 43$, $T_{\max} = 58$, $T \in [43, 58]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

²¹[1931] $1(12)12231513221(12)12231513221(12)122^1 31513 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 55$, $\Delta = -115$, $\det = 64$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.94 \rightarrow 43$, $T_{\max} = 55$, $T \in [43, 55]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

²¹[1930] $1(12)12^1 231513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 55$, $\Delta = -115$, $\det = 64$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.94 \rightarrow 43$, $T_{\max} = 55$, $T \in [43, 55]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

^{315132^1 21}[1934] $1(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 55$, $\Delta = -115$, $\det = 64$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.94 \rightarrow 43$, $T_{\max} = 55$, $T \in [43, 55]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

¹[1958] $71231513221(12)12231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

¹[1971] $812231513221(12)122315132^1 21(12)122315132217 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

¹[1986] $413221(12)12231513221(12)1223151321812^1 32181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 52$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 42.84 \rightarrow 43, \quad T_{\max} = 52, \quad T \in [43, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1987] & 413221(12)122^1 31513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 38, \quad T_{\max} = 52, \quad \Delta = -112, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 38, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, \quad T_{\max} = 52, \quad T \in [43, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1988] & 61316131513221(12)12231513221(12)12^1 231513216 + \hat{1} \rightarrow 2 \quad T = 38, \quad T_{\max} = 50, \quad \Delta = -109, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 38, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, \quad T_{\max} = 50, \quad T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1989] & 61316131513221(12)122315132^1 21(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 38, \quad T_{\max} = 50, \quad \Delta = -109, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 38, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, \quad T_{\max} = 50, \quad T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1990] & 41413221(12)12231513221(12)1223151321812^1 3217 + \hat{1} \rightarrow 2 \quad T = 38, \quad T_{\max} = 52, \quad \Delta = -112, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 38, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, \quad T_{\max} = 52, \quad T \in [43, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[1991] & 4141322^1 1(12)12231513221(12)12231513218123217 + \hat{1} \rightarrow 2 \quad T = 38, \quad T_{\max} = 52, \quad \Delta = -112, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 38, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, \quad T_{\max} = 52, \quad T \in [43, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1992] & 41413221(12)122^1 31513221(12)12231513218123217 + \hat{1} \rightarrow 2 \quad T = 38, \quad T_{\max} = 52, \quad \Delta = -112, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 38, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, \quad T_{\max} = 52, \quad T \in [43, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1948] & 1 \begin{smallmatrix} 1513221 \\ (12) \end{smallmatrix} 122^1 31513221 \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 12231513221 \begin{smallmatrix} 21 \\ (12) \end{smallmatrix} 12 + \hat{1} \rightarrow 2 \quad T = 38, \quad T_{\max} = 50, \quad \Delta = -107, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 38, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, \quad T_{\max} = 50, \quad T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1949] & 1 \begin{smallmatrix} 2231513221 \\ (12) \end{smallmatrix} 12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 38, \quad T_{\max} = 58, \quad \Delta = -116, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 38, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, \quad T_{\max} = 58, \quad T \in [43, 58] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1947] & 1 \begin{smallmatrix} 221 \\ (12) \end{smallmatrix} 12231513221 \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 122315132^1 21 \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 1223151 + \hat{1} \rightarrow 2 \quad T = 38, \quad T_{\max} = 50, \quad \Delta = -107, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 38, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, \quad T_{\max} = 50, \quad T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1944] & 21 \begin{smallmatrix} 513221 \\ (12) \end{smallmatrix} 12231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 38, \quad T_{\max} = 50, \quad \Delta = -107, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 38, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, \quad T_{\max} = 50, \quad T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1945] & 21 \begin{smallmatrix} 513221 \\ (12) \end{smallmatrix} 122315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 38, \quad T_{\max} = 50, \quad \Delta = -107, \quad \det = 81, \quad (n_+, n_-, n_0) = \\
& (1, 38, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, \quad T_{\max} = 50, \quad T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1946] $1(12)12^1 231513221(12)12231513221(12)122315132+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 58$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 58$, $T \in [43, 58]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1950] $1(12)12231513221(12)12231513221^1(12)12+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 58$, $\Delta = -116$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 58$, $T \in [43, 58]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1951] $1(12)122^1 31513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1952] $1(12)12231513221^1(12)12231513221(12)122315+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1953] $1(12)12231513221(12)122^1 31513221(12)122315+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1954] $21(12)12231513221^1(12)12231513221(12)122315+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1955] $21(12)12231513221(12)122^1 31513221(12)122315+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1956] $71232^1 181231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

[1957] $71232181231513221(12)122315132^1 21(12)1+ \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[1959] $71231513221(12)122315132^1 21(12)122315132181+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1960] $321812^1 32181231513221(12)12231513221(12)12231+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 43$, $\Delta = -93$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 43$, $T \in [43, 43]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1961] $181231513221(12)12231513221(12)12^1 2315132171+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) =$

(1, 38, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1962] $181231513221(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1963] $71223151322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1964] $712231513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1965] $71231513221(12)12231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1966] $71231513221(12)122315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1967] $812231513221(12)12231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1968] $812231513221(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1969] $71231513221(12)12231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1970] $71231513221(12)122315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1984] $41413221(12)122^1 31513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 51$, $\Delta = -111$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 51$, $T \in [43, 51]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1983] $61316131513221(12)12231513221(12)12^1 231513161 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1982] & 61316131513221(12)122315132^1 21(12)12231513161+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1981] & 1513218123218123218123218123151322^1 1(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1980] & 5123218123218123218123218123151322^1 1(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1979] & 3151321812321812321812321812315132^1 21(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 54, \Delta = -114, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 54, T \in [43, 54] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1978] & 41413221(12)122^1 31513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1977] & 4141322^1 1(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1976] & 4141322^1 1(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1975] & 41413221(12)122^1 31513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1974] & 1(12)12^1 231513218123218123218123218123151+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1973] & 1(12)12^1 231513218123218123218123218123215+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1972] & 812231513221(12)12231513221(12)12^1 2315132217+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 81, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.84 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1985] $4141322^1 1(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 51$, $\Delta = -111$, $\det = 81$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.84 \rightarrow 43$, $T_{\max} = 51$, $T \in [43, 51]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [2018] $41413221(12)12231513221(12)122^1 31513221(12)1222+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 51$, $\Delta = -111$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43$, $T_{\max} = 51$, $T \in [43, 51]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2017] $(12)122^1 3151321812321812321812321812315132+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 55$, $\Delta = -115$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43$, $T_{\max} = 55$, $T \in [43, 55]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2016] $151321812321812^1 32181232181231513221(12)12+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2015] $512321812321812^1 32181232181231513221(12)12+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2014] $31513218123218123218123218123151322^1 1(12)1+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 54$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43$, $T_{\max} = 54$, $T \in [43, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [2013] $3151321812^1 32181232181232181231513221(12)1+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 54$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43$, $T_{\max} = 54$, $T \in [43, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2012] $41413221(12)12231513221(12)122^1 31513221(12)12+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 51$, $\Delta = -111$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43$, $T_{\max} = 51$, $T \in [43, 51]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2011] $41413221(12)12231513221(12)122^1 31513221(12)1+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 51$, $\Delta = -111$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43$, $T_{\max} = 51$, $T \in [43, 51]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2010] $1(12)122315132181232181232^1 18123218123151+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2009] $1(12)122315132181232181232^1 18123218123215+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2008] $8122315132^1 21(12)12231513221(12)122315132217+\hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2007] $712315132^1 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2006] $8122315132^1 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2005] $712315132^1 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2004] $712231513221(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2003] $1812315132^1 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2002] $32181232181231513221(12)122^1 31513221(12)12231 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 43$, $\Delta = -93$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.76 \rightarrow 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2001] $712315132^1 21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2000] $21(12)12231513221(12)12231513221(12)12^1 2315 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1999] $1(12)12231513221(12)12231513221(12)12^1 2315 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1998] $1(12)12231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1997] $1(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 58$, $\Delta = -116$, $\det = 100$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.76 \rightarrow 43, T_{\max} = 58, T \in [43, 58]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

- [1996] $1^{1513221}(12)12231513221(12)122^131513221(12)12+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -107, \det = 100, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1995] $1^{221}(12)122315132^121(12)12231513221(12)1223151+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -107, \det = 100, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1994] $1^{3221}(12)12231513221(12)12231513221(12)12^12315+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 58, \Delta = -116, \det = 100, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43, T_{\max} = 58, T \in [43, 58]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1993] $21^{51322^11}(12)12231513221(12)12231513221(12)12222+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -107, \det = 100, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.76 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [2064] $(12)12^123151321812321812321812321812315132+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 55, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.69 \rightarrow 43, T_{\max} = 55, T \in [43, 55]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [2051] $812231513221(12)1223151322^11(12)122315132217+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2056] $41413221(12)122315132^121(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 51, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.69 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [2054] $41413221(12)12^1231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 51, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.69 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2067] $413221(12)12^1231513221(12)122315132181232181+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 52, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.69 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2068] $413221(12)122315132^121(12)122315132181232181+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 52, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.69 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [2069] $61316131513221(12)122^131513221(12)12231513216+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [2070] $61316131513221(12)1223151322^11(12)12231513216+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2071] & 41413221(12)122315132^1 21(12)12231513218123217+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 52, \Delta = -112, \det = 121, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 52, T \in [43, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2072] & 41413221(12)12^1 231513221(12)12231513218123217+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 52, \Delta = -112, \det = 121, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 52, T \in [43, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2053] & 1(12)1223151321812321812^1 3218123218123151+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2048] & 71231513221(12)122^1 31513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2066] & 41413221(12)122315132^1 21(12)12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 121, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2055] & 41413221(12)122315132^1 21(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 121, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2057] & 41413221(12)12^1 231513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 121, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2058] & 315132181232^1 181232181232181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 54, \Delta = -114, \det = 121, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 54, T \in [43, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2059] & 51232181232181232^1 181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 121, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2060] & 15132181232181232^1 181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2061] & 61316131513221(12)122^1 31513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 121, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[2062] & 61316131513221(12)1223151322^1 1(12)12231513161+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 121, (n_+, n_-, n_0) =
\end{aligned}$$

$$(1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[2063] (12)1223151321812321812321812^1 315132 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 55, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 55, T \in [43, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2047] 812231513221^1 (12)1223151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2046] 812231513221^1 (12)122^1 31513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2045] 71231513221^1 (12)1223151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2044] 71231513221^1 (12)122^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2043] 712231513221^1 (12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2042] 712231513221^1 (12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2041] 181231513221^1 (12)122^1 31513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2040] 181231513221^1 (12)1223151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2039] 71231513221^1 (12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2038] 71231513221^1 (12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2037] \quad 71232181231513221(12)1223151322^{\overset{1513221}{1}}1(12) \quad 1 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2036] \quad 71232181231513221(12)122^{\overset{1513221}{1}}31513221(12) \quad 1 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2035] \quad 21(12)12231513221(12)122315132^{\overset{21}{1}}21(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2034] \quad 21(12)12231513221(12)12^{\overset{221}{1}}231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2033] \quad 1(12)12231513221(12)12^{\overset{2221}{1}}231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2032] \quad 1(12)12231513221(12)122315132^{\overset{21}{1}}21(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2031] \quad 1(12) \quad 12^{\overset{1513221}{1}}231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2030] \quad 1(12) \quad 122315132^{\overset{1513221}{1}}21(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2029] \quad 1(12) \quad 12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 55, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 55, T \in [43, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2028] \quad 1(12) \quad 12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 58, T \in [43, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2027] \quad 1(12) \quad 122315132^{\overset{1513221}{1}}21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2026] & 1(12) \overset{1513221}{12} \overset{1}{12} \overset{21}{231513221}(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[2025] & 1(12)12231513221(12)122 \overset{221}{131513221}(12)1223151 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2024] & 1(12)12231513221(12)1223151322 \overset{1}{1}(12)1223151 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2023] & 1(12)12231513221(12)12231513221(12)12 \overset{21}{1231513} + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 55, \Delta = -115, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 55, T \in [43, 55] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2022] & 1(12)122 \overset{3221}{131513221}(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2021] & 1(12)12231513221(12)12231513221(12)122 \overset{1}{1315132} + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2020] & 21(12) \overset{513221}{1223151322} \overset{1}{1}(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[2019] & 21(12) \overset{513221}{122} \overset{1}{131513221}(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[2065] & 41413221(12)12 \overset{1}{1231513221}(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2052] & 1(12)1223151321812321812 \overset{1}{13218123218123215} + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2049] & 71231513221(12)1223151322 \overset{1}{1}(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]
\end{aligned}$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2050] $812231513221(12)122^1 31513221(12)122315132217+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.69 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2073] $32181232^1 181231513221(12)12231513221(12)12231+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 43, \Delta = -93, \det = 144, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.64 \rightarrow 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 12 & 2 \\ 12 & 203 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[2102] $41413221(12)12231513221(12)122^1 31513218123217+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 52, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2074] $1(12)12^{3221} 231513221(12)12231513221(12)122315+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 58, \Delta = -116, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 58, T \in [43, 58]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2075] $1(12)122315132^1 21(12)12231513221(12)12231513+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 55, \Delta = -115, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 55, T \in [43, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2076] $1(12)1223151322^1 1(12)12231513221(12)1223151+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -107, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2077] $1(12)12231513221(12)12^1 231513221(12)12+ \hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -107, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[2078] $1(12)122^{31513221} 31513221(12)12231513221(12)122+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 55, \Delta = -115, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 55, T \in [43, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2079] $1(12)12231513221(12)122^1 31513221(12)122+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 55, \Delta = -115, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 55, T \in [43, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2080] $1(12)12231513221(12)12^1 231513221(12)1222+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -107, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2081] $712321812315132^1 21(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -108, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

- [2082] $7123151322^1 1(12)12231513221(12)122315132181+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2083] $32181232181231513221(12)12^1 231513221(12)12231+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 43, \Delta = -93, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.59 \rightarrow 43, T_{\max} = 43, T \in [43, 43]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [2084] $18123151322^1 1(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2085] $712231513221(12)12231513221(12)12^1 2315132181+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 20 & 3 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2086] $7123151322^1 1(12)12231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2087] $81223151322^1 1(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2088] $7123151322^1 1(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2089] $81223151322^1 1(12)12231513221(12)122315132217+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2090] $1(12)122315132181232^1 18123218123218123215+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -108, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2091] $1(12)122315132181232^1 18123218123218123151+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2092] $41413221(12)12231513221(12)12^1 231513221(12)1+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 51, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.59 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2093] $41413221(12)12231513221(12)12^1 231513221(12)12+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 51, \Delta = -111, \det = 169, (n_+, n_-, n_0) =$

(1, 38, 0), $c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2094] 3151321812321812¹32181232181231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 38, T_{\max} = 54, \Delta = -114, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 54, T \in [43, 54]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[2095] 512321812321812321812¹32181231513221(12)12+ $\hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -108, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[2096] 151321812321812321812¹32181231513221(12)12+ $\hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[2097] 613161315132¹21(12)12231513221(12)12231513161+ $\hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -108, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[2098] (12)122315132181232181232181232¹1812315132+ $\hat{1} \rightarrow 2$ $T = 38, T_{\max} = 55, \Delta = -115, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 55, T \in [43, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2099] 41413221(12)12231513221¹(12)12¹231513221(12)1222+ $\hat{1} \rightarrow 2$ $T = 38, T_{\max} = 51, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 51, T \in [43, 51]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2100] 413221¹(12)12231513221(12)122¹315132181232181+ $\hat{1} \rightarrow 2$ $T = 38, T_{\max} = 52, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2101] 613161315132¹21(12)12231513221(12)12231513216+ $\hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -109, \det = 169, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.59 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2117] 812231513221¹(12)12¹231513221(12)122315132171+ $\hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2114] 181231513221¹(12)12¹231513221(12)122315132171+ $\hat{1} \rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2130] 413221¹(12)1223151322¹1(12)122315132181232181+ $\hat{1} \rightarrow 2$ $T = 38, T_{\max} = 52, \Delta = -112, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 42.55 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2132] $41413221(12)1223151322^1 1(12)12231513218123217+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 52, \Delta = -112, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2118] $71231513221(12)12^1 231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2119] $812231513221(12)12^1 231513221(12)122315132217+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2120] $1(12)1223151321812^1 3218123218123218123215+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -108, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[2121] $1(12)1223151321812^1 3218123218123218123151+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[2113] $321812321812315132^1 21(12)12231513221(12)12231+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 43, \Delta = -93, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43, T_{\max} = 43, T \in [43, 43]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

[2112] $71231513221(12)12^1 231513221(12)122315132181+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2111] $71232181231513221(12)12^1 231513221(12)1+1+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -108, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2110] $21(12)12231513221(12)1223151322^1 1(12)122315+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -107, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2109] $1(12)12231513221(12)1223151322^1 1(12)122315+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -107, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2108] $1(12)1223151322^1 1(12)12231513221(12)1222+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -107, \det = 196, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2107] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 58, \Delta = -116, \det = 196, (n_+, n_-, n_0) =$

$(1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 58$, $T \in [43, 58]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

$[2106]$ $1^{1513221}(12)1223151322^11(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 196$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

$[2105]$ $1^{221}(12)12231513221(12)12^1231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 196$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[2104]$ $1^{3221}(12)12231513221(12)122315132^121(12)122315 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 58$, $\Delta = -116$, $\det = 196$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 58$, $T \in [43, 58]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

$[2103]$ $21^{513221}(12)12^1231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -107$, $\det = 196$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

$[2122]$ $41413221(12)1223151322^11(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 51$, $\Delta = -111$, $\det = 196$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 51$, $T \in [43, 51]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[2123]$ $41413221(12)1223151322^11(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 51$, $\Delta = -111$, $\det = 196$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 51$, $T \in [43, 51]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[2124]$ $315132181232181232^1181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 54$, $\Delta = -114$, $\det = 196$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 54$, $T \in [43, 54]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$

$[2125]$ $51232181232181232181232^1181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 196$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$

$[2126]$ $15132181232181232181232^1181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -106$, $\det = 196$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$

$[2127]$ $61316131513221(12)12^1231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 50$, $\Delta = -108$, $\det = 196$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 50$, $T \in [43, 50]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

$[2128]$ $(12)1223151321812321812321812^1321812315132 + \hat{1} \rightarrow 2$ $T = 38$, $T_{\max} = 55$, $\Delta = -115$, $\det = 196$, $(n_+, n_-, n_0) = (1, 38, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 42.55 \rightarrow 43$, $T_{\max} = 55$, $T \in [43, 55]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2131] & 61316131513221(12)12^1 231513221(12)12231513216+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -109, \det = 196, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2116] & 71231513221(12)12^1 231513221(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 196, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2115] & 712231513221(12)1223151322^1 1(12)122315132181+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 196, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.55 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2129] & 41413221(12)1223151322^1 1(12)12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 51, \Delta = -111, \det = 196, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.55 \rightarrow 43, T_{\max} = 51, T \in [43, 51] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2136] & 1 \begin{pmatrix} 31513221 \\ 12 \end{pmatrix} 122315132^1 21(12)12231513221(12)122+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 55, \Delta = -115, \det = 225, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.52 \rightarrow 43, T_{\max} = 55, T \in [43, 55] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2135] & 1 \begin{pmatrix} 23151322^1 1 \\ 12 \end{pmatrix} 12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 225, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.52 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2134] & 1 \begin{pmatrix} 21 \\ 12 \end{pmatrix} 12231513221(12)122^1 31513221(12)12231513+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 55, \Delta = -115, \det = 225, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.52 \rightarrow 43, T_{\max} = 55, T \in [43, 55] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2133] & 1(12)12231513221(12)12231513221(12)12^1 2315132+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 225, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.52 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2137] & 712321812^1 31513221(12)12231513221 \begin{pmatrix} 1513221 \\ 12 \end{pmatrix} 1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 225, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.52 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2138] & 413221(12)12231513221(12)122315132^1 181232181+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 52, \Delta = -112, \det = 225, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.52 \rightarrow 43, T_{\max} = 52, T \in [43, 52] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2139] & 41413221(12)12231513221(12)122315132^1 18123217+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 52, \Delta = -112, \det = 225, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.52 \rightarrow 43, T_{\max} = 52, T \in [43, 52] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2140] $1(12)122315132^1 21(12)12231513221(12)122315132+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 58, \Delta = -116, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.48 \rightarrow 43, T_{\max} = 58, T \in [43, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2141] $1(12)12231513221(12)1223151322^1 1(12)12231513+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 55, \Delta = -115, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.48 \rightarrow 43, T_{\max} = 55, T \in [43, 55]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[2142] $1(12)12231513221(12)122^1 31513221(12)12+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 58, \Delta = -116, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.48 \rightarrow 43, T_{\max} = 58, T \in [43, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2143] $1(12)12^1 231513221(12)12231513221(12)122+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 55, \Delta = -115, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.48 \rightarrow 43, T_{\max} = 55, T \in [43, 55]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[2144] $7123218123151322^1 1(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -108, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.48 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[2145] $1(12)122315132^1 18123218123218123218123215+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -108, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.48 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2146] $1(12)122315132^1 18123218123218123218123151+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -106, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.48 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2147] $3151321812321812321812^1 32181231513221(12)1+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 54, \Delta = -114, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.48 \rightarrow 43, T_{\max} = 54, T \in [43, 54]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[2154] $41413221(12)12231513221(12)12^1 231513218123217+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 52, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.48 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

[2153] $6131613151322^1 1(12)12231513221(12)12231513216+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 50, \Delta = -109, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.48 \rightarrow 43, T_{\max} = 50, T \in [43, 50]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[2152] $413221(12)12231513221(12)12^1 2315132181232181+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 52, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.48 \rightarrow 43, T_{\max} = 52, T \in [43, 52]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[2151] $(12)122315132181232181232^1 1812321812315132+\hat{1}\rightarrow 2$ $T = 38, T_{\max} = 55, \Delta = -115, \det = 256, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 42.48 \rightarrow 43, T_{\max} = 55, T \in [43, 55] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[2150] & 6131613151322^1 1(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 256, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.48 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[2149] & 151321812321812321812321812^1 31513221(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -106, \det = 256, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.48 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[2148] & 512321812321812321812321812^1 31513221(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 50, \Delta = -108, \det = 256, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.48 \rightarrow 43, T_{\max} = 50, T \in [43, 50] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[2156] & 1(12)12231513221(12)1223151322^1 1(12)122315+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 289, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.46 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 101 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2155] & 1(12)12231513221(12)122315132^1 21(12)122315132+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 289, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.46 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[2157] & 1(12)122315132^1 21(12)12231513221(12)122315+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 289, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.46 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 118 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[2158] & 1(12)1223151322^1 1(12)12231513221(12)12231513+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 55, \Delta = -115, \det = 289, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.46 \rightarrow 43, T_{\max} = 55, T \in [43, 55] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[2159] & 1(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 289, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.46 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[2160] & 1(12)12231513221(12)12^1 231513221(12)122+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 55, \Delta = -115, \det = 289, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.46 \rightarrow 43, T_{\max} = 55, T \in [43, 55] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[2161] & 3218123218123151322^1 1(12)12231513221(12)12231+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 43, \Delta = -93, \det = 289, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.46 \rightarrow 43, T_{\max} = 43, T \in [43, 43] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 390 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0} \\
[2162] & 315132181232181232181232^1 181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 54, \Delta = -114, \det = 289, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.46 \rightarrow 43, T_{\max} = 54, T \in [43, 54] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2163] $(12)1223151321812321812^1 321812321812315132+\hat{1} \rightarrow 2$ $T = 38, T_{\max} = 55, \Delta = -115, \det = 289, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.46 \rightarrow 43, T_{\max} = 55, T \in [43, 55]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [2164] $1(12)12231513221(12)122^1 31513221(12)122315+\hat{1} \rightarrow 2$ $T = 38, T_{\max} = 58, \Delta = -116, \det = 324, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.43 \rightarrow 43, T_{\max} = 58, T \in [43, 58]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 125 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [2165] $321812321812^1 31513221(12)12231513221(12)12231+\hat{1} \rightarrow 2$ $T = 38, T_{\max} = 43, \Delta = -93, \det = 324, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.43 \rightarrow 43, T_{\max} = 43, T \in [43, 43]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 449 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [2170] $1(12)122315132^1 21(12)12231513221(12)12+\hat{1} \rightarrow 2$ $T = 38, T_{\max} = 58, \Delta = -116, \det = 361, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.41 \rightarrow 43, T_{\max} = 58, T \in [43, 58]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [2173] $(12)122315132181232^1 1812321812321812315132+\hat{1} \rightarrow 2$ $T = 38, T_{\max} = 55, \Delta = -115, \det = 361, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.41 \rightarrow 43, T_{\max} = 55, T \in [43, 55]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [2172] $3151321812321812321812321812^1 31513221(12)1+\hat{1} \rightarrow 2$ $T = 38, T_{\max} = 54, \Delta = -114, \det = 361, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.41 \rightarrow 43, T_{\max} = 54, T \in [43, 54]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [2171] $1(12)1223151322^1 1(12)12231513221(12)122+\hat{1} \rightarrow 2$ $T = 38, T_{\max} = 55, \Delta = -115, \det = 361, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.41 \rightarrow 43, T_{\max} = 55, T \in [43, 55]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [2169] $1(12)1223151322^1 1(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 38, T_{\max} = 58, \Delta = -116, \det = 361, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.41 \rightarrow 43, T_{\max} = 58, T \in [43, 58]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [2168] $1(12)12231513221(12)122^1 31513221(12)1+\hat{1} \rightarrow 2$ $T = 38, T_{\max} = 58, \Delta = -116, \det = 361, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.41 \rightarrow 43, T_{\max} = 58, T \in [43, 58]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [2167] $1(12)12231513221(12)12^1 231513221(12)12231513+\hat{1} \rightarrow 2$ $T = 38, T_{\max} = 55, \Delta = -115, \det = 361, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.41 \rightarrow 43, T_{\max} = 55, T \in [43, 55]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [2166] $1(12)12231513221(12)122^1 31513221(12)122315132+\hat{1} \rightarrow 2$ $T = 38, T_{\max} = 58, \Delta = -116, \det = 361, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 42.41 \rightarrow 43, T_{\max} = 58, T \in [43, 58]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [2174] $(12)1223151321812^1 321812321812321812315132+\hat{1} \rightarrow 2$ $T = 38, T_{\max} = 55, \Delta = -115, \det = 400, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 42.39 \rightarrow 43, T_{\max} = 55, T \in [43, 55] \\
& \begin{pmatrix} 0 & 20 & 2 \\ 20 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[2175] & 1 \overset{1}{(12)} 1223151322^1 1 \overset{1}{(12)} 12231513221 \overset{1}{(12)} 122315132 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 441, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.37 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[2176] & 1 \overset{2231513221}{(12)} 122^1 31513221 \overset{1}{(12)} 12231513221 \overset{1}{(12)} 1 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 441, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.37 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[2177] & 1 \overset{231513221}{(12)} 12^1 231513221 \overset{1}{(12)} 12231513221 \overset{1}{(12)} 12 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 441, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.37 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[2178] & 1 \overset{231513221}{(12)} 12231513221 \overset{1}{(12)} 12^1 231513221 \overset{1}{(12)} 12 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 441, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.37 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[2179] & 1 \overset{3221}{(12)} 1223151322^1 1 \overset{1}{(12)} 12231513221 \overset{1}{(12)} 122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 484, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.36 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 197 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[2180] & (12) 122315132^1 1812321812321812321812315132 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 55, \Delta = -115, \det = 484, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.36 \rightarrow 43, T_{\max} = 55, T \in [43, 55] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[2181] & 1 \overset{3221}{(12)} 12231513221 \overset{1}{(12)} 12^1 231513221 \overset{1}{(12)} 122315 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 529, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.34 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 206 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[2182] & 1 \overset{2231513221}{(12)} 122315132^1 21 \overset{1}{(12)} 12231513221 \overset{1}{(12)} 1 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 529, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.34 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0} \\
[2183] & 1 \overset{1}{(12)} 12231513221 \overset{1}{(12)} 12^1 231513221 \overset{1}{(12)} 122315132 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 576, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.33 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[2184] & 1 \overset{231513221}{(12)} 1223151322^1 1 \overset{1}{(12)} 12231513221 \overset{1}{(12)} 12 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 576, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.33 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[2185] & 1 \overset{2231513221}{(12)} 12231513221 \overset{1}{(12)} 12^1 231513221 \overset{1}{(12)} 1 + \hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 625, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.31 \rightarrow 43, T_{\max} = 58, T \in [43, 58]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0} \\
[2186] & 1 \quad (12) \quad 12^1 231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 676, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.30 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0} \\
[2187] & 1 \quad (12) \quad 12231513221^1 1(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 38, T_{\max} = 58, \Delta = -116, \det = 841, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 42.27 \rightarrow 43, T_{\max} = 58, T \in [43, 58] \\
& \begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0} \\
T = 39 & [2188] \quad 81231513221(12)12231513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 55, \Delta = -114, \det = \\
& -9, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.22 \rightarrow 46, T_{\max} = 55, T \in [46, 55] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[2190] & 1612^1 32181232181232181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -16, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.75 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2189] & 1(12)122315132181232181232181232181232^1 161+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -16, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.75 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2200] & 61316131513221(12)12231513221(12)122315132^1 217+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.44 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[2199] & 61316131513221(12)12231513221(12)122315132^1 171+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -25, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.44 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2198] & 812231513221(12)12231513221(12)122315132^1 2181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -25, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.44 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2197] & 812231513221(12)12231513221(12)122315132^1 1812+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -25, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.44 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[2196] & 181231513221(12)12231513221(12)122315132^1 2181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -25, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.44 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[2195] & 181231513221(12)12231513221(12)122315132^1 1812+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -25, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.44 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2194] $812231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.44 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2193] $1812^1 31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.44 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [2192] $4132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 56, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.44 \rightarrow 45, T_{\max} = 56, T \in [45, 56]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [2191] $1^{15132^1 21}(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.44 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2201] $1(12)1223151321812321812321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.44 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [2202] $161232^1 181232181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.44 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [2206] $1812^1 31513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.22 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [2207] $181231513221(12)12231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.22 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2209] $8122^1 31513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.22 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2205] $8122^1 31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.22 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2204] $181231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.22 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [2203] $1^{1513221}(12)12231513221(12)122315132181232^1 181 + \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -107, \det = -36, (n_+, n_-, n_0) =$

$$(1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.22 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2208] \text{ } 1812^1 31513221(12)12231513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.22 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2210] \text{ } 812231513221(12)12231513221(12)1223151322^1 181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.22 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2211] \text{ } 8122^1 31513221(12)12231513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.22 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2212] \text{ } 61316131513221(12)12231513221(12)1223151322^1 17+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.22 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2213] \text{ } 41413221(12)12231513221(12)122315132181232^1 181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 52, \Delta = -111, \det = -36, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.22 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2224] \text{ } 812231513221(12)12231513221(12)122^1 3151321812+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2232] \text{ } 31612^1 32181232181232181232181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 54, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.06 \rightarrow 45, T_{\max} = 54, T \in [45, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[2214] \text{ } 1^{151322^1 1}(12)12231513221(12)122315132181232181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[2215] \text{ } 812^1 32181231513221(12)12231513221^{31513221}(12)1+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 56, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.06 \rightarrow 45, T_{\max} = 56, T \in [45, 56]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2216] \text{ } 41322^1 1(12)12231513221(12)12231513221^{513221}(12)1+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 56, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.06 \rightarrow 45, T_{\max} = 56, T \in [45, 56]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2217] \text{ } 413221(12)12231513221(12)12231513221^{5132^1 21}(12)1+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 56, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.06 \rightarrow 45, T_{\max} = 56, T \in [45, 56]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[2218] & \begin{matrix} 1 \\ 1812315132^1 21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -49, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2219] & \begin{matrix} 1 \\ 812231513221(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -49, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[2220] & \begin{matrix} 1 \\ 812^1 231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -49, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2221] & \begin{matrix} 1 \\ 181231513221(12)12231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -49, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2222] & \begin{matrix} 1 \\ 181231513221(12)12231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -49, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2223] & \begin{matrix} 1 \\ 812^1 31513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 39, T_{\max} = 55, \Delta = -114, \det = -49, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.06 \rightarrow 45, T_{\max} = 55, T \in [45, 55] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[2227] & \begin{matrix} 1 \\ 812^1 231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -49, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2225] & \begin{matrix} 1 \\ 812^1 231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -49, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[2228] & \begin{matrix} 1 \\ 61316131513221(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[2229] & \begin{matrix} 1 \\ 61316131513221(12)12231513221(12)122^1 315132217 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -49, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[2230] & \begin{matrix} 1 \\ 1(12)122315132181232181232^1 181232161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -49, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[2231] $1612321812^1 32181232181232181231513221(12)12+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[2233] $414132^1 21(12)12231513221(12)122315132181232181+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 52, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.06 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$

[2226] $812231513221^1(12)12231513221(12)122^1 3151322181+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.06 \rightarrow 45, T_{\max} = 51, T \in [45, 51]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2234] $81232181231513221(12)12231513221^1(12)1+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 56, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.94 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[2235] $1(12)122^1 315132181232181232181232181232161+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -64, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.94 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2236] $1(12)1223151321812321812321812^1 32181232161+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -64, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.94 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

[2237] $161232181232^1 181232181232181231513221(12)12+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -64, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.94 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2239] $3161232^1 181232181232181232181231513221(12)1+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 54, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.94 \rightarrow 44, T_{\max} = 54, T \in [44, 54]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2238] $1612321812321812321812321812315132^1 21(12)12+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -64, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.94 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2250] $181231513221^1(12)12231513221(12)12^1 23151322181+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[2256] $812231513221^1(12)12231513221(12)12^1 23151322181+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

[2240] $21(12)122315132^1 21(12)12231513221(12)122313+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 44, \Delta = -95, \det = -81, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 44, T \in [44, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 513221 \\ [2241] \end{matrix} 21 (12) 12231513221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 44, \Delta = -95, \det = -81, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 44, T \in [44, 44] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1513221 \\ [2242] \end{matrix} 1 (12) 122^1 31513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1513221 \\ [2243] \end{matrix} 1 (12) 12231513221(12)1223151321812^1 32181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -81, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [2244] \end{matrix} 18123151322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -81, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [2245] \end{matrix} 181231513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -81, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [2246] \end{matrix} 812231513221(12)12231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -81, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [2247] \end{matrix} 812231513221(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -81, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [2248] \end{matrix} 181231513221(12)122315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -81, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
& [2262] 16123218123218123218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -81, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& [2263] 31612321812321812321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 54, \Delta = -113, \det = -81, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 54, T \in [44, 54] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& [2264] 41413221(12)122^1 31513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 52, \Delta = -111, \det = -81, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.84 \rightarrow 44, T_{\max} = 52, T \in [44, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2265] 41413221(12)12231513221(12)1223151321812¹32181+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 52$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 52, T \in [44, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2266] 4141322¹1(12)12231513221(12)122315132181232181+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 52$, $\Delta = -111$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 52, T \in [44, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2261] 1(12)12¹2315132181232181232181232161+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 152 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[2260] 61316131513221(12)12231513221(12)12¹2315132217+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2249] 181231513221(12)12231513221(12)12¹23151321812+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2252] 81231513221(12)12231513221(12)122315132¹18123+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 55$, $\Delta = -114$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 55, T \in [44, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2253] 812231513221(12)122315132¹21(12)1223151321812+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2254] 812231513221(12)12231513221(12)12¹23151321812+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2255] 812231513221(12)122315132¹21(12)1223151322181+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2259] 61316131513221(12)122315132¹21(12)122315132217+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2258] 61316131513221(12)122315132¹21(12)122315132171+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2257] 61316131513221(12)12231513221(12)12¹2315132171+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2251] $181231513221^1(12)122315132^121(12)1223151322181+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.84 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2273] $8122315132^121(12)12231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.76 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2274] $8122315132^121(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.76 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2275] $1(12)122315132181232181232^1181232181232161+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.76 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2276] $1612321812321812^132181232181231513221(12)12+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.76 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2277] $31612321812^132181232181232181231513221(12)1+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 54, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.76 \rightarrow 44, T_{\max} = 54, T \in [44, 54]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2278] $316123218123218123218123218123151322^11(12)1+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 54, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.76 \rightarrow 44, T_{\max} = 54, T \in [44, 54]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2272] $1812315132^121(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.76 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2271] $1812315132^121(12)12231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.76 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2270] $8122315132^121(12)12231513221(12)122315132181+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.76 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2269] $181231513221(12)12231513221(12)122^1315132181+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.76 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2268] $413221(12)12231513221(12)12231513221$ (12) $1+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 56, \Delta = -115, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.76 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[2267] $21(12)12231513221(12)12231513221(12)122313+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 44, \Delta = -95, \det = -100, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.76 \rightarrow 44, T_{\max} = 44, T \in [44, 44]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[2302] $61316131513221(12)122^131513221(12)122315132217+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2279] $21(12)122^131513221(12)12231513221(12)122313+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 44, \Delta = -95, \det = -121, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.69 \rightarrow 44, T_{\max} = 44, T \in [44, 44]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2305] $3161232181232^1181232181232181231513221(12)1+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 54, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.69 \rightarrow 44, T_{\max} = 54, T \in [44, 54]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[2306] $41413221(12)12^1231513221(12)122315132181232181+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 52, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.69 \rightarrow 44, T_{\max} = 52, T \in [44, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[2307] $41413221(12)122315132^121(12)122315132181232181+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 52, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.69 \rightarrow 44, T_{\max} = 52, T \in [44, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[2281] $1(12)12^1231513221(12)122315132181232181+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[2282] $1(12)122315132^121(12)122315132181232181+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[2283] $81232^1181231513221(12)12231513221$ (12) $1+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 56, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.69 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[2284] $81232181231513221(12)12231513221$ (12) $1+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 56, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.69 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2301] $61316131513221(12)1223151322^11(12)122315132217+\hat{1}\rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2300] & 61316131513221(12)1223151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[2303] & 1(12)1223151321812321812^1 32181232181232161 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[2285] & 413221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 56, \Delta = -115, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 56, T \in [44, 56] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2286] & 181231513221(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2287] & 181231513221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2288] & 812231513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2289] & 812231513221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2290] & 181231513221(12)122^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2291] & 181231513221(12)1223151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2292] & 181231513221(12)122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2293] & 181231513221(12)1223151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2304] & 161232181232181232^1 181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2294] & 81231513221(12)12231513221(12)122^1 31513218123+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 55, \Delta = -114, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 55, T \in [44, 55] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2295] & 812231513221(12)1223151322^1 1(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2296] & 812231513221(12)122^1 31513221(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2297] & 812231513221(12)122^1 31513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2298] & 812231513221(12)1223151322^1 1(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2299] & 61316131513221(12)122^1 31513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[2280] & 21(12)1223151322^1 1(12)12231513221(12)122313+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 44, \Delta = -95, \det = -121, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.69 \rightarrow 44, T_{\max} = 44, T \in [44, 44] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[2324] & 41413221(12)12231513221(12)122^1 315132181232181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 52, \Delta = -111, \det = -169, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.59 \rightarrow 44, T_{\max} = 52, T \in [44, 52] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[2308] & 1(12)12231513221(12)122^1 315132181232181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -169, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.59 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[2309] & 81232181231513221(12)122315132^1 21(12)1+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 56, \Delta = -115, \det = -169, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.59 \rightarrow 44, T_{\max} = 56, T \in [44, 56] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2310] $413221(12)12^1 231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 56, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.59 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2311] $181231513221(12)12231513221(12)12^1 2315132181+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.59 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2312] $812231513221^1(12)12231513221(12)122315132181+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.59 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2313] $181231513221^1(12)12231513221(12)1223151321812+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.59 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2314] $181231513221^1(12)12231513221(12)1223151322181+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.59 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2315] $812315132^1 21(12)12231513221(12)12231513218123+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 55, \Delta = -114, \det = -169, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.59 \rightarrow 44, T_{\max} = 55, T \in [44, 55]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2316] $81231513221(12)122315132^1 21(12)12231513218123+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 55, \Delta = -114, \det = -169, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.59 \rightarrow 44, T_{\max} = 55, T \in [44, 55]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2317] $812231513221^1(12)12231513221(12)1223151321812+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.59 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2318] $812231513221^1(12)12231513221(12)1223151322181+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.59 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2319] $613161315132^1 21(12)12231513221(12)122315132171+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.59 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [2320] $613161315132^1 21(12)12231513221(12)122315132217+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 43.59 \rightarrow 44, T_{\max} = 51, T \in [44, 51]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2321] $1(12)122315132181232^1 181232181232181232161+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 51, \Delta = -105, \det = -169, (n_+, n_-, n_0) =$

(1, 39, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 43.59 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$$

[2322] 1612321812321812321812¹32181231513221(12)12+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -169$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.59 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[2323] 31612321812321812¹32181232181231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 54$, $\Delta = -113$, $\det = -169$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.59 \rightarrow 44$, $T_{\max} = 54$, $T \in [44, 54]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[2332] 81231513221(12)¹12231513221(12)12¹231513218123+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 55$, $\Delta = -114$, $\det = -196$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.55 \rightarrow 44$, $T_{\max} = 55$, $T \in [44, 55]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2325] 21 (12) 12¹231513221(12)12231513221(12)122313+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 44$, $\Delta = -95$, $\det = -196$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.55 \rightarrow 44$, $T_{\max} = 44$, $T \in [44, 44]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[2326] 1¹⁵¹³²²¹(12) 1223151322¹1(12)122315132181232181+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -107$, $\det = -196$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.55 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[2327] 413221(12)12231513221(12)¹122315132¹21 (12) 1+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 56$, $\Delta = -115$, $\det = -196$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.55 \rightarrow 44$, $T_{\max} = 56$, $T \in [44, 56]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2328] 181231513221(12)1223151322¹1(12)122315132181+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -196$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.55 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2329] 812231513221(12)12¹231513221(12)122315132181+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -196$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.55 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2330] 181231513221(12)12¹231513221(12)1223151321812+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -196$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.55 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2340] 41413221(12)1223151322¹1(12)122315132181232181+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 52$, $\Delta = -111$, $\det = -196$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.55 \rightarrow 44$, $T_{\max} = 52$, $T \in [44, 52]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[2339] 3161232181232181232¹181232181231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 54$, $\Delta = -113$, $\det = -196$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.55 \rightarrow 44$, $T_{\max} = 54$, $T \in [44, 54]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[2338] & 161232181232181232181232^1 181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -196, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.55 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
& \begin{pmatrix} 1 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[2337] & 1(12)1223151321812^1 32181232181232181232161+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -196, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.55 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0} \\
& \begin{pmatrix} 1 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0} \\
[2331] & 181231513221(12)12^1 231513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -196, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.55 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2335] & 61316131513221(12)12^1 231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -196, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.55 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
& \begin{pmatrix} 1 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[2334] & 812231513221(12)12^1 231513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -196, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.55 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& \begin{pmatrix} 1 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2333] & 812231513221(12)12^1 231513221(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -196, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.55 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& \begin{pmatrix} 1 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2336] & 61316131513221(12)12^1 231513221(12)122315132217+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -196, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.55 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2344] & 41413221(12)12231513221(12)122315132^1 181232181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 52, \Delta = -111, \det = -225, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.52 \rightarrow 44, T_{\max} = 52, T \in [44, 52] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
& \begin{pmatrix} 1 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[2343] & 81231513221(12)122^1 31513221(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 55, \Delta = -114, \det = -225, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.52 \rightarrow 44, T_{\max} = 55, T \in [44, 55] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
& \begin{pmatrix} 1 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2342] & 81232181231513221(12)122^1 31513221 \quad (12) \quad 1+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 56, \Delta = -115, \det = -225, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.52 \rightarrow 44, T_{\max} = 56, T \in [44, 56] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
& \begin{pmatrix} 1 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2341] & 1 \quad (12) \quad 12231513221(12)122315132^1 181232181+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -107, \det = -225, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 43.52 \rightarrow 44, T_{\max} = 51, T \in [44, 51] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
& \begin{pmatrix} 1 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[2351] & 1612321812321812321812321812^1 31513221(12)12+\hat{1} \rightarrow 2 \quad T = 39, T_{\max} = 51, \Delta = -105, \det = -256, (n_+, n_-, n_0) =
\end{aligned}$$

(1, 39, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 43.48 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[2345] $1^{1513221}(12)12231513221(12)12^1 2315132181232181 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -107$, $\det = -256$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.48 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[2346] $81232181231513221(12)12231513221^{31513221}1(12)1 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 56$, $\Delta = -115$, $\det = -256$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.48 \rightarrow 44$, $T_{\max} = 56$, $T \in [44, 56]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2347] $81231513221(12)1^{1223151322}1(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 55$, $\Delta = -114$, $\det = -256$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.48 \rightarrow 44$, $T_{\max} = 55$, $T \in [44, 55]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2348] $6131613151322^11(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -107$, $\det = -256$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.48 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[2349] $6131613151322^11(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -107$, $\det = -256$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.48 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2350] $1^{122315132}181232181232181232181232161 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 51$, $\Delta = -105$, $\det = -256$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.48 \rightarrow 44$, $T_{\max} = 51$, $T \in [44, 51]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 463 & 43 \\ 2 & 43 & 3 \end{pmatrix}_{b_0}$$

[2352] $31612321812321812321812^132181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 54$, $\Delta = -113$, $\det = -256$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.48 \rightarrow 44$, $T_{\max} = 54$, $T \in [44, 54]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[2353] $41413221(12)12231513221(12)12^1 2315132181232181 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 52$, $\Delta = -111$, $\det = -256$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.48 \rightarrow 44$, $T_{\max} = 52$, $T \in [44, 52]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[2357] $8123151322^11(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 55$, $\Delta = -114$, $\det = -289$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.46 \rightarrow 44$, $T_{\max} = 55$, $T \in [44, 55]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[2354] $812321812315132^121(12)12231513221^{31513221}(12)1 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 56$, $\Delta = -115$, $\det = -289$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.46 \rightarrow 44$, $T_{\max} = 56$, $T \in [44, 56]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[2355] $413221(12)12231513221(12)1223151322^11(12)1 + \hat{1} \rightarrow 2$ $T = 39$, $T_{\max} = 56$, $\Delta = -115$, $\det = -289$, $(n_+, n_-, n_0) = (1, 39, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 43.46 \rightarrow 44$, $T_{\max} = 56$, $T \in [44, 56]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 101 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2356] $413221(12)122315132^1 21(12)12231513221(12)1+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 56, \Delta = -115, \det = -289, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.46 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 118 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[2358] $3161232181232181232181232^1 181231513221(12)1+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 54, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.46 \rightarrow 44, T_{\max} = 54, T \in [44, 54]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[2359] $413221(12)12231513221(12)122^1 31513221(12)1+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 56, \Delta = -115, \det = -324, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.43 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 125 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[2363] $31612321812321812321812321812^1 31513221(12)1+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 54, \Delta = -113, \det = -361, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.41 \rightarrow 44, T_{\max} = 54, T \in [44, 54]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[2360] $81232181231513221(12)12^1 231513221(12)1+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 56, \Delta = -115, \det = -361, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.41 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[2361] $812321812^1 31513221(12)12231513221(12)1+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 56, \Delta = -115, \det = -361, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.41 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[2362] $81231513221(12)12^1 231513221(12)12231513218123+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 55, \Delta = -114, \det = -361, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.41 \rightarrow 44, T_{\max} = 55, T \in [44, 55]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[2364] $8123218123151322^1 1(12)12231513221(12)1+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 56, \Delta = -115, \det = -441, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.37 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[2365] $413221(12)1223151322^1 1(12)12231513221(12)1+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 56, \Delta = -115, \det = -484, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.36 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 197 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[2366] $413221(12)12231513221(12)12^1 231513221(12)1+ \hat{1} \rightarrow 2$ $T = 39, T_{\max} = 56, \Delta = -115, \det = -529, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 43.34 \rightarrow 44, T_{\max} = 56, T \in [44, 56]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 206 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$$

$T = 40$ [2367] $1(12)12231513221(12)12231513221(12)1223151315+ \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47, T_{\max} = 52, T \in [47, 52]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[2368] $22^1 21(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47, T_{\max} = 52, T \in [47, 52]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[2369] $222^1 1(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 46.22 \rightarrow 47, T_{\max} = 52, T \in [47, 52]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[2370] $32^1 171232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 54, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 46.22 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[2374] $21712^1 32181232181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 16, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 45.75 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2373] $1(12)122315132181232181232181232181232^1 1712 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 16, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 45.75 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2372] $1712^1 32181232181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 16, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 45.75 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2371] $1712^1 32181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 16, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 45.75 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2377] $61316131513221(12)12231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 45.44 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2376] $61316131513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 45.44 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2379] $171232^1 181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 45.44 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2375] $1(12)12231513221(12)12231513221(12)122^1 3151 + \hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 45.44 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2380] $\overset{1}{171232^1 181232181232181232181231513221(12)12+\hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.44 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[2381] $\overset{1}{1(12)1223151321812321812321812321812^1 321712+\hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.44 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[2382] $\overset{1}{2171232^1 181232181232181232181231513221(12)12+\hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.44 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[2383] $\overset{1}{(12)1223151321812321812321812321812321712^1 32+\hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 56, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.44 \rightarrow 46, T_{\max} = 56, T \in [46, 56]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

[2378] $\overset{1}{61316131513221(12)12231513221(12)122315132^1 2181+\hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 52, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.44 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2388] $\overset{1}{61316131513221(12)12231513221(12)1223151322^1 181+\hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 52, \Delta = -106, \det = 36, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.22 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[2387] $\overset{1}{712^1 32181231513221(12)12231513221(12)1223151315+\hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 52, \Delta = -110, \det = 36, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.22 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2386] $\overset{1}{2221(12)122^1 31513221(12)12231513221(12)1223151315 + \hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 52, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.22 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2385] $\overset{21}{21(12)122^1 31513221(12)12231513221(12)1223151315+\hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 52, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.22 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2384] $\overset{221}{1(12)122^1 31513221(12)12231513221(12)1223151315+\hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 52, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.22 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2399] $\overset{1}{61316131513221(12)12231513221(12)122^1 3151321812+\hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 52, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[2398] $\overset{1}{61316131513221(12)12231513221(12)122^1 315132181+\hat{1}\rightarrow 2}$ $T = 40, T_{\max} = 52, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2397] $71232181231513221(12)12231513221(12)122^1 3151315 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2396] $2221(12)12231513221(12)12231513221(12)122^1 3151315 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2395] $2221(12)12^1 231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2394] $21(12)12231513221(12)12231513221(12)122^1 3151315 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2393] $21(12)12^1 231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2400] $61316131513221(12)12231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -106$, $\det = 49$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2391] $1(12)12231513221(12)12231513221(12)12^1 23151 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2390] $1(12)12231513221(12)12231513221(12)122^1 3151315 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2389] $1(12)12^1 231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -109$, $\det = 49$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2401] $1712321812^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2402] $1712321812^1 32181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& [2403] \ 1(12)122315132181232181232181232^1 1812321712+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 49, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
& [2404] \ 321712^1 32181232181232181232181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 54, \Delta = -112, \det = 49, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.06 \rightarrow 46, T_{\max} = 54, T \in [46, 54] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
& [2405] \ 21712321812^1 32181232181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 49, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
& [2392] \ 1 \begin{smallmatrix} 5132^1 21 \\ (12) \end{smallmatrix} 12231513221(12)12231513221(12)1223151+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -110, \det = 49, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.06 \rightarrow 46, T_{\max} = 52, T \in [46, 52] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0} \\
& [2406] \ 1712321812321812321812321812315132^1 21(12)1+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 64, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.94 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [2411] \ 1(12)1223151321812321812321812^1 321812321712+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 64, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.94 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
& [2410] \ 1(12)122^1 3151321812321812321812321812321712+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 64, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.94 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& [2412] \ 32171232^1 181232181232181232181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 54, \Delta = -112, \det = 64, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.94 \rightarrow 45, T_{\max} = 54, T \in [45, 54] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& [2413] \ 2171232181232^1 181232181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 64, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.94 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& [2409] \ 171232181232^1 181232181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 64, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.94 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& [2414] \ 21712321812321812321812321812315132^1 21(12)12+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 64, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.94 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [2408] \ 1712321812321812321812321812315132^1 21(12)12+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 64, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.94 \rightarrow 45, T_{\max} = 52, T \in [45, 52]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2407] & \begin{matrix} 1 \\ 171232181232^1 181232181232181231513221(12)1+\hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 64, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.94 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2426] & \begin{matrix} 1 \\ 61316131513221(12)122315132^1 21(12)1223151321812+\hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -106, \det = 81, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2434] & \begin{matrix} 1 \\ 217123218123218123218123218123151322^1 1(12)12+\hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 81, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2415] & \begin{matrix} 221 \\ 1(12)12231513221(12)122315132^1 21(12)1223151315+\hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -109, \det = 81, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2416] & \begin{matrix} 221 \\ 1(12)12231513221(12)12231513221(12)12^1 23151315+\hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -109, \det = 81, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2417] & \begin{matrix} 21 \\ 21(12)12231513221(12)122315132^1 21(12)1223151315+\hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -109, \det = 81, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2418] & \begin{matrix} 21 \\ 21(12)12231513221(12)12231513221(12)12^1 23151315+\hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -109, \det = 81, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2419] & \begin{matrix} 1 \\ 2221(12)12231513221(12)122315132^1 21(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -109, \det = \\
& 81, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2420] & \begin{matrix} 1 \\ 2221(12)12231513221(12)12231513221(12)12^1 23151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -109, \det = \\
& 81, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2421] & \begin{matrix} 1 \\ 71232181231513221(12)122315132^1 21(12)1223151315+\hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -110, \det = 81, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2422] & \begin{matrix} 1 \\ 71232^1 181231513221(12)12231513221(12)1223151315+\hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -110, \det = 81, (n_+, n_-, n_0) = \\
& (1, 40, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2423] & \begin{matrix} 1 \\ 71232181231513221(12)12231513221(12)12^1 23151315+\hat{1} \rightarrow 2 \end{matrix} \quad T = 40, T_{\max} = 52, \Delta = -110, \det = 81, (n_+, n_-, n_0) =
\end{aligned}$$

$$(1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2424] \text{ 61316131513221(12)12231513221(12)12}^1\text{2315132181}+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2425] \text{ 61316131513221(12)122315132}^1\text{21(12)122315132181}+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2427] \text{ 61316131513221(12)12231513221(12)12}^1\text{23151321812}+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2428] \text{ 61316131513221(12)122315132}^1\text{21(12)1223151322181}+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2429] \text{ 61316131513221(12)12231513221(12)12}^1\text{23151322181}+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2430] \text{ 17123218123218123218123218123151322}^1\text{1(12)1}+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2432] \text{ 1(12)12}^1\text{23151321812321812321812321812321712}+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 152 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[2433] \text{ 321712321812321812321812321812315132}^1\text{21(12)1}+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 54, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.84 \rightarrow 45, T_{\max} = 54, T \in [45, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2431] \text{ 17123218123218123218123218123151322}^1\text{1(12)12}+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 52, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.84 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2443] \text{ 321712321812}^1\text{32181232181232181231513221(12)1}+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 54, \Delta = -112, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.76 \rightarrow 45, T_{\max} = 54, T \in [45, 54]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2442] \text{ 3217123218123218123218123218123151322}^1\text{1(12)1}+\hat{1} \rightarrow 2 \quad T = 40, T_{\max} = 54, \Delta = -112, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 44.76 \rightarrow 45, T_{\max} = 54, T \in [45, 54]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2441] $1(12)122315132181232181232^1 1812321812321712 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.76 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[2440] $1712321812321812^1 32181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.76 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2439] $1712321812321812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.76 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2438] $2221(12)122315132^1 21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -109, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.76 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2437] $21(12)122315132^1 21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -109, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.76 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2436] $1(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -110, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.76 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2435] $1(12)122315132^1 21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -109, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.76 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2444] $21712321812321812^1 32181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.76 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2445] $(12)122315132181232181232181232181232^1 171232 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 56, \Delta = -114, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.76 \rightarrow 45, T_{\max} = 56, T \in [45, 56]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2446] $(12)122^1 315132181232181232181232181232171232 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 56, \Delta = -114, \det = 100, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.76 \rightarrow 45, T_{\max} = 56, T \in [45, 56]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2462] $171232181232181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.69 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[2468] $(12)1223151321812321812321812321812^1 32171232 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 56, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 56, \quad T \in [45, 56] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& [2448] \overset{221}{1}(12)12231513221 \overset{1}{(12)1223151322^1 1}(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 40, \quad T_{\max} = 52, \quad \Delta = -109, \quad \det = 121, \quad (n_+, n_-, n_0) = \\
& (1, 40, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 52, \quad T \in [45, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [2449] \overset{513221}{1}(12)12231513221 \overset{1}{(12)122315132^1 21}(12)1223151 + \hat{1} \rightarrow 2 \quad T = 40, \quad T_{\max} = 52, \quad \Delta = -110, \quad \det = 121, \quad (n_+, n_-, n_0) = \\
& (1, 40, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 52, \quad T \in [45, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [2450] \overset{21}{21}(12)12231513221 \overset{1}{(12)122^1 31513221}(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 40, \quad T_{\max} = 52, \quad \Delta = -109, \quad \det = 121, \quad (n_+, n_-, n_0) = \\
& (1, 40, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 52, \quad T \in [45, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
& [2463] \overset{1}{17}1232181232181232^1 181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 40, \quad T_{\max} = 52, \quad \Delta = -103, \quad \det = 121, \quad (n_+, n_-, n_0) = \\
& (1, 40, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 52, \quad T \in [45, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
& [2451] \overset{21}{21}(12)12231513221 \overset{1}{(12)1223151322^1 1}(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 40, \quad T_{\max} = 52, \quad \Delta = -109, \quad \det = 121, \quad (n_+, n_-, n_0) = \\
& (1, 40, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 52, \quad T \in [45, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [2452] \overset{1}{2221}(12)12231513221 \overset{1}{(12)1223151322^1 1}(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 40, \quad T_{\max} = 52, \quad \Delta = -109, \quad \det = \\
& 121, \quad (n_+, n_-, n_0) = (1, 40, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 52, \quad T \in [45, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [2464] \overset{1}{1}(12)1223151321812321812^1 321812321812321712 + \hat{1} \rightarrow 2 \quad T = 40, \quad T_{\max} = 52, \quad \Delta = -103, \quad \det = 121, \quad (n_+, n_-, n_0) = \\
& (1, 40, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 52, \quad T \in [45, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
& [2467] (12)12^1 2315132181232181232181232171232 + \hat{1} \rightarrow 2 \quad T = 40, \quad T_{\max} = 56, \quad \Delta = -114, \quad \det = 121, \quad (n_+, n_-, n_0) = \\
& (1, 40, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 56, \quad T \in [45, 56] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
& [2465] \overset{1}{3217}1232181232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 40, \quad T_{\max} = 54, \quad \Delta = -112, \quad \det = 121, \quad (n_+, n_-, n_0) = \\
& (1, 40, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 54, \quad T \in [45, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
& [2453] \overset{1}{2221}(12)12231513221 \overset{1}{(12)122^1 31513221}(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 40, \quad T_{\max} = 52, \quad \Delta = -109, \quad \det = \\
& 121, \quad (n_+, n_-, n_0) = (1, 40, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 52, \quad T \in [45, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
& [2454] \overset{1}{71}232181231513221(12)122^1 31513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 40, \quad T_{\max} = 52, \quad \Delta = -110, \quad \det = 121, \quad (n_+, n_-, n_0) = \\
& (1, 40, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 44.69 \rightarrow 45, \quad T_{\max} = 52, \quad T \in [45, 52] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2455] $71232181231513221(12)1223151322^11(12)1223151315+\hat{1}\rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.69 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2456] $61316131513221(12)122^131513221(12)122315132181+\hat{1}\rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.69 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2457] $61316131513221(12)1223151322^11(12)122315132181+\hat{1}\rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.69 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[2458] $61316131513221(12)122^131513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.69 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2459] $61316131513221(12)1223151322^11(12)1223151321812+\hat{1}\rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.69 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[2460] $61316131513221(12)1223151322^11(12)1223151322181+\hat{1}\rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.69 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[2461] $61316131513221(12)122^131513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.69 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2466] $2171232181232181232^1181232181231513221(12)12+\hat{1}\rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.69 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[2447] $1^{221}(12)12231513221(12)122^131513221(12)1223151315+\hat{1}\rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -109$, $\det = 121$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.69 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2470] $1^{513221}(12)12231513221(12)1223151322^11(12)1223151+\hat{1}\rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -110$, $\det = 169$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[2480] $321712321812321812^132181232181231513221(12)1+\hat{1}\rightarrow 2$ $T = 40$, $T_{\max} = 54$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45$, $T_{\max} = 54$, $T \in [45, 54]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2482] $(12)122315132181232181232181232^1181232171232+\hat{1}\rightarrow 2$ $T = 40$, $T_{\max} = 56$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45$, $T_{\max} = 56$, $T \in [45, 56]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2469] $1(12)1223151322^1 1(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -109, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2481] $21712321812321812321812^1 32181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2471] $21(12)1223151322^1 1(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -109, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2472] $2221(12)1223151322^1 1(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -109, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2473] $712321812315132^1 21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -110, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2474] $613161315132^1 21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[2475] $613161315132^1 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[2476] $613161315132^1 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[2477] $1712321812321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2478] $1712321812321812321812^1 32181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[2479] $1(12)122315132181232^1 1812321812321812321712 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 52, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.59 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$

[2496] $(12)1223151321812321812321812^1 32181232171232 + \hat{1} \rightarrow 2$ $T = 40, T_{\max} = 56, \Delta = -114, \det = 196, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 44.55 \rightarrow 45, T_{\max} = 56, T \in [45, 56]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[2494] $32171232181232181232^1 181232181231513221(12)1+\hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 54$, $\Delta = -112$, $\det = 196$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 54, T \in [45, 54]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[2483] $1(12)12231513221(12)12^{221}231513221(12)1223151315+\hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -109$, $\det = 196$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2484] $1(12)122^131513221(12)12231513221(12)1223151+\hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -110$, $\det = 196$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2485] $21(12)12231513221(12)12^{21}231513221(12)1223151315+\hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -109$, $\det = 196$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2486] $2221(12)12231513221(12)12^1231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -109$, $\det = 196$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2487] $71232181231513221(12)12^1231513221(12)1223151315+\hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -110$, $\det = 196$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2488] $61316131513221(12)12^1231513221(12)122315132181+\hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -106$, $\det = 196$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[2489] $61316131513221(12)12^1231513221(12)1223151321812+\hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -106$, $\det = 196$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[2490] $61316131513221(12)12^1231513221(12)1223151322181+\hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -106$, $\det = 196$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[2491] $171232181232181232181232^1 181231513221(12)1+\hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -103$, $\det = 196$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[2492] $171232181232181232181232^1 181231513221(12)12+\hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -103$, $\det = 196$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 44.55 \rightarrow 45, T_{\max} = 52, T \in [45, 52]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[2495] $2171232181232181232181232^1 181231513221(12)12+\hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -103$, $\det = 196$, $(n_+, n_-, n_0) =$

(1, 40, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 44.55 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[2493] $1(12)1223151321812^1 321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -103$, $\det = 196$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 44.55 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$$

[2497] $712321812^1 31513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -110$, $\det = 225$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 44.52 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[2505] $321712321812321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 54$, $\Delta = -112$, $\det = 256$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 44.48 \rightarrow 45$, $T_{\max} = 54$, $T \in [45, 54]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[2502] $1712321812321812321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 44.48 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[2501] $6131613151322^1 1(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -106$, $\det = 256$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 44.48 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[2503] $1712321812321812321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 44.48 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[2498] $7123218123151322^1 1(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -110$, $\det = 256$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 44.48 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2499] $6131613151322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -106$, $\det = 256$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 44.48 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[2500] $6131613151322^1 1(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -106$, $\det = 256$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 44.48 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[2504] $1(12)122315132^1 1812321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 44.48 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 463 & 43 \\ 2 & 43 & 3 \end{pmatrix}_{b_0}$$

[2506] $21712321812321812321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 44.48 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

- [2507] $(12)122315132181232181232^1 181232181232171232 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 56$, $\Delta = -114$, $\det = 256$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.48 \rightarrow 45$, $T_{\max} = 56$, $T \in [45, 56]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [2508] $1 \overset{513221}{(12)} 12231513221 \overset{1}{(12)} 122^1 31513221 (12)1223151 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -110$, $\det = 289$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.46 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [2509] $1 \overset{513221}{(12)} 12^1 231513221 \overset{1}{(12)} 12231513221 (12)1223151 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -110$, $\det = 289$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.46 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [2510] $32171232181232181232181232^1 181231513221 (12)1 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 54$, $\Delta = -112$, $\det = 289$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.46 \rightarrow 45$, $T_{\max} = 54$, $T \in [45, 54]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [2511] $(12)1223151321812321812^1 32181232181232171232 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 56$, $\Delta = -114$, $\det = 289$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.46 \rightarrow 45$, $T_{\max} = 56$, $T \in [45, 56]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [2512] $1 \overset{513221}{(12)} 122315132^1 21 \overset{1}{(12)} 12231513221 (12)1223151 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -110$, $\det = 324$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.43 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [2513] $321712321812321812321812321812^1 31513221 (12)1 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 54$, $\Delta = -112$, $\det = 361$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.41 \rightarrow 45$, $T_{\max} = 54$, $T \in [45, 54]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [2514] $(12)122315132181232^1 181232181232181232171232 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 56$, $\Delta = -114$, $\det = 361$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.41 \rightarrow 45$, $T_{\max} = 56$, $T \in [45, 56]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [2515] $(12)1223151321812^1 32181232181232181232171232 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 56$, $\Delta = -114$, $\det = 400$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.39 \rightarrow 45$, $T_{\max} = 56$, $T \in [45, 56]$
 $\begin{pmatrix} 0 & 20 & 2 \\ 20 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [2517] $(12)122315132^1 181232181232181232181232171232 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 56$, $\Delta = -114$, $\det = 484$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.36 \rightarrow 45$, $T_{\max} = 56$, $T \in [45, 56]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [2516] $1 \overset{513221}{(12)} 12231513221 \overset{1}{(12)} 12^1 231513221 (12)1223151 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -110$, $\det = 484$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.36 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [2518] $1 \overset{513221}{(12)} 1223151322^1 1 \overset{1}{(12)} 12231513221 (12)1223151 + \hat{1} \rightarrow 2$ $T = 40$, $T_{\max} = 52$, $\Delta = -110$, $\det = 529$, $(n_+, n_-, n_0) = (1, 40, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 44.34 \rightarrow 45$, $T_{\max} = 52$, $T \in [45, 52]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$T = 41 \quad [2519] \quad 1^{22^1 1} (12)12231513221(12)^1 12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2520] \quad 61231513221(12)12231513221(12)^1 12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2521] \quad 6131513221(12)^1 12231513221(12)12231513221(12)^{22^1 21} 1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2522] \quad 6131513221(12)^1 12231513221(12)12231513221(12)^{22^1 1} 12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2523] \quad 22^1 21(12)12231513221(12)^1 12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2524] \quad 22^1 1(12)12231513221(12)^1 12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48, T_{\max} = 57, T \in [48, 57]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2525] \quad 61231513221(12)^1 12231513221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2526] \quad 6131513221(12)^1 12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2527] \quad 1(12)^1 1223151321812321812321812321812321812^1 2 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -4, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[2528] \quad 22^1 181232181232181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -4, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[2529] \quad 6131513221(12)^1 12231513221(12)12231513221(12)^{32^1 21} 1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 58, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[2530] $6131513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[2531] $222^1 1(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[2532] $61231513221(12)12231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -107, \det = -9, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[2535] $32^1 2181232181232181232181232181231513221(12)1+ \hat{1} \rightarrow 2$ $T = 41, T_{\max} = 54, \Delta = -111, \det = -9, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[2534] $1^1 8^1 1232181232181232181232181231513221(12)1+ \hat{1} \rightarrow 2$ $T = 41, T_{\max} = 54, \Delta = -111, \det = -9, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 14 & 6 \\ 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[2533] $6131513221(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 53, T \in [48, 53]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[2538] $1(12)122315132181232181232181232181232^1 18122+ \hat{1} \rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -102, \det = -16, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 46.75 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2540] $221812^1 32181232181232181232181231513221(12)12+ \hat{1} \rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -102, \det = -16, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 46.75 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2536] $6131513221(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 46.75 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2537] $1^1 8^1 12^1 32181232181232181232181231513221(12)1+ \hat{1} \rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -102, \det = -16, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 46.75 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2539] $1^1 8^1 12^1 32181232181232181232181231513221(12)12+ \hat{1} \rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -102, \det = -16, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 46.75 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2541] $6131513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 58, \Delta = -115, \det = -25, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.44 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[2542] $612^1 31513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.44 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[2543] $612^1 31513221(12)12231513221(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.44 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[2544] $6131513221(12)12231513221(12)122315132^1 21(12)1+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.44 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[2545] $6131513221(12)12231513221(12)122315132^1 21(12)12+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.44 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[2546] $612^1 31513221(12)12231513221(12)12231513221(12)1222 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.44 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

[2547] $6131513221(12)12231513221(12)122315132^1 21(12)1222 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -25, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.44 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[2548] $612^1 31513221(12)12231513221(12)12231513218123217+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -108, \det = -25, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.44 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[2549] $1^1 81232^1 181232181232181232181231513221(12)1+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -102, \det = -25, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.44 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[2550] $1(12)1223151321812321812321812321812^1 3218122+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -102, \det = -25, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.44 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[2551] $1^1 81232^1 181232181232181232181231513221(12)12+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -102, \det = -25, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.44 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[2552] $(12)1223151321812321812321812321812321^1 81+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 56, \Delta = -113, \det = -25, (n_+, n_-, n_0) =$

(1, 41, 0), $c_{\text{ext}} = 15.0$

$T_{\min} = 46.44 \rightarrow 47$, $T_{\max} = 56$, $T \in [47, 56]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[2553] $322^1 181232181232181232181232181231513221(12)1+\hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 54$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.44 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2554] $22181232^1 181232181232181232181231513221(12)12+\hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.44 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2555] $1(12)122^1 31513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.22 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2556] $21(12)122^1 31513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.22 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2557] $61231513221(12)12231513221(12)122315132^1 21(12)1+\hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.22 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2558] $61231513221(12)12231513221(12)122315132^1 21(12)12+\hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.22 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2559] $6131513221(12)12231513221(12)1223151322^1 1(12)1+\hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.22 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2560] $6131513221(12)12231513221(12)1223151322^1 1(12)12+\hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.22 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2561] $1812^1 32181231513221(12)12231513221(12)1223151315+\hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -109$, $\det = -36$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.22 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2562] $2221(12)122^1 31513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.22 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2563] $61231513221(12)12231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.22 \rightarrow 47$, $T_{\max} = 53$, $T \in [47, 53]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2564] $6131513221(12)^1 12231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.22 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[2565] $712^1 32181231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -107$, $\det = -36$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.22 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2566] $61231513221(12)12231513221(12)122315132181232^1 17 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -108$, $\det = -36$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.22 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[2573] $61231513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2592] $221812321812^1 32181232181232181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[2569] $21(12)^{21} 12^1 231513221(12)^1 12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2570] $21(12)^{21} 12231513221(12)^1 12231513221(12)122^1 31513161 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2571] $1(12)^{21} 122^1 31513221(12)^1 12231513221(12)122315131613 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 57$, $\Delta = -114$, $\det = -49$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.06 \rightarrow 47, T_{\max} = 57, T \in [47, 57]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[2572] $612315132^1 21(12)12231513221(12)^1 12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2567] $1(12)^{221} 12231513221(12)^1 12231513221(12)122^1 31513161 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2574] $612315132^1 21(12)12231513221(12)^1 12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[2575] $61231513221(12)12231513221(12)1223151322^11(12)12+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[2576] $6131513221(12)12231513221(12)122^131513221(12)1+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2577] $6131513221(12)12231513221(12)122^131513221(12)12+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2578] $181232181231513221(12)12231513221(12)122^13151315+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[2579] $2221(12)12231513221(12)12231513221(12)122^131513161 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[2580] $2221(12)12^1231513221(12)12231513221(12)12231513161 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[2581] $221(12)122^131513221(12)12231513221(12)122315131613 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 57, \Delta = -114, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.06 \rightarrow 47, T_{\max} = 57, T \in [47, 57]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[2582] $61231513221(12)12231513221(12)1223151322^11(12)1222 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[2583] $612315132^121(12)12231513221(12)12231513221(12)1222 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[2584] $6131513221(12)12231513221(12)122^131513221(12)12222 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[2585] $71232181231513221(12)12231513221(12)122^131513161+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.06 \rightarrow 47, T_{\max} = 53, T \in [47, 53]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[2586] $612315132^121(12)12231513221(12)12231513218123217+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 46.06 \rightarrow 47, \quad T_{\max} = 53, \quad T \in [47, 53] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
& [2587] \quad 1 \quad 8 \quad 12321812^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 53, \quad \Delta = -102, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.06 \rightarrow 47, \quad T_{\max} = 53, \quad T \in [47, 53] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
& [2588] \quad 1 \quad 8 \quad 12^1 32181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 54, \quad \Delta = -111, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.06 \rightarrow 47, \quad T_{\max} = 54, \quad T \in [47, 54] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
& [2589] \quad 1(12)122315132181232181232181232^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 53, \quad \Delta = -102, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.06 \rightarrow 47, \quad T_{\max} = 53, \quad T \in [47, 53] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
& [2590] \quad 1 \quad 8 \quad 12321812^1 32181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 53, \quad \Delta = -102, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.06 \rightarrow 47, \quad T_{\max} = 53, \quad T \in [47, 53] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
& [2591] \quad 3221812^1 32181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 54, \quad \Delta = -111, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.06 \rightarrow 47, \quad T_{\max} = 54, \quad T \in [47, 54] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
& [2568] \quad 1(12)12^1 231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 53, \quad \Delta = -106, \quad \det = -49, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.06 \rightarrow 47, \quad T_{\max} = 53, \quad T \in [47, 53] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
& [2593] \quad 1(12)12^1 231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 57, \quad \Delta = -114, \quad \det = -64, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, \quad T_{\max} = 57, \quad T \in [46, 57] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
& [2594] \quad 221(12)12^1 231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 57, \quad \Delta = -114, \quad \det = \\
& -64, \quad (n_+, n_-, n_0) = (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, \quad T_{\max} = 57, \quad T \in [46, 57] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
& [2595] \quad 1 \quad 8 \quad 12321812321812321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 53, \quad \Delta = -102, \quad \det = -64, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, \quad T_{\max} = 53, \quad T \in [46, 53] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
& [2596] \quad 1 \quad 8 \quad 1232181232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 53, \quad \Delta = -102, \quad \det = -64, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, \quad T_{\max} = 53, \quad T \in [46, 53] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& [2597] \quad 1 \quad 8 \quad 1232^1 181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 54, \quad \Delta = -111, \quad \det = -64, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, \quad T_{\max} = 54, \quad T \in [46, 54]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2598] & \overset{1}{1}(12)1223151321812321812321812^1 3218123218122 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2599] & \overset{1}{1}(12)122^1 31513218123218123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2600] & \overset{21}{1}81232181232^1 181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2601] & \overset{21}{1}812321812321812321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2602] & 322181232^1 181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -64, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, T_{\max} = 54, T \in [46, 54] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2603] & 221812321812321812321812321812315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2604] & 22181232181232^1 181232181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.94 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2616] & \overset{1}{6}131513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2637] & 2218123218123218123218123218123151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2624] & \overset{1}{6}131513221(12)12231513221(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2623] & \overset{1}{6}1231513221(12)122^1 31513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[2622] & \overset{1}{6}123151322^1 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det =
\end{aligned}$$

$$-81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2621] \quad 2221(12)12231513221(12)12231513221(12)12^1 231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2620] \quad 2221(12)12231513221(12)122315132^1 21(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2619] \quad 181232181231513221(12)12231513221(12)12^1 23151315 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2618] \quad 181232^1 181231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2615] \quad 6131513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2626] \quad 71232^1 181231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[2617] \quad 181232181231513221(12)122315132^1 21(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2614] \quad 6131513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2613] \quad 6131513221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2612] \quad 6123151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2611] \quad 61231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2610] \ 61231513221(12)122^1 31513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2609] \ 6123151322^1 1(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2608] \ 21(12)12231513221(12)122315132^1 21(12)12231513161+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2607] \ 21(12)12231513221(12)12231513221(12)12^1 231513161+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2606] \ 1(12)12231513221(12)122315132^1 21(12)12231513161+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2605] \ 1(12)12231513221(12)12231513221(12)12^1 231513161+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2627] \ 71232181231513221(12)12231513221(12)12^1 231513161+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2628] \ 71232181231513221(12)122315132^1 21(12)12231513161+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2629] \ 6123151322^1 1(12)12231513221(12)12231513218123217+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2630] \ 61231513221(12)122^1 31513221(12)12231513218123217+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2631] \ 61231513221(12)12231513221(12)1223151321812^1 3217+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 17 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2632] \ 1^1 8123218123218123218123151322^1 1(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -81, (n_+, n_-, n_0) =$$

$$(1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2633] \overset{321}{1 \ 8 \ 12321812321812321812321812315132^1 21(12)1+\hat{1} \rightarrow 2} \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 54, T \in [46, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2634] \overset{1}{1(12)12^1 231513218123218123218123218123218122+\hat{1} \rightarrow 2} \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 152 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[2635] \overset{21}{1 \ 8 \ 123218123218123218123218123151322^1 1(12)12+\hat{1} \rightarrow 2} \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2636] \overset{321}{3221812321812321812321812321812315132^1 21(12)1+\hat{1} \rightarrow 2} \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 54, T \in [46, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2625] \overset{1}{6131513221(12)122315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2} \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.84 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2649] \overset{321}{1 \ 8 \ 123218123218123218123218123151322^1 1(12)1+\hat{1} \rightarrow 2} \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.76 \rightarrow 46, T_{\max} = 54, T \in [46, 54]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[2639] \overset{21}{21(12)122315132^1 21(12)12231513221(12)12231513161+\hat{1} \rightarrow 2} \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2640] \overset{3221}{61315132^1 21(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2} \quad T = 41, T_{\max} = 58, \Delta = -115, \det = -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.76 \rightarrow 46, T_{\max} = 58, T \in [46, 58]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2641] \overset{1}{61231513221(12)12231513221(12)122^1 31513221(12)1+\hat{1} \rightarrow 2} \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2642] \overset{1}{61231513221(12)12231513221(12)122^1 31513221(12)12+\hat{1} \rightarrow 2} \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2643] \overset{2221}{61315132^1 21(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2} \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2644] & 61315132^1 21(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -100, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2645] & 2221(12)122315132^1 21(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = \\
& -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2646] & 61231513221(12)12231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = \\
& -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2647] & 61315132^1 21(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = \\
& -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2648] & 1^21 8 12321812321812^1 32181232181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -100, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2638] & 1(12)122315132^1 21(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -100, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2650] & 1^321 8 12321812^1 32181232181232181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -100, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 54, T \in [46, 54] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2651] & 1(12)122315132181232181232^1 18123218123218122+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -100, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[2652] & 1^21 8 12321812321812^1 32181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -100, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2653] & (12)122315132181232181232181232181232^1 1^2321 8 1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 56, \Delta = -113, \det = -100, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 56, T \in [46, 56] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2654] & (12)122^1 3151321812321812321812321812321^2321 8 1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 56, \Delta = -113, \det = -100, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 56, T \in [46, 56]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2655] & 3221812321812^1 32181232181232181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 54, T \in [46, 54] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2656] & 32218123218123218123218123218123151322^1 1(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 54, T \in [46, 54] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2657] & 221812321812321812^1 32181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -100, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.76 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2666] & 61231513221(12)122315132^1 21(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2690] & (12)12^1 23151321812321812321812321812321^1 8 \quad T = 41, T_{\max} = 56, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 56, T \in [46, 56] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2660] & 21(12)12231513221(12)1223151322^1 1(12)12231513161+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2661] & 21(12)12231513221(12)122^1 31513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2662] & 1(12)12231513221(12)12231513221(12)122^1 315131613+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 57, T \in [46, 57] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2663] & 6131513221(12)12231513221(12)122315132^1 21(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 58, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 58, T \in [46, 58] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2664] & 61231513221(12)12^1 231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2665] & 61231513221(12)122315132^1 21(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2658] $1^{221}(12)12231513221(12)1223151322^11(12)12231513161+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2667] $61231513221(12)12^1231513221(12)12231513221(12)12+ \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2668] $6131513221(12)1223151322^11(12)12231513221(12)1+ \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [2669] $6131513221(12)122^131513221(12)12231513221(12)1+ \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [2670] $6131513221(12)122^131513221(12)12231513221(12)12+ \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [2671] $6131513221(12)1223151322^11(12)12231513221(12)12+ \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [2672] $181232181231513221(12)1223151322^11(12)1223151315+ \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -109, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2673] $181232181231513221(12)122^131513221(12)1223151315+ \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -109, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [2674] $2221(12)12231513221(12)122^131513221(12)12231513161 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [2675] $2221(12)12231513221(12)1223151322^11(12)12231513161 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2676] $221(12)12231513221(12)12231513221(12)122^1315131613 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 57, \Delta = -114, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.69 \rightarrow 46, T_{\max} = 57, T \in [46, 57]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [2677] $61231513221(12)12^1231513221(12)12231513221(12)1222 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -107, \det =$

$$-121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2678] \quad 61231513221(12)122315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2679] \quad 6131513221(12)1223151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2680] \quad 6131513221(12)122^1 31513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2681] \quad 71232181231513221(12)122^1 31513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[2682] \quad 71232181231513221(12)1223151322^1 1(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[2683] \quad 61231513221(12)12^1 231513221(12)12231513218123217 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2684] \quad 61231513221(12)122315132^1 21(12)12231513218123217 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2685] \quad 1^1 8 1232181232181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[2686] \quad 1^1 8 1232181232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 54, T \in [46, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[2687] \quad 1(12)1223151321812321812^1 3218123218123218122 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[2688] \quad 1^1 8 1232181232181232^1 181232181231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2691] & 322181232181232^1 181232181232181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 54, T \in [46, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2692] & 22181232181232181232^1 181232181231513221(12)12+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2689] & (12)1223151321812321812321812321812^1 321 \overset{2321}{8} 1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 56, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 56, T \in [46, 56] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2659] & 1(12)12231513221(12)122^1 31513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.69 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2701] & 613151322^1 1(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2703] & 1812321812315132^1 21(12)12231513221(12)1223151315+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -109, \det = -169, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2694] & 21(12)1223151322^1 1(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2695] & 1(12)122315132^1 21(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = -169, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 57, T \in [46, 57] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2704] & 2221(12)1223151322^1 1(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2705] & 221(12)12231513221(12)122315132^1 21(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = -169, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 57, T \in [46, 57] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2706] & 221(12)122315132^1 21(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = -169, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 57, T \in [46, 57] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2707] $61231513221(12)12231513221(12)12^1231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -107$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.59 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [2708] $613151322^11(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.59 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2709] $712321812315132^121(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -107$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.59 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [2710] $61231513221(12)12231513221(12)122^131513218123217 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -108$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.59 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2711] $1^21812321812321812^132181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -102$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.59 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [2712] $1^321812321812^132181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 54$, $\Delta = -111$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.59 \rightarrow 46$, $T_{\max} = 54$, $T \in [46, 54]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [2713] $1^1(12)122315132181232^118123218123218123218122 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -102$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.59 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$
- [2714] $1^21812321812321812^132181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -102$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.59 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [2715] $(12)122315132181232181232181232^11812321^81 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 56$, $\Delta = -113$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.59 \rightarrow 46$, $T_{\max} = 56$, $T \in [46, 56]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [2716] $3221812321812321812^132181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 54$, $\Delta = -111$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.59 \rightarrow 46$, $T_{\max} = 54$, $T \in [46, 54]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [2717] $221812321812321812321812^132181231513221(12)12 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -102$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 45.59 \rightarrow 46$, $T_{\max} = 53$, $T \in [46, 53]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [2702] $613151322^11(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 41$, $T_{\max} = 53$, $\Delta = -106$, $\det = -169$, $(n_+, n_-, n_0) = (1, 41, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2700] & 61231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -169, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2696] & 1(12)12231513221(12)122315132^1 21(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = -169, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 57, T \in [46, 57] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2697] & 613151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 58, \Delta = -115, \det = -169, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 58, T \in [46, 58] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2698] & 6131513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 58, \Delta = -115, \det = -169, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 58, T \in [46, 58] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[2699] & 61231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -169, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2693] & 1(12)1223151322^1 1(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -169, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.59 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2724] & 6131513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -196, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2728] & 221(12)12231513221(12)12231513221(12)12^1 2315131613 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = \\
& -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.55 \rightarrow 46, T_{\max} = 57, T \in [46, 57] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2723] & 61231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -196, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2722] & 61231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -196, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2726] & 181232181231513221(12)12^1 231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -109, \det = -196, (n_+, n_-, n_0) = \\
& (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]
\end{aligned}$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2727] \quad 2221(12)12231513221(12)12^{1231513221}(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2739] \quad 22181232181232181232181232^{1181231513221}(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[2719] \quad 21(12)12231513221(12)12^{1231513221}(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2718] \quad 1(12)12231513221(12)12^{1231513221}(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -106, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2738] \quad 322181232181232181232^{1181232181231513221}(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.55 \rightarrow 46, T_{\max} = 54, T \in [46, 54]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[2737] \quad (12)1223151321812321812321812^{1321812321} \quad 8 \quad 1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 56, \Delta = -113, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.55 \rightarrow 46, T_{\max} = 56, T \in [46, 56]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[2736] \quad 1 \quad 8 \quad 1232181232181232181232^{1181231513221}(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[2735] \quad 1(12)1223151321812^{13218123218123218123218122} + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$$

$$[2734] \quad 1 \quad 8 \quad 1232181232181232^{1181232181231513221}(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.55 \rightarrow 46, T_{\max} = 54, T \in [46, 54]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[2733] \quad 1 \quad 8 \quad 1232181232181232^{1181231513221}(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[2732] \quad 61231513221(12)1223151322^{11}(12)12231513218123217 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -108, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2720] $1(12)^{21}12231513221(12)^112231513221(12)12^12315131613+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 57, \Delta = -114, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.55 \rightarrow 46, T_{\max} = 57, T \in [46, 57]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2731] $71232181231513221(12)12^1231513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -107, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[2721] $6131513221(12)122^131513221(12)^112231513221(12)^{3221}1+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 58, \Delta = -115, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.55 \rightarrow 46, T_{\max} = 58, T \in [46, 58]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2730] $6131513221(12)12^1231513221(12)12231513221(12)12222 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2729] $61231513221(12)1223151322^11(12)12231513221(12)1222 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -107, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2725] $6131513221(12)^112^1231513221(12)12231513221(12)12+ \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -106, \det = -196, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.55 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2740] $1(12)^{21}12231513221(12)^1122^131513221(12)122315131613+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 57, \Delta = -114, \det = -225, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.52 \rightarrow 46, T_{\max} = 57, T \in [46, 57]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[2741] $1812321812^131513221(12)12231513221(12)1223151315+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -109, \det = -225, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.52 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[2742] $221(12)12231513221(12)^1122^131513221(12)122315131613 + \hat{1}\rightarrow 2$ $T = 41, T_{\max} = 57, \Delta = -114, \det = -225, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.52 \rightarrow 46, T_{\max} = 57, T \in [46, 57]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[2743] $712321812^131513221(12)12231513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -107, \det = -225, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.52 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$

[2744] $61231513221(12)12231513221(12)122315132^118123217+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -108, \det = -225, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.52 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2746] $18123218123151322^11(12)12231513221(12)1223151315+\hat{1}\rightarrow 2$ $T = 41, T_{\max} = 53, \Delta = -109, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 45.48 \rightarrow 46, T_{\max} = 53, T \in [46, 53]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[2745] & 1(12)12231513221(12)1223151322^1 1(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.48 \rightarrow 46, T_{\max} = 57, T \in [46, 57] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[2747] & 221(12)12231513221(12)1223151322^1 1(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.48 \rightarrow 46, T_{\max} = 57, T \in [46, 57] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[2756] & 221812321812321812321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.48 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[2755] & 3221812321812321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.48 \rightarrow 46, T_{\max} = 54, T \in [46, 54] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[2754] & (12)122315132181232181232^1 1812321812321^{2321} 8 \quad 1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 56, \Delta = -113, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.48 \rightarrow 46, T_{\max} = 56, T \in [46, 56] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[2753] & 1812321812321812321812321812^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.48 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[2752] & 1(12)122315132^1 18123218123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.48 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 463 & 43 \\ 2 & 43 & 3 \end{pmatrix}_{b_0} \\
[2751] & 1812321812321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.48 \rightarrow 46, T_{\max} = 54, T \in [46, 54] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[2750] & 1812321812321812321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -102, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.48 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[2749] & 61231513221(12)12231513221(12)12^1 231513218123217 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -108, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.48 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2748] & 7123218123151322^1 1(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 53, \Delta = -107, \det = -256, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.48 \rightarrow 46, T_{\max} = 53, T \in [46, 53] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[2763] & 322181232181232181232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -289, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$(1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.46 \rightarrow 46, T_{\max} = 54, T \in [46, 54]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[2762] \quad (12)1223151321812321812^{2321}8 \ 1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 56, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.46 \rightarrow 46, T_{\max} = 56, T \in [46, 56]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[2761] \quad 1 \ 8 \ 12321812321812321812321^{321}181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -289, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.46 \rightarrow 46, T_{\max} = 54, T \in [46, 54]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[2760] \quad 221(12)12231513221^{11}(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = -289, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.46 \rightarrow 46, T_{\max} = 57, T \in [46, 57]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[2759] \quad 6131513221(12)12^{1231513221}(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 58, \Delta = -115, \det = -289, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.46 \rightarrow 46, T_{\max} = 58, T \in [46, 58]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[2758] \quad 6131513221(12)12231513221(12)122^{131513221}(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 58, \Delta = -115, \det = -289, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.46 \rightarrow 46, T_{\max} = 58, T \in [46, 58]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[2757] \quad 1(12)12231513221^{21}1(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = -289, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.46 \rightarrow 46, T_{\max} = 57, T \in [46, 57]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[2764] \quad 6131513221(12)122315132^{121}(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 58, \Delta = -115, \det = -324, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.43 \rightarrow 46, T_{\max} = 58, T \in [46, 58]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[2769] \quad 3221812321812321812321812^{131513221}(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -361, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.41 \rightarrow 46, T_{\max} = 54, T \in [46, 54]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

$$[2768] \quad (12)122315132181232^{11812321812321812321}8 \ 1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 56, \Delta = -113, \det = -361, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.41 \rightarrow 46, T_{\max} = 56, T \in [46, 56]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[2767] \quad 1 \ 8 \ 12321812321812321812^{321}31513221(12)1+\hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 54, \Delta = -111, \det = -361, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 45.41 \rightarrow 46, T_{\max} = 54, T \in [46, 54]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

$$[2766] \quad 221(12)12231513221(12)12^{1231513221}(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 41, T_{\max} = 57, \Delta = -114, \det = -361, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 45.41 \rightarrow 46, \quad T_{\max} = 57, \quad T \in [46, 57] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[2765] & 1(12)12231513221(12)12^{21}231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 57, \quad \Delta = -114, \quad \det = -361, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.41 \rightarrow 46, \quad T_{\max} = 57, \quad T \in [46, 57] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[2770] & (12)1223151321812^{2321}321812321812321812321 \quad 8 \quad 1 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 56, \quad \Delta = -113, \quad \det = -400, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.39 \rightarrow 46, \quad T_{\max} = 56, \quad T \in [46, 56] \\
& \begin{pmatrix} 0 & 20 & 2 \\ 20 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[2771] & 6131513221(12)12231513221(12)12^{3221}231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 58, \quad \Delta = -115, \quad \det = -484, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.36 \rightarrow 46, \quad T_{\max} = 58, \quad T \in [46, 58] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[2772] & (12)122315132^{181}1812321812321812321812321 \quad 8 \quad 1 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 56, \quad \Delta = -113, \quad \det = -484, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.36 \rightarrow 46, \quad T_{\max} = 56, \quad T \in [46, 56] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[2773] & 6131513221(12)1223151322^{11}1(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 41, \quad T_{\max} = 58, \quad \Delta = -115, \quad \det = -529, \quad (n_+, n_-, n_0) = \\
& (1, 41, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 45.34 \rightarrow 46, \quad T_{\max} = 58, \quad T \in [46, 58] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
\\
T = 42 \quad [3] & 71231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 42, \quad T_{\max} = 54, \quad \Delta = -105, \quad \det = \\
& 4, \quad (n_+, n_-, n_0) = (1, 42, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 49, \quad T_{\max} = 54, \quad T \in [49, 54] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[0] & 1(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, \quad T_{\max} = 54, \quad \Delta = -105, \quad \det = 4, \quad (n_+, n_-, n_0) = \\
& (1, 42, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 49, \quad T_{\max} = 54, \quad T \in [49, 54] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[6] & 71231513221(12)12231513221(12)12231513221(12)1222^12 + \hat{1} \rightarrow 2 \quad T = 42, \quad T_{\max} = 54, \quad \Delta = -105, \quad \det = 4, \quad (n_+, n_-, n_0) = \\
& (1, 42, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 49, \quad T_{\max} = 54, \quad T \in [49, 54] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[5] & 712231513221(12)12231513221(12)12231513221(12)122^12 + \hat{1} \rightarrow 2 \quad T = 42, \quad T_{\max} = 54, \quad \Delta = -105, \quad \det = 4, \quad (n_+, n_-, n_0) = \\
& (1, 42, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 49, \quad T_{\max} = 54, \quad T \in [49, 54] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[4] & 22^121(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, \quad T_{\max} = 54, \quad \Delta = -105, \quad \det = 4, \quad (n_+, n_-, n_0) = \\
& (1, 42, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 49, \quad T_{\max} = 54, \quad T \in [49, 54] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[2] & 71231513221(12)12231513221(12)12231513221(12)122^121 + \hat{1} \rightarrow 2 \quad T = 42, \quad T_{\max} = 54, \quad \Delta = -105, \quad \det = 4, \quad (n_+, n_-, n_0) = \\
& (1, 42, 0), \quad c_{\text{ext}} = 15.0
\end{aligned}$$

$$T_{\min} = 49, T_{\max} = 54, T \in [49, 54]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 712231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49, T_{\max} = 54, T \in [49, 54]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 712231513221(12)12231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 54, T \in [49, 54]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[9] \quad 222^1 1(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 54, T \in [49, 54]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 71231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 54, T \in [49, 54]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 71231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 71231513221(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 9, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 54, T \in [49, 54]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 71231513221(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 16, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.75 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[24] \quad 612^1 31513221(12)12231513221(12)122315132181232181+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -107, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[19] \quad 71231513221(12)12231513221(12)122315132^1 21(12)1+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[25] \quad 71232181231513221(12)12231513221(12)122315132^1 171+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[26] \quad 4132^1 21(12)12231513221(12)12231513221(12)1223151316 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.44 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[27] & 71232181231513221(12)12231513221(12)122315132^1 217 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[20] & 71231513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[22] & 7122^1 31513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[21] & 2221(12)12231513221(12)12231513221(12)122315132^1 171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[16] & 71231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.44 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[18] & 7122^1 31513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[17] & 7122^1 31513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[15] & 21(12)12231513221(12)12231513221(12)122315132^1 171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[23] & 71231513221(12)12231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[14] & 1(12)12231513221(12)12231513221(12)122315132^1 171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[13] & 4132^1 181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.44 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[44] 16131513221(12)12231513221(12)122315132181232¹181+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[41] 712¹231513221(12)12231513221(12)12231513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[42] 712¹31513221(12)12231513221(12)12231513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[43] 71231513221(12)12231513221(12)1223151322¹1(12)1222 + $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[40] 712231513221(12)12231513221(12)122315132¹21(12)1222 + $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[45] 61231513221(12)12231513221(12)122315132181232¹181+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[46] 712¹32181231513221(12)12231513221(12)122315132171+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[47] 71232181231513221(12)12231513221(12)1223151322¹17+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[48] 712¹32181231513221(12)12231513221(12)122315132217+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[32] 712231513221(12)12231513221(12)122315132¹21(12)1+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[31] 712¹231513221(12)12231513221(12)12231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[39] 2221(12)122¹31513221(12)12231513221(12)122315132171 + $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[38] & 71231513221(12)^1 12231513221(12) 1223151322^1 1(12)^{221} 12 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[33] & 712231513221(12)^1 12231513221(12) 122315132^1 21(12)^{21} 12 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[34] & 712^1 231513221(12)^1 12231513221(12) 12231513221(12)^{21} 12 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[30] & 712^1 31513221(12)^1 12231513221(12) 12231513221(12)^{3221} 1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.22 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[28] & 1(12)^{221} 122^1 31513221(12)^1 12231513221(12) 122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[35] & 712^1 31513221(12)^1 12231513221(12) 12231513221(12)^{2221} 1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[36] & 71231513221(12)^1 12231513221(12) 1223151322^1 1(12)^{2221} 1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[37] & 712^1 31513221(12)^1 12231513221(12) 12231513221(12)^{221} 12 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[29] & 21(12)^{21} 122^1 31513221(12)^1 12231513221(12) 122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 36, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.22 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[59] & 71231513221(12)^1 12231513221(12) 122^1 31513221(12)^{221} 12 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[58] & 71231513221(12)^1 12231513221(12) 122^1 31513221(12)^{2221} 1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[61] $2221(12)12^1 231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[62] $7122315132^1 21(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[63] $712231513221(12)12231513221(12)1223151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[64] $71231513221(12)12231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[65] $161315132^1 21(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[66] $812^1 32181231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[67] $612315132^1 21(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[68] $71232181231513221(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[69] $41322^1 1(12)12231513221(12)12231513221(12)1223151316 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[70] $71232181231513221(12)12231513221(12)122^1 315132217 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[57] $7122315132^1 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[56] $712231513221(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) =$

$$(1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[55] \quad 7122315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[54] \quad 712231513221(12)12231513221(12)12231513221^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[53] \quad 21(12)12231513221(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[52] \quad 21(12)12^1 231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[51] \quad 1(12)12231513221(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[50] \quad 1(12)12^1 231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[49] \quad 41321812^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -110, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[60] \quad 2221(12)12231513221(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.06 \rightarrow 48, T_{\max} = 54, T \in [48, 54]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[71] \quad 4132181232^1 181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -110, \det = 64, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.94 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[74] \quad 1(12)12231513221(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[104] \quad 71232181231513221(12)122315132^1 21(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[83] $71231513221(12)^1 122315132^1 21(12) 12231513221(12)^{221} 12 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[84] $71231513221(12)^1 12231513221(12) 12^1 231513221(12)^{221} 12 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[85] $2221(12) 12231513221(12)^1 12231513221(12) 12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[86] $2221(12) 12231513221(12)^1 122315132^1 21(12) 122315132171 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[87] $712231513221(12) 122^1 31513221(12)^1 12231513221(12) 1222 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[88] $71223151322^1 1(12) 12231513221(12)^1 12231513221(12) 1222 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[89] $6131513221(12)^1 122315132^1 21(12) 12231513221(12) 122313 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 47$, $\Delta = -94$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 47, T \in [47, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[90] $6131513221(12)^1 12231513221(12) 12^1 231513221(12) 122313 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 47$, $\Delta = -94$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 47, T \in [47, 47]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[91] $71231513221(12)^1 12231513221(12) 12^1 231513221(12) 12222 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[81] $71231513221(12)^1 12231513221(12) 12^1 231513221(12)^{2221} 1 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[92] $71231513221(12)^1 122315132^1 21(12) 12231513221(12) 12222 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[93] 16131513221(12)12231513221(12)1223151321812¹32181+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[94] 16131513221(12)122¹31513221(12)122315132181232181+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[95] 1613151322¹1(12)12231513221(12)122315132181232181+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[72] 41321812321812321812321812321812315132¹21(12)1+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -110$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[73] 1²²¹(12)12231513221(12)12231513221(12)12¹2315132171+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[75] 21²¹(12)12231513221(12)12231513221(12)12¹2315132171+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[76] 21²¹(12)12231513221(12)122315132¹21(12)122315132171+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[77] 71223151322¹1(12)12231513221(12)12231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[78] 712231513221(12)122¹31513221(12)12231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[79] 712231513221(12)122¹31513221(12)12231513221(12)12+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[80] 71223151322¹1(12)12231513221(12)12231513221(12)12+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.84 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[96] 61231513221(12)12231513221(12)1223151321812¹32181+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) =$

$(1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.84 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[97] $61231513221(12)122^1 31513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.84 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[98] $6123151322^1 1(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.84 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[99] $71232^1 181231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.84 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[100] $71232181231513221(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.84 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[101] $71232181231513221(12)12231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.84 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[102] $71232^1 181231513221(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.84 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 62 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[103] $71232181231513221(12)12231513221(12)12^1 2315132217 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.84 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[82] $71231513221(12)^1 122315132^1 21(12)12231513221(12)^{2221} 1 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.84 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[111] $712231513221(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[118] $413221(12)12231513221(12)12231513221(12)122^1 3151316 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 58$, $T \in [47, 58]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[117] $712315132^1 21(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[116] $61315132^1 21(12)12231513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 47$, $\Delta = -94$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 47$, $T \in [47, 47]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[115] $712231513221(12)12231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[114] $2221(12)122315132^1 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[113] $712315132^1 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[105] $41321812321812^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -110$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[106] $413218123218123218123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -110$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

[107] $1(12)122315132^1 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[108] $21(12)122315132^1 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[112] $712315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[109] $712315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 58$, $T \in [47, 58]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[110] $712231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 100$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.76 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[130] $71231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 121$, $(n_+, n_-, n_0) =$

$$(1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[149] \quad 413221(12)122^1 31513221(12)12231513221(12)1223151316 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[124] \quad 71231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[125] \quad 712231513221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[126] \quad 712231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[127] \quad 712231513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[128] \quad 712231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[129] \quad 71231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[119] \quad 4132181232181232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[120] \quad 1(12)12231513221(12)1223151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[121] \quad 1(12)12231513221(12)122^1 31513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[122] \quad 21(12)12231513221(12)122^1 31513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[131] & 71231513221(12)122^1 31513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 121, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[132] & 71231513221(12)1223151322^1 1(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 121, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[133] & 2221(12)12231513221(12)122^1 31513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = \\
& 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[134] & 2221(12)12231513221(12)1223151322^1 1(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = \\
& 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[135] & 712231513221(12)12^1 231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = \\
& 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[136] & 712231513221(12)122315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = \\
& 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[137] & 6131513221(12)122^1 31513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 47, \Delta = -94, \det = \\
& 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 47, T \in [47, 47] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[139] & 71231513221(12)1223151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = \\
& 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[140] & 71231513221(12)122^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = \\
& 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[138] & 6131513221(12)1223151322^1 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 47, \Delta = -94, \det = \\
& 121, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 47, T \in [47, 47] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[141] & 16131513221(12)122315132^1 21(12)122315132181232181+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -106, \det = 121, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[151] & 71232181231513221(12)122^1 31513221(12)122315132217+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -106, \det = 121, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[150] & 71232181231513221(12)1223151322^1 1(12)122315132217+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -106, \det = 121, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[142] & 16131513221(12)12^1 231513221(12)122315132181232181+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -106, \det = 121, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[143] & 81232^1 181231513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 121, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 58, T \in [47, 58] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[144] & 81232181231513221(12)12231513221(12)122^1 315131613+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 121, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 58, T \in [47, 58] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[145] & 61231513221(12)122315132^1 21(12)122315132181232181+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -107, \det = 121, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[146] & 61231513221(12)12^1 231513221(12)122315132181232181+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -107, \det = 121, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[147] & 71232181231513221(12)122^1 31513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -106, \det = 121, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[148] & 71232181231513221(12)1223151322^1 1(12)122315132171+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -106, \det = 121, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[123] & \begin{matrix} 21 & & 1 \\ 21(12)12231513221(12)1223151322^1 1(12)122315132171+\hat{1} \rightarrow 2 \end{matrix} \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 121, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.69 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[170] & 413221(12)12^1 231513221(12)12231513221(12)1223151316 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = \\
& 169, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 58, T \in [47, 58] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[171] & 712321812315132^1 21(12)12231513221(12)122315132217+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -106, \det = 169, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 54, T \in [47, 54]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[152] & 41321812321812321812^1 32181232181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -110, \det = 169, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[153] & 1^{221}(12)1223151322^1 1^{12231513221}(12)122315132171+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 169, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[154] & 21^{21}(12)1223151322^1 1^{12231513221}(12)122315132171+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 169, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[155] & 71231513221(12)12231513221(12)1223151322^1 1^{3221}(12)1+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 58, T \in [47, 58] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[156] & 7123151322^1 1^{12231513221}(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 169, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 58, T \in [47, 58] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[157] & 712231513221(12)12231513221(12)12^1 231513221(12)1+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 169, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[158] & 712231513221(12)12231513221(12)12^1 231513221(12)12+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 169, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[159] & 7123151322^1 1^{12231513221}(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 169, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[160] & 7123151322^1 1^{12231513221}(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 169, (n_+, n_-, n_0) = \\
& (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[161] & 2221(12)1223151322^1 1^{12231513221}(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = \\
& 169, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[162] & 712231513221(12)12231513221(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = \\
& 169, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.59 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [169] $413221(12)12231513221(12)12231513221(12)12^1 23151316 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.59 \rightarrow 47$, $T_{\max} = 58$, $T \in [47, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [168] $712321812315132^1 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 169$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.59 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [167] $61231513221(12)12231513221(12)122^1 315132181232181 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -107$, $\det = 169$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.59 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [166] $81232181231513221(12)122315132^1 21(12)122315131613 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.59 \rightarrow 47$, $T_{\max} = 58$, $T \in [47, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [165] $16131513221(12)12231513221(12)122^1 315132181232181 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 169$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.59 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [164] $7123151322^1 1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 169$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.59 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [163] $613151322^1 1(12)12231513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 47$, $\Delta = -94$, $\det = 169$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.59 \rightarrow 47$, $T_{\max} = 47$, $T \in [47, 47]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [179] $71231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -105$, $\det = 196$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.55 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [188] $413221(12)12231513221(12)122315132^1 21(12)1223151316 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 196$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.55 \rightarrow 47$, $T_{\max} = 58$, $T \in [47, 58]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [189] $71232181231513221(12)12^1 231513221(12)122315132217 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 196$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.55 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [187] $71232181231513221(12)12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -106$, $\det = 196$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.55 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [186] $61231513221(12)1223151322^1 1(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -107$, $\det = 196$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 46.55 \rightarrow 47$, $T_{\max} = 54$, $T \in [47, 54]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[185] & 81232181231513221(12)12231513221(12)12^1 2315131613 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.55 \rightarrow 47, T_{\max} = 58, T \in [47, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[184] & 16131513221(12)1223151322^1 1(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -106, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.55 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[183] & 71231513221(12)12^1 231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.55 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[182] & 6131513221(12)12^1 231513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 47, \Delta = -94, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.55 \rightarrow 47, T_{\max} = 47, T \in [47, 47] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[181] & 712231513221(12)1223151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.55 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[180] & 2221(12)12231513221(12)12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.55 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[178] & 71231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.55 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[172] & 4132181232181232181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -110, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.55 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[173] & 1(12)12231513221(12)12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.55 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[174] & 21(12)12231513221(12)12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 54, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.55 \rightarrow 47, T_{\max} = 54, T \in [47, 54] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[175] & 71231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 46.55 \rightarrow 47, T_{\max} = 58, T \in [47, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[176] $712231513221(12)1223151322^11(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 54, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.55 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[177] $712231513221(12)1223151322^11(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 54, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.55 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[192] $61231513221(12)12231513221(12)122315132^1181232181+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 54, \Delta = -107, \det = 225, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.52 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$

[190] $16131513221(12)12231513221(12)122315132^1181232181+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 54, \Delta = -106, \det = 225, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.52 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$

[191] $81232181231513221(12)122^131513221(12)122315131613+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 58, \Delta = -114, \det = 225, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.52 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[194] $712321812^131513221(12)12231513221(12)122315132217+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 54, \Delta = -106, \det = 225, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.52 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[193] $712321812^131513221(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 54, \Delta = -106, \det = 225, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.52 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$

[196] $16131513221(12)12231513221(12)12^12315132181232181+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 54, \Delta = -106, \det = 256, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.48 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[195] $41321812321812321812321812^132181231513221(12)1+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 54, \Delta = -110, \det = 256, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.48 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[199] $7123218123151322^11(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 54, \Delta = -106, \det = 256, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.48 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[200] $7123218123151322^11(12)12231513221(12)122315132217+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 54, \Delta = -106, \det = 256, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.48 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[197] $81232181231513221(12)1223151322^11(12)122315131613+\hat{1}\rightarrow 2$ $T = 42, T_{\max} = 58, \Delta = -114, \det = 256, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 46.48 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[198] 61231513221(12)12231513221(12)12¹2315132181232181+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -107$, $\det = 256$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.48 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[201] 4132181232181232181232181232¹181231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -110$, $\det = 289$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.46 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[202] 71231513221(12)12¹231513221(12)12231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 289$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.46 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[203] 71231513221(12)12231513221(12)122¹31513221(12)1+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 289$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.46 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[204] 812321812315132¹21(12)12231513221(12)122315131613+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 289$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.46 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[205] 413221(12)12231513221(12)1223151322¹1(12)1223151316 + $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 289$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.46 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 101 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[206] 413221(12)122315132¹21(12)12231513221(12)1223151316 + $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 289$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.46 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 118 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[207] 71231513221(12)122315132¹21(12)12231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 324$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.43 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[208] 413221(12)12231513221(12)122¹31513221(12)1223151316 + $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 324$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.43 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 125 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[210] 812321812¹31513221(12)12231513221(12)122315131613+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 361$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.41 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[211] 81232181231513221(12)12¹231513221(12)122315131613+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 58$, $\Delta = -114$, $\det = 361$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.41 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[209] 41321812321812321812321812¹31513221(12)1+ $\hat{1} \rightarrow 2$ $T = 42$, $T_{\max} = 54$, $\Delta = -110$, $\det = 361$, $(n_+, n_-, n_0) = (1, 42, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 46.41 \rightarrow 47, T_{\max} = 54, T \in [47, 54]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

$$[212] \ 8123218123151322^1 1(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 441, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.37 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[213] \ 71231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 484, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.36 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[214] \ 413221(12)1223151322^1 1(12)12231513221(12)1223151316 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 484, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.36 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 197 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[215] \ 71231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 529, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.34 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[216] \ 413221(12)12231513221(12)12^1 231513221(12)1223151316 + \hat{1} \rightarrow 2 \quad T = 42, T_{\max} = 58, \Delta = -114, \det = 529, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 46.34 \rightarrow 47, T_{\max} = 58, T \in [47, 58]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 206 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$$

$$T = 43 \quad [229] \ 22^1 21(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 50, T_{\max} = 55, T \in [50, 55]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[230] \ 812231513221(12)12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 50, T_{\max} = 55, T \in [50, 55]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[217] \ 1(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 50, T_{\max} = 55, T \in [50, 55]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[218] \ 1(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 50, T_{\max} = 55, T \in [50, 55]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[219] \ 1(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 50, T_{\max} = 55, T \in [50, 55]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[220] \ 181231513221(12)12231513221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 50, T_{\max} = 55, T \in [50, 55]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[221] & 1(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -4, (n_+, n_-, n_0) = \\
& (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 50, T_{\max} = 55, T \in [50, 55] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[222] & 21(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -4, (n_+, n_-, n_0) = \\
& (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 50, T_{\max} = 55, T \in [50, 55] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[223] & 81231513221(12)12231513221(12)12231513221(12)12^1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -4, (n_+, n_-, n_0) = \\
& (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[224] & 812231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -4, (n_+, n_-, n_0) = \\
& (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 50, T_{\max} = 55, T \in [50, 55] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[225] & 812231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -4, (n_+, n_-, n_0) = \\
& (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 50, T_{\max} = 55, T \in [50, 55] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[226] & 181231513221(12)12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = \\
& -4, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 50, T_{\max} = 55, T \in [50, 55] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[227] & 22^1 1(12)12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = \\
& -4, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 50, T_{\max} = 58, T \in [50, 58] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[228] & 22^1 21(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = \\
& -4, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 50, T_{\max} = 55, T \in [50, 55] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[241] & 222^1 1(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = \\
& -9, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.22 \rightarrow 50, T_{\max} = 55, T \in [50, 55] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[242] & 812231513221(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = \\
& -9, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.22 \rightarrow 50, T_{\max} = 55, T \in [50, 55] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[231] & 1(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -9, (n_+, n_-, n_0) = \\
& (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.22 \rightarrow 50, T_{\max} = 58, T \in [50, 58]
\end{aligned}$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[243] \quad 181231513221(12)12231513221(12)12231513221(12)12^1222 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.75 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[273] \quad 51232^1181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -107, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.44 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[272] \quad 4132^121(12)12231513221(12)12231513221(12)12231513217 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 58, \quad \Delta = -113, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.44 \rightarrow 49, \quad T_{\max} = 58, \quad T \in [49, 58]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[246] \quad 1(12)12231513221(12)12231513221(12)122315132^1181 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.44 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[247] \quad 21(12)12231513221(12)12231513221(12)122315132^1181 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.44 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[248] \quad 1(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 58, \quad \Delta = -113, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.44 \rightarrow 49, \quad T_{\max} = 58, \quad T \in [49, 58]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[249] \quad 1(12)12231513221(12)12231513221(12)1223151321812^132 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 60, \quad \Delta = -115, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.44 \rightarrow 49, \quad T_{\max} = 60, \quad T \in [49, 60]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[250] \quad 1(12)12231513221(12)12231513221(12)122315132^11812 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.44 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[251] \quad 21(12)12231513221(12)12231513221(12)122315132^11812 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.44 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[252] \quad 1(12)12231513221(12)12231513221(12)122315132^12181 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.44 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[253] \quad 21(12)12231513221(12)12231513221(12)122315132^12181 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.44 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

- [254] $812231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [255] $1812^1 31513221(12)12231513221(12)12231513221(12)1222 + \hat{1}\rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [256] $1(12)122^1 31513221(12)12231513221(12)122315132181 + \hat{1}\rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [257] $21(12)122^1 31513221(12)12231513221(12)122315132181 + \hat{1}\rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [258] $81231513221(12)12231513221(12)12231513221(12)12 + \hat{1}\rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [259] $812231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1}\rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [260] $812231513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1}\rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [261] $181231513221(12)12231513221(12)122315132^1 21(12)12222 + \hat{1}\rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [262] $21(12)12231513221(12)12231513221(12)1223151321812^1 32 + \hat{1}\rightarrow 2$ $T = 43$, $T_{\max} = 60$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 60$, $T \in [49, 60]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [263] $2221(12)12231513221(12)12231513221(12)122315132^1 1812 + \hat{1}\rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [264] $2221(12)12231513221(12)12231513221(12)122315132^1 2181 + \hat{1}\rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$

- [265] $6131513221(12)12231513221(12)12231513221(12)122^13151 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [266] $812231513221(12)12231513221(12)122315132^121(12)12222 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [267] $71232181231513221(12)12231513221(12)122315132^1181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [268] $1712^131513221(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [269] $71232181231513221(12)12231513221(12)122315132^11812 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [270] $7122^131513221(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$
- [271] $71232181231513221(12)12231513221(12)122315132^12181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [274] $15132^1181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.44 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [307] $71232181231513221(12)12231513221(12)1223151322^1181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.22 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [306] $712^132181231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.22 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [275] $1(12)122^131513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.22 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [276] $21(12)122^131513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.22 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[277] \quad 1(12)12231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[278] \quad 1(12)122^1 31513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[279] \quad 21(12)122^1 31513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[280] \quad 1(12)122^1 31513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[282] \quad 21(12)12231513221(12)12231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[281] \quad 1(12)12231513221(12)12231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[308] \quad 413221(12)12231513221(12)12231513221(12)122315132^1 17 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[284] \quad 8122^1 31513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[285] \quad 181231513221(12)12231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[286] \quad 1(12)12231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[287] \quad 1(12)12^1 231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [288] $21(12)^{221}12^1 231513221(12)12231513221(12)^1 122315132181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [289] $21(12)^{221}12231513221(12)12231513221(12)^1 122315132^1 181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [290] $8122^1 31513221(12)^1 12231513221(12)12231513221(12)^{221} 1 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [291] $812231513221(12)^1 12231513221(12)1223151322^1 1(12)^{221} 1 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [292] $812231513221(12)^1 12231513221(12)1223151322^1 1(12)^{221} 12 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [293] $8122^1 31513221(12)^1 12231513221(12)12231513221(12)^{221} 12 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [294] $181231513221(12)^1 12231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [295] $1812^1 31513221(12)^1 12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [296] $2221(12)122^1 31513221(12)^1 12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [297] $2221(12)122^1 31513221(12)^1 12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$
- $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
 [298] $2221(12)12231513221(12)^1 12231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[299] $812231513221(12)12231513221(12)12231513221^1(12)12222 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[300] $8122^131513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[301] $712^132181231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[302] $171231513221(12)12231513221(12)122315132181232^1181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[303] $712^132181231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[304] $712^1231513221(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[305] $712231513221(12)12231513221(12)122315132181232^1181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[283] $21(12)122^131513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.22 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[348] $151321812^132181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -49$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[347] $512321812^132181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -107$, $\det = -49$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[309] $1(12)12^1231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[310] $1(12)12231513221(12)12231513221(12)122^1315132181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[322] \quad 812^1 231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[323] \quad 181231513221(12)12231513221(12)12231513221^1(12)1222 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[324] \quad 1812315132^1 21(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[325] \quad 1(12)122315132^1 21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[326] \quad 21(12)122315132^1 21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[327] \quad 812^1 31513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -49, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[328] \quad 812^1 231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[329] \quad 812231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[330] \quad 812^1 231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[331] \quad 812231513221(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[332] \quad 181231513221(12)12231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.06 \rightarrow 49, T_{\max} = 55, T \in [49, 55]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[333] & 221(12)122^1 31513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 58, \quad \Delta = -113, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.06 \rightarrow 49, \quad T_{\max} = 58, \quad T \in [49, 58] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[334] & 2221(12)12^1 231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.06 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[335] & 2221(12)12231513221(12)12231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.06 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[336] & 2221(12)12^1 231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.06 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[337] & 2221(12)12231513221(12)12231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.06 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[338] & 6131513221(12)12231513221(12)12231513221(12)12^1 23151 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -109, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.06 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[339] & 812231513221(12)12231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.06 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[340] & 812^1 231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -104, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.06 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[341] & 71232181231513221(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -105, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.06 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[342] & 1712315132^1 21(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -105, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.06 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[343] & 71232181231513221(12)12231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 43, \quad T_{\max} = 55, \quad \Delta = -105, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 43, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.06 \rightarrow 49, \quad T_{\max} = 55, \quad T \in [49, 55] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[344] $7122315132^1 21(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -49$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.06 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[345] $71232181231513221(12)12231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -49$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.06 \rightarrow 49$, $T_{\max} = 55$, $T \in [49, 55]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[346] $41322^1 1(12)12231513221(12)12231513221(12)12231513217 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -49$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.06 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[350] $1(12)12^1 231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 47.94 \rightarrow 48$, $T_{\max} = 58$, $T \in [48, 58]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[355] $(12)1223151321812321812321812321812321812^1 31513 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 47.94 \rightarrow 48$, $T_{\max} = 58$, $T \in [48, 58]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[349] $1(12)122^1 31513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 60$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 47.94 \rightarrow 48$, $T_{\max} = 60$, $T \in [48, 60]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[354] $15132181232^1 181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -64$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 47.94 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[351] $221(12)12^1 231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -64$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 47.94 \rightarrow 48$, $T_{\max} = 58$, $T \in [48, 58]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[352] $21(12)122^1 31513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 60$, $\Delta = -115$, $\det = -64$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 47.94 \rightarrow 48$, $T_{\max} = 60$, $T \in [48, 60]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[353] $51232181232^1 181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -107$, $\det = -64$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 47.94 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[380] $812231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -81$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 47.84 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[409] $151321812321812321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 47.84 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[382] $181231513221(12)12231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[383] $221(12)12231513221(12)12231513221(12)122315132^1 18123 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[384] $21(12)12^1 231513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 60, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 60, T \in [48, 60]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[379] $812231513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[378] $812231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[377] $812231513221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[376] $81231513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[375] $21(12)1223151322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[374] $21(12)12231513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[373] $1(12)1223151322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[372] $1(12)12231513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[360] & 1(12)12^1 231513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 60, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.84 \rightarrow 48, T_{\max} = 60, T \in [48, 60] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[359] & 21(12)12231513221(12)12231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[358] & 21(12)12231513221(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[357] & 1(12)12231513221(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[356] & 1(12)12231513221(12)12231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[402] & 71223151322^1 1(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[385] & 2221(12)12231513221(12)122315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[386] & 2221(12)12231513221(12)12231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[387] & 2221(12)12231513221(12)12231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[388] & 2221(12)12231513221(12)122315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[389] & 71231513221(12)12231513221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 48, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.84 \rightarrow 48, T_{\max} = 48, T \in [48, 48] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [390] $71231513221(12)122315132^1 21(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 48, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 48, T \in [48, 48]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [391] $812231513221(12)12231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [392] $812231513221(12)122315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [393] $71232181231513221(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [394] $71232181231513221(12)12231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [395] $71232^1 181231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [396] $17123151322^1 1(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [397] $171231513221(12)12231513221(12)1223151321812^1 32181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [398] $171231513221(12)122^1 31513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [399] $71232181231513221(12)12231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [400] $71232181231513221(12)122315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [401] $71232^1 181231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[403] 712231513221(12)122¹31513221(12)122315132181232181 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[404] 712231513221(12)12231513221(12)1223151321812¹32181 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[405] 71232181231513221(12)12231513221(12)12¹23151322181 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[406] 71232¹181231513221(12)12231513221(12)1223151322181 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[407] 71232181231513221(12)122315132¹21(12)1223151322181 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[408] 512321812321812321812321812321812315132¹21(12)1 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -107$, $\det = -81$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[381] 181231513221(12)122315132¹21(12)12231513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -81$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.84 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[422] 8122315132¹21(12)12231513221(12)12231513221²²¹(12)12 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[410] 1²²¹(12)122315132¹21(12)12231513221(12)122315132181 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[433] 1513218123218123218123218123218123151322¹1(12)1 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[432] 151321812321812¹32181232181232181231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[431] 512321812321812¹32181232181232181231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[430] 5123218123218123218123218123218123151322¹1(12)1+ $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -107$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[429] 413221(12)12231513221¹(12)12231513221(12)122¹31513217 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[428] 8122315132¹21(12)12231513221(12)12231513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[427] 712315132¹21(12)12231513221(12)12231513221(12)122313 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 48$, $\Delta = -93$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 48, T \in [48, 48]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[426] 61315132¹21(12)12231513221(12)12231513221(12)1223151 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -109$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[425] 2221(12)122315132¹21(12)12231513221(12)1223151322181 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[424] 2221(12)122315132¹21(12)12231513221(12)1223151321812 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[423] 1812315132¹21(12)12231513221(12)12231513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[435] (12)122315132181232181232181232181232¹181231513+ $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[421] 8122315132¹21(12)12231513221(12)12231513221²²²¹(12)1+ $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[420] 21²²¹(12)12231513221(12)12231513221¹(12)122¹315132181+ $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.76 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[419] $1(12)12231513221(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.76 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[418] $181231513221(12)12231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.76 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[417] $8122315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.76 \rightarrow 48$, $T_{\max} = 58$, $T \in [48, 58]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[416] $21(12)122315132^1 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.76 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[415] $1(12)122315132^1 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.76 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[414] $21(12)122315132^1 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.76 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[413] $1(12)122315132^1 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.76 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[412] $1(12)12231513221(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.76 \rightarrow 48$, $T_{\max} = 58$, $T \in [48, 58]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[411] $21(12)122315132^1 21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.76 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[434] $(12)122^1 315132181232181232181232181231513 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.76 \rightarrow 48$, $T_{\max} = 58$, $T \in [48, 58]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[449] $21(12)12231513221(12)122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -121$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[488] $(12)12^1 2315132181232181232181232181231513 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -121$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$(1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[438] \quad 21(12)12231513221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[439] \quad 21(12)12231513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[440] \quad 1(12)122^1 31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[441] \quad 1(12)12231513221(12)12231513221(12)122^1 31513218123 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[442] \quad 1(12)12231513221(12)122^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[443] \quad 1(12)12231513221(12)1223151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[489] \quad (12)1223151321812321812321812321812^1 32181231513 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[444] \quad 21(12)12231513221(12)122^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[445] \quad 21(12)12231513221(12)1223151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[446] \quad 1(12)12231513221(12)122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[447] \quad 1(12)12231513221(12)1223151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[448] & \begin{matrix} 21 & & 1 \\ 21(12)12231513221(12)1223151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 55, \Delta = -104, \det = \\ -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[436] & \begin{matrix} 221 & & 1 \\ 1(12)12231513221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = \\ (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[450] & \begin{matrix} & & 1 & & 3221 \\ 812231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 58, \Delta = -113, \det = -121, (n_+, n_-, n_0) = \\ (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.69 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[451] & \begin{matrix} 1 & & 1 \\ 181231513221(12)12^1 231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 55, \Delta = -104, \det = \\ -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[452] & \begin{matrix} 1 & & 1 \\ 181231513221(12)122315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 55, \Delta = -104, \det = \\ -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[453] & \begin{matrix} & & 1 \\ 1(12)12231513221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = \\ (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[454] & \begin{matrix} & & 1 \\ 1(12)12231513221(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = \\ (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[455] & \begin{matrix} & & 1 \\ 21(12)12231513221(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = \\ (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[456] & \begin{matrix} & & 1 \\ 21(12)12231513221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = \\ (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[457] & \begin{matrix} & & 1 & & 3221 \\ 81231513221(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 58, \Delta = -113, \det = -121, (n_+, n_-, n_0) = \\ (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.69 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[458] & \begin{matrix} & & 1 & & 3221 \\ 81231513221(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 58, \Delta = -113, \det = -121, (n_+, n_-, n_0) = \\ (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.69 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \end{matrix}
\end{aligned}$$

- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [459] $812231513221(12)122^1 31513221(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
 [460] $812231513221(12)1223151322^1 1(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
 [461] $812231513221(12)1223151322^1 1(12)12231513221(12)12+\hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [462] $812231513221(12)122^1 31513221(12)12231513221(12)12+\hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [463] $181231513221(12)1223151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [464] $181231513221(12)122^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
 [465] $221(12)12231513221(12)12231513221(12)122^1 31513218123 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 58, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$
- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
 [466] $2221(12)12231513221(12)1223151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [467] $2221(12)12231513221(12)122^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
 [468] $2221(12)12231513221(12)122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
 [469] $2221(12)12231513221(12)1223151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$

$$-121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[482] \quad 712231513221(12)122315132^1 21(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[483] \quad 71232181231513221(12)122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[484] \quad 71232181231513221(12)1223151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[485] \quad 413221(12)122^1 31513221(12)12231513221(12)12231513217 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[486] \quad 51232181232181232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -107, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[487] \quad 15132181232181232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[437] \quad 1(12)12231513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.69 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[508] \quad 81223151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[490] \quad 1(12)1223151322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[529] \quad (12)122315132181232181232^1 181232181231513 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[528] \quad 151321812321812321812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[527] 512321812321812321812¹32181232181231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -107$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[526] 413221(12)12¹231513221(12)12231513221(12)12231513217 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[525] 413221(12)12231513221(12)12231513221(12)12¹231513217 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[524] 712321812315132¹21(12)12231513221(12)1223151322181 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[523] 712231513221(12)12231513221(12)122¹315132181232181 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[522] 712321812315132¹21(12)12231513221(12)1223151321812 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[521] 171231513221(12)12231513221(12)122¹315132181232181 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[520] 712321812315132¹21(12)12231513221(12)122315132181 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[519] 81223151322¹1(12)12231513221(12)12231513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -104$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[518] 7123151322¹1(12)12231513221(12)12231513221(12)122313 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 48$, $\Delta = -93$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 48, T \in [48, 48]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[517] 6131513221(12)12231513221(12)1223151322¹1(12)1223151 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -109$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[516] 613151322¹1(12)12231513221(12)12231513221(12)1223151 + $\hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -109$, $\det = -169$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[502] \quad 81223151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[501] \quad 812231513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[500] \quad 21(12)1223151322^1 1(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[499] \quad 1(12)1223151322^1 1(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[498] \quad 21(12)1223151322^1 1(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[497] \quad 1(12)1223151322^1 1(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[496] \quad 1(12)12231513221(12)122315132^1 21(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[495] \quad 1(12)122315132^1 21(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[494] \quad 1(12)12231513221(12)12231513221(12)122315132^1 181232 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 60, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 60, T \in [48, 60]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[493] \quad 1(12)12^1 231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[492] \quad 1(12)12231513221(12)12231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 47.59 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[491] & 21^{21}(12)1223151322^1 1^{12}(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.59 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[506] & 812315132^1 21^{3221}(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.59 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[534] & 1^{221}(12)12231513221(12)12^1 231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[557] & 413221(12)12231513221(12)122315132^1 21(12)12231513217 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[536] & 1^{221}(12)12231513221(12)12^1 231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[531] & 21^{21}(12)12231513221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[535] & 21^{21}(12)12231513221(12)12^1 231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[537] & 21^{21}(12)12231513221(12)12^1 231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[538] & 812231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[539] & 1^{18}1231513221(12)1223151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[540] & 1^{2221}(12)12231513221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55]
\end{aligned}$$

- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 [541] $21(12)^{221}12231513221(12)1223151322^11(12)122315132181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [542] $81231513221(12)^112231513221(12)12^1231513221(12)12 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 58, \Delta = -113, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.55 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$
- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 [543] $812231513221(12)12^1231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 [544] $812231513221(12)^112^1231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 [545] $181231513221(12)^112^1231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [546] $221(12)12231513221(12)^112^1231513218123 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 58, \Delta = -113, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.55 \rightarrow 48, T_{\max} = 58, T \in [48, 58]$
- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [547] $2221(12)12231513221(12)^112^1231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [548] $2221(12)12231513221(12)^112^1231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [549] $6131513221(12)122^131513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -109, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$
- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [550] $71231513221(12)^112^1231513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 48, \Delta = -93, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.55 \rightarrow 48, T_{\max} = 48, T \in [48, 48]$
- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
 [551] $812231513221(12)^112^1231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[530] & 1(12)12231513221(12)12^{11}231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[552] & 71232181231513221(12)12^{11}231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[532] & 1(12)12231513221(12)122315132^{11}21(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[553] & 171231513221(12)1223151322^{11}1(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[554] & 71232181231513221(12)12^{11}231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[555] & 712231513221(12)1223151322^{11}1(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[556] & 71232181231513221(12)12^{11}231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[533] & 1(12)12231513221(12)12231513221(12)12^{11}231513218123 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[560] & (12)1223151321812321812321812^{11}32181232181231513 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[559] & 15132181232181232181232^{11}181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[558] & 51232181232181232181232^{11}181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -107, \det = -196, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.55 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[564] & 221(12)12231513221(12)122^{11}31513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -225, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 47.52 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[569] & 712231513221(12)12231513221(12)122315132^1 181232181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = \\
& -225, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.52 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[570] & 712321812^1 31513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = \\
& -225, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.52 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[561] & 1(12)12231513221(12)12231513221(12)122^1 315132181232 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 60, \Delta = -115, \det = \\
& -225, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.52 \rightarrow 48, T_{\max} = 60, T \in [48, 60] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[562] & 1(12)12231513221(12)122^1 31513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = \\
& -225, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.52 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[563] & 81231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -225, (n_+, n_-, n_0) = \\
& (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.52 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[568] & 712321812^1 31513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = \\
& -225, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.52 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[567] & 171231513221(12)12231513221(12)122315132^1 181232181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = \\
& -225, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.52 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[566] & 712321812^1 31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -225, (n_+, n_-, n_0) = \\
& (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.52 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[565] & 21(12)12231513221(12)12231513221(12)122^1 315132181232 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 60, \Delta = -115, \det = \\
& -225, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.52 \rightarrow 48, T_{\max} = 60, T \in [48, 60] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[572] & 1(12)12231513221(12)1223151322^1 1(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = \\
& -256, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.48 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[571] & 1(12)122315132^1 21(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 60, \Delta = -115, \det = \\
& -256, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.48 \rightarrow 48, T_{\max} = 60, T \in [48, 60] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [574] $221(12)12231513221(12)1223151322^11(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -256$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.48 \rightarrow 48$, $T_{\max} = 58$, $T \in [48, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [575] $21(12)122315132^121(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 60$, $\Delta = -115$, $\det = -256$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.48 \rightarrow 48$, $T_{\max} = 60$, $T \in [48, 60]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [576] $7123218123151322^11(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -256$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.48 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [577] $171231513221(12)12231513221(12)12^12315132181232181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -256$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.48 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [578] $7123218123151322^11(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -256$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.48 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [579] $712231513221(12)12231513221(12)12^12315132181232181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -256$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.48 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [580] $7123218123151322^11(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -256$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.48 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [581] $512321812321812321812321812^132181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -107$, $\det = -256$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.48 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [582] $151321812321812321812321812^132181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -256$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.48 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [583] $(12)122315132181232181232^1181232181232181231513 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -256$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.48 \rightarrow 48$, $T_{\max} = 58$, $T \in [48, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [573] $81231513221(12)1223151322^11(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -256$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.48 \rightarrow 48$, $T_{\max} = 58$, $T \in [48, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [584] $1(12)122315132^121(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -289$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.46 \rightarrow 48$, $T_{\max} = 58$, $T \in [48, 58]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[585] & \begin{matrix} 3221 & 1 \\ 1(12)12231513221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 58, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.46 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \end{matrix} \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[586] & \begin{matrix} 1 & 1 \\ 1(12)12231513221(12)122315132^1 21(12)122315132181232 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 60, \Delta = -115, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.46 \rightarrow 48, T_{\max} = 60, T \in [48, 60] \end{matrix} \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[587] & \begin{matrix} 21 & 1 \\ 1(12)1223151322^1 1(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 58, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.46 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \end{matrix} \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[588] & \begin{matrix} 1 & 3221 \\ 812231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 58, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.46 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \end{matrix} \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[589] & \begin{matrix} 1 & 3221 \\ 812231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 58, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.46 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \end{matrix} \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[590] & \begin{matrix} 1 & 3221 \\ 8123151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 58, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.46 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \end{matrix} \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[591] & \begin{matrix} 1 \\ 221(12)1223151322^1 1(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 58, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.46 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \end{matrix} \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[592] & \begin{matrix} 1 \\ 21(12)12231513221(12)122315132^1 21(12)122315132181232 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 60, \Delta = -115, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.46 \rightarrow 48, T_{\max} = 60, T \in [48, 60] \end{matrix} \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[593] & \begin{matrix} 1 \\ 6131513221(12)12231513221(12)122^1 31513221(12)1223151 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 55, \Delta = -109, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.46 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \end{matrix} \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[594] & \begin{matrix} 1 \\ 6131513221(12)12^1 231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 55, \Delta = -109, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.46 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \end{matrix} \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[595] & \begin{matrix} 1 \\ 413221(12)12231513221(12)1223151322^1 1(12)12231513217 + \hat{1} \rightarrow 2 & T = 43, T_{\max} = 58, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 47.46 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \end{matrix}
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 101 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[596] & 413221(12)122315132^1 21(12)12231513221(12)12231513217 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.46 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 118 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[597] & 51232181232181232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -107, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.46 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[598] & 15132181232181232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -105, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.46 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[599] & (12)1223151321812321812^1 32181232181232181231513 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.46 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[603] & 413221(12)12231513221(12)122^1 31513221(12)12231513217 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -324, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.43 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 125 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[600] & 1(12)12231513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -324, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.43 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[601] & 812231513221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -324, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.43 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[602] & 6131513221(12)122315132^1 21(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -109, \det = -324, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.43 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[606] & 1(12)12231513221(12)12^1 231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -361, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.41 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[605] & 1(12)12231513221(12)12231513221(12)12^1 2315132181232 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 60, \Delta = -115, \det = -361, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.41 \rightarrow 48, T_{\max} = 60, T \in [48, 60] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[604] & 1(12)12231513221(12)122^1 31513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 60, \Delta = -115, \det = -361, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.41 \rightarrow 48, T_{\max} = 60, T \in [48, 60] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [607] $81231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -361$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.41 \rightarrow 48$, $T_{\max} = 58$, $T \in [48, 58]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [608] $221(12)12231513221(12)12^1 231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -361$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.41 \rightarrow 48$, $T_{\max} = 58$, $T \in [48, 58]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [609] $21(12)12231513221(12)12231513221(12)12^1 2315132181232 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 60$, $\Delta = -115$, $\det = -361$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.41 \rightarrow 48$, $T_{\max} = 60$, $T \in [48, 60]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [610] $21(12)12231513221(12)122^1 31513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 60$, $\Delta = -115$, $\det = -361$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.41 \rightarrow 48$, $T_{\max} = 60$, $T \in [48, 60]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [611] $512321812321812321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -107$, $\det = -361$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.41 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [612] $151321812321812321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 55$, $\Delta = -105$, $\det = -361$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.41 \rightarrow 48$, $T_{\max} = 55$, $T \in [48, 55]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [613] $(12)122315132181232^1 181232181232181232181231513 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -361$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.41 \rightarrow 48$, $T_{\max} = 58$, $T \in [48, 58]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [614] $(12)1223151321812^1 32181232181232181232181231513 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 58$, $\Delta = -113$, $\det = -400$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.39 \rightarrow 48$, $T_{\max} = 58$, $T \in [48, 58]$
 $\begin{pmatrix} 0 & 20 & 2 \\ 20 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [618] $21(12)12231513221(12)1223151322^1 1(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 60$, $\Delta = -115$, $\det = -441$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.37 \rightarrow 48$, $T_{\max} = 60$, $T \in [48, 60]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [615] $1(12)1223151322^1 1(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 60$, $\Delta = -115$, $\det = -441$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.37 \rightarrow 48$, $T_{\max} = 60$, $T \in [48, 60]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$
- [616] $1(12)12231513221(12)1223151322^1 1(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 60$, $\Delta = -115$, $\det = -441$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 47.37 \rightarrow 48$, $T_{\max} = 60$, $T \in [48, 60]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [617] $21(12)1223151322^1 1(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 43$, $T_{\max} = 60$, $\Delta = -115$, $\det = -441$, $(n_+, n_-, n_0) = (1, 43, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 47.37 \rightarrow 48, T_{\max} = 60, T \in [48, 60] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[619] & 1^{3221}(12)1223151322^1 1^{12}(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -484, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.36 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[620] & 812231513221(12)12231513221(12)12^{1231513221} 1^{3221}(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -484, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.36 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[621] & 6131513221(12)12231513221(12)12^{1231513221} 1^{12}(12)1223151 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -109, \det = -484, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.36 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[622] & 413221(12)1223151322^1 1^{12}(12)12231513221(12)12231513217 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -484, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.36 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 197 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[623] & (12)122315132^1 181232181232181232181232181231513 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -484, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.36 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[627] & 413221(12)12231513221(12)12^{1231513221} 1^{12}(12)12231513217 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -529, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.34 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 206 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[626] & 6131513221(12)1223151322^1 1^{12}(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 55, \Delta = -109, \det = -529, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.34 \rightarrow 48, T_{\max} = 55, T \in [48, 55] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[625] & 812231513221(12)1223151322^1 1^{12}(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -529, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.34 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[624] & 1^{3221}(12)12231513221(12)12^{1231513221} 1^{12}(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 58, \Delta = -113, \det = -529, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.34 \rightarrow 48, T_{\max} = 58, T \in [48, 58] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[629] & 21(12)12231513221(12)12^{1231513221} 1^{12}(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 60, \Delta = -115, \det = -576, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.33 \rightarrow 48, T_{\max} = 60, T \in [48, 60] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[628] & 1^{12}(12)12231513221(12)12^{1231513221} 1^{12}(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 43, T_{\max} = 60, \Delta = -115, \det = -576, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 47.33 \rightarrow 48, T_{\max} = 60, T \in [48, 60]
\end{aligned}$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$$T = 44 \quad [632] \quad 61316131513221(12)12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -105, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 51, \quad T_{\max} = 56, \quad T \in [51, 56]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[631] \quad 61316131513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -105, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 51, \quad T_{\max} = 56, \quad T \in [51, 56]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[630] \quad 61316131513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -105, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 51, \quad T_{\max} = 56, \quad T \in [51, 56]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[635] \quad 81232181231513221(12)12231513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 59, \quad \Delta = -113, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 50.22 \rightarrow 51, \quad T_{\max} = 59, \quad T \in [51, 59]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[634] \quad 61316131513221(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -105, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 50.22 \rightarrow 51, \quad T_{\max} = 56, \quad T \in [51, 56]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[633] \quad 61316131513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -105, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 50.22 \rightarrow 51, \quad T_{\max} = 56, \quad T \in [51, 56]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[636] \quad 1612^1 32181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -104, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.75 \rightarrow 50, \quad T_{\max} = 56, \quad T \in [50, 56]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[637] \quad 61316131513221(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -105, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.75 \rightarrow 50, \quad T_{\max} = 56, \quad T \in [50, 56]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[638] \quad 161232^1 181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -104, \quad \det = 25, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.44 \rightarrow 50, \quad T_{\max} = 56, \quad T \in [50, 56]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[639] \quad 71231513221(12)12231513221(12)12231513221(12)122^1 3151 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -108, \quad \det = 25, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.44 \rightarrow 50, \quad T_{\max} = 56, \quad T \in [50, 56]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[640] \quad 1812^1 31513221(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -104, \quad \det = 25, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.44 \rightarrow 50, \quad T_{\max} = 56, \quad T \in [50, 56]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

- [641] $61316131513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.44 \rightarrow 50$, $T_{\max} = 56$, $T \in [50, 56]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [642] $61316131513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.44 \rightarrow 50$, $T_{\max} = 56$, $T \in [50, 56]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [643] $181232181231513221(12)12231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 25$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.44 \rightarrow 50$, $T_{\max} = 56$, $T \in [50, 56]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [644] $181232181231513221(12)12231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 25$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.44 \rightarrow 50$, $T_{\max} = 56$, $T \in [50, 56]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [645] $61316131513221(12)12231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 25$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.44 \rightarrow 50$, $T_{\max} = 56$, $T \in [50, 56]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [646] $4132^1 21(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.44 \rightarrow 50$, $T_{\max} = 58$, $T \in [50, 58]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [647] $4132^1 21(12)12231513221(12)12231513221(12)12231513218 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.44 \rightarrow 50$, $T_{\max} = 58$, $T \in [50, 58]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [648] $4132^1 21(12)12231513221(12)12231513221(12)122315132218 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 25$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.44 \rightarrow 50$, $T_{\max} = 58$, $T \in [50, 58]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [654] $181232181231513221(12)12231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 36$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.22 \rightarrow 50$, $T_{\max} = 56$, $T \in [50, 56]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [651] $61316131513221(12)12231513221(12)122315132^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 36$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.22 \rightarrow 50$, $T_{\max} = 56$, $T \in [50, 56]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [658] $413221(12)12231513221(12)12231513221(12)122315132^1 218 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 36$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.22 \rightarrow 50$, $T_{\max} = 58$, $T \in [50, 58]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [657] $413221(12)12231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 36$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 49.22 \rightarrow 50, \quad T_{\max} = 58, \quad T \in [50, 58] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[656] & 61316131513221(12)12231513221(12)1223151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -105, \quad \det = \\
& 36, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.22 \rightarrow 50, \quad T_{\max} = 56, \quad T \in [50, 56] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[655] & 1812^1 32181231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -104, \quad \det = \\
& 36, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.22 \rightarrow 50, \quad T_{\max} = 56, \quad T \in [50, 56] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[649] & 712^1 31513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -108, \quad \det = \\
& 36, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.22 \rightarrow 50, \quad T_{\max} = 56, \quad T \in [50, 56] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[650] & 181231513221(12)12231513221(12)122315132181232^1 181 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -104, \quad \det = \\
& 36, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.22 \rightarrow 50, \quad T_{\max} = 56, \quad T \in [50, 56] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[653] & 1812^1 32181231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -104, \quad \det = \\
& 36, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.22 \rightarrow 50, \quad T_{\max} = 56, \quad T \in [50, 56] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[652] & 61316131513221(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -105, \quad \det = \\
& 36, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.22 \rightarrow 50, \quad T_{\max} = 56, \quad T \in [50, 56] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[670] & 413221(12)12231513221(12)12231513221(12)122315132^1 18 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 58, \quad \Delta = -112, \quad \det = \\
& 49, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.06 \rightarrow 50, \quad T_{\max} = 58, \quad T \in [50, 58] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[669] & 41322^1 1(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 58, \quad \Delta = -112, \quad \det = \\
& 49, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.06 \rightarrow 50, \quad T_{\max} = 58, \quad T \in [50, 58] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[668] & 812^1 32181231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 59, \quad \Delta = -113, \quad \det = \\
& 49, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.06 \rightarrow 50, \quad T_{\max} = 59, \quad T \in [50, 59] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[667] & 61316131513221(12)12231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -105, \quad \det = \\
& 49, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.06 \rightarrow 50, \quad T_{\max} = 56, \quad T \in [50, 56] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[666] & 181232181231513221(12)12231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -104, \quad \det = \\
& 49, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.06 \rightarrow 50, \quad T_{\max} = 56, \quad T \in [50, 56] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[665] & 181232181231513221(12)12231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2 \quad T = 44, \quad T_{\max} = 56, \quad \Delta = -104, \quad \det =
\end{aligned}$$

$$49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.06 \rightarrow 50, T_{\max} = 56, T \in [50, 56]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[664] \quad 61316131513221(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.06 \rightarrow 50, T_{\max} = 56, T \in [50, 56]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[673] \quad 41322^1 1(12)12231513221(12)12231513221(12)122315132218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.06 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[672] \quad 413221(12)12231513221(12)12231513221(12)1223151322^1 18 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.06 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[671] \quad 41322^1 1(12)12231513221(12)12231513221(12)12231513218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.06 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[659] \quad 1612321812^1 32181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.06 \rightarrow 50, T_{\max} = 56, T \in [50, 56]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[660] \quad (12)122315132181232181232181232181232181232^1 1613 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.06 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[663] \quad 61316131513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.06 \rightarrow 50, T_{\max} = 56, T \in [50, 56]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[662] \quad 1812315132^1 21(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.06 \rightarrow 50, T_{\max} = 56, T \in [50, 56]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[661] \quad 71231513221(12)12231513221(12)12231513221(12)12^1 23151 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.06 \rightarrow 50, T_{\max} = 56, T \in [50, 56]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[675] \quad (12)1223151321812321812321812321812321812^1 321613 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.94 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[674] \quad 161232181232^1 181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 64, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.94 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[690] 61316131513221(12)12231513221(12)12¹231513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.84 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[692] 81232181231513221(12)12231513221(12)122315132¹18123 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 59$, $\Delta = -113$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.84 \rightarrow 49, T_{\max} = 59, T \in [49, 59]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[691] 61316131513221(12)122315132¹21(12)12231513221(12)12222 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.84 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[693] 413221(12)122¹31513221(12)12231513221(12)12231513218 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.84 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[676] 1612321812321812321812321812321812315132¹21(12)1 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.84 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[677] 18123151322¹1(12)12231513221(12)122315132181232181 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.84 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[678] 181231513221(12)122¹31513221(12)122315132181232181 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.84 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[679] 181231513221(12)12231513221(12)1223151321812¹32181 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.84 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[680] 61316131513221(12)122315132¹21(12)12231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.84 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[681] 61316131513221(12)12231513221(12)12¹231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.84 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[682] 61316131513221(12)12231513221(12)12¹231513221(12)12 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 48.84 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[683] 61316131513221(12)122315132¹21(12)12231513221(12)12 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det =$

81, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.84 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[684] 181232181231513221(12)12231513221(12)12¹23151321812 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.84 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[685] 181232181231513221(12)122315132¹21(12)1223151321812 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.84 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[686] 181232¹181231513221(12)12231513221(12)1223151321812 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.84 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[687] 181232¹181231513221(12)12231513221(12)1223151322181 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.84 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[688] 181232181231513221(12)12231513221(12)12¹23151322181 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.84 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[689] 181232181231513221(12)122315132¹21(12)1223151322181 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 81$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.84 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[694] 16123218123218123218123218123218123151322¹1(12)1 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 100$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.76 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[696] (12)122315132181232181232181232181232¹1812321613 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 100$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.76 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[697] (12)122¹3151321812321812321812321812321812321613 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 100$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.76 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[698] 712315132¹21(12)12231513221(12)12231513221(12)1223151 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -108$, $\det = 100$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.76 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[699] 413221(12)12231513221(12)12231513221(12)122¹315132181 + $\hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 100$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.76 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[700] $413221(12)12231513221(12)12231513221(12)122^1315132218 + \hat{1} \rightarrow 2$ $T = 44, T_{\max} = 58, \Delta = -112, \det = 100, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.76 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[695] $1612321812321812^132181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 44, T_{\max} = 56, \Delta = -104, \det = 100, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.76 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[718] $81232181231513221(12)12231513221(12)122^131513218123 + \hat{1} \rightarrow 2$ $T = 44, T_{\max} = 59, \Delta = -113, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.69 \rightarrow 49, T_{\max} = 59, T \in [49, 59]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[701] $161232181232181232^1181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 44, T_{\max} = 56, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.69 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[704] $71231513221(12)12231513221(12)122315132^121(12)1223151 + \hat{1} \rightarrow 2$ $T = 44, T_{\max} = 56, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.69 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[705] $181231513221(12)122315132^121(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 44, T_{\max} = 56, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.69 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[706] $181231513221(12)12^1231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 44, T_{\max} = 56, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.69 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[707] $61316131513221(12)1223151322^11(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 44, T_{\max} = 56, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.69 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[708] $61316131513221(12)122^131513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 44, T_{\max} = 56, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.69 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[709] $61316131513221(12)1223151322^11(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 44, T_{\max} = 56, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.69 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[710] $61316131513221(12)122^131513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 44, T_{\max} = 56, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.69 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[711] $181232181231513221(12)1223151322^11(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 44, T_{\max} = 56, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[712] & 181232181231513221(12)122^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[713] & 181232181231513221(12)122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[714] & 181232181231513221(12)1223151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[715] & 61316131513221(12)1223151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[716] & 61316131513221(12)122^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[717] & 81232^1 181231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 59, \Delta = -113, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 59, T \in [49, 59] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[702] & (12)12^1 23151321812321812321812321812321613 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[719] & 413221(12)122^1 31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[720] & 413221(12)12^1 231513221(12)12231513221(12)12231513218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[721] & 413221(12)122315132^1 21(12)12231513221(12)12231513218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[722] & 413221(12)122^1 31513221(12)12231513221(12)122315132218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.69 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[703] & (12)1223151321812321812321812321812^1 321812321613 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0
\end{aligned}$$

$(1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.69 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[731] $1812321812315132^1 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[725] $71231513221(12)12231513221(12)1223151322^1 1(12)1223151 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -108$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[726] $7123151322^1 1(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -108$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[724] $(12)122315132181232181232181232^1 1812321812321613 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[728] $613161315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[729] $613161315132^1 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[730] $1812321812315132^1 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[723] $1612321812321812321812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[732] $613161315132^1 21(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[733] $81232181231513221(12)122315132^1 21(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 59$, $\Delta = -113$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 59$, $T \in [49, 59]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[734] $413221(12)12231513221(12)12231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

- [735] $413221(12)12^1 231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [736] $413221(12)12231513221(12)12231513221(12)122^1 31513218 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [737] $413221(12)12231513221(12)122^1 31513221(12)12231513218 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [738] $413221(12)12^1 231513221(12)12231513221(12)122315132218 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [739] $413221(12)12231513221(12)12231513221(12)12^1 2315132218 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [727] $181231513221(12)12231513221(12)122^1 315132181232181 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 169$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.59 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [741] $(12)1223151321812321812321812^1 321812321812321613 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 58$, $\Delta = -112$, $\det = 196$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.55 \rightarrow 49$, $T_{\max} = 58$, $T \in [49, 58]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [740] $161232181232181232181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 196$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.55 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [743] $181231513221(12)1223151322^1 1(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 196$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.55 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [744] $61316131513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 196$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.55 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [745] $61316131513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -105$, $\det = 196$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 48.55 \rightarrow 49$, $T_{\max} = 56$, $T \in [49, 56]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [746] $181232181231513221(12)12^1 231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 44$, $T_{\max} = 56$, $\Delta = -104$, $\det = 196$, $(n_+, n_-, n_0) = (1, 44, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 48.55 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[747] & 181232181231513221(12)12^1 231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = \\
& 196, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.55 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[748] & 61316131513221(12)12^1 231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -105, \det = \\
& 196, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.55 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[749] & 81232181231513221(12)12231513221(12)12^1 231513218123 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 59, \Delta = -113, \det = \\
& 196, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.55 \rightarrow 49, T_{\max} = 59, T \in [49, 59] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[750] & 413221(12)12231513221(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = \\
& 196, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.55 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[751] & 413221(12)1223151322^1 1(12)12231513221(12)12231513218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = \\
& 196, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.55 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[752] & 413221(12)12231513221(12)122315132^1 21(12)122315132218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = \\
& 196, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.55 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[742] & 71231513221(12)122^1 31513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -108, \det = \\
& 196, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.55 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[753] & 181231513221(12)12231513221(12)122315132^1 181232181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = \\
& 225, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.52 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[754] & 1812321812^1 31513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = \\
& 225, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.52 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[755] & 1812321812^1 31513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = \\
& 225, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.52 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[756] & 81232181231513221(12)122^1 31513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 59, \Delta = -113, \det = \\
& 225, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.52 \rightarrow 49, T_{\max} = 59, T \in [49, 59] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[757] $413221(12)12231513221(12)122315132^1 21(12)12231513218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 225, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.52 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[762] $6131613151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -105, \det = 256, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.48 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[760] $181231513221(12)12231513221(12)12^1 2315132181232181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 256, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.48 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[761] $6131613151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -105, \det = 256, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.48 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[759] $(12)122315132181232181232^1 1812321812321812321613 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.48 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[758] $1612321812321812321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 256, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.48 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[764] $18123218123151322^1 1(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 256, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.48 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[765] $6131613151322^1 1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -105, \det = 256, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.48 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[766] $81232181231513221(12)1223151322^1 1(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 59, \Delta = -113, \det = 256, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.48 \rightarrow 49, T_{\max} = 59, T \in [49, 59]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[767] $413221(12)12231513221(12)12^1 231513221(12)12231513218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.48 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

[763] $18123218123151322^1 1(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 256, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.48 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[769] $(12)1223151321812321812^1 321812321812321812321613 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.46 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[770] & 71231513221(12)12231513221(12)122^1 31513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -108, \det = 289, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.46 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[771] & 71231513221(12)12^1 231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -108, \det = 289, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.46 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[772] & 812321812315132^1 21(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 59, \Delta = -113, \det = 289, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.46 \rightarrow 49, T_{\max} = 59, T \in [49, 59] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[773] & 413221(12)12231513221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.46 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[774] & 413221(12)122315132^1 21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.46 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[775] & 413221(12)12231513221(12)12231513221(12)12^1 231513218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.46 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[776] & 413221(12)12231513221(12)1223151322^1 1(12)122315132218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.46 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 101 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[777] & 413221(12)122315132^1 21(12)12231513221(12)122315132218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.46 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 118 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[768] & 161232181232181232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 289, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.46 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[778] & 71231513221(12)122315132^1 21(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -108, \det = 324, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.43 \rightarrow 49, T_{\max} = 56, T \in [49, 56] \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[779] & 413221(12)12231513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 324, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 48.43 \rightarrow 49, T_{\max} = 58, T \in [49, 58] \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[780] $413221(12)12231513221(12)122^131513221(12)122315132218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 324, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.43 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 125 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[781] $1612321812321812321812321812^131513221(12)1 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.41 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$

[782] $(12)122315132181232^11812321812321812321812321613 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 361, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.41 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[783] $81232181231513221(12)12^1231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 59, \Delta = -113, \det = 361, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.41 \rightarrow 49, T_{\max} = 59, T \in [49, 59]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

[784] $812321812^131513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 59, \Delta = -113, \det = 361, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.41 \rightarrow 49, T_{\max} = 59, T \in [49, 59]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$

[785] $413221(12)12231513221(12)1223151322^11(12)12231513218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 361, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.41 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$

[786] $(12)1223151321812^1321812321812321812321812321613 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 400, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.39 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$
 $\begin{pmatrix} 0 & 20 & 2 \\ 20 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[787] $8123218123151322^11(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 59, \Delta = -113, \det = 441, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.37 \rightarrow 49, T_{\max} = 59, T \in [49, 59]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[788] $(12)122315132^11812321812321812321812321812321613 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 484, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.36 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$

[789] $71231513221(12)12231513221(12)12^1231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -108, \det = 484, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.36 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[790] $413221(12)1223151322^11(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 484, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.36 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[791] $413221(12)1223151322^11(12)12231513221(12)122315132218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 484, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 48.36 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 197 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[792] \quad 71231513221(12)12231513221^1(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 56, \Delta = -108, \det = 529, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.34 \rightarrow 49, T_{\max} = 56, T \in [49, 56]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[793] \quad 413221(12)12231513221(12)12^1231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 529, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.34 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[794] \quad 413221(12)12231513221(12)12^1231513221(12)122315132218 + \hat{1} \rightarrow 2 \quad T = 44, T_{\max} = 58, \Delta = -112, \det = 529, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 48.34 \rightarrow 49, T_{\max} = 58, T \in [49, 58]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 206 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$$

$$T = 45 \quad [795] \quad (12)1223151321812321812321812321812321812321712^13 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 58, \Delta = -111, \det = -9, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.22 \rightarrow 52, T_{\max} = 58, T \in [52, 58]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[796] \quad 1712^132181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -102, \det = -16, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 50.75 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[797] \quad 21712^132181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -102, \det = -16, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 50.75 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[798] \quad 171232^1181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -102, \det = -25, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 50.44 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[799] \quad 2171232^1181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -102, \det = -25, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 50.44 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[800] \quad 81231513221(12)12231513221(12)12231513221^{15132^121}(12)1 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 50.44 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[801] \quad 15132^121(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 50.44 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[802] \quad 812231513221(12)12231513221(12)12231513221(12)122^13151 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -107, \det = -25, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 50.44 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

- [804] $8122^1 31513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 50.22 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [803] $1513221(12)12231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 50.22 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [809] $812^1 31513221(12)12231513221(12)12231513221^{1513221} (12) 1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 50.06 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [807] $(12)122315132181232181232181232181232^1 17123 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 50.06 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [808] $81231513221(12)12231513221(12)12231513221^{1513221} (12) 1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 50.06 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [806] $21712321812^1 32181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -102, \det = -49, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 50.06 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [810] $151322^1 1(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 50.06 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [811] $812^1 31513221(12)12231513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 62, \Delta = -115, \det = -49, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 50.06 \rightarrow 51, T_{\max} = 62, T \in [51, 62]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [812] $812231513221(12)12231513221(12)12231513221(12)12^1 23151 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 50.06 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [813] $812^1 231513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -49, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 50.06 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [814] $812^1 31513221(12)12231513221(12)12231513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det = -49, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 50.06 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [815] $81231513221(12)12231513221(12)12231513221(12)122^1 31414 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det =$

$$-49, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 50.06 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[805] \overset{1}{1712321812^1 32181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2} \quad T = 45, T_{\max} = 57, \Delta = -102, \det = -49, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 50.06 \rightarrow 51, T_{\max} = 57, T \in [51, 57]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[816] \overset{1}{171232181232^1 181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2} \quad T = 45, T_{\max} = 57, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.94 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[817] 2171232181232^1 181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.94 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[818] (12)1223151321812321812321812321812321812^1 3217123 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 58, \Delta = -111, \det = -64, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.94 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[819] 81231513221(12)12231513221(12)12231513221(12)122^1 31513 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 62, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.94 \rightarrow 50, T_{\max} = 62, T \in [50, 62]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[820] \overset{1}{1712321812321812321812321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2} \quad T = 45, T_{\max} = 57, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.84 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[821] 21712321812321812321812321812321812315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.84 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[822] 81231513221(12)12231513221(12)122315132^1 21 \overset{1513221}{(12) 1 + \hat{1} \rightarrow 2} \quad T = 45, T_{\max} = 57, \Delta = -107, \det = -81, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.84 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[823] 61316131513221(12)12231513221(12)12^1 231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 50, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.84 \rightarrow 50, T_{\max} = 50, T \in [50, 50]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[824] 61316131513221(12)122315132^1 21(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 50, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.84 \rightarrow 50, T_{\max} = 50, T \in [50, 50]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[825] \overset{1}{81231513221(12)12231513221(12)122315132^1 21(12)12231414 + \hat{1} \rightarrow 2} \quad T = 45, T_{\max} = 58, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.84 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[826] $81231513221(12)12231513221(12)12231513221(12)12^1 231414 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.84 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[828] $17123218123218123218123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -102, \det = -100, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.76 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

[829] $21712321812321812^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -102, \det = -100, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.76 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[830] $217123218123218123218123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -102, \det = -100, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.76 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

[831] $(12)122^1 31513218123218123218123218123218123217123 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det = -100, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.76 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[832] $(12)122315132181232181232181232181232^1 18123217123 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det = -100, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.76 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[833] $1513221(12)12231513221(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.76 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[834] $8122315132^1 21(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -100, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.76 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[827] $1712321812321812^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -102, \det = -100, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.76 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[835] $171232181232181232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.69 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[836] $2171232181232181232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.69 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[837] $(12)12^1 231513218123218123218123218123218123217123 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.69 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[838] $(12)1223151321812321812321812321812^1 3218123217123 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 58$, $\Delta = -111$, $\det = -121$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 49.69 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[839] $81231513221(12)12231513221(12)122^1 31513221$ $(12) 1 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 49.69 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[840] $81231513221(12)12231513221(12)1223151322^1 1$ $(12) 1 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 49.69 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[841] $1513221(12)122^1 31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 49.69 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[842] $81231513221(12)12231513221(12)12231513221(12)12^1 231513 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 62$, $\Delta = -115$, $\det = -121$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 49.69 \rightarrow 50, T_{\max} = 62, T \in [50, 62]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[843] $812231513221(12)12231513221(12)122315132^1 21(12)1223151 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -121$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 49.69 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[844] $61316131513221(12)1223151322^1 1(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 50$, $\Delta = -93$, $\det = -121$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 49.69 \rightarrow 50, T_{\max} = 50, T \in [50, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[845] $61316131513221(12)122^1 31513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 50$, $\Delta = -93$, $\det = -121$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 49.69 \rightarrow 50, T_{\max} = 50, T \in [50, 50]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[846] $81231513221(12)12231513221(12)1223151322^1 1(12)12231414 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 58$, $\Delta = -111$, $\det = -121$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 49.69 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[847] $81231513221(12)12231513221(12)122^1 31513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 58$, $\Delta = -111$, $\det = -121$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 49.69 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[851] $81231513221(12)122315132^1 21(12)12231513221$ $(12) 1 + \hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -169$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 49.59 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

- [852] $812315132^1 21(12)12231513221(12)12231513221^{1513221} (12) 1+\hat{1}\rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.59 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [849] $21712321812321812321812^1 32181232181231513221(12)1+\hat{1}\rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -102, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.59 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [848] $1712321812321812321812^1 32181232181231513221(12)1+\hat{1}\rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -102, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.59 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [853] $1513221(12)12^1 231513221(12)12231513221(12)122315132181 + \hat{1}\rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.59 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [854] $1513221(12)12231513221(12)12231513221(12)12^1 2315132181 + \hat{1}\rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.59 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [855] $812315132^1 21(12)12231513221(12)12231513221(12)12231513 + \hat{1}\rightarrow 2$ $T = 45, T_{\max} = 62, \Delta = -115, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.59 \rightarrow 50, T_{\max} = 62, T \in [50, 62]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [856] $81223151322^1 1(12)12231513221(12)12231513221(12)1223151 + \hat{1}\rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.59 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [857] $812231513221(12)12231513221(12)1223151322^1 1(12)1223151 + \hat{1}\rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.59 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [858] $613161315132^1 21(12)12231513221(12)12231513221(12)122313 + \hat{1}\rightarrow 2$ $T = 45, T_{\max} = 50, \Delta = -93, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.59 \rightarrow 50, T_{\max} = 50, T \in [50, 50]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [859] $81231513221(12)122315132^1 21(12)12231513221(12)12231414 + \hat{1}\rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.59 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [860] $812315132^1 21(12)12231513221(12)12231513221(12)12231414 + \hat{1}\rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.59 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [850] $(12)122315132181232181232181232^1 18123218123217123+\hat{1}\rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det = -169, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 49.59 \rightarrow 50, T_{\max} = 58, T \in [50, 58] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[862] & 2171232181232181232181232^1 181232181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.55 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[863] & (12)1223151321812321812321812^1 3218123218123217123+\hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 58, \Delta = -111, \det = -196, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.55 \rightarrow 50, T_{\max} = 58, T \in [50, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[864] & 81231513221(12)12231513221(12)12^1 231513221 \overset{1513221}{(12)} 1+\hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -107, \det = -196, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.55 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[865] & 1513221(12)12231513221(12)122315132^1 21(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -107, \det = -196, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.55 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[866] & 812231513221(12)122^1 31513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -107, \det = -196, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.55 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[867] & 61316131513221(12)12^1 231513221(12)12231513221(12)122313 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 50, \Delta = -93, \det = -196, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.55 \rightarrow 50, T_{\max} = 50, T \in [50, 50] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[868] & 81231513221(12)12231513221(12)12^1 231513221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 58, \Delta = -111, \det = -196, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.55 \rightarrow 50, T_{\max} = 58, T \in [50, 58] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[861] & 171232181232181232181232^1 181232181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.55 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[869] & 81231513221(12)122^1 31513221(12)12231513221 \overset{1513221}{(12)} 1+\hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -107, \det = -225, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.52 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[870] & 81231513221(12)122^1 31513221(12)12231513221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 58, \Delta = -111, \det = -225, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.52 \rightarrow 50, T_{\max} = 58, T \in [50, 58] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[871] & 1712321812321812321812321812^1 32181231513221(12)1+\hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -102, \det = -256, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.48 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [872] 21712321812321812321812321812¹32181231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.48 \rightarrow 50$, $T_{\max} = 57$, $T \in [50, 57]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [873] (12)122315132181232181232¹18123218123218123217123+ $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 58$, $\Delta = -111$, $\det = -256$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.48 \rightarrow 50$, $T_{\max} = 58$, $T \in [50, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [874] 81231513221¹(12)1223151322¹1(12)12231513221¹⁵¹³²²¹(12) 1+ $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -256$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.48 \rightarrow 50$, $T_{\max} = 57$, $T \in [50, 57]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [875] 81231513221¹(12)12231513221(12)122315132¹21(12)12231513 + $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 62$, $\Delta = -115$, $\det = -256$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.48 \rightarrow 50$, $T_{\max} = 62$, $T \in [50, 62]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [877] 81231513221¹(12)1223151322¹1(12)12231513221(12)12231414 + $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 58$, $\Delta = -111$, $\det = -256$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.48 \rightarrow 50$, $T_{\max} = 58$, $T \in [50, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [876] 6131613151322¹1(12)12231513221(12)12231513221(12)122313 + $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 50$, $\Delta = -93$, $\det = -256$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.48 \rightarrow 50$, $T_{\max} = 50$, $T \in [50, 50]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [878] 171232181232181232181232181232¹181231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -102$, $\det = -289$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.46 \rightarrow 50$, $T_{\max} = 57$, $T \in [50, 57]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [879] 2171232181232181232181232181232¹181231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -102$, $\det = -289$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.46 \rightarrow 50$, $T_{\max} = 57$, $T \in [50, 57]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [880] (12)1223151321812321812¹3218123218123218123217123+ $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 58$, $\Delta = -111$, $\det = -289$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.46 \rightarrow 50$, $T_{\max} = 58$, $T \in [50, 58]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [881] 8123151322¹1(12)12231513221(12)12231513221¹⁵¹³²²¹(12) 1+ $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -289$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.46 \rightarrow 50$, $T_{\max} = 57$, $T \in [50, 57]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [882] 1513221(12)122315132¹21(12)12231513221(12)122315132181 + $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -289$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.46 \rightarrow 50$, $T_{\max} = 57$, $T \in [50, 57]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [883] 1513221¹(12)12231513221(12)1223151322¹1(12)122315132181 + $\hat{1} \rightarrow 2$ $T = 45$, $T_{\max} = 57$, $\Delta = -107$, $\det = -289$, $(n_+, n_-, n_0) = (1, 45, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 49.46 \rightarrow 50$, $T_{\max} = 57$, $T \in [50, 57]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[884] & 8123151322^1 1(12)12231513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 62, \Delta = -115, \det = -289, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.46 \rightarrow 50, T_{\max} = 62, T \in [50, 62] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[885] & 812231513221(12)12^1 231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -107, \det = -289, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.46 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[886] & 812231513221(12)12231513221(12)122^1 31513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -107, \det = -289, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.46 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[887] & 8123151322^1 1(12)12231513221(12)12231513221(12)12231414 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 58, \Delta = -111, \det = -289, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.46 \rightarrow 50, T_{\max} = 58, T \in [50, 58] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[888] & 1513221(12)12231513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -107, \det = -324, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.43 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[889] & 812231513221(12)122315132^1 21(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -107, \det = -324, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.43 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[892] & (12)122315132181232^1 18123218123218123218123217123 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 58, \Delta = -111, \det = -361, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.41 \rightarrow 50, T_{\max} = 58, T \in [50, 58] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[891] & 21712321812321812321812321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.41 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[893] & 81231513221(12)12^1 231513221(12)12231513221(12)1513221(12)1 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 57, \Delta = -107, \det = -361, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.41 \rightarrow 50, T_{\max} = 57, T \in [50, 57] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[894] & 81231513221(12)12231513221(12)1223151322^1 1(12)12231513 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 62, \Delta = -115, \det = -361, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.41 \rightarrow 50, T_{\max} = 62, T \in [50, 62] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 246 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[895] & 81231513221(12)122^1 31513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 62, \Delta = -115, \det = -361, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 49.41 \rightarrow 50, T_{\max} = 62, T \in [50, 62] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [896] $81231513221(12)12^1 231513221(12)12231513221(12)12231414 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det = -361, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.41 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [890] $1712321812321812321812321812321812^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.41 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [897] $(12)1223151321812^1 3218123218123218123218123217123 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det = -400, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.39 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$
 $\begin{pmatrix} 0 & 20 & 2 \\ 20 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [898] $(12)122315132^1 18123218123218123218123218123217123 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 58, \Delta = -111, \det = -484, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.36 \rightarrow 50, T_{\max} = 58, T \in [50, 58]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [900] $812231513221(12)12231513221(12)12^1 231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -484, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.36 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [899] $1513221(12)1223151322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -484, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.36 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [901] $1513221(12)12231513221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -529, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.34 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [902] $81231513221(12)12^1 231513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 62, \Delta = -115, \det = -529, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.34 \rightarrow 50, T_{\max} = 62, T \in [50, 62]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 390 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0}$
- [903] $812231513221(12)1223151322^1 1(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 57, \Delta = -107, \det = -529, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.34 \rightarrow 50, T_{\max} = 57, T \in [50, 57]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [904] $81231513221(12)12231513221(12)122^1 31513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 62, \Delta = -115, \det = -576, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.33 \rightarrow 50, T_{\max} = 62, T \in [50, 62]$
 $\begin{pmatrix} 0 & 24 & 2 \\ 24 & 407 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [905] $81231513221(12)122315132^1 21(12)12231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 62, \Delta = -115, \det = -625, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 49.31 \rightarrow 50, T_{\max} = 62, T \in [50, 62]$
 $\begin{pmatrix} 0 & 25 & 2 \\ 25 & 449 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [906] $81231513221(12)12231513221(12)12^1 231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 45, T_{\max} = 62, \Delta = -115, \det =$

$$-961, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.25 \rightarrow 50, T_{\max} = 62, T \in [50, 62]$$

$$\begin{pmatrix} 0 & 31 & 2 \\ 31 & 681 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

$$[907] \quad 81231513221(12)1223151322^1 1(12)12231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 45, T_{\max} = 62, \Delta = -115, \det = -1024, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 49.25 \rightarrow 50, T_{\max} = 62, T \in [50, 62]$$

$$\begin{pmatrix} 0 & 32 & 2 \\ 32 & 735 & 53 \\ 2 & 53 & 3 \end{pmatrix}_{b_0}$$

$$T = 46 \quad [908] \quad 22^1 181232181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 46, T_{\max} = 58, \Delta = -101, \det = 4, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53, T_{\max} = 58, T \in [53, 58]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[909] \quad (12)1223151321812321812321812321812321812321812321 \quad 8^{32^1 1} 1 + \hat{1} \rightarrow 2 \quad T = 46, T_{\max} = 58, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.22 \rightarrow 53, T_{\max} = 58, T \in [53, 58]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[910] \quad (12)12231513218123218123218123218123218123218122^1 3 + \hat{1} \rightarrow 2 \quad T = 46, T_{\max} = 58, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.22 \rightarrow 53, T_{\max} = 58, T \in [53, 58]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[911] \quad 18 \quad 12^{21} 32181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 46, T_{\max} = 58, \Delta = -101, \det = 16, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.75 \rightarrow 52, T_{\max} = 58, T \in [52, 58]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[912] \quad 221812^1 32181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 46, T_{\max} = 58, \Delta = -101, \det = 16, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.75 \rightarrow 52, T_{\max} = 58, T \in [52, 58]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[913] \quad 18 \quad 1232^1 181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 46, T_{\max} = 58, \Delta = -101, \det = 25, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.44 \rightarrow 52, T_{\max} = 58, T \in [52, 58]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[915] \quad (12)1223151321812321812321812321812321812321812321812^1 23 + \hat{1} \rightarrow 2 \quad T = 46, T_{\max} = 58, \Delta = -110, \det = 25, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.44 \rightarrow 52, T_{\max} = 58, T \in [52, 58]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[914] \quad 22181232^1 181232181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 46, T_{\max} = 58, \Delta = -101, \det = 25, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.44 \rightarrow 52, T_{\max} = 58, T \in [52, 58]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 39 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[916] \quad 18 \quad 12321812^1 32181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 46, T_{\max} = 58, \Delta = -101, \det = 49, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.06 \rightarrow 52, T_{\max} = 58, T \in [52, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[918] \quad 221812321812^1 32181232181232181232181231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 46, T_{\max} = 58, \Delta = -101, \det = 49, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.06 \rightarrow 52, T_{\max} = 58, T \in [52, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[919] (12)122315132181232181232181232181232181232¹181223 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.06 \rightarrow 52, T_{\max} = 58, T \in [52, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[917] (12)122315132181232181232181232181232181232¹1³²¹8 1 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 49$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.06 \rightarrow 52, T_{\max} = 58, T \in [52, 58]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[920] ²¹18 1232181232¹181232181232181232181231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 64$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.94 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[921] (12)1223151321812321812321812321812321812¹321³²¹8 1 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 64$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.94 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[922] 22181232181232¹181232181232181232181231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 64$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.94 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[923] (12)1223151321812321812321812321812321812¹32181223 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 64$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.94 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[924] ²¹18 12321812321812321812321812321812315132¹21(12)1 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 81$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.84 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[925] 221812321812321812321812321812321812315132¹21(12)1 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 81$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.84 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[926] ²¹18 123218123218123218123218123218123151322¹1(12)1 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 100$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.76 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[927] ²¹18 12321812321812¹32181232181232181231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 100$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.76 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[928] (12)122¹3151321812321812321812321812321812321³²¹8 1 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 100$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.76 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[929] (12)122315132181232181232181232181232¹1812321³²¹8 1 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 100$, $(n_+, n_-, n_0) =$

$(1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.76 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[930] $2218123218123218123218123218123218123151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 100$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.76 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[931] $221812321812321812^1 32181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 100$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.76 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[932] $(12)122^1 315132181232181232181232181232181223 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 100$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.76 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[933] $(12)122315132181232181232181232181232^1 181232181223 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 100$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.76 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[937] $22181232181232181232^1 181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 121$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.69 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[936] $(12)1223151321812321812321812321812^1 321812321 \overset{321}{8} 1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.69 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[935] $(12)12^1 23151321812321812321812321812321 \overset{321}{8} 1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.69 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[934] $18 \overset{21}{1232181232181232^1} 181232181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 121$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.69 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[939] $(12)12^1 2315132181232181232181232181232181223 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.69 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[938] $(12)1223151321812321812321812321812^1 32181232181223 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.69 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[941] $(12)122315132181232181232181232^1 1812321812321 \overset{321}{8} 1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 169$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 50.59 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

- [942] $221812321812321812321812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 169$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.59 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [943] $(12)122315132181232181232181232^1 181232181232181223 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 169$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.59 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [940] $1812321812321812321812^1 32181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 169$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.59 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [944] $181232181232181232181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 196$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.55 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [945] $(12)1223151321812321812321812^1 321812321812321^{321} 81 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 196$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.55 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [946] $22181232181232181232181232^1 181232181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 196$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.55 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [947] $(12)1223151321812321812321812^1 32181232181232181223 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 196$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.55 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$
- [948] $1812321812321812321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 256$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.48 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [949] $(12)122315132181232181232^1 1812321812321812321^{321} 81 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 256$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.48 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [950] $221812321812321812321812321812^1 32181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 256$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.48 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [951] $(12)122315132181232181232^1 181232181232181232181223 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 256$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.48 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [952] $181232181232181232181232181232^1 181231513221(12)1 + \hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 289$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 50.46 \rightarrow 51$, $T_{\max} = 58$, $T \in [51, 58]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[954] 22181232181232181232181232181232¹181231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 289$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.46 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[955] (12)1223151321812321812¹32181232181232181232181223 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 289$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.46 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[953] (12)1223151321812321812¹321812321812321812321³²¹8 1 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 289$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.46 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[956] ²¹18 12321812321812321812321812321812¹31513221(12)1 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 361$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.41 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[957] (12)122315132181232¹1812321812321812321812321³²¹8 1 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 361$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.41 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[958] 221812321812321812321812321812321812¹31513221(12)1 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -101$, $\det = 361$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.41 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[959] (12)122315132181232¹181232181232181232181232181223 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 361$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.41 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[960] (12)1223151321812¹321812321812321812321812321³²¹8 1 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 400$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.39 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 20 & 2 \\ 20 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[961] (12)1223151321812¹32181232181232181232181232181223 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 400$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.39 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 20 & 2 \\ 20 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[962] (12)122315132¹1812321812321812321812321812321³²¹8 1 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 484$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.36 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[963] (12)122315132¹181232181232181232181232181232181223 + $\hat{1} \rightarrow 2$ $T = 46$, $T_{\max} = 58$, $\Delta = -110$, $\det = 484$, $(n_+, n_-, n_0) = (1, 46, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 50.36 \rightarrow 51, T_{\max} = 58, T \in [51, 58]$$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& T = 47 \quad [984] \quad 71232181231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54, T_{\max} = 59, T \in [54, 59] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
& [966] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54, T_{\max} = 59, T \in [54, 59] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
& [967] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
& [968] \quad 21(12)12231513221(12)12231513221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54, T_{\max} = 59, T \in [54, 59] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
& [969] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54, T_{\max} = 59, T \in [54, 59] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
& [970] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54, T_{\max} = 59, T \in [54, 59] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
& [971] \quad 21(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54, T_{\max} = 59, T \in [54, 59] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
& [972] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
& [973] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54, T_{\max} = 59, T \in [54, 59] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
& [974] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54, T_{\max} = 59, T \in [54, 59] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
& [975] \quad 21(12)12231513221(12)12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54, T_{\max} = 59, T \in [54, 59] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

[976] $1^{22^1 21} (12) 12231513221(12) 12231513221(12) 12231513221(12) 1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[977] $1^{2221} (12) 12231513221(12) 12231513221(12) 12231513221(12) 122^1 2 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[978] $21^{3221} (12) 12231513221(12) 12231513221(12) 12231513221(12) 12^1 2 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[979] $3221^{2^1 1} (12) 12231513221(12) 12231513221(12) 12231513221(12) 122 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$

[980] $21^{22^1 1} (12) 12231513221(12) 12231513221(12) 12231513221(12) 1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[965] $1^{221} (12) 12231513221(12) 12231513221(12) 12231513221(12) 122^1 2 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[982] $22^1 21 (12) 12231513221(12) 12231513221(12) 12231513221(12) 12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[983] $2221^{1} (12) 12231513221(12) 12231513221(12) 12231513221(12) 1222^1 2 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[964] $1^{3221} (12) 12231513221(12) 12231513221(12) 12231513221(12) 1 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[985] $71232181231513221(12) 12231513221(12) 12231513221(12) 12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[986] $71232181231513221(12) 12231513221(12) 12231513221(12) 1222^1 2 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[981] $21^{221}(12)12231513221(12)12231513221(12)12231513221(12)122^12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[996] $1^{2221}(12)12231513221(12)12231513221(12)12231513221(12)12^122 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[989] $1^{21}(12)12231513221(12)12231513221(12)12231513221(12)122^13 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[990] $1^{32^121}(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[991] $21^{21}(12)12231513221(12)12231513221(12)12231513221(12)12^122 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[992] $1^{32^121}(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[993] $1^{3221}(12)12231513221(12)12231513221(12)12231513221(12)12^122 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

[994] $1^{221}(12)12231513221(12)12231513221(12)12231513221(12)122^122 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[995] $21^{21}(12)12231513221(12)12231513221(12)12231513221(12)122^122 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[987] $1^{32^121}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$

[997] $1^{222^11}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -9, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54, T_{\max} = 59, T \in [54, 59]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$

- [998] $^{32^1 21} 21(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 61$, $\Delta = -112$, $\det = -9$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [999] $^{221} 21(12)12231513221(12)12231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -9$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54$, $T_{\max} = 59$, $T \in [54, 59]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1000] $^{1} 2221(12)12231513221(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -9$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54$, $T_{\max} = 59$, $T \in [54, 59]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1001] $^{1} 222^1 1(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -9$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54$, $T_{\max} = 59$, $T \in [54, 59]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$
- [1002] $^{222^1 1} 71232181231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -104$, $\det = -9$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54$, $T_{\max} = 59$, $T \in [54, 59]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [1003] $71232181231513221(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -104$, $\det = -9$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54$, $T_{\max} = 59$, $T \in [54, 59]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [988] $^{221} 1(12)12231513221(12)12231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -9$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.22 \rightarrow 54$, $T_{\max} = 59$, $T \in [54, 59]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [1004] $^{221} 1(12)12231513221(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -16$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.75 \rightarrow 53$, $T_{\max} = 59$, $T \in [53, 59]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [1005] $^{21} 21(12)12231513221(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -16$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.75 \rightarrow 53$, $T_{\max} = 59$, $T \in [53, 59]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [1006] $^{1} 2221(12)12231513221(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -16$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.75 \rightarrow 53$, $T_{\max} = 59$, $T \in [53, 59]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [1007] $71232181231513221(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -104$, $\det = -16$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.75 \rightarrow 53$, $T_{\max} = 59$, $T \in [53, 59]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1008] $^{1} 4132^1 181232181232181232181232181232181231513221(12) + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -109$, $\det = -25$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 52.44 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 52.44 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 52.44 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 52.44 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 52.44 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 52.44 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 52.44 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 52.44 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 52.44 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 52.44 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& T_{\min} = 52.44 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1020] & \begin{matrix} 322^1 1 \\ 1 \end{matrix} 21(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det = \\
& -25, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.44 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1021] & \begin{matrix} 221 \\ 1 \end{matrix} 21(12)122^1 31513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = \\
& -25, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.44 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1022] & \begin{matrix} 1 \\ 1 \end{matrix} 2221(12)12231513221(12)12231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = \\
& -25, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.44 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1023] & \begin{matrix} 2221 \\ 1 \end{matrix} 71232181231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = \\
& -25, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.44 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1024] & \begin{matrix} 221 \\ 1 \end{matrix} 71232181231513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = \\
& -25, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.44 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1025] & \begin{matrix} 221 \\ 1 \end{matrix} 71232181231513221(12)12231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = \\
& -25, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.44 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1046] & \begin{matrix} 1 \\ 1 \end{matrix} 2221(12)122^1 31513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = \\
& -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1045] & \begin{matrix} 1 \\ 1 \end{matrix} 2221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = \\
& -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1035] & \begin{matrix} 21 \\ 1 \end{matrix} 21(12)122^1 31513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = \\
& -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1047] & \begin{matrix} 2221 \\ 1 \end{matrix} 712^1 32181231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = \\
& -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1036] & \begin{matrix} 3221 \\ 1 \end{matrix} 1(12)12231513221(12)12231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det = \\
& -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 61, T \in [53, 61]
\end{aligned}$$

1043

$$\begin{aligned}
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1043] & 21^{221}(12)12231513221(12)12231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1044] & 21^{221}(12)12^1 231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1032] & 1^{221}(12)12231513221(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1048] & 71232181231513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1049] & 712^1 32181231513221(12)12231513221(12)12231513221^{221}(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1050] & 71232181231513221(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1051] & 71232181231513221(12)12231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1052] & 712^1 32181231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1033] & 1^{221}(12)122^1 31513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1031] & 21^{21}(12)122^1 31513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1034] & 21^{21}(12)12231513221(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -36, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.22 \rightarrow 53, T_{\max} = 59, T \in [53, 59] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[1069] $21(12)12^{21}231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1066] $1(12)12^{21}231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1065] $1(12)12231513221(12)12231513221(12)1223151322^11(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1070] $1(12)122315132^{221}21(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1063] $21(12)12231513221(12)12231513221(12)122^131513221(12)12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[1062] $1(12)12^{21}231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1061] $1(12)12231513221(12)12231513221(12)122^131513221(12)12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[1060] $21(12)12^{21}231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1059] $21(12)12231513221(12)12231513221(12)122^131513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

[1058] $1(12)12231513221(12)12231513221(12)1223151322^11(12)12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1057] $1(12)122^131513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

- [1056] $1(12)12231513221(12)12231513221(12)122^131513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -49$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53$, $T_{\max} = 59$, $T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1055] $1(12)12^1231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -49$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53$, $T_{\max} = 59$, $T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1054] $1(12)12231513221(12)12231513221(12)1223151322^11(12)1 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 61$, $\Delta = -112$, $\det = -49$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53$, $T_{\max} = 61$, $T \in [53, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1053] $41321812^132181232181232181232181232181231513221(12) + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53$, $T_{\max} = 59$, $T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1071] $1(12)12231513221(12)12231513221(12)1223151322^11(12)1222 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -49$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53$, $T_{\max} = 59$, $T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1072] $21(12)12231513221(12)12231513221(12)122315132^121(12)122 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 61$, $\Delta = -112$, $\det = -49$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53$, $T_{\max} = 61$, $T \in [53, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1073] $21(12)122315132^121(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -49$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53$, $T_{\max} = 59$, $T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1074] $21(12)12231513221(12)12231513221(12)1223151322^11(12)1222 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -49$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53$, $T_{\max} = 59$, $T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [1075] $2221(12)12^1231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -49$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53$, $T_{\max} = 59$, $T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1076] $2221(12)12231513221(12)12231513221(12)122^131513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -49$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53$, $T_{\max} = 59$, $T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1077] $812^131513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -109$, $\det = -49$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.06 \rightarrow 53$, $T_{\max} = 59$, $T \in [53, 59]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1078] $81231513221(12)12231513221(12)12231513221(12)122^13151315 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -109$, $\det =$

$$-49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[1079] \quad 71232181231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[1080] \quad 71232181231513221(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[1081] \quad 71232181231513221(12)12231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[1067] \quad 1(12)12231513221(12)12231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[1068] \quad 21(12)12231513221(12)12231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[1064] \quad 21(12)12^1 231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.06 \rightarrow 53, T_{\max} = 59, T \in [53, 59]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1082] \quad 4132181232^1 181232181232181232181232181231513221(12) + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -109, \det = -64, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.94 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 23 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1083] \quad 1(12)12^1 231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.94 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1084] \quad 21(12)12231513221(12)12231513221(12)1223151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.94 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1093] \quad 21(12)12231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1095] \quad 1(12)12231513221(12)122315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1101] & \begin{matrix} 221 \\ 21(12)12231513221(12)122^1 31513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1102] & \begin{matrix} 221 \\ 21(12)1223151322^1 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1103] & \begin{matrix} 1 \\ 2221(12)12231513221(12)122315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1104] & \begin{matrix} 1 \\ 2221(12)12231513221(12)12231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1105] & \begin{matrix} 1 \\ 81231513221(12)12231513221(12)122315132^1 21(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1106] & \begin{matrix} 1 \\ 81231513221(12)12231513221(12)12231513221(12)12^1 23151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -109, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1107] & \begin{matrix} 2221 \\ 71232181231513221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1108] & \begin{matrix} 2221 \\ 71232^1 181231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[1109] & \begin{matrix} 2221 \\ 71232181231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1110] & \begin{matrix} 221 \\ 71232181231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1111] & \begin{matrix} 221 \\ 71232^1 181231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59]
\end{aligned}$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[1112] $71232181231513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -104$, $\det = -81$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1113] $71232181231513221(12)122315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -104$, $\det = -81$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1114] $71232^1 181231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -104$, $\det = -81$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[1115] $71232181231513221(12)12231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -104$, $\det = -81$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1091] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -81$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1092] $21(12)12231513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -81$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1090] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -81$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.84 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1118] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 61$, $\Delta = -112$, $\det = -100$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.76 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1120] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 61$, $\Delta = -112$, $\det = -100$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.76 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1121] $21(12)122315132^1 21(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -100$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.76 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1122] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47$, $T_{\max} = 59$, $\Delta = -103$, $\det = -100$, $(n_+, n_-, n_0) = (1, 47, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 51.76 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

- [1123] $21^{21}(12)122315132^1 21^1(12)12231513221(12)12231513221^{221}(12)12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -100, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.76 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1124] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -100, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.76 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1125] $1(12)122315132^1 21^1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -100, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.76 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1126] $21^{21}(12)122315132^1 21^1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -100, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.76 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1127] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -100, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.76 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1128] $21^{221}(12)12231513221(12)12231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -100, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.76 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1129] $2221(12)122315132^1 21^1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -100, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.76 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1119] $1(12)122315132^1 21^1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -100, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.76 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1116] $41321812321812321812321812321812321812315132^1 21(12) + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -109, \det = -100, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.76 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 9 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [1117] $41321812321812^1 32181232181232181232181231513221(12) + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -109, \det = -100, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.76 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1157] $81231513221(12)12231513221(12)122^1 31513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -109, \det = -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.69 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1134] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det =$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[1132] & \begin{matrix} 3221 & 1 & 221 \\ 1(12)122^1 31513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 61, \Delta = -112, \det = \\ -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.69 \rightarrow 52, T_{\max} = 61, T \in [52, 61] \end{matrix} \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1131] & \begin{matrix} 413218123218123218123218123218123218123151322^1 1(12) + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 59, \Delta = -109, \det = \\ -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.69 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \end{matrix} \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 10 & 10 \\ 2 & 10 & 3 \end{pmatrix}_{b_0} \\
[1159] & \begin{matrix} 71232181231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 59, \Delta = -104, \det = \\ -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.69 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \end{matrix} \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1160] & \begin{matrix} 71232181231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 59, \Delta = -104, \det = \\ -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.69 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \end{matrix} \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[1161] & \begin{matrix} 71232181231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 59, \Delta = -104, \det = \\ -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.69 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \end{matrix} \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1162] & \begin{matrix} 71232181231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 59, \Delta = -104, \det = \\ -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.69 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \end{matrix} \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[1163] & \begin{matrix} 71232181231513221(12)122^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 59, \Delta = -104, \det = \\ -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.69 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \end{matrix} \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1164] & \begin{matrix} 71232181231513221(12)1223151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 59, \Delta = -104, \det = \\ -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.69 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \end{matrix} \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[1156] & \begin{matrix} 2221(12)12231513221(12)122^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 59, \Delta = -103, \det = \\ -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.69 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \end{matrix} \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[1158] & \begin{matrix} 81231513221(12)12231513221(12)1223151322^1 1(12)1223151315 + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 59, \Delta = -109, \det = \\ -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.69 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \end{matrix} \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1130] & \begin{matrix} 4132181232181232^1 181232181232181232181231513221(12) + \hat{1} \rightarrow 2 & T = 47, T_{\max} = 59, \Delta = -109, \det = \\ -121, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 51.69 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \end{matrix} \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 32 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1179] $21^{21}(12)1223151322^1 1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1169] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1170] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1171] $1(12)12^{3221} 231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[1168] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1173] $21^{21}(12)1223151322^1 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1174] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1175] $21^{21}(12)1223151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1176] $1(12)12231513221(12)12231513221(12)12^{3221} 231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1177] $1(12)12^{3221} 231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[1178] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

- [1167] $1(12)12^{3221}1^{231513221}(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [1180] $1(12)12231513221(12)12231513221(12)12^{2221}1^{231513221}(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1181] $21(12)12231513221(12)12231513221(12)122^{3221}1^{31513221}(12)122 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1182] $21(12)12231513221(12)122^{3221}1^{31513221}(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1183] $21(12)12231513221(12)12231513221(12)12^{2221}1^{231513221}(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1184] $2221(12)1223151322^{11}(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1185] $81231513221(12)122315132^{11}21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -109, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1186] $812315132^{11}21(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -109, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1187] $712321812315132^{11}21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [1188] $712321812315132^{11}21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [1189] $712321812315132^{11}21(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [1165] $41321812321812321812^{11}32181232181232181231513221(12) + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -109, \det =$

$-169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

$[1166] \overset{3221}{1}(12)12231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det =$
 $-169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

$[1172] \overset{3221}{1}(12)12231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det =$
 $-169, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.59 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

$[1197] \overset{21}{21}(12)12231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det =$
 $-196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[1194] \overset{3221}{1}(12)12231513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det =$
 $-196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.55 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[1195] \overset{21}{21}(12)12231513221(12)12^1 231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det =$
 $-196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[1196] \overset{221}{1}(12)12231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det =$
 $-196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[1192] \overset{221}{1}(12)12231513221(12)12^1 231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det =$
 $-196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[1198] \overset{3221}{1}(12)12231513221(12)122315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det =$
 $-196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.55 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[1199] \overset{221}{1}(12)12231513221(12)12^1 231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det =$
 $-196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[1200] \overset{21}{21}(12)12231513221(12)12^1 231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det =$
 $-196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[1201] \overset{2221}{1}(12)12231513221(12)1223151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -103, \det =$
 $-196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1202] & \begin{matrix} 3221 \\ 21(12)1223151322^1 1(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.55 \rightarrow 52, T_{\max} = 61, T \in [52, 61] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1203] & \begin{matrix} 221 \\ 21(12)12231513221(12)1223151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1204] & \begin{matrix} 1 \\ 2221(12)12231513221(12)12^1 231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -103, \det = -196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1205] & \begin{matrix} 1 \\ 81231513221(12)12231513221(12)12^1 231513221(12)1223151315 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -109, \det = -196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1206] & \begin{matrix} 2221 \\ 71232181231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1207] & \begin{matrix} 221 \\ 71232181231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1208] & \begin{matrix} 1 \\ 71232181231513221(12)12^1 231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1193] & \begin{matrix} 21 \\ 1(12)12231513221(12)12231513221(12)12^1 231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.55 \rightarrow 52, T_{\max} = 61, T \in [52, 61] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1191] & \begin{matrix} 3221 \\ 1(12)12231513221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.55 \rightarrow 52, T_{\max} = 61, T \in [52, 61] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1190] & \begin{matrix} 221 \\ 4132181232181232181232^1 181232181232181231513221(12) + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 59, \Delta = -109, \det = -196, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.55 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 41 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1210] & \begin{matrix} 3221 \\ 21(12)12231513221(12)122315132^1 21(12)12231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -225, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.52 \rightarrow 52, T_{\max} = 61, T \in [52, 61]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[1211] & \text{81231513221}(12)122^1 131513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -109, \det = -225, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.52 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[1212] & \text{712321812}^1 131513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -225, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.52 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[1213] & \text{712321812}^1 131513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -225, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.52 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[1214] & \text{712321812}^1 131513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -225, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.52 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[1209] & \text{1}(12)12231513221(12)122^1 131513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -225, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.52 \rightarrow 52, T_{\max} = 61, T \in [52, 61] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[1215] & \text{41321812321812321812321812}^1 32181232181231513221(12) + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -109, \det = -256, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.48 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 47 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1216] & \text{1}(12)12231513221(12)1223151322^1 1(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -256, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.48 \rightarrow 52, T_{\max} = 61, T \in [52, 61] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1217] & \text{21}(12)12231513221(12)12^1 231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -256, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.48 \rightarrow 52, T_{\max} = 61, T \in [52, 61] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1218] & \text{81231513221}(12)1223151322^1 1(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -109, \det = -256, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.48 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1219] & \text{7123218123151322}^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -256, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.48 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[1220] & \text{7123218123151322}^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -104, \det = -256, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.48 \rightarrow 52, T_{\max} = 59, T \in [52, 59] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1221] $7123218123151322^1 1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -104, \det = -256, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.48 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [1231] $8123151322^1 1(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -109, \det = -289, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.46 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [1230] $21(12)12231513221(12)12231513221(12)12^1 231513221(12)122 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -289, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.46 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [1227] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -289, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.46 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1228] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -289, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.46 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [1229] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -289, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.46 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1225] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -289, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.46 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [1224] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -289, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.46 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1226] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -289, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.46 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [1222] $4132181232181232181232181232^1 181232181231513221(12) + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 59, \Delta = -109, \det = -289, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.46 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 50 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [1223] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det = -289, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.46 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [1232] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 47, T_{\max} = 61, \Delta = -112, \det =$

$-324, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.43 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

$[1233] \overset{3221}{1}(12)12231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det =$
 $-324, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.43 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

$[1234] \overset{3221}{1}(12)12231513221(12)122^1 31513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det =$
 $-324, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.43 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

$[1235] 41321812321812321812321812321812^1 32181231513221(12) + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -109, \det =$
 $-361, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.41 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 56 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

$[1236] \overset{21}{1}(12)12231513221(12)12^1 231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det =$
 $-361, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.41 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

$[1237] \overset{3221}{21}(12)12231513221(12)1223151322^1 1(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det =$
 $-361, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.41 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

$[1238] \overset{1}{81}231513221(12)12^1 231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -109, \det =$
 $-361, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.41 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

$[1239] 4132181232181232181232181232181232^1 181231513221(12) + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -109, \det =$
 $-400, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.39 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 20 & 2 \\ 20 & 59 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

$[1240] 41321812321812321812321812321812321812^1 31513221(12) + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 59, \Delta = -109, \det =$
 $-484, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.36 \rightarrow 52, T_{\max} = 59, T \in [52, 59]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 65 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

$[1241] \overset{3221}{1}(12)1223151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det =$
 $-484, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.36 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

$[1242] \overset{3221}{1}(12)1223151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det =$
 $-484, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.36 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

$[1243] \overset{3221}{1}(12)1223151322^1 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 47, T_{\max} = 61, \Delta = -112, \det =$
 $-484, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 51.36 \rightarrow 52, T_{\max} = 61, T \in [52, 61]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[1244] & \begin{matrix} 3221 & 1 & 221 \\ 1(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -529, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.34 \rightarrow 52, T_{\max} = 61, T \in [52, 61] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1246] & \begin{matrix} 3221 & 1 & 221 \\ 1(12)12231513221(12)12^1 231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -529, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.34 \rightarrow 52, T_{\max} = 61, T \in [52, 61] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1245] & \begin{matrix} 3221 & 1 & 21 \\ 1(12)12231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 47, T_{\max} = 61, \Delta = -112, \det = -529, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 51.34 \rightarrow 52, T_{\max} = 61, T \in [52, 61] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
T = 48 & [1248] \begin{matrix} 22^1 1 \\ 21(12)12231513221(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55, T_{\max} = 60, T \in [55, 60] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1249] & \begin{matrix} 3221 \\ 81232181231513221(12)12231513221(12)12231513221(12)12^1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 62, \Delta = -112, \det = 4, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54, T_{\max} = 62, T \in [54, 62] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[1250] & \begin{matrix} 1 & 22^1 1 \\ 413221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55, T_{\max} = 61, T \in [55, 61] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1251] & \begin{matrix} 1 & 22^1 1 \\ 181232181231513221(12)12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55, T_{\max} = 60, T \in [55, 60] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[1252] & \begin{matrix} 1 & 1 & 22^1 1 \\ 413221(12)12231513221(12)12231513221(12)12231513221(12)12^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55, T_{\max} = 61, T \in [55, 61] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1253] & \begin{matrix} 1 & 1 & 22^1 1 \\ 413221(12)12231513221(12)12231513221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55, T_{\max} = 61, T \in [55, 61] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[1247] & \begin{matrix} 22^1 21 \\ 1(12)12231513221(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55, T_{\max} = 60, T \in [55, 60] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 9 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[1254] & \begin{matrix} 222^1 1 \\ 1(12)12231513221(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 9, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.22 \rightarrow 55, T_{\max} = 60, T \in [55, 60]
\end{aligned}$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 20 & 8 \\ 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[1255] $81232181231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 62, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.22 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[1256] $181232181231513221(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 9, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.22 \rightarrow 55, T_{\max} = 60, T \in [55, 60]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[1257] $413221(12)12231513221(12)12231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.22 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[1258] $(12)122315132181232181232181232181232181232181232^1 15 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 16, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.75 \rightarrow 54, T_{\max} = 60, T \in [54, 60]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[1259] $181232181231513221(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 16, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.75 \rightarrow 54, T_{\max} = 60, T \in [54, 60]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1260] $(12)122315132181232181232181232181232181232181232^1 3215 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.44 \rightarrow 54, T_{\max} = 60, T \in [54, 60]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1264] $81232181231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 62, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.44 \rightarrow 54, T_{\max} = 62, T \in [54, 62]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1265] $4132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.44 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1266] $4132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.44 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1267] $4132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.44 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1268] $181232181231513221(12)12231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.44 \rightarrow 54, T_{\max} = 60, T \in [54, 60]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1269] $4132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det =$

25, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.44 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1270] $4132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det = 25$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.44 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1271] $81231513221^1(12)12231513221(12)12231513221(12)122315132^1 16 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -107$, $\det = 25$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.44 \rightarrow 54$, $T_{\max} = 60$, $T \in [54, 60]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1262] $1(12)122^1 31513221(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det = 25$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.44 \rightarrow 54$, $T_{\max} = 60$, $T \in [54, 60]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[1263] $21(12)122^1 31513221(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det = 25$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.44 \rightarrow 54$, $T_{\max} = 60$, $T \in [54, 60]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 44 & 12 \\ 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[1261] $(12)1223151321812321812321812321812321812^1 3151 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -104$, $\det = 25$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.44 \rightarrow 54$, $T_{\max} = 60$, $T \in [54, 60]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1274] $21(12)12^1 231513221(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.22 \rightarrow 54$, $T_{\max} = 60$, $T \in [54, 60]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1276] $413221(12)12231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.22 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1277] $413221(12)12231513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.22 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[1278] $181232181231513221(12)12231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.22 \rightarrow 54$, $T_{\max} = 60$, $T \in [54, 60]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[1279] $1812^1 32181231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.22 \rightarrow 54$, $T_{\max} = 60$, $T \in [54, 60]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[1280] $413221(12)12231513221(12)12231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det = 36$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 53.22 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1272] & \begin{matrix} 2221 \\ 1(12)12231513221(12)12231513221(12)122315132181232^1 181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 36, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.22 \rightarrow 54, T_{\max} = 60, T \in [54, 60] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1273] & \begin{matrix} 2221 \\ 1(12)12^1 231513221(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 36, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.22 \rightarrow 54, T_{\max} = 60, T \in [54, 60] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 65 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1275] & \begin{matrix} 221 \\ 21(12)12231513221(12)12231513221(12)122315132181232^1 181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 36, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.22 \rightarrow 54, T_{\max} = 60, T \in [54, 60] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1297] & \begin{matrix} 1 \\ 812^1 31513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 60, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 60, T \in [54, 60] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1285] & \begin{matrix} 3221 \\ 812^1 32181231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 62, \Delta = -112, \det = 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 62, T \in [54, 62] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1286] & \begin{matrix} 1 & 1 & 21 \\ 413221(12)12231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1287] & \begin{matrix} 1 & 1 & 21 \\ 41322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1288] & \begin{matrix} 1 & 221 \\ 413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1289] & \begin{matrix} 1 & 221 \\ 41322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1290] & \begin{matrix} 1 & 21 \\ 413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1291] & \begin{matrix} 1 & 21 \\ 41322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 61, T \in [54, 61]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1292] & 181232181231513221(12)12231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -103, \det = \\
& 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 60, T \in [54, 60] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[1293] & 41322^1 1(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1294] & 413221(12)12231513221(12)12231513221(12)122315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1295] & 413221(12)12231513221(12)12231513221(12)122315132^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1296] & 41322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1284] & 21^{221}(12)122315132^1 21(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -103, \det = \\
& 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 60, T \in [54, 60] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[1298] & 81231513221(12)12231513221(12)12231513221(12)122^1 31513161 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -106, \det = \\
& 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 60, T \in [54, 60] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1299] & 812^1 31513221(12)12231513221(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -107, \det = \\
& 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 60, T \in [54, 60] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1300] & 81231513221(12)12231513221(12)12231513221(12)122^1 31513216 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -107, \det = \\
& 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 60, T \in [54, 60] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1281] & (12)122315132181232181232181232181232^1 18123215 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -106, \det = \\
& 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 60, T \in [54, 60] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1282] & (12)122315132181232181232181232181232^1 18123151 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -104, \det = \\
& 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.06 \rightarrow 54, T_{\max} = 60, T \in [54, 60] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1283] $1(12)122315132^1 21(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.06 \rightarrow 54, T_{\max} = 60, T \in [54, 60]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1303] $413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 64, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.94 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [1304] $413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)122 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 64, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.94 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [1301] $(12)1223151321812321812321812321812^1 3218123215 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 64, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.94 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1302] $(12)1223151321812321812321812321812^1 3218123151 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -104, \det = 64, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.94 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1305] $1(12)12231513221(12)12231513221(12)1223151321812^1 32181 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [1309] $21(12)12231513221(12)122^1 31513221(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1310] $21(12)12231513221(12)12231513221(12)1223151321812^1 32181 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [1311] $81232181231513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 62, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.84 \rightarrow 53, T_{\max} = 62, T \in [53, 62]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1312] $413221(12)122^1 31513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.84 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1313] $181232181231513221(12)12231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1314] $181232181231513221(12)122315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1315] & 181232^1 181231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 143 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[1316] & 413221^1(12)122^1 31513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.84 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1317] & 81231513221^1(12)12231513221(12)12231513221(12)12^1 231513161 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1318] & 81231513221^1(12)12231513221(12)122315132^1 21(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1319] & 81231513221^1(12)12231513221(12)122315132^1 21(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -107, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1320] & 81231513221^1(12)12231513221(12)12231513221(12)12^1 231513216 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -107, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1307] & 1^{2221}(12)1223151322^1 1(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1306] & 1^{2221}(12)12231513221(12)122^1 31513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1308] & 21^{221}(12)1223151322^1 1(12)12231513221(12)122315132181232181 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.84 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1321] & (12)122315132181232181232181232181232^1 18123218123215 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -106, \det = 100, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.76 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1322] & (12)122^1 31513218123218123218123218123218123215 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -106, \det = 100, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.76 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1323] (12)122¹31513218123218123218123218123218123218123151 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -104$, $\det = 100$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.76 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1324] (12)12231513218123218123218123218123218123218123218123151 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -104$, $\det = 100$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.76 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1325] ¹413221(12)12231513221(12)12231513221(12)122¹31513221(12)1 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det = 100$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.76 \rightarrow 53$, $T_{\max} = 61$, $T \in [53, 61]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1326] ¹413221(12)12231513221(12)12231513221(12)122¹31513221(12)12 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det = 100$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.76 \rightarrow 53$, $T_{\max} = 61$, $T \in [53, 61]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1327] ¹413221(12)12231513221(12)12231513221(12)122¹31513221(12)1222 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det = 100$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.76 \rightarrow 53$, $T_{\max} = 61$, $T \in [53, 61]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1348] ¹81231513221(12)12231513221(12)122¹31513221(12)12231513161 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.69 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [1332] ²²²¹1(12)12231513221(12)122315132¹21(12)122315132181232181 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.69 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [1333] ²²²¹1(12)12231513221(12)12¹231513221(12)122315132181232181 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.69 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1334] ²²¹21(12)12231513221(12)122315132¹21(12)122315132181232181 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.69 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [1335] ²²¹21(12)12231513221(12)12¹231513221(12)122315132181232181 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.69 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1336] ³²²¹81232¹181231513221(12)12231513221(12)12231513221(12)12 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 62$, $\Delta = -112$, $\det = 121$, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 52.69 \rightarrow 53$, $T_{\max} = 62$, $T \in [53, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [1337] ³²²¹81232181231513221(12)12231513221(12)122¹31513221(12)12 + $\hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 62$, $\Delta = -112$, $\det =$

$$121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 62, T \in [53, 62]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1338] \quad 81232181231513221(12)12231513221(12)12231513221^{3221}1(12)12 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 62, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 62, T \in [53, 62]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1339] \quad 413221(12)122315132^{121}(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1340] \quad 413221(12)122^{121}231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1341] \quad 413221(12)122^{131513221}(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1342] \quad 413221(12)122^{131513221}(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1343] \quad 181232181231513221(12)12231513221^{121}(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[1344] \quad 181232181231513221(12)122^{131513221}(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1345] \quad 413221(12)12^{121}231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1346] \quad 413221(12)122315132^{121}(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1347] \quad 413221(12)122^{131513221}(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1331] \quad (12)1223151321812321812321812321812^{13218123218123151} + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1349] $81231513221(12)12231513221(12)1223151322^11(12)12231513161 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[1350] $81231513221(12)12231513221(12)122^131513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1351] $81231513221(12)12231513221(12)1223151322^11(12)12231513216 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1328] $(12)1223151321812321812321812321812^13218123218123215 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1330] $(12)12^1231513218123218123218123218123218123218123151 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1329] $(12)12^1231513218123218123218123218123218123218123215 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.69 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1354] $1(12)12231513221(12)12231513221(12)122^1315132181232181 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.59 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1356] $81232181231513221(12)122315132^121(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 62, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.59 \rightarrow 53, T_{\max} = 62, T \in [53, 62]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1357] $413221(12)12231513221(12)12231513221(12)122^131513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.59 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1358] $413221(12)12231513221(12)122^131513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.59 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1359] $413221(12)12231513221(12)12231513221(12)12^1231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.59 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

- [1360] $413221(12)12^1 231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.59 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [1361] $413221(12)12231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.59 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1362] $413221(12)12^1 231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.59 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [1363] $1812321812315132^1 21(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.59 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [1364] $413221(12)12231513221(12)122^1 31513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.59 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1365] $413221(12)12231513221(12)12231513221(12)122^1 31513221(12)122 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.59 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1366] $413221(12)12231513221(12)12231513221(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.59 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1367] $413221(12)12^1 231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.59 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [1368] $812315132^1 21(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.59 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [1369] $81231513221(12)122315132^1 21(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.59 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1370] $81231513221(12)122315132^1 21(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -107, \det = 169, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.59 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1371] $812315132^1 21(12)12231513221(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -107, \det =$

169, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.59 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1352] $(12)122315132181232181232181232^1 18123218123218123215 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -106$, $\det =$

169, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.59 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1353] $(12)122315132181232181232181232^1 18123218123218123151 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -104$, $\det =$

169, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.59 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1355] $21^{221}(12)12231513221(12)12231513221(12)122^1 315132181232181 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det =$

169, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.59 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1382] $413221(12)12231513221(12)122315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det =$

196, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.55 \rightarrow 53$, $T_{\max} = 61$, $T \in [53, 61]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1376] $81232181231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 62$, $\Delta = -112$, $\det =$

196, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.55 \rightarrow 53$, $T_{\max} = 62$, $T \in [53, 62]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1377] $413221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det =$

196, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.55 \rightarrow 53$, $T_{\max} = 61$, $T \in [53, 61]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1378] $413221(12)12231513221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det =$

196, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.55 \rightarrow 53$, $T_{\max} = 61$, $T \in [53, 61]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1379] $413221(12)12231513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det =$

196, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.55 \rightarrow 53$, $T_{\max} = 61$, $T \in [53, 61]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1380] $181232181231513221(12)12^1 231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det =$

196, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.55 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[1381] $413221(12)1223151322^1 1(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 61$, $\Delta = -111$, $\det =$

196, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.55 \rightarrow 53$, $T_{\max} = 61$, $T \in [53, 61]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1375] $21^{221}(12)12231513221(12)1223151322^1 1(12)122315132181232181 + \hat{1} \rightarrow 2$ $T = 48$, $T_{\max} = 60$, $\Delta = -103$, $\det =$

196, $(n_+, n_-, n_0) = (1, 48, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 52.55 \rightarrow 53$, $T_{\max} = 60$, $T \in [53, 60]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[1383] & \begin{matrix} 1 \\ 81231513221(12)12231513221(12)12^1 231513221(12)12231513161 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 60, \Delta = -106, \det = 196, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.55 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1384] & \begin{matrix} 1 \\ 81231513221(12)12231513221(12)12^1 231513221(12)12231513216 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 60, \Delta = -107, \det = 196, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.55 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1374] & \begin{matrix} 2221 \\ 1(12)12231513221(12)1223151322^1 1(12)122315132181232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 196, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.55 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[1372] & (12)1223151321812321812321812^1 3218123218123218123215 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -106, \det = 196, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.55 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[1373] & (12)1223151321812321812321812^1 3218123218123218123151 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -104, \det = 196, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.55 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[1385] & \begin{matrix} 2221 \\ 1(12)12231513221(12)12231513221(12)122315132^1 181232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.52 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[1386] & \begin{matrix} 221 \\ 21(12)12231513221(12)12231513221(12)122315132^1 181232181 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.52 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[1387] & \begin{matrix} 3221 \\ 81232181231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 62, \Delta = -112, \det = 225, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.52 \rightarrow 53, T_{\max} = 62, T \in [53, 62] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[1388] & \begin{matrix} 1 & & 21 \\ 413221(12)12231513221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 225, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.52 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[1389] & 1812321812^1 31513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.52 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[1390] & \begin{matrix} 1 & & 1 \\ 413221(12)12231513221(12)122315132^1 21(12)12231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 225, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.52 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1391] $81231513221(12)122^1 31513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 225, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.52 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1392] $81231513221(12)122^1 31513221(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -107, \det = 225, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.52 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [1393] $(12)122315132181232181232^1 18123218123218123218123215 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 256, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.48 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [1397] $81232181231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 62, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.48 \rightarrow 53, T_{\max} = 62, T \in [53, 62]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1398] $413221(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 256, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.48 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1399] $18123218123151322^1 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.48 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [1400] $413221(12)12231513221(12)12^1 231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 256, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.48 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1401] $81231513221(12)1223151322^1 1(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 256, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.48 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1402] $81231513221(12)1223151322^1 1(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -107, \det = 256, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.48 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1394] $(12)122315132181232181232^1 18123218123218123218123151 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -104, \det = 256, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.48 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [1396] $21(12)12231513221(12)12231513221(12)12^1 2315132181232181 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.48 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [1395] $1(12)12231513221(12)12231513221(12)12^1 2315132181232181 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 52.48 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1414] & 8123151322^1 1(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -106, \det = \\
& 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.46 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1407] & 413221(12)122315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.46 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1408] & 413221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.46 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[1409] & 413221(12)122315132^1 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.46 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1410] & 413221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.46 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[1411] & 413221(12)12231513221(12)12231513221(12)12^1 231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.46 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[1412] & 413221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.46 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[1413] & 413221(12)122315132^1 21(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.46 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1406] & 413221(12)12231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = \\
& 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.46 \rightarrow 53, T_{\max} = 61, T \in [53, 61] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[1415] & 8123151322^1 1(12)12231513221(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -107, \det = \\
& 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.46 \rightarrow 53, T_{\max} = 60, T \in [53, 60] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[1403] & (12)1223151321812321812^1 3218123218123218123218123215 + \hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 60, \Delta = -106, \det = \\
& 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 52.46 \rightarrow 53, T_{\max} = 60, T \in [53, 60]
\end{aligned}$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[1405] $812321812315132^1 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 62, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.46 \rightarrow 53, T_{\max} = 62, T \in [53, 62]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1404] $(12)1223151321812321812^1 3218123218123218123218123151 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -104, \det = 289, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.46 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 237 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[1418] $413221(12)12231513221(12)122^1 31513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 324, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.43 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1417] $413221(12)12231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 324, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.43 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1416] $413221(12)12231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 324, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.43 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1420] $(12)122315132181232^1 18123218123218123218123218123151 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.41 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[1421] $81232181231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 62, \Delta = -112, \det = 361, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.41 \rightarrow 53, T_{\max} = 62, T \in [53, 62]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[1422] $812321812^1 31513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 62, \Delta = -112, \det = 361, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.41 \rightarrow 53, T_{\max} = 62, T \in [53, 62]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[1423] $413221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 361, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.41 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[1424] $413221(12)12231513221(12)1223151322^1 1(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 361, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.41 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[1425] $81231513221(12)12^1 231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 361, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 52.41 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1426] $81231513221(12)12^1 231513221(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -107, \det = 361, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.41 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

[1419] $(12)122315132181232^1 18123218123218123218123218123215 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 361, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.41 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[1427] $(12)1223151321812^1 3218123218123218123218123218123215 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 400, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.39 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 20 & 2 \\ 20 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[1428] $(12)1223151321812^1 3218123218123218123218123218123151 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -104, \det = 400, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.39 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 20 & 2 \\ 20 & 339 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[1429] $8123218123151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 62, \Delta = -112, \det = 441, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.37 \rightarrow 53, T_{\max} = 62, T \in [53, 62]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[1430] $(12)122315132^1 18123218123218123218123218123218123215 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -106, \det = 484, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.36 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$

[1431] $(12)122315132^1 18123218123218123218123218123218123151 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 60, \Delta = -104, \det = 484, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.36 \rightarrow 53, T_{\max} = 60, T \in [53, 60]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$

[1432] $413221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 484, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.36 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[1433] $413221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 484, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.36 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[1434] $413221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 484, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.36 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[1435] $413221(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 529, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.34 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[1436] $413221(12)12231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 48, T_{\max} = 61, \Delta = -111, \det = 529, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 52.34 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[1437] \quad 413221(12)12231513221(12)12^{11}231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 48, T_{\max} = 61, \Delta = -111, \det = 529, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 52.34 \rightarrow 53, T_{\max} = 61, T \in [53, 61]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$T = 49 \quad [1438] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56, T_{\max} = 61, T \in [56, 61]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1439] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12^{11}2 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56, T_{\max} = 61, T \in [56, 61]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1440] \quad (12)12231513221(12)12231513221(12)12231513221(12)12^{21}31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56, T_{\max} = 66, T \in [56, 66]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[1441] \quad 22^{11}31513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56, T_{\max} = 66, T \in [56, 66]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[1442] \quad 31513221(12)12231513221(12)12231513221(12)12231513221(12)12^{11}2 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56, T_{\max} = 65, T \in [56, 65]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1443] \quad 1513221(12)12231513221(12)12231513221(12)12231513221(12)122^{11}2 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56, T_{\max} = 61, T \in [56, 61]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1444] \quad 41413221(12)12231513221(12)12231513221(12)12231513221(12)12^{11}2 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56, T_{\max} = 61, T \in [56, 61]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1447] \quad 1513221(12)12231513221(12)12231513221(12)12231513221(12)12^{11}22 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -9, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 55.22 \rightarrow 56, T_{\max} = 61, T \in [56, 61]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[1445] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 55.22 \rightarrow 56, T_{\max} = 66, T \in [56, 66]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1446] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122^{11}3 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -9, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 55.22 \rightarrow 56, T_{\max} = 66, T \in [56, 66]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[1448] $(12)122315132181232181232181232181232181232181232^1 161 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -103$, $\det = -16$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.75 \rightarrow 55$, $T_{\max} = 61$, $T \in [55, 61]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1458] $81231513221(12)12231513221(12)12231513221(12)122315132^1 217 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.44 \rightarrow 55$, $T_{\max} = 61$, $T \in [55, 61]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1459] $4132^1 21(12)12231513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 62$, $\Delta = -111$, $\det = -25$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.44 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[1449] $(12)1223151321812321812321812321812321812321812321812^1 32161 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -103$, $\det = -25$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.44 \rightarrow 55$, $T_{\max} = 61$, $T \in [55, 61]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1450] $1(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.44 \rightarrow 55$, $T_{\max} = 61$, $T \in [55, 61]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1451] $1(12)12231513221(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.44 \rightarrow 55$, $T_{\max} = 66$, $T \in [55, 66]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[1452] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1 23 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -25$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.44 \rightarrow 55$, $T_{\max} = 66$, $T \in [55, 66]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1453] $1(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.44 \rightarrow 55$, $T_{\max} = 61$, $T \in [55, 61]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1454] $1(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.44 \rightarrow 55$, $T_{\max} = 61$, $T \in [55, 61]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1455] $21(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.44 \rightarrow 55$, $T_{\max} = 61$, $T \in [55, 61]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1456] $15132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.44 \rightarrow 55$, $T_{\max} = 61$, $T \in [55, 61]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

- [1457] $81231513221(12)12231513221(12)12231513221(12)122315132^1 171 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -105$, $\det = -25$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.44 \rightarrow 55$, $T_{\max} = 61$, $T \in [55, 61]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1462] $1513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.22 \rightarrow 55$, $T_{\max} = 61$, $T \in [55, 61]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1463] $81231513221(12)12231513221(12)12231513221(12)1223151322^1 17 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -105$, $\det = -36$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.22 \rightarrow 55$, $T_{\max} = 61$, $T \in [55, 61]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [1460] $1(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.22 \rightarrow 55$, $T_{\max} = 61$, $T \in [55, 61]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1461] $21(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -36$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.22 \rightarrow 55$, $T_{\max} = 61$, $T \in [55, 61]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [1487] $41413221(12)12231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55$, $T_{\max} = 61$, $T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1486] $41413221(12)12231513221(12)12231513221(12)122315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55$, $T_{\max} = 61$, $T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1464] $(12)122315132181232181232181232181232181232^1 181232161 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -103$, $\det = -49$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55$, $T_{\max} = 61$, $T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$
- [1465] $1(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55$, $T_{\max} = 61$, $T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [1466] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -49$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55$, $T_{\max} = 61$, $T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [1467] $1(12)12231513221(12)12231513221(12)12231513221(12)122^1 315 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -49$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55$, $T_{\max} = 66$, $T \in [55, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 13 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [1468] $1(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det =$

$-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 66, T \in [55, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

$[1469] \overset{21}{1}(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det =$
 $-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 65, T \in [55, 65]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

$[1470] \overset{221}{1}(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det =$
 $-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

$[1471] \overset{221}{1}(12)12^1 231513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det =$
 $-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

$[1472] \overset{1513221}{1}(12)12231513221(12)12231513221(12)122315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det =$
 $-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

$[1473] \overset{151322^1 1}{1}(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det =$
 $-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

$[1474] \overset{21}{21}(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det =$
 $-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

$[1475] \overset{21}{21}(12)12^1 231513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det =$
 $-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

$[1476] \overset{1}{31513221}(12)12231513221(12)12231513221(12)122315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det =$
 $-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 65, T \in [55, 65]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

$[1477] \overset{1}{1513221}(12)12231513221(12)12231513221(12)1223151322^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det =$
 $-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

$[1478] \overset{1}{151322^1 1}(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det =$
 $-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$

$[1479] \overset{1}{81231513221}(12)12231513221(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det =$
 $-49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1480] & 812^1 31513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = \\
& -49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1481] & 812^1 31513221(12)12231513221(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = \\
& -49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[1482] & 81231513221(12)12231513221(12)12231513221(12)122^1 315132217 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = \\
& -49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1483] & 41322^1 1(12)12231513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 62, \Delta = -111, \det = \\
& -49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.06 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1484] & 413221(12)12231513221(12)12231513221(12)122315132181232^1 18 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 62, \Delta = -111, \det = \\
& -49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.06 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1485] & 414132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = \\
& -49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1488] & 414132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = \\
& -49, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.06 \rightarrow 55, T_{\max} = 61, T \in [55, 61] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[1502] & 41413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = \\
& -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1503] & 41413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = \\
& -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1489] & (12)1223151321812321812321812321812^1 32181232161 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -103, \det = \\
& -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1490] & 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = \\
& -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 65, T \in [54, 65]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1491] & 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1492] & 1(12)12^1 231513221(12)12231513221(12)12231513221(12)1223151+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1493] & 1(12)122^1 31513221(12)12231513221(12)12231513221(12)122315132+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1494] & 1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1495] & 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1496] & 1 \begin{pmatrix} 31513221 \\ (12) \end{pmatrix} 12231513221(12)12231513221(12)122315132^1 21(12)12+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1497] & 1 \begin{pmatrix} 315132^1 21 \\ (12) \end{pmatrix} 12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1498] & 1 \begin{pmatrix} 1513221 \\ (12) \end{pmatrix} 12231513221(12)12231513221(12)1223151322^1 1(12)122+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1499] & 231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1500] & 315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1501] & 31513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)122+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -64, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.94 \rightarrow 54, T_{\max} = 65, T \in [54, 65]
\end{aligned}$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1519] \quad 41413221(12)122^1 31513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1507] \quad 1^1 (12) \quad 12231513221(12)12231513221(12)122315132^1 21(12)1+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1504] \quad 1^1 (12)12^1 231513221(12)12231513221(12)12231513221(12)12231513+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1506] \quad 1^1 (12)12^1 231513221(12)12231513221(12)12231513221(12)122315132+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1505] \quad 1^1 (12)12231513221(12)12231513221(12)122315132^1 21(12)1223151+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1520] \quad 4141322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1508] \quad 1^1 (12) \quad 12231513221(12)12231513221(12)1223151322^1 1(12)12+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1509] \quad 1^1 (12) \quad 122^1 31513221(12)12231513221(12)12231513221(12)122+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1510] \quad 2231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1511] \quad 231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1512] \quad 81231513221(12)12231513221(12)122315132^1 21(12)122315132171+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1513] \quad 81231513221(12)12231513221(12)12231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1514] \quad 81231513221(12)12231513221(12)12231513221(12)12^1 2315132217 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1515] \quad 81231513221(12)12231513221(12)122315132^1 21(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1516] \quad 413221(12)122^1 31513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 62, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 62, T \in [54, 62]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1517] \quad 41413221(12)122^1 31513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1518] \quad 4141322^1 1(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.84 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1529] \quad 2231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -100, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.76 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1530] \quad 1513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -100, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.76 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1521] \quad (12)122315132181232181232181232181232^1 181232181232161 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -103, \det = -100, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.76 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[1522] \quad (12)122^1 315132181232181232181232181232181232181232161 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -103, \det = -100, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.76 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 189 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1523] \quad 1^{231513221} (12) 12231513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -100, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.76 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1524] $1(12)12231513221(12)12231513221(12)12231513221(12)12^12315 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -100, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.76 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1525] $1(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -100, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.76 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1526] $1(12)122315132^121(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -100, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.76 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1527] $21(12)122315132^121(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -100, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.76 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1528] $(12)122^131513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -100, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.76 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$

[1551] $81231513221(12)12231513221(12)1223151322^11(12)122315132217 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1557] $41413221(12)122315132^121(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1531] $(12)12^12315132181232181232181232181232181232161 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -103, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 230 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$

[1532] $(12)1223151321812321812321812321812^132181232181232161 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -103, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[1533] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1231513 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.69 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1534] $1(12)12231513221(12)12231513221(12)122^131513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1535] $1(12)12231513221(12)12231513221(12)1223151322^11(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det =$

$$-121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1536] \overset{1}{1}(12)12231513221(12)12231513221(12)12231513221(12)122^1315132+\hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1537] \overset{2315132^1 21}{1} (12) \overset{1}{12231513221}(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1538] \overset{3221}{1}(12)122^131513221(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1539] \overset{21}{1}(12)12231513221(12)12231513221(12)12231513221(12)12^1231513 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1540] \overset{3151322^1 1}{1} (12) \overset{1}{12231513221}(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1541] \overset{221}{1}(12)12231513221(12)12231513221(12)122315132^121(12)1223151 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1542] \overset{1513221}{1} (12) \overset{1}{12}^1231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1543] \overset{1513221}{1} (12) \overset{1}{122315132^1 21}(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1544] \overset{21}{21}(12)12231513221(12)12231513221(12)122315132^121(12)1223151 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1545] (12)12^1231513221(12)12231513221(12)12231513221 \overset{2231513221}{(12)} 1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1546] \overset{1}{2315132^1 21}(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1547] \quad 3151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)122+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1548] \quad 1513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1222+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1549] \quad 81231513221(12)12231513221(12)122^1 31513221(12)122315132171+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1550] \quad 81231513221(12)12231513221(12)1223151322^1 1(12)122315132171+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1559] \quad 41413221(12)122315132^1 21(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1552] \quad 81231513221(12)12231513221(12)122^1 31513221(12)122315132217+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1558] \quad 41413221(12)12^1 231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1554] \quad 413221(12)122315132^1 21(12)12231513221(12)12231513218123218+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 62, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 62, T \in [54, 62]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1555] \quad 413221(12)12^1 231513221(12)12231513221(12)12231513218123218+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 62, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 62, T \in [54, 62]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1556] \quad 41413221(12)12^1 231513221(12)12231513221(12)12231513221(12)122+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1553] \quad 413221(12)12231513221(12)12231513221(12)1223151321812^1 3218+ \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 62, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.69 \rightarrow 54, T_{\max} = 62, T \in [54, 62]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 21 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1585] $41413221(12)12231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -110$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1584] $41413221(12)12231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -110$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1560] $(12)122315132181232181232181232^1 181232181232181232161 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -103$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$$

[1561] $1(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.59 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1562] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1563] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1564] $1(12)12^1 231513221(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.59 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1565] $1(12)12231513221(12)12231513221(12)122315132^1 21(12)1223 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.59 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1566] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.59 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1567] $1(12)122^1 31513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.59 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1568] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1569] $1^{221}(12)12231513221(12)12231513221(12)1223151322^11(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[1570] $1^{1513221}(12)12231513221(12)122^131513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1571] $1^{1513221}(12)12231513221(12)12231513221(12)122^131513221(12)122 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1572] $21^{21}(12)1223151322^11(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1573] $21^{21}(12)12231513221(12)12231513221(12)1223151322^11(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[1574] $31513221(12)12231513221(12)12231513221(12)122^131513221(12)122 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -169, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1575] $1513221(12)12231513221(12)12231513221(12)12^1231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[1576] $1513221(12)12^1231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$

[1577] $81231513221(12)122315132^121(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1578] $812315132^121(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

[1579] $81231513221(12)122315132^121(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

- [1580] $812315132^1 21(12)12231513221(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -105$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1581] $413221(12)12231513221(12)122^1 31513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 62$, $\Delta = -111$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54$, $T_{\max} = 62$, $T \in [54, 62]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1582] $41413221(12)12231513221(12)12231513221(12)122^1 31513221(12)122 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -110$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1583] $41413221(12)12231513221(12)122^1 31513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -110$, $\det = -169$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.59 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1586] $(12)1223151321812321812321812^1 32181232181232181232161 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -103$, $\det = -196$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.55 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$
- [1598] $41413221(12)1223151322^1 1(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -110$, $\det = -196$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.55 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1599] $41413221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -110$, $\det = -196$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.55 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1587] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -196$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.55 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1588] $1(12)122^1 31513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -196$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.55 \rightarrow 54$, $T_{\max} = 66$, $T \in [54, 66]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1589] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -196$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.55 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1590] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -196$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.55 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [1591] $21(12)12231513221(12)122^1 31513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det =$

$$-196, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.55 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1592] \quad (12)12231513221(12)12231513221(12)12231513221 \overset{1}{(12)} \overset{22315132^1 21}{1} + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -196, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.55 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[1593] \quad 22315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -196, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.55 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 153 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[1594] \quad 1513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -196, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.55 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1595] \quad 81231513221(12)12231513221(12)12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = -196, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.55 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1596] \quad 81231513221(12)12231513221(12)12^1 231513221(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = -196, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.55 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1597] \quad 413221(12)1223151322^1 1(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 62, \Delta = -111, \det = -196, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.55 \rightarrow 54, T_{\max} = 62, T \in [54, 62]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1611] \quad 41413221(12)12231513221(12)122315132^1 21(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = -225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.52 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[1612] \quad 41413221(12)12231513221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = -225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.52 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[1600] \quad 1(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.52 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[1601] \quad 1(12)12231513221(12)122^1 31513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 53.52 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[1602] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12^1 2315132 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$$

$$\begin{aligned}
& T_{\min} = 53.52 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 231513221^1 \\ [1603] \end{matrix} 1 \begin{matrix} (12) \\ (12) \end{matrix} 12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = \\
& -225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.52 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 513221 \\ [1604] \end{matrix} 1 \begin{matrix} (12) \\ (12) \end{matrix} 12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = \\
& -225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.52 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 31513221 \\ [1605] \end{matrix} 1 \begin{matrix} (12) \\ (12) \end{matrix} 122315132^1 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = \\
& -225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.52 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1513221 \\ [1606] \end{matrix} 1 \begin{matrix} (12) \\ (12) \end{matrix} 12231513221(12)122315132^1 21(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = \\
& -225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.52 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1607] \end{matrix} 231513221^1 1(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = \\
& -225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.52 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1608] \end{matrix} 81231513221(12)122^1 31513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = \\
& -225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.52 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1609] \end{matrix} 81231513221(12)122^1 31513221(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = \\
& -225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.52 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1610] \end{matrix} 413221(12)12231513221(12)122315132^1 21(12)12231513218123218 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 62, \Delta = -111, \det = \\
& -225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.52 \rightarrow 54, T_{\max} = 62, T \in [54, 62] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1627] \end{matrix} 41413221(12)12231513221(12)12^1 231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = \\
& -256, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.48 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1 \\ [1628] \end{matrix} 41413221(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = \\
& -256, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.48 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 21 \\ [1613] \end{matrix} (12)122315132181232181232^1 181232181232181232181232161 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -103, \det = \\
& -256, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.48 \rightarrow 54, T_{\max} = 61, T \in [54, 61]
\end{aligned}$$

[1614] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513+ \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.48 \rightarrow 54$, $T_{\max} = 65$, $T \in [54, 65]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1615] $1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513+ \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.48 \rightarrow 54$, $T_{\max} = 65$, $T \in [54, 65]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1616] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)1223151+ \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.48 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1617] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)122315132+ \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.48 \rightarrow 54$, $T_{\max} = 66$, $T \in [54, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1618] $1(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513+ \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.48 \rightarrow 54$, $T_{\max} = 65$, $T \in [54, 65]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1619] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)12+ \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.48 \rightarrow 54$, $T_{\max} = 65$, $T \in [54, 65]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1620] $1(12)12^1 231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.48 \rightarrow 54$, $T_{\max} = 65$, $T \in [54, 65]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1621] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)122+ \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.48 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1622] $231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12+ \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.48 \rightarrow 54$, $T_{\max} = 66$, $T \in [54, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$

[1623] $31513221(12)122^1 31513221(12)12231513221(12)12231513221(12)122+ \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.48 \rightarrow 54$, $T_{\max} = 65$, $T \in [54, 65]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1624] $81231513221(12)1223151322^1 1(12)12231513221(12)122315132171+ \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -105$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.48 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1625] $81231513221(12)12231513221^1(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -105$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.48 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1626] $413221(12)12231513221(12)12^1231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 62$, $\Delta = -111$, $\det = -256$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.48 \rightarrow 54, T_{\max} = 62, T \in [54, 62]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1649] $41413221(12)12231513221(12)12231513221(12)12^1231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -110$, $\det = -289$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1650] $41413221(12)12231513221(12)12231513221(12)12^1231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -110$, $\det = -289$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1629] $(12)1223151321812321812^132181232181232181232181232161 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -103$, $\det = -289$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 526 & 46 \\ 2 & 46 & 3 \end{pmatrix}_{b_0}$$

[1630] $1(12)12231513221(12)122315132^121(12)12231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -289$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.46 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1631] $1(12)1223151322^11(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -289$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1632] $1^{231513221}(12)122^131513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -289$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.46 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1633] $1(12)12231513221(12)12231513221(12)122315132^121(12)122315 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -289$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.46 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 101 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1634] $1(12)122315132^121(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -289$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.46 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 118 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1635] $1(12)12^1231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -289$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 53.46 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[1636] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[1637] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[1638] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1639] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)1223151+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[1640] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)122+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[1641] $21(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1642] $21(12)12231513221(12)12^1 231513221(12)12231513221(12)1223151+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[1643] $31513221(12)12231513221(12)12231513221(12)12^1 231513221(12)122+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[1644] $1513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1222+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$

[1645] $1513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1222+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1646] $8123151322^1 1(12)12231513221(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -105, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

- [1647] $8123151322^1 1(12)12231513221(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -105, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [1648] $413221(12)12231513221(12)12231513221(12)122^1 31513218123218 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 62, \Delta = -111, \det = -289, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.46 \rightarrow 54, T_{\max} = 62, T \in [54, 62]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [1653] $21(12)12231513221(12)122315132^1 21(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -324, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.43 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [1654] $1513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -324, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.43 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [1651] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -324, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.43 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [1652] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -324, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.43 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [1676] $41413221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -110, \det = -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.41 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [1655] $(12)122315132181232^1 181232181232181232181232181232161 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -103, \det = -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.41 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 664 & 52 \\ 2 & 52 & 3 \end{pmatrix}_{b_0}$
- [1656] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.41 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [1657] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.41 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [1658] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 61, \Delta = -106, \det = -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.41 \rightarrow 54, T_{\max} = 61, T \in [54, 61]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [1659] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det =$

$$\begin{aligned}
& T_{\min} = 53.41 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[1671] & 81231513221(12)12^1 231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.41 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[1672] & 81231513221(12)12^1 231513221(12)12231513221(12)122315132217 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -105, \det = -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.41 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[1673] & 413221(12)12231513221(12)12231513221(12)122315132^1 18123218 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 62, \Delta = -111, \det = -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.41 \rightarrow 54, T_{\max} = 62, T \in [54, 62] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 75 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1674] & 413221(12)12231513221(12)1223151322^1 1(12)12231513218123218 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 62, \Delta = -111, \det = -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.41 \rightarrow 54, T_{\max} = 62, T \in [54, 62] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[1675] & 41413221(12)12231513221(12)1223151322^1 1(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -110, \det = -361, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.41 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[1678] & 1(12)12231513221(12)12231513221(12)1223151322^1 1(12)122315 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -400, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.39 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 20 & 2 \\ 20 & 139 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1677] & (12)1223151321812^1 32181232181232181232181232181232161 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -103, \det = -400, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.39 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 20 & 2 \\ 20 & 739 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0} \\
[1680] & 1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -441, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.37 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1681] & 1(12)12231513221(12)12231513221(12)122315132^1 21(12)122315132 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -441, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.37 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1679] & 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -441, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.37 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[1682] & 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)122315132 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -441, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.37 \rightarrow 54, T_{\max} = 66, T \in [54, 66]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[1691] & (12)122315132^1 181232181232181232181232181232181232161 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -103, \det = -484, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.36 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 901 & 61 \\ 2 & 61 & 3 \end{pmatrix}_{b_0} \\
[1695] & \begin{matrix} 21 \\ 12 \end{matrix} 12231513221(12)12231513221(12)12^1 231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -484, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.36 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[1696] & (12)122315132^1 21(12)12231513221(12)12231513221 \begin{matrix} 2231513221 \\ (12) \end{matrix} 1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -484, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.36 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[1697] & 1513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -484, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.36 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[1698] & \begin{matrix} 231513221 \\ 1 \end{matrix} (12) \begin{matrix} 12231513221 \\ (12) \end{matrix} 122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -529, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.34 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0} \\
[1699] & \begin{matrix} 3221 \\ 1 \end{matrix} (12)12231513221(12)122^1 31513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -529, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.34 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 206 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[1700] & \begin{matrix} 513221 \\ 1 \end{matrix} (12) \begin{matrix} 12231513221 \\ (12) \end{matrix} 12231513221(12)122^1 31513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -529, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.34 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[1701] & \begin{matrix} 513221 \\ 1 \end{matrix} (12) \begin{matrix} 1223151322^1 \\ (12) \end{matrix} 1(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 66, \Delta = -115, \det = -529, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.34 \rightarrow 54, T_{\max} = 66, T \in [54, 66] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1702] & \begin{matrix} 21 \\ 1 \end{matrix} (12)12231513221(12)12^1 231513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 65, \Delta = -114, \det = -529, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.34 \rightarrow 54, T_{\max} = 65, T \in [54, 65] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 390 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0} \\
[1703] & \begin{matrix} 221 \\ 1 \end{matrix} (12)12231513221(12)1223151322^1 1(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -529, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.34 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1704] & \begin{matrix} 21 \\ 21 \end{matrix} (12)12231513221(12)1223151322^1 1(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 49, T_{\max} = 61, \Delta = -106, \det = -529, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 53.34 \rightarrow 54, T_{\max} = 61, T \in [54, 61] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1705] $(12)12231513221(12)12231513221(12)122^131513221(12)1 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -529$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.34 \rightarrow 54$, $T_{\max} = 66$, $T \in [54, 66]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$
- [1706] $31513221(12)12231513221(12)1223151322^11(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -529$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.34 \rightarrow 54$, $T_{\max} = 65$, $T \in [54, 65]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 390 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0}$
- [1707] $1513221(12)12231513221(12)12^1231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 61$, $\Delta = -106$, $\det = -529$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.34 \rightarrow 54$, $T_{\max} = 61$, $T \in [54, 61]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [1712] $1(12)12231513221(12)1223151322^11(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -576$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.33 \rightarrow 54$, $T_{\max} = 65$, $T \in [54, 65]$
 $\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [1709] $1(12)12231513221(12)122^131513221(12)12231513221(12)122315132 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -576$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.33 \rightarrow 54$, $T_{\max} = 66$, $T \in [54, 66]$
 $\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [1710] $1(12)1223151322^11(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -576$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.33 \rightarrow 54$, $T_{\max} = 66$, $T \in [54, 66]$
 $\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [1711] $1(12)12231513221(12)12231513221(12)122^131513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -576$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.33 \rightarrow 54$, $T_{\max} = 65$, $T \in [54, 65]$
 $\begin{pmatrix} 0 & 24 & 2 \\ 24 & 407 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [1708] $1(12)12231513221(12)12^1231513221(12)12231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -576$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.33 \rightarrow 54$, $T_{\max} = 65$, $T \in [54, 65]$
 $\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [1713] $231513221(12)12231513221(12)122315132^121(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -576$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.33 \rightarrow 54$, $T_{\max} = 66$, $T \in [54, 66]$
 $\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [1714] $31513221(12)122315132^121(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 65$, $\Delta = -114$, $\det = -576$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.33 \rightarrow 54$, $T_{\max} = 65$, $T \in [54, 65]$
 $\begin{pmatrix} 0 & 24 & 2 \\ 24 & 407 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [1715] $1(12)12231513221(12)12231513221(12)1223151322^11(12)122315132 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -625$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.31 \rightarrow 54$, $T_{\max} = 66$, $T \in [54, 66]$
 $\begin{pmatrix} 0 & 25 & 2 \\ 25 & 474 & 42 \\ 2 & 42 & 3 \end{pmatrix}_{b_0}$

[1716] $1^{231513221}(12)12231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -625, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.31 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$

[1717] $1^{21}(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -625, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.31 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 25 & 2 \\ 25 & 449 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[1718] $(12)12231513221(12)122315132^1 21(12)12231513221^{2231513221}(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -625, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.31 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$

[1719] $2231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -625, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.31 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$

[1720] $231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -625, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.31 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 25 & 2 \\ 25 & 474 & 42 \\ 2 & 42 & 3 \end{pmatrix}_{b_0}$

[1721] $31513221(12)12231513221(12)122^1 31513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -625, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.31 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 25 & 2 \\ 25 & 449 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[1722] $1^{231513221}(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -676, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.30 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$

[1723] $1^{513221}(12)12231513221(12)122315132^1 21(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -676, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.30 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 26 & 2 \\ 26 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$

[1724] $(12)12231513221(12)12231513221(12)12231513221^1 (12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -676, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.30 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$

[1725] $2231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -676, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.30 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$

[1726] $1^{3221}(12)12231513221(12)12231513221(12)122^1 31513221(12)122315 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -729, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.29 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 27 & 2 \\ 27 & 269 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[1727] $1^{513221}(12)12231513221(12)12^1 231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -729, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.29 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 27 & 2 \\ 27 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[1728] $(12)12231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -729, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.29 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 27 & 2 \\ 27 & 620 & 48 \\ 2 & 48 & 3 \end{pmatrix}_{b_0}$

[1729] $1^{3221}(12)12231513221(12)12^1 231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -784, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.28 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 28 & 2 \\ 28 & 307 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$

[1730] $1^{12231513221}(12)12^1 231513221(12)12231513221(12)122315132 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -841, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.27 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 29 & 2 \\ 29 & 666 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$

[1731] $1^{231513221}(12)12231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -841, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.27 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$

[1732] $1^{3221}(12)12231513221(12)122315132^1 21(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -841, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.27 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 29 & 2 \\ 29 & 318 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0}$

[1733] $(12)1223151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -841, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.27 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 29 & 2 \\ 29 & 724 & 52 \\ 2 & 52 & 3 \end{pmatrix}_{b_0}$

[1734] $(12)12231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -841, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.27 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$

[1735] $2231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -841, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.27 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$

[1736] $231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -841, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.27 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 29 & 2 \\ 29 & 666 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$

[1737] $1^{513221}(12)12231513221(12)12231513221(12)12^1 231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -900, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.26 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 30 & 2 \\ 30 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$

[1738] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)122315132+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -961, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.25 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 31 & 2 \\ 31 & 743 & 53 \\ 2 & 53 & 3 \end{pmatrix}_{b_0}$

[1739] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -961, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.25 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 31 & 2 \\ 31 & 681 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$

[1740] $(12)12231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1}\rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -961, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.25 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 31 & 2 \\ 31 & 805 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0}$

[1741] $2231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -961, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.25 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 31 & 2 \\ 31 & 774 & 54 \\ 2 & 54 & 3 \end{pmatrix}_{b_0}$

[1742] $231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -961, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.25 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 31 & 2 \\ 31 & 743 & 53 \\ 2 & 53 & 3 \end{pmatrix}_{b_0}$

[1743] $31513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)122+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -961, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.25 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 31 & 2 \\ 31 & 681 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$

[1745] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -1024, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.25 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 32 & 2 \\ 32 & 735 & 53 \\ 2 & 53 & 3 \end{pmatrix}_{b_0}$

[1744] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)122315132+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1024, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.25 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 32 & 2 \\ 32 & 799 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0}$

[1746] $231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1024, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.25 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 32 & 2 \\ 32 & 799 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0}$

[1747] $31513221(12)12231513221(12)12^1 231513221(12)12231513221(12)122+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 65, \Delta = -114, \det = -1024, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.25 \rightarrow 54, T_{\max} = 65, T \in [54, 65]$
 $\begin{pmatrix} 0 & 32 & 2 \\ 32 & 735 & 53 \\ 2 & 53 & 3 \end{pmatrix}_{b_0}$

[1748] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)1223+\hat{1}\rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1089, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.24 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 33 & 2 \\ 33 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$

[1749] $(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1156, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.23 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 34 & 2 \\ 34 & 985 & 61 \\ 2 & 61 & 3 \end{pmatrix}_{b_0}$

[1750] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.23 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 35 & 2 \\ 35 & 454 & 46 \\ 2 & 46 & 3 \end{pmatrix}_{b_0}$

[1751] $2231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1225, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.23 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 35 & 2 \\ 35 & 1014 & 62 \\ 2 & 62 & 3 \end{pmatrix}_{b_0}$

[1752] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1369, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.21 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 37 & 2 \\ 37 & 517 & 49 \\ 2 & 49 & 3 \end{pmatrix}_{b_0}$

[1753] $2231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1444, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.21 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 38 & 2 \\ 38 & 1177 & 67 \\ 2 & 67 & 3 \end{pmatrix}_{b_0}$

[1754] $2231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1521, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.20 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 39 & 2 \\ 39 & 1247 & 69 \\ 2 & 69 & 3 \end{pmatrix}_{b_0}$

[1755] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)122315132 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1600, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.20 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 40 & 2 \\ 40 & 1239 & 69 \\ 2 & 69 & 3 \end{pmatrix}_{b_0}$

[1756] $231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1600, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.20 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 40 & 2 \\ 40 & 1239 & 69 \\ 2 & 69 & 3 \end{pmatrix}_{b_0}$

[1757] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)122315132 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1681, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.19 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 41 & 2 \\ 41 & 1311 & 71 \\ 2 & 71 & 3 \end{pmatrix}_{b_0}$

[1758] $231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -1681, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.19 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 41 & 2 \\ 41 & 1311 & 71 \\ 2 & 71 & 3 \end{pmatrix}_{b_0}$

[1759] $2231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49, T_{\max} = 66, \Delta = -115, \det = -2401, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 53.16 \rightarrow 54, T_{\max} = 66, T \in [54, 66]$
 $\begin{pmatrix} 0 & 49 & 2 \\ 49 & 1959 & 87 \\ 2 & 87 & 3 \end{pmatrix}_{b_0}$

[1760] $2231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 49$, $T_{\max} = 66$, $\Delta = -115$, $\det = -2500$, $(n_+, n_-, n_0) = (1, 49, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 53.16 \rightarrow 54$, $T_{\max} = 66$, $T \in [54, 66]$
 $\begin{pmatrix} 0 & 50 & 2 \\ 50 & 2049 & 89 \\ 2 & 89 & 3 \end{pmatrix}_{b_0}$

$T = 50$ [1761] $(12)122315132181232181232181232181232181232181232^1 171 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 16$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 55.75 \rightarrow 56$, $T_{\max} = 62$, $T \in [56, 62]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1762] $(12)122315132181232181232181232181232181232181232^1 1712 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 16$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 55.75 \rightarrow 56$, $T_{\max} = 62$, $T \in [56, 62]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1763] $(12)1223151321812321812321812321812321812321812^1 32171 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 25$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 55.44 \rightarrow 56$, $T_{\max} = 62$, $T \in [56, 62]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[1764] $(12)1223151321812321812321812321812321812321812^1 321712 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 25$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 55.44 \rightarrow 56$, $T_{\max} = 62$, $T \in [56, 62]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[1765] $81232181231513221(12)12231513221(12)12231513221^{15132^1 21} (12) 1 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -106$, $\det = 25$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 55.44 \rightarrow 56$, $T_{\max} = 62$, $T \in [56, 62]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1766] $81231513221(12)12231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 25$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 55.44 \rightarrow 56$, $T_{\max} = 62$, $T \in [56, 62]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1767] $4132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 66$, $\Delta = -114$, $\det = 25$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 55.44 \rightarrow 56$, $T_{\max} = 66$, $T \in [56, 66]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$

[1768] $81231513221(12)12231513221(12)12231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 25$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 55.44 \rightarrow 56$, $T_{\max} = 62$, $T \in [56, 62]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1769] $81231513221(12)12231513221(12)12231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 25$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 55.44 \rightarrow 56$, $T_{\max} = 62$, $T \in [56, 62]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1770] $81231513221(12)12231513221(12)12231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 36$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 55.22 \rightarrow 56$, $T_{\max} = 62$, $T \in [56, 62]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

$$\begin{aligned}
& T_{\min} = 55.06 \rightarrow 56, T_{\max} = 62, T \in [56, 62] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1783] & 81231513221(12)12231513221(12)12231513221(12)122^13151322181 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.06 \rightarrow 56, T_{\max} = 62, T \in [56, 62] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1784] & 812^131513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.06 \rightarrow 56, T_{\max} = 62, T \in [56, 62] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[1785] & (12)1223151321812321812321812321812321812^132181232171 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -101, \det = 64, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.94 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1786] & (12)1223151321812321812321812321812321812^1321812321712 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -101, \det = 64, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.94 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1787] & 41413221(12)122^131513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.84 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1788] & 4141322^11(12)12231513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.84 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1789] & 81232181231513221(12)12231513221(12)122315132^121(12)1 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.84 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1790] & 81231513221(12)12231513221(12)12231513221(12)12^12315132181 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.84 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1791] & 81231513221(12)12231513221(12)122315132^121(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.84 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1792] & 81231513221(12)12231513221(12)12231513221(12)12^123151321812 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.84 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1793] & 81231513221(12)12231513221(12)122315132^121(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.84 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1794] $81231513221(12)12231513221(12)12231513221(12)12^123151322181+\hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 81$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.84 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [1795] $81231513221(12)12231513221(12)122315132^121(12)1223151322181+\hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 81$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.84 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1796] $(12)122315132181232181232181232^1181232181232171 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 100$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.76 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1797] $(12)122^1315132181232181232181232181232181232171 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 100$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.76 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 189 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [1798] $(12)122315132181232181232181232^11812321812321712 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 100$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.76 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [1799] $(12)122^13151321812321812321812321812321812321712 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 100$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.76 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 189 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [1800] $413221(12)12231513221(12)12231513221(12)12231513221(12)12^12315+\hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 66$, $\Delta = -114$, $\det = 100$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.76 \rightarrow 55$, $T_{\max} = 66$, $T \in [55, 66]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 29 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [1801] $(12)12^12315132181232181232181232181232181232171 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 230 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$
- [1802] $(12)1223151321812321812321812321812^132181232181232171 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [1803] $(12)1223151321812321812321812321812^1321812321812321712 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [1804] $(12)12^123151321812321812321812321812321812321712 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 230 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$
- [1805] $41413221(12)12^1231513221(12)12231513221(12)12231513218123218+\hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
 [1806] 41413221(12)122315132¹21(12)12231513221(12)12231513218123218+ $\hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [1807] 41413221(12)12231513221(12)12231513221(12)1223151321812¹3218+ $\hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -110$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 21 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [1808] 81232181231513221(12)12231513221(12)122¹31513221¹⁵¹³²²¹(12) 1 + $\hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
 [1809] 81232181231513221(12)12231513221(12)1223151322¹1¹⁵¹³²²¹(12) 1 + $\hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
 [1810] 81232¹181231513221(12)12231513221(12)12231513221¹⁵¹³²²¹(12) 1 + $\hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
 [1811] 81231513221¹(12)12231513221(12)122¹31513221(12)122315132181¹+ $\hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [1812] 81231513221¹(12)12231513221(12)1223151322¹1(12)122315132181¹+ $\hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
 [1813] 413221(12)122¹31513221(12)12231513221¹(12)12231513221(12)122315+ $\hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 66$, $\Delta = -114$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 66$, $T \in [55, 66]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
 [1814] 81231513221¹(12)12231513221(12)1223151322¹1(12)1223151321812+ $\hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
 [1815] 81231513221¹(12)12231513221(12)122¹31513221(12)1223151321812+ $\hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [1816] 81231513221¹(12)12231513221(12)1223151322¹1(12)1223151322181+ $\hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$

- [1817] $81231513221(12)12231513221(12)122^131513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 121$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.69 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [1818] $(12)122315132181232181232181232^1181232181232181232171 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 169$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.59 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$
- [1819] $(12)122315132181232181232181232^11812321812321812321712 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 169$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.59 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$
- [1820] $41413221(12)12231513221(12)122^131513221(12)12231513218123218+\hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -110$, $\det = 169$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.59 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1821] $81232181231513221(12)122315132^121(12)12231513221(12)1 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -106$, $\det = 169$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.59 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [1822] $812315132^121(12)12231513221(12)12231513221(12)122315132181 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 169$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.59 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [1823] $81231513221(12)122315132^121(12)12231513221(12)122315132181 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 169$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.59 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1824] $413221(12)12^1231513221(12)12231513221(12)12231513221(12)122315 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 66$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.59 \rightarrow 55$, $T_{\max} = 66$, $T \in [55, 66]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 77 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [1825] $812315132^121(12)12231513221(12)12231513221(12)1223151321812 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 169$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.59 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [1826] $81231513221(12)122315132^121(12)12231513221(12)1223151321812 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 169$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.59 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1827] $812315132^121(12)12231513221(12)12231513221(12)1223151322181 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 169$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.59 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [1828] $81231513221(12)122315132^121(12)12231513221(12)1223151322181 + \hat{1}\rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det =$

169, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.59 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1829] $(12)1223151321812321812321812^1 32181232181232181232171 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 196$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.55 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$$

[1830] $(12)1223151321812321812321812^1 321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 196$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.55 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$$

[1831] $41413221(12)1223151322^1 1(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -110$, $\det = 196$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.55 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1832] $81232181231513221(12)12231513221(12)12^1 231513221 (12) 1 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -106$, $\det = 196$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.55 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1833] $81231513221(12)12231513221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 196$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.55 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1834] $81231513221(12)12231513221(12)12^1 231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 196$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.55 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1835] $81231513221(12)12231513221(12)12^1 231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 196$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.55 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1836] $41413221(12)12231513221(12)122315132^1 21(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -110$, $\det = 225$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.52 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1837] $81232181231513221(12)122^1 31513221(12)12231513221 (12) 1 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -106$, $\det = 225$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.52 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[1838] $81231513221(12)122^1 31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 225$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.52 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1839] $81231513221(12)122^1 31513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 225$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.52 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1840] $81231513221(12)122^1 31513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 225$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.52 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1841] $(12)122315132181232181232^1 181232181232181232181232171 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 256$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.48 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 463 & 43 \\ 2 & 43 & 3 \end{pmatrix}_{b_0}$$

[1842] $(12)122315132181232181232^1 1812321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 256$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.48 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 463 & 43 \\ 2 & 43 & 3 \end{pmatrix}_{b_0}$$

[1843] $41413221(12)12231513221(12)12^1 231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -110$, $\det = 256$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.48 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1844] $81232181231513221(12)1223151322^1 1(12)12231513221^{1513221} (12) 1 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -106$, $\det = 256$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.48 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1845] $81231513221(12)1223151322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 256$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.48 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1846] $81231513221(12)1223151322^1 1(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 256$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.48 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1847] $81231513221(12)1223151322^1 1(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 256$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.48 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1848] $(12)1223151321812321812^1 32181232181232181232181232171 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 289$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.46 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 526 & 46 \\ 2 & 46 & 3 \end{pmatrix}_{b_0}$$

[1849] $(12)1223151321812321812^1 321812321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 289$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.46 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 526 & 46 \\ 2 & 46 & 3 \end{pmatrix}_{b_0}$$

[1850] $41413221(12)12231513221(12)12231513221(12)122^1 31513218123218 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -110$, $\det = 289$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 54.46 \rightarrow 55, T_{\max} = 62, T \in [55, 62]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1851] $812321812315132^1 21(12)12231513221(12)12231513221^{1513221} (12) 1 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -106$, $\det =$

289, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.46 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1852] $8123151322^1 1(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 289$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.46 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1853] $413221(12)122315132^1 21(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 66$, $\Delta = -114$, $\det = 289$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.46 \rightarrow 55$, $T_{\max} = 66$, $T \in [55, 66]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 118 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1854] $413221(12)12231513221(12)12231513221(12)122315132^1 21(12)122315 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 66$, $\Delta = -114$, $\det = 289$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.46 \rightarrow 55$, $T_{\max} = 66$, $T \in [55, 66]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 101 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1855] $8123151322^1 1(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 289$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.46 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1856] $8123151322^1 1(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -104$, $\det = 289$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.46 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1857] $(12)122315132181232^1 181232181232181232181232181232171 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 361$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.41 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 664 & 52 \\ 2 & 52 & 3 \end{pmatrix}_{b_0}$$

[1858] $(12)122315132181232^1 1812321812321812321812321812321712 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -101$, $\det = 361$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.41 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 664 & 52 \\ 2 & 52 & 3 \end{pmatrix}_{b_0}$$

[1859] $41413221(12)12231513221(12)12231513221(12)122315132^1 18123218 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -110$, $\det = 361$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.41 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 75 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1860] $41413221(12)12231513221(12)1223151322^1 1(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -110$, $\det = 361$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.41 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1861] $81232181231513221(12)12^1 231513221(12)12231513221(12)1513221(12)1 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -106$, $\det = 361$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.41 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[1862] $812321812^1 31513221(12)12231513221(12)12231513221(12)1513221(12)1 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 62$, $\Delta = -106$, $\det = 361$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 54.41 \rightarrow 55$, $T_{\max} = 62$, $T \in [55, 62]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[1863] & 81231513221(12)12^1 231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.41 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[1864] & 81231513221(12)12^1 231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.41 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[1865] & 81231513221(12)12^1 231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.41 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[1866] & (12)1223151321812^1 32181232181232181232181232181232171 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -101, \det = 400, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.39 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 20 & 2 \\ 20 & 739 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0} \\
[1867] & (12)1223151321812^1 321812321812321812321812321812321712 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -101, \det = 400, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.39 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 20 & 2 \\ 20 & 739 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0} \\
[1868] & 413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)122315 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 66, \Delta = -114, \det = 400, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.39 \rightarrow 55, T_{\max} = 66, T \in [55, 66] \\
& \begin{pmatrix} 0 & 20 & 2 \\ 20 & 139 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1869] & 41413221(12)12231513221(12)12231513221(12)12^1 231513218123218 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -110, \det = 441, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.37 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[1870] & 8123218123151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -106, \det = 441, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.37 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[1871] & (12)122315132^1 181232181232181232181232181232181232171 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -101, \det = 484, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.36 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 901 & 61 \\ 2 & 61 & 3 \end{pmatrix}_{b_0} \\
[1872] & (12)122315132^1 1812321812321812321812321812321812321712 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 62, \Delta = -101, \det = 484, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.36 \rightarrow 55, T_{\max} = 62, T \in [55, 62] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 901 & 61 \\ 2 & 61 & 3 \end{pmatrix}_{b_0} \\
[1873] & 413221(12)1223151322^1 1(12)12231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2 \quad T = 50, T_{\max} = 66, \Delta = -114, \det = 484, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 54.36 \rightarrow 55, T_{\max} = 66, T \in [55, 66] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 197 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1874] $413221(12)12231513221(12)122^1 31513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 66$, $\Delta = -114$, $\det = 529$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.34 \rightarrow 55$, $T_{\max} = 66$, $T \in [55, 66]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 206 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$

[1875] $413221(12)12231513221(12)12231513221(12)122^1 31513221(12)122315 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 66$, $\Delta = -114$, $\det = 729$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.29 \rightarrow 55$, $T_{\max} = 66$, $T \in [55, 66]$
 $\begin{pmatrix} 0 & 27 & 2 \\ 27 & 269 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[1876] $413221(12)12231513221(12)12^1 231513221(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 66$, $\Delta = -114$, $\det = 784$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.28 \rightarrow 55$, $T_{\max} = 66$, $T \in [55, 66]$
 $\begin{pmatrix} 0 & 28 & 2 \\ 28 & 307 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$

[1877] $413221(12)12231513221(12)122315132^1 21(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 66$, $\Delta = -114$, $\det = 841$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.27 \rightarrow 55$, $T_{\max} = 66$, $T \in [55, 66]$
 $\begin{pmatrix} 0 & 29 & 2 \\ 29 & 318 & 38 \\ 2 & 38 & 3 \end{pmatrix}_{b_0}$

[1878] $413221(12)12231513221(12)12231513221(12)12^1 231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 66$, $\Delta = -114$, $\det = 1225$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.23 \rightarrow 55$, $T_{\max} = 66$, $T \in [55, 66]$
 $\begin{pmatrix} 0 & 35 & 2 \\ 35 & 454 & 46 \\ 2 & 46 & 3 \end{pmatrix}_{b_0}$

[1879] $413221(12)12231513221(12)1223151322^1 1(12)12231513221(12)122315 + \hat{1} \rightarrow 2$ $T = 50$, $T_{\max} = 66$, $\Delta = -114$, $\det = 1369$, $(n_+, n_-, n_0) = (1, 50, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 54.21 \rightarrow 55$, $T_{\max} = 66$, $T \in [55, 66]$
 $\begin{pmatrix} 0 & 37 & 2 \\ 37 & 517 & 49 \\ 2 & 49 & 3 \end{pmatrix}_{b_0}$

$T = 51$ [1880] $(12)1223151321812321812321812321812321812321812321812^1 2 + \hat{1} \rightarrow 2$ $T = 51$, $T_{\max} = 63$, $\Delta = -100$, $\det = -4$, $(n_+, n_-, n_0) = (1, 51, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58$, $T_{\max} = 63$, $T \in [58, 63]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$

[1881] $22^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 51$, $T_{\max} = 63$, $\Delta = -108$, $\det = -4$, $(n_+, n_-, n_0) = (1, 51, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58$, $T_{\max} = 63$, $T \in [58, 63]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[1882] $(12)122315132181232181232181232181232181232181232^1 181 + \hat{1} \rightarrow 2$ $T = 51$, $T_{\max} = 63$, $\Delta = -100$, $\det = -16$, $(n_+, n_-, n_0) = (1, 51, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.75 \rightarrow 57$, $T_{\max} = 63$, $T \in [57, 63]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1883] $(12)122315132181232181232181232181232181232181232^1 18122 + \hat{1} \rightarrow 2$ $T = 51$, $T_{\max} = 63$, $\Delta = -100$, $\det = -16$, $(n_+, n_-, n_0) = (1, 51, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.75 \rightarrow 57$, $T_{\max} = 63$, $T \in [57, 63]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1884] $(12)1223151321812321812321812321812321812321812^1 32181 + \hat{1} \rightarrow 2$ $T = 51$, $T_{\max} = 63$, $\Delta = -100$, $\det = -25$, $(n_+, n_-, n_0) = (1, 51, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.44 \rightarrow 57$, $T_{\max} = 63$, $T \in [57, 63]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$

[1885] $(12)1223151321812321812321812321812321812321812^1 3218122 + \hat{1} \rightarrow 2$ $T = 51$, $T_{\max} = 63$, $\Delta = -100$, $\det =$

$$-25, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.44 \rightarrow 57, T_{\max} = 63, T \in [57, 63]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[1886] \quad 513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -109, \det = -25, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.44 \rightarrow 57, T_{\max} = 63, T \in [57, 63]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1887] \quad (12)122315132181232181232181232181232181232^1 1812321 8 1 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -100, \det = -49, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.06 \rightarrow 57, T_{\max} = 63, T \in [57, 63]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[1888] \quad (12)122315132181232181232181232181232181232^1 18123218122 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -100, \det = -49, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.06 \rightarrow 57, T_{\max} = 63, T \in [57, 63]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 76 & 16 \\ 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[1889] \quad 1(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.06 \rightarrow 57, T_{\max} = 63, T \in [57, 63]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1890] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.06 \rightarrow 57, T_{\max} = 63, T \in [57, 63]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[1891] \quad 221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.06 \rightarrow 57, T_{\max} = 63, T \in [57, 63]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[1892] \quad 221(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.06 \rightarrow 57, T_{\max} = 63, T \in [57, 63]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1893] \quad 5132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.06 \rightarrow 57, T_{\max} = 63, T \in [57, 63]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 34 & 10 \\ 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[1894] \quad 513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -109, \det = -49, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.06 \rightarrow 57, T_{\max} = 63, T \in [57, 63]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[1895] \quad (12)1223151321812321812321812321812321812^1 321812321 8 1 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -100, \det = -64, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 55.94 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1896] \quad (12)1223151321812321812321812321812321812^1 3218123218122 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -100, \det = -64, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 55.94 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 103 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1897] & 1(12)12^{21}1^{1231513221}(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -64, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.94 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1898] & 221(12)12^{21}1^{1231513221}(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -64, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.94 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1899] & 1(12)12231513221(12)12231513221(12)122315132^{21}21(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.84 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1900] & 1(12)12231513221(12)12231513221(12)12231513221(12)12^{21}23151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.84 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1901] & 221(12)12231513221(12)12231513221(12)12231513221(12)12^{21}23151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.84 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[1902] & 221(12)12231513221(12)12231513221(12)122315132^{21}21(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.84 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1903] & (12)122315132181232181232181232181232^{21}181232181232181 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -100, \det = -100, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.76 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1904] & (12)122^{21}315132181232181232181232181232181232181232181232181 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -100, \det = -100, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.76 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 189 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1905] & (12)122^{21}31513218123218123218123218123218123218123218122 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -100, \det = -100, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.76 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 189 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[1906] & (12)122315132181232181232181232181232^{21}18123218123218122 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -100, \det = -100, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.76 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 169 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1907] & 51322^{21}1(12)12231513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -109, \det = -100, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 55.76 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1908] $(12)1223151321812321812321812321812^{1\rightarrow 2}32181232181232181^{21} + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -100, \det = -121, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.69 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [1909] $(12)12^{1\rightarrow 2}2315132181232181232181232181232181232181232181^{21} + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -100, \det = -121, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.69 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 230 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$
- [1910] $(12)1223151321812321812321812321812^{1\rightarrow 2}3218123218123218122 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -100, \det = -121, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.69 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 208 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [1911] $(12)12^{1\rightarrow 2}231513218123218123218123218123218123218123218122 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -100, \det = -121, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.69 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 230 & 30 \\ 2 & 30 & 3 \end{pmatrix}_{b_0}$
- [1912] $1^{21}(12)12231513221(12)12231513221(12)122^{1\rightarrow 2}31513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.69 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1913] $1^{21}(12)12231513221(12)12231513221(12)1223151322^{1\rightarrow 2}1(12)1223151315 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.69 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1914] $221(12)12231513221(12)12231513221(12)122^{1\rightarrow 2}31513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.69 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1915] $221(12)12231513221(12)12231513221(12)1223151322^{1\rightarrow 2}1(12)1223151315 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.69 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1916] $513221(12)12231513221(12)12231513221(12)122315132^{1\rightarrow 2}21(12)1223151 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -109, \det = -121, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.69 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [1917] $(12)122315132181232181232181232^{1\rightarrow 2}181232181232181232181^{21} + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -100, \det = -169, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.59 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$
- [1918] $(12)122315132181232181232181232^{1\rightarrow 2}18123218123218123218122 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -100, \det = -169, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 55.59 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$
- [1919] $1^{21}(12)12231513221(12)122315132^{1\rightarrow 2}21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 51, T_{\max} = 63, \Delta = -108, \det = -169, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& T_{\min} = 55.59 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1920] & \begin{matrix} 21 & 1 \\ 1(12)122315132^1 21(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 & T = 51, T_{\max} = 63, \Delta = -108, \det = \\ -169, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 55.59 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1921] & \begin{matrix} 1 \\ 221(12)12231513221(12)122315132^1 21(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 & T = 51, T_{\max} = 63, \Delta = \\ -108, \det = -169, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 55.59 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1922] & \begin{matrix} 1 \\ 221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 & T = 51, T_{\max} = 63, \Delta = \\ -108, \det = -169, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 55.59 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1923] & \begin{matrix} 1 \\ 513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1223151 + \hat{1} \rightarrow 2 & T = 51, T_{\max} = 63, \Delta = \\ -109, \det = -169, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 55.59 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[1924] & \begin{matrix} 21 \\ (12)1223151321812321812321812^1 321812321812321812321 8 1 + \hat{1} \rightarrow 2 & T = 51, T_{\max} = 63, \Delta = -100, \det = \\ -196, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 55.55 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0} \\
[1925] & \begin{matrix} 1 \\ (12)1223151321812321812321812^1 3218123218123218123218122 + \hat{1} \rightarrow 2 & T = 51, T_{\max} = 63, \Delta = -100, \det = \\ -196, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 55.55 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 349 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0} \\
[1926] & \begin{matrix} 21 & 1 \\ 1(12)12231513221(12)12231513221(12)12^1 231513221(12)1223151315 + \hat{1} \rightarrow 2 & T = 51, T_{\max} = 63, \Delta = -108, \det = \\ -196, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 55.55 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1927] & \begin{matrix} 1 \\ 221(12)12231513221(12)12231513221(12)12^1 231513221(12)1223151315 + \hat{1} \rightarrow 2 & T = 51, T_{\max} = 63, \Delta = \\ -108, \det = -196, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 55.55 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1928] & \begin{matrix} 21 & 1 \\ 1(12)12231513221(12)122^1 31513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 & T = 51, T_{\max} = 63, \Delta = -108, \det = \\ -225, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 55.52 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[1929] & \begin{matrix} 1 \\ 221(12)12231513221(12)122^1 31513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 & T = 51, T_{\max} = 63, \Delta = \\ -108, \det = -225, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 55.52 \rightarrow 56, T_{\max} = 63, T \in [56, 63] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[1930] & \begin{matrix} 21 \\ (12)122315132181232181232^1 1812321812321812321812321 8 1 + \hat{1} \rightarrow 2 & T = 51, T_{\max} = 63, \Delta = -100, \det = \\ -256, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0 \end{matrix} \\
& T_{\min} = 55.48 \rightarrow 56, T_{\max} = 63, T \in [56, 63]
\end{aligned}$$

$$T_{\min} = 55.27 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 521 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[1954] \quad 513221(12)12231513221^1(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -109, \det = -1225, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 55.23 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$$

$$\begin{pmatrix} 0 & 35 & 2 \\ 35 & 769 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0}$$

$$[1955] \quad 513221(12)12231513221(12)12^1231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 51, T_{\max} = 63, \Delta = -109, \det = -1369, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 55.21 \rightarrow 56, T_{\max} = 63, T \in [56, 63]$$

$$\begin{pmatrix} 0 & 37 & 2 \\ 37 & 850 & 58 \\ 2 & 58 & 3 \end{pmatrix}_{b_0}$$

$$T = 52 \quad [0] \quad 22^1(12)12231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 59, T_{\max} = 64, T \in [59, 64]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 61231513221(12)12231513221(12)12231513221(12)12231513221(12)12^12 + \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 59, T_{\max} = 64, T \in [59, 64]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 6131513221(12)12231513221(12)12231513221(12)12231513221(12)122^13 + \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 68, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.22 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 612^131513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.44 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 6131513221(12)12231513221(12)12231513221(12)12231513221(12)12^123 + \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 68, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.44 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 612^131513221(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -106, \det = 25, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.44 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122^131513161 + \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.06 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1(12)122^131513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.06 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[8] \quad 61231513221(12)12231513221(12)12231513221(12)122315132^121(12)1 + \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.06 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[9] \text{ } 612315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.06 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[10] \text{ } 812^1 32181231513221(12)12231513221(12)12231513221(12)1223151315+ \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.06 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[11] \text{ } 81232181231513221(12)12231513221(12)12231513221(12)122^1 3151315+ \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -108, \det = 49, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.06 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[12] \text{ } 221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513161+ \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.06 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[13] \text{ } 221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513161+ \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.06 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[14] \text{ } 612315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122+ \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.06 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[15] \text{ } 61231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)122+ \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -106, \det = 49, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.06 \rightarrow 58, T_{\max} = 64, T \in [58, 64]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[16] \text{ } 1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513161+ \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.94 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[17] \text{ } 1(12)12^1 31513221(12)12231513221(12)12231513221(12)122315131613+ \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 67, \Delta = -113, \det = 64, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.94 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[18] \text{ } 61231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1+ \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -106, \det = 64, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.94 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[19] \text{ } 221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513161+ \hat{1} \rightarrow 2 \quad T = 52, T_{\max} = 64, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 56.94 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[20] $21(12)122^1 31513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 64$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 56.94 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[21] $61231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)122+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 64$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 56.94 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[22] $1(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 56.84 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[23] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 56.84 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[24] $1(12)12^1 231513221(12)12231513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 81$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 56.84 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[25] $61231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 56.84 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[26] $6123151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 81$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 56.84 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[27] $81232181231513221(12)12231513221(12)122315132^1 21(12)1223151315+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 56.84 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[28] $81232181231513221(12)12231513221(12)12231513221(12)12^1 23151315+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -108$, $\det = 81$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 56.84 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[29] $221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 56.84 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[30] $221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 81$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 56.84 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

- [31] $21(12)12^1 231513221(12)12231513221(12)12231513221(12)122315131613+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 67, \Delta = -113, \det = 81, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.84 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [32] $6123151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)122+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.84 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [33] $61231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)122+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -106, \det = 81, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.84 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [34] $61315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 68, \Delta = -114, \det = 100, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.76 \rightarrow 57, T_{\max} = 68, T \in [57, 68]$
 $\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [35] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [36] $1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513161+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [37] $1(12)12231513221(12)12231513221(12)12231513221(12)122^1 315131613+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 67, \Delta = -113, \det = 121, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [38] $61231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [39] $61231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -106, \det = 121, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [40] $81232^1 181231513221(12)12231513221(12)12231513221(12)1223151315+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 98 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [41] $81232181231513221(12)12231513221(12)1223151322^1 1(12)1223151315+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [42] $81232181231513221(12)12231513221(12)122^1 31513221(12)1223151315+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -108, \det = 121, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 56.69 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [43] $221(12)12231513221(12)12231513221(12)12231513221^1(12)12231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 121$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [44] $221(12)12231513221(12)12231513221(12)122^131513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 121$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [45] $21(12)12231513221(12)12231513221(12)12231513221(12)122^1315131613+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 121$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [46] $61231513221(12)12^1231513221(12)12231513221(12)12231513221(12)122+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [47] $61231513221(12)122315132^121(12)12231513221(12)12231513221(12)122+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 121$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.69 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [48] $1(12)12231513221(12)122315132^121(12)12231513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 169$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.59 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [49] $1(12)122315132^121(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 169$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.59 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [50] $1(12)12231513221(12)12231513221(12)122315132^121(12)122315131613+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 169$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.59 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [51] $61231513221(12)12231513221(12)12231513221(12)122^131513221(12)1+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 169$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.59 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [52] $61231513221(12)12231513221(12)122^131513221(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 169$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.59 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [53] $81232181231513221(12)122315132^121(12)12231513221(12)1223151315+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -108$, $\det = 169$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.59 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[54] $221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -105, \det = 169, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 56.59 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[55] $221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -105, \det = 169, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 56.59 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[56] $21(12)12231513221(12)12231513221(12)122315132^1 21(12)122315131613+\hat{1} \rightarrow 2$ $T = 52, T_{\max} = 67, \Delta = -113, \det = 169, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 56.59 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[57] $6131513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223+\hat{1} \rightarrow 2$ $T = 52, T_{\max} = 68, \Delta = -114, \det = 169, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 56.59 \rightarrow 57, T_{\max} = 68, T \in [57, 68]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[58] $613151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223+\hat{1} \rightarrow 2$ $T = 52, T_{\max} = 68, \Delta = -114, \det = 169, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 56.59 \rightarrow 57, T_{\max} = 68, T \in [57, 68]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[59] $61231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)122+\hat{1} \rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 56.59 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[60] $61231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)122+\hat{1} \rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -106, \det = 169, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 56.59 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[61] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 56.55 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[62] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1 2315131613+\hat{1} \rightarrow 2$ $T = 52, T_{\max} = 67, \Delta = -113, \det = 196, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 56.55 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[63] $61231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -106, \det = 196, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 56.55 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[64] $81232181231513221(12)12231513221(12)12^1 231513221(12)1223151315+\hat{1} \rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -108, \det = 196, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 56.55 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

- [65] $221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.55 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [66] $21(12)12231513221(12)12231513221(12)12231513221(12)12^1 2315131613+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 67, \Delta = -113, \det = 196, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.55 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [67] $6131513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1223+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 68, \Delta = -114, \det = 196, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.55 \rightarrow 57, T_{\max} = 68, T \in [57, 68]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [68] $61231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)122+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -106, \det = 196, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.55 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [69] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -105, \det = 225, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.52 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [70] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)122315131613+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 67, \Delta = -113, \det = 225, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.52 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [71] $61231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1+ \hat{1}\rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -106, \det = 225, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.52 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [72] $81232181231513221(12)122^1 31513221(12)12231513221(12)1223151315+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -108, \det = 225, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.52 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [73] $221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -105, \det = 225, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.52 \rightarrow 57, T_{\max} = 64, T \in [57, 64]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [74] $21(12)12231513221(12)12231513221(12)122^1 31513221(12)122315131613+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 67, \Delta = -113, \det = 225, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.52 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [75] $6131513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1223+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 68, \Delta = -114, \det = 225, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.52 \rightarrow 57, T_{\max} = 68, T \in [57, 68]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [76] $61231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)122+\hat{1}\rightarrow 2$ $T = 52, T_{\max} = 64, \Delta = -106, \det =$

225, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 56.52 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[77] $1(12)12231513221(12)12231513221^1(12)12231513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 256$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 56.48 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[78] $1(12)122315132^121(12)12231513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 256$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 56.48 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[79] $1(12)12231513221(12)12231513221(12)12231513221^1(12)122315131613+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 256$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 56.48 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[80] $61231513221(12)12231513221(12)12^1231513221(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 256$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 56.48 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[81] $81232181231513221(12)12231513221^1(12)12231513221(12)1223151315+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -108$, $\det = 256$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 56.48 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[82] $221(12)12231513221(12)12231513221^1(12)12231513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 256$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 56.48 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[83] $21(12)122315132^121(12)12231513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 256$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 56.48 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[84] $21(12)12231513221(12)12231513221(12)12231513221^1(12)122315131613+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 256$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 56.48 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[85] $61231513221(12)12231513221(12)12^1231513221(12)12231513221(12)122+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 256$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 56.48 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[86] $1(12)12231513221^1(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 289$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 56.46 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[87] $1(12)12231513221(12)122315132^121(12)12231513221(12)122315131613+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 289$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 56.46 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [88] $61231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 289$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.46 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [89] $812321812315132^1 21(12)12231513221(12)12231513221(12)1223151315+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -108$, $\det = 289$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.46 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [90] $221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 289$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.46 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [91] $21(12)12231513221(12)122315132^1 21(12)12231513221(12)122315131613+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 289$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.46 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [92] $6131513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 68$, $\Delta = -114$, $\det = 289$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.46 \rightarrow 57$, $T_{\max} = 68$, $T \in [57, 68]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [93] $61231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)122+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 289$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.46 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [94] $6131513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 68$, $\Delta = -114$, $\det = 324$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.43 \rightarrow 57$, $T_{\max} = 68$, $T \in [57, 68]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [95] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 361$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.41 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [96] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)122315131613+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 361$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.41 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [97] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 361$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.41 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [98] $61231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 361$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.41 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$

- $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
 [99] 812321812¹31513221(12)12231513221(12)12231513221(12)1223151315+ $\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -108$, $\det = 361$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.41 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [100] 81232181231513221(12)12¹231513221(12)12231513221(12)1223151315+ $\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -108$, $\det = 361$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.41 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [101] 221(12)12231513221(12)12¹231513221(12)12231513221(12)12231513161 + $\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -105$, $\det = 361$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.41 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [102] 21(12)12231513221(12)12231513221(12)12¹231513221(12)122315131613 + $\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 361$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.41 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [103] 21(12)12231513221(12)122¹31513221(12)12231513221(12)122315131613 + $\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 361$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.41 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [104] 61231513221(12)12231513221(12)1223151322¹1(12)12231513221(12)122 + $\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -106$, $\det = 361$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.41 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [107] 8123218123151322¹1(12)12231513221(12)12231513221(12)1223151315+ $\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 64$, $\Delta = -108$, $\det = 441$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.37 \rightarrow 57$, $T_{\max} = 64$, $T \in [57, 64]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [109] 21(12)12231513221(12)1223151322¹1(12)12231513221(12)122315131613 + $\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 441$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.37 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [108] 21(12)1223151322¹1(12)12231513221(12)12231513221(12)122315131613 + $\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 441$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.37 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$
- [106] 1(12)12231513221(12)1223151322¹1(12)12231513221(12)122315131613+ $\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 441$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.37 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [105] 1(12)1223151322¹1(12)12231513221(12)12231513221(12)122315131613+ $\hat{1} \rightarrow 2$ $T = 52$, $T_{\max} = 67$, $\Delta = -113$, $\det = 441$, $(n_+, n_-, n_0) = (1, 52, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 56.37 \rightarrow 57$, $T_{\max} = 67$, $T \in [57, 67]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$

[110] $6131513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 52, T_{\max} = 68, \Delta = -114, \det = 484, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.36 \rightarrow 57, T_{\max} = 68, T \in [57, 68]$
 $\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[111] $6131513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223 + \hat{1} \rightarrow 2$ $T = 52, T_{\max} = 68, \Delta = -114, \det = 529, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.34 \rightarrow 57, T_{\max} = 68, T \in [57, 68]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$

[112] $6131513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 52, T_{\max} = 68, \Delta = -114, \det = 529, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.34 \rightarrow 57, T_{\max} = 68, T \in [57, 68]$
 $\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[113] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2$ $T = 52, T_{\max} = 67, \Delta = -113, \det = 576, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.33 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$
 $\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[114] $21(12)12231513221(12)12^1 231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2$ $T = 52, T_{\max} = 67, \Delta = -113, \det = 576, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.33 \rightarrow 57, T_{\max} = 67, T \in [57, 67]$
 $\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[115] $6131513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 52, T_{\max} = 68, \Delta = -114, \det = 676, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.30 \rightarrow 57, T_{\max} = 68, T \in [57, 68]$
 $\begin{pmatrix} 0 & 26 & 2 \\ 26 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$

[116] $6131513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 52, T_{\max} = 68, \Delta = -114, \det = 729, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.29 \rightarrow 57, T_{\max} = 68, T \in [57, 68]$
 $\begin{pmatrix} 0 & 27 & 2 \\ 27 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$

[117] $6131513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 52, T_{\max} = 68, \Delta = -114, \det = 900, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.26 \rightarrow 57, T_{\max} = 68, T \in [57, 68]$
 $\begin{pmatrix} 0 & 30 & 2 \\ 30 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$

[118] $6131513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 52, T_{\max} = 68, \Delta = -114, \det = 1089, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 56.24 \rightarrow 57, T_{\max} = 68, T \in [57, 68]$
 $\begin{pmatrix} 0 & 33 & 2 \\ 33 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$

$T = 53$ [119] $22^1 1(12)12231513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60, T_{\max} = 65, T \in [60, 65]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

[120] $712231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60, T_{\max} = 65, T \in [60, 65]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$

- [121] $71231513221(12)12231513221(12)12231513221(12)12231513221(12)122^13 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 68$, $\Delta = -113$, $\det = -9$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.22 \rightarrow 60$, $T_{\max} = 68$, $T \in [60, 68]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [122] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1171 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.44 \rightarrow 59$, $T_{\max} = 65$, $T \in [59, 65]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [123] $7122^131513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.44 \rightarrow 59$, $T_{\max} = 65$, $T \in [59, 65]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [124] $221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1171 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.44 \rightarrow 59$, $T_{\max} = 65$, $T \in [59, 65]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [125] $71231513221(12)12231513221(12)12231513221(12)12231513221(12)12^123 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 68$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.44 \rightarrow 59$, $T_{\max} = 68$, $T \in [59, 68]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [126] $7122^131513221(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -25$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.44 \rightarrow 59$, $T_{\max} = 65$, $T \in [59, 65]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [127] $612^131513221(12)12231513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -106$, $\det = -25$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.44 \rightarrow 59$, $T_{\max} = 65$, $T \in [59, 65]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [128] $712^1231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.22 \rightarrow 59$, $T_{\max} = 65$, $T \in [59, 65]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [129] $712^131513221(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 68$, $\Delta = -113$, $\det = -36$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.22 \rightarrow 59$, $T_{\max} = 68$, $T \in [59, 68]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [130] $712^1231513221(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -36$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.22 \rightarrow 59$, $T_{\max} = 65$, $T \in [59, 65]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [131] $1(12)12231513221(12)12231513221(12)12231513221(12)122^1315132171 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -49$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.06 \rightarrow 59$, $T_{\max} = 65$, $T \in [59, 65]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [132] $1(12)122^131513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det =$

$$-49, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.06 \rightarrow 59, T_{\max} = 65, T \in [59, 65]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[133] \quad 7122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.06 \rightarrow 59, T_{\max} = 65, T \in [59, 65]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[134] \quad 712231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1+ \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.06 \rightarrow 59, T_{\max} = 65, T \in [59, 65]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[135] \quad 221(12)12231513221(12)12231513221(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.06 \rightarrow 59, T_{\max} = 65, T \in [59, 65]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[136] \quad 221(12)122^1 31513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.06 \rightarrow 59, T_{\max} = 65, T \in [59, 65]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[137] \quad 712231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.06 \rightarrow 59, T_{\max} = 65, T \in [59, 65]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[138] \quad 7122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.06 \rightarrow 59, T_{\max} = 65, T \in [59, 65]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[139] \quad 812^1 32181231513221(12)12231513221(12)12231513221(12)12231513161+ \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.06 \rightarrow 59, T_{\max} = 65, T \in [59, 65]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[140] \quad 81232181231513221(12)12231513221(12)12231513221(12)122^1 31513161+ \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.06 \rightarrow 59, T_{\max} = 65, T \in [59, 65]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[141] \quad 612315132^1 21(12)12231513221(12)12231513221(12)12231513218123218+ \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.06 \rightarrow 59, T_{\max} = 65, T \in [59, 65]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[142] \quad 61231513221(12)12231513221(12)12231513221(12)122315132181232^1 18+ \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -106, \det = -49, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.06 \rightarrow 59, T_{\max} = 65, T \in [59, 65]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[143] \quad 1(12)12^1 231513221(12)12231513221(12)12231513221(12)122315132171+ \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.94 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[144] \quad 712231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.94 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[145] \quad 221(12)12^1 231513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.94 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[146] \quad 712231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)122 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.94 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[147] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[148] \quad 1(12)12231513221(12)12231513221(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[149] \quad 71223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[150] \quad 712231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[151] \quad 221(12)12231513221(12)12231513221(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[152] \quad 221(12)12231513221(12)12231513221(12)12231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[153] \quad 712231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[154] \quad 71223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[155] 81232181231513221(12)12231513221(12)122315132¹21(12)12231513161+ $\hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[156] 81232181231513221(12)12231513221(12)12231513221(12)12¹231513161+ $\hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[157] 61231513221(12)122¹31513221(12)12231513221(12)12231513218123218+ $\hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -106$, $\det = -81$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[158] 6123151322¹1(12)12231513221(12)12231513221(12)12231513218123218+ $\hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -106$, $\det = -81$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.84 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[159] 712315132¹21(12)12231513221(12)12231513221(12)12231513221(12)1223 + $\hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 68$, $\Delta = -113$, $\det = -100$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.76 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[160] 1²¹(12)12231513221(12)12231513221(12)122¹31513221(12)122315132171+ $\hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -121$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[161] 1²¹(12)12231513221(12)12231513221(12)1223151322¹1(12)122315132171+ $\hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -121$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[162] 712231513221(12)122315132¹21(12)12231513221(12)12231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -121$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[163] 712231513221(12)12¹231513221(12)12231513221(12)12231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -121$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[164] 221(12)12231513221(12)12231513221(12)1223151322¹1(12)122315132171 + $\hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -121$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[165] 221(12)12231513221(12)12231513221(12)122¹31513221(12)122315132171 + $\hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -121$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

- [166] $712231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [167] $712231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [168] $81232181231513221(12)12231513221(12)122^1 31513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [169] $81232^1 181231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 219 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [170] $81232181231513221(12)12231513221(12)1223151322^1 1(12)12231513161 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [171] $61231513221(12)12^1 231513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [172] $61231513221(12)12231513221(12)12231513221(12)1223151321812^1 3218 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 21 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [173] $61231513221(12)122315132^1 21(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.69 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [174] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.59 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [175] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.59 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [176] $712231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.59 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [177] $712231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.59 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[178] $221(12)122315132^1 21(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 57.59 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[179] $221(12)12231513221(12)122315132^1 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 57.59 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[180] $71231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 68, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 57.59 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[181] $7123151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 68, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 57.59 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[182] $712231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)122 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 57.59 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[183] $712231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 57.59 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[184] $81232181231513221(12)122315132^1 21(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 57.59 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[185] $61231513221(12)12231513221(12)122^1 31513221(12)12231513218123218 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 57.59 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[186] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 57.55 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[187] $712231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 57.55 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[188] $221(12)12231513221(12)12231513221(12)12^1 231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 57.55 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

- [189] $71231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 68$, $\Delta = -113$, $\det = -196$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 57.55 \rightarrow 58$, $T_{\max} = 68$, $T \in [58, 68]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [190] $712231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -196$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 57.55 \rightarrow 58$, $T_{\max} = 65$, $T \in [58, 65]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [191] $81232181231513221(12)12231513221(12)12^1 231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -105$, $\det = -196$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 57.55 \rightarrow 58$, $T_{\max} = 65$, $T \in [58, 65]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [192] $61231513221(12)1223151322^1 1(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -106$, $\det = -196$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 57.55 \rightarrow 58$, $T_{\max} = 65$, $T \in [58, 65]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [193] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -225$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 57.52 \rightarrow 58$, $T_{\max} = 65$, $T \in [58, 65]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [194] $712231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -225$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 57.52 \rightarrow 58$, $T_{\max} = 65$, $T \in [58, 65]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [195] $221(12)12231513221(12)122^1 31513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -225$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 57.52 \rightarrow 58$, $T_{\max} = 65$, $T \in [58, 65]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [196] $71231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1223 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 68$, $\Delta = -113$, $\det = -225$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 57.52 \rightarrow 58$, $T_{\max} = 68$, $T \in [58, 68]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [197] $712231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -225$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 57.52 \rightarrow 58$, $T_{\max} = 65$, $T \in [58, 65]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [198] $81232181231513221(12)122^1 31513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -105$, $\det = -225$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 57.52 \rightarrow 58$, $T_{\max} = 65$, $T \in [58, 65]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [199] $61231513221(12)12231513221(12)122315132^1 21(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -106$, $\det = -225$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 57.52 \rightarrow 58$, $T_{\max} = 65$, $T \in [58, 65]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [200] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 53$, $T_{\max} = 65$, $\Delta = -104$, $\det = -256$, $(n_+, n_-, n_0) = (1, 53, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 57.48 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[201] \quad 712231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -256, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.48 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[202] \quad 221(12)12231513221(12)12231513221^1 1(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -256, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.48 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[203] \quad 712231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -256, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.48 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[204] \quad 81232181231513221(12)12231513221^1 1(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -105, \det = -256, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.48 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$$[205] \quad 61231513221(12)12231513221(12)12^1 231513221(12)12231513218123218 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -106, \det = -256, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.48 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[206] \quad 1(12)12231513221^1 1(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -289, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.46 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[207] \quad 712231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -289, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.46 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[208] \quad 221(12)12231513221^1 1(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -289, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.46 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[209] \quad 71231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 68, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.46 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[210] \quad 712231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -104, \det = -289, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.46 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[211] \quad 812321812315132^1 21(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 65, \Delta = -105, \det = -289, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 57.46 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

- [212] $61231513221(12)12231513221(12)12231513221(12)122^131513218123218+\hat{1}\rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -106, \det = -289, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.46 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [213] $71231513221(12)122315132^121(12)12231513221(12)12231513221(12)1223 + \hat{1}\rightarrow 2$ $T = 53, T_{\max} = 68, \Delta = -113, \det = -324, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.43 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$
 $\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [214] $1(12)12231513221(12)12^1231513221(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -104, \det = -361, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.41 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [215] $712231513221(12)12231513221(12)1223151322^11(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -104, \det = -361, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.41 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [216] $221(12)12231513221(12)12^1231513221(12)12231513221(12)122315132171 + \hat{1}\rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -104, \det = -361, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.41 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [217] $712231513221(12)12231513221(12)1223151322^11(12)12231513221(12)122 + \hat{1}\rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -104, \det = -361, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.41 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [218] $812321812^131513221(12)12231513221(12)12231513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -105, \det = -361, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.41 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 645 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$
- [219] $81232181231513221(12)12^1231513221(12)12231513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -105, \det = -361, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.41 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$
- [220] $61231513221(12)12231513221(12)12231513221(12)122315132^118123218+\hat{1}\rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -106, \det = -361, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.41 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 75 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [221] $61231513221(12)12231513221(12)1223151322^11(12)12231513218123218+\hat{1}\rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -106, \det = -361, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.41 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$
- [222] $8123218123151322^11(12)12231513221(12)12231513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -105, \det = -441, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.37 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 776 & 56 \\ 2 & 56 & 3 \end{pmatrix}_{b_0}$
- [223] $61231513221(12)12231513221(12)12231513221(12)12^1231513218123218+\hat{1}\rightarrow 2$ $T = 53, T_{\max} = 65, \Delta = -106, \det = -441, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.37 \rightarrow 58, T_{\max} = 65, T \in [58, 65]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[224] $71231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 68, \Delta = -113, \det = -484, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.36 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[225] $71231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 68, \Delta = -113, \det = -529, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.34 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$$

[226] $71231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 68, \Delta = -113, \det = -529, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.34 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[227] $71231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 68, \Delta = -113, \det = -676, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.30 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[228] $71231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 68, \Delta = -113, \det = -729, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.29 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$

$$\begin{pmatrix} 0 & 27 & 2 \\ 27 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[229] $71231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 68, \Delta = -113, \det = -900, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.26 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$

$$\begin{pmatrix} 0 & 30 & 2 \\ 30 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

[230] $71231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 53, T_{\max} = 68, \Delta = -113, \det = -1089, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 57.24 \rightarrow 58, T_{\max} = 68, T \in [58, 68]$

$$\begin{pmatrix} 0 & 33 & 2 \\ 33 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$$

$T = 54$ [231] $181231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 2 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 61, T_{\max} = 66, T \in [61, 66]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[232] $81231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 61, T_{\max} = 66, T \in [61, 66]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[233] $81231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 61, T_{\max} = 66, T \in [61, 66]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[234] $22^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 61, T_{\max} = 66, T \in [61, 66]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[235] $22^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 61, T_{\max} = 66, T \in [61, 66]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[236] $81231513221(12)12231513221(12)12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 61, T_{\max} = 66, T \in [61, 66]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[237] $1(12)12231513221(12)12231513221(12)12231513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60.22 \rightarrow 61, T_{\max} = 68, T \in [61, 68]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[238] $81231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 9, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60.22 \rightarrow 61, T_{\max} = 66, T \in [61, 66]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[239] $32^1 21(12)12231513221(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60.22 \rightarrow 61, T_{\max} = 68, T \in [61, 68]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[240] $21(12)12231513221(12)12231513221(12)12231513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60.22 \rightarrow 61, T_{\max} = 68, T \in [61, 68]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[241] $812231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 9, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60.22 \rightarrow 61, T_{\max} = 68, T \in [61, 68]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[242] $81231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 9, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60.22 \rightarrow 61, T_{\max} = 66, T \in [61, 66]$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[243] $81231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 16, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.75 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[244] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[245] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$

- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
 [246] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [247] $1812^1 131513221(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [248] $81231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [249] $81231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [250] $322^1 1(12)12231513221(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 68, T \in [60, 68]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [251] $1(12)12231513221(12)12231513221(12)12231513221(12)1223151321812^1 32 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 70, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 70, T \in [60, 70]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
 [252] $221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
 [253] $221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$
 [254] $812231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 23 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 68, T \in [60, 68]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
 [255] $81231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12222 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
 [256] $81232181231513221(12)12231513221(12)12231513221(12)122315132^1 171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60, T_{\max} = 66, T \in [60, 66]$
- $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

- [257] $7122^1 31513221(12)12231513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -104$, $\det = 25$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.44 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [258] $1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.22 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [259] $81231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.22 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [260] $81231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.22 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [261] $3221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 68$, $\Delta = -112$, $\det = 36$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.22 \rightarrow 60$, $T_{\max} = 68$, $T \in [60, 68]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [262] $221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.22 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [263] $8122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 68$, $\Delta = -112$, $\det = 36$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.22 \rightarrow 60$, $T_{\max} = 68$, $T \in [60, 68]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [264] $81231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12222 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 36$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.22 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [265] $712^1 231513221(12)12231513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -104$, $\det = 36$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.22 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [266] $1(12)12231513221(12)12231513221(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [267] $1(12)122^1 31513221(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
- [268] $1(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta =$

-103, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[269] $1(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[270] $1(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[271] $1(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[272] $181231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)122 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[273] $1812315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[274] $812^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[275] $81231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[276] $81231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[277] $812^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[278] $221(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[279] $221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151321812 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

- $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [280] $221(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [281] $221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151322181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [282] $812^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 68$, $\Delta = -112$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 68$, $T \in [60, 68]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [283] $812^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [284] $81231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [285] $81232181231513221(12)12231513221(12)12231513221(12)122^1 315132171 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -104$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [286] $812^1 32181231513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -104$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$
- [287] $7122315132^1 21(12)12231513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -104$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [288] $712231513221(12)12231513221(12)12231513221(12)122315132181232^1 18 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -104$, $\det = 49$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 59.06 \rightarrow 60$, $T_{\max} = 66$, $T \in [60, 66]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [289] $1(12)12^1 231513221(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 64$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.94 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [290] $1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 68$, $\Delta = -112$, $\det = 64$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.94 \rightarrow 59$, $T_{\max} = 68$, $T \in [59, 68]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$

- [291] $1(12)12^{21}1^{23}1513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.94 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [292] $1(12)12^{21}1^{23}1513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.94 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [293] $181231513221(12)12231513221(12)12231513221(12)12231513221^{11}(12)122 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.94 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 7 & 7 \\ 2 & 7 & 3 \end{pmatrix}_{b_0}$
- [294] $21(12)122^{21}1^{31}513221(12)12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.94 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [295] $221(12)12^{21}1^{23}1513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.94 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [296] $221(12)12^{21}1^{23}1513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.94 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [297] $1(12)12231513221(12)12231513221(12)12231513221(12)12^{21}1^{23}15132181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [298] $1(12)12231513221(12)12231513221(12)122315132^{21}21(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [299] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^{21}18123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [300] $1(12)12^{21}1^{23}1513221(12)12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [301] $1(12)12231513221(12)12231513221(12)12231513221(12)12^{21}1^{23}151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$

- [302] $1(12)^{21}12231513221(12)^112231513221(12)122315132^121(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [303] $1(12)^{21}12231513221(12)^112231513221(12)122315132^121(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [304] $1(12)^{21}12231513221(12)^112231513221(12)12231513221(12)12^123151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [305] $181231513221(12)122^131513221(12)12231513221(12)^112231513221(12)122 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [306] $18123151322^11(12)12231513221(12)12231513221(12)^112231513221(12)122 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [307] $81231513221(12)12231513221(12)122315132^121(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [308] $81231513221(12)12231513221(12)12231513221(12)12^1231513221(12)^{2221}1 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [309] $81231513221(12)12231513221(12)12231513221(12)12^1231513221(12)^{221}12 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [310] $81231513221(12)12231513221(12)122315132^121(12)12231513221(12)^{221}12 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [311] $1(12)122^131513221(12)12231513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 70, \Delta = -114, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 70, T \in [59, 70]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [312] $21(12)12231513221(12)12231513221(12)12231513221(12)122315132^118123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$

- [313] $21(12)12^1 231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [314] $221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [315] $221(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [316] $221(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [317] $221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [318] $81231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [319] $81231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [320] $81232181231513221(12)12231513221(12)122315132^1 21(12)122315132171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [321] $81232181231513221(12)12231513221(12)12231513221(12)12^1 2315132171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [322] $71223151322^1 1(12)12231513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [323] $712231513221(12)122^1 31513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.84 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [324] $3221(12)12231513221(12)12231513221(12)12231513221(12)122^1 315132181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 100, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.76 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[325] \quad 1(12)12^1 231513221(12)12231513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 70, \Delta = -114, \det = 100, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.76 \rightarrow 59, T_{\max} = 70, T \in [59, 70]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[326] \quad 8122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 100, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.76 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[327] \quad 1(12)12231513221(12)12231513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[328] \quad 1(12)12231513221(12)12231513221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[329] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513218123 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.69 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[330] \quad 1(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[331] \quad 1(12)12231513221(12)12231513221(12)1223151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[332] \quad 1(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[333] \quad 1(12)12231513221(12)12231513221(12)1223151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[334] \quad 181231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[335] \quad 181231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

- [336] $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
 ${}^1 81231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [337] ${}^1 81231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [338] ${}^1 81231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12+\hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [339] ${}^1 81231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12+\hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [340] ${}^1 21(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.69 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [341] ${}^1 221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [342] ${}^1 221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [343] ${}^1 221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [344] ${}^1 221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [345] ${}^1 81231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
- [346] ${}^1 81231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$

- $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [347] $81232^1 181231513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 219 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
 [348] $81232181231513221(12)12231513221(12)122^1 31513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
 [349] $81232181231513221(12)12231513221(12)1223151322^1 1(12)122315132171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
 [350] $712231513221(12)12231513221(12)12231513221(12)1223151321812^1 3218 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 21 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
 [351] $712231513221(12)122315132^1 21(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
 [352] $712231513221(12)12^1 231513221(12)12231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.69 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$
 [353] $1^{21}(12)12231513221(12)122315132^1 21(12)12231513221(12)122315132181^1 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
 [354] $1^{21}(12)122315132^1 21(12)12231513221(12)12231513221(12)122315132181^1 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$
 [355] $1^{21}(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.59 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
 [356] $1^{21}(12)122315132^1 21(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [357] $1^{21}(12)12231513221(12)122315132^1 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$
 [358] $1^{21}(12)122315132^1 21(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta =$

-103 , $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 58.59 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[359] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 58.59 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[360] $181231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 58.59 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[361] $181231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)122 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 58.59 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[362] $812315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 58.59 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[363] $81231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 58.59 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[364] $81231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 58.59 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[365] $812315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 58.59 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[366] $3221(12)122^1 31513221(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 68$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 58.59 \rightarrow 59$, $T_{\max} = 68$, $T \in [59, 68]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[367] $3221(12)12231513221(12)12231513221(12)12231513221(12)12^1 2315132181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 68$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 58.59 \rightarrow 59$, $T_{\max} = 68$, $T \in [59, 68]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[368] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 181232 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 70$, $\Delta = -114$, $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 58.59 \rightarrow 59$, $T_{\max} = 70$, $T \in [59, 70]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[369] $21(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 68$, $\Delta = -112$, $\det = 169$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 58.59 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[370] \quad 221(12)12231513221(12)122315132^1 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[371] \quad 221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[372] \quad 221(12)12231513221(12)122315132^1 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[373] \quad 221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[374] \quad 81223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.59 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[375] \quad 812231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.59 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[376] \quad 81231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[377] \quad 812315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 38 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[378] \quad 81232181231513221(12)122315132^1 21(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -104, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[379] \quad 712231513221(12)12231513221(12)122^1 31513221(12)12231513218123218 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -104, \det = 169, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.59 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[380] \quad 1(12)12231513221(12)12231513221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 196, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 58.55 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

- $$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$
- [381] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1231513218123 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 196, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.55 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$
- [382] $1(12)12231513221(12)12231513221(12)12^1231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 196, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.55 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$
- [383] $1(12)12231513221(12)12231513221(12)12^1231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 196, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.55 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$
- [384] $181231513221(12)12231513221^1(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 196, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.55 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$
- [385] $81231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 196, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.55 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$
- [386] $81231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 196, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.55 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$
- [387] $3221(12)12231513221(12)12231513221(12)122315132^121(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 196, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.55 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$
- [388] $21(12)12231513221(12)12231513221(12)12231513221(12)12^1231513218123 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 196, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.55 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$
- [389] $221(12)12231513221(12)12231513221(12)12^1231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 196, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.55 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$
- [390] $221(12)12231513221(12)12231513221(12)12^1231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -103, \det = 196, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.55 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$
- [391] $812231513221(12)122^131513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 196, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.55 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$

- [403] $1(12)12231513221(12)12231513221(12)122^1315132181232 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 70, \Delta = -114, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 70, T \in [59, 70]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [404] $21(12)12231513221(12)12231513221(12)122^131513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [405] $221(12)12231513221(12)122^131513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [406] $221(12)12231513221(12)122^131513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [407] $812231513221(12)12231513221(12)12231513221(12)1223151322^11(12)1223 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [408] $81231513221(12)122^131513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [409] $81232181231513221(12)122^131513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [410] $712231513221(12)12231513221(12)122315132^121(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 225, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.52 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [424] $81232181231513221(12)1223151322^11(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 256, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [412] $1(12)122315132^121(12)12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [413] $1(12)12231513221(12)12231513221(12)1223151322^11(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 256, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [414] $1(12)12231513221(12)1223151322^11(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

- $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [415] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [416] $181231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [417] $81231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [418] $81231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [419] $21(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 68$, $\Delta = -112$, $\det = 256$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 68$, $T \in [59, 68]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [420] $21(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 68$, $\Delta = -112$, $\det = 256$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 68$, $T \in [59, 68]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [421] $221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [422] $221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
- [423] $81231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- [411] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -103$, $\det = 256$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [425] $712231513221(12)12231513221(12)12^1 231513221(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 54$, $T_{\max} = 66$, $\Delta = -104$, $\det = 256$, $(n_+, n_-, n_0) = (1, 54, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 58.48 \rightarrow 59$, $T_{\max} = 66$, $T \in [59, 66]$

- $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
 [426] $1(12)12231513221^1 1(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.46 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
 [427] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.46 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
- $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
 [428] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.46 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
- $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
 [429] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.46 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
- $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
 [430] $181231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)122 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.46 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
- $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
 [431] $8123151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.46 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
- $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
 [432] $8123151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.46 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
- $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
 [433] $3221(12)12231513221(12)12231513221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.46 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
- $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
 [434] $1(12)12231513221(12)12231513221(12)122315132^1 21(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 70, \Delta = -114, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.46 \rightarrow 59, T_{\max} = 70, T \in [59, 70]$
- $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
 [435] $21(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.46 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
- $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
 [436] $221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.46 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[437] $221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.46 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[438] $812231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.46 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[439] $8123151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.46 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[440] $812321812315132^1 21(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.46 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

[441] $712231513221(12)12231513221(12)12231513221(12)122^1 31513218123218 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 289, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.46 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[442] $3221(12)12231513221(12)12231513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 324, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.43 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[443] $812231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 324, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.43 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[444] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$$

[445] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.41 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[446] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.41 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[447] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

- [448] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [449] $181231513221(12)12231513221(12)12231513221^1(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$
- [450] $81231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$
- [451] $81231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$
- [452] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 70, \Delta = -114, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 70, T \in [59, 70]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [453] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 70, \Delta = -114, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 70, T \in [59, 70]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [454] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1 2315132181232 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 70, \Delta = -114, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 70, T \in [59, 70]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [455] $21(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [456] $21(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [457] $221(12)12231513221(12)12^1 231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [458] $221(12)12231513221(12)12^1 231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

- [459] $81231513221(12)^1 12^1 231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$
- [460] $81232181231513221(12)12^1 231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$
- [461] $812321812^1 31513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 645 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$
- [462] $712231513221(12)12231513221(12)1223151322^1 1(12)12231513218123218 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$
- [463] $712231513221(12)12231513221(12)12231513221(12)122315132^1 18123218 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.41 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 75 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [464] $1(12)12231513221(12)^1 1223151322^1 1(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 441, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.37 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [465] $1(12)1223151322^1 1(12)^1 12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 441, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.37 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$
- [466] $1(12)12231513221(12)^1 12231513221(12)1223151322^1 1(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 70, \Delta = -114, \det = 441, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.37 \rightarrow 59, T_{\max} = 70, T \in [59, 70]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [467] $1(12)12231513221(12)^1 122315132^1 21(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 70, \Delta = -114, \det = 441, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.37 \rightarrow 59, T_{\max} = 70, T \in [59, 70]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$
- [468] $21(12)1223151322^1 1(12)^1 12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 441, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.37 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$
- [469] $21(12)12231513221(12)^1 1223151322^1 1(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 441, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.37 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [470] $8123218123151322^1 1(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 66, \Delta = -104, \det = 441, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.37 \rightarrow 59, T_{\max} = 66, T \in [59, 66]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 776 & 56 \\ 2 & 56 & 3 \end{pmatrix}_{b_0} \\
[471] & 712231513221(12)12231513221(12)12231513221(12)12^1 231513218123218 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 66, \Delta = -104, \det = 441, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.37 \rightarrow 59, T_{\max} = 66, T \in [59, 66] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[472] & 3221(12)12231513221(12)122315132^1 21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 484, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.36 \rightarrow 59, T_{\max} = 68, T \in [59, 68] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[473] & 812231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 484, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.36 \rightarrow 59, T_{\max} = 68, T \in [59, 68] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[474] & 3221(12)12231513221(12)12231513221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 529, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.34 \rightarrow 59, T_{\max} = 68, T \in [59, 68] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[475] & 3221(12)122315132^1 21(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 529, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.34 \rightarrow 59, T_{\max} = 68, T \in [59, 68] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[476] & 1(12)12231513221(12)122^1 31513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 70, \Delta = -114, \det = 529, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.34 \rightarrow 59, T_{\max} = 70, T \in [59, 70] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0} \\
[477] & 812231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 529, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.34 \rightarrow 59, T_{\max} = 68, T \in [59, 68] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[478] & 812231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 529, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.34 \rightarrow 59, T_{\max} = 68, T \in [59, 68] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[479] & 1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 576, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.33 \rightarrow 59, T_{\max} = 68, T \in [59, 68] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[480] & 1(12)12231513221(12)12231513221(12)12^1 231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 70, \Delta = -114, \det = 576, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.33 \rightarrow 59, T_{\max} = 70, T \in [59, 70] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[481] & 21(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 54, T_{\max} = 68, \Delta = -112, \det = 576, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 58.33 \rightarrow 59, T_{\max} = 68, T \in [59, 68] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [482] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 70, \Delta = -114, \det = 625, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.31 \rightarrow 59, T_{\max} = 70, T \in [59, 70]$
 $\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$
- [483] $3221(12)12231513221(12)122^1 31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 676, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.30 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 26 & 2 \\ 26 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [484] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 70, \Delta = -114, \det = 676, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.30 \rightarrow 59, T_{\max} = 70, T \in [59, 70]$
 $\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$
- [485] $812231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 676, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.30 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 26 & 2 \\ 26 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [486] $3221(12)12231513221(12)1223151322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 729, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.29 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 27 & 2 \\ 27 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [487] $812231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 729, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.29 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 27 & 2 \\ 27 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [488] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 70, \Delta = -114, \det = 841, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.27 \rightarrow 59, T_{\max} = 70, T \in [59, 70]$
 $\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$
- [489] $3221(12)1223151322^1 1(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 900, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.26 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 30 & 2 \\ 30 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$
- [490] $812231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 900, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.26 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 30 & 2 \\ 30 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$
- [491] $3221(12)12231513221(12)12^1 231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 1089, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.24 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 33 & 2 \\ 33 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$
- [492] $812231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 54, T_{\max} = 68, \Delta = -112, \det = 1089, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 58.24 \rightarrow 59, T_{\max} = 68, T \in [59, 68]$
 $\begin{pmatrix} 0 & 33 & 2 \\ 33 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$

$T = 55$ [493] $81232181231513221(12)12231513221(12)12231513221(12)122315132^1181 + \hat{1} \rightarrow 2$ $T = 55, T_{\max} = 67, \Delta = -103, \det = -25, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60.44 \rightarrow 61, T_{\max} = 67, T \in [61, 67]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[494] $81232181231513221(12)12231513221(12)12231513221(12)122315132^11812 + \hat{1} \rightarrow 2$ $T = 55, T_{\max} = 67, \Delta = -103, \det = -25, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60.44 \rightarrow 61, T_{\max} = 67, T \in [61, 67]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[495] $81232181231513221(12)12231513221(12)12231513221(12)122315132^12181 + \hat{1} \rightarrow 2$ $T = 55, T_{\max} = 67, \Delta = -103, \det = -25, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60.44 \rightarrow 61, T_{\max} = 67, T \in [61, 67]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$

[496] $81232181231513221(12)12231513221(12)12231513221(12)122315132^1181 + \hat{1} \rightarrow 2$ $T = 55, T_{\max} = 67, \Delta = -103, \det = -36, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60.22 \rightarrow 61, T_{\max} = 67, T \in [61, 67]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$

[497] $812^132181231513221(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 55, T_{\max} = 67, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60.06 \rightarrow 61, T_{\max} = 67, T \in [61, 67]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$

[498] $81232181231513221(12)12231513221(12)12231513221(12)122^1315132181 + \hat{1} \rightarrow 2$ $T = 55, T_{\max} = 67, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60.06 \rightarrow 61, T_{\max} = 67, T \in [61, 67]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[499] $812^132181231513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 55, T_{\max} = 67, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60.06 \rightarrow 61, T_{\max} = 67, T \in [61, 67]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$

[500] $81232181231513221(12)12231513221(12)12231513221(12)122^13151321812 + \hat{1} \rightarrow 2$ $T = 55, T_{\max} = 67, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60.06 \rightarrow 61, T_{\max} = 67, T \in [61, 67]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[501] $812^132181231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 55, T_{\max} = 67, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60.06 \rightarrow 61, T_{\max} = 67, T \in [61, 67]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$

[502] $81232181231513221(12)12231513221(12)12231513221(12)122^13151322181 + \hat{1} \rightarrow 2$ $T = 55, T_{\max} = 67, \Delta = -103, \det = -49, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 60.06 \rightarrow 61, T_{\max} = 67, T \in [61, 67]$
 $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$

[503] $81232181231513221(12)12231513221(12)122315132^121(12)122315132181 + \hat{1} \rightarrow 2$ $T = 55, T_{\max} = 67, \Delta = -103, \det = -81, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.84 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[504] $81232181231513221(12)12231513221(12)12231513221(12)12^12315132181 + \hat{1} \rightarrow 2$ $T = 55, T_{\max} = 67, \Delta = -103, \det = -81, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.84 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$

- [517] $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$
 $81232^1 181231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -121, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.69 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 219 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [518] $81232181231513221(12)122315132^1 21(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.59 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [519] $81232181231513221(12)122315132^1 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.59 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [520] $81232181231513221(12)122315132^1 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -169, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.59 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$
- [521] $81232181231513221(12)12231513221(12)12^1 231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -196, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.55 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [522] $81232181231513221(12)12231513221(12)12^1 231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -196, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.55 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [523] $81232181231513221(12)12231513221(12)12^1 231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -196, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.55 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$
- [524] $81232181231513221(12)122^1 31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -225, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.52 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [525] $81232181231513221(12)122^1 31513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -225, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.52 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [526] $81232181231513221(12)122^1 31513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -225, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.52 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- [527] $81232181231513221(12)1223151322^1 1(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -256, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.48 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [528] $81232181231513221(12)1223151322^1 1(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -256, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.48 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$

- [529] $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
 $81232181231513221(12)1223151322^11(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -256, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.48 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
- [530] $812321812315132^121(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -289, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.46 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$
- [531] $812321812315132^121(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -289, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.46 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$
- [532] $812321812315132^121(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -289, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.46 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$
- [533] $81232181231513221(12)12^1231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -361, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.41 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$
- [534] $812321812^131513221(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -361, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.41 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 645 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$
- [535] $812321812^131513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -361, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.41 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 645 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$
- [536] $81232181231513221(12)12^1231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -361, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.41 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$
- [537] $81232181231513221(12)12^1231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -361, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.41 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$
- [538] $812321812^131513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -361, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.41 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 645 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$
- [539] $8123218123151322^11(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -441, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.37 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 776 & 56 \\ 2 & 56 & 3 \end{pmatrix}_{b_0}$
- [540] $8123218123151322^11(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 55, T_{\max} = 67, \Delta = -103, \det = -441, (n_+, n_-, n_0) = (1, 55, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 59.37 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 776 & 56 \\ 2 & 56 & 3 \end{pmatrix}_{b_0}$$

[541] $8123218123151322^1 1(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 55$, $T_{\max} = 67$, $\Delta = -103$, $\det = -441$, $(n_+, n_-, n_0) = (1, 55, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 59.37 \rightarrow 60, T_{\max} = 67, T \in [60, 67]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 776 & 56 \\ 2 & 56 & 3 \end{pmatrix}_{b_0}$$

$T = 58$ [542] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 71$, $\Delta = -111$, $\det = 4$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[543] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 4$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[544] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 4$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[545] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 4$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[546] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 71$, $\Delta = -111$, $\det = 4$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 65, T_{\max} = 71, T \in [65, 71]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[547] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 4$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[548] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 4$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[549] $3221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 71$, $\Delta = -111$, $\det = 4$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[550] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 2 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 4$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[551] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 4$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[552] \quad 3221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 2 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 65, T_{\max} = 71, T \in [65, 71]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[553] \quad 22^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[554] \quad 221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[555] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.22 \rightarrow 65, T_{\max} = 71, T \in [65, 71]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[556] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 9, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.22 \rightarrow 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[557] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 9, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.22 \rightarrow 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[558] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.22 \rightarrow 65, T_{\max} = 71, T \in [65, 71]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[559] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 9, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.22 \rightarrow 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[560] \quad 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.22 \rightarrow 65, T_{\max} = 71, T \in [65, 71]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[561] \quad 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.22 \rightarrow 65, T_{\max} = 71, T \in [65, 71]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[562] \quad 3221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.22 \rightarrow 65, T_{\max} = 71, T \in [65, 71]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[563] \quad 32^{\overset{1}{1}}21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 9, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.22 \rightarrow 65, T_{\max} = 71, T \in [65, 71]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[564] \quad 221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^{\overset{1}{1}}22+\hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 9, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.22 \rightarrow 65, T_{\max} = 70, T \in [65, 70]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[565] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^{\overset{21}{1}}222+\hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 16, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.75 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[566] \quad 221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^{\overset{1}{1}}222+\hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 16, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.75 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[577] \quad 221(12)12231513221(12)12231513221(12)12231513221(12)122315132^{\overset{1}{1}}21(12)12222+\hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 25, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.44 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[568] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^{\overset{1}{1}}21(12)1222+\hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 25, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.44 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[569] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^{\overset{221}{1}}21(12)12+\hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 25, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.44 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[570] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^{\overset{1}{1}}21(12)12222+\hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 25, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.44 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[571] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^{\overset{221}{1}}23+\hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.44 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[572] \quad 1(12)122^{\overset{221}{1}}31513221(12)12231513221(12)12231513221(12)12231513221(12)122+\hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 25, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.44 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[573] \quad 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^{\overset{1}{1}}23+\hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 25, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.44 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

1176

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[585] \quad 3221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 36, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.22 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[586] \quad 221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 1(12)1222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 36, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.22 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[598] \quad 21(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[588] \quad 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[589] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[590] \quad 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[591] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[592] \quad 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[593] \quad 1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.06 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[594] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)122 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 6 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[595] \quad 1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[596] & 21(12)12^{21}1^{23}1513221(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.06 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[597] & 21(12)122315132^{21}21(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[587] & 1(12)12231513221(12)12231513221(12)12231513221(12)122^{21}31513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[599] & 3221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^{21}1(12)1222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.06 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[600] & 221(12)122^{21}31513221(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[601] & 221(12)12231513221(12)12231513221(12)12231513221(12)122^{21}31513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 49, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.06 \rightarrow 64, T_{\max} = 70, T \in [64, 70] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[608] & 21(12)12231513221(12)12231513221(12)12231513221(12)1223151322^{21}1(12)122 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 64, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.94 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 7 & 7 \\ 2 & 7 & 3 \end{pmatrix}_{b_0} \\
[603] & 1(12)12^{21}1^{23}1513221(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 64, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.94 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[604] & 1(12)12^{21}1^{23}1513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 64, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.94 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[605] & 1(12)12^{21}1^{23}1513221(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 64, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.94 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[606] & 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^{21}1(12)122 + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 64, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.94 \rightarrow 63, T_{\max} = 70, T \in [63, 70]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 7 & 7 \\ 2 & 7 & 3 \end{pmatrix}_{b_0} \\
[607] & \begin{matrix} 3221 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 64, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.94 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[602] & \begin{matrix} 1 \\ 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 64, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.94 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[609] & \begin{matrix} 1 \\ 221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 64, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.94 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[616] & \begin{matrix} 21 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[611] & \begin{matrix} 1 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[612] & \begin{matrix} 21 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[613] & \begin{matrix} 21 \\ 1(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[614] & \begin{matrix} 21 \\ 1(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[615] & \begin{matrix} 21 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[610] & \begin{matrix} 1 \\ 1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[617] & \begin{matrix} 21 \\ 1(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 70, T \in [63, 70]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[618] & \begin{matrix} 2221 \\ 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[619] & \begin{matrix} 2221 \\ 1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[620] & \begin{matrix} 3221 \\ 21(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[621] & \begin{matrix} 3221 \\ 21(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[622] & \begin{matrix} 221 \\ 21(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[623] & \begin{matrix} 221 \\ 21(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[624] & \begin{matrix} 1 \\ 221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[625] & \begin{matrix} 1 \\ 221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 81, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.84 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[626] & \begin{matrix} 221 \\ 1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 100, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.76 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[627] & \begin{matrix} 21 \\ 21(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 100, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.76 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 69 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[628] & \begin{matrix} 1 \\ 3221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 100, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.76 \rightarrow 63, T_{\max} = 71, T \in [63, 71]
\end{aligned}$$

1181

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[643] $221(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 121$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 62.69 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[644] $221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 121$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 62.69 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[630] $1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 71$, $\Delta = -111$, $\det = 121$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 62.69 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[634] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 121$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 62.69 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[629] $1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1223 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 71$, $\Delta = -111$, $\det = 121$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 62.69 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[658] $21(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 71$, $\Delta = -111$, $\det = 169$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 62.59 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[649] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 169$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 62.59 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$$

[650] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 169$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 62.59 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[651] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 70$, $\Delta = -102$, $\det = 169$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 62.59 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[652] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 71$, $\Delta = -111$, $\det = 169$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 62.59 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[653] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 58$, $T_{\max} = 71$, $\Delta = -111$, $\det = 169$, $(n_+, n_-, n_0) = (1, 58, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 62.59 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

1183

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 90 & 18 \\ 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[662] $3221(12)12231513221(12)12231513221(12)12231513221(12)12^{1231513221}(12)1222 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.59 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[663] $221(12)12231513221(12)122315132^{121}(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 169, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.59 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[664] $221(12)122315132^{121}(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 169, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.59 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 129 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[666] $1(12)12231513221(12)12231513221(12)12^{1231513221}(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 196, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.55 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[665] $1(12)12231513221(12)12231513221(12)12231513221(12)12^{1231513221}(12)1223 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 196, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.55 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[667] $1(12)12231513221(12)12231513221(12)12^{1231513221}(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 196, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.55 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[668] $1(12)12231513221(12)12231513221(12)12^{1231513221}(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 196, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.55 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[669] $1(12)12231513221(12)122^{131513221}(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 196, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.55 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[670] $1(12)12231513221(12)1223151322^{11}(12)12231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 196, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.55 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[671] $21(12)12231513221(12)122^{131513221}(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 196, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.55 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[672] $21(12)1223151322^{13221}(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 196, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.55 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

- $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [673] $\overset{221}{21(12)12231513221(12)1223151322^11(12)12231513221(12)122} + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 196, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.55 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- $\overset{1}{3221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)1222} + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 196, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.55 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- $\overset{1}{221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12222} + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 196, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.55 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
 $\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- $\overset{1}{1(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)1223} + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 225, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.52 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- $\overset{21}{1(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)1222} + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 225, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.52 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- $\overset{21}{1(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12} + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 225, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.52 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$
- $\overset{21}{1(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12222} + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 225, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.52 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
- $\overset{221}{1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)1223} + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 225, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.52 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- $\overset{2221}{1(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)122} + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 225, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.52 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- $\overset{21}{21(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)1223} + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 225, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.52 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
- $\overset{3221}{21(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12} + \hat{1} \rightarrow 2 \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 225, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.52 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$

- $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
 [684] $21(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 225, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.52 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
 $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$
 [685] $3221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 225, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.52 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$
- $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
 [686] $221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 225, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.52 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
- $\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$
 [687] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 256, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.48 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
 [688] $1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 256, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.48 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$
- $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
 [689] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 256, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.48 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
- $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
 [690] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 256, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.48 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
- $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$
 [691] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 256, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.48 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
- $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
 [692] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 256, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.48 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$
 $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$
- $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
 [693] $21(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 256, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.48 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$
- $\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$
 [694] $21(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 256, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 62.48 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[695] & \begin{matrix} 221 \\ 21(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 256, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.48 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[696] & \begin{matrix} 1 \\ 221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 256, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.48 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[697] & \begin{matrix} 1 \\ 1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 289, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.46 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[698] & \begin{matrix} 21 \\ 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 289, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.46 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0} \\
[699] & \begin{matrix} 21 \\ 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 289, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.46 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0} \\
[700] & \begin{matrix} 21 \\ 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 289, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.46 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[701] & \begin{matrix} 221 \\ 1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 289, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.46 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[702] & \begin{matrix} 2221 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 289, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.46 \rightarrow 63, T_{\max} = 70, T \in [63, 70] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[703] & \begin{matrix} 21 \\ 21(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 289, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.46 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[704] & \begin{matrix} 3221 \\ 21(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 289, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.46 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[705] & \begin{matrix} 221 \\ 21(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)122 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 70, \Delta = -102, \det = 289, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.46 \rightarrow 63, T_{\max} = 70, T \in [63, 70]
\end{aligned}$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[706] $3221(12)12231513221(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 289, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.46 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[707] $221(12)12231513221(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 289, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.46 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[708] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223+\hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 324, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.43 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[709] $21(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223+\hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 324, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.43 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[710] $3221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 324, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.43 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 18 & 2 \\ 18 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[715] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 361, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.41 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[712] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1223+\hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 361, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.41 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[713] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 361, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.41 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$$

[714] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 361, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.41 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$$

[711] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1223+\hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 361, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.41 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[716] $1(12)12231513221(12)12231513221(12)12231513221(12)122+\hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 361, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.41 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[720] $221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 361, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.41 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[719] $21(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)122 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 70, \Delta = -102, \det = 361, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.41 \rightarrow 63, T_{\max} = 70, T \in [63, 70]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[717] $21(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 361, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.41 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[718] $21(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 361, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.41 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[724] $21(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 441, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.37 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[723] $21(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 441, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.37 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[722] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 441, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.37 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[721] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 441, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.37 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[725] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 484, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.36 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[726] $21(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 484, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.36 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[727] $3221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2$ $T = 58, T_{\max} = 71, \Delta = -111, \det = 484, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 62.36 \rightarrow 63, T_{\max} = 71, T \in [63, 71]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[728] & \begin{matrix} 221 \\ 1(12)12231513221(12)1223151322^1 1(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 529, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.34 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[729] & \begin{matrix} 221 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 529, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.34 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[731] & \begin{matrix} 21 \\ 21(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 529, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.34 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[732] & \begin{matrix} 1 \\ 3221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 529, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.34 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[733] & \begin{matrix} 1 \\ 3221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 529, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.34 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 35 & 3 \end{pmatrix}_{b_0} \\
[730] & \begin{matrix} 21 \\ 21(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 529, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.34 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 298 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[734] & \begin{matrix} 1 \\ 1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 576, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.33 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[735] & \begin{matrix} 3221 \\ 21(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 576, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.33 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[736] & \begin{matrix} 221 \\ 1(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 676, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.30 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 26 & 2 \\ 26 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[737] & \begin{matrix} 21 \\ 21(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 676, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.30 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 26 & 2 \\ 26 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[738] & \begin{matrix} 1 \\ 3221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 676, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.30 \rightarrow 63, T_{\max} = 71, T \in [63, 71]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 26 & 2 \\ 26 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[739] & \begin{matrix} 221 \\ 1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 729, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.29 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 27 & 2 \\ 27 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[740] & \begin{matrix} 21 \\ 21(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 729, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.29 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 27 & 2 \\ 27 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[741] & \begin{matrix} 1 \\ 3221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 729, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.29 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 27 & 2 \\ 27 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[743] & \begin{matrix} 21 \\ 21(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 900, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.26 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 30 & 2 \\ 30 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0} \\
[742] & \begin{matrix} 221 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 900, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.26 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 30 & 2 \\ 30 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0} \\
[744] & \begin{matrix} 1 \\ 3221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 900, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.26 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 30 & 2 \\ 30 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0} \\
[746] & \begin{matrix} 21 \\ 21(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 1089, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.24 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 33 & 2 \\ 33 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0} \\
[747] & \begin{matrix} 1 \\ 3221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1222 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 1089, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.24 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 33 & 2 \\ 33 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0} \\
[745] & \begin{matrix} 221 \\ 1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1223 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 58, T_{\max} = 71, \Delta = -111, \det = 1089, (n_+, n_-, n_0) = (1, 58, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 62.24 \rightarrow 63, T_{\max} = 71, T \in [63, 71] \\
& \begin{pmatrix} 0 & 33 & 2 \\ 33 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0} \\
T = 59 & [749] \begin{matrix} 22^1 1 \\ 81232181231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -4, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 66, T_{\max} = 71, T \in [66, 71] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[748] & \begin{matrix} 22^1 21 \\ 81232181231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \end{matrix} \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -4, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 66, T_{\max} = 71, T \in [66, 71]
\end{aligned}$$

- $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
 [750] 81232181231513221(12)12231513221(12)12231513221(12)12231513221(12)1222¹2+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -4$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 66$, $T_{\max} = 71$, $T \in [66, 71]$
 $\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$
- [752] 81232181231513221(12)12231513221(12)12231513221(12)12231513221(12)122¹22+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -9$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 65.22 \rightarrow 66$, $T_{\max} = 71$, $T \in [66, 71]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [751] 81232181231513221(12)12231513221(12)12231513221(12)12231513221(12)1²²²1+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -9$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 65.22 \rightarrow 66$, $T_{\max} = 71$, $T \in [66, 71]$
 $\begin{pmatrix} 0 & 3 & 2 \\ 3 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [753] 81232181231513221(12)12231513221(12)12231513221(12)12231513221(12)12¹222+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -16$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 64.75 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$
 $\begin{pmatrix} 0 & 4 & 2 \\ 4 & 11 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [754] 4132¹21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 64.44 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [755] 4132¹21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -25$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 64.44 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$
- [756] 81232181231513221(12)12231513221(12)12231513221(12)122315132²²²¹21(12)1+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 64.44 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [757] 81232181231513221(12)12231513221(12)12231513221(12)122315132²²¹21(12)12+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 64.44 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [758] 81232181231513221(12)12231513221(12)12231513221(12)122315132¹21(12)12222+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 64.44 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$
 $\begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [759] 81232181231513221(12)12231513221(12)12231513221(12)1223151322²²²¹1(12)1+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -36$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 64.22 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [760] 81232181231513221(12)12231513221(12)12231513221(12)1223151322²²¹1(12)12+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -36$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 64.22 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$
 $\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [761] 81232181231513221(12)12231513221(12)12231513221(12)1223151322¹1(12)12222+ $\hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta =$

-102, $\det = -36$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 64.22 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[763] $41322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 64.06 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[764] $812^1 32181231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 64.06 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[765] $81232181231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 64.06 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[766] $812^1 32181231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 64.06 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[767] $81232181231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 64.06 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[768] $812^1 32181231513221(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 64.06 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 90 & 18 \\ 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[769] $81232181231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12222 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -49$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 64.06 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[762] $41322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -49$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 64.06 \rightarrow 65$, $T_{\max} = 71$, $T \in [65, 71]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[770] $413221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -64$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.94 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[771] $413221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -64$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.94 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[776] $81232181231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -81$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.84 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[775] & 413221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.84 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[777] & 81232181231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.84 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[778] & 81232181231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.84 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[779] & 81232181231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.84 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[780] & 81232181231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12222+ \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.84 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[781] & 81232181231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12222+ \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.84 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[772] & 413221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.84 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[773] & 413221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12+ \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.84 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[774] & 413221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1+ \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.84 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[785] & 413221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.69 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[786] & 81232^1 181231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.69 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 219 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 63.59 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[796] & 413221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -110, \det = -169, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.59 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[797] & 81232181231513221(12)122315132^121(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -169, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.59 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[798] & 81232181231513221(12)122315132^121(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -169, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.59 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[799] & 81232181231513221(12)122315132^121(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -169, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.59 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[804] & 81232181231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.55 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[801] & 413221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -110, \det = -196, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.55 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[802] & 81232181231513221(12)12231513221(12)12^1231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.55 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[803] & 81232181231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.55 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[800] & 413221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -110, \det = -196, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.55 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[805] & 413221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -110, \det = -225, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.52 \rightarrow 64, T_{\max} = 71, T \in [64, 71] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[807] & 81232181231513221(12)122^131513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 59, T_{\max} = 71, \Delta = -102, \det = -225, (n_+, n_-, n_0) = (1, 59, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 63.52 \rightarrow 64, T_{\max} = 71, T \in [64, 71]
\end{aligned}$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[808] $81232181231513221(12)122^131513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -225$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 63.52 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[809] $81232181231513221(12)122^131513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -225$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 63.52 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[806] $413221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -225$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 63.52 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[815] $81232181231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 63.48 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[816] $81232181231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 63.48 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[810] $413221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -256$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 63.48 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[811] $413221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -256$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 63.48 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[812] $413221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -256$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 63.48 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[813] $413221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -256$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 63.48 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[814] $81232181231513221(12)1223151322^11(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 63.48 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[820] $812321812315132^121(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -289$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 63.46 \rightarrow 64, T_{\max} = 71, T \in [64, 71]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

- [818] $413221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -289$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 63.46 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [817] $413221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -289$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 63.46 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [819] $812321812315132^121(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -289$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 63.46 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$
- [821] $812321812315132^121(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -289$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 63.46 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$
- [822] $413221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -361$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 63.41 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [823] $413221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -361$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 63.41 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [824] $413221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -361$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 63.41 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$
- [825] $413221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -361$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 63.41 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [826] $812321812^131513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 63.41 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 645 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$
- [827] $81232181231513221(12)12^1231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 63.41 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$
- [828] $81232181231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 63.41 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$
- [829] $812321812^131513221(12)12231513221(12)12231513221(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta =$

-102, $\det = -361$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.41 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 645 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[830] $812321812^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.41 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 645 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[831] $81232181231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.41 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 626 & 50 \\ 2 & 50 & 3 \end{pmatrix}_{b_0}$$

[832] $413221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -441$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.37 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[833] $413221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -441$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.37 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[834] $413221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -441$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.37 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[835] $413221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -441$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.37 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[836] $8123218123151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -441$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.37 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 776 & 56 \\ 2 & 56 & 3 \end{pmatrix}_{b_0}$$

[837] $8123218123151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -441$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.37 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 776 & 56 \\ 2 & 56 & 3 \end{pmatrix}_{b_0}$$

[838] $8123218123151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12222 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -102$, $\det = -441$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.37 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 776 & 56 \\ 2 & 56 & 3 \end{pmatrix}_{b_0}$$

[840] $413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -576$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.33 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[839] $413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 59$, $T_{\max} = 71$, $\Delta = -110$, $\det = -576$, $(n_+, n_-, n_0) = (1, 59, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 63.33 \rightarrow 64$, $T_{\max} = 71$, $T \in [64, 71]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$$T = 60 \quad [842] \quad 1 \quad (12) \quad \overset{15132^1 21}{12231513221(12)12231513221(12)12231513221(12)12231513221(12)12} + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 65.44 \rightarrow 66, T_{\max} = 72, T \in [66, 72]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[841] \quad 1 \quad (12) \quad \overset{1}{12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151} + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 25, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 65.44 \rightarrow 66, T_{\max} = 72, T \in [66, 72]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[843] \quad 414132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 65.06 \rightarrow 66, T_{\max} = 72, T \in [66, 72]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[846] \quad 1 \quad (12) \quad \overset{151322^1 1}{12231513221(12)12231513221(12)12231513221(12)12231513221(12)12} + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 65.06 \rightarrow 66, T_{\max} = 72, T \in [66, 72]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[845] \quad 1 \quad (12) \quad \overset{1}{12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151} + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 49, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 65.06 \rightarrow 66, T_{\max} = 72, T \in [66, 72]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[844] \quad 414132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 65.06 \rightarrow 66, T_{\max} = 72, T \in [66, 72]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[848] \quad 41413221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 64, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.94 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[847] \quad 41413221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 64, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.94 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[849] \quad 1 \quad (12) \quad \overset{1}{122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1223151} + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.94 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[850] \quad 1 \quad (12) \quad \overset{315132^1 21}{12231513221(12)12231513221(12)12231513221(12)12231513221(12)1} + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 75, \Delta = -113, \det = 64, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.94 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[851] \quad 1 \quad (12) \quad \overset{1513221}{12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12} + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 64, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 64.94 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[852] $41413221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.84 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[853] $4141322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.84 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[854] $41413221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.84 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[855] $41413221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.84 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[856] $41413221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.84 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[857] $4141322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.84 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[858] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223151 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.84 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[859] $1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.84 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[860] $1 \begin{smallmatrix} 31513221 \\ (12) \end{smallmatrix} 12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 75, \Delta = -113, \det = 81, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.84 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[861] $1 \begin{smallmatrix} 1513221 \\ (12) \end{smallmatrix} 122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.84 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[862] $1 \begin{smallmatrix} 1513221 \\ (12) \end{smallmatrix} 12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.84 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[864] $(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 76, \Delta = -114, \det = 100, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.76 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[863] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 75, \Delta = -113, \det = 100, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.76 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[874] $(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 76, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.69 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[865] $41413221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 72, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.69 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[866] $41413221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 72, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.69 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[867] $41413221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 72, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.69 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[868] $41413221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 72, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.69 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[869] $1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 72, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.69 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[870] $1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1223151 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 72, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.69 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[871] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 75, \Delta = -113, \det = 121, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.69 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[875] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 21(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 76, \Delta = -114, \det = 121, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.69 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 87 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[872] & 1 \text{ (12) } 12^{\overline{1}} 231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.69 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[873] & 1 \text{ (12) } 122315132^{\overline{1}} 21(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 121, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.69 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[879] & 1 \text{ (12) } 122^{\overline{1}} 31513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 75, \Delta = -113, \det = 169, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.59 \rightarrow 65, T_{\max} = 75, T \in [65, 75] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[880] & 1 \text{ (12) } 12231513221(12)122^{\overline{1}} 31513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 169, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.59 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[876] & 41413221(12)12231513221(12)122^{\overline{1}} 31513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 169, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.59 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[877] & 41413221(12)12231513221(12)122^{\overline{1}} 31513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 169, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.59 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[878] & 1(12)12231513221(12)12231513221(12)122315132^{\overline{1}} 21(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 169, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.59 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[882] & 41413221(12)1223151322^{\overline{1}} 1(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 196, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.55 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[884] & 1 \text{ (12) } 1223151322^{\overline{1}} 1(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.55 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[881] & 41413221(12)1223151322^{\overline{1}} 1(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 196, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.55 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[883] & 1(12)12231513221(12)12231513221(12)12231513221(12)12^{\overline{1}} 231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.55 \rightarrow 65, T_{\max} = 72, T \in [65, 72]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[889] & \begin{matrix} 1513221 \\ 1 \end{matrix} (12) 12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 225, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.52 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[887] & \begin{matrix} 1 \\ 1 \end{matrix} (12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 225, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.52 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[888] & \begin{matrix} 31513221 \\ 1 \end{matrix} (12) 122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 75, \Delta = -113, \det = 225, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.52 \rightarrow 65, T_{\max} = 75, T \in [65, 75] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[890] & \begin{matrix} 1 \\ 1 \end{matrix} (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 76, \Delta = -114, \det = 225, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.52 \rightarrow 65, T_{\max} = 76, T \in [65, 76] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[885] & 41413221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 225, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.52 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[886] & 41413221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 225, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.52 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[899] & \begin{matrix} 1513221 \\ 1 \end{matrix} (12) 12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 256, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.48 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[895] & \begin{matrix} 1 \\ 1 \end{matrix} (12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 256, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.48 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[896] & \begin{matrix} 1 \\ 1 \end{matrix} (12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 256, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.48 \rightarrow 65, T_{\max} = 72, T \in [65, 72] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[897] & \begin{matrix} 31513221 \\ 1 \end{matrix} (12) 12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 75, \Delta = -113, \det = 256, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.48 \rightarrow 65, T_{\max} = 75, T \in [65, 75] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[898] & \begin{matrix} 1513221 \\ 1 \end{matrix} (12) 12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 256, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 64.48 \rightarrow 65, T_{\max} = 72, T \in [65, 72]
\end{aligned}$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[894] $41413221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -109$, $\det = 256$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.48 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[892] $41413221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -109$, $\det = 256$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.48 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[893] $41413221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -109$, $\det = 256$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.48 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[891] $41413221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -109$, $\det = 256$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.48 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[902] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -105$, $\det = 289$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.46 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[904] $1 \begin{smallmatrix} 1513221 \\ (12) \end{smallmatrix} 12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -105$, $\det = 289$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.46 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[905] $(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21 \begin{smallmatrix} 231513221 \\ (12) \end{smallmatrix} 1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 76$, $\Delta = -114$, $\det = 289$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.46 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[900] $41413221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -109$, $\det = 289$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.46 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[901] $41413221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -109$, $\det = 289$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.46 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[903] $1 \begin{smallmatrix} 31513221 \\ (12) \end{smallmatrix} 12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 75$, $\Delta = -113$, $\det = 289$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.46 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[906] $41413221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -109$, $\det = 361$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[907] $41413221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 361, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[908] $41413221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 361, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[909] $41413221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -109, \det = 361, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[910] $1(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 361, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[911] $1(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 361, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[912] $1 \begin{smallmatrix} 31513221 \\ (12) \end{smallmatrix} 12231513221(12)12231513221(12)12231513221(12)122^131513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 75, \Delta = -113, \det = 361, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[913] $1 \begin{smallmatrix} 31513221 \\ (12) \end{smallmatrix} 1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 75, \Delta = -113, \det = 361, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[914] $1 \begin{smallmatrix} 31513221 \\ (12) \end{smallmatrix} 12231513221(12)122315132^121(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 75, \Delta = -113, \det = 361, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[915] $1 \begin{smallmatrix} 1513221 \\ (12) \end{smallmatrix} 12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 361, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[916] $1 \begin{smallmatrix} 1513221 \\ (12) \end{smallmatrix} 12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 72, \Delta = -105, \det = 361, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[917] $(12)12231513221(12)12231513221(12)12231513221(12)122^131513221 \begin{smallmatrix} 231513221 \\ (12) \end{smallmatrix} 1 + \hat{1} \rightarrow 2 \quad T = 60, T_{\max} = 76, \Delta = -114, \det = 361, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.41 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[921] $41413221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -109$, $\det = 441$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.37 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[930] $(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 76$, $\Delta = -114$, $\det = 484$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.36 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[932] $(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 76$, $\Delta = -114$, $\det = 529$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.34 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

[931] $1(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 75$, $\Delta = -113$, $\det = 529$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.34 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

[933] $41413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -109$, $\det = 576$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.33 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[935] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223151 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -105$, $\det = 576$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.33 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[936] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 75$, $\Delta = -113$, $\det = 576$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.33 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[937] $1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -105$, $\det = 576$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.33 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[938] $(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 76$, $\Delta = -114$, $\det = 576$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.33 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[934] $41413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 72$, $\Delta = -109$, $\det = 576$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.33 \rightarrow 65, T_{\max} = 72, T \in [65, 72]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[939] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60$, $T_{\max} = 75$, $\Delta = -113$, $\det = 625$, $(n_+, n_-, n_0) = (1, 60, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 64.31 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$$

[940] $(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 76, \Delta = -114, \det = 625, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.31 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$$

[941] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 75, \Delta = -113, \det = 676, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.30 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

[942] $(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 76, \Delta = -114, \det = 676, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.30 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

[943] $(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 76, \Delta = -114, \det = 729, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.29 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 27 & 2 \\ 27 & 620 & 48 \\ 2 & 48 & 3 \end{pmatrix}_{b_0}$$

[945] $(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 76, \Delta = -114, \det = 841, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.27 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 724 & 52 \\ 2 & 52 & 3 \end{pmatrix}_{b_0}$$

[946] $(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 76, \Delta = -114, \det = 841, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.27 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[944] $1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 75, \Delta = -113, \det = 841, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.27 \rightarrow 65, T_{\max} = 75, T \in [65, 75]$$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[947] $(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 76, \Delta = -114, \det = 961, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.25 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 31 & 2 \\ 31 & 805 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0}$$

[948] $(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 60, T_{\max} = 76, \Delta = -114, \det = 1156, (n_+, n_-, n_0) = (1, 60, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 64.23 \rightarrow 65, T_{\max} = 76, T \in [65, 76]$$

$$\begin{pmatrix} 0 & 34 & 2 \\ 34 & 985 & 61 \\ 2 & 61 & 3 \end{pmatrix}_{b_0}$$

$T = 62$ [0] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.06 \rightarrow 68, T_{\max} = 74, T \in [68, 74]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[1] $21(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 49, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.06 \rightarrow 68, T_{\max} = 74, T \in [68, 74]$$

- $\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$
 [2] $1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 64, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.94 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [3] $21(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 64, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.94 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$
 $\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [4] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 81, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.84 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [5] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 81, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.84 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [6] $1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 81, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.84 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [7] $21(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 81, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.84 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [8] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 81, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.84 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [9] $21(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 81, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.84 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$
 $\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [10] $1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1223151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.69 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [11] $1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.69 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$
 $\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [12] $21(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151315 + \hat{1} \rightarrow 2$ $T = 62, T_{\max} = 74, \Delta = -107, \det = 121, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.69 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$

$$\begin{array}{l} \text{[13]} \quad 21(12)12231513221(12)12231513221(12)12231513221(12)1223151322\hat{\mathbf{1}}1(12)1223151315 + \hat{\mathbf{1}} \rightarrow \mathbf{2} \quad T = 62, \quad T_{\max} = \\ 74, \quad \Delta = -107, \quad \det = 121, \quad (n_+, n_-, n_0) = (1, 62, 0), \quad c_{\text{ext}} = 15.0 \end{array}$$

- $$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$
- [24] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 62, T_{\max} = 74, \Delta = -107, \det = 289, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.46 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$
- [25] $21(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 62, T_{\max} = 74, \Delta = -107, \det = 289, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.46 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$
- [26] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 62, T_{\max} = 74, \Delta = -107, \det = 361, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.41 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$
- [27] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 62, T_{\max} = 74, \Delta = -107, \det = 361, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.41 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$
- [28] $21(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 62, T_{\max} = 74, \Delta = -107, \det = 361, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.41 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$
- [29] $21(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 62, T_{\max} = 74, \Delta = -107, \det = 361, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.41 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$
- [30] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 62, T_{\max} = 74, \Delta = -107, \det = 441, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.37 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$
- [31] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 62, T_{\max} = 74, \Delta = -107, \det = 441, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.37 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$
- [32] $21(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 62, T_{\max} = 74, \Delta = -107, \det = 441, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.37 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$
- [33] $21(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 62, T_{\max} = 74, \Delta = -107, \det = 441, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.37 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$
- [34] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 62, T_{\max} = 74, \Delta = -107, \det = 576, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 66.33 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[35] $21(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 62, T_{\max} = 74, \Delta = -107, \det = 576, (n_+, n_-, n_0) = (1, 62, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 66.33 \rightarrow 67, T_{\max} = 74, T \in [67, 74]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$T = 63$ [36] $612^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 68.44 \rightarrow 69, T_{\max} = 75, T \in [69, 75]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[37] $612^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -25, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 68.44 \rightarrow 69, T_{\max} = 75, T \in [69, 75]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[40] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 68.06 \rightarrow 69, T_{\max} = 75, T \in [69, 75]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[39] $612315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 68.06 \rightarrow 69, T_{\max} = 75, T \in [69, 75]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[41] $612315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -49, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 68.06 \rightarrow 69, T_{\max} = 75, T \in [69, 75]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[38] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -49, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 68.06 \rightarrow 69, T_{\max} = 75, T \in [69, 75]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[43] $61231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -64, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.94 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[44] $1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.94 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[45] $61231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -64, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.94 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[42] $21(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -64, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.94 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

[57] $6123151322^{\hat{1}}1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

[58] $61231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221^1(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -81, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

[59] $1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -100, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

[61] $21(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

[62] $\overset{1}{1}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^{\color{red}1}315131613 + \hat{\color{red}1} \rightarrow \color{red}{2} \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -121, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

[63] $61231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -121, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

[64] 61231513221(12)122315132¹21(12)12231513221(12)12231513221¹(12)12231513221(12)12 + ¹ $\hat{1} \rightarrow 2$ $T = 63$, $T_{\max} = 75$, $\Delta = -105$, $\det = -121$, $(n_+, n_-, n_0) = (1, 63, 0)$, $c_{\text{ext}} = 15.0$

[65] $\overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)12231513221(12)122\overset{1}{1}31513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, \quad T_{\max} = 75, \quad \Delta = -104, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 63, 0), \quad c_{\text{ext}} = 15.0$

[66] $\overset{1}{1}(12)12231513221\overset{1}{(12)}12231513221(12)12231513221(12)1223151322\overset{1}{1}(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, \quad T_{\max} = 75, \quad \Delta = -104, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 63, 0), \quad c_{\text{ext}} = 15.0$

[67] $61231513221(12)122315132^{\hat{1}}21(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 63$, $T_{\max} = 75$, $\Delta = -105$, $\det = -121$, $(n_+, n_-, n_0) = (1, 63, 0)$, $c_{\text{ext}} = 15.0$

[68] $61231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, \quad T_{\max} = 75, \quad \Delta = -105, \quad \det = -121, \quad (n_+, n_-, n_0) = (1, 63, 0), \quad c_{\text{ext}} = 15.0$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[60] $21(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.69 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[69] $21(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.59 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[70] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -169, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.59 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[71] $61231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.59 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[72] $1(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.59 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[73] $61231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.59 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[74] $21(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.55 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[75] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^12315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -196, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.55 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[76] $61231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -196, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.55 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[77] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.55 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[78] $61231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -196, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.55 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[81] $61231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 63, T_{\max} = 75, \Delta = -105, \det = -225, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.52 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[80] $1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)122315131613 + \hat{1} \rightarrow 2$ $T = 63, T_{\max} = 77, \Delta = -112, \det = -225, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.52 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[79] $21(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 63, T_{\max} = 75, \Delta = -104, \det = -225, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.52 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[83] $61231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 63, T_{\max} = 75, \Delta = -105, \det = -225, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.52 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[82] $1(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 63, T_{\max} = 75, \Delta = -104, \det = -225, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.52 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[85] $21(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 63, T_{\max} = 75, \Delta = -104, \det = -256, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.48 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[86] $1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)122315131613 + \hat{1} \rightarrow 2$ $T = 63, T_{\max} = 77, \Delta = -112, \det = -256, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.48 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[87] $61231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 63, T_{\max} = 75, \Delta = -105, \det = -256, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.48 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[88] $61231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12 + \hat{1} \rightarrow 2$ $T = 63, T_{\max} = 75, \Delta = -105, \det = -256, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.48 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[89] $1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 63, T_{\max} = 75, \Delta = -104, \det = -256, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.48 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[90] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 63, T_{\max} = 75, \Delta = -104, \det = -256, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.48 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

[91] $61231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -256, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.48 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[92] $61231513221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -256, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.48 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[84] $21(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -256, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.48 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[93] $21(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -289, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.46 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[94] $1(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -289, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.46 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[95] 61231513221(12)12231513221(12)12231513221(12)122¹31513221(12)12231513221(12)12 + $\hat{1} \rightarrow 2$ $T = 63$, $T_{\max} = 75$, $\Delta = -105$, $\det = -289$, $(n_+, n_-, n_0) = (1, 63, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 67.46 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[96] $\overset{1}{1}(12)12231513221(\overset{1}{12})122315132\overset{1}{2}1(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, \quad T_{\max} = 75, \quad \Delta = -104, \quad \det = -289, \quad (n_+, n_-, n_0) = (1, 63, 0), \quad c_{\text{ext}} = 15.0$

$$T_{\min} = 67.46 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[97] $61231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -289, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.46 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[98] $21(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -361, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.41 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[99] $21(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -361, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 67.41 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[100] 1(12)122315132¹21(12)12231513221(12)12231513221(12)12231513221(12)122315131613 + $\hat{1} \rightarrow 2$ $T = 63$, $T_{\max} = 77$, $\Delta = -112$, $\det = -361$, $(n_+, n_-, n_0) = (1, 63, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 67.41 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[101] \quad 1(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -361, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 67.41 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[102] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -361, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 67.41 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[103] \quad 61231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -361, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 67.41 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[104] \quad 61231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -361, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 67.41 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[105] \quad 1(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -361, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 67.41 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[106] \quad 1(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -361, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 67.41 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[107] \quad 61231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -361, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 67.41 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[108] \quad 61231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -361, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 67.41 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[114] \quad 61231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -441, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 67.37 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[111] \quad 1(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -441, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 67.37 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[110] \quad 21(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -441, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 67.37 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[113] $61231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -441, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.37 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[109] $21(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -441, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.37 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[115] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -441, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.37 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[116] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -441, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.37 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[117] $61231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -441, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.37 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[118] $61231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -441, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.37 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[112] $1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -441, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.37 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[119] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -529, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.34 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

[120] $21(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -576, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.33 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[121] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -576, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.33 \rightarrow 68, T_{\max} = 77, T \in [68, 77]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[122] $61231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -576, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 67.33 \rightarrow 68, T_{\max} = 75, T \in [68, 75]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[123] & \overset{1}{1}(12)12231513221(12)12^{\overset{1}{1}}231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -104, \det = -576, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 67.33 \rightarrow 68, T_{\max} = 75, T \in [68, 75] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[124] & 61231513221(12)12231513221(12)12231513221(12)1223151322^{\overset{1}{1}}1(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 75, \Delta = -105, \det = -576, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 67.33 \rightarrow 68, T_{\max} = 75, T \in [68, 75] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[125] & \overset{1}{1}(12)1223151322^{\overset{1}{1}}1(12)12231513221(12)12231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -625, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 67.31 \rightarrow 68, T_{\max} = 77, T \in [68, 77] \\
& \begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0} \\
[126] & \overset{1}{1}(12)12231513221(12)1223151322^{\overset{1}{1}}1(12)12231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -676, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 67.30 \rightarrow 68, T_{\max} = 77, T \in [68, 77] \\
& \begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0} \\
[127] & \overset{1}{1}(12)12231513221(12)12^{\overset{1}{1}}231513221(12)12231513221(12)12231513221(12)122315131613 + \hat{1} \rightarrow 2 \quad T = 63, T_{\max} = 77, \Delta = -112, \det = -841, (n_+, n_-, n_0) = (1, 63, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 67.27 \rightarrow 68, T_{\max} = 77, T \in [68, 77] \\
& \begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0} \\
T = 64 & [128] \overset{1}{21}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^{\overset{1}{1}}171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.44 \rightarrow 70, T_{\max} = 76, T \in [70, 76] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[129] & 7122^{\overset{1}{1}}31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.44 \rightarrow 70, T_{\max} = 76, T \in [70, 76] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[130] & \overset{1}{1}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^{\overset{1}{1}}171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.44 \rightarrow 70, T_{\max} = 76, T \in [70, 76] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[131] & 7122^{\overset{1}{1}}31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 25, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.44 \rightarrow 70, T_{\max} = 76, T \in [70, 76] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[132] & 712^{\overset{1}{1}}231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 36, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.22 \rightarrow 70, T_{\max} = 76, T \in [70, 76] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[133] & 712^{\overset{1}{1}}231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 36, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.22 \rightarrow 70, T_{\max} = 76, T \in [70, 76]
\end{aligned}$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[134] \quad 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 69.06 \rightarrow 70, T_{\max} = 76, T \in [70, 76]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[135] \quad 7122315132^121(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 69.06 \rightarrow 70, T_{\max} = 76, T \in [70, 76]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[136] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 69.06 \rightarrow 70, T_{\max} = 76, T \in [70, 76]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[137] \quad 7122315132^121(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 69.06 \rightarrow 70, T_{\max} = 76, T \in [70, 76]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[138] \quad 21(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 68.94 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[139] \quad 712231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 68.94 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[140] \quad 1(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 68.94 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[141] \quad 712231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)1 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 64, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 68.94 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[151] \quad 712231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)1 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 68.84 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[144] \quad 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^12315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 68.84 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[145] \quad 71223151322^11(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 68.84 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

[157] 712231513221(12)12¹231513221(12)12231513221(12)12231513221¹(12)12231513221(12)12+ $\hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

[158] $\overset{1}{1}(12)1223151322\overset{1}{1}(12)1223151322\overset{1}{1}(12)1223151322\overset{1}{1}(12)1223151322^{\color{red}1}\overset{1}{1}(12)12231513217\overset{1}{1} + \hat{\textcolor{red}{1}} \rightarrow \textcolor{red}{2}$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

[159] $1(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$

[160] 712231513221(12)122315132¹21(12)12231513221(12)12231513221¹(12)12231513221¹(12)1 + $\hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

[161] 712231513221(12)12¹231513221(12)12231513221(12)12231513221¹(12)12231513221¹(12)1 + $\hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

[154] $21(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$

[162] $21(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$

[163] 712231513221(12)12231513221(12)122¹31513221(12)12231513221¹(12)12231513221(12)12+ $\hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 169$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

[164] $1(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$

[165] 712231513221(12)12231513221(12)122¹31513221(12)12231513221¹(12)12231513221¹(12)1 + $\hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 169$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

[166] $21(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 196, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[167] $712231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 196, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 68.55 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[168] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)122315132171+ \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 196, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 68.55 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[169] $712231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 196, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 68.55 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[170] $21(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)122315132171+ \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 68.52 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[171] $712231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 68.52 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[172] $1(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)122315132171+ \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 68.52 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[173] $712231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 68.52 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[174] $21(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)122315132171+ \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 68.48 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[175] $21(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)122315132171+ \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 68.48 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[176] $712231513221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12+ \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 68.48 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[177] $712231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 68.48 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[178] & 1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 68.48 \rightarrow 69, T_{\max} = 76, T \in [69, 76] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[179] & 1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 68.48 \rightarrow 69, T_{\max} = 76, T \in [69, 76] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[180] & 712231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 68.48 \rightarrow 69, T_{\max} = 76, T \in [69, 76] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[181] & 712231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 68.48 \rightarrow 69, T_{\max} = 76, T \in [69, 76] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[183] & 712231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 68.46 \rightarrow 69, T_{\max} = 76, T \in [69, 76] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[182] & 21(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 68.46 \rightarrow 69, T_{\max} = 76, T \in [69, 76] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[185] & 712231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 68.46 \rightarrow 69, T_{\max} = 76, T \in [69, 76] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[184] & 1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 68.46 \rightarrow 69, T_{\max} = 76, T \in [69, 76] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[189] & 712231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 68.41 \rightarrow 69, T_{\max} = 76, T \in [69, 76] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[186] & 21(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 68.41 \rightarrow 69, T_{\max} = 76, T \in [69, 76] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[187] & 21(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2 \quad T = 64, T_{\max} = 76, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 64, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 68.41 \rightarrow 69, T_{\max} = 76, T \in [69, 76]
\end{aligned}$$

1227

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[200] $712231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 441$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.37 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[201] $712231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 441$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.37 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[205] $712231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 576$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.33 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[204] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 576$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.33 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[202] $21(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)122315132171 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 576$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.33 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[203] $712231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 64$, $T_{\max} = 76$, $\Delta = -103$, $\det = 576$, $(n_+, n_-, n_0) = (1, 64, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 68.33 \rightarrow 69, T_{\max} = 76, T \in [69, 76]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$T = 65$ [206] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812^1 3 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 78$, $\Delta = -111$, $\det = -9$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 71.22 \rightarrow 72, T_{\max} = 78, T \in [72, 78]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[207] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 1812 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 70.44 \rightarrow 71, T_{\max} = 77, T \in [71, 77]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[208] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 2181 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 70.44 \rightarrow 71, T_{\max} = 77, T \in [71, 77]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[209] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812^1 32 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 80$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 70.44 \rightarrow 71, T_{\max} = 80, T \in [71, 80]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 14 & 6 \\ 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[210] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 181 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 70.44 \rightarrow 71, T_{\max} = 77, T \in [71, 77]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[224] & 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.94 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[225] & 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.94 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[222] & 21(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.94 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[226] & 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.94 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[227] & 181231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.94 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 7 & 7 \\ 2 & 7 & 3 \end{pmatrix}_{b_0} \\
[223] & 21(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -64, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.94 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[228] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 18123 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 78, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.84 \rightarrow 70, T_{\max} = 78, T \in [70, 78] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[230] & 21(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.84 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[231] & 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.84 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[232] & 21(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.84 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[233] & 21(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.84 \rightarrow 70, T_{\max} = 77, T \in [70, 77]
\end{aligned}$$

[234] $21(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -81$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

[235] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^{\hat{1}}23151322181+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -81$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

[236] $\overset{1}{1}(12)12231513221\overset{1}{(12)}12231513221(12)12231513221(12)122315132^{\textcolor{red}{1}}21(12)122315132181 + \textcolor{red}{\hat{1} \rightarrow 2} \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$

[237] $\overset{1}{1}(12)12231513221\overset{1}{(12)}12231513221(12)12231513221(12)12231513221(12)12\overset{1}{1}23151321\overset{1}{81} + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$

[238] $\overset{1}{1}(12)12\overset{1}{1}231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$

[239] $\overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)12231513221(12)122315132\overset{1}{2}1(12)1223151321812+\hat{1}\rightarrow\hat{2}$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -81$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

[240] $1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -81$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

[241] $\overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)12231513221(12)12231513221(12)12^123151321812+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -81$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

[242] $\overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)12231513221(12)12231513221(12)12^123151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$

[243] $\overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)12231513221(12)122315132^{\color{red}1}21(12)1223151322181 + \hat{\color{red}1} \rightarrow \color{red}2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$

[244] $1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -81$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[245] & \begin{matrix} 1 \\ 181231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221^1(12)12+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.84 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 88 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[246] & \begin{matrix} 1 \\ 181231513221^1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.84 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[247] & \begin{matrix} 1 \\ 181231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.84 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[229] & \begin{matrix} 1 \\ 1(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)12231513218123+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 65, T_{\max} = 78, \Delta = -111, \det = -81, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.84 \rightarrow 70, T_{\max} = 78, T \in [70, 78] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[248] & \begin{matrix} 1 \\ 1(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513218123+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 65, T_{\max} = 78, \Delta = -111, \det = -100, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.76 \rightarrow 70, T_{\max} = 78, T \in [70, 78] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[249] & \begin{matrix} 1 \\ (12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)122315132181232+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 65, T_{\max} = 80, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.76 \rightarrow 70, T_{\max} = 80, T \in [70, 80] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[258] & \begin{matrix} 1 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221^1(12)1223151321812+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[252] & \begin{matrix} 1 \\ 21(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)1223151321812+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[253] & \begin{matrix} 1 \\ 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221^1(12)1223151322181+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[254] & \begin{matrix} 1 \\ 21(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)1223151322181+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[255] & \begin{matrix} 1 \\ (12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132181232+ \hat{1} \rightarrow 2 \end{matrix} \quad T = 65, T_{\max} = 80, \Delta = -113, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 80, T \in [70, 80]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[251] & 21(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[257] & 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[250] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513218123 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 78, \Delta = -111, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 78, T \in [70, 78] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[259] & 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[260] & 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[261] & 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[262] & 181231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[263] & 181231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[256] & 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.69 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[265] & 21(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -169, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.59 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[266] & 21(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -169, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.59 \rightarrow 70, T_{\max} = 77, T \in [70, 77]
\end{aligned}$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[267] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1181232+\hat{1}\rightarrow 2$ $T = 65, T_{\max} = 80, \Delta = -113, \det = -169, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 69.59 \rightarrow 70, T_{\max} = 80, T \in [70, 80]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[268] $1(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)122315132181+\hat{1}\rightarrow 2$ $T = 65, T_{\max} = 77, \Delta = -102, \det = -169, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 69.59 \rightarrow 70, T_{\max} = 77, T \in [70, 77]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 285 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[269] $1(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 65, T_{\max} = 77, \Delta = -102, \det = -169, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 69.59 \rightarrow 70, T_{\max} = 77, T \in [70, 77]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[264] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513218123+\hat{1}\rightarrow 2$ $T = 65, T_{\max} = 78, \Delta = -111, \det = -169, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 69.59 \rightarrow 70, T_{\max} = 78, T \in [70, 78]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[271] $181231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 65, T_{\max} = 77, \Delta = -102, \det = -169, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 69.59 \rightarrow 70, T_{\max} = 77, T \in [70, 77]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[270] $1(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 65, T_{\max} = 77, \Delta = -102, \det = -169, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 69.59 \rightarrow 70, T_{\max} = 77, T \in [70, 77]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[272] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513218123+\hat{1}\rightarrow 2$ $T = 65, T_{\max} = 78, \Delta = -111, \det = -196, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 69.55 \rightarrow 70, T_{\max} = 78, T \in [70, 78]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[273] $21(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 65, T_{\max} = 77, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 69.55 \rightarrow 70, T_{\max} = 77, T \in [70, 77]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[274] $21(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 65, T_{\max} = 77, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 69.55 \rightarrow 70, T_{\max} = 77, T \in [70, 77]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[275] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)122315132181+\hat{1}\rightarrow 2$ $T = 65, T_{\max} = 77, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 69.55 \rightarrow 70, T_{\max} = 77, T \in [70, 77]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 321 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[276] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 65, T_{\max} = 77, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 69.55 \rightarrow 70, T_{\max} = 77, T \in [70, 77]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[277] & \overset{1}{1}(12)12231513221(12)12231513221(12)12231513221(12)12^{\hat{1}}231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.55 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[278] & 181231513221(12)1223151322^{\hat{1}}1(12)12231513221(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.55 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[279] & \overset{1}{1}(12)12231513221(12)12231513221(12)12231513221(12)122^{\hat{1}}31513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 78, \Delta = -111, \det = -225, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.52 \rightarrow 70, T_{\max} = 78, T \in [70, 78] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[280] & \overset{1}{21}(12)12231513221(12)12231513221(12)122^{\hat{1}}31513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -225, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.52 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[281] & 21(12)12231513221(12)12231513221(12)122^{\hat{1}}31513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -225, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.52 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[282] & (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^{\hat{1}}315132181232 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 80, \Delta = -113, \det = -225, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.52 \rightarrow 70, T_{\max} = 80, T \in [70, 80] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[283] & \overset{1}{1}(12)12231513221(12)12231513221(12)122^{\hat{1}}31513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -225, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.52 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 389 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[284] & \overset{1}{1}(12)12231513221(12)12231513221(12)122^{\hat{1}}31513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -225, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.52 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[285] & \overset{1}{1}(12)12231513221(12)12231513221(12)122^{\hat{1}}31513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -225, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.52 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[286] & 181231513221(12)12231513221(12)122315132^{\hat{1}}21(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -225, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.52 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[296] & \overset{1}{1}(12)12231513221(12)12231513221(12)1223151322^{\hat{1}}1(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -256, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.48 \rightarrow 70, T_{\max} = 77, T \in [70, 77]
\end{aligned}$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[289] $21(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.48 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[290] $21(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.48 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[291] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.48 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[292] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132181+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.48 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[293] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122315132181+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.48 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 463 & 43 \\ 2 & 43 & 3 \end{pmatrix}_{b_0}$$

[294] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.48 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[295] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.48 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[288] $21(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.48 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[287] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513218123+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 78$, $\Delta = -111$, $\det = -256$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.48 \rightarrow 70$, $T_{\max} = 78$, $T \in [70, 78]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[298] $181231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.48 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[299] $181231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -256$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.48 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 47 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[297] & 1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -256, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.48 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[301] & 21(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -289, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.46 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[302] & 21(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -289, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.46 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[303] & (12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 80, \Delta = -113, \det = -289, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.46 \rightarrow 70, T_{\max} = 80, T \in [70, 80] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[304] & 1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -289, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.46 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0} \\
[305] & 1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -289, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.46 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[306] & 1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -289, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.46 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[307] & 181231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -289, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.46 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[300] & 1(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 78, \Delta = -111, \det = -289, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.46 \rightarrow 70, T_{\max} = 78, T \in [70, 78] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[308] & 1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 78, \Delta = -111, \det = -361, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.41 \rightarrow 70, T_{\max} = 78, T \in [70, 78] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[309] & 1(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 78, \Delta = -111, \det = -361, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.41 \rightarrow 70, T_{\max} = 78, T \in [70, 78]
\end{aligned}$$

[310] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513218123+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 78$, $\Delta = -111$, $\det = -361$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

[311] 21(12)12231513221¹(12)12231513221(12)12¹231513221(12)12231513221(12)1223151321812+ $\hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

[312] 21(12)12231513221(12)122¹31513221(12)12231513221(12)12231513221(12)1223151321812+¹~~1~~→2 $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

[314] $21(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

[315] $(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)122315132181232+\hat{1}\rightarrow 2 \quad T = 65, T_{\max} = 80, \Delta = -113, \det = -361, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$

[316] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^{\hat{1}}2315132181232+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 80$, $\Delta = -113$, $\det = -361$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

[317] $1(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$

[318] $\overset{1}{1}(12)12231513221(\overset{1}{12})122^{\textcolor{red}{1}}31513221(12)12231513221(12)12231513221(12)122315132181 + \hat{\textcolor{red}{1}} \rightarrow \textcolor{red}{2} \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$

[319] $1(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

[320] $\overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)12^{\color{red}1}231513221(12)12231513221(12)1223151321812 + \hat{\color{red}1} \rightarrow \color{red}2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$

[321] $\overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)12^{\color{red}1}231513221(12)12231513221(12)1223151322181 + \hat{\color{red}1} \rightarrow \color{red}2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0$

$$\begin{aligned}
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[322] & 1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.41 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[323] & 181231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.41 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[324] & 181231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.41 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 75 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[313] & 21(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.41 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[335] & 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[327] & 21(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[328] & 21(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[326] & 1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 78, \Delta = -111, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 78, T \in [70, 78] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[330] & 21(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[331] & (12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 80, \Delta = -113, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 80, T \in [70, 80] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[332] & (12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 80, \Delta = -113, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 80, T \in [70, 80]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[333] & 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 797 & 57 \\ 2 & 57 & 3 \end{pmatrix}_{b_0} \\
[334] & 1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 776 & 56 \\ 2 & 56 & 3 \end{pmatrix}_{b_0} \\
[325] & 1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 78, \Delta = -111, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 78, T \in [70, 78] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[336] & 1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[337] & 1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[338] & 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[339] & 181231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 83 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[340] & 181231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[329] & 21(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 77, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.37 \rightarrow 70, T_{\max} = 77, T \in [70, 77] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[341] & (12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 80, \Delta = -113, \det = -484, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.36 \rightarrow 70, T_{\max} = 80, T \in [70, 80] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[343] & (12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2 \quad T = 65, T_{\max} = 80, \Delta = -113, \det = -529, (n_+, n_-, n_0) = (1, 65, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 69.34 \rightarrow 70, T_{\max} = 80, T \in [70, 80]
\end{aligned}$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

[342] $1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 78$, $\Delta = -111$, $\det = -529$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.34 \rightarrow 70$, $T_{\max} = 78$, $T \in [70, 78]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

[348] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)122315132181 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -576$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.33 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 1031 & 65 \\ 2 & 65 & 3 \end{pmatrix}_{b_0}$$

[346] $21(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -576$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.33 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[347] $(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 80$, $\Delta = -113$, $\det = -576$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.33 \rightarrow 70$, $T_{\max} = 80$, $T \in [70, 80]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[345] $21(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -576$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.33 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[349] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223151321812 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -576$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.33 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[350] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -576$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.33 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[351] $181231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 77$, $\Delta = -102$, $\det = -576$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.33 \rightarrow 70$, $T_{\max} = 77$, $T \in [70, 77]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 119 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[344] $1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 78$, $\Delta = -111$, $\det = -576$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.33 \rightarrow 70$, $T_{\max} = 78$, $T \in [70, 78]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[352] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513218123 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 78$, $\Delta = -111$, $\det = -625$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.31 \rightarrow 70$, $T_{\max} = 78$, $T \in [70, 78]$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$$

[353] $(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)122315132181232 + \hat{1} \rightarrow 2$ $T = 65$, $T_{\max} = 80$, $\Delta = -113$, $\det = -625$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 69.31 \rightarrow 70$, $T_{\max} = 80$, $T \in [70, 80]$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$$

[354] $1(12)12231513221(12)1223151322^11(12)12231513221(12)12231513218123+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 78$, $\Delta = -111$, $\det = -676$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 69.30 \rightarrow 70, T_{\max} = 78, T \in [70, 78]$$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

[355] $(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)122315132181232+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 80$, $\Delta = -113$, $\det = -676$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 69.30 \rightarrow 70, T_{\max} = 80, T \in [70, 80]$$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

[356] $(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)122315132181232+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 80$, $\Delta = -113$, $\det = -729$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 69.29 \rightarrow 70, T_{\max} = 80, T \in [70, 80]$$

$$\begin{pmatrix} 0 & 27 & 2 \\ 27 & 620 & 48 \\ 2 & 48 & 3 \end{pmatrix}_{b_0}$$

[357] $1(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12231513218123+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 78$, $\Delta = -111$, $\det = -841$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 69.27 \rightarrow 70, T_{\max} = 78, T \in [70, 78]$$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[359] $(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)122315132181232+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 80$, $\Delta = -113$, $\det = -841$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 69.27 \rightarrow 70, T_{\max} = 80, T \in [70, 80]$$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 724 & 52 \\ 2 & 52 & 3 \end{pmatrix}_{b_0}$$

[358] $(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)122315132181232+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 80$, $\Delta = -113$, $\det = -841$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 69.27 \rightarrow 70, T_{\max} = 80, T \in [70, 80]$$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[360] $(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)122315132181232+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 80$, $\Delta = -113$, $\det = -961$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 69.25 \rightarrow 70, T_{\max} = 80, T \in [70, 80]$$

$$\begin{pmatrix} 0 & 31 & 2 \\ 31 & 805 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0}$$

[361] $(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)122315132181232+\hat{1}\rightarrow 2$ $T = 65$, $T_{\max} = 80$, $\Delta = -113$, $\det = -1156$, $(n_+, n_-, n_0) = (1, 65, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 69.23 \rightarrow 70, T_{\max} = 80, T \in [70, 80]$$

$$\begin{pmatrix} 0 & 34 & 2 \\ 34 & 985 & 61 \\ 2 & 61 & 3 \end{pmatrix}_{b_0}$$

$T = 69$ [362] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222^12+\hat{1}\rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -4$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 76, T_{\max} = 81, T \in [76, 81]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[363] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^12+\hat{1}\rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -4$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 76, T_{\max} = 81, T \in [76, 81]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[364] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+^{22}1\hat{1}\rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -4$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 76, T_{\max} = 81, T \in [76, 81]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[365] & \overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222^1 2 + \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 76, T_{\max} = 81, T \in [76, 81] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0} \\
[366] & \overset{2^1}{3221}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(\overset{1}{12})12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[367] & \overset{22^1}{1}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(\overset{1}{12})12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 76, T_{\max} = 81, T \in [76, 81] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[368] & \overset{22^1}{21}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(\overset{1}{12})12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 76, T_{\max} = 81, T \in [76, 81] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[371] & \overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -9, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.22 \rightarrow 76, T_{\max} = 81, T \in [76, 81] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[373] & \overset{222^1}{1}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(\overset{1}{12})12231513221(12)12 + \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -9, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.22 \rightarrow 76, T_{\max} = 81, T \in [76, 81] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[370] & \overset{1}{1}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(\overset{1}{12})12231513221(12)12^1 22 + \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -9, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.22 \rightarrow 76, T_{\max} = 81, T \in [76, 81] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[369] & \overset{1}{21}(12)12231513221(\overset{1}{12})12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 22 + \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -9, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.22 \rightarrow 76, T_{\max} = 81, T \in [76, 81] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[372] & \overset{32^1}{21}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(\overset{1}{12})12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -110, \det = -9, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.22 \rightarrow 76, T_{\max} = 81, T \in [76, 81] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[375] & \overset{1}{1}(12)12231513221(\overset{1}{12})12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -16, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.75 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[374] & \overset{1}{21}(12)12231513221(\overset{1}{12})12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 222 + \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -16, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.75 \rightarrow 75, T_{\max} = 81, T \in [75, 81]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[382] & 21(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -25, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.44 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[381] & 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -25, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.44 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[380] & 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -110, \det = -25, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.44 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[379] & 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1222+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -25, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.44 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[378] & 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -25, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.44 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[377] & 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1222+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -25, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.44 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[376] & 21(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1222+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -25, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.44 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[388] & 21(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -36, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.22 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[387] & 1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -36, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.22 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[386] & 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1222+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -36, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.22 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[385] & 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -36, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.22 \rightarrow 75, T_{\max} = 81, T \in [75, 81]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[384] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -36, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.22 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 41 & 11 \\ 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[383] & 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -36, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.22 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[393] & 1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -49, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.06 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[389] & 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1222+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -49, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.06 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[392] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1222+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -49, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.06 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[390] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1222+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -49, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.06 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[391] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -49, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.06 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 69 & 15 \\ 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[394] & 21(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -49, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.06 \rightarrow 75, T_{\max} = 81, T \in [75, 81] \\
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[400] & 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -64, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.94 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 7 & 7 \\ 2 & 7 & 3 \end{pmatrix}_{b_0} \\
[399] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -64, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.94 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 7 & 7 \\ 2 & 7 & 3 \end{pmatrix}_{b_0} \\
[398] & 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -64, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.94 \rightarrow 74, T_{\max} = 81, T \in [74, 81]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[397] & 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -64, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.94 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[396] & 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -64, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.94 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 119 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[395] & 21(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -64, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.94 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 8 & 2 \\ 8 & 55 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[408] & 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -81, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.84 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[412] & 1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -81, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.84 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[413] & 21(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.84 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[414] & 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -110, \det = -81, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.84 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[415] & 1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -81, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.84 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[416] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -81, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.84 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 8 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[417] & 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -81, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.84 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[418] & 21(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -81, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.84 \rightarrow 74, T_{\max} = 81, T \in [74, 81]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[407] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^{221}231513221(12)12+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -81, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.84 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[421] & 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)1+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -110, \det = -100, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.76 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[426] & 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -121, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.69 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[423] & 21(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -121, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.69 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[424] & 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -121, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.69 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 186 & 26 \\ 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[425] & 1(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -121, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.69 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[427] & 1(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -121, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.69 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 197 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[428] & 1(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -121, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.69 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[429] & 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -121, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.69 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[430] & 21(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.69 \rightarrow 74, T_{\max} = 81, T \in [74, 81] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[431] & 21(12)12^{3221}231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -110, \det = -121, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 73.69 \rightarrow 74, T_{\max} = 81, T \in [74, 81]
\end{aligned}$$

1249

1251

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 47 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[457] $21(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -256$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 73.48 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 207 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[458] $21(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12222+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -256$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 73.48 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[459] $1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -256$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 73.48 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[460] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -256$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 73.48 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 463 & 43 \\ 2 & 43 & 3 \end{pmatrix}_{b_0}$$

[461] $1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -256$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 73.48 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 431 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[462] $1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -256$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 73.48 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 463 & 43 \\ 2 & 43 & 3 \end{pmatrix}_{b_0}$$

[470] $21(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -289$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 73.46 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[471] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -289$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 73.46 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

[472] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -289$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 73.46 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 509 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

[473] $1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -101$, $\det = -289$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 73.46 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[474] $21(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2$ $T = 69$, $T_{\max} = 81$, $\Delta = -110$, $\det = -289$, $(n_+, n_-, n_0) = (1, 69, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 73.46 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$$

- $$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 776 & 56 \\ 2 & 56 & 3 \end{pmatrix}_{b_0}$$
- [498] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -441, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 73.37 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$
- [499] $1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -441, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 73.37 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$
- [500] $21(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -110, \det = -441, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 73.37 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$
- [501] $21(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -110, \det = -441, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 73.37 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$
- [502] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -441, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 73.37 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$
- [503] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -441, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 73.37 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 83 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$
- [504] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -441, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 73.37 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 83 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$
- [505] $21(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -441, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 73.37 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$
- [492] $21(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -441, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 73.37 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$
- [506] $21(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -110, \det = -529, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 73.34 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$
- [508] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1222+\hat{1} \rightarrow 2 \quad T = 69, T_{\max} = 81, \Delta = -101, \det = -576, (n_+, n_-, n_0) = (1, 69, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 73.33 \rightarrow 74, T_{\max} = 81, T \in [74, 81]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[519] & 413221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1+\hat{1} \rightarrow 2 \quad T = 70, T_{\max} = 82, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 70, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.84 \rightarrow 75, T_{\max} = 82, T \in [75, 82] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[520] & 413221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 70, T_{\max} = 82, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 70, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.84 \rightarrow 75, T_{\max} = 82, T \in [75, 82] \\
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[521] & 413221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1+\hat{1} \rightarrow 2 \quad T = 70, T_{\max} = 82, \Delta = -109, \det = 100, (n_+, n_-, n_0) = (1, 70, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.76 \rightarrow 75, T_{\max} = 82, T \in [75, 82] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[522] & 413221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 70, T_{\max} = 82, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 70, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.69 \rightarrow 75, T_{\max} = 82, T \in [75, 82] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[523] & 413221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 70, T_{\max} = 82, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 70, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.69 \rightarrow 75, T_{\max} = 82, T \in [75, 82] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[524] & 413221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 70, T_{\max} = 82, \Delta = -109, \det = 169, (n_+, n_-, n_0) = (1, 70, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.59 \rightarrow 75, T_{\max} = 82, T \in [75, 82] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[525] & 413221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 70, T_{\max} = 82, \Delta = -109, \det = 196, (n_+, n_-, n_0) = (1, 70, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.55 \rightarrow 75, T_{\max} = 82, T \in [75, 82] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[526] & 413221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 70, T_{\max} = 82, \Delta = -109, \det = 225, (n_+, n_-, n_0) = (1, 70, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.52 \rightarrow 75, T_{\max} = 82, T \in [75, 82] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[527] & 413221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 70, T_{\max} = 82, \Delta = -109, \det = 256, (n_+, n_-, n_0) = (1, 70, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.48 \rightarrow 75, T_{\max} = 82, T \in [75, 82] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[528] & 413221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 70, T_{\max} = 82, \Delta = -109, \det = 289, (n_+, n_-, n_0) = (1, 70, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.46 \rightarrow 75, T_{\max} = 82, T \in [75, 82] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[531] & 413221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 70, T_{\max} = 82, \Delta = -109, \det = 361, (n_+, n_-, n_0) = (1, 70, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.41 \rightarrow 75, T_{\max} = 82, T \in [75, 82]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[530] & \begin{pmatrix} 1 \\ 413221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \end{pmatrix} \quad T = 70, \quad T_{\max} = 82, \quad \Delta = -109, \quad \det = 361, \quad (n_+, n_-, n_0) = (1, 70, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.41 \rightarrow 75, \quad T_{\max} = 82, \quad T \in [75, 82] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[529] & \begin{pmatrix} 1 \\ 413221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1+ \hat{1} \rightarrow 2 \end{pmatrix} \quad T = 70, \quad T_{\max} = 82, \quad \Delta = -109, \quad \det = 361, \quad (n_+, n_-, n_0) = (1, 70, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.41 \rightarrow 75, \quad T_{\max} = 82, \quad T \in [75, 82] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[533] & \begin{pmatrix} 1 \\ 413221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \end{pmatrix} \quad T = 70, \quad T_{\max} = 82, \quad \Delta = -109, \quad \det = 441, \quad (n_+, n_-, n_0) = (1, 70, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.37 \rightarrow 75, \quad T_{\max} = 82, \quad T \in [75, 82] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[532] & \begin{pmatrix} 1 \\ 413221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \end{pmatrix} \quad T = 70, \quad T_{\max} = 82, \quad \Delta = -109, \quad \det = 441, \quad (n_+, n_-, n_0) = (1, 70, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.37 \rightarrow 75, \quad T_{\max} = 82, \quad T \in [75, 82] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[534] & \begin{pmatrix} 1 \\ 413221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1+ \hat{1} \rightarrow 2 \end{pmatrix} \quad T = 70, \quad T_{\max} = 82, \quad \Delta = -109, \quad \det = 529, \quad (n_+, n_-, n_0) = (1, 70, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.34 \rightarrow 75, \quad T_{\max} = 82, \quad T \in [75, 82] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0} \\
[535] & \begin{pmatrix} 1 \\ 413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \end{pmatrix} \quad T = 70, \quad T_{\max} = 82, \quad \Delta = -109, \quad \det = 576, \quad (n_+, n_-, n_0) = (1, 70, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.33 \rightarrow 75, \quad T_{\max} = 82, \quad T \in [75, 82] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[536] & \begin{pmatrix} 1 \\ 413221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1+ \hat{1} \rightarrow 2 \end{pmatrix} \quad T = 70, \quad T_{\max} = 82, \quad \Delta = -109, \quad \det = 625, \quad (n_+, n_-, n_0) = (1, 70, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.31 \rightarrow 75, \quad T_{\max} = 82, \quad T \in [75, 82] \\
& \begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0} \\
[537] & \begin{pmatrix} 1 \\ 413221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \end{pmatrix} \quad T = 70, \quad T_{\max} = 82, \quad \Delta = -109, \quad \det = 676, \quad (n_+, n_-, n_0) = (1, 70, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.30 \rightarrow 75, \quad T_{\max} = 82, \quad T \in [75, 82] \\
& \begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0} \\
[538] & \begin{pmatrix} 1 \\ 413221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1+ \hat{1} \rightarrow 2 \end{pmatrix} \quad T = 70, \quad T_{\max} = 82, \quad \Delta = -109, \quad \det = 841, \quad (n_+, n_-, n_0) = (1, 70, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 74.27 \rightarrow 75, \quad T_{\max} = 82, \quad T \in [75, 82] \\
& \begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0} \\
T = 71 & \begin{pmatrix} 15132^1 21 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \end{pmatrix} \quad T = 71, \quad T_{\max} = 83, \quad \Delta = -104, \quad \det = -25, \quad (n_+, n_-, n_0) = (1, 71, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 76.44 \rightarrow 77, \quad T_{\max} = 83, \quad T \in [77, 83] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[541] & \begin{pmatrix} 151322^1 1 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \end{pmatrix} \quad T = 71, \quad T_{\max} = 83, \quad \Delta = -104, \quad \det = -49, \quad (n_+, n_-, n_0) = (1, 71, 0), \quad c_{\text{ext}} = 15.0 \\
& T_{\min} = 76.06 \rightarrow 77, \quad T_{\max} = 83, \quad T \in [77, 83]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[540] $414132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -108, \det = -49, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 76.06 \rightarrow 77, T_{\max} = 83, T \in [77, 83]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[542] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 85, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 75.94 \rightarrow 76, T_{\max} = 85, T \in [76, 85]$

$$\begin{pmatrix} 0 & 8 & 2 \\ 8 & 39 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[547] $1 \begin{pmatrix} 1513221 \\ (12) \end{pmatrix} 12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 75.84 \rightarrow 76, T_{\max} = 83, T \in [76, 83]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[546] $1 \begin{pmatrix} 1513221 \\ (12) \end{pmatrix} 12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -104, \det = -81, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 75.84 \rightarrow 76, T_{\max} = 83, T \in [76, 83]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[545] $41413221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 75.84 \rightarrow 76, T_{\max} = 83, T \in [76, 83]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[544] $41413221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 75.84 \rightarrow 76, T_{\max} = 83, T \in [76, 83]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[543] $41413221^1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -108, \det = -81, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 75.84 \rightarrow 76, T_{\max} = 83, T \in [76, 83]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[550] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 85, \Delta = -112, \det = -100, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 75.76 \rightarrow 76, T_{\max} = 85, T \in [76, 85]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[549] $1 \begin{pmatrix} 1513221 \\ (12) \end{pmatrix} 12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -104, \det = -100, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 75.76 \rightarrow 76, T_{\max} = 83, T \in [76, 83]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[548] $41413221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -108, \det = -100, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 75.76 \rightarrow 76, T_{\max} = 83, T \in [76, 83]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[554] $1 \begin{pmatrix} 1513221 \\ (12) \end{pmatrix} 12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 75.69 \rightarrow 76, T_{\max} = 83, T \in [76, 83]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[553] & 1 \begin{pmatrix} 1513221 \\ (12) \end{pmatrix} 12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -104, \det = -121, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.69 \rightarrow 76, T_{\max} = 83, T \in [76, 83] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[555] & (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 85, \Delta = -112, \det = -121, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.69 \rightarrow 76, T_{\max} = 85, T \in [76, 85] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[556] & (12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 85, \Delta = -112, \det = -121, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.69 \rightarrow 76, T_{\max} = 85, T \in [76, 85] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[551] & 41413221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.69 \rightarrow 76, T_{\max} = 83, T \in [76, 83] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[552] & 41413221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -108, \det = -121, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.69 \rightarrow 76, T_{\max} = 83, T \in [76, 83] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[557] & 41413221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -108, \det = -169, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.59 \rightarrow 76, T_{\max} = 83, T \in [76, 83] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[558] & 1 \begin{pmatrix} 1513221 \\ (12) \end{pmatrix} 12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -104, \det = -169, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.59 \rightarrow 76, T_{\max} = 83, T \in [76, 83] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[559] & (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 85, \Delta = -112, \det = -169, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.59 \rightarrow 76, T_{\max} = 85, T \in [76, 85] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[560] & 41413221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -108, \det = -196, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.55 \rightarrow 76, T_{\max} = 83, T \in [76, 83] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[561] & 1 \begin{pmatrix} 1513221 \\ (12) \end{pmatrix} 1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -104, \det = -196, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.55 \rightarrow 76, T_{\max} = 83, T \in [76, 83] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[562] & 41413221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -108, \det = -225, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 75.52 \rightarrow 76, T_{\max} = 83, T \in [76, 83]
\end{aligned}$$

1261

[577] $(12)12231513221(12)12231513221(12)12231513221(12)122^{\overline{1}}31513221(12)12231513221$ $(12)1+\hat{1}\rightarrow 2$ $T = 71$, $T_{\max} = 85$, $\Delta = -112$, $\det = -361$, $(n_+, n_-, n_0) = (1, 71, 0)$, $c_{\text{ext}} = 15.0$

[576] $\overset{1513221}{1} (12) 12231513221(12)12231513221(12)12231513221(12)12231513221(12)122\overset{1}{1}31513221(12)1+\overset{1}{\hat{1}}\rightarrow\overset{2}{2} \quad T = 71, T_{\max} = 83, \Delta = -104, \det = -361, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$

[575] $\overset{1513221}{1} (12) 12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -104, \det = -361, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$

[574] $\overset{1513221}{1} (12) 12231513221(12)1223151322\overset{1}{1}(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow\overset{1}{2} \quad T = 71, T_{\max} = 83, \Delta = -104, \det = -361, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$

[573] 41413221(12)12231513221(12)12231513221(12)122315132¹21(12)12231513221(12)12231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 71$, $T_{\max} = 83$, $\Delta = -108$, $\det = -361$, $(n_+, n_-, n_0) = (1, 71, 0)$, $c_{\text{ext}} = 15.0$

[584] $(12)12231513221(12)12231513221(12)122315132^{\color{red}1}21(12)12231513221(12)12231513221$ $\overset{31513221}{(12)}1+\hat{\color{red}1}\rightarrow\color{red}2$ $T = 71$, $T_{\max} = 85$, $\Delta = -112$, $\det = -441$, $(n_+, n_-, n_0) = (1, 71, 0)$, $c_{\text{ext}} = 15.0$

[583] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221^11(12)12231513221$ $(12)1+1\hat{\rightarrow}2$ $T = 71$, $T_{\max} = 85$, $\Delta = -112$, $\det = -441$, $(n_+, n_-, n_0) = (1, 71, 0)$, $c_{\text{ext}} = 15.0$

[582] $\overset{1513221}{1} (12) 12231513221(12)12231513221(12)12^{\color{red}{1}}231513221(12)12231513221(12)12231513221(12)1+\hat{\color{red}{1}}\rightarrow\color{red}{2} \quad T = 71, T_{\max} = 83, \Delta = -104, \det = -441, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$

[581] $1^{1513221} (12) 12231513221 (12) 12231513221 (12) 12231513221 (12) 122^1 31513221 (12) 12231513221 (12) 1 + \hat{1} \rightarrow 2 \quad T = 71, T_{\max} = 83, \Delta = -104, \det = -441, (n_+, n_-, n_0) = (1, 71, 0), c_{\text{ext}} = 15.0$

[580] 41413221(12)12231513221(12)12231513221(12)12231513221(12)122¹31513221(12)12231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 71$, $T_{\max} = 83$, $\Delta = -108$, $\det = -441$, $(n_+, n_-, n_0) = (1, 71, 0)$, $c_{\text{ext}} = 15.0$

[579] 41413221(12)12231513221(12)12231513221(12)12¹231513221(12)12231513221(12)¹12231513221(12)1 + $\hat{1} \rightarrow 2$ $T = 71$, $T_{\max} = 83$, $\Delta = -108$, $\det = -441$, $(n_+, n_-, n_0) = (1, 71, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^123151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.84 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -81, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.84 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -100, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.76 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.69 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -121, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.69 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -169, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.59 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -196, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.55 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -225, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.52 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -256, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.48 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -289, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.46 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -361, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.41 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -361, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.41 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -361, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.41 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -441, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.37 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 1(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -441, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.37 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -529, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.34 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 1(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -576, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.33 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$$[19] \quad 1(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -625, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.31 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 1(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -676, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.30 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[21] \quad 1(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315 + \hat{1} \rightarrow 2 \quad T = 73, T_{\max} = 85, \Delta = -106, \det = -841, (n_+, n_-, n_0) = (1, 73, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 77.27 \rightarrow 78, T_{\max} = 85, T \in [78, 85]$$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

$$T = 74 \quad [22] \quad 612^131513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1 + \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 25, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.44 \rightarrow 80, T_{\max} = 86, T \in [80, 86]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[23] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^131513161 + \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 49, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.06 \rightarrow 80, T_{\max} = 86, T \in [80, 86]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[24] $612315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 49, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 79.06 \rightarrow 80, T_{\max} = 86, T \in [80, 86]$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[25] $1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.84 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[26] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513161+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.84 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[27] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513161+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 81, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.84 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[28] $61231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.84 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[29] $6123151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.84 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[30] $61231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 81, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.84 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[32] $1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 100, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.76 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[33] $61231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 100, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.76 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[31] $(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315131613+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 100, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.76 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[38] $61231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.69 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[39] $61231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 121, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.69 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[36] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513161+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.69 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[35] $(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315131613+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.69 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[34] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 315131613+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 121, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.69 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[37] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513161+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 121, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.69 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[40] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)122315131613+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 169, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.59 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[42] $61231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 169, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.59 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[41] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513161+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 169, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.59 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[43] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 2315131613+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 196, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.55 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[44] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513161+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 196, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.55 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[45] $61231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 196, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.55 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[46] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)122315131613+\hat{1}\rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 225, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.52 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[47] $1(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513161+\hat{1}\rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.52 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[48] $61231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 225, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.52 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[51] $61231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 256, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.48 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[49] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)122315131613+\hat{1}\rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 256, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.48 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[50] $1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513161+\hat{1}\rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.48 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[54] $61231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 289, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.46 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[53] $1(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513161+\hat{1}\rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.46 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[52] $(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)122315131613+\hat{1}\rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 289, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.46 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[55] $(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)122315131613+\hat{1}\rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 361, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.41 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[56] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)122315131613+\hat{1}\rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 361, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 78.41 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[57] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12^{1231513221}(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.41 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[58] \quad 1(12)122315132^{121}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.41 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[59] \quad 1(12)12231513221(12)12231513221(12)122^{131513221}(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 361, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.41 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 61231513221(12)12231513221(12)1223151322^{11}(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.41 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 61231513221(12)12231513221(12)12231513221(12)122315132^{121}(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.41 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 61231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^{131513221}(12)1+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 361, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.41 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[67] \quad 61231513221(12)12231513221(12)12231513221(12)12^{1231513221}(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 441, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.37 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[63] \quad (12)12231513221(12)12231513221(12)122315132^{121}(12)12231513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 441, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.37 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[64] \quad (12)12231513221(12)12231513221(12)12231513221(12)1223151322^{11}(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 441, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.37 \rightarrow 79, T_{\max} = 87, T \in [79, 87]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[68] \quad 61231513221(12)12231513221(12)12231513221(12)12231513221(12)122^{131513221}(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 441, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.37 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[65] \quad 1(12)12231513221(12)12231513221(12)1223151322^{11}(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 441, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 78.37 \rightarrow 79, T_{\max} = 86, T \in [79, 86]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[66] & 1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 441, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 78.37 \rightarrow 79, T_{\max} = 86, T \in [79, 86] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[69] & (12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 484, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 78.36 \rightarrow 79, T_{\max} = 87, T \in [79, 87] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[71] & 1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 529, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 78.34 \rightarrow 79, T_{\max} = 86, T \in [79, 86] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0} \\
[70] & (12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 529, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 78.34 \rightarrow 79, T_{\max} = 87, T \in [79, 87] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0} \\
[72] & 61231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 529, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 78.34 \rightarrow 79, T_{\max} = 86, T \in [79, 86] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0} \\
[73] & (12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 576, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 78.33 \rightarrow 79, T_{\max} = 87, T \in [79, 87] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[74] & 1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 576, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 78.33 \rightarrow 79, T_{\max} = 86, T \in [79, 86] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[75] & 61231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 576, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 78.33 \rightarrow 79, T_{\max} = 86, T \in [79, 86] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[76] & (12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122315131613+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 87, \Delta = -111, \det = 625, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 78.31 \rightarrow 79, T_{\max} = 87, T \in [79, 87] \\
& \begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0} \\
[77] & 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -103, \det = 625, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 78.31 \rightarrow 79, T_{\max} = 86, T \in [79, 86] \\
& \begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0} \\
[78] & 61231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1+\hat{1} \rightarrow 2 \quad T = 74, T_{\max} = 86, \Delta = -104, \det = 625, (n_+, n_-, n_0) = (1, 74, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 78.31 \rightarrow 79, T_{\max} = 86, T \in [79, 86]
\end{aligned}$$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$$

[79] $(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)122315131613+\hat{1}\rightarrow 2$ $T = 74$, $T_{\max} = 87$, $\Delta = -111$, $\det = 676$, $(n_+, n_-, n_0) = (1, 74, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 78.30 \rightarrow 79$, $T_{\max} = 87$, $T \in [79, 87]$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

[80] $1(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 74$, $T_{\max} = 86$, $\Delta = -103$, $\det = 676$, $(n_+, n_-, n_0) = (1, 74, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 78.30 \rightarrow 79$, $T_{\max} = 86$, $T \in [79, 86]$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

[81] $61231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 74$, $T_{\max} = 86$, $\Delta = -104$, $\det = 676$, $(n_+, n_-, n_0) = (1, 74, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 78.30 \rightarrow 79$, $T_{\max} = 86$, $T \in [79, 86]$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

[82] $(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)122315131613+\hat{1}\rightarrow 2$ $T = 74$, $T_{\max} = 87$, $\Delta = -111$, $\det = 729$, $(n_+, n_-, n_0) = (1, 74, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 78.29 \rightarrow 79$, $T_{\max} = 87$, $T \in [79, 87]$

$$\begin{pmatrix} 0 & 27 & 2 \\ 27 & 620 & 48 \\ 2 & 48 & 3 \end{pmatrix}_{b_0}$$

[83] $(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315131613+\hat{1}\rightarrow 2$ $T = 74$, $T_{\max} = 87$, $\Delta = -111$, $\det = 841$, $(n_+, n_-, n_0) = (1, 74, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 78.27 \rightarrow 79$, $T_{\max} = 87$, $T \in [79, 87]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 724 & 52 \\ 2 & 52 & 3 \end{pmatrix}_{b_0}$$

[84] $(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)122315131613+\hat{1}\rightarrow 2$ $T = 74$, $T_{\max} = 87$, $\Delta = -111$, $\det = 841$, $(n_+, n_-, n_0) = (1, 74, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 78.27 \rightarrow 79$, $T_{\max} = 87$, $T \in [79, 87]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[85] $1(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161+\hat{1}\rightarrow 2$ $T = 74$, $T_{\max} = 86$, $\Delta = -103$, $\det = 841$, $(n_+, n_-, n_0) = (1, 74, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 78.27 \rightarrow 79$, $T_{\max} = 86$, $T \in [79, 86]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[86] $61231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 74$, $T_{\max} = 86$, $\Delta = -104$, $\det = 841$, $(n_+, n_-, n_0) = (1, 74, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 78.27 \rightarrow 79$, $T_{\max} = 86$, $T \in [79, 86]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[87] $(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)122315131613+\hat{1}\rightarrow 2$ $T = 74$, $T_{\max} = 87$, $\Delta = -111$, $\det = 961$, $(n_+, n_-, n_0) = (1, 74, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 78.25 \rightarrow 79$, $T_{\max} = 87$, $T \in [79, 87]$

$$\begin{pmatrix} 0 & 31 & 2 \\ 31 & 805 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0}$$

[88] $(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)122315131613+\hat{1}\rightarrow 2$ $T = 74$, $T_{\max} = 87$, $\Delta = -111$, $\det = 1156$, $(n_+, n_-, n_0) = (1, 74, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 78.23 \rightarrow 79$, $T_{\max} = 87$, $T \in [79, 87]$

$$\begin{pmatrix} 0 & 34 & 2 \\ 34 & 985 & 61 \\ 2 & 61 & 3 \end{pmatrix}_{b_0}$$

$T = 75$ [89] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1171+\hat{1}\rightarrow 2$ $T = 75$, $T_{\max} = 87$, $\Delta = -102$, $\det = -25$, $(n_+, n_-, n_0) = (1, 75, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.44 \rightarrow 81$, $T_{\max} = 87$, $T \in [81, 87]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[90] \ 7122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -25, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.44 \rightarrow 81, T_{\max} = 87, T \in [81, 87]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[91] \ 712^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -36, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.22 \rightarrow 81, T_{\max} = 87, T \in [81, 87]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[93] \ 7122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -49, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.06 \rightarrow 81, T_{\max} = 87, T \in [81, 87]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[92] \ 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 315132171+ \hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -49, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.06 \rightarrow 81, T_{\max} = 87, T \in [81, 87]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[99] \ 712231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1+ \hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.84 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[98] \ 712231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.84 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[97] \ 71223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.84 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[96] \ 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 2315132171+ \hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.84 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[95] \ 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+ \hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.84 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[94] \ 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)122315132171+ \hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.84 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[101] \ 712231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1+ \hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -100, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.76 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[100] & 1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -100, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.76 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[104] & 712231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.69 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[102] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.69 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[103] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.69 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[105] & 712231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.69 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[106] & 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -169, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.59 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[107] & 712231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -169, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.59 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[108] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.55 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[109] & 712231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -196, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.55 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[110] & 1(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -225, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.52 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[111] & 712231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -225, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.52 \rightarrow 80, T_{\max} = 87, T \in [80, 87]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[112] & 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -256, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.48 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[113] & 712231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -256, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.48 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[115] & 712231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -289, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.46 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[114] & 1(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -289, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.46 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[121] & 712231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.41 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[120] & 712231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.41 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[119] & 712231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.41 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[118] & 1(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.41 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[117] & 1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.41 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[116] & 1(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -361, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.41 \rightarrow 80, T_{\max} = 87, T \in [80, 87] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[125] & 712231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 79.37 \rightarrow 80, T_{\max} = 87, T \in [80, 87]
\end{aligned}$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[124] \text{ } 712231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.37 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[123] \text{ } 1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.37 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[122] \text{ } 1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -441, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.37 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[126] \text{ } 1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -529, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.34 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

$$[127] \text{ } 712231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -529, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.34 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

$$[129] \text{ } 712231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -576, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.33 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$$[128] \text{ } 1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -576, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.33 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$$[130] \text{ } 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -625, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.31 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$$

$$[131] \text{ } 712231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -625, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.31 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$$

$$[132] \text{ } 1(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -676, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.30 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[133] \text{ } 712231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -676, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.30 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[134] \quad 1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -841, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.27 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

$$[135] \quad 712231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 75, T_{\max} = 87, \Delta = -102, \det = -841, (n_+, n_-, n_0) = (1, 75, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 79.27 \rightarrow 80, T_{\max} = 87, T \in [80, 87]$$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

$$T = 76 \quad [136] \quad (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812^1 3+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 82.22 \rightarrow 83, T_{\max} = 88, T \in [83, 88]$$

$$\begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[137] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 1812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 25, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 81.44 \rightarrow 82, T_{\max} = 88, T \in [82, 88]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[138] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 2181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 25, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 81.44 \rightarrow 82, T_{\max} = 88, T \in [82, 88]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[139] \quad 1812^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 25, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 81.44 \rightarrow 82, T_{\max} = 88, T \in [82, 88]$$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[140] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 36, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 81.22 \rightarrow 82, T_{\max} = 88, T \in [82, 88]$$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[141] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 49, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 81.06 \rightarrow 82, T_{\max} = 88, T \in [82, 88]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[142] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 3151322181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 49, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 81.06 \rightarrow 82, T_{\max} = 88, T \in [82, 88]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 20 & 8 \\ 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[143] \quad 1812315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 49, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 81.06 \rightarrow 82, T_{\max} = 88, T \in [82, 88]$$

$$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[147] \quad 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 81, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.84 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[149] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 81, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.84 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[150] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151322181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 81, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.84 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[146] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 81, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.84 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[145] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 23151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 81, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.84 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[144] \quad (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 18123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 81, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.84 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[151] \quad 18123151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 81, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.84 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[152] \quad 181231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 81, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.84 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[153] \quad 181231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 81, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.84 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 8 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[148] \quad 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 81, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.84 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[154] \quad (12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 100, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.76 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[155] \quad 1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 100, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.76 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[156] \quad 1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 100, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.76 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[157] \quad 181231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221^1(12)1+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 100, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.76 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 9 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[165] \quad 181231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 121, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.69 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[164] \quad 181231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 121, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.69 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 43 & 13 \\ 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[158] \quad (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.69 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[159] \quad (12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 121, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.69 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[160] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221^1(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 121, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.69 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[161] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 121, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.69 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[162] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 121, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.69 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[163] \quad 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221^1(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 121, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.69 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[166] \quad (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 169, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 80.59 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[192] & 1(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 361, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 80.41 \rightarrow 81, T_{\max} = 88, T \in [81, 88] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[191] & 1(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 361, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 80.41 \rightarrow 81, T_{\max} = 88, T \in [81, 88] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[190] & 1(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 361, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 80.41 \rightarrow 81, T_{\max} = 88, T \in [81, 88] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[195] & 181231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 361, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 80.41 \rightarrow 81, T_{\max} = 88, T \in [81, 88] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 75 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[188] & 1(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 361, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 80.41 \rightarrow 81, T_{\max} = 88, T \in [81, 88] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[187] & (12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 361, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 80.41 \rightarrow 81, T_{\max} = 88, T \in [81, 88] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[186] & (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 361, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 80.41 \rightarrow 81, T_{\max} = 88, T \in [81, 88] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[196] & 181231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)1+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 361, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 80.41 \rightarrow 81, T_{\max} = 88, T \in [81, 88] \\
& \begin{pmatrix} 0 & 19 & 2 \\ 19 & 56 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[200] & 1(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 441, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 80.37 \rightarrow 81, T_{\max} = 88, T \in [81, 88] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[198] & (12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 441, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 80.37 \rightarrow 81, T_{\max} = 88, T \in [81, 88] \\
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[199] & 1(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 441, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 80.37 \rightarrow 81, T_{\max} = 88, T \in [81, 88]
\end{aligned}$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[197] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221^1(12)12231513221(12)12231513218123+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -110$, $\det = 441$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.37 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[201] $1(12)12231513221(12)12231513221(12)12231513221^1(12)12231513221(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 441$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.37 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[202] $1(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 441$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.37 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[203] $181231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 441$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.37 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[204] $181231513221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 441$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.37 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 83 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[205] $(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513218123+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -110$, $\det = 484$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.36 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[206] $(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513218123+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -110$, $\det = 529$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.34 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

[207] $1(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 529$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.34 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

[208] $1(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 529$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.34 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$

[209] $181231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 529$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.34 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 91 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[213] $181231513221(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 576$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 80.33 \rightarrow 81$, $T_{\max} = 88$, $T \in [81, 88]$

[212] $1(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 576$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$

[211] $1(12)12231513221(12)12231513221(12)12^{\hat{1}}231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 576$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$

[210] (12)12231513221(12)12231513221(12)12231513221(12)12¹231513221(12)12231513221(12)12231513218123+ $\hat{1} \rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -110$, $\det = 576$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$

[215] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 625$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$

[216] $1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 625, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$

[217] $\overset{1}{18}1231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12\overset{1}{1}231513221(12)1+\hat{1}\rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 625, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$

[214] $(12)12231513221(12)122315132^{\color{red}1}21(12)12231513221(12)12231513221(12)12231513221(12)12231513218123+\hat{\color{red}1}\rightarrow\color{red}2$ $T = 76$, $T_{\max} = 88$, $\Delta = -110$, $\det = 625$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$

[218] $(12)12231513221(12)12231513221(12)1223151322^{\color{red}1}1(12)12231513221(12)12231513221(12)12231513218123+\hat{\color{red}1}\rightarrow\color{red}2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 676, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$

[220] $1(12)12231513221(12)1223151322^1(12)12231513221(12)12231513221(12)12231513221(12)1223151322181+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 676$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$

[219] $1(12)12231513221(12)1223151322^1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1}\rightarrow 2$ $T = 76$, $T_{\max} = 88$, $\Delta = -101$, $\det = 676$, $(n_+, n_-, n_0) = (1, 76, 0)$, $c_{\text{ext}} = 15.0$

[221] $\overset{1}{18}1231513221(12)12231513221(12)12231513221(12)12231513221(12)12\overset{1}{\color{red}1}231513221(12)12231513221(12)1+\overset{1}{\color{red}\hat{1}}\rightarrow\color{red}{2} \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 676, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 129 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[222] $(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 729, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 80.29 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$

$$\begin{pmatrix} 0 & 27 & 2 \\ 27 & 620 & 48 \\ 2 & 48 & 3 \end{pmatrix}_{b_0}$$

[223] $(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 841, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 80.27 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 724 & 52 \\ 2 & 52 & 3 \end{pmatrix}_{b_0}$$

[224] $(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 841, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 80.27 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[225] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151321812+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 841, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 80.27 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[226] $1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322181+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 841, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 80.27 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[227] $181231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -101, \det = 841, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 80.27 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 144 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[228] $(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 961, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 80.25 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$

$$\begin{pmatrix} 0 & 31 & 2 \\ 31 & 805 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0}$$

[229] $(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513218123+\hat{1} \rightarrow 2 \quad T = 76, T_{\max} = 88, \Delta = -110, \det = 1156, (n_+, n_-, n_0) = (1, 76, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 80.23 \rightarrow 81, T_{\max} = 88, T \in [81, 88]$

$$\begin{pmatrix} 0 & 34 & 2 \\ 34 & 985 & 61 \\ 2 & 61 & 3 \end{pmatrix}_{b_0}$$

$T = 80$ [233] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 3221+\hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86, T_{\max} = 92, T \in [86, 92]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[230] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1222^1 2+\hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 87, T_{\max} = 92, T \in [87, 92]$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 1 & 1 \\ 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$22^1 21$ [231] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 87, T_{\max} = 92, T \in [87, 92]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[232] & 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 87, T_{\max} = 92, T \in [87, 92] \\
& \begin{pmatrix} 0 & 2 & 2 \\ 2 & 5 & 3 \\ 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[234] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 22+ \\
& \hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 9, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 86.22 \rightarrow 87, T_{\max} = 92, T \in [87, 92] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[235] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 9, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 86.22 \rightarrow 87, T_{\max} = 92, T \in [87, 92] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 11 & 5 \\ 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[236] & (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+^1 12+\hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -109, \det = 9, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 86.22 \rightarrow 87, T_{\max} = 92, T \in [87, 92] \\
& \begin{pmatrix} 0 & 3 & 2 \\ 3 & 2 & 2 \\ 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[237] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 222+ \\
& \hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 16, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 85.75 \rightarrow 86, T_{\max} = 92, T \in [86, 92] \\
& \begin{pmatrix} 0 & 4 & 2 \\ 4 & 3 & 3 \\ 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[238] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12222+ \\
& \hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 25, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 85.44 \rightarrow 86, T_{\max} = 92, T \in [86, 92] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 4 & 4 \\ 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[239] & 1(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 25, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 85.44 \rightarrow 86, T_{\max} = 92, T \in [86, 92] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[240] & 21(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 25, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 85.44 \rightarrow 86, T_{\max} = 92, T \in [86, 92] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[241] & (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+^1 12+\hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -109, \det = 25, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 85.44 \rightarrow 86, T_{\max} = 92, T \in [86, 92] \\
& \begin{pmatrix} 0 & 5 & 2 \\ 5 & 9 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[242] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12222+ \\
& \hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 36, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 85.22 \rightarrow 86, T_{\max} = 92, T \in [86, 92] \\
& \begin{pmatrix} 0 & 6 & 2 \\ 6 & 5 & 5 \\ 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[243] & 1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 36, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 85.22 \rightarrow 86, T_{\max} = 92, T \in [86, 92]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 9 & 2 \\ 9 & 71 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[254] & \begin{matrix} 221 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)1+\hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -100, \det = 81, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.84 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 9 & 2 \\ 9 & 8 & 8 \\ 2 & 8 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[250] & \begin{matrix} 1 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12222+ \\ \hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 81, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.84 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[251] & \begin{matrix} 2221 \\ 1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -100, \det = 81, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.84 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[258] & \begin{matrix} 1 \\ 1(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+ \\ \hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 100, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.76 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[259] & \begin{matrix} 2221 \\ 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1+\hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -100, \det = 100, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.76 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 10 & 2 \\ 10 & 9 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[260] & \begin{matrix} 221 \\ 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)1+\hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -100, \det = 100, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.76 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 10 & 2 \\ 10 & 9 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[261] & \begin{matrix} 1 \\ (12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -109, \det = 100, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.76 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[264] & \begin{matrix} 2221 \\ 1(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -100, \det = 121, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.69 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 54 & 14 \\ 2 & 14 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[270] & \begin{matrix} 1 \\ (12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+\hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.69 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[269] & \begin{matrix} 1 \\ (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12+\hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.69 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\ \begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0} \end{matrix} \\
[268] & \begin{matrix} 1 \\ (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12+\hat{1} \rightarrow 2 \\ T = 80, T_{\max} = 92, \Delta = -109, \det = 121, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\ T_{\min} = 84.69 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \end{matrix}
\end{aligned}$$

[267] $\overset{221}{21(12)12231513221(12)12^{\color{red}1}231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\color{red}\hat{1}\rightarrow\color{red}2}$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 121$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$

[266] $\overset{221}{21}(12)12231513221(12)122315132^{\color{red}1}21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\overset{1}{\color{red}\hat{1}}\rightarrow \color{red}{2}$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 121$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$

[265] $\overset{2221}{1}(12)12231513221(12)12231513\overset{1}{2}1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\overset{1}{\hat{1}}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 121, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$

[263] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+$
 $\hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 121, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$

[262] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^{\mathbf{1}}31513221(12)12231513221(12)12222 + \mathbf{\hat{1} \rightarrow 2}$ $T = 80, T_{\max} = 92, \Delta = -100, \det = 121, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$

[273] $\overset{221}{21}(12)12231513221(12)12231513221(12)122\overset{1}{3}1513221(12)12231513221(12)12231513221(12)12231513221(12)1+\overset{1}{\hat{1}}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 169$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$

[272] $\overset{2221}{1}(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)\overset{1}{1}2231513221(12)1+ \hat{\textcolor{red}{1}}\rightarrow \textcolor{red}{2}$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 169$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$

[271] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12222+$
 $\hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 169, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$

[274] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12+1\hat{\rightarrow}2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -109$, $\det = 169$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$

[278] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12+1\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -109, \det = 196, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$

[277] $21^{221}(12)12231513221(12)1223151322^{\color{red}1}1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{\color{red}1}\rightarrow\color{red}2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 196, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$

[287] $1(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 289$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 84.46 \rightarrow 85$, $T_{\max} = 92$, $T \in [85, 92]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[288] $1(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 289$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 84.46 \rightarrow 85$, $T_{\max} = 92$, $T \in [85, 92]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$

[290] $(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -109$, $\det = 289$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 84.46 \rightarrow 85$, $T_{\max} = 92$, $T \in [85, 92]$
 $\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$

[301] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -109$, $\det = 361$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 84.41 \rightarrow 85$, $T_{\max} = 92$, $T \in [85, 92]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

[300] $(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -109$, $\det = 361$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 84.41 \rightarrow 85$, $T_{\max} = 92$, $T \in [85, 92]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$

[297] $21(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 361$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 84.41 \rightarrow 85$, $T_{\max} = 92$, $T \in [85, 92]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$

[299] $21(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 361$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 84.41 \rightarrow 85$, $T_{\max} = 92$, $T \in [85, 92]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 75 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[298] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)1+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 361$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 84.41 \rightarrow 85$, $T_{\max} = 92$, $T \in [85, 92]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 56 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[295] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)1+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 361$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 84.41 \rightarrow 85$, $T_{\max} = 92$, $T \in [85, 92]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 56 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$

[296] $1(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 361$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 84.41 \rightarrow 85$, $T_{\max} = 92$, $T \in [85, 92]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 75 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$

[294] $1(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 361$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 84.41 \rightarrow 85$, $T_{\max} = 92$, $T \in [85, 92]$

[293] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12222+\hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 361, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.41 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$

[292] $1(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 361, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.41 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$

[291] $1(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 361, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.41 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$
 $\begin{pmatrix} 0 & 19 & 2 \\ 19 & 303 & 33 \\ 2 & 33 & 3 \end{pmatrix}_{b_0}$

[303] $1(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 441, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.37 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$

[304] $1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)1+\hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 441, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.37 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 83 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[305] $1(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 441, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.37 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[306] $21(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 441, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.37 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$

[307] $21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)1+\hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 441, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.37 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 83 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$

[308] $(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)12+\hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -109, \det = 441, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.37 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$

[309] $(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12+\hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -109, \det = 441, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.37 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$
 $\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$

[302] $1(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)12231513221(12)12222+\hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 441, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 84.37 \rightarrow 85, T_{\max} = 92, T \in [85, 92]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[310] & (12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -109, \det = 484, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 84.36 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\
& \begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0} \\
[314] & (12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -109, \det = 529, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 84.34 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0} \\
[313] & 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1+ \hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 529, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 84.34 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 91 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[312] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)1+ \hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 529, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 84.34 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 91 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[311] & 1(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+ \hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 529, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 84.34 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\
& \begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0} \\
[318] & (12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -109, \det = 576, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 84.33 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[317] & 21(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 576, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 84.33 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 119 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[315] & 1(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+ \hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 576, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 84.33 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[316] & 1(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)1+ \hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 576, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 84.33 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\
& \begin{pmatrix} 0 & 24 & 2 \\ 24 & 119 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[319] & 1(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+ \hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 625, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 84.31 \rightarrow 85, T_{\max} = 92, T \in [85, 92] \\
& \begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0} \\
[320] & 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1+ \hat{1} \rightarrow 2 \\
& T = 80, T_{\max} = 92, \Delta = -100, \det = 625, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 84.31 \rightarrow 85, T_{\max} = 92, T \in [85, 92]
\end{aligned}$$

[321] $\overset{221}{21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)1+\hat{1}\rightarrow 2}$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 625, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$

[322] $(12)12231513221(12)122315132^{\hat{1}}21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -109, \det = 625, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$

[323] $\overset{1}{1}(12)12231513221(12)1223151322\overset{1}{1}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+$
 $\hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 676, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$

[324] $1^{2221}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -100, \det = 676, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$

[325] $21^{221}(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)1+\hat{1}\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 676$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$

[326] $(12)12231513221(12)12231513221(12)12231513221^11(12)12231513221(12)12231513221(12)12231513221(12)12+\hat{1}\rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -109, \det = 676, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$

[327] $(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{1} \rightarrow 2$
 $T = 80, T_{\max} = 92, \Delta = -109, \det = 729, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$

[332] $(12)1223151322^{\overline{1}}1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12+1\rightarrow 2$
 $T = 80$, $T_{\max} = 92$, $\Delta = -109$, $\det = 841$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$

[331] $(12)12231513221(12)12231513221(12)12^{\mathbf{1}}231513221(12)12231513221(12)12231513221(12)12231513221(12)12+ \hat{\mathbf{1}} \rightarrow \mathbf{2}$
 $T = 80, T_{\max} = 92, \Delta = -109, \det = 841, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$

[328] $1(12)12231513221(12)12^1231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12222+$
 $\hat{1} \rightarrow 2 \quad T = 80, T_{\max} = 92, \Delta = -100, \det = 841, (n_+, n_-, n_0) = (1, 80, 0), c_{\text{ext}} = 15.0$

[329] $\overset{2221}{1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^{\color{red}1}1(12)12231513221(12)1+\hat{\textcolor{violet}{1}}\rightarrow\textcolor{red}{2}}$
 $T = 80$, $T_{\max} = 92$, $\Delta = -100$, $\det = 841$, $(n_+, n_-, n_0) = (1, 80, 0)$, $c_{\text{ext}} = 15.0$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 51 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[343] $413221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)+$
 $\hat{1} \rightarrow 2 \quad T = 81, T_{\max} = 93, \Delta = -108, \det = -196, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.55 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 69 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[344] $413221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)+$
 $\hat{1} \rightarrow 2 \quad T = 81, T_{\max} = 93, \Delta = -108, \det = -225, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.52 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 59 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[345] $413221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)+$
 $\hat{1} \rightarrow 2 \quad T = 81, T_{\max} = 93, \Delta = -108, \det = -256, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.48 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 79 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[346] $413221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)+$
 $\hat{1} \rightarrow 2 \quad T = 81, T_{\max} = 93, \Delta = -108, \det = -289, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.46 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[348] $413221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)+$
 $\hat{1} \rightarrow 2 \quad T = 81, T_{\max} = 93, \Delta = -108, \det = -361, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.41 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[347] $413221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231513221(12)+$
 $\hat{1} \rightarrow 2 \quad T = 81, T_{\max} = 93, \Delta = -108, \det = -361, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.41 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 75 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[350] $413221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)+$
 $\hat{1} \rightarrow 2 \quad T = 81, T_{\max} = 93, \Delta = -108, \det = -441, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.37 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 83 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[349] $413221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)+$
 $\hat{1} \rightarrow 2 \quad T = 81, T_{\max} = 93, \Delta = -108, \det = -441, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.37 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[351] $413221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)+$
 $\hat{1} \rightarrow 2 \quad T = 81, T_{\max} = 93, \Delta = -108, \det = -484, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.36 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 65 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[352] $413221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)+$
 $\hat{1} \rightarrow 2 \quad T = 81, T_{\max} = 93, \Delta = -108, \det = -529, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.34 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 91 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[353] $413221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231513221(12)12231513221(12)+$
 $\hat{1} \rightarrow 2 \quad T = 81, T_{\max} = 93, \Delta = -108, \det = -576, (n_+, n_-, n_0) = (1, 81, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 85.33 \rightarrow 86, T_{\max} = 93, T \in [86, 93]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[3] (12)12231513221¹(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132¹21(12)12231414+
 $\hat{1} \rightarrow 2$ $T = 82$, $T_{\max} = 94$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) = (1, 82, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 86.84 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[4] (12)12231513221¹(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12¹231414+
 $\hat{1} \rightarrow 2$ $T = 82$, $T_{\max} = 94$, $\Delta = -107$, $\det = 81$, $(n_+, n_-, n_0) = (1, 82, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 86.84 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[6] (12)122¹31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231414+
 $\hat{1} \rightarrow 2$ $T = 82$, $T_{\max} = 94$, $\Delta = -107$, $\det = 100$, $(n_+, n_-, n_0) = (1, 82, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 86.76 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[7] (12)122¹31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221¹⁵¹³²²¹(12)1+ $\hat{1} \rightarrow 2$
 $T = 82$, $T_{\max} = 94$, $\Delta = -103$, $\det = 100$, $(n_+, n_-, n_0) = (1, 82, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 86.76 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[9] (12)12231513221¹(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322¹1(12)12231414+
 $\hat{1} \rightarrow 2$ $T = 82$, $T_{\max} = 94$, $\Delta = -107$, $\det = 121$, $(n_+, n_-, n_0) = (1, 82, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 86.69 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[8] (12)12231513221¹(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122¹31513221(12)12231414+
 $\hat{1} \rightarrow 2$ $T = 82$, $T_{\max} = 94$, $\Delta = -107$, $\det = 121$, $(n_+, n_-, n_0) = (1, 82, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 86.69 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[13] (12)12¹231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221¹⁵¹³²²¹(12)1+ $\hat{1} \rightarrow 2$
 $T = 82$, $T_{\max} = 94$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 82, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 86.69 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[12] (12)12231513221¹(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322¹⁵¹³²²¹1(12)1+ $\hat{1} \rightarrow 2$
 $T = 82$, $T_{\max} = 94$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 82, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 86.69 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[11] (12)12231513221¹(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122¹31513221¹⁵¹³²²¹(12)1+ $\hat{1} \rightarrow 2$
 $T = 82$, $T_{\max} = 94$, $\Delta = -103$, $\det = 121$, $(n_+, n_-, n_0) = (1, 82, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 86.69 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[10] (12)12¹231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231414+
 $\hat{1} \rightarrow 2$ $T = 82$, $T_{\max} = 94$, $\Delta = -107$, $\det = 121$, $(n_+, n_-, n_0) = (1, 82, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 86.69 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[15] (12)12231513221¹(12)12231513221(12)12231513221(12)12231513221(12)122315132¹⁵¹³²²¹21(12)12231513221¹⁵¹³²²¹(12)1+ $\hat{1} \rightarrow 2$
 $T = 82$, $T_{\max} = 94$, $\Delta = -103$, $\det = 169$, $(n_+, n_-, n_0) = (1, 82, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 86.59 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$$

- $$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$
- [14] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231414+$
 $\hat{1} \rightarrow 2 \quad T = 82, T_{\max} = 94, \Delta = -107, \det = 169, (n_+, n_-, n_0) = (1, 82, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.59 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$
- [17] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)1+1\hat{1} \rightarrow 2$
 $T = 82, T_{\max} = 94, \Delta = -103, \det = 196, (n_+, n_-, n_0) = (1, 82, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.55 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$
- [16] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231414+$
 $\hat{1} \rightarrow 2 \quad T = 82, T_{\max} = 94, \Delta = -107, \det = 196, (n_+, n_-, n_0) = (1, 82, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.55 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$
- [19] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)1+1\hat{1} \rightarrow 2$
 $T = 82, T_{\max} = 94, \Delta = -103, \det = 225, (n_+, n_-, n_0) = (1, 82, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.52 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$
- [18] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231414+$
 $\hat{1} \rightarrow 2 \quad T = 82, T_{\max} = 94, \Delta = -107, \det = 225, (n_+, n_-, n_0) = (1, 82, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.52 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$
- [20] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)12231414+$
 $\hat{1} \rightarrow 2 \quad T = 82, T_{\max} = 94, \Delta = -107, \det = 256, (n_+, n_-, n_0) = (1, 82, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.48 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$
- [21] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513221(12)1+1\hat{1} \rightarrow 2$
 $T = 82, T_{\max} = 94, \Delta = -103, \det = 256, (n_+, n_-, n_0) = (1, 82, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.48 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$
- [23] $(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)1+1\hat{1} \rightarrow 2$
 $T = 82, T_{\max} = 94, \Delta = -103, \det = 289, (n_+, n_-, n_0) = (1, 82, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.46 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$
- [22] $(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513221(12)12231513221(12)12231414+$
 $\hat{1} \rightarrow 2 \quad T = 82, T_{\max} = 94, \Delta = -107, \det = 289, (n_+, n_-, n_0) = (1, 82, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.46 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$
- [24] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513221(12)12231513221(12)12231414+$
 $\hat{1} \rightarrow 2 \quad T = 82, T_{\max} = 94, \Delta = -107, \det = 361, (n_+, n_-, n_0) = (1, 82, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.41 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$
- [25] $(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513221(12)12231513221(12)12231414+$
 $\hat{1} \rightarrow 2 \quad T = 82, T_{\max} = 94, \Delta = -107, \det = 361, (n_+, n_-, n_0) = (1, 82, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 86.41 \rightarrow 87, T_{\max} = 94, T \in [87, 94]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[59] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)1223151315+$
 $\hat{1} \rightarrow 2 \quad T = 84, T_{\max} = 96, \Delta = -105, \det = 169, (n_+, n_-, n_0) = (1, 84, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 88.59 \rightarrow 89, T_{\max} = 96, T \in [89, 96]$

$$\begin{pmatrix} 0 & 13 & 2 \\ 13 & 116 & 20 \\ 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[60] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)1223151315+$
 $\hat{1} \rightarrow 2 \quad T = 84, T_{\max} = 96, \Delta = -105, \det = 196, (n_+, n_-, n_0) = (1, 84, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 88.55 \rightarrow 89, T_{\max} = 96, T \in [89, 96]$

$$\begin{pmatrix} 0 & 14 & 2 \\ 14 & 125 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[61] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)1223151315+$
 $\hat{1} \rightarrow 2 \quad T = 84, T_{\max} = 96, \Delta = -105, \det = 225, (n_+, n_-, n_0) = (1, 84, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 88.52 \rightarrow 89, T_{\max} = 96, T \in [89, 96]$

$$\begin{pmatrix} 0 & 15 & 2 \\ 15 & 164 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[62] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)1223151315+$
 $\hat{1} \rightarrow 2 \quad T = 84, T_{\max} = 96, \Delta = -105, \det = 256, (n_+, n_-, n_0) = (1, 84, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 88.48 \rightarrow 89, T_{\max} = 96, T \in [89, 96]$

$$\begin{pmatrix} 0 & 16 & 2 \\ 16 & 175 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[63] $(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)1223151315+$
 $\hat{1} \rightarrow 2 \quad T = 84, T_{\max} = 96, \Delta = -105, \det = 289, (n_+, n_-, n_0) = (1, 84, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 88.46 \rightarrow 89, T_{\max} = 96, T \in [89, 96]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 220 & 28 \\ 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[64] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)1223151315+$
 $\hat{1} \rightarrow 2 \quad T = 84, T_{\max} = 96, \Delta = -105, \det = 361, (n_+, n_-, n_0) = (1, 84, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 88.41 \rightarrow 89, T_{\max} = 96, T \in [89, 96]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[65] $(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)1223151315+$
 $\hat{1} \rightarrow 2 \quad T = 84, T_{\max} = 96, \Delta = -105, \det = 361, (n_+, n_-, n_0) = (1, 84, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 88.41 \rightarrow 89, T_{\max} = 96, T \in [89, 96]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[66] $(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)1223151315+$
 $\hat{1} \rightarrow 2 \quad T = 84, T_{\max} = 96, \Delta = -105, \det = 441, (n_+, n_-, n_0) = (1, 84, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 88.37 \rightarrow 89, T_{\max} = 96, T \in [89, 96]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[67] $(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)1223151315+$
 $\hat{1} \rightarrow 2 \quad T = 84, T_{\max} = 96, \Delta = -105, \det = 441, (n_+, n_-, n_0) = (1, 84, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 88.37 \rightarrow 89, T_{\max} = 96, T \in [89, 96]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[68] $(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315+$
 $\hat{1} \rightarrow 2 \quad T = 84, T_{\max} = 96, \Delta = -105, \det = 484, (n_+, n_-, n_0) = (1, 84, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 88.36 \rightarrow 89, T_{\max} = 96, T \in [89, 96]$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[69] $(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)1223151315+$
 $\hat{1} \rightarrow 2 \quad T = 84, T_{\max} = 96, \Delta = -105, \det = 529, (n_+, n_-, n_0) = (1, 84, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 88.34 \rightarrow 89, T_{\max} = 96, T \in [89, 96]$

- $$\begin{pmatrix} 0 & 7 & 2 \\ 7 & 27 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$
- [84] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513161 + \hat{1} \rightarrow 2$ $T = 85, T_{\max} = 97, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 85, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 89.84 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$
- [81] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513216 + \hat{1} \rightarrow 2$ $T = 85, T_{\max} = 97, \Delta = -103, \det = -81, (n_+, n_-, n_0) = (1, 85, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 89.84 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 35 & 11 \\ 2 & 11 & 3 \end{pmatrix}_{b_0}$$
- [82] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132^1 21(12)12231513216 + \hat{1} \rightarrow 2$ $T = 85, T_{\max} = 97, \Delta = -103, \det = -81, (n_+, n_-, n_0) = (1, 85, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 89.84 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$
- [83] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1 231513161 + \hat{1} \rightarrow 2$ $T = 85, T_{\max} = 97, \Delta = -102, \det = -81, (n_+, n_-, n_0) = (1, 85, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 89.84 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$

$$\begin{pmatrix} 0 & 9 & 2 \\ 9 & 44 & 12 \\ 2 & 12 & 3 \end{pmatrix}_{b_0}$$
- [86] $(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 85, T_{\max} = 97, \Delta = -102, \det = -100, (n_+, n_-, n_0) = (1, 85, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 89.76 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 9 & 9 \\ 2 & 9 & 3 \end{pmatrix}_{b_0}$$
- [85] $(12)122^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 85, T_{\max} = 97, \Delta = -103, \det = -100, (n_+, n_-, n_0) = (1, 85, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 89.76 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$

$$\begin{pmatrix} 0 & 10 & 2 \\ 10 & 89 & 17 \\ 2 & 17 & 3 \end{pmatrix}_{b_0}$$
- [87] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 85, T_{\max} = 97, \Delta = -103, \det = -121, (n_+, n_-, n_0) = (1, 85, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 89.69 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 76 & 16 \\ 2 & 16 & 3 \end{pmatrix}_{b_0}$$
- [88] $(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513216 + \hat{1} \rightarrow 2$ $T = 85, T_{\max} = 97, \Delta = -103, \det = -121, (n_+, n_-, n_0) = (1, 85, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 89.69 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 109 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$
- [89] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322^1 1(12)12231513216 + \hat{1} \rightarrow 2$ $T = 85, T_{\max} = 97, \Delta = -103, \det = -121, (n_+, n_-, n_0) = (1, 85, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 89.69 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 65 & 15 \\ 2 & 15 & 3 \end{pmatrix}_{b_0}$$
- [90] $(12)12^1 231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 85, T_{\max} = 97, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 85, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 89.69 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$

$$\begin{pmatrix} 0 & 11 & 2 \\ 11 & 10 & 10 \\ 2 & 10 & 3 \end{pmatrix}_{b_0}$$
- [91] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122^1 31513221(12)12231513161 + \hat{1} \rightarrow 2$ $T = 85, T_{\max} = 97, \Delta = -102, \det = -121, (n_+, n_-, n_0) = (1, 85, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 89.69 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[106] (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12¹231513221(12)12231513221(12)12231513161+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 89.41 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[105] (12)12231513221(12)12231513221(12)12231513221(12)122¹31513221(12)12231513221(12)12231513221(12)12231513161+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -102$, $\det = -361$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 89.41 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 75 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[104] (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12¹231513221(12)12231513221(12)12231513216+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -103$, $\det = -361$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 89.41 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[103] (12)12231513221(12)12231513221(12)12231513221(12)122¹31513221(12)12231513221(12)12231513221(12)12231513216+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -103$, $\det = -361$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 89.41 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[107] (12)12231513221(12)12231513221(12)12231513221(12)1223151322¹1(12)12231513221(12)12231513221(12)12231513216+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -103$, $\det = -441$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 89.37 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[108] (12)12231513221(12)12231513221(12)122315132¹21(12)12231513221(12)12231513221(12)12231513221(12)12231513216+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -103$, $\det = -441$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 89.37 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[109] (12)12231513221(12)12231513221(12)12231513221(12)1223151322¹1(12)12231513221(12)12231513221(12)12231513161+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -102$, $\det = -441$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 89.37 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[110] (12)12231513221(12)12231513221(12)122315132¹21(12)12231513221(12)12231513221(12)12231513221(12)12231513161+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -102$, $\det = -441$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 89.37 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 83 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[111] (12)122315132¹21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513216+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -103$, $\det = -484$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 89.36 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[112] (12)122315132¹21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -102$, $\det = -484$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 89.36 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 65 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[113] (12)12231513221(12)12231513221(12)122¹31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513216+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -103$, $\det = -529$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 89.34 \rightarrow 90, T_{\max} = 97, T \in [90, 97]$$

- $$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 436 & 40 \\ 2 & 40 & 3 \end{pmatrix}_{b_0}$$
- [114] (12)12231513221(12)12231513221(12)122¹31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -102$, $\det = -529$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 89.34 \rightarrow 90$, $T_{\max} = 97$, $T \in [90, 97]$

$$\begin{pmatrix} 0 & 23 & 2 \\ 23 & 91 & 25 \\ 2 & 25 & 3 \end{pmatrix}_{b_0}$$
- [115] (12)12231513221(12)12231513221(12)12231513221(12)12¹231513221(12)12231513221(12)12231513221(12)12231513216+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -103$, $\det = -576$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 89.33 \rightarrow 90$, $T_{\max} = 97$, $T \in [90, 97]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 455 & 41 \\ 2 & 41 & 3 \end{pmatrix}_{b_0}$$
- [116] (12)12231513221(12)12231513221(12)12231513221(12)12¹231513221(12)12231513221(12)12231513221(12)12231513161+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -102$, $\det = -576$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 89.33 \rightarrow 90$, $T_{\max} = 97$, $T \in [90, 97]$

$$\begin{pmatrix} 0 & 24 & 2 \\ 24 & 119 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$
- [117] (12)12231513221(12)122315132¹21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513216+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -103$, $\det = -625$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 89.31 \rightarrow 90$, $T_{\max} = 97$, $T \in [90, 97]$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 524 & 44 \\ 2 & 44 & 3 \end{pmatrix}_{b_0}$$
- [118] (12)12231513221(12)122315132¹21(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -102$, $\det = -625$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 89.31 \rightarrow 90$, $T_{\max} = 97$, $T \in [90, 97]$

$$\begin{pmatrix} 0 & 25 & 2 \\ 25 & 99 & 27 \\ 2 & 27 & 3 \end{pmatrix}_{b_0}$$
- [120] (12)12231513221(12)12231513221(12)1223151322¹1(12)12231513221(12)12231513221(12)12231513221(12)12231513161+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -102$, $\det = -676$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 89.30 \rightarrow 90$, $T_{\max} = 97$, $T \in [90, 97]$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 129 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$
- [119] (12)12231513221(12)12231513221(12)1223151322¹1(12)12231513221(12)12231513221(12)12231513221(12)12231513216+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -103$, $\det = -676$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 89.30 \rightarrow 90$, $T_{\max} = 97$, $T \in [90, 97]$

$$\begin{pmatrix} 0 & 26 & 2 \\ 26 & 545 & 45 \\ 2 & 45 & 3 \end{pmatrix}_{b_0}$$
- [122] (12)12231513221(12)122¹31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -102$, $\det = -729$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 89.29 \rightarrow 90$, $T_{\max} = 97$, $T \in [90, 97]$

$$\begin{pmatrix} 0 & 27 & 2 \\ 27 & 107 & 29 \\ 2 & 29 & 3 \end{pmatrix}_{b_0}$$
- [121] (12)12231513221(12)122¹31513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513216+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -103$, $\det = -729$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 89.29 \rightarrow 90$, $T_{\max} = 97$, $T \in [90, 97]$

$$\begin{pmatrix} 0 & 27 & 2 \\ 27 & 620 & 48 \\ 2 & 48 & 3 \end{pmatrix}_{b_0}$$
- [123] (12)1223151322¹1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513216+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -103$, $\det = -841$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 89.27 \rightarrow 90$, $T_{\max} = 97$, $T \in [90, 97]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 724 & 52 \\ 2 & 52 & 3 \end{pmatrix}_{b_0}$$
- [125] (12)1223151322¹1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513161+
 $\hat{1} \rightarrow 2$ $T = 85$, $T_{\max} = 97$, $\Delta = -102$, $\det = -841$, $(n_+, n_-, n_0) = (1, 85, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 89.27 \rightarrow 90$, $T_{\max} = 97$, $T \in [90, 97]$

$$\begin{pmatrix} 0 & 17 & 2 \\ 17 & 67 & 19 \\ 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[158] $(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)122315132217+\hat{1}\rightarrow 2$ $T = 86, T_{\max} = 98, \Delta = -101, \det = 361, (n_+, n_-, n_0) = (1, 86, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 90.41 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 284 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[159] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)122315132217+\hat{1}\rightarrow 2$ $T = 86, T_{\max} = 98, \Delta = -101, \det = 361, (n_+, n_-, n_0) = (1, 86, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 90.41 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 265 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[160] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 86, T_{\max} = 98, \Delta = -101, \det = 361, (n_+, n_-, n_0) = (1, 86, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 90.41 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 94 & 22 \\ 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[161] $(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 86, T_{\max} = 98, \Delta = -101, \det = 361, (n_+, n_-, n_0) = (1, 86, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 90.41 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$

$$\begin{pmatrix} 0 & 19 & 2 \\ 19 & 75 & 21 \\ 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[162] $(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)122315132217+\hat{1}\rightarrow 2$ $T = 86, T_{\max} = 98, \Delta = -101, \det = 441, (n_+, n_-, n_0) = (1, 86, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 90.37 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 335 & 35 \\ 2 & 35 & 3 \end{pmatrix}_{b_0}$$

[164] $(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 86, T_{\max} = 98, \Delta = -101, \det = 441, (n_+, n_-, n_0) = (1, 86, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 90.37 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 83 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[163] $(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)122315132217+\hat{1}\rightarrow 2$ $T = 86, T_{\max} = 98, \Delta = -101, \det = 441, (n_+, n_-, n_0) = (1, 86, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 90.37 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 356 & 36 \\ 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[165] $(12)12231513221(12)12231513221(12)12231513221(12)1223151322^11(12)12231513221(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 86, T_{\max} = 98, \Delta = -101, \det = 441, (n_+, n_-, n_0) = (1, 86, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 90.37 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$

$$\begin{pmatrix} 0 & 21 & 2 \\ 21 & 104 & 24 \\ 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[167] $(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 86, T_{\max} = 98, \Delta = -101, \det = 484, (n_+, n_-, n_0) = (1, 86, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 90.36 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 65 & 23 \\ 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[166] $(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132217+\hat{1}\rightarrow 2$ $T = 86, T_{\max} = 98, \Delta = -101, \det = 484, (n_+, n_-, n_0) = (1, 86, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 90.36 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$

$$\begin{pmatrix} 0 & 22 & 2 \\ 22 & 417 & 39 \\ 2 & 39 & 3 \end{pmatrix}_{b_0}$$

[169] $(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+\hat{1}\rightarrow 2$ $T = 86, T_{\max} = 98, \Delta = -101, \det = 529, (n_+, n_-, n_0) = (1, 86, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 90.34 \rightarrow 91, T_{\max} = 98, T \in [91, 98]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 695 & 51 \\ 2 & 51 & 3 \end{pmatrix}_{b_0}$$

[181] (12)12231513221(12)12231513221(12)12¹231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 841$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 90.27 \rightarrow 91$, $T_{\max} = 98$, $T \in [91, 98]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 144 & 32 \\ 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[180] (12)1223151322¹1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 841$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 90.27 \rightarrow 91$, $T_{\max} = 98$, $T \in [91, 98]$

$$\begin{pmatrix} 0 & 29 & 2 \\ 29 & 115 & 31 \\ 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[183] (12)12231513221(12)1223151322¹1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 961$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 90.25 \rightarrow 91$, $T_{\max} = 98$, $T \in [91, 98]$

$$\begin{pmatrix} 0 & 31 & 2 \\ 31 & 154 & 34 \\ 2 & 34 & 3 \end{pmatrix}_{b_0}$$

[182] (12)12231513221(12)1223151322¹1(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132217+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 961$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 90.25 \rightarrow 91$, $T_{\max} = 98$, $T \in [91, 98]$

$$\begin{pmatrix} 0 & 31 & 2 \\ 31 & 805 & 55 \\ 2 & 55 & 3 \end{pmatrix}_{b_0}$$

[185] (12)12231513221(12)12¹231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 1156$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 90.23 \rightarrow 91$, $T_{\max} = 98$, $T \in [91, 98]$

$$\begin{pmatrix} 0 & 34 & 2 \\ 34 & 169 & 37 \\ 2 & 37 & 3 \end{pmatrix}_{b_0}$$

[184] (12)12231513221(12)12¹231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132217+
 $\hat{1} \rightarrow 2$ $T = 86$, $T_{\max} = 98$, $\Delta = -101$, $\det = 1156$, $(n_+, n_-, n_0) = (1, 86, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 90.23 \rightarrow 91$, $T_{\max} = 98$, $T \in [91, 98]$

$$\begin{pmatrix} 0 & 34 & 2 \\ 34 & 985 & 61 \\ 2 & 61 & 3 \end{pmatrix}_{b_0}$$

$T = 87$ [187] (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132171+
 $\hat{1} \rightarrow 2$ $T = 87$, $T_{\max} = 99$, $\Delta = -100$, $\det = -25$, $(n_+, n_-, n_0) = (1, 87, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 92.44 \rightarrow 93$, $T_{\max} = 99$, $T \in [93, 99]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[188] (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132¹181+
 $\hat{1} \rightarrow 2$ $T = 87$, $T_{\max} = 99$, $\Delta = -100$, $\det = -25$, $(n_+, n_-, n_0) = (1, 87, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 92.44 \rightarrow 93$, $T_{\max} = 99$, $T \in [93, 99]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[186] (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122315132¹1812+
 $\hat{1} \rightarrow 2$ $T = 87$, $T_{\max} = 99$, $\Delta = -100$, $\det = -25$, $(n_+, n_-, n_0) = (1, 87, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 92.44 \rightarrow 93$, $T_{\max} = 99$, $T \in [93, 99]$

$$\begin{pmatrix} 0 & 5 & 2 \\ 5 & 19 & 7 \\ 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[189] (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)1223151322¹181+
 $\hat{1} \rightarrow 2$ $T = 87$, $T_{\max} = 99$, $\Delta = -100$, $\det = -36$, $(n_+, n_-, n_0) = (1, 87, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 92.22 \rightarrow 93$, $T_{\max} = 99$, $T \in [93, 99]$

$$\begin{pmatrix} 0 & 6 & 2 \\ 6 & 29 & 9 \\ 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[190] (12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)122¹3151321812+
 $\hat{1} \rightarrow 2$ $T = 87$, $T_{\max} = 99$, $\Delta = -100$, $\det = -49$, $(n_+, n_-, n_0) = (1, 87, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 92.06 \rightarrow 93$, $T_{\max} = 99$, $T \in [93, 99]$

[illegible]

[illegible]

[224] $(12)12231513221(12)12231513221(12)12231513221(12)122315132^{\color{red}1}21(12)12231513221(12)12231513221(12)1223151322181 + \color{red}{\hat{1}} \rightarrow 2$ $T = 87$, $T_{\max} = 99$, $\Delta = -100$, $\det = -289$, $(n_+, n_-, n_0) = (1, 87, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 91.46 \rightarrow 92$, $T_{\max} = 99$, $T \in [92, 99]$

[225] $(12)12231513221(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)122315132181 +$
 $\hat{1} \rightarrow 2 \quad T = 87, T_{\max} = 99, \Delta = -100, \det = -289, (n_+, n_-, n_0) = (1, 87, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 91.46 \rightarrow 92, T_{\max} = 99, T \in [92, 99]$

[226] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)1223151321812 + \hat{\mathbf{1}} \rightarrow 2$ $T = 87$, $T_{\max} = 99$, $\Delta = -100$, $\det = -361$, $(n_+, n_-, n_0) = (1, 87, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 91.41 \rightarrow 92$, $T_{\max} = 99$, $T \in [92, 99]$

[227] $(12)12231513221(12)12231513221(12)12231513221(12)122^{\overset{1}{\rightarrow}1}31513221(12)12231513221(12)12231513221(12)1223151321812 + \overset{1}{\rightarrow}1 \rightarrow 2$ $T = 87$, $T_{\max} = 99$, $\Delta = -100$, $\det = -361$, $(n_+, n_-, n_0) = (1, 87, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 91.41 \rightarrow 92$, $T_{\max} = 99$, $T \in [92, 99]$

[228] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^{\color{red}1}231513221(12)12231513221(12)1223151322181 + \color{red}{\hat{1}} \rightarrow 2$ $T = 87$, $T_{\max} = 99$, $\Delta = -100$, $\det = -361$, $(n_+, n_-, n_0) = (1, 87, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 91.41 \rightarrow 92$, $T_{\max} = 99$, $T \in [92, 99]$

[229] $(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$ $T = 87$, $T_{\max} = 99$, $\Delta = -100$, $\det = -361$, $(n_+, n_-, n_0) = (1, 87, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 91.41 \rightarrow 92$, $T_{\max} = 99$, $T \in [92, 99]$

[230] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221(12)12^1231513221(12)12231513221(12)122315132181 +$
 $\hat{1} \rightarrow 2 \quad T = 87, T_{\max} = 99, \Delta = -100, \det = -361, (n_+, n_-, n_0) = (1, 87, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 91.41 \rightarrow 92, T_{\max} = 99, T \in [92, 99]$

[231] $(12)12231513221(12)12231513221(12)12231513221(12)122^131513221(12)12231513221(12)12231513221(12)122315132181 +$
 $\hat{1} \rightarrow 2 \quad T = 87, T_{\max} = 99, \Delta = -100, \det = -361, (n_+, n_-, n_0) = (1, 87, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 91.41 \rightarrow 92, T_{\max} = 99, T \in [92, 99]$

[234] $(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)1223151322181 +$
 $\hat{1} \rightarrow 2 \quad T = 87, T_{\max} = 99, \Delta = -100, \det = -441, (n_+, n_-, n_0) = (1, 87, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 91.37 \rightarrow 92, T_{\max} = 99, T \in [92, 99]$

[235] $(12)12231513221(12)12231513221(12)12231513221(12)12231513221^1(12)12231513221(12)12231513221(12)1223151322181 + \hat{1} \rightarrow 2$
 $T = 87, T_{\max} = 99, \Delta = -100, \det = -441, (n_+, n_-, n_0) = (1, 87, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 91.37 \rightarrow 92, T_{\max} = 99, T \in [92, 99]$

[236] $(12)12231513221(12)12231513221(12)122315132^121(12)12231513221(12)12231513221(12)12231513221(12)122315132181 +$
 $\hat{1} \rightarrow 2 \quad T = 87, T_{\max} = 99, \Delta = -100, \det = -441, (n_+, n_-, n_0) = (1, 87, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 91.37 \rightarrow 92, T_{\max} = 99, T \in [92, 99]$

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

[illegible]

1.9 $\mathfrak{so}_8 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$

$$T = 4 \quad [0] \quad 1^{11}1^1 + \mathfrak{so}_8 + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 106, \Delta = -240, \det = 16, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12, T_{\max} = 106, T \in [12, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & -1 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [1] \quad 1^{11}21^1 + \mathfrak{so}_8 + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 100, \Delta = -211, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12, T_{\max} = 100, T \in [12, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 2 & 0 & 0 \\ 1 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [2] \quad 1^{11}321^1 + \mathfrak{so}_8 + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 100, \Delta = -162, \det = -3, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 1 & 0 & 2 & 2 \\ 2 & 0 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

1.10 $\mathfrak{su}_2 + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$

$$T = 4 \quad [0] \quad 1^{11}1^1 + \mathfrak{su}_2 + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 170, \Delta = -261, \det = 12, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 170, T \in [13, 170]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & -4 & 0 & -2 \\ 1 & 0 & -1 & -1 \\ 3 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}1^1 + \mathfrak{su}_2 + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 170, \Delta = -261, \det = 17, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 170, T \in [13, 170]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -6 & 0 & -4 \\ 2 & 0 & 1 & 1 \\ 3 & -4 & 1 & -2 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [2] \quad 1^{11}21^1 + \mathfrak{su}_2 + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 101, \Delta = -232, \det = -4, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13, T_{\max} = 101, T \in [13, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & -4 & 0 & -2 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}21^1 + \mathfrak{su}_2 + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 101, \Delta = -232, \det = -14, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 101, T \in [13, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -6 & 0 & -4 \\ 1 & 2 & 0 & 1 & 1 \\ 2 & 2 & -4 & 1 & -2 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [4] \quad 1^{11}221^1 + \mathfrak{su}_2 + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 100, \Delta = -203, \det = 11, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -6 & 0 & -4 \\ 1 & 3 & 0 & 2 & 2 \\ 2 & 3 & -4 & 2 & -2 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [5] \quad 1^{11}2221^1 + \mathfrak{su}_2 + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 100, \Delta = -174, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 0 & 4 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [6] \quad 1^{11}22221^1 + \mathfrak{su}_2 + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 98, T \in [15, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 0 & 5 & 5 \\ 1 & 0 & -6 & 0 & -4 \\ 1 & 5 & 0 & 4 & 4 \\ 2 & 5 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [7] \quad 1^{11}22221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_6 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 18, T_{\max} = 29, T \in [18, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 4 & 0 & 6 & 6 \\ 1 & 0 & -6 & 0 & -4 \\ 1 & 6 & 0 & 5 & 5 \\ 2 & 6 & -4 & 5 & -3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11}223131^1 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_6 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 93, T \in [16, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 0 & 5 & 5 \\ 1 & 0 & -6 & 0 & -4 \\ 1 & 5 & 0 & 4 & 4 \\ 2 & 5 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [9] \quad 1^{11}2231321^1 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_6 + \hat{1} \rightarrow 1 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 19, T_{\max} = 22, T \in [19, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 4 & 0 & 6 & 6 \\ 1 & 0 & -6 & 0 & -4 \\ 1 & 6 & 0 & 5 & 5 \\ 2 & 6 & -4 & 5 & -3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [10] \quad 1^131413221^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{e}_6 + \hat{1} \rightarrow 1 \quad T = 11, T_{\max} = 23, \Delta = -103, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 20, T_{\max} = 23, T \in [20, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 2 \\ 2 & 3 & -1 & 2 & -3 \end{pmatrix}_{b_0}$$

1.11 $su_2 + e_7 + \hat{1} \rightarrow (-1)$

$$T = 4 \quad [0] \quad 1^{11}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_7 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 188, \Delta = -316, \det = 16, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.50 \rightarrow 15, T_{\max} = 188, T \in [15, 188]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & -6 & 0 & -4 \\ 1 & 0 & -1 & -1 \\ 3 & -4 & -1 & -4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_7 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 188, \Delta = -316, \det = 23, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.52 \rightarrow 15, T_{\max} = 188, T \in [15, 188]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -8 & 0 & -6 \\ 2 & 0 & 1 & 1 \\ 3 & -6 & 1 & -4 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [2] \quad 1^{11}21^1 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_7 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 186, \Delta = -287, \det = -6, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 186, T \in [15, 186]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & -6 & 0 & -4 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & -4 & -1 & -4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}21^1 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_7 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 186, \Delta = -287, \det = -20, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 186, T \in [15, 186]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -8 & 0 & -6 \\ 1 & 2 & 0 & 1 & 1 \\ 2 & 2 & -6 & 1 & -4 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [4] \quad 1^{11}221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_7 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 170, \Delta = -258, \det = 17, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 170, T \in [15, 170]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -8 & 0 & -6 \\ 1 & 3 & 0 & 2 & 2 \\ 2 & 3 & -6 & 2 & -3 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [5] \quad 1^{11}2221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_7 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 102, \Delta = -229, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 0 & 4 & 4 \\ 1 & 0 & -8 & 0 & -6 \\ 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & -6 & 3 & -2 \end{pmatrix}_{b_0}$$

$T = 8$ [6] $1^{11}22221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_7 + \hat{1} \rightarrow 1$ $T = 8, T_{\max} = 102, \Delta = -200, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 15.09 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 0 & 5 & 5 \\ 1 & 0 & -8 & 0 & -6 \\ 1 & 5 & 0 & 4 & 4 \\ 2 & 5 & -6 & 4 & -2 \end{pmatrix}_{b_0}$$

$T = 9$ [7] $1^{11}222221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_7 + \hat{1} \rightarrow 1$ $T = 9, T_{\max} = 102, \Delta = -171, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 15.50 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 4 & 0 & 6 & 6 \\ 1 & 0 & -8 & 0 & -6 \\ 1 & 6 & 0 & 5 & 5 \\ 2 & 6 & -6 & 5 & -1 \end{pmatrix}_{b_0}$$

$T = 10$ [8] $1^{11}22313^1 21^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_7 + \hat{1} \rightarrow 1$ $T = 10, T_{\max} = 103, \Delta = -162, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19, T_{\max} = 103, T \in [19, 103]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 10 & 0 & 8 & 8 \\ 1 & 0 & -8 & 0 & -6 \\ 1 & 8 & 0 & 5 & 5 \\ 2 & 8 & -6 & 5 & -3 \end{pmatrix}_{b_0}$$

[9] $1^{11}2222221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_7 + \hat{1} \rightarrow 1$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 16.40 \rightarrow 17, T_{\max} = 100, T \in [17, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 5 & 0 & 7 & 7 \\ 1 & 0 & -8 & 0 & -6 \\ 1 & 7 & 0 & 6 & 6 \\ 2 & 7 & -6 & 6 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{11}2231321^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_7 + \hat{1} \rightarrow 1$ $T = 10, T_{\max} = 102, \Delta = -160, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 102, T \in [17, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 4 & 0 & 6 & 6 \\ 1 & 0 & -8 & 0 & -6 \\ 1 & 6 & 0 & 5 & 5 \\ 2 & 6 & -6 & 5 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ [11] $1^{11}22222221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_7 + \hat{1} \rightarrow 1$ $T = 11, T_{\max} = 26, \Delta = -113, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 20, T_{\max} = 26, T \in [20, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 6 & 0 & 8 & 8 \\ 1 & 0 & -8 & 0 & -6 \\ 1 & 8 & 0 & 7 & 7 \\ 2 & 8 & -6 & 7 & -3 \end{pmatrix}_{b_0}$$

[12] $1^{11}22313221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_7 + \hat{1} \rightarrow 1$ $T = 11, T_{\max} = 94, \Delta = -131, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 17.40 \rightarrow 18, T_{\max} = 94, T \in [18, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 5 & 0 & 7 & 7 \\ 1 & 0 & -8 & 0 & -6 \\ 1 & 7 & 0 & 6 & 6 \\ 2 & 7 & -6 & 6 & -1 \end{pmatrix}_{b_0}$$

$T = 12$ [13] $1^1 231413221^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{c}_7 + \hat{1} \rightarrow 1$ $T = 12, T_{\max} = 95, \Delta = -130, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 18.40 \rightarrow 19, T_{\max} = 95, T \in [19, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 3 & 0 & 2 & 2 \\ 2 & 3 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$T = 13$ [14] $1^{11} 5^{2231} 1321^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_7 + \hat{1} \rightarrow 1$ $T = 13, T_{\max} = 88, \Delta = -125, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.40 \rightarrow 20, T_{\max} = 88, T \in [20, 88]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -4 & 1 & -2 \\ 1 & 5 & 1 & 4 & 4 \\ 2 & 5 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[15] $1^{11}2231512321^1 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_7 + \hat{1} \rightarrow 1$ $T = 13, T_{\max} = 93, \Delta = -127, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19.40 \rightarrow 20, T_{\max} = 93, T \in [20, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 5 & 0 & 7 & 7 \\ 1 & 0 & -8 & 0 & -6 \\ 1 & 7 & 0 & 6 & 6 \\ 2 & 7 & -6 & 6 & -1 \end{pmatrix}_{b_0}$$

1.12 $\mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$

- $T = 4$ [0] $1^{11}1^1 + \mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 4, T_{\max} = 185, \Delta = -288, \det = 14, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 185, T \in [14, 185]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & -5 & 0 & -3 \\ 1 & 0 & -1 & -1 \\ 3 & -3 & -1 & -3 \end{pmatrix}_{b_0}$
- [1] $1^{11}1^1 + \mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 4, T_{\max} = 185, \Delta = -288, \det = 20, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.60 \rightarrow 14, T_{\max} = 185, T \in [14, 185]$
 $\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -7 & 0 & -5 \\ 2 & 0 & 1 & 1 \\ 3 & -5 & 1 & -3 \end{pmatrix}_{b_0}$
- $T = 5$ [2] $1^{11}21^1 + \mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 169, \Delta = -259, \det = -5, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 169, T \in [14, 169]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & -5 & 0 & -3 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & -3 & -1 & -3 \end{pmatrix}_{b_0}$
- [3] $1^{11}21^1 + \mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 169, \Delta = -259, \det = -17, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.71 \rightarrow 14, T_{\max} = 169, T \in [14, 169]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -7 & 0 & -5 \\ 1 & 2 & 0 & 1 & 1 \\ 2 & 2 & -5 & 1 & -3 \end{pmatrix}_{b_0}$
- $T = 6$ [4] $1^{11}221^1 + \mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 101, \Delta = -230, \det = 14, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.86 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -7 & 0 & -5 \\ 1 & 3 & 0 & 2 & 2 \\ 2 & 3 & -5 & 2 & -2 \end{pmatrix}_{b_0}$
- $T = 7$ [5] $1^{11}2221^1 + \mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 101, \Delta = -201, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.09 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 0 & 4 & 4 \\ 1 & 0 & -7 & 0 & -5 \\ 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & -5 & 3 & -2 \end{pmatrix}_{b_0}$
- $T = 8$ [6] $1^{11}22221^1 + \mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 8, T_{\max} = 101, \Delta = -172, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.50 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 0 & 5 & 5 \\ 1 & 0 & -7 & 0 & -5 \\ 1 & 5 & 0 & 4 & 4 \\ 2 & 5 & -5 & 4 & -1 \end{pmatrix}_{b_0}$
- $T = 9$ [7] $1^{11}222221^1 + \mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 9, T_{\max} = 99, \Delta = -143, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 15.40 \rightarrow 16, T_{\max} = 99, T \in [16, 99]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 4 & 0 & 6 & 6 \\ 1 & 0 & -7 & 0 & -5 \\ 1 & 6 & 0 & 5 & 5 \\ 2 & 6 & -5 & 5 & -1 \end{pmatrix}_{b_0}$
- $T = 10$ [8] $1^{11}2222221^1 + \mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 19, T_{\max} = 26, T \in [19, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 5 & 0 & 7 & 7 \\ 1 & 0 & -7 & 0 & -5 \\ 1 & 7 & 0 & 6 & 6 \\ 2 & 7 & -5 & 6 & -3 \end{pmatrix}_{b_0}$
- [9] $1^{11}2231321^1 + \mathfrak{su}_2 + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 10, T_{\max} = 94, \Delta = -132, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 16.40 \rightarrow 17, T_{\max} = 94, T \in [17, 94]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 4 & 0 & 6 & 6 \\ 1 & 0 & -7 & 0 & -5 \\ 1 & 6 & 0 & 5 & 5 \\ 2 & 6 & -5 & 5 & -1 \end{pmatrix}_{b_0}$

$$T = 11 \quad [10] \quad 1^{\hat{1}1} 22313221^{\hat{1}} + \textcolor{red}{su}_2 + \textcolor{blue}{c}'_7 + \hat{1} \rightarrow 1 \quad T = 11, T_{\max} = 23, \Delta = -103, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 20, T_{\max} = 23, T \in [20, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 5 & 0 & 7 & 7 \\ 1 & 0 & -7 & 0 & -5 \\ 1 & 7 & 0 & 6 & 6 \\ 2 & 7 & -5 & 6 & -3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [11] \quad 1^{\hat{1}} 231413221^{\hat{1}1} + \textcolor{red}{su}_2 + \textcolor{blue}{c}'_7 + \hat{1} \rightarrow 1 \quad T = 12, T_{\max} = 24, \Delta = -102, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 21, T_{\max} = 24, T \in [21, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 2 \\ 2 & 3 & -1 & 2 & -3 \end{pmatrix}_{b_0}$$

1.13 $su_2 + \mathfrak{c}_8 + \hat{1} \rightarrow (-1)$

$$T = 4 \quad [0] \quad 1^{\hat{1}1} 1^{\hat{1}} + \textcolor{red}{su}_2 + \textcolor{blue}{c}_8 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 193, \Delta = -431, \det = 35, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.34 \rightarrow 19, T_{\max} = 193, T \in [19, 193]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -12 & 0 & -10 \\ 2 & 0 & 1 & 1 \\ 3 & -10 & 1 & -8 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [1] \quad 1^{\hat{1}1} 21^{\hat{1}} + \textcolor{red}{su}_2 + \textcolor{blue}{c}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 193, \Delta = -402, \det = -32, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.38 \rightarrow 19, T_{\max} = 193, T \in [19, 193]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -12 & 0 & -10 \\ 1 & 2 & 0 & 1 & 1 \\ 2 & 2 & -10 & 1 & -8 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [2] \quad 1^{\hat{1}1} 221^{\hat{1}} + \textcolor{red}{su}_2 + \textcolor{blue}{c}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 193, \Delta = -373, \det = 29, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.41 \rightarrow 19, T_{\max} = 193, T \in [19, 193]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -12 & 0 & -10 \\ 1 & 3 & 0 & 2 & 2 \\ 2 & 3 & -10 & 2 & -7 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [3] \quad 1^{\hat{1}1} 2221^{\hat{1}} + \textcolor{red}{su}_2 + \textcolor{blue}{c}_8 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 193, \Delta = -344, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.46 \rightarrow 19, T_{\max} = 193, T \in [19, 193]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 0 & 4 & 4 \\ 1 & 0 & -12 & 0 & -10 \\ 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & -10 & 3 & -6 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [4] \quad 1^{\hat{1}1} 22221^{\hat{1}} + \textcolor{red}{su}_2 + \textcolor{blue}{c}_8 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 192, \Delta = -315, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.52 \rightarrow 19, T_{\max} = 192, T \in [19, 192]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 0 & 5 & 5 \\ 1 & 0 & -12 & 0 & -10 \\ 1 & 5 & 0 & 4 & 4 \\ 2 & 5 & -10 & 4 & -5 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \quad 1^{\hat{1}1} 222221^{\hat{1}} + \textcolor{red}{su}_2 + \textcolor{blue}{c}_8 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 190, \Delta = -286, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.60 \rightarrow 19, T_{\max} = 190, T \in [19, 190]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 4 & 0 & 6 & 6 \\ 1 & 0 & -12 & 0 & -10 \\ 1 & 6 & 0 & 5 & 5 \\ 2 & 6 & -10 & 5 & -4 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [6] \quad 1^{\hat{1}1} 2222221^{\hat{1}} + \textcolor{red}{su}_2 + \textcolor{blue}{c}_8 + \hat{1} \rightarrow 1 \quad T = 10, T_{\max} = 173, \Delta = -257, \det = 17, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.71 \rightarrow 19, T_{\max} = 173, T \in [19, 173]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 5 & 0 & 7 & 7 \\ 1 & 0 & -12 & 0 & -10 \\ 1 & 7 & 0 & 6 & 6 \\ 2 & 7 & -10 & 6 & -3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \quad 1^{11}2222221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_8 + \hat{1} \rightarrow 1 \quad T = 11, T_{\max} = 106, \Delta = -228, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.86 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 6 & 0 & 8 & 8 \\ 1 & 0 & -12 & 0 & -10 \\ 1 & 8 & 0 & 7 & 7 \\ 2 & 8 & -10 & 7 & -2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11}22313221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_8 + \hat{1} \rightarrow 1 \quad T = 11, T_{\max} = 151, \Delta = -246, \det = -17, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 19.71 \rightarrow 20, T_{\max} = 151, T \in [20, 151]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 5 & 0 & 7 & 7 \\ 1 & 0 & -12 & 0 & -10 \\ 1 & 7 & 0 & 6 & 6 \\ 2 & 7 & -10 & 6 & -3 \end{pmatrix}_{b_0}$$

1.14 $\textcolor{red}{su}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow (-1)$

$$T = 4 \quad [0] \quad 1^{11}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 102, \Delta = -235, \det = 10, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 102, T \in [12, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & -3 & 0 & -1 \\ 1 & 0 & -1 & -1 \\ 3 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 102, \Delta = -235, \det = 14, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 102, T \in [12, 102]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -5 & 0 & -3 \\ 2 & 0 & 1 & 1 \\ 3 & -3 & 1 & -1 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [2] \quad 1^{11}21^1 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 99, \Delta = -206, \det = -3, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 99, T \in [13, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}21^1 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 99, \Delta = -206, \det = -11, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 99, T \in [13, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -5 & 0 & -3 \\ 1 & 2 & 0 & 1 & 1 \\ 2 & 2 & -3 & 1 & -2 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [4] \quad 1^{11}221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 99, \Delta = -177, \det = 8, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 99, T \in [13, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -5 & 0 & -3 \\ 1 & 3 & 0 & 2 & 2 \\ 2 & 3 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [5] \quad 1^{11}2221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 98, \Delta = -148, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 0 & 4 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [6] \quad 1^{11}22221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 45, \Delta = -119, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 17, T_{\max} = 45, T \in [17, 45]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 0 & 5 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 1 & 5 & 0 & 4 & 4 \\ 2 & 5 & -3 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11}3221^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 89, \Delta = -128, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 16.67 \rightarrow 17, T_{\max} = 89, T \in [17, 89]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 6 & 0 & 5 & 6 \\ 1 & 0 & -5 & 0 & -3 \\ 1 & 5 & 0 & 3 & 3 \\ 2 & 6 & -3 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^13221^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 89, \Delta = -128, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 89, T \in [15, 89]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 1 & -1 \\ 1 & 4 & 1 & 3 & 3 \\ 2 & 4 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [9] \quad 1^{11}223131^1 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 0 & 5 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 1 & 5 & 0 & 4 & 4 \\ 2 & 5 & -3 & 4 & -3 \end{pmatrix}_{b_0}$$

1.15 $su_2 + \hat{1} \rightarrow (-1)$

$$T = 3 \quad [0] \quad 1^{11}1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 3, T_{\max} = 98, \Delta = -212, \det = -2, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 98, T \in [10, 98]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 11^{11} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 3, T_{\max} = 98, \Delta = -212, \det = -3, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 1 \\ 3 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 3, T_{\max} = 98, \Delta = -212, \det = -8, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 98, T \in [10, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 1 & 1 \\ 3 & 1 & 1 \end{pmatrix}_{b_0}$$

$$T = 4 \quad [3] \quad 121^{11} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -183, \det = 3, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 98, T \in [12, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 \\ 2 & 2 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11}21^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -183, \det = 7, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 98, T \in [10, 98]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 1 & 1 \\ 2 & 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [5] \quad 1221^{11} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -154, \det = -3, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 96, T \in [13, 96]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 2 & 2 \\ 2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11}221^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -154, \det = -6, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 3 & 3 \\ 2 & 3 & 2 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [7] \quad 2^1231^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 91, \Delta = -134, \det = 2, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 91, T \in [13, 91]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 2 & 2 \\ 2 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 12221^{11} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 81, \Delta = -125, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 81, T \in [14, 81]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 4 & 4 \\ 1 & 4 & 3 & 3 \\ 2 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11}2221^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 81, \Delta = -125, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 81, T \in [10, 81]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 2 & 4 & 4 \\ 2 & 4 & 3 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [10] \quad 1^1 3^1 221^{11} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 16, T_{\max} = 19, T \in [16, 19]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 6 & 5 & 6 \\ 1 & 5 & 3 & 3 \\ 2 & 6 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[12] \quad 213221^{11} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -3, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 19, T \in [15, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 4 & 4 \\ 1 & 4 & 3 & 3 \\ 2 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[11] \quad 13221^{11} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -3, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 19, T \in [15, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 4 & 4 \\ 1 & 4 & 3 & 3 \\ 2 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11} 22221^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 13, \Delta = -96, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 13, T \in [10, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 3 & 5 & 5 \\ 2 & 5 & 4 & 4 \\ 2 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [14] \quad 1^{11} 222221^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 10, \Delta = -67, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 4 & 6 & 6 \\ 2 & 6 & 5 & 5 \\ 2 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [15] \quad 1^{11} 2222221^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 5 & 7 & 7 \\ 2 & 7 & 6 & 6 \\ 2 & 7 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11} 22313^1 21^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 10 & 8 & 8 \\ 2 & 8 & 5 & 5 \\ 2 & 8 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[17] \quad 2^2 313221^{11} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 6 & 6 \\ 2 & 6 & 5 & 5 \\ 2 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [18] \quad 31^{11} 5^{231} 13^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 7 & 7 & 7 \\ 2 & 7 & 5 & 5 \\ 2 & 7 & 5 & 5 \end{pmatrix}_{b_0}$$

1.16 $\textcolor{red}{su}_2 + \hat{1} \rightarrow (-1) + \hat{1} \rightarrow (-1)$

$$T = 4 \quad [0] \quad 1^{11} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -210, \det = 2, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -1 & 0 & -1 \\ 2 & 0 & 1 & 1 \\ 3 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

1.17 $\textcolor{red}{su}_2 + \hat{1} \rightarrow (-2)$

$$T = 4 \quad [0] \quad 12^1 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 4, T_{\max} = 94, \Delta = -147, \det = 2, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 11, T_{\max} = 94, T \in [11, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 0 \\ 2 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

- $T = 5$ [2] $1^2 3 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 5, T_{\max} = 84, \Delta = -126, \det = -4, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13, T_{\max} = 84, T \in [13, 84]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 12 & 0 & 6 \\ 1 & 0 & -1 & -1 \\ 2 & 6 & -1 & -1 \end{pmatrix}_{b_0}$
- [1] $1 3 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 5, T_{\max} = 84, \Delta = -126, \det = -4, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 12, T_{\max} = 84, T \in [12, 84]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 2 & 2 & -1 & 0 \end{pmatrix}_{b_0}$
- [3] $1^1 3 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 5, T_{\max} = 84, \Delta = -126, \det = -5, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 12.20 \rightarrow 13, T_{\max} = 84, T \in [13, 84]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -1 & -1 \\ 2 & 4 & -1 & -1 \end{pmatrix}_{b_0}$
- $T = 6$ [8] $2 1^2 4 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17, T_{\max} = 29, T \in [17, 29]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 20 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -4 \end{pmatrix}_{b_0}$
- [9] $1 2 3 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 6, T_{\max} = 13, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 2 & 2 & -1 & 0 \end{pmatrix}_{b_0}$
- [7] $2 1 4 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13, T_{\max} = 29, T \in [13, 29]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$
- [6] $1^2 1^1 4 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13, T_{\max} = 29, T \in [13, 29]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 2 & 2 & -1 & 0 \end{pmatrix}_{b_0}$
- [5] $1^2 1^1 4 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14, T_{\max} = 29, T \in [14, 29]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 12 & 0 & 6 \\ 1 & 0 & -1 & -1 \\ 2 & 6 & -1 & -1 \end{pmatrix}_{b_0}$
- [4] $1^2 1^1 4 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17, T_{\max} = 29, T \in [17, 29]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 32 & 0 & 10 \\ 1 & 0 & -1 & -1 \\ 2 & 10 & -1 & -4 \end{pmatrix}_{b_0}$
- [10] $1^1 1^2 4 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 29, T \in [15, 29]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 20 & 0 & 8 \\ 1 & 0 & -1 & -1 \\ 2 & 8 & -1 & -2 \end{pmatrix}_{b_0}$
- [11] $1^2 1^1 4 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.20 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -1 & -1 \\ 2 & 4 & -1 & -1 \end{pmatrix}_{b_0}$
- [12] $1^1 1^2 4 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 6, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.67 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 0 & 6 \\ 1 & 0 & -1 & -1 \\ 2 & 6 & -1 & -1 \end{pmatrix}_{b_0}$
- [13] $2 1^1 4 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 6, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.67 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 4 & 6 \\ 1 & 4 & 1 & 1 \\ 2 & 6 & 1 & -1 \end{pmatrix}_{b_0}$$

[19] $1^{\overline{1}} 4^{\overline{2}} 1^{\overline{1}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 20 & 0 & 8 \\ 1 & 0 & -1 & -1 \\ 2 & 8 & -1 & -1 \end{pmatrix}_{b_0}$

[15] $1^{\overline{3}} 4^{\overline{2}} 1^{\overline{3}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.25 \rightarrow 16, T_{\max} = 29, T \in [16, 29]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 84 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -3 \end{pmatrix}_{b_0}$

[16] $21^{\overline{3}} 41^{\overline{2}} 2^{\overline{1}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.25 \rightarrow 16, T_{\max} = 29, T \in [16, 29]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 64 & 9 & 14 \\ 1 & 9 & 1 & 1 \\ 2 & 14 & 1 & -3 \end{pmatrix}_{b_0}$

[17] $21^{\overline{1}} 41^{\overline{2}} 2^{\overline{1}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 20 & 0 & 8 \\ 1 & 0 & -1 & -1 \\ 2 & 8 & -1 & -1 \end{pmatrix}_{b_0}$

[18] $2^{\overline{2}} 3^{\overline{1}} 2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 18, \Delta = -100, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 18, T \in [12, 18]$
 $\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 8 & 5 & 6 \\ 2 & 5 & 2 & 2 \\ 2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$

[14] $1^{\overline{1}} 4^{\overline{2}} 1^{\overline{3}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14, T_{\max} = 29, T \in [14, 29]$
 $\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 48 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -1 \end{pmatrix}_{b_0}$

[20] $1^{\overline{1}} 4^{\overline{2}} 1^{\overline{2}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 17, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.53 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$
 $\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 32 & 0 & 10 \\ 1 & 0 & -1 & -1 \\ 2 & 10 & -1 & -1 \end{pmatrix}_{b_0}$

[21] $1^{\overline{2}} 4^{\overline{2}} 1^{\overline{3}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 17, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.47 \rightarrow 15, T_{\max} = 29, T \in [15, 29]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 64 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -2 \end{pmatrix}_{b_0}$

[22] $1^{\overline{2}} 4^{\overline{2}} 1^{\overline{2}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 18, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.89 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$
 $\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 46 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -1 \end{pmatrix}_{b_0}$

[23] $21^{\overline{2}} 41^{\overline{2}} 2^{\overline{1}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 18, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.89 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$
 $\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 46 & 8 & 12 \\ 1 & 8 & 1 & 1 \\ 2 & 12 & 1 & -1 \end{pmatrix}_{b_0}$

[24] $1^{\overline{2}} 4^{\overline{1}} 1^{\overline{3}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 36, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14, T_{\max} = 29, T \in [14, 29]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 108 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$

[25] $1^{\overline{2}} 4^{\overline{1}} 1^{\overline{3}} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 36, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.44 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 64 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -1 \end{pmatrix}_{b_0}$

- [26] $21^{241^3 2^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 29$, $T \in [14, 29]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 64 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -1 \end{pmatrix}_{b_0}$
- [27] $21^{441^3 2^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.78 \rightarrow 15$, $T_{\max} = 29$, $T \in [15, 29]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 132 & 13 & 20 \\ 1 & 13 & 1 & 1 \\ 2 & 20 & 1 & -2 \end{pmatrix}_{b_0}$
- [28] $1^4 4^{21^3} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 36$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.78 \rightarrow 15$, $T_{\max} = 29$, $T \in [15, 29]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 160 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$
- [29] $1^2 4^{21^3} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 37$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.68 \rightarrow 14$, $T_{\max} = 29$, $T \in [14, 29]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 84 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -1 \end{pmatrix}_{b_0}$
- [30] $1^3 4^{21^3} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 37$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.32 \rightarrow 15$, $T_{\max} = 29$, $T \in [15, 29]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 132 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -2 \end{pmatrix}_{b_0}$
- [31] $1^3 4^{21^3} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 38$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.95 \rightarrow 14$, $T_{\max} = 29$, $T \in [14, 29]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 106 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$
- [32] $21^{341^3 2^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 38$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.95 \rightarrow 14$, $T_{\max} = 29$, $T \in [14, 29]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 106 & 12 & 18 \\ 1 & 12 & 1 & 1 \\ 2 & 18 & 1 & -1 \end{pmatrix}_{b_0}$
- [34] $21^{541^4 2^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.56 \rightarrow 15$, $T_{\max} = 29$, $T \in [15, 29]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 224 & 17 & 26 \\ 1 & 17 & 1 & 1 \\ 2 & 26 & 1 & -2 \end{pmatrix}_{b_0}$
- [33] $1^3 4^{21^4} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.56 \rightarrow 14$, $T_{\max} = 29$, $T \in [14, 29]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 132 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$
- [35] $1^3 4^{21^4} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14$, $T_{\max} = 29$, $T \in [14, 29]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 192 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$
- [36] $1^5 4^{21^4} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.56 \rightarrow 15$, $T_{\max} = 29$, $T \in [15, 29]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 260 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$
- [37] $21^{341^4 2^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 64$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.56 \rightarrow 14$, $T_{\max} = 29$, $T \in [14, 29]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 132 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$
- [38] $1^4 4^{21^4} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 65$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.25 \rightarrow 15$, $T_{\max} = 29$, $T \in [15, 29]$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 224 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -2 \end{pmatrix}_{b_0}$$

[39] $1^3 \overset{2^1 1^4}{4} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 65, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.75 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 160 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$$

[40] $1^4 \overset{2^1 1^4}{4} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 66, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.97 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 190 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$$

[41] $21^4 41^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 66, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.97 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 190 & 16 & 24 \\ 1 & 16 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$$

[43] $21^4 41^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 100, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.64 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 224 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -1 \end{pmatrix}_{b_0}$$

[42] $1^4 \overset{2^1 1^5}{4} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 100, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.64 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 224 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -1 \end{pmatrix}_{b_0}$$

[44] $1^4 \overset{2^1 1^5}{4} 1^5 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 101, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 260 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -1 \end{pmatrix}_{b_0}$$

[46] $21^5 41^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 102, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.98 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 298 & 20 & 30 \\ 1 & 20 & 1 & 1 \\ 2 & 30 & 1 & -1 \end{pmatrix}_{b_0}$$

[45] $1^5 \overset{2^1 1^5}{4} 1^5 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 102, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.98 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 298 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -1 \end{pmatrix}_{b_0}$$

$T = 7$ [53] $1^1 \overset{2^1 1^1}{5} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 2 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

[52] $1^1 \overset{2^1 1^1}{5} 12 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 2 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

[51] $22151^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

[50] $1^2 \overset{1^2}{2} 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 12 & 0 & 6 \\ 1 & 0 & -1 & -1 \\ 2 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1 \overset{1^2}{\underset{2^1 1^1}{5}} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 12 & 0 & 6 \\ 1 & 0 & -1 & -1 \\ 2 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1 \overset{1^2}{\underset{1^2}{5}} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 32 & 0 & 10 \\ 1 & 0 & -1 & -1 \\ 2 & 10 & -1 & -4 \end{pmatrix}_{b_0}$$

$$[47] \quad 1 \overset{21^2}{\underset{1^1}{5}} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 32 & 0 & 10 \\ 1 & 0 & -1 & -1 \\ 2 & 10 & -1 & -4 \end{pmatrix}_{b_0}$$

$$[58] \quad 1 \overset{1^2}{\underset{1^1}{5}} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.80 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 20 & 0 & 8 \\ 1 & 0 & -1 & -1 \\ 2 & 8 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[56] \quad 21 \overset{1^1}{\underset{1^1}{5}} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -1 & -1 \\ 2 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[55] \quad 1 \overset{1^1}{\underset{2^1 1^1}{5}} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -1 & -1 \\ 2 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[54] \quad 21 \overset{1^2}{\underset{1^1}{5}} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 44 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -5 \end{pmatrix}_{b_0}$$

$$[57] \quad 1 \overset{1^2}{\underset{1^2}{5}} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 44 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -5 \end{pmatrix}_{b_0}$$

$$[62] \quad 1 \overset{1^2}{\underset{1^1}{5}} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.67 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 30 & 0 & 10 \\ 1 & 0 & -1 & -1 \\ 2 & 10 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[61] \quad 1 \overset{2^1 1^1}{\underset{1^1}{5}} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 4 & 6 \\ 1 & 4 & 1 & 1 \\ 2 & 6 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[60] \quad 1 \overset{1^1}{\underset{1^1}{5}} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 0 & 6 \\ 1 & 0 & -1 & -1 \\ 2 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

- [59] $21^{1^2 1^1} 5^1 1^2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -6$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.67 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 30 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -3 \end{pmatrix}_{b_0}$
- [64] $221^1 51^1 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -7$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.29 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -2 \end{pmatrix}_{b_0}$
- [65] $1^1 51^1 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -7$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.29 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 0 & 8 \\ 1 & 0 & -1 & -1 \\ 2 & 8 & -1 & -2 \end{pmatrix}_{b_0}$
- [63] $21^{1^2 1^1} 5^1 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -7$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.29 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -2 \end{pmatrix}_{b_0}$
- [66] $1^1 51^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -12$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 19.33 \rightarrow 20$, $T_{\max} = 21$, $T \in [20, 21]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 132 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -6 \end{pmatrix}_{b_0}$
- [67] $21^4 51^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -13$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 20.23 \rightarrow 21$, $T_{\max} = 21$, $T \in [21, 21]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 156 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -7 \end{pmatrix}_{b_0}$
- [68] $21^2 5^1 1^5 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -13$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 20.23 \rightarrow 21$, $T_{\max} = 21$, $T \in [21, 21]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 156 & 13 & 22 \\ 1 & 13 & 1 & 1 \\ 2 & 22 & 1 & -7 \end{pmatrix}_{b_0}$
- [69] $1^2 51^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -13$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 20.23 \rightarrow 21$, $T_{\max} = 21$, $T \in [21, 21]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 156 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -7 \end{pmatrix}_{b_0}$
- [70] $1^1 51^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -13$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 20.23 \rightarrow 21$, $T_{\max} = 21$, $T \in [21, 21]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 156 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -7 \end{pmatrix}_{b_0}$
- [71] $21^{1^2 1^2} 5^1 1^4 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -16$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.25 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 104 & 11 & 18 \\ 1 & 11 & 1 & 1 \\ 2 & 18 & 1 & -3 \end{pmatrix}_{b_0}$
- [72] $1^1 51^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -16$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.25 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 84 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -3 \end{pmatrix}_{b_0}$
- [73] $21^3 51^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -16$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 20.25 \rightarrow 21$, $T_{\max} = 21$, $T \in [21, 21]$

- $$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 180 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -7 \end{pmatrix}_{b_0}$$
- [74] $1 \overset{2^1 1^2}{5} 1^2 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.25 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 32 & 7 & 10 \\ 1 & 7 & 1 & 1 \\ 2 & 10 & 1 & -1 \end{pmatrix}_{b_0}$$
- [75] $1 \overset{1^4}{5} \overset{1^2}{1^3} 2^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 18, T_{\max} = 21, T \in [18, 21]$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 128 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -4 \end{pmatrix}_{b_0}$$
- [76] $1 \overset{1^3}{5} \overset{1^2}{1^1} 2^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 48 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -1 \end{pmatrix}_{b_0}$$
- [77] $1 \overset{21^3}{5} \overset{1^2}{1^1} 2^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.25 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 84 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -3 \end{pmatrix}_{b_0}$$
- [78] $1 \overset{1^3}{5} \overset{1^2}{1^3} 2^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.25 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 84 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -3 \end{pmatrix}_{b_0}$$
- [79] $1 \overset{1^1}{5} \overset{1^2}{1^1} 2^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.25 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 20 & 0 & 8 \\ 1 & 0 & -1 & -1 \\ 2 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$
- [80] $1 \overset{1^4}{3} \overset{1^2}{5} \overset{1^2}{1^3} 2^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 20.25 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 180 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -7 \end{pmatrix}_{b_0}$$
- [81] $1 \overset{1^3}{5} \overset{1^2}{1^2} 2^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.47 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 64 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -2 \end{pmatrix}_{b_0}$$
- [82] $1 \overset{1^4}{1^2} \overset{1^2}{5} \overset{1^2}{1^2} 2^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.88 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$

$$\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 104 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -3 \end{pmatrix}_{b_0}$$
- [83] $1 \overset{2^1}{1^2} \overset{1^4}{5} \overset{1^2}{1^3} 2^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 18.76 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 152 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -5 \end{pmatrix}_{b_0}$$
- [84] $1 \overset{1^2}{5} \overset{1^2}{1^1} 2^1 + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.53 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 32 & 0 & 10 \\ 1 & 0 & -1 & -1 \\ 2 & 10 & -1 & -1 \end{pmatrix}_{b_0}$$

- [85] $1^2 \overset{1^4}{5} 1^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -18$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.56 \rightarrow 18$, $T_{\max} = 21$, $T \in [18, 21]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 126 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -4 \end{pmatrix}_{b_0}$
- [86] $1^2 \overset{2^1 2^2}{5} 1^2 2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -18$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.89 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 46 & 8 & 12 \\ 1 & 8 & 1 & 1 \\ 2 & 12 & 1 & -1 \end{pmatrix}_{b_0}$
- [87] $21^2 \overset{2^1 2^2}{5} 1^4 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -18$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.56 \rightarrow 18$, $T_{\max} = 21$, $T \in [18, 21]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 126 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -4 \end{pmatrix}_{b_0}$
- [88] $1^2 \overset{1^2}{5} 1^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -18$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.89 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 46 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -1 \end{pmatrix}_{b_0}$
- [98] $1^1 \overset{1^1}{5} 1^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.47 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 30 & 0 & 10 \\ 1 & 0 & -1 & -1 \\ 2 & 10 & -1 & -1 \end{pmatrix}_{b_0}$
- [89] $221^1 \overset{1^2 2^1}{5} 1^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.47 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 44 & 8 & 12 \\ 1 & 8 & 1 & 1 \\ 2 & 12 & 1 & -1 \end{pmatrix}_{b_0}$
- [90] $21^1 \overset{2^1 2^2}{5} 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.47 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 44 & 8 & 12 \\ 1 & 8 & 1 & 1 \\ 2 & 12 & 1 & -1 \end{pmatrix}_{b_0}$
- [91] $221^3 \overset{1^2 2^1}{5} 1^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 18.26 \rightarrow 19$, $T_{\max} = 21$, $T \in [19, 21]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 124 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -5 \end{pmatrix}_{b_0}$
- [92] $1^1 \overset{1^3}{5} 1^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.32 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 62 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -2 \end{pmatrix}_{b_0}$
- [93] $21^3 \overset{1^1}{5} 1^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.58 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 102 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -3 \end{pmatrix}_{b_0}$
- [94] $21^1 \overset{2^1 2^2}{5} 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.32 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 80 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -2 \end{pmatrix}_{b_0}$
- [95] $1^3 \overset{1^3}{5} 1^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 18.26 \rightarrow 19$, $T_{\max} = 21$, $T \in [19, 21]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 150 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -5 \end{pmatrix}_{b_0}$

- [96] $1^1 \overset{1^3}{5} 1^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.58 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 102 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -3 \end{pmatrix}_{b_0}$
- [97] $21^3 \overset{1^3}{5} 1^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 18.26 \rightarrow 19$, $T_{\max} = 21$, $T \in [19, 21]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 150 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -5 \end{pmatrix}_{b_0}$
- [103] $21^1 \overset{2^1}{5} 1^2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.80 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 60 & 9 & 14 \\ 1 & 9 & 1 & 1 \\ 2 & 14 & 1 & -1 \end{pmatrix}_{b_0}$
- [99] $1^1 \overset{1^2}{5} 1^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.80 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 44 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -1 \end{pmatrix}_{b_0}$
- [100] $21^3 \overset{1^2}{5} 1^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.20 \rightarrow 18$, $T_{\max} = 21$, $T \in [18, 21]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 124 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -4 \end{pmatrix}_{b_0}$
- [101] $1^2 \overset{1^3}{5} 1^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.20 \rightarrow 18$, $T_{\max} = 21$, $T \in [18, 21]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 124 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -4 \end{pmatrix}_{b_0}$
- [102] $1^1 \overset{1^3}{1^2} 5 1^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.80 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 80 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -2 \end{pmatrix}_{b_0}$
- [107] $1^2 \overset{1^3}{1^2} 5 1^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.33 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 100 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -3 \end{pmatrix}_{b_0}$
- [105] $21^2 \overset{2^1}{5} 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.19 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 60 & 9 & 14 \\ 1 & 9 & 1 & 1 \\ 2 & 14 & 1 & -2 \end{pmatrix}_{b_0}$
- [106] $21^2 \overset{2^1}{5} 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.33 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 100 & 11 & 18 \\ 1 & 11 & 1 & 1 \\ 2 & 18 & 1 & -3 \end{pmatrix}_{b_0}$
- [104] $1^1 \overset{1^2}{1^2} 5 1^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.19 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 60 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -2 \end{pmatrix}_{b_0}$
- [108] $1^2 \overset{1^2}{1^2} 5 1^2 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -22$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.64 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 78 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -2 \end{pmatrix}_{b_0}$$

[109] $21^{2^{1,2}} 5^{1^2} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.64 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 78 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -2 \end{pmatrix}_{b_0}$$

[110] $221^{2^1 5^1 2^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.64 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 78 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -2 \end{pmatrix}_{b_0}$$

[111] $21^{1^{2^1 1^3}} 5^{1^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.26 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 82 & 11 & 16 \\ 1 & 11 & 1 & 1 \\ 2 & 16 & 1 & -1 \end{pmatrix}_{b_0}$$

[112] $1^{1^5} 5^{1^3} 2^{1^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.46 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 254 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -4 \end{pmatrix}_{b_0}$$

[113] $21^{5^{1^5}} 5^{1^3} 2^{1^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 20.43 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 406 & 0 & 36 \\ 1 & 0 & -1 & -1 \\ 2 & 36 & -1 & -7 \end{pmatrix}_{b_0}$$

[114] $21^{5^{1^1}} 5^{1^3} 2^{1^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.46 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 254 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -4 \end{pmatrix}_{b_0}$$

[115] $221^{1^1 5^1 3^1 2^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.26 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 82 & 11 & 16 \\ 1 & 11 & 1 & 1 \\ 2 & 16 & 1 & -1 \end{pmatrix}_{b_0}$$

[116] $221^{5^1 5^1 3^1 2^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 20.43 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 322 & 19 & 32 \\ 1 & 19 & 1 & 1 \\ 2 & 32 & 1 & -7 \end{pmatrix}_{b_0}$$

[117] $1^{1^1} 5^{1^1} 3^1 2^{1^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.26 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 46 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -1 \end{pmatrix}_{b_0}$$

[118] $1^{5^{1^5}} 5^{1^3} 2^{1^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 20.43 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 406 & 0 & 36 \\ 1 & 0 & -1 & -1 \\ 2 & 36 & -1 & -7 \end{pmatrix}_{b_0}$$

[119] $1^{1^4} 5^{1^3} 2^{1^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 108 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$$

[120] $1^{1^2} 5^{1^3} 2^{1^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.44 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 64 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[121] \begin{matrix} 1^4 \\ 1^4 \end{matrix} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.78 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 160 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[122] \begin{matrix} 2^1 1^3 \\ 1^4 \end{matrix} 1^2 2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.44 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 84 & 11 & 16 \\ 1 & 11 & 1 & 1 \\ 2 & 16 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[123] \begin{matrix} 2^1 1^3 \\ 1^4 \end{matrix} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.78 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 160 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[125] \begin{matrix} 1^3 \\ 1^2 \end{matrix} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.68 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 84 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[124] \begin{matrix} 1^4 \\ 1^3 \end{matrix} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.32 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 132 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[128] 21^5 \begin{matrix} 1^2 \\ 1^2 \end{matrix} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.79 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 286 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -4 \end{pmatrix}_{b_0}$$

$$[137] 1^1 \begin{matrix} 1^4 \\ 1^1 \end{matrix} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.95 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 106 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[127] 1^4 \begin{matrix} 1^5 \\ 1^5 \end{matrix} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 19.16 \rightarrow 20, T_{\max} = 21, T \in [20, 21]$$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 362 & 0 & 34 \\ 1 & 0 & -1 & -1 \\ 2 & 34 & -1 & -6 \end{pmatrix}_{b_0}$$

$$[126] \begin{matrix} 1^3 \\ 1^3 \end{matrix} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.95 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 106 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[129] 1^1 \begin{matrix} 1^5 \\ 1^4 \end{matrix} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.63 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 218 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[130] 21^1 \begin{matrix} 2^1 1^3 \\ 1^4 \end{matrix} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.95 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 154 & 14 & 22 \\ 1 & 14 & 1 & 1 \\ 2 & 22 & 1 & -1 \end{pmatrix}_{b_0}$$

- [131] $1 \overset{21^3}{5} 1^3 2 + \textcolor{red}{\mathfrak{su}_2} + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.95 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 106 & 12 & 18 \\ 1 & 12 & 1 & 1 \\ 2 & 18 & 1 & -1 \end{pmatrix}_{b_0}$
- [132] $21 \overset{21^3}{1} 5 1^2 + \textcolor{red}{\mathfrak{su}_2} + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.42 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 102 & 12 & 18 \\ 1 & 12 & 1 & 1 \\ 2 & 18 & 1 & -1 \end{pmatrix}_{b_0}$
- [133] $1 \overset{1^5}{5} 1^3 2^1 + \textcolor{red}{\mathfrak{su}_2} + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.68 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 158 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$
- [134] $21 \overset{1^4}{5} 1^3 2^1 + \textcolor{red}{\mathfrak{su}_2} + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 19.16 \rightarrow 20, T_{\max} = 21, T \in [20, 21]$
 $\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 362 & 0 & 34 \\ 1 & 0 & -1 & -1 \\ 2 & 34 & -1 & -6 \end{pmatrix}_{b_0}$
- [135] $1 \overset{1^5}{2} 5 1^3 2^1 + \textcolor{red}{\mathfrak{su}_2} + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.79 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 286 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -4 \end{pmatrix}_{b_0}$
- [136] $1 \overset{1^2}{5} 1^3 2^1 + \textcolor{red}{\mathfrak{su}_2} + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.42 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 62 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -1 \end{pmatrix}_{b_0}$
- [139] $21 \overset{1^3}{5} 1^3 2^1 + \textcolor{red}{\mathfrak{su}_2} + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 18.33 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 322 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -5 \end{pmatrix}_{b_0}$
- [138] $1 \overset{1^5}{3} 5 1^3 2^1 + \textcolor{red}{\mathfrak{su}_2} + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 18.33 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 322 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -5 \end{pmatrix}_{b_0}$
- [140] $1 \overset{1^5}{1^3} 5 1^3 2^1 + \textcolor{red}{\mathfrak{su}_2} + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.08 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 186 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -3 \end{pmatrix}_{b_0}$
- [141] $21 \overset{21^3}{1} 5 1^3 + \textcolor{red}{\mathfrak{su}_2} + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.64 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 126 & 13 & 20 \\ 1 & 13 & 1 & 1 \\ 2 & 20 & 1 & -1 \end{pmatrix}_{b_0}$
- [142] $1 \overset{1^3}{1} 5 1^3 2^1 + \textcolor{red}{\mathfrak{su}_2} + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.64 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 82 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -1 \end{pmatrix}_{b_0}$
- [151] $21 \overset{21^3}{2} 5 1^1 + \textcolor{red}{\mathfrak{su}_2} + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.61 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 102 & 12 & 18 \\ 1 & 12 & 1 & 1 \\ 2 & 18 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[143] \ 1^1 \overset{1^2}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -41, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.61 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 80 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[144] \ 1^1 \overset{1^5}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -41, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.12 \rightarrow 19, \ T_{\max} = 21, \ T \in [19, 21]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 320 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -5 \end{pmatrix}_{b_0}$$

$$[145] \ 21^4 \overset{1^1}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -41, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.98 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 184 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[146] \ 1^2 \overset{1^5}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -41, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.95 \rightarrow 17, \ T_{\max} = 21, \ T \in [17, 21]$$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 248 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[147] \ 21^2 \overset{2^1}{5} 1^5 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -41, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.98 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 214 & 16 & 26 \\ 1 & 16 & 1 & 1 \\ 2 & 26 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[148] \ 1^2 \overset{1^5}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -41, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.98 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 184 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[149] \ 1^1 \overset{1^4}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -41, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.20 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 128 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[152] \ 21^4 \overset{1^5}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -41, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.12 \rightarrow 19, \ T_{\max} = 21, \ T \in [19, 21]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 320 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -5 \end{pmatrix}_{b_0}$$

$$[150] \ 1^1 \overset{1^4}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -41, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.98 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 184 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[154] \ 1^1 \overset{1^4}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -42, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.52 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 154 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[155] \ 1^3 \overset{1^5}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -42, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.43 \rightarrow 18, \ T_{\max} = 21, \ T \in [18, 21]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 282 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -4 \end{pmatrix}_{b_0}$$

- [156] $1^1 \overset{1^3}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -42$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.86 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 102 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$
- [153] $1^2 \overset{1^5}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -42$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.38 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 214 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -3 \end{pmatrix}_{b_0}$
- [159] $1^1 \overset{1^3}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -43$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.14 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 126 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -2 \end{pmatrix}_{b_0}$
- [157] $21^3 \overset{2^1 1^3}{5} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -43$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.14 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 126 & 13 & 20 \\ 1 & 13 & 1 & 1 \\ 2 & 20 & 1 & -2 \end{pmatrix}_{b_0}$
- [158] $21^3 \overset{2^1 1^3}{5} 1^5 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -43$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.81 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 246 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -3 \end{pmatrix}_{b_0}$
- [160] $1^3 \overset{1^5}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -43$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.81 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 246 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -3 \end{pmatrix}_{b_0}$
- [170] $1^2 \overset{1^2}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.82 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 100 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$
- [161] $21^4 \overset{1^4}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.27 \rightarrow 18$, $T_{\max} = 21$, $T \in [18, 21]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 280 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -4 \end{pmatrix}_{b_0}$
- [162] $21^2 \overset{2^1 1^3}{5} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.82 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 124 & 13 & 20 \\ 1 & 13 & 1 & 1 \\ 2 & 20 & 1 & -1 \end{pmatrix}_{b_0}$
- [163] $1^2 \overset{1^4}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.45 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 152 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$
- [164] $21^4 \overset{1^2}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.27 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 212 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -3 \end{pmatrix}_{b_0}$
- [165] $221^2 \overset{1^2}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.82 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 124 & 13 & 20 \\ 1 & 13 & 1 & 1 \\ 2 & 20 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[166] \ 1^2 \overset{1^4}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.27 \rightarrow 17, \ T_{\max} = 21, \ T \in [17, 21]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 212 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[167] \ 21^2 \overset{2^1 1^3}{5} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.45 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 180 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[168] \ 1^4 \overset{1^4}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.27 \rightarrow 18, \ T_{\max} = 21, \ T \in [18, 21]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 280 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -4 \end{pmatrix}_{b_0}$$

$$[169] \ 221^4 \overset{1^4}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.27 \rightarrow 18, \ T_{\max} = 21, \ T \in [18, 21]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 244 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -4 \end{pmatrix}_{b_0}$$

$$[171] \ 1^3 \overset{1^4}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -45, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.69 \rightarrow 17, \ T_{\max} = 21, \ T \in [17, 21]$$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 244 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[172] \ 1^2 \overset{1^4}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -45, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.80 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 180 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[173] \ 21^4 \overset{1^3}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -45, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.69 \rightarrow 17, \ T_{\max} = 21, \ T \in [17, 21]$$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 244 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[174] \ 21^2 \overset{2^1 1^3}{5} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -45, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.09 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 150 & 14 & 22 \\ 1 & 14 & 1 & 1 \\ 2 & 22 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[175] \ 1^2 \overset{1^3}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -45, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.09 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 124 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[176] \ 21^3 \overset{2^1 1^3}{5} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -46, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.39 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 150 & 14 & 22 \\ 1 & 14 & 1 & 1 \\ 2 & 22 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[177] \ 1^2 \overset{1^3}{5} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -46, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.39 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 150 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[178] \ 21^3 \overset{2^1 1^3}{5} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -46, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.17 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 210 & 16 & 26 \\ 1 & 16 & 1 & 1 \\ 2 & 26 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[179] \ 1^{\overset{14}{\underset{13}{3}}} 5 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.17 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 210 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[180] \ 1^{\overset{13}{\underset{13}{3}}} 5 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.72 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 178 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[181] \ 21^{\overset{21}{\underset{13}{3}}} 5 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.72 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 178 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[182] \ 221^{\overset{3}{\underset{13}{5}}} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.72 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 178 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[184] \ 1^{\overset{12}{\underset{11}{3}}} 5 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.27 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 84 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[183] \ 21^{\overset{21}{\underset{11}{3}}} 5 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.27 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 156 & 15 & 22 \\ 1 & 15 & 1 & 1 \\ 2 & 22 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[185] \ 21^{\overset{21}{\underset{11}{3}}} 5 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 184 & 16 & 24 \\ 1 & 16 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[186] \ 1^{\overset{13}{\underset{11}{3}}} 5 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 106 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[189] \ 1^{\overset{21}{\underset{11}{3}}} 5 1^3 2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.56 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 160 & 15 & 22 \\ 1 & 15 & 1 & 1 \\ 2 & 22 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[187] \ 1^{\overset{21}{\underset{15}{5}}} 5 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.56 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 260 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[188] \ 1^{\overset{5}{\underset{15}{5}}} 5 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.56 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 260 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$$

- [191] $21^{1^{21^4}} 5^{1^4} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -64$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.56 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 216 & 17 & 26 \\ 1 & 17 & 1 & 1 \\ 2 & 26 & 1 & -1 \end{pmatrix}_{b_0}$
- [192] $15^{1^3} 1^{42^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -64$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.56 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 132 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$
- [193] $1^{1^4} 15^{1^4} 2^{1^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -64$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.56 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 132 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$
- [190] $15^{1^5} 1^{42^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -64$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 192 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$
- [194] $15^{1^5} 1^{42^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -65$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.25 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 224 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -2 \end{pmatrix}_{b_0}$
- [195] $1^{1^2} 15^{1^4} 2^{1^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -65$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.38 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 104 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$
- [196] $15^{1^4} 1^{42^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -65$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.75 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 160 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$
- [197] $21^{2^{21^4}} 5^{1^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -65$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.38 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 156 & 15 & 22 \\ 1 & 15 & 1 & 1 \\ 2 & 22 & 1 & -1 \end{pmatrix}_{b_0}$
- [198] $1^{2^{21^4}} 5^{1^4} 2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -66$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.97 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 190 & 16 & 24 \\ 1 & 16 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$
- [199] $15^{1^4} 1^{42^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -66$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.97 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 190 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$
- [200] $1^{1^5} 15^{1^4} 2^{1^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -68$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.94 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 188 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$
- [201] $1^{1^3} 15^{1^4} 2^{1^1} + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -68$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.53 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 128 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[202] \ 1^1 \overset{1^4}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -69, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.71 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 156 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[205] \ 21^2 \overset{2^1}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -70, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.51 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 182 & 16 & 24 \\ 1 & 16 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[203] \ 1^2 \overset{1^2}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -70, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.51 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 126 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[204] \ 221^2 \overset{2^1}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -70, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.51 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 182 & 16 & 24 \\ 1 & 16 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[208] \ 1^1 \overset{1^3}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -71, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.69 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 154 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[206] \ 21^5 \overset{1^1}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -71, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.70 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 290 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[207] \ 21^3 \overset{2^1}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -71, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.69 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 184 & 16 & 24 \\ 1 & 16 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[209] \ 1^1 \overset{1^5}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -71, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.14 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 218 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[210] \ 1^1 \overset{1^5}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -71, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.70 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 290 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[211] \ 1^1 \overset{1^5}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -72, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.39 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 252 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[212] \ 1^1 \overset{1^4}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, \ T_{\max} = 21, \ \Delta = -112, \ \det = -72, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.89 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 184 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$$

- [213] $1^2 \overset{1^5}{5} 1^4 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -73$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.11 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 216 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -2 \end{pmatrix}_{b_0}$
- [214] $21^4 \overset{2^1}{5} 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -73$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.11 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 216 & 17 & 26 \\ 1 & 17 & 1 & 1 \\ 2 & 26 & 1 & -2 \end{pmatrix}_{b_0}$
- [215] $1^1 \overset{1^4}{5} 1^4 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -73$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.11 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 216 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -2 \end{pmatrix}_{b_0}$
- [216] $1^2 \overset{1^3}{5} 1^4 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -73$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 152 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$
- [217] $21^2 \overset{2^1}{5} 1^5 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -73$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.11 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 284 & 19 & 30 \\ 1 & 19 & 1 & 1 \\ 2 & 30 & 1 & -2 \end{pmatrix}_{b_0}$
- [218] $21^2 \overset{2^1}{5} 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -73$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 212 & 17 & 26 \\ 1 & 17 & 1 & 1 \\ 2 & 26 & 1 & -1 \end{pmatrix}_{b_0}$
- [219] $1^2 \overset{1^4}{5} 1^4 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -74$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.86 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 182 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$
- [220] $21^2 \overset{2^1}{5} 1^4 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -74$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.86 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 246 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -1 \end{pmatrix}_{b_0}$
- [221] $1^2 \overset{1^3}{5} 1^4 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -76$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.84 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 180 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$
- [222] $1^2 \overset{1^5}{5} 1^4 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -76$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.32 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 248 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$
- [223] $1^2 \overset{1^5}{5} 1^4 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -76$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.89 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 324 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -2 \end{pmatrix}_{b_0}$
- [224] $21^3 \overset{2^1}{5} 1^2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -76$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.84 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 212 & 17 & 26 \\ 1 & 17 & 1 & 1 \\ 2 & 26 & 1 & -1 \end{pmatrix}_{b_0}$$

[225] $21 \overset{1^2}{5} \overset{1^4}{1} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.89 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$
 $\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 324 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -2 \end{pmatrix}_{b_0}$

[226] $1 \overset{1^4}{2} \overset{1^5}{5} \overset{1^4}{1} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -77, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.05 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 212 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -2 \end{pmatrix}_{b_0}$

[227] $1 \overset{1^5}{2} \overset{1^4}{5} \overset{1^4}{1} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -77, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.57 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 284 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -2 \end{pmatrix}_{b_0}$

[229] $1 \overset{1^4}{2} \overset{1^5}{5} \overset{1^4}{1} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.28 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 246 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$

[228] $21 \overset{2^1}{4} \overset{1^4}{5} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.28 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 246 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -2 \end{pmatrix}_{b_0}$

[239] $1 \overset{1^5}{5} \overset{1^4}{1} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.85 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 0 & 23 & 1 & 2 \\ 23 & 450 & 0 & 38 \\ 1 & 0 & -1 & -1 \\ 2 & 38 & -1 & -3 \end{pmatrix}_{b_0}$

[230] $21 \overset{2^1}{3} \overset{1^4}{5} 1^5 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.53 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 320 & 20 & 32 \\ 1 & 20 & 1 & 1 \\ 2 & 32 & 1 & -2 \end{pmatrix}_{b_0}$

[231] $1 \overset{1^3}{3} \overset{1^5}{5} \overset{1^4}{1} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.03 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 210 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -2 \end{pmatrix}_{b_0}$

[232] $21 \overset{1^5}{5} \overset{1^4}{1} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.85 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 0 & 23 & 1 & 2 \\ 23 & 450 & 0 & 38 \\ 1 & 0 & -1 & -1 \\ 2 & 38 & -1 & -3 \end{pmatrix}_{b_0}$

[233] $221 \overset{1^3}{3} \overset{1^5}{5} \overset{1^4}{1} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.03 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 244 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -2 \end{pmatrix}_{b_0}$

[234] $1 \overset{1^5}{3} \overset{1^4}{5} \overset{1^4}{1} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.14 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 362 & 0 & 34 \\ 1 & 0 & -1 & -1 \\ 2 & 34 & -1 & -3 \end{pmatrix}_{b_0}$

[235] $21 \overset{2^1}{3} \overset{1^4}{5} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.03 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 244 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[236] \ 21^{\overset{13}{\underset{13}{5}}1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.14 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 362 & 0 & 34 \\ 1 & 0 & -1 & -1 \\ 2 & 34 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[237] \ 1^{\overset{15}{\underset{13}{3}}51^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.53 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 282 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[238] \ 221^{\overset{5}{\underset{5}{5}}1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.85 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 23 & 1 & 2 \\ 23 & 404 & 22 & 36 \\ 1 & 22 & 1 & 1 \\ 2 & 36 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[242] \ 1^{\overset{15}{\underset{14}{3}}51^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.80 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 320 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[240] \ 1^{\overset{15}{\underset{15}{4}}51^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.45 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 404 & 0 & 36 \\ 1 & 0 & -1 & -1 \\ 2 & 36 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[241] \ 21^{\overset{14}{\underset{5}{5}}1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.45 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 404 & 0 & 36 \\ 1 & 0 & -1 & -1 \\ 2 & 36 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[243] \ 21^{\overset{21}{\underset{3}{3}}51^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 280 & 19 & 30 \\ 1 & 19 & 1 & 1 \\ 2 & 30 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[244] \ 1^{\overset{14}{\underset{13}{3}}51^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 244 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[245] \ 1^{\overset{14}{\underset{14}{3}}51^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.49 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 280 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[246] \ 1^{\overset{15}{\underset{14}{4}}51^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.09 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 360 & 0 & 34 \\ 1 & 0 & -1 & -1 \\ 2 & 34 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[247] \ 21^{\overset{21}{\underset{4}{4}}51^5 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.09 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 360 & 21 & 34 \\ 1 & 21 & 1 & 1 \\ 2 & 34 & 1 & -3 \end{pmatrix}_{b_0}$$

[248] $21^4 \overset{21^4}{5} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -81$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.49 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 280 & 19 & 30 \\ 1 & 19 & 1 & 1 \\ 2 & 30 & 1 & -2 \end{pmatrix}_{b_0}$

[249] $221^4 \overset{4}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -82$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.76 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 318 & 20 & 32 \\ 1 & 20 & 1 & 1 \\ 2 & 32 & 1 & -2 \end{pmatrix}_{b_0}$

[251] $1^4 \overset{1^4}{5} 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -82$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.76 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 318 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -2 \end{pmatrix}_{b_0}$

[250] $21^4 \overset{21^4}{5} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -82$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.76 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 318 & 20 & 32 \\ 1 & 20 & 1 & 1 \\ 2 & 32 & 1 & -2 \end{pmatrix}_{b_0}$

[252] $1^1 \overset{1^3}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -98$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.37 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 158 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$

[253] $1^1 \overset{1^4}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -100$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.64 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 224 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -1 \end{pmatrix}_{b_0}$

[254] $1^2 \overset{1^2}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -100$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.36 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 156 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$

[257] $1^2 \overset{21^5}{5} 1^4 2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -100$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.64 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 260 & 19 & 28 \\ 1 & 19 & 1 & 1 \\ 2 & 28 & 1 & -1 \end{pmatrix}_{b_0}$

[256] $221^2 \overset{2}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -100$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.36 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 252 & 19 & 28 \\ 1 & 19 & 1 & 1 \\ 2 & 28 & 1 & -1 \end{pmatrix}_{b_0}$

[255] $21^2 \overset{21^5}{5} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -100$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.36 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 252 & 19 & 28 \\ 1 & 19 & 1 & 1 \\ 2 & 28 & 1 & -1 \end{pmatrix}_{b_0}$

[259] $1^1 \overset{1^4}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -101$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.49 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 188 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$

[258] $1^5 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -101$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.80 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 260 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -1 \end{pmatrix}_{b_0}$

- [262] $1^1 \overset{1^5}{5} 1^5 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -102$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.63 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 222 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -1 \end{pmatrix}_{b_0}$
- [261] $1^1 \overset{1^5}{5} 1^5 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -102$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.98 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 298 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -1 \end{pmatrix}_{b_0}$
- [260] $1^2 \overset{1^5}{5} 1^5 2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -102$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.98 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 298 & 20 & 30 \\ 1 & 20 & 1 & 1 \\ 2 & 30 & 1 & -1 \end{pmatrix}_{b_0}$
- [264] $1^1 \overset{1^3}{5} 1^5 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -103$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.48 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 186 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$
- [263] $21^3 \overset{2^1}{5} 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -103$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.48 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 254 & 19 & 28 \\ 1 & 19 & 1 & 1 \\ 2 & 28 & 1 & -1 \end{pmatrix}_{b_0}$
- [265] $1^2 \overset{1^3}{5} 1^5 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -105$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.47 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 184 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$
- [266] $21^2 \overset{2^1}{5} 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -105$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.47 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 286 & 20 & 30 \\ 1 & 20 & 1 & 1 \\ 2 & 30 & 1 & -1 \end{pmatrix}_{b_0}$
- [267] $1^1 \overset{1^4}{5} 1^5 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -106$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.60 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 218 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -1 \end{pmatrix}_{b_0}$
- [268] $1^1 \overset{1^5}{5} 1^5 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -107$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.76 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 254 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -1 \end{pmatrix}_{b_0}$
- [270] $1^2 \overset{1^4}{5} 1^5 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -108$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.59 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 216 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -1 \end{pmatrix}_{b_0}$
- [269] $21^2 \overset{2^1}{5} 1^4 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -108$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.59 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 324 & 21 & 32 \\ 1 & 21 & 1 & 1 \\ 2 & 32 & 1 & -1 \end{pmatrix}_{b_0}$
- [271] $1^1 \overset{1^4}{5} 1^5 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -109$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 252 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[272] \ 1^2 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 252 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[273] \ 21^4 \overset{2^1}{5} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 290 & 20 & 30 \\ 1 & 20 & 1 & 1 \\ 2 & 30 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[274] \ 21^2 \overset{2^1}{5} 1^5 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 366 & 22 & 34 \\ 1 & 22 & 1 & 1 \\ 2 & 34 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[277] \ 1^1 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -110, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 290 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[275] \ 1^2 \overset{1^3}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -110, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.58 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 214 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[276] \ 21^3 \overset{2^1}{5} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -110, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.58 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 286 & 20 & 30 \\ 1 & 20 & 1 & 1 \\ 2 & 30 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[279] \ 1^1 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.09 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 330 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[278] \ 21^5 \overset{2^1}{5} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.09 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 330 & 21 & 32 \\ 1 & 21 & 1 & 1 \\ 2 & 32 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[280] \ 1^2 \overset{1^4}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -113, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 248 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[281] \ 1^2 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 286 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[284] \ 1^3 \overset{1^3}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 246 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[283] \ 221^3 \overset{1^3}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 322 & 21 & 32 \\ 1 & 21 & 1 & 1 \\ 2 & 32 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[282] \ 21^3 \overset{21^5}{5} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 322 & 21 & 32 \\ 1 & 21 & 1 & 1 \\ 2 & 32 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[286] \ 1^2 \overset{1^4}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 284 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[285] \ 21^4 \overset{21^5}{5} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 324 & 21 & 32 \\ 1 & 21 & 1 & 1 \\ 2 & 32 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[287] \ 1^2 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.03 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 324 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[291] \ 21^5 \overset{21^5}{5} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.22 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 366 & 22 & 34 \\ 1 & 22 & 1 & 1 \\ 2 & 34 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[290] \ 1^2 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.22 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 366 & 0 & 34 \\ 1 & 0 & -1 & -1 \\ 2 & 34 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[289] \ 21^3 \overset{21^5}{5} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.85 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 362 & 22 & 34 \\ 1 & 22 & 1 & 1 \\ 2 & 34 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[288] \ 1^3 \overset{1^4}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.85 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 282 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[293] \ 21^3 \overset{21^5}{5} 1^5 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -119, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 406 & 23 & 36 \\ 1 & 23 & 1 & 1 \\ 2 & 36 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[292] \ 1^3 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -119, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 322 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[295] \ 1^3 \overset{1^4}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -121, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 320 & 0 & 32 \\ 1 & 0 & -1 & -1 \\ 2 & 32 & -1 & -1 \end{pmatrix}_{b_0}$$

- [294] $21^4 \overset{21^5}{5} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -121$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 362 & 22 & 34 \\ 1 & 22 & 1 & 1 \\ 2 & 34 & 1 & -1 \end{pmatrix}_{b_0}$
- [296] $1^3 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -122$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.18 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 362 & 0 & 34 \\ 1 & 0 & -1 & -1 \\ 2 & 34 & -1 & -2 \end{pmatrix}_{b_0}$
- [297] $1^3 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -123$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.37 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 23 & 1 & 2 \\ 23 & 406 & 0 & 36 \\ 1 & 0 & -1 & -1 \\ 2 & 36 & -1 & -2 \end{pmatrix}_{b_0}$
- [298] $21^5 \overset{21^5}{5} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -123$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.37 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 23 & 1 & 2 \\ 23 & 406 & 23 & 36 \\ 1 & 23 & 1 & 1 \\ 2 & 36 & 1 & -2 \end{pmatrix}_{b_0}$
- [301] $21^4 \overset{21^5}{5} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -124$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.16 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 404 & 23 & 36 \\ 1 & 23 & 1 & 1 \\ 2 & 36 & 1 & -2 \end{pmatrix}_{b_0}$
- [299] $221^4 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -124$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.16 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 404 & 23 & 36 \\ 1 & 23 & 1 & 1 \\ 2 & 36 & 1 & -2 \end{pmatrix}_{b_0}$
- [300] $1^4 \overset{1^4}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -124$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.16 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 360 & 0 & 34 \\ 1 & 0 & -1 & -1 \\ 2 & 34 & -1 & -2 \end{pmatrix}_{b_0}$
- [302] $1^4 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -125$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.35 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 23 & 1 & 2 \\ 23 & 404 & 0 & 36 \\ 1 & 0 & -1 & -1 \\ 2 & 36 & -1 & -2 \end{pmatrix}_{b_0}$
- [303] $21^4 \overset{21^5}{5} 1^5 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -125$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.35 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 23 & 1 & 2 \\ 23 & 450 & 24 & 38 \\ 1 & 24 & 1 & 1 \\ 2 & 38 & 1 & -2 \end{pmatrix}_{b_0}$
- [304] $21^5 \overset{21^5}{5} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -126$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.56 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 24 & 1 & 2 \\ 24 & 450 & 24 & 38 \\ 1 & 24 & 1 & 1 \\ 2 & 38 & 1 & -2 \end{pmatrix}_{b_0}$
- [305] $1^4 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -126$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.56 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$
 $\begin{pmatrix} 0 & 24 & 1 & 2 \\ 24 & 450 & 0 & 38 \\ 1 & 0 & -1 & -1 \\ 2 & 38 & -1 & -2 \end{pmatrix}_{b_0}$
- [306] $1^5 \overset{1^5}{5} 1^5 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 7$, $T_{\max} = 21$, $\Delta = -112$, $\det = -127$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.77 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$

$$\begin{pmatrix} 0 & 25 & 1 & 2 \\ 25 & 498 & 0 & 40 \\ 1 & 0 & -1 & -1 \\ 2 & 40 & -1 & -2 \end{pmatrix}_{b_0}$$

[307] $21^{5^{21}5} 1^5 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.77 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$

$$\begin{pmatrix} 0 & 25 & 1 & 2 \\ 25 & 498 & 25 & 40 \\ 1 & 25 & 1 & 1 \\ 2 & 40 & 1 & -2 \end{pmatrix}_{b_0}$$

[308] $221^{551^5} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 7, T_{\max} = 21, \Delta = -112, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.77 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$

$$\begin{pmatrix} 0 & 25 & 1 & 2 \\ 25 & 498 & 25 & 40 \\ 1 & 25 & 1 & 1 \\ 2 & 40 & 1 & -2 \end{pmatrix}_{b_0}$$

$T = 8$ [309] $1^{21^2} 6^{21^1} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 19, T_{\max} = 20, T \in [19, 20]$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 20 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -4 \end{pmatrix}_{b_0}$$

[310] $221^{21^1} 6^{1^2} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 20 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -1 \end{pmatrix}_{b_0}$$

[311] $1^{21^1} 6^{1^2} 12 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 20 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -1 \end{pmatrix}_{b_0}$$

[312] $1^{21} 6^{21^1} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

[313] $21^{21^1} 6^{1^2} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 19, T_{\max} = 20, T \in [19, 20]$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 44 & 7 & 12 \\ 1 & 7 & 1 & 1 \\ 2 & 12 & 1 & -4 \end{pmatrix}_{b_0}$$

[314] $222161^{1^2} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

[315] $1^{21^1} 6^{1^2} 122 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

[316] $313151^{1^2} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 16, \Delta = -99, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

[317] $21^{21^1} 6^{1^2} 12 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

[318] $21^{2^2} 6^{21^1} 12 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 19, T_{\max} = 20, T \in [19, 20]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 44 & 7 & 12 \\ 1 & 7 & 1 & 1 \\ 2 & 12 & 1 & -4 \end{pmatrix}_{b_0} \\
[321] & 1 \overset{2^1 1^1}{\underset{1^1}{6}} 12 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 15.20 \rightarrow 16, T_{\max} = 20, T \in [16, 20] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 4 & 6 \\ 1 & 4 & 1 & 1 \\ 2 & 6 & 1 & -1 \end{pmatrix}_{b_0} \\
[322] & 221 \overset{2^1 1^1}{\underset{1^1}{6}} 1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 15.20 \rightarrow 16, T_{\max} = 20, T \in [16, 20] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 4 & 6 \\ 1 & 4 & 1 & 1 \\ 2 & 6 & 1 & -1 \end{pmatrix}_{b_0} \\
[320] & 1 \overset{2^1 2}{\underset{2^1 1^1}{6}} 1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 30 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -5 \end{pmatrix}_{b_0} \\
[319] & 21 \overset{2^1 1^1}{\underset{1^2}{6}} 1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 16.80 \rightarrow 17, T_{\max} = 20, T \in [17, 20] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 30 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -2 \end{pmatrix}_{b_0} \\
[325] & 1 \overset{2^1 1^1}{\underset{2^1 1^1}{6}} 1 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 15.67 \rightarrow 16, T_{\max} = 20, T \in [16, 20] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 4 & 6 \\ 1 & 4 & 1 & 1 \\ 2 & 6 & 1 & -1 \end{pmatrix}_{b_0} \\
[326] & 1 \overset{2^1 1^1}{\underset{2^1 1^1}{6}} 1^2 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 17.67 \rightarrow 18, T_{\max} = 20, T \in [18, 20] \\
& \begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 30 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -3 \end{pmatrix}_{b_0} \\
[323] & 21 \overset{2^1 1^1}{\underset{1^1}{6}} 12 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 15.67 \rightarrow 16, T_{\max} = 20, T \in [16, 20] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -1 \end{pmatrix}_{b_0} \\
[324] & 21 \overset{2^1 1^1}{\underset{1^1}{6}} 1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 15.67 \rightarrow 16, T_{\max} = 20, T \in [16, 20] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -1 \end{pmatrix}_{b_0} \\
[328] & 1 \overset{2^1 1^1}{\underset{2^1 1^1}{6}} 1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 16.29 \rightarrow 17, T_{\max} = 20, T \in [17, 20] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -2 \end{pmatrix}_{b_0} \\
[329] & 1 \overset{221^1}{\underset{1^1}{6}} 1^2 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 16.29 \rightarrow 17, T_{\max} = 20, T \in [17, 20] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -2 \end{pmatrix}_{b_0} \\
[330] & 21 \overset{2^1 1^1}{\underset{1^2}{6}} 1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 18.57 \rightarrow 19, T_{\max} = 20, T \in [19, 20] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 42 & 7 & 12 \\ 1 & 7 & 1 & 1 \\ 2 & 12 & 1 & -4 \end{pmatrix}_{b_0}
\end{aligned}$$

[327] $221^{1^{2^1 1^1}} 6^1 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 18.57 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$
 $\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 42 & 7 & 12 \\ 1 & 7 & 1 & 1 \\ 2 & 12 & 1 & -4 \end{pmatrix}_{b_0}$

[332] $221^{1^{2^1 1^1}} 6^1 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -2 \end{pmatrix}_{b_0}$

[333] $21^{1^{2^1 1^1}} 6^1 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -2 \end{pmatrix}_{b_0}$

[334] $2221^{1^6 1^1} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -2 \end{pmatrix}_{b_0}$

[331] $21^{1^{2^1 1^1}} 6^1 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -2 \end{pmatrix}_{b_0}$

[337] $21^{1^{2^1 1^2}} 6^1 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.25 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 44 & 8 & 12 \\ 1 & 8 & 1 & 1 \\ 2 & 12 & 1 & -1 \end{pmatrix}_{b_0}$

[342] $1^{2^1 1^2} 6^1 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.25 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 64 & 9 & 14 \\ 1 & 9 & 1 & 1 \\ 2 & 14 & 1 & -3 \end{pmatrix}_{b_0}$

[341] $21^{1^{2^1 1^2}} 6^1 12 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.25 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 44 & 8 & 12 \\ 1 & 8 & 1 & 1 \\ 2 & 12 & 1 & -1 \end{pmatrix}_{b_0}$

[340] $1^{2^1 1^2} 6^1 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.25 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 84 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -3 \end{pmatrix}_{b_0}$

[339] $21^{1^{2^1 1^2}} 6^1 12 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.25 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 124 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -3 \end{pmatrix}_{b_0}$

[338] $21^{1^{2^1 1^2}} 6^1 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.25 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 124 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -3 \end{pmatrix}_{b_0}$

[343] $31^{1^{2^1 1^1}} 5^1 13 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 16, \Delta = -99, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$
 $\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 8 & 5 & 6 \\ 2 & 5 & 2 & 2 \\ 2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$

- [335] $21 \begin{smallmatrix} 2^1 1^2 \\ 1^3 \end{smallmatrix} 6 \begin{smallmatrix} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \end{smallmatrix} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 80 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -1 \end{pmatrix}_{b_0}$
- [336] $1 \begin{smallmatrix} 2^1 1^2 \\ 21^1 \end{smallmatrix} 6 \begin{smallmatrix} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \end{smallmatrix} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.25 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 20 & 0 & 8 \\ 1 & 0 & -1 & -1 \\ 2 & 8 & -1 & -1 \end{pmatrix}_{b_0}$
- [345] $21 \begin{smallmatrix} 2^1 1^2 \\ 1^3 \end{smallmatrix} 6 \begin{smallmatrix} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \end{smallmatrix} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.47 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 100 & 11 & 18 \\ 1 & 11 & 1 & 1 \\ 2 & 18 & 1 & -2 \end{pmatrix}_{b_0}$
- [344] $21 \begin{smallmatrix} 2^1 1^2 \\ 1^2 \end{smallmatrix} 6 \begin{smallmatrix} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \end{smallmatrix} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.53 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 60 & 9 & 14 \\ 1 & 9 & 1 & 1 \\ 2 & 14 & 1 & -1 \end{pmatrix}_{b_0}$
- [346] $21 \begin{smallmatrix} 2^1 1^2 \\ 1^2 \end{smallmatrix} 6 \begin{smallmatrix} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \end{smallmatrix} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.89 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 78 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -1 \end{pmatrix}_{b_0}$
- [347] $1 \begin{smallmatrix} 21^2 \\ 21^2 \end{smallmatrix} 6 \begin{smallmatrix} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \end{smallmatrix} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.89 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 46 & 8 & 12 \\ 1 & 8 & 1 & 1 \\ 2 & 12 & 1 & -1 \end{pmatrix}_{b_0}$
- [349] $21 \begin{smallmatrix} 2^1 1^2 \\ 1^2 \end{smallmatrix} 6 \begin{smallmatrix} 12 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \end{smallmatrix} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.89 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 78 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -1 \end{pmatrix}_{b_0}$
- [348] $1 \begin{smallmatrix} 21^2 \\ 21^2 \end{smallmatrix} 6 \begin{smallmatrix} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \end{smallmatrix} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 18.56 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 126 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -4 \end{pmatrix}_{b_0}$
- [358] $1 \begin{smallmatrix} 221^1 \\ 1^2 \end{smallmatrix} 6 \begin{smallmatrix} 1^2 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \end{smallmatrix} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.47 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 44 & 8 & 12 \\ 1 & 8 & 1 & 1 \\ 2 & 12 & 1 & -1 \end{pmatrix}_{b_0}$
- [357] $21 \begin{smallmatrix} 2^1 1^2 \\ 1^4 \end{smallmatrix} 6 \begin{smallmatrix} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \end{smallmatrix} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 19.26 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 150 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -5 \end{pmatrix}_{b_0}$
- [356] $221 \begin{smallmatrix} 2^1 1^2 \\ 1^1 \end{smallmatrix} 6 \begin{smallmatrix} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \end{smallmatrix} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.58 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 124 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -3 \end{pmatrix}_{b_0}$
- [355] $1 \begin{smallmatrix} 2^1 1^2 \\ 21^1 \end{smallmatrix} 6 \begin{smallmatrix} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \end{smallmatrix} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.47 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 30 & 0 & 10 \\ 1 & 0 & -1 & -1 \\ 2 & 10 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[354] \quad 1 \overset{21^3}{\underset{21^{12}}{6}} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 19.26 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 124 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -5 \end{pmatrix}_{b_0}$$

$$[353] \quad 21 \overset{21^2}{\underset{1^4}{1}} \overset{21^2}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.58 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 102 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[352] \quad 1 \overset{21^3}{\underset{21^{12}}{6}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.58 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 80 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[351] \quad 1 \overset{221^3}{\underset{1^4}{6}} 1^{221} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 19.26 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 124 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -5 \end{pmatrix}_{b_0}$$

$$[350] \quad 1 \overset{21^2}{\underset{21^1}{6}} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.32 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 62 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[360] \quad 1 \overset{21^3}{\underset{21^{12}}{6}} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.20 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 100 & 11 & 18 \\ 1 & 11 & 1 & 1 \\ 2 & 18 & 1 & -4 \end{pmatrix}_{b_0}$$

$$[361] \quad 21 \overset{21^2}{\underset{1^4}{1}} \overset{21^2}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.20 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 124 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -4 \end{pmatrix}_{b_0}$$

$$[359] \quad 1 \overset{21^2}{\underset{21^1}{6}} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.80 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 44 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[362] \quad 1 \overset{21^2}{\underset{21^{12}}{6}} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.33 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 100 & 11 & 18 \\ 1 & 11 & 1 & 1 \\ 2 & 18 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[363] \quad 1 \overset{21^2}{\underset{21^{12}}{6}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.19 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 60 & 9 & 14 \\ 1 & 9 & 1 & 1 \\ 2 & 14 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[364] \quad 21 \overset{21^2}{\underset{1^4}{2}} \overset{21^2}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.86 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 148 & 13 & 22 \\ 1 & 13 & 1 & 1 \\ 2 & 22 & 1 & -4 \end{pmatrix}_{b_0}$$

- [371] $221^3 \overset{21^2}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 19.55 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 146 & 13 & 22 \\ 1 & 13 & 1 & 1 \\ 2 & 22 & 1 & -5 \end{pmatrix}_{b_0}$
- [369] $21^1 \overset{21^2}{6} 1^1 2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.73 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 42 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -1 \end{pmatrix}_{b_0}$
- [366] $21^3 \overset{21^2}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.91 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 98 & 11 & 18 \\ 1 & 11 & 1 & 1 \\ 2 & 18 & 1 & -3 \end{pmatrix}_{b_0}$
- [367] $2221^1 \overset{161^2}{2} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.73 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 58 & 9 & 14 \\ 1 & 9 & 1 & 1 \\ 2 & 14 & 1 & -1 \end{pmatrix}_{b_0}$
- [377] $21^1 \overset{21^3}{6} 1^2 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.91 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 98 & 11 & 18 \\ 1 & 11 & 1 & 1 \\ 2 & 18 & 1 & -3 \end{pmatrix}_{b_0}$
- [376] $1 \overset{221^2}{6} 1^2 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.64 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 78 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -2 \end{pmatrix}_{b_0}$
- [378] $221^2 \overset{21^2}{6} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 19.55 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 174 & 14 & 24 \\ 1 & 14 & 1 & 1 \\ 2 & 24 & 1 & -5 \end{pmatrix}_{b_0}$
- [373] $21^1 \overset{21^2}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.64 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 78 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -2 \end{pmatrix}_{b_0}$
- [372] $1 \overset{21^2}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.64 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 78 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -2 \end{pmatrix}_{b_0}$
- [365] $21^2 \overset{21^2}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 19.55 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 174 & 14 & 24 \\ 1 & 14 & 1 & 1 \\ 2 & 24 & 1 & -5 \end{pmatrix}_{b_0}$
- [375] $21^3 \overset{21^2}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 19.55 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 146 & 13 & 22 \\ 1 & 13 & 1 & 1 \\ 2 & 22 & 1 & -5 \end{pmatrix}_{b_0}$
- [368] $21^1 \overset{21^2}{6} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.91 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 122 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -3 \end{pmatrix}_{b_0}$$

[374] $221 \overset{2^1 1^2}{1} 6 \overset{1^3}{1^3} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.64 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 98 & 11 & 18 \\ 1 & 11 & 1 & 1 \\ 2 & 18 & 1 & -2 \end{pmatrix}_{b_0}$$

[370] $21 \overset{2^1 1^2}{1} 6 \overset{1^1}{1^1} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.73 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 42 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -1 \end{pmatrix}_{b_0}$$

[379] $221 \overset{2^1 1^2}{1} 6 \overset{1^1}{1^1} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.73 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 58 & 9 & 14 \\ 1 & 9 & 1 & 1 \\ 2 & 14 & 1 & -1 \end{pmatrix}_{b_0}$$

[383] $21 \overset{2^1 1^2}{1} 6 \overset{1^2}{1^2} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.13 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$

$$\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 98 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -3 \end{pmatrix}_{b_0}$$

[382] $21 \overset{2^1 1^2}{1} 6 \overset{1^1}{1^1} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 18.52 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 120 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -4 \end{pmatrix}_{b_0}$$

[381] $21 \overset{2^1 1^2}{1} 6 \overset{1^1}{1^1} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.09 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 58 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -2 \end{pmatrix}_{b_0}$$

[380] $221 \overset{2^1 1^2}{1} 6 \overset{1^2}{1^2} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.09 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 76 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -2 \end{pmatrix}_{b_0}$$

[390] $21 \overset{2^1 1^2}{1} 6 \overset{1^2}{1^2} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 76 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -2 \end{pmatrix}_{b_0}$$

[384] $21 \overset{2^1 1^2}{1} 6 \overset{1^1}{1^1} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.67 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 120 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -3 \end{pmatrix}_{b_0}$$

[385] $21 \overset{2^1 1^2}{1} 6 \overset{1^2 2^1}{1^2 2^1} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 76 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -2 \end{pmatrix}_{b_0}$$

[388] $21 \overset{2^1 1^2}{1} 6 \overset{1^3}{1^3} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 19.17 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 172 & 14 & 24 \\ 1 & 14 & 1 & 1 \\ 2 & 24 & 1 & -5 \end{pmatrix}_{b_0}$$

- [387] $21^3 \overset{21^2}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 24$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 19.17 \rightarrow 20$, $T_{\max} = 20$, $T \in [20, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 144 & 13 & 22 \\ 1 & 13 & 1 & 1 \\ 2 & 22 & 1 & -5 \end{pmatrix}_{b_0}$
- [386] $21^2 \overset{21^3}{6} 1^{22^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 24$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 19.17 \rightarrow 20$, $T_{\max} = 20$, $T \in [20, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 144 & 13 & 22 \\ 1 & 13 & 1 & 1 \\ 2 & 22 & 1 & -5 \end{pmatrix}_{b_0}$
- [389] $21^2 \overset{21^2}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 24$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.50 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 76 & 10 & 16 \\ 1 & 10 & 1 & 1 \\ 2 & 16 & 1 & -2 \end{pmatrix}_{b_0}$
- [394] $221^2 \overset{21^2}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.96 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 96 & 11 & 18 \\ 1 & 11 & 1 & 1 \\ 2 & 18 & 1 & -2 \end{pmatrix}_{b_0}$
- [393] $21^2 \overset{21^2}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.96 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 96 & 11 & 18 \\ 1 & 11 & 1 & 1 \\ 2 & 18 & 1 & -2 \end{pmatrix}_{b_0}$
- [392] $221^2 \overset{21^2}{6} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 18.24 \rightarrow 19$, $T_{\max} = 20$, $T \in [19, 20]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 144 & 13 & 22 \\ 1 & 13 & 1 & 1 \\ 2 & 22 & 1 & -4 \end{pmatrix}_{b_0}$
- [391] $21^2 \overset{21^2}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 25$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 18.24 \rightarrow 19$, $T_{\max} = 20$, $T \in [19, 20]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 144 & 13 & 22 \\ 1 & 13 & 1 & 1 \\ 2 & 22 & 1 & -4 \end{pmatrix}_{b_0}$
- [397] $221^2 \overset{21^2}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 26$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.46 \rightarrow 18$, $T_{\max} = 20$, $T \in [18, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 118 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -3 \end{pmatrix}_{b_0}$
- [395] $2221^2 \overset{21^2}{6} 1^{22^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 26$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.46 \rightarrow 18$, $T_{\max} = 20$, $T \in [18, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 118 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -3 \end{pmatrix}_{b_0}$
- [398] $21^2 \overset{21^2}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 26$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.46 \rightarrow 18$, $T_{\max} = 20$, $T \in [18, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 118 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -3 \end{pmatrix}_{b_0}$
- [396] $21^2 \overset{21^2}{6} 1^{22^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 26$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.46 \rightarrow 18$, $T_{\max} = 20$, $T \in [18, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 118 & 12 & 20 \\ 1 & 12 & 1 & 1 \\ 2 & 20 & 1 & -3 \end{pmatrix}_{b_0}$
- [399] $1 \overset{21^3}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 35$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.26 \rightarrow 16$, $T_{\max} = 20$, $T \in [16, 20]$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 46 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 2 & 12 & -1 & -1 \end{pmatrix}_{b_0}$$

[400] $1 \overset{221^1}{6} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.26 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 82 & 11 & 16 \\ 1 & 11 & 1 & 1 \\ 2 & 16 & 1 & -1 \end{pmatrix}_{b_0}$$

[401] $21 \overset{2^1 1^3}{4} 6 12 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.78 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 244 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -2 \end{pmatrix}_{b_0}$$

[402] $1 \overset{2^1 1^3}{21^2} 6 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.44 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 64 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -1 \end{pmatrix}_{b_0}$$

[403] $21 \overset{2^1 1^3}{1^2} 6 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.44 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 124 & 13 & 20 \\ 1 & 13 & 1 & 1 \\ 2 & 20 & 1 & -1 \end{pmatrix}_{b_0}$$

[404] $21 \overset{2^1 1^3}{2^2} 6 12 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.44 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 124 & 13 & 20 \\ 1 & 13 & 1 & 1 \\ 2 & 20 & 1 & -1 \end{pmatrix}_{b_0}$$

[405] $1 \overset{21^4}{2^1 1^3} 6 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.78 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 132 & 13 & 20 \\ 1 & 13 & 1 & 1 \\ 2 & 20 & 1 & -2 \end{pmatrix}_{b_0}$$

[406] $21 \overset{2^1 1^3}{1^4} 6 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.78 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 244 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -2 \end{pmatrix}_{b_0}$$

[407] $21 \overset{2^1 1^3}{1^4} 6 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 180 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$$

[408] $21 \overset{2^1 1^3}{1^4} 6 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.32 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 210 & 16 & 26 \\ 1 & 16 & 1 & 1 \\ 2 & 26 & 1 & -2 \end{pmatrix}_{b_0}$$

[409] $21 \overset{2^1 1^3}{1^3} 6 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.68 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 150 & 14 & 22 \\ 1 & 14 & 1 & 1 \\ 2 & 22 & 1 & -1 \end{pmatrix}_{b_0}$$

[412] $1 \overset{2^1 1^3}{21^1} 6 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.42 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 62 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -1 \end{pmatrix}_{b_0}$$

- [413] $1 \overset{21^3}{\underset{21^3}{6}} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.95 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 106 & 12 & 18 \\ 1 & 12 & 1 & 1 \\ 2 & 18 & 1 & -1 \end{pmatrix}_{b_0}$
- [414] $21 \overset{21^3}{\underset{1^3}{6}} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.95 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 178 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$
- [410] $1 \overset{21^3}{\underset{21^1}{6}} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.95 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 106 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$
- [411] $21 \overset{21^3}{\underset{1^3}{6}} 12 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.95 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 178 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$
- [415] $1 \overset{21^3}{\underset{21^1}{6}} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.64 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 82 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -1 \end{pmatrix}_{b_0}$
- [417] $21 \overset{21^3}{\underset{1^1}{6}} 1^1 2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.40 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 60 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -1 \end{pmatrix}_{b_0}$
- [419] $21 \overset{21^3}{\underset{1^1}{6}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.40 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 60 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 2 & 14 & -1 & -1 \end{pmatrix}_{b_0}$
- [418] $221 \overset{21^3}{\underset{1^1}{6}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.40 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 100 & 12 & 18 \\ 1 & 12 & 1 & 1 \\ 2 & 18 & 1 & -1 \end{pmatrix}_{b_0}$
- [416] $2221 \overset{21^3}{\underset{1^1}{6}} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.40 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 100 & 12 & 18 \\ 1 & 12 & 1 & 1 \\ 2 & 18 & 1 & -1 \end{pmatrix}_{b_0}$
- [420] $1 \overset{21^3}{\underset{21^2}{6}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.61 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 80 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -1 \end{pmatrix}_{b_0}$
- [421] $1 \overset{21^4}{\underset{21^3}{6}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.98 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 154 & 14 & 22 \\ 1 & 14 & 1 & 1 \\ 2 & 22 & 1 & -2 \end{pmatrix}_{b_0}$
- [422] $1 \overset{21^3}{\underset{21^3}{6}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.14 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 126 & 13 & 20 \\ 1 & 13 & 1 & 1 \\ 2 & 20 & 1 & -2 \end{pmatrix}_{b_0}$$

[423] $221 \overset{21^3}{1} 6 \overset{1^2}{1^2} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.58 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 122 & 13 & 20 \\ 1 & 13 & 1 & 1 \\ 2 & 20 & 1 & -1 \end{pmatrix}_{b_0}$$

[424] $21 \overset{21^3}{1} 6 \overset{1^1}{1^1} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.58 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 78 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

[425] $21 \overset{21^3}{1} 6 \overset{1^1}{1^1} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.14 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 126 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -2 \end{pmatrix}_{b_0}$$

[426] $221 \overset{21^3}{1} 6 \overset{1^4}{1^4} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.14 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 178 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -2 \end{pmatrix}_{b_0}$$

[434] $1 \overset{221^4}{6} 1 \overset{32^1}{1^3} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 18.27 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 244 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -4 \end{pmatrix}_{b_0}$$

[432] $221 \overset{21^3}{1} 6 \overset{1^3}{1^3} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.82 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 148 & 14 & 22 \\ 1 & 14 & 1 & 1 \\ 2 & 22 & 1 & -1 \end{pmatrix}_{b_0}$$

[431] $1 \overset{21^4}{6} 1 \overset{1^4}{1^4} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 18.27 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 244 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -4 \end{pmatrix}_{b_0}$$

[430] $1 \overset{21^3}{6} \overset{1^2}{1^2} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.82 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 100 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$$

[427] $21 \overset{21^3}{1} 6 \overset{1^1}{1^1} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.82 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 100 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$$

[428] $1 \overset{21^3}{6} 1 \overset{1^4}{1^4} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.45 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 152 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$$

[429] $1 \overset{21^4}{6} 1 \overset{1^2}{1^2} + \text{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.27 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 180 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -3 \end{pmatrix}_{b_0}$$

- [433] $1 \overset{221^2}{6} 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.82 \rightarrow 16$, $T_{\max} = 20$, $T \in [16, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 124 & 13 & 20 \\ 1 & 13 & 1 & 1 \\ 2 & 20 & 1 & -1 \end{pmatrix}_{b_0}$
- [435] $1 \overset{21^4}{6} 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.69 \rightarrow 18$, $T_{\max} = 20$, $T \in [18, 20]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 210 & 16 & 26 \\ 1 & 16 & 1 & 1 \\ 2 & 26 & 1 & -3 \end{pmatrix}_{b_0}$
- [436] $1 \overset{21^3}{21^2} 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.09 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 124 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -2 \end{pmatrix}_{b_0}$
- [441] $21 \overset{21^3}{1^4} 6 1^2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.39 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 150 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$
- [444] $21 \overset{21^3}{1^4} 6 1^2 2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.78 \rightarrow 16$, $T_{\max} = 20$, $T \in [16, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 98 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$
- [445] $1 \overset{21^3}{21^3} 6 1^2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.39 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 150 & 14 & 22 \\ 1 & 14 & 1 & 1 \\ 2 & 22 & 1 & -2 \end{pmatrix}_{b_0}$
- [440] $21 \overset{21^4}{1^4} 6 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.17 \rightarrow 18$, $T_{\max} = 20$, $T \in [18, 20]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 178 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -3 \end{pmatrix}_{b_0}$
- [439] $21 \overset{21^3}{1^4} 6 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.17 \rightarrow 18$, $T_{\max} = 20$, $T \in [18, 20]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 178 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -3 \end{pmatrix}_{b_0}$
- [442] $21 \overset{21^3}{1^4} 6 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.78 \rightarrow 16$, $T_{\max} = 20$, $T \in [16, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 98 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$
- [438] $21 \overset{21^3}{1^4} 6 1^4 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.17 \rightarrow 18$, $T_{\max} = 20$, $T \in [18, 20]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 210 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -3 \end{pmatrix}_{b_0}$
- [437] $1 \overset{21^3}{21^3} 6 1^4 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.17 \rightarrow 18$, $T_{\max} = 20$, $T \in [18, 20]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 210 & 16 & 26 \\ 1 & 16 & 1 & 1 \\ 2 & 26 & 1 & -3 \end{pmatrix}_{b_0}$
- [443] $21 \overset{21^3}{1^2} 6 1^2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.78 \rightarrow 16$, $T_{\max} = 20$, $T \in [16, 20]$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 98 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[448] \quad 1 \overset{21^3}{\underset{21^3}{6}} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.72 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 178 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[449] \quad 1 \overset{221^3}{\underset{21^3}{6}} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.72 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 178 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[447] \quad 21 \overset{21^3}{\underset{1^4}{1}} \overset{21^3}{\underset{1^4}{6}} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.72 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 178 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[446] \quad 21 \overset{21^3}{\underset{1^3}{1}} \overset{21^3}{\underset{1^3}{6}} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.04 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 122 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[451] \quad 21 \overset{21^3}{\underset{1^1}{3}} \overset{21^3}{\underset{1^1}{6}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.33 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 148 & 14 & 22 \\ 1 & 14 & 1 & 1 \\ 2 & 22 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[450] \quad 21 \overset{21^3}{\underset{1^3}{1}} \overset{21^3}{\underset{1^3}{6}} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.33 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 148 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[452] \quad 21 \overset{21^3}{\underset{1^3}{1}} \overset{21^3}{\underset{1^3}{6}} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.33 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 148 & 14 & 22 \\ 1 & 14 & 1 & 1 \\ 2 & 22 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[453] \quad 221 \overset{21^3}{\underset{1^4}{2}} \overset{21^3}{\underset{1^4}{6}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 146 & 14 & 22 \\ 1 & 14 & 1 & 1 \\ 2 & 22 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[454] \quad 21 \overset{21^3}{\underset{1^4}{2}} \overset{21^3}{\underset{1^4}{6}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.65 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 176 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[455] \quad 21 \overset{21^3}{\underset{1^2}{4}} \overset{21^3}{\underset{1^2}{6}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.47 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 206 & 16 & 26 \\ 1 & 16 & 1 & 1 \\ 2 & 26 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[456] \quad 221 \overset{21^3}{\underset{1^4}{4}} \overset{21^3}{\underset{1^4}{6}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.45 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 274 & 18 & 30 \\ 1 & 18 & 1 & 1 \\ 2 & 30 & 1 & -4 \end{pmatrix}_{b_0}$$

- [457] $21^4 \overset{2^1 1^3}{\underset{1^4}{6}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 18.45 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 274 & 18 & 30 \\ 1 & 18 & 1 & 1 \\ 2 & 30 & 1 & -4 \end{pmatrix}_{b_0}$
- [458] $21^2 \overset{2^1 1^3}{\underset{1^2}{6}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 120 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$
- [459] $21^2 \overset{2^1 1^3}{\underset{1^3}{6}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.28 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 146 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$
- [460] $21^4 \overset{2^1 1^3}{\underset{1^3}{6}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.88 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 238 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -3 \end{pmatrix}_{b_0}$
- [461] $21^3 \overset{2^1 1^3}{\underset{1^2}{6}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.59 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 174 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -2 \end{pmatrix}_{b_0}$
- [462] $21^3 \overset{2^1 1^3}{\underset{1^4}{6}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.37 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 238 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -3 \end{pmatrix}_{b_0}$
- [478] $21^4 \overset{2^1 1^3}{\underset{1^4}{6}} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 18.77 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$
 $\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 308 & 19 & 32 \\ 1 & 19 & 1 & 1 \\ 2 & 32 & 1 & -4 \end{pmatrix}_{b_0}$
- [470] $21^2 \overset{2^1 1^3}{\underset{1^2}{6}} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.23 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 144 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$
- [471] $21^4 \overset{2^1 1^3}{\underset{1^4}{6}} 1^{3^2 1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 19.92 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 388 & 21 & 36 \\ 1 & 21 & 1 & 1 \\ 2 & 36 & 1 & -5 \end{pmatrix}_{b_0}$
- [472] $21^2 \overset{2^1 1^3}{\underset{1^2}{6}} 1^2 2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.23 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 144 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -2 \end{pmatrix}_{b_0}$
- [473] $221^4 \overset{2^1 1^3}{\underset{1^4}{6}} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 19.92 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 388 & 21 & 36 \\ 1 & 21 & 1 & 1 \\ 2 & 36 & 1 & -5 \end{pmatrix}_{b_0}$
- [474] $221^3 \overset{2^1 1^3}{\underset{1^3}{6}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.92 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 204 & 16 & 26 \\ 1 & 16 & 1 & 1 \\ 2 & 26 & 1 & -2 \end{pmatrix}_{b_0}$$

[475] $21^4 \overset{21^3}{6} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 19.92 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 388 & 21 & 36 \\ 1 & 21 & 1 & 1 \\ 2 & 36 & 1 & -5 \end{pmatrix}_{b_0}$$

[476] $21^3 \overset{21^3}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.92 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 204 & 16 & 26 \\ 1 & 16 & 1 & 1 \\ 2 & 26 & 1 & -2 \end{pmatrix}_{b_0}$$

[477] $21^4 \overset{21^3}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.77 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 236 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -3 \end{pmatrix}_{b_0}$$

[469] $2221^4 \overset{61^3}{2} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 19.92 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 388 & 21 & 36 \\ 1 & 21 & 1 & 1 \\ 2 & 36 & 1 & -5 \end{pmatrix}_{b_0}$$

[479] $21^2 \overset{21^3}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.92 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 204 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -2 \end{pmatrix}_{b_0}$$

[463] $221^2 \overset{21^3}{6} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.92 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 236 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -2 \end{pmatrix}_{b_0}$$

[468] $21^2 \overset{21^4}{6} 1^3 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.77 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 236 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -3 \end{pmatrix}_{b_0}$$

[464] $21^2 \overset{21^3}{6} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 17.77 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 272 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -3 \end{pmatrix}_{b_0}$$

[465] $221^4 \overset{21^3}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 18.77 \rightarrow 19, T_{\max} = 20, T \in [19, 20]$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 308 & 19 & 32 \\ 1 & 19 & 1 & 1 \\ 2 & 32 & 1 & -4 \end{pmatrix}_{b_0}$$

[466] $2221^2 \overset{61^3}{2} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.23 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 172 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -2 \end{pmatrix}_{b_0}$$

[467] $221^2 \overset{21^3}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.23 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 172 & 15 & 24 \\ 1 & 15 & 1 & 1 \\ 2 & 24 & 1 & -2 \end{pmatrix}_{b_0}$$

[481] $21^2 \overset{21^3}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.53 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 172 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[485] \ 21^4 \overset{21^3}{6} \overset{1^2}{1^2} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 53, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.19 \rightarrow 19, \ T_{\max} = 20, \ T \in [19, 20]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 270 & 18 & 30 \\ 1 & 18 & 1 & 1 \\ 2 & 30 & 1 & -4 \end{pmatrix}_{b_0}$$

$$[484] \ 221^2 \overset{21^3}{6} \overset{1^3}{1^3} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 53, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.53 \rightarrow 17, \ T_{\max} = 20, \ T \in [17, 20]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 202 & 16 & 26 \\ 1 & 16 & 1 & 1 \\ 2 & 26 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[483] \ 221^4 \overset{21^3}{6} \overset{1^3}{1^3} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 53, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 19.25 \rightarrow 20, \ T_{\max} = 20, \ T \in [20, 20]$$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 346 & 20 & 34 \\ 1 & 20 & 1 & 1 \\ 2 & 34 & 1 & -5 \end{pmatrix}_{b_0}$$

$$[482] \ 21^4 \overset{21^3}{6} \overset{1^3}{1^3} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 53, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 19.25 \rightarrow 20, \ T_{\max} = 20, \ T \in [20, 20]$$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 346 & 20 & 34 \\ 1 & 20 & 1 & 1 \\ 2 & 34 & 1 & -5 \end{pmatrix}_{b_0}$$

$$[480] \ 21^2 \overset{21^3}{6} \overset{1^3}{1^3} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 53, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.28 \rightarrow 18, \ T_{\max} = 20, \ T \in [18, 20]$$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 236 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[487] \ 21^3 \overset{21^3}{6} \overset{1^2}{1^2} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 54, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.67 \rightarrow 18, \ T_{\max} = 20, \ T \in [18, 20]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 270 & 18 & 30 \\ 1 & 18 & 1 & 1 \\ 2 & 30 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[488] \ 21^3 \overset{21^3}{6} \overset{1^4}{1^4} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 54, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.63 \rightarrow 19, \ T_{\max} = 20, \ T \in [19, 20]$$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 346 & 20 & 34 \\ 1 & 20 & 1 & 1 \\ 2 & 34 & 1 & -4 \end{pmatrix}_{b_0}$$

$$[489] \ 21^2 \overset{21^3}{6} \overset{1^3}{1^3} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 54, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.85 \rightarrow 17, \ T_{\max} = 20, \ T \in [17, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 202 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[490] \ 21^3 \overset{21^4}{6} \overset{1^3 2^1}{1^3 2^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 54, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.63 \rightarrow 19, \ T_{\max} = 20, \ T \in [19, 20]$$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 306 & 19 & 32 \\ 1 & 19 & 1 & 1 \\ 2 & 32 & 1 & -4 \end{pmatrix}_{b_0}$$

$$[491] \ 21^3 \overset{21^3}{6} \overset{1^2}{1^2} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 54, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.85 \rightarrow 17, \ T_{\max} = 20, \ T \in [17, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 202 & 16 & 26 \\ 1 & 16 & 1 & 1 \\ 2 & 26 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[492] \ 21^2 \overset{21^3}{6} \overset{1^3 2^1}{1^3 2^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 54, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.85 \rightarrow 17, \ T_{\max} = 20, \ T \in [17, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 202 & 16 & 26 \\ 1 & 16 & 1 & 1 \\ 2 & 26 & 1 & -2 \end{pmatrix}_{b_0}$$

- [486] $21^4 \overset{21^3}{6} 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 54$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 18.63 \rightarrow 19$, $T_{\max} = 20$, $T \in [19, 20]$
 $\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 306 & 19 & 32 \\ 1 & 19 & 1 & 1 \\ 2 & 32 & 1 & -4 \end{pmatrix}_{b_0}$
- [493] $221^3 \overset{21^3}{6} 1^4 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 55$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 18.07 \rightarrow 19$, $T_{\max} = 20$, $T \in [19, 20]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 306 & 19 & 32 \\ 1 & 19 & 1 & 1 \\ 2 & 32 & 1 & -4 \end{pmatrix}_{b_0}$
- [494] $21^3 \overset{21^3}{6} 1^2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 55$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.20 \rightarrow 18$, $T_{\max} = 20$, $T \in [18, 20]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 234 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -3 \end{pmatrix}_{b_0}$
- [495] $221^3 \overset{21^3}{6} 1^2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 55$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.20 \rightarrow 18$, $T_{\max} = 20$, $T \in [18, 20]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 234 & 17 & 28 \\ 1 & 17 & 1 & 1 \\ 2 & 28 & 1 & -3 \end{pmatrix}_{b_0}$
- [496] $21^3 \overset{21^3}{6} 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 55$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 18.07 \rightarrow 19$, $T_{\max} = 20$, $T \in [19, 20]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 306 & 19 & 32 \\ 1 & 19 & 1 & 1 \\ 2 & 32 & 1 & -4 \end{pmatrix}_{b_0}$
- [500] $2221^3 61^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.57 \rightarrow 18$, $T_{\max} = 20$, $T \in [18, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 268 & 18 & 30 \\ 1 & 18 & 1 & 1 \\ 2 & 30 & 1 & -3 \end{pmatrix}_{b_0}$
- [497] $21^3 \overset{21^3}{6} 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.57 \rightarrow 18$, $T_{\max} = 20$, $T \in [18, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 268 & 18 & 30 \\ 1 & 18 & 1 & 1 \\ 2 & 30 & 1 & -3 \end{pmatrix}_{b_0}$
- [498] $221^3 \overset{21^3}{6} 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.57 \rightarrow 18$, $T_{\max} = 20$, $T \in [18, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 268 & 18 & 30 \\ 1 & 18 & 1 & 1 \\ 2 & 30 & 1 & -3 \end{pmatrix}_{b_0}$
- [499] $21^3 \overset{21^3}{6} 1^3 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 17.57 \rightarrow 18$, $T_{\max} = 20$, $T \in [18, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 268 & 18 & 30 \\ 1 & 18 & 1 & 1 \\ 2 & 30 & 1 & -3 \end{pmatrix}_{b_0}$
- [501] $1^{\overset{21^4}{21^1}} 6 1^2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 60$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.27 \rightarrow 16$, $T_{\max} = 20$, $T \in [16, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 84 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -1 \end{pmatrix}_{b_0}$
- [504] $21^1 \overset{21^4}{6} 1^1 2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 62$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.26 \rightarrow 16$, $T_{\max} = 20$, $T \in [16, 20]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 82 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -1 \end{pmatrix}_{b_0}$
- [502] $21^1 \overset{21^4}{6} 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 62$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.26 \rightarrow 16$, $T_{\max} = 20$, $T \in [16, 20]$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 82 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 2 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

[505] $2221^1 61^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.26 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 154 & 15 & 22 \\ 1 & 15 & 1 & 1 \\ 2 & 22 & 1 & -1 \end{pmatrix}_{b_0}$$

[503] $221^1 6^1 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.26 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 154 & 15 & 22 \\ 1 & 15 & 1 & 1 \\ 2 & 22 & 1 & -1 \end{pmatrix}_{b_0}$$

[506] $1^1 6^1 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.40 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 106 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$$

[508] $1^1 6^1 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.56 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 132 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$$

[509] $21^1 6^1 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.56 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 244 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -1 \end{pmatrix}_{b_0}$$

[510] $1^1 6^1 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.56 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 132 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$$

[507] $21^3 6^1 12 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.56 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 244 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -1 \end{pmatrix}_{b_0}$$

[512] $1^1 6^1 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.38 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 104 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$$

[511] $21^1 6^1 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.75 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 280 & 19 & 30 \\ 1 & 19 & 1 & 1 \\ 2 & 30 & 1 & -1 \end{pmatrix}_{b_0}$$

[514] $1^1 6^1 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.97 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 190 & 16 & 24 \\ 1 & 16 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$$

[515] $21^1 6^1 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.97 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 318 & 20 & 32 \\ 1 & 20 & 1 & 1 \\ 2 & 32 & 1 & -1 \end{pmatrix}_{b_0}$$

- [513] $21^4 \overset{2^1 1^4}{6} 12 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.97 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 318 & 20 & 32 \\ 1 & 20 & 1 & 1 \\ 2 & 32 & 1 & -1 \end{pmatrix}_{b_0}$
- [516] $21^1 \overset{2^1 1^4}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.37 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 102 & 0 & 18 \\ 1 & 0 & -1 & -1 \\ 2 & 18 & -1 & -1 \end{pmatrix}_{b_0}$
- [517] $221^1 \overset{2^1 1^4}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.37 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 180 & 16 & 24 \\ 1 & 16 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$
- [518] $21^1 \overset{2^1 1^4}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.51 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 126 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$
- [519] $221^1 \overset{2^1 1^4}{6} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.51 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 210 & 17 & 26 \\ 1 & 17 & 1 & 1 \\ 2 & 26 & 1 & -1 \end{pmatrix}_{b_0}$
- [520] $1 \overset{221^2}{6} 1^4 \textcolor{red}{2} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.51 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 182 & 16 & 24 \\ 1 & 16 & 1 & 1 \\ 2 & 24 & 1 & -1 \end{pmatrix}_{b_0}$
- [521] $1 \overset{2^1 1^4}{6} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.51 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 126 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$
- [523] $1 \overset{2^1 1^4}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.69 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 154 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$
- [524] $21^1 \overset{2^1 1^4}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.69 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 154 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$
- [522] $221^1 \overset{2^1 1^4}{6} 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.69 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 244 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -1 \end{pmatrix}_{b_0}$
- [525] $21^1 \overset{2^1 1^4}{6} 1^2 \textcolor{red}{2} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.50 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 124 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$
- [526] $21^2 \overset{2^1 1^4}{6} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.50 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 124 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[527] \ 21 \overset{2^1 1^4}{1^2} 6 \ 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 72, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.50 \rightarrow 16, \ T_{\max} = 20, \ T \in [16, 20]$$

$$\begin{pmatrix} 0 & 14 & 1 & 2 \\ 14 & 124 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 2 & 20 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[528] \ 21 \overset{2^1 1^4}{1^{1^4}} 6 \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 73, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.11 \rightarrow 17, \ T_{\max} = 20, \ T \in [17, 20]$$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 216 & 17 & 26 \\ 1 & 17 & 1 & 1 \\ 2 & 26 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[529] \ 1 \overset{2^1 1^4}{21^2} 6 \ 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 73, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.67 \rightarrow 16, \ T_{\max} = 20, \ T \in [16, 20]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 152 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[530] \ 1 \overset{2^1 1^4}{21^2} 6 \ 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 74, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.86 \rightarrow 16, \ T_{\max} = 20, \ T \in [16, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 182 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[531] \ 21 \overset{2^1 1^4}{1^3} 6 \ 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.65 \rightarrow 16, \ T_{\max} = 20, \ T \in [16, 20]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 150 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[533] \ 21 \overset{2^1 1^4}{1^4} 6 \ 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 76, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.84 \rightarrow 16, \ T_{\max} = 20, \ T \in [16, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 180 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[532] \ 1 \overset{2^1 1^4}{21^3} 6 \ 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 76, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.84 \rightarrow 16, \ T_{\max} = 20, \ T \in [16, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 180 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[535] \ 21 \overset{2^1 1^4}{1^2} 6 \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 77, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.64 \rightarrow 16, \ T_{\max} = 20, \ T \in [16, 20]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 148 & 0 & 22 \\ 1 & 0 & -1 & -1 \\ 2 & 22 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[534] \ 221 \overset{2^1 1^4}{2^2} 6 \ 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 77, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.64 \rightarrow 16, \ T_{\max} = 20, \ T \in [16, 20]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 208 & 17 & 26 \\ 1 & 17 & 1 & 1 \\ 2 & 26 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[537] \ 21 \overset{2^1 1^4}{1^1} 6 \ 1^3 2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 78, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.82 \rightarrow 16, \ T_{\max} = 20, \ T \in [16, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 178 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[538] \ 21 \overset{2^1 1^4}{1^3} 6 \ 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 78, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.82 \rightarrow 16, \ T_{\max} = 20, \ T \in [16, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 178 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$$

- [539] $21 \overset{21^4}{\underset{1^1}{3}} 6 \overset{1^1}{1^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.82 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 178 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$
- [536] $1 \overset{21^4}{\underset{21^4}{6}} \overset{1^2}{1^2} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.28 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 246 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -2 \end{pmatrix}_{b_0}$
- [540] $21 \overset{21^4}{\underset{1^4}{1}} \overset{1^3}{6} \overset{1^3}{1^3} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.03 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 210 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -2 \end{pmatrix}_{b_0}$
- [541] $1 \overset{221^3}{\underset{1^4}{6}} \overset{1^4}{1^4} \overset{21^1}{21^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.03 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 244 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -2 \end{pmatrix}_{b_0}$
- [542] $1 \overset{21^4}{\underset{21^3}{6}} \overset{1^3}{1^3} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.03 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 210 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -2 \end{pmatrix}_{b_0}$
- [543] $21 \overset{21^4}{\underset{1^4}{1}} \overset{1^4}{6} \overset{1^4}{1^4} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.25 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 244 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$
- [544] $21 \overset{21^4}{\underset{1^1}{4}} \overset{1^1}{6} \overset{1^1}{1^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.25 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 244 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -2 \end{pmatrix}_{b_0}$
- [545] $1 \overset{21^4}{\underset{21^3}{6}} \overset{1^4}{1^4} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.25 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 244 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$
- [546] $21 \overset{21^4}{\underset{1^1}{1}} \overset{1^4}{6} \overset{1^4}{1^4} \overset{21^1}{21^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.25 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 244 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -2 \end{pmatrix}_{b_0}$
- [547] $21 \overset{21^4}{\underset{1^3}{2}} \overset{1^1}{6} \overset{1^1}{1^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 15.80 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$
 $\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 176 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$
- [548] $1 \overset{21^4}{\underset{21^4}{6}} \overset{1^3}{1^3} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 16.49 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 280 & 19 & 30 \\ 1 & 19 & 1 & 1 \\ 2 & 30 & 1 & -2 \end{pmatrix}_{b_0}$
- [549] $21 \overset{21^4}{\underset{1^4}{2}} \overset{1^1}{6} \overset{1^1}{1^1} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 208 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[554] \quad 2221^2 61^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.78 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 238 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[555] \quad 21^2 6^{2^1 1^4} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.78 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 174 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[553] \quad 221^4 1^6 1^4 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.76 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 318 & 20 & 32 \\ 1 & 20 & 1 & 1 \\ 2 & 32 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[551] \quad 221^2 6^{2^1 1^4} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.78 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 238 & 18 & 28 \\ 1 & 18 & 1 & 1 \\ 2 & 28 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[552] \quad 21^2 6^{2^1 1^4} 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.78 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 174 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 2 & 24 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[550] \quad 21^4 1^6 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.76 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 318 & 20 & 32 \\ 1 & 20 & 1 & 1 \\ 2 & 32 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[556] \quad 21^3 6^{2^1 1^4} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.98 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 206 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[559] \quad 221^2 6^{2^1 1^4} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.95 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 272 & 19 & 30 \\ 1 & 19 & 1 & 1 \\ 2 & 30 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[558] \quad 21^4 6^{2^1 1^4} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.42 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 276 & 19 & 30 \\ 1 & 19 & 1 & 1 \\ 2 & 30 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[557] \quad 21^2 6^{2^1 1^4} 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.95 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 204 & 0 & 26 \\ 1 & 0 & -1 & -1 \\ 2 & 26 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[560] \quad 21^2 6^{2^1 1^4} 1^4 1^2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.16 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 238 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$$

- [561] $21^3 \overset{2^1 1^4}{6} 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 86$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.16 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 238 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$
- [562] $221^2 \overset{2^1 1^4}{6} 1^4 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 86$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.16 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 310 & 20 & 32 \\ 1 & 20 & 1 & 1 \\ 2 & 32 & 1 & -2 \end{pmatrix}_{b_0}$
- [563] $221^3 \overset{2^1 1^4}{6} 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 86$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.16 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 274 & 19 & 30 \\ 1 & 19 & 1 & 1 \\ 2 & 30 & 1 & -2 \end{pmatrix}_{b_0}$
- [564] $21^3 \overset{2^1 1^4}{6} 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 87$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.39 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 274 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -2 \end{pmatrix}_{b_0}$
- [565] $21^2 \overset{2^1 1^4}{6} 1^3 2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 88$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.14 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 236 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$
- [566] $21^4 \overset{2^1 1^4}{6} 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 88$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.64 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$
 $\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 312 & 20 & 32 \\ 1 & 20 & 1 & 1 \\ 2 & 32 & 1 & -2 \end{pmatrix}_{b_0}$
- [567] $21^2 \overset{2^1 1^4}{6} 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 88$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.14 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 236 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$
- [568] $21^3 \overset{2^1 1^4}{6} 1^2 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 88$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.14 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$
 $\begin{pmatrix} 0 & 18 & 1 & 2 \\ 18 & 236 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 2 & 28 & -1 & -2 \end{pmatrix}_{b_0}$
- [569] $221^4 \overset{2^1 1^4}{6} 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 89$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.90 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$
 $\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 352 & 21 & 34 \\ 1 & 21 & 1 & 1 \\ 2 & 34 & 1 & -2 \end{pmatrix}_{b_0}$
- [570] $21^2 \overset{2^1 1^4}{6} 1^3 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 89$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.36 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$
 $\begin{pmatrix} 0 & 19 & 1 & 2 \\ 19 & 272 & 0 & 30 \\ 1 & 0 & -1 & -1 \\ 2 & 30 & -1 & -2 \end{pmatrix}_{b_0}$
- [571] $21^4 \overset{2^1 1^4}{6} 1^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 89$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.90 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$
 $\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 352 & 21 & 34 \\ 1 & 21 & 1 & 1 \\ 2 & 34 & 1 & -2 \end{pmatrix}_{b_0}$
- [572] $21^2 \overset{2^1 1^4}{6} 1^4 2^1 + \text{su}_2 + \hat{1} \rightarrow 2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 90$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 16.60 \rightarrow 17$, $T_{\max} = 20$, $T \in [17, 20]$

$$T_{\min} = 16.53 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 20 & 1 & 2 \\ 20 & 346 & 21 & 34 \\ 1 & 21 & 1 & 1 \\ 2 & 34 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[585] \ 221 \overset{2^1 1^4}{\underset{1^4}{3}} 6 \ 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.78 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 388 & 22 & 36 \\ 1 & 22 & 1 & 1 \\ 2 & 36 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[586] \ 21 \overset{2^1 1^4}{\underset{1^4}{3}} 6 \ 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.78 \rightarrow 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 21 & 1 & 2 \\ 21 & 346 & 0 & 34 \\ 1 & 0 & -1 & -1 \\ 2 & 34 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[587] \ 21 \overset{2^1 1^4}{\underset{1^3}{4}} 6 \ 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.04 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 388 & 22 & 36 \\ 1 & 22 & 1 & 1 \\ 2 & 36 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[589] \ 21 \overset{2^1 1^4}{\underset{1^4}{3}} 6 \ 1^{421} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.04 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 388 & 22 & 36 \\ 1 & 22 & 1 & 1 \\ 2 & 36 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[588] \ 21 \overset{2^1 1^4}{\underset{1^4}{3}} 6 \ 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.04 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 22 & 1 & 2 \\ 22 & 388 & 0 & 36 \\ 1 & 0 & -1 & -1 \\ 2 & 36 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[590] \ 21 \overset{2^1 1^4}{\underset{1^4}{4}} 6 \ 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.32 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 23 & 1 & 2 \\ 23 & 432 & 23 & 38 \\ 1 & 23 & 1 & 1 \\ 2 & 38 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[591] \ 221 \overset{2^1 1^4}{\underset{1^4}{4}} 6 \ 1^3 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.32 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 23 & 1 & 2 \\ 23 & 432 & 23 & 38 \\ 1 & 23 & 1 & 1 \\ 2 & 38 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[595] \ 2221 \overset{2^1 1^4}{\underset{1^4}{4}} 61 \overset{2^1 1^4}{\underset{1^4}{4}} 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.61 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 24 & 1 & 2 \\ 24 & 478 & 24 & 40 \\ 1 & 24 & 1 & 1 \\ 2 & 40 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[592] \ 21 \overset{2^1 1^4}{\underset{1^4}{4}} 6 \ 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.61 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 24 & 1 & 2 \\ 24 & 478 & 24 & 40 \\ 1 & 24 & 1 & 1 \\ 2 & 40 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[593] \ 21 \overset{2^1 1^4}{\underset{1^4}{4}} 6 \ 1^{421} + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.61 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 24 & 1 & 2 \\ 24 & 478 & 24 & 40 \\ 1 & 24 & 1 & 1 \\ 2 & 40 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[594] \ 221 \overset{2^1 1^4}{\underset{1^4}{4}} 6 \ 1^4 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.61 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 0 & 24 & 1 & 2 \\ 24 & 478 & 24 & 40 \\ 1 & 24 & 1 & 1 \\ 2 & 40 & 1 & -3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& T = 9 \quad [600] \quad 1 \overset{221}{\underset{2^1 1^1}{7}} 1 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 16, T_{\max} = 21, T \in [16, 21] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0} \\
& [596] \quad 1 \overset{2^1 1^1}{\underset{1^1}{7}} 1222 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 16, T_{\max} = 21, T \in [16, 21] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0} \\
& [597] \quad 1 \overset{21}{\underset{2^1 1^1}{7}} 12 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 16, T_{\max} = 21, T \in [16, 21] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0} \\
& [598] \quad 2222171 \overset{1}{2} \overset{1}{2} + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 16, T_{\max} = 21, T \in [16, 21] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0} \\
& [599] \quad 31 \overset{2^1 1^1}{\underset{1^1}{6}} 13 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 2 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
& [601] \quad 21 \overset{2^1 1^1}{\underset{1^1}{7}} 122 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 16, T_{\max} = 21, T \in [16, 21] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0} \\
& [602] \quad 21 \overset{2^1 1^1}{\underset{1^1}{7}} 12 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 16.20 \rightarrow 17, T_{\max} = 21, T \in [17, 21] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 4 & 6 \\ 1 & 4 & 1 & 1 \\ 2 & 6 & 1 & -1 \end{pmatrix}_{b_0} \\
& [603] \quad 2221 \overset{2^1 1^1}{\underset{1^1}{7}} 1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 16.20 \rightarrow 17, T_{\max} = 21, T \in [17, 21] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 4 & 6 \\ 1 & 4 & 1 & 1 \\ 2 & 6 & 1 & -1 \end{pmatrix}_{b_0} \\
& [604] \quad 1 \overset{2^1 1^1}{\underset{1^1}{7}} 122 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 16.20 \rightarrow 17, T_{\max} = 21, T \in [17, 21] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 4 & 6 \\ 1 & 4 & 1 & 1 \\ 2 & 6 & 1 & -1 \end{pmatrix}_{b_0} \\
& [605] \quad 1 \overset{2^1 1^1}{\underset{2^1 1^1}{7}} 12 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 16.67 \rightarrow 17, T_{\max} = 21, T \in [17, 21] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -1 \end{pmatrix}_{b_0} \\
& [606] \quad 21 \overset{2^1 1^1}{\underset{1^1}{7}} 122 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 16.67 \rightarrow 17, T_{\max} = 21, T \in [17, 21] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -1 \end{pmatrix}_{b_0} \\
& [607] \quad 221 \overset{2^1 1^1}{\underset{1^1}{7}} 1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 16.67 \rightarrow 17, T_{\max} = 21, T \in [17, 21]
\end{aligned}$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[608] \quad 1 \overset{221}{\underset{2^1 1^1}{7}} 1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.29 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[609] \quad 21 \overset{221}{\underset{2^1 1^1}{7}} 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.29 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 5 & 8 \\ 1 & 5 & 1 & 1 \\ 2 & 8 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[610] \quad 21 \overset{2^1 1^1}{\underset{1^1}{7}} 12 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.29 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 28 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[611] \quad 1 \overset{21^1}{\underset{21^1}{7}} 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[612] \quad 1 \overset{2221^1}{\underset{2^1 1^1}{7}} 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[613] \quad 1 \overset{221^1}{\underset{2^1 1^1}{7}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 6 & 10 \\ 1 & 6 & 1 & 1 \\ 2 & 10 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[614] \quad 21 \overset{221^1}{\underset{2^1 1^1}{7}} 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.78 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 40 & 7 & 12 \\ 1 & 7 & 1 & 1 \\ 2 & 12 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[615] \quad 221 \overset{2^1 1^1}{\underset{1^1}{7}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.78 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 40 & 7 & 12 \\ 1 & 7 & 1 & 1 \\ 2 & 12 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[616] \quad 22221 \overset{2^1 1^1}{\underset{2^1 1^1}{7}} 1^1 2^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.78 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 40 & 7 & 12 \\ 1 & 7 & 1 & 1 \\ 2 & 12 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[617] \quad 21 \overset{2^1 1^1}{\underset{2^1 1^1}{7}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.78 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 40 & 7 & 12 \\ 1 & 7 & 1 & 1 \\ 2 & 12 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[618] \quad 2221 \overset{2^1 1^1}{\underset{2^1 1^1}{7}} 1^1 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.78 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 40 & 7 & 12 \\ 1 & 7 & 1 & 1 \\ 2 & 12 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[619] \quad 1 \overset{2^1 1^1}{\underset{31}{6}} 13 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 2 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 8 & 5 & 6 \\ 2 & 5 & 2 & 2 \\ 2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[620] \quad 2^{11} 1^1 321 \quad 6 \quad 13 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 8 & 5 & 6 \\ 2 & 5 & 2 & 2 \\ 2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [621] \quad 2^{11} 1^1 2221 \quad 1 \quad 8 \quad 1 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 2 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[622] \quad 2^{11} 1^1 21 \quad 8 \quad 12 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[623] \quad 2^{11} 1^1 221 \quad 8 \quad 122 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[624] \quad 2^{11} 1^1 21 \quad 8 \quad 1222 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[625] \quad 2^{11} 1^1 1 \quad 8 \quad 12222 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[626] \quad 2^{11} 1^1 22222181 \quad 1^1 2^1 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[627] \quad 2^{11} 1^1 221 \quad 1 \quad 8 \quad 12 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 3 & 4 \\ 1 & 3 & 1 & 1 \\ 2 & 4 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[628] \quad 2^{11} 1^1 321 \quad 7 \quad 123 + \textcolor{red}{su}_2 + \textcolor{blue}{\hat{1} \rightarrow 2} \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 8 & 5 & 6 \\ 2 & 5 & 2 & 2 \\ 2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

1.18 $\textcolor{red}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{\hat{1} \rightarrow (-1)}$

$$T = 4 \quad [0] \quad 1^{11} 1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 + \textcolor{green}{\hat{1} \rightarrow 1} \quad T = 4, T_{\max} = 99, \Delta = -211, \det = 8, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11, T_{\max} = 99, T \in [11, 99]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & -2 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 3 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 + \textcolor{green}{\hat{1} \rightarrow 1} \quad T = 4, T_{\max} = 99, \Delta = -211, \det = 11, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -4 & 0 & -2 \\ 2 & 0 & 1 & 1 \\ 3 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [2] \quad 1^{11}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -182, \det = -2, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, T_{\max} = 98, T \in [12, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -182, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 98, T \in [12, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 2 & 0 & 1 & 1 \\ 2 & 2 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [4] \quad 1^{11}221^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 97, \Delta = -153, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 3 & 0 & 2 & 2 \\ 2 & 3 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [5] \quad 1^{11}2221^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 80, \Delta = -124, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16, T_{\max} = 80, T \in [16, 80]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & -2 & 3 & -3 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [6] \quad 1^1 1^1 3221^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 17, T_{\max} = 20, T \in [17, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 4 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & -3 \end{pmatrix}_{b_0}$$

1.19 $su_2 + so_8 + so_8 + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \quad 1^{11}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 + \textcolor{green}{so}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 99, \Delta = -210, \det = -40, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 \\ 1 & -4 & 0 & 0 & -2 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & -2 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \quad 1^{11}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 + \textcolor{green}{so}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 99, \Delta = -181, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 2 & 2 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 2 & 0 & 0 & 1 & 1 \\ 2 & 2 & -2 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

1.20 $su_2 + su_2 + e_6 + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \quad 1^{11}1^{21} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{e}_6 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 100, \Delta = -231, \det = -10, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & -2 & 0 & 0 & 0 \\ 1 & 0 & -6 & 0 & -4 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & 0 & -4 & 1 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{e}_6 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 100, \Delta = -231, \det = -24, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & -4 & 0 & -2 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{11}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{e}_6 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 100$, $\Delta = -231$, $\det = -28$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 12.71 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$

$$\begin{pmatrix} 1 & 1 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 \\ 2 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 0 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

[3] $1^{11}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{e}_6 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 100$, $\Delta = -231$, $\det = -52$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 12.69 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 0 & 2 & 0 & 2 \\ 2 & 2 & -4 & 0 & -2 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$T = 6$ [4] $1^{11}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{e}_6 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 100$, $\Delta = -202$, $\det = 4$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 14$, $T_{\max} = 100$, $T \in [14, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 0 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -2 & -1 & -3 \end{pmatrix}_{b_0}$$

[5] $1^{11}1^{11}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{e}_6 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 100$, $\Delta = -202$, $\det = 16$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13$, $T_{\max} = 100$, $T \in [13, 100]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 0 & 2 \\ 1 & 2 & 2 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

1.21 $su_2 + su_2 + e_7 + \hat{1} \rightarrow (-1)$

$T = 5$ [0] $1^{11}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{e}_7 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 186$, $\Delta = -286$, $\det = -38$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.53 \rightarrow 15$, $T_{\max} = 186$, $T \in [15, 186]$

$$\begin{pmatrix} 1 & 1 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 \\ 2 & 0 & -6 & 0 & -4 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 0 & -4 & -1 & -4 \end{pmatrix}_{b_0}$$

$T = 6$ [1] $1^{11}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{e}_7 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 169$, $\Delta = -257$, $\det = 8$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15$, $T_{\max} = 169$, $T \in [15, 169]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 0 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -6 & 0 & -4 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -4 & -1 & -4 \end{pmatrix}_{b_0}$$

1.22 $su_2 + su_2 + e'_7 + \hat{1} \rightarrow (-1)$

$T = 5$ [0] $1^{11}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{e}'_7 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 169$, $\Delta = -258$, $\det = -33$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.61 \rightarrow 14$, $T_{\max} = 169$, $T \in [14, 169]$

$$\begin{pmatrix} 1 & 1 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 \\ 2 & 0 & -5 & 0 & -3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 0 & -3 & -1 & -3 \end{pmatrix}_{b_0}$$

$T = 6$ [1] $1^{11}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{e}'_7 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 101$, $\Delta = -229$, $\det = 6$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.33 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 0 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -5 & 0 & -3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -3 & -1 & -4 \end{pmatrix}_{b_0}$$

1.23 $su_2 + su_2 + f_4 + \hat{1} \rightarrow (-1)$

$T = 5$ [0] $1^{11}1^{21} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 99$, $\Delta = -205$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.7$
 $T_{\min} = 12$, $T_{\max} = 99$, $T \in [12, 99]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & -2 & 0 & 0 & 0 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & 0 & -3 & 1 & -1 \end{pmatrix}_{b_0}$$

- [1] $1^{11}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 99$, $\Delta = -205$, $\det = -20$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 99$, $T \in [12, 99]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$
- [3] $1^{13}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 99$, $\Delta = -205$, $\det = -20$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 99$, $T \in [12, 99]$
 $\begin{pmatrix} 1 & 6 & 2 & 1 & 3 \\ 6 & 16 & 6 & 0 & 6 \\ 2 & 6 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & -1 & -1 & -1 \end{pmatrix}_{b_0}$
- [2] $1^{12}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 99$, $\Delta = -205$, $\det = -20$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.7$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 99$, $T \in [12, 99]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 6 & 4 & 0 & 4 \\ 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & -1 & -1 & -1 \end{pmatrix}_{b_0}$
- [4] $1^{11}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 99$, $\Delta = -205$, $\det = -23$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 99$, $T \in [12, 99]$
 $\begin{pmatrix} 1 & 1 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 \\ 2 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 0 & -1 & -1 & -1 \end{pmatrix}_{b_0}$
- [5] $1^{11}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 99$, $\Delta = -205$, $\det = -43$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 99$, $T \in [12, 99]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 0 & 2 & 0 & 2 \\ 2 & 2 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$
- [6] $1^{12}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 99$, $\Delta = -205$, $\det = -63$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.7$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 99$, $T \in [12, 99]$
 $\begin{pmatrix} 1 & 5 & 2 & 1 & 3 \\ 5 & 6 & 4 & 0 & 4 \\ 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & -1 & -1 & -1 \end{pmatrix}_{b_0}$
- [7] $1^{13}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 99$, $\Delta = -205$, $\det = -83$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 99$, $T \in [12, 99]$
 $\begin{pmatrix} 1 & 7 & 2 & 1 & 3 \\ 7 & 16 & 6 & 0 & 6 \\ 2 & 6 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & -1 & -1 & -1 \end{pmatrix}_{b_0}$
- $T = 6$ [8] $1^{12}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 99$, $\Delta = -176$, $\det = 2$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.7$
 $T_{\min} = 15$, $T_{\max} = 99$, $T \in [15, 99]$
 $\begin{pmatrix} 0 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 4 & 2 & 0 & 2 \\ 3 & 4 & 6 & 4 & 0 & 4 \\ 1 & 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 4 & -1 & -1 & -4 \end{pmatrix}_{b_0}$
- [9] $1^{11}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 99$, $\Delta = -176$, $\det = 2$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 15$, $T_{\max} = 99$, $T \in [15, 99]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 0 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -1 & -1 & -4 \end{pmatrix}_{b_0}$
- [10] $1^{11}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 99$, $\Delta = -176$, $\det = 12$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.33 \rightarrow 13$, $T_{\max} = 99$, $T \in [13, 99]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 0 & 2 \\ 1 & 2 & 2 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 2 & -1 & -1 & -2 \end{pmatrix}_{b_0}$

1.24 $\textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow (-1)$

- $T = 4$ [0] $1^{11}1^2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1$ $T = 4$, $T_{\max} = 97$, $\Delta = -182$, $\det = 2$, $(n_+, n_-, n_0) = (1, 4, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 11$, $T_{\max} = 97$, $T \in [11, 97]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & -2 & 0 & 0 \\ 2 & 0 & 1 & 1 \\ 3 & 0 & 1 & 0 \end{pmatrix}_{b_0}$

$$[1] \ 1^{15}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 4, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 10.1$$

$$T_{\min} = 10, T_{\max} = 97, T \in [10, 97]$$

$$\begin{pmatrix} 1 & 10 & 1 & 3 \\ 10 & 48 & 0 & 10 \\ 1 & 0 & -1 & -1 \\ 3 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{14}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 4, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10, T_{\max} = 97, T \in [10, 97]$$

$$\begin{pmatrix} 1 & 8 & 1 & 3 \\ 8 & 30 & 0 & 8 \\ 1 & 0 & -1 & -1 \\ 3 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{17}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 4, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10, T_{\max} = 97, T \in [10, 97]$$

$$\begin{pmatrix} 1 & 14 & 1 & 3 \\ 14 & 96 & 0 & 14 \\ 1 & 0 & -1 & -1 \\ 3 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 4, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 97, T \in [10, 97]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 0 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 3 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{13}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 4, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10, T_{\max} = 97, T \in [10, 97]$$

$$\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 16 & 0 & 6 \\ 1 & 0 & -1 & -1 \\ 3 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{12}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 4, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10, T_{\max} = 97, T \in [10, 97]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 6 & 0 & 4 \\ 1 & 0 & -1 & -1 \\ 3 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{16}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 4, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 10, T_{\max} = 97, T \in [10, 97]$$

$$\begin{pmatrix} 1 & 12 & 1 & 3 \\ 12 & 70 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 3 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 5, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.40 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -2 & 0 & 0 \\ 2 & 0 & 1 & 1 \\ 3 & 0 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 9, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.22 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 0 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 3 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{12}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 13, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.15 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$$

$$\begin{pmatrix} 1 & 5 & 1 & 3 \\ 5 & 6 & 0 & 4 \\ 1 & 0 & -1 & -1 \\ 3 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 15, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 97, T \in [10, 97]$$

$$\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -2 & 0 & 0 \\ 3 & 0 & 1 & 1 \\ 3 & 0 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{13}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 17, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.12 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$$

$$\begin{pmatrix} 1 & 7 & 1 & 3 \\ 7 & 16 & 0 & 6 \\ 1 & 0 & -1 & -1 \\ 3 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{14}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 97, \Delta = -182, \det = 21, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$$

$$\begin{pmatrix} 1 & 9 & 1 & 3 \\ 9 & 30 & 0 & 8 \\ 1 & 0 & -1 & -1 \\ 3 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [15] \ 1^{12}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -153, \det = -2, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 14, T_{\max} = 96, T \in [14, 96]$$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 4 & 0 & 2 \\ 3 & 4 & 6 & 0 & 4 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & 4 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11}21^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -153, \det = -2, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 14, T_{\max} = 96, T \in [14, 96]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 1 \\ 2 & 2 & 0 & 1 & -3 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{11}21^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -153, \det = -4, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 96, T \in [12, 96]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{11}21^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -153, \det = -10, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 2 & 0 & 1 & 1 \\ 2 & 2 & 0 & 1 & 1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [18] \ 1^{11}221^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 79, \Delta = -124, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 79, T \in [10, 79]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

1.25 $\mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow (-1) + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \ 1^{11}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -180, \det = -3, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 1 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 \\ 2 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 0 & 1 & -1 & 0 \end{pmatrix}_{b_0}$$

1.26 $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \ 1^{11}1^{21} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{so}}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -181, \det = -6, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 98, T \in [12, 98]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & -2 & 0 & 0 & 0 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & 0 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{15}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{so}}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -181, \det = -16, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.1$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 10 & 2 & 1 & 3 \\ 10 & 48 & 10 & 0 & 10 \\ 2 & 10 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 10 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{16}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{so}}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -181, \det = -16, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 12 & 2 & 1 & 3 \\ 12 & 70 & 12 & 0 & 12 \\ 2 & 12 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 12 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{17}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{so}}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -181, \det = -16, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.3$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 14 & 2 & 1 & 3 \\ 14 & 96 & 14 & 0 & 14 \\ 2 & 14 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 14 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{so}}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -181, \det = -16, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{13}1^{11} + \text{su}_2 + \text{su}_2 + \text{so}_8 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 98$, $\Delta = -181$, $\det = -16$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 11$, $T_{\max} = 98$, $T \in [11, 98]$

$$\begin{pmatrix} 1 & 6 & 2 & 1 & 3 \\ 6 & 16 & 6 & 0 & 6 \\ 2 & 6 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{12}1^{11} + \text{su}_2 + \text{su}_2 + \text{so}_8 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 98$, $\Delta = -181$, $\det = -16$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 11$, $T_{\max} = 98$, $T \in [11, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 6 & 4 & 0 & 4 \\ 2 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{14}1^{11} + \text{su}_2 + \text{su}_2 + \text{so}_8 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 98$, $\Delta = -181$, $\det = -16$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 11$, $T_{\max} = 98$, $T \in [11, 98]$

$$\begin{pmatrix} 1 & 8 & 2 & 1 & 3 \\ 8 & 30 & 8 & 0 & 8 \\ 2 & 8 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{11}1^{11} + \text{su}_2 + \text{su}_2 + \text{so}_8 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 98$, $\Delta = -181$, $\det = -18$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 11.11 \rightarrow 12$, $T_{\max} = 98$, $T \in [12, 98]$

$$\begin{pmatrix} 1 & 1 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 \\ 2 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 0 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{11}1^{11} + \text{su}_2 + \text{su}_2 + \text{so}_8 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 98$, $\Delta = -181$, $\det = -34$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 98$, $T \in [12, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 0 & 2 & 0 & 2 \\ 2 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{12}1^{11} + \text{su}_2 + \text{su}_2 + \text{so}_8 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 98$, $\Delta = -181$, $\det = -50$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 11.04 \rightarrow 12$, $T_{\max} = 98$, $T \in [12, 98]$

$$\begin{pmatrix} 1 & 5 & 2 & 1 & 3 \\ 5 & 6 & 4 & 0 & 4 \\ 2 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{13}1^{11} + \text{su}_2 + \text{su}_2 + \text{so}_8 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 98$, $\Delta = -181$, $\det = -66$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 11.03 \rightarrow 12$, $T_{\max} = 98$, $T \in [12, 98]$

$$\begin{pmatrix} 1 & 7 & 2 & 1 & 3 \\ 7 & 16 & 6 & 0 & 6 \\ 2 & 6 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{14}1^{11} + \text{su}_2 + \text{su}_2 + \text{so}_8 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 98$, $\Delta = -181$, $\det = -82$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 11.02 \rightarrow 12$, $T_{\max} = 98$, $T \in [12, 98]$

$$\begin{pmatrix} 1 & 9 & 2 & 1 & 3 \\ 9 & 30 & 8 & 0 & 8 \\ 2 & 8 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$T = 6$ [13] $1^{11}1^{21}1^{11} + \text{su}_2 + \text{su}_2 + \text{so}_8 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 97$, $\Delta = -152$, $\det = 8$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12$, $T_{\max} = 97$, $T \in [12, 97]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 \\ 2 & 2 & 0 & 2 & 0 \\ 1 & 2 & 2 & -2 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

1.27 $\text{su}_2 + \text{su}_2 + \text{su}_2 + \text{e}_6 + \hat{1} \rightarrow (-1)$

$T = 6$ [0] $1^{11}1^{11} + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{e}_6 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 100$, $\Delta = -201$, $\det = 44$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 0 & 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

1.28 $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$

$T = 6$ [0] $1^{121}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 99$, $\Delta = -175$, $\det = 6$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.7$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 99, T \in [13, 99]$$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 4 & 0 & 6 & 4 & 0 & 4 \\ 2 & 0 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 0 & 4 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{111}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 99$, $\Delta = -175$, $\det = 36$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 0 & 2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

1.29 $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow (-1)$

$T = 5$ [0] $1^{134}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -2$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.8$

$$T_{\min} = 12, T_{\max} = 96, T \in [12, 96]$$

$$\begin{pmatrix} 1 & 7 & 8 & 1 & 3 \\ 7 & 16 & 24 & 0 & 6 \\ 8 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{121}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -2$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 10.5$

$$T_{\min} = 12, T_{\max} = 96, T \in [12, 96]$$

$$\begin{pmatrix} 1 & 1 & 4 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 \\ 4 & 0 & 6 & 0 & 4 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 0 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{122}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$$

$$\begin{pmatrix} 1 & 4 & 4 & 1 & 3 \\ 4 & 6 & 8 & 0 & 4 \\ 4 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{123}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.3$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$$

$$\begin{pmatrix} 1 & 5 & 6 & 1 & 3 \\ 5 & 6 & 12 & 0 & 4 \\ 6 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{126}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.8$

$$T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$$

$$\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 6 & 24 & 0 & 4 \\ 12 & 24 & 70 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{166}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 12.5$

$$T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$$

$$\begin{pmatrix} 1 & 12 & 12 & 1 & 3 \\ 12 & 70 & 72 & 0 & 12 \\ 12 & 72 & 70 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 12 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{133}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.6$

$$T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$$

$$\begin{pmatrix} 1 & 6 & 6 & 1 & 3 \\ 6 & 16 & 18 & 0 & 6 \\ 6 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{147}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 12.3$

$$T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$$

$$\begin{pmatrix} 1 & 8 & 14 & 1 & 3 \\ 8 & 30 & 56 & 0 & 8 \\ 14 & 56 & 96 & 0 & 14 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{145}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 12.1$

$$T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$$

- [27] $\begin{pmatrix} 1 & 8 & 10 & 1 & 3 \\ 8 & 30 & 40 & 0 & 8 \\ 10 & 40 & 48 & 0 & 10 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 10 & -1 & 1 \end{pmatrix}_{b_0}$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 14 & 14 & 1 & 3 \\ 14 & 96 & 98 & 0 & 14 \\ 14 & 98 & 96 & 0 & 14 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 14 & 14 & -1 & 1 \end{pmatrix}_{b_0}$
- [26] $1^{123}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 6 & 12 & 0 & 4 \\ 6 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & -1 & 1 \end{pmatrix}_{b_0}$
- [25] $1^{144}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 8 & 8 & 1 & 3 \\ 8 & 30 & 32 & 0 & 8 \\ 8 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$
- [2] $1^{177}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 2 & 14 & 1 & 3 \\ 2 & 0 & 14 & 0 & 2 \\ 14 & 14 & 96 & 0 & 14 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 14 & -1 & 1 \end{pmatrix}_{b_0}$
- [10] $1^{146}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 8 & 12 & 1 & 3 \\ 8 & 30 & 48 & 0 & 8 \\ 12 & 48 & 70 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$
- [9] $1^{111}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & -1 & 1 \end{pmatrix}_{b_0}$
- [8] $1^{166}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 2 & 12 & 1 & 3 \\ 2 & 0 & 12 & 0 & 2 \\ 12 & 12 & 70 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 12 & -1 & 1 \end{pmatrix}_{b_0}$
- [3] $1^{167}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 12 & 14 & 1 & 3 \\ 12 & 70 & 84 & 0 & 12 \\ 14 & 84 & 96 & 0 & 14 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 12 & 14 & -1 & 1 \end{pmatrix}_{b_0}$
- [4] $1^{155}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 10 & 10 & 1 & 3 \\ 10 & 48 & 50 & 0 & 10 \\ 10 & 50 & 48 & 0 & 10 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 10 & 10 & -1 & 1 \end{pmatrix}_{b_0}$
- [24] $1^{156}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 10 & 12 & 1 & 3 \\ 10 & 48 & 60 & 0 & 10 \\ 12 & 60 & 70 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 10 & 12 & -1 & 1 \end{pmatrix}_{b_0}$
- [23] $1^{135}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.9$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$
 $\begin{pmatrix} 1 & 6 & 10 & 1 & 3 \\ 6 & 16 & 30 & 0 & 6 \\ 10 & 30 & 48 & 0 & 10 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 10 & -1 & 1 \end{pmatrix}_{b_0}$
- [22] $1^{157}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5, T_{\max} = 96, \Delta = -152, \det = -8, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 10, T_{\max} = 96, T \in [10, 96]$

- [11] $1^{11}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.50 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$

$$\begin{pmatrix} 1 & 10 & 14 & 1 & 3 \\ 10 & 48 & 70 & 0 & 10 \\ 14 & 70 & 96 & 0 & 14 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 10 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{11}31^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 10$, $T_{\max} = 96$, $T \in [10, 96]$

$$\begin{pmatrix} 1 & 2 & 6 & 1 & 3 \\ 2 & 0 & 6 & 0 & 2 \\ 6 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 0 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$
- [5] $1^{12}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 10$, $T_{\max} = 96$, $T \in [10, 96]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 0 & 4 & 0 & 2 \\ 4 & 4 & 6 & 0 & 4 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$
- [13] $1^{12}71^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 10$, $T_{\max} = 96$, $T \in [10, 96]$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 3 \\ 4 & 6 & 28 & 0 & 4 \\ 14 & 28 & 96 & 0 & 14 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [14] $1^{13}41^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 10$, $T_{\max} = 96$, $T \in [10, 96]$

$$\begin{pmatrix} 1 & 6 & 8 & 1 & 3 \\ 6 & 16 & 24 & 0 & 6 \\ 8 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [15] $1^{12}51^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.6$
 $T_{\min} = 10$, $T_{\max} = 96$, $T \in [10, 96]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 3 \\ 4 & 6 & 20 & 0 & 4 \\ 10 & 20 & 48 & 0 & 10 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [16] $1^{11}41^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 10$, $T_{\max} = 96$, $T \in [10, 96]$

$$\begin{pmatrix} 1 & 2 & 8 & 1 & 3 \\ 2 & 0 & 8 & 0 & 2 \\ 8 & 8 & 30 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [17] $1^{13}71^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 12.1$
 $T_{\min} = 10$, $T_{\max} = 96$, $T \in [10, 96]$

$$\begin{pmatrix} 1 & 6 & 14 & 1 & 3 \\ 6 & 16 & 42 & 0 & 6 \\ 14 & 42 & 96 & 0 & 14 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [18] $1^{12}41^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 10$, $T_{\max} = 96$, $T \in [10, 96]$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 3 \\ 4 & 6 & 16 & 0 & 4 \\ 8 & 16 & 30 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [19] $1^{13}61^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 12.1$
 $T_{\min} = 10$, $T_{\max} = 96$, $T \in [10, 96]$

$$\begin{pmatrix} 1 & 6 & 12 & 1 & 3 \\ 6 & 16 & 36 & 0 & 6 \\ 12 & 36 & 70 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{11}51^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -8$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.1$
 $T_{\min} = 10$, $T_{\max} = 96$, $T \in [10, 96]$

$$\begin{pmatrix} 1 & 2 & 10 & 1 & 3 \\ 2 & 0 & 10 & 0 & 2 \\ 10 & 10 & 48 & 0 & 10 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [32] $1^{11}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 96$, $\Delta = -152$, $\det = -10$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 10.40 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 0 & 4 & 0 & 2 \\ 4 & 4 & 6 & 0 & 4 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{14}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -152, \det = -10, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

$$\begin{pmatrix} 1 & 8 & 9 & 1 & 3 \\ 8 & 30 & 32 & 0 & 8 \\ 9 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{13}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -152, \det = -16, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.6$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

$$\begin{pmatrix} 1 & 6 & 7 & 1 & 3 \\ 6 & 16 & 18 & 0 & 6 \\ 7 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{11}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -152, \det = -16, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 0 & 2 & 0 & 2 \\ 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{12}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -152, \det = -18, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 6 & 8 & 0 & 4 \\ 5 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{11}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -152, \det = -24, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

$$\begin{pmatrix} 1 & 2 & 5 & 1 & 3 \\ 2 & 0 & 4 & 0 & 2 \\ 5 & 4 & 6 & 0 & 4 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{13}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -152, \det = -24, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

$$\begin{pmatrix} 1 & 6 & 9 & 1 & 3 \\ 6 & 16 & 24 & 0 & 6 \\ 9 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{12}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -152, \det = -26, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

$$\begin{pmatrix} 1 & 4 & 7 & 1 & 3 \\ 4 & 6 & 12 & 0 & 4 \\ 7 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{11}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -152, \det = -32, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

$$\begin{pmatrix} 1 & 2 & 7 & 1 & 3 \\ 2 & 0 & 6 & 0 & 2 \\ 7 & 6 & 16 & 0 & 6 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{12}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -152, \det = -34, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

$$\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 6 & 16 & 0 & 4 \\ 9 & 16 & 30 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{11}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 96, \Delta = -152, \det = -40, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

$$\begin{pmatrix} 1 & 2 & 9 & 1 & 3 \\ 2 & 0 & 8 & 0 & 2 \\ 9 & 8 & 30 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

1.30 $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$

$T = 6 \quad [0] 1^{12}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 97, \Delta = -151, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 2 & 7 & 2 & 1 & 3 \\ 2 & 0 & 6 & 2 & 0 & 2 \\ 7 & 6 & 16 & 6 & 0 & 6 \\ 2 & 2 & 6 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{1141}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 97, \Delta = -151, \det = 156, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 11.03 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 2 & 9 & 2 & 1 & 3 \\ 2 & 0 & 8 & 2 & 0 & 2 \\ 9 & 8 & 30 & 8 & 0 & 8 \\ 2 & 2 & 8 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

1.31 $su_2 + su_2 + su_2 + su_2 + \hat{1} \rightarrow (-1)$

$T = 6 \quad [0] \quad 1^{1244}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 70, \Delta = -122, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 12, T_{\max} = 70, T \in [12, 70]$

$$\begin{pmatrix} 1 & 4 & 8 & 9 & 1 & 3 \\ 4 & 6 & 16 & 16 & 0 & 4 \\ 8 & 16 & 30 & 32 & 0 & 8 \\ 9 & 16 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 8 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{1122}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 70, \Delta = -122, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12, T_{\max} = 70, T \in [12, 70]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 1 & 3 \\ 3 & 0 & 4 & 4 & 0 & 2 \\ 4 & 4 & 6 & 8 & 0 & 4 \\ 4 & 4 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{111}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 70, \Delta = -122, \det = 12, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 70, T \in [11, 70]$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 0 & 2 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{1334}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 70, \Delta = -122, \det = 12, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 70, T \in [11, 70]$

$$\begin{pmatrix} 1 & 6 & 6 & 9 & 1 & 3 \\ 6 & 16 & 18 & 24 & 0 & 6 \\ 6 & 18 & 16 & 24 & 0 & 6 \\ 9 & 24 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 6 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{1123}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 70, \Delta = -122, \det = 12, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 70, T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 5 & 6 & 1 & 3 \\ 2 & 0 & 4 & 6 & 0 & 2 \\ 5 & 4 & 6 & 12 & 0 & 4 \\ 6 & 6 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

[84] $1^{1224}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 4 & 8 & 1 & 3 \\ 4 & 6 & 8 & 16 & 0 & 4 \\ 4 & 8 & 6 & 16 & 0 & 4 \\ 8 & 16 & 16 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{1223}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 1 & 3 \\ 4 & 6 & 8 & 12 & 0 & 4 \\ 4 & 8 & 6 & 12 & 0 & 4 \\ 6 & 12 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{1122}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 4 & 4 & 1 & 3 \\ 2 & 0 & 4 & 4 & 0 & 2 \\ 4 & 4 & 6 & 8 & 0 & 4 \\ 4 & 4 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{1123}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

- [89] $1^{1235}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.4$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 4 & 6 & 1 & 3 \\ 2 & 0 & 4 & 6 & 0 & 2 \\ 4 & 4 & 6 & 12 & 0 & 4 \\ 6 & 6 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$
- [44] $1^{1147}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 8 & 14 & 1 & 3 \\ 2 & 0 & 8 & 14 & 0 & 2 \\ 8 & 8 & 30 & 56 & 0 & 8 \\ 14 & 14 & 56 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [43] $1^{1116}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 1 & 3 \\ 2 & 0 & 2 & 12 & 0 & 2 \\ 2 & 2 & 0 & 12 & 0 & 2 \\ 12 & 12 & 12 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [42] $1^{1556}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 10 & 10 & 12 & 1 & 3 \\ 10 & 48 & 50 & 60 & 0 & 10 \\ 10 & 50 & 48 & 60 & 0 & 10 \\ 12 & 60 & 60 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 10 & 10 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [41] $1^{1577}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 10 & 14 & 14 & 1 & 3 \\ 10 & 48 & 70 & 70 & 0 & 10 \\ 14 & 70 & 96 & 98 & 0 & 14 \\ 14 & 70 & 98 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 10 & 14 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [39] $1^{1155}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 10 & 10 & 1 & 3 \\ 2 & 0 & 10 & 10 & 0 & 2 \\ 10 & 10 & 48 & 50 & 0 & 10 \\ 10 & 10 & 50 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 10 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [48] $1^{1225}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 4 & 10 & 1 & 3 \\ 4 & 6 & 8 & 20 & 0 & 4 \\ 4 & 8 & 6 & 20 & 0 & 4 \\ 10 & 20 & 20 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [49] $1^{1112}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 70, T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 1 & 3 \\ 2 & 0 & 2 & 4 & 0 & 2 \\ 3 & 2 & 0 & 4 & 0 & 2 \\ 4 & 4 & 4 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$
- [50] $1^{1677}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.9$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 12 & 14 & 14 & 1 & 3 \\ 12 & 70 & 84 & 84 & 0 & 12 \\ 14 & 84 & 96 & 98 & 0 & 14 \\ 14 & 84 & 98 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 12 & 14 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [51] $1^{1166}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 12 & 12 & 1 & 3 \\ 2 & 0 & 12 & 12 & 0 & 2 \\ 12 & 12 & 70 & 72 & 0 & 12 \\ 12 & 12 & 72 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 12 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [38] $1^{1222}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

- [13] $1^{1666}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 4 & 4 & 1 & 3 \\ 4 & 6 & 8 & 8 & 0 & 4 \\ 4 & 8 & 6 & 8 & 0 & 4 \\ 4 & 8 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$
- [86] $1^{1145}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 12 & 12 & 12 & 1 & 3 \\ 12 & 70 & 72 & 72 & 0 & 12 \\ 12 & 72 & 70 & 72 & 0 & 12 \\ 12 & 72 & 72 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 12 & 12 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [85] $1^{1555}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.4$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 8 & 10 & 1 & 3 \\ 2 & 0 & 8 & 10 & 0 & 2 \\ 8 & 8 & 30 & 40 & 0 & 8 \\ 10 & 10 & 40 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [15] $1^{1356}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 10 & 10 & 10 & 1 & 3 \\ 10 & 48 & 50 & 50 & 0 & 10 \\ 10 & 50 & 48 & 50 & 0 & 10 \\ 10 & 50 & 50 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 10 & 10 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{1236}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 6 & 12 & 1 & 3 \\ 4 & 6 & 12 & 24 & 0 & 4 \\ 6 & 12 & 16 & 36 & 0 & 6 \\ 12 & 24 & 36 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [16] $1^{1567}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.7$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 10 & 12 & 14 & 1 & 3 \\ 10 & 48 & 60 & 70 & 0 & 10 \\ 12 & 60 & 70 & 84 & 0 & 12 \\ 14 & 70 & 84 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 10 & 12 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [17] $1^{1136}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 6 & 12 & 1 & 3 \\ 2 & 0 & 6 & 12 & 0 & 2 \\ 6 & 6 & 16 & 36 & 0 & 6 \\ 12 & 12 & 36 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [18] $1^{1144}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 8 & 8 & 1 & 3 \\ 2 & 0 & 8 & 8 & 0 & 2 \\ 8 & 8 & 30 & 32 & 0 & 8 \\ 8 & 8 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [19] $1^{1177}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 1 & 3 \\ 2 & 0 & 2 & 14 & 0 & 2 \\ 2 & 2 & 0 & 14 & 0 & 2 \\ 14 & 14 & 14 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [20] $1^{1346}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.1$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 8 & 12 & 1 & 3 \\ 6 & 16 & 24 & 36 & 0 & 6 \\ 8 & 24 & 30 & 48 & 0 & 8 \\ 12 & 36 & 48 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [21] $1^{1367}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.4$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

- [22] $1^{1167}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 12 & 14 & 1 & 3 \\ 6 & 16 & 36 & 42 & 0 & 6 \\ 12 & 36 & 70 & 84 & 0 & 12 \\ 14 & 42 & 84 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 12 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [23] $1^{1115}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.1$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 1 & 3 \\ 2 & 0 & 2 & 10 & 0 & 2 \\ 2 & 2 & 0 & 10 & 0 & 2 \\ 10 & 10 & 10 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [24] $1^{1233}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 6 & 6 & 1 & 3 \\ 4 & 6 & 12 & 12 & 0 & 4 \\ 6 & 12 & 16 & 18 & 0 & 6 \\ 6 & 12 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$
- [25] $1^{1446}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 8 & 8 & 12 & 1 & 3 \\ 8 & 30 & 32 & 48 & 0 & 8 \\ 8 & 32 & 30 & 48 & 0 & 8 \\ 12 & 48 & 48 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 8 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [26] $1^{1125}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 4 & 10 & 1 & 3 \\ 2 & 0 & 4 & 10 & 0 & 2 \\ 4 & 4 & 6 & 20 & 0 & 4 \\ 10 & 10 & 20 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [27] $1^{1366}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.3$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 12 & 12 & 1 & 3 \\ 6 & 16 & 36 & 36 & 0 & 6 \\ 12 & 36 & 70 & 72 & 0 & 12 \\ 12 & 36 & 72 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 12 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [28] $1^{1477}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.7$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 8 & 14 & 14 & 1 & 3 \\ 8 & 30 & 56 & 56 & 0 & 8 \\ 14 & 56 & 96 & 98 & 0 & 14 \\ 14 & 56 & 98 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 8 & 14 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [29] $1^{1447}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.3$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 8 & 8 & 14 & 1 & 3 \\ 8 & 30 & 32 & 56 & 0 & 8 \\ 8 & 32 & 30 & 56 & 0 & 8 \\ 14 & 56 & 56 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 8 & 8 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [14] $1^{1144}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 70, T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 8 & 9 & 1 & 3 \\ 2 & 0 & 8 & 8 & 0 & 2 \\ 8 & 8 & 30 & 32 & 0 & 8 \\ 9 & 8 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{1246}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 8 & 12 & 1 & 3 \\ 4 & 6 & 16 & 24 & 0 & 4 \\ 8 & 16 & 30 & 48 & 0 & 8 \\ 12 & 24 & 48 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [87] $1^{1347}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.1$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

- [88] $1^{1227}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 8 & 14 & 1 & 3 \\ 6 & 16 & 24 & 42 & 0 & 6 \\ 8 & 24 & 30 & 56 & 0 & 8 \\ 14 & 42 & 56 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 8 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [83] $1^{1226}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 4 & 14 & 1 & 3 \\ 4 & 6 & 8 & 28 & 0 & 4 \\ 4 & 8 & 6 & 28 & 0 & 4 \\ 14 & 28 & 28 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [37] $1^{1237}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 6 & 14 & 1 & 3 \\ 4 & 6 & 12 & 28 & 0 & 4 \\ 6 & 12 & 16 & 42 & 0 & 6 \\ 14 & 28 & 42 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [52] $1^{1234}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 6 & 8 & 1 & 3 \\ 4 & 6 & 12 & 16 & 0 & 4 \\ 6 & 12 & 16 & 24 & 0 & 6 \\ 8 & 16 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [53] $1^{1267}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.1$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 12 & 14 & 1 & 3 \\ 4 & 6 & 24 & 28 & 0 & 4 \\ 12 & 24 & 70 & 84 & 0 & 12 \\ 14 & 28 & 84 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 12 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [82] $1^{1566}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.6$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 10 & 12 & 12 & 1 & 3 \\ 10 & 48 & 60 & 60 & 0 & 10 \\ 12 & 60 & 70 & 72 & 0 & 12 \\ 12 & 60 & 72 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 10 & 12 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [81] $1^{1256}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.9$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 10 & 12 & 1 & 3 \\ 4 & 6 & 20 & 24 & 0 & 4 \\ 10 & 20 & 48 & 60 & 0 & 10 \\ 12 & 24 & 60 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 10 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [54] $1^{1467}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.6$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 8 & 12 & 14 & 1 & 3 \\ 8 & 30 & 48 & 56 & 0 & 8 \\ 12 & 48 & 70 & 84 & 0 & 12 \\ 14 & 56 & 84 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 8 & 12 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [55] $1^{1335}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 6 & 10 & 1 & 3 \\ 6 & 16 & 18 & 30 & 0 & 6 \\ 6 & 18 & 16 & 30 & 0 & 6 \\ 10 & 30 & 30 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 6 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [56] $1^{127}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 4 & 14 & 1 & 3 \\ 2 & 0 & 4 & 14 & 0 & 2 \\ 4 & 4 & 6 & 28 & 0 & 4 \\ 14 & 14 & 28 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [57] $1^{1257}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

- [58] $1^{1112}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 10 & 14 & 1 & 3 \\ 4 & 6 & 20 & 28 & 0 & 4 \\ 10 & 20 & 48 & 70 & 0 & 10 \\ 14 & 28 & 70 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 10 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [59] $1^{1177}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 14 & 14 & 1 & 3 \\ 2 & 0 & 14 & 14 & 0 & 2 \\ 14 & 14 & 96 & 98 & 0 & 14 \\ 14 & 14 & 98 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 14 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{1233}1^{11} + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 70, T \in [11, 70]$

$$\begin{pmatrix} 1 & 4 & 6 & 7 & 1 & 3 \\ 4 & 6 & 12 & 12 & 0 & 4 \\ 6 & 12 & 16 & 18 & 0 & 6 \\ 7 & 12 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{1247}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 8 & 14 & 1 & 3 \\ 4 & 6 & 16 & 28 & 0 & 4 \\ 8 & 16 & 30 & 56 & 0 & 8 \\ 14 & 28 & 56 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 8 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [8] $1^{1134}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 6 & 8 & 1 & 3 \\ 2 & 0 & 6 & 8 & 0 & 2 \\ 6 & 6 & 16 & 24 & 0 & 6 \\ 8 & 8 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{1377}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 14 & 14 & 1 & 3 \\ 6 & 16 & 42 & 42 & 0 & 6 \\ 14 & 42 & 96 & 98 & 0 & 14 \\ 14 & 42 & 98 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 14 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [6] $1^{1357}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.3$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 10 & 14 & 1 & 3 \\ 6 & 16 & 30 & 42 & 0 & 6 \\ 10 & 30 & 48 & 70 & 0 & 10 \\ 14 & 42 & 70 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 10 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [5] $1^{1111}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 3 \\ 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$
- [60] $1^{1557}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.6$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 10 & 10 & 14 & 1 & 3 \\ 10 & 48 & 50 & 70 & 0 & 10 \\ 10 & 50 & 48 & 70 & 0 & 10 \\ 14 & 70 & 70 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 10 & 10 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [61] $1^{1137}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 6 & 14 & 1 & 3 \\ 2 & 0 & 6 & 14 & 0 & 2 \\ 6 & 6 & 16 & 42 & 0 & 6 \\ 14 & 14 & 42 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [62] $1^{1255}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

- [63] $1^{1457}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 10 & 10 & 1 & 3 \\ 4 & 6 & 20 & 20 & 0 & 4 \\ 10 & 20 & 48 & 50 & 0 & 10 \\ 10 & 20 & 50 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 10 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [64] $1^{1344}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 8 & 10 & 14 & 1 & 3 \\ 8 & 30 & 40 & 56 & 0 & 8 \\ 10 & 40 & 48 & 70 & 0 & 10 \\ 14 & 56 & 70 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 8 & 10 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [65] $1^{1157}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 10 & 14 & 1 & 3 \\ 2 & 0 & 10 & 14 & 0 & 2 \\ 10 & 10 & 48 & 70 & 0 & 10 \\ 14 & 14 & 70 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 10 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [66] $1^{1244}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 8 & 8 & 1 & 3 \\ 4 & 6 & 16 & 16 & 0 & 4 \\ 8 & 16 & 30 & 32 & 0 & 8 \\ 8 & 16 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [80] $1^{1355}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.1$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 10 & 10 & 1 & 3 \\ 6 & 16 & 30 & 30 & 0 & 6 \\ 10 & 30 & 48 & 50 & 0 & 10 \\ 10 & 30 & 50 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 10 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [79] $1^{1455}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.3$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 8 & 10 & 10 & 1 & 3 \\ 8 & 30 & 40 & 40 & 0 & 8 \\ 10 & 40 & 48 & 50 & 0 & 10 \\ 10 & 40 & 50 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 8 & 10 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [30] $1^{1133}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 6 & 6 & 1 & 3 \\ 2 & 0 & 6 & 6 & 0 & 2 \\ 6 & 6 & 16 & 18 & 0 & 6 \\ 6 & 6 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$
- [31] $1^{1466}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 8 & 12 & 12 & 1 & 3 \\ 8 & 30 & 48 & 48 & 0 & 8 \\ 12 & 48 & 70 & 72 & 0 & 12 \\ 12 & 48 & 72 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 8 & 12 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [32] $1^{1277}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 14 & 14 & 1 & 3 \\ 4 & 6 & 28 & 28 & 0 & 4 \\ 14 & 28 & 96 & 98 & 0 & 14 \\ 14 & 28 & 98 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 14 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [33] $1^{1667}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 12 & 12 & 14 & 1 & 3 \\ 12 & 70 & 72 & 84 & 0 & 12 \\ 12 & 72 & 70 & 84 & 0 & 12 \\ 14 & 84 & 84 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 12 & 12 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [34] $1^{1445}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.1$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

- [40] $1^{11241}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 8 & 8 & 10 & 1 & 3 \\ 8 & 30 & 32 & 40 & 0 & 8 \\ 8 & 32 & 30 & 40 & 0 & 8 \\ 10 & 40 & 40 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 8 & 8 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [90] $1^{1146}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 8 & 12 & 1 & 3 \\ 2 & 0 & 8 & 12 & 0 & 2 \\ 8 & 8 & 30 & 48 & 0 & 8 \\ 12 & 12 & 48 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [78] $1^{1135}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 6 & 10 & 1 & 3 \\ 2 & 0 & 6 & 10 & 0 & 2 \\ 6 & 6 & 16 & 30 & 0 & 6 \\ 10 & 10 & 30 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [67] $1^{1113}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 1 & 3 \\ 2 & 0 & 2 & 6 & 0 & 2 \\ 2 & 2 & 0 & 6 & 0 & 2 \\ 6 & 6 & 6 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$
- [68] $1^{1444}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 8 & 8 & 8 & 1 & 3 \\ 8 & 30 & 32 & 32 & 0 & 8 \\ 8 & 32 & 30 & 32 & 0 & 8 \\ 8 & 32 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 8 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [77] $1^{1345}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.9$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 8 & 10 & 1 & 3 \\ 6 & 16 & 24 & 30 & 0 & 6 \\ 8 & 24 & 30 & 40 & 0 & 8 \\ 10 & 30 & 40 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 8 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [76] $1^{1114}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 1 & 3 \\ 2 & 0 & 2 & 8 & 0 & 2 \\ 2 & 2 & 0 & 8 & 0 & 2 \\ 8 & 8 & 8 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [35] $1^{1337}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.9$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 6 & 14 & 1 & 3 \\ 6 & 16 & 18 & 42 & 0 & 6 \\ 6 & 18 & 16 & 42 & 0 & 6 \\ 14 & 42 & 42 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 6 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [75] $1^{1333}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.4$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 6 & 6 & 1 & 3 \\ 6 & 16 & 18 & 18 & 0 & 6 \\ 6 & 18 & 16 & 18 & 0 & 6 \\ 6 & 18 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 6 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$
- [74] $1^{1245}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 8 & 10 & 1 & 3 \\ 4 & 6 & 16 & 20 & 0 & 4 \\ 8 & 16 & 30 & 40 & 0 & 8 \\ 10 & 20 & 40 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 8 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [73] $1^{1266}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

- [69] $1^{1334}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 4 & 12 & 12 & 1 & 3 \\ 4 & 6 & 24 & 24 & 0 & 4 \\ 12 & 24 & 70 & 72 & 0 & 12 \\ 12 & 24 & 72 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 12 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [72] $1^{1126}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 6 & 8 & 1 & 3 \\ 6 & 16 & 18 & 24 & 0 & 6 \\ 6 & 18 & 16 & 24 & 0 & 6 \\ 8 & 24 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 6 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [71] $1^{1456}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.4$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 8 & 10 & 12 & 1 & 3 \\ 8 & 30 & 40 & 48 & 0 & 8 \\ 10 & 40 & 48 & 60 & 0 & 10 \\ 12 & 48 & 60 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 8 & 10 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [70] $1^{1336}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 6 & 6 & 12 & 1 & 3 \\ 6 & 16 & 18 & 36 & 0 & 6 \\ 6 & 18 & 16 & 36 & 0 & 6 \\ 12 & 36 & 36 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 6 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [47] $1^{1156}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.4$
 $T_{\min} = 10, T_{\max} = 70, T \in [10, 70]$

$$\begin{pmatrix} 1 & 2 & 10 & 12 & 1 & 3 \\ 2 & 0 & 10 & 12 & 0 & 2 \\ 10 & 10 & 48 & 60 & 0 & 10 \\ 12 & 12 & 60 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 10 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [91] $1^{1222}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 20, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 70, T \in [11, 70]$

$$\begin{pmatrix} 1 & 4 & 4 & 5 & 1 & 3 \\ 4 & 6 & 8 & 8 & 0 & 4 \\ 4 & 8 & 6 & 8 & 0 & 4 \\ 5 & 8 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$
- [92] $1^{1111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 28, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 70, T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 1 & 3 \\ 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 0 & 2 \\ 3 & 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$
- [93] $1^{1133}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 28, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 70, T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 6 & 7 & 1 & 3 \\ 2 & 0 & 6 & 6 & 0 & 2 \\ 6 & 6 & 16 & 18 & 0 & 6 \\ 7 & 6 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$
- [94] $1^{1234}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 32, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 70, T \in [11, 70]$

$$\begin{pmatrix} 1 & 4 & 6 & 9 & 1 & 3 \\ 4 & 6 & 12 & 16 & 0 & 4 \\ 6 & 12 & 16 & 24 & 0 & 6 \\ 9 & 16 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$
- [95] $1^{1122}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 32, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 70, T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 4 & 5 & 1 & 3 \\ 2 & 0 & 4 & 4 & 0 & 2 \\ 4 & 4 & 6 & 8 & 0 & 4 \\ 5 & 4 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$
- [96] $1^{1223}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6, T_{\max} = 70, \Delta = -122, \det = 36, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 70, T \in [11, 70]$

$$\begin{pmatrix} 1 & 4 & 4 & 7 & 1 & 3 \\ 4 & 6 & 8 & 12 & 0 & 4 \\ 4 & 8 & 6 & 12 & 0 & 4 \\ 7 & 12 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

[97] $1^{1112}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 70$, $\Delta = -122$, $\det = 44$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 10.18 \rightarrow 11$, $T_{\max} = 70$, $T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 1 & 3 \\ 2 & 0 & 2 & 4 & 0 & 2 \\ 2 & 2 & 0 & 4 & 0 & 2 \\ 5 & 4 & 4 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

[98] $1^{1134}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 70$, $\Delta = -122$, $\det = 44$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 10.18 \rightarrow 11$, $T_{\max} = 70$, $T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 6 & 9 & 1 & 3 \\ 2 & 0 & 6 & 8 & 0 & 2 \\ 6 & 6 & 16 & 24 & 0 & 6 \\ 9 & 8 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[99] $1^{1123}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 70$, $\Delta = -122$, $\det = 48$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.3$
 $T_{\min} = 10.17 \rightarrow 11$, $T_{\max} = 70$, $T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 4 & 7 & 1 & 3 \\ 2 & 0 & 4 & 6 & 0 & 2 \\ 4 & 4 & 6 & 12 & 0 & 4 \\ 7 & 6 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

[100] $1^{1224}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 70$, $\Delta = -122$, $\det = 52$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 10.15 \rightarrow 11$, $T_{\max} = 70$, $T \in [11, 70]$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 1 & 3 \\ 4 & 6 & 8 & 16 & 0 & 4 \\ 4 & 8 & 6 & 16 & 0 & 4 \\ 9 & 16 & 16 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[101] $1^{1113}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 70$, $\Delta = -122$, $\det = 60$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 10.13 \rightarrow 11$, $T_{\max} = 70$, $T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 1 & 3 \\ 2 & 0 & 2 & 6 & 0 & 2 \\ 2 & 2 & 0 & 6 & 0 & 2 \\ 7 & 6 & 6 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

[102] $1^{1124}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 70$, $\Delta = -122$, $\det = 64$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 10.12 \rightarrow 11$, $T_{\max} = 70$, $T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 4 & 9 & 1 & 3 \\ 2 & 0 & 4 & 8 & 0 & 2 \\ 4 & 4 & 6 & 16 & 0 & 4 \\ 9 & 8 & 16 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[103] $1^{1114}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 70$, $\Delta = -122$, $\det = 76$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 10.11 \rightarrow 11$, $T_{\max} = 70$, $T \in [11, 70]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 1 & 3 \\ 2 & 0 & 2 & 8 & 0 & 2 \\ 2 & 2 & 0 & 8 & 0 & 2 \\ 9 & 8 & 8 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

1.32 $\textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{black}{so}_8 + \hat{1} \rightarrow (-1)$

$T = 7$ [0] $1^{1111}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{black}{so}_8 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 66$, $\Delta = -121$, $\det = -40$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 11.20 \rightarrow 12$, $T_{\max} = 66$, $T \in [12, 66]$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 2 & 2 & 0 & 2 \\ 2 & 0 & 2 & 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 0 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 0 & 2 & 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{1111}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{black}{so}_8 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 66$, $\Delta = -121$, $\det = -64$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 11$, $T_{\max} = 66$, $T \in [11, 66]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 1 & 3 \\ 2 & 0 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{1111}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{black}{so}_8 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 66$, $\Delta = -121$, $\det = -104$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 11.08 \rightarrow 12, T_{\max} = 66, T \in [12, 66]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 2 & 1 & 3 \\ 2 & 0 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 0 & 2 \\ 3 & 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

1.33 $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow (-1)$

$$T = 7 \quad [0] \quad 1^{\hat{1}1\hat{1}23}1^{\hat{1}1} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.3$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 6 & 1 & 3 \\ 2 & 0 & 2 & 4 & 6 & 0 & 2 \\ 2 & 2 & 0 & 4 & 6 & 0 & 2 \\ 5 & 4 & 4 & 6 & 12 & 0 & 4 \\ 6 & 6 & 6 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\hat{1}1\hat{1}1}1^{\hat{1}1} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 2 & 2 & 0 & 2 \\ 2 & 0 & 0 & 2 & 2 & 0 & 2 \\ 2 & 0 & 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 0 & 2 & 2 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\hat{1}1334}1^{\hat{1}1} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.6$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 6 & 6 & 9 & 1 & 3 \\ 2 & 0 & 6 & 6 & 8 & 0 & 2 \\ 6 & 6 & 16 & 18 & 24 & 0 & 6 \\ 6 & 6 & 18 & 16 & 24 & 0 & 6 \\ 9 & 8 & 24 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 6 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\hat{1}1144}1^{\hat{1}1} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 9 & 1 & 3 \\ 2 & 0 & 2 & 8 & 8 & 0 & 2 \\ 2 & 2 & 0 & 8 & 8 & 0 & 2 \\ 8 & 8 & 8 & 30 & 32 & 0 & 8 \\ 9 & 8 & 8 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 8 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\hat{1}1233}1^{\hat{1}1} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.1$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 6 & 7 & 1 & 3 \\ 2 & 0 & 4 & 6 & 6 & 0 & 2 \\ 4 & 4 & 6 & 12 & 12 & 0 & 4 \\ 6 & 6 & 12 & 16 & 18 & 0 & 6 \\ 7 & 6 & 12 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 6 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\hat{1}1112}1^{\hat{1}1} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 4 & 1 & 3 \\ 2 & 0 & 2 & 2 & 4 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 & 0 & 2 \\ 3 & 2 & 2 & 0 & 4 & 0 & 2 \\ 4 & 4 & 4 & 4 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\hat{1}1336}1^{\hat{1}1} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 6 & 6 & 12 & 1 & 3 \\ 2 & 0 & 6 & 6 & 12 & 0 & 2 \\ 6 & 6 & 16 & 18 & 36 & 0 & 6 \\ 6 & 6 & 18 & 16 & 36 & 0 & 6 \\ 12 & 12 & 36 & 36 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 6 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{\hat{1}1335}1^{\hat{1}1} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.7$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 6 & 6 & 10 & 1 & 3 \\ 2 & 0 & 6 & 6 & 10 & 0 & 2 \\ 6 & 6 & 16 & 18 & 30 & 0 & 6 \\ 6 & 6 & 18 & 16 & 30 & 0 & 6 \\ 10 & 10 & 30 & 30 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 6 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$

[39] $1^{11222}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 4 & 4 & 1 & 3 \\ 2 & 0 & 4 & 4 & 4 & 0 & 2 \\ 4 & 4 & 8 & 8 & 0 & 0 & 4 \\ 4 & 4 & 8 & 8 & 0 & 0 & 4 \\ 4 & 4 & 8 & 8 & 0 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 4 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{12233}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.6$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 6 & 1 & 3 \\ 4 & 6 & 8 & 12 & 12 & 0 & 4 \\ 4 & 8 & 6 & 12 & 12 & 0 & 4 \\ 6 & 12 & 12 & 16 & 18 & 0 & 6 \\ 6 & 12 & 12 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 6 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{11177}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.7$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 14 & 1 & 3 \\ 2 & 0 & 2 & 14 & 14 & 0 & 2 \\ 2 & 2 & 0 & 14 & 14 & 0 & 2 \\ 14 & 14 & 14 & 96 & 98 & 0 & 14 \\ 14 & 14 & 14 & 98 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 14 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{11257}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 10 & 14 & 1 & 3 \\ 2 & 0 & 4 & 10 & 14 & 0 & 2 \\ 4 & 4 & 6 & 20 & 28 & 0 & 4 \\ 10 & 10 & 20 & 48 & 70 & 0 & 10 \\ 14 & 14 & 28 & 70 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 10 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{11236}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.6$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 6 & 12 & 1 & 3 \\ 2 & 0 & 4 & 6 & 12 & 0 & 2 \\ 4 & 4 & 6 & 12 & 24 & 0 & 4 \\ 6 & 6 & 12 & 16 & 36 & 0 & 6 \\ 12 & 12 & 24 & 36 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 6 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{11147}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.3$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 14 & 1 & 3 \\ 2 & 0 & 2 & 8 & 14 & 0 & 2 \\ 2 & 2 & 0 & 8 & 14 & 0 & 2 \\ 8 & 8 & 8 & 30 & 56 & 0 & 8 \\ 14 & 14 & 14 & 56 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 8 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{12227}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 4 & 14 & 1 & 3 \\ 4 & 6 & 8 & 8 & 28 & 0 & 4 \\ 4 & 8 & 6 & 8 & 28 & 0 & 4 \\ 4 & 8 & 8 & 6 & 28 & 0 & 4 \\ 14 & 28 & 28 & 28 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 4 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{11246}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 8 & 12 & 1 & 3 \\ 2 & 0 & 4 & 8 & 12 & 0 & 2 \\ 4 & 4 & 6 & 16 & 24 & 0 & 4 \\ 8 & 8 & 16 & 30 & 48 & 0 & 8 \\ 12 & 12 & 24 & 48 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{11116}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 12 & 1 & 3 \\ 2 & 0 & 2 & 2 & 12 & 0 & 2 \\ 2 & 2 & 0 & 2 & 12 & 0 & 2 \\ 2 & 2 & 2 & 0 & 12 & 0 & 2 \\ 12 & 12 & 12 & 12 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{11345}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, \ T_{\max} = 11, \ \Delta = -92, \ \det = -32, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 14.9$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 6 & 8 & 10 & 1 & 3 \\ 2 & 0 & 6 & 8 & 10 & 0 & 2 \\ 6 & 6 & 16 & 24 & 30 & 0 & 6 \\ 8 & 8 & 24 & 30 & 40 & 0 & 8 \\ 10 & 10 & 30 & 40 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 8 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{11235}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, \ T_{\max} = 11, \ \Delta = -92, \ \det = -32, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 4 & 6 & 1 & 3 \\ 2 & 0 & 4 & 4 & 6 & 0 & 2 \\ 4 & 4 & 6 & 8 & 12 & 0 & 4 \\ 4 & 4 & 8 & 6 & 12 & 0 & 4 \\ 6 & 6 & 12 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 4 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{11127}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, \ T_{\max} = 11, \ \Delta = -92, \ \det = -32, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 14 & 1 & 3 \\ 2 & 0 & 2 & 4 & 14 & 0 & 2 \\ 2 & 2 & 0 & 4 & 14 & 0 & 2 \\ 4 & 4 & 4 & 6 & 28 & 0 & 4 \\ 14 & 14 & 14 & 28 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{11136}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, \ T_{\max} = 11, \ \Delta = -92, \ \det = -32, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 14.1$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 12 & 1 & 3 \\ 2 & 0 & 2 & 6 & 12 & 0 & 2 \\ 2 & 2 & 0 & 6 & 12 & 0 & 2 \\ 6 & 6 & 6 & 16 & 36 & 0 & 6 \\ 12 & 12 & 12 & 36 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 6 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{11113}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, \ T_{\max} = 11, \ \Delta = -92, \ \det = -32, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 1 & 3 \\ 2 & 0 & 2 & 6 & 0 & 2 \\ 2 & 2 & 0 & 6 & 0 & 2 \\ 2 & 2 & 0 & 6 & 0 & 2 \\ 6 & 6 & 6 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{12225}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, \ T_{\max} = 11, \ \Delta = -92, \ \det = -32, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 14.6$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 4 & 10 & 1 & 3 \\ 4 & 6 & 8 & 8 & 20 & 0 & 4 \\ 4 & 8 & 6 & 8 & 20 & 0 & 4 \\ 4 & 8 & 8 & 6 & 20 & 0 & 4 \\ 10 & 20 & 20 & 20 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 4 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{11133}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, \ T_{\max} = 11, \ \Delta = -92, \ \det = -32, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 13.6$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 6 & 1 & 3 \\ 2 & 0 & 2 & 6 & 6 & 0 & 2 \\ 2 & 2 & 0 & 6 & 6 & 0 & 2 \\ 6 & 6 & 6 & 16 & 18 & 0 & 6 \\ 6 & 6 & 6 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 6 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11234}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, \ T_{\max} = 11, \ \Delta = -92, \ \det = -32, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 14.3$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 6 & 8 & 1 & 3 \\ 2 & 0 & 4 & 6 & 8 & 0 & 2 \\ 4 & 4 & 6 & 12 & 16 & 0 & 4 \\ 6 & 6 & 12 & 16 & 24 & 0 & 6 \\ 8 & 8 & 16 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 6 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{12234}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, \ T_{\max} = 11, \ \Delta = -92, \ \det = -32, \ (n_+, n_-, n_0) =$$

$$(1, 7, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 8 & 1 & 3 \\ 4 & 6 & 8 & 12 & 16 & 0 & 4 \\ 4 & 8 & 6 & 12 & 16 & 0 & 4 \\ 6 & 12 & 12 & 16 & 24 & 0 & 6 \\ 8 & 16 & 16 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 6 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11337}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.9$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 6 & 6 & 14 & 1 & 3 \\ 2 & 0 & 6 & 6 & 14 & 0 & 2 \\ 6 & 6 & 16 & 18 & 42 & 0 & 6 \\ 6 & 6 & 18 & 16 & 42 & 0 & 6 \\ 14 & 14 & 42 & 42 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 6 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{11224}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 4 & 8 & 1 & 3 \\ 2 & 0 & 4 & 4 & 8 & 0 & 2 \\ 4 & 4 & 6 & 8 & 16 & 0 & 4 \\ 4 & 4 & 8 & 6 & 16 & 0 & 4 \\ 8 & 8 & 16 & 16 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{12223}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.3$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 4 & 6 & 1 & 3 \\ 4 & 6 & 8 & 8 & 12 & 0 & 4 \\ 4 & 8 & 6 & 8 & 12 & 0 & 4 \\ 4 & 8 & 8 & 6 & 12 & 0 & 4 \\ 6 & 12 & 12 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 4 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{11157}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 14 & 1 & 3 \\ 2 & 0 & 2 & 10 & 14 & 0 & 2 \\ 2 & 2 & 0 & 10 & 14 & 0 & 2 \\ 10 & 10 & 10 & 48 & 70 & 0 & 10 \\ 14 & 14 & 14 & 70 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 10 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{11225}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.1$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 4 & 10 & 1 & 3 \\ 2 & 0 & 4 & 4 & 10 & 0 & 2 \\ 4 & 4 & 6 & 8 & 20 & 0 & 4 \\ 4 & 4 & 8 & 6 & 20 & 0 & 4 \\ 10 & 10 & 20 & 20 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 4 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{11247}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 8 & 14 & 1 & 3 \\ 2 & 0 & 4 & 8 & 14 & 0 & 2 \\ 4 & 4 & 6 & 16 & 28 & 0 & 4 \\ 8 & 8 & 16 & 30 & 56 & 0 & 8 \\ 14 & 14 & 28 & 56 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 8 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{11135}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.9$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 10 & 1 & 3 \\ 2 & 0 & 2 & 6 & 10 & 0 & 2 \\ 2 & 2 & 0 & 6 & 10 & 0 & 2 \\ 6 & 6 & 6 & 16 & 30 & 0 & 6 \\ 10 & 10 & 10 & 30 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 6 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11145}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.1$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 10 & 1 & 3 \\ 2 & 0 & 2 & 8 & 10 & 0 & 2 \\ 2 & 2 & 0 & 8 & 10 & 0 & 2 \\ 8 & 8 & 8 & 30 & 40 & 0 & 8 \\ 10 & 10 & 10 & 40 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 8 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$

- [14] $1^{11117}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.3$
 $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 14 & 1 & 3 \\ 2 & 0 & 2 & 2 & 14 & 0 & 2 \\ 2 & 2 & 0 & 2 & 14 & 0 & 2 \\ 2 & 2 & 2 & 0 & 14 & 0 & 2 \\ 14 & 14 & 14 & 14 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [13] $1^{11114}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 8 & 1 & 3 \\ 2 & 0 & 2 & 2 & 8 & 0 & 2 \\ 2 & 2 & 0 & 2 & 8 & 0 & 2 \\ 2 & 2 & 2 & 0 & 8 & 0 & 2 \\ 8 & 8 & 8 & 8 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{11155}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.3$
 $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 10 & 10 & 1 & 3 \\ 2 & 0 & 2 & 2 & 10 & 10 & 0 & 2 \\ 2 & 2 & 0 & 2 & 10 & 10 & 0 & 2 \\ 10 & 10 & 10 & 48 & 50 & 0 & 10 \\ 10 & 10 & 10 & 50 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 10 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{11127}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.3$
 $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 4 & 4 & 14 & 1 & 3 \\ 2 & 0 & 4 & 4 & 14 & 0 & 2 \\ 4 & 4 & 6 & 8 & 28 & 0 & 4 \\ 4 & 4 & 8 & 6 & 28 & 0 & 4 \\ 14 & 14 & 28 & 28 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 4 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{12222}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$

$$\begin{pmatrix} 1 & 4 & 4 & 4 & 4 & 1 & 3 \\ 4 & 6 & 8 & 8 & 8 & 0 & 4 \\ 4 & 8 & 6 & 8 & 8 & 0 & 4 \\ 4 & 8 & 8 & 6 & 8 & 0 & 4 \\ 4 & 8 & 8 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 4 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{11137}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.1$
 $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 6 & 14 & 1 & 3 \\ 2 & 0 & 2 & 2 & 6 & 14 & 0 & 2 \\ 2 & 2 & 0 & 2 & 6 & 14 & 0 & 2 \\ 6 & 6 & 6 & 16 & 42 & 0 & 6 \\ 14 & 14 & 14 & 42 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 6 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$
- [8] $1^{11166}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 12 & 12 & 1 & 3 \\ 2 & 0 & 2 & 2 & 12 & 12 & 0 & 2 \\ 2 & 2 & 0 & 2 & 12 & 12 & 0 & 2 \\ 12 & 12 & 12 & 70 & 72 & 0 & 12 \\ 12 & 12 & 12 & 72 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 12 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [64] $1^{12224}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$

$$\begin{pmatrix} 1 & 4 & 4 & 4 & 8 & 1 & 3 \\ 4 & 6 & 8 & 8 & 16 & 0 & 4 \\ 4 & 8 & 6 & 8 & 16 & 0 & 4 \\ 4 & 8 & 8 & 6 & 16 & 0 & 4 \\ 8 & 16 & 16 & 16 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$
- [6] $1^{11167}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.6$
 $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 12 & 1 & 3 \\ 2 & 0 & 2 & 4 & 12 & 0 & 2 \\ 2 & 2 & 0 & 4 & 12 & 0 & 2 \\ 4 & 4 & 4 & 6 & 24 & 0 & 4 \\ 12 & 12 & 12 & 24 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & 12 & -1 & 1 \end{pmatrix}$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 10 & 12 & 1 & 3 \\ 2 & 0 & 4 & 10 & 12 & 0 & 2 \\ 4 & 4 & 6 & 20 & 24 & 0 & 4 \\ 10 & 10 & 20 & 48 & 60 & 0 & 10 \\ 12 & 12 & 24 & 60 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 10 & 12 & -1 & 1 \end{pmatrix}$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 9 & 1 & 3 \\ 4 & 6 & 8 & 12 & 16 & 0 & 4 \\ 4 & 8 & 6 & 12 & 16 & 0 & 4 \\ 6 & 12 & 12 & 16 & 24 & 0 & 6 \\ 9 & 16 & 16 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 6 & 8 & -1 & 0 \end{pmatrix}_b$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 8 & 1 & 3 \\ 2 & 0 & 2 & 8 & 8 & 0 & 2 \\ 2 & 2 & 0 & 8 & 8 & 0 & 2 \\ 8 & 8 & 8 & 30 & 32 & 0 & 8 \\ 8 & 8 & 8 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 8 & 8 & -1 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 12 & 12 & 1 & 3 \\ 2 & 0 & 4 & 12 & 12 & 0 & 2 \\ 4 & 4 & 6 & 24 & 24 & 0 & 4 \\ 12 & 12 & 24 & 70 & 72 & 0 & 12 \\ 12 & 12 & 24 & 72 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 12 & 12 & -1 & 1 \end{pmatrix}$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 12 & 1 & 3 \\ 2 & 0 & 2 & 10 & 12 & 0 & 2 \\ 2 & 2 & 0 & 10 & 12 & 0 & 2 \\ 10 & 10 & 10 & 48 & 60 & 0 & 10 \\ 12 & 12 & 12 & 60 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 10 & 12 & -1 & 1 \end{pmatrix}$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 10 & 1 & 3 \\ 2 & 0 & 2 & 2 & 10 & 0 & 2 \\ 2 & 2 & 0 & 2 & 10 & 0 & 2 \\ 2 & 2 & 2 & 0 & 10 & 0 & 2 \\ 10 & 10 & 10 & 10 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 10 & -1 & 1 \end{pmatrix}$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 4 & 12 & 1 & 3 \\ 2 & 0 & 4 & 4 & 12 & 0 & 2 \\ 4 & 4 & 6 & 8 & 24 & 0 & 4 \\ 4 & 4 & 8 & 6 & 24 & 0 & 4 \\ 12 & 12 & 24 & 24 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 4 & 12 & -1 & 1 \end{pmatrix}$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 6 & 8 & 8 & 1 & 3 \\ 2 & 0 & 6 & 8 & 8 & 0 & 2 \\ 6 & 6 & 16 & 24 & 24 & 0 & 6 \\ 8 & 8 & 24 & 30 & 32 & 0 & 8 \\ 8 & 8 & 24 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 8 & 8 & -1 & 1 \end{pmatrix} b_{\text{max}}$$

1420

$$(1, 7, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 1 & 3 \\ 2 & 0 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{12235}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.9$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 10 & 1 & 3 \\ 4 & 6 & 8 & 12 & 20 & 0 & 4 \\ 4 & 8 & 6 & 12 & 20 & 0 & 4 \\ 6 & 12 & 12 & 16 & 30 & 0 & 6 \\ 10 & 20 & 20 & 30 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 6 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{11237}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.6$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 6 & 14 & 1 & 3 \\ 2 & 0 & 4 & 6 & 14 & 0 & 2 \\ 4 & 4 & 6 & 12 & 28 & 0 & 4 \\ 6 & 6 & 12 & 16 & 42 & 0 & 6 \\ 14 & 14 & 28 & 42 & 96 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 6 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{12333}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.9$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 4 & 6 & 6 & 6 & 1 & 3 \\ 4 & 6 & 12 & 12 & 12 & 0 & 4 \\ 6 & 12 & 16 & 18 & 18 & 0 & 6 \\ 6 & 12 & 18 & 16 & 18 & 0 & 6 \\ 6 & 12 & 18 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & 6 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{11146}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 12 & 1 & 3 \\ 2 & 0 & 2 & 8 & 12 & 0 & 2 \\ 2 & 2 & 0 & 8 & 12 & 0 & 2 \\ 8 & 8 & 8 & 30 & 48 & 0 & 8 \\ 12 & 12 & 12 & 48 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{11124}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 8 & 1 & 3 \\ 2 & 0 & 2 & 4 & 8 & 0 & 2 \\ 2 & 2 & 0 & 4 & 8 & 0 & 2 \\ 4 & 4 & 4 & 6 & 16 & 0 & 4 \\ 8 & 8 & 8 & 16 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{11233}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.1$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 6 & 6 & 1 & 3 \\ 2 & 0 & 4 & 6 & 6 & 0 & 2 \\ 4 & 4 & 6 & 12 & 12 & 0 & 4 \\ 6 & 6 & 12 & 16 & 18 & 0 & 6 \\ 6 & 6 & 12 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 6 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{12244}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 8 & 8 & 1 & 3 \\ 4 & 6 & 8 & 16 & 16 & 0 & 4 \\ 4 & 8 & 6 & 16 & 16 & 0 & 4 \\ 8 & 16 & 16 & 30 & 32 & 0 & 8 \\ 8 & 16 & 16 & 32 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{12226}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 4 & 12 & 1 & 3 \\ 4 & 6 & 8 & 8 & 24 & 0 & 4 \\ 4 & 8 & 6 & 8 & 24 & 0 & 4 \\ 4 & 8 & 8 & 6 & 24 & 0 & 4 \\ 12 & 24 & 24 & 24 & 70 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 4 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{11235}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.4$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 4 & 6 & 10 & 1 & 3 \\ 2 & 0 & 4 & 6 & 10 & 0 & 2 \\ 4 & 4 & 6 & 12 & 20 & 0 & 4 \\ 6 & 6 & 12 & 16 & 30 & 0 & 6 \\ 10 & 10 & 20 & 30 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 6 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$

[67] $1^{11222}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 4 & 4 & 5 & 1 & 3 \\ 2 & 0 & 4 & 4 & 0 & 0 & 2 \\ 4 & 4 & 6 & 8 & 8 & 0 & 4 \\ 4 & 4 & 8 & 6 & 8 & 0 & 4 \\ 5 & 4 & 8 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 4 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{11334}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.6$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 6 & 6 & 8 & 1 & 3 \\ 2 & 0 & 6 & 6 & 8 & 0 & 2 \\ 6 & 6 & 16 & 18 & 24 & 0 & 6 \\ 6 & 6 & 18 & 16 & 24 & 0 & 6 \\ 8 & 8 & 24 & 24 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 6 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{11123}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 6 & 1 & 3 \\ 2 & 0 & 2 & 4 & 6 & 0 & 2 \\ 2 & 2 & 0 & 4 & 6 & 0 & 2 \\ 4 & 4 & 4 & 6 & 12 & 0 & 4 \\ 6 & 6 & 6 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{11125}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 10 & 1 & 3 \\ 2 & 0 & 2 & 4 & 10 & 0 & 2 \\ 2 & 2 & 0 & 4 & 10 & 0 & 2 \\ 4 & 4 & 4 & 6 & 20 & 0 & 4 \\ 10 & 10 & 10 & 20 & 48 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{12223}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.3$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 4 & 4 & 7 & 1 & 3 \\ 4 & 6 & 8 & 8 & 12 & 0 & 4 \\ 4 & 8 & 6 & 8 & 12 & 0 & 4 \\ 4 & 8 & 8 & 6 & 12 & 0 & 4 \\ 7 & 12 & 12 & 12 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 4 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{11111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 & 1 & 3 \\ 2 & 0 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 0 & 2 \\ 3 & 2 & 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{11133}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 7 & 1 & 3 \\ 2 & 0 & 2 & 6 & 6 & 0 & 2 \\ 2 & 2 & 0 & 6 & 6 & 0 & 2 \\ 6 & 6 & 6 & 16 & 18 & 0 & 6 \\ 7 & 6 & 6 & 18 & 16 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 6 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{11234}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 11, \Delta = -92, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.3$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 9 & 1 & 3 \\ 2 & 0 & 2 & 4 & 8 & 0 & 2 \\ 2 & 2 & 0 & 4 & 8 & 0 & 2 \\ 4 & 4 & 4 & 6 & 16 & 0 & 4 \\ 9 & 8 & 8 & 16 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{11114}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 11, \Delta = -92, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 10.11 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 9 & 1 & 3 \\ 2 & 0 & 2 & 2 & 8 & 0 & 2 \\ 2 & 2 & 0 & 2 & 8 & 0 & 2 \\ 2 & 2 & 0 & 2 & 8 & 0 & 2 \\ 2 & 2 & 2 & 0 & 8 & 0 & 2 \\ 9 & 8 & 8 & 8 & 30 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

$$\mathbf{1.34} \quad \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{so}_8 + \hat{1} \rightarrow (-1)$$

$$T = 8 \quad [0] \quad 1^{11111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 12, \Delta = -91, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 2 & 2 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 2 & 2 & 2 & 0 & 2 \\ 2 & 0 & 0 & 2 & 2 & 2 & 0 & 2 \\ 2 & 0 & 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 0 & 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 2 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 0 & 2 & 2 & 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$\mathbf{1.35} \quad \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow (-1)$$

$$T = 8 \quad [0] \quad 1^{11111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 10, \Delta = -62, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 2 & 1 & 3 \\ 2 & 0 & 2 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 2 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{1.36} \quad \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow (-1)$$

$$T = 9 \quad [0] \quad 1^{1111111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 10, \Delta = -32, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 2 & 2 & 1 & 3 \\ 2 & 0 & 2 & 2 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 2 & 2 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{1.37} \quad \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow (-1)$$

$$T = 10 \quad [0] \quad 1^{11111111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \hat{1} \rightarrow 1 \quad T = 10, T_{\max} = 10, \Delta = -2, \det = 256, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 1 & 3 \\ 2 & 0 & 2 & 2 & 2 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 0 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 & 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{1.38} \quad \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow (-1)$$

$$T = 7 \quad [1] \quad 1^{11221}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13, T_{\max} = 19, T \in [13, 19]$$

$$\begin{pmatrix} 5 & 7 & 12 & 12 & 1 & 5 \\ 7 & 4 & 12 & 12 & 0 & 4 \\ 12 & 12 & 22 & 24 & 0 & 8 \\ 12 & 12 & 24 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

[0] $1^{1112}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -4$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 4 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 2 & 4 & 0 & 2 \\ 2 & 0 & 2 & 0 & 4 & 0 & 2 \\ 4 & 0 & 4 & 4 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 0 & 2 & 2 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{1113}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -12$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 6 & 1 & 3 \\ 2 & 0 & 2 & 2 & 6 & 0 & 2 \\ 2 & 2 & 0 & 2 & 6 & 0 & 2 \\ 3 & 2 & 2 & 0 & 6 & 0 & 2 \\ 6 & 6 & 6 & 6 & 15 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{11122}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -16$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.7$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 4 & 1 & 3 \\ 2 & 0 & 2 & 4 & 4 & 0 & 2 \\ 3 & 2 & 0 & 4 & 4 & 0 & 2 \\ 4 & 4 & 4 & 6 & 8 & 0 & 4 \\ 4 & 4 & 4 & 8 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{11231}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.3$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 13 & 18 & 1 & 5 \\ 6 & 4 & 12 & 18 & 0 & 4 \\ 13 & 12 & 22 & 36 & 0 & 8 \\ 18 & 18 & 36 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{1111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 1 & 6 & 6 & 1 & 5 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 6 & 0 & 4 & 6 & 0 & 4 \\ 6 & 0 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 0 & 4 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11441}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -40$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 24 & 25 & 1 & 5 \\ 6 & 4 & 24 & 24 & 0 & 4 \\ 24 & 24 & 94 & 96 & 0 & 16 \\ 25 & 24 & 96 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 16 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{11121}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -40$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.5$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 7 & 12 & 1 & 5 \\ 6 & 4 & 6 & 12 & 0 & 4 \\ 7 & 6 & 4 & 12 & 0 & 4 \\ 12 & 12 & 12 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{11171}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.3$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 6 & 42 & 1 & 5 \\ 6 & 4 & 6 & 42 & 0 & 4 \\ 6 & 6 & 4 & 42 & 0 & 4 \\ 42 & 42 & 42 & 292 & 0 & 28 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 28 & -1 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{12221}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.5$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 12 & 12 & 12 & 1 & 5 \\ 12 & 22 & 24 & 24 & 0 & 8 \\ 12 & 24 & 22 & 24 & 0 & 8 \\ 12 & 24 & 24 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 8 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{11231}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.3$

$T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 12 & 18 & 1 & 5 \\ 6 & 4 & 12 & 18 & 0 & 4 \\ 12 & 12 & 22 & 36 & 0 & 8 \\ 18 & 18 & 36 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{11261}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.8$

$T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 12 & 36 & 1 & 5 \\ 6 & 4 & 12 & 36 & 0 & 4 \\ 12 & 12 & 22 & 72 & 0 & 8 \\ 36 & 36 & 72 & 214 & 0 & 24 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & 24 & -1 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{11241}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.5$

$T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 12 & 24 & 1 & 5 \\ 6 & 4 & 12 & 24 & 0 & 4 \\ 12 & 12 & 22 & 48 & 0 & 8 \\ 24 & 24 & 48 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{11122}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.7$

$T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 4 & 1 & 3 \\ 2 & 0 & 2 & 4 & 4 & 0 & 2 \\ 2 & 2 & 0 & 4 & 4 & 0 & 2 \\ 4 & 4 & 4 & 6 & 8 & 0 & 4 \\ 4 & 4 & 4 & 8 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{11111}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 6 & 6 & 1 & 5 \\ 6 & 4 & 6 & 6 & 0 & 4 \\ 6 & 6 & 4 & 6 & 0 & 4 \\ 6 & 6 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{12231}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.8$

$T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 12 & 12 & 18 & 1 & 5 \\ 12 & 22 & 24 & 36 & 0 & 8 \\ 12 & 24 & 22 & 36 & 0 & 8 \\ 18 & 36 & 36 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 8 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{11131}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 6 & 18 & 1 & 5 \\ 6 & 4 & 6 & 18 & 0 & 4 \\ 6 & 6 & 4 & 18 & 0 & 4 \\ 18 & 18 & 18 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{11351}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.9$

$T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 18 & 30 & 1 & 5 \\ 6 & 4 & 18 & 30 & 0 & 4 \\ 18 & 18 & 52 & 90 & 0 & 12 \\ 30 & 30 & 90 & 148 & 0 & 20 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 12 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{11221}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 12 & 12 & 1 & 5 \\ 6 & 4 & 12 & 12 & 0 & 4 \\ 12 & 12 & 22 & 24 & 0 & 8 \\ 12 & 12 & 24 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

- [9] $1^{11161}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 19, \Delta = -101, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 6 & 36 & 1 & 5 \\ 6 & 4 & 6 & 36 & 0 & 4 \\ 6 & 6 & 4 & 36 & 0 & 4 \\ 36 & 36 & 36 & 214 & 0 & 24 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 24 & -1 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{12241}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 19, \Delta = -101, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 5 & 12 & 12 & 24 & 1 & 5 \\ 12 & 22 & 24 & 48 & 0 & 8 \\ 12 & 24 & 22 & 48 & 0 & 8 \\ 24 & 48 & 48 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 8 & 8 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{11151}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 19, \Delta = -101, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.1$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 6 & 30 & 1 & 5 \\ 6 & 4 & 6 & 30 & 0 & 4 \\ 6 & 6 & 4 & 30 & 0 & 4 \\ 30 & 30 & 30 & 148 & 0 & 20 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{11251}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 19, \Delta = -101, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.6$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 12 & 30 & 1 & 5 \\ 6 & 4 & 12 & 30 & 0 & 4 \\ 12 & 12 & 22 & 60 & 0 & 8 \\ 30 & 30 & 60 & 148 & 0 & 20 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$
- [13] $1^{11441}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 19, \Delta = -101, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 24 & 24 & 1 & 5 \\ 6 & 4 & 24 & 24 & 0 & 4 \\ 24 & 24 & 94 & 96 & 0 & 16 \\ 24 & 24 & 96 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 16 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$
- [14] $1^{11141}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 19, \Delta = -101, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 6 & 24 & 1 & 5 \\ 6 & 4 & 6 & 24 & 0 & 4 \\ 6 & 6 & 4 & 24 & 0 & 4 \\ 24 & 24 & 24 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$
- [15] $1^{11271}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 19, \Delta = -101, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 12 & 42 & 1 & 5 \\ 6 & 4 & 12 & 42 & 0 & 4 \\ 12 & 12 & 22 & 84 & 0 & 8 \\ 42 & 42 & 84 & 292 & 0 & 28 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & 28 & -1 & 1 \end{pmatrix}_{b_0}$$
- [16] $1^{11112}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 19, \Delta = -101, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 4 & 1 & 3 \\ 2 & 0 & 2 & 2 & 4 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 & 0 & 2 \\ 2 & 2 & 2 & 0 & 4 & 0 & 2 \\ 4 & 4 & 4 & 4 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [17] $1^{11341}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 19, \Delta = -101, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 18 & 24 & 1 & 5 \\ 6 & 4 & 18 & 24 & 0 & 4 \\ 18 & 18 & 52 & 72 & 0 & 12 \\ 24 & 24 & 72 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 12 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$
- [18] $1^{11113}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7, T_{\max} = 19, \Delta = -101, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 6 & 1 & 3 \\ 2 & 0 & 2 & 2 & 6 & 0 & 2 \\ 2 & 2 & 0 & 2 & 6 & 0 & 2 \\ 2 & 2 & 2 & 0 & 6 & 0 & 2 \\ 6 & 6 & 6 & 6 & 15 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{11\overline{33}1}1^{\overline{1}} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.6$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 18 & 18 & 1 & 5 \\ 6 & 4 & 18 & 18 & 0 & 4 \\ 18 & 18 & 52 & 54 & 0 & 12 \\ 18 & 18 & 54 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 12 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{11\overline{12}1}1^{\overline{1}} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 6 & 12 & 1 & 5 \\ 6 & 4 & 6 & 12 & 0 & 4 \\ 6 & 6 & 4 & 12 & 0 & 4 \\ 12 & 12 & 12 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{11\overline{13}2}1^{\overline{1}} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 4 & 1 & 3 \\ 2 & 0 & 2 & 6 & 4 & 0 & 2 \\ 2 & 2 & 0 & 6 & 4 & 0 & 2 \\ 6 & 6 & 6 & 16 & 12 & 0 & 6 \\ 4 & 4 & 4 & 12 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 6 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{11\overline{11}2}1^{\overline{11}} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 4 & 1 & 3 \\ 2 & 0 & 2 & 2 & 4 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 & 0 & 2 \\ 3 & 2 & 2 & 0 & 4 & 0 & 2 \\ 4 & 4 & 4 & 4 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 2 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{12\overline{2}21}1^{\overline{11}} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 10.54 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 12 & 12 & 13 & 1 & 5 \\ 12 & 22 & 24 & 24 & 0 & 8 \\ 12 & 24 & 22 & 24 & 0 & 8 \\ 13 & 24 & 24 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 8 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{11\overline{11}1}1^{\overline{11}} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 10.47 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 6 & 7 & 1 & 5 \\ 6 & 4 & 6 & 6 & 0 & 4 \\ 6 & 6 & 4 & 6 & 0 & 4 \\ 7 & 6 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{11\overline{33}1}1^{\overline{11}} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.6$$

$$T_{\min} = 10.47 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 18 & 19 & 1 & 5 \\ 6 & 4 & 18 & 18 & 0 & 4 \\ 18 & 18 & 52 & 54 & 0 & 12 \\ 19 & 18 & 54 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 12 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{11\overline{2}21}1^{\overline{11}} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 12 & 13 & 1 & 5 \\ 6 & 4 & 12 & 12 & 0 & 4 \\ 12 & 12 & 22 & 24 & 0 & 8 \\ 13 & 12 & 24 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{12\overline{2}31}1^{\overline{11}} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 5 & 12 & 12 & 19 & 1 & 5 \\ 12 & 22 & 24 & 36 & 0 & 8 \\ 12 & 24 & 22 & 36 & 0 & 8 \\ 19 & 36 & 36 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 8 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{11}1^{122}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -100$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.7$

$T_{\min} = 10.36 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 4 & 1 & 3 \\ 2 & 0 & 2 & 4 & 4 & 0 & 2 \\ 2 & 2 & 0 & 4 & 4 & 0 & 2 \\ 5 & 4 & 4 & 6 & 8 & 0 & 4 \\ 4 & 4 & 4 & 8 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 4 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{11}1^{341}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -124$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.8$

$T_{\min} = 10.42 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 18 & 25 & 1 & 5 \\ 6 & 4 & 18 & 24 & 0 & 4 \\ 18 & 18 & 52 & 72 & 0 & 12 \\ 25 & 24 & 72 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 12 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

[39] $1^{11}1^{121}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -124$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 10.42 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 6 & 13 & 1 & 5 \\ 6 & 4 & 6 & 12 & 0 & 4 \\ 6 & 6 & 4 & 12 & 0 & 4 \\ 13 & 12 & 12 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{11}1^{231}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -136$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.3$

$T_{\min} = 10.41 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 12 & 19 & 1 & 5 \\ 6 & 4 & 12 & 18 & 0 & 4 \\ 12 & 12 & 22 & 36 & 0 & 8 \\ 19 & 18 & 36 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{11}1^{132}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -148$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 10.35 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 4 & 1 & 3 \\ 2 & 0 & 2 & 6 & 4 & 0 & 2 \\ 2 & 2 & 0 & 6 & 4 & 0 & 2 \\ 7 & 6 & 6 & 16 & 12 & 0 & 6 \\ 4 & 4 & 4 & 12 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 6 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{12}1^{241}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -148$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 10.41 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 12 & 12 & 25 & 1 & 5 \\ 12 & 22 & 24 & 48 & 0 & 8 \\ 12 & 24 & 22 & 48 & 0 & 8 \\ 25 & 48 & 48 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 8 & 8 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{11}1^{131}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -172$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 10.40 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 6 & 19 & 1 & 5 \\ 6 & 4 & 6 & 18 & 0 & 4 \\ 6 & 6 & 4 & 18 & 0 & 4 \\ 19 & 18 & 18 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{11}1^{241}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -184$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.5$

$T_{\min} = 10.39 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 12 & 25 & 1 & 5 \\ 6 & 4 & 12 & 24 & 0 & 4 \\ 12 & 12 & 22 & 48 & 0 & 8 \\ 25 & 24 & 48 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{11}1^{141}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -220$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 10.38 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 6 & 6 & 25 & 1 & 5 \\ 6 & 4 & 6 & 24 & 0 & 4 \\ 6 & 6 & 4 & 24 & 0 & 4 \\ 25 & 24 & 24 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

1.39 $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$

$$T = 8 \quad [0] \quad 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 11, \Delta = -80, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 1 & 6 & 6 & 1 & 5 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 6 & 0 & 4 & 6 & 0 & 4 \\ 6 & 0 & 6 & 4 & 6 & 0 \\ 6 & 0 & 6 & 6 & 3 & 0 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 0 & 4 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 11, \Delta = -80, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 6 & 6 & 6 & 1 & 5 \\ 6 & 4 & 6 & 6 & 0 & 4 \\ 6 & 6 & 4 & 6 & 0 & 4 \\ 6 & 6 & 6 & 4 & 6 & 0 \\ 6 & 6 & 6 & 6 & 3 & 0 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 11, \Delta = -80, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 6 & 6 & 7 & 6 & 1 & 5 \\ 6 & 4 & 6 & 6 & 6 & 0 & 4 \\ 6 & 6 & 4 & 6 & 6 & 0 & 4 \\ 6 & 6 & 6 & 4 & 6 & 0 & 4 \\ 7 & 6 & 6 & 4 & 6 & 0 & 4 \\ 6 & 6 & 6 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & 4 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

1.40 $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [1] \quad 1^{\textcolor{red}{1}\textcolor{blue}{3}\textcolor{red}{4}\textcolor{blue}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 2, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13, T_{\max} = 89, T \in [13, 89]$$

$$\begin{pmatrix} 5 & 19 & 24 & 1 & 5 \\ 19 & 52 & 72 & 0 & 12 \\ 24 & 72 & 94 & 0 & 16 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 12 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{2}\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 2, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 13, T_{\max} = 89, T \in [13, 89]$$

$$\begin{pmatrix} 5 & 1 & 12 & 1 & 5 \\ 1 & -2 & 0 & 0 & 0 \\ 12 & 0 & 22 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 0 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\textcolor{red}{1}\textcolor{blue}{2}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 8, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 1 & 1 & 2 & 4 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 0 & 0 & 4 & 0 & 2 \\ 4 & 0 & 4 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 0 & 2 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\textcolor{red}{1}\textcolor{blue}{2}\textcolor{red}{3}\textcolor{blue}{2}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 8, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 1 & 5 & 6 & 4 & 1 & 3 \\ 5 & 6 & 12 & 8 & 0 & 4 \\ 6 & 12 & 16 & 12 & 0 & 6 \\ 4 & 8 & 12 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{3}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 12, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 1 & 2 & 3 & 6 & 1 & 3 \\ 2 & 0 & 2 & 6 & 0 & 3 \\ 3 & 2 & 0 & 6 & 0 & 2 \\ 6 & 6 & 6 & 15 & 0 & 5 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{2}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 14, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

- [6] $1^{1223}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 18$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 1 & 3 \\ 3 & 0 & 4 & 4 & 0 & 2 \\ 4 & 4 & 6 & 8 & 0 & 4 \\ 4 & 4 & 8 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{1231}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 20$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.3$
 $T_{\min} = 10.60 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 13 & 18 & 1 & 5 \\ 13 & 22 & 36 & 0 & 8 \\ 18 & 36 & 52 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 20$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 10.60 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 1 & 6 & 1 & 5 \\ 1 & -2 & 0 & 0 & 0 \\ 6 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 0 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$
- [29] $1^{1113}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 1 & 3 \\ 2 & 0 & 2 & 6 & 0 & 2 \\ 2 & 2 & 0 & 6 & 0 & 2 \\ 6 & 6 & 6 & 15 & 0 & 5 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$
- [15] $1^{1661}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 36 & 36 & 1 & 5 \\ 36 & 214 & 216 & 0 & 24 \\ 36 & 216 & 214 & 0 & 24 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 24 & 24 & -1 & 1 \end{pmatrix}_{b_0}$$
- [45] $1^{1123}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 4 & 6 & 1 & 3 \\ 2 & 0 & 4 & 6 & 0 & 2 \\ 4 & 4 & 6 & 12 & 0 & 4 \\ 6 & 6 & 12 & 15 & 0 & 5 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$
- [44] $1^{1342}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 6 & 8 & 4 & 1 & 3 \\ 6 & 16 & 24 & 12 & 0 & 6 \\ 8 & 24 & 30 & 16 & 0 & 8 \\ 4 & 12 & 16 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 8 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [57] $1^{1252}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 4 & 10 & 4 & 1 & 3 \\ 4 & 6 & 20 & 8 & 0 & 4 \\ 10 & 20 & 48 & 20 & 0 & 10 \\ 4 & 8 & 20 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 10 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [56] $1^{1112}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 1 & 3 \\ 2 & 0 & 2 & 4 & 0 & 2 \\ 2 & 2 & 0 & 4 & 0 & 2 \\ 4 & 4 & 4 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [55] $1^{1371}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.1$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 18 & 42 & 1 & 5 \\ 18 & 52 & 126 & 0 & 12 \\ 42 & 126 & 292 & 0 & 28 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 12 & 28 & -1 & 1 \end{pmatrix}_{b_0}$$
- [54] $1^{1271}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 12 & 42 & 1 & 5 \\ 12 & 22 & 84 & 0 & 8 \\ 42 & 84 & 292 & 0 & 28 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 8 & 28 & -1 & 1 \end{pmatrix}_{b_0}$$

- [53] $1^{1161}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 6 & 36 & 1 & 5 \\ 6 & 4 & 36 & 0 & 4 \\ 36 & 36 & 214 & 0 & 24 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 24 & -1 & 1 \end{pmatrix}_{b_0}$$
- [43] $1^{1233}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 1 & 3 \\ 4 & 6 & 8 & 12 & 0 & 4 \\ 4 & 8 & 6 & 12 & 0 & 4 \\ 6 & 12 & 12 & 15 & 0 & 5 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$
- [42] $1^{1332}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 6 & 6 & 4 & 1 & 3 \\ 6 & 16 & 18 & 12 & 0 & 6 \\ 6 & 18 & 16 & 12 & 0 & 6 \\ 4 & 12 & 12 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 6 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [41] $1^{1151}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.1$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 6 & 30 & 1 & 5 \\ 6 & 4 & 30 & 0 & 4 \\ 30 & 30 & 148 & 0 & 20 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$
- [52] $1^{1172}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 14 & 4 & 1 & 3 \\ 2 & 0 & 14 & 4 & 0 & 2 \\ 14 & 14 & 96 & 28 & 0 & 14 \\ 4 & 4 & 28 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 14 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [40] $1^{1441}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 24 & 24 & 1 & 5 \\ 24 & 94 & 96 & 0 & 16 \\ 24 & 96 & 94 & 0 & 16 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 16 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$
- [39] $1^{1143}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 8 & 6 & 1 & 3 \\ 2 & 0 & 8 & 6 & 0 & 2 \\ 8 & 8 & 30 & 24 & 0 & 8 \\ 6 & 6 & 24 & 15 & 0 & 5 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$
- [38] $1^{1341}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 18 & 24 & 1 & 5 \\ 18 & 52 & 72 & 0 & 12 \\ 24 & 72 & 94 & 0 & 16 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 12 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$
- [37] $1^{1461}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 24 & 36 & 1 & 5 \\ 24 & 94 & 144 & 0 & 16 \\ 36 & 144 & 214 & 0 & 24 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 16 & 24 & -1 & 1 \end{pmatrix}_{b_0}$$
- [36] $1^{1142}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 8 & 4 & 1 & 3 \\ 2 & 0 & 8 & 4 & 0 & 2 \\ 8 & 8 & 30 & 16 & 0 & 8 \\ 4 & 4 & 16 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [35] $1^{1551}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.3$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 30 & 30 & 1 & 5 \\ 30 & 148 & 150 & 0 & 20 \\ 30 & 150 & 148 & 0 & 20 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 20 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$
- [34] $1^{1141}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 6 & 24 & 1 & 5 \\ 6 & 4 & 24 & 0 & 4 \\ 24 & 24 & 94 & 0 & 16 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{11111}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 6 & 6 & 1 & 5 \\ 6 & 4 & 6 & 0 & 4 \\ 6 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{12421}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.7$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 4 & 8 & 4 & 1 & 3 \\ 4 & 6 & 16 & 8 & 0 & 4 \\ 8 & 16 & 30 & 16 & 0 & 8 \\ 4 & 8 & 16 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 8 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{12311}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.3$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 22 & 36 & 0 & 8 \\ 18 & 36 & 52 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{15711}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 30 & 42 & 1 & 5 \\ 30 & 148 & 210 & 0 & 20 \\ 42 & 210 & 292 & 0 & 28 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 20 & 28 & -1 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{11221}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 4 & 4 & 1 & 3 \\ 2 & 0 & 4 & 4 & 0 & 2 \\ 4 & 4 & 6 & 8 & 0 & 4 \\ 4 & 4 & 8 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{11321}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 6 & 4 & 1 & 3 \\ 2 & 0 & 6 & 4 & 0 & 2 \\ 6 & 6 & 16 & 12 & 0 & 6 \\ 4 & 4 & 12 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{11211}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 6 & 12 & 1 & 5 \\ 6 & 4 & 12 & 0 & 4 \\ 12 & 12 & 22 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{11311}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 6 & 18 & 1 & 5 \\ 6 & 4 & 18 & 0 & 4 \\ 18 & 18 & 52 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{12221}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 4 & 4 & 4 & 1 & 3 \\ 4 & 6 & 8 & 8 & 0 & 4 \\ 4 & 8 & 6 & 8 & 0 & 4 \\ 4 & 8 & 8 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{14711}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.3$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 24 & 42 & 1 & 5 \\ 24 & 94 & 168 & 0 & 16 \\ 42 & 168 & 292 & 0 & 28 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 16 & 28 & -1 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{12511}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.6$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 12 & 30 & 1 & 5 \\ 12 & 22 & 60 & 0 & 8 \\ 30 & 60 & 148 & 0 & 20 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 8 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{15611}1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.4$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 5 & 30 & 36 & 1 & 5 \\ 30 & 148 & 180 & 0 & 20 \\ 36 & 180 & 214 & 0 & 24 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 20 & 24 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{1162}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 1 & 2 & 12 & 4 & 1 & 3 \\ 2 & 0 & 12 & 4 & 0 & 2 \\ 12 & 12 & 70 & 24 & 0 & 12 \\ 4 & 4 & 24 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 12 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{1361}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.1$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 5 & 18 & 36 & 1 & 5 \\ 18 & 52 & 108 & 0 & 12 \\ 36 & 108 & 214 & 0 & 24 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 12 & 24 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1232}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 1 & 4 & 6 & 4 & 1 & 3 \\ 4 & 6 & 12 & 8 & 0 & 4 \\ 6 & 12 & 16 & 12 & 0 & 6 \\ 4 & 8 & 12 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1771}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.7$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 5 & 42 & 42 & 1 & 5 \\ 42 & 292 & 294 & 0 & 28 \\ 42 & 294 & 292 & 0 & 28 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 28 & 28 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{1133}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 1 & 2 & 6 & 6 & 1 & 3 \\ 2 & 0 & 6 & 6 & 0 & 2 \\ 6 & 6 & 16 & 18 & 0 & 6 \\ 6 & 6 & 18 & 15 & 0 & 5 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1261}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 5 & 12 & 36 & 1 & 5 \\ 12 & 22 & 72 & 0 & 8 \\ 36 & 72 & 214 & 0 & 24 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 8 & 24 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{1241}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 5 & 12 & 24 & 1 & 5 \\ 12 & 22 & 48 & 0 & 8 \\ 24 & 48 & 94 & 0 & 16 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 8 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{1152}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.3$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 1 & 2 & 10 & 4 & 1 & 3 \\ 2 & 0 & 10 & 4 & 0 & 2 \\ 10 & 10 & 48 & 20 & 0 & 10 \\ 4 & 4 & 20 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 10 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{1671}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.6$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 5 & 36 & 42 & 1 & 5 \\ 36 & 214 & 252 & 0 & 24 \\ 42 & 252 & 292 & 0 & 28 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 24 & 28 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1451}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.1$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 5 & 24 & 30 & 1 & 5 \\ 24 & 94 & 120 & 0 & 16 \\ 30 & 120 & 148 & 0 & 20 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 16 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1262}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 89, \Delta = -131, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.9$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 89, T \in [11, 89]$$

$$\begin{pmatrix} 1 & 4 & 12 & 4 & 1 & 3 \\ 4 & 6 & 24 & 8 & 0 & 4 \\ 12 & 24 & 70 & 24 & 0 & 12 \\ 4 & 8 & 24 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 12 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

- [10] $1^{1351}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.9$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$
 $\begin{pmatrix} 5 & 18 & 30 & 1 & 5 \\ 18 & 52 & 90 & 0 & 12 \\ 30 & 90 & 148 & 0 & 20 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 12 & 20 & -1 & 1 \end{pmatrix}_{b_0}$
- [11] $1^{1151}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$
 $\begin{pmatrix} 1 & 2 & 2 & 10 & 1 & 3 \\ 2 & 0 & 2 & 10 & 0 & 2 \\ 2 & 2 & 0 & 10 & 0 & 2 \\ 10 & 10 & 10 & 47 & 0 & 9 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 9 & -1 & 0 \end{pmatrix}_{b_0}$
- [12] $1^{1141}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.6$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$
 $\begin{pmatrix} 1 & 2 & 2 & 8 & 1 & 3 \\ 2 & 0 & 2 & 8 & 0 & 2 \\ 2 & 2 & 0 & 8 & 0 & 2 \\ 8 & 8 & 8 & 29 & 0 & 7 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 7 & -1 & 0 \end{pmatrix}_{b_0}$
- [13] $1^{1331}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.6$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$
 $\begin{pmatrix} 5 & 18 & 18 & 1 & 5 \\ 18 & 52 & 54 & 0 & 12 \\ 18 & 54 & 52 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 12 & 12 & -1 & 1 \end{pmatrix}_{b_0}$
- [14] $1^{1221}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$
 $\begin{pmatrix} 5 & 12 & 12 & 1 & 5 \\ 12 & 22 & 24 & 0 & 8 \\ 12 & 24 & 22 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$
- [30] $1^{1171}1^1 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.3$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$
 $\begin{pmatrix} 5 & 6 & 42 & 1 & 5 \\ 6 & 4 & 42 & 0 & 4 \\ 42 & 42 & 292 & 0 & 28 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 28 & -1 & 1 \end{pmatrix}_{b_0}$
- [58] $1^{1121}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 26$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 10.54 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$
 $\begin{pmatrix} 5 & 7 & 12 & 1 & 5 \\ 7 & 4 & 12 & 0 & 4 \\ 12 & 12 & 22 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$
- [59] $1^{1441}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 26$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 10.54 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$
 $\begin{pmatrix} 5 & 24 & 25 & 1 & 5 \\ 24 & 94 & 96 & 0 & 16 \\ 25 & 96 & 94 & 0 & 16 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 16 & 16 & -1 & 1 \end{pmatrix}_{b_0}$
- [60] $1^{1112}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 32$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 10.38 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$
 $\begin{pmatrix} 1 & 2 & 3 & 4 & 1 & 3 \\ 2 & 0 & 2 & 4 & 0 & 2 \\ 3 & 2 & 0 & 4 & 0 & 2 \\ 4 & 4 & 4 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 2 & 3 & -1 & 0 \end{pmatrix}_{b_0}$
- [61] $1^{1332}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 32$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 10.37 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$
 $\begin{pmatrix} 1 & 6 & 7 & 4 & 1 & 3 \\ 6 & 16 & 18 & 12 & 0 & 6 \\ 7 & 18 & 16 & 12 & 0 & 6 \\ 4 & 12 & 12 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 6 & 3 & -1 & 0 \end{pmatrix}_{b_0}$
- [62] $1^{1123}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 36$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$
 $\begin{pmatrix} 1 & 2 & 5 & 6 & 1 & 3 \\ 2 & 0 & 4 & 6 & 0 & 2 \\ 5 & 4 & 6 & 12 & 0 & 4 \\ 6 & 6 & 12 & 15 & 0 & 5 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 5 & -1 & 0 \end{pmatrix}_{b_0}$
- [63] $1^{1222}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 38$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 10.37 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 4 & 5 & 4 & 1 & 3 \\ 4 & 6 & 8 & 8 & 0 & 4 \\ 5 & 8 & 6 & 8 & 0 & 4 \\ 4 & 8 & 8 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{1111}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 44$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 6 & 7 & 1 & 5 \\ 6 & 4 & 6 & 0 & 4 \\ 7 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{1331}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 44$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.6$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 18 & 19 & 1 & 5 \\ 18 & 52 & 54 & 0 & 12 \\ 19 & 54 & 52 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 12 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{1221}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 50$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 10.44 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 12 & 13 & 1 & 5 \\ 12 & 22 & 24 & 0 & 8 \\ 13 & 24 & 22 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 8 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

[67] $1^{1122}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 56$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 10.36 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 5 & 4 & 1 & 3 \\ 2 & 0 & 4 & 4 & 0 & 2 \\ 5 & 4 & 6 & 8 & 0 & 4 \\ 4 & 4 & 8 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 4 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{1342}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 56$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 10.36 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 6 & 9 & 4 & 1 & 3 \\ 6 & 16 & 24 & 12 & 0 & 6 \\ 9 & 24 & 30 & 16 & 0 & 8 \\ 4 & 12 & 16 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 6 & 8 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{1133}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 60$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 7 & 6 & 1 & 3 \\ 2 & 0 & 6 & 6 & 0 & 2 \\ 7 & 6 & 16 & 18 & 0 & 6 \\ 6 & 6 & 18 & 15 & 0 & 5 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{1232}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 62$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 10.35 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 4 & 7 & 4 & 1 & 3 \\ 4 & 6 & 12 & 8 & 0 & 4 \\ 7 & 12 & 16 & 12 & 0 & 6 \\ 4 & 8 & 12 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 6 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{1341}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 68$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 10.41 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 18 & 25 & 1 & 5 \\ 18 & 52 & 72 & 0 & 12 \\ 25 & 72 & 94 & 0 & 16 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 12 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{1121}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 68$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 10.41 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 6 & 13 & 1 & 5 \\ 6 & 4 & 12 & 0 & 4 \\ 13 & 12 & 22 & 0 & 8 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{1231}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 74$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.3$
 $T_{\min} = 10.41 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 12 & 19 & 1 & 5 \\ 12 & 22 & 36 & 0 & 8 \\ 19 & 36 & 52 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 8 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{1132}1^{11} + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 80$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 10.35 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 7 & 4 & 1 & 3 \\ 2 & 0 & 6 & 4 & 0 & 2 \\ 7 & 6 & 16 & 12 & 0 & 6 \\ 4 & 4 & 12 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 6 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

- [75] $1^{1143}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 84$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 9 & 6 & 1 & 3 \\ 2 & 0 & 8 & 6 & 0 & 2 \\ 9 & 8 & 30 & 24 & 0 & 8 \\ 6 & 6 & 24 & 15 & 0 & 5 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$
- [76] $1^{1242}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 86$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.7$
 $T_{\min} = 10.35 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 4 & 9 & 4 & 1 & 3 \\ 4 & 6 & 16 & 8 & 0 & 4 \\ 9 & 16 & 30 & 16 & 0 & 8 \\ 4 & 8 & 16 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 4 & 8 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [77] $1^{1131}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 92$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 10.39 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 6 & 19 & 1 & 5 \\ 6 & 4 & 18 & 0 & 4 \\ 19 & 18 & 52 & 0 & 12 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$
- [78] $1^{1241}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 98$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 10.39 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 12 & 25 & 1 & 5 \\ 12 & 22 & 48 & 0 & 8 \\ 25 & 48 & 94 & 0 & 16 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 8 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$
- [79] $1^{1142}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 104$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 10.35 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 1 & 2 & 9 & 4 & 1 & 3 \\ 2 & 0 & 8 & 4 & 0 & 2 \\ 9 & 8 & 30 & 16 & 0 & 8 \\ 4 & 4 & 16 & 5 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 3 & 2 & 8 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$
- [80] $1^{1141}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 89$, $\Delta = -131$, $\det = 116$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 10.38 \rightarrow 11$, $T_{\max} = 89$, $T \in [11, 89]$

$$\begin{pmatrix} 5 & 6 & 25 & 1 & 5 \\ 6 & 4 & 24 & 0 & 4 \\ 25 & 24 & 94 & 0 & 16 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

1.41 $\textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$

- $T = 7$ [0] $1^{1111}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 90$, $\Delta = -130$, $\det = -68$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 11.35 \rightarrow 12$, $T_{\max} = 90$, $T \in [12, 90]$

$$\begin{pmatrix} 5 & 1 & 6 & 6 & 1 & 5 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 6 & 0 & 4 & 6 & 0 & 4 \\ 6 & 0 & 6 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 0 & 4 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

1.42 $\textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \hat{1} \rightarrow (-1)$

- $T = 7$ [0] $1^{1112}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -110$, $\det = -12$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 10.67 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 1 & 6 & 12 & 1 & 5 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 6 & 0 & 4 & 12 & 0 & 4 \\ 12 & 0 & 12 & 21 & 0 & 7 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 0 & 4 & 7 & -1 & 1 \end{pmatrix}_{b_0}$$
- [1] $1^{1111}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -110$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 5 & 1 & 6 & 6 & 1 & 5 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 6 & 0 & 4 & 6 & 0 & 4 \\ 6 & 0 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 0 & 4 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$
- [2] $1^{1121}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 19$, $\Delta = -110$, $\det = -66$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 10.73 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$[12] \ 1^{11411}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, \ T_{\max} = 19, \ \Delta = -110, \ \det = -336, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 10.68 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 5 & 6 & 25 & 6 & 1 & 5 \\ 6 & 4 & 24 & 6 & 0 & 4 \\ 25 & 24 & 94 & 24 & 0 & 16 \\ 6 & 6 & 24 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 16 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{1.43} \quad \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{su}_3 + \textcolor{red}{su}_3 + \hat{1} \rightarrow (-1)$$

$$T = 8 \quad [0] \ 1^{11111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 8, \ T_{\max} = 12, \ \Delta = -89, \ \det = 108, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 5 & 1 & 6 & 6 & 1 & 5 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 6 & 0 & 4 & 6 & 0 & 4 \\ 6 & 0 & 6 & 3 & 0 & 3 \\ 6 & 0 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 0 & 4 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{1.44} \quad \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \hat{1} \rightarrow (-1)$$

$$T = 5 \quad [0] \ 1^{11}1^{21} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, \ T_{\max} = 97, \ \Delta = -161, \ \det = -4, \ (n_+, n_-, n_0) = (1, 5, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 11, \ T_{\max} = 97, \ T \in [11, 97]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & -2 & 0 & 0 & 0 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & 0 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1241}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, \ T_{\max} = 97, \ \Delta = -161, \ \det = -7, \ (n_+, n_-, n_0) = (1, 5, 0), \ c_{\text{ext}} = 14.1$$

$$T_{\min} = 10.43 \rightarrow 11, \ T_{\max} = 97, \ T \in [11, 97]$$

$$\begin{pmatrix} 1 & 5 & 8 & 1 & 3 \\ 5 & 6 & 16 & 0 & 4 \\ 8 & 16 & 29 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 7 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{121}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, \ T_{\max} = 97, \ \Delta = -161, \ \det = -7, \ (n_+, n_-, n_0) = (1, 5, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 10.43 \rightarrow 11, \ T_{\max} = 97, \ T \in [11, 97]$$

$$\begin{pmatrix} 1 & 1 & 4 & 1 & 3 \\ 1 & -2 & 0 & 0 & 0 \\ 4 & 0 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 0 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{113}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, \ T_{\max} = 97, \ \Delta = -161, \ \det = -9, \ (n_+, n_-, n_0) = (1, 5, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 97, \ T \in [11, 97]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 0 & 6 & 0 & 2 \\ 6 & 6 & 15 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{141}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, \ T_{\max} = 97, \ \Delta = -161, \ \det = -12, \ (n_+, n_-, n_0) = (1, 5, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 97, \ T \in [11, 97]$$

$$\begin{pmatrix} 5 & 24 & 1 & 5 \\ 24 & 94 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 5 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{134}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, \ T_{\max} = 97, \ \Delta = -161, \ \det = -12, \ (n_+, n_-, n_0) = (1, 5, 0), \ c_{\text{ext}} = 14.4$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 97, \ T \in [11, 97]$$

$$\begin{pmatrix} 1 & 6 & 8 & 1 & 3 \\ 6 & 16 & 24 & 0 & 6 \\ 8 & 24 & 29 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 7 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{123}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, \ T_{\max} = 97, \ \Delta = -161, \ \det = -12, \ (n_+, n_-, n_0) = (1, 5, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 97, \ T \in [11, 97]$$

$$\begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 6 & 12 & 0 & 4 \\ 6 & 12 & 15 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{173}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, \ T_{\max} = 97, \ \Delta = -161, \ \det = -12, \ (n_+, n_-, n_0) = (1, 5, 0), \ c_{\text{ext}} = 14.3$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 97, \ T \in [11, 97]$$

$$\begin{pmatrix} 1 & 14 & 6 & 1 & 3 \\ 14 & 96 & 42 & 0 & 14 \\ 6 & 42 & 15 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 14 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{124}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, \ T_{\max} = 97, \ \Delta = -161, \ \det = -12, \ (n_+, n_-, n_0) = (1, 5, 0), \ c_{\text{ext}} = 14.1$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 97, \ T \in [11, 97]$$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 3 \\ 4 & 6 & 16 & 0 & 4 \\ 8 & 16 & 29 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 7 & -1 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{132}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 6 & 4 & 1 & 3 \\ 6 & 16 & 12 & 0 & 6 \\ 4 & 12 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{143}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 8 & 6 & 1 & 3 \\ 8 & 30 & 24 & 0 & 8 \\ 6 & 24 & 15 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{153}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.1$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 10 & 6 & 1 & 3 \\ 10 & 48 & 30 & 0 & 10 \\ 6 & 30 & 15 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 10 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{164}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 12 & 8 & 1 & 3 \\ 12 & 70 & 48 & 0 & 12 \\ 8 & 48 & 29 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 12 & 7 & -1 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{174}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.9$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 14 & 8 & 1 & 3 \\ 14 & 96 & 56 & 0 & 14 \\ 8 & 56 & 29 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 14 & 7 & -1 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{114}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.6$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 2 & 8 & 1 & 3 \\ 2 & 0 & 8 & 0 & 2 \\ 8 & 8 & 29 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 7 & -1 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{115}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 2 & 10 & 1 & 3 \\ 2 & 0 & 10 & 0 & 2 \\ 10 & 10 & 47 & 0 & 9 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 9 & -1 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{126}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 6 & 24 & 0 & 4 \\ 12 & 24 & 69 & 0 & 11 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 11 & -1 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{133}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 6 & 6 & 1 & 3 \\ 6 & 16 & 18 & 0 & 6 \\ 6 & 18 & 15 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{135}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 6 & 10 & 1 & 3 \\ 6 & 16 & 30 & 0 & 6 \\ 10 & 30 & 47 & 0 & 9 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 9 & -1 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{171}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 12.3$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 5 & 42 & 1 & 5 \\ 42 & 292 & 0 & 28 \\ 1 & 0 & -1 & -1 \\ 5 & 28 & -1 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{145}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 8 & 10 & 1 & 3 \\ 8 & 30 & 40 & 0 & 8 \\ 10 & 40 & 47 & 0 & 9 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 9 & -1 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{121}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 22 & 0 & 8 \\ 1 & 0 & -1 & -1 \\ 5 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{142}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 8 & 4 & 1 & 3 \\ 8 & 30 & 16 & 0 & 8 \\ 4 & 16 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{122}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 4 & 4 & 1 & 3 \\ 4 & 6 & 8 & 0 & 4 \\ 4 & 8 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{162}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.4$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 12 & 4 & 1 & 3 \\ 12 & 70 & 24 & 0 & 12 \\ 4 & 24 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 12 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{131}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 5 & 18 & 1 & 5 \\ 18 & 52 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 5 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{144}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.6$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 8 & 8 & 1 & 3 \\ 8 & 30 & 32 & 0 & 8 \\ 8 & 32 & 29 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 7 & -1 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{117}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.6$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 2 & 14 & 1 & 3 \\ 2 & 0 & 14 & 0 & 2 \\ 14 & 14 & 95 & 0 & 13 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 13 & -1 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{161}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 5 & 36 & 1 & 5 \\ 36 & 214 & 0 & 24 \\ 1 & 0 & -1 & -1 \\ 5 & 24 & -1 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{112}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 0 & 4 & 0 & 2 \\ 4 & 4 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{151}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.1$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 5 & 30 & 1 & 5 \\ 30 & 148 & 0 & 20 \\ 1 & 0 & -1 & -1 \\ 5 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{172}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 14 & 4 & 1 & 3 \\ 14 & 96 & 28 & 0 & 14 \\ 4 & 28 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 14 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{125}1^1 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 3 \\ 4 & 6 & 20 & 0 & 4 \\ 10 & 20 & 47 & 0 & 9 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 9 & -1 & 0 \end{pmatrix}_{b_0}$$

- [29] $1^{113}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
 $\begin{pmatrix} 1 & 2 & 6 & 1 & 3 \\ 2 & 0 & 6 & 0 & 2 \\ 6 & 6 & 15 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 5 & -1 & 0 \end{pmatrix}_{b_0}$
- [28] $1^{111}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
 $\begin{pmatrix} 5 & 6 & 1 & 5 \\ 6 & 4 & 0 & 4 \\ 1 & 0 & -1 & -1 \\ 5 & 4 & -1 & 1 \end{pmatrix}_{b_0}$
- [19] $1^{163}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
 $\begin{pmatrix} 1 & 12 & 6 & 1 & 3 \\ 12 & 70 & 36 & 0 & 12 \\ 6 & 36 & 15 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 12 & 5 & -1 & 0 \end{pmatrix}_{b_0}$
- [18] $1^{154}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.7$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
 $\begin{pmatrix} 1 & 10 & 8 & 1 & 3 \\ 10 & 48 & 40 & 0 & 10 \\ 8 & 40 & 29 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 10 & 7 & -1 & 0 \end{pmatrix}_{b_0}$
- [37] $1^{152}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.3$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
 $\begin{pmatrix} 1 & 10 & 4 & 1 & 3 \\ 10 & 48 & 20 & 0 & 10 \\ 4 & 20 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 10 & 3 & -1 & 0 \end{pmatrix}_{b_0}$
- [30] $1^{116}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.3$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
 $\begin{pmatrix} 1 & 2 & 12 & 1 & 3 \\ 2 & 0 & 12 & 0 & 2 \\ 12 & 12 & 69 & 0 & 11 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 11 & -1 & 0 \end{pmatrix}_{b_0}$
- [39] $1^{111}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -13$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 10.54 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
 $\begin{pmatrix} 5 & 1 & 1 & 5 \\ 1 & -2 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 5 & 0 & -1 & 1 \end{pmatrix}_{b_0}$
- [40] $1^{145}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -13$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 10.54 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
 $\begin{pmatrix} 1 & 9 & 10 & 1 & 3 \\ 9 & 30 & 40 & 0 & 8 \\ 10 & 40 & 47 & 0 & 9 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 9 & -1 & 0 \end{pmatrix}_{b_0}$
- [41] $1^{112}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -19$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 10.37 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
 $\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 0 & 4 & 0 & 2 \\ 4 & 4 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 2 & 3 & -1 & 0 \end{pmatrix}_{b_0}$
- [42] $1^{134}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -19$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.4$
 $T_{\min} = 10.37 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
 $\begin{pmatrix} 1 & 7 & 8 & 1 & 3 \\ 7 & 16 & 24 & 0 & 6 \\ 8 & 24 & 29 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 7 & -1 & 0 \end{pmatrix}_{b_0}$
- [43] $1^{123}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -21$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
 $\begin{pmatrix} 1 & 5 & 6 & 1 & 3 \\ 5 & 6 & 12 & 0 & 4 \\ 6 & 12 & 15 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 5 & -1 & 0 \end{pmatrix}_{b_0}$
- [44] $1^{111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -25$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 10.44 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$
 $\begin{pmatrix} 5 & 7 & 1 & 5 \\ 7 & 4 & 0 & 4 \\ 1 & 0 & -1 & -1 \\ 5 & 4 & -1 & 1 \end{pmatrix}_{b_0}$
- [45] $1^{122}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 97$, $\Delta = -161$, $\det = -31$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 12.7$
 $T_{\min} = 10.35 \rightarrow 11$, $T_{\max} = 97$, $T \in [11, 97]$

$$\begin{pmatrix} 1 & 5 & 4 & 1 & 3 \\ 5 & 6 & 8 & 0 & 4 \\ 4 & 8 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{144}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -31, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.6$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 9 & 8 & 1 & 3 \\ 9 & 30 & 32 & 0 & 8 \\ 8 & 32 & 29 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 7 & -1 & 0 \end{pmatrix}_{b_0}$$

[47] $1^{133}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -33, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 7 & 6 & 1 & 3 \\ 7 & 16 & 18 & 0 & 6 \\ 6 & 18 & 15 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[48] $1^{121}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -37, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 5 & 13 & 1 & 5 \\ 13 & 22 & 0 & 8 \\ 1 & 0 & -1 & -1 \\ 5 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{132}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -43, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 7 & 4 & 1 & 3 \\ 7 & 16 & 12 & 0 & 6 \\ 4 & 12 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 6 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{143}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -45, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 9 & 6 & 1 & 3 \\ 9 & 30 & 24 & 0 & 8 \\ 6 & 24 & 15 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{131}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -49, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 5 & 19 & 1 & 5 \\ 19 & 52 & 0 & 12 \\ 1 & 0 & -1 & -1 \\ 5 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{142}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -55, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 9 & 4 & 1 & 3 \\ 9 & 30 & 16 & 0 & 8 \\ 4 & 16 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 8 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{141}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 97, \Delta = -161, \det = -61, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 5 & 25 & 1 & 5 \\ 25 & 94 & 0 & 16 \\ 1 & 0 & -1 & -1 \\ 5 & 16 & -1 & 1 \end{pmatrix}_{b_0}$$

$T = 6$ [54] $1^{111}21^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 90, \Delta = -132, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13, T_{\max} = 90, T \in [13, 90]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 4 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

1.45 $\textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{so}}_8 + \hat{1} \rightarrow (-1)$

$T = 6$ [0] $1^{11}1^{211} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{so}}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 98, \Delta = -160, \det = 10, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.5$

$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 98, T \in [12, 98]$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 2 & 3 \\ 2 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 0 & 0 & 0 & 1 & 1 \\ 3 & 0 & -1 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{so}}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 98, \Delta = -160, \det = 46, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 98, T \in [12, 98]$

$$\begin{pmatrix} 5 & 1 & 6 & 1 & 5 \\ 1 & -2 & 0 & 0 & 0 \\ 6 & 0 & 2 & 0 & 2 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 0 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{1111}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 98$, $\Delta = -160$, $\det = 48$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 98$, $T \in [12, 98]$

$$\begin{pmatrix} 5 & 6 & 6 & 1 & 5 \\ 6 & 4 & 6 & 0 & 4 \\ 6 & 6 & 2 & 0 & 2 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{1111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 98$, $\Delta = -160$, $\det = 94$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 11.34 \rightarrow 12$, $T_{\max} = 98$, $T \in [12, 98]$

$$\begin{pmatrix} 5 & 7 & 6 & 1 & 5 \\ 7 & 4 & 6 & 0 & 4 \\ 6 & 6 & 2 & 0 & 2 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

1.46 $\textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow (-1)$

$T = 6$ [0] $1^{11}1^{211} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 96$, $\Delta = -140$, $\det = 6$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 11$, $T_{\max} = 96$, $T \in [11, 96]$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 2 & 3 \\ 2 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & 1 & 1 \\ 3 & 0 & -1 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{112}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 96$, $\Delta = -140$, $\det = 15$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 10.67 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$

$$\begin{pmatrix} 5 & 1 & 12 & 1 & 5 \\ 1 & -2 & 0 & 0 & 0 \\ 12 & 0 & 21 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 0 & 7 & -1 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{1113}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 96$, $\Delta = -140$, $\det = 21$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 10.76 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$

$$\begin{pmatrix} 5 & 7 & 18 & 1 & 5 \\ 7 & 4 & 18 & 0 & 4 \\ 18 & 18 & 51 & 0 & 11 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 11 & -1 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}1^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 96$, $\Delta = -140$, $\det = 33$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 10.73 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$

$$\begin{pmatrix} 5 & 1 & 6 & 1 & 5 \\ 1 & -2 & 0 & 0 & 0 \\ 6 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 0 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{1311}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 96$, $\Delta = -140$, $\det = 36$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 10.67 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$

$$\begin{pmatrix} 5 & 18 & 6 & 1 & 5 \\ 18 & 52 & 18 & 0 & 12 \\ 6 & 18 & 3 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 12 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{1112}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 96$, $\Delta = -140$, $\det = 36$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 10.67 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$

$$\begin{pmatrix} 5 & 6 & 12 & 1 & 5 \\ 6 & 4 & 12 & 0 & 4 \\ 12 & 12 & 21 & 0 & 7 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 4 & 7 & -1 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{1411}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 96$, $\Delta = -140$, $\det = 36$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 10.67 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$

$$\begin{pmatrix} 5 & 24 & 6 & 1 & 5 \\ 24 & 94 & 24 & 0 & 16 \\ 6 & 24 & 3 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 16 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{1211}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 96$, $\Delta = -140$, $\det = 36$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 10.67 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 22 & 12 & 0 & 8 \\ 6 & 12 & 3 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 8 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{1312}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 96$, $\Delta = -140$, $\det = 36$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 10.67 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$

$$\begin{pmatrix} 5 & 19 & 6 & 1 & 5 \\ 19 & 52 & 18 & 0 & 12 \\ 6 & 18 & 3 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 12 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{1411}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 96$, $\Delta = -140$, $\det = 177$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 10.68 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$

$$\begin{pmatrix} 5 & 25 & 6 & 1 & 5 \\ 25 & 94 & 24 & 0 & 16 \\ 6 & 24 & 3 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 16 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

1.47 $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$

$T = 7$ [0] $1^{1111}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 44$, $\Delta = -119$, $\det = -81$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 11$, $T_{\max} = 44$, $T \in [11, 44]$

$$\begin{pmatrix} 5 & 1 & 6 & 6 & 1 & 5 \\ 1 & -2 & 0 & 0 & 0 & 0 \\ 6 & 0 & 3 & 6 & 0 & 3 \\ 6 & 0 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 0 & 3 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{1111}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 44$, $\Delta = -119$, $\det = -108$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 11$, $T_{\max} = 44$, $T \in [11, 44]$

$$\begin{pmatrix} 5 & 6 & 6 & 6 & 1 & 5 \\ 6 & 4 & 6 & 6 & 0 & 4 \\ 6 & 6 & 3 & 6 & 0 & 3 \\ 6 & 6 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 3 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{1111}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1$ $T = 7$, $T_{\max} = 44$, $\Delta = -119$, $\det = -189$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 11$, $T_{\max} = 44$, $T \in [11, 44]$

$$\begin{pmatrix} 5 & 7 & 6 & 6 & 1 & 5 \\ 7 & 4 & 6 & 6 & 0 & 4 \\ 6 & 6 & 3 & 6 & 0 & 3 \\ 6 & 6 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 5 & 4 & 3 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

1.48 $\mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow (-1)$

$T = 12$ [0] $31^{1^{11}2231}13^1 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) + \hat{1} \rightarrow 1$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 8$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 9.0$

$T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 7 & 7 \\ 3 & 7 & 6 & 6 \\ 3 & 7 & 6 & 6 \end{pmatrix}_{b_0}$$

1.49 $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$

$T = 5$ [0] $1^{11}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{green}{\mathfrak{e}}_6 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 107$, $\Delta = -240$, $\det = -45$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13$, $T_{\max} = 107$, $T \in [13, 107]$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 \\ 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & -6 & 0 & -4 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & -1 & -4 & 1 & -2 \end{pmatrix}_{b_0}$$

$T = 6$ [1] $1^{11}21^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{green}{\mathfrak{e}}_6 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 101$, $\Delta = -211$, $\det = 18$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13$, $T_{\max} = 101$, $T \in [13, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 2 & 2 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 2 & 0 & 0 & 1 & 1 \\ 2 & 2 & -1 & -4 & 1 & -2 \end{pmatrix}_{b_0}$$

1.50 $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$

$T = 5$ [0] $1^{11}1^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{green}{\mathfrak{f}}_4 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 100$, $\Delta = -214$, $\det = -37$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.14 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 \\ 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & -1 & -3 & 1 & -2 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \ 1^{11}21^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{green}{f}_4 + \hat{1} \rightarrow 1 \quad T = 6, \ T_{\max} = 100, \ \Delta = -185, \ \det = 13, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.15 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 0 & 0 & 2 & 2 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 0 & 0 & 1 & 1 \\ 2 & 2 & -1 & -3 & 1 & -2 \end{pmatrix}_{b_0}$$

1.51 $su_2 + su_3 + \hat{1} \rightarrow (-1)$

$$T = 4 \quad [0] \ 1^{11}1^2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, \ T_{\max} = 98, \ \Delta = -191, \ \det = 5, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 10.40 \rightarrow 11, \ T_{\max} = 98, \ T \in [11, 98]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & -3 & 0 & -1 \\ 2 & 0 & 1 & 1 \\ 3 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{14}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, \ T_{\max} = 98, \ \Delta = -191, \ \det = 6, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 12.6$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 98, \ T \in [11, 98]$$

$$\begin{pmatrix} 1 & 8 & 1 & 3 \\ 8 & 29 & 0 & 7 \\ 1 & 0 & -1 & -1 \\ 3 & 7 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{13}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, \ T_{\max} = 98, \ \Delta = -191, \ \det = 6, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 98, \ T \in [11, 98]$$

$$\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 15 & 0 & 5 \\ 1 & 0 & -1 & -1 \\ 3 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{12}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, \ T_{\max} = 98, \ \Delta = -191, \ \det = 6, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 98, \ T \in [11, 98]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 5 & 0 & 3 \\ 1 & 0 & -1 & -1 \\ 3 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{15}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, \ T_{\max} = 98, \ \Delta = -191, \ \det = 6, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 98, \ T \in [11, 98]$$

$$\begin{pmatrix} 1 & 10 & 1 & 3 \\ 10 & 47 & 0 & 9 \\ 1 & 0 & -1 & -1 \\ 3 & 9 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{16}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, \ T_{\max} = 98, \ \Delta = -191, \ \det = 6, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 13.3$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 98, \ T \in [11, 98]$$

$$\begin{pmatrix} 1 & 12 & 1 & 3 \\ 12 & 69 & 0 & 11 \\ 1 & 0 & -1 & -1 \\ 3 & 11 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{17}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, \ T_{\max} = 98, \ \Delta = -191, \ \det = 6, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 13.6$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 98, \ T \in [11, 98]$$

$$\begin{pmatrix} 1 & 14 & 1 & 3 \\ 14 & 95 & 0 & 13 \\ 1 & 0 & -1 & -1 \\ 3 & 13 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, \ T_{\max} = 98, \ \Delta = -191, \ \det = 6, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 98, \ T \in [11, 98]$$

$$\begin{pmatrix} 5 & 1 & 5 \\ 1 & -1 & -1 \\ 5 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11}1^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, \ T_{\max} = 98, \ \Delta = -191, \ \det = 8, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.50 \rightarrow 11, \ T_{\max} = 98, \ T \in [11, 98]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -3 & 0 & -1 \\ 2 & 0 & 1 & 1 \\ 3 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [9] \ 1^{11}21^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, \ T_{\max} = 98, \ \Delta = -162, \ \det = -5, \ (n_+, n_-, n_0) = (1, 5, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.40 \rightarrow 12, \ T_{\max} = 98, \ T \in [12, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 1 \\ 2 & 2 & -1 & 1 & -1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [10] \ 1^{11}221^1 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, \ T_{\max} = 90, \ \Delta = -133, \ \det = 2, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 15, T_{\max} = 90, T \in [15, 90]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 2 \\ 2 & 3 & -1 & 2 & -3 \end{pmatrix}_{b_0}$$

1.52 $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \quad 1^{11}1^{21} + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 99, \Delta = -190, \det = -17, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 11.35 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & -3 & 0 & 0 & -1 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & -1 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 99, \Delta = -190, \det = -24, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 5 & 6 & 1 & 5 \\ 6 & 2 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 5 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 99, \Delta = -190, \det = -29, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 11.34 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 \\ 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & -1 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [3] \quad 1^{11}21^{11} + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 98, \Delta = -161, \det = 8, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 98, T \in [12, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 2 & 2 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 2 & 0 & 0 & 1 & 1 \\ 2 & 2 & -1 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

1.53 $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \quad 1^{11}1^{22} + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -170, \det = -3, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & -3 & 0 & 0 & -1 \\ 2 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & -1 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}1^{12} + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -170, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 3 \\ 1 & -3 & 0 & 0 & -1 \\ 2 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & -1 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -170, \det = -18, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 4 & 4 & 1 & 3 \\ 4 & 5 & 8 & 0 & 3 \\ 4 & 8 & 5 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 3 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -170, \det = -18, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 5 & 18 & 1 & 5 \\ 18 & 51 & 0 & 11 \\ 1 & 0 & -1 & -1 \\ 5 & 11 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{15}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -170, \det = -18, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 5 & 30 & 1 & 5 \\ 30 & 147 & 0 & 19 \\ 1 & 0 & -1 & -1 \\ 5 & 19 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11}1^{11} + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -170, \det = -18, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 21 & 0 & 7 \\ 1 & 0 & -1 & -1 \\ 5 & 7 & -1 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{111}1^1 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{green}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -170, \det = -18, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$

$$\begin{pmatrix} 5 & 6 & 1 & 5 \\ 6 & 3 & 0 & 3 \\ 1 & 0 & -1 & -1 \\ 5 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{114}1^1 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{green}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -170, \det = -18, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.6$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$

$$\begin{pmatrix} 5 & 24 & 1 & 5 \\ 24 & 93 & 0 & 15 \\ 1 & 0 & -1 & -1 \\ 5 & 15 & -1 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{111}1^1 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{green}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -170, \det = -21, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 \\ 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 1 & 1 \\ 3 & -1 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

$T = 6$ [9] $1^{11}21^{11} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{green}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 96, \Delta = -141, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12, T_{\max} = 96, T \in [12, 96]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 2 & 2 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 0 & 1 & 1 \\ 2 & 2 & -1 & -1 & 1 & -1 \end{pmatrix}_{b_0}$$

1.54 $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$

$T = 6$ [0] $1^{11}1^{111} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 99, \Delta = -169, \det = 75, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 2 & 3 \\ 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 0 & 0 & 0 & 1 & 1 \\ 3 & -1 & -1 & -2 & 1 & -1 \end{pmatrix}_{b_0}$$

1.55 $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$

$T = 6$ [0] $1^{11}1^{112} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 97, \Delta = -149, \det = 27, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 11.11 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 0 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & 1 & 1 \\ 3 & -1 & -1 & -1 & 1 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{11}1^{111} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 97, \Delta = -149, \det = 54, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 2 & 3 \\ 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & 1 & 1 \\ 3 & -1 & -1 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111}1^1 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 97, \Delta = -149, \det = 54, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 11, T_{\max} = 97, T \in [11, 97]$

$$\begin{pmatrix} 5 & 6 & 6 & 1 & 5 \\ 6 & 3 & 6 & 0 & 3 \\ 6 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 5 & 3 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

1.56 $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$

$T = 7$ [0] $1^{11}1^{1111} + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 88, \Delta = -128, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 88, T \in [12, 88]$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 2 & 3 \\ 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & 0 & 1 & 1 \\ 3 & -1 & -1 & -1 & -1 & 1 & -1 \end{pmatrix}_{b_0}$$

1.57 $\mathfrak{su}_2 + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [0] \quad 1^{11}2^{11}1^1 + \mathfrak{su}_2 + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 96, \Delta = -142, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 96, T \in [14, 96]$$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 0 & 7 \\ 1 & 0 & -1 & -1 \\ 5 & 7 & -1 & -1 \end{pmatrix}_{b_0}$$

1.58 $\mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [0] \quad 1^{11}2^{11}1^1 + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 101, \Delta = -212, \det = 12, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 2 & 1 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -2 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [1] \quad 1^{11}22^{11}1^1 + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 101, \Delta = -183, \det = -13, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 1 & 0 & 2 & 2 \\ 2 & 3 & 0 & -4 & 2 & -2 \end{pmatrix}_{b_0}$$

1.59 $\mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [0] \quad 1^{11}2^{11}1^1 + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{f}_4 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 100, \Delta = -186, \det = 9, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 2 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [1] \quad 1^{11}22^{11}1^1 + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{f}_4 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 99, \Delta = -157, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 99, T \in [13, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 1 & 0 & 2 & 2 \\ 2 & 3 & 0 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

1.60 $\mathfrak{su}_2 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \quad 1^{12}2^{11}1^1 + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -163, \det = -3, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \quad 12^{12}21^{11} + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 91, \Delta = -134, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 91, T \in [13, 91]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 3 & 1 & 2 & 2 \\ 2 & 3 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}22^{11}1^1 + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 91, \Delta = -134, \det = 14, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 91, T \in [11, 91]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 3 & 1 & 2 & 2 \\ 2 & 3 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

1.61 $\mathfrak{su}_2 \rightarrow (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [0] \quad 1^{11}2^{11}1^1 + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{(-3) \rightarrow (-2)} + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\hat{1} \rightarrow 1} \quad T = 6, \quad T_{\max} = 99, \quad \Delta = -162, \quad \det = 6, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 99, \quad T \in [13, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 2 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & 0 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [1] \quad 1^{11}22^{11}1^1 + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{(-3) \rightarrow (-2)} + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\hat{1} \rightarrow 1} \quad T = 7, \quad T_{\max} = 91, \quad \Delta = -133, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13, \quad T_{\max} = 91, \quad T \in [13, 91]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 1 & 0 & 2 & 2 \\ 2 & 3 & 0 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

1.62 $\mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$

$$T = 8 \quad [0] \quad 1^{12}23^{11}131^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{e}_6} + \textcolor{green}{\hat{1} \rightarrow 1} \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -165, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, \quad T_{\max} = 101, \quad T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -6 & 0 & -4 \\ 1 & 2 & 0 & 3 & 3 \\ 2 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

1.63 $\mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_7 + \hat{1} \rightarrow (-1)$

$$T = 9 \quad [0] \quad 1^{12}2313^{12}21^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{e}_7} + \textcolor{green}{\hat{1} \rightarrow 1} \quad T = 9, \quad T_{\max} = 103, \quad \Delta = -192, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 16.33 \rightarrow 17, \quad T_{\max} = 103, \quad T \in [17, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [1] \quad 1^{12}22313^{12}21^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{e}_7} + \textcolor{green}{\hat{1} \rightarrow 1} \quad T = 10, \quad T_{\max} = 103, \quad \Delta = -163, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 17, \quad T_{\max} = 103, \quad T \in [17, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \quad 1^{12}23^{11}141321^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{e}_7} + \textcolor{green}{\hat{1} \rightarrow 1} \quad T = 11, \quad T_{\max} = 104, \quad \Delta = -162, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 18, \quad T_{\max} = 104, \quad T \in [18, 104]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -6 & 0 & -4 \\ 1 & 2 & 0 & 3 & 3 \\ 2 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

1.64 $\mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$

$$T = 9 \quad [0] \quad 1^{12}2313^{12}21^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{e}'_7} + \textcolor{green}{\hat{1} \rightarrow 1} \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -164, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 16, \quad T_{\max} = 102, \quad T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

1.65 $\mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_8 + \hat{1} \rightarrow (-1)$

$$T = 10 \quad [0] \quad 1^1 22313^1 21^1 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_8 + \hat{1} \rightarrow 1 \quad T = 10, \quad T_{\max} = 190, \quad \Delta = -278, \quad \det = 6, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 19.67 \rightarrow 20, \quad T_{\max} = 190, \quad T \in [20, 190]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & -6 & 0 & -4 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & -4 & -1 & -4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1] \quad 1^1 23^1 1413221^1 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_8 + \hat{1} \rightarrow 1 \quad T = 12, \quad T_{\max} = 158, \quad \Delta = -248, \quad \det = 5, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 20.80 \rightarrow 21, \quad T_{\max} = 158, \quad T \in [21, 158]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -9 & 0 & -7 \\ 1 & 2 & 0 & 3 & 3 \\ 2 & 2 & -7 & 3 & -3 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [0] \quad 1^1 23^1 141413221^1 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_8 + \hat{1} \rightarrow 1 \quad T = 14, \quad T_{\max} = 109, \quad \Delta = -218, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 22, \quad T_{\max} = 109, \quad T \in [22, 109]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & -9 & 0 & -7 \\ 1 & 3 & 0 & 4 & 4 \\ 2 & 3 & -7 & 4 & -2 \end{pmatrix}_{b_0}$$

1.66 $\mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow (-1)$

$$T = 7 \quad [0] \quad 2^3 3^1 21^1 + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 7, \quad T_{\max} = 30, \quad \Delta = -117, \quad \det = -5, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 30, \quad T \in [11, 30]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 0 & 0 \\ 2 & 0 & 3 & 3 \\ 2 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [1] \quad 2^2 3^1 1^1 21^1 + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 8, \quad T_{\max} = 12, \quad \Delta = -90, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 4 & 4 \\ 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 2313^1 21^1 + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 8, \quad T_{\max} = 12, \quad \Delta = -88, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 2 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \quad 1^1 22313^1 21^1 + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -59, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 5 & 5 \\ 2 & 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \quad 1^1 23^1 141321^1 + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 5 & 5 \\ 2 & 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[5] \quad 31^1 5^1 13^1 + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -88, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, \quad T_{\max} = 14, \quad T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[6] \quad 2^1 23^1 141^1 3^1 + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -60, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 2 & 4 & 5 & 5 \\ 2 & 4 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \quad 1^{123^1} 1413221^1 + \textcolor{red}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 6 & 6 \\ 2 & 2 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[8] \quad 31^{1^{12231}} 5^{13^1} + \textcolor{red}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 11, \quad T_{\max} = 13, \quad \Delta = -59, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [9] \quad 2^{1^{123^1} 14141} 3^2 + \textcolor{red}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -30, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 \\ 2 & 5 & 6 & 6 \\ 2 & 5 & 6 & 6 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [0] \quad 321^{3^{121}} 8^{12321^1} + \textcolor{red}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 13, \quad T_{\max} = 16, \quad \Delta = -87, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 13, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 16, \quad T_{\max} = 16, \quad T \in [16, 16]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [1] \quad 2^{1^{123^1} 1414141} 3^2 + \textcolor{red}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 14, \quad T_{\max} = 14, \quad \Delta = 0, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 6 & 6 \\ 2 & 6 & 7 & 7 \\ 2 & 6 & 7 & 7 \end{pmatrix}_{b_0}$$

1.67 $\textcolor{red}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow (-2)$

$$T = 5 \quad [0] \quad 13^{12^1} 1^1 + \textcolor{red}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 2 \quad T = 5, \quad T_{\max} = 86, \quad \Delta = -128, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 5, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 12, \quad T_{\max} = 86, \quad T \in [12, 86]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \quad 123^{12^1} 1^1 + \textcolor{red}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 2 \quad T = 6, \quad T_{\max} = 18, \quad \Delta = -101, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 13, \quad T_{\max} = 18, \quad T \in [13, 18]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [2] \quad 31^{2^{11}} 5^{13^1} + \textcolor{red}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 2 \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -101, \quad \det = 6, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 13, \quad T_{\max} = 20, \quad T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \quad 31^{3^{121}} 6^{12^1} + \textcolor{red}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -100, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \quad 321^{3^{121}} 7^{12^1} + \textcolor{red}{su}_2 \rightarrow (-3) + \hat{1} \rightarrow 2 \quad T = 10, \quad T_{\max} = 18, \quad \Delta = -99, \quad \det = 6, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 15, \quad T_{\max} = 18, \quad T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [5] \quad 1 \overset{2^1 1}{8} 123^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \hat{1} \rightarrow 2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, T_{\max} = 18, T \in [16, 18]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 3221 \overset{3^1 2^1}{8} 12^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \hat{1} \rightarrow 2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, T_{\max} = 18, T \in [16, 18]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 0 & 0 \\ 2 & 0 & 2 & 2 \\ 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [7] \quad 41321 \overset{3^1 2^1}{8} 12^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \hat{1} \rightarrow 2 \quad T = 12, T_{\max} = 18, \Delta = -97, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 0 & 0 \\ 2 & 0 & 2 & 2 \\ 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [0] \quad 512321 \overset{3^1 2^1}{8} 12^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \hat{1} \rightarrow 2 \quad T = 13, T_{\max} = 18, \Delta = -94, \det = -6, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$\mathbf{1.68} \quad \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \hat{1} \rightarrow (-1)$$

$$T = 8 \quad [0] \quad 2 \overset{3^1 1^1}{3} 21^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2 \rightarrow (-3)} + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 12, \Delta = -90, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 3 & 3 \\ 2 & 0 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [1] \quad 1^1 23^1 13^1 21^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2 \rightarrow (-3)} + \hat{1} \rightarrow 1 \quad T = 9, T_{\max} = 11, \Delta = -61, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 3 & 4 & 4 \\ 2 & 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \quad 1^1 23^1 1413^1 21^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2 \rightarrow (-3)} + \hat{1} \rightarrow 1 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[3] \quad 3^1 1^1 \overset{1^1 2^1 3^1}{5} 13^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2 \rightarrow (-3)} + \hat{1} \rightarrow 1 \quad T = 11, T_{\max} = 13, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -2 & 0 & 0 \\ 3 & 0 & 5 & 5 \\ 3 & 0 & 5 & 5 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [0] \quad 1^1 23^1 14141413^1 21^1 + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2 \rightarrow (-3)} + \hat{1} \rightarrow 1 \quad T = 14, T_{\max} = 14, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 9.0$$

$$\begin{aligned}
[1] \quad & 1^1 2321^1 8^1 123^1 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) + \hat{1} \rightarrow 1 \quad T = 14, T_{\max} = 16, \Delta = -60, \det = 3, (n_+, n_-, n_0) = \\
& (1, 14, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -2 & 0 & 0 \\ 3 & 0 & 5 & 5 \\ 3 & 0 & 5 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

1.69 $\mathfrak{su}_3 + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$

$$\begin{aligned}
T = 4 \quad & [0] \quad 1^1 1^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{e}}_6 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 179, \Delta = -270, \det = 18, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 13, T_{\max} = 179, T \in [13, 179] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -5 & 0 & -3 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -3 & -1 & -3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
T = 5 \quad & [1] \quad 1^1 1^1 21^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{e}}_6 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 112, \Delta = -241, \det = -9, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 13, T_{\max} = 112, T \in [13, 112] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 0 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 3 & 0 & -1 & -1 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
T = 8 \quad & [2] \quad 1^1 2321^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{e}}_6 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 101, \Delta = -165, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0 \\
& T_{\min} = 15, T_{\max} = 101, T \in [15, 101] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & -3 & 1 & -1 \\ 1 & 0 & 1 & 2 & 2 \\ 2 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}
\end{aligned}$$

1.70 $\mathfrak{su}_3 + \mathfrak{e}_7 + \hat{1} \rightarrow (-1)$

$$\begin{aligned}
T = 8 \quad & [0] \quad 1^1 2321^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{e}}_7 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 103, \Delta = -220, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 16, T_{\max} = 103, T \in [16, 103] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & -5 & 1 & -3 \\ 1 & 0 & 1 & 2 & 2 \\ 2 & -1 & -3 & 2 & -2 \end{pmatrix}_{b_0}
\end{aligned}$$

1.71 $\mathfrak{su}_3 + \mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$

$$\begin{aligned}
T = 8 \quad & [0] \quad 1^1 2321^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{e}}'_7 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 102, \Delta = -192, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0 \\
& T_{\min} = 15.33 \rightarrow 16, T_{\max} = 102, T \in [16, 102] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & -4 & 1 & -2 \\ 1 & 0 & 1 & 2 & 2 \\ 2 & -1 & -2 & 2 & -2 \end{pmatrix}_{b_0}
\end{aligned}$$

1.72 $\mathfrak{su}_3 + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$

$$\begin{aligned}
T = 4 \quad & [0] \quad 1^1 1^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{f}}_4 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 133, \Delta = -244, \det = 15, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 14.2 \\
& T_{\min} = 12.13 \rightarrow 13, T_{\max} = 133, T \in [13, 133] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -4 & 0 & -2 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -2 & -1 & -3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
T = 5 \quad & [1] \quad 1^1 1^1 21^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{f}}_4 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 100, \Delta = -215, \det = -7, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.2 \\
& T_{\min} = 12.14 \rightarrow 13, T_{\max} = 100, T \in [13, 100] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 1 & 0 & 1 \\ 1 & 0 & -1 & -1 \\ 3 & 1 & -1 & -1 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
T = 7 \quad & [2] \quad 1^1 321^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{f}}_4 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 100, \Delta = -166, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.2 \\
& T_{\min} = 14, T_{\max} = 100, T \in [14, 100]
\end{aligned}$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -5 & 0 & -3 \\ 2 & 0 & 3 & 3 \\ 3 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

1.73 $\mathfrak{su}_3 + \hat{1} \rightarrow (-1)$

$$T = 3 \quad [1] \quad 1^2 1^1 + \mathfrak{su}_3 + \hat{1} \rightarrow 1 \quad T = 3, T_{\max} = 98, \Delta = -221, \det = -3, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ 1 & 0 & -1 & -1 \\ 2 & 1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^1 1^1 + \mathfrak{su}_3 + \hat{1} \rightarrow 1 \quad T = 3, T_{\max} = 98, \Delta = -221, \det = -3, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 4 \quad [2] \quad 1^1 21^1 + \mathfrak{su}_3 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -192, \det = 2, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [3] \quad 2^1 231^1 + \mathfrak{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 96, \Delta = -143, \det = 6, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 96, T \in [11, 96]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [4] \quad 1^1 2^1 321^1 + \mathfrak{su}_3 + \hat{1} \rightarrow 1 \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 27, T \in [11, 27]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -3 & 0 & -1 \\ 2 & 0 & 7 & 5 \\ 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [5] \quad 2^1 3^1 21^1 + \mathfrak{su}_3 + \hat{1} \rightarrow 1 \quad T = 8, T_{\max} = 25, \Delta = -115, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 25, T \in [12, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [6] \quad 1^1 231^1 51321^1 + \mathfrak{su}_3 + \hat{1} \rightarrow 1 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -3 & 0 & -1 \\ 2 & 0 & 9 & 7 \\ 2 & -1 & 7 & 5 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [7] \quad 1^1 231^1 513221^1 + \mathfrak{su}_3 + \hat{1} \rightarrow 1 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -3 & 0 & -1 \\ 2 & 0 & 10 & 8 \\ 2 & -1 & 8 & 6 \end{pmatrix}_{b_0}$$

1.74 $\mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$

$$T = 4 \quad [1] \quad 1^1 1^1 1^1 + \mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 99, \Delta = -220, \det = 12, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^1 1^1 1^1 + \mathfrak{su}_3 + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 99, \Delta = -220, \det = 12, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 1 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [2] \quad 1^{11}1^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{so}}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 99, \Delta = -191, \det = -5, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 2 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 3 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

1.75 $\mathfrak{su}_3 + \mathfrak{su}_3 + \hat{1} \rightarrow (-1)$

$$T = 4 \quad [1] \quad 1^{11}1^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -200, \det = 9, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{14}1^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -200, \det = 9, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 13.6$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 4 & 0 & 0 \\ 4 & 4 & 13 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{16}1^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -200, \det = 9, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 14.3$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 6 & 1 & 2 \\ 1 & -2 & 6 & 0 & 0 \\ 6 & 6 & 33 & 0 & 5 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & 5 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{23}1^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -200, \det = 9, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 2 & 3 & 1 & 2 \\ 2 & 1 & 6 & 0 & 1 \\ 3 & 6 & 6 & 0 & 2 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 1 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{24}1^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -200, \det = 9, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 13.6$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 2 & 4 & 1 & 2 \\ 2 & 1 & 8 & 0 & 1 \\ 4 & 8 & 13 & 0 & 3 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 1 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{13}1^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -200, \det = 9, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & -2 & 3 & 0 & 0 \\ 3 & 3 & 6 & 0 & 2 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{22}1^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -200, \det = 9, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 0 & 1 \\ 2 & 4 & 1 & 0 & 1 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 1 & 1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{12}1^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -200, \det = 9, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & -2 & 2 & 0 & 0 \\ 2 & 2 & 1 & 0 & 1 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & 1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{17}1^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -200, \det = 9, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 14.6$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 7 & 1 & 2 \\ 1 & -2 & 7 & 0 & 0 \\ 7 & 7 & 46 & 0 & 6 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{15}1^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -200, \det = 9, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 5 & 1 & 2 \\ 1 & -2 & 5 & 0 & 0 \\ 5 & 5 & 22 & 0 & 4 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [10] \quad 1^{11}21^1 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -171, \det = -3, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 3 & 0 & 3 \\ 1 & 0 & -1 & -1 \\ 3 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

1.76 $\textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{green}{so}_8 + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \quad 1^{11}1^1 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{green}{so}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 99, \Delta = -199, \det = -36, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 1 & 0 & 0 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 0 & 0 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \quad 1^{11}21^1 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{green}{so}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 99, \Delta = -170, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 3 & 6 & 0 & 3 \\ 3 & 6 & 2 & 0 & 2 \\ 1 & 0 & 0 & -1 & -1 \\ 3 & 3 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

1.77 $\textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{green}{su}_3 + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [1] \quad 1^{11}21^1 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -179, \det = -27, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 2 & 0 & 0 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 2 & 2 & 2 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 0 & 0 & 1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{11}31^1 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -179, \det = -27, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 3 & 0 & 0 \\ 1 & 1 & -2 & 3 & 0 & 0 \\ 3 & 3 & 3 & 6 & 0 & 2 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 0 & 0 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}1^1 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{green}{su}_3 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -179, \det = -27, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 1 & 0 & 0 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 0 & 0 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

1.78 $\textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{green}{su}_3 + \textcolor{red}{su}_3 + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [0] \quad 1^{11}1^1 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{green}{su}_3 + \textcolor{red}{su}_3 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 98, \Delta = -158, \det = 81, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 98, T \in [12, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & -2 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -1 & -1 \\ 2 & 0 & 0 & 0 & 0 & -1 & -2 \end{pmatrix}_{b_0}$$

1.79 $\textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [0] \quad 1^{11}21^1 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 97, \Delta = -151, \det = 9, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 12 & 0 & 6 \\ 1 & 0 & -1 & -1 \\ 5 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

1.80 $\mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \quad 1^1 2^1 1^1 + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -172, \det = -6, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 98, T \in [12, 98]$$

$$\begin{pmatrix} 5 & 1 & 5 \\ 1 & -1 & -1 \\ 5 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \quad 1^1 2^2 1^1 + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 96, \Delta = -143, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 96, T \in [14, 96]$$

$$\begin{pmatrix} 0 & 1 & 1 & 0 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -1 & -2 \end{pmatrix}_{b_0}$$

1.81 $\mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [0] \quad 1^1 2^1 1^1 + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} + \textcolor{green}{\mathfrak{so}_8} + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 99, \Delta = -171, \det = 15, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 11 & 0 & 5 \\ 1 & 0 & -1 & -1 \\ 5 & 5 & -1 & 1 \end{pmatrix}_{b_0}$$

1.82 $(-3) \rightarrow (-2) + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \quad 1^1 2^1 1^1 + \textcolor{red}{(-3)} \rightarrow \textcolor{red}{(-2)} + \textcolor{blue}{\mathfrak{e}_6} + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 123, \Delta = -242, \det = -15, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 123, T \in [14, 123]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -4 & 0 & -2 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -2 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \quad 1^1 2^1 21^1 + \textcolor{red}{(-3)} \rightarrow \textcolor{red}{(-2)} + \textcolor{blue}{\mathfrak{e}_6} + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 101, \Delta = -213, \det = 7, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 1 & 0 & 1 \\ 1 & 0 & -1 & -1 \\ 3 & 1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 2^1 21^1 + \textcolor{red}{(-3)} \rightarrow \textcolor{red}{(-2)} + \textcolor{blue}{\mathfrak{e}_6} + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 101, \Delta = -213, \det = 12, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -6 & 0 & -4 \\ 3 & 0 & 6 & 4 \\ 3 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

1.83 $(-3) \rightarrow (-2) + \mathfrak{e}_7 + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \quad 1^1 2^1 1^1 + \textcolor{red}{(-3)} \rightarrow \textcolor{red}{(-2)} + \textcolor{blue}{\mathfrak{e}_7} + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 187, \Delta = -297, \det = -21, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 187, T \in [15, 187]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -6 & 0 & -4 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -4 & -1 & -4 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \quad 1^1 2^1 21^1 + \textcolor{red}{(-3)} \rightarrow \textcolor{red}{(-2)} + \textcolor{blue}{\mathfrak{e}_7} + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 180, \Delta = -268, \det = 11, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 180, T \in [15, 180]$$

$$\begin{pmatrix} 10 & 1 & 0 \\ 1 & -1 & -1 \\ 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\textcolor{blue}{1}}2^{\textcolor{red}{1}}21^{\textcolor{blue}{1}} + (-\textcolor{red}{3}) \rightarrow (-\textcolor{red}{2}) + \textcolor{blue}{e}_7 + \hat{\textcolor{green}{1}} \rightarrow \textcolor{green}{1} \quad T = 6, \ T_{\max} = 180, \ \Delta = -268, \ \det = 18, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15, \ T_{\max} = 180, \ T \in [15, 180]$$

$$\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -8 & 0 & -6 \\ 3 & 0 & 6 & 4 \\ 3 & -6 & 4 & -2 \end{pmatrix}_{b_0}$$

1.84 $(-3) \rightarrow (-2) + \textcolor{blue}{e}'_7 + \hat{\textcolor{green}{1}} \rightarrow (-1)$

$$T = 5 \quad [0] \ 1^{\textcolor{blue}{1}}2^{\textcolor{red}{1}}1^{\textcolor{blue}{1}} + (-\textcolor{red}{3}) \rightarrow (-\textcolor{red}{2}) + \textcolor{blue}{e}'_7 + \hat{\textcolor{green}{1}} \rightarrow \textcolor{green}{1} \quad T = 5, \ T_{\max} = 179, \ \Delta = -269, \ \det = -18, \ (n_+, n_-, n_0) = (1, 5, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14, \ T_{\max} = 179, \ T \in [14, 179]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -5 & 0 & -3 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -3 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \ 1^{\textcolor{blue}{1}}2^{\textcolor{red}{1}}21^{\textcolor{blue}{1}} + (-\textcolor{red}{3}) \rightarrow (-\textcolor{red}{2}) + \textcolor{blue}{e}'_7 + \hat{\textcolor{green}{1}} \rightarrow \textcolor{green}{1} \quad T = 6, \ T_{\max} = 108, \ \Delta = -240, \ \det = 9, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14, \ T_{\max} = 108, \ T \in [14, 108]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 0 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 3 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\textcolor{blue}{1}}2^{\textcolor{red}{1}}21^{\textcolor{blue}{1}} + (-\textcolor{red}{3}) \rightarrow (-\textcolor{red}{2}) + \textcolor{blue}{e}'_7 + \hat{\textcolor{green}{1}} \rightarrow \textcolor{green}{1} \quad T = 6, \ T_{\max} = 108, \ \Delta = -240, \ \det = 15, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.13 \rightarrow 15, \ T_{\max} = 108, \ T \in [15, 108]$$

$$\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -7 & 0 & -5 \\ 3 & 0 & 6 & 4 \\ 3 & -5 & 4 & -2 \end{pmatrix}_{b_0}$$

1.85 $(-3) \rightarrow (-2) + \textcolor{blue}{f}_4 + \hat{\textcolor{green}{1}} \rightarrow (-1)$

$$T = 5 \quad [0] \ 1^{\textcolor{blue}{1}}2^{\textcolor{red}{1}}1^{\textcolor{blue}{1}} + (-\textcolor{red}{3}) \rightarrow (-\textcolor{red}{2}) + \textcolor{blue}{f}_4 + \hat{\textcolor{green}{1}} \rightarrow \textcolor{green}{1} \quad T = 5, \ T_{\max} = 100, \ \Delta = -216, \ \det = -12, \ (n_+, n_-, n_0) = (1, 5, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.33 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \ 1^{\textcolor{blue}{1}}2^{\textcolor{red}{1}}21^{\textcolor{blue}{1}} + (-\textcolor{red}{3}) \rightarrow (-\textcolor{red}{2}) + \textcolor{blue}{f}_4 + \hat{\textcolor{green}{1}} \rightarrow \textcolor{green}{1} \quad T = 6, \ T_{\max} = 100, \ \Delta = -187, \ \det = 5, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 2 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 3 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\textcolor{blue}{1}}2^{\textcolor{red}{1}}21^{\textcolor{blue}{1}} + (-\textcolor{red}{3}) \rightarrow (-\textcolor{red}{2}) + \textcolor{blue}{f}_4 + \hat{\textcolor{green}{1}} \rightarrow \textcolor{green}{1} \quad T = 6, \ T_{\max} = 100, \ \Delta = -187, \ \det = 9, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.67 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -5 & 0 & -3 \\ 3 & 0 & 6 & 4 \\ 3 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

1.86 $(-3) \rightarrow (-2) + \hat{\textcolor{green}{1}} \rightarrow (-1)$

$$T = 4 \quad [0] \ 12^{\textcolor{blue}{1}}1^{\textcolor{blue}{1}} + (-\textcolor{red}{3}) \rightarrow (-\textcolor{red}{2}) + \hat{\textcolor{blue}{1}} \rightarrow \textcolor{blue}{1} \quad T = 4, \ T_{\max} = 98, \ \Delta = -193, \ \det = 3, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.33 \rightarrow 12, \ T_{\max} = 98, \ T \in [12, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 2 & 0 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\textcolor{blue}{1}}2^{\textcolor{red}{1}}1^{\textcolor{blue}{1}} + (-\textcolor{red}{3}) \rightarrow (-\textcolor{red}{2}) + \hat{\textcolor{blue}{1}} \rightarrow \textcolor{blue}{1} \quad T = 4, \ T_{\max} = 98, \ \Delta = -193, \ \det = 11, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.36 \rightarrow 11, \ T_{\max} = 98, \ T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 1 \\ 2 & 0 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 21^{\textcolor{blue}{1}}2^{\textcolor{red}{1}} + (-\textcolor{red}{3}) \rightarrow (-\textcolor{red}{2}) + \hat{\textcolor{blue}{1}} \rightarrow \textcolor{blue}{1} \quad T = 4, \ T_{\max} = 98, \ \Delta = -193, \ \det = 11, \ (n_+, n_-, n_0) = (1, 4, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.36 \rightarrow 11, \ T_{\max} = 98, \ T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 1 \\ 2 & 0 & 1 & 0 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& T = 5 \quad [3] \quad 12^1 21^1 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -164, \det = -2, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12, T_{\max} = 98, T \in [12, 98] \\
& \begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0} \\
& [4] \quad 122^1 1^1 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -164, \det = -3, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12.33 \rightarrow 13, T_{\max} = 98, T \in [13, 98] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 4 \\ 3 & 4 & 0 \end{pmatrix}_{b_0} \\
& [5] \quad 1^1 2^1 21^1 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -164, \det = -10, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.40 \rightarrow 11, T_{\max} = 98, T \in [11, 98] \\
& \begin{pmatrix} 1 & 4 & 3 \\ 4 & 6 & 4 \\ 3 & 4 & 2 \end{pmatrix}_{b_0} \\
\\
& T = 14 \quad [0] \quad 2^1 1^1 \quad 8^1 \quad 12321^1 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0} \\
\\
& \mathbf{1.87} \quad (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow (-1) \\
& T = 5 \quad [0] \quad 1^1 2^1 1^1 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 99, \Delta = -192, \det = -9, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.0 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 99, T \in [12, 99] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & 0 & -1 & -1 \end{pmatrix}_{b_0} \\
\\
& T = 6 \quad [1] \quad 1^1 2^1 21^1 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 99, \Delta = -163, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.0 \\
& T_{\min} = 12, T_{\max} = 99, T \in [12, 99] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 3 & 0 & 3 \\ 1 & 0 & -1 & -1 \\ 3 & 3 & -1 & 1 \end{pmatrix}_{b_0} \\
& [2] \quad 1^1 2^1 21^1 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 99, \Delta = -163, \det = 6, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.0 \\
& T_{\min} = 12.33 \rightarrow 13, T_{\max} = 99, T \in [13, 99] \\
& \begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -4 & 0 & -2 \\ 3 & 0 & 6 & 4 \\ 3 & -2 & 4 & 0 \end{pmatrix}_{b_0}
\end{aligned}$$

2 $\mathfrak{su}_8 \rightarrow \mathfrak{so}_{16}$ External (7 combos, 12 bases)

2.1 $\mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$

$$T = 3 \quad [0] \quad 212^2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8 \quad T = 3, T_{\max} = 110, \Delta = -241, \det = -4, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 110, T \in [11, 110]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 12^2 1 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8 \quad T = 3, T_{\max} = 110, \Delta = -241, \det = -4, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 110, T \in [11, 110]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}_{b_0}$$

2.2 $\mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$

$$T = 4 \quad [0] \quad 12^2 1^1 + \mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8 \quad T = 4, T_{\max} = 99, \Delta = -211, \det = 8, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 99, T \in [11, 99]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 0 \\ 2 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

2.3 $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$

$$T = 5 \quad [0] \quad 1^1 2^1 1^1 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8 \quad T = 5, T_{\max} = 98, \Delta = -181, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

2.4 $\mathfrak{su}_2 + \mathfrak{su}_8 \rightarrow \mathfrak{so}_{16}$

$$T = 7 \quad [0] \quad 4^2 12231^1 + \mathfrak{su}_2 + \mathfrak{su}_8 \rightarrow \mathfrak{so}_{16} \quad T = 7, T_{\max} = 91, \Delta = -133, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 2.5$$

$$T_{\min} = 14, T_{\max} = 91, T \in [14, 91]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 0 & 0 \\ 2 & 0 & 6 & 4 \\ 2 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

2.5 $(-3) \rightarrow (-2) + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$

$$T = 4 \quad [0] \quad 2^1 12^2 + (-3) \rightarrow (-2) + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8 \quad T = 4, T_{\max} = 99, \Delta = -221, \det = 12, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 0 & 0 \\ 2 & 0 & -4 & -2 \\ 2 & 0 & -2 & -1 \end{pmatrix}_{b_0}$$

2.6 $\mathfrak{su}_8 \rightarrow \mathfrak{so}_{16}$

$$T = 5 \quad [4] \quad 22214^2 + \mathfrak{su}_8 \rightarrow \mathfrak{so}_{16} \quad T = 5, T_{\max} = 99, \Delta = -183, \det = -4, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 1.5$$

$$T_{\min} = 11, T_{\max} = 99, T \in [11, 99]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 214^2 12 + \mathfrak{su}_8 \rightarrow \mathfrak{so}_{16} \quad T = 5, T_{\max} = 99, \Delta = -183, \det = -4, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 1.5$$

$$T_{\min} = 11, T_{\max} = 99, T \in [11, 99]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 14^2 122 + \mathfrak{su}_8 \rightarrow \mathfrak{so}_{16} \quad T = 5, T_{\max} = 99, \Delta = -183, \det = -4, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 1.5$$

$$T_{\min} = 11, T_{\max} = 99, T \in [11, 99]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 14^2 1 + \mathfrak{su}_8 \rightarrow \mathfrak{so}_{16} \quad T = 5, T_{\max} = 99, \Delta = -183, \det = -4, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 1.5$$

$$T_{\min} = 11, T_{\max} = 99, T \in [11, 99]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[0] \ 14^{\overset{1}{\underset{1}{2}}}1 + \textcolor{red}{\mathfrak{su}}_8 \rightarrow \textcolor{red}{\mathfrak{so}}_{16} \quad T = 5, \ T_{\max} = 99, \ \Delta = -183, \ \det = -4, \ (n_+, n_-, n_0) = (1, 5, 0), \ c_{\text{ext}} = 1.5 \\
T_{\min} = 11, \ T_{\max} = 99, \ T \in [11, 99] \\
\begin{pmatrix} 0 & 2 & 2 \\ 2 & 2 & 2 \\ 2 & 2 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

2.7 $\mathfrak{su}_8 \rightarrow \mathfrak{so}_{16} + (-3) \rightarrow (-2)$

$$\begin{aligned}
T = 6 \quad [0] \ 14^{\textcolor{red}{2}}122^{\textcolor{blue}{1}} + \textcolor{red}{\mathfrak{su}}_8 \rightarrow \textcolor{red}{\mathfrak{so}}_{16} + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} \quad T = 6, \ T_{\max} = 99, \ \Delta = -163, \ \det = 10, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 3.5 \\
T_{\min} = 11.40 \rightarrow 12, \ T_{\max} = 99, \ T \in [12, 99] \\
\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 2 \\ 1 & 0 & -3 & -1 \\ 2 & 2 & -1 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

3 $\mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$ External (67 combos, 581 bases)

3.1 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \quad 1^{11} 31^{111} 2^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 1219, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 4 & 6 & 8 & 3 & 1 & 4 \\ 6 & 4 & 9 & 3 & 0 & 4 \\ 8 & 9 & 9 & 4 & 0 & 5 \\ 3 & 3 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 4 & 5 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

3.2 $\text{NHC}(-2)(-2)(-3) + \mathfrak{f}_4 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \quad 1^{11} 31^{111} 2^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 952, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 31, T \in [12, 31]$$

$$\begin{pmatrix} 4 & 6 & 8 & 3 & 1 & 4 \\ 6 & 4 & 9 & 3 & 0 & 4 \\ 8 & 9 & 9 & 4 & 0 & 5 \\ 3 & 3 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 4 & 5 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

3.3 $\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \quad 1^{11} 31^{111} 2^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 12 & 20 & 1 & 7 \\ 12 & 10 & 17 & 0 & 6 \\ 20 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 6 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{111} 2^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 12 & 16 & 1 & 6 \\ 12 & 13 & 18 & 0 & 7 \\ 16 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 7 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{111} 2^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 15 & 17 & 1 & 7 \\ 15 & 13 & 18 & 0 & 7 \\ 17 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 7 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 31^{111} 2^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 685, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 15 & 20 & 1 & 7 \\ 15 & 13 & 21 & 0 & 7 \\ 20 & 21 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 7 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \quad 1^{11} 4^{211} 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 10 & 1 & 5 \\ 9 & 13 & 15 & 0 & 7 \\ 10 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{211} 4^{11} 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -313, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 7 & 10 & 1 & 5 \\ 7 & 5 & 11 & 0 & 5 \\ 10 & 11 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

3.4 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + (-3) \rightarrow (-2)$

$$T = 10 \quad [0] \quad 1^{11} 5^{11} 1^1 2^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 1 & 4 & 2 & 0 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 3 & 4 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 5^{2^1 2^1 1^1} 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 413, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 6 & 2 & 1 & 3 \\ 5 & 8 & 13 & 4 & 0 & 6 \\ 6 & 13 & 13 & 5 & 0 & 7 \\ 2 & 4 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & 7 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

3.5 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{11} 5^{11} 1^1 2^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 1 & 4 & 2 & 0 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 4 \\ 2 & 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 3 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

3.6 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^1 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 925, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 1 & 4 \\ 8 & 9 & 4 & 8 & 0 & 5 \\ 3 & 4 & -3 & 3 & 0 & -1 \\ 6 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & -1 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

3.7 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2)$

$$T = 9 \quad [0] \quad 1^{11} 23^{11} 1^1 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 36, \Delta = -118, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, T_{\max} = 36, T \in [13, 36]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 1 & 4 \\ 5 & 5 & 5 & 3 & 0 & 5 \\ 3 & 5 & -2 & 2 & 0 & 0 \\ 3 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

3.8 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + (-3) \rightarrow (-2)$

$$T = 7 \quad [0] \quad 1^{11} 3^{11} 1^1 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 100, \Delta = -173, \det = -350, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [1] \quad 1^{11} 23^{11} 1^1 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 4 \\ 5 & 5 & 5 & 0 & 5 \\ 3 & 5 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1}}231^{\overline{11}}2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 98, \ \Delta = -145, \ \det = 275, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \ 2^{\overline{1}}1^{\overline{1}}41^{\overline{1}}31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + (-3) \rightarrow (-2) \quad T = 9, \ T_{\max} = 99, \ \Delta = -143, \ \det = -91, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.81 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 1 & 6 & 0 & 3 \\ 5 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 3 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \ 1^{\overline{1}}5^{\overline{211}}131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 100, \ \Delta = -138, \ \det = 124, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.61 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [5] \ 1^{\overline{11}}5^{\overline{1231}}12^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + (-3) \rightarrow (-2) \quad T = 11, \ T_{\max} = 23, \ \Delta = -110, \ \det = -3, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 21, \ T_{\max} = 23, \ T \in [21, 23]$$

$$\begin{pmatrix} 5 & 15 & 1 & 5 \\ 15 & 42 & 0 & 12 \\ 1 & 0 & -3 & -1 \\ 5 & 12 & -1 & -3 \end{pmatrix}_{b_0}$$

3.9 NHC(-2)(-3) + $\mathfrak{e}_6$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{\overline{1}}31^{\overline{111}}2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -143, \ \det = 925, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 6 & 4 \\ 8 & 9 & 4 & 0 & 8 & 5 \\ 3 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 8 & 3 & 0 & 4 & 4 \\ 4 & 5 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

3.10 NHC(-2)(-3) + $\mathfrak{e}_6$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_2 \rightarrow$ (-3)

$$T = 9 \quad [0] \ 1^{\overline{1}}231^{\overline{11}}1^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 36, \ \Delta = -118, \ \det = -8, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, \ T_{\max} = 36, \ T \in [13, 36]$$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 3 & 4 \\ 5 & 5 & 5 & 0 & 3 & 5 \\ 3 & 5 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 3 & 2 & 0 & 0 & 2 \\ 4 & 5 & 0 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

3.11 NHC(-2)(-3) + $\mathfrak{e}_7$ + (-3) $\rightarrow$ (-2)

$$T = 8 \quad [0] \ 1^{\overline{1}}231^{\overline{11}}2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 102, \ \Delta = -200, \ \det = 425, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.69 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & -4 & 0 & -2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

3.12 NHC(-2)(-3) + $\mathfrak{e}'_7$ + (-3) $\rightarrow$ (-2)

$$T = 8 \quad [0] \ 1^{\overline{1}}231^{\overline{11}}2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 101, \ \Delta = -172, \ \det = 350, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.79 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

3.13 NHC(-2)(-3) + f₄ + su₂ + (-3)→(-2)

$$T = 8 \quad [0] \quad 1^{\bar{1}} 3^{\bar{1}} 1^{\bar{1}} 2^{\bar{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 725, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 31, T \in [12, 31]$$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 1 & 4 \\ 8 & 9 & 4 & 8 & 0 & 5 \\ 3 & 4 & -2 & 3 & 0 & 0 \\ 6 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

3.14 NHC(-2)(-3) + f₄ + su₂→(-3) + (-3)→(-2)

$$T = 8 \quad [0] \quad 1^{\bar{1}} 3^{\bar{1}} 1^{\bar{1}} 2^{\bar{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 45, \Delta = -119, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 45, T \in [12, 45]$$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 1 & 4 \\ 8 & 9 & 4 & 4 & 0 & 5 \\ 3 & 4 & -2 & 2 & 0 & 0 \\ 3 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [1] \quad 2^{\bar{1}} 1^{\bar{1}} 4^{\bar{1}} 3^{\bar{1}} 1^{\bar{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 1 & 3 \\ 3 & 1 & 6 & 0 & 0 & 3 \\ 5 & 6 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 6 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \quad 1^{\bar{1}} 5^{\bar{1}} 13^{\bar{1}} 1^{\bar{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 10 & 8 & 1 & 5 \\ 10 & 13 & 12 & 0 & 7 \\ 8 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

3.15 NHC(-2)(-3) + f₄ + (-3)→(-2)

$$T = 7 \quad [0] \quad 1^{\bar{1}} 3^{\bar{1}} 1^{\bar{1}} 2^{\bar{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 97, \Delta = -147, \det = -275, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [1] \quad 1^{\bar{1}} 23^{\bar{1}} 1^{\bar{1}} 2^{\bar{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 45, \Delta = -119, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 45, T \in [13, 45]$$

$$\begin{pmatrix} 13 & 20 & 1 & 7 \\ 20 & 25 & 0 & 9 \\ 1 & 0 & -3 & -1 \\ 7 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \quad 2^{\bar{1}} 1^{\bar{1}} 4^{\bar{1}} 13^{\bar{1}} 1^{\bar{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 1 & 6 & 0 & 3 \\ 5 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 3 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [3] \quad 1^{\overline{1}} 5^{\overline{1}} 13 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 13 & 0 & 7 \\ 1 & 0 & -3 & -1 \\ 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

3.16 NHC(-2)(-3) + $\mathfrak{f}_4$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^{\overline{1}} 3 1^{\overline{1}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 725, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 31, T \in [12, 31]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 6 & 4 \\ 8 & 9 & 4 & 0 & 8 & 5 \\ 3 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 8 & 3 & 0 & 4 & 4 \\ 4 & 5 & 0 & -1 & 4 & 1 \end{pmatrix}_{b_0}$$

3.17 NHC(-2)(-3) + $\mathfrak{f}_4$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_2 \rightarrow$ (-3)

$$T = 8 \quad [0] \quad 1^{\overline{1}} 3 1^{\overline{1}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 45, \Delta = -119, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 45, T \in [12, 45]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 3 & 4 \\ 8 & 9 & 4 & 0 & 4 & 5 \\ 3 & 4 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 2 & 0 & 0 & 2 \\ 4 & 5 & 0 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [1] \quad 2^{\overline{1}} 1^{\overline{1}} 4 1^{\overline{1}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 1 & 3 \\ 3 & 1 & 6 & 0 & 0 & 3 \\ 5 & 6 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 6 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \quad 1^{\overline{1}} 5^{\overline{1}} 13 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 10 & 1 & 8 & 5 \\ 10 & 13 & 0 & 12 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 12 & 0 & 10 & 6 \\ 5 & 7 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

3.18 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2$ + (-3) $\rightarrow$ (-2)

$$T = 8 \quad [0] \quad 1^{\overline{1}} 1^{\overline{1}} 3 1^{\overline{1}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 12 & 1 & 7 \\ 17 & 20 & 16 & 0 & 8 \\ 12 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1}} 3 1^{\overline{1}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1}} 3 1^{\overline{1}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 27 & 1 & 7 \\ 17 & 20 & 33 & 0 & 8 \\ 27 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^{111}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 325, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 13 & 20 & 27 & 1 & 7 \\ 20 & 25 & 36 & 0 & 9 \\ 27 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{11}31^{14}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 434, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 13 & 17 & 72 & 1 & 7 \\ 17 & 20 & 84 & 0 & 8 \\ 72 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{11}31^{12}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 13 & 17 & 42 & 1 & 7 \\ 17 & 20 & 50 & 0 & 8 \\ 42 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{11}31^{13}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 499, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 13 & 17 & 57 & 1 & 7 \\ 17 & 20 & 67 & 0 & 8 \\ 57 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[7] $1^{13}1^{11}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 525, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 13 & 20 & 15 & 1 & 7 \\ 20 & 25 & 20 & 0 & 9 \\ 15 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [8] $1^{12}1^{11}4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 10 & 7 & 1 & 5 \\ 10 & 13 & 10 & 0 & 7 \\ 7 & 10 & 5 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

[9] $1^{11}1^{11}4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 10 & 16 & 1 & 5 \\ 10 & 13 & 24 & 0 & 7 \\ 16 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[10] $1^{12}1^{12}4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 10 & 14 & 1 & 5 \\ 10 & 13 & 20 & 0 & 7 \\ 14 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[11] $1^{12}1^{13}4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -589, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 10 & 21 & 1 & 5 \\ 10 & 13 & 30 & 0 & 7 \\ 21 & 30 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[12] $1^{11}1^{12}4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -709, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 10 & 23 & 1 & 5 \\ 10 & 13 & 34 & 0 & 7 \\ 23 & 34 & 77 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1^{2^1 1^{14}}} 4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -904, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.83 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 28 & 1 & 5 \\ 10 & 13 & 40 & 0 & 7 \\ 28 & 40 & 110 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 20 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1^{2^1 1^{13}}} 4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -1164, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.80 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 30 & 1 & 5 \\ 10 & 13 & 44 & 0 & 7 \\ 30 & 44 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1^{2^1 1^{14}}} 4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.81 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 37 & 1 & 5 \\ 10 & 13 & 54 & 0 & 7 \\ 37 & 54 & 197 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

3.19 NHC(-2)(-3) + so₈ + su₂→(-3) + (-3)→(-2)

$$T = 8 \quad [0] \quad 1^{1^{13} 1^{1^2} 2^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 8, \quad T_{\max} = 13, \quad \Delta = -95, \quad \det = 130, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.22 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 9 & 1 & 7 \\ 17 & 20 & 11 & 0 & 8 \\ 9 & 11 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1^3} 1^{1^{1^2} 2^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 8, \quad T_{\max} = 13, \quad \Delta = -95, \quad \det = 325, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.22 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 9 & 1 & 7 \\ 20 & 25 & 12 & 0 & 9 \\ 9 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

3.20 NHC(-2)(-3) + so₈ + su₂n3mix + (-3)→(-2)

$$T = 8 \quad [0] \quad 1^{1^{13} 3^{1^2} 1^{1^2} 2^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, \quad T_{\max} = 13, \quad \Delta = -95, \quad \det = 145, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.17 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 24 & 1 & 7 \\ 17 & 20 & 28 & 0 & 8 \\ 24 & 28 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$

3.21 NHC(-2)(-3) + so₈ + su₃ + (-3)→(-2)

$$T = 8 \quad [0] \quad 1^{1^{13} 31^{1^2} 2^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -102, \quad \det = 146, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.49 \rightarrow 12, \quad T_{\max} = 20, \quad T \in [12, 20]$$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1^{13} 31^{1^2} 2^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -102, \quad \det = 200, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.56 \rightarrow 12, \quad T_{\max} = 20, \quad T \in [12, 20]$$

$$\begin{pmatrix} 13 & 20 & 12 & 1 & 7 \\ 20 & 25 & 16 & 0 & 9 \\ 12 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1^{13} 31^{1^2} 2^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -102, \quad \det = 209, \quad (n_+, n_-, n_0) =$$

$$(1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 16 & 12 & 1 & 6 \\ 16 & 20 & 17 & 0 & 8 \\ 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \quad 1^{11} 4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 13 & 14 & 9 & 1 & 7 \\ 14 & 13 & 10 & 0 & 7 \\ 9 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{21} 1^{11} 4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 10 & 7 & 1 & 5 \\ 10 & 13 & 10 & 0 & 7 \\ 7 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

3.22 NHC(-2)(-3) + so₈ + (-3)→(-2)

$$T = 7 \quad [0] \quad 1^{11} 31^{12} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$$

$$\begin{pmatrix} 13 & 17 & 1 & 7 \\ 17 & 20 & 0 & 8 \\ 1 & 0 & -3 & -1 \\ 7 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{13} 1^{11} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.35 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$$

$$\begin{pmatrix} 10 & 16 & 1 & 6 \\ 16 & 20 & 0 & 8 \\ 1 & 0 & -3 & -1 \\ 6 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{13} 1^{11} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$$

$$\begin{pmatrix} 13 & 20 & 1 & 7 \\ 20 & 25 & 0 & 9 \\ 1 & 0 & -3 & -1 \\ 7 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [3] \quad 1^{11} 412^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -3 & -1 \\ 5 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{21} 1^1 4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 13 & 14 & 1 & 7 \\ 14 & 13 & 0 & 7 \\ 1 & 0 & -3 & -1 \\ 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{21} 1^1 4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 13 & 0 & 7 \\ 1 & 0 & -3 & -1 \\ 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{1231} 1^{11} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 125, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{141} 1^{11} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 417, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 1 & 2 \\ 3 & 4 & 7 & 2 & 0 & 4 \\ 3 & 7 & 3 & 2 & 0 & 3 \\ 1 & 2 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \quad 2^1 1^4 1^1 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 1 & 6 & 0 & 3 \\ 5 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [9] \quad 1^1 1^1 5^1 12^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^1 1^1 4^1 14^1 1^1 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 1 & 4 & 0 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^1 1^2 1^1 5^1 13^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

3.23 NHC(-2)(-3) + $\mathfrak{so}_8$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^1 1^1 3 1^1 12^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 12 & 7 \\ 17 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 16 & 0 & 10 & 6 \\ 7 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 1^1 3 1^1 1^1 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 10 & 16 & 1 & 12 & 6 \\ 16 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 17 & 0 & 13 & 7 \\ 6 & 8 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 1^1 3 1^1 1^1 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 15 & 7 \\ 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 17 & 0 & 13 & 7 \\ 7 & 8 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^1 1^1 3 1^1 1^1 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 27 & 7 \\ 17 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 27 & 33 & 0 & 49 & 13 \\ 7 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^1 1^1 3 1^1 1^1 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 325, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{aligned}
& \begin{pmatrix} 13 & 20 & 1 & 27 & 7 \\ 20 & 25 & 0 & 36 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 27 & 36 & 0 & 49 & 13 \\ 7 & 9 & -1 & 13 & 2 \end{pmatrix}_{b_0} \\
[5] \quad & 1^{111}31^{14}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 13, \quad \Delta = -93, \quad \det = 434, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.17 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 13 & 17 & 1 & 72 & 7 \\ 17 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 72 & 84 & 0 & 346 & 34 \\ 7 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0} \\
[6] \quad & 1^{111}31^{12}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 13, \quad \Delta = -93, \quad \det = 454, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.5 \\
& T_{\min} = 11.15 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 13 & 17 & 1 & 42 & 7 \\ 17 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 42 & 50 & 0 & 118 & 20 \\ 7 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0} \\
[7] \quad & 1^{111}31^{13}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 13, \quad \Delta = -93, \quad \det = 499, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.8 \\
& T_{\min} = 11.15 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 13 & 17 & 1 & 57 & 7 \\ 17 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 57 & 67 & 0 & 217 & 27 \\ 7 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0} \\
[8] \quad & 1^{131}1^{11}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 13, \quad \Delta = -93, \quad \det = 525, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.20 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 13 & 20 & 1 & 15 & 7 \\ 20 & 25 & 0 & 20 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 20 & 0 & 13 & 7 \\ 7 & 9 & -1 & 7 & 2 \end{pmatrix}_{b_0} \\
\\
T = 9 \quad [9] \quad & 1^{11}4 \quad 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -69, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.78 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 1 & 9 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 14 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 3 \end{pmatrix}_{b_0} \\
[10] \quad & 1^{12}4 \quad 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -229, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.79 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 1 & 7 & 5 \\ 10 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 10 & 0 & 5 & 5 \\ 5 & 7 & -1 & 5 & 3 \end{pmatrix}_{b_0} \\
[11] \quad & 1^{11}4 \quad 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -344, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.79 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 1 & 16 & 5 \\ 10 & 13 & 0 & 24 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 24 & 0 & 38 & 12 \\ 5 & 7 & -1 & 12 & 3 \end{pmatrix}_{b_0} \\
[12] \quad & 1^{12}4 \quad 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -364, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.5 \\
& T_{\min} = 11.81 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 1 & 14 & 5 \\ 10 & 13 & 0 & 20 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 20 & 0 & 26 & 10 \\ 5 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0} \\
[13] \quad & 1^{12}4 \quad 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -589, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.8 \\
& T_{\min} = 11.83 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 1 & 21 & 5 \\ 10 & 13 & 0 & 30 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 30 & 0 & 61 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0} \\
[14] \quad & 1^{11}4 \quad 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0
\end{aligned}$$

$$(1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 23 & 5 \\ 10 & 13 & 0 & 34 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 34 & 0 & 77 & 17 \\ 5 & 7 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11} 4 \ 1^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -904, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 28 & 5 \\ 10 & 13 & 0 & 40 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 28 & 40 & 0 & 110 & 20 \\ 5 & 7 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{11} 4 \ 1^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -1164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 30 & 5 \\ 10 & 13 & 0 & 44 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 30 & 44 & 0 & 130 & 22 \\ 5 & 7 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{11} 4 \ 1^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -1709, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 37 & 5 \\ 10 & 13 & 0 & 54 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 37 & 54 & 0 & 197 & 27 \\ 5 & 7 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

3.24 NHC(-2)(-3) + $\mathfrak{so}_8$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_2 \rightarrow$ (-3)

$$T = 8 \quad [0] \ 1^{\textcolor{teal}{1}} 3^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} 2^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.22 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 9 & 7 \\ 17 & 20 & 0 & 11 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 11 & 0 & 4 & 4 \\ 7 & 8 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\textcolor{teal}{1}} 3^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} 2^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 325, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.22 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 1 & 9 & 7 \\ 20 & 25 & 0 & 12 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 12 & 0 & 4 & 4 \\ 7 & 9 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

3.25 NHC(-2)(-3) + $\mathfrak{so}_8$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_2 \mathfrak{n} 3 \text{mix}$

$$T = 8 \quad [0] \ 1^{\textcolor{teal}{1}} 3^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} 2^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 24 & 7 \\ 17 & 20 & 0 & 28 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 28 & 0 & 37 & 11 \\ 7 & 8 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

3.26 NHC(-2)(-3) + $\mathfrak{so}_8$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{\textcolor{teal}{1}} 3^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} 2^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 17 & 1 & 15 & 7 \\ 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 17 & 0 & 12 & 6 \\ 7 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\textcolor{teal}{1}} 3^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} 2^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 20 & 1 & 12 & 7 \\ 20 & 25 & 0 & 16 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 16 & 0 & 9 & 5 \\ 7 & 9 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{11}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 16 & 1 & 12 & 6 \\ 16 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 17 & 0 & 12 & 6 \\ 6 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [3] $1^{11}4^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 13 & 14 & 1 & 9 & 7 \\ 14 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 10 & 0 & 4 & 4 \\ 7 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{21}4^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 5 & 10 & 1 & 7 & 5 \\ 10 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 10 & 0 & 4 & 4 \\ 5 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

3.27 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + (-3) \rightarrow (-2)$

$T = 8$ [0] $1^{11}31^{11}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 925, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 1 & 4 \\ 8 & 9 & 8 & 4 & 0 & 5 \\ 6 & 8 & 4 & 3 & 0 & 4 \\ 3 & 4 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

3.28 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + (-3) \rightarrow (-2)$

$T = 8$ [0] $1^{11}31^{11}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 725, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 31, T \in [12, 31]$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 1 & 4 \\ 8 & 9 & 8 & 4 & 0 & 5 \\ 6 & 8 & 4 & 3 & 0 & 4 \\ 3 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 4 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

3.29 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + (-3) \rightarrow (-2)$

$T = 8$ [0] $1^{111}31^{12}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 13 & 17 & 12 & 1 & 7 \\ 17 & 20 & 16 & 0 & 8 \\ 12 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{11}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 13 & 17 & 27 & 1 & 7 \\ 17 & 20 & 33 & 0 & 8 \\ 27 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^{111}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 325, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 13 & 20 & 27 & 1 & 7 \\ 20 & 25 & 36 & 0 & 9 \\ 27 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 13 & -1 & 2 \end{pmatrix}_{b_0}$

[4] $1^{11}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 434, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 13 & 17 & 72 & 1 & 7 \\ 17 & 20 & 84 & 0 & 8 \\ 72 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$

[5] $1^{11}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 13 & 17 & 42 & 1 & 7 \\ 17 & 20 & 50 & 0 & 8 \\ 42 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$

[6] $1^{11}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 499, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 13 & 17 & 57 & 1 & 7 \\ 17 & 20 & 67 & 0 & 8 \\ 57 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$

[7] $1^{13}1^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 525, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 13 & 20 & 15 & 1 & 7 \\ 20 & 25 & 20 & 0 & 9 \\ 15 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 7 & -1 & 2 \end{pmatrix}_{b_0}$

$T = 9$ [8] $1^{12}4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 10 & 7 & 1 & 5 \\ 10 & 13 & 10 & 0 & 7 \\ 7 & 10 & 5 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$

[9] $1^{11}4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 10 & 16 & 1 & 5 \\ 10 & 13 & 24 & 0 & 7 \\ 16 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 12 & -1 & 3 \end{pmatrix}_{b_0}$

[10] $1^{12}4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 10 & 14 & 1 & 5 \\ 10 & 13 & 20 & 0 & 7 \\ 14 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 10 & -1 & 3 \end{pmatrix}_{b_0}$

[11] $1^{12}4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -589, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 10 & 21 & 1 & 5 \\ 10 & 13 & 30 & 0 & 7 \\ 21 & 30 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 15 & -1 & 3 \end{pmatrix}_{b_0}$

[12] $1^{11}4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -709, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 10 & 23 & 1 & 5 \\ 10 & 13 & 34 & 0 & 7 \\ 23 & 34 & 77 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{\overline{1}^1} 4^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -904, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.83 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 28 & 1 & 5 \\ 10 & 13 & 40 & 0 & 7 \\ 28 & 40 & 110 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 20 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{\overline{1}^1} 4^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -1164, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.80 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 30 & 1 & 5 \\ 10 & 13 & 44 & 0 & 7 \\ 30 & 44 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{\overline{1}^1} 4^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -84, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.81 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 37 & 1 & 5 \\ 10 & 13 & 54 & 0 & 7 \\ 37 & 54 & 197 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

3.30 NHC(-2)(-3) + $\textcolor{red}{su}_2 + \textcolor{red}{su}_3 + (-3) \rightarrow (-2)$

$$T = 10 \quad [0] \quad 1^{\overline{1}^1} 5^{\overline{1}^1} 1^{\overline{2}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 13, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 3 & 3 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1}^1} 5^{\overline{1}^1} 1^{\overline{2}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1}^1} 5^{\overline{1}^1} 1^{\overline{2}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 60, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 10 & 3 & 0 & 4 \\ 8 & 10 & 21 & 6 & 0 & 9 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1}^1} 5^{\overline{1}^1} 1^{\overline{2}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 85, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 7 & 3 & 0 & 4 \\ 5 & 7 & 8 & 4 & 0 & 6 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1}^1} 5^{\overline{1}^1} 1^{\overline{2}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 141, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 11 & 3 & 0 & 4 \\ 7 & 11 & 21 & 6 & 0 & 9 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{13} \underset{1^1}{5} 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 181, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.99 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 15 & 3 & 0 & 4 \\ 9 & 15 & 40 & 8 & 0 & 12 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{12} \underset{1^1}{5} 1^3 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 181, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 1 & 3 \\ 3 & 4 & 17 & 3 & 0 & 4 \\ 13 & 17 & 61 & 10 & 0 & 15 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{12} \underset{1^1}{5} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 196, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 14 & 3 & 0 & 4 \\ 10 & 14 & 38 & 8 & 0 & 12 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{2^1 21^{11}} \underset{1^1}{5} \ 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 285, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 5 & 2 & 1 & 3 \\ 6 & 13 & 12 & 5 & 0 & 7 \\ 5 & 12 & 8 & 4 & 0 & 6 \\ 2 & 5 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 6 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{13} \underset{1^1}{5} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 316, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 18 & 3 & 0 & 4 \\ 12 & 18 & 61 & 10 & 0 & 15 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{13} \underset{1^1}{5} 1^3 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 381, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 1 & 3 \\ 3 & 4 & 21 & 3 & 0 & 4 \\ 15 & 21 & 88 & 12 & 0 & 18 \\ 2 & 3 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1^2 21^{12}} \underset{1^1}{5} \ 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 516, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 10 & 2 & 1 & 3 \\ 6 & 13 & 24 & 5 & 0 & 7 \\ 10 & 24 & 38 & 8 & 0 & 12 \\ 2 & 5 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 12 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1^2 21^{13}} \underset{1^1}{5} \ 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 901, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 15 & 2 & 1 & 3 \\ 6 & 13 & 36 & 5 & 0 & 7 \\ 15 & 36 & 88 & 12 & 0 & 18 \\ 2 & 5 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 18 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

3.31 NHC(-2)(-3) + $\mathfrak{su}_2$ + (-3)→(-2)

$$T = 7 \quad [0] \quad 1^{11}31^{12}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 4 \\ 5 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}31^{12}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 9 & 12 & 0 & 5 \\ 6 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 7 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}312^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 9 & 8 & 0 & 5 \\ 3 & 8 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{13}31^{11}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 6 & 1 & 4 \\ 5 & 4 & 5 & 0 & 4 \\ 6 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{13}31^{11}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 15 & 1 & 4 \\ 5 & 4 & 17 & 0 & 4 \\ 15 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{13}31^{12}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 5 & 12 & 1 & 4 \\ 5 & 4 & 10 & 0 & 4 \\ 12 & 10 & 22 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{12}31^{11}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 14 & 0 & 4 \\ 9 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{13}31^{11}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{112}31^{11}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 9 & 20 & 0 & 5 \\ 9 & 20 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{13}31^{13}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 5 & 18 & 1 & 4 \\ 5 & 4 & 15 & 0 & 4 \\ 18 & 15 & 52 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11}31^{11}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 9 & 1 & 4 \\ 5 & 4 & 9 & 0 & 4 \\ 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12}31^{11}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 4 & 5 & 12 & 1 & 4 \\ 5 & 4 & 13 & 0 & 4 \\ 12 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 10 & -1 & 2 \end{pmatrix}_{b_0} \\
[12] \quad & 1^{11}31^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 4 & 9 & 0 & 4 \\ 6 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -1 & 2 \end{pmatrix}_{b_0} \\
[13] \quad & 1^{14}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 4 & 28 & 0 & 4 \\ 18 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 2 \end{pmatrix}_{b_0} \\
[14] \quad & 1^{13}1^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 4 & 8 & 0 & 4 \\ 6 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 2 \end{pmatrix}_{b_0} \\
[15] \quad & 1^{11}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -149, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 9 & 32 & 0 & 5 \\ 15 & 32 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 18 & -1 & 1 \end{pmatrix}_{b_0} \\
[16] \quad & 1^{13}1^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 8 & 9 & 1 & 4 \\ 8 & 9 & 12 & 0 & 5 \\ 9 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 7 & -1 & 1 \end{pmatrix}_{b_0} \\
[17] \quad & 1^{11}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 12 & 28 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 15 & -1 & 1 \end{pmatrix}_{b_0} \\
[18] \quad & 131^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 8 & 6 & 1 & 4 \\ 8 & 9 & 8 & 0 & 5 \\ 6 & 8 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 4 & -1 & 2 \end{pmatrix}_{b_0} \\
[19] \quad & 1^{11}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -201, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 15 & 1 & 4 \\ 5 & 4 & 14 & 0 & 4 \\ 15 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 11 & -1 & 2 \end{pmatrix}_{b_0} \\
[20] \quad & 1^{11}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -209, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 21 & 1 & 3 \\ 4 & 9 & 44 & 0 & 5 \\ 21 & 44 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 25 & -1 & 1 \end{pmatrix}_{b_0} \\
[21] \quad & 1^{13}1^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 12 & 0 & 4 \\ 9 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[22] $1^{114}31^{12^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 7, T_{\max} = 12, \Delta = -94, \det = -249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.19 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 15 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 19 & -1 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{113}31^{12^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 7, T_{\max} = 12, \Delta = -94, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 13 & 0 & 4 \\ 9 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{13}31^{12^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 7, T_{\max} = 12, \Delta = -94, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 23 & 0 & 4 \\ 15 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{114}31^{12^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 7, T_{\max} = 12, \Delta = -94, \det = -284, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 18 & 40 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 22 & -1 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{12}31^{12^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 7, T_{\max} = 12, \Delta = -94, \det = -286, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 18 & 1 & 4 \\ 5 & 4 & 18 & 0 & 4 \\ 18 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 14 & -1 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{12}31^{12^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 7, T_{\max} = 12, \Delta = -94, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 4 & 18 & 0 & 4 \\ 12 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 14 & -1 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{114}31^{12^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 7, T_{\max} = 12, \Delta = -94, \det = -306, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 24 & 1 & 4 \\ 5 & 4 & 26 & 0 & 4 \\ 24 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[28] $1^{11}31^{13^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 7, T_{\max} = 12, \Delta = -94, \det = -306, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 21 & 1 & 4 \\ 5 & 4 & 19 & 0 & 4 \\ 21 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 15 & -1 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{13}31^{12^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 7, T_{\max} = 12, \Delta = -94, \det = -321, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 21 & 1 & 4 \\ 5 & 4 & 22 & 0 & 4 \\ 21 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 17 & -1 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{13}31^{14^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 7, T_{\max} = 12, \Delta = -94, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 4 & 16 & 0 & 4 \\ 12 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 2 \end{pmatrix}_{b_0}$$

[32] $31^{111}22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 7, T_{\max} = 12, \Delta = -94, \det = -409, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$

$$\begin{aligned}
& T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 3 & 1 & 3 \\ 3 & 4 & 0 & 3 & 1 & 2 \\ 3 & 3 & 3 & 1 & 1 & 3 \\ 1 & 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 3 & 0 & 1 \end{pmatrix}_{b_0} \\
[33] \quad & 1^{11}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -421, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.72 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 27 & 1 & 4 \\ 5 & 4 & 24 & 0 & 4 \\ 27 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 19 & -1 & 2 \end{pmatrix}_{b_0} \\
[34] \quad & 22^1 1^{11}3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 7 & 7 & 7 & 4 \\ 7 & 5 & 8 & 7 & 5 \\ 7 & 8 & 4 & 7 & 4 \\ 7 & 7 & 7 & 5 & 5 \\ 4 & 5 & 4 & 5 & 2 \end{pmatrix}_{b_0} \\
[35] \quad & 1^{11}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -444, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 4 & 17 & 0 & 4 \\ 12 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 2 \end{pmatrix}_{b_0} \\
[36] \quad & 1^{12}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -471, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 24 & 1 & 4 \\ 5 & 4 & 23 & 0 & 4 \\ 24 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 18 & -1 & 2 \end{pmatrix}_{b_0} \\
[37] \quad & 1^{14}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -564, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 21 & 1 & 3 \\ 4 & 4 & 32 & 0 & 4 \\ 21 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 2 \end{pmatrix}_{b_0} \\
[38] \quad & 1^{13}31^{15}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -564, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.1 \\
& T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 20 & 0 & 4 \\ 15 & 20 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 2 \end{pmatrix}_{b_0} \\
[39] \quad & 1^{12}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -564, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 22 & 0 & 4 \\ 15 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 17 & -1 & 2 \end{pmatrix}_{b_0} \\
[40] \quad & 1^{13}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -586, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 27 & 1 & 4 \\ 5 & 4 & 27 & 0 & 4 \\ 27 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 21 & -1 & 2 \end{pmatrix}_{b_0} \\
[41] \quad & 1^{13}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -604, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 4 & 27 & 0 & 4 \\ 18 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 2 \end{pmatrix}_{b_0} \\
[42] \quad & 1^{14}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -651, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 4 & 5 & 30 & 1 & 4 \\ 5 & 4 & 31 & 0 & 4 \\ 30 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 24 & -1 & 2 \end{pmatrix}_{b_0} \\
[43] \quad & 1^{11}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -664, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 21 & 0 & 4 \\ 15 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 2 \end{pmatrix}_{b_0} \\
[44] \quad & 1^{12}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -666, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 30 & 1 & 4 \\ 5 & 4 & 28 & 0 & 4 \\ 30 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 22 & -1 & 2 \end{pmatrix}_{b_0} \\
[45] \quad & 31^{11}22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -724, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 3 & 3 & 6 & 1 & 2 \\ 3 & 1 & 4 & 6 & 1 & 3 \\ 3 & 4 & 0 & 6 & 1 & 2 \\ 6 & 6 & 6 & 10 & 2 & 6 \\ 1 & 1 & 1 & 2 & -2 & 0 \\ 2 & 3 & 2 & 6 & 0 & 1 \end{pmatrix}_{b_0} \\
[46] \quad & 1^{13}31^{16}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -784, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2 \\
& T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 4 & 24 & 0 & 4 \\ 18 & 24 & 106 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 2 \end{pmatrix}_{b_0} \\
[47] \quad & 1^{13}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -861, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 33 & 1 & 4 \\ 5 & 4 & 32 & 0 & 4 \\ 33 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 25 & -1 & 2 \end{pmatrix}_{b_0} \\
[48] \quad & 1^{12}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -864, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 4 & 26 & 0 & 4 \\ 18 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 20 & -1 & 2 \end{pmatrix}_{b_0} \\
[49] \quad & 1^{13}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -984, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 21 & 1 & 3 \\ 4 & 4 & 31 & 0 & 4 \\ 21 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 2 \end{pmatrix}_{b_0} \\
[50] \quad & 1^{14}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -1006, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 36 & 1 & 4 \\ 5 & 4 & 36 & 0 & 4 \\ 36 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 28 & -1 & 2 \end{pmatrix}_{b_0} \\
[51] \quad & 1^{14}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -1024, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 24 & 1 & 3 \\ 4 & 4 & 36 & 0 & 4 \\ 24 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 2 \end{pmatrix}_{b_0} \\
[52] \quad & 1^{13}31^{17}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -1044, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 4 & 21 & 1 & 3 \\ 4 & 4 & 28 & 0 & 4 \\ 21 & 28 & 145 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 2 \end{pmatrix}_{b_0}$$

[53] $31^{113}22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -1249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 3 & 3 & 9 & 1 & 2 \\ 3 & 1 & 4 & 9 & 1 & 3 \\ 3 & 4 & 0 & 9 & 1 & 2 \\ 9 & 9 & 9 & 25 & 3 & 9 \\ 1 & 1 & 1 & 3 & -2 & 0 \\ 2 & 3 & 2 & 9 & 0 & 1 \end{pmatrix}_{b_0}$$

[54] $31^{114}22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -1984, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 3 & 3 & 12 & 1 & 2 \\ 3 & 1 & 4 & 12 & 1 & 3 \\ 3 & 4 & 0 & 12 & 1 & 2 \\ 12 & 12 & 12 & 46 & 4 & 12 \\ 1 & 1 & 1 & 4 & -2 & 0 \\ 2 & 3 & 2 & 12 & 0 & 1 \end{pmatrix}_{b_0}$$

[55] $31^{115}22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -2929, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.1$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 3 & 3 & 15 & 1 & 2 \\ 3 & 1 & 4 & 15 & 1 & 3 \\ 3 & 4 & 0 & 15 & 1 & 2 \\ 15 & 15 & 15 & 73 & 5 & 15 \\ 1 & 1 & 1 & 5 & -2 & 0 \\ 2 & 3 & 2 & 15 & 0 & 1 \end{pmatrix}_{b_0}$$

$T = 8$ [56] $1^{11}4^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[57] $1^{11}4^{12}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

[58] $1^{11}4^{12}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[59] $21^{14}1^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 4 \\ 8 & 13 & 12 & 6 & 7 \\ 6 & 12 & 8 & 5 & 6 \\ 3 & 6 & 5 & 0 & 2 \\ 4 & 7 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[60] $22^{11}4^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 6 & 3 & 3 & 3 \\ 6 & 13 & 8 & 6 & 7 \\ 3 & 8 & 4 & 3 & 4 \\ 3 & 6 & 3 & 0 & 2 \\ 3 & 7 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[62] $1^{11}4^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 6 & 8 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[61] $1^{11}4^{11}22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 6 & 3 & 1 & 3 \\ 6 & 13 & 8 & 0 & 7 \\ 3 & 8 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

- [63] $1^{11} 4^{1^1} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 8 & 10 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$
- [64] $1^{11} 4^{1^1} 12^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 5 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -1 & 3 \end{pmatrix}_{b_0}$
- [65] $1^{11} 4^{1^1} 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0}$
- [66] $1^{12} 4^{1^1} 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 11 & 0 & 4 \\ 7 & 11 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$
- [67] $1^1 4^{1^1} 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 4 & 7 & 0 & 4 \\ 5 & 7 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0}$
- [68] $1^{11} 4^{1^1} 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 8 & 12 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 3 \end{pmatrix}_{b_0}$
- [69] $21^{11} 41^{11} 12^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 4 & 8 & 12 & 3 & 4 \\ 8 & 13 & 20 & 6 & 7 \\ 12 & 20 & 26 & 8 & 10 \\ 3 & 6 & 8 & 0 & 2 \\ 4 & 7 & 10 & 2 & 3 \end{pmatrix}_{b_0}$
- [72] $1^{13} 4^{1^1} 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 9 & 15 & 40 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$
- [70] $1^{11} 4^{1^1} 1^{13} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 14 & 18 & 70 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$
- [71] $1^{12} 4^{1^1} 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 10 & 14 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$
- [74] $1^{11} 4^{1^1} 1^{12} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 11 & 15 & 45 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 3 \end{pmatrix}_{b_0}$$

[73] $1^{12} 2^{13} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 13 & 17 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[75] $1^{11} 2^{111} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 3 \\ 4 & 4 & 12 & 0 & 4 \\ 8 & 12 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[76] $1^{211} 2^{12} 4^1 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 3 \\ 4 & 4 & 14 & 0 & 4 \\ 10 & 14 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[77] $1^{12} 2^{11} 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 10 & 16 & 45 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 3 \end{pmatrix}_{b_0}$$

[78] $1^{141} 1^{122} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 6 & 5 & 1 & 3 \\ 6 & 13 & 12 & 0 & 7 \\ 5 & 12 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

[79] $1^{1141} 1^{122} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 3 & 4 & 1 & 2 \\ 1 & 0 & 3 & 4 & 0 & 2 \\ 3 & 3 & 3 & 8 & 0 & 3 \\ 4 & 4 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[80] $22^1 1^{1141} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 6 & 5 & 3 & 3 \\ 6 & 13 & 12 & 6 & 7 \\ 5 & 12 & 8 & 5 & 6 \\ 3 & 6 & 5 & 0 & 2 \\ 3 & 7 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[81] $1^{13} 2^{12} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 12 & 18 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[82] $22^1 1^{1141} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 6 & 8 & 3 & 3 \\ 6 & 13 & 20 & 6 & 7 \\ 8 & 20 & 26 & 8 & 10 \\ 3 & 6 & 8 & 0 & 2 \\ 3 & 7 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[83] $21^{1141} 1^{142} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 8 & 30 & 3 & 4 \\ 8 & 13 & 56 & 6 & 7 \\ 30 & 56 & 212 & 23 & 28 \\ 3 & 6 & 23 & 0 & 2 \\ 4 & 7 & 28 & 2 & 3 \end{pmatrix}_{b_0}$$

[84] $21^{11}41^{12}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 8 & 18 & 3 & 4 \\ 8 & 13 & 32 & 6 & 7 \\ 18 & 32 & 68 & 13 & 16 \\ 3 & 6 & 13 & 0 & 2 \\ 4 & 7 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[85] $1^{11}41^{11}22^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 6 & 8 & 1 & 3 \\ 6 & 13 & 20 & 0 & 7 \\ 8 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[86] $1^{13}4^{21}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 222, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 12 & 20 & 70 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

[87] $1^{12}4^{21}1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 13 & 19 & 68 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

[88] $21^{11}41^{13}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 8 & 24 & 3 & 4 \\ 8 & 13 & 44 & 6 & 7 \\ 24 & 44 & 130 & 18 & 22 \\ 3 & 6 & 18 & 0 & 2 \\ 4 & 7 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[89] $1^{13}4^{21}1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 15 & 21 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[90] $1^{13}4^{21}1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 18 & 24 & 121 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[91] $1^{12}4^{21}1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 16 & 22 & 97 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

[92] $1^{14}4^{21}1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 14 & 22 & 90 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[93] $1^{12}4^{21}1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 25 & 0 & 4 \\ 19 & 25 & 132 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

[94] $1^{1^2 1^1 3} 4 1^3 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 21 & 0 & 4 \\ 15 & 21 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$

[95] $1^{1^2 1^1 2} 4 1^2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 13 & 1 & 3 \\ 4 & 4 & 19 & 0 & 4 \\ 13 & 19 & 68 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$

[96] $1^{1^4 2^1 1} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 281, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 101 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}$

[97] $1^{1^4 1^2 2^1} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 308, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 6 & 10 & 1 & 3 \\ 6 & 13 & 24 & 0 & 7 \\ 10 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 12 & -1 & 3 \end{pmatrix}_{b_0}$

[98] $22^1 1^{1^2 4^1} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 322, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 6 & 10 & 3 & 3 \\ 6 & 13 & 24 & 6 & 7 \\ 10 & 24 & 38 & 10 & 12 \\ 3 & 6 & 10 & 0 & 2 \\ 3 & 7 & 12 & 2 & 3 \end{pmatrix}_{b_0}$

[99] $1^{1^3 2^1 2} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 327, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 15 & 23 & 97 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}$

[100] $1^{1^4 2^1 3} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 351, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 25 & 0 & 4 \\ 17 & 25 & 121 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$

[101] $1^{1^3 2^1 3} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 402, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 18 & 26 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0}$

[102] $1^{1^4 2^1 4} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 20 & 28 & 158 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$

[103] $1^{1^4 2^1 2} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 426, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 17 & 27 & 132 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0}$

[104] $1^{13} 4^{21} 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 447, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 29 & 0 & 4 \\ 21 & 29 & 169 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 3 \end{pmatrix}_{b_0}$

[105] $1^{12} 4^{14} 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 20 & 1 & 3 \\ 4 & 4 & 28 & 0 & 4 \\ 20 & 28 & 158 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$

[106] $22^1 1^{12} 41^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 475, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 6 & 13 & 3 & 3 \\ 6 & 13 & 32 & 6 & 7 \\ 13 & 32 & 68 & 13 & 16 \\ 3 & 6 & 13 & 0 & 2 \\ 3 & 7 & 16 & 2 & 3 \end{pmatrix}_{b_0}$

[107] $1^{11} 4^{13} 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 4 & 26 & 0 & 4 \\ 18 & 26 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0}$

[108] $1^{11} 41^{12} 22^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 481, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 6 & 13 & 1 & 3 \\ 6 & 13 & 32 & 0 & 7 \\ 13 & 32 & 68 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 16 & -1 & 3 \end{pmatrix}_{b_0}$

[109] $1^{14} 4^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 541, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 20 & 30 & 169 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 3 \end{pmatrix}_{b_0}$

[110] $1^{14} 41^{13} 22^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 553, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 6 & 15 & 1 & 3 \\ 6 & 13 & 36 & 0 & 7 \\ 15 & 36 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 18 & -1 & 3 \end{pmatrix}_{b_0}$

[111] $22^1 1^{13} 41^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 567, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 6 & 15 & 3 & 3 \\ 6 & 13 & 36 & 6 & 7 \\ 15 & 36 & 88 & 15 & 18 \\ 3 & 6 & 15 & 0 & 2 \\ 3 & 7 & 18 & 2 & 3 \end{pmatrix}_{b_0}$

[112] $1^{14} 4^{14} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 626, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 23 & 1 & 3 \\ 3 & 4 & 33 & 0 & 4 \\ 23 & 33 & 212 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 3 \end{pmatrix}_{b_0}$

[113] $1^{11} 4^{14} 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 728, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 23 & 1 & 3 \\ 4 & 4 & 33 & 0 & 4 \\ 23 & 33 & 212 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 3 \end{pmatrix}_{b_0} \\
[114] \quad & 22^1 1^{13} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 846, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 18 & 3 & 3 \\ 6 & 13 & 44 & 6 & 7 \\ 18 & 44 & 130 & 18 & 22 \\ 3 & 6 & 18 & 0 & 2 \\ 3 & 7 & 22 & 2 & 3 \end{pmatrix}_{b_0} \\
[115] \quad & 1^{11} 41^{13} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 852, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 18 & 1 & 3 \\ 6 & 13 & 44 & 0 & 7 \\ 18 & 44 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 22 & -1 & 3 \end{pmatrix}_{b_0} \\
[116] \quad & 1^1 41^{14} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 896, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 20 & 1 & 3 \\ 6 & 13 & 48 & 0 & 7 \\ 20 & 48 & 158 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 24 & -1 & 3 \end{pmatrix}_{b_0} \\
[117] \quad & 22^1 1^{14} 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 910, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 20 & 3 & 3 \\ 6 & 13 & 48 & 6 & 7 \\ 20 & 48 & 158 & 20 & 24 \\ 3 & 6 & 20 & 0 & 2 \\ 3 & 7 & 24 & 2 & 3 \end{pmatrix}_{b_0} \\
[118] \quad & 22^1 1^{14} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1315, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 23 & 3 & 3 \\ 6 & 13 & 56 & 6 & 7 \\ 23 & 56 & 212 & 23 & 28 \\ 3 & 6 & 23 & 0 & 2 \\ 3 & 7 & 28 & 2 & 3 \end{pmatrix}_{b_0} \\
[119] \quad & 1^{11} 41^{14} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1321, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 23 & 1 & 3 \\ 6 & 13 & 56 & 0 & 7 \\ 23 & 56 & 212 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 28 & -1 & 3 \end{pmatrix}_{b_0} \\
[120] \quad & 1^1 41^{15} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1337, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.1 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 25 & 1 & 3 \\ 6 & 13 & 60 & 0 & 7 \\ 25 & 60 & 248 & 0 & 30 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 30 & -1 & 3 \end{pmatrix}_{b_0} \\
[121] \quad & 1^1 41^{16} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1876, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.2 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 30 & 1 & 3 \\ 6 & 13 & 72 & 0 & 7 \\ 30 & 72 & 358 & 0 & 36 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 36 & -1 & 3 \end{pmatrix}_{b_0} \\
[122] \quad & 1^1 41^{17} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 2513, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 35 & 1 & 3 \\ 6 & 13 & 84 & 0 & 7 \\ 35 & 84 & 488 & 0 & 42 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 42 & -1 & 3 \end{pmatrix}_{b_0} \\
T = 9 \quad & [123] \quad 22^1 1^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 5 & 6 & 8 & 5 & 6 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[124] \quad & 21^{1^{21}1^1} 5^{1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 5 & 9 & 3 & 4 \\ 5 & 4 & 10 & 3 & 4 \\ 9 & 10 & 19 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 4 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0} \\
[125] \quad & 1^{1^{14}} 5^{1^1} 1^1 2^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 13 & 19 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0} \\
[127] \quad & 15^{1^1} 1^1 2^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0} \\
[126] \quad & 22^{1^1} 15^{1^1} 1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[129] \quad & 1^{1^{21}1^1} 5^{1^1} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 5 & 6 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0} \\
[128] \quad & 1^{1^1} 5^{1^1} 1^1 2^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 5 & 6 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0} \\
[130] \quad & 1^{1^1} 5^{1^1} 1^1 2^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0} \\
[131] \quad & 1^{1^1} 5^{1^2} 1^1 2^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -17, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 6 & 10 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0} \\
[132] \quad & 1^{1^1} 5^{1^1} 12^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0} \\
[133] \quad & 1^{1^1} 5^{1^1} 12^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0} \\
[134] & 21^{11} 5^{12} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 4 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[135] & 1^{11} 5^{12} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 4 & 7 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0} \\
[136] & 1^{12} 5^{12} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 10 & 17 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0} \\
[139] & 21^{11} 51^{11} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 2 & 3 & 1 & 2 \\ 3 & 4 & 5 & 6 & 0 & 4 \\ 2 & 5 & 3 & 4 & 0 & 3 \\ 3 & 6 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & 4 & -1 & 2 \end{pmatrix}_{b_0} \\
[137] & 1^{12} 5^{12} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 11 & 0 & 4 \\ 6 & 11 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0} \\
[138] & 1^{14} 5^{12} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 16 & 22 & 117 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0} \\
[140] & 1^{11} 5^{12} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 8 & 10 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0} \\
[141] & 1^{11} 5^{13} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 13 & 16 & 59 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0} \\
[142] & 1^{11} 5^{13} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 11 & 16 & 59 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0} \\
[143] & 1^{12} 5^{12} 12^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0} \\
[144] & 1^{11} 5^{11} 12^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0} \\
[145] & 1^{11} 5^{14} 12^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 25 & 0 & 4 \\ 19 & 25 & 152 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0} \\
[146] & 1^{11} 5^{21} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 6 & 2 & 1 & 3 \\ 6 & 13 & 5 & 0 & 7 \\ 2 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 3 & -1 & 3 \end{pmatrix}_{b_0} \\
[147] & 22^{11} 1^{11} 5^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -49, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 10 & 3 & 4 \\ 7 & 10 & 19 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0} \\
[148] & 22^{11} 1^{13} 5^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -49, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 17 & 22 & 115 & 17 & 21 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0} \\
[149] & 21^{11} 5^{14} 12^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[150] & 1^{13} 5^{14} 12^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 121 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0} \\
[151] & 1^{11} 5^{12} 1^{32} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 15 & 19 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[152] $1^{11}51^{12}2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -62$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.55 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 9 & 13 & 36 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[154] $1^{11}51^{12}2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 7 & 10 & 19 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[153] $1^{11}2^{21}51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 17 & 22 & 115 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[157] $22^11^{12}51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 12 & 16 & 57 & 12 & 15 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[156] $1^{11}51^{13}2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 17 & 22 & 115 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[155] $1^{11}2^{21}51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 7 & 10 & 19 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[158] $21^{11}51^{21}2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 7 & 2 & 3 \\ 4 & 4 & 10 & 3 & 4 \\ 7 & 10 & 19 & 7 & 9 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[160] $1^{11}51^{13}2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -68$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 14 & 19 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[159] $1^{11}51^{12}2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -68$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 10 & 13 & 36 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[161] $22^{11}1^{11}5^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -72$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.61 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 9 & 14 & 36 & 9 & 12 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[163] $22^{11}1^{11}5^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -73$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 11 & 18 & 59 & 11 & 15 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[162] $1^{13}5^{12}1^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -73$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 10 & 18 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[164] $1^{13}5^{12}1^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -76$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 12 & 21 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[166] $1^{11}5^{12}1^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -77$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 12 & 16 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[165] $1^{11}5^{21}2^{21}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -77$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 12 & 16 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[167] $1^{11}5^{11}1^{12}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 11 & 0 & 4 \\ 7 & 11 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[168] $21^{12}5^{11}1^{12}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -86$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 3 \\ 4 & 4 & 8 & 3 & 4 \\ 6 & 8 & 10 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[169] $1^{12}5^{14}1^{12}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -107$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 15 & 23 & 117 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[170] $1^{12} 5^1 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 9 & 14 & 36 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[171] $1^{12} 5^1 1^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 9 & 14 & 36 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[173] $1^{13} 5^1 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 9 & 15 & 40 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[172] $1^{12} 5^1 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 10 & 14 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[174] $1^{12} 5^1 1^1 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 13 & 17 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[175] $21^{12} 5^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 14 & 2 & 3 \\ 4 & 4 & 20 & 3 & 4 \\ 14 & 20 & 82 & 14 & 18 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[176] $1^{12} 5^1 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 13 & 20 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[177] $1^{12} 5^1 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 11 & 17 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[178] $21^{13} 5^1 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 4 & 12 & 3 & 4 \\ 9 & 12 & 25 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[179] \ 22^{1^1} 1^{2^1} 5^1 1^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -142, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 20 & 3 & 4 \\ 14 & 20 & 82 & 14 & 18 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[180] \ 1^{1^2} 1^{2^1} 5^1 1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -153, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.80 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 7 & 1 & 3 \\ 6 & 13 & 17 & 0 & 7 \\ 7 & 17 & 19 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[181] \ 1^{2^1} 1^{2^1} 5^1 1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -161, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.81 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 5 & 1 & 3 \\ 6 & 13 & 12 & 0 & 7 \\ 5 & 12 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[182] \ 1^{1^2} 5^1 1^{2^1} 1^{2^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -161, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.66 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 12 & 17 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[183] \ 1^{1^3} 5^1 1^{2^1} 1^{2^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -167, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.74 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 11 & 18 & 59 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[184] \ 1^{1^3} 5^1 1^{2^1} 1^{2^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -167, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.74 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 11 & 18 & 59 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[185] \ 21^{1^2} 5^1 1^{2^1} 1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.61 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 10 & 2 & 3 \\ 4 & 4 & 14 & 3 & 4 \\ 10 & 14 & 36 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[186] \ 1^{1^2} 5^1 1^{3^1} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.70 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 15 & 20 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[187] \ 1^{1^2} 5^1 1^{2^1} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -182, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.58 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 18 & 26 & 150 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[188] $22^{1^1}1^35^{1^1}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 19 & 26 & 148 & 19 & 24 \\ 3 & 3 & 19 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

[189] $1^{1^2}5^{1^2}1^{2^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 14 & 20 & 82 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[190] $1^{1^2}5^{1^2}1^{2^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 14 & 20 & 82 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[191] $1^{1^3}5^{1^2}1^{2^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -193, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 16 & 23 & 113 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[192] $1^{1^1}5^{1^2}1^{2^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -193, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 12 & 18 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[193] $1^{1^3}5^{1^2}1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 13 & 21 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[194] $22^{1^1}1^25^{1^1}1^{1^3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 16 & 24 & 113 & 16 & 21 \\ 3 & 3 & 16 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[195] $1^{1^3}5^{1^4}1^{1^1}1^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 17 & 27 & 152 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[196] $1^{1^3}5^{1^3}1^{1^1}1^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -211, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 15 & 24 & 115 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[197] $1^{12} 5^{1^2} 1^3 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -211, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 17 & 23 & 113 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[198] $21^{13} 5^{2^1 1^3} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 21 & 2 & 3 \\ 4 & 4 & 30 & 3 & 4 \\ 21 & 30 & 187 & 21 & 27 \\ 2 & 3 & 21 & 0 & 2 \\ 3 & 4 & 27 & 2 & 3 \end{pmatrix}_{b_0}$$

[199] $1^{12} 5^{1^4} 1^3 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 29 & 0 & 4 \\ 21 & 29 & 189 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

[200] $1^{12} 5^{2^1 21^3} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 19 & 26 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[201] $1^{12} 5^{1^3} 1^3 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 19 & 26 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[202] $1^{1^1} 5^{1^3} 1^3 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 15 & 21 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[203] $1^{12} 5^{2^1 21^4} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 24 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 24 & 32 & 234 & 0 & 30 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 30 & -1 & 3 \end{pmatrix}_{b_0}$$

[204] $1^{13} 5^{1^1} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -252, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 14 & 21 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[205] $1^{2^1 21^{12}} 5^{1^2} 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 14 & 1 & 3 \\ 6 & 13 & 34 & 0 & 7 \\ 14 & 34 & 82 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[206] $1^{13}5^{12}1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 16 & 24 & 113 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[207] $1^{13}5^{21}1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 16 & 24 & 113 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[208] $22^{13}5^{13}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -297, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 21 & 3 & 3 \\ 3 & 4 & 30 & 3 & 4 \\ 21 & 30 & 187 & 21 & 27 \\ 3 & 3 & 21 & 0 & 2 \\ 3 & 4 & 27 & 2 & 3 \end{pmatrix}_{b_0}$$

[209] $21^{13}5^{21}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -306, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 13 & 2 & 3 \\ 4 & 4 & 18 & 3 & 4 \\ 13 & 18 & 59 & 11 & 15 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[210] $1^{13}5^{13}1^{32} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 17 & 24 & 115 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[211] $1^{21}21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 10 & 1 & 3 \\ 6 & 13 & 24 & 0 & 7 \\ 10 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[212] $1^{13}5^{13}1^{22} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 18 & 27 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[213] $1^{13}5^{14}1^{22} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 20 & 30 & 189 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

[214] $21^{13}5^{21}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 17 & 2 & 3 \\ 4 & 4 & 24 & 3 & 4 \\ 17 & 24 & 113 & 16 & 21 \\ 2 & 3 & 16 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[215] $1^{\overline{1}^{2^1 21^{12}}}$ $5^{\overline{1}^1}$ $1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -356$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 12 & 1 & 3 \\ 6 & 13 & 29 & 0 & 7 \\ 12 & 29 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[216] $1^{\overline{1}^{1^2} 5^{\overline{1}^3} 2^{\overline{1}^1}}$ $1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -358$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 19 & 27 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[217] $1^{\overline{1}^{1^3} 5^{\overline{1}^3} 2^{\overline{1}^1}}$ $1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -391$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 21 & 30 & 187 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

[218] $1^{\overline{1}^{1^3} 5^{\overline{1}^2} 21^{\overline{1}^3}}$ $1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -391$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 21 & 30 & 187 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

[219] $1^{\overline{1}^{1^4} 5^{\overline{1}^3} 2^{\overline{1}^1}}$ $1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -406$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.61 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 23 & 1 & 3 \\ 3 & 4 & 33 & 0 & 4 \\ 23 & 33 & 232 & 0 & 30 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 30 & -1 & 3 \end{pmatrix}_{b_0}$$

[220] $1^{\overline{1}^{1^3} 5^{\overline{1}^2} 21^{\overline{1}^4}}$ $1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -461$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 26 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 26 & 36 & 281 & 0 & 33 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 33 & -1 & 3 \end{pmatrix}_{b_0}$$

[221] $1^{\overline{1}^{2^1 21^{13}}}$ $5^{\overline{1}^3}$ $1^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -481$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 21 & 1 & 3 \\ 6 & 13 & 51 & 0 & 7 \\ 21 & 51 & 187 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

[222] $1^{\overline{1}^{2^1 21^{13}}}$ $5^{\overline{1}^1}$ $1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -553$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 15 & 1 & 3 \\ 6 & 13 & 36 & 0 & 7 \\ 15 & 36 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[223] $1^{\overline{1}^{2^1 21^{13}}}$ $5^{\overline{1}^1}$ $1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -633$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 19 & 1 & 3 \\ 6 & 13 & 46 & 0 & 7 \\ 19 & 46 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[224] $1^{\overline{1}^{21}1^{13}} 5^{\overline{1}^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -657, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 17 & 1 & 3 \\ 6 & 13 & 41 & 0 & 7 \\ 17 & 41 & 115 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

$T = 10$ [225] $1^{\overline{1}^{11}31} 5^{\overline{1}^1} 12^{\overline{1}^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 1 & 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[226] $1^{\overline{1}^{11}31} 5^{\overline{1}^1} 12^{\overline{1}^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 2 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

[227] $1^{\overline{1}^{11}31} 5^{\overline{1}^1} 12^{\overline{1}^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[228] $1^{\overline{1}^{21}1^1} 5^{\overline{1}^1} 131^{\overline{1}^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 3 & 2 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

[229] $1^{\overline{1}^{21}1^1} 5^{\overline{1}^1} 131^{\overline{1}^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 5 & 8 & 0 & 5 \\ 5 & 8 & 9 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 7 & -1 & 4 \end{pmatrix}_{b_0}$$

[230] $1^{\overline{1}^{11}31} 5^{\overline{1}^1} 12^{\overline{1}^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 3 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

[231] $1^{\overline{1}^{21}1^1} 5^{\overline{1}^1} 131^{\overline{1}^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 5 & 13 & 0 & 5 \\ 7 & 13 & 25 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 11 & -1 & 4 \end{pmatrix}_{b_0}$$

[232] $1^{\overline{1}^{21}1^1} 5^{\overline{1}^1} 131^{\overline{1}^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 62, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 5 & 18 & 0 & 5 \\ 9 & 18 & 49 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 15 & -1 & 4 \end{pmatrix}_{b_0}$$

[233] $1^{11} 5^{2^{11}} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 4 & 0 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 \\ 4 & 4 & 5 & 11 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[234] $1^{11} 5^{2^{11}} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 101, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 5 & 0 & 2 \\ 1 & 2 & 0 & 8 & 0 & 2 \\ 5 & 5 & 8 & 24 & 0 & 10 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[235] $1^{12} 5^{2^{11}} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 \\ 1 & 2 & 0 & 6 & 0 & 2 \\ 5 & 6 & 6 & 21 & 0 & 9 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[236] $1^1 5^{2^{12}} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 122, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 5 & 21 & 0 & 5 \\ 12 & 21 & 70 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 18 & -1 & 4 \end{pmatrix}_{b_0}$$

[237] $1^{11} 5^{2^{12}} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 141, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 \\ 7 & 6 & 9 & 36 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[238] $1^1 5^{2^{12}} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 5 & 26 & 0 & 5 \\ 14 & 26 & 106 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 22 & -1 & 4 \end{pmatrix}_{b_0}$$

[239] $1^{12} 5^{2^{11}} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 168, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 \\ 6 & 7 & 9 & 36 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[240] $1^{11} 5^{2^{12}} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 189, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 12 & 0 & 2 \\ 8 & 7 & 12 & 57 & 0 & 15 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[241] $1^{12} 5^{2^{11}} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 204, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 12 & 0 & 2 \\ 7 & 8 & 12 & 57 & 0 & 15 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[242] $1^{12} 5^{21^2} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 10 & 0 & 2 \\ 8 & 8 & 10 & 50 & 0 & 14 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 14 & -1 & 3 \end{pmatrix}_{b_0}$$

[243] $1^{11} 5^{21^3} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 229, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 13 & 0 & 2 \\ 10 & 8 & 13 & 75 & 0 & 17 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

[244] $1^{13} 5^{21^1} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 229, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 10 & 0 & 2 \\ 7 & 9 & 10 & 52 & 0 & 14 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 14 & -1 & 3 \end{pmatrix}_{b_0}$$

[245] $1^{13} 5^{21^1} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 293, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 13 & 0 & 2 \\ 8 & 10 & 13 & 75 & 0 & 17 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

[246] $1^{11} 5^{21^3} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 293, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 16 & 0 & 2 \\ 11 & 9 & 16 & 104 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 20 & -1 & 3 \end{pmatrix}_{b_0}$$

[247] $1^{12} 5^{21^2} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 304, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 13 & 0 & 2 \\ 9 & 9 & 13 & 73 & 0 & 17 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

[248] $1^{13} 5^{21^2} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 309, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 \\ 9 & 10 & 11 & 68 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

[249] $1^{12} 5^{21^2} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 16 & 0 & 2 \\ 10 & 10 & 16 & 102 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 20 & -1 & 3 \end{pmatrix}_{b_0}$$

[250] $1^{12} 5^{21^3} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 364, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 14 & 0 & 2 \\ 11 & 10 & 14 & 93 & 0 & 19 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[251] \quad 1^{13} 5^{2^{1^2}} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 413, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 \\ 1 & 2 & 0 & 14 & 0 & 2 \\ 10 & 11 & 14 & 93 & 0 & 19 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[252] \quad 1^{12} 5^{2^{1^3}} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 456, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 \\ 1 & 2 & 0 & 17 & 0 & 2 \\ 12 & 11 & 17 & 124 & 0 & 22 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[253] \quad 1^{13} 5^{2^{1^2}} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 2 \\ 1 & 2 & 0 & 17 & 0 & 2 \\ 11 & 12 & 17 & 124 & 0 & 22 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[254] \quad 1^{13} 5^{2^{1^3}} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 2 \\ 1 & 2 & 0 & 15 & 0 & 2 \\ 12 & 12 & 15 & 115 & 0 & 21 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[255] \quad 1^{13} 5^{2^{1^3}} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 613, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 1 & 2 \\ 1 & 0 & 2 & 13 & 0 & 2 \\ 1 & 2 & 0 & 18 & 0 & 2 \\ 13 & 13 & 18 & 148 & 0 & 24 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

3.32 NHC(-2)(-3) + $\mathfrak{su}_2$ + (-3) $\rightarrow$ (-2) + $\mathfrak{e}_6$

$$T = 8 \quad [0] \quad 1^{13} 1^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 925, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 4 & 8 & 6 & 1 & 3 & 4 \\ 8 & 9 & 8 & 0 & 4 & 5 \\ 6 & 8 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 0 & -3 & -1 \\ 4 & 5 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

3.33 NHC(-2)(-3) + $\mathfrak{su}_2$ + (-3) $\rightarrow$ (-2) + $\mathfrak{f}_4$

$$T = 8 \quad [0] \quad 1^{13} 1^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{f}_4 \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 725, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 31, T \in [12, 31]$$

$$\begin{pmatrix} 4 & 8 & 6 & 1 & 3 & 4 \\ 8 & 9 & 8 & 0 & 4 & 5 \\ 6 & 8 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 0 & -2 & 0 \\ 4 & 5 & 4 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

3.34 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8$

$$T = 8 \quad [0] \quad 1^{111}31^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 12 & 1 & 7 \\ 17 & 20 & 16 & 0 & 8 \\ 12 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 27 & 1 & 7 \\ 17 & 20 & 33 & 0 & 8 \\ 27 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}31^{111}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 325, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 27 & 1 & 7 \\ 20 & 25 & 36 & 0 & 9 \\ 27 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}31^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 434, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 72 & 1 & 7 \\ 17 & 20 & 84 & 0 & 8 \\ 72 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111}31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 42 & 1 & 7 \\ 17 & 20 & 50 & 0 & 8 \\ 42 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111}31^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 499, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 57 & 1 & 7 \\ 17 & 20 & 67 & 0 & 8 \\ 57 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{131}1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 525, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 15 & 1 & 7 \\ 20 & 25 & 20 & 0 & 9 \\ 15 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

3.35 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{211}5 \quad 131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 3 & 0 & 4 & 2 \\ 1 & 1 & 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 2 & 4 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & 3 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{122}5 \quad 131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -12, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 7 & 2 \\ 1 & 2 & 0 & 3 & 0 & 9 & 2 \\ 1 & 1 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 7 & 9 & 9 & 0 & 36 & 12 \\ 2 & 2 & 2 & 3 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12} 5^{2^{1^2}} 131^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 9 & 2 \\ 1 & 2 & 0 & 3 & 0 & 13 & 2 \\ 1 & 1 & 3 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 9 & 9 & 13 & 13 & 0 & 73 & 17 \\ 2 & 2 & 2 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12} 5^{2^{1^3}} 131^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 1 & 0 & 11 & 2 \\ 1 & 2 & 0 & 3 & 0 & 17 & 2 \\ 1 & 1 & 3 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 12 & 11 & 17 & 17 & 0 & 124 & 22 \\ 2 & 2 & 2 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{12} 5^{2^{1^1}} 131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 5 & 5 & 0 & 13 & 5 \\ 2 & 5 & 2 & 0 & 10 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 7 & 13 & 10 & 0 & 25 & 11 \\ 3 & 5 & 4 & -1 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{12} 5^{2^{1^1}} 131^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -53, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 5 & 5 & 0 & 18 & 5 \\ 2 & 5 & 2 & 0 & 14 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 18 & 14 & 0 & 49 & 15 \\ 3 & 5 & 4 & -1 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{12} 5^{2^{1^2}} 131^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 5 & 5 & 0 & 26 & 5 \\ 2 & 5 & 2 & 0 & 20 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 26 & 20 & 0 & 106 & 22 \\ 3 & 5 & 4 & -1 & 22 & 4 \end{pmatrix}_{b_0}$$

3.36 NHC(-2)(-3) + $\mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 12 \quad [0] \ 1^{12} 5^{2^{1^1}} 131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 5 & 5 & 0 & 13 & 4 & 5 \\ 2 & 5 & 2 & 0 & 10 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 7 & 13 & 10 & 0 & 25 & 8 & 11 \\ 2 & 4 & 3 & 0 & 8 & 1 & 3 \\ 3 & 5 & 4 & -1 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12} 5^{2^{1^2}} 131^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 122, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 2 & 3 \\ 3 & 5 & 5 & 0 & 26 & 4 & 5 \\ 2 & 5 & 2 & 0 & 20 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 14 & 26 & 20 & 0 & 106 & 16 & 22 \\ 2 & 4 & 3 & 0 & 16 & 1 & 3 \\ 3 & 5 & 4 & -1 & 22 & 3 & 4 \end{pmatrix}_{b_0}$$

3.37 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \quad 1^{\overline{1}1131} \quad 5^{\overline{1}} \quad 12^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 4 & 5 \\ 2 & 5 & 2 & 0 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 5 & 4 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

[1] $1^{\textcolor{red}{1}}1^{\textcolor{red}{1}}5^{13^11^1}12^{\textcolor{brown}{1}} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (\textcolor{brown}{-3}) \rightarrow (\textcolor{brown}{-2}) + \textcolor{red}{s}\textcolor{red}{u}_2 \rightarrow (\textcolor{brown}{-3}) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[2] $1^{\textcolor{red}{2}}1^{\textcolor{teal}{1}}5\ 13^{\textcolor{red}{1}}1^{\textcolor{teal}{1}} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{s}\textcolor{teal}{u}_2 + (-3) \rightarrow (-2) + \textcolor{red}{s}\textcolor{red}{u}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 & 2 \\ 3 & 2 & 4 & 5 & 0 & 3 & 5 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 3 & 0 & 0 & 2 \\ 2 & 2 & 2 & 5 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[3] $1^{\textcolor{blue}{2}}1^{\textcolor{blue}{1}}5\ 13^{\textcolor{blue}{1}}1^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2) + \textcolor{red}{s}\textcolor{blue}{u}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 5 & 8 & 0 & 4 & 5 \\ 5 & 8 & 9 & 0 & 5 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 5 & 0 & 1 & 3 \\ 3 & 5 & 7 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

[5] $1^{\textcolor{red}{1}\textcolor{blue}{1}} 1^{\textcolor{red}{2}\textcolor{blue}{1}\textcolor{green}{1}} 5 \ 13^{\textcolor{red}{1}\textcolor{blue}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (\textcolor{red}{-3}) \rightarrow (\textcolor{red}{-2}) + \textcolor{red}{s}\textcolor{blue}{u}_2 \rightarrow (\textcolor{red}{-3}) \quad T = 11, \ T_{\max} = 12, \ \Delta = -31, \ \det = -76, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 0 & 2 & 4 & 0 & 1 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 & 2 \\ 4 & 4 & 5 & 11 & 0 & 4 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 4 & 0 & 0 & 2 \\ 2 & 2 & 2 & 7 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[4] $1^{\textcolor{blue}{2}} 1^{\textcolor{blue}{1}} 5^{\textcolor{blue}{1}} 13^{\textcolor{blue}{1}} 1^{\textcolor{red}{12}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{s}\textcolor{blue}{u}_2 + (\textcolor{red}{-3}) \rightarrow (\textcolor{red}{-2}) + \textcolor{red}{s}\textcolor{blue}{u}_2 \rightarrow (\textcolor{red}{-3})$ $T = 11$, $T_{\max} = 12$, $\Delta = -31$, $\det = -76$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 5 & 13 & 0 & 4 & 5 \\ 7 & 13 & 25 & 0 & 8 & 11 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 8 & 0 & 1 & 3 \\ 3 & 5 & 11 & -1 & 3 & 4 \end{pmatrix} b_0$$

[6] $1^{12} 5^{211} 13^{1111} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{u}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -91, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 1 & 2 \\ 1 & 2 & 0 & 6 & 0 & 2 & 2 \\ 5 & 6 & 6 & 21 & 0 & 5 & 9 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 5 & 0 & 0 & 2 \\ 2 & 2 & 2 & 9 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[7] $1^{\textcolor{blue}{2}}1^{\textcolor{green}{5}}13^{\textcolor{red}{1}}1^{\textcolor{blue}{13}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{s}\textcolor{blue}{u}_2 + (\textcolor{brown}{-3}) \rightarrow (\textcolor{brown}{-2}) + \textcolor{red}{s}\textcolor{blue}{u}_2 \rightarrow (\textcolor{red}{-3})$ $T = 11$, $T_{\max} = 12$, $\Delta = -31$, $\det = -91$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 5 & 18 & 0 & 4 & 5 \\ 9 & 18 & 49 & 0 & 11 & 15 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 11 & 0 & 1 & 3 \\ 3 & 5 & 15 & -1 & 3 & 4 \end{pmatrix}_{b_1}$$

[9] $1^{\mathbf{2}^1\mathbf{1}^2}5\ 13^{\mathbf{1}^1\mathbf{1}^3} + [\mathbf{s}u_2 \cdot \mathbf{g}_2] + \mathbf{s}u_2 + (-3) \rightarrow (-2) + \mathbf{s}u_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -151, (n_+, n_-, n_0) =$

$$(1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 5 & 21 & 0 & 4 & 5 \\ 12 & 21 & 70 & 0 & 13 & 18 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 13 & 0 & 1 & 3 \\ 3 & 5 & 18 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[8] 1^{11} 5^{2^1 1^2} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (\textcolor{orange}{-3}) \rightarrow (\textcolor{orange}{-2}) + \textcolor{red}{su}_2 \rightarrow (\textcolor{red}{-3}) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -151, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 1 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 & 2 \\ 7 & 6 & 9 & 36 & 0 & 7 & 12 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 7 & 0 & 0 & 2 \\ 2 & 2 & 2 & 12 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[11] 1^{11} 5^{2^1 1^2} 13^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (\textcolor{orange}{-3}) \rightarrow (\textcolor{orange}{-2}) + \textcolor{red}{su}_2 \rightarrow (\textcolor{red}{-3}) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -226, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 5 & 26 & 0 & 4 & 5 \\ 14 & 26 & 106 & 0 & 16 & 22 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 16 & 0 & 1 & 3 \\ 3 & 5 & 22 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[10] 1^{12} 5^{2^1 1^2} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (\textcolor{orange}{-3}) \rightarrow (\textcolor{orange}{-2}) + \textcolor{red}{su}_2 \rightarrow (\textcolor{red}{-3}) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -226, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 1 & 2 \\ 1 & 2 & 0 & 10 & 0 & 2 & 2 \\ 8 & 8 & 10 & 50 & 0 & 8 & 14 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 8 & 0 & 0 & 2 \\ 2 & 2 & 2 & 14 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[12] 1^{13} 5^{2^1 1^2} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (\textcolor{orange}{-3}) \rightarrow (\textcolor{orange}{-2}) + \textcolor{red}{su}_2 \rightarrow (\textcolor{red}{-3}) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -271, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 1 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 & 2 \\ 9 & 10 & 11 & 68 & 0 & 9 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 9 & 0 & 0 & 2 \\ 2 & 2 & 2 & 16 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[13] 1^{12} 5^{2^1 1^3} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (\textcolor{orange}{-3}) \rightarrow (\textcolor{orange}{-2}) + \textcolor{red}{su}_2 \rightarrow (\textcolor{red}{-3}) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -371, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 1 & 2 \\ 1 & 2 & 0 & 14 & 0 & 2 & 2 \\ 11 & 10 & 14 & 93 & 0 & 11 & 19 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 11 & 0 & 0 & 2 \\ 2 & 2 & 2 & 19 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[14] 1^{13} 5^{2^1 1^3} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (\textcolor{orange}{-3}) \rightarrow (\textcolor{orange}{-2}) + \textcolor{red}{su}_2 \rightarrow (\textcolor{red}{-3}) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -476, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 1 & 2 \\ 1 & 2 & 0 & 15 & 0 & 2 & 2 \\ 12 & 12 & 15 & 115 & 0 & 12 & 21 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 12 & 0 & 0 & 2 \\ 2 & 2 & 2 & 21 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

3.38 NHC(-2)(-3) + $\textcolor{blue}{su}_2$ + (-3) $\rightarrow$ (-2) + $\textcolor{blue}{su}_2 \rightarrow$ (-3) + $\textcolor{blue}{su}_2$

$$T = 12 \quad [0] 1^{2^1 1^1} 5^{12} 13^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (\textcolor{orange}{-3}) \rightarrow (\textcolor{orange}{-2}) + \textcolor{red}{su}_2 \rightarrow (\textcolor{red}{-3}) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 5 & 5 & 0 & 4 & 13 & 5 \\ 2 & 5 & 2 & 0 & 3 & 10 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 8 & 3 \\ 7 & 13 & 10 & 0 & 8 & 25 & 11 \\ 3 & 5 & 4 & -1 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1] 1^{2^1 1^2} 5^{13} 13^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (\textcolor{orange}{-3}) \rightarrow (\textcolor{orange}{-2}) + \textcolor{red}{su}_2 \rightarrow (\textcolor{red}{-3}) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 122, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 5 & 5 & 0 & 4 & 26 & 5 \\ 2 & 5 & 2 & 0 & 3 & 20 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 16 & 3 \\ 14 & 26 & 20 & 0 & 16 & 106 & 22 \\ 3 & 5 & 4 & -1 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

3.39 NHC(-2)(-3) + $\mathfrak{su}_2$ + (-3)→(-2) + su2n3mix

$$T = 11 \quad [0] \quad 1^{\underline{1} \overset{2}{1} \overset{1}{1}} \overset{1}{5} \overset{1}{13} \overset{1}{1} \overset{1}{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 5 & 5 & 0 & 17 & 5 \\ 2 & 5 & 2 & 0 & 13 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 13 & 0 & 44 & 14 \\ 3 & 5 & 4 & -1 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\underline{1} \overset{2}{1} \overset{2}{2}} \overset{1}{5} \overset{1}{13} \overset{1}{1} \overset{1}{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -131, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 5 & 5 & 0 & 30 & 5 \\ 2 & 5 & 2 & 0 & 23 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 16 & 30 & 23 & 0 & 141 & 25 \\ 3 & 5 & 4 & -1 & 25 & 4 \end{pmatrix}_{b_0}$$

3.40 NHC(-2)(-3) + $\mathfrak{su}_2$ + (-3)→(-2) + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{\underline{1} \overset{1}{1} \overset{1}{2}} \overset{1}{5} \overset{1}{1} \overset{1}{2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\underline{1} \overset{1}{1} \overset{1}{2}} \overset{1}{5} \overset{1}{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\underline{1} \overset{1}{1} \overset{1}{2}} \overset{1}{5} \overset{1}{1} \overset{1}{2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 10 & 0 & 3 & 4 \\ 8 & 10 & 21 & 0 & 6 & 9 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 9 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\underline{1} \overset{1}{1} \overset{1}{2}} \overset{1}{5} \overset{1}{1} \overset{1}{2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 7 & 0 & 3 & 4 \\ 5 & 7 & 8 & 0 & 4 & 6 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 6 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\underline{1} \overset{1}{1} \overset{1}{2}} \overset{1}{5} \overset{1}{1} \overset{1}{2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 141, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 11 & 0 & 3 & 4 \\ 7 & 11 & 21 & 0 & 6 & 9 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 9 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[5] $1^{13}5^{11}2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 15 & 0 & 3 & 4 \\ 9 & 15 & 40 & 0 & 8 & 12 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 3 & 4 & 12 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[6] $1^{12}5^{13}2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 3 & 4 \\ 13 & 17 & 61 & 0 & 10 & 15 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 3 & 4 & 15 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[7] $1^{12}5^{12}2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 196, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 4 \\ 10 & 14 & 38 & 0 & 8 & 12 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 3 & 4 & 12 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{21}5^{21}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 285, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 5 & 1 & 2 & 3 \\ 6 & 13 & 12 & 0 & 5 & 7 \\ 5 & 12 & 8 & 0 & 4 & 6 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 5 & 4 & 0 & 0 & 2 \\ 3 & 7 & 6 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[9] $1^{13}5^{12}2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 316, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 3 & 4 \\ 12 & 18 & 61 & 0 & 10 & 15 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 3 & 4 & 15 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[10] $1^{13}5^{13}2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 381, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 3 & 4 \\ 15 & 21 & 88 & 0 & 12 & 18 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 12 & 0 & 0 & 2 \\ 3 & 4 & 18 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[11] $1^{21}5^{21}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 516, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 10 & 1 & 2 & 3 \\ 6 & 13 & 24 & 0 & 5 & 7 \\ 10 & 24 & 38 & 0 & 8 & 12 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 5 & 8 & 0 & 0 & 2 \\ 3 & 7 & 12 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[12] $1^{12}5^{21}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 901, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 15 & 1 & 2 & 3 \\ 6 & 13 & 36 & 0 & 5 & 7 \\ 15 & 36 & 88 & 0 & 12 & 18 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 5 & 12 & 0 & 0 & 2 \\ 3 & 7 & 18 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

3.41 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \quad 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 20 & 12 & 1 & 7 \\ 20 & 25 & 16 & 0 & 9 \\ 12 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 16 & 12 & 1 & 6 \\ 16 & 20 & 17 & 0 & 8 \\ 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \quad 1^{11} 4^{211} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 13 & 14 & 9 & 1 & 7 \\ 14 & 13 & 10 & 0 & 7 \\ 9 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{211} 4^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 10 & 7 & 1 & 5 \\ 10 & 13 & 10 & 0 & 7 \\ 7 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

3.42 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2)$

$$T = 10 \quad [0] \quad 1^{11} 51^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 3 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 51^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 4 & 0 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 51^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 10 & 0 & 4 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 8 & 10 & 6 & 21 & 0 & 9 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 51^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 7 & 0 & 4 \\ 2 & 3 & 0 & 4 & 0 & 2 \\ 5 & 7 & 4 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{12} \underset{1^1}{5} 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 141, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 3 & 11 & 0 & 4 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 7 & 11 & 6 & 21 & 0 & 9 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{13} \underset{1^1}{5} 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 3 & 15 & 0 & 4 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 9 & 15 & 8 & 40 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12} \underset{1^1}{5} 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 3 & 17 & 0 & 4 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 13 & 17 & 10 & 61 & 0 & 15 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{12} \underset{1^1}{5} 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 196, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 3 & 14 & 0 & 4 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 10 & 14 & 8 & 38 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{2^1 21^{11}} \underset{1^1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 285, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 5 & 1 & 3 \\ 6 & 13 & 5 & 12 & 0 & 7 \\ 2 & 5 & 0 & 4 & 0 & 2 \\ 5 & 12 & 4 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 2 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{13} \underset{1^1}{5} 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 316, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 3 & 18 & 0 & 4 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 12 & 18 & 10 & 61 & 0 & 15 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{13} \underset{1^1}{5} 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 381, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 1 & 3 \\ 3 & 4 & 3 & 21 & 0 & 4 \\ 2 & 3 & 0 & 12 & 0 & 2 \\ 15 & 21 & 12 & 88 & 0 & 18 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{2^1 21^{12}} \underset{1^1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 516, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 10 & 1 & 3 \\ 6 & 13 & 5 & 24 & 0 & 7 \\ 2 & 5 & 0 & 8 & 0 & 2 \\ 10 & 24 & 8 & 38 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 2^{12} 1^{13} 5^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 901, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 15 & 1 & 3 \\ 6 & 13 & 5 & 36 & 0 & 7 \\ 2 & 5 & 0 & 12 & 0 & 2 \\ 15 & 36 & 12 & 88 & 0 & 18 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 2 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

3.43 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \ 1^{11} 31^{13} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 1 & 3 \\ 4 & 4 & 5 & 12 & 0 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 \\ 9 & 12 & 9 & 24 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 1 & 4 \\ 8 & 9 & 4 & 4 & 0 & 5 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 2 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 108, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.67 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 1 & 3 \\ 4 & 4 & 4 & 14 & 0 & 4 \\ 3 & 4 & 0 & 9 & 0 & 2 \\ 9 & 14 & 9 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 10 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 31^{12} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 108, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 10.74 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 1 & 3 \\ 4 & 4 & 5 & 8 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 6 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 153, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 1 & 3 \\ 4 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 237, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.73 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 9 & 1 & 4 \\ 5 & 4 & 4 & 9 & 0 & 4 \\ 3 & 4 & 0 & 6 & 0 & 2 \\ 9 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 4 & 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 453, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.75 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 1 & 3 \\ 4 & 4 & 4 & 9 & 0 & 4 \\ 3 & 4 & 0 & 6 & 0 & 2 \\ 6 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

3.44 NHC(-2)(-3) + su₃ + (-3)→(-2)

$$T = 7 \quad [0] \quad 1^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 9 & 12 & 0 & 5 \\ 6 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 19, T \in [13, 19]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 9 & 8 & 0 & 5 \\ 3 & 8 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 4 \\ 5 & 4 & 4 & 0 & 4 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 4 & 5 & 0 & 4 \\ 3 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 4 & 5 & 6 & 1 & 4 \\ 5 & 4 & 5 & 0 & 4 \\ 6 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{13}31^{11}2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 15 & 1 & 4 \\ 5 & 4 & 17 & 0 & 4 \\ 15 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 12 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{13}31^{11}2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.62 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 4 & 4 & 0 & 4 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{12}31^{11}2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.62 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 14 & 0 & 4 \\ 9 & 14 & 36 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{13}31^{11}2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -125, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11}31^{11}2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.63 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 9 & 1 & 4 \\ 5 & 4 & 9 & 0 & 4 \\ 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{12}31^{11}2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -134, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 12 & 1 & 4 \\ 5 & 4 & 13 & 0 & 4 \\ 12 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{11}31^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10.46 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 4 & 9 & 0 & 4 \\ 6 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[12] $1^{11}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 4 & 8 & 0 & 4 \\ 6 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{11}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 12 & 0 & 4 \\ 9 & 12 & 24 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

[14] $1^{11}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -296, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 13 & 0 & 4 \\ 9 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{12}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -336, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 4 & 18 & 0 & 4 \\ 12 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{11}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -476, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.42 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 4 & 17 & 0 & 4 \\ 12 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 2 \end{pmatrix}_{b_0}$$

[17] $31^{11}22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -561, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 3 & 1 \\ 3 & 4 & 0 & 3 & 1 \\ 3 & 3 & 3 & 0 & 1 \\ 1 & 1 & 1 & 1 & -2 \\ 2 & 3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{12}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -596, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.34 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 22 & 0 & 4 \\ 15 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 2 \end{pmatrix}_{b_0}$$

$T = 8$ [19] $1^{12}4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[20] $21^{11}41^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 4 & 8 & 6 & 1 & 4 \\ 8 & 13 & 12 & 0 & 7 \\ 6 & 12 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 7 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{11}4^{12}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[22] $1^{11}4^{12}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 4 & 5 & 6 & 1 & 4 \\ 5 & 4 & 7 & 0 & 4 \\ 6 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{11}4^{12}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[24] $1^{21}4^{11}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 4 \\ 5 & 4 & 4 & 0 & 4 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{21}4^{11}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 4 \\ 5 & 4 & 3 & 0 & 4 \\ 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[26] $22^{11}4^{11}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 7 & 5 & 5 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

[27] $21^{14}1^{12}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 4 \\ 8 & 13 & 12 & 6 & 7 \\ 6 & 12 & 7 & 5 & 5 \\ 3 & 6 & 5 & 0 & 2 \\ 4 & 7 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

[28] $1^{21}4^{11}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 7 & 1 & 3 \\ 4 & 4 & 10 & 0 & 4 \\ 7 & 10 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{11}4^{11}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{11}4^{12}22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{21}4^{11}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{\overline{2}1^{\overline{1}1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 4 & 4 & 0 & 4 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{\overline{2}1^{\overline{1}1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{\overline{1}} 41^{\overline{1}1} 22^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.19 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 6 & 5 & 1 & 3 \\ 6 & 13 & 12 & 0 & 7 \\ 5 & 12 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[35] $22^{\overline{1}} 1^{\overline{1}} 41^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 6 & 5 & 3 & 3 \\ 6 & 13 & 12 & 6 & 7 \\ 5 & 12 & 7 & 5 & 5 \\ 3 & 6 & 5 & 0 & 2 \\ 3 & 7 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{\overline{1}} 41^{\overline{1}1} 22^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 3 \\ 5 & 8 & 11 & 0 & 6 \\ 5 & 11 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 6 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [37] $21^{\overline{1}} 51^{\overline{1}2^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 4 & 1 & 3 \\ 4 & 4 & 6 & 0 & 4 \\ 4 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

[38] $1^{\overline{1}} 51^{\overline{1}2^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 4 & 1 & 3 \\ 4 & 4 & 6 & 0 & 4 \\ 4 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

[39] $1^{\overline{1}} 5^{\overline{2}1^{\overline{2}1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 5 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{\overline{1}} 5^{\overline{1}} 12^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[41] $21^{\overline{1}} 51^{\overline{1}2^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 1 & 5 & 0 & 3 \\ 4 & 5 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[42] \quad 1 \overset{1}{5} 1^1 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[43] \quad 1 \overset{2}{1} \overset{2}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 5 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 5 & 5 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[44] \quad 1 \overset{2}{1} \overset{2}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 6 & 2 & 1 & 3 \\ 6 & 13 & 5 & 0 & 7 \\ 2 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[45] \quad 22^1 1^1 \overset{1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 6 & 2 & 3 & 3 \\ 6 & 13 & 5 & 6 & 7 \\ 2 & 5 & 0 & 2 & 2 \\ 3 & 6 & 2 & 0 & 2 \\ 3 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [46] \quad 1 \overset{2}{1} \overset{1}{6} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 11, T_{\max} = 13, \Delta = -65, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [47] \quad 1 \overset{1}{1} \overset{1}{5} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 6 & 2 & 6 \\ 2 & 0 & 2 \\ 6 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[48] \quad 1 \overset{1}{1} \overset{1}{5} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -13, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 14.85 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

3.45 NHC(-2)(-3) + $\mathfrak{su}_3$ + (-3) $\rightarrow$ (-2) + $\mathfrak{so}_8$

$$T = 8 \quad [0] \quad 1^{11} 31^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 20 & 12 & 1 & 7 \\ 20 & 25 & 16 & 0 & 9 \\ 12 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3 1^{\overline{11}} 2^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -102, \ \det = 209, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 10 & 16 & 12 & 1 & 6 \\ 16 & 20 & 17 & 0 & 8 \\ 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \ 1^{\overline{11}} 4 \ 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 14, \ \Delta = -93, \ \det = -129, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 13 & 14 & 9 & 1 & 7 \\ 14 & 13 & 10 & 0 & 7 \\ 9 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{2^1 1^1}} 4 \ 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 14, \ \Delta = -93, \ \det = -321, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.18 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 5 & 10 & 7 & 1 & 5 \\ 10 & 13 & 10 & 0 & 7 \\ 7 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

3.46 NHC(-2)(-3) + $\mathfrak{su}_3$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{1^1}} 5 1^{\overline{11}} 2^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 13, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 3 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 5 1^{\overline{1}} 2^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 40, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 5 1^{\overline{2^1}} 2^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 60, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 3 & 0 & 10 & 4 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 10 & 6 & 0 & 21 & 9 \\ 3 & 4 & 2 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 5 1^{\overline{1}} 2^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 85, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 3 & 0 & 7 & 4 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 7 & 4 & 0 & 8 & 6 \\ 3 & 4 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{12}} 5 1^{\overline{1}} 2^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 141, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 3 & 0 & 11 & 4 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 7 & 11 & 6 & 0 & 21 & 9 \\ 3 & 4 & 2 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{13}} 5 1^{\overline{1}} 2^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 181, \ (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 3 & 0 & 15 & 4 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 15 & 8 & 0 & 40 & 12 \\ 3 & 4 & 2 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{12} \overset{1}{5} 1^{32} \overset{1}{2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 181, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 3 & 0 & 17 & 4 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 13 & 17 & 10 & 0 & 61 & 15 \\ 3 & 4 & 2 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{12} \overset{1}{5} 1^{22} \overset{1}{2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 196, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 3 & 0 & 14 & 4 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 14 & 8 & 0 & 38 & 12 \\ 3 & 4 & 2 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{21} \overset{1}{5} 1^{11} \overset{1}{2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 285, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 1 & 5 & 3 \\ 6 & 13 & 5 & 0 & 12 & 7 \\ 2 & 5 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 4 & 0 & 8 & 6 \\ 3 & 7 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{13} \overset{1}{5} 1^{22} \overset{1}{2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 316, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 3 & 0 & 18 & 4 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 18 & 10 & 0 & 61 & 15 \\ 3 & 4 & 2 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{13} \overset{1}{5} 1^{32} \overset{1}{2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 381, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 3 & 0 & 21 & 4 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 21 & 12 & 0 & 88 & 18 \\ 3 & 4 & 2 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{21} \overset{1}{5} 1^{12} \overset{1}{2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 516, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 1 & 10 & 3 \\ 6 & 13 & 5 & 0 & 24 & 7 \\ 2 & 5 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 24 & 8 & 0 & 38 & 12 \\ 3 & 7 & 2 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{21} \overset{1}{5} 1^{13} \overset{1}{2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 901, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 1 & 15 & 3 \\ 6 & 13 & 5 & 0 & 36 & 7 \\ 2 & 5 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 36 & 12 & 0 & 88 & 18 \\ 3 & 7 & 2 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [13] \ 1^{21} \overset{1}{6} 131 \overset{1}{5} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 10, (n_+, n_-, n_0) =$$

$$(1, 14, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 6 & 2 & 12 & 6 \\ 2 & 0 & 5 & 2 \\ 12 & 5 & 20 & 12 \\ 6 & 2 & 12 & 6 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 1^{11} 5^{131} 6^{12^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 2 & 3 \\ 2 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 3 \\ 3 & 3 & 0 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

3.47 NHC(-2)(-3) + $\textcolor{blue}{su}_3$ + (-3) $\rightarrow$ (-2) + $\textcolor{blue}{su}_3$

$$T = 8 \quad [0] \quad 1^{11} 31^{132^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 9 & 3 \\ 4 & 4 & 5 & 0 & 12 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 12 & 9 & 0 & 24 & 8 \\ 3 & 4 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{112^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 3 & 4 \\ 8 & 9 & 4 & 0 & 4 & 5 \\ 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 4 & 5 & 2 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12} 31^{111} 2^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 9 & 3 \\ 4 & 4 & 4 & 0 & 14 & 4 \\ 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 14 & 9 & 0 & 36 & 10 \\ 3 & 4 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 31^{122^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 6 & 3 \\ 4 & 4 & 5 & 0 & 8 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 31^{112^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 & 3 \\ 4 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 31^{112^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 237, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 9 & 4 \\ 5 & 4 & 4 & 0 & 9 & 4 \\ 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 9 & 6 & 0 & 12 & 6 \\ 4 & 4 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 31^{111} 2^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 453, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 6 & 3 \\ 4 & 4 & 4 & 0 & 9 & 4 \\ 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

3.48 $\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \quad 1^{\textcolor{teal}{1}} 31^{\textcolor{teal}{1}} 2^{\textcolor{teal}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 17 & 1 & 7 \\ 20 & 25 & 23 & 0 & 9 \\ 17 & 23 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} 31^{\textcolor{teal}{1}} 2^{\textcolor{teal}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 17 & 1 & 7 \\ 17 & 20 & 22 & 0 & 8 \\ 17 & 22 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \quad 1^{\textcolor{teal}{1}} 2^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} 4 \quad 1^{\textcolor{teal}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 10 & 1 & 5 \\ 10 & 13 & 15 & 0 & 7 \\ 10 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

3.49 $\mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$

$$T = 3 \quad [0] \quad 212^{\textcolor{red}{2}} + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8 \quad T = 3, T_{\max} = 110, \Delta = -241, \det = -4, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 110, T \in [11, 110]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 12^{\textcolor{red}{2}} 1 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8 \quad T = 3, T_{\max} = 110, \Delta = -241, \det = -4, (n_+, n_-, n_0) = (1, 3, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 110, T \in [11, 110]$$

$$\begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 0 \\ 2 & 0 & 0 \end{pmatrix}_{b_0}$$

3.50 $\mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$

$$T = 4 \quad [0] \quad 12^{\textcolor{teal}{2}} 1^{\textcolor{teal}{1}} + \mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8 \quad T = 4, T_{\max} = 99, \Delta = -211, \det = 8, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 99, T \in [11, 99]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 0 \\ 2 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

3.51 $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$

$$T = 5 \quad [0] \quad 1^{\textcolor{teal}{1}} 2^{\textcolor{teal}{2}} 1^{\textcolor{teal}{1}} + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8 \quad T = 5, T_{\max} = 98, \Delta = -181, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

3.52 $\mathfrak{su}_2 + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [0] \quad 1^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} 2^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} + \mathfrak{su}_2 + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 96, \Delta = -142, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 96, T \in [14, 96]$$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 0 & 7 \\ 1 & 0 & -1 & -1 \\ 5 & 7 & -1 & -1 \end{pmatrix}_{b_0}$$

3.53 $\mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [0] \quad 1^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} 2^{\textcolor{teal}{1}} 1^{\textcolor{teal}{1}} + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{e}_6 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 101, \Delta = -212, \det = 12, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 2 & 1 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -2 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [1] \quad 1^{11}22^11^1 + \textcolor{red}{su}_2 + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} + \textcolor{violet}{c}_6 + \textcolor{orange}{\hat{1}} \rightarrow \textcolor{orange}{1} \quad T = 7, \quad T_{\max} = 101, \quad \Delta = -183, \quad \det = -13, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.15 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 1 & 0 & 2 & 2 \\ 2 & 3 & 0 & -4 & 2 & -2 \end{pmatrix}_{b_0}$$

3.54 $\textcolor{red}{su}_2 + (-3) \rightarrow (-2) + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [0] \quad 1^{11}2^11^1 + \textcolor{red}{su}_2 + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} + \textcolor{violet}{f}_4 + \textcolor{orange}{\hat{1}} \rightarrow \textcolor{orange}{1} \quad T = 6, \quad T_{\max} = 100, \quad \Delta = -186, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.67 \rightarrow 13, \quad T_{\max} = 100, \quad T \in [13, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 2 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [1] \quad 1^{11}22^11^1 + \textcolor{red}{su}_2 + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} + \textcolor{violet}{f}_4 + \textcolor{orange}{\hat{1}} \rightarrow \textcolor{orange}{1} \quad T = 7, \quad T_{\max} = 99, \quad \Delta = -157, \quad \det = -8, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 99, \quad T \in [13, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 1 & 0 & 2 & 2 \\ 2 & 3 & 0 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

3.55 $\textcolor{red}{su}_2 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \quad 1^{12}2^11^1 + \textcolor{red}{su}_2 + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} + \textcolor{green}{\hat{1}} \rightarrow \textcolor{green}{1} \quad T = 5, \quad T_{\max} = 98, \quad \Delta = -163, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 5, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 98, \quad T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 & -1 \\ 2 & 2 & 0 & -1 & -2 & -2 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \quad 12^121^{11} + \textcolor{red}{su}_2 + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} + \textcolor{green}{\hat{1}} \rightarrow \textcolor{green}{1} \quad T = 6, \quad T_{\max} = 91, \quad \Delta = -134, \quad \det = 5, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.40 \rightarrow 13, \quad T_{\max} = 91, \quad T \in [13, 91]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 3 & 1 & 2 & 2 & 2 \\ 2 & 3 & 0 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}22^11^1 + \textcolor{red}{su}_2 + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} + \textcolor{green}{\hat{1}} \rightarrow \textcolor{green}{1} \quad T = 6, \quad T_{\max} = 91, \quad \Delta = -134, \quad \det = 14, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.43 \rightarrow 11, \quad T_{\max} = 91, \quad T \in [11, 91]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 2 & 3 & 1 & 2 & 2 & 2 \\ 2 & 3 & 0 & 2 & 1 & -1 \end{pmatrix}_{b_0}$$

3.56 $\textcolor{red}{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$

$$T = 6 \quad [0] \quad 1^{11}2^11^1 + \textcolor{red}{su}_2 + \textcolor{blue}{(-3)} \rightarrow \textcolor{blue}{(-2)} + \textcolor{violet}{so}_8 + \textcolor{orange}{\hat{1}} \rightarrow \textcolor{orange}{1} \quad T = 6, \quad T_{\max} = 99, \quad \Delta = -162, \quad \det = 6, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 99, \quad T \in [13, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 2 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & 0 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [1] \quad 1^{11}22^11^1 + \textcolor{red}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) + \textcolor{green}{\mathfrak{so}}_8 + \hat{1} \rightarrow 1 \quad T = 7, \quad T_{\max} = 91, \quad \Delta = -133, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13, \quad T_{\max} = 91, \quad T \in [13, 91]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 1 & 0 & 2 & 2 \\ 2 & 3 & 0 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$\mathbf{3.57} \quad \mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$$

$$T = 6 \quad [0] \quad 1^{11}2^11^1 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 6, \quad T_{\max} = 97, \quad \Delta = -151, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.33 \rightarrow 12, \quad T_{\max} = 97, \quad T \in [12, 97]$$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 12 & 0 & 6 \\ 1 & 0 & -1 & -1 \\ 5 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{3.58} \quad \mathfrak{su}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$$

$$T = 5 \quad [0] \quad 1^{12}2^11^1 + \textcolor{red}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 5, \quad T_{\max} = 98, \quad \Delta = -172, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 5, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.33 \rightarrow 12, \quad T_{\max} = 98, \quad T \in [12, 98]$$

$$\begin{pmatrix} 5 & 1 & 5 \\ 1 & -1 & -1 \\ 5 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \quad 1^{12}2^11^1 + \textcolor{red}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 6, \quad T_{\max} = 96, \quad \Delta = -143, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.33 \rightarrow 14, \quad T_{\max} = 96, \quad T \in [14, 96]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 2 & 0 & -1 & -2 \end{pmatrix}_{b_0}$$

$$\mathbf{3.59} \quad \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$$

$$T = 6 \quad [0] \quad 1^{11}2^11^1 + \textcolor{red}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{green}{\mathfrak{so}}_8 + \hat{1} \rightarrow 1 \quad T = 6, \quad T_{\max} = 99, \quad \Delta = -171, \quad \det = 15, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 11.73 \rightarrow 12, \quad T_{\max} = 99, \quad T \in [12, 99]$$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 11 & 0 & 5 \\ 1 & 0 & -1 & -1 \\ 5 & 5 & -1 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{3.60} \quad (-3) \rightarrow (-2) + \mathfrak{e}_6 + \hat{1} \rightarrow (-1)$$

$$T = 5 \quad [0] \quad 1^{12}2^11^1 + (-3) \rightarrow (-2) + \textcolor{blue}{\mathfrak{e}}_6 + \hat{1} \rightarrow 1 \quad T = 5, \quad T_{\max} = 123, \quad \Delta = -242, \quad \det = -15, \quad (n_+, n_-, n_0) = (1, 5, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 123, \quad T \in [14, 123]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -4 & 0 & -2 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -2 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \quad 1^{12}2^121^1 + (-3) \rightarrow (-2) + \textcolor{blue}{\mathfrak{e}}_6 + \hat{1} \rightarrow 1 \quad T = 6, \quad T_{\max} = 101, \quad \Delta = -213, \quad \det = 7, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 1 & 0 & 1 \\ 1 & 0 & -1 & -1 \\ 3 & 1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12}2^121^1 + (-3) \rightarrow (-2) + \textcolor{blue}{\mathfrak{e}}_6 + \hat{1} \rightarrow 1 \quad T = 6, \quad T_{\max} = 101, \quad \Delta = -213, \quad \det = 12, \quad (n_+, n_-, n_0) = (1, 6, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.33 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -6 & 0 & -4 \\ 3 & 0 & 6 & 4 \\ 3 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

3.61 $(-3) \rightarrow (-2) + \mathfrak{e}_7 + \hat{1} \rightarrow (-1)$

$T = 5$ [0] $1^1 2^1 1^1 + (-3) \rightarrow (-2) + \mathfrak{e}_7 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 187$, $\Delta = -297$, $\det = -21$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 187, T \in [15, 187]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -6 & 0 & -4 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -4 & -1 & -4 \end{pmatrix}_{b_0}$$

$T = 6$ [1] $1^1 2^1 21^1 + (-3) \rightarrow (-2) + \mathfrak{e}_7 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 180$, $\Delta = -268$, $\det = 11$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 180, T \in [15, 180]$$

$$\begin{pmatrix} 10 & 1 & 0 \\ 1 & -1 & -1 \\ 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[2] $1^1 2^1 21^1 + (-3) \rightarrow (-2) + \mathfrak{e}_7 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 180$, $\Delta = -268$, $\det = 18$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 15, T_{\max} = 180, T \in [15, 180]$$

$$\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -8 & 0 & -6 \\ 3 & 0 & 6 & 4 \\ 3 & -6 & 4 & -2 \end{pmatrix}_{b_0}$$

3.62 $(-3) \rightarrow (-2) + \mathfrak{e}'_7 + \hat{1} \rightarrow (-1)$

$T = 5$ [0] $1^1 2^1 1^1 + (-3) \rightarrow (-2) + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 179$, $\Delta = -269$, $\det = -18$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 14, T_{\max} = 179, T \in [14, 179]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -5 & 0 & -3 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -3 & -1 & -3 \end{pmatrix}_{b_0}$$

$T = 6$ [1] $1^1 2^1 21^1 + (-3) \rightarrow (-2) + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 108$, $\Delta = -240$, $\det = 9$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 14, T_{\max} = 108, T \in [14, 108]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 0 & 0 & 0 \\ 1 & 0 & -1 & -1 \\ 3 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[2] $1^1 2^1 21^1 + (-3) \rightarrow (-2) + \mathfrak{e}'_7 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 108$, $\Delta = -240$, $\det = 15$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 14.13 \rightarrow 15, T_{\max} = 108, T \in [15, 108]$$

$$\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -7 & 0 & -5 \\ 3 & 0 & 6 & 4 \\ 3 & -5 & 4 & -2 \end{pmatrix}_{b_0}$$

3.63 $(-3) \rightarrow (-2) + \mathfrak{f}_4 + \hat{1} \rightarrow (-1)$

$T = 5$ [0] $1^1 2^1 1^1 + (-3) \rightarrow (-2) + \mathfrak{f}_4 + \hat{1} \rightarrow 1$ $T = 5$, $T_{\max} = 100$, $\Delta = -216$, $\det = -12$, $(n_+, n_-, n_0) = (1, 5, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$T = 6$ [1] $1^1 2^1 21^1 + (-3) \rightarrow (-2) + \mathfrak{f}_4 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 100$, $\Delta = -187$, $\det = 5$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 2 & 0 & 2 \\ 1 & 0 & -1 & -1 \\ 3 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

[2] $1^1 2^1 21^1 + (-3) \rightarrow (-2) + \mathfrak{f}_4 + \hat{1} \rightarrow 1$ $T = 6$, $T_{\max} = 100$, $\Delta = -187$, $\det = 9$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -5 & 0 & -3 \\ 3 & 0 & 6 & 4 \\ 3 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

3.64 $(-3) \rightarrow (-2) + \hat{1} \rightarrow (-1)$

$$T = 4 \quad [0] \quad 12^1 1^1 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -193, \det = 3, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 98, T \in [12, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 2 & 0 & 1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 2^1 1^1 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -193, \det = 11, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.36 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 1 \\ 2 & 0 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 21^1 2^1 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 4, T_{\max} = 98, \Delta = -193, \det = 11, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.36 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 1 \\ 2 & 0 & 1 & 0 \end{pmatrix}_{b_0}$$

$$T = 5 \quad [3] \quad 12^1 21^1 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -164, \det = -2, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 98, T \in [12, 98]$$

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & -1 \\ 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 122^1 1^1 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -164, \det = -3, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 4 \\ 3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^1 2^1 21^1 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 98, \Delta = -164, \det = -10, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.40 \rightarrow 11, T_{\max} = 98, T \in [11, 98]$$

$$\begin{pmatrix} 1 & 4 & 3 \\ 4 & 6 & 4 \\ 3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [0] \quad 2^1 1^1 8^1 12321^1 + (-3) \rightarrow (-2) + \hat{1} \rightarrow 1 \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0}$$

3.65 $(-3) \rightarrow (-2) + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8$

$$T = 4 \quad [0] \quad 2^1 12^2 + (-3) \rightarrow (-2) + \mathfrak{so}_{16} \rightarrow \mathfrak{su}_8 \quad T = 4, T_{\max} = 99, \Delta = -221, \det = 12, (n_+, n_-, n_0) = (1, 4, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 0 & 0 \\ 2 & 0 & -4 & -2 \\ 2 & 0 & -2 & -1 \end{pmatrix}_{b_0}$$

3.66 $(-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow (-1)$

$$T = 5 \quad [0] \quad 1^1 2^1 1^1 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 5, T_{\max} = 99, \Delta = -192, \det = -9, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -1 & -1 \\ 2 & 0 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 6 \quad [1] \quad 1^1 2^1 21^1 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 99, \Delta = -163, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 3 & 0 & 3 \\ 1 & 0 & -1 & -1 \\ 3 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 2^1 21^1 + (-3) \rightarrow (-2) + \mathfrak{so}_8 + \hat{1} \rightarrow 1 \quad T = 6, T_{\max} = 99, \Delta = -163, \det = 6, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 99, T \in [13, 99]$$

$$\begin{pmatrix} 1 & 1 & 3 & 3 \\ 1 & -4 & 0 & -2 \\ 3 & 0 & 6 & 4 \\ 3 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$\mathbf{3.67} \quad \mathfrak{su}_8 \rightarrow \mathfrak{so}_{16} + (-3) \rightarrow (-2)$$

$$T=6 \quad [0] \, 14^{\textcolor{red}{2}} 122^{\textcolor{blue}{1}} + \textcolor{red}{su}_8 \rightarrow \textcolor{red}{so}_{16} + (\textcolor{blue}{-3}) \rightarrow (\textcolor{blue}{-2}) \quad T=6, \, T_{\max}=99, \, \Delta=-163, \, \det=10, \, (n_+, n_-, n_0)=(1, 6, 0), \, c_{\text{ext}}=3.5$$

$$T_{\min}=11.40 \rightarrow 12, \, T_{\max}=99, \, T \in [12, 99]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 2 \\ 1 & 0 & -3 & -1 \\ 2 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

4 NHC $(-2)(-3)$ External (1227 combos, 49200 bases)

4.1 NHC $(-2)(-3)$

$$T = 9 \quad [247] \quad 1^1 2231321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 10, \Delta = -39, \det = -1, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 9, T_{\max} = 10, T \in [9, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[248] \quad 1^1 2231321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 10, \Delta = -39, \det = -1, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 9, T_{\max} = 10, T \in [9, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$T = 10 \quad [365] \quad 1^1 31413221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 11, \Delta = -37, \det = 1, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[366] \quad 1^1 31413221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 11, \Delta = -37, \det = 1, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[367] \quad 1^1 23141321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 11, \Delta = -38, \det = 1, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 11 \quad [532] \quad 1^1 314141321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -36, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[533] \quad 1^1 5^{12231} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[524] \quad 1^1 314141321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -36, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[531] \quad 1^1 41413221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 11, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[525] \quad 1^1 231512321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -35, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[526] \quad 1^1 223151231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[527] \quad 1^1 231512321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -35, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[528] \quad 1^1 41413221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 11, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$$

$$\begin{aligned}
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9 \\
[529] & 1^{\overline{1}2231} 1^{\overline{1}} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9 \\
[530] & 1^{\overline{1}2231} 1^{\overline{1}} 1321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9 \\
[534] & 1^{\overline{1}223151231} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9 \\
T = 12 & [712] 1^{\overline{1}4131513221} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10} \\
[707] & 1^{\overline{1}2321612321} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11} \\
[708] & 1^{\overline{1}3141414131} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10} \\
[709] & 1^{\overline{1}} 1414141321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 12, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11} \\
[710] & 1^{\overline{1}4131513221} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11} \\
[711] & 1^{\overline{1}} 1414141321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 12, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10} \\
T = 13 & [881] 1^{\overline{1}} 15131513221^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 13, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 8 & 8 \\ 3 & 8 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12} \\
[880] & 1^{\overline{1}22315132151} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 13, \Delta = -27, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11} \\
[879] & 1^{\overline{1}} 14141414131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 13, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}
\end{aligned}$$

$$[878] \ 1^1 22315132151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 13, \Delta = -27, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$[877] \ 1^1 23151315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -34, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$[876] \ 1^1 23151315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -34, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$[875] \ 1^1 2321 \ 12321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -32, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[874] \ 1^1 14141414131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 13, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[873] \ 1^1 15131513221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 13, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 3 & 5 \\ 2 & 2 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$T = 14 \quad [1047] \ 1^1 316123151321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[1058] \ 1^1 316123151321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[1059] \ 1^1 2315131 \ 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[1060] \ 1^1 232161315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -33, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[1061] \ 1^1 141414141 \ 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 14, \Delta = -16, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 8 & 8 \\ 3 & 8 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$$

$$[1057] \ 21 \ 8 \ 12321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$$

$$[1056] \ 1^1 232161315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -33, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$$

$$[1055] \begin{smallmatrix} 1 \\ 6 \\ 1 \\ 1 \end{smallmatrix} 131513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 14, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 3 & 5 \\ 2 & 2 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

[1054] $1^1 6 1 2 2 3 1 5 1 3 2 2 1^1 + [\text{su}_2 \cdot \text{g}_2]$ $T = 14, T_{\max} = 14, \Delta = -24, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[1053] \ 216^{1^1}131513221^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 14, \ T_{\max} = 14, \ \Delta = -23, \ \det = 1, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

[1052] $1_{12321}^8 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 14, T_{\max} = 15, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[1051] \ 1^1 321613151321^1 + [\textcolor{red}{\mathfrak{s}}\mathfrak{u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[1050] \ 1 \overset{1}{\underset{6}{\text{---}}} \overset{1}{\text{---}} \overset{2315131}{\text{---}} 131 \overset{1}{\text{---}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

[1049] $16^1 1231513221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 14$, $T_{\max} = 14$, $\Delta = -24$, $\det = 1$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 3.0$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

[1048] $1^1 3 2 1 6 1 3 1 5 1 3 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 14, T_{\max} = 15, \Delta = -32, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$T = 15$ [1181] $1^1 2321712315131^1 + [\text{5u}_2 \cdot \text{g}_2]$ $T = 15$, $T_{\max} = 16$, $\Delta = -32$, $\det = -1$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 30$

$$T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

[1184] $17^1 12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 15, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$$

[1185] $1^1 3217123151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 15, T_{\max} = 16, \Delta = -31, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$$

[1186] $1316^{1}131513221^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 15, \Delta = -2, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$$

[1173] $1^1 3216131612321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 15, T_{\max} = 16, \Delta = -31, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14} \\
[1174] & 1 \overset{1}{7} 1231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 15, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 3 & 5 \\ 2 & 2 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13} \\
[1175] & 1^1 3216131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -31, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13} \\
[1176] & 1^1 4131613151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13} \\
[1177] & 1 \overset{1}{6} \overset{1}{31} 131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14} \\
[1178] & 1 \overset{1}{6} \overset{1}{31} 131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13} \\
[1179] & 1^1 4131613151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13} \\
[1180] & 1^1 2321712315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -32, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13} \\
[1182] & 1^1 3217123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -31, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13} \\
[1183] & 21 \overset{1}{7} 1231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 15, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13} \\
T = 16 & [1308] 1 \overset{1}{8} 12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 16, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 3 & 5 \\ 2 & 2 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14} \\
[1306] & 221 \overset{1}{8} 1231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 16, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14} \\
[1307] & 1 \overset{1}{8} \overset{1}{3151321} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 17, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17]
\end{aligned}$$

[1296] $1^1 31612321712321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 16, T_{\max} = 17, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$

[1297] $1^1 31612321712321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 16, T_{\max} = 17, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$

[1298] $1^1 21812231513221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 16, T_{\max} = 16, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$

[1299] $1^1 15131613151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 16, T_{\max} = 16, \Delta = -25, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$

[1302] $1^1 23218122315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 16, T_{\max} = 17, \Delta = -31, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$

$T = 17$ [1401] $1^1 151316131612321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 17, T_{\max} = 17, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$

[1394] $1^1 321613161315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 17, T_{\max} = 18, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$

[1402] $1^1 321613161315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 17, T_{\max} = 18, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$

[1403] $1^{131} 613161315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 17, T_{\max} = 18, \Delta = -29, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$

[1399] $1^{12321} 8123216131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 17, T_{\max} = 18, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$

[1404] $1^1 413218123151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 17, T_{\max} = 18, \Delta = -29, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$

[1405] $1^1 321712321712321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 17, T_{\max} = 18, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$

[1398] $1^{12321} 8123216131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 17, T_{\max} = 18, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$

[1406] $1^{1321712321712321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 17, T_{\max} = 18, \Delta = -30, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$

[1407] $1^{131} 1^6 13161315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 17, T_{\max} = 18, \Delta = -29, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$

[1400] $1^1 1^5 1316131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 17, T_{\max} = 17, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$

[1397] $1^{1413218123151321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 17, T_{\max} = 18, \Delta = -29, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 18, T \in [17, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$

[1395] $1^{1321} 8^{1231513221^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 17, T_{\max} = 17, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$

[1396] $1^{1216131613151321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 17, T_{\max} = 17, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$

$T = 18$ [1514] $1^{13216131613161231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 18, T_{\max} = 18, \Delta = -28, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$

[1508] $1^{1513218123151321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 18, T_{\max} = 18, \Delta = -24, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$

[1526] $1^{1321} 1^8 12321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -29, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 19, T \in [18, 19]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$

[1525] $1^{131} 1^6 13161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 18, T_{\max} = 18, \Delta = -26, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$

[1524] $1^{1316131613151321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 18, T_{\max} = 18, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$

[1523] $1^{12161316131612321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 18, T_{\max} = 18, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$

[1522] $1^{12321712321812231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -29, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 19, T \in [18, 19]$

[illegible]

$$T = 19 \quad [1636] \quad 13161316131612321^{1+} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 19, T_{\max} = 19, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{aligned}
[1627] \quad & 1^{131} 6^{131} 1316131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 19, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17} \\
[1626] \quad & 1^{141} 41316131613161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 19, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17} \\
[1625] \quad & 1^{131} 6^{131} 1316131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 19, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17} \\
[1624] \quad & 1^{131} 5^{131} 131613161315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 19, \Delta = -23, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18} \\
T = 20 \quad [1732] \quad & 1^{131} 5^{131} 1316131613161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 20, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19} \\
[1733] \quad & 1^{131} 21^{131} 7^{131} 123218123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 20, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18} \\
[1734] \quad & 1^{141} 413161316131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 20, \Delta = -24, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18} \\
[1735] \quad & 1^{12321} 8^{12321} 123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 20, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18} \\
[1736] \quad & 1^{12321} 8^{12321} 123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 20, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18} \\
[1737] \quad & 1^{131} 6^{131} 1316131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 20, \Delta = -20, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18} \\
[1738] \quad & 1^{131} 6^{131} 1316131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -29, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 21, T \in [20, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18} \\
[1739] \quad & 1^{131} 21^{131} 6^{131} 131613161315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 20, \Delta = -20, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19} \\
[1740] \quad & 1^{131} 5^{131} 1321812321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 20, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$$

$$T = 21 \quad [1844] \quad 1^1 3221812321812315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 21, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$$

$$[1836] \quad 21 \quad (12) \quad 1223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 21, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$$

$$[1837] \quad 1^1 3218123151321812321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 22, \Delta = -29, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 21, T_{\max} = 22, T \in [21, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$$

$$[1838] \quad 1^1 3218123151321812321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 22, \Delta = -29, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 21, T_{\max} = 22, T \in [21, 22]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

$$[1839] \quad 1^1 513161316131613141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 21, \Delta = -19, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

$$[1840] \quad 21 6131613161316131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 21, \Delta = -17, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$$

$$[1841] \quad 1^1 3221812321812315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 21, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

$$[1842] \quad 1^1 612321812321712321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 21, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

$$[1843] \quad 1^1 3161232181232171231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 21, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$$

$$[1835] \quad 1^1 513161316131613141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 21, \Delta = -19, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$$

$$[1845] \quad 1^1 81232181232151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 21, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{19}$$

$$[1846] \quad 21 61316131613161231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 21, \Delta = -18, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

[1920] $1^1 21 6 13161316131613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -16, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[1921] $1^1 5 13221 21 (12) 1223151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$

[1922] $1^1 31513221 21 (12) 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -30, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 23, T \in [22, 23]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$

[1923] $1^1 41321812321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 3 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[1924] $1^1 21 7 12321812321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -20, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$

[1925] $1^1 32171232181232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[1926] $1^1 1321 8 123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[1927] $1^1 5 131613161316131 5 1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -14, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[1928] $1^1 1 (12) 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -30, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $1^1 31513221$
 $T_{\min} = 22, T_{\max} = 23, T \in [22, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[1929] $1^1 16 12321812321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[1930] $1^1 31612321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$

[1931] $1^1 31612321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[1932] $1^{1231513221} (12) 122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[1913] $1^{141321812321812315131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$

[1914] $1^{1231513221} (12) 122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[1916] $1^{123151321} (12) 1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

[1915] $1^{1321} 8^{123218123216131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$

$T = 23$ [1993] $1^{1321712321812321812231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$

[1996] $221^{1812321812321712321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -19, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$

[1997] $1^{1321} 8^{1232181232171231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$

[1998] $21^{1712321812321812321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -19, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$

[1999] $1^{1321} 8^{1232181232171231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$

[2000] $1^{1315132181232181232151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 23, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$

[2001] $1^1 315132181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 23, T_{\max} = 23, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$

[2002] $1^1 4141413221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$

[2003] $1^1 4141413221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$

[2004] $1^1 216131613161316131^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 23, T_{\max} = 23, \Delta = -11, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$

[2005] $1^1 151321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 23, T_{\max} = 23, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$

[2006] $1^1 151321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 23, T_{\max} = 23, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$

[2007] $1^1 321712321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$

[1995] $1^1 321812321712321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 23, T_{\max} = 23, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$

[1991] $1^1 413218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$

[1994] $1^1 321812321712321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 23, T_{\max} = 23, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$

[1992] $1^1 413218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 23, T_{\max} = 23, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$

$$\begin{aligned}
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22} \\
[2066] & 1^{12321812321} 8^{1232181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]} \quad T = 24, T_{\max} = 24, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22} \\
[2067] & 1^{15131513221} (12)^{1223151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]} \quad T = 24, T_{\max} = 24, \Delta = -25, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23} \\
[2068] & 1^{13161232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]} \quad T = 24, T_{\max} = 24, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22} \\
[2069] & 1^{13161232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]} \quad T = 24, T_{\max} = 24, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23} \\
\\
T = 25 & [2134] 1^{141321812321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]} \quad T = 25, T_{\max} = 25, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
[2135] & 1^{132171232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]} \quad T = 25, T_{\max} = 25, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23} \\
[2136] & 1^{132171232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]} \quad T = 25, T_{\max} = 25, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
[2137] & 1^{15132181232181232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]} \quad T = 25, T_{\max} = 25, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23} \\
[2138] & 1^{141321812321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]} \quad T = 25, T_{\max} = 25, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23} \\
[2132] & 21^{1712321812321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]} \quad T = 25, T_{\max} = 25, \Delta = -19, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
[2139] & 1^{15132181232181232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]} \quad T = 25, T_{\max} = 25, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
[2130] & \begin{matrix} 1^1 & & 1 \\ 16 & 131513221 & (12)1223151321^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 25, \quad \Delta = -22, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23} \\
[2140] & \begin{matrix} 1^1 321 \\ 1 & 8 & 123218123218123141^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 25, \quad \Delta = -24, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23} \\
[2141] & \begin{matrix} 1^1 31513221 \\ 1 & (12) & 1223151316131^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 26, \quad \Delta = -29, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, \quad T_{\max} = 26, \quad T \in [25, 26] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
[2142] & \begin{matrix} 1^1 321 \\ 1 & 8 & 123218123218123141^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 25, \quad \Delta = -24, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23} \\
[2143] & \begin{matrix} 1^1 \\ 16 & 123218123218123216131^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 25, \quad \Delta = -20, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23} \\
[2144] & \begin{matrix} 1^1 31513221 \\ 1 & (12) & 1223151316131^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 26, \quad \Delta = -29, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, \quad T_{\max} = 26, \quad T \in [25, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23} \\
[2129] & \begin{matrix} 1^1 3221 \\ 21 & (12) & 1223151321812321^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 25, \quad \Delta = -28, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
[2131] & \begin{matrix} 1^1 231513221 \\ 1 & (12) & 122315132161^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 25, \quad \Delta = -23, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23} \\
[2133] & \begin{matrix} 1^1 3221 \\ 21 & (12) & 1223151321812321^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, \quad T_{\max} = 25, \quad \Delta = -28, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 25, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{23} \\
T = 26 & [2191] \begin{matrix} 1^1 321 & & 1^1 \\ 1 & 8 & 123218123218123151^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, \quad T_{\max} = 26, \quad \Delta = -19, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, \quad T_{\max} = 26, \quad T \in [26, 26]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
[2192] & 1 \overset{1}{8} \overset{1}{1232181232181232151} + [\text{su}_2 \cdot \text{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25} \\
[2189] & 1 \overset{1}{8} \overset{1}{1232181232181231} \overset{1}{51} + [\text{su}_2 \cdot \text{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -19, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25} \\
[2188] & 1 \overset{1}{61232181232181232171231} + [\text{su}_2 \cdot \text{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -19, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
[2187] & 21 \overset{1}{7123218123218123216131} + [\text{su}_2 \cdot \text{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -18, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
[2186] & 1 \overset{1}{322181232181232181232151} + [\text{su}_2 \cdot \text{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25} \\
[2193] & 1 \overset{1}{51321812321812321812231} + [\text{su}_2 \cdot \text{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -19, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25} \\
[2194] & 1 \overset{1}{8} \overset{1}{1232181232181232151} + [\text{su}_2 \cdot \text{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
[2195] & 221 \overset{1}{812321812321812315131} + [\text{su}_2 \cdot \text{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -18, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25} \\
[2196] & 1 \overset{1}{71231513221} \overset{1}{(12)1223151321} + [\text{su}_2 \cdot \text{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
[2197] & 1 \overset{1}{(12)1223141} + [\text{su}_2 \cdot \text{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24} \\
[2198] & 1 \overset{1}{413218123218123218123141} + [\text{su}_2 \cdot \text{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$$

$$[2199] \ 1^1 5 1321812321812321812231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -19, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$$

$$[2200] \ 1^1 322181232181232181232151^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$$

$$[2201] \ 1^1 231513221 (12) \ 1223151322171^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$$

$$[2190] \ 1^1 232181231513221 (12) \ 1223141^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$$

$$T = 27 \quad [2250] \ 221^1 8 123218123218123216131^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -17, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$$

$$[2243] \ 1^1 51316131513221(12)1223151321^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$$

$$[2258] \ 1^1 41413221(12)1223151321812321^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$$

$$[2257] \ 1^1 4132181232181232181232151^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$$

$$[2256] \ 1^1 4132181232181232181232151^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$$

$$[2255] \ 1^1 61232181232181232181231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -18, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$$

$$[2254] \ 1^1 812231513221(12)1223151321^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$$

$$\begin{aligned}
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25} \\
[2253] & \overset{1}{1} 8 1231513221 \overset{1}{(12)} 1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25} \\
[2252] & \overset{1}{1} 6 12321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -18, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25} \\
[2251] & \overset{1}{1} 21 7 1232181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -17, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25} \\
[2241] & \overset{1}{1} 5 13218123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -18, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25} \\
[2249] & \overset{1}{1} \overset{1}{5} 13221 (12) 1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26} \\
[2248] & \overset{1}{1} \overset{1}{5} 13221 (12) 1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25} \\
[2247] & \overset{1}{1} \overset{1}{5} 1316131513221 (12) 1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25} \\
[2246] & \overset{1}{1} \overset{1}{3} 1513221 (12) 122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25} \\
[2245] & \overset{1}{1} \overset{1}{3} 1513221 (12) 122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -28, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26} \\
[2244] & \overset{1}{1} \overset{1}{4} 1413221 (12) 1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 27, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{25}$$

$$[2242] \quad 15 \overset{1}{1} 13218123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, \quad T_{\max} = 27, \quad \Delta = -18, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$$

$$T = 28 \quad [2300] \quad 15 \overset{1}{1} 132181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, \quad T_{\max} = 28, \quad \Delta = -15, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$$

$$[2302] \quad 15 \overset{1}{1} 132181232181232181231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, \quad T_{\max} = 28, \quad \Delta = -13, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$$

$$[2299] \quad 1^1 51232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, \quad T_{\max} = 28, \quad \Delta = -17, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$$

$$[2292] \quad 21 \overset{1}{7} 1232181232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, \quad T_{\max} = 28, \quad \Delta = -16, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$$

$$[2293] \quad 1^1 3161316131513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, \quad T_{\max} = 28, \quad \Delta = -28, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27}$$

$$[2294] \quad 1^1 3161316131513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, \quad T_{\max} = 28, \quad \Delta = -28, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$$

$$[2295] \quad 1^1 316131513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, \quad T_{\max} = 28, \quad \Delta = -28, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$$

$$[2303] \quad 16 \overset{1}{1} 1316131513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, \quad T_{\max} = 28, \quad \Delta = -21, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$$

$$[2296] \quad 21 \overset{1}{7} 12321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, \quad T_{\max} = 28, \quad \Delta = -16, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 28, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$$

$$[2297] \quad 221 \overset{1}{8} 1232181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, \quad T_{\max} = 28, \quad \Delta = -16, \quad \det = 1, \quad (n_+, n_-, n_0) =$$

$$(1, 28, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$$

$$[2298] \quad 16123218123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -17, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{26}$$

$$[2301] \quad 15132181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -15, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27}$$

$$T = 29 \quad [2344] \quad 217123218123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -15, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27}$$

$$[2343] \quad 1^1 5132181231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -23, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27}$$

$$[2342] \quad 1^1 5132181231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -23, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27}$$

$$[2341] \quad 161232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -14, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27}$$

$$[2340] \quad 16123218123218123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -12, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27}$$

$$[2339] \quad 22181232181232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -15, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27}$$

$$[2336] \quad 221812321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -15, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27}$$

$$[2337] \quad 1^1 232181231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{27} \\
[2338] & 1^1 232181231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, T_{\max} = 29, T \in [29, 29] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{28} \\
& T = 30 \quad [2381] 1^1 6131513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 30, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 30, T_{\max} = 30, T \in [30, 30] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29} \\
& [2374] 2218123218123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 30, \Delta = -14, \det = 1, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 30, T_{\max} = 30, T \in [30, 30] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{28} \\
[2377] & 1^1 316131513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 30, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 30, T_{\max} = 30, T \in [30, 30] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{28} \\
[2378] & 1^1 316131513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 30, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 30, T_{\max} = 30, T \in [30, 30] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29} \\
[2379] & 1^1 61231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 30, \Delta = -22, \det = 1, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 30, T_{\max} = 30, T \in [30, 30] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29} \\
[2380] & 1^1 61232181231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 30, \Delta = -22, \det = 1, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 30, T_{\max} = 30, T \in [30, 30] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{28} \\
[2375] & 1^1 2171232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 30, \Delta = -12, \det = 1, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 30, T_{\max} = 30, T \in [30, 30] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{28} \\
[2382] & 1^1 3221 1(12)1223151321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 30, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 30, T_{\max} = 30, T \in [30, 30] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{28} \\
[2383] & 1^1 3221 1(12)1223151321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 30, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 30, T_{\max} = 30, T \in [30, 30] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29} \\
[2376] & 1^1 217123218123218123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\max} = 30, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{28}$$

$$T = 31 \quad [2420] \quad 1 \overset{1}{7} 1232181231513221(12)1223151321^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29}$$

$$[2421] \quad 1 \overset{1}{3} 1513221 \overset{21}{(12)} 12231513221(12)12231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -27, \det = -1, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29}$$

$$[2422] \quad 1 \overset{1}{7} 12231513221(12)1223151321812321^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30}$$

$$[2423] \quad 221 \overset{1}{8} 1232181232181232181232151^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -11, \det = -1, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29}$$

$$[2418] \quad 1 \overset{1}{4} 13221(12)1223151321812321812321^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29}$$

$$[2417] \quad 1 \overset{1}{7} 1231513221(12)1223151321812321^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30}$$

$$[2416] \quad 1 \overset{1}{3} 1513221 \overset{21}{(12)} 12231513221(12)12231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -27, \det = -1, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 3 & 2 & 5 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29}$$

$$[2415] \quad 221 \overset{1}{8} 1232181232181232181231 \overset{1}{5} 1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -9, \det = -1, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{29}$$

$$[2419] \quad 1 \overset{1}{4} 13221(12)1223151321812321812321^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30}$$

$$T = 32 \quad [2458] \quad 1 \overset{1}{5} 13221(12)1223151321812321812321^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31} \\
[2456] & \begin{matrix} 1 \\ 1513221(12)1223151321812321812321 \end{matrix}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30} \\
[2457] & \begin{matrix} 1 \\ 2181232181231513221(12)1223151321 \end{matrix}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -19, \det = 1, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30} \\
[2455] & \begin{matrix} 1 \\ 1812231513221(12)1223151321812321 \end{matrix}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -19, \det = 1, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31} \\
[2459] & \begin{matrix} 1 \\ 132181232181231513221(12)122315131 \end{matrix}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31} \\
[2460] & \begin{matrix} 1 \\ 132181231513221(12)122315132181231 \end{matrix}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -26, \det = 1, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30} \\
[2461] & \begin{matrix} 1 \\ 1 \end{matrix} \begin{matrix} 131513221 \\ (12) \end{matrix} \begin{matrix} 1 \\ 12231513221(12)1223141 \end{matrix}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -26, \det = 1, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30} \\
[2462] & \begin{matrix} 1 \\ 1 \end{matrix} \begin{matrix} 131513221 \\ (12) \end{matrix} \begin{matrix} 1 \\ 12231513221(12)1223141 \end{matrix}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -26, \det = 1, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30} \\
[2463] & \begin{matrix} 1 \\ 2181231513221(12)1223151321812321 \end{matrix}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -19, \det = 1, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31} \\
[2454] & \begin{matrix} 1 \\ 132181232181231513221(12)122315131 \end{matrix}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, T_{\max} = 32, \Delta = -27, \det = 1, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 32, T_{\max} = 32, T \in [32, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30} \\
T = 33 & [2494] \begin{matrix} 1 \\ 1 \end{matrix} \begin{matrix} 131513221 \\ (12) \end{matrix} \begin{matrix} 1 \\ 12231513221(12)1223151 \end{matrix}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -21, \det = -1, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 33, T_{\max} = 33, T \in [33, 33] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31} \\
[2496] & \begin{matrix} 1 \\ 21 \end{matrix} \begin{matrix} 1513221 \\ (12) \end{matrix} \begin{matrix} 1 \\ 12231513221(12)1223151321 \end{matrix}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -22, \det = -1, (n_+, n_-, n_0) =
\end{aligned}$$

$$(1, 33, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31}$$

$$[2495] \quad 1^1 (12) \quad 12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -22, \det = -1, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31}$$

$$[2492] \quad 1^1 (12) \quad 12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31}$$

$$[2493] \quad 1^1 (12) \quad 12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31}$$

$$[2491] \quad 1^1 (12) \quad 12231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -21, \det = -1, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{31}$$

$$T = 34 \quad [2513] \quad 21 (12) \quad 12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 34, \Delta = -26, \det = 1, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 34, T_{\max} = 34, T \in [34, 34]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{33}$$

$$[2512] \quad 21 (12) \quad 12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 34, \Delta = -26, \det = 1, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 34, T_{\max} = 34, T \in [34, 34]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{32}$$

$$T = 35 \quad [2536] \quad 1^1 (12) \quad 12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 35, \Delta = -23, \det = -1, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 35, T_{\max} = 35, T \in [35, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{33}$$

$$[2535] \quad 1^1 (12) \quad 12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 35, \Delta = -23, \det = -1, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 35, T_{\max} = 35, T \in [35, 35]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34}$$

$$[2537] \quad 1^1 (12) \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, T_{\max} = 35, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 35, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 35, T_{\max} = 35, T \in [35, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{33}$$

$$\begin{aligned}
& \text{[2538]} \quad 1 \quad \overset{1}{1316131513221(12)12231513221} \quad (12) \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, \quad T_{\max} = 35, \quad \Delta = -25, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 35, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, \quad T_{\max} = 35, \quad T \in [35, 35] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{33} \\
& \text{[2539]} \quad 1 \quad \overset{1}{232181232181232181231513221(12)12231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, \quad T_{\max} = 35, \quad \Delta = -26, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 35, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, \quad T_{\max} = 35, \quad T \in [35, 35] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{33} \\
& \text{[2540]} \quad 1 \quad \overset{1}{232181232181232181231513221(12)12231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, \quad T_{\max} = 35, \quad \Delta = -26, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 35, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, \quad T_{\max} = 35, \quad T \in [35, 35] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34} \\
& T = 36 \quad \text{[2569]} \quad 1 \quad \overset{1}{41413221(12)12231513221(12)1223151316131^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, \quad T_{\max} = 36, \quad \Delta = -24, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 36, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 36, \quad T_{\max} = 36, \quad T \in [36, 36] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34} \\
& \text{[2570]} \quad 21 \quad \overset{1}{13221} \quad (12) \quad 12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, \quad T_{\max} = 36, \quad \Delta = -25, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 36, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 36, \quad T_{\max} = 36, \quad T \in [36, 36] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34} \\
& \text{[2571]} \quad 1 \quad \overset{1}{1513221} \quad (12) \quad 12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, \quad T_{\max} = 36, \quad \Delta = -20, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 36, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 36, \quad T_{\max} = 36, \quad T \in [36, 36] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35} \\
& \text{[2572]} \quad 1 \quad \overset{1}{231513221(12)12231513221} \quad (12) \quad 12231513161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, \quad T_{\max} = 36, \quad \Delta = -21, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 36, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 36, \quad T_{\max} = 36, \quad T \in [36, 36] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34} \\
& \text{[2573]} \quad 1 \quad \overset{1}{6131513221(12)12231513221(12)122315131^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, \quad T_{\max} = 36, \quad \Delta = -20, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 36, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 36, \quad T_{\max} = 36, \quad T \in [36, 36] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34} \\
& \text{[2565]} \quad 1 \quad \overset{1}{41413221(12)12231513221(12)1223151316131^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, \quad T_{\max} = 36, \quad \Delta = -24, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 36, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 36, \quad T_{\max} = 36, \quad T \in [36, 36] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35} \\
& \text{[2566]} \quad 1 \quad \overset{1}{1513221} \quad (12) \quad 12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, \quad T_{\max} = 36, \quad \Delta = -20, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 36, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 36, \quad T_{\max} = 36, \quad T \in [36, 36] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34} \\
& \text{[2567]} \quad 21 \quad \overset{1}{13221} \quad (12) \quad 12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, \quad T_{\max} = 36, \quad \Delta = -25, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 36, 0), \quad c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35} \\
& \begin{matrix} 1^1 31513221 \\ [2568] \end{matrix} 1 \quad (12) \quad 12231513221(12)122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34} \\
& \begin{matrix} 1^1 231513221(12)12231513221 \\ T = 37 \quad [2591] \end{matrix} 1 \quad (12) \quad 122315132171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, T_{\max} = 37, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35} \\
& \begin{matrix} 1^1 31513221 \\ [2592] \end{matrix} 1 \quad (12) \quad 12231513221(12)1223151322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, T_{\max} = 37, \Delta = -19, \det = -1, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35} \\
& \begin{matrix} 1^1 32181231513221(12)12231513221 \\ [2593] \end{matrix} 1 \quad (12) \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, T_{\max} = 37, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35} \\
& \begin{matrix} 1^1 \\ [2594] \end{matrix} 1^1 7 1231513221(12)12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, T_{\max} = 37, \Delta = -19, \det = -1, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35} \\
& \begin{matrix} 1^1 32181231513221(12)12231513221 \\ [2590] \end{matrix} 1 \quad (12) \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 37, T_{\max} = 37, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35} \\
& T = 38 \quad [2623] \quad 1^1 41413221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{37} \\
& \begin{matrix} 1^1 \\ [2614] \end{matrix} 21 8 1231513221(12)12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -18, \det = 1, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{36} \\
& \begin{matrix} 1^1 \\ [2615] \end{matrix} 1^1 41413221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{36} \\
& \begin{matrix} 1^1 \\ [2616] \end{matrix} 1^1 8 12231513221(12)12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, T_{\max} = 38, \Delta = -18, \det = 1, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 1 & 3 \\ 2 & 4 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{36} \\
& \begin{matrix} 1^1 231513221(12)12231513221 \\ [2617] \ 1 \quad (12) \quad 1223151322181^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \end{matrix} \quad T = 38, T_{\max} = 38, \Delta = -19, \det = 1, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{36} \\
& \begin{matrix} 1^1 1513221 \\ [2618] \ 1 \quad (12) \quad 12231513221(12)122315132181231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \end{matrix} \quad T = 38, T_{\max} = 38, \Delta = -19, \det = 1, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{37} \\
& \begin{matrix} 1^1 1513221 \\ [2619] \ 1 \quad (12) \quad 12231513221(12)122315132181231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \end{matrix} \quad T = 38, T_{\max} = 38, \Delta = -19, \det = 1, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{36} \\
& \begin{matrix} 1^1 181231513221(12)12231513221(12)1223151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \\ [2620] \end{matrix} \quad T = 38, T_{\max} = 38, \Delta = -19, \det = 1, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{36} \\
& \begin{matrix} 1^1 316131513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \\ [2621] \end{matrix} \quad T = 38, T_{\max} = 38, \Delta = -22, \det = 1, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{37} \\
& \begin{matrix} 1^1 316131513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \\ [2622] \end{matrix} \quad T = 38, T_{\max} = 38, \Delta = -22, \det = 1, (n_+, n_-, n_0) = \\
& (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{36} \\
& T = 39 \quad \begin{matrix} 1^1 316131513221(12)12231513221(12)122315132161^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \\ [2631] \end{matrix} \quad T = 39, T_{\max} = 39, \Delta = -20, \det = \\
& -1, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 39, T_{\max} = 39, T \in [39, 39] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{37} \\
& \begin{matrix} 1^1 61316131513221(12)12231513221(12)1223151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \\ [2632] \end{matrix} \quad T = 39, T_{\max} = 39, \Delta = -20, \det = \\
& -1, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 39, T_{\max} = 39, T \in [39, 39] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{37} \\
& \begin{matrix} 1^1 16131513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \\ [2630] \end{matrix} \quad T = 39, T_{\max} = 39, \Delta = -19, \det = -1, (n_+, n_-, n_0) = \\
& (1, 39, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 39, T_{\max} = 39, T \in [39, 39] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{37} \\
& T = 40 \quad \begin{matrix} 1^1 3221 \\ [2640] \end{matrix} \quad \begin{matrix} 21 \\ (12) \end{matrix} \quad 12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 40, T_{\max} = 40, \Delta = -24, \det = \\
& 1, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 40, T_{\max} = 40, T \in [40, 40]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{38} \\
[2642] & \text{}^1 1^3 2181231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 40, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 40, T_{\max} = 40, T \in [40, 40] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{38} \\
[2643] & \text{}^1 1^7 1231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 40, \Delta = -18, \det = 1, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 40, T_{\max} = 40, T \in [40, 40] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{38} \\
[2644] & \text{}^1 1^3 16131513221(12)12231513221(12)1223151322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 40, \Delta = -18, \det = 1, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 40, T_{\max} = 40, T \in [40, 40] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{38} \\
[2641] & \text{}^1 1^3 2181231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 40, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 40, T_{\max} = 40, T \in [40, 40] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{39} \\
T = 41 & [2656] \text{}^1 1^6 131513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 41, \Delta = -18, \det = -1, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{39} \\
[2657] & \text{}^1 1^6 1231513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 41, \Delta = -19, \det = -1, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{39} \\
[2658] & \text{}^1 21^8 1231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 41, \Delta = -17, \det = -1, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{39} \\
[2659] & \text{}^1 1^3 221^1 21(12)12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 41, \Delta = -23, \det = -1, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{39} \\
[2660] & \text{}^1 1^8 12231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 41, \Delta = -17, \det = -1, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{39} \\
[2661] & \text{}^1 1^3 221^1 21(12)12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 41, \Delta = -23, \det = -1, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{39}
\end{aligned}$$

$$\begin{aligned}
& T = 42 \quad [2683] \quad 21 \quad (12) \quad 12231513221(12)12231513221(12)122314141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -22, \det = 1, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 42, T_{\max} = 42, T \in [42, 42] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{40} \\
& [2674] \quad 1^1 712231513221(12)12231513221(12)122315132181231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -17, \det = 1, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 42, T_{\max} = 42, T \in [42, 42] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{40} \\
& [2675] \quad 21 \quad (12) \quad 12231513221(12)12231513221(12)122314141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -22, \det = 1, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 42, T_{\max} = 42, T \in [42, 42] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{40} \\
& [2676] \quad 1^1 71232181231513221(12)12231513221(12)1223151321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -19, \det = 1, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 42, T_{\max} = 42, T \in [42, 42] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{40} \\
& [2677] \quad 1^1 171231513221(12)12231513221(12)122315132181231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -17, \det = 1, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 42, T_{\max} = 42, T \in [42, 42] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{40} \\
& [2678] \quad 1^1 413221(12)12231513221(12)12231513221(12)1223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -22, \det = 1, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 42, T_{\max} = 42, T \in [42, 42] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{40} \\
& [2679] \quad 1^1 3221 \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)122315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -26, \det = 1, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 42, T_{\max} = 42, T \in [42, 42] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{40} \\
& [2680] \quad 1^1 3221 \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)122315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -26, \det = 1, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 42, T_{\max} = 42, T \in [42, 42] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{40} \\
& [2681] \quad 1^1 513221 \quad 21 \quad (12) \quad 12231513221(12)12231513221(12)12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -18, \det = 1, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 42, T_{\max} = 42, T \in [42, 42] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{40} \\
& [2682] \quad 1^1 513221 \quad 21 \quad (12) \quad 12231513221(12)12231513221(12)12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -18, \det = 1, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 42, T_{\max} = 42, T \in [42, 42] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{40}
\end{aligned}$$

$$T = 43 \quad [2703] \quad 1 \quad \begin{matrix} 1^1 \\ 1^5 13221 \end{matrix} (12) \quad 12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 43, T_{\max} = 43, \Delta = -17, \det = -1, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{41}$$

$$[2695] \quad 1^1 \quad 18 \quad 12231513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 43, T_{\max} = 43, \Delta = -16, \det = -1, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{41}$$

$$[2696] \quad 1^1 \quad 21 \quad 8 \quad 1231513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 43, T_{\max} = 43, \Delta = -16, \det = -1, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{41}$$

$$[2697] \quad 1^1 \quad 18 \quad 1232181231513221(12)12231513221(12)1223151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 43, T_{\max} = 43, \Delta = -18, \det = -1, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{41}$$

$$[2698] \quad 1^1 \quad 41413221(12)12231513221(12)12231513221 \quad (12) \quad 1223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 43, T_{\max} = 43, \Delta = -21, \det = -1, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{41}$$

$$[2699] \quad 1^1 \quad 41413221(12)12231513221(12)12231513221 \quad (12) \quad 1223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 43, T_{\max} = 43, \Delta = -21, \det = -1, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{41}$$

$$[2700] \quad 1^1 \quad 413221(12)12231513221 \quad (12) \quad 12231513221(12)122315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 43, T_{\max} = 43, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{42}$$

$$[2701] \quad 1^1 \quad 413221(12)12231513221 \quad (12) \quad 12231513221(12)122315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 43, T_{\max} = 43, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{41}$$

$$[2702] \quad 1^1 \quad 1^5 13221 \quad (12) \quad 12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 43, T_{\max} = 43, \Delta = -17, \det = -1, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{41}$$

$$T = 44 \quad [2713] \quad 21 \quad \begin{matrix} 1^1 \\ 1^5 13221 \end{matrix} (12) \quad 12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 44, T_{\max} = 44, \Delta = -20, \det = 1, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 44, T_{\max} = 44, T \in [44, 44]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{42} \\
& [2714] \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -16, \det = \\
& 1, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 44, T_{\max} = 44, T \in [44, 44] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{42} \\
& [2715] \quad 21 \quad (12) \quad 12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -20, \det = \\
& 1, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 44, T_{\max} = 44, T \in [44, 44] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{43} \\
& [2716] \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -16, \det = \\
& 1, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 44, T_{\max} = 44, T \in [44, 44] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{42} \\
& [2717] \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)122315141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -12, \det = 1, (n_+, n_-, n_0) = \\
& (1, 44, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 44, T_{\max} = 44, T \in [44, 44] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{42} \\
& [2718] \quad 1^1 41413221(12)12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -20, \det = \\
& 1, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 44, T_{\max} = 44, T \in [44, 44] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{42} \\
& [2719] \quad 1513221(12)12231513221(12)12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -20, \det = \\
& 1, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 44, T_{\max} = 44, T \in [44, 44] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{43} \\
& [2720] \quad 1513221(12)12231513221(12)12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -20, \det = \\
& 1, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 44, T_{\max} = 44, T \in [44, 44] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{42} \\
& T = 45 \quad [2725] \quad 21 \quad (12) \quad 12231513221(12)12231513221(12)122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 45, T_{\max} = 45, \Delta = \\
& -18, \det = -1, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 45, T_{\max} = 45, T \in [45, 45] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{43} \\
& [2727] \quad 1 \quad (12) \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 45, T_{\max} = 45, \Delta = -19, \det = -1, (n_+, n_-, n_0) = \\
& (1, 45, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 45, T_{\max} = 45, T \in [45, 45] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 2 & 4 \\ 2 & 3 & 4 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{43} \\
& [2726] \quad 1 \quad (12) \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 45, T_{\max} = 45, \Delta = -19, \det = -1, (n_+, n_-, n_0) = \\
& (1, 45, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 45, T_{\max} = 45, T \in [45, 45]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{43}$$

$$[2724] \quad 1 \overset{1}{6} 131513221(12)12231513221(12)12231513221(12)12231^{\overset{21}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 45, T_{\max} = 45, \Delta = -17, \det = -1, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 45, T_{\max} = 45, T \in [45, 45]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{43}$$

$$T = 46 \quad [2742] \quad 1 \overset{1}{6} 131513221(12)12231513221(12)12231513221(12)1223141^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 46, T_{\max} = 46, \Delta = -16, \det = 1, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 46, T_{\max} = 46, T \in [46, 46]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{44}$$

$$[2739] \quad 1 \overset{1}{15} 13221(12)12231513221(12)12231513221(12)12231513151^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 46, T_{\max} = 46, \Delta = -14, \det = 1, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 46, T_{\max} = 46, T \in [46, 46]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{44}$$

$$[2741] \quad 1^{\overset{1}{1}} 41413221(12)12231513221(12)12231513221(12)12231513151^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 46, T_{\max} = 46, \Delta = -18, \det = 1, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 46, T_{\max} = 46, T \in [46, 46]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{44}$$

$$[2735] \quad 1 \overset{1}{6} 1231513221(12)12231513221(12)12231513221(12)1223141^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 46, T_{\max} = 46, \Delta = -17, \det = 1, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 46, T_{\max} = 46, T \in [46, 46]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{44}$$

$$[2736] \quad 1 \overset{1}{15} 13221(12)12231513221(12)12231513221(12)12231513151^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 46, T_{\max} = 46, \Delta = -14, \det = 1, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 46, T_{\max} = 46, T \in [46, 46]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45}$$

$$[2737] \quad 1 \overset{1}{7} 1231513221(12)12231513221(12)12231513221(12)12231^{\overset{21}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 46, T_{\max} = 46, \Delta = -16, \det = 1, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 46, T_{\max} = 46, T \in [46, 46]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{44}$$

$$[2738] \quad 21 \overset{1}{1} 3221(12)12231513221(12)12231513221(12)1223151322171^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 46, T_{\max} = 46, \Delta = -16, \det = 1, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 46, T_{\max} = 46, T \in [46, 46]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{44}$$

$$[2740] \quad 1^{\overset{1}{1}} 41413221(12)12231513221(12)12231513221(12)12231513151^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 46, T_{\max} = 46, \Delta = -18, \det = 1, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 46, T_{\max} = 46, T \in [46, 46]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 5 \\ 3 & 5 & 8 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45}$$

$$T = 47 \quad [2755] \quad 21 \overset{1}{8} 1231513221(12)12231513221(12)12231513221(12)12231^{\overset{21}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 47, T_{\max} = 47, \Delta = -15, \det = -1, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 47, T_{\max} = 47, T \in [47, 47]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45} \\
& \begin{matrix} 1^1 7 12231513221(12)12231513221(12)12231513221 \\ [2753] \ 1 \quad (12) \quad 1223141^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 47, T_{\max} = 47, \Delta = -15, \det = -1, (n_+, n_-, n_0) = \\ (1, 47, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 47, T_{\max} = 47, T \in [47, 47] \\ \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45} \end{matrix} \\
& \begin{matrix} [2754] \ 1^1 41413221(12)12231513221(12)12231513221(12)122315132161^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 47, T_{\max} = 47, \Delta = -16, \det = \\ -1, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 47, T_{\max} = 47, T \in [47, 47] \\ \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45} \end{matrix} \\
& \begin{matrix} [2760] \ 1^1 6 131513221(12)12231513221(12)12231513221(12)122314141^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 47, T_{\max} = 47, \Delta = -15, \det = \\ -1, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 47, T_{\max} = 47, T \in [47, 47] \\ \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45} \end{matrix} \\
& \begin{matrix} [2759] \ 1^1 5 13221 \quad (12) \quad 12231513221(12)12231513221(12)122315132161^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 47, T_{\max} = 47, \Delta = -12, \det = \\ -1, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 47, T_{\max} = 47, T \in [47, 47] \\ \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45} \end{matrix} \\
& \begin{matrix} [2758] \ 1^1 8 12231513221(12)12231513221(12)12231513221(12)12231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 47, T_{\max} = 47, \Delta = -15, \det = \\ -1, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 47, T_{\max} = 47, T \in [47, 47] \\ \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45} \end{matrix} \\
& \begin{matrix} [2757] \ 1^1 7 1231513221(12)12231513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 47, T_{\max} = 47, \Delta = -15, \det = \\ -1, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 47, T_{\max} = 47, T \in [47, 47] \\ \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45} \end{matrix} \\
& \begin{matrix} [2756] \ 1^1 6 131513221(12)12231513221(12)12231513221(12)12231^1 5 1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 47, T_{\max} = 47, \Delta = -11, \det = \\ -1, (n_+, n_-, n_0) = (1, 47, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 47, T_{\max} = 47, T \in [47, 47] \\ \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{45} \end{matrix} \\
& T = 48 \quad [2771] \ 1^1 5131513221(12)12231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 48, T_{\max} = 48, \Delta = \\ -16, \det = 1, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 48, T_{\max} = 48, T \in [48, 48] \\ \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{46} \\
& [2772] \ 1^1 7 1231513221(12)12231513221(12)12231513221(12)122314141^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -14, \det = \\ 1, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 48, T_{\max} = 48, T \in [48, 48] \\ \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{46} \\
& [2766] \ 1^1 8 12231513221(12)12231513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -14, \det = \\ 1, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 48, T_{\max} = 48, T \in [48, 48]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{46} \\
& [2770] \quad 1^{\overline{1}5} 12231513221(12)12231513221(12)1223151322171^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -10, \det = \\
& 1, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 48, T_{\max} = 48, T \in [48, 48] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{46} \\
& [2769] \quad 1^{\overline{1}4} 141413221(12)12231513221(12)12231513221(12)1223151322171^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -14, \det = \\
& 1, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 48, T_{\max} = 48, T \in [48, 48] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 5 \\ 2 & 0 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{46} \\
& [2768] \quad 1^{\overline{1}7} 1231513221(12)12231513221(12)12231513221(12)12231513221^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -10, \det = \\
& 1, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 48, T_{\max} = 48, T \in [48, 48] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{46} \\
& [2767] \quad 21^{\overline{1}8} 1231513221(12)12231513221(12)12231513221(12)1223141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -14, \det = \\
& 1, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 48, T_{\max} = 48, T \in [48, 48] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{46} \\
& T = 49 \quad [2784] \quad 1^{\overline{1}6} 1231513221(12)12231513221(12)12231513221(12)12231513151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 49, T_{\max} = 49, \Delta = \\
& -14, \det = -1, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 49, T_{\max} = 49, T \in [49, 49] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{47} \\
& [2786] \quad 1^{\overline{1}23} 1513221(12)12231513221(12)12231513221 \quad 122315132181^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -18, \det = \\
& -1, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 49, T_{\max} = 49, T \in [49, 49] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 2 & 2 & 3 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{47} \\
& [2787] \quad 1^{\overline{1}8} 12231513221(12)12231513221(12)12231513221(12)122314141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -13, \det = \\
& -1, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 49, T_{\max} = 49, T \in [49, 49] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{47} \\
& [2788] \quad 1^{\overline{1}8} 12231513221(12)12231513221(12)12231513221(12)12231513221^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -9, \det = \\
& -1, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 49, T_{\max} = 49, T \in [49, 49] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{47} \\
& [2789] \quad 21^{\overline{1}8} 1231513221(12)12231513221(12)12231513221(12)12231513221^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -9, \det = \\
& -1, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 49, T_{\max} = 49, T \in [49, 49] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 2 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{47} \\
& [2785] \quad 21^{\overline{1}8} 1231513221(12)12231513221(12)12231513221(12)122314141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -13, \det = \\
& -1, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 49, T_{\max} = 49, T \in [49, 49]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{47}$$

$$[2783] \quad 1^1 6 131513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -13, \det = -1, (n_+, n_-, n_0) = (1, 49, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 49, T_{\max} = 49, T \in [49, 49]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{47}$$

$$T = 50 \quad [2792] \quad 1^1 7 12231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 50, T_{\max} = 50, \Delta = -12, \det = 1, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 50, T_{\max} = 50, T \in [50, 50]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{48}$$

$$[2793] \quad 1^1 7 1231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 50, T_{\max} = 50, \Delta = -12, \det = 1, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 50, T_{\max} = 50, T \in [50, 50]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{48}$$

$$T = 51 \quad [2803] \quad 1^1 8 12231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 51, \Delta = -11, \det = -1, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 51, T_{\max} = 51, T \in [51, 51]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{49}$$

$$[2802] \quad 1^1 21 8 1231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 51, \Delta = -11, \det = -1, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 51, T_{\max} = 51, T \in [51, 51]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{49}$$

$$T = 53 \quad [2827] \quad 1^1 31513221(12)12231513221(12)12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 53, T_{\max} = 53, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 53, T_{\max} = 53, T \in [53, 53]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{51}$$

$$[2828] \quad 1^1 31513221(12)12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 53, T_{\max} = 53, \Delta = -25, \det = -1, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 53, T_{\max} = 53, T \in [53, 53]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{51}$$

$$T = 54 \quad [2837] \quad 1^1 8 1232181231513221(12)12231513221(12)12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 54, T_{\max} = 54, \Delta = -17, \det = 1, (n_+, n_-, n_0) = (1, 54, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 54, T_{\max} = 54, T \in [54, 54]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{52}$$

4.2 $\text{NHC}(-2)(-3) + \mathfrak{e}_7$

$$T = 10 \quad [13] \quad 1^1 1^1 2231321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 1, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 58, T_{\max} = 100, T \in [58, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 6 & 0 & -2 & 0 \\ 2 & 6 & -1 & 0 & -42 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$T = 11 \quad [15] \quad 1^{11} 31413221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 \quad T = 11, T_{\max} = 101, \Delta = -140, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 59, T_{\max} = 101, T \in [59, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 3 & 0 & -5 & -3 \\ 2 & 3 & 2 & -3 & -42 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$T = 12 \quad [18] \quad 1^{11} 41413221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 \quad T = 12, T_{\max} = 95, \Delta = -131, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 60, T_{\max} = 95, T \in [60, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 3 & 0 & -5 & -3 \\ 2 & 3 & 2 & -3 & -42 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[19] \quad 1^{11} 5^{2231} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 \quad T = 12, T_{\max} = 98, \Delta = -135, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 60, T_{\max} = 98, T \in [60, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 2 & 5 & 5 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 5 & 1 & -3 & -1 \\ 2 & 5 & 2 & -1 & -42 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[20] \quad 1^{11} 223151231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 \quad T = 12, T_{\max} = 102, \Delta = -136, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 60, T_{\max} = 102, T \in [60, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 6 & 0 & -2 & 0 \\ 2 & 6 & -1 & 0 & -42 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 13 \quad [22] \quad 1^{11} 4131513221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 \quad T = 13, T_{\max} = 98, \Delta = -134, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 61, T_{\max} = 98, T \in [61, 98]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 3 & 4 & 5 \\ 1 & 4 & -4 & -2 \\ 3 & 5 & -2 & -41 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$T = 14 \quad [23] \quad 1^{11} 22315132151^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 \quad T = 14, T_{\max} = 97, \Delta = -130, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 62, T_{\max} = 97, T \in [62, 97]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 4 & 1 & 6 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 6 & 0 & -2 & 0 \\ 2 & 6 & -1 & 0 & -42 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$[24] \quad 1^{11} 5131513221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 \quad T = 14, T_{\max} = 95, \Delta = -129, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 62, T_{\max} = 95, T \in [62, 95]$$

$$\begin{pmatrix} 5 & 19 & 5 \\ 19 & 72 & 16 \\ 5 & 16 & -40 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$$

4.3 $\text{NHC}(-2)(-3) + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_2$

$$T = 9 \quad [31] \quad 1^{11} 3^{11} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -1, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 17, T_{\max} = 93, T \in [17, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 2 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & -1 & 6 & -3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_5$$

4.4 $\text{NHC}(-2)(-3) + \textcolor{red}{e}'_7 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_3$

$$T = 10 \quad [3] \quad 1^{11} 23^{11} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 1, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 66, T_{\max} = 80, T \in [66, 80]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 2 & 0 & -1 \\ 1 & 0 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & -1 & -1 & 2 & -1 & -52 \end{pmatrix}_{b_0} \rightarrow \text{bd}_5$$

4.5 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [3] \quad 1^{12} 3^{11} 13^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)] \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 1, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 66, T_{\max} = 80, T \in [66, 80]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 2 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 2 & 0 & 0 & 2 \\ 2 & 0 & -1 & -1 & -1 & 2 & -52 \end{pmatrix}_{b_0} \rightarrow \text{bd}_5$$

4.6 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 12 \quad [1] \quad 1^{13} 5^{11} 13^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 11 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 9 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 7 & 10 & 11 & 9 & 0 & 68 & 9 & 16 \\ 1 & 1 & 2 & 1 & 0 & 9 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 16 & 2 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$[3] \quad 1^{11} 5^{11} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 4 & 3 & 0 & 5 & 3 & 5 \\ 1 & 1 & 2 & 1 & 0 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 5 & 2 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$[2] \quad 1^{11} 5^{11} 13^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 7 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 5 & 6 & 9 & 7 & 0 & 36 & 7 & 12 \\ 1 & 1 & 2 & 1 & 0 & 7 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 12 & 2 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$[4] \quad 1^{12} 5^{11} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 5 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 6 & 5 & 0 & 21 & 5 & 9 \\ 1 & 1 & 2 & 1 & 0 & 5 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 9 & 2 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$[5] \quad 1^{11} 5^{11} 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 1 & 0 & 0 & 1 & 2 \\ 1 & 1 & 2 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 2 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$[0] \quad 1^{12} 5^{11} 13^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) =$$

4.8 NHC(-2)(-3) + $\mathfrak{so}_8$

$$T = 7 \quad [9] \quad 1^{11} 3^{11} 21^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -1, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 4 & 0 & 14 & 6 \\ 2 & 2 & -1 & 6 & -2 \end{pmatrix}_{b_0} \rightarrow \text{bd}_4$$

4.9 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2$

$$T = 13 \quad [1132] \quad 1^{11} 31^1 5^{11} 13221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 6 & 5 \\ 3 & 0 & 6 & 4 & 6 \\ 3 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[1131] \quad 1^{12} 31^1 5^{11} 13221^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 15 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 26 & 5 \\ 15 & 0 & 26 & 148 & 30 \\ 3 & 0 & 5 & 30 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

4.10 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 13 \quad [13] \quad 1^{12} 3^1 1^1 5^{11} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 11 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 & 3 \\ 3 & 0 & 3 & 4 & 14 & 3 & 4 \\ 11 & 0 & 12 & 14 & 42 & 10 & 14 \\ 2 & 0 & 3 & 3 & 10 & 1 & 3 \\ 3 & 0 & 3 & 4 & 14 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[12] \quad 1^{12} 3^1 1^1 5^{11} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 9 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 & 3 \\ 3 & 0 & 3 & 4 & 14 & 3 & 4 \\ 9 & 0 & 12 & 14 & 42 & 10 & 14 \\ 2 & 0 & 3 & 3 & 10 & 1 & 3 \\ 3 & 0 & 3 & 4 & 14 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[11] \quad 1^1 3^1 1^1 5^{11} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 1 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 0 & 3 & 3 \\ 3 & 0 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 3 & 3 & 0 & 1 & 3 \\ 3 & 0 & 3 & 4 & 0 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

4.11 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 13 \quad [12] \quad 1^{12} 3^1 1^1 5^{11} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 2 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 3 & 12 & 3 \\ 3 & 0 & 3 & 4 & 3 & 14 & 4 \\ 2 & 0 & 3 & 3 & 1 & 10 & 3 \\ 9 & 0 & 12 & 14 & 10 & 42 & 14 \\ 3 & 0 & 3 & 4 & 3 & 14 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[13] \quad 1^1 3^1 1^1 5^{11} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 3 & 0 & 3 \\ 3 & 0 & 3 & 4 & 3 & 0 & 4 \\ 2 & 0 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 0 & 3 & 4 & 3 & 0 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[111] \ 1^{12}3^1 1^1 5^1 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 13, \ T_{\max} = 13, \ \Delta = -1, \ \det = -1, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 2 & 11 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 3 & 12 & 3 \\ 3 & 0 & 3 & 4 & 3 & 14 & 4 \\ 2 & 0 & 3 & 3 & 1 & 10 & 3 \\ 11 & 0 & 12 & 14 & 10 & 42 & 14 \\ 3 & 0 & 3 & 4 & 3 & 14 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

4.12 NHC(-2)(-3) + $\textcolor{blue}{su}_2$

$$T = 11 \quad [3662] \ 1^{11} 31^1 5^1 1321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -1, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 3 & 0 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$T = 12 \quad [3784] \ 1^{11} 31^1 5^1 13221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 2 & 5 & 5 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 5 & 1 & 3 & 5 \\ 2 & 5 & 2 & 5 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[3782] \ 1^{23} 31^1 5^1 13221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 7 & 4 \\ 4 & 0 & 7 & 19 & 11 \\ 2 & 0 & 4 & 11 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[3781] \ 1^{12321} 6^1 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & 0 & 2 \\ 3 & 4 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[3783] \ 1^{43} 31^1 5^1 13221^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 19 & 4 \\ 8 & 0 & 19 & 115 & 27 \\ 2 & 0 & 4 & 27 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 13 \quad [3986] \ 1^{11} 4^1 14^1 14^1 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, \ T_{\max} = 13, \ \Delta = -15, \ \det = -1, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 7 & 6 & 7 \\ 3 & 6 & 4 & 6 \\ 3 & 7 & 6 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$[3987] \ 1^{12321} 7^1 1231^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, \ T_{\max} = 13, \ \Delta = -28, \ \det = -1, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & 0 & 2 \\ 3 & 4 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$T = 14 \quad [4039] \ 1^{1315131} 6^1 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 14, \ T_{\max} = 14, \ \Delta = -28, \ \det = 1, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

[4198] $1\overset{1}{5}1321\overset{1}{8}12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = -21, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & 0 & 2 \\ 3 & 4 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$

[4196] $1\overset{1}{1}\overset{1}{6}131613141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = -24, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 3 \\ 1 & 1 & -3 & 0 \\ 1 & 3 & 0 & 2 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$

$T = 17$ [4234] $1\overset{1}{6}12321\overset{1}{8}12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 17, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 0 & 0 & -2 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$

[4235] $1\overset{1}{5}1316131\overset{1}{6}131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 17, \Delta = -19, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 1 & 2 & 0 & 1 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$

[4236] $1\overset{1}{1}\overset{1}{8}123216131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 17, \Delta = -26, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 1 & 2 & 0 & 1 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$

$T = 18$ [4274] $1\overset{1}{1}\overset{1}{8}1232171231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 18, T_{\max} = 18, \Delta = -25, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 1 & 2 & 0 & 1 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$

[4273] $21\overset{1}{7}12321\overset{1}{8}12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 18, T_{\max} = 18, \Delta = -18, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 0 \\ 1 & 0 & 0 & -2 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$

$T = 19$ [4299] $1\overset{1}{1}\overset{1}{8}12321\overset{1}{8}1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 19, T_{\max} = 19, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 \\ 1 & 2 & 1 & 1 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$

[4300] $1\overset{1}{1}\overset{1}{5}13161316131\overset{1}{5}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 19, T_{\max} = 19, \Delta = -13, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$

[4301] $1\overset{1}{1}\overset{1}{8}12321812231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 19, T_{\max} = 19, \Delta = -24, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 4 & 0 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$[4298] \quad 221 \overset{1^1}{8} 12321 \overset{1^1}{8} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 19, T_{\max} = 19, \Delta = -17, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 3 & 0 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$T = 20 \quad [4317] \quad 21 \overset{1^1}{6} 13161316131 \overset{1^1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 20, T_{\max} = 20, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 1 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 6 & 0 & 13 & 9 \\ 2 & 4 & 0 & 9 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$$

$$[4318] \quad 1 \overset{1^1}{8} \overset{1^1}{321} 123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 20, T_{\max} = 20, \Delta = -23, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 1 & 3 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 5 & -1 & 4 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$$

$$T = 21 \quad [4329] \quad 1 \overset{1^1}{8} \overset{1^1}{321} 1232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 21, T_{\max} = 21, \Delta = -20, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 4 & 0 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$$

$$[4328] \quad 1 \overset{1^1}{5} 1321812321 \overset{1^1}{8} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 21, T_{\max} = 21, \Delta = -18, \det = -1, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 4 & 0 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$$

$$T = 23 \quad [4350] \quad 1 \overset{1^1}{5} \overset{1^1}{132181232181231} \overset{1^1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\max} = 23, \Delta = -12, \det = -1, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 3 & 5 \\ 2 & 4 & 0 & 5 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

$$T = 24 \quad [4361] \quad 1 \overset{1^1}{6} 1232181232181231 \overset{1^1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 24, \Delta = -11, \det = 1, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 3 & 5 \\ 2 & 4 & 0 & 5 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$$

$$T = 25 \quad [4372] \quad 21 \overset{1^1}{7} 1232181232181231 \overset{1^1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 25, T_{\max} = 25, \Delta = -9, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 1 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 6 & 0 & 13 & 9 \\ 2 & 4 & 0 & 9 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{22}$$

$$T = 33 \quad [4415] \quad 1 \quad \begin{smallmatrix} 1^1 1^1 \\ 5 \end{smallmatrix} 13221 \quad \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 12231513221(12)12231 \begin{smallmatrix} 1^1 \\ 5 \end{smallmatrix} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 33, \quad T_{\max} = 33, \quad \Delta = -11, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 33, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 33, \quad T_{\max} = 33, \quad T \in [33, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 4 & 3 & 7 & 7 \\ 2 & 4 & 2 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{30}$$

$$T = 36 \quad [4420] \quad 1 \quad \begin{smallmatrix} 1^1 1^1 \\ 5 \end{smallmatrix} 13221 \quad \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 12231513221(12)122315131 \begin{smallmatrix} 1^1 \\ 6 \end{smallmatrix} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 36, \quad T_{\max} = 36, \quad \Delta = -10, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 36, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 36, \quad T_{\max} = 36, \quad T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 4 & 3 & 7 & 7 \\ 2 & 4 & 2 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{33}$$

$$T = 37 \quad [4422] \quad 1 \quad \begin{smallmatrix} 1^1 1^1 \\ 5 \end{smallmatrix} 13221 \quad \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 12231513221(12)1223151321 \begin{smallmatrix} 1^1 \\ 7 \end{smallmatrix} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 37, \quad T_{\max} = 37, \quad \Delta = -9, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 37, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 37, \quad T_{\max} = 37, \quad T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 4 & 3 & 7 & 7 \\ 2 & 4 & 2 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{34}$$

$$T = 38 \quad [4426] \quad 21 \begin{smallmatrix} 1^1 \\ 8 \end{smallmatrix} 1231513221(12)12231513221(12)12231 \begin{smallmatrix} 1^1 \\ 5 \end{smallmatrix} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 38, \quad T_{\max} = 38, \quad \Delta = -8, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 38, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 38, \quad T_{\max} = 38, \quad T \in [38, 38]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 2 & 7 & 4 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 7 & 4 & 19 & 11 \\ 2 & 4 & 2 & 11 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35}$$

$$[4425] \quad 1 \quad \begin{smallmatrix} 1^1 1^1 \\ 5 \end{smallmatrix} 13221 \quad \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} 12231513221(12)12231513221 \begin{smallmatrix} 1^1 \\ 8 \end{smallmatrix} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 38, \quad T_{\max} = 38, \quad \Delta = -8, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 38, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 38, \quad T_{\max} = 38, \quad T \in [38, 38]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 4 & 3 & 7 & 7 \\ 2 & 4 & 2 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{35}$$

$$T = 44 \quad [4432] \quad 1 \quad \begin{smallmatrix} 1^1 1^1 \\ 5 \end{smallmatrix} 13221(12)12231513221 \quad \begin{smallmatrix} 1^1 \\ (12) \end{smallmatrix} 12231513221(12)12231 \begin{smallmatrix} 1^1 \\ 5 \end{smallmatrix} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 44, \quad T_{\max} = 44, \quad \Delta = -10, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 44, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 44, \quad T_{\max} = 44, \quad T \in [44, 44]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 2 & 3 & 4 \\ 3 & 3 & 4 & 6 \\ 3 & 4 & 6 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{42}$$

4.13 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 12 \quad [1] \quad 1 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 5 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 13 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 0 & 4 & 1 & 0 & 2 & 2 \\ 2 & 2 & 4 & 5 & 3 & 0 & 3 & 5 \\ 1 & 1 & 1 & 3 & 0 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 1 & -3 & 0 & -1 \\ 1 & 1 & 2 & 3 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 5 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$[0] \quad 1 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 5 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 13 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 3 & 4 \\ 9 & 14 & 42 & 0 & 12 & 10 & 14 \\ 1 & 0 & 0 & -2 & 1 & 0 & 0 \\ 2 & 3 & 12 & 1 & 1 & 3 & 3 \\ 2 & 3 & 10 & 0 & 3 & 1 & 3 \\ 3 & 4 & 14 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[2] \ 1^{13} 5^{1^{21}} 13^{11} 1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 2 & 3 \\ 3 & 4 & 21 & 0 & 3 & 3 & 4 \\ 14 & 21 & 97 & 0 & 18 & 15 & 21 \\ 1 & 0 & 0 & -2 & 1 & 0 & 0 \\ 2 & 3 & 18 & 1 & 1 & 3 & 3 \\ 2 & 3 & 15 & 0 & 3 & 1 & 3 \\ 3 & 4 & 21 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[3] \ 1^{11} 5^{1^{21}} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 0 & 3 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 1 & 0 & 0 \\ 2 & 3 & 0 & 1 & 1 & 3 & 3 \\ 2 & 3 & 0 & 0 & 3 & 1 & 3 \\ 3 & 4 & 0 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[4] \ 1^{11} 5^{1^{21}} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 2 & 3 \\ 3 & 4 & 7 & 0 & 3 & 3 & 4 \\ 6 & 7 & 9 & 0 & 6 & 5 & 7 \\ 1 & 0 & 0 & -2 & 1 & 0 & 0 \\ 2 & 3 & 6 & 1 & 1 & 3 & 3 \\ 2 & 3 & 5 & 0 & 3 & 1 & 3 \\ 3 & 4 & 7 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[5] \ 1^{12} 5^{1^{31}} 13^{11} 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 3 & 4 \\ 11 & 14 & 42 & 0 & 12 & 10 & 14 \\ 1 & 0 & 0 & -2 & 1 & 0 & 0 \\ 2 & 3 & 12 & 1 & 1 & 3 & 3 \\ 2 & 3 & 10 & 0 & 3 & 1 & 3 \\ 3 & 4 & 14 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.14 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [1187] \ 1^{11} 5^{1^{21}} 13^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 11, \ \Delta = -21, \ \det = -1, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$T = 12 \quad [1232] \ 1^{11} 4^{11} 14^{11} 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -15, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 6 & 5 & 5 & 6 \\ 3 & 5 & 3 & 5 & 5 \\ 2 & 5 & 5 & 3 & 5 \\ 3 & 6 & 5 & 5 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[1233] \ 1^{11} 31^{11} 5^{11} 13^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 2 & 2 & 5 & 9 & 4 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 5 & 4 & 7 & 14 & 7 \\ 4 & 9 & 5 & 14 & 21 & 11 \\ 2 & 4 & 2 & 7 & 11 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[1234] \ 1^{11} 31^{11} 5^{11} 13^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 1 & 5 & 6 & 3 \\ 2 & 4 & 5 & 7 & 13 & 7 \\ 4 & 7 & 6 & 13 & 18 & 10 \\ 2 & 3 & 3 & 7 & 10 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[1235] \ 1^{\overline{1}} \overset{1}{6} \ 131^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -25, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 2 & 0 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[1236] \ 1^{\overline{1}} \overset{1}{6} \ 12321^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -20, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[1237] \ 1^{\overline{1}} \overset{1}{6} \overset{1}{3} \overset{1}{5} \ 1321^{\overline{1} \overline{2}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 2 & 2 & 5 & 9 & 4 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 2 & 5 & 4 & 7 & 14 & 7 \\ 5 & 9 & 5 & 14 & 21 & 11 \\ 2 & 4 & 2 & 7 & 11 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[1238] \ 1^{\overline{1} \overline{2}} \overset{1}{3} \overset{1}{5} \ 1321^{\overline{1} \overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 5 & 6 & 3 \\ 2 & 0 & 5 & 7 & 13 & 7 \\ 4 & 0 & 6 & 13 & 18 & 10 \\ 2 & 0 & 3 & 7 & 10 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$T = 14 \quad [1352] \ 1^{\overline{1}} \overset{1}{8} \ 1231^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 14, \ T_{\max} = 14, \ \Delta = -24, \ \det = 1, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 2 & 0 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[1353] \ 1^{\overline{1}} \overset{1}{6} \ 1315131^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 14, \ T_{\max} = 14, \ \Delta = -19, \ \det = 1, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[1354] \ 1^{\overline{1}} \overset{1}{6} \overset{1}{5} \overset{1}{13} \ 131^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 14, \ T_{\max} = 14, \ \Delta = -19, \ \det = 1, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 2 & 2 & 1 & 4 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 4 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$T = 16 \quad [1384] \ 1^{\overline{1}} \overset{1}{8} \overset{1}{5} \overset{1}{13} \ 1231^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 16, \ T_{\max} = 16, \ \Delta = -18, \ \det = 1, \ (n_+, n_-, n_0) = (1, 16, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 2 & 2 & 1 & 4 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 4 & 3 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$[1386] \ 1^{\overline{1}} \overset{1}{8} \ 122315131^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 16, \ T_{\max} = 16, \ \Delta = -17, \ \det = 1, \ (n_+, n_-, n_0) = (1, 16, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[1385] \ 1^{\overset{1}{\color{red}1}} \overset{1}{\color{red}5} \ 1316131 \overset{1}{\color{red}5} 1^{\overset{1}{\color{red}1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 16, T_{\max} = 16, \Delta = -13, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 2 & 3 & 4 \\ 1 & 2 & 0 & 3 & 1 & 3 \\ 2 & 3 & 0 & 4 & 3 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$T = 17 \quad [1391] \ 1^{\overset{1}{\color{red}1}} \overset{1}{\color{red}6} 1232181231^{\overset{1}{\color{red}1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -17, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$$

$$T = 18 \quad [1393] \ 1^{\overset{1}{\color{red}1}} \overset{1}{\color{red}5} \ 132181231^{\overset{1}{\color{red}1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 18, T_{\max} = 18, \Delta = -12, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 2 & 3 & 4 \\ 1 & 2 & 0 & 3 & 1 & 3 \\ 2 & 3 & 0 & 4 & 3 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$T = 19 \quad [1398] \ 1^{\overset{1}{\color{red}1}} \overset{1}{\color{red}6} \ 1232181231^{\overset{1}{\color{red}1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 19, T_{\max} = 19, \Delta = -11, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 & 3 \\ 2 & 2 & 0 & 3 & 2 & 4 \\ 2 & 3 & 0 & 3 & 4 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$$

$$T = 22 \quad [1400] \ 1 \overset{1}{(12)} \ 12231^{\overset{1}{\color{red}1}} \overset{1}{\color{red}5} 1^{\overset{1}{\color{red}1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 22, T_{\max} = 22, \Delta = -11, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & 1 & 4 & 3 \\ 2 & 3 & 3 & 4 & 6 & 6 \\ 2 & 3 & 2 & 3 & 6 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$$

$$[1401] \ 21 \overset{1}{(12)} \ 12231^{\overset{1}{\color{red}1}} \overset{1}{\color{red}5} 1^{\overset{1}{\color{red}1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 22, T_{\max} = 22, \Delta = -11, \det = 1, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 & 3 \\ 2 & 2 & 0 & 3 & 2 & 4 \\ 2 & 3 & 0 & 3 & 4 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{18}$$

$$T = 25 \quad [1404] \ 1 \overset{1}{(12)} \ 122315131^{\overset{1}{\color{red}1}} \overset{1}{\color{red}6} 1^{\overset{1}{\color{red}1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 25, T_{\max} = 25, \Delta = -10, \det = -1, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & 1 & 4 & 3 \\ 2 & 3 & 3 & 4 & 6 & 6 \\ 2 & 3 & 2 & 3 & 6 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{21}$$

$$T = 28 \quad [1406] \quad 1^{1^{11}} 6^{11} 1316131513221(12)1223151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 28, T_{\max} = 28, \Delta = -9, \det = 1, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 2 & 3 & 4 \\ 1 & 2 & 0 & 3 & 1 & 3 \\ 2 & 3 & 0 & 4 & 3 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{24}$$

4.15 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [18] \quad 1^{111} 231^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 1, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 4 & 7 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 0 & 4 & 0 & 2 & 7 & 4 \\ 4 & 0 & 7 & 0 & 7 & 14 & 8 \\ 2 & 0 & 3 & 0 & 4 & 8 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_5$$

$$[17] \quad 1^{111} 231^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 1, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 2 & 1 & 4 & 7 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 4 & 0 & 2 & 7 & 4 & 4 \\ 3 & 7 & 0 & 7 & 14 & 7 & 8 \\ 2 & 4 & 0 & 4 & 7 & 2 & 4 \\ 2 & 4 & -1 & 4 & 8 & 4 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_5$$

$$T = 11 \quad [28] \quad 1^{111} 31^{111} 41^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & 1 & 3 & 4 & 6 & 3 \\ 1 & 1 & -2 & 0 & 0 & 1 & 0 \\ 1 & 3 & 0 & 1 & 4 & 6 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 5 \\ 4 & 6 & 1 & 6 & 9 & 14 & 8 \\ 2 & 3 & 0 & 3 & 5 & 8 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$[27] \quad 1^{111} 31^{111} 4131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 1 & 0 \\ 1 & 3 & 0 & 1 & 3 & 6 & 3 \\ 2 & 3 & 1 & 3 & 2 & 7 & 4 \\ 4 & 6 & 1 & 6 & 7 & 14 & 8 \\ 2 & 3 & 0 & 3 & 4 & 8 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$[26] \quad 1^{111} 3141^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 3 & 4 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 & 0 \\ 1 & 3 & 0 & 1 & 3 & 4 & 3 \\ 2 & 3 & 1 & 3 & 2 & 4 & 4 \\ 3 & 4 & 0 & 4 & 4 & 5 & 5 \\ 2 & 3 & 0 & 3 & 4 & 5 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$T = 14 \quad [32] \quad 1^{11} 6^{11} 13151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 3 & 0 & 1 & 2 \\ 2 & 2 & 3 & 5 & 0 & 3 & 5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 1 & 1 & 3 & 0 & 0 & 2 \\ 2 & 2 & 2 & 5 & 0 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[30] \quad 1^{11} 6^{11} 13151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 & 1 & 0 & 2 \\ 3 & 2 & 3 & 5 & 3 & 0 & 5 \\ 1 & 1 & 1 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 2 & 2 & 5 & 2 & 0 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[34] \quad 1^{11} 6^{11} 13151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 & 2 \\ 1 & 2 & 0 & 2 & 1 & 3 & 2 \\ 1 & 1 & 2 & 1 & 1 & 4 & 3 \\ 1 & 1 & 1 & 1 & 0 & 3 & 2 \\ 3 & 2 & 3 & 4 & 3 & 5 & 5 \\ 2 & 2 & 2 & 3 & 2 & 5 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[33] \ 1^{\overset{1}{\underset{1}{6}}} 1^{\overset{1}{\underset{1}{6}}} 131 \overset{1}{5} 1^{\overset{1}{\underset{1}{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 14, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 0 & 2 \\ 1 & 2 & 0 & 2 & 1 & 0 & 2 \\ 1 & 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[31] \ 1^{\overset{1}{\underset{1}{6}}} 1^{\overset{1}{\underset{1}{6}}} 131 \overset{1}{5} 1^{\overset{1}{\underset{1}{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 14, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 & 2 & 1 & 2 \\ 2 & 2 & 3 & 5 & 4 & 3 & 5 \\ 2 & 1 & 2 & 4 & 1 & 1 & 3 \\ 1 & 1 & 1 & 3 & 1 & 0 & 2 \\ 2 & 2 & 2 & 5 & 3 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 15 \quad [35] \ 1^{\overset{1}{\underset{1}{7}}} 1^{\overset{1}{\underset{1}{7}}} 1231 \overset{1}{5} 1^{\overset{1}{\underset{1}{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = -9, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 0 & 2 \\ 1 & 2 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 & 2 & 1 & 2 \\ 1 & 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$T = 16 \quad [36] \ 1^{\overset{1}{\underset{1}{8}}} 1^{\overset{1}{\underset{1}{8}}} 12231 \overset{1}{5} 1^{\overset{1}{\underset{1}{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = -8, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 0 & 2 \\ 1 & 2 & 0 & 2 & 1 & 0 & 2 \\ 1 & 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

4.16 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [260] \ 1^{\overset{1}{\underset{1}{13}}} 3^{\overset{1}{\underset{1}{1}}} 1^{\overset{1}{\underset{1}{1}}} 1^{\overset{1}{\underset{1}{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 10, \Delta = -8, \det = 1, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 6 & 5 \\ 3 & 0 & 6 & 4 & 6 \\ 3 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$T = 11 \quad [276] \ 1^{\overset{1}{\underset{1}{11}}} 3^{\overset{1}{\underset{1}{1}}} 1^{\overset{1}{\underset{1}{1}}} 141321^{\overset{1}{\underset{1}{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 11, \Delta = -7, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 5 & 2 & 4 \\ 4 & 5 & 7 & 2 & 7 \\ 1 & 2 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$T = 12 \quad [347] \ 1^{\overset{1}{\underset{1}{13}}} 3^{\overset{1}{\underset{1}{1}}} 1^{\overset{1}{\underset{1}{1}}} 5^{\overset{1}{\underset{1}{1}}} 1321^{\overset{1}{\underset{1}{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & 2 & 1 & 15 & 1 & 3 \\ 10 & 12 & 15 & 79 & 9 & 21 \\ 1 & 2 & 1 & 9 & 0 & 2 \\ 2 & 3 & 3 & 21 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[349] \quad 1^{11}2315123^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 6 & 5 \\ 3 & 0 & 6 & 4 & 6 \\ 3 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[350] \quad 1^{13}3^{11}5^{12}15^{13}1231^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & 2 & 1 & 15 & 1 & 3 \\ 8 & 12 & 15 & 79 & 9 & 21 \\ 1 & 2 & 1 & 9 & 0 & 2 \\ 2 & 3 & 3 & 21 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[351] \quad 1^{13}3^{11}5^{12}15^{13}1231^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 3 \\ 1 & 2 & 1 & 0 & 1 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 3 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[352] \quad 1^{11}2321513^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 6 & 5 \\ 3 & 0 & 6 & 4 & 6 \\ 3 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[353] \quad 1^{12}3^{11}5^{12}15^{13}1231^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 2 & 8 & 2 & 3 \\ 1 & 2 & 1 & 10 & 1 & 3 \\ 5 & 8 & 10 & 34 & 6 & 14 \\ 1 & 2 & 1 & 6 & 0 & 2 \\ 2 & 3 & 3 & 14 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[345] \quad 1^{11}231^{11}5^{12}13^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 4 & 2 & 3 \\ 1 & 2 & 1 & 5 & 1 & 3 \\ 2 & 4 & 5 & 7 & 3 & 7 \\ 1 & 2 & 1 & 3 & 0 & 2 \\ 2 & 3 & 3 & 7 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[344] \quad 1^{12}3^{11}5^{12}15^{13}1231^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 6 & 1 & 3 \\ 4 & 0 & 6 & 18 & 5 & 10 \\ 1 & 0 & 1 & 5 & 0 & 2 \\ 2 & 0 & 3 & 10 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[343] \quad 1^{11}3^{11}5^{12}15^{13}1231^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 2 & 4 & 2 & 3 \\ 1 & 2 & 1 & 5 & 1 & 3 \\ 4 & 4 & 5 & 7 & 3 & 7 \\ 1 & 2 & 1 & 3 & 0 & 2 \\ 2 & 3 & 3 & 7 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[342] \quad 1^{14}3^{11}5^{12}15^{13}1231^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 16 & 1 & 3 \\ 8 & 0 & 16 & 106 & 11 & 24 \\ 1 & 0 & 1 & 11 & 0 & 2 \\ 2 & 0 & 3 & 24 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[341] \ 1^{13} 3^{11} 5^{13} 1321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -2, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 5 & 1 & 3 \\ 2 & 0 & 5 & 7 & 3 & 7 \\ 1 & 0 & 1 & 3 & 0 & 2 \\ 2 & 0 & 3 & 7 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[340] \ 1^{13} 3^{11} 1414131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -5, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 2 & 1 & 4 & 3 & 4 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 2 & 4 & 1 & 3 & 3 & 5 \\ 1 & 3 & 0 & 3 & 1 & 3 \\ 2 & 4 & 0 & 5 & 3 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[348] \ 1^{14} 3^{11} 5^{13} 1321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -2, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 1 & 2 & 16 & 2 & 3 \\ 1 & 2 & 1 & 20 & 1 & 3 \\ 11 & 16 & 20 & 142 & 12 & 28 \\ 1 & 2 & 1 & 12 & 0 & 2 \\ 2 & 3 & 3 & 28 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[346] \ 1^{12} 3^{11} 5^{13} 1321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -2, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 1 & 2 & 8 & 2 & 3 \\ 1 & 2 & 1 & 10 & 1 & 3 \\ 7 & 8 & 10 & 34 & 6 & 14 \\ 1 & 2 & 1 & 6 & 0 & 2 \\ 2 & 3 & 3 & 14 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$T = 13 \quad [361] \ 1^{13} 3^{11} 21612321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 13, \ T_{\max} = 13, \ \Delta = -3, \ \det = -1, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 5 & 2 & 4 \\ 4 & 5 & 7 & 2 & 7 \\ 1 & 2 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$T = 14 \quad [362] \ 1^{12} 7^{12} 123^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 14, \ T_{\max} = 14, \ \Delta = -1, \ \det = 1, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 2 & 1 & 2 & 4 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 3 & 4 & 2 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$[363] \ 1^{13} 3^{11} 151315131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 14, \ T_{\max} = 14, \ \Delta = -3, \ \det = 1, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 3 & 1 & 5 & 4 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 3 & 4 & 5 \\ 1 & 4 & 0 & 4 & 2 & 4 \\ 2 & 5 & -1 & 5 & 4 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$T = 15 \quad [364] \ 1^{12} 8^{12} 123^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 15, \ T_{\max} = 15, \ \Delta = 0, \ \det = -1, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$[365] \ 1^{13} 3161^{12} 231513^{11} 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 15, \ T_{\max} = 15, \ \Delta = -1, \ \det = -1, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 12 & 3 \\ 1 & 1 & 0 & 12 & 2 \\ 15 & 12 & 12 & 103 & 27 \\ 3 & 3 & 2 & 27 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[366] \quad 1^{11}3^{16}1^{12}3^{15}1^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 9 & 6 & 6 & 31 & 15 \\ 3 & 3 & 2 & 15 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[367] \quad 1^{11}3^{21}6^{13}1^{15}1^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 3 & 3 & 4 & 6 \\ 3 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[368] \quad 1^{11}3^{12}6^{13}1^{15}1^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 15, \Delta = -2, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 5 & 2 & 4 \\ 4 & 5 & 7 & 2 & 7 \\ 1 & 2 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[369] \quad 1^{11}3^{16}1^{12}3^{15}1^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 3 & 3 & 4 & 6 \\ 3 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$T = 17 \quad [371] \quad 1^{11}3^{15}1^{13}2^{11}8^{12}1^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 2 & 1 & 2 & 4 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 3 & 4 & 2 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

4.17 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 12 \quad [495] \quad 1^{12}3^{11}5^{13}1^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 1 & 7 & 3 \\ 1 & 2 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 4 & 2 \\ 4 & 7 & 5 & 0 & 4 & 17 & 9 \\ 2 & 3 & 2 & 0 & 2 & 9 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[494] \quad 1^{11}2^{13}3^{11}5^{13}1^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 5 & 2 \\ 1 & 1 & 2 & 4 & 2 & 7 & 3 \\ 1 & 2 & 0 & 4 & 1 & 5 & 2 \\ 2 & 4 & 4 & 6 & 3 & 12 & 6 \\ 1 & 2 & 1 & 3 & 0 & 5 & 2 \\ 5 & 7 & 5 & 12 & 5 & 17 & 9 \\ 2 & 3 & 2 & 6 & 2 & 9 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.18 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 12 \quad [74] \quad 1^{14}3^{11}5^{13}1^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 1 & 2 \\ 1 & 1 & 2 & 14 & 1 & 2 & 3 \\ 1 & 2 & 0 & 10 & 2 & 1 & 2 \\ 9 & 14 & 10 & 74 & 8 & 10 & 18 \\ 1 & 1 & 2 & 8 & 0 & 1 & 2 \\ 1 & 2 & 1 & 10 & 1 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[73] \quad 1^{\overline{1}3^{\overline{1}1}5^{\overline{1}3^{\overline{1}1}1^{\overline{1}}} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{green}{u}_2 + \textcolor{orange}{s}\textcolor{orange}{u}_2 \rightarrow (-3) + \textcolor{red}{s}\textcolor{red}{u}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 1 & 2 & 3 \\ 1 & 2 & 0 & 0 & 2 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 1 & 0 & 2 \\ 2 & 3 & 2 & 0 & 2 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[72] \ 1^{12}3^11513^{11}1^{11} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{brown}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -2, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 1 & 2 & 7 & 1 & 2 & 3 \\ 1 & 2 & 0 & 5 & 2 & 1 & 2 \\ 4 & 7 & 5 & 17 & 4 & 5 & 9 \\ 1 & 1 & 2 & 4 & 0 & 1 & 2 \\ 1 & 2 & 1 & 5 & 1 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[71] \ 1^{\overline{1}^3} 3^{\overline{1}} 1^{\overline{1}} 5 13^{\overline{1}^1} 1^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 1 & 2 \\ 1 & 1 & 2 & 7 & 1 & 2 & 3 \\ 1 & 2 & 0 & 5 & 2 & 1 & 2 \\ 6 & 7 & 5 & 17 & 4 & 5 & 9 \\ 1 & 1 & 2 & 4 & 0 & 1 & 2 \\ 1 & 2 & 1 & 5 & 1 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.19 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_{2 \rightarrow (-3)} + \text{su}2\text{n}3\text{mix}$

$$T = 12 \quad [307] \quad 1^{\overline{1}1} 3^{\overline{1}1} 5^{\overline{1}1} 13^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 1 & 3 & 2 \\ 2 & 0 & 4 & 6 & 3 & 7 & 6 \\ 1 & 0 & 1 & 3 & 0 & 3 & 2 \\ 2 & 0 & 3 & 7 & 3 & 5 & 5 \\ 2 & 0 & 2 & 6 & 2 & 5 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$\begin{aligned}
[308] \quad & 1^{11}3^11^513^11^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 6 & 1 & 3 & 2 \\ 6 & 0 & 6 & 34 & 7 & 15 & 12 \\ 1 & 0 & 1 & 7 & 0 & 3 & 2 \\ 2 & 0 & 3 & 15 & 3 & 5 & 5 \\ 2 & 0 & 2 & 12 & 2 & 5 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7
\end{aligned}$$

$$\begin{aligned}
[309] \quad & 1^{12} 3^1 1^5 13^1 1^{31} + [\textcolor{red}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s} \textcolor{green}{u}_2 + \textcolor{brown}{s} \textcolor{brown}{u}_2 \rightarrow (-3) + \textcolor{red}{s} \textcolor{red}{u}_2 \quad T = 12, \, T_{\max} = 12, \, \Delta = -2, \, \det = 1, \, (n_+, n_-, n_0) = \\
& (1, 12, 0), \, c_{\text{ext}} = 6.5 \\
& T_{\min} = 12, \, T_{\max} = 12, \, T \in [12, 12] \\
& \left(\begin{array}{cccccc} 0 & 1 & 1 & 6 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 9 & 1 & 3 & 2 \\ 6 & 0 & 9 & 49 & 8 & 18 & 15 \\ 1 & 0 & 1 & 8 & 0 & 3 & 2 \\ 2 & 0 & 3 & 18 & 3 & 5 & 5 \\ 2 & 0 & 2 & 15 & 2 & 5 & 4 \end{array} \right)_{b_0} \rightarrow \text{bd}_7
\end{aligned}$$

$$[310] \ 1^{\textcolor{blue}{3}}1^{\textcolor{blue}{1}}5 \ 1^{\textcolor{blue}{3}}1^{\textcolor{blue}{1}}1^{\textcolor{blue}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -2, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 3 & 2 \\ 2 & 0 & 1 & 1 & 2 & 4 & 3 \\ 1 & 0 & 1 & 2 & 0 & 3 & 2 \\ 2 & 0 & 3 & 4 & 3 & 5 & 5 \\ 2 & 0 & 2 & 3 & 2 & 5 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.20 `NHC(-2)(-3) + su2 + su2n3mix + su2`

$$T = 12 \quad [804] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{3}} 3^{\textcolor{red}{1}} 1^{\textcolor{blue}{1}} 5^{\textcolor{red}{1}} 13^{\textcolor{blue}{1}} 1^{\textcolor{red}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & 2 & 3 & 13 & 4 & 3 \\ 1 & 2 & 0 & 1 & 10 & 4 & 2 \\ 1 & 3 & 1 & 1 & 13 & 4 & 3 \\ 9 & 13 & 10 & 13 & 67 & 23 & 17 \\ 2 & 4 & 4 & 4 & 23 & 6 & 6 \\ 2 & 3 & 2 & 3 & 17 & 6 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

[803] $1^{\overline{1111}1}3^{\overline{11}1}5^{\overline{11}1}13^{\overline{11}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 2 & 3 & 4 & 4 & 3 \\ 1 & 2 & 0 & 1 & 3 & 4 & 2 \\ 1 & 3 & 1 & 1 & 4 & 4 & 3 \\ 3 & 4 & 3 & 4 & 5 & 7 & 5 \\ 2 & 4 & 4 & 4 & 7 & 6 & 6 \\ 2 & 3 & 2 & 3 & 5 & 6 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

[802] $1^{\overline{1111}1}3^{\overline{11}1}5^{\overline{11}1}13^{\overline{11}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 2 & 3 & 6 & 4 & 3 \\ 1 & 2 & 0 & 1 & 5 & 4 & 2 \\ 1 & 3 & 1 & 1 & 6 & 4 & 3 \\ 4 & 6 & 5 & 6 & 14 & 11 & 8 \\ 2 & 4 & 4 & 4 & 11 & 6 & 6 \\ 2 & 3 & 2 & 3 & 8 & 6 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.21 NHC(-2)(-3) + $\mathfrak{su}_2$ + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$

$T = 12$ [310] $1^{\overline{11}1}3^{\overline{11}1}5^{\overline{11}1}13^{\overline{11}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 3 & 1 & 2 \\ 2 & 0 & 4 & 6 & 7 & 3 & 6 \\ 2 & 0 & 3 & 7 & 5 & 3 & 5 \\ 1 & 0 & 1 & 3 & 3 & 0 & 2 \\ 2 & 0 & 2 & 6 & 5 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

[309] $1^{\overline{12}3}1^{\overline{11}1}5^{\overline{11}1}13^{\overline{11}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 9 & 3 & 1 & 2 \\ 6 & 0 & 9 & 49 & 18 & 8 & 15 \\ 2 & 0 & 3 & 18 & 5 & 3 & 5 \\ 1 & 0 & 1 & 8 & 3 & 0 & 2 \\ 2 & 0 & 2 & 15 & 5 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

[308] $1^{\overline{1}3}1^{\overline{11}1}5^{\overline{11}1}13^{\overline{11}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 1 & 2 \\ 2 & 0 & 1 & 1 & 4 & 2 & 3 \\ 2 & 0 & 3 & 4 & 5 & 3 & 5 \\ 1 & 0 & 1 & 2 & 3 & 0 & 2 \\ 2 & 0 & 2 & 3 & 5 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

[307] $1^{\overline{11}3}1^{\overline{12}1}5^{\overline{11}1}13^{\overline{11}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 6 & 3 & 1 & 2 \\ 6 & 0 & 6 & 34 & 15 & 7 & 12 \\ 2 & 0 & 3 & 15 & 5 & 3 & 5 \\ 1 & 0 & 1 & 7 & 3 & 0 & 2 \\ 2 & 0 & 2 & 12 & 5 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.22 NHC(-2)(-3) + $\mathfrak{su}_2$ + su2n3mix + su2n3mix

$T = 12$ [336] $1^{\overline{111}3}1^{\overline{11}1}5^{\overline{11}1}13^{\overline{11}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 2 & 3 & 6 & 11 & 3 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 1 & 3 & 1 & 1 & 6 & 11 & 3 \\ 4 & 6 & 5 & 6 & 14 & 27 & 8 \\ 8 & 11 & 8 & 11 & 27 & 46 & 14 \\ 2 & 3 & 2 & 3 & 8 & 14 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[335] \ 1^{\overline{11}13}3^{\overline{1}13}1^{\overline{1}12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -2, \ \det = 1, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 3 & 4 & 13 & 3 \\ 1 & 2 & 0 & 1 & 3 & 10 & 2 \\ 1 & 3 & 1 & 1 & 4 & 13 & 3 \\ 3 & 4 & 3 & 4 & 5 & 21 & 5 \\ 9 & 13 & 10 & 13 & 21 & 67 & 17 \\ 2 & 3 & 2 & 3 & 5 & 17 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.23 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 16 \quad [1646] \ 1^{\overline{1}131}6^{\overline{1}1}1316^{\overline{1}1}131^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 16, \ T_{\max} = 16, \ \Delta = -4, \ \det = 1, \ (n_+, n_-, n_0) = (1, 16, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 2 & 0 & 4 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 2 & 3 & -1 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$T = 17 \quad [1647] \ 1^{\overline{1}1}5^{\overline{1}1}1321^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 17, \ T_{\max} = 17, \ \Delta = 0, \ \det = -1, \ (n_+, n_-, n_0) = (1, 17, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 17, \ T_{\max} = 17, \ T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 2 & 1 & 0 & 4 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 0 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

4.24 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [42] \ 1^{\overline{1}111}4^{\overline{1}1}1^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 11, \ \Delta = -3, \ \det = -1, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 5 & 5 & 4 \\ 3 & 1 & 1 & 4 & 3 & 3 \\ 2 & 1 & 0 & 2 & 2 & 2 \\ 5 & 4 & 2 & 6 & 5 & 6 \\ 5 & 3 & 2 & 5 & 3 & 5 \\ 4 & 3 & 2 & 6 & 5 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.25 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [6] \ 1^{\overline{1}1}4^{\overline{1}1}13^{\overline{1}1}1^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 11, \ \Delta = -5, \ \det = -1, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 & 3 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 4 & 3 & 0 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

4.26 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 11 \quad [6] \ 1^{\overline{1}1}4^{\overline{1}1}13^{\overline{1}1}1^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 11, \ \Delta = -5, \ \det = -1, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 1 & 2 & 3 \\ 2 & 1 & 1 & 3 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 3 & 0 & 1 & 3 \\ 3 & 3 & 0 & 4 & 0 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

4.27 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{\overline{1}1}4^{\overline{1}1}1^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 11, \ \Delta = -3, \ \det = -1, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 & 0 & 4 \\ 2 & 3 & 1 & 3 & 0 & 2 & 0 & 3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 2 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & -1 & 3 & 0 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_5$$

4.28 NHC(-2)(-3) + $\mathfrak{su}_3$

$$T = 12 \quad [811] \quad 1^{\overline{1}} 31 \overline{5} 13221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 12, T_{\max} = 12, \Delta = -10, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 2 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 4 & -1 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[812] \quad 1^{\overline{1}} 231 \overline{5} 1321^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 12, T_{\max} = 12, \Delta = -11, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 0 & 4 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & -1 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 15 \quad [932] \quad 1^{\overline{1}} 2315131 \overline{6} 131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 15, T_{\max} = 15, \Delta = -9, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & 0 & 2 \\ 3 & 4 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$$

$$T = 16 \quad [951] \quad 1^{\overline{1}} 31 \overline{6} 131612321^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 16, T_{\max} = 16, \Delta = -8, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 3 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$T = 18 \quad [977] \quad 1^{\overline{1}} 31 \overline{6} 13161315131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 18, T_{\max} = 18, \Delta = -7, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 3 & 2 & 5 \\ 1 & 2 & 0 & 2 \\ 3 & 5 & 2 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$T = 19 \quad [984] \quad 1^{\overline{1}} 31 \overline{6} 131613161231^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 19, T_{\max} = 19, \Delta = -5, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 1 & 3 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 3 & 0 & 1 & 3 \\ 2 & 5 & 0 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$[985] \quad 1^{\overline{1}} 31 \overline{6} 13161316131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 19, T_{\max} = 19, \Delta = -4, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 2 & 3 & 5 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 3 & 1 & 1 & 3 \\ 2 & 5 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$T = 20 \quad [992] \quad 1^{\overline{1}} 31 \overline{6} 1316131613141^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 20, T_{\max} = 20, \Delta = -3, \det = 1, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 4 & 1 & 4 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 4 & 0 & 2 & 4 \\ 2 & 6 & -1 & 4 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$$

4.29 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 11 \quad [1628] \quad 1^{11} 4^{11} 13221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -6, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 3 & 3 & 4 & 6 \\ 3 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$T = 12 \quad [1690] \quad 1^{11} 5^{14} 13221^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 3 & 20 & 5 \\ 12 & 9 & 20 & 67 & 21 \\ 3 & 3 & 5 & 21 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[1691] \quad 1^{11} 4^{11} 141321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -5, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 2 & 2 & 5 & 4 \\ 1 & 2 & 0 & 2 & 2 \\ 4 & 5 & 2 & 7 & 7 \\ 3 & 4 & 2 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[1692] \quad 1^{11} 5^{11} 13221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 3 & 8 & 5 \\ 4 & 5 & 8 & 11 & 9 \\ 3 & 3 & 5 & 9 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 13 \quad [1751] \quad 1^{11} 4^{11} 1414131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 13, \quad T_{\max} = 13, \quad \Delta = -3, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 13, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 3 & 4 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 1 & 3 & 3 & 5 \\ 2 & 4 & 0 & 3 & 5 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 14 \quad [1761] \quad 1^{11} 5^{11} 13151321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 14, \quad T_{\max} = 14, \quad \Delta = -3, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 2 & 0 & 6 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 6 & 0 & 7 & 7 \\ 2 & 2 & 4 & -1 & 7 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$T = 15 \quad [1787] \quad 1^{11} 6^{11} 123151321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 15, \quad T_{\max} = 15, \quad \Delta = -1, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 15, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 3 & 3 & 4 & 6 \\ 3 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[1788] \quad 1^{11} 6^{12} 23151321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 15, \quad T_{\max} = 15, \quad \Delta = -1, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 15, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 9 & 6 & 6 & 31 & 15 \\ 3 & 3 & 2 & 15 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

[1789] $1 \overset{1}{\underset{11}{6}} 13151321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 3 & 8 & 5 \\ 4 & 5 & 8 & 11 & 9 \\ 3 & 3 & 5 & 9 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$

[1790] $1 \overset{1}{\underset{11}{6}} 13151321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 3 & 20 & 5 \\ 12 & 9 & 20 & 67 & 21 \\ 3 & 3 & 5 & 21 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$

[1791] $1 \overset{1}{\underset{112321}{8}} 1231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -6, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$

[1792] $1 \overset{1}{\underset{1112321}{8}} 1231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -6, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 1 & 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & 1 & 0 & 4 & 2 \\ 3 & 2 & 5 & 4 & 7 & 7 \\ 2 & 2 & 3 & 2 & 7 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$

[1793] $1 \overset{1}{\underset{11}{2321613151}} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -2, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 4 & 6 \\ 3 & 4 & 0 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$

$T = 16$ [1803] $1 \overset{1}{\underset{11}{6}} 1316131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 16, T_{\max} = 16, \Delta = -4, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 0 & 2 & 4 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & 1 & 3 \\ 2 & 4 & 2 & -1 & 3 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$

[1802] $1 \overset{1}{\underset{11231}{6}} 1316131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 16, T_{\max} = 16, \Delta = -4, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 2 & 2 & 0 & 11 & 4 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 11 & 6 & 0 & 34 & 14 \\ 2 & 4 & 2 & -1 & 14 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$

[1801] $1 \overset{1}{\underset{11}{23217123151}} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 16, T_{\max} = 16, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 4 & 6 \\ 3 & 4 & 0 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$

$T = 17$ [1815] $1 \overset{1}{\underset{11}{315131613151}} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 4 & 6 \\ 3 & 4 & 0 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$

$$[1820] \ 1^{11} 1^{11} 5^{1321} 8^{12321} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 4 & 6 \\ 3 & 4 & 0 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$[1819] \ 1^{11} 1^{11} 5^{1321} 8^{12321} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 2 & 0 & 1 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 3 & 4 & 0 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$[1818] \ 1^{11} 6^{11} 1^{12321712321} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 2 & 2 & 11 & 4 \\ 1 & 2 & 0 & 7 & 2 \\ 10 & 11 & 7 & 50 & 18 \\ 3 & 4 & 2 & 18 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$[1817] \ 1^{11} 6^{11} 1^{12321712321} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 2 & 2 & 5 & 4 \\ 1 & 2 & 0 & 2 & 2 \\ 4 & 5 & 2 & 7 & 7 \\ 3 & 4 & 2 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$[1816] \ 1^{12} 232181^{122315} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 2 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 13 & 0 & 52 & 18 \\ 3 & 4 & 0 & 18 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$[1814] \ 1^{11} 23218122315 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 4 & 6 \\ 3 & 4 & 0 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$[1813] \ 1^{11} 1^{12} 5^{1321} 8^{12321} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 2 & 0 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 11 & 0 & 28 & 14 \\ 3 & 4 & 0 & 14 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$T = 19 \quad [1833] \ 1^{11} 3151321812315 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 19, T_{\max} = 19, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 4 & 6 \\ 3 & 4 & 0 & 6 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

4.30 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [42] \ 1^{1111} 4^{31} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 5 & 5 & 4 \\ 3 & 1 & 1 & 4 & 3 & 3 \\ 2 & 1 & 0 & 2 & 2 & 2 \\ 5 & 4 & 2 & 6 & 5 & 6 \\ 5 & 3 & 2 & 5 & 3 & 5 \\ 4 & 3 & 2 & 6 & 5 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.31 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [6] \quad 1^{11} 4 \quad 13^1 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{cyan}{\mathfrak{su}_2} \rightarrow (-3) \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 & 3 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 4 & 3 & 0 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

4.32 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 15 \quad [76] \quad 1^{11} 3^1 15^1 13^1 2^1 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \rightarrow (-3) \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 10 & 2 & 4 \\ 3 & 2 & 8 & 1 & 4 \\ 10 & 8 & 25 & 3 & 13 \\ 2 & 1 & 3 & 0 & 2 \\ 4 & 4 & 13 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

4.33 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 11 \quad [6] \quad 1^{11} 4 \quad 13^1 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \rightarrow (-3) + \textcolor{cyan}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 1 & 2 & 3 \\ 2 & 1 & 1 & 3 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 3 & 0 & 1 & 3 \\ 3 & 3 & 0 & 4 & 0 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

4.34 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{11} 4 \quad 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{cyan}{\mathfrak{su}_2} + \textcolor{magenta}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 & 0 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 0 & 2 \\ 2 & 3 & 3 & 1 & 0 & 2 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 2 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 3 & -1 & 3 & 0 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_5$$

4.35 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 15 \quad [127] \quad 1^{11} 3^1 15^1 13^1 2^1 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} \rightarrow (-3) + \textcolor{cyan}{\mathfrak{su}_2} \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 2 & 10 & 4 \\ 3 & 2 & 1 & 8 & 4 \\ 2 & 1 & 0 & 3 & 2 \\ 10 & 8 & 3 & 25 & 13 \\ 4 & 4 & 2 & 13 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

4.36 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 11 \quad [6] \quad 1^{11} 4 \quad 13^1 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} \rightarrow (-3) + \textcolor{red}{\mathfrak{su}_2} + \textcolor{cyan}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 & 2 & 3 \\ 2 & 1 & 1 & 0 & 3 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 1 & 3 \\ 3 & 3 & 0 & 0 & 4 & 3 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

4.37 NHC(-2)(-3) + $\mathfrak{su}_3$ + su2n3mix + $\mathfrak{su}_2$

$$T = 15 \quad [98] \quad 1^{11} 6^{11} 131513^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 4 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 2 & 4 & 7 & 4 \\ 3 & 2 & 4 & 2 & 6 & 4 \\ 4 & 5 & 7 & 6 & 10 & 8 \\ 3 & 3 & 4 & 4 & 8 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

4.38 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 12 \quad [10] \quad 1^{11} 4^{11} 1^{11} 4^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -3, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 3 & 3 \\ 2 & 2 & 4 & 0 & 7 & 4 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 5 & 7 & 6 & 0 & 14 & 7 & 8 \\ 3 & 4 & 3 & 0 & 7 & 2 & 4 \\ 3 & 4 & 3 & -1 & 8 & 4 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[11] \quad 1^{11} 4^{11} 1^{11} 4^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -3, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 5 & 0 & 4 & 4 & 5 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 4 & 2 & 3 & 4 \\ 2 & 5 & 0 & 3 & 3 & 2 & 4 \\ 3 & 5 & -1 & 3 & 4 & 4 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

4.39 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{11} 4^{11} 1^{11} 4^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -1, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 & 0 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 & 0 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 2 & 0 & 3 \\ 2 & 4 & 2 & 0 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & -1 & 3 & 3 & 0 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_5$$

4.40 NHC(-2)(-3)

$$T = 5 \quad [0] \quad 1^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 5, T_{\max} = 95, \Delta = -145, \det = -5, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 9, T_{\max} = 95, T \in [9, 95]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 2131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 5, T_{\max} = 95, \Delta = -145, \det = -7, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 95, T \in [11, 95]$$

$$\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 5, T_{\max} = 95, \Delta = -145, \det = -7, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 95, T \in [11, 95]$$

$$\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 21^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 5, T_{\max} = 95, \Delta = -145, \det = -9, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 9.44 \rightarrow 10, T_{\max} = 95, T \in [10, 95]$$

$$\begin{aligned}
& \begin{pmatrix} 4 & 5 & 4 \\ 5 & 4 & 4 \\ 4 & 4 & 3 \end{pmatrix}_{b_0} \\
[4] \quad 21^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 5, T_{\max} = 95, \Delta = -145, \det = -12, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 9.33 \rightarrow 10, T_{\max} = 95, T \in [10, 95] \\
& \begin{pmatrix} 1 & 4 & 3 \\ 4 & 4 & 4 \\ 3 & 4 & 3 \end{pmatrix}_{b_0} \\
[5] \quad 131^{11} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 5, T_{\max} = 95, \Delta = -145, \det = -28, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 10.29 \rightarrow 11, T_{\max} = 95, T \in [11, 95] \\
& \begin{pmatrix} 4 & 8 & 4 \\ 8 & 9 & 5 \\ 4 & 5 & 2 \end{pmatrix}_{b_0} \\
[6] \quad 221^{11} 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 5, T_{\max} = 95, \Delta = -145, \det = -63, (n_+, n_-, n_0) = (1, 5, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 9.73 \rightarrow 10, T_{\max} = 95, T \in [10, 95] \\
& \begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
\\
T = 6 \quad [9] \quad 1^1 3131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 6, T_{\max} = 79, \Delta = -124, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 9, T_{\max} = 79, T \in [9, 79] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0} \\
[8] \quad 1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 9, T_{\max} = 29, T \in [9, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[7] \quad 1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 4, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 9, T_{\max} = 29, T \in [9, 29] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0} \\
[13] \quad 2141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 6, T_{\max} = 96, \Delta = -136, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 10, T_{\max} = 96, T \in [10, 96] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0} \\
[12] \quad 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 10, T_{\max} = 29, T \in [10, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -3 & -1 \\ 2 & 2 & -1 & 3 \end{pmatrix}_{b_0} \\
[11] \quad 141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 6, T_{\max} = 96, \Delta = -136, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 10, T_{\max} = 96, T \in [10, 96] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0} \\
[10] \quad 21^1 321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 10, T_{\max} = 29, T \in [10, 29] \\
& \begin{pmatrix} 4 & 5 & 4 \\ 5 & 5 & 5 \\ 4 & 5 & 4 \end{pmatrix}_{b_0} \\
[15] \quad 1412 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 6, T_{\max} = 96, \Delta = -136, \det = 7, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 96, T \in [12, 96] \\
& \begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0} \\
[14] \quad 1231 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 6, T_{\max} = 29, \Delta = -117, \det = 7, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 29, T \in [12, 29] \\
& \begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0} \\
[16] \quad 141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 6, T_{\max} = 96, \Delta = -136, \det = 7, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 96, T \in [12, 96] \\
& \begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [17] $13131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 79, \Delta = -124, \det = 7, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 79, T \in [12, 79]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [18] $31231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 81, \Delta = -125, \det = 7, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 81, T \in [12, 81]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [19] $22141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 96, \Delta = -136, \det = 7, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 96, T \in [12, 96]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [20] $1231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 7, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 29, T \in [12, 29]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [21] $21321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 7, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 29, T \in [12, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [22] $1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 7, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 29, T \in [12, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [24] $21^1 321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 8, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9.50 \rightarrow 10, T_{\max} = 29, T \in [10, 29]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [23] $2231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 8, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9.50 \rightarrow 10, T_{\max} = 29, T \in [10, 29]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [25] $1^1 4^1 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 96, \Delta = -136, \det = 9, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$
 $\begin{pmatrix} 4 & 5 & 4 \\ 5 & 4 & 4 \\ 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [26] $131^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 79, \Delta = -124, \det = 9, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 79, T \in [11, 79]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 \\ 2 & 4 & 3 & 3 \\ 2 & 3 & 3 & 2 \end{pmatrix}_{b_0}$
- [27] $2231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 29, \Delta = -117, \det = 11, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 9.45 \rightarrow 10, T_{\max} = 29, T \in [10, 29]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [29] $131^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 79, \Delta = -124, \det = 12, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 79, T \in [11, 79]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 4 & 4 \\ 1 & 4 & 0 & 2 \\ 2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [30] $31^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 81, \Delta = -125, \det = 12, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 81, T \in [11, 81]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [31] $21^1 41^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6, T_{\max} = 96, \Delta = -136, \det = 12, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$

- $\begin{pmatrix} 4 & 8 & 4 \\ 8 & 13 & 7 \\ 4 & 7 & 3 \end{pmatrix}_{b_0}$
 [28] $1 \overset{21}{4} 1^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6$, $T_{\max} = 96$, $\Delta = -136$, $\det = 12$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 4 & 4 \\ 3 & 4 & 3 \end{pmatrix}_{b_0}$
 [32] $221^{\overset{1}{1}} 41^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6$, $T_{\max} = 96$, $\Delta = -136$, $\det = 15$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10.40 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 3 & 2 \\ 3 & 3 & 3 & 3 \\ 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
 [34] $31^{\overset{1}{1}} 231^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6$, $T_{\max} = 81$, $\Delta = -125$, $\det = 20$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10$, $T_{\max} = 81$, $T \in [10, 81]$
 $\begin{pmatrix} 1 & 5 & 3 \\ 5 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$
 [33] $232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6$, $T_{\max} = 43$, $\Delta = -119$, $\det = 20$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 43$, $T \in [10, 43]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
 [35] $221^{\overset{1}{1}} 41^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6$, $T_{\max} = 96$, $\Delta = -136$, $\det = 23$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10.26 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$
 $\begin{pmatrix} 1 & 6 & 3 \\ 6 & 13 & 7 \\ 3 & 7 & 3 \end{pmatrix}_{b_0}$
 [36] $231^{\overset{1}{1}} 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6$, $T_{\max} = 81$, $\Delta = -125$, $\det = 27$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10.37 \rightarrow 11$, $T_{\max} = 81$, $T \in [11, 81]$
 $\begin{pmatrix} 0 & 2 & 3 & 2 \\ 2 & 1 & 3 & 3 \\ 3 & 3 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
 [37] $1231^{\overset{1}{1}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 28$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 29$, $T \in [12, 29]$
 $\begin{pmatrix} 4 & 8 & 4 \\ 8 & 9 & 5 \\ 4 & 5 & 2 \end{pmatrix}_{b_0}$
 [40] $2 \overset{2}{3} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6$, $T_{\max} = 43$, $\Delta = -119$, $\det = 28$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10.29 \rightarrow 11$, $T_{\max} = 43$, $T \in [11, 43]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
 [41] $1 \overset{21}{4} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6$, $T_{\max} = 96$, $\Delta = -136$, $\det = 28$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 96$, $T \in [12, 96]$
 $\begin{pmatrix} 4 & 8 & 4 \\ 8 & 9 & 5 \\ 4 & 5 & 2 \end{pmatrix}_{b_0}$
 [42] $21^{\overset{1}{1}} 412 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6$, $T_{\max} = 96$, $\Delta = -136$, $\det = 28$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 96$, $T \in [12, 96]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 7 & 4 \\ 2 & 7 & 3 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix}_{b_0}$
 [43] $131^{\overset{1}{1}} 31 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6$, $T_{\max} = 79$, $\Delta = -124$, $\det = 28$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 79$, $T \in [12, 79]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 7 & 4 \\ 2 & 7 & 3 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix}_{b_0}$
 [39] $2 \overset{1}{3} 13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6$, $T_{\max} = 81$, $\Delta = -125$, $\det = 28$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10.29 \rightarrow 11$, $T_{\max} = 81$, $T \in [11, 81]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
 [38] $13 \overset{1}{2} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6$, $T_{\max} = 29$, $\Delta = -117$, $\det = 28$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10.29 \rightarrow 11$, $T_{\max} = 29$, $T \in [11, 29]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[44] $231^1 1^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6$, $T_{\max} = 81$, $\Delta = -125$, $\det = 32$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10.12 \rightarrow 11$, $T_{\max} = 81$, $T \in [11, 81]$

$$\begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[45] $31^1 31^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6$, $T_{\max} = 90$, $\Delta = -132$, $\det = 36$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10.44 \rightarrow 11$, $T_{\max} = 90$, $T \in [11, 90]$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 4 & 4 & 4 \\ 3 & 4 & 0 & 2 \\ 2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

[46] $1321^1 1^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6$, $T_{\max} = 81$, $\Delta = -125$, $\det = 63$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10.73 \rightarrow 11$, $T_{\max} = 81$, $T \in [11, 81]$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[47] $141^1 1^1 22 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6$, $T_{\max} = 96$, $\Delta = -136$, $\det = 63$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10.73 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 4 & 7 & 4 \\ 3 & 7 & 3 & 3 \\ 2 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[48] $31^1 1^1 313 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6$, $T_{\max} = 90$, $\Delta = -132$, $\det = 63$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10.73 \rightarrow 11$, $T_{\max} = 90$, $T \in [11, 90]$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 4 & 7 & 4 \\ 3 & 7 & 3 & 3 \\ 2 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[49] $2^1 3^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6$, $T_{\max} = 43$, $\Delta = -119$, $\det = 112$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 43$, $T \in [10, 43]$

$$\begin{pmatrix} 0 & 4 & 4 & 2 \\ 4 & 8 & 11 & 6 \\ 4 & 11 & 7 & 5 \\ 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[50] $41^1 1^1 222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6$, $T_{\max} = 96$, $\Delta = -136$, $\det = 112$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10.39 \rightarrow 11$, $T_{\max} = 96$, $T \in [11, 96]$

$$\begin{pmatrix} 0 & 4 & 4 & 2 \\ 4 & 10 & 13 & 6 \\ 4 & 13 & 9 & 5 \\ 2 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[51] $231^1 1^1 23 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 6$, $T_{\max} = 85$, $\Delta = -126$, $\det = 175$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 10.17 \rightarrow 11$, $T_{\max} = 85$, $T \in [11, 85]$

$$\begin{pmatrix} 0 & 5 & 5 & 2 \\ 5 & 8 & 11 & 6 \\ 5 & 11 & 7 & 5 \\ 2 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

$T = 7$ [53] $13221^1 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -3$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 9$, $T_{\max} = 11$, $T \in [9, 11]$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0}$$

[56] $1^1 23131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7$, $T_{\max} = 13$, $\Delta = -96$, $\det = -3$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 9$, $T_{\max} = 13$, $T \in [9, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

[55] $1^1 3^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -90$, $\det = -3$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 9$, $T_{\max} = 11$, $T \in [9, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

[54] $13221^1 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -3$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 9$, $T_{\max} = 11$, $T \in [9, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

[52] $1^1 23131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7$, $T_{\max} = 13$, $\Delta = -96$, $\det = -3$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 9$, $T_{\max} = 13$, $T \in [9, 13]$

- [57] $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$ $21^1 3221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 11, \Delta = -88, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 \end{pmatrix}_{b_0}$
- [58] $1^1 41231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 27, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 27, T \in [10, 27]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 3 & 2 & 5 \end{pmatrix}_{b_0}$
- [59] $1^1 41231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 27, \Delta = -116, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 27, T \in [10, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [60] $12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 11, \Delta = -90, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$
- [61] $12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 11, \Delta = -90, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$
- [62] $14131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 24, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 24, T \in [10, 24]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 3 & 2 & 5 \end{pmatrix}_{b_0}$
- [63] $14131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 24, \Delta = -115, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10, T_{\max} = 24, T \in [10, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [65] $131^1 321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 13, \Delta = -96, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 13, T \in [11, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 2 & 4 & 4 \\ 2 & 4 & 3 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [67] $21^1 4131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 24, \Delta = -115, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 24, T \in [11, 24]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$
- [66] $1^1 5^1 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 90, \Delta = -131, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 90, T \in [11, 90]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$
- [68] $1^1 5^1 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 90, \Delta = -131, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 90, T \in [11, 90]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$
- [64] $131^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 24, \Delta = -115, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 24, T \in [11, 24]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$
- [69] $12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 7, T_{\max} = 11, \Delta = -88, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

- $$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -3 & -1 \\ 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$
- [70] $221\bar{5}1^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 90, \Delta = -131, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11, T_{\max} = 90, T \in [11, 90]$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$$
- [71] $312321^{\bar{1}\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 14, \Delta = -98, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$
- [72] $131321^{\bar{1}\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 13, \Delta = -96, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 5 & 4 \\ 1 & 5 & 1 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$
- [73] $1^{\bar{1}\bar{1}}\bar{4}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 24, T \in [13, 24]$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$$
- [74] $123131^{\bar{1}\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 13, \Delta = -96, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$
- [75] $1^{\bar{1}\bar{1}}\bar{4}131 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 24, T \in [13, 24]$

$$\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$$
- [76] $2^{\bar{1}\bar{1}}\bar{3}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 27, T \in [13, 27]$

$$\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$$
- [77] $141231^{\bar{1}\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 27, T \in [13, 27]$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$$
- [78] $412231^{\bar{1}\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 27, T \in [13, 27]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$
- [79] $313141^{\bar{1}\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 77, T \in [13, 77]$

$$\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$$
- [80] $132141^{\bar{1}\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 27, T \in [13, 27]$

$$\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$$
- [81] $214131^{\bar{1}\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 24, T \in [13, 24]$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$$
- [83] $2\bar{3}141^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 27, T \in [11, 27]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$
- [82] $131^{\bar{1}}\bar{3}21^{\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 7, T_{\max} = 13, \Delta = -96, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$

- [84] $21^1 4131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7, T_{\max} = 24, \Delta = -115, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 4 & 4 \\ 1 & 4 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [85] $1231^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7, T_{\max} = 13, \Delta = -96, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [86] $21^1 51^1 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7, T_{\max} = 90, \Delta = -131, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 90, T \in [12, 90]$
 $\begin{pmatrix} 4 & 5 & 4 \\ 5 & 4 & 4 \\ 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [87] $1^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7, T_{\max} = 90, \Delta = -131, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 90, T \in [12, 90]$
 $\begin{pmatrix} 4 & 5 & 4 \\ 5 & 4 & 4 \\ 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [88] $31^1 2321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7, T_{\max} = 14, \Delta = -98, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 14, T \in [11, 14]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [89] $23141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7, T_{\max} = 27, \Delta = -116, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 27, T \in [11, 27]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [90] $1231^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7, T_{\max} = 13, \Delta = -96, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 2 \end{pmatrix}_{b_0}$
- [91] $1^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7, T_{\max} = 90, \Delta = -131, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 90, T \in [12, 90]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 4 & 4 \\ 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [92] $1^1 31^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7, T_{\max} = 24, \Delta = -115, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 2 \end{pmatrix}_{b_0}$
- [93] $21^1 41^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7, T_{\max} = 24, \Delta = -115, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 0 & 3 & 2 \\ 2 & 3 & 3 & 3 \\ 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [94] $141^1 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7, T_{\max} = 27, \Delta = -116, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 27, T \in [12, 27]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 4 & 4 \\ 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [95] $31^1 3141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7, T_{\max} = 77, \Delta = -123, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [96] $1^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7, T_{\max} = 24, \Delta = -115, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 4 & 4 \\ 3 & 4 & 3 \end{pmatrix}_{b_0}$

- [97] $21^1 51^1 22 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7$, $T_{\max} = 90$, $\Delta = -131$, $\det = -12$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 90$, $T \in [12, 90]$
 $\begin{pmatrix} 0 & 2 & 3 & 2 \\ 2 & 4 & 5 & 4 \\ 3 & 5 & 3 & 3 \\ 2 & 4 & 3 & 2 \end{pmatrix}_{b_0}$
- [98] $21^1 51^1 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7$, $T_{\max} = 90$, $\Delta = -131$, $\det = -12$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 90$, $T \in [12, 90]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 4 & 4 \\ 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [99] $1^1 51^1 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7$, $T_{\max} = 90$, $\Delta = -131$, $\det = -12$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 90$, $T \in [12, 90]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 0 & 3 & 2 \\ 2 & 3 & 3 & 3 \\ 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [100] $2141^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -12$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 2 \end{pmatrix}_{b_0}$
- [102] $41^1 2231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7$, $T_{\max} = 27$, $\Delta = -116$, $\det = -15$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 27$, $T \in [12, 27]$
 $\begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 \\ 4 & 1 & 25 & 9 \\ 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$
- [101] $1321^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7$, $T_{\max} = 27$, $\Delta = -116$, $\det = -15$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 27$, $T \in [12, 27]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 3 & 2 \\ 3 & 3 & 3 & 3 \\ 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [103] $1^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7$, $T_{\max} = 90$, $\Delta = -131$, $\det = -15$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 90$, $T \in [12, 90]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 3 & 2 \\ 3 & 3 & 3 & 3 \\ 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [104] $21^1 51^1 22 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7$, $T_{\max} = 90$, $\Delta = -131$, $\det = -17$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 90$, $T \in [12, 90]$
 $\begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 4 & 5 & 4 \\ 2 & 5 & 3 & 3 \\ 2 & 4 & 3 & 2 \end{pmatrix}_{b_0}$
- [105] $141^1 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7$, $T_{\max} = 27$, $\Delta = -116$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11$, $T_{\max} = 27$, $T \in [11, 27]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [106] $31^1 3141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7$, $T_{\max} = 77$, $\Delta = -123$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11$, $T_{\max} = 77$, $T \in [11, 77]$
 $\begin{pmatrix} 1 & 5 & 3 \\ 5 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [107] $3131^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7$, $T_{\max} = 77$, $\Delta = -123$, $\det = -23$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 77$, $T \in [12, 77]$
 $\begin{pmatrix} 1 & 6 & 3 \\ 6 & 13 & 7 \\ 3 & 7 & 3 \end{pmatrix}_{b_0}$
- [108] $41^1 231^1 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7$, $T_{\max} = 80$, $\Delta = -124$, $\det = -23$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 80$, $T \in [12, 80]$
 $\begin{pmatrix} 0 & 3 & 4 & 2 \\ 3 & 1 & 5 & 3 \\ 4 & 5 & 9 & 5 \\ 2 & 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [109] $1^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7$, $T_{\max} = 90$, $\Delta = -131$, $\det = -23$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 90$, $T \in [12, 90]$
 $\begin{pmatrix} 1 & 6 & 3 \\ 6 & 13 & 7 \\ 3 & 7 & 3 \end{pmatrix}_{b_0}$
- [110] $2^1 3^1 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 7$, $T_{\max} = 27$, $\Delta = -116$, $\det = -23$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 27$, $T \in [12, 27]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 6 & 3 \\ 6 & 13 & 7 \\ 3 & 7 & 3 \end{pmatrix}_{b_0} \\
[111] & 2221^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 90, \Delta = -131, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11.26 \rightarrow 12, T_{\max} = 90, T \in [12, 90] \\
& \begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 \\ 4 & 1 & 17 & 7 \\ 2 & 0 & 7 & 2 \end{pmatrix}_{b_0} \\
[112] & 1321^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11.26 \rightarrow 12, T_{\max} = 27, T \in [12, 27] \\
& \begin{pmatrix} 1 & 6 & 3 \\ 6 & 13 & 7 \\ 3 & 7 & 3 \end{pmatrix}_{b_0} \\
[113] & 31^{1\bar{1}} 21^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 14, \Delta = -98, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 10.57 \rightarrow 11, T_{\max} = 14, T \in [11, 14] \\
& \begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 \\ 2 & 3 & 3 & 3 \end{pmatrix}_{b_0} \\
[114] & 2321^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 14, \Delta = -98, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 10.29 \rightarrow 11, T_{\max} = 14, T \in [11, 14] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[115] & 41^{1\bar{1}} 232 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 34, \Delta = -118, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11.29 \rightarrow 12, T_{\max} = 34, T \in [12, 34] \\
& \begin{pmatrix} 0 & 2 & 4 & 2 \\ 2 & 1 & 5 & 3 \\ 4 & 5 & 9 & 5 \\ 2 & 3 & 5 & 2 \end{pmatrix}_{b_0} \\
[116] & 41232 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 34, \Delta = -118, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11.29 \rightarrow 12, T_{\max} = 34, T \in [12, 34] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[117] & 21^{1\bar{1}} 4131 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.29 \rightarrow 13, T_{\max} = 24, T \in [13, 24] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 7 & 4 \\ 2 & 7 & 3 & 3 \\ 2 & 4 & 3 & 1 \end{pmatrix}_{b_0} \\
[118] & 31^{1\bar{1}} 413 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11.29 \rightarrow 12, T_{\max} = 77, T \in [12, 77] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[119] & 141^{1\bar{1}} 32 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11.29 \rightarrow 12, T_{\max} = 27, T \in [12, 27] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[120] & 23131^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 14, \Delta = -98, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 10.29 \rightarrow 11, T_{\max} = 14, T \in [11, 14] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[121] & 41^{1\bar{1}} 231^{1\bar{3}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 80, \Delta = -124, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11.33 \rightarrow 12, T_{\max} = 80, T \in [12, 80] \\
& \begin{pmatrix} 0 & 4 & 3 & 2 \\ 4 & 10 & 5 & 6 \\ 3 & 5 & 0 & 2 \\ 2 & 6 & 2 & 2 \end{pmatrix}_{b_0} \\
[122] & 41^{1\bar{1}} 2231^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 27, T \in [11, 27] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 26 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 2 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[123] & 141^{1\bar{1}} 32 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11.12 \rightarrow 12, T_{\max} = 27, T \in [12, 27]
\end{aligned}$$

$$\begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 0 & 2 \\ 2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

[124] $31^{\overline{1}}413 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$

$$\begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 0 & 2 \\ 2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

[125] $141^{\overline{1}}31^{\overline{1}}3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 4 & 4 & 4 \\ 3 & 4 & 0 & 2 \\ 2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

[126] $2221^{\overline{1}}51^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 90, \Delta = -131, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 90, T \in [12, 90]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -3 & -1 \\ 2 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

[127] $41^{\overline{1}}232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 34, \Delta = -118, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 34, T \in [11, 34]$

$$\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 10 & 5 & 6 \\ 2 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[128] $321^{\overline{1}}41^{\overline{1}}3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 80, \Delta = -124, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 80, T \in [12, 80]$

$$\begin{pmatrix} 0 & 3 & 5 & 2 \\ 3 & 1 & 6 & 3 \\ 5 & 6 & 12 & 6 \\ 2 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[129] $2^{\overline{1}}3^{\overline{1}}2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 14, \Delta = -98, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 14, T \in [11, 14]$

$$\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 8 & 6 & 6 \\ 2 & 6 & 1 & 3 \\ 2 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

[130] $41^{\overline{1}}322 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -62, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 27, T \in [11, 27]$

$$\begin{pmatrix} 0 & 4 & 3 & 2 \\ 4 & 10 & 9 & 6 \\ 3 & 9 & 4 & 4 \\ 2 & 6 & 4 & 3 \end{pmatrix}_{b_0}$$

[131] $14131^{\overline{1}}3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[132] $12321^{\overline{1}}3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 14, \Delta = -98, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[133] $41^{\overline{1}}322 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 27, T \in [11, 27]$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 5 & 8 & 5 \\ 3 & 8 & 4 & 4 \\ 2 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[134] $41^{\overline{1}}31^{\overline{1}}32 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 80, \Delta = -124, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 80, T \in [12, 80]$

$$\begin{pmatrix} 0 & 4 & 5 & 2 \\ 4 & 10 & 13 & 6 \\ 5 & 13 & 12 & 6 \\ 2 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[135] $41^{\overline{1}}2231 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 27, \Delta = -116, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 27, T \in [12, 27]$

$$\begin{pmatrix} 0 & 4 & 4 & 2 \\ 4 & 26 & 29 & 10 \\ 4 & 29 & 25 & 9 \\ 2 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

$T = 8$ [136] $1^{\overline{1}}223131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 10, \Delta = -67, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 9, T_{\max} = 10, T \in [9, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
[137] & 1^1 231321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 10, \Delta = -68, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 9, T_{\max} = 10, T \in [9, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0} \\
[138] & 31^1 3221^1 5 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 45, \Delta = -119, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 45, T \in [15, 45] \\
& \begin{pmatrix} 0 & 3 & 5 & 2 \\ 3 & 4 & 4 & 4 \\ 5 & 4 & 2 & 4 \\ 2 & 4 & 4 & 0 \end{pmatrix}_{b_0} \\
[139] & 1^1 223131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 10, \Delta = -67, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 9, T_{\max} = 10, T \in [9, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0} \\
[143] & 1^1 141321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 10, T_{\max} = 11, T \in [10, 11] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[140] & 1^1 412321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 10, T_{\max} = 12, T \in [10, 12] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[142] & 1^1 141321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 10, T_{\max} = 11, T \in [10, 11] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[144] & 1^1 314131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 10, T_{\max} = 13, T \in [10, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[141] & 1^1 412321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 10, T_{\max} = 12, T \in [10, 12] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[146] & 1^1 322151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 21, T \in [11, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0} \\
[152] & 1^1 413141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 24, T \in [11, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0} \\
[150] & 21^1 5131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 20, T \in [11, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[148] & 1^1 5131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 20, T \in [11, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[149] & 1^1 5131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 20, T \in [11, 20]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0} \\
[145] & 21^1 41321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0} \\
[151] & 1^1 151231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 21, T \in [11, 21] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0} \\
[153] & 1^1 14141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 20, T \in [11, 20] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0} \\
[154] & 1^1 151231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 21, T \in [11, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[147] & 1^1 215131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 20, T \in [11, 20] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0} \\
[155] & 1^1 322151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 21, T \in [11, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[156] & 1^1 14141^1 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -106, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 4 & 5 & 4 \\ 5 & 5 & 5 \\ 4 & 5 & 4 \end{pmatrix}_{b_0} \\
[157] & 21^1 15131^1 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 35, T \in [12, 35] \\
& \begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 3 \\ 2 & 3 & 3 & 3 \end{pmatrix}_{b_0} \\
[158] & 1^1 14141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -106, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[159] & 21^1 1^1 5131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 4 & 5 & 4 \\ 5 & 5 & 5 \\ 4 & 5 & 4 \end{pmatrix}_{b_0} \\
[160] & 3131^1 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 35, T \in [12, 35] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0} \\
[161] & 21^1 151231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 21, T \in [12, 21] \\
& \begin{pmatrix} 4 & 5 & 4 \\ 5 & 5 & 5 \\ 4 & 5 & 4 \end{pmatrix}_{b_0} \\
[162] & 1^1 132151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 21, T \in [12, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -3 & -1 \\ 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [163] $131\overset{1}{5}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -110, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$
- [164] $131\overset{2}{5}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 21, \Delta = -111, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 21, T \in [12, 21]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$
- [165] $2141\overset{1}{4}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -106, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [166] $221\overset{1}{5}131 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -110, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 20, T \in [13, 20]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 1 & 3 & 3 \\ 3 & 3 & 3 & 3 \\ 2 & 3 & 3 & 2 \end{pmatrix}_{b_0}$
- [167] $1\overset{1}{4}141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 20, T \in [14, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [168] $1\overset{1}{5}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -110, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 20, T \in [14, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [169] $1\overset{1}{5}12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -110, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 20, T \in [14, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [170] $2\overset{1}{3}21 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 \end{pmatrix}_{b_0}$
- [171] $2215131 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -110, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 20, T \in [14, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [172] $2151231 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 21, \Delta = -111, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 21, T \in [14, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [173] $4123141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 25, \Delta = -115, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 25, T \in [14, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$
- [174] $3214141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 25, \Delta = -115, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 25, T \in [14, 25]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [175] $1512231 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 21, \Delta = -111, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 21, T \in [14, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$

- [176] $151231^{11} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 21, \Delta = -111, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 21, T \in [14, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [177] $1413141^{11} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 24, \Delta = -114, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 24, T \in [14, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [178] $14141^{31} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 24, \Delta = -114, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 24, T \in [14, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$
- [179] $1^1 4 1412 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 20, T \in [14, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [180] $21^1 5132 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 21, \Delta = -111, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 21, T \in [12, 21]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 3 & 3 \\ 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$
- [181] $21^1 5132 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 21, \Delta = -111, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [182] $232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 23, \Delta = -113, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [183] $131^1 4131 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [184] $314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 24, \Delta = -114, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 4 & 4 \\ 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$
- [185] $15132^{11} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 21, \Delta = -111, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [186] $21^1 51231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 21, \Delta = -111, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [187] $14141^1 2 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -106, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [188] $1^1 5 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -110, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

- [189] $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
 $131\bar{5}1^12 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.44 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$
 $\begin{pmatrix} 4 & 5 & 4 \\ 5 & 4 & 4 \\ 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [190] $2\bar{3}21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 6 & 6 \\ 3 & 6 & 5 \end{pmatrix}_{b_0}$
- [194] $2\bar{3}2151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [191] $1\bar{5}1\bar{3}2 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [192] $31\bar{4}141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 3 \end{pmatrix}_{b_0}$
- [193] $141^12321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [195] $31^132151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 45, \Delta = -119, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 45, T \in [13, 45]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [197] $1\bar{5}131^13 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [198] $13141^131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 4 & 4 \\ 1 & 4 & 0 & 2 \\ 2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [196] $131\bar{5}1^12 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 4 & 4 \\ 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [199] $223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 3 & 2 \\ 3 & 3 & 4 & 4 \\ 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$
- [200] $221^15131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [201] $41^123141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 25, \Delta = -115, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

- [202] $\begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 \\ 4 & 1 & 25 & 9 \\ 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$ $151^1 2231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 21, \Delta = -111, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & 1 & 6 & 3 \\ 4 & 6 & 17 & 7 \\ 2 & 3 & 7 & 2 \end{pmatrix}_{b_0}$
- [203] $1321^1 51^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 21, \Delta = -111, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 4 & 11 & 4 \\ 11 & 26 & 10 \\ 4 & 10 & 3 \end{pmatrix}_{b_0}$
- [204] $31^1 3221^1 5 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 45, \Delta = -119, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 45, T \in [13, 45]$
 $\begin{pmatrix} 0 & 5 & 3 & 2 \\ 5 & 3 & 4 & 5 \\ 3 & 4 & 3 & 3 \\ 2 & 5 & 3 & 2 \end{pmatrix}_{b_0}$
- [205] $1^1 4 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 24, \Delta = -114, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 24, T \in [12, 24]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 2 & 7 & -1 & 3 \end{pmatrix}_{b_0}$
- [206] $31^1 32151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 45, \Delta = -119, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 45, T \in [12, 45]$
 $\begin{pmatrix} 1 & 5 & 3 \\ 5 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [207] $2^1 3^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 2 & 4 & 3 & 4 \end{pmatrix}_{b_0}$
- [208] $1^1 5^1 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 21, \Delta = -111, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 21, T \in [12, 21]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 5 & 6 \\ 1 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [209] $31^1 5^1 13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 35, \Delta = -118, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [210] $1^1 5^1 131^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 35, \Delta = -118, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 35, T \in [12, 35]$
 $\begin{pmatrix} 1 & 5 & 3 \\ 5 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [211] $141^1 3141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 24, \Delta = -114, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 24, T \in [12, 24]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 2 & 7 & -1 & 3 \end{pmatrix}_{b_0}$
- [212] $2151^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 21, \Delta = -111, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 21, T \in [12, 21]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 5 & 6 \\ 1 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [213] $223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 21, \Delta = -111, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$
 $\begin{pmatrix} 1 & 6 & 3 \\ 6 & 14 & 8 \\ 3 & 8 & 4 \end{pmatrix}_{b_0}$
- [215] $1321^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 8, T_{\max} = 21, \Delta = -111, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 7 & 3 \\ 7 & 26 & 10 \\ 3 & 10 & 3 \end{pmatrix}_{b_0} \\
[214] & 41231^{11}41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 25, \Delta = -115, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.26 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 2 \end{pmatrix}_{b_0} \\
[216] & 14131^{11}41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.26 \rightarrow 13, T_{\max} = 24, T \in [13, 24] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 6 & 7 \\ 1 & 6 & 0 & 2 \\ 2 & 7 & 2 & 2 \end{pmatrix}_{b_0} \\
[217] & 321^{11}4141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 25, \Delta = -115, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.26 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 1 & 5 & 2 \\ 1 & 0 & 3 & 2 \\ 5 & 3 & 7 & 5 \\ 2 & 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[218] & 13221^{11}51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.26 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 \\ 4 & 1 & 17 & 7 \\ 2 & 0 & 7 & 2 \end{pmatrix}_{b_0} \\
[219] & 41^{11}231^{11}41 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 25, \Delta = -115, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.26 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 3 & 4 & 2 \\ 3 & 10 & 17 & 6 \\ 4 & 17 & 25 & 9 \\ 2 & 6 & 9 & 2 \end{pmatrix}_{b_0} \\
[220] & 1^{11}2^{11}3^11 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.29 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[221] & 231^{11}3151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 45, \Delta = -119, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.14 \rightarrow 13, T_{\max} = 45, T \in [13, 45] \\
& \begin{pmatrix} 0 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 \\ 5 & 1 & 32 & 10 \\ 2 & 0 & 10 & 2 \end{pmatrix}_{b_0} \\
[222] & 1512^13^12 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.29 \rightarrow 13, T_{\max} = 23, T \in [13, 23] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[223] & 141321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[224] & 1^14^1132 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[225] & 2151^{11}3^12 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.29 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[226] & 21^{11}41412 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -106, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.29 \rightarrow 14, T_{\max} = 20, T \in [14, 20] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 8 & 11 & 6 \\ 2 & 11 & 7 & 5 \\ 2 & 6 & 5 & 1 \end{pmatrix}_{b_0} \\
[227] & 31^14^1141 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.29 \rightarrow 13, T_{\max} = 24, T \in [13, 24]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 5 & 8 & 5 \\ 2 & 8 & 4 & 4 \\ 2 & 5 & 4 & 2 \end{pmatrix}_{b_0} \\
[228] & 14141^{11}2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -106, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.29 \rightarrow 14, T_{\max} = 20, T \in [14, 20] \\
& \begin{pmatrix} 4 & 8 & 4 \\ 8 & 9 & 5 \\ 4 & 5 & 2 \end{pmatrix}_{b_0} \\
[229] & 151^{11}2231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 21, T \in [12, 21] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 6 & 8 \\ 1 & 6 & 0 & 2 \\ 2 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[230] & 41^{11}23141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 25, \Delta = -115, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 25, T \in [12, 25] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 26 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 2 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[231] & 151^{11}231^13 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 45, \Delta = -119, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.33 \rightarrow 13, T_{\max} = 45, T \in [13, 45] \\
& \begin{pmatrix} 0 & 4 & 3 & 2 \\ 4 & 10 & 5 & 6 \\ 3 & 5 & 0 & 2 \\ 2 & 6 & 2 & 2 \end{pmatrix}_{b_0} \\
[232] & 31^{11}4141 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.12 \rightarrow 13, T_{\max} = 24, T \in [13, 24] \\
& \begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 13 & 10 & 7 \\ 2 & 10 & 4 & 4 \\ 2 & 7 & 4 & 2 \end{pmatrix}_{b_0} \\
[235] & 31321^{11}51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 45, \Delta = -119, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.16 \rightarrow 13, T_{\max} = 45, T \in [13, 45] \\
& \begin{pmatrix} 1 & 8 & 3 \\ 8 & 26 & 10 \\ 3 & 10 & 3 \end{pmatrix}_{b_0} \\
[233] & 1321^2151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.16 \rightarrow 13, T_{\max} = 23, T \in [13, 23] \\
& \begin{pmatrix} 1 & 8 & 3 \\ 8 & 26 & 10 \\ 3 & 10 & 3 \end{pmatrix}_{b_0} \\
[234] & 13221^{11}51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.16 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -3 & -1 \\ 2 & 8 & -1 & 2 \end{pmatrix}_{b_0} \\
[236] & 151^{11}232 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 23, T \in [12, 23] \\
& \begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 10 & 5 & 6 \\ 2 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[237] & 321^{11}4141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 25, \Delta = -115, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11.79 \rightarrow 12, T_{\max} = 25, T \in [12, 25] \\
& \begin{pmatrix} 1 & 8 & 3 \\ 8 & 17 & 9 \\ 3 & 9 & 4 \end{pmatrix}_{b_0} \\
[238] & 231^{11}3151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 45, \Delta = -119, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 45, T \in [12, 45] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 1 & 11 \\ 1 & 1 & -3 & -1 \\ 2 & 11 & -1 & 3 \end{pmatrix}_{b_0} \\
[242] & 215131^{11}3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.73 \rightarrow 13, T_{\max} = 35, T \in [13, 35] \\
& \begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[239] & 1^{3111}5^{11}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.73 \rightarrow 13, T_{\max} = 35, T \in [13, 35] \\
& \begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[240] \quad 41 \quad 4 \quad 132 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 25, \Delta = -115, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 25, T \in [12, 25]$$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 5 & 8 & 5 \\ 3 & 8 & 4 & 4 \\ 2 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[241] \quad 151 \quad 3 \quad 22 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 5 & 8 & 5 \\ 3 & 8 & 4 & 4 \\ 2 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[243] \quad 31 \quad 5 \quad 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 45, \Delta = -119, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 45, T \in [12, 45]$$

$$\begin{pmatrix} 0 & 5 & 2 & 2 \\ 5 & 8 & 5 & 6 \\ 2 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[244] \quad 41 \quad 4 \quad 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 25, \Delta = -115, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 25, T \in [12, 25]$$

$$\begin{pmatrix} 0 & 5 & 3 & 2 \\ 5 & 13 & 10 & 7 \\ 3 & 10 & 4 & 4 \\ 2 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[245] \quad 21 \quad 11 \quad 4132 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 4 & 4 & 2 \\ 4 & 12 & 15 & 8 \\ 4 & 15 & 11 & 7 \\ 2 & 8 & 7 & 4 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [246] \quad 1^1 22313221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 10, \Delta = -39, \det = 0, (n_+, n_-, n_0) = (1, 9, 1), c_{\text{ext}} = 3.0$$

$$[252] \quad 1^1 3141321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 11, \Delta = -66, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[251] \quad 1^1 3141321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 11, \Delta = -66, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[249] \quad 1413221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[253] \quad 141^1 3221^1 5 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 3 & 5 & 2 \\ 3 & 4 & 4 & 4 \\ 5 & 4 & 2 & 4 \\ 2 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[250] \quad 1413221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[264] \quad 1^1 3151231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 13, T \in [11, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[263] \quad 1^1 3151231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 13, T \in [11, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$$

$$[262] \quad 1414131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[261] & 21 \overset{1}{5} 1321^1 + [\textcolor{red}{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0} \\
[260] & 21 \overset{1}{5} 1321^1 + [\textcolor{red}{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0} \\
[259] & 1 \overset{1}{5} 1321^1 + [\textcolor{red}{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0} \\
[258] & 1 \overset{1}{5} 1321^1 + [\textcolor{red}{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0} \\
[257] & 1 \overset{1}{5} 12321^1 + [\textcolor{red}{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0} \\
[256] & 1 \overset{1}{5} 12321^1 + [\textcolor{red}{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0} \\
[255] & 1 \overset{1}{5} 131^1 + [\textcolor{red}{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 13, T \in [11, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[254] & 1 \overset{1}{4} 14131^1 + [\textcolor{red}{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[270] & 1 \overset{1}{4} 132151^1 + [\textcolor{red}{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 21, T \in [12, 21] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[269] & 1 \overset{1}{4} 132151^1 + [\textcolor{red}{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 21, T \in [12, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[268] & 1 \overset{21}{6} 131^1 + [\textcolor{red}{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 21, T \in [12, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[267] & 1 \overset{1}{6} 1231^1 + [\textcolor{red}{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12, T_{\max} = 21, T \in [12, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

- [266] $131\overset{1}{5}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 13, \Delta = -89, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$
- [265] $21^1\overset{1}{5}12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 \end{pmatrix}_{b_0}$
- [271] $1\overset{1}{5}1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 \end{pmatrix}_{b_0}$
- [272] $1\overset{1}{4}141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 21, T \in [12, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [273] $1\overset{1}{5}13141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 21, T \in [12, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [274] $1\overset{1}{5}13141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -109, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 21, T \in [12, 21]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 3 & 2 & 5 \end{pmatrix}_{b_0}$
- [275] $1\overset{1}{6}12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 21, T \in [12, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [276] $2\overset{1}{3}15131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 13, \Delta = -90, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$
- [277] $221\overset{1}{6}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -107, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 21, T \in [12, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [278] $21\overset{1}{6}1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 21, T \in [12, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [279] $131\overset{1}{5}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 13, \Delta = -89, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$
- [280] $21^1\overset{1}{4}14131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 4 \end{pmatrix}_{b_0}$
- [284] $14131\overset{1}{5}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -109, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0} \\
[281] & 21^1 513141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 4 & 3 & 4 \\ 3 & 1 & 3 \\ 4 & 3 & 4 \end{pmatrix}_{b_0} \\
[282] & 141^1 3151^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 4 & 5 & 4 \\ 5 & 5 & 5 \\ 4 & 5 & 4 \end{pmatrix}_{b_0} \\
[283] & 41231^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -3 & -1 \\ 2 & 2 & -1 & 3 \end{pmatrix}_{b_0} \\
[293] & 15123141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 21, T \in [15, 21] \\
& \begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0} \\
[301] & 22161231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 21, T \in [15, 21] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[298] & 41314141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 24, T \in [15, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[297] & 22216131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 21, T \in [15, 21] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[296] & 51223141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 21, T \in [15, 21] \\
& \begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0} \\
[295] & 1^1 \begin{smallmatrix} 6 \\ 1^{11} 31 \end{smallmatrix} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 21, T \in [15, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[294] & 21513141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 21, T \in [15, 21] \\
& \begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0} \\
[299] & 31315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[292] & 1^1 \begin{smallmatrix} 6 \\ 1^{11} 31 \end{smallmatrix} 122 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 21, T \in [15, 21] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[291] & 21^1 \begin{smallmatrix} 6 \\ 1^{11} 31 \end{smallmatrix} 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 21, T \in [15, 21] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[290] & 1414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 21, T \in [15, 21]
\end{aligned}$$

- [289] $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$ $1^{11} 21612231 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [288] $1^{11} 3221$ $1^{11} 6^1 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [287] $1^{11} 321$ $1^{11} 6^1 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [286] $1^{11} 4131$ $1^{11} 5^1 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -109$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [285] $1^{11} 321$ $1^{11} 6^1 12 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 15$, $T_{\max} = 21$, $T \in [15, 21]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [300] $2315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$
- [309] $21^1 61232 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -110$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [302] $1^1 1^1 161232 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -110$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [303] $1^1 1414141 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [304] $21^1 6^1 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -107$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [305] $21^1 1^1 31$ $21^1 6^1 12 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -107$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [306] $21^1 612231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [307] $21^1 1^1 6^1 1231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[308] & 21^1 513141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 1 & 3 \\ 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[310] & 31^1 315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.45 \rightarrow 13, T_{\max} = 15, T \in [13, 15] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[311] & 141^1 4141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.45 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0} \\
[314] & 31^1 322161^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.33 \rightarrow 14, T_{\max} = 29, T \in [14, 29] \\
& \begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[312] & 16^1 131^1 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.33 \rightarrow 14, T_{\max} = 26, T \in [14, 26] \\
& \begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 4 & 2 \\ 3 & 4 & 12 & 6 \\ 2 & 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[313] & 16^1 1231^1 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.33 \rightarrow 14, T_{\max} = 29, T \in [14, 29] \\
& \begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[315] & 216^1 131^1 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.33 \rightarrow 14, T_{\max} = 26, T \in [14, 26] \\
& \begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[318] & 1321^1 5131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 6 & 7 \\ 1 & 6 & 1 & 3 \\ 2 & 7 & 3 & 3 \end{pmatrix}_{b_0} \\
[317] & 1^1 6^1 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[316] & 221^1 61231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[319] & 41^1 232151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.40 \rightarrow 14, T_{\max} = 23, T \in [14, 23] \\
& \begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 \\ 4 & 1 & 25 & 9 \\ 2 & 0 & 9 & 2 \end{pmatrix}_{b_0} \\
[320] & 41^1 23151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.40 \rightarrow 14, T_{\max} = 21, T \in [14, 21] \\
& \begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & 0 & 5 & 2 \\ 4 & 5 & 25 & 9 \\ 2 & 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[322] & 51^1 223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 13.56 \rightarrow 14, T_{\max} = 21, T \in [14, 21] \\
& \begin{pmatrix} 0 & 1 & 5 & 2 \\ 1 & 0 & 2 & 2 \\ 5 & 2 & 2 & 4 \\ 2 & 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[321] & 161^1 2231^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.56 \rightarrow 14, T_{\max} = 29, T \in [14, 29] \\
& \begin{pmatrix} 0 & 5 & 3 & 2 \\ 5 & 18 & 6 & 8 \\ 3 & 6 & 0 & 2 \\ 2 & 8 & 2 & 2 \end{pmatrix}_{b_0} \\
[326] & 13151^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 2 & 7 & -1 & 3 \end{pmatrix}_{b_0} \\
[327] & 141^1 32151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[328] & 22161^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[325] & 141^1 31^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[324] & 1^1 6^1 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[323] & 1^1 6^1 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 5 & 6 \\ 1 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[329] & 31^1 413151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 32, T \in [13, 32] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[330] & 2221^1 6131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -22, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0} \\
[331] & 1^1 5^1 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.26 \rightarrow 14, T_{\max} = 21, T \in [14, 21] \\
& \begin{pmatrix} 1 & 6 & 3 \\ 6 & 13 & 7 \\ 3 & 7 & 3 \end{pmatrix}_{b_0} \\
[332] & 215131^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.26 \rightarrow 14, T_{\max} = 21, T \in [14, 21] \\
& \begin{pmatrix} 1 & 6 & 3 \\ 6 & 13 & 7 \\ 3 & 7 & 3 \end{pmatrix}_{b_0} \\
[333] & 512231^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.26 \rightarrow 14, T_{\max} = 21, T \in [14, 21] \\
& \begin{pmatrix} 1 & 6 & 3 \\ 6 & 13 & 7 \\ 3 & 7 & 3 \end{pmatrix}_{b_0} \\
[334] & 151231^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.26 \rightarrow 14, T_{\max} = 21, T \in [14, 21]
\end{aligned}$$

- [335] $\begin{pmatrix} 1 & 6 & 3 \\ 6 & 13 & 7 \\ 3 & 7 & 3 \end{pmatrix}_{b_0}$
 11 41 4 141 $^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 24, \Delta = -113, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$
 $\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [340] 111 21612 3 2 + $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -110, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.29 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [337] 11 16 131 $^1 32 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 29, \Delta = -116, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$
 $\begin{pmatrix} 0 & 1 & 5 & 2 \\ 1 & 0 & 4 & 2 \\ 5 & 4 & 12 & 6 \\ 2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [339] 111 141 4 141 + $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -105, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.29 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [338] 111 $^2 3 21$ 1 $^6 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -110, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.29 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [336] 111 31 31 $^5 13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 15, \Delta = -97, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [342] 111 3221 1 $^6 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [341] 311 321 1 $^6 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 29, \Delta = -116, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$
 $\begin{pmatrix} 0 & 4 & 3 & 2 \\ 4 & 10 & 5 & 6 \\ 3 & 5 & 0 & 2 \\ 2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$
- [348] 11 41 $^1 232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 23, \Delta = -112, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 26 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 2 & 10 & -1 & 3 \end{pmatrix}_{b_0}$
- [347] 11 41 $^1 23151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -110, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 1 & 9 & 3 \\ 9 & 51 & 15 \\ 3 & 15 & 4 \end{pmatrix}_{b_0}$
- [346] 11 2161 1231 $^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 29, \Delta = -116, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$
 $\begin{pmatrix} 0 & 4 & 3 & 2 \\ 4 & 10 & 5 & 6 \\ 3 & 5 & 0 & 2 \\ 2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$
- [345] 11 41 $^1 314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 24, \Delta = -113, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 26 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 2 & 10 & -1 & 3 \end{pmatrix}_{b_0}$
- [344] 11 151 $^1 23141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9, T_{\max} = 21, \Delta = -110, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$

- $T_{\min} = 12.80 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 6 & 3 \\ 6 & 6 & 6 \\ 3 & 6 & 4 \end{pmatrix}_{b_0}$
- [343] $2161^1 2231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -30$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.80 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [350] $412321^1 51^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -38$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 13.16 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -3 & -1 \\ 2 & 8 & -1 & 2 \end{pmatrix}_{b_0}$
- [349] $141321^1 51^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -110$, $\det = -38$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 13.16 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 2 \end{pmatrix}_{b_0}$
- $1^{11} 2321^1$
 [353] $1^1 6^1 1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -110$, $\det = -40$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 10 & 5 & 6 \\ 2 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- $1^{11} 31513$
 [352] $41^1 31513 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -40$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$
 $\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 10 & 5 & 6 \\ 2 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- $1^{11} 2161^1 232$
 [351] $2161^1 232 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -110$, $\det = -40$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 10 & 5 & 6 \\ 2 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- $1^{11} 51^1 223141^1$
 [354] $51^1 223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 21$, $\Delta = -110$, $\det = -42$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.67 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 7 & 3 \\ 7 & 7 & 7 \\ 3 & 7 & 4 \end{pmatrix}_{b_0}$
- $1^{11} 4131^1 4141^1$
 [355] $4131^1 4141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 24$, $\Delta = -113$, $\det = -47$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 24$, $T \in [13, 24]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$
- $1^{11} 2231^1 3161^1$
 [356] $2231^1 3161^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 29$, $\Delta = -116$, $\det = -52$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 29$, $T \in [13, 29]$
 $\begin{pmatrix} 0 & 1 & 7 & 2 \\ 1 & 0 & 6 & 2 \\ 7 & 6 & 32 & 10 \\ 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$
- $1^{11} 3141^1 3151^1$
 [357] $3141^1 3151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -52$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- $1^{11} 1315131^1$
 [361] $1315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 15$, $T \in [14, 15]$
 $\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- $1^{11} 41^1 4^1 1413$
 [360] $41^1 4^1 1413 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 24$, $\Delta = -113$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 24$, $T \in [13, 24]$
 $\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 5 & 8 & 5 \\ 3 & 8 & 4 & 4 \\ 2 & 5 & 4 & 3 \end{pmatrix}_{b_0}$
- $1^{11} 2216131^1$
 [359] $2216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[358] \quad 1 \overset{31}{\underset{31}{1}} \overset{11}{6} 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[362] \quad 1 \overset{1}{\underset{31}{1}} \overset{11}{6} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 13 & 16 & 7 \\ 3 & 16 & 12 & 6 \\ 2 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [363] \quad 1^1 231413221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 11, \Delta = -38, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 3.0$$

$$[364] \quad 1^1 231413221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 11, \Delta = -38, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 3.0$$

$$[380] \quad 1^1 23215131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -63, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$$

$$[368] \quad 1^1 4141321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 11, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[369] \quad 1^1 513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 11, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 7 & 7 \\ 3 & 7 & 7 \end{pmatrix}_{b_0}$$

$$[370] \quad 1^1 23215131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -63, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[371] \quad 1^1 5131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -61, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[372] \quad 1^1 4141321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 11, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[373] \quad 1^1 31414131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -64, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[374] \quad 1^1 23151231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$$

$$[375] \quad 1^1 23151231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[376] \quad 1^1 5131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -61, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[377] \ 21 \overset{1}{5} 13221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 11, \ \Delta = -53, \ \det = 2, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 2 & 4 \\ 2 & 2 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[378] \ 21 \overset{1}{5} 13221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 11, \ \Delta = -53, \ \det = 2, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 7 & 7 \\ 3 & 7 & 7 \end{pmatrix}_{b_0}$$

$$[379] \ 1 \overset{1}{5} 13221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 11, \ \Delta = -53, \ \det = 2, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 2 & 4 \\ 2 & 2 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[392] \ 1^1 32161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 14, \ \Delta = -88, \ \det = 3, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, \ T_{\max} = 14, \ T \in [12, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[383] \ 1 \overset{1}{6} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 13, \ \Delta = -86, \ \det = 3, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[381] \ 1^1 31612231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 3, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[395] \ 23 \overset{1}{15} 1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 12, \ \Delta = -62, \ \det = 3, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[394] \ 1 \overset{1}{6} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 3, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[393] \ 1 \overset{1}{6} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 3, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[382] \ 131 \overset{1}{5} 1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 12, \ \Delta = -61, \ \det = 3, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$$

$$[391] \ 1414141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 12, \ \Delta = -76, \ \det = 3, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[390] \ 21 \overset{1}{6} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 13, \ \Delta = -86, \ \det = 3, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

[389] $1\overset{1}{6}12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$

[388] $1^141315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 14, T \in [12, 14]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[387] $1^131612231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[386] $1^141315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 14, T \in [12, 14]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[385] $21\overset{1}{6}12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[384] $131\overset{1}{5}1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 12, \Delta = -61, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[397] $1^151232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$

[411] $1\overset{1}{5}132151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[396] $1321\overset{1}{6}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[398] $1^141322161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -107, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

[399] $21\overset{1}{6}13141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$

[400] $3131\overset{1}{6}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 15, \Delta = -94, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 15, T \in [13, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[401] $1\overset{21}{7}1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

- [402] $1^{\overline{21}} 6 12321^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10$, $T_{\max} = 13$, $\Delta = -81$, $\det = 4$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0}$
- [403] $1^{\overline{1}} 7 12231^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10$, $T_{\max} = 22$, $\Delta = -106$, $\det = 4$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- [404] $131^{\overline{1}} 6 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10$, $T_{\max} = 13$, $\Delta = -86$, $\det = 4$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- [405] $221^{\overline{1}} 7 1231^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10$, $T_{\max} = 22$, $\Delta = -106$, $\det = 4$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [406] $21^{\overline{1}} 7 12231^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10$, $T_{\max} = 22$, $\Delta = -106$, $\det = 4$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [407] $1^{\overline{1}} 41413151^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10$, $T_{\max} = 22$, $\Delta = -108$, $\det = 4$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 3 & 2 & 5 \end{pmatrix}_{b_0}$
- [408] $1^{\overline{1}} 41413151^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10$, $T_{\max} = 22$, $\Delta = -108$, $\det = 4$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [409] $1^{\overline{1}} 6 123141^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10$, $T_{\max} = 22$, $\Delta = -107$, $\det = 4$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$
- [410] $1^{\overline{1}} 5 131^{\overline{1}} 5 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10$, $T_{\max} = 22$, $\Delta = -104$, $\det = 4$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0}$
- [414] $23216131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10$, $T_{\max} = 14$, $\Delta = -89$, $\det = 4$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 13$, $T_{\max} = 14$, $T \in [13, 14]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$
- [413] $131^{\overline{1}} 6 1231^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 4$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$
- [412] $1^{\overline{1}} 5 132151^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 4$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0}$
- [415] $512231^{\overline{1}} 5 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 5$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 4 & 4 \\ 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[416] $151231\bar{5}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -3 & -1 \\ 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[420] $21^1\bar{5}132151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 7 & 5 \\ 2 & 0 & 5 & 3 \end{pmatrix}_{b_0}$$

[419] $1\bar{5}13151^1\bar{2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 4 & 5 & 4 \\ 5 & 5 & 5 \\ 4 & 5 & 4 \end{pmatrix}_{b_0}$$

[418] $21\bar{5}131\bar{5}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[417] $1\bar{5}131\bar{5}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[421] $1^1\bar{1}^{11}321\bar{1}22 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[423] $1^1\bar{1}^{11}3221\bar{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[424] $512314141^{\bar{1}\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -109, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$

$$\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[425] $1\bar{6}13141^{\bar{1}\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

[426] $151314141^{\bar{1}\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$

$$\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[427] $221613141^{\bar{1}\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[428] $221712231^{\bar{1}\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[429] $222171231^{\bar{1}\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$$

- [430] $1^{11} 7^{3221} 12 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [431] $216123141^{11} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -107, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [432] $1^{21} 7^{11} 1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [433] $23216131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 14, \Delta = -89, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$
- [434] $161223141^{11} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -107, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [435] $41414141^{11} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -104, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$
 $\begin{pmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [436] $16123141^{11} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -107, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$
- [422] $21^{11} 7^{321} 12 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [446] $21^1 7^{11} 1232 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -108, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [443] $21^1 5132151^{11} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 8 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 3 \end{pmatrix}_{b_0}$
- [444] $1^{21} 6^{11} 13141^{11} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 8 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 3 \end{pmatrix}_{b_0}$
- [447] $31^1 513151^{11} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [448] $1^{21} 7^{11} 1232 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -108, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[442] & 1 \overset{1}{5} 13151^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[441] & 1 \overset{1}{7} 1232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 4 & 4 \\ 2 & 2 & 4 & 3 \end{pmatrix}_{b_0} \\
[440] & 21 \overset{1}{7} \overset{1}{321} 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[439] & 221 \overset{1}{7} 1232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 10 & 4 & 6 \\ 2 & 4 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[438] & 1 \overset{21}{7} 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[437] & 21 \overset{1}{6} 123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[445] & 1 \overset{21}{7} \overset{1}{321} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[454] & 141 \overset{1}{3} 15131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.45 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0} \\
[450] & 31 \overset{1}{5} 13151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.45 \rightarrow 14, T_{\max} = 24, T \in [14, 24] \\
& \begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0} \\
[451] & 31 \overset{1}{3} 161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.45 \rightarrow 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[452] & 31 \overset{1}{3} 216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.45 \rightarrow 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[453] & 41 \overset{1}{4} 14141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.45 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 1 & 3 \\ 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [449] $1 \overset{31}{6} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 15, \Delta = -94, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [458] $131 \overset{1}{6} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 15, \Delta = -94, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [457] $21 \overset{1}{7} 1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [456] $1 \overset{1}{7} 1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 4 & 2 \\ 3 & 4 & 12 & 6 \\ 2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [455] $1 \overset{1}{7} 12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [459] $41 \overset{1}{2} 315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 14, \Delta = -89, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 42 & 1 & 14 \\ 1 & 1 & -3 & -1 \\ 2 & 14 & -1 & 3 \end{pmatrix}_{b_0}$
- [462] $1 \overset{221}{7} 1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [463] $221 \overset{1}{6} 13141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [460] $1321 \overset{1}{6} 1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [461] $221 \overset{1}{7} 12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [464] $41 \overset{1}{2} 321 \overset{1}{6} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 22, \Delta = -109, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$
 $\begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & 0 & 5 & 2 \\ 4 & 5 & 25 & 9 \\ 2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$
- [465] $31 \overset{1}{6} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 15, \Delta = -94, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 15, T \in [13, 15]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 \\ 2 & 3 & 3 & 4 \end{pmatrix}_{b_0}$
- [466] $1 \overset{31}{7} 13221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.56 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

- [467] $\begin{pmatrix} 0 & 5 & 3 & 2 \\ 5 & 18 & 6 & 8 \\ 3 & 6 & 0 & 2 \\ 2 & 8 & 2 & 2 \end{pmatrix}_{b_0}$ $2171^1 2231^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.56 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$
 $\begin{pmatrix} 0 & 5 & 3 & 2 \\ 5 & 18 & 6 & 8 \\ 3 & 6 & 0 & 2 \\ 2 & 8 & 2 & 2 \end{pmatrix}_{b_0}$
- [468] $1^1 15 13221^1 5 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.56 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$
 $\begin{pmatrix} 0 & 1 & 5 & 2 \\ 1 & 0 & 2 & 2 \\ 5 & 2 & 2 & 4 \\ 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [469] $1^1 41 414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -104, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.26 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 1 & 5 & 3 \\ 5 & 6 & 6 \\ 3 & 6 & 4 \end{pmatrix}_{b_0}$
- [470] $313161^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 15, \Delta = -95, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 5 & 6 \\ 1 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [471] $132161^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 5 & 6 \\ 1 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [472] $2^1 3^1 4^1 141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 \\ 2 & 5 & 2 & 4 \\ 2 & 5 & 4 & 5 \end{pmatrix}_{b_0}$
- [473] $2221^1 71231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 22, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$
- [474] $31321^1 6131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 15, \Delta = -95, \det = 22, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 4 & 6 \\ 1 & 4 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [475] $2^1 3^1 13161^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 14, \Delta = -89, \det = 22, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 4 & 6 \\ 1 & 4 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [476] $2161231^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -107, \det = 23, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.26 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 6 & 7 \\ 1 & 6 & 0 & 2 \\ 2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$
- [477] $21^1 16131^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 23, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.26 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$
- [478] $1^1 161231^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -107, \det = 23, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.26 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$
- [479] $2216131^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 23, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.26 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$

- [480] $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 6 & 7 \\ 1 & 6 & 0 & 2 \\ 2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$ $1612231^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -107, \det = 23, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.26 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$
- [481] $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 6 & 7 \\ 1 & 6 & 0 & 2 \\ 2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$ $231^{1316131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 15, \Delta = -95, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$
- [482] $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 13 & 4 & 7 \\ 1 & 4 & 0 & 2 \\ 2 & 7 & 2 & 3 \end{pmatrix}_{b_0}$ $321^3 6131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 15, \Delta = -95, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.29 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$
- [487] $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$ $31^3 161231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 15, \Delta = -95, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.29 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$
- [486] $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$ $221712 \ 3 \ 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -108, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.29 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$
- [485] $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 5 & 8 & 5 \\ 2 & 8 & 4 & 4 \\ 2 & 5 & 4 & 2 \end{pmatrix}_{b_0}$ $1^7 \ 2 \ 3 \ 21 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -108, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.29 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$
- [484] $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$ $31^1 6131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 15, \Delta = -94, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.29 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$
- [483] $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$ $1^2 \ 3 \ 21 \ 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -108, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.29 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$
- [495] $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$ $1^{13221} 1^7 \ 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
- [494] $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 6 & 8 \\ 1 & 6 & 0 & 2 \\ 2 & 8 & 2 & 3 \end{pmatrix}_{b_0}$ $1^{13221} 1^7 \ 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
- [493] $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 26 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 2 & 10 & -1 & 3 \end{pmatrix}_{b_0}$ $1^{141321} 1^6 \ 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -107, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
- [492] $22171^1 2231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[491] \quad & 2161^1 23141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.80 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 6 & 8 \\ 1 & 6 & 0 & 2 \\ 2 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[490] \quad & 13161^1 231^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.33 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 4 & 3 & 2 \\ 4 & 10 & 5 & 6 \\ 3 & 5 & 0 & 2 \\ 2 & 6 & 2 & 2 \end{pmatrix}_{b_0} \\
[489] \quad & 151^1 231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.80 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 1 & 9 & 3 \\ 9 & 51 & 15 \\ 3 & 15 & 4 \end{pmatrix}_{b_0} \\
[488] \quad & 151^1 232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.80 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 6 & 8 \\ 1 & 6 & 0 & 2 \\ 2 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[497] \quad & 151321^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 38, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.16 \rightarrow 15, T_{\max} = 22, T \in [15, 22] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 26 & 8 & 10 \\ 1 & 8 & 0 & 2 \\ 2 & 10 & 2 & 2 \end{pmatrix}_{b_0} \\
[496] \quad & 1512321^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 38, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.16 \rightarrow 15, T_{\max} = 22, T \in [15, 22] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -3 & -1 \\ 2 & 8 & -1 & 2 \end{pmatrix}_{b_0} \\
[498] \quad & 2151321^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 38, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.16 \rightarrow 15, T_{\max} = 22, T \in [15, 22] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -3 & -1 \\ 2 & 8 & -1 & 2 \end{pmatrix}_{b_0} \\
[499] \quad & 13161^1 232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 10 & 5 & 6 \\ 2 & 5 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[500] \quad & 1513221^1 5 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 42, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.67 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 1 & 7 & 3 \\ 7 & 7 & 7 \\ 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[501] \quad & 51^1 2314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -109, \det = 42, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.67 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 1 & 7 & 3 \\ 7 & 7 & 7 \\ 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[502] \quad & 161^1 223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 42, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.67 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 28 & 7 & 10 \\ 1 & 7 & 0 & 2 \\ 2 & 10 & 2 & 3 \end{pmatrix}_{b_0} \\
[503] \quad & 321^1 6131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 43, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.49 \rightarrow 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 5 & 2 & 2 \\ 5 & 13 & 6 & 7 \\ 2 & 6 & 1 & 3 \\ 2 & 7 & 3 & 3 \end{pmatrix}_{b_0} \\
[504] \quad & 15131^1 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 47, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.79 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 1 & 8 & 3 \\ 8 & 17 & 9 \\ 3 & 9 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[505] \quad 321^{11}513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 47, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[506] \quad 51231^{11}4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -109, \det = 47, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 8 & 3 \\ 8 & 17 & 9 \\ 3 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[507] \quad 31^151321^{11}5 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 5 & 2 & 2 \\ 5 & 3 & 3 & 5 \\ 2 & 3 & 0 & 2 \\ 2 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[508] \quad 14141^{11}3151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[509] \quad 13216131^{11}3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[510] \quad 41^1414141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 2 & 5 & 4 \\ 3 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[511] \quad 1^{31}6^{11}131 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[512] \quad 31^{11}3161313 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 13 & 16 & 7 \\ 3 & 16 & 12 & 6 \\ 2 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[513] \quad 31^{11}321^{11}13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 6 & 2 & 2 \\ 6 & 19 & 8 & 9 \\ 2 & 8 & 1 & 3 \\ 2 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[514] \quad 3151^{11}23151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 93, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 68 & 1 & 16 \\ 1 & 1 & -3 & -1 \\ 2 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[515] \quad 21^{11}414132^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 4 & 2 \\ 4 & 16 & 19 & 10 \\ 4 & 19 & 15 & 9 \\ 2 & 10 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[516] \quad 31^{31}6^{11}13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 252, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 6 & 6 & 2 \\ 6 & 22 & 25 & 10 \\ 6 & 25 & 21 & 9 \\ 2 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [517] \quad 1^{11}3141413221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -36, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 3.0$$

$$[518] \quad 1^{11}3141413221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -36, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 3.0$$

$$[519] \quad 1^{11}2231512321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -35, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 3.0$$

[520] $1^1 2231512321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 12, \Delta = -35, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 3.0$

[521] $1^1 2314141321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 12, \Delta = -37, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 3.0$

[522] $1^1 5^{12321} 1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 12, \Delta = -33, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 3.0$

[523] $1^1 5^{12321} 1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 12, \Delta = -33, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 3.0$

[535] $1^1 413151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -60, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[536] $1^1 141414131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 12, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[537] $1^1 6^{12321} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -60, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[538] $1^1 6^{12321} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -60, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[539] $1^1 321612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -61, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[540] $1^1 321612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -61, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[541] $1^1 413151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -60, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[542] $1^1 141414131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 12, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[543] $131^1 5^{13221} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$

[544] $131^1 5^{13221} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$

[559] $1^1 7^{1321} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[546] $221^1 7^{12321} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

- [547] $1^1 413161231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [548] $1^1 51315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -83, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0}$
- [549] $1^1 321712231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [550] $1^1 321712231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [551] $1^1 613141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -85, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$
- [552] $1^1 613141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -85, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$
- [553] $1^{21} 712321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$
- [554] $1^1 413216131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [555] $1^1 51315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -83, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$
- [556] $1^1 413216131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [557] $1^{131} 131^1 6131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -65, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$
- [558] $1^1 131612321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 13, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [545] $1^1 315132151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$
- [560] $1^1 315132151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 13, T_{\max} = 14, T \in [13, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[561] & 1^1 1^1 413161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 14, T \in [13, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[569] & 1^1 1^1 71223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 23, T \in [14, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0} \\
[582] & 21^1 1^1 51315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 \end{pmatrix}_{b_0} \\
[564] & 21^1 613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 23, T \in [14, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[565] & 221^1 812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 23, T \in [14, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[566] & 1^1 1^1 61232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 23, T \in [14, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[567] & 2221^1 81231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 23, T \in [14, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[568] & 1^1 1^1 132171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -86, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[562] & 2^1 32171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 15, \Delta = -88, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[570] & 1^1 1^1 414132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -106, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 23, T \in [14, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[571] & 1^1 1^1 61314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 23, T \in [14, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0} \\
[572] & 1^1 1^1 6123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 23, T \in [14, 23]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[573] $21 \overset{1}{1} \overset{1}{7} \overset{2321}{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 3 & 3 \\ 3 & 3 & 3 \end{pmatrix}_{b_0}$

[574] $21 \overset{21}{1} \overset{1}{8} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 23, T \in [14, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[575] $1 \overset{1}{5} 1322161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 23, T \in [14, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[576] $1 \overset{221}{1} \overset{1}{8} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 23, T \in [14, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[577] $131 \overset{1}{6} 13141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -85, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$

[578] $1 \overset{21}{1} \overset{1}{7} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 \end{pmatrix}_{b_0}$

[579] $1 \overset{1}{6} 131 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 23, T \in [14, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[580] $1 \overset{21}{1} \overset{1}{8} 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 23, T \in [14, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[581] $14131 \overset{1}{6} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -85, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 1 & 3 \\ 2 & 3 & 3 & 4 \end{pmatrix}_{b_0}$

[563] $21 \overset{1}{7} 123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14, T_{\max} = 23, T \in [14, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$

[583] $1 \overset{1}{6} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -101, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $\overset{1}{1} \overset{1}{5} 131$
 $T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$

[584] $2161231 \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 15, T_{\max} = 23, T \in [15, 23] \\
& \left(\begin{smallmatrix} 5 & 5 \\ 5 & 5 \end{smallmatrix} \right)_{b_0} \\
[585] & 1 \overset{1}{\overset{1}{\overset{1}{5}}} 6^{131} 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -101, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 23, T \in [15, 23] \\
& \left(\begin{smallmatrix} 5 & 5 \\ 5 & 5 \end{smallmatrix} \right)_{b_0} \\
[586] & 2216131 \overset{1}{\overset{1}{\overset{1}{5}}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -101, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 23, T \in [15, 23] \\
& \left(\begin{smallmatrix} 5 & 5 \\ 5 & 5 \end{smallmatrix} \right)_{b_0} \\
[587] & 21 \overset{1}{\overset{1}{\overset{1}{5}}} 1322161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 23, T \in [15, 23] \\
& \left(\begin{smallmatrix} 4 & 3 & 4 \\ 3 & 1 & 3 \\ 4 & 3 & 4 \end{smallmatrix} \right)_{b_0} \\
[588] & 1 \overset{1}{\overset{1}{\overset{1}{5}}} 6^{1321} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 23, T \in [15, 23] \\
& \left(\begin{smallmatrix} 5 & 5 \\ 5 & 5 \end{smallmatrix} \right)_{b_0} \\
[589] & 21 \overset{1}{\overset{1}{\overset{1}{6}}} 13151^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -101, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 23, T \in [15, 23] \\
& \left(\begin{smallmatrix} 4 & 3 & 4 \\ 3 & 1 & 3 \\ 4 & 3 & 4 \end{smallmatrix} \right)_{b_0} \\
[590] & 1 \overset{1}{\overset{1}{\overset{1}{6}}} 13151^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -101, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 23, T \in [15, 23] \\
& \left(\begin{smallmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 7 & 5 \\ 2 & 2 & 5 & 3 \end{smallmatrix} \right)_{b_0} \\
[591] & 1 \overset{1}{\overset{1}{\overset{1}{6}}} 123151^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 23, T \in [15, 23] \\
& \left(\begin{smallmatrix} 4 & 3 & 4 \\ 3 & 1 & 3 \\ 4 & 3 & 4 \end{smallmatrix} \right)_{b_0} \\
[592] & 1612231 \overset{1}{\overset{1}{\overset{1}{5}}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 23, T \in [15, 23] \\
& \left(\begin{smallmatrix} 5 & 5 \\ 5 & 5 \end{smallmatrix} \right)_{b_0} \\
[611] & 6122314141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -106, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 23, T \in [17, 23] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{smallmatrix} \right)_{b_0} \\
[610] & 1 \overset{21}{\overset{21}{\overset{21}{7}}} 123141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 23, T \in [17, 23] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{smallmatrix} \right)_{b_0} \\
[593] & 1 \overset{1}{\overset{1}{\overset{1}{2}}} 32171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 15, \Delta = -88, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13.57 \rightarrow 14, T_{\max} = 15, T \in [14, 15] \\
& \left(\begin{smallmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{smallmatrix} \right)_{b_0} \\
[594] & 5131414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 23, T \in [17, 23] \\
& \left(\begin{smallmatrix} 1 & 4 & 3 \\ 4 & 9 & 5 \\ 3 & 5 & 2 \end{smallmatrix} \right)_{b_0} \\
[595] & 2161314141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 23, T \in [17, 23] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{smallmatrix} \right)_{b_0}
\end{aligned}$$

- [609] $2222181231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11$, $T_{\max} = 23$, $\Delta = -105$, $\det = -7$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 17$, $T_{\max} = 23$, $T \in [17, 23]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [630] $1 \overset{21^1}{6} 131 \overset{1^1}{5} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11$, $T_{\max} = 23$, $\Delta = -101$, $\det = -8$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 14.50 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [628] $21^1 61314141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11$, $T_{\max} = 23$, $\Delta = -105$, $\det = -8$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 14.50 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [629] $21^1 \overset{1^1}{8} 12 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11$, $T_{\max} = 23$, $\Delta = -107$, $\det = -8$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 14.50 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [613] $21^1 61231 \overset{1^1}{5} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11$, $T_{\max} = 23$, $\Delta = -102$, $\det = -8$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 14.50 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [614] $1 \overset{21^1}{7} 123141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11$, $T_{\max} = 23$, $\Delta = -105$, $\det = -8$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 14.50 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 8 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 3 \end{pmatrix}_{b_0}$
- [615] $21^1 61232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11$, $T_{\max} = 23$, $\Delta = -104$, $\det = -8$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 14.50 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [616] $31 \overset{1^1}{5} 132161^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11$, $T_{\max} = 23$, $\Delta = -110$, $\det = -8$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 14.50 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [617] $1 \overset{221^1}{8} 1232 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11$, $T_{\max} = 23$, $\Delta = -107$, $\det = -8$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 14.50 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$
 $\begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 10 & 4 & 6 \\ 2 & 4 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [618] $221 \overset{1^1}{8} 1232 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11$, $T_{\max} = 23$, $\Delta = -107$, $\det = -8$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 14.50 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [619] $21^1 71223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11$, $T_{\max} = 23$, $\Delta = -105$, $\det = -8$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 14.50 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [620] $21^1 6131 \overset{1^1}{5} 13 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11$, $T_{\max} = 23$, $\Delta = -109$, $\det = -8$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 14.50 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [621] $21^1 \overset{1^1}{8} 122 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11$, $T_{\max} = 23$, $\Delta = -105$, $\det = -8$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 14.50 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[622] & 1 \overset{21}{8} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[623] & 1 \overset{21}{8} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[624] & 21^1 \overset{1}{8} 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[625] & 1 \overset{21}{8} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 0 & 3 & 2 \\ 2 & 3 & 4 & 4 \\ 2 & 2 & 4 & 3 \end{pmatrix}_{b_0} \\
[626] & 1 \overset{21}{8} 1232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 4 & 4 \\ 2 & 2 & 4 & 3 \end{pmatrix}_{b_0} \\
[627] & 1 \overset{1}{6} 131513 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[612] & 2221^1 81232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 18 & 5 & 8 \\ 2 & 5 & 0 & 2 \\ 2 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[632] & 31^1 32171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 16, \Delta = -94, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.45 \rightarrow 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[633] & 31 \overset{1}{6} 13151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.45 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0} \\
[634] & 31^1 31613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.45 \rightarrow 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[635] & 31 \overset{1}{6} 123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.45 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0} \\
[636] & 321 \overset{1}{6} 13151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.45 \rightarrow 15, T_{\max} = 23, T \in [15, 23]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0} \\
[631] \quad & 31^1 41316131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.45 \rightarrow 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 3 \end{pmatrix}_{b_0} \\
[637] \quad & 1^1 81231^1 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 26, \Delta = -113, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.33 \rightarrow 16, T_{\max} = 26, T \in [16, 26] \\
& \begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 4 & 2 \\ 3 & 4 & 12 & 6 \\ 2 & 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[638] \quad & 221^1 81231^1 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 26, \Delta = -113, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.33 \rightarrow 16, T_{\max} = 26, T \in [16, 26] \\
& \begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[639] \quad & 21^1 812231^1 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 26, \Delta = -113, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.33 \rightarrow 16, T_{\max} = 26, T \in [16, 26] \\
& \begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[640] \quad & 1^1 812231^1 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 26, \Delta = -113, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.33 \rightarrow 16, T_{\max} = 26, T \in [16, 26] \\
& \begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 4 & 2 \\ 3 & 4 & 12 & 6 \\ 2 & 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[641] \quad & 221^1 1^1 812231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.57 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[642] \quad & 221^1 1^1 6131^1 51 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -101, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.57 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[643] \quad & 41^1 1^1 23216131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 15, \Delta = -88, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 26 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0} \\
[644] \quad & 231^1 513161^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.57 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 3 & 2 \\ 3 & 3 & 4 & 4 \\ 2 & 2 & 4 & 3 \end{pmatrix}_{b_0} \\
[645] \quad & 221^1 1^1 81 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.57 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[646] \quad & 221^1 21^1 81231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.57 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[647] \quad & 221^1 1^1 7123141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.57 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[648] \ 41^1 2321^1 71 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 23, \ \Delta = -107, \ \det = -15, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.40 \rightarrow 16, \ T_{\max} = 23, \ T \in [16, 23]$$

$$\begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & 0 & 5 & 2 \\ 4 & 5 & 25 & 9 \\ 2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[649] \ 41^1 2321^1 712 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 23, \ \Delta = -107, \ \det = -15, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.40 \rightarrow 16, \ T_{\max} = 23, \ T \in [16, 23]$$

$$\begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & 0 & 5 & 2 \\ 4 & 5 & 25 & 9 \\ 2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[650] \ 1 \begin{smallmatrix} 8 \\ 31^1 3221^1 \end{smallmatrix} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 26, \ \Delta = -113, \ \det = -18, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.56 \rightarrow 16, \ T_{\max} = 26, \ T \in [16, 26]$$

$$\begin{pmatrix} 0 & 5 & 3 & 2 \\ 5 & 28 & 19 & 10 \\ 3 & 19 & 12 & 6 \\ 2 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[651] \ 1 \begin{smallmatrix} 8 \\ 31^1 3221^1 \end{smallmatrix} 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 26, \ \Delta = -113, \ \det = -18, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.56 \rightarrow 16, \ T_{\max} = 26, \ T \in [16, 26]$$

$$\begin{pmatrix} 0 & 5 & 3 & 2 \\ 5 & 18 & 6 & 8 \\ 3 & 6 & 0 & 2 \\ 2 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[652] \ 22181^1 2231^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 26, \ \Delta = -113, \ \det = -18, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.56 \rightarrow 16, \ T_{\max} = 26, \ T \in [16, 26]$$

$$\begin{pmatrix} 0 & 5 & 3 & 2 \\ 5 & 18 & 6 & 8 \\ 3 & 6 & 0 & 2 \\ 2 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[655] \ 1 \begin{smallmatrix} 8 \\ 2221^1 \end{smallmatrix} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 23, \ \Delta = -105, \ \det = -22, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 14.55 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[653] \ 2221^1 812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 23, \ \Delta = -105, \ \det = -22, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 14.55 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[654] \ 412321^1 6131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 15, \ \Delta = -88, \ \det = -22, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 14.55 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[658] \ 21712231^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 23, \ \Delta = -105, \ \det = -23, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.26 \rightarrow 16, \ T_{\max} = 23, \ T \in [16, 23]$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 6 & 7 \\ 1 & 6 & 0 & 2 \\ 2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[656] \ 171231^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 23, \ \Delta = -105, \ \det = -23, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.26 \rightarrow 16, \ T_{\max} = 23, \ T \in [16, 23]$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[657] \ 1712231^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 23, \ \Delta = -105, \ \det = -23, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.26 \rightarrow 16, \ T_{\max} = 23, \ T \in [16, 23]$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[660] \ 22171231^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 23, \ \Delta = -105, \ \det = -23, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.26 \rightarrow 16, \ T_{\max} = 23, \ T \in [16, 23]$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 6 & 7 \\ 1 & 6 & 0 & 2 \\ 2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[659] \ 31316131^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 16, \ \Delta = -93, \ \det = -23, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.26 \rightarrow 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 6 & 7 \\ 1 & 6 & 0 & 2 \\ 2 & 7 & 2 & 2 \end{pmatrix}_{b_0} \\
[661] & 41 \overset{31}{1} 316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 10 & 5 & 6 \\ 2 & 5 & 1 & 3 \\ 2 & 6 & 3 & 3 \end{pmatrix}_{b_0} \\
[662] & 3141 \overset{1}{1} 316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.48 \rightarrow 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 1 & 11 \\ 1 & 1 & -2 & 0 \\ 2 & 11 & 0 & 3 \end{pmatrix}_{b_0} \\
[668] & 1 \overset{21}{8} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& \overset{11}{2} 3 21 \\
& T_{\min} = 15.29 \rightarrow 16, T_{\max} = 23, T \in [16, 23] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 5 & 8 & 5 \\ 2 & 8 & 4 & 4 \\ 2 & 5 & 4 & 2 \end{pmatrix}_{b_0} \\
[667] & 2221812 \overset{11}{3} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.29 \rightarrow 16, T_{\max} = 23, T \in [16, 23] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[666] & 31613141 \overset{31}{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.29 \rightarrow 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[665] & 41316131 \overset{31}{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.29 \rightarrow 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[664] & 1 \overset{21}{8} 122 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& \overset{11}{2} 3 21 \\
& T_{\min} = 15.29 \rightarrow 16, T_{\max} = 23, T \in [16, 23] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[663] & 21 \overset{21}{8} 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& \overset{11}{2} 3 21 \\
& T_{\min} = 15.29 \rightarrow 16, T_{\max} = 23, T \in [16, 23] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 \\ 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[676] & 2161 \overset{1}{1} 23151 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -30, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.80 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[675] & 1 \overset{1}{5} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -104, \det = -30, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.80 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[674] & 21 \overset{1}{8} 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -30, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.80 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[673] & 2161 \overset{1}{1} 232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -104, \det = -30, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.80 \rightarrow 15, T_{\max} = 23, T \in [15, 23]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[672] & 1 \overset{1}{5} \overset{1321}{6} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -30, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.80 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[671] & 1 \overset{21}{13221} 8 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -30, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.80 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 6 & 8 \\ 1 & 6 & 0 & 2 \\ 2 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[670] & 41 \overset{1}{31513151} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -30, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.80 \rightarrow 15, T_{\max} = 24, T \in [15, 24] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 26 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 2 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[669] & 1 \overset{13221}{8} 122 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -30, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.80 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[677] & 222181 \overset{1}{2231} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -30, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.80 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[679] & 22221 \overset{1}{81231} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0} \\
[678] & 21 \overset{15131}{1} 6 13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[680] & 1 \overset{151}{1} \overset{2321}{6} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -104, \det = -38, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.16 \rightarrow 16, T_{\max} = 23, T \in [16, 23] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 2 \end{pmatrix}_{b_0} \\
[681] & 21612321 \overset{1}{51} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -104, \det = -38, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.16 \rightarrow 16, T_{\max} = 23, T \in [16, 23] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 2 \end{pmatrix}_{b_0} \\
[682] & 1 \overset{1413221}{7} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -42, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.67 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 43 & 1 & 13 \\ 1 & 1 & -3 & -1 \\ 2 & 13 & -1 & 3 \end{pmatrix}_{b_0} \\
[683] & 161 \overset{1}{2314141} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -106, \det = -42, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.67 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 28 & 7 & 10 \\ 1 & 7 & 0 & 2 \\ 2 & 10 & 2 & 3 \end{pmatrix}_{b_0} \\
[684] & 161 \overset{1}{2231} 5 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -42, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.67 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0} \\
[685] & 2171 \overset{1}{223141} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -42, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 14.67 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 28 & 7 & 10 \\ 1 & 7 & 0 & 2 \\ 2 & 10 & 2 & 3 \end{pmatrix}_{b_0} \\
[686] & \begin{matrix} 321^1 \\ 1 & 6 & 13151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.79 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 13 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 3 \end{pmatrix}_{b_0} \\
[687] & \begin{matrix} 612231^1 4141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 11, T_{\max} = 23, \Delta = -106, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.79 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 2 & 12 & -1 & 3 \end{pmatrix}_{b_0} \\
[688] & \begin{matrix} 321^1 6123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.79 \rightarrow 15, T_{\max} = 24, T \in [15, 24] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 13 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 3 \end{pmatrix}_{b_0} \\
[689] & \begin{matrix} 1 \\ 16131^1 4141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.79 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 2 & 12 & -1 & 3 \end{pmatrix}_{b_0} \\
[690] & \begin{matrix} 216131^1 4141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.79 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 8 & 11 \\ 1 & 8 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0} \\
[691] & \begin{matrix} 161231^1 4141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 11, T_{\max} = 23, \Delta = -106, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.79 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 8 & 11 \\ 1 & 8 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0} \\
[692] & \begin{matrix} 1 \\ 161^1 231^1 513 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -48, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.75 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 5 & 2 & 2 \\ 5 & 18 & 6 & 8 \\ 2 & 6 & 0 & 2 \\ 2 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[693] & \begin{matrix} 1 \\ 61^1 2231^1 513 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -56, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.64 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 6 & 2 & 2 \\ 6 & 28 & 7 & 10 \\ 2 & 7 & 0 & 2 \\ 2 & 10 & 2 & 3 \end{pmatrix}_{b_0} \\
[694] & \begin{matrix} 1 \\ 61^1 22314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 11, T_{\max} = 23, \Delta = -106, \det = -56, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.57 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 64 & 1 & 16 \\ 1 & 1 & -3 & -1 \\ 2 & 16 & -1 & 3 \end{pmatrix}_{b_0} \\
[695] & \begin{matrix} 141316131^1 3 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -63, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.73 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 2 \\ 2 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[696] & \begin{matrix} 1 \\ 61^1 231^1 5132 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.67 \rightarrow 15, T_{\max} = 24, T \in [15, 24] \\
& \begin{pmatrix} 0 & 6 & 3 & 2 \\ 6 & 28 & 13 & 10 \\ 3 & 13 & 4 & 4 \\ 2 & 10 & 4 & 3 \end{pmatrix}_{b_0} \\
[697] & \begin{matrix} 1 \\ 5131^1 414141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -79, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.57 \rightarrow 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 1 & 10 & 3 \\ 10 & 21 & 11 \\ 3 & 11 & 4 \end{pmatrix}_{b_0} \\
[698] & \begin{matrix} 1 \\ 3221^1 613151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \end{matrix} \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -86, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.65 \rightarrow 15, T_{\max} = 23, T \in [15, 23]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 26 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[699] $1^{31} 316131^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -92, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.09 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{pmatrix} 0 & 6 & 2 & 2 \\ 6 & 22 & 9 & 10 \\ 2 & 9 & 1 & 3 \\ 2 & 10 & 3 & 3 \end{pmatrix}_{b_0}$$

[701] $1^{151321^1} 6^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -93, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.74 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 19 & 1 & 9 \\ 1 & 1 & -2 & 0 \\ 2 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

[700] $32161^1 23151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -93, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.74 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 19 & 1 & 9 \\ 1 & 1 & -2 & 0 \\ 2 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

[702] $3161^1 223151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -146, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.63 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$T = 12$ [703] $1^1 31414141321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 13, \Delta = -35, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 3.0$

[704] $1^1 31414141321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 13, \Delta = -35, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 3.0$

[705] $1^1 414141413221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 12, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 3.0$

[706] $1^1 414141413221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 12, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 3.0$

[720] $1^1 2321712231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 14, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

[716] $1^1 2321712231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 14, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

[715] $1^1 1513151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 13, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 2 & 4 \\ 2 & 2 & 4 & 6 \end{pmatrix}_{b_0}$$

[714] $1^1 3151315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 14, \Delta = -62, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$$

[713] $1^1 2315132151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 13, \Delta = -56, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$$

[717] $1^1 2315132151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 13, \Delta = -56, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$$

[718] $1^{12321} 7^{1231^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 14, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

[719] $1^{12321} 7^{1231^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 14, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 13, T_{\max} = 14, T \in [13, 14] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[721] & \overset{1}{1} 5 13151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 7 & 7 \\ 3 & 7 & 7 \end{pmatrix}_{b_0} \\
[722] & \overset{1}{1} 4 14141^1 41 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 7 & 7 \\ 3 & 7 & 7 \end{pmatrix}_{b_0} \\
[723] & \overset{1}{1} 1^1 4131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 14, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 14, T \in [13, 14] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[724] & \overset{1}{1} 1^1 4131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 14, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 13, T_{\max} = 14, T \in [13, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[726] & \overset{1}{1} 1^1 4132171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -85, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[751] & \overset{221}{1} 8 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 14, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0} \\
[727] & \overset{1}{1} 6 1315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0} \\
[728] & \overset{1}{1} 1^1 4141316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -84, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[729] & \overset{1}{1} 8 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -85, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[730] & \overset{1}{1} 1^1 4131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -84, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[731] & \overset{1}{1} 1^1 4141316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -84, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[732] & \overset{1}{1} 1^1 3161232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -83, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[733] & \overset{1}{1} 1^1 3161232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -83, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15]
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[747] & 1^1 4132171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -85, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[748] & 21^1 8^{1321} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -85, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[749] & 23^1 21712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 14, \Delta = -61, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0} \\
[750] & 21^1 61315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[725] & 1^1 612315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[762] & 31321^1 81231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 17, \Delta = -93, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 17, T \in [15, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[779] & 1^1 4141413161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 24, T \in [15, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[760] & 1^1 71232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 24, T \in [15, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[759] & 1^1 5131513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -106, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 24, T \in [15, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0} \\
[758] & 1^1 1321^1 812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -85, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[757] & 21^1 513161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[756] & 21^1 1^1 61315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \end{pmatrix}_{b_0} \\
[755] & 1^1 4141322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[772] \quad 1 \overset{1}{7} 12231 \overset{1}{5} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[773] \quad 1 \overset{1}{7} 12314141 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[774] \quad 1 \overset{1}{7} 1231 \overset{1}{5} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[775] \quad 2 \overset{1}{1} \overset{1}{3} \overset{1}{2} \overset{1}{8} \overset{1}{12} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 15, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[776] \quad 21 \overset{1}{8} 1223141 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[777] \quad 31 \overset{1}{5} 1316131 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[778] \quad 1 \overset{1}{2} \overset{1}{3} \overset{1}{2} 1231 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 15, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[761] \quad 21 \overset{1}{5} 13216131 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[780] \quad 1 \overset{1}{7} \overset{1}{5} 1321 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[781] \quad 31316131 \overset{1}{5} 1 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0}$$

$$[782] \quad 22171231 \overset{1}{5} 1 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0}$$

$$[783] \quad 1 \overset{1}{7} \overset{1}{5} 13221 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{aligned}
& \begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0} \\
[784] & 21 \overset{1}{7} 123151^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 24, T \in [16, 24] \\
& \begin{pmatrix} 4 & 3 & 4 \\ 3 & 1 & 3 \\ 4 & 3 & 4 \end{pmatrix}_{b_0} \\
[785] & 21712231 \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 24, T \in [16, 24] \\
& \begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0} \\
[786] & 1 \overset{1}{5} \overset{1}{7} \overset{1}{13} 21 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 24, T \in [16, 24] \\
& \begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0} \\
[787] & 1 \overset{1}{7} 123151^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 24, T \in [16, 24] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 7 & 5 \\ 2 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[788] & 1 \overset{1}{7} 1223151^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 24, T \in [16, 24] \\
& \begin{pmatrix} 4 & 3 & 4 \\ 3 & 1 & 3 \\ 4 & 3 & 4 \end{pmatrix}_{b_0} \\
[789] & 21 \overset{21}{8} 123141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 24, T \in [18, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[790] & 1 \overset{1}{2} \overset{1}{3} \overset{21}{8} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -87, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[791] & 22218123141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 24, T \in [18, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[792] & 21712314141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 24, T \in [18, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[793] & 61231414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 24, T \in [18, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[794] & 1 \overset{221}{8} 123141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 24, T \in [18, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[798] & 17122314141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 24, T \in [18, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[799] & 31 \overset{1}{5} 1316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.57 \rightarrow 15, T_{\max} = 16, T \in [15, 16]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[800] & \begin{matrix} 1 \\ 1712314141 \end{matrix}^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 24, T \in [18, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[801] & \begin{matrix} 21 \\ 181223141 \end{matrix}^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 24, T \in [18, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[802] & \begin{matrix} 21 \\ 22181223141 \end{matrix}^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 24, T \in [18, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[797] & \begin{matrix} 1 \\ 2321812231 \end{matrix}^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 15, \Delta = -87, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[795] & \begin{matrix} 1 \\ 31 \end{matrix}^{15131} \begin{matrix} 6 \\ 131 \end{matrix}^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14.57 \rightarrow 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 4 \\ 3 & 4 & 5 \end{pmatrix}_{b_0} \\
[796] & \begin{matrix} 1 \\ 16131414141 \end{matrix}^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 24, T \in [18, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[810] & \begin{matrix} 1 \\ 21 \end{matrix}^{1712314141}^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[803] & \begin{matrix} 1 \\ 21 \end{matrix}^{171231513}^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[804] & \begin{matrix} 1 \\ 31 \end{matrix}^{151322171}^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[805] & \begin{matrix} 21 \\ 21 \end{matrix}^{18123141}^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 8 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 3 \end{pmatrix}_{b_0} \\
[806] & \begin{matrix} 21 \\ 1 \end{matrix}^{17123151}^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[807] & \begin{matrix} 1 \\ 21 \end{matrix}^{171223151}^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[808] \quad 1^1 7 1231 \bar{5} 13 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[809] \quad 21^1 1 \bar{8} 1223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 8 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[811] \quad 21^1 1 \bar{7} 1232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -102, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[812] \quad 31^1 \bar{6} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 10, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[818] \quad 321^1 7 123151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -109, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.45 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[817] \quad 31^1 316131 \bar{5} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.45 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[816] \quad 4131 \bar{6} 13151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.45 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[815] \quad 31^1 321812231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 12, T_{\max} = 17, \Delta = -93, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.45 \rightarrow 16, T_{\max} = 17, T \in [16, 17]$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[814] \quad 1^1 \bar{8} 1231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 12, T_{\max} = 17, \Delta = -93, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.45 \rightarrow 16, T_{\max} = 17, T \in [16, 17]$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[813] \quad 31^1 322181231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 12, T_{\max} = 17, \Delta = -93, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.45 \rightarrow 16, T_{\max} = 17, T \in [16, 17]$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[819] \quad 1321 \bar{8} 1231^1 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 12, T_{\max} = 17, \Delta = -93, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 16.33 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[822] \quad 221^1 71231 \bar{5} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -100, \det = 14, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.57 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[823] \quad 221^1 71232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 12, T_{\max} = 24, \Delta = -102, \det = 14, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.57 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[820] $1 \overset{221}{8} 123141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 14, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.57 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 19 & 1 & 9 \\ 1 & 1 & -3 & -1 \\ 2 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[824] $231 \overset{1}{5} 132171^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -109, \det = 14, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.57 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 3 & 2 \\ 3 & 3 & 4 & 4 \\ 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[821] $221 \overset{1}{8} 1223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 14, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.57 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[825] $41 \overset{1}{1} 23218122 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -106, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.40 \rightarrow 17, T_{\max} = 24, T \in [17, 24]$

$$\begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & 0 & 5 & 2 \\ 4 & 5 & 25 & 9 \\ 2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[826] $132181 \overset{1}{2} 231^1 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 17, \Delta = -93, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.56 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 5 & 3 & 2 \\ 5 & 18 & 6 & 8 \\ 3 & 6 & 0 & 2 \\ 2 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

[827] $2 \overset{1}{1} \overset{1}{4} 14141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 6 & 6 \\ 2 & 6 & 3 & 5 \\ 2 & 6 & 5 & 6 \end{pmatrix}_{b_0}$$

[828] $31 \overset{1}{1} \overset{1}{5} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 2 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[830] $321 \overset{1}{5} 1316131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 16, \Delta = -89, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.55 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 8 & 3 & 6 \\ 1 & 3 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[829] $2221 \overset{1}{8} 123141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.55 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 4 & 6 \\ 1 & 4 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[835] $221812231 \overset{1}{4} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.26 \rightarrow 17, T_{\max} = 24, T \in [17, 24]$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 6 & 7 \\ 1 & 6 & 0 & 2 \\ 2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[834] $21 \overset{21}{8} 1231^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.26 \rightarrow 17, T_{\max} = 24, T \in [17, 24]$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[833] $1 \overset{21}{8} 12231^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.26 \rightarrow 17, T_{\max} = 24, T \in [17, 24]$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[832] $1 \overset{221}{8} 1231^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 24, \Delta = -104, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.26 \rightarrow 17, T_{\max} = 24, T \in [17, 24]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$

[831] $222181231^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 24, \Delta = -104, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.26 \rightarrow 17, T_{\max} = 24, T \in [17, 24]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 6 & 7 \\ 1 & 6 & 0 & 2 \\ 2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$

[836] $231^1 32181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 17, \Delta = -94, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.48 \rightarrow 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 13 & 4 & 7 \\ 1 & 4 & 0 & 2 \\ 2 & 7 & 2 & 3 \end{pmatrix}_{b_0}$

[839] $321 \overset{321}{8} 1231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 17, \Delta = -94, \det = 28, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.29 \rightarrow 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$

[838] $31 \overset{15131}{6} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 16, \Delta = -88, \det = 28, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.29 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$

[837] $1^{11} 2321812313 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 17, \Delta = -95, \det = 28, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.29 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 5 & 8 & 5 \\ 2 & 8 & 4 & 4 \\ 2 & 5 & 4 & 2 \end{pmatrix}_{b_0}$

[843] $41^1 316123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 24, \Delta = -109, \det = 30, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.80 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$

[842] $41^1 321613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 24, \Delta = -109, \det = 30, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.80 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$

[841] $41^1 31 \overset{1^1 5131}{6} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 24, \Delta = -108, \det = 30, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.80 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$

[840] $3132181^1 2231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 17, \Delta = -93, \det = 30, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.80 \rightarrow 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 6 & 8 \\ 1 & 6 & 0 & 2 \\ 2 & 8 & 2 & 3 \end{pmatrix}_{b_0}$

[844] $313221^1 81231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 17, \Delta = -93, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.50 \rightarrow 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 5 & 8 \\ 1 & 5 & 0 & 2 \\ 2 & 8 & 2 & 3 \end{pmatrix}_{b_0}$

[845] $1 \overset{1^1 51^1 2321}{7} 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 24, \Delta = -102, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.16 \rightarrow 17, T_{\max} = 24, T \in [17, 24]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 2 \end{pmatrix}_{b_0}$

[846] $1 \overset{1}{1^1 51^1 2321} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 12, T_{\max} = 24, \Delta = -102, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.16 \rightarrow 17, T_{\max} = 24, T \in [17, 24]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 26 & 8 & 10 \\ 1 & 8 & 0 & 2 \\ 2 & 10 & 2 & 2 \end{pmatrix}_{b_0}$

[847] $221712321^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 12, T_{\max} = 24, \Delta = -102, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.16 \rightarrow 17, T_{\max} = 24, T \in [17, 24]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 2 \end{pmatrix}_{b_0}$

[848] $22181^1 223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 12, T_{\max} = 24, \Delta = -104, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.67 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 28 & 7 & 10 \\ 1 & 7 & 0 & 2 \\ 2 & 10 & 2 & 3 \end{pmatrix}_{b_0}$

[849] $1^1 5^1 13221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 12, T_{\max} = 24, \Delta = -100, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.67 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$

[850] $1^1 413221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 12, T_{\max} = 24, \Delta = -104, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.67 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 43 & 1 & 13 \\ 1 & 1 & -3 & -1 \\ 2 & 13 & -1 & 3 \end{pmatrix}_{b_0}$

[851] $51^1 231513151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 12, T_{\max} = 24, \Delta = -107, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.67 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$
 $\begin{pmatrix} 1 & 7 & 3 \\ 7 & 7 & 7 \\ 3 & 7 & 4 \end{pmatrix}_{b_0}$

[852] $2171^1 2231^1 51 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 12, T_{\max} = 24, \Delta = -100, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.67 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$

[854] $2171231^1 4141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 12, T_{\max} = 24, \Delta = -104, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.79 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 8 & 11 \\ 1 & 8 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0}$

[855] $1712231^1 4141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 12, T_{\max} = 24, \Delta = -104, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.79 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 8 & 11 \\ 1 & 8 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0}$

[853] $171231^1 4141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 12, T_{\max} = 24, \Delta = -104, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.79 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$

[858] $171^1 2231^1 513 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 12, T_{\max} = 24, \Delta = -108, \det = 56, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.64 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$
 $\begin{pmatrix} 0 & 6 & 2 & 2 \\ 6 & 28 & 7 & 10 \\ 2 & 7 & 0 & 2 \\ 2 & 10 & 2 & 3 \end{pmatrix}_{b_0}$

[857] $61^1 231414141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 12, T_{\max} = 24, \Delta = -105, \det = 56, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.57 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 64 & 1 & 16 \\ 1 & 1 & -3 & -1 \\ 2 & 16 & -1 & 3 \end{pmatrix}_{b_0}$

[856] $171^1 22314141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 12, T_{\max} = 24, \Delta = -104, \det = 56, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.57 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 40 & 8 & 12 \\ 1 & 8 & 0 & 2 \\ 2 & 12 & 2 & 3 \end{pmatrix}_{b_0}$

[859] $71^1 2231^1 5132 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 12, T_{\max} = 24, \Delta = -109, \det = 74, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.59 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$
 $\begin{pmatrix} 0 & 7 & 3 & 2 \\ 7 & 40 & 15 & 12 \\ 3 & 15 & 4 & 4 \\ 2 & 12 & 4 & 3 \end{pmatrix}_{b_0}$

$$[861] \ 16131^1 414141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 79, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.57 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 61 & 10 & 15 \\ 1 & 10 & 0 & 2 \\ 2 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[860] \ 61231^1 414141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 79, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.57 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 82 & 1 & 18 \\ 1 & 1 & -3 & -1 \\ 2 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[863] \ 41321^1 613151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -109, \det = 86, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.65 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 26 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[862] \ 3221^1 7123151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -109, \det = 86, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.65 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 26 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[864] \ 413161^1 23151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -109, \det = 93, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.74 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 19 & 1 & 9 \\ 1 & 1 & -2 & 0 \\ 2 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[865] \ 21^1 41414132 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 112, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 4 & 4 & 2 \\ 4 & 20 & 23 & 12 \\ 4 & 23 & 19 & 11 \\ 2 & 12 & 11 & 6 \end{pmatrix}_{b_0}$$

$$[866] \ 32171^1 223151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 24, \Delta = -109, \det = 146, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.63 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [867] \ 1^1 231513151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 14, \Delta = -35, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 3.0$$

$$[869] \ 1^1 223151315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 14, \Delta = -34, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 3.0$$

$$[870] \ 1^1 41414141321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 13, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 3.0$$

$$[872] \ 1^1 314141414131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 14, \Delta = -33, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 3.0$$

$$[871] \ 1^1 41414141321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 13, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 3.0$$

$$[868] \ 1^1 223151315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 14, \Delta = -34, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 3.0$$

$$[896] \ 1^1 315131 \quad 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[901] \ 21^1 613151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 14, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[900] \ 1^1 32161315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 15, \Delta = -60, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[899] \ 1^1 32161315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 15, \Delta = -60, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[898] \ 1^1 3221 \quad 1^1 8 \quad 12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[897] \quad 1^{1^1 3221} \quad 8 \quad 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[895] \quad 21^{1^1 2321} \quad 8 \quad 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[894] \quad 16^{1^1} 123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[893] \quad 21^{1^1 2321} \quad 8 \quad 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[892] \quad 1^{1^1} 31612315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -60, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[891] \quad 1^{1^1 315131} \quad 6 \quad 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[890] \quad 1^{1^1} 61223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[889] \quad 1^{1^1} \quad 8 \quad 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[888] \quad 1^{1^1} 41321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[887] \quad 16^{1^1} 13151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 2 & 4 \\ 2 & 2 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[886] \quad 1^{1^1} 41321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[884] \quad 1^{1^1} 31612315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -60, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[883] \quad 15^{1^1} 131612321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$$

$$\begin{aligned}
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[882] & 1 \overset{1}{\underset{1}{2321}} 8 \overset{1}{1231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[885] & 1 \overset{1}{15131612321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[916] & 1 \overset{1}{17122315131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[924] & 1 \overset{1}{1321812321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0} \\
[925] & 1 \overset{1}{15132171231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[915] & 1 \overset{1}{15131613141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[923] & 1 \overset{1}{13161315131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[917] & 1 \overset{1}{\overset{2321}{812321}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0} \\
[902] & 1 \overset{1}{132171232151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -82, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[903] & 1 \overset{1}{15132171231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[904] & 1 \overset{1}{132171232151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -82, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[905] & 1 \overset{1}{31321812321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -66, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

- [906] $21\overset{1}{6}13161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 15, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [907] $1\overset{1}{8}123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$
- [908] $1\overset{1}{7}12315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$
- [909] $1\overset{1}{6}123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [910] $2\overset{1}{3}12321\overset{1}{8}12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 15, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$
- [911] $1\overset{1}{8}123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$
- [912] $1\overset{1}{5}131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [913] $21\overset{1}{6}1316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 15, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$
- [914] $1\overset{1}{4}1321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [918] $1\overset{1}{4}1322181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [919] $1\overset{1}{4}1322181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [920] $13221\overset{1}{8}12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0}$
- [921] $1\overset{1}{4}1321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -84, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[922] & 21 \overset{1}{7} 12315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[927] & 21 \overset{1}{8} 12314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 25, T \in [16, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0} \\
[935] & 1^1 \overset{1}{5} 1321613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 25, T \in [16, 25] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[949] & 1 \overset{1}{8} 122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 25, T \in [16, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0} \\
[945] & 21^1 \overset{1}{5} 132171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[944] & 221 \overset{1}{8} 1231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 21, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 21, T \in [16, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[943] & 141321 \overset{1}{8} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -84, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 1 & 3 \\ 2 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[942] & 221 \overset{1}{8} 1232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 25, T \in [16, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[948] & 23 \overset{1}{2} 18123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -86, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[941] & 21^1 \overset{1}{5} 131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[940] & 1321 \overset{1}{8} 123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -84, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0} \\
[939] & 16 \overset{1}{1} 31513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 25, T \in [16, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[938] & 1 \overset{1}{8} 12231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 21, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 16, T_{\max} = 21, T \in [16, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[937] & 31 \overset{1}{6} 13161^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -85, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 0 & 3 & 2 \\ 2 & 3 & 5 & 5 \\ 2 & 2 & 5 & 4 \end{pmatrix}_{b_0} \\
[936] & 21^1 6123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 4 \end{pmatrix}_{b_0} \\
[926] & \overset{21}{1} 7 12315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0} \\
[928] & 412321 \overset{1}{8} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -86, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[929] & 21 \overset{1}{8} 12231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 21, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 21, T \in [16, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[930] & \overset{21}{1} 8 1232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 25, T \in [16, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[947] & 1^1 61231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 25, T \in [16, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[931] & 21^1 7122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \\ 3 & 5 & 5 \end{pmatrix}_{b_0} \\
[932] & 1^1 \overset{1}{6} 13151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 25, T \in [16, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[933] & \overset{21}{1} 8 1231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 21, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 21, T \in [16, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[934] & 1^1 51321613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -104, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 25, T \in [16, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[946] & 1^1 41414132171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 25, T \in [16, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

- [960] $1\overset{1}{8}1223151\overset{1}{2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 7 & 5 \\ 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$
- [959] $221\overset{1}{8}123151\overset{1}{2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 4 & 3 & 4 \\ 3 & 1 & 3 \\ 4 & 3 & 4 \end{pmatrix}_{b_0}$
- [958] $222181231\overset{1}{5}1\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0}$
- [957] $21\overset{1}{1}\overset{1}{5}1321\overset{1}{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0}$
- [956] $1\overset{1}{8}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [955] $21\overset{1}{8}1223151\overset{1}{2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 4 & 3 & 4 \\ 3 & 1 & 3 \\ 4 & 3 & 4 \end{pmatrix}_{b_0}$
- [954] $221812231\overset{1}{5}1\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0}$
- [953] $1\overset{1}{1}\overset{1}{5}13221\overset{1}{8}12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0}$
- [952] $1\overset{21}{8}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [951] $1\overset{1}{1}\overset{1}{5}1321\overset{1}{8}122 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0}$
- [950] $1\overset{21}{8}123151\overset{1}{2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 7 & 5 \\ 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$
- [965] $321\overset{1}{6}1316131\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 16, \Delta = -86, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 15.57 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 4 \end{pmatrix}_{b_0}$
- [970] $171231414141\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -103, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 25, T \in [19, 25]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[961] & \begin{matrix} 1^{131} & 1^1 \\ 31 & 6 & 131 & 61 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -85, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.57 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 5 & 5 \\ 2 & 2 & 5 & 4 \end{pmatrix}_{b_0} \\
[962] & \begin{matrix} 1 \\ 18122314141 \end{matrix} \begin{matrix} 1^1 \\ 1^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 25, T \in [19, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[963] & \begin{matrix} 21 \\ 1812314141 \end{matrix} \begin{matrix} 1^1 \\ 1^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 25, T \in [19, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[964] & \begin{matrix} 1^1 \\ 23218123141 \end{matrix} \begin{matrix} 1^1 \\ 1^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -86, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15.57 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[966] & \begin{matrix} 1^1 \\ 613141414141 \end{matrix} \begin{matrix} 1^1 \\ 1^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 25, T \in [19, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[967] & \begin{matrix} 1^1 \\ 221812314141 \end{matrix} \begin{matrix} 1^1 \\ 1^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 25, T \in [19, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[968] & \begin{matrix} 1^1 \\ 712231414141 \end{matrix} \begin{matrix} 1^1 \\ 1^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 25, T \in [19, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[969] & \begin{matrix} 1^1 \\ 218122314141 \end{matrix} \begin{matrix} 1^1 \\ 1^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 25, T \in [19, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[983] & \begin{matrix} 21^1 \\ 1 \\ 151321 \end{matrix} \begin{matrix} 1^1 \\ 8 \\ 1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 21, \Delta = -99, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.50 \rightarrow 17, T_{\max} = 21, T \in [17, 21] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[974] & \begin{matrix} 21^1 & 1^1 \\ 1 & 8 & 1231 & 513 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[973] & \begin{matrix} 1^1 \\ 21^1 & 812231 & 513 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[972] & \begin{matrix} 1^1 & 1^1 \\ 1812231 & 513 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [971] $1 \overset{21^1}{8} 12314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -103, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 8 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 3 \end{pmatrix}_{b_0}$
- [975] $1 \overset{1}{8} 1231 \overset{1^1}{5} 13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 4 & 4 \\ 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$
- [976] $221^1 81231 \overset{1^1}{5} 13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 3 & 2 & 2 \\ 3 & 10 & 4 & 6 \\ 2 & 4 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [978] $21^1 \overset{1^1}{5} 12321 8^1 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -101, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [979] $21^1 8122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -103, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [980] $1 \overset{21^1}{8} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -101, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [981] $21^1 \overset{1^1}{5} 1321 8^1 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [982] $21^1 6131513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -103, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [985] $21^1 81231 \overset{1^1}{5} 132 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -108, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 2 & 3 & 2 \\ 2 & 4 & 5 & 4 \\ 3 & 5 & 4 & 4 \\ 2 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [984] $1 \overset{21^1}{8} 12231 \overset{1^1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [977] $21^1 81231 \overset{1^1}{5} 13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -107, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [988] $31^1 4132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 17, \Delta = -92, \det = -11, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.45 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 3 \end{pmatrix}_{b_0}$
- [991] $31^1 3218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 13, T_{\max} = 17, \Delta = -92, \det = -11, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 16.45 \rightarrow 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[990] & \begin{matrix} 1 \\ 32181223151 \end{matrix} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -108, \det = -11, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.45 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0} \\
[986] & \begin{matrix} 1 \\ 32218123151 \end{matrix} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -108, \det = -11, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.45 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0} \\
[989] & \begin{matrix} 1 \\ 15131613151 \end{matrix} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -11, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.45 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0} \\
[987] & \begin{matrix} 1 \\ 3218123151 \end{matrix} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -108, \det = -11, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.45 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0} \\
[995] & \begin{matrix} 1 \\ 22181223151 \end{matrix} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 21, \Delta = -99, \det = -14, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.57 \rightarrow 17, T_{\max} = 21, T \in [17, 21] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[997] & \begin{matrix} 1 \\ 1812315132 \end{matrix} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -108, \det = -14, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.57 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 3 & 2 \\ 3 & 3 & 4 & 4 \\ 2 & 2 & 4 & 3 \end{pmatrix}_{b_0} \\
[994] & \begin{matrix} 1 \\ 221812314141 \end{matrix} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -14, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.57 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[993] & \begin{matrix} 1 \\ 23151322181 \end{matrix} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -108, \det = -14, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.57 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 1 & 3 & 2 \\ 1 & 0 & 3 & 2 \\ 3 & 3 & 4 & 4 \\ 2 & 2 & 4 & 3 \end{pmatrix}_{b_0} \\
[992] & \begin{matrix} 221 \\ 18123151 \end{matrix} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 21, \Delta = -99, \det = -14, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.57 \rightarrow 17, T_{\max} = 21, T \in [17, 21] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[996] & \begin{matrix} 221 \\ 181232151 \end{matrix} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -101, \det = -14, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.57 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[998] & \begin{matrix} 31 \\ 2161316131 \end{matrix} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -85, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 24 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 4 \end{pmatrix}_{b_0} \\
[1001] & \begin{matrix} 1 \\ 22218123151 \end{matrix} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 13, T_{\max} = 21, \Delta = -99, \det = -22, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.55 \rightarrow 17, T_{\max} = 21, T \in [17, 21]
\end{aligned}$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

[1000] $2231 \overset{1}{5} 132181^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -108, \det = -22, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.55 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 1 & 4 & 2 \\ 1 & 0 & 4 & 2 \\ 4 & 4 & 10 & 6 \\ 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[999] $2221^1 81232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -101, \det = -22, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.55 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

[1002] $3141^1 32181231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 17, \Delta = -92, \det = -27, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.48 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 1 & 11 \\ 1 & 1 & -2 & 0 \\ 2 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

[1003] $41^1 3217123151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -108, \det = -30, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.80 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[1004] $21^1 \overset{1}{5} 1321^1 8^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -108, \det = -32, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[1008] $21^1 \overset{1}{5} 1^{12321} 8^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -101, \det = -38, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.16 \rightarrow 18, T_{\max} = 25, T \in [18, 25]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

[1007] $1^1 \overset{21}{8} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -101, \det = -38, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.16 \rightarrow 18, T_{\max} = 25, T \in [18, 25]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 26 & 8 & 10 \\ 1 & 8 & 0 & 2 \\ 2 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[1006] $1^1 \overset{1}{5} 1^{12321} 8^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -101, \det = -38, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.16 \rightarrow 18, T_{\max} = 25, T \in [18, 25]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

[1005] $2221812321^1 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -101, \det = -38, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.16 \rightarrow 18, T_{\max} = 25, T \in [18, 25]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

[1013] $22181^1 2231 \overset{1}{5} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -42, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.67 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

[1012] $1^1 \overset{1}{5} 13221^1 8^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 21, \Delta = -99, \det = -42, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.67 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

[1011] $161^1 231513151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -104, \det = -42, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.67 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

[1010] $51^1 2321613151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -106, \det = -42, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 16.67 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0} \\
[1009] & 1 \overset{1}{8} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 21, \Delta = -99, \det = -42, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& \overset{1}{1} \overset{1}{5} 13221^1 \\
& T_{\min} = 16.67 \rightarrow 17, T_{\max} = 21, T \in [17, 21] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 28 & 7 & 10 \\ 1 & 7 & 0 & 2 \\ 2 & 10 & 2 & 3 \end{pmatrix}_{b_0} \\
[1015] & 1812231^1 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -47, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.79 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 2 & 12 & -1 & 3 \end{pmatrix}_{b_0} \\
[1016] & 181231^1 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -47, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.79 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 2 & 12 & -1 & 3 \end{pmatrix}_{b_0} \\
[1014] & 21812231^1 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -47, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.79 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 8 & 11 \\ 1 & 8 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0} \\
[1017] & 22181231^1 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -47, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.79 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 8 & 11 \\ 1 & 8 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0} \\
[1018] & 2181^1 2231 \overset{1}{5} 13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -56, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.64 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 6 & 2 & 2 \\ 6 & 28 & 7 & 10 \\ 2 & 7 & 0 & 2 \\ 2 & 10 & 2 & 3 \end{pmatrix}_{b_0} \\
[1019] & 61^1 2231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -104, \det = -56, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.57 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 64 & 1 & 16 \\ 1 & 1 & -3 & -1 \\ 2 & 16 & -1 & 3 \end{pmatrix}_{b_0} \\
[1020] & 1^1 41413221^1 8 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -56, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.57 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 64 & 1 & 16 \\ 1 & 1 & -3 & -1 \\ 2 & 16 & -1 & 3 \end{pmatrix}_{b_0} \\
[1021] & 2181^1 22314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -56, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.57 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 40 & 8 & 12 \\ 1 & 8 & 0 & 2 \\ 2 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[1022] & 1^1 \overset{1}{31} \overset{1}{5} 13221^1 8 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -56, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.64 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 6 & 2 & 2 \\ 6 & 28 & 7 & 10 \\ 2 & 7 & 0 & 2 \\ 2 & 10 & 2 & 3 \end{pmatrix}_{b_0} \\
[1023] & 71^1 2231414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -72, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 89 & 1 & 19 \\ 1 & 1 & -3 & -1 \\ 2 & 19 & -1 & 3 \end{pmatrix}_{b_0} \\
[1024] & 181^1 2231 \overset{1}{5} 132 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 13, T_{\max} = 25, \Delta = -108, \det = -74, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16.59 \rightarrow 17, T_{\max} = 25, T \in [17, 25]
\end{aligned}$$

$$\begin{pmatrix} 0 & 7 & 3 & 2 \\ 7 & 40 & 15 & 12 \\ 3 & 15 & 4 & 4 \\ 2 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

[1025] $171231^1 414141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -103, \det = -79, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 61 & 10 & 15 \\ 1 & 10 & 0 & 2 \\ 2 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1026] $712231^1 414141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -103, \det = -79, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 82 & 1 & 18 \\ 1 & 1 & -3 & -1 \\ 2 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[1028] $1^{3221} 8^{123151} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -108, \det = -86, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.65 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 26 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[1027] $151321^1 613151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -104, \det = -86, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.65 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 26 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[1029] $512321^1 613151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -106, \det = -86, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.65 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 26 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[1030] $3221^1 81223151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -108, \det = -86, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.65 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 26 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[1031] $1513161^1 23151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -104, \det = -93, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.74 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 1 & 12 & 3 \\ 12 & 51 & 15 \\ 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[1032] $81^1 223151322 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -108, \det = -96, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.54 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 8 & 4 & 2 \\ 8 & 54 & 25 & 14 \\ 4 & 25 & 10 & 6 \\ 2 & 14 & 6 & 3 \end{pmatrix}_{b_0}$$

[1033] $4141^1 31613151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -107, \det = -106, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 38 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

[1034] $6131^1 41414141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -103, \det = -119, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.45 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 142 & 1 & 24 \\ 1 & 1 & -3 & -1 \\ 2 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[1036] $322181^1 223151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -108, \det = -146, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.63 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

[1035] $1^{1513221} 8^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 13, T_{\max} = 25, \Delta = -108, \det = -146, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.63 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$T = 14$ [1037] $1^{12321613151321} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 15, \Delta = -34, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 3.0$

[1039] $1^{12321613151321} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 15, \Delta = -34, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[1071] & 1 \overset{131}{6} 13161231^{\textcolor{red}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1062] & 1^{\textcolor{red}{1}} 321613161231^{\textcolor{blue}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1063] & 1^{\textcolor{red}{1}} 413161315131^{\textcolor{blue}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1064] & 1^{\textcolor{red}{1}} 413161315131^{\textcolor{red}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[1065] & 1^{\textcolor{red}{1}} 232171232151^{\textcolor{blue}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1066] & 1^{\textcolor{red}{1}} 7123151321^{\textcolor{blue}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 2 & 4 \\ 2 & 2 & 4 & 6 \end{pmatrix}_{b_0} \\
[1067] & 1^{\textcolor{red}{1}} 8^{12321} 123141^{\textcolor{blue}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0} \\
[1068] & 1^{\textcolor{red}{1}} 8^{12321} 123141^{\textcolor{red}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
[1069] & 1^{\textcolor{red}{1}} 6^{131} 13161231^{\textcolor{blue}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1070] & 1^{\textcolor{red}{1}} 321712315131^{\textcolor{blue}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 16, T \in [15, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1078] & 1^{\textcolor{red}{1}} 413221812321^{\textcolor{blue}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1094] & 1^{\textcolor{red}{1}} 51321812231^{\textcolor{red}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1095] & 1^{\textcolor{red}{1}} 51321812231^{\textcolor{blue}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1093] & 1^1 322181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1097] & 412321^1 812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1098] & 1^1 51316131^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1099] & 21^1 6131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[1100] & 21^1 8122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1101] & 1^1 414132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -83, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1102] & 1^1 8122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0} \\
[1103] & 1^1 512321^1 81231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1104] & 1^1 512321^1 81231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1105] & 1^1 51322181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1106] & 1^1 51322181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1107] & 221^1 812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1108] & 1^1 51321^1 81231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1092] & 1^1 322181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1091] & 1^{131} 31 \overset{1}{6} 1316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -64, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0} \\
[1090] & 1^1 217123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1089] & 1^1 316131513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -82, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1088] & 1^1 316131513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -82, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1087] & 1^1 414132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -83, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1086] & 1^1 413218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -83, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[1085] & 1^{15} 8^{1321} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1084] & 1^1 161232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1096] & 1^{21} 1812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& 1^1 T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 5 \end{pmatrix}_{b_0} \\
[1120] & 21^1 51322181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[1109] & 21^1 51321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[1117] & 1^1 171231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 26, T \in [17, 26]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1122] & \begin{matrix} 1^1 \\ 23218123151 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -81, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \end{pmatrix}_{b_0} \\
[1123] & \begin{matrix} 21^1 51321 \\ 1^1 8^{1^1} 1231^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[1124] & \begin{matrix} 1^1 \\ 215132181231^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[1125] & \begin{matrix} 1^1 \\ 1^1 712231513151^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 26, T \in [17, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1126] & \begin{matrix} 1^1 \\ 1^1 414141322181^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 26, T \in [17, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[1127] & \begin{matrix} 1^1 \\ 1^1 513217123151^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 26, T \in [17, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1128] & \begin{matrix} 1^1 \\ 13218123151 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1114] & \begin{matrix} 1^1 \\ 161316123151^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -101, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 26, T \in [17, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1113] & \begin{matrix} 1^1 \\ 1^1 612321613151^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -103, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 26, T \in [17, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1112] & \begin{matrix} 1^1 \\ 21^1 61232171231^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 4 \end{pmatrix}_{b_0} \\
[1111] & \begin{matrix} 1^1 321^1 8^{1^1} 1231 \\ 1^1 5^{1^1} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[1110] & \begin{matrix} 1^1 \\ 232181232151^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -83, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \end{pmatrix}_{b_0} \\
[1118] & \begin{matrix} 1^1 \\ 151232181231^1 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -81, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1121] \quad 1^1 8 1231414141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 17, T_{\max} = 26, T \in [17, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[1116] \quad 1^1 132181232151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 17, \Delta = -81, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[1119] \quad 1^1 315132181231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 17, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1115] \quad 1^1 16131613151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 26, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 17, T_{\max} = 26, T \in [17, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[1129] \quad 1^{131} 31 \ 6 \ 1316131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 16, \Delta = -64, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 15.67 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 6 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[1137] \quad 1^1 315132181231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 17, \Delta = -87, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1136] \quad 2181231414141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 20, T_{\max} = 26, T \in [20, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1135] \quad 1^1 181231414141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 20, T_{\max} = 26, T \in [20, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[1133] \quad 7123141414141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 20, T_{\max} = 26, T \in [20, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[1132] \quad 1^1 23218123151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 17, \Delta = -81, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[1131] \quad 1^1 232181232151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 17, \Delta = -83, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[1130] \quad 1812231414141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 20, T_{\max} = 26, T \in [20, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[1134] $413161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

[1140] $21^1 71231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.50 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

[1138] $21^1 81231414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.50 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[1141] $21^1 61316123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -101, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.50 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

[1139] $1^1 6^{15131} 13161^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -100, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.50 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[1145] $216131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.45 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 4 & 5 \\ 1 & 4 & 0 & 2 \\ 2 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[1144] $16131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.45 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[1143] $31^1 32181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 18, \Delta = -89, \det = 11, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.45 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

[1142] $413218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 11, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.45 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[1146] $31^{31} 161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 16, \Delta = -64, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 2 & 4 & 2 & 4 \\ 2 & 4 & 4 & 5 \end{pmatrix}_{b_0}$$

[1147] $321^1 5132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 18, \Delta = -88, \det = 22, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.55 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$

$$\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 8 & 3 & 6 \\ 1 & 3 & 0 & 2 \\ 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[1149] $41^1 32218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 30, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.80 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[1148] $41^1 321^1 8^{151321} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 30, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 17.80 \rightarrow 18, T_{\max} = 26, T \in [18, 26] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[1150] & 41^1 32181223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 30, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.80 \rightarrow 18, T_{\max} = 26, T \in [18, 26] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[1151] & 161^1 2321613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -103, \det = 42, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.67 \rightarrow 18, T_{\max} = 26, T \in [18, 26] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0} \\
[1152] & 51^1 23217123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -105, \det = 42, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.67 \rightarrow 18, T_{\max} = 26, T \in [18, 26] \\
& \begin{pmatrix} 1 & 7 & 3 \\ 7 & 7 & 7 \\ 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[1153] & 3151^1 232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 18, \Delta = -89, \det = 44, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.45 \rightarrow 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 26 & 5 & 10 \\ 1 & 5 & 0 & 2 \\ 2 & 10 & 2 & 3 \end{pmatrix}_{b_0} \\
[1154] & 171^1 2231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 56, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.57 \rightarrow 18, T_{\max} = 26, T \in [18, 26] \\
& \begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0} \\
[1155] & 181^1 2231414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 72, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.50 \rightarrow 18, T_{\max} = 26, T \in [18, 26] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 54 & 9 & 14 \\ 1 & 9 & 0 & 2 \\ 2 & 14 & 2 & 3 \end{pmatrix}_{b_0} \\
[1158] & 181231^1 414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 79, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.57 \rightarrow 18, T_{\max} = 26, T \in [18, 26] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 82 & 1 & 18 \\ 1 & 1 & -3 & -1 \\ 2 & 18 & -1 & 3 \end{pmatrix}_{b_0} \\
[1157] & 1812231^1 414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 79, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.57 \rightarrow 18, T_{\max} = 26, T \in [18, 26] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 61 & 10 & 15 \\ 1 & 10 & 0 & 2 \\ 2 & 15 & 2 & 3 \end{pmatrix}_{b_0} \\
[1156] & 2181231^1 414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 79, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.57 \rightarrow 18, T_{\max} = 26, T \in [18, 26] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 61 & 10 & 15 \\ 1 & 10 & 0 & 2 \\ 2 & 15 & 2 & 3 \end{pmatrix}_{b_0} \\
[1159] & 1612321^1 613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -103, \det = 86, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.65 \rightarrow 18, T_{\max} = 26, T \in [18, 26] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 26 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0} \\
[1160] & 1613161^1 23151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -101, \det = 93, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.74 \rightarrow 18, T_{\max} = 26, T \in [18, 26] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 19 & 1 & 9 \\ 1 & 1 & -2 & 0 \\ 2 & 9 & 0 & 3 \end{pmatrix}_{b_0} \\
[1161] & 21613161^1 23151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -101, \det = 93, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.74 \rightarrow 18, T_{\max} = 26, T \in [18, 26] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 19 & 1 & 9 \\ 1 & 1 & -2 & 0 \\ 2 & 9 & 0 & 3 \end{pmatrix}_{b_0} \\
[1162] & 71231^1 41414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 119, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.45 \rightarrow 18, T_{\max} = 26, T \in [18, 26] \\
& \begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 142 & 1 & 24 \\ 1 & 1 & -3 & -1 \\ 2 & 24 & -1 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1163] $413221^1 8123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 26, \Delta = -107, \det = 137, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.56 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 43 & 1 & 13 \\ 1 & 1 & -2 & 0 \\ 2 & 13 & 0 & 3 \end{pmatrix}_{b_0}$

[1164] $4132181^1 223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 14, T_{\max} = 26, \Delta = -107, \det = 146, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17.63 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$

$T = 15$ [1171] $1^1 41316131513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$

[1172] $1^1 23216131612321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -33, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$

[1170] $1^1 41316131513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$

[1169] $1^1 4141414141^1 41 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 15, \Delta = -15, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$

[1168] $1^1 32171231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$

[1167] $1^1 32171231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$

[1165] $1^1 23217123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -33, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$

[1166] $1^1 23217123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -33, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 3.0$

[1213] $1^1 1316131612321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1194] $1^1 3161232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 17, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1199] $1^1 3221812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 17, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1200] $1^1 4131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1201] $1^1 4131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1202] $1^1 151321 8^{15} 12321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1203] $1^1 3221812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 17, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[1204] $1^1 1513221812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1205] $1^1 2218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

- [1206] $1 \overset{1}{\overset{31}{6}} 131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$
- [1207] $1 \overset{1}{\overset{31}{6}} 131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$
- [1208] $21 \overset{1}{8} 1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$
- [1209] $1 \overset{1}{\overset{15}{8}} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
- [1210] $1 \overset{1}{\overset{3151321}{8}} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 17, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [1211] $1 \overset{1}{\overset{3151321}{8}} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 17, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$
- [1212] $1 \overset{1}{\overset{512321}{8}} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [1197] $1^1 3218122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 17, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$
- [1214] $1 \overset{1}{\overset{513161315131}{15}} 13161315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [1196] $1^1 3218122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 17, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [1187] $1 \overset{1}{\overset{513221812321}{15}} 13221812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
- [1188] $1 \overset{1}{\overset{512321}{8}} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
- [1189] $1 \overset{1}{\overset{513161315131}{15}} 13161315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 15, T_{\max} = 16, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1190] & 1^1 4141321812321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1195] & 1^1 1612321712321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1191] & 1^1 4141321812321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[1192] & 1^1 181223151321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 2 & 4 \\ 2 & 2 & 4 & 6 \end{pmatrix}_{b_0} \\
[1193] & 1^1 3161232171231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -58, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 17, T \in [16, 17] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1198] & 1^1 18123151321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 16, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 2 & 4 \\ 2 & 2 & 4 & 6 \end{pmatrix}_{b_0} \\
[1224] & 1^1 4132181232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1225] & 1^1 1321812315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1226] & 1^1 6123151316131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1227] & 1^1 2218123216131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1228] & 1^1 161232181231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1229] & 1^1 1613151316131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1230] & 1^1 216131613151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -71, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1231] & \overset{1}{2}321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0} \\
[1232] & \overset{1}{1}612321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1223] & \overset{1}{1}4132181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1222] & \overset{1}{1}513218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1221] & \overset{1}{1}513218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1220] & \overset{1}{1}5131 \quad 1 \quad \overset{1}{6}1316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1219] & \overset{1}{1}3161316123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1218] & \overset{1}{3}151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0} \\
[1217] & \overset{1}{1}3161316123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1216] & \overset{1}{1}5131 \quad 1 \quad \overset{1}{6}1316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 18, T \in [17, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1215] & \overset{1}{2}171232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 17, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1236] & \overset{1}{1}6123217123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -102, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 27, T \in [18, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1241] & \overset{1}{2}1^1613151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 4 \end{pmatrix}_{b_0} \\
[1243] & 1^1 5132218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -102, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 27, T \in [18, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1242] & 1^1 5132218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -102, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 27, T \in [18, 27] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & 2 \\ 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1234] & 1^1 61232181232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -80, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[1233] & 1^1 51321 8 123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -102, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 27, T \in [18, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1235] & 1^1 812231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 27, T \in [18, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1237] & 1^1 4141414132181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 27, T \in [18, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[1238] & 1^1 712321613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 27, T \in [18, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1239] & 1^1 2181231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 27, T \in [18, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1240] & 1^1 81231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 27, T \in [18, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[1254] & 1^{11} 18123141414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[1249] & 1^{11} 2221 (12) 1222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[1257] & 1^1 3161232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -86, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.57 \rightarrow 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
[1244] & 1^{11} 21 (12) 12222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[1256] & \begin{matrix} 221 \\ 1^{11}3221 \end{matrix} 21 \ (12) \ 122 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[1255] & 22222221(12)12231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[1253] & \begin{matrix} 21 \\ 1^{11}3221 \end{matrix} 21 \ (12) \ 1222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[1245] & 81223141414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[1252] & \begin{matrix} 221 \\ 1^{11}3221 \end{matrix} 1 \ (12) \ 1222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[1251] & \begin{matrix} 1^1 \\ 1^{11}3221 \end{matrix} 1513161316131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 18, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.57 \rightarrow 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
[1250] & \begin{matrix} 1^{11}3221 \\ 1^{11}3221 \end{matrix} 221 \ (12) \ 12222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[1246] & \begin{matrix} 1 \\ 1^{11}3221 \end{matrix} 1 \ (12) \ 122222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[1247] & \begin{matrix} 21 \\ 1^{11}3221 \end{matrix} 1 \ (12) \ 12222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[1248] & \begin{matrix} 1^{11}3221 \\ 1^{11}3221 \end{matrix} 1 \ (12) \ 1222222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 27, T \in [21, 27] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 2 \\ 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[1259] & 21^{11}812231513151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -8, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.50 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[1258] & 21^{11}712321613151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -8, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.50 \rightarrow 19, T_{\max} = 27, T \in [19, 27]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[1260] & 1 \overset{21}{8} 1231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -8, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.50 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[1261] & 512321 \overset{1}{8} 123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -11, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.45 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0} \\
[1262] & 151321 \overset{1}{8} 123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -102, \det = -11, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.45 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 1 & 4 & 3 \\ 4 & 5 & 5 \\ 3 & 5 & 4 \end{pmatrix}_{b_0} \\
[1263] & 221 \overset{1}{8} 1231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -14, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.57 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[1264] & 161 \overset{1}{23217} 123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -102, \det = -42, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.67 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0} \\
[1265] & 1 \overset{51}{8} \overset{2321}{123151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -42, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.67 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 1 & 7 & 3 \\ 7 & 7 & 7 \\ 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[1266] & 51 \overset{1}{232181223151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -104, \det = -42, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.67 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 1 & 7 & 3 \\ 7 & 7 & 7 \\ 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[1267] & 2181 \overset{1}{2231513151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -56, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.57 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0} \\
[1268] & 1 \overset{1}{5131513221^1} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -56, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.57 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0} \\
[1270] & 1712321 \overset{1}{613151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -86, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.65 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 26 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0} \\
[1269] & 21712321 \overset{1}{613151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -86, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.65 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 68 & 11 & 16 \\ 1 & 11 & 0 & 2 \\ 2 & 16 & 2 & 3 \end{pmatrix}_{b_0} \\
[1271] & 81 \overset{1}{223141414141^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -90, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.44 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 118 & 1 & 22 \\ 1 & 1 & -3 & -1 \\ 2 & 22 & -1 & 3 \end{pmatrix}_{b_0} \\
[1272] & 4141 \overset{1}{3218123151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 15, T_{\max} = 27, \Delta = -106, \det = -106, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.57 \rightarrow 19, T_{\max} = 27, T \in [19, 27] \\
& \begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 38 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1274] $181231^1 41414141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 27, \Delta = -101, \det = -119, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18.45 \rightarrow 19, T_{\max} = 27, T \in [19, 27]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 97 & 12 & 19 \\ 1 & 12 & 0 & 2 \\ 2 & 19 & 2 & 3 \end{pmatrix}_{b_0}$

[1273] $812231^1 41414141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 27, \Delta = -101, \det = -119, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18.45 \rightarrow 19, T_{\max} = 27, T \in [19, 27]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 142 & 1 & 24 \\ 1 & 1 & -3 & -1 \\ 2 & 24 & -1 & 3 \end{pmatrix}_{b_0}$

[1275] $1513221^1 8123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 27, \Delta = -102, \det = -137, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18.56 \rightarrow 19, T_{\max} = 27, T \in [19, 27]$
 $\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 43 & 1 & 13 \\ 1 & 1 & -2 & 0 \\ 2 & 13 & 0 & 3 \end{pmatrix}_{b_0}$

[1277] $51232181^1 223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 27, \Delta = -104, \det = -146, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18.63 \rightarrow 19, T_{\max} = 27, T \in [19, 27]$
 $\begin{pmatrix} 1 & 15 & 3 \\ 15 & 79 & 19 \\ 3 & 19 & 4 \end{pmatrix}_{b_0}$

[1276] $15132181^1 223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 15, T_{\max} = 27, \Delta = -102, \det = -146, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18.63 \rightarrow 19, T_{\max} = 27, T \in [19, 27]$
 $\begin{pmatrix} 1 & 15 & 3 \\ 15 & 79 & 19 \\ 3 & 19 & 4 \end{pmatrix}_{b_0}$

$T = 16$ [1286] $1^{123151321} 8^{12321} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1278] $1^1 322181231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1279] $1^1 321812231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1290] $1^{123151321} 8^{1231} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1281] $1^1 322181231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1282] $1^1 232181223151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1283] $1^{123151321} 8^{12321} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1284] $1^1 51316131513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 16, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1289] $1^1 315131613151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -33, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1285] $1^1 51316131513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 16, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1288] $1^{123151321} 8^{1231} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1287] $1^1 315131613151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -33, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1291] $1^1 321812231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1280] $1^1 232181223151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 3.0$

[1313] $1^1 31612321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1314] $1^1 41321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[1315] $13218^{123151321} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -30, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[1316] $16^{12321812321} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 17, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$

$$\begin{aligned}
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1343] & \overset{1}{1}5132181231\overset{1}{5}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -73, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1341] & \overset{221}{1}(12)12231\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 19, \Delta = -84, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& \overset{1}{1}3221 \\
& T_{\min} = 18, T_{\max} = 19, T \in [18, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1340] & \overset{1}{1}31613151322171\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[1339] & \overset{1}{21}712321812231\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1338] & \overset{1}{31}612321812321\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0} \\
[1337] & \overset{1}{2}3218123216131\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0} \\
[1336] & \overset{1}{1}32181231513151\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 19, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 19, T \in [18, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1335] & \overset{1}{1}32181231513151\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 19, \Delta = -81, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 19, T \in [18, 19] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1334] & \overset{1}{1}6123218123141\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[1333] & \overset{1}{1}31613161232161\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 19, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 19, T \in [18, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[1332] & \overset{1}{1}613161316131\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1331] & \overset{1}{221}81232171231\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

[1330] $1\bar{5}132181232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 18, \Delta = -75, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1344] $21(12)12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 19, \Delta = -84, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 19, T \in [18, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[1345] $2221(12)12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 19, \Delta = -84, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 19, T \in [18, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1347] $1^1\bar{5}1232181232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 18, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1348] $1\bar{5}132181232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 18, \Delta = -75, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1342] $21\bar{7}1232181231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 18, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[1349] $1^1\bar{6}1315131613151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 28, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 28, T \in [19, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[1350] $21\bar{8}12321613151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 28, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 28, T \in [19, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[1351] $1\bar{7}123217123151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 28, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 28, T \in [19, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[1352] $1^1\bar{8}1232161^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 28, \Delta = -101, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 28, T \in [19, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

[1353] $1^1\bar{6}1232181223151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 28, \Delta = -101, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 28, T \in [19, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[1357] $2222221(12)1223141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 28, \Delta = -103, \det = 7, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 28, T \in [22, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$

[1359] $21(12)1223141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 16, T_{\max} = 28, \Delta = -103, \det = 7, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 28, T \in [22, 28]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[1358] & 812314141414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 28, \Delta = -100, \det = 7, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 28, T \in [22, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
& 2221 \\
[1354] & 221(12)1223141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 7, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 28, T \in [22, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
& 1^1 \\
[1355] & 32171232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 19, \Delta = -85, \det = 7, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18.57 \rightarrow 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
& 222221 \\
[1356] & 1(12)1223141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 7, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 28, T \in [22, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[1361] & 21^1 7123217123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 28, \Delta = -100, \det = 8, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19.50 \rightarrow 20, T_{\max} = 28, T \in [20, 28] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
& 21^1 \\
[1360] & 1812321613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 28, \Delta = -100, \det = 8, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19.50 \rightarrow 20, T_{\max} = 28, T \in [20, 28] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
& 31^1 3221 \\
[1364] & 221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 20, \Delta = -92, \det = 11, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19.45 \rightarrow 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
& 21 \\
[1363] & 1(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 20, \Delta = -92, \det = 11, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& 31^1 3221 \\
& T_{\min} = 19.45 \rightarrow 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 7 & 2 & 3 \end{pmatrix}_{b_0} \\
& 1^1 \\
[1362] & 16123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 28, \Delta = -101, \det = 11, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19.45 \rightarrow 20, T_{\max} = 28, T \in [20, 28] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 4 & 5 \\ 1 & 4 & 0 & 2 \\ 2 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[1365] & 221^1 812321613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 28, \Delta = -100, \det = 14, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19.57 \rightarrow 20, T_{\max} = 28, T \in [20, 28] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
& 1^1 \\
[1366] & 51316131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 28, \Delta = -102, \det = 19, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19.26 \rightarrow 20, T_{\max} = 28, T \in [20, 28] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 3 \end{pmatrix}_{b_0} \\
& 31 \\
[1367] & 316131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 16, T_{\max} = 18, \Delta = -58, \det = 20, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 17.80 \rightarrow 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 \\ 2 & 5 & 2 & 4 \\ 2 & 5 & 4 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1371] $^{2221}221(12)12231^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 23$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 20.26 \rightarrow 21$, $T_{\max} = 28$, $T \in [21, 28]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$
- [1370] $^{2221}21(12)12231^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 23$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 20.26 \rightarrow 21$, $T_{\max} = 28$, $T \in [21, 28]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$
- [1369] $^{222221}2222221(12)12231^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 23$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 20.26 \rightarrow 21$, $T_{\max} = 28$, $T \in [21, 28]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 6 & 7 \\ 1 & 6 & 0 & 2 \\ 2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$
- [1368] $^{222221}1(12)12231^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 23$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 20.26 \rightarrow 21$, $T_{\max} = 28$, $T \in [21, 28]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$
- [1373] $^{231^1 3221}21(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16$, $T_{\max} = 21$, $\Delta = -93$, $\det = 27$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 19.48 \rightarrow 20$, $T_{\max} = 21$, $T \in [20, 21]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 13 & 4 & 7 \\ 1 & 4 & 0 & 2 \\ 2 & 7 & 2 & 3 \end{pmatrix}_{b_0}$
- [1372] $^{1 231^1 3221}1(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16$, $T_{\max} = 21$, $\Delta = -93$, $\det = 27$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 19.48 \rightarrow 20$, $T_{\max} = 21$, $T \in [20, 21]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 6 & 11 \\ 1 & 6 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0}$
- [1374] $^{1 3221}3221(12)12231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16$, $T_{\max} = 21$, $\Delta = -93$, $\det = 28$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 19.29 \rightarrow 20$, $T_{\max} = 21$, $T \in [20, 21]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [1376] $^{1 151321}161^1 232181223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16$, $T_{\max} = 28$, $\Delta = -101$, $\det = 42$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 19.67 \rightarrow 20$, $T_{\max} = 28$, $T \in [20, 28]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 18 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$
- [1375] $^{1 151321}1^1 8^1 12321^1 61 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16$, $T_{\max} = 28$, $\Delta = -101$, $\det = 42$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 19.67 \rightarrow 20$, $T_{\max} = 28$, $T \in [20, 28]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 28 & 7 & 10 \\ 1 & 7 & 0 & 2 \\ 2 & 10 & 2 & 3 \end{pmatrix}_{b_0}$
- [1377] $^{2231^1 3221}1(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16$, $T_{\max} = 21$, $\Delta = -93$, $\det = 51$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 19.41 \rightarrow 20$, $T_{\max} = 21$, $T \in [20, 21]$
 $\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 33 & 6 & 11 \\ 1 & 6 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0}$
- [1378] $^{1 2315131613151^1}61^1 2315131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16$, $T_{\max} = 28$, $\Delta = -103$, $\det = 56$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 19.57 \rightarrow 20$, $T_{\max} = 28$, $T \in [20, 28]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$
- [1379] $^{1 2315131613151^1}221812321^1 613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16$, $T_{\max} = 28$, $\Delta = -100$, $\det = 86$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 19.65 \rightarrow 20$, $T_{\max} = 28$, $T \in [20, 28]$
 $\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 68 & 11 & 16 \\ 1 & 11 & 0 & 2 \\ 2 & 16 & 2 & 3 \end{pmatrix}_{b_0}$
- [1380] $^{21}1812321^1 613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 16$, $T_{\max} = 28$, $\Delta = -100$, $\det = 86$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 19.65 \rightarrow 20, T_{\max} = 28, T \in [20, 28] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 26 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0} \\
[1381] & 161232181^1 223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 16, T_{\max} = 28, \Delta = -101, \det = 146, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19.63 \rightarrow 20, T_{\max} = 28, T \in [20, 28] \\
& \begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0} \\
[1382] & 51321^1 6131613151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 16, T_{\max} = 28, \Delta = -103, \det = 152, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19.45 \rightarrow 20, T_{\max} = 28, T \in [20, 28] \\
& \begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 68 & 1 & 16 \\ 1 & 1 & -2 & 0 \\ 2 & 16 & 0 & 3 \end{pmatrix}_{b_0} \\
[1383] & 81231^1 4141414141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 16, T_{\max} = 28, \Delta = -100, \det = 167, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19.37 \rightarrow 20, T_{\max} = 28, T \in [20, 28] \\
& \begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 218 & 1 & 30 \\ 1 & 1 & -3 & -1 \\ 2 & 30 & -1 & 3 \end{pmatrix}_{b_0} \\
\\
T = 17 & [1393] 1^1 4132181231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0 \\
[1392] & 1^1 4132181231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0 \\
[1391] & 1^1 2321613161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0 \\
[1390] & 1^1 3216131613151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0 \\
[1389] & 1^1 2321613161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0 \\
[1388] & 21^1 61316131513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 17, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0 \\
[1387] & 1^1 2321712321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0 \\
[1386] & 1^1 3216131613151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0 \\
[1385] & 1^1 31 6131613151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0 \\
[1384] & 1^1 31 6131613151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 3.0 \\
[1411] & 1^1 321712321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1412] & 1^1 51321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1413] & 1^1 51321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1410] & 21^1 613161316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[1409] & 1^1 321 81232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1414] & 1^1 413218123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1415] & 1^1 413218123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18]
\end{aligned}$$

[1408]	$1^1 315132181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$	$T = 17, T_{\max} = 18, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
	$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$	
	$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$	
[1416]	$1^1 232181231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$	$T = 17, T_{\max} = 18, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
	$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$	
	$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$	
[1417]	$1^1 232181231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$	$T = 17, T_{\max} = 18, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
	$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$	
	$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$	
[1418]	$21^1 712321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$	$T = 17, T_{\max} = 18, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
	$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$	
	$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$	
[1419]	$1^1 151316131613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$	$T = 17, T_{\max} = 18, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
	$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$	
	$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$	
[1427]	$1^1 321712321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$	$T = 17, T_{\max} = 18, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
	$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$	
	$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$	
[1426]	$1^1 151316131613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$	$T = 17, T_{\max} = 18, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
	$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$	
	$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$	
[1425]	$21^1 6131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$	$T = 17, T_{\max} = 18, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
	$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$	
	$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$	
[1424]	$1^1 315132181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$	$T = 17, T_{\max} = 18, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
	$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$	
	$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$	
[1423]	$1^1 316131^1 1316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$	$T = 17, T_{\max} = 18, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
	$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$	
	$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$	
[1422]	$221^1 812321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$	$T = 17, T_{\max} = 18, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
	$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$	
	$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$	
[1421]	$1^1 131613161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$	$T = 17, T_{\max} = 18, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
	$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$	
	$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$	
[1420]	$1^1 81232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$	$T = 17, T_{\max} = 18, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
	$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$	
	$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$	

$$\begin{aligned}
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1440] & 1^1 513161232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 20, \Delta = -80, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 20, T \in [19, 20] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1441] & 1^1 61232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 19, \Delta = -74, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1443] & 1^1 232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 19, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0} \\
[1444] & 1^1 3221 221 (12) 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 20, \Delta = -83, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 20, T \in [19, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[1445] & 1^1 51316131 13161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 25, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 25, T \in [20, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1446] & 1^1 71232181223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 25, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 25, T \in [20, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1449] & 1^1 513161315132171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 29, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 29, T \in [20, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[1448] & 1^1 7123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 25, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 25, T \in [20, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1447] & 1^1 218123217123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 25, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 25, T \in [20, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1452] & 1^1 513221 21 (12) 12222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -5, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 24, T \in [21, 24] \\
& \begin{pmatrix} 5 & 5 \\ 5 & 5 \end{pmatrix}_{b_0} \\
[1451] & 1^1 513221 1 (12) 1222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -5, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 24, T \in [21, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[1453] & 1^1 513221 221 (12) 1222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -5, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 24, T \in [21, 24]
\end{aligned}$$

$$\begin{aligned}
& \left(\begin{smallmatrix} 5 & 5 \\ 5 & 5 \end{smallmatrix} \right)_{b_0} \\
[1450] \quad & 1 \quad \begin{smallmatrix} 221 \\ (12) \end{smallmatrix} \quad 122 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -5, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& \begin{smallmatrix} 1^1 1^1 \\ 5 \end{smallmatrix} 13221 \\
& T_{\min} = 21, T_{\max} = 24, T \in [21, 24] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{smallmatrix} \right)_{b_0} \\
[1454] \quad & 1 \quad \begin{smallmatrix} 1^1 1^1 \\ 5 \end{smallmatrix} 13221 \\
& 122222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -5, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 24, T \in [21, 24] \\
& \left(\begin{smallmatrix} 5 & 5 \\ 5 & 5 \end{smallmatrix} \right)_{b_0} \\
[1455] \quad & 1 \quad \begin{smallmatrix} 1 \\ (12) \end{smallmatrix} \quad 12222 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -5, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& \begin{smallmatrix} 1^1 1^1 \\ 5 \end{smallmatrix} 13221 \\
& T_{\min} = 21, T_{\max} = 24, T \in [21, 24] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{smallmatrix} \right)_{b_0} \\
[1456] \quad & 2222221(12)12231 \begin{smallmatrix} 1^1 \\ 5 \end{smallmatrix} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -5, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 24, T \in [21, 24] \\
& \left(\begin{smallmatrix} 5 & 5 \\ 5 & 5 \end{smallmatrix} \right)_{b_0} \\
[1457] \quad & 21 \quad \begin{smallmatrix} 221 \\ (12) \end{smallmatrix} \quad 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 24, \Delta = -98, \det = -5, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& \begin{smallmatrix} 1^1 1^1 \\ 5 \end{smallmatrix} 13221 \\
& T_{\min} = 21, T_{\max} = 24, T \in [21, 24] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{smallmatrix} \right)_{b_0} \\
[1458] \quad & 222221(12)122314141 \begin{smallmatrix} 1^1 \\ 5 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 29, \Delta = -102, \det = -7, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 29, T \in [23, 29] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{smallmatrix} \right)_{b_0} \\
[1459] \quad & 221 \begin{smallmatrix} 221 \\ (12) \end{smallmatrix} 122314141 \begin{smallmatrix} 1^1 \\ 5 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 29, \Delta = -102, \det = -7, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 29, T \in [23, 29] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{smallmatrix} \right)_{b_0} \\
[1460] \quad & 21 \begin{smallmatrix} 2221 \\ (12) \end{smallmatrix} 122314141 \begin{smallmatrix} 1^1 \\ 5 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 29, \Delta = -102, \det = -7, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 29, T \in [23, 29] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{smallmatrix} \right)_{b_0} \\
[1461] \quad & 3221 \begin{smallmatrix} 1^1 \\ 8 \end{smallmatrix} 1232181231 \begin{smallmatrix} 1^1 \\ 5 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 20, \Delta = -84, \det = -7, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19.57 \rightarrow 20, T_{\max} = 20, T \in [20, 20] \\
& \left(\begin{smallmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 4 \end{smallmatrix} \right)_{b_0} \\
[1462] \quad & 1 \begin{smallmatrix} 22221 \\ (12) \end{smallmatrix} 122314141 \begin{smallmatrix} 1^1 \\ 5 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 29, \Delta = -102, \det = -7, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 29, T \in [23, 29] \\
& \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{smallmatrix} \right)_{b_0} \\
[1463] \quad & 1 \quad \begin{smallmatrix} 1^1 51321 \\ 8 \end{smallmatrix} \quad 1232171 \begin{smallmatrix} 1^1 \\ 5 \end{smallmatrix} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 25, \Delta = -99, \det = -8, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.50 \rightarrow 21, T_{\max} = 25, T \in [21, 25] \\
& \left(\begin{smallmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{smallmatrix} \right)_{b_0} \\
[1464] \quad & 1 \begin{smallmatrix} 21^1 \\ 8 \end{smallmatrix} 123217123151 \begin{smallmatrix} 1^1 \\ 5 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 25, \Delta = -99, \det = -8, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.50 \rightarrow 21, T_{\max} = 25, T \in [21, 25]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[1465] & 21^{1^1} 71232181223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 25, \Delta = -99, \det = -8, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.50 \rightarrow 21, T_{\max} = 25, T \in [21, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[1466] & 21^{1^1} 7123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 25, \Delta = -99, \det = -11, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.45 \rightarrow 21, T_{\max} = 25, T \in [21, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 4 & 5 \\ 1 & 4 & 0 & 2 \\ 2 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[1467] & 21^{31^1 3221} (12) 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 21, \Delta = -91, \det = -11, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.45 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 22 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 2 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[1468] & 1712321^{1^1} 8123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 25, \Delta = -99, \det = -11, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.45 \rightarrow 21, T_{\max} = 25, T \in [21, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0} \\
[1469] & 21^{1^1 3221} (12) 1223141^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 21, \Delta = -91, \det = -11, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.45 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[1470] & 221^{1^1} 8123217123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 25, \Delta = -99, \det = -14, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.57 \rightarrow 21, T_{\max} = 25, T \in [21, 25] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[1471] & 21^{1^1 321} 81232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 20, \Delta = -84, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 24 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 4 \end{pmatrix}_{b_0} \\
[1472] & 161316^{1^1} 131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 25, \Delta = -99, \det = -19, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.26 \rightarrow 21, T_{\max} = 25, T \in [21, 25] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 3 \end{pmatrix}_{b_0} \\
[1473] & 41^{3221} 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 21, \Delta = -92, \det = -24, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.50 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 4 & 2 & 2 \\ 4 & 10 & 5 & 6 \\ 2 & 5 & 1 & 3 \\ 2 & 6 & 3 & 3 \end{pmatrix}_{b_0} \\
[1474] & 21^{1^1 3221} (12) 12231^1 413 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 21, \Delta = -91, \det = -27, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.48 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 13 & 4 & 7 \\ 1 & 4 & 0 & 2 \\ 2 & 7 & 2 & 3 \end{pmatrix}_{b_0} \\
[1475] & 1^{231^1 3221} (12) 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 21, \Delta = -92, \det = -27, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.48 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 58 & 1 & 16 \\ 1 & 1 & -3 & -1 \\ 2 & 16 & -1 & 3 \end{pmatrix}_{b_0} \\
[1476] & 1^{1^1 3221} (12) 1223141^1 32 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 17, T_{\max} = 21, \Delta = -92, \det = -27, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20.48 \rightarrow 21, T_{\max} = 21, T \in [21, 21]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 13 & 4 & 7 \\ 1 & 4 & 0 & 2 \\ 2 & 7 & 2 & 3 \end{pmatrix}_{b_0} \\
[1477] & \begin{matrix} 3221 \\ 3221(12)1223141^{11} + [\text{su}_2 \cdot \text{g}_2] \quad T = 17, T_{\max} = 21, \Delta = -92, \det = -28, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 20.29 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \end{matrix} \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[1478] & \begin{matrix} 3221 \\ 413221(12)12231^{11} + [\text{su}_2 \cdot \text{g}_2] \quad T = 17, T_{\max} = 21, \Delta = -92, \det = -28, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 20.29 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \end{matrix} \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 5 & 4 \\ 2 & 5 & 1 & 3 \\ 2 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[1479] & \begin{matrix} 22221 \\ 1(12)12231^1 4141^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 17, T_{\max} = 29, \Delta = -102, \det = -47, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 20.79 \rightarrow 21, T_{\max} = 29, T \in [21, 29] \end{matrix} \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 2 & 12 & -1 & 3 \end{pmatrix}_{b_0} \\
[1480] & \begin{matrix} 221 \\ 221(12)12231^1 4141^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 17, T_{\max} = 29, \Delta = -102, \det = -47, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 20.79 \rightarrow 21, T_{\max} = 29, T \in [21, 29] \end{matrix} \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 2 & 12 & -1 & 3 \end{pmatrix}_{b_0} \\
[1481] & \begin{matrix} 2221 \\ 21(12)12231^1 4141^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 17, T_{\max} = 29, \Delta = -102, \det = -47, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 20.79 \rightarrow 21, T_{\max} = 29, T \in [21, 29] \end{matrix} \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 2 & 12 & -1 & 3 \end{pmatrix}_{b_0} \\
[1482] & \begin{matrix} 222221 \\ 222221(12)12231^1 4141^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 17, T_{\max} = 29, \Delta = -102, \det = -47, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 20.79 \rightarrow 21, T_{\max} = 29, T \in [21, 29] \end{matrix} \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 8 & 11 \\ 1 & 8 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0} \\
[1483] & \begin{matrix} 2231^1 413221(12)12231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 17, T_{\max} = 21, \Delta = -92, \det = -51, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 20.41 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \end{matrix} \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 61 & 1 & 15 \\ 1 & 1 & -2 & 0 \\ 2 & 15 & 0 & 3 \end{pmatrix}_{b_0} \\
[1484] & \begin{matrix} 2231^1 3221(12)1223141^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 17, T_{\max} = 21, \Delta = -92, \det = -51, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 20.41 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \end{matrix} \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 33 & 6 & 11 \\ 1 & 6 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0} \\
[1485] & \begin{matrix} 1^1 3221 \\ 1(12)12231^1 4132 + [\text{su}_2 \cdot \text{g}_2] \quad T = 17, T_{\max} = 21, \Delta = -92, \det = -51, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 20.41 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \end{matrix} \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 33 & 6 & 11 \\ 1 & 6 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0} \\
[1486] & \begin{matrix} 61^1 23216131613151^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 17, T_{\max} = 29, \Delta = -102, \det = -56, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 20.57 \rightarrow 21, T_{\max} = 29, T \in [21, 29] \end{matrix} \\
& \begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0} \\
[1487] & \begin{matrix} 71^1 22315131613151^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 17, T_{\max} = 29, \Delta = -101, \det = -72, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 20.50 \rightarrow 21, T_{\max} = 29, T \in [21, 29] \end{matrix} \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 40 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0} \\
[1488] & \begin{matrix} 223141^1 3221(12)12231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 17, T_{\max} = 21, \Delta = -92, \det = -83, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 20.35 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \end{matrix}
\end{aligned}$$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 97 & 1 & 19 \\ 1 & 1 & -2 & 0 \\ 2 & 19 & 0 & 3 \end{pmatrix}_{b_0}$$

[1489] $3221(12)12231^1 41^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 17, T_{\max} = 21, \Delta = -92, \det = -92, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.09 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$

$$\begin{pmatrix} 0 & 6 & 2 & 2 \\ 6 & 22 & 9 & 10 \\ 2 & 9 & 1 & 3 \\ 2 & 10 & 3 & 3 \end{pmatrix}_{b_0}$$

[1490] $171232181^1 223151^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 17, T_{\max} = 25, \Delta = -99, \det = -146, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.63 \rightarrow 21, T_{\max} = 25, T \in [21, 25]$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

[1491] $2171232181^1 223151^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 17, T_{\max} = 25, \Delta = -99, \det = -146, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.63 \rightarrow 21, T_{\max} = 25, T \in [21, 25]$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

[1492] $612321^1 6131613151^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 17, T_{\max} = 29, \Delta = -102, \det = -152, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.45 \rightarrow 21, T_{\max} = 29, T \in [21, 29]$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 68 & 1 & 16 \\ 1 & 1 & -2 & 0 \\ 2 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

$T = 18$ [1494] $1^1 41316131613151321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0$

[1493] $1^1 41316131613151321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0$

[1495] $13161316131513221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 18, T_{\max} = 18, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0$

[1496] $1^1 32161316131612321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0$

[1497] $1^1 8^{12321} 12321712321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0$

[1498] $1^1 6^{131} 1316131612321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0$

[1499] $1^1 32161316131612321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0$

[1500] $1^1 8^{12321} 12321712321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0$

[1501] $1^1 6^{131} 1316131612321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0$

[1507] $1^1 22315132181232151^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 18, T_{\max} = 18, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0$

[1506] $1^1 22315132181232151^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 18, T_{\max} = 18, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0$

[1505] $1^1 31513218123151321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0$

[1504] $1^1 5^{131} 132181231513221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 18, T_{\max} = 18, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0$

[1503] $1^1 31513218123151321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -32, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0$

[1502] $1^1 5^{131} 132181231513221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 18, T_{\max} = 18, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 3.0$

[1530] $1^1 5^{131} 13218123216131^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

[1529] $1^1 5^{131} 13218123216131^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

[1547] $1^1 6123151321812321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

[1546] $1^1 4132181232171231^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

- [1533] $1^1 5131612321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [1545] $13218^{11} 12321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [1544] $1^1 3221812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
- [1543] $1^{1321} 1^8 12321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [1542] $1^1 3161315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [1541] $1^1 3161315132181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
- [1540] $1^{1321} 1^8 12321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
- [1539] $216^{11} 1316131613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$
- [1538] $1^1 51316131613151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
- [1537] $1^{1321} 1^8 1232181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$
- [1536] $1^1 5131612321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$
- [1532] $1^1 3161232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
- [1535] $2218^{11} 12321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [1534] $1^1 1613151321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1527] & 1^1 3161232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1528] & 1^1 4132181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1531] & 1^1 1612321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1550] & 1^{21} (12) 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 20, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& 1^1 1513221 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1564] & 1^{21} (12) 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 20, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& 1^1 1513221 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1549] & 1^1 3221 21 (12) 122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -82, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 21, T \in [20, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[1548] & 1^1 2171232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 20, \Delta = -72, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1553] & 1^1 2218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 20, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[1554] & 1^1 2181232161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 20, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1555] & 1^1 1712315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 20, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1556] & 1^1 1613161232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 20, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1551] & 1^1 217123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 20, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1557] & \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 413221 \quad 21 \quad (12) \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -82, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 21, T \in [20, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1558] & \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 513221 \quad 221 \quad (12) \quad 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 20, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1559] & \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 3221 \quad 21 \quad (12) \quad 122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -82, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 21, T \in [20, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[1560] & \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 3161316131513161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 20, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[1561] & \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 5132171232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 21, T \in [20, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1552] & \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 7122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 20, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1562] & \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 5132171232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -79, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 21, T \in [20, 21] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1563] & \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 513221 \quad 221 \quad (12) \quad 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 20, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1572] & \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 5131613151322181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 30, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 30, T \in [21, 30] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[1566] & \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 81232181223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 25, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 25, T \in [21, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1565] & \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 5131613161232171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 30, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 30, T \in [21, 30] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[1567] & \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} \quad 1 \quad (12) \quad 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -86, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0 \\
& \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 31513221 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 3 & 3 \\ 2 & 3 & 4 & 4 \\ 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[1568] $21 \begin{smallmatrix} 1 \\ 31 \ 5 \ 13221 \end{smallmatrix} (12) \ 12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 18, T_{\max} = 21, \Delta = -86, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$

[1569] $18 \begin{smallmatrix} 1 \\ 12315131613151 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 18, T_{\max} = 30, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 30, T \in [21, 30]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[1570] $1^1 6131513218123151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 18, T_{\max} = 30, \Delta = -101, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 30, T \in [21, 30]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[1571] $21 \begin{smallmatrix} 1 \\ 18 \ 123218123151 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 18, T_{\max} = 25, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 25, T \in [21, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[1578] $21 \begin{smallmatrix} 1 \\ 31 \ 5 \ 13221 \end{smallmatrix} (12) \ 12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 18, T_{\max} = 21, \Delta = -86, \det = 7, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.57 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$

[1577] $21 \begin{smallmatrix} 221 \\ (12)12231414141 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 18, T_{\max} = 30, \Delta = -101, \det = 7, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$

[1576] $22221(12)12231414141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 18, T_{\max} = 30, \Delta = -101, \det = 7, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$

[1575] $3221 \begin{smallmatrix} 1 \\ 1^1 513221 \end{smallmatrix} (12) \ 12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 18, T_{\max} = 21, \Delta = -87, \det = 7, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.57 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 2 & 4 \\ 3 & 4 & 5 \end{pmatrix}_{b_0}$

[1574] $1 \begin{smallmatrix} 2221 \\ (12)12231414141 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 18, T_{\max} = 30, \Delta = -101, \det = 7, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 30, T \in [24, 30]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$

[1573] $413218 \begin{smallmatrix} 1 \\ 1232181231 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 18, T_{\max} = 21, \Delta = -83, \det = 7, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.57 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$

[1579] $1 \begin{smallmatrix} 1 \\ (12) \ 12231 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 18, T_{\max} = 21, \Delta = -86, \det = 7, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.57 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 3 & 5 \\ 1 & 3 & 0 & 2 \\ 2 & 5 & 2 & 4 \end{pmatrix}_{b_0}$

[1582] $18 \begin{smallmatrix} 21 \\ 1232181223151 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 18, T_{\max} = 25, \Delta = -98, \det = 8, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.50 \rightarrow 22, T_{\max} = 25, T \in [22, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$

- [1581] $21^1 812315131613151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 30, \Delta = -100, \det = 8, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.50 \rightarrow 22, T_{\max} = 30, T \in [22, 30]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [1580] $21^1 1^1 8123218123151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 25, \Delta = -98, \det = 8, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.50 \rightarrow 22, T_{\max} = 25, T \in [22, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [1583] $1^1 513221 (12) 12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 21, \Delta = -87, \det = 10, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.60 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 4 \end{pmatrix}_{b_0}$
- [1585] $221812321^1 8123151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 25, \Delta = -98, \det = 11, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.45 \rightarrow 22, T_{\max} = 25, T \in [22, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 4 & 5 \\ 1 & 4 & 0 & 2 \\ 2 & 5 & 2 & 3 \end{pmatrix}_{b_0}$
- [1584] $31^1 413221 (12) 1223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 22, \Delta = -90, \det = 11, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.45 \rightarrow 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 4 & 7 \\ 1 & 4 & 0 & 2 \\ 2 & 7 & 2 & 3 \end{pmatrix}_{b_0}$
- [1586] $1^1 3221 1 (12) 122314141^1 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 22, \Delta = -90, \det = 11, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.45 \rightarrow 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [1587] $31^1 3221 1 (12) 122314141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 22, \Delta = -90, \det = 11, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.45 \rightarrow 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 22 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 2 & 10 & -1 & 3 \end{pmatrix}_{b_0}$
- [1588] $21^1 1812321^1 8123151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 25, \Delta = -98, \det = 11, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.45 \rightarrow 22, T_{\max} = 25, T \in [22, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 3 \end{pmatrix}_{b_0}$
- [1589] $231^1 513221 1 (12) 12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 21, \Delta = -87, \det = 13, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.77 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 5 & 3 & 5 \\ 1 & 3 & 0 & 2 \\ 2 & 5 & 2 & 4 \end{pmatrix}_{b_0}$
- [1590] $1^1 51321 1^1 81232181^1 22 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 25, \Delta = -98, \det = 14, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.57 \rightarrow 22, T_{\max} = 25, T \in [22, 25]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [1591] $221^1 81232181232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 25, \Delta = -98, \det = 14, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.57 \rightarrow 22, T_{\max} = 25, T \in [22, 25]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [1592] $31^1 3161316131^1 6131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 19, \Delta = -34, \det = 16, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 \\ 2 & 5 & 3 & 5 \\ 2 & 5 & 5 & 6 \end{pmatrix}_{b_0}$

- [1593] $3221 \overset{1}{\overset{1}{5}} 13221 (12) 1223 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -87, \det = 20, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.80 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 2 & 4 & 3 & 4 \end{pmatrix}_{b_0}$
- [1594] $2231 \overset{1}{\overset{1}{5}} 13221 (12) 12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 21, \Delta = -87, \det = 21, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 20.86 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 11 & 4 & 7 \\ 1 & 4 & 0 & 2 \\ 2 & 7 & 2 & 4 \end{pmatrix}_{b_0}$
- [1595] $231^1 413221 (12) 1223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 22, \Delta = -91, \det = 27, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.48 \rightarrow 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 13 & 4 & 7 \\ 1 & 4 & 0 & 2 \\ 2 & 7 & 2 & 3 \end{pmatrix}_{b_0}$
- [1596] $3141^1 3221 (12) 1223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 22, \Delta = -90, \det = 27, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.48 \rightarrow 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 6 & 11 \\ 1 & 6 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0}$
- [1597] $231^1 41413221 (12) 12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 22, \Delta = -91, \det = 27, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.48 \rightarrow 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 1 & 11 \\ 1 & 1 & -2 & 0 \\ 2 & 11 & 0 & 3 \end{pmatrix}_{b_0}$
- [1598] $231^1 3221 (12) 122314141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 22, \Delta = -91, \det = 27, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.48 \rightarrow 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 13 & 4 & 7 \\ 1 & 4 & 0 & 2 \\ 2 & 7 & 2 & 3 \end{pmatrix}_{b_0}$
- [1599] $41^1 3221 \overset{1}{\overset{1}{5}} 13221 (12) 122 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 30, \Delta = -106, \det = 30, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.80 \rightarrow 22, T_{\max} = 30, T \in [22, 30]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [1600] $2313221 (12) 12231^1 4141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 22, \Delta = -91, \det = 47, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.79 \rightarrow 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 8 & 11 \\ 1 & 8 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0}$
- [1601] $1 \overset{3}{1} 3221 (12) 12231^1 4141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 22, \Delta = -90, \det = 47, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.79 \rightarrow 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$
- [1602] $23141^1 3221 (12) 1223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 22, \Delta = -91, \det = 51, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.41 \rightarrow 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 33 & 6 & 11 \\ 1 & 6 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0}$
- [1603] $1 \overset{1}{\overset{1}{3}} 3221 (12) 12231^1 41413 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 22, \Delta = -90, \det = 51, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.41 \rightarrow 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 33 & 6 & 11 \\ 1 & 6 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0}$
- [1604] $61^1 231513218123151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 30, \Delta = -102, \det = 56, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.57 \rightarrow 22, T_{\max} = 30, T \in [22, 30]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$
- [1605] $1 \overset{2321}{\overset{1}{\overset{1}{5}}} 13221 (12) 12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 22, \Delta = -89, \det = 58, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.41 \rightarrow 22, T_{\max} = 22, T \in [22, 22]$

$$\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 38 & 6 & 12 \\ 1 & 6 & 0 & 2 \\ 2 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[1606] $181^1 22315131613151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 30, \Delta = -100, \det = 72, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.50 \rightarrow 22, T_{\max} = 30, T \in [22, 30]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 54 & 9 & 14 \\ 1 & 9 & 0 & 2 \\ 2 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

[1607] $22221(12)12231^1 414141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 30, \Delta = -101, \det = 79, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 21.57 \rightarrow 22, T_{\max} = 30, T \in [22, 30]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 61 & 10 & 15 \\ 1 & 10 & 0 & 2 \\ 2 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1609] $21^{221}(12)12231^1 414141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 30, \Delta = -101, \det = 79, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.57 \rightarrow 22, T_{\max} = 30, T \in [22, 30]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 82 & 1 & 18 \\ 1 & 1 & -3 & -1 \\ 2 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[1608] $1^{221}(12)12231^1 414141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 30, \Delta = -101, \det = 79, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.57 \rightarrow 22, T_{\max} = 30, T \in [22, 30]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 82 & 1 & 18 \\ 1 & 1 & -3 & -1 \\ 2 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[1610] $2314141^1 3221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 22, \Delta = -91, \det = 83, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.35 \rightarrow 22, T_{\max} = 22, T \in [22, 22]$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 97 & 1 & 19 \\ 1 & 1 & -2 & 0 \\ 2 & 19 & 0 & 3 \end{pmatrix}_{b_0}$$

[1612] $22181232181^1 223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 25, \Delta = -98, \det = 146, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.63 \rightarrow 22, T_{\max} = 25, T \in [22, 25]$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

[1611] $18^{21}1232181^1 223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 25, \Delta = -98, \det = 146, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.63 \rightarrow 22, T_{\max} = 25, T \in [22, 25]$

$$\begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

[1613] $1712321^1 6131613151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 30, \Delta = -100, \det = 152, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.45 \rightarrow 22, T_{\max} = 30, T \in [22, 30]$

$$\begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 128 & 14 & 22 \\ 1 & 14 & 0 & 2 \\ 2 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1614] $513161^1 23218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 18, T_{\max} = 30, \Delta = -103, \det = 189, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21.45 \rightarrow 22, T_{\max} = 30, T \in [22, 30]$

$$\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 75 & 1 & 17 \\ 1 & 1 & -2 & 0 \\ 2 & 17 & 0 & 3 \end{pmatrix}_{b_0}$$

$T = 19$ [1621] $1^1 151316131613151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 19, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0$

[1619] $1^1 413161316131612321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 19, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0$

[1623] $1^1 151316131613151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 19, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0$

[1622] $1^1 8^{12321} 12321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 20, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0$

[1615] $1^1 316123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 20, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0$

[1616] $1^1 161232181231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 19, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0$

[1620] $1^1 316123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 20, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0$

[1617] $1^1 413161316131612321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 19, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0$

[1618] $1^1 315131613161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 20, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 3.0$

[1653] $221^{1^1 31513221}(12) 12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 20, \Delta = -57, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1658] & 1^1 16131612321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1659] & 1^{21} (12) 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -57, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& 1^{131513221} \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1660] & 1^1 2161316131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1661] & 1^1 71223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1662] & 1^1 51321712321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1663] & 1^{131513221} 221 (12) 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -57, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[1642] & 1^{21} (12) 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -57, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& 1^{131513221} \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[1643] & 1^1 51321712321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 20, \Delta = -52, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 20, T_{\max} = 20, T \in [20, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[1667] & 1^{141413221} (12) 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 22, \Delta = -81, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 22, T \in [21, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1666] & 1^1 18122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 21, \Delta = -76, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1668] & 1^1 2218123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 21, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1669] & 1^{151321} 1^1 8 1232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 21, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1670] $1^{\overline{151321}} 8^{1232181231^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 21, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [1671] $21^{1^{\overline{1513221}}} (12)^{1223141^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 21, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$
- [1672] $1^{\overline{161232171232181231^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 21, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [1673] $218^{1^{\overline{12315132181231^1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 21, \Delta = -76, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [1674] $1^{\overline{131613161315132171^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 21, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$
- [1675] $1^{\overline{132181232181223151^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 21, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [1676] $1^{\overline{132181232181223151^1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 21, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$
- [1665] $21^{1^{\overline{1513221}}} (12)^{1223141^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 21, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$
- [1664] $1^{1^{\overline{141413221}}} (12)^{1223141^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 22, \Delta = -81, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 22, T \in [21, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$
- [1677] $2218^{1^{\overline{1232181232151^1}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 21, \Delta = -71, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$
- [1680] $1^{1^{\overline{13221}}} (12)^{12231414141^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 22, \Delta = -81, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 22, T \in [21, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$
- [1679] $1^{1^{\overline{13221}}} (12)^{12231414141^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 22, \Delta = -81, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 22, T \in [21, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$
- [1678] $1^{\overline{13161316131}} 6^{13161^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 19, T_{\max} = 21, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1686] & \begin{matrix} 31 & 5 & 13221 \\ 1 & (12) & 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 22, \Delta = -85, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 8 & 6 \\ 2 & 0 & 6 & 4 \end{pmatrix}_{b_0} \\
[1685] & \begin{matrix} 1 & 513221 \\ 221 & (12) & 1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 31, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 31, T \in [22, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1684] & \begin{matrix} 1 & 161316123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 31, T \in [22, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1683] & \begin{matrix} 1 & 171231513218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 31, T \in [22, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1682] & \begin{matrix} 21 \\ 1 & (12) & 1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 31, \Delta = -101, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& \begin{matrix} 1 & 513221 \\ T_{\min} = 22, T_{\max} = 31, T \in [22, 31] \end{matrix} \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1681] & \begin{matrix} 1 \\ 1 & 8123216131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 27, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 27, T \in [22, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1689] & \begin{matrix} 2221(12)1223141414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -7, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[1687] & \begin{matrix} 1 \\ 51232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 22, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.57 \rightarrow 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
[1691] & \begin{matrix} 31 & 5 & 13221 \\ 1 & (12) & 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 22, \Delta = -85, \det = -7, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.57 \rightarrow 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 9 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 2 & 7 & -1 & 4 \end{pmatrix}_{b_0} \\
[1688] & \begin{matrix} 221 \\ 1(12)1223141414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -7, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[1690] & \begin{matrix} 21 \\ 21(12)1223141414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -7, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 31, T \in [25, 31] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
[1692] & \begin{matrix} 21 & 18123216131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \end{matrix} \quad T = 19, T_{\max} = 27, \Delta = -99, \det = -8, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.50 \rightarrow 23, T_{\max} = 27, T \in [23, 27]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[1694] \quad & 31^1 4141413221(12)12231^{\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 23, \Delta = -89, \det = -11, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.45 \rightarrow 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 3 \end{pmatrix}_{b_0} \\
[1695] \quad & 31^1 3221(12)12231414141^{\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 23, \Delta = -89, \det = -11, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.45 \rightarrow 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[1696] \quad & 31^1 413221(12)122314141^{\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 23, \Delta = -89, \det = -11, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.45 \rightarrow 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[1693] \quad & 31^1 41413221(12)1223141^{\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 23, \Delta = -89, \det = -11, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.45 \rightarrow 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 22 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 2 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[1697] \quad & 231^1 513221(12)1223141^{\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 22, \Delta = -86, \det = -13, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21.77 \rightarrow 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 5 & 3 & 5 \\ 1 & 3 & 0 & 2 \\ 2 & 5 & 2 & 4 \end{pmatrix}_{b_0} \\
[1698] \quad & 3141^1 3221(12)122314141^{\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 23, \Delta = -89, \det = -27, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.48 \rightarrow 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 13 & 4 & 7 \\ 1 & 4 & 0 & 2 \\ 2 & 7 & 2 & 3 \end{pmatrix}_{b_0} \\
[1699] \quad & 31413221(12)12231^1 4141^{\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 23, \Delta = -89, \det = -47, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.79 \rightarrow 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 8 & 11 \\ 1 & 8 & 0 & 2 \\ 2 & 11 & 2 & 3 \end{pmatrix}_{b_0} \\
[1700] \quad & 314141^1 3221(12)1223141^{\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 23, \Delta = -89, \det = -51, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.41 \rightarrow 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 110 & 1 & 22 \\ 1 & 1 & -3 & -1 \\ 2 & 22 & -1 & 3 \end{pmatrix}_{b_0} \\
[1701] \quad & 2321^1 513221(12)1223141^{\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 23, \Delta = -88, \det = -58, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.41 \rightarrow 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 38 & 6 & 12 \\ 1 & 6 & 0 & 2 \\ 2 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[1702] \quad & 71^1 2231513218123151^{\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -72, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.50 \rightarrow 23, T_{\max} = 31, T \in [23, 31] \\
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 40 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0} \\
[1703] \quad & 313221(12)12231^1 414141^{\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 23, \Delta = -89, \det = -79, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.57 \rightarrow 23, T_{\max} = 23, T \in [23, 23]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 61 & 10 & 15 \\ 1 & 10 & 0 & 2 \\ 2 & 15 & 2 & 3 \end{pmatrix}_{b_0} \\
[1704] & 31414141^1 3221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 23, \Delta = -89, \det = -83, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.35 \rightarrow 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 97 & 1 & 19 \\ 1 & 1 & -2 & 0 \\ 2 & 19 & 0 & 3 \end{pmatrix}_{b_0} \\
[1705] & 4141^1 3221^{1^1 513221} (12) 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 31, \Delta = -105, \det = -106, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.57 \rightarrow 23, T_{\max} = 31, T \in [23, 31] \\
& \begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 38 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0} \\
[1706] & 21^{21} (12)12231^1 41414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -119, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.45 \rightarrow 23, T_{\max} = 31, T \in [23, 31] \\
& \begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 142 & 1 & 24 \\ 1 & 1 & -3 & -1 \\ 2 & 24 & -1 & 3 \end{pmatrix}_{b_0} \\
[1707] & 2221(12)12231^1 41414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -119, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.45 \rightarrow 23, T_{\max} = 31, T \in [23, 31] \\
& \begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 97 & 12 & 19 \\ 1 & 12 & 0 & 2 \\ 2 & 19 & 2 & 3 \end{pmatrix}_{b_0} \\
[1708] & 1^{221} (12)12231^1 41414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -119, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.45 \rightarrow 23, T_{\max} = 31, T \in [23, 31] \\
& \begin{pmatrix} 0 & 9 & 1 & 2 \\ 9 & 142 & 1 & 24 \\ 1 & 1 & -3 & -1 \\ 2 & 24 & -1 & 3 \end{pmatrix}_{b_0} \\
[1709] & 1812321^1 6131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 27, \Delta = -99, \det = -152, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.45 \rightarrow 23, T_{\max} = 27, T \in [23, 27] \\
& \begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 68 & 1 & 16 \\ 1 & 1 & -2 & 0 \\ 2 & 16 & 0 & 3 \end{pmatrix}_{b_0} \\
[1710] & 21812321^1 6131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 27, \Delta = -99, \det = -152, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.45 \rightarrow 23, T_{\max} = 27, T \in [23, 27] \\
& \begin{pmatrix} 0 & 10 & 1 & 2 \\ 10 & 128 & 14 & 22 \\ 1 & 14 & 0 & 2 \\ 2 & 22 & 2 & 3 \end{pmatrix}_{b_0} \\
[1711] & 1613161^1 23218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 19, T_{\max} = 31, \Delta = -100, \det = -189, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22.45 \rightarrow 23, T_{\max} = 31, T \in [23, 31] \\
& \begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 75 & 1 & 17 \\ 1 & 1 & -2 & 0 \\ 2 & 17 & 0 & 3 \end{pmatrix}_{b_0} \\
T = 20 & [1716] 1^1 3216131613161315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 3.0 \\
[1712] & 21^1 71232181231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 20, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 3.0 \\
[1718] & 1^1 3217123218123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 3.0 \\
[1719] & 1^{21} (12) 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 3.0 \\
& 1^{1^1 2231513221} \\
[1720] & 221^{1^1 2231513221} (12) 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 3.0 \\
[1721] & 1^{1^1 31} 6 131613161315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 20, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 3.0 \\
[1722] & 1^{1^1 31} 6 131613161315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 20, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 3.0 \\
[1723] & 1^1 2321712321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 3.0
\end{aligned}$$

[1724] $^{1^1 2231513221} 221 \quad (12) \quad 12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 3.0$

[1725] $^{1^1 2321712321812315131^1} 1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 3.0$

[1726] $^{1^1} 21^6 1316131613151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 20, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 3.0$

[1727] $^{1^1} 15^1 13161316131612321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 20, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 3.0$

[1713] $^{21} 1 \quad (12) \quad 12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 3.0$

[1717] $^{1^1 2231513221} 1^1 3216131613161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 3.0$

[1714] $^{1^1} 15^1 13161316131612321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 20, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 3.0$

[1715] $^{1^1} 1^1 3217123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 3.0$

[1768] $^{1^1} 16^1 1232181232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1755] $^{1^1 612321712321812321^1} 1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1756] $^{1^1 321} 1 \quad 8^1 1232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1757] $^{1^1 321} 1 \quad 8^1 1232181231^1 5^1 1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1758] $^{1^1 321} 1 \quad 8^1 1232181231^1 5^1 1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1759] $^{1^1} 15^1 1321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1760] $^{1^1} 15^1 1321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1761] $^{1^1} 1^1 232181232181223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1762] $^{1^1} 1^1 232181232181223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[1763] $^{1^1 31513221} 21 \quad (12) \quad 1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -56, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$

[1764] $^{1^1 31513221} 21 \quad (12) \quad 1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -56, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0} \\
[1765] & 1^{\overline{151321}} 8^{12321812321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1766] & 1^{\overline{151321}} 8^{12321812321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[1767] & 2218^{12321812315131^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1754] & 1^{\overline{1321}} 8^{1232181232151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1769] & 1^{\overline{1321812315132181231^1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -56, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1770] & 218^{123151321812321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1771] & 217^{123218123216131^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1750] & 1^{\overline{1322181232181232151^1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1751] & 18^{1223151321812321^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1752] & 1^{\overline{1322181232181232151^1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1753] & 1^{\overline{1413218123218123141^1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 21, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0} \\
[1780] & 1^{\overline{14141413221(12)1223141^1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 23, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 23, T \in [22, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[1775] & 1^{\overline{1513221}} (12)^{122314141^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 22, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[1778] & 1 \quad \begin{matrix} 1 \\ 1^1 5131513221 \end{matrix} (12) \quad 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 23, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 23, T \in [22, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1779] & 1^1 41413221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 23, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 23, T \in [22, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[1776] & \begin{matrix} 1^1 \\ 1^1 513221 \end{matrix} 1 \quad (12) \quad 122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 22, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[1781] & 1^1 4141413221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 23, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 23, T \in [22, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[1782] & 1^1 414141413221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 23, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 23, T \in [22, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1783] & \begin{matrix} 1^1 \\ 1^1 5131513221 \end{matrix} 21 \quad (12) \quad 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 23, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 23, T \in [22, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1784] & 1 \quad \begin{matrix} 1 \\ 1^1 513221 \end{matrix} (12) \quad 12231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 22, \Delta = -72, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1785] & \begin{matrix} 1^1 \\ 1^1 71232171232181231 \end{matrix} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 22, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1786] & \begin{matrix} 1^1 \\ 1^1 81231513161316131 \end{matrix} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 22, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1787] & 1^1 414141413221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 23, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 23, T \in [22, 23] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1788] & \begin{matrix} 1^1 \\ 1^1 321812321 \end{matrix} 8 \quad 1232161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 20, T_{\max} = 22, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}
\end{aligned}$$

[1789] $1^1 316131613151322181^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 22, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[1772] $1^1 (12) 12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 23, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 23, T \in [22, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[1777] $21^1 (12) 12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 23, \Delta = -80, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 23, T \in [22, 23]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1773] $21^1 (12) 12231^1 51 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 22, \Delta = -72, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1774] $1^1 3161316131613161232171^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 22, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[1791] $1^1 812231513218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 28, \Delta = -99, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[1794] $31^1 513221(12)12231^1 513 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 24, \Delta = -88, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 24, T \in [23, 24]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 5 & 3 & 5 \\ 2 & 3 & 0 & 2 \\ 2 & 5 & 2 & 4 \end{pmatrix}_{b_0}$

[1793] $31^1 513221(12)122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 23, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$

[1792] $1^1 (12) 12231^1 51 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 23, \Delta = -80, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$

[1790] $1^1 81231513218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 28, \Delta = -99, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 28, T \in [23, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[1795] $1^1 (12) 12231^1 51 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 23, \Delta = -80, \det = 7, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22.57 \rightarrow 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$

[1796] $1(12)122314141414141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20, T_{\max} = 28, \Delta = -99, \det = 7, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 28, T \in [26, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$

[1797] $31\overset{1}{5}13221(12)122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20$, $T_{\max} = 23$, $\Delta = -84$, $\det = 7$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 22.57 \rightarrow 23$, $T_{\max} = 23$, $T \in [23, 23]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$

[1798] $221(12)122314141414141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20$, $T_{\max} = 28$, $\Delta = -99$, $\det = 7$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 26$, $T_{\max} = 28$, $T \in [26, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$

[1799] $21\overset{1}{8}1231513218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20$, $T_{\max} = 28$, $\Delta = -99$, $\det = 8$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 23.50 \rightarrow 24$, $T_{\max} = 28$, $T \in [24, 28]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$

[1800] $1\overset{1}{5}131513221\overset{1}{3} + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20$, $T_{\max} = 24$, $\Delta = -88$, $\det = 11$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 23.45 \rightarrow 24$, $T_{\max} = 24$, $T \in [24, 24]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 13 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 3 \end{pmatrix}_{b_0}$

[1801] $1\overset{1}{5}13221(12)12231\overset{1}{5}132 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20$, $T_{\max} = 23$, $\Delta = -81$, $\det = 13$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 22.77 \rightarrow 23$, $T_{\max} = 23$, $T \in [23, 23]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 5 & 3 & 5 \\ 1 & 3 & 0 & 2 \\ 2 & 5 & 2 & 4 \end{pmatrix}_{b_0}$

[1802] $51321\overset{1}{8}123218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20$, $T_{\max} = 32$, $\Delta = -101$, $\det = 19$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 23.26 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}$

[1803] $231\overset{1}{3}221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20$, $T_{\max} = 24$, $\Delta = -89$, $\det = 27$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 23.48 \rightarrow 24$, $T_{\max} = 24$, $T \in [24, 24]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 33 & 1 & 11 \\ 1 & 1 & -2 & 0 \\ 2 & 11 & 0 & 3 \end{pmatrix}_{b_0}$

[1804] $6131\overset{1}{6}1316131613151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20$, $T_{\max} = 27$, $\Delta = -98$, $\det = 29$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 23.17 \rightarrow 24$, $T_{\max} = 27$, $T \in [24, 27]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 11 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 3 \end{pmatrix}_{b_0}$

[1805] $61\overset{1}{2}3217123218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20$, $T_{\max} = 32$, $\Delta = -101$, $\det = 56$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 23.57 \rightarrow 24$, $T_{\max} = 32$, $T \in [24, 32]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$

[1806] $181\overset{1}{2}231513218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20$, $T_{\max} = 28$, $\Delta = -99$, $\det = 72$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 23.50 \rightarrow 24$, $T_{\max} = 28$, $T \in [24, 28]$
 $\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 40 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$

[1807] $1\overset{21}{(12)}12231\overset{1}{4}141414141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20$, $T_{\max} = 28$, $\Delta = -99$, $\det = 167$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 23.37 \rightarrow 24$, $T_{\max} = 28$, $T \in [24, 28]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 218 & 1 & 30 \\ 1 & 1 & -3 & -1 \\ 2 & 30 & -1 & 3 \end{pmatrix}_{b_0}$

[1808] $221(12)12231\overset{1}{4}141414141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 20$, $T_{\max} = 28$, $\Delta = -99$, $\det = 167$, $(n_+, n_-, n_0) = (1, 20, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 23.37 \rightarrow 24$, $T_{\max} = 28$, $T \in [24, 28]$
 $\begin{pmatrix} 0 & 11 & 1 & 2 \\ 11 & 141 & 14 & 23 \\ 1 & 14 & 0 & 2 \\ 2 & 23 & 2 & 3 \end{pmatrix}_{b_0}$

[1809] $414141^1 3221^{1513221} (12) 1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 32, \Delta = -104, \det = 230, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23.45 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 0 & 12 & 1 & 2 \\ 12 & 82 & 1 & 18 \\ 1 & 1 & -2 & 0 \\ 2 & 18 & 0 & 3 \end{pmatrix}_{b_0}$

[1810] $513221^1 8123218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 20, T_{\max} = 32, \Delta = -101, \det = 249, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23.37 \rightarrow 24, T_{\max} = 32, T \in [24, 32]$
 $\begin{pmatrix} 1 & 18 & 3 \\ 18 & 75 & 21 \\ 3 & 21 & 4 \end{pmatrix}_{b_0}$

$T = 21$ [1822] $1^1 6^{131} 1316131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[1816] $1^1 8^{1321} 123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[1817] $1^1 8^{1321} 123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[1818] $1^1 8^{12321} 12321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[1819] $1^1 32218123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[1815] $1^1 6^{131} 1316131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[1821] $1^1 6^{131} 13161316131613151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 21, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[1814] $1^1 8^{12321} 12321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[1823] $1^1 32218123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[1824] $21^{12231513221} (12) 1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[1825] $1^1 41316131613161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[1826] $1^1 31612321812321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[1827] $221^1 8^{12321} 1232181231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[1828] $1^1 31612321812321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[1829] $1^1 32161316131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[1830] $21^{12231513221} (12) 1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[1813] $1^1 41316131613161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[1812] $21^1 6^{131} 13161316131612321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[1811] $1^1 6^{131} 131613161316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 21, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[1820] $1^1 23218123151321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 3.0$

[1859] $1^{131513221} (12) 122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$

[1854] $1^{1513221} (12) 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -51, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1857] $1^{12321812321} 8^{1232161} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 21, T_{\max} = 22, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
[1853] & \begin{matrix} 1 \\ 2218123218123216131 \end{matrix}^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 22, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1869] & \begin{matrix} 1 \\ 1513221(12)12231414141 \end{matrix}^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 23, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1872] & \begin{matrix} 1 \\ 1 \end{matrix}^{15131513221} (12) \quad 1223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 24, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 24, T \in [23, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[1873] & \begin{matrix} 31 \\ 1 \end{matrix}^{1513221} (12) \quad 122315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 23, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0} \\
[1874] & \begin{matrix} 1 \\ 171232181232181231 \end{matrix}^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 23, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[1875] & \begin{matrix} 1 \\ 21 \end{matrix}^{13221} (12) \quad 122315132161^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 23, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[1876] & \begin{matrix} 1 \\ 16131513221 \end{matrix}^{21} (12) \quad 12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 23, \Delta = -77, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1871] & \begin{matrix} 1 \\ 1812321613161316131 \end{matrix}^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 23, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1870] & \begin{matrix} 1 \\ 1 \end{matrix}^{15131513221} (12) \quad 1223141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 24, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 24, T \in [23, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[1868] & \begin{matrix} 1 \\ 1513221(12)12231414141 \end{matrix}^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 23, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1867] & \begin{matrix} 1 \\ 2181232171232181231 \end{matrix}^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 23, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[1877] & \begin{matrix} 1 \\ 17123217123218123151 \end{matrix}^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 21, T_{\max} = 29, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} =
\end{aligned}$$

3.0

$$T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[1878] \ 1^1 5131613161315132181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 29, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24, T_{\max} = 29, T \in [24, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1879] \ 1(12)12231414141414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 28, \Delta = -98, \det = -7, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 28, T \in [27, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[1880] \ 21(12)12231414141414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 28, \Delta = -98, \det = -7, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 27, T_{\max} = 28, T \in [27, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1881] \ 31^1 6131513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 24, \Delta = -85, \det = -7, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 23.57 \rightarrow 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 3 & 5 \\ 1 & 3 & 0 & 2 \\ 2 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1882] \ 612321^1 8123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 33, \Delta = -100, \det = -19, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24.26 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[1883] \ 61^1 23218123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 33, \Delta = -100, \det = -56, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24.57 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[1884] \ 2321^1 6131513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 25, \Delta = -88, \det = -58, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24.41 \rightarrow 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 86 & 1 & 18 \\ 1 & 1 & -2 & 0 \\ 2 & 18 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[1885] \ 81^1 223151316131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 29, \Delta = -99, \det = -90, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24.44 \rightarrow 25, T_{\max} = 29, T \in [25, 29]$$

$$\begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 54 & 1 & 14 \\ 1 & 1 & -2 & 0 \\ 2 & 14 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[1886] \ 1(12)12231^1 414141414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 28, \Delta = -98, \det = -223, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24.31 \rightarrow 25, T_{\max} = 28, T \in [25, 28]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 310 & 1 & 36 \\ 1 & 1 & -3 & -1 \\ 2 & 36 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1887] \ 21(12)12231^1 414141414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 21, T_{\max} = 28, \Delta = -98, \det = -223, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 24.31 \rightarrow 25, T_{\max} = 28, T \in [25, 28]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 193 & 16 & 27 \\ 1 & 16 & 0 & 2 \\ 2 & 27 & 2 & 3 \end{pmatrix}_{b_0}$$

[1888] $712321^1 61316131613151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 29, \Delta = -99, \det = -236, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 24.34 \rightarrow 25, T_{\max} = 29, T \in [25, 29]$$

$$\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 128 & 1 & 22 \\ 1 & 1 & -2 & 0 \\ 2 & 22 & 0 & 3 \end{pmatrix}_{b_0}$$

[1889] $41414141^1 3221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 21, T_{\max} = 33, \Delta = -103, \det = -402, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 24.39 \rightarrow 25, T_{\max} = 33, T \in [25, 33]$$

$$\begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 142 & 1 & 24 \\ 1 & 1 & -2 & 0 \\ 2 & 24 & 0 & 3 \end{pmatrix}_{b_0}$$

$T = 22$ [1891] $21 \overset{1}{15} 13221(12) 12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

[1892] $1^1 413218123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

[1896] $1^1 15 1316131613161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

[1897] $1^1 6 13161316131613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

[1898] $1 \overset{1}{(12)} 12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

$$\overset{1}{15} 13221$$

[1899] $1 \overset{1}{8} 123218123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

[1900] $1 \overset{1}{(12)} 122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

[1901] $1 \overset{1}{(12)} 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

$$\overset{1}{1231513221}$$

[1902] $1 \overset{1}{8} 123218123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

[1903] $1321 \overset{1}{8} 1232181231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

[1904] $1^1 413218123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

[1905] $1^1 413161316131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

[1906] $1 \overset{1}{(12)} 12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

$$\overset{1}{15} 13221$$

[1907] $1^1 321712321812321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

[1908] $1 \overset{1}{(12)} 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

$$\overset{1}{1231513221}$$

[1909] $1^1 321712321812321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

[1910] $21 \overset{1}{(12)} 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

$$\overset{1}{1231513221}$$

[1911] $21 \overset{1}{(12)} 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

[1912] $1^1 413161316131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

[1890] $21 \overset{1}{(12)} 12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

$$\overset{1}{15} 13221$$

[1895] $1^1 6 13161316131613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

[1893] $1^1 15 1316131613161315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

$$\overset{1}{1231513221}$$

[1894] $1 \overset{1}{(12)} 122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 22, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 3.0$

[1946] $1^1 51232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1938] & 1^1 6 123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 23, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0} \\
[1939] & 1^1 7 12321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 23, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[1940] & 1^1 5 13218123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 23, \Delta = -43, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1941] & 1^1 5 132181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 23, \Delta = -45, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1942] & 1^1 32181232171232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 23, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1943] & 1^1 3221 21 (12) 1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 23, \Delta = -56, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1944] & 1^1 21 7 12321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 23, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1945] & 1^1 221 8 1232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 23, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1937] & 1^1 21 7 1232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 23, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[1947] & 1^1 3221 21 (12) 1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 23, \Delta = -56, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[1933] & 1^1 21 8 12321712321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 23, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[1934] & 1^1 4141413221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 23, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 23, T_{\max} = 23, T \in [23, 23] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}
\end{aligned}$$

[1936] $1^1 4141413221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[1935] $1^1 5132181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 23, \Delta = -45, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[1957] $1^1 81232181232181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 24, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1953] $1^1 5131513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 24, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1954] $1^1 3221(12)1223151322171^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 24, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[1955] $1^1 61315132181232181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 24, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1956] $1^1 5131513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 24, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1951] $1^1 41413221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 24, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[1952] $1^1 61231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 24, \Delta = -77, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[1950] $1^1 41413221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 24, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1949] $1^1 6131513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 24, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[1948] $1^1 71231513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22, T_{\max} = 24, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[1958] & 21 \quad {}^{1^1}513221(12)12231513161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 34, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 34, T \in [25, 34] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[1959] & {}^{1^1}51316131613161232181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 29, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 29, T \in [25, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[1960] & {}^{1^1}31513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 25, \Delta = -82, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 1 & 3 & 3 \\ 3 & 5 & 5 \end{pmatrix}_{b_0} \\
[1961] & {}^{1^1}513218123211 \quad 8 \quad 1232171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 29, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 29, T \in [25, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1962] & {}^{1^1}18123217123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 29, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 29, T \in [25, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1963] & {}^{1^1}1(12)1223141414141414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 28, \Delta = -97, \det = 7, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[1965] & {}^{1^1}316131513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 25, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24.57 \rightarrow 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 9 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 2 & 7 & -1 & 4 \end{pmatrix}_{b_0} \\
[1964] & {}^{1^1}31513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 25, \Delta = -82, \det = 7, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24.57 \rightarrow 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
[1966] & {}^{1^1}21^18123217123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 29, \Delta = -98, \det = 8, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25.50 \rightarrow 26, T_{\max} = 29, T \in [26, 29] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[1967] & {}^{31}316131613161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -4, \det = 16, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 22, T_{\max} = 22, T \in [22, 22] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 \\ 2 & 4 & 6 & 6 \\ 2 & 6 & 4 & 6 \\ 2 & 6 & 6 & 7 \end{pmatrix}_{b_0} \\
[1968] & {}^{1^1}17123218123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 22, T_{\max} = 29, \Delta = -98, \det = 19, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25.26 \rightarrow 26, T_{\max} = 29, T \in [26, 29] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1969] $61^1 231513221^{1513221} (12) 12 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 34$, $\Delta = -101$, $\det = 56$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 25.57 \rightarrow 26$, $T_{\max} = 34$, $T \in [26, 34]$
 $\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$
- [1970] $1812321^1 61316131613151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 29$, $\Delta = -98$, $\det = 236$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 25.34 \rightarrow 26$, $T_{\max} = 29$, $T \in [26, 29]$
 $\begin{pmatrix} 0 & 13 & 1 & 2 \\ 13 & 206 & 17 & 28 \\ 1 & 17 & 0 & 2 \\ 2 & 28 & 2 & 3 \end{pmatrix}_{b_0}$
- [1971] $1(12)12231^1 41414141414141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 22$, $T_{\max} = 28$, $\Delta = -97$, $\det = 287$, $(n_+, n_-, n_0) = (1, 22, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 25.27 \rightarrow 26$, $T_{\max} = 28$, $T \in [26, 28]$
 $\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 253 & 18 & 31 \\ 1 & 18 & 0 & 2 \\ 2 & 31 & 2 & 3 \end{pmatrix}_{b_0}$
- $T = 23$ [1980] $1^1 4141413221(12)12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -26$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$
- [1977] $1^1 2315132181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -25$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$
- [1978] $1^1 6^{131} 13161316131613151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -19$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$
- [1979] $1^1 6^{131} 13161316131613151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -19$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$
- [1973] $1^1 5^{131} 13161316131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -20$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$
- [1981] $1^1 4141413221(12)12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -26$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$
- [1982] $1^1 3151321812321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 24$, $\Delta = -30$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$
- [1983] $1^1 2321812321712321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 24$, $\Delta = -30$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$
- [1984] $1^1 2321712321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -28$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$
- [1985] $1^1 8^{1321} 12321812321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -28$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$
- [1986] $1^1 8^{1321} 12321812321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -28$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$
- [1987] $1^1 8^{12321} 1232181232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -28$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$
- [1988] $1^1 8^{12321} 1232181232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -28$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$
- [1989] $1^1 2321712321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -28$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$
- [1990] $1^1 5^{131} 13161316131613161231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -20$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$
- [1974] $1^1 5^{1321} 13218123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -23$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$
- [1976] $1^1 2315132181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -25$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$
- [1972] $1^1 4131613161316131613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -23$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$
- [1975] $1^1 5^{1321} 13218123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 23$, $\Delta = -23$, $\det = 0$, $(n_+, n_-, n_0) = (1, 23, 1)$, $c_{\text{ext}} = 3.0$
- [2008] $1^1 5131513221 (12) 122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 23$, $T_{\max} = 24$, $\Delta = -53$, $\det = -2$, $(n_+, n_-, n_0) = (1, 23, 0)$, $c_{\text{ext}} = 3.0$
 $T_{\min} = 24$, $T_{\max} = 24$, $T \in [24, 24]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2010] & 1^{1^1 5131513221} (12) \quad 122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2011] & 1^{1^1 321812321} 8 \quad 1232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2012] & 1^1 161232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -44, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2013] & 1^1 217123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0} \\
[2014] & 1^1 221812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2015] & 1^1 16123218123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -42, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2016] & 1^{1^1 316131513221} (12) \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[2017] & 1^{1^1 316131513221} (12) \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2018] & 1^1 21812321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[2019] & 1^1 1613151321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2009] & 1^1 22181232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 24, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 24, T_{\max} = 24, T \in [24, 24]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[2030] & 1^1 712315132181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -76, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2025] & 1^1 613161232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -76, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2026] & 1^1 712231513221(12) 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[2027] & 1^1 41413221(12)122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -76, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[2028] & 1^1 71231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[2029] & 1^1 3221(12) 122315131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 26, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 26, T \in [25, 26] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2024] & 1^1 6131513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[2031] & 1^1 3221(12) 122315131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 26, \Delta = -79, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 26, T \in [25, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2032] & 1^1 812231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[2020] & 1^1 81231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 25, T_{\max} = 25, T \in [25, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[2022] & 1^1 316131613161315132181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 25, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

[2041] $^{11}232181231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 26, \Delta = -86, \det = -13, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 25.77 \rightarrow 26, T_{\max} = 26, T \in [26, 26]$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 11 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 4 \end{pmatrix}_{b_0}$$

[2042] $^{11}218123218123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 29, \Delta = -97, \det = -19, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 26.26 \rightarrow 27, T_{\max} = 29, T \in [27, 29]$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

[2043] $^{11}18123218123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 29, \Delta = -97, \det = -19, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 26.26 \rightarrow 27, T_{\max} = 29, T \in [27, 29]$

$$\begin{pmatrix} 0 & 3 & 1 & 2 \\ 3 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

[2044] $^{11}71^{11}2231513221(12)12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 31, \Delta = -99, \det = -72, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 26.50 \rightarrow 27, T_{\max} = 31, T \in [27, 31]$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 40 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

[2045] $(12)12231^{11}4141414141414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 23, T_{\max} = 29, \Delta = -96, \det = -359, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 26.24 \rightarrow 27, T_{\max} = 29, T \in [27, 29]$

$$\begin{pmatrix} 0 & 17 & 1 & 2 \\ 17 & 542 & 1 & 48 \\ 1 & 1 & -3 & -1 \\ 2 & 48 & -1 & 3 \end{pmatrix}_{b_0}$$

$T = 24$ [2050] $^{11}21613161316131613161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0$

[2049] $^{11}1^{11}8^{11}2321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0$

[2052] $^{11}1^{11}41321812321812321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0$

[2053] $^{11}1^{11}41321812321812321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0$

[2054] $^{11}1^{11}2321812321812321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0$

[2055] $^{11}1^{11}31612321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0$

[2056] $^{11}1^{11}31612321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0$

[2057] $^{11}15131613161316131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0$

[2058] $^{11}15131613161316131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0$

[2059] $^{11}1^{11}8^{11}2321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0$

[2060] $^{11}1^{11}8^{11}2321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0$

[2061] $^{11}16123218123218123151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0$

[2048] $^{11}2161316131613161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0$

3.0

$$[2047] \quad 1 \quad \begin{smallmatrix} 1^1 5131513221 \\ (12) \end{smallmatrix} \quad 12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0$$

$$[2046] \quad 1 \quad \begin{smallmatrix} 1^1 5131513221 \\ (12) \end{smallmatrix} \quad 12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0$$

$$[2051] \quad 1 \quad \begin{smallmatrix} 1^1 2321 \\ 8 \end{smallmatrix} \quad 12321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 24, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 3.0$$

$$[2089] \quad 1 \quad \begin{smallmatrix} 1^1 \\ 15 \end{smallmatrix} \begin{smallmatrix} 13221 \\ (12) \end{smallmatrix} \quad 1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -50, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[2078] \quad 1^1 41413221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[2080] \quad 21 \quad \begin{smallmatrix} 1^1 3221 \\ (12) \end{smallmatrix} \quad 122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[2081] \quad 21 \quad \begin{smallmatrix} 1^1 3221 \\ (12) \end{smallmatrix} \quad 122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[2082] \quad 221 \quad \begin{smallmatrix} 1^1 \\ 8 \end{smallmatrix} \quad 123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$$

$$[2083] \quad 1^1 41413221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[2084] \quad 1^1 6131612321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[2085] \quad 1 \quad \begin{smallmatrix} 1^1 31513221 \\ (12) \end{smallmatrix} \quad 122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[2086] \quad 16 \quad \begin{smallmatrix} 1^1 \\ 16 \end{smallmatrix} \begin{smallmatrix} 131513221 \\ (12) \end{smallmatrix} \quad 122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -50, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[2087] \quad 21 \quad \begin{smallmatrix} 1^1 \\ 7 \end{smallmatrix} \quad 1232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 24, T_{\max} = 25, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

[2079] $1^1 7123151321812321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 24, T_{\max} = 25, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2088] $21^1 71232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 24, T_{\max} = 25, \Delta = -42, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[2090] $1^1 513221 (12) 1223151316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 24, T_{\max} = 25, \Delta = -50, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[2092] $31^1 6131513221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 24, T_{\max} = 26, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 1 & 3 & 3 \\ 3 & 6 & 6 \\ 3 & 6 & 6 \end{pmatrix}_{b_0}$

[2091] $31^1 513221(12)1223151316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 24, T_{\max} = 26, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 5 \end{pmatrix}_{b_0}$

[2093] $1^1 3161316131613161232181^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 24, T_{\max} = 26, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[2094] $1^1 812315132181232181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 24, T_{\max} = 26, \Delta = -75, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[2095] $1^1 3218123218123151322181^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 24, T_{\max} = 26, \Delta = -75, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[2096] $21^1 81231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 24, T_{\max} = 26, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[2097] $1^1 413221(12)122315131613151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 24, T_{\max} = 26, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[2098] $1^1 413221(12)122315131613151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 24, T_{\max} = 26, \Delta = -78, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[2099] $1\overset{1}{7}1231513221(12)12231\overset{1}{5}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 24, T_{\max} = 26, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[2100] $1\overset{1}{4}1413221(12)1223151322171\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 24, T_{\max} = 26, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[2101] $1\overset{1}{6}1316131513221(12)12231\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 24, T_{\max} = 26, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[2102] $1\overset{1}{5}131513221(12)12231513151\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 24, T_{\max} = 26, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[2103] $1\overset{1}{7}1231513221(12)122314141\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 24, T_{\max} = 26, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$

[2104] $1\overset{1}{5}13221(12)1223151322171\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 24, T_{\max} = 26, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[2105] $1\overset{1}{8}12231513221(12)1223141\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 24, T_{\max} = 26, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[2106] $1\overset{1}{8}1231513221(12)1223151\overset{21}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 24, T_{\max} = 31, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 31, T \in [27, 31]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[2107] $1\overset{1}{5}13221(12)1223151322181\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 24, T_{\max} = 31, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 31, T \in [27, 31]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

[2108] $1\overset{1}{3}161316131513221(12)12231\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 24, T_{\max} = 27, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26.57 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 4 \end{pmatrix}_{b_0}$

[2109] $32181231513221(12)1223141\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 24, T_{\max} = 27, \Delta = -83, \det = 7, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 26.57 \rightarrow 27, T_{\max} = 27, T \in [27, 27]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 9 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 2 & 7 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[2110] \quad 1^1 513221 \quad 21 \quad (12) \quad 122315132181^1 2 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 24, T_{\max} = 31, \Delta = -98, \det = 8, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 27.50 \rightarrow 28, T_{\max} = 31, T \in [28, 31]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2111] \quad 1^1 513221 \quad 21 \quad (12) \quad 12231513221^1 81 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 24, T_{\max} = 31, \Delta = -98, \det = 72, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 27.50 \rightarrow 28, T_{\max} = 31, T \in [28, 31]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 \\ 7 & 54 & 9 & 14 \\ 1 & 9 & 0 & 2 \\ 2 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2112] \quad 613161^1 23218123218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 24, T_{\max} = 32, \Delta = -99, \det = 317, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 3.0 \\ T_{\min} = 27.32 \rightarrow 28, T_{\max} = 32, T \in [28, 32]$$

$$\begin{pmatrix} 0 & 15 & 1 & 2 \\ 15 & 163 & 1 & 25 \\ 1 & 1 & -2 & 0 \\ 2 & 25 & 0 & 3 \end{pmatrix}_{b_0}$$

$$T = 25 \quad [2123] \quad 216131613161316131613141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -15, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$$

$$[2115] \quad 1^1 232171232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$$

$$[2125] \quad 1^1 2231513221 \quad 1 \quad (12) \quad 122315132161^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$$

$$[2124] \quad 1^1 2321 \quad 1 \quad 8 \quad 123218123218123141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$$

$$[2127] \quad 1^1 16131513221(12)12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$$

$$[2122] \quad 1^1 321712321812321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$$

$$[2121] \quad 1^1 321712321812321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$$

$$[2120] \quad 1^1 2321 \quad 1 \quad 8 \quad 123218123218123141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$$

$$[2119] \quad 21 \quad 7 \quad 123218123218123151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$$

$$[2118] \quad 1^1 231513221 \quad 1 \quad (12) \quad 1223151316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$$

$$[2117] \quad 1^1 316123218123218123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$$

$$[2116] \quad 1^1 51321812321812321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$$

$$[2128] \quad 1^1 231513221 \quad 1 \quad (12) \quad 1223151316131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$$

$$[2126] \quad 1^1 51316131613161316131^1 51 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -13, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$$

$$[2114] \quad 1^1 232171232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$$

$$[2113] \quad 1^1 51321812321812321712321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 25, T_{\max} = 25, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 3.0$$

$$[2153] \quad 1^1 171231513221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2149] & 221 \overset{1}{8} 1232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2150] & 1 \overset{1}{(12)} 12231513161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2151] & 1 \overset{1}{(12)} 12231513161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2152] & 1 \overset{1}{(12)} 1223151322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
[2148] & 1 \overset{1}{23218123218123151322181}^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
[2154] & 1 \overset{1}{32181231513221} (12) 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2146] & 1 \overset{1}{32181231513221} (12) 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[2147] & 221 \overset{1}{8} 1232181232181231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2145] & 1 \overset{1}{8} 123151321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 26, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2158] & 1 \overset{1}{16} 1316131513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 25, T_{\max} = 27, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

[2159] $1^1 71232171232181232181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 27, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[2160] $1^1 3221 1 (12) 12231513218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 27, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[2157] $1^1 3221 1 (12) 12231513218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 27, \Delta = -78, \det = -3, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[2162] $1^1 1513221(12)122315131613151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 27, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[2163] $1^1 1513221(12)122315131613151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 27, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[2164] $1^1 1812231513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 27, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[2165] $1^1 2181231513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 27, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[2166] $1^1 61231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 27, \Delta = -74, \det = -3, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[2156] $1^1 2181231513221(12)122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 27, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$

[2155] $1^1 1812231513221(12)122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 27, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$

[2161] $1^1 16131513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 25, T_{\max} = 27, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0$

$$\begin{aligned}
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2167] & 1^{151321812321812315132181^1} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 25, T_{\max} = 32, \Delta = -98, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 32, T \in [28, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[2168] & 1^{1513221} (12) 122315131613161^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 25, T_{\max} = 33, \Delta = -99, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, T_{\max} = 33, T \in [28, 33] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[2169] & 81^{12231513218123218123151^1} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 25, T_{\max} = 32, \Delta = -98, \det = -90, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28.44 \rightarrow 29, T_{\max} = 32, T \in [29, 32] \\
& \begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 54 & 1 & 14 \\ 1 & 1 & -2 & 0 \\ 2 & 14 & 0 & 3 \end{pmatrix}_{b_0} \\
[2170] & 812321^{1613161316131613151^1} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 25, T_{\max} = 31, \Delta = -97, \det = -338, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 28.27 \rightarrow 29, T_{\max} = 31, T \in [29, 31] \\
& \begin{pmatrix} 0 & 16 & 1 & 2 \\ 16 & 206 & 1 & 28 \\ 1 & 1 & -2 & 0 \\ 2 & 28 & 0 & 3 \end{pmatrix}_{b_0} \\
T = 26 & [2183] 1^{12321} 8^{1232181232181232151^1} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0 \\
[2176] & 221^{11} 8^{123218123218123151321^1} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0 \\
[2171] & 1^{1321} 8^{12321812321812315131^1} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0 \\
[2172] & 1^{12321} 8^{1232181232181232151^1} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0 \\
[2173] & 1^{13221812321812321812315131^1} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0 \\
[2181] & 1^{11} 7^{1231513221(12)12231513221^1} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0 \\
[2175] & 1^{12231513221} (12) 1223151322171^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0 \\
[2174] & 1^{1321} 8^{12321812321812315131^1} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0 \\
[2178] & 1^{13161232181232181232171231^1} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0 \\
[2179] & 1^{11} 216^{131613161316131613151^1} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -10, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0 \\
[2180] & 1^{12321} 8^{123218123218123151^1} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0 \\
[2182] & 1^{13161232181232181232171231^1} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0 \\
[2184] & 1^{11} 16^{12321812321812321712321^1} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0 \\
[2185] & 1^{13221812321812321812315131^1} + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0
\end{aligned}$$

[2177] $1^{12321} 1^8 1232181232181231 5 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 26, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 3.0$

[2210] $21 8^{11231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$

[2213] $1^{1413221(12)12231513161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[2214] $1^{141413221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2206] $1^{1812231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$

[2205] $1^{1513221} 1^{12} 122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[2202] $1^{1413221(12)12231513161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2203] $1^{1712321712321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2204] $1^{1513221} 1^{12} 122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2207] $1^{151316131513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[2208] $1^{151316131513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2209] $1^{141413221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, T_{\max} = 27, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 27, T_{\max} = 27, T \in [27, 27]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2212] & 1^1 316131513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, \quad T_{\max} = 27, \quad \Delta = -52, \quad \det = 2, \quad (n_+, n_-, n_0) = \\
& (1, 26, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2211] & 1^1 316131513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, \quad T_{\max} = 27, \quad \Delta = -52, \quad \det = 2, \quad (n_+, n_-, n_0) = \\
& (1, 26, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 27, \quad T_{\max} = 27, \quad T \in [27, 27] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2219] & 1^1 513161316131513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, \quad T_{\max} = 28, \quad \Delta = -78, \quad \det = 3, \quad (n_+, n_-, n_0) = \\
& (1, 26, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2220] & 1^1 513161316131513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, \quad T_{\max} = 28, \quad \Delta = -78, \quad \det = 3, \quad (n_+, n_-, n_0) = \\
& (1, 26, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2221] & 1^{13221} 1 (12) 122315132181232161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, \quad T_{\max} = 28, \quad \Delta = -77, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = \\
& 3.0 \\
& T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[2222] & 1^{1321812321812321} 1^1 8^{1232171} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, \quad T_{\max} = 28, \quad \Delta = -74, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2223] & 1^1 712231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, \quad T_{\max} = 28, \quad \Delta = -72, \quad \det = 3, \quad (n_+, n_-, n_0) = \\
& (1, 26, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2224] & 1^1 8 1232171232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, \quad T_{\max} = 28, \quad \Delta = -74, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = \\
& 3.0 \\
& T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2215] & 1^1 413221(12)12231513218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, \quad T_{\max} = 28, \quad \Delta = -77, \quad \det = 3, \quad (n_+, n_-, n_0) = \\
& (1, 26, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2216] & 1^1 6 1316131513221(12)12231^1 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, \quad T_{\max} = 28, \quad \Delta = -70, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 26, 0), \quad c_{\text{ext}} = \\
& 3.0 \\
& T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2217] & 1^1 413221(12)12231513218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 26, \quad T_{\max} = 28, \quad \Delta = -77, \quad \det = 3, \quad (n_+, n_-, n_0) = \\
& (1, 26, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

[2218] $1 \overset{1}{7} 1231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 26, T_{\max} = 28, \Delta = -72, \det = 3, (n_+, n_-, n_0) = (1, 26, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

$T = 27$ [2235] $21 \overset{1}{7} 12321812321812321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 27, T_{\max} = 27, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 3.0$

[2236] $1 \overset{1}{8} 12231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 27, T_{\max} = 27, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 3.0$

[2237] $1 \overset{1}{3} 1513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 27, T_{\max} = 28, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 3.0$

[2238] $1 \overset{1}{8} 123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 27, T_{\max} = 27, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 3.0$

[2239] $1 \overset{1}{8} 123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 27, T_{\max} = 27, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 3.0$

[2240] $1 \overset{1}{2} 31513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 27, T_{\max} = 28, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 3.0$

[2234] $1 \overset{1}{5} 1316131513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 27, T_{\max} = 27, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 3.0$

[2233] $21 \overset{1}{8} 1231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 27, T_{\max} = 27, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 3.0$

[2232] $1 \overset{1}{3} 1612321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 27, T_{\max} = 27, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 3.0$

[2231] $1 \overset{1}{4} 1321812321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 27, T_{\max} = 27, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 3.0$

[2230] $1 \overset{1}{3} 1513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 27, T_{\max} = 28, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 3.0$

[2229] $1 \overset{1}{3} 2171232181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 27, T_{\max} = 27, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 3.0$

[2228] $1 \overset{1}{3} 1612321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 27, T_{\max} = 27, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 3.0$

[2227] $1 \overset{1}{5} 1316131513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 27, T_{\max} = 27, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 3.0$

[2226] $1 \overset{1}{2} 31513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 27, T_{\max} = 28, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 3.0$

[2225] $1 \overset{1}{6} 12321812321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 27, T_{\max} = 27, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 3.0$

[2259] $1 \overset{1}{3} 16131513221(12)122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 27, T_{\max} = 28, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$

[2262] $1 \overset{1}{5} 13221(12)12231513161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 27, T_{\max} = 28, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[2263] $1 \overset{1}{5} 13221(12)12231513161316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 27, T_{\max} = 28, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2264] & 16131513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, \quad T_{\max} = 28, \quad \Delta = -49, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2265] & 161316131513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, \quad T_{\max} = 28, \quad \Delta = -49, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2261] & 1^{12321812321812321} 8^{1232171} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, \quad T_{\max} = 28, \quad \Delta = -47, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[2260] & 1812321712321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, \quad T_{\max} = 28, \quad \Delta = -47, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 28, \quad T_{\max} = 28, \quad T \in [28, 28] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2266] & 1513221(12)12231513218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, \quad T_{\max} = 29, \quad \Delta = -72, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, \quad T_{\max} = 29, \quad T \in [29, 29] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2267] & 1513221(12)12231513218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, \quad T_{\max} = 29, \quad \Delta = -72, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, \quad T_{\max} = 29, \quad T \in [29, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2268] & 2181231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, \quad T_{\max} = 29, \quad \Delta = -71, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, \quad T_{\max} = 29, \quad T \in [29, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2269] & 181232181232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, \quad T_{\max} = 29, \quad \Delta = -73, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, \quad T_{\max} = 29, \quad T \in [29, 29] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2270] & 1613161316131513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, \quad T_{\max} = 29, \quad \Delta = -75, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, \quad T_{\max} = 29, \quad T \in [29, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2273] & 171232181231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, \quad T_{\max} = 29, \quad \Delta = -75, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 27, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 29, \quad T_{\max} = 29, \quad T \in [29, 29] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[2272] & 1^{1413221}(12)122315132181232161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, \quad T_{\max} = 29, \quad \Delta = -76, \quad \det = -3, \quad (n_+, n_-, n_0) =
\end{aligned}$$

$$(1, 27, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[2271] \quad 1^1 1812231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 29, \Delta = -71, \det = -3, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[2274] \quad 1^1 8123217123218123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 33, \Delta = -97, \det = -4, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, T_{\max} = 33, T \in [30, 33]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[2275] \quad 712321^1 8123218123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 33, \Delta = -97, \det = -29, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 30.17 \rightarrow 31, T_{\max} = 33, T \in [31, 33]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 11 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[2276] \quad 61^1 232181231513221^{1^1 513221} (12) 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 27, T_{\max} = 39, \Delta = -100, \det = -56, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 30.57 \rightarrow 31, T_{\max} = 39, T \in [31, 39]$$

$$\begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

$$T = 28 \quad [2288] \quad 1^1 315132181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$$

$$[2280] \quad 1^1 413218123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$$

$$[2289] \quad 1^1 315132181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$$

$$[2287] \quad 1^1 8^{1^1 321} 1232181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$$

$$[2286] \quad 221^1 812321812321812321712321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$$

$$[2291] \quad 161316131513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$$

$$[2282] \quad 151321812321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$$

$$[2279] \quad 1^1 413218123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$$

$$[2278] \quad 1^1 321712321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$$

$$[2277] \quad 1^1 321712321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$$

$$[2290] \quad 1^1 8^{1^1 321} 1232181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$$

$$[2283] \quad 1^1 3161316131513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$$

$$[2284] \quad 1^1 3161316131513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$$

$$[2285] \quad 21712321812321812321^1 812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$$

[2281] $1^1 151321812321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 28, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 3.0$

[2311] $1^1 31613161316131513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[2310] $1^1 5132181231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[2309] $1^1 5132181231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2308] $1^1 32181231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2307] $1^1 812321812321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[2306] $1^1 32181231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -51, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[2305] $1^1 316131513221(12)1223151322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$

[2304] $1^1 171231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[2312] $1^1 31613161316131513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2313] $1^1 3221^{21} 21(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 29, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$

[2314] $1^1 2181232181231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 30, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[2316] & 1^1 513221(12)122315132181232161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 30, \Delta = -71, \det = 3, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 30, T_{\max} = 30, T \in [30, 30] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[2315] & 1^1 171232181231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 30, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 30, T_{\max} = 30, T \in [30, 30] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
[2317] & 1^1 1613161316131513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 35, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, T_{\max} = 35, T \in [31, 35] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[2318] & 1^1 51321812321812321 8 1232181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 34, \Delta = -96, \det = 4, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, T_{\max} = 34, T \in [31, 34] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[2319] & 1^1 513221 1 (12) 122315132181232171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 35, \Delta = -98, \det = 4, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 31, T_{\max} = 35, T \in [31, 35] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[2320] & 1^1 32181232181231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 31, \Delta = -83, \det = 7, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 30.57 \rightarrow 31, T_{\max} = 31, T \in [31, 31] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 2 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
[2321] & 1^1 18123218123218123218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 28, T_{\max} = 34, \Delta = -96, \det = 29, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 31.17 \rightarrow 32, T_{\max} = 34, T \in [32, 34] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 \\ 4 & 11 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 2 & 7 & 0 & 3 \end{pmatrix}_{b_0} \\
T = 29 & [2328] 1^1 4132181232181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0 \\
[2322] & 1^1 3161232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0 \\
[2323] & 1^1 3161232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0 \\
[2324] & 1^1 1513218123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0 \\
[2325] & 1^1 1513218123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0 \\
[2326] & 1^1 5132181231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0 \\
[2327] & 1^1 1612321812321812321812315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0
\end{aligned}$$

[2330] $1^{1^1}5132181231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$

[2331] $1^{1^1}8^{1^1}12321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$

[2332] $1^{1^1}8^{1^1}12321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$

[2333] $1^{1^1}3221812321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$

[2334] $1^{1^1}8^{1^1}1232181232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$

[2335] $1^{1^1}4132181232181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$

[2329] $2218^{1^1}12321812321812321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 29, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 3.0$

[2347] $218^{1^1}1231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 30, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[2351] $1^{1^1}12^{1^1}122315132181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 30, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2350] $1^{1^1}61232181231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 30, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2349] $21^{1^1}12^{1^1}12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 30, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$

[2348] $18^{1^1}12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 30, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[2353] $21^{1^1}12^{1^1}12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 30, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$

[2346] $16^{1^1}131513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 30, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[2345] $1^{1^1}61231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 29, T_{\max} = 30, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[2352] \quad 1^{1^1 3221} (12) 122315132181232181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 29, \quad T_{\max} = 30, \quad \Delta = -54, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 29, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 30, \quad T_{\max} = 30, \quad T \in [30, 30]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[2354] \quad 1^{1^1} 81232181231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 29, \quad T_{\max} = 31, \quad \Delta = -73, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 29, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, \quad T_{\max} = 31, \quad T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$$

$$[2356] \quad 1^{1^1} 71232181231513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 29, \quad T_{\max} = 31, \quad \Delta = -69, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 29, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, \quad T_{\max} = 31, \quad T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[2357] \quad 1^{1^1 3221} (12) 12231513221^{21} (12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 29, \quad T_{\max} = 31, \quad \Delta = -77, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 29, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, \quad T_{\max} = 31, \quad T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[2358] \quad 1^{1^1 3221} (12) 12231513221^{21} (12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 29, \quad T_{\max} = 31, \quad \Delta = -77, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 29, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, \quad T_{\max} = 31, \quad T \in [31, 31]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[2355] \quad 1^{1^1} 321812321812321812315132181^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 29, \quad T_{\max} = 31, \quad \Delta = -74, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 29, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 31, \quad T_{\max} = 31, \quad T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[2359] \quad 1^{1^1} 81232181231513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 29, \quad T_{\max} = 35, \quad \Delta = -97, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 29, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 32, \quad T_{\max} = 35, \quad T \in [32, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[2360] \quad 1^{1^1 513221} (12) 122315132181232181^1 2 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 29, \quad T_{\max} = 35, \quad \Delta = -97, \quad \det = -8, \quad (n_+, n_-, n_0) = (1, 29, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 32.50 \rightarrow 33, \quad T_{\max} = 35, \quad T \in [33, 35]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 30 \quad [2361] \quad 1^{1^1} 6123218123218123218123216131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, \quad T_{\max} = 30, \quad \Delta = -19, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 30, 1), \quad c_{\text{ext}} = 3.0$$

$$[2362] \quad 21^{1^1} 712321812321812321812315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, \quad T_{\max} = 30, \quad \Delta = -18, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 30, 1), \quad c_{\text{ext}} = 3.0$$

$$[2363] \quad 1^{1^1} 41321812321812321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, \quad T_{\max} = 30, \quad \Delta = -23, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 30, 1), \quad c_{\text{ext}} = 3.0$$

$$[2364] \quad 1^{1^1} 41321812321812321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, \quad T_{\max} = 30, \quad \Delta = -23, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 30, 1), \quad c_{\text{ext}} = 3.0$$

$$[2365] \quad 1^{1^1 321} 8123218123218123218123141^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 30, \quad T_{\max} = 30, \quad \Delta = -23, \quad \det = 0, \quad (n_+, n_-, n_0) =$$

$(1, 30, 1), c_{\text{ext}} = 3.0$
 $[2366] \ 1^{\overline{1}1321} \ 8 \ 123218123218123218123141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\text{max}} = 30, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 $[2367] \ 1^{\overline{1}} \ 15132181232181232181232171231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\text{max}} = 30, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 $[2368] \ 1^{\overline{1}} \ 61232181231513221(12)12231513221^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\text{max}} = 30, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 $[2369] \ 1^{\overline{1}} \ 316131513221(12)1223151321812321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\text{max}} = 31, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 $[2370] \ 1^{\overline{1}} \ 32171232181232181232181232151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\text{max}} = 30, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 $[2371] \ 1^{\overline{1}} \ 32171232181232181232181232151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\text{max}} = 30, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 $[2372] \ 1^{\overline{1}} \ 316131513221(12)1223151321812321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\text{max}} = 31, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 $[2373] \ 1^{\overline{1}} \ 15132181232181232181232171231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\text{max}} = 30, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 3.0$
 $[2385] \ 1^{\overline{1} \ 5 \ 13221} \ (12) \ 12231513221(12)12231^{\overline{21}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\text{max}} = 31, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 31, T_{\text{max}} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$
 $[2384] \ 1^{\overline{1}} \ 413221(12)122315132181232181231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\text{max}} = 31, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 31, T_{\text{max}} = 31, T \in [31, 31]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$
 $[2387] \ 1^{\overline{1}} \ 2321812321812321812315132181^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\text{max}} = 31, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 31, T_{\text{max}} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$
 $[2388] \ 1^{\overline{1}} \ 413221(12)122315132181232181231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\text{max}} = 31, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 31, T_{\text{max}} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$
 $[2389] \ 1^{\overline{1} \ 413221(12)12231513221} \ (12) \ 1223141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\text{max}} = 31, \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 31, T_{\text{max}} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$
 $[2390] \ 1^{\overline{1} \ 5 \ 13221} \ (12) \ 12231513221(12)12231^{\overline{21}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\text{max}} = 31, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 31, T_{\text{max}} = 31, T \in [31, 31]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$
 $[2391] \ 1^{\overline{1}} \ 171232181231513221(12)122315131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, T_{\text{max}} = 31, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 31, T_{\text{max}} = 31, T \in [31, 31]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

$$\begin{aligned}
& [2392] \quad 21 \quad (12) \quad 12231513221(12)122314141^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, \quad T_{\max} = 31, \quad \Delta = -52, \quad \det = 2, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 31, \quad T_{\max} = 31, \quad T \in [31, 31] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0} \\
& [2393] \quad 1^1 712231513221(12)122315132181231^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, \quad T_{\max} = 31, \quad \Delta = -47, \quad \det = 2, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 31, \quad T_{\max} = 31, \quad T \in [31, 31] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
& [2394] \quad 21 \quad (12) \quad 12231513221(12)122314141^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, \quad T_{\max} = 31, \quad \Delta = -52, \quad \det = 2, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 31, \quad T_{\max} = 31, \quad T \in [31, 31] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
& [2386] \quad 1^1 71231513221(12)122315132181231^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, \quad T_{\max} = 31, \quad \Delta = -47, \quad \det = 2, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 31, \quad T_{\max} = 31, \quad T \in [31, 31] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
& [2396] \quad 1^1 5132181232181231513221(12)12231^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, \quad T_{\max} = 32, \quad \Delta = -77, \quad \det = 3, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 32, \quad T_{\max} = 32, \quad T \in [32, 32] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& [2397] \quad 21 \quad (12) \quad 12231513221(12)1223141^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, \quad T_{\max} = 32, \quad \Delta = -76, \quad \det = 3, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 32, \quad T_{\max} = 32, \quad T \in [32, 32] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
& [2398] \quad 21 \quad 8 \quad 1232181231513221(12)12231^1 5 \quad 1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, \quad T_{\max} = 32, \quad \Delta = -68, \quad \det = 3, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 32, \quad T_{\max} = 32, \quad T \in [32, 32] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
& [2399] \quad 1^1 5132181232181231513221(12)12231^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, \quad T_{\max} = 32, \quad \Delta = -77, \quad \det = 3, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 32, \quad T_{\max} = 32, \quad T \in [32, 32] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
& [2395] \quad 21 \quad (12) \quad 12231513221(12)1223141^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 30, \quad T_{\max} = 32, \quad \Delta = -76, \quad \det = 3, \quad (n_+, n_-, n_0) = \\
& (1, 30, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 32, \quad T_{\max} = 32, \quad T \in [32, 32] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0} \\
& T = 31 \quad [2400] \quad 1^1 51321812321812321812321812231^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, \quad T_{\max} = 31, \quad \Delta = -18, \quad \det = 0, \quad (n_+, n_-, n_0) = \\
& (1, 31, 1), \quad c_{\text{ext}} = 3.0 \\
& [2402] \quad 1^1 161232181232181232181232171231^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, \quad T_{\max} = 31, \quad \Delta = -18, \quad \det = 0, \quad (n_+, n_-, n_0) = \\
& (1, 31, 1), \quad c_{\text{ext}} = 3.0 \\
& [2403] \quad 21 \quad 7 \quad 123218123218123218123216131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, \quad T_{\max} = 31, \quad \Delta = -17, \quad \det = 0, \quad (n_+, n_-, n_0) = \\
& (1, 31, 1), \quad c_{\text{ext}} = 3.0
\end{aligned}$$

[2404] $1^{1^1}413218123218123218123218123141^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$

[2405] $1^{1^1}51321812321812321812321812231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$

[2406] $1^{1^1}8^{1^1}123218123218123218123151^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$

[2407] $1^{1^1}322181232181232181232181232151^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$

[2408] $1^{1^1}322181232181232181232181232151^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$

[2409] $1^{1^1}231513221^{21}(12)12231513221^{21}(12)12231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$

[2410] $1^{1^1}231513221^{21}(12)12231513221^{21}(12)12231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$

[2411] $1^{1^1}8^{1^1}123218123218123218123151^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$

[2412] $1^{1^1}8^{1^1}1232181232181232181232151^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$

[2413] $1^{1^1}8^{1^1}1232181232181232181232151^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$

[2414] $1^{1^1}71232181231513221(12)12231513221^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$

[2401] $2218^{1^1}12321812321812321812315131^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 31, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 3.0$

[2424] $1^{1^1}41413221(12)12231513221^{1^1}1223141^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 32, \Delta = -51, \det = -2, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

[2426] $1^{1^1}513221^{1^1}(12)12231513221^{1^1}(12)1223141^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 32, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$$

[2427] $1^{1^1}812231513221(12)122315132181231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 32, \Delta = -46, \det = -2, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

[2428] $1^{1^1}513221^{1^1}(12)12231513221^{1^1}(12)1223141^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 32, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

[2429] $1^{1^1}513221(12)122315132181232181231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 31, T_{\max} = 32, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 3.0$

$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& [2430] \quad 1 \overset{1}{5} 13221(12)122315132181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, \quad T_{\max} = 32, \quad \Delta = -48, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 31, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 32, \quad T_{\max} = 32, \quad T \in [32, 32] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
& [2431] \quad 21 \overset{1}{8} 1232181231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, \quad T_{\max} = 32, \quad \Delta = -47, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 31, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 32, \quad T_{\max} = 32, \quad T \in [32, 32] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
& [2432] \quad 21 \overset{1}{8} 1231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, \quad T_{\max} = 32, \quad \Delta = -46, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 31, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 32, \quad T_{\max} = 32, \quad T \in [32, 32] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
& [2425] \quad 1 \overset{1}{4} 1413221(12)12231513221 \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, \quad T_{\max} = 32, \quad \Delta = -51, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 31, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 32, \quad T_{\max} = 32, \quad T \in [32, 32] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& [2433] \quad 1^1 \overset{1}{6} 1232181232181231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, \quad T_{\max} = 33, \quad \Delta = -76, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 31, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 33, \quad T_{\max} = 33, \quad T \in [33, 33] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
& [2434] \quad 21 \overset{1}{5} 13221(12)12231513221(12)12231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, \quad T_{\max} = 33, \quad \Delta = -71, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 31, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 33, \quad T_{\max} = 33, \quad T \in [33, 33] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& [2435] \quad 21 \overset{1}{5} 13221(12)12231513221(12)12231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, \quad T_{\max} = 33, \quad \Delta = -71, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 31, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 33, \quad T_{\max} = 33, \quad T \in [33, 33] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
& [2436] \quad 1^1 \overset{1}{8} 1232171232181232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 31, \quad T_{\max} = 33, \quad \Delta = -73, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 31, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 33, \quad T_{\max} = 33, \quad T \in [33, 33] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
& T = 32 \quad [2437] \quad 1^1 \overset{1}{4} 132181232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, \quad T_{\max} = 32, \quad \Delta = -19, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 32, 1), \quad c_{\text{ext}} = 3.0 \\
& [2438] \quad 221 \overset{1}{8} 123218123218123218123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, \quad T_{\max} = 32, \quad \Delta = -16, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 32, 1), \quad c_{\text{ext}} = 3.0 \\
& [2439] \quad 1^1 \overset{1}{4} 132181232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, \quad T_{\max} = 32, \quad \Delta = -19, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 32, 1), \quad c_{\text{ext}} = 3.0 \\
& [2441] \quad 1^1 \overset{1}{3} 2181232181231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, \quad T_{\max} = 32, \quad \Delta = -28, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 32, 1), \quad c_{\text{ext}} = 3.0 \\
& [2442] \quad 1 \overset{1}{2} 31513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, \quad T_{\max} = 32, \quad \Delta = -27, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 32, 1), \quad c_{\text{ext}} = 3.0 \\
& [2443] \quad 1^1 \overset{1}{3} 2181231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, \quad T_{\max} = 32, \quad \Delta = -28, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 32, 1), \quad c_{\text{ext}} = 3.0
\end{aligned}$$

[2444] $1^1 32181231513221(12)1223151321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 32, T_{\max} = 32, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0$

[2445] $1^1 1513218123218123218123218123141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 32, T_{\max} = 32, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0$

[2446] $1^1 232181232181231513221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 32, T_{\max} = 33, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0$

[2447] $1^1 1513218123218123218123218123141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 32, T_{\max} = 32, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0$

[2448] $1^1 1612321812321812321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 32, T_{\max} = 32, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0$

[2449] $1^1 231513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 32, T_{\max} = 32, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0$

[2450] $1^1 161232181232181232181232181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 32, T_{\max} = 32, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0$

[2451] $1^1 232181232181231513221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 32, T_{\max} = 33, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0$

[2452] $1^1 2171232181232181232181232171231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 32, T_{\max} = 32, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0$

[2453] $1^1 2181232181231513221(12)12231513221^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 32, T_{\max} = 32, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0$

[2440] $1^1 32181232181231513221(12)1223151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 32, T_{\max} = 32, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 32, 1), c_{\text{ext}} = 3.0$

[2464] $1^1 812321712321812321812321812321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 32, T_{\max} = 33, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2466] $1^1 21(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 32, T_{\max} = 33, \Delta = -50, \det = 2, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2467] $1^1 1513221(12)12231513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 32, T_{\max} = 33, \Delta = -42, \det = 2, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$

[2468] $1^1 41413221(12)12231513221(12)122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 32, T_{\max} = 33, \Delta = -50, \det = 2, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$

[2469] $1^1 21(12)12231513221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 32, T_{\max} = 33, \Delta = -50, \det = 2, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2470] $1^1 21(12)12231513221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 32, T_{\max} = 33, \Delta = -50, \det = 2, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

$$\begin{aligned}
& [2471] \quad 1 \quad \overset{1^1}{15} (12) \quad 12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, \quad T_{\max} = 33, \quad \Delta = -46, \quad \det = 2, \quad (n_+, n_-, n_0) = \\
& (1, 32, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 33, \quad T_{\max} = 33, \quad T \in [33, 33] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0} \\
& [2472] \quad 1 \quad \overset{1^1}{15} (12) \quad 12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, \quad T_{\max} = 33, \quad \Delta = -46, \quad \det = 2, \quad (n_+, n_-, n_0) = \\
& (1, 32, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 33, \quad T_{\max} = 33, \quad T \in [33, 33] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
& [2465] \quad 21 \quad \overset{1^1}{15} (12) \quad 12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, \quad T_{\max} = 33, \quad \Delta = -50, \quad \det = 2, \quad (n_+, n_-, n_0) = \\
& (1, 32, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 33, \quad T_{\max} = 33, \quad T \in [33, 33] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
& [2475] \quad 1 \quad \overset{1^1}{15} (12) \quad 12231513221(12)12231513161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, \quad T_{\max} = 34, \quad \Delta = -76, \quad \det = 3, \quad (n_+, n_-, n_0) = \\
& (1, 32, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 34, \quad T_{\max} = 34, \quad T \in [34, 34] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& [2473] \quad 1 \quad \overset{1^1}{321812321812321812321} \quad 1232181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, \quad T_{\max} = 34, \quad \Delta = -72, \quad \det = 3, \quad (n_+, n_-, n_0) = \\
& (1, 32, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 34, \quad T_{\max} = 34, \quad T \in [34, 34] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& [2474] \quad 1 \quad \overset{1^1}{71232181232181231513221(12)12231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, \quad T_{\max} = 34, \quad \Delta = -74, \quad \det = 3, \quad (n_+, n_-, n_0) = \\
& (1, 32, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 34, \quad T_{\max} = 34, \quad T \in [34, 34] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
& [2476] \quad 61 \quad \overset{1^1}{232181232181231513221(12)1223151^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 32, \quad T_{\max} = 40, \quad \Delta = -99, \quad \det = 56, \quad (n_+, n_-, n_0) = \\
& (1, 32, 0), \quad c_{\text{ext}} = 3.0 \\
& T_{\min} = 35.57 \rightarrow 36, \quad T_{\max} = 40, \quad T \in [36, 40] \\
& \begin{pmatrix} 0 & 6 & 1 & 2 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -2 & 0 \\ 2 & 10 & 0 & 3 \end{pmatrix}_{b_0} \\
& T = 33 \quad [2477] \quad \overset{1^1}{15} 132181232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, \quad T_{\max} = 33, \quad \Delta = -14, \quad \det = 0, \quad (n_+, n_-, n_0) = \\
& (1, 33, 1), \quad c_{\text{ext}} = 3.0 \\
& [2478] \quad 1 \quad \overset{1^1}{231513221} \quad \overset{1}{(12)} \quad 12231513221(12)12231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, \quad T_{\max} = 33, \quad \Delta = -22, \quad \det = 0, \quad (n_+, n_-, n_0) = \\
& (1, 33, 1), \quad c_{\text{ext}} = 3.0 \\
& [2479] \quad \overset{1^1}{15} 132181232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, \quad T_{\max} = 33, \quad \Delta = -14, \quad \det = 0, \quad (n_+, n_-, n_0) = \\
& (1, 33, 1), \quad c_{\text{ext}} = 3.0 \\
& [2481] \quad \overset{1^1}{15} 1232181232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, \quad T_{\max} = 33, \quad \Delta = -16, \quad \det = 0, \quad (n_+, n_-, n_0) = \\
& (1, 33, 1), \quad c_{\text{ext}} = 3.0 \\
& [2482] \quad 1 \quad \overset{1^1}{31513221} \quad \overset{1}{(12)} \quad 12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, \quad T_{\max} = 34, \quad \Delta = -29, \quad \det = 0, \quad (n_+, n_-, n_0) = \\
& (1, 33, 1), \quad c_{\text{ext}} = 3.0 \\
& [2483] \quad 1 \quad \overset{1^1}{231513221} \quad \overset{1}{(12)} \quad 12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, \quad T_{\max} = 33, \quad \Delta = -26, \quad \det = 0, \quad (n_+, n_-, n_0) = \\
& (1, 33, 1), \quad c_{\text{ext}} = 3.0 \\
& [2484] \quad \overset{1^1}{15} 132181232181232181232181231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, \quad T_{\max} = 33, \quad \Delta = -12, \quad \det = 0, \quad (n_+, n_-, n_0) = \\
& (1, 33, 1), \quad c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
[2485] & \quad 1^1 22181232181232181232171231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -15, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[2486] & \quad 1^1 16123218123218123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[2487] & \quad 1^1 231513221 (12) 12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[2488] & \quad 1^1 21712321812321812321812321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -15, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[2489] & \quad 1^1 231513221 (12) 12231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[2490] & \quad 1^1 513221 21 (12) 12231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[2480] & \quad 1^1 513221 21 (12) 12231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 33, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 33, 1), c_{\text{ext}} = 3.0 \\
[2500] & \quad 1^1 5131513221(12)12231513221 (12) 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 34, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 34, T_{\max} = 34, T \in [34, 34] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[2501] & \quad 1^1 5131513221(12)12231513221 (12) 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 34, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 34, T_{\max} = 34, T \in [34, 34] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 3 \\ 2 & 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2498] & \quad 1^1 16131513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 34, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 34, T_{\max} = 34, T \in [34, 34] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[2497] & \quad 1^1 2321812321812321812321 8 1232181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 34, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 34, T_{\max} = 34, T \in [34, 34] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[2499] & \quad 1^1 3221 21 (12) 12231513221(12)122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 34, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 34, T_{\max} = 34, T \in [34, 34] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
[2502] & \quad 1^1 413221(12)12231513221 (12) 12231513161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 35, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2503] & \quad 1^1 3221 1 (12) 12231513221(12)122315132171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 33, T_{\max} = 35, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}
\end{aligned}$$

[2504] $1^1 2181232181232181231513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 33, T_{\max} = 35, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 35, T_{\max} = 35, T \in [35, 35]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[2505] $1^1 71232181232181231513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 33, T_{\max} = 39, \Delta = -97, \det = -4, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 36, T_{\max} = 39, T \in [36, 39]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[2506] $1^1 8123218123218123218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 33, T_{\max} = 38, \Delta = -95, \det = -41, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 36.12 \rightarrow 37, T_{\max} = 38, T \in [37, 38]$
 $\begin{pmatrix} 0 & 5 & 1 & 2 \\ 5 & 19 & 1 & 9 \\ 1 & 1 & -2 & 0 \\ 2 & 9 & 0 & 3 \end{pmatrix}_{b_0}$

$T = 34$ [2507] $1^1 161232181232181232181232181232151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 34, T_{\max} = 34, \Delta = -13, \det = 0, (n_+, n_-, n_0) = (1, 34, 1), c_{\text{ext}} = 3.0$

[2508] $1^1 16123218123218123218123218123151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 34, T_{\max} = 34, \Delta = -11, \det = 0, (n_+, n_-, n_0) = (1, 34, 1), c_{\text{ext}} = 3.0$

[2509] $1^1 22181232181232181232181232181231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 34, T_{\max} = 34, \Delta = -14, \det = 0, (n_+, n_-, n_0) = (1, 34, 1), c_{\text{ext}} = 3.0$

[2510] $1^1 221812321812321812321812321812231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 34, T_{\max} = 34, \Delta = -14, \det = 0, (n_+, n_-, n_0) = (1, 34, 1), c_{\text{ext}} = 3.0$

[2511] $1^1 217123218123218123218123218123141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 34, T_{\max} = 34, \Delta = -14, \det = 0, (n_+, n_-, n_0) = (1, 34, 1), c_{\text{ext}} = 3.0$

[2522] $1^1 32181232181232181231513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 34, T_{\max} = 35, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 35, T_{\max} = 35, T \in [35, 35]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[2523] $1^1 32181232181232181231513221(12)12231^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 34, T_{\max} = 35, \Delta = -53, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 35, T_{\max} = 35, T \in [35, 35]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2515] $1^1 16131513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 34, T_{\max} = 35, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 35, T_{\max} = 35, T \in [35, 35]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$

[2516] $1^1 1513221 12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 34, T_{\max} = 35, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 35, T_{\max} = 35, T \in [35, 35]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2517] $1^1 1513221 12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 34, T_{\max} = 35, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 35, T_{\max} = 35, T \in [35, 35]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

$$\begin{aligned}
& [2518] \quad 1^{13221} 21 (12) 12231513221(12)1223151322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 35, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
& [2519] \quad 1^{141413221}(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 35, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
& [2514] \quad 1^{161231513221}(12)12231513221 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 35, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
& [2521] \quad 1^{171231513221}(12)12231513221(12)12231^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 35, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
& [2520] \quad 1^{141413221}(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 35, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 35, T_{\max} = 35, T \in [35, 35] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
& [2524] \quad 1^{13221} 1 (12) 12231513221(12)1223151322181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 36, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& [2525] \quad 1^{1413221}(12)12231513221 122315132171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 36, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& [2526] \quad 1^{181231513221}(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 36, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
& [2527] \quad 1^{1513221}(12)12231513221(12)12231513161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 36, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& [2528] \quad 1^{181232181232181231513221}(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 40, \Delta = -96, \det = 4, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 37, T_{\max} = 40, T \in [37, 40] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
& [2529] \quad 21^{181232181232181231513221}(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 34, T_{\max} = 40, \Delta = -96, \det = 8, (n_+, n_-, n_0) = (1, 34, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 37.50 \rightarrow 38, T_{\max} = 40, T \in [38, 40]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

$$T = 35 \quad [2534] \quad 221 \overset{1}{8} 123218123218123218123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, \quad T_{\max} = 35, \quad \Delta = -13, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 35, 1), \quad c_{\text{ext}} = 3.0$$

$$[2530] \quad 1 \quad \overset{1}{(12)} \quad 12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, \quad T_{\max} = 35, \quad \Delta = -24, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 35, 1), \quad c_{\text{ext}} = 3.0$$

$$[2531] \quad 1 \quad \overset{1}{(12)} \quad 12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, \quad T_{\max} = 35, \quad \Delta = -24, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 35, 1), \quad c_{\text{ext}} = 3.0$$

$$[2532] \quad 21 \overset{1}{7} 1232181232181232181232181231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, \quad T_{\max} = 35, \quad \Delta = -9, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 35, 1), \quad c_{\text{ext}} = 3.0$$

$$[2533] \quad 21 \overset{1}{7} 1232181232181232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, \quad T_{\max} = 35, \quad \Delta = -11, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 35, 1), \quad c_{\text{ext}} = 3.0$$

$$[2549] \quad 1 \overset{1}{7} 1231513221(12)12231513221(\overset{1}{12})1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, \quad T_{\max} = 36, \quad \Delta = -45, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 35, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 36, \quad T_{\max} = 36, \quad T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[2541] \quad 1 \overset{1}{6} 131513221(12)12231513221(\overset{1}{12})12231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, \quad T_{\max} = 36, \quad \Delta = -41, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 35, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 36, \quad T_{\max} = 36, \quad T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[2542] \quad 1 \quad \overset{1}{(12)} \quad 12231513161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, \quad T_{\max} = 36, \quad \Delta = -49, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 35, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 36, \quad T_{\max} = 36, \quad T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[2543] \quad 1 \overset{1}{6} 131513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, \quad T_{\max} = 36, \quad \Delta = -45, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 35, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 36, \quad T_{\max} = 36, \quad T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$$

$$[2544] \quad 1 \quad \overset{1}{(12)} \quad 12231513221(12)122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, \quad T_{\max} = 36, \quad \Delta = -42, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 35, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 36, \quad T_{\max} = 36, \quad T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[2545] \quad 1 \quad \overset{1}{(12)} \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, \quad T_{\max} = 36, \quad \Delta = -45, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 35, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 36, \quad T_{\max} = 36, \quad T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[2546] \quad 1 \overset{1}{4} 1413221(12)12231513221(12)122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, \quad T_{\max} = 36, \quad \Delta = -46, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 35, 0), \quad c_{\text{ext}} = 3.0$$

$$T_{\min} = 36, \quad T_{\max} = 36, \quad T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[2547] \quad 21 \overset{1}{8} 1231513221(12)12231513221(\overset{21}{12})12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 35, \quad T_{\max} = 36, \quad \Delta = -45, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 35, 0), \quad c_{\text{ext}} = 3.0$$

$$\begin{aligned}
& T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[2548] & \begin{matrix} 1^1 & & & 21 \\ 1 & 8 & 1231513221(12)12231513221(12)12231^1 + [\text{su}_2 \cdot \text{g}_2] \end{matrix} \quad T = 35, T_{\max} = 36, \Delta = -45, \det = -2, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 36, T_{\max} = 36, T \in [36, 36] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[2551] & \begin{matrix} 1^1 & & & 1 \\ 1 & 8 & 1231513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \text{g}_2] \end{matrix} \quad T = 35, T_{\max} = 37, \Delta = -73, \det = -3, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2553] & \begin{matrix} 1^1 3221 \\ 1 & (12) & 12231513221(12)122315131613161^1 + [\text{su}_2 \cdot \text{g}_2] \end{matrix} \quad T = 35, T_{\max} = 37, \Delta = -75, \det = -3, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[2552] & \begin{matrix} 1^1 413221(12)12231513221 \\ 1 & (12) & 1223151322181^1 + [\text{su}_2 \cdot \text{g}_2] \end{matrix} \quad T = 35, T_{\max} = 37, \Delta = -73, \det = -3, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2550] & \begin{matrix} 1^1 \\ 1 & 5 & 13221(12)12231513221(12)122315132171^1 + [\text{su}_2 \cdot \text{g}_2] \end{matrix} \quad T = 35, T_{\max} = 37, \Delta = -69, \det = -3, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
[2554] & \begin{matrix} 1^1 513221 \\ 1 & (12) & 12231513221(12)122315132181^1 + [\text{su}_2 \cdot \text{g}_2] \end{matrix} \quad T = 35, T_{\max} = 41, \Delta = -97, \det = -4, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38, T_{\max} = 41, T \in [38, 41] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[2555] & \begin{matrix} 1^1 513221 \\ 81^1 2231513221(12)12231513221(12)1^1 + [\text{su}_2 \cdot \text{g}_2] \end{matrix} \quad T = 35, T_{\max} = 41, \Delta = -97, \det = -90, (n_+, n_-, n_0) = \\
& (1, 35, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 38.44 \rightarrow 39, T_{\max} = 41, T \in [39, 41] \\
& \begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 54 & 1 & 14 \\ 1 & 1 & -2 & 0 \\ 2 & 14 & 0 & 3 \end{pmatrix}_{b_0} \\
T = 36 & [2560] \begin{matrix} 1^1 31513221 \\ 1 & (12) & 12231513221(12)1223151316131^1 + [\text{su}_2 \cdot \text{g}_2] \end{matrix} \quad T = 36, T_{\max} = 36, \Delta = -28, \det = \\
& 0, (n_+, n_-, n_0) = (1, 36, 1), c_{\text{ext}} = 3.0 \\
[2563] & \begin{matrix} 1^1 231513221 \\ 1 & (12) & 12231513221(12)122315132161^1 + [\text{su}_2 \cdot \text{g}_2] \end{matrix} \quad T = 36, T_{\max} = 36, \Delta = -22, \det = 0, (n_+, n_-, n_0) = \\
& (1, 36, 1), c_{\text{ext}} = 3.0 \\
[2564] & \begin{matrix} 1^1 \\ 1 & 6 & 131513221(12)12231513221(12)1223151321^1 + [\text{su}_2 \cdot \text{g}_2] \end{matrix} \quad T = 36, T_{\max} = 36, \Delta = -21, \det = 0, (n_+, n_-, n_0) = \\
& (1, 36, 1), c_{\text{ext}} = 3.0 \\
[2561] & \begin{matrix} 1^1 31513221 \\ 1 & (12) & 12231513221(12)1223151316131^1 + [\text{su}_2 \cdot \text{g}_2] \end{matrix} \quad T = 36, T_{\max} = 36, \Delta = -28, \det = 0, (n_+, n_-, n_0) = \\
& (1, 36, 1), c_{\text{ext}} = 3.0 \\
[2559] & \begin{matrix} 1^1 2231513221(12)12231513221 \\ 1 & (12) & 12231513161^1 + [\text{su}_2 \cdot \text{g}_2] \end{matrix} \quad T = 36, T_{\max} = 36, \Delta = -21, \det = 0, (n_+, n_-, n_0) = \\
& (1, 36, 1), c_{\text{ext}} = 3.0 \\
[2556] & \begin{matrix} 1^1 \\ 221 & 8 & 123218123218123218123218123151^1 + [\text{su}_2 \cdot \text{g}_2] \end{matrix} \quad T = 36, T_{\max} = 36, \Delta = -8, \det = 0, (n_+, n_-, n_0) = \\
& (1, 36, 1), c_{\text{ext}} = 3.0
\end{aligned}$$

$$\begin{aligned}
& [2557] \quad 21 \quad (12) \quad 12231513221(12)1223151321812321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 36, 1), c_{\text{ext}} = 3.0 \\
& [2558] \quad 221 \quad 81232181232181232181232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -10, \det = 0, (n_+, n_-, n_0) = (1, 36, 1), c_{\text{ext}} = 3.0 \\
& [2562] \quad 21 \quad (12) \quad 12231513221(12)1223151321812321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\max} = 36, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 36, 1), c_{\text{ext}} = 3.0 \\
& [2580] \quad 1 \quad (12) \quad 122315132171^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\max} = 37, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
& [2581] \quad 1 \quad 812231513221(12)12231513221(12)1223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\max} = 37, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
& [2574] \quad 1 \quad 41413221(12)12231513221(12)1223151322171^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\max} = 37, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
& [2575] \quad 21 \quad 81231513221(12)12231513221(12)1223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\max} = 37, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
& [2576] \quad 1 \quad 71231513221(12)12231513221(12)1223151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\max} = 37, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
& [2577] \quad 1 \quad 5131513221(12)12231513221(12)12231513151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\max} = 37, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
& [2578] \quad 1 \quad 71231513221(12)12231513221(12)122314141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\max} = 37, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0} \\
& [2579] \quad 1 \quad (12) \quad 12231513221(12)1223151322171^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\max} = 37, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 37, T_{\max} = 37, T \in [37, 37] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 4 \\ 2 & 0 & 4 & 6 \end{pmatrix}_{b_0} \\
& [2582] \quad 1 \quad 81231513221(12)12231513221(12)1223151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 36, T_{\max} = 38, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 38, T_{\max} = 38, T \in [38, 38] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}
\end{aligned}$$

[2583] $1^1 413221(12)12231513221(12)122315131613161^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 36, T_{\max} = 38, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[2584] $1^1 1513221(12)12231513221(12)1223151322181^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 36, T_{\max} = 38, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 36, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

$T = 37$ [2589] $1^1 231513221(12)12231513221(12)1223151322171^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 37, T_{\max} = 37, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 37, 1), c_{\text{ext}} = 3.0$

[2585] $1^1 2231513221(12)12231513221(12)122315132171^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 37, T_{\max} = 37, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 37, 1), c_{\text{ext}} = 3.0$

[2586] $1^1 232181231513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 37, T_{\max} = 37, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 37, 1), c_{\text{ext}} = 3.0$

[2587] $1^1 71231513221(12)12231513221(12)1223151321^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 37, T_{\max} = 37, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 37, 1), c_{\text{ext}} = 3.0$

[2588] $1^1 232181231513221(12)12231513221(12)1223141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 37, T_{\max} = 37, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 37, 1), c_{\text{ext}} = 3.0$

[2599] $1^1 812231513221(12)12231513221(12)122314141^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 37, T_{\max} = 38, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$

[2600] $1^1 812231513221(12)12231513221(12)1223151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 37, T_{\max} = 38, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$

[2601] $1^1 61231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 37, T_{\max} = 38, \Delta = -44, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[2602] $1^1 81231513221(12)12231513221(12)122315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 37, T_{\max} = 38, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[2595] $1^1 31513221(12)12231513221(12)1223151322181^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 37, T_{\max} = 38, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

[2596] $1^1 6131513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \mathfrak{g}_2]$ $T = 37, T_{\max} = 38, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[2597] $\overset{1}{21}81231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 37, T_{\max} = 38, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 2 & 5 & -1 & 6 \end{pmatrix}_{b_0}$

[2598] $\overset{1}{21}81231513221(12)12231513221(12)12231\overset{1}{5}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 37, T_{\max} = 38, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 38, T_{\max} = 38, T \in [38, 38]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0}$

[2603] $\overset{1}{1}513221(12)12231513221(12)122315131613161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 37, T_{\max} = 39, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 37, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 39, T_{\max} = 39, T \in [39, 39]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

$T = 38$ [2609] $\overset{1}{1}812231513221(12)12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 38, T_{\max} = 38, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 38, 1), c_{\text{ext}} = 3.0$

[2610] $\overset{1}{1}41413221(12)12231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 38, T_{\max} = 38, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 38, 1), c_{\text{ext}} = 3.0$

[2607] $\overset{1}{1}31513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 38, T_{\max} = 38, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 38, 1), c_{\text{ext}} = 3.0$

[2613] $\overset{1}{21}81231513221(12)12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 38, T_{\max} = 38, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 38, 1), c_{\text{ext}} = 3.0$

[2612] $\overset{1}{1}513221(12)12231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 38, T_{\max} = 38, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 38, 1), c_{\text{ext}} = 3.0$

[2604] $\overset{1}{1}81231513221(12)12231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 38, T_{\max} = 38, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 38, 1), c_{\text{ext}} = 3.0$

[2605] $\overset{1}{1}2231513221(12)12231513221(12)1223151322181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 38, T_{\max} = 38, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 38, 1), c_{\text{ext}} = 3.0$

[2606] $\overset{1}{1}513221(12)12231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 38, T_{\max} = 38, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 38, 1), c_{\text{ext}} = 3.0$

[2611] $\overset{1}{1}41413221(12)12231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 38, T_{\max} = 38, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 38, 1), c_{\text{ext}} = 3.0$

[2608] $\overset{1}{1}31513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 38, T_{\max} = 38, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 38, 1), c_{\text{ext}} = 3.0$

[2626] $\overset{1}{1}712231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 38, T_{\max} = 39, \Delta = -42, \det = 2, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 39, T_{\max} = 39, T \in [39, 39]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[2625] $\overset{1}{1}71231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 38, T_{\max} = 39, \Delta = -42, \det = 2, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 39, T_{\max} = 39, T \in [39, 39]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[2624] $\overset{1}{1}61316131513221(12)12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 38, T_{\max} = 39, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 39, T_{\max} = 39, T \in [39, 39]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2627] & 1^{13221} (12) 12231513221(12)122315132181232171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 38, T_{\max} = 40, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 38, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 40, T_{\max} = 40, T \in [40, 40] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
T = 39 & [2628] 1^{1316131513221}(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 39, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 39, 1), c_{\text{ext}} = 3.0 \\
[2629] & 1^{161316131513221}(12)12231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 39, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 39, 1), c_{\text{ext}} = 3.0 \\
[2635] & 1^{18} 12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 40, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 40, T_{\max} = 40, T \in [40, 40] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2634] & 1^{13221} (12) 12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 40, \Delta = -53, \det = -2, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 40, T_{\max} = 40, T \in [40, 40] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
[2633] & 1^{218} 1231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 40, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 40, T_{\max} = 40, T \in [40, 40] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 2 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
[2636] & 1^{18} 1232181231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 41, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
[2637] & 1^{1413221}(12)12231513221(12)122315132181232171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 39, T_{\max} = 41, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 39, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
T = 40 & [2639] 1^{1232181231513221}(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 40, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 40, 1), c_{\text{ext}} = 3.0 \\
[2638] & 1^{1232181231513221}(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 40, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 40, 1), c_{\text{ext}} = 3.0 \\
[2645] & 1^{13221} (12) 12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 41, \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
[2646] & 1^{13221} (12) 12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 41, \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 41, T_{\max} = 41, T \in [41, 41] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
[2649] & 1^{15} 13221(12)12231513221(12)122315132181232171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 42, \Delta = -68, \det = 3, (n_+, n_-, n_0) =
\end{aligned}$$

$$(1, 40, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 42, T_{\max} = 42, T \in [42, 42]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[2647] \quad 1^1 181232181231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 42, \Delta = -72, \det = 3, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 42, T_{\max} = 42, T \in [42, 42]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$$

$$[2648] \quad 1^1 1^1 513221 (12) 12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 42, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 42, T_{\max} = 42, T \in [42, 42]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[2650] \quad 1^1 513221 (12) 12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 42, \Delta = -76, \det = 3, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 42, T_{\max} = 42, T \in [42, 42]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[2651] \quad 1^1 513221 (12) 12231513221(12)122315132181232181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 40, T_{\max} = 46, \Delta = -96, \det = 4, (n_+, n_-, n_0) = (1, 40, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 43, T_{\max} = 46, T \in [43, 46]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 41 \quad [2654] \quad 1^1 61231513221(12)12231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 41, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 41, 1), c_{\text{ext}} = 3.0$$

$$[2652] \quad 1^1 316131513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 41, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 41, 1), c_{\text{ext}} = 3.0$$

$$[2653] \quad 1^1 316131513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 41, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 41, 1), c_{\text{ext}} = 3.0$$

$$[2655] \quad 1^1 16131513221(12)12231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 41, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 41, 1), c_{\text{ext}} = 3.0$$

$$[2662] \quad 1^1 71232181231513221(12)12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 42, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 42, T_{\max} = 42, T \in [42, 42]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[2663] \quad 1^1 413221(12)12231513221 (12) 12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 42, \Delta = -51, \det = -2, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 42, T_{\max} = 42, T \in [42, 42]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[2664] \quad 1^1 3221 (12) 12231513221(12)12231513221(12)12231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 42, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 42, T_{\max} = 42, T \in [42, 42]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[2665] \quad 1^1 3221 (12) 12231513221(12)12231513221(12)12231^1 51 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 42, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 42, T_{\max} = 42, T \in [42, 42]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& [2666] \overset{1}{1} 8 1232181231513221(12)12231513221(12)12231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 41, T_{\max} = 43, \Delta = -67, \det = -3, (n_+, n_-, n_0) = (1, 41, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 43, T_{\max} = 43, T \in [43, 43] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
& T = 42 \quad [2671] \overset{1}{1} (12) \overset{1}{12231513221(12)12231513221(12)1223151321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 42, 1), c_{\text{ext}} = 3.0 \\
& [2673] \overset{1}{1} 7 1231513221(12)12231513221(12)1223151321812321 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 42, 1), c_{\text{ext}} = 3.0 \\
& [2672] \overset{1}{1} (12) \overset{1}{12231513221(12)12231513221(12)1223151321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 42, 1), c_{\text{ext}} = 3.0 \\
& [2667] \overset{1}{1} \overset{1}{31513221} (12) \overset{1}{12231513221(12)12231513221(12)12231} \overset{21}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 42, 1), c_{\text{ext}} = 3.0 \\
& [2668] \overset{1}{1} \overset{1}{31513221} (12) \overset{1}{12231513221(12)12231513221(12)12231} \overset{21}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 42, 1), c_{\text{ext}} = 3.0 \\
& [2669] \overset{1}{1} 7 12231513221(12)12231513221(12)1223151321812321 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 42, 1), c_{\text{ext}} = 3.0 \\
& [2670] \overset{1}{1} 7 1232181231513221(12)12231513221(12)12231513221 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 42, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 42, 1), c_{\text{ext}} = 3.0 \\
& [2685] \overset{1}{1} 5 13221(12)12231513221(12)12231513221(12)1223141 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 43, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 43, T_{\max} = 43, T \in [43, 43] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
& [2686] \overset{1}{1} 5 13221(12)12231513221(12)12231513221(12)1223141 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 43, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 43, T_{\max} = 43, T \in [43, 43] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0} \\
& [2684] \overset{1}{1} 8 1232181231513221(12)12231513221(12)122315131 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 42, T_{\max} = 43, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 3.0 \\
& T_{\min} = 43, T_{\max} = 43, T \in [43, 43] \\
& \begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0} \\
& T = 43 \quad [2687] \overset{1}{1} 8 12231513221(12)12231513221(12)1223151321812321 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 43, 1), c_{\text{ext}} = 3.0 \\
& [2688] \overset{1}{1} 3 2181231513221(12)12231513221(12)122315132181231 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 43, 1), c_{\text{ext}} = 3.0 \\
& [2689] \overset{1}{1} 8 1232181231513221(12)12231513221(12)12231513221 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 43, 1), c_{\text{ext}} = 3.0 \\
& [2690] \overset{1}{1} \overset{1}{31513221} (12) \overset{1}{12231513221(12)12231513221(12)1223141} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 43, 1), c_{\text{ext}} = 3.0 \\
& [2691] \overset{1}{1} \overset{1}{413221(12)12231513221} (12) \overset{1}{12231513221(12)1223151321} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 43, 1), c_{\text{ext}} = 3.0 \\
& [2692] \overset{1}{1} \overset{1}{413221(12)12231513221} (12) \overset{1}{12231513221(12)1223151321} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 43, 1), c_{\text{ext}} = 3.0 \\
& [2693] \overset{1}{1} \overset{1}{31513221} (12) \overset{1}{12231513221(12)12231513221(12)1223141} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 43, 1), c_{\text{ext}} = 3.0
\end{aligned}$$

[2694] $21 \overset{1}{8} 1231513221(12)12231513221(12)1223151321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 43, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 43, 1), c_{\text{ext}} = 3.0$

[2704] $1 \overset{1}{5} 13221(12)12231513221(12)12231513221(12)12231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 44, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 44, T_{\max} = 44, T \in [44, 44]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

[2705] $1 \overset{1}{6} 131513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 43, T_{\max} = 45, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 43, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 45, T_{\max} = 45, T \in [45, 45]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

$T = 44$ [2707] $1 \overset{1}{3} 1513221(12)12231513221(12)12231513221(12)12231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 3.0$

[2706] $1 \overset{1}{3} 1513221(12)12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 3.0$

[2708] $1 \overset{1}{3} 1513221(12)12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 3.0$

[2709] $1 \overset{1}{5} 13221(12)12231513221(12)12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 3.0$

[2710] $1 \overset{1}{5} 13221(12)12231513221(12)12231513221(12)1223151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 3.0$

[2711] $1 \overset{1}{3} 1513221(12)12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 3.0$

[2712] $1 \overset{1}{3} 1513221(12)12231513221(12)12231513221(12)12231 \overset{1}{5} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 44, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 3.0$

[2721] $1 \overset{1}{7} 1231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 44, T_{\max} = 46, \Delta = -74, \det = 3, (n_+, n_-, n_0) = (1, 44, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 46, T_{\max} = 46, T \in [46, 46]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

$T = 45$ [2722] $21 \overset{1}{3} 13221(12)12231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 45, T_{\max} = 45, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 45, 1), c_{\text{ext}} = 3.0$

[2723] $21 \overset{1}{3} 13221(12)12231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 45, T_{\max} = 45, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 45, 1), c_{\text{ext}} = 3.0$

[2729] $1 \overset{1}{3} 13221(12)12231513221(12)12231513221(12)122315132181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 45, T_{\max} = 47, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 47, T_{\max} = 47, T \in [47, 47]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[2728] $1 \overset{1}{8} 12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 45, T_{\max} = 47, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 47, T_{\max} = 47, T \in [47, 47]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$

[2730] $1 \overset{1}{8} 1231513221(12)12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 45, T_{\max} = 47, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 45, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 47, T_{\max} = 47, T \in [47, 47]$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& T = 46 \quad [2732] \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 46, \quad T_{\max} = 46, \quad \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 46, 1), \quad c_{\text{ext}} = 3.0 \\
& [2733] \quad 1 \quad (12) \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 46, \quad T_{\max} = 46, \quad \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 46, 1), \quad c_{\text{ext}} = 3.0 \\
& [2731] \quad 1 \quad (12) \quad 1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 46, \quad T_{\max} = 46, \quad \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 46, 1), \quad c_{\text{ext}} = 3.0 \\
& [2734] \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 46, \quad T_{\max} = 46, \quad \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 46, 1), \quad c_{\text{ext}} = 3.0 \\
& [2743] \quad 1 \quad (12) \quad 122315132181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 46, \quad T_{\max} = 48, \quad \Delta = -72, \det = 3, (n_+, n_-, n_0) = (1, 46, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 48, \quad T_{\max} = 48, \quad T \in [48, 48] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& [2744] \quad 1 \quad (12) \quad 12231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 46, \quad T_{\max} = 52, \quad \Delta = -96, \det = 4, (n_+, n_-, n_0) = (1, 46, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 49, \quad T_{\max} = 52, \quad T \in [49, 52] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
& [2745] \quad 81^1 2231513221(12)12231513221(12)12231513221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 46, \quad T_{\max} = 52, \quad \Delta = -96, \det = 90, (n_+, n_-, n_0) = (1, 46, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 49.44 \rightarrow 50, \quad T_{\max} = 52, \quad T \in [50, 52] \\
& \quad \begin{pmatrix} 0 & 8 & 1 & 2 \\ 8 & 54 & 1 & 14 \\ 1 & 1 & -2 & 0 \\ 2 & 14 & 0 & 3 \end{pmatrix}_{b_0} \\
& T = 47 \quad [2751] \quad 1^1 41413221(12)12231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 47, \quad T_{\max} = 47, \quad \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 47, 1), \quad c_{\text{ext}} = 3.0 \\
& [2750] \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 47, \quad T_{\max} = 47, \quad \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 47, 1), \quad c_{\text{ext}} = 3.0 \\
& [2749] \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 47, \quad T_{\max} = 47, \quad \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 47, 1), \quad c_{\text{ext}} = 3.0 \\
& [2748] \quad 16^1 131513221(12)12231513221(12)12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 47, \quad T_{\max} = 47, \quad \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 47, 1), \quad c_{\text{ext}} = 3.0 \\
& [2747] \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)122315132161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 47, \quad T_{\max} = 47, \quad \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 47, 1), \quad c_{\text{ext}} = 3.0 \\
& [2746] \quad 21 \quad (12) \quad 12231513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 47, \quad T_{\max} = 47, \quad \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 47, 1), \quad c_{\text{ext}} = 3.0 \\
& [2752] \quad 21 \quad (12) \quad 12231513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 47, \quad T_{\max} = 47, \quad \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 47, 1), \quad c_{\text{ext}} = 3.0 \\
& [2761] \quad 15^1 13221(12)12231513221(12)12231513221(12)122315132181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 47, \quad T_{\max} = 49, \quad \Delta = -67, \det = -3, (n_+, n_-, n_0) = (1, 47, 0), \quad c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 49, \quad T_{\max} = 49, \quad T \in [49, 49] \\
& \quad \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{pmatrix}_{b_0} \\
& T = 48 \quad [2762] \quad 17^1 1231513221(12)12231513221(12)12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 48, \quad T_{\max} = 48, \quad \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 48, 1), \quad c_{\text{ext}} = 3.0
\end{aligned}$$

$[2763] \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)1223151322171^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 48, 1), c_{\text{ext}} = 3.0$

$[2764] \quad 1 \quad (12) \quad 1223141^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 48, 1), c_{\text{ext}} = 3.0$

$[2765] \quad 1 \quad (12) \quad 1223141^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 48, T_{\max} = 48, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 48, 1), c_{\text{ext}} = 3.0$

$[2773] \quad 1 \quad (12) \quad 122315132181^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 48, T_{\max} = 49, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 48, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 49, T_{\max} = 49, T \in [49, 49]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

$T = 49 \quad [2775] \quad 1^1 41413221(12)12231513221(12)12231513221(12)122315132181231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), c_{\text{ext}} = 3.0$

$[2776] \quad 1^1 41413221(12)12231513221(12)12231513221(12)122315132181231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), c_{\text{ext}} = 3.0$

$[2777] \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)122315132181231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), c_{\text{ext}} = 3.0$

$[2778] \quad 21^1 81231513221(12)12231513221(12)12231513221(12)122315131^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), c_{\text{ext}} = 3.0$

$[2779] \quad 1^1 81231513221(12)12231513221(12)12231513221(12)122315131^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), c_{\text{ext}} = 3.0$

$[2780] \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)122315132181231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), c_{\text{ext}} = 3.0$

$[2781] \quad 1 \quad (12) \quad 122315132181^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), c_{\text{ext}} = 3.0$

$[2782] \quad 1^1 316131513221(12)12231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), c_{\text{ext}} = 3.0$

$[2774] \quad 1^1 316131513221(12)12231513221(12)12231513221(12)12231513151^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 49, T_{\max} = 49, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), c_{\text{ext}} = 3.0$

$T = 50 \quad [2790] \quad 1^1 6131513221(12)12231513221(12)12231513221(12)1223151316131^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 50, T_{\max} = 50, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 50, 1), c_{\text{ext}} = 3.0$

$[2791] \quad 1^1 316131513221(12)12231513221(12)12231513221(12)122315132161^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 50, T_{\max} = 50, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 50, 1), c_{\text{ext}} = 3.0$

$[2794] \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)12231513221(12)12231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 50, T_{\max} = 51, \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 51, T_{\max} = 51, T \in [51, 51]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$

$[2795] \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)12231513221(12)12231^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 50, T_{\max} = 51, \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 51, T_{\max} = 51, T \in [51, 51]$
 $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$

$[2796] \quad 1 \quad (12) \quad 12231513221(12)12231513221(12)122315132181232181^1 + [\text{su}_2 \cdot \text{g}_2] \quad T = 50, T_{\max} = 52, \Delta = -72, \det = 3, (n_+, n_-, n_0) = (1, 50, 0), c_{\text{ext}} = 3.0$
 $T_{\min} = 52, T_{\max} = 52, T \in [52, 52]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

$$\begin{aligned}
& T = 51 \quad [2801] \quad 1^{1316131513221(12)12231513221(12)12231513221(12)1223151322171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 51, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 51, 1), c_{\text{ext}} = 3.0 \\
& [2797] \quad 1^{171231513221(12)12231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 51, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 51, 1), c_{\text{ext}} = 3.0 \\
& [2799] \quad 1^{132181231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 51, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 51, 1), c_{\text{ext}} = 3.0 \\
& [2800] \quad 1^{13221} 21 (12) 12231513221(12)12231513221(12)12231513221(12)12231^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 51, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 51, 1), c_{\text{ext}} = 3.0 \\
& [2798] \quad 1^{132181231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 51, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 51, 1), c_{\text{ext}} = 3.0 \\
& [2805] \quad 1^{1413221(12)12231513221(12)12231513221} 1 (12) 12231513221(12)12231^{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 52, \Delta = -51, \det = -2, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 52, T_{\max} = 52, T \in [52, 52] \\
& \quad \left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{smallmatrix} \right)_{b_0} \\
& [2804] \quad 1^{1413221(12)12231513221(12)12231513221} 1 (12) 12231513221(12)12231^{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 52, \Delta = -51, \det = -2, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 52, T_{\max} = 52, T \in [52, 52] \\
& \quad \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{smallmatrix} \right)_{b_0} \\
& [2808] \quad 1^{1413221(12)12231513221(12)12231513221(12)122315132181232181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 53, \Delta = -71, \det = -3, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 53, T_{\max} = 53, T \in [53, 53] \\
& \quad \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 3 \\ 2 & 0 & 3 & 5 \end{smallmatrix} \right)_{b_0} \\
& [2807] \quad 1^{13221(12)12231513221(12)12231513221} 1 (12) 12231513221(12)1223151^{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 53, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 53, T_{\max} = 53, T \in [53, 53] \\
& \quad \left(\begin{smallmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{smallmatrix} \right)_{b_0} \\
& [2806] \quad 1^{13221(12)12231513221(12)12231513221} 1 (12) 12231513221(12)1223151^{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 51, T_{\max} = 53, \Delta = -75, \det = -3, (n_+, n_-, n_0) = (1, 51, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 53, T_{\max} = 53, T \in [53, 53] \\
& \quad \left(\begin{smallmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 3 & 3 & 6 \end{smallmatrix} \right)_{b_0} \\
& T = 52 \quad [2814] \quad 1^{13221} 21 (12) 12231513221(12)12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 52, T_{\max} = 52, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 52, 1), c_{\text{ext}} = 3.0 \\
& [2813] \quad 1^{13221} 21 (12) 12231513221(12)12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 52, T_{\max} = 52, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 52, 1), c_{\text{ext}} = 3.0 \\
& [2812] \quad 1^{12181231513221(12)12231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 52, T_{\max} = 52, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 52, 1), c_{\text{ext}} = 3.0 \\
& [2811] \quad 1^{11812231513221(12)12231513221(12)12231513221(12)1223151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 52, T_{\max} = 52, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 52, 1), c_{\text{ext}} = 3.0 \\
& [2810] \quad 1^{116131513221(12)12231513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 52, T_{\max} = 52, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 52, 1), c_{\text{ext}} = 3.0 \\
& [2809] \quad 1^{161231513221(12)12231513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 52, T_{\max} = 52, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 52, 1), c_{\text{ext}} = 3.0 \\
& [2816] \quad 1^{11513221(12)12231513221(12)12231513221(12)12231513221(12)12231^{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 52, T_{\max} = 53, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 3.0 \\
& \quad T_{\min} = 53, T_{\max} = 53, T \in [53, 53]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[2815] \quad 1^1 \bar{5} 13221(12)12231513221(12)12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 52, T_{\max} = 53, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 53, T_{\max} = 53, T \in [53, 53]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[2817] \quad 1^1 \bar{5} 13221(12)12231513221(12)12231513221(12)122315132181232181^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 52, T_{\max} = 54, \Delta = -66, \det = 3, (n_+, n_-, n_0) = (1, 52, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 54, T_{\max} = 54, T \in [54, 54]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$T = 53 \quad [2825] \quad 1^1 \bar{5} 13221(12)12231513221(12)12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 53, T_{\max} = 53, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$$

$$[2824] \quad 1^1 \bar{7} 12231513221(12)12231513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 53, T_{\max} = 53, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$$

$$[2823] \quad 1^1 \bar{2} 31513221(12)12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 53, T_{\max} = 53, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$$

$$[2826] \quad 1^1 \bar{5} 13221(12)12231513221(12)12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 53, T_{\max} = 53, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$$

$$[2822] \quad 1^1 \bar{2} 31513221(12)12231513221(12)12231513221(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 53, T_{\max} = 53, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$$

$$[2821] \quad 1^1 \bar{3} 221(12)12231513221(12)12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 53, T_{\max} = 53, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$$

$$[2820] \quad 1^1 \bar{3} 221(12)12231513221(12)12231513221(12)12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 53, T_{\max} = 53, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$$

$$[2819] \quad 1^1 \bar{4} 13221(12)12231513221(12)12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 53, T_{\max} = 53, \Delta = -21, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$$

$$[2818] \quad 1^1 \bar{7} 1231513221(12)12231513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 53, T_{\max} = 53, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 53, 1), c_{\text{ext}} = 3.0$$

$$[2829] \quad 1^1 \bar{8} 1232181231513221(12)12231513221(12)12231513221(12)122315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 53, T_{\max} = 54, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 53, 0), c_{\text{ext}} = 3.0$$

$$T_{\min} = 54, T_{\max} = 54, T \in [54, 54]$$

$$\begin{pmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$T = 54 \quad [2836] \quad 1^1 \bar{8} 12231513221(12)12231513221(12)12231513221(12)122315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 54, T_{\max} = 54, \Delta = -15, \det = 0, (n_+, n_-, n_0) = (1, 54, 1), c_{\text{ext}} = 3.0$$

$$[2835] \quad 1^1 \bar{5} 13221(12)12231513221(12)12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 54, T_{\max} = 54, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 54, 1), c_{\text{ext}} = 3.0$$

$$[2834] \quad 1^1 \bar{8} 1232181231513221(12)12231513221(12)12231513221(12)12231513221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 54, T_{\max} = 54, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 54, 1), c_{\text{ext}} = 3.0$$

$$[2833] \quad 1^1 \bar{4} 1413221(12)12231513221(12)12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 54, T_{\max} = 54, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 54, 1), c_{\text{ext}} = 3.0$$

$$[2832] \quad 1^1 \bar{5} 13221(12)12231513221(12)12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 54, T_{\max} = 54, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 54, 1), c_{\text{ext}} = 3.0$$

$$[2831] \quad 1^1 \bar{4} 1413221(12)12231513221(12)12231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 54, T_{\max} = 54, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 54, 1), c_{\text{ext}} = 3.0$$

[2830] $21\ 8\ 1231513221(12)12231513221(12)12231513221(12)122315132181231^{\mathbf{1}}$ + $[\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 54$, $T_{\max} = 54$, $\Delta = -15$, $\det = 0$, $(n_+, n_-, n_0) = (1, 54, 1)$, $c_{\text{ext}} = 3.0$

[2839] $\overset{1}{1} \overset{1}{5} 13221 (12) 12231513221 (12) 12231513221 (12) 12231513221 (12) 12231 \overset{1}{5} 1 + [\textcolor{red}{5} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 55, \quad T_{\max} = 55, \quad \Delta = -11, \det = 0, (n_+, n_-, n_0) = (1, 55, 1), \quad c_{\text{ext}} = 3.0$

$$[2842] \quad 21(12)12231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 55, T_{\max} = 55, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 55, 1), c_{\text{ext}} = 3.0$$

$$\begin{array}{l} \text{[2841]} \quad \overset{1}{1} \overset{1}{5} \overset{1}{13} \overset{2}{22} \overset{1}{1} \quad (12) \quad 12231513221(12)12231513221(12)12231513221(12)122314141^{\textcolor{red}{1}} + [\textcolor{red}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 55, \quad T_{\max} = 55, \quad \Delta = \\ -15, \det = 0, (n_+, n_-, n_0) = (1, 55, 1), \quad c_{\text{ext}} = 3.0 \end{array}$$

$$[2845] \overset{1^1}{16} 131513221(12)12231513221(12)12231513221(12)12231513221(12)12231^{\overset{21}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] \quad T = 56, T_{\max} = 56, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 56, 1), c_{\text{ext}} = 3.0$$

$$1^1 5131513221(12)12231513221(12)12231513221(12)12231513221$$

$$[2847] \quad 1 \quad (12) \quad 1223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 56, T_{\max} = 56, \Delta = -18, \det =$$

$$0, (n_+, n_-, n_0) = (1, 56, 1), c_{\text{ext}} = 3.0$$

[2854] $1 \overset{1}{\underset{1 \ 5 \ 13221}{(12)}} 12231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 57, T_{\max} = 57, \Delta = -13, \det = 0, (n_+, n_-, n_0) = (1, 57, 1), c_{\text{ext}} = 3.0$

$$\begin{array}{l} \text{[2851]} \quad 21 \text{ (12) } 12231513221(12)12231513221(12)12231513221(12)1223151322171^1 + [\text{su}_2 \cdot \mathfrak{g}_2] \quad T = 57, T_{\max} = 57, \Delta = \\ -15, \det = 0, (n_+, n_-, n_0) = (1, 57, 1), c_{\text{ext}} = 3.0 \end{array}$$

$$[2849] \quad 1^{16} 1231513221(12)12231513221(12)12231513221(12)12231513221(12) \quad T = 57, \quad T_{\max} = 57, \quad \Delta = -16, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 57, 1), \quad c_{\text{ext}} = 3.0$$

[2855] $1^1 41413221(12)12231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 57, \quad T_{\max} = 57, \quad \Delta = -17, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 57, 1), \quad c_{\text{ext}} = 3.0$

1770

$$[2878] \ 1^1 8 \ 12231513221(12)12231513221(12)12231513221(12)12231513221(12)12231513151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 62, \ T_{\max} = 62, \ \Delta = -10, \ \det = 0, \ (n_+, n_-, n_0) = (1, 62, 1), \ c_{\text{ext}} = 3.0$$

4.41 $\text{NHC}(-2)(-3) + \mathfrak{e}_6$

$$T = 6 \quad [1] \ 1^1 1^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 6, \ T_{\max} = 100, \ \Delta = -193, \ \det = 20, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 4 & 5 \\ 3 & 4 & -3 & -1 \\ 3 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[0] \ 21^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 6, \ T_{\max} = 100, \ \Delta = -193, \ \det = 20, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 4 & 5 \\ 3 & 4 & -3 & -1 \\ 3 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^1 1^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 6, \ T_{\max} = 100, \ \Delta = -193, \ \det = 26, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -6 & -4 \\ 3 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^1 1^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 6, \ T_{\max} = 100, \ \Delta = -193, \ \det = 33, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.91 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & -3 & -1 \\ 3 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 21^1 1^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 6, \ T_{\max} = 100, \ \Delta = -193, \ \det = 131, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 4 & 8 & 3 & 4 \\ 8 & 9 & 4 & 5 \\ 3 & 4 & -3 & -1 \\ 4 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [5] \ 2^1 2^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -165, \ \det = -13, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.54 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 4 & 5 \\ 3 & 4 & -2 & 0 \\ 3 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 21^1 321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -165, \ \det = -15, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 5 & 5 & 5 \\ 3 & 5 & -2 & 0 \\ 4 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^1 1^1 321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -165, \ \det = -21, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.14 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -3 & -1 \\ 2 & 2 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^1 1^1 321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -165, \ \det = -23, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 13.35 \rightarrow 14, \ T_{\max} = 100, \ T \in [14, 100]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & -2 & 0 \\ 3 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^1 1^1 3131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -172, \ \det = -26, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 13.08 \rightarrow 14, \ T_{\max} = 100, \ T \in [14, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -6 & -4 \\ 2 & 3 & 2 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^1 1^1 3 \ 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -165, \ \det = -26, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 13.08 \rightarrow 14, \ T_{\max} = 100, \ T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & -2 & 0 \\ 3 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^1 31^1 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathfrak{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 \\ 2 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & -6 & -4 \\ 2 & 3 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^1 31^1 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathfrak{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[13] \quad 2^1 2^1 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathfrak{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -165, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & -2 & 0 \\ 3 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 21^1 1^1 321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathfrak{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -165, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 4 & 8 & 3 & 4 \\ 8 & 9 & 4 & 5 \\ 3 & 4 & -2 & 0 \\ 4 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^1 31^1 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathfrak{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.69 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 7 & 3 & 4 \\ 2 & 7 & 3 & 3 & 3 \\ 2 & 3 & 3 & -3 & -1 \\ 2 & 4 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 131^1 1^1 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathfrak{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -172, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 4 & 7 & 3 & 4 \\ 2 & 7 & 3 & 3 & 3 \\ 1 & 3 & 3 & -3 & -1 \\ 2 & 4 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [17] \quad 131^1 321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 2 & 4 & 4 & 4 \\ 2 & 4 & 3 & 3 & 3 \\ 1 & 4 & 3 & -2 & 0 \\ 2 & 4 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^1 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 3 & 1 & -3 & -1 \\ 2 & 3 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^1 1^1 23131^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^1 3221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 5 & 15 & 5 \\ 15 & 42 & 12 \\ 5 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^1 1^1 2321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -3 & -1 \\ 2 & 3 & -1 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^1 1^1 2321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

- [23] $1^1 3221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6$ $T = 8, T_{\max} = 98, \Delta = -136, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$
- [24] $1^1 23131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6$ $T = 8, T_{\max} = 98, \Delta = -144, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & 0 & 0 \end{pmatrix}_{b_0}$
- [25] $1^{11} 23131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6$ $T = 8, T_{\max} = 98, \Delta = -144, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 5 & 0 & 4 \\ 1 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & -6 & -4 \\ 2 & 4 & 3 & -4 & 0 \end{pmatrix}_{b_0}$
- [26] $1^1 231^{131} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6$ $T = 8, T_{\max} = 98, \Delta = -144, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 15, T_{\max} = 98, T \in [15, 98]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 2 & 4 & 0 & 4 \\ 2 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & -6 & -4 \\ 2 & 4 & 3 & -4 & -1 \end{pmatrix}_{b_0}$
- [27] $1^1 4131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6$ $T = 8, T_{\max} = 101, \Delta = -163, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -5 & -3 \\ 3 & 4 & -3 & 1 \end{pmatrix}_{b_0}$
- [28] $1^1 12321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6$ $T = 8, T_{\max} = 98, \Delta = -138, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14.35 \rightarrow 15, T_{\max} = 98, T \in [15, 98]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & -3 & -1 \\ 2 & 3 & -1 & -1 & -1 \end{pmatrix}_{b_0}$
- [29] $1^{11} 12321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6$ $T = 8, T_{\max} = 98, \Delta = -138, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 2 \\ 2 & 3 & -1 & 2 & 1 \end{pmatrix}_{b_0}$
- [30] $1^{11} 231^{131} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6$ $T = 8, T_{\max} = 98, \Delta = -144, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 2 & 4 & 4 & 4 \\ 2 & 4 & 3 & 3 & 3 \\ 2 & 4 & 3 & -2 & 0 \\ 2 & 4 & 3 & 0 & 1 \end{pmatrix}_{b_0}$
- [31] $1^1 4131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6$ $T = 8, T_{\max} = 101, \Delta = -163, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14.08 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$
 $\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 9 & 0 & 5 \\ 1 & 0 & -5 & -3 \\ 3 & 5 & -3 & 0 \end{pmatrix}_{b_0}$
- [32] $1^1 231^{131} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6$ $T = 8, T_{\max} = 98, \Delta = -144, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.12 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 3 & 3 \\ 1 & 0 & 3 & -2 & 0 \\ 2 & 0 & 3 & 0 & 0 \end{pmatrix}_{b_0}$
- [33] $1^1 231^{131} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6$ $T = 8, T_{\max} = 98, \Delta = -144, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.22 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -6 & -4 \\ 2 & 4 & 3 & -4 & 0 \end{pmatrix}_{b_0}$
- [34] $1^1 231^{131} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6$ $T = 8, T_{\max} = 98, \Delta = -144, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 \\ 2 & 4 & -1 & 0 & 1 \end{pmatrix}_{b_0}$

[35] $21^{1^1 41^1 31^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 8, T_{\max} = 101, \Delta = -163, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 2 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{1^1 31^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 8, T_{\max} = 101, \Delta = -163, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.71 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 4 & 0 & 4 \\ 1 & 0 & -5 & -3 \\ 3 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

[37] $21^{1^1 4131^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 8, T_{\max} = 101, \Delta = -163, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.53 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 4 & 7 & 0 & 4 \\ 2 & 7 & 3 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 4 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[38] $31^{1^1 2321^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 8, T_{\max} = 98, \Delta = -146, \det = 215, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -3 & -1 \\ 2 & 3 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [39] $1^{1^1 223131^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 16.14 \rightarrow 17, T_{\max} = 26, T \in [17, 26]$

$$\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 3 & 10 & -1 & -1 \end{pmatrix}_{b_0}$$

[40] $1^{1^1 231321^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 16.14 \rightarrow 17, T_{\max} = 29, T \in [17, 29]$

$$\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 3 & 10 & -1 & -1 \end{pmatrix}_{b_0}$$

[41] $1^{1^1 412321^{1^1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 99, \Delta = -137, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & -4 & -2 \\ 3 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{1^1} 141321^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & -4 & -2 \\ 3 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{1^1} 15131^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14.14 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 \\ 2 & 3 & -2 & 0 \\ 3 & 4 & 0 & 1 \end{pmatrix}_{b_0}$$

[44] $21^{1^1 5131^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14.22 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 3 & 4 \\ 3 & 3 & -2 & 0 \\ 4 & 4 & 0 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{1^1} 31^{1^1} 4131^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14.22 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 3 & 0 & -3 & -1 \\ 2 & 4 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{1^1} 31^{1^1} 4131^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.82 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[47] $21^{1^1 5131^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 14.15 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 4 & 3 & 4 \\ 2 & 3 & -2 & 0 \\ 3 & 4 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[48] \quad 2 \overset{1}{3} 21^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -137, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 10 & 22 & 6 \\ 22 & 42 & 12 \\ 6 & 12 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [50] \quad 1^{\overset{1}{1}} 3141321^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 17.14 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[49] \quad 1413221^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 17.14 \rightarrow 18, T_{\max} = 22, T \in [18, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 \\ 1 & 3 & -3 & -1 \\ 3 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{\overset{1}{1}} 315131^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 10, T_{\max} = 100, \Delta = -137, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -3 & -1 \\ 2 & 3 & -1 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{\overset{1}{1}} 51321^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 10, T_{\max} = 93, \Delta = -130, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 16, T_{\max} = 93, T \in [16, 93]$$

$$\begin{pmatrix} 5 & 15 & 5 \\ 15 & 42 & 12 \\ 5 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{\overset{1}{1}} 512321^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 10, T_{\max} = 94, \Delta = -132, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 16, T_{\max} = 94, T \in [16, 94]$$

$$\begin{pmatrix} 5 & 15 & 5 \\ 15 & 42 & 12 \\ 5 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[51] \quad 21^{\overset{1}{1}} 51321^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 10, T_{\max} = 93, \Delta = -130, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 16, T_{\max} = 93, T \in [16, 93]$$

$$\begin{pmatrix} 5 & 15 & 5 \\ 15 & 42 & 12 \\ 5 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{\overset{1}{1}} 315131^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 10, T_{\max} = 100, \Delta = -137, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{\overset{1}{1}} 3151^{\overset{1}{1}} 231^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 10, T_{\max} = 100, \Delta = -138, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 13 & 1 & 5 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 5 & 0 & -2 & 0 \\ 2 & 7 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[57] \quad 31^{\overset{1}{1}} 1315131^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 10, T_{\max} = 100, \Delta = -145, \det = 215, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 14.92 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -3 & -1 \\ 2 & 3 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [58] \quad 1^{\overset{1}{1}} 513221^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 11, T_{\max} = 23, \Delta = -101, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 18.14 \rightarrow 19, T_{\max} = 23, T \in [19, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 7 & 7 & 7 \\ 2 & 7 & 0 & 2 \\ 3 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{\overset{1}{1}} 23215131^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 18.14 \rightarrow 19, T_{\max} = 24, T \in [19, 24]$$

$$\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 3 & 10 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{11} 21513221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 11, T_{\max} = 23, \Delta = -101, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 18.14 \rightarrow 19, T_{\max} = 23, T \in [19, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 7 & 7 & 7 \\ 2 & 7 & 0 & 2 \\ 3 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{11} 5^{11} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 18.14 \rightarrow 19, T_{\max} = 23, T \in [19, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 4 & 1 & -2 & 0 \\ 2 & 4 & 2 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{11} 23151231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 18.14 \rightarrow 19, T_{\max} = 23, T \in [19, 23]$$

$$\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 3 & 10 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[63] \quad 14141321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 11, T_{\max} = 23, \Delta = -105, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 18.14 \rightarrow 19, T_{\max} = 23, T \in [19, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^1 3151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 1 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^1 3151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[66] \quad 13151321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{11} 6^{11} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 11, T_{\max} = 97, \Delta = -135, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 97, T \in [16, 97]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & -3 & -1 \\ 2 & 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{11} 6^{11} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 11, T_{\max} = 96, \Delta = -134, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 96, T \in [16, 96]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & -3 & -1 \\ 2 & 2 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^1 3151321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 5 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 3 \\ 2 & 4 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[70] \quad 31^1 316131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 \quad T = 11, T_{\max} = 101, \Delta = -142, \det = -44, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 15.82 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 3 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & -3 & -1 \\ 2 & 2 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

[71] $31^1 321^1 6131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 11, T_{\max} = 101, \Delta = -143, \det = -110, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 15.82 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 0 & 4 & 3 & 1 & 2 \\ 4 & 10 & 5 & 3 & 6 \\ 3 & 5 & 0 & 1 & 2 \\ 1 & 3 & 1 & -3 & -1 \\ 2 & 6 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

[72] $2321^1 6131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 11, T_{\max} = 101, \Delta = -137, \det = -151, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 15.62 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 0 & 4 & 2 & 1 & 2 \\ 4 & 10 & 5 & 3 & 6 \\ 2 & 5 & 0 & 1 & 2 \\ 1 & 3 & 1 & -3 & -1 \\ 2 & 6 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{311} 6^{31} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 11, T_{\max} = 101, \Delta = -142, \det = -215, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 15.92 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -3 & -1 \\ 2 & 3 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$T = 12$ [74] $1^1 31^1 513221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

[75] $1^{11} 6^{11} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 19.14 \rightarrow 20, T_{\max} = 24, T \in [20, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

[76] $1^1 413151321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 19.14 \rightarrow 20, T_{\max} = 24, T \in [20, 24]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 \\ 1 & 3 & -3 & -1 \\ 3 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

[77] $1^1 321612321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 12, T_{\max} = 24, \Delta = -109, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 19.14 \rightarrow 20, T_{\max} = 24, T \in [20, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

[78] $1^{11} 6^{11} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 15.45 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 3 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[79] $1^{131} 6^{131} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 12, T_{\max} = 27, \Delta = -113, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 15.45 \rightarrow 16, T_{\max} = 27, T \in [16, 27]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & -3 & 2 \end{pmatrix}_{b_0}$$

[80] $1^1 31^1 513221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 6 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 4 & 4 \\ 2 & 5 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[81] $1^{131} 6^{131} 13141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 12, T_{\max} = 96, \Delta = -133, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 16.25 \rightarrow 17, T_{\max} = 96, T \in [17, 96]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & -2 & 0 \\ 2 & 3 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [86] \quad 1^{11} 2315132151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 13, T_{\max} = 25, \Delta = -104, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 20.14 \rightarrow 21, T_{\max} = 25, T \in [21, 25]$$

$$\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 3 & 10 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^1 4131612321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 20.14 \rightarrow 21, T_{\max} = 25, T \in [21, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^1 1513151321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 20.14 \rightarrow 21, T_{\max} = 25, T \in [21, 25]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 7 & 7 & 7 \\ 2 & 7 & 0 & 2 \\ 3 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{11} 2321712231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 20.14 \rightarrow 21, T_{\max} = 25, T \in [21, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{11} 2321^1 7^1 1231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 20.14 \rightarrow 21, T_{\max} = 25, T \in [21, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^1 151316131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 13, T_{\max} = 94, \Delta = -128, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 17.25 \rightarrow 18, T_{\max} = 94, T \in [18, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & -3 & -1 \\ 2 & 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [90] \quad 1^1 41321712321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 14, T_{\max} = 26, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 21.14 \rightarrow 22, T_{\max} = 26, T \in [22, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^1 8^1 1231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 14, T_{\max} = 26, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 21.14 \rightarrow 22, T_{\max} = 26, T \in [22, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 4 & 1 & -2 & 0 \\ 2 & 4 & 2 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^1 15131612321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 14, T_{\max} = 26, \Delta = -102, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 21.14 \rightarrow 22, T_{\max} = 26, T \in [22, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[91] \quad 21^1 8^1 1231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 14, T_{\max} = 26, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 21.14 \rightarrow 22, T_{\max} = 26, T \in [22, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^1 8^1 12321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 14, T_{\max} = 26, \Delta = -106, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 21.14 \rightarrow 22, T_{\max} = 26, T \in [22, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[93] \ 1^{\overline{1}1^{1315131}} \ 6 \ 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 11, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 17.45 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 3 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [94] \ 1^{\overline{1}1^{2315131}} \ 6 \ 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 15, T_{\max} = 18, \Delta = -79, \det = -6, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 1 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[98] \ 1^{\overline{1}1^{2321}} \ 8 \ 123141^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 15, T_{\max} = 27, \Delta = -105, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 22.14 \rightarrow 23, T_{\max} = 27, T \in [23, 27]$$

$$\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 28 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 3 & 10 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[97] \ 1^{\overline{1}1^{232171232151}} \ 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 15, T_{\max} = 27, \Delta = -103, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 22.14 \rightarrow 23, T_{\max} = 27, T \in [23, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[96] \ 1^{\overline{1}1^{413221812321}} \ 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 15, T_{\max} = 27, \Delta = -105, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 22.14 \rightarrow 23, T_{\max} = 27, T \in [23, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[95] \ 1^{\overline{1}} \ 15 \ 1321712321^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 15, T_{\max} = 27, \Delta = -101, \det = -7, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 22.14 \rightarrow 23, T_{\max} = 27, T \in [23, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$T = 16 \quad [102] \ 1^{\overline{1}1^{512321}} \ 8 \ 12321^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 16, T_{\max} = 28, \Delta = -102, \det = 7, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 23.14 \rightarrow 24, T_{\max} = 28, T \in [24, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[101] \ 1^{\overline{1}1^{51321}} \ 8 \ 12321^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 16, T_{\max} = 28, \Delta = -100, \det = 7, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 23.14 \rightarrow 24, T_{\max} = 28, T \in [24, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[100] \ 1^{\overline{1}1^{4141321812321}} \ 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 16, T_{\max} = 28, \Delta = -104, \det = 7, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 23.14 \rightarrow 24, T_{\max} = 28, T \in [24, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[99] \ 1^{\overline{1}} \ 15 \ 13221812321^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 16, T_{\max} = 28, \Delta = -100, \det = 7, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 23.14 \rightarrow 24, T_{\max} = 28, T \in [24, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$T = 18 \quad [103] \quad 1^{11}232181231513151^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 18, T_{\max} = 30, \Delta = -102, \det = 7, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 25.14 \rightarrow 26, T_{\max} = 30, T \in [26, 30]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$T = 19 \quad [104] \quad 1^{11}5131612321812321^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 19, T_{\max} = 31, \Delta = -101, \det = -7, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 26.14 \rightarrow 27, T_{\max} = 31, T \in [27, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

$$T = 20 \quad [105] \quad 1^{11}51321712321812321^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 20, T_{\max} = 32, \Delta = -100, \det = 7, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 27.14 \rightarrow 28, T_{\max} = 32, T \in [28, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

$$T = 21 \quad [107] \quad 1^{11}232181232181223151^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 21, T_{\max} = 29, \Delta = -99, \det = -7, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 28.14 \rightarrow 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[106] \quad 1^{11}51321 \quad 12321812321^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 \quad T = 21, T_{\max} = 29, \Delta = -99, \det = -7, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 28.14 \rightarrow 29, T_{\max} = 29, T \in [29, 29]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

4.42 NHC(-2)(-3) + $\textcolor{violet}{e}_6$ + $\textcolor{violet}{e}_6$

$$T = 7 \quad [0] \quad 1^{11}31^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{violet}{e}_6 \quad T = 7, T_{\max} = 125, \Delta = -242, \det = -155, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.06 \rightarrow 16, T_{\max} = 125, T \in [16, 125]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & -1 & -4 & -3 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [1] \quad 1^{11}31^{11}31^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{violet}{e}_6 \quad T = 8, T_{\max} = 103, \Delta = -221, \det = 612, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.35 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 3 & -1 & -4 & -3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \quad 1^{11}231^131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{violet}{e}_6 \quad T = 9, T_{\max} = 103, \Delta = -193, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.50 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 2 & 4 & 4 & 0 & 4 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 3 & 0 & -4 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}2321^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{violet}{e}_6 \quad T = 9, T_{\max} = 103, \Delta = -187, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 3 & 0 & 1 & -3 & -1 \\ 2 & 3 & -1 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [4] \quad 1^{11} 315^{11} 1321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{e}}_6 \quad T = 12, \quad T_{\max} = 104, \quad \Delta = -158, \quad \det = 33, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 17.45 \rightarrow 18, \quad T_{\max} = 104, \quad T \in [18, 104]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 4 & 0 & 1 & -2 & 0 \\ 2 & 4 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

4.43 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{e}_6 + \mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^{11} 1^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 103, \quad \Delta = -212, \quad \det = 385, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.06 \rightarrow 16, \quad T_{\max} = 103, \quad T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -4 & 4 & -3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [5] \quad 1^{11} 2^3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \quad T_{\max} = 102, \quad \Delta = -157, \quad \det = 20, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.40 \rightarrow 17, \quad T_{\max} = 102, \quad T \in [17, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 0 & -1 \\ 1 & 3 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & -1 & -1 & 0 & -3 \end{pmatrix}_{b_0}$$

4.44 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{e}_7$

$$T = 9 \quad [0] \quad 1^{11} 2^3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{e}}_7 \quad T = 9, \quad T_{\max} = 127, \quad \Delta = -242, \quad \det = -126, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.14 \rightarrow 18, \quad T_{\max} = 127, \quad T \in [18, 127]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 1 & -1 \\ 1 & 1 & 0 & 1 & -5 & -3 \\ 2 & 3 & -1 & -1 & -3 & -4 \end{pmatrix}_{b_0}$$

4.45 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{e}'_7$

$$T = 9 \quad [0] \quad 1^{11} 2^3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{e}}'_7 \quad T = 9, \quad T_{\max} = 104, \quad \Delta = -214, \quad \det = -103, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.24 \rightarrow 17, \quad T_{\max} = 104, \quad T \in [17, 104]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 1 & -1 \\ 1 & 1 & 0 & 1 & -4 & -2 \\ 2 & 3 & -1 & -1 & -2 & -3 \end{pmatrix}_{b_0}$$

4.46 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{f}_4$

$$T = 7 \quad [0] \quad 1^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 7, \quad T_{\max} = 102, \quad \Delta = -216, \quad \det = -122, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.20 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & -1 & -3 & -2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [1] \quad 1^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{f}}_4 \quad T = 8, \quad T_{\max} = 102, \quad \Delta = -188, \quad \det = 80, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & -2 & 0 & 0 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 102, \Delta = -188, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 102, \Delta = -195, \det = 481, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.51 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 3 & -1 & -3 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \ 1^{11}231^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{f}_4 \quad T = 9, T_{\max} = 102, \Delta = -167, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 3 \\ 5 & 18 & 16 & 0 & 8 \\ 5 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}231^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{f}_4 \quad T = 9, T_{\max} = 102, \Delta = -167, \det = -25, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 2 & 4 & 4 & 0 & 4 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 3 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11}3221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{f}_4 \quad T = 9, T_{\max} = 101, \Delta = -159, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 5 & 15 & 1 & 5 \\ 15 & 42 & 0 & 12 \\ 1 & 0 & -5 & -3 \\ 5 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11}2321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{f}_4 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.68 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 1 & -1 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11}2321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{f}_4 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.48 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 3 & 0 & 1 & -2 & 0 \\ 2 & 3 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [9] \ 1^{11}3151321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{f}_4 \quad T = 12, T_{\max} = 96, \Delta = -132, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 17, T_{\max} = 96, T \in [17, 96]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 1 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 4 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11}3151321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{f}_4 \quad T = 12, T_{\max} = 96, \Delta = -132, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 16.50 \rightarrow 17, T_{\max} = 96, T \in [17, 96]$$

$$\begin{pmatrix} 1 & 5 & 1 & 2 & 3 \\ 5 & 18 & 1 & 5 & 8 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 \\ 3 & 8 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

4.47 NHC(-2)(-3) + e₆ + f₄ + su₂

$$T = 8 \quad [0] \ 1^{111} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -3 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{112} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.7$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & -1 & -3 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{113} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 739, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 9 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & -3 & 12 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \ 1^{11} 3 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -3 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 7 & 0 & 14 & 8 \\ 3 & 5 & 0 & -3 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.34 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -3 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [8] \ 1^{11} 2 3 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 16.40 \rightarrow 17, T_{\max} = 93, T \in [17, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 0 & -1 \\ 1 & 3 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & -1 & -1 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 2 3 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 16.15 \rightarrow 17, T_{\max} = 93, T \in [17, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 1 & 0 & -1 \\ 1 & 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & -1 & -1 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3 2 2 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 91, \Delta = -129, \det = 127, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.46 \rightarrow 15, T_{\max} = 91, T \in [15, 91]$$

$$\begin{pmatrix} 5 & 15 & 1 & 21 & 5 \\ 15 & 42 & 0 & 59 & 12 \\ 1 & 0 & -5 & 0 & -3 \\ 21 & 59 & 0 & 79 & 17 \\ 5 & 12 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

4.48 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3)$

$$T = 8 \quad [0] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 \\ 2 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 \\ 3 & 5 & 0 & 4 & 3 & 3 \\ 3 & 4 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \ 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 21 & 27 & 1 & 9 \\ 27 & 33 & 1 & 11 \\ 1 & 1 & -3 & -1 \\ 9 & 11 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 4 & 0 \\ 4 & 8 & 0 & 4 & 9 & 5 \\ 3 & 4 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.36 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 4 & 0 & 2 \\ 2 & 3 & 3 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -265, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 4 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [7] \ 1^{11} 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 1 & 3 & 0 & 1 & 1 & 0 & 2 \\ 2 & 3 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{11}2^3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 1 & 0 & 3 & 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{11}4^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.93 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11}231^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 205, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.32 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 0 & 0 \\ 2 & 6 & 0 & 3 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 2 & 4 & -1 & 0 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{11}4^{11} 1^1 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 273, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.91 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 3 \\ 4 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 11$ [13] $1^{11}5^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 23, \Delta = -108, \det = -93, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.61 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 4 & 0 & 4 \\ 2 & 3 & -2 & 2 & 0 & 0 \\ 2 & 4 & 2 & 1 & 1 & 3 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 0 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{11}31^{11} 4131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 15, \Delta = -92, \det = -205, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.32 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 4 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

4.49 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \text{NHC}(-2)(-3)$

$T = 10$ [0] $1^{1111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 5 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 4 & 0 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 4 \\ 3 & 5 & 0 & 3 & 3 & 4 & 3 \\ 3 & 4 & -4 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

4.50 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{so}_8$

$T = 9$ [0] $1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 10 & 13 & 6 \\ 9 & 17 & 0 & 13 & 12 & 6 \\ 5 & 8 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

4.51 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -210, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 6 & 3 \\ 2 & 4 & 0 & 4 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -298, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -312, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 0 & 5 & 4 & 19 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 21 & 4 \\ 2 & 4 & 0 & 4 & 0 & 12 & 2 \\ 11 & 19 & 0 & 21 & 12 & 73 & 15 \\ 3 & 4 & -4 & 4 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -452, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 4 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & -4 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -507, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 0 & 5 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 15 & 4 \\ 2 & 4 & 0 & 4 & 0 & 9 & 2 \\ 8 & 14 & 0 & 15 & 9 & 37 & 11 \\ 3 & 4 & -4 & 4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{112} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -992, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 0 & 5 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 18 & 4 \\ 2 & 4 & 0 & 4 & 0 & 12 & 2 \\ 10 & 18 & 0 & 18 & 12 & 58 & 14 \\ 3 & 4 & -4 & 4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{112} 3^1 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -1212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 0 & 5 & 4 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 30 & 4 \\ 2 & 4 & 0 & 4 & 0 & 18 & 2 \\ 16 & 28 & 0 & 30 & 18 & 154 & 22 \\ 3 & 4 & -4 & 4 & 2 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{112} 3^1 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -1227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 0 & 5 & 4 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 24 & 4 \\ 2 & 4 & 0 & 4 & 0 & 15 & 2 \\ 13 & 23 & 0 & 24 & 15 & 100 & 18 \\ 3 & 4 & -4 & 4 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{113} 3^1 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -1892, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 0 & 5 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 4 & 0 & 18 & 2 \\ 15 & 27 & 0 & 27 & 18 & 133 & 21 \\ 3 & 4 & -4 & 4 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{113} 3^1 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -2307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 0 & 5 & 4 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 33 & 4 \\ 2 & 4 & 0 & 4 & 0 & 21 & 2 \\ 18 & 32 & 0 & 33 & 21 & 193 & 25 \\ 3 & 4 & -4 & 4 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{114} 3^1 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -3152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 0 & 5 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 36 & 4 \\ 2 & 4 & 0 & 4 & 0 & 24 & 2 \\ 20 & 36 & 0 & 36 & 24 & 238 & 28 \\ 3 & 4 & -4 & 4 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [13] \ 1^{111} 3 \ 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 21 & 27 & 1 & 27 & 9 \\ 27 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 35 & 0 & 33 & 11 \\ 9 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 31^{112} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 170, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 4 & 12 & 4 \\ 1 & 3 & 3 & 0 & 4 & 0 & 6 & 2 \\ 4 & 12 & 6 & 0 & 12 & 6 & 22 & 8 \\ 2 & 4 & 2 & -4 & 4 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3 \ 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 236, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 21 & 27 & 1 & 52 & 9 \\ 27 & 33 & 1 & 65 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 52 & 65 & 0 & 123 & 21 \\ 9 & 11 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[16] $1^{111} 3^{111} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 21 & 27 & 1 & 28 & 9 \\ 27 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 35 & 0 & 33 & 11 \\ 9 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[17] $1^{11} 31^{111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 440, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 4 & 6 & 4 \\ 1 & 3 & 3 & 0 & 4 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 6 & 3 & 4 & 4 \\ 2 & 4 & 2 & -4 & 4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

4.52 $\text{NHC}(-2)(-3) + \text{e}_6 + \text{NHC}(-2)(-3) + \text{su}_2 + \text{su}_2$

$T = 10 \quad [0] 1^{1111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 3 & 3 & 4 \\ 2 & 4 & 0 & 4 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & -4 & 4 & 2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{1111} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 429, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 4 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 9 & 6 & 13 & 7 & 7 \\ 3 & 4 & -4 & 4 & 2 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{1111} 31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 1264, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 9 & 9 & 4 \\ 2 & 4 & 0 & 4 & 0 & 6 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 9 & 6 & 15 & 13 & 7 \\ 3 & 4 & -4 & 4 & 2 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

4.53 $\text{NHC}(-2)(-3) + \text{e}_6 + \text{NHC}(-2)(-3) + \text{su}_2 + \text{su}_3$

$T = 10 \quad [0] 1^{1111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 & 3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.54 $\text{NHC}(-2)(-3) + \text{e}_6 + \text{NHC}(-2)(-3) + \text{su}_2 \rightarrow (-3)$

$T = 10 \quad [0] 1^{11} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 21 & 27 & 1 & 16 & 9 \\ 27 & 33 & 1 & 20 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 20 & 0 & 10 & 6 \\ 9 & 11 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

4.55 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{\textcolor{red}{11}1} 3 1^{\textcolor{blue}{11}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{blue}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 4 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 4 & -4 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\textcolor{red}{11}2} 3 1^{\textcolor{blue}{11}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{blue}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 0 & 4 & 5 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 6 & 3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 3 \\ 4 & 8 & 0 & 6 & 6 & 9 & 5 \\ 3 & 4 & -4 & 3 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\textcolor{red}{11}1} 3 1^{\textcolor{blue}{11}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{blue}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\textcolor{red}{11}1} 3 1^{\textcolor{blue}{11}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{blue}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\textcolor{red}{11}1} 3 1^{\textcolor{blue}{11}1} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{blue}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -588, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 4 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 12 & 6 \\ 3 & 4 & -4 & 4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [5] \quad 1^{\textcolor{red}{11}1} 3 2 1^{\textcolor{blue}{11}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{blue}{\mathfrak{su}_3} \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 1 & 3 \\ 3 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 4 & 0 & 0 \\ 4 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & 0 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\textcolor{blue}{11}} 3 1^{\textcolor{red}{11}1} 3 1^{\textcolor{blue}{11}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{blue}{\mathfrak{su}_3} \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 705, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 6 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 4 & 6 \\ 1 & 3 & 3 & 0 & 4 & 0 & 3 \\ 2 & 6 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 2 & -4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

4.56 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{1111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 & 3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

4.57 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{1111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 3 & 6 & 4 \\ 2 & 4 & 0 & 4 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 3 & 3 & 3 \\ 3 & 4 & -4 & 4 & 2 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 825, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 & 3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.58 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8$

$$T = 7 \quad [0] \ 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 101, \Delta = -192, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 101, \Delta = -192, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 1 & 5 \\ 11 & 20 & 0 & 8 \\ 1 & 0 & -6 & -4 \\ 5 & 8 & -4 & 0 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [2] \ 1^{11} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.80 \rightarrow 17, T_{\max} = 101, T \in [17, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & -4 & 2 & -4 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -164, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.68 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & -2 & 0 & 0 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -164, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.57 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -164, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 2 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11}31^{11}31^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{s}08 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 125, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 10 & 13 & 0 & 6 \\ 5 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11}31^{11}31^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{s}08 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 350, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 3 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \ 1^{11}231^{11}31^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{s}08 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 2 & 4 & 4 & 0 & 4 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 3 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11}3221^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{s}08 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14, T_{\max} = 95, T \in [14, 95]$$

$$\begin{pmatrix} 5 & 15 & 1 & 5 \\ 15 & 42 & 0 & 12 \\ 1 & 0 & -4 & -2 \\ 5 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11}231^{11}31^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{s}08 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 4 & 0 & 0 & 2 \\ 2 & 4 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11}2321^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{s}08 \quad T = 9, T_{\max} = 99, \Delta = -137, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11}2321^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{s}08 \quad T = 9, T_{\max} = 99, \Delta = -137, \det = -37, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 3 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11}2321^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{s}08 \quad T = 9, T_{\max} = 99, \Delta = -137, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.09 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 9 & 4 & 5 \\ 2 & 0 & 4 & -2 & 0 \\ 3 & 0 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11}2321^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{s}08 \quad T = 9, T_{\max} = 99, \Delta = -137, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 10 & 1 & 4 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 \\ 3 & 6 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11}3221^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{s}08 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 2 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{11}4^{11}131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 \quad T = 9, T_{\max} = 102, \Delta = -162, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -5 & -3 \\ 5 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{11}231^{11}131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.87 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 1 & 0 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 0 & 3 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{11}4^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 \quad T = 9, T_{\max} = 102, \Delta = -162, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.42 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 13 & 0 & 7 \\ 1 & 0 & -5 & -3 \\ 5 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [19] $1^{11}4^{11}1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 \quad T = 10, T_{\max} = 95, \Delta = -134, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 5 & 4 & 1 & 5 \\ 4 & 2 & 2 & 4 \\ 1 & 2 & -4 & -2 \\ 5 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$T = 11$ [20] $1^{11}31^{11}5^{11}131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 \quad T = 11, T_{\max} = 101, \Delta = -136, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.60 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 10 & 1 & 4 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 \\ 3 & 6 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$T = 12$ [21] $1^{11}31^{11}5^{11}1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 22, T_{\max} = 24, T \in [22, 24]$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 28 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 5 & 10 & -1 & -3 \end{pmatrix}_{b_0}$$

[22] $1^{11}31^{11}5^{11}1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 10, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 17.40 \rightarrow 18, T_{\max} = 24, T \in [18, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 1 & 0 & 4 & 1 & 0 & 2 \\ 2 & 0 & 3 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

[23] $1^{11}31^{11}5^{11}1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 15.60 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 4 & 0 & 1 & 0 & 2 \\ 2 & 4 & -1 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 13$ [24] $1^{11}6^{11}131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -10, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 18.40 \rightarrow 19, T_{\max} = 25, T \in [19, 25]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 2 & 0 & 2 & 2 \\ 3 & 3 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{11}31^6 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8$ $T = 13$, $T_{\max} = 27$, $\Delta = -112$, $\det = -30$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 17.20 \rightarrow 18$, $T_{\max} = 27$, $T \in [18, 27]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 10 & 5 & 0 & 6 \\ 2 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

4.59 $\text{NHC}(-2)(-3) + \textcolor{violet}{e}_6 + \textcolor{orange}{so}_8 + \text{NHC}(-2)(-3)$

$T = 9$ $[0] 1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 9$, $T_{\max} = 99$, $\Delta = -142$, $\det = -500$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.76 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 10 & 13 & 6 \\ 9 & 17 & 0 & 13 & 12 & 6 \\ 5 & 8 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

4.60 $\text{NHC}(-2)(-3) + \textcolor{violet}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$

$T = 8$ $[0] 1^{11}3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 6$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 14.33 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 139$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 9 & 17 & 0 & 13 & 7 \\ 5 & 8 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 140$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.71 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 1 & 44 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 44 & 84 & 0 & 346 & 34 \\ 5 & 8 & -4 & 34 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{113}3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 141$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 13.50 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 213$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 220$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.53 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 241$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{112}3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 318$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.5$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 379, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 1 & 35 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 35 & 67 & 0 & 217 & 27 \\ 5 & 8 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 17 & 33 & 0 & 49 & 13 \\ 5 & 8 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{112} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 430, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 26 & 50 & 0 & 118 & 20 \\ 5 & 8 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 493, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 9 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 546, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 738, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{113} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 745, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 921, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1005, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[18] $1^{115} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1053$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.1$
 $T_{\min} = 13.52 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & -3 & 0 & 15 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 40 & 15 & 0 & 148 & 20 \\ 3 & 5 & -1 & -2 & 20 & -1 \end{pmatrix}_{b_0}$$

[19] $1^{114} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1186$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{114} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1366$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

[21] $1^{114} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1590$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

[22] $1^{114} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1650$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

$T = 9$ [23] $1^{111} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$

[24] $1^3 31^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 33$, $T_{\max} = 99$, $T \in [33, 99]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 1 & 0 & 0 & 0 & 0 & -2 \\ 2 & 4 & 3 & -1 & -2 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{111} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -9$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 16.11 \rightarrow 17$, $T_{\max} = 94$, $T \in [17, 94]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 7 & 0 & 14 & 8 \\ 3 & 5 & 0 & -2 & 8 & -3 \end{pmatrix}_{b_0}$$

[26] $1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -13$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 15.15 \rightarrow 16$, $T_{\max} = 94$, $T \in [16, 94]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -2 & 4 & -2 \end{pmatrix}_{b_0}$$

- [27] $1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -81$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.64 \rightarrow 14$, $T_{\max} = 94$, $T \in [14, 94]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 7 & 0 & 14 & 8 \\ 3 & 5 & 0 & -2 & 8 & 0 \end{pmatrix}_{b_0}$$
- [28] $1^{111}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 4 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 12 & 0 & 10 & 6 \\ 3 & 6 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$
- [29] $1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -111$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.63 \rightarrow 14$, $T_{\max} = 94$, $T \in [14, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 0 \\ 1 & 2 & 0 & 1 & -2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$
- [30] $1^{111}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -151$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 15 & 0 & 13 & 7 \\ 3 & 6 & 6 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$
- [31] $1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -165$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.61 \rightarrow 14$, $T_{\max} = 94$, $T \in [14, 94]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$
- [32] $1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -189$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.57 \rightarrow 14$, $T_{\max} = 94$, $T \in [14, 94]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$
- [33] $1^{111}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -571$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 9 & 3 \\ 4 & 10 & 13 & 0 & 24 & 6 \\ 5 & 13 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 24 & 27 & 0 & 49 & 13 \\ 3 & 6 & 6 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$
- $T = 10$ [34] $1^{112}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -107$, $\det = 6$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 17.67 \rightarrow 18$, $T_{\max} = 22$, $T \in [18, 22]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$
- [35] $1^{112}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -107$, $\det = 18$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 16.11 \rightarrow 17$, $T_{\max} = 22$, $T \in [17, 22]$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 \\ 2 & 0 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 0 & 5 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$
- [36] $1^{11}3221^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 48$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 5 & 15 & 1 & 13 & 5 \\ 15 & 42 & 0 & 36 & 12 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 36 & 0 & 28 & 10 \\ 5 & 12 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{111}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 4 & 0 \\ 4 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{111}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 10 & 1 & 4 & 16 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 8 & 0 \\ 6 & 16 & 0 & 8 & 22 & 10 \\ 3 & 6 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[39] $1^1231^131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & 0 & -2 & 0 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{11}3^1221^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 15 & 1 & 21 & 5 \\ 15 & 42 & 0 & 59 & 12 \\ 1 & 0 & -4 & 0 & -2 \\ 21 & 59 & 0 & 79 & 17 \\ 5 & 12 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{111}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 70, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.63 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 7 & 3 \\ 4 & 10 & 1 & 4 & 16 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 8 & 0 \\ 7 & 16 & 0 & 8 & 22 & 10 \\ 3 & 6 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{11}3^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 83, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.83 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{111}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 4 & 0 \\ 6 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{11}4^11^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -132, \det = 133, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.45 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 14 & 0 & 13 & 7 \\ 5 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{114}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 180, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 18 & 3 \\ 4 & 10 & 1 & 4 & 52 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 20 & 0 \\ 18 & 52 & 0 & 20 & 226 & 28 \\ 3 & 6 & -1 & 0 & 28 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{112}2^321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 180, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 10 & 3 \\ 4 & 10 & 1 & 4 & 28 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 12 & 0 \\ 10 & 28 & 0 & 12 & 66 & 16 \\ 3 & 6 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{111}2^321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 190, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.61 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 4 & 0 \\ 5 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{113}2^321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 14 & 3 \\ 4 & 10 & 1 & 4 & 40 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 16 & 0 \\ 14 & 40 & 0 & 16 & 134 & 22 \\ 3 & 6 & -1 & 0 & 22 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{112}2^321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 310, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 13.66 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 9 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 8 & 0 \\ 9 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{114}2^321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 310, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 17 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 16 & 0 \\ 17 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & 0 & 24 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{113}2^321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 350, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 13 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 12 & 0 \\ 13 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & 0 & 18 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{131}4^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -132, \det = 405, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.42 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 5 & 10 & 1 & 7 & 5 \\ 10 & 13 & 0 & 10 & 7 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 10 & 0 & 5 & 5 \\ 5 & 7 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[53] $1^{112}2^321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 430, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 11 & 3 \\ 4 & 10 & 1 & 4 & 28 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 12 & 0 \\ 11 & 28 & 0 & 12 & 66 & 16 \\ 3 & 6 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[55] $1^{112}2^321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 460, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 10 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 8 & 0 \\ 10 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{112} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 460, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 12 & 3 \\ 4 & 10 & 1 & 4 & 28 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 12 & 0 \\ 12 & 28 & 0 & 12 & 66 & 16 \\ 3 & 6 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{11} 4^{131} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -132, \det = 628, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.42 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 5 & 10 & 1 & 16 & 5 \\ 10 & 13 & 0 & 24 & 7 \\ 1 & 0 & -5 & 0 & -3 \\ 16 & 24 & 0 & 38 & 12 \\ 5 & 7 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{113} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 710, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.63 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 15 & 3 \\ 4 & 10 & 1 & 4 & 40 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 16 & 0 \\ 15 & 40 & 0 & 16 & 134 & 22 \\ 3 & 6 & -1 & 0 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{113} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 740, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 14 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 12 & 0 \\ 14 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & 0 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{114} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 910, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.65 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 19 & 3 \\ 4 & 10 & 1 & 4 & 52 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 20 & 0 \\ 19 & 52 & 0 & 20 & 226 & 28 \\ 3 & 6 & -1 & 0 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{114} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 940, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.68 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 18 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 16 & 0 \\ 18 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & 0 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{113} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 980, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.61 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 16 & 3 \\ 4 & 10 & 1 & 4 & 40 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 16 & 0 \\ 16 & 40 & 0 & 16 & 134 & 22 \\ 3 & 6 & -1 & 0 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{114} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 1420, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 20 & 3 \\ 4 & 10 & 1 & 4 & 52 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 20 & 0 \\ 20 & 52 & 0 & 20 & 226 & 28 \\ 3 & 6 & -1 & 0 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [63] \quad 1^{111} 31^1 5131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 24, \quad \Delta = -106, \quad \det = 60, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.73 \rightarrow 16, \quad T_{\max} = 24, \quad T \in [16, 24]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 4 & 0 \\ 4 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

4.61 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -28, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.71 \rightarrow 15, \quad T_{\max} = 93, \quad T \in [15, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 0 & 6 & -1 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 16 & 6 & 0 & 0 & 22 & 8 \\ 3 & 5 & -1 & -2 & 0 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -114, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.89 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 12 & 7 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -132, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.64 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -372, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.46 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 6 & 4 \\ 4 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -432, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.58 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 35 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 35 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -4 & 7 & 27 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -496, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.46 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 6 & 4 \\ 3 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -512, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.52 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}} 3 \ 1^{\textcolor{red}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -608, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 3 & 0 & 4 & 6 & 4 \\ 4 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3 \ 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{2}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -618, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.51 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 26 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 26 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -4 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [9] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}} 3 \ 2 \ 1^{\textcolor{red}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -104, \ \det = 104, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 0 & 0 & -1 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [10] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}} 3 \ 1^{\textcolor{red}{1}\textcolor{blue}{1}} 4 \ 1^{\textcolor{red}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -102, \ \det = -68, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.71 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 7 & 5 \\ 10 & 13 & 0 & 14 & 10 & 7 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 14 & 0 & 13 & 9 & 7 \\ 7 & 10 & 0 & 9 & 5 & 5 \\ 5 & 7 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

4.62 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}} 3 \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 372, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 27 & 7 \\ 9 & 17 & 0 & 15 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 27 & 49 & 13 \\ 5 & 8 & -4 & 7 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

4.63 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}} 3 \ 1^{\textcolor{red}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -115, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, \ T_{\max} = 99, \ T \in [15, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -3 & 0 & 3 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -2 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}} 3 \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -271, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}} 3 \ 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -334, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} =$$

16.0

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -4 & 13 & 6 & 0 \end{pmatrix}_{b_0}$$

4.64 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] 1^{11} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 99, \Delta = -137, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.69 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 3 & 5 \\ 2 & 5 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 5 & 0 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] 1^{11} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 99, \Delta = -137, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & 0 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] 1^{11} 3^1 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 & 3 \\ 4 & 10 & 13 & 0 & 8 & 6 \\ 5 & 13 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 9 & 0 & 4 & 4 \\ 3 & 6 & 6 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [3] 1^{11} 23^1 1^1 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 30, \Delta = -116, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 30, T \in [15, 30]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 \\ 1 & 4 & 1 & 0 & 4 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 \\ 1 & 3 & 4 & 0 & 0 & 2 \\ 2 & 4 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [4] 1^{11} 4^1 13^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 5 & 4 & 1 & 1 & 5 \\ 4 & 2 & 2 & 0 & 4 \\ 1 & 2 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 4 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [5] 1^{11} 3^1 1^1 5^1 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 95, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.68 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 10 & 1 & 4 & 8 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 3 & 0 \\ 3 & 8 & 0 & 3 & 4 & 4 \\ 3 & 6 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

4.65 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{so}_8$ + $\mathfrak{su}_3$

$$T = 8 \quad [1] \ 1^{11} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 101, \ \Delta = -171, \ \det = 95, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.96 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 101, \ \Delta = -171, \ \det = 245, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 9 & 17 & 0 & 12 & 6 \\ 5 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 101, \ \Delta = -171, \ \det = 302, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.86 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 101, \ \Delta = -171, \ \det = 330, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.68 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \ 1^{11} 3 \ 2 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 99, \ \Delta = -143, \ \det = -70, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.83 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 3 & 5 & 0 & -2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3 \ 2 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 99, \ \Delta = -143, \ \det = -196, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.88 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 0 & -1 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -1 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^1 3 \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 100, \ \Delta = -150, \ \det = -354, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.07 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 0 & 3 \\ 1 & 3 & 3 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & -1 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [8] \ 1^{11} 2 \ 3 \ 2 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 30, \ \Delta = -116, \ \det = 62, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.10 \rightarrow 15, \ T_{\max} = 30, \ T \in [15, 30]$$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 \\ 2 & 0 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 0 & 5 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 2 \ 3 \ 2 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 30, \ \Delta = -116, \ \det = 110, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.49 \rightarrow 15, \ T_{\max} = 30, \ T \in [15, 30]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 4 & 0 \\ 4 & 12 & 0 & 4 & 9 & 5 \\ 3 & 6 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{11} 3^{11} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 147$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 14.12 \rightarrow 15$, $T_{\max} = 26$, $T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & 0 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

4.66 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$T = 9$ [0] $1^{11} 3^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 14.23 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 0 & 3 & -1 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 3 & 0 & 0 & 4 & 4 \\ 3 & 5 & -1 & -2 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 15 & 6 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -4 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -334$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -931$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

4.67 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$

$T = 7$ [0] $1^{11} 3^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -14$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.0$

$T_{\min} = 15.71 \rightarrow 16$, $T_{\max} = 100$, $T \in [16, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -3 & 3 & -1 \\ 5 & 8 & 3 & 4 & 4 \\ 3 & 5 & -1 & 4 & -3 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -23$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.0$

$T_{\min} = 13.87 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.5$

$T_{\min} = 13.33 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 6 & 10 & 0 & 22 & 8 \\ 3 & 4 & -4 & 8 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{1}}31^{\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 3 & 4 & -3 & 3 & -1 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & -1 & 4 & 0 \end{pmatrix}_{b_0}$

[3] $1^{114}31^1 + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{brown}{5}\textcolor{blue}{u}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 3 & 4 & -3 & 12 & -1 \\ 12 & 32 & 12 & 94 & 16 \\ 3 & 5 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{112}3^11 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$
 $\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 3 & 4 & -3 & 6 & -1 \\ 6 & 16 & 6 & 22 & 8 \\ 3 & 5 & -1 & 8 & 0 \end{pmatrix}_{b_0}$

[8] $21^1 31^{14} + [\textcolor{red}{\mathfrak{s}}\textcolor{red}{u}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{c}}_6 + \textcolor{orange}{\mathfrak{s}}\textcolor{orange}{u}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$
 $\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 3 & 4 & -3 & 12 & -1 \\ 12 & 32 & 12 & 94 & 16 \\ 3 & 5 & -1 & 16 & 0 \end{pmatrix}_{b_0}$

[5] $21^1 31^{111} + [\textcolor{red}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{c}_6 + \textcolor{brown}{s} \textcolor{brown}{u}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 3 & 4 & -3 & 3 & -1 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & -1 & 4 & 0 \end{pmatrix}_{b_0}$

[9] $1^{115}31^1 + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{green}{\mathbf{e}}_6 + \textcolor{red}{\mathbf{su}}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.1$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$
 $\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 9 & 4 & 40 & 5 \\ 3 & 4 & -3 & 15 & -1 \\ 15 & 40 & 15 & 148 & 20 \\ 3 & 5 & -1 & 20 & 0 \end{pmatrix}_{b_0}$

$$[11] \ 1^{11} 3^1 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}_6 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 100, \ \Delta = -163, \ \det = -40, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 3 & 4 & -3 & 9 & -1 \\ 9 & 24 & 9 & 52 & 12 \\ 3 & 5 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[12] $21^1 31^{115} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.1$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 9 & 4 & 40 & 5 \\ 3 & 4 & -3 & 15 & -1 \\ 15 & 40 & 15 & 148 & 20 \\ 3 & 5 & -1 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 21^1 31^{113} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{brown}{s}u_2 \quad T = 7, \, T_{\max} = 100, \, \Delta = -163, \, \det = -40, \, (n_+, n_-, n_0) = (1, 7, 0), \, c_{\text{ext}} = 10.8$$

$$T_{\min} = 12.40 \rightarrow 13, \, T_{\max} = 100, \, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 3 & 4 & -3 & 9 & -1 \\ 9 & 24 & 9 & 52 & 12 \\ 3 & 5 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[7] $21^1 31^{112} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{brown}{s}u_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$
 $\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 3 & 4 & -3 & 6 & -1 \\ 6 & 16 & 6 & 22 & 8 \\ 3 & 5 & -1 & 8 & 0 \end{pmatrix}_{b_0}$

[13] $1^1 \bar{1}^1 3^1 \bar{1}^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 \\ 3 & 4 & -4 & 4 & 0 \end{pmatrix}_{b_0}$

[14] $1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.15 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 4 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{112}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -59$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.95 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & -3 & 6 & -1 \\ 9 & 16 & 6 & 22 & 8 \\ 3 & 5 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{12}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -60$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.33 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 4 & 8 & 0 & 10 & 6 \\ 3 & 4 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{13}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -70$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 6 & 12 & 0 & 25 & 9 \\ 3 & 4 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{14}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -84$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 8 & 16 & 0 & 46 & 12 \\ 3 & 4 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

[19] $1^113^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -86$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.02 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -3 & 3 & -1 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{111}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -93$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -3 & 3 & -1 \\ 4 & 8 & 3 & 4 & 4 \\ 3 & 5 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{15}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -102$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.1$
 $T_{\min} = 13.02 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 10 & 20 & 0 & 73 & 15 \\ 3 & 4 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

[22] $1^{11}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -115$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.27 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 5 & 9 & 0 & 13 & 7 \\ 3 & 4 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{16}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -124$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 13.19 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 12 & 24 & 0 & 106 & 18 \\ 3 & 4 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

[24] $1^113^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -146$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.18 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & -3 & 6 & -1 \\ 6 & 16 & 6 & 22 & 8 \\ 3 & 5 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$

[25] $1^{17}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 100$, $\Delta = -163$, $\det = -150$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.3$
 $T_{\min} = 13.33 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 14 & 28 & 0 & 145 & 21 \\ 3 & 4 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

[26] $1^{113}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -158, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & -3 & 9 & -1 \\ 13 & 24 & 9 & 52 & 12 \\ 3 & 5 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

[27] $1^{113}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -162, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 8 & 14 & 0 & 37 & 11 \\ 3 & 4 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{112}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -174, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & -3 & 6 & -1 \\ 8 & 16 & 6 & 22 & 8 \\ 3 & 5 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{12}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -189, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.31 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 7 & 13 & 0 & 28 & 10 \\ 3 & 4 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{11}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -195, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 11 & 19 & 0 & 73 & 15 \\ 3 & 4 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{112}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -203, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & -3 & 6 & -1 \\ 7 & 16 & 6 & 22 & 8 \\ 3 & 5 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{11}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 14 & 24 & 0 & 121 & 19 \\ 3 & 4 & -4 & 19 & 0 \end{pmatrix}_{b_0}$$

[33] $1^1 13^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -246, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 13.27 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & -3 & 9 & -1 \\ 9 & 24 & 9 & 52 & 12 \\ 3 & 5 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

[34] $1^{13}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -267, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 9 & 17 & 0 & 49 & 13 \\ 3 & 4 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{12}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 10 & 18 & 0 & 58 & 14 \\ 3 & 4 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{113}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -309, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & -3 & 9 & -1 \\ 12 & 24 & 9 & 52 & 12 \\ 3 & 5 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{14}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -349, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 11 & 21 & 0 & 76 & 16 \\ 3 & 4 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[38] $21^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -352, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 4 \\ 8 & 9 & 4 & 8 & 5 \\ 3 & 4 & -3 & 3 & -1 \\ 6 & 8 & 3 & 4 & 4 \\ 4 & 5 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -353, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & -3 & 9 & -1 \\ 10 & 24 & 9 & 52 & 12 \\ 3 & 5 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{113}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -374, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & -3 & 9 & -1 \\ 11 & 24 & 9 & 52 & 12 \\ 3 & 5 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[41] $1^113^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -386, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.32 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & -3 & 12 & -1 \\ 12 & 32 & 12 & 94 & 16 \\ 3 & 5 & -1 & 16 & -1 \end{pmatrix}_{b_0}$$

[42] $1^{12}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -405, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.44 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 13 & 23 & 0 & 100 & 18 \\ 3 & 4 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{13}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -450, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 12 & 22 & 0 & 85 & 17 \\ 3 & 4 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{12}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -492, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.53 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 16 & 28 & 0 & 154 & 22 \\ 3 & 4 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{114}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -498, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & -3 & 12 & -1 \\ 16 & 32 & 12 & 94 & 16 \\ 3 & 5 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{114}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -543, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & -3 & 12 & -1 \\ 13 & 32 & 12 & 94 & 16 \\ 3 & 5 & -1 & 16 & -1 \end{pmatrix}_{b_0}$$

[47] $1^113^11^{15} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -566, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.1$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 9 & 4 & 40 & 5 \\ 2 & 4 & -3 & 15 & -1 \\ 15 & 40 & 15 & 148 & 20 \\ 3 & 5 & -1 & 20 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{14}3^11^3 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -599, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & -3 & 12 & -1 \\ 15 & 32 & 12 & 94 & 16 \\ 3 & 5 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{14}3^11^2 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 14 & 26 & 0 & 118 & 20 \\ 3 & 4 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{14}3^11^2 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -614, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & -3 & 12 & -1 \\ 14 & 32 & 12 & 94 & 16 \\ 3 & 5 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{13}3^11^3 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -619, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 15 & 27 & 0 & 133 & 21 \\ 3 & 4 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[52] $21^{112}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -622, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 4 & 8 & 3 & 12 & 4 \\ 8 & 9 & 4 & 16 & 5 \\ 3 & 4 & -3 & 6 & -1 \\ 12 & 16 & 6 & 22 & 8 \\ 4 & 5 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{13}3^11^4 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -774, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.44 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 18 & 32 & 0 & 193 & 25 \\ 3 & 4 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{14}3^11^3 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -837, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 17 & 31 & 0 & 172 & 24 \\ 3 & 4 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{14}3^11^4 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -1060, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 20 & 36 & 0 & 238 & 28 \\ 3 & 4 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

[56] $21^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -1072, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 4 & 8 & 3 & 18 & 4 \\ 8 & 9 & 4 & 24 & 5 \\ 3 & 4 & -3 & 9 & -1 \\ 18 & 24 & 9 & 52 & 12 \\ 4 & 5 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[57] $21^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -1702, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 4 & 8 & 3 & 24 & 4 \\ 8 & 9 & 4 & 32 & 5 \\ 3 & 4 & -3 & 12 & -1 \\ 24 & 32 & 12 & 94 & 16 \\ 4 & 5 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[58] $21^{115}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -2512, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.1$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 4 & 8 & 3 & 30 & 4 \\ 8 & 9 & 4 & 40 & 5 \\ 3 & 4 & -3 & 15 & -1 \\ 30 & 40 & 15 & 148 & 20 \\ 4 & 5 & -1 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [59] \quad 21^{111} 321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.67 \rightarrow 16, T_{\max} = 94, T \in [16, 94]$$

$$\begin{pmatrix} 4 & 8 & 3 & 9 & 4 \\ 8 & 9 & 4 & 12 & 5 \\ 3 & 4 & -2 & 7 & 0 \\ 9 & 12 & 7 & 14 & 8 \\ 4 & 5 & 0 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^1 3 \quad 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.14 \rightarrow 16, T_{\max} = 94, T \in [16, 94]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 2 & 5 & -2 & 3 & 0 \\ 3 & 5 & 3 & 4 & 4 \\ 3 & 5 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^1 3 \quad 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.82 \rightarrow 16, T_{\max} = 94, T \in [16, 94]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{111} 3 \quad 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.82 \rightarrow 16, T_{\max} = 94, T \in [16, 94]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 \\ 5 & 8 & 3 & 4 & 4 \\ 3 & 5 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[63] \quad 2^1 2 \quad 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.15 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 5 & 5 & 5 & 5 \\ 3 & 5 & -2 & 3 & 0 \\ 3 & 5 & 3 & 4 & 4 \\ 3 & 5 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{111} 3 \quad 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.08 \rightarrow 16, T_{\max} = 94, T \in [16, 94]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 2 & 4 & -2 & 7 & 0 \\ 6 & 12 & 7 & 14 & 8 \\ 3 & 5 & 0 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^1 3 \quad 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 94, T \in [13, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 2 & 1 & 0 & 3 & 1 & 3 \\ 2 & 2 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^1 3 \quad 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13.67 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{11} 3 \quad 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.11 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[68] \quad 2^1 2 \quad 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 3 & 4 & -2 & 3 & 0 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^1 3 \quad 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 2 & 4 & -2 & 7 & 0 \\ 5 & 12 & 7 & 14 & 8 \\ 3 & 5 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

- [70] $1^{1^{12}} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 36$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.33 \rightarrow 14$, $T_{\max} = 94$, $T \in [14, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 3 & 1 & 0 & 3 & 1 & 3 \\ 2 & 2 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$
- [71] $1^{1^{14}} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 36$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 16$, $T_{\max} = 94$, $T \in [16, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 5 & 1 & 0 & 3 & 1 & 3 \\ 2 & 2 & -1 & -1 & 3 & -3 \end{pmatrix}_{b_0}$$
- [72] $1^{1^{11}} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 36$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.33 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$
- [73] $1^{1^1} 1^1 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 37$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 2 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$
- [74] $1^{1^{13}} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 42$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.14 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 4 & 1 & 0 & 3 & 1 & 3 \\ 2 & 2 & -1 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$
- [75] $1^{1^1} 1^1 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 43$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$
- [76] $1^{1^4} 3^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 43$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 15.02 \rightarrow 16$, $T_{\max} = 98$, $T \in [16, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 15 & 18 & 0 & 76 & 16 \\ 2 & 3 & 3 & -4 & 16 & -3 \end{pmatrix}_{b_0}$$
- [77] $1^{1^1} 1^1 3^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 43$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.35 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 3 & 3 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$
- [78] $1^{1^1} 1^1 3^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$
- [79] $1^{1^2} 1^2 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.48 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 2 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$
- [80] $1^{1^3} 1^3 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 51$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{14}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 58, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.21 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 4 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{111}31^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 58, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.21 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 6 & 6 & 0 & 13 & 7 \\ 2 & 3 & 2 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[83] \ 1^{111}31^121^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{111}3131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 3 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{15}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.1$$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 5 & 10 & 0 & 5 & 48 & 10 \\ 2 & 2 & -1 & -1 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[86] \ 1^{111}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 \\ 4 & 8 & 3 & 4 & 4 \\ 3 & 5 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[87] \ 1^{112}31^231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 12 & 0 & 58 & 14 \\ 2 & 3 & 2 & -4 & 14 & -1 \end{pmatrix}_{b_0}$$

$$[88] \ 1^{16}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 6 & 12 & 0 & 6 & 70 & 12 \\ 2 & 2 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[89] \ 1^131^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.82 \rightarrow 15, T_{\max} = 98, T \in [15, 98]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 3 & -1 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[90] \ 1^{12}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 3 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & -1 & -1 & 7 & 0 \end{pmatrix}_{b_0}$$

- [91] $1^{11}31^{11}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 91$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.34 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 6 & 0 & 4 & 4 \\ 2 & 3 & 3 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$
- [92] $1^{11}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 91$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 2 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$
- [93] $1^{17}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 91$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.3$
 $T_{\min} = 14.22 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 7 & 14 & 0 & 7 & 96 & 14 \\ 2 & 2 & -1 & -1 & 14 & -2 \end{pmatrix}_{b_0}$$
- [94] $1^{11}331^{11}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 106$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.72 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 4 & 0 & 18 & 2 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 18 & 18 & 0 & 133 & 21 \\ 2 & 3 & 2 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$
- [95] $1^{11}23131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 112$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 6 & 0 & 10 & 6 \\ 2 & 3 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$
- [96] $1^{11}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 121$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 3 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$
- [97] $1^{11}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 127$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 3 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$
- [98] $1^{13}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 141$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.51 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 4 & 7 & 0 & 6 & 25 & 9 \\ 2 & 2 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$
- [99] $1^{11}431^431^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.89 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 1 & 4 & 0 & 24 & 3 \\ 1 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 24 & 24 & 0 & 238 & 28 \\ 2 & 3 & 2 & -4 & 28 & -1 \end{pmatrix}_{b_0}$$
- [100] $1^{12}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 3 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$
- [101] $1^{13}1^{11}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 4 \\ 2 & 4 & 2 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

[102] $1^{11} 3^{13} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 4 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[103] $1^{11} 3^{14} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 5 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

[104] $1^{13} 31^{11} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.23 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 15 & 0 & 49 & 13 \\ 2 & 3 & 3 & -4 & 13 & -1 \end{pmatrix}_{b_0}$$

[105] $1^{12} 31^{11} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 9 & 0 & 28 & 10 \\ 2 & 4 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[106] $1^{13} 3131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 9 & 0 & 25 & 9 \\ 2 & 3 & 2 & -4 & 9 & -1 \end{pmatrix}_{b_0}$$

[107] $1^{12} 31^{11} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.08 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 9 & 9 & 0 & 28 & 10 \\ 2 & 3 & 2 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

[108] $1^{11} 3^{12} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.52 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 4 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

[109] $1^{14} 321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 5 & 9 & 0 & 7 & 41 & 11 \\ 2 & 2 & -1 & -1 & 11 & 0 \end{pmatrix}_{b_0}$$

[110] $1^{11} 31^{11} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 6 & 9 & 0 & 13 & 7 \\ 2 & 3 & 3 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

[111] $1^{13} 3^{11} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

- [112] $1^{12}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 223$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.07 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 9 & 12 & 0 & 28 & 10 \\ 2 & 3 & 3 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$
- [113] $1^{11}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 226$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$
- [114] $1^{12}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 226$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.42 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 4 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & -1 & -1 & 7 & 0 \end{pmatrix}_{b_0}$$
- [115] $1^{12}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 238$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 4 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$
- [116] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 253$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 5 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$
- [117] $1^{11}31^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 256$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 6 & 0 & 22 & 8 \\ 2 & 4 & 2 & -4 & 8 & 0 \end{pmatrix}_{b_0}$$
- [118] $1^{14}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 268$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 5 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$
- [119] $21^{11}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 276$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.54 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 4 \\ 8 & 9 & 4 & 8 & 5 \\ 3 & 4 & -2 & 3 & 0 \\ 6 & 8 & 3 & 4 & 4 \\ 4 & 5 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [120] $1^{114}3131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 292$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.62 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 12 & 0 & 46 & 12 \\ 2 & 3 & 2 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$
- [121] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 301$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 6 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$
- [122] $1^{12}321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 316$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 5 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0} \\
[123] & 1^{11} 3^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 13.09 \rightarrow 14, T_{\max} = 98, T \in [14, 98] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 15 & 15 & 0 & 85 & 17 \\ 2 & 3 & 2 & -4 & 17 & -1 \end{pmatrix}_{b_0} \\
[124] & 1^{11} 3^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 333, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 12.43 \rightarrow 13, T_{\max} = 94, T \in [13, 94] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 5 & 7 & 0 & 6 & 25 & 9 \\ 2 & 2 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0} \\
[125] & 1^{11} 3^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 351, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 12.50 \rightarrow 13, T_{\max} = 94, T \in [13, 94] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0} \\
[126] & 1^{14} 31^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 18 & 0 & 118 & 20 \\ 2 & 4 & 2 & -4 & 20 & 0 \end{pmatrix}_{b_0} \\
[127] & 1^{12} 3^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 12.45 \rightarrow 13, T_{\max} = 94, T \in [13, 94] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 5 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & -1 & -1 & 7 & 0 \end{pmatrix}_{b_0} \\
[128] & 1^{11} 3^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 358, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 13.11 \rightarrow 14, T_{\max} = 98, T \in [14, 98] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 12 & 0 & 49 & 13 \\ 2 & 3 & 2 & -4 & 13 & -1 \end{pmatrix}_{b_0} \\
[129] & 1^{12} 3^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 12.69 \rightarrow 13, T_{\max} = 94, T \in [13, 94] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 6 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0} \\
[130] & 1^{11} 5^3 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 427, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.1 \\
& T_{\min} = 13.65 \rightarrow 14, T_{\max} = 98, T \in [14, 98] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 15 & 0 & 73 & 15 \\ 2 & 3 & 2 & -4 & 15 & -1 \end{pmatrix}_{b_0} \\
[131] & 1^{11} 31^{13} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 9 & 0 & 52 & 12 \\ 2 & 4 & 2 & -4 & 12 & 0 \end{pmatrix}_{b_0} \\
[132] & 1^{14} 3^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 442, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 12.46 \rightarrow 13, T_{\max} = 94, T \in [13, 94] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 6 & 9 & 0 & 7 & 41 & 11 \\ 2 & 2 & -1 & -1 & 11 & 0 \end{pmatrix}_{b_0}
\end{aligned}$$

- [133] $1^{12} 3^{13} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 466$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.51 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 6 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & -1 & -1 & 7 & 0 \end{pmatrix}_{b_0}$$
- [134] $1^{14} 3^{12} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 466$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.51 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 6 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$
- [135] $1^{11} 31^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 478$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 9 & 0 & 37 & 11 \\ 2 & 4 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$
- [136] $1^{11} 431^3 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 481$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.13 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 4 & 0 & 21 & 3 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 21 & 21 & 0 & 172 & 24 \\ 2 & 3 & 2 & -4 & 24 & -1 \end{pmatrix}_{b_0}$$
- [137] $1^{13} 3^{12} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 483$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.53 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 6 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$
- [138] $1^{13} 3^{12} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 513$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.44 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 6 & 7 & 0 & 6 & 25 & 9 \\ 2 & 2 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$
- [139] $1^{13} 31^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 526$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 15 & 0 & 85 & 17 \\ 2 & 4 & 2 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$
- [140] $1^{11} 431^3 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 553$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.16 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 15 & 15 & 0 & 76 & 16 \\ 2 & 3 & 2 & -4 & 16 & -1 \end{pmatrix}_{b_0}$$
- [142] $1^{12} 3^{14} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 568$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 7 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & -1 & -1 & 7 & 0 \end{pmatrix}_{b_0}$$
- [141] $1^{12} 31^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 568$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 12 & 0 & 58 & 14 \\ 2 & 4 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$
- [143] $1^{13} 3^{14} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 94$, $\Delta = -135$, $\det = 603$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 94$, $T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 7 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[144] $1^{11}4^131^231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 616, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.08 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 7 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[145] $1^{14}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 652, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.52 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 7 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[146] $1^{14}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 676, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 7 & 9 & 0 & 7 & 41 & 11 \\ 2 & 2 & -1 & -1 & 11 & 0 \end{pmatrix}_{b_0}$$

[147] $1^{13}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 681, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.48 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 7 & 7 & 0 & 6 & 25 & 9 \\ 2 & 2 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

[148] $1^{13}1^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 688, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 12 & 0 & 94 & 16 \\ 2 & 4 & 2 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[149] $1^{11}31^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 802, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 12 & 0 & 73 & 15 \\ 2 & 4 & 2 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

[150] $1^{14}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 826, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 8 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[151] $1^{13}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 837, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.52 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 8 & 7 & 0 & 6 & 25 & 9 \\ 2 & 2 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

[152] $1^{14}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 898, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.47 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 8 & 9 & 0 & 7 & 41 & 11 \\ 2 & 2 & -1 & -1 & 11 & 0 \end{pmatrix}_{b_0}$$

[153] $1^{13}1^{15}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1012, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.1$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 15 & 0 & 148 & 20 \\ 2 & 4 & 2 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

- [154] $1^{12}31^{13}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1036, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 15 & 0 & 100 & 18 \\ 2 & 4 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$
- [155] $1^{14}31^{13}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 21 & 0 & 172 & 24 \\ 2 & 4 & 2 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$
- [156] $1^{14}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 1108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 9 & 9 & 0 & 7 & 41 & 11 \\ 2 & 2 & -1 & -1 & 11 & 0 \end{pmatrix}_{b_0}$$
- [157] $1^{13}31^{13}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1138, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 18 & 0 & 133 & 21 \\ 2 & 4 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$
- [158] $1^{11}31^{14}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 15 & 0 & 121 & 19 \\ 2 & 4 & 2 & -4 & 19 & 0 \end{pmatrix}_{b_0}$$
- [159] $1^{13}1^{16}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1408, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 36 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 36 & 18 & 0 & 214 & 24 \\ 2 & 4 & 2 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$
- [160] $1^{12}31^{14}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1576, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 18 & 0 & 154 & 22 \\ 2 & 4 & 2 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$
- [161] $1^{13}31^{14}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1822, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 33 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 33 & 21 & 0 & 193 & 25 \\ 2 & 4 & 2 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$
- [162] $1^{13}1^{17}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1876, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 14 & 2 \\ 2 & 4 & 4 & 0 & 42 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 42 & 21 & 0 & 292 & 28 \\ 2 & 4 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$
- [163] $1^{14}31^{14}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1936, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 36 & 4 \\ 1 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 36 & 24 & 0 & 238 & 28 \\ 2 & 4 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$
- $T = 9$ [164] $1^{12}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 19, T_{\max} = 21, T \in [19, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 2 & 0 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & -5 \end{pmatrix}_{b_0}$$

[166] $1^{11} 2^3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 16.14 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & -1 & 0 & -3 \end{pmatrix}_{b_0}$$

[165] $1^{11} 2^3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 16.14 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 3 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & -1 & 0 & -3 \end{pmatrix}_{b_0}$$

[167] $1^{11} 2^3 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

[170] $1^{13} 3 221^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 20.45 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 19 & 0 \\ 10 & 10 & 0 & 19 & 104 & 22 \\ 2 & 2 & -1 & 0 & 22 & -7 \end{pmatrix}_{b_0}$$

[169] $1^{13} 2^3 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 20.45 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 4 & 0 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & -7 \end{pmatrix}_{b_0}$$

[168] $1^{13} 3 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 16.09 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 4 & 1 & 0 & 4 & 2 & 4 \\ 2 & 2 & -1 & 0 & 4 & -3 \end{pmatrix}_{b_0}$$

[174] $1^{14} 3 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 19, T_{\max} = 21, T \in [19, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 5 & 1 & 0 & 4 & 2 & 4 \\ 2 & 2 & -1 & 0 & 4 & -5 \end{pmatrix}_{b_0}$$

[171] $1^{12} 2^3 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[172] $1^1 3 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 19, T_{\max} = 21, T \in [19, 21]$

$$\begin{pmatrix} 5 & 15 & 12 & 5 \\ 15 & 42 & 36 & 12 \\ 12 & 36 & 28 & 10 \\ 5 & 12 & 10 & -3 \end{pmatrix}_{b_0}$$

[173] $1^{12} 3 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[175] $1^{12} 2^3 3^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 16.33 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & -3 \end{pmatrix}_{b_0}$$

[176] $1^{11} 2^3 3^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 1 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[177] $1^{11} 2^3 3^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 2 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{12} 2^3 3^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

[179] $1^{13} 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[180] $1^{13} 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 17, T_{\max} = 93, T \in [17, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 11 & 13 & 0 & 40 & 12 \\ 3 & 4 & -3 & 12 & -2 \end{pmatrix}_{b_0}$$

[183] $1^{11} 4^1 13^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.11 \rightarrow 16, T_{\max} = 93, T \in [16, 93]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 5 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

[182] $1^{14} 2^3 3^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 19.56 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 2 & 5 & 0 & 22 & 4 \\ 1 & 5 & 1 & 0 & 22 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 22 & 22 & 0 & 134 & 24 \\ 2 & 4 & 3 & -4 & 24 & -6 \end{pmatrix}_{b_0}$$

[181] $1^{11} 2^3 3^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 19.56 \rightarrow 20, T_{\max} = 25, T \in [20, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 10 & 0 \\ 8 & 0 & 0 & 10 & 38 & 12 \\ 2 & 0 & -1 & 0 & 12 & -6 \end{pmatrix}_{b_0}$$

[184] $1^{11} 2^3 3^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

- [185] $1^{11}3221^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -22$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 16.73 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 5 & 15 & 22 & 5 \\ 15 & 42 & 59 & 12 \\ 22 & 59 & 79 & 17 \\ 5 & 12 & 17 & -1 \end{pmatrix}_{b_0}$
- [186] $1^{11}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -26$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 2 & 4 & 0 & 4 & 6 & 6 \\ 2 & 3 & -1 & -1 & 6 & 1 \end{pmatrix}_{b_0}$
- [190] $1^{11}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -27$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.48 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$
- [189] $1^{11}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -27$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.48 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 0 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$
- [188] $1^{13}231^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -27$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 16.93 \rightarrow 17$, $T_{\max} = 25$, $T \in [17, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 18 & 0 & 94 & 20 \\ 2 & 4 & 3 & -4 & 20 & -3 \end{pmatrix}_{b_0}$
- [187] $1^{13}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -27$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 4 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -1 & 9 & 0 \end{pmatrix}_{b_0}$
- [194] $1^{16}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 16.75 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 6 & 12 & 0 & 6 & 70 & 12 \\ 2 & 2 & -1 & 0 & 12 & -3 \end{pmatrix}_{b_0}$
- [192] $1^{17}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.3$
 $T_{\min} = 18.37 \rightarrow 19$, $T_{\max} = 21$, $T \in [19, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 7 & 14 & 0 & 7 & 96 & 14 \\ 2 & 2 & -1 & 0 & 14 & -5 \end{pmatrix}_{b_0}$
- [191] $1^{11}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$
- [193] $1^{12}31^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 1 & 0 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & 0 & 0 & 0 \end{pmatrix}_{b_0}$
- [195] $1^{15}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.1$
 $T_{\min} = 15.38 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 5 & 10 & 0 & 5 & 48 & 10 \\ 2 & 2 & -1 & 0 & 10 & -2 \end{pmatrix}_{b_0}$$

[196] $1^{14}3221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.25 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 4 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

[197] $1^{13}3221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

[198] $1^{12}3221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[199] $1^{11}231^131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.21 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 2 & 5 & 0 & 11 & 4 \\ 1 & 5 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 11 & 11 & 0 & 32 & 12 \\ 2 & 4 & 3 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$

[200] $1^{12}31^131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[201] $1^{12}3221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 5 & 6 & 0 & 10 & 30 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[203] $1^{11}4131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 \\ 3 & 4 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[202] $1^{11}41^{13}1^2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 11 & 0 & 28 & 10 \\ 3 & 4 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

[204] $1^{12}31^131^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.59 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 3 & 0 \\ 3 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[205] $1^{11}3221^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 5 & 15 & 13 & 5 \\ 15 & 42 & 36 & 12 \\ 13 & 36 & 28 & 10 \\ 5 & 12 & 10 & 2 \end{pmatrix}_{b_0}$$

[206] $1^{11}4^{11}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.23 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 4 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[209] $1^{14}3^{12}221^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.09 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 20 & 0 \\ 10 & 12 & 0 & 20 & 126 & 24 \\ 2 & 2 & -1 & 0 & 24 & -1 \end{pmatrix}_{b_0}$$

[208] $1^{131^{11}}4^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 14.36 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 1 & 4 & 1 & 7 & 3 \\ 4 & 4 & 0 & 10 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 10 & 0 & 21 & 9 \\ 3 & 4 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

[207] $1^{12}3^{11}221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 3 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[211] $1^{12}4^{11}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 4 & 8 & 0 & 10 & 6 \\ 3 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[210] $1^{131^{11}}4^{11}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.74 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 \\ 4 & 4 & 0 & 5 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 \\ 3 & 4 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[214] $21^141^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.58 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 \\ 2 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 2 & 2 & 0 & 0 & 2 \\ 2 & 2 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[213] $1^{11}23131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.58 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 5 & 0 & 4 & 4 \\ 1 & 5 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 4 & 4 & 0 & 2 & 4 \\ 2 & 4 & 3 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

[212] $1^123131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.58 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[216] $1^1231^{231^{112}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 14.08 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 6 & 0 \\ 6 & 0 & 0 & 6 & 22 & 8 \\ 2 & 0 & -1 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

[215] $1^{131^{11}}4^{11}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.08 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 3 & 0 & 1 & 3 \\ 3 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

- [217] $1^{13}4^{11}131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.82 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 12 & 0 & 25 & 9 \\ 3 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$
- [218] $1^{11}231^331^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 16.96 \rightarrow 17$, $T_{\max} = 25$, $T \in [17, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 13 & 0 \\ 11 & 0 & 0 & 13 & 74 & 16 \\ 2 & 0 & -1 & 0 & 16 & -3 \end{pmatrix}_{b_0}$
- [219] $1^{12}2321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 3 & 7 & 0 & 5 & 17 & 9 \\ 2 & 3 & -1 & -1 & 9 & 1 \end{pmatrix}_{b_0}$
- [220] $1^{14}2321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 5 & 13 & 0 & 7 & 57 & 15 \\ 2 & 3 & -1 & -1 & 15 & 1 \end{pmatrix}_{b_0}$
- [221] $1^{14}4^{11}131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -58$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.21 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 16 & 0 & 46 & 12 \\ 3 & 4 & -3 & 12 & 0 \end{pmatrix}_{b_0}$
- [222] $1^13221^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -58$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14$, $T_{\max} = 21$, $T \in [14, 21]$
 $\begin{pmatrix} 5 & 15 & 21 & 5 \\ 15 & 42 & 59 & 12 \\ 21 & 59 & 79 & 17 \\ 5 & 12 & 17 & 2 \end{pmatrix}_{b_0}$
- [223] $1^{11}23131^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -60$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 4 & 0 \\ 3 & 0 & 0 & 4 & 2 & 4 \\ 2 & 0 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$
- [224] $1^14^{131^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 8 & 0 & 13 & 7 \\ 3 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$
- [226] $1^1231^331^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -65$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 14.60 \rightarrow 15$, $T_{\max} = 25$, $T \in [15, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 9 & 0 \\ 9 & 0 & 0 & 9 & 52 & 12 \\ 2 & 0 & -1 & 0 & 12 & -1 \end{pmatrix}_{b_0}$
- [225] $1^{11}2^321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -65$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 2 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & -1 & 3 & 1 \end{pmatrix}_{b_0}$
- [227] $1^{13}2321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 4 & 10 & 0 & 6 & 34 & 12 \\ 2 & 3 & -1 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$

[232] $1^{11} 3^{11} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 3 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[231] $1^{12} 3^{11} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 3 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[230] $1^{11} 3^{11} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 2 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[229] $1^{15} 4^{11} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.1$
 $T_{\min} = 14.58 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 10 & 20 & 0 & 73 & 15 \\ 3 & 4 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

[228] $1^{11} 2^1 3^{11} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 2 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[233] $1^{11} 2^1 3^{11} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.08 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 3 & 4 & 0 & 4 & 6 & 6 \\ 2 & 3 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[237] $1^1 231^{14} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 21.22 \rightarrow 22, T_{\max} = 25, T \in [22, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 12 & 0 & 94 & 16 \\ 2 & 4 & 3 & -4 & 16 & -8 \end{pmatrix}_{b_0}$$

[236] $1^1 231^{13} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 17.72 \rightarrow 18, T_{\max} = 25, T \in [18, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 9 & 0 & 52 & 12 \\ 2 & 4 & 3 & -4 & 12 & -4 \end{pmatrix}_{b_0}$$

[235] $1^1 231^{11} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.72 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 4 \\ 2 & 4 & 3 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

[234] $1^1 231^{12} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 15.22 \rightarrow 16, T_{\max} = 25, T \in [16, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 6 & 0 & 22 & 8 \\ 2 & 4 & 3 & -4 & 8 & -2 \end{pmatrix}_{b_0}$$

- [239] $1^{12}3131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -78$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.90 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$
- [240] $1^{112}23131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -78$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.90 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 5 & 0 & 8 & 4 \\ 1 & 5 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 8 & 8 & 0 & 14 & 8 \\ 2 & 4 & 3 & -4 & 8 & 0 \end{pmatrix}_{b_0}$$
- [238] $1^{16}4131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -78$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 14.92 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 12 & 24 & 0 & 106 & 18 \\ 3 & 4 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$
- [241] $1^{11}41^{131^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -79$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.58 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 10 & 0 & 21 & 9 \\ 3 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$
- [242] $1^{112}2321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.15 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 3 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$
- [243] $1^{14}3221^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.70 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 12 & 0 \\ 6 & 10 & 0 & 12 & 62 & 16 \\ 2 & 2 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$
- [244] $1^{11}3221^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 6 & 4 & 0 & 9 & 20 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$
- [245] $1^{12}3221^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.12 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 14 & 0 \\ 8 & 7 & 0 & 14 & 54 & 16 \\ 2 & 2 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$
- [246] $1^{112}2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -86$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.77 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 4 & 4 & 0 & 4 & 6 & 6 \\ 2 & 3 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$
- [247] $1^{12}31^431^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -86$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 15.02 \rightarrow 16$, $T_{\max} = 25$, $T \in [16, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 12 & 0 \\ 12 & 0 & 0 & 12 & 94 & 16 \\ 2 & 0 & -1 & 0 & 16 & -2 \end{pmatrix}_{b_0}$$
- [248] $1^{11}41^{131^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -88$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.32 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 5 & 7 & 0 & 8 & 6 \\ 3 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[249] $1^{11}2^321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 3 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[250] $1^{11}4^{11}131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 5 & 9 & 0 & 13 & 7 \\ 3 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[251] $1^{17}4^{11}131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 15.22 \rightarrow 16, T_{\max} = 93, T \in [16, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 14 & 28 & 0 & 145 & 21 \\ 3 & 4 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[252] $1^{13}3^{11}221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.52 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 4 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[253] $1^{11}3^{12}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 3 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[255] $1^{14}2^321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 5 & 12 & 0 & 4 & 46 & 12 \\ 2 & 3 & -1 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$

[254] $1^{12}2^321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 3 & 6 & 0 & 2 & 10 & 6 \\ 2 & 3 & -1 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[256] $1^{13}2^321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -105, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.54 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 4 & 9 & 0 & 3 & 25 & 9 \\ 2 & 3 & -1 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[257] $1^{11}3^{13}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 4 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

[258] $1^{12}3^{11}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 3 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[260] $1^{11} 2^3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -112$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 3 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[259] $1^{14} 3^2 221^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -112$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.54 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 5 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

[262] $1^{14} 3^2 221^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 16 & 0 \\ 8 & 11 & 0 & 16 & 90 & 20 \\ 2 & 2 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[261] $1^{14} 2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 13 & 2 \\ 1 & 0 & 1 & -3 & 7 & -1 \\ 6 & 0 & 13 & 7 & 57 & 15 \\ 2 & 0 & 2 & -1 & 15 & 1 \end{pmatrix}_{b_0}$$

[263] $1^1 231^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 0 \\ 2 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[264] $1^{11} 231^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -127$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.24 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 7 & 0 & 14 & 8 \\ 2 & 4 & 3 & -4 & 8 & 0 \end{pmatrix}_{b_0}$$

[269] $1^{13} 23131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -128$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 14.09 \rightarrow 15$, $T_{\max} = 25$, $T \in [15, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 2 & 5 & 0 & 12 & 4 \\ 1 & 5 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 12 & 12 & 0 & 34 & 12 \\ 2 & 4 & 3 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$

[268] $1^1 231^{11} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -128$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[267] $1^{11} 231^{12} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -128$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 10 & 0 \\ 5 & 15 & 0 & 10 & 38 & 12 \\ 2 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[266] $1^{11} 3^2 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -128$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.53 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 4 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

- [265] $1^{12}3131^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -128$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 14.09 \rightarrow 15$, $T_{\max} = 25$, $T \in [15, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$
- [270] $1^{11}31^{11}4^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -130$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.03 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 1 & 4 & 1 & 10 & 3 \\ 4 & 4 & 0 & 15 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 10 & 15 & 0 & 45 & 13 \\ 3 & 4 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$
- [271] $1^{12}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -139$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 5 & -1 \\ 4 & 0 & 7 & 5 & 17 & 9 \\ 2 & 0 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$
- [272] $1^{14}3221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 5 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$
- [273] $1^{12}3131^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -141$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 3 & 0 \\ 4 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$
- [274] $1^{11}231^{11}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -143$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 7 & 0 \\ 6 & 9 & 0 & 7 & 14 & 8 \\ 2 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [275] $1^{13}3221^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -144$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.69 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 15 & 0 \\ 8 & 9 & 0 & 15 & 70 & 18 \\ 2 & 2 & -1 & 0 & 18 & 1 \end{pmatrix}_{b_0}$$
- [276] $1^{13}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -147$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 6 & -1 \\ 5 & 0 & 10 & 6 & 34 & 12 \\ 2 & 0 & 2 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$
- [277] $1^{13}3221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -147$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$
- [278] $1^{11}2321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -147$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 4 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$
- [279] $1^{12}4131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -148$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 13 & 0 & 28 & 10 \\ 3 & 4 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[280] $1^{\overline{1}31^{\overline{11}}}$ $4^{\overline{1}}$ $1^{\overline{1}}$ $+$ $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -148$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.78 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 \\ 4 & 4 & 0 & 4 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 3 & 4 & 0 & 1 & 3 \\ 3 & 4 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[281] $1^{\overline{11}}231^{\overline{11}}31^{\overline{11}}$ $+$ $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -151$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 7 & 0 \\ 4 & 9 & 0 & 7 & 14 & 8 \\ 2 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[282] $1^{\overline{12}}3^{\overline{11}}221^{\overline{11}}$ $+$ $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -155$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 4 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[283] $1^{\overline{1}31^{\overline{11}}}$ $4^{\overline{1}}$ $1^{\overline{1}}$ $+$ $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -158$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.72 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 4 & 0 & 7 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 5 & 7 & 0 & 8 & 6 \\ 3 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[284] $1^{\overline{11}}3^{\overline{14}}221^{\overline{12}}$ $+$ $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -160$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 7 & 4 & 0 & 9 & 20 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[285] $1^{\overline{11}}2^{\overline{12}}3^{\overline{11}}21^{\overline{11}}$ $+$ $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -160$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.65 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 4 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[286] $1^{\overline{12}}2^{\overline{11}}3^{\overline{11}}21^{\overline{11}}$ $+$ $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -161$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.70 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 4 & 7 & 0 & 5 & 17 & 9 \\ 2 & 3 & -1 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[287] $1^{\overline{12}}3^{\overline{1}31^{\overline{11}}}$ $4^{\overline{1}}$ $1^{\overline{1}}$ $+$ $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -163$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.36 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 11 & 0 & 21 & 9 \\ 3 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[288] $1^{\overline{11}}3^{\overline{1}31^{\overline{14}}}$ $4^{\overline{1}}$ $1^{\overline{11}}$ $+$ $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -163$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.09 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 17 & 21 & 0 & 101 & 19 \\ 3 & 4 & -3 & 19 & 0 \end{pmatrix}_{b_0}$$

[289] $1^{\overline{11}}23131^{\overline{113}}$ $+$ $[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -170$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.39 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 4 & 0 \\ 4 & 0 & 0 & 4 & 2 & 4 \\ 2 & 0 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

- [290] $1^{12}2^3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -170$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 4 & 6 & 0 & 2 & 10 & 6 \\ 2 & 3 & -1 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$
- [291] $1^{11}231^1 31^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -171$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 7 & 0 \\ 6 & 0 & 0 & 7 & 14 & 8 \\ 2 & 0 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$
- [292] $1^{13}3^2 221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -171$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 6 & 8 & 0 & 11 & 44 & 14 \\ 2 & 2 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$
- [297] $1^{12}3^2 221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -172$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.53 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 6 & 6 & 0 & 10 & 30 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$
- [296] $1^{11}2^3 3^2 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -172$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 5 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$
- [295] $1^{12}2^3 3^2 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -172$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.53 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 4 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$
- [294] $1^{14}2^3 3^2 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -172$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 6 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$
- [293] $1^{12}3^2 3^2 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -172$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 4 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [298] $1^{113}231^1 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -173$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.39 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 2 & 5 & 0 & 15 & 4 \\ 1 & 5 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 15 & 0 & 58 & 16 \\ 2 & 4 & 3 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$
- [299] $1^{11}231^{11} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -175$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.85 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [300] $1^{113}3^2 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -179$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 5 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[301] \ 1^{14} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 5 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[303] \ 1^{11} 1^{31^1} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 12 & 0 & 26 & 10 \\ 3 & 4 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[302] \ 1^{12} 1^{31^4} 4 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 16 & 20 & 0 & 90 & 18 \\ 3 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[304] \ 1^{13} 2 \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 5 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[306] \ 1^1 23131^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[305] \ 1^{14} 23131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 2 & 5 & 0 & 16 & 4 \\ 1 & 5 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 16 & 16 & 0 & 62 & 16 \\ 2 & 4 & 3 & -4 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[307] \ 1^1 231^{12} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 6 & 0 \\ 4 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[308] \ 1^{11} 231^{11} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 7 & 0 \\ 5 & 9 & 0 & 7 & 14 & 8 \\ 2 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[309] \ 1^{13} 4 \ 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -207, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 13.49 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 17 & 0 & 49 & 13 \\ 3 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[310] \ 1^{11} 2 \ 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 5 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

- [311] $1^{11} 3^{14} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -220$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.80 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 6 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$
- [312] $1^{12} 4^{131^2} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -226$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.42 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 10 & 14 & 0 & 38 & 12 \\ 3 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$
- [313] $1^{11} 4^{131^3} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -226$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.63 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 14 & 18 & 0 & 70 & 16 \\ 3 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$
- [314] $1^{131^{11}} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -226$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 4 & 1 & 6 & 3 \\ 4 & 4 & 0 & 9 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 9 & 0 & 13 & 7 \\ 3 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$
- [315] $21^{11} 41^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -226$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 4 & 2 \\ 2 & 0 & 3 & 0 & 4 & 2 \\ 2 & 3 & 3 & 0 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 4 & 8 & 0 & 10 & 6 \\ 2 & 2 & 3 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$
- [316] $1^{12} 3^{13} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -227$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 5 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [317] $1^{13} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -231$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 5 & 10 & 0 & 6 & 34 & 12 \\ 2 & 3 & -1 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$
- [318] $1^{14} 231^{131^3} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -232$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.78 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 1 & 0 & 25 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 25 & 0 & 188 & 28 \\ 2 & 4 & 3 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$
- [319] $1^{12} 4^{131^3} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -235$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.55 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 13 & 17 & 0 & 61 & 15 \\ 3 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$
- [320] $1^{11} 4^{131^2} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -235$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 11 & 15 & 0 & 45 & 13 \\ 3 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$
- [321] $1^{12} 3^{12} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -236$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 5 & 7 & 0 & 5 & 17 & 9 \\ 2 & 3 & -1 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[322] $1^{12} 3^{14} 221^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 14 & 0 \\ 9 & 7 & 0 & 14 & 54 & 16 \\ 2 & 2 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[325] $1^{13} 3^{14} 221^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 19 & 0 \\ 11 & 10 & 0 & 19 & 104 & 22 \\ 2 & 2 & -1 & 0 & 22 & 1 \end{pmatrix}_{b_0}$$

[327] $1^{13} 4^{1311} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 15 & 0 & 40 & 12 \\ 3 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[326] $1^{11} 4^{13111} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 4 & 1 & 8 & 3 \\ 4 & 4 & 0 & 12 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 12 & 0 & 26 & 10 \\ 3 & 4 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[323] $1^{13} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 5 & 9 & 0 & 3 & 25 & 9 \\ 2 & 3 & -1 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[324] $1^1 231^{231113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 6 & 0 \\ 7 & 0 & 0 & 6 & 22 & 8 \\ 2 & 0 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

[328] $1^{13} 3^{111} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 5 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[329] $1^1 231^{123114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 6 & 0 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[330] $1^{13} 3^{12} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -252, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[331] $1^{12} 3^{12} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 5 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0} \\
[332] & 1^{12} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -261, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 13.41 \rightarrow 14, T_{\max} = 25, T \in [14, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 3 & 0 \\ 5 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0} \\
[333] & 1^{12} 2 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -267, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 5 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0} \\
[336] & 1^{14} 4 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 13.55 \rightarrow 14, T_{\max} = 93, T \in [14, 93] \\
& \begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 11 & 21 & 0 & 76 & 16 \\ 3 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0} \\
[335] & 1^{11} 2 3 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 12.88 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 6 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0} \\
[334] & 1^{12} 2 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 5 & -1 \\ 5 & 0 & 7 & 5 & 17 & 9 \\ 2 & 0 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0} \\
[337] & 1^{12} 3 221^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 6 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0} \\
[338] & 1^{14} 2 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -281, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 6 & 13 & 0 & 7 & 57 & 15 \\ 2 & 3 & -1 & -1 & 15 & 1 \end{pmatrix}_{b_0} \\
[341] & 1^{12} 3131^{15} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.1 \\
& T_{\min} = 14.26 \rightarrow 15, T_{\max} = 25, T \in [15, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0} \\
[340] & 1^{11} 231^2 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 13.56 \rightarrow 14, T_{\max} = 25, T \in [14, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 10 & 0 \\ 9 & 0 & 0 & 10 & 38 & 12 \\ 2 & 0 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0} \\
[339] & 1^{15} 23131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.1 \\
& T_{\min} = 14.26 \rightarrow 15, T_{\max} = 25, T \in [15, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 2 & 5 & 0 & 20 & 4 \\ 1 & 5 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 20 & 20 & 0 & 98 & 20 \\ 2 & 4 & 3 & -4 & 20 & -1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [342] $1^{14}2^321^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -290, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 6 & 12 & 0 & 4 & 46 & 12 \\ 2 & 3 & -1 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$
- [343] $1^{12}231^{12}31^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 14 & 0 & 62 & 16 \\ 2 & 4 & 3 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$
- [344] $1^{12}3^1221^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 7 & 6 & 0 & 10 & 30 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$
- [345] $1^{11}23131^{114} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 4 & 0 \\ 5 & 0 & 0 & 4 & 2 & 4 \\ 2 & 0 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$
- [346] $1^{11}231^{12}31^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.29 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 10 & 0 & 38 & 12 \\ 2 & 4 & 3 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$
- [347] $1^{14}3^1221^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 12 & 0 \\ 7 & 10 & 0 & 12 & 62 & 16 \\ 2 & 2 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$
- [348] $1^{11}231^{13}31^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 13.01 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 13 & 0 \\ 7 & 21 & 0 & 13 & 74 & 16 \\ 2 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$
- [349] $1^{14}4^11^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 11 & 19 & 0 & 65 & 15 \\ 3 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$
- [350] $1^{12}4^11^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 10 & 16 & 0 & 45 & 13 \\ 3 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$
- [351] $1^{14}3^1221^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -323, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.66 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 20 & 0 \\ 11 & 12 & 0 & 20 & 126 & 24 \\ 2 & 2 & -1 & 0 & 24 & 1 \end{pmatrix}_{b_0}$$
- [352] $1^{14}3^1221^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.52 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 6 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[353] $1^{14}3^{12}221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -332$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 6 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[354] $1^{14}231^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -333$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 2 & 5 & 0 & 19 & 4 \\ 1 & 5 & 1 & 0 & 19 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 19 & 19 & 0 & 92 & 20 \\ 2 & 4 & 3 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[355] $1^{13}2^3 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -336$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 6 & -1 \\ 6 & 0 & 10 & 6 & 34 & 12 \\ 2 & 0 & 2 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$

[356] $1^1 231^{13} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -340$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.06 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 9 & 0 \\ 6 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

[357] $1^1 231^3 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -345$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 9 & 0 \\ 10 & 0 & 0 & 9 & 52 & 12 \\ 2 & 0 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

[358] $1^{13}3^{13}221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -347$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 6 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[359] $1^{13}2^3 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -347$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 6 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[360] $1^{12}3^{13}221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -347$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 6 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[361] $1^{13}3^{13}221^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -347$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 15 & 0 \\ 9 & 9 & 0 & 15 & 70 & 18 \\ 2 & 2 & -1 & 0 & 18 & 1 \end{pmatrix}_{b_0}$$

[362] $1^{13}3^{12}221^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -348$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.48 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 7 & 8 & 0 & 11 & 44 & 14 \\ 2 & 2 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

[363] $1^{12} 2^3 3 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.66 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 6 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[364] $1^{14} 2^3 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 13 & 2 \\ 1 & 0 & 1 & -3 & 7 & -1 \\ 7 & 0 & 13 & 7 & 57 & 15 \\ 2 & 0 & 2 & -1 & 15 & 1 \end{pmatrix}_{b_0}$$

[365] $1^{12} 2^3 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -366, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 6 & 10 & 0 & 6 & 34 & 12 \\ 2 & 3 & -1 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$

[366] $1^{14} 3 221^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.52 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 16 & 0 \\ 9 & 11 & 0 & 16 & 90 & 20 \\ 2 & 2 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[367] $1^1 231^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -371, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 6 & 0 \\ 5 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[368] $1^{11} 231^1 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -371, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 7 & 0 \\ 7 & 0 & 0 & 7 & 14 & 8 \\ 2 & 0 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

[369] $1^{13} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -375, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 12 & 18 & 0 & 61 & 15 \\ 3 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[370] $1^{12} 2^3 3 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.51 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 6 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[371] $1^{12} 2^3 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -387, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 5 & -1 \\ 6 & 0 & 7 & 5 & 17 & 9 \\ 2 & 0 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[372] $1^{14} 2^3 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -387, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 7 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & -1 & 12 & 1 \end{pmatrix}_{b_0} \\
[373] & 1^{1231^{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 12.86 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 6 & 0 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0} \\
[374] & 1^{11231^{12}31^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 12.86 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 10 & 0 \\ 6 & 15 & 0 & 10 & 38 & 12 \\ 2 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0} \\
[375] & 1^{116}23131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -398, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 14.30 \rightarrow 15, T_{\max} = 25, T \in [15, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 2 & 5 & 0 & 24 & 4 \\ 1 & 5 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 24 & 24 & 0 & 142 & 24 \\ 2 & 4 & 3 & -4 & 24 & -1 \end{pmatrix}_{b_0} \\
[376] & 1^{1231^{13}1^{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 13.48 \rightarrow 14, T_{\max} = 25, T \in [14, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 3 & 0 \\ 6 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0} \\
[377] & 1^{12}3^{14}221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 8 & 6 & 0 & 10 & 30 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0} \\
[378] & 1^{12}3^{14}221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -428, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 7 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0} \\
[379] & 1^{13}3^{14}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 12.84 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 7 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0} \\
[380] & 1^{1231^{12}31^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 12.87 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 6 & 0 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0} \\
[381] & 1^{12}4^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 13.39 \rightarrow 14, T_{\max} = 93, T \in [14, 93] \\
& \begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 13 & 19 & 0 & 68 & 16 \\ 3 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0} \\
[382] & 1^{1231^{23}1^{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -450, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 13.37 \rightarrow 14, T_{\max} = 25, T \in [14, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 6 & 0 \\ 8 & 0 & 0 & 6 & 22 & 8 \\ 2 & 0 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}
\end{aligned}$$

- [383] $1^{13} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -450$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.39 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 12 & 20 & 0 & 70 & 16 \\ 3 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0}$
- [384] $1^{13} 4 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -456$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$
 $\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 15 & 21 & 0 & 88 & 18 \\ 3 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$
- [385] $1^{13} 3 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -467$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 7 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$
- [386] $1^{12} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -476$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.65 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 7 & 13 & 0 & 7 & 57 & 15 \\ 2 & 3 & -1 & -1 & 15 & 1 \end{pmatrix}_{b_0}$
- [387] $1^{13} 4 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -483$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.53 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$
 $\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 18 & 24 & 0 & 121 & 21 \\ 3 & 4 & -3 & 21 & 1 \end{pmatrix}_{b_0}$
- [388] $1^{11} 231^{13} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -487$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 13 & 0 & 74 & 16 \\ 2 & 4 & 3 & -4 & 16 & 0 \end{pmatrix}_{b_0}$
- [389] $1^{13} 2 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -492$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 7 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$
- [390] $1^{12} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -496$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.65 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 5 & -1 \\ 7 & 0 & 7 & 5 & 17 & 9 \\ 2 & 0 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$
- [391] $1^{12} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -499$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$
 $\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 16 & 22 & 0 & 97 & 19 \\ 3 & 4 & -3 & 19 & 1 \end{pmatrix}_{b_0}$
- [392] $1^{12} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -508$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 13.55 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$
 $\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 19 & 25 & 0 & 132 & 22 \\ 3 & 4 & -3 & 22 & 1 \end{pmatrix}_{b_0}$
- [393] $1^{12} 3 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -512$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.53 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 7 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[394] $1^{13} 3 221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -515, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.51 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 8 & 8 & 0 & 11 & 44 & 14 \\ 2 & 2 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

[395] $1^{13} 3 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -515, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 7 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[396] $1^{13} 2 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -515, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 6 & -1 \\ 7 & 0 & 10 & 6 & 34 & 12 \\ 2 & 0 & 2 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$

[397] $1^{14} 3 221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 12 & 0 \\ 8 & 10 & 0 & 12 & 62 & 16 \\ 2 & 2 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[398] $1^{14} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -526, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.41 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 14 & 22 & 0 & 90 & 18 \\ 3 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[399] $1^{117} 23131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 2 & 5 & 0 & 28 & 4 \\ 1 & 5 & 1 & 0 & 28 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 28 & 28 & 0 & 194 & 28 \\ 2 & 4 & 3 & -4 & 28 & -1 \end{pmatrix}_{b_0}$$

[400] $1^1 231^{14} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 \\ 8 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

[401] $1^{13} 3 221^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -540, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 15 & 0 \\ 10 & 9 & 0 & 15 & 70 & 18 \\ 2 & 2 & -1 & 0 & 18 & 1 \end{pmatrix}_{b_0}$$

[402] $1^{11} 231^{12} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -556, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 10 & 0 \\ 7 & 15 & 0 & 10 & 38 & 12 \\ 2 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[404] $1^{11} 231^{14} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.03 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 16 & 0 \\ 9 & 27 & 0 & 16 & 122 & 20 \\ 2 & 4 & -1 & 0 & 20 & 0 \end{pmatrix}_{b_0}$$

- [403] $1^{11}231^{12}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -560$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 10 & 0 \\ 9 & 15 & 0 & 10 & 38 & 12 \\ 2 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$
- [405] $1^{13}231^{13}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -567$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 21 & 0 & 142 & 24 \\ 2 & 4 & 3 & -4 & 24 & 0 \end{pmatrix}_{b_0}$
- [406] $31^{11}2321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 29$, $\Delta = -116$, $\det = -580$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 29$, $T \in [13, 29]$
 $\begin{pmatrix} 0 & 3 & 3 & 1 & 3 & 2 \\ 3 & 1 & 4 & 1 & 3 & 3 \\ 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 3 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$
- [407] $1^{14}4^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -583$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.41 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$
 $\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 14 & 24 & 0 & 101 & 19 \\ 3 & 4 & -3 & 19 & 1 \end{pmatrix}_{b_0}$
- [408] $1^{14}2^3 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -592$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 8 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & -1 & 12 & 1 \end{pmatrix}_{b_0}$
- [410] $1^{14}3^2 221^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -592$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 20 & 0 \\ 12 & 12 & 0 & 20 & 126 & 24 \\ 2 & 2 & -1 & 0 & 24 & 1 \end{pmatrix}_{b_0}$
- [409] $1^{14}3^2 221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -592$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.80 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 8 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$
- [411] $1^{14}2^3 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -603$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 13 & 2 \\ 1 & 0 & 1 & -3 & 7 & -1 \\ 8 & 0 & 13 & 7 & 57 & 15 \\ 2 & 0 & 2 & -1 & 15 & 1 \end{pmatrix}_{b_0}$
- [413] $1^{14}3^2 221^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -611$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.51 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 16 & 0 \\ 10 & 11 & 0 & 16 & 90 & 20 \\ 2 & 2 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$
- [412] $1^{11}231^{12}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -611$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 10 & 0 \\ 8 & 15 & 0 & 10 & 38 & 12 \\ 2 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$
- [414] $1^{11}231^{13}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -623$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 9 & 0 \\ 7 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[415] $1^{13} 3^{14} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -636, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 8 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[416] $1^{13} 4^{131^2} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -639, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 15 & 23 & 0 & 97 & 19 \\ 3 & 4 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

[417] $1^{12} 231^{13} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -652, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 13.24 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 17 & 0 & 104 & 20 \\ 2 & 4 & 3 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[418] $1^{11} 231^{14} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -667, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.64 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 16 & 0 & 122 & 20 \\ 2 & 4 & 3 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[419] $1^{13} 3^{14} 221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -672, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.54 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 9 & 8 & 0 & 11 & 44 & 14 \\ 2 & 2 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

[420] $1^{14} 4^{131^3} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -679, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.42 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 17 & 25 & 0 & 121 & 21 \\ 3 & 4 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[421] $1^{14} 3^{13} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -683, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 8 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[422] $1^{13} 2^3 3^{14} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -684, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 6 & -1 \\ 8 & 0 & 10 & 6 & 34 & 12 \\ 2 & 0 & 2 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$

[423] $1^{11} 231^{13} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -695, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 13 & 0 \\ 8 & 21 & 0 & 13 & 74 & 16 \\ 2 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[424] $1^{14} 3^{13} 221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -731, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 12 & 0 \\ 9 & 10 & 0 & 12 & 62 & 16 \\ 2 & 2 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

- [425] $1^{13} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -774$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.41 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$
 $\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 18 & 26 & 0 & 130 & 22 \\ 3 & 4 & -3 & 22 & 1 \end{pmatrix}_{b_0}$
- [426] $1^{14} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -778$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$
 $\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 20 & 28 & 0 & 158 & 24 \\ 3 & 4 & -3 & 24 & 1 \end{pmatrix}_{b_0}$
- [427] $1^1 231^{15} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -788$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.1$
 $T_{\min} = 13.11 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 15 & 0 \\ 10 & 30 & 0 & 15 & 148 & 20 \\ 2 & 4 & -1 & 0 & 20 & 0 \end{pmatrix}_{b_0}$
- [428] $1^1 231^{13} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -800$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 9 & 0 \\ 8 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$
- [429] $1^{14} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -832$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.56 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 13 & 2 \\ 1 & 0 & 1 & -3 & 7 & -1 \\ 9 & 0 & 13 & 7 & 57 & 15 \\ 2 & 0 & 2 & -1 & 15 & 1 \end{pmatrix}_{b_0}$
- [430] $1^1 231^{13} 31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -836$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.87 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 9 & 0 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$
- [433] $1^{14} 3 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -844$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 9 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$
- [431] $1^{14} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -844$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.39 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$
 $\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 17 & 27 & 0 & 132 & 22 \\ 3 & 4 & -3 & 22 & 1 \end{pmatrix}_{b_0}$
- [432] $1^{14} 3 221^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -844$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.52 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 16 & 0 \\ 11 & 11 & 0 & 16 & 90 & 20 \\ 2 & 2 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$
- [434] $1^{13} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -855$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$
 $\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 21 & 29 & 0 & 169 & 25 \\ 3 & 4 & -3 & 25 & 1 \end{pmatrix}_{b_0}$
- [435] $1^1 231^{13} 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -871$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 9 & 0 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[436] $1^{14} 3^{22} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -928, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.53 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 12 & 0 \\ 10 & 10 & 0 & 12 & 62 & 16 \\ 2 & 2 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[437] $1^{14} 231^{14} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 \\ 9 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[438] $1^{14} 231^{14} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -952, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 36 & 4 \\ 1 & 4 & 1 & 0 & 28 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 36 & 28 & 0 & 254 & 32 \\ 2 & 4 & 3 & -4 & 32 & 0 \end{pmatrix}_{b_0}$$

[439] $1^{11} 231^{13} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -968, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 13 & 0 \\ 9 & 21 & 0 & 13 & 74 & 16 \\ 2 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[440] $1^{12} 231^{14} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1012, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.31 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 20 & 0 & 158 & 24 \\ 2 & 4 & 3 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[441] $1^{14} 4^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -1051, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 20 & 30 & 0 & 169 & 25 \\ 3 & 4 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[442] $1^{11} 231^{14} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1051, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 16 & 0 \\ 10 & 27 & 0 & 16 & 122 & 20 \\ 2 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[443] $1^{13} 231^{14} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.23 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 33 & 4 \\ 1 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 33 & 24 & 0 & 202 & 28 \\ 2 & 4 & 3 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

[444] $1^{11} 231^{13} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 13 & 0 \\ 10 & 21 & 0 & 13 & 74 & 16 \\ 2 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[445] $1^{11} 231^{13} 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 13 & 0 \\ 11 & 21 & 0 & 13 & 74 & 16 \\ 2 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[446] $1^{14} 4^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -1204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.42 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 33 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 23 & 33 & 0 & 212 & 28 \\ 3 & 4 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[447] $1^1 231^{14} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1220$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.91 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 \\ 10 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[448] $1^1 231^{14} 31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1403$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 \\ 11 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[449] $1^{11} 231^{14} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1436$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 16 & 0 \\ 11 & 27 & 0 & 16 & 122 & 20 \\ 2 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[450] $1^1 231^{14} 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1480$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[451] $1^{11} 231^{14} 31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1715$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.87 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 16 & 0 \\ 12 & 27 & 0 & 16 & 122 & 20 \\ 2 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[452] $1^{11} 231^{14} 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1888$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 16 & 0 \\ 13 & 27 & 0 & 16 & 122 & 20 \\ 2 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [453] $1^1 31^{11} 41^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 8$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$

$T_{\min} = 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 8 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 4 & 0 & 10 & 6 \\ 2 & 4 & 2 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[454] $1^{11} 5 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 91$, $\Delta = -128$, $\det = 25$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$

$T_{\min} = 14.16 \rightarrow 15$, $T_{\max} = 91$, $T \in [15, 91]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[455] $1^{12} 5^{14} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 91$, $\Delta = -128$, $\det = 30$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.5$

$T_{\min} = 14.20 \rightarrow 15$, $T_{\max} = 91$, $T \in [15, 91]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 2 & 3 & -2 & 4 & 0 \\ 8 & 8 & 4 & 10 & 6 \\ 3 & 4 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[456] $1^{11} 31^1 4131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 35$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$

$T_{\min} = 13.83 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

- [457] $1^{11} 5^{12} 13^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 91$, $\Delta = -128$, $\det = 39$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.15 \rightarrow 15$, $T_{\max} = 91$, $T \in [15, 91]$
 $\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & -2 & 2 & 0 \\ 4 & 4 & 2 & 1 & 3 \\ 3 & 4 & 0 & 3 & 1 \end{pmatrix}_{b_0}$
- [458] $1^{14} 31^{11} 41^3 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 40$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.40 \rightarrow 15$, $T_{\max} = 24$, $T \in [15, 24]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 24 & 18 & 0 & 130 & 22 \\ 2 & 4 & 2 & -3 & 22 & 0 \end{pmatrix}_{b_0}$
- [459] $1^1 31^{11} 41^{23} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 40$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.50 \rightarrow 15$, $T_{\max} = 24$, $T \in [15, 24]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 10 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & -3 & 5 & -1 \\ 6 & 10 & 0 & 5 & 20 & 8 \\ 2 & 4 & 0 & -1 & 8 & 0 \end{pmatrix}_{b_0}$
- [460] $1^1 5^{11} 13^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 91$, $\Delta = -128$, $\det = 43$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.23 \rightarrow 15$, $T_{\max} = 91$, $T \in [15, 91]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & -2 & 2 & 0 \\ 2 & 4 & 2 & 1 & 3 \\ 3 & 4 & 0 & 3 & 1 \end{pmatrix}_{b_0}$
- [461] $21^1 5^{11} 13^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 91$, $\Delta = -128$, $\det = 43$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.23 \rightarrow 15$, $T_{\max} = 91$, $T \in [15, 91]$
 $\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 3 & 4 & 4 \\ 3 & 3 & -2 & 2 & 0 \\ 3 & 4 & 2 & 1 & 3 \\ 4 & 4 & 0 & 3 & 1 \end{pmatrix}_{b_0}$
- [462] $1^1 5^{12} 13^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 91$, $\Delta = -128$, $\det = 46$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 14.48 \rightarrow 15$, $T_{\max} = 91$, $T \in [15, 91]$
 $\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 2 & 3 & -2 & 4 & 0 \\ 4 & 8 & 4 & 10 & 6 \\ 3 & 4 & 0 & 6 & 1 \end{pmatrix}_{b_0}$
- [463] $1^1 5^{13} 13^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 91$, $\Delta = -128$, $\det = 51$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 14.82 \rightarrow 15$, $T_{\max} = 91$, $T \in [15, 91]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & -2 & 6 & 0 \\ 6 & 12 & 6 & 25 & 9 \\ 3 & 4 & 0 & 9 & 1 \end{pmatrix}_{b_0}$
- [464] $1^{11} 5^{11} 13^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 91$, $\Delta = -128$, $\det = 58$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.17 \rightarrow 15$, $T_{\max} = 91$, $T \in [15, 91]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & -2 & 2 & 0 \\ 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 0 & 3 & 1 \end{pmatrix}_{b_0}$
- [465] $1^1 31^{11} 4131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 72$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 14.72 \rightarrow 15$, $T_{\max} = 24$, $T \in [15, 24]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & -3 & 3 & -1 \\ 2 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$
- [466] $1^{12} 5^{11} 13^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 91$, $\Delta = -128$, $\det = 85$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 14.19 \rightarrow 15$, $T_{\max} = 91$, $T \in [15, 91]$
 $\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 2 & 3 & -2 & 4 & 0 \\ 7 & 8 & 4 & 10 & 6 \\ 3 & 4 & 0 & 6 & 1 \end{pmatrix}_{b_0}$
- [467] $1^{12} 5^{11} 13^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 91$, $\Delta = -128$, $\det = 93$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 14.26 \rightarrow 15$, $T_{\max} = 91$, $T \in [15, 91]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 2 & 3 & -2 & 4 & 0 \\ 5 & 8 & 4 & 10 & 6 \\ 3 & 4 & 0 & 6 & 1 \end{pmatrix}_{b_0} \\
[468] & 1^1 31^{11} 41^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 99, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 14.59 \rightarrow 15, T_{\max} = 24, T \in [15, 24] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 3 & 0 & -3 & 4 & -1 \\ 5 & 8 & 1 & 4 & 11 & 7 \\ 2 & 4 & 0 & -1 & 7 & 0 \end{pmatrix}_{b_0} \\
[469] & 1^{12} 5^{12} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 91, \Delta = -128, \det = 106, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 14.21 \rightarrow 15, T_{\max} = 91, T \in [15, 91] \\
& \begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 2 & 3 & -2 & 4 & 0 \\ 6 & 8 & 4 & 10 & 6 \\ 3 & 4 & 0 & 6 & 1 \end{pmatrix}_{b_0} \\
[470] & 1^1 31^{11} 4131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 13.90 \rightarrow 14, T_{\max} = 24, T \in [14, 24] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 4 \\ 2 & 4 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0} \\
[471] & 21^{11} 5^{11} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 91, \Delta = -128, \det = 120, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 14.17 \rightarrow 15, T_{\max} = 91, T \in [15, 91] \\
& \begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 4 & 3 & 4 & 4 \\ 2 & 3 & -2 & 2 & 0 \\ 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 0 & 3 & 1 \end{pmatrix}_{b_0} \\
[472] & 1^1 31^{11} 4131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 127, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 14.24 \rightarrow 15, T_{\max} = 24, T \in [15, 24] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 3 & 0 & -3 & 3 & -1 \\ 3 & 6 & 1 & 3 & 5 & 5 \\ 2 & 4 & 0 & -1 & 5 & 0 \end{pmatrix}_{b_0} \\
[474] & 1^{12} 31^{11} 41^{23} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 16 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 11 & 0 & 52 & 14 \\ 2 & 4 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0} \\
[473] & 1^{12} 31^{11} 4131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 9 & 0 & 28 & 10 \\ 2 & 4 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0} \\
[475] & 1^{13} 5^{11} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 91, \Delta = -128, \det = 130, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 14.35 \rightarrow 15, T_{\max} = 91, T \in [15, 91] \\
& \begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & -2 & 6 & 0 \\ 7 & 12 & 6 & 25 & 9 \\ 3 & 4 & 0 & 9 & 1 \end{pmatrix}_{b_0} \\
[476] & 1^1 31^{11} 41^{13} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 14.24 \rightarrow 15, T_{\max} = 24, T \in [15, 24] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & -3 & 4 & -1 \\ 4 & 8 & 0 & 4 & 10 & 6 \\ 2 & 4 & 0 & -1 & 6 & 0 \end{pmatrix}_{b_0} \\
[477] & 1^{13} 5^{14} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 91, \Delta = -128, \det = 163, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 14.21 \rightarrow 15, T_{\max} = 91, T \in [15, 91] \\
& \begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & -2 & 6 & 0 \\ 10 & 12 & 6 & 25 & 9 \\ 3 & 4 & 0 & 9 & 1 \end{pmatrix}_{b_0} \\
[478] & 1^{11} 31^{11} 4131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 175, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0
\end{aligned}$$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[479] \ 1^{13} 5^{12} 131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 91, \Delta = -128, \det = 175, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 14.26 \rightarrow 15, T_{\max} = 91, T \in [15, 91]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & -2 & 6 & 0 \\ 8 & 12 & 6 & 25 & 9 \\ 3 & 4 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[480] \ 1^{14} 31^{11} 41^1 31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 20 & 4 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 16 & 0 & 90 & 18 \\ 2 & 4 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[481] \ 1^{13} 5^{12} 131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 91, \Delta = -128, \det = 186, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 91, T \in [15, 91]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & -2 & 6 & 0 \\ 9 & 12 & 6 & 25 & 9 \\ 3 & 4 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[482] \ 21^{12} 5^{11} 131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 91, \Delta = -128, \det = 198, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 14.19 \rightarrow 15, T_{\max} = 91, T \in [15, 91]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 3 \\ 4 & 4 & 3 & 8 & 4 \\ 2 & 3 & -2 & 4 & 0 \\ 6 & 8 & 4 & 10 & 6 \\ 3 & 4 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[483] \ 1^{11} 31^{11} 41^1 31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 211, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 11 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 11 & 7 & 0 & 21 & 9 \\ 2 & 4 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[484] \ 1^{13} 31^{11} 41^2 31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 263, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 19 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 19 & 14 & 0 & 77 & 17 \\ 2 & 4 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[485] \ 1^{14} 31^{11} 41^2 31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 292, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 22 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 22 & 17 & 0 & 108 & 20 \\ 2 & 4 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[486] \ 1^{13} 31^{11} 41^1 31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 299, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 17 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 17 & 13 & 0 & 61 & 15 \\ 2 & 4 & 2 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[487] \ 1^{12} 31^{11} 41^1 31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 308, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 14 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 14 & 10 & 0 & 38 & 12 \\ 2 & 4 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[488] \ 21^{13} 5^{11} 131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 91, \Delta = -128, \det = 328, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 91, T \in [15, 91]$$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 3 \\ 4 & 4 & 3 & 12 & 4 \\ 2 & 3 & -2 & 6 & 0 \\ 9 & 12 & 6 & 25 & 9 \\ 3 & 4 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [490] \quad 1^{11} 31 \bar{5} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 1 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[489] \quad 1^{12} 31 \bar{5} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 16.33 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[491] \quad 1^{11} 31 \bar{5} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[492] \quad 1^{11} 31 \bar{5} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 15.37 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 4 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[493] \quad 1^{11} 31 \bar{5} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.48 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[494] \quad 1^{12} 31 \bar{5} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 14.55 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 3 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[495] \quad 1^{11} 31 \bar{5} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 2 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[496] \quad 1^{12} 31 \bar{5} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -112, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 3 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[497] \quad 1^{13} 31 \bar{5} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -147, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 4 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[498] $1^{12}31^1 5 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -172, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 14.53 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 4 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[499] $1^{14}31^1 5 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -172, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 6 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$

[500] $1^{13}31^1 5 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -192, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.8$

$$T_{\min} = 14.58 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 5 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[501] $1^{12}31^1 5 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -267, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 14.59 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 5 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[502] $1^{13}31^1 5 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -347, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.8$

$$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 6 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[503] $1^{14}31^1 5 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -387, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 14.59 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 7 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$

$T = 12$ [504] $1^{11}6^{14} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 19.33 \rightarrow 20, T_{\max} = 24, T \in [20, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 3 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 6 & 3 & 4 & 2 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & -3 \end{pmatrix}_{b_0}$$

[505] $2^1 3 21^1 6 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -107, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 19.14 \rightarrow 20, T_{\max} = 24, T \in [20, 24]$$

$$\begin{pmatrix} 0 & 4 & 2 & 1 & 2 & 2 \\ 4 & 10 & 5 & 3 & 4 & 6 \\ 2 & 5 & 0 & 1 & 3 & 2 \\ 1 & 3 & 1 & -3 & 1 & -1 \\ 2 & 4 & 3 & 1 & 1 & 3 \\ 2 & 6 & 2 & -1 & 3 & -3 \end{pmatrix}_{b_0}$$

[506] $1^{11}31^1 5 1321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.5$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 8 & 3 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 3 & 0 & 8 & 2 & 14 & 8 \\ 2 & 0 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 3 & 1 & 3 & 1 & 1 & 3 \\ 2 & 2 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 5 & 3 & 1 & 5 \\ 3 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 5 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 1 & 1 & 3 & 1 & 1 & 3 \\ 2 & 2 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 4 & 2 & 6 & 2 & 10 & 6 \\ 2 & 2 & 2 & -1 & 6 & 1 \end{pmatrix}_b$$

$$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 1 \end{pmatrix}$$

$$T_{\min} = 15.48 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 2 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 15.48 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 1 & 2 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 15.48 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & -1 & 2 & 1 \end{pmatrix}$$

$$T_{\min} = 16.37 \rightarrow 17, T_{\max} = 24, T \in [17, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 5 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 1 & 1 & -3 & 4 & -1 \\ 4 & 5 & 10 & 4 & 33 & 11 \\ 2 & 2 & 2 & -1 & 11 & 0 \end{pmatrix}$$

[515] $1^{11} \overset{1^{13}321}{6} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 16.37 \rightarrow 17, T_{\max} = 24, T \in [17, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 4 & 2 & 3 & 1 & 9 & 5 \\ 2 & 2 & 0 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

[517] $1^{11} \overset{1^1}{6} 131^{\overset{1^{11}}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 28, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.29 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 1 & 3 & 1 & 1 & 3 \\ 2 & 2 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[518] $1^{11} 1^{31} 5 1321^{\overset{1^{11}}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.56 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 2 & 0 & 4 & 1 & 2 & 4 \\ 2 & 0 & 3 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

[519] $1^{1^{31}} \overset{1^{11}}{6} 131^{\overset{1^{12}}{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 67, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.55 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & 1 & -3 & 3 & -1 \\ 3 & 4 & 7 & 3 & 16 & 8 \\ 2 & 2 & 2 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

[520] $1^{1^{31}} \overset{1^{11}}{6} 131^{\overset{1^{11}}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 67, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.43 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 3 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 2 & 3 & 4 & 2 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[521] $1^{11} \overset{1^{11}321}{6} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 67, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.43 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 2 & 2 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[522] $1^{11} \overset{1^{12}321}{6} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 67, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.55 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 3 & 2 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & -1 & 4 & 1 \end{pmatrix}_{b_0}$$

[523] $1^{11} \overset{1^3}{6} 131^{\overset{1^{11}}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 67, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.43 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 3 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 5 & 3 & 4 & 2 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[524] $31^{\overset{1^1}{11}} 1^{31} 6 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 26, \Delta = -112, \det = 67, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 17.22 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 1 & 2 \\ 3 & 1 & 4 & 1 & 1 & 3 \\ 3 & 4 & 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 1 & 1 & 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

[525] $1^{1^{31}} \overset{1^{12}}{5} 1321^{\overset{1^{13}}{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 68, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 14.59 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 5 & 0 & 12 & 3 & 34 & 12 \\ 2 & 0 & 3 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[526] $1^{11} \overset{1}{6} 131^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 75, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 15.81 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 6 & 2 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 1 & 1 & 1 & -3 & 5 & -1 \\ 6 & 6 & 13 & 5 & 56 & 14 \\ 2 & 2 & 2 & -1 & 14 & 1 \end{pmatrix}_{b_0}$$

[527] $31^{11} 316131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 26, \Delta = -112, \det = 76, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 15.89 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 3 & 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 3 & 1 & 3 & 1 & 1 & 3 \\ 2 & 2 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[528] $1^{11} 31 \overset{12}{5} 1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 76, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 14.84 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 3 & 0 & 4 & 1 & 2 & 4 \\ 2 & 0 & 3 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

[529] $1^{11} \overset{1}{6} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 92, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 15.52 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 4 & 4 & 0 & 3 & 10 & 6 \\ 2 & 2 & 0 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[530] $1^{11} \overset{1}{6} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 99, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 15.37 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 3 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 3 & 3 & 4 & 2 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[531] $1^{11} \overset{2}{6} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 99, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 15.37 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 3 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 4 & 3 & 4 & 2 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[532] $1^{11} 31 \overset{12}{5} 1321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 106, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.5$

$T_{\min} = 14.62 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 8 & 3 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 4 & 0 & 8 & 2 & 14 & 8 \\ 2 & 0 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[533] $1^{11} \overset{1}{6} 131^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 123, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 5 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 1 & 1 & -3 & 4 & -1 \\ 5 & 5 & 10 & 4 & 33 & 11 \\ 2 & 2 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

[535] $1^{11} \overset{4}{6} 131^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 131, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & 1 & -3 & 3 & -1 \\ 7 & 4 & 7 & 3 & 16 & 8 \\ 2 & 2 & 2 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

[534] $1^{11} \overset{1}{6} 131^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 131, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & 1 & -3 & 3 & -1 \\ 4 & 4 & 7 & 3 & 16 & 8 \\ 2 & 2 & 2 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[536] \ 1^{11} \overset{1^2}{6} \overset{1^2}{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 163, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.37 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & 1 & -3 & 3 & -1 \\ 5 & 4 & 7 & 3 & 16 & 8 \\ 2 & 2 & 2 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[537] \ 1^{11} \overset{1^3}{6} \overset{1^2}{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 163, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.37 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & 1 & -3 & 3 & -1 \\ 6 & 4 & 7 & 3 & 16 & 8 \\ 2 & 2 & 2 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[538] \ 1^{13} \overset{1^3}{31} \overset{1^4}{5} \overset{1^2}{1321} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 171, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.61 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 16 & 3 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 7 & 0 & 16 & 4 & 62 & 16 \\ 2 & 0 & 3 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[539] \ 1^{11} \overset{1^2}{6} \overset{1^4}{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 171, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.50 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 6 & 2 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 1 & 1 & 1 & -3 & 5 & -1 \\ 7 & 6 & 13 & 5 & 56 & 14 \\ 2 & 2 & 2 & -1 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[540] \ 1^{13} \overset{1^3}{31} \overset{1^2}{5} \overset{1^2}{1321} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 183, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 14.77 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 8 & 3 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 5 & 0 & 8 & 2 & 14 & 8 \\ 2 & 0 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[541] \ 1^{11} \overset{1^2}{6} \overset{1^3}{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 187, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.41 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 5 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 1 & 1 & -3 & 4 & -1 \\ 6 & 5 & 10 & 4 & 33 & 11 \\ 2 & 2 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[542] \ 1^{11} \overset{1^3}{321} \overset{1^1}{6} \overset{1^1}{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.44 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 5 & 4 & 1 & 3 & 11 & 7 \\ 2 & 2 & 0 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[543] \ 31 \overset{1^{11}}{1^1} \overset{1^1}{31} \overset{1^1}{6} \overset{1^1}{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 26, \Delta = -112, \det = 211, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.82 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 3 & 2 \\ 3 & 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 4 & 3 & 4 & 2 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[544] \ 1^{13} \overset{1^3}{31} \overset{1^3}{5} \overset{1^3}{1321} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 211, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 6 & 0 & 12 & 3 & 34 & 12 \\ 2 & 0 & 3 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[545] $1^{11} \overset{1^3}{6} 131^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 219, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.39 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 5 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 1 & 1 & -3 & 4 & -1 \\ 7 & 5 & 10 & 4 & 33 & 11 \\ 2 & 2 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

[546] $1^{11} \overset{1^4}{6} 131^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 219, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.39 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 5 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 1 & 1 & -3 & 4 & -1 \\ 8 & 5 & 10 & 4 & 33 & 11 \\ 2 & 2 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

[547] $1^{11} \overset{1^3}{6} 131^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 235, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.43 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 6 & 2 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 1 & 1 & 1 & -3 & 5 & -1 \\ 8 & 6 & 13 & 5 & 56 & 14 \\ 2 & 2 & 2 & -1 & 14 & 1 \end{pmatrix}_{b_0}$$

[548] $1^{11} \overset{1^{14}321^1}{6} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 252, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.57 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 8 & 4 & 4 & 3 & 26 & 10 \\ 2 & 2 & 0 & -1 & 10 & 1 \end{pmatrix}_{b_0}$$

[549] $1^{11} \overset{1^{12}321^1}{6} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 252, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.44 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 6 & 4 & 2 & 3 & 14 & 8 \\ 2 & 2 & 0 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

[550] $1^{11} \overset{1^4}{6} 131^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -104, \det = 267, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.41 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 6 & 2 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 1 & 1 & 1 & -3 & 5 & -1 \\ 9 & 6 & 13 & 5 & 56 & 14 \\ 2 & 2 & 2 & -1 & 14 & 1 \end{pmatrix}_{b_0}$$

[551] $1^{11} \overset{1^{13}321^1}{6} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -105, \det = 272, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.49 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 7 & 4 & 3 & 3 & 19 & 9 \\ 2 & 2 & 0 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

$T = 13$ [552] $1^{11} \overset{1^{13}1}{31} 6 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.50 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & -3 & 3 & 2 \end{pmatrix}_{b_0}$$

[553] $1^1 \overset{1^{13}1}{6} 131^1 41^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 16.33 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 4 & 0 \\ 3 & 4 & 0 & 4 & 10 & 6 \\ 2 & 3 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[554] $1^{11} 6^{131} 13141^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 16.33 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[555] $1^{11} 6^{131} 131^{141} 41^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 4 & 0 \\ 4 & 4 & 0 & 4 & 10 & 6 \\ 2 & 3 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[556] $1^{11} 6^{131} 13141^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -27, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 16.48 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 2 & 0 \\ 1 & 3 & 0 & 2 & 1 & 3 \\ 2 & 3 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[557] $1^{11} 31^{131} 6^{131} 131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -46, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 15.65 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & -3 & 6 & 2 \end{pmatrix}_{b_0}$$

[558] $1^{13} 31^{131} 6^{131} 131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -62, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.8$

$T_{\min} = 15.74 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 10 & 6 & 0 & 34 & 12 \\ 2 & 3 & 2 & -3 & 12 & 2 \end{pmatrix}_{b_0}$$

[559] $1^{11} 6^{131} 13141^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -67, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 16.43 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 1 & 3 \\ 2 & 3 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[560] $1^{12} 31^{131} 6^{131} 131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -68, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$

$T_{\min} = 15.65 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 7 & 5 & 0 & 17 & 9 \\ 2 & 3 & 2 & -3 & 9 & 2 \end{pmatrix}_{b_0}$$

[561] $1^{11} 6^{131} 131^{441} 41^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -75, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 16.81 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 2 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 18 & 0 \\ 13 & 19 & 0 & 18 & 225 & 27 \\ 2 & 3 & -1 & 0 & 27 & 1 \end{pmatrix}_{b_0}$$

[562] $1^{11} 6^{131} 131^{141} 41^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -99, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 16.37 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 2 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 6 & 0 \\ 4 & 7 & 0 & 6 & 21 & 9 \\ 2 & 3 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[563] $1^{12}31^{1^131}6^{131}12+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -118, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 15.76 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 8 & 3 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 8 & 0 & 30 & 12 \\ 2 & 3 & 2 & -3 & 12 & 2 \end{pmatrix}_{b_0}$$

[564] $1^{11}6^{1^131}131^341^1+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -123, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.8$

$$T_{\min} = 16.50 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 2 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 14 & 0 \\ 10 & 15 & 0 & 14 & 133 & 21 \\ 2 & 3 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

[565] $1^{11}6^{1^131}131^241^1+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -131, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 16.40 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 2 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 10 & 0 \\ 7 & 11 & 0 & 10 & 65 & 15 \\ 2 & 3 & -1 & 0 & 15 & 1 \end{pmatrix}_{b_0}$$

[566] $1^{11}6^{1^131}131^141^{11}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -163, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 16.37 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 2 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 6 & 0 \\ 5 & 7 & 0 & 6 & 21 & 9 \\ 2 & 3 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[567] $1^{13}31^{1^131}6^{131}12+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -176, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.8$

$$T_{\min} = 15.70 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 11 & 9 & 0 & 49 & 15 \\ 2 & 3 & 2 & -3 & 15 & 2 \end{pmatrix}_{b_0}$$

[568] $1^{14}31^{1^131}6^{131}12+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -206, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 15.71 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 14 & 3 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 14 & 10 & 0 & 74 & 18 \\ 2 & 3 & 2 & -3 & 18 & 2 \end{pmatrix}_{b_0}$$

[569] $1^{11}6^{1^131}131^441^{11}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -211, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 16.52 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 2 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 18 & 0 \\ 14 & 19 & 0 & 18 & 225 & 27 \\ 2 & 3 & -1 & 0 & 27 & 1 \end{pmatrix}_{b_0}$$

[570] $1^{11}6^{1^131}131^241^{11}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -219, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 16.39 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 2 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 10 & 0 \\ 8 & 11 & 0 & 10 & 65 & 15 \\ 2 & 3 & -1 & 0 & 15 & 1 \end{pmatrix}_{b_0}$$

[571] $1^{11}6^{1^131}131^341^{11}+[\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 25, \Delta = -103, \det = -235, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} =$

10.8

$$T_{\min} = 16.43 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 2 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 14 & 0 \\ 11 & 15 & 0 & 14 & 133 & 21 \\ 2 & 3 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

[572] $1^{13}31 \overset{1^{131}}{6} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -238, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.8$

$$T_{\min} = 15.80 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 12 & 3 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 12 & 0 & 70 & 18 \\ 2 & 3 & 2 & -3 & 18 & 2 \end{pmatrix}_{b_0}$$

[573] $1^{13}31 \overset{1^{131}}{6} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -248, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.8$

$$T_{\min} = 15.98 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 13 & 3 \\ 1 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 15 & 0 & 97 & 21 \\ 2 & 3 & 2 & -3 & 21 & 2 \end{pmatrix}_{b_0}$$

[574] $1^{14}31 \overset{1^{131}}{6} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -332, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 15.73 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 15 & 3 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 15 & 13 & 0 & 97 & 21 \\ 2 & 3 & 2 & -3 & 21 & 2 \end{pmatrix}_{b_0}$$

[575] $1^{14}31 \overset{1^{131}}{6} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -406, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 15.81 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 16 & 3 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 16 & 0 & 126 & 24 \\ 2 & 3 & 2 & -3 & 24 & 2 \end{pmatrix}_{b_0}$$

$T = 14$ [576] $1^{11} \overset{1^1}{5} 131 \overset{1^1}{6} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 21, \Delta = -98, \det = 11, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 17.45 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & -3 & -1 \\ 3 & 2 & 0 & -1 & 2 \end{pmatrix}_{b_0}$$

[577] $1^{11} \overset{1^1}{5} 131 \overset{1^1}{6} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 21, \Delta = -98, \det = 19, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 17.26 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 3 & 2 & 0 & 1 & 4 & 4 \\ 2 & 2 & 0 & -1 & 4 & 1 \end{pmatrix}_{b_0}$$

[578] $1^1 \overset{1^{11}}{5} 131 \overset{1^{11}}{6} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 21, \Delta = -98, \det = 27, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 17.48 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[579] $1^{11} \overset{1^1}{5} 131 \overset{1^1}{6} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 21, \Delta = -98, \det = 44, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 17.27 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 2 & 2 & 0 & 1 & 1 & 3 \\ 2 & 2 & 0 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

4.68 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{e}_6$

$$T = 8 \quad [0] \ 1^{\textcolor{brown}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{blue}{1}} 1^{\textcolor{blue}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{e}}_6 \quad T = 8, \ T_{\max} = 103, \ \Delta = -212, \ \det = 385, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.06 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & -1 & 4 & -4 & -3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [5] \ 1^{\textcolor{brown}{1}\textcolor{blue}{1}\textcolor{red}{1}} 2^{\textcolor{blue}{1}} 3^{\textcolor{blue}{1}} 21^{\textcolor{blue}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{e}}_6 \quad T = 10, \ T_{\max} = 102, \ \Delta = -157, \ \det = 20, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.40 \rightarrow 17, \ T_{\max} = 102, \ T \in [17, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 0 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & -3 & -1 \\ 2 & 3 & -1 & -1 & 0 & -1 & -3 \end{pmatrix}_{b_0}$$

4.69 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{f}_4$

$$T = 8 \quad [0] \ 1^{\textcolor{brown}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{blue}{1}} 1^{\textcolor{blue}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{f}}_4 \quad T = 8, \ T_{\max} = 102, \ \Delta = -186, \ \det = 299, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.20 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & -1 & 4 & -3 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\textcolor{brown}{1}\textcolor{blue}{1}\textcolor{red}{1}} 2^{\textcolor{blue}{1}} 3^{\textcolor{blue}{1}} 1^{\textcolor{blue}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{f}}_4 \quad T = 8, \ T_{\max} = 102, \ \Delta = -186, \ \det = 464, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 15.7$$

$$T_{\min} = 14.21 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -3 & 6 & 0 & -1 \\ 6 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & -1 & 8 & -3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\textcolor{brown}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{blue}{1}} 1^{\textcolor{blue}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{f}}_4 \quad T = 8, \ T_{\max} = 102, \ \Delta = -186, \ \det = 739, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.21 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -3 & 9 & 0 & -1 \\ 9 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & -1 & 12 & -3 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \ 1^{\textcolor{brown}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{blue}{1}} 21^{\textcolor{blue}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{f}}_4 \quad T = 9, \ T_{\max} = 101, \ \Delta = -158, \ \det = -20, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 15.40 \rightarrow 16, \ T_{\max} = 101, \ T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 2 & 5 & -2 & 3 & 0 & 0 \\ 3 & 5 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 4 & -3 & -2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\textcolor{brown}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{blue}{1}} 21^{\textcolor{blue}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{f}}_4 \quad T = 9, \ T_{\max} = 101, \ \Delta = -158, \ \det = -112, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.43 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & -2 & 7 & 0 & 0 \\ 5 & 12 & 7 & 14 & 0 & 8 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 8 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\textcolor{brown}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{blue}{1}} 21^{\textcolor{blue}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{f}}_4 \quad T = 9, \ T_{\max} = 101, \ \Delta = -158, \ \det = -232, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.34 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 4 & -3 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [8] \quad 1^{11} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 10, \quad T_{\max} = 93, \quad \Delta = -131, \quad \det = 5, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.2$$

$$T_{\min} = 16.40 \rightarrow 17, \quad T_{\max} = 93, \quad T \in [17, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 3 & 0 & 1 & 0 & -2 & 0 \\ 2 & 3 & -1 & -1 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 10, \quad T_{\max} = 93, \quad \Delta = -131, \quad \det = 13, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.2$$

$$T_{\min} = 16.15 \rightarrow 17, \quad T_{\max} = 93, \quad T \in [17, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 0 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & -2 & 0 \\ 2 & 3 & -1 & -1 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 3^{11} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 10, \quad T_{\max} = 91, \quad \Delta = -129, \quad \det = 127, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.46 \rightarrow 15, \quad T_{\max} = 91, \quad T \in [15, 91]$$

$$\begin{pmatrix} 5 & 15 & 21 & 1 & 5 \\ 15 & 42 & 59 & 0 & 12 \\ 21 & 59 & 79 & 0 & 17 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 12 & 17 & -3 & 1 \end{pmatrix}_{b_0}$$

4.70 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, \quad T_{\max} = 24, \quad \Delta = -113, \quad \det = -147, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 24, \quad T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 3 & 5 & 0 & 6 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, \quad T_{\max} = 24, \quad \Delta = -113, \quad \det = -210, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.05 \rightarrow 14, \quad T_{\max} = 24, \quad T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 4 & 3 \\ 3 & 5 & 0 & 3 & 4 & 3 & 3 \\ 3 & 4 & -4 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, \quad T_{\max} = 24, \quad \Delta = -113, \quad \det = -288, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.18 \rightarrow 14, \quad T_{\max} = 24, \quad T \in [14, 24]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 3 & 0 & -1 \\ 4 & 8 & 3 & 4 & 6 & 0 & 4 \\ 3 & 8 & 3 & 6 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & -1 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, \quad T_{\max} = 24, \quad \Delta = -113, \quad \det = -298, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, \quad T_{\max} = 24, \quad T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 4 & 3 \\ 2 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, \quad T_{\max} = 24, \quad \Delta = -113, \quad \det = -312, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.81 \rightarrow 13, \quad T_{\max} = 24, \quad T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 2 & 3 \\ 3 & 4 & 0 & 19 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 11 & 19 & 0 & 73 & 21 & 12 & 15 \\ 3 & 5 & 0 & 21 & 4 & 4 & 4 \\ 2 & 4 & 0 & 12 & 4 & 0 & 2 \\ 3 & 4 & -4 & 15 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, \quad T_{\max} = 24, \quad \Delta = -113, \quad \det = -452, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} =$$

13.0

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 9 & 6 & 7 \\ 3 & 5 & 0 & 9 & 4 & 4 & 4 \\ 2 & 4 & 0 & 6 & 4 & 0 & 2 \\ 3 & 4 & -4 & 7 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + [\text{su}_2 \cdot \text{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -507, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 & 2 & 3 \\ 3 & 4 & 0 & 14 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 15 & 9 & 11 \\ 3 & 5 & 0 & 15 & 4 & 4 & 4 \\ 2 & 4 & 0 & 9 & 4 & 0 & 2 \\ 3 & 4 & -4 & 11 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{121}31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + [\text{su}_2 \cdot \text{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -992, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 & 2 & 3 \\ 3 & 4 & 0 & 18 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 10 & 18 & 0 & 58 & 18 & 12 & 14 \\ 3 & 5 & 0 & 18 & 4 & 4 & 4 \\ 2 & 4 & 0 & 12 & 4 & 0 & 2 \\ 3 & 4 & -4 & 14 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{121}31^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + [\text{su}_2 \cdot \text{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -1212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 & 2 & 3 \\ 3 & 4 & 0 & 28 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 16 & 28 & 0 & 154 & 30 & 18 & 22 \\ 3 & 5 & 0 & 30 & 4 & 4 & 4 \\ 2 & 4 & 0 & 18 & 4 & 0 & 2 \\ 3 & 4 & -4 & 22 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{121}31^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + [\text{su}_2 \cdot \text{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -1227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 & 2 & 3 \\ 3 & 4 & 0 & 23 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 13 & 23 & 0 & 100 & 24 & 15 & 18 \\ 3 & 5 & 0 & 24 & 4 & 4 & 4 \\ 2 & 4 & 0 & 15 & 4 & 0 & 2 \\ 3 & 4 & -4 & 18 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{131}31^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + [\text{su}_2 \cdot \text{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -1892, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 & 2 & 3 \\ 3 & 4 & 0 & 27 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 15 & 27 & 0 & 133 & 27 & 18 & 21 \\ 3 & 5 & 0 & 27 & 4 & 4 & 4 \\ 2 & 4 & 0 & 18 & 4 & 0 & 2 \\ 3 & 4 & -4 & 21 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{131}31^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + [\text{su}_2 \cdot \text{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -2307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 & 2 & 3 \\ 3 & 4 & 0 & 32 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 18 & 32 & 0 & 193 & 33 & 21 & 25 \\ 3 & 5 & 0 & 33 & 4 & 4 & 4 \\ 2 & 4 & 0 & 21 & 4 & 0 & 2 \\ 3 & 4 & -4 & 25 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{141}31^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + [\text{su}_2 \cdot \text{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -3152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 & 2 & 3 \\ 3 & 4 & 0 & 36 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 20 & 36 & 0 & 238 & 36 & 24 & 28 \\ 3 & 5 & 0 & 36 & 4 & 4 & 4 \\ 2 & 4 & 0 & 24 & 4 & 0 & 2 \\ 3 & 4 & -4 & 28 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [13] \quad 1^{111} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 21 & 27 & 1 & 1 & 9 \\ 27 & 33 & 1 & 0 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 9 & 11 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 31^{121} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 170, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 12 & 6 & 0 & 22 & 12 & 6 & 8 \\ 2 & 6 & 3 & 0 & 12 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 6 & 4 & 0 & 2 \\ 2 & 4 & 2 & -4 & 8 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{111} 3 \quad 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 236, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 21 & 27 & 1 & 52 & 9 \\ 27 & 33 & 1 & 65 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 52 & 65 & 0 & 123 & 21 \\ 9 & 11 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 21 & 27 & 1 & 28 & 9 \\ 27 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 35 & 0 & 33 & 11 \\ 9 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 31^{111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 440, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 6 & 3 & 4 \\ 2 & 6 & 3 & 0 & 6 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 3 & 4 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

4.71 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 3 & 6 & 4 \\ 3 & 5 & 0 & 6 & 4 & 4 & 6 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 6 & 3 & 4 & 4 \\ 3 & 4 & -4 & 4 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 429, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 3 & 9 & 4 \\ 3 & 5 & 0 & 6 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 9 & 6 & 13 & 7 \\ 3 & 4 & -4 & 4 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 1264, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 9 & 6 & 15 & 7 \\ 3 & 5 & 0 & 9 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 6 & 4 & 0 & 6 & 2 \\ 5 & 9 & 0 & 15 & 9 & 6 & 13 & 7 \\ 3 & 4 & -4 & 7 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

4.72 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 3 & 4 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.73 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 0 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 9 & 17 & 0 & 13 & 7 \\ 5 & 8 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.71 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 1 & 44 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 44 & 84 & 0 & 346 & 34 \\ 5 & 8 & -4 & 34 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{113}} 3^{\overline{1}} 1^{\overline{41}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -3 & 9 & 0 & -1 \\ 13 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 12 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.53 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 4 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 318, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -3 & 6 & 0 & -1 \\ 6 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 8 & -2 & -1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 40 & 0 & 5 \\ 2 & 4 & -3 & 15 & 0 & -1 \\ 15 & 40 & 15 & 148 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 20 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[19] 1^{114} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & -1 \\ 16 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 16 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[20] 1^{114} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & -1 \\ 13 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 16 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[21] 1^{114} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1590, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & -1 \\ 15 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 16 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[22] 1^{114} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1650, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & -1 \\ 14 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 16 & -2 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [23] 1^{131} 3^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 33, T_{\max} = 99, T \in [33, 99]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 1 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & -1 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[24] 1^{111} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 0 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[25] 1^{111} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 16.11 \rightarrow 17, T_{\max} = 94, T \in [17, 94]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & -2 & 7 & 0 & 0 \\ 6 & 12 & 7 & 14 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 8 & -2 & -3 \end{pmatrix}_{b_0}$$

$$[26] 1^{111} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.15 \rightarrow 16, T_{\max} = 94, T \in [16, 94]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 2 & 5 & -2 & 3 & 0 & 0 \\ 3 & 5 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 4 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[27] 1^{111} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -81, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & -2 & 7 & 0 & 0 \\ 5 & 12 & 7 & 14 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 8 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{11}1^131^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -94, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.74 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 4 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 12 & 0 & 10 & 6 \\ 3 & 6 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{11}1^13 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 94, \ \Delta = -134, \ \det = -111, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.63 \rightarrow 14, \ T_{\max} = 94, \ T \in [14, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{11}1^131^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -151, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 15 & 0 & 13 & 7 \\ 3 & 6 & 6 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{11}1^13 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 94, \ \Delta = -134, \ \det = -165, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.61 \rightarrow 14, \ T_{\max} = 94, \ T \in [14, 94]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{11}1^13 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 94, \ \Delta = -134, \ \det = -189, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.57 \rightarrow 14, \ T_{\max} = 94, \ T \in [14, 94]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 4 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{11}1^131^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -571, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 9 & 3 \\ 4 & 10 & 13 & 0 & 24 & 6 \\ 5 & 13 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 24 & 27 & 0 & 49 & 13 \\ 3 & 6 & 6 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [34] \ 1^{11}2^13 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, \ T_{\max} = 22, \ \Delta = -107, \ \det = 6, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 17.67 \rightarrow 18, \ T_{\max} = 22, \ T \in [18, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{11}2^13 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, \ T_{\max} = 22, \ \Delta = -107, \ \det = 18, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 16.11 \rightarrow 17, \ T_{\max} = 22, \ T \in [17, 22]$$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 \\ 2 & 0 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 0 & 5 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{11}3 \ 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 48, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 15 & 13 & 1 & 5 \\ 15 & 42 & 36 & 0 & 12 \\ 13 & 36 & 28 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 12 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{11}2^13 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, \ T_{\max} = 22, \ \Delta = -107, \ \det = 60, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 10 & 1 & 4 & 16 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 8 & 0 \\ 6 & 16 & 0 & 8 & 22 & 10 \\ 3 & 6 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{111} 2^1 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 4 & 0 \\ 4 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{1231} 1^1 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 0 & 3 & 0 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{11} 3^1 221^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 15 & 21 & 1 & 5 \\ 15 & 42 & 59 & 0 & 12 \\ 21 & 59 & 79 & 0 & 17 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 12 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{111} 2^1 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 70, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.63 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 7 & 3 \\ 4 & 10 & 1 & 4 & 16 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 8 & 0 \\ 7 & 16 & 0 & 8 & 22 & 10 \\ 3 & 6 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{11} 3^1 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 83, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{111} 2^1 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 4 & 0 \\ 6 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{11} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 94, \Delta = -132, \det = 133, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 14 & 0 & 13 & 7 \\ 5 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{141} 2^1 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 180, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 18 & 3 \\ 4 & 10 & 1 & 4 & 52 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 20 & 0 \\ 18 & 52 & 0 & 20 & 226 & 28 \\ 3 & 6 & -1 & 0 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{121} 2^1 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 180, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 10 & 3 \\ 4 & 10 & 1 & 4 & 28 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 12 & 0 \\ 10 & 28 & 0 & 12 & 66 & 16 \\ 3 & 6 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{111}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 190, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.61 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 4 & 0 \\ 5 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{131}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 14 & 3 \\ 4 & 10 & 1 & 4 & 40 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 16 & 0 \\ 14 & 40 & 0 & 16 & 134 & 22 \\ 3 & 6 & -1 & 0 & 22 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{121}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 310, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 13.66 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 9 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 8 & 0 \\ 9 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{141}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 310, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 17 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 16 & 0 \\ 17 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & 0 & 24 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{131}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 350, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 13 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 12 & 0 \\ 13 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & 0 & 18 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{131}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 94, \Delta = -132, \det = 405, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.42 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 5 & 10 & 1 & 7 & 5 \\ 10 & 13 & 0 & 10 & 7 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 10 & 0 & 5 & 5 \\ 5 & 7 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[53] $1^{121}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 430, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 11 & 3 \\ 4 & 10 & 1 & 4 & 28 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 12 & 0 \\ 11 & 28 & 0 & 12 & 66 & 16 \\ 3 & 6 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[54] $1^{121}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 460, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 10 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 8 & 0 \\ 10 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[55] $1^{121}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 460, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 12 & 3 \\ 4 & 10 & 1 & 4 & 28 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 12 & 0 \\ 12 & 28 & 0 & 12 & 66 & 16 \\ 3 & 6 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{11} 1^{131} 1^{11} 4 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 94, \Delta = -132, \det = 628, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.42 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 5 & 10 & 1 & 16 & 5 \\ 10 & 13 & 0 & 24 & 7 \\ 1 & 0 & -5 & 0 & -3 \\ 16 & 24 & 0 & 38 & 12 \\ 5 & 7 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{131} 2 \ 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 710, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.63 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 15 & 3 \\ 4 & 10 & 1 & 4 & 40 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 16 & 0 \\ 15 & 40 & 0 & 16 & 134 & 22 \\ 3 & 6 & -1 & 0 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{131} 2 \ 3 \ 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 740, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 14 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 12 & 0 \\ 14 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & 0 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{141} 2 \ 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 910, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.65 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 19 & 3 \\ 4 & 10 & 1 & 4 & 52 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 20 & 0 \\ 19 & 52 & 0 & 20 & 226 & 28 \\ 3 & 6 & -1 & 0 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{141} 2 \ 3 \ 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 940, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.68 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 18 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 16 & 0 \\ 18 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & 0 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{131} 2 \ 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 980, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.61 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 16 & 3 \\ 4 & 10 & 1 & 4 & 40 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 16 & 0 \\ 16 & 40 & 0 & 16 & 134 & 22 \\ 3 & 6 & -1 & 0 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{141} 2 \ 3 \ 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 1420, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 20 & 3 \\ 4 & 10 & 1 & 4 & 52 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 20 & 0 \\ 20 & 52 & 0 & 20 & 226 & 28 \\ 3 & 6 & -1 & 0 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [63] \ 1^{111} 31 \ 5 \ 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 12, T_{\max} = 24, \Delta = -106, \det = 60, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.73 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 4 & 0 \\ 4 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

4.74 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{112} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 0 & 6 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 7 & 16 & 6 & 0 & 0 & 22 & 8 \\ 3 & 5 & -1 & 0 & -2 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 12 & 7 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 4 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -3 & 3 & 0 & 3 & -1 \\ 3 & 8 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 6 & 0 & 4 & 4 \\ 3 & 5 & -1 & 4 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 35 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 35 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -4 & 7 & 27 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1111} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -3 & 3 & 0 & 3 & -1 \\ 3 & 8 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 6 & 0 & 4 & 4 \\ 3 & 5 & -1 & 4 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{1111} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 4 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -3 & 3 & 0 & 3 & -1 \\ 4 & 8 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 6 & 0 & 4 & 4 \\ 3 & 5 & -1 & 4 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{111} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -618, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

15.0

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 26 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 26 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -4 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [9] \quad 1^{11} 3^{21} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [10] \quad 1^{11} 3^{11} 4^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -68, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 7 & 5 \\ 10 & 13 & 0 & 14 & 10 & 7 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 14 & 0 & 13 & 9 & 7 \\ 7 & 10 & 0 & 9 & 5 & 5 \\ 5 & 7 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

4.75 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 1^{111} 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 372, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 27 & 7 \\ 9 & 17 & 0 & 15 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 27 & 49 & 13 \\ 5 & 8 & -4 & 7 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

4.76 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -4 & 13 & 6 & 0 \end{pmatrix}_{b_0}$$

4.77 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{132} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$$

$$T_{\min} = 15.11 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 12 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 25 & 27 & 9 \\ 7 & 13 & 0 & 27 & 28 & 10 \\ 3 & 4 & -4 & 9 & 10 & -3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 21.15 \rightarrow 22, T_{\max} = 92, T \in [22, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & -1 & 0 & 8 & -9 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 & 3 \\ 3 & 4 & 0 & 10 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 10 & 0 & 22 & 12 & 8 \\ 3 & 5 & 0 & 12 & 4 & 4 \\ 3 & 4 & -4 & 8 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15.78 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 4 & 3 \\ 4 & 9 & 4 & 8 & 8 & 5 \\ 2 & 4 & -3 & 3 & 3 & -1 \\ 5 & 8 & 3 & 4 & 6 & 4 \\ 4 & 8 & 3 & 6 & 4 & 4 \\ 3 & 5 & -1 & 4 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & 0 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{144} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 42, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 11 & 3 \\ 3 & 4 & 0 & 16 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 46 & 60 & 12 \\ 11 & 21 & 0 & 60 & 76 & 16 \\ 3 & 4 & -4 & 12 & 16 & -2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{133} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 15 & 3 \\ 3 & 4 & 0 & 17 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 81 & 13 \\ 15 & 27 & 0 & 81 & 133 & 21 \\ 3 & 4 & -4 & 13 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -4 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$

- [9] $1^{121}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -4 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{124}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 17 & 3 \\ 3 & 4 & 0 & 8 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 42 & 6 \\ 17 & 31 & 0 & 42 & 172 & 24 \\ 3 & 4 & -4 & 6 & 24 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{131}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.47 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 3 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -4 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{1111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 8 & 5 \\ 3 & 4 & -3 & 3 & 3 & -1 \\ 3 & 8 & 3 & 4 & 6 & 4 \\ 3 & 8 & 3 & 6 & 4 & 4 \\ 3 & 5 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$
- [13] $21^{131}31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 8 & 5 \\ 3 & 4 & -3 & 3 & 3 & -1 \\ 3 & 8 & 3 & 4 & 6 & 4 \\ 3 & 8 & 3 & 6 & 4 & 4 \\ 3 & 5 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$
- [14] $1^{112}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & -1 & 0 & 8 & -2 \end{pmatrix}_{b_0}$$
- [15] $1^{113}31^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 10 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 10 & 0 & 22 & 30 & 8 \\ 8 & 14 & 0 & 30 & 37 & 11 \\ 3 & 4 & -4 & 8 & 11 & -1 \end{pmatrix}_{b_0}$$
- [16] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$
- [17] $1^{133}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 9 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -4 & 4 & 13 & 0 \end{pmatrix}_{b_0}$$
- [18] $1^{112}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 4 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$
- [19] $1^{113}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 15.23 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & 0 & 12 & -3 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{122} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 4 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & -1 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 6 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -4 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.49 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 8 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & -4 & 3 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{111} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 9 & 24 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 42 & 7 \\ 14 & 24 & 0 & 42 & 121 & 19 \\ 3 & 4 & -4 & 7 & 19 & -1 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{123} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 6 & 12 & 0 & 18 & 25 & 9 \\ 3 & 4 & -4 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{133} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 9 & 3 \\ 3 & 4 & 0 & 12 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 25 & 36 & 9 \\ 9 & 17 & 0 & 36 & 49 & 13 \\ 3 & 4 & -4 & 9 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.34 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 8 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -4 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{133} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 0 & 12 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 25 & 27 & 9 \\ 6 & 12 & 0 & 27 & 25 & 9 \\ 3 & 4 & -4 & 9 & 9 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{1111}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.17 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 4 & 3 \\ 4 & 9 & 4 & 8 & 8 & 5 \\ 2 & 4 & -3 & 3 & 3 & -1 \\ 3 & 8 & 3 & 4 & 6 & 4 \\ 4 & 8 & 3 & 6 & 4 & 4 \\ 3 & 5 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[31] $1^{124}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 8 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 24 & 6 \\ 8 & 16 & 0 & 24 & 46 & 12 \\ 3 & 4 & -4 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{12}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 7 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -4 & 4 & 10 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{111}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.28 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{114}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 190, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 15.48 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 16 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & 0 & 16 & -3 \end{pmatrix}_{b_0}$$

[35] $1^{112}3^11^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.44 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 7 & 3 \\ 3 & 4 & 0 & 14 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 33 & 11 \\ 7 & 13 & 0 & 33 & 28 & 10 \\ 3 & 4 & -4 & 11 & 10 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{123}3^11^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 8 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 30 & 6 \\ 12 & 22 & 0 & 30 & 85 & 17 \\ 3 & 4 & -4 & 6 & 17 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{134}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 12 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 25 & 36 & 9 \\ 8 & 16 & 0 & 36 & 46 & 12 \\ 3 & 4 & -4 & 9 & 12 & -1 \end{pmatrix}_{b_0}$$

[38] $1^{115}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.1$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 10 & 20 & 0 & 15 & 73 & 15 \\ 3 & 4 & -4 & 3 & 15 & -1 \end{pmatrix}_{b_0}$$

[39] $13^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 8 & 5 \\ 2 & 4 & -3 & 3 & 3 & -1 \\ 3 & 8 & 3 & 4 & 6 & 4 \\ 3 & 8 & 3 & 6 & 4 & 4 \\ 3 & 5 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[40] $1^{122} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -4 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 215, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 14.28 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & 0 & 12 & -2 \end{pmatrix}_{b_0}$$

[42] $1^{125} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 8 & 20 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 30 & 6 \\ 10 & 20 & 0 & 30 & 73 & 15 \\ 3 & 4 & -4 & 6 & 15 & -1 \end{pmatrix}_{b_0}$$

[43] $1^{144} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 3 \\ 3 & 4 & 0 & 16 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 46 & 48 & 12 \\ 8 & 16 & 0 & 48 & 46 & 12 \\ 3 & 4 & -4 & 12 & 12 & -1 \end{pmatrix}_{b_0}$$

[44] $1^{135} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 13.17 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 10 & 3 \\ 3 & 4 & 0 & 12 & 20 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 25 & 45 & 9 \\ 10 & 20 & 0 & 45 & 73 & 15 \\ 3 & 4 & -4 & 9 & 15 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 4 & 3 \\ 4 & 9 & 4 & 8 & 8 & 5 \\ 2 & 4 & -3 & 3 & 3 & -1 \\ 4 & 8 & 3 & 4 & 6 & 4 \\ 4 & 8 & 3 & 6 & 4 & 4 \\ 3 & 5 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{116} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.21 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 12 & 24 & 0 & 18 & 106 & 18 \\ 3 & 4 & -4 & 3 & 18 & -1 \end{pmatrix}_{b_0}$$

[47] $1^{113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.88 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{126} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.24 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 8 & 24 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 36 & 6 \\ 12 & 24 & 0 & 36 & 106 & 18 \\ 3 & 4 & -4 & 6 & 18 & -1 \end{pmatrix}_{b_0}$$

[49] $1^{11} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 5 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 8 & 14 & 0 & 15 & 37 & 11 \\ 3 & 4 & -4 & 4 & 11 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{145} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 10 & 3 \\ 3 & 4 & 0 & 16 & 20 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 46 & 60 & 12 \\ 10 & 20 & 0 & 60 & 73 & 15 \\ 3 & 4 & -4 & 12 & 15 & -1 \end{pmatrix}_{b_0}$$

[51] $1^{112} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 16 & 28 & 0 & 18 & 154 & 22 \\ 3 & 4 & -4 & 3 & 22 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.81 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & 0 & 16 & -2 \end{pmatrix}_{b_0}$$

[53] $1^{136} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 284, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.30 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 12 & 3 \\ 3 & 4 & 0 & 12 & 24 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 25 & 54 & 9 \\ 12 & 24 & 0 & 54 & 106 & 18 \\ 3 & 4 & -4 & 9 & 18 & -1 \end{pmatrix}_{b_0}$$

[54] $1^{155} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 10 & 3 \\ 3 & 4 & 0 & 20 & 20 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 10 & 20 & 0 & 73 & 75 & 15 \\ 10 & 20 & 0 & 75 & 73 & 15 \\ 3 & 4 & -4 & 15 & 15 & -1 \end{pmatrix}_{b_0}$$

[55] $1^{117} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 14 & 28 & 0 & 21 & 145 & 21 \\ 3 & 4 & -4 & 3 & 21 & -1 \end{pmatrix}_{b_0}$$

[56] $1^{146} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 4 & 0 & 16 & 24 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 46 & 72 & 12 \\ 12 & 24 & 0 & 72 & 106 & 18 \\ 3 & 4 & -4 & 12 & 18 & -1 \end{pmatrix}_{b_0}$$

[57] $1^{127} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 8 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 42 & 6 \\ 14 & 28 & 0 & 42 & 145 & 21 \\ 3 & 4 & -4 & 6 & 21 & -1 \end{pmatrix}_{b_0}$$

[58] $1^{132} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 318, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 10 & 3 \\ 3 & 4 & 0 & 17 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 54 & 13 \\ 10 & 18 & 0 & 54 & 58 & 14 \\ 3 & 4 & -4 & 13 & 14 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{134} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 329, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 11 & 3 \\ 3 & 4 & 0 & 12 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 25 & 45 & 9 \\ 11 & 21 & 0 & 45 & 76 & 16 \\ 3 & 4 & -4 & 9 & 16 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{11} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.71 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 19 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 21 & 4 \\ 11 & 19 & 0 & 21 & 73 & 15 \\ 3 & 4 & -4 & 4 & 15 & 0 \end{pmatrix}_{b_0}$$

[61] $1^{11} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 19 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 33 & 7 \\ 11 & 19 & 0 & 33 & 73 & 15 \\ 3 & 4 & -4 & 7 & 15 & 0 \end{pmatrix}_{b_0}$$

[62] $1^{137} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 14 & 3 \\ 3 & 4 & 0 & 12 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 25 & 63 & 9 \\ 14 & 28 & 0 & 63 & 145 & 21 \\ 3 & 4 & -4 & 9 & 21 & -1 \end{pmatrix}_{b_0}$$

[63] $1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 337, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.32 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{156} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 348, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.4$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 12 & 3 \\ 3 & 4 & 0 & 20 & 24 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 10 & 20 & 0 & 73 & 90 & 15 \\ 12 & 24 & 0 & 90 & 106 & 18 \\ 3 & 4 & -4 & 15 & 18 & -1 \end{pmatrix}_{b_0}$$

[65] $1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

[66] $1^{122} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 13 & 3 \\ 3 & 4 & 0 & 13 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 54 & 10 \\ 13 & 23 & 0 & 54 & 100 & 18 \\ 3 & 4 & -4 & 10 & 18 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{147} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 14 & 3 \\ 3 & 4 & 0 & 16 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 46 & 84 & 12 \\ 14 & 28 & 0 & 84 & 145 & 21 \\ 3 & 4 & -4 & 12 & 21 & -1 \end{pmatrix}_{b_0}$$

[68] $1^{11} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 5 & 24 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 27 & 4 \\ 14 & 24 & 0 & 27 & 121 & 19 \\ 3 & 4 & -4 & 4 & 19 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{143} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 12 & 3 \\ 3 & 4 & 0 & 21 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 0 & 76 & 81 & 16 \\ 12 & 22 & 0 & 81 & 85 & 17 \\ 3 & 4 & -4 & 16 & 17 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{123} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 24 & 6 \\ 9 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{114} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.97 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & 0 & 16 & -1 \end{pmatrix}_{b_0}$$

[71] $1^{166} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 12 & 3 \\ 3 & 4 & 0 & 24 & 24 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 12 & 24 & 0 & 106 & 108 & 18 \\ 12 & 24 & 0 & 108 & 106 & 18 \\ 3 & 4 & -4 & 18 & 18 & -1 \end{pmatrix}_{b_0}$$

[74] $1^{113} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 4 & 0 & 14 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 57 & 11 \\ 12 & 22 & 0 & 57 & 85 & 17 \\ 3 & 4 & -4 & 11 & 17 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{157} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 14 & 3 \\ 3 & 4 & 0 & 20 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 10 & 20 & 0 & 73 & 105 & 15 \\ 14 & 28 & 0 & 105 & 145 & 21 \\ 3 & 4 & -4 & 15 & 21 & -1 \end{pmatrix}_{b_0}$$

[75] $1^{111} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 405, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 9 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 24 & 7 \\ 8 & 14 & 0 & 24 & 37 & 11 \\ 3 & 4 & -4 & 7 & 11 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{112} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 409, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 13 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & -4 & 3 & 18 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{12} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 414, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 5 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 10 & 18 & 0 & 18 & 58 & 14 \\ 3 & 4 & -4 & 4 & 14 & 0 \end{pmatrix}_{b_0}$$

[78] $1^{114} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 33 & 7 \\ 11 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -4 & 7 & 16 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{114} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 422, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.93 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 14 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & 0 & 16 & -1 \end{pmatrix}_{b_0}$$

[80] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 10 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

[81] $1^{167}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 444, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 14 & 3 \\ 3 & 4 & 0 & 24 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 12 & 24 & 0 & 106 & 126 & 18 \\ 14 & 28 & 0 & 126 & 145 & 21 \\ 3 & 4 & -4 & 18 & 21 & -1 \end{pmatrix}_{b_0}$$

[82] $1^{112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 459, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.34 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 21 & 7 \\ 7 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

[83] $1^{14}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 466, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 5 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 14 & 26 & 0 & 24 & 118 & 20 \\ 3 & 4 & -4 & 4 & 20 & 0 \end{pmatrix}_{b_0}$$

[84] $1^{113}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 27 & 7 \\ 9 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

[85] $1^{13}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 481, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 5 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 21 & 4 \\ 12 & 22 & 0 & 21 & 85 & 17 \\ 3 & 4 & -4 & 4 & 17 & 0 \end{pmatrix}_{b_0}$$

[86] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 493, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -4 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

[87] $1^{177}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 496, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.60 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 14 & 3 \\ 3 & 4 & 0 & 28 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 14 & 28 & 0 & 145 & 147 & 21 \\ 14 & 28 & 0 & 147 & 145 & 21 \\ 3 & 4 & -4 & 21 & 21 & -1 \end{pmatrix}_{b_0}$$

[88] $1^{124}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 496, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 8 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 36 & 6 \\ 14 & 26 & 0 & 36 & 118 & 20 \\ 3 & 4 & -4 & 6 & 20 & 0 \end{pmatrix}_{b_0}$$

[89] $1^{124}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 534, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 0 & 8 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 30 & 6 \\ 11 & 21 & 0 & 30 & 76 & 16 \\ 3 & 4 & -4 & 6 & 16 & 0 \end{pmatrix}_{b_0}$$

[90] $1^{111} 31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 3 \\ 3 & 4 & 0 & 14 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 39 & 11 \\ 8 & 14 & 0 & 39 & 37 & 11 \\ 3 & 4 & -4 & 11 & 11 & 0 \end{pmatrix}_{b_0}$$

[91] $1^{12} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 616, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 13 & 3 \\ 3 & 4 & 0 & 5 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 13 & 23 & 0 & 24 & 100 & 18 \\ 3 & 4 & -4 & 4 & 18 & 0 \end{pmatrix}_{b_0}$$

[92] $1^{112} 31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 648, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 10 & 3 \\ 3 & 4 & 0 & 14 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 48 & 11 \\ 10 & 18 & 0 & 48 & 58 & 14 \\ 3 & 4 & -4 & 11 & 14 & 0 \end{pmatrix}_{b_0}$$

[93] $1^{122} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 652, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 30 & 10 \\ 7 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

[94] $1^{114} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 657, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 11 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -4 & 3 & 16 & 0 \end{pmatrix}_{b_0}$$

[95] $1^{123} 31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 697, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 18 & 3 \\ 3 & 4 & 0 & 13 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 75 & 10 \\ 18 & 32 & 0 & 75 & 193 & 25 \\ 3 & 4 & -4 & 10 & 25 & 0 \end{pmatrix}_{b_0}$$

[96] $1^{122} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 702, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 0 & 13 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 42 & 10 \\ 10 & 18 & 0 & 42 & 58 & 14 \\ 3 & 4 & -4 & 10 & 14 & 0 \end{pmatrix}_{b_0}$$

[97] $1^{113} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 12 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

[98] $1^{112} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 734, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 9 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 30 & 7 \\ 10 & 18 & 0 & 30 & 58 & 14 \\ 3 & 4 & -4 & 7 & 14 & 0 \end{pmatrix}_{b_0}$$

[99] $1^{123} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 763, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 39 & 10 \\ 9 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -4 & 10 & 13 & 0 \end{pmatrix}_{b_0}$$

[100] $1^{12}3^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 790$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 16 & 3 \\ 3 & 4 & 0 & 5 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 16 & 28 & 0 & 30 & 154 & 22 \\ 3 & 4 & -4 & 4 & 22 & 0 \end{pmatrix}_{b_0}$$

[101] $1^{124}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 792$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 0 & 13 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 48 & 10 \\ 11 & 21 & 0 & 48 & 76 & 16 \\ 3 & 4 & -4 & 10 & 16 & 0 \end{pmatrix}_{b_0}$$

[102] $1^{13}3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 819$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 12.42 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 3 \\ 3 & 4 & 0 & 5 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 27 & 4 \\ 15 & 27 & 0 & 27 & 133 & 21 \\ 3 & 4 & -4 & 4 & 21 & 0 \end{pmatrix}_{b_0}$$

[103] $1^{113}3^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 832$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 12.49 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 18 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & -4 & 3 & 25 & 0 \end{pmatrix}_{b_0}$$

[104] $1^{113}3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 837$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 12.41 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 15 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & -4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

[105] $21^{111}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 884$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 6 & 4 \\ 8 & 9 & 4 & 8 & 8 & 5 \\ 3 & 4 & -3 & 3 & 3 & -1 \\ 6 & 8 & 3 & 4 & 6 & 4 \\ 6 & 8 & 3 & 6 & 4 & 4 \\ 4 & 5 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[106] $1^{112}3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 891$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.44 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 9 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 39 & 7 \\ 13 & 23 & 0 & 39 & 100 & 18 \\ 3 & 4 & -4 & 7 & 18 & 0 \end{pmatrix}_{b_0}$$

[107] $1^{133}3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 925$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.6$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 12 & 3 \\ 3 & 4 & 0 & 17 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 66 & 13 \\ 12 & 22 & 0 & 66 & 85 & 17 \\ 3 & 4 & -4 & 13 & 17 & 0 \end{pmatrix}_{b_0}$$

[108] $1^{112}3^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 930$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.54 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 16 & 3 \\ 3 & 4 & 0 & 9 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 48 & 7 \\ 16 & 28 & 0 & 48 & 154 & 22 \\ 3 & 4 & -4 & 7 & 22 & 0 \end{pmatrix}_{b_0}$$

[109] $1^{14}3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 940$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 3 \\ 3 & 4 & 0 & 5 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 17 & 31 & 0 & 30 & 172 & 24 \\ 3 & 4 & -4 & 4 & 24 & 0 \end{pmatrix}_{b_0}$$

[110] $1^{134}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 964$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.6$
 $T_{\min} = 12.39 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 0 & 17 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 51 & 13 \\ 9 & 17 & 0 & 51 & 49 & 13 \\ 3 & 4 & -4 & 13 & 13 & 0 \end{pmatrix}_{b_0}$$

[111] $1^{134}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 979$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.39 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 17 & 3 \\ 3 & 4 & 0 & 17 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 93 & 13 \\ 17 & 31 & 0 & 93 & 172 & 24 \\ 3 & 4 & -4 & 13 & 24 & 0 \end{pmatrix}_{b_0}$$

[112] $1^{113}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 981$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 9 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 36 & 7 \\ 12 & 22 & 0 & 36 & 85 & 17 \\ 3 & 4 & -4 & 7 & 17 & 0 \end{pmatrix}_{b_0}$$

[113] $1^{124}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 984$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 14 & 3 \\ 3 & 4 & 0 & 18 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 0 & 58 & 84 & 14 \\ 14 & 26 & 0 & 84 & 118 & 20 \\ 3 & 4 & -4 & 14 & 20 & 0 \end{pmatrix}_{b_0}$$

[114] $1^{114}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 1024$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 14 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & -4 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

[115] $1^{144}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 1074$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 14 & 3 \\ 3 & 4 & 0 & 21 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 0 & 76 & 96 & 16 \\ 14 & 26 & 0 & 96 & 118 & 20 \\ 3 & 4 & -4 & 16 & 20 & 0 \end{pmatrix}_{b_0}$$

[116] $1^{134}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 1083$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.41 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 11 & 3 \\ 3 & 4 & 0 & 17 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 63 & 13 \\ 11 & 21 & 0 & 63 & 76 & 16 \\ 3 & 4 & -4 & 13 & 16 & 0 \end{pmatrix}_{b_0}$$

[117] $1^{123}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 1107$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.3$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 15 & 3 \\ 3 & 4 & 0 & 13 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 63 & 10 \\ 15 & 27 & 0 & 63 & 133 & 21 \\ 3 & 4 & -4 & 10 & 21 & 0 \end{pmatrix}_{b_0}$$

[118] $1^{122}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 1112$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 10 & 3 \\ 3 & 4 & 0 & 18 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 0 & 58 & 60 & 14 \\ 10 & 18 & 0 & 60 & 58 & 14 \\ 3 & 4 & -4 & 14 & 14 & 0 \end{pmatrix}_{b_0}$$

[119] $1^{123}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$

$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 12 & 3 \\ 3 & 4 & 0 & 13 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 51 & 10 \\ 12 & 22 & 0 & 51 & 85 & 17 \\ 3 & 4 & -4 & 10 & 17 & 0 \end{pmatrix}_{b_0}$$

[120] $1^{13}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 18 & 3 \\ 3 & 4 & 0 & 5 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 33 & 4 \\ 18 & 32 & 0 & 33 & 193 & 25 \\ 3 & 4 & -4 & 4 & 25 & 0 \end{pmatrix}_{b_0}$$

[121] $1^{114}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 9 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 42 & 7 \\ 14 & 26 & 0 & 42 & 118 & 20 \\ 3 & 4 & -4 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

[122] $1^{123}31^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$

$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 12 & 3 \\ 3 & 4 & 0 & 18 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 0 & 58 & 72 & 14 \\ 12 & 22 & 0 & 72 & 85 & 17 \\ 3 & 4 & -4 & 14 & 17 & 0 \end{pmatrix}_{b_0}$$

[123] $1^{114}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1273, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 17 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

[124] $1^{144}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 11 & 3 \\ 3 & 4 & 0 & 21 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 0 & 76 & 78 & 16 \\ 11 & 21 & 0 & 78 & 76 & 16 \\ 3 & 4 & -4 & 16 & 16 & 0 \end{pmatrix}_{b_0}$$

[125] $1^{113}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 4 & 0 & 9 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 45 & 7 \\ 15 & 27 & 0 & 45 & 133 & 21 \\ 3 & 4 & -4 & 7 & 21 & 0 \end{pmatrix}_{b_0}$$

[126] $1^{14}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1386, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 20 & 3 \\ 3 & 4 & 0 & 5 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 36 & 4 \\ 20 & 36 & 0 & 36 & 238 & 28 \\ 3 & 4 & -4 & 4 & 28 & 0 \end{pmatrix}_{b_0}$$

[127] $1^{114}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 24 & 3 \\ 20 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & -4 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

[128] $1^{134}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 14 & 3 \\ 3 & 4 & 0 & 17 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 78 & 13 \\ 14 & 26 & 0 & 78 & 118 & 20 \\ 3 & 4 & -4 & 13 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{124} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1474, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 14 & 3 \\ 3 & 4 & 0 & 13 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 60 & 10 \\ 14 & 26 & 0 & 60 & 118 & 20 \\ 3 & 4 & -4 & 10 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[130] \ 1^{113} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.44 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 18 & 3 \\ 3 & 4 & 0 & 9 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 54 & 7 \\ 18 & 32 & 0 & 54 & 193 & 25 \\ 3 & 4 & -4 & 7 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{124} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1674, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 20 & 3 \\ 3 & 4 & 0 & 13 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 84 & 10 \\ 20 & 36 & 0 & 84 & 238 & 28 \\ 3 & 4 & -4 & 10 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{133} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1696, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 12 & 3 \\ 3 & 4 & 0 & 22 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 0 & 85 & 87 & 17 \\ 12 & 22 & 0 & 87 & 85 & 17 \\ 3 & 4 & -4 & 17 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{114} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1755, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 17 & 3 \\ 3 & 4 & 0 & 9 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 51 & 7 \\ 17 & 31 & 0 & 51 & 172 & 24 \\ 3 & 4 & -4 & 7 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{124} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 17 & 3 \\ 3 & 4 & 0 & 13 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 72 & 10 \\ 17 & 31 & 0 & 72 & 172 & 24 \\ 3 & 4 & -4 & 10 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[135] \ 1^{134} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1816, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 14 & 3 \\ 3 & 4 & 0 & 22 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 0 & 85 & 102 & 17 \\ 14 & 26 & 0 & 102 & 118 & 20 \\ 3 & 4 & -4 & 17 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[136] \ 1^{114} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 2246, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 20 & 3 \\ 3 & 4 & 0 & 9 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 60 & 7 \\ 20 & 36 & 0 & 60 & 238 & 28 \\ 3 & 4 & -4 & 7 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{144} 31^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 2296, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.39 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 14 & 3 \\ 3 & 4 & 0 & 26 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 14 & 26 & 0 & 118 & 120 & 20 \\ 14 & 26 & 0 & 120 & 118 & 20 \\ 3 & 4 & -4 & 20 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [138] \ 1^{111} 31^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -8, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 19.50 \rightarrow 20, \ T_{\max} = 23, \ T \in [20, 23]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 1 & 3 & 3 & 0 & 3 & 1 \\ 2 & 3 & 3 & -4 & 3 & -7 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{111} 31^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -11, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 20.45 \rightarrow 21, \ T_{\max} = 23, \ T \in [21, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 & 3 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 6 & 0 & 13 & 6 & 7 \\ 1 & 3 & 3 & 0 & 6 & 1 & 3 \\ 2 & 3 & 2 & -4 & 7 & 3 & -8 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{111} 321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -19, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.16 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 12 & 7 & 0 & 14 & 8 \\ 3 & 5 & 0 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[141] \ 1^{111} 321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -24, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 0 & -1 \\ 2 & 1 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{111} 31^{11} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -29, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.41 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 6 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & -4 & 3 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[143] \ 1^{133} 321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -30, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.6$$

$$T_{\min} = 13.20 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 6 & -1 \\ 3 & 6 & 0 & 3 & 16 & 21 & 6 \\ 4 & 7 & 0 & 6 & 21 & 25 & 9 \\ 2 & 2 & -1 & -1 & 6 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[144] \ 1^{111} 321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -31, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 15.52 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[146] \ 1^{114} 31^{12} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.75 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 24 & 4 \\ 1 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 18 & 3 \\ 8 & 24 & 18 & 0 & 18 & 118 & 20 \\ 2 & 4 & 2 & -4 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

[145] $1^{122}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 8 & 2 \\ 2 & 4 & 4 & 0 & 12 & 24 & 4 \\ 1 & 4 & 0 & 0 & 9 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 9 & 0 & 28 & 54 & 10 \\ 8 & 24 & 15 & 0 & 54 & 100 & 18 \\ 2 & 4 & 2 & -4 & 10 & 18 & -2 \end{pmatrix}_{b_0}$$

[147] $1^{111}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

[148] $1^{111}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 14.44 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

[149] $1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 1 & -1 \\ 2 & 3 & 0 & 4 & 5 & 3 & 5 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -1 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

[150] $1^{121}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -1 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[151] $1^{124}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 7 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -1 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

[152] $1^{111}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[153] $1^{122}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 5 & -1 \\ 2 & 4 & 0 & 2 & 6 & 10 & 4 \\ 3 & 5 & 0 & 5 & 10 & 13 & 7 \\ 2 & 2 & -1 & -1 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

[154] $1^{14}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 0 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 7 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 7 & 0 & 41 & 11 \\ 2 & 2 & -1 & -1 & 0 & 11 & 0 \end{pmatrix}_{b_0}$$

[155] $1^{112}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 9 & 3 \\ 4 & 12 & 9 & 0 & 9 & 28 & 10 \\ 2 & 4 & 2 & -4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

[156] $1^{13} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -87$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 12.52 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 16 & 6 \\ 2 & 2 & -1 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

[157] $1^{111} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -88$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.32 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[158] $1^{112} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[159] $1^{111} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 4 & 3 & 0 & 4 & 3 & 5 & 5 \\ 2 & 2 & -1 & -1 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

[160] $1^{122} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -100$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 2 & 4 & 0 & 2 & 6 & 8 & 4 \\ 2 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[162] $1^{113} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -104$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 3 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[161] $1^{111} 31^{14} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -104$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 15 & 3 \\ 9 & 27 & 15 & 0 & 15 & 121 & 19 \\ 2 & 4 & 2 & -4 & 3 & 19 & -2 \end{pmatrix}_{b_0}$$

[163] $1^{123} 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, T

11.0

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[165] $1^{133} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.6$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 16 & 18 & 6 \\ 4 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[166] $1^{114} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 4 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[167] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 3 & 3 & 0 & 4 & 3 & 5 & 5 \\ 2 & 2 & -1 & -1 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

[168] $1^{133} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.6$

$$T_{\min} = 13.30 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 16 & 18 & 6 \\ 3 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[169] $1^{124} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 4 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -1 & 4 & 8 & -1 \end{pmatrix}_{b_0}$$

[171] $1^{14} 31^{113} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 10 & 2 \\ 2 & 4 & 4 & 0 & 6 & 30 & 4 \\ 1 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 30 & 4 \\ 10 & 30 & 21 & 0 & 30 & 172 & 24 \\ 2 & 4 & 2 & -4 & 4 & 24 & 0 \end{pmatrix}_{b_0}$$

[170] $1^{1111} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 13.31 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 6 & 3 \\ 1 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 6 & 0 & 13 & 15 & 7 \\ 3 & 6 & 6 & 0 & 15 & 13 & 7 \\ 2 & 3 & 2 & -4 & 7 & 7 & -1 \end{pmatrix}_{b_0}$$

[172] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 2 & 2 & 0 & 1 & 0 & 3 & 2 \\ 3 & 3 & 0 & 4 & 3 & 5 & 5 \\ 2 & 2 & -1 & -1 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

[173] $1^{113}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -131$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 14.05 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 3 & 12 & 3 \\ 2 & 4 & 3 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 12 & 3 \\ 5 & 12 & 15 & 0 & 12 & 49 & 13 \\ 2 & 3 & 3 & -4 & 3 & 13 & -2 \end{pmatrix}_{b_0}$$

[174] $1^{134}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -134$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.58 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 4 & -1 \\ 3 & 6 & 0 & 3 & 16 & 24 & 6 \\ 4 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -1 & 6 & 8 & -1 \end{pmatrix}_{b_0}$$

[175] $1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -135$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

[176] $1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -135$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 4 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 3 & 0 & 4 & 0 & 5 & 5 \\ 2 & 2 & -1 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$

[177] $1^{144}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 7 & -1 \\ 5 & 8 & 0 & 4 & 30 & 36 & 8 \\ 6 & 9 & 0 & 7 & 36 & 41 & 11 \\ 2 & 2 & -1 & -1 & 8 & 11 & 0 \end{pmatrix}_{b_0}$$

[178] $1^{115}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.1$

$T_{\min} = 13.62 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 5 & 10 & 0 & 5 & 10 & 48 & 10 \\ 2 & 2 & -1 & -1 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[179] $1^{134}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 7 & -1 \\ 3 & 6 & 0 & 3 & 16 & 27 & 6 \\ 5 & 9 & 0 & 7 & 27 & 41 & 11 \\ 2 & 2 & -1 & -1 & 6 & 11 & 0 \end{pmatrix}_{b_0}$$

[180] $1^{125}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -142$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.6$

$T_{\min} = 13.72 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 5 & -1 \\ 2 & 4 & 0 & 2 & 6 & 20 & 4 \\ 5 & 10 & 0 & 5 & 20 & 48 & 10 \\ 2 & 2 & -1 & -1 & 4 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[181] \ 1^{12} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.44 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 2 & 0 & 6 & 4 \\ 2 & 2 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[182] \ 1^{112} 3^{21} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -148, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.46 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 3 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & -1 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[183] \ 1^{144} 3^{21} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -148, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.81 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 4 & -1 \\ 4 & 8 & 0 & 4 & 30 & 32 & 8 \\ 4 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[184] \ 1^{111} 3^{21} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -149, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[185] \ 1^{123} 3^{21} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -150, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.3$$

$$T_{\min} = 12.51 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 5 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -1 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[186] \ 1^{135} 3^{21} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -152, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.9$$

$$T_{\min} = 13.87 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 5 & -1 \\ 3 & 6 & 0 & 3 & 16 & 30 & 6 \\ 5 & 10 & 0 & 5 & 30 & 48 & 10 \\ 2 & 2 & -1 & -1 & 6 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[187] \ 1^{116} 3^{21} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -158, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.95 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 12 & 2 \\ 6 & 12 & 0 & 6 & 12 & 70 & 12 \\ 2 & 2 & -1 & -1 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[188] \ 1^{13} 3^{21} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -159, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.43 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 0 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 7 & 0 & 6 & 0 & 25 & 9 \\ 2 & 2 & -1 & -1 & 0 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[191] \ 1^{122} 3^{21} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -164, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.51 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 2 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[189] $1^{1111}3131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -164$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 13.32 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 3 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 & 3 \\ 1 & 3 & 3 & 0 & 3 & 1 & 3 \\ 2 & 3 & 2 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[190] $1^{126}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -164$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.02 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 6 & -1 \\ 2 & 4 & 0 & 2 & 6 & 24 & 4 \\ 6 & 12 & 0 & 6 & 24 & 70 & 12 \\ 2 & 2 & -1 & -1 & 4 & 12 & -2 \end{pmatrix}_{b_0}$$

[192] $1^{145}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -166$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.1$

$T_{\min} = 14.05 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 5 & -1 \\ 4 & 8 & 0 & 4 & 30 & 40 & 8 \\ 5 & 10 & 0 & 5 & 40 & 48 & 10 \\ 2 & 2 & -1 & -1 & 8 & 10 & -2 \end{pmatrix}_{b_0}$$

[193] $1^{123}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -170$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.3$

$T_{\min} = 12.56 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 6 & -1 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 4 & 7 & 0 & 6 & 14 & 25 & 9 \\ 2 & 2 & -1 & -1 & 4 & 9 & 0 \end{pmatrix}_{b_0}$$

[195] $1^{136}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -174$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.1$

$T_{\min} = 14.14 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 6 & -1 \\ 3 & 6 & 0 & 3 & 16 & 36 & 6 \\ 6 & 12 & 0 & 6 & 36 & 70 & 12 \\ 2 & 2 & -1 & -1 & 6 & 12 & -2 \end{pmatrix}_{b_0}$$

[194] $1^{123}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -174$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.3$

$T_{\min} = 12.48 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 6 & -1 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 5 & 7 & 0 & 6 & 14 & 25 & 9 \\ 2 & 2 & -1 & -1 & 4 & 9 & 0 \end{pmatrix}_{b_0}$$

[196] $1^{111}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -179$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 6 & 3 \\ 3 & 6 & 9 & 0 & 6 & 13 & 7 \\ 2 & 3 & 3 & -4 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

[197] $1^{12}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -180$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$

$T_{\min} = 12.42 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 5 & 0 & 5 & 0 & 13 & 7 \\ 2 & 2 & -1 & -1 & 0 & 7 & 0 \end{pmatrix}_{b_0}$$

[198] $1^{122}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -184$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

12.0

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 5 & -1 \\ 3 & 4 & 0 & 2 & 6 & 10 & 4 \\ 4 & 5 & 0 & 5 & 10 & 13 & 7 \\ 2 & 2 & -1 & -1 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[199] \ 1^{117} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.3$$

$$T_{\min} = 14.24 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 14 & 2 \\ 7 & 14 & 0 & 7 & 14 & 96 & 14 \\ 2 & 2 & -1 & -1 & 2 & 14 & -2 \end{pmatrix}_{b_0}$$

$$[200] \ 1^{111} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -185, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 6 & 3 \\ 3 & 9 & 6 & 0 & 6 & 13 & 7 \\ 2 & 4 & 2 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[201] \ 1^{133} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.6$$

$$T_{\min} = 12.52 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 6 & -1 \\ 4 & 6 & 0 & 3 & 16 & 21 & 6 \\ 5 & 7 & 0 & 6 & 21 & 25 & 9 \\ 2 & 2 & -1 & -1 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[202] \ 1^{127} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 2 & 4 & 0 & 2 & 6 & 28 & 4 \\ 7 & 14 & 0 & 7 & 28 & 96 & 14 \\ 2 & 2 & -1 & -1 & 4 & 14 & -2 \end{pmatrix}_{b_0}$$

$$[203] \ 1^{124} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 7 & 9 & 0 & 7 & 18 & 41 & 11 \\ 2 & 2 & -1 & -1 & 4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[204] \ 1^{137} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.1$$

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 7 & -1 \\ 3 & 6 & 0 & 3 & 16 & 42 & 6 \\ 7 & 14 & 0 & 7 & 42 & 96 & 14 \\ 2 & 2 & -1 & -1 & 6 & 14 & -2 \end{pmatrix}_{b_0}$$

$$[205] \ 1^{11} 31^{112} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 4 & 2 \\ 2 & 4 & 4 & 0 & 9 & 12 & 4 \\ 1 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 18 & 7 \\ 4 & 12 & 6 & 0 & 18 & 22 & 8 \\ 2 & 4 & 2 & -4 & 7 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[206] \ 1^{112} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13.11 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 6 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[207] $1^{112}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -233$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 3 & 9 & 3 \\ 2 & 4 & 3 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 9 & 3 \\ 4 & 9 & 12 & 0 & 9 & 28 & 10 \\ 2 & 3 & 3 & -4 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

[208] $1^{134}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -239$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 4 & -1 \\ 3 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -1 & 6 & 8 & 0 \end{pmatrix}_{b_0}$$

[209] $1^{113}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -254$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 12.52 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 4 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & -1 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

[210] $1^{111}31^{13}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -257$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 13.20 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 12 & 3 \\ 7 & 21 & 12 & 0 & 12 & 73 & 15 \\ 2 & 4 & 2 & -4 & 3 & 15 & -1 \end{pmatrix}_{b_0}$$

[211] $1^{112}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -263$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$

$T_{\min} = 12.44 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[212] $1^{111}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -280$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 12.44 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 2 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[213] $1^{124}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -280$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 5 & 9 & 0 & 7 & 18 & 41 & 11 \\ 2 & 2 & -1 & -1 & 4 & 11 & 0 \end{pmatrix}_{b_0}$$

[214] $1^{123}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -282$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.3$

$T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -1 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

[215] $1^{111}31^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -284$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$

$T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 9 & 3 \\ 5 & 15 & 9 & 0 & 9 & 37 & 11 \\ 2 & 4 & 2 & -4 & 3 & 11 & 0 \end{pmatrix}_{b_0}$$

[216] $1^{112}3^{111}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -305$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$

$T_{\min} = 12.42 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 2 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & -1 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[217] $1^{112}3^{114}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -308$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$

$T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 7 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & -1 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[218] $1^{114}3^{111}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -319$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 2 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[219] $1^{112}3^{113}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -319$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$

$T_{\min} = 12.64 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 5 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[220] $1^{112}3^{111}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -329$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$

$T_{\min} = 12.42 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & -1 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[221] $1^{112}3^{112}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -338$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$

$T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 4 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[222] $1^{112}3^{111}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -359$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$

$T_{\min} = 12.46 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[223] $1^{114} 3^{21} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -364$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 5 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & -1 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

[224] $1^{131} 1^{111} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -368$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 6 & 4 \\ 2 & 6 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & 2 & -4 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[225] $1^{113} 3^{21} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -372$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 12.49 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 2 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[227] $1^{113} 3^1 1^{231} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -380$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$

$T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 15 & 3 \\ 7 & 21 & 15 & 0 & 15 & 85 & 17 \\ 2 & 4 & 2 & -4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

[226] $1^{124} 3^{21} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -380$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.54 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 6 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -1 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

[228] $1^{113} 3^{21} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -381$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 12.51 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[229] $1^{113} 1^{111} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -389$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 9 & 4 \\ 3 & 9 & 6 & 0 & 9 & 13 & 7 \\ 2 & 4 & 2 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

[230] $1^{124} 3^{21} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -392$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.51 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 6 & 9 & 0 & 7 & 18 & 41 & 11 \\ 2 & 2 & -1 & -1 & 4 & 11 & 0 \end{pmatrix}_{b_0}$$

[231] $1^{114}31^{112}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 8 & 2 \\ 2 & 4 & 4 & 0 & 9 & 24 & 4 \\ 1 & 4 & 0 & 0 & 6 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 42 & 7 \\ 8 & 24 & 18 & 0 & 42 & 118 & 20 \\ 2 & 4 & 2 & -4 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

[232] $1^{124}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -1 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

[233] $1^{112}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -409, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 6 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & -1 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[234] $1^{113}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -414, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 2 & 2 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & -1 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

[235] $1^{112}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 5 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & -1 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[236] $1^{123}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -423, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.3$

$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 6 & -1 \\ 3 & 4 & 0 & 2 & 6 & 14 & 4 \\ 5 & 7 & 0 & 6 & 14 & 25 & 9 \\ 2 & 2 & -1 & -1 & 4 & 9 & 0 \end{pmatrix}_{b_0}$$

[237] $1^{121}31^{111}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -437, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 3 & 2 \\ 2 & 4 & 4 & 0 & 12 & 9 & 4 \\ 1 & 4 & 0 & 0 & 9 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 9 & 0 & 28 & 21 & 10 \\ 3 & 9 & 6 & 0 & 21 & 13 & 7 \\ 2 & 4 & 2 & -4 & 10 & 7 & 0 \end{pmatrix}_{b_0}$$

[238] $1^{114}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -457, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 2 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & -1 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

[239] $1^{112}31^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 18 & 4 \\ 1 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 12 & 3 \\ 6 & 18 & 12 & 0 & 12 & 58 & 14 \\ 2 & 4 & 2 & -4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[240] \ 1^{122} 31^{111} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 9 & 0 & 28 & 30 & 10 \\ 4 & 12 & 9 & 0 & 30 & 28 & 10 \\ 2 & 4 & 2 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[241] \ 1^{134} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -489, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.49 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 7 & -1 \\ 4 & 6 & 0 & 3 & 16 & 27 & 6 \\ 6 & 9 & 0 & 7 & 27 & 41 & 11 \\ 2 & 2 & -1 & -1 & 6 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[242] \ 1^{114} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -503, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[243] \ 1^{122} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -508, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 3 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[244] \ 1^{113} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -543, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.44 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & -1 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[245] \ 1^{113} 3 \ 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 5 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[246] \ 1^1 31^{112} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 4 & 0 & 6 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 12 & 4 \\ 4 & 12 & 6 & 0 & 12 & 22 & 8 \\ 2 & 4 & 2 & -4 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[247] \ 1^{123} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -591, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.3$$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 3 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -1 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

[248] $1^{124} 1^{111} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -596, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$

$T_{\min} = 12.46 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 3 & 4 & 0 & 2 & 6 & 18 & 4 \\ 6 & 9 & 0 & 7 & 18 & 41 & 11 \\ 2 & 2 & -1 & -1 & 4 & 11 & 0 \end{pmatrix}_{b_0}$$

[249] $1^{124} 1^{111} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -608, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$

$T_{\min} = 12.55 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 3 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -1 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

[250] $1^{113} 1^{14} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -648, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$

$T_{\min} = 12.64 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 7 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[251] $1^{113} 1^{13} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -653, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$

$T_{\min} = 12.54 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 6 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[252] $1^{123} 1^{112} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -668, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$

$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 6 & 2 \\ 2 & 4 & 4 & 0 & 6 & 18 & 4 \\ 1 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 18 & 4 \\ 6 & 18 & 12 & 0 & 18 & 58 & 14 \\ 2 & 4 & 2 & -4 & 4 & 14 & 0 \end{pmatrix}_{b_0}$$

[253] $1^{111} 3 1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -680, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$

$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 15 & 7 \\ 3 & 9 & 6 & 0 & 15 & 13 & 7 \\ 2 & 4 & 2 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

[254] $1^{11} 3 1^{122} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -704, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$

$T_{\min} = 12.90 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 5 & 2 \\ 2 & 4 & 4 & 0 & 12 & 15 & 4 \\ 1 & 4 & 0 & 0 & 6 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 6 & 0 & 22 & 30 & 8 \\ 5 & 15 & 9 & 0 & 30 & 37 & 11 \\ 2 & 4 & 2 & -4 & 8 & 11 & 0 \end{pmatrix}_{b_0}$$

[255] $1^{122} 3 1^{112} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -716, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$

$T_{\min} = 12.75 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 6 & 2 \\ 2 & 4 & 4 & 0 & 12 & 18 & 4 \\ 1 & 4 & 0 & 0 & 9 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 9 & 0 & 28 & 42 & 10 \\ 6 & 18 & 12 & 0 & 42 & 58 & 14 \\ 2 & 4 & 2 & -4 & 10 & 14 & 0 \end{pmatrix}_{b_0}$$

[256] $1^{112}31^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -725, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 24 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 15 & 3 \\ 8 & 24 & 15 & 0 & 15 & 100 & 18 \\ 2 & 4 & 2 & -4 & 3 & 18 & 0 \end{pmatrix}_{b_0}$$

[257] $1^{112}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -738, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.44 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 6 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & -1 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

[258] $1^{133}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.6$

$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 16 & 18 & 6 \\ 4 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[259] $1^{114}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -761, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & -1 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

[260] $1^{114}321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -794, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 6 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[261] $1^131^{122}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -800, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 12 & 4 \\ 1 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 6 & 0 & 22 & 24 & 8 \\ 4 & 12 & 6 & 0 & 24 & 22 & 8 \\ 2 & 4 & 2 & -4 & 8 & 8 & 0 \end{pmatrix}_{b_0}$$

[262] $1^{134}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -831, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 4 & -1 \\ 4 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -1 & 6 & 8 & 0 \end{pmatrix}_{b_0}$$

[263] $1^{113}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -839, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 7 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & -1 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

[264] $1^{113}321^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -846, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 8 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & -1 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[265] \ 1^{112}31^{14}31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -860, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 30 & 4 \\ 1 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 18 & 3 \\ 10 & 30 & 18 & 0 & 18 & 154 & 22 \\ 2 & 4 & 2 & -4 & 3 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[266] \ 1^{114}31^{13}31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -869, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 30 & 4 \\ 1 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 21 & 3 \\ 10 & 30 & 21 & 0 & 21 & 172 & 24 \\ 2 & 4 & 2 & -4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[267] \ 1^{11}31^{112}31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -893, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 5 & 2 \\ 2 & 4 & 4 & 0 & 6 & 15 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 15 & 4 \\ 5 & 15 & 9 & 0 & 15 & 37 & 11 \\ 2 & 4 & 2 & -4 & 4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[268] \ 1^{113}31^{13}31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -929, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 18 & 3 \\ 9 & 27 & 18 & 0 & 18 & 133 & 21 \\ 2 & 4 & 2 & -4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[269] \ 1^131^{113}31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 6 & 2 \\ 2 & 4 & 4 & 0 & 6 & 18 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 18 & 4 \\ 6 & 18 & 9 & 0 & 18 & 52 & 12 \\ 2 & 4 & 2 & -4 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[270] \ 1^{144}3^121^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -988, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 4 & -1 \\ 5 & 8 & 0 & 4 & 30 & 32 & 8 \\ 5 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[271] \ 1^{114}3^121^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -991, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.52 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 7 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[272] \ 1^{111}31^{112}31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1049, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 5 & 2 \\ 2 & 4 & 4 & 0 & 9 & 15 & 4 \\ 1 & 4 & 0 & 0 & 6 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 24 & 7 \\ 5 & 15 & 9 & 0 & 24 & 37 & 11 \\ 2 & 4 & 2 & -4 & 7 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[273] \ 1^{114} 3^{21} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -1064, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.45 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 7 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & -1 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[274] \ 1^{114} 3^{21} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -1094, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.56 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 8 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[275] \ 1^{13} 31^{113} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1133, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 9 & 2 \\ 2 & 4 & 4 & 0 & 6 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 27 & 4 \\ 9 & 27 & 18 & 0 & 27 & 133 & 21 \\ 2 & 4 & 2 & -4 & 4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[276] \ 1^{13} 31^{112} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1145, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 7 & 2 \\ 2 & 4 & 4 & 0 & 9 & 21 & 4 \\ 1 & 4 & 0 & 0 & 6 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 36 & 7 \\ 7 & 21 & 15 & 0 & 36 & 85 & 17 \\ 2 & 4 & 2 & -4 & 7 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[277] \ 1^{114} 3^{21} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -105, \ \det = -1273, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.47 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 8 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & -1 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[278] \ 1^{11} 131^{113} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1292, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 7 & 2 \\ 2 & 4 & 4 & 0 & 9 & 21 & 4 \\ 1 & 4 & 0 & 0 & 6 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 33 & 7 \\ 7 & 21 & 12 & 0 & 33 & 73 & 15 \\ 2 & 4 & 2 & -4 & 7 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[279] \ 1^{12} 31^{122} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1328, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 6 & 2 \\ 2 & 4 & 4 & 0 & 15 & 18 & 4 \\ 1 & 4 & 0 & 0 & 9 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 15 & 9 & 0 & 37 & 48 & 11 \\ 6 & 18 & 12 & 0 & 48 & 58 & 14 \\ 2 & 4 & 2 & -4 & 11 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[280] \ 1^{113} 31^{14} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1352, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.82 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 3 & 33 & 4 \\ 1 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 21 & 3 \\ 11 & 33 & 21 & 0 & 21 & 193 & 25 \\ 2 & 4 & 2 & -4 & 3 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[281] \ 1^{12} 31^{123} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1364, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.75 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 6 & 2 \\ 2 & 4 & 4 & 0 & 9 & 18 & 4 \\ 1 & 4 & 0 & 0 & 6 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 30 & 7 \\ 6 & 18 & 12 & 0 & 30 & 58 & 14 \\ 2 & 4 & 2 & -4 & 7 & 14 & 0 \end{pmatrix}_{b_0}$$

[282] $1^{114}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -1388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 9 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & -1 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

[283] $1^{111}31^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1409, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 9 & 2 \\ 2 & 4 & 4 & 0 & 9 & 27 & 4 \\ 1 & 4 & 0 & 0 & 6 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 42 & 7 \\ 9 & 27 & 15 & 0 & 42 & 121 & 19 \\ 2 & 4 & 2 & -4 & 7 & 19 & 0 \end{pmatrix}_{b_0}$$

[284] $1^131^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1448, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 8 & 2 \\ 2 & 4 & 4 & 0 & 6 & 24 & 4 \\ 1 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 24 & 4 \\ 8 & 24 & 12 & 0 & 24 & 94 & 16 \\ 2 & 4 & 2 & -4 & 4 & 16 & 0 \end{pmatrix}_{b_0}$$

[285] $1^{11}31^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1541, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 7 & 2 \\ 2 & 4 & 4 & 0 & 6 & 21 & 4 \\ 1 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 21 & 4 \\ 7 & 21 & 12 & 0 & 21 & 73 & 15 \\ 2 & 4 & 2 & -4 & 4 & 15 & 0 \end{pmatrix}_{b_0}$$

[286] $1^{114}31^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1580, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 3 & 36 & 4 \\ 1 & 4 & 0 & 0 & 3 & 24 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 24 & 3 \\ 12 & 36 & 24 & 0 & 24 & 238 & 28 \\ 2 & 4 & 2 & -4 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

[287] $1^{12}31^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1604, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 8 & 2 \\ 2 & 4 & 4 & 0 & 6 & 24 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 24 & 4 \\ 8 & 24 & 15 & 0 & 24 & 100 & 18 \\ 2 & 4 & 2 & -4 & 4 & 18 & 0 \end{pmatrix}_{b_0}$$

[288] $1^{111}31^{122}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1688, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 15 & 4 \\ 1 & 4 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 15 & 9 & 0 & 37 & 39 & 11 \\ 5 & 15 & 9 & 0 & 39 & 37 & 11 \\ 2 & 4 & 2 & -4 & 11 & 11 & 0 \end{pmatrix}_{b_0}$$

[289] $1^{14}31^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1784, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 12 & 2 \\ 2 & 4 & 4 & 0 & 6 & 36 & 4 \\ 1 & 4 & 0 & 0 & 3 & 24 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 36 & 4 \\ 12 & 36 & 24 & 0 & 36 & 238 & 28 \\ 2 & 4 & 2 & -4 & 4 & 28 & 0 \end{pmatrix}_{b_0}$$

[290] $1^{122}31^{122}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2048, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 18 & 4 \\ 1 & 4 & 0 & 0 & 12 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 18 & 12 & 0 & 58 & 60 & 14 \\ 6 & 18 & 12 & 0 & 60 & 58 & 14 \\ 2 & 4 & 2 & -4 & 14 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[291] \quad 1^{112}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2165, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 8 & 2 \\ 2 & 4 & 4 & 0 & 9 & 24 & 4 \\ 1 & 4 & 0 & 0 & 6 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 39 & 7 \\ 8 & 24 & 15 & 0 & 39 & 100 & 18 \\ 2 & 4 & 2 & -4 & 7 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[292] \quad 1^{114}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2309, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 10 & 2 \\ 2 & 4 & 4 & 0 & 9 & 30 & 4 \\ 1 & 4 & 0 & 0 & 6 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 51 & 7 \\ 10 & 30 & 21 & 0 & 51 & 172 & 24 \\ 2 & 4 & 2 & -4 & 7 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[293] \quad 1^{113}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 9 & 2 \\ 2 & 4 & 4 & 0 & 6 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 27 & 4 \\ 9 & 27 & 15 & 0 & 27 & 121 & 19 \\ 2 & 4 & 2 & -4 & 4 & 19 & 0 \end{pmatrix}_{b_0}$$

$$[294] \quad 1^{13}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2501, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 11 & 2 \\ 2 & 4 & 4 & 0 & 6 & 33 & 4 \\ 1 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 33 & 4 \\ 11 & 33 & 21 & 0 & 33 & 193 & 25 \\ 2 & 4 & 2 & -4 & 4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[295] \quad 1^{113}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 9 & 2 \\ 2 & 4 & 4 & 0 & 9 & 27 & 4 \\ 1 & 4 & 0 & 0 & 6 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 45 & 7 \\ 9 & 27 & 18 & 0 & 45 & 133 & 21 \\ 2 & 4 & 2 & -4 & 7 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[296] \quad 1^{123}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2684, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 10 & 2 \\ 2 & 4 & 4 & 0 & 6 & 30 & 4 \\ 1 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 30 & 4 \\ 10 & 30 & 18 & 0 & 30 & 154 & 22 \\ 2 & 4 & 2 & -4 & 4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[297] \quad 1^{112}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2840, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 10 & 2 \\ 2 & 4 & 4 & 0 & 9 & 30 & 4 \\ 1 & 4 & 0 & 0 & 6 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 48 & 7 \\ 10 & 30 & 18 & 0 & 48 & 154 & 22 \\ 2 & 4 & 2 & -4 & 7 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[298] \quad 1^{113}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -3737, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 11 & 2 \\ 2 & 4 & 4 & 0 & 9 & 33 & 4 \\ 1 & 4 & 0 & 0 & 6 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 54 & 7 \\ 11 & 33 & 21 & 0 & 54 & 193 & 25 \\ 2 & 4 & 2 & -4 & 7 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[299] \quad 1^{114}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -4100, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 12 & 2 \\ 2 & 4 & 4 & 0 & 9 & 36 & 4 \\ 1 & 4 & 0 & 0 & 6 & 24 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 60 & 7 \\ 12 & 36 & 24 & 0 & 60 & 238 & 28 \\ 2 & 4 & 2 & -4 & 7 & 28 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [300] \quad 1^{111} 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 19.60 \rightarrow 20, T_{\max} = 22, T \in [20, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 10 & 3 \\ 11 & 15 & 0 & 10 & 45 & 13 \\ 3 & 4 & -3 & 3 & 13 & -5 \end{pmatrix}_{b_0}$$

$$[301] \quad 1^{113} 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 23 & 7 \\ 9 & 15 & 0 & 23 & 40 & 12 \\ 3 & 4 & -3 & 7 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[302] \quad 1^{11} 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 5 & 7 & 0 & 6 & 8 & 6 \\ 3 & 4 & -3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[303] \quad 1^{12} 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 3 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 0 & 1 & 7 & 3 \\ 7 & 11 & 0 & 7 & 21 & 9 \\ 3 & 4 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[304] \quad 1^{11} 231^{111} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 3 & 2 \\ 2 & 4 & 1 & 3 & 9 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 7 & 3 & 0 \\ 4 & 9 & 0 & 7 & 14 & 9 & 8 \\ 3 & 6 & 0 & 3 & 9 & 4 & 4 \\ 2 & 4 & -1 & 0 & 8 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[305] \quad 1^{131} 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 & 3 \\ 4 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -3 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[307] \quad 1^{11} 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 5 & 3 \\ 4 & 4 & 0 & 5 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 5 & 7 & 0 & 6 & 8 & 6 \\ 3 & 4 & -3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[306] \quad 1^{112} 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 17 & 7 \\ 7 & 11 & 0 & 17 & 21 & 9 \\ 3 & 4 & -3 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

[308] $1^{11} 4^{11} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -3 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

[309] $1^{11} 4^{11} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.10 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 3 \\ 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & -3 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

[312] $1^{11} 231^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 7 & 0 & 14 & 8 \\ 2 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[311] $1^1 231^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[310] $1^{11} 4^{11} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 0 & 1 & 6 & 3 \\ 6 & 8 & 0 & 6 & 13 & 7 \\ 3 & 4 & -3 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[313] $1^{121} 4^{11} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

[314] $1^{11} 4^{11} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 59, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 0 & 1 & 5 & 3 \\ 5 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & -3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[315] $1^{11} 231^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 9 & 0 & 7 & 0 & 14 & 8 \\ 2 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[316] $1^{11} 4^{11} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 74, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 0 & 1 & 8 & 3 \\ 8 & 12 & 0 & 8 & 26 & 10 \\ 3 & 4 & -3 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[317] $1^{11} 1^{11} 4^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 13.49 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 8 & 0 & 13 & 11 & 7 \\ 5 & 7 & 0 & 11 & 8 & 6 \\ 3 & 4 & -3 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

[318] $1^{122} 1^{131} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 82, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.49 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 16 & 6 \\ 7 & 11 & 0 & 16 & 21 & 9 \\ 3 & 4 & -3 & 6 & 9 & 1 \end{pmatrix}_{b_0}$$

[319] $1^{111} 4^1 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 88, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -3 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[320] $1^{112} 4^1 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 4 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[321] $1^{122} 4^1 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 4 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[322] $1^{113} 4^1 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 6 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[323] $1^{11} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 109, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 1 & 2 & 3 \\ 3 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -3 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[324] $1^{111} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 115, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 3 \\ 5 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & -3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[325] $1^{114} 4^1 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 118, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 8 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -3 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[326] 1^{112} 1^{131^3} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 16 & 22 & 0 & 15 & 97 & 19 \\ 3 & 4 & -3 & 3 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[327] 1^{11} 1^{131^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 129, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 1 & 5 & 3 \\ 5 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & -3 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[328] 1^{11} 231^{111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 133, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 7 & 0 \\ 2 & 6 & 0 & 3 & 4 & 9 & 4 \\ 4 & 9 & 0 & 7 & 9 & 14 & 8 \\ 2 & 4 & -1 & 0 & 4 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[329] 1^1 231^{11} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[330] 1^{111} 1^{11} 4 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 149, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -3 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[331] 1^{11} 231^{11} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 159, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 7 & 0 & 14 & 8 \\ 2 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[332] 1^{114} 1^{131^{11}} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 163, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 7 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 8 & 25 & 6 \\ 11 & 19 & 0 & 25 & 65 & 15 \\ 3 & 4 & -3 & 6 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[333] 1^{122} 1^{11} 4 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 164, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -3 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

- [334] $1^{11} 1^{131^{11}} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 176, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.23 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 1 & 5 & 3 \\ 6 & 9 & 0 & 5 & 13 & 7 \\ 3 & 4 & -3 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [335] $1^{111} 1^{131^1} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 194, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 8 & 3 \\ 8 & 12 & 0 & 8 & 26 & 10 \\ 3 & 4 & -3 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$
- [336] $1^{121} 1^{131^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 194, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 8 & 3 \\ 3 & 4 & 0 & 11 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 11 & 0 & 21 & 24 & 9 \\ 8 & 12 & 0 & 24 & 26 & 10 \\ 3 & 4 & -3 & 9 & 10 & 1 \end{pmatrix}_{b_0}$$
- [337] $1^{11} 231^{111} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 199, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 5 & 2 \\ 2 & 4 & 1 & 3 & 6 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 7 & 0 \\ 3 & 6 & 0 & 3 & 4 & 9 & 4 \\ 5 & 9 & 0 & 7 & 9 & 14 & 8 \\ 2 & 4 & -1 & 0 & 4 & 8 & 1 \end{pmatrix}_{b_0}$$
- [338] $1^{112} 1^{131^2} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 202, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.44 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 10 & 3 \\ 10 & 14 & 0 & 10 & 38 & 12 \\ 3 & 4 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$
- [339] $1^{11} 1^{131^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 204, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.16 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 8 & 3 \\ 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 1 & 8 & 3 \\ 8 & 12 & 0 & 8 & 26 & 10 \\ 3 & 4 & -3 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$
- [340] $1^1 231^{1231^{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 216, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [341] $1^{112} 4 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 263, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.44 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -3 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$
- [342] $1^1 231^{1231^{113}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 264, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[343] $1^{112}1^{131^1}4^11^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 265, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 8 & 3 \\ 7 & 11 & 0 & 8 & 21 & 9 \\ 3 & 4 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[344] $1^{111}1^{131^{11}}4^11^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 268, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 7 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 8 & 10 & 6 \\ 5 & 7 & 0 & 10 & 8 & 6 \\ 3 & 4 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[345] $1^1231^{111}31^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 288, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 3 & 0 \\ 2 & 6 & 0 & 3 & 4 & 6 & 4 \\ 2 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[346] $1^1231^{111}31^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 301, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 3 & 0 \\ 2 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[347] $1^{113}1^{131^4}4^11^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 309, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 13.70 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 21 & 3 \\ 3 & 4 & 0 & 4 & 29 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 20 & 3 \\ 21 & 29 & 0 & 20 & 169 & 25 \\ 3 & 4 & -3 & 3 & 25 & 1 \end{pmatrix}_{b_0}$$

[348] $1^{111}1^{131^{11}}4^11^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 320, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 8 & 16 & 6 \\ 8 & 12 & 0 & 16 & 26 & 10 \\ 3 & 4 & -3 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

[349] $1^{113}1^{131^{11}}4^11^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 320, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 7 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 8 & 20 & 6 \\ 9 & 15 & 0 & 20 & 40 & 12 \\ 3 & 4 & -3 & 6 & 12 & 1 \end{pmatrix}_{b_0}$$

[350] $1^{114}1^{131^{11}}4^11^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 323, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 7 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 8 & 31 & 6 \\ 14 & 24 & 0 & 31 & 101 & 19 \\ 3 & 4 & -3 & 6 & 19 & 1 \end{pmatrix}_{b_0}$$

[351] $1^1231^{111}31^{121} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 325, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[352] \ 1^1 231^{11} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 336, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[353] \ 1^1 231^{12} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 346, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[354] \ 1^{113} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 347, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 15 & 21 & 0 & 15 & 88 & 18 \\ 3 & 4 & -3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[355] \ 1^{112} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 355, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 7 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 8 & 15 & 6 \\ 7 & 11 & 0 & 15 & 21 & 9 \\ 3 & 4 & -3 & 6 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[356] \ 1^{11} 231^{12} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 376, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 10 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 15 & 0 & 10 & 0 & 38 & 12 \\ 2 & 4 & -1 & 0 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[357] \ 1^{113} 4^1 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 381, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -3 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[358] \ 1^{11} 231^{12} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 384, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 10 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 0 & 10 & 0 & 38 & 12 \\ 2 & 4 & -1 & 0 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[359] \ 1^{112} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 407, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 13 & 3 \\ 13 & 19 & 0 & 13 & 68 & 16 \\ 3 & 4 & -3 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

[360] $1^{113} 1^{131} 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 419$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 13.41 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 11 & 3 \\ 9 & 15 & 0 & 11 & 40 & 12 \\ 3 & 4 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[361] $1^{112} 1^{131} 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 443$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.5$

$T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 8 & 21 & 6 \\ 10 & 16 & 0 & 21 & 45 & 13 \\ 3 & 4 & -3 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

[362] $1^{113} 1^{131} 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 444$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 13.39 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 7 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 8 & 26 & 6 \\ 12 & 20 & 0 & 26 & 70 & 16 \\ 3 & 4 & -3 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

[363] $1^{112} 1^{131} 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 452$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.5$

$T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 11 & 3 \\ 10 & 16 & 0 & 11 & 45 & 13 \\ 3 & 4 & -3 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

[364] $1^{11} 231^{12} 31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 486$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.5$

$T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 10 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 15 & 0 & 10 & 0 & 38 & 12 \\ 2 & 4 & -1 & 0 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[365] $1^{11} 231^{13} 31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 495$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 0 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 13 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 21 & 0 & 13 & 0 & 74 & 16 \\ 2 & 4 & -1 & 0 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[366] $1^{113} 1^{131} 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 500$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 13 & 3 \\ 12 & 18 & 0 & 13 & 61 & 15 \\ 3 & 4 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

[367] $1^{114} 4^1 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 503$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.57 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 11 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -3 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

[368] $1^1 231^{11} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 524$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[369] $1^{122} 1^{131^{11}} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 533$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 0 & 11 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 11 & 0 & 21 & 32 & 9 \\ 10 & 16 & 0 & 32 & 45 & 13 \\ 3 & 4 & -3 & 9 & 13 & 1 \end{pmatrix}_{b_0}$$

[370] $1^{1231} 1^{131^{112}} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 549$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & 0 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[371] $1^{114} 1^{131^4} 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 550$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.53 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 20 & 3 \\ 20 & 28 & 0 & 20 & 158 & 24 \\ 3 & 4 & -3 & 3 & 24 & 1 \end{pmatrix}_{b_0}$$

[372] $1^{122} 1^{131^{11}} 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 568$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.39 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 11 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 11 & 0 & 21 & 23 & 9 \\ 7 & 11 & 0 & 23 & 21 & 9 \\ 3 & 4 & -3 & 9 & 9 & 1 \end{pmatrix}_{b_0}$$

[373] $1^{114} 1^{131^1} 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 577$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 11 & 19 & 0 & 14 & 65 & 15 \\ 3 & 4 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

[374] $1^{1231} 1^{131^{114}} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 621$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 12.87 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & 0 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[375] $1^{113} 1^{131^3} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 678$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 18 & 26 & 0 & 18 & 130 & 22 \\ 3 & 4 & -3 & 3 & 22 & 1 \end{pmatrix}_{b_0}$$

[376] $1^{1231} 1^{131^{113}} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 691$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & 0 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[377] $1^{113} 1^{131^1} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 714$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} =$

11.8

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 12 & 20 & 0 & 14 & 70 & 16 \\ 3 & 4 & -3 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

[378] $1^{114} 1^{131^3} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 793, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 25 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 17 & 25 & 0 & 18 & 121 & 21 \\ 3 & 4 & -3 & 3 & 21 & 1 \end{pmatrix}_{b_0}$$

[379] $1^{114} 1^{131^2} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 802, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 16 & 3 \\ 14 & 22 & 0 & 16 & 90 & 18 \\ 3 & 4 & -3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

[380] $1^{113} 1^{131^2} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 813, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 16 & 3 \\ 15 & 23 & 0 & 16 & 97 & 19 \\ 3 & 4 & -3 & 3 & 19 & 1 \end{pmatrix}_{b_0}$$

[381] $1^{11} 231^{13} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 829, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 0 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 13 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 21 & 0 & 13 & 0 & 74 & 16 \\ 2 & 4 & -1 & 0 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[382] $1^{11} 231^{13} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 951, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 1 & 3 & 0 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 13 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 11 & 21 & 0 & 13 & 0 & 74 & 16 \\ 2 & 4 & -1 & 0 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[383] $1^{114} 1^{131^1} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 980, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 17 & 3 \\ 14 & 24 & 0 & 17 & 101 & 19 \\ 3 & 4 & -3 & 3 & 19 & 1 \end{pmatrix}_{b_0}$$

[384] $1^{114} 1^{131^4} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 1007, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 23 & 3 \\ 3 & 4 & 0 & 4 & 33 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 23 & 3 \\ 23 & 33 & 0 & 23 & 212 & 28 \\ 3 & 4 & -3 & 3 & 28 & 1 \end{pmatrix}_{b_0}$$

[385] $1^1 231^{14} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 1020, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 11 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & 0 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[386] \quad 1^{11}231^{14}31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 1074, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.92 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 12 & 2 \\ 2 & 4 & 1 & 3 & 0 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 16 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 12 & 27 & 0 & 16 & 0 & 122 & 20 \\ 2 & 4 & -1 & 0 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[387] \quad 1^{1231}1431^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 1174, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.89 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 12 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 12 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & 0 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[388] \quad 1^{114}4^{131^2}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -103, \quad \det = 1223, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.39 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 19 & 3 \\ 17 & 27 & 0 & 19 & 132 & 22 \\ 3 & 4 & -3 & 3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[389] \quad 1^{114}4^{131^3}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -103, \quad \det = 1232, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.40 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 30 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 20 & 30 & 0 & 21 & 169 & 25 \\ 3 & 4 & -3 & 3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[390] \quad 1^{11}231^{14}31^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 1420, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.88 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 13 & 2 \\ 2 & 4 & 1 & 3 & 0 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 16 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 13 & 27 & 0 & 16 & 0 & 122 & 20 \\ 2 & 4 & -1 & 0 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [391] \quad 1^{11}5^{11}131^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 11, \quad T_{\max} = 18, \quad \Delta = -98, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.25 \rightarrow 15, \quad T_{\max} = 18, \quad T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 3 \\ 3 & 4 & 3 & 0 & 0 & 4 \\ 2 & 3 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[392] \quad 1^{11}5^{12}131^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 11, \quad T_{\max} = 18, \quad \Delta = -98, \quad \det = -31, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.16 \rightarrow 15, \quad T_{\max} = 18, \quad T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 2 & 3 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 4 & 2 & 0 & 1 & 3 \\ 3 & 4 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[393] \quad 1^{11}5^{11}131^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 11, \quad T_{\max} = 18, \quad \Delta = -98, \quad \det = -39, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.33 \rightarrow 15, \quad T_{\max} = 18, \quad T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 2 & 3 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 4 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[394] \quad 1^{111}31^{11}41^{131^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -82, \quad \det = -40, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.90 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 11 & 4 \\ 1 & 4 & 0 & 0 & 3 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 7 & 3 \\ 5 & 11 & 7 & 0 & 7 & 21 & 9 \\ 2 & 4 & 2 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[395] $1^{11} 5^{12} 31^{11} 4131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 9 & 3 \\ 4 & 12 & 9 & 0 & 9 & 28 & 10 \\ 2 & 4 & 2 & -3 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[396] $1^{11} 5^{11} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -69, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 14.19 \rightarrow 15, T_{\max} = 18, T \in [15, 18]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 2 & 3 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 2 & 0 & 1 & 3 \\ 3 & 4 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[397] $1^{12} 5^{13} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -84, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 14.19 \rightarrow 15, T_{\max} = 18, T \in [15, 18]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 3 & 0 & 8 & 4 \\ 2 & 3 & -2 & 0 & 4 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 8 & 4 & 0 & 10 & 6 \\ 3 & 4 & 0 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[398] $1^{12} 5^{11} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 14.32 \rightarrow 15, T_{\max} = 18, T \in [15, 18]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 3 & 0 & 8 & 4 \\ 2 & 3 & -2 & 0 & 4 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 8 & 4 & 0 & 10 & 6 \\ 3 & 4 & 0 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[399] $1^{13} 5^{11} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -109, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 14.56 \rightarrow 15, T_{\max} = 18, T \in [15, 18]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 3 & 0 & 12 & 4 \\ 2 & 3 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 6 & 0 & 25 & 9 \\ 3 & 4 & 0 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[400] $1^{12} 5^{12} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 18, \Delta = -98, \det = -126, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 18, T \in [15, 18]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 3 & 0 & 8 & 4 \\ 2 & 3 & -2 & 0 & 4 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 8 & 4 & 0 & 10 & 6 \\ 3 & 4 & 0 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[401] $1^{11} 31^{11} 4131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -139, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 6 & 3 \\ 3 & 9 & 6 & 0 & 6 & 13 & 7 \\ 2 & 4 & 2 & -3 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[402] $1^{13} 31^{11} 412^1 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -171, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 19 & 4 \\ 1 & 4 & 0 & 0 & 3 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 14 & 3 \\ 9 & 19 & 14 & 0 & 14 & 77 & 17 \\ 2 & 4 & 2 & -3 & 3 & 17 & 1 \end{pmatrix}_{b_0}$$

[403] $1^{13} \overset{1^{12}}{5} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 18, \Delta = -98, \det = -199, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 18, T \in [15, 18]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 3 & 0 & 12 & 4 \\ 2 & 3 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 12 & 6 & 0 & 25 & 9 \\ 3 & 4 & 0 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[404] $1^{113} 31^{11} 41^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 14, \Delta = -82, \det = -216, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 17 & 4 \\ 1 & 4 & 0 & 0 & 3 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 13 & 3 \\ 7 & 17 & 13 & 0 & 13 & 61 & 15 \\ 2 & 4 & 2 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

[405] $1^{13} \overset{1^{13}}{5} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 18, \Delta = -98, \det = -221, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 14.24 \rightarrow 15, T_{\max} = 18, T \in [15, 18]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 3 & 0 & 12 & 4 \\ 2 & 3 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 12 & 6 & 0 & 25 & 9 \\ 3 & 4 & 0 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[406] $1^{114} 31^{11} 41^2 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 14, \Delta = -82, \det = -229, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 22 & 4 \\ 1 & 4 & 0 & 0 & 3 & 17 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 17 & 3 \\ 10 & 22 & 17 & 0 & 17 & 108 & 20 \\ 2 & 4 & 2 & -3 & 3 & 20 & 1 \end{pmatrix}_{b_0}$$

[407] $1^{112} 31^{11} 41^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 14, \Delta = -82, \det = -234, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 14 & 4 \\ 1 & 4 & 0 & 0 & 3 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 10 & 3 \\ 6 & 14 & 10 & 0 & 10 & 38 & 12 \\ 2 & 4 & 2 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

4.78 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$

$T = 10$ [0] $1^{11} \overset{1^1}{3} 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 3 & 5 & 0 & 6 & 4 & 6 & 3 & 4 \\ 3 & 5 & 0 & 6 & 6 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 4 & 4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111} \overset{1^1}{3} 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -83, \det = 429, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 9 & 6 & 3 & 4 \\ 5 & 9 & 0 & 9 & 13 & 9 & 6 & 7 \\ 3 & 5 & 0 & 6 & 9 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 6 & 4 & 0 & 2 \\ 3 & 4 & -4 & 4 & 7 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{1111} \overset{1^1}{3} 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -83, \det = 1264, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 9 & 9 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 15 & 9 & 6 & 7 \\ 5 & 9 & 0 & 15 & 13 & 9 & 6 & 7 \\ 3 & 5 & 0 & 9 & 9 & 4 & 4 & 4 \\ 2 & 4 & 0 & 6 & 6 & 4 & 0 & 2 \\ 3 & 4 & -4 & 7 & 7 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

4.79 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{2}} 3 \ 1^{\textcolor{red}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 1 & 3 \\ 4 & 9 & 4 & 0 & 16 & 0 & 5 \\ 2 & 4 & -3 & 0 & 6 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 16 & 6 & 0 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 0 & 8 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3 \ 1^{\textcolor{red}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 12 & 7 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}} 3 \ 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3 \ 1^{\textcolor{red}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 4 & 6 & 0 & 4 \\ 4 & 8 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 4 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}} 3 \ 1^{\textcolor{red}{1}\textcolor{blue}{3}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 35 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 35 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -4 & 7 & 27 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3 \ 1^{\textcolor{red}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 4 & 6 & 0 & 4 \\ 3 & 8 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 4 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}} 3 \ 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3 \ 1^{\textcolor{red}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 3 & 0 & -1 \\ 4 & 8 & 3 & 4 & 6 & 0 & 4 \\ 4 & 8 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 4 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}} 3 \ 1^{\textcolor{red}{1}\textcolor{blue}{2}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -618, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

15.0

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 26 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 26 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -4 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [9] \quad 1^{11} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [10] \quad 1^{11} 3^{11} 4^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -68, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 7 & 5 \\ 10 & 13 & 0 & 14 & 10 & 7 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 14 & 0 & 13 & 9 & 7 \\ 7 & 10 & 0 & 9 & 5 & 5 \\ 5 & 7 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

4.80 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 372, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 27 & 7 \\ 9 & 17 & 0 & 15 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 27 & 49 & 13 \\ 5 & 8 & -4 & 7 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

4.81 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [1] \quad 1^{11} 2^{11} 3^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 18.25 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 4 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 \\ 13 & 23 & 0 & 15 & 15 & 100 & 18 \\ 3 & 4 & -4 & 3 & 3 & 18 & -6 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{11} 3^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12} 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 12 & 6 \\ 5 & 9 & 0 & 12 & 13 & 15 & 7 \\ 5 & 9 & 0 & 12 & 15 & 13 & 7 \\ 3 & 4 & -4 & 6 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{13} 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 0 & 5 & 17 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 15 & 9 & 4 \\ 9 & 17 & 0 & 15 & 49 & 27 & 13 \\ 5 & 9 & 0 & 9 & 27 & 13 & 7 \\ 3 & 4 & -4 & 4 & 13 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1123} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 12 & 3 \\ 3 & 4 & 0 & 4 & 8 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 15 & 3 \\ 4 & 8 & 0 & 6 & 10 & 30 & 6 \\ 12 & 22 & 0 & 15 & 30 & 85 & 17 \\ 3 & 4 & -4 & 3 & 6 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1211} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 6 & 6 \\ 5 & 9 & 0 & 12 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 6 & 1 & 3 \\ 3 & 4 & -4 & 6 & 7 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 17.66 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 0 & 5 & 4 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 30 & 4 \\ 2 & 4 & 0 & 3 & 1 & 18 & 3 \\ 16 & 28 & 0 & 30 & 18 & 154 & 22 \\ 3 & 4 & -4 & 4 & 3 & 22 & -5 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{113} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 5 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 15 & 4 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 \\ 9 & 17 & 0 & 15 & 12 & 49 & 13 \\ 3 & 4 & -4 & 4 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16.50 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 8 & 3 & 0 & 4 & 6 & 4 \\ 3 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & -1 & 0 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 0 & 5 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 15 & 4 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 8 & 14 & 0 & 15 & 9 & 37 & 11 \\ 3 & 4 & -4 & 4 & 3 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1113} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 12 & 3 \\ 3 & 4 & 0 & 4 & 14 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 15 & 3 \\ 8 & 14 & 0 & 9 & 37 & 57 & 11 \\ 12 & 22 & 0 & 15 & 57 & 85 & 17 \\ 3 & 4 & -4 & 3 & 11 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1222} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 8 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 18 & 6 \\ 4 & 8 & 0 & 12 & 10 & 18 & 6 \\ 7 & 13 & 0 & 18 & 18 & 28 & 10 \\ 3 & 4 & -4 & 6 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -124, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.81 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 6 & 4 \\ 3 & 5 & 0 & 6 & 4 & 6 & 4 \\ 3 & 5 & 0 & 6 & 4 & 4 & 4 \\ 3 & 4 & -4 & 4 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -128, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.47 \rightarrow 15, \quad T_{\max} = 21, \quad T \in [15, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 8 & 3 & 0 & 4 & 6 & 4 \\ 4 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & -1 & 0 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{\overline{1214}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -150, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 11 & 3 \\ 3 & 4 & 0 & 8 & 9 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 30 & 6 \\ 5 & 9 & 0 & 12 & 13 & 33 & 7 \\ 11 & 21 & 0 & 30 & 33 & 76 & 16 \\ 3 & 4 & -4 & 6 & 7 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -170, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.35 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & -4 & 4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{\overline{1213}} 3^{\overline{1112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -194, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 12 & 3 \\ 3 & 4 & 0 & 8 & 9 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 30 & 6 \\ 5 & 9 & 0 & 12 & 13 & 36 & 7 \\ 12 & 22 & 0 & 30 & 36 & 85 & 17 \\ 3 & 4 & -4 & 6 & 7 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{\overline{1112}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -196, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 12 & 4 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 7 & 13 & 0 & 12 & 9 & 28 & 10 \\ 3 & 4 & -4 & 4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{\overline{111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.48 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 5 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 9 & 4 \\ 3 & 5 & 0 & 6 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 9 & 13 & 7 \\ 3 & 4 & -4 & 4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{\overline{1212}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -230, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 7 & 3 \\ 3 & 4 & 0 & 8 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 18 & 6 \\ 5 & 9 & 0 & 12 & 13 & 21 & 7 \\ 7 & 13 & 0 & 18 & 21 & 28 & 10 \\ 3 & 4 & -4 & 6 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1113} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -232, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.10 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 4 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 21 & 3 \\ 2 & 4 & 0 & 3 & 1 & 21 & 3 \\ 18 & 32 & 0 & 21 & 21 & 193 & 25 \\ 3 & 4 & -4 & 3 & 3 & 25 & -1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1111} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -232, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.28 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{1113} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -248, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 9 & 3 \\ 3 & 4 & 0 & 4 & 12 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 12 & 3 \\ 6 & 12 & 0 & 9 & 25 & 36 & 9 \\ 9 & 17 & 0 & 12 & 36 & 49 & 13 \\ 3 & 4 & -4 & 3 & 9 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{1112} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -256, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.44 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 4 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 4 & 8 & 0 & 6 & 6 & 10 & 6 \\ 3 & 4 & -4 & 3 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{1213} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -272, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.44 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 9 & 3 \\ 3 & 4 & 0 & 8 & 9 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 24 & 6 \\ 5 & 9 & 0 & 12 & 13 & 27 & 7 \\ 9 & 17 & 0 & 24 & 27 & 49 & 13 \\ 3 & 4 & -4 & 6 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{133} 31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -280, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.44 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 9 & 3 \\ 3 & 4 & 0 & 5 & 17 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 15 & 15 & 4 \\ 9 & 17 & 0 & 15 & 49 & 51 & 13 \\ 9 & 17 & 0 & 15 & 51 & 49 & 13 \\ 3 & 4 & -4 & 4 & 13 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{1122} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -280, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.57 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 6 & 3 \\ 4 & 8 & 0 & 6 & 10 & 12 & 6 \\ 4 & 8 & 0 & 6 & 12 & 10 & 6 \\ 3 & 4 & -4 & 3 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{1114} 31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -280, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.44 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 9 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 15 & 33 & 7 \\ 5 & 9 & 0 & 15 & 13 & 33 & 7 \\ 11 & 21 & 0 & 33 & 33 & 76 & 16 \\ 3 & 4 & -4 & 7 & 7 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1431}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -296, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.45 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 12 & 2 & 3 \\ 3 & 4 & 0 & 21 & 22 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 11 & 21 & 0 & 76 & 81 & 15 & 16 \\ 12 & 22 & 0 & 81 & 85 & 15 & 17 \\ 2 & 4 & 0 & 15 & 15 & 1 & 3 \\ 3 & 4 & -4 & 16 & 17 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1113}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -296, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.65 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 6 & 12 & 0 & 9 & 9 & 25 & 9 \\ 3 & 4 & -4 & 3 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -296, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.28 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1222}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -304, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.68 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 12 & 6 \\ 4 & 8 & 0 & 12 & 10 & 12 & 6 \\ 4 & 8 & 0 & 12 & 12 & 10 & 6 \\ 3 & 4 & -4 & 6 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1123}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -320, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 6 & 3 \\ 3 & 4 & 0 & 4 & 8 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 9 & 3 \\ 4 & 8 & 0 & 6 & 10 & 18 & 6 \\ 6 & 12 & 0 & 9 & 18 & 25 & 9 \\ 3 & 4 & -4 & 3 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{132}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -328, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 0 & 5 & 17 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 15 & 12 & 4 \\ 9 & 17 & 0 & 15 & 49 & 39 & 13 \\ 7 & 13 & 0 & 12 & 39 & 28 & 10 \\ 3 & 4 & -4 & 4 & 13 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1112}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -338, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.21 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 16 & 3 \\ 3 & 4 & 0 & 4 & 9 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 18 & 3 \\ 5 & 9 & 0 & 6 & 13 & 48 & 7 \\ 16 & 28 & 0 & 18 & 48 & 154 & 22 \\ 3 & 4 & -4 & 3 & 7 & 22 & -1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{112}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -340, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 0 & 5 & 4 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 24 & 4 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 \\ 13 & 23 & 0 & 24 & 15 & 100 & 18 \\ 3 & 4 & -4 & 4 & 3 & 18 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{1223}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 6 & 3 \\ 3 & 4 & 0 & 8 & 8 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 18 & 6 \\ 4 & 8 & 0 & 12 & 10 & 18 & 6 \\ 6 & 12 & 0 & 18 & 18 & 25 & 9 \\ 3 & 4 & -4 & 6 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{1122}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -346$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.39 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 7 & 3 \\ 3 & 4 & 0 & 4 & 8 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 9 & 3 \\ 4 & 8 & 0 & 6 & 10 & 18 & 6 \\ 7 & 13 & 0 & 9 & 18 & 28 & 10 \\ 3 & 4 & -4 & 3 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{1114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -352$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 \\ 8 & 16 & 0 & 12 & 12 & 46 & 12 \\ 3 & 4 & -4 & 3 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{1133}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -360$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.6$

$T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 6 & 3 \\ 3 & 4 & 0 & 4 & 12 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 9 & 3 \\ 6 & 12 & 0 & 9 & 25 & 27 & 9 \\ 6 & 12 & 0 & 9 & 27 & 25 & 9 \\ 3 & 4 & -4 & 3 & 9 & 9 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{1111}31^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -368$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.49 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 5 & 3 \\ 3 & 4 & 0 & 4 & 14 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 6 & 3 \\ 8 & 14 & 0 & 9 & 37 & 24 & 11 \\ 5 & 9 & 0 & 6 & 24 & 13 & 7 \\ 3 & 4 & -4 & 3 & 11 & 7 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{1111}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -376$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.68 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 9 & 3 \\ 8 & 14 & 0 & 9 & 37 & 39 & 11 \\ 8 & 14 & 0 & 9 & 39 & 37 & 11 \\ 3 & 4 & -4 & 3 & 11 & 11 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{1124}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -376$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 8 & 3 \\ 3 & 4 & 0 & 4 & 8 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 12 & 3 \\ 4 & 8 & 0 & 6 & 10 & 24 & 6 \\ 8 & 16 & 0 & 12 & 24 & 46 & 12 \\ 3 & 4 & -4 & 3 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{112}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -386$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.44 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 0 & 5 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 18 & 4 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 \\ 10 & 18 & 0 & 18 & 12 & 58 & 14 \\ 3 & 4 & -4 & 4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{1223}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -400$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 8 & 3 \\ 3 & 4 & 0 & 8 & 8 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 24 & 6 \\ 4 & 8 & 0 & 12 & 10 & 24 & 6 \\ 8 & 16 & 0 & 24 & 24 & 46 & 12 \\ 3 & 4 & -4 & 6 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{1223}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -412$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.53 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 9 & 3 \\ 3 & 4 & 0 & 8 & 8 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 24 & 6 \\ 4 & 8 & 0 & 12 & 10 & 24 & 6 \\ 9 & 17 & 0 & 24 & 24 & 49 & 13 \\ 3 & 4 & -4 & 6 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{1134}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.04 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 8 & 3 \\ 3 & 4 & 0 & 4 & 12 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 12 & 3 \\ 6 & 12 & 0 & 9 & 25 & 36 & 9 \\ 8 & 16 & 0 & 12 & 36 & 46 & 12 \\ 3 & 4 & -4 & 3 & 9 & 12 & -1 \end{pmatrix}_{b_0}$$

[47] $1^{1115}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -424$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.1$

$T_{\min} = 13.06 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 4 & 20 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 \\ 10 & 20 & 0 & 15 & 15 & 73 & 15 \\ 3 & 4 & -4 & 3 & 3 & 15 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{112}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -440$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.35 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 0 & 5 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 9 & 12 & 4 \\ 5 & 9 & 0 & 9 & 13 & 21 & 7 \\ 7 & 13 & 0 & 12 & 21 & 28 & 10 \\ 3 & 4 & -4 & 4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{1125}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -448$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.6$

$T_{\min} = 13.11 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 10 & 3 \\ 3 & 4 & 0 & 4 & 8 & 20 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 15 & 3 \\ 4 & 8 & 0 & 6 & 10 & 30 & 6 \\ 10 & 20 & 0 & 15 & 30 & 73 & 15 \\ 3 & 4 & -4 & 3 & 6 & 15 & -1 \end{pmatrix}_{b_0}$$

[50] $1^{1144}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -472$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.15 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 8 & 3 \\ 3 & 4 & 0 & 4 & 16 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 12 & 3 \\ 8 & 16 & 0 & 12 & 46 & 48 & 12 \\ 8 & 16 & 0 & 12 & 48 & 46 & 12 \\ 3 & 4 & -4 & 3 & 12 & 12 & -1 \end{pmatrix}_{b_0}$$

[51] $1^{1135}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -488$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.9$

$T_{\min} = 13.18 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 10 & 3 \\ 3 & 4 & 0 & 4 & 12 & 20 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 15 & 3 \\ 6 & 12 & 0 & 9 & 25 & 45 & 9 \\ 10 & 20 & 0 & 15 & 45 & 75 & 15 \\ 3 & 4 & -4 & 3 & 9 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{114}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -490, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 0 & 5 & 4 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 24 & 4 \\ 2 & 4 & 0 & 3 & 1 & 18 & 3 \\ 14 & 26 & 0 & 24 & 18 & 118 & 20 \\ 3 & 4 & -4 & 4 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1232}31^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -490, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 7 & 3 \\ 3 & 4 & 0 & 8 & 22 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 30 & 18 & 6 \\ 12 & 22 & 0 & 30 & 85 & 51 & 17 \\ 7 & 13 & 0 & 18 & 51 & 28 & 10 \\ 3 & 4 & -4 & 6 & 17 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -504, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 9 & 9 & 4 \\ 5 & 9 & 0 & 9 & 13 & 15 & 7 \\ 5 & 9 & 0 & 9 & 15 & 13 & 7 \\ 3 & 4 & -4 & 4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1116}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -512, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.22 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 4 & 24 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 1 & 18 & 3 \\ 12 & 24 & 0 & 18 & 18 & 106 & 18 \\ 3 & 4 & -4 & 3 & 3 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1112}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -512, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 \\ 10 & 18 & 0 & 12 & 12 & 58 & 14 \\ 3 & 4 & -4 & 3 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{113}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -520, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 0 & 5 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 21 & 4 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 \\ 12 & 22 & 0 & 21 & 15 & 85 & 17 \\ 3 & 4 & -4 & 4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1214}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 14 & 3 \\ 3 & 4 & 0 & 8 & 9 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 36 & 6 \\ 5 & 9 & 0 & 12 & 13 & 42 & 7 \\ 14 & 26 & 0 & 36 & 42 & 118 & 20 \\ 3 & 4 & -4 & 6 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{1126}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -536, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.25 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 12 & 3 \\ 3 & 4 & 0 & 4 & 8 & 24 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 18 & 3 \\ 4 & 8 & 0 & 6 & 10 & 36 & 6 \\ 12 & 24 & 0 & 18 & 36 & 106 & 18 \\ 3 & 4 & -4 & 3 & 6 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{1111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -540, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & -4 & 3 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{122} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -556, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.35 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 0 & 5 & 13 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 12 & 12 & 4 \\ 7 & 13 & 0 & 12 & 28 & 30 & 10 \\ 7 & 13 & 0 & 12 & 30 & 28 & 10 \\ 3 & 4 & -4 & 4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{1134} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -576, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.62 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 11 & 3 \\ 3 & 4 & 0 & 4 & 12 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 15 & 3 \\ 6 & 12 & 0 & 9 & 25 & 45 & 9 \\ 11 & 21 & 0 & 15 & 45 & 76 & 16 \\ 3 & 4 & -4 & 3 & 9 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{1112} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -584, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.49 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 10 & 3 \\ 3 & 4 & 0 & 4 & 14 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 12 & 3 \\ 8 & 14 & 0 & 9 & 37 & 48 & 11 \\ 10 & 18 & 0 & 12 & 48 & 58 & 14 \\ 3 & 4 & -4 & 3 & 11 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{1112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -592, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 7 & 13 & 0 & 9 & 9 & 28 & 10 \\ 3 & 4 & -4 & 3 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{1117} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -616, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 13.35 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 4 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 21 & 3 \\ 2 & 4 & 0 & 3 & 1 & 21 & 3 \\ 14 & 28 & 0 & 21 & 21 & 145 & 21 \\ 3 & 4 & -4 & 3 & 3 & 21 & -1 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{1222} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -616, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 18 & 18 & 6 \\ 7 & 13 & 0 & 18 & 28 & 30 & 10 \\ 7 & 13 & 0 & 18 & 30 & 28 & 10 \\ 3 & 4 & -4 & 6 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{1127} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -640, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.38 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 14 & 3 \\ 3 & 4 & 0 & 4 & 8 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 21 & 3 \\ 4 & 8 & 0 & 6 & 10 & 42 & 6 \\ 14 & 28 & 0 & 21 & 42 & 145 & 21 \\ 3 & 4 & -4 & 3 & 6 & 21 & -1 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{1124}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -640, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.40 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 14 & 3 \\ 3 & 4 & 0 & 4 & 8 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 18 & 3 \\ 4 & 8 & 0 & 6 & 10 & 36 & 6 \\ 14 & 26 & 0 & 18 & 36 & 118 & 20 \\ 3 & 4 & -4 & 3 & 6 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{1123}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -658, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 9 & 3 \\ 3 & 4 & 0 & 4 & 8 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 12 & 3 \\ 4 & 8 & 0 & 6 & 10 & 24 & 6 \\ 9 & 17 & 0 & 12 & 24 & 49 & 13 \\ 3 & 4 & -4 & 3 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{1114}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -666, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 11 & 3 \\ 3 & 4 & 0 & 4 & 9 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 15 & 3 \\ 5 & 9 & 0 & 6 & 13 & 33 & 7 \\ 11 & 21 & 0 & 15 & 33 & 76 & 16 \\ 3 & 4 & -4 & 3 & 7 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{113}31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -736, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.66 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 0 & 5 & 4 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 33 & 4 \\ 2 & 4 & 0 & 3 & 1 & 21 & 3 \\ 18 & 32 & 0 & 33 & 21 & 193 & 25 \\ 3 & 4 & -4 & 4 & 3 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{1224}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -740, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.54 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 11 & 3 \\ 3 & 4 & 0 & 8 & 8 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 30 & 6 \\ 4 & 8 & 0 & 12 & 10 & 30 & 6 \\ 11 & 21 & 0 & 30 & 30 & 76 & 16 \\ 3 & 4 & -4 & 6 & 6 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{1112}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -746, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.34 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 7 & 3 \\ 3 & 4 & 0 & 4 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 9 & 3 \\ 5 & 9 & 0 & 6 & 13 & 21 & 7 \\ 7 & 13 & 0 & 9 & 21 & 28 & 10 \\ 3 & 4 & -4 & 3 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{113}31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -746, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.47 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 0 & 5 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 27 & 4 \\ 2 & 4 & 0 & 3 & 1 & 18 & 3 \\ 15 & 27 & 0 & 27 & 18 & 133 & 21 \\ 3 & 4 & -4 & 4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{1113}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -788, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 9 & 3 \\ 3 & 4 & 0 & 4 & 9 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 12 & 3 \\ 5 & 9 & 0 & 6 & 13 & 27 & 7 \\ 9 & 17 & 0 & 12 & 27 & 49 & 13 \\ 3 & 4 & -4 & 3 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{1223} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -838, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.41 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 9 & 3 \\ 3 & 4 & 0 & 8 & 13 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 18 & 24 & 6 \\ 7 & 13 & 0 & 18 & 28 & 39 & 10 \\ 9 & 17 & 0 & 24 & 39 & 49 & 13 \\ 3 & 4 & -4 & 6 & 10 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{1113} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -852, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.36 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 9 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 15 & 27 & 7 \\ 5 & 9 & 0 & 15 & 13 & 27 & 7 \\ 9 & 17 & 0 & 27 & 27 & 49 & 13 \\ 3 & 4 & -4 & 7 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{1113} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -872, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 1 & 18 & 3 \\ 15 & 27 & 0 & 18 & 18 & 133 & 21 \\ 3 & 4 & -4 & 3 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{1112} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -890, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 13 & 3 \\ 3 & 4 & 0 & 4 & 9 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 15 & 3 \\ 5 & 9 & 0 & 6 & 13 & 39 & 7 \\ 13 & 23 & 0 & 15 & 39 & 100 & 18 \\ 3 & 4 & -4 & 3 & 7 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{1224} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 11 & 3 \\ 3 & 4 & 0 & 8 & 13 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 18 & 30 & 6 \\ 7 & 13 & 0 & 18 & 28 & 48 & 10 \\ 11 & 21 & 0 & 30 & 48 & 76 & 16 \\ 3 & 4 & -4 & 6 & 10 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{1113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -904, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 12 & 49 & 13 \\ 3 & 4 & -4 & 3 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{1142} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -936, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.41 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 7 & 3 \\ 3 & 4 & 0 & 9 & 21 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 33 & 21 & 7 \\ 11 & 21 & 0 & 33 & 76 & 48 & 16 \\ 7 & 13 & 0 & 21 & 48 & 28 & 10 \\ 3 & 4 & -4 & 7 & 16 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{1122} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -962, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 10 & 3 \\ 3 & 4 & 0 & 4 & 13 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 12 & 3 \\ 7 & 13 & 0 & 9 & 28 & 42 & 10 \\ 10 & 18 & 0 & 12 & 42 & 58 & 14 \\ 3 & 4 & -4 & 3 & 10 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{1111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -964, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.35 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 15 & 15 & 7 \\ 5 & 9 & 0 & 15 & 13 & 15 & 7 \\ 5 & 9 & 0 & 15 & 15 & 13 & 7 \\ 3 & 4 & -4 & 7 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{1124}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -986, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 11 & 3 \\ 3 & 4 & 0 & 4 & 8 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 15 & 3 \\ 4 & 8 & 0 & 6 & 10 & 30 & 6 \\ 11 & 21 & 0 & 15 & 30 & 76 & 16 \\ 3 & 4 & -4 & 3 & 6 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{114}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -988, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.40 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 0 & 5 & 4 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 30 & 4 \\ 2 & 4 & 0 & 3 & 1 & 21 & 3 \\ 17 & 31 & 0 & 30 & 21 & 172 & 24 \\ 3 & 4 & -4 & 4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{1112}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1026, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 10 & 3 \\ 3 & 4 & 0 & 4 & 9 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 12 & 3 \\ 5 & 9 & 0 & 6 & 13 & 30 & 7 \\ 10 & 18 & 0 & 12 & 30 & 58 & 14 \\ 3 & 4 & -4 & 3 & 7 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{1134}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1066, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 17 & 3 \\ 3 & 4 & 0 & 4 & 17 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 21 & 3 \\ 9 & 17 & 0 & 12 & 49 & 93 & 13 \\ 17 & 31 & 0 & 21 & 93 & 172 & 24 \\ 3 & 4 & -4 & 3 & 13 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{1112}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1080, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.35 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 15 & 21 & 7 \\ 5 & 9 & 0 & 15 & 13 & 21 & 7 \\ 7 & 13 & 0 & 21 & 21 & 28 & 10 \\ 3 & 4 & -4 & 7 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{1113}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1096, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.36 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 \\ 12 & 22 & 0 & 15 & 15 & 85 & 17 \\ 3 & 4 & -4 & 3 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{1122}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1132, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.35 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 9 & 3 \\ 7 & 13 & 0 & 9 & 28 & 30 & 10 \\ 7 & 13 & 0 & 9 & 30 & 28 & 10 \\ 3 & 4 & -4 & 3 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

[92] $1^{1224}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1180$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.39 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 14 & 3 \\ 3 & 4 & 0 & 8 & 13 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 18 & 36 & 6 \\ 7 & 13 & 0 & 18 & 28 & 60 & 10 \\ 14 & 26 & 0 & 36 & 60 & 118 & 20 \\ 3 & 4 & -4 & 6 & 10 & 20 & 0 \end{pmatrix}_{b_0}$$

[93] $1^{1144}31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1216$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.46 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 11 & 3 \\ 3 & 4 & 0 & 9 & 21 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 33 & 33 & 7 \\ 11 & 21 & 0 & 33 & 76 & 78 & 16 \\ 11 & 21 & 0 & 33 & 78 & 76 & 16 \\ 3 & 4 & -4 & 7 & 16 & 16 & 0 \end{pmatrix}_{b_0}$$

[94] $1^{1114}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1232$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.42 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 \\ 11 & 21 & 0 & 15 & 15 & 76 & 16 \\ 3 & 4 & -4 & 3 & 3 & 16 & 0 \end{pmatrix}_{b_0}$$

[95] $1^{1143}31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1248$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.42 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 9 & 3 \\ 3 & 4 & 0 & 9 & 21 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 33 & 27 & 7 \\ 11 & 21 & 0 & 33 & 76 & 63 & 16 \\ 9 & 17 & 0 & 27 & 63 & 49 & 13 \\ 3 & 4 & -4 & 7 & 16 & 13 & 0 \end{pmatrix}_{b_0}$$

[96] $1^{114}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1250$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.48 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 0 & 5 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 36 & 4 \\ 2 & 4 & 0 & 3 & 1 & 24 & 3 \\ 20 & 36 & 0 & 36 & 24 & 238 & 28 \\ 3 & 4 & -4 & 4 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

[97] $1^{1124}31^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 14 & 3 \\ 3 & 4 & 0 & 4 & 18 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 18 & 3 \\ 10 & 18 & 0 & 12 & 58 & 84 & 14 \\ 14 & 26 & 0 & 18 & 84 & 118 & 20 \\ 3 & 4 & -4 & 3 & 14 & 20 & 0 \end{pmatrix}_{b_0}$$

[98] $1^{1123}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1322$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 12.41 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 15 & 3 \\ 3 & 4 & 0 & 4 & 13 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 18 & 3 \\ 7 & 13 & 0 & 9 & 28 & 63 & 10 \\ 15 & 27 & 0 & 18 & 63 & 133 & 21 \\ 3 & 4 & -4 & 3 & 10 & 21 & 0 \end{pmatrix}_{b_0}$$

[99] $1^{1123}31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1328$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 9 & 3 \\ 3 & 4 & 0 & 9 & 13 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 21 & 27 & 7 \\ 7 & 13 & 0 & 21 & 28 & 39 & 10 \\ 9 & 17 & 0 & 27 & 39 & 49 & 13 \\ 3 & 4 & -4 & 7 & 10 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[100] \ 1^{1123} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -1354, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 9 & 3 \\ 3 & 4 & 0 & 4 & 13 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 12 & 3 \\ 7 & 13 & 0 & 9 & 28 & 39 & 10 \\ 9 & 17 & 0 & 12 & 39 & 49 & 13 \\ 3 & 4 & -4 & 3 & 10 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[101] \ 1^{1123} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -1376, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.36 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 21 & 21 & 7 \\ 7 & 13 & 0 & 21 & 28 & 30 & 10 \\ 7 & 13 & 0 & 21 & 30 & 28 & 10 \\ 3 & 4 & -4 & 7 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[102] \ 1^{1114} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -1376, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.45 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 24 & 3 \\ 2 & 4 & 0 & 3 & 1 & 24 & 3 \\ 20 & 36 & 0 & 24 & 24 & 238 & 28 \\ 3 & 4 & -4 & 3 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{1133} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -1408, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 12 & 3 \\ 3 & 4 & 0 & 4 & 17 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 15 & 3 \\ 9 & 17 & 0 & 12 & 49 & 66 & 13 \\ 12 & 22 & 0 & 15 & 66 & 85 & 17 \\ 3 & 4 & -4 & 3 & 13 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{1124} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -1412, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.41 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 11 & 3 \\ 3 & 4 & 0 & 4 & 13 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 15 & 3 \\ 7 & 13 & 0 & 9 & 28 & 48 & 10 \\ 11 & 21 & 0 & 15 & 48 & 76 & 16 \\ 3 & 4 & -4 & 3 & 10 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{1133} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -1460, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.39 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 9 & 3 \\ 3 & 4 & 0 & 9 & 17 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 27 & 27 & 7 \\ 9 & 17 & 0 & 27 & 49 & 51 & 13 \\ 9 & 17 & 0 & 27 & 51 & 49 & 13 \\ 3 & 4 & -4 & 7 & 13 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{1122} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -1512, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 12 & 3 \\ 10 & 18 & 0 & 12 & 58 & 60 & 14 \\ 10 & 18 & 0 & 12 & 60 & 58 & 14 \\ 3 & 4 & -4 & 3 & 14 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[107] \ 1^{1113} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -1520, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.37 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 12 & 3 \\ 3 & 4 & 0 & 4 & 9 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 15 & 3 \\ 5 & 9 & 0 & 6 & 13 & 36 & 7 \\ 12 & 22 & 0 & 15 & 36 & 85 & 17 \\ 3 & 4 & -4 & 3 & 7 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[108] \quad 1^{1114} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1696, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 4 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 1 & 18 & 3 \\ 14 & 26 & 0 & 18 & 18 & 118 & 20 \\ 3 & 4 & -4 & 3 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[109] \quad 1^{1144} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1706, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.41 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 14 & 3 \\ 3 & 4 & 0 & 4 & 21 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 18 & 3 \\ 11 & 21 & 0 & 15 & 76 & 96 & 16 \\ 14 & 26 & 0 & 18 & 96 & 118 & 20 \\ 3 & 4 & -4 & 3 & 16 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[111] \quad 1^{1113} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1736, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.49 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 18 & 3 \\ 3 & 4 & 0 & 4 & 9 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 21 & 3 \\ 5 & 9 & 0 & 6 & 13 & 54 & 7 \\ 18 & 32 & 0 & 21 & 54 & 193 & 25 \\ 3 & 4 & -4 & 3 & 7 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[110] \quad 1^{1123} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1736, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 12 & 3 \\ 3 & 4 & 0 & 4 & 18 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 15 & 3 \\ 10 & 18 & 0 & 12 & 58 & 72 & 14 \\ 12 & 22 & 0 & 15 & 72 & 85 & 17 \\ 3 & 4 & -4 & 3 & 14 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[112] \quad 1^{1114} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1744, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 4 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 21 & 3 \\ 2 & 4 & 0 & 3 & 1 & 21 & 3 \\ 17 & 31 & 0 & 21 & 21 & 172 & 24 \\ 3 & 4 & -4 & 3 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[113] \quad 1^{1133} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1756, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 12 & 3 \\ 9 & 17 & 0 & 12 & 49 & 51 & 13 \\ 9 & 17 & 0 & 12 & 51 & 49 & 13 \\ 3 & 4 & -4 & 3 & 13 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[114] \quad 1^{1224} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1772, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.40 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 11 & 3 \\ 3 & 4 & 0 & 13 & 13 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 30 & 48 & 10 \\ 7 & 13 & 0 & 30 & 28 & 48 & 10 \\ 11 & 21 & 0 & 48 & 48 & 76 & 16 \\ 3 & 4 & -4 & 10 & 10 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[115] \quad 1^{1123} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1816, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 12 & 3 \\ 3 & 4 & 0 & 4 & 13 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 15 & 3 \\ 7 & 13 & 0 & 9 & 28 & 51 & 10 \\ 12 & 22 & 0 & 15 & 51 & 85 & 17 \\ 3 & 4 & -4 & 3 & 10 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[116] \quad 1^{1124}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1826, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 20 & 3 \\ 3 & 4 & 0 & 4 & 13 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 24 & 3 \\ 7 & 13 & 0 & 9 & 28 & 84 & 10 \\ 20 & 36 & 0 & 24 & 84 & 238 & 28 \\ 3 & 4 & -4 & 3 & 10 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[117] \quad 1^{1113}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1836, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.41 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 15 & 3 \\ 3 & 4 & 0 & 4 & 9 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 18 & 3 \\ 5 & 9 & 0 & 6 & 13 & 45 & 7 \\ 15 & 27 & 0 & 18 & 45 & 133 & 21 \\ 3 & 4 & -4 & 3 & 7 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[118] \quad 1^{1114}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1850, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 14 & 3 \\ 3 & 4 & 0 & 4 & 9 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 18 & 3 \\ 5 & 9 & 0 & 6 & 13 & 42 & 7 \\ 14 & 26 & 0 & 18 & 42 & 118 & 20 \\ 3 & 4 & -4 & 3 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[119] \quad 1^{1222}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1852, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 30 & 30 & 10 \\ 7 & 13 & 0 & 30 & 28 & 30 & 10 \\ 7 & 13 & 0 & 30 & 30 & 28 & 10 \\ 3 & 4 & -4 & 10 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[120] \quad 1^{1223}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1984, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 9 & 3 \\ 3 & 4 & 0 & 13 & 13 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 30 & 39 & 10 \\ 7 & 13 & 0 & 30 & 28 & 39 & 10 \\ 9 & 17 & 0 & 39 & 39 & 49 & 13 \\ 3 & 4 & -4 & 10 & 10 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[121] \quad 1^{1134}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1994, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.42 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 11 & 3 \\ 3 & 4 & 0 & 4 & 17 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 15 & 3 \\ 9 & 17 & 0 & 12 & 49 & 63 & 13 \\ 11 & 21 & 0 & 15 & 63 & 76 & 16 \\ 3 & 4 & -4 & 3 & 13 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[122] \quad 1^{1144}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2412, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 11 & 3 \\ 3 & 4 & 0 & 4 & 21 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 15 & 3 \\ 11 & 21 & 0 & 15 & 76 & 78 & 16 \\ 11 & 21 & 0 & 15 & 78 & 76 & 16 \\ 3 & 4 & -4 & 3 & 16 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[123] \quad 1^{1134}31^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2458, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 14 & 3 \\ 3 & 4 & 0 & 4 & 17 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 18 & 3 \\ 9 & 17 & 0 & 12 & 49 & 78 & 13 \\ 14 & 26 & 0 & 18 & 78 & 118 & 20 \\ 3 & 4 & -4 & 3 & 13 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[124] \quad 1^{1124}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2506, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 14 & 3 \\ 3 & 4 & 0 & 4 & 13 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 18 & 3 \\ 7 & 13 & 0 & 9 & 28 & 60 & 10 \\ 14 & 26 & 0 & 18 & 60 & 118 & 20 \\ 3 & 4 & -4 & 3 & 10 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[125] \quad 1^{1114}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2618, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 17 & 3 \\ 3 & 4 & 0 & 4 & 9 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 21 & 3 \\ 5 & 9 & 0 & 6 & 13 & 51 & 7 \\ 17 & 31 & 0 & 21 & 51 & 172 & 24 \\ 3 & 4 & -4 & 3 & 7 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[126] \quad 1^{1124}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2644, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 17 & 3 \\ 3 & 4 & 0 & 4 & 13 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 21 & 3 \\ 7 & 13 & 0 & 9 & 28 & 72 & 10 \\ 17 & 31 & 0 & 21 & 72 & 172 & 24 \\ 3 & 4 & -4 & 3 & 10 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[127] \quad 1^{1133}31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2680, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 12 & 3 \\ 3 & 4 & 0 & 4 & 22 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 15 & 3 \\ 12 & 22 & 0 & 15 & 85 & 87 & 17 \\ 12 & 22 & 0 & 15 & 87 & 85 & 17 \\ 3 & 4 & -4 & 3 & 17 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[128] \quad 1^{1134}31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2920, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 14 & 3 \\ 3 & 4 & 0 & 4 & 22 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 18 & 3 \\ 12 & 22 & 0 & 15 & 85 & 102 & 17 \\ 14 & 26 & 0 & 18 & 102 & 118 & 20 \\ 3 & 4 & -4 & 3 & 17 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[129] \quad 1^{1114}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2970, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.42 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 20 & 3 \\ 3 & 4 & 0 & 4 & 9 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 24 & 3 \\ 5 & 9 & 0 & 6 & 13 & 60 & 7 \\ 20 & 36 & 0 & 24 & 60 & 238 & 28 \\ 3 & 4 & -4 & 3 & 7 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[130] \quad 1^{1144}31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -3880, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 14 & 3 \\ 3 & 4 & 0 & 4 & 26 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 18 & 18 & 3 \\ 14 & 26 & 0 & 18 & 118 & 120 & 20 \\ 14 & 26 & 0 & 18 & 120 & 118 & 20 \\ 3 & 4 & -4 & 3 & 20 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [131] \quad 1^{1111}31^{112}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 484, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 15 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 6 & 9 & 3 \\ 3 & 9 & 6 & 0 & 6 & 13 & 24 & 7 \\ 5 & 15 & 9 & 0 & 9 & 24 & 37 & 11 \\ 2 & 4 & 2 & -4 & 3 & 7 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[132] \quad 1^{1111}31^{1111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 556, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 6 & 6 & 3 \\ 3 & 9 & 6 & 0 & 6 & 13 & 15 & 7 \\ 3 & 9 & 6 & 0 & 6 & 15 & 13 & 7 \\ 2 & 4 & 2 & -4 & 3 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[133] \quad 1^{11}31^{1111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 652, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 6 & 9 & 4 \\ 2 & 6 & 3 & 0 & 6 & 4 & 9 & 4 \\ 3 & 9 & 6 & 0 & 9 & 9 & 13 & 7 \\ 2 & 4 & 2 & -4 & 4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[134] \quad 1^{11}31^{1111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 880, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 6 & 6 & 4 \\ 2 & 6 & 3 & 0 & 6 & 4 & 6 & 4 \\ 2 & 6 & 3 & 0 & 6 & 6 & 4 & 4 \\ 2 & 4 & 2 & -4 & 4 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[135] \quad 1^{1111}31^{122}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 952, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 15 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 9 & 9 & 3 \\ 5 & 15 & 9 & 0 & 9 & 37 & 39 & 11 \\ 5 & 15 & 9 & 0 & 9 & 39 & 37 & 11 \\ 2 & 4 & 2 & -4 & 3 & 11 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[136] \quad 1^{1111}31^{1111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 964, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 9 & 9 & 4 \\ 3 & 9 & 6 & 0 & 9 & 13 & 15 & 7 \\ 3 & 9 & 6 & 0 & 9 & 15 & 13 & 7 \\ 2 & 4 & 2 & -4 & 4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[137] \quad 1^{1111}31^{1111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1816, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 15 & 15 & 7 \\ 3 & 9 & 6 & 0 & 15 & 13 & 15 & 7 \\ 3 & 9 & 6 & 0 & 15 & 15 & 13 & 7 \\ 2 & 4 & 2 & -4 & 7 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

4.82 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$

$$T = 10 \quad [0] \quad 1^{1111}31^{1111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 372, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 27 & 7 \\ 9 & 17 & 0 & 15 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 27 & 49 & 13 \\ 5 & 8 & -4 & 7 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

4.83 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] 1^{\overline{1111}} 3 1^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 148, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 9 & 3 & 4 \\ 3 & 5 & 0 & 6 & 4 & 9 & 3 & 4 \\ 5 & 9 & 0 & 3 & 9 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & -4 & 4 & 4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] 1^{\overline{1111}} 3 1^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 9 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 9 & 6 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & -4 & 4 & 3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] 1^{\overline{1111}} 3 1^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 14 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 6 & 3 & 3 \\ 8 & 14 & 0 & 9 & 37 & 24 & 9 & 11 \\ 5 & 9 & 0 & 6 & 24 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 9 & 6 & 0 & 2 \\ 3 & 4 & -4 & 3 & 11 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] 1^{\overline{1121}} 3 1^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 274, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 10 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 18 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 18 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 & 3 \\ 10 & 18 & 0 & 18 & 12 & 58 & 12 & 14 \\ 2 & 4 & 0 & 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & -4 & 4 & 3 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] 1^{\overline{1131}} 3 1^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 454, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 15 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 27 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 27 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 18 & 3 & 3 \\ 15 & 27 & 0 & 27 & 18 & 133 & 18 & 21 \\ 2 & 4 & 0 & 3 & 3 & 18 & 0 & 2 \\ 3 & 4 & -4 & 4 & 3 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & -4 & 3 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[6] 1^{\overline{1131}} 3 1^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 12 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 22 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 21 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 & 3 \\ 12 & 22 & 0 & 21 & 15 & 85 & 15 & 17 \\ 2 & 4 & 0 & 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & -4 & 4 & 3 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] 1^{\overline{1141}} 3 1^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 706, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$(1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 6 & 3 & 3 \\ 5 & 9 & 0 & 6 & 13 & 15 & 6 & 7 \\ 5 & 9 & 0 & 6 & 15 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 6 & 0 & 2 \\ 3 & 4 & -4 & 3 & 7 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 2368, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 9 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 15 & 15 & 6 & 7 \\ 5 & 9 & 0 & 15 & 13 & 15 & 6 & 7 \\ 5 & 9 & 0 & 15 & 15 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 6 & 6 & 0 & 2 \\ 3 & 4 & -4 & 7 & 7 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

4.84 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{111} 3^1 1^{11} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 329, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 6 & 2 & 3 \\ 3 & 9 & 6 & 0 & 6 & 13 & 5 & 7 \\ 1 & 3 & 2 & 0 & 2 & 5 & 0 & 2 \\ 2 & 4 & 2 & -4 & 3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{12} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 464, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 9 & 2 & 3 \\ 5 & 15 & 9 & 0 & 9 & 37 & 8 & 11 \\ 1 & 3 & 2 & 0 & 2 & 8 & 0 & 2 \\ 2 & 4 & 2 & -4 & 3 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{13} 1^{111} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 604, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 6 & 3 & 4 \\ 2 & 6 & 3 & 0 & 6 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 3 & 3 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{111} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 669, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 9 & 3 & 4 \\ 3 & 9 & 6 & 0 & 9 & 13 & 5 & 7 \\ 1 & 3 & 2 & 0 & 3 & 5 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{111} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 1184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 6 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 15 & 5 & 7 \\ 3 & 9 & 6 & 0 & 15 & 13 & 5 & 7 \\ 1 & 3 & 2 & 0 & 5 & 5 & 0 & 2 \\ 2 & 4 & 2 & -4 & 7 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

4.85 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{11} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.86 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 3 & 5 & 0 & 6 & 3 & 3 & 3 \\ 3 & 4 & -4 & 4 & 3 & 3 & -4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -56, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.71 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 3 & 5 & 0 & 6 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -56, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.71 \rightarrow 17, \ T_{\max} = 23, \ T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 8 & 14 & 0 & 9 & 9 & 36 & 10 \\ 3 & 4 & -4 & 3 & 3 & 10 & -4 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -65, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 4 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1211}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -90, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.07 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 12 & 6 \\ 5 & 9 & 0 & 12 & 13 & 15 & 7 \\ 5 & 9 & 0 & 12 & 15 & 12 & 6 \\ 3 & 4 & -4 & 6 & 7 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -92, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 & 3 \\ 8 & 14 & 0 & 9 & 37 & 9 & 11 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -4 & 3 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -101, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 0 & 5 & 19 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 21 & 3 & 4 \\ 11 & 19 & 0 & 21 & 73 & 12 & 15 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -4 & 4 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{113}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -107, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.32 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 5 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 15 & 4 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 \\ 9 & 17 & 0 & 15 & 12 & 48 & 12 \\ 3 & 4 & -4 & 4 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1111}} 3^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 2 & 3 \\ 3 & 4 & 0 & 9 & 19 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 33 & 6 & 7 \\ 11 & 19 & 0 & 33 & 73 & 12 & 15 \\ 2 & 4 & 0 & 6 & 12 & 0 & 2 \\ 3 & 4 & -4 & 7 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1211} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.34 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 6 & 6 \\ 5 & 9 & 0 & 12 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 6 & 0 & 2 \\ 3 & 4 & -4 & 6 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 4 & -4 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1121} 3^1 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -137, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 7 & 2 & 3 \\ 3 & 4 & 0 & 14 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 33 & 9 & 11 \\ 7 & 13 & 0 & 33 & 28 & 9 & 10 \\ 2 & 4 & 0 & 9 & 9 & 0 & 2 \\ 3 & 4 & -4 & 11 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{131} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -173, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 15 & 3 & 4 \\ 9 & 17 & 0 & 15 & 49 & 12 & 13 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -4 & 4 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -174, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.85 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 5 & 3 \\ 3 & 4 & 0 & 4 & 8 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 6 & 3 \\ 4 & 8 & 0 & 6 & 10 & 12 & 6 \\ 5 & 9 & 0 & 6 & 12 & 12 & 6 \\ 3 & 4 & -4 & 3 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -191, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 4 & 6 & 0 & 4 \\ 4 & 8 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1221} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -200, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 13 & 2 & 3 \\ 3 & 4 & 0 & 13 & 23 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 54 & 9 & 10 \\ 13 & 23 & 0 & 54 & 100 & 15 & 18 \\ 2 & 4 & 0 & 9 & 15 & 0 & 2 \\ 3 & 4 & -4 & 10 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -219, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 9 & 6 & 12 & 6 \\ 3 & 4 & -4 & 4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -245, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 12 & 4 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 7 & 13 & 0 & 12 & 9 & 27 & 9 \\ 3 & 4 & -4 & 4 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -284, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 4 & 6 & 0 & 4 \\ 3 & 8 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -341, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 0 & 9 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & -4 & 4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -344, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.81 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -353, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 0 & 5 & 14 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 15 & 3 & 4 \\ 8 & 14 & 0 & 15 & 37 & 9 & 11 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -4 & 4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -359, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 3 \\ 3 & 4 & 0 & 4 & 9 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 6 & 13 & 9 & 7 \\ 3 & 5 & 0 & 3 & 9 & 3 & 3 \\ 3 & 4 & -4 & 3 & 7 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -368, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 3 & 0 & -1 \\ 4 & 8 & 3 & 4 & 6 & 0 & 4 \\ 4 & 8 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{121} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -380, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 12 & 3 & 4 \\ 7 & 13 & 0 & 12 & 28 & 9 & 10 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -4 & 4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{1121}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 & 3 \\ 4 & 8 & 0 & 6 & 10 & 6 & 6 \\ 2 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & -4 & 3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{1111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -408, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & -4 & 3 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{114}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -413, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 & 3 \\ 3 & 4 & 0 & 4 & 21 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 3 & 3 \\ 11 & 21 & 0 & 15 & 76 & 18 & 16 \\ 3 & 5 & 0 & 3 & 18 & 3 & 3 \\ 3 & 4 & -4 & 3 & 16 & 3 & -1 \end{pmatrix}_{b_0}$$

[28] $1^{1221}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 6 & 6 \\ 4 & 8 & 0 & 12 & 10 & 6 & 6 \\ 2 & 4 & 0 & 6 & 6 & 0 & 2 \\ 3 & 4 & -4 & 6 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

[29] $1^{1121}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -425, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 2 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 3 & 3 \\ 13 & 23 & 0 & 15 & 100 & 15 & 18 \\ 2 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & -4 & 3 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{1131}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 2 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 & 3 \\ 6 & 12 & 0 & 9 & 25 & 9 & 9 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -4 & 3 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

[31] $1^{1221}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 5 & 3 \\ 3 & 4 & 0 & 8 & 13 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 18 & 12 & 6 \\ 7 & 13 & 0 & 18 & 28 & 21 & 10 \\ 5 & 9 & 0 & 12 & 21 & 12 & 6 \\ 3 & 4 & -4 & 6 & 10 & 6 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{1111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -485, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & -4 & 3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{1131}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -488$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.87 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 2 & 3 \\ 3 & 4 & 0 & 14 & 22 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 57 & 9 & 11 \\ 12 & 22 & 0 & 57 & 85 & 15 & 17 \\ 2 & 4 & 0 & 9 & 15 & 0 & 2 \\ 3 & 4 & -4 & 11 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{1141}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -524$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.34 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 3 & 3 \\ 8 & 16 & 0 & 12 & 46 & 12 & 12 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -4 & 3 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

[35] $1^{112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -533$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.03 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 & 3 \\ 3 & 4 & 0 & 4 & 13 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 & 3 \\ 7 & 13 & 0 & 9 & 28 & 12 & 10 \\ 3 & 5 & 0 & 3 & 12 & 3 & 3 \\ 3 & 4 & -4 & 3 & 10 & 3 & -1 \end{pmatrix}_{b_0}$$

[36] $1^{113}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -551$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.06 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 & 3 \\ 3 & 4 & 0 & 4 & 17 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 3 & 3 \\ 9 & 17 & 0 & 12 & 49 & 15 & 13 \\ 3 & 5 & 0 & 3 & 15 & 3 & 3 \\ 3 & 4 & -4 & 3 & 13 & 3 & -1 \end{pmatrix}_{b_0}$$

[37] $1^{1221}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -560$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 2 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 18 & 6 & 6 \\ 7 & 13 & 0 & 18 & 28 & 9 & 10 \\ 2 & 4 & 0 & 6 & 9 & 0 & 2 \\ 3 & 4 & -4 & 6 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{1111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -729$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 15 & 12 & 6 \\ 3 & 4 & -4 & 3 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{1111}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -773$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 2 & 3 \\ 3 & 4 & 0 & 9 & 14 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 24 & 6 & 7 \\ 8 & 14 & 0 & 24 & 37 & 9 & 11 \\ 2 & 4 & 0 & 6 & 9 & 0 & 2 \\ 3 & 4 & -4 & 7 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{121}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -800$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 0 & 5 & 18 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 18 & 3 & 4 \\ 10 & 18 & 0 & 18 & 58 & 12 & 14 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -4 & 4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{121} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -848, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 0 & 5 & 28 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 30 & 3 & 4 \\ 16 & 28 & 0 & 30 & 154 & 18 & 22 \\ 2 & 4 & 0 & 3 & 18 & 0 & 2 \\ 3 & 4 & -4 & 4 & 22 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 15 & 6 & 7 \\ 5 & 9 & 0 & 15 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 6 & 0 & 2 \\ 3 & 4 & -4 & 7 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1111} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -920, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 2 & 3 \\ 3 & 4 & 0 & 14 & 14 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 39 & 9 & 11 \\ 8 & 14 & 0 & 39 & 37 & 9 & 11 \\ 2 & 4 & 0 & 9 & 9 & 0 & 2 \\ 3 & 4 & -4 & 11 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1121} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -929, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 & 3 \\ 7 & 13 & 0 & 9 & 28 & 9 & 10 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -4 & 3 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1121} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 3 & 3 \\ 10 & 18 & 0 & 12 & 58 & 12 & 14 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -4 & 3 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{121} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 0 & 5 & 23 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 24 & 3 & 4 \\ 13 & 23 & 0 & 24 & 100 & 15 & 18 \\ 2 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & -4 & 4 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{141} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.91 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 0 & 5 & 26 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 24 & 3 & 4 \\ 14 & 26 & 0 & 24 & 118 & 18 & 20 \\ 2 & 4 & 0 & 3 & 18 & 0 & 2 \\ 3 & 4 & -4 & 4 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{131} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1001, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 0 & 5 & 22 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 21 & 3 & 4 \\ 12 & 22 & 0 & 21 & 85 & 15 & 17 \\ 2 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & -4 & 4 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1131^1} 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 2 & 3 \\ 3 & 4 & 0 & 4 & 32 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 21 & 3 & 3 \\ 18 & 32 & 0 & 21 & 193 & 21 & 25 \\ 2 & 4 & 0 & 3 & 21 & 0 & 2 \\ 3 & 4 & -4 & 3 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1121^1} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1083, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 5 & 3 \\ 3 & 4 & 0 & 4 & 13 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 6 & 3 \\ 7 & 13 & 0 & 9 & 28 & 21 & 10 \\ 5 & 9 & 0 & 6 & 21 & 12 & 6 \\ 3 & 4 & -4 & 3 & 10 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1141^1} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1085, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 2 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 33 & 6 & 7 \\ 11 & 21 & 0 & 33 & 76 & 15 & 16 \\ 2 & 4 & 0 & 6 & 15 & 0 & 2 \\ 3 & 4 & -4 & 7 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1121^1} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1205, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 2 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 21 & 6 & 7 \\ 7 & 13 & 0 & 21 & 28 & 9 & 10 \\ 2 & 4 & 0 & 6 & 9 & 0 & 2 \\ 3 & 4 & -4 & 7 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1121^1} 31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 10 & 2 & 3 \\ 3 & 4 & 0 & 14 & 18 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 48 & 9 & 11 \\ 10 & 18 & 0 & 48 & 58 & 12 & 14 \\ 2 & 4 & 0 & 9 & 12 & 0 & 2 \\ 3 & 4 & -4 & 11 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1121^1} 31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 16 & 2 & 3 \\ 3 & 4 & 0 & 9 & 28 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 48 & 6 & 7 \\ 16 & 28 & 0 & 48 & 154 & 18 & 22 \\ 2 & 4 & 0 & 6 & 18 & 0 & 2 \\ 3 & 4 & -4 & 7 & 22 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1131^1} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 2 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 27 & 6 & 7 \\ 9 & 17 & 0 & 27 & 49 & 12 & 13 \\ 2 & 4 & 0 & 6 & 12 & 0 & 2 \\ 3 & 4 & -4 & 7 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1131^1} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1281, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 5 & 3 \\ 3 & 4 & 0 & 4 & 17 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 6 & 3 \\ 9 & 17 & 0 & 12 & 49 & 27 & 13 \\ 5 & 9 & 0 & 6 & 27 & 12 & 6 \\ 3 & 4 & -4 & 3 & 13 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1141} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1323, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 5 & 3 \\ 3 & 4 & 0 & 4 & 21 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 6 & 3 \\ 11 & 21 & 0 & 15 & 76 & 33 & 16 \\ 5 & 9 & 0 & 6 & 33 & 12 & 6 \\ 3 & 4 & -4 & 3 & 16 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1131} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1397, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 3 & 3 \\ 9 & 17 & 0 & 12 & 49 & 12 & 13 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -4 & 3 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{131} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1565, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 0 & 5 & 27 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 27 & 3 & 4 \\ 15 & 27 & 0 & 27 & 133 & 18 & 21 \\ 2 & 4 & 0 & 3 & 18 & 0 & 2 \\ 3 & 4 & -4 & 4 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{1221} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 2 & 3 \\ 3 & 4 & 0 & 13 & 18 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 42 & 9 & 10 \\ 10 & 18 & 0 & 42 & 58 & 12 & 14 \\ 2 & 4 & 0 & 9 & 12 & 0 & 2 \\ 3 & 4 & -4 & 10 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{1131} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 2 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 18 & 3 & 3 \\ 15 & 27 & 0 & 18 & 133 & 18 & 21 \\ 2 & 4 & 0 & 3 & 18 & 0 & 2 \\ 3 & 4 & -4 & 3 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{1121} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1760, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 2 & 3 \\ 3 & 4 & 0 & 9 & 18 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 30 & 6 & 7 \\ 10 & 18 & 0 & 30 & 58 & 12 & 14 \\ 2 & 4 & 0 & 6 & 12 & 0 & 2 \\ 3 & 4 & -4 & 7 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{1121} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1781, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 2 & 3 \\ 3 & 4 & 0 & 9 & 23 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 39 & 6 & 7 \\ 13 & 23 & 0 & 39 & 100 & 15 & 18 \\ 2 & 4 & 0 & 6 & 15 & 0 & 2 \\ 3 & 4 & -4 & 7 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{1221} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1784, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 30 & 9 & 10 \\ 7 & 13 & 0 & 30 & 28 & 9 & 10 \\ 2 & 4 & 0 & 9 & 9 & 0 & 2 \\ 3 & 4 & -4 & 10 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{1131^1}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1820, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$

$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 2 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 3 & 3 \\ 12 & 22 & 0 & 15 & 85 & 15 & 17 \\ 2 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & -4 & 3 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

[66] $1^{131^1}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1865, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$

$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 0 & 5 & 32 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 33 & 3 & 4 \\ 18 & 32 & 0 & 33 & 193 & 21 & 25 \\ 2 & 4 & 0 & 3 & 21 & 0 & 2 \\ 3 & 4 & -4 & 4 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{1141^1}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1889, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$

$T_{\min} = 12.86 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 3 & 3 \\ 11 & 21 & 0 & 15 & 76 & 15 & 16 \\ 2 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & -4 & 3 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{141^1}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1928, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$

$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 0 & 5 & 31 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 30 & 3 & 4 \\ 17 & 31 & 0 & 30 & 172 & 21 & 24 \\ 2 & 4 & 0 & 3 & 21 & 0 & 2 \\ 3 & 4 & -4 & 4 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{1131^1}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2501, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$

$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 2 & 3 \\ 3 & 4 & 0 & 9 & 22 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 36 & 6 & 7 \\ 12 & 22 & 0 & 36 & 85 & 15 & 17 \\ 2 & 4 & 0 & 6 & 15 & 0 & 2 \\ 3 & 4 & -4 & 7 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{1221^1}31^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2624, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$

$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 10 & 2 & 3 \\ 3 & 4 & 0 & 18 & 18 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 10 & 18 & 0 & 58 & 60 & 12 & 14 \\ 10 & 18 & 0 & 60 & 58 & 12 & 14 \\ 2 & 4 & 0 & 12 & 12 & 0 & 2 \\ 3 & 4 & -4 & 14 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{141^1}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2636, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 20 & 2 & 3 \\ 3 & 4 & 0 & 5 & 36 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 36 & 3 & 4 \\ 20 & 36 & 0 & 36 & 238 & 24 & 28 \\ 2 & 4 & 0 & 3 & 24 & 0 & 2 \\ 3 & 4 & -4 & 4 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{1141^1}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2720, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$

$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 18 & 3 & 3 \\ 14 & 26 & 0 & 18 & 118 & 18 & 20 \\ 2 & 4 & 0 & 3 & 18 & 0 & 2 \\ 3 & 4 & -4 & 3 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{1141} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2780, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 2 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 24 & 3 & 3 \\ 20 & 36 & 0 & 24 & 238 & 24 & 28 \\ 2 & 4 & 0 & 3 & 24 & 0 & 2 \\ 3 & 4 & -4 & 3 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{1141} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2996, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 2 & 3 \\ 3 & 4 & 0 & 9 & 26 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 42 & 6 & 7 \\ 14 & 26 & 0 & 42 & 118 & 18 & 20 \\ 2 & 4 & 0 & 6 & 18 & 0 & 2 \\ 3 & 4 & -4 & 7 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{1141} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3017, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 2 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 21 & 3 & 3 \\ 17 & 31 & 0 & 21 & 172 & 21 & 24 \\ 2 & 4 & 0 & 3 & 21 & 0 & 2 \\ 3 & 4 & -4 & 3 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{1131} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 2 & 3 \\ 3 & 4 & 0 & 9 & 27 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 45 & 6 & 7 \\ 15 & 27 & 0 & 45 & 133 & 18 & 21 \\ 2 & 4 & 0 & 6 & 18 & 0 & 2 \\ 3 & 4 & -4 & 7 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{1131} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3365, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 18 & 2 & 3 \\ 3 & 4 & 0 & 9 & 32 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 54 & 6 & 7 \\ 18 & 32 & 0 & 54 & 193 & 21 & 25 \\ 2 & 4 & 0 & 6 & 21 & 0 & 2 \\ 3 & 4 & -4 & 7 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{1141} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -4373, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 17 & 2 & 3 \\ 3 & 4 & 0 & 9 & 31 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 51 & 6 & 7 \\ 17 & 31 & 0 & 51 & 172 & 21 & 24 \\ 2 & 4 & 0 & 6 & 21 & 0 & 2 \\ 3 & 4 & -4 & 7 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{1141} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -5216, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 20 & 2 & 3 \\ 3 & 4 & 0 & 9 & 36 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 60 & 6 & 7 \\ 20 & 36 & 0 & 60 & 238 & 24 & 28 \\ 2 & 4 & 0 & 6 & 24 & 0 & 2 \\ 3 & 4 & -4 & 7 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [80] \quad 1^{111} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 44, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{1111} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 72, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{1111} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 82, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.90 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 3 \\ 3 & 4 & 1 & 3 & 9 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 3 & 0 \\ 5 & 9 & 0 & 7 & 14 & 9 & 8 \\ 3 & 6 & 0 & 3 & 9 & 4 & 4 \\ 3 & 4 & -1 & 0 & 8 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[83] \ 1^{1111} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 111, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.62 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 6 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 6 & 3 & 3 \\ 3 & 6 & 9 & 0 & 6 & 13 & 6 & 7 \\ 1 & 3 & 3 & 0 & 3 & 6 & 0 & 2 \\ 2 & 3 & 3 & -4 & 3 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{11} 31^{112} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 138, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.35 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 12 & 3 & 4 \\ 4 & 12 & 6 & 0 & 12 & 22 & 6 & 8 \\ 1 & 3 & 3 & 0 & 3 & 6 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{1111} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 224, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.89 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 7 & 0 \\ 4 & 6 & 0 & 3 & 4 & 9 & 4 \\ 6 & 9 & 0 & 7 & 9 & 14 & 8 \\ 3 & 4 & -1 & 0 & 4 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[86] \ 1^{111} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 239, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.82 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 6 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 7 & 0 & 14 & 8 \\ 3 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[87] \ 1^{1111} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 268, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.97 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[88] \ 1^{111} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 269, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.86 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{112}1^{11}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 273$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 9 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 12 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 9 & 3 & 3 \\ 4 & 9 & 12 & 0 & 9 & 28 & 9 & 10 \\ 1 & 3 & 3 & 0 & 3 & 9 & 0 & 2 \\ 2 & 3 & 3 & -4 & 3 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

[90] $1^{121}1^{11}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 296$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 12.99 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{11}31^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 300$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.16 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 6 & 3 & 4 \\ 2 & 6 & 3 & 0 & 6 & 4 & 3 & 4 \\ 1 & 3 & 3 & 0 & 3 & 3 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

[92] $1^{111}1^{11}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 320$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{111}31^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 363$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.22 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 9 & 3 & 4 \\ 3 & 9 & 6 & 0 & 9 & 13 & 6 & 7 \\ 1 & 3 & 3 & 0 & 3 & 6 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[94] $1^{121}1^{11}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 374$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 9 & 3 \\ 3 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 10 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 0 & 10 & 0 & 38 & 12 \\ 3 & 4 & -1 & 0 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{111}1^{11}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 381$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[96] $1^{111}1^{11}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 704$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{111}31^{112}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 1065, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 15 & 3 & 4 \\ 5 & 15 & 9 & 0 & 15 & 37 & 9 & 11 \\ 1 & 3 & 3 & 0 & 3 & 9 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 11 & 2 & -1 \end{pmatrix}_{b_0}$$

[98] $1^{111}31^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 1659, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 21 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 12 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 21 & 3 & 4 \\ 7 & 21 & 12 & 0 & 21 & 73 & 12 & 15 \\ 1 & 3 & 3 & 0 & 3 & 12 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 15 & 2 & -1 \end{pmatrix}_{b_0}$$

[99] $1^{111}31^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 2145, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 9 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 27 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 27 & 3 & 4 \\ 9 & 27 & 15 & 0 & 27 & 121 & 15 & 19 \\ 1 & 3 & 3 & 0 & 3 & 15 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 19 & 2 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ [100] $1^{111}4^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 14, \Delta = -82, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 5 & 7 & 0 & 6 & 8 & 5 & 6 \\ 2 & 4 & 0 & 3 & 5 & 0 & 2 \\ 3 & 4 & -3 & 4 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[101] $1^{11}4^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 14, \Delta = -82, \det = -48, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 5 & 2 & 3 \\ 4 & 4 & 0 & 5 & 7 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 5 & 7 & 0 & 6 & 8 & 5 & 6 \\ 2 & 3 & 0 & 3 & 5 & 0 & 2 \\ 3 & 4 & -3 & 4 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

4.86 NHC(-2)(-3) + e₆ + su₂ + su₂ + su₃ + su₂

$T = 10$ [0] $1^{111}31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 148, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 3 & 9 & 4 \\ 3 & 5 & 0 & 6 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 9 & 6 & 13 & 7 \\ 3 & 4 & -4 & 4 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{1111}31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 4 & 3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

- [2] $1^{11111}31^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 & 6 & 3 \\ 8 & 14 & 0 & 9 & 37 & 9 & 24 & 11 \\ 2 & 4 & 0 & 3 & 9 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 24 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 11 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$
- [3] $1^{1112}31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 274, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 18 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 12 & 2 \\ 10 & 18 & 0 & 18 & 12 & 12 & 58 & 14 \\ 3 & 4 & -4 & 4 & 3 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$
- [4] $1^{1113}31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 454, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 27 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 18 & 2 \\ 15 & 27 & 0 & 27 & 18 & 18 & 133 & 21 \\ 3 & 4 & -4 & 4 & 3 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$
- [5] $1^{11111}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 6 & 6 & 12 & 6 & 6 \\ 2 & 4 & 0 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & -4 & 3 & 3 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$
- [6] $1^{1113}31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 21 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 15 & 2 \\ 12 & 22 & 0 & 21 & 15 & 15 & 85 & 17 \\ 3 & 4 & -4 & 4 & 3 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$
- [7] $1^{1114}31^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 706, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 20 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 36 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 & 24 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 24 & 2 \\ 20 & 36 & 0 & 36 & 24 & 24 & 238 & 28 \\ 3 & 4 & -4 & 4 & 3 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{11111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 712, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & -4 & 3 & 3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{11111}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 724, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 6 \\ 5 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 3 & 2 & 7 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1112} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 796, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.84 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 9 & 3 & 12 & 4 \\ 5 & 9 & 0 & 9 & 13 & 6 & 21 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 9 & 2 \\ 7 & 13 & 0 & 12 & 21 & 9 & 28 & 10 \\ 3 & 4 & -4 & 4 & 7 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 988, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 9 & 3 & 9 & 4 \\ 5 & 9 & 0 & 9 & 13 & 6 & 15 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 15 & 6 & 13 & 7 \\ 3 & 4 & -4 & 4 & 7 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1032, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 12 & 15 & 6 \\ 5 & 9 & 0 & 6 & 6 & 15 & 13 & 7 \\ 3 & 4 & -4 & 3 & 3 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1212} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1144, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.87 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 12 & 3 & 12 & 4 \\ 7 & 13 & 0 & 12 & 28 & 9 & 30 & 10 \\ 2 & 4 & 0 & 3 & 9 & 0 & 9 & 2 \\ 7 & 13 & 0 & 12 & 30 & 9 & 28 & 10 \\ 3 & 4 & -4 & 4 & 10 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1114} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1180, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 30 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 & 21 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 21 & 2 \\ 17 & 31 & 0 & 30 & 21 & 21 & 172 & 24 \\ 3 & 4 & -4 & 4 & 3 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1276, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 6 & 15 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 15 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 7 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 2368, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 15 & 6 & 15 & 7 \\ 5 & 9 & 0 & 15 & 13 & 6 & 15 & 7 \\ 2 & 4 & 0 & 6 & 6 & 0 & 6 & 2 \\ 5 & 9 & 0 & 15 & 15 & 6 & 13 & 7 \\ 3 & 4 & -4 & 7 & 7 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

4.87 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} \ 3^{\overline{1111}} \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 48, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 4 & 1 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 0 & 5 \\ 2 & 4 & -3 & 3 & 3 & 0 & 0 & -1 \\ 4 & 8 & 3 & 4 & 6 & 0 & 0 & 4 \\ 4 & 8 & 3 & 6 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 & 4 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} \ 3^{\overline{1111}} \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 228, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} \ 3^{\overline{1111}} \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 231, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 6 & 12 & 6 \\ 3 & 4 & -4 & 4 & 3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} \ 3^{\overline{1111}} \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 507, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 9 & 3 & 3 & 4 \\ 5 & 9 & 0 & 9 & 13 & 6 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 6 & 3 & 0 & 2 \\ 3 & 4 & -4 & 4 & 7 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} \ 3^{\overline{1111}} \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 651, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 & 3 & 3 \\ 5 & 9 & 0 & 6 & 13 & 9 & 6 & 7 \\ 3 & 5 & 0 & 3 & 9 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 6 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 7 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1141}} \ 3^{\overline{1111}} \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 813, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 21 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 3 & 3 & 3 \\ 11 & 21 & 0 & 15 & 76 & 18 & 15 & 16 \\ 3 & 5 & 0 & 3 & 18 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 15 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 16 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1121}} \ 3^{\overline{1111}} \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1173, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 13 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 & 3 & 3 \\ 7 & 13 & 0 & 9 & 28 & 12 & 9 & 10 \\ 3 & 5 & 0 & 3 & 12 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 9 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 10 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1131}} \ 3^{\overline{1111}} \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1227, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 17 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 3 & 3 & 3 \\ 9 & 17 & 0 & 12 & 49 & 15 & 12 & 13 \\ 3 & 5 & 0 & 3 & 15 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 12 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 13 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.88 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{13} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 0 \\ 6 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & 0 & 9 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 16.67 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 & 3 & 5 \\ 2 & 5 & -2 & 3 & 2 & 0 \\ 3 & 5 & 3 & 4 & 3 & 4 \\ 2 & 3 & 2 & 3 & 0 & 2 \\ 3 & 5 & 0 & 4 & 2 & -3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 17.60 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 3 & 5 \\ 2 & 5 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 5 & 0 & 0 & 2 & -4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{12} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 0 \\ 3 & 2 & 0 & 4 & 2 & 4 \\ 3 & 3 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.14 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & 0 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 0 \\ 4 & 2 & 0 & 4 & 2 & 4 \\ 3 & 3 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$(1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 2 & 0 \\ 3 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{13} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -170, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 10 & 0 \\ 8 & 11 & 0 & 10 & 41 & 13 \\ 3 & 3 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{14} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 14 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 12 & 0 \\ 10 & 14 & 0 & 12 & 66 & 16 \\ 3 & 3 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 2 & 0 \\ 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -210, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 1 & 1 & 2 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 0 \\ 5 & 5 & 0 & 6 & 9 & 7 \\ 3 & 3 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{12} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 0 \\ 5 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{13} 3^1 1^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -257, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{13} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -282, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 0 \\ 7 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{12} 3^1 1^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -297, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 9 & 0 & 28 & 7 & 10 \\ 1 & 3 & 2 & 0 & 7 & 0 & 2 \\ 2 & 4 & 2 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{14}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -312, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.59 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 8 & 0 \\ 9 & 12 & 0 & 8 & 46 & 12 \\ 3 & 3 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{11}3^{12}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -322, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.62 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 0 \\ 6 & 5 & 0 & 6 & 9 & 7 \\ 3 & 3 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{12}3^{12}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -372, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.59 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 0 \\ 6 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{12}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.55 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 3 \\ 2 & 1 & 1 & 2 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 8 & 0 \\ 7 & 8 & 0 & 8 & 22 & 10 \\ 3 & 3 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{13}3^{11}2^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -392, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.90 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 6 & 0 & 22 & 6 & 8 \\ 1 & 3 & 2 & 0 & 6 & 0 & 2 \\ 2 & 4 & 2 & -4 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{11}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -402, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 5 & 7 \\ 1 & 3 & 2 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{13}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -522, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.55 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 10 & 0 \\ 9 & 11 & 0 & 10 & 41 & 13 \\ 3 & 3 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{13}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -522, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.58 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 0 \\ 8 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{14}3^{11}1^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -552, \quad (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 0 & 24 & 3 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 24 & 18 & 0 & 118 & 14 & 20 \\ 1 & 3 & 2 & 0 & 14 & 0 & 2 \\ 2 & 4 & 2 & -4 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{12} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -588, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 8 & 0 \\ 8 & 8 & 0 & 8 & 22 & 10 \\ 3 & 3 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{13} 3^1 1^{13} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -617, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 18 & 3 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 18 & 9 & 0 & 52 & 9 & 12 \\ 1 & 3 & 2 & 0 & 9 & 0 & 2 \\ 2 & 4 & 2 & -4 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{14} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -648, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 11 & 3 \\ 2 & 1 & 1 & 2 & 14 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 12 & 0 \\ 11 & 14 & 0 & 12 & 66 & 16 \\ 3 & 3 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{12} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -652, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 8 & 0 \\ 10 & 12 & 0 & 8 & 46 & 12 \\ 3 & 3 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{11} 3^1 1^{12} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -807, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 15 & 9 & 0 & 37 & 8 & 11 \\ 1 & 3 & 2 & 0 & 8 & 0 & 2 \\ 2 & 4 & 2 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{13} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -834, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 10 & 0 \\ 10 & 11 & 0 & 10 & 41 & 13 \\ 3 & 3 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{13} 3^1 1^{12} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -887, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 21 & 3 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 21 & 15 & 0 & 85 & 12 & 17 \\ 1 & 3 & 2 & 0 & 12 & 0 & 2 \\ 2 & 4 & 2 & -4 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{13} 3^1 1^{14} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -932, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 0 & 24 & 3 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 24 & 12 & 0 & 94 & 12 & 16 \\ 1 & 3 & 2 & 0 & 12 & 0 & 2 \\ 2 & 4 & 2 & -4 & 16 & 2 & 0 \end{pmatrix}_{b_0} \\
[42] \quad & 1^{12}3^{11}1^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -972, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5 \\
& T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 18 & 3 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 18 & 12 & 0 & 58 & 10 & 14 \\ 1 & 3 & 2 & 0 & 10 & 0 & 2 \\ 2 & 4 & 2 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0} \\
[43] \quad & 1^{14}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1060, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 2 & 1 & 2 & 12 & 3 \\ 2 & 1 & 1 & 2 & 14 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 12 & 0 \\ 12 & 14 & 0 & 12 & 66 & 16 \\ 3 & 3 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0} \\
[44] \quad & 1^{11}3^{11}1^{13}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8 \\
& T_{\min} = 12.85 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 21 & 3 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 21 & 12 & 0 & 73 & 11 & 15 \\ 1 & 3 & 2 & 0 & 11 & 0 & 2 \\ 2 & 4 & 2 & -4 & 15 & 2 & 0 \end{pmatrix}_{b_0} \\
[45] \quad & 1^{13}1^{15}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1337, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.1 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 0 & 30 & 3 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 30 & 15 & 0 & 148 & 15 & 20 \\ 1 & 3 & 2 & 0 & 15 & 0 & 2 \\ 2 & 4 & 2 & -4 & 20 & 2 & 0 \end{pmatrix}_{b_0} \\
[46] \quad & 1^{12}3^{11}1^{13}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1737, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8 \\
& T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 0 & 24 & 3 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 24 & 15 & 0 & 100 & 13 & 18 \\ 1 & 3 & 2 & 0 & 13 & 0 & 2 \\ 2 & 4 & 2 & -4 & 18 & 2 & 0 \end{pmatrix}_{b_0} \\
[47] \quad & 1^{14}3^{11}1^{13}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1857, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8 \\
& T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 0 & 30 & 3 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 30 & 21 & 0 & 172 & 17 & 24 \\ 1 & 3 & 2 & 0 & 17 & 0 & 2 \\ 2 & 4 & 2 & -4 & 24 & 2 & 0 \end{pmatrix}_{b_0} \\
[48] \quad & 1^{11}3^{11}1^{14}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1887, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 12.87 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 0 & 27 & 3 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 27 & 15 & 0 & 121 & 14 & 19 \\ 1 & 3 & 2 & 0 & 14 & 0 & 2 \\ 2 & 4 & 2 & -4 & 19 & 2 & 0 \end{pmatrix}_{b_0} \\
[49] \quad & 1^{13}3^{11}1^{13}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1922, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8 \\
& T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 0 & 27 & 3 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 27 & 18 & 0 & 133 & 15 & 21 \\ 1 & 3 & 2 & 0 & 15 & 0 & 2 \\ 2 & 4 & 2 & -4 & 21 & 2 & 0 \end{pmatrix}_{b_0} \\
[50] \quad & 1^{12}3^{11}1^{14}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -2592, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0
\end{aligned}$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 0 & 30 & 3 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 30 & 18 & 0 & 154 & 16 & 22 \\ 1 & 3 & 2 & 0 & 16 & 0 & 2 \\ 2 & 4 & 2 & -4 & 22 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{13}3^{11}4^{14}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -3047, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 1 & 2 \\ 2 & 4 & 4 & 0 & 33 & 3 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 33 & 21 & 0 & 193 & 18 & 25 \\ 1 & 3 & 2 & 0 & 18 & 0 & 2 \\ 2 & 4 & 2 & -4 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{14}3^{11}4^{14}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -3252, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 1 & 2 \\ 2 & 4 & 4 & 0 & 36 & 3 & 4 \\ 1 & 4 & 0 & 0 & 24 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 36 & 24 & 0 & 238 & 20 & 28 \\ 1 & 3 & 2 & 0 & 20 & 0 & 2 \\ 2 & 4 & 2 & -4 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [53] \quad 1^{12}231^{13}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 1 & 5 \\ 14 & 33 & 1 & 0 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 11 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{11}231^{11}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 201, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 29 & 5 \\ 14 & 33 & 1 & 65 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 65 & 0 & 123 & 21 \\ 5 & 11 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{12}231^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 201, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 14 & 5 \\ 14 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 35 & 0 & 33 & 11 \\ 5 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{11}231^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 221, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 27 & 5 \\ 14 & 33 & 1 & 65 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 65 & 0 & 123 & 21 \\ 5 & 11 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{12}231^{11}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 221, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 16 & 5 \\ 14 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 35 & 0 & 33 & 11 \\ 5 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{11}23^{11}1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 234, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 7 & 2 & 0 \\ 4 & 9 & 0 & 7 & 14 & 5 & 8 \\ 1 & 3 & 0 & 2 & 5 & 0 & 2 \\ 2 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{12}23^{11}1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 234, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 2 & 0 \\ 3 & 6 & 0 & 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{11}231^{11}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 311, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 28 & 5 \\ 14 & 33 & 1 & 65 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 65 & 0 & 123 & 21 \\ 5 & 11 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{12}231^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 311, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 15 & 5 \\ 14 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 35 & 0 & 33 & 11 \\ 5 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{12}231^{12}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 504, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 6 & 2 & 0 \\ 6 & 12 & 0 & 6 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{11}231^{12}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 792, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 10 & 2 & 0 \\ 7 & 15 & 0 & 10 & 38 & 8 & 12 \\ 1 & 3 & 0 & 2 & 8 & 0 & 2 \\ 2 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [64] \quad 1^{11}3^{11}1^{12}4131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -66, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 10 & 2 & 0 \\ 1 & 3 & 0 & 2 & 8 & 0 & 2 \\ 2 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [65] \quad 1^{11}3^{11}1^{12}5131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 1 & 0 & 3 & 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 1 & 2 & 0 & 2 \\ 2 & 0 & 2 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{13}3^{11}1^{12}5131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.52 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 2 & 1 & 0 & 0 & 2 \\ 2 & 0 & 2 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{11}3^{11}1^{12}5131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 37, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.59 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 3 & -1 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{13}3^{11}5^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 15, \ \Delta = -79, \ \det = 423, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 14.61 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 & 2 \\ 1 & 0 & 1 & -3 & 3 & 1 & -1 \\ 6 & 0 & 9 & 3 & 25 & 6 & 9 \\ 1 & 0 & 2 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[77] \ 1^{14}3^{11}5^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 15, \ \Delta = -79, \ \det = 503, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.64 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 & 2 \\ 1 & 0 & 1 & -3 & 4 & 1 & -1 \\ 7 & 0 & 12 & 4 & 46 & 8 & 12 \\ 1 & 0 & 2 & 1 & 8 & 0 & 2 \\ 2 & 0 & 2 & -1 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

4.89 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 329, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.84 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 2 & 6 & 3 \\ 1 & 3 & 2 & 0 & 2 & 0 & 5 & 2 \\ 3 & 9 & 6 & 0 & 6 & 5 & 13 & 7 \\ 2 & 4 & 2 & -4 & 3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 464, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.92 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 2 & 9 & 3 \\ 1 & 3 & 2 & 0 & 2 & 0 & 8 & 2 \\ 5 & 15 & 9 & 0 & 9 & 8 & 37 & 11 \\ 2 & 4 & 2 & -4 & 3 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{13}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 604, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.87 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 6 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 6 & 3 & 4 & 4 \\ 2 & 4 & 2 & -4 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 669, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.85 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 9 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 5 & 2 \\ 3 & 9 & 6 & 0 & 9 & 5 & 13 & 7 \\ 2 & 4 & 2 & -4 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 1184, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.82 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 5 & 15 & 7 \\ 1 & 3 & 2 & 0 & 5 & 0 & 5 & 2 \\ 3 & 9 & 6 & 0 & 15 & 5 & 13 & 7 \\ 2 & 4 & 2 & -4 & 7 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

4.90 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 164, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 2 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & -4 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{13} 1^{11} 1^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 436, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12} 3^{11} 1^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 500, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 9 & 0 & 28 & 7 & 10 \\ 1 & 3 & 2 & 0 & 7 & 0 & 2 \\ 2 & 4 & 2 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{13} 1^{11} 1^{12} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 544, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 6 & 0 & 22 & 6 & 8 \\ 1 & 3 & 2 & 0 & 6 & 0 & 2 \\ 2 & 4 & 2 & -4 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{13} 3^{11} 1^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 704, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 5 & 7 \\ 1 & 3 & 2 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{13} 3^{11} 1^{12} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 1316, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 15 & 9 & 0 & 37 & 8 & 11 \\ 1 & 3 & 2 & 0 & 8 & 0 & 2 \\ 2 & 4 & 2 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{12} 3^{11} 1^{12} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 1616, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 18 & 3 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 18 & 12 & 0 & 58 & 10 & 14 \\ 1 & 3 & 2 & 0 & 10 & 0 & 2 \\ 2 & 4 & 2 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

4.91 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + su2n3mix

$$T = 11 \quad [0] \ 1^{12} 3^{11} 1^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 1 & 9 & 5 \\ 14 & 33 & 1 & 0 & 20 & 11 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 20 & 0 & 0 & 10 & 6 \\ 5 & 11 & -1 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

4.92 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 0 & 2 & 0 & 2 \\ 1 & 3 & 3 & 0 & 3 & 2 & 0 \\ 2 & 3 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 2 & 3 \\ 1 & 3 & 2 & 0 & 2 & 0 & 3 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 2 & -4 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 130, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 1 & 2 & 5 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 2 & 0 \\ 4 & 5 & 0 & 6 & 9 & 5 & 7 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 3 & -1 & 0 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 146, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 3 & -1 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 226, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 3 & -1 & 0 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 3 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 2 & 2 \\ 1 & 3 & 3 & 0 & 3 & 2 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 328, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 2 & 3 \\ 2 & 1 & 1 & 2 & 8 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 8 & 2 & 0 \\ 6 & 8 & 0 & 8 & 22 & 8 & 10 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 3 & -1 & 0 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 386, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$T_{\min} = 14.97 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 1 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 1 & 3 & 0 & 1 & 3 & 2 & 0 \\ 2 & 3 & -1 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

4.93 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + su2n3mix

$$T = 9 \quad [0] \quad 1^{122} 3^1 21^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 6 & -1 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 6 & 12 & 20 & 8 \\ 2 & 2 & -1 & -1 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 21^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 2 & -1 \\ 2 & 3 & 0 & 4 & 5 & 5 & 5 \\ 2 & 3 & 0 & 2 & 5 & 3 & 3 \\ 2 & 2 & -1 & -1 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{132} 3^1 21^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & 1 & 7 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 3 & -1 \\ 6 & 7 & 0 & 6 & 25 & 18 & 9 \\ 3 & 5 & 0 & 3 & 18 & 11 & 5 \\ 2 & 2 & -1 & -1 & 9 & 5 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 21^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 3 & 4 \\ 3 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 2 & 0 & -2 & 2 & 0 \\ 3 & 3 & 0 & 2 & 1 & 3 \\ 4 & 3 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 21^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 3 & 2 & 0 & 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & -1 & -1 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^1 31^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 5 & 3 \\ 2 & 4 & 3 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 5 & 3 \\ 2 & 5 & 6 & 0 & 5 & 7 & 5 \\ 2 & 3 & 3 & -4 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{113} 3^1 21^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.70 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 2 & 2 & 0 & 1 & 0 & 7 & 2 \\ 4 & 7 & 0 & 4 & 7 & 23 & 7 \\ 2 & 2 & -1 & -1 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 3^1 21^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 3 & 3 \\ 2 & 2 & -1 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

- [10] $1^{142}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -68$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.35 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 3 & -1 \\ 4 & 8 & 0 & 4 & 30 & 20 & 8 \\ 3 & 5 & 0 & 3 & 20 & 11 & 5 \\ 2 & 2 & -1 & -1 & 8 & 5 & -1 \end{pmatrix}_{b_0}$$
- [9] $1^{122}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -68$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 13.35 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 3 & -1 \\ 4 & 5 & 0 & 5 & 13 & 13 & 7 \\ 3 & 5 & 0 & 3 & 13 & 11 & 5 \\ 2 & 2 & -1 & -1 & 7 & 5 & -1 \end{pmatrix}_{b_0}$$
- [8] $1^{131}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -68$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 2 & -1 \\ 3 & 6 & 0 & 3 & 16 & 9 & 6 \\ 2 & 3 & 0 & 2 & 9 & 3 & 3 \\ 2 & 2 & -1 & -1 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{124}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -72$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.56 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 8 & -1 \\ 2 & 4 & 0 & 2 & 6 & 20 & 4 \\ 8 & 10 & 0 & 8 & 20 & 52 & 12 \\ 2 & 2 & -1 & -1 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$
- [13] $1^{123}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -78$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.67 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 6 & 7 & 0 & 4 & 14 & 23 & 7 \\ 2 & 2 & -1 & -1 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{113}3^11^{11}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -78$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 5 & 3 \\ 3 & 9 & 5 & 0 & 5 & 12 & 6 \\ 2 & 4 & 2 & -4 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$
- [14] $1^{122}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -92$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 3 & -1 \\ 3 & 5 & 0 & 5 & 13 & 13 & 7 \\ 3 & 5 & 0 & 3 & 13 & 11 & 5 \\ 2 & 2 & -1 & -1 & 7 & 5 & 0 \end{pmatrix}_{b_0}$$
- [15] $1^{111}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.85 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 5 & 4 & 10 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$
- [16] $1^{123}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 6 & 8 & 0 & 7 & 16 & 34 & 10 \\ 2 & 2 & -1 & -1 & 4 & 10 & 0 \end{pmatrix}_{b_0}$$

- [17] $1^{11} 1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -103$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 2 & -1 \\ 3 & 3 & 0 & 4 & 5 & 5 & 5 \\ 2 & 3 & 0 & 2 & 5 & 3 & 3 \\ 2 & 2 & -1 & -1 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$
- [18] $1^{121} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -118$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.64 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 \\ 2 & 2 & -1 & -1 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$
- [19] $1^{124} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -120$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 5 & -1 \\ 3 & 4 & 0 & 2 & 6 & 18 & 4 \\ 5 & 9 & 0 & 5 & 18 & 39 & 9 \\ 2 & 2 & -1 & -1 & 4 & 9 & -1 \end{pmatrix}_{b_0}$$
- [20] $1^{121} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 2 & -1 \\ 3 & 5 & 0 & 5 & 13 & 8 & 7 \\ 2 & 3 & 0 & 2 & 8 & 3 & 3 \\ 2 & 2 & -1 & -1 & 7 & 3 & 0 \end{pmatrix}_{b_0}$$
- [21] $1^{132} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -138$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 12.87 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 16 & 15 & 6 \\ 3 & 5 & 0 & 3 & 15 & 11 & 5 \\ 2 & 2 & -1 & -1 & 6 & 5 & 0 \end{pmatrix}_{b_0}$$
- [22] $1^{122} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 2 & 4 & 0 & 2 & 6 & 10 & 4 \\ 4 & 5 & 0 & 3 & 10 & 11 & 5 \\ 2 & 2 & -1 & -1 & 4 & 5 & 0 \end{pmatrix}_{b_0}$$
- [23] $1^{111} 3^1 1^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -143$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 6 & 4 \\ 1 & 4 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 11 & 7 \\ 2 & 6 & 5 & 0 & 11 & 7 & 5 \\ 2 & 4 & 2 & -4 & 7 & 5 & 0 \end{pmatrix}_{b_0}$$
- [24] $1^{111} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -148$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & -1 & -1 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$
- [25] $1^{112} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -158$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

- [34] $1^{112}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -218$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.70 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 3 & 5 & 0 & 3 & 5 & 11 & 5 \\ 2 & 2 & -1 & -1 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$
- [35] $1^{131}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -218$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 2 & -1 \\ 4 & 7 & 0 & 6 & 25 & 11 & 9 \\ 2 & 3 & 0 & 2 & 11 & 3 & 3 \\ 2 & 2 & -1 & -1 & 9 & 3 & 0 \end{pmatrix}_{b_0}$$
- [36] $1^{111}3^11^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -242$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 8 & 3 \\ 4 & 12 & 8 & 0 & 8 & 25 & 9 \\ 2 & 4 & 2 & -4 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$
- [37] $1^{123}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -247$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 3 & 4 & 0 & 2 & 6 & 14 & 4 \\ 4 & 7 & 0 & 4 & 14 & 23 & 7 \\ 2 & 2 & -1 & -1 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$
- [38] $1^{132}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -258$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 16 & 15 & 6 \\ 3 & 5 & 0 & 3 & 15 & 11 & 5 \\ 2 & 2 & -1 & -1 & 6 & 5 & 0 \end{pmatrix}_{b_0}$$
- [39] $1^{112}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -260$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.65 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 6 & 6 & 20 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$
- [40] $1^{123}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -262$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 4 & 7 & 0 & 4 & 14 & 23 & 7 \\ 2 & 2 & -1 & -1 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$
- [41] $1^{141}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -280$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 2 & -1 \\ 5 & 9 & 0 & 7 & 41 & 14 & 11 \\ 2 & 3 & 0 & 2 & 14 & 3 & 3 \\ 2 & 2 & -1 & -1 & 11 & 3 & 0 \end{pmatrix}_{b_0}$$
- [42] $1^{113}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -292$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 4 & 7 & 0 & 4 & 7 & 23 & 7 \\ 2 & 2 & -1 & -1 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{12} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -298, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 4 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 \\ 2 & 2 & -1 & -1 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -303, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 3 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 \\ 2 & 2 & -1 & -1 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 3 & 4 & 0 & 2 & 6 & 10 & 4 \\ 3 & 5 & 0 & 3 & 10 & 11 & 5 \\ 2 & 2 & -1 & -1 & 4 & 5 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & 1 & 9 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 3 & -1 \\ 5 & 9 & 0 & 7 & 41 & 23 & 11 \\ 3 & 5 & 0 & 3 & 23 & 11 & 5 \\ 2 & 2 & -1 & -1 & 11 & 5 & 0 \end{pmatrix}_{b_0}$$

[47] $1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 5 & 7 & 0 & 4 & 14 & 23 & 7 \\ 2 & 2 & -1 & -1 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

[48] $1^{11} 3^1 1^{14} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -323, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.77 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 30 & 4 \\ 1 & 4 & 0 & 0 & 3 & 17 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 17 & 3 \\ 10 & 30 & 17 & 0 & 17 & 151 & 21 \\ 2 & 4 & 2 & -4 & 3 & 21 & -1 \end{pmatrix}_{b_0}$$

[49] $1^{14} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -340, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 5 & -1 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 5 & 9 & 0 & 5 & 18 & 39 & 9 \\ 2 & 2 & -1 & -1 & 4 & 9 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -345, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 2 & -1 \\ 4 & 5 & 0 & 5 & 13 & 8 & 7 \\ 2 & 3 & 0 & 2 & 8 & 3 & 3 \\ 2 & 2 & -1 & -1 & 7 & 3 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{12} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

11.5

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 2 & -1 \\ 5 & 5 & 0 & 5 & 13 & 8 & 7 \\ 2 & 3 & 0 & 2 & 8 & 3 & 3 \\ 2 & 2 & -1 & -1 & 7 & 3 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{11} 1^{132} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -366, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & 1 & 7 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 3 & -1 \\ 5 & 7 & 0 & 6 & 25 & 18 & 9 \\ 3 & 5 & 0 & 3 & 18 & 11 & 5 \\ 2 & 2 & -1 & -1 & 9 & 5 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{11} 1^{114} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -370, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 5 & 9 & 0 & 5 & 9 & 39 & 9 \\ 2 & 2 & -1 & -1 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{12} 1^{124} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 5 & -1 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 7 & 9 & 0 & 5 & 18 & 39 & 9 \\ 2 & 2 & -1 & -1 & 4 & 9 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{11} 1^{124} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 8 & -1 \\ 2 & 4 & 0 & 2 & 6 & 20 & 4 \\ 6 & 10 & 0 & 8 & 20 & 52 & 12 \\ 2 & 2 & -1 & -1 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

[56] $1^{11} 1^{124} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 8 & -1 \\ 2 & 4 & 0 & 2 & 6 & 20 & 4 \\ 7 & 10 & 0 & 8 & 20 & 52 & 12 \\ 2 & 2 & -1 & -1 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{11} 1^{131} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -393, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 2 & -1 \\ 4 & 6 & 0 & 3 & 16 & 9 & 6 \\ 2 & 3 & 0 & 2 & 9 & 3 & 3 \\ 2 & 2 & -1 & -1 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

[58] $1^{11} 1^{113} 3^1 1^{12} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 18 & 4 \\ 1 & 4 & 0 & 0 & 3 & 11 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 11 & 3 \\ 6 & 18 & 11 & 0 & 11 & 55 & 13 \\ 2 & 4 & 2 & -4 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{11} 1^{112} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 3 & 5 & 11 & 5 \\ 2 & 2 & -1 & -1 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{11}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 4 & 0 & 6 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 12 & 4 \\ 4 & 12 & 8 & 0 & 12 & 25 & 9 \\ 2 & 4 & 2 & -4 & 4 & 9 & 0 \end{pmatrix}_{b_0}$$

[61] $1^{11}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -422, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 13.19 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 24 & 4 \\ 1 & 4 & 0 & 0 & 3 & 14 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 14 & 3 \\ 8 & 24 & 14 & 0 & 14 & 97 & 17 \\ 2 & 4 & 2 & -4 & 3 & 17 & -1 \end{pmatrix}_{b_0}$$

[62] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -430, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 7 & 8 & 34 & 10 \\ 2 & 2 & -1 & -1 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[63] $1^{11}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -438, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 12.68 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 16 & 15 & 6 \\ 3 & 5 & 0 & 3 & 15 & 11 & 5 \\ 2 & 2 & -1 & -1 & 6 & 5 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -449, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 5 & 6 & 0 & 6 & 6 & 20 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{11}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -463, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 2 & -1 \\ 5 & 8 & 0 & 4 & 30 & 12 & 8 \\ 2 & 3 & 0 & 2 & 12 & 3 & 3 \\ 2 & 2 & -1 & -1 & 8 & 3 & 0 \end{pmatrix}_{b_0}$$

[66] $1^{11}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -473, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 9 & 4 \\ 3 & 9 & 5 & 0 & 9 & 12 & 6 \\ 2 & 4 & 2 & -4 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{11}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -478, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 5 & 5 & 0 & 3 & 5 & 11 & 5 \\ 2 & 2 & -1 & -1 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{11}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -483, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

11.0

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 14 & 7 \\ 3 & 9 & 5 & 0 & 14 & 12 & 6 \\ 2 & 4 & 2 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{1^{12}} 3^{1^{12}} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 1 & 1 & 9 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 3 & -1 \\ 7 & 9 & 0 & 7 & 41 & 23 & 11 \\ 3 & 5 & 0 & 3 & 23 & 11 & 5 \\ 2 & 2 & -1 & -1 & 11 & 5 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{1^{11}} 3^{1^{11}} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -508, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 5 & -1 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 6 & 9 & 0 & 5 & 18 & 39 & 9 \\ 2 & 2 & -1 & -1 & 4 & 9 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{1^{12}} 3^{1^{12}} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 2 & -1 \\ 5 & 6 & 0 & 3 & 16 & 9 & 6 \\ 2 & 3 & 0 & 2 & 9 & 3 & 3 \\ 2 & 2 & -1 & -1 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{1^{12}} 3^{1^{12}} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 6 & 6 & 0 & 6 & 6 & 20 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{1^{12}} 3^{1^{12}} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -548, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 3 & -1 \\ 6 & 8 & 0 & 4 & 30 & 20 & 8 \\ 3 & 5 & 0 & 3 & 20 & 11 & 5 \\ 2 & 2 & -1 & -1 & 8 & 5 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{1^{11}} 3^{1^{11}} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -548, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 3 & -1 \\ 5 & 8 & 0 & 4 & 30 & 20 & 8 \\ 3 & 5 & 0 & 3 & 20 & 11 & 5 \\ 2 & 2 & -1 & -1 & 8 & 5 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{1^{11}} 3^{1^{11}} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -567, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 2 & -1 \\ 5 & 7 & 0 & 6 & 25 & 11 & 9 \\ 2 & 3 & 0 & 2 & 11 & 3 & 3 \\ 2 & 2 & -1 & -1 & 9 & 3 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{1^{11}} 3^{1^{11}} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -577, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 4 & 7 & 23 & 7 \\ 2 & 2 & -1 & -1 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{114}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{c}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -604, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 8 & -1 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 6 & 10 & 0 & 8 & 10 & 52 & 12 \\ 2 & 2 & -1 & -1 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[78] $1^{142}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{c}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -644, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & 1 & 9 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 3 & -1 \\ 6 & 9 & 0 & 7 & 41 & 23 & 11 \\ 3 & 5 & 0 & 3 & 23 & 11 & 5 \\ 2 & 2 & -1 & -1 & 11 & 5 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{131}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{c}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -726, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 2 & -1 \\ 6 & 7 & 0 & 6 & 25 & 11 & 9 \\ 2 & 3 & 0 & 2 & 11 & 3 & 3 \\ 2 & 2 & -1 & -1 & 9 & 3 & 0 \end{pmatrix}_{b_0}$$

[80] $1^{113}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{c}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -727, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 6 & 8 & 0 & 7 & 8 & 34 & 10 \\ 2 & 2 & -1 & -1 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[81] $1^{141}3^121^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{c}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -738, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 2 & -1 \\ 6 & 8 & 0 & 4 & 30 & 12 & 8 \\ 2 & 3 & 0 & 2 & 12 & 3 & 3 \\ 2 & 2 & -1 & -1 & 8 & 3 & 0 \end{pmatrix}_{b_0}$$

[82] $1^{114}3^121^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{c}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -763, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 5 & 9 & 39 & 9 \\ 2 & 2 & -1 & -1 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

[83] $1^{113}3^121^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{c}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -768, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 6 & 7 & 0 & 4 & 7 & 23 & 7 \\ 2 & 2 & -1 & -1 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[84] $1^{141}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{c}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -769, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 2 & -1 \\ 6 & 9 & 0 & 7 & 41 & 14 & 11 \\ 2 & 3 & 0 & 2 & 14 & 3 & 3 \\ 2 & 2 & -1 & -1 & 11 & 3 & 0 \end{pmatrix}_{b_0}$$

[85] $1^{111}3^11^{111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -782, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 4 & 2 \\ 2 & 4 & 4 & 0 & 9 & 12 & 4 \\ 1 & 4 & 0 & 0 & 6 & 8 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 20 & 7 \\ 4 & 12 & 8 & 0 & 20 & 25 & 9 \\ 2 & 4 & 2 & -4 & 7 & 9 & 0 \end{pmatrix}_{b_0}$$

[86] $1^{112}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -930, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 7 & 8 & 0 & 7 & 8 & 34 & 10 \\ 2 & 2 & -1 & -1 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[87] $1^{111}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1009, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 8 & -1 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 7 & 10 & 0 & 8 & 10 & 52 & 12 \\ 2 & 2 & -1 & -1 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[88] $1^{112}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1062, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 7 & 9 & 0 & 5 & 9 & 39 & 9 \\ 2 & 2 & -1 & -1 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

[89] $1^{112}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1068, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 2 & -1 \\ 7 & 9 & 0 & 7 & 41 & 14 & 11 \\ 2 & 3 & 0 & 2 & 14 & 3 & 3 \\ 2 & 2 & -1 & -1 & 11 & 3 & 0 \end{pmatrix}_{b_0}$$

[90] $1^{112}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 8 & -1 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 8 & 10 & 0 & 8 & 10 & 52 & 12 \\ 2 & 2 & -1 & -1 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [91] $1^{11}231^13^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 5 & 0 & 5 & 5 \\ 2 & 4 & -1 & 0 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

[92] $1^{11}23^11^{11}31^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 12 & 0 & 9 & 0 & 26 & 10 \\ 2 & 4 & -1 & 0 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{123}1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 2 & 4 & -1 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{123}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 101, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 2 & 0 \\ 2 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{1231}1^{11}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 13 & 21 & 1 & 22 & 7 \\ 21 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 35 & 0 & 33 & 11 \\ 7 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[96] $1^{11231}1^{11}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 11 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 8 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 11 & 0 & 8 & 0 & 21 & 9 \\ 2 & 4 & -1 & 0 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{1123}1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 157, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 & 2 \\ 2 & 4 & 1 & 3 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 7 & 2 & 0 \\ 4 & 9 & 0 & 7 & 14 & 5 & 8 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 2 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

[98] $1^{1231}1^{11}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 4 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 8 & 0 & 4 & 0 & 9 & 5 \\ 2 & 4 & -1 & 0 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

[99] $1^{123}1^{11}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 209, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 5 & 0 & 12 & 6 \\ 2 & 4 & -1 & 0 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[100] $1^{1123}1^{11}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 215, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 12 & 0 & 9 & 0 & 26 & 10 \\ 2 & 4 & -1 & 0 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[101] $1^{123}1^{11}31^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 221, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 2 & 0 \\ 4 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[102] \quad 1^{11}2^{13}1^{11}3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -87, \quad \det = 229, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 9 & 0 & 26 & 10 \\ 2 & 4 & -1 & 0 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[103] $1^{11}23^11^{11}31^{131} + [\textcolor{blue}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{e}_6 + \textcolor{blue}{s}u_2 + \textcolor{red}{s}u_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -87, \quad \det = 245, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.0$
 $T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 & 2 \\ 2 & 4 & 1 & 3 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 7 & 2 & 0 \\ 6 & 9 & 0 & 7 & 14 & 5 & 8 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 2 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

[104] $1^{11}23^11^{12}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -87, \quad \det = 254, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.5$
 $T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 & 2 \\ 2 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 10 & 2 & 0 \\ 6 & 15 & 0 & 10 & 38 & 8 & 12 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 2 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix} b_0$$

[105] $1^1 2^3 1^1 3^1 1^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{c}_6 + \textcolor{blue}{s}u_2 + \textcolor{red}{s}u_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 271, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\min_{\mathbf{b}_0} \left(\begin{array}{cccccc} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 2 & 0 \\ 3 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & -1 & 0 & 4 & 2 & 1 \end{array} \right)_{b_0}$$

$$[106] \ 1^{11} 23^{11} 1^{11} 31^{121} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{brown}{s}u_2 + \textcolor{red}{s}u_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 311, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.0$$

$$\left(\begin{array}{cccccc} 0 & 2 & 1 & 1 & 5 & 2 & 2 \\ 2 & 4 & 1 & 3 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 7 & 2 & 0 \\ 5 & 9 & 0 & 7 & 14 & 5 & 8 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 2 & 4 & -1 & 0 & 8 & 2 & 1 \end{array} \right)_{b_1}$$

[107] $1^{12} 2^3 1^1 1^{12} 3^1 1^{14} + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{c}_6 + \textcolor{blue}{s} \textcolor{red}{u}_2 + \textcolor{red}{s} \textcolor{blue}{u}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 334$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\text{min}} = 12$ $92 \rightarrow 13$ $T_{\text{max}} = 13$ $T \in [13, 13]$

$$I_{\min} = 12.52 \rightarrow 13, I_{\max} = 13, I \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 8 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 0 & 8 & 0 & 36 & 10 \end{pmatrix}.$$

[108] $1^1 2^3 1^1 1^2 3^1 1^{11} + [\textcolor{blue}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{c}_6 + \textcolor{blue}{s}u_2 + \textcolor{red}{s}u_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 6 & 2 & 0 \\ 5 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

[109] $1^{11}23^{11}1^{12}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 376, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$I_{\min} = 12.55, I_{\max} = 18, I \in [16, 18]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 18 & 0 & 12 & 0 & 56 & 14 \\ 2 & 4 & -1 & 0 & 0 & 14 & 0 \end{pmatrix},$$

[110] $1^1 2^3 1^1 1^{12} 3^1 1^{41} + [\textcolor{blue}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{c}_6 + \textcolor{blue}{s}u_2 + \textcolor{red}{s}u_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 386, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$

$$(1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 & 2 \\ 2 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 10 & 2 & 0 \\ 8 & 15 & 0 & 10 & 38 & 8 & 12 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 2 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [120] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} 4^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} 3^{\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 8 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 6 & 3 \\ 4 & 8 & 6 & 0 & 6 & 13 & 7 \\ 2 & 4 & 2 & -3 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [121] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} 5^{\textcolor{red}{1}} 13^{\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 37, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.59 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 3 & 0 & 1 & 1 & 5 & 3 \\ 2 & 5 & 0 & 2 & 5 & 7 & 5 \\ 2 & 3 & -1 & -1 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[122] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{2}} 5^{\textcolor{red}{1}} 13^{\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 44, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.82 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & -3 & 1 & 2 & -1 \\ 1 & 0 & 3 & 1 & 1 & 5 & 3 \\ 4 & 0 & 5 & 2 & 5 & 7 & 5 \\ 2 & 0 & 2 & -1 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[123] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} 5^{\textcolor{red}{1}} 13^{\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 68, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & -3 & 1 & 2 & -1 \\ 1 & 0 & 3 & 1 & 1 & 5 & 3 \\ 3 & 0 & 5 & 2 & 5 & 7 & 5 \\ 2 & 0 & 2 & -1 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

4.94 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{2}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{3}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 1 & 9 & 5 \\ 14 & 33 & 1 & 0 & 20 & 11 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 20 & 0 & 0 & 10 & 6 \\ 5 & 11 & -1 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

4.95 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} 3^{\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 494, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 5 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 9 & 3 & 4 \\ 3 & 9 & 5 & 0 & 9 & 12 & 5 & 6 \\ 1 & 3 & 3 & 0 & 3 & 5 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

4.96 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{14} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.6$$

$$T_{\min} = 24, T_{\max} = 98, T \in [24, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 11 & 21 & 0 & 18 & 75 & 15 \\ 3 & 4 & -4 & 4 & 15 & -11 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16, T_{\max} = 98, T \in [16, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & -1 & 0 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & -1 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{132} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 12 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 25 & 27 & 9 \\ 7 & 13 & 0 & 27 & 27 & 9 \\ 3 & 4 & -4 & 9 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.03 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 121, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 3 & 5 & 0 & 6 & 3 & 3 \\ 3 & 4 & -4 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 0 & 12 & 12 & 6 \\ 3 & 4 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 4 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{13}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 9 & 17 & 0 & 15 & 48 & 12 \\ 3 & 4 & -4 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^11 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{112}3^11 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.21 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -3 & 6 & 0 & -1 \\ 6 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{112}3^11 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.07 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 4 & 8 & 0 & 6 & 9 & 5 \\ 3 & 4 & -4 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{112}3^11 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.17 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 12 & 7 \\ 4 & 8 & 0 & 12 & 9 & 5 \\ 3 & 4 & -4 & 7 & 5 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{121}3^11 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 6 & 6 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -4 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}3^11^3 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 11 & 19 & 0 & 73 & 12 & 15 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -4 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{122}3^11 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 4 & 8 & 0 & 12 & 9 & 5 \\ 3 & 4 & -4 & 6 & 5 & -1 \end{pmatrix}_{b_0}$$

[18] $1^{113}3^11 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 6 & 12 & 0 & 9 & 24 & 8 \\ 3 & 4 & -4 & 3 & 8 & -1 \end{pmatrix}_{b_0}$$

[19] $1^{111}3^11^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 8 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -4 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{122}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.54 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 4 & 3 \\ 3 & 4 & 0 & 18 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 0 & 58 & 24 & 14 \\ 4 & 8 & 0 & 24 & 9 & 5 \\ 3 & 4 & -4 & 14 & 5 & -1 \end{pmatrix}_{b_0}$$

[21] $1^{131}3^11 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 25 & 9 & 9 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -4 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

[22] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 12 & 6 \\ 3 & 4 & -4 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{132}3^11 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 4 & 3 \\ 3 & 4 & 0 & 12 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 25 & 18 & 9 \\ 4 & 8 & 0 & 18 & 9 & 5 \\ 3 & 4 & -4 & 9 & 5 & -1 \end{pmatrix}_{b_0}$$

[24] $1^{123}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 7 & 13 & 0 & 12 & 27 & 9 \\ 3 & 4 & -4 & 4 & 9 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & -4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{113}3^11 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -3 & 9 & 0 & -1 \\ 9 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 12 & -1 & -1 \end{pmatrix}_{b_0}$$

[27] $1^{132}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.98 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 4 & 3 \\ 3 & 4 & 0 & 27 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 15 & 27 & 0 & 133 & 36 & 21 \\ 4 & 8 & 0 & 36 & 9 & 5 \\ 3 & 4 & -4 & 21 & 5 & -1 \end{pmatrix}_{b_0}$$

[28] $1^{112}3^11^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -3 & 6 & 0 & -1 \\ 8 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{141}3^11 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.31 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 46 & 12 & 12 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -4 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{122}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 0 & 18 & 27 & 9 \\ 3 & 4 & -4 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{11}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 9 & 7 \\ 3 & 5 & 0 & 9 & 3 & 3 \\ 3 & 4 & -4 & 7 & 3 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{151}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 10 & 20 & 0 & 73 & 15 & 15 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -4 & 15 & 2 & -1 \end{pmatrix}_{b_0}$$

[34] $1^{111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 9 & 11 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{14}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 313, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 21 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 0 & 76 & 18 & 16 \\ 3 & 5 & 0 & 18 & 3 & 3 \\ 3 & 4 & -4 & 16 & 3 & -1 \end{pmatrix}_{b_0}$$

[36] $1^{112}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -3 & 6 & 0 & -1 \\ 7 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{114}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & -1 \\ 12 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

[38] $1^{12}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 373, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 12 & 10 \\ 3 & 5 & 0 & 12 & 3 & 3 \\ 3 & 4 & -4 & 10 & 3 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{161}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.64 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 12 & 24 & 0 & 106 & 18 & 18 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -4 & 18 & 2 & -1 \end{pmatrix}_{b_0}$$

[40] $1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 15 & 13 \\ 3 & 5 & 0 & 15 & 3 & 3 \\ 3 & 4 & -4 & 13 & 3 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 391, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 0 & 9 & 27 & 9 \\ 3 & 4 & -4 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{122} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 403, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 18 & 10 \\ 4 & 8 & 0 & 18 & 9 & 5 \\ 3 & 4 & -4 & 10 & 5 & -1 \end{pmatrix}_{b_0}$$

[43] $1^{113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -3 & 9 & 0 & -1 \\ 12 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{171} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 13.76 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 28 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 14 & 28 & 0 & 145 & 21 & 21 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & -4 & 21 & 2 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 0 & 15 & 12 & 6 \\ 3 & 4 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{112} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 486, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 10 & 18 & 0 & 12 & 57 & 13 \\ 3 & 4 & -4 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

[47] $1^{115} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 487, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 40 & 0 & 5 \\ 2 & 4 & -3 & 15 & 0 & -1 \\ 15 & 40 & 15 & 148 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 20 & -1 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 526, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 9 & 10 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 547, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 48 & 12 \\ 3 & 4 & -4 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 568, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -3 & 9 & 0 & -1 \\ 10 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 574, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 21 & 7 \\ 7 & 13 & 0 & 21 & 27 & 9 \\ 3 & 4 & -4 & 7 & 9 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 595, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 27 & 7 \\ 9 & 17 & 0 & 27 & 48 & 12 \\ 3 & 4 & -4 & 7 & 12 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{113} 3^1 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 631, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -3 & 9 & 0 & -1 \\ 11 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{132} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 0 & 17 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 24 & 13 \\ 4 & 8 & 0 & 24 & 9 & 5 \\ 3 & 4 & -4 & 13 & 5 & -1 \end{pmatrix}_{b_0}$$

[55] $1^{132} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 4 & 3 \\ 3 & 4 & 0 & 22 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 0 & 85 & 30 & 17 \\ 4 & 8 & 0 & 30 & 9 & 5 \\ 3 & 4 & -4 & 17 & 5 & -1 \end{pmatrix}_{b_0}$$

[56] $1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 648, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 21 & 10 \\ 5 & 9 & 0 & 21 & 12 & 6 \\ 3 & 4 & -4 & 10 & 6 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{114} 3^1 1^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 688, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & -1 \\ 16 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[58] $1^{121} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 736, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{131} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 747, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 27 & 13 \\ 5 & 9 & 0 & 27 & 12 & 6 \\ 3 & 4 & -4 & 13 & 6 & 0 \end{pmatrix}_{b_0}$$

[61] $1^{131} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 760, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 12 & 13 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -4 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{121} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 760, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 28 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 16 & 28 & 0 & 154 & 18 & 22 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -4 & 22 & 2 & 0 \end{pmatrix}_{b_0}$$

[62] $1^{141} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 5 & 3 \\ 3 & 4 & 0 & 21 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 0 & 76 & 33 & 16 \\ 5 & 9 & 0 & 33 & 12 & 6 \\ 3 & 4 & -4 & 16 & 6 & 0 \end{pmatrix}_{b_0}$$

[63] $1^{113} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 778, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.66 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 12 & 22 & 0 & 15 & 84 & 16 \\ 3 & 4 & -4 & 3 & 16 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{121} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 814, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 23 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 13 & 23 & 0 & 100 & 15 & 18 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -4 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{114} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 823, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & -1 \\ 13 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

[66] $1^{122} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 841, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 30 & 10 \\ 7 & 13 & 0 & 30 & 27 & 9 \\ 3 & 4 & -4 & 10 & 9 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{114} 31^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 991, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & -1 \\ 15 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{141} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1006, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 0 & 76 & 15 & 16 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -4 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{132} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1030, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 7 & 3 \\ 3 & 4 & 0 & 17 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 39 & 13 \\ 7 & 13 & 0 & 39 & 27 & 9 \\ 3 & 4 & -4 & 13 & 9 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{114} 3 1^{1+} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1036, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & -1 \\ 14 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{131} 3 1^{12+} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 0 & 85 & 15 & 17 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -4 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{131} 3 1^{13+} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 27 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 15 & 27 & 0 & 133 & 18 & 21 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -4 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{131} 3 1^{14+} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1606, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 0 & 32 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 18 & 32 & 0 & 193 & 21 & 25 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & -4 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{141} 3 1^{12+} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1624, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 26 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 14 & 26 & 0 & 118 & 18 & 20 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -4 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{141} 3 1^{13+} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 2110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 31 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 17 & 31 & 0 & 172 & 21 & 24 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & -4 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{141} 3 1^{14+} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 2464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 0 & 36 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 20 & 36 & 0 & 238 & 24 & 28 \\ 2 & 4 & 0 & 24 & 0 & 2 \\ 3 & 4 & -4 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [77] $1^{111} 3 2 1^{11+} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 0 & -1 \\ 2 & 3 & 0 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -1 & 5 & -1 & -4 \end{pmatrix}_{b_0}$$

[78] $1^{111} 3 2 1^{11+} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 16.67 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & -2 & 3 & 0 & 0 \\ 3 & 5 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 4 & -1 & -3 \end{pmatrix}_{b_0}$$

[79] $1^{111} 3 1^{131+} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 16, T_{\max} = 68, T \in [16, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 & 3 \\ 1 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & -4 & 3 & 2 & -3 \end{pmatrix}_{b_0}$$

$$[80] \ 1^{11} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16.40 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 3 & 5 & 0 & 0 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{121} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 16.33 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 13 & 4 \\ 5 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 10 & 0 \\ 13 & 15 & 0 & 10 & 38 & 12 \\ 4 & 4 & -1 & 0 & 12 & -3 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{12} 31^{111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.17 \rightarrow 16, T_{\max} = 68, T \in [16, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 4 & 0 & 6 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 12 & 4 \\ 4 & 12 & 9 & 0 & 12 & 27 & 9 \\ 2 & 4 & 2 & -4 & 4 & 9 & -3 \end{pmatrix}_{b_0}$$

$$[83] \ 1^{12} 31^{111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 18.58 \rightarrow 19, T_{\max} = 68, T \in [19, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 & 2 \\ 2 & 4 & 4 & 0 & 12 & 6 & 4 \\ 1 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 9 & 0 & 28 & 12 & 10 \\ 2 & 6 & 3 & 0 & 12 & 3 & 3 \\ 2 & 4 & 2 & -4 & 10 & 3 & -6 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{111} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & -2 & 7 & 0 & 0 \\ 5 & 12 & 7 & 14 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{111} 31^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 & 3 \\ 1 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[86] \ 1^{11} 31^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 0 & 4 & 3 & 4 \\ 1 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[88] \ 1^{11} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[87] \ 1^{11} 31^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 68, T \in [15, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 2 & -4 & 3 & 3 & -2 \end{pmatrix}_{b_0}$$

- [89] $1^{111} 3^{14} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 9 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [90] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -82$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.12 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$
- [91] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 5 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [93] $1^{11} 31^{12} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 6 & 0 & 22 & 6 & 8 \\ 1 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 4 & 2 & -4 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$
- [92] $1^{121} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 6 & 0 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [94] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.22 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$
- [95] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.10 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 4 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$
- [96] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -131$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [97] $1^{112} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -137$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.29 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -1 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$
- [98] $1^{111} 3^{13} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

- [99] $1^{121} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 13.11 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$
- [100] $1^{112} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -141$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -1 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$
- [101] $1^{131} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -155$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$
- [102] $1^{141} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -176$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.77 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$
- [103] $1^{113} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -177$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$
- [104] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [105] $1^{151} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -203$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.1$
 $T_{\min} = 14.11 \rightarrow 15$, $T_{\max} = 25$, $T \in [15, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 15 & 0 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & 0 & 20 & -1 \end{pmatrix}_{b_0}$$
- [106] $1^{111} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -231$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.08 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 6 & 3 \\ 3 & 9 & 6 & 0 & 6 & 12 & 6 \\ 2 & 4 & 2 & -4 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$
- [108] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -236$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 8 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [107] $1^{121} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -236$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

12.5

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 8 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[110] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[109] $1^{111} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 6 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[111] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{111} 3^{11} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -291, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 7 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[113] $1^{131} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -303, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.26 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 5 & 1 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 12 & 15 & 0 & 49 & 12 & 13 \\ 1 & 3 & 3 & 0 & 12 & 0 & 2 \\ 2 & 3 & 3 & -4 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

[114] $1^{112} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -306, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -1 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[115] $1^{111} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -375, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 & 3 \\ 2 & 4 & 3 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 9 & 0 & 13 & 6 & 7 \\ 1 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 3 & 3 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[116] $1^{131} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 11 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[118] $1^{111}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -411$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.13 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 6 & 7 \\ 1 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 4 & 2 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[117] $1^{121}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -411$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.91 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[119] $1^{121}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -456$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.16 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 4 & 1 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 9 & 12 & 0 & 28 & 9 & 10 \\ 1 & 3 & 3 & 0 & 9 & 0 & 2 \\ 2 & 3 & 3 & -4 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

[120] $1^{131}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -516$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.29 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 6 & 4 \\ 2 & 6 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 2 & -4 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

[121] $1^{111}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -537$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.04 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 9 & 4 \\ 3 & 9 & 6 & 0 & 9 & 12 & 6 \\ 2 & 4 & 2 & -4 & 4 & 6 & -1 \end{pmatrix}_{b_0}$$

[122] $1^{121}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -555$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.85 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 9 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[123] $1^{141}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -560$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.03 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 14 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 0 \end{pmatrix}_{b_0}$$

[124] $1^{121}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -576$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[125] $1^{131}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -588$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{121} 3^{14} 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -588, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{11} 31^{111} 31^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -615, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 6 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 9 & 7 \\ 2 & 6 & 3 & 0 & 9 & 3 & 3 \\ 2 & 4 & 2 & -4 & 7 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{121} 3^{13} 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -635, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 9 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{111} 31^{12} 31^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -762, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 15 & 9 & 0 & 37 & 9 & 11 \\ 1 & 3 & 3 & 0 & 9 & 0 & 2 \\ 2 & 4 & 2 & -4 & 11 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[130] \ 1^{121} 3^{12} 21^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -768, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 10 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{141} 3^{11} 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -771, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.02 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 13 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{121} 3^{13} 21^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -875, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 11 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{131} 3^{11} 21^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -876, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 12 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{121}1^{14}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -876, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 12 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[135] $1^{111}31^{111}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -906, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 15 & 7 \\ 3 & 9 & 6 & 0 & 15 & 12 & 6 \\ 2 & 4 & 2 & -4 & 7 & 6 & -1 \end{pmatrix}_{b_0}$$

[136] $1^{131}1^{12}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -915, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 11 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{111}31^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -1059, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 21 & 3 & 4 \\ 1 & 4 & 0 & 0 & 12 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 21 & 12 & 0 & 73 & 12 & 15 \\ 1 & 3 & 3 & 0 & 12 & 0 & 2 \\ 2 & 4 & 2 & -4 & 15 & 2 & -1 \end{pmatrix}_{b_0}$$

[138] $1^{131}1^{13}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 12 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{141}1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 15 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{131}1^{14}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[141] $1^{131}1^{12}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 13 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[142] $1^{141}1^{12}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[143] $1^{111}31^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -1302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 0 & 27 & 3 & 4 \\ 1 & 4 & 0 & 0 & 15 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 27 & 15 & 0 & 121 & 15 & 19 \\ 1 & 3 & 3 & 0 & 15 & 0 & 2 \\ 2 & 4 & 2 & -4 & 19 & 2 & -1 \end{pmatrix}_{b_0}$$

[144] $1^{131}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1520, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 14 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[145] $1^{141}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1643, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[146] $1^{131}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1683, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 15 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{141}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1740, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 16 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{141}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1920, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[149] $1^{141}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -2171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 17 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[150] $1^{141}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -2496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 18 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& T = 10 \quad [151] \ 1^{\overline{11}} \ 4 \ 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 3, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0 \\
& \quad T_{\min} = 17.33 \rightarrow 18, \ T_{\max} = 24, \ T \in [18, 24] \\
& \quad \begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 8 & 0 & 13 & 5 & 7 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & -3 \end{pmatrix}_{b_0} \\
& [152] \ 1^{\overline{11}} \ 4 \ 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 12, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0 \\
& \quad T_{\min} = 17.33 \rightarrow 18, \ T_{\max} = 24, \ T \in [18, 24] \\
& \quad \begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 3 \\ 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & -3 \end{pmatrix}_{b_0} \\
& [153] \ 1^{\overline{1}} \ 4 \ 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 26, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0 \\
& \quad T_{\min} = 14.23 \rightarrow 15, \ T_{\max} = 24, \ T \in [15, 24] \\
& \quad \begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0} \\
& [154] \ 1^{\overline{1}} \ 4 \ 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 27, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0 \\
& \quad T_{\min} = 14.67 \rightarrow 15, \ T_{\max} = 24, \ T \in [15, 24] \\
& \quad \begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0} \\
& [155] \ 1^{\overline{1}} \ 231^{\overline{1}} \ 31^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 15, \ \Delta = -93, \ \det = 32, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0 \\
& \quad T_{\min} = 13.37 \rightarrow 14, \ T_{\max} = 15, \ T \in [14, 15] \\
& \quad \begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 3 & 0 & 0 & -1 & 0 \end{pmatrix}_{b_0} \\
& [156] \ 1^{\overline{11}} \ 2321^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 34, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0 \\
& \quad T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 1 & -1 \\ 2 & 4 & 0 & 4 & 6 & 4 & 6 \\ 1 & 3 & 0 & 1 & 4 & 0 & 2 \\ 2 & 3 & -1 & -1 & 6 & 2 & 1 \end{pmatrix}_{b_0} \\
& [157] \ 1^{\overline{11}} \ 2321^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 40, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0 \\
& \quad T_{\min} = 12.90 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
& [158] \ 1^{\overline{11}} \ 4 \ 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 48, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0 \\
& \quad T_{\min} = 14.08 \rightarrow 15, \ T_{\max} = 24, \ T \in [15, 24] \\
& \quad \begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 5 & 7 & 0 & 6 & 7 & 5 \\ 3 & 4 & -3 & 4 & 5 & 0 \end{pmatrix}_{b_0} \\
& [159] \ 1^{\overline{11}} \ 3 \ 221^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -85, \ \det = 56, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0 \\
& \quad T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13] \\
& \quad \begin{pmatrix} 5 & 9 & 1 & 1 & 5 \\ 9 & 13 & 1 & 0 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 7 & -1 & 0 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[160] $1^{1231^{11}31^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 6 & 0 & 3 & 0 & 3 & 3 \\ 2 & 4 & -1 & 0 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

[161] $1^{1112^1321^1} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 1 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[162] $1^{1231^{12}31^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 74, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 6 & 0 & 0 \\ 4 & 12 & 0 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[163] $1^{1114131^1} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[164] $1^{1111^{13}31^1} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 78, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 11 & 7 \\ 5 & 7 & 0 & 11 & 7 & 5 \\ 3 & 4 & -3 & 7 & 5 & 0 \end{pmatrix}_{b_0}$$

[165] $1^{1111^{13}31^1} 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 87, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 3 \\ 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[167] $1^{1211^13221^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 30 & 5 \\ 9 & 13 & 1 & 48 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 48 & 0 & 160 & 24 \\ 5 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[166] $1^{1411^13221^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 60 & 5 \\ 9 & 13 & 1 & 96 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 60 & 96 & 0 & 646 & 48 \\ 5 & 7 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

[168] $1^{1112^1321^1} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 101, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 3 & 1 & 0 & 0 & 2 \\ 2 & 0 & 2 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[169] $1^{11}1^{11}31^{11}4^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 10 & 2 & 3 \\ 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 15 & 0 & 45 & 11 & 13 \\ 2 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & -3 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

[170] $1^{112}2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -3 & 1 & 2 & -1 \\ 1 & 0 & 3 & 1 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 9 & 5 \\ 2 & 0 & 2 & -1 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[171] $1^{121}2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 1 & 7 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 1 & -1 \\ 3 & 7 & 0 & 5 & 17 & 7 & 9 \\ 1 & 3 & 0 & 1 & 7 & 0 & 2 \\ 2 & 3 & -1 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[172] $1^{111}2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 125, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 2 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[173] $1^{111}4^{131}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[174] $1^{123}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 137, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.30 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 0 & 0 \\ 2 & 6 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

[175] $1^{131}2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 154, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 10 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 1 & -1 \\ 4 & 10 & 0 & 6 & 34 & 10 & 12 \\ 1 & 3 & 0 & 1 & 10 & 0 & 2 \\ 2 & 3 & -1 & -1 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[176] $1^{111}2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 157, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & -3 & 4 & 1 & -1 \\ 3 & 0 & 4 & 4 & 6 & 4 & 6 \\ 2 & 0 & 3 & 1 & 4 & 0 & 2 \\ 2 & 0 & 2 & -1 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[177] $1^{111} 4^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 158$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.84 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 3 \\ 5 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & -3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{111} 23131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 15$, $\Delta = -93$, $\det = 164$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.32 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 5 & 0 & 4 & 4 & 4 \\ 1 & 5 & 1 & 0 & 4 & 4 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 4 & 4 & 0 & 2 & 4 & 4 \\ 1 & 4 & 4 & 0 & 4 & 1 & 3 \\ 2 & 4 & 3 & -4 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

[179] $1^1 231^{11} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 15$, $\Delta = -93$, $\det = 174$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.21 \rightarrow 14$, $T_{\max} = 15$, $T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 2 & 4 & -1 & 0 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[180] $1^{111} 4^1 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 203$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.81 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 8 & 5 & 6 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[181] $1^{11} 231^{11} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 15$, $\Delta = -93$, $\det = 212$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.34 \rightarrow 14$, $T_{\max} = 15$, $T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 7 & 0 & 0 \\ 4 & 9 & 0 & 7 & 14 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[182] $1^{121} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 215$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.91 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 3 & 6 & 0 & 2 & 10 & 6 & 6 \\ 1 & 3 & 0 & 1 & 6 & 0 & 2 \\ 2 & 3 & -1 & -1 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[183] $1^{121} 2^3 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 221$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -3 & 2 & 1 & -1 \\ 3 & 0 & 6 & 2 & 10 & 6 & 6 \\ 2 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -1 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[184] $1^{112} 3^1 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -85$, $\det = 221$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 6 & 0 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 6 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[185] $1^{121} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 7 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 1 & -1 \\ 5 & 7 & 0 & 5 & 17 & 7 & 9 \\ 1 & 3 & 0 & 1 & 7 & 0 & 2 \\ 2 & 3 & -1 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[186] $1^{111} 4^1 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[187] $1^{111} 4^1 131^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 15 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 15 & 0 & 45 & 10 & 13 \\ 2 & 4 & 0 & 10 & 0 & 2 \\ 3 & 4 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

[188] $1^{131} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 245, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 1 & -1 \\ 4 & 9 & 0 & 3 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 9 & 0 & 2 \\ 2 & 3 & -1 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[189] $1^{131} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 245, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 3 & 2 \\ 1 & 0 & 1 & -3 & 6 & 1 & -1 \\ 5 & 0 & 10 & 6 & 34 & 10 & 12 \\ 2 & 0 & 3 & 1 & 10 & 0 & 2 \\ 2 & 0 & 2 & -1 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[190] $1^{111} 2^3 21^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 254, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & -3 & 4 & 1 & -1 \\ 4 & 0 & 4 & 4 & 6 & 4 & 6 \\ 2 & 0 & 3 & 1 & 4 & 0 & 2 \\ 2 & 0 & 2 & -1 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[191] $1^{131} 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 257, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 3 \\ 4 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 0 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[192] $1^{121} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 269, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 7 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 1 & -1 \\ 4 & 7 & 0 & 5 & 17 & 7 & 9 \\ 1 & 3 & 0 & 1 & 7 & 0 & 2 \\ 2 & 3 & -1 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[193] $1^{111} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 271, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[194] \ 1^{1^{11}31^{11}} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 287, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 2 & 3 \\ 4 & 4 & 0 & 7 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 8 & 5 & 6 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[195] \ 1^{1^{12}2^3 21^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 290, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & 1 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 4 & 6 & 0 & 2 & 10 & 6 & 6 \\ 1 & 3 & 0 & 1 & 6 & 0 & 2 \\ 2 & 3 & -1 & -1 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[196] \ 1^{1^{12}2^3 3 21^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 311, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 7 & 3 & 2 \\ 1 & 0 & 1 & -3 & 5 & 1 & -1 \\ 4 & 0 & 7 & 5 & 17 & 7 & 9 \\ 2 & 0 & 3 & 1 & 7 & 0 & 2 \\ 2 & 0 & 2 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[197] \ 1^{1^{12}31^{11}31^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 314, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.31 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 0 & 0 \\ 3 & 6 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[198] \ 1^{1^{12}1^{11}3 221^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 344, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 31 & 5 \\ 9 & 13 & 1 & 48 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 48 & 0 & 160 & 24 \\ 5 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[199] \ 1^{1^{11}1^{12}2^3 21^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[200] \ 1^{1^{11}1^{13}1^{11}} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 384, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 12 & 0 & 26 & 8 & 10 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 3 & 4 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[201] \ 1^{1^{14}1^{12}2^3 21^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 386, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 3 & 2 \\ 1 & 0 & 1 & -3 & 4 & 1 & -1 \\ 6 & 0 & 12 & 4 & 46 & 12 & 12 \\ 2 & 0 & 3 & 1 & 12 & 0 & 2 \\ 2 & 0 & 2 & -1 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[202] $1^{11} 1^{11} 31^{11} 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 402, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 6 & 2 & 3 \\ 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 9 & 0 & 13 & 5 & 7 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[203] $1^{11} 1^{11} 31^{11} 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 444, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 8 & 2 & 3 \\ 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 12 & 0 & 26 & 8 & 10 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

[204] $1^{11} 231^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 449, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 10 & 0 & 0 \\ 6 & 15 & 0 & 10 & 38 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

[205] $1^{131} 2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 479, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 10 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 1 & -1 \\ 5 & 10 & 0 & 6 & 34 & 10 & 12 \\ 1 & 3 & 0 & 1 & 10 & 0 & 2 \\ 2 & 3 & -1 & -1 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[206] $1^{131} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 500, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 1 & -1 \\ 5 & 9 & 0 & 3 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 9 & 0 & 2 \\ 2 & 3 & -1 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{121} 3^2 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 32 & 5 \\ 9 & 13 & 1 & 48 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 32 & 48 & 0 & 160 & 24 \\ 5 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[208] $1^{121} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 586, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -3 & 2 & 1 & -1 \\ 4 & 0 & 6 & 2 & 10 & 6 & 6 \\ 2 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -1 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[209] $1^{1231} 1231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 587, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 6 & 0 & 0 \\ 5 & 12 & 0 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[210] \ 1^{131} 2^1 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 596, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 3 & 2 \\ 1 & 0 & 1 & -3 & 3 & 1 & -1 \\ 5 & 0 & 9 & 3 & 25 & 9 & 9 \\ 2 & 0 & 3 & 1 & 9 & 0 & 2 \\ 2 & 0 & 2 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[211] \ 1^{121} 3 \ 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -85, \ \det = 600, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 33 & 5 \\ 9 & 13 & 1 & 48 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 33 & 48 & 0 & 160 & 24 \\ 5 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[212] \ 1^{121} 3 \ 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -85, \ \det = 608, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 34 & 5 \\ 9 & 13 & 1 & 48 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 48 & 0 & 160 & 24 \\ 5 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[213] \ 1^{131} 2^1 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 614, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.90 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 1 & 10 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 1 & -1 \\ 6 & 10 & 0 & 6 & 34 & 10 & 12 \\ 1 & 3 & 0 & 1 & 10 & 0 & 2 \\ 2 & 3 & -1 & -1 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[214] \ 1^{141} 2^1 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 629, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.91 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 1 & 13 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 1 & -1 \\ 6 & 13 & 0 & 7 & 57 & 13 & 15 \\ 1 & 3 & 0 & 1 & 13 & 0 & 2 \\ 2 & 3 & -1 & -1 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[215] \ 1^{141} 3 \ 221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -85, \ \det = 632, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 61 & 5 \\ 9 & 13 & 1 & 96 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 61 & 96 & 0 & 646 & 48 \\ 5 & 7 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

$$[216] \ 1^{11} 231^{13} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 15, \ \Delta = -93, \ \det = 644, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.35 \rightarrow 14, \ T_{\max} = 15, \ T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 1 & 3 & 21 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 13 & 0 & 0 \\ 8 & 21 & 0 & 13 & 74 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[217] \ 1^{141} 2^1 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 650, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 1 & -1 \\ 6 & 12 & 0 & 4 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 12 & 0 & 2 \\ 2 & 3 & -1 & -1 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[218] \ 1^{121} 2^1 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 688, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 7 & 3 & 2 \\ 1 & 0 & 1 & -3 & 5 & 1 & -1 \\ 5 & 0 & 7 & 5 & 17 & 7 & 9 \\ 2 & 0 & 3 & 1 & 7 & 0 & 2 \\ 2 & 0 & 2 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[219] $1^{11}231^{14}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 797, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 \\ 2 & 4 & 1 & 3 & 27 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 16 & 0 & 0 \\ 10 & 27 & 0 & 16 & 122 & 0 & 20 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 20 & -1 & 0 \end{pmatrix}_{b_0}$$

[220] $1^1231^{13}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 818, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 1 & 3 & 18 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 9 & 0 & 0 \\ 7 & 18 & 0 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

[221] $1^{141}2^321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 896, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 13 & 3 & 2 \\ 1 & 0 & 1 & -3 & 7 & 1 & -1 \\ 7 & 0 & 13 & 7 & 57 & 13 & 15 \\ 2 & 0 & 3 & 1 & 13 & 0 & 2 \\ 2 & 0 & 2 & -1 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[222] $1^{131}2^321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 902, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 3 & 2 \\ 1 & 0 & 1 & -3 & 6 & 1 & -1 \\ 6 & 0 & 10 & 6 & 34 & 10 & 12 \\ 2 & 0 & 3 & 1 & 10 & 0 & 2 \\ 2 & 0 & 2 & -1 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[223] $1^{141}2^321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 1 & 1 & 1 & 13 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 1 & -1 \\ 7 & 13 & 0 & 7 & 57 & 13 & 15 \\ 1 & 3 & 0 & 1 & 13 & 0 & 2 \\ 2 & 3 & -1 & -1 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[224] $1^1231^{14}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 1007, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 1 & 3 & 24 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 & 0 \\ 9 & 24 & 0 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[225] $1^{141}3221^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 1088, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 62 & 5 \\ 9 & 13 & 1 & 96 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 62 & 96 & 0 & 646 & 48 \\ 5 & 7 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

[226] $1^{141}3221^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 1464, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 63 & 5 \\ 9 & 13 & 1 & 96 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 63 & 96 & 0 & 646 & 48 \\ 5 & 7 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

[227] $1^{141} 3^{221} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 1760, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 64 & 5 \\ 9 & 13 & 1 & 96 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 64 & 96 & 0 & 646 & 48 \\ 5 & 7 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

$T = 11$ [228] $1^{131} 5^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 23, \Delta = -107, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 18, T_{\max} = 23, T \in [18, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & 3 & 12 & 0 & 4 \\ 2 & 3 & -2 & 6 & 0 & 0 \\ 6 & 12 & 6 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 0 & 9 & -1 & -2 \end{pmatrix}_{b_0}$$

[229] $1^{111} 3^{141} 1^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 15, \Delta = -91, \det = -26, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.46 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 5 & 4 \\ 1 & 4 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 4 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 2 & -3 & 3 & 4 & 0 \end{pmatrix}_{b_0}$$

[230] $1^{121} 5^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 23, \Delta = -107, \det = -52, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 14.54 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 8 & 0 & 4 \\ 2 & 3 & -2 & 4 & 0 & 0 \\ 4 & 8 & 4 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 0 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

[231] $1^{131} 1^{111} 4^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 15, \Delta = -91, \det = -80, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 7 & 4 \\ 6 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 5 & 0 & -2 & 5 & 0 \\ 7 & 10 & 1 & 5 & 9 & 7 \\ 4 & 6 & 0 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

[232] $1^{111} 5^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 23, \Delta = -107, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.49 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 4 & 0 & 4 \\ 2 & 3 & -2 & 2 & 0 & 0 \\ 2 & 4 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 0 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[233] $1^{111} 3^{111} 4^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 15, \Delta = -91, \det = -137, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[234] $1^{111} 3^{111} 4^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 15, \Delta = -91, \det = -251, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.47 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 11 & 3 & 4 \\ 1 & 4 & 0 & 0 & 7 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 11 & 7 & 0 & 21 & 7 & 9 \\ 1 & 3 & 3 & 0 & 7 & 0 & 2 \\ 2 & 4 & 2 & -3 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

[235] $1^{111} 3^{111} 4^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 15, \Delta = -91, \det = -314, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 6 & 7 \\ 1 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 4 & 2 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [236] \quad 1^{111} 31^1 5131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -86, \det = 40, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [237] \quad 1^{11} 6131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 1 & 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[238] \quad 1^{11} 6131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -32, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & 1 & -1 \\ 1 & 1 & 3 & 1 & 1 & 3 & 3 \\ 2 & 1 & 3 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[239] \quad 1^{11} 6131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.76 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 1 & 4 & 0 & 2 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 1 & 1 & 1 & -3 & 3 & 0 & -1 \\ 3 & 4 & 7 & 3 & 16 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[240] \quad 1^{11} 6131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -50, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.84 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 2 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[241] \quad 1^{11} 6131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -98, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.76 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 3 & 0 & 2 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 1 & 1 & 1 & -3 & 2 & 0 & -1 \\ 2 & 3 & 4 & 2 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 5 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[242] \quad 1^{11} 6131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -112, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.82 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 3 & 1 & 2 \\ 1 & 2 & 0 & 1 & 4 & 3 & 2 \\ 1 & 1 & 1 & -3 & 2 & 1 & -1 \\ 2 & 3 & 4 & 2 & 5 & 4 & 5 \\ 2 & 1 & 3 & 1 & 4 & 0 & 2 \\ 2 & 2 & 2 & -1 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$

4.97 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 4 \\ 3 & 5 & 0 & 3 & 3 & 4 & 3 \\ 3 & 4 & -4 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

4.98 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 4 & -1 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -4 & 13 & 6 & 0 \end{pmatrix}_{b_0}$$

4.99 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.86 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 6 & 3 \\ 3 & 5 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -4 & 3 & 3 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.71 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 9 & 3 & 3 \\ 8 & 14 & 0 & 9 & 36 & 9 & 10 \\ 2 & 4 & 0 & 3 & 9 & 1 & 3 \\ 3 & 4 & -4 & 3 & 10 & 3 & -4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & -4 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1211} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 12 & 6 \\ 5 & 9 & 0 & 12 & 12 & 15 & 6 \\ 5 & 9 & 0 & 12 & 15 & 13 & 7 \\ 3 & 4 & -4 & 6 & 6 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1111} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 9 & 37 & 11 \\ 3 & 4 & -4 & 3 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 0 & 5 & 4 & 19 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 21 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 11 & 19 & 0 & 21 & 12 & 73 & 15 \\ 3 & 4 & -4 & 4 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{131} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 15 & 3 & 4 \\ 9 & 17 & 0 & 15 & 48 & 12 & 12 \\ 2 & 4 & 0 & 3 & 12 & 1 & 3 \\ 3 & 4 & -4 & 4 & 12 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1211} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 6 & 12 & 6 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 0 & 12 & 6 & 13 & 7 \\ 3 & 4 & -4 & 6 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3^1 1^{31} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 5 & 3 \\ 3 & 4 & 0 & 19 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 11 & 19 & 0 & 73 & 12 & 33 & 15 \\ 2 & 4 & 0 & 12 & 0 & 6 & 2 \\ 5 & 9 & 0 & 33 & 6 & 13 & 7 \\ 3 & 4 & -4 & 15 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1112} 3^1 1^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -137, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 7 & 3 \\ 3 & 4 & 0 & 14 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 9 & 33 & 11 \\ 2 & 4 & 0 & 9 & 0 & 9 & 2 \\ 7 & 13 & 0 & 33 & 9 & 28 & 10 \\ 3 & 4 & -4 & 11 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{113} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -173, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.39 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 5 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 15 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 9 & 17 & 0 & 15 & 12 & 49 & 13 \\ 3 & 4 & -4 & 4 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1211} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -174, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.85 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 6 & 6 \\ 5 & 9 & 0 & 12 & 12 & 6 & 6 \\ 2 & 4 & 0 & 6 & 6 & 1 & 3 \\ 3 & 4 & -4 & 6 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1111} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -191, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 4 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -3 & 3 & 0 & 3 & -1 \\ 3 & 8 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 6 & 0 & 4 & 4 \\ 3 & 5 & -1 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1212} 3^1 1^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 13 & 3 \\ 3 & 4 & 0 & 13 & 4 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 9 & 54 & 10 \\ 2 & 4 & 0 & 9 & 0 & 15 & 2 \\ 13 & 23 & 0 & 54 & 15 & 100 & 18 \\ 3 & 4 & -4 & 10 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1111} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -219, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 0 & 9 & 12 & 6 & 6 \\ 2 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & -4 & 4 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{1211} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -245, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 12 & 3 & 4 \\ 7 & 13 & 0 & 12 & 27 & 9 & 9 \\ 2 & 4 & 0 & 3 & 9 & 1 & 3 \\ 3 & 4 & -4 & 4 & 9 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1111} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -284, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -3 & 3 & 0 & 3 & -1 \\ 3 & 8 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 3 & 6 & 0 & 4 & 4 \\ 3 & 5 & -1 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{111} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -341, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & -4 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{111} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -344, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.81 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111} 31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -353, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 0 & 5 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 15 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 8 & 14 & 0 & 15 & 9 & 37 & 11 \\ 3 & 4 & -4 & 4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{111} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -359, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 3 & 5 & 0 & 3 & 3 & 9 & 3 \\ 5 & 9 & 0 & 6 & 9 & 13 & 7 \\ 3 & 4 & -4 & 3 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -368, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 4 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -3 & 3 & 0 & 3 & -1 \\ 4 & 8 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 6 & 0 & 4 & 4 \\ 3 & 5 & -1 & 4 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{112} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -380, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 12 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 7 & 13 & 0 & 12 & 9 & 28 & 10 \\ 3 & 4 & -4 & 4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{112} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -380, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.96 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 4 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 4 & 8 & 0 & 6 & 6 & 10 & 6 \\ 3 & 4 & -4 & 3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -408, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 6 & 12 & 6 & 6 \\ 2 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & -4 & 3 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -413, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 11 & 3 \\ 3 & 4 & 0 & 4 & 5 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 15 & 3 \\ 3 & 5 & 0 & 3 & 3 & 18 & 3 \\ 11 & 21 & 0 & 15 & 18 & 76 & 16 \\ 3 & 4 & -4 & 3 & 3 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1212} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 4 & 3 \\ 3 & 4 & 0 & 8 & 4 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 6 & 12 & 6 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 4 & 8 & 0 & 12 & 6 & 10 & 6 \\ 3 & 4 & -4 & 6 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1112} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -425, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 4 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 13 & 23 & 0 & 15 & 15 & 100 & 18 \\ 3 & 4 & -4 & 3 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1113} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -440, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.15 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 6 & 12 & 0 & 9 & 9 & 25 & 9 \\ 3 & 4 & -4 & 3 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1212} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -444, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 7 & 3 \\ 3 & 4 & 0 & 8 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 12 & 18 & 6 \\ 5 & 9 & 0 & 12 & 12 & 21 & 6 \\ 7 & 13 & 0 & 18 & 21 & 28 & 10 \\ 3 & 4 & -4 & 6 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -485, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1113} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -488, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.87 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 12 & 3 \\ 3 & 4 & 0 & 14 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 9 & 57 & 11 \\ 2 & 4 & 0 & 9 & 0 & 15 & 2 \\ 12 & 22 & 0 & 57 & 15 & 85 & 17 \\ 3 & 4 & -4 & 11 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1114} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 8 & 16 & 0 & 12 & 12 & 46 & 12 \\ 3 & 4 & -4 & 3 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -533, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.03 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 7 & 3 \\ 3 & 4 & 0 & 4 & 5 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 \\ 3 & 5 & 0 & 3 & 3 & 12 & 3 \\ 7 & 13 & 0 & 9 & 12 & 28 & 10 \\ 3 & 4 & -4 & 3 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -551, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 9 & 3 \\ 3 & 4 & 0 & 4 & 5 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 12 & 3 \\ 3 & 5 & 0 & 3 & 3 & 15 & 3 \\ 9 & 17 & 0 & 12 & 15 & 49 & 13 \\ 3 & 4 & -4 & 3 & 3 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1212} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -560, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 7 & 3 \\ 3 & 4 & 0 & 8 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 10 & 6 & 18 & 6 \\ 2 & 4 & 0 & 6 & 0 & 9 & 2 \\ 7 & 13 & 0 & 18 & 9 & 28 & 10 \\ 3 & 4 & -4 & 6 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -729, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 6 & 3 \\ 5 & 9 & 0 & 6 & 12 & 15 & 6 \\ 5 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & -4 & 3 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1111} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -773, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 8 & 3 \\ 3 & 4 & 0 & 9 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 24 & 7 \\ 2 & 4 & 0 & 6 & 0 & 9 & 2 \\ 8 & 14 & 0 & 24 & 9 & 37 & 11 \\ 3 & 4 & -4 & 7 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{112} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -800, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 0 & 5 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 18 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 10 & 18 & 0 & 18 & 12 & 58 & 14 \\ 3 & 4 & -4 & 4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{112} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -848, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 0 & 5 & 4 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 30 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 16 & 28 & 0 & 30 & 18 & 154 & 22 \\ 3 & 4 & -4 & 4 & 2 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 15 & 7 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 0 & 15 & 6 & 13 & 7 \\ 3 & 4 & -4 & 7 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1111}31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -920, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 8 & 3 \\ 3 & 4 & 0 & 14 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 9 & 39 & 11 \\ 2 & 4 & 0 & 9 & 0 & 9 & 2 \\ 8 & 14 & 0 & 39 & 9 & 37 & 11 \\ 3 & 4 & -4 & 11 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1112}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -929, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 7 & 13 & 0 & 9 & 9 & 28 & 10 \\ 3 & 4 & -4 & 3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1112}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 10 & 18 & 0 & 12 & 12 & 58 & 14 \\ 3 & 4 & -4 & 3 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{114}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.91 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 0 & 5 & 4 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 24 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 14 & 26 & 0 & 24 & 18 & 118 & 20 \\ 3 & 4 & -4 & 4 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{112}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 0 & 5 & 4 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 24 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 13 & 23 & 0 & 24 & 15 & 100 & 18 \\ 3 & 4 & -4 & 4 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{113}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1001, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 0 & 5 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 21 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 12 & 22 & 0 & 21 & 15 & 85 & 17 \\ 3 & 4 & -4 & 4 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1113}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 4 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 21 & 3 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 18 & 32 & 0 & 21 & 21 & 193 & 25 \\ 3 & 4 & -4 & 3 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1112}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1083, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 7 & 3 \\ 3 & 4 & 0 & 4 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 9 & 3 \\ 5 & 9 & 0 & 6 & 12 & 21 & 6 \\ 7 & 13 & 0 & 9 & 21 & 28 & 10 \\ 3 & 4 & -4 & 3 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1085, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 11 & 3 \\ 3 & 4 & 0 & 9 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 33 & 7 \\ 2 & 4 & 0 & 6 & 0 & 15 & 2 \\ 11 & 21 & 0 & 33 & 15 & 76 & 16 \\ 3 & 4 & -4 & 7 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1205, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 7 & 3 \\ 3 & 4 & 0 & 9 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 21 & 7 \\ 2 & 4 & 0 & 6 & 0 & 9 & 2 \\ 7 & 13 & 0 & 21 & 9 & 28 & 10 \\ 3 & 4 & -4 & 7 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1112} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 10 & 3 \\ 3 & 4 & 0 & 14 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 9 & 48 & 11 \\ 2 & 4 & 0 & 9 & 0 & 12 & 2 \\ 10 & 18 & 0 & 48 & 12 & 58 & 14 \\ 3 & 4 & -4 & 11 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 9 & 3 \\ 3 & 4 & 0 & 9 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 27 & 7 \\ 2 & 4 & 0 & 6 & 0 & 12 & 2 \\ 9 & 17 & 0 & 27 & 12 & 49 & 13 \\ 3 & 4 & -4 & 7 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1112} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 16 & 3 \\ 3 & 4 & 0 & 9 & 4 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 48 & 7 \\ 2 & 4 & 0 & 6 & 0 & 18 & 2 \\ 16 & 28 & 0 & 48 & 18 & 154 & 22 \\ 3 & 4 & -4 & 7 & 2 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1281, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 9 & 3 \\ 3 & 4 & 0 & 4 & 9 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 12 & 3 \\ 5 & 9 & 0 & 6 & 12 & 27 & 6 \\ 9 & 17 & 0 & 12 & 27 & 49 & 13 \\ 3 & 4 & -4 & 3 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1323, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 11 & 3 \\ 3 & 4 & 0 & 4 & 9 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 15 & 3 \\ 5 & 9 & 0 & 6 & 12 & 33 & 6 \\ 11 & 21 & 0 & 15 & 33 & 76 & 16 \\ 3 & 4 & -4 & 3 & 6 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1397, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 9 & 17 & 0 & 12 & 12 & 49 & 13 \\ 3 & 4 & -4 & 3 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{113}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1565, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$

$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 0 & 5 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 27 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 15 & 27 & 0 & 27 & 18 & 133 & 21 \\ 3 & 4 & -4 & 4 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{1212}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 10 & 3 \\ 3 & 4 & 0 & 13 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 9 & 42 & 10 \\ 2 & 4 & 0 & 9 & 0 & 12 & 2 \\ 10 & 18 & 0 & 42 & 12 & 58 & 14 \\ 3 & 4 & -4 & 10 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

[61] $1^{1113}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$

$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 15 & 27 & 0 & 18 & 18 & 133 & 21 \\ 3 & 4 & -4 & 3 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

[62] $1^{1112}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1760, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$

$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 10 & 3 \\ 3 & 4 & 0 & 9 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 30 & 7 \\ 2 & 4 & 0 & 6 & 0 & 12 & 2 \\ 10 & 18 & 0 & 30 & 12 & 58 & 14 \\ 3 & 4 & -4 & 7 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

[63] $1^{1112}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1781, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$

$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 13 & 3 \\ 3 & 4 & 0 & 9 & 4 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 39 & 7 \\ 2 & 4 & 0 & 6 & 0 & 15 & 2 \\ 13 & 23 & 0 & 39 & 15 & 100 & 18 \\ 3 & 4 & -4 & 7 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{1212}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1784, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$

$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 9 & 30 & 10 \\ 2 & 4 & 0 & 9 & 0 & 9 & 2 \\ 7 & 13 & 0 & 30 & 9 & 28 & 10 \\ 3 & 4 & -4 & 10 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{1113}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1820, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$

$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 12 & 22 & 0 & 15 & 15 & 85 & 17 \\ 3 & 4 & -4 & 3 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

[66] $1^{113}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1865, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$

$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 0 & 5 & 4 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 33 & 4 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 18 & 32 & 0 & 33 & 21 & 193 & 25 \\ 3 & 4 & -4 & 4 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{1114} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1889, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.86 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 11 & 21 & 0 & 15 & 15 & 76 & 16 \\ 3 & 4 & -4 & 3 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{114} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1928, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 0 & 5 & 4 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 30 & 4 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 17 & 31 & 0 & 30 & 21 & 172 & 24 \\ 3 & 4 & -4 & 4 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{1113} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2501, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 12 & 3 \\ 3 & 4 & 0 & 9 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 36 & 7 \\ 2 & 4 & 0 & 6 & 0 & 15 & 2 \\ 12 & 22 & 0 & 36 & 15 & 85 & 17 \\ 3 & 4 & -4 & 7 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{1212} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2624, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 10 & 3 \\ 3 & 4 & 0 & 18 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 10 & 18 & 0 & 58 & 12 & 60 & 14 \\ 2 & 4 & 0 & 12 & 0 & 12 & 2 \\ 10 & 18 & 0 & 60 & 12 & 58 & 14 \\ 3 & 4 & -4 & 14 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{114} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2636, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 0 & 5 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 36 & 4 \\ 2 & 4 & 0 & 3 & 0 & 24 & 2 \\ 20 & 36 & 0 & 36 & 24 & 238 & 28 \\ 3 & 4 & -4 & 4 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{1114} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2720, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 4 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 14 & 26 & 0 & 18 & 18 & 118 & 20 \\ 3 & 4 & -4 & 3 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{1114} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2780, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 24 & 3 \\ 2 & 4 & 0 & 3 & 0 & 24 & 2 \\ 20 & 36 & 0 & 24 & 24 & 238 & 28 \\ 3 & 4 & -4 & 3 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{1114} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2996, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 14 & 3 \\ 3 & 4 & 0 & 9 & 4 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 42 & 7 \\ 2 & 4 & 0 & 6 & 0 & 18 & 2 \\ 14 & 26 & 0 & 42 & 18 & 118 & 20 \\ 3 & 4 & -4 & 7 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{1114} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3017, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 4 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 21 & 3 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 17 & 31 & 0 & 21 & 21 & 172 & 24 \\ 3 & 4 & -4 & 3 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{1113} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 15 & 3 \\ 3 & 4 & 0 & 9 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 45 & 7 \\ 2 & 4 & 0 & 6 & 0 & 18 & 2 \\ 15 & 27 & 0 & 45 & 18 & 133 & 21 \\ 3 & 4 & -4 & 7 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{1113} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3365, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 18 & 3 \\ 3 & 4 & 0 & 9 & 4 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 54 & 7 \\ 2 & 4 & 0 & 6 & 0 & 21 & 2 \\ 18 & 32 & 0 & 54 & 21 & 193 & 25 \\ 3 & 4 & -4 & 7 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{1114} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -4373, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 17 & 3 \\ 3 & 4 & 0 & 9 & 4 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 51 & 7 \\ 2 & 4 & 0 & 6 & 0 & 21 & 2 \\ 17 & 31 & 0 & 51 & 21 & 172 & 24 \\ 3 & 4 & -4 & 7 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{1114} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -5216, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 20 & 3 \\ 3 & 4 & 0 & 9 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 60 & 7 \\ 2 & 4 & 0 & 6 & 0 & 24 & 2 \\ 20 & 36 & 0 & 60 & 24 & 238 & 28 \\ 3 & 4 & -4 & 7 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [80] \quad 1^{11} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 44, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{1111} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{1111} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 82, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.90 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 3 \\ 3 & 4 & 1 & 3 & 9 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 3 & 0 \\ 5 & 9 & 0 & 7 & 14 & 9 & 8 \\ 3 & 6 & 0 & 3 & 9 & 4 & 4 \\ 3 & 4 & -1 & 0 & 8 & 4 & 1 \end{pmatrix}_{b_0}$$

[83] $1^{1111}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 111, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 & 6 & 3 \\ 1 & 3 & 3 & 0 & 3 & 0 & 6 & 2 \\ 3 & 6 & 9 & 0 & 6 & 6 & 13 & 7 \\ 2 & 3 & 3 & -4 & 3 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

[84] $1^{11}31^{121}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 138, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 12 & 6 & 0 & 22 & 6 & 12 & 8 \\ 1 & 3 & 3 & 0 & 6 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 12 & 3 & 4 & 4 \\ 2 & 4 & 2 & -4 & 8 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[85] $1^{1111}3 \ 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 4 & 3 \\ 3 & 4 & 1 & 3 & 9 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 3 & 0 \\ 6 & 9 & 0 & 7 & 14 & 9 & 8 \\ 4 & 6 & 0 & 3 & 9 & 4 & 4 \\ 3 & 4 & -1 & 0 & 8 & 4 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{111}3 \ 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 239, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 6 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 7 & 0 & 14 & 8 \\ 3 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{1111}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 268, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{111}3 \ 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 269, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{1112}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 273, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 9 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 & 9 & 3 \\ 1 & 3 & 3 & 0 & 3 & 0 & 9 & 2 \\ 4 & 9 & 12 & 0 & 9 & 9 & 28 & 10 \\ 2 & 3 & 3 & -4 & 3 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[90] $1^{112}3 \ 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{111}31^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 300, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 6 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 6 \\ 1 & 3 & 3 & 0 & 3 & 0 & 3 \\ 2 & 6 & 3 & 0 & 6 & 3 & 4 \\ 2 & 4 & 2 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{111}3^{111}3^{111}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 320, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.90 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{111}31^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 363, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 3 \\ 2 & 4 & 4 & 0 & 6 & 3 & 9 \\ 1 & 4 & 0 & 0 & 3 & 3 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 9 \\ 1 & 3 & 3 & 0 & 3 & 0 & 6 \\ 3 & 9 & 6 & 0 & 9 & 6 & 13 \\ 2 & 4 & 2 & -4 & 4 & 2 & 7 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{112}3^{111}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 374, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.88 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 9 & 3 \\ 3 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 10 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 0 & 10 & 0 & 38 & 12 \\ 3 & 4 & -1 & 0 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{111}3^{111}3^{111}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 381, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.90 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[96] \quad 1^{111}3^{111}3^{111}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 704, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.86 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[97] \quad 1^{111}31^{112}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 1065, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 5 \\ 2 & 4 & 4 & 0 & 6 & 3 & 15 \\ 1 & 4 & 0 & 0 & 3 & 3 & 9 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 15 \\ 1 & 3 & 3 & 0 & 3 & 0 & 9 \\ 5 & 15 & 9 & 0 & 15 & 9 & 37 \\ 2 & 4 & 2 & -4 & 4 & 2 & 11 \end{pmatrix}_{b_0}$$

$$[98] \quad 1^{111}31^{113}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 1659, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 21 & 4 \\ 1 & 3 & 3 & 0 & 3 & 0 & 12 & 2 \\ 7 & 21 & 12 & 0 & 21 & 12 & 73 & 15 \\ 2 & 4 & 2 & -4 & 4 & 2 & 15 & -1 \end{pmatrix}_{b_0}$$

[99] $1^{111}31^{114}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 2145, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 27 & 4 \\ 1 & 3 & 3 & 0 & 3 & 0 & 15 & 2 \\ 9 & 27 & 15 & 0 & 27 & 15 & 121 & 19 \\ 2 & 4 & 2 & -4 & 4 & 2 & 19 & -1 \end{pmatrix}_{b_0}$$

$T = 11 \quad [100] \quad 1^{111}4 \quad 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 4 & 0 & 3 & 0 & 5 & 2 \\ 5 & 7 & 0 & 6 & 5 & 8 & 6 \\ 3 & 4 & -3 & 4 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[101] $1^{111}31^{11}4 \quad 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -48, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 5 & 3 \\ 4 & 4 & 0 & 5 & 3 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 3 & 0 & 3 & 0 & 5 & 2 \\ 5 & 7 & 0 & 6 & 5 & 8 & 6 \\ 3 & 4 & -3 & 4 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

4.100 $\text{NHC}(-2)(-3) + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2$

$T = 10 \quad [0] \quad 1^{111}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 148, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 6 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 6 & 9 & 13 & 7 \\ 3 & 4 & -4 & 4 & 2 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{1111}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 9 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 4 & 2 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{1111}31^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 9 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 9 & 6 & 2 \\ 8 & 14 & 0 & 9 & 9 & 37 & 24 & 11 \\ 5 & 9 & 0 & 6 & 6 & 24 & 13 & 7 \\ 3 & 4 & -4 & 3 & 2 & 11 & 7 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{1112}31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 274, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 18 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 12 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 12 & 3 \\ 10 & 18 & 0 & 18 & 12 & 12 & 58 & 14 \\ 3 & 4 & -4 & 4 & 2 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

- [4] $1^{\overline{1113}} 3 1^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 454, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 27 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 18 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 18 & 3 \\ 15 & 27 & 0 & 27 & 18 & 18 & 133 & 21 \\ 3 & 4 & -4 & 4 & 2 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$
- [5] $1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 & 3 & 3 \\ 5 & 9 & 0 & 6 & 12 & 6 & 6 & 6 \\ 2 & 4 & 0 & 3 & 6 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 6 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 6 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$
- [6] $1^{\overline{1113}} 3 1^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 21 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 15 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 15 & 3 \\ 12 & 22 & 0 & 21 & 15 & 15 & 85 & 17 \\ 3 & 4 & -4 & 4 & 2 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$
- [7] $1^{\overline{1114}} 3 1^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 706, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 20 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 36 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 24 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 24 & 3 \\ 20 & 36 & 0 & 36 & 24 & 24 & 238 & 28 \\ 3 & 4 & -4 & 4 & 2 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{\overline{1111}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 712, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & -4 & 3 & 2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 724, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 2 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{\overline{1112}} 3 1^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 796, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 9 & 12 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 9 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 21 & 7 \\ 7 & 13 & 0 & 12 & 9 & 21 & 28 & 10 \\ 3 & 4 & -4 & 4 & 2 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{\overline{1111}} 3 1^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 9 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 9 & 6 & 15 & 13 & 7 \\ 3 & 4 & -4 & 4 & 2 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{11111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1032, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 6 & 3 & 6 & 3 \\ 5 & 9 & 0 & 6 & 12 & 6 & 15 & 6 \\ 2 & 4 & 0 & 3 & 6 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 15 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 6 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{1122}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 12 & 12 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 9 & 2 \\ 7 & 13 & 0 & 12 & 9 & 28 & 30 & 10 \\ 7 & 13 & 0 & 12 & 9 & 30 & 28 & 10 \\ 3 & 4 & -4 & 4 & 2 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{1114}31^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1180, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 30 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 21 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 21 & 3 \\ 17 & 31 & 0 & 30 & 21 & 21 & 172 & 24 \\ 3 & 4 & -4 & 4 & 2 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{11111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 6 & 6 & 2 \\ 5 & 9 & 0 & 6 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 6 & 15 & 13 & 7 \\ 3 & 4 & -4 & 3 & 2 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{11111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 2368, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 15 & 15 & 7 \\ 2 & 4 & 0 & 6 & 0 & 6 & 6 & 2 \\ 5 & 9 & 0 & 15 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 15 & 6 & 15 & 13 & 7 \\ 3 & 4 & -4 & 7 & 2 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

4.101 NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 10$ [0] $1^{11111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & 3 & 0 & -1 \\ 4 & 8 & 3 & 4 & 0 & 6 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 8 & 3 & 6 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 231, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 9 & 6 & 6 & 12 & 6 \\ 3 & 4 & -4 & 4 & 2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 507, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 9 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 3 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & -4 & 4 & 2 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 651, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 9 & 3 & 3 \\ 5 & 9 & 0 & 6 & 9 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{1141}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 813, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 11 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 21 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 15 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 18 & 3 & 3 \\ 11 & 21 & 0 & 15 & 18 & 76 & 15 & 16 \\ 2 & 4 & 0 & 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 16 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{1121}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 1173, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 7 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 12 & 3 & 3 \\ 7 & 13 & 0 & 9 & 12 & 28 & 9 & 10 \\ 2 & 4 & 0 & 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\overline{1131}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 1227, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 9 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 17 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 12 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 15 & 3 & 3 \\ 9 & 17 & 0 & 12 & 15 & 49 & 12 & 13 \\ 2 & 4 & 0 & 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

4.102 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \quad 1^{\overline{111}} 31^{\overline{111}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 86, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.56 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 & 2 & 3 \\ 1 & 3 & 3 & 0 & 3 & 0 & 2 & 2 \\ 1 & 2 & 3 & 0 & 2 & 2 & 0 & 2 \\ 2 & 3 & 3 & -4 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 15, \ \Delta = -93, \ \det = 126, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.43 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 & 2 & 3 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 & 3 \\ 1 & 3 & 2 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & -4 & 3 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{121}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 130, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 1 & 3 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 2 & 0 \\ 5 & 9 & 0 & 7 & 14 & 5 & 8 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{121}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 146, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 4 & -1 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{121}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 226, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 2 & 0 \\ 3 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 15, \ \Delta = -93, \ \det = 296, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 15, \ T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 3 & 4 \\ 1 & 3 & 3 & 0 & 3 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 3 & 2 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{121}3^{121}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 328, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 2 & 0 \\ 8 & 15 & 0 & 10 & 38 & 8 & 12 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111}3^{121}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 386, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 2 & 0 \\ 5 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111}3^{121}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 416, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 2 & 3 \\ 3 & 4 & 1 & 3 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 2 & 0 \\ 6 & 9 & 0 & 7 & 14 & 5 & 8 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{111}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 416, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 2 & 0 \\ 4 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{111}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 482, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 1 & 3 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 2 & 0 \\ 7 & 9 & 0 & 7 & 14 & 5 & 8 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{121}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 626, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 2 & 0 \\ 7 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{121}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 896, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 2 & 0 \\ 8 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{121}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 914, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 2 & 0 \\ 9 & 15 & 0 & 10 & 38 & 8 & 12 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{121}3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 1280, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 2 & 0 \\ 10 & 15 & 0 & 10 & 38 & 8 & 12 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [15] \quad 1^{111}3^{115}131 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -69, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.97 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 & 0 \\ 2 & 3 & -1 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

4.103 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2\mathfrak{n}3\mathfrak{mix}$

$$T = 10 \quad [0] \quad 1^{111}3^{111}31 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 494, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 5 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 9 & 4 \\ 1 & 3 & 3 & 0 & 3 & 0 & 5 & 2 \\ 3 & 9 & 5 & 0 & 9 & 5 & 12 & 6 \\ 2 & 4 & 2 & -4 & 4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

4.104 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{\overline{112}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 & 3 \\ 3 & 4 & 0 & 9 & 8 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 12 & 9 & 7 \\ 4 & 8 & 0 & 12 & 9 & 6 & 5 \\ 3 & 5 & 0 & 9 & 6 & 3 & 3 \\ 3 & 4 & -4 & 7 & 5 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{114}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & 0 & -1 \\ 13 & 32 & 12 & 94 & 0 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 16 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & 0 & -1 \\ 4 & 8 & 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 3 & 3 \\ 3 & 4 & -4 & 4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{112}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 1 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 0 & 5 \\ 2 & 4 & -3 & 6 & 0 & 0 & -1 \\ 7 & 16 & 6 & 22 & 0 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 8 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{113}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

14.8

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 1 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 0 & 5 \\ 2 & 4 & -3 & 9 & 0 & 0 & -1 \\ 10 & 24 & 9 & 52 & 0 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 12 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{1111}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 9 & 36 & 10 \\ 3 & 4 & -4 & 3 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{122}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -267, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.7$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 4 & 3 \\ 3 & 4 & 0 & 13 & 5 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 12 & 18 & 10 \\ 3 & 5 & 0 & 12 & 3 & 6 & 3 \\ 4 & 8 & 0 & 18 & 6 & 9 & 5 \\ 3 & 4 & -4 & 10 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{132}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 4 & 3 \\ 3 & 4 & 0 & 17 & 5 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 15 & 24 & 13 \\ 3 & 5 & 0 & 15 & 3 & 6 & 3 \\ 4 & 8 & 0 & 24 & 6 & 9 & 5 \\ 3 & 4 & -4 & 13 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{1111}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -402, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 9 & 9 & 11 \\ 2 & 4 & 0 & 9 & 0 & 3 & 2 \\ 2 & 4 & 0 & 9 & 3 & 0 & 2 \\ 3 & 4 & -4 & 11 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{113}3^11^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -420, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 0 & 5 \\ 2 & 4 & -3 & 9 & 0 & 0 & -1 \\ 11 & 24 & 9 & 52 & 0 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 12 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{111}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 12 & 6 \\ 3 & 4 & -4 & 4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{1111}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[15] $1^{114}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -555, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & 0 & -1 \\ 15 & 32 & 12 & 94 & 0 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 16 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -645, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 9 & 6 & 7 \\ 3 & 5 & 0 & 9 & 3 & 3 & 3 \\ 2 & 4 & 0 & 6 & 3 & 0 & 2 \\ 3 & 4 & -4 & 7 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -651, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & -4 & 3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{114} 3^1 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -690, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 0 & 5 \\ 2 & 4 & -3 & 12 & 0 & 0 & -1 \\ 14 & 32 & 12 & 94 & 0 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 16 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{141} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -726, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 2 & 3 \\ 3 & 4 & 0 & 21 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 11 & 21 & 0 & 76 & 18 & 15 & 16 \\ 3 & 5 & 0 & 18 & 3 & 3 & 3 \\ 2 & 4 & 0 & 15 & 3 & 0 & 2 \\ 3 & 4 & -4 & 16 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -789, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 6 & 7 \\ 2 & 4 & 0 & 6 & 0 & 3 & 2 \\ 2 & 4 & 0 & 6 & 3 & 0 & 2 \\ 3 & 4 & -4 & 7 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -906, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 2 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 28 & 12 & 9 & 10 \\ 3 & 5 & 0 & 12 & 3 & 3 & 3 \\ 2 & 4 & 0 & 9 & 3 & 0 & 2 \\ 3 & 4 & -4 & 10 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{131} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -933, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 2 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 9 & 17 & 0 & 49 & 15 & 12 & 13 \\ 3 & 5 & 0 & 15 & 3 & 3 & 3 \\ 2 & 4 & 0 & 12 & 3 & 0 & 2 \\ 3 & 4 & -4 & 13 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -1200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 15 & 7 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 0 & 15 & 6 & 12 & 6 \\ 3 & 4 & -4 & 7 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \quad 1^{111} 3 \quad 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 0 & 5 \\ 2 & 4 & -2 & 7 & 0 & 0 & 0 \\ 5 & 12 & 7 & 14 & 0 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 8 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{111} 3 \quad 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 & 0 \\ 4 & 8 & 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{111} 3 \quad 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & 0 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{111} 3 \quad 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 182, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.26 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 & 0 \\ 3 & 6 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & 0 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 477, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 6 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 3 & 6 \\ 1 & 3 & 3 & 0 & 3 & 0 & 3 \\ 2 & 6 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 2 & -4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{111} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 486, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 9 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 9 & 0 & 13 & 6 & 6 & 7 \\ 1 & 3 & 3 & 0 & 6 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 6 & 3 & 0 & 2 \\ 2 & 3 & 3 & -4 & 7 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{111} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 774, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 13 & 6 & 9 & 7 \\ 1 & 3 & 3 & 0 & 6 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 9 & 3 & 3 & 3 \\ 2 & 4 & 2 & -4 & 7 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [31] \quad 1^{11} 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 3 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 0 & 2 & 2 \\ 3 & 4 & -3 & 4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{11} 4 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -108, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 2 & 3 \\ 4 & 4 & 0 & 5 & 3 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 3 & 4 & -3 & 4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{11} 4 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -210, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 2 & 3 \\ 4 & 4 & 0 & 4 & 3 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 4 & 0 & 1 & 2 & 2 & 3 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 3 & 4 & -3 & 3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{11} 4 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -378, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 6 & 2 & 2 & 3 \\ 4 & 4 & 0 & 9 & 3 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 6 & 9 & 0 & 13 & 5 & 5 & 7 \\ 2 & 3 & 0 & 5 & 0 & 3 & 2 \\ 2 & 3 & 0 & 5 & 3 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

4.105 $\text{NHC}(-2)(-3) + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2$

$$T = 10 \quad [0] \ 1^{1111} 3 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{violet}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 1 & 4 & 3 \\ 4 & 9 & 4 & 8 & 0 & 0 & 8 & 5 \\ 2 & 4 & -3 & 3 & 0 & 0 & 3 & -1 \\ 4 & 8 & 3 & 4 & 0 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 6 & 0 & 0 & 4 & 4 \\ 3 & 5 & -1 & 4 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{violet}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{violet}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 231, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 9 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 3 & 2 \\ 5 & 9 & 0 & 9 & 6 & 12 & 6 & 6 \\ 2 & 4 & 0 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & -4 & 4 & 2 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{violet}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 507, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 4 & 2 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 651, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 6 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 9 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 3 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1114}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 813, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 15 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 15 & 2 \\ 11 & 21 & 0 & 15 & 18 & 15 & 76 & 16 \\ 3 & 4 & -4 & 3 & 3 & 2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1112}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1173, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 9 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 9 & 2 \\ 7 & 13 & 0 & 9 & 12 & 9 & 28 & 10 \\ 3 & 4 & -4 & 3 & 3 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1113}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1227, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 12 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 12 & 2 \\ 9 & 17 & 0 & 12 & 15 & 12 & 49 & 13 \\ 3 & 4 & -4 & 3 & 3 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

4.106 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 459, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 & 2 \\ 3 & 4 & -4 & 3 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 1296, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 9 & 6 & 6 & 7 \\ 3 & 5 & 0 & 9 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 6 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 6 & 3 & 0 & 2 & 2 \\ 3 & 4 & -4 & 7 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.107 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \ 1^{\overline{1}} 3 1^{\overline{111}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 98, \ \Delta = -143, \ \det = 925, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 1 & 4 \\ 8 & 9 & 4 & 8 & 0 & 5 \\ 3 & 4 & -3 & 3 & 0 & -1 \\ 6 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & -1 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

4.108 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \quad 21^1 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -138, \quad \det = 22, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, \quad T_{\max} = 98, \quad T \in [13, 98]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 5 & 5 & 3 & 5 \\ 3 & 5 & -2 & 2 & 0 \\ 3 & 3 & 2 & 0 & 2 \\ 4 & 5 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 13^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -138, \quad \det = 34, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.35 \rightarrow 14, \quad T_{\max} = 98, \quad T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 5 & 5 & 3 & 5 \\ 2 & 5 & -2 & 2 & 0 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 5 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 13^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -138, \quad \det = 44, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.09 \rightarrow 14, \quad T_{\max} = 98, \quad T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 2 & 4 & -2 & 2 & 0 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 5 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -144, \quad \det = 44, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.09 \rightarrow 14, \quad T_{\max} = 98, \quad T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 2 & 2 & 0 & 0 & 2 \\ 2 & 3 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -138, \quad \det = 46, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.48 \rightarrow 13, \quad T_{\max} = 98, \quad T \in [13, 98]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 2 & 0 & -2 & 0 \\ 3 & 3 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^1 3^1 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -144, \quad \det = 81, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 98, \quad T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 2 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^1 3^1 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -144, \quad \det = 106, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.81 \rightarrow 13, \quad T_{\max} = 98, \quad T \in [13, 98]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 2 & 4 & 2 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 21^1 1^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -138, \quad \det = 170, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.60 \rightarrow 13, \quad T_{\max} = 98, \quad T \in [13, 98]$$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 4 \\ 8 & 9 & 4 & 4 & 5 \\ 3 & 4 & -2 & 2 & 0 \\ 3 & 4 & 2 & 0 & 2 \\ 4 & 5 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \quad 1^{11} 23^1 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 32, \quad \Delta = -117, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 16, \quad T_{\max} = 32, \quad T \in [16, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 5 & 0 & 2 & 4 \\ 1 & 5 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 2 & 2 & 0 & 0 & 2 \\ 2 & 4 & 3 & -4 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^1 23^1 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 32, \quad \Delta = -117, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 16, \quad T_{\max} = 32, \quad T \in [16, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & -1 & 0 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 2313^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 29, \quad \Delta = -116, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$\begin{aligned}
& T_{\min} = 19, T_{\max} = 29, T \in [19, 29] \\
& \begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 28 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 5 & 10 & -1 & -3 \end{pmatrix}_{b_0} \\
[11] \quad & 1^{11}23^{11}131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -22, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 14, T_{\max} = 32, T \in [14, 32] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 2 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 4 & 0 & -2 & 2 & 0 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 4 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0} \\
[12] \quad & 1^{11}2313^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 13.38 \rightarrow 14, T_{\max} = 29, T \in [14, 29] \\
& \begin{pmatrix} 5 & 1 & 1 & 5 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 5 & 0 & -1 & 2 \end{pmatrix}_{b_0} \\
[13] \quad & 1^{11}23^{11}1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 13.39 \rightarrow 14, T_{\max} = 32, T \in [14, 32] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 2 & 4 & 3 & -4 & 2 & 0 \end{pmatrix}_{b_0} \\
[14] \quad & 1^{11}23^{11}1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 13, T_{\max} = 32, T \in [13, 32] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 0 \\ 1 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0} \\
[15] \quad & 1^{11}231^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 12.93 \rightarrow 13, T_{\max} = 29, T \in [13, 29] \\
& \begin{pmatrix} 5 & 14 & 1 & 5 \\ 14 & 33 & 1 & 11 \\ 1 & 1 & -3 & -1 \\ 5 & 11 & -1 & 3 \end{pmatrix}_{b_0} \\
[16] \quad & 1^{11}2321^{11}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 36, \Delta = -118, \det = -375, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 36, T \in [13, 36] \\
& \begin{pmatrix} 0 & 3 & 3 & 1 & 1 & 2 \\ 3 & 1 & 4 & 1 & 0 & 3 \\ 3 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0} \\
& T = 10 \quad [17] \quad 14^{11}13^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 22, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 15, T_{\max} = 22, T \in [15, 22] \\
& \begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 2 & 0 & 3 \\ 1 & 2 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & -2 & 0 & 1 \end{pmatrix}_{b_0} \\
[18] \quad & 1^{11}31^{11}413^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 14.37 \rightarrow 15, T_{\max} = 26, T \in [15, 26] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 0 & -1 & 0 & 0 \end{pmatrix}_{b_0} \\
[19] \quad & 1^{11}3^{11}1^{11}4131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 83, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 13.93 \rightarrow 14, T_{\max} = 26, T \in [14, 26] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 2 & 4 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

$$T = 11 \quad [20] \quad 1^{\overline{1}3}1^{\overline{1}5}131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.48 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{\overline{1}3}1^{\overline{1}5}131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -29, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.59 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 3 & -1 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{\overline{1}3}151^{\overline{1}23}1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -115, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 5 & 22 & 1 & 5 \\ 22 & 86 & 1 & 18 \\ 1 & 1 & -3 & -1 \\ 5 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [23] \quad 1^{\overline{1}3}1^{\overline{1}21}6^{\overline{1}}131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.48 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{\overline{1}3}1^{\overline{1}31}6^{\overline{1}}131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 28, \Delta = -114, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.94 \rightarrow 16, T_{\max} = 28, T \in [16, 28]$$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 3 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [25] \quad 1^{\overline{1}3}1^{\overline{1}5}15131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 18, \Delta = -79, \det = -8, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 17.50 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 1 & 0 & 0 & 2 \\ 3 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

4.109 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \quad 1^{\overline{1}1}1^{\overline{1}3}1^{\overline{1}21}1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 2 & 3 & 0 & 2 & 1 & 0 & 2 \\ 3 & 3 & -1 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

4.110 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\overline{1}1}1^{\overline{1}3}1^{\overline{1}21}1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -137, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.69 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 5 & 5 & 3 & 0 & 5 \\ 2 & 5 & -2 & 2 & 0 & 0 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 2 & -2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^{11}21^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{violet}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -137$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -2 & 2 & 0 & 0 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 2 & -2 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^{11}1^{11}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{violet}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -143$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 & 3 \\ 4 & 10 & 13 & 0 & 8 & 6 \\ 5 & 13 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 9 & 0 & 4 & 4 \\ 3 & 6 & 6 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [3] $1^{11}23^{11}1^{11}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{violet}{\mathfrak{so}}_8$ $T = 10$, $T_{\max} = 30$, $\Delta = -116$, $\det = 32$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 30, T \in [15, 30]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 \\ 1 & 4 & 1 & 0 & 2 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 1 & 3 & 4 & 0 & 2 & 0 \\ 2 & 4 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ [4] $1^{11}4^{11}13^{11}21^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{violet}{\mathfrak{so}}_8$ $T = 11$, $T_{\max} = 23$, $\Delta = -107$, $\det = -24$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 5 & 4 & 1 & 1 & 5 \\ 4 & 2 & 2 & 0 & 4 \\ 1 & 2 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 4 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$T = 12$ [5] $1^{11}3^{11}1^{11}5^{11}131^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{violet}{\mathfrak{so}}_8$ $T = 12$, $T_{\max} = 24$, $\Delta = -108$, $\det = 95$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 15.68 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 10 & 1 & 4 & 8 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 3 & 0 \\ 3 & 8 & 0 & 3 & 4 & 4 \\ 3 & 6 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

4.111 $\text{NHC}(-2)(-3) + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{brown}{\mathfrak{su}}_2$

$T = 9$ [0] $1^{11}3^{11}21^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{brown}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -2$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$

$$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 0 \\ 6 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & 0 & 9 & -2 \end{pmatrix}_{b_0}$$

[1] $13^{11}21^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{brown}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -6$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 16.67 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 5 & 5 & 3 & 5 & 5 \\ 2 & 5 & -2 & 2 & 3 & 0 \\ 2 & 3 & 2 & 0 & 3 & 2 \\ 3 & 5 & 3 & 3 & 4 & 4 \\ 3 & 5 & 0 & 2 & 4 & -3 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^{11}21^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{brown}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -10$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 17.60 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 5 & 5 & 3 & 0 & 5 \\ 2 & 5 & -2 & 2 & 0 & 0 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & 2 & 0 & -4 \end{pmatrix}_{b_0}$$

[3] $1^{12}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -12$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 0 \\ 3 & 2 & 0 & 4 & 2 & 4 \\ 3 & 3 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -28$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 14.14 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -2 & 2 & 0 & 0 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & 2 & 0 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{12}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -36$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 0 \\ 4 & 2 & 0 & 4 & 2 & 4 \\ 3 & 3 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{12}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -52$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 0 \\ 4 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -58$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 0 \\ 4 & 5 & 0 & 6 & 9 & 7 \\ 3 & 3 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -72$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11}3^{11}1^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -82$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 2 & 3 & 0 & 0 & 2 \\ 1 & 3 & 3 & 0 & 2 & 1 \\ 2 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

- [10] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -82$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.49 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$
- [12] $1^{11}3^{11}1^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -87$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 1 & 3 & 3 & 0 & 2 & 1 \\ 2 & 4 & 2 & -4 & 2 & 3 \end{pmatrix}_{b_0}$
- [13] $1^{111}3^{11}1^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.22 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 2 & 6 & 3 \\ 1 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 2 & 2 & 0 & 0 & 5 & 2 \\ 3 & 6 & 6 & 0 & 5 & 13 & 7 \\ 2 & 3 & 2 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$
- [14] $1^{111}3^{11}1^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.22 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 & 3 \\ 1 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 2 & 2 & 0 & 0 & 2 & 2 \\ 1 & 3 & 3 & 0 & 2 & 1 & 3 \\ 2 & 3 & 2 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$
- [15] $1^{12}3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -124$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 8 & 0 \\ 6 & 8 & 0 & 8 & 22 & 10 \\ 3 & 3 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$
- [16] $1^{111}3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -162$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.54 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 2 & 0 \\ 3 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$
- [17] $1^{13}3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -170$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 12.56 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 10 & 0 \\ 8 & 11 & 0 & 10 & 41 & 13 \\ 3 & 3 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$
- [18] $1^{14}3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -196$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 14 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 12 & 0 \\ 10 & 14 & 0 & 12 & 66 & 16 \\ 3 & 3 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$
- [19] $1^{112}3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -202$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 2 & 0 \\ 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -210, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.57 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 1 & 1 & 2 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 0 \\ 5 & 5 & 0 & 6 & 9 & 7 \\ 3 & 3 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{12} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.55 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 0 \\ 5 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{13} 3^1 1^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -257, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.84 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & -4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{13} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -282, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.57 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 0 \\ 7 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{12} 3^1 1^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -297, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.81 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 2 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 7 & 2 \\ 4 & 12 & 9 & 0 & 7 & 28 & 10 \\ 2 & 4 & 2 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{14} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -312, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.59 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 8 & 0 \\ 9 & 12 & 0 & 8 & 46 & 12 \\ 3 & 3 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{12} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -322, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.62 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 0 \\ 6 & 5 & 0 & 6 & 9 & 7 \\ 3 & 3 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{12} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -372, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.59 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 0 \\ 6 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{11}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 3 \\ 2 & 1 & 1 & 2 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 8 & 0 \\ 7 & 8 & 0 & 8 & 22 & 10 \\ 3 & 3 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{13}3^11^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 6 & 2 \\ 4 & 12 & 6 & 0 & 6 & 22 & 8 \\ 2 & 4 & 2 & -4 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{11}3^11^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -402, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 5 & 2 \\ 3 & 9 & 6 & 0 & 5 & 13 & 7 \\ 2 & 4 & 2 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{12}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -522, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 0 \\ 8 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{11}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -522, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 10 & 0 \\ 9 & 11 & 0 & 10 & 41 & 13 \\ 3 & 3 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{14}3^11^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -552, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 24 & 4 \\ 1 & 4 & 0 & 0 & 2 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 14 & 2 \\ 8 & 24 & 18 & 0 & 14 & 118 & 20 \\ 2 & 4 & 2 & -4 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{12}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -588, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 8 & 0 \\ 8 & 8 & 0 & 8 & 22 & 10 \\ 3 & 3 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{13}3^11^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -617, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 18 & 4 \\ 1 & 4 & 0 & 0 & 2 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 9 & 2 \\ 6 & 18 & 9 & 0 & 9 & 52 & 12 \\ 2 & 4 & 2 & -4 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{14}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -648, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 30 & 4 \\ 1 & 4 & 0 & 0 & 2 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 15 & 2 \\ 10 & 30 & 15 & 0 & 15 & 148 & 20 \\ 2 & 4 & 2 & -4 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{12}3^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1737, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 24 & 4 \\ 1 & 4 & 0 & 0 & 2 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 13 & 2 \\ 8 & 24 & 15 & 0 & 13 & 100 & 18 \\ 2 & 4 & 2 & -4 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{14}3^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1857, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 30 & 4 \\ 1 & 4 & 0 & 0 & 2 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 17 & 2 \\ 10 & 30 & 21 & 0 & 17 & 172 & 24 \\ 2 & 4 & 2 & -4 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{11}3^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1887, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 2 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 14 & 2 \\ 9 & 27 & 15 & 0 & 14 & 121 & 19 \\ 2 & 4 & 2 & -4 & 2 & 19 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{13}3^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1922, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 2 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 15 & 2 \\ 9 & 27 & 18 & 0 & 15 & 133 & 21 \\ 2 & 4 & 2 & -4 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{12}3^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -2592, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 30 & 4 \\ 1 & 4 & 0 & 0 & 2 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 16 & 2 \\ 10 & 30 & 18 & 0 & 16 & 154 & 22 \\ 2 & 4 & 2 & -4 & 2 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{13}3^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -3047, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 3 & 33 & 4 \\ 1 & 4 & 0 & 0 & 2 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 18 & 2 \\ 11 & 33 & 21 & 0 & 18 & 193 & 25 \\ 2 & 4 & 2 & -4 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{14}3^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -3252, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 3 & 36 & 4 \\ 1 & 4 & 0 & 0 & 2 & 24 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 20 & 2 \\ 12 & 36 & 24 & 0 & 20 & 238 & 28 \\ 2 & 4 & 2 & -4 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [53] \quad 1^1 231^1 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 1 & 5 \\ 14 & 33 & 1 & 0 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 11 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{11} 231^1 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 201, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 29 & 5 \\ 14 & 33 & 1 & 65 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 65 & 0 & 123 & 21 \\ 5 & 11 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[55] \ 1^{11}231^{11}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 201, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 14 & 5 \\ 14 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 35 & 0 & 33 & 11 \\ 5 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{11}231^{11}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 221, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 16 & 5 \\ 14 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 35 & 0 & 33 & 11 \\ 5 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{11}231^{11}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 221, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 27 & 5 \\ 14 & 33 & 1 & 65 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 65 & 0 & 123 & 21 \\ 5 & 11 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{11}231^{11}31^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 234, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 7 & 0 \\ 1 & 3 & 0 & 2 & 0 & 5 & 2 \\ 4 & 9 & 0 & 7 & 5 & 14 & 8 \\ 2 & 4 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{12}3^{11}1^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 234, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 3 & 0 \\ 1 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 3 & 4 & 4 \\ 2 & 4 & -1 & 0 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{12}31^{11}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 311, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 15 & 5 \\ 14 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 35 & 0 & 33 & 11 \\ 5 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{11}231^{11}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 311, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 28 & 5 \\ 14 & 33 & 1 & 65 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 65 & 0 & 123 & 21 \\ 5 & 11 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{12}3^{11}1^{12}31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 504, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 6 & 0 \\ 1 & 3 & 0 & 2 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{11}231^{11}231^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 792, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 10 & 0 \\ 1 & 3 & 0 & 2 & 0 & 8 & 2 \\ 7 & 15 & 0 & 10 & 8 & 38 & 12 \\ 2 & 4 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [64] \quad 1^{11} 3^1 1^1 4131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -66, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 1 & 3 & 3 & 0 & 2 & 1 \\ 2 & 4 & 2 & -3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [65] \quad 1^{11} 3^1 1^1 5131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 1 & 0 & 3 & 1 & 2 & 1 & 3 \\ 2 & 0 & 2 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{11} 3^1 1^1 5131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.52 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 0 & 1 & -3 & 1 & 0 & -1 \\ 1 & 0 & 2 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -1 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{11} 3^1 1^1 5131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 37, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.59 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 3 & 0 & 1 & 2 & 1 & 3 \\ 2 & 3 & -1 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{11} 3^1 1^1 5131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 59, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 4 & 0 & 3 & 1 & 2 & 1 & 3 \\ 2 & 0 & 2 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{11} 3^1 1^1 5131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 95, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.55 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 2 & 0 & 3 & 1 & 2 & 1 & 3 \\ 2 & 0 & 2 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{12} 3^1 1^1 5131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 99, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & -3 & 1 & 2 & -1 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 3 & 0 & 6 & 2 & 4 & 10 & 6 \\ 2 & 0 & 2 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{11} 3^1 1^1 5131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.59 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 3 & 0 & 3 & 1 & 2 & 1 & 3 \\ 2 & 0 & 2 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{12}3^1 1^5 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 218, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & -3 & 1 & 2 & -1 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 4 & 0 & 6 & 2 & 4 & 10 & 6 \\ 2 & 0 & 2 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{14}3^1 1^5 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 234, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 1 & -3 & 1 & 4 & -1 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 6 & 0 & 12 & 4 & 8 & 46 & 12 \\ 2 & 0 & 2 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{13}3^1 1^5 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 260, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 1 & -3 & 1 & 3 & -1 \\ 1 & 0 & 2 & 1 & 0 & 6 & 2 \\ 5 & 0 & 9 & 3 & 6 & 25 & 9 \\ 2 & 0 & 2 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{12}3^1 1^5 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 275, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.61 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & -3 & 1 & 2 & -1 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 5 & 0 & 6 & 2 & 4 & 10 & 6 \\ 2 & 0 & 2 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{13}3^1 1^5 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 423, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 14.61 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 1 & -3 & 1 & 3 & -1 \\ 1 & 0 & 2 & 1 & 0 & 6 & 2 \\ 6 & 0 & 9 & 3 & 6 & 25 & 9 \\ 2 & 0 & 2 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{14}3^1 1^5 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 503, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 1 & -3 & 1 & 4 & -1 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 7 & 0 & 12 & 4 & 8 & 46 & 12 \\ 2 & 0 & 2 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

4.112 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{111}3^1 1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 329, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 5 & 2 \\ 1 & 3 & 3 & 0 & 2 & 1 & 6 & 3 \\ 3 & 9 & 6 & 0 & 5 & 6 & 13 & 7 \\ 2 & 4 & 2 & -4 & 2 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^1 1^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 464, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 15 \\ 1 & 4 & 0 & 0 & 2 & 3 & 9 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 \\ 1 & 3 & 2 & 0 & 0 & 2 & 8 \\ 5 & 3 & 3 & 0 & 2 & 1 & 9 \\ 1 & 15 & 9 & 0 & 8 & 9 & 37 \\ 2 & 4 & 2 & -4 & 2 & 3 & 11 \end{pmatrix}$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 3 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 6 & 4 \\ 2 & 6 & 3 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & 2 & -4 & 2 & 4 & 4 & 0 \end{pmatrix} b_0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 9 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 3 & 5 \\ 2 & 6 & 3 & 0 & 3 & 4 & 9 \\ 3 & 9 & 6 & 0 & 5 & 9 & 13 \\ 2 & 4 & 2 & -4 & 2 & 4 & 7 \end{pmatrix}_b$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 2 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 5 & 5 & 2 \\ 3 & 9 & 6 & 0 & 5 & 13 & 15 & 7 \\ 3 & 9 & 6 & 0 & 5 & 13 & 15 & 7 \\ 2 & 4 & 2 & -4 & 2 & 7 & 7 & 0 \end{pmatrix}$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 \\ 1 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 3 & 3 & 0 & 2 & 1 & 2 \\ 1 & 3 & 2 & 0 & 2 & 2 & 0 \\ 2 & 4 & 2 & -4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 3 & 2 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & -4 & 2 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 9 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 7 & 2 & 2 \\ 4 & 12 & 9 & 0 & 7 & 28 & 7 & 10 \\ 1 & 3 & 2 & 0 & 2 & 7 & 0 & 2 \\ 2 & 4 & 2 & -4 & 2 & 10 & 2 & 0 \end{pmatrix}$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 6 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 6 & 2 & 2 \\ 4 & 12 & 6 & 0 & 6 & 22 & 6 & 8 \\ 1 & 3 & 2 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & 2 & -4 & 2 & 8 & 2 & 0 \end{pmatrix}$$

2052

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 6 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 5 & 2 & 2 \\ 3 & 9 & 6 & 0 & 5 & 13 & 5 & 7 \\ 1 & 3 & 2 & 0 & 2 & 5 & 0 & 2 \\ 2 & 4 & 2 & -4 & 2 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}1^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 1316, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 9 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 8 & 2 & 2 \\ 5 & 15 & 9 & 0 & 8 & 37 & 8 & 11 \\ 1 & 3 & 2 & 0 & 2 & 8 & 0 & 2 \\ 2 & 4 & 2 & -4 & 2 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{12}3^{11}1^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 1616, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 18 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 12 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 10 & 2 & 2 \\ 6 & 18 & 12 & 0 & 10 & 58 & 10 & 14 \\ 1 & 3 & 2 & 0 & 2 & 10 & 0 & 2 \\ 2 & 4 & 2 & -4 & 2 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

4.114 NHC(-2)(-3) + e₆ + su₂→(-3) + su₂ + su_{2n3mix}

$$T = 11 \quad [0] \ 1^{12}3^{11}1^{13}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 1 & 9 & 5 \\ 14 & 33 & 1 & 0 & 20 & 11 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 20 & 0 & 0 & 10 & 6 \\ 5 & 11 & -1 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

4.115 NHC(-2)(-3) + e₆ + su₂→(-3) + su₂ + su₃

$$T = 10 \quad [0] \ 1^{11}3^{11}1^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 1 & 3 & 3 & 0 & 2 & 1 & 3 & 3 \\ 1 & 3 & 3 & 0 & 2 & 3 & 0 & 2 \\ 2 & 3 & 3 & -4 & 2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 3 & 2 \\ 1 & 3 & 3 & 0 & 2 & 1 & 3 & 3 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 2 & -4 & 2 & 3 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 130, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 1 & 2 & 5 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 6 & 2 & 0 \\ 4 & 5 & 0 & 6 & 9 & 5 & 7 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 3 & -1 & 0 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 146, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 2 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 2 & 0 \\ 5 & 6 & 0 & 4 & 10 & 6 & 6 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 3 & -1 & 0 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{121} 3^{121} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 896, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 2 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 2 & 0 \\ 6 & 6 & 0 & 4 & 10 & 6 & 6 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 3 & -1 & 0 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111} 3^{111} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 914, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 2 & 3 \\ 2 & 1 & 1 & 2 & 8 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 8 & 2 & 0 \\ 7 & 8 & 0 & 8 & 22 & 8 & 10 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 3 & -1 & 0 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{121} 3^{121} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 1280, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 2 & 3 \\ 2 & 1 & 1 & 2 & 8 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 8 & 2 & 0 \\ 8 & 8 & 0 & 8 & 22 & 8 & 10 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 3 & -1 & 0 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [15] \quad 1^{111} 3^{111} 15^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -69, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.97 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 3 & 0 & 1 & 2 & 1 & 3 \\ 1 & 3 & 0 & 1 & 2 & 3 & 0 \\ 2 & 3 & -1 & -1 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

4.116 NHC(-2)(-3) + e₆ + su₂ → (-3) + su₂ → (-3)

$$T = 9 \quad [0] \quad 1^{113} 1^{113} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.11 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 2 & 3 \\ 1 & 4 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 2 & 2 & 0 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 2 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{13} 1^{11} 1^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & -4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [2] \quad 1^{123} 1^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 1 & 1 & 6 & 5 & 0 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 & 0 \\ 6 & 0 & 0 & 4 & 4 & 0 \\ 5 & 0 & -1 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{123} 1^{11} 1^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 130, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 14 & 1 & 8 & 5 \\ 14 & 33 & 1 & 20 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 20 & 0 & 10 & 6 \\ 5 & 11 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1231}1^{1311}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 156, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 14 & 1 & 6 & 5 \\ 14 & 33 & 1 & 14 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 14 & 0 & 4 & 4 \\ 5 & 11 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

4.117 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{113}1^{11}1^{131}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 164, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 & 2 \\ 1 & 3 & 3 & 0 & 2 & 2 & 1 \\ 2 & 4 & 2 & -4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{13}1^{11}1^{131}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 436, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 6 \\ 1 & 4 & 0 & 0 & 2 & 2 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 3 \\ 1 & 3 & 2 & 0 & 2 & 0 & 3 \\ 2 & 6 & 3 & 0 & 3 & 3 & 4 \\ 2 & 4 & 2 & -4 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{123}1^{11}1^{131}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 500, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 7 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 & 7 & 2 \\ 4 & 12 & 9 & 0 & 7 & 7 & 28 & 10 \\ 2 & 4 & 2 & -4 & 2 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{13}1^{11}1^{231}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 544, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 6 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 & 6 & 2 \\ 4 & 12 & 6 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & 2 & -4 & 2 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{113}1^{11}1^{131}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 704, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 5 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 & 5 & 2 \\ 3 & 9 & 6 & 0 & 5 & 5 & 13 & 7 \\ 2 & 4 & 2 & -4 & 2 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{113}1^{11}1^{231}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 1316, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 8 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 & 8 & 2 \\ 5 & 15 & 9 & 0 & 8 & 8 & 37 & 11 \\ 2 & 4 & 2 & -4 & 2 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{123}1^{11}1^{231}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 1616, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 18 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 10 & 2 \\ 1 & 3 & 2 & 0 & 0 & 2 & 10 & 2 \\ 6 & 18 & 12 & 0 & 10 & 10 & 58 & 14 \\ 2 & 4 & 2 & -4 & 2 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

4.118 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2 \rightarrow (-3)$ + su2n3mix

$$T = 9 \quad [0] \quad 1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 4 & 2 \\ 2 & 6 & 5 & 0 & 4 & 7 & 5 \\ 2 & 4 & 2 & -4 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{13}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -389, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 5 & 2 \\ 3 & 9 & 5 & 0 & 5 & 12 & 6 \\ 2 & 4 & 2 & -4 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -445, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 2 & 11 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 9 & 2 \\ 5 & 15 & 11 & 0 & 9 & 44 & 12 \\ 2 & 4 & 2 & -4 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -542, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 2 & 8 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 7 & 2 \\ 4 & 12 & 8 & 0 & 7 & 25 & 9 \\ 2 & 4 & 2 & -4 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{13}3^{11}1^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -668, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 2 & 8 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 8 & 2 \\ 5 & 15 & 8 & 0 & 8 & 36 & 10 \\ 2 & 4 & 2 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11}3^{11}1^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -1091, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 18 & 4 \\ 1 & 4 & 0 & 0 & 2 & 11 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 10 & 2 \\ 6 & 18 & 11 & 0 & 10 & 55 & 13 \\ 2 & 4 & 2 & -4 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12}3^{11}1^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -1264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 2 & 14 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 12 & 2 \\ 7 & 21 & 14 & 0 & 12 & 80 & 16 \\ 2 & 4 & 2 & -4 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [7] \quad 1^{11}231^{13}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 14 & 1 & 19 & 5 \\ 14 & 33 & 1 & 44 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 44 & 0 & 58 & 14 \\ 5 & 11 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{1231}1^31^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 14 & 1 & 7 & 5 \\ 14 & 33 & 1 & 14 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 14 & 0 & 4 & 4 \\ 5 & 11 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{1231}1^31^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 85, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 1 & 1 & 9 & 5 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 0 & 0 & 10 & 6 \\ 5 & 0 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{1231}1^31^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 14 & 1 & 9 & 5 \\ 14 & 33 & 1 & 20 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 20 & 0 & 10 & 6 \\ 5 & 11 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

4.119 $\text{NHC}(-2)(-3) + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2\text{n3mix} + \textcolor{blue}{\mathfrak{su}}_2$

$$T = 11 \quad [0] \quad 1^{1231}1^31^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 9 & 1 & 5 \\ 14 & 33 & 1 & 20 & 0 & 11 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 9 & 20 & 0 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 11 & -1 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

4.120 $\text{NHC}(-2)(-3) + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_3$

$$T = 9 \quad [0] \quad 1^{131}1^321^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.36 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 5 & 5 & 3 & 0 & 5 \\ 2 & 5 & -2 & 2 & 0 & 0 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1131}1^321^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.11 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -2 & 2 & 0 & 0 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1231}1^321^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 0 \\ 4 & 6 & 0 & 4 & 9 & 5 \\ 3 & 3 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1131}1^321^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 3 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1131}1^31^31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.21 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 2 & 3 & 0 & 0 & 2 \\ 1 & 3 & 3 & 0 & 2 & 0 \\ 2 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{11}3^{11}1^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -193, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.21 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 2 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 1 & 3 & 3 & 0 & 2 & 0 \\ 2 & 4 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{13}1^{11}1^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -363, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 \\ 1 & 4 & 0 & 0 & 2 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 3 \\ 2 & 6 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{11}3^{11}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -508, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 5 & 2 \\ 3 & 9 & 6 & 0 & 5 & 12 & 6 \\ 2 & 4 & 2 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$T = 10$ [8] $1^{12}3^{11}1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 106, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.30 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 0 & 0 \\ 1 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{12}31^{13}1^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 149, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 5 & 14 & 1 & 1 & 5 \\ 14 & 33 & 1 & 0 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 11 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

$T = 11$ [10] $1^{11}3^{11}1^{14}1^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.24 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 5 & 4 \\ 1 & 4 & 0 & 0 & 2 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 3 & 2 \\ 3 & 5 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{11}3^{11}1^{14}1^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -149, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 3 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$T = 12$ [12] $1^{11}3^{11}1^{15}1^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 15.45 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 1 & 0 & 3 & 1 & 2 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11}3^{11}15131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 66, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.94 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 3 & 0 & 1 & 2 & 0 & 2 \\ 2 & 3 & -1 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.121 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11}3^{11}15131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 3 & 3 & 0 & 2 & 0 & 3 \\ 1 & 3 & 3 & 0 & 2 & 3 & 1 \\ 2 & 3 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}15131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 3 & 2 & 2 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 & 3 \\ 1 & 3 & 3 & 0 & 2 & 3 & 1 & 3 \\ 2 & 4 & 2 & -4 & 2 & 3 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 130, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 3 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 2 & 6 & 0 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 4 & 5 & 0 & 6 & 5 & 9 & 7 \\ 3 & 3 & -1 & 0 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 146, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & 0 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 226, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 3 & 1 & 3 \\ 3 & 3 & -1 & 0 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11}3^{11}15131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 3 \\ 1 & 3 & 3 & 0 & 2 & 0 & 3 \\ 2 & 6 & 3 & 0 & 3 & 3 & 4 \\ 2 & 4 & 2 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 328, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 3 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 2 & 8 & 0 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 8 & 8 & 0 & 8 & 8 & 22 & 10 \\ 3 & 3 & -1 & 0 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [15] \quad 1^{11}3^{11}15^{13}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 13, \quad T_{\max} = 15, \quad \Delta = -58, \quad \det = -69, \quad (n_+, n_-, n_0) = (1, 13, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.97 \rightarrow 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 3 & 0 & 1 & 2 & 0 & 3 \\ 1 & 3 & 0 & 1 & 2 & 3 & 1 \\ 2 & 3 & -1 & -1 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

4.122 $\text{NHC}(-2)(-3) + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3$

$$T = 10 \quad [0] \quad 1^{11}3^{11}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 10, \quad T_{\max} = 16, \quad \Delta = -96, \quad \det = 36, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.33 \rightarrow 14, \quad T_{\max} = 16, \quad T \in [14, 16]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 0 & 5 \\ 2 & 4 & -2 & 2 & 0 & 0 & 0 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 2 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^{11}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -102, \quad \det = 249, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.42 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 3 & 3 & 0 & 2 & 0 & 3 \\ 1 & 3 & 3 & 0 & 2 & 3 & 0 \\ 2 & 3 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}1^{11}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -102, \quad \det = 489, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.39 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 6 \\ 1 & 4 & 0 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 0 & 0 & 2 & 3 \\ 1 & 3 & 3 & 0 & 2 & 0 & 3 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 2 & -4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

4.123 $\text{NHC}(-2)(-3) + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + (-3) \rightarrow (-2)$

$$T = 9 \quad [0] \quad 1^{11}23^{11}1^{12}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 9, \quad T_{\max} = 36, \quad \Delta = -118, \quad \det = -8, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, \quad T_{\max} = 36, \quad T \in [13, 36]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 1 & 4 \\ 5 & 5 & 5 & 3 & 0 & 5 \\ 3 & 5 & -2 & 2 & 0 & 0 \\ 3 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

4.124 $\text{NHC}(-2)(-3) + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \text{n3mix}$

$$T = 8 \quad [0] \quad 1^{11}3^{11}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -138, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 19, \quad T_{\max} = 98, \quad T \in [19, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 2 & 2 & 0 & 4 & 4 & 4 \\ 2 & 2 & -1 & -1 & 4 & -6 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^{11}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -144, \quad \det = 44, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 98, \quad T \in [13, 98]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 5 & 0 & 7 & 5 \\ 2 & 4 & 2 & -4 & 5 & 0 \end{pmatrix}_{b_0}$$

- [2] $1^{11}3^11^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.15 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 8 & 3 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 8 & 0 & 25 & 9 \\ 2 & 3 & 2 & -4 & 9 & -1 \end{pmatrix}_{b_0}$$
- [3] $1^{11}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 58, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.21 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 3 & 2 & 0 & 4 & 4 & 4 \\ 2 & 2 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$
- [6] $1^{11}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 4 \\ 3 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 2 & 0 & -2 & 0 \\ 4 & 3 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$
- [4] $1^{13}3^11^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 14 & 0 & 69 & 15 \\ 2 & 4 & 2 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$
- [5] $1^{11}3^11^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 5 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 5 & 6 & 0 & 7 & 5 \\ 2 & 3 & 3 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$
- [7] $1^{11}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 9 & 4 & 1 & 4 & 5 \\ 4 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 2 & 0 & -2 & 0 \\ 5 & 3 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{11}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 2 & 3 & 0 & 2 & 3 & 3 \\ 2 & 2 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{11}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 3 & 4 & 0 & 5 & 10 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{11}3^11^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.23 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 5 & 3 \\ 1 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 5 & 5 & 0 & 7 & 5 \\ 2 & 3 & 2 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$
- [11] $1^{11}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.24 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 4 & 2 & 0 & 4 & 4 & 4 \\ 2 & 2 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$
- [12] $1^{11}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.68 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 5 & 8 & 0 & 7 & 34 & 10 \\ 2 & 2 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{12} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 289, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\ T_{\min} = 12.68 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 5 & 4 & 0 & 5 & 10 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{11} 3^{11} 1^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\ T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 8 & 0 & 25 & 9 \\ 2 & 4 & 2 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{12} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 330, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\ T_{\min} = 12.56 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 5 & 6 & 0 & 6 & 20 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{13} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\ T_{\min} = 12.61 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 5 & 7 & 0 & 4 & 23 & 7 \\ 2 & 2 & -1 & -1 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{14} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\ T_{\min} = 12.65 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & -1 \\ 6 & 10 & 0 & 8 & 52 & 12 \\ 2 & 2 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{12} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 354, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\ T_{\min} = 12.63 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 5 & 5 & 0 & 3 & 11 & 5 \\ 2 & 2 & -1 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{13} 3^{11} 1^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 394, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\ T_{\min} = 12.95 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 8 & 0 & 36 & 10 \\ 2 & 4 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{14} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\ T_{\min} = 12.63 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 6 & 9 & 0 & 5 & 39 & 9 \\ 2 & 2 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{13} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 469, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\ T_{\min} = 12.55 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 6 & 8 & 0 & 7 & 34 & 10 \\ 2 & 2 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{12} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\ T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 6 & 6 & 0 & 6 & 20 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{13} 3^{12} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 499, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 6 & 7 & 0 & 4 & 23 & 7 \\ 2 & 2 & -1 & -1 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{14} 3^{11} 1^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 506, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 20 & 0 & 148 & 22 \\ 2 & 4 & 2 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{14} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 610, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & -1 \\ 7 & 10 & 0 & 8 & 52 & 12 \\ 2 & 2 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{11} 3^{11} 1^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 620, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 11 & 0 & 55 & 13 \\ 2 & 4 & 2 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{14} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 7 & 9 & 0 & 5 & 39 & 9 \\ 2 & 2 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{13} 3^{11} 1^{13} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 11 & 0 & 72 & 14 \\ 2 & 4 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{13} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 673, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 7 & 8 & 0 & 7 & 34 & 10 \\ 2 & 2 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{13} 3^{11} 1^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 676, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 17 & 0 & 111 & 19 \\ 2 & 4 & 2 & -4 & 19 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{12} 3^{11} 1^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 714, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 14 & 0 & 80 & 16 \\ 2 & 4 & 2 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{14} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 868, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & -1 \\ 8 & 10 & 0 & 8 & 52 & 12 \\ 2 & 2 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[43] $1^1 3^1 1^1 1^4 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 970, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 14 & 0 & 120 & 18 \\ 2 & 4 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

[44] $1^1 1^3 1^1 1^3 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 1016, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 14 & 0 & 97 & 17 \\ 2 & 4 & 2 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[45] $1^1 2^3 1^1 1^3 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 1254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 17 & 0 & 128 & 20 \\ 2 & 4 & 2 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[46] $1^1 4^3 1^1 1^3 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 1334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 33 & 4 \\ 1 & 4 & 0 & 0 & 23 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 33 & 23 & 0 & 208 & 26 \\ 2 & 4 & 2 & -4 & 26 & 0 \end{pmatrix}_{b_0}$$

[47] $1^1 3^3 1^1 1^3 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 1360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 20 & 0 & 165 & 23 \\ 2 & 4 & 2 & -4 & 23 & 0 \end{pmatrix}_{b_0}$$

[48] $1^1 1^1 3^1 1^1 1^4 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 1484, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 17 & 0 & 151 & 21 \\ 2 & 4 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[49] $1^1 2^3 1^1 1^4 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 1866, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 33 & 4 \\ 1 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 33 & 20 & 0 & 188 & 24 \\ 2 & 4 & 2 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[50] $1^1 3^3 1^1 1^4 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 2116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 36 & 4 \\ 1 & 4 & 0 & 0 & 23 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 36 & 23 & 0 & 231 & 27 \\ 2 & 4 & 2 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

[51] $1^1 4^3 1^1 1^4 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 2234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 39 & 4 \\ 1 & 4 & 0 & 0 & 26 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 13 & 39 & 26 & 0 & 280 & 30 \\ 2 & 4 & 2 & -4 & 30 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [52] $1^1 2^3 1^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 17.29 \rightarrow 18, T_{\max} = 32, T \in [18, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 13 & 0 & 48 & 14 \\ 2 & 4 & 3 & -4 & 14 & -4 \end{pmatrix}_{b_0}$$

[53] $1^1 1^1 2^3 1^1 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13, T_{\max} = 29, T \in [13, 29]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 5 & 0 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 2 & 4 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

- [54] $1^{1231}1^31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13, T_{\max} = 29, T \in [13, 29]$
 $\begin{pmatrix} 13 & 21 & 1 & 7 \\ 21 & 33 & 1 & 11 \\ 1 & 1 & -3 & -1 \\ 7 & 11 & -1 & 3 \end{pmatrix}_{b_0}$
- [55] $1^{1231}1^31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 4 & 0 \\ 6 & 8 & 0 & 4 & 9 & 5 \\ 2 & 4 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$
- [56] $1^{1231}1^31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -37, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.54 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 4 & 0 \\ 4 & 0 & 0 & 4 & 9 & 5 \\ 2 & 0 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$
- [57] $1^{111}23^1131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 5 & 0 & 6 & 4 \\ 1 & 5 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 6 & 0 & 8 & 6 \\ 2 & 4 & 3 & -4 & 6 & 0 \end{pmatrix}_{b_0}$
- [58] $1^{123}131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 2 & 0 \\ 2 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$
- [59] $1^{11}23^1131^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.12 \rightarrow 15, T_{\max} = 32, T \in [15, 32]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 9 & 0 \\ 7 & 0 & 0 & 9 & 26 & 10 \\ 2 & 0 & -1 & 0 & 10 & -1 \end{pmatrix}_{b_0}$
- [60] $1^{12313}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.56 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$
 $\begin{pmatrix} 13 & 1 & 1 & 7 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 7 & 0 & -1 & 2 \end{pmatrix}_{b_0}$
- [61] $1^{123}1^131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$
- [62] $1^{11231}1^31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 11 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 8 & 0 \\ 8 & 11 & 0 & 8 & 21 & 9 \\ 2 & 4 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$
- [63] $1^{113}23^1131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 2 & 5 & 0 & 17 & 4 \\ 1 & 5 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 17 & 17 & 0 & 78 & 18 \\ 2 & 4 & 3 & -4 & 18 & 0 \end{pmatrix}_{b_0}$
- [64] $1^{11}231^{11}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 11 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 8 & 0 \\ 4 & 11 & 0 & 8 & 21 & 9 \\ 2 & 4 & -1 & 0 & 9 & 0 \end{pmatrix}_{b_0}$
- [65] $1^{123}1^131^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -97, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.41 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

- [66] $1^{123}1^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.07 \rightarrow 15, T_{\max} = 32, T \in [15, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 5 & 0 \\ 5 & 0 & 0 & 5 & 12 & 6 \\ 2 & 0 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$
- [68] $1^{112}23^1131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -122, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.51 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 1 & 2 & 4 & 0 & 9 & 4 \\ 1 & 4 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 5 & 0 & 12 & 6 \\ 2 & 4 & 3 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$
- [67] $1^{123}1^{11}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -122, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.51 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 2 & 5 & 0 & 10 & 4 \\ 1 & 5 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 10 & 10 & 0 & 24 & 10 \\ 2 & 4 & 3 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$
- [69] $1^{112}23^11^{12}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 \\ 6 & 18 & 0 & 12 & 56 & 14 \\ 2 & 4 & -1 & 0 & 14 & 0 \end{pmatrix}_{b_0}$$
- [70] $1^{112}23^11^{23}31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.09 \rightarrow 15, T_{\max} = 32, T \in [15, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 12 & 0 \\ 10 & 0 & 0 & 12 & 56 & 14 \\ 2 & 0 & -1 & 0 & 14 & -1 \end{pmatrix}_{b_0}$$
- [71] $1^{123}1^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.15 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 5 & 0 \\ 3 & 9 & 0 & 5 & 12 & 6 \\ 2 & 4 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$
- [72] $1^{123}1^{112}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.91 \rightarrow 15, T_{\max} = 32, T \in [15, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 8 & 0 & 36 & 10 \\ 2 & 4 & 3 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$
- [73] $1^{112}23^1131^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 6 & 0 \\ 5 & 0 & 0 & 6 & 8 & 6 \\ 2 & 0 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$
- [74] $1^{123}1^{12}31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.54 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 8 & 0 \\ 8 & 0 & 0 & 8 & 36 & 10 \\ 2 & 0 & -1 & 0 & 10 & 0 \end{pmatrix}_{b_0}$$
- [75] $1^{13}23^11^{12}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 20 & 0 & 120 & 22 \\ 2 & 4 & 3 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$
- [76] $1^{1231}1^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$

[77] $1^{11}23^{11}1^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.24 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 4 & 0 \\ 3 & 8 & 0 & 4 & 9 & 5 \\ 2 & 4 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{11}23^{11}1^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -183, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 9 & 0 \\ 5 & 12 & 0 & 9 & 26 & 10 \\ 2 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{12}231^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -183, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 9 & 0 \\ 5 & 8 & 0 & 4 & 9 & 5 \\ 2 & 4 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

[80] $1^{12}231^{11}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -183, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 5 & 0 \\ 6 & 9 & 0 & 5 & 12 & 6 \\ 2 & 4 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{11}23^{11}1^{11}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 9 & 0 \\ 8 & 12 & 0 & 9 & 26 & 10 \\ 2 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{12}231^{13}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.68 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 11 & 0 \\ 11 & 0 & 0 & 11 & 72 & 14 \\ 2 & 0 & -1 & 0 & 14 & 0 \end{pmatrix}_{b_0}$$

[84] $1^{12}231^{13}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.62 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 2 & 0 \\ 4 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

[83] $1^{113}23^{11}131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.62 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 2 & 5 & 0 & 14 & 4 \\ 1 & 5 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 14 & 14 & 0 & 48 & 14 \\ 2 & 4 & 3 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[85] $1^{12}231^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 4 & 0 \\ 4 & 8 & 0 & 4 & 9 & 5 \\ 2 & 4 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{12}231^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.19 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 8 & 0 \\ 5 & 15 & 0 & 8 & 36 & 10 \\ 2 & 4 & -1 & 0 & 10 & 0 \end{pmatrix}_{b_0}$$

[87] $1^{11}231^{11}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$

[88] $1^{11}231^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 29$, $\Delta = -116$, $\det = -252$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 29$, $T \in [13, 29]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 11 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 8 & 0 \\ 7 & 11 & 0 & 8 & 21 & 9 \\ 2 & 4 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{12}231^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -255$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.96 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 5 & 0 \\ 4 & 9 & 0 & 5 & 12 & 6 \\ 2 & 4 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{12}231^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -257$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 5 & 0 \\ 6 & 0 & 0 & 5 & 12 & 6 \\ 2 & 0 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

[91] $1^{12}231^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -272$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 5 & 0 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 2 & 4 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[92] $1^{114}231^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -295$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 2 & 5 & 0 & 21 & 4 \\ 1 & 5 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 21 & 0 & 116 & 22 \\ 2 & 4 & 3 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

[93] $1^{11}231^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -296$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 9 & 0 \\ 6 & 12 & 0 & 9 & 26 & 10 \\ 2 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{11}231^{11}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -299$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.41 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 9 & 0 \\ 8 & 0 & 0 & 9 & 26 & 10 \\ 2 & 0 & -1 & 0 & 10 & 0 \end{pmatrix}_{b_0}$$

[95] $1^{11}231^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 29$, $\Delta = -116$, $\det = -303$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 29$, $T \in [13, 29]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 11 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 8 & 0 \\ 6 & 11 & 0 & 8 & 21 & 9 \\ 2 & 4 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[96] $1^{11}231^{11}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -303$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 9 & 0 \\ 7 & 12 & 0 & 9 & 26 & 10 \\ 2 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{11}231^{11}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -316$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.39 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 6 & 0 \\ 6 & 0 & 0 & 6 & 8 & 6 \\ 2 & 0 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

[98] $1^{12}231^{11}31^{114} + [\text{su}_2 \cdot \text{g}_2] +$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 \\ 8 & 18 & 0 & 12 & 56 & 14 \\ 2 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

[110] $1^{11}23^11^{12}31^{13}+[\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -803, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 \\ 9 & 18 & 0 & 12 & 56 & 14 \\ 2 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

[111] $1^{11}23^11^{12}31^{14}+[\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 \\ 10 & 18 & 0 & 12 & 56 & 14 \\ 2 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [112] $1^{11}3^11^14131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 5 & 0 & 7 & 5 \\ 2 & 4 & 2 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[113] $1^{12}3^11^141^131^1+[\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.82 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 11 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 11 & 9 & 0 & 28 & 10 \\ 2 & 4 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[114] $1^{11}3^11^141^131^1+[\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 68, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.82 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 8 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[115] $1^131^{11}413^11^{11}+[\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 76, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.37 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 3 & 0 & -3 & 3 & -1 \\ 4 & 6 & 1 & 3 & 5 & 5 \\ 2 & 4 & 0 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

[116] $1^131^{11}41^{11}31^1+[\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 115, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.23 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & -3 & 4 & -1 \\ 5 & 8 & 0 & 4 & 10 & 6 \\ 2 & 4 & 0 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[117] $1^131^{11}413^11^1+[\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 135, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.34 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$T = 11$ [118] $1^{11}3^11^15131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 18.33 \rightarrow 19, T_{\max} = 23, T \in [19, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 2 & 0 & 5 & 2 & 7 & 5 \\ 2 & 0 & 2 & -1 & 5 & -3 \end{pmatrix}_{b_0}$$

[119] $1^{12}3^11^15131^1+[\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 3 & 8 & 0 & 3 & 20 & 8 \\ 2 & 3 & -1 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

[120] $1^{11}3^{11}1^15^{11}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -29, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.59 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 2 & 5 & 0 & 2 & 7 & 5 \\ 2 & 3 & -1 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[121] $1^{13}3^{11}1^15^{11}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -48, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.58 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 5 & 0 & 11 & 4 & 39 & 11 \\ 2 & 0 & 2 & -1 & 11 & 0 \end{pmatrix}_{b_0}$$

[122] $1^{13}3^{11}1^15^{11}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -60, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 2 & 0 & 2 & 1 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[123] $1^{13}3^{11}1^15^{11}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -83, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.93 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 3 & 0 & 2 & 1 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[124] $1^{12}3^{11}1^15^{11}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -92, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.78 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 4 & 0 & 8 & 3 & 20 & 8 \\ 2 & 0 & 2 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

[126] $1^{11}3^{11}1^15^{11}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.62 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 3 & 0 & 5 & 2 & 7 & 5 \\ 2 & 0 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[125] $1^{13}3^{11}1^15^{11}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.29 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 4 & 0 & 2 & 1 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[127] $1^{11}3^{11}1^15^{11}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -179, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.65 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 4 & 0 & 5 & 2 & 7 & 5 \\ 2 & 0 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[128] $1^{14}3^{11}1^15^{11}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -227, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.83 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 14 & 2 \\ 1 & 0 & 1 & -3 & 5 & -1 \\ 7 & 0 & 14 & 5 & 64 & 14 \\ 2 & 0 & 2 & -1 & 14 & 1 \end{pmatrix}_{b_0}$$

[129] $1^{12}3^11^5131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -109$, $\det = -235$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 14.63 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 5 & 0 & 8 & 3 & 20 & 8 \\ 2 & 0 & 2 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

[130] $1^{13}3^11^5131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -109$, $\det = -251$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 14.69 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 6 & 0 & 11 & 4 & 39 & 11 \\ 2 & 0 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

[131] $1^{11}3^11^5131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -109$, $\det = -252$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 5 & 0 & 5 & 2 & 7 & 5 \\ 2 & 0 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{12}3^11^5131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -109$, $\det = -368$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 6 & 0 & 8 & 3 & 20 & 8 \\ 2 & 0 & 2 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{13}3^11^5131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -109$, $\det = -444$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 14.63 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -3 & 4 & -1 \\ 7 & 0 & 11 & 4 & 39 & 11 \\ 2 & 0 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{14}3^11^5131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -109$, $\det = -480$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 14.66 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 14 & 2 \\ 1 & 0 & 1 & -3 & 5 & -1 \\ 8 & 0 & 14 & 5 & 64 & 14 \\ 2 & 0 & 2 & -1 & 14 & 1 \end{pmatrix}_{b_0}$$

$T = 12$ [135] $1^{11}3^{121}6^1131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 24$, $\Delta = -108$, $\det = 3$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 19.33 \rightarrow 20, T_{\max} = 24, T \in [20, 24]$$

$$\begin{pmatrix} 13 & 4 & 1 & 7 \\ 4 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 7 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

[136] $1^{13}3^{121}6^1131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 24$, $\Delta = -108$, $\det = 44$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 15.82 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 4 & 2 & 0 & 2 & 4 & 4 \\ 2 & 2 & 0 & -1 & 4 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{11}3^{11}6^1131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 28$, $\Delta = -114$, $\det = 55$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 15.82 \rightarrow 16, T_{\max} = 28, T \in [16, 28]$$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 3 & 2 & 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 2 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[138] $1^{11} 6^{1^2 3^1 2^1} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 60, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.67 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 2 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{11} 6^{1^2 3^1 2^1} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 76, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.89 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 6 & 2 & 2 & 2 & 8 & 6 \\ 2 & 2 & 0 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{11} 6^{1^2 3^1 2^1} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 80, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.70 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 5 & 2 & 1 & 2 & 5 & 5 \\ 2 & 2 & 0 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[141] $1^{11} 6^{1^2 3^1 2^1} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 92, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.78 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 4 & 2 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & -1 & 4 & 1 \end{pmatrix}_{b_0}$$

[142] $1^{11} 6^{1^2 3^1 2^1} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 96, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.62 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 3 & 2 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[143] $1^{11} 6^{1^2 3^1 2^1} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 147, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.63 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 5 & 4 & 0 & 3 & 10 & 6 \\ 2 & 2 & 0 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[144] $1^{11} 6^{1^2 3^1 2^1} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 243, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.55 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 6 & 4 & 1 & 3 & 11 & 7 \\ 2 & 2 & 0 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

[145] $1^{11} 6^{1^2 3^1 2^1} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 299, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.55 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 7 & 4 & 2 & 3 & 14 & 8 \\ 2 & 2 & 0 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

$T = 15$ [146] $1^{11} 6^{1^2 3^1 15 131} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \text{su}_2 \quad T = 15, T_{\max} = 18, \Delta = -79, \det = -41, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 17.63 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & -3 & 1 & -1 \\ 6 & 5 & 1 & 14 & 8 \\ 3 & 3 & -1 & 8 & 3 \end{pmatrix}_{b_0}$$

[147] $1^{11} 3^{11} 15 13 1$ $131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 15, T_{\max} = 18, \Delta = -79, \det = -41, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 17.83 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 1 & -1 \\ 3 & 3 & 1 & 2 & 4 \\ 3 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

4.125 NHC(-2)(-3) + e₆ + su₂n₃mix + su₂

$T = 9$ $[0] 1^{12} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 2 & -1 \\ 4 & 6 & 0 & 6 & 20 & 12 & 8 \\ 2 & 4 & 0 & 2 & 12 & 6 & 4 \\ 2 & 2 & -1 & -1 & 8 & 4 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 3 & 0 & 2 & 3 & 5 & 3 \\ 2 & 3 & 0 & 4 & 5 & 5 & 5 \\ 2 & 2 & -1 & -1 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{12} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 6 & -1 \\ 3 & 5 & 0 & 3 & 11 & 18 & 5 \\ 6 & 7 & 0 & 6 & 18 & 25 & 9 \\ 2 & 2 & -1 & -1 & 5 & 9 & -2 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 3 & 4 \\ 3 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 2 & 0 & -2 & 2 & 0 \\ 3 & 3 & 0 & 2 & 1 & 3 \\ 4 & 3 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 3 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3^1 1^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 5 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 5 & 6 & 0 & 7 & 5 & 5 \\ 1 & 3 & 3 & 0 & 5 & 1 & 3 \\ 2 & 3 & 3 & -4 & 5 & 3 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13.70 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 1 & -1 \\ 4 & 7 & 0 & 4 & 23 & 7 & 7 \\ 2 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & -1 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 0 & -1 \\ 2 & 3 & 0 & 2 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{124}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.35 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 4 & -1 \\ 3 & 5 & 0 & 3 & 11 & 20 & 5 \\ 4 & 8 & 0 & 4 & 20 & 30 & 8 \\ 2 & 2 & -1 & -1 & 5 & 8 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{113}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 2 & 3 & 0 & 2 & 3 & 9 & 3 \\ 3 & 6 & 0 & 3 & 9 & 16 & 6 \\ 2 & 2 & -1 & -1 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{122}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.35 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 5 & -1 \\ 3 & 5 & 0 & 3 & 11 & 13 & 5 \\ 4 & 5 & 0 & 5 & 13 & 13 & 7 \\ 2 & 2 & -1 & -1 & 5 & 7 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{142}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & 2 & -1 \\ 8 & 10 & 0 & 8 & 52 & 20 & 12 \\ 2 & 4 & 0 & 2 & 20 & 6 & 4 \\ 2 & 2 & -1 & -1 & 12 & 4 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{132}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 2 & -1 \\ 6 & 7 & 0 & 4 & 23 & 14 & 7 \\ 2 & 4 & 0 & 2 & 14 & 6 & 4 \\ 2 & 2 & -1 & -1 & 7 & 4 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{113}3^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.44 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 5 & 0 & 12 & 5 & 6 \\ 1 & 3 & 3 & 0 & 5 & 1 & 3 \\ 2 & 4 & 2 & -4 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{122}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 5 & -1 \\ 3 & 5 & 0 & 3 & 11 & 13 & 5 \\ 3 & 5 & 0 & 5 & 13 & 13 & 7 \\ 2 & 2 & -1 & -1 & 5 & 7 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{132}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 2 & -1 \\ 6 & 8 & 0 & 7 & 34 & 16 & 10 \\ 2 & 4 & 0 & 2 & 16 & 6 & 4 \\ 2 & 2 & -1 & -1 & 10 & 4 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 1 & -1 \\ 3 & 4 & 0 & 5 & 10 & 4 & 6 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -1 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{111}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 3 & 0 & 2 & 3 & 5 & 3 \\ 3 & 3 & 0 & 4 & 5 & 5 & 5 \\ 2 & 2 & -1 & -1 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{112}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 2 & -1 & -1 & 3 & 4 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{142}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 2 & -1 \\ 5 & 9 & 0 & 5 & 39 & 18 & 9 \\ 3 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & -1 & 9 & 4 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{112}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 5 & -1 \\ 2 & 3 & 0 & 2 & 3 & 8 & 3 \\ 3 & 5 & 0 & 5 & 8 & 13 & 7 \\ 2 & 2 & -1 & -1 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{123}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -138, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 3 & -1 \\ 3 & 5 & 0 & 3 & 11 & 15 & 5 \\ 3 & 6 & 0 & 3 & 15 & 16 & 6 \\ 2 & 2 & -1 & -1 & 5 & 6 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{122}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 2 & -1 \\ 4 & 5 & 0 & 3 & 11 & 10 & 5 \\ 2 & 4 & 0 & 2 & 10 & 6 & 4 \\ 2 & 2 & -1 & -1 & 5 & 4 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{111}3^{11}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -143, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 5 & 0 & 7 & 11 & 5 \\ 3 & 9 & 6 & 0 & 11 & 13 & 7 \\ 2 & 4 & 2 & -4 & 5 & 7 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{111}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{121}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -158, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 1 & -1 \\ 3 & 5 & 0 & 3 & 11 & 5 & 5 \\ 2 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & -1 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1^{12}} 1^{111} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 1 & -1 \\ 5 & 4 & 0 & 5 & 10 & 4 & 6 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -1 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1^{11}} 1^{111} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 1 & -1 \\ 4 & 4 & 0 & 5 & 10 & 4 & 6 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -1 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1^{12}} 1^{122} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 2 & -1 \\ 3 & 5 & 0 & 3 & 11 & 10 & 5 \\ 2 & 4 & 0 & 2 & 10 & 6 & 4 \\ 2 & 2 & -1 & -1 & 5 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1^{12}} 1^{111} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 4 & 3 & 0 & 2 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1^{11}} 1^{111} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -193, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1^{12}} 1^{132} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 2 & -1 \\ 5 & 8 & 0 & 7 & 34 & 16 & 10 \\ 2 & 4 & 0 & 2 & 16 & 6 & 4 \\ 2 & 2 & -1 & -1 & 10 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1^{12}} 1^{123} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -210, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 6 & -1 \\ 3 & 5 & 0 & 3 & 11 & 18 & 5 \\ 4 & 7 & 0 & 6 & 18 & 25 & 9 \\ 2 & 2 & -1 & -1 & 5 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1^{11}} 1^{111} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -217, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 3 & 3 & 0 & 2 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{121}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 1 & -1 \\ 3 & 5 & 0 & 3 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & -1 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{113}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 3 & 11 & 3 \\ 4 & 7 & 0 & 6 & 11 & 25 & 9 \\ 2 & 2 & -1 & -1 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{111}3^{11}1^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 8 & 0 & 25 & 8 & 9 \\ 1 & 3 & 3 & 0 & 8 & 1 & 3 \\ 2 & 4 & 2 & -4 & 9 & 3 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{132}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -247, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 2 & -1 \\ 4 & 7 & 0 & 4 & 23 & 14 & 7 \\ 3 & 4 & 0 & 2 & 14 & 6 & 4 \\ 2 & 2 & -1 & -1 & 7 & 4 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{123}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 3 & -1 \\ 3 & 5 & 0 & 3 & 11 & 15 & 5 \\ 5 & 6 & 0 & 3 & 15 & 16 & 6 \\ 2 & 2 & -1 & -1 & 5 & 6 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{121}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 1 & -1 \\ 4 & 6 & 0 & 6 & 20 & 6 & 8 \\ 1 & 2 & 0 & 1 & 6 & 0 & 2 \\ 2 & 2 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{132}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -262, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 2 & -1 \\ 4 & 7 & 0 & 4 & 23 & 14 & 7 \\ 2 & 4 & 0 & 2 & 14 & 6 & 4 \\ 2 & 2 & -1 & -1 & 7 & 4 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{114}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 2 & 3 & 0 & 2 & 3 & 14 & 3 \\ 5 & 9 & 0 & 7 & 14 & 41 & 11 \\ 2 & 2 & -1 & -1 & 3 & 11 & 0 \end{pmatrix}_{b_0}$$

- [42] $1^{131}3^{121}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -292$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 1 & -1 \\ 4 & 7 & 0 & 4 & 23 & 7 & 7 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & -1 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$
- [43] $1^{112}3^{121}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -298$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 \\ 4 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 2 & -1 & -1 & 3 & 4 & 0 \end{pmatrix}_{b_0}$$
- [44] $1^{112}3^{121}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -303$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 \\ 3 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 2 & -1 & -1 & 3 & 4 & 0 \end{pmatrix}_{b_0}$$
- [46] $1^{122}3^{121}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -308$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.69 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 2 & -1 \\ 3 & 5 & 0 & 3 & 11 & 10 & 5 \\ 3 & 4 & 0 & 2 & 10 & 6 & 4 \\ 2 & 2 & -1 & -1 & 5 & 4 & 0 \end{pmatrix}_{b_0}$$
- [45] $1^{124}3^{121}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -308$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.69 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 7 & -1 \\ 3 & 5 & 0 & 3 & 11 & 23 & 5 \\ 5 & 9 & 0 & 7 & 23 & 41 & 11 \\ 2 & 2 & -1 & -1 & 5 & 11 & 0 \end{pmatrix}_{b_0}$$
- [47] $1^{132}3^{121}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -322$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 2 & -1 \\ 5 & 7 & 0 & 4 & 23 & 14 & 7 \\ 2 & 4 & 0 & 2 & 14 & 6 & 4 \\ 2 & 2 & -1 & -1 & 7 & 4 & 0 \end{pmatrix}_{b_0}$$
- [48] $1^{111}3^{114}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -323$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.77 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 0 & 30 & 3 & 4 \\ 1 & 4 & 0 & 0 & 17 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 30 & 17 & 0 & 151 & 17 & 21 \\ 1 & 3 & 3 & 0 & 17 & 1 & 3 \\ 2 & 4 & 2 & -4 & 21 & 3 & -1 \end{pmatrix}_{b_0}$$
- [49] $1^{142}3^{121}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -340$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 2 & -1 \\ 5 & 9 & 0 & 5 & 39 & 18 & 9 \\ 2 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & -1 & 9 & 4 & 0 \end{pmatrix}_{b_0}$$
- [50] $1^{112}3^{121}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -345$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 5 & -1 \\ 2 & 3 & 0 & 2 & 3 & 8 & 3 \\ 4 & 5 & 0 & 5 & 8 & 13 & 7 \\ 2 & 2 & -1 & -1 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1^{12}} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 5 & -1 \\ 2 & 3 & 0 & 2 & 3 & 8 & 3 \\ 5 & 5 & 0 & 5 & 8 & 13 & 7 \\ 2 & 2 & -1 & -1 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1^{11}} 1^{23} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -366, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 6 & -1 \\ 3 & 5 & 0 & 3 & 11 & 18 & 5 \\ 5 & 7 & 0 & 6 & 18 & 25 & 9 \\ 2 & 2 & -1 & -1 & 5 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1^{141}} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -370, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 1 & -1 \\ 5 & 9 & 0 & 5 & 39 & 9 & 9 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & -1 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1^{12}} 1^{42} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 2 & -1 \\ 7 & 9 & 0 & 5 & 39 & 18 & 9 \\ 2 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & -1 & 9 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1^{142}} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & 2 & -1 \\ 6 & 10 & 0 & 8 & 52 & 20 & 12 \\ 2 & 4 & 0 & 2 & 20 & 6 & 4 \\ 2 & 2 & -1 & -1 & 12 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1^{11}} 1^{42} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & 2 & -1 \\ 7 & 10 & 0 & 8 & 52 & 20 & 12 \\ 2 & 4 & 0 & 2 & 20 & 6 & 4 \\ 2 & 2 & -1 & -1 & 12 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1^{11}} 1^{13} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -393, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 2 & 3 & 0 & 2 & 3 & 9 & 3 \\ 4 & 6 & 0 & 3 & 9 & 16 & 6 \\ 2 & 2 & -1 & -1 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1^{11}} 1^{21} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 1 & -1 \\ 4 & 5 & 0 & 3 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & -1 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{111}3^11^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -395, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 18 & 3 & 4 \\ 1 & 4 & 0 & 0 & 11 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 18 & 11 & 0 & 55 & 11 & 13 \\ 1 & 3 & 3 & 0 & 11 & 1 & 3 \\ 2 & 4 & 2 & -4 & 13 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{11}3^11^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -412, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.82 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 & 2 \\ 2 & 4 & 4 & 0 & 12 & 6 & 4 \\ 1 & 4 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 8 & 0 & 25 & 12 & 9 \\ 2 & 6 & 3 & 0 & 12 & 4 & 4 \\ 2 & 4 & 2 & -4 & 9 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{111}3^11^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -422, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.19 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 0 & 24 & 3 & 4 \\ 1 & 4 & 0 & 0 & 14 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 24 & 14 & 0 & 97 & 14 & 17 \\ 1 & 3 & 3 & 0 & 14 & 1 & 3 \\ 2 & 4 & 2 & -4 & 17 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{131}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -430, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.63 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 1 & -1 \\ 5 & 8 & 0 & 7 & 34 & 8 & 10 \\ 1 & 2 & 0 & 1 & 8 & 0 & 2 \\ 2 & 2 & -1 & -1 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{123}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -438, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.68 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 3 & -1 \\ 3 & 5 & 0 & 3 & 11 & 15 & 5 \\ 4 & 6 & 0 & 3 & 15 & 16 & 6 \\ 2 & 2 & -1 & -1 & 5 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{121}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -449, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.57 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 1 & -1 \\ 5 & 6 & 0 & 6 & 20 & 6 & 8 \\ 1 & 2 & 0 & 1 & 6 & 0 & 2 \\ 2 & 2 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{114}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -463, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.63 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 3 & 0 & 2 & 3 & 12 & 3 \\ 5 & 8 & 0 & 4 & 12 & 30 & 8 \\ 2 & 2 & -1 & -1 & 3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{13}3^11^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -473, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.92 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 6 & 4 \\ 1 & 4 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 5 & 0 & 12 & 9 & 6 \\ 2 & 6 & 3 & 0 & 9 & 4 & 4 \\ 2 & 4 & 2 & -4 & 6 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[67] \ 1^{121}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -478, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.64 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 1 & -1 \\ 5 & 5 & 0 & 3 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & -1 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{11}3^{11}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -483, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 5 & 0 & 12 & 14 & 6 \\ 3 & 9 & 6 & 0 & 14 & 13 & 7 \\ 2 & 4 & 2 & -4 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{12}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 7 & -1 \\ 3 & 5 & 0 & 3 & 11 & 23 & 5 \\ 7 & 9 & 0 & 7 & 23 & 41 & 11 \\ 2 & 2 & -1 & -1 & 5 & 11 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -508, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 2 & -1 \\ 6 & 9 & 0 & 5 & 39 & 18 & 9 \\ 2 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & -1 & 9 & 4 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 2 & 3 & 0 & 2 & 3 & 9 & 3 \\ 5 & 6 & 0 & 3 & 9 & 16 & 6 \\ 2 & 2 & -1 & -1 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{12}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & 1 & -1 \\ 6 & 6 & 0 & 6 & 20 & 6 & 8 \\ 1 & 2 & 0 & 1 & 6 & 0 & 2 \\ 2 & 2 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{12}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -548, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 4 & -1 \\ 3 & 5 & 0 & 3 & 11 & 20 & 5 \\ 6 & 8 & 0 & 4 & 20 & 30 & 8 \\ 2 & 2 & -1 & -1 & 5 & 8 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -548, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 4 & -1 \\ 3 & 5 & 0 & 3 & 11 & 20 & 5 \\ 5 & 8 & 0 & 4 & 20 & 30 & 8 \\ 2 & 2 & -1 & -1 & 5 & 8 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -567, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 3 & 11 & 3 \\ 5 & 7 & 0 & 6 & 11 & 25 & 9 \\ 2 & 2 & -1 & -1 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

- [76] $1^{1^{11}} 1^{131} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -577$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 1 & -1 \\ 5 & 7 & 0 & 4 & 23 & 7 & 7 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & -1 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$
- [77] $1^{1^{11}} 1^{141} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -604$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.65 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & 1 & -1 \\ 6 & 10 & 0 & 8 & 52 & 10 & 12 \\ 1 & 2 & 0 & 1 & 10 & 0 & 2 \\ 2 & 2 & -1 & -1 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$
- [78] $1^{1^{11}} 1^{124} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -644$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.70 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 7 & -1 \\ 3 & 5 & 0 & 3 & 11 & 23 & 5 \\ 6 & 9 & 0 & 7 & 23 & 41 & 11 \\ 2 & 2 & -1 & -1 & 5 & 11 & 0 \end{pmatrix}_{b_0}$$
- [79] $1^{1^{12}} 1^{113} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -726$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 3 & 11 & 3 \\ 6 & 7 & 0 & 6 & 11 & 25 & 9 \\ 2 & 2 & -1 & -1 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$
- [80] $1^{1^{11}} 1^{131} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -727$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 1 & -1 \\ 6 & 8 & 0 & 7 & 34 & 8 & 10 \\ 1 & 2 & 0 & 1 & 8 & 0 & 2 \\ 2 & 2 & -1 & -1 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$
- [81] $1^{1^{12}} 1^{114} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -738$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 3 & 0 & 2 & 3 & 12 & 3 \\ 6 & 8 & 0 & 4 & 12 & 30 & 8 \\ 2 & 2 & -1 & -1 & 3 & 8 & 0 \end{pmatrix}_{b_0}$$
- [82] $1^{1^{11}} 1^{141} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -763$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.64 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 1 & -1 \\ 6 & 9 & 0 & 5 & 39 & 9 & 9 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & -1 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$
- [83] $1^{1^{12}} 1^{131} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -768$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 1 & -1 \\ 6 & 7 & 0 & 4 & 23 & 7 & 7 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & -1 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$
- [84] $1^{1^{11}} 1^{114} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -769$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 2 & 3 & 0 & 2 & 3 & 14 & 3 \\ 6 & 9 & 0 & 7 & 14 & 41 & 11 \\ 2 & 2 & -1 & -1 & 3 & 11 & 0 \end{pmatrix}_{b_0}$$

[85] $1^{111}3^{11}1^{111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -782, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 3 & 2 \\ 2 & 4 & 4 & 0 & 12 & 9 & 4 \\ 1 & 4 & 0 & 0 & 8 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 8 & 0 & 25 & 20 & 9 \\ 3 & 9 & 6 & 0 & 20 & 13 & 7 \\ 2 & 4 & 2 & -4 & 9 & 7 & 0 \end{pmatrix}_{b_0}$$

[86] $1^{112}1^{131}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -930, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & 1 & -1 \\ 7 & 8 & 0 & 7 & 34 & 8 & 10 \\ 1 & 2 & 0 & 1 & 8 & 0 & 2 \\ 2 & 2 & -1 & -1 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

[87] $1^{111}1^{141}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1009, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & 1 & -1 \\ 7 & 10 & 0 & 8 & 52 & 10 & 12 \\ 1 & 2 & 0 & 1 & 10 & 0 & 2 \\ 2 & 2 & -1 & -1 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

[88] $1^{112}1^{141}3^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1062, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & 1 & -1 \\ 7 & 9 & 0 & 5 & 39 & 9 & 9 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & -1 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

[89] $1^{112}1^{114}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1068, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 2 & 3 & 0 & 2 & 3 & 14 & 3 \\ 7 & 9 & 0 & 7 & 14 & 41 & 11 \\ 2 & 2 & -1 & -1 & 3 & 11 & 0 \end{pmatrix}_{b_0}$$

[90] $1^{112}1^{141}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & 1 & -1 \\ 8 & 10 & 0 & 8 & 52 & 10 & 12 \\ 1 & 2 & 0 & 1 & 10 & 0 & 2 \\ 2 & 2 & -1 & -1 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [91] $1^{11}231^{13}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 5 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 5 & 0 & 0 \\ 3 & 5 & 0 & 5 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & 0 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

[92] $1^{11}23^{11}1^{11}31^{141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

[illegible]

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 8 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 6 & 0 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 8 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 8 & 0 & 0 \\ 7 & 15 & 0 & 8 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & 0 & 10 & 0 & 1 \end{pmatrix}$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 8 & 0 & 0 \\ 8 & 15 & 0 & 8 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & 0 & 10 & 0 & 1 \end{pmatrix}$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 6 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 6 & 0 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & 0 & 2 & 8 & 1 \end{pmatrix} b_c$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 7 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 6 & 0 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 7 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & 0 & 2 & 8 & 1 \end{pmatrix} b_C$$

$$T_{\min} = 12.90 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 1 & 3 & 18 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 & 0 \\ 9 & 18 & 0 & 12 & 56 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & 0 & 14 & 0 & 1 \end{pmatrix}$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 & \backslash \\ 2 & 4 & 1 & 3 & 18 & 0 & 4 & \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 & \\ 1 & 3 & 0 & -2 & 12 & 0 & 0 & \\ 10 & 18 & 0 & 12 & 56 & 0 & 14 & \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & \\ 2 & 4 & -1 & 0 & 14 & 0 & 1 & / \end{pmatrix}$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 7 & 2 \\ 2 & 4 & 1 & 3 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 10 & 0 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 7 & 15 & 0 & 10 & 8 & 38 & 12 \\ 2 & 4 & -1 & 0 & 2 & 12 & 1 \end{pmatrix} b_C$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 9 & 2 \\ 2 & 4 & 1 & 3 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 10 & 0 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 9 & 15 & 0 & 10 & 8 & 38 & 12 \\ 2 & 4 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_C}$$

$$[119] \quad 1^{11}23^11^{12}31^{113} + [\textcolor{blue}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{s}u_2 + \textcolor{red}{s}u_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -87, \quad \det = 902, \quad (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 8 & 2 \\ 2 & 4 & 1 & 3 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 10 & 0 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 8 & 15 & 0 & 10 & 8 & 38 & 12 \\ 2 & 4 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [120] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{blue}{1}} 1^{\textcolor{red}{1}} 4 1^{\textcolor{blue}{1}} 3 1^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 8 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 13 & 6 & 7 \\ 1 & 3 & 3 & 0 & 6 & 1 & 3 \\ 2 & 4 & 2 & -3 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [121] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{blue}{1}} 1^{\textcolor{blue}{1}} 5 1 3 1^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 37, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.59 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 5 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 2 & 5 & 0 & 2 & 7 & 5 & 5 \\ 1 & 3 & 0 & 1 & 5 & 1 & 3 \\ 2 & 3 & -1 & -1 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[122] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{blue}{1}} 1^{\textcolor{blue}{1}} 5 1 3 1^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 44, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.82 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & -3 & 2 & 1 & -1 \\ 4 & 0 & 5 & 2 & 7 & 5 & 5 \\ 1 & 0 & 3 & 1 & 5 & 1 & 3 \\ 2 & 0 & 2 & -1 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[123] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{blue}{1}} 1^{\textcolor{blue}{1}} 5 1 3 1^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 68, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & -3 & 2 & 1 & -1 \\ 3 & 0 & 5 & 2 & 7 & 5 & 5 \\ 1 & 0 & 3 & 1 & 5 & 1 & 3 \\ 2 & 0 & 2 & -1 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

4.126 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{su2n3mix} + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \quad 1^{\textcolor{red}{2}} 3^{\textcolor{blue}{1}} 1^{\textcolor{red}{1}} 3^{\textcolor{blue}{1}} 1^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 9 & 1 & 5 \\ 14 & 33 & 1 & 20 & 0 & 11 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 9 & 20 & 0 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 11 & -1 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

4.127 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{su2n3mix} + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{\textcolor{blue}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{blue}{1}} 3 1^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 494, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 5 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 9 & 5 & 0 & 12 & 9 & 5 & 6 \\ 2 & 6 & 3 & 0 & 9 & 4 & 3 & 4 \\ 1 & 3 & 3 & 0 & 5 & 3 & 0 & 2 \\ 2 & 4 & 2 & -4 & 6 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

4.128 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{\textcolor{blue}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{blue}{1}} 3 1^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 5 & 0 & 7 & 4 & 5 \\ 1 & 3 & 2 & 0 & 4 & 0 & 2 \\ 2 & 4 & 2 & -4 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{13}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -389, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 5 & 0 & 12 & 5 & 6 \\ 1 & 3 & 2 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & -4 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -445, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 15 & 11 & 0 & 44 & 9 & 12 \\ 1 & 3 & 2 & 0 & 9 & 0 & 2 \\ 2 & 4 & 2 & -4 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -542, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 8 & 0 & 25 & 7 & 9 \\ 1 & 3 & 2 & 0 & 7 & 0 & 2 \\ 2 & 4 & 2 & -4 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{13}3^{11}1^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -668, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 15 & 8 & 0 & 36 & 8 & 10 \\ 1 & 3 & 2 & 0 & 8 & 0 & 2 \\ 2 & 4 & 2 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}1^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -1091, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 18 & 3 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 18 & 11 & 0 & 55 & 10 & 13 \\ 1 & 3 & 2 & 0 & 10 & 0 & 2 \\ 2 & 4 & 2 & -4 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{12}3^{11}1^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -1264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 21 & 3 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 21 & 14 & 0 & 80 & 12 & 16 \\ 1 & 3 & 2 & 0 & 12 & 0 & 2 \\ 2 & 4 & 2 & -4 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [7] \ 1^{11}231^13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 14 & 1 & 19 & 5 \\ 14 & 33 & 1 & 44 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 44 & 0 & 58 & 14 \\ 5 & 11 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{12}31^13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 13 & 21 & 1 & 9 & 7 \\ 21 & 33 & 1 & 14 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 14 & 0 & 4 & 4 \\ 7 & 11 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{12}31^13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 85, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 1 & 1 & 9 & 5 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 0 & 0 & 10 & 6 \\ 5 & 0 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{12}3^{11}1^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 14 & 1 & 9 & 5 \\ 14 & 33 & 1 & 20 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 20 & 0 & 10 & 6 \\ 5 & 11 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

4.129 NHC(-2)(-3) + $\mathfrak{e}_6$ + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{12}3^{11}1^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 14 & 1 & 9 & 5 \\ 14 & 33 & 1 & 20 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 20 & 0 & 10 & 6 \\ 5 & 11 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

4.130 NHC(-2)(-3) + $\mathfrak{e}_6$ + su2n3mix + su2n3mix

$$T = 10 \quad [0] \ 1^{12}3^{11}1^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 1 & 3 & 5 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 5 & 2 & 0 \\ 3 & 5 & 0 & 5 & 5 & 3 & 5 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & -1 & 0 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12}3^{11}1^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 13 & 21 & 1 & 13 & 7 \\ 21 & 33 & 1 & 20 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 20 & 0 & 10 & 6 \\ 7 & 11 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

4.131 NHC(-2)(-3) + $\mathfrak{e}_6$ + su2n3mix + $\mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 3 & 4 \\ 3 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 2 & 0 & -2 & 2 & 0 \\ 3 & 3 & 0 & 2 & 0 & 2 \\ 4 & 3 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 5 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 5 & 6 & 0 & 7 & 5 & 5 \\ 1 & 3 & 3 & 0 & 5 & 0 & 2 \\ 2 & 3 & 3 & -4 & 5 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 7 & 12 & 0 & 9 & 26 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.04 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

- [4] $1^{11}3^{11}1^{14}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -265, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.15 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 5 & 0 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$
- [5] $1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 0 & 27 & 3 & 4 \\ 1 & 4 & 0 & 0 & 14 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 27 & 14 & 0 & 120 & 14 & 18 \\ 1 & 3 & 3 & 0 & 14 & 0 & 2 \\ 2 & 4 & 2 & -4 & 18 & 2 & -1 \end{pmatrix}_{b_0}$$
- [6] $1^{11}3^{11}1^{13}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -328, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.12 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 21 & 3 & 4 \\ 1 & 4 & 0 & 0 & 11 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 21 & 11 & 0 & 72 & 11 & 14 \\ 1 & 3 & 3 & 0 & 11 & 0 & 2 \\ 2 & 4 & 2 & -4 & 14 & 2 & -1 \end{pmatrix}_{b_0}$$
- [7] $1^{121}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -332, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 8 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & 0 & 10 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{11}3^{11}1^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -337, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.12 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 15 & 8 & 0 & 36 & 8 & 10 \\ 1 & 3 & 3 & 0 & 8 & 0 & 2 \\ 2 & 4 & 2 & -4 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$
- [10] $1^{111}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 8 & 12 & 0 & 9 & 26 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{111}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 5 & 0 \\ 6 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{112}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -403, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 5 & 0 \\ 7 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{121}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -428, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 10 & 18 & 0 & 12 & 56 & 14 \\ 3 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{1^{12}} 1^{111} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -451, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 9 & 12 & 0 & 9 & 26 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{1^{11}} 1^{121} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -635, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 9 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{1^{11}} 1^{121} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -779, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 11 & 18 & 0 & 12 & 56 & 14 \\ 3 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{1^{12}} 1^{121} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -832, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 10 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{1^{12}} 1^{121} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -1024, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 12 & 18 & 0 & 12 & 56 & 14 \\ 3 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [18] $1^{12} 23^1 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 5 & 0 & 0 \\ 3 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{1111} 23^1 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 2 & 5 & 0 & 6 & 4 & 4 \\ 1 & 5 & 1 & 0 & 6 & 4 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 6 & 0 & 8 & 6 & 6 \\ 1 & 4 & 4 & 0 & 6 & 1 & 3 \\ 2 & 4 & 3 & -4 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{12} 23^1 1^1 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{11} 23^1 1^{11} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.30 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 9 & 0 & 0 \\ 5 & 12 & 0 & 9 & 26 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 10 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11}231^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 226, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.30 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 4 & 0 & 0 \\ 3 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{11}231^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 385, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.31 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 1 & 3 & 18 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 12 & 0 & 0 \\ 7 & 18 & 0 & 12 & 56 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 14 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11}231^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 430, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 5 & 0 & 0 \\ 4 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{11}231^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 679, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 8 & 0 & 0 \\ 6 & 15 & 0 & 8 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & 10 & -1 & 0 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [26] \ 1^{11}3^{11}1^{11}5^{11}131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3 \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 56, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.14 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 2 & 0 & 2 & 1 & 0 & 2 & 2 \\ 1 & 0 & 3 & 1 & 2 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{111}3^{11}1^{11}5^{11}131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3 \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 66, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.12 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 5 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 2 & 5 & 0 & 2 & 7 & 5 & 5 \\ 1 & 3 & 0 & 1 & 5 & 0 & 2 \\ 2 & 3 & -1 & -1 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

4.132 NHC(-2)(-3) + $\textcolor{violet}{e}_6$ + su2n3mix + $\textcolor{red}{su}_3$ + $\textcolor{blue}{su}_2$

$$T = 10 \quad [0] \ 1^{11}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 494, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 5 & 0 & 12 & 5 & 6 \\ 1 & 3 & 3 & 0 & 5 & 0 & 2 \\ 2 & 6 & 3 & 0 & 9 & 3 & 4 \\ 2 & 4 & 2 & -4 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

4.133 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$

- $T = 7$ [0] $1^{111}31^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -47$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.91 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -3 & 3 & -1 \\ 5 & 8 & 3 & 3 & 3 \\ 3 & 5 & -1 & 3 & 0 \end{pmatrix}_{b_0}$
- [1] $1^{11}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -50$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 14.32 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 6 & 10 & 0 & 21 & 7 \\ 3 & 4 & -4 & 7 & -2 \end{pmatrix}_{b_0}$
- [2] $1^{11}31^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -56$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 0 \end{pmatrix}_{b_0}$
- [3] $1^{111}31^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -60$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 3 & 4 & -3 & 3 & -1 \\ 3 & 8 & 3 & 3 & 3 \\ 3 & 5 & -1 & 3 & 0 \end{pmatrix}_{b_0}$
- [4] $21^{131}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -60$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 3 & 4 & -3 & 3 & -1 \\ 3 & 8 & 3 & 3 & 3 \\ 3 & 5 & -1 & 3 & 0 \end{pmatrix}_{b_0}$
- [5] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -71$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 \\ 3 & 4 & -4 & 3 & 0 \end{pmatrix}_{b_0}$
- [6] $1^{11}31^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -80$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.70 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 4 & -4 & 2 & 0 \end{pmatrix}_{b_0}$
- [7] $1^{12}31^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -86$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 13.02 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 4 & 8 & 0 & 9 & 5 \\ 3 & 4 & -4 & 5 & -1 \end{pmatrix}_{b_0}$
- [8] $1^{13}31^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -96$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.33 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 6 & 12 & 0 & 24 & 8 \\ 3 & 4 & -4 & 8 & -1 \end{pmatrix}_{b_0}$
- [9] $1^{14}31^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -110$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.6$
 $T_{\min} = 13.60 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 8 & 16 & 0 & 45 & 11 \\ 3 & 4 & -4 & 11 & -1 \end{pmatrix}_{b_0}$
- [10] $131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -119$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.57 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -3 & 3 & -1 \\ 3 & 8 & 3 & 3 & 3 \\ 3 & 5 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 7, \ T_{\max} = 100, \ \Delta = -172, \ \det = -126, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 11.0 \\ T_{\min} = 13.05 \rightarrow 14, \ T_{\max} = 100, \ T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -3 & 3 & -1 \\ 4 & 8 & 3 & 3 & 3 \\ 3 & 5 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{15} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 7, \ T_{\max} = 100, \ \Delta = -172, \ \det = -128, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 14.0 \\ T_{\min} = 13.81 \rightarrow 14, \ T_{\max} = 100, \ T \in [14, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 10 & 20 & 0 & 72 & 14 \\ 3 & 4 & -4 & 14 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 7, \ T_{\max} = 100, \ \Delta = -172, \ \det = -141, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 11.0 \\ T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 5 & 9 & 0 & 12 & 6 \\ 3 & 4 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 7, \ T_{\max} = 100, \ \Delta = -172, \ \det = -188, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 11.0 \\ T_{\min} = 12.91 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 8 & 14 & 0 & 36 & 10 \\ 3 & 4 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{12} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 7, \ T_{\max} = 100, \ \Delta = -172, \ \det = -215, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 12.2 \\ T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 7 & 13 & 0 & 27 & 9 \\ 3 & 4 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{13} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 7, \ T_{\max} = 100, \ \Delta = -172, \ \det = -293, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 13.0 \\ T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 9 & 17 & 0 & 48 & 12 \\ 3 & 4 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{12} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 7, \ T_{\max} = 100, \ \Delta = -172, \ \det = -330, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 12.2 \\ T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 10 & 18 & 0 & 57 & 13 \\ 3 & 4 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{14} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 7, \ T_{\max} = 100, \ \Delta = -172, \ \det = -375, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 13.6 \\ T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 11 & 21 & 0 & 75 & 15 \\ 3 & 4 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{13} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 7, \ T_{\max} = 100, \ \Delta = -172, \ \det = -476, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 13.0 \\ T_{\min} = 12.65 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 12 & 22 & 0 & 84 & 16 \\ 3 & 4 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 21^{11} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 7, \ T_{\max} = 100, \ \Delta = -172, \ \det = -483, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 11.0 \\ T_{\min} = 12.69 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 4 \\ 8 & 9 & 4 & 8 & 5 \\ 3 & 4 & -3 & 3 & -1 \\ 6 & 8 & 3 & 3 & 3 \\ 4 & 5 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{14} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 7, \ T_{\max} = 100, \ \Delta = -172, \ \det = -626, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 13.6 \\ T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -6 & 0 & -4 \\ 14 & 26 & 0 & 117 & 19 \\ 3 & 4 & -4 & 19 & 0 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [22] \quad 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 2 & 5 & -2 & 3 & 0 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{\overline{12}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 2 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{\overline{13}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.67 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{\overline{11}} 31^{\overline{11}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{\overline{11}} 31^{\overline{11}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 6 & 6 & 0 & 12 & 6 \\ 2 & 3 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[31] \quad 2^{\overline{11}} 2^{\overline{11}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 5 & 5 & 5 & 5 \\ 3 & 5 & -2 & 3 & 0 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

- [32] $1^{11}3131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 8$, $T_{\max} = 99$, $\Delta = -151$, $\det = 93$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.81 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$
- [33] $1^{11}31^131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 8$, $T_{\max} = 99$, $\Delta = -151$, $\det = 102$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.12 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$
- [34] $1^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -144$, $\det = 112$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 4 \\ 5 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & -2 & 0 \\ 4 & 4 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$
- [35] $1^{12}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -144$, $\det = 169$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 3 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$
- [36] $1^131^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 8$, $T_{\max} = 99$, $\Delta = -151$, $\det = 204$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.27 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$
- [37] $1^131^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 8$, $T_{\max} = 99$, $\Delta = -151$, $\det = 219$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$
- [39] $1^{12}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 8$, $T_{\max} = 99$, $\Delta = -151$, $\det = 228$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.04 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 9 & 0 & 27 & 9 \\ 2 & 4 & 2 & -4 & 9 & -1 \end{pmatrix}_{b_0}$$
- [38] $1^{13}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -144$, $\det = 228$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$
- [40] $1^{11}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 8$, $T_{\max} = 99$, $\Delta = -151$, $\det = 282$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.04 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 0 & 12 & 6 \\ 2 & 4 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$
- $T = 9$ [41] $1^13221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -3$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 19$, $T_{\max} = 26$, $T \in [19, 26]$

$$\begin{pmatrix} 5 & 15 & 1 & 5 \\ 15 & 42 & 0 & 12 \\ 1 & 0 & -3 & -1 \\ 5 & 12 & -1 & -3 \end{pmatrix}_{b_0}$$
- [42] $1^{12}2321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -5$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 15.40 \rightarrow 16$, $T_{\max} = 32$, $T \in [16, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 2 & 6 & 0 & 2 & 9 & 5 \\ 2 & 3 & -1 & -1 & 5 & -2 \end{pmatrix}_{b_0}$$

[43] $1^{11}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 16.82 \rightarrow 17, T_{\max} = 32, T \in [17, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 1 & -1 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 2 & 3 & -1 & -1 & 2 & -3 \end{pmatrix}_{b_0}$$

[44] $1^{11}3^1221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 2 & 3 & 0 & 5 & 5 & 5 \\ 2 & 2 & -1 & 0 & 5 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{11}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -22, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 3 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[47] $1^{11}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 1 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{12}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 15.14 \rightarrow 16, T_{\max} = 32, T \in [16, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

[48] $1^{11}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 3 & 0 & 1 & 0 & 2 \\ 2 & 3 & -1 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{131^1}4^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{11}3^1221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.23 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{13}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.35 \rightarrow 15, T_{\max} = 32, T \in [15, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 4 & 0 & 9 & 3 & 24 & 8 \\ 2 & 0 & 2 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$

[50] $1^{12}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.23 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{11}4131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.09 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{11}3221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13, T_{\max} = 26, T \in [13, 26]$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[55] $1^{12}3221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.58 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[56] $1^{13}3221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & 0 & 5 & -1 \end{pmatrix}_{b_0}$$

[57] $1^{112}231^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 2 & 5 & 0 & 11 & 4 \\ 1 & 5 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 11 & 11 & 0 & 31 & 11 \\ 2 & 4 & 3 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[58] $1^{123}1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 3 & 0 \\ 3 & 0 & 0 & 3 & 3 & 3 \\ 2 & 0 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{12}3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.07 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 3 & 5 & 0 & 6 & 13 & 7 \\ 2 & 2 & -1 & 0 & 7 & 0 \end{pmatrix}_{b_0}$$

[61] $1^{11}4131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{123}1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 1 & 0 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 3 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[63] $1^{123}131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.22 \rightarrow 15, T_{\max} = 79, T \in [15, 79]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & -1 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[62] $1^{111}23131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 79, T \in [15, 79]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 5 & 0 & 4 \\ 1 & 5 & 1 & 0 & 4 \\ 1 & 0 & 0 & -6 & 0 \\ 1 & 4 & 4 & 0 & 1 \\ 2 & 4 & 3 & -4 & 3 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{11} 2^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -79, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{11} 2^3 3^{11} 2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 2 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[67] \ 1^{12} 2^3 3^{11} 2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 3 & 0 & 6 & 2 & 9 & 5 \\ 2 & 0 & 2 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{11} 3^{11} 2^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 13 & 14 & 1 & 7 \\ 14 & 13 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 7 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{11} 3^{11} 2^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 3 & 3 & 0 & 5 & 5 & 5 \\ 2 & 2 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{11} 2^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -97, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 79, T \in [15, 79]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 5 & 0 & 8 & 4 \\ 1 & 5 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 8 & 8 & 0 & 13 & 7 \\ 2 & 4 & 3 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{13} 3^{11} 2^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 4 & 7 & 0 & 7 & 25 & 9 \\ 2 & 2 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{11} 2^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.56 \rightarrow 15, T_{\max} = 79, T \in [15, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 5 & 7 & 0 & 7 & 5 \\ 3 & 4 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

- [74] $1^{12} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.15 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 3 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$
- [75] $1^{11} 2^3 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.03 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 3 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$
- [76] $1^{12} 231^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 0 \\ 2 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$
- [78] $1^{13} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.21 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$
- [77] $1^{11} 231^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 7 & 0 & 13 & 7 \\ 2 & 4 & 3 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$
- [79] $1^{12} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 4 & 5 & 0 & 6 & 13 & 7 \\ 2 & 2 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$
- [80] $1^{12} 231^{11} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.23 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$
- [81] $1^{12} 2^3 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 2 & -1 \\ 4 & 0 & 6 & 2 & 9 & 5 \\ 2 & 0 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$
- [82] $1^{13} 2^3 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 3 & -1 \\ 5 & 0 & 9 & 3 & 24 & 8 \\ 2 & 0 & 2 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$
- [83] $1^{12} 231^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.19 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$
- [84] $1^{13} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -259, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 5 & 7 & 0 & 7 & 25 & 9 \\ 2 & 2 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [85] \quad 1^1 1^1 4 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 20, T_{\max} = 26, T \in [20, 26]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 2 & 1 & 3 \\ 1 & 2 & -4 & 0 & -2 \\ 2 & 1 & 0 & 0 & 2 \\ 3 & 3 & -2 & 2 & -4 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^1 31^1 41^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 4 \\ 6 & 8 & 1 & 5 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 5 & 0 & -2 & 0 \\ 4 & 6 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{12} 31^1 41^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 8 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 7 & 0 & 15 & 7 \\ 2 & 4 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^1 1^1 31^1 41^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 43, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 5 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 4 & 0 & 4 & 4 \\ 2 & 4 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^1 5 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -137, \det = 46, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.48 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^1 1^1 31^1 4131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 79, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.28 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [91] \quad 1^1 5 12321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 21, T_{\max} = 24, T \in [21, 24]$$

$$\begin{pmatrix} 5 & 15 & 1 & 5 \\ 15 & 42 & 0 & 12 \\ 1 & 0 & -3 & -1 \\ 5 & 12 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^1 5 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 21, T_{\max} = 23, T \in [21, 23]$$

$$\begin{pmatrix} 5 & 15 & 6 & 5 \\ 15 & 42 & 15 & 12 \\ 6 & 15 & 4 & 4 \\ 5 & 12 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{12} 31^1 5 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 11, T_{\max} = 31, \Delta = -116, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 17.40 \rightarrow 18, T_{\max} = 31, T \in [18, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 2 & 6 & 0 & 2 & 9 & 5 \\ 2 & 3 & -1 & -1 & 5 & -2 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^1 1^1 31^1 5 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 \quad T = 11, T_{\max} = 31, \Delta = -116, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15.14 \rightarrow 16, T_{\max} = 31, T \in [16, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & -1 \\ 1 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[95] $1^{11}315131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 11$, $T_{\max} = 31$, $\Delta = -116$, $\det = -35$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.83 \rightarrow 15$, $T_{\max} = 31$, $T \in [15, 31]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 3 & 0 & 1 & 0 & 2 \\ 2 & 3 & -1 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[96] $1^13151^{1231} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 11$, $T_{\max} = 34$, $\Delta = -117$, $\det = -70$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 15.14 \rightarrow 16$, $T_{\max} = 34$, $T \in [16, 34]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 1 & 2 \\ 3 & 13 & 1 & 5 & 0 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 5 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 7 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$T = 12$ [97] $1^13151321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 12$, $T_{\max} = 16$, $\Delta = -88$, $\det = 6$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 15$, $T_{\max} = 16$, $T \in [15, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 4 & 0 & 1 & 1 & 3 \\ 2 & 4 & -1 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$

[98] $1^16131^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 12$, $T_{\max} = 27$, $\Delta = -113$, $\det = 16$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 17.50 \rightarrow 18$, $T_{\max} = 27$, $T \in [18, 27]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 3 & 2 & 6 & 2 & 9 & 5 \\ 2 & 2 & 2 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

[99] $1^16131^{1321} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 12$, $T_{\max} = 28$, $\Delta = -114$, $\det = 28$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 16.14 \rightarrow 17$, $T_{\max} = 28$, $T \in [17, 28]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

[100] $1^16131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 12$, $T_{\max} = 27$, $\Delta = -113$, $\det = 28$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 16.14 \rightarrow 17$, $T_{\max} = 27$, $T \in [17, 27]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 1 & 1 & 3 & 1 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[101] $1^16131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 12$, $T_{\max} = 27$, $\Delta = -113$, $\det = 32$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 15.75 \rightarrow 16$, $T_{\max} = 27$, $T \in [16, 27]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & -1 & 1 \end{pmatrix}_{b_0}$$

[102] $1^16131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 12$, $T_{\max} = 27$, $\Delta = -113$, $\det = 44$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 15.82 \rightarrow 16$, $T_{\max} = 27$, $T \in [16, 27]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 1 & 3 & 1 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[103] $31^{11}316131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3$ $T = 12$, $T_{\max} = 71$, $\Delta = -121$, $\det = 282$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 16.02 \rightarrow 17$, $T_{\max} = 71$, $T \in [17, 71]$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 3 \\ 4 & 2 & 5 & 2 & 4 \\ 4 & 5 & 1 & 2 & 3 \\ 2 & 2 & 2 & -2 & 0 \\ 3 & 4 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [104] \quad 1^{11}31 \quad 6 \quad 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 17, \Delta = -92, \det = -19, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 16.26 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 1 & 0 & 0 & 2 \\ 2 & 3 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[105] \quad 1^{11}31 \quad 6 \quad 131^{14}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 27, \Delta = -112, \det = -28, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 17.14 \rightarrow 18, T_{\max} = 27, T \in [18, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 4 & 0 \\ 3 & 4 & 0 & 4 & 9 & 5 \\ 2 & 3 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[106] \quad 1^{11}31 \quad 6 \quad 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 17, \Delta = -92, \det = -57, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 16.16 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 0 & 5 & 5 \\ 2 & 3 & 2 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[107] \quad 1^{13}31 \quad 6 \quad 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 17, \Delta = -92, \det = -73, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.23 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 10 & 6 & 0 & 33 & 11 \\ 2 & 3 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[108] \quad 1^{12}31 \quad 6 \quad 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 17, \Delta = -92, \det = -79, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 16.06 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 7 & 5 & 0 & 16 & 8 \\ 2 & 3 & 2 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[109] \quad 1^{12}31 \quad 6 \quad 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 17, \Delta = -92, \det = -129, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 16.13 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 8 & 3 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 8 & 0 & 29 & 11 \\ 2 & 3 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[110] \quad 1^{13}31 \quad 6 \quad 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 17, \Delta = -92, \det = -187, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.98 \rightarrow 16, T_{\max} = 17, T \in [16, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 11 & 9 & 0 & 48 & 14 \\ 2 & 3 & 2 & -3 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[111] \quad 1^{14}31 \quad 6 \quad 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 17, \Delta = -92, \det = -217, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 13.6$$

$$T_{\min} = 15.97 \rightarrow 16, T_{\max} = 17, T \in [16, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 14 & 3 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 14 & 10 & 0 & 73 & 17 \\ 2 & 3 & 2 & -3 & 17 & 2 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [112] \quad 1^{11}5^{13}31 \quad 6 \quad 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 14, T_{\max} = 26, \Delta = -107, \det = 27, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 17.48 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

4.134 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -3 & 0 & 3 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & -1 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\textcolor{red}{1}\textcolor{blue}{2}\textcolor{red}{1}} 3^{\textcolor{red}{1}\textcolor{blue}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 & 3 \\ 3 & 4 & 0 & 8 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 9 & 6 & 6 & 5 \\ 2 & 4 & 0 & 6 & 1 & 4 & 3 \\ 3 & 5 & 0 & 6 & 4 & 3 & 3 \\ 3 & 4 & -4 & 5 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}} 3^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 6 & 3 & 3 \\ 3 & 5 & 0 & 6 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 3 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}\textcolor{blue}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 4 & 3 \\ 3 & 5 & 0 & 3 & 4 & 3 & 3 \\ 3 & 4 & -4 & 2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -588, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 9 & 6 & 6 \\ 3 & 5 & 0 & 9 & 4 & 4 & 4 \\ 2 & 4 & 0 & 6 & 4 & 0 & 2 \\ 3 & 4 & -4 & 6 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [5] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}\textcolor{blue}{1}} 2^{\textcolor{red}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & 0 & -1 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}} 3^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 705, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 3 & 6 & 3 & 3 \\ 2 & 6 & 3 & 0 & 6 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 3 & 4 & 0 & 2 \\ 2 & 4 & 2 & -4 & 3 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

4.135 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 4 & 3 \\ 3 & 5 & 0 & 0 & 3 & 4 & 3 & 3 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 1 \\ 3 & 4 & -4 & 2 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

4.136 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 6 & 3 & 3 & 3 \\ 3 & 5 & 0 & 6 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 825, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 4 & 3 \\ 3 & 5 & 0 & 0 & 3 & 4 & 3 & 3 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 0 \\ 3 & 4 & -4 & 2 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.137 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 8 \quad [1] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 9 & 17 & 0 & 12 & 6 \\ 5 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 302, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 3 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 330, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.68 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 4 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 3 & -2 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \ 1^{\overline{11}} 3^{\overline{1}} 3^{\overline{1}} 2^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 2 & 5 & -2 & 3 & 0 & 0 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 3 & -2 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3^{11} 2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 99, \Delta = -143, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.88 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3^1 1^{11} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 100, \Delta = -150, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.07 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 0 & 3 \\ 1 & 3 & 3 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 3 & -1 & -1 & -2 & -2 \end{pmatrix}_{b_0}$$

$T = 10$ [8] $1^{11} 2^3 3^1 2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{so}_8$ $T = 10, T_{\max} = 30, \Delta = -116, \det = 62, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.10 \rightarrow 15, T_{\max} = 30, T \in [15, 30]$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 \\ 2 & 0 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 0 & 5 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{11} 2^3 2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{so}_8$ $T = 10, T_{\max} = 30, \Delta = -116, \det = 110, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.49 \rightarrow 15, T_{\max} = 30, T \in [15, 30]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 4 & 0 \\ 4 & 12 & 0 & 4 & 9 & 5 \\ 3 & 6 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{11} 3^1 2^1 2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{so}_8$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 147, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

4.138 NHC(-2)(-3) + e₆ + su₃ + so₈ + su₂

$T = 9$ [0] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 99, \Delta = -141, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 0 & 0 & 4 & 4 \\ 3 & 5 & -1 & -1 & -2 & 4 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 15 & 6 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -4 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -931$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

4.139 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$T = 8$ [0] $1^{14}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 9$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.6$

$T_{\min} = 24$, $T_{\max} = 98$, $T \in [24, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 21 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 0 & 75 & 18 & 15 \\ 3 & 5 & 0 & 18 & 4 & 4 \\ 3 & 4 & -4 & 15 & 4 & -11 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 64$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 16$, $T_{\max} = 98$, $T \in [16, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & -1 & 3 & 0 & -3 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 78$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.87 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 4 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & -1 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{123}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 79$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 0 & 13 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 27 & 27 & 9 \\ 6 & 12 & 0 & 27 & 25 & 9 \\ 3 & 4 & -4 & 9 & 9 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{11}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 94$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.70 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{11}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 103$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.03 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{11}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 121$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.22 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 6 & 3 \\ 3 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -4 & 3 & 4 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{112}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 126$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 12 & 6 \\ 4 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111}3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{111}3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{13}3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 48 & 15 & 12 \\ 3 & 5 & 0 & 15 & 4 & 4 \\ 3 & 4 & -4 & 12 & 4 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{112}3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.21 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & -1 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{121}3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.07 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 9 & 6 & 5 \\ 2 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -4 & 5 & 3 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{121}3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.17 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 9 & 12 & 5 \\ 5 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -4 & 5 & 7 & -1 \end{pmatrix}_{b_0}$$

[15] $1^{112}3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 4 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -4 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{111}3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 11 & 19 & 0 & 12 & 73 & 15 \\ 3 & 4 & -4 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{122}3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 9 & 12 & 5 \\ 4 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -4 & 5 & 6 & -1 \end{pmatrix}_{b_0}$$

[18] $1^{131} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 0 & 24 & 9 & 8 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -4 & 8 & 3 & -1 \end{pmatrix}_{b_0}$$

[19] $1^{111} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 0 & 36 & 9 & 10 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -4 & 10 & 3 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{122} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.54 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 8 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 9 & 24 & 5 \\ 10 & 18 & 0 & 24 & 58 & 14 \\ 3 & 4 & -4 & 5 & 14 & -1 \end{pmatrix}_{b_0}$$

[21] $1^{113} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 6 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -4 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

[22] $1^{113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 9 & 6 \\ 3 & 5 & 0 & 9 & 4 & 4 \\ 3 & 4 & -4 & 6 & 4 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{123} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 9 & 18 & 5 \\ 6 & 12 & 0 & 18 & 25 & 9 \\ 3 & 4 & -4 & 5 & 9 & -1 \end{pmatrix}_{b_0}$$

[24] $1^{123} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 27 & 12 & 9 \\ 3 & 5 & 0 & 12 & 4 & 4 \\ 3 & 4 & -4 & 9 & 4 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 6 & 6 \\ 2 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -4 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{113} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 9 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

[27] $1^{123} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.98 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 15 & 3 \\ 3 & 4 & 0 & 8 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 9 & 36 & 5 \\ 15 & 27 & 0 & 36 & 133 & 21 \\ 3 & 4 & -4 & 5 & 21 & -1 \end{pmatrix}_{b_0}$$

[29] $1^{114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.31 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 8 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -4 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[28] $1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{122} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 27 & 18 & 9 \\ 4 & 8 & 0 & 18 & 10 & 6 \\ 3 & 4 & -4 & 9 & 6 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 9 & 3 \\ 5 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{111} 3^1 1^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & -4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{115} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 10 & 20 & 0 & 15 & 73 & 15 \\ 3 & 4 & -4 & 2 & 15 & -1 \end{pmatrix}_{b_0}$$

[35] $1^{143} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 313, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 18 & 3 \\ 11 & 21 & 0 & 18 & 76 & 16 \\ 3 & 4 & -4 & 3 & 16 & -1 \end{pmatrix}_{b_0}$$

[36] $1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 7 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & -1 & 16 & -1 \end{pmatrix}_{b_0}$$

[38] $1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 373, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 12 & 3 \\ 7 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{116} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.64 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 12 & 24 & 0 & 18 & 106 & 18 \\ 3 & 4 & -4 & 2 & 18 & -1 \end{pmatrix}_{b_0}$$

[40] $1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 15 & 3 \\ 9 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -4 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 391, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 27 & 9 & 9 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -4 & 9 & 3 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{122} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 403, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 9 & 18 & 5 \\ 7 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -4 & 5 & 10 & -1 \end{pmatrix}_{b_0}$$

[43] $1^{113} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{117} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 13.76 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 14 & 28 & 0 & 21 & 145 & 21 \\ 3 & 4 & -4 & 2 & 21 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 15 & 6 \\ 5 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -4 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{121} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 486, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 0 & 57 & 12 & 13 \\ 2 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -4 & 13 & 3 & 0 \end{pmatrix}_{b_0}$$

[47] $1^{115} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 487, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & -3 & 0 & 15 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 40 & 15 & 0 & 148 & 20 \\ 3 & 5 & -1 & -1 & 20 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 526, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 7 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{131} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 547, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 48 & 12 & 12 \\ 2 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -4 & 12 & 3 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{113} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 568, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{121} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 574, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 27 & 21 & 9 \\ 5 & 9 & 0 & 21 & 13 & 7 \\ 3 & 4 & -4 & 9 & 7 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{131} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 595, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 48 & 27 & 12 \\ 5 & 9 & 0 & 27 & 13 & 7 \\ 3 & 4 & -4 & 12 & 7 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{113} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 631, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 11 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{123} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 9 & 24 & 5 \\ 9 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -4 & 5 & 13 & -1 \end{pmatrix}_{b_0}$$

[55] $1^{123} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 8 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 0 & 9 & 30 & 5 \\ 12 & 22 & 0 & 30 & 85 & 17 \\ 3 & 4 & -4 & 5 & 17 & -1 \end{pmatrix}_{b_0}$$

[56] $1^{112} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 648, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 21 & 6 \\ 7 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -4 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{114} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 688, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 16 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[58] $1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 736, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 10 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{113} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 747, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 27 & 6 \\ 9 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 760, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 9 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -4 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

[61] $1^{112} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 760, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 16 & 28 & 0 & 18 & 154 & 22 \\ 3 & 4 & -4 & 2 & 22 & 0 \end{pmatrix}_{b_0}$$

[62] $1^{114} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 33 & 6 \\ 11 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -4 & 6 & 16 & 0 \end{pmatrix}_{b_0}$$

[63] $1^{131} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 778, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.66 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 0 & 84 & 15 & 16 \\ 2 & 4 & 0 & 15 & 1 & 3 \\ 3 & 4 & -4 & 16 & 3 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{112} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 814, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 13 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & -4 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 823, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 13 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & -1 & 16 & -1 \end{pmatrix}_{b_0}$$

[66] $1^{122} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 841, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 27 & 30 & 9 \\ 7 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & -4 & 9 & 10 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{114} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 991, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1006, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 11 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -4 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{123} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1030, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 0 & 27 & 39 & 9 \\ 9 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -4 & 9 & 13 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{114} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1036, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{113} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 12 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -4 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{113} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 15 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & -4 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{113} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1606, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 18 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & -4 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1624, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 14 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & -4 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{114} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 2110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 17 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -4 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{114} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 2464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 20 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & -4 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [77] $1^{11} 3 2 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 4 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 4 & 0 & 5 & 5 \\ 2 & 2 & -1 & -1 & -1 & 5 & -4 \end{pmatrix}_{b_0}$$

$$[78] \ 1^{11} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16.67 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -1 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[79] \ 1^{11} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16, T_{\max} = 68, T \in [16, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 3 & 1 & 3 \\ 2 & 4 & 2 & -4 & 2 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[80] \ 1^{11} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16.40 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 2 & 5 & -2 & 3 & 0 & 0 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & 3 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{11} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 16.33 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 13 & 4 \\ 5 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 10 & 0 \\ 13 & 15 & 0 & 10 & 38 & 12 \\ 4 & 4 & -1 & 0 & 12 & -3 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{12} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.17 \rightarrow 16, T_{\max} = 68, T \in [16, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 & 2 \\ 2 & 4 & 4 & 0 & 12 & 6 & 4 \\ 1 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 9 & 0 & 27 & 12 & 9 \\ 2 & 6 & 3 & 0 & 12 & 4 & 4 \\ 2 & 4 & 2 & -4 & 9 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[83] \ 1^{12} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 18.58 \rightarrow 19, T_{\max} = 68, T \in [19, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 4 & 0 & 6 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 3 & 12 & 3 \\ 4 & 12 & 9 & 0 & 12 & 28 & 10 \\ 2 & 4 & 2 & -4 & 3 & 10 & -6 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{11} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 7 & 0 & 14 & 8 \\ 3 & 5 & 0 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{11} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 3 & 1 & 3 \\ 2 & 3 & 3 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[86] \ 1^{11} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 6 & 0 & 3 & 4 & 4 \\ 2 & 3 & 3 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

- [89] $1^{111} 3^{14} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 9 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$
- [87] $1^{11} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 68$, $T \in [15, 68]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 \\ 1 & 3 & 3 & 0 & 3 & 1 & 3 \\ 2 & 4 & 2 & -4 & 3 & 3 & -2 \end{pmatrix}_{b_0}$
- [88] $1^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$
- [90] $1^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -82$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.12 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$
- [91] $1^{111} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 5 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$
- [92] $1^{112} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 6 & 0 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$
- [93] $1^{11} 31^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 4 & 12 & 6 & 0 & 6 & 22 & 8 \\ 2 & 4 & 2 & -4 & 2 & 8 & -1 \end{pmatrix}_{b_0}$
- [94] $1^{111} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.22 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$
- [95] $1^{111} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.10 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$
- [96] $1^{111} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -131$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$

- [97] $1^{121} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -137$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.29 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 2 & 4 & 0 & 2 & 5 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -1 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$
- [99] $1^{112} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 13.11 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$
- [98] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [100] $1^{121} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -141$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 2 & 4 & 0 & 2 & 5 & 4 & 3 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -1 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$
- [101] $1^{113} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -155$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$
- [102] $1^{114} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -176$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.77 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$
- [103] $1^{113} 1^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -177$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$
- [104] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [105] $1^{115} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -203$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.1$
 $T_{\min} = 14.11 \rightarrow 15$, $T_{\max} = 25$, $T \in [15, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 15 & 0 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & 0 & 20 & -1 \end{pmatrix}_{b_0}$$
- [106] $1^{111} 1^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -231$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.08 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 12 & 6 & 6 \\ 1 & 3 & 3 & 0 & 6 & 1 & 3 \\ 2 & 4 & 2 & -4 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

[108] $1^{111} 3^{113} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -236$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 8 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[107] $1^{112} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -236$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 8 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[110] $1^{111} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -240$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 6 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[109] $1^{111} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -240$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.85 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[111] $1^{111} 3^{112} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -243$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{111} 3^{112} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -291$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 7 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[113] $1^{113} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -303$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.26 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 3 & 12 & 3 \\ 2 & 4 & 3 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 12 & 2 \\ 5 & 12 & 15 & 0 & 12 & 49 & 13 \\ 2 & 3 & 3 & -4 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

[114] $1^{121} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -306$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 3 & 4 & 0 & 2 & 5 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[115] $1^{111} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -375$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

12.0

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 3 & 6 & 9 & 0 & 6 & 13 & 7 \\ 2 & 3 & 3 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

[116] $1^{113} 3^{21} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 11 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[117] $1^{112} 3^{21} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -411, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[118] $1^{111} 3^{11} 3^{11} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -411, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 3 & 9 & 6 & 0 & 6 & 13 & 7 \\ 2 & 4 & 2 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

[119] $1^{112} 3^{11} 3^{11} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 3 & 9 & 3 \\ 2 & 4 & 3 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 9 & 2 \\ 4 & 9 & 12 & 0 & 9 & 28 & 10 \\ 2 & 3 & 3 & -4 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[120] $1^{11} 3^{11} 3^{11} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 3 & 6 & 3 \\ 2 & 6 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & 2 & -4 & 3 & 4 & -1 \end{pmatrix}_{b_0}$$

[121] $1^{11} 3^{11} 3^{11} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -537, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.04 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 6 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 12 & 9 & 6 \\ 2 & 6 & 3 & 0 & 9 & 4 & 4 \\ 2 & 4 & 2 & -4 & 6 & 4 & -1 \end{pmatrix}_{b_0}$$

[122] $1^{112} 3^{21} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -555, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 9 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[123] $1^{114} 3^{21} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.03 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 14 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{112} 3^{12} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -576, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{112} 3^{14} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -588, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.86 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{113} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -588, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.97 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{113} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -615, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.35 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 3 & 9 & 3 \\ 3 & 9 & 6 & 0 & 9 & 13 & 7 \\ 2 & 4 & 2 & -4 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{112} 3^{13} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -635, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.86 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 9 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{111} 31^{12} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -762, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 9 & 2 \\ 5 & 15 & 9 & 0 & 9 & 37 & 11 \\ 2 & 4 & 2 & -4 & 2 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[130] \ 1^{112} 3^{12} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -768, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 10 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{114} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -771, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.02 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 13 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{112} 3^{13} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -875, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 11 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{113} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -876$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 12 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{112} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -876$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 12 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[135] $1^{111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -906$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 0 & 12 & 15 & 6 \\ 3 & 9 & 6 & 0 & 15 & 13 & 7 \\ 2 & 4 & 2 & -4 & 6 & 7 & -1 \end{pmatrix}_{b_0}$$

[136] $1^{113} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -915$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 11 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{111} 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -1059$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 12 & 2 \\ 7 & 21 & 12 & 0 & 12 & 73 & 15 \\ 2 & 4 & 2 & -4 & 2 & 15 & -1 \end{pmatrix}_{b_0}$$

[138] $1^{113} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 12 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{114} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1203$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 15 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[141] $1^{113} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1251$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{113} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1251$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

12.8

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 13 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[142] $1^{114} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -1260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[143] $1^{111} 31^{14} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -1302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 15 & 2 \\ 9 & 27 & 15 & 0 & 15 & 121 & 19 \\ 2 & 4 & 2 & -4 & 2 & 19 & -1 \end{pmatrix}_{b_0}$$

[144] $1^{113} 3^{13} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -1520, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 14 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[145] $1^{114} 3^{13} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -1643, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[146] $1^{113} 3^{14} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -1683, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 15 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{114} 3^{12} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -1740, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 16 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{114} 3^{14} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -1920, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[149] $1^{114} 3^{13} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -2171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 17 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[150] $1^{114} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -2496$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.85 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 18 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [151] $1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 3$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 17.33 \rightarrow 18$, $T_{\max} = 24$, $T \in [18, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 6 & 8 & 0 & 5 & 13 & 7 \\ 3 & 4 & -3 & 2 & 7 & -3 \end{pmatrix}_{b_0}$$

[152] $1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 12$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 17.33 \rightarrow 18$, $T_{\max} = 24$, $T \in [18, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -3 & 2 & 3 & -3 \end{pmatrix}_{b_0}$$

[153] $1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 26$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.23 \rightarrow 15$, $T_{\max} = 24$, $T \in [15, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[154] $1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 27$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 24$, $T \in [15, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[155] $1^{11} 231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 15$, $\Delta = -93$, $\det = 32$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 15$, $T \in [14, 15]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & 0 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

[156] $1^{11} 2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 34$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 3 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 4 & 4 & 6 & 6 \\ 2 & 3 & -1 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[157] $1^{11} 2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 40$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[158] $1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 48$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.08 \rightarrow 15$, $T_{\max} = 24$, $T \in [15, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 7 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 7 & 6 & 5 \\ 3 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -3 & 5 & 4 & 0 \end{pmatrix}_{b_0}$$

[159] $1^{11} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 5 \\ 9 & 13 & 1 & 0 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 7 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

[160] $1^1 231^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.43 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 0 & 0 \\ 2 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & 0 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

[161] $1^{11} 2^1 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 2 & 0 & 3 & 1 & 0 & 3 & 2 \\ 1 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[162] $1^1 231^{12} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 74, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & 0 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[163] $1^{11} 4^1 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

[164] $1^{11} 4^1 1^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 78, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.23 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 7 & 11 & 5 \\ 5 & 9 & 0 & 11 & 13 & 7 \\ 3 & 4 & -3 & 5 & 7 & 0 \end{pmatrix}_{b_0}$$

[165] $1^{11} 4^1 1^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 87, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

[167] $1^{11} 4^1 3^{11} 221^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 60 & 5 \\ 9 & 13 & 1 & 96 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 60 & 96 & 0 & 646 & 48 \\ 5 & 7 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

[166] $1^{11} 4^1 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 30 & 5 \\ 9 & 13 & 1 & 48 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 48 & 0 & 160 & 24 \\ 5 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[168] $1^{11}2^13^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 101, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 0 & 1 & -3 & 1 & 0 & -1 \\ 2 & 0 & 3 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -1 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

[169] $1^{11}1^{31}4^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 10 & 3 \\ 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 10 & 15 & 0 & 11 & 45 & 13 \\ 3 & 4 & -3 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

[170] $1^{12}2^13^{12}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -3 & 2 & 1 & -1 \\ 4 & 0 & 6 & 2 & 9 & 6 & 5 \\ 1 & 0 & 3 & 1 & 6 & 1 & 3 \\ 2 & 0 & 2 & -1 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[171] $1^{11}2^12^13^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 3 & 0 & 1 & 0 & 7 & 2 \\ 3 & 7 & 0 & 5 & 7 & 17 & 9 \\ 2 & 3 & -1 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[172] $1^{11}1^{11}2^13^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 125, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[173] $1^{11}1^{11}4^{11}131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[174] $1^1231^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 137, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.30 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[175] $1^{11}3^12^13^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 154, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 3 & 0 & 1 & 0 & 10 & 2 \\ 4 & 10 & 0 & 6 & 10 & 34 & 12 \\ 2 & 3 & -1 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[176] $1^{111}2^321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 157$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.96 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & -3 & 1 & 4 & -1 \\ 2 & 0 & 3 & 1 & 0 & 4 & 2 \\ 3 & 0 & 4 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[177] $1^{111}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 158$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.84 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 0 & 7 & 5 & 5 \\ 2 & 4 & 0 & 5 & 1 & 3 \\ 3 & 4 & -3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{111}23131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 15$, $\Delta = -93$, $\det = 164$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.32 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 5 & 0 & 4 & 4 & 4 \\ 1 & 5 & 1 & 0 & 4 & 4 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 4 & 4 & 0 & 1 & 4 & 3 \\ 1 & 4 & 4 & 0 & 4 & 2 & 4 \\ 2 & 4 & 3 & -4 & 3 & 4 & -1 \end{pmatrix}_{b_0}$$

[179] $1^1231^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 15$, $\Delta = -93$, $\det = 174$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.21 \rightarrow 14$, $T_{\max} = 15$, $T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 3 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & 0 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

[180] $1^{111}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 203$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.81 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 5 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[181] $1^{11}231^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 15$, $\Delta = -93$, $\det = 212$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.34 \rightarrow 14$, $T_{\max} = 15$, $T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 9 & 0 & 7 & 0 & 14 & 8 \\ 2 & 4 & -1 & 0 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[182] $1^{112}2^321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 215$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.91 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 3 & 0 & 1 & 0 & 6 & 2 \\ 3 & 6 & 0 & 2 & 6 & 10 & 6 \\ 2 & 3 & -1 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[183] $1^{112}2^321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 221$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -3 & 1 & 2 & -1 \\ 2 & 0 & 3 & 1 & 0 & 6 & 2 \\ 3 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[184] $1^{121}3^1221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -85$, $\det = 221$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 1 & 0 \\ 4 & 5 & 0 & 6 & 13 & 5 & 7 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & 0 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[185] $1^{112}2^13^121^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}_6+\textcolor{orange}{su}_3+\textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 3 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 5 & 7 & 17 & 9 \\ 2 & 3 & -1 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[186] $1^{111}4^1131^1+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}_6+\textcolor{orange}{su}_3+\textcolor{red}{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[187] $1^{111}4^1131^2+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}_6+\textcolor{orange}{su}_3+\textcolor{red}{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 10 & 2 \\ 11 & 15 & 0 & 10 & 45 & 13 \\ 3 & 4 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[188] $1^{113}2^13^121^1+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}_6+\textcolor{orange}{su}_3+\textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 245, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 3 & 0 & 1 & 0 & 9 & 2 \\ 4 & 9 & 0 & 3 & 9 & 25 & 9 \\ 2 & 3 & -1 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[189] $1^{113}2^13^121^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}_6+\textcolor{orange}{su}_3+\textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 245, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 10 & 2 \\ 1 & 0 & 1 & -3 & 1 & 6 & -1 \\ 2 & 0 & 3 & 1 & 0 & 10 & 2 \\ 5 & 0 & 10 & 6 & 10 & 34 & 12 \\ 2 & 0 & 2 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[190] $1^{111}2^13^121^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}_6+\textcolor{orange}{su}_3+\textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 254, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & -3 & 1 & 4 & -1 \\ 2 & 0 & 3 & 1 & 0 & 4 & 2 \\ 4 & 0 & 4 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[191] $1^{1131}4^11^1+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}_6+\textcolor{orange}{su}_3+\textcolor{red}{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 257, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 3 \\ 3 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[192] $1^{112}2^13^121^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{e}_6+\textcolor{orange}{su}_3+\textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 269, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 3 & 0 & 1 & 0 & 7 & 2 \\ 4 & 7 & 0 & 5 & 7 & 17 & 9 \\ 2 & 3 & -1 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[193] $1^{111}2^13^121^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 271, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 2 & 0 & 3 & 1 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[194] $1^{11}31^{11}4^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 287, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 5 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[195] $1^{112}2^13^121^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 290, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 3 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 2 & 6 & 10 & 6 \\ 2 & 3 & -1 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[196] $1^{112}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 311, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 7 & 2 \\ 1 & 0 & 1 & -3 & 1 & 5 & -1 \\ 2 & 0 & 3 & 1 & 0 & 7 & 2 \\ 4 & 0 & 7 & 5 & 7 & 17 & 9 \\ 2 & 0 & 2 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[197] $1^{1231}1131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 314, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.31 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[198] $1^{112}3^1221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 344, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 31 & 5 \\ 9 & 13 & 1 & 48 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 48 & 0 & 160 & 24 \\ 5 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[199] $1^{111}2^13^121^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 2 & 0 & 3 & 1 & 0 & 3 & 2 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[200] $1^{111}4^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 384, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 8 & 12 & 0 & 8 & 26 & 10 \\ 3 & 4 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[201] $1^{114}2^13^121^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 386, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -3 & 1 & 4 & -1 \\ 2 & 0 & 3 & 1 & 0 & 12 & 2 \\ 6 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[202] $1^{11}1^{131}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 402, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 6 & 9 & 0 & 5 & 13 & 7 \\ 3 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[203] $1^{11}1^{131}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 444, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 8 & 3 \\ 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 8 & 12 & 0 & 8 & 26 & 10 \\ 3 & 4 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[204] $1^{11}231^{12}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 449, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 10 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 15 & 0 & 10 & 0 & 38 & 12 \\ 2 & 4 & -1 & 0 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[205] $1^{113}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 479, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 3 & 0 & 1 & 0 & 10 & 2 \\ 5 & 10 & 0 & 6 & 10 & 34 & 12 \\ 2 & 3 & -1 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[206] $1^{113}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 500, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 3 & 0 & 1 & 0 & 9 & 2 \\ 5 & 9 & 0 & 3 & 9 & 25 & 9 \\ 2 & 3 & -1 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{112}3^1221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 32 & 5 \\ 9 & 13 & 1 & 48 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 32 & 48 & 0 & 160 & 24 \\ 5 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[208] $1^{112}2^13^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 586, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -3 & 1 & 2 & -1 \\ 2 & 0 & 3 & 1 & 0 & 6 & 2 \\ 4 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[209] $1^1231^{12}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 587, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & 0 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[210] $1^{113}2^13^{12}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 596, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -3 & 1 & 3 & -1 \\ 2 & 0 & 3 & 1 & 0 & 9 & 2 \\ 5 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[211] $1^{113}2^13^{13}221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 600, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 33 & 5 \\ 9 & 13 & 1 & 48 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 33 & 48 & 0 & 160 & 24 \\ 5 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[212] $1^{112}2^13^{14}221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 608, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 34 & 5 \\ 9 & 13 & 1 & 48 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 48 & 0 & 160 & 24 \\ 5 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[213] $1^{113}2^13^{12}23^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 614, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 3 & 0 & 1 & 0 & 10 & 2 \\ 6 & 10 & 0 & 6 & 10 & 34 & 12 \\ 2 & 3 & -1 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[214] $1^{114}2^13^{11}23^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 629, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 3 & 0 & 1 & 0 & 13 & 2 \\ 6 & 13 & 0 & 7 & 13 & 57 & 15 \\ 2 & 3 & -1 & -1 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[215] $1^{114}2^13^{11}221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 632, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 61 & 5 \\ 9 & 13 & 1 & 96 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 61 & 96 & 0 & 646 & 48 \\ 5 & 7 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

[216] $1^{11}231^{13}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 644, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 0 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 13 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 21 & 0 & 13 & 0 & 74 & 16 \\ 2 & 4 & -1 & 0 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[217] $1^{114}2^13^{12}23^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 650, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 3 & 0 & 1 & 0 & 12 & 2 \\ 6 & 12 & 0 & 4 & 12 & 46 & 12 \\ 2 & 3 & -1 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[218] $1^{112}2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 688, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 7 & 2 \\ 1 & 0 & 1 & -3 & 1 & 5 & -1 \\ 2 & 0 & 3 & 1 & 0 & 7 & 2 \\ 5 & 0 & 7 & 5 & 7 & 17 & 9 \\ 2 & 0 & 2 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[219] $1^{11}231^{14}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 797, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 0 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 16 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 10 & 27 & 0 & 16 & 0 & 122 & 20 \\ 2 & 4 & -1 & 0 & -1 & 20 & 0 \end{pmatrix}_{b_0}$$

[220] $1^{11}231^{13}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 818, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & 0 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[221] $1^{114}2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 896, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 13 & 2 \\ 1 & 0 & 1 & -3 & 1 & 7 & -1 \\ 2 & 0 & 3 & 1 & 0 & 13 & 2 \\ 7 & 0 & 13 & 7 & 13 & 57 & 15 \\ 2 & 0 & 2 & -1 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[222] $1^{113}2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 902, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 10 & 2 \\ 1 & 0 & 1 & -3 & 1 & 6 & -1 \\ 2 & 0 & 3 & 1 & 0 & 10 & 2 \\ 6 & 0 & 10 & 6 & 10 & 34 & 12 \\ 2 & 0 & 2 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[223] $1^{114}2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 1 & 3 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 3 & 0 & 1 & 0 & 13 & 2 \\ 7 & 13 & 0 & 7 & 13 & 57 & 15 \\ 2 & 3 & -1 & -1 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[224] $1^{11}231^{14}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 15, \Delta = -93, \det = 1007, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 9 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & 0 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[225] $1^{114}3^1 221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 1088, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 62 & 5 \\ 9 & 13 & 1 & 96 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 62 & 96 & 0 & 646 & 48 \\ 5 & 7 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

[226] $1^{114}3^1 221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -85, \det = 1464, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 63 & 5 \\ 9 & 13 & 1 & 96 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 63 & 96 & 0 & 646 & 48 \\ 5 & 7 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

[227] $1^{114} 3^{221} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -85$, $\det = 1760$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 64 & 5 \\ 9 & 13 & 1 & 96 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 64 & 96 & 0 & 646 & 48 \\ 5 & 7 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

$T = 11$ [228] $1^{13} 5^{11} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -107$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 18$, $T_{\max} = 23$, $T \in [18, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 3 & 0 & 12 & 4 \\ 2 & 3 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 6 & 0 & 25 & 9 \\ 3 & 4 & 0 & -1 & 9 & -2 \end{pmatrix}_{b_0}$$

[229] $1^{111} 31^1 41^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -26$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.46 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 4 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 4 & 1 & 3 \\ 2 & 4 & 2 & -3 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

[230] $1^{12} 5^{11} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -107$, $\det = -52$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.54 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 3 & 0 & 8 & 4 \\ 2 & 3 & -2 & 0 & 4 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 4 & 0 & 10 & 6 \\ 3 & 4 & 0 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[231] $1^{13} 1^{11} 41^1 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -80$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.35 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 7 & 4 \\ 6 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 5 & 0 & -2 & 5 & 0 \\ 7 & 10 & 1 & 5 & 9 & 7 \\ 4 & 6 & 0 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

[232] $1^{11} 5^{11} 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -107$, $\det = -82$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.49 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 2 & 3 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 4 & 0 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[233] $1^{11} 31^{11} 4131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -137$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.30 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[234] $1^{111} 31^{11} 41^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -251$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.47 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 11 & 4 \\ 1 & 4 & 0 & 0 & 3 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 7 & 2 \\ 5 & 11 & 7 & 0 & 7 & 21 & 9 \\ 2 & 4 & 2 & -3 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

[235] $1^{111} 31^{11} 4131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -314$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 3 & 9 & 6 & 0 & 6 & 13 & 7 \\ 2 & 4 & 2 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [236] \quad 1^{111} 31^1 5 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 15, \Delta = -86, \det = 40, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [237] \quad 1^{11} 6^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 1 & 1 & 1 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 3 & 1 & 0 & 1 & 3 \\ 2 & 2 & 2 & -1 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[238] \quad 1^{11} 6^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -32, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & 1 & -1 \\ 2 & 1 & 3 & 1 & 0 & 3 & 2 \\ 1 & 1 & 3 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[239] \quad 1^{11} 6^{11} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.76 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 1 & 1 & 1 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 7 & 3 & 0 & 16 & 8 \\ 2 & 2 & 2 & -1 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[240] \quad 1^{11} 6^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -50, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.84 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & -1 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[241] \quad 1^{11} 6^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -98, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.76 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 1 & 1 & 1 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 4 & 2 & 0 & 5 & 5 \\ 2 & 2 & 2 & -1 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[242] \quad 1^{11} 6^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -112, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.82 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 1 & 3 & 2 \\ 1 & 2 & 0 & 1 & 3 & 4 & 2 \\ 1 & 1 & 1 & -3 & 1 & 2 & -1 \\ 2 & 1 & 3 & 1 & 0 & 4 & 2 \\ 2 & 3 & 4 & 2 & 4 & 5 & 5 \\ 2 & 2 & 2 & -1 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

4.140 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 4 \\ 3 & 5 & 0 & 3 & 3 & 4 & 3 \\ 3 & 4 & -4 & 2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

4.141 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -3 & 0 & 3 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & -1 & 4 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 15 & 6 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -4 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

4.142 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.86 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 3 & 5 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & -4 & 3 & 3 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 4 \\ 3 & 5 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -4 & 2 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{12}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.71 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 36 & 9 & 9 & 10 \\ 2 & 4 & 0 & 9 & 1 & 3 & 3 \\ 2 & 4 & 0 & 9 & 3 & 1 & 3 \\ 3 & 4 & -4 & 10 & 3 & 3 & -4 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 2 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1121} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 5 & 3 \\ 3 & 4 & 0 & 9 & 8 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 12 & 15 & 6 \\ 4 & 8 & 0 & 12 & 10 & 12 & 6 \\ 5 & 9 & 0 & 15 & 12 & 13 & 7 \\ 3 & 4 & -4 & 6 & 6 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1111} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 8 & 14 & 0 & 9 & 9 & 37 & 11 \\ 3 & 4 & -4 & 2 & 3 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 11 & 3 \\ 3 & 4 & 0 & 4 & 5 & 19 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 3 & 5 & 0 & 3 & 4 & 21 & 4 \\ 11 & 19 & 0 & 12 & 21 & 73 & 15 \\ 3 & 4 & -4 & 2 & 4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{131} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 2 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 9 & 17 & 0 & 48 & 15 & 12 & 12 \\ 3 & 5 & 0 & 15 & 4 & 3 & 4 \\ 2 & 4 & 0 & 12 & 3 & 1 & 3 \\ 3 & 4 & -4 & 12 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1121} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 5 & 3 \\ 3 & 4 & 0 & 4 & 8 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 4 & 8 & 0 & 6 & 10 & 12 & 6 \\ 5 & 9 & 0 & 6 & 12 & 13 & 7 \\ 3 & 4 & -4 & 2 & 6 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3^1 1^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 5 & 3 \\ 3 & 4 & 0 & 4 & 19 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 12 & 6 & 2 \\ 11 & 19 & 0 & 12 & 73 & 33 & 15 \\ 5 & 9 & 0 & 6 & 33 & 13 & 7 \\ 3 & 4 & -4 & 2 & 15 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1112} 3^1 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -137, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 7 & 3 \\ 3 & 4 & 0 & 4 & 14 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 9 & 2 \\ 8 & 14 & 0 & 9 & 37 & 33 & 11 \\ 7 & 13 & 0 & 9 & 33 & 28 & 10 \\ 3 & 4 & -4 & 2 & 11 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{113} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -173, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.39 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 9 & 3 \\ 3 & 4 & 0 & 4 & 5 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 3 & 5 & 0 & 3 & 4 & 15 & 4 \\ 9 & 17 & 0 & 12 & 15 & 49 & 13 \\ 3 & 4 & -4 & 2 & 4 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -174, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.85 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 2 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 12 & 6 & 6 \\ 4 & 8 & 0 & 12 & 10 & 6 & 6 \\ 2 & 4 & 0 & 6 & 6 & 1 & 3 \\ 3 & 4 & -4 & 6 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -191, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 6 & 4 \\ 4 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1122} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 13 & 3 \\ 3 & 4 & 0 & 4 & 13 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 15 & 2 \\ 7 & 13 & 0 & 9 & 28 & 54 & 10 \\ 13 & 23 & 0 & 15 & 54 & 100 & 18 \\ 3 & 4 & -4 & 2 & 10 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -219, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 9 & 6 & 6 \\ 3 & 5 & 0 & 9 & 4 & 3 & 4 \\ 2 & 4 & 0 & 6 & 3 & 1 & 3 \\ 3 & 4 & -4 & 6 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{121} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -245, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 2 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 0 & 27 & 12 & 9 & 9 \\ 3 & 5 & 0 & 12 & 4 & 3 & 4 \\ 2 & 4 & 0 & 9 & 3 & 1 & 3 \\ 3 & 4 & -4 & 9 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -284, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 6 & 4 \\ 3 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -341, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 4 \\ 5 & 9 & 0 & 6 & 9 & 13 & 7 \\ 3 & 4 & -4 & 2 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -344, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.81 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -353, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 8 & 3 \\ 3 & 4 & 0 & 4 & 5 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 3 & 5 & 0 & 3 & 4 & 15 & 4 \\ 8 & 14 & 0 & 9 & 15 & 37 & 11 \\ 3 & 4 & -4 & 2 & 4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -359, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -368, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 6 & 4 \\ 4 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -380, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 7 & 3 \\ 3 & 4 & 0 & 4 & 5 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 3 & 5 & 0 & 3 & 4 & 12 & 4 \\ 7 & 13 & 0 & 9 & 12 & 28 & 10 \\ 3 & 4 & -4 & 2 & 4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{112} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -380, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 4 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 4 & 8 & 0 & 6 & 6 & 10 & 6 \\ 3 & 4 & -4 & 2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -408, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 6 & 6 & 6 \\ 2 & 4 & 0 & 6 & 1 & 3 & 3 \\ 2 & 4 & 0 & 6 & 3 & 1 & 3 \\ 3 & 4 & -4 & 6 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -413, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 0 & 5 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 \\ 11 & 21 & 0 & 18 & 15 & 76 & 16 \\ 3 & 4 & -4 & 3 & 3 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1122} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 4 & 8 & 0 & 6 & 10 & 12 & 6 \\ 4 & 8 & 0 & 6 & 12 & 10 & 6 \\ 3 & 4 & -4 & 2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1112} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -425, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 4 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 \\ 13 & 23 & 0 & 15 & 15 & 100 & 18 \\ 3 & 4 & -4 & 2 & 3 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1113} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -440, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.15 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 12 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 6 & 12 & 0 & 9 & 9 & 25 & 9 \\ 3 & 4 & -4 & 2 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1122} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -444, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 7 & 3 \\ 3 & 4 & 0 & 9 & 8 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 12 & 21 & 6 \\ 4 & 8 & 0 & 12 & 10 & 18 & 6 \\ 7 & 13 & 0 & 21 & 18 & 28 & 10 \\ 3 & 4 & -4 & 6 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -485, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 2 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1113} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -488, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.87 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 12 & 3 \\ 3 & 4 & 0 & 4 & 14 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 15 & 2 \\ 8 & 14 & 0 & 9 & 37 & 57 & 11 \\ 12 & 22 & 0 & 15 & 57 & 85 & 17 \\ 3 & 4 & -4 & 2 & 11 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1114} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 16 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 \\ 8 & 16 & 0 & 12 & 12 & 46 & 12 \\ 3 & 4 & -4 & 2 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{112} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -533, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.03 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 7 & 13 & 0 & 12 & 9 & 28 & 10 \\ 3 & 4 & -4 & 3 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{113} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -551, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 5 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 \\ 9 & 17 & 0 & 15 & 12 & 49 & 13 \\ 3 & 4 & -4 & 3 & 3 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -560, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 7 & 3 \\ 3 & 4 & 0 & 4 & 8 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 9 & 2 \\ 4 & 8 & 0 & 6 & 10 & 18 & 6 \\ 7 & 13 & 0 & 9 & 18 & 28 & 10 \\ 3 & 4 & -4 & 2 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -729, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 6 & 15 & 6 \\ 2 & 4 & 0 & 6 & 1 & 6 & 3 \\ 5 & 9 & 0 & 15 & 6 & 13 & 7 \\ 3 & 4 & -4 & 6 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{111} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -773, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 8 & 3 \\ 3 & 4 & 0 & 4 & 9 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 9 & 2 \\ 5 & 9 & 0 & 6 & 13 & 24 & 7 \\ 8 & 14 & 0 & 9 & 24 & 37 & 11 \\ 3 & 4 & -4 & 2 & 7 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{112} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -800, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 10 & 3 \\ 3 & 4 & 0 & 4 & 5 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 3 & 5 & 0 & 3 & 4 & 18 & 4 \\ 10 & 18 & 0 & 12 & 18 & 58 & 14 \\ 3 & 4 & -4 & 2 & 4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{112} 3^1 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -848, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 16 & 3 \\ 3 & 4 & 0 & 4 & 5 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 3 & 5 & 0 & 3 & 4 & 30 & 4 \\ 16 & 28 & 0 & 18 & 30 & 154 & 22 \\ 3 & 4 & -4 & 2 & 4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 5 & 9 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & -4 & 2 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1111} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -920, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 9 & 2 \\ 8 & 14 & 0 & 9 & 37 & 39 & 11 \\ 8 & 14 & 0 & 9 & 39 & 37 & 11 \\ 3 & 4 & -4 & 2 & 11 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -929, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 9 \\ 7 & 13 & 0 & 9 & 9 & 28 & 10 \\ 3 & 4 & -4 & 2 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 12 \\ 10 & 18 & 0 & 12 & 12 & 58 & 14 \\ 3 & 4 & -4 & 2 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{112} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 13 & 3 \\ 3 & 4 & 0 & 4 & 5 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 3 & 5 & 0 & 3 & 4 & 24 & 4 \\ 13 & 23 & 0 & 15 & 24 & 100 & 18 \\ 3 & 4 & -4 & 2 & 4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{114} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.91 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 14 & 3 \\ 3 & 4 & 0 & 4 & 5 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 3 & 5 & 0 & 3 & 4 & 24 & 4 \\ 14 & 26 & 0 & 18 & 24 & 118 & 20 \\ 3 & 4 & -4 & 2 & 4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{113} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1001, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 12 & 3 \\ 3 & 4 & 0 & 4 & 5 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 3 & 5 & 0 & 3 & 4 & 21 & 4 \\ 12 & 22 & 0 & 15 & 21 & 85 & 17 \\ 3 & 4 & -4 & 2 & 4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1113} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 4 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 21 \\ 18 & 32 & 0 & 21 & 21 & 193 & 25 \\ 3 & 4 & -4 & 2 & 3 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1112} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1083, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 7 & 3 \\ 3 & 4 & 0 & 9 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 6 & 21 & 6 \\ 2 & 4 & 0 & 6 & 1 & 9 & 3 \\ 7 & 13 & 0 & 21 & 9 & 28 & 10 \\ 3 & 4 & -4 & 6 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1085, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 11 & 3 \\ 3 & 4 & 0 & 4 & 9 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 15 & 2 \\ 5 & 9 & 0 & 6 & 13 & 33 & 7 \\ 11 & 21 & 0 & 15 & 33 & 76 & 16 \\ 3 & 4 & -4 & 2 & 7 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1205, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 7 & 3 \\ 3 & 4 & 0 & 4 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 9 & 2 \\ 5 & 9 & 0 & 6 & 13 & 21 & 7 \\ 7 & 13 & 0 & 9 & 21 & 28 & 10 \\ 3 & 4 & -4 & 2 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1112} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 10 & 3 \\ 3 & 4 & 0 & 4 & 14 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 12 & 2 \\ 8 & 14 & 0 & 9 & 37 & 48 & 11 \\ 10 & 18 & 0 & 12 & 48 & 58 & 14 \\ 3 & 4 & -4 & 2 & 11 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 9 & 3 \\ 3 & 4 & 0 & 4 & 9 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 12 & 2 \\ 5 & 9 & 0 & 6 & 13 & 27 & 7 \\ 9 & 17 & 0 & 12 & 27 & 49 & 13 \\ 3 & 4 & -4 & 2 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1112} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 16 & 3 \\ 3 & 4 & 0 & 4 & 9 & 28 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 18 & 2 \\ 5 & 9 & 0 & 6 & 13 & 48 & 7 \\ 16 & 28 & 0 & 18 & 48 & 154 & 22 \\ 3 & 4 & -4 & 2 & 7 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1281, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 9 & 3 \\ 3 & 4 & 0 & 9 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 6 & 27 & 6 \\ 2 & 4 & 0 & 6 & 1 & 12 & 3 \\ 9 & 17 & 0 & 27 & 12 & 49 & 13 \\ 3 & 4 & -4 & 6 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1323, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 11 & 3 \\ 3 & 4 & 0 & 9 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 6 & 33 & 6 \\ 2 & 4 & 0 & 6 & 1 & 15 & 3 \\ 11 & 21 & 0 & 33 & 15 & 76 & 16 \\ 3 & 4 & -4 & 6 & 3 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1397, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 12 & 49 & 13 \\ 3 & 4 & -4 & 2 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{113}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -1565$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 15 & 3 \\ 3 & 4 & 0 & 4 & 5 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 3 & 5 & 0 & 3 & 4 & 27 & 4 \\ 15 & 27 & 0 & 18 & 27 & 133 & 21 \\ 3 & 4 & -4 & 2 & 4 & 21 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{1122}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -1664$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 10 & 3 \\ 3 & 4 & 0 & 4 & 13 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 12 & 2 \\ 7 & 13 & 0 & 9 & 28 & 42 & 10 \\ 10 & 18 & 0 & 12 & 42 & 58 & 14 \\ 3 & 4 & -4 & 2 & 10 & 14 & 0 \end{pmatrix}_{b_0}$$

[61] $1^{1113}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -1709$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 2 & 4 & 0 & 3 & 1 & 18 & 3 \\ 15 & 27 & 0 & 18 & 18 & 133 & 21 \\ 3 & 4 & -4 & 2 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

[62] $1^{1112}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -1760$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 10 & 3 \\ 3 & 4 & 0 & 4 & 9 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 12 & 2 \\ 5 & 9 & 0 & 6 & 13 & 30 & 7 \\ 10 & 18 & 0 & 12 & 30 & 58 & 14 \\ 3 & 4 & -4 & 2 & 7 & 14 & 0 \end{pmatrix}_{b_0}$$

[63] $1^{1112}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -1781$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 13 & 3 \\ 3 & 4 & 0 & 4 & 9 & 23 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 15 & 2 \\ 5 & 9 & 0 & 6 & 13 & 39 & 7 \\ 13 & 23 & 0 & 15 & 39 & 100 & 18 \\ 3 & 4 & -4 & 2 & 7 & 18 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{1122}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -1784$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 9 & 2 \\ 7 & 13 & 0 & 9 & 28 & 30 & 10 \\ 7 & 13 & 0 & 9 & 30 & 28 & 10 \\ 3 & 4 & -4 & 2 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{1113}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -1820$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 \\ 12 & 22 & 0 & 15 & 15 & 85 & 17 \\ 3 & 4 & -4 & 2 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

[66] $1^{1113}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -1865$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 18 & 3 \\ 3 & 4 & 0 & 4 & 5 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 3 & 5 & 0 & 3 & 4 & 33 & 4 \\ 18 & 32 & 0 & 21 & 33 & 193 & 25 \\ 3 & 4 & -4 & 2 & 4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{1114} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1889, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.86 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 15 \\ 11 & 21 & 0 & 15 & 15 & 76 & 16 \\ 3 & 4 & -4 & 2 & 3 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{114} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1928, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 17 & 3 \\ 3 & 4 & 0 & 4 & 5 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 3 & 5 & 0 & 3 & 4 & 30 & 4 \\ 17 & 31 & 0 & 21 & 30 & 172 & 24 \\ 3 & 4 & -4 & 2 & 4 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{1113} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2501, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 12 & 3 \\ 3 & 4 & 0 & 4 & 9 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 15 & 2 \\ 5 & 9 & 0 & 6 & 13 & 36 & 7 \\ 12 & 22 & 0 & 15 & 36 & 85 & 17 \\ 3 & 4 & -4 & 2 & 7 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{1122} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2624, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 12 & 12 & 2 \\ 10 & 18 & 0 & 12 & 58 & 60 & 14 \\ 10 & 18 & 0 & 12 & 60 & 58 & 14 \\ 3 & 4 & -4 & 2 & 14 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{114} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2636, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 20 & 3 \\ 3 & 4 & 0 & 4 & 5 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 24 & 2 \\ 3 & 5 & 0 & 3 & 4 & 36 & 4 \\ 20 & 36 & 0 & 24 & 36 & 238 & 28 \\ 3 & 4 & -4 & 2 & 4 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{1114} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2720, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 4 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 18 \\ 14 & 26 & 0 & 18 & 18 & 118 & 20 \\ 3 & 4 & -4 & 2 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{1114} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2780, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 24 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 24 \\ 20 & 36 & 0 & 24 & 24 & 238 & 28 \\ 3 & 4 & -4 & 2 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{1114} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2996, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 14 & 3 \\ 3 & 4 & 0 & 4 & 9 & 26 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 18 & 2 \\ 5 & 9 & 0 & 6 & 13 & 42 & 7 \\ 14 & 26 & 0 & 18 & 42 & 118 & 20 \\ 3 & 4 & -4 & 2 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{1114} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3017, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 4 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 21 \\ 17 & 31 & 0 & 21 & 21 & 172 & 24 \\ 3 & 4 & -4 & 2 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{1113} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 15 & 3 \\ 3 & 4 & 0 & 4 & 9 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 18 & 2 \\ 5 & 9 & 0 & 6 & 13 & 45 & 7 \\ 15 & 27 & 0 & 18 & 45 & 133 & 21 \\ 3 & 4 & -4 & 2 & 7 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{1113} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3365, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 18 & 3 \\ 3 & 4 & 0 & 4 & 9 & 32 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 21 & 2 \\ 5 & 9 & 0 & 6 & 13 & 54 & 7 \\ 18 & 32 & 0 & 21 & 54 & 193 & 25 \\ 3 & 4 & -4 & 2 & 7 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{1114} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -4373, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 17 & 3 \\ 3 & 4 & 0 & 4 & 9 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 21 & 2 \\ 5 & 9 & 0 & 6 & 13 & 51 & 7 \\ 17 & 31 & 0 & 21 & 51 & 172 & 24 \\ 3 & 4 & -4 & 2 & 7 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{1114} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -5216, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 20 & 3 \\ 3 & 4 & 0 & 4 & 9 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 24 & 2 \\ 5 & 9 & 0 & 6 & 13 & 60 & 7 \\ 20 & 36 & 0 & 24 & 60 & 238 & 28 \\ 3 & 4 & -4 & 2 & 7 & 28 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [80] \quad 1^{111} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 44, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{1111} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{1111} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 82, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.90 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 3 \\ 3 & 4 & 1 & 3 & 9 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 3 & 0 \\ 5 & 9 & 0 & 7 & 14 & 9 & 8 \\ 3 & 6 & 0 & 3 & 9 & 4 & 4 \\ 3 & 4 & -1 & 0 & 8 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{1111}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 111, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.62 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 3 & 3 & 0 & 3 & 1 & 6 & 3 \\ 3 & 6 & 9 & 0 & 6 & 6 & 13 & 7 \\ 2 & 3 & 3 & -4 & 2 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{11}31^{121}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 138, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.35 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 6 & 3 & 2 \\ 4 & 12 & 6 & 0 & 6 & 22 & 12 & 8 \\ 2 & 6 & 3 & 0 & 3 & 12 & 4 & 4 \\ 2 & 4 & 2 & -4 & 2 & 8 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{1111}3 \quad 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 224, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.89 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 7 & 0 \\ 4 & 6 & 0 & 3 & 4 & 9 & 4 \\ 6 & 9 & 0 & 7 & 9 & 14 & 8 \\ 3 & 4 & -1 & 0 & 4 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{111}3 \quad 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 239, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 6 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 7 & 0 & 14 & 8 \\ 3 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{1111}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 268, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.97 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{111}3 \quad 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 269, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.86 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{1112}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 273, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.38 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 9 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 9 & 2 \\ 1 & 3 & 3 & 0 & 3 & 1 & 9 & 3 \\ 4 & 9 & 12 & 0 & 9 & 9 & 28 & 10 \\ 2 & 3 & 3 & -4 & 2 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{112}3 \quad 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 296, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.99 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{11}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 300$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.16 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 4 \\ 2 & 6 & 3 & 0 & 3 & 6 & 4 \\ 2 & 4 & 2 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[92] $1^{111}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 320$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{11}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 363$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.22 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 6 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 9 & 4 \\ 3 & 9 & 6 & 0 & 6 & 9 & 13 & 7 \\ 2 & 4 & 2 & -4 & 2 & 4 & 7 & -1 \end{pmatrix}_{b_0}$$

[94] $1^{112}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 374$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 9 & 3 \\ 3 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 10 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 0 & 10 & 0 & 38 & 12 \\ 3 & 4 & -1 & 0 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{111}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 381$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[96] $1^{111}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 704$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{11}31^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 1065$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 15 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 9 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 15 & 4 \\ 5 & 15 & 9 & 0 & 9 & 15 & 37 & 11 \\ 2 & 4 & 2 & -4 & 2 & 4 & 11 & -1 \end{pmatrix}_{b_0}$$

[98] $1^{11}31^{13}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 1659$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 21 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 12 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 21 & 4 \\ 7 & 21 & 12 & 0 & 12 & 21 & 73 & 15 \\ 2 & 4 & 2 & -4 & 2 & 4 & 15 & -1 \end{pmatrix}_{b_0}$$

[99] $1^{111}31^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 2145$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 15 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 27 & 4 \\ 9 & 27 & 15 & 0 & 15 & 27 & 121 & 19 \\ 2 & 4 & 2 & -4 & 2 & 4 & 19 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ $[100] 1^{111}4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -82$, $\det = -24$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.83 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 5 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 4 \\ 5 & 7 & 0 & 5 & 6 & 8 & 6 \\ 3 & 4 & -3 & 2 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

[101] $1^{111}31^{11}4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -82$, $\det = -48$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.92 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 5 & 3 \\ 4 & 4 & 0 & 3 & 5 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 5 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 4 \\ 5 & 7 & 0 & 5 & 6 & 8 & 6 \\ 3 & 4 & -3 & 2 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

4.143 NHC(-2)(-3) + e₆ + su₃ + su₂ + su₂ + su₂

$T = 10$ $[0] 1^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 148$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 4 \\ 3 & 5 & 0 & 3 & 6 & 9 & 4 \\ 5 & 9 & 0 & 6 & 9 & 13 & 7 \\ 3 & 4 & -4 & 2 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{1111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 166$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 4 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 9 & 13 & 7 \\ 3 & 4 & -4 & 2 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{1111}31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 220$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.85 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 6 & 2 \\ 2 & 4 & 0 & 3 & 1 & 9 & 6 & 3 \\ 8 & 14 & 0 & 9 & 9 & 37 & 24 & 11 \\ 5 & 9 & 0 & 6 & 6 & 24 & 13 & 7 \\ 3 & 4 & -4 & 2 & 3 & 11 & 7 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{1112}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 274$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.5$

$T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 18 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 12 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 18 & 4 \\ 2 & 4 & 0 & 3 & 3 & 1 & 12 & 3 \\ 10 & 18 & 0 & 12 & 18 & 12 & 58 & 14 \\ 3 & 4 & -4 & 2 & 4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{1113} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 454$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.8$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 27 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 18 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 27 & 4 \\ 2 & 4 & 0 & 3 & 3 & 1 & 18 & 3 \\ 15 & 27 & 0 & 18 & 27 & 18 & 133 & 21 \\ 3 & 4 & -4 & 2 & 4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{11111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 6 & 6 & 6 & 6 \\ 2 & 4 & 0 & 6 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 6 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 6 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 6 & 3 & 3 & 3 & 0 \end{pmatrix} b_0$$

$$[6] \quad 1^{1113} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 676, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 22 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 15 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 21 & 4 \\ 2 & 4 & 0 & 3 & 3 & 1 & 15 & 3 \\ 12 & 22 & 0 & 15 & 21 & 15 & 85 & 17 \\ 3 & 4 & -4 & 2 & 4 & 3 & 17 & 0 \end{pmatrix} b_0$$

$$[7] \quad 1^{1114} 3^1 1^{14} + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{c}_6 + \textcolor{brown}{5}\textcolor{blue}{u}_3 + \textcolor{red}{5}\textcolor{blue}{u}_2 + \textcolor{blue}{5}\textcolor{blue}{u}_2 + \textcolor{violet}{5}\textcolor{blue}{u}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 706, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 36 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 0 & 3 & 3 & 24 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 36 & 4 \\ 2 & 4 & 0 & 3 & 3 & 1 & 24 & 3 \\ 20 & 36 & 0 & 24 & 36 & 24 & 238 & 28 \\ 3 & 4 & -4 & 2 & 4 & 3 & 28 & 0 \end{pmatrix} b_0$$

[8] $1^{11111}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 712, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 2 & 3 & 3 & 3 & 0 \end{pmatrix} b_0$$

[9] $1^{11111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 724, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 5 & 9 & 0 & 6 & 6 & 6 & 7 \\ 3 & 4 & -4 & 2 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{1112} 3^1 1^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 796, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 7 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 9 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 12 & 4 \\ 5 & 9 & 0 & 6 & 9 & 13 & 21 & 7 \\ 7 & 13 & 0 & 9 & 12 & 21 & 28 & 10 \\ 3 & 4 & -4 & 2 & 4 & 7 & 10 & 0 \end{pmatrix} b_0$$

$$[11] \quad 1^{\text{red}} 1^{\text{blue}} 1^{\text{green}} 3 1^{\text{red}} 1^{\text{blue}} 1^{\text{green}} + [\text{red} \mathfrak{su}_2 \cdot \text{blue} \mathfrak{g}_2] + \text{green} \mathfrak{e}_6 + \text{blue} \mathfrak{su}_3 + \text{red} \mathfrak{su}_2 + \text{blue} \mathfrak{su}_2 + \text{red} \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 6 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 9 & 4 \\ 5 & 9 & 0 & 6 & 9 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 9 & 15 & 13 & 7 \\ 3 & 4 & -4 & 2 & 4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1032, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 12 & 6 & 6 & 15 & 6 \\ 2 & 4 & 0 & 6 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 6 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 15 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 6 & 3 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1122} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 7 & 7 & 3 \\ 3 & 4 & 0 & 4 & 5 & 13 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 9 & 2 \\ 3 & 5 & 0 & 3 & 4 & 12 & 12 & 4 \\ 7 & 13 & 0 & 9 & 12 & 28 & 30 & 10 \\ 7 & 13 & 0 & 9 & 12 & 30 & 28 & 10 \\ 3 & 4 & -4 & 2 & 4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1114} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1180, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 31 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 21 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 30 & 4 \\ 2 & 4 & 0 & 3 & 3 & 1 & 21 & 3 \\ 17 & 31 & 0 & 21 & 30 & 21 & 172 & 24 \\ 3 & 4 & -4 & 2 & 4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 6 & 2 \\ 2 & 4 & 0 & 3 & 1 & 6 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 6 & 15 & 13 & 7 \\ 3 & 4 & -4 & 2 & 3 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 2368, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 6 & 2 \\ 5 & 9 & 0 & 6 & 13 & 15 & 15 & 7 \\ 5 & 9 & 0 & 6 & 15 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 15 & 15 & 13 & 7 \\ 3 & 4 & -4 & 2 & 7 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

4.144 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{11111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 4 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 0 & 5 \\ 2 & 4 & -3 & 0 & 3 & 3 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 6 & 0 & 4 \\ 4 & 8 & 3 & 0 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{1111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 231$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.13 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 6 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 9 & 6 & 12 & 6 \\ 3 & 4 & -4 & 2 & 4 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{1111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 507$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.92 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 3 & 4 \\ 5 & 9 & 0 & 6 & 9 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & -4 & 2 & 4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{1111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 651$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.17 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 9 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 9 & 6 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{1141}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 813$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.27 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 21 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 18 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 & 3 \\ 11 & 21 & 0 & 18 & 15 & 76 & 15 & 16 \\ 2 & 4 & 0 & 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 16 & 2 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{1121}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 1173$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.5$

$T_{\min} = 13.15 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 12 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 & 3 \\ 7 & 13 & 0 & 12 & 9 & 28 & 9 & 10 \\ 2 & 4 & 0 & 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{1131}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 1227$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.8$

$T_{\min} = 13.17 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 17 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 15 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 & 3 \\ 9 & 17 & 0 & 15 & 12 & 49 & 12 & 13 \\ 2 & 4 & 0 & 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

4.145 NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 10$ [0] $1^{111}3^11^131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 15$, $\Delta = -93$, $\det = 86$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.56 \rightarrow 14$, $T_{\max} = 15$, $T \in [14, 15]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 3 & 3 & 0 & 3 & 1 & 2 & 3 \\ 1 & 2 & 3 & 0 & 2 & 2 & 0 & 2 \\ 2 & 3 & 3 & -4 & 2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 15, \ \Delta = -93, \ \det = 126, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.43 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 & 3 \\ 1 & 3 & 3 & 0 & 3 & 1 & 2 & 3 \\ 1 & 3 & 2 & 0 & 3 & 2 & 0 & 2 \\ 2 & 4 & 2 & -4 & 3 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{11}2^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 130, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 1 & 3 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 2 & 0 \\ 5 & 9 & 0 & 7 & 14 & 5 & 8 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}2^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 146, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 4 & -1 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{11}2^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 226, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 2 & 0 \\ 3 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 15, \ \Delta = -93, \ \det = 296, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 15, \ T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & -4 & 2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112}3^{11}2^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 328, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 2 & 0 \\ 8 & 15 & 0 & 10 & 38 & 8 & 12 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112}3^{11}2^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 386, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 2 & 0 \\ 5 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111}3^{11}2^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 416, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 2 & 3 \\ 3 & 4 & 1 & 3 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 2 & 0 \\ 6 & 9 & 0 & 7 & 14 & 5 & 8 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 416, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 2 & 0 \\ 4 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 482, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 1 & 3 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 2 & 0 \\ 7 & 9 & 0 & 7 & 14 & 5 & 8 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{112}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 626, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 2 & 0 \\ 7 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{112}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 896, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 2 & 0 \\ 8 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{112}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 914, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 2 & 0 \\ 9 & 15 & 0 & 10 & 38 & 8 & 12 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{112}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -87, \ \det = 1280, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 2 & 0 \\ 10 & 15 & 0 & 10 & 38 & 8 & 12 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [15] \ 1^{111}3^{11}15^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 13, \ T_{\max} = 15, \ \Delta = -58, \ \det = -69, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.97 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 & 2 & 0 & 2 \\ 2 & 3 & -1 & -1 & 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

4.146 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \text{n3mix}$

$$T = 10 \quad [0] \ 1^{111}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 15, \ \Delta = -93, \ \det = 494, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, \ T_{\max} = 15, \ T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 5 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 5 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 9 & 4 \\ 3 & 9 & 5 & 0 & 5 & 9 & 12 & 6 \\ 2 & 4 & 2 & -4 & 2 & 4 & 6 & -1 \end{pmatrix}_{b_0}$$

4.147 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{121} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 & 3 \\ 3 & 4 & 0 & 8 & 9 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 4 & 8 & 0 & 9 & 12 & 6 & 5 \\ 5 & 9 & 0 & 12 & 13 & 9 & 7 \\ 3 & 5 & 0 & 6 & 9 & 3 & 3 \\ 3 & 4 & -4 & 5 & 7 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{114} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 1 & 3 \\ 4 & 9 & 4 & 0 & 32 & 0 & 5 \\ 2 & 4 & -3 & 0 & 12 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 13 & 32 & 12 & 0 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -3 & 0 & 3 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 4 \\ 3 & 5 & 0 & 3 & 6 & 3 & 3 \\ 3 & 4 & -4 & 2 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{113} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 1 & 3 \\ 4 & 9 & 4 & 0 & 24 & 0 & 5 \\ 2 & 4 & -3 & 0 & 9 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 10 & 24 & 9 & 0 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 12 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{112} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

14.5

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 1 & 3 \\ 4 & 9 & 4 & 0 & 16 & 0 & 5 \\ 2 & 4 & -3 & 0 & 6 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 7 & 16 & 6 & 0 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{1111}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 9 \\ 8 & 14 & 0 & 9 & 9 & 36 & 10 \\ 3 & 4 & -4 & 2 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{122}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -267, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.7$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 0 & 5 & 13 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 12 & 6 & 3 \\ 7 & 13 & 0 & 12 & 28 & 18 & 10 \\ 4 & 8 & 0 & 6 & 18 & 9 & 5 \\ 3 & 4 & -4 & 3 & 10 & 5 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{132}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 0 & 5 & 17 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 15 & 6 & 3 \\ 9 & 17 & 0 & 15 & 49 & 24 & 13 \\ 4 & 8 & 0 & 6 & 24 & 9 & 5 \\ 3 & 4 & -4 & 3 & 13 & 5 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{1111}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -402, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 3 & 2 \\ 8 & 14 & 0 & 9 & 37 & 9 & 11 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -4 & 2 & 11 & 2 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{113}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -420, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 1 & 3 \\ 4 & 9 & 4 & 0 & 24 & 0 & 5 \\ 2 & 4 & -3 & 0 & 9 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 11 & 24 & 9 & 0 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 12 & -1 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{111}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 4 \\ 5 & 9 & 0 & 6 & 9 & 12 & 6 \\ 3 & 4 & -4 & 2 & 4 & 6 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{1111}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[15] $1^{114}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -555, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 1 & 3 \\ 4 & 9 & 4 & 0 & 32 & 0 & 5 \\ 2 & 4 & -3 & 0 & 12 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 15 & 32 & 12 & 0 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -645, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 9 & 3 & 3 \\ 5 & 9 & 0 & 9 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & -4 & 3 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -651, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & -4 & 2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{114} 3^1 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -690, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 1 & 3 \\ 4 & 9 & 4 & 0 & 32 & 0 & 5 \\ 2 & 4 & -3 & 0 & 12 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 14 & 32 & 12 & 0 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{141} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -726, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 18 & 3 & 3 \\ 11 & 21 & 0 & 18 & 76 & 15 & 16 \\ 2 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & -4 & 3 & 16 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -789, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 5 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & -4 & 2 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{121} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -906, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 12 & 3 & 3 \\ 7 & 13 & 0 & 12 & 28 & 9 & 10 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -4 & 3 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{131} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -933, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 15 & 3 & 3 \\ 9 & 17 & 0 & 15 & 49 & 12 & 13 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -4 & 3 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -1200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 5 & 9 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 15 & 12 & 6 \\ 3 & 4 & -4 & 2 & 7 & 6 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \quad 1^{111} 3 \quad 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 1 & 3 \\ 4 & 9 & 4 & 0 & 12 & 0 & 5 \\ 2 & 4 & -2 & 0 & 7 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 5 & 12 & 7 & 0 & 14 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{111} 3 \quad 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{111} 3 \quad 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & 0 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{111} 3 \quad 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 182, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.26 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 477, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 6 & 4 \\ 2 & 6 & 3 & 0 & 3 & 6 & 3 & 3 \\ 2 & 4 & 2 & -4 & 2 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{111} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 486, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 6 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 6 & 3 & 2 \\ 3 & 6 & 9 & 0 & 6 & 13 & 6 & 7 \\ 1 & 3 & 3 & 0 & 3 & 6 & 0 & 2 \\ 2 & 3 & 3 & -4 & 2 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{111} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 774, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 6 & 3 & 2 \\ 3 & 9 & 6 & 0 & 6 & 13 & 9 & 7 \\ 2 & 6 & 3 & 0 & 3 & 9 & 3 & 3 \\ 2 & 4 & 2 & -4 & 2 & 7 & 3 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [31] \quad 1^{11} 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 4 \\ 2 & 4 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{11} 1^{13} 1^{11} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -108, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 2 & 3 \\ 4 & 4 & 0 & 3 & 5 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{11} 1^{13} 1^{11} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -210, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 2 & 3 \\ 4 & 4 & 0 & 3 & 5 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 3 & 2 \\ 3 & 4 & 0 & 2 & 1 & 2 & 3 \\ 2 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 2 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{11} 1^{13} 1^{11} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -378, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 2 & 3 \\ 4 & 4 & 0 & 3 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 3 & 2 \\ 6 & 9 & 0 & 5 & 13 & 5 & 7 \\ 2 & 3 & 0 & 3 & 5 & 0 & 2 \\ 3 & 4 & -3 & 2 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

4.148 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{1111} 1^{1111} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 6 & 0 & 4 & 4 \\ 3 & 5 & -1 & -1 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111} 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1111} 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 231, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 3 & 4 \\ 5 & 9 & 0 & 6 & 9 & 12 & 6 & 6 \\ 2 & 4 & 0 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & -4 & 2 & 4 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1111} 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 507, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 6 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 9 & 6 & 13 & 7 \\ 3 & 4 & -4 & 2 & 4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 651, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 3 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1114}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 813, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 15 & 2 \\ 11 & 21 & 0 & 18 & 15 & 15 & 76 & 16 \\ 3 & 4 & -4 & 3 & 3 & 2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1112}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1173, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 9 & 2 \\ 7 & 13 & 0 & 12 & 9 & 9 & 28 & 10 \\ 3 & 4 & -4 & 3 & 3 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1113}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1227, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 12 & 2 \\ 9 & 17 & 0 & 15 & 12 & 12 & 49 & 13 \\ 3 & 4 & -4 & 3 & 3 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

4.149 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 459, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 & 2 \\ 3 & 4 & -4 & 3 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 1296, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 9 & 3 & 3 & 3 \\ 5 & 9 & 0 & 9 & 13 & 6 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 6 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 7 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.150 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{\overline{131}} 2 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -44, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.36 \rightarrow 14, \ T_{\max} = 32, \ T \in [14, 32]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 3 & 5 \\ 2 & 5 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 5 & 0 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{12}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -72, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.11 \rightarrow 14, \ T_{\max} = 32, \ T \in [14, 32]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & 0 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -98, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.22 \rightarrow 14, \ T_{\max} = 32, \ T \in [14, 32]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -2 & 4 & 0 \\ 4 & 6 & 0 & 4 & 9 & 5 \\ 3 & 3 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{12}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 32, \ T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 79, \ \Delta = -123, \ \det = -163, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.21 \rightarrow 14, \ T_{\max} = 79, \ T \in [14, 79]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 2 & 3 & 0 & 2 & 0 \\ 2 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}1^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 79, \ \Delta = -123, \ \det = -193, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.21 \rightarrow 14, \ T_{\max} = 79, \ T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 \\ 2 & 4 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{13}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 79, \ \Delta = -123, \ \det = -363, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.29 \rightarrow 14, \ T_{\max} = 79, \ T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 \\ 2 & 4 & 4 & 0 & 6 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 \\ 2 & 6 & 3 & 0 & 3 & 3 \\ 1 & 3 & 2 & 0 & 3 & 0 \\ 2 & 4 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 79, \ \Delta = -123, \ \det = -508, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.08 \rightarrow 14, \ T_{\max} = 79, \ T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 \\ 2 & 4 & 4 & 0 & 9 & 3 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 \\ 3 & 9 & 6 & 0 & 12 & 5 \\ 1 & 3 & 2 & 0 & 5 & 0 \\ 2 & 4 & 2 & -4 & 6 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [8] \ 1^{123}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 16, \ \Delta = -96, \ \det = 106, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.30 \rightarrow 14, \ T_{\max} = 16, \ T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 \\ 1 & 3 & 0 & -2 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 \\ 1 & 3 & 0 & 2 & 0 & 0 \\ 2 & 4 & -1 & 0 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{123}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 15, \ \Delta = -95, \ \det = 149, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.38 \rightarrow 14, \ T_{\max} = 15, \ T \in [14, 15]$$

$$\begin{pmatrix} 5 & 14 & 1 & 1 & 5 \\ 14 & 33 & 1 & 0 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 11 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [10] \quad 1^{11}3^{11}1^141^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.24 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 4 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11}3^{11}1^14131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -149, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [12] \quad 1^{11}3^{11}1^15131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.45 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 1 & 0 & 3 & 1 & 0 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11}3^{11}1^15131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 66, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.94 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 3 & -1 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.151 NHC(-2)(-3) + $\textcolor{violet}{e}_6$ + $\textcolor{brown}{su}_3$ + $\textcolor{blue}{su}_2 \rightarrow (-3)$ + $\textcolor{blue}{su}_2$

$$T = 10 \quad [0] \quad 1^{11}3^{11}1^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 2 & 3 \\ 1 & 2 & 3 & 0 & 2 & 0 & 2 \\ 1 & 3 & 3 & 0 & 3 & 2 & 1 \\ 2 & 3 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 \\ 1 & 3 & 2 & 0 & 3 & 0 & 2 \\ 1 & 3 & 3 & 0 & 3 & 2 & 1 \\ 2 & 4 & 2 & -4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 130, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 7 & 0 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 5 & 9 & 0 & 7 & 5 & 14 & 8 \\ 3 & 4 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 146, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 6 & 0 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 7 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{112}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 896, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 6 & 0 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 8 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 914, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 10 & 0 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 9 & 15 & 0 & 10 & 8 & 38 & 12 \\ 3 & 4 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{112}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 1280, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 10 & 0 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 10 & 15 & 0 & 10 & 8 & 38 & 12 \\ 3 & 4 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [15] \quad 1^{111}3^{11}15^{11}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -69, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.97 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 0 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 1 & 3 & 0 & 1 & 3 & 2 & 1 \\ 2 & 3 & -1 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.152 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{111}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 4 & 0 & 5 \\ 2 & 4 & -2 & 0 & 2 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 2 & 3 \\ 1 & 2 & 3 & 0 & 2 & 0 & 2 \\ 1 & 3 & 3 & 0 & 3 & 2 & 0 \\ 2 & 3 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^{11}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 489, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 2 & 3 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 2 & -4 & 2 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

4.153 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + su2n3mix

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 32, \quad \Delta = -117, \quad \det = -107, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 32, \quad T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 0 \\ 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 79, \quad \Delta = -123, \quad \det = -112, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 79, \quad T \in [14, 79]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 5 & 3 \\ 2 & 4 & 3 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 5 & 2 \\ 2 & 5 & 6 & 0 & 5 & 7 & 5 \\ 2 & 3 & 3 & -4 & 2 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 32, \quad \Delta = -117, \quad \det = -179, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.97 \rightarrow 13, \quad T_{\max} = 32, \quad T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 7 & 12 & 0 & 9 & 26 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 32, \quad \Delta = -117, \quad \det = -227, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.04 \rightarrow 14, \quad T_{\max} = 32, \quad T \in [14, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 5 & 0 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 1^1 4^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 79, \quad \Delta = -123, \quad \det = -265, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.15 \rightarrow 14, \quad T_{\max} = 79, \quad T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 14 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 14 & 2 \\ 9 & 27 & 14 & 0 & 14 & 120 & 18 \\ 2 & 4 & 2 & -4 & 2 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 79, \quad \Delta = -123, \quad \det = -292, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 79, \quad T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 5 & 2 \\ 3 & 9 & 5 & 0 & 5 & 12 & 6 \\ 2 & 4 & 2 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 3^{11} 1^1 3^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 79, \quad \Delta = -123, \quad \det = -328, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 79, \quad T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 3 & 11 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 11 & 2 \\ 7 & 21 & 11 & 0 & 11 & 72 & 14 \\ 2 & 4 & 2 & -4 & 2 & 14 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 32, \quad \Delta = -117, \quad \det = -332, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 32, \quad T \in [14, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 8 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & 0 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 3^{11} 1^1 2^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 79, \quad \Delta = -123, \quad \det = -337, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} =$$

12.0

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 8 & 2 \\ 5 & 15 & 8 & 0 & 8 & 36 & 10 \\ 2 & 4 & 2 & -4 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[9] 1^{111} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 8 & 12 & 0 & 9 & 26 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[10] 1^{111} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 5 & 0 \\ 6 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[11] 1^{112} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -403, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 5 & 0 \\ 7 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[12] 1^{112} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -428, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 10 & 18 & 0 & 12 & 56 & 14 \\ 3 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[13] 1^{112} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -451, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 9 & 12 & 0 & 9 & 26 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[14] 1^{112} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -635, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 9 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[15] 1^{111} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -779, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 11 & 18 & 0 & 12 & 56 & 14 \\ 3 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[16] 1^{112} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -832, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 10 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[17] 1^{112} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -1024, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 12 & 18 & 0 & 12 & 56 & 14 \\ 3 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [18] \quad 1^{123}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 9 & 0 & 5 & 0 & 12 & 6 \\ 2 & 4 & -1 & 0 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{111}23^{11}131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 5 & 0 & 4 & 6 & 4 \\ 1 & 5 & 1 & 0 & 4 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 4 & 4 & 0 & 1 & 6 & 3 \\ 2 & 6 & 6 & 0 & 6 & 8 & 6 \\ 2 & 4 & 3 & -4 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{123}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 2 & 4 & -1 & 0 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{112}23^{11}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.30 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 0 & 9 & 0 & 26 & 10 \\ 2 & 4 & -1 & 0 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{123}1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 226, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.30 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 4 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 0 & 4 & 0 & 9 & 5 \\ 2 & 4 & -1 & 0 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{112}23^{11}1^{23}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 385, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.31 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 18 & 0 & 12 & 0 & 56 & 14 \\ 2 & 4 & -1 & 0 & -1 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{123}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 430, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 9 & 0 & 5 & 0 & 12 & 6 \\ 2 & 4 & -1 & 0 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{123}1^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 679, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 8 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 15 & 0 & 8 & 0 & 36 & 10 \\ 2 & 4 & -1 & 0 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [26] \quad 1^{11} 3^{11} 1^5 131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 56, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.14 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 1 & 0 & 3 & 1 & 0 & 2 & 2 \\ 2 & 0 & 2 & 1 & 2 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{11} 3^{11} 1^5 131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 66, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.12 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 3 & 0 & 1 & 0 & 5 & 2 \\ 2 & 5 & 0 & 2 & 5 & 7 & 5 \\ 2 & 3 & -1 & -1 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

4.154 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + su2n3mix + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 1^{11} 31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 494, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 5 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 5 & 3 & 2 \\ 3 & 9 & 5 & 0 & 5 & 12 & 9 & 6 \\ 2 & 6 & 3 & 0 & 3 & 9 & 4 & 4 \\ 2 & 4 & 2 & -4 & 2 & 6 & 4 & -1 \end{pmatrix}_{b_0}$$

4.155 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 5 & 8 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 6 & 3 \\ 4 & 8 & 0 & 6 & 9 & 5 \\ 3 & 4 & -4 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.11 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 246, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 3 & 4 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.02 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -4 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \ 1^{11} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.20 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 2 & 5 & -2 & 3 & 0 & 0 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{12} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 5 & 3 \\ 2 & 2 & -1 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^1 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 0 & 3 \\ 1 & 3 & 3 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111} 31^1 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.24 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{11}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -279, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 2 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{111}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.21 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [17] $1^{11}2321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.26 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 0 & 1 & -3 & 1 & 0 & -1 \\ 1 & 0 & 3 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -1 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{12}3221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 13.59 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 5 & 3 \\ 2 & 2 & -1 & 0 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{11}4131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 9 & 0 & 4 & 4 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 5 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{11}41^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{111}2321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[22] $1^1231^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 3 & 0 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{11}3221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 5 \\ 9 & 13 & 1 & 0 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{111} 2^1 3^{12} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.26 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 3 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{111} 3^{11} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 5 & 9 & 1 & 22 & 5 \\ 9 & 13 & 1 & 33 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 33 & 0 & 78 & 16 \\ 5 & 7 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{11} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 132, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.18 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{111} 2^1 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -3 & 1 & 1 & -1 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{111} 3^{11} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 5 & 9 & 1 & 9 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 12 & 6 \\ 5 & 7 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{11} 231^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 216, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 3 & 3 \\ 2 & 4 & -1 & 0 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{111} 3^{11} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 5 & 9 & 1 & 10 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 12 & 6 \\ 5 & 7 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [31] \quad 1^{111} 31^1 5^{11} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 12, T_{\max} = 17, \Delta = -95, \det = 75, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.60 \rightarrow 16, T_{\max} = 17, T \in [16, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [32] \quad 1^1 \overset{1}{6} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 \quad T = 13, T_{\max} = 17, \Delta = -92, \det = -40, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.20 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 & 3 & 0 \\ 1 & 1 & 1 & -3 & 1 & 0 \\ 1 & 1 & 3 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 \\ 2 & 2 & 2 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

4.156 $\text{NHC}(-2)(-3) + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \quad 1^{111} \overset{1}{3} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + [\text{su}_2 \cdot \text{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 6 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 3 & 2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} \overset{1}{3} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + [\text{su}_2 \cdot \text{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 825, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 4 & 3 \\ 3 & 5 & 0 & 3 & 3 & 4 & 3 & 3 \\ 3 & 4 & -4 & 2 & 2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

4.157 $\text{NHC}(-2)(-3) + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2$

$$T = 9 \quad [0] \quad 1^{121} \overset{1}{3} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 0 & 5 & 8 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 6 & 9 & 3 \\ 4 & 8 & 0 & 6 & 9 & 12 & 5 \\ 5 & 9 & 0 & 9 & 12 & 13 & 7 \\ 3 & 4 & -4 & 3 & 5 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{114} \overset{1}{3} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 0 & 12 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 13 & 32 & 12 & 0 & 0 & 94 & 16 \\ 3 & 5 & -1 & -1 & -1 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} \overset{1}{3} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 0 & 4 & 4 \\ 3 & 5 & -1 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} \overset{1}{3} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & -4 & 3 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} \overset{1}{3} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

14.0

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 0 & 3 & 0 & 2 \\ 3 & 5 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -4 & 2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{113} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 0 & 9 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 10 & 24 & 9 & 0 & 0 & 52 & 12 \\ 3 & 5 & -1 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{112} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 0 & 16 & 5 \\ 2 & 4 & -3 & 0 & 0 & 6 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 16 & 6 & 0 & 0 & 22 & 8 \\ 3 & 5 & -1 & -1 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{111} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{111} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 3 & 2 \\ 8 & 14 & 0 & 9 & 36 & 9 & 10 \\ 2 & 4 & 0 & 3 & 9 & 1 & 3 \\ 3 & 4 & -4 & 2 & 10 & 3 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{122} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -267, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.7$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 0 & 5 & 8 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 6 & 12 & 3 \\ 4 & 8 & 0 & 6 & 9 & 18 & 5 \\ 7 & 13 & 0 & 12 & 18 & 28 & 10 \\ 3 & 4 & -4 & 3 & 5 & 10 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{123} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 0 & 5 & 8 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 6 & 15 & 3 \\ 4 & 8 & 0 & 6 & 9 & 24 & 5 \\ 9 & 17 & 0 & 15 & 24 & 49 & 13 \\ 3 & 4 & -4 & 3 & 5 & 13 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{111} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -402, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 9 & 3 & 2 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 9 & 37 & 11 \\ 3 & 4 & -4 & 2 & 2 & 11 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{113} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -420, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 0 & 24 & 5 \\ 2 & 4 & -3 & 0 & 0 & 9 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 11 & 24 & 9 & 0 & 0 & 52 & 12 \\ 3 & 5 & -1 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 3 \\ 3 & 4 & 0 & 4 & 9 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 5 & 9 & 0 & 6 & 12 & 9 & 6 \\ 3 & 5 & 0 & 3 & 9 & 4 & 4 \\ 3 & 4 & -4 & 2 & 6 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 2 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{114} 3^1 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -555, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 0 & 12 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 15 & 32 & 12 & 0 & 0 & 94 & 16 \\ 3 & 5 & -1 & -1 & -1 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -645, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -651, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 5 & 9 & 0 & 6 & 12 & 6 & 6 \\ 2 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & -4 & 2 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{114} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -690, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 0 & 32 & 5 \\ 2 & 4 & -3 & 0 & 0 & 12 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 14 & 32 & 12 & 0 & 0 & 94 & 16 \\ 3 & 5 & -1 & -1 & -1 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{114} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -726, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 0 & 5 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 11 & 21 & 0 & 18 & 15 & 76 & 16 \\ 3 & 4 & -4 & 3 & 2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -789, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 2 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{112} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -906, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 7 & 13 & 0 & 12 & 9 & 28 & 10 \\ 3 & 4 & -4 & 3 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{113} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -933, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 5 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 9 & 17 & 0 & 15 & 12 & 49 & 13 \\ 3 & 4 & -4 & 3 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -1200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 5 & 9 & 0 & 6 & 12 & 15 & 6 \\ 5 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & -4 & 2 & 6 & 7 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \quad 1^{111} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 21, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.43 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 0 & 7 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 7 & 0 & 0 & 14 & 8 \\ 3 & 5 & 0 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{111} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 39, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 0 & 4 & 4 \\ 3 & 5 & 0 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{111} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 126, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.05 \rightarrow 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{111} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 182, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.26 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11} 3^1 1^{11} 3^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -100, \quad \det = 477, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.37 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 3 & 6 & 3 \\ 2 & 6 & 3 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & 2 & -4 & 2 & 3 & 4 & -1 \end{pmatrix}_{b_0}$$

[29] $1^{1111}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -100$, $\det = 486$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 3 & 3 & 0 & 3 & 0 & 6 & 2 \\ 3 & 6 & 9 & 0 & 6 & 6 & 13 & 7 \\ 2 & 3 & 3 & -4 & 2 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{1111}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -100$, $\det = 774$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 6 & 2 \\ 2 & 6 & 3 & 0 & 3 & 3 & 9 & 3 \\ 3 & 9 & 6 & 0 & 6 & 9 & 13 & 7 \\ 2 & 4 & 2 & -4 & 2 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ [31] $1^{11}4^{11}31^{11}4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -36$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 3 & 0 & 0 & 2 & 3 & 2 \\ 2 & 4 & 0 & 2 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -3 & 2 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{11}4^{11}31^{11}4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -108$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.22 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 & 3 \\ 4 & 4 & 0 & 3 & 3 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 3 & 2 \\ 2 & 3 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -3 & 2 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{11}4^{11}31^{11}4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -210$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.23 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 & 3 \\ 4 & 4 & 0 & 3 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 2 & 3 & 0 & 3 & 0 & 2 & 3 \\ 3 & 4 & 0 & 2 & 2 & 1 & 3 \\ 3 & 4 & -3 & 2 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{11}4^{11}31^{11}4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -378$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.25 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 6 & 3 \\ 4 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 5 & 2 \\ 2 & 3 & 0 & 3 & 0 & 5 & 2 \\ 6 & 9 & 0 & 5 & 5 & 13 & 7 \\ 3 & 4 & -3 & 2 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

4.158 NHC(-2)(-3) + $\textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$

$T = 10$ [0] $1^{1111}3^{1111}3^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 48$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 4 & 4 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 8 & 5 \\ 2 & 4 & -3 & 0 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 8 & 3 & 0 & 0 & 4 & 6 & 4 \\ 4 & 8 & 3 & 0 & 0 & 6 & 4 & 4 \\ 3 & 5 & -1 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 228, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & -4 & 3 & 2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 231, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 6 & 3 & 3 & 2 \\ 5 & 9 & 0 & 6 & 12 & 9 & 6 & 6 \\ 3 & 5 & 0 & 3 & 9 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 6 & 3 & 1 & 3 \\ 3 & 4 & -4 & 2 & 6 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 507, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 3 & 5 & 0 & 3 & 3 & 4 & 9 & 4 \\ 5 & 9 & 0 & 6 & 6 & 9 & 13 & 7 \\ 3 & 4 & -4 & 2 & 2 & 4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 651, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 9 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 2 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1114}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 813, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 21 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 15 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 15 & 3 \\ 11 & 21 & 0 & 18 & 15 & 15 & 76 & 16 \\ 3 & 4 & -4 & 3 & 2 & 3 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1112}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1173, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 13 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 9 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 9 & 3 \\ 7 & 13 & 0 & 12 & 9 & 9 & 28 & 10 \\ 3 & 4 & -4 & 3 & 2 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1113}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1227, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 17 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 12 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 12 & 3 \\ 9 & 17 & 0 & 15 & 12 & 12 & 49 & 13 \\ 3 & 4 & -4 & 3 & 2 & 3 & 13 & -1 \end{pmatrix}_{b_0}$$

4.159 NHC(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 459, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 1296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 9 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 6 & 3 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & -4 & 3 & 2 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

4.160 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 0 & 4 & 5 \\ 2 & 4 & -2 & 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 3 & 5 & 0 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 3 & 0 & 2 \\ 1 & 2 & 3 & 0 & 2 & 2 & 0 \\ 2 & 3 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 489, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 & 3 \\ 1 & 3 & 2 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & -4 & 2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.161 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.11 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 3 & 0 & 0 & 3 & 3 \\ 3 & 5 & -1 & -1 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 0 & 3 & 3 \\ 3 & 5 & -1 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -405, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [3] \quad 1^{111} 3^{11} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 321, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & 0 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

4.162 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\textcolor{brown}{su}_3$ + $\textcolor{red}{su}_3$ + $\textcolor{blue}{su}_3$ + $\textcolor{blue}{su}_2$

$$T = 10 \quad [0] \quad 1^{1111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 459, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -4 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 1296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 2 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

4.163 NHC(-2)(-3) + $\mathfrak{e}_6$ + (-3)→(-2)

$$T = 7 \quad [0] \quad 1^1 3^1 1^{11} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 100, \Delta = -173, \det = -350, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [1] \quad 1^{11} 2^{11} 3^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 4 \\ 5 & 5 & 5 & 0 & 5 \\ 3 & 5 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 2^{11} 3^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 275, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \quad 2^1 1^1 4^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_6 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 1 & 6 & 0 & 3 \\ 5 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 3 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \quad 1^{1^2 1^1} 5^{131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 10, T_{\max} = 100, \Delta = -138, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.61 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [5] \quad 1^{1^{11} 2^{31}} 5^{12^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 21, T_{\max} = 23, T \in [21, 23]$$

$$\begin{pmatrix} 5 & 15 & 1 & 5 \\ 15 & 42 & 0 & 12 \\ 1 & 0 & -3 & -1 \\ 5 & 12 & -1 & -3 \end{pmatrix}_{b_0}$$

4.164 $\text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{e}}_6 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2$

$$T = 8 \quad [0] \quad 1^{131} 1^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 925, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 6 & 4 \\ 8 & 9 & 4 & 0 & 8 & 5 \\ 3 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 8 & 3 & 0 & 4 & 4 \\ 4 & 5 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

4.165 $\text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{e}}_6 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{1^2 2^{31}} 1^{12^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_6 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{\mathfrak{su}}_2 \rightarrow \textcolor{red}{(-3)} \quad T = 9, T_{\max} = 36, \Delta = -118, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, T_{\max} = 36, T \in [13, 36]$$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 3 & 4 \\ 5 & 5 & 5 & 0 & 3 & 5 \\ 3 & 5 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 3 & 2 & 0 & 0 & 2 \\ 4 & 5 & 0 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

4.166 $\text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{e}}_7$

$$T = 7 \quad [0] \quad 2^{1^2} 3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 \quad T = 7, T_{\max} = 102, \Delta = -220, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 4 & 5 \\ 3 & 4 & -4 & -2 \\ 3 & 5 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1^2} 3^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 \quad T = 7, T_{\max} = 102, \Delta = -220, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.03 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -5 & -3 \\ 2 & 2 & -1 & -3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1^2} 3^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 \quad T = 7, T_{\max} = 102, \Delta = -220, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.80 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & -4 & -2 \\ 3 & 5 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 21^{1^2} 3^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 \quad T = 7, T_{\max} = 102, \Delta = -220, \det = -159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 4 & 8 & 3 & 4 \\ 8 & 9 & 4 & 5 \\ 3 & 4 & -4 & -2 \\ 4 & 5 & -2 & -1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [4] \quad 1^3 2^{21} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 \quad T = 8, T_{\max} = 102, \Delta = -191, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 6 & 6 & 6 \\ 2 & 6 & -3 & -1 \\ 3 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^1 2^1 3^1 2^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 \quad T = 8, \ T_{\max} = 102, \ \Delta = -193, \ \det = 23, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.35 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -5 & -3 \\ 2 & 3 & -1 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^1 2^1 3^1 2^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 \quad T = 8, \ T_{\max} = 102, \ \Delta = -193, \ \det = 24, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.17 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 1 & -5 & -3 \\ 2 & 0 & 2 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^1 3^1 2^1 2^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 \quad T = 8, \ T_{\max} = 102, \ \Delta = -191, \ \det = 26, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.08 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -4 & -2 \\ 2 & 2 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^1 2^1 3^1 1^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 \quad T = 8, \ T_{\max} = 102, \ \Delta = -199, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.91 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 0 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^1 2^1 3^1 1^1 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 \quad T = 8, \ T_{\max} = 102, \ \Delta = -199, \ \det = 50, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 3 & 3 \\ 1 & 0 & 3 & -4 & -2 \\ 2 & 0 & 3 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^1 2^1 3^1 1^1 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 \quad T = 8, \ T_{\max} = 102, \ \Delta = -199, \ \det = 68, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.65 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & -4 & -2 \\ 2 & 4 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 3^1 1^1 2^1 3^1 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 \quad T = 8, \ T_{\max} = 102, \ \Delta = -201, \ \det = 341, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.71 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -5 & -3 \\ 2 & 3 & 2 & -3 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [12] \ 1^1 1^1 2^1 2^1 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 \quad T = 9, \ T_{\max} = 102, \ \Delta = -170, \ \det = -11, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 17.91 \rightarrow 18, \ T_{\max} = 102, \ T \in [18, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 5 & 0 & -3 & -1 \\ 2 & 5 & -1 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [14] \ 1^1 4^1 1^1 3^1 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 \quad T = 10, \ T_{\max} = 102, \ \Delta = -161, \ \det = 11, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 18.91 \rightarrow 19, \ T_{\max} = 102, \ T \in [19, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 \\ 1 & 3 & -5 & -3 \\ 3 & 4 & -3 & -2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [16] \ 1^1 5^1 1^1 3^1 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 \quad T = 11, \ T_{\max} = 102, \ \Delta = -156, \ \det = -11, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 19.91 \rightarrow 20, \ T_{\max} = 102, \ T \in [20, 102]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 7 & 7 & 7 \\ 2 & 7 & -2 & 0 \\ 3 & 7 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[17] \ 2^1 1^1 5^1 1^1 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 \quad T = 11, \ T_{\max} = 102, \ \Delta = -156, \ \det = -11, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 19.91 \rightarrow 20, \ T_{\max} = 102, \ T \in [20, 102]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 7 & 7 & 7 \\ 2 & 7 & -2 & 0 \\ 3 & 7 & 0 & -2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [21] \quad 131\overset{1}{5}13221^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 \quad T = 12, T_{\max} = 98, \Delta = -135, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 20.91 \rightarrow 21, T_{\max} = 98, T \in [21, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 5 & 0 & -3 & -1 \\ 2 & 5 & -1 & -1 & -3 \end{pmatrix}_{b_0}$$

4.167 $\text{NHC}(-2)(-3) + \textcolor{violet}{e}_7 + \textcolor{violet}{e}_6$

$$T = 9 \quad [0] \quad 1^{11}2\overset{1}{3}21^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{e}_6 \quad T = 9, T_{\max} = 127, \Delta = -242, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 17.14 \rightarrow 18, T_{\max} = 127, T \in [18, 127]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 1 & 3 & 0 & 1 & -3 & -1 \\ 2 & 3 & -1 & -3 & -1 & -4 \end{pmatrix}_{b_0}$$

4.168 $\text{NHC}(-2)(-3) + \textcolor{violet}{e}_7 + \textcolor{violet}{f}_4$

$$T = 8 \quad [0] \quad 1^{11}3\overset{1}{2}21^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 131, \Delta = -243, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 16.11 \rightarrow 17, T_{\max} = 131, T \in [17, 131]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -4 & 0 & -2 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & -2 & -3 & -3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [1] \quad 1^{11}3\overset{1}{2}221^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{f}_4 \quad T = 9, T_{\max} = 104, \Delta = -214, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 16.41 \rightarrow 17, T_{\max} = 104, T \in [17, 104]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 6 & 6 & 0 & 6 \\ 2 & 6 & -3 & 0 & -1 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 6 & -1 & -3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}2\overset{1}{3}221^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{f}_4 \quad T = 9, T_{\max} = 104, \Delta = -216, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 16.24 \rightarrow 17, T_{\max} = 104, T \in [17, 104]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 1 & 3 & 0 & 1 & -2 & 0 \\ 2 & 3 & -1 & -3 & 0 & -3 \end{pmatrix}_{b_0}$$

4.169 $\text{NHC}(-2)(-3) + \textcolor{violet}{e}_7 + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \quad 1^{11}3\overset{1}{2}21^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 102, \Delta = -170, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 5 \\ 7 & 8 & 1 & 5 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 5 & 0 & -3 & -1 \\ 5 & 6 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3\overset{1}{2}21^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 102, \Delta = -170, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 4 & -2 \\ 4 & 8 & 0 & 4 & 9 & 5 \\ 3 & 4 & -1 & -2 & 5 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [2] \quad 1^{111}2\overset{1}{3}221^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 100, \Delta = -143, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.10 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 4 & 3 \\ 1 & 3 & 0 & 1 & 4 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

[3] $1^{11}2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + [\text{su}_2 \cdot \text{g}_2] \quad T = 10, T_{\max} = 100, \Delta = -143, \det = 34, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 15.35 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 1 & 3 & 0 & 1 & 1 & 0 & 2 \\ 2 & 3 & -1 & -3 & 0 & 2 & -2 \end{pmatrix}_{b_0}$$

[4] $1^{11}2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + [\text{su}_2 \cdot \text{g}_2] \quad T = 10, T_{\max} = 100, \Delta = -143, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.91 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 1 & 0 & 3 & 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & -3 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{11}3^1 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + [\text{su}_2 \cdot \text{g}_2] \quad T = 10, T_{\max} = 100, \Delta = -141, \det = 136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.79 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 5 \\ 7 & 8 & 1 & 5 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 5 & 0 & -2 & 0 \\ 5 & 6 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{111}3^1 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + [\text{su}_2 \cdot \text{g}_2] \quad T = 10, T_{\max} = 100, \Delta = -141, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 15.06 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 4 & -1 \\ 4 & 8 & 0 & 4 & 9 & 5 \\ 3 & 4 & -1 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

[7] $1^1 231^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + [\text{su}_2 \cdot \text{g}_2] \quad T = 10, T_{\max} = 101, \Delta = -149, \det = 325, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.05 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 0 & -2 \\ 2 & 6 & 0 & 3 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 2 & 4 & -1 & -2 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

4.170 $\text{NHC}(-2)(-3) + \text{e}_7 + \text{NHC}(-2)(-3) + \text{su}_2$

$T = 10 \quad [0] \quad 1^{111}3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 178, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 7 & 5 \\ 7 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -3 & 5 & -1 \\ 7 & 10 & 0 & 5 & 8 & 6 \\ 5 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 548, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 8 & 5 \\ 7 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -3 & 5 & -1 \\ 8 & 10 & 0 & 5 & 8 & 6 \\ 5 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$T = 11 \quad [2] \quad 1^{1111}2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 11, T_{\max} = 26, \Delta = -113, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 4 & 3 & 3 \\ 1 & 3 & 0 & 1 & 4 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 3 & 2 & 3 & -12 \end{pmatrix}_{b_0}$$

- [3] $1^{1112}2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 11, T_{\max} = 26, \Delta = -113, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 15.45 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 1 & 4 & 6 & 3 \\ 1 & 3 & 0 & 1 & 4 & 0 & 6 & 2 \\ 3 & 6 & 0 & 2 & 6 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 3 & 2 & 6 & -2 \end{pmatrix}_{b_0}$$
- [4] $1^{111}3^{111}221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 11, T_{\max} = 24, \Delta = -111, \det = -68, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 15.29 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 12 & 5 & 6 \\ 7 & 8 & 1 & 5 & 15 & 6 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 11 & 0 & 0 \\ 12 & 15 & 0 & 11 & 24 & 12 & 12 \\ 5 & 6 & -1 & 0 & 12 & 0 & 0 \end{pmatrix}_{b_0}$$
- [5] $1^{1114}2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 11, T_{\max} = 26, \Delta = -113, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 16.24 \rightarrow 17, T_{\max} = 26, T \in [17, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 4 & 12 & 3 \\ 1 & 3 & 0 & 1 & 4 & 0 & 12 & 2 \\ 6 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 2 & 12 & -3 \end{pmatrix}_{b_0}$$
- [6] $1^{111}3^{111}221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 11, T_{\max} = 24, \Delta = -111, \det = -147, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 7 & 5 & 6 \\ 7 & 8 & 1 & 5 & 10 & 6 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 5 & 0 & 0 \\ 7 & 10 & 0 & 5 & 8 & 6 & 6 \\ 5 & 6 & -1 & 0 & 6 & 1 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{111}2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 11, T_{\max} = 26, \Delta = -113, \det = -155, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.94 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 1 & 1 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 0 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 0 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$
- [8] $1^{111}2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 11, T_{\max} = 26, \Delta = -113, \det = -190, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.95 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 1 & 1 & 0 & 3 & 2 \\ 3 & 0 & 3 & 1 & 0 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 0 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$
- [9] $1^{1113}2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 11, T_{\max} = 26, \Delta = -113, \det = -227, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 15.10 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 1 & 4 & 9 & 3 \\ 1 & 3 & 0 & 1 & 4 & 0 & 9 & 2 \\ 4 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 3 & 2 & 9 & -2 \end{pmatrix}_{b_0}$$
- [10] $1^{1114}2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 11, T_{\max} = 26, \Delta = -113, \det = -267, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 15.17 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 4 & 12 & 3 \\ 1 & 3 & 0 & 1 & 4 & 0 & 12 & 2 \\ 5 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 2 & 12 & -2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{111} 3^{221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 24, \quad \Delta = -111, \quad \det = -452, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.80 \rightarrow 15, \quad T_{\max} = 24, \quad T \in [15, 24]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 8 & 5 \\ 7 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 5 & 0 \\ 8 & 10 & 0 & 5 & 8 & 6 \\ 5 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{1231} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 48, \quad \Delta = -119, \quad \det = -540, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.07 \rightarrow 16, \quad T_{\max} = 48, \quad T \in [16, 48]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 0 & 3 & -2 \\ 2 & 6 & 0 & 3 & 4 & 1 & 6 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 6 & 0 & 4 & 4 \\ 2 & 4 & -1 & -2 & 4 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

4.171 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \text{NHC}(-2)(-3) + \mathfrak{su}_{2 \rightarrow (-3)}$

$$T = 10 \quad [0] \quad 1^{11} 3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_{2 \rightarrow (-3)} \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -143, \quad \det = 304, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.79 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 4 & 5 \\ 7 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -3 & 3 & -1 \\ 4 & 5 & 0 & 3 & 1 & 3 \\ 5 & 6 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \quad 1^{1231} 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_{2 \rightarrow (-3)} \quad T = 11, \quad T_{\max} = 75, \quad \Delta = -122, \quad \det = -440, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.09 \rightarrow 16, \quad T_{\max} = 75, \quad T \in [16, 75]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 0 & 2 & -2 \\ 2 & 6 & 0 & 3 & 4 & 1 & 3 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 3 & 0 & 0 & 2 \\ 2 & 4 & -1 & -2 & 4 & -1 & 2 & -2 \end{pmatrix}_{b_0}$$

4.172 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{111} 3^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 101, \quad \Delta = -149, \quad \det = 103, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.03 \rightarrow 16, \quad T_{\max} = 101, \quad T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 1 & 3 \\ 3 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 4 & 0 & -2 \\ 4 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -2 & 5 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \quad 1^{1111} 2^3 2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 11, \quad T_{\max} = 75, \quad \Delta = -122, \quad \det = -33, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.27 \rightarrow 19, \quad T_{\max} = 75, \quad T \in [19, 75]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 4 & 3 & 3 \\ 1 & 3 & 0 & 1 & 4 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 2 & 2 & -5 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^{221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 11, \quad T_{\max} = 59, \quad \Delta = -120, \quad \det = -82, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.12 \rightarrow 16, \quad T_{\max} = 59, \quad T \in [16, 59]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 1 & 3 \\ 3 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 4 & 0 & -1 \\ 4 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -1 & 5 & -1 & -1 \end{pmatrix}_{b_0}$$

4.173 NHC(-2)(-3) + e₇ + s₀₈

$$T = 8 \quad [0] \ 1^{11} 3 2 1^1 + [\textcolor{blue}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{s}_{08} \quad T = 8, \ T_{\max} = 103, \ \Delta = -219, \ \det = 108, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.33 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -4 & 0 & -2 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & -2 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 2 1^1 + [\textcolor{blue}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{s}_{08} \quad T = 8, \ T_{\max} = 103, \ \Delta = -219, \ \det = 127, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.39 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 1 & 2 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & -3 & 0 & -3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \ 1^{11} 3 2 2 1^{11} + [\textcolor{blue}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{s}_{08} \quad T = 9, \ T_{\max} = 103, \ \Delta = -190, \ \det = -30, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 6 & 6 & 0 & 6 \\ 2 & 6 & -3 & 0 & -1 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 6 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 2 3 2 1^1 + [\textcolor{blue}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{s}_{08} \quad T = 9, \ T_{\max} = 103, \ \Delta = -192, \ \det = -68, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.82 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 9 & 4 & 5 \\ 2 & 0 & 4 & -4 & -2 \\ 3 & 0 & 5 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 2 3 2 1^1 + [\textcolor{blue}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{s}_{08} \quad T = 9, \ T_{\max} = 103, \ \Delta = -192, \ \det = -80, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.40 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 10 & 1 & 4 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & -4 & -2 \\ 3 & 6 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 2 2 1^1 + [\textcolor{blue}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{s}_{08} \quad T = 9, \ T_{\max} = 103, \ \Delta = -190, \ \det = -106, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.47 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 1 & 2 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 2 3 1^{11} 3 1^{11} + [\textcolor{blue}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{s}_{08} \quad T = 9, \ T_{\max} = 103, \ \Delta = -198, \ \det = -154, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.64 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 1 & 0 & 3 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 0 & 3 & -2 & -2 & -2 \end{pmatrix}_{b_0}$$

4.174 NHC(-2)(-3) + e₇ + s₀₈ + s_{u2}

$$T = 9 \quad [0] \ 1^{11} 3 2 1^1 + [\textcolor{blue}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{s}_{08} + \textcolor{red}{s}_{u2} \quad T = 9, \ T_{\max} = 103, \ \Delta = -189, \ \det = -8, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & -2 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 2 1^1 + [\textcolor{blue}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{s}_{08} + \textcolor{red}{s}_{u2} \quad T = 9, \ T_{\max} = 103, \ \Delta = -189, \ \det = -167, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.43 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 0 & -3 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -3 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 103$, $\Delta = -189$, $\det = -261$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.34 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -2 & -2 & 4 & -2 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 103$, $\Delta = -189$, $\det = -293$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.33 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -2 & -2 & 4 & -2 \end{pmatrix}_{b_0}$$

$T = 10$ [4] $1^{11} 2 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 103$, $\Delta = -162$, $\det = 28$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.71 \rightarrow 18$, $T_{\max} = 103$, $T \in [18, 103]$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 \\ 2 & 0 & 4 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 0 & 5 & -2 & 0 & -3 \end{pmatrix}_{b_0}$$

[5] $1^{111} 2 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 103$, $\Delta = -162$, $\det = 112$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.43 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & -2 \\ 4 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{11} 2 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 103$, $\Delta = -168$, $\det = 124$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.68 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & -2 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 102$, $\Delta = -160$, $\det = 139$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.51 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 0 & -2 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

[8] $1^{111} 2 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 103$, $\Delta = -162$, $\det = 160$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.40 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & -2 \\ 6 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{111} 2 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 103$, $\Delta = -162$, $\det = 308$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.40 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & -2 \\ 5 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{112} 2 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 103$, $\Delta = -162$, $\det = 532$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} =$

15.5

$$T_{\min} = 15.41 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 9 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 8 & -2 \\ 9 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{113} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 103, \Delta = -162, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.8$

$$T_{\min} = 15.43 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 13 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 12 & -2 \\ 13 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & -2 & 18 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{114} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 103, \Delta = -162, \det = 692, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.46 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 17 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 16 & -2 \\ 17 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & -2 & 24 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{112} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 103, \Delta = -162, \det = 752, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 10 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 8 & -2 \\ 10 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{113} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 103, \Delta = -162, \det = 1248, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.8$

$$T_{\min} = 15.41 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 14 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 12 & -2 \\ 14 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & -2 & 18 & -1 \end{pmatrix}_{b_0}$$

[15] $1^{114} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2 \quad T = 10, T_{\max} = 103, \Delta = -162, \det = 1648, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.42 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 18 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 16 & -2 \\ 18 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & -2 & 24 & -1 \end{pmatrix}_{b_0}$$

4.175 NHC(-2)(-3) + e₇ + so₈ + su₂ + su₂

$T = 10 \quad [0] \quad 1^{111} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 102, \Delta = -159, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.57 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 0 & -3 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 \\ 2 & 2 & -1 & -3 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

$T = 11 \quad [1] \quad 1^{111} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 95, \Delta = -132, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.50 \rightarrow 16, T_{\max} = 95, T \in [16, 95]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & 4 & -2 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 4 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{21} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 94, \ \Delta = -130, \ \det = -132, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.64 \rightarrow 16, \ T_{\max} = 94, \ T \in [16, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 0 & 0 & -2 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 95, \ \Delta = -132, \ \det = -200, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.44 \rightarrow 16, \ T_{\max} = 95, \ T \in [16, 95]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & 4 & -2 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 5 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

4.176 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{11} 3^{21} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 103, \ \Delta = -192, \ \det = -180, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.40 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & -2 & -2 & 2 & -2 \end{pmatrix}_{b_0}$$

4.177 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{11} 3^{21} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 103, \ \Delta = -198, \ \det = -294, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.63 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 0 & -3 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -3 & 0 & -1 & -3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [1] \ 1^{11} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 103, \ \Delta = -171, \ \det = 96, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.83 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 \\ 2 & 0 & 4 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 0 & 5 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 103, \ \Delta = -171, \ \det = 192, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & -2 \\ 4 & 12 & 0 & 4 & 9 & 5 \\ 3 & 6 & -1 & -2 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{21} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 103, \ \Delta = -169, \ \det = 245, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.73 \rightarrow 16, \ T_{\max} = 103, \ T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 0 & -2 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -2 & 0 & -1 & -2 \end{pmatrix}_{b_0}$$

4.178 NHC(-2)(-3) + e₇ + su₂

$$T = 8 \quad [0] 1^{111} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 17.65 \rightarrow 18, T_{\max} = 102, T \in [18, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -4 & 3 & -2 \\ 5 & 8 & 3 & 4 & 4 \\ 3 & 5 & -2 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[1] 1^{11} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15.71 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[2] 1^{112} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 15.13 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & -3 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -3 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[3] 1^{11} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.19 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -3 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[4] 2^{11} 23^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.34 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 3 & 4 & -4 & 3 & -2 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[5] 1^{11} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.09 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[6] 1^{12} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.24 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & -3 \\ 2 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -3 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[7] 1^{13} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & -3 \\ 3 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -3 & 6 & -2 \end{pmatrix}_{b_0}$$

$$[8] 1^{111} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -4 & 3 & -2 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[9] 1^{14} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & -3 \\ 4 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -3 & 8 & -2 \end{pmatrix}_{b_0}$$

[10] $1^{11}1^1321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.83 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$
 $\begin{pmatrix} 1 & 4 & 2 & 4 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -4 & 3 & -2 \\ 4 & 8 & 3 & 4 & 4 \\ 3 & 5 & -2 & 4 & -1 \end{pmatrix}_{b_0}$

$$[11] \ 1^{15} 3^{21} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{c}_7 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 102, \ \Delta = -190, \ \det = 137, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.1$$

$$T_{\min} = 14.74 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 5 & -3 \\ 5 & 10 & 0 & 5 & 48 & 10 \\ 2 & 2 & -1 & -3 & 10 & -2 \end{pmatrix}_{b_0}$$

[12] $1^{11} 3^{21} 1^1 + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, \, T_{\max} = 102, \, \Delta = -190, \, \det = 139, \, (n_+, n_-, n_0) = (1, 8, 0), \, c_{\text{ext}} = 11.0$
 $T_{\min} = 14.20 \rightarrow 15, \, T_{\max} = 102, \, T \in [15, 102]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 2 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & -2 \end{pmatrix}_{b_0}$

[13] $1^{16}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 170, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 14.86 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 6 & -3 \\ 6 & 12 & 0 & 6 & 70 & 12 \\ 2 & 2 & -1 & -3 & 12 & -2 \end{pmatrix}_{b_0}$

[14] $1^{11} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 197, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.34 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 3 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & -2 \end{pmatrix}_{b_0}$

[15] $1^{17}3^{21}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}_7 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 14.95 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 7 & -3 \\ 7 & 14 & 0 & 7 & 96 & 14 \\ 2 & 2 & -1 & -3 & 14 & -2 \end{pmatrix}_{b_0}$

[16] $1^{12} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 230, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.23 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & -3 \\ 3 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -3 & 4 & -2 \end{pmatrix}_{b_0}$

[17] $1^{11} 1^{13} 3^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 239, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.50 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 4 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & -2 \end{pmatrix}_{b_0}$

[18] $1^{11} 3^{14} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 8, T_{\max} = 102, \Delta = -190, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.72 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 5 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & -2 \end{pmatrix}_{b_0}$

[19] $1^{13}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 8, T_{\max} = 102, \Delta = -190, \det = 327, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 14.27 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & -3 \\ 4 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -3 & 6 & -2 \end{pmatrix}_{b_0}$

$$T_{\min} = 14.32 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & -3 \\ 8 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -3 & 8 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [31] \quad 1^{11} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & -2 \\ 2 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & -2 & 6 & -2 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{12} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 19, T_{\max} = 101, T \in [19, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 3 & 1 & 0 & 4 & 2 & 4 \\ 2 & 2 & -1 & -2 & 4 & -5 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{11} 2^3 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{14} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 21, T_{\max} = 101, T \in [21, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 5 & 1 & 0 & 4 & 2 & 4 \\ 2 & 2 & -1 & -2 & 4 & -7 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{12} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 15.80 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & -3 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -3 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{13} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 17.95 \rightarrow 18, T_{\max} = 101, T \in [18, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 4 & 1 & 0 & 4 & 2 & 4 \\ 2 & 2 & -1 & -2 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{11} 2^3 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15.35 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & -3 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{12} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & -2 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{11} 2^3 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

- [40] $1^{11} 1^2 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -163$, $\det = -41$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.37 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -3 & 0 & -1 \end{pmatrix}_{b_0}$$
- [41] $1^{11} 1^3 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -45$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.24 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$
- [42] $1^1 231^1 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -54$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 15.33 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 1 & 0 & 3 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$
- [43] $1^{11} 1^3 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -54$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.15 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$
- [44] $1^1 231^1 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -57$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.68 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$
- [45] $1^{12} 1^3 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -60$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 14.33 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 2 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$
- [46] $1^{13} 1^2 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -163$, $\det = -65$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.60 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -5 & 3 & -3 \\ 4 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -3 & 9 & -1 \end{pmatrix}_{b_0}$$
- [47] $1^{13} 1^3 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -70$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.57 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$
- [48] $1^1 231^1 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 3 & -2 \\ 3 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$
- [49] $1^{12} 1^3 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -76$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 14.37 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 6 & -2 \\ 3 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$
- [50] $1^{14} 1^3 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 4 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{11} 23131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{15} 3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -102, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.1$$

$$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & -2 \\ 5 & 10 & 0 & 5 & 48 & 10 \\ 2 & 2 & -1 & -2 & 10 & -2 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{11} 231^{231} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -102, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 15.18 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 6 & -2 \\ 6 & 0 & 0 & 6 & 22 & 8 \\ 2 & 0 & -1 & -2 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{11} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.53 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 2 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[55] \ 1^{11} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -105, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{12} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -113, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -5 & 2 & -3 \\ 3 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{11} 3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.27 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 2 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{11} 3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.34 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & -2 \\ 3 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{16} 3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 15.19 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 6 & -2 \\ 6 & 12 & 0 & 6 & 70 & 12 \\ 2 & 2 & -1 & -2 & 12 & -2 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{11} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

- [61] $1^{12}3131^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -146$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 15.18 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 3 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & -3 & 3 & -2 \end{pmatrix}_{b_0}$$
- [62] $1^{12}3131^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -147$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 15.33 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$
- [64] $1^{13}3221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -150$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.37 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 7 & -2 \\ 4 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$
- [63] $1^{17}3221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -150$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.3$
 $T_{\min} = 15.33 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 7 & -2 \\ 7 & 14 & 0 & 7 & 96 & 14 \\ 2 & 2 & -1 & -2 & 14 & -2 \end{pmatrix}_{b_0}$$
- [65] $1^{12}2321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -163$, $\det = -159$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 14.45 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & -3 \\ 3 & 6 & 0 & 2 & 10 & 6 \\ 2 & 3 & -1 & -3 & 6 & -1 \end{pmatrix}_{b_0}$$
- [66] $1^{11}3221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -162$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.42 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 3 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$
- [67] $1^{11}2321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -163$, $\det = -176$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & -3 \\ 3 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$
- [68] $1^{12}3131^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -180$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.71 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & -2 \\ 2 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$
- [69] $1^{12}3221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -189$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 14.31 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 3 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$
- [70] $1^{13}2321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -163$, $\det = -191$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & -3 \\ 4 & 9 & 0 & 3 & 25 & 9 \\ 2 & 3 & -1 & -3 & 9 & -1 \end{pmatrix}_{b_0}$$

[71] $1^{11} 3^{13} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.61 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 4 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

[72] $1^{14} 2^3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & -3 \\ 5 & 12 & 0 & 4 & 46 & 12 \\ 2 & 3 & -1 & -3 & 12 & -1 \end{pmatrix}_{b_0}$$

[74] $1^{11} 3^{12} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -210, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & -2 \\ 4 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[73] $1^{14} 231^{43} 31^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -210, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 15.43 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 12 & -2 \\ 12 & 0 & 0 & 12 & 94 & 16 \\ 2 & 0 & -1 & -2 & 16 & -2 \end{pmatrix}_{b_0}$$

[75] $1^{11} 3^{14} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -214, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.86 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 5 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

[76] $1^{14} 231^{11} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.65 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & -2 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[77] $1^{14} 3^{11} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 8 & -2 \\ 5 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[78] $1^{11} 2^3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -233, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.55 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & -3 \\ 4 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

[79] $1^{14} 231^{13} 31^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.91 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 3 & -2 \\ 4 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[80] $1^{14} 23131^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -246, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 15.27 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

[81] $1^{12} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -253, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.30 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 6 & -2 \\ 4 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[82] $1^{13} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -267, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 14.36 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 4 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[83] $1^{12} 2^3 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.63 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & -3 \\ 4 & 6 & 0 & 2 & 10 & 6 \\ 2 & 3 & -1 & -3 & 6 & -1 \end{pmatrix}_{b_0}$$

[84] $1^{12} 2^3 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.36 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -5 & 2 & -3 \\ 4 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -3 & 6 & -1 \end{pmatrix}_{b_0}$$

[85] $1^{11} 2^3 3^{14} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.71 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & -3 \\ 5 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

[86] $1^1 231^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -277, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.66 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[87] $1^{11} 3^{13} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -291, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.49 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & -2 \\ 5 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[88] $1^{12} 3^{12} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.37 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 4 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[89] $1^{14} 2^3 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.43 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -5 & 4 & -3 \\ 6 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & -3 & 12 & -1 \end{pmatrix}_{b_0}$$

[90] $1^1 231^{12} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -312, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.79 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & -2 \\ 4 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[91] $1^{13} 2^3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 14.38 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -5 & 3 & -3 \\ 5 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -3 & 9 & -1 \end{pmatrix}_{b_0}$$

[92] $1^{14} 3^2 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -349, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 5 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[93] $1^{11} 3^2 221^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -358, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & -2 \\ 6 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[94] $1^{14} 23131^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 15.32 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

[95] $1^{13} 3^2 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 14.30 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 7 & -2 \\ 5 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

[96] $1^{13} 2^3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -398, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 14.52 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & -3 \\ 5 & 9 & 0 & 3 & 25 & 9 \\ 2 & 3 & -1 & -3 & 9 & -1 \end{pmatrix}_{b_0}$$

[97] $1^{12} 3^2 221^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -405, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.44 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 5 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[98] $1^{12} 3^2 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 6 & -2 \\ 5 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[99] $1^{12} 231^{231} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -422, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.92 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 6 & -2 \\ 7 & 0 & 0 & 6 & 22 & 8 \\ 2 & 0 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[100] $1^{12} 2^3 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -425, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -5 & 2 & -3 \\ 5 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[101] \ 1^{12} 1^{13} 3 \ 221^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, \ T_{\max} = 101, \ \Delta = -161, \ \det = -450, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 14.36 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 5 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[102] \ 1^1 231^1 31^{13} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, \ T_{\max} = 102, \ \Delta = -169, \ \det = -455, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.94 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 3 & -2 \\ 5 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[103] \ 1^1 231^{12} 31^{14} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, \ T_{\max} = 102, \ \Delta = -169, \ \det = -456, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.65 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & -2 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{12} 1^{14} 3 \ 221^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, \ T_{\max} = 101, \ \Delta = -161, \ \det = -492, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.53 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 6 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{14} 2^3 \ 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, \ T_{\max} = 102, \ \Delta = -163, \ \det = -498, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.47 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & -3 \\ 6 & 12 & 0 & 4 & 46 & 12 \\ 2 & 3 & -1 & -3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[106] \ 1^1 231^{13} 31^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, \ T_{\max} = 102, \ \Delta = -169, \ \det = -532, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 14.84 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 9 & -2 \\ 6 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[107] \ 1^{14} 1^{11} 3 \ 221^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, \ T_{\max} = 101, \ \Delta = -161, \ \det = -541, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.32 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 8 & -2 \\ 6 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[108] \ 1^{12} 2^3 \ 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, \ T_{\max} = 102, \ \Delta = -163, \ \det = -560, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.47 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -5 & 2 & -3 \\ 6 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{13} 1^{13} 2^3 \ 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, \ T_{\max} = 102, \ \Delta = -163, \ \det = -561, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 14.38 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -5 & 3 & -3 \\ 6 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{12} 1^{13} 3 \ 221^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, \ T_{\max} = 101, \ \Delta = -161, \ \det = -565, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.37 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 6 & -2 \\ 6 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0} \\
[111] & 1^1 23131^{115} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -566, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.1 \\
& T_{\min} = 15.34 \rightarrow 16, T_{\max} = 102, T \in [16, 102] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0} \\
[112] & 1^1 231^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -585, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5 \\
& T_{\min} = 14.70 \rightarrow 15, T_{\max} = 102, T \in [15, 102] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & -2 \\ 5 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0} \\
[113] & 1^{14} 3 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -600, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 14.37 \rightarrow 15, T_{\max} = 101, T \in [15, 101] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 6 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0} \\
[114] & 1^1 231^3 31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -617, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8 \\
& T_{\min} = 14.94 \rightarrow 15, T_{\max} = 102, T \in [15, 102] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 9 & -2 \\ 10 & 0 & 0 & 9 & 52 & 12 \\ 2 & 0 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0} \\
[115] & 1^{13} 3 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -619, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8 \\
& T_{\min} = 14.40 \rightarrow 15, T_{\max} = 101, T \in [15, 101] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 6 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0} \\
[116] & 1^{13} 3 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8 \\
& T_{\min} = 14.31 \rightarrow 15, T_{\max} = 101, T \in [15, 101] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 7 & -2 \\ 6 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0} \\
[117] & 1^{14} 2 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -641, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 14.39 \rightarrow 15, T_{\max} = 102, T \in [15, 102] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -5 & 4 & -3 \\ 7 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & -3 & 12 & -1 \end{pmatrix}_{b_0} \\
[118] & 1^1 231^{12} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -657, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5 \\
& T_{\min} = 14.66 \rightarrow 15, T_{\max} = 102, T \in [15, 102] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & -2 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0} \\
[119] & 1^{12} 3 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -700, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5 \\
& T_{\min} = 14.43 \rightarrow 15, T_{\max} = 101, T \in [15, 101] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 6 & -2 \\ 7 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0} \\
[120] & 1^1 231^{12} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -700, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5 \\
& T_{\min} = 14.67 \rightarrow 15, T_{\max} = 102, T \in [15, 102] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & -2 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [121] $1^{12}31^{13}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -705$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.97 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 3 & -2 \\ 6 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$
- [122] $1^{13}3221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -774$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.44 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 7 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$
- [123] $1^{12}31^{13}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -782$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 14.91 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 6 & -2 \\ 8 & 0 & 0 & 6 & 22 & 8 \\ 2 & 0 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$
- [124] $1^{13}2321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -163$, $\det = -788$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -5 & 3 & -3 \\ 7 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -3 & 9 & -1 \end{pmatrix}_{b_0}$$
- [125] $1^{14}3221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -837$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.39 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 7 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$
- [127] $1^{12}31^{13}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -840$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.87 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 12 & -2 \\ 8 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$
- [126] $1^{14}3221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -840$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.31 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 8 & -2 \\ 7 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$
- [128] $1^{13}3221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -843$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.34 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 7 & -2 \\ 7 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$
- [129] $31^{11}2321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -171$, $\det = -922$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.72 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 3 & 2 \\ 3 & 1 & 4 & 1 & 3 & 3 \\ 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -5 & 1 & -3 \\ 3 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$
- [130] $1^{14}2321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -163$, $\det = -960$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.39 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -5 & 4 & -3 \\ 8 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & -3 & 12 & -1 \end{pmatrix}_{b_0}$$
- [131] $1^{12}31^{13}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -981$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.72 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 9 & -2 \\ 7 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[132] $1^{13} 3^{14} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -1046$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.37 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 7 & -2 \\ 8 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

[133] $1^{14} 3^{14} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -1060$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 8 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[134] $1^{14} 3^{13} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -1125$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.33 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 8 & -2 \\ 8 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[135] $1^1 231^{15} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -1236$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.1$
 $T_{\min} = 14.88 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 15 & -2 \\ 10 & 30 & 0 & 15 & 148 & 20 \\ 2 & 4 & -1 & -2 & 20 & -1 \end{pmatrix}_{b_0}$$

[136] $1^1 231^{13} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -1272$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.69 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 9 & -2 \\ 8 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[137] $1^1 231^{13} 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -1380$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.66 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 9 & -2 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[138] $1^{14} 3^{14} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -161$, $\det = -1396$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.35 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 8 & -2 \\ 9 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[139] $1^1 231^{13} 31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -1405$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 9 & -2 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[140] $1^1 231^{14} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -1465$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.75 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 12 & -2 \\ 9 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

[141] $1^1 231^{14} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -1932$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.71 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 12 & -2 \\ 10 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

[142] $1^1 231^{14} 31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -2241$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 14.68 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 12 & -2 \\ 11 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[143] \ 1^1 231^{14} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -2392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 12 & -2 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

4.179 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \ 1^{11} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 178, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 1 & 5 \\ 7 & 8 & 1 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 6 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 548, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 8 & 5 \\ 7 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -3 & 5 & -1 \\ 8 & 10 & 0 & 5 & 8 & 6 \\ 5 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \ 1^{111} 23 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 26, \Delta = -113, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 4 \\ 1 & 3 & 0 & 1 & 3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{121} 23 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 26, \Delta = -113, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 15.45 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & -3 \\ 3 & 6 & 0 & 2 & 10 & 6 & 6 \\ 1 & 3 & 0 & 1 & 6 & 1 & 4 \\ 1 & 3 & 0 & 1 & 6 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3 \ 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -68, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.29 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 12 & 5 \\ 7 & 8 & 1 & 5 & 15 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 11 & 0 \\ 12 & 15 & 0 & 11 & 24 & 12 \\ 5 & 6 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{141} 23 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 26, \Delta = -113, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 16.24 \rightarrow 17, T_{\max} = 26, T \in [17, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 1 & 1 & -3 \\ 6 & 12 & 0 & 4 & 46 & 12 & 12 & 12 \\ 1 & 3 & 0 & 1 & 12 & 1 & 4 & 3 \\ 1 & 3 & 0 & 1 & 12 & 4 & 0 & 2 \\ 2 & 3 & -1 & -3 & 12 & 3 & 2 & -3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -147, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.80 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 1 & 5 \\ 7 & 8 & 1 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 6 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 26, \Delta = -113, \det = -155, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.94 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 0 & 1 & -3 \\ 2 & 0 & 3 & 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 1 & 0 \\ 1 & 0 & 3 & 1 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{111} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 26, \Delta = -113, \det = -190, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.95 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 0 & 1 & -3 \\ 3 & 0 & 3 & 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 1 & 0 \\ 1 & 0 & 3 & 1 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{131} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 26, \Delta = -113, \det = -227, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 15.10 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 1 & 1 & -3 \\ 4 & 9 & 0 & 3 & 25 & 9 & 9 & 9 \\ 1 & 3 & 0 & 1 & 9 & 1 & 4 & 3 \\ 1 & 3 & 0 & 1 & 9 & 4 & 0 & 2 \\ 2 & 3 & -1 & -3 & 9 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{141} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 26, \Delta = -113, \det = -267, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.17 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 1 & 1 & -3 \\ 5 & 12 & 0 & 4 & 46 & 12 & 12 & 12 \\ 1 & 3 & 0 & 1 & 12 & 1 & 4 & 3 \\ 1 & 3 & 0 & 1 & 12 & 4 & 0 & 2 \\ 2 & 3 & -1 & -3 & 12 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{111} 3^{221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -452, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.80 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 8 & 5 \\ 7 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 5 & 0 \\ 8 & 10 & 0 & 5 & 8 & 6 \\ 5 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{1231} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 48, \Delta = -119, \det = -540, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 48, T \in [16, 48]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 3 & 0 & -2 \\ 2 & 6 & 0 & 3 & 4 & 6 & 0 & 4 \\ 2 & 6 & 0 & 3 & 6 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 0 & 1 & -3 & -1 \\ 2 & 4 & -1 & -2 & 4 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

4.180 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 103, \Delta = -189, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -2 & 0 & -2 & -2 \end{pmatrix}_{b_0}$$

- [1] $1^{11} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 9$, $T_{\max} = 103$, $\Delta = -189$, $\det = -167$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.43 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & -3 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$
- [2] $1^{11} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 9$, $T_{\max} = 103$, $\Delta = -189$, $\det = -261$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.34 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -4 & 3 & 0 & -2 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -2 & 4 & -2 & -2 \end{pmatrix}_{b_0}$$
- [3] $1^{11} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 9$, $T_{\max} = 103$, $\Delta = -189$, $\det = -293$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.33 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -4 & 3 & 0 & -2 \\ 4 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -2 & 4 & -2 & -2 \end{pmatrix}_{b_0}$$
- $T = 10$ [4] $1^{11} 2 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10$, $T_{\max} = 103$, $\Delta = -162$, $\det = 28$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 17.71 \rightarrow 18$, $T_{\max} = 103$, $T \in [18, 103]$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 \\ 2 & 0 & 4 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 0 & 5 & -2 & 0 & -3 \end{pmatrix}_{b_0}$$
- [5] $1^{11} 2 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10$, $T_{\max} = 103$, $\Delta = -162$, $\det = 112$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.43 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & -2 \\ 4 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$
- [6] $1^{11} 2 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10$, $T_{\max} = 103$, $\Delta = -168$, $\det = 124$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.68 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 0 & 3 & -2 & 0 & -2 & -2 \end{pmatrix}_{b_0}$$
- [7] $1^{11} 3 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10$, $T_{\max} = 102$, $\Delta = -160$, $\det = 139$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.51 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$
- [8] $1^{11} 2 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10$, $T_{\max} = 103$, $\Delta = -162$, $\det = 160$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.40 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & -2 \\ 6 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$
- [9] $1^{11} 2 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 10$, $T_{\max} = 103$, $\Delta = -162$, $\det = 308$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.40 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & -2 \\ 5 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{121}2^13^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 103, \Delta = -162, \det = 532, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$

$$T_{\min} = 15.41 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 9 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 8 & -2 \\ 9 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{131}2^13^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 103, \Delta = -162, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.8$

$$T_{\min} = 15.43 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 13 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 12 & -2 \\ 13 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & -2 & 18 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{141}2^13^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 103, \Delta = -162, \det = 692, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.46 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 17 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 16 & -2 \\ 17 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & -2 & 24 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{121}2^13^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 103, \Delta = -162, \det = 752, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 10 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 8 & -2 \\ 10 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{131}2^13^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 103, \Delta = -162, \det = 1248, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.8$

$$T_{\min} = 15.41 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 14 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 12 & -2 \\ 14 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & -2 & 18 & -1 \end{pmatrix}_{b_0}$$

[15] $1^{141}2^13^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 103, \Delta = -162, \det = 1648, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.42 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 18 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 16 & -2 \\ 18 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & -2 & 24 & -1 \end{pmatrix}_{b_0}$$

4.181 $\text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2$

$T = 10 \quad [0] \quad 1^{111}3^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 102, \Delta = -159, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.58 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 1 & 0 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -3 & 0 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

$T = 11 \quad [1] \quad 1^{1111}2^13^121^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 95, \Delta = -132, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.50 \rightarrow 16, T_{\max} = 95, T \in [16, 95]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & 4 & -2 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 4 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 1^{11} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 94, \ \Delta = -130, \ \det = -132, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.64 \rightarrow 16, \ T_{\max} = 94, \ T \in [16, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 1^{11} 2 \ 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 95, \ \Delta = -132, \ \det = -200, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.44 \rightarrow 16, \ T_{\max} = 95, \ T \in [16, 95]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & 4 & -2 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 5 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

4.182 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{132} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 101, \ \Delta = -160, \ \det = -19, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.3$$

$$T_{\min} = 15.42 \rightarrow 16, \ T_{\max} = 101, \ T \in [16, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 2 & -3 \\ 3 & 6 & 0 & 3 & 16 & 12 & 6 \\ 3 & 4 & 0 & 2 & 12 & 6 & 4 \\ 2 & 2 & -1 & -3 & 6 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 1^{11} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 101, \ \Delta = -160, \ \det = -51, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 16.73 \rightarrow 17, \ T_{\max} = 101, \ T \in [17, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -2 & 0 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 1^{11} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 101, \ \Delta = -160, \ \det = -59, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.19 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -3 & 0 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 1^{11} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 101, \ \Delta = -160, \ \det = -67, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 16.07 \rightarrow 17, \ T_{\max} = 101, \ T \in [17, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -2 & 0 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{144} 1^{11} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 101, \ \Delta = -160, \ \det = -76, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.53 \rightarrow 16, \ T_{\max} = 101, \ T \in [16, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 4 & -3 \\ 4 & 8 & 0 & 4 & 30 & 32 & 8 \\ 5 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -3 & 8 & 8 & -3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{121} 1^{11} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 101, \ \Delta = -160, \ \det = -82, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.24 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & -3 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -3 & 4 & 2 & -2 \end{pmatrix}_{b_0}$$

[15] $1^{11} 1^{11} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -207, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.27 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -3 & 0 & 2 & -2 \end{pmatrix}_{b_0}$$

[16] $1^{133} 1^{11} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -213, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.6$

$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 3 & -3 \\ 3 & 6 & 0 & 3 & 16 & 18 & 6 \\ 4 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -3 & 6 & 6 & -2 \end{pmatrix}_{b_0}$$

[17] $1^{12} 1^{11} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 2 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 2 & 0 & 6 & 4 \\ 2 & 2 & -1 & -3 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$

[18] $1^{114} 1^1 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.62 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & -3 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 4 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -3 & 2 & 8 & -2 \end{pmatrix}_{b_0}$$

[19] $1^{111} 1^{11} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

[20] $1^{133} 1^1 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.6$

$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 3 & -3 \\ 3 & 6 & 0 & 3 & 16 & 18 & 6 \\ 3 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -3 & 6 & 6 & -2 \end{pmatrix}_{b_0}$$

[21] $1^{123} 1^{12} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.3$

$T_{\min} = 14.28 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 3 & -3 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 5 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -3 & 4 & 6 & -2 \end{pmatrix}_{b_0}$$

[22] $1^{124} 1^1 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 4 & -3 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 4 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -3 & 4 & 8 & -2 \end{pmatrix}_{b_0}$$

[23] $1^{122} 1^{11} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.27 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 2 & -3 \\ 2 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -3 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

[24] $1^{134}321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -274$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 4 & -3 \\ 3 & 6 & 0 & 3 & 16 & 24 & 6 \\ 4 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -3 & 6 & 8 & -2 \end{pmatrix}_{b_0}$$

[25] $1^{115}321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -280$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.1$
 $T_{\min} = 14.76 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 5 & -3 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 5 & 10 & 0 & 5 & 10 & 48 & 10 \\ 2 & 2 & -1 & -3 & 2 & 10 & -2 \end{pmatrix}_{b_0}$$

[26] $1^{125}321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -298$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.6$
 $T_{\min} = 14.79 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 5 & -3 \\ 2 & 4 & 0 & 2 & 6 & 20 & 4 \\ 5 & 10 & 0 & 5 & 20 & 48 & 10 \\ 2 & 2 & -1 & -3 & 4 & 10 & -2 \end{pmatrix}_{b_0}$$

[27] $1^{144}321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -316$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 14.82 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 4 & -3 \\ 4 & 8 & 0 & 4 & 30 & 32 & 8 \\ 4 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -3 & 8 & 8 & -2 \end{pmatrix}_{b_0}$$

[28] $1^{135}321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -328$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.9$
 $T_{\min} = 14.84 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 5 & -3 \\ 3 & 6 & 0 & 3 & 16 & 30 & 6 \\ 5 & 10 & 0 & 5 & 30 & 48 & 10 \\ 2 & 2 & -1 & -3 & 6 & 10 & -2 \end{pmatrix}_{b_0}$$

[29] $1^{112}321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -330$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.76 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 6 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -3 & 2 & 4 & -2 \end{pmatrix}_{b_0}$$

[30] $1^{116}321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -346$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 14.87 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 6 & -3 \\ 1 & 2 & 0 & 1 & 0 & 12 & 2 \\ 6 & 12 & 0 & 6 & 12 & 70 & 12 \\ 2 & 2 & -1 & -3 & 2 & 12 & -2 \end{pmatrix}_{b_0}$$

[31] $1^{126}321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -364$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 14.89 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 6 & -3 \\ 2 & 4 & 0 & 2 & 6 & 24 & 4 \\ 6 & 12 & 0 & 6 & 24 & 70 & 12 \\ 2 & 2 & -1 & -3 & 4 & 12 & -2 \end{pmatrix}_{b_0}$$

[32] $1^{145}321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -370$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.1$
 $T_{\min} = 14.90 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 5 & -3 \\ 4 & 8 & 0 & 4 & 30 & 40 & 8 \\ 5 & 10 & 0 & 5 & 40 & 48 & 10 \\ 2 & 2 & -1 & -3 & 8 & 10 & -2 \end{pmatrix}_{b_0}$$

[33] $1^{136} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -394$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.1$
 $T_{\min} = 14.92 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 6 & -3 \\ 3 & 6 & 0 & 3 & 16 & 36 & 6 \\ 6 & 12 & 0 & 6 & 36 & 70 & 12 \\ 2 & 2 & -1 & -3 & 6 & 12 & -2 \end{pmatrix}_{b_0}$$

[34] $1^{112} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -411$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 14.24 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -3 & 2 & 4 & -2 \end{pmatrix}_{b_0}$$

[35] $1^{134} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -419$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 14.43 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 4 & -3 \\ 3 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -3 & 6 & 8 & -2 \end{pmatrix}_{b_0}$$

[36] $1^{117} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -424$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.3$
 $T_{\min} = 14.95 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 7 & -3 \\ 1 & 2 & 0 & 1 & 0 & 14 & 2 \\ 7 & 14 & 0 & 7 & 14 & 96 & 14 \\ 2 & 2 & -1 & -3 & 2 & 14 & -2 \end{pmatrix}_{b_0}$$

[37] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -432$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.25 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 2 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

[38] $1^{127} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -442$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 14.97 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 7 & -3 \\ 2 & 4 & 0 & 2 & 6 & 28 & 4 \\ 7 & 14 & 0 & 7 & 28 & 96 & 14 \\ 2 & 2 & -1 & -3 & 4 & 14 & -2 \end{pmatrix}_{b_0}$$

[39] $1^{123} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -458$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.3$
 $T_{\min} = 14.31 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 3 & -3 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -3 & 4 & 6 & -2 \end{pmatrix}_{b_0}$$

[40] $1^{137} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 101$, $\Delta = -160$, $\det = -472$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.1$
 $T_{\min} = 14.99 \rightarrow 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 7 & -3 \\ 3 & 6 & 0 & 3 & 16 & 42 & 6 \\ 7 & 14 & 0 & 7 & 42 & 96 & 14 \\ 2 & 2 & -1 & -3 & 6 & 14 & -2 \end{pmatrix}_{b_0}$$

[49] $1^{122} 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -796, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.27 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 2 & -3 \\ 3 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -3 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

[50] $1^{114} 1^{11} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -811, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & -3 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -3 & 2 & 8 & -2 \end{pmatrix}_{b_0}$$

[51] $1^{113} 1^{12} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -884, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 14.28 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & -3 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 5 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -3 & 2 & 6 & -2 \end{pmatrix}_{b_0}$$

[52] $1^{123} 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -935, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.3$

$T_{\min} = 14.28 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 3 & -3 \\ 3 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -3 & 4 & 6 & -2 \end{pmatrix}_{b_0}$$

[53] $1^{124} 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -976, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 4 & -3 \\ 3 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -3 & 4 & 8 & -2 \end{pmatrix}_{b_0}$$

[54] $1^{113} 1^{14} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -1016, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & -3 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 7 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -3 & 2 & 6 & -2 \end{pmatrix}_{b_0}$$

[55] $1^{113} 1^{13} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -1021, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 14.32 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & -3 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 6 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -3 & 2 & 6 & -2 \end{pmatrix}_{b_0}$$

[56] $1^{133} 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -1184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.6$

$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 3 & -3 \\ 4 & 6 & 0 & 3 & 16 & 18 & 6 \\ 4 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -3 & 6 & 6 & -2 \end{pmatrix}_{b_0}$$

[57] $1^{114} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -1254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & -3 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 6 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -3 & 2 & 8 & -2 \end{pmatrix}_{b_0}$$

[58] $1^{134} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -1335, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 4 & -3 \\ 4 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -3 & 6 & 8 & -2 \end{pmatrix}_{b_0}$$

[59] $1^{114} 3^{13} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -1555, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & -3 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 7 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -3 & 2 & 8 & -2 \end{pmatrix}_{b_0}$$

[60] $1^{144} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -1596, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.32 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 4 & -3 \\ 5 & 8 & 0 & 4 & 30 & 32 & 8 \\ 5 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -3 & 8 & 8 & -2 \end{pmatrix}_{b_0}$$

[61] $1^{114} 3^{14} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -160, \det = -1714, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & -3 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 8 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -3 & 2 & 8 & -2 \end{pmatrix}_{b_0}$$

$T = 10$ [62] $1^{132} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$T_{\min} = 17.11 \rightarrow 18, T_{\max} = 93, T \in [18, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 2 & -2 \\ 3 & 6 & 0 & 3 & 16 & 12 & 6 \\ 3 & 4 & 0 & 2 & 12 & 6 & 4 \\ 2 & 2 & -1 & -2 & 6 & 4 & -4 \end{pmatrix}_{b_0}$$

[63] $1^1 231^1 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 19.50 \rightarrow 20, T_{\max} = 100, T \in [20, 100]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & -2 & 0 & 0 & -6 \end{pmatrix}_{b_0}$$

[64] $1^1 231^1 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 25, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 18, T_{\max} = 100, T \in [18, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 3 & 0 & -2 \\ 3 & 0 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -2 & 4 & 0 & -4 \end{pmatrix}_{b_0}$$

- [65] $1^1 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 26$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 17.31 \rightarrow 18$, $T_{\max} = 94$, $T \in [18, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & 0 & -3 \\ 2 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -3 & 0 & 0 & -4 \end{pmatrix}_{b_0}$$
- [66] $1^{11} 3^1 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 36$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 15.33 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 5 & -2 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 4 & 3 & 0 & 5 & 3 & 6 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -2 \end{pmatrix}_{b_0}$$
- [67] $1^{14} 3^1 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 36$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 14.44 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 0 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 8 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 8 & 0 & 42 & 12 \\ 2 & 2 & -1 & -2 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$
- [68] $1^1 231^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -139$, $\det = 37$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 15.41 \rightarrow 16$, $T_{\max} = 100$, $T \in [16, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -2 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$
- [69] $1^{144} 3^1 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 42$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 16.81 \rightarrow 17$, $T_{\max} = 93$, $T \in [17, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 4 & -2 \\ 4 & 8 & 0 & 4 & 30 & 32 & 8 \\ 5 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & 8 & -3 \end{pmatrix}_{b_0}$$
- [70] $1^{111} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 44$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.36 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$
- [71] $1^{11} 3^1 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 49$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.24 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -2 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$
- [72] $1^{134} 3^1 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 60$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 8 & -2 \\ 3 & 6 & 0 & 3 & 16 & 27 & 6 \\ 5 & 9 & 0 & 8 & 27 & 42 & 12 \\ 2 & 2 & -1 & -2 & 6 & 12 & -1 \end{pmatrix}_{b_0}$$
- [73] $1^{111} 3^1 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 61$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.46 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 5 & -2 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 3 & 3 & 0 & 5 & 3 & 6 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[74] $1^{121} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

[75] $1^{12} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 68, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & 0 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -3 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[76] $1^{124} 3^{113} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 70, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 7 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -2 & 4 & 8 & -1 \end{pmatrix}_{b_0}$$

[77] $1^{112} 2^1 3^{13} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 74, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 1 & 0 & 3 & 1 & 1 & 6 & 3 \\ 5 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[78] $1^{111} 2^1 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 74, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[79] $1^{111} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 76, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.47 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[80] $1^{112} 2^1 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 90, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 14.42 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 1 & 0 & 3 & 1 & 1 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[81] $1^{112} 3^{123} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 90, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 16.02 \rightarrow 17, T_{\max} = 94, T \in [17, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & -3 \\ 2 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -3 & 0 & 3 & -3 \end{pmatrix}_{b_0}$$

[82] $1^{1231^{12}31^{114}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 6 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -2 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

[83] $1^{111} 3^{221^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 99, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.34 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 5 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 3 & 0 & 5 & 0 & 6 & 6 \\ 2 & 2 & -1 & -2 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

[84] $1^{111} 2^3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 102, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.69 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 2 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[85] $1^{112} 3^{221^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 6 & -2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 3 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[86] $1^{1231^{11}31^{112}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 109, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

[87] $1^{111} 3^{221^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[88] $1^{123} 3^{221^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 7 & -2 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 5 & 7 & 0 & 7 & 14 & 26 & 10 \\ 2 & 2 & -1 & -2 & 4 & 10 & -1 \end{pmatrix}_{b_0}$$

[89] $1^{13} 3^{221^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 115, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 16 & 6 \\ 2 & 2 & -1 & -2 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

[90] $1^{11} 2^3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$

12.0

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -3 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

[91] $1^{112} 3^{11} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[92] $1^{123} 3^{11} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 7 & -2 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 4 & 7 & 0 & 7 & 14 & 26 & 10 \\ 2 & 2 & -1 & -2 & 4 & 10 & -1 \end{pmatrix}_{b_0}$$

[93] $1^{1231} 3^{14} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 132, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 17, T_{\max} = 100, T \in [17, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 12 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & -2 & 0 & 16 & -3 \end{pmatrix}_{b_0}$$

[94] $1^{122} 3^{11} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.53 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & -2 \\ 2 & 4 & 0 & 2 & 6 & 8 & 4 \\ 2 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[95] $1^{13} 3^{111} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 0 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 7 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 7 & 0 & 7 & 0 & 26 & 10 \\ 2 & 2 & -1 & -2 & 0 & 10 & -1 \end{pmatrix}_{b_0}$$

[96] $1^{113} 3^{11} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.61 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 3 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[97] $1^{124} 3^{112} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 8 & -2 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 7 & 9 & 0 & 8 & 18 & 42 & 12 \\ 2 & 2 & -1 & -2 & 4 & 12 & -1 \end{pmatrix}_{b_0}$$

[98] $1^{134} 3^{111} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 147, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 8 & -2 \\ 4 & 6 & 0 & 3 & 16 & 27 & 6 \\ 6 & 9 & 0 & 8 & 27 & 42 & 12 \\ 2 & 2 & -1 & -2 & 6 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[99] \ 1^{111} 1^{12} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 148, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.49 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[100] \ 1^{123} 1^1 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 156, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 3 & -2 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 3 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -2 & 4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[101] \ 1^{123} 1^{11} 3 \ 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 6 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 5 & 0 & 6 & 0 & 14 & 8 \\ 2 & 2 & -1 & -2 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[102] \ 1^{133} 1^{11} 2 \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 164, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.6$$

$$T_{\min} = 14.85 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 9 & 2 \\ 1 & 0 & 1 & -5 & 3 & 3 & -3 \\ 4 & 0 & 9 & 3 & 25 & 27 & 9 \\ 4 & 0 & 9 & 3 & 27 & 25 & 9 \\ 2 & 0 & 2 & -3 & 9 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{133} 1^{11} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.6$$

$$T_{\min} = 14.61 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 16 & 18 & 6 \\ 4 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{1231} 1^{11} 31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 168, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 3 & 3 & -2 \\ 3 & 0 & 0 & 3 & 4 & 6 & 4 \\ 3 & 0 & 0 & 3 & 6 & 4 & 4 \\ 2 & 0 & -1 & -2 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{114} 1^{11} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 171, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.34 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -2 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{114} 1^1 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.84 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 4 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[107] $1^{133} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 176$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.6$

$T_{\min} = 14.86 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 16 & 18 & 6 \\ 3 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[108] $1^{112} 3^{124} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 176$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 15.82 \rightarrow 16$, $T_{\max} = 94$, $T \in [16, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & -3 \\ 2 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -3 & 0 & 3 & -2 \end{pmatrix}_{b_0}$$

[109] $1^{112} 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 176$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.36 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 1 & 0 & 3 & 1 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[110] $1^{132} 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 180$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 14.60 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 6 & 2 \\ 1 & 0 & 1 & -5 & 3 & 2 & -3 \\ 4 & 0 & 9 & 3 & 25 & 18 & 9 \\ 3 & 0 & 6 & 2 & 18 & 10 & 6 \\ 2 & 0 & 2 & -3 & 9 & 6 & -1 \end{pmatrix}_{b_0}$$

[111] $1^{12} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 184$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.33 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 2 & 0 & 6 & 4 \\ 2 & 2 & -1 & -2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

[112] $1^{124} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 184$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.91 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 4 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -2 & 4 & 8 & -1 \end{pmatrix}_{b_0}$$

[113] $1^{111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 189$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.28 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[114] $1^{112} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 195$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.30 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 6 & -2 \\ 2 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[115] $1^{123} 3^{12} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 196$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 14.37 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 3 & -2 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 5 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -2 & 4 & 6 & -1 \end{pmatrix}_{b_0}$$

[116] $1^{134} 3^{11} 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 204$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 15.02 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 4 & -2 \\ 3 & 6 & 0 & 3 & 16 & 24 & 6 \\ 4 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -2 & 6 & 8 & -2 \end{pmatrix}_{b_0}$$

[117] $1^{112} 2^3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 204$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 15.02 \rightarrow 16$, $T_{\max} = 94$, $T \in [16, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 1 & 6 & 3 \\ 4 & 6 & 0 & 2 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 3 & 6 & -2 \end{pmatrix}_{b_0}$$

[118] $1^{115} 3^{11} 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 208$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.1$

$T_{\min} = 15.04 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 5 & -2 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 5 & 10 & 0 & 5 & 10 & 48 & 10 \\ 2 & 2 & -1 & -2 & 2 & 10 & -2 \end{pmatrix}_{b_0}$$

[119] $1^{123} 3^{111} 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 211$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 14.31 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 7 & -2 \\ 3 & 4 & 0 & 2 & 6 & 14 & 4 \\ 5 & 7 & 0 & 7 & 14 & 26 & 10 \\ 2 & 2 & -1 & -2 & 4 & 10 & -1 \end{pmatrix}_{b_0}$$

[120] $1^{123} 131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -139$, $\det = 212$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 15.09 \rightarrow 16$, $T_{\max} = 100$, $T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -2 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

[122] $1^{122} 3^{11} 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 214$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.36 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & -2 \\ 2 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[121] $1^{112} 2^3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 214$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.48 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 1 & 6 & 3 \\ 3 & 6 & 0 & 2 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[123] $1^{12} 2^1 3^{112} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 218$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.39 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 6 & 2 \\ 1 & 0 & 1 & -5 & 0 & 2 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 0 & 6 & 2 & 0 & 10 & 6 \\ 2 & 0 & 2 & -3 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

[124] $1^{125} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 220$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.6$

$T_{\min} = 15.09 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 5 & -2 \\ 2 & 4 & 0 & 2 & 6 & 20 & 4 \\ 5 & 10 & 0 & 5 & 20 & 48 & 10 \\ 2 & 2 & -1 & -2 & 4 & 10 & -2 \end{pmatrix}_{b_0}$$

[125] $1^{231} 1^{11} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -139$, $\det = 231$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.70 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

[126] $1^{144} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 232$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.14 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 4 & -2 \\ 4 & 8 & 0 & 4 & 30 & 32 & 8 \\ 4 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & 8 & -2 \end{pmatrix}_{b_0}$$

[128] $1^{114} 2^1 3^{12} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 240$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.53 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 4 & -3 \\ 1 & 0 & 3 & 1 & 1 & 12 & 3 \\ 6 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[127] $1^{113} 2^1 3^{14} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 240$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.53 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 1 & 0 & 3 & 1 & 1 & 9 & 3 \\ 7 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[129] $1^{135} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 240$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.9$

$T_{\min} = 15.17 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 5 & -2 \\ 3 & 6 & 0 & 3 & 16 & 30 & 6 \\ 5 & 10 & 0 & 5 & 30 & 48 & 10 \\ 2 & 2 & -1 & -2 & 6 & 10 & -2 \end{pmatrix}_{b_0}$$

[130] $1^{116} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 252$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 15.21 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 6 & -2 \\ 1 & 2 & 0 & 1 & 0 & 12 & 2 \\ 6 & 12 & 0 & 6 & 12 & 70 & 12 \\ 2 & 2 & -1 & -2 & 2 & 12 & -2 \end{pmatrix}_{b_0}$$

[131] $1^{113} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 256$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.38 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 7 & -2 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 4 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[132] $1^{112} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 258$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.49 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -3 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

[133] $1^{113} 2^{11} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 260$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.66 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 \\ 4 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[134] $1^{112} 2^{11} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 260$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.35 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 2 & 0 & 3 & 1 & 1 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[135] $1^{113} 2^{11} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 264$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.39 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 1 & 0 & 3 & 1 & 1 & 9 & 3 \\ 5 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[136] $1^{126} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 264$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 15.24 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 6 & -2 \\ 2 & 4 & 0 & 2 & 6 & 24 & 4 \\ 6 & 12 & 0 & 6 & 24 & 70 & 12 \\ 2 & 2 & -1 & -2 & 4 & 12 & -2 \end{pmatrix}_{b_0}$$

[137] $1^{112} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 268$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.90 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 6 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[138] $1^{145} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 268$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.1$

$T_{\min} = 15.25 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 5 & -2 \\ 4 & 8 & 0 & 4 & 30 & 40 & 8 \\ 5 & 10 & 0 & 5 & 40 & 48 & 10 \\ 2 & 2 & -1 & -2 & 8 & 10 & -2 \end{pmatrix}_{b_0}$$

[139] $1^{113}2^13^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 278$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.42 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 1 & 9 & 3 \\ 4 & 9 & 0 & 3 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[140] $1^{124}3^{11}221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 280$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.43 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 8 & -2 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 5 & 9 & 0 & 8 & 18 & 42 & 12 \\ 2 & 2 & -1 & -2 & 4 & 12 & -1 \end{pmatrix}_{b_0}$$

[141] $1^{136}3^{11}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 284$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.1$

$T_{\min} = 15.30 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 6 & -2 \\ 3 & 6 & 0 & 3 & 16 & 36 & 6 \\ 6 & 12 & 0 & 6 & 36 & 70 & 12 \\ 2 & 2 & -1 & -2 & 6 & 12 & -2 \end{pmatrix}_{b_0}$$

[142] $1^{114}2^13^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 294$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.40 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 12 & 3 \\ 5 & 12 & 0 & 4 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[143] $1^{122}2^{124}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 296$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.68 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & -5 & 2 & 2 & -3 \\ 4 & 0 & 6 & 2 & 10 & 12 & 6 \\ 6 & 0 & 6 & 2 & 12 & 10 & 6 \\ 2 & 0 & 2 & -3 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[144] $1^{117}3^{11}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 304$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 15.34 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 7 & -2 \\ 1 & 2 & 0 & 1 & 0 & 14 & 2 \\ 7 & 14 & 0 & 7 & 14 & 96 & 14 \\ 2 & 2 & -1 & -2 & 2 & 14 & -2 \end{pmatrix}_{b_0}$$

[146] $1^{112}3^{114}221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 316$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.73 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 6 & -2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 7 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[145] $1^{127}3^{11}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 316$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 15.37 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 7 & -2 \\ 2 & 4 & 0 & 2 & 6 & 28 & 4 \\ 7 & 14 & 0 & 7 & 28 & 96 & 14 \\ 2 & 2 & -1 & -2 & 4 & 14 & -2 \end{pmatrix}_{b_0}$$

[147] $1^{1231}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 319, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 15.16 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 9 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & -2 & 0 & 12 & -2 \end{pmatrix}_{b_0}$$

[148] $1^{111}2^1 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 324, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[149] $1^{134}3^1 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 329, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 4 & -2 \\ 3 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -2 & 6 & 8 & -1 \end{pmatrix}_{b_0}$$

[150] $1^{134}2^1 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 330, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 12 & 2 \\ 1 & 0 & 1 & -5 & 3 & 4 & -3 \\ 4 & 0 & 9 & 3 & 25 & 36 & 9 \\ 6 & 0 & 12 & 4 & 36 & 46 & 12 \\ 2 & 0 & 2 & -3 & 9 & 12 & -1 \end{pmatrix}_{b_0}$$

[151] $1^{137}3^1 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 336, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.1$

$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 93, T \in [16, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 7 & -2 \\ 3 & 6 & 0 & 3 & 16 & 42 & 6 \\ 7 & 14 & 0 & 7 & 42 & 96 & 14 \\ 2 & 2 & -1 & -2 & 6 & 14 & -2 \end{pmatrix}_{b_0}$$

[152] $1^{112}3^1 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 337, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 14.32 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[153] $1^{112}3^1 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 337, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 6 & -2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[154] $1^{113}3^1 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 344, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 7 & -2 \\ 2 & 2 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[155] $1^{113}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 346, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 1 & 0 & 3 & 1 & 1 & 9 & 3 \\ 6 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[156] $1^{1231}12^131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 352, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.83 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 6 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -2 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

[158] $1^{122}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & -5 & 2 & 2 & -3 \\ 3 & 0 & 6 & 2 & 10 & 12 & 6 \\ 3 & 0 & 6 & 2 & 12 & 10 & 6 \\ 2 & 0 & 2 & -3 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[157] $1^{111}1^13^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 2 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[159] $1^{122}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 362, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & -5 & 2 & 2 & -3 \\ 3 & 0 & 6 & 2 & 10 & 12 & 6 \\ 4 & 0 & 6 & 2 & 12 & 10 & 6 \\ 2 & 0 & 2 & -3 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[160] $1^{123}1^13^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 370, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 3 & -2 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -2 & 4 & 6 & -1 \end{pmatrix}_{b_0}$$

[161] $1^{113}1^12^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 372, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.63 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -3 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

[162] $1^{111}1^12^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 386, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[163] $1^{13}2^1 3^{113} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 388$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.40 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 9 & 2 \\ 1 & 0 & 1 & -5 & 0 & 3 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 0 & 9 & 3 & 0 & 25 & 9 \\ 2 & 0 & 2 & -3 & 0 & 9 & -1 \end{pmatrix}_{b_0}$$

[164] $1^{124} 3^{111} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 406$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.35 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 8 & -2 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 6 & 9 & 0 & 8 & 18 & 42 & 12 \\ 2 & 2 & -1 & -2 & 4 & 12 & -1 \end{pmatrix}_{b_0}$$

[165] $1^{112} 3^{113} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 409$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.50 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 5 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[166] $1^{114} 2^1 3^{112} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 410$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.44 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 4 & -3 \\ 2 & 0 & 3 & 1 & 1 & 12 & 3 \\ 6 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[167] $1^{114} 3^{111} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 411$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.32 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 8 & -2 \\ 2 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[168] $1^{114} 3^{111} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 412$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 8 & -2 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 5 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[169] $1^{114} 3^{111} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 419$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 2 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[170] $1^{112} 3^{112} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 432$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.37 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 4 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[171] $1^{112} 3^{13} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 441$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.44 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 6 & -2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 6 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[172] $1^{124} 3^{111} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 448$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.32 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 8 & -2 \\ 3 & 4 & 0 & 2 & 6 & 18 & 4 \\ 6 & 9 & 0 & 8 & 18 & 42 & 12 \\ 2 & 2 & -1 & -2 & 4 & 12 & -1 \end{pmatrix}_{b_0}$$

[173] $1^{1231} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -139$, $\det = 448$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.75 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 3 & -2 \\ 2 & 6 & 0 & 3 & 4 & 6 & 4 \\ 2 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[174] $1^{112} 3^{12} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 448$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.34 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 6 & -2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 5 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[175] $1^{113} 2^3 21^{12} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 452$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.62 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 1 & 9 & 3 \\ 5 & 9 & 0 & 3 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[176] $1^{112} 3^1 21^{114} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 458$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.82 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -3 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

[177] $1^{112} 3^1 221^{111} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 459$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.34 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[178] $1^{231} 31^{111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -139$, $\det = 477$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.71 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 3 & -2 \\ 2 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[179] $1^{113} 1^{111} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 93, \ \Delta = -131, \ \det = 480, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.8$

$$T_{\min} = 14.36 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 2 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[180] $1^{112} 2^1 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 94, \ \Delta = -133, \ \det = 488, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$

$$T_{\min} = 14.41 \rightarrow 15, \ T_{\max} = 94, \ T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 3 & 0 & 3 & 1 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[181] $1^{111} 2^1 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 94, \ \Delta = -133, \ \det = 488, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$

$$T_{\min} = 14.80 \rightarrow 15, \ T_{\max} = 94, \ T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 \\ 5 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[182] $1^{113} 1^{11} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 93, \ \Delta = -131, \ \det = 493, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.8$

$$T_{\min} = 14.37 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[183] $1^{1231} 1^{231} 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 496, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$

$$T_{\min} = 14.69 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 6 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -2 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

[184] $1^{124} 1^{12} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 93, \ \Delta = -131, \ \det = 496, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$

$$T_{\min} = 14.39 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 6 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -2 & 4 & 8 & -1 \end{pmatrix}_{b_0}$$

[185] $1^{114} 2^1 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 94, \ \Delta = -133, \ \det = 506, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$

$$T_{\min} = 14.39 \rightarrow 15, \ T_{\max} = 94, \ T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 4 & -3 \\ 1 & 0 & 3 & 1 & 1 & 12 & 3 \\ 7 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[186] $1^{122} 1^{113} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 94, \ \Delta = -133, \ \det = 516, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$

$$T_{\min} = 14.46 \rightarrow 15, \ T_{\max} = 94, \ T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 6 & 2 \\ 1 & 0 & 1 & -5 & 0 & 2 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 0 & 6 & 2 & 0 & 10 & 6 \\ 2 & 0 & 2 & -3 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

[187] $1^{144}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 522$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.44 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 12 & 2 \\ 1 & 0 & 1 & -5 & 4 & 4 & -3 \\ 6 & 0 & 12 & 4 & 46 & 48 & 12 \\ 7 & 0 & 12 & 4 & 48 & 46 & 12 \\ 2 & 0 & 2 & -3 & 12 & 12 & -1 \end{pmatrix}_{b_0}$$

[188] $1^{123}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 532$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 14.38 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 9 & 2 \\ 1 & 0 & 1 & -5 & 2 & 3 & -3 \\ 3 & 0 & 6 & 2 & 10 & 18 & 6 \\ 6 & 0 & 9 & 3 & 18 & 25 & 9 \\ 2 & 0 & 2 & -3 & 6 & 9 & -1 \end{pmatrix}_{b_0}$$

[189] $1^{124}3^{11}221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 534$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.45 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -2 & 4 & 8 & -1 \end{pmatrix}_{b_0}$$

[190] $1^{1231}1^{11}31^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -139$, $\det = 549$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.66 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 3 & -2 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[191] $1^{1231}1^{12}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -139$, $\det = 582$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.71 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 6 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -2 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

[192] $1^{114}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 584$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.40 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 4 & -3 \\ 1 & 0 & 3 & 1 & 1 & 12 & 3 \\ 8 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[193] $1^{1231}1^{11}31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -139$, $\det = 592$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.65 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 3 & -2 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[194] $1^{111}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 608$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.45 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[195] $1^{113} 3^{11} 221 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 621$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.30 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 7 & -2 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[196] $1^{142} 3^{114} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 626$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 12 & 2 \\ 1 & 0 & 1 & -5 & 0 & 4 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 0 & 12 & 4 & 0 & 46 & 12 \\ 2 & 0 & 2 & -3 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

[197] $1^{1112} 3^{123} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 642$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.55 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 \\ 4 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[199] $1^{122} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 652$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.35 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & -2 \\ 3 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[198] $1^{1142} 3^{112} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 652$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.52 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 12 & 3 \\ 6 & 12 & 0 & 4 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[200] $1^{114} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 657$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[201] $1^{1132} 3^{123} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 658$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 3 & 0 & 3 & 1 & 1 & 9 & 3 \\ 6 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[202] $1^{1122} 3^{112} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 666$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.37 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 2 & 0 & 3 & 1 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[203] $1^{113}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 674$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.38 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 2 & 0 & 3 & 1 & 1 & 9 & 3 \\ 5 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[204] $1^{123}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 690$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 14.39 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 9 & 2 \\ 1 & 0 & 1 & -5 & 2 & 3 & -3 \\ 3 & 0 & 6 & 2 & 10 & 18 & 6 \\ 5 & 0 & 9 & 3 & 18 & 25 & 9 \\ 2 & 0 & 2 & -3 & 6 & 9 & -1 \end{pmatrix}_{b_0}$$

[205] $1^{113}3^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 724$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.36 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 5 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[206] $1^{124}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 746$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.44 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 12 & 2 \\ 1 & 0 & 1 & -5 & 2 & 4 & -3 \\ 3 & 0 & 6 & 2 & 10 & 24 & 6 \\ 6 & 0 & 12 & 4 & 24 & 46 & 12 \\ 2 & 0 & 2 & -3 & 6 & 12 & -1 \end{pmatrix}_{b_0}$$

[207] $1^{123}3^1221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 763$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 14.37 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 3 & -2 \\ 3 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -2 & 4 & 6 & -1 \end{pmatrix}_{b_0}$$

[208] $1^{124}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 770$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.39 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 12 & 2 \\ 1 & 0 & 1 & -5 & 2 & 4 & -3 \\ 3 & 0 & 6 & 2 & 10 & 24 & 6 \\ 8 & 0 & 12 & 4 & 24 & 46 & 12 \\ 2 & 0 & 2 & -3 & 6 & 12 & -1 \end{pmatrix}_{b_0}$$

[209] $1^{123}2^13^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 776$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 14.38 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 9 & 2 \\ 1 & 0 & 1 & -5 & 2 & 3 & -3 \\ 4 & 0 & 6 & 2 & 10 & 18 & 6 \\ 5 & 0 & 9 & 3 & 18 & 25 & 9 \\ 2 & 0 & 2 & -3 & 6 & 9 & -1 \end{pmatrix}_{b_0}$$

[210] $1^{122}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 786$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.53 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 6 & 2 \\ 1 & 0 & 1 & -5 & 0 & 2 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 0 & 6 & 2 & 0 & 10 & 6 \\ 2 & 0 & 2 & -3 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

[211] $1^{124} 1^{111} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 93, \ \Delta = -131, \ \det = 792, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$

$T_{\min} = 14.40 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 3 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -2 & 4 & 8 & -1 \end{pmatrix}_{b_0}$$

[212] $1^{113} 1^{14} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 93, \ \Delta = -131, \ \det = 832, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.8$

$T_{\min} = 14.49 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 7 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[213] $1^{231} 1^{113} 1^{111} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 836, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$

$T_{\min} = 14.67 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[214] $1^{113} 1^{13} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 93, \ \Delta = -131, \ \det = 837, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.8$

$T_{\min} = 14.41 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 6 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[215] $1^{143} 1^{122} 2 \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 94, \ \Delta = -133, \ \det = 840, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.8$

$T_{\min} = 14.43 \rightarrow 15, \ T_{\max} = 94, \ T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 9 & 2 \\ 1 & 0 & 1 & -5 & 4 & 3 & -3 \\ 6 & 0 & 12 & 4 & 46 & 36 & 12 \\ 5 & 0 & 9 & 3 & 36 & 25 & 9 \\ 2 & 0 & 2 & -3 & 12 & 9 & -1 \end{pmatrix}_{b_0}$$

[216] $1^{132} 1^{114} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 94, \ \Delta = -133, \ \det = 842, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.8$

$T_{\min} = 14.44 \rightarrow 15, \ T_{\max} = 94, \ T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 9 & 2 \\ 1 & 0 & 1 & -5 & 0 & 3 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 0 & 9 & 3 & 0 & 25 & 9 \\ 2 & 0 & 2 & -3 & 0 & 9 & -1 \end{pmatrix}_{b_0}$$

[217] $1^{113} 1^{12} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 93, \ \Delta = -131, \ \det = 868, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.8$

$T_{\min} = 14.31 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 7 & -2 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 6 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[218] $1^{112} 1^{113} 2 \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 94, \ \Delta = -133, \ \det = 884, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$

$T_{\min} = 14.41 \rightarrow 15, \ T_{\max} = 94, \ T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 2 & 0 & 3 & 1 & 1 & 6 & 3 \\ 5 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[219] $1^{114}2^13^{124}21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 896, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 4 & -3 \\ 3 & 0 & 3 & 1 & 1 & 12 & 3 \\ 8 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[220] $1^{1231}13^{112}31^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 901, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.78 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 9 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & -2 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

[221] $1^{114}3^{111}221^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 913, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.32 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 8 & -2 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[222] $1^{112}2^13^{114}21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 914, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.49 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 2 & 0 & 3 & 1 & 1 & 6 & 3 \\ 6 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[223] $1^{133}2^13^{123}21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 946, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.6$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 9 & 2 \\ 1 & 0 & 1 & -5 & 3 & 3 & -3 \\ 5 & 0 & 9 & 3 & 25 & 27 & 9 \\ 6 & 0 & 9 & 3 & 27 & 25 & 9 \\ 2 & 0 & 2 & -3 & 9 & 9 & -1 \end{pmatrix}_{b_0}$$

[224] $1^{133}3^{111}221^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 964, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.6$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 3 & -2 \\ 4 & 6 & 0 & 3 & 16 & 18 & 6 \\ 4 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[225] $1^{122}2^13^{123}21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 986, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & -5 & 2 & 2 & -3 \\ 4 & 0 & 6 & 2 & 10 & 12 & 6 \\ 5 & 0 & 6 & 2 & 12 & 10 & 6 \\ 2 & 0 & 2 & -3 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[226] $1^{113}3^{113}221^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 997, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 7 & -2 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 7 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[227] $1^{122}2^13^121^1+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{violet}{e}_7+\textcolor{brown}{su}_2+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 1008$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.38 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & -5 & 2 & 2 & -3 \\ 4 & 0 & 6 & 2 & 10 & 12 & 6 \\ 4 & 0 & 6 & 2 & 12 & 10 & 6 \\ 2 & 0 & 2 & -3 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[228] $1^{113}3^1221^1+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{violet}{e}_7+\textcolor{brown}{su}_2+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 1008$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 7 & -2 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 8 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[229] $1^{114}3^1221^1+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{violet}{e}_7+\textcolor{brown}{su}_2+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 1024$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.38 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 6 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[230] $1^{112}2^13^121^1+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{violet}{e}_7+\textcolor{brown}{su}_2+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 1026$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.43 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 3 & 0 & 3 & 1 & 1 & 6 & 3 \\ 5 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[231] $1^{231}1^{431}1^{12}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{violet}{e}_7+\textcolor{brown}{su}_2+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -139$, $\det = 1066$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.88 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 12 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & -2 & 0 & 16 & -1 \end{pmatrix}_{b_0}$$

[232] $1^{134}3^1221^1+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{violet}{e}_7+\textcolor{brown}{su}_2+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 1083$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 4 & -2 \\ 4 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -2 & 6 & 8 & -1 \end{pmatrix}_{b_0}$$

[233] $1^{124}2^13^121^1+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{violet}{e}_7+\textcolor{brown}{su}_2+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 1092$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.39 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 12 & 2 \\ 1 & 0 & 1 & -5 & 2 & 4 & -3 \\ 3 & 0 & 6 & 2 & 10 & 24 & 6 \\ 7 & 0 & 12 & 4 & 24 & 46 & 12 \\ 2 & 0 & 2 & -3 & 6 & 12 & -1 \end{pmatrix}_{b_0}$$

[234] $1^{231}1^{331}1^{14}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{violet}{e}_7+\textcolor{brown}{su}_2+\textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -139$, $\det = 1117$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.69 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 9 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & -2 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

[235] $1^{144}2^13^{122}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 12 & 2 \\ 1 & 0 & 1 & -5 & 4 & 4 & -3 \\ 6 & 0 & 12 & 4 & 46 & 48 & 12 \\ 6 & 0 & 12 & 4 & 48 & 46 & 12 \\ 2 & 0 & 2 & -3 & 12 & 12 & -1 \end{pmatrix}_{b_0}$$

[236] $1^{231}1^{13}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 1167, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 9 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & -2 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

[237] $1^{133}2^13^{122}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.6$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 9 & 2 \\ 1 & 0 & 1 & -5 & 3 & 3 & -3 \\ 5 & 0 & 9 & 3 & 25 & 27 & 9 \\ 5 & 0 & 9 & 3 & 27 & 25 & 9 \\ 2 & 0 & 2 & -3 & 9 & 9 & -1 \end{pmatrix}_{b_0}$$

[238] $1^{134}2^13^{124}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 12 & 2 \\ 1 & 0 & 1 & -5 & 3 & 4 & -3 \\ 5 & 0 & 9 & 3 & 25 & 36 & 9 \\ 8 & 0 & 12 & 4 & 36 & 46 & 12 \\ 2 & 0 & 2 & -3 & 9 & 12 & -1 \end{pmatrix}_{b_0}$$

[239] $1^{113}2^13^{113}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1236, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 2 & 0 & 3 & 1 & 1 & 9 & 3 \\ 6 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[240] $1^{114}1^{13}3^{113}221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 1273, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 7 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[241] $1^{144}1^{111}3^{111}221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 1292, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.42 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 4 & -2 \\ 5 & 8 & 0 & 4 & 30 & 32 & 8 \\ 5 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & 8 & -1 \end{pmatrix}_{b_0}$$

[242] $1^{114}1^{12}3^{112}221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 1296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 8 & -2 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 7 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[243] $1^{114}2^1 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1316, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 4 & -3 \\ 2 & 0 & 3 & 1 & 1 & 12 & 3 \\ 7 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[244] $1^{112}2^1 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1376, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.48 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 3 & 0 & 3 & 1 & 1 & 6 & 3 \\ 6 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[245] $1^{114}3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 1404, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.42 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 8 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[246] $1^{124}2^1 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1418, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 12 & 2 \\ 1 & 0 & 1 & -5 & 2 & 4 & -3 \\ 4 & 0 & 6 & 2 & 10 & 24 & 6 \\ 7 & 0 & 12 & 4 & 24 & 46 & 12 \\ 2 & 0 & 2 & -3 & 6 & 12 & -1 \end{pmatrix}_{b_0}$$

[247] $1^{113}2^1 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 3 & 0 & 3 & 1 & 1 & 9 & 3 \\ 7 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[248] $1^{114}3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 1561, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 8 & -2 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 8 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[249] $1^{123}2^1 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1578, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 9 & 2 \\ 1 & 0 & 1 & -5 & 2 & 3 & -3 \\ 4 & 0 & 6 & 2 & 10 & 18 & 6 \\ 6 & 0 & 9 & 3 & 18 & 25 & 9 \\ 2 & 0 & 2 & -3 & 6 & 9 & -1 \end{pmatrix}_{b_0}$$

[250] $1^{113}2^1 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1610, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 2 & 0 & 3 & 1 & 1 & 9 & 3 \\ 7 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[251] $1^{1231}1^{1431}1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 1684, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 12 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 11 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & -2 & 0 & 16 & -1 \end{pmatrix}_{b_0}$$

[252] $1^{114}1^{14}3^{221}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 1708, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.36 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 8 & -2 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 9 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[253] $1^{123}2^{124}3^{21}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1712, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 9 & 2 \\ 1 & 0 & 1 & -5 & 2 & 3 & -3 \\ 4 & 0 & 6 & 2 & 10 & 18 & 6 \\ 7 & 0 & 9 & 3 & 18 & 25 & 9 \\ 2 & 0 & 2 & -3 & 6 & 9 & -1 \end{pmatrix}_{b_0}$$

[254] $1^{134}2^{123}3^{21}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1746, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 12 & 2 \\ 1 & 0 & 1 & -5 & 3 & 4 & -3 \\ 5 & 0 & 9 & 3 & 25 & 36 & 9 \\ 7 & 0 & 12 & 4 & 36 & 46 & 12 \\ 2 & 0 & 2 & -3 & 9 & 12 & -1 \end{pmatrix}_{b_0}$$

[255] $1^{1231}1^{1431}1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 1986, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 12 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 12 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 12 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & -2 & 0 & 16 & -1 \end{pmatrix}_{b_0}$$

[256] $1^{114}1^{14}3^{21}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 2034, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 4 & -3 \\ 2 & 0 & 3 & 1 & 1 & 12 & 3 \\ 8 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[257] $1^{124}2^{124}3^{21}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 2376, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 12 & 2 \\ 1 & 0 & 1 & -5 & 2 & 4 & -3 \\ 4 & 0 & 6 & 2 & 10 & 24 & 6 \\ 8 & 0 & 12 & 4 & 24 & 46 & 12 \\ 2 & 0 & 2 & -3 & 6 & 12 & -1 \end{pmatrix}_{b_0}$$

4.183 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$

$T = 10 \quad [0] 1^{11}1^{111}3^{21}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 102, \Delta = -159, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.58 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -3 & 0 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \quad 1^{1111} 2^3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 11, T_{\max} = 95, \Delta = -132, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.50 \rightarrow 16, T_{\max} = 95, T \in [16, 95]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & 4 & -2 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 4 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 1^{11} 3 \quad 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 11, T_{\max} = 94, \Delta = -130, \det = -132, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.64 \rightarrow 16, T_{\max} = 94, T \in [16, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 1^{11} 2^3 \quad 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 11, T_{\max} = 95, \Delta = -132, \det = -200, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.44 \rightarrow 16, T_{\max} = 95, T \in [16, 95]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 4 & 4 & -2 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 5 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

4.184 NHC(-2)(-3) + $\textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$

$$T = 10 \quad [1] \quad 1^{111} 1^{11} 3 \quad 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -130, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 1 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & -1 & -3 & 0 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{1112} 1^{13} 3 \quad 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -130, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 19.80 \rightarrow 20, T_{\max} = 93, T \in [20, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 4 & 2 \\ 5 & 4 & 0 & 2 & 4 & 4 & 6 & 4 \\ 2 & 2 & -1 & -3 & 2 & 2 & 4 & -7 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1211} 1^{11} 3 \quad 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -130, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & 1 & -3 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 4 & 2 & 0 & 2 \\ 2 & 2 & -1 & -3 & 4 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1211} 1^{11} 3 \quad 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -130, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$[11] \quad 1^{1122} 3^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{e}_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{teal}{su}_2 \quad T = 10, \quad T_{\max} = 93, \quad \Delta = -130, \quad \det = 356, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 4 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 8 & 4 \\ 2 & 4 & 0 & 2 & 4 & 8 & 6 & 4 \\ 2 & 2 & -1 & -3 & 2 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

[12] $1^{1111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 93, \quad \Delta = -130, \quad \det = 360, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & 0 & 1 & 2 & 2 & 0 \\ 2 & 2 & -1 & -3 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1113} 3^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{teal}{su}_2 \quad T = 10, \quad T_{\max} = 93, \quad \Delta = -130, \quad \det = 380, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.48 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 6 \\ 1 & 2 & 0 & 1 & 2 & 0 & 6 \\ 3 & 6 & 0 & 3 & 6 & 6 & 16 \\ 2 & 2 & -1 & -3 & 2 & 2 & 6 \end{pmatrix}_{b_0}$$

[14] $1^{1122} 3^{21} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 93, \Delta = -130, \det = 430, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 4 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 4 & 8 & 6 & 4 \\ 2 & 2 & -1 & -3 & 2 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

[15] $1^{1114}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 93, \quad \Delta = -130, \quad \det = 464, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$

$$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 4 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -3 & 2 & 2 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1115} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 93, \quad \Delta = -130, \quad \det = 572, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.1$$

$$T_{\min} = 14.77 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 5 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 10 & 2 \\ 5 & 10 & 0 & 5 & 10 & 10 & 48 & 10 \\ 2 & 2 & -1 & -3 & 2 & 2 & 10 & -2 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{1112} 3^{21} 1^{12} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{s}u_2 + \textcolor{red}{s}u_2 + \textcolor{blue}{s}u_2 \quad T = 10, \quad T_{\max} = 93, \quad \Delta = -130, \quad \det = 624, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 4 & 2 \\ 4 & 4 & 0 & 2 & 4 & 4 & 6 & 4 \\ 2 & 2 & -1 & -3 & 2 & 2 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1111} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 93, \ \Delta = -130, \ \det = 656, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

[26] $1^{1122} 3^{21} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{c}_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 93, \Delta = -130, \det = 1384, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.27 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 4 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 4 & 8 & 6 & 4 \\ 2 & 2 & -1 & -3 & 2 & 4 & 4 & -2 \end{pmatrix} b_0$$

[27] $1^{1114} 3^{21} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 93, \Delta = -130, \det = 1524, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.32 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -3 & 2 & 2 & 8 & -2 \end{pmatrix} b_0$$

[28] $1^{1114} 3^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 93, \Delta = -130, \det = 1680, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 \\ 8 & 8 & 0 & 4 & 8 & 8 & 30 \\ 2 & 2 & -1 & -3 & 2 & 2 & 8 \end{pmatrix} b_0$$

[29] $1^{1114} 3^{21} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 93, \Delta = -130, \det = 2080, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 \\ 6 & 8 & 0 & 4 & 8 & 8 & 30 \\ 2 & 2 & -1 & -3 & 2 & 2 & 8 \end{pmatrix} b_0$$

[30] $1^{1114} 3^{21} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 93, \Delta = -130, \det = 2132, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 \\ 7 & 8 & 0 & 4 & 8 & 8 & 30 \\ 2 & 2 & -1 & -3 & 2 & 2 & 8 \end{pmatrix} b_0$$

$$T = 11 \quad [32] \ 1^{1112} 3^{13} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -16, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 20.25 \rightarrow 21, T_{\max} = 23, T \in [21, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 \\ 1 & 2 & 0 & 1 & 2 & 0 & 4 \\ 5 & 4 & 0 & 2 & 4 & 4 & 6 \\ 2 & 2 & -1 & -2 & 2 & 2 & -7 \end{pmatrix} b_0$$

[31] $1^{111}3^{221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -101$, $\det = -16$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 \\ 2 & 2 & -1 & -2 & 0 & 2 & -1 \end{pmatrix}_{b_2}$$

[34] $1^1 2^3 3^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 22.33 \rightarrow 23, T_{\max} = 23, T \in [23, 23]$$

$$[41] \ 1^{1112} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -88, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.45 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 2 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 & 6 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{12} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -108, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.07 \rightarrow 16, \ T_{\max} = 23, \ T \in [16, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & 0 & 0 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -3 & 0 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{1112} 3 \ 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -128, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.50 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 6 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 5 & 2 \\ 5 & 5 & 0 & 6 & 5 & 5 & 14 & 8 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{1121} 3 \ 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -128, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.38 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 6 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 & 2 \\ 3 & 5 & 0 & 6 & 5 & 5 & 14 & 8 \\ 1 & 2 & 0 & 1 & 2 & 5 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{113} 3 \ 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -130, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.31 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & 7 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 7 & 2 \\ 5 & 7 & 0 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 0 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{111} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -170, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.35 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 0 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{112} 3 \ 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -172, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.30 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & 6 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 5 & 2 \\ 4 & 5 & 0 & 6 & 0 & 5 & 14 & 8 \\ 2 & 2 & -1 & -2 & 0 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{112} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -196, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.33 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

[56] $1^{1112}3221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -256, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 14.44 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[57] $1^{1122}3221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -280, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 8 & 4 \\ 2 & 4 & 0 & 2 & 4 & 8 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[58] $1^{1122}2^1321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -280, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.49 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & 2 & -3 \\ 1 & 0 & 3 & 1 & 1 & 6 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 10 & 12 & 6 \\ 3 & 0 & 6 & 2 & 6 & 12 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[60] $1^{1113}3221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -296, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 14.65 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 6 & 2 \\ 3 & 6 & 0 & 3 & 6 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[59] $1^{1111}3221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -296, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.28 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[61] $1^{1112}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -300, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.39 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 0 & 3 & 1 & 0 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 0 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 0 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[62] $1^{112}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -328, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.66 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 0 & 3 & 1 & 0 & 0 & 1 & 3 \\ 2 & 0 & 2 & -3 & 0 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

[63] $1^{1112}3221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -336, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 14.30 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 6 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 5 & 2 \\ 4 & 5 & 0 & 6 & 5 & 5 & 14 & 8 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[64] $1^{1122} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -346, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 4 & 8 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[65] $1^{1113} 2^1 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -348, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 9 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 9 & 3 \\ 4 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[66] $1^{1114} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -352, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 4 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[67] $1^{1114} 2^1 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -380, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 12 & 3 \\ 5 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[68] $1^{1113} 2^1 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -388, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 3 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 & 9 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 9 & 3 \\ 7 & 0 & 9 & 3 & 9 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[69] $1^{1114} 2^1 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -408, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.49 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 4 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 & 12 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 12 & 3 \\ 6 & 0 & 12 & 4 & 12 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[70] $1^{1112} 2^1 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -420, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 2 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 & 6 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[73] $1^{1115}3221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -424, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.1$
 $T_{\min} = 15.06 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 5 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 10 & 2 \\ 5 & 10 & 0 & 5 & 10 & 10 & 48 & 10 \\ 2 & 2 & -1 & -2 & 2 & 2 & 10 & -2 \end{pmatrix}_{b_0}$$

[72] $1^{1113}3221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -424, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 14.38 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 7 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 7 & 2 \\ 4 & 7 & 0 & 7 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 2 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[71] $1^{1112}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -424, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.51 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 0 & 3 & 1 & 0 & 1 & 3 & 3 \\ 3 & 0 & 3 & 1 & 0 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 0 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[74] $1^{1113}2321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -436, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 14.39 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 3 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 & 9 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 9 & 3 \\ 5 & 0 & 9 & 3 & 9 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[75] $1^{1112}2321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -504, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & 2 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 0 & 3 & 1 & 0 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 0 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 0 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[76] $1^{1112}3221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -512, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 14.38 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 4 & 2 \\ 4 & 4 & 0 & 2 & 4 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[77] $1^{1113}3221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -540, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 14.30 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 7 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 7 & 2 \\ 5 & 7 & 0 & 7 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 2 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[78] $1^{1111}3221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -540, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[79] $1^{1121} 2^1 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -588, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.36 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & 1 & -3 \\ 2 & 0 & 3 & 1 & 1 & 6 & 3 & 3 \\ 3 & 0 & 6 & 2 & 6 & 10 & 6 & 6 \\ 2 & 0 & 3 & 1 & 3 & 6 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

[80] $1^{1112} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -592, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[82] $1^{1113} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -616, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 7 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 7 & 2 \\ 7 & 7 & 0 & 7 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -2 & 2 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[81] $1^{1114} 2^1 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -616, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 12 & 3 \\ 6 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[83] $1^{1122} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -620, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & 2 & -3 \\ 2 & 0 & 3 & 1 & 1 & 6 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 10 & 12 & 6 \\ 3 & 0 & 6 & 2 & 6 & 12 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[84] $1^{1122} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -624, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & 2 & -3 \\ 1 & 0 & 3 & 1 & 1 & 6 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 10 & 12 & 6 \\ 4 & 0 & 6 & 2 & 6 & 12 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[85] $1^{1114} 3^1 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -666, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[86] $1^{1114} 3^{221} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -674, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.32 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 8 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 9 & 2 \\ 6 & 9 & 0 & 8 & 9 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[87] $1^{1231} 1^{11} 31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -109, \det = -696, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.71 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & -1 & -2 & 0 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[88] $1^{1114} 3^{221} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -736, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.41 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 8 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 9 & 2 \\ 5 & 9 & 0 & 8 & 9 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[89] $1^{1112} 3^{221} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -746, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 14.34 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[90] $1^{1113} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -760, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 14.39 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 3 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 & 9 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 9 & 3 \\ 6 & 0 & 9 & 3 & 9 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[91] $1^{1113} 3^{221} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -101, \det = -788, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.8$
 $T_{\min} = 14.37 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[92] $1^{1122} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -792, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.37 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & 2 & -3 \\ 2 & 0 & 3 & 1 & 1 & 6 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 10 & 12 & 6 \\ 4 & 0 & 6 & 2 & 6 & 12 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[93] $1^{1111} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -840, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.41 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$[101] \ 1^{1122} 3^{111} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_7 + \textcolor{orange}{s}\textcolor{blue}{u}_2 + \textcolor{red}{s}\textcolor{blue}{u}_2 + \textcolor{blue}{s}\textcolor{blue}{u}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -1132, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 4 & 8 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & 4 & -1 \end{pmatrix} b_0$$

$$[102] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{4}} 3^{\textcolor{blue}{1}\textcolor{blue}{1}} 2 2 1^{\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}\textcolor{blue}{7} + \textcolor{blue}{s}\textcolor{red}{u}_2 + \textcolor{red}{s}\textcolor{blue}{u}_2 + \textcolor{blue}{s}\textcolor{blue}{u}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -1232, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.42 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 & -1 \end{pmatrix} b_0$$

$$[103] \ 1^{1114} 2^1 3^{114} 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -1236, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 4 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 & 12 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 12 & 3 \\ 8 & 0 & 12 & 4 & 12 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{1114} 3 \ 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -1248, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 8 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 9 & 2 \\ 9 & 9 & 0 & 8 & 9 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 2 & 12 & -1 \end{pmatrix} b_0$$

$$[105] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{4}} 3^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{4}} 221^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}\textcolor{blue}{g} + \textcolor{red}{s}\textcolor{blue}{u}_2 + \textcolor{red}{s}\textcolor{blue}{u}_2 + \textcolor{red}{s}\textcolor{blue}{u}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -1376, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 8 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 & -1 \end{pmatrix} b_0$$

$$[106] \ 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{3}} 2^{\textcolor{red}{1}} 3^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{2}} 21^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}\textcolor{blue}{g}_7 + \textcolor{red}{s}\textcolor{blue}{u}_2 + \textcolor{red}{s}\textcolor{blue}{u}_2 + \textcolor{blue}{s}\textcolor{blue}{u}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -1416, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 3 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 & 9 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 9 & 3 \\ 5 & 0 & 9 & 3 & 9 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 3 & 3 & 9 & -1 \end{pmatrix} b_0$$

$$[107] \ 1^{1114} 3^{221} 1^{11} + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{orange}{s} \textcolor{blue}{u}_2 + \textcolor{red}{s} \textcolor{blue}{u}_2 + \textcolor{blue}{s} \textcolor{blue}{u}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -1488, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 8 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 9 & 2 \\ 6 & 9 & 0 & 8 & 9 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 2 & 12 & -1 \end{pmatrix} b_0$$

$$[108] \ 1^{\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{2}} 2^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}\textcolor{blue}{2}} 3^{\textcolor{red}{1}} 21^{\textcolor{red}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -1560, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 2 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 & 6 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{1114} 3^{12} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -1696, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.38 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 6 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{1114} 3^{13} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -1744, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.39 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 7 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{1114} 3^{13} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -1744, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.33 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 8 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 9 & 2 \\ 8 & 9 & 0 & 8 & 9 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{1114} 3^{12} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -101, \ \det = -1824, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.32 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 8 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 9 & 2 \\ 7 & 9 & 0 & 8 & 9 & 9 & 42 & 12 \\ 2 & 2 & -1 & -2 & 2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

4.185 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{1112} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -100, \ \det = -64, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.25 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 1 & 1 & 2 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 0 & 4 & 4 & 6 & 4 \\ 2 & 2 & -1 & -3 & 0 & 2 & 2 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -100, \ \det = -524, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.22 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & 2 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1111} 3^{11} 321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -100, \ \det = -592, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.24 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 5 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 6 & 3 \\ 5 & 6 & 0 & 4 & 6 & 6 & 10 & 6 \\ 3 & 3 & -1 & -2 & 3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

4.187 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 11 \quad [0] \quad 1^{1111} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 16, T_{\max} = 23, T \in [16, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 & 0 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 & 0 \\ 1 & 2 & 0 & 1 & 2 & 2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -3 & 2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -202, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.85 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [2] \quad 1^{1111} 23 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 3 & 0 \\ 2 & 3 & -1 & -3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 3 \quad 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 166, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.95 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11121} 2 \quad 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 244, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 2 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 6 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 6 & 3 & 3 \\ 3 & 6 & 0 & 2 & 6 & 6 & 10 & 6 & 6 \\ 1 & 3 & 0 & 1 & 3 & 3 & 6 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 3 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11141} 2 \quad 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 472, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 12 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 12 & 3 & 3 \\ 6 & 12 & 0 & 4 & 12 & 12 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 3 & 3 & 12 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 3 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11131} 2 \quad 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 628, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.8$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 3 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 9 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 9 & 3 & 3 \\ 4 & 9 & 0 & 3 & 9 & 9 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 3 & 3 & 9 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 3 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11141} 2^3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 724, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.77 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 12 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 12 & 3 & 3 \\ 5 & 12 & 0 & 4 & 12 & 12 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 3 & 3 & 12 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 3 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

4.188 NHC(-2)(-3) + $\mathbf{e}_7$ + $\mathbf{su}_2$ + $\mathbf{su}_2$ + $\mathbf{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{113} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.50 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 1 & 9 & 3 \\ 6 & 9 & 0 & 6 & 9 & 25 & 9 \\ 3 & 3 & -1 & -2 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{121} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 2 & -2 \\ 4 & 6 & 0 & 4 & 10 & 6 & 6 \\ 3 & 3 & 0 & 2 & 6 & 1 & 3 \\ 3 & 3 & -1 & -2 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{122} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 4 & -2 \\ 4 & 6 & 0 & 4 & 10 & 12 & 6 \\ 4 & 6 & 0 & 4 & 12 & 10 & 6 \\ 3 & 3 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 \\ 3 & 3 & -1 & -2 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 1 & 6 & 3 \\ 4 & 6 & 0 & 4 & 6 & 10 & 6 \\ 3 & 3 & -1 & -2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.36 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 10 & 3 \\ 2 & 1 & 1 & 2 & 6 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 8 & -2 \\ 4 & 6 & 0 & 4 & 10 & 24 & 6 \\ 10 & 12 & 0 & 8 & 24 & 46 & 12 \\ 3 & 3 & -1 & -2 & 6 & 12 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{113}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 808, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 1 & 9 & 3 \\ 7 & 9 & 0 & 6 & 9 & 25 & 9 \\ 3 & 3 & -1 & -2 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[15] $1^{112}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 824, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.42 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 1 & 6 & 3 \\ 6 & 6 & 0 & 4 & 6 & 10 & 6 \\ 3 & 3 & -1 & -2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{114}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 8 & -2 \\ 2 & 3 & 0 & 2 & 1 & 12 & 3 \\ 9 & 12 & 0 & 8 & 12 & 46 & 12 \\ 3 & 3 & -1 & -2 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[17] $1^{113}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1312, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 1 & 9 & 3 \\ 8 & 9 & 0 & 6 & 9 & 25 & 9 \\ 3 & 3 & -1 & -2 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[18] $1^{114}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1752, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 8 & -2 \\ 2 & 3 & 0 & 2 & 1 & 12 & 3 \\ 10 & 12 & 0 & 8 & 12 & 46 & 12 \\ 3 & 3 & -1 & -2 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ [19] $1^1231^{113}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -51, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 15.43 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 4 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & -1 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

[20] $1^1231^{113}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -192, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 25, T \in [16, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & -1 & -2 & 0 & 4 & 2 & -2 \end{pmatrix}_{b_0}$$

[21] $1^1231^{123}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -368, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 15.30 \rightarrow 16, T_{\max} = 25, T \in [16, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 6 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 12 & 0 & 6 & 0 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 0 & 6 & 0 & 2 \\ 2 & 4 & -1 & -2 & 0 & 8 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{12}3^{11}1^{12}3^{11}2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 25, \ \Delta = -112, \ \det = -408, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 15.29 \rightarrow 16, \ T_{\max} = 25, \ T \in [16, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 6 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 12 & 0 & 6 & 0 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 0 & 6 & 0 & 2 \\ 2 & 4 & -1 & -2 & 0 & 8 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{12}3^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 24, \ \Delta = -111, \ \det = -411, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.78 \rightarrow 15, \ T_{\max} = 24, \ T \in [15, 24]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 5 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

4.189 $\text{NHC}(-2)(-3) + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$

$$T = 11 \quad [0] \ 1^{11}3^{11}2^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -16, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 3 & 1 & 3 \\ 3 & 3 & -1 & -2 & 0 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}2^{11}3^{11}2^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -256, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.38 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 4 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 4 & 2 & -2 \\ 2 & 3 & 0 & 2 & 1 & 6 & 3 & 3 \\ 4 & 6 & 0 & 4 & 6 & 10 & 6 & 6 \\ 2 & 3 & 0 & 2 & 3 & 6 & 1 & 3 \\ 3 & 3 & -1 & -2 & 3 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}2^{11}3^{11}2^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -400, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.36 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & 2 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 3 & 1 & 3 \\ 3 & 3 & -1 & -2 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}2^{11}3^{11}2^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -576, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.39 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & 2 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 3 & 1 & 3 \\ 3 & 3 & -1 & -2 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}2^{11}3^{11}2^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -960, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.43 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 6 & 3 \\ 6 & 6 & 0 & 4 & 6 & 6 & 10 & 6 \\ 3 & 3 & -1 & -2 & 3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{1112} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -1040, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 5 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 6 & 3 \\ 5 & 6 & 0 & 4 & 6 & 6 & 10 & 6 \\ 3 & 3 & -1 & -2 & 3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

4.190 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$

$T = 11$ [0] $1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 3 & -1 & -2 & 0 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.191 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \text{n3mix}$

$T = 10$ [0] $1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 94, \Delta = -133, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.44 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 1 & 2 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 3 & 3 \\ 2 & 2 & -1 & -3 & 0 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{1211} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 94, \Delta = -133, \det = 174, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & 2 & -3 \\ 2 & 4 & 0 & 2 & 6 & 4 & 6 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 & 3 \\ 2 & 2 & -1 & -3 & 4 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{1121} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 94, \Delta = -133, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 4 & 3 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 6 & 4 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 & 3 \\ 2 & 2 & -1 & -3 & 2 & 4 & 3 & -2 \end{pmatrix}_{b_0}$$

[3] $1^{1112} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 94, \Delta = -133, \det = 384, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.54 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 3 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 5 & 2 \\ 3 & 5 & 0 & 3 & 5 & 5 & 11 & 5 \\ 2 & 2 & -1 & -3 & 2 & 2 & 5 & -2 \end{pmatrix}_{b_0}$$

[4] $1^{1111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 94, \Delta = -133, \det = 408, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 3 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 5 & 2 \\ 4 & 5 & 0 & 3 & 5 & 5 & 11 & 5 \\ 2 & 2 & -1 & -3 & 2 & 2 & 5 & -2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [12] \quad 1^{12311} 3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -290, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & -1 & -2 & 0 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

4.192 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{111} 3^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 68, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & -1 & -2 & 0 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 79, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.84 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & -1 & -2 & 0 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 3^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 & 1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 156, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.18 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 0 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -3 & 2 & 2 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{131} 3^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 221, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 15.18 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 9 & -2 & 1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 & 12 \\ 3 & 4 & -1 & -2 & 0 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 413, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 & 1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1111} 3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 420, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.71 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{121} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 464, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.73 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 6 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & -2 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 496, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.69 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -4 & 3 & 3 & -2 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1111} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 589, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.67 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1111} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 1088, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.65 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [11] \ 1^{111} 3 \ 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, \ T_{\max} = 23, \ \Delta = -110, \ \det = -56, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.71 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111} 3 \ 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, \ T_{\max} = 23, \ \Delta = -110, \ \det = -65, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.94 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1111} 23 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, \ T_{\max} = 25, \ \Delta = -112, \ \det = -84, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.71 \rightarrow 15, \ T_{\max} = 25, \ T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 2 \\ 2 & 3 & -1 & -3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

- [14] $1^{1111} 3^{12} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -92, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.83 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$
- [15] $1^{1111} 2^1 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.73 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 1 & 3 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$
- [16] $1^{111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 15.22 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 0 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -2 & 2 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$
- [17] $1^{111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -149, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.70 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 6 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 8 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 8 & 0 & 15 & 9 \\ 3 & 4 & -1 & -1 & 0 & 9 & 0 \end{pmatrix}_{b_0}$$
- [18] $1^{131} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -173, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 15.39 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 3 & 4 & -1 & -1 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$
- [19] $1^{1111} 2^1 3^{112} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -246, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 15.15 \rightarrow 16, T_{\max} = 25, T \in [16, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 & 3 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 & 3 \\ 1 & 0 & 3 & 1 & 3 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$
- [20] $1^{131} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -269, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 14.90 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 12 & 3 \\ 3 & 4 & 1 & 3 & 0 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 14 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 12 & 21 & 0 & 14 & 0 & 75 & 17 \\ 3 & 4 & -1 & -1 & 0 & 17 & 0 \end{pmatrix}_{b_0}$$
- [21] $1^{1111} 2^1 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -290, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.74 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{121} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -332, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 9 & 3 & 4 \\ 3 & 4 & 1 & 3 & 0 & 15 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 11 & -1 & 1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 15 & 0 & 11 & 0 & 39 & 13 & 13 \\ 3 & 4 & -1 & -1 & 0 & 13 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -341, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 & 1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{111} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -344, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 & 1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{123} 231^{111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 38, \Delta = -118, \det = -368, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.05 \rightarrow 16, T_{\max} = 38, T \in [16, 38]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 3 & 0 & -2 \\ 2 & 6 & 0 & 3 & 4 & 6 & 0 & 4 \\ 2 & 6 & 0 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 4 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{121} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -380, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.83 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 6 & -1 & 1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 & 8 \\ 3 & 4 & -1 & -1 & 0 & 8 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{111} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -408, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.77 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 3 & 3 & -1 & 1 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 & 4 \\ 4 & 4 & -1 & -1 & 4 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{111} 2^{122} 3^{111} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -432, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.96 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 & 3 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 & 3 \\ 1 & 0 & 3 & 1 & 3 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{112} 2^{111} 3^{111} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -434, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 6 & 3 & 3 \\ 3 & 6 & 0 & 2 & 6 & 10 & 6 & 6 \\ 1 & 3 & 0 & 1 & 3 & 6 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{1231}1^{11}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, \quad T_{\max} = 38, \quad \Delta = -118, \quad \det = -455, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$

[30] $1^{1231}1^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, \quad T_{\max} = 38, \quad \Delta = -118, \quad \det = -455, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$

$$T_{\min} = 15.09 \rightarrow 16, T_{\max} = 38, T \in [16, 38]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 3 & 0 & -2 \\ 2 & 6 & 0 & 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 4 & 4 & -1 & -2 \end{pmatrix} b_0$$

$$[31] \quad 1^{1111} 3^{221} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, \quad T_{\max} = 23, \quad \Delta = -110, \quad \det = -485, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix} b_0$$

[32] $1^{1111}2^321^1 + [\textcolor{blue}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{e}7 + \textcolor{brown}{s}u_2 + \textcolor{brown}{s}u_2 + \textcolor{brown}{s}u_3 \quad T = 11, \quad T_{\max} = 25, \quad \Delta = -112, \quad \det = -540, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 & 3 \\ 1 & 0 & 3 & 1 & 3 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[33] $1^{1131}2^33^{21}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -626, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 9 & 3 & 3 \\ 4 & 9 & 0 & 3 & 9 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 3 & 9 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 9 & 2 & -1 \end{pmatrix} b_0$$

[34] $1^{1131}2^33^{21}1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -668, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 9 & 3 & 3 \\ 5 & 9 & 0 & 3 & 9 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 3 & 9 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 9 & 2 & -1 \end{pmatrix} b_0$$

[35] $1^{1141}2^33^{21}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -674, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 12 & 3 & 3 \\ 5 & 12 & 0 & 4 & 12 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 3 & 12 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 12 & 2 & -1 \end{pmatrix} b_0$$

[36] $1^{1111}3^{221} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{c}7 + \textcolor{orange}{s}u_2 + \textcolor{red}{s}u_2 + \textcolor{blue}{s}u_3 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -896, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix} b_0$$

[37] $1^{114}2^33^{21}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -1268, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 12 & 3 & 3 \\ 6 & 12 & 0 & 4 & 12 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 3 & 12 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

4.193 NHC(-2)(-3) + e₇ + su₂ + su₂ + su₃ + su₂

$$T = 11 \quad [0] \quad 1^{1111} 3 \quad 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 16, T_{\max} = 23, T \in [16, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 0 & -3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & 0 \\ 1 & 2 & 0 & 1 & 2 & 2 & 0 & 0 \\ 2 & 2 & -1 & -3 & 2 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 3 \quad 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -202, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.85 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [2] \quad 1^{1111} 23 \quad 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 3 \quad 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 166, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.95 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1112} 23 \quad 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 244, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 6 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 6 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 6 \\ 1 & 3 & 0 & 2 & 6 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1114} 23 \quad 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 472, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 12 & 2 \\ 6 & 12 & 0 & 4 & 12 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 3 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{1113} 23 \quad 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 628, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 15.8$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 9 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 9 & 2 \\ 4 & 9 & 0 & 3 & 9 & 9 & 25 & 9 & 9 \\ 2 & 3 & -1 & -3 & 3 & 3 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11114} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 724, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.77 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 12 & 2 \\ 5 & 12 & 0 & 4 & 12 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 3 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$4.194 \quad \text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 11 \quad [0] \ 1^{11113} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 0 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$4.195 \quad \text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$$

$$T = 11 \quad [0] \ 1^{11113} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 38, \Delta = -118, \det = -284, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 38, T \in [16, 38]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & 0 & -2 \\ 3 & 6 & 0 & 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1] \ 1^{11111} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 12, T_{\max} = 16, \Delta = -91, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11111} 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 12, T_{\max} = 16, \Delta = -89, \det = 228, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -1 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$4.196 \quad \text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 9 \quad [0] \ 1^{112} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

[9] $1^{11} 1^{13} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 14.39 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 6 & -2 \\ 7 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & -2 & 9 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{11} 1^{14} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.39 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 8 & -2 \\ 9 & 12 & 0 & 8 & 46 & 12 \\ 3 & 3 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{12} 1^{12} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & -2 \\ 6 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{12} 1^{13} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -828, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 6 & -2 \\ 8 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & -2 & 9 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{12} 1^{14} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -1048, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 8 & -2 \\ 10 & 12 & 0 & 8 & 46 & 12 \\ 3 & 3 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

$T = 10$ [14] $1^{12} 23^1 131^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 18, T_{\max} = 100, T \in [18, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 2 & 0 & 0 & 2 \\ 2 & 0 & -1 & -2 & 0 & 2 & -4 \end{pmatrix}_{b_0}$$

[15] $1^{12} 23^1 131^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 16.20 \rightarrow 17, T_{\max} = 100, T \in [17, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 9 & 2 & -2 \\ 10 & 0 & 0 & 9 & 52 & 9 & 12 \\ 1 & 0 & 0 & 2 & 9 & 0 & 2 \\ 2 & 0 & -1 & -2 & 12 & 2 & -3 \end{pmatrix}_{b_0}$$

[16] $1^{12} 23^1 131^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 15.20 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 3 & 0 & 2 & 0 & 0 & 2 \\ 2 & 4 & -1 & -2 & 0 & 2 & -2 \end{pmatrix}_{b_0}$$

[17] $1^{12} 23^1 131^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 88, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 2 & -2 \\ 2 & 6 & 0 & 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

[27] $1^1 231^{11} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{brown}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -141, \quad \det = 313, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$

[27] $1^1 231^{11} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{c}}_7 + \textcolor{brown}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -141, \quad \det = 313, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$

$$T_{\min} = 14.77 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 5 & -1 \\ 4 & 10 & 0 & 5 & 8 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{1231}1^{113}1^{12} + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}\textcolor{teal}{g}_7 + \textcolor{brown}{s}\textcolor{brown}{u}_2 + \textcolor{brown}{s}\textcolor{brown}{u}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -141, \quad \det = 373, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 5 & -1 \\ 6 & 10 & 0 & 5 & 8 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{12}3^{11}1^{11}31^{11} + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}\textcolor{teal}{g}_7 + \textcolor{brown}{s}\textcolor{brown}{u}_2 + \textcolor{brown}{s}\textcolor{brown}{u}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -142, \quad \det = 380, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 2 & -2 \\ 3 & 6 & 0 & 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

[30] $1^1 2 3^1 1^2 3 1^1 + [\textcolor{blue}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{red}{\mathbf{c}}_7 + \textcolor{brown}{\mathbf{su}}_2 + \textcolor{blue}{\mathbf{su}}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -142, \quad \det = 480, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.5$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & 2 & -2 \\ 4 & 12 & 0 & 6 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}$$

[31] $1^1 2 3 1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -141, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 5 & -1 \\ 5 & 10 & 0 & 5 & 8 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{123}1^{12}31^{13} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{teal}{\mathbf{e}}_7 + \textcolor{brown}{\mathbf{su}}_2 + \textcolor{red}{\mathbf{su}}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -142, \quad \det = 540, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.5$

$$T_{\min} = 14.96 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & 2 & -2 \\ 7 & 12 & 0 & 6 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}$$

[33] $1^{12}23^11^{12}31^{11} + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}\textcolor{teal}{g} + \textcolor{brown}{s}\textcolor{brown}{u}_2 + \textcolor{brown}{s}\textcolor{brown}{u}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 828, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & 2 & -2 \\ 5 & 12 & 0 & 6 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}$$

[34] $1^{12}23^11^{12}31^{12} + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}\textcolor{teal}{g}_7 + \textcolor{brown}{s}\textcolor{brown}{u}_2 + \textcolor{brown}{s}\textcolor{brown}{u}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 848, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & 2 & -2 \\ 6 & 12 & 0 & 6 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}$$

4.197 NHC(-2)(-3) + e₇ + su₂ + su₂→(-3) + su₂

$$T = 10 \quad [0] \ 1^{131}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.50 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 2 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 6 & 2 & -2 \\ 6 & 9 & 0 & 6 & 25 & 9 & 9 \\ 2 & 3 & 0 & 2 & 9 & 1 & 3 \\ 3 & 3 & -1 & -2 & 9 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{122}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 4 & -2 \\ 4 & 6 & 0 & 4 & 10 & 12 & 6 \\ 4 & 6 & 0 & 4 & 12 & 10 & 6 \\ 3 & 3 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{121}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 2 & -2 \\ 4 & 6 & 0 & 4 & 10 & 6 & 6 \\ 3 & 3 & 0 & 2 & 6 & 1 & 3 \\ 3 & 3 & -1 & -2 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 \\ 3 & 3 & -1 & -2 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{121}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 2 & -2 \\ 4 & 6 & 0 & 4 & 10 & 6 & 6 \\ 2 & 3 & 0 & 2 & 6 & 1 & 3 \\ 3 & 3 & -1 & -2 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.36 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 3 \\ 3 & 3 & -1 & -2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{123}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 280, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.3$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 8 & 3 \\ 2 & 1 & 1 & 2 & 6 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 6 & -2 \\ 4 & 6 & 0 & 4 & 10 & 18 & 6 \\ 8 & 9 & 0 & 6 & 18 & 25 & 9 \\ 3 & 3 & -1 & -2 & 6 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 288, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.50 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 1 & 6 & 3 \\ 6 & 6 & 0 & 4 & 6 & 10 & 6 \\ 3 & 3 & -1 & -2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{11} 1^{11} 3^{11} 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 944$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.39 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 8 & -2 \\ 2 & 3 & 0 & 2 & 1 & 12 & 3 \\ 9 & 12 & 0 & 8 & 12 & 46 & 12 \\ 3 & 3 & -1 & -2 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

[17] $1^{12} 1^{12} 3^{11} 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 1312$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.40 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 1 & 9 & 3 \\ 8 & 9 & 0 & 6 & 9 & 25 & 9 \\ 3 & 3 & -1 & -2 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

[18] $1^{12} 1^{12} 3^{11} 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 1752$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.40 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 8 & -2 \\ 2 & 3 & 0 & 2 & 1 & 12 & 3 \\ 10 & 12 & 0 & 8 & 12 & 46 & 12 \\ 3 & 3 & -1 & -2 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ [19] $1^{12} 231^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 11$, $T_{\max} = 24$, $\Delta = -111$, $\det = -51$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 15.43 \rightarrow 16$, $T_{\max} = 24$, $T \in [16, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 4 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & -1 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{12} 231^{11} 3^{11} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 11$, $T_{\max} = 25$, $\Delta = -112$, $\det = -192$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 15.25 \rightarrow 16$, $T_{\max} = 25$, $T \in [16, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & 0 & 2 & 4 & -2 \end{pmatrix}_{b_0}$$

[21] $1^{12} 231^{11} 3^{11} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 11$, $T_{\max} = 25$, $\Delta = -112$, $\det = -368$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 15.30 \rightarrow 16$, $T_{\max} = 25$, $T \in [16, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & 6 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & 2 & 0 & 0 & 6 & 2 \\ 5 & 12 & 0 & 6 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 0 & 2 & 8 & -2 \end{pmatrix}_{b_0}$$

[22] $1^{12} 231^{11} 3^{11} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 11$, $T_{\max} = 25$, $\Delta = -112$, $\det = -408$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 15.29 \rightarrow 16$, $T_{\max} = 25$, $T \in [16, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & 6 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & 2 & 0 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 0 & 2 & 8 & -2 \end{pmatrix}_{b_0}$$

[23] $1^{12} 231^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 11$, $T_{\max} = 24$, $\Delta = -111$, $\det = -411$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.78 \rightarrow 15$, $T_{\max} = 24$, $T \in [15, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 5 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & -1 & 0 & 6 & 0 \end{pmatrix}$$

4.198 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_{2 \rightarrow (-3)} + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 11, \quad T_{\max} = 23, \quad \Delta = -103, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 3 & 1 & 3 \\ 3 & 3 & -1 & -2 & 0 & 3 & 3 & -1 \end{pmatrix} b_0$$

[1] $1^{1211}3^{1211} + [\text{su}_2 \cdot \text{g}_2] + \text{c}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -256, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 2 & 2 & -2 \\ 4 & 6 & 0 & 4 & 10 & 6 & 6 & 6 \\ 2 & 3 & 0 & 2 & 6 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 6 & 3 & 1 & 3 \\ 3 & 3 & -1 & -2 & 6 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{1111}3^{1211} + [\text{su}_2 \cdot \text{g}_2] + \text{c}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -400, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.36 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 3 \\ 2 & 3 & 0 & 2 & 3 & 3 & 1 \\ 3 & 3 & -1 & -2 & 3 & 3 & -1 \end{pmatrix} b_0$$

[3] $1^{1111}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -576, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 3 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & 2 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 3 & 3 \\ 3 & 3 & 0 & 2 & 3 & 3 & 1 & 3 \\ 3 & 3 & -1 & -2 & 3 & 3 & 3 & -1 \end{pmatrix} b_0$$

[4] $1^{1112}3^12^11+[\textcolor{blue}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{red}{\mathfrak{e}}_7+\textcolor{brown}{\mathfrak{su}}_2+\textcolor{brown}{\mathfrak{su}}_2\rightarrow(-3)+\textcolor{teal}{\mathfrak{su}}_2+\textcolor{violet}{\mathfrak{su}}_2$ $T=11$, $T_{\max}=23$, $\Delta=-103$, $\det=-960$, $(n_+, n_-, n_0)=(1, 11, 0)$, $c_{\text{ext}}=14.5$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 6 & 3 \\ 6 & 6 & 0 & 4 & 6 & 6 & 10 & 6 \\ 3 & 3 & -1 & -2 & 3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{1112} 3^1 2^1 1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -1040, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 5 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 6 & 3 \\ 5 & 6 & 0 & 4 & 6 & 6 & 10 & 6 \\ 3 & 3 & -1 & -2 & 3 & 3 & 6 & -1 \end{pmatrix} b_0$$

4.199 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_{2 \rightarrow (-3)} + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 11 \quad [0] \quad 1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 11, \quad T_{\max} = 25, \quad \Delta = -112, \quad \det = -128, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 3 & -1 & -2 & 0 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.200 NHC(-2)(-3) + e₇ + su₂ + su₂→(-3) + su₂→(-3)

$$T = 11 \quad [0] \quad 1^{12311}3^{111}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 27, \Delta = -114, \det = -108, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.22 \rightarrow 16, T_{\max} = 27, T \in [16, 27]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 2 & 3 \\ 4 & 8 & 1 & 5 & 0 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 0 & 1 & 3 \\ 3 & 6 & -1 & -1 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{12311}3^{111}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 27, \Delta = -114, \det = -204, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 27, T \in [15, 27]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 2 & 3 \\ 4 & 8 & 1 & 5 & 0 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 0 & 0 & 2 \\ 3 & 6 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

4.201 NHC(-2)(-3) + e₇ + su₂ + su₂→(-3) + su₂n3mix

$$T = 11 \quad [0] \quad 1^{12311}3^{111}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, T_{\max} = 27, \Delta = -114, \det = -168, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 27, T \in [15, 27]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 & 3 \\ 4 & 8 & 1 & 5 & 0 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 3 & 0 & 1 & 3 \\ 3 & 6 & -1 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

4.202 NHC(-2)(-3) + e₇ + su₂ + su₂→(-3) + su₃

$$T = 10 \quad [0] \quad 1^{111}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 3 & -1 & -2 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1211}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.80 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 2 & -2 \\ 4 & 6 & 0 & 4 & 10 & 6 & 6 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 3 & -1 & -2 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 348, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 3 & -1 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1112}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 604, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & -2 \\ 4 & 3 & 0 & 2 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 3 & -1 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 1^{11} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & -2 \\ 3 & 3 & 0 & 2 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 3 & -1 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{12} 1^{11} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 2 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 2 & -2 \\ 5 & 6 & 0 & 4 & 10 & 6 & 6 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 3 & -1 & -2 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12} 1^{12} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 1408, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 2 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 2 & -2 \\ 6 & 6 & 0 & 4 & 10 & 6 & 6 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 3 & -1 & -2 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \quad 1^{12} 3^1 1^{12} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -152, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 15.16 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & 2 & 0 & -2 \\ 4 & 12 & 0 & 6 & 22 & 6 & 0 & 8 \\ 1 & 3 & 0 & 2 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 8 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{12} 3^1 1^{11} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -296, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.08 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 2 & 0 & -2 \\ 2 & 6 & 0 & 3 & 4 & 3 & 0 & 4 \\ 1 & 3 & 0 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 4 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{12} 3^1 1^{11} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 11, T_{\max} = 59, \Delta = -120, \det = -364, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.15 \rightarrow 16, T_{\max} = 59, T \in [16, 59]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 1 & 3 \\ 4 & 8 & 1 & 5 & 10 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 5 & 0 & -1 \\ 4 & 10 & 0 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{12} 3^1 1^{11} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -572, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 2 & 0 & -2 \\ 3 & 6 & 0 & 3 & 4 & 3 & 0 & 4 \\ 1 & 3 & 0 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 4 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12} 3^1 1^{12} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -1196, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & 2 & 0 & -2 \\ 5 & 12 & 0 & 6 & 22 & 6 & 0 & 8 \\ 1 & 3 & 0 & 2 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 8 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

4.203 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 25, \quad \Delta = -112, \quad \det = -128, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.88 \rightarrow 15, \quad T_{\max} = 25, \quad T \in [15, 25]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 1 & 3 \\ 3 & 3 & -1 & -2 & 0 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

4.204 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \mathfrak{n} 3 \text{mix}$

$$T = 9 \quad [0] \quad 1^{141} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -46, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.70 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 2 & 2 & -3 \\ 4 & 8 & 0 & 4 & 30 & 12 & 8 & 8 \\ 2 & 3 & 0 & 2 & 12 & 3 & 3 & 3 \\ 2 & 2 & -1 & -3 & 8 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -65, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.09 \rightarrow 16, \quad T_{\max} = 102, \quad T \in [16, 102]$$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 3 & 4 & 4 \\ 3 & 1 & 1 & 2 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 2 & 0 & -4 & 2 & 2 & -2 \\ 3 & 3 & 0 & 2 & 1 & 3 & 3 \\ 4 & 3 & -1 & -2 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{112} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -78, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.41 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & 2 & -3 \\ 3 & 2 & 0 & 1 & 0 & 3 & 2 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 & 3 \\ 2 & 2 & -1 & -3 & 2 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -95, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.41 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 2 & 2 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 3 & 3 & 3 \\ 2 & 2 & -1 & -3 & 0 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{113} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -95, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.17 \rightarrow 16, \quad T_{\max} = 102, \quad T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & 2 & -3 \\ 2 & 2 & 0 & 1 & 0 & 7 & 2 & 2 \\ 4 & 7 & 0 & 4 & 7 & 23 & 7 & 7 \\ 2 & 2 & -1 & -3 & 2 & 7 & -3 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{131} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -130, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 14.46 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 6 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 2 & 2 & -3 \\ 3 & 6 & 0 & 3 & 16 & 9 & 6 & 6 \\ 2 & 3 & 0 & 2 & 9 & 3 & 3 & 3 \\ 2 & 2 & -1 & -3 & 6 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{123} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -134, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.54 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 4 & -3 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 6 & 7 & 0 & 4 & 14 & 23 & 7 \\ 2 & 2 & -1 & -3 & 4 & 7 & -2 \end{pmatrix}_{b_0}$$

[7] $1^{142}3^{121}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 3 & -3 \\ 4 & 8 & 0 & 4 & 30 & 20 & 8 \\ 3 & 5 & 0 & 3 & 20 & 11 & 5 \\ 2 & 2 & -1 & -3 & 8 & 5 & -2 \end{pmatrix}_{b_0}$$

[8] $1^{121}3^{121}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 14.42 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 2 & -3 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 \\ 2 & 2 & -1 & -3 & 4 & 3 & -2 \end{pmatrix}_{b_0}$$

[9] $1^{124}3^{121}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 15.16 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 5 & -3 \\ 3 & 4 & 0 & 2 & 6 & 18 & 4 \\ 5 & 9 & 0 & 5 & 18 & 39 & 9 \\ 2 & 2 & -1 & -3 & 4 & 9 & -3 \end{pmatrix}_{b_0}$$

[10] $1^{111}3^{121}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & -1 & -3 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

[11] $1^{122}3^{121}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 3 & -3 \\ 2 & 4 & 0 & 2 & 6 & 10 & 4 \\ 4 & 5 & 0 & 3 & 10 & 11 & 5 \\ 2 & 2 & -1 & -3 & 4 & 5 & -2 \end{pmatrix}_{b_0}$$

[12] $1^{132}3^{121}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 14.51 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 3 & -3 \\ 3 & 6 & 0 & 3 & 16 & 15 & 6 \\ 3 & 5 & 0 & 3 & 15 & 11 & 5 \\ 2 & 2 & -1 & -3 & 6 & 5 & -2 \end{pmatrix}_{b_0}$$

[13] $1^{112}3^{121}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.51 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & -3 \\ 2 & 2 & 0 & 1 & 0 & 5 & 2 \\ 3 & 5 & 0 & 3 & 5 & 11 & 5 \\ 2 & 2 & -1 & -3 & 2 & 5 & -2 \end{pmatrix}_{b_0}$$

[14] $1^{111}3^{121}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.68 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & -3 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 3 & 5 & 11 & 5 \\ 2 & 2 & -1 & -3 & 2 & 5 & -2 \end{pmatrix}_{b_0}$$

[31] $1^{1^{11}} 1^{132} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -623, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 2 & -3 \\ 4 & 6 & 0 & 3 & 16 & 9 & 6 \\ 2 & 3 & 0 & 2 & 9 & 3 & 3 \\ 2 & 2 & -1 & -3 & 6 & 3 & -2 \end{pmatrix}_{b_0}$$

[32] $1^{1^{11}} 1^{132} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -692, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.46 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 3 & -3 \\ 4 & 6 & 0 & 3 & 16 & 15 & 6 \\ 3 & 5 & 0 & 3 & 15 & 11 & 5 \\ 2 & 2 & -1 & -3 & 6 & 5 & -2 \end{pmatrix}_{b_0}$$

[33] $1^{1^{12}} 1^{112} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & -3 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 5 & 5 & 0 & 3 & 5 & 11 & 5 \\ 2 & 2 & -1 & -3 & 2 & 5 & -2 \end{pmatrix}_{b_0}$$

[34] $1^{1^{11}} 1^{141} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -751, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 2 & -3 \\ 5 & 8 & 0 & 4 & 30 & 12 & 8 \\ 2 & 3 & 0 & 2 & 12 & 3 & 3 \\ 2 & 2 & -1 & -3 & 8 & 3 & -2 \end{pmatrix}_{b_0}$$

[35] $1^{1^{11}} 1^{124} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -820, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 5 & -3 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 6 & 9 & 0 & 5 & 18 & 39 & 9 \\ 2 & 2 & -1 & -3 & 4 & 9 & -2 \end{pmatrix}_{b_0}$$

[36] $1^{1^{12}} 1^{131} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -830, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 2 & -3 \\ 5 & 6 & 0 & 3 & 16 & 9 & 6 \\ 2 & 3 & 0 & 2 & 9 & 3 & 3 \\ 2 & 2 & -1 & -3 & 6 & 3 & -2 \end{pmatrix}_{b_0}$$

[37] $1^{1^{12}} 1^{142} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -864, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.48 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 3 & -3 \\ 6 & 8 & 0 & 4 & 30 & 20 & 8 \\ 3 & 5 & 0 & 3 & 20 & 11 & 5 \\ 2 & 2 & -1 & -3 & 8 & 5 & -2 \end{pmatrix}_{b_0}$$

[38] $1^{1^{11}} 1^{142} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -880, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 3 & -3 \\ 5 & 8 & 0 & 4 & 30 & 20 & 8 \\ 3 & 5 & 0 & 3 & 20 & 11 & 5 \\ 2 & 2 & -1 & -3 & 8 & 5 & -2 \end{pmatrix}_{b_0}$$

[39] $1^{113}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -899, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & -3 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 4 & 7 & 23 & 7 \\ 2 & 2 & -1 & -3 & 2 & 7 & -2 \end{pmatrix}_{b_0}$$

[40] $1^{141}3^{12}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -1170, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 2 & -3 \\ 6 & 8 & 0 & 4 & 30 & 12 & 8 \\ 2 & 3 & 0 & 2 & 12 & 3 & 3 \\ 2 & 2 & -1 & -3 & 8 & 3 & -2 \end{pmatrix}_{b_0}$$

[41] $1^{113}3^{12}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -1190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & -3 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 6 & 7 & 0 & 4 & 7 & 23 & 7 \\ 2 & 2 & -1 & -3 & 2 & 7 & -2 \end{pmatrix}_{b_0}$$

[42] $1^{114}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -1201, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 5 & -3 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 5 & 9 & 39 & 9 \\ 2 & 2 & -1 & -3 & 2 & 9 & -2 \end{pmatrix}_{b_0}$$

[43] $1^{114}3^{12}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -1656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 5 & -3 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 7 & 9 & 0 & 5 & 9 & 39 & 9 \\ 2 & 2 & -1 & -3 & 2 & 9 & -2 \end{pmatrix}_{b_0}$$

$T = 10$ [44] $1^{123}1^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 2 & 4 & -1 & -2 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{123}1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -141, \det = 97, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 4 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 8 & 0 & 4 & 0 & 9 & 5 \\ 2 & 4 & -1 & -2 & 0 & 5 & -2 \end{pmatrix}_{b_0}$$

[46] $1^{123}1^{11}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 109, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 5 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 5 & 0 & 12 & 6 \\ 2 & 4 & -1 & -2 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

[47] $1^1 23^1 1^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 157, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 2 & -2 \\ 2 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

[48] $1^1 23^1 1^{11} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 163, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 15.17 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 3 & 2 & -2 \\ 3 & 0 & 0 & 3 & 4 & 3 & 4 \\ 2 & 0 & 0 & 2 & 3 & 0 & 2 \\ 2 & 0 & -1 & -2 & 4 & 2 & -2 \end{pmatrix}_{b_0}$$

[49] $1^1 23^1 131^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 164, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 2 & 0 & 0 & 2 \\ 2 & 0 & -1 & -2 & 0 & 2 & -2 \end{pmatrix}_{b_0}$$

[50] $1^1 231^{11} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -141, \det = 182, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 7 & 4 & 4 \\ 6 & 8 & 1 & 5 & 10 & 6 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & -3 & 5 & -1 & 6 \\ 7 & 10 & 0 & 5 & 8 & 6 & 6 \\ 4 & 6 & -1 & -1 & 6 & 0 & 6 \end{pmatrix}_{b_0}$$

[51] $1^1 23^1 1^{11} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 285, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 5 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 9 & 0 & 5 & 0 & 12 & 6 \\ 2 & 4 & -1 & -2 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

[52] $1^1 231^{11} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -141, \det = 303, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.77 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 4 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 8 & 0 & 4 & 0 & 9 & 5 \\ 2 & 4 & -1 & -2 & 0 & 5 & -1 \end{pmatrix}_{b_0}$$

[53] $1^1 23^1 1^{12} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 322, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 15.19 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 8 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 15 & 0 & 8 & 0 & 36 & 10 \\ 2 & 4 & -1 & -2 & 0 & 10 & -2 \end{pmatrix}_{b_0}$$

[54] $1^1 23^1 1^{11} 31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 355, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 5 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 5 & 0 & 12 & 6 \\ 2 & 4 & -1 & -2 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

[55] $1^1 23^1 1^{11} 31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 373, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

[illegible]

4.205 NHC(-2)(-3) + e7 + su2 + su2n3mix + su2

$$T = 10 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}} 3^{\textcolor{red}{1}\textcolor{blue}{1}} 21^{\textcolor{red}{1}} + [\textcolor{red}{s}u_{\textcolor{red}{2}} \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}g_7 + \textcolor{red}{s}u_2 + \textcolor{red}{s}u_2 + \textcolor{blue}{s}u_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.44 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 2 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 0 & 3 & 0 & 2 \\ 2 & 2 & -1 & -3 & 0 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1211} 3^{1211} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{s}u_2 + \textcolor{red}{s}u_2 + \textcolor{blue}{s}u_2 \quad T = 10, \quad T_{\max} = 94, \quad \Delta = -133, \quad \det = 174, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 2 & 1 & -3 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 & 4 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 & 3 \\ 2 & 2 & 0 & 1 & 4 & 3 & 0 & 2 \\ 2 & 2 & -1 & -3 & 4 & 3 & 2 & -2 \end{pmatrix} b_0$$

$$[2] \quad 1^{1211} 3^{1211} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 94, \quad \Delta = -133, \quad \det = 356, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 2 & 1 & -3 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 & 4 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 4 & 3 & 0 & 2 \\ 2 & 2 & -1 & -3 & 4 & 3 & 2 & -2 \end{pmatrix} b_0$$

$$[3] \quad 1^{1121} 3^{1211} + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{c} \textcolor{red}{g}_7 + \textcolor{red}{s} \textcolor{blue}{u}_2 + \textcolor{red}{s} \textcolor{blue}{u}_2 + \textcolor{blue}{s} \textcolor{blue}{u}_2 \quad T = 10, \quad T_{\max} = 94, \quad \Delta = -133, \quad \det = 384, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.54 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & 1 & -3 \\ 2 & 2 & 0 & 1 & 0 & 5 & 2 & 2 \\ 3 & 5 & 0 & 3 & 5 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 2 & 5 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & 5 & 2 & -2 \end{pmatrix} b_0$$

[4] $1^{11111}3^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 94, \Delta = -133, \det = 408, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.48 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 0 & 2 & 3 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 & 3 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & 3 & 2 & -2 \end{pmatrix} b_0$$

$$[5] \quad 1^{1111} 3^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 94, \quad \Delta = -133, \quad \det = 428, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 & 3 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & 3 & 2 & -2 \end{pmatrix} b_0$$

$$[6] \quad 1^{1111} 3^{1211} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_7 + \textcolor{orange}{s}u_2 + \textcolor{red}{s}u_2 + \textcolor{blue}{s}u_2 \quad T = 10, \quad T_{\max} = 94, \quad \Delta = -133, \quad \det = 486, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 & 3 \\ 2 & 2 & 0 & 1 & 2 & 3 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & 3 & 2 & -2 \end{pmatrix} b_0$$

$$[7] \quad 1^{1112} 3^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 94, \quad \Delta = -133, \quad \det = 588, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 3 & 4 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 6 & 3 \\ 4 & 4 & 0 & 2 & 4 & 6 & 6 & 4 \\ 2 & 2 & -1 & -3 & 2 & 3 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{1121} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 656, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.46 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 & 2 \\ 3 & 5 & 0 & 3 & 5 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 2 & 5 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & 5 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{1121} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 768, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.50 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 & 2 \\ 5 & 5 & 0 & 3 & 5 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 2 & 5 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & 5 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{1112} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 838, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 3 & 4 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 6 & 3 \\ 3 & 4 & 0 & 2 & 4 & 6 & 6 & 4 \\ 2 & 2 & -1 & -3 & 2 & 3 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1121} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 964, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 & 2 \\ 4 & 5 & 0 & 3 & 5 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 2 & 5 & 0 & 2 \\ 2 & 2 & -1 & -3 & 2 & 5 & 2 & -2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [12] \quad 1^1 23^1 1^{11} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -290, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & 0 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

4.206 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{su}_{2 \rightarrow (-3)}$

$$T = 11 \quad [0] \quad 1^1 23^1 1^{13} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_{2 \rightarrow (-3)} \quad T = 11, T_{\max} = 27, \Delta = -114, \det = -168, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 27, T \in [15, 27]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 & 3 \\ 4 & 8 & 1 & 5 & 0 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 3 & 0 & 1 & 3 \\ 3 & 6 & -1 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

4.207 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.13 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 5 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 5 & 0 & 12 & 6 \\ 3 & 4 & -1 & -2 & 0 & 6 & -2 \end{pmatrix}_{b_0}$$

4.208 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -113, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.63 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.97 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{121} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -4 & 6 & -2 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -4 & 3 & 0 & -2 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -2 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.84 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -4 & 3 & 0 & -2 \\ 4 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -2 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -201, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{112} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -3 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{112} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 15.05 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -3 & 2 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{121} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.79 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & -2 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.63 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{131} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -273, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 9 & -2 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.68 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -4 & 3 & -2 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{141} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -377, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.63 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{151} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -417, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.1$$

$$T_{\min} = 15.19 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 15 & -2 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & -2 & 20 & -2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{112} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.62 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -3 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{121} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -641, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

14.0

$$T_{\min} = 14.69 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

[27] $1^{141} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -2577, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

[28] $1^{141} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -3008, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.66 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

$T = 10$ [29] $1^{111} 2321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 24, T_{\max} = 100, T \in [24, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 & 3 \\ 1 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 2 & -10 \end{pmatrix}_{b_0}$$

[30] $1^{122} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 32.40 \rightarrow 33, T_{\max} = 100, T \in [33, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 0 & 2 \\ 1 & 0 & 1 & -5 & 2 & 0 & -3 \\ 3 & 0 & 6 & 2 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 6 & -1 & -19 \end{pmatrix}_{b_0}$$

[31] $1^{112} 2321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 17.67 \rightarrow 18, T_{\max} = 100, T \in [18, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 1 & 6 & 3 \\ 2 & 6 & 0 & 2 & 6 & 9 & 5 \\ 2 & 3 & -1 & -3 & 3 & 5 & -4 \end{pmatrix}_{b_0}$$

[32] $1^{1231} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 16.15 \rightarrow 17, T_{\max} = 101, T \in [17, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 0 & -1 & -3 \end{pmatrix}_{b_0}$$

[33] $1^{111} 2 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 34, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 15.35 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 3 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

[34] $1^{121} 2 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 46, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 18.30 \rightarrow 19, T_{\max} = 100, T \in [19, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & 0 & -3 \\ 2 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 0 & -1 & -5 \end{pmatrix}_{b_0}$$

[35] $1^{1312} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 59, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 18.17 \rightarrow 19, T_{\max} = 100, T \in [19, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 3 & 2 \\ 1 & 0 & 1 & -5 & 3 & 1 & -3 \\ 4 & 0 & 9 & 3 & 25 & 9 & 9 \\ 1 & 0 & 3 & 1 & 9 & 0 & 2 \\ 2 & 0 & 2 & -3 & 9 & 2 & -5 \end{pmatrix}_{b_0}$$

[36] $1^{1222} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.7$

$T_{\min} = 15.38 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 1 & 1 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 2 & -3 \\ 3 & 6 & 0 & 2 & 10 & 12 & 6 \\ 2 & 6 & 0 & 2 & 12 & 9 & 5 \\ 2 & 3 & -1 & -3 & 6 & 5 & -2 \end{pmatrix}_{b_0}$$

[37] $1^{1231} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 70, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.34 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 3 & -2 & 0 & -1 & -2 \end{pmatrix}_{b_0}$$

[38] $1^{111} 23 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 78, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.69 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[41] $1^{11} 3 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{111} 2 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[39] $1^{1231} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 16.40 \rightarrow 17, T_{\max} = 101, T \in [17, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 6 & 0 & 3 & 0 & 3 & 3 \\ 2 & 4 & -1 & -2 & 0 & 3 & -3 \end{pmatrix}_{b_0}$$

[42] $1^{11} 3 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 103, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.03 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 0 & -2 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -2 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

[43] $1^{111} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 109, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.43 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & 0 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 0 & -1 & -2 \end{pmatrix}_{b_0}$$

[44] $1^{112} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 110, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 1 & 0 & 3 & 1 & 1 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 9 & 5 \\ 2 & 0 & 2 & -3 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

[46] $1^{142} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 118, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 15.68 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 1 & 1 & 12 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 2 & -3 \\ 6 & 12 & 0 & 4 & 46 & 24 & 12 \\ 2 & 6 & 0 & 2 & 24 & 9 & 5 \\ 2 & 3 & -1 & -3 & 12 & 5 & -2 \end{pmatrix}_{b_0}$$

[45] $1^{1231} 12^3 11 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 118, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 15.14 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & 0 & -2 \\ 4 & 12 & 0 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 8 & -1 & -2 \end{pmatrix}_{b_0}$$

[47] $1^{121} 3^2 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 6 & -1 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[48] $1^{111} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.95 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 \\ 1 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[49] $1^{132} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 157, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 15.10 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 1 & 1 & 9 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 2 & -3 \\ 4 & 9 & 0 & 3 & 25 & 18 & 9 \\ 2 & 6 & 0 & 2 & 18 & 9 & 5 \\ 2 & 3 & -1 & -3 & 9 & 5 & -2 \end{pmatrix}_{b_0}$$

[50] $1^{111} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 157, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 3 & 1 & 0 & 0 & 2 \\ 2 & 0 & 2 & -3 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

[51] $1^{111} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{111} 3^{11} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 176, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 6 & 9 & 0 & 8 & 15 & 9 \\ 3 & 4 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{112} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 178, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -2 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

[54] $1^{111} 2^3 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 179, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 \\ 1 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[55] $1^{142} 2^3 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$

$$T_{\min} = 15.17 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & 1 & 1 & 12 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 2 & -3 \\ 5 & 12 & 0 & 4 & 46 & 24 & 12 \\ 2 & 6 & 0 & 2 & 24 & 9 & 5 \\ 2 & 3 & -1 & -3 & 12 & 5 & -2 \end{pmatrix}_{b_0}$$

[56] $1^{112} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 15.17 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -2 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

[57] $1^{111} 3^{113} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 8 & 9 & 0 & 8 & 15 & 9 \\ 3 & 4 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

[58] $1^{121} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{111} 3^{13} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{121} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 11 & -1 \\ 8 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & -1 & 13 & 0 \end{pmatrix}_{b_0}$$

[61] $1^{112} 2^1 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 196, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 14.69 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 1 & 0 & 3 & 1 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 9 & 5 \\ 2 & 0 & 2 & -3 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

[62] $1^{121} 2^1 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 203, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 15.21 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -5 & 2 & 1 & -3 \\ 3 & 0 & 6 & 2 & 10 & 6 & 6 \\ 1 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -3 & 6 & 2 & -2 \end{pmatrix}_{b_0}$$

[63] $1^{111} 2^1 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 205, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 2 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[64] $1^{131} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 214, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 15.11 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

[65] $1^1 231^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 217, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 15.03 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 0 & -2 \\ 2 & 6 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

[66] $1^{111} 2^1 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 221, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 15.10 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 0 & 1 & -5 & 1 & 0 & -3 \\ 2 & 0 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 3 & -1 & -2 \end{pmatrix}_{b_0}$$

[67] $1^{13}2^13^{121}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 226, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 15.62 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 0 & 2 \\ 1 & 0 & 1 & -5 & 3 & 0 & -3 \\ 5 & 0 & 9 & 3 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 9 & -1 & -2 \end{pmatrix}_{b_0}$$

[68] $1^{111}3^{111}221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 243, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.77 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{111}3^{112}221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 246, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 7 & 9 & 0 & 8 & 15 & 9 \\ 3 & 4 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{141}3^{11}221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 256, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 15.31 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & -1 \end{pmatrix}_{b_0}$$

[71] $1^{1231}1^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 288, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 2 & 4 & -1 & -2 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

[72] $1^{111}3^{111}221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 304, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{151}3^{11}221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 310, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.1$

$T_{\min} = 15.49 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 15 & -1 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & -1 & 20 & -1 \end{pmatrix}_{b_0}$$

[74] $1^{111}3^{112}221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 310, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{121}2^13^{14}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 344, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.98 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -5 & 2 & 1 & -3 \\ 6 & 0 & 6 & 2 & 10 & 6 & 6 \\ 1 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -3 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

[76] $1^{112}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 365, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 14.84 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 2 & 0 & 3 & 1 & 1 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 9 & 5 \\ 2 & 0 & 2 & -3 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

[78] $1^{121}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 373, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & -3 \\ 3 & 6 & 0 & 2 & 10 & 6 & 6 \\ 1 & 3 & 0 & 1 & 6 & 0 & 2 \\ 2 & 3 & -1 & -3 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

[77] $1^{121}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 373, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -5 & 2 & 1 & -3 \\ 3 & 0 & 6 & 2 & 10 & 6 & 6 \\ 2 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -3 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

[79] $1^{112}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 391, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -2 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[80] $1^{131}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 406, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.78 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 14 & -1 \\ 1 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & -1 & 17 & 0 \end{pmatrix}_{b_0}$$

[81] $1^{111}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 429, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[82] $1^{112}3^12^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 434, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 15.14 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 0 & 1 & -5 & 1 & 0 & -3 \\ 3 & 0 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 3 & -1 & -2 \end{pmatrix}_{b_0}$$

[83] $1^{121}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 452, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.94 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -5 & 2 & 1 & -3 \\ 4 & 0 & 6 & 2 & 10 & 6 & 6 \\ 1 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -3 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

[84] $1^{131}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 469, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 1 & -3 \\ 4 & 9 & 0 & 3 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 9 & 0 & 2 \\ 2 & 3 & -1 & -3 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

[85] $1^{121}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 478, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & -3 \\ 4 & 6 & 0 & 2 & 10 & 6 & 6 \\ 1 & 3 & 0 & 1 & 6 & 0 & 2 \\ 2 & 3 & -1 & -3 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

[86] $1^{141}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 1 & -3 \\ 5 & 12 & 0 & 4 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 12 & 0 & 2 \\ 2 & 3 & -1 & -3 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

[87] $1^{121}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 494, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 15.11 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 0 & 2 \\ 1 & 0 & 1 & -5 & 2 & 0 & -3 \\ 4 & 0 & 6 & 2 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 6 & -1 & -2 \end{pmatrix}_{b_0}$$

[88] $1^{121}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 499, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -5 & 2 & 1 & -3 \\ 5 & 0 & 6 & 2 & 10 & 6 & 6 \\ 1 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -3 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

[89] $1^{1231}1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 508, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 0 & -2 \\ 3 & 6 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

[90] $1^{121}3^2 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 526, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[91] $1^{141}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 548, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 15.36 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 3 & 2 \\ 1 & 0 & 1 & -5 & 4 & 1 & -3 \\ 6 & 0 & 12 & 4 & 46 & 12 & 12 \\ 1 & 0 & 3 & 1 & 12 & 0 & 2 \\ 2 & 0 & 2 & -3 & 12 & 2 & -2 \end{pmatrix}_{b_0}$$

[92] $1^{111}2^13^121^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 562, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[93] $1^{131}2^13^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 584, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 3 & 2 \\ 1 & 0 & 1 & -5 & 3 & 1 & -3 \\ 5 & 0 & 9 & 3 & 25 & 9 & 9 \\ 1 & 0 & 3 & 1 & 9 & 0 & 2 \\ 2 & 0 & 2 & -3 & 9 & 2 & -2 \end{pmatrix}_{b_0}$$

[94] $1^{121}3^1221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 590, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 11 & -1 \\ 9 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & -1 & 13 & 0 \end{pmatrix}_{b_0}$$

[95] $1^{141}3^1221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.83 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 17 & -1 \\ 14 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & -1 & 21 & 0 \end{pmatrix}_{b_0}$$

[96] $1^{141}2^13^121^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 718, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 3 & 2 \\ 1 & 0 & 1 & -5 & 4 & 1 & -3 \\ 6 & 0 & 12 & 4 & 46 & 12 & 12 \\ 2 & 0 & 3 & 1 & 12 & 0 & 2 \\ 2 & 0 & 2 & -3 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

[97] $1^{121}3^1221^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 736, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[98] $1^{131}3^1221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 760, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[99] $1^{121}3^1221^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 760, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[100] $1^{121} 3^{13} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 814$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.73 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 9 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[101] $1^{131} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 850$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.73 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 1 & -3 \\ 5 & 9 & 0 & 3 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 9 & 0 & 2 \\ 2 & 3 & -1 & -3 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

[102] $1^{121} 3^{12} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 864$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.70 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 11 & -1 \\ 10 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & -1 & 13 & 0 \end{pmatrix}_{b_0}$$

[103] $1^{131} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 907$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.94 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 3 & 2 \\ 1 & 0 & 1 & -5 & 3 & 1 & -3 \\ 6 & 0 & 9 & 3 & 25 & 9 & 9 \\ 1 & 0 & 3 & 1 & 9 & 0 & 2 \\ 2 & 0 & 2 & -3 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

[104] $1^{1231} 2^{1231} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -148$, $\det = 937$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 15.02 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & 0 & -2 \\ 5 & 12 & 0 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 8 & -1 & -2 \end{pmatrix}_{b_0}$$

[105] $1^{121} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 942$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -5 & 2 & 1 & -3 \\ 4 & 0 & 6 & 2 & 10 & 6 & 6 \\ 2 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -3 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

[106] $1^{131} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 994$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 3 & 2 \\ 1 & 0 & 1 & -5 & 3 & 1 & -3 \\ 5 & 0 & 9 & 3 & 25 & 9 & 9 \\ 2 & 0 & 3 & 1 & 9 & 0 & 2 \\ 2 & 0 & 2 & -3 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

[107] $1^{121} 3^{13} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 1006$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.70 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 11 & -1 \\ 11 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & -1 & 13 & 0 \end{pmatrix}_{b_0}$$

[108] $1^{141} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 1006$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 14.84 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 13 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[109] \quad 1^{121} 3^{14} 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 1016, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 11 & -1 \\ 12 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & -1 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[110] \quad 1^{131} 3^{11} 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 1016, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 14 & -1 \\ 12 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & -1 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[111] \quad 1^{131} 2^3 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 1028, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 3 & 2 \\ 1 & 0 & 1 & -5 & 3 & 1 & -3 \\ 7 & 0 & 9 & 3 & 25 & 9 & 9 \\ 1 & 0 & 3 & 1 & 9 & 0 & 2 \\ 2 & 0 & 2 & -3 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[112] \quad 1^{141} 2^3 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 1147, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 3 & 2 \\ 1 & 0 & 1 & -5 & 4 & 1 & -3 \\ 7 & 0 & 12 & 4 & 46 & 12 & 12 \\ 1 & 0 & 3 & 1 & 12 & 0 & 2 \\ 2 & 0 & 2 & -3 & 12 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[113] \quad 1^{141} 2^3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 1150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 1 & -3 \\ 6 & 12 & 0 & 4 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 12 & 0 & 2 \\ 2 & 3 & -1 & -3 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[114] \quad 1^{131} 3^{12} 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 1174, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 11 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[115] \quad 1^{1231} 3^{13} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 1300, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 1 & 3 & 18 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 9 & 0 & -2 \\ 7 & 18 & 0 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 12 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[116] \quad 1^{141} 3^{11} 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 1454, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 17 & -1 \\ 15 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & -1 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[117] \quad 1^{131} 3^{11} 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1456, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.74 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 12 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[118] \quad 1^{131} 3^{11} 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1494, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.72 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 14 & -1 \\ 13 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & -1 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[119] \quad 1^{141} 2^3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -142, \quad \det = 1544, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.94 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 3 & 2 \\ 1 & 0 & 1 & -5 & 4 & 1 & -3 \\ 8 & 0 & 12 & 4 & 46 & 12 & 12 \\ 1 & 0 & 3 & 1 & 12 & 0 & 2 \\ 2 & 0 & 2 & -3 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[120] \quad 1^{231} 1^{431} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \quad T_{\max} = 101, \quad \Delta = -148, \quad \det = 1597, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.02 \rightarrow 16, \quad T_{\max} = 101, \quad T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 1 & 3 & 24 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 12 & 0 & -2 \\ 9 & 24 & 0 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 16 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[121] \quad 1^{131} 3^{11} 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1606, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.74 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[122] \quad 1^{141} 3^{11} 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1624, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.78 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[123] \quad 1^{131} 3^{11} 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1840, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.71 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 14 & -1 \\ 14 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & -1 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[124] \quad 1^{131} 3^{11} 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 2054, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.71 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 14 & -1 \\ 15 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & -1 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[125] \quad 1^{141} 3^{11} 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 2110, \quad (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{141} 3^{112} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 2136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 17 & -1 \\ 16 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & -1 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{141} 3^{114} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 2464, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{141} 3^{113} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 2686, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 17 & -1 \\ 17 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & -1 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{141} 3^{114} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 3104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 17 & -1 \\ 18 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & -1 & 21 & 0 \end{pmatrix}_{b_0}$$

4.209 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{111} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 68, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 79, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.84 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1111} 3^{112} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 156, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.18 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 0 & -3 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 \\ 1 & 2 & 0 & 1 & 2 & 0 & 0 \\ 2 & 2 & -1 & -3 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 221, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 15.18 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 9 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 3 & 4 & -1 & -2 & 0 & 12 & -2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 413, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 420, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 464, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 6 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & -2 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 496, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.69 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -4 & 3 & 3 & -2 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 589, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 1088, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [11] \quad 1^{\textcolor{red}{11}} 1^{\textcolor{blue}{11}} 3 \quad 221^{\textcolor{green}{1}} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}7 + \textcolor{orange}{s}u_2 + \textcolor{red}{s}u_3 + \textcolor{blue}{s}u_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -56, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111} 3^{221} + [\textcolor{red}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_7 + \textcolor{orange}{s} \textcolor{orange}{u}_2 + \textcolor{red}{s} \textcolor{red}{u}_3 + \textcolor{blue}{s} \textcolor{blue}{u}_2 \quad T = 11, \quad T_{\max} = 23, \quad \Delta = -110, \quad \det = -65, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.94 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1111} 2^3 21^1 + [\textcolor{red}{5} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_7 + \textcolor{orange}{5} \textcolor{orange}{u}_2 + \textcolor{red}{5} \textcolor{red}{u}_3 + \textcolor{blue}{5} \textcolor{blue}{u}_2 \quad T = 11, \quad T_{\max} = 25, \quad \Delta = -112, \quad \det = -84, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 \\ 2 & 3 & -1 & -3 & 3 & 2 & -1 \end{pmatrix} b_0$$

[14] $1^{1111} 3^{221} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -92, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.83 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix} b_0$$

[15] $1^{11} 1^{11} 2^3 3^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 \\ 2 & 3 & 0 & 1 & 3 & 3 & 1 \\ 2 & 3 & -1 & -3 & 3 & 2 & -1 \end{pmatrix} b_0$$

[16] $1^{11} 1^{11} 3^{22} 1^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{s}u_2 + \textcolor{red}{s}u_3 + \textcolor{blue}{s}u_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 15.22 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 0 & -2 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & -1 & -2 \end{pmatrix} b_0$$

$$[17] \quad 1^{\overline{111}} 3 \, 221^{\overline{11}} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_7 + \textcolor{orange}{s}u_2 + \textcolor{red}{s}u_3 + \textcolor{blue}{s}u_2 \quad T = 11, \, T_{\max} = 23, \, \Delta = -110, \, \det = -149, \, (n_+, n_-, n_0) = (1, 11, 0), \, c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 6 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 8 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 8 & 0 & 15 & 9 \\ 3 & 4 & -1 & -1 & 0 & 9 & 0 \end{pmatrix} b_0$$

[18] $1^{113} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -173, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 15.39 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 3 & 4 & -1 & -1 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

[19] $1^{1111}2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -246, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 15.15 \rightarrow 16, T_{\max} = 25, T \in [16, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 & 3 \\ 1 & 0 & 3 & 1 & 3 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

[20] $1^{111}3^1 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -269, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$

$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 12 & 3 \\ 3 & 4 & 1 & 3 & 0 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 14 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 12 & 21 & 0 & 14 & 0 & 75 & 17 \\ 3 & 4 & -1 & -1 & 0 & 17 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{1111}2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -290, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 1 & 3 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[22] $1^{112}3^1 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -332, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 9 & 3 \\ 3 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 11 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 0 & 11 & 0 & 39 & 13 \\ 3 & 4 & -1 & -1 & 0 & 13 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{111}3^1 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -341, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{1111}3^1 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -344, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{1231}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 38, \Delta = -118, \det = -368, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 15.05 \rightarrow 16, T_{\max} = 38, T \in [16, 38]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 0 & 3 & -2 \\ 2 & 6 & 0 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 6 & 0 & 4 & 4 \\ 2 & 4 & -1 & -2 & 4 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

[26] $1^{112}3^1 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -380, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 14.83 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{1111}2^1 3^1 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -408, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.77 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{1111}2^1 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -432, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.96 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 & 3 \\ 1 & 0 & 3 & 1 & 3 & 0 & 3 & 2 \\ 3 & 0 & 3 & 1 & 3 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[29] $1^{1112}2^1 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -434, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 6 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 6 & 2 \\ 3 & 6 & 0 & 2 & 6 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[30] $1^1 231^{111} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 38, \Delta = -118, \det = -455, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 15.09 \rightarrow 16, T_{\max} = 38, T \in [16, 38]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 0 & 3 & -2 \\ 2 & 6 & 0 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 6 & 0 & 4 & 4 \\ 2 & 4 & -1 & -2 & 4 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

[31] $1^{1111}2^1 3^1 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -485, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{1111}2^1 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -540, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 & 3 \\ 1 & 0 & 3 & 1 & 3 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[33] $1^{1113}2^1 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -626, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$

$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 9 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 9 & 2 \\ 4 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 3 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

[34] $1^{1113}2^1 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -668, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 9 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 9 & 2 \\ 5 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 3 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{111} 4^1 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -674, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 12 & 2 \\ 5 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{111} 1^1 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -896, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{111} 4^1 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -1268, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 12 & 2 \\ 6 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

4.210 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{111} 1^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 16, T_{\max} = 23, T \in [16, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 0 & 1 & -3 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 & 0 \\ 2 & 2 & -1 & -3 & 2 & -1 & 2 & -3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 1^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -202, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.85 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [2] \ 1^{111} 1^1 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 1^1 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 166, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.95 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11112} 2^3 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 244, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 6 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 6 & 0 & 2 & 6 & 6 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 3 & 2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11114} 2^3 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 472, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 12 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 12 & 3 \\ 6 & 12 & 0 & 4 & 12 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 2 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11113} 2^3 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 628, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 15.8$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 9 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 9 & 3 \\ 4 & 9 & 0 & 3 & 9 & 9 & 25 & 9 & 9 \\ 2 & 3 & -1 & -3 & 3 & 2 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11114} 2^3 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 724, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.77 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 12 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 12 & 3 \\ 5 & 12 & 0 & 4 & 12 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 3 & 2 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

4.211 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \quad 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 0 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

4.212 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 11 \quad [0] \quad 1^{1111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 38, \Delta = -118, \det = -284, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 38, T \in [16, 38]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 0 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1] \ 1^{\overline{11111}} 2 \overline{3} 21^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_3 \quad T = 12, T_{\max} = 16, \Delta = -91, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3 \overline{2} 21^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_3 \quad T = 12, T_{\max} = 16, \Delta = -89, \det = 228, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 0 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

4.213 NHC(-2)(-3) + e₇ + su₂ + su₃ + su₂ → (-3)

$$T = 10 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 4 & -1 & -2 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{121}} 3^{\overline{1}} 21^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.80 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & 2 & -2 \\ 6 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1}} 21^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 348, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 2 & -2 \\ 3 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{112}} 3^{\overline{1}} 21^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 604, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 2 & -2 \\ 5 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{1}} 21^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 2 & -2 \\ 4 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{121}} 3^{\overline{1}} 21^{\overline{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & 2 & -2 \\ 7 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{12} 3^{11} 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 100, \ \Delta = -142, \ \det = 1408, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.75 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & 2 & -2 \\ 8 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \ 1^{12} 3^{11} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 70, \ \Delta = -121, \ \det = -152, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 15.16 \rightarrow 16, \ T_{\max} = 70, \ T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & 0 & 2 & -2 \\ 4 & 12 & 0 & 6 & 22 & 0 & 6 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 6 & 0 & 0 & 2 \\ 2 & 4 & -1 & -2 & 8 & -1 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{12} 3^{11} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 70, \ \Delta = -121, \ \det = -296, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.08 \rightarrow 16, \ T_{\max} = 70, \ T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 0 & 2 & -2 \\ 2 & 6 & 0 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 3 & 0 & 0 & 2 \\ 2 & 4 & -1 & -2 & 4 & -1 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{12} 3^{11} 3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 59, \ \Delta = -120, \ \det = -364, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.15 \rightarrow 16, \ T_{\max} = 59, \ T \in [16, 59]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 1 & 3 \\ 4 & 8 & 1 & 5 & 10 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 5 & 0 & -1 \\ 4 & 10 & 0 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{12} 3^{11} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 70, \ \Delta = -121, \ \det = -572, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.07 \rightarrow 16, \ T_{\max} = 70, \ T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 0 & 2 & -2 \\ 3 & 6 & 0 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 3 & 0 & 0 & 2 \\ 2 & 4 & -1 & -2 & 4 & -1 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{12} 3^{11} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 70, \ \Delta = -121, \ \det = -1196, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 15.07 \rightarrow 16, \ T_{\max} = 70, \ T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 6 & 0 & 2 & -2 \\ 5 & 12 & 0 & 6 & 22 & 0 & 6 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 6 & 0 & 0 & 2 \\ 2 & 4 & -1 & -2 & 8 & -1 & 2 & -2 \end{pmatrix}_{b_0}$$

4.214 $\text{NHC}(-2)(-3) + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$

$$T = 11 \quad [0] \ 1^{11} 3^{11} 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, \ T_{\max} = 25, \ \Delta = -112, \ \det = -128, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.88 \rightarrow 15, \ T_{\max} = 25, \ T \in [15, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 2 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

4.215 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + su2n3mix

$$T = 10 \quad [0] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.13 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 5 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 5 & 0 & 12 & 6 \\ 3 & 4 & -1 & -2 & 0 & 6 & -2 \end{pmatrix}_{b_0}$$

4.216 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$

$$T = 10 \quad [1] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.05 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 0 & 5 \\ 2 & 4 & -4 & 3 & 0 & 0 & -2 \\ 4 & 8 & 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -2 & 4 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.75 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -2 & 0 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 280, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 0 & -2 \\ 3 & 6 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -2 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [4] \quad 1^{111} 2^1 3^{121} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.43 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 0 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 0 & -3 \\ 3 & 0 & 3 & 1 & 1 & 3 & 0 & 3 \\ 1 & 0 & 3 & 1 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 3 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 2^1 3^{121} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -95, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.11 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 0 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 0 & -3 \\ 2 & 0 & 3 & 1 & 1 & 3 & 0 & 3 \\ 1 & 0 & 3 & 1 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 3 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 48, \Delta = -119, \det = -159, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.87 \rightarrow 16, T_{\max} = 48, T \in [16, 48]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -1 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 2^1 3^{121} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -162, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{121} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -194, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 15.11 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 0 & 2 \\ 1 & 0 & 1 & -5 & 2 & 1 & 0 & -3 \\ 4 & 0 & 6 & 2 & 10 & 6 & 0 & 6 \\ 1 & 0 & 3 & 1 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 6 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 11, T_{\max} = 48, \Delta = -119, \det = -231, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.15 \rightarrow 16, T_{\max} = 48, T \in [16, 48]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -303, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.30 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 2 & 3 & 0 & 1 & 1 & 3 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 3 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{121} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -402, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 15.49 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & 1 & -3 \\ 4 & 6 & 0 & 2 & 10 & 6 & 6 & 6 \\ 1 & 3 & 0 & 1 & 6 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 6 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 6 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{121} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -807, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & 1 & -3 \\ 3 & 6 & 0 & 2 & 10 & 6 & 6 & 6 \\ 1 & 3 & 0 & 1 & 6 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 6 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 6 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{131} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -1095, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.8$$

$$T_{\min} = 15.21 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 1 & 1 & -3 \\ 4 & 9 & 0 & 3 & 25 & 9 & 9 & 9 \\ 1 & 3 & 0 & 1 & 9 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 9 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 9 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{141} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -1167, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.26 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 1 & 1 & -3 \\ 5 & 12 & 0 & 4 & 46 & 12 & 12 & 12 \\ 1 & 3 & 0 & 1 & 12 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 12 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 12 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

[15] $1^{1311}2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 11, T_{\max} = 70, \Delta = -121, \det = -1518, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.8$

$T_{\min} = 15.23 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 1 & 1 & -3 \\ 5 & 9 & 0 & 3 & 25 & 9 & 9 & 9 \\ 1 & 3 & 0 & 1 & 9 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 9 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 9 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

[16] $1^{1411}2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 11, T_{\max} = 70, \Delta = -121, \det = -2418, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 1 & 1 & -3 \\ 6 & 12 & 0 & 4 & 46 & 12 & 12 & 12 \\ 1 & 3 & 0 & 1 & 12 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 12 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 12 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

4.217 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$

$T = 11$ [0] $1^{1111}3^2 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 38, \Delta = -118, \det = -284, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 38, T \in [16, 38]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 0 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

$T = 12$ [1] $1^{1111}2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 12, T_{\max} = 16, \Delta = -91, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 3 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{1111}3^2 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 12, T_{\max} = 16, \Delta = -89, \det = 228, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 0 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

4.218 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 8$ [1] $1^{11}3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 102, \Delta = -193, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 14.82 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 2 & 4 & -4 & 2 & -2 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 5 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

[0] $1^{11}3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 102, \Delta = -193, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 2 & 0 & -4 & -2 \\ 3 & 3 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

[2] $21^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 102, \Delta = -193, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.39 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 4 \\ 8 & 9 & 4 & 4 & 5 \\ 3 & 4 & -4 & 2 & -2 \\ 3 & 4 & 2 & 0 & 2 \\ 4 & 5 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$T = 9 \quad [3] \quad 1^{12}3^{11}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 2 & -2 \\ 1 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{12}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 102, \Delta = -171, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 0 & 0 & -3 & -1 \\ 3 & 0 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{12}3^{11}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & -2 \\ 1 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{12}3^{11}1^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 102, \Delta = -171, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 8 & 1 & 5 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{12}3^{11}1^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 102, \Delta = -173, \det = -595, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 1 & 2 \\ 3 & 1 & 4 & 1 & 0 & 3 \\ 3 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -5 & 0 & -3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & -3 & 0 & -1 \end{pmatrix}_{b_0}$$

4.219 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \text{NHC}(-2)(-3)$

$T = 10 \quad [0] \quad 1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 100, \Delta = -143, \det = 304, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.79 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 2 & 3 & 0 & 2 & 1 & 0 & 2 \\ 3 & 3 & -1 & -2 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$T = 11 \quad [1] \quad 1^{12}3^{11}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 75, \Delta = -122, \det = -440, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 15.09 \rightarrow 16, T_{\max} = 75, T \in [16, 75]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 3 & 0 & -2 \\ 1 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 2 & 6 & 0 & 3 & 3 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 0 & 1 & -3 & -1 \\ 2 & 4 & -1 & -2 & 2 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

4.220 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, T_{\max} = 103, \Delta = -192, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -4 & 2 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -2 & 2 & -2 & -2 \end{pmatrix}_{b_0}$$

4.221 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & -2 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{13} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 6 & -2 \\ 6 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & -2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.73 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -4 & 2 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & -2 & 2 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{12} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.36 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & -2 \\ 4 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & -2 \\ 3 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{12} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -316, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.44 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & -2 \\ 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -364, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.38 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & -2 \\ 5 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -456, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 14.39 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 6 & -2 \\ 7 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & -2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.39 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 8 & -2 \\ 9 & 12 & 0 & 8 & 46 & 12 \\ 3 & 3 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -584, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.41 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & -2 \\ 6 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{12} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -828, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 14.40 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 6 & -2 \\ 8 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & -2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{12} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 102, \quad \Delta = -163, \quad \det = -1048, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.40 \rightarrow 15, \quad T_{\max} = 102, \quad T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 8 & -2 \\ 10 & 12 & 0 & 8 & 46 & 12 \\ 3 & 3 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [14] \quad 1^1 23^1 131^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -142, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 18, \quad T_{\max} = 100, \quad T \in [18, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 2 & 0 & -2 \\ 1 & 0 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -2 & 2 & 0 & -4 \end{pmatrix}_{b_0}$$

[24] $1^{123}1^{131}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 168, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 15.29 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 2 & 3 & -2 \\ 1 & 0 & 0 & 2 & 0 & 3 & 2 \\ 6 & 0 & 0 & 3 & 3 & 4 & 4 \\ 2 & 0 & -1 & -2 & 2 & 4 & -2 \end{pmatrix}_{b_0}$$

[25] $1^{123}1^{231}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 280, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 2 & 6 & -2 \\ 1 & 0 & 0 & 2 & 0 & 6 & 2 \\ 8 & 0 & 0 & 6 & 6 & 22 & 8 \\ 2 & 0 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[26] $1^{123}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 288, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.89 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 3 & -2 \\ 1 & 3 & 0 & 2 & 0 & 3 & 2 \\ 2 & 6 & 0 & 3 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[27] $1^{1231}1^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -141, \det = 313, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.77 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 5 & -1 \\ 4 & 10 & 0 & 5 & 8 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{1231}1^{13}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -141, \det = 373, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 5 & -1 \\ 6 & 10 & 0 & 5 & 8 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{123}1^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 380, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.86 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 3 & -2 \\ 1 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{123}1^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 480, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 14.93 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 6 & -2 \\ 1 & 3 & 0 & 2 & 0 & 6 & 2 \\ 4 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[31] $1^{1231}1^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -141, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 5 & -1 \\ 5 & 10 & 0 & 5 & 8 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{123}1^{12}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 540, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 14.96 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 6 & -2 \\ 1 & 3 & 0 & 2 & 0 & 6 & 2 \\ 7 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{123}1^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 828, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 6 & -2 \\ 1 & 3 & 0 & 2 & 0 & 6 & 2 \\ 5 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{123}1^{12}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 848, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 6 & -2 \\ 1 & 3 & 0 & 2 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

4.222 NHC(-2)(-3) + e₇ + su₂ → (-3) + su₂ + su₂

$$T = 10 \quad [0] \quad 1^{123}1^{12}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.50 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 2 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 6 & 2 & -2 \\ 6 & 9 & 0 & 6 & 25 & 9 & 9 \\ 2 & 3 & 0 & 2 & 9 & 1 & 3 \\ 3 & 3 & -1 & -2 & 9 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{122}1^{12}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 4 & -2 \\ 4 & 6 & 0 & 4 & 10 & 12 & 6 \\ 4 & 6 & 0 & 4 & 12 & 10 & 6 \\ 3 & 3 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{121}1^{12}3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 2 & -2 \\ 4 & 6 & 0 & 4 & 10 & 6 & 6 \\ 3 & 3 & 0 & 2 & 6 & 1 & 3 \\ 3 & 3 & -1 & -2 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}1^{12}3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 \\ 3 & 3 & -1 & -2 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{121}1^{12}3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 2 & -2 \\ 4 & 6 & 0 & 4 & 10 & 6 & 6 \\ 2 & 3 & 0 & 2 & 6 & 1 & 3 \\ 3 & 3 & -1 & -2 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111}1^{12}3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.36 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 10 & 3 \\ 2 & 1 & 1 & 2 & 6 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 8 & -2 \\ 4 & 6 & 0 & 4 & 10 & 24 & 6 \\ 10 & 12 & 0 & 8 & 24 & 46 & 12 \\ 3 & 3 & -1 & -2 & 6 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 1^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 808, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 1 & 9 & 3 \\ 7 & 9 & 0 & 6 & 9 & 25 & 9 \\ 3 & 3 & -1 & -2 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{12} 1^{12} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 824, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.42 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 1 & 6 & 3 \\ 6 & 6 & 0 & 4 & 6 & 10 & 6 \\ 3 & 3 & -1 & -2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11} 1^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 8 & -2 \\ 2 & 3 & 0 & 2 & 1 & 12 & 3 \\ 9 & 12 & 0 & 8 & 12 & 46 & 12 \\ 3 & 3 & -1 & -2 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{12} 1^{12} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1312, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 1 & 9 & 3 \\ 8 & 9 & 0 & 6 & 9 & 25 & 9 \\ 3 & 3 & -1 & -2 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{12} 1^{12} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 1752, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 8 & -2 \\ 2 & 3 & 0 & 2 & 1 & 12 & 3 \\ 10 & 12 & 0 & 8 & 12 & 46 & 12 \\ 3 & 3 & -1 & -2 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [19] \quad 1^{12} 231^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -51, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.43 \rightarrow 16, T_{\max} = 24, T \in [16, 24]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 4 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & -1 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{12} 231^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -192, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.25 \rightarrow 16, T_{\max} = 25, T \in [16, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 0 & 3 & -2 \\ 1 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -2 & 2 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{12} 231^{11} 231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -368, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 15.30 \rightarrow 16, T_{\max} = 25, T \in [16, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 0 & 6 & -2 \\ 1 & 3 & 0 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 12 & 0 & 6 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -2 & 2 & 0 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{123}1^{12}3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 25, \ \Delta = -112, \ \det = -408, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 15.29 \rightarrow 16, \ T_{\max} = 25, \ T \in [16, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 0 & 6 & -2 \\ 1 & 3 & 0 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 12 & 0 & 6 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -2 & 2 & 0 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{1231}1^{13}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 24, \ \Delta = -111, \ \det = -411, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.78 \rightarrow 15, \ T_{\max} = 24, \ T \in [15, 24]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 5 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

4.223 NHC(-2)(-3) + $\textcolor{green}{e}_7$ + $\textcolor{blue}{su}_2 \rightarrow (-3)$ + $\textcolor{blue}{su}_2$ + $\textcolor{blue}{su}_2$ + $\textcolor{blue}{su}_2$

$$T = 11 \quad [0] \ 1^{111}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -16, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 3 & 1 & 3 \\ 3 & 3 & -1 & -2 & 0 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1211}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -256, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.38 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & 2 & 2 & -2 \\ 4 & 6 & 0 & 4 & 10 & 6 & 6 & 6 \\ 2 & 3 & 0 & 2 & 6 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 6 & 3 & 1 & 3 \\ 3 & 3 & -1 & -2 & 6 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1111}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -400, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.36 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & 2 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 3 & 1 & 3 \\ 3 & 3 & -1 & -2 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1111}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -576, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.39 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & 2 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 3 & 1 & 3 \\ 3 & 3 & -1 & -2 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1112}3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -103, \ \det = -960, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.43 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 6 & 3 \\ 6 & 6 & 0 & 4 & 6 & 6 & 10 & 6 \\ 3 & 3 & -1 & -2 & 3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{11} 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 23, \Delta = -103, \det = -1040, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 5 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 2 & 3 & 1 & 6 & 3 \\ 5 & 6 & 0 & 4 & 6 & 6 & 10 & 6 \\ 3 & 3 & -1 & -2 & 3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

4.224 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$T = 11$ [0] $1^{11} 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 3 & -1 & -2 & 0 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.225 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$

$T = 11$ [0] $1^{11} 23^{11} 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 27, \Delta = -114, \det = -108, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.22 \rightarrow 16, T_{\max} = 27, T \in [16, 27]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & -1 & -1 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11} 231^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 27, \Delta = -114, \det = -204, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 27, T \in [15, 27]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 2 & 3 \\ 4 & 8 & 1 & 5 & 0 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 0 & 0 & 2 \\ 3 & 6 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

4.226 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$

$T = 11$ [0] $1^{11} 23^{11} 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 27, \Delta = -114, \det = -168, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 27, T \in [15, 27]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 & 3 \\ 4 & 8 & 1 & 5 & 0 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 3 & 0 & 1 & 3 \\ 3 & 6 & -1 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

4.227 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$T = 10$ [0] $1^{11} 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 3 & -1 & -2 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1231^{11}3^11^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 11, T_{\max} = 59, \Delta = -120, \det = -364, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.15 \rightarrow 16, T_{\max} = 59, T \in [16, 59]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 1 & 3 \\ 4 & 8 & 1 & 5 & 10 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 5 & 0 & -1 \\ 4 & 10 & 0 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{123^11^{11}3^11^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -572, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 3 & 0 & -2 \\ 1 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 6 & 0 & 3 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 2 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{123^11^{12}3^11^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -1196, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 6 & 0 & -2 \\ 1 & 3 & 0 & 2 & 0 & 6 & 0 & 2 \\ 5 & 12 & 0 & 6 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 2 & 8 & -1 & -2 \end{pmatrix}_{b_0}$$

4.228 NHC(-2)(-3) + $\mathbf{e}_7$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{su}_2$ + $\mathbf{su}_3$ + $\mathbf{su}_2$

$$T = 11 \quad [0] \ 1^{1^{11}1^{11}3^12^11^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 0 & 2 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 1 & 3 \\ 3 & 3 & -1 & -2 & 0 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

4.229 NHC(-2)(-3) + $\mathbf{e}_7$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{123^113^11^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -144, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 0 & 0 & -3 & 3 & -1 \\ 2 & 0 & 0 & 3 & 1 & 3 \\ 3 & 0 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12313^{11}1^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -144, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.96 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 0 & 0 & -3 & 2 & -1 \\ 2 & 0 & 0 & 2 & 0 & 2 \\ 3 & 0 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{123^11^{13}1^11^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -144, \det = 204, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & -1 \\ 2 & 5 & 0 & 3 & 1 & 3 \\ 3 & 6 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1231^{13}1^{11}1^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -144, \det = 244, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.84 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 8 & 1 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 2 & -1 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 6 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

4.230 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^1 23^1 1^1 3^1 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 27, \Delta = -114, \det = -108, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.22 \rightarrow 16, T_{\max} = 27, T \in [16, 27]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & -1 & -1 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 231^1 3^1 1^1 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 27, \Delta = -114, \det = -204, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 27, T \in [15, 27]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 4 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 2 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & -1 & -1 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

4.231 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \quad 1^1 23^1 13^1 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 34, \Delta = -117, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15, T_{\max} = 34, T \in [15, 34]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 2 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 0 & 0 & -3 & 3 & 2 & -1 \\ 2 & 0 & 0 & 3 & 1 & 2 & 3 \\ 2 & 0 & 0 & 2 & 2 & 0 & 2 \\ 3 & 0 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 23^1 1^1 3^1 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 34, \Delta = -117, \det = -368, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.96 \rightarrow 15, T_{\max} = 34, T \in [15, 34]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 2 & 3 \\ 4 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & 2 & -1 \\ 2 & 5 & 0 & 3 & 1 & 2 & 3 \\ 2 & 4 & 0 & 2 & 2 & 0 & 2 \\ 3 & 6 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

4.232 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix}$

$$T = 11 \quad [0] \quad 1^1 23^1 1^1 3^1 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 34, \Delta = -117, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15, T_{\max} = 34, T \in [15, 34]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 & 3 \\ 4 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & 2 & -1 \\ 2 & 5 & 0 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 2 & 2 & 0 & 2 \\ 3 & 6 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 23^1 1^1 3^1 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 34, \Delta = -117, \det = -420, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 34, T \in [15, 34]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 & 3 \\ 4 & 8 & 1 & 5 & 4 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 2 & 3 & -1 \\ 2 & 4 & 0 & 2 & 0 & 2 & 2 \\ 3 & 5 & 0 & 3 & 2 & 1 & 3 \\ 3 & 6 & -1 & -1 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

4.233 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$

$$T = 11 \quad [0] \quad 1^1 23^1 1^1 3^1 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 11, T_{\max} = 81, \Delta = -123, \det = -312, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.13 \rightarrow 16, T_{\max} = 81, T \in [16, 81]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

4.234 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix}$

$T = 10$ [0] $1^1 23^1 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 100, \Delta = -144, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 3 & 4 \\ 6 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & -3 & 3 & -1 \\ 3 & 5 & 0 & 3 & 1 & 3 \\ 4 & 6 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

[1] $1^1 23^1 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 100, \Delta = -144, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 4 & 1 & 1 & 3 & 3 & 4 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 0 & 0 & -3 & 3 & -1 \\ 3 & 0 & 0 & 3 & 1 & 3 \\ 4 & 0 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

[2] $1^1 23^1 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 100, \Delta = -144, \det = 106, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 8 & 1 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 2 & -1 \\ 3 & 4 & 0 & 2 & 0 & 2 \\ 3 & 6 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[3] $1^1 23^1 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 100, \Delta = -144, \det = 151, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.97 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 0 & 0 & -3 & 3 & -1 \\ 3 & 0 & 0 & 3 & 1 & 3 \\ 3 & 0 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

[4] $1^1 23^1 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 100, \Delta = -144, \det = 234, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.79 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & -1 \\ 3 & 5 & 0 & 3 & 1 & 3 \\ 3 & 6 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

4.235 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix} + \mathfrak{su}_2$

$T = 11$ [0] $1^1 23^1 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 27, \Delta = -114, \det = -168, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 27, T \in [15, 27]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & 0 & -1 \\ 3 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & -1 & -1 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

4.236 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$

$T = 11$ [0] $1^1 23^1 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 34, \Delta = -117, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15, T_{\max} = 34, T \in [15, 34]$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 3 & 3 & 4 \\ 6 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & -3 & 3 & 2 & -1 \\ 3 & 5 & 0 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 2 & 2 & 0 & 2 \\ 4 & 6 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^1 23^1 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 34, \Delta = -117, \det = -420, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 34, T \in [15, 34]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 2 & 3 \\ 4 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & 2 & -1 \\ 3 & 5 & 0 & 3 & 1 & 2 & 3 \\ 2 & 4 & 0 & 2 & 2 & 0 & 2 \\ 3 & 6 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

4.237 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -4 & 2 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -2 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & -2 \\ 4 & 6 & 0 & 4 & 9 & 5 \\ 3 & 3 & -1 & -2 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & -2 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 3 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [3] \ 1^{12} 23^{11} 131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -151, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 18, T_{\max} = 101, T \in [18, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 2 & 0 & -2 \\ 1 & 0 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & -1 & -2 & 2 & -1 & -4 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{12} 3^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -151, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 0 & -2 \\ 1 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{12} 3^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -150, \det = 237, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.10 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

4.238 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 0 & -2 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & -2 & 2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.80 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 4 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & -1 & -1 & 6 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{123}1^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 70$, $\Delta = -121$, $\det = -572$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.07 \rightarrow 16$, $T_{\max} = 70$, $T \in [16, 70]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 0 & -2 \\ 1 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & 0 & 3 & 3 & 0 & 4 \\ 2 & 4 & -1 & -2 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

[11] $1^{123}1^{12}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 70$, $\Delta = -121$, $\det = -1196$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.5$

$T_{\min} = 15.07 \rightarrow 16$, $T_{\max} = 70$, $T \in [16, 70]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 12 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 0 & 6 \\ 1 & 3 & 0 & 2 & 0 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 5 & 12 & 0 & 6 & 6 & 0 & 22 \\ 2 & 4 & -1 & -2 & 2 & -1 & 8 \end{pmatrix}_{b_0}$$

4.239 NHC(-2)(-3) + e₇ + su₂→(-3) + su₃ + su₂ + su₂

$T = 11$ [0] $1^{111}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11$, $T_{\max} = 25$, $\Delta = -112$, $\det = -128$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 14.88 \rightarrow 15$, $T_{\max} = 25$, $T \in [15, 25]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 2 & 0 & 2 & -2 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 3 & 0 & 1 & 3 \\ 3 & 3 & -1 & -2 & 2 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

4.240 NHC(-2)(-3) + e₇ + su₂→(-3) + su₃ + su₂→(-3)

$T = 11$ [0] $1^{123}1^{13}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_3 + \text{su}_2 \rightarrow (-3)$ $T = 11$, $T_{\max} = 81$, $\Delta = -123$, $\det = -312$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.13 \rightarrow 16$, $T_{\max} = 81$, $T \in [16, 81]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

4.241 NHC(-2)(-3) + e₇ + su₂→(-3) + su₃ + su₃

$T = 10$ [0] $1^{111}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_3 + \text{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -151$, $\det = 60$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 15.20 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 0 & 5 \\ 2 & 4 & -4 & 2 & 0 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -2 & 2 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

4.242 NHC(-2)(-3) + e₇ + su₂n₃mix

$T = 8$ [0] $1^{13}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 8$, $T_{\max} = 102$, $\Delta = -193$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 14.63 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 4 \\ 3 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 2 & 0 & -4 & -2 \\ 4 & 3 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{13}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 8$, $T_{\max} = 102$, $\Delta = -193$, $\det = 108$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 15.04 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 5 & -3 \\ 6 & 9 & 0 & 5 & 39 & 9 \\ 2 & 2 & -1 & -3 & 9 & -2 \end{pmatrix}_{b_0}$$

$$[12] \text{ } 1^{12} 3^{12} 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 102, \Delta = -193, \det = 771, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & -3 \\ 6 & 7 & 0 & 4 & 23 & 7 \\ 2 & 2 & -1 & -3 & 7 & -2 \end{pmatrix}_{b_0}$$

$$[13] \text{ } 1^{12} 3^{12} 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 102, \Delta = -193, \det = 1004, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 5 & -3 \\ 7 & 9 & 0 & 5 & 39 & 9 \\ 2 & 2 & -1 & -3 & 9 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [14] \text{ } 1^{12} 23^{11} 1^{11} 31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 5 & -2 \\ 7 & 9 & 0 & 5 & 12 & 6 \\ 2 & 4 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[15] \text{ } 1^{12} 231^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 102, \Delta = -171, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 4 \\ 6 & 8 & 1 & 5 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 5 & 0 & -3 & -1 \\ 4 & 6 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[16] \text{ } 1^{12} 231^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 102, \Delta = -171, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.94 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 4 & -2 \\ 4 & 0 & 0 & 4 & 9 & 5 \\ 2 & 0 & -1 & -2 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[17] \text{ } 1^{12} 23^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.96 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 2 & -2 \\ 2 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[18] \text{ } 1^{12} 231^{11} 3^{11} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 102, \Delta = -171, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 4 & -2 \\ 6 & 8 & 0 & 4 & 9 & 5 \\ 2 & 4 & -1 & -2 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[19] \text{ } 1^{12} 2313^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 102, \Delta = -171, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15.05 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 4 & 1 & 1 & 3 & 4 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 0 & 0 & -3 & -1 \\ 4 & 0 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[20] \text{ } 1^{12} 23^{11} 1^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & -2 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[21] \text{ } 1^{12} 23^{11} 1^{11} 31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 11 & -2 \\ 11 & 0 & 0 & 11 & 72 & 14 \\ 2 & 0 & -1 & -2 & 14 & -1 \end{pmatrix}_{b_0}$$

[33] $1^1 23^1 1^1 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -432$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.71 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 5 & -2 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 2 & 4 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[34] $1^1 23^1 1^1 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -449$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.92 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 5 & -2 \\ 6 & 0 & 0 & 5 & 12 & 6 \\ 2 & 0 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[35] $1^1 23^1 131^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -572$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 15.14 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 2 & -2 \\ 5 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & -1 & -2 & 2 & -2 \end{pmatrix}_{b_0}$$

[36] $1^1 23^1 1^2 31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -704$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.91 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 8 & -2 \\ 9 & 0 & 0 & 8 & 36 & 10 \\ 2 & 0 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

[37] $1^1 23^1 1^2 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -753$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.78 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 8 & -2 \\ 6 & 15 & 0 & 8 & 36 & 10 \\ 2 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

[38] $1^1 23^1 1^1 31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -759$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.95 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 5 & -2 \\ 7 & 0 & 0 & 5 & 12 & 6 \\ 2 & 0 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[39] $1^1 23^1 1^2 31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -912$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.68 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 8 & -2 \\ 9 & 15 & 0 & 8 & 36 & 10 \\ 2 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

[40] $1^1 23^1 1^2 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -964$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.73 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 8 & -2 \\ 7 & 15 & 0 & 8 & 36 & 10 \\ 2 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

[41] $1^1 23^1 1^2 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -1017$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.70 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 8 & -2 \\ 8 & 15 & 0 & 8 & 36 & 10 \\ 2 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

4.243 NHC(-2)(-3) + e₇ + su2n3mix + su₂

$T = 9$ [0] $1^{1143^1} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -163$, $\det = -46$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.70 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 4 & -3 \\ 2 & 3 & 0 & 2 & 3 & 12 & 3 \\ 4 & 8 & 0 & 4 & 12 & 30 & 8 \\ 2 & 2 & -1 & -3 & 3 & 8 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 15.09 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 3 & 4 \\ 3 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 2 & 0 & -4 & 2 & -2 \\ 3 & 3 & 0 & 2 & 1 & 3 \\ 4 & 3 & -1 & -2 & 3 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{112}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & -3 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 3 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -3 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

[4] $1^{11}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 0 & -3 \\ 2 & 3 & 0 & 2 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -3 & 3 & 0 & -2 \end{pmatrix}_{b_0}$$

[3] $1^{131}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 15.17 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 1 & -3 \\ 4 & 7 & 0 & 4 & 23 & 7 & 7 \\ 2 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & -3 & 7 & 2 & -3 \end{pmatrix}_{b_0}$$

[5] $1^{11}3^{11}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 14.46 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 3 & -3 \\ 2 & 3 & 0 & 2 & 3 & 9 & 3 \\ 3 & 6 & 0 & 3 & 9 & 16 & 6 \\ 2 & 2 & -1 & -3 & 3 & 6 & -2 \end{pmatrix}_{b_0}$$

[6] $1^{112}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 14.54 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 2 & -3 \\ 6 & 7 & 0 & 4 & 23 & 14 & 7 \\ 2 & 4 & 0 & 2 & 14 & 6 & 4 \\ 2 & 2 & -1 & -3 & 7 & 4 & -2 \end{pmatrix}_{b_0}$$

[7] $1^{124}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 4 & -3 \\ 3 & 5 & 0 & 3 & 11 & 20 & 5 \\ 4 & 8 & 0 & 4 & 20 & 30 & 8 \\ 2 & 2 & -1 & -3 & 5 & 8 & -2 \end{pmatrix}_{b_0}$$

[8] $1^{11}3^{11}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 14.42 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 2 & -3 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 2 & -1 & -3 & 3 & 4 & -2 \end{pmatrix}_{b_0}$$

[9] $1^{142}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 15.16 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 5 & 2 & -3 \\ 5 & 9 & 0 & 5 & 39 & 18 & 9 \\ 3 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & -3 & 9 & 4 & -3 \end{pmatrix}_{b_0}$$

[10] $1^{11}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & -3 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -3 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

[11] $1^{122}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 2 & -3 \\ 4 & 5 & 0 & 3 & 11 & 10 & 5 \\ 2 & 4 & 0 & 2 & 10 & 6 & 4 \\ 2 & 2 & -1 & -3 & 5 & 4 & -2 \end{pmatrix}_{b_0}$$

[13] $1^{121}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.51 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 1 & -3 \\ 3 & 5 & 0 & 3 & 11 & 5 & 5 \\ 2 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & -3 & 5 & 2 & -2 \end{pmatrix}_{b_0}$$

[12] $1^{123}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.51 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 3 & -3 \\ 3 & 5 & 0 & 3 & 11 & 15 & 5 \\ 3 & 6 & 0 & 3 & 15 & 16 & 6 \\ 2 & 2 & -1 & -3 & 5 & 6 & -2 \end{pmatrix}_{b_0}$$

[14] $1^{111}3^{12}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.68 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & -3 \\ 4 & 3 & 0 & 2 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -3 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

[15] $1^{111}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & -3 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -3 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

[16] $1^{122}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.47 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 2 & -3 \\ 3 & 5 & 0 & 3 & 11 & 10 & 5 \\ 2 & 4 & 0 & 2 & 10 & 6 & 4 \\ 2 & 2 & -1 & -3 & 5 & 4 & -2 \end{pmatrix}_{b_0}$$

[17] $1^{111}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.44 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & -3 \\ 3 & 3 & 0 & 2 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -3 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

[18] $1^{123}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -340, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.46 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 1 & -3 \\ 3 & 5 & 0 & 3 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & -3 & 5 & 2 & -2 \end{pmatrix}_{b_0}$$

[19] $1^{132}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -391, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.61 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 2 & -3 \\ 4 & 7 & 0 & 4 & 23 & 14 & 7 \\ 3 & 4 & 0 & 2 & 14 & 6 & 4 \\ 2 & 2 & -1 & -3 & 7 & 4 & -2 \end{pmatrix}_{b_0}$$

[20] $1^{123}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.54 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 3 & -3 \\ 3 & 5 & 0 & 3 & 11 & 15 & 5 \\ 5 & 6 & 0 & 3 & 15 & 16 & 6 \\ 2 & 2 & -1 & -3 & 5 & 6 & -2 \end{pmatrix}_{b_0}$$

[21] $1^{132}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -430, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.52 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 2 & -3 \\ 4 & 7 & 0 & 4 & 23 & 14 & 7 \\ 2 & 4 & 0 & 2 & 14 & 6 & 4 \\ 2 & 2 & -1 & -3 & 7 & 4 & -2 \end{pmatrix}_{b_0}$$

[23] $1^{131}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -466, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.51 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 1 & -3 \\ 4 & 7 & 0 & 4 & 23 & 7 & 7 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & -3 & 7 & 2 & -2 \end{pmatrix}_{b_0}$$

[22] $1^{112}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -466, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 2 & -3 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 \\ 4 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 2 & -1 & -3 & 3 & 4 & -2 \end{pmatrix}_{b_0}$$

[24] $1^{112}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -471, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 2 & -3 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 \\ 3 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 2 & -1 & -3 & 3 & 4 & -2 \end{pmatrix}_{b_0}$$

[25] $1^{122}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -480, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.47 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 2 & -3 \\ 3 & 5 & 0 & 3 & 11 & 10 & 5 \\ 3 & 4 & 0 & 2 & 10 & 6 & 4 \\ 2 & 2 & -1 & -3 & 5 & 4 & -2 \end{pmatrix}_{b_0}$$

[26] $1^{1^{11}} 1^{132} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -518, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 2 & -3 \\ 5 & 7 & 0 & 4 & 23 & 14 & 7 \\ 2 & 4 & 0 & 2 & 14 & 6 & 4 \\ 2 & 2 & -1 & -3 & 7 & 4 & -2 \end{pmatrix}_{b_0}$$

[27] $1^{1^{12}} 1^{142} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -568, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.56 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 5 & 2 & -3 \\ 5 & 9 & 0 & 5 & 39 & 18 & 9 \\ 2 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & -3 & 9 & 4 & -2 \end{pmatrix}_{b_0}$$

[28] $1^{1^{12}} 1^{142} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -600, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 5 & 2 & -3 \\ 7 & 9 & 0 & 5 & 39 & 18 & 9 \\ 2 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & -3 & 9 & 4 & -2 \end{pmatrix}_{b_0}$$

[29] $1^{1^{11}} 1^{141} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -604, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.55 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 5 & 1 & -3 \\ 5 & 9 & 0 & 5 & 39 & 9 & 9 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & -3 & 9 & 2 & -2 \end{pmatrix}_{b_0}$$

[30] $1^{1^{11}} 1^{121} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -609, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 1 & -3 \\ 4 & 5 & 0 & 3 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & -3 & 5 & 2 & -2 \end{pmatrix}_{b_0}$$

[31] $1^{1^{11}} 1^{113} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -623, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 3 & -3 \\ 2 & 3 & 0 & 2 & 3 & 9 & 3 \\ 4 & 6 & 0 & 3 & 9 & 16 & 6 \\ 2 & 2 & -1 & -3 & 3 & 6 & -2 \end{pmatrix}_{b_0}$$

[32] $1^{1^{11}} 1^{121} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -692, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.46 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 3 & -3 \\ 3 & 5 & 0 & 3 & 11 & 15 & 5 \\ 4 & 6 & 0 & 3 & 15 & 16 & 6 \\ 2 & 2 & -1 & -3 & 5 & 6 & -2 \end{pmatrix}_{b_0}$$

[33] $1^{1^{12}} 1^{121} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 1 & -3 \\ 5 & 5 & 0 & 3 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & -3 & 5 & 2 & -2 \end{pmatrix}_{b_0}$$

[34] $1^{114}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -751, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 4 & -3 \\ 2 & 3 & 0 & 2 & 3 & 12 & 3 \\ 5 & 8 & 0 & 4 & 12 & 30 & 8 \\ 2 & 2 & -1 & -3 & 3 & 8 & -2 \end{pmatrix}_{b_0}$$

[35] $1^{142}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -820, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 5 & 2 & -3 \\ 6 & 9 & 0 & 5 & 39 & 18 & 9 \\ 2 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & -3 & 9 & 4 & -2 \end{pmatrix}_{b_0}$$

[36] $1^{112}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -830, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 3 & -3 \\ 2 & 3 & 0 & 2 & 3 & 9 & 3 \\ 5 & 6 & 0 & 3 & 9 & 16 & 6 \\ 2 & 2 & -1 & -3 & 3 & 6 & -2 \end{pmatrix}_{b_0}$$

[37] $1^{124}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -864, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.48 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 4 & -3 \\ 3 & 5 & 0 & 3 & 11 & 20 & 5 \\ 6 & 8 & 0 & 4 & 20 & 30 & 8 \\ 2 & 2 & -1 & -3 & 5 & 8 & -2 \end{pmatrix}_{b_0}$$

[38] $1^{124}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -880, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 4 & -3 \\ 3 & 5 & 0 & 3 & 11 & 20 & 5 \\ 5 & 8 & 0 & 4 & 20 & 30 & 8 \\ 2 & 2 & -1 & -3 & 5 & 8 & -2 \end{pmatrix}_{b_0}$$

[39] $1^{131}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -899, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 1 & -3 \\ 5 & 7 & 0 & 4 & 23 & 7 & 7 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & -3 & 7 & 2 & -2 \end{pmatrix}_{b_0}$$

[40] $1^{114}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -1170, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 4 & -3 \\ 2 & 3 & 0 & 2 & 3 & 12 & 3 \\ 6 & 8 & 0 & 4 & 12 & 30 & 8 \\ 2 & 2 & -1 & -3 & 3 & 8 & -2 \end{pmatrix}_{b_0}$$

[41] $1^{131}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -1190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.38 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 4 & 1 & -3 \\ 6 & 7 & 0 & 4 & 23 & 7 & 7 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & -3 & 7 & 2 & -2 \end{pmatrix}_{b_0}$$

[42] $1^{141}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -1201, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 5 & 1 & -3 \\ 6 & 9 & 0 & 5 & 39 & 9 & 9 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & -3 & 9 & 2 & -2 \end{pmatrix}_{b_0}$$

[43] $1^{141}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -163, \det = -1656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 5 & 1 & -3 \\ 7 & 9 & 0 & 5 & 39 & 9 & 9 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & -3 & 9 & 2 & -2 \end{pmatrix}_{b_0}$$

$T = 10$ [44] $1^{123}1^{131}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 0 & -2 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 2 & 0 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{123}1^{131}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -141, \det = 97, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 4 & 0 & -2 \\ 3 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 5 & 0 & -2 \end{pmatrix}_{b_0}$$

[46] $1^{123}1^{131}1^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 109, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 5 & 0 & -2 \\ 6 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 6 & 0 & -1 \end{pmatrix}_{b_0}$$

[47] $1^{123}1^{131}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 157, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 2 & 6 & 0 & 3 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{123}1^{131}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 163, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 15.17 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 2 & 3 & -2 \\ 2 & 0 & 0 & 2 & 0 & 3 & 2 \\ 3 & 0 & 0 & 3 & 3 & 4 & 4 \\ 2 & 0 & -1 & -2 & 2 & 4 & -2 \end{pmatrix}_{b_0}$$

[49] $1^{123}1^{131}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 164, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 2 & 0 & -2 \\ 2 & 0 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -2 & 2 & 0 & -2 \end{pmatrix}_{b_0}$$

[50] $1^{1231}1^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -141, \det = 182, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 7 & 4 \\ 6 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & -3 & 5 & -1 \\ 7 & 10 & 0 & 5 & 8 & 6 \\ 4 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{1231}1^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 285, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.88 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 5 & 0 & -2 \\ 4 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 6 & 0 & -1 \end{pmatrix}_{b_0}$$

[52] $1^{1231}1^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -141, \det = 303, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.77 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 4 & 0 & -2 \\ 4 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 5 & 0 & -1 \end{pmatrix}_{b_0}$$

[53] $1^{1231}1^{1231}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 322, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 15.19 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 8 & 0 & -2 \\ 6 & 15 & 0 & 8 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 10 & 0 & -2 \end{pmatrix}_{b_0}$$

[54] $1^{1231}1^{11}31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 355, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 5 & 0 & -2 \\ 5 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 6 & 0 & -1 \end{pmatrix}_{b_0}$$

[55] $1^{1231}1^{11}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 373, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 4 & 6 & 0 & 3 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[56] $1^{1231}1^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 429, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[57] $1^{1231}1^{1231}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 562, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 5 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 5 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[58] $1^{1231}1^{1231}1^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 8 & 0 & -2 \\ 9 & 15 & 0 & 8 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 10 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{123}1^{12}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 718, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 8 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 8 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{123}1^{12}31^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 744, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 8 & 0 & -2 \\ 7 & 15 & 0 & 8 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 10 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{123}1^{12}31^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 850, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 8 & 0 & -2 \\ 8 & 15 & 0 & 8 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 10 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{123}1^{12}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 942, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 6 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{123}1^{12}31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 994, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 7 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 7 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

4.244 NHC(-2)(-3) + $\textcolor{green}{e}_7$ + su2n3mix + $\textcolor{red}{su}_2$ + $\textcolor{red}{su}_2$

$$T = 10 \quad [0] \ 1^{11}1^{11}3^{121}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.44 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 0 & 1 & -3 \\ 2 & 3 & 0 & 2 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 & 0 & 0 & 2 \\ 2 & 2 & -1 & -3 & 3 & 0 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{112}1^{12}3^{121}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 174, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 2 & 1 & -3 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 & 3 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 & 4 \\ 2 & 2 & 0 & 1 & 3 & 4 & 0 & 2 \\ 2 & 2 & -1 & -3 & 3 & 4 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112}1^{12}3^{121}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 94, \Delta = -133, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 1 & 1 & -3 \\ 5 & 5 & 0 & 3 & 11 & 5 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 & 0 & 2 \\ 2 & 2 & -1 & -3 & 5 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

[10] $1^{1112}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 838$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & 2 & -3 \\ 2 & 3 & 0 & 2 & 3 & 3 & 6 & 3 \\ 1 & 2 & 0 & 1 & 3 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 6 & 4 & 6 & 4 \\ 2 & 2 & -1 & -3 & 3 & 2 & 4 & -2 \end{pmatrix}_{b_0}$$

[11] $1^{1211}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 94$, $\Delta = -133$, $\det = 964$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.39 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & 1 & 1 & -3 \\ 4 & 5 & 0 & 3 & 11 & 5 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 & 0 & 2 \\ 2 & 2 & -1 & -3 & 5 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

$T = 11$ [12] $1^{1231}1^{11}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 25$, $\Delta = -112$, $\det = -290$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 25$, $T \in [15, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 0 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -2 & 2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

4.245 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2\mathfrak{n}3\text{mix} + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 11$ [0] $1^{1231}1^{11}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 11$, $T_{\max} = 27$, $\Delta = -114$, $\det = -168$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.81 \rightarrow 15$, $T_{\max} = 27$, $T \in [15, 27]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & 0 & -1 \\ 3 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & -1 & -1 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

4.246 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2\mathfrak{n}3\text{mix} + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 10$ [0] $1^{1112}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 181$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.13 \rightarrow 16$, $T_{\max} = 100$, $T \in [16, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 1 & 3 \\ 3 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & 0 & -2 \\ 5 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 6 & 0 & -2 \end{pmatrix}_{b_0}$$

4.247 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2\mathfrak{n}3\text{mix} + \mathfrak{su}_2 \rightarrow (-3)$

$T = 10$ [0] $1^{1231}1^{11}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 100$, $\Delta = -144$, $\det = 16$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 3 & 4 \\ 6 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & -3 & 3 & -1 \\ 3 & 5 & 0 & 3 & 1 & 3 \\ 4 & 6 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{1231}1^{11}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 100$, $\Delta = -144$, $\det = 36$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 15.33 \rightarrow 16$, $T_{\max} = 100$, $T \in [16, 100]$

$$\begin{pmatrix} 4 & 1 & 1 & 3 & 3 & 4 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 0 & 0 & -3 & 3 & -1 \\ 3 & 0 & 0 & 3 & 1 & 3 \\ 4 & 0 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{1231}1^31^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -144, \det = 106, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 3 & 4 \\ 6 & 8 & 1 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & -3 & 2 & -1 \\ 3 & 4 & 0 & 2 & 0 & 2 \\ 4 & 6 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{1231}1^31^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -144, \det = 151, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.97 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 0 & 0 & -3 & 3 & -1 \\ 3 & 0 & 0 & 3 & 1 & 3 \\ 3 & 0 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{1231}1^31^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -144, \det = 234, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.79 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & -1 \\ 3 & 5 & 0 & 3 & 1 & 3 \\ 3 & 6 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

4.248 $\text{NHC}(-2)(-3) + \text{e}_7 + \text{su2n3mix} + \text{su}_2 \rightarrow (-3) + \text{su}_2$

$T = 11 \quad [0] \quad 1^{1231}1^31^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, T_{\max} = 27, \Delta = -114, \det = -168, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 27, T \in [15, 27]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & 0 & -1 \\ 3 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & -1 & -1 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

4.249 $\text{NHC}(-2)(-3) + \text{e}_7 + \text{su2n3mix} + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$

$T = 11 \quad [0] \quad 1^{1231}1^31^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 34, \Delta = -117, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15, T_{\max} = 34, T \in [15, 34]$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 3 & 3 & 4 \\ 6 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & -3 & 3 & 2 & -1 \\ 3 & 5 & 0 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 2 & 2 & 0 & 2 \\ 4 & 6 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{1231}1^31^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 34, \Delta = -117, \det = -420, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 34, T \in [15, 34]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 2 & 3 \\ 4 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 3 & 2 & -1 \\ 3 & 5 & 0 & 3 & 1 & 2 & 3 \\ 2 & 4 & 0 & 2 & 2 & 0 & 2 \\ 3 & 6 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

4.250 $\text{NHC}(-2)(-3) + \text{e}_7 + \text{su2n3mix} + \text{su2n3mix}$

$T = 10 \quad [0] \quad 1^{1231}1^31^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -144, \det = 106, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 4 & 4 \\ 6 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & -3 & 3 & -1 \\ 4 & 5 & 0 & 3 & 1 & 3 \\ 4 & 6 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

4.251 NHC(-2)(-3) + e₇ + su₂n₃mix + su₃

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 3 & 4 \\ 3 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 2 & 0 & -4 & 2 & -2 \\ 3 & 3 & 0 & 2 & 0 & 2 \\ 4 & 3 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -345, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & -2 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 8 & -2 \\ 8 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & -2 \\ 6 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -617, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & -2 \\ 7 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -977, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 8 & -2 \\ 9 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -1280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 8 & -2 \\ 10 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [7] \quad 1^{12} 3^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -151, \det = 103, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.03 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 5 & 0 & -2 \\ 3 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 6 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{12} 3^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -151, \det = 223, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.23 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 2 & 0 & -2 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

[9] $1^{12}31^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -150$, $\det = 358$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 15.03 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 4 & 0 & -2 \\ 3 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 5 & -1 & -2 \end{pmatrix}_{b_0}$$

[10] $1^{12}3^{11}31^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -151$, $\det = 682$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 15.05 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 5 & 0 & -2 \\ 4 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 6 & -1 & -2 \end{pmatrix}_{b_0}$$

[11] $1^{12}3^{11}31^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -151$, $\det = 1075$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 15.03 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 8 & 0 & -2 \\ 6 & 15 & 0 & 8 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & 10 & -1 & -2 \end{pmatrix}_{b_0}$$

4.252 NHC(-2)(-3) + e₇ + su_{2n3mix} + su₃ + su₂

$T = 10$ [0] $1^{111}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 181$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.13 \rightarrow 16$, $T_{\max} = 100$, $T \in [16, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 1 & 3 \\ 3 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & 0 & -2 \\ 5 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 6 & 0 & -2 \end{pmatrix}_{b_0}$$

4.253 NHC(-2)(-3) + e₇ + su₃

$T = 8$ [0] $1^{111}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 8$, $T_{\max} = 102$, $\Delta = -199$, $\det = 68$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.82 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -4 & 0 & -2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 8$, $T_{\max} = 102$, $\Delta = -199$, $\det = 85$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.87 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & -3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -3 & -1 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 8$, $T_{\max} = 102$, $\Delta = -199$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.62 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & -4 & -2 \\ 3 & 4 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{12}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 8$, $T_{\max} = 102$, $\Delta = -199$, $\det = 105$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 14.90 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & -3 \\ 2 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -3 & 3 & -2 \end{pmatrix}_{b_0}$$

- [4] $1^{13} 3 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 8$, $T_{\max} = 102$, $\Delta = -199$, $\det = 120$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 15.13 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & -3 \\ 3 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -3 & 5 & -3 \end{pmatrix}_{b_0}$$
- [5] $1^{11} 3 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 8$, $T_{\max} = 102$, $\Delta = -199$, $\det = 170$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 4 \\ 5 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & -4 & -2 \\ 4 & 4 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$
- [6] $1^{12} 3 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 8$, $T_{\max} = 102$, $\Delta = -199$, $\det = 261$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 14.62 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & -3 \\ 3 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -3 & 3 & -2 \end{pmatrix}_{b_0}$$
- [7] $1^{13} 3 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 8$, $T_{\max} = 102$, $\Delta = -199$, $\det = 358$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 14.61 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 3 & -3 \\ 4 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -3 & 5 & -2 \end{pmatrix}_{b_0}$$
- $T = 9$ [8] $1^{13} 3 2 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -170$, $\det = -9$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 18.33 \rightarrow 19$, $T_{\max} = 102$, $T \in [19, 102]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 6 & 6 & 0 & 6 \\ 2 & 6 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & -4 \end{pmatrix}_{b_0}$$
- [9] $1^{12} 2 3 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -21$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 15.10 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & -3 \\ 2 & 6 & 0 & 2 & 9 & 5 \\ 2 & 3 & -1 & -3 & 5 & -2 \end{pmatrix}_{b_0}$$
- [11] $1^{11} 2 3 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.91 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$
- [10] $1^{12} 2 3 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 16.73 \rightarrow 17$, $T_{\max} = 102$, $T \in [17, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & -3 \\ 2 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & -1 & -3 \end{pmatrix}_{b_0}$$
- [12] $1^{11} 2 3 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -57$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.68 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 2 \\ 2 & 3 & -1 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$
- [13] $1^{12} 2 3 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -65$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 15.09 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & -3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & -1 & -2 \end{pmatrix}_{b_0}$$

- [14] $1^{11} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -170$, $\det = -71$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.94 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$
- [15] $1^{11} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -170$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.70 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & -3 & -1 \\ 3 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$
- [16] $1^{12} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -170$, $\det = -86$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 15.02 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 2 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -2 & 3 & -2 \end{pmatrix}_{b_0}$$
- [17] $1^{13} 2^3 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -89$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 15.34 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -5 & 3 & -3 \\ 4 & 0 & 9 & 3 & 24 & 8 \\ 2 & 0 & 2 & -3 & 8 & -2 \end{pmatrix}_{b_0}$$
- [18] $1^{13} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -170$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 15.33 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -2 & 5 & -2 \end{pmatrix}_{b_0}$$
- [19] $1^{12} 231^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -178$, $\det = -104$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 15.08 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 1 & 0 & 3 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 3 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$
- [20] $1^{12} 231^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -178$, $\det = -108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 3 & -2 \\ 3 & 0 & 0 & 3 & 3 & 3 \\ 2 & 0 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$
- [21] $1^{12} 231^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -178$, $\det = -119$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 15.57 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & -1 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$
- [22] $1^{12} 231^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -178$, $\det = -125$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 15.02 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$
- [23] $1^{11} 2^3 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -129$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & -3 \\ 2 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$
- [24] $1^{12} 2^3 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -172$, $\det = -137$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 14.81 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -5 & 2 & -3 \\ 3 & 0 & 6 & 2 & 9 & 5 \\ 2 & 0 & 2 & -3 & 5 & -1 \end{pmatrix}_{b_0}$$

[25] $1^{11} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -170, \det = -141, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.76 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 4 \\ 5 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & -3 & -1 \\ 4 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{12} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.82 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & -3 \\ 3 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

[27] $1^{12} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -170, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 14.72 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 3 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

[28] $1^1 231^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -178, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 15.19 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & -2 \\ 2 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & -1 & -2 & 3 & -2 \end{pmatrix}_{b_0}$$

[29] $1^1 231^{11} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -178, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.92 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & -2 \\ 4 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{13} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -170, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.72 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 4 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -2 & 5 & -1 \end{pmatrix}_{b_0}$$

[31] $1^{12} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -5 & 2 & -3 \\ 4 & 0 & 6 & 2 & 9 & 5 \\ 2 & 0 & 2 & -3 & 5 & -1 \end{pmatrix}_{b_0}$$

[32] $1^{13} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.66 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -5 & 3 & -3 \\ 5 & 0 & 9 & 3 & 24 & 8 \\ 2 & 0 & 2 & -3 & 8 & -1 \end{pmatrix}_{b_0}$$

[33] $1^1 231^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -178, \det = -345, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.95 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

4.254 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$

$T = 10 \quad [0] 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 101, \Delta = -149, \det = 103, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 15.03 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -4 & 0 & 3 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & -2 & -1 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \quad 1^{1111} 2321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 75, \Delta = -122, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.27 \rightarrow 19, T_{\max} = 75, T \in [19, 75]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 4 \\ 1 & 3 & 0 & 1 & 3 & 4 & 0 \\ 2 & 3 & -1 & -3 & 2 & 3 & -5 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 59, \Delta = -120, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.12 \rightarrow 16, T_{\max} = 59, T \in [16, 59]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 4 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 0 & 4 & 0 & 9 & 5 \\ 3 & 4 & -1 & -1 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

4.255 NHC(-2)(-3) + $\textcolor{green}{e}_7$ + $\textcolor{blue}{su}_3$ + $\textcolor{black}{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 103, \Delta = -198, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.63 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & -3 & -1 & 0 & -3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [1] \quad 1^{11} 2321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 103, \Delta = -171, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.83 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 & 5 \\ 2 & 0 & 4 & -4 & 0 & -2 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 & -1 \\ 3 & 0 & 5 & -2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 2321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 103, \Delta = -171, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 & -1 \\ 2 & 4 & 0 & -4 & 4 & -2 & -2 \\ 4 & 12 & 0 & 4 & 9 & 5 & 5 \\ 3 & 6 & -1 & -2 & 5 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 103, \Delta = -169, \det = 245, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.73 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & -1 & 0 & -2 \end{pmatrix}_{b_0}$$

4.256 NHC(-2)(-3) + $\textcolor{green}{e}_7$ + $\textcolor{blue}{su}_3$ + $\textcolor{blue}{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -113, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.63 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 9 & -2 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.68 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -4 & 3 & -2 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{114} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -377, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.63 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{115} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -417, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.1$$

$$T_{\min} = 15.19 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 15 & -2 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & -2 & 20 & -2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{121} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.62 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & -3 \\ 3 & 4 & 0 & 2 & 5 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -3 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{112} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -641, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & -2 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{112} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & -2 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{112} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -932, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

13.5

$$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & -2 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[19] $1^{113} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -932, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 9 & -2 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[21] $1^{112} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -993, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & -2 \\ 9 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[22] $1^{114} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -1241, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 13 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

[23] $1^{113} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -1433, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 9 & -2 \\ 11 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[24] $1^{113} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -1776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.66 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 9 & -2 \\ 12 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[25] $1^{113} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -1961, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 9 & -2 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[26] $1^{114} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -1988, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.69 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

[27] $1^{113} 3 \ 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -2577, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

[28] $1^{114} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -3008$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.66 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

$T = 10$ [29] $1^{111} 23 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 4$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 24$, $T_{\max} = 100$, $T \in [24, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 1 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & 3 & -10 \end{pmatrix}_{b_0}$$

[30] $1^{122} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 5$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 32.40 \rightarrow 33$, $T_{\max} = 100$, $T \in [33, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 6 & 2 \\ 1 & 0 & 1 & -5 & 0 & 2 & -3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 0 & 6 & 2 & 0 & 10 & 6 \\ 2 & 0 & 2 & -3 & -1 & 6 & -19 \end{pmatrix}_{b_0}$$

[31] $1^{121} 23 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 6$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 17.67 \rightarrow 18$, $T_{\max} = 100$, $T \in [18, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & -3 \\ 2 & 6 & 0 & 2 & 9 & 6 & 5 \\ 1 & 3 & 0 & 1 & 6 & 1 & 3 \\ 2 & 3 & -1 & -3 & 5 & 3 & -4 \end{pmatrix}_{b_0}$$

[32] $1^{123} 13 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 101$, $\Delta = -148$, $\det = 13$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 16.15 \rightarrow 17$, $T_{\max} = 101$, $T \in [17, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & -1 & 0 & -3 \end{pmatrix}_{b_0}$$

[33] $1^{111} 2 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 34$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 15.35 \rightarrow 16$, $T_{\max} = 100$, $T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 3 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

[34] $1^{12} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 46$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 18.30 \rightarrow 19$, $T_{\max} = 100$, $T \in [19, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & 0 & -3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -3 & -1 & 0 & -5 \end{pmatrix}_{b_0}$$

[35] $1^{113} 2 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 59$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 18.17 \rightarrow 19$, $T_{\max} = 100$, $T \in [19, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 1 & 0 & 3 & 1 & 0 & 9 & 2 \\ 4 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 2 & 9 & -5 \end{pmatrix}_{b_0}$$

[36] $1^{122}2^13^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.7$

$T_{\min} = 15.38 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 1 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 2 & -3 \\ 2 & 6 & 0 & 2 & 9 & 12 & 5 \\ 3 & 6 & 0 & 2 & 12 & 10 & 6 \\ 2 & 3 & -1 & -3 & 5 & 6 & -2 \end{pmatrix}_{b_0}$$

[37] $1^1231^131^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 70, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.34 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & -2 & -1 & 0 & -2 \end{pmatrix}_{b_0}$$

[38] $1^{111}2^13^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 78, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.69 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[40] $1^1231^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 16.40 \rightarrow 17, T_{\max} = 101, T \in [17, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 0 & -2 \\ 2 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 3 & 0 & -3 \end{pmatrix}_{b_0}$$

[39] $1^{11}3^1221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{111}2^13^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 0 & 3 & 2 \\ 1 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[42] $1^{11}3^1221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 103, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.03 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -2 & -1 & 2 & -2 \end{pmatrix}_{b_0}$$

[43] $1^12^13^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 109, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.43 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & 0 & -3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -3 & -1 & 0 & -2 \end{pmatrix}_{b_0}$$

[44] $1^{121}2^13^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 110, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -5 & 2 & 1 & -3 \\ 3 & 0 & 6 & 2 & 9 & 6 & 5 \\ 1 & 0 & 3 & 1 & 6 & 1 & 3 \\ 2 & 0 & 2 & -3 & 5 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{1231} 1^{12} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 118, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 15.14 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 6 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -2 & -1 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{124} 2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 118, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 15.68 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 1 & 6 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 4 & -3 \\ 2 & 6 & 0 & 2 & 9 & 24 & 5 \\ 6 & 12 & 0 & 4 & 24 & 46 & 12 \\ 2 & 3 & -1 & -3 & 5 & 12 & -2 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{112} 3^{11} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 6 & -1 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{111} 2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.95 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{111} 2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 157, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 0 & 1 & -5 & 1 & 0 & -3 \\ 2 & 0 & 3 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -3 & 2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{123} 2^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 157, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.10 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 1 & 6 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 3 & -3 \\ 2 & 6 & 0 & 2 & 9 & 18 & 5 \\ 4 & 9 & 0 & 3 & 18 & 25 & 9 \\ 2 & 3 & -1 & -3 & 5 & 9 & -2 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{111} 3^{11} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{111} 3^{11} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 176, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 6 & 9 & 0 & 8 & 15 & 9 \\ 3 & 4 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{121} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 178, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 2 & 4 & 0 & 2 & 5 & 4 \\ 1 & 2 & 0 & 1 & 4 & 0 \\ 2 & 2 & -1 & -2 & 3 & -2 \end{pmatrix}_{b_0}$$

[54] $1^{111} 2^3 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 179, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[55] $1^{121} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 15.17 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 2 & 4 & 0 & 2 & 5 & 4 & 3 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -2 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

[56] $1^{124} 2^3 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$

$$T_{\min} = 15.17 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 1 & 6 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 4 & -3 \\ 2 & 6 & 0 & 2 & 9 & 24 & 5 \\ 5 & 12 & 0 & 4 & 24 & 46 & 12 \\ 2 & 3 & -1 & -3 & 5 & 12 & -2 \end{pmatrix}_{b_0}$$

[60] $1^{112} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 11 & -1 \\ 8 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & -1 & 13 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{112} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[58] $1^{111} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{111} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 8 & 9 & 0 & 8 & 15 & 9 \\ 3 & 4 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

- [61] $1^{121}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 196$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 14.69 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -5 & 2 & 1 & -3 \\ 4 & 0 & 6 & 2 & 9 & 6 & 5 \\ 1 & 0 & 3 & 1 & 6 & 1 & 3 \\ 2 & 0 & 2 & -3 & 5 & 3 & -1 \end{pmatrix}_{b_0}$$
- [62] $1^{112}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 203$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 15.21 \rightarrow 16$, $T_{\max} = 100$, $T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 1 & 0 & 3 & 1 & 0 & 6 & 2 \\ 3 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 2 & 6 & -2 \end{pmatrix}_{b_0}$$
- [63] $1^{111}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 205$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 14.72 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$
- [64] $1^{113}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 214$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 15.11 \rightarrow 16$, $T_{\max} = 100$, $T \in [16, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$
- [65] $1^1231^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 101$, $\Delta = -148$, $\det = 217$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 15.03 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -2 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$
- [66] $1^{111}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 221$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 15.10 \rightarrow 16$, $T_{\max} = 100$, $T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -3 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$
- [67] $1^{132}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 226$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 15.62 \rightarrow 16$, $T_{\max} = 100$, $T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 9 & 2 \\ 1 & 0 & 1 & -5 & 0 & 3 & -3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 0 & 9 & 3 & 0 & 25 & 9 \\ 2 & 0 & 2 & -3 & -1 & 9 & -2 \end{pmatrix}_{b_0}$$
- [68] $1^{111}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 243$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 14.77 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$
- [69] $1^{111}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 246$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 7 & 9 & 0 & 8 & 15 & 9 \\ 3 & 4 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{114} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 256, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.31 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{1231} 1^{11} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 288, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 0 & -2 \\ 3 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -2 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{111} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 304, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{115} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 310, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.1$$

$$T_{\min} = 15.49 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 15 & -1 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & -1 & 20 & -1 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{111} 3^{112} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 310, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{112} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 344, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.98 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 1 & 0 & 3 & 1 & 0 & 6 & 2 \\ 6 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{121} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 365, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.84 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -5 & 2 & 1 & -3 \\ 3 & 0 & 6 & 2 & 9 & 6 & 5 \\ 2 & 0 & 3 & 1 & 6 & 1 & 3 \\ 2 & 0 & 2 & -3 & 5 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[78] \ 1^{112} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 373, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 0 & 6 & 2 \\ 3 & 6 & 0 & 2 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[77] $1^{112}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 373$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 2 & 0 & 3 & 1 & 0 & 6 & 2 \\ 3 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[79] $1^{121}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 391$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 14.72 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 3 & 4 & 0 & 2 & 5 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[80] $1^{113}3^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 406$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.78 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 14 & -1 \\ 11 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & -1 & 17 & 0 \end{pmatrix}_{b_0}$$

[81] $1^{112}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 429$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[82] $1^{112}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 434$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 15.14 \rightarrow 16$, $T_{\max} = 100$, $T \in [16, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & -3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -3 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$

[83] $1^{112}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 452$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.94 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 1 & 0 & 3 & 1 & 0 & 6 & 2 \\ 4 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[84] $1^{113}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 469$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.72 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 0 & 9 & 2 \\ 4 & 9 & 0 & 3 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

[85] $1^{112}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 478$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.75 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 2 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[86] $1^{114}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 0 & 12 & 2 \\ 5 & 12 & 0 & 4 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[87] $1^{12}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 494, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 15.11 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 6 & 2 \\ 1 & 0 & 1 & -5 & 0 & 2 & -3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 0 & 6 & 2 & 0 & 10 & 6 \\ 2 & 0 & 2 & -3 & -1 & 6 & -2 \end{pmatrix}_{b_0}$$

[88] $1^{112}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 499, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 1 & 0 & 3 & 1 & 0 & 6 & 2 \\ 5 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[89] $1^{1231}1^{131}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 508, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 15.02 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -2 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

[90] $1^{112}2^13^1221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 526, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[91] $1^{114}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 548, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 15.36 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 4 & -3 \\ 1 & 0 & 3 & 1 & 0 & 12 & 2 \\ 6 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 2 & 12 & -2 \end{pmatrix}_{b_0}$$

[92] $1^{111}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 562, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 0 & 3 & 2 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[93] $1^{111}2^13^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 584, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 1 & 0 & 3 & 1 & 0 & 9 & 2 \\ 5 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 2 & 9 & -2 \end{pmatrix}_{b_0}$$

[94] $1^{112} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 590$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.71 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 11 & -1 \\ 9 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & -1 & 13 & 0 \end{pmatrix}_{b_0}$$

[95] $1^{114} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 640$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.82 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 17 & -1 \\ 14 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & -1 & 21 & 0 \end{pmatrix}_{b_0}$$

[96] $1^{114} 2^1 3^{112} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 718$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 4 & -3 \\ 2 & 0 & 3 & 1 & 0 & 12 & 2 \\ 6 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[97] $1^{112} 3^{112} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 736$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[98] $1^{113} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 760$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.81 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[99] $1^{112} 3^{114} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 760$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.73 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[100] $1^{112} 3^{113} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 814$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.73 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 9 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[101] $1^{113} 2^1 3^{112} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 850$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.73 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 0 & 9 & 2 \\ 5 & 9 & 0 & 3 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

[102] $1^{112} 3^{11} 221 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 864$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.70 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 11 & -1 \\ 10 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & -1 & 13 & 0 \end{pmatrix}_{b_0}$$

[103] $1^{113} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 907$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.94 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 1 & 0 & 3 & 1 & 0 & 9 & 2 \\ 6 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

[104] $1^{123} 1^{12} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 101$, $\Delta = -148$, $\det = 937$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 15.02 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 6 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -2 & -1 & 8 & -2 \end{pmatrix}_{b_0}$$

[105] $1^{112} 2^3 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 942$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & -3 \\ 2 & 0 & 3 & 1 & 0 & 6 & 2 \\ 4 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[106] $1^{113} 2^3 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -142$, $\det = 994$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.74 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 2 & 0 & 3 & 1 & 0 & 9 & 2 \\ 5 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

[107] $1^{114} 3 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 1006$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.84 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 13 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[108] $1^{112} 3 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 1006$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.70 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 11 & -1 \\ 11 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & -1 & 13 & 0 \end{pmatrix}_{b_0}$$

[109] $1^{113} 3 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 1016$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.73 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 14 & -1 \\ 12 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & -1 & 17 & 0 \end{pmatrix}_{b_0}$$

[110] $1^{112} 3 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 100$, $\Delta = -140$, $\det = 1016$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 14.70 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 11 & -1 \\ 12 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & -1 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[111] \quad 1^{113} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -142, \quad \det = 1028, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.91 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -5 & 1 & 3 & -3 \\ 1 & 0 & 3 & 1 & 0 & 9 & 2 \\ 7 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -3 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[112] \quad 1^{114} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -142, \quad \det = 1147, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.02 \rightarrow 16, \quad T_{\max} = 100, \quad T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 4 & -3 \\ 1 & 0 & 3 & 1 & 0 & 12 & 2 \\ 7 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 2 & 12 & -2 \end{pmatrix}_{b_0}$$

$$[113] \quad 1^{112} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -142, \quad \det = 1150, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.73 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 0 & 12 & 2 \\ 6 & 12 & 0 & 4 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[114] \quad 1^{113} 3^1 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1174, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.76 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 & 4 \\ 3 & 4 & 1 & 3 & 18 & 4 & 1 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 & 11 \\ 11 & 18 & 0 & 9 & 52 & 12 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[115] \quad 1^{1231} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 101, \quad \Delta = -148, \quad \det = 1300, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 15.02 \rightarrow 16, \quad T_{\max} = 101, \quad T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 9 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & -2 & -1 & 12 & -2 \end{pmatrix}_{b_0}$$

$$[116] \quad 1^{114} 3^1 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1454, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 & 4 \\ 3 & 4 & 1 & 3 & 27 & 4 & 1 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 17 & -1 & 15 \\ 15 & 27 & 0 & 17 & 123 & 21 & 3 \\ 3 & 4 & -1 & -1 & 21 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[117] \quad 1^{113} 3^1 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1456, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.74 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 & 4 \\ 3 & 4 & 1 & 3 & 18 & 4 & 1 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 & 12 \\ 12 & 18 & 0 & 9 & 52 & 12 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[118] \quad 1^{113} 3^1 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1494, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.72 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 & 4 \\ 3 & 4 & 1 & 3 & 21 & 4 & 1 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 14 & -1 & 13 \\ 13 & 21 & 0 & 14 & 75 & 17 & 3 \\ 3 & 4 & -1 & -1 & 17 & 0 & 0 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[119] \quad & 1^{114} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -142, \quad \det = 1544, \quad (n_+, n_-, n_0) = \\
& (1, 10, 0), \quad c_{\text{ext}} = 14.0 \\
& T_{\min} = 14.94 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -5 & 1 & 4 & -3 \\ 1 & 0 & 3 & 1 & 0 & 12 & 2 \\ 8 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -3 & 2 & 12 & -1 \end{pmatrix}_{b_0} \\
[120] \quad & 1^{1231} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 101, \quad \Delta = -148, \quad \det = 1597, \quad (n_+, n_-, n_0) = \\
& (1, 10, 0), \quad c_{\text{ext}} = 14.0 \\
& T_{\min} = 15.02 \rightarrow 16, \quad T_{\max} = 101, \quad T \in [16, 101] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 12 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 9 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & -2 & -1 & 16 & -2 \end{pmatrix}_{b_0} \\
[121] \quad & 1^{113} 3^2 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1606, \quad (n_+, n_-, n_0) = \\
& (1, 10, 0), \quad c_{\text{ext}} = 13.8 \\
& T_{\min} = 14.74 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0} \\
[122] \quad & 1^{114} 3^2 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1624, \quad (n_+, n_-, n_0) = \\
& (1, 10, 0), \quad c_{\text{ext}} = 14.0 \\
& T_{\min} = 14.78 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0} \\
[123] \quad & 1^{113} 3^2 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 1840, \quad (n_+, n_-, n_0) = \\
& (1, 10, 0), \quad c_{\text{ext}} = 13.8 \\
& T_{\min} = 14.71 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 14 & -1 \\ 14 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & -1 & 17 & 0 \end{pmatrix}_{b_0} \\
[124] \quad & 1^{113} 3^2 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 2054, \quad (n_+, n_-, n_0) = \\
& (1, 10, 0), \quad c_{\text{ext}} = 13.8 \\
& T_{\min} = 14.71 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 14 & -1 \\ 15 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & -1 & 17 & 0 \end{pmatrix}_{b_0} \\
[125] \quad & 1^{114} 3^2 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 2110, \quad (n_+, n_-, n_0) = \\
& (1, 10, 0), \quad c_{\text{ext}} = 14.0 \\
& T_{\min} = 14.76 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0} \\
[126] \quad & 1^{114} 3^2 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 2136, \quad (n_+, n_-, n_0) = \\
& (1, 10, 0), \quad c_{\text{ext}} = 14.0 \\
& T_{\min} = 14.73 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 17 & -1 \\ 16 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & -1 & 21 & 0 \end{pmatrix}_{b_0} \\
[127] \quad & 1^{114} 3^2 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 10, \quad T_{\max} = 100, \quad \Delta = -140, \quad \det = 2464, \quad (n_+, n_-, n_0) = \\
& (1, 10, 0), \quad c_{\text{ext}} = 14.0 \\
& T_{\min} = 14.75 \rightarrow 15, \quad T_{\max} = 100, \quad T \in [15, 100]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{114} 3^{113} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -140, \ \det = 2686, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.72 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 17 & -1 \\ 17 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & -1 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{114} 3^{114} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -140, \ \det = 3104, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.72 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 17 & -1 \\ 18 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & -1 & 21 & 0 \end{pmatrix}_{b_0}$$

4.257 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{111} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 68, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.65 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 79, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.84 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^{112} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 112, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 156, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.18 \rightarrow 16, \ T_{\max} = 100, \ T \in [16, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 1 & 1 & -3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & -1 & -3 & -1 & 2 & 2 & -3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{113} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 221, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.8$$

$$T_{\min} = 15.18 \rightarrow 16, \ T_{\max} = 100, \ T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 9 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 3 & 4 & -1 & -2 & 0 & 12 & -2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -139, \ \det = 413, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1111} 3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 420, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 464, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 6 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & -2 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 496, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.69 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -4 & 3 & 3 & -2 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1111} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 589, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1111} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 1088, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [11] \ 1^{11} 3 \ 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -56, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111} 3 \ 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -65, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.94 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1111} 2321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -84, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 \\ 2 & 3 & -1 & -3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111} 3 \ 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -92, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.83 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1111} 2 \ 3 \ 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & 0 & 1 & 3 & 3 & 1 \\ 2 & 3 & -1 & -3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1111} 3 \ 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.22 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 \\ 2 & 2 & -1 & -2 & -1 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{1111} 3 \ 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -149, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 6 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 8 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 8 & 0 & 15 & 9 \\ 3 & 4 & -1 & -1 & 0 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1111} 3 \ 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -173, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 15.39 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 3 & 4 & -1 & -1 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{1111} 2 \ 3 \ 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -246, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.15 \rightarrow 16, T_{\max} = 25, T \in [16, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 3 & 3 & 2 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & 3 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1111} 3 \ 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -269, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 12 & 3 \\ 3 & 4 & 1 & 3 & 0 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 14 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 12 & 21 & 0 & 14 & 0 & 75 & 17 \\ 3 & 4 & -1 & -1 & 0 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1111} 2 \ 3 \ 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -290, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 2 & 0 & 3 & 1 & 0 & 3 & 3 & 2 \\ 1 & 0 & 3 & 1 & 3 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{112} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -332, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 9 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 15 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 11 & -1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 15 & 0 & 11 & 0 & 39 & 13 & 13 \\ 3 & 4 & -1 & -1 & 0 & 13 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -341, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{111} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -344, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{123} 3^{111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 38, \Delta = -118, \det = -368, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.05 \rightarrow 16, T_{\max} = 38, T \in [16, 38]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 2 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & -1 & -2 & -1 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{112} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -380, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.83 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 6 & -1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 & 8 \\ 3 & 4 & -1 & -1 & 0 & 8 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{111} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -408, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.77 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 3 & 3 & -1 & -1 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 & 4 \\ 4 & 4 & -1 & -1 & 4 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{111} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -432, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.96 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 3 & 3 & 2 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 & 3 \\ 3 & 0 & 3 & 1 & 3 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{111} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -434, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 6 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 6 & 3 \\ 3 & 6 & 0 & 2 & 6 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{1231} 1^{11} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 38, \Delta = -118, \det = -455, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.09 \rightarrow 16, T_{\max} = 38, T \in [16, 38]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & -1 & -2 & -1 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{1111} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -485, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{1111} 2^1 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -540, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 3 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 3 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{1113} 2^1 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -626, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 9 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 9 & 3 \\ 4 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 2 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{1113} 2^1 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -668, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 9 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 9 & 3 \\ 5 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 2 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{1114} 2^1 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -674, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 12 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 12 & 3 \\ 5 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 2 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{1111} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -896, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{\overline{111}4} 2 \ 3 \ 21^{\overline{1}12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 25, \ \Delta = -112, \ \det = -1268, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.73 \rightarrow 15, \ T_{\max} = 25, \ T \in [15, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 12 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 12 & 3 \\ 6 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 2 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$4.258 \quad \text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$$

$$T = 11 \quad [0] \ 1^{\overline{111}1} 3 \ 21^{\overline{1}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -109, \ \det = -128, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 16, \ T_{\max} = 23, \ T \in [16, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 1 & 1 & -3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & -1 & -3 & -1 & 2 & 2 & -3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}1} 3 \ 21^{\overline{1}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -109, \ \det = -202, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.85 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [2] \ 1^{\overline{111}1} 2 \ 3 \ 21^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 15, \ \Delta = -82, \ \det = 24, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}1} 3 \ 221^{\overline{1}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 15, \ \Delta = -80, \ \det = 166, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.95 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}12} 2 \ 3 \ 21^{\overline{1}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 15, \ \Delta = -82, \ \det = 244, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.72 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 6 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 6 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 6 & 0 & 2 & 6 & 6 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 2 & 3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}14} 2 \ 3 \ 21^{\overline{1}12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 15, \ \Delta = -82, \ \det = 472, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.73 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 12 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 12 & 3 \\ 6 & 12 & 0 & 4 & 12 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 2 & 3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11113} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 15, \ \Delta = -82, \ \det = 628, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 14.74 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 3 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 9 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 9 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 9 & 3 \\ 4 & 9 & 0 & 3 & 9 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -3 & 2 & 3 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11114} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 15, \ \Delta = -82, \ \det = 724, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.77 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 12 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 12 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 12 & 3 \\ 5 & 12 & 0 & 4 & 12 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 2 & 3 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$4.259 \quad \text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 11 \quad [0] \ 1^{11113} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 25, \ \Delta = -112, \ \det = -128, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.87 \rightarrow 15, \ T_{\max} = 25, \ T \in [15, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 0 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$4.260 \quad \text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$$

$$T = 11 \quad [0] \ 1^{11111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, \ T_{\max} = 38, \ \Delta = -118, \ \det = -284, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.20 \rightarrow 16, \ T_{\max} = 38, \ T \in [16, 38]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & -1 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1] \ 1^{11111} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 12, \ T_{\max} = 16, \ \Delta = -91, \ \det = 108, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 2 & 3 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11111} 3^1 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 12, \ T_{\max} = 16, \ \Delta = -89, \ \det = 228, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.25 \rightarrow 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

4.261 NHC(-2)(-3) + e₇ + su₃ + su₂ + su₂→(-3)

$$T = 10 \quad [0] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 4 & -1 & -2 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.80 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & 2 & -2 \\ 6 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 348, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 2 & -2 \\ 3 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 604, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 2 & -2 \\ 5 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 2 & -2 \\ 4 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & 2 & -2 \\ 7 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 1408, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & 2 & -2 \\ 8 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \quad 1^{12} 3^1 1^{12} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -152, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 15.16 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 6 & 2 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 12 & 0 & 6 & 0 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 0 & 6 & 0 & 2 \\ 2 & 4 & -1 & -2 & -1 & 8 & 2 & -2 \end{pmatrix}_{b_0}$$

[8] $1^{123}1^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 70, \Delta = -121, \det = -296, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 15.08 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & 2 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & -1 & -2 & -1 & 4 & 2 & -2 \end{pmatrix}_{b_0}$$

[9] $1^{1231}1^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 59, \Delta = -120, \det = -364, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 15.15 \rightarrow 16, T_{\max} = 59, T \in [16, 59]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 4 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & -1 & -1 & 6 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{123}1^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 70, \Delta = -121, \det = -572, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & 2 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & -1 & -2 & -1 & 4 & 2 & -2 \end{pmatrix}_{b_0}$$

[11] $1^{123}1^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 70, \Delta = -121, \det = -1196, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 6 & 2 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 5 & 12 & 0 & 6 & 0 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 0 & 6 & 0 & 2 \\ 2 & 4 & -1 & -2 & -1 & 8 & 2 & -2 \end{pmatrix}_{b_0}$$

4.262 $\text{NHC}(-2)(-3) + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$

$T = 11$ [0] $1^{111}3^{11}2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 25, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 2 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

4.263 $\text{NHC}(-2)(-3) + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$

$T = 10$ [0] $1^{111}3^{11}2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 15.13 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 5 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 5 & 0 & 12 & 6 \\ 3 & 4 & -1 & -2 & 0 & 6 & -2 \end{pmatrix}_{b_0}$$

4.264 $\text{NHC}(-2)(-3) + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_3$

$T = 10$ [1] $1^{111}3^{11}2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 101, \Delta = -148, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 15.05 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -4 & 0 & 3 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -2 & -1 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3 \, 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \quad T_{\max} = 101, \quad \Delta = -148, \quad \det = 192, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.75 \rightarrow 16, \quad T_{\max} = 101, \quad T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -2 & 0 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3 \, 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \quad T_{\max} = 101, \quad \Delta = -148, \quad \det = 280, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.07 \rightarrow 16, \quad T_{\max} = 101, \quad T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [4] \quad 1^{111} 2 \, 3 \, 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, \quad T_{\max} = 70, \quad \Delta = -121, \quad \det = -14, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.43 \rightarrow 16, \quad T_{\max} = 70, \quad T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 0 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 0 & -3 \\ 1 & 0 & 3 & 1 & 0 & 3 & 0 & 2 \\ 3 & 0 & 3 & 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 2 & 3 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 2 \, 3 \, 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, \quad T_{\max} = 70, \quad \Delta = -121, \quad \det = -95, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.11 \rightarrow 16, \quad T_{\max} = 70, \quad T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 0 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & 0 & -3 \\ 1 & 0 & 3 & 1 & 0 & 3 & 0 & 2 \\ 2 & 0 & 3 & 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 2 & 3 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3 \, 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, \quad T_{\max} = 48, \quad \Delta = -119, \quad \det = -159, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.87 \rightarrow 16, \quad T_{\max} = 48, \quad T \in [16, 48]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -1 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 2 \, 3 \, 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, \quad T_{\max} = 70, \quad \Delta = -121, \quad \det = -162, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.33 \rightarrow 16, \quad T_{\max} = 70, \quad T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 2 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{112} 2 \, 3 \, 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, \quad T_{\max} = 70, \quad \Delta = -121, \quad \det = -194, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.5$$

$$T_{\min} = 15.11 \rightarrow 16, \quad T_{\max} = 70, \quad T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 0 & 2 \\ 1 & 0 & 1 & -5 & 1 & 2 & 0 & -3 \\ 1 & 0 & 3 & 1 & 0 & 6 & 0 & 2 \\ 4 & 0 & 6 & 2 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 2 & 6 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{111} 3 \, 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, \quad T_{\max} = 48, \quad \Delta = -119, \quad \det = -231, \quad (n_+, n_-, n_0) =$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 4 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 12 & 3 & 2 \\ 6 & 12 & 0 & 4 & 12 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 3 & 12 & 0 & 2 \\ 2 & 3 & -1 & -3 & 2 & 12 & 2 & -2 \end{pmatrix}_{b_0}$$

4.265 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{\overline{1111}} 3 \ 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 38, \ \Delta = -118, \ \det = -284, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.20 \rightarrow 16, \ T_{\max} = 38, \ T \in [16, 38]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & -1 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1] \ 1^{\overline{1111}} 2 \ 321^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 16, \ \Delta = -91, \ \det = 108, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 2 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 16, \ \Delta = -89, \ \det = 228, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.25 \rightarrow 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

4.266 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 102, \ \Delta = -172, \ \det = -112, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.86 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & -2 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{12}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 102, \ \Delta = -172, \ \det = -156, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.90 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -4 & 4 & -2 \\ 4 & 6 & 0 & 4 & 9 & 5 \\ 3 & 3 & -1 & -2 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 102, \ \Delta = -172, \ \det = -192, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.75 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & -2 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [3] \ 1^{\overline{1}} 23^{\overline{1}} 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 101, \ \Delta = -151, \ \det = 16, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 18, \ T_{\max} = 101, \ T \in [18, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 2 & 0 & 0 & 2 \\ 2 & 0 & -1 & -2 & -1 & 2 & -4 \end{pmatrix}_{b_0}$$

[4] $1^{12}3^11^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 101, \Delta = -151, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 2 \\ 2 & 4 & -1 & -2 & -1 & 2 & -4 \end{pmatrix}_{b_0}$$

[5] $1^{12}3^11^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 101, \Delta = -150, \det = 237, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.10 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

4.267 NHC(-2)(-3) + $\mathfrak{e}_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$T = 10$ [0] $1^{11}3^11^{11}2^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 0 & -2 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 2 & 0 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^11^{11}2^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 14.80 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^11^{11}2^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 348, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{11}3^11^{11}2^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 604, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 5 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{11}3^11^{11}2^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 4 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{11}3^11^{11}2^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 7 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112}3^{12}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 1408, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 8 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \ 1^{123}1^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -152, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 15.16 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & 6 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 6 & 2 \\ 4 & 12 & 0 & 6 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & -1 & 2 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{123}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -296, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.08 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 2 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & -1 & 2 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{123}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 59, \Delta = -120, \det = -364, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.15 \rightarrow 16, T_{\max} = 59, T \in [16, 59]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 4 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & -1 & -1 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{123}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -572, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -2 & -1 & 2 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{123}1^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 70, \Delta = -121, \det = -1196, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & 6 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 6 & 2 \\ 5 & 12 & 0 & 6 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -2 & -1 & 2 & 8 & -2 \end{pmatrix}_{b_0}$$

$$4.268 \quad \text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$$

$$T = 11 \quad [0] \ 1^{111}3^{12}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 0 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & 2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

4.269 NHC(-2)(-3) + e₇ + su₃ + su₂→(-3) + su₂→(-3)

$$T = 11 \quad [0] \quad 1^{11} 23^1 1^1 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 81, \Delta = -123, \det = -312, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.13 \rightarrow 16, T_{\max} = 81, T \in [16, 81]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 2 & 3 \\ 4 & 8 & 1 & 5 & 0 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 0 & 1 & 3 \\ 3 & 6 & -1 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

4.270 NHC(-2)(-3) + e₇ + su₃ + su₂→(-3) + su₃

$$T = 10 \quad [0] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -151, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 4 & 0 & 5 \\ 2 & 4 & -4 & 0 & 2 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -2 & -1 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

4.271 NHC(-2)(-3) + e₇ + su₃ + su₂n3mix

$$T = 9 \quad [0] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & -2 \\ 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -345, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & -2 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 8 & -2 \\ 8 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & -2 \\ 6 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -617, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & -2 \\ 7 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -977, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 8 & -2 \\ 9 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{12}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 102, \ \Delta = -172, \ \det = -1280, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.70 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 8 & -2 \\ 10 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [7] \ 1^{\overline{1}} 23^{\overline{1}} 1^{\overline{1}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 101, \ \Delta = -151, \ \det = 103, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.03 \rightarrow 16, \ T_{\max} = 101, \ T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 5 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 9 & 0 & 5 & 0 & 12 & 6 \\ 2 & 4 & -1 & -2 & -1 & 6 & -2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1}} 23^{\overline{1}} 1^{\overline{1}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 101, \ \Delta = -151, \ \det = 223, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.23 \rightarrow 16, \ T_{\max} = 101, \ T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 2 & 4 & -1 & -2 & -1 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1}} 23^{\overline{1}} 1^{\overline{1}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 101, \ \Delta = -150, \ \det = 358, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.03 \rightarrow 16, \ T_{\max} = 101, \ T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 4 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 0 & 4 & 0 & 9 & 5 \\ 2 & 4 & -1 & -2 & -1 & 5 & -2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1}} 23^{\overline{1}} 1^{\overline{1}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 101, \ \Delta = -151, \ \det = 682, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.05 \rightarrow 16, \ T_{\max} = 101, \ T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 5 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 9 & 0 & 5 & 0 & 12 & 6 \\ 2 & 4 & -1 & -2 & -1 & 6 & -2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1}} 23^{\overline{1}} 1^{\overline{1}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 101, \ \Delta = -151, \ \det = 1075, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.03 \rightarrow 16, \ T_{\max} = 101, \ T \in [16, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 8 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 15 & 0 & 8 & 0 & 36 & 10 \\ 2 & 4 & -1 & -2 & -1 & 10 & -2 \end{pmatrix}_{b_0}$$

4.272 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_{2n3\text{mix}} + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 100, \ \Delta = -142, \ \det = 181, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.13 \rightarrow 16, \ T_{\max} = 100, \ T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 1 & 3 \\ 3 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & 0 & -2 \\ 5 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 6 & 0 & -2 \end{pmatrix}_{b_0}$$

4.273 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 102, \ \Delta = -178, \ \det = -48, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{12} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -178$, $\det = -71$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 14.94 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 0 & 2 & -3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 5 & 3 \\ 2 & 2 & -1 & -3 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -178$, $\det = -209$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 15.03 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -178$, $\det = -297$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 15.18 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & -2 & 3 & -2 \end{pmatrix}_{b_0}$$

[4] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -178$, $\det = -464$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 14.95 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 4 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

$T = 10$ [5] $1^{121} 23^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -151$, $\det = 27$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 17.78 \rightarrow 18$, $T_{\max} = 101$, $T \in [18, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 2 & 1 & -3 \\ 2 & 6 & 0 & 2 & 9 & 6 & 5 \\ 1 & 3 & 0 & 1 & 6 & 0 & 2 \\ 2 & 3 & -1 & -3 & 5 & 2 & -4 \end{pmatrix}_{b_0}$$

[6] $1^{11} 2^3 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -151$, $\det = 31$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 15.10 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 0 & 1 & -5 & 1 & 0 & -3 \\ 1 & 0 & 3 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -3 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

[7] $1^{111} 2^3 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -151$, $\det = 48$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 18.25 \rightarrow 19$, $T_{\max} = 101$, $T \in [19, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 1 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 2 & 2 & -5 \end{pmatrix}_{b_0}$$

[8] $1^{12} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -149$, $\det = 57$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 15.05 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 5 & 3 \\ 2 & 2 & -1 & -2 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$

[9] $1^{111} 2^3 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -151$, $\det = 135$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.20 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -3 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

[11] $1^{11} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -149$, $\det = 174$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.11 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 \\ 3 & 4 & 1 & 3 & 0 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 & -1 \\ 3 & 4 & -1 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[10] $1^1 231^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 102$, $\Delta = -157$, $\det = 174$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.28 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 3 & -2 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

[12] $1^{11} 1^1 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -151$, $\det = 196$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15$, $T_{\max} = 101$, $T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 3 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{11} 1^1 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -151$, $\det = 223$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.23 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -3 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

[14] $1^{11} 1^1 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -149$, $\det = 246$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.29 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 4 \\ 3 & 4 & 1 & 3 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 & -1 \\ 3 & 6 & 0 & 3 & 3 & 3 & 3 \\ 3 & 4 & -1 & -1 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

[15] $1^1 231^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 102$, $\Delta = -157$, $\det = 342$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.27 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 3 & 3 \\ 2 & 4 & -1 & -2 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$

[16] $1^{11} 1^1 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 101$, $\Delta = -149$, $\det = 384$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.05 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 & 4 \\ 3 & 4 & 1 & 3 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 & -1 \\ 4 & 6 & 0 & 3 & 3 & 3 & 3 \\ 3 & 4 & -1 & -1 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

4.274 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$

$T = 10$ [1] $1^{11} 1^1 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 101$, $\Delta = -148$, $\det = 63$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 15.05 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -4 & 0 & 0 & 3 & -2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 0 & 4 & 4 \\ 3 & 5 & -2 & -1 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 101, \ \Delta = -148, \ \det = 192, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.75 \rightarrow 16, \ T_{\max} = 101, \ T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 0 & -2 \\ 3 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 3 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 101, \ \Delta = -148, \ \det = 280, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.07 \rightarrow 16, \ T_{\max} = 101, \ T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [4] \ 1^{111} 2 \ 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 70, \ \Delta = -121, \ \det = -14, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.43 \rightarrow 16, \ T_{\max} = 70, \ T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 0 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 0 & 3 & 1 & 3 & 0 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 2 \ 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 70, \ \Delta = -121, \ \det = -95, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.11 \rightarrow 16, \ T_{\max} = 70, \ T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 0 & 1 & -5 & 1 & 0 & 1 & -3 \\ 1 & 0 & 3 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 0 & 3 & 1 & 3 & 0 & 1 & 3 \\ 2 & 0 & 2 & -3 & 2 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3 \ 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 48, \ \Delta = -119, \ \det = -159, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.87 \rightarrow 16, \ T_{\max} = 48, \ T \in [16, 48]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 2 \ 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 70, \ \Delta = -121, \ \det = -162, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.33 \rightarrow 16, \ T_{\max} = 70, \ T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 2 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{112} 2 \ 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 70, \ \Delta = -121, \ \det = -194, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 15.11 \rightarrow 16, \ T_{\max} = 70, \ T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 6 & 2 \\ 1 & 0 & 1 & -5 & 1 & 0 & 2 & -3 \\ 1 & 0 & 3 & 1 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 0 & 6 & 2 & 6 & 0 & 10 & 6 \\ 2 & 0 & 2 & -3 & 2 & -1 & 6 & -2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3 \ 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 48, \ \Delta = -119, \ \det = -231, \ (n_+, n_-, n_0) =$$

$(1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 15.15 \rightarrow 16, T_{\max} = 48, T \in [16, 48]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1111} 2^3 21^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_7 + \textcolor{orange}{s}u_3 + \textcolor{red}{s}u_3 + \textcolor{blue}{s}u_2 \quad T = 11, \ T_{\max} = 70, \ \Delta = -121, \ \det = -303, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.30 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 2 \\ 2 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 2 & 3 & -1 & -3 & 2 & 2 & 3 & -2 \end{pmatrix} b_0$$

$$[11] \ 1^{1112} 2^3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 70, \ \Delta = -121, \ \det = -402, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 15.49 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 6 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 6 & 2 \\ 4 & 6 & 0 & 2 & 6 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 2 & 2 & 6 & -2 \end{pmatrix} b_0$$

[12] $1^{1112} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_7 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 70$, $\Delta = -121$, $\det = -807$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.5$

$$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 2 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 6 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 6 & 2 \\ 3 & 6 & 0 & 2 & 6 & 6 & 10 & 6 \\ 2 & 3 & -1 & -3 & 2 & 2 & 6 & -2 \end{pmatrix} b_0$$

[13] $1^{1113}2^321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + \textcolor{red}{su}_3 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 70, \Delta = -121, \det = -1095, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.8$

$$T_{\min} = 15.21 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 9 \\ 1 & 3 & 0 & 1 & 3 & 0 & 9 \\ 4 & 9 & 0 & 3 & 9 & 9 & 25 \\ 2 & 3 & -1 & -3 & 2 & 2 & 9 \end{pmatrix} b_0$$

[14] $1^{1114}2^33^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 70, \Delta = -121, \det = -1167, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.26 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 12 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 12 & 2 \\ 5 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 2 & 2 & 12 & -2 \end{pmatrix}_{b_0}$$

[15] $1^{1113} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 70$, $\Delta = -121$, $\det = -1518$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.8$

$$T_{\min} = 15.23 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 9 \\ 1 & 3 & 0 & 1 & 3 & 0 & 9 \\ 5 & 9 & 0 & 3 & 9 & 9 & 25 \\ 2 & 3 & -1 & -3 & 2 & 2 & 9 \end{pmatrix} b_0$$

[16] $1^{1114} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_7 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 70, \Delta = -121, \det = -2418, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 70, T \in [16, 70]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 4 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 12 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 12 & 2 \\ 6 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -3 & 2 & 2 & 12 & -2 \end{pmatrix}_{b_0}$$

4.275 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{\overline{1111}} 3 \ 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 38, \ \Delta = -118, \ \det = -284, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.20 \rightarrow 16, \ T_{\max} = 38, \ T \in [16, 38]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & -1 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1] \ 1^{\overline{1111}} 2 \ 3 \ 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 16, \ \Delta = -91, \ \det = 108, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 3 & 1 \\ 2 & 3 & -1 & -3 & 2 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 16, \ \Delta = -89, \ \det = 228, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.25 \rightarrow 16, \ T_{\max} = 16, \ T \in [16, 16]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

4.276 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 101, \ \Delta = -151, \ \det = 60, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.20 \rightarrow 16, \ T_{\max} = 101, \ T \in [16, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 0 & 4 & 5 \\ 2 & 4 & -4 & 0 & 0 & 2 & -2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 3 & 5 & -2 & -1 & -1 & 2 & -2 \end{pmatrix}_{b_0}$$

4.277 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3 \ 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 102, \ \Delta = -157, \ \det = 489, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.28 \rightarrow 16, \ T_{\max} = 102, \ T \in [16, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -2 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \ 1^{\overline{111}} 2 \ 3 \ 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, \ T_{\max} = 94, \ \Delta = -130, \ \det = -297, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.09 \rightarrow 17, \ T_{\max} = 94, \ T \in [17, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -5 & 1 & 1 & -3 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 \\ 2 & 3 & -1 & -3 & 2 & 2 & -3 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 11, T_{\max} = 92, \Delta = -128, \det = -405, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 15.39 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

4.278 NHC(-2)(-3) + e₇ + (-3)→(-2)

$T = 8$ [0] $1^1 231^{11} 2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 102, \Delta = -200, \det = 425, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.69 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & -4 & 0 & -2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

4.279 NHC(-2)(-3) + e'₇

$T = 7$ [0] $2^1 231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7$ $T = 7, T_{\max} = 101, \Delta = -192, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 4 & 5 \\ 3 & 4 & -3 & -1 \\ 3 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11} 321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7$ $T = 7, T_{\max} = 101, \Delta = -192, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -4 & -2 \\ 2 & 2 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{11} 321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7$ $T = 7, T_{\max} = 101, \Delta = -192, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & -3 & -1 \\ 3 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[3] $21^{11} 321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7$ $T = 7, T_{\max} = 101, \Delta = -192, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 4 & 8 & 3 & 4 \\ 8 & 9 & 4 & 5 \\ 3 & 4 & -3 & -1 \\ 4 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$T = 8$ [4] $1^1 13221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7$ $T = 8, T_{\max} = 101, \Delta = -163, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 6 & 6 & 6 \\ 2 & 6 & -2 & 0 \\ 3 & 6 & 0 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{12} 321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7$ $T = 8, T_{\max} = 101, \Delta = -165, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -4 & -2 \\ 2 & 3 & -1 & -2 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{12} 321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7$ $T = 8, T_{\max} = 101, \Delta = -165, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 13.20 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 1 & -4 & -2 \\ 2 & 0 & 2 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^1 \bar{3} 2 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 \quad T = 8, \ T_{\max} = 101, \ \Delta = -163, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.14 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -3 & -1 \\ 2 & 2 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^1 \bar{2} 3 1 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 \quad T = 8, \ T_{\max} = 101, \ \Delta = -171, \ \det = 26, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.08 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 0 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^1 \bar{2} 3 1^1 \bar{3} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 \quad T = 8, \ T_{\max} = 101, \ \Delta = -171, \ \det = 41, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.05 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 3 & 3 \\ 1 & 0 & 3 & -3 & -1 \\ 2 & 0 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^1 \bar{2} 3 1^1 \bar{3} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 \quad T = 8, \ T_{\max} = 101, \ \Delta = -171, \ \det = 56, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.71 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 3 1^1 \bar{2} 3 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 \quad T = 8, \ T_{\max} = 101, \ \Delta = -173, \ \det = 278, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.79 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -4 & -2 \\ 2 & 3 & 2 & -2 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [12] \ 1^1 \bar{1} \bar{2} 2 3 1 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 \quad T = 9, \ T_{\max} = 99, \ \Delta = -142, \ \det = -9, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 17, \ T_{\max} = 99, \ T \in [17, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 5 & 0 & -2 & 0 \\ 2 & 5 & -1 & 0 & -2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [13] \ 1^1 \bar{4} \bar{1} 3 2 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 \quad T = 10, \ T_{\max} = 94, \ \Delta = -133, \ \det = 9, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 18, \ T_{\max} = 94, \ T \in [18, 94]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 \\ 1 & 3 & -4 & -2 \\ 3 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [14] \ 1^1 \bar{5} \bar{1} 3 2 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 \quad T = 11, \ T_{\max} = 92, \ \Delta = -128, \ \det = -9, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 19, \ T_{\max} = 92, \ T \in [19, 92]$$

$$\begin{pmatrix} 5 & 17 & 5 \\ 17 & 56 & 14 \\ 5 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 2 1^1 \bar{5} \bar{1} 3 2 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 \quad T = 11, \ T_{\max} = 92, \ \Delta = -128, \ \det = -9, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 19, \ T_{\max} = 92, \ T \in [19, 92]$$

$$\begin{pmatrix} 5 & 17 & 5 \\ 17 & 56 & 14 \\ 5 & 14 & 0 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [16] \ 1 3 1^1 \bar{5} \bar{1} 3 2 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 \quad T = 12, \ T_{\max} = 24, \ \Delta = -107, \ \det = 9, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 20, \ T_{\max} = 24, \ T \in [20, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 5 & 0 & -2 & 0 \\ 2 & 5 & -1 & 0 & -2 \end{pmatrix}_{b_0}$$

4.280 $\text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{brown}{\mathfrak{e}}_6$

$$T = 9 \quad [0] \ 1^1 \bar{1} \bar{2} \bar{3} 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{brown}{\mathfrak{e}}_6 \quad T = 9, \ T_{\max} = 104, \ \Delta = -214, \ \det = -103, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 16.24 \rightarrow 17, T_{\max} = 104, T \in [17, 104]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 1 & 3 & 0 & 1 & -3 & -1 \\ 2 & 3 & -1 & -2 & -1 & -3 \end{pmatrix}_{b_0}$$

4.281 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{f}_4$

$$T = 8 \quad [0] \ 1^{11} 3 2 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 103, \Delta = -215, \det = 122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & -1 & -3 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [1] \ 1^{11} 3 2 2 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 103, \Delta = -186, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 15.43 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 6 & 6 & 0 & 6 \\ 2 & 6 & -2 & 0 & 0 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 6 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 2 3 2 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 103, \Delta = -188, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 1 & 3 & 0 & 1 & -2 & 0 \\ 2 & 3 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

4.282 NHC(-2)(-3) + $\mathfrak{e}'_7$ + NHC(-2)(-3)

$$T = 9 \quad [0] \ 1^{11} 3 2 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 5 \\ 7 & 8 & 1 & 5 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 5 & 0 & -2 & 0 \\ 5 & 6 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 2 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.06 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 4 & -1 \\ 4 & 8 & 0 & 4 & 9 & 5 \\ 3 & 4 & -1 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [2] \ 1^{11} 2 3 2 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 27, \Delta = -115, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.56 \rightarrow 15, T_{\max} = 27, T \in [15, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 4 & 3 \\ 1 & 3 & 0 & 1 & 4 & 0 & 2 \\ 2 & 3 & -1 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 2 3 2 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 27, \Delta = -115, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 27, T \in [15, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 1 & 3 & 0 & 1 & 1 & 0 & 2 \\ 2 & 3 & -1 & -2 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 2 3 2 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 27, \Delta = -115, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14, T_{\max} = 27, T \in [14, 27]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 1 & 0 & 3 & 1 & 1 & 0 & 2 \\ 2 & 0 & 2 & -2 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}} 3 \ 221^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, \ T_{\max} = 25, \ \Delta = -113, \ \det = 108, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0 \\ T_{\min} = 13.93 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 21 & 27 & 1 & 9 \\ 27 & 33 & 1 & 11 \\ 1 & 1 & -3 & -1 \\ 9 & 11 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3 \ 221^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, \ T_{\max} = 25, \ \Delta = -113, \ \det = 112, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0 \\ T_{\min} = 14.25 \rightarrow 15, \ T_{\max} = 25, \ T \in [15, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 4 & 0 \\ 4 & 8 & 0 & 4 & 9 & 5 \\ 3 & 4 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1}} 231^{\overline{11}} 31^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, \ T_{\max} = 69, \ \Delta = -121, \ \det = 265, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0 \\ T_{\min} = 14.15 \rightarrow 15, \ T_{\max} = 69, \ T \in [15, 69]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 2 & 4 & -1 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

4.283 $\text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{e}}'_7 + \text{NHC}(-2)(-3) + \textcolor{blue}{\mathfrak{su}}_2$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3 \ 21^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 147, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.80 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 7 & 5 \\ 7 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 5 & 0 \\ 7 & 10 & 0 & 5 & 8 & 6 \\ 5 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3 \ 21^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 452, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.80 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 8 & 5 \\ 7 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 5 & 0 \\ 8 & 10 & 0 & 5 & 8 & 6 \\ 5 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \ 1^{\overline{111}} 3 \ 221^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -83, \ \det = -116, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.93 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 21 & 27 & 1 & 27 & 9 \\ 27 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 35 & 0 & 33 & 11 \\ 9 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3 \ 221^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -83, \ \det = -356, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.93 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 21 & 27 & 1 & 28 & 9 \\ 27 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 35 & 0 & 33 & 11 \\ 9 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1231}} 1^{\overline{11}} 31^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 15, \ \Delta = -91, \ \det = -440, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.18 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 0 & 3 & -1 \\ 2 & 6 & 0 & 3 & 4 & 1 & 6 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 6 & 0 & 4 & 4 \\ 2 & 4 & -1 & -1 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

4.284 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 27, \Delta = -115, \det = 252, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 27, T \in [14, 27]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 4 & 5 \\ 7 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 3 & 0 \\ 4 & 5 & 0 & 3 & 1 & 3 \\ 5 & 6 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \quad 1^{11} 23^1 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 16, \Delta = -94, \det = -355, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.18 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 0 & 2 & -1 \\ 2 & 6 & 0 & 3 & 4 & 1 & 3 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 3 & 0 & 0 & 2 \\ 2 & 4 & -1 & -1 & 4 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

4.285 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 82, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 1 & 3 \\ 3 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 4 & 0 & -1 \\ 4 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -1 & 5 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \quad 1^{111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -92, \det = -61, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.28 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 1 & 3 \\ 3 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 4 & 0 & 0 \\ 4 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & 0 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

4.286 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{so}_8$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 102, \Delta = -191, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 102, \Delta = -191, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.47 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 & -2 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 2 & 2 & -1 & -2 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \quad 1^{11} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 102, \Delta = -162, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 6 & 6 & 0 & 6 \\ 2 & 6 & -2 & 0 & 0 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 6 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 23^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 102, \Delta = -164, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 9 & 4 & 5 \\ 2 & 0 & 4 & -3 & -1 \\ 3 & 0 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{11} 2^1 3^1 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 102, \Delta = -164, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.48 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 10 & 1 & 4 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3^1 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 102, \Delta = -162, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 2 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{11} 231^1 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 102, \Delta = -170, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 14.73 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 1 & 0 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 0 & 3 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

4.287 NHC(-2)(-3) + $\textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$

$T = 9 \quad [0] \quad 1^{11} 3^1 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^1 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.51 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 0 & -2 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3^1 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -213, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.45 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^1 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -241, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.43 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$T = 10 \quad [4] \quad 1^{11} 2^1 3^1 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 95, \Delta = -134, \det = 23, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 16.87 \rightarrow 17, T_{\max} = 95, T \in [17, 95]$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 \\ 2 & 0 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 0 & 5 & -1 & 0 & -2 \end{pmatrix}_{b_0}$$

[5] $1^{111} 2^1 3^1 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 95, \Delta = -134, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.53 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

- [6] $1^{11}231^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{so}_8 + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -140, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & -1 \\ 4 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$
- [7] $1^{11}3^{11}221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{so}_8 + \text{su}_2$ $T = 10, T_{\max} = 94, \Delta = -132, \det = 111, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.63 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 1 & 0 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & -1 & -2 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{111}2^3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{so}_8 + \text{su}_2$ $T = 10, T_{\max} = 95, \Delta = -134, \det = 130, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.48 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & -1 \\ 6 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{111}2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{so}_8 + \text{su}_2$ $T = 10, T_{\max} = 95, \Delta = -134, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 14.48 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & -1 \\ 5 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{112}2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{so}_8 + \text{su}_2$ $T = 10, T_{\max} = 95, \Delta = -134, \det = 421, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$
 $T_{\min} = 14.50 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 9 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 8 & -1 \\ 9 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{114}2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{so}_8 + \text{su}_2$ $T = 10, T_{\max} = 95, \Delta = -134, \det = 501, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 17 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 16 & -1 \\ 17 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & -1 & 24 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{113}2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{so}_8 + \text{su}_2$ $T = 10, T_{\max} = 95, \Delta = -134, \det = 505, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.8$
 $T_{\min} = 14.54 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 13 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 12 & -1 \\ 13 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & -1 & 18 & 0 \end{pmatrix}_{b_0}$$
- [13] $1^{112}2^3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{so}_8 + \text{su}_2$ $T = 10, T_{\max} = 95, \Delta = -134, \det = 606, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$
 $T_{\min} = 14.49 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 10 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 8 & -1 \\ 10 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$
- [14] $1^{113}2^3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{so}_8 + \text{su}_2$ $T = 10, T_{\max} = 95, \Delta = -134, \det = 994, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.8$
 $T_{\min} = 14.50 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 14 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 12 & -1 \\ 14 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & -1 & 18 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{114}2^13^121^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 10$, $T_{\max} = 95$, $\Delta = -134$, $\det = 1294$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 14.51 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 18 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 16 & -1 \\ 18 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & -1 & 24 & 0 \end{pmatrix}_{b_0}$$

4.288 $\text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$

$T = 10$ [0] $1^{111}3^121^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 10$, $T_{\max} = 93$, $\Delta = -131$, $\det = 132$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 0 & 0 & -2 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$T = 11$ [1] $1^{1111}2^13^121^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -104$, $\det = -84$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & 4 & -1 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 4 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^1221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -102$, $\det = -104$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 0 & 0 & -1 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{1111}2^13^121^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 11$, $T_{\max} = 23$, $\Delta = -104$, $\det = -150$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 14.56 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & 4 & -1 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 5 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

4.289 $\text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{11}3^121^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 102$, $\Delta = -164$, $\det = -148$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 14.49 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.290 $\text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$

$T = 9$ [0] $1^{11}3^121^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3$ $T = 9$, $T_{\max} = 102$, $\Delta = -170$, $\det = -245$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 0 & -2 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -2 & 0 & -1 & -2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [1] \quad 1^{11} 2^1 3^1 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 100, \Delta = -143, \det = 79, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.94 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 \\ 2 & 0 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 0 & 5 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 2^1 3^1 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 100, \Delta = -143, \det = 151, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.18 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & -1 \\ 4 & 12 & 0 & 4 & 9 & 5 \\ 3 & 6 & -1 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 221^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 100, \Delta = -141, \det = 196, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 0 & -1 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -1 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

4.291 $\text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{red}{\mathfrak{su}}_2$

$$T = 8 \quad [0] \quad 1^{111} 3^1 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 16.71 \rightarrow 17, T_{\max} = 101, T \in [17, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -3 & 3 & -1 \\ 5 & 8 & 3 & 4 & 4 \\ 3 & 5 & -1 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{112} 3^1 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & -2 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 2^1 2^1 31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 3 & 4 & -3 & 3 & -1 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.24 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^1 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{12} 3 2 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 2 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{13} 3 2 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 13.57 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{14} 3 2 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 4 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3 \ 2 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.02 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -3 & 3 & -1 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3 2 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -3 & 3 & -1 \\ 4 & 8 & 3 & 4 & 4 \\ 3 & 5 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{15} 3 2 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 102, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.1$$

$$T_{\min} = 14.02 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & -2 \\ 5 & 10 & 0 & 5 & 48 & 10 \\ 2 & 2 & -1 & -2 & 10 & -2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 3 \ 2 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 2 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{16} 3 2 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 14.19 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 6 & -2 \\ 6 & 12 & 0 & 6 & 70 & 12 \\ 2 & 2 & -1 & -2 & 12 & -2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{17} 3 2 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 7 & -2 \\ 7 & 14 & 0 & 7 & 96 & 14 \\ 2 & 2 & -1 & -2 & 14 & -2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11} 3 \ 2 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 6 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[26] $1^{14} 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 600, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 6 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[27] $1^{13} 3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 619, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 6 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[28] $1^{13} 3 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 774, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.44 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 7 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

[29] $1^{14} 3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 837, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.39 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 7 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{14} 3 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1060, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.41 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 8 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$T = 9$ [32] $1^{12} 3 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 20, T_{\max} = 93, T \in [20, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 3 & 1 & 0 & 4 & 2 & 4 \\ 2 & 2 & -1 & -1 & 4 & -6 \end{pmatrix}_{b_0}$$

[34] $1^{13} 3 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 17, T_{\max} = 93, T \in [17, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 4 & 1 & 0 & 4 & 2 & 4 \\ 2 & 2 & -1 & -1 & 4 & -3 \end{pmatrix}_{b_0}$$

[35] $1^{14} 3 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 20, T_{\max} = 93, T \in [20, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 5 & 1 & 0 & 4 & 2 & 4 \\ 2 & 2 & -1 & -1 & 4 & -6 \end{pmatrix}_{b_0}$$

[36] $1^{11} 2321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & -2 \\ 1 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{12} 2^1 3^1 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{e}}'_7 + \textcolor{brown}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 95, \Delta = -135, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 15, T_{\max} = 95, T \in [15, 95]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -4 & 0 & -2 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -2 & 0 & -1 \end{pmatrix}_{b_0}$

[37] $1^{12} 1^3 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{c}}'_7 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -133, \ \det = -18, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$
 $T_{\min} = 14.67 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$

[38] $1^{11}2^13^121^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{c}}'_7 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 9, \, T_{\max} = 95, \, \Delta = -135, \, \det = -19, \, (n_+, n_-, n_0) = (1, 9, 0), \, c_{\text{ext}} = 11.0$
 $T_{\min} = 14.37 \rightarrow 15, \, T_{\max} = 95, \, T \in [15, 95]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & -2 & 0 & -1 \end{pmatrix}_{b_0}$

[39] $1^{11}2^321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & -2 & 3 & 0 \end{pmatrix}_{b_0}$

[40] $1^{11} 2^1 3^1 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{e}}'_7 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -34, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$
 $T_{\min} = 13.41 \rightarrow 14, \ T_{\max} = 95, \ T \in [14, 95]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -2 & 0 & 0 \end{pmatrix}_{b_0}$

[41] $1^{11} 3^{221} + [\textcolor{red}{5u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{e}'_7 + \textcolor{brown}{5u}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$

[42] $1^1 2 3 1^1 3 1^{11} + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}'_7 + \textcolor{red}{s} \textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.35 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 1 & 0 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[43] $1^{11} 1^1 3^{221} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.23 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$

[46] $1^{13}2^3 2^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.83 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -4 & 3 & -2 \\ 4 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -2 & 9 & 0 \end{pmatrix}_{b_0}$

[45] $1^{1231}1^{31}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -46, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$
 $T_{\min} = 13.74 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

- [44] $1^{12}3^{11}221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 2 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$
- [47] $1^{13}3^{11}221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.82 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$
- [48] $1^{1231}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.21 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 3 & -1 \\ 3 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$
- [49] $1^{14}3^{11}221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.21 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 4 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$
- [50] $1^{12}3^{11}221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 3 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$
- [51] $1^{15}3^{11}221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.1$
 $T_{\min} = 14.58 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 5 & 10 & 0 & 5 & 48 & 10 \\ 2 & 2 & -1 & -1 & 10 & -1 \end{pmatrix}_{b_0}$$
- [52] $1^{123131}^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.22 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$
- [53] $1^{1231}231^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.47 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 6 & -1 \\ 6 & 0 & 0 & 6 & 22 & 8 \\ 2 & 0 & -1 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$
- [54] $1^{16}3^{11}221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 14.92 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 6 & 12 & 0 & 6 & 70 & 12 \\ 2 & 2 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$
- [55] $1^{112}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 2 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & -2 & 3 & 0 \end{pmatrix}_{b_0}$$
- [56] $1^{112}3^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & -2 \\ 2 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{12} 2^3 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -4 & 2 & -2 \\ 3 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{17} 3 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.3$$

$$T_{\min} = 15.22 \rightarrow 16, T_{\max} = 93, T \in [16, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 7 & 14 & 0 & 7 & 96 & 14 \\ 2 & 2 & -1 & -1 & 14 & -2 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{11} 3 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 2 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{11} 3 221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 3 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{12} 2^3 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.07 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 3 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[62] \ 1^1 231^3 31^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 9 & -1 \\ 9 & 0 & 0 & 9 & 52 & 12 \\ 2 & 0 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[63] \ 1^1 23131^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{13} 3 221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -121, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 4 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{12} 3 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 3 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{12} 2^3 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13.49 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 3 & 6 & 0 & 2 & 10 & 6 \\ 2 & 3 & -1 & -2 & 6 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{11}2^3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & -2 \\ 3 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{14}2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 5 & 12 & 0 & 4 & 46 & 12 \\ 2 & 3 & -1 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{12}3^1 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 3 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{12}31^{14} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.89 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 12 & -1 \\ 12 & 0 & 0 & 12 & 94 & 16 \\ 2 & 0 & -1 & -1 & 16 & -1 \end{pmatrix}_{b_0}$$

[71] $1^{13}2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.46 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 4 & 9 & 0 & 3 & 25 & 9 \\ 2 & 3 & -1 & -2 & 9 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{12}31^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.78 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & -1 \\ 2 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{11}3^1 221^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.78 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 4 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{11}3^1 221^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.09 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & -1 \\ 5 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

[75] $1^{11}3^1 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -169, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 4 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{12}31^{11} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.72 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{14}3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & -1 \\ 5 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[78] $1^{12}3131^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 14.55 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[79] $1^{11}2321^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.62 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & -2 \\ 4 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 0 \end{pmatrix}_{b_0}$$

[80] $1^{12}31^{131}1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -193, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.08 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 3 & -1 \\ 4 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[81] $1^{12}3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.35 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 4 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[82] $1^{13}3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -207, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[83] $1^{12}2321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.68 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 4 & 6 & 0 & 2 & 10 & 6 \\ 2 & 3 & -1 & -2 & 6 & 0 \end{pmatrix}_{b_0}$$

[85] $1^{11}2321^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & -2 \\ 5 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 0 \end{pmatrix}_{b_0}$$

[84] $1^{12}2321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -4 & 2 & -2 \\ 4 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -2 & 6 & 0 \end{pmatrix}_{b_0}$$

[86] $1^{12}31^{131}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[87] $1^{11} 3^{13} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 5 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[88] $1^{12} 3^{12} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 4 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[89] $1^{14} 2^3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.53 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -4 & 4 & -2 \\ 6 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$

[91] $1^1 231^{12} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.88 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & -1 \\ 4 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[90] $1^{13} 2^3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -4 & 3 & -2 \\ 5 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -2 & 9 & 0 \end{pmatrix}_{b_0}$$

[92] $1^{14} 3^{11} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 5 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[93] $1^{11} 3^{14} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -289, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.68 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 5 & -1 \\ 6 & 3 & 0 & 5 & 6 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[94] $1^1 23131^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.62 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[95] $1^{12} 3^{13} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.58 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 5 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[96] $1^{13} 2^3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 5 & 9 & 0 & 3 & 25 & 9 \\ 2 & 3 & -1 & -2 & 9 & 0 \end{pmatrix}_{b_0}$$

[97] $1^{13} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 5 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

[98] $1^{12} 231^{231} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.09 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 6 & -1 \\ 7 & 0 & 0 & 6 & 22 & 8 \\ 2 & 0 & -1 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$

[99] $1^{12} 3^{12} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.39 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 5 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[100] $1^{12} 3^{13} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -4 & 2 & -2 \\ 5 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -2 & 6 & 0 \end{pmatrix}_{b_0}$$

[101] $1^{13} 3^{12} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -351, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[102] $1^{12} 231^{12} 31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & -1 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[103] $1^{12} 231^{13} 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -358, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.11 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 3 & -1 \\ 5 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[104] $1^{12} 3^{14} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -382, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.69 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 6 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[105] $1^{12} 2^{14} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.52 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & -2 \\ 6 & 12 & 0 & 4 & 46 & 12 \\ 2 & 3 & -1 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$

[106] $1^{12} 23131^{115} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -427, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.1$
 $T_{\min} = 14.65 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0} \\
[107] & 1^{14} 231^{13} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8 \\
& T_{\min} = 13.93 \rightarrow 14, T_{\max} = 99, T \in [14, 99] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 9 & -1 \\ 6 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0} \\
[108] & 1^{14} 3 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 13.39 \rightarrow 14, T_{\max} = 93, T \in [14, 93] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & -1 \\ 6 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0} \\
[109] & 1^{13} 2 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -454, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8 \\
& T_{\min} = 13.45 \rightarrow 14, T_{\max} = 95, T \in [14, 95] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -4 & 3 & -2 \\ 6 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -2 & 9 & 0 \end{pmatrix}_{b_0} \\
[110] & 1^{13} 3 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5 \\
& T_{\min} = 13.45 \rightarrow 14, T_{\max} = 93, T \in [14, 93] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 6 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0} \\
[111] & 1^{12} 2 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5 \\
& T_{\min} = 13.54 \rightarrow 14, T_{\max} = 95, T \in [14, 95] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -4 & 2 & -2 \\ 6 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & -2 & 6 & 0 \end{pmatrix}_{b_0} \\
[112] & 1^{12} 3 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -466, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 13.51 \rightarrow 14, T_{\max} = 93, T \in [14, 93] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 6 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0} \\
[113] & 1^{12} 231^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -478, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5 \\
& T_{\min} = 13.77 \rightarrow 14, T_{\max} = 99, T \in [14, 99] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & -1 \\ 5 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0} \\
[114] & 1^{12} 231^{13} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -481, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8 \\
& T_{\min} = 14.13 \rightarrow 15, T_{\max} = 99, T \in [15, 99] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 9 & -1 \\ 10 & 0 & 0 & 9 & 52 & 12 \\ 2 & 0 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0} \\
[115] & 1^{13} 3 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -483, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8 \\
& T_{\min} = 13.53 \rightarrow 14, T_{\max} = 93, T \in [14, 93] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 6 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0} \\
[116] & 1^{13} 3 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -505, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8 \\
& T_{\min} = 13.39 \rightarrow 14, T_{\max} = 93, T \in [14, 93]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 6 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

[117] $1^{14} 2^3 3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -514, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.46 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -4 & 4 & -2 \\ 7 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$

[118] $1^1 231^{12} 31^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -526, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & -1 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[119] $1^1 231^1 31^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -553, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.16 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 3 & -1 \\ 6 & 0 & 0 & 3 & 4 & 4 \\ 2 & 0 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[120] $1^{12} 3 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.51 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 6 & -1 \\ 7 & 5 & 0 & 6 & 14 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[121] $1^1 231^{12} 31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -568, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.75 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & -1 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[122] $1^{13} 3 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -603, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.59 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 7 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[123] $1^1 231^{23} 31^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -616, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.08 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 6 & -1 \\ 8 & 0 & 0 & 6 & 22 & 8 \\ 2 & 0 & -1 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$

[124] $1^{13} 2^3 3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -640, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -4 & 3 & -2 \\ 7 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & -2 & 9 & 0 \end{pmatrix}_{b_0}$$

[125] $1^{14} 3 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -652, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.52 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 7 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[126] $1^{14} 3 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -676, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & -1 \\ 7 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

- [127] $1^{13} 3^{13} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -679$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 13.42 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 7 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$
- [128] $1^1 231^{14} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -688$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.95 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 12 & -1 \\ 8 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$
- [129] $31^{11} 2321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -143$, $\det = -751$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.80 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 3 & 2 \\ 3 & 1 & 4 & 1 & 3 & 3 \\ 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -4 & 1 & -2 \\ 3 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & -2 & 3 & 0 \end{pmatrix}_{b_0}$$
- [130] $1^{14} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2$ $T = 9$, $T_{\max} = 95$, $\Delta = -135$, $\det = -776$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 95$, $T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -4 & 4 & -2 \\ 8 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$
- [131] $1^1 231^{13} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -802$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 13.81 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 9 & -1 \\ 7 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$
- [132] $1^{14} 3^{14} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -826$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.55 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & -1 \\ 8 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$
- [133] $1^{13} 3^{14} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -841$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 7 & -1 \\ 8 & 7 & 0 & 7 & 26 & 10 \\ 2 & 2 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$
- [134] $1^{14} 3^{13} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -904$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.42 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & -1 \\ 8 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$
- [135] $1^1 231^{15} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -1012$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.1$
 $T_{\min} = 13.97 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 15 & -1 \\ 10 & 30 & 0 & 15 & 148 & 20 \\ 2 & 4 & -1 & -1 & 20 & 0 \end{pmatrix}_{b_0}$$
- [136] $1^1 231^{13} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -1036$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 13.77 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 9 & -1 \\ 8 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$
- [137] $1^1 231^{13} 31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -1108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 9 & -1 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[138] $1^{14} 3 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -1120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.44 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 8 & -1 \\ 9 & 9 & 0 & 8 & 42 & 12 \\ 2 & 2 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[139] $1^1 231^{13} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -1138, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.75 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 9 & -1 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 2 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[140] $1^1 231^{14} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -1198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.83 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 12 & -1 \\ 9 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[141] $1^1 231^{14} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -1576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.79 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 12 & -1 \\ 10 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[142] $1^1 231^{14} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -1822, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.77 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 12 & -1 \\ 11 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[143] $1^1 231^{14} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -1936, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.75 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 12 & -1 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 2 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

4.292 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$

$T = 10 \quad [0] 1^{111} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 147, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 1 & 5 \\ 7 & 8 & 1 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 6 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 452, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 8 & 5 \\ 7 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 5 & 0 \\ 8 & 10 & 0 & 5 & 8 & 6 \\ 5 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$T = 11 \quad [2] 1^{111} 3 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -116, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.93 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 21 & 27 & 1 & 1 & 9 \\ 27 & 33 & 1 & 0 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 9 & 11 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3 \ 221^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, \ T_{\max} = 14, \ \Delta = -83, \ \det = -356, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.93 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 21 & 27 & 1 & 28 & 9 \\ 27 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 35 & 0 & 33 & 11 \\ 9 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1231} 1^{11} 31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, \ T_{\max} = 15, \ \Delta = -91, \ \det = -440, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.18 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 4 & 6 & 0 & 4 \\ 2 & 6 & 0 & 3 & 6 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 0 & 1 & -3 & -1 \\ 2 & 4 & -1 & -1 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

4.293 $\text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8$

$$T = 9 \quad [0] \ 1^{11} 3 \ 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 101, \ \Delta = -161, \ \det = -6, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.33 \rightarrow 16, \ T_{\max} = 101, \ T \in [16, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 0 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 101, \ \Delta = -161, \ \det = -139, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.51 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 & 1 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 & 0 \\ 2 & 2 & -1 & -2 & 0 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3 \ 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 101, \ \Delta = -161, \ \det = -213, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.45 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3 \ 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 101, \ \Delta = -161, \ \det = -241, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.43 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 4 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \ 1^{11} 2 \ 3 \ 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 10, \ T_{\max} = 95, \ \Delta = -134, \ \det = 23, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 16.87 \rightarrow 17, \ T_{\max} = 95, \ T \in [17, 95]$$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 \\ 2 & 0 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 0 & 5 & -1 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 2 \ 3 \ 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 10, \ T_{\max} = 95, \ \Delta = -134, \ \det = 86, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.53 \rightarrow 15, \ T_{\max} = 95, \ T \in [15, 95]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & -1 \\ 4 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1231} 1^{11} 31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 10, \ T_{\max} = 100, \ \Delta = -140, \ \det = 94, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.74 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 0 & 3 & -1 & 0 & -2 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3^{11} 2^{11} 2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{so}_8$ $T = 10$, $T_{\max} = 94$, $\Delta = -132$, $\det = 111$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 14.63 \rightarrow 15$, $T_{\max} = 94$, $T \in [15, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{11} 2^{12} 3^{11} 2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{so}_8$ $T = 10$, $T_{\max} = 95$, $\Delta = -134$, $\det = 130$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 14.48 \rightarrow 15$, $T_{\max} = 95$, $T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & -1 \\ 6 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{11} 2^{11} 3^{11} 2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{so}_8$ $T = 10$, $T_{\max} = 95$, $\Delta = -134$, $\det = 249$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 14.48 \rightarrow 15$, $T_{\max} = 95$, $T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & -1 \\ 5 & 12 & 0 & 4 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{12} 2^{11} 3^{11} 2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{so}_8$ $T = 10$, $T_{\max} = 95$, $\Delta = -134$, $\det = 421$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.5$
 $T_{\min} = 14.50 \rightarrow 15$, $T_{\max} = 95$, $T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 9 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 8 & -1 \\ 9 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{14} 2^{11} 3^{11} 2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{so}_8$ $T = 10$, $T_{\max} = 95$, $\Delta = -134$, $\det = 501$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.60 \rightarrow 15$, $T_{\max} = 95$, $T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 17 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 16 & -1 \\ 17 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & -1 & 24 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{13} 2^{11} 3^{11} 2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{so}_8$ $T = 10$, $T_{\max} = 95$, $\Delta = -134$, $\det = 505$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.8$
 $T_{\min} = 14.54 \rightarrow 15$, $T_{\max} = 95$, $T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 13 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 12 & -1 \\ 13 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & -1 & 18 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{12} 2^{11} 3^{11} 2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{so}_8$ $T = 10$, $T_{\max} = 95$, $\Delta = -134$, $\det = 606$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.5$
 $T_{\min} = 14.49 \rightarrow 15$, $T_{\max} = 95$, $T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 10 & 3 \\ 4 & 10 & 1 & 4 & 24 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 8 & -1 \\ 10 & 24 & 0 & 8 & 46 & 12 \\ 3 & 6 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{13} 2^{11} 3^{11} 2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{so}_8$ $T = 10$, $T_{\max} = 95$, $\Delta = -134$, $\det = 994$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.8$
 $T_{\min} = 14.50 \rightarrow 15$, $T_{\max} = 95$, $T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 14 & 3 \\ 4 & 10 & 1 & 4 & 36 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 12 & -1 \\ 14 & 36 & 0 & 12 & 106 & 18 \\ 3 & 6 & -1 & -1 & 18 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{14} 2^{11} 3^{11} 2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{so}_8$ $T = 10$, $T_{\max} = 95$, $\Delta = -134$, $\det = 1294$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.51 \rightarrow 15$, $T_{\max} = 95$, $T \in [15, 95]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 18 & 3 \\ 4 & 10 & 1 & 4 & 48 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 16 & -1 \\ 18 & 48 & 0 & 16 & 190 & 24 \\ 3 & 6 & -1 & -1 & 24 & 0 \end{pmatrix}_{b_0}$$

4.294 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 132, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \quad 1^{11} 23 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -104, \det = -84, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & 4 & -1 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 4 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 2 \quad 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -104, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 2 \quad 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -104, \det = -150, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.56 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & 4 & -1 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 5 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

4.295 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{132} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.3$$

$$T_{\min} = 16.11 \rightarrow 17, T_{\max} = 93, T \in [17, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 2 & -2 \\ 3 & 6 & 0 & 3 & 16 & 12 & 6 \\ 3 & 4 & 0 & 2 & 12 & 6 & 4 \\ 2 & 2 & -1 & -2 & 6 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16.02 \rightarrow 17, T_{\max} = 93, T \in [17, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & 0 & 4 & -3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{144} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.81 \rightarrow 16, T_{\max} = 93, T \in [16, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 4 & -2 \\ 4 & 8 & 0 & 4 & 30 & 32 & 8 \\ 5 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & 8 & -3 \end{pmatrix}_{b_0}$$

[3] $1^{\textcolor{red}{1}\textcolor{blue}{1}} 3 21^{\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}2 \cdot \textcolor{blue}{g}2] + \textcolor{green}{c}'_7 + \textcolor{orange}{s}\textcolor{blue}{u}2 + \textcolor{red}{s}\textcolor{blue}{u}2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -49, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.24 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -2 & 0 & 2 & -1 \end{pmatrix}_{b_0}$

$$[4] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3 \ 21^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{c}'_7 + \textcolor{blue}{s}\textcolor{red}{u}_2 + \textcolor{blue}{s}\textcolor{red}{u}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -55, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.22 \rightarrow 16, \ T_{\max} = 93, \ T \in [16, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{121} 3 \ 21^1 + [\textcolor{blue}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}'_7 + \textcolor{brown}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -66, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.33 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{124} 3^{21} 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{c}'_7 + \textcolor{brown}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -70, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.43 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 7 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -2 & 4 & 8 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{111} 3^{211} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{green}{\mathbf{e}}'_7 + \textcolor{orange}{\mathbf{su}}_2 + \textcolor{red}{\mathbf{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{\textcolor{brown}{1}\textcolor{red}{1}\textcolor{blue}{1}}32\textcolor{green}{1} + [\textcolor{red}{\text{su}}_2 \cdot \textcolor{blue}{\text{g}}_2] + \textcolor{teal}{\text{c}}'_7 + \textcolor{brown}{\text{su}}_2 + \textcolor{red}{\text{su}}_2 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -112, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$
 $T_{\min} = 13.21 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{13} 3^{21} 1^{111} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{teal}{\mathbf{e}}'_7 + \textcolor{orange}{\mathbf{su}}_2 + \textcolor{red}{\mathbf{su}}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -115, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.37 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 16 & 6 \\ 2 & 2 & -1 & -2 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{112}3^{21}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}'_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.39 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{122} 3^{21} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{c}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -136, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.53 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & -2 \\ 2 & 4 & 0 & 2 & 6 & 8 & 4 \\ 2 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{113}3^{21}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}'_7 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.61 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 6 & -2 \\ 2 & 4 & 0 & 2 & 6 & 24 & 4 \\ 6 & 12 & 0 & 6 & 24 & 70 & 12 \\ 2 & 2 & -1 & -2 & 4 & 12 & -2 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{112} 3^{14} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -268, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.90 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 6 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{145} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -268, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.1$$

$$T_{\min} = 14.25 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 5 & -2 \\ 4 & 8 & 0 & 4 & 30 & 40 & 8 \\ 5 & 10 & 0 & 5 & 40 & 48 & 10 \\ 2 & 2 & -1 & -2 & 8 & 10 & -2 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{136} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -284, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.1$$

$$T_{\min} = 14.30 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 6 & -2 \\ 3 & 6 & 0 & 3 & 16 & 36 & 6 \\ 6 & 12 & 0 & 6 & 36 & 70 & 12 \\ 2 & 2 & -1 & -2 & 6 & 12 & -2 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{117} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -304, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.3$$

$$T_{\min} = 14.34 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 7 & -2 \\ 1 & 2 & 0 & 1 & 0 & 14 & 2 \\ 7 & 14 & 0 & 7 & 14 & 96 & 14 \\ 2 & 2 & -1 & -2 & 2 & 14 & -2 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{127} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -316, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.37 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 7 & -2 \\ 2 & 4 & 0 & 2 & 6 & 28 & 4 \\ 7 & 14 & 0 & 7 & 28 & 96 & 14 \\ 2 & 2 & -1 & -2 & 4 & 14 & -2 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{134} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -329, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.57 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 4 & -2 \\ 3 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -2 & 6 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{137} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -336, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.1$$

$$T_{\min} = 14.40 \rightarrow 15, \ T_{\max} = 93, \ T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 7 & -2 \\ 3 & 6 & 0 & 3 & 16 & 42 & 6 \\ 7 & 14 & 0 & 7 & 42 & 96 & 14 \\ 2 & 2 & -1 & -2 & 6 & 14 & -2 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{112} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -337, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.32 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -356, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.33 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$T_{\min} = 13.45 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -2 & 4 & 8 & -1 \end{pmatrix}_{b_0}$$

[49] $1^{122} 3^{111} 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -652, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.35 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & -2 \\ 3 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[50] $1^{11} 1^{14} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -657$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_b$$

[51] $1^{113} 3^{21} 1^{12} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{teal}{\mathbf{e}}'_7 + \textcolor{orange}{\mathbf{su}}_2 + \textcolor{red}{\mathbf{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -724, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 13.36 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 5 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -1 \end{pmatrix}_b$$

[52] $1^{123} 3 \ 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -763, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.3$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 3 & -2 \\ 3 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -2 & 4 & 6 & -1 \end{pmatrix}$$

[53] $1^{124} 3^{111} 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -792$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 3 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -2 & 4 & 8 & -1 \end{pmatrix}$$

[54] $1^{113} 3^{21} 1^{14} [\textcolor{red}{\mathbf{su}_2} \cdot \textcolor{blue}{\mathbf{g}_2}] + \textcolor{green}{\mathbf{e}}'_7 + \textcolor{orange}{\mathbf{su}_2} + \textcolor{red}{\mathbf{su}_2} \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -832, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 13.49 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 7 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -1 \end{pmatrix}_b$$

[55] $1^{113} 3^{21} 1^{13} [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -837$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 6 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 2 & 6 & -1 \end{pmatrix}_b$$

[56] $1^{133} 3^{111} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 93, \Delta = -132, \det = -964, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.6$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 3 & -2 \\ 4 & 6 & 0 & 3 & 16 & 18 & 6 \\ 4 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -2 & 6 & 6 & -1 \end{pmatrix}$$

$$[57] \ 1^{114} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -1024, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.38 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 6 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{134} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -1083, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.41 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 4 & -2 \\ 4 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -2 & 6 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{114} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -1273, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 7 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{144} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -1292, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.42 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 4 & -2 \\ 5 & 8 & 0 & 4 & 30 & 32 & 8 \\ 5 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -2 & 8 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{114} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -1404, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.42 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 8 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [62] \ 1^{1231} 1^1 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 8, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 20.50 \rightarrow 21, \ T_{\max} = 23, \ T \in [21, 23]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & -1 & 0 & 0 & -7 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{134} 3 \ 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -103, \ \det = 8, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 17, \ T_{\max} = 22, \ T \in [17, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 8 & -1 \\ 3 & 6 & 0 & 3 & 16 & 27 & 6 \\ 5 & 9 & 0 & 8 & 27 & 42 & 12 \\ 2 & 2 & -1 & -1 & 6 & 12 & -3 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{1231} 1^1 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 11, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 21.45 \rightarrow 22, \ T_{\max} = 23, \ T \in [22, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 3 & 0 & -1 \\ 3 & 0 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -1 & 4 & 0 & -8 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{134} 3 \ 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -103, \ \det = 12, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} =$$

13.8

$$T_{\min} = 16.33 \rightarrow 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 8 & -1 \\ 4 & 6 & 0 & 3 & 16 & 27 & 6 \\ 6 & 9 & 0 & 8 & 27 & 42 & 12 \\ 2 & 2 & -1 & -1 & 6 & 12 & -3 \end{pmatrix}_{b_0}$$

[66] $1^{111} 3^{12} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.56 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 4 & 3 & 0 & 5 & 3 & 6 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[67] $1^{121} 2^1 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 16.80 \rightarrow 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -4 & 0 & 0 & -2 \\ 2 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -2 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

[68] $1^{14} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 0 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 8 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 8 & 0 & 42 & 12 \\ 2 & 2 & -1 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{1231} 1^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -1 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

[70] $1^{111} 2^1 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{1231} 1^{12} 31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{111} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 37, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 3 & 3 & 0 & 5 & 3 & 6 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{111} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{121} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -1 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{124} 3^{113} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.54 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 7 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -1 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{112} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.71 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -2 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{123} 3^{111} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 58, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 5 & 7 & 0 & 7 & 14 & 26 & 10 \\ 2 & 2 & -1 & -1 & 4 & 10 & 0 \end{pmatrix}_{b_0}$$

[78] $1^{112} 2^1 3^{113} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 59, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 1 & 0 & 3 & 1 & 1 & 6 & 3 \\ 5 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{111} 2^1 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 59, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.36 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 1 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[80] $1^{111} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[81] $1^{112} 2^1 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 67, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 1 & 0 & 3 & 1 & 1 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[82] $1^{124} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 8 & -1 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 7 & 9 & 0 & 8 & 18 & 42 & 12 \\ 2 & 2 & -1 & -1 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

[83] $1^{111} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 3 & 0 & 5 & 0 & 6 & 6 \\ 2 & 2 & -1 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

[84] $1^{112} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 15.22 \rightarrow 16, T_{\max} = 22, T \in [16, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -4 & 0 & 1 & -2 \\ 2 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -2 & 0 & 3 & -2 \end{pmatrix}_{b_0}$$

[85] $1^{123} 31^{11} 2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 77, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[86] $1^{112} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 3 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[87] $1^{111} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 81, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.70 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 2 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[88] $1^{123} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 82, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$$T_{\min} = 13.49 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 4 & 7 & 0 & 7 & 14 & 26 & 10 \\ 2 & 2 & -1 & -1 & 4 & 10 & 0 \end{pmatrix}_{b_0}$$

[89] $1^{13} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 87, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 16 & 6 \\ 2 & 2 & -1 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

- [90] $1^{111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 88$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.32 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$
- [91] $1^{112} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 94$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 13.55 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$
- [92] $1^{112} 3^{11} 2^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 95$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -2 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$
- [93] $1^{122} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 100$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.76 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 2 & 4 & 0 & 2 & 6 & 8 & 4 \\ 2 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$
- [96] $1^{123} 14^1 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 104$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 16$, $T_{\max} = 23$, $T \in [16, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & -1 & 0 & 16 & -2 \end{pmatrix}_{b_0}$$
- [94] $1^{133} 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 104$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.6$
 $T_{\min} = 14.31 \rightarrow 15$, $T_{\max} = 22$, $T \in [15, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 9 & 2 \\ 1 & 0 & 1 & -4 & 3 & 3 & -2 \\ 4 & 0 & 9 & 3 & 25 & 27 & 9 \\ 4 & 0 & 9 & 3 & 27 & 25 & 9 \\ 2 & 0 & 2 & -2 & 9 & 9 & -1 \end{pmatrix}_{b_0}$$
- [95] $1^{113} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 104$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 13.88 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 3 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$
- [98] $1^{13} 3^1 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 108$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 0 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 7 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 7 & 0 & 7 & 0 & 26 & 10 \\ 2 & 2 & -1 & -1 & 0 & 10 & 0 \end{pmatrix}_{b_0}$$
- [97] $1^{123} 3^1 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 108$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$
 $T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 3 & 4 & 0 & 2 & 6 & 14 & 4 \\ 5 & 7 & 0 & 7 & 14 & 26 & 10 \\ 2 & 2 & -1 & -1 & 4 & 10 & 0 \end{pmatrix}_{b_0}$$

[99] $1^{123} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 110, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 3 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -1 & 4 & 6 & -1 \end{pmatrix}_{b_0}$$

[100] $1^{111} 3^{112} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 3 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[101] $1^{133} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 117, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.6$

$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 16 & 18 & 6 \\ 4 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[102] $1^{114} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 118, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 4 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

[103] $1^{133} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 120, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.6$

$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 16 & 18 & 6 \\ 3 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

[104] $1^{12} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 123, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 5 & 0 & 6 & 0 & 14 & 8 \\ 2 & 2 & -1 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

[105] $1^{124} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 4 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -1 & 4 & 8 & -1 \end{pmatrix}_{b_0}$$

[106] $1^{132} 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 127, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 6 & 2 \\ 1 & 0 & 1 & -4 & 3 & 2 & -2 \\ 4 & 0 & 9 & 3 & 25 & 18 & 9 \\ 3 & 0 & 6 & 2 & 18 & 10 & 6 \\ 2 & 0 & 2 & -2 & 9 & 6 & 0 \end{pmatrix}_{b_0}$$

[107] $1^{1231}1^{11}31^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 3 & 3 & -1 \\ 3 & 0 & 0 & 3 & 4 & 6 & 4 \\ 3 & 0 & 0 & 3 & 6 & 4 & 4 \\ 2 & 0 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[108] $1^{112}1^{111}3 \ 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 2 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[109] $1^{134}1^13 \ 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 134, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.58 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 4 & -1 \\ 3 & 6 & 0 & 3 & 16 & 24 & 6 \\ 4 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -1 & 6 & 8 & -1 \end{pmatrix}_{b_0}$$

[110] $1^{111}1^13 \ 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 135, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

[111] $1^{115}1^13 \ 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.1$

$T_{\min} = 14.62 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 5 & -1 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 5 & 10 & 0 & 5 & 10 & 48 & 10 \\ 2 & 2 & -1 & -1 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

[112] $1^{112}1^{124}3 \ 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 15.03 \rightarrow 16, T_{\max} = 22, T \in [16, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -4 & 0 & 1 & -2 \\ 2 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -2 & 0 & 3 & -2 \end{pmatrix}_{b_0}$$

[113] $1^{112}1^{12}2 \ 3 \ 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 1 & 0 & 3 & 1 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[114] $1^{125}1^13 \ 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 142, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.6$

$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 5 & -1 \\ 2 & 4 & 0 & 2 & 6 & 20 & 4 \\ 5 & 10 & 0 & 5 & 20 & 48 & 10 \\ 2 & 2 & -1 & -1 & 4 & 10 & -1 \end{pmatrix}_{b_0}$$

[115] $1^{12} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 144$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 2 & 0 & 6 & 4 \\ 2 & 2 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[116] $1^{14} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 148$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.81 \rightarrow 15$, $T_{\max} = 22$, $T \in [15, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 4 & -1 \\ 4 & 8 & 0 & 4 & 30 & 32 & 8 \\ 4 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 8 & -1 \end{pmatrix}_{b_0}$$

[117] $1^{11} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 149$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[118] $1^{12} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 150$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 5 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -1 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

[119] $1^{13} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 152$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.9$

$T_{\min} = 14.87 \rightarrow 15$, $T_{\max} = 22$, $T \in [15, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 5 & -1 \\ 3 & 6 & 0 & 3 & 16 & 30 & 6 \\ 5 & 10 & 0 & 5 & 30 & 48 & 10 \\ 2 & 2 & -1 & -1 & 6 & 10 & -1 \end{pmatrix}_{b_0}$$

[120] $1^{16} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 158$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 14.95 \rightarrow 15$, $T_{\max} = 22$, $T \in [15, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 12 & 2 \\ 6 & 12 & 0 & 6 & 12 & 70 & 12 \\ 2 & 2 & -1 & -1 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[121] $1^{12} 2^3 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 162$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.04 \rightarrow 15$, $T_{\max} = 22$, $T \in [15, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 3 & 0 & 1 & 1 & 6 & 3 \\ 4 & 6 & 0 & 2 & 6 & 10 & 6 \\ 2 & 3 & -1 & -2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[122] $1^{16} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 164$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 15.02 \rightarrow 16$, $T_{\max} = 22$, $T \in [16, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 6 & -1 \\ 2 & 4 & 0 & 2 & 6 & 24 & 4 \\ 6 & 12 & 0 & 6 & 24 & 70 & 12 \\ 2 & 2 & -1 & -1 & 4 & 12 & -2 \end{pmatrix}_{b_0}$$

[123] $1^{122} 3^{11} 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 164$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 2 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[124] $1^1 23131^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 164$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 14.32 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[125] $1^{145} 3^{11} 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 166$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.1$

$T_{\min} = 15.05 \rightarrow 16$, $T_{\max} = 22$, $T \in [16, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 5 & -1 \\ 4 & 8 & 0 & 4 & 30 & 40 & 8 \\ 5 & 10 & 0 & 5 & 40 & 48 & 10 \\ 2 & 2 & -1 & -1 & 8 & 10 & -2 \end{pmatrix}_{b_0}$$

[126] $1^{112} 2^3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 167$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 3 & 0 & 1 & 1 & 6 & 3 \\ 3 & 6 & 0 & 2 & 6 & 10 & 6 \\ 2 & 3 & -1 & -2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[127] $1^{114} 2^3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 172$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.72 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -4 & 1 & 4 & -2 \\ 1 & 0 & 3 & 1 & 1 & 12 & 3 \\ 6 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -2 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[128] $1^{136} 3^{11} 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 174$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.1$

$T_{\min} = 15.14 \rightarrow 16$, $T_{\max} = 22$, $T \in [16, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 6 & -1 \\ 3 & 6 & 0 & 3 & 16 & 36 & 6 \\ 6 & 12 & 0 & 6 & 36 & 70 & 12 \\ 2 & 2 & -1 & -1 & 6 & 12 & -2 \end{pmatrix}_{b_0}$$

[129] $1^{122} 3^{112} 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 176$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 6 & 2 \\ 1 & 0 & 1 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 0 & 6 & 2 & 0 & 10 & 6 \\ 2 & 0 & 2 & -2 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

[130] $1^{117} 3^{11} 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 184$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 15.24 \rightarrow 16$, $T_{\max} = 22$, $T \in [16, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 14 & 2 \\ 7 & 14 & 0 & 7 & 14 & 96 & 14 \\ 2 & 2 & -1 & -1 & 2 & 14 & -2 \end{pmatrix}_{b_0}$$

[131] $1^{12}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 185$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.76 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[132] $1^{12}7^13221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 190$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 15.29 \rightarrow 16$, $T_{\max} = 22$, $T \in [16, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 7 & -1 \\ 2 & 4 & 0 & 2 & 6 & 28 & 4 \\ 7 & 14 & 0 & 7 & 28 & 96 & 14 \\ 2 & 2 & -1 & -1 & 4 & 14 & -2 \end{pmatrix}_{b_0}$$

[133] $1^{11}3^12^1321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 192$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.58 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 1 & 0 & 3 & 1 & 1 & 9 & 3 \\ 7 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[134] $1^{13}7^13221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 200$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.1$

$T_{\min} = 15.38 \rightarrow 16$, $T_{\max} = 22$, $T \in [16, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 7 & -1 \\ 3 & 6 & 0 & 3 & 16 & 42 & 6 \\ 7 & 14 & 0 & 7 & 42 & 96 & 14 \\ 2 & 2 & -1 & -1 & 6 & 14 & -2 \end{pmatrix}_{b_0}$$

[135] $1^{11}3^13221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 202$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 4 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -1 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[136] $1^{11}3^12^1321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 204$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.49 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 1 & 0 & 3 & 1 & 1 & 9 & 3 \\ 5 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[137] $1^{11}2^13221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 206$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 14.11 \rightarrow 15$, $T_{\max} = 22$, $T \in [15, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 6 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[139] $1^{11}4^12^1321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 207$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 3 & 0 & 1 & 1 & 12 & 3 \\ 5 & 12 & 0 & 4 & 12 & 46 & 12 \\ 2 & 3 & -1 & -2 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[138] $1^{112}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 207$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 2 & 0 & 3 & 1 & 1 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[140] $1^{124}3^1221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 208$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.54 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 8 & -1 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 5 & 9 & 0 & 8 & 18 & 42 & 12 \\ 2 & 2 & -1 & -1 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

[141] $1^{113}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 209$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 3 & 0 & 1 & 1 & 9 & 3 \\ 4 & 9 & 0 & 3 & 9 & 25 & 9 \\ 2 & 3 & -1 & -2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[142] $1^{112}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 211$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.55 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -2 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[143] $1^{113}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 212$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 \\ 4 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[144] $1^{134}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 232$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.83 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 12 & 2 \\ 1 & 0 & 1 & -4 & 3 & 4 & -2 \\ 4 & 0 & 9 & 3 & 25 & 36 & 9 \\ 6 & 0 & 12 & 4 & 36 & 46 & 12 \\ 2 & 0 & 2 & -2 & 9 & 12 & 0 \end{pmatrix}_{b_0}$$

[145] $1^{134}3^1221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 239$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.82 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 4 & -1 \\ 3 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -1 & 6 & 8 & 0 \end{pmatrix}_{b_0}$$

[146] $1^{122}2^13^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 240$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & -4 & 2 & 2 & -2 \\ 4 & 0 & 6 & 2 & 10 & 12 & 6 \\ 6 & 0 & 6 & 2 & 12 & 10 & 6 \\ 2 & 0 & 2 & -2 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[147] $1^{112} 3^{14} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 248$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.77 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 7 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[148] $1^{113} 3^{111} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 253$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 2 & 2 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -1 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[149] $1^{123} 3^{111} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 257$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 14.20 \rightarrow 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & -1 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

[151] $1^{112} 3^{11} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 263$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.35 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[150] $1^{112} 3^{11} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 263$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[152] $1^{111} 2^3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 264$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.39 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[153] $1^{113} 2^3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 275$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 1 & 0 & 3 & 1 & 1 & 9 & 3 \\ 6 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[155] $1^{111} 3^{11} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 280$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 1 & -1 \\ 2 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -1 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[154] $1^{122}2^{111}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 280$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.49 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & -4 & 2 & 2 & -2 \\ 3 & 0 & 6 & 2 & 10 & 12 & 6 \\ 3 & 0 & 6 & 2 & 12 & 10 & 6 \\ 2 & 0 & 2 & -2 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[156] $1^{123}3^{111}221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 282$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 13.57 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -1 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

[157] $1^{1231}231^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 284$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.89 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

[158] $1^{122}2^{112}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 288$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & -4 & 2 & 2 & -2 \\ 3 & 0 & 6 & 2 & 10 & 12 & 6 \\ 4 & 0 & 6 & 2 & 12 & 10 & 6 \\ 2 & 0 & 2 & -2 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[159] $1^{124}3^{111}221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 292$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 8 & -1 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 6 & 9 & 0 & 8 & 18 & 42 & 12 \\ 2 & 2 & -1 & -1 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

[160] $1^{112}3^{113}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 303$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.72 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -2 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[161] $1^{124}3^{111}221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 307$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 8 & -1 \\ 3 & 4 & 0 & 2 & 6 & 18 & 4 \\ 6 & 9 & 0 & 8 & 18 & 42 & 12 \\ 2 & 2 & -1 & -1 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

[162] $1^{132}3^{113}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 311$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.47 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 9 & 2 \\ 1 & 0 & 1 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 0 & 9 & 3 & 0 & 25 & 9 \\ 2 & 0 & 2 & -2 & 0 & 9 & 0 \end{pmatrix}_{b_0}$$

[163] $1^{114}2^13^{112}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 312$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.56 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -4 & 1 & 4 & -2 \\ 2 & 0 & 3 & 1 & 1 & 12 & 3 \\ 6 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -2 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[164] $1^{114}3^{111}221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 312$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 8 & -1 \\ 2 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -1 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[165] $1^{111}2^13^{112}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 315$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.47 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[166] $1^{114}3^{111}221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 319$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.57 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 2 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[167] $1^{112}3^{113}221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 319$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.64 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 5 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[168] $1^{114}3^{11}221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 328$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.49 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 8 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 5 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -1 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[169] $1^{112}3^{112}221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 338$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.50 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 4 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[170] $1^{112}3^{113}221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 347$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.50 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 6 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[171] \ 1^{112} 3^{12} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -103, \ \det = 352, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.40 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 6 & -1 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 5 & 5 & 0 & 6 & 5 & 14 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{113} 2^3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 356, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.65 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 3 & 0 & 1 & 1 & 9 & 3 \\ 5 & 9 & 0 & 3 & 9 & 25 & 9 \\ 2 & 3 & -1 & -2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[173] \ 1^{112} 3^{111} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -103, \ \det = 359, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.46 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[174] \ 1^{1231} 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 368, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.83 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 3 & -1 \\ 2 & 6 & 0 & 3 & 4 & 6 & 4 \\ 2 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[175] \ 1^{112} 3^{114} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 371, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.92 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -2 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[176] \ 1^{113} 3^{111} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -103, \ \det = 372, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.49 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 2 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[177] \ 1^{1231} 1^{1231} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 380, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.75 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[178] \ 1^{124} 3^{112} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -103, \ \det = 380, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.54 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 6 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -1 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

[179] $1^{113} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 381$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[180] $1^{144} 2^{123} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 388$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.57 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 12 & 2 \\ 1 & 0 & 1 & -4 & 4 & 4 & -2 \\ 6 & 0 & 12 & 4 & 46 & 48 & 12 \\ 7 & 0 & 12 & 4 & 48 & 46 & 12 \\ 2 & 0 & 2 & -2 & 12 & 12 & 0 \end{pmatrix}_{b_0}$$

[181] $1^{1231} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 389$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.78 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 3 & -1 \\ 2 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[182] $1^{114} 2^{13} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 395$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -4 & 1 & 4 & -2 \\ 1 & 0 & 3 & 1 & 1 & 12 & 3 \\ 7 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -2 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[183] $1^{111} 2^{124} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 396$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.90 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 \\ 5 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[184] $1^{112} 2^{122} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 396$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.47 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 3 & 0 & 3 & 1 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[185] $1^{124} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 404$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.62 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -1 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

[186] $1^{122} 3^{113} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 420$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.52 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 6 & 2 \\ 1 & 0 & 1 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 0 & 6 & 2 & 0 & 10 & 6 \\ 2 & 0 & 2 & -2 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

[187] $1^{123}2^13^121^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 423, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 9 & 2 \\ 1 & 0 & 1 & -4 & 2 & 3 & -2 \\ 3 & 0 & 6 & 2 & 10 & 18 & 6 \\ 6 & 0 & 9 & 3 & 18 & 25 & 9 \\ 2 & 0 & 2 & -2 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

[188] $1^{1231}1^{11}31^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 437, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[189] $1^{1231}1^{11}31^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 464, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.72 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[190] $1^{114}2^13^121^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 464, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -4 & 1 & 4 & -2 \\ 1 & 0 & 3 & 1 & 1 & 12 & 3 \\ 8 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -2 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[191] $1^{1231}1^{23}31^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 464, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

[192] $1^{113}3^1221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -1 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[193] $1^{111}2^13^121^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 496, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[194] $1^{142}3^121^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 500, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 12 & 2 \\ 1 & 0 & 1 & -4 & 0 & 4 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 0 & 12 & 4 & 0 & 46 & 12 \\ 2 & 0 & 2 & -2 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

[195] $1^{114} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 503$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.57 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[196] $1^{114} 2^1 3^{112} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 506$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.55 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 3 & 0 & 1 & 1 & 12 & 3 \\ 6 & 12 & 0 & 4 & 12 & 46 & 12 \\ 2 & 3 & -1 & -2 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[197] $1^{122} 1^{111} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 508$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 2 & -1 \\ 3 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[198] $1^{111} 2^1 3^{123} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 523$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.62 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 \\ 4 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[199] $1^{113} 2^1 3^{123} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 531$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 3 & 0 & 3 & 1 & 1 & 9 & 3 \\ 6 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[200] $1^{113} 2^1 3^{112} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 539$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 2 & 0 & 3 & 1 & 1 & 9 & 3 \\ 5 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[201] $1^{112} 2^1 3^{112} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 540$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 2 & 0 & 3 & 1 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[202] $1^{123} 2^1 3^{112} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 547$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 9 & 2 \\ 1 & 0 & 1 & -4 & 2 & 3 & -2 \\ 3 & 0 & 6 & 2 & 10 & 18 & 6 \\ 5 & 0 & 9 & 3 & 18 & 25 & 9 \\ 2 & 0 & 2 & -2 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

[203] $1^{113} 3^{112} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 564$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.50 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 5 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[204] $1^{124} 2^{112} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 580$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.54 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 12 & 2 \\ 1 & 0 & 1 & -4 & 2 & 4 & -2 \\ 3 & 0 & 6 & 2 & 10 & 24 & 6 \\ 6 & 0 & 12 & 4 & 24 & 46 & 12 \\ 2 & 0 & 2 & -2 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

[205] $1^{123} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 591$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 3 & -1 \\ 3 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -1 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

[206] $1^{124} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 608$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.55 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 4 & -1 \\ 3 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -1 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

[207] $1^{124} 3^{114} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 612$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 12 & 2 \\ 1 & 0 & 1 & -4 & 2 & 4 & -2 \\ 3 & 0 & 6 & 2 & 10 & 24 & 6 \\ 8 & 0 & 12 & 4 & 24 & 46 & 12 \\ 2 & 0 & 2 & -2 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

[208] $1^{123} 2^{122} 3^{114} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 620$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 9 & 2 \\ 1 & 0 & 1 & -4 & 2 & 3 & -2 \\ 4 & 0 & 6 & 2 & 10 & 18 & 6 \\ 5 & 0 & 9 & 3 & 18 & 25 & 9 \\ 2 & 0 & 2 & -2 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

[209] $1^{122} 3^{114} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 640$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.60 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 6 & 2 \\ 1 & 0 & 1 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 0 & 6 & 2 & 0 & 10 & 6 \\ 2 & 0 & 2 & -2 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

[210] $1^{113} 3^{114} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 648$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.64 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 7 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[211] $1^{143}2^13^{122}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 652$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.53 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 9 & 2 \\ 1 & 0 & 1 & -4 & 4 & 3 & -2 \\ 6 & 0 & 12 & 4 & 46 & 36 & 12 \\ 5 & 0 & 9 & 3 & 36 & 25 & 9 \\ 2 & 0 & 2 & -2 & 12 & 9 & 0 \end{pmatrix}_{b_0}$$

[212] $1^{113}3^{113}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 653$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.54 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 3 & -1 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 6 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[213] $1^{123}1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 680$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[214] $1^{132}3^{114}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 683$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 9 & 2 \\ 1 & 0 & 1 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 0 & 9 & 3 & 0 & 25 & 9 \\ 2 & 0 & 2 & -2 & 0 & 9 & 0 \end{pmatrix}_{b_0}$$

[215] $1^{113}3^{112}221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 690$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.39 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 6 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -1 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[216] $1^{112}2^13^{113}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 719$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 2 & 0 & 3 & 1 & 1 & 6 & 3 \\ 5 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[217] $1^{114}2^13^{124}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 720$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -4 & 1 & 4 & -2 \\ 3 & 0 & 3 & 1 & 1 & 12 & 3 \\ 8 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -2 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[218] $1^1231^{13}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 725$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.84 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

[219] $1^{114} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 727$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 8 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -1 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[221] $1^{112} 2^{114} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 744$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 13.56 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 2 & 0 & 3 & 1 & 1 & 6 & 3 \\ 6 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[220] $1^{133} 3^{111} 2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 744$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.6$

$T_{\min} = 13.53 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 16 & 18 & 6 \\ 4 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -1 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[222] $1^{133} 2^{123} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 755$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.6$

$T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 9 & 2 \\ 1 & 0 & 1 & -4 & 3 & 3 & -2 \\ 5 & 0 & 9 & 3 & 25 & 27 & 9 \\ 6 & 0 & 9 & 3 & 27 & 25 & 9 \\ 2 & 0 & 2 & -2 & 9 & 9 & 0 \end{pmatrix}_{b_0}$$

[223] $1^{113} 3^{11} 2^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 793$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.42 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 7 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -1 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[224] $1^{114} 3^{112} 2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 794$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 6 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[225] $1^{122} 2^{123} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 800$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & -4 & 2 & 2 & -2 \\ 4 & 0 & 6 & 2 & 10 & 12 & 6 \\ 5 & 0 & 6 & 2 & 12 & 10 & 6 \\ 2 & 0 & 2 & -2 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[226] $1^{113} 3^{114} 2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 802$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 13.49 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 7 & -1 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 8 & 7 & 0 & 7 & 7 & 26 & 10 \\ 2 & 2 & -1 & -1 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[227] \ 1^{122} 2^1 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 816, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.45 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & -4 & 2 & 2 & -2 \\ 4 & 0 & 6 & 2 & 10 & 12 & 6 \\ 4 & 0 & 6 & 2 & 12 & 10 & 6 \\ 2 & 0 & 2 & -2 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[228] \ 1^{134} 2^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -103, \ \det = 831, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.56 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & 4 & -1 \\ 4 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -1 & 6 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[229] \ 1^{122} 2^1 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 835, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.50 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 3 & 0 & 3 & 1 & 1 & 6 & 3 \\ 5 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[230] \ 1^1 231^{14} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 860, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.93 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[231] \ 1^{124} 2^1 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 868, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.47 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 12 & 2 \\ 1 & 0 & 1 & -4 & 2 & 4 & -2 \\ 3 & 0 & 6 & 2 & 10 & 24 & 6 \\ 7 & 0 & 12 & 4 & 24 & 46 & 12 \\ 2 & 0 & 2 & -2 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[232] \ 1^1 231^{13} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 869, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.76 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[233] \ 1^{144} 2^1 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 880, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.56 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 12 & 2 \\ 1 & 0 & 1 & -4 & 4 & 4 & -2 \\ 6 & 0 & 12 & 4 & 46 & 48 & 12 \\ 6 & 0 & 12 & 4 & 48 & 46 & 12 \\ 2 & 0 & 2 & -2 & 12 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[234] \ 1^1 231^{13} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 929, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.78 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

[235] $1^{133}2^1 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.6$

$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 9 & 2 \\ 1 & 0 & 1 & -4 & 3 & 3 & -2 \\ 5 & 0 & 9 & 3 & 25 & 27 & 9 \\ 5 & 0 & 9 & 3 & 27 & 25 & 9 \\ 2 & 0 & 2 & -2 & 9 & 9 & 0 \end{pmatrix}_{b_0}$$

[236] $1^{134}2^1 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 12 & 2 \\ 1 & 0 & 1 & -4 & 3 & 4 & -2 \\ 5 & 0 & 9 & 3 & 25 & 36 & 9 \\ 8 & 0 & 12 & 4 & 36 & 46 & 12 \\ 2 & 0 & 2 & -2 & 9 & 12 & 0 \end{pmatrix}_{b_0}$$

[237] $1^{144}1^{111} 3 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.59 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 4 & 4 & -1 \\ 5 & 8 & 0 & 4 & 30 & 32 & 8 \\ 5 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -1 & 8 & 8 & 0 \end{pmatrix}_{b_0}$$

[238] $1^{114}1^{13} 3 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 991, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 7 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[239] $1^{113}1^{13} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 1000, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 2 & 0 & 3 & 1 & 1 & 9 & 3 \\ 6 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[240] $1^{114}1^{12} 3 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 1032, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 8 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 7 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -1 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[241] $1^{114}1^{13} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 1055, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -4 & 1 & 4 & -2 \\ 2 & 0 & 3 & 1 & 1 & 12 & 3 \\ 7 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -2 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[242] $1^{114}1^{14} 3 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -103, \det = 1094, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 4 & -1 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 8 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[243] $1^{112}2^13^121^1+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 1120, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 3 & 0 & 3 & 1 & 1 & 6 & 3 \\ 6 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[244] $1^{124}2^13^121^1+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 1136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 12 & 2 \\ 1 & 0 & 1 & -4 & 2 & 4 & -2 \\ 4 & 0 & 6 & 2 & 10 & 24 & 6 \\ 7 & 0 & 12 & 4 & 24 & 46 & 12 \\ 2 & 0 & 2 & -2 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

[245] $1^{113}2^13^121^1+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 1228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 3 & 0 & 3 & 1 & 1 & 9 & 3 \\ 7 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[246] $1^{114}3^1221^{11}+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 1243, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 8 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 8 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -1 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[247] $1^{123}2^13^121^1+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 1276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 9 & 2 \\ 1 & 0 & 1 & -4 & 2 & 3 & -2 \\ 4 & 0 & 6 & 2 & 10 & 18 & 6 \\ 6 & 0 & 9 & 3 & 18 & 25 & 9 \\ 2 & 0 & 2 & -2 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

[248] $1^{113}2^13^121^1+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 1307, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 2 & 0 & 3 & 1 & 1 & 9 & 3 \\ 7 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[249] $1^{1231}1431^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 1352, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 11 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

[250] $1^{114}3^1221^{11}+[\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 1360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 8 & -1 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 9 & 9 & 0 & 8 & 9 & 42 & 12 \\ 2 & 2 & -1 & -1 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[251] $1^{123}2^1 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 1388, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.3$

$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 9 & 2 \\ 1 & 0 & 1 & -4 & 2 & 3 & -2 \\ 4 & 0 & 6 & 2 & 10 & 18 & 6 \\ 7 & 0 & 9 & 3 & 18 & 25 & 9 \\ 2 & 0 & 2 & -2 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

[252] $1^{134}2^1 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 1395, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 12 & 2 \\ 1 & 0 & 1 & -4 & 3 & 4 & -2 \\ 5 & 0 & 9 & 3 & 25 & 36 & 9 \\ 7 & 0 & 12 & 4 & 36 & 46 & 12 \\ 2 & 0 & 2 & -2 & 9 & 12 & 0 \end{pmatrix}_{b_0}$$

[253] $1^{1231}14^1 31^{114} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 1580, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 12 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 12 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

[254] $1^{114}2^1 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 1644, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -4 & 1 & 4 & -2 \\ 2 & 0 & 3 & 1 & 1 & 12 & 3 \\ 8 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -2 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[255] $1^{124}2^1 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 1920, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 12 & 2 \\ 1 & 0 & 1 & -4 & 2 & 4 & -2 \\ 4 & 0 & 6 & 2 & 10 & 24 & 6 \\ 8 & 0 & 12 & 4 & 24 & 46 & 12 \\ 2 & 0 & 2 & -2 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

4.296 $\text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8$

$T = 10 \quad [0] 1^{111} 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 132, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & -2 \\ 2 & 2 & -1 & -2 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$T = 11 \quad [1] 1^{111} 23 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 11, T_{\max} = 23, \Delta = -104, \det = -84, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & 4 & -1 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 4 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 11, \ T_{\max} = 23, \ \Delta = -102, \ \det = -104, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.73 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 2 \ 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 11, \ T_{\max} = 23, \ \Delta = -104, \ \det = -150, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.56 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 5 & 3 \\ 4 & 10 & 1 & 4 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & 4 & -1 \\ 4 & 12 & 0 & 4 & 10 & 12 & 6 \\ 5 & 12 & 0 & 4 & 12 & 10 & 6 \\ 3 & 6 & -1 & -1 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$4.297 \quad \text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$$

$$T = 10 \quad [0] \ 1^{111} 2 \ 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 16, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 19.25 \rightarrow 20, \ T_{\max} = 22, \ T \in [20, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 4 & 2 \\ 5 & 4 & 0 & 2 & 4 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 2 & 4 & -7 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 16, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.25 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 0 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{121} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 24, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.33 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & 1 & -2 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 4 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{121} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 50, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.48 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & 1 & -2 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{113} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 58, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.48 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 0 & 6 & 16 & 6 \\ 2 & 2 & -1 & -2 & 0 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 170, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$[13] \quad 1^{1111} 3^{211} + [\text{su}_2 \cdot \text{g}_2] + \text{c}'_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -102, \quad \det = 296, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 2 & -1 \end{pmatrix} b_0$$

[14] $1^{112} 3^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}'_7 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 22, \Delta = -102, \det = 346, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 4 & 8 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 4 & 4 & -1 \end{pmatrix} b_0$$

$$[15] \quad 1^{1114} 3^{21} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -102, \quad \det = 352, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 4 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 & -1 \end{pmatrix} b_0$$

$$[16] \quad 1^{1115} 3^{121} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -102, \quad \det = 424, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.1$$

$$T_{\min} = 14.06 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 5 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 10 & 2 \\ 5 & 10 & 0 & 5 & 10 & 10 & 48 & 10 \\ 2 & 2 & -1 & -2 & 2 & 2 & 10 & -2 \end{pmatrix} b_0$$

$$[17] \quad 1^{1112} 3^{21} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -102, \quad \det = 512, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 \\ 1 & 2 & 0 & 1 & 2 & 0 & 4 \\ 4 & 4 & 0 & 2 & 4 & 4 & 6 \\ 2 & 2 & -1 & -2 & 2 & 2 & 4 \end{pmatrix} b_0$$

$$[18] \quad 1^{1111} 3 \, 21^1 + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{brown}{s} \textcolor{blue}{u}_2 + \textcolor{red}{s} \textcolor{blue}{u}_2 + \textcolor{blue}{s} \textcolor{blue}{u}_2 \quad T = 10, \, T_{\max} = 22, \, \Delta = -102, \, \det = 540, \, (n_+, n_-, n_0) = (1, 10, 0), \, c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 2 & 2 & -1 \end{pmatrix} b_0$$

$$[19] \quad 1^{1112} 3^{21} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -102, \quad \det = 592, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 2 & 4 & -1 \end{pmatrix} b_0$$

$$[20] \quad 1^{1114} 3 \, 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \, T_{\max} = 22, \, \Delta = -102, \, \det = 666, \, (n_+, n_-, n_0) = (1, 10, 0), \, c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$[28] \ 1^{1114} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 1376, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.45 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 8 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{1114} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 1696, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.38 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 6 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{1114} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 1744, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 7 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -2 & 2 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [31] \ 1^1 231^{111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -81, \ \det = -556, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.77 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & -1 & -1 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

4.298 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 11 \quad [0] \ 1^{1111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, \ T_{\max} = 14, \ \Delta = -81, \ \det = -166, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.95 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

4.299 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{113} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 8, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.50 \rightarrow 15, \ T_{\max} = 22, \ T \in [15, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 1 & 9 & 3 \\ 6 & 9 & 0 & 6 & 9 & 25 & 9 \\ 3 & 3 & -1 & -1 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{122} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 52, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.54 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & 4 & -1 \\ 4 & 6 & 0 & 4 & 10 & 12 & 6 \\ 4 & 6 & 0 & 4 & 12 & 10 & 6 \\ 3 & 3 & -1 & -1 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{112} 1^{114} 3^{12} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 1418, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.47 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 8 & -1 \\ 2 & 3 & 0 & 2 & 1 & 12 & 3 \\ 10 & 12 & 0 & 8 & 12 & 46 & 12 \\ 3 & 3 & -1 & -1 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [19] \ 1^{1231} 1^{11} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 14, \ \Delta = -83, \ \det = -329, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.84 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 5 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & 0 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$4.300 \quad \text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$$

$$T = 11 \quad [0] \ 1^{111} 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 11, \ T_{\max} = 14, \ \Delta = -84, \ \det = -105, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.97 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 0 & 2 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 3 & -1 & -1 & 0 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$4.301 \quad \text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \text{su2n3mix}$$

$$T = 10 \quad [0] \ 1^{111} 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 71, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.52 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 3 & 3 \\ 2 & 2 & -1 & -2 & 0 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1211} 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 140, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.54 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & 2 & -2 \\ 2 & 4 & 0 & 2 & 6 & 4 & 6 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 & 3 \\ 2 & 2 & -1 & -2 & 4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1121} 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 286, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.52 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 3 & 2 \\ 2 & 4 & 0 & 2 & 4 & 6 & 6 & 4 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 & 3 \\ 2 & 2 & -1 & -2 & 2 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1112} 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 315, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.64 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$[11] \ 1^{1112} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 792, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.47 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 3 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 5 & 2 \\ 4 & 5 & 0 & 3 & 5 & 5 & 11 & 5 \\ 2 & 2 & -1 & -2 & 2 & 2 & 5 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [12] \ 1^{1231} 1^{11} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -84, \ \det = -231, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.81 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & -1 & -1 & 0 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$4.302 \quad \text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$$

$$T = 10 \quad [0] \ 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 56, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.71 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 65, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.94 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 92, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.83 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 128, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.22 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 0 & -2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -2 & 2 & 2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1311} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 173, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.8$$

$$T_{\min} = 14.39 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 9 & 9 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 & 12 \\ 3 & 4 & -1 & -1 & 0 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 23, \ \Delta = -111, \ \det = 341, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{1111}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 344$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.81 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{121}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 380$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.5$

$T_{\min} = 13.83 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{1111}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 408$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.77 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{1111}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 485$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.76 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{1111}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 896$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$T = 11$ [11] $1^{111}3221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 14$, $\Delta = -82$, $\det = -44$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.82 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{1111}2321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 14$, $\Delta = -84$, $\det = -52$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.85 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 \\ 2 & 3 & -1 & -2 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{1111}2321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 14$, $\Delta = -84$, $\det = -69$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 1 & 3 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & -2 & 3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1111} 3^{112} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1111} 3^{111} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -101, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 6 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 8 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 8 & 0 & 15 & 9 \\ 3 & 4 & -1 & 0 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{131} 3^{111} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -209, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 12 & 3 \\ 3 & 4 & 1 & 3 & 0 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 14 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 12 & 21 & 0 & 14 & 0 & 75 & 17 \\ 3 & 4 & -1 & 0 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{1111} 2^1 3^{111} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -231, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & 1 & -2 \\ 1 & 0 & 3 & 1 & 1 & 3 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 3 & 0 & 2 \\ 2 & 0 & 2 & -2 & 3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{121} 3^{111} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -254, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 9 & 3 \\ 3 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 11 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 0 & 11 & 0 & 39 & 13 \\ 3 & 4 & -1 & 0 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1111} 3^{11} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -268, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1111} 3^{111} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -269, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{121} 3^{111} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -296, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.99 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{1231} 1^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -90, \det = -300, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 4 & 6 & 0 & 4 \\ 2 & 6 & 0 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{1111} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -320, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{1121} 2^1 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -327, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 6 & 3 & 3 \\ 3 & 6 & 0 & 2 & 6 & 10 & 6 & 6 \\ 1 & 3 & 0 & 1 & 3 & 6 & 0 & 2 \\ 2 & 3 & -1 & -2 & 3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{1231} 1^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -90, \det = -363, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{1111} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -381, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{1141} 2^1 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -447, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 12 & 3 & 3 \\ 5 & 12 & 0 & 4 & 12 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 3 & 12 & 0 & 2 \\ 2 & 3 & -1 & -2 & 3 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{1131} 2^1 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -453, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 9 & 3 & 3 \\ 4 & 9 & 0 & 3 & 9 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 3 & 9 & 0 & 2 \\ 2 & 3 & -1 & -2 & 3 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{1131} 2^1 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -504, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 9 & 3 & 3 \\ 5 & 9 & 0 & 3 & 9 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 3 & 9 & 0 & 2 \\ 2 & 3 & -1 & -2 & 3 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1111} 3 \quad 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -704, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1141} 2 \quad 3 \quad 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -954, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 12 & 3 & 3 \\ 6 & 12 & 0 & 4 & 12 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 3 & 12 & 0 & 2 \\ 2 & 3 & -1 & -2 & 3 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

$$4.303 \quad \text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$$

$$T = 11 \quad [0] \quad 1^{1111} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -166, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.95 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$4.304 \quad \text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 11 \quad [0] \quad 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & -1 & 0 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$4.305 \quad \text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$$

$$T = 11 \quad [0] \quad 1^{1111} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -90, \det = -228, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -1 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$4.306 \quad \text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 9 \quad [0] \quad 1^{13} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.67 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$$

[9] $1^{11}3^{13}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -369, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$
 $T_{\min} = 13.46 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 6 & -1 \\ 7 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{11}3^{14}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -418, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$
 $T_{\min} = 13.46 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 8 & -1 \\ 9 & 12 & 0 & 8 & 46 & 12 \\ 3 & 3 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{12}3^{12}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -478, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$
 $T_{\min} = 13.48 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & -1 \\ 6 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{12}3^{13}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -675, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$
 $T_{\min} = 13.47 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 6 & -1 \\ 8 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{12}3^{14}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 95, \quad \Delta = -135, \quad \det = -850, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$
 $T_{\min} = 13.47 \rightarrow 14, \quad T_{\max} = 95, \quad T \in [14, 95]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 8 & -1 \\ 10 & 12 & 0 & 8 & 46 & 12 \\ 3 & 3 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [14] $1^{12}3^{11}23^131^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 26, \quad \Delta = -114, \quad \det = 42, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.5$
 $T_{\min} = 15.10 \rightarrow 16, \quad T_{\max} = 26, \quad T \in [16, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 6 & 2 & -1 \\ 7 & 0 & 0 & 6 & 22 & 6 & 8 \\ 1 & 0 & 0 & 2 & 6 & 0 & 2 \\ 2 & 0 & -1 & -1 & 8 & 2 & -2 \end{pmatrix}_{b_0}$$

[15] $1^{12}3^{11}23^131^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 26, \quad \Delta = -114, \quad \det = 47, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$
 $T_{\min} = 14.21 \rightarrow 15, \quad T_{\max} = 26, \quad T \in [15, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 3 & 0 & 2 & 0 & 0 & 2 \\ 2 & 4 & -1 & -1 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{12}3^{11}23^131^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 26, \quad \Delta = -114, \quad \det = 51, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$
 $T_{\min} = 14.43 \rightarrow 15, \quad T_{\max} = 26, \quad T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 3 & 2 & -1 \\ 4 & 0 & 0 & 3 & 4 & 3 & 4 \\ 1 & 0 & 0 & 2 & 3 & 0 & 2 \\ 2 & 0 & -1 & -1 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

[17] $1^{12}3^{11}23^131^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 26, \quad \Delta = -114, \quad \det = 71, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$
 $T_{\min} = 15.49 \rightarrow 16, \quad T_{\max} = 26, \quad T \in [16, 26]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 5 & 0 \\ 6 & 10 & 0 & 5 & 8 & 6 \\ 3 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{123}1^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 307, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 2 & -1 \\ 3 & 6 & 0 & 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & -1 & -1 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{123}1^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & 2 & -1 \\ 4 & 12 & 0 & 6 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{1231}1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 402, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 5 & 0 \\ 5 & 10 & 0 & 5 & 8 & 6 \\ 3 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{123}1^{12}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 407, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 13.99 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & 2 & -1 \\ 7 & 12 & 0 & 6 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{123}1^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 671, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & 2 & -1 \\ 5 & 12 & 0 & 6 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{123}1^{12}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & 2 & -1 \\ 6 & 12 & 0 & 6 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

4.307 $\text{NHC}(-2)(-3) + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$

$T = 10 \quad [0] \quad 1^{123}1^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.50 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 2 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 6 & 2 & -1 \\ 6 & 9 & 0 & 6 & 25 & 9 & 9 \\ 2 & 3 & 0 & 2 & 9 & 1 & 3 \\ 3 & 3 & -1 & -1 & 9 & 3 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{122}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.54 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 1 & 9 & 3 \\ 8 & 9 & 0 & 6 & 9 & 25 & 9 \\ 3 & 3 & -1 & -1 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{112}3^{12}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 1418$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.47 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 8 & -1 \\ 2 & 3 & 0 & 2 & 1 & 12 & 3 \\ 10 & 12 & 0 & 8 & 12 & 46 & 12 \\ 3 & 3 & -1 & -1 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

$T = 11$ [19] $1^{1231}3^{11}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -83$, $\det = -329$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.84 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 5 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & 0 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

4.308 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 11$ [0] $1^{111}3^{12}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 14$, $\Delta = -84$, $\det = -105$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.97 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 \\ 3 & 3 & -1 & -1 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

4.309 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$

$T = 11$ [0] $1^{1231}3^{11}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$ $T = 11$, $T_{\max} = 14$, $\Delta = -86$, $\det = -164$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.95 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 2 & 3 \\ 4 & 8 & 1 & 5 & 0 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 0 & 0 & 2 \\ 3 & 6 & -1 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

4.310 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \mathfrak{n}3\text{mix}$

$T = 11$ [0] $1^{1231}3^{11}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -86$, $\det = -134$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.88 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 & 3 \\ 4 & 8 & 1 & 5 & 0 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 3 & 0 & 1 & 3 \\ 3 & 6 & -1 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

4.311 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$

$T = 10$ [0] $1^{111}3^{12}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 183$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.84 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 3 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1231^{11}3^11^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 11, \ T_{\max} = 15, \ \Delta = -92, \ \det = -296, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.27 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 1 & 3 \\ 4 & 8 & 1 & 5 & 10 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 5 & 0 & 0 \\ 4 & 10 & 0 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & 0 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1231^{11}3^11^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 11, \ T_{\max} = 16, \ \Delta = -93, \ \det = -452, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.16 \rightarrow 15, \ T_{\max} = 16, \ T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 2 & -1 \\ 3 & 6 & 0 & 3 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 & 3 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1231^{12}3^11^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 11, \ T_{\max} = 16, \ \Delta = -93, \ \det = -959, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.16 \rightarrow 15, \ T_{\max} = 16, \ T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & 2 & -1 \\ 5 & 12 & 0 & 6 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 6 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

4.312 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{1^{11}3^12^11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -84, \ \det = -105, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.97 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 1 \\ 3 & 3 & -1 & -1 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

4.313 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2\mathfrak{n}3\text{mix}$

$$T = 9 \quad [0] \ 1^{1^{11}3^12^11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -22, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.36 \rightarrow 15, \ T_{\max} = 95, \ T \in [15, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 2 & -2 \\ 4 & 8 & 0 & 4 & 30 & 12 & 8 \\ 2 & 3 & 0 & 2 & 12 & 3 & 3 \\ 2 & 2 & -1 & -2 & 8 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1^{11}3^12^11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -54, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.15 \rightarrow 15, \ T_{\max} = 95, \ T \in [15, 95]$$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 3 & 4 \\ 3 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 2 & 0 & -3 & 2 & -1 \\ 3 & 3 & 0 & 2 & 1 & 3 \\ 4 & 3 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1^{11}3^12^11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -63, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.48 \rightarrow 14, \ T_{\max} = 95, \ T \in [14, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 3 & 2 & 0 & 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & -1 & -2 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1^{11}3^12^11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -76, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.37 \rightarrow 15, \ T_{\max} = 95, \ T \in [15, 95]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 2 & 2 & 0 & 1 & 0 & 7 & 2 \\ 4 & 7 & 0 & 4 & 7 & 23 & 7 \\ 2 & 2 & -1 & -2 & 2 & 7 & -2 \end{pmatrix}_{b_0}$$

[4] $1^{11}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -79, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 3 & 3 \\ 2 & 2 & -1 & -2 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{131}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.59 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 2 & -2 \\ 3 & 6 & 0 & 3 & 16 & 9 & 6 \\ 2 & 3 & 0 & 2 & 9 & 3 & 3 \\ 2 & 2 & -1 & -2 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{123}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.58 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 6 & 7 & 0 & 4 & 14 & 23 & 7 \\ 2 & 2 & -1 & -2 & 4 & 7 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{142}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.82 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 3 & -2 \\ 4 & 8 & 0 & 4 & 30 & 20 & 8 \\ 3 & 5 & 0 & 3 & 20 & 11 & 5 \\ 2 & 2 & -1 & -2 & 8 & 5 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{121}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.51 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & -2 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 \\ 2 & 2 & -1 & -2 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{124}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 14.37 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 5 & -2 \\ 3 & 4 & 0 & 2 & 6 & 18 & 4 \\ 5 & 9 & 0 & 5 & 18 & 39 & 9 \\ 2 & 2 & -1 & -2 & 4 & 9 & -2 \end{pmatrix}_{b_0}$$

[10] $1^{122}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 3 & -2 \\ 2 & 4 & 0 & 2 & 6 & 10 & 4 \\ 4 & 5 & 0 & 3 & 10 & 11 & 5 \\ 2 & 2 & -1 & -2 & 4 & 5 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & -1 & -2 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{132}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.64 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 4 & 7 & 23 & 7 \\ 2 & 2 & -1 & -2 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

[40] $1^{141}3^{12}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -954, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 2 & -2 \\ 6 & 8 & 0 & 4 & 30 & 12 & 8 \\ 2 & 3 & 0 & 2 & 12 & 3 & 3 \\ 2 & 2 & -1 & -2 & 8 & 3 & -1 \end{pmatrix}_{b_0}$$

[41] $1^{113}3^{12}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -979, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 6 & 7 & 0 & 4 & 7 & 23 & 7 \\ 2 & 2 & -1 & -2 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

[42] $1^{114}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -982, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 5 & -2 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 5 & 9 & 39 & 9 \\ 2 & 2 & -1 & -2 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

[43] $1^{114}3^{12}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -1359, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 5 & -2 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 7 & 9 & 0 & 5 & 9 & 39 & 9 \\ 2 & 2 & -1 & -2 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

$T = 10$ [44] $1^{123}1^{11}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 70, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 5 & 0 & 12 & 6 \\ 2 & 4 & -1 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{123}1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 2 & 4 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{1231}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 78, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.44 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 4 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 8 & 0 & 4 & 0 & 9 & 5 \\ 2 & 4 & -1 & -1 & 0 & 5 & -1 \end{pmatrix}_{b_0}$$

[47] $1^{123}1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.23 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 2 & 0 & 0 & 2 \\ 2 & 0 & -1 & -1 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{123}1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 3 & 2 & -1 \\ 3 & 0 & 0 & 3 & 4 & 3 & 4 \\ 2 & 0 & 0 & 2 & 3 & 0 & 2 \\ 2 & 0 & -1 & -1 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

[49] $1^1 2^3 3^1 1^1 3^1 1^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{c}'_7 + \textcolor{orange}{s}u_2 + \textcolor{violet}{s}u_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 129, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 7 & 4 \\ 6 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & -2 & 5 & 0 \\ 7 & 10 & 0 & 5 & 8 & 6 \\ 4 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{\text{1}}23^{\text{1}}1^{\text{11}}31^{\text{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_7^{\text{c}} + \text{su}_2^{\text{c}} + \text{su}_2^{\text{c}} \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 230, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 4 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 8 & 0 & 4 & 0 & 9 & 5 \\ 2 & 4 & -1 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{123}1^{123}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{c}'_7 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 259, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 \\ 1 & 1 & -3 & 0 & 0 & 4 \\ 1 & 3 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 5 & 0 \\ 5 & 9 & 0 & 0 & -2 & -1 \\ 2 & 4 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{123}1^{11}31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 297, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.81 \rightarrow 14, \quad T_{\max} = 26, \quad T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 2 & -1 \\ 3 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & -1 & -1 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[57] \ 1^1 23^1 1^{12} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 459, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.81 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & 2 & -1 \\ 5 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[58] \ 1^1 23^1 1^{12} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 487, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.77 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 8 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 0 & 8 & 0 & 36 & 10 \\ 2 & 4 & -1 & -1 & 0 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[59] \ 1^1 23^1 1^{12} 31^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 552, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.81 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & 2 & -1 \\ 8 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[60] \ 1^1 23^1 1^{12} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 599, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.87 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 8 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 15 & 0 & 8 & 0 & 36 & 10 \\ 2 & 4 & -1 & -1 & 0 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[61] \ 1^1 23^1 1^{12} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 675, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.80 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 8 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 15 & 0 & 8 & 0 & 36 & 10 \\ 2 & 4 & -1 & -1 & 0 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[62] \ 1^1 23^1 1^{12} 31^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 764, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.81 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & 2 & -1 \\ 6 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[63] \ 1^1 23^1 1^{12} 31^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 795, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.81 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 & 2 \\ 2 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & 2 & -1 \\ 7 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 2 & 4 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

4.314 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{111} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 71, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.52 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 2 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 0 & 3 & 0 & 2 \\ 2 & 2 & -1 & -2 & 0 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1211} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -105, \ \det = 140, \ (n_+, n_-, n_0) =$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 & 2 \\ 3 & 5 & 0 & 3 & 5 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 2 & 5 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 5 & 2 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{112}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 632, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 & 2 \\ 5 & 5 & 0 & 3 & 5 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 2 & 5 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 5 & 2 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{1112}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 687, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 3 & 4 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 6 & 3 \\ 3 & 4 & 0 & 2 & 4 & 6 & 6 & 4 \\ 2 & 2 & -1 & -2 & 2 & 3 & 4 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 792, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & 1 & -2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 & 2 \\ 4 & 5 & 0 & 3 & 5 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 2 & 5 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & 5 & 2 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ [12] $1^{123}1^{11}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -231, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 2 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -1 & 0 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

4.315 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_2$ + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$

$T = 11$ [0] $1^{123}1^{13}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 14, \Delta = -86, \det = -134, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 & 3 \\ 4 & 8 & 1 & 5 & 0 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 3 & 0 & 1 & 3 \\ 3 & 6 & -1 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

4.316 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_2$ + su2n3mix + $\mathfrak{su}_3$

$T = 10$ [0] $1^{111}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 5 & 0 & 12 & 6 \\ 3 & 4 & -1 & -1 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

4.317 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.70 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.03 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{121} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 6 & -1 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.02 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.95 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & -1 \\ 4 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{112} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -178, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.07 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 \\ 2 & 4 & 0 & 2 & 4 & 5 \\ 2 & 2 & -1 & -2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{112} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.17 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 \\ 2 & 2 & 0 & 1 & 0 & 4 \\ 2 & 4 & 0 & 2 & 4 & 5 \\ 2 & 2 & -1 & -2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{121} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{111} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$T_{\min} = 14.11 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & -1 \end{pmatrix} b_0$$

$$T_{\min} = 13.72 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix} b_0$$

$$T_{\min} = 14.49 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 15 & -1 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & -1 & 20 & -1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 13.72 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$T_{\min} = 13.72 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 5 & 3 \end{pmatrix}$$

$$T = [13, 76] \cup [14, T] \cup [00, T] \cup [14, 00]$$

$$\min = 15.70 \rightarrow 14, \quad I_m$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 7 & 12 & 0 & 6 & 22 & 8 \end{pmatrix}$$

$$T_{\text{min}} = 13.74 \text{ K}, \quad T_{\text{max}} = 14 \text{ K}, \quad T_{\text{c}} \in [14, 00]$$

$$\min = 15.74 \rightarrow 14, \quad I_{\text{m}} \begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 8 & 12 & 0 & 6 & 22 & 8 \end{pmatrix}$$

$$1 \quad 3 \quad 21 \quad + \quad \textcolor{red}{542} \cdot \textcolor{blue}{92}$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 9 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix} b_0$$

4.0

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24.5 \rightarrow 25$$

$$23] \ 1^{131} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 18.75 \rightarrow 19$$

$$\begin{pmatrix} 5 & 4 & -1 & -1 & 12 & 0 & 7 \\ 1 & 13 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} b_0$$

$$1^{1\text{red}}3^{1\text{green}}21 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$$

$$T_{\min} = 13.74 \rightarrow 14, \quad T_{\max} = 14.26 \rightarrow 15$$

$$\begin{pmatrix} 2 & 3 & 0 & -3 & 9 & -1 \\ 12 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_I$$

$$1^{131} 3^{14} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$$

$$\begin{pmatrix} 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 3 & 0 & 3 & 3 & 1 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

$$1^{141} 3^{12} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 1 & 0 & 24 & 4 \\ 1 & 3 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix} b_0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix} b_0$$

4.0

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [29] \quad 1^{1231} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 17, T_{\max} = 58, T \in [17, 58]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 0 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{131} 2^3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 21.84 \rightarrow 22, T_{\max} = 26, T \in [22, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 3 & 2 \\ 1 & 0 & 1 & -4 & 3 & 1 & -2 \\ 4 & 0 & 9 & 3 & 25 & 9 & 9 \\ 1 & 0 & 3 & 1 & 9 & 0 & 2 \\ 2 & 0 & 2 & -2 & 9 & 2 & -8 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{111} 2^3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 3 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{122} 2^3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.7$$

$$T_{\min} = 15.21 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 1 & 1 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & -2 \\ 3 & 6 & 0 & 2 & 10 & 12 & 6 \\ 2 & 6 & 0 & 2 & 12 & 9 & 5 \\ 2 & 3 & -1 & -2 & 6 & 5 & -2 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{12} 2^3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 17.78 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -4 & 0 & 0 & -2 \\ 2 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & 0 & -1 & -4 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1231} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 3 & -1 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{142} 2^3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 16.19 \rightarrow 17, T_{\max} = 26, T \in [17, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 1 & 1 & 12 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 2 & -2 \\ 6 & 12 & 0 & 4 & 46 & 24 & 12 \\ 2 & 6 & 0 & 2 & 24 & 9 & 5 \\ 2 & 3 & -1 & -2 & 12 & 5 & -3 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{111} 2^3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 59, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -2 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{11}231^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.41 \rightarrow 16, T_{\max} = 58, T \in [16, 58]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 6 & 0 & 3 & 0 & 3 & 3 \\ 2 & 4 & -1 & -1 & 0 & 3 & -2 \end{pmatrix}_{b_0}$$

[39] $1^{11}3221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{11}12321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 1 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -2 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{11}3221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 82, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -1 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

[41] $1^{11}12321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 83, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 1 & 0 & 3 & 1 & 1 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 9 & 5 \\ 2 & 0 & 2 & -2 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

[42] $1^{11}2321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 90, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.51 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[43] $1^{11}1422321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 93, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & 1 & 1 & 12 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 2 & -2 \\ 5 & 12 & 0 & 4 & 46 & 24 & 12 \\ 2 & 6 & 0 & 2 & 24 & 9 & 5 \\ 2 & 3 & -1 & -2 & 12 & 5 & -1 \end{pmatrix}_{b_0}$$

[46] $1^{11}1213221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 6 & 0 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{11}231^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$

13.5

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & 0 & -1 \\ 4 & 12 & 0 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{111} 2^3 3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 14.54 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 1 & 1 & 9 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 2 & -2 \\ 4 & 9 & 0 & 3 & 25 & 18 & 9 \\ 2 & 6 & 0 & 2 & 18 & 9 & 5 \\ 2 & 3 & -1 & -2 & 9 & 5 & -1 \end{pmatrix}_{b_0}$$

[47] $1^{112} 2^3 3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.03 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 \\ 1 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{111} 3 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 6 & 9 & 0 & 8 & 15 & 9 \\ 3 & 4 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{112} 2^3 3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 129, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 3 & 1 & 0 & 0 & 2 \\ 2 & 0 & 2 & -2 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{111} 3 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{112} 3 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 137, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -1 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[55] $1^{111} 3 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 8 & 9 & 0 & 8 & 15 & 9 \\ 3 & 4 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[54] $1^{121} 3 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 11 & 0 \\ 8 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[53] $1^{121} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.11 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{111} 3^{113} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{111} 3^{111} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 141, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -1 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[57] $1^{111} 2^1 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 145, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.03 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 \\ 1 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[58] $1^{131} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 155, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{112} 3^{112} 2^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 156, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 1 & 0 & 3 & 1 & 1 & 6 & 3 \\ 4 & 0 & 6 & 2 & 6 & 9 & 5 \\ 2 & 0 & 2 & -2 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{121} 2^1 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 157, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -4 & 2 & 1 & -2 \\ 3 & 0 & 6 & 2 & 10 & 6 & 6 \\ 1 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -2 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

[61] $1^{111} 2^1 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 2 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -2 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[62] $1^{13}2^1 3^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 171$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.62 \rightarrow 15$, $T_{\max} = 26$, $T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 0 & 2 \\ 1 & 0 & 1 & -4 & 3 & 0 & -2 \\ 5 & 0 & 9 & 3 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & 9 & -1 & -1 \end{pmatrix}_{b_0}$$

[63] $1^{141}3^1 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 176$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.77 \rightarrow 15$, $T_{\max} = 24$, $T \in [15, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{1231}1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 58$, $\Delta = -120$, $\det = 177$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.14 \rightarrow 15$, $T_{\max} = 58$, $T \in [15, 58]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

[65] $1^{11}2^1 3^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 181$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.15 \rightarrow 15$, $T_{\max} = 26$, $T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 0 & 1 & -4 & 1 & 0 & -2 \\ 2 & 0 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

[66] $1^{111}3^1 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 187$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.84 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 7 & 9 & 0 & 8 & 15 & 9 \\ 3 & 4 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[67] $1^{111}3^1 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 192$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.90 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[68] $1^{151}3^1 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 203$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.1$

$T_{\min} = 15.11 \rightarrow 16$, $T_{\max} = 24$, $T \in [16, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 15 & 0 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & 0 & 20 & -1 \end{pmatrix}_{b_0}$$

[69] $1^{1231}1^{11}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 58$, $\Delta = -120$, $\det = 231$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.08 \rightarrow 15$, $T_{\max} = 58$, $T \in [15, 58]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 2 & 4 & -1 & -1 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

- [70] $1^{111} 3^{11} 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 240$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.85 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [71] $1^{111} 3^{112} 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 243$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.84 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [72] $1^{121} 2^1 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 280$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 14.06 \rightarrow 15$, $T_{\max} = 26$, $T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -4 & 2 & 1 & -2 \\ 6 & 0 & 6 & 2 & 10 & 6 & 6 \\ 1 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -2 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$
- [73] $1^{112} 2^1 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 293$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.96 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 2 & 1 & -2 \\ 2 & 0 & 3 & 1 & 1 & 6 & 3 \\ 3 & 0 & 6 & 2 & 6 & 9 & 5 \\ 2 & 0 & 2 & -2 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$
- [74] $1^{121} 2^1 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 294$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 13.79 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 3 & 6 & 0 & 2 & 10 & 6 & 6 \\ 1 & 3 & 0 & 1 & 6 & 0 & 2 \\ 2 & 3 & -1 & -2 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$
- [75] $1^{121} 2^1 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 297$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 13.81 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -4 & 2 & 1 & -2 \\ 3 & 0 & 6 & 2 & 10 & 6 & 6 \\ 2 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -2 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$
- [76] $1^{112} 3^1 221^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 306$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.88 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 1 & 2 & -1 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -1 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$
- [77] $1^{131} 3^1 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 323$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 13.88 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 14 & 0 \\ 11 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$
- [78] $1^{111} 2^1 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 350$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 2 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -2 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[79] \ 1^{141} 2^3 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 354, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 1 & -2 \\ 5 & 12 & 0 & 4 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 12 & 0 & 2 \\ 2 & 3 & -1 & -2 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[80] \ 1^{112} 2^3 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 355, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 0 & 1 & -4 & 1 & 0 & -2 \\ 3 & 0 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{131} 2^3 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 357, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 1 & -2 \\ 4 & 9 & 0 & 3 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 9 & 0 & 2 \\ 2 & 3 & -1 & -2 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{121} 2^3 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 364, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.04 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -4 & 2 & 1 & -2 \\ 4 & 0 & 6 & 2 & 10 & 6 & 6 \\ 1 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -2 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[83] \ 1^{121} 2^3 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 384, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 4 & 6 & 0 & 2 & 10 & 6 & 6 \\ 1 & 3 & 0 & 1 & 6 & 0 & 2 \\ 2 & 3 & -1 & -2 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{121} 2^3 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 400, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 0 & 2 \\ 1 & 0 & 1 & -4 & 2 & 0 & -2 \\ 4 & 0 & 6 & 2 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{121} 2^3 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 405, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -4 & 2 & 1 & -2 \\ 5 & 0 & 6 & 2 & 10 & 6 & 6 \\ 1 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -2 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[87] \ 1^{1231} 1^{11} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 411, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.13 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

[86] $1^{121} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 411, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{141} 2^3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 412, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.62 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 3 & 2 \\ 1 & 0 & 1 & -4 & 4 & 1 & -2 \\ 6 & 0 & 12 & 4 & 46 & 12 & 12 \\ 1 & 0 & 3 & 1 & 12 & 0 & 2 \\ 2 & 0 & 2 & -2 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

[89] $1^{111} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 459, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 3 & 0 & 3 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -2 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[90] $1^{131} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 460, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 3 & 2 \\ 1 & 0 & 1 & -4 & 3 & 1 & -2 \\ 5 & 0 & 9 & 3 & 25 & 9 & 9 \\ 1 & 0 & 3 & 1 & 9 & 0 & 2 \\ 2 & 0 & 2 & -2 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

[91] $1^{121} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 467, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 11 & 0 \\ 9 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[92] $1^{141} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 17 & 0 \\ 14 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{141} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 552, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 3 & 2 \\ 1 & 0 & 1 & -4 & 4 & 1 & -2 \\ 6 & 0 & 12 & 4 & 46 & 12 & 12 \\ 2 & 0 & 3 & 1 & 12 & 0 & 2 \\ 2 & 0 & 2 & -2 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

[94] $1^{121} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 576, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{121} 1^{14} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 588, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[96] $1^{131} 1^{11} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 588, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{121} 1^{13} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 635, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 9 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[98] $1^{131} 1^{12} 2 \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 675, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 1 & -2 \\ 5 & 9 & 0 & 3 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 9 & 0 & 2 \\ 2 & 3 & -1 & -2 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

[99] $1^{121} 1^{12} 3 \ 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 688, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 11 & 0 \\ 10 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[100] $1^{131} 1^{13} 2 \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 729, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 3 & 2 \\ 1 & 0 & 1 & -4 & 3 & 1 & -2 \\ 6 & 0 & 9 & 3 & 25 & 9 & 9 \\ 1 & 0 & 3 & 1 & 9 & 0 & 2 \\ 2 & 0 & 2 & -2 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

[101] $1^{1231} 1^{231} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 762, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & 0 & -1 \\ 5 & 12 & 0 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

[102] $1^{121} 1^{121} 2 \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 764, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -4 & 2 & 1 & -2 \\ 4 & 0 & 6 & 2 & 10 & 6 & 6 \\ 2 & 0 & 3 & 1 & 6 & 0 & 2 \\ 2 & 0 & 2 & -2 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

[103] $1^{141} 3^{221} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 771, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.02 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 13 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

[104] $1^{131} 2^3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 795, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 3 & 2 \\ 1 & 0 & 1 & -4 & 3 & 1 & -2 \\ 5 & 0 & 9 & 3 & 25 & 9 & 9 \\ 2 & 0 & 3 & 1 & 9 & 0 & 2 \\ 2 & 0 & 2 & -2 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

[105] $1^{121} 3^{221} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 803, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 11 & 0 \\ 11 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[106] $1^{121} 3^{221} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 812, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 11 & 0 \\ 12 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[107] $1^{131} 3^{221} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 812, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 14 & 0 \\ 12 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

[108] $1^{131} 2^3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 832, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.00 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 3 & 2 \\ 1 & 0 & 1 & -4 & 3 & 1 & -2 \\ 7 & 0 & 9 & 3 & 25 & 9 & 9 \\ 1 & 0 & 3 & 1 & 9 & 0 & 2 \\ 2 & 0 & 2 & -2 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

[109] $1^{141} 2^3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 900, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 1 & -2 \\ 6 & 12 & 0 & 4 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 12 & 0 & 2 \\ 2 & 3 & -1 & -2 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

[110] $1^{141} 2^3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 909, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 3 & 2 \\ 1 & 0 & 1 & -4 & 4 & 1 & -2 \\ 7 & 0 & 12 & 4 & 46 & 12 & 12 \\ 1 & 0 & 3 & 1 & 12 & 0 & 2 \\ 2 & 0 & 2 & -2 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

[111] $1^{131} 3^{221} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 915, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 11 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{131} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 1059, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 1 & 3 & 18 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 9 & 0 & -1 \\ 7 & 18 & 0 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 12 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{131} 3^{221} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 1136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 12 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[114] \ 1^{141} 3^{221} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 1163, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 17 & 0 \\ 15 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{131} 3^{221} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 1195, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 14 & 0 \\ 13 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{141} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 1240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 3 & 2 \\ 1 & 0 & 1 & -4 & 4 & 1 & -2 \\ 8 & 0 & 12 & 4 & 46 & 12 & 12 \\ 1 & 0 & 3 & 1 & 12 & 0 & 2 \\ 2 & 0 & 2 & -2 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{131} 3^{221} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 1251, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{141} 3^{221} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 1260, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{231} 14^1 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 1302, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 1 & 3 & 24 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 12 & 0 & -1 \\ 9 & 24 & 0 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[120] 1^{131} 3^{221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 1472, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 14 & 0 \\ 14 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[122] 1^{141} 3^{221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 1643, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[121] 1^{131} 3^{221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 1643, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 14 & 0 \\ 15 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[123] 1^{141} 3^{221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 1708, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 17 & 0 \\ 16 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[124] 1^{141} 3^{221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 1920, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[125] 1^{141} 3^{221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 2147, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 17 & 0 \\ 17 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[126] 1^{141} 3^{221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 2480, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 17 & 0 \\ 18 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

4.318 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] 1^{111} 3^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.71 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] 1^{111} 3^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 65, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$$

[1] $T_{\min} = 13.94 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{1111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 92$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 13.83 \rightarrow 14$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 128$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 14.22 \rightarrow 15$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 0 & 1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{113} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 173$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.8$

$$T_{\min} = 14.39 \rightarrow 15$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 3 & 4 & -1 & -1 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 341$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 13.73 \rightarrow 14$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{1111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 344$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 13.81 \rightarrow 14$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{112} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 380$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.5$

$$T_{\min} = 13.83 \rightarrow 14$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{1111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 408$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 13.77 \rightarrow 14$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{1111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 485$, $(n_+, n_-, n_0) =$

$$(1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 896, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [11] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -44, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & -1 & 0 & 0 & 1 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1111} 2321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -52, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 \\ 2 & 3 & -1 & -2 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1111} 2321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -69, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & 0 & 1 & 3 & 3 & 1 \\ 2 & 3 & -1 & -2 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3 \ 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -101, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 6 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 8 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 8 & 0 & 15 & 9 \\ 3 & 4 & -1 & 0 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{113} 3 \ 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -209, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 12 & 3 \\ 3 & 4 & 1 & 3 & 0 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 14 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 12 & 21 & 0 & 14 & 0 & 75 & 17 \\ 3 & 4 & -1 & 0 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{\overline{111}1} 2^{\overline{1}} 3^{\overline{1}} 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -84, \ \det = -231, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.81 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & 1 & -2 \\ 1 & 0 & 3 & 1 & 1 & 3 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{\overline{112}} 3^{\overline{111}} 221^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -82, \ \det = -254, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.83 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 9 & 3 \\ 3 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 11 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 0 & 11 & 0 & 39 & 13 \\ 3 & 4 & -1 & 0 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{\overline{111}} 3^{\overline{1}} 221^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -82, \ \det = -268, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.97 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{\overline{111}} 3^{\overline{1}} 221^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -82, \ \det = -269, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.86 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{\overline{112}} 3^{\overline{111}} 221^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -82, \ \det = -296, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.99 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{\overline{1231}} 3^{\overline{111}} 31^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 15, \ \Delta = -90, \ \det = -300, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.16 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 0 & 3 & -1 \\ 2 & 6 & 0 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 6 & 0 & 4 & 4 \\ 2 & 4 & -1 & -1 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{\overline{111}} 3^{\overline{111}} 221^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -82, \ \det = -320, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.90 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{\overline{1112}} 2^{\overline{11}} 3^{\overline{1}} 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -84, \ \det = -327, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.80 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 2 & -2 \\ 1 & 3 & 0 & 1 & 1 & 3 & 6 & 3 \\ 1 & 3 & 0 & 1 & 3 & 0 & 6 & 2 \\ 3 & 6 & 0 & 2 & 6 & 6 & 10 & 6 \\ 2 & 3 & -1 & -2 & 3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$4.320 \quad \text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 11 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -84, \quad \det = -105, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.97 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & -1 & 0 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$4.321 \quad \text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$$

$$T = 11 \quad [0] \quad 1^{11} 3^{11} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, \quad T_{\max} = 15, \quad \Delta = -90, \quad \det = -228, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.25 \rightarrow 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 0 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$4.322 \quad \text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 26, \quad \Delta = -114, \quad \det = 183, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.84 \rightarrow 14, \quad T_{\max} = 26, \quad T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 4 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{12} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 26, \quad \Delta = -114, \quad \det = 188, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.91 \rightarrow 14, \quad T_{\max} = 26, \quad T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & 2 & -1 \\ 6 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 26, \quad \Delta = -114, \quad \det = 287, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.84 \rightarrow 14, \quad T_{\max} = 26, \quad T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 2 & -1 \\ 3 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -1 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{12} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 26, \quad \Delta = -114, \quad \det = 495, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.83 \rightarrow 14, \quad T_{\max} = 26, \quad T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 2 & -1 \\ 5 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -1 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 26, \quad \Delta = -114, \quad \det = 528, \quad (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 2 & -1 \\ 4 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -1 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{121} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 807, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & 2 & -1 \\ 7 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{122} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & 2 & -1 \\ 8 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \ 1^{123} 1^{12} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -122, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.26 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & 0 & 2 & -1 \\ 4 & 12 & 0 & 6 & 22 & 0 & 6 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 6 & 0 & 0 & 2 \\ 2 & 4 & -1 & -1 & 8 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{123} 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -239, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.17 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 0 & 2 & -1 \\ 2 & 6 & 0 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 3 & 0 & 0 & 2 \\ 2 & 4 & -1 & -1 & 4 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{123} 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 15, \Delta = -92, \det = -296, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.27 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 1 & 3 \\ 4 & 8 & 1 & 5 & 10 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 5 & 0 & 0 \\ 4 & 10 & 0 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & 0 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{123} 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -452, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 0 & 2 & -1 \\ 3 & 6 & 0 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 3 & 0 & 0 & 2 \\ 2 & 4 & -1 & -1 & 4 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{123} 1^{12} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -959, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 12 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 6 & 0 & 2 & -1 \\ 5 & 12 & 0 & 6 & 22 & 0 & 6 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 6 & 0 & 0 & 2 \\ 2 & 4 & -1 & -1 & 8 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

4.323 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 2 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

4.324 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \text{n3mix}$

$$T = 10 \quad [0] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 5 & 0 & 12 & 6 \\ 3 & 4 & -1 & -1 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

4.325 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & 0 & -1 \\ 4 & 8 & 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 159, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -1 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 231, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.15 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [3] \quad 1^{111} 2^3 3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 0 \\ 1 & 0 & 1 & -4 & 1 & 1 & 0 \\ 3 & 0 & 3 & 1 & 1 & 3 & 0 \\ 1 & 0 & 3 & 1 & 3 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 \\ 2 & 0 & 2 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 2^3 3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -74, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$(1, 11, 0), c_{\text{ext}} = 15.8$$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 1 & 1 & -2 \\ 4 & 9 & 0 & 3 & 25 & 9 & 9 & 9 \\ 1 & 3 & 0 & 1 & 9 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 9 & 3 & 0 & 2 \\ 2 & 3 & -1 & -2 & 9 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1311} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -1179, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.8$$

$$T_{\min} = 14.36 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 1 & 1 & -2 \\ 5 & 9 & 0 & 3 & 25 & 9 & 9 & 9 \\ 1 & 3 & 0 & 1 & 9 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 9 & 3 & 0 & 2 \\ 2 & 3 & -1 & -2 & 9 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1411} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 + \text{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -1854, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.34 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 1 & 1 & -2 \\ 6 & 12 & 0 & 4 & 46 & 12 & 12 & 12 \\ 1 & 3 & 0 & 1 & 12 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 12 & 3 & 0 & 2 \\ 2 & 3 & -1 & -2 & 12 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$4.326 \quad \text{NHC}(-2)(-3) + \text{e}'_7 + \text{su}_2 + \text{su}_3 + \text{su}_3 + \text{su}_2$$

$$T = 11 \quad [0] \ 1^{1111} 3^2 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_3 + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 15, \Delta = -90, \det = -228, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 0 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$4.327 \quad \text{NHC}(-2)(-3) + \text{e}'_7 + \text{su}_2 \rightarrow (-3)$$

$$T = 8 \quad [0] \ 1^{11} 13^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 101, \Delta = -165, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.93 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 2 & 4 & -3 & 2 & -1 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 5 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 13^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 101, \Delta = -165, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 2 & 0 & -3 & -1 \\ 3 & 3 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 21^{11} 13^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 101, \Delta = -165, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 4 \\ 8 & 9 & 4 & 4 & 5 \\ 3 & 4 & -3 & 2 & -1 \\ 3 & 4 & 2 & 0 & 2 \\ 4 & 5 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \ 1^{123} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 2 & -1 \\ 1 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{11}2313^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 99, \Delta = -143, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 14.09 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 0 & 0 & -2 & 0 \\ 3 & 0 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{11}23^{11}1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 99, \Delta = -144, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & -1 \\ 1 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{11}231^{11}31^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 99, \Delta = -143, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 8 & 1 & 5 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 \\ 3 & 6 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11}2321^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 99, \Delta = -145, \det = -485, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 1 & 2 \\ 3 & 1 & 4 & 1 & 0 & 3 \\ 3 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & -2 & 0 & 0 \end{pmatrix}_{b_0}$$

4.328 $\text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \text{NHC}(-2)(-3)$

$T = 10$ [0] $1^{11}1^{11}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 27, \Delta = -115, \det = 252, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 27, T \in [14, 27]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 2 & 3 & 0 & 2 & 1 & 0 & 2 \\ 3 & 3 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$T = 11$ [1] $1^{11}23^{11}1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + [\mathbf{su}_2 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 16, \Delta = -94, \det = -355, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.18 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 2 & 6 & 0 & 3 & 3 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 0 & 1 & -3 & -1 \\ 2 & 4 & -1 & -1 & 2 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

4.329 $\text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8$

$T = 9$ [0] $1^{11}1^{11}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8$ $T = 9, T_{\max} = 102, \Delta = -164, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.49 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -3 & 2 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & -1 & 2 & -2 & -1 \end{pmatrix}_{b_0}$$

4.330 $\text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$

$T = 9$ [0] $1^{11}1^{11}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 95, \Delta = -135, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 13.67 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 6 & -1 \\ 6 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12} 3^{12} 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -18, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.67 \rightarrow 15, \ T_{\max} = 95, \ T \in [15, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{12} 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -36, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.89 \rightarrow 15, \ T_{\max} = 95, \ T \in [15, 95]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -3 & 2 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & -1 & 2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12} 3^{12} 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -70, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.43 \rightarrow 14, \ T_{\max} = 95, \ T \in [14, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & -1 \\ 4 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^{12} 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -90, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.47 \rightarrow 14, \ T_{\max} = 95, \ T \in [14, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^{12} 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -103, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.41 \rightarrow 14, \ T_{\max} = 95, \ T \in [14, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & -1 \\ 2 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^{12} 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -205, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.45 \rightarrow 14, \ T_{\max} = 95, \ T \in [14, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & -1 \\ 3 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{12} 3^{12} 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -259, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.52 \rightarrow 14, \ T_{\max} = 95, \ T \in [14, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & -1 \\ 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{12} 3^{12} 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -298, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.45 \rightarrow 14, \ T_{\max} = 95, \ T \in [14, 95]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & -1 \\ 5 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{11} 1^{13} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9$, $T_{\max} = 95$, $\Delta = -135$, $\det = -369$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 95$, $T \in [14, 95]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 6 & -1 \\ 7 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{11} 1^{14} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9$, $T_{\max} = 95$, $\Delta = -135$, $\det = -418$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 95$, $T \in [14, 95]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 8 & -1 \\ 9 & 12 & 0 & 8 & 46 & 12 \\ 3 & 3 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{12} 1^{12} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9$, $T_{\max} = 95$, $\Delta = -135$, $\det = -478$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 95$, $T \in [14, 95]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & -1 \\ 6 & 6 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{12} 1^{13} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9$, $T_{\max} = 95$, $\Delta = -135$, $\det = -675$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 13.47 \rightarrow 14$, $T_{\max} = 95$, $T \in [14, 95]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 6 & -1 \\ 8 & 9 & 0 & 6 & 25 & 9 \\ 3 & 3 & -1 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{12} 1^{14} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9$, $T_{\max} = 95$, $\Delta = -135$, $\det = -850$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.47 \rightarrow 14$, $T_{\max} = 95$, $T \in [14, 95]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 8 & -1 \\ 10 & 12 & 0 & 8 & 46 & 12 \\ 3 & 3 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [14] $1^{12} 23^1 1^2 31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 42$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 15.10 \rightarrow 16$, $T_{\max} = 26$, $T \in [16, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 2 & 6 & -1 \\ 1 & 0 & 0 & 2 & 0 & 6 & 2 \\ 7 & 0 & 0 & 6 & 6 & 22 & 8 \\ 2 & 0 & -1 & -1 & 2 & 8 & -2 \end{pmatrix}_{b_0}$$

[15] $1^{12} 23^1 1^1 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 47$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.21 \rightarrow 15$, $T_{\max} = 26$, $T \in [15, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -1 & 2 & 0 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{12} 23^1 1^1 31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 51$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.43 \rightarrow 15$, $T_{\max} = 26$, $T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 2 & 3 & -1 \\ 1 & 0 & 0 & 2 & 0 & 3 & 2 \\ 4 & 0 & 0 & 3 & 3 & 4 & 4 \\ 2 & 0 & -1 & -1 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[17] $1^{12} 23^1 1^1 31^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 71$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 15.49 \rightarrow 16$, $T_{\max} = 26$, $T \in [16, 26]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 5 & 0 \\ 6 & 10 & 0 & 5 & 8 & 6 \\ 3 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{123}1^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 307, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 3 & -1 \\ 1 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 3 & 4 & 4 \\ 2 & 4 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{123}1^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 6 & -1 \\ 1 & 3 & 0 & 2 & 0 & 6 & 2 \\ 4 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{1231}1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 25, \Delta = -113, \det = 402, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 5 & 0 \\ 5 & 10 & 0 & 5 & 8 & 6 \\ 3 & 6 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{123}1^{12}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 407, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 13.99 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 6 & -1 \\ 1 & 3 & 0 & 2 & 0 & 6 & 2 \\ 7 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{123}1^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 671, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 6 & -1 \\ 1 & 3 & 0 & 2 & 0 & 6 & 2 \\ 5 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{123}1^{12}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 6 & -1 \\ 1 & 3 & 0 & 2 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 6 & 22 & 8 \\ 2 & 4 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

4.331 $\text{NHC}(-2)(-3) + \text{e}'_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$

$T = 10$ [0] $1^{13}1^{13}1^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.50 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 2 & 3 \\ 2 & 1 & 1 & 2 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 6 & 2 & -1 \\ 6 & 9 & 0 & 6 & 25 & 9 & 9 \\ 2 & 3 & 0 & 2 & 9 & 1 & 3 \\ 3 & 3 & -1 & -1 & 9 & 3 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{122}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.54 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 1 & 9 & 3 \\ 8 & 9 & 0 & 6 & 9 & 25 & 9 \\ 3 & 3 & -1 & -1 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{112}3^{121}1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -105$, $\det = 1418$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.47 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 8 & -1 \\ 2 & 3 & 0 & 2 & 1 & 12 & 3 \\ 10 & 12 & 0 & 8 & 12 & 46 & 12 \\ 3 & 3 & -1 & -1 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

$T = 11$ [19] $1^{1231}1^{113}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -83$, $\det = -329$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.84 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 5 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & 0 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

4.332 $\text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$

$T = 11$ [0] $1^{111}3^{121}1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 11$, $T_{\max} = 14$, $\Delta = -84$, $\det = -105$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.97 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 3 & -1 & -1 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

4.333 $\text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3)$

$T = 11$ [0] $1^{1231}1^{113}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3)$ $T = 11$, $T_{\max} = 14$, $\Delta = -86$, $\det = -164$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.95 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 2 & 3 \\ 4 & 8 & 1 & 5 & 0 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 0 & 0 & 2 \\ 3 & 6 & -1 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

4.334 $\text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \mathbf{n}3 \mathbf{mix}$

$T = 11$ [0] $1^{1231}1^{113}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -86$, $\det = -134$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.88 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 & 3 \\ 4 & 8 & 1 & 5 & 0 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 3 & 0 & 1 & 3 \\ 3 & 6 & -1 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

4.335 $\text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3$

$T = 10$ [0] $1^{111}3^{121}1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 183$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.84 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 3 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1231^{11}3^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 11, \ T_{\max} = 15, \ \Delta = -92, \ \det = -296, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.27 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 1 & 3 \\ 4 & 8 & 1 & 5 & 10 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 5 & 0 & 0 \\ 4 & 10 & 0 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & 0 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1231^{11}3^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 11, \ T_{\max} = 16, \ \Delta = -93, \ \det = -452, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.16 \rightarrow 15, \ T_{\max} = 16, \ T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 6 & 0 & 3 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 2 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1231^{12}3^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 11, \ T_{\max} = 16, \ \Delta = -93, \ \det = -959, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.16 \rightarrow 15, \ T_{\max} = 16, \ T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 12 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 6 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 6 & 0 & 2 \\ 5 & 12 & 0 & 6 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 2 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

4.336 NHC(-2)(-3) + $\mathbf{e}'_7$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{su}_2$ + $\mathbf{su}_3$ + $\mathbf{su}_2$

$$T = 11 \quad [0] \ 1^{1111}3^{11}2^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -84, \ \det = -105, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.97 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 0 & 2 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 1 & 3 \\ 3 & 3 & -1 & -1 & 0 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

4.337 NHC(-2)(-3) + $\mathbf{e}'_7$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{1231^{13}1^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 30, \ \Delta = -116, \ \det = 8, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.50 \rightarrow 15, \ T_{\max} = 30, \ T \in [15, 30]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 0 & 0 & -2 & 3 & 0 \\ 2 & 0 & 0 & 3 & 1 & 3 \\ 3 & 0 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12313^{11}1^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 30, \ \Delta = -116, \ \det = 72, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.11 \rightarrow 15, \ T_{\max} = 30, \ T \in [15, 30]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 0 & 0 & -2 & 2 & 0 \\ 2 & 0 & 0 & 2 & 0 & 2 \\ 3 & 0 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1231^{13}1^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 30, \ \Delta = -116, \ \det = 167, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.96 \rightarrow 14, \ T_{\max} = 30, \ T \in [14, 30]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 0 \\ 2 & 5 & 0 & 3 & 1 & 3 \\ 3 & 6 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1231^{13}1^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 30, \ \Delta = -116, \ \det = 200, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.92 \rightarrow 14, \ T_{\max} = 30, \ T \in [14, 30]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 8 & 1 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 6 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

4.338 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^1 23^1 1^3 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -86, \det = -164, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.95 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 4 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & -1 & 0 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

4.339 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \quad 1^1 23^1 1^3 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 15, \Delta = -89, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 2 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 0 & 0 & -2 & 3 & 2 & 0 \\ 2 & 0 & 0 & 3 & 1 & 2 & 3 \\ 2 & 0 & 0 & 2 & 2 & 0 & 2 \\ 3 & 0 & -1 & 0 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 23^1 1^3 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 15, \Delta = -89, \det = -300, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.03 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 2 & 3 \\ 4 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 2 & 0 \\ 2 & 5 & 0 & 3 & 1 & 2 & 3 \\ 2 & 4 & 0 & 2 & 2 & 0 & 2 \\ 3 & 6 & -1 & 0 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

4.340 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \mathfrak{n}3\text{mix}$

$$T = 11 \quad [0] \quad 1^1 23^1 1^3 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 15, \Delta = -89, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 & 3 \\ 4 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 2 & 0 \\ 2 & 5 & 0 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 2 & 2 & 0 & 2 \\ 3 & 6 & -1 & 0 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 23^1 1^3 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 15, \Delta = -89, \det = -344, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.95 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 & 3 \\ 4 & 8 & 1 & 5 & 4 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 2 & 3 & 0 \\ 2 & 4 & 0 & 2 & 0 & 2 & 2 \\ 3 & 5 & 0 & 3 & 2 & 1 & 3 \\ 3 & 6 & -1 & 0 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

4.341 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$

$$T = 11 \quad [0] \quad 1^1 23^1 1^3 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -95, \det = -251, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 0 & 0 \\ 2 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & 0 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

4.342 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \mathfrak{n}3\text{mix}$

$$T = 10 \quad [0] \quad 1^1 23^1 1^3 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 30, \Delta = -116, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14, T_{\max} = 30, T \in [14, 30]$$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 3 & 4 \\ 6 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & -2 & 3 & 0 \\ 3 & 5 & 0 & 3 & 1 & 3 \\ 4 & 6 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{123}1^{13}1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 30, \ \Delta = -116, \ \det = 18, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.22 \rightarrow 16, \ T_{\max} = 30, \ T \in [16, 30]$$

$$\begin{pmatrix} 4 & 1 & 1 & 3 & 3 & 4 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 0 & 0 & -2 & 3 & 0 \\ 3 & 0 & 0 & 3 & 1 & 3 \\ 4 & 0 & -1 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{123}1^{13}1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 30, \ \Delta = -116, \ \det = 83, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.93 \rightarrow 14, \ T_{\max} = 30, \ T \in [14, 30]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 8 & 1 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 2 & 0 \\ 3 & 4 & 0 & 2 & 0 & 2 \\ 3 & 6 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{123}1^{13}1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 30, \ \Delta = -116, \ \det = 118, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.12 \rightarrow 15, \ T_{\max} = 30, \ T \in [15, 30]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 0 & 0 & -2 & 3 & 0 \\ 3 & 0 & 0 & 3 & 1 & 3 \\ 3 & 0 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{123}1^{13}1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 30, \ \Delta = -116, \ \det = 192, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.88 \rightarrow 14, \ T_{\max} = 30, \ T \in [14, 30]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 0 \\ 3 & 5 & 0 & 3 & 1 & 3 \\ 3 & 6 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

4.343 $\text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \text{mix} + \textcolor{red}{\mathfrak{su}}_2$

$$T = 11 \quad [0] \ 1^{123}1^{13}1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -86, \ \det = -134, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.88 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 0 & 0 \\ 3 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & -1 & 0 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

4.344 $\text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \text{mix} + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \ 1^{123}1^{13}1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 15, \ \Delta = -89, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14, \ T_{\max} = 15, \ T \in [14, 15]$$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 3 & 3 & 4 \\ 6 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & -2 & 3 & 2 & 0 \\ 3 & 5 & 0 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 2 & 2 & 0 & 2 \\ 4 & 6 & -1 & 0 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{123}1^{13}1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 15, \ \Delta = -89, \ \det = -344, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.95 \rightarrow 14, \ T_{\max} = 15, \ T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 2 & 3 \\ 4 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 2 & 0 \\ 3 & 5 & 0 & 3 & 1 & 2 & 3 \\ 2 & 4 & 0 & 2 & 2 & 0 & 2 \\ 3 & 6 & -1 & 0 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

4.345 $\text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_3$

$$T = 9 \quad [0] \ 1^{11}3^{121}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 99, \ \Delta = -144, \ \det = -92, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.96 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -3 & 2 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12}3^{11}21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}'_7} + \textcolor{orange}{\mathfrak{su}_2} \rightarrow (-3) + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, \ T_{\max} = 99, \ \Delta = -144, \ \det = -127, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.02 \rightarrow 15, \ T_{\max} = 99, \ T \in [15, 99]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & -1 \\ 4 & 6 & 0 & 4 & 9 & 5 \\ 3 & 3 & -1 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}'_7} + \textcolor{orange}{\mathfrak{su}_2} \rightarrow (-3) + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, \ T_{\max} = 99, \ \Delta = -144, \ \det = -160, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.82 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & -1 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 3 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \ 1^{12}3^{11}1^{13}1^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}'_7} + \textcolor{orange}{\mathfrak{su}_2} \rightarrow (-3) + \textcolor{red}{\mathfrak{su}_3} \quad T = 10, \ T_{\max} = 80, \ \Delta = -123, \ \det = 139, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.16 \rightarrow 15, \ T_{\max} = 80, \ T \in [15, 80]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 0 \\ 1 & 3 & 0 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{23}1^{13}1^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}'_7} + \textcolor{orange}{\mathfrak{su}_2} \rightarrow (-3) + \textcolor{red}{\mathfrak{su}_3} \quad T = 10, \ T_{\max} = 74, \ \Delta = -122, \ \det = 193, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.21 \rightarrow 15, \ T_{\max} = 74, \ T \in [15, 74]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

4.346 $\text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{e}'_7} + \textcolor{orange}{\mathfrak{su}_2} \rightarrow (-3) + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_2}$

$$T = 10 \quad [0] \ 1^{11}3^{11}21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}'_7} + \textcolor{orange}{\mathfrak{su}_2} \rightarrow (-3) + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 183, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.84 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 0 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & -1 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}'_7} + \textcolor{orange}{\mathfrak{su}_2} \rightarrow (-3) + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 188, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.91 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 4 & -1 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 4 & 6 & 0 & 4 & 6 & 10 & 6 \\ 3 & 3 & -1 & -1 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}'_7} + \textcolor{orange}{\mathfrak{su}_2} \rightarrow (-3) + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 287, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.84 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 2 & -1 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 3 & 1 & 3 \\ 3 & 3 & -1 & -1 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}'_7} + \textcolor{orange}{\mathfrak{su}_2} \rightarrow (-3) + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 495, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 0 & 6 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 0 & 6 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -1 & 2 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$

4.347 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -84, \quad \det = -105, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.97 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 2 & 0 & 2 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 3 & 0 & 1 & 3 \\ 3 & 3 & -1 & -1 & 2 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

4.348 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \quad 1^{12} 3^{11} 1^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 16, \quad \Delta = -95, \quad \det = -251, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.23 \rightarrow 15, \quad T_{\max} = 16, \quad T \in [15, 16]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 0 & 0 \\ 2 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & 0 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

4.349 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 80, \quad \Delta = -123, \quad \det = 48, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.25 \rightarrow 15, \quad T_{\max} = 80, \quad T \in [15, 80]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 0 & 5 \\ 2 & 4 & -3 & 2 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 & -1 \\ 3 & 5 & -1 & 2 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

4.350 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_2 \mathfrak{n} 3 \text{mix}$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -165, \quad \det = 80, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.70 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 4 \\ 3 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 2 & 0 & -3 & -1 \\ 4 & 3 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{12} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -165, \quad \det = 89, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.16 \rightarrow 15, \quad T_{\max} = 101, \quad T \in [15, 101]$$

$$\begin{pmatrix} 9 & 4 & 1 & 4 & 5 \\ 4 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 2 & 0 & -3 & -1 \\ 5 & 3 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -165, \quad \det = 99, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.47 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 2 & 3 & 0 & 2 & 3 & 3 \\ 2 & 2 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{12} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -165, \quad \det = 145, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.54 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & -2 \\ 7 & 9 & 0 & 5 & 39 & 9 \\ 2 & 2 & -1 & -2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [14] \quad 1^1 23^1 1^{11} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 5 & -1 \\ 7 & 9 & 0 & 5 & 12 & 6 \\ 2 & 4 & -1 & -1 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^1 231^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 4 \\ 6 & 8 & 1 & 5 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 5 & 0 & -2 & 0 \\ 4 & 6 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^1 231^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.15 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 4 & -1 \\ 4 & 0 & 0 & 4 & 9 & 5 \\ 2 & 0 & -1 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^1 231^{11} 3^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 4 & -1 \\ 6 & 8 & 0 & 4 & 9 & 5 \\ 2 & 4 & -1 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^1 23^1 131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.11 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 2 & -1 \\ 2 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^1 2313^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 4 & 1 & 1 & 3 & 4 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 0 & 0 & -2 & 0 \\ 4 & 0 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^1 23^1 1^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & -1 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^1 23^1 1^1 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -138, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.09 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 5 & -1 \\ 5 & 0 & 0 & 5 & 12 & 6 \\ 2 & 0 & -1 & -1 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^1 23^1 131^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 2 & -1 \\ 3 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^1 23^1 1^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 5 & -1 \\ 3 & 9 & 0 & 5 & 12 & 6 \\ 2 & 4 & -1 & -1 & 6 & -1 \end{pmatrix}_{b_0}$$

- [35] $1^{123}1^{131}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -144$, $\det = -447$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.35 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 2 & -1 \\ 5 & 0 & 0 & 2 & 0 & 2 \\ 2 & 0 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$
- [36] $1^{123}1^{1231}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -144$, $\det = -551$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.08 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 8 & -1 \\ 9 & 0 & 0 & 8 & 36 & 10 \\ 2 & 0 & -1 & -1 & 10 & -1 \end{pmatrix}_{b_0}$$
- [37] $1^{123}1^{131}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -144$, $\det = -598$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 14.12 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 5 & -1 \\ 7 & 0 & 0 & 5 & 12 & 6 \\ 2 & 0 & -1 & -1 & 6 & -1 \end{pmatrix}_{b_0}$$
- [38] $1^{123}1^{1231}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -144$, $\det = -618$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.86 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 8 & -1 \\ 6 & 15 & 0 & 8 & 36 & 10 \\ 2 & 4 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$
- [39] $1^{123}1^{1231}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -144$, $\det = -732$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.77 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 8 & -1 \\ 9 & 15 & 0 & 8 & 36 & 10 \\ 2 & 4 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$
- [40] $1^{123}1^{1231}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -144$, $\det = -788$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.81 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 8 & -1 \\ 7 & 15 & 0 & 8 & 36 & 10 \\ 2 & 4 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$
- [41] $1^{123}1^{1231}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -144$, $\det = -826$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.78 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 8 & -1 \\ 8 & 15 & 0 & 8 & 36 & 10 \\ 2 & 4 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

4.351 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_{2n3\text{mix}} + \mathfrak{su}_2$

- $T = 9$ $[0] 1^{114}3^{121}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 95$, $\Delta = -135$, $\det = -22$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 14.36 \rightarrow 15$, $T_{\max} = 95$, $T \in [15, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 3 & 12 & 3 \\ 4 & 8 & 0 & 4 & 12 & 30 & 8 \\ 2 & 2 & -1 & -2 & 3 & 8 & -2 \end{pmatrix}_{b_0}$$
- [1] $1^{111}3^{121}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 95$, $\Delta = -135$, $\det = -54$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.15 \rightarrow 15$, $T_{\max} = 95$, $T \in [15, 95]$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 3 & 4 \\ 3 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 2 & 0 & -3 & 2 & -1 \\ 3 & 3 & 0 & 2 & 1 & 3 \\ 4 & 3 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$
- [2] $1^{111}3^{121}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 95$, $\Delta = -135$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.48 \rightarrow 14$, $T_{\max} = 95$, $T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 3 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & -2 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{131}3^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.37 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 1 & -2 \\ 4 & 7 & 0 & 4 & 23 & 7 & 7 \\ 2 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & -2 & 7 & 2 & -2 \end{pmatrix}_{b_0}$$

[4] $1^{11}3^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -79, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 0 & -2 \\ 2 & 3 & 0 & 2 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{113}3^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.59 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 3 & -2 \\ 2 & 3 & 0 & 2 & 3 & 9 & 3 \\ 3 & 6 & 0 & 3 & 9 & 16 & 6 \\ 2 & 2 & -1 & -2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{132}3^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.58 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 2 & -2 \\ 6 & 7 & 0 & 4 & 23 & 14 & 7 \\ 2 & 4 & 0 & 2 & 14 & 6 & 4 \\ 2 & 2 & -1 & -2 & 7 & 4 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{124}3^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.82 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 4 & -2 \\ 3 & 5 & 0 & 3 & 11 & 20 & 5 \\ 4 & 8 & 0 & 4 & 20 & 30 & 8 \\ 2 & 2 & -1 & -2 & 5 & 8 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{112}3^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.51 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & -2 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 2 & -1 & -2 & 3 & 4 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{142}3^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 14.37 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & 2 & -2 \\ 5 & 9 & 0 & 5 & 39 & 18 & 9 \\ 3 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & -2 & 9 & 4 & -2 \end{pmatrix}_{b_0}$$

[10] $1^{122}3^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 2 & -2 \\ 4 & 5 & 0 & 3 & 11 & 10 & 5 \\ 2 & 4 & 0 & 2 & 10 & 6 & 4 \\ 2 & 2 & -1 & -2 & 5 & 4 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 4 & -2 \\ 3 & 5 & 0 & 3 & 11 & 20 & 5 \\ 5 & 8 & 0 & 4 & 20 & 30 & 8 \\ 2 & 2 & -1 & -2 & 5 & 8 & -1 \end{pmatrix}_{b_0}$$

[39] $1^{1^{11}} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -738, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 1 & -2 \\ 5 & 7 & 0 & 4 & 23 & 7 & 7 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & -2 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[40] $1^{1^{12}} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -954, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 3 & 0 & 2 & 3 & 12 & 3 \\ 6 & 8 & 0 & 4 & 12 & 30 & 8 \\ 2 & 2 & -1 & -2 & 3 & 8 & -1 \end{pmatrix}_{b_0}$$

[41] $1^{1^{12}} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -979, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 4 & 1 & -2 \\ 6 & 7 & 0 & 4 & 23 & 7 & 7 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & -2 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[42] $1^{1^{11}} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -982, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & 1 & -2 \\ 6 & 9 & 0 & 5 & 39 & 9 & 9 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & -2 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

[43] $1^{1^{12}} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -1359, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 5 & 1 & -2 \\ 7 & 9 & 0 & 5 & 39 & 9 & 9 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & -2 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

$T = 10$ [44] $1^{12} 3^1 1^{11} 31^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 70, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 5 & 0 & -1 \\ 6 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -1 & 6 & 0 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{12} 3^1 1^{11} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 0 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -1 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{12} 3^1 1^{11} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 78, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.44 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 4 & 0 & -1 \\ 3 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -1 & 5 & 0 & -1 \end{pmatrix}_{b_0}$$

[47] $1^{12} 3^1 131^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$

12.0

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 2 & 0 & -1 \\ 2 & 0 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & -1 & -1 & 2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[48] \ 1^1 23^1 1^1 31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 2 & 3 & -1 \\ 2 & 0 & 0 & 2 & 0 & 3 & 2 \\ 3 & 0 & 0 & 3 & 3 & 4 & 4 \\ 2 & 0 & -1 & -1 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[49] \ 1^1 23^1 1^1 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 129, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 3 & -1 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 2 & 6 & 0 & 3 & 3 & 4 & 4 \\ 2 & 4 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[50] \ 1^1 231^{11} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 143, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 7 & 4 & 4 \\ 6 & 8 & 1 & 5 & 10 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & -2 & 5 & 0 & 0 \\ 7 & 10 & 0 & 5 & 8 & 6 & 6 \\ 4 & 6 & -1 & 0 & 6 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[51] \ 1^1 231^{11} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 230, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 5 & 0 & -1 \\ 4 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -1 & 6 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[52] \ 1^1 231^{11} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 242, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 4 & 0 & -1 \\ 4 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -1 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[53] \ 1^1 23^1 1^1 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 259, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 8 & 0 & -1 \\ 6 & 15 & 0 & 8 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -1 & 10 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[54] \ 1^1 23^1 1^1 31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 282, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 5 & 0 & -1 \\ 5 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -1 & 6 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[55] \ 1^1 23^1 1^1 31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 297, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 3 & -1 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 4 & 6 & 0 & 3 & 3 & 4 & 4 \\ 2 & 4 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & 2 & -2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 6 & 3 \\ 1 & 2 & 0 & 1 & 3 & 0 & 4 & 2 \\ 4 & 4 & 0 & 2 & 6 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 3 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{1^1} 1^{1211} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 536, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 1 & 1 & -2 \\ 3 & 5 & 0 & 3 & 11 & 5 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 5 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{1^2} 1^{1211} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 632, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 1 & 1 & -2 \\ 5 & 5 & 0 & 3 & 11 & 5 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 5 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{111} 1^{1211} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 687, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & 2 & -2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 6 & 3 \\ 1 & 2 & 0 & 1 & 3 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 6 & 4 & 6 & 4 \\ 2 & 2 & -1 & -2 & 3 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{1211} 1^{1211} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 22, \Delta = -105, \det = 792, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & 1 & 1 & -2 \\ 4 & 5 & 0 & 3 & 11 & 5 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 & 0 & 2 \\ 2 & 2 & -1 & -2 & 5 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ [12] $1^{1231} 1^{11} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -231, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 0 & 3 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -1 & 2 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

4.353 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \text{su2n3mix} + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 11$ [0] $1^{1231} 1^{131} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 11, T_{\max} = 14, \Delta = -86, \det = -134, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 0 & 0 \\ 3 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & -1 & 0 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

4.354 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \text{su2n3mix} + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 10$ [0] $1^{111} 1^{131} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 26, \Delta = -114, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 1 & 3 \\ 3 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & 0 & -1 \\ 5 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 6 & 0 & -1 \end{pmatrix}_{b_0}$$

4.355 NHC(-2)(-3) + $\mathfrak{e}'_7$ + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \quad 1^{\overline{1}} 23^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 30, \Delta = -116, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14, T_{\max} = 30, T \in [14, 30]$$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 3 & 4 \\ 6 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & -2 & 3 & 0 \\ 3 & 5 & 0 & 3 & 1 & 3 \\ 4 & 6 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1}} 23^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 30, \Delta = -116, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.22 \rightarrow 16, T_{\max} = 30, T \in [16, 30]$$

$$\begin{pmatrix} 4 & 1 & 1 & 3 & 3 & 4 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 0 & 0 & -2 & 3 & 0 \\ 3 & 0 & 0 & 3 & 1 & 3 \\ 4 & 0 & -1 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1}} 23^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 30, \Delta = -116, \det = 83, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.93 \rightarrow 14, T_{\max} = 30, T \in [14, 30]$$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 3 & 4 \\ 6 & 8 & 1 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & -2 & 2 & 0 \\ 3 & 4 & 0 & 2 & 0 & 2 \\ 4 & 6 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1}} 23^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 30, \Delta = -116, \det = 118, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 30, T \in [15, 30]$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 0 & 0 & -2 & 3 & 0 \\ 3 & 0 & 0 & 3 & 1 & 3 \\ 3 & 0 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1}} 23^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 30, \Delta = -116, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 30, T \in [14, 30]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 0 \\ 3 & 5 & 0 & 3 & 1 & 3 \\ 3 & 6 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

4.356 NHC(-2)(-3) + $\mathfrak{e}'_7$ + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{\overline{1}} 23^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -86, \det = -134, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 1 & 3 \\ 4 & 8 & 1 & 5 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 0 & 0 \\ 3 & 5 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & -1 & 0 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

4.357 NHC(-2)(-3) + $\mathfrak{e}'_7$ + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \quad 1^{\overline{1}} 23^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 15, \Delta = -89, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 3 & 3 & 4 \\ 6 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & -2 & 3 & 2 & 0 \\ 3 & 5 & 0 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 2 & 2 & 0 & 2 \\ 4 & 6 & -1 & 0 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1}} 23^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 15, \Delta = -89, \det = -344, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.95 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 2 & 3 \\ 4 & 8 & 1 & 5 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 2 & 0 \\ 3 & 5 & 0 & 3 & 1 & 2 & 3 \\ 2 & 4 & 0 & 2 & 2 & 0 & 2 \\ 3 & 6 & -1 & 0 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

4.358 NHC(-2)(-3) + $\mathfrak{e}'_7$ + su2n3mix + su2n3mix

$$T = 10 \quad [0] \quad 1^{11} 23^1 1^3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \mathfrak{e}'_7 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 30, \Delta = -116, \det = 83, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.93 \rightarrow 14, T_{\max} = 30, T \in [14, 30]$$

$$\begin{pmatrix} 4 & 6 & 1 & 3 & 4 & 4 \\ 6 & 8 & 1 & 5 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & -2 & 3 & 0 \\ 4 & 5 & 0 & 3 & 1 & 3 \\ 4 & 6 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

4.359 NHC(-2)(-3) + $\mathfrak{e}'_7$ + su2n3mix + su₃

$$T = 9 \quad [0] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \mathfrak{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 3 & 4 \\ 3 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 2 & 0 & -3 & 2 & -1 \\ 3 & 3 & 0 & 2 & 0 & 2 \\ 4 & 3 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \mathfrak{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -286, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & -1 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \mathfrak{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 8 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \mathfrak{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & -1 \\ 6 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \mathfrak{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & -1 \\ 7 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{12} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \mathfrak{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -806, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 9 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \mathfrak{e}'_7 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -1056, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 10 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [7] \quad 1^{1231^{11}31^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 82, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 5 & 0 & -1 \\ 3 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{1231^{11}31^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 2 & 0 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{1231^{11}31^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 74, \Delta = -122, \det = 292, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.14 \rightarrow 15, T_{\max} = 74, T \in [15, 74]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 4 & 0 & -1 \\ 3 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 5 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{1231^{11}31^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 556, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 5 & 0 & -1 \\ 4 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1231^{11}31^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 877, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.13 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 1 & 3 & 15 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 8 & 0 & -1 \\ 6 & 15 & 0 & 8 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & 10 & -1 & -1 \end{pmatrix}_{b_0}$$

4.360 $\text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_{2\text{n}3\text{mix}} + \mathbf{su}_3 + \mathbf{su}_2$

$$T = 10 \quad [0] \quad 1^{111}3^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 1 & 3 \\ 3 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & 0 & -1 \\ 5 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 6 & 0 & -1 \end{pmatrix}_{b_0}$$

4.361 $\text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_3$

$$T = 8 \quad [0] \quad 1^{11}3^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.93 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.70 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & -3 & -1 \\ 3 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{brown}{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 14.02 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 2 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{13} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{brown}{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -2 & 5 & -2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{brown}{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 4 \\ 5 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & -3 & -1 \\ 4 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{12} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{brown}{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 215, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.72 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 3 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{13} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{brown}{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 293, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.72 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 3 & -2 \\ 4 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -2 & 5 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \ 1^{13} 3 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{brown}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 18, T_{\max} = 99, T \in [18, 99]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 6 & 6 & 0 & 6 \\ 2 & 6 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 6 & 0 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{12} 2 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{brown}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 14.54 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & -2 \\ 2 & 6 & 0 & 2 & 9 & 5 \\ 2 & 3 & -1 & -2 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 2 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{brown}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & -2 \\ 1 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{12} 2 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{brown}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 15.89 \rightarrow 16, T_{\max} = 99, T \in [16, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -4 & 0 & -2 \\ 2 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 2 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{brown}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & -2 \\ 1 & 3 & 0 & 1 & 0 & 2 \\ 2 & 3 & -1 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T_{\min} = 14.15 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -4 & 3 & -2 \\ 4 & 0 & 9 & 3 & 24 & 8 \\ 2 & 0 & 2 & -2 & 8 & -1 \end{pmatrix} b_0$$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 2 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -1 & 5 & -1 \end{pmatrix} b_0$$

$$T_{\min} = 14.14 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 1 & 0 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 3 & -1 & -1 & -1 \end{pmatrix} b_0$$

$$T_{\min} = 14.14 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 3 & -1 \\ 3 & 0 & 0 & 3 & 3 & 3 \\ 2 & 0 & -1 & -1 & 3 & -1 \end{pmatrix} b_0$$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[23] $1^{11}2^3 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & -2 \\ 2 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{12}2^3 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.93 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -4 & 2 & -2 \\ 3 & 0 & 6 & 2 & 9 & 5 \\ 2 & 0 & 2 & -2 & 5 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{11}3 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.89 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 4 \\ 5 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & -2 & 0 \\ 4 & 4 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{11}2^3 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.90 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & -2 \\ 3 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{12}3 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -169, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.88 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & -1 \\ 3 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

[28] $1^1 231^{11} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_3 \quad T = 9, T_{\max} = 100, \Delta = -150, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.27 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & -1 \\ 2 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

[29] $1^1 231^{11} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_3 \quad T = 9, T_{\max} = 100, \Delta = -150, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.04 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{13}3 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.89 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{12}2^3 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.75 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -4 & 2 & -2 \\ 4 & 0 & 6 & 2 & 9 & 5 \\ 2 & 0 & 2 & -2 & 5 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{13}2^3 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \text{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.77 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -4 & 3 & -2 \\ 5 & 0 & 9 & 3 & 24 & 8 \\ 2 & 0 & 2 & -2 & 8 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{11}231^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 100$, $\Delta = -150$, $\det = -282$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.04 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

4.362 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$

$T = 10$ $[0] 1^{111}3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 10$, $T_{\max} = 69$, $\Delta = -121$, $\det = 82$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 14.12 \rightarrow 15$, $T_{\max} = 69$, $T \in [15, 69]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -3 & 0 & 3 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & -1 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ $[1] 1^{111}3 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11$, $T_{\max} = 15$, $\Delta = -92$, $\det = -61$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 14.28 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 4 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 0 & 4 & 0 & 9 & 5 \\ 3 & 4 & -1 & 0 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

4.363 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{so}_8$

$T = 9$ $[0] 1^{11}3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 102$, $\Delta = -170$, $\det = -245$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 14.73 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & -2 & -1 & 0 & -2 \end{pmatrix}_{b_0}$$

$T = 10$ $[1] 1^{11}2 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 10$, $T_{\max} = 100$, $\Delta = -143$, $\det = 79$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 14.94 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 1 & 4 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 9 & 4 & 0 & 5 \\ 2 & 0 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 0 & 5 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}23 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 10$, $T_{\max} = 100$, $\Delta = -143$, $\det = 151$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 15.18 \rightarrow 16$, $T_{\max} = 100$, $T \in [16, 100]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 10 & 1 & 4 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 4 & -1 \\ 4 & 12 & 0 & 4 & 9 & 5 \\ 3 & 6 & -1 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{11}3 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 10$, $T_{\max} = 100$, $\Delta = -141$, $\det = 196$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 14.88 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

4.364 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -94, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.70 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -103, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.03 \rightarrow 15, \quad T_{\max} = 99, \quad T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -2 & -1 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{112} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -126, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.81 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 6 & -1 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -127, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.02 \rightarrow 15, \quad T_{\max} = 99, \quad T \in [15, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -148, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.95 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.76 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{121} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -178, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.07 \rightarrow 15, \quad T_{\max} = 99, \quad T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 2 & 4 & 0 & 2 & 5 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -2 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{121} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -181, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.17 \rightarrow 15, \quad T_{\max} = 99, \quad T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 2 & 4 & 0 & 2 & 5 & 4 & 3 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -2 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{112} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.91 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{111} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{11} 1^3 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{c}}'_7 + \textcolor{brown}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -760, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{112}1^{13}321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{c}}'_7 + \textcolor{brown}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -814, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 9 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix} b_0$$

[22] $1^{114} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{c}'_7 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -1006$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 13 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{113} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}'_7 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -1174$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 11 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{113} 3^{13} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -1456$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 12 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{113} 3^{14} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 99, \Delta = -141, \det = -1606, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{114} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 99, \Delta = -141, \det = -1624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix} b_0$$

[27] $1^{114} 3^{13} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -2110$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix} b_0$$

[28] $1^{114} 3^{21} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 99, \Delta = -141, \det = -2464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [29] \quad 1^{1231} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 17, T_{\max} = 58, T \in [17, 58]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -1 & -1 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{113} 2^3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 25, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 21.84 \rightarrow 22, T_{\max} = 26, T \in [22, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 1 & 0 & 3 & 1 & 0 & 9 & 2 \\ 4 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 2 & 9 & -8 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1112} 2^3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 3 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -2 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{122} 2^3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.7$$

$$T_{\min} = 15.21 \rightarrow 16, T_{\max} = 26, T \in [16, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 1 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 2 & -2 \\ 2 & 6 & 0 & 2 & 9 & 12 & 5 \\ 3 & 6 & 0 & 2 & 12 & 10 & 6 \\ 2 & 3 & -1 & -2 & 5 & 6 & -2 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{112} 2^3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 17.78 \rightarrow 18, T_{\max} = 26, T \in [18, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -2 & -1 & 0 & -4 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1231} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1242} 2^3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 16.19 \rightarrow 17, T_{\max} = 26, T \in [17, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 1 & 6 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 6 & 0 & 2 & 9 & 24 & 5 \\ 6 & 12 & 0 & 4 & 24 & 46 & 12 \\ 2 & 3 & -1 & -2 & 5 & 12 & -3 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{111} 2^3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 59, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -2 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{11}231^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15.41 \rightarrow 16, T_{\max} = 58, T \in [16, 58]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -1 & 3 & 0 & -2 \end{pmatrix}_{b_0}$$

[38] $1^{111}2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 2 & 0 & 3 & 1 & 0 & 3 & 2 \\ 1 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{11}3^1 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[40] $1^{11}3^1 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 82, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

[41] $1^{121}2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 83, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -4 & 2 & 1 & -2 \\ 3 & 0 & 6 & 2 & 9 & 6 & 5 \\ 1 & 0 & 3 & 1 & 6 & 1 & 3 \\ 2 & 0 & 2 & -2 & 5 & 3 & -1 \end{pmatrix}_{b_0}$$

[42] $1^{12}3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 90, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 14.51 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -2 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[43] $1^{124}2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 93, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 1 & 6 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 4 & -2 \\ 2 & 6 & 0 & 2 & 9 & 24 & 5 \\ 5 & 12 & 0 & 4 & 24 & 46 & 12 \\ 2 & 3 & -1 & -2 & 5 & 12 & -1 \end{pmatrix}_{b_0}$$

[46] $1^{123}2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.54 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 1 & 6 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 3 & -2 \\ 2 & 6 & 0 & 2 & 9 & 18 & 5 \\ 4 & 9 & 0 & 3 & 18 & 25 & 9 \\ 2 & 3 & -1 & -2 & 5 & 9 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{1231}1^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$

13.5

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -1 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$

[44] $1^{112} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 6 & 0 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{112} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.03 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{111} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 6 & 9 & 0 & 8 & 15 & 9 \\ 3 & 4 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{112} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 129, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 0 & 1 & -4 & 1 & 0 & -2 \\ 2 & 0 & 3 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & -2 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{111} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{121} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 137, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 2 & 4 & 0 & 2 & 5 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -1 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[54] $1^{112} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 11 & 0 \\ 8 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[53] $1^{112} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.11 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{111} 3^{13} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{111} 3^{13} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 8 & 9 & 0 & 8 & 15 & 9 \\ 3 & 4 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{121} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 141, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 2 & 4 & 0 & 2 & 5 & 4 & 3 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -1 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[57] $1^{111} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 145, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.03 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[58] $1^{113} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 155, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{121} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 156, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -4 & 2 & 1 & -2 \\ 4 & 0 & 6 & 2 & 9 & 6 & 5 \\ 1 & 0 & 3 & 1 & 6 & 1 & 3 \\ 2 & 0 & 2 & -2 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{112} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 157, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 1 & 0 & 3 & 1 & 0 & 6 & 2 \\ 3 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[61] $1^{111} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 1 & 3 & 1 & 3 \\ 2 & 3 & -1 & -2 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[62] $1^{13}2^1 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 171$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.62 \rightarrow 15$, $T_{\max} = 26$, $T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 9 & 2 \\ 1 & 0 & 1 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 0 & 9 & 3 & 0 & 25 & 9 \\ 2 & 0 & 2 & -2 & -1 & 9 & -1 \end{pmatrix}_{b_0}$$

[63] $1^{114}3^1 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 176$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.77 \rightarrow 15$, $T_{\max} = 24$, $T \in [15, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

[64] $1^1 231^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 58$, $\Delta = -120$, $\det = 177$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.14 \rightarrow 15$, $T_{\max} = 58$, $T \in [15, 58]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[65] $1^{111}2^1 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 181$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.15 \rightarrow 15$, $T_{\max} = 26$, $T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -2 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

[66] $1^{111}3^1 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 187$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.84 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 7 & 9 & 0 & 8 & 15 & 9 \\ 3 & 4 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[67] $1^{111}3^1 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 192$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.90 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[68] $1^{115}3^1 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 203$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.1$

$T_{\min} = 15.11 \rightarrow 16$, $T_{\max} = 24$, $T \in [16, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 15 & 0 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & 0 & 20 & -1 \end{pmatrix}_{b_0}$$

[69] $1^1 231^{11} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 58$, $\Delta = -120$, $\det = 231$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.08 \rightarrow 15$, $T_{\max} = 58$, $T \in [15, 58]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & -1 & -1 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

[70] $1^{111} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{111} 3^{112} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 243, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{112} 3^{114} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 280, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.06 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 1 & 0 & 3 & 1 & 0 & 6 & 2 \\ 6 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[73] $1^{121} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 293, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & -4 & 2 & 1 & -2 \\ 3 & 0 & 6 & 2 & 9 & 6 & 5 \\ 2 & 0 & 3 & 1 & 6 & 1 & 3 \\ 2 & 0 & 2 & -2 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{112} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 294, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 3 & 0 & 1 & 0 & 6 & 2 \\ 3 & 6 & 0 & 2 & 6 & 10 & 6 \\ 2 & 3 & -1 & -2 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{112} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 297, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 2 & 0 & 3 & 1 & 0 & 6 & 2 \\ 3 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{121} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 306, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 2 & 1 & -1 \\ 3 & 4 & 0 & 2 & 5 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{113} 3^{11} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 323, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 14 & 0 \\ 11 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{111} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 350, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 2 & 0 & 3 & 1 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[79] \ 1^{114} 2^3 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 354, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 3 & 0 & 1 & 0 & 12 & 2 \\ 5 & 12 & 0 & 4 & 12 & 46 & 12 \\ 2 & 3 & -1 & -2 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[80] \ 1^{112} 2^3 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 355, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -4 & 0 & 1 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 0 & 3 & 1 & 0 & 1 & 3 \\ 2 & 0 & 2 & -2 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{113} 2^3 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 357, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 3 & 0 & 1 & 0 & 9 & 2 \\ 4 & 9 & 0 & 3 & 9 & 25 & 9 \\ 2 & 3 & -1 & -2 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{112} 2^3 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 364, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.04 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 1 & 0 & 3 & 1 & 0 & 6 & 2 \\ 4 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[83] \ 1^{112} 2^3 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 384, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 2 & -2 \\ 1 & 3 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 2 & 6 & 10 & 6 \\ 2 & 3 & -1 & -2 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{12} 2^3 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 400, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 6 & 2 \\ 1 & 0 & 1 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 0 & 6 & 2 & 0 & 10 & 6 \\ 2 & 0 & 2 & -2 & -1 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{112} 2^3 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 405, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 1 & 0 & 3 & 1 & 0 & 6 & 2 \\ 5 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[86] \ 1^{11} 2^3 3 221^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 411, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{1231}1^{1131}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 411, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 14.13 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 2 & 4 & -1 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[88] 1^{1142}1^{112}3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 412, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.62 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -4 & 1 & 4 & -2 \\ 1 & 0 & 3 & 1 & 0 & 12 & 2 \\ 6 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[89] 1^{1112}1^{112}3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 459, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 2 & 0 & 3 & 1 & 0 & 3 & 2 \\ 3 & 0 & 3 & 1 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[90] 1^{1132}1^{12}3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 460, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 1 & 0 & 3 & 1 & 0 & 9 & 2 \\ 5 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[91] 1^{112}1^{11}3 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 467, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 11 & 0 \\ 9 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[92] 1^{114}1^{11}3 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 17 & 0 \\ 14 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[93] 1^{1142}1^{112}3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 552, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -4 & 1 & 4 & -2 \\ 2 & 0 & 3 & 1 & 0 & 12 & 2 \\ 6 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -2 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[94] 1^{112}1^{12}3 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 576, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[96] $1^{113} 3^{11} 221^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7'} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 588, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{112} 3^{14} 221^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7'} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 588, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{112} 3^{13} 221^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7'} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 635, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 9 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[98] $1^{112} 2^3 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7'} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 675, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & -2 \\ 1 & 3 & 0 & 1 & 0 & 9 & 2 \\ 5 & 9 & 0 & 3 & 9 & 25 & 9 \\ 2 & 3 & -1 & -2 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

[99] $1^{112} 3^{12} 221^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7'} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 688, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 11 & 0 \\ 10 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[100] $1^{113} 2^3 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7'} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 729, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 1 & 0 & 3 & 1 & 0 & 9 & 2 \\ 6 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

[101] $1^{1231} 1^{231} 1^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7'} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 762, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 0 & 6 & 0 & 22 & 8 \\ 2 & 4 & -1 & -1 & -1 & 8 & -1 \end{pmatrix}_{b_0}$$

[102] $1^{112} 2^3 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7'} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 764, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 2 & -2 \\ 2 & 0 & 3 & 1 & 0 & 6 & 2 \\ 4 & 0 & 6 & 2 & 6 & 10 & 6 \\ 2 & 0 & 2 & -2 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[103] $1^{114} 3^{11} 221^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7'} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 771, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.02 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 13 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

[104] $1^{113}2^13^{112}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 795$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.81 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 2 & 0 & 3 & 1 & 0 & 9 & 2 \\ 5 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

[105] $1^{112}3^{113}221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 803$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.81 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 11 & 0 \\ 11 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[106] $1^{112}3^{114}221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 812$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.81 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 11 & 0 \\ 12 & 15 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[107] $1^{113}3^{111}221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 812$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.84 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 14 & 0 \\ 12 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

[108] $1^{113}2^13^{114}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 832$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 14.00 \rightarrow 15$, $T_{\max} = 26$, $T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -4 & 1 & 3 & -2 \\ 1 & 0 & 3 & 1 & 0 & 9 & 2 \\ 7 & 0 & 9 & 3 & 9 & 25 & 9 \\ 2 & 0 & 2 & -2 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

[109] $1^{114}2^13^{112}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 900$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.80 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & -2 \\ 1 & 3 & 0 & 1 & 0 & 12 & 2 \\ 6 & 12 & 0 & 4 & 12 & 46 & 12 \\ 2 & 3 & -1 & -2 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[110] $1^{114}2^13^{113}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 26$, $\Delta = -114$, $\det = 909$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 14.16 \rightarrow 15$, $T_{\max} = 26$, $T \in [15, 26]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -4 & 1 & 4 & -2 \\ 1 & 0 & 3 & 1 & 0 & 12 & 2 \\ 7 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[111] $1^{113}3^{112}221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 915$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.91 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 11 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{1231^{13}31^{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 58, \ \Delta = -120, \ \det = 1059, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.12 \rightarrow 15, \ T_{\max} = 58, \ T \in [15, 58]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 18 & 0 & 9 & 0 & 52 & 12 \\ 2 & 4 & -1 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{113} 3 \ 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 1136, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.88 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 12 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[114] \ 1^{114} 3 \ 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 1163, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.86 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 17 & 0 \\ 15 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{113} 3 \ 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 1195, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.83 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 14 & 0 \\ 13 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{114} 2 \ 3 \ 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 1240, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.05 \rightarrow 15, \ T_{\max} = 26, \ T \in [15, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 1 & -4 & 1 & 4 & -2 \\ 1 & 0 & 3 & 1 & 0 & 12 & 2 \\ 8 & 0 & 12 & 4 & 12 & 46 & 12 \\ 2 & 0 & 2 & -2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{113} 3 \ 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 1251, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.87 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{114} 3 \ 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 1260, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.94 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{231^{14}31^{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 58, \ \Delta = -120, \ \det = 1302, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.12 \rightarrow 15, \ T_{\max} = 58, \ T \in [15, 58]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 1 & 3 & 0 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 12 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 9 & 24 & 0 & 12 & 0 & 94 & 16 \\ 2 & 4 & -1 & -1 & -1 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{113} 3 \ 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 1472, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.82 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 14 & 0 \\ 14 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

[121] $1^{114} 3^{13} 221^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7'} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2}$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 1643, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[122] $1^{113} 3^{14} 221^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7'} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2}$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 1643, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 14 & 0 \\ 15 & 21 & 0 & 14 & 75 & 17 \\ 3 & 4 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

[123] $1^{114} 3^{12} 221^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7'} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2}$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 1708, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 17 & 0 \\ 16 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

[124] $1^{114} 3^{14} 221^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7'} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2}$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 1920, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[125] $1^{114} 3^{13} 221^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7'} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2}$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 2147, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 17 & 0 \\ 17 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

[126] $1^{114} 3^{14} 221^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7'} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2}$ $T = 10, T_{\max} = 24, \Delta = -112, \det = 2480, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 17 & 0 \\ 18 & 27 & 0 & 17 & 123 & 21 \\ 3 & 4 & -1 & 0 & 21 & 1 \end{pmatrix}_{b_0}$$

4.365 $\text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{e}_7'} + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2}$

$T = 10$ $[0] 1^{111} 3^{11} 21^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7'} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2}$ $T = 10, T_{\max} = 23, \Delta = -111, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.71 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3^{11} 21^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{e}_7'} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2}$ $T = 10, T_{\max} = 23, \Delta = -111, \det = 65, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{1111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 23, \Delta = -111, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 23, \Delta = -111, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & -1 & -2 & -1 & 2 & -2 \end{pmatrix}_{b_0}$$

[4] $1^{113} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 23, \Delta = -111, \det = 173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 3 & 4 & -1 & -1 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 23, \Delta = -111, \det = 341, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{1111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 23, \Delta = -111, \det = 344, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{112} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 23, \Delta = -111, \det = 380, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{1111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 23, \Delta = -111, \det = 408, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{1111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 23, \Delta = -111, \det = 485, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{1111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 23, \quad \Delta = -111, \quad \det = 896, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.73 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [11] \quad 1^{111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -82, \quad \det = -44, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.82 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{1111} 2321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -84, \quad \det = -52, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.85 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 \\ 2 & 3 & -1 & -2 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1111} 2321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -84, \quad \det = -69, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.78 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 3 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 \\ 2 & 3 & -1 & -2 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -82, \quad \det = -72, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.94 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -82, \quad \det = -101, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.85 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 6 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 8 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 8 & 0 & 15 & 9 \\ 3 & 4 & -1 & 0 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -82, \quad \det = -209, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.97 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 12 & 3 \\ 3 & 4 & 1 & 3 & 0 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 14 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 12 & 21 & 0 & 14 & 0 & 75 & 17 \\ 3 & 4 & -1 & 0 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{1111} 2321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -84, \quad \det = -231, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & 1 & -2 \\ 2 & 0 & 3 & 1 & 0 & 3 & 3 & 2 \\ 1 & 0 & 3 & 1 & 3 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 & 3 & 1 & 3 \\ 2 & 0 & 2 & -2 & 2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{112} 3^{111} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -254, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 9 & 3 \\ 3 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 11 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 0 & 11 & 0 & 39 & 13 \\ 3 & 4 & -1 & 0 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{111} 3^{111} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -268, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111} 3^{111} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -269, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{112} 3^{111} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -296, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.99 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{1231} 3^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 15, \Delta = -90, \det = -300, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 2 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & -1 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111} 3^{111} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -320, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{1112} 3^{111} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -327, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 2 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 6 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 6 & 3 \\ 3 & 6 & 0 & 2 & 6 & 6 & 10 & 6 \\ 2 & 3 & -1 & -2 & 2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{1231} 3^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 15, \Delta = -90, \det = -363, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & -1 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{1111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -381, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{1114} 2 \ 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -447, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 12 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 12 & 3 \\ 5 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -2 & 2 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{1113} 2 \ 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -453, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 3 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 9 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 9 & 3 \\ 4 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -2 & 2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{1113} 2 \ 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -504, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 3 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 9 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 9 & 3 \\ 5 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -2 & 2 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{1111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -704, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{1114} 2 \ 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -954, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 12 & 2 \\ 1 & 3 & 0 & 1 & 3 & 1 & 12 & 3 \\ 6 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -2 & 2 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

4.366 NHC(-2)(-3) + $\mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{1111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -166, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.95 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$4.367 \quad \text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 11 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & -1 & 0 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$4.368 \quad \text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$$

$$T = 11 \quad [0] \quad 1^{11} 3^{11} 3 \cdot 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -90, \det = -228, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$4.369 \quad \text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 183, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 4 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 188, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & 2 & -1 \\ 6 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 287, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 2 & -1 \\ 3 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -1 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 495, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 2 & -1 \\ 5 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -1 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 528, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 2 & -1 \\ 4 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -1 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{112} 3^{121} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 807, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & 2 & -1 \\ 7 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112} 3^{121} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & 2 & -1 \\ 8 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \ 1^{123} 1^{123} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -122, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.26 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 6 & 2 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 12 & 0 & 6 & 0 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 0 & 6 & 0 & 2 \\ 2 & 4 & -1 & -1 & -1 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{123} 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -239, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.17 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 6 & 2 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & -1 & -1 & -1 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{123} 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 15, \Delta = -92, \det = -296, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.27 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 4 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & 0 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{123} 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -452, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 6 & 2 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & -1 & -1 & -1 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{123} 1^{123} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -959, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 6 & 2 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 5 & 12 & 0 & 6 & 0 & 22 & 6 & 8 \\ 1 & 3 & 0 & 2 & 0 & 6 & 0 & 2 \\ 2 & 4 & -1 & -1 & -1 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

4.370 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 2 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

4.371 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \mathfrak{n} 3 \text{mix}$

$$T = 10 \quad [0] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 5 & 0 & 12 & 6 \\ 3 & 4 & -1 & -1 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

4.372 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -3 & 0 & 3 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 159, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -1 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 231, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.15 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [3] \quad 1^{111} 2^1 3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 0 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & 0 & -2 \\ 1 & 0 & 3 & 1 & 0 & 3 & 0 & 2 \\ 3 & 0 & 3 & 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & 2 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 2^1 3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -74, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$(1, 11, 0), c_{\text{ext}} = 15.8$$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & 1 & -2 \\ 1 & 3 & 0 & 1 & 0 & 9 & 3 & 2 \\ 4 & 9 & 0 & 3 & 9 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 3 & 9 & 0 & 2 \\ 2 & 3 & -1 & -2 & 2 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{113} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -1179, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.8$$

$$T_{\min} = 14.36 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 3 & 1 & -2 \\ 1 & 3 & 0 & 1 & 0 & 9 & 3 & 2 \\ 5 & 9 & 0 & 3 & 9 & 25 & 9 & 9 \\ 1 & 3 & 0 & 1 & 3 & 9 & 0 & 2 \\ 2 & 3 & -1 & -2 & 2 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{114} 2^3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -1854, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.34 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 4 & 1 & -2 \\ 1 & 3 & 0 & 1 & 0 & 12 & 3 & 2 \\ 6 & 12 & 0 & 4 & 12 & 46 & 12 & 12 \\ 1 & 3 & 0 & 1 & 3 & 12 & 0 & 2 \\ 2 & 3 & -1 & -2 & 2 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

$$4.373 \quad \text{NHC}(-2)(-3) + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_3 + \text{su}_2$$

$$T = 11 \quad [0] \ 1^{111} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 15, \Delta = -90, \det = -228, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$4.374 \quad \text{NHC}(-2)(-3) + \text{e}'_7 + \text{su}_3 + \text{su}_2 \rightarrow (-3)$$

$$T = 9 \quad [0] \ 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.02 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & -3 & 4 & -1 \\ 4 & 6 & 0 & 4 & 9 & 5 \\ 3 & 3 & -1 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & -1 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \ 1^1 23^1 1^1 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{e}'_7 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 \\ 1 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & -1 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{12}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 74, \Delta = -122, \det = 193, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 74, T \in [15, 74]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 8 & 1 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

4.375 NHC(-2)(-3) + $\mathbf{e}'_7$ + $\mathbf{su}_3$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{su}_2$

$$T = 10 \quad [0] \ 1^{11}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 183, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 0 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 188, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 287, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 3 & -1 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 495, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 3 & -1 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 5 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 528, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 3 & -1 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 4 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 807, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 7 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112}3^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 8 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \ 1^{123}1^{123}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -122, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.26 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 2 & 6 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 6 & 2 \\ 4 & 12 & 0 & 6 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -1 & -1 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{123}1^{11}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -239, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.17 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 2 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 2 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -1 & -1 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{123}1^{11}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 15, \Delta = -92, \det = -296, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.27 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 4 & 3 \\ 4 & 8 & 1 & 5 & 0 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 10 & 0 & 5 & 0 & 8 & 6 \\ 3 & 6 & -1 & 0 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{123}1^{11}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -452, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 2 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & -1 & -1 & -1 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{123}1^{12}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -959, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 2 & 6 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 2 & 0 & 0 & 6 & 2 \\ 5 & 12 & 0 & 6 & 0 & 6 & 22 & 8 \\ 2 & 4 & -1 & -1 & -1 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$\mathbf{4.376} \quad \text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$$

$$T = 11 \quad [0] \ 1^{111}3^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 0 & 3 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 2 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$\mathbf{4.377} \quad \text{NHC}(-2)(-3) + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3)$$

$$T = 11 \quad [0] \ 1^{123}1^{13}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{e}'_7 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 16, \Delta = -95, \det = -251, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 2 & 3 \\ 4 & 8 & 1 & 5 & 0 & 5 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 0 & 1 & 3 \\ 3 & 6 & -1 & 0 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$4.378 \quad \text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 4 & 0 & 5 \\ 2 & 4 & -3 & 0 & 2 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$4.379 \quad \text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \text{n3mix}$$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & -1 \\ 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -286, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & -1 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 8 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & -1 \\ 6 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & -1 \\ 7 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -806, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 9 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -1056, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 10 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [7] \quad 1^{12}3^{11}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 82, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 9 & 0 & 5 & 0 & 12 & 6 \\ 2 & 4 & -1 & -1 & -1 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{12}3^{11}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 2 & 4 & -1 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{12}31^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 74, \Delta = -122, \det = 292, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.14 \rightarrow 15, T_{\max} = 74, T \in [15, 74]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 4 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 0 & 4 & 0 & 9 & 5 \\ 2 & 4 & -1 & -1 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{12}3^{11}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 556, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 9 & 0 & 5 & 0 & 12 & 6 \\ 2 & 4 & -1 & -1 & -1 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12}3^{11}1^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 877, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.13 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 8 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 15 & 0 & 8 & 0 & 36 & 10 \\ 2 & 4 & -1 & -1 & -1 & 10 & -1 \end{pmatrix}_{b_0}$$

4.380 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11}1^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 1 & 3 \\ 3 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & 0 & -1 \\ 5 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 6 & 0 & -1 \end{pmatrix}_{b_0}$$

4.381 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11}1^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 100, \Delta = -150, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.08 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 & 5 \\ 4 & 9 & 4 & 0 & 0 & 5 & 1 \\ 2 & 4 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & -1 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{12}1^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 100, \Delta = -150, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 5 & 3 \\ 2 & 2 & -1 & -2 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 100, \Delta = -150, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.11 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 & -1 \\ 3 & 4 & -1 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 100, \Delta = -150, \det = -246, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 & -1 \\ 3 & 6 & 0 & 3 & 3 & 3 & 3 \\ 3 & 4 & -1 & -1 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 100, \Delta = -150, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 & -1 \\ 4 & 6 & 0 & 3 & 3 & 3 & 3 \\ 3 & 4 & -1 & -1 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [5] \ 1^{12} 2 \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 27.67 \rightarrow 28, T_{\max} = 80, T \in [28, 80]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 2 & 1 & -2 \\ 2 & 6 & 0 & 2 & 9 & 6 & 5 \\ 1 & 3 & 0 & 1 & 6 & 0 & 2 \\ 2 & 3 & -1 & -2 & 5 & 2 & -14 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 2 \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 25, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 0 & 1 & -4 & 1 & 0 & -2 \\ 1 & 0 & 3 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & -2 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 2 \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 18, T_{\max} = 80, T \in [18, 80]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 1 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -2 & 2 & 2 & -4 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{12} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 43, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 5 & 3 \\ 2 & 2 & -1 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 2 \ 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 105, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.34 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{12} 3 \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}'_7 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 91, \Delta = -129, \det = 135, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 91, T \in [15, 91]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 3 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 3 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10, T_{\max} = 69, \Delta = -121, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.24 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{111} 2^3 3^{12} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10, T_{\max} = 80, \Delta = -123, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.10 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 3 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{111} 2^3 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10, T_{\max} = 80, \Delta = -123, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.35 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 1 & -2 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & -2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10, T_{\max} = 69, \Delta = -121, \det = 195, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.46 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[15] $1^1 231^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10, T_{\max} = 91, \Delta = -129, \det = 279, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.37 \rightarrow 15, T_{\max} = 91, T \in [15, 91]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 0 & 3 & 3 \\ 2 & 4 & -1 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{111} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10, T_{\max} = 69, \Delta = -121, \det = 304, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

4.382 NHC(-2)(-3) + $\mathfrak{e}'_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$T = 10$ [0] $1^{111} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 58, \Delta = -120, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 0 & 4 & 4 \\ 3 & 5 & -1 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 58, \Delta = -120, \det = 159, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 58$, $\Delta = -120$, $\det = 231$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 14.15 \rightarrow 15$, $T_{\max} = 58$, $T \in [15, 58]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ [3] $1^{1112} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 16$, $\Delta = -93$, $\det = -11$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 14.45 \rightarrow 15$, $T_{\max} = 16$, $T \in [15, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 0 & 1 & -2 \\ 1 & 0 & 3 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 0 & 3 & 1 & 3 & 0 & 1 & 3 \\ 2 & 0 & 2 & -2 & 2 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{111} 2 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 16$, $\Delta = -93$, $\det = -74$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 14.16 \rightarrow 15$, $T_{\max} = 16$, $T \in [15, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 0 & 1 & -4 & 1 & 0 & 1 & -2 \\ 1 & 0 & 3 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 0 & 3 & 1 & 3 & 0 & 1 & 3 \\ 2 & 0 & 2 & -2 & 2 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{1111} 2 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 16$, $\Delta = -93$, $\det = -111$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 14.62 \rightarrow 15$, $T_{\max} = 16$, $T \in [15, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 2 & 3 & -1 & -2 & 2 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{112} 2 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 16$, $\Delta = -93$, $\det = -146$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.5$

$T_{\min} = 14.16 \rightarrow 15$, $T_{\max} = 16$, $T \in [15, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 0 & 6 & 2 \\ 1 & 0 & 1 & -4 & 1 & 0 & 2 & -2 \\ 1 & 0 & 3 & 1 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 0 & 6 & 2 & 6 & 0 & 10 & 6 \\ 2 & 0 & 2 & -2 & 2 & -1 & 6 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{111} 3 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 15$, $\Delta = -91$, $\det = -182$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 14.26 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111} 2 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 16$, $\Delta = -93$, $\det = -234$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 14.46 \rightarrow 15$, $T_{\max} = 16$, $T \in [15, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 3 & 2 \\ 2 & 3 & 0 & 1 & 3 & 3 & 1 & 3 \\ 2 & 3 & -1 & -2 & 2 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1112} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 16, \ \Delta = -93, \ \det = -306, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.71 \rightarrow 15, \ T_{\max} = 16, \ T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 2 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 6 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 6 & 2 \\ 4 & 6 & 0 & 2 & 6 & 6 & 10 & 6 \\ 2 & 3 & -1 & -2 & 2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1112} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 16, \ \Delta = -93, \ \det = -621, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.34 \rightarrow 15, \ T_{\max} = 16, \ T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 2 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 6 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 6 & 2 \\ 3 & 6 & 0 & 2 & 6 & 6 & 10 & 6 \\ 2 & 3 & -1 & -2 & 2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1114} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 16, \ \Delta = -93, \ \det = -801, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.51 \rightarrow 15, \ T_{\max} = 16, \ T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 12 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 12 & 2 \\ 5 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -2 & 2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1113} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 16, \ \Delta = -93, \ \det = -810, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 14.39 \rightarrow 15, \ T_{\max} = 16, \ T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 3 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 9 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 9 & 2 \\ 4 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -2 & 2 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1113} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 16, \ \Delta = -93, \ \det = -1179, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 14.36 \rightarrow 15, \ T_{\max} = 16, \ T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 3 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 9 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 9 & 2 \\ 5 & 9 & 0 & 3 & 9 & 9 & 25 & 9 \\ 2 & 3 & -1 & -2 & 2 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1114} 2^3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 16, \ \Delta = -93, \ \det = -1854, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.34 \rightarrow 15, \ T_{\max} = 16, \ T \in [15, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & 4 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 12 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 12 & 2 \\ 6 & 12 & 0 & 4 & 12 & 12 & 46 & 12 \\ 2 & 3 & -1 & -2 & 2 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

4.383 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{1111} 3^2 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 15, \ \Delta = -90, \ \det = -228, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.25 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

4.384 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \quad 1^{11} 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 0 & 4 & 5 \\ 2 & 4 & -3 & 0 & 0 & 2 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 3 & 5 & -1 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

4.385 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{11} 1^{11} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 91, \Delta = -129, \det = 405, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 91, T \in [15, 91]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \quad 1^{11} 1^{11} 2321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -216, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.50 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -4 & 1 & 1 & -2 \\ 1 & 3 & 0 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 1 & 3 & 0 & 1 & 3 & 0 & 2 \\ 2 & 3 & -1 & -2 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 1^{11} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 23, \Delta = -100, \det = -321, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.55 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & 0 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

4.386 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \quad 1^1 231^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 101, \Delta = -172, \det = 350, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

4.387 $\text{NHC}(-2)(-3) + \mathfrak{e}_8$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 \quad T = 8, T_{\max} = 191, \Delta = -306, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 17.96 \rightarrow 18, T_{\max} = 191, T \in [18, 191]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -8 & -6 \\ 2 & 2 & -1 & -6 & -4 \end{pmatrix}_{b_0}$$

4.388 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 172, \Delta = -256, \det = 248, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 18.56 \rightarrow 19, T_{\max} = 172, T \in [19, 172]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 5 \\ 7 & 8 & 1 & 5 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 5 & 0 & -6 & -4 \\ 5 & 6 & -1 & -4 & -3 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3^{221} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_8 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 172, \Delta = -256, \det = 272, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 18.74 \rightarrow 19, T_{\max} = 172, T \in [19, 172]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 4 & -5 \\ 4 & 8 & 0 & 4 & 9 & 5 \\ 3 & 4 & -1 & -5 & 5 & -4 \end{pmatrix}_{b_0}$$

4.389 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$

$T = 11 \quad [0] 1^{111} 3^{221} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_8 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 106, \Delta = -226, \det = -271, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 18.56 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 7 & 5 \\ 7 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -6 & 5 & -4 \\ 7 & 10 & 0 & 5 & 8 & 6 \\ 5 & 6 & -1 & -4 & 6 & -3 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3^{221} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_8 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 106, \Delta = -226, \det = -836, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 18.56 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 8 & 5 \\ 7 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -6 & 5 & -4 \\ 8 & 10 & 0 & 5 & 8 & 6 \\ 5 & 6 & -1 & -4 & 6 & -3 \end{pmatrix}_{b_0}$$

4.390 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$

$T = 11 \quad [0] 1^{111} 3^{221} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_8 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 109, \Delta = -235, \det = -166, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 18.89 \rightarrow 19, T_{\max} = 109, T \in [19, 109]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 1 & 3 \\ 3 & 4 & 1 & 3 & 8 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 4 & 0 & -5 \\ 4 & 8 & 0 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -5 & 5 & -1 & -4 \end{pmatrix}_{b_0}$$

4.391 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{so}_8$

$T = 9 \quad [0] 1^{11} 3^{221} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_8 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 192, \Delta = -305, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 19.24 \rightarrow 20, T_{\max} = 192, T \in [20, 192]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & -6 \\ 1 & 2 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & -6 & 0 & -6 \end{pmatrix}_{b_0}$$

4.392 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{so}_8 + \mathfrak{su}_2$

$T = 10 \quad [0] 1^{11} 3^{221} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_8 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 190, \Delta = -275, \det = 251, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 19.30 \rightarrow 20, T_{\max} = 190, T \in [20, 190]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 0 & -6 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -6 & 0 & 0 & -6 \end{pmatrix}_{b_0}$$

4.393 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_2$

$T = 9 \quad [0] 1^{112} 3^{221} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 189, \Delta = -276, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & -6 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -6 & 0 & -5 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & -6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -6 & 0 & -5 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -6 & 2 & -4 \end{pmatrix} b_0$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & -6 \\ 2 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & -6 & 4 & -5 \end{pmatrix} b_0$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & -6 \\ 3 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -6 & 6 & -5 \end{pmatrix} b_0$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 4 & -6 \\ 4 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -6 & 8 & -5 \end{pmatrix} b_0$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & -6 \\ 2 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -6 & 2 & -5 \end{pmatrix} b_0$$

$$1^{15} 3^{22} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]_{\min} = 18.40 \rightarrow 19, T_{\min}$$

$$\min = 18.40 \rightarrow 19, \quad I_{\text{m}} \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 5 & -6 \\ 5 & 10 & 0 & 5 & 48 & 10 \\ 2 & 2 & 1 & 6 & 10 & 5 \end{pmatrix},$$

$$\begin{pmatrix} 5 & 10 & 0 & 5 & 48 & 10 \\ 2 & 2 & -1 & -6 & 10 & -5 \end{pmatrix} b_c$$

$$t_{\min} = 18.20 \rightarrow 19, T_{\min} = \begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & -6 \\ 3 & 2 & 0 & 1 & 2 & 0 \\ 2 & 2 & -1 & -6 & 0 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 & 0 & -8 & 1 & -6 \\ 3 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -6 & 2 & -5 \end{pmatrix}_{b_0}$$

$$T_{\min} = 18.45 \rightarrow 19, \quad T_{\max} = 19.5 \rightarrow 20$$

$$\begin{pmatrix} 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 6 & -6 \\ 6 & 12 & 0 & 6 & 70 & 12 \\ 2 & 2 & -1 & -6 & 12 & -5 \end{pmatrix} b_0$$

$$1^{+-} 3^{--} 221^{+} + [\mathfrak{su}_2 \cdot \mathfrak{g}]$$

$$T_{\min} = 18.15 \rightarrow 19, T_{\max} = 189, T \in [19, 189]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 4 & -6 \\ 6 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -6 & 8 & -5 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{13} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 189, \Delta = -276, \det = -1163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 18.17 \rightarrow 19, T_{\max} = 189, T \in [19, 189]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & -6 \\ 6 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -6 & 6 & -5 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{13} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 189, \Delta = -276, \det = -1458, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 18.21 \rightarrow 19, T_{\max} = 189, T \in [19, 189]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & -6 \\ 7 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & -6 & 6 & -5 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{14} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 189, \Delta = -276, \det = -1577, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 18.16 \rightarrow 19, T_{\max} = 189, T \in [19, 189]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 4 & -6 \\ 7 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -6 & 8 & -5 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{14} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 189, \Delta = -276, \det = -1996, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 18.18 \rightarrow 19, T_{\max} = 189, T \in [19, 189]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 4 & -6 \\ 8 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & -6 & 8 & -5 \end{pmatrix}_{b_0}$$

4.394 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$

$$T = 11 \quad [0] \ 1^{11} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 106, \Delta = -226, \det = -271, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.56 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 1 & 5 \\ 7 & 8 & 1 & 5 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -6 & 0 & -4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 6 & -1 & -4 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 106, \Delta = -226, \det = -836, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.56 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 8 & 5 \\ 7 & 8 & 1 & 5 & 10 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -6 & 5 & -4 \\ 8 & 10 & 0 & -5 & 8 & 6 \\ 5 & 6 & -1 & -4 & 6 & -3 \end{pmatrix}_{b_0}$$

4.395 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 10 \quad [0] \ 1^{11} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 190, \Delta = -275, \det = 251, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 19.30 \rightarrow 20, T_{\max} = 190, T \in [20, 190]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 1 & -6 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & -6 & 0 & 0 & -6 \end{pmatrix}_{b_0}$$

4.396 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{13} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 49, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.3$$

$$T_{\min} = 18.49 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & 2 & -6 \\ 3 & 6 & 0 & 3 & 16 & 12 & 6 \\ 3 & 4 & 0 & 2 & 12 & 6 & 4 \\ 2 & 2 & -1 & -6 & 6 & 4 & -5 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 89, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 18.09 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 1 & -6 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -6 & 0 & 2 & -5 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{121} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 130, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 18.11 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 1 & -6 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -6 & 4 & 2 & -5 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 18.26 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 0 & -6 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -6 & 0 & 0 & -5 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{124} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 142, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 18.27 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 4 & -6 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 7 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -6 & 4 & 8 & -5 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{144} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 178, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.62 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 4 & 4 & -6 \\ 4 & 8 & 0 & 4 & 30 & 32 & 8 \\ 5 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -6 & 8 & 8 & -5 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 208, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 18.04 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -6 & 2 & 2 & -5 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{13} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 227, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 18.13 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 3 & -6 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 16 & 6 \\ 2 & 2 & -1 & -6 & 0 & 6 & -5 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{112} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 244, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 18.13 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & 3 & -6 \\ 3 & 6 & 0 & 3 & 16 & 18 & 6 \\ 4 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -6 & 6 & 6 & -5 \end{pmatrix}_{b_0}$$

[17] $1^{123} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 150, \ \Delta = -246, \ \det = 380, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.3$

$$T_{\min} = 18.15 \rightarrow 19, \ T_{\max} = 150, \ T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 3 & -6 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 5 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -6 & 4 & 6 & -5 \end{pmatrix}_{b_0}$$

[18] $1^{114} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 150, \ \Delta = -246, \ \det = 388, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$

$$T_{\min} = 18.33 \rightarrow 19, \ T_{\max} = 150, \ T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 4 & -6 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 4 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -6 & 2 & 8 & -5 \end{pmatrix}_{b_0}$$

[19] $1^{133} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 150, \ \Delta = -246, \ \det = 400, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.6$

$$T_{\min} = 18.34 \rightarrow 19, \ T_{\max} = 150, \ T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & 3 & -6 \\ 3 & 6 & 0 & 3 & 16 & 18 & 6 \\ 3 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -6 & 6 & 6 & -5 \end{pmatrix}_{b_0}$$

[20] $1^{122} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 150, \ \Delta = -246, \ \det = 414, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$

$$T_{\min} = 18.13 \rightarrow 19, \ T_{\max} = 150, \ T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 2 & -6 \\ 2 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -6 & 4 & 4 & -5 \end{pmatrix}_{b_0}$$

[21] $1^{124} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 150, \ \Delta = -246, \ \det = 424, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.5$

$$T_{\min} = 18.36 \rightarrow 19, \ T_{\max} = 150, \ T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 4 & -6 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 4 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -6 & 4 & 8 & -5 \end{pmatrix}_{b_0}$$

[22] $1^{134} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 150, \ \Delta = -246, \ \det = 484, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.8$

$$T_{\min} = 18.40 \rightarrow 19, \ T_{\max} = 150, \ T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & 4 & -6 \\ 3 & 6 & 0 & 3 & 16 & 24 & 6 \\ 4 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -6 & 6 & 8 & -5 \end{pmatrix}_{b_0}$$

[23] $1^{115} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 150, \ \Delta = -246, \ \det = 496, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.1$

$$T_{\min} = 18.40 \rightarrow 19, \ T_{\max} = 150, \ T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 5 & -6 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 5 & 10 & 0 & 5 & 10 & 48 & 10 \\ 2 & 2 & -1 & -6 & 2 & 10 & -5 \end{pmatrix}_{b_0}$$

[24] $1^{112} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 150, \ \Delta = -246, \ \det = 516, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$

$$T_{\min} = 18.56 \rightarrow 19, \ T_{\max} = 150, \ T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 2 & -6 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 6 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -6 & 2 & 4 & -5 \end{pmatrix}_{b_0}$$

[25] $1^{125} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 532, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.6$

$$T_{\min} = 18.42 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 5 & -6 \\ 2 & 4 & 0 & 2 & 6 & 20 & 4 \\ 5 & 10 & 0 & 5 & 20 & 48 & 10 \\ 2 & 2 & -1 & -6 & 4 & 10 & -5 \end{pmatrix}_{b_0}$$

[26] $1^{144} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 568, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.44 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 4 & 4 & -6 \\ 4 & 8 & 0 & 4 & 30 & 32 & 8 \\ 4 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -6 & 8 & 8 & -5 \end{pmatrix}_{b_0}$$

[27] $1^{135} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 592, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.9$

$$T_{\min} = 18.45 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & 5 & -6 \\ 3 & 6 & 0 & 3 & 16 & 30 & 6 \\ 5 & 10 & 0 & 5 & 30 & 48 & 10 \\ 2 & 2 & -1 & -6 & 6 & 10 & -5 \end{pmatrix}_{b_0}$$

[28] $1^{116} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 628, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 18.46 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 6 & -6 \\ 1 & 2 & 0 & 1 & 0 & 12 & 2 \\ 6 & 12 & 0 & 6 & 12 & 70 & 12 \\ 2 & 2 & -1 & -6 & 2 & 12 & -5 \end{pmatrix}_{b_0}$$

[29] $1^{112} 3^{111} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 633, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 18.11 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 2 & -6 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -6 & 2 & 4 & -5 \end{pmatrix}_{b_0}$$

[30] $1^{111} 3^{111} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 18.13 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & -6 \\ 2 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & -6 & 2 & 2 & -5 \end{pmatrix}_{b_0}$$

[31] $1^{126} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 664, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 18.47 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 6 & -6 \\ 2 & 4 & 0 & 2 & 6 & 24 & 4 \\ 6 & 12 & 0 & 6 & 24 & 70 & 12 \\ 2 & 2 & -1 & -6 & 4 & 12 & -5 \end{pmatrix}_{b_0}$$

[32] $1^{145} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.1$

$$T_{\min} = 18.47 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 4 & 5 & -6 \\ 4 & 8 & 0 & 4 & 30 & 40 & 8 \\ 5 & 10 & 0 & 5 & 40 & 48 & 10 \\ 2 & 2 & -1 & -6 & 8 & 10 & -5 \end{pmatrix}_{b_0}$$

[33] $1^{134} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 150, \ \Delta = -246, \ \det = 689, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.8$

$$T_{\min} = 18.22 \rightarrow 19, \ T_{\max} = 150, \ T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & 4 & -6 \\ 3 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -6 & 6 & 8 & -5 \end{pmatrix}_{b_0}$$

[34] $1^{123} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 150, \ \Delta = -246, \ \det = 722, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.3$

$$T_{\min} = 18.15 \rightarrow 19, \ T_{\max} = 150, \ T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 3 & -6 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -6 & 4 & 6 & -5 \end{pmatrix}_{b_0}$$

[35] $1^{136} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 150, \ \Delta = -246, \ \det = 724, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.1$

$$T_{\min} = 18.49 \rightarrow 19, \ T_{\max} = 150, \ T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & 6 & -6 \\ 3 & 6 & 0 & 3 & 16 & 36 & 6 \\ 6 & 12 & 0 & 6 & 36 & 70 & 12 \\ 2 & 2 & -1 & -6 & 6 & 12 & -5 \end{pmatrix}_{b_0}$$

[36] $1^{112} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 150, \ \Delta = -246, \ \det = 769, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$

$$T_{\min} = 18.26 \rightarrow 19, \ T_{\max} = 150, \ T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 2 & -6 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 5 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -6 & 2 & 4 & -5 \end{pmatrix}_{b_0}$$

[37] $1^{117} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 150, \ \Delta = -246, \ \det = 784, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.3$

$$T_{\min} = 18.50 \rightarrow 19, \ T_{\max} = 150, \ T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 7 & -6 \\ 1 & 2 & 0 & 1 & 0 & 14 & 2 \\ 7 & 14 & 0 & 7 & 14 & 96 & 14 \\ 2 & 2 & -1 & -6 & 2 & 14 & -5 \end{pmatrix}_{b_0}$$

[38] $1^{112} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 150, \ \Delta = -246, \ \det = 808, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$

$$T_{\min} = 18.16 \rightarrow 19, \ T_{\max} = 150, \ T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 2 & -6 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 4 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -6 & 2 & 4 & -5 \end{pmatrix}_{b_0}$$

[39] $1^{114} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 150, \ \Delta = -246, \ \det = 819, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$

$$T_{\min} = 18.15 \rightarrow 19, \ T_{\max} = 150, \ T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 4 & -6 \\ 2 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -6 & 2 & 8 & -5 \end{pmatrix}_{b_0}$$

[40] $1^{127} 3 \ 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 150, \ \Delta = -246, \ \det = 820, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.8$

$$T_{\min} = 18.51 \rightarrow 19, \ T_{\max} = 150, \ T \in [19, 150]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 7 & -6 \\ 2 & 4 & 0 & 2 & 6 & 28 & 4 \\ 7 & 14 & 0 & 7 & 28 & 96 & 14 \\ 2 & 2 & -1 & -6 & 4 & 14 & -5 \end{pmatrix}_{b_0}$$

[41] $1^{112} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 859, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 18.13 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 2 & -6 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & -6 & 2 & 4 & -5 \end{pmatrix}_{b_0}$$

[42] $1^{137} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 880, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.1$

$T_{\min} = 18.52 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & 7 & -6 \\ 3 & 6 & 0 & 3 & 16 & 42 & 6 \\ 7 & 14 & 0 & 7 & 42 & 96 & 14 \\ 2 & 2 & -1 & -6 & 6 & 14 & -5 \end{pmatrix}_{b_0}$$

[43] $1^{113} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 912, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 18.14 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 3 & -6 \\ 2 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -6 & 2 & 6 & -5 \end{pmatrix}_{b_0}$$

[44] $1^{113} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 941, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 18.14 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 3 & -6 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -6 & 2 & 6 & -5 \end{pmatrix}_{b_0}$$

[45] $1^{124} 3^{112} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 960, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 18.15 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 4 & -6 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 6 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -6 & 4 & 8 & -5 \end{pmatrix}_{b_0}$$

[46] $1^{124} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 1054, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 18.17 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 4 & -6 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -6 & 4 & 8 & -5 \end{pmatrix}_{b_0}$$

[47] $1^{122} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 1228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 18.13 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 2 & -6 \\ 3 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & -6 & 4 & 4 & -5 \end{pmatrix}_{b_0}$$

[48] $1^{114} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 1273, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 18.16 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 4 & -6 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -6 & 2 & 8 & -5 \end{pmatrix}_{b_0}$$

[49] $1^{113} 3^{12} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 1364, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 18.15 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 3 & -6 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 5 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -6 & 2 & 6 & -5 \end{pmatrix}_{b_0}$$

[50] $1^{123} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 1451, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.3$

$T_{\min} = 18.14 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 3 & -6 \\ 3 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & -6 & 4 & 6 & -5 \end{pmatrix}_{b_0}$$

[51] $1^{124} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 1528, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 18.15 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 4 & -6 \\ 3 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & -6 & 4 & 8 & -5 \end{pmatrix}_{b_0}$$

[52] $1^{113} 3^{14} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 1568, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 18.25 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 3 & -6 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 7 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -6 & 2 & 6 & -5 \end{pmatrix}_{b_0}$$

[53] $1^{113} 3^{13} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 1573, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 18.18 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 3 & -6 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 6 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & -6 & 2 & 6 & -5 \end{pmatrix}_{b_0}$$

[54] $1^{133} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 1844, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.6$

$T_{\min} = 18.15 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & 3 & -6 \\ 4 & 6 & 0 & 3 & 16 & 18 & 6 \\ 4 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & -6 & 6 & 6 & -5 \end{pmatrix}_{b_0}$$

[55] $1^{114} 3^{12} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 1944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 18.15 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 4 & -6 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 6 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -6 & 2 & 8 & -5 \end{pmatrix}_{b_0}$$

[56] $1^{134} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{c}_8 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 150, \Delta = -246, \det = 2091, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$T_{\min} = 18.16 \rightarrow 19, T_{\max} = 150, T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & 4 & -6 \\ 4 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & -6 & 6 & 8 & -5 \end{pmatrix}_{b_0}$$

[57] $1^{114} 3^{113} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 150$, $\Delta = -246$, $\det = 2401$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 18.16 \rightarrow 19$, $T_{\max} = 150$, $T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 4 & -6 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 7 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -6 & 2 & 8 & -5 \end{pmatrix}_{b_0}$$

[58] $1^{144} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 150$, $\Delta = -246$, $\det = 2508$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 18.17 \rightarrow 19$, $T_{\max} = 150$, $T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 4 & 4 & -6 \\ 5 & 8 & 0 & 4 & 30 & 32 & 8 \\ 5 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & -6 & 8 & 8 & -5 \end{pmatrix}_{b_0}$$

[59] $1^{114} 3^{114} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 150$, $\Delta = -246$, $\det = 2644$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 18.19 \rightarrow 19$, $T_{\max} = 150$, $T \in [19, 150]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 4 & -6 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 8 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & -6 & 2 & 8 & -5 \end{pmatrix}_{b_0}$$

4.397 NHC(-2)(-3) + $\mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$T = 11$ [0] $1^{112} 3^{113} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 106$, $\Delta = -216$, $\det = -32$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.5$

$T_{\min} = 23.12 \rightarrow 24$, $T_{\max} = 106$, $T \in [24, 106]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 2 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 4 & 2 \\ 5 & 4 & 0 & 2 & 4 & 4 & 6 & 4 \\ 2 & 2 & -1 & -6 & 2 & 2 & 4 & -10 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 106$, $\Delta = -216$, $\det = -32$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 18.13 \rightarrow 19$, $T_{\max} = 106$, $T \in [19, 106]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 1 & 1 & -6 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & -1 & -6 & 0 & 2 & 2 & -5 \end{pmatrix}_{b_0}$$

[2] $1^{121} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 106$, $\Delta = -216$, $\det = -56$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.5$

$T_{\min} = 18.14 \rightarrow 19$, $T_{\max} = 106$, $T \in [19, 106]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 1 & 1 & -6 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 4 & 2 & 0 & 2 \\ 2 & 2 & -1 & -6 & 4 & 2 & 2 & -5 \end{pmatrix}_{b_0}$$

[3] $1^{121} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 106$, $\Delta = -216$, $\det = -114$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.5$

$T_{\min} = 18.14 \rightarrow 19$, $T_{\max} = 106$, $T \in [19, 106]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 1 & 1 & -6 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 & 0 & 2 \\ 2 & 2 & -1 & -6 & 4 & 2 & 2 & -5 \end{pmatrix}_{b_0}$$

[4] $1^{113} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 11, \quad T_{\max} = 106, \quad \Delta = -216, \quad \det = -138, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.8$

$$T_{\min} = 18.14 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 1 & 3 & -6 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 0 & 6 & 16 & 6 \\ 2 & 2 & -1 & -6 & 0 & 2 & 6 & -5 \end{pmatrix} b_0$$

[5] $1^{111} 3^{221} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 106, \Delta = -216, \det = -314, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 18.17 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 1 & 1 & -6 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & -1 & -6 & 0 & 2 & 2 & -5 \end{pmatrix} b_0$$

[6] $1^{112} 3^{221} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 11, \quad T_{\max} = 106, \quad \Delta = -216, \quad \det = -372, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.5$

$$T_{\min} = 18.13 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 1 & 2 & -6 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 0 & 4 & 6 & 4 \\ 2 & 2 & -1 & -6 & 0 & 2 & 4 & -5 \end{pmatrix} b_0$$

[7] $1^{1111}3^{221}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 11$, $T_{\max} = 106$, $\Delta = -216$, $\det = -440$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 18.07 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 & 2 \\ 2 & 2 & -1 & -6 & 2 & 2 & -5 \end{pmatrix} b_0$$

$$[8] \quad 1^{1113} 3^{221} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 11, \quad T_{\max} = 106, \quad \Delta = -216, \quad \det = -440, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 18.80 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 3 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 6 & 2 \\ 7 & 6 & 0 & 3 & 6 & 6 & 16 & 6 \\ 2 & 2 & -1 & -6 & 2 & 2 & 6 & -5 \end{pmatrix}_{b_0}$$

[9] $1^{1212} 3^{221} + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{s} \textcolor{blue}{u}_2 + \textcolor{red}{s} \textcolor{blue}{u}_2 + \textcolor{blue}{s} \textcolor{blue}{u}_2$ $T = 11, T_{\max} = 106, \Delta = -216, \det = -454, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.13 \rightarrow 19, \quad T_{\max} = 106, \quad T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 1 & 2 & -6 \\ 2 & 4 & 0 & 2 & 6 & 4 & 8 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 8 & 4 & 6 & 4 \\ 2 & 2 & -1 & -6 & 4 & 2 & 4 & -5 \end{pmatrix} b_0$$

$$[10] \quad 1^{1112} 3^{221} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \quad T_{\max} = 106, \quad \Delta = -216, \quad \det = -512, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 18.16 \rightarrow 19, \quad T_{\max} = 106, \quad T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 4 & 0 & 2 & 4 & 4 & 4 \\ 2 & 2 & -1 & -6 & 2 & 2 & -5 \end{pmatrix} b_0$$

$$[111] \text{ } 1^{1111} 3 \text{ } 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \quad T_{\max} = 106, \quad \Delta = -216, \quad \det = -552, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 18.09 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

[19] $1^{1112}3^{221}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{violet}{\mathfrak{e}}_8 + \textcolor{brown}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 106, \Delta = -216, \det = -1120, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 18.12 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 2 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 4 & 6 & 4 \\ 2 & 2 & -1 & -6 & 2 & 2 & 4 & -5 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1114} 3^{221} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \quad T_{\max} = 106, \quad \Delta = -216, \quad \det = -1322, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.16 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 4 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 8 & 30 & 8 \\ 2 & 2 & -1 & -6 & 2 & 2 & 8 & -5 \end{pmatrix} b_0$$

$$[21] \ 1^{1112} 3 \ 221^1 + [\textcolor{red}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{c}_8 + \textcolor{brown}{s} \textcolor{red}{u}_2 + \textcolor{red}{s} \textcolor{red}{u}_2 + \textcolor{blue}{s} \textcolor{blue}{u}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -216, \ \det = -1402, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 18.13 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 2 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 4 & 6 & 4 \\ 2 & 2 & -1 & -6 & 2 & 2 & 4 & -5 \end{pmatrix} b_0$$

[22] $1^{1113}3^{221}1^{111} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{c}_8 + \textcolor{blue}{s}u_2 + \textcolor{red}{s}u_2 + \textcolor{blue}{s}u_2$ $T = 11, T_{\max} = 106, \Delta = -216, \det = -1508, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 18.14 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 3 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 6 & 16 & 6 \\ 2 & 2 & -1 & -6 & 2 & 2 & 6 & -5 \end{pmatrix} b_0$$

[23] $1^{1113}3^{221}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11$, $T_{\max} = 106$, $\Delta = -216$, $\det = -1640$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.8$

$$T_{\min} = 18.21 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 6 \\ 1 & 2 & 0 & 1 & 2 & 0 & 6 \\ 6 & 6 & 0 & 3 & 6 & 6 & 16 \\ 2 & 2 & -1 & -6 & 2 & 2 & 6 \end{pmatrix} b_0$$

[24] $1^{1113}3^{221}1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 106, \Delta = -216, \det = -1736, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 18.14 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 3 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 6 & 16 & 6 \\ 2 & 2 & -1 & -6 & 2 & 2 & 6 & -5 \end{pmatrix} b_0$$

[25] $1^{1113}3^{221}1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 11$, $T_{\max} = 106$, $\Delta = -216$, $\det = -2072$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 14.8$

$$T_{\min} = 18.15 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 6 \\ 1 & 2 & 0 & 1 & 2 & 0 & 6 \\ 5 & 6 & 0 & 3 & 6 & 6 & 16 \\ 2 & 2 & -1 & -6 & 2 & 2 & 6 \end{pmatrix} b_0$$

[26] $1^{1122} 3^{221} 1^{111} + [\textcolor{blue}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{c}_8 + \textcolor{brown}{s} \textcolor{blue}{u}_2 + \textcolor{red}{s} \textcolor{blue}{u}_2 + \textcolor{blue}{s} \textcolor{blue}{u}_2$ $T = 11, T_{\max} = 106, \Delta = -216, \det = -2140, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.13 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 1 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 & 2 & 0 \\ 2 & 2 & -1 & -6 & 2 & 2 & 2 & -5 \end{pmatrix}_{b_0}$$

[3] $1^{1111} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 106$, $\Delta = -186$, $\det = 1376$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 18.13 \rightarrow 19$, $T_{\max} = 106$, $T \in [19, 106]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 1 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 & 2 & 0 \\ 2 & 2 & -1 & -6 & 2 & 2 & 2 & -5 \end{pmatrix}_{b_0}$$

4.399 NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$T = 12$ [0] $1^{1111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 12$, $T_{\max} = 106$, $\Delta = -195$, $\det = 212$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 20$, $T_{\max} = 106$, $T \in [20, 106]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 1 & -6 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 & 2 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -6 & 2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{1111} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 12$, $T_{\max} = 106$, $\Delta = -195$, $\det = 310$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 18.69 \rightarrow 19$, $T_{\max} = 106$, $T \in [19, 106]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 0 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

4.400 NHC(-2)(-3) + $\mathfrak{e}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$T = 11$ [0] $1^{111} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 106$, $\Delta = -225$, $\det = -104$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 18.54 \rightarrow 19$, $T_{\max} = 106$, $T \in [19, 106]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 0 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & -1 & -5 & 0 & 0 & -4 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 106$, $\Delta = -225$, $\det = -121$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 18.67 \rightarrow 19$, $T_{\max} = 106$, $T \in [19, 106]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 0 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 & 4 \\ 3 & 4 & -1 & -5 & 0 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{1111} 3^{112} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 106$, $\Delta = -225$, $\det = -172$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 18.63 \rightarrow 19$, $T_{\max} = 106$, $T \in [19, 106]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 3 & -5 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^{111} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 106$, $\Delta = -225$, $\det = -240$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 19.12 \rightarrow 20, T_{\max} = 106, T \in [20, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 0 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 \\ 1 & 2 & 0 & 1 & 2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 \\ 2 & 2 & -1 & -6 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{131} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 106, \Delta = -225, \det = -365, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.8$$

$$T_{\min} = 18.87 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 9 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 3 & 4 & -1 & -5 & 0 & 12 & -4 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 106, \Delta = -225, \det = -629, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.52 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -5 & 0 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 106, \Delta = -225, \det = -648, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.56 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 3 & -5 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{121} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 106, \Delta = -225, \det = -716, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 18.58 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 6 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & -5 & 0 & 8 & -4 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 106, \Delta = -225, \det = -760, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.56 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -7 & 3 & 3 & -5 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & -5 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 106, \Delta = -225, \det = -901, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.53 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 3 & -5 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 106, \Delta = -225, \det = -1664, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 3 & -5 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

4.401 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 12 \quad [0] \quad 1^{1111} 3 \quad 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 106, \Delta = -195, \det = 212, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 20, T_{\max} = 106, T \in [20, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 1 & 0 & 1 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & 0 \\ 1 & 2 & 0 & 1 & 2 & 2 & 0 & 0 \\ 2 & 2 & -1 & -6 & 2 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 3 \quad 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 106, \Delta = -195, \det = 310, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.69 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 0 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

4.402 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{11} 3 \quad 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 170, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -5 & 0 & -4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3 \quad 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 187, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 18.87 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 0 & -6 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -6 & 2 & -1 & -5 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{121} 3 \quad 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 246, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 18.64 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -7 & 6 & -5 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & -5 & 8 & -4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3 \quad 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 306, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 18.53 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & -5 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{112} 3 \quad 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 341, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 18.84 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 2 & -6 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -6 & 2 & 3 & -5 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{112} 3 \quad 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 342, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 18.71 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 2 & -6 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -6 & 2 & 3 & -5 \end{pmatrix}_{b_0}$$

[6] $1^{121} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 18.60 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 6 & -5 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -5 & 8 & -4 \end{pmatrix}_{b_0}$$

[7] $1^{111} 3^{13} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 18.50 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & -5 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & -4 \end{pmatrix}_{b_0}$$

[8] $1^{111} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 447, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 18.54 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -7 & 3 & -5 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & -5 & 4 & -4 \end{pmatrix}_{b_0}$$

[9] $1^{131} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 450, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 18.68 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 9 & -5 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -5 & 12 & -4 \end{pmatrix}_{b_0}$$

[10] $1^{111} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 560, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & -5 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & -4 \end{pmatrix}_{b_0}$$

[11] $1^{141} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 576, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.75 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 12 & -5 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -5 & 16 & -4 \end{pmatrix}_{b_0}$$

[12] $1^{111} 3^{12} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 578, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 18.50 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & -5 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & -4 \end{pmatrix}_{b_0}$$

[13] $1^{112} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 731, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$

$$T_{\min} = 18.46 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 2 & -6 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & -6 & 2 & 3 & -5 \end{pmatrix}_{b_0}$$

[14] $1^{151} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 738, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.1$

$T_{\min} = 18.80 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 15 & -5 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & -5 & 20 & -4 \end{pmatrix}_{b_0}$$

[15] $1^{121} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 986, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 18.53 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 6 & -5 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -5 & 8 & -4 \end{pmatrix}_{b_0}$$

[16] $1^{121} 3^{112} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 1376, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 6 & -5 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -5 & 8 & -4 \end{pmatrix}_{b_0}$$

[17] $1^{121} 3^{114} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 1448, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 18.50 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 6 & -5 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -5 & 8 & -4 \end{pmatrix}_{b_0}$$

[18] $1^{131} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 1448, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$T_{\min} = 18.54 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 9 & -5 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -5 & 12 & -4 \end{pmatrix}_{b_0}$$

[19] $1^{121} 3^{113} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 1530, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 6 & -5 \\ 9 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -5 & 8 & -4 \end{pmatrix}_{b_0}$$

[20] $1^{141} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 1946, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 18.56 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 12 & -5 \\ 13 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -5 & 16 & -4 \end{pmatrix}_{b_0}$$

[21] $1^{131} 3^{112} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 2210, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$T_{\min} = 18.52 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 9 & -5 \\ 11 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -5 & 12 & -4 \end{pmatrix}_{b_0}$$

[22] $1^{131} 3^{113} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 2736, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 9 & -5 \\ 12 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -5 & 12 & -4 \end{pmatrix}_{b_0}$$

[23] $1^{131} 3^{14} 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 3026, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 9 & -5 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -5 & 12 & -4 \end{pmatrix}_{b_0}$$

[24] $1^{141} 3^{12} 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 3080, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.53 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 12 & -5 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -5 & 16 & -4 \end{pmatrix}_{b_0}$$

[25] $1^{141} 3^{13} 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 3978, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.52 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 12 & -5 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -5 & 16 & -4 \end{pmatrix}_{b_0}$$

[26] $1^{141} 3^{14} 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 4640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 12 & -5 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -5 & 16 & -4 \end{pmatrix}_{b_0}$$

4.403 NHC(-2)(-3) + $\mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$T = 11 \quad [0] 1^{111} 3^{111} 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 106, \Delta = -225, \det = -104, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.54 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 0 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -5 & 0 & 0 & -4 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3^{111} 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 106, \Delta = -225, \det = -121, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.67 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -5 & 0 & 4 & -4 \end{pmatrix}_{b_0}$$

[2] $1^{1111} 3^{12} 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 106, \Delta = -225, \det = -172, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.63 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 3 & -5 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^{111} 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 106, \Delta = -225, \det = -240, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 19.12 \rightarrow 20, T_{\max} = 106, T \in [20, 106]$$

4.404 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 12 \quad [0] \ 1^{\overline{1111}} 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 106, \Delta = -195, \det = 212, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 20, T_{\max} = 106, T \in [20, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 1 & 0 & 1 & -6 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 & 2 \\ 2 & 2 & -1 & -6 & 2 & -1 & 2 & -6 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 106, \Delta = -195, \det = 310, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.69 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 0 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

4.405 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 11 \quad [1] \ 1^{\overline{111}} 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 108, \Delta = -234, \det = -291, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 19.56 \rightarrow 20, T_{\max} = 108, T \in [20, 108]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -5 & 0 & 3 & -5 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 108, \Delta = -234, \det = -427, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.95 \rightarrow 19, T_{\max} = 108, T \in [19, 108]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 0 & -5 \\ 3 & 6 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -5 & 4 & -1 & -4 \end{pmatrix}_{b_0}$$

4.406 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 190, \Delta = -285, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 18.75 \rightarrow 19, T_{\max} = 190, T \in [19, 190]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & -6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -6 & -1 & -5 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 190, \Delta = -285, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 18.50 \rightarrow 19, T_{\max} = 190, T \in [19, 190]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & -7 & -5 \\ 3 & 4 & -1 & -5 & -4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{12}} 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 190, \Delta = -285, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 18.69 \rightarrow 19, T_{\max} = 190, T \in [19, 190]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & -6 \\ 2 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -6 & 3 & -5 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{13}} 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 190, \Delta = -285, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.83 \rightarrow 19, T_{\max} = 190, T \in [19, 190]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & -6 \\ 3 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -6 & 5 & -5 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 190$, $\Delta = -285$, $\det = -257$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 18.53 \rightarrow 19$, $T_{\max} = 190$, $T \in [19, 190]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 4 \\ 5 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & -7 & -5 \\ 4 & 4 & -1 & -5 & -4 \end{pmatrix}_{b_0}$$

[5] $1^{12} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 190$, $\Delta = -285$, $\det = -399$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 18.46 \rightarrow 19$, $T_{\max} = 190$, $T \in [19, 190]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & -6 \\ 3 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -6 & 3 & -5 \end{pmatrix}_{b_0}$$

[6] $1^{13} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 190$, $\Delta = -285$, $\det = -553$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.43 \rightarrow 19$, $T_{\max} = 190$, $T \in [19, 190]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 3 & -6 \\ 4 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & -6 & 5 & -5 \end{pmatrix}_{b_0}$$

4.407 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$

$T = 11$ [0] $1^{11} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2]$ $T = 11$, $T_{\max} = 109$, $\Delta = -235$, $\det = -166$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 18.89 \rightarrow 19$, $T_{\max} = 109$, $T \in [19, 109]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 4 & -5 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 0 & 4 & 0 & 9 & 5 \\ 3 & 4 & -1 & -5 & -1 & 5 & -4 \end{pmatrix}_{b_0}$$

4.408 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$

$T = 10$ [0] $1^{11} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 170$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 18.51 \rightarrow 19$, $T_{\max} = 171$, $T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -5 & 0 & -4 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 187$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 18.87 \rightarrow 19$, $T_{\max} = 171$, $T \in [19, 171]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 1 & -6 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & -6 & -1 & 2 & -5 \end{pmatrix}_{b_0}$$

[2] $1^{112} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 246$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.5$

$T_{\min} = 18.64 \rightarrow 19$, $T_{\max} = 171$, $T \in [19, 171]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -7 & 6 & -5 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & -5 & 8 & -4 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 306$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 18.53 \rightarrow 19$, $T_{\max} = 171$, $T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & -5 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & -4 \end{pmatrix}_{b_0}$$

[4] $1^{121} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 341$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 18.84 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 1 & -6 \\ 2 & 4 & 0 & 2 & 5 & 4 & 3 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -6 & 3 & 2 & -5 \end{pmatrix}_{b_0}$$

[5] $1^{121} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 342$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 18.71 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 1 & -6 \\ 2 & 4 & 0 & 2 & 5 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -6 & 3 & 2 & -5 \end{pmatrix}_{b_0}$$

[6] $1^{111} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 360$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 18.50 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & -5 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & -4 \end{pmatrix}_{b_0}$$

[7] $1^{112} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 360$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.5$

$$T_{\min} = 18.60 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 6 & -5 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -5 & 8 & -4 \end{pmatrix}_{b_0}$$

[8] $1^{111} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 447$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 18.54 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -7 & 3 & -5 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & -5 & 4 & -4 \end{pmatrix}_{b_0}$$

[9] $1^{113} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 450$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.8$

$$T_{\min} = 18.68 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 9 & -5 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -5 & 12 & -4 \end{pmatrix}_{b_0}$$

[10] $1^{111} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 560$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & -5 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & -4 \end{pmatrix}_{b_0}$$

[11] $1^{114} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 576$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 18.75 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 12 & -5 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -5 & 16 & -4 \end{pmatrix}_{b_0}$$

- [12] $1^{111} 3^{12} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 578$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 18.50 \rightarrow 19$, $T_{\max} = 171$, $T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & -5 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & -4 \end{pmatrix}_{b_0}$$
- [13] $1^{121} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 731$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 18.46 \rightarrow 19$, $T_{\max} = 171$, $T \in [19, 171]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 2 & 1 & -6 \\ 3 & 4 & 0 & 2 & 5 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & -6 & 3 & 2 & -5 \end{pmatrix}_{b_0}$$
- [14] $1^{115} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 738$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.1$
 $T_{\min} = 18.80 \rightarrow 19$, $T_{\max} = 171$, $T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 15 & -5 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & -5 & 20 & -4 \end{pmatrix}_{b_0}$$
- [15] $1^{112} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 986$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 18.53 \rightarrow 19$, $T_{\max} = 171$, $T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 6 & -5 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -5 & 8 & -4 \end{pmatrix}_{b_0}$$
- [16] $1^{112} 3^{12} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 1376$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 18.51 \rightarrow 19$, $T_{\max} = 171$, $T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 6 & -5 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -5 & 8 & -4 \end{pmatrix}_{b_0}$$
- [18] $1^{112} 3^{14} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 1448$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 18.50 \rightarrow 19$, $T_{\max} = 171$, $T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 6 & -5 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -5 & 8 & -4 \end{pmatrix}_{b_0}$$
- [17] $1^{113} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 1448$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 18.54 \rightarrow 19$, $T_{\max} = 171$, $T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 9 & -5 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -5 & 12 & -4 \end{pmatrix}_{b_0}$$
- [19] $1^{112} 3^{13} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 1530$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 18.51 \rightarrow 19$, $T_{\max} = 171$, $T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 6 & -5 \\ 9 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -5 & 8 & -4 \end{pmatrix}_{b_0}$$
- [20] $1^{114} 3^{11} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 171$, $\Delta = -255$, $\det = 1946$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 18.56 \rightarrow 19$, $T_{\max} = 171$, $T \in [19, 171]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 12 & -5 \\ 13 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -5 & 16 & -4 \end{pmatrix}_{b_0}$$

[21] $1^{113} 3^{12} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 2210, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 18.52 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 9 & -5 \\ 11 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -5 & 12 & -4 \end{pmatrix}_{b_0}$$

[22] $1^{113} 3^{13} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 2736, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 9 & -5 \\ 12 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -5 & 12 & -4 \end{pmatrix}_{b_0}$$

[23] $1^{113} 3^{14} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 3026, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 9 & -5 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -5 & 12 & -4 \end{pmatrix}_{b_0}$$

[24] $1^{114} 3^{12} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 3080, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.53 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 12 & -5 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -5 & 16 & -4 \end{pmatrix}_{b_0}$$

[25] $1^{114} 3^{13} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 3978, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.52 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 12 & -5 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -5 & 16 & -4 \end{pmatrix}_{b_0}$$

[26] $1^{114} 3^{14} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 171, \Delta = -255, \det = 4640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 171, T \in [19, 171]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 12 & -5 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -5 & 16 & -4 \end{pmatrix}_{b_0}$$

4.409 $\text{NHC}(-2)(-3) + \text{e}_8 + \text{su}_3 + \text{su}_2 + \text{su}_2$

$T = 11 \quad [0] 1^{111} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 106, \Delta = -225, \det = -104, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.54 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 0 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -5 & 0 & 0 & -4 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3^{111} 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 106, \Delta = -225, \det = -121, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 18.67 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -5 & 0 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1111} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -172, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.63 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 3 & -5 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -240, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.12 \rightarrow 20, \ T_{\max} = 106, \ T \in [20, 106]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 1 & -6 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & -1 & -6 & -1 & 2 & -6 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{113} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -365, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 18.87 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 9 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 3 & 4 & -1 & -5 & 0 & 12 & -4 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -629, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.52 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -5 & 0 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1111} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -648, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.56 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 3 & -5 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -716, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 18.58 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 6 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & -5 & 0 & 8 & -4 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -760, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.56 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -7 & 3 & 3 & -5 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & -5 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1111} 3 \ 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{e}_8 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 11, \ T_{\max} = 106, \ \Delta = -225, \ \det = -901, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.53 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 3 & -5 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

[10] $1^{1111} 3^{221^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 106, \Delta = -225, \det = -1664, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 18.51 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 3 & -5 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

4.410 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$T = 12$ [0] $1^{1111} 3^{221^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 106, \Delta = -195, \det = 212, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 20, T_{\max} = 106, T \in [20, 106]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -8 & 0 & 1 & 1 & -6 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 & 0 \\ 2 & 2 & -1 & -6 & -1 & 2 & 2 & -6 \end{pmatrix}_{b_0}$$

[1] $1^{1111} 3^{221^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 106, \Delta = -195, \det = 310, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 18.69 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -5 & 0 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

4.411 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 11$ [1] $1^{1111} 3^{221^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11, T_{\max} = 108, \Delta = -234, \det = -291, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 19.56 \rightarrow 20, T_{\max} = 108, T \in [20, 108]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & 3 & -5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & -5 & 0 & 3 & -5 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{1111} 3^{221^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11, T_{\max} = 108, \Delta = -234, \det = -427, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 18.95 \rightarrow 19, T_{\max} = 108, T \in [19, 108]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & 3 & -5 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 & 4 \\ 3 & 4 & -1 & -5 & -1 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

4.412 $\text{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_3$

$T = 10$ [1] $1^{1111} 3^{221^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10, T_{\max} = 181, \Delta = -264, \det = 314, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 18.89 \rightarrow 19, T_{\max} = 181, T \in [19, 181]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & -5 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -5 & -1 & 4 & -4 \end{pmatrix}_{b_0}$$

[2] $1^{1111} 3^{221^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10, T_{\max} = 181, \Delta = -264, \det = 450, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 19, T_{\max} = 181, T \in [19, 181]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & -5 \\ 3 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & -5 & 3 & -4 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 181, \Delta = -264, \det = 704, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.78 \rightarrow 19, T_{\max} = 181, T \in [19, 181]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & -5 \\ 4 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & -5 & 3 & -4 \end{pmatrix}_{b_0}$$

4.413 NHC(-2)(-3) + $\mathfrak{e}_8$ + $\textcolor{blue}{su}_3$ + $\textcolor{blue}{su}_3$ + $\textcolor{blue}{su}_2$

$$T = 11 \quad [1] \ 1^{111} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 108, \Delta = -234, \det = -291, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 19.56 \rightarrow 20, T_{\max} = 108, T \in [20, 108]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 3 & 0 & -5 \\ 3 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -5 & 3 & 0 & -5 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 108, \Delta = -234, \det = -427, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.95 \rightarrow 19, T_{\max} = 108, T \in [19, 108]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 3 & -5 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -5 & -1 & 4 & -4 \end{pmatrix}_{b_0}$$

4.414 NHC(-2)(-3) + $\mathfrak{f}_4$

$$T = 6 \quad [1] \ 1^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 6, T_{\max} = 99, \Delta = -167, \det = 13, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 4 & 5 \\ 3 & 4 & -2 & 0 \\ 3 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[0] \ 21^1 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 6, T_{\max} = 99, \Delta = -167, \det = 13, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 4 & 5 \\ 3 & 4 & -2 & 0 \\ 3 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 6, T_{\max} = 99, \Delta = -167, \det = 21, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -5 & -3 \\ 3 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 6, T_{\max} = 99, \Delta = -167, \det = 26, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 99, T \in [13, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & -2 & 0 \\ 3 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 21^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 6, T_{\max} = 99, \Delta = -167, \det = 103, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 11.53 \rightarrow 12, T_{\max} = 99, T \in [12, 99]$$

$$\begin{pmatrix} 4 & 8 & 3 & 4 \\ 8 & 9 & 4 & 5 \\ 3 & 4 & -2 & 0 \\ 4 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [5] \ 2231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 10 & 16 & 6 \\ 16 & 25 & 9 \\ 6 & 9 & 3 \end{pmatrix}_{b_0}$$

- [6] $1^{11}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -139$, $\det = -6$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 12$, $T_{\max} = 97$, $T \in [12, 97]$
 $\begin{pmatrix} 10 & 16 & 6 \\ 16 & 25 & 9 \\ 6 & 9 & 3 \end{pmatrix}_{b_0}$
- [7] $21^1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -139$, $\det = -10$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 12$, $T_{\max} = 97$, $T \in [12, 97]$
 $\begin{pmatrix} 13 & 20 & 7 \\ 20 & 30 & 10 \\ 7 & 10 & 3 \end{pmatrix}_{b_0}$
- [8] $1^{11}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -139$, $\det = -15$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -5 & -3 \\ 3 & 5 & -3 & 2 \end{pmatrix}_{b_0}$
- [9] $1^{11}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -139$, $\det = -16$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 1 & 0 & 0 & -5 \\ 2 & 2 & 2 & -3 \end{pmatrix}_{b_0}$
- [10] $1^{11}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -139$, $\det = -16$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$
- [11] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -139$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 97$, $T \in [13, 97]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 3 \\ 1 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -3 & -1 \\ 2 & 3 & 2 & -1 & 0 \end{pmatrix}_{b_0}$
- [14] $1^{11}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -139$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 97$, $T \in [13, 97]$
 $\begin{pmatrix} 5 & 12 & 5 \\ 12 & 25 & 9 \\ 5 & 9 & 2 \end{pmatrix}_{b_0}$
- [13] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -139$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 97$, $T \in [13, 97]$
 $\begin{pmatrix} 5 & 13 & 5 \\ 13 & 30 & 10 \\ 5 & 10 & 2 \end{pmatrix}_{b_0}$
- [12] $1^{11}3131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 97$, $T \in [13, 97]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 \\ 1 & 4 & 0 & 2 \\ 1 & 0 & 0 & -5 \\ 2 & 3 & 2 & -3 \end{pmatrix}_{b_0}$
- [15] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -139$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$
 $\begin{pmatrix} 10 & 18 & 6 \\ 18 & 30 & 10 \\ 6 & 10 & 3 \end{pmatrix}_{b_0}$
- [16] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -139$, $\det = -26$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 13.08 \rightarrow 14$, $T_{\max} = 97$, $T \in [14, 97]$
 $\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 9 & 0 & 5 \\ 1 & 0 & -5 & -3 \\ 3 & 5 & -3 & 0 \end{pmatrix}_{b_0}$
- [17] $1^{11}31^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 97$, $T \in [13, 97]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 \\ 2 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 3 & 3 & -3 & 0 \end{pmatrix}_{b_0}$
- [18] $1^{11}31^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[19] $2 \overset{11}{2} 2 \overset{1}{3} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 10 & 19 & 6 \\ 19 & 30 & 10 \\ 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[20] $21 \overset{11}{1} 3 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 13 & 20 & 7 \\ 20 & 25 & 9 \\ 7 & 9 & 3 \end{pmatrix}_{b_0}$$

[21] $1 \overset{11}{3} 3 1^1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 7 & 3 & 4 \\ 2 & 7 & 3 & 3 & 3 \\ 2 & 3 & 3 & -2 & 0 \\ 2 & 4 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

[22] $2 \overset{11}{2} 2 \overset{1}{3} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 1 & 3 \\ 2 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -3 & -1 \\ 2 & 3 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

[23] $131 \overset{11}{1} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 12.53 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 4 & 7 & 3 & 4 \\ 2 & 7 & 3 & 3 & 3 \\ 1 & 3 & 3 & -2 & 0 \\ 2 & 4 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

[24] $31 \overset{11}{1} 2 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -147, \det = -215, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -3 & -1 \\ 2 & 3 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$T = 8$ [25] $2 \overset{11}{3} 1 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 10 & 16 & 6 \\ 16 & 25 & 9 \\ 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[26] $1 \overset{11}{3} 2 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 1 & 6 & 3 \\ 5 & 6 & 17 & 7 \\ 3 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[27] $1 \overset{11}{1} 3 2 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 6 & 0 & 6 \\ 1 & 0 & -5 & -3 \\ 3 & 6 & -3 & 3 \end{pmatrix}_{b_0}$$

[28] $131 \overset{11}{1} 3 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 5 & 5 & 3 \\ 5 & 18 & 16 & 8 \\ 5 & 16 & 12 & 6 \\ 3 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[29] $1 \overset{11}{1} 3 2 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 2 & 3 & -3 & 2 \end{pmatrix}_{b_0}$$

[30] $1 \overset{11}{1} 2 3 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 3 & -1 & 0 & 2 \end{pmatrix}_{b_0}$$

[31] $1^1 3^1 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 3 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[37] $1^1 13^1 221^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 6 & 6 & 6 \\ 2 & 6 & 0 & 2 \\ 3 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[36] $1^1 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 3 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{11} 23131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 5 & 1 & 3 \\ 5 & 18 & 1 & 8 \\ 1 & 1 & -3 & -1 \\ 3 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

[34] $1^1 1^1 2321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 0 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

[33] $1^1 23131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -3 & -1 \\ 3 & 0 & -1 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{11} 23131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 5 & 0 & 4 \\ 1 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 4 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[38] $1^1 1^1 3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 4 & 3 & 9 & 4 \\ 3 & 1 & 6 & 3 \\ 9 & 6 & 17 & 7 \\ 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[39] $1^1 231^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 14, T_{\max} = 35, T \in [14, 35]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 2 & 4 & 0 & 4 \\ 2 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 4 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[40] $1^1 1^1 4131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -137, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -4 & -2 \\ 3 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

[41] $1^1 1^1 3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -110, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 6 & 6 & 6 \\ 3 & 6 & 0 & 2 \\ 3 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[42] $1^1 1^1 4131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -137, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 9 & 0 & 5 \\ 1 & 0 & -4 & -2 \\ 3 & 5 & -2 & 1 \end{pmatrix}_{b_0} \\
[43] \quad 1^{11}41231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2 \\
T_{\min} = 13.37 \rightarrow 14, T_{\max} = 98, T \in [14, 98] \\
& \begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 9 & 0 & 5 \\ 1 & 0 & -4 & -2 \\ 3 & 5 & -2 & 1 \end{pmatrix}_{b_0} \\
[44] \quad 1^{11}12321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2 \\
T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & -2 & 0 \\ 2 & 3 & -1 & 0 & 0 \end{pmatrix}_{b_0} \\
[45] \quad 1^{11}12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2 \\
T_{\min} = 11.80 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 3 & 3 \\ 2 & 3 & -1 & 3 & 2 \end{pmatrix}_{b_0} \\
[46] \quad 1^{11}231^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2 \\
T_{\min} = 12, T_{\max} = 35, T \in [12, 35] \\
& \begin{pmatrix} 4 & 9 & 8 & 4 \\ 9 & 18 & 16 & 8 \\ 8 & 16 & 12 & 6 \\ 4 & 8 & 6 & 3 \end{pmatrix}_{b_0} \\
[47] \quad 1^1231^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2 \\
T_{\min} = 12.26 \rightarrow 13, T_{\max} = 35, T \in [13, 35] \\
& \begin{pmatrix} 1 & 1 & 5 & 3 \\ 1 & -2 & 1 & 0 \\ 5 & 1 & 12 & 6 \\ 3 & 0 & 6 & 2 \end{pmatrix}_{b_0} \\
[48] \quad 1^1231^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2 \\
T_{\min} = 12.29 \rightarrow 13, T_{\max} = 35, T \in [13, 35] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 4 & 3 & -3 & 1 \end{pmatrix}_{b_0} \\
[49] \quad 1^1231^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2 \\
T_{\min} = 12, T_{\max} = 35, T \in [12, 35] \\
& \begin{pmatrix} 1 & 5 & 1 & 3 \\ 5 & 13 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 3 & 7 & -1 & 3 \end{pmatrix}_{b_0} \\
[50] \quad 21^141^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -137, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2 \\
T_{\min} = 12.56 \rightarrow 13, T_{\max} = 98, T \in [13, 98] \\
& \begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 2 & 3 & -2 & 1 \end{pmatrix}_{b_0} \\
[51] \quad 1^14^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -137, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2 \\
T_{\min} = 12.82 \rightarrow 13, T_{\max} = 98, T \in [13, 98] \\
& \begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 4 & 0 & 4 \\ 1 & 0 & -4 & -2 \\ 3 & 4 & -2 & 2 \end{pmatrix}_{b_0} \\
[52] \quad 1^141^1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2 \\
T_{\min} = 12.82 \rightarrow 13, T_{\max} = 98, T \in [13, 98] \\
& \begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 3 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 2 & 3 & -2 & 1 \end{pmatrix}_{b_0} \\
[53] \quad 21^{11}4131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -137, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2 \\
T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98] \\
& \begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 4 & 7 & 0 & 4 \\ 2 & 7 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 4 & 3 & -2 & 1 \end{pmatrix}_{b_0} \\
[54] \quad 1^141^1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2 \\
T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98] \\
& \begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 13 & 0 & 7 \\ 1 & 0 & -4 & -2 \\ 3 & 7 & -2 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[55] \quad 2 \overset{1}{\overset{1}{\overset{1}{4}}} 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 10 & 24 & 6 \\ 24 & 49 & 13 \\ 6 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[56] \quad 31 \overset{1}{\overset{1}{\overset{1}{2}}} 321^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 56, \Delta = -120, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 56, T \in [13, 56]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[57] \quad 41 \overset{1}{\overset{1}{\overset{1}{2}}} 231^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 26 & 29 & 0 & 10 \\ 4 & 29 & 25 & 0 & 9 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 10 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [58] \quad 1^1 412321^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 \quad T = 9, T_{\max} = 22, \Delta = -111, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & -3 & -1 \\ 3 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^1 141321^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & -3 & -1 \\ 3 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[60] \quad 2^1 31^1 51^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 10 & 12 & 6 \\ 12 & 13 & 7 \\ 6 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^1 1^1 5131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 5 \\ 9 & 13 & 7 \\ 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^1 1^1 51231^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[63] \quad 21^1 1^1 5131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 13 & 14 & 7 \\ 14 & 13 & 7 \\ 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^1 31^1 4131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 3 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^1 31^1 4131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 13, T_{\max} = 29, T \in [13, 29]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[66] \quad 21^1 1^1 5131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 10 & 5 \\ 10 & 13 & 7 \\ 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[67] \quad 2 \overset{1}{\overset{1}{\overset{1}{4}}} 3 \overset{1}{\overset{1}{\overset{1}{2}}} 1^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 \quad T = 9, T_{\max} = 22, \Delta = -111, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$$

- $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 6 & 6 & 6 \\ 3 & 6 & 0 & 2 \\ 3 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [68] $21^1 51^1 231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -55$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$
 $\begin{pmatrix} 13 & 35 & 7 \\ 35 & 90 & 18 \\ 7 & 18 & 3 \end{pmatrix}_{b_0}$
- [69] $1^1 3221^1 51^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -76$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 13.58 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$
 $\begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 17 & 5 & 7 \\ 1 & 0 & 5 & -2 & 0 \\ 2 & 0 & 7 & 0 & 1 \end{pmatrix}_{b_0}$
- [70] $1^1 5^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -79$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 13.58 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$
 $\begin{pmatrix} 5 & 23 & 5 \\ 23 & 90 & 18 \\ 5 & 18 & 3 \end{pmatrix}_{b_0}$
- [71] $1^1 3221^1 51^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -126$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$
 $\begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 18 & 1 & 5 & 8 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 5 & 0 & -2 & 0 \\ 2 & 8 & -1 & 0 & 1 \end{pmatrix}_{b_0}$
- $T = 10$ [72] $1^1 31^1 5131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 11$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 3 & -1 & 0 & 2 \end{pmatrix}_{b_0}$
- [74] $1^1 512321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 10$, $T_{\max} = 22$, $\Delta = -106$, $\det = 12$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 15$, $T_{\max} = 22$, $T \in [15, 22]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 6 & 6 & 6 \\ 2 & 6 & 0 & 2 \\ 3 & 6 & 2 & 2 \end{pmatrix}_{b_0}$
- [76] $1^1 51321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 10$, $T_{\max} = 22$, $\Delta = -104$, $\det = 12$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 15$, $T_{\max} = 22$, $T \in [15, 22]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 6 & 6 & 6 \\ 2 & 6 & 0 & 2 \\ 3 & 6 & 2 & 2 \end{pmatrix}_{b_0}$
- [75] $21^1 51321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 10$, $T_{\max} = 22$, $\Delta = -104$, $\det = 12$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 15$, $T_{\max} = 22$, $T \in [15, 22]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 6 & 6 & 6 \\ 2 & 6 & 0 & 2 \\ 3 & 6 & 2 & 2 \end{pmatrix}_{b_0}$
- [73] $1^1 31^1 5131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 10$, $T_{\max} = 23$, $\Delta = -111$, $\det = 12$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 13.33 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 0 & 2 & 0 & 2 \end{pmatrix}_{b_0}$
- [77] $1^1 3151^1 231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 47$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 14.13 \rightarrow 15$, $T_{\max} = 24$, $T \in [15, 24]$
 $\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 13 & 6 & 0 & 7 \\ 1 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 7 & 3 & -3 & 1 \end{pmatrix}_{b_0}$
- [78] $1^1 3151^1 231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 10$, $T_{\max} = 24$, $\Delta = -112$, $\det = 55$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 13.82 \rightarrow 14$, $T_{\max} = 24$, $T \in [14, 24]$
 $\begin{pmatrix} 1 & 8 & 1 & 3 \\ 8 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 3 & 12 & -1 & 3 \end{pmatrix}_{b_0}$
- [79] $31^{11} 315131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4$ $T = 10$, $T_{\max} = 47$, $\Delta = -119$, $\det = 152$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 14.16 \rightarrow 15$, $T_{\max} = 47$, $T \in [15, 47]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

[80] $1^{11} 31^1 51321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 11, T_{\max} = 14, \Delta = -83, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 1 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 4 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

[81] $1^{11} 31^1 51321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 11, T_{\max} = 14, \Delta = -83, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 0 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[82] $1^{11} 6^{1321} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 11, T_{\max} = 23, \Delta = -109, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{11} 6^{1321} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 11, T_{\max} = 23, \Delta = -109, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

[84] $1^{11} 6^{131} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 11, T_{\max} = 23, \Delta = -108, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 14.33 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

[85] $1^{11} 31^1 51321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 11, T_{\max} = 14, \Delta = -83, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 5 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 4 & 4 \\ 2 & 4 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[86] $31^{11} 316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 11, T_{\max} = 31, \Delta = -116, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 15, T_{\max} = 31, T \in [15, 31]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 3 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

[87] $1^{11} 321^{161231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 15.13 \rightarrow 16, T_{\max} = 23, T \in [16, 23]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 10 & 1 & 4 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 4 & 0 & -2 & 0 \\ 2 & 6 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{11} 321^{161231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 11, T_{\max} = 23, \Delta = -110, \det = -55, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 14.82 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 10 & 5 & 0 & 6 \\ 1 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 6 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

[89] $31^{11} 321^{16131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 11, T_{\max} = 34, \Delta = -117, \det = -80, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 15, T_{\max} = 34, T \in [15, 34]$

$$\begin{pmatrix} 0 & 4 & 3 & 1 & 2 \\ 4 & 10 & 5 & 3 & 6 \\ 3 & 5 & 0 & 1 & 2 \\ 1 & 3 & 1 & -2 & 0 \\ 2 & 6 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

[90] $2321^{16131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 11, T_{\max} = 24, \Delta = -111, \det = -111, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 14.77 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 4 & 2 & 1 & 2 \\ 4 & 10 & 5 & 3 & 6 \\ 2 & 5 & 0 & 1 & 2 \\ 1 & 3 & 1 & -2 & 0 \\ 2 & 6 & 2 & 0 & 2 \end{pmatrix}_{b_0} \\
[92] & 31^{11} 3161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 11, T_{\max} = 34, \Delta = -117, \det = -152, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2 \\
& T_{\min} = 15.16 \rightarrow 16, T_{\max} = 34, T \in [16, 34] \\
& \begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 1 \end{pmatrix}_{b_0} \\
[91] & 1^{31} 6^{11} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 11, T_{\max} = 31, \Delta = -116, \det = -152, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2 \\
& T_{\min} = 15.16 \rightarrow 16, T_{\max} = 31, T \in [16, 31] \\
& \begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 1 \end{pmatrix}_{b_0} \\
T = 12 & [94] 1^{11} 31^{11} 6^{11} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 12, T_{\max} = 15, \Delta = -87, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.2 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -2 & 3 \end{pmatrix}_{b_0} \\
[93] & 1^{11} 6^{11} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.2 \\
& T_{\min} = 14.50 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0} \\
[95] & 1^{11} 6^{11} 13141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 12, T_{\max} = 24, \Delta = -107, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.2 \\
& T_{\min} = 15.33 \rightarrow 16, T_{\max} = 24, T \in [16, 24] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 3 & 5 & -1 & 3 \end{pmatrix}_{b_0} \\
[96] & 1^{11} 7^{11} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.2 \\
& T_{\min} = 15.33 \rightarrow 16, T_{\max} = 24, T \in [16, 24] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & 2 \end{pmatrix}_{b_0} \\
T = 13 & [97] 1^{11} 7^{11} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 13, T_{\max} = 16, \Delta = -81, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.2 \\
& T_{\min} = 15.50 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0} \\
[101] & 1^{11} 8^{11} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.2 \\
& T_{\min} = 16.33 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 2 \end{pmatrix}_{b_0} \\
[100] & 1^{11} 8^{11} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.2 \\
& T_{\min} = 16.33 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & 2 \end{pmatrix}_{b_0} \\
[99] & 1^{11} 8^{11} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 13, T_{\max} = 25, \Delta = -107, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.2 \\
& T_{\min} = 16.33 \rightarrow 17, T_{\max} = 25, T \in [17, 25] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0} \\
[98] & 1^1 1^1 151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 \quad T = 13, T_{\max} = 25, \Delta = -102, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.2 \\
& T_{\min} = 16.33 \rightarrow 17, T_{\max} = 25, T \in [17, 25]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[102] $31^1 1^1 321^1 81231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 13$, $T_{\max} = 30$, $\Delta = -115$, $\det = -32$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 17$, $T_{\max} = 30$, $T \in [17, 30]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 3 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

[103] $31^1 1^1 3221^1 81231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 13$, $T_{\max} = 30$, $\Delta = -115$, $\det = -47$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 17.13 \rightarrow 18$, $T_{\max} = 30$, $T \in [18, 30]$

$$\begin{pmatrix} 0 & 5 & 3 & 1 & 2 \\ 5 & 18 & 6 & 4 & 8 \\ 3 & 6 & 0 & 1 & 2 \\ 1 & 4 & 1 & -2 & 0 \\ 2 & 8 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$T = 14$ [106] $1^1 8^1 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 14$, $T_{\max} = 17$, $\Delta = -80$, $\det = 8$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 16.50 \rightarrow 17$, $T_{\max} = 17$, $T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 2 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[105] $1^1 1^1 315131^1 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 14$, $T_{\max} = 17$, $\Delta = -81$, $\det = 8$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 16.50 \rightarrow 17$, $T_{\max} = 17$, $T \in [17, 17]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[104] $1^1 1^1 3221^1 8^1 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 14$, $T_{\max} = 17$, $\Delta = -80$, $\det = 8$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 16.50 \rightarrow 17$, $T_{\max} = 17$, $T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 2 & 3 & -2 & 3 \end{pmatrix}_{b_0}$$

[107] $1^1 1^1 321^1 8^1 123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 14$, $T_{\max} = 26$, $\Delta = -106$, $\det = 12$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 17.33 \rightarrow 18$, $T_{\max} = 26$, $T \in [18, 26]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 3 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

$T = 15$ [109] $1^1 1^1 512321^1 8^1 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 15$, $T_{\max} = 27$, $\Delta = -103$, $\det = -12$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 18.33 \rightarrow 19$, $T_{\max} = 27$, $T \in [19, 27]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[108] $1^1 1^1 151321^1 8^1 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 15$, $T_{\max} = 27$, $\Delta = -101$, $\det = -12$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 18.33 \rightarrow 19$, $T_{\max} = 27$, $T \in [19, 27]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

$T = 16$ [110] $1^1 1^1 3151321^1 8^1 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 16$, $T_{\max} = 19$, $\Delta = -80$, $\det = 8$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 18.50 \rightarrow 19$, $T_{\max} = 19$, $T \in [19, 19]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$T = 18$ [111] $1^1 1^1 3221^1 (12) 12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4$ $T = 18$, $T_{\max} = 21$, $\Delta = -86$, $\det = 8$, $(n_+, n_-, n_0) = (1, 18, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 20.50 \rightarrow 21$, $T_{\max} = 21$, $T \in [21, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -2 & 3 \end{pmatrix}_{b_0}$$

4.415 NHC(-2)(-3) + f₄ + e₆

$$T = 7 \quad [0] \ 1^{11} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 102, \Delta = -216, \det = -122, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 0 & -4 & -2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [1] \ 1^{11} 3^1 2^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 102, \Delta = -188, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 5 & 0 & -2 & 0 \\ 3 & 5 & -3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 2^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 102, \Delta = -188, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 4 & 0 & -2 & 0 \\ 3 & 5 & -3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 102, \Delta = -195, \det = 481, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.51 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 3 & 0 & -4 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \ 1^{11} 2^1 3^1 1^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{e}_6 \quad T = 9, T_{\max} = 102, \Delta = -167, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 3 \\ 5 & 18 & 16 & 0 & 8 \\ 5 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 2^1 3^1 1^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{e}_6 \quad T = 9, T_{\max} = 102, \Delta = -167, \det = -25, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 2 & 4 & 0 & 4 & 4 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 4 & 3 & 0 & -2 & 0 \\ 2 & 4 & 3 & -3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 2^1 2^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{e}_6 \quad T = 9, T_{\max} = 101, \Delta = -159, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 5 & 15 & 1 & 5 \\ 15 & 42 & 0 & 12 \\ 1 & 0 & -5 & -3 \\ 5 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 2^1 3^1 2^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{e}_6 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.68 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 3 & 0 & 1 & -3 & -1 \\ 2 & 3 & -1 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 2^1 3^1 2^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{e}_6 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.48 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & -3 & -1 \\ 2 & 3 & -1 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [9] \quad 1^{11} 3^1 5^1 13 21^{11} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{e}}_6 \quad T = 12, T_{\max} = 96, \Delta = -132, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 17, T_{\max} = 96, T \in [17, 96]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 4 & 0 & 1 & -2 & 0 \\ 2 & 4 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

4.416 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{e}_6$ + $\mathfrak{s}\mathfrak{u}_2$

$$T = 8 \quad [0] \quad 1^{111} 3^1 1^1 + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -4 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{112} 3^1 1^1 + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.7$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 0 & -4 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{113} 3^1 1^1 + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 739, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 9 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & -4 & 12 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \quad 1^{11} 3^1 21^{11} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & -2 & 3 & 0 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & -3 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3^1 21^{11} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 9 & 0 & 4 & 12 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & -2 & 7 & 0 \\ 5 & 12 & 0 & 7 & 14 & 8 \\ 3 & 5 & -3 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 3^1 21^{11} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.34 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & -2 & 3 & 0 \\ 3 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -3 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [8] \quad 1^{112} 2^3 21^{11} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 16.40 \rightarrow 17, T_{\max} = 93, T \in [17, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & 0 & -1 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} 2^3 21^{11} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 16.15 \rightarrow 17, T_{\max} = 93, T \in [17, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 & 0 \\ 1 & 3 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & 0 & -1 & 0 & -3 \end{pmatrix}_{b_0}$$

[10] $1^{11}3221^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{e}_6 + \text{su}_2$ $T = 10$, $T_{\max} = 91$, $\Delta = -129$, $\det = 127$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 14.46 \rightarrow 15$, $T_{\max} = 91$, $T \in [15, 91]$
 $\begin{pmatrix} 5 & 15 & 1 & 21 & 5 \\ 15 & 42 & 0 & 59 & 12 \\ 1 & 0 & -5 & 0 & -3 \\ 21 & 59 & 0 & 79 & 17 \\ 5 & 12 & -3 & 17 & 1 \end{pmatrix}_{b_0}$

4.417 NHC(-2)(-3) + f₄ + e₇

$T = 8$ [0] $1^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{e}_7$ $T = 8$, $T_{\max} = 131$, $\Delta = -243$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 16.11 \rightarrow 17$, $T_{\max} = 131$, $T \in [17, 131]$
 $\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 4 & 0 & -4 & -2 \\ 3 & 5 & -3 & -2 & -3 \end{pmatrix}_{b_0}$

$T = 9$ [1] $1^{11}3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{e}_7$ $T = 9$, $T_{\max} = 104$, $\Delta = -214$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 16.41 \rightarrow 17$, $T_{\max} = 104$, $T \in [17, 104]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 6 & 6 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 6 & 0 & -3 & -1 \\ 3 & 6 & -3 & -1 & -2 \end{pmatrix}_{b_0}$

[2] $1^{11}2321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{e}_7$ $T = 9$, $T_{\max} = 104$, $\Delta = -216$, $\det = -103$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 16.24 \rightarrow 17$, $T_{\max} = 104$, $T \in [17, 104]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & -5 & -3 \\ 2 & 3 & -1 & 0 & -3 & -3 \end{pmatrix}_{b_0}$

4.418 NHC(-2)(-3) + f₄ + e'₇

$T = 8$ [0] $1^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{e}'_7$ $T = 8$, $T_{\max} = 103$, $\Delta = -215$, $\det = 122$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 15.20 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$
 $\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 4 & 0 & -3 & -1 \\ 3 & 5 & -3 & -1 & -2 \end{pmatrix}_{b_0}$

$T = 9$ [1] $1^{11}3221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{e}'_7$ $T = 9$, $T_{\max} = 103$, $\Delta = -186$, $\det = -42$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 15.43 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 6 & 6 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 6 & 0 & -2 & 0 \\ 3 & 6 & -3 & 0 & -1 \end{pmatrix}_{b_0}$

[2] $1^{11}2321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{e}'_7$ $T = 9$, $T_{\max} = 103$, $\Delta = -188$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 15.33 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & -4 & -2 \\ 2 & 3 & -1 & 0 & -2 & -2 \end{pmatrix}_{b_0}$

4.419 NHC(-2)(-3) + f₄ + f₄

$T = 7$ [0] $1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{f}_4$ $T = 7$, $T_{\max} = 101$, $\Delta = -190$, $\det = -96$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.4$
 $T_{\min} = 13.33 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & -3 & -1 \end{pmatrix}_{b_0}$

$$\begin{aligned}
T = 8 \quad [1] \ 1^{\overline{1}} 3 2 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.4 \\
T_{\min} = 13.40 \rightarrow 14, T_{\max} = 101, T \in [14, 101] \\
\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 \\ 1 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & -1 & -3 & -1 \end{pmatrix}_{b_0} \\
[2] \ 1^{\overline{1}} 3 2 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.4 \\
T_{\min} = 13.48 \rightarrow 14, T_{\max} = 101, T \in [14, 101] \\
\begin{pmatrix} 5 & 13 & 1 & 5 \\ 13 & 30 & 0 & 10 \\ 1 & 0 & -5 & -3 \\ 5 & 10 & -3 & 1 \end{pmatrix}_{b_0} \\
[3] \ 1^{\overline{11}} 3 2 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.4 \\
T_{\min} = 13.57 \rightarrow 14, T_{\max} = 101, T \in [14, 101] \\
\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -5 & -3 \\ 5 & 9 & -3 & 1 \end{pmatrix}_{b_0} \\
[4] \ 1^{\overline{1}} 3^{\overline{11}} 1^{\overline{1}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 101, \Delta = -169, \det = 378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.4 \\
T_{\min} = 13.67 \rightarrow 14, T_{\max} = 101, T \in [14, 101] \\
\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 3 & 0 & -3 & -1 \end{pmatrix}_{b_0} \\
T = 9 \quad [5] \ 1^{\overline{11}} 2 3 1^{\overline{1}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4 \\
T_{\min} = 14, T_{\max} = 99, T \in [14, 99] \\
\begin{pmatrix} 1 & 5 & 5 & 1 & 3 \\ 5 & 18 & 16 & 0 & 8 \\ 5 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0} \\
[6] \ 1^{\overline{1}} 3 2 2 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4 \\
T_{\min} = 15.10 \rightarrow 16, T_{\max} = 93, T \in [16, 93] \\
\begin{pmatrix} 1 & 2 & 5 & 1 & 3 \\ 2 & 1 & 6 & 0 & 7 \\ 5 & 6 & 17 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 3 & 7 & -3 & -1 \end{pmatrix}_{b_0} \\
[7] \ 1^{\overline{1}} 3 2 2 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -133, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4 \\
T_{\min} = 13.50 \rightarrow 14, T_{\max} = 93, T \in [14, 93] \\
\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 6 & 6 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 6 & 0 & 0 & 2 \\ 3 & 6 & -3 & 2 & 1 \end{pmatrix}_{b_0} \\
[8] \ 1^{\overline{11}} 2 3 2 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4 \\
T_{\min} = 13.74 \rightarrow 14, T_{\max} = 95, T \in [14, 95] \\
\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 3 & 0 & 1 & -2 & 0 \\ 2 & 3 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0} \\
T = 12 \quad [9] \ 1^{\overline{1}} 3 1^{\overline{11}} 5 1 3 2 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{f}_4 \quad T = 12, T_{\max} = 24, \Delta = -106, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 13.4 \\
T_{\min} = 16, T_{\max} = 24, T \in [16, 24] \\
\begin{pmatrix} 5 & 15 & 1 & 5 \\ 15 & 43 & 1 & 13 \\ 1 & 1 & -3 & -1 \\ 5 & 13 & -1 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

4.420 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_2$

$$\begin{aligned}
T = 8 \quad [0] \ 1^{\overline{111}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 100, \Delta = -160, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.4 \\
T_{\min} = 13.34 \rightarrow 14, T_{\max} = 100, T \in [14, 100] \\
\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -3 & 4 & -1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1] $1^{112}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 100$, $\Delta = -160$, $\det = 352$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.9$
 $T_{\min} = 13.36 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 0 & -3 & 8 & -1 \end{pmatrix}_{b_0}$$
- [2] $1^{113}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 100$, $\Delta = -160$, $\det = 552$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 9 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & -3 & 12 & -1 \end{pmatrix}_{b_0}$$
- [3] $1^{114}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 100$, $\Delta = -160$, $\det = 832$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.4$
 $T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & -3 & 16 & -1 \end{pmatrix}_{b_0}$$
- [4] $1^{115}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 100$, $\Delta = -160$, $\det = 1192$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.5$
 $T_{\min} = 13.39 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & -2 & 0 & 15 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 40 & 15 & 0 & 148 & 20 \\ 3 & 5 & 0 & -3 & 20 & -1 \end{pmatrix}_{b_0}$$
- $T = 9$ [5] $1^{11}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -5$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.4$
 $T_{\min} = 15.40 \rightarrow 16$, $T_{\max} = 93$, $T \in [16, 93]$

$$\begin{pmatrix} 5 & 13 & 1 & 9 & 5 \\ 13 & 30 & 0 & 20 & 10 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 10 & -3 & 7 & -1 \end{pmatrix}_{b_0}$$
- [6] $1^{111}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -65$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.4$
 $T_{\min} = 13.92 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 5 & 12 & 1 & 19 & 5 \\ 12 & 25 & 0 & 40 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 19 & 40 & 0 & 63 & 15 \\ 5 & 9 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{13}21^{116} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.6$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 1 & 0 & 18 & 3 \\ 1 & 4 & 0 & 1 & 0 & 18 & 2 \\ 1 & 1 & 1 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 18 & 6 & 0 & 106 & 18 \\ 2 & 3 & 2 & -1 & -3 & 18 & -1 \end{pmatrix}_{b_0}$$
- [8] $1^{13}21^{117} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.7$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 1 & 0 & 21 & 3 \\ 1 & 4 & 0 & 1 & 0 & 21 & 2 \\ 1 & 1 & 1 & -3 & 0 & 7 & -1 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 21 & 7 & 0 & 145 & 21 \\ 2 & 3 & 2 & -1 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$
- [9] $1^{13}21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.4$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 & 3 \\ 1 & 4 & 0 & 1 & 0 & 3 & 2 \\ 1 & 1 & 1 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & -1 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$
- [10] $1^{13}21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.9$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 0 & 6 & 3 \\ 1 & 4 & 0 & 1 & 0 & 6 & 2 \\ 1 & 1 & 1 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 6 & 2 & 0 & 10 & 6 \\ 2 & 3 & 2 & -1 & -3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{\overline{1}} \overline{3} 2 1^{\overline{113}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 1 & 0 & 9 & 3 \\ 1 & 4 & 0 & 1 & 0 & 9 & 2 \\ 1 & 1 & 1 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 9 & 3 & 0 & 25 & 9 \\ 2 & 3 & 2 & -1 & -3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{\overline{1}} \overline{3} 2 1^{\overline{114}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 1 & 0 & 12 & 3 \\ 1 & 4 & 0 & 1 & 0 & 12 & 2 \\ 1 & 1 & 1 & -3 & 0 & 4 & -1 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 12 & 4 & 0 & 46 & 12 \\ 2 & 3 & 2 & -1 & -3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{\overline{1}} \overline{3} 2 1^{\overline{115}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 1 & 0 & 15 & 3 \\ 1 & 4 & 0 & 1 & 0 & 15 & 2 \\ 1 & 1 & 1 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 15 & 5 & 0 & 73 & 15 \\ 2 & 3 & 2 & -1 & -3 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{\overline{111}} \overline{3} 2 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -165, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.61 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 9 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [15] \quad 1^{\overline{111}} \overline{2} \overline{3} 2 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 19, T_{\max} = 22, T \in [19, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 & 0 \\ 1 & 3 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & 0 & 0 & 0 & -5 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{\overline{11}} \overline{3} 2 2 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 88, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 6 & 0 & 6 & 11 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 0 & 8 & 2 \\ 5 & 11 & 0 & 8 & 15 & 9 \\ 3 & 6 & -3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{\overline{11}} \overline{3} 2 2 1^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 103, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.63 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 6 & 0 & 6 & 17 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 0 & 13 & 2 \\ 7 & 17 & 0 & 13 & 36 & 14 \\ 3 & 6 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{\overline{14}} \overline{3} 2 2 1^{\overline{113}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 103, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.63 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 6 & 0 & 6 & 38 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 0 & 27 & 2 \\ 18 & 38 & 0 & 27 & 211 & 31 \\ 3 & 6 & -3 & 2 & 31 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{\overline{12}} \overline{3} 2 2 1^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 208, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.9$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 6 & 0 & 6 & 22 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 0 & 16 & 2 \\ 10 & 22 & 0 & 16 & 66 & 18 \\ 3 & 6 & -3 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{12} 3^{11} 2^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 343$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.9$
 $T_{\min} = 13.54 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 6 & 0 & 6 & 28 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 0 & 21 & 2 \\ 12 & 28 & 0 & 21 & 103 & 23 \\ 3 & 6 & -3 & 2 & 23 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{13} 3^{11} 2^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 408$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 13.50 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 6 & 0 & 6 & 33 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 0 & 24 & 2 \\ 15 & 33 & 0 & 24 & 151 & 27 \\ 3 & 6 & -3 & 2 & 27 & 1 \end{pmatrix}_{b_0}$$

4.421 $\text{NHC}(-2)(-3) + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$

$T = 9$ [0] $1^{111} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 92$, $\Delta = -130$, $\det = -544$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.4$
 $T_{\min} = 13.35 \rightarrow 14$, $T_{\max} = 92$, $T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 3 & 0 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 8 & 3 & 0 & 4 & 6 & 4 \\ 3 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & 0 & -3 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$T = 10$ [2] $1^{11} 3^{21} 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -102$, $\det = 160$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.4$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 4 & 0 & 1 & 0 & 3 & 3 & 2 \\ 1 & 1 & 1 & -3 & 0 & 1 & 1 & -1 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 1 & 0 & 1 & 3 & 3 \\ 1 & 3 & 3 & 1 & 0 & 3 & 1 & 3 \\ 2 & 3 & 2 & -1 & -3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3^{21} 1^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -102$, $\det = 160$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.9$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 4 & 0 & 1 & 0 & 3 & 6 & 2 \\ 1 & 1 & 1 & -3 & 0 & 1 & 2 & -1 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 1 & 0 & 1 & 6 & 3 \\ 2 & 6 & 6 & 2 & 0 & 6 & 10 & 6 \\ 2 & 3 & 2 & -1 & -3 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

4.422 $\text{NHC}(-2)(-3) + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{111} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -320$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.4$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 2 & 3 \\ 4 & 9 & 4 & 0 & 8 & 4 & 5 \\ 2 & 4 & -2 & 0 & 3 & 2 & 0 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 8 & 3 & 0 & 4 & 3 & 4 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 3 & 5 & 0 & -3 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

4.423 $\text{NHC}(-2)(-3) + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3)$

$T = 8$ [0] $1^{11} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3)$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 160$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.4$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & 0 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.424 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -320, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.40 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 & 3 \\ 4 & 9 & 4 & 0 & 4 & 8 & 5 \\ 2 & 4 & -2 & 0 & 2 & 3 & 0 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 3 & 8 & 3 & 0 & 3 & 4 & 4 \\ 3 & 5 & 0 & -3 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

4.425 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_2 \mathfrak{n}3 \text{mix}$

$$T = 8 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 101, \ \Delta = -162, \ \det = 360, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.40 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 & 5 \\ 2 & 4 & -2 & 0 & 5 & 0 & 0 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 12 & 5 & 0 & 12 & 6 & 6 \\ 3 & 5 & 0 & -3 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

4.426 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 101, \ \Delta = -169, \ \det = 328, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 & 0 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 8 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 0 & -3 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [1] \ 1^{\overline{11}} 3^{\overline{1}} 2^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -70, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.83 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 13 & 1 & 9 & 5 & 5 \\ 13 & 30 & 0 & 20 & 10 & 10 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 20 & 0 & 12 & 6 & 6 \\ 5 & 10 & -3 & 6 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1}} 3^{\overline{1}} 2^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -120, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 & 3 \\ 1 & 4 & 0 & 1 & 0 & 3 & 2 \\ 1 & 1 & 1 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 1 & 0 & 0 & 2 \\ 2 & 3 & 2 & -1 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.427 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \text{NHC}(-2)(-3)$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, \ T_{\max} = 31, \ \Delta = -117, \ \det = 108, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 31, \ T \in [12, 31]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 4 & 0 & 2 & 2 \\ 3 & 4 & -3 & 4 & 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, \ T_{\max} = 31, \ \Delta = -117, \ \det = 112, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.25 \rightarrow 13, \ T_{\max} = 31, \ T \in [13, 31]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 4 & 0 & 2 & 2 \\ 3 & 4 & -3 & 3 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, \ T_{\max} = 31, \ \Delta = -117, \ \det = 115, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 31, T \in [13, 31]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 4 & 3 \\ 3 & 5 & 0 & 4 & 3 & 3 \\ 3 & 4 & -3 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 4 \\ 2 & 5 & 0 & 4 & 1 & 3 \\ 3 & 5 & -3 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 0 & 1 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 2 & 2 \\ 1 & 1 & 3 & 0 & 2 & 0 & 2 \\ 2 & 2 & 2 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 12 & 5 \\ 9 & 13 & 1 & 20 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 20 & 0 & 25 & 9 \\ 5 & 7 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 4 & 5 \\ 9 & 13 & 1 & 5 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 5 & 0 & 0 & 2 \\ 5 & 7 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}} 31^{\overline{11}} 31^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 6 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 4 & 0 & 2 \\ 2 & 3 & 3 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{11}} 31^{\overline{11}} 31^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 4 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [9] \ 1^{\overline{11}} 31^{\overline{1}} 4^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [10] \ 1^{\overline{11}} 5^{\overline{1}} 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -69, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 5 \\ 9 & 13 & 10 & 0 & 7 \\ 6 & 10 & 5 & 1 & 5 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

4.428 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \text{NHC}(-2)(-3) + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 10 & 13 & 6 \\ 9 & 17 & 0 & 13 & 12 & 6 \\ 5 & 8 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

4.429 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -3 & 4 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{3}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 0 & 5 & 4 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 21 & 4 \\ 2 & 4 & 0 & 4 & 0 & 12 & 2 \\ 11 & 19 & 0 & 21 & 12 & 73 & 15 \\ 3 & 4 & -3 & 4 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 4 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & -3 & 4 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{2}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -396, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 0 & 5 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 15 & 4 \\ 2 & 4 & 0 & 4 & 0 & 9 & 2 \\ 8 & 14 & 0 & 15 & 9 & 37 & 11 \\ 3 & 4 & -3 & 4 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{2}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{2}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 0 & 5 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 18 & 4 \\ 2 & 4 & 0 & 4 & 0 & 12 & 2 \\ 10 & 18 & 0 & 18 & 12 & 58 & 14 \\ 3 & 4 & -3 & 4 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{2}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{4}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -936, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 0 & 5 & 4 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 30 & 4 \\ 2 & 4 & 0 & 4 & 0 & 18 & 2 \\ 16 & 28 & 0 & 30 & 18 & 154 & 22 \\ 3 & 4 & -3 & 4 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{2}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{3}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -956, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 0 & 5 & 4 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 24 & 4 \\ 2 & 4 & 0 & 4 & 0 & 15 & 2 \\ 13 & 23 & 0 & 24 & 15 & 100 & 18 \\ 3 & 4 & -3 & 4 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{113} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -87, \ \det = -1476, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 0 & 5 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 4 & 0 & 18 & 2 \\ 15 & 27 & 0 & 27 & 18 & 133 & 21 \\ 3 & 4 & -3 & 4 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{113} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -87, \ \det = -1796, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 0 & 5 & 4 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 33 & 4 \\ 2 & 4 & 0 & 4 & 0 & 21 & 2 \\ 18 & 32 & 0 & 33 & 21 & 193 & 25 \\ 3 & 4 & -3 & 4 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{114} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -87, \ \det = -2456, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 0 & 5 & 4 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 36 & 4 \\ 2 & 4 & 0 & 4 & 0 & 24 & 2 \\ 20 & 36 & 0 & 36 & 24 & 238 & 28 \\ 3 & 4 & -3 & 4 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

4.430 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 200, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 2 \\ 2 & 3 & 0 & 3 & 2 & 2 & 0 \\ 3 & 4 & -3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 600, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 9 & 3 & 4 \\ 2 & 4 & 0 & 4 & 0 & 6 & 2 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 5 & 7 \\ 2 & 3 & 0 & 3 & 2 & 5 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.431 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -89, \ \det = -160, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.25 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 4 & 2 & 3 \\ 2 & 4 & 0 & 4 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -89, \ \det = -200, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 4 & 0 & 2 & 2 \\ 2 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.432 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{\overline{111}3^{\overline{11}}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 4 & 0 & 2 & 2 \\ 2 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}3^{\overline{11}}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 600, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 4 & 0 & 2 & 6 & 2 \\ 2 & 3 & 0 & 3 & 2 & 0 & 5 & 2 \\ 5 & 9 & 0 & 9 & 6 & 5 & 13 & 7 \\ 3 & 4 & -3 & 4 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.433 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \mathfrak{n}3 \text{mix}$

$$T = 9 \quad [0] \quad 1^{\overline{111}3^{\overline{11}}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 6 & 4 \\ 2 & 4 & 0 & 4 & 0 & 5 & 2 \\ 4 & 7 & 0 & 6 & 5 & 7 & 5 \\ 3 & 4 & -3 & 4 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}3^{\overline{11}}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 12 & 4 \\ 2 & 4 & 0 & 4 & 0 & 8 & 2 \\ 7 & 12 & 0 & 12 & 8 & 25 & 9 \\ 3 & 4 & -3 & 4 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \quad 1^{\overline{11}1^{\overline{13}}1^{\overline{11}}4^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 3 & 3 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 4 & 5 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -2 & 3 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

4.434 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{\overline{111}3^{\overline{11}}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 4 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 4 & -3 & 3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{112}3^{\overline{11}}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 0 & 4 & 5 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 4 & 6 & 3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 3 \\ 4 & 8 & 0 & 6 & 6 & 9 & 5 \\ 3 & 4 & -3 & 3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 4 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 12 & 6 \\ 3 & 4 & -3 & 4 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

4.435 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$

$$T = 7 \quad [0] \ 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 100, \Delta = -166, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 100, \Delta = -166, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 5 & 11 & 1 & 5 \\ 11 & 20 & 0 & 8 \\ 1 & 0 & -5 & -3 \\ 5 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [2] \ 1^{11} 3 2 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 \\ 1 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -1 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3 2 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 2 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 4 & 5 & 9 & 0 & 5 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 3 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 2 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 13 & 1 & 5 \\ 13 & 30 & 0 & 10 \\ 1 & 0 & -4 & -2 \\ 5 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{11}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 5 & 0 & 0 & 2 \\ 3 & 5 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -4 & -2 \\ 5 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{11}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.12 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 1 & 0 & -2 & 0 \\ 2 & 2 & 2 & -3 & 0 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{11}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 2 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 10 & 13 & 0 & 6 \\ 5 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 3 & 3 \\ 2 & 2 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 275, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 3 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [13] $1^{11}3221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 1 & 4 & 0 & 0 & 2 \\ 2 & 2 & 3 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{11}231^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 4 & 0 & 0 & 2 \\ 2 & 4 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{11}3221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 6 & 6 & 0 & 6 \\ 2 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 6 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

- [16] $1^1 1^1 3 2 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -109$, $\det = -13$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 15.15 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$

$$\begin{pmatrix} 1 & 2 & 5 & 1 & 3 \\ 2 & 1 & 6 & 0 & 3 \\ 5 & 6 & 17 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 3 & 7 & -2 & -1 \end{pmatrix}_{b_0}$$
- [17] $1^1 1^1 3 2 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -109$, $\det = -15$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 13 & 12 & 24 & 7 \\ 12 & 10 & 21 & 6 \\ 24 & 21 & 42 & 12 \\ 7 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$
- [18] $1^1 1^1 3 2 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -109$, $\det = -15$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 6 & 6 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 6 & 0 & 1 & 3 \\ 3 & 6 & -3 & 3 & 2 \end{pmatrix}_{b_0}$$
- [19] $1^1 1^1 2 3 2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 22$, $\Delta = -111$, $\det = -27$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 18 & 1 & 8 \\ 1 & 1 & -3 & -1 \\ 5 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$
- [20] $1^1 1^1 2 3 2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 22$, $\Delta = -111$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 13.41 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 \\ 1 & 2 & 0 & 1 & 0 \\ 1 & 3 & 1 & -2 & 0 \\ 1 & 0 & 0 & 0 & -4 \\ 2 & 3 & 2 & 0 & -2 \end{pmatrix}_{b_0}$$
- [21] $1^1 1^1 2 3 2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 22$, $\Delta = -111$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 5 & 1 & 12 & 5 \\ 1 & -2 & 1 & 0 \\ 12 & 1 & 25 & 9 \\ 5 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$
- [22] $1^1 1^1 3 2 2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -109$, $\det = -33$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 14.45 \rightarrow 15$, $T_{\max} = 21$, $T \in [15, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 1 \\ 1 & 0 & 0 & -5 & 0 \\ 1 & 2 & 1 & 0 & -2 \end{pmatrix}_{b_0}$$
- [23] $1^1 1^1 2 3 2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 22$, $\Delta = -111$, $\det = -35$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 26 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 5 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$
- [24] $1^1 1^1 3 2 2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -109$, $\det = -43$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 13.23 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 5 & 8 & 1 & 5 \\ 8 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 5 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$
- [25] $1^1 1^1 4 1 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -136$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 13.82 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -4 & -2 \\ 5 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$
- [26] $1^1 1^1 3 2 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -109$, $\det = -65$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 5 & 0 & 4 & 4 \\ 2 & 3 & 3 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$
- [27] $1^1 2 3 1^1 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -67$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 1 & 1 & 5 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 5 & 1 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 0 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^1 1^{11} 31^1 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -136, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 13 & 0 & 7 \\ 1 & 0 & -4 & -2 \\ 5 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [29] \ 1^1 1^{11} 41321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 4 & 1 & 5 \\ 4 & 2 & 2 & 4 \\ 1 & 2 & -3 & -1 \\ 5 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [30] \ 1^{11} 31^1 5131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 11, T_{\max} = 23, \Delta = -110, \det = -35, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 14.83 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 26 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 5 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [31] \ 1^1 31^1 51321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 4 & 0 & 1 & 0 & 2 \\ 2 & 4 & -1 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

4.436 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8 + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \ 1^{11} 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 10 & 13 & 6 \\ 9 & 17 & 0 & 13 & 12 & 6 \\ 5 & 8 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

4.437 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{11} 1^{11} 31^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 1^{11} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 44 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 44 & 84 & 0 & 346 & 34 \\ 5 & 8 & -3 & 34 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 1^{11} 31^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.63 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 17 & 0 & 13 & 7 \\ 5 & 8 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.68 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{111}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{112}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 0 & -2 & 8 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 291, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 1 & 35 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 35 & 67 & 0 & 217 & 27 \\ 5 & 8 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{111}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 17 & 33 & 0 & 49 & 13 \\ 5 & 8 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{112}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 0 & -2 & 8 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{113}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 365, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.71 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 9 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{111}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 26 & 50 & 0 & 118 & 20 \\ 5 & 8 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -136, \quad \det = 424, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 13.7$
 $T_{\min} = 12.58 \rightarrow 13, \quad T_{\max} = 98, \quad T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 0 & -2 & 8 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{14} 3^1 1^1 + [\textcolor{red}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{s} \textcolor{brown}{o}_8 + \textcolor{red}{s} \textcolor{red}{u}_2 \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -136, \quad \det = 540, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.2$
 $T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 98, \quad T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & -2 & 16 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{113} 3^1 1^{\textcolor{red}{13}} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{s}o_8 + \textcolor{red}{s}u_2 \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -136, \quad \det = 581, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.0$
 $T_{\min} = 12.57 \rightarrow 13, \quad T_{\max} = 98, \quad T \in [13, 98]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & -2 & 12 & 0 \end{pmatrix}_{b_0}$

[16] $1^{113} 3^1 1^1 + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 709, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$
 $T_{\min} = 12.60 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & -2 & 12 & 0 \end{pmatrix}_{b_0}$

[17] $1^{11}5^13^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 765$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.3$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & -2 & 0 & 15 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 40 & 15 & 0 & 148 & 20 \\ 3 & 5 & 0 & -2 & 20 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{113} 3^1 1^{12} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -136, \quad \det = 781, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.0$
 $T_{\min} = 12.58 \rightarrow 13, \quad T_{\max} = 98, \quad T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{14} 3^1 1^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 924, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.2$
 $T_{\min} = 12.57 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & -2 & 16 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{11}4^13^11^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{s}o_8 + \textcolor{red}{s}u_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 1044, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.2$
 $T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & -2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{14} 3^{13} 1^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}o_8 + \textcolor{red}{s}u_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 1236, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.58 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & -2 & 16 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{114} 3^1 1^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{teal}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 1276, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.2$
 $T_{\min} = 12.59 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & -2 & 16 & 0 \end{pmatrix}_{b_0}$

$$\begin{aligned}
& T = 9 \quad [24] \quad 1^{11} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2 \\
& T_{\min} = 18, T_{\max} = 21, T \in [18, 21] \\
& \begin{pmatrix} 5 & 13 & 1 & 9 & 5 \\ 13 & 30 & 0 & 20 & 10 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 10 & -2 & 7 & -3 \end{pmatrix}_{b_0} \\
& [23] \quad 1^{11} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2 \\
& T_{\min} = 16, T_{\max} = 21, T \in [16, 21] \\
& \begin{pmatrix} 5 & 12 & 1 & 1 & 5 \\ 12 & 25 & 0 & 0 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 9 & -2 & 0 & -1 \end{pmatrix}_{b_0} \\
& [25] \quad 1^{11} 3 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7 \\
& T_{\min} = 16, T_{\max} = 21, T \in [16, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 0 & 6 & 3 \\ 1 & 4 & 0 & 1 & 0 & 6 & 2 \\ 1 & 1 & 1 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 6 & 6 & 2 & 0 & 10 & 6 \\ 2 & 3 & 2 & -1 & -2 & 6 & -3 \end{pmatrix}_{b_0} \\
& [27] \quad 1^{11} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2 \\
& T_{\min} = 15.11 \rightarrow 16, T_{\max} = 21, T \in [16, 21] \\
& \begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 3 \\ 3 & 3 & 5 & -3 & 3 & -2 \end{pmatrix}_{b_0} \\
& [26] \quad 1^{111} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2 \\
& T_{\min} = 16, T_{\max} = 21, T \in [16, 21] \\
& \begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 9 & 0 & 4 & 12 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 7 & 2 \\ 5 & 12 & 0 & 7 & 14 & 8 \\ 3 & 5 & -3 & 2 & 8 & -2 \end{pmatrix}_{b_0} \\
& [28] \quad 1^{11} 3 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2 \\
& T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 & 3 \\ 1 & 4 & 0 & 1 & 0 & 3 & 2 \\ 1 & 1 & 1 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & -1 & -2 & 3 & 0 \end{pmatrix}_{b_0} \\
& [29] \quad 1^{111} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2 \\
& T_{\min} = 13.13 \rightarrow 14, T_{\max} = 21, T \in [14, 21] \\
& \begin{pmatrix} 5 & 12 & 1 & 19 & 5 \\ 12 & 25 & 0 & 40 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 19 & 40 & 0 & 63 & 15 \\ 5 & 9 & -2 & 15 & 1 \end{pmatrix}_{b_0} \\
& [30] \quad 1^{11} 3 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2 \\
& T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 4 & 2 \\ 2 & 5 & 0 & 4 & 2 & 4 \\ 3 & 5 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0} \\
& [31] \quad 1^{111} 3 1^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2 \\
& T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26] \\
& \begin{pmatrix} 1 & 4 & 5 & 1 & 4 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 12 & 0 & 10 & 6 \\ 3 & 6 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0} \\
& [32] \quad 1^{11} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2 \\
& T_{\min} = 12.83 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 & 0 & 0 \end{pmatrix}_{b_0}
\end{aligned}$$

- [33] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.22 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 3 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$
- [34] $1^{14} 3 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 5 & 0 & 5 & 30 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 20 & 2 \\ 16 & 30 & 0 & 20 & 158 & 24 \\ 3 & 5 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$
- [35] $1^{12} 3 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 10 & 2 \\ 8 & 15 & 0 & 10 & 38 & 12 \\ 3 & 5 & -3 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$
- [36] $1^{111} 3 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -102, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 1 & 0 & 6 & 3 \\ 1 & 4 & 0 & 1 & 0 & 6 & 2 \\ 1 & 1 & 1 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 6 & 2 & 0 & 10 & 6 \\ 2 & 3 & 2 & -1 & -2 & 6 & 0 \end{pmatrix}_{b_0}$$
- [37] $1^{11} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 26, \Delta = -115, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 15 & 0 & 13 & 7 \\ 3 & 6 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$
- [38] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 9 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$
- [39] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -137, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 12 & 1 & 10 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 20 & 0 & 13 & 7 \\ 5 & 9 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$
- [40] $1^{11} 3 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 5 & 0 & 5 & 10 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 7 & 2 \\ 5 & 10 & 0 & 7 & 14 & 8 \\ 3 & 5 & -3 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$
- [41] $1^{11} 3 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 1 & 0 & 9 & 3 \\ 1 & 4 & 0 & 1 & 0 & 9 & 2 \\ 1 & 1 & 1 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 9 & 9 & 3 & 0 & 25 & 9 \\ 2 & 3 & 2 & -1 & -2 & 9 & 0 \end{pmatrix}_{b_0}$$
- [42] $1^{11} 3 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 1 & 0 & 12 & 3 \\ 1 & 4 & 0 & 1 & 0 & 12 & 2 \\ 1 & 1 & 1 & -3 & 0 & 4 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 12 & 4 & 0 & 46 & 12 \\ 2 & 3 & 2 & -1 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$

- [44] $1^{11}321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -310$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.66 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 11 & 2 \\ 7 & 15 & 0 & 11 & 32 & 12 \\ 3 & 5 & -3 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$
- [43] $1^{11}321^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -310$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 19 & 2 \\ 11 & 25 & 0 & 19 & 92 & 20 \\ 3 & 5 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$
- [45] $1^{11}321^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -350$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 5 & 0 & 5 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 15 & 2 \\ 9 & 20 & 0 & 15 & 58 & 16 \\ 3 & 5 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$
- [46] $1^{13}321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -390$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 17 & 2 \\ 13 & 25 & 0 & 17 & 104 & 20 \\ 3 & 5 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$
- [47] $1^{111}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -435$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 9 & 3 \\ 4 & 10 & 13 & 0 & 24 & 6 \\ 5 & 13 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 24 & 27 & 0 & 49 & 13 \\ 3 & 6 & 6 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$
- [48] $1^{12}321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -460$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 5 & 0 & 5 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 14 & 2 \\ 10 & 20 & 0 & 14 & 62 & 16 \\ 3 & 5 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$
- [49] $1^{12}321^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -740$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.64 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 18 & 2 \\ 12 & 25 & 0 & 18 & 94 & 20 \\ 3 & 5 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$
- [50] $1^{14}321^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -860$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 5 & 0 & 5 & 35 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 24 & 2 \\ 18 & 35 & 0 & 24 & 202 & 28 \\ 3 & 5 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$
- [51] $1^{13}321^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -910$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 5 & 0 & 5 & 30 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 21 & 2 \\ 15 & 30 & 0 & 21 & 142 & 24 \\ 3 & 5 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$
- [52] $1^{12}321^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -940$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.68 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 5 & 0 & 5 & 30 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 22 & 2 \\ 14 & 30 & 0 & 22 & 134 & 24 \\ 3 & 5 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

- [53] $1^{13}3^{11}21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -1350, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 5 & 0 & 5 & 35 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 25 & 2 \\ 17 & 35 & 0 & 25 & 188 & 28 \\ 3 & 5 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$
- [54] $1^{14}3^{11}21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -1540, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 5 & 0 & 5 & 40 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 28 & 2 \\ 20 & 40 & 0 & 28 & 254 & 32 \\ 3 & 5 & -3 & 2 & 32 & 1 \end{pmatrix}_{b_0}$$
- $T = 10$ [55] $1^{11}3^{11}221^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 6 & 0 & 6 & 6 & 6 \\ 2 & 6 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 5 & 0 & 3 & 5 \\ 3 & 6 & 2 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$
- [56] $1^{111}2^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 34, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 22 & 5 \\ 12 & 26 & 1 & 48 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 48 & 0 & 86 & 18 \\ 5 & 10 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$
- [57] $1^{11}3^{11}221^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 6 & 6 & 0 & 11 & 6 \\ 2 & 6 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 11 & 8 & 0 & 15 & 9 \\ 3 & 6 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$
- [58] $1^{11}3^{11}221^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 49, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 6 & 0 & 6 & 11 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 1 & 8 & 3 \\ 5 & 11 & 0 & 8 & 15 & 9 \\ 3 & 6 & -3 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$
- [59] $1^{11}3^{11}221^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.69 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 6 & 0 & 6 & 17 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 1 & 13 & 3 \\ 7 & 17 & 0 & 13 & 36 & 14 \\ 3 & 6 & -3 & 3 & 14 & 2 \end{pmatrix}_{b_0}$$
- [60] $1^{111}2^{112}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 70, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 14 & 5 \\ 12 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 28 & 0 & 26 & 10 \\ 5 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$
- [61] $1^{114}2^{11}3^{11}21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 78 & 5 \\ 12 & 26 & 1 & 172 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 78 & 172 & 0 & 1106 & 64 \\ 5 & 10 & -1 & 64 & 3 \end{pmatrix}_{b_0}$$
- [62] $1^{11}4^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 101, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.56 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 14 & 0 & 13 & 7 \\ 5 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{12}3^{11}221^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 104$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 6 & 6 & 0 & 22 & 6 \\ 2 & 6 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 22 & 16 & 0 & 66 & 18 \\ 3 & 6 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[64] $1^{12}3^{11}221^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 106$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 6 & 0 & 6 & 22 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 1 & 16 & 3 \\ 10 & 22 & 0 & 16 & 66 & 18 \\ 3 & 6 & -3 & 3 & 18 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{11}3^{11}221^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 110$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 6 & 6 & 0 & 17 & 6 \\ 2 & 6 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 17 & 13 & 0 & 36 & 14 \\ 3 & 6 & 2 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{112}23^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -81$, $\det = 126$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.87 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 34 & 5 \\ 12 & 26 & 1 & 76 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 76 & 0 & 210 & 28 \\ 5 & 10 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[67] $1^{111}2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -81$, $\det = 131$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.85 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 13 & 5 \\ 12 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 28 & 0 & 26 & 10 \\ 5 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[68] $1^{113}23^{11}21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -81$, $\det = 146$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 56 & 5 \\ 12 & 26 & 1 & 124 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 56 & 124 & 0 & 570 & 46 \\ 5 & 10 & -1 & 46 & 3 \end{pmatrix}_{b_0}$$

[69] $1^{12}3^{11}221^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 154$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 6 & 0 & 6 & 34 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 1 & 26 & 3 \\ 14 & 34 & 0 & 26 & 150 & 28 \\ 3 & 6 & -3 & 3 & 28 & 2 \end{pmatrix}_{b_0}$$

[70] $1^{112}2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -81$, $\det = 199$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.98 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 25 & 5 \\ 12 & 26 & 1 & 56 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 56 & 0 & 110 & 20 \\ 5 & 10 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[71] $1^{13}3^{11}221^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 201$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 6 & 0 & 6 & 33 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 1 & 24 & 3 \\ 15 & 33 & 0 & 24 & 151 & 27 \\ 3 & 6 & -3 & 3 & 27 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{13}3^{11}221^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 204$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 6 & 6 & 0 & 33 & 6 \\ 2 & 6 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 33 & 24 & 0 & 151 & 27 \\ 3 & 6 & 2 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{12} 3^{221} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 214$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.64 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 6 & 0 & 6 & 28 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 1 & 21 & 3 \\ 12 & 28 & 0 & 21 & 103 & 23 \\ 3 & 6 & -3 & 3 & 23 & 2 \end{pmatrix}_{b_0}$$

[74] $1^{12} 3^{221} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 230$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 6 & 6 & 0 & 28 & 6 \\ 2 & 6 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 28 & 21 & 0 & 103 & 23 \\ 3 & 6 & 2 & -2 & 23 & 2 \end{pmatrix}_{b_0}$$

[75] $1^{114} 2^3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -81$, $\det = 294$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 68 & 5 \\ 12 & 26 & 1 & 152 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 68 & 152 & 0 & 846 & 56 \\ 5 & 10 & -1 & 56 & 3 \end{pmatrix}_{b_0}$$

[76] $1^{112} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -81$, $\det = 299$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 35 & 5 \\ 12 & 26 & 1 & 76 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 35 & 76 & 0 & 210 & 28 \\ 5 & 10 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[78] $1^{112} 2^3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -81$, $\det = 314$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.87 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 26 & 5 \\ 12 & 26 & 1 & 56 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 56 & 0 & 110 & 20 \\ 5 & 10 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[77] $1^{112} 2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -81$, $\det = 314$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 36 & 5 \\ 12 & 26 & 1 & 76 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 36 & 76 & 0 & 210 & 28 \\ 5 & 10 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[79] $1^{131} 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -106$, $\det = 317$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.56 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 5 & 10 & 1 & 7 & 5 \\ 10 & 13 & 0 & 10 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 10 & 0 & 5 & 5 \\ 5 & 7 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{14} 3^{221} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 334$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 6 & 0 & 6 & 44 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 1 & 32 & 3 \\ 20 & 44 & 0 & 32 & 270 & 36 \\ 3 & 6 & -3 & 3 & 36 & 2 \end{pmatrix}_{b_0}$$

[81] $1^{113} 2^3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -81$, $\det = 395$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 57 & 5 \\ 12 & 26 & 1 & 124 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 57 & 124 & 0 & 570 & 46 \\ 5 & 10 & -1 & 46 & 3 \end{pmatrix}_{b_0}$$

[82] $1^{13} 3^{221} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 405$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.64 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 6 & 0 & 6 & 39 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 1 & 29 & 3 \\ 17 & 39 & 0 & 29 & 204 & 32 \\ 3 & 6 & -3 & 3 & 32 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{114} 2^3 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 419, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 79 & 5 \\ 12 & 26 & 1 & 172 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 79 & 172 & 0 & 1106 & 64 \\ 5 & 10 & -1 & 64 & 3 \end{pmatrix}_{b_0}$$

[84] $1^{113} 2^3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 486, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 38 & 5 \\ 12 & 26 & 1 & 84 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 38 & 84 & 0 & 250 & 30 \\ 5 & 10 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[85] $1^{11} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 486, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 1 & 16 & 5 \\ 10 & 13 & 0 & 24 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 24 & 0 & 38 & 12 \\ 5 & 7 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[86] $1^{113} 2^3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 486, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 58 & 5 \\ 12 & 26 & 1 & 124 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 58 & 124 & 0 & 570 & 46 \\ 5 & 10 & -1 & 46 & 3 \end{pmatrix}_{b_0}$$

[87] $1^{113} 2^3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 495, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 47 & 5 \\ 12 & 26 & 1 & 104 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 47 & 104 & 0 & 390 & 38 \\ 5 & 10 & -1 & 38 & 3 \end{pmatrix}_{b_0}$$

[88] $1^{114} 2^3 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 586, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 80 & 5 \\ 12 & 26 & 1 & 172 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 80 & 172 & 0 & 1106 & 64 \\ 5 & 10 & -1 & 64 & 3 \end{pmatrix}_{b_0}$$

[89] $1^{114} 2^3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 619, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 59 & 5 \\ 12 & 26 & 1 & 132 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 59 & 132 & 0 & 626 & 48 \\ 5 & 10 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

[90] $1^{113} 2^3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 686, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 48 & 5 \\ 12 & 26 & 1 & 104 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 48 & 104 & 0 & 390 & 38 \\ 5 & 10 & -1 & 38 & 3 \end{pmatrix}_{b_0}$$

[91] $1^{114} 2^3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 719, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 69 & 5 \\ 12 & 26 & 1 & 152 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 69 & 152 & 0 & 846 & 56 \\ 5 & 10 & -1 & 56 & 3 \end{pmatrix}_{b_0}$$

[92] $1^{114}2^13^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 986, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 70 & 5 \\ 12 & 26 & 1 & 152 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 70 & 152 & 0 & 846 & 56 \\ 5 & 10 & -1 & 56 & 3 \end{pmatrix}_{b_0}$$

[93] $1^{114}2^13^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 986, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 60 & 5 \\ 12 & 26 & 1 & 132 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 60 & 132 & 0 & 626 & 48 \\ 5 & 10 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

4.438 NHC(-2)(-3) + f₄ + so₈ + su₂ + su₂

$T = 9$ [0] $1^{112}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.7$

$T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 0 & 6 & 0 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 16 & 6 & 0 & 0 & 22 & 8 \\ 3 & 5 & 0 & -2 & 0 & 8 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 12 & 7 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -3 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -3 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{1111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 6 & 4 \\ 4 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & 0 & -2 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^11^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 35 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 35 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -3 & 7 & 27 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{1111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 6 & 4 \\ 3 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & 0 & -2 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

14.2

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -3 & 7 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1111}} 3^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 4 & 8 & 3 & 0 & 6 & 4 \\ 3 & 5 & 0 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -482, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 26 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 26 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -3 & 7 & 20 & 1 \end{pmatrix}_{b_0}$$

4.439 NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 9 & 7 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -3 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 2 & 3 \\ 4 & 9 & 4 & 0 & 8 & 4 & 5 \\ 2 & 4 & -2 & 0 & 3 & 2 & 0 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 3 & 4 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 3 & 5 & 0 & -2 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 0 & 8 & 4 & 5 \\ 2 & 4 & -2 & 0 & 3 & 2 & 0 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 3 & 0 & 4 & 3 & 4 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 3 & 5 & 0 & -2 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

4.440 NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \text{ n3mix}$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 15 & 5 \\ 11 & 20 & 0 & 17 & 28 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 24 & 7 \\ 15 & 28 & 0 & 24 & 37 & 11 \\ 5 & 8 & -3 & 7 & 11 & 1 \end{pmatrix}_{b_0}$$

4.441 NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -2 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -215$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -3 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -264$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -735$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -3 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

4.442 NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 8$ [0] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 8$, $T_{\max} = 98$, $\Delta = -138$, $\det = 116$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & 0 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 8$, $T_{\max} = 98$, $\Delta = -138$, $\det = 174$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.68 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 5 \\ 11 & 20 & 0 & 11 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 11 & 0 & 4 & 4 \\ 5 & 8 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [2] $1^{13}3^12^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 22$, $\Delta = -111$, $\det = -59$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$

$$\begin{pmatrix} 5 & 13 & 1 & 6 & 5 \\ 13 & 30 & 0 & 13 & 10 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 13 & 0 & 4 & 4 \\ 5 & 10 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11}3^12^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 22$, $\Delta = -111$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 5 \\ 12 & 25 & 0 & 12 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 12 & 0 & 4 & 4 \\ 5 & 9 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{11}3^13^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -112$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 3 \\ 4 & 10 & 13 & 0 & 4 & 6 \\ 5 & 13 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 0 & 0 & 2 \\ 3 & 6 & 6 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^11^131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -187, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 32, \ T \in [13, 32]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 & 3 \\ 4 & 10 & 13 & 0 & 8 & 6 \\ 5 & 13 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 9 & 0 & 4 & 4 \\ 3 & 6 & 6 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [6] \ 1^{11}4^{11}13^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 22, \ \Delta = -108, \ \det = 2, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 17, \ T_{\max} = 22, \ T \in [17, 22]$$

$$\begin{pmatrix} 5 & 12 & 1 & 1 & 5 \\ 12 & 25 & 0 & 0 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 9 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11}23^{11}1^131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 14, \ \Delta = -90, \ \det = 20, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 \\ 1 & 4 & 1 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 \\ 1 & 3 & 4 & 0 & 0 & 2 \\ 2 & 4 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11}3^{11}4^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 22, \ \Delta = -108, \ \det = 202, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.63 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 1 & 1 & 5 \\ 10 & 13 & 0 & 0 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 7 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [9] \ 1^{11}4^{11}13^121^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 14, \ \Delta = -81, \ \det = -14, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 5 & 4 & 1 & 1 & 5 \\ 4 & 2 & 2 & 0 & 4 \\ 1 & 2 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 4 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [10] \ 1^{11}3^{11}1^15^{11}131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 15, \ \Delta = -82, \ \det = 66, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 14.94 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 26 & 1 & 20 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 10 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

4.443 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -218, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 13 & 7 \\ 5 & 8 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -222, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 & 3 \\ 4 & 9 & 4 & 0 & 4 & 8 & 5 \\ 2 & 4 & -2 & 0 & 2 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 3 & 8 & 3 & 0 & 3 & 4 & 4 \\ 3 & 5 & 0 & -2 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -302, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 4 & 3 \\ 4 & 9 & 4 & 0 & 4 & 8 & 5 \\ 2 & 4 & -2 & 0 & 2 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 4 & 8 & 3 & 0 & 3 & 4 & 4 \\ 3 & 5 & 0 & -2 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

4.444 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -392, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, \ T_{\max} = 32, \ T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 12 & 6 \\ 5 & 8 & -3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

4.445 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \mathfrak{n}3\text{mix}$

$$T = 8 \quad [0] \ 1^{11}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -138, \ \det = 50, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.60 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 24 & 5 \\ 11 & 20 & 0 & 45 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 24 & 45 & 0 & 100 & 18 \\ 5 & 8 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -138, \ \det = 171, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 5 \\ 11 & 20 & 0 & 28 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 15 & 28 & 0 & 37 & 11 \\ 5 & 8 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -138, \ \det = 221, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.60 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 5 & 0 & 12 & 6 \\ 3 & 5 & 0 & -2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -138, \ \det = 261, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.66 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 5 & 0 & 12 & 6 \\ 3 & 5 & 0 & -2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \ 1^{11}31^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -240, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.90 \rightarrow 13, \ T_{\max} = 32, \ T \in [13, 32]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 6 & 3 \\ 4 & 10 & 13 & 0 & 16 & 6 \\ 5 & 13 & 12 & 0 & 17 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 16 & 17 & 0 & 20 & 8 \\ 3 & 6 & 6 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [5] \ 1^{11}4^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -108, \ \det = 233, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.59 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 1 & 10 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 14 & 0 & 13 & 7 \\ 5 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [6] \quad 1^{111} 3^1 5^1 13^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, \quad T_{\max} = 15, \quad \Delta = -82, \quad \det = 94, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 14.87 \rightarrow 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 5 & 12 & 1 & 20 & 5 \\ 12 & 26 & 1 & 44 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 44 & 0 & 70 & 16 \\ 5 & 10 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

4.446 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -212, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.64 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 9 & 5 \\ 11 & 20 & 0 & 28 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 28 & 0 & 37 & 24 & 11 \\ 9 & 17 & 0 & 24 & 13 & 7 \\ 5 & 8 & -3 & 11 & 7 & 1 \end{pmatrix}_{b_0}$$

4.447 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + $\mathfrak{su}_3$

$$T = 8 \quad [0] \quad 1^{11} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -145, \quad \det = 19, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 15.4$$

$$T_{\min} = 14.21 \rightarrow 15, \quad T_{\max} = 98, \quad T \in [15, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 18 & 5 \\ 11 & 20 & 0 & 34 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 18 & 34 & 0 & 57 & 13 \\ 5 & 8 & -3 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -145, \quad \det = 74, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.05 \rightarrow 14, \quad T_{\max} = 98, \quad T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -145, \quad \det = 196, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.88 \rightarrow 13, \quad T_{\max} = 98, \quad T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 17 & 0 & 12 & 6 \\ 5 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -145, \quad \det = 235, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 98, \quad T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 0 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -145, \quad \det = 259, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.84 \rightarrow 13, \quad T_{\max} = 98, \quad T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 0 & -2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \quad 1^{11} 3^1 2^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, \quad T_{\max} = 32, \quad \Delta = -117, \quad \det = -23, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.4$$

$$T_{\min} = 14.83 \rightarrow 15, \quad T_{\max} = 32, \quad T \in [15, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 0 & 6 & 3 \\ 1 & 4 & 0 & 1 & 0 & 6 & 2 \\ 1 & 1 & 1 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 6 & 6 & 2 & 0 & 9 & 5 \\ 2 & 3 & 2 & -1 & -2 & 5 & -2 \end{pmatrix}_{b_0}$$

[6] $1^{11}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -40$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{11}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -47$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.13 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 0 & 1 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & -2 & 1 & 0 \\ 1 & 1 & 3 & 0 & 1 & 0 & 2 \\ 2 & 2 & 2 & -3 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{11}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 5 & 13 & 1 & 9 & 5 \\ 13 & 30 & 0 & 20 & 10 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 20 & 0 & 12 & 6 \\ 5 & 10 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{11}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -53$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.23 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 & 3 \\ 1 & 4 & 0 & 1 & 0 & 3 & 2 \\ 1 & 1 & 1 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 3 & 1 & 0 & 0 & 2 \\ 2 & 3 & 2 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{11}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -62$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.10 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 3 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{11}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -110$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.49 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 4 & 2 \\ 2 & 5 & 0 & 4 & 1 & 3 \\ 3 & 5 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{11}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -147$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.12 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & 0 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{11}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -240$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 5 & 0 & 5 & 10 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 0 & 7 & 2 \\ 5 & 10 & 0 & 7 & 13 & 7 \\ 3 & 5 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 82$, $\Delta = -124$, $\det = -277$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 82$, $T \in [14, 82]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 0 & 3 \\ 1 & 3 & 3 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & 0 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

$T = 10$ [15] $1^{11}2^321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{so}_8 + \text{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -90$, $\det = 45$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 5 & 1 & 12 & 1 & 5 \\ 1 & -2 & 1 & 0 & 0 \\ 12 & 1 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 0 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{11} 3^{11} 221^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 6 & 0 & 6 & 11 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 1 & 8 & 3 \\ 5 & 11 & 0 & 8 & 14 & 8 \\ 3 & 6 & -3 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{11} 3^{11} 221^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 98, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.61 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 5 & 8 & 1 & 1 & 5 \\ 8 & 10 & 1 & 0 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 6 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

4.448 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$

$T = 9$ $[0] 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 0 & 3 & 0 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 3 & 0 & 0 & 4 & 4 \\ 3 & 5 & 0 & -2 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 15 & 6 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -3 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

4.449 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9$ $[0] 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 9 & 6 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -3 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

4.450 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2$

$T = 7$ $[0] 21^1 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 7, T_{\max} = 97, \Delta = -137, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 3 & 4 & -2 & 6 & 0 \\ 6 & 16 & 6 & 22 & 8 \\ 3 & 5 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{112}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 3 & 4 & -2 & 6 & 0 \\ 6 & 16 & 6 & 22 & 8 \\ 3 & 5 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.82 \rightarrow 15, T_{\max} = 97, T \in [15, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 \\ 5 & 8 & 3 & 4 & 4 \\ 3 & 5 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$

[4] $1^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 97, T \in [14, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 10 & 0 & 22 & 8 \\ 3 & 4 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

[6] $21^131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 3 & 4 & -2 & 3 & 0 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 3 & 4 & -2 & 3 & 0 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 \\ 3 & 4 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 4 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{12}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 4 & 8 & 0 & 10 & 6 \\ 3 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{112}31^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 97, T \in [14, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & -2 & 6 & 0 \\ 9 & 16 & 6 & 22 & 8 \\ 3 & 5 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{13}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 12 & 0 & 25 & 9 \\ 3 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{14}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 16 & 0 & 46 & 12 \\ 3 & 4 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{13}1^{11}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{15}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 10 & 20 & 0 & 73 & 15 \\ 3 & 4 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 \\ 4 & 8 & 3 & 4 & 4 \\ 3 & 5 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{16}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.4$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 12 & 24 & 0 & 106 & 18 \\ 3 & 4 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 5 & 9 & 0 & 13 & 7 \\ 3 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{17}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.22 \rightarrow 14, T_{\max} = 97, T \in [14, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 14 & 28 & 0 & 145 & 21 \\ 3 & 4 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[19] $1^{13}1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & -2 & 6 & 0 \\ 6 & 16 & 6 & 22 & 8 \\ 3 & 5 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{113}3^11^4 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & -2 & 9 & 0 \\ 13 & 24 & 9 & 52 & 12 \\ 3 & 5 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{11}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 14 & 0 & 37 & 11 \\ 3 & 4 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{112}3^11^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & -2 & 6 & 0 \\ 8 & 16 & 6 & 22 & 8 \\ 3 & 5 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{12}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 97, \ \Delta = -137, \ \det = -148, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 97, \ T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 13 & 0 & 28 & 10 \\ 3 & 4 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11}3^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 97, \ \Delta = -137, \ \det = -151, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.78 \rightarrow 12, \ T_{\max} = 97, \ T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 11 & 19 & 0 & 73 & 15 \\ 3 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{112}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 97, \ \Delta = -137, \ \det = -158, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.7$$

$$T_{\min} = 12.10 \rightarrow 13, \ T_{\max} = 97, \ T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & -2 & 6 & 0 \\ 7 & 16 & 6 & 22 & 8 \\ 3 & 5 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{11}3^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 97, \ \Delta = -137, \ \det = -163, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.09 \rightarrow 13, \ T_{\max} = 97, \ T \in [13, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 14 & 24 & 0 & 121 & 19 \\ 3 & 4 & -3 & 19 & 0 \end{pmatrix}_{b_0}$$

$$[27] \ 13^{11}3^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 97, \ \Delta = -137, \ \det = -187, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.55 \rightarrow 13, \ T_{\max} = 97, \ T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & -2 & 9 & 0 \\ 9 & 24 & 9 & 52 & 12 \\ 3 & 5 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{13}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 97, \ \Delta = -137, \ \det = -207, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.49 \rightarrow 12, \ T_{\max} = 97, \ T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 17 & 0 & 49 & 13 \\ 3 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{12}3^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 97, \ \Delta = -137, \ \det = -238, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.50 \rightarrow 12, \ T_{\max} = 97, \ T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 10 & 18 & 0 & 58 & 14 \\ 3 & 4 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{113}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 97, \ \Delta = -137, \ \det = -241, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 97, \ T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & -2 & 9 & 0 \\ 12 & 24 & 9 & 52 & 12 \\ 3 & 5 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{14}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 97, \ \Delta = -137, \ \det = -268, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.55 \rightarrow 12, \ T_{\max} = 97, \ T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 11 & 21 & 0 & 76 & 16 \\ 3 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{113}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 97, \ \Delta = -137, \ \det = -273, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.17 \rightarrow 13, \ T_{\max} = 97, \ T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & -2 & 9 & 0 \\ 10 & 24 & 9 & 52 & 12 \\ 3 & 5 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[33] \ 21^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 97, \ \Delta = -137, \ \det = -276, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.54 \rightarrow 12, \ T_{\max} = 97, \ T \in [12, 97]$$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & -2 & 12 & 0 \\ 14 & 32 & 12 & 94 & 16 \\ 3 & 5 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{13} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -483, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.53 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 15 & 27 & 0 & 133 & 21 \\ 3 & 4 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 21^{112} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -486, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 4 & 8 & 3 & 12 & 4 \\ 8 & 9 & 4 & 16 & 5 \\ 3 & 4 & -2 & 6 & 0 \\ 12 & 16 & 6 & 22 & 8 \\ 4 & 5 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{13} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -603, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 18 & 32 & 0 & 193 & 25 \\ 3 & 4 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{14} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -652, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.52 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 17 & 31 & 0 & 172 & 24 \\ 3 & 4 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{14} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -826, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 20 & 36 & 0 & 238 & 28 \\ 3 & 4 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 21^{113} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -836, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 4 & 8 & 3 & 18 & 4 \\ 8 & 9 & 4 & 24 & 5 \\ 3 & 4 & -2 & 9 & 0 \\ 18 & 24 & 9 & 52 & 12 \\ 4 & 5 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 21^{114} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -1326, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 4 & 8 & 3 & 24 & 4 \\ 8 & 9 & 4 & 32 & 5 \\ 3 & 4 & -2 & 12 & 0 \\ 24 & 32 & 12 & 94 & 16 \\ 4 & 5 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 21^{115} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -1956, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 4 & 8 & 3 & 30 & 4 \\ 8 & 9 & 4 & 40 & 5 \\ 3 & 4 & -2 & 15 & 0 \\ 30 & 40 & 15 & 148 & 20 \\ 4 & 5 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [55] \quad 22^1 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 25 & 20 & 9 \\ 12 & 20 & 13 & 7 \\ 6 & 9 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{111} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 25 & 20 & 9 \\ 12 & 20 & 13 & 7 \\ 6 & 9 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{111} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 10 & 16 & 26 & 6 \\ 16 & 25 & 40 & 9 \\ 26 & 40 & 63 & 15 \\ 6 & 9 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{\overline{11}} 1^{\overline{1}} 3 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$$

$$\begin{pmatrix} 5 & 13 & 9 & 5 \\ 13 & 30 & 20 & 10 \\ 9 & 20 & 13 & 7 \\ 5 & 10 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[57] \quad 2^{\overline{11}} 2^{\overline{1}} 3 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 10 & 19 & 12 & 6 \\ 19 & 30 & 20 & 10 \\ 12 & 20 & 13 & 7 \\ 6 & 10 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{\overline{14}} 3 1^{\overline{11}} 3 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 18.57 \rightarrow 19, T_{\max} = 28, T \in [19, 28]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 15 & 18 & 0 & 76 & 16 \\ 2 & 3 & 3 & -3 & 16 & -6 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{\overline{11}} 1^{\overline{1}} 3 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 18 & 12 & 6 \\ 18 & 30 & 20 & 10 \\ 12 & 20 & 13 & 7 \\ 6 & 10 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{\overline{1}} 1^{\overline{11}} 3 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 5 & 13 & 1 & 5 \\ 13 & 30 & 0 & 10 \\ 1 & 0 & -2 & 0 \\ 5 & 10 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{\overline{11}} 1^{\overline{12}} 3 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 5 & 12 & 11 & 5 \\ 12 & 25 & 20 & 9 \\ 11 & 20 & 13 & 7 \\ 5 & 9 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{\overline{1}} 1^{\overline{12}} 3 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 2 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{\overline{1}} 1^{\overline{1}} 3 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 3 \\ 2 & 2 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{\overline{11}} 1^{\overline{1}} 3 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 2 & 1 & 0 & 3 & 1 & 3 \\ 2 & 2 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{\overline{12}} 1^{\overline{1}} 3 21^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 4 & 6 & 0 & 8 & 26 & 10 \\ 2 & 2 & -1 & 0 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{\overline{12}} 1^{\overline{1}} 3 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 2 & 0 & 6 & 4 \\ 2 & 2 & 2 & -3 & 4 & -1 \end{pmatrix}_{b_0}$$

- [67] $1^{11} 3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 13$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13.54 \rightarrow 14$, $T_{\max} = 20$, $T \in [14, 20]$
 $\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -2 & 0 \\ 5 & 9 & 0 & 1 \end{pmatrix}_{b_0}$
- [68] $1^{11} 3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 15$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 \\ 3 & 5 & -3 & 4 & 1 \end{pmatrix}_{b_0}$
- [70] $1^{111} 1 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 18$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.56 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$
 $\begin{pmatrix} 5 & 12 & 19 & 5 \\ 12 & 25 & 40 & 9 \\ 19 & 40 & 63 & 15 \\ 5 & 9 & 15 & 2 \end{pmatrix}_{b_0}$
- [69] $1^{12} 3 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 18$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 15$, $T_{\max} = 20$, $T \in [15, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 1 & 6 & 3 \\ 1 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 4 & 6 & 6 & 2 & 10 & 6 \\ 2 & 3 & 2 & -1 & 6 & -2 \end{pmatrix}_{b_0}$
- [71] $1^1 1^1 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 5 & 0 & 2 & 4 \\ 3 & 5 & -3 & 4 & 2 \end{pmatrix}_{b_0}$
- [72] $1^{11} 3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$
- [74] $1^{11} 3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$
- [75] $1^{11} 3 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 5 & 10 & 0 & 33 & 11 \\ 2 & 2 & 2 & -3 & 11 & 0 \end{pmatrix}_{b_0}$
- [76] $1^{11} 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 4 & 4 & 0 & 7 & 16 & 8 \\ 2 & 2 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$
- [77] $1^{112} 1^1 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 3 & 1 & 0 & 3 & 1 & 3 \\ 2 & 2 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$
- [78] $1^{14} 1^1 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 15.33 \rightarrow 16$, $T_{\max} = 20$, $T \in [16, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 1 & 0 & 3 & 1 & 3 \\ 2 & 2 & -1 & 0 & 3 & -3 \end{pmatrix}_{b_0}$$

[73] $1^{11} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 2 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{16} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.4$
 $T_{\min} = 15.75 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 6 & 12 & 0 & 6 & 70 & 12 \\ 2 & 2 & -1 & 0 & 12 & -3 \end{pmatrix}_{b_0}$$

[80] $1^{11} 31^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 3 & 3 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[81] $1^{15} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 14.38 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 5 & 10 & 0 & 5 & 48 & 10 \\ 2 & 2 & -1 & 0 & 10 & -2 \end{pmatrix}_{b_0}$$

[82] $1^{14} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 4 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

[83] $1^{13} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

[84] $1^{12} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{11} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{13} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 4 & 1 & 0 & 3 & 1 & 3 \\ 2 & 2 & -1 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

[87] $1^{12} 3 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 6 & 7 & 0 & 11 & 45 & 13 \\ 2 & 2 & -1 & 0 & 13 & -1 \end{pmatrix}_{b_0}$$

[88] $1^{17}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 17.37 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 7 & 14 & 0 & 7 & 96 & 14 \\ 2 & 2 & -1 & 0 & 14 & -5 \end{pmatrix}_{b_0}$$

[89] $1^{11}321^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 3 & 3 \\ 1 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

[90] $1^{11}31^{131} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{111}31^{131} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 6 & 0 & 13 & 7 \\ 2 & 3 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[92] $1^{111}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.91 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 9 & 0 & 12 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 5 & 12 & 0 & 14 & 8 \\ 3 & 5 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

[93] $1^{111}3131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 3 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[94] $1321^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 3 & 3 \\ 1 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 1 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

[95] $1^{111}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 5 & 12 & 9 & 5 \\ 12 & 25 & 20 & 9 \\ 9 & 20 & 13 & 7 \\ 5 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{112}31^{231} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 13.08 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 12 & 0 & 58 & 14 \\ 2 & 3 & 2 & -3 & 14 & -1 \end{pmatrix}_{b_0}$$

[97] $1^{111}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 5 & 12 & 10 & 5 \\ 12 & 25 & 20 & 9 \\ 10 & 20 & 13 & 7 \\ 5 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[98] $1^{113}31^{331} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.60 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 4 & 0 & 18 & 3 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 18 & 18 & 0 & 133 & 21 \\ 2 & 3 & 2 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[99] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 5 & 0 & 10 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 5 & 10 & 0 & 14 & 8 \\ 3 & 5 & -3 & 8 & 2 \end{pmatrix}_{b_0}$$

[100] $1^{13}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 5 & 0 & 20 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 11 & 20 & 0 & 74 & 16 \\ 3 & 5 & -3 & 16 & 0 \end{pmatrix}_{b_0}$$

[101] $1231^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.22 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 \\ 4 & 9 & 0 & 8 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 3 & 8 & 0 & 4 & 4 \\ 3 & 5 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[102] $1^{11}3^{12}21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 4 & 7 & 0 & 16 & 8 \\ 2 & 2 & 2 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[103] $1^{12}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 3 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

[104] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 2 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[105] $1^{11}3^{12}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[106] $1^{13}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.91 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 3 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[107] $1^{13}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 6 & 0 & 4 & 4 \\ 2 & 3 & 3 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[108] $1321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 6 & 3 \\ 1 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 2 & 6 & 6 & 2 & 10 & 6 \\ 2 & 3 & 2 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

- [109] $1^{112}3131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 78$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 6 & 0 & 10 & 6 \\ 2 & 3 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$
- [110] $1^{12}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 80$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 5 & 0 & 15 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 15 & 0 & 38 & 12 \\ 3 & 5 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$
- [111] $1^{114}31^431^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 86$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 14.02 \rightarrow 15$, $T_{\max} = 28$, $T \in [15, 28]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 1 & 4 & 0 & 24 & 3 \\ 1 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 24 & 24 & 0 & 238 & 28 \\ 2 & 3 & 2 & -3 & 28 & -2 \end{pmatrix}_{b_0}$$
- [115] $1^{13}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 92$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 5 & 8 & 0 & 9 & 40 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$
- [112] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 92$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 3 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$
- [113] $1^{11}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 92$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 3 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$
- [114] $1^{11}321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 92$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 7 & 5 & 0 & 10 & 33 & 11 \\ 2 & 2 & -1 & 0 & 11 & 0 \end{pmatrix}_{b_0}$$
- [117] $1^{11}321^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 95$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 5 & 0 & 25 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 11 & 25 & 0 & 92 & 20 \\ 3 & 5 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$
- [116] $1^{11}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 95$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 5 & 0 & 15 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 15 & 0 & 32 & 12 \\ 3 & 5 & -3 & 12 & 2 \end{pmatrix}_{b_0}$$
- [118] $1^{11}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 103$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 1 & 6 & 3 \\ 1 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 3 & 6 & 6 & 2 & 10 & 6 \\ 2 & 3 & 2 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$
- [119] $1^{11}321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 105$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 5 & 0 & 20 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 20 & 0 & 58 & 16 \\ 3 & 5 & -3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{12} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 107, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 3 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{11} 3^{13} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 107, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.09 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 4 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{14} 3^{12} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 107, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.09 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 0 \\ 7 & 11 & 0 & 13 & 81 & 17 \\ 2 & 2 & -1 & 0 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{13} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 107, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 4 & 7 & 0 & 6 & 25 & 9 \\ 2 & 2 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{11} 3^{12} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 112, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.82 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 5 & 4 & 0 & 7 & 16 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{11} 3^{14} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 112, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.54 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 5 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{12} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 112, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 0 & 13 & 7 \\ 2 & 2 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[127] \ 1^1 3^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 116, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 28, \ T \in [12, 28]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 4 \\ 2 & 4 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{13} 3^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 119, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.25 \rightarrow 13, \ T_{\max} = 28, \ T \in [13, 28]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 15 & 0 & 49 & 13 \\ 2 & 3 & 3 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{11} 3^{13} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 128, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.09 \rightarrow 14, \ T_{\max} = 28, \ T \in [14, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 9 & 0 & 25 & 9 \\ 2 & 3 & 2 & -3 & 9 & -1 \end{pmatrix}_{b_0}$$

- [130] $1^1 1^1 3 2 1^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 1 & 9 & 3 \\ 1 & 4 & 0 & 1 & 9 & 2 \\ 1 & 1 & 1 & -3 & 3 & -1 \\ 3 & 9 & 9 & 3 & 25 & 9 \\ 2 & 3 & 2 & -1 & 9 & -1 \end{pmatrix}_{b_0}$$
- [131] $1^{12} 3 1^{11} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 9 & 0 & 28 & 10 \\ 2 & 4 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$
- [132] $1^{112} 3 1^{11} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 9 & 9 & 0 & 28 & 10 \\ 2 & 3 & 2 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$
- [133] $1^{112} 3 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 4 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$
- [134] $1^{14} 3 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 5 & 9 & 0 & 7 & 41 & 11 \\ 2 & 2 & -1 & 0 & 11 & 1 \end{pmatrix}_{b_0}$$
- [135] $1^{13} 3 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$
- [136] $1^{13} 3 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 0 & 25 & 9 \\ 2 & 2 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$
- [137] $1^{12} 3 2 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 5 & 6 & 0 & 8 & 26 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$
- [138] $1^{12} 3 2 1^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 1 & 9 & 3 \\ 1 & 4 & 0 & 1 & 9 & 2 \\ 1 & 1 & 1 & -3 & 3 & -1 \\ 5 & 9 & 9 & 3 & 25 & 9 \\ 2 & 3 & 2 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$
- [139] $1^{11} 3 1^{11} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.15 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 9 & 0 & 13 & 7 \\ 2 & 3 & 3 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [140] $1^{12} 3 1^{11} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 167, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 9 & 12 & 0 & 28 & 10 \\ 2 & 3 & 3 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

[141] $1^{12}321^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 170, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 5 & 0 & 20 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 10 & 20 & 0 & 62 & 16 \\ 3 & 5 & -3 & 16 & 2 \end{pmatrix}_{b_0}$$

[142] $1^{14}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 7 & 0 & 41 & 11 \\ 2 & 2 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[143] $1^{12}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 4 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

[144] $1^{12}321^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 8 & 14 & 0 & 70 & 16 \\ 2 & 2 & 2 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[145] $1^{12}321^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 0 & 26 & 10 \\ 2 & 2 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[146] $1^{14}321^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 6 & 10 & 0 & 10 & 58 & 14 \\ 2 & 2 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{12}321^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 4 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{11}31^{11}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[149] $1^{13}321^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.64 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 1 & 12 & 3 \\ 1 & 4 & 0 & 1 & 12 & 2 \\ 1 & 1 & 1 & -3 & 4 & -1 \\ 7 & 12 & 12 & 4 & 46 & 12 \\ 2 & 3 & 2 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[150] $1^{12}321^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 7 & 7 & 0 & 11 & 45 & 13 \\ 2 & 2 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[151] \ 1^{11} 3^{14} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 187, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.22 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 8 & 5 & 0 & 10 & 33 & 11 \\ 2 & 2 & -1 & 0 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[152] \ 1^{11} 3^{13} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 187, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.84 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 6 & 4 & 0 & 7 & 16 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[153] \ 1^{14} 3^{21} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 187, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.84 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 5 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[154] \ 1^{12} 3^{21} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 192, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.58 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 7 & 11 & 0 & 45 & 13 \\ 2 & 2 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[155] \ 1^{13} 3^{21} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 192, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 12 & 0 \\ 7 & 9 & 0 & 12 & 61 & 15 \\ 2 & 2 & -1 & 0 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[156] \ 1^{11} 3^{21} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 192, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 5 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[157] \ 1^{13} 3^{21} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 193, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.41 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 1 & 9 & 3 \\ 1 & 4 & 0 & 1 & 9 & 2 \\ 1 & 1 & 1 & -3 & 3 & -1 \\ 4 & 9 & 9 & 3 & 25 & 9 \\ 2 & 3 & 2 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[158] \ 1^1 3^{21} 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 198, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.20 \rightarrow 14, \ T_{\max} = 20, \ T \in [14, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 1 & 12 & 3 \\ 1 & 4 & 0 & 1 & 12 & 2 \\ 1 & 1 & 1 & -3 & 4 & -1 \\ 4 & 12 & 12 & 4 & 46 & 12 \\ 2 & 3 & 2 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[159] \ 1^{14} 3^{13} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 198, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.20 \rightarrow 14, \ T_{\max} = 28, \ T \in [14, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 12 & 0 & 46 & 12 \\ 2 & 3 & 2 & -3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[161] \ 1^1 3^{13} 21^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 200, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.7$$

$$T_{\min} = 12, \ T_{\max} = 28, \ T \in [12, 28]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 6 & 0 & 22 & 8 \\ 2 & 4 & 2 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[160] $21^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 13 & 20 & 15 & 7 \\ 20 & 25 & 20 & 9 \\ 15 & 20 & 13 & 7 \\ 7 & 9 & 7 & 3 \end{pmatrix}_{b_0}$$

[162] $1^{13}321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 215, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.03 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 5 & 0 & 25 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 13 & 25 & 0 & 104 & 20 \\ 3 & 5 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[163] $1^{11}321^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 227, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 6 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

[164] $1^{12}321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 227, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 5 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[165] $1^{14}321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 230, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.44 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 5 & 0 & 30 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 16 & 30 & 0 & 158 & 24 \\ 3 & 5 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[167] $1^{12}321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 5 & 0 & 25 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 12 & 25 & 0 & 94 & 20 \\ 3 & 5 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$

[166] $1^{11}31^{23}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 15 & 15 & 0 & 85 & 17 \\ 2 & 3 & 2 & -3 & 17 & 0 \end{pmatrix}_{b_0}$$

[168] $1^{14}31^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 18 & 0 & 118 & 20 \\ 2 & 4 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[171] $1^{13}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 5 & 7 & 0 & 6 & 25 & 9 \\ 2 & 2 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[170] $1^{13}321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[169] $1^{11}321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 7 & 4 & 0 & 7 & 16 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[172] \quad 1^{111} 2^2 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 \\ 2 & 1 & 4 & 1 & 3 & 3 \\ 2 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[173] \quad 1^{113} 31^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.41 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 12 & 0 & 49 & 13 \\ 2 & 3 & 2 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[176] \quad 1^{12} 3^2 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 5 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[175] \quad 1^{13} 3^2 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 9 & 0 & 40 & 12 \\ 2 & 2 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[174] \quad 1^{13} 3^2 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 6 & 8 & 0 & 9 & 40 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[178] \quad 1^{12} 3^2 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 6 & 6 & 0 & 8 & 26 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[177] \quad 1^{12} 3^2 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 6 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[179] \quad 1^{115} 3131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 13.26 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 15 & 0 & 73 & 15 \\ 2 & 3 & 2 & -3 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[180] \quad 1^1 1321^{115} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 13.26 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 1 & 15 & 3 \\ 1 & 4 & 0 & 1 & 15 & 2 \\ 1 & 1 & 1 & -3 & 5 & -1 \\ 5 & 15 & 15 & 5 & 73 & 15 \\ 2 & 3 & 2 & -1 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[181] \quad 1^{12} 3^2 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 5 & 0 & 30 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 14 & 30 & 0 & 134 & 24 \\ 3 & 5 & -3 & 24 & 2 \end{pmatrix}_{b_0}$$

[182] $1^{\overline{1}^2} 3 21^{\overline{1}4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 298, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 1 & 12 & 3 \\ 1 & 4 & 0 & 1 & 12 & 2 \\ 1 & 1 & 1 & -3 & 4 & -1 \\ 6 & 12 & 12 & 4 & 46 & 12 \\ 2 & 3 & 2 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[183] $1^{\overline{1}^1} 3 21^{\overline{1}4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 303, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 1 & 12 & 3 \\ 1 & 4 & 0 & 1 & 12 & 2 \\ 1 & 1 & 1 & -3 & 4 & -1 \\ 5 & 12 & 12 & 4 & 46 & 12 \\ 2 & 3 & 2 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[184] $1^{\overline{1}^{11}} 3 21^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 6 & 9 & 0 & 7 & 41 & 11 \\ 2 & 2 & -1 & 0 & 11 & 1 \end{pmatrix}_{b_0}$$

[185] $1^{\overline{1}^{12}} 3 21^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 6 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[186] $1^{\overline{1}^{13}} 3 21^{\overline{1}3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 8 & 7 & 0 & 11 & 45 & 13 \\ 2 & 2 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[187] $1^{\overline{1}^{13}} 31^{\overline{1}3} 31^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 9 & 0 & 52 & 12 \\ 2 & 4 & 2 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

[188] $1^{\overline{1}^1} 3^{\overline{1}3} 21^{\overline{1}3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 345, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 5 & 0 & 30 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 15 & 30 & 0 & 142 & 24 \\ 3 & 5 & -3 & 24 & 2 \end{pmatrix}_{b_0}$$

[189] $1^{\overline{1}^{14}} 31^{\overline{1}3} 31^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 345, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 4 & 0 & 21 & 3 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 21 & 21 & 0 & 172 & 24 \\ 2 & 3 & 2 & -3 & 24 & 0 \end{pmatrix}_{b_0}$$

[190] $1^{\overline{1}^{13}} 3^{\overline{1}3} 21^{\overline{1}3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 347, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 12 & 0 & 61 & 15 \\ 2 & 2 & 2 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[191] $1^{\overline{1}^{13}} 3^{\overline{1}3} 21^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 347, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 6 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

- [194] $1^{14}3^{11}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 352$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.66 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 0 \\ 8 & 11 & 0 & 13 & 81 & 17 \\ 2 & 2 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$
- [193] $1^{12}3^{13}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 352$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 11.66 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 6 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$
- [192] $1^{14}3^{12}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 352$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.66 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 10 & 10 & 0 & 58 & 14 \\ 2 & 2 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$
- [195] $1^{11}3^{12}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 371$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 9 & 0 & 37 & 11 \\ 2 & 4 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$
- [196] $1^{12}3^{13}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 387$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 7 & 6 & 0 & 8 & 26 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$
- [200] $1^{13}3^{14}21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 387$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 10 & 15 & 0 & 88 & 18 \\ 2 & 2 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$
- [199] $1^{13}3^{12}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 387$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 6 & 7 & 0 & 6 & 25 & 9 \\ 2 & 2 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$
- [198] $1^{13}3^{12}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 387$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 12 & 0 \\ 8 & 9 & 0 & 12 & 61 & 15 \\ 2 & 2 & -1 & 0 & 15 & 1 \end{pmatrix}_{b_0}$$
- [197] $1^{14}3^{11}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 387$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 7 & 10 & 0 & 10 & 58 & 14 \\ 2 & 2 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$
- [201] $1^{13}3^{12}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 395$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 15 & 0 & 85 & 17 \\ 2 & 4 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$
- [202] $1^13^{16}21^{16} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 398$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.4$

$$T_{\min} = 13.30 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 1 & 18 & 3 \\ 1 & 4 & 0 & 1 & 18 & 2 \\ 1 & 1 & 1 & -3 & 6 & -1 \\ 6 & 18 & 18 & 6 & 106 & 18 \\ 2 & 3 & 2 & -1 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[203] \ 1^{14} 31^{13} 1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 401, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 15 & 15 & 0 & 76 & 16 \\ 2 & 3 & 2 & -3 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[204] \ 1^{14} 321^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 420, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 5 & 0 & 35 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 18 & 35 & 0 & 202 & 28 \\ 3 & 5 & -3 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[205] \ 1^{12} 321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 427, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 7 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[207] \ 1^{13} 321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 7 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[206] \ 1^{12} 321^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 7 & 8 & 0 & 9 & 40 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[208] \ 1^{12} 31^{12} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 12 & 0 & 58 & 14 \\ 2 & 4 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[209] \ 1^{14} 31^{23} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 0 & 18 & 3 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 18 & 18 & 0 & 118 & 20 \\ 2 & 3 & 2 & -3 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[210] \ 2^2 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 452, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 4 & 2 \\ 2 & 1 & 4 & 1 & 6 & 3 \\ 2 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 4 & 6 & 6 & 2 & 10 & 6 \\ 2 & 3 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[211] \ 1^{13} 321^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 455, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 5 & 0 & 35 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 17 & 35 & 0 & 188 & 28 \\ 3 & 5 & -3 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[212] \ 1^{14} 321^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 467, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 7 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[213] \ 1^{12} 3^{14} 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 467, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 9 & 7 & 0 & 11 & 45 & 13 \\ 2 & 2 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[214] \ 1^{12} 3^{14} 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 492, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 8 & 6 & 0 & 8 & 26 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[215] \ 1^{14} 3^{13} 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 492, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 13 & 0 & 81 & 17 \\ 2 & 2 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[216] \ 1^{14} 3^{12} 3^{11} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 507, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.61 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 7 & 9 & 0 & 7 & 41 & 11 \\ 2 & 2 & -1 & 0 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[217] \ 1^{13} 3^{13} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 512, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 7 & 7 & 0 & 6 & 25 & 9 \\ 2 & 2 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[218] \ 1^1 13^{14} 21^{17} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 528, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 13.33 \rightarrow 14, \ T_{\max} = 20, \ T \in [14, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 1 & 21 & 3 \\ 1 & 4 & 0 & 1 & 21 & 2 \\ 1 & 1 & 1 & -3 & 7 & -1 \\ 7 & 21 & 21 & 7 & 145 & 21 \\ 2 & 3 & 2 & -1 & 21 & -1 \end{pmatrix}_{b_0}$$

$$[219] \ 1^{13} 31^{14} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 536, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.09 \rightarrow 13, \ T_{\max} = 28, \ T \in [13, 28]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 12 & 0 & 94 & 16 \\ 2 & 4 & 2 & -3 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[220] \ 1^{13} 3^{13} 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 572, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 12 & 0 \\ 9 & 9 & 0 & 12 & 61 & 15 \\ 2 & 2 & -1 & 0 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[221] \ 31^{11} 231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, \ T_{\max} = 31, \ \Delta = -117, \ \det = 580, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 31, \ T \in [12, 31]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 3 & 2 \\ 3 & 1 & 4 & 1 & 3 & 3 \\ 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 3 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[223] \ 1^{12} 3^{12} 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 587, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.61 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 0 \\ 9 & 11 & 0 & 13 & 81 & 17 \\ 2 & 2 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

[222] $1^{13} 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 587, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 8 & 8 & 0 & 9 & 40 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[224] $1^{11} 3 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 590, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 5 & 0 & 40 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 20 & 40 & 0 & 254 & 32 \\ 3 & 5 & -3 & 32 & 2 \end{pmatrix}_{b_0}$$

[227] $1^{14} 3 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 592, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 16 & 0 & 110 & 20 \\ 2 & 2 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[226] $1^{14} 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 592, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 8 & 10 & 0 & 10 & 58 & 14 \\ 2 & 2 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

[225] $1^{14} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 592, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 8 & 8 & 0 & 4 & 30 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[228] $1^{11} 31^{13} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 623, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 12 & 0 & 73 & 15 \\ 2 & 4 & 2 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[229] $1^{13} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 627, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 8 & 7 & 0 & 6 & 25 & 9 \\ 2 & 2 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[230] $1^{14} 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 8 & 9 & 0 & 7 & 41 & 11 \\ 2 & 2 & -1 & 0 & 11 & 1 \end{pmatrix}_{b_0}$$

[231] $1^{14} 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 732, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 9 & 8 & 0 & 9 & 40 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[232] $1^{13} 3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 747, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 12 & 0 \\ 10 & 9 & 0 & 12 & 61 & 15 \\ 2 & 2 & -1 & 0 & 15 & 1 \end{pmatrix}_{b_0}$$

[233] $2^1 2^{113} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 777, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 6 & 2 \\ 2 & 1 & 4 & 1 & 9 & 3 \\ 2 & 4 & 0 & 1 & 9 & 2 \\ 1 & 1 & 1 & -3 & 3 & -1 \\ 6 & 9 & 9 & 3 & 25 & 9 \\ 2 & 3 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[234] $1^{14} 3^{113} 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 787, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 9 & 10 & 0 & 10 & 58 & 14 \\ 2 & 2 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

[235] $1^1 31^{15} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 788, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 15 & 0 & 148 & 20 \\ 2 & 4 & 2 & -3 & 20 & 0 \end{pmatrix}_{b_0}$$

[236] $1^{12} 31^{13} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 15 & 0 & 100 & 18 \\ 2 & 4 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[237] $1^{14} 3^{113} 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 812, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 0 \\ 10 & 11 & 0 & 13 & 81 & 17 \\ 2 & 2 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

[238] $1^{14} 3^{114} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 827, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 9 & 9 & 0 & 7 & 41 & 11 \\ 2 & 2 & -1 & 0 & 11 & 1 \end{pmatrix}_{b_0}$$

[239] $1^{14} 31^{13} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 836, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 21 & 0 & 172 & 24 \\ 2 & 4 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[240] $1^{13} 31^{13} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 871, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 18 & 0 & 133 & 21 \\ 2 & 4 & 2 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[241] $1^{11} 31^{14} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 931, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 15 & 0 & 121 & 19 \\ 2 & 4 & 2 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

[242] $1^{14} 3^{114} 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 972, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 10 & 10 & 0 & 10 & 58 & 14 \\ 2 & 2 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

[243] $1^{14} 3^{21} 1^3 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 1027, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 0 \\ 11 & 11 & 0 & 13 & 81 & 17 \\ 2 & 2 & -1 & 0 & 17 & 1 \end{pmatrix}_{b_0}$$

[244] $1^1 31^{16} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8, T_{\max} = 28, \Delta = -116, \det = 1096, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.4$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 36 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 36 & 18 & 0 & 214 & 24 \\ 2 & 4 & 2 & -3 & 24 & 0 \end{pmatrix}_{b_0}$$

[245] $1^{12} 31^{14} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8, T_{\max} = 28, \Delta = -116, \det = 1220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 18 & 0 & 154 & 22 \\ 2 & 4 & 2 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[246] $2^1 2^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 1232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 8 & 2 \\ 2 & 1 & 4 & 1 & 12 & 3 \\ 2 & 4 & 0 & 1 & 12 & 2 \\ 1 & 1 & 1 & -3 & 4 & -1 \\ 8 & 12 & 12 & 4 & 46 & 12 \\ 2 & 3 & 2 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$

[247] $1^{13} 31^{14} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8, T_{\max} = 28, \Delta = -116, \det = 1403, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 33 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 33 & 21 & 0 & 193 & 25 \\ 2 & 4 & 2 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[248] $1^1 31^{17} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8, T_{\max} = 28, \Delta = -116, \det = 1460, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 14 & 2 \\ 2 & 4 & 4 & 0 & 42 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 42 & 21 & 0 & 292 & 28 \\ 2 & 4 & 2 & -3 & 28 & 0 \end{pmatrix}_{b_0}$$

[249] $1^{14} 31^{14} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8, T_{\max} = 28, \Delta = -116, \det = 1480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 36 & 4 \\ 1 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 36 & 24 & 0 & 238 & 28 \\ 2 & 4 & 2 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[250] $2^1 2^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 1817, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 10 & 2 \\ 2 & 1 & 4 & 1 & 15 & 3 \\ 2 & 4 & 0 & 1 & 15 & 2 \\ 1 & 1 & 1 & -3 & 5 & -1 \\ 10 & 15 & 15 & 5 & 73 & 15 \\ 2 & 3 & 2 & -1 & 15 & 1 \end{pmatrix}_{b_0}$$

[251] $2^1 2^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 2532, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.4$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 12 & 2 \\ 2 & 1 & 4 & 1 & 18 & 3 \\ 2 & 4 & 0 & 1 & 18 & 2 \\ 1 & 1 & 1 & -3 & 6 & -1 \\ 12 & 18 & 18 & 6 & 106 & 18 \\ 2 & 3 & 2 & -1 & 18 & 1 \end{pmatrix}_{b_0}$$

[252] $2^1 2^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 3377, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 14 & 2 \\ 2 & 1 & 4 & 1 & 21 & 3 \\ 2 & 4 & 0 & 1 & 21 & 2 \\ 1 & 1 & 1 & -3 & 7 & -1 \\ 14 & 21 & 21 & 7 & 145 & 21 \\ 2 & 3 & 2 & -1 & 21 & 1 \end{pmatrix}_{b_0}$$

[253] $1^{11} 2^1 3^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

[254] $1^{12} 3^{11} 221^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 16 & 0 & 130 & 22 \\ 3 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[255] $1^{11} 3^{11} 221^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 4 & 9 & 0 & 20 & 10 \\ 2 & 2 & 3 & -3 & 10 & 2 \end{pmatrix}_{b_0}$$

[256] $1^{11} 23^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 4 & 0 \\ 2 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[257] $1^{11} 23^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 1 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[258] $1^{11} 4^{131^3} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 16.09 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 13 & 0 & 40 & 12 \\ 3 & 4 & -2 & 12 & -2 \end{pmatrix}_{b_0}$$

[259] $1^{11} 4^{131^3} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[260] $1^{11} 23^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[261] $1^{11} 4^{131^3} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.54 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 5 & -2 & 3 & 0 \end{pmatrix}_{b_0}$$

[262] $1^{11} 3^{11} 221^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 6 & 6 & 6 & 6 \\ 3 & 6 & 0 & 5 & 2 \\ 2 & 6 & 5 & 3 & 5 \\ 3 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[263] $1^{11} 23^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 2 & 4 & 0 & 4 & 6 & 6 \\ 2 & 3 & -1 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

- [264] $1^{11} 3^{11} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$
- [265] $1^{11} 2^1 3^1 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$
- [266] $1^1 231^1 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 5 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 5 & 1 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 0 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$
- [267] $1^{11} 3^{11} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 1 & 3 \\ 3 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$
- [268] $1^{12} 231^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 21 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 60 & 0 & 258 & 30 \\ 3 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$
- [269] $1^1 231^1 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 1 & 3 \\ 5 & 13 & 1 & 0 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 7 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$
- [271] $1^{11} 4^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 11 & 0 & 28 & 10 \\ 3 & 4 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$
- [270] $1^{11} 4^1 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 5 & 0 & 4 & 4 \\ 3 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$
- [272] $1^{12} 3^{11} 221^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 11 & 0 & 50 & 14 \\ 3 & 3 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$
- [278] $1^{13} 4^1 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 12 & 0 & 25 & 9 \\ 3 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$
- [274] $1^{14} 4^1 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 14.25 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 16 & 0 & 46 & 12 \\ 3 & 4 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$
- [275] $1^{15} 4^1 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.3$

$$T_{\min} = 15.38 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 20 & 0 & 73 & 15 \\ 3 & 4 & -2 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[276] \ 1^{16} 4^{13} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.4$$

$$T_{\min} = 16.75 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 24 & 0 & 106 & 18 \\ 3 & 4 & -2 & 18 & -2 \end{pmatrix}_{b_0}$$

$$[277] \ 1^{17} 4^{13} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 18.37 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 28 & 0 & 145 & 21 \\ 3 & 4 & -2 & 21 & -4 \end{pmatrix}_{b_0}$$

$$[273] \ 1^{12} 4^{13} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 4 & 8 & 0 & 10 & 6 \\ 3 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[279] \ 1^{13} 4^{11} 1^2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 4 & 1 & 7 & 3 \\ 4 & 4 & 0 & 10 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 10 & 0 & 21 & 9 \\ 3 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[280] \ 1^{11} 4^{11} 2^{31} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 & 2 \\ 3 & 3 & 3 & 0 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 2 & 2 & 0 & 0 & 2 \\ 2 & 2 & 3 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[281] \ 1^{11} 4^{13} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[282] \ 1^{14} 2^3 2^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 5 & 13 & 0 & 7 & 57 & 15 \\ 2 & 3 & -1 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[283] \ 1^{13} 4^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 \\ 4 & 4 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 5 & 0 & 4 & 4 \\ 3 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[284] \ 2^{14} 1^{41} 1^{13} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 \\ 2 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 2 & 2 & 0 & 0 & 2 \\ 2 & 2 & 3 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[285] \ 1^{13} 4^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 3 & 0 & 1 & 3 \\ 3 & 4 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[286] \ 1^{12} 2^3 2^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 3 & 7 & 0 & 5 & 17 & 9 \\ 2 & 3 & -1 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[287] \ 1^{13} 3^{11} 221^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 6 & 6 & 27 & 6 \\ 3 & 6 & 0 & 19 & 2 \\ 13 & 27 & 19 & 108 & 22 \\ 3 & 6 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[288] \ 1^{11} 3^{13} 221^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 6 & 0 & 17 & 6 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 17 & 0 & 36 & 14 \\ 3 & 6 & -3 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[289] \ 2^{11} 3^{41} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 10 & 24 & 12 & 6 \\ 24 & 49 & 26 & 13 \\ 12 & 26 & 13 & 7 \\ 6 & 13 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[291] \ 1^{11} 3^{13} 221^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 13 & 0 & 101 & 19 \\ 3 & 3 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[290] \ 1^{11} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & 1 & 3 \\ 3 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[294] \ 1^{11} 3^{11} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 5 & 0 & 6 & 6 \\ 2 & 2 & 3 & -3 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[293] \ 1^{11} 3^{11} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 8 & 0 & 31 & 11 \\ 3 & 3 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[292] \ 1^{14} 2^3 3^{21} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 5 & 12 & 0 & 4 & 46 & 12 \\ 2 & 3 & -1 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[295] \ 1^{12} 2^3 3^{21} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 2 & 0 \\ 3 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[299] \ 1^{11} 2^3 3^{21} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 2 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

- [298] $1^{14}3^{221}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 22 & 0 & 206 & 28 \\ 3 & 3 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$
- [297] $1^{14}3^{221}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 24 & 3 \\ 2 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 27 & 0 & 346 & 36 \\ 3 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$
- [296] $1^{13}2^321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 4 & 10 & 0 & 6 & 34 & 12 \\ 2 & 3 & -1 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$
- [302] $1^{11}41^{1231}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 6 & 1 & 3 & 3 \\ 6 & 13 & 0 & 8 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 8 & 0 & 4 & 4 \\ 3 & 7 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$
- [300] $1^{11}2^321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 2 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$
- [301] $1^{11}4^11^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 8 & 0 & 13 & 7 \\ 3 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$
- [303] $1^{1231}1^{131}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 4 \\ 2 & 4 & 3 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$
- [304] $1^{11}3^{221}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 6 & 6 & 11 & 6 \\ 3 & 6 & 0 & 8 & 2 \\ 5 & 11 & 8 & 15 & 9 \\ 3 & 6 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$
- [305] $1^{11}4^11^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 10 & 0 & 21 & 9 \\ 3 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$
- [306] $1^{13}2^321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 5 & 11 & 0 & 9 & 49 & 15 \\ 2 & 3 & -1 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$
- [307] $1^{13}2^321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 4 & 9 & 0 & 3 & 25 & 9 \\ 2 & 3 & -1 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[310] $1^{13} 3^{12} 221^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 23 & 3 \\ 2 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 24 & 0 & 295 & 33 \\ 3 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

[309] $1^{13} 3^{11} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 14 & 0 & 75 & 17 \\ 3 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[308] $1^{12} 2^1 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 3 & 6 & 0 & 2 & 10 & 6 \\ 2 & 3 & -1 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[312] $1^{11} 4^{11} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 9 & 0 & 13 & 7 \\ 3 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[311] $1^{11} 4^{131} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 7 & 0 & 8 & 6 \\ 3 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[314] $1^{11} 2^1 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 4 & 0 \\ 3 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[313] $1^{12} 3^{11} 221^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 7 & 14 & 0 & 54 & 16 \\ 2 & 2 & 3 & -3 & 16 & 2 \end{pmatrix}_{b_0}$$

[315] $1^{12} 3^{11} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 6 & 0 & 14 & 8 \\ 2 & 2 & 3 & -3 & 8 & 2 \end{pmatrix}_{b_0}$$

[316] $1^{11} 2^1 3^{12} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 3 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[317] $1^{11} 3^{11} 221^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -81, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 6 & 6 & 23 & 6 \\ 3 & 6 & 0 & 18 & 2 \\ 9 & 23 & 18 & 67 & 19 \\ 3 & 6 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

- [318] $1^{1231^{11}31^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 5 & 3 \\ 5 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 15 & 0 & 13 & 7 \\ 3 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$
- [319] $1^{122321^{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -2 & 8 & 0 \\ 5 & 0 & 8 & 8 & 30 & 12 \\ 2 & 0 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$
- [320] $1^{11231^{12}31^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 15 & 3 \\ 5 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 45 & 0 & 138 & 22 \\ 3 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$
- [321] $1^{1231^{11}31^{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 7 & 3 \\ 5 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 15 & 0 & 13 & 7 \\ 3 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$
- [322] $1^{13231^{12}31^{14}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 29 & 3 \\ 5 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 75 & 0 & 418 & 38 \\ 3 & 7 & -1 & 38 & 2 \end{pmatrix}_{b_0}$$
- [323] $1^{1113221^{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 8 & 0 & 31 & 11 \\ 3 & 3 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$
- [324] $1^{1231^{13}31^{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -89, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 1 & 7 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 0 & 0 & 13 & 7 \\ 3 & 0 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$
- [325] $1^{1113221^{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 6 & 6 & 17 & 6 \\ 3 & 6 & 0 & 13 & 2 \\ 7 & 17 & 13 & 36 & 14 \\ 3 & 6 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$
- [326] $1^{11131^{11}41^2} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 1 & 4 & 1 & 10 & 3 \\ 4 & 4 & 0 & 15 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 15 & 0 & 45 & 13 \\ 3 & 4 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$
- [327] $1^{143221^{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 17 & 0 & 106 & 20 \\ 3 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$
- [328] $1^{111231^{11}31^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 7 & 0 & 14 & 8 \\ 2 & 4 & 3 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$
- [329] $1^{142321^{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 6 & 14 & 0 & 10 & 74 & 18 \\ 2 & 3 & -1 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

[331] $1^{11} 3^{14} 221^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 13 & 0 & 101 & 19 \\ 3 & 3 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

[330] $1^{13} 3^{14} 221^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 30 & 3 \\ 2 & 1 & 1 & 29 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 29 & 0 & 465 & 41 \\ 3 & 3 & -1 & 41 & 3 \end{pmatrix}_{b_0}$$

[336] $1^{11} 23131^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 1 & 8 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 0 & 0 & 18 & 8 \\ 3 & 0 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[335] $1^{11} 231^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 11 & 3 \\ 5 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 30 & 0 & 63 & 15 \\ 3 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[334] $1^{11} 231^{11} 31^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 13 & 3 \\ 5 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 30 & 0 & 63 & 15 \\ 3 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[333] $1^{13} 2^3 21^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 4 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[332] $1^{12} 2^3 21^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -2 & 5 & 0 \\ 4 & 0 & 7 & 5 & 17 & 9 \\ 2 & 0 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[338] $1^{12} 3^{11} 221^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 11 & 0 & 50 & 14 \\ 3 & 3 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

[337] $1^{12} 3^{11} 221^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 3 \\ 1 & 2 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 10 & 0 & 30 & 12 \\ 2 & 2 & 3 & -3 & 12 & 2 \end{pmatrix}_{b_0}$$

[339] $1^{12} 3^{12} 221^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 16 & 0 & 130 & 22 \\ 3 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[341] $1^{12} 4^{11} 131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 13 & 0 & 28 & 10 \\ 3 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[340] $1^{13}2^13^{11}221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 19 & 0 & 165 & 25 \\ 3 & 3 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

[342] $1^{13}2^13^{11}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 6 & 0 \\ 5 & 0 & 10 & 6 & 34 & 12 \\ 2 & 0 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[343] $1^{13}2^13^{11}221^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 6 & 0 & 34 & 6 \\ 1 & 0 & -5 & 0 & -3 \\ 14 & 34 & 0 & 150 & 28 \\ 3 & 6 & -3 & 28 & 3 \end{pmatrix}_{b_0}$$

[344] $1^{131}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 \\ 4 & 4 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 4 & 0 & 1 & 3 \\ 3 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[345] $1^{12}2^13^{11}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 4 & 7 & 0 & 5 & 17 & 9 \\ 2 & 3 & -1 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[346] $1^{12}2^13^{11}2321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 4 & 6 & 0 & 2 & 10 & 6 \\ 2 & 3 & -1 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[348] $1^{12}3^{11}221^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 6 & 6 & 22 & 6 \\ 3 & 6 & 0 & 16 & 2 \\ 10 & 22 & 16 & 66 & 18 \\ 3 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[349] $1^{11}2^13^{11}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 4 & 0 \\ 4 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[347] $1^{12}2^13^{11}2321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 2 & 0 \\ 4 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[350] $1^{131}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 4 & 0 & 7 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 7 & 0 & 8 & 6 \\ 3 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[351] $1^{11} 1^{13} 31^4 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.15 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 21 & 0 & 101 & 19 \\ 3 & 4 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[352] $1^{12} 1^{13} 31^1 4 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 11 & 0 & 21 & 9 \\ 3 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[353] $1^{11} 1^{13} 3 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 8 & 0 & 31 & 11 \\ 3 & 3 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[354] $1^{13} 231^{14} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 36 & 3 \\ 5 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 36 & 105 & 0 & 778 & 52 \\ 3 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[355] $1^{14} 231^{11} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 6 & 3 \\ 5 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 15 & 0 & 13 & 7 \\ 3 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[356] $1^{13} 2 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -2 & 9 & 0 \\ 6 & 0 & 11 & 9 & 49 & 15 \\ 2 & 0 & 2 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[357] $1^{13} 2 3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 3 & 0 \\ 5 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[358] $1^{12} 3 221^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 23 & 3 \\ 2 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 21 & 0 & 250 & 30 \\ 3 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[359] $1^{14} 3 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 6 & 6 & 38 & 6 \\ 3 & 6 & 0 & 27 & 2 \\ 18 & 38 & 27 & 211 & 31 \\ 3 & 6 & 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[360] $1^{14} 231^{13} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 1 & 0 & 25 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 25 & 0 & 188 & 28 \\ 2 & 4 & 3 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[361] $1^{14} 2 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 14 & 2 \\ 1 & 0 & 1 & -2 & 10 & 0 \\ 7 & 0 & 14 & 10 & 74 & 18 \\ 2 & 0 & 2 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

- [364] $1^{12}231^{14}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -144$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.22 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 30 & 3 \\ 5 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 90 & 0 & 558 & 44 \\ 3 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$
- [366] $1^{13}231^{13}31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -144$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 32 & 3 \\ 5 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 32 & 90 & 0 & 583 & 45 \\ 3 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$
- [365] $1^1231^{12}31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -144$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 12.22 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 14 & 3 \\ 5 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 30 & 0 & 58 & 14 \\ 3 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$
- [367] $1^{12}4^11^{131^4} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -107$, $\det = -144$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 20 & 0 & 90 & 18 \\ 3 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$
- [363] $1^1231^{12}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -144$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 12.22 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 10 & 3 \\ 5 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 30 & 0 & 58 & 14 \\ 3 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$
- [362] $1^{11}231^{11}31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -144$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 12 & 3 \\ 5 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 30 & 0 & 63 & 15 \\ 3 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$
- [368] $1^{11}4^11^{131^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -107$, $\det = -144$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 12 & 0 & 26 & 10 \\ 3 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$
- [369] $1^{13}4^1131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -107$, $\det = -147$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 49 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$
- [370] $1^{13}2^33^121^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -161$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 9 & 0 & 3 & 25 & 9 \\ 2 & 3 & -1 & 0 & 9 & 2 \end{pmatrix}_{b_0}$
- [371] $1^{14}3^1221^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -162$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.68 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 31 & 3 \\ 2 & 1 & 1 & 32 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 32 & 0 & 526 & 44 \\ 3 & 3 & -1 & 44 & 3 \end{pmatrix}_{b_0}$
- [372] $1^{11}2^33^121^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -164$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 4 & 0 \\ 5 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$
- [374] $1^{13}3^1221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -167$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 14 & 0 & 75 & 17 \\ 3 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[373] \ 1^{13} 2^3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 5 & 10 & 0 & 6 & 34 & 12 \\ 2 & 3 & -1 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[375] \ 21^{11} 41^{11} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -170, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 4 & 2 \\ 2 & 0 & 3 & 0 & 4 & 2 \\ 2 & 3 & 3 & 0 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 4 & 8 & 0 & 10 & 6 \\ 2 & 2 & 3 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[376] \ 1^{13} 3^{221} 1^4 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 1 & 0 & 19 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 10 & 19 & 0 & 104 & 22 \\ 2 & 2 & 3 & -3 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[378] \ 1^{11} 4^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 18 & 0 & 70 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[377] \ 1^{12} 4^{13} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 14 & 0 & 38 & 12 \\ 3 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[379] \ 1^{12} 2^3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 5 & 7 & 0 & 5 & 17 & 9 \\ 2 & 3 & -1 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[380] \ 1^{13} 1^{11} 4^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 1 & 6 & 3 \\ 4 & 4 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 9 & 0 & 13 & 7 \\ 3 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[381] \ 1^{12} 3^{221} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 11 & 0 & 50 & 14 \\ 3 & 3 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[384] \ 1^{13} 4^{13} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 15 & 0 & 40 & 12 \\ 3 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[383] \ 1^{12} 4^{13} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 17 & 0 & 61 & 15 \\ 3 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

- [382] $1^{11} 1^{13} 31^{12} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 15 & 0 & 45 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$
- [385] $1^{13} 2 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 6 & 11 & 0 & 9 & 49 & 15 \\ 2 & 3 & -1 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$
- [386] $1^{13} 3 221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -183, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 11 & 0 & 44 & 14 \\ 2 & 2 & 3 & -3 & 14 & 2 \end{pmatrix}_{b_0}$$
- [387] $1^{11} 1^{13} 31^{11} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 4 & 1 & 8 & 3 \\ 4 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 12 & 0 & 26 & 10 \\ 3 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$
- [388] $1^{14} 2 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 6 & 12 & 0 & 4 & 46 & 12 \\ 2 & 3 & -1 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$
- [389] $1^{14} 4 131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 21 & 0 & 76 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$
- [390] $1^{12} 2 3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 2 & 0 \\ 5 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$
- [391] $1^{14} 2 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 6 & 13 & 0 & 7 & 57 & 15 \\ 2 & 3 & -1 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$
- [392] $1^{12} 3 221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 16 & 0 & 130 & 22 \\ 3 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$
- [394] $1^{11} 2 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 4 & 0 \\ 6 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$
- [395] $1^{12} 2 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -2 & 8 & 0 \\ 6 & 0 & 8 & 8 & 30 & 12 \\ 2 & 0 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[393] $1^{12}2^3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -2 & 5 & 0 \\ 5 & 0 & 7 & 5 & 17 & 9 \\ 2 & 0 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[396] $1^{14}3^1 221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 19 & 3 \\ 2 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 22 & 0 & 206 & 28 \\ 3 & 3 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[397] $1^{12}3^1 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -207, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 6 & 6 & 28 & 6 \\ 3 & 6 & 0 & 21 & 2 \\ 12 & 28 & 21 & 103 & 23 \\ 3 & 6 & 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[398] $1^{14}231^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -209, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 2 & 5 & 0 & 19 & 4 \\ 1 & 5 & 1 & 0 & 19 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 19 & 19 & 0 & 92 & 20 \\ 2 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[399] $1^{13}3^1 221^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -211, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 24 & 3 \\ 2 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 24 & 0 & 295 & 33 \\ 3 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

[400] $1^{13}3^1 221^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -211, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 15 & 0 & 70 & 18 \\ 2 & 2 & 3 & -3 & 18 & 2 \end{pmatrix}_{b_0}$$

[401] $1^{12}231^{12} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 14 & 0 & 62 & 16 \\ 2 & 4 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[403] $1^{11}231^{13} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 20 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 60 & 0 & 243 & 29 \\ 3 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[402] $1^{12}231^{12} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 22 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 60 & 0 & 258 & 30 \\ 3 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[404] $1^{13}3^1 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -225, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 6 & 6 & 33 & 6 \\ 3 & 6 & 0 & 24 & 2 \\ 15 & 33 & 24 & 151 & 27 \\ 3 & 6 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[405] $1^{14}3^1 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

- [406] $1^{14} 41^{11} 231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 17 & 0 & 106 & 20 \\ 3 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0}$
- [407] $1^{14} 3^{12} 221^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 25 & 3 \\ 2 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 27 & 0 & 346 & 36 \\ 3 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0}$
- [408] $1^{13} 3^{12} 221^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -233, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 19 & 0 & 165 & 25 \\ 3 & 3 & -1 & 25 & 3 \end{pmatrix}_{b_0}$
- [409] $1^{14} 4^{131} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 19 & 0 & 65 & 15 \\ 3 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$
- [410] $1^{11} 231^{12} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 10 & 0 & 38 & 12 \\ 2 & 4 & 3 & -3 & 12 & 1 \end{pmatrix}_{b_0}$
- [411] $1^{12} 3^{13} 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 11 & 0 & 50 & 14 \\ 3 & 3 & -1 & 14 & 3 \end{pmatrix}_{b_0}$
- [412] $1^{14} 3^{11} 221^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 10 & 12 & 0 & 62 & 16 \\ 2 & 2 & 3 & -3 & 16 & 2 \end{pmatrix}_{b_0}$
- [413] $1^{14} 2^{11} 3^{13} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 0 \\ 8 & 15 & 0 & 13 & 97 & 21 \\ 2 & 3 & -1 & 0 & 21 & 2 \end{pmatrix}_{b_0}$
- [414] $1^{12} 4^{131} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.52 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 16 & 0 & 45 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$
- [415] $1^{13} 2^{13} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 3 & 0 \\ 6 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$
- [416] $1^{12} 231^{13} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 15 & 3 \\ 5 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 45 & 0 & 133 & 21 \\ 3 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[417] $1^{12} 2^3 3 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 2 & 0 \\ 6 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[419] $1^{12} 2^3 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 6 & 0 \\ 6 & 0 & 10 & 6 & 34 & 12 \\ 2 & 0 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[418] $1^{12} 2^3 3 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 11 & 0 \\ 8 & 0 & 9 & 11 & 49 & 15 \\ 2 & 0 & 2 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[420] $1^{12} 2^3 3 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -2 & 12 & 0 \\ 8 & 0 & 12 & 12 & 70 & 18 \\ 2 & 0 & 2 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

[421] $1^{12} 2^3 3 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -2 & 4 & 0 \\ 7 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[422] $1^{12} 3^2 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 14 & 0 & 75 & 17 \\ 3 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[423] $1^{12} 2^3 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 7 & 14 & 0 & 10 & 74 & 18 \\ 2 & 3 & -1 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

[424] $1^{12} 2^3 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 7 & 0 \\ 7 & 0 & 13 & 7 & 57 & 15 \\ 2 & 0 & 2 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[425] $1^{12} 231^{12} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 13 & 3 \\ 5 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 30 & 0 & 58 & 14 \\ 3 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[426] $1^{12} 231^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 11 & 3 \\ 5 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 30 & 0 & 58 & 14 \\ 3 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[427] $1^{13}2^3 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -266, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 6 & 10 & 0 & 6 & 34 & 12 \\ 2 & 3 & -1 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[428] $1^{13}2^3 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -281, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 7 & 11 & 0 & 9 & 49 & 15 \\ 2 & 3 & -1 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[429] $1^{12}3^3 221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -282, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 16 & 0 & 130 & 22 \\ 3 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[433] $1^{13}4^1 1^{131^2} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.54 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 18 & 0 & 61 & 15 \\ 3 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[432] $1^{12}231^{13}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 26 & 3 \\ 5 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 75 & 0 & 393 & 37 \\ 3 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[431] $1^{12}31^{231^{114}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 1 & 14 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 0 & 0 & 58 & 14 \\ 3 & 0 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[430] $1^{11}231^{12}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 16 & 3 \\ 5 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 45 & 0 & 138 & 22 \\ 3 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[434] $1^{13}2^3 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -2 & 9 & 0 \\ 7 & 0 & 11 & 9 & 49 & 15 \\ 2 & 0 & 2 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[435] $1^{12}2^3 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -2 & 5 & 0 \\ 6 & 0 & 7 & 5 & 17 & 9 \\ 2 & 0 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[436] $1^{12}3^3 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 11 & 0 & 50 & 14 \\ 3 & 3 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

[437] $1^{11}41^{11}231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -298, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 6 & 1 & 8 & 3 \\ 6 & 13 & 0 & 20 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 20 & 0 & 26 & 10 \\ 3 & 7 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

- [438] $1^{1231^{12}31^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -304$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 12.11 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 12 & 3 \\ 5 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 30 & 0 & 58 & 14 \\ 3 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$
- [439] $1^{122}321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -308$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -2 & 8 & 0 \\ 7 & 0 & 8 & 8 & 30 & 12 \\ 2 & 0 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$
- [440] $1^{12}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -107$, $\det = -331$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 12.52 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 19 & 0 & 68 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$
- [441] $1^{14}3221^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -332$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 16 & 0 & 90 & 20 \\ 2 & 2 & 3 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$
- [442] $1^{13}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -107$, $\det = -336$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 20 & 0 & 70 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$
- [443] $1^{142}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -341$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 7 & 13 & 0 & 7 & 57 & 15 \\ 2 & 3 & -1 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$
- [444] $1^{132}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 3 & 0 \\ 7 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$
- [447] $1^{12231^{12}31^{13}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 23 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 60 & 0 & 258 & 30 \\ 3 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$
- [446] $1^{142}321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 15 & 2 \\ 1 & 0 & 1 & -2 & 13 & 0 \\ 9 & 0 & 15 & 13 & 97 & 21 \\ 2 & 0 & 2 & 0 & 21 & 2 \end{pmatrix}_{b_0}$$
- [445] $1^{142}321^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 14 & 2 \\ 1 & 0 & 1 & -2 & 10 & 0 \\ 8 & 0 & 14 & 10 & 74 & 18 \\ 2 & 0 & 2 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$
- [448] $1^{13}41^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -107$, $\det = -347$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 21 & 0 & 88 & 18 \\ 3 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[449] $1^{14} 3^{12} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -348, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 17 & 0 & 106 & 20 \\ 3 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[452] $1^{13} 3^{12} 221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -351, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 19 & 3 \\ 2 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 19 & 0 & 165 & 25 \\ 3 & 3 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

[451] $1^{14} 3^{12} 221^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -351, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 25 & 3 \\ 2 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 24 & 0 & 295 & 33 \\ 3 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

[450] $1^{13} 3^{12} 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -351, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 14 & 0 & 75 & 17 \\ 3 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[455] $1^{14} 3^{12} 221^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -358, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 20 & 0 & 126 & 24 \\ 2 & 2 & 3 & -3 & 24 & 2 \end{pmatrix}_{b_0}$$

[454] $1^{14} 3^{12} 221^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -358, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 32 & 3 \\ 2 & 1 & 1 & 32 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 32 & 32 & 0 & 526 & 44 \\ 3 & 3 & -1 & 44 & 3 \end{pmatrix}_{b_0}$$

[453] $1^{14} 3^{12} 221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -358, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 22 & 0 & 206 & 28 \\ 3 & 3 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[458] $1^{12} 2^3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 11 & 0 \\ 9 & 0 & 9 & 11 & 49 & 15 \\ 2 & 0 & 2 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[456] $1^{12} 2^3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -2 & 5 & 0 \\ 7 & 0 & 7 & 5 & 17 & 9 \\ 2 & 0 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[457] $1^{13} 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 24 & 0 & 121 & 21 \\ 3 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[459] $1^{11} 231^{13} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 13 & 0 & 74 & 16 \\ 2 & 4 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[461] \ 1^{13} 2^3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 6 & 0 \\ 7 & 0 & 10 & 6 & 34 & 12 \\ 2 & 0 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[460] \ 1^{12} 4^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 22 & 0 & 97 & 19 \\ 3 & 4 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[462] \ 1^{14} 2^3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 0 \\ 9 & 15 & 0 & 13 & 97 & 21 \\ 2 & 3 & -1 & 0 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[463] \ 1^{12} 231^{12} 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 5 & 1 & 24 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 60 & 0 & 258 & 30 \\ 3 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[464] \ 1^{12} 231^{14} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 1 & 20 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 60 & 0 & 238 & 28 \\ 3 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[465] \ 1^{12} 4^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -387, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 19 & 25 & 0 & 132 & 22 \\ 3 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[466] \ 1^{14} 4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -396, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 22 & 0 & 90 & 18 \\ 3 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[467] \ 1^{14} 3^1 221^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -402, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 26 & 3 \\ 2 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 27 & 0 & 346 & 36 \\ 3 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[468] \ 1^{11} 231^{12} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 1 & 17 & 3 \\ 5 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 45 & 0 & 138 & 22 \\ 3 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[469] \ 1^{11} 231^{14} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 1 & 25 & 3 \\ 5 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 75 & 0 & 378 & 36 \\ 3 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

$$[470] \ 1^{11} 231^{12} 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 1 & 19 & 3 \\ 5 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 45 & 0 & 138 & 22 \\ 3 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[471] $1^{14}2^3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -406, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 8 & 14 & 0 & 10 & 74 & 18 \\ 2 & 3 & -1 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

[473] $1^{12}2^3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -408, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -2 & 8 & 0 \\ 8 & 0 & 8 & 8 & 30 & 12 \\ 2 & 0 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[472] $1^{14}2^3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -408, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -2 & 4 & 0 \\ 8 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[474] $31^{11}2321^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -409, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 3 & 2 \\ 3 & 1 & 4 & 1 & 3 & 3 \\ 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 3 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[475] $1^{13}231^{13}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 21 & 0 & 142 & 24 \\ 2 & 4 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[476] $1^{13}2^3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -420, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -2 & 12 & 0 \\ 9 & 0 & 12 & 12 & 70 & 18 \\ 2 & 0 & 2 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

[477] $1^{13}231^{13}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 33 & 3 \\ 5 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 33 & 90 & 0 & 583 & 45 \\ 3 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

[478] $1^{13}3^2 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -431, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 14 & 0 & 75 & 17 \\ 3 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[479] $1^{14}4^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 24 & 0 & 101 & 19 \\ 3 & 4 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

[480] $1^{14}1^{12}231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -438, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 6 & 1 & 10 & 3 \\ 6 & 13 & 0 & 24 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 24 & 0 & 38 & 12 \\ 3 & 7 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[482] $1^{13}2^3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -2 & 9 & 0 \\ 8 & 0 & 11 & 9 & 49 & 15 \\ 2 & 0 & 2 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[481] $1^{14}2^3 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 7 & 0 \\ 8 & 0 & 13 & 7 & 57 & 15 \\ 2 & 0 & 2 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[484] $1^{12}31^{13}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 16 & 3 \\ 5 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 45 & 0 & 133 & 21 \\ 3 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[483] $1^{11}231^{12}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 18 & 3 \\ 5 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 45 & 0 & 138 & 22 \\ 3 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[485] $1^{13}3^2 221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -461, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 19 & 0 & 165 & 25 \\ 3 & 3 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

[486] $1^{14}3^2 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -462, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 17 & 0 & 106 & 20 \\ 3 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[487] $1^{13}4^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -483, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.52 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 23 & 0 & 97 & 19 \\ 3 & 4 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

[488] $1^{12}231^{13}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.31 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 17 & 0 & 104 & 20 \\ 2 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[489] $1^{14}3^2 221^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -502, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 21 & 3 \\ 2 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 22 & 0 & 206 & 28 \\ 3 & 3 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[491] $1^{11}231^{13}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 21 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 60 & 0 & 243 & 29 \\ 3 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[490] $1^{13}2^3 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 6 & 0 \\ 8 & 0 & 10 & 6 & 34 & 12 \\ 2 & 0 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[492] $1^{14}4^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -515, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 25 & 0 & 121 & 21 \\ 3 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[493] $1^{11}231^{14}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 16 & 0 & 122 & 20 \\ 2 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[494] $1^{14}2^3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -540, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 14 & 2 \\ 1 & 0 & 1 & -2 & 10 & 0 \\ 9 & 0 & 14 & 10 & 74 & 18 \\ 2 & 0 & 2 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

[495] $1^{14}2^3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 15 & 2 \\ 1 & 0 & 1 & -2 & 13 & 0 \\ 10 & 0 & 15 & 13 & 97 & 21 \\ 2 & 0 & 2 & 0 & 21 & 2 \end{pmatrix}_{b_0}$$

[500] $1^{12}231^{13}31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 27 & 3 \\ 5 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 75 & 0 & 393 & 37 \\ 3 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[499] $1^1 231^{13} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 17 & 3 \\ 5 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 45 & 0 & 133 & 21 \\ 3 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[498] $1^1 231^{13} 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 19 & 3 \\ 5 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 45 & 0 & 133 & 21 \\ 3 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[497] $1^{13} 231^{14} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 37 & 3 \\ 5 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 37 & 105 & 0 & 778 & 52 \\ 3 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[496] $1^1 231^{15} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 25 & 3 \\ 5 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 75 & 0 & 373 & 35 \\ 3 & 7 & -1 & 35 & 2 \end{pmatrix}_{b_0}$$

[502] $1^{14} 3^2 221^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -568, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 27 & 3 \\ 2 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 27 & 0 & 346 & 36 \\ 3 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

[501] $1^{14} 3^2 221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -568, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 17 & 0 & 106 & 20 \\ 3 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[505] $1^{13} 2^3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -2 & 12 & 0 \\ 10 & 0 & 12 & 12 & 70 & 18 \\ 2 & 0 & 2 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

[504] $1^{13}2^3 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -2 & 9 & 0 \\ 9 & 0 & 11 & 9 & 49 & 15 \\ 2 & 0 & 2 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[503] $1^{12}231^{14}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 31 & 3 \\ 5 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 90 & 0 & 558 & 44 \\ 3 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

[506] $1^{13}4^3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -588, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.54 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 26 & 0 & 130 & 22 \\ 3 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[507] $1^{14}4^3 1^3 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -592, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 20 & 28 & 0 & 158 & 24 \\ 3 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[508] $1^1 231^{13}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -604, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 18 & 3 \\ 5 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 45 & 0 & 133 & 21 \\ 3 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[509] $1^{14}2^3 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 7 & 0 \\ 9 & 0 & 13 & 7 & 57 & 15 \\ 2 & 0 & 2 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[510] $1^{13}231^{13}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 34 & 3 \\ 5 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 90 & 0 & 583 & 45 \\ 3 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

[511] $1^{14}4^3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -635, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.54 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 27 & 0 & 132 & 22 \\ 3 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[512] $1^{14}3^3 221^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -638, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 22 & 3 \\ 2 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 22 & 0 & 206 & 28 \\ 3 & 3 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[513] $1^{13}4^3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -651, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 21 & 29 & 0 & 169 & 25 \\ 3 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[514] $1^1 231^{14}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -664, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.19 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 21 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 60 & 0 & 238 & 28 \\ 3 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[515] $1^{11}41^{12}231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -688, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 6 & 1 & 13 & 3 \\ 6 & 13 & 0 & 32 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 32 & 0 & 68 & 16 \\ 3 & 7 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[516] $1^{14}231^{14}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 36 & 4 \\ 1 & 4 & 1 & 0 & 28 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 36 & 28 & 0 & 254 & 32 \\ 2 & 4 & 3 & -3 & 32 & 1 \end{pmatrix}_{b_0}$$

[517] $1^{11}231^{13}31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 22 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 60 & 0 & 243 & 29 \\ 3 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[518] $1^{14}2^3 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -728, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 14 & 2 \\ 1 & 0 & 1 & -2 & 10 & 0 \\ 10 & 0 & 14 & 10 & 74 & 18 \\ 2 & 0 & 2 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

[520] $1^{12}231^{13}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -764, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 28 & 3 \\ 5 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 75 & 0 & 393 & 37 \\ 3 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[519] $1^{11}231^{14}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -764, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 26 & 3 \\ 5 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 75 & 0 & 378 & 36 \\ 3 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

[521] $1^{14}2^3 3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -768, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 15 & 2 \\ 1 & 0 & 1 & -2 & 13 & 0 \\ 11 & 0 & 15 & 13 & 97 & 21 \\ 2 & 0 & 2 & 0 & 21 & 2 \end{pmatrix}_{b_0}$$

[522] $1^{12}231^{14}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 20 & 0 & 158 & 24 \\ 2 & 4 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[523] $1^{14}41^{13}231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -788, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 6 & 1 & 15 & 3 \\ 6 & 13 & 0 & 36 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 36 & 0 & 88 & 18 \\ 3 & 7 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[524] $1^{14}4^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -796, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.53 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 20 & 30 & 0 & 169 & 25 \\ 3 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[525] $1^{11}231^{13}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -824, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 23 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 60 & 0 & 243 & 29 \\ 3 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

- [526] $1^{13}231^{14}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -836, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 33 & 4 \\ 1 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 33 & 24 & 0 & 202 & 28 \\ 2 & 4 & 3 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$
- [528] $1^1231^{14}31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -864, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 22 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 60 & 0 & 238 & 28 \\ 3 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$
- [527] $1^{11}231^{13}31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -864, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 24 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 60 & 0 & 243 & 29 \\ 3 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$
- [529] $1^{12}231^{13}31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -884, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 29 & 3 \\ 5 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 75 & 0 & 393 & 37 \\ 3 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$
- [530] $1^{14}41^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -107, \det = -915, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 33 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 23 & 33 & 0 & 212 & 28 \\ 3 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$
- [531] $1^{13}231^{14}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -924, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 38 & 3 \\ 5 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 38 & 105 & 0 & 778 & 52 \\ 3 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$
- [532] $1^{12}231^{14}31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 32 & 3 \\ 5 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 32 & 90 & 0 & 558 & 44 \\ 3 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$
- [533] $1^1231^{14}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -984, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 23 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 60 & 0 & 238 & 28 \\ 3 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$
- [534] $1^1231^{14}31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1024, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 24 & 3 \\ 5 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 60 & 0 & 238 & 28 \\ 3 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$
- [535] $1^{11}231^{14}31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1044, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 27 & 3 \\ 5 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 75 & 0 & 378 & 36 \\ 3 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$
- [536] $1^{13}231^{14}31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 39 & 3 \\ 5 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 105 & 0 & 778 & 52 \\ 3 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$
- [537] $1^{11}41^{13}231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 6 & 1 & 18 & 3 \\ 6 & 13 & 0 & 44 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 44 & 0 & 130 & 22 \\ 3 & 7 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$
- [538] $1^{12}231^{14}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 1 & 33 & 3 \\ 5 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 33 & 90 & 0 & 558 & 44 \\ 3 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

$$[539] 1^{11}231^{14}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 1 & 28 & 3 \\ 5 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 75 & 0 & 378 & 36 \\ 3 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

$$[540] 1^{14}41^{14}231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1278, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 6 & 1 & 20 & 3 \\ 6 & 13 & 0 & 48 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 20 & 48 & 0 & 158 & 24 \\ 3 & 7 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[541] 1^{11}231^{14}31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 1 & 29 & 3 \\ 5 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 75 & 0 & 378 & 36 \\ 3 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

$$[542] 1^{12}231^{14}31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 1 & 34 & 3 \\ 5 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 90 & 0 & 558 & 44 \\ 3 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

$$[543] 1^{11}41^{14}231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1888, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 6 & 1 & 23 & 3 \\ 6 & 13 & 0 & 56 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 23 & 56 & 0 & 212 & 28 \\ 3 & 7 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [544] 1^{11}51^{11}231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 3 & 3 & 0 & 4 & 10 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[545] 1^{12}51^{14}131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 16 & 5 \\ 9 & 13 & 20 & 7 \\ 16 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[546] 231^{11}51^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 10 & 12 & 8 & 6 \\ 12 & 13 & 10 & 7 \\ 8 & 10 & 5 & 5 \\ 6 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[547] 1^{11}51^{11}131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.26 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 0 & 7 \\ 1 & 0 & -2 & 0 \\ 5 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[548] 1^{11}31^{14}131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[549] 1^{11}51^{12}131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 8 & 5 \\ 9 & 13 & 10 & 7 \\ 8 & 10 & 5 & 5 \\ 5 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[557] \ 1 \overset{1}{\underset{1^{11}}{5}} 131^{\overset{1}{\text{g}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 32, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.38 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 6 & 5 \\ 9 & 13 & 10 & 7 \\ 6 & 10 & 5 & 5 \\ 5 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[556] \ 1^{\overset{1}{\text{g}}} 3221^{\overset{1}{\text{g}}} 51^{\overset{11}{\text{g}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -103, \ \det = 32, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 14.37 \rightarrow 15, \ T_{\max} = 22, \ T \in [15, 22]$$

$$\begin{pmatrix} 0 & 1 & 4 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 4 & 1 & 17 & 5 & 0 & 7 \\ 1 & 0 & 5 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 7 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[555] \ 21^{\overset{1}{\text{g}}} 5 131^{\overset{1}{\text{g}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 32, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.38 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 13 & 14 & 9 & 7 \\ 14 & 13 & 10 & 7 \\ 9 & 10 & 5 & 5 \\ 7 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[554] \ 1^{\overset{1}{\text{g}}} 5 1231^{\overset{1}{\text{g}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -103, \ \det = 32, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.37 \rightarrow 15, \ T_{\max} = 22, \ T \in [15, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[553] \ 1^{\overset{1}{\text{g}}} 5 1231^{\overset{1}{\text{g}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -103, \ \det = 32, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.7$$

$$T_{\min} = 13.75 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[552] \ 1 \overset{1}{\underset{1^{12}}{5}} 131^{\overset{1}{\text{g}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 32, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.7$$

$$T_{\min} = 13.75 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 12 & 5 \\ 9 & 13 & 20 & 7 \\ 12 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[551] \ 1 \overset{1}{\underset{1^{13}}{5}} 131^{\overset{1}{\text{g}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 32, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 14.37 \rightarrow 15, \ T_{\max} = 22, \ T \in [15, 22]$$

$$\begin{pmatrix} 5 & 9 & 18 & 5 \\ 9 & 13 & 30 & 7 \\ 18 & 30 & 61 & 15 \\ 5 & 7 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[550] \ 1^{\overset{1}{\text{g}}} 5 1231^{\overset{1}{\text{g}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -103, \ \det = 32, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.38 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[558] \ 1^{\overset{1}{\text{g}}} 11^{\overset{1}{\text{g}}} 5 131^{\overset{1}{\text{g}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 43, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.30 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 7 & 5 \\ 9 & 13 & 10 & 7 \\ 7 & 10 & 5 & 5 \\ 5 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[559] \ 1^{\overset{1}{\text{g}}} 12^{\overset{3}{\text{g}}} 5 131^{\overset{1}{\text{g}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 59, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.7$$

$$T_{\min} = 13.36 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 15 & 5 \\ 9 & 13 & 20 & 7 \\ 15 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[560] \ 1^{\overset{1}{\text{g}}} 12^{\overset{3}{\text{g}}} 5 131^{\overset{1}{\text{g}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 67, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.7$$

$$T_{\min} = 13.43 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 13 & 5 \\ 9 & 13 & 20 & 7 \\ 13 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[561] $1^{11}5^{12}1^{12}231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 68, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.82 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 5 & 23 & 1 & 5 \\ 23 & 90 & 0 & 18 \\ 1 & 0 & -2 & 0 \\ 5 & 18 & 0 & 3 \end{pmatrix}_{b_0}$

[562] $1^{12}5^{12}131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 76, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 5 & 9 & 14 & 5 \\ 9 & 13 & 20 & 7 \\ 14 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0}$

[563] $21^{11}5^{12}131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 88, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.32 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 5 & 10 & 7 & 5 \\ 10 & 13 & 10 & 7 \\ 7 & 10 & 5 & 5 \\ 5 & 7 & 5 & 3 \end{pmatrix}_{b_0}$

[564] $1^{13}5^{12}131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 91, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.58 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 5 & 9 & 19 & 5 \\ 9 & 13 & 30 & 7 \\ 19 & 30 & 61 & 15 \\ 5 & 7 & 15 & 3 \end{pmatrix}_{b_0}$

[565] $1^{11}5^{11}1^{12}231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.35 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 4 & 5 & 0 & 5 & 18 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$

[566] $1^{11}5^{11}1^{13}231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.52 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 0 \\ 10 & 11 & 0 & 13 & 126 & 20 \\ 2 & 2 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$

[567] $1^{11}5^{12}231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 7 & 8 & 0 & 9 & 60 & 14 \\ 2 & 2 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$

[568] $1^{13}5^{14}131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 5 & 9 & 22 & 5 \\ 9 & 13 & 30 & 7 \\ 22 & 30 & 61 & 15 \\ 5 & 7 & 15 & 3 \end{pmatrix}_{b_0}$

[569] $1^{13}5^{12}131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 5 & 9 & 20 & 5 \\ 9 & 13 & 30 & 7 \\ 20 & 30 & 61 & 15 \\ 5 & 7 & 15 & 3 \end{pmatrix}_{b_0}$

[570] $1^{14}31^{11}41^{13}131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 20 & 4 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 20 & 16 & 0 & 90 & 18 \\ 2 & 4 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0}$

[571] $1^{13}5^{13}131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 9 & 21 & 5 \\ 9 & 13 & 30 & 7 \\ 21 & 30 & 61 & 15 \\ 5 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[572] $21^{12}5^{11}131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 142, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 10 & 14 & 5 \\ 10 & 13 & 20 & 7 \\ 14 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[573] $1^{11}31^{11}41^131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 11 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 11 & 7 & 0 & 21 & 9 \\ 2 & 4 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[574] $1^{12}5^{11}231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 5 & 7 & 0 & 6 & 30 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[575] $1^{13}31^{11}41^131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 216, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 17 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 17 & 13 & 0 & 61 & 15 \\ 2 & 4 & 2 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[576] $1^{12}31^{11}41^131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 14 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 14 & 10 & 0 & 38 & 12 \\ 2 & 4 & 2 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[577] $21^{13}5^{11}131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 232, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 10 & 21 & 5 \\ 10 & 13 & 30 & 7 \\ 21 & 30 & 61 & 15 \\ 5 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[578] $1^{13}5^{11}231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 252, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.44 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 6 & 9 & 0 & 7 & 46 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[579] $1^{12}5^{11}2231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 272, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 8 & 10 & 0 & 10 & 78 & 16 \\ 2 & 2 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[580] $1^{12}5^{11}3231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 332, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 14 & 0 \\ 11 & 13 & 0 & 14 & 150 & 22 \\ 2 & 2 & -1 & 0 & 22 & 1 \end{pmatrix}_{b_0}$$

[581] $1^13221^{11}51^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 382, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 4 & 2 \\ 4 & 18 & 1 & 5 & 20 & 8 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 5 & 0 & -2 & 5 & 0 \\ 4 & 20 & 0 & 5 & 18 & 8 \\ 2 & 8 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[582] $1^{13}5^{11}2231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 432, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 9 & 12 & 0 & 11 & 100 & 18 \\ 2 & 2 & -1 & 0 & 18 & 1 \end{pmatrix}_{b_0}$$

[583] $1^{13}51^3231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 572$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.39 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 15 & 0 \\ 12 & 15 & 0 & 15 & 178 & 24 \\ 2 & 2 & -1 & 0 & 24 & 1 \end{pmatrix}_{b_0}$$

[584] $1^13221^{12}51^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 772$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 8 & 2 \\ 4 & 18 & 1 & 5 & 40 & 8 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 5 & 0 & -2 & 10 & 0 \\ 8 & 40 & 0 & 10 & 78 & 16 \\ 2 & 8 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[585] $1^13221^{13}51^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -103$, $\det = 1422$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 12 & 2 \\ 4 & 18 & 1 & 5 & 60 & 8 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 5 & 0 & -2 & 15 & 0 \\ 12 & 60 & 0 & 15 & 178 & 24 \\ 2 & 8 & -1 & 0 & 24 & 1 \end{pmatrix}_{b_0}$$

$T = 11$ [586] $1^{11}31^5131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -81$, $\det = -8$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13.50 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 1 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[587] $1^{11}31^5131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -81$, $\det = -13$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 3 & 0 & 1 & 1 & 3 \\ 2 & 3 & -1 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[588] $1^{11}31^5131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -81$, $\det = -20$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13.60 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[589] $1^{12}31^5131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -81$, $\det = -44$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 13.82 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 2 & 0 \\ 3 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[590] $1^{11}31^5131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -81$, $\det = -48$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13.58 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 2 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[591] $1^{11}31^5131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -81$, $\det = -80$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13.75 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 3 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[592] $1^{13}31^5131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -81$, $\det = -104$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13.96 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 4 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[593] $1^{12}31^1 5 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 11, T_{\max} = 14, \Delta = -81, \det = -120, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 2 & 0 \\ 4 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[594] $1^{13}31^1 5 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 11, T_{\max} = 14, \Delta = -81, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.84 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 3 & 0 \\ 5 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[595] $1^{12}31^1 5 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 11, T_{\max} = 14, \Delta = -81, \det = -188, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 13.79 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 2 & 0 \\ 5 & 0 & 6 & 2 & 10 & 6 \\ 2 & 0 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[596] $1^{13}31^1 5 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 11, T_{\max} = 14, \Delta = -81, \det = -240, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.78 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 3 & 0 \\ 6 & 0 & 9 & 3 & 25 & 9 \\ 2 & 0 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[597] $1^{14}31^1 5 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 11, T_{\max} = 14, \Delta = -81, \det = -260, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.85 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -2 & 4 & 0 \\ 7 & 0 & 12 & 4 & 46 & 12 \\ 2 & 0 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

$T = 12$ [599] $1^{11}6^{1321} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 12, T_{\max} = 15, \Delta = -79, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$

$T_{\min} = 14.50 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 5 & 3 & 1 & 5 \\ 3 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 5 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[598] $1^{11}6^{1321} 131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 12, T_{\max} = 15, \Delta = -79, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.50 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 1 & 3 & 0 & 1 & 3 \\ 2 & 2 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[601] $1^{11}6^{1321} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 12, T_{\max} = 15, \Delta = -79, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[600] $1^{11}6^{1321} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 12, T_{\max} = 15, \Delta = -79, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[603] $1^{11}6^{1321} 131^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 12, T_{\max} = 15, \Delta = -79, \det = 44, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.82 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 7 & 0 & 16 & 8 \\ 2 & 2 & 2 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[602] $1^{11} 6^{1^{12}321} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 44, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.82 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 3 & 2 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

[606] $1^{11} 6^{1^{1321}} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 48, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.58 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 3 & 5 & 0 & 11 & 7 \\ 2 & 2 & 2 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[605] $1^{11} 6^{1^{1321}} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 48, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.58 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[604] $1^{11} 6^{1^{11321}} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 48, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.58 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 2 & 2 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[607] $1^{11} 6^{1^{1321}} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 68, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.65 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 4 & 4 & 0 & 3 & 10 & 6 \\ 2 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[608] $1^{11} 6^{1^{1321}} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 80, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.75 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 5 & 11 & 0 & 43 & 13 \\ 2 & 2 & 2 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[609] $1^{11} 6^{1^{1321}} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 80, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 4 & 8 & 0 & 24 & 10 \\ 2 & 2 & 2 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[610] $1^{11} 6^{1^{1321}} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.59 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 5 & 6 & 0 & 19 & 9 \\ 2 & 2 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[611] $1^{11} 6^{1^{11321}} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 140, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 4 & 1 & 3 & 11 & 7 \\ 2 & 2 & 0 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[612] $31^{11} 31^6 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -86, \det = 152, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 3 & 2 \\ 3 & 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & -2 & 2 & 0 \\ 4 & 3 & 4 & 2 & 5 & 5 \\ 2 & 2 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[613] $1^{11} 1^{14} 321^1 6 \quad 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 164, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.85 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 8 & 4 & 4 & 3 & 26 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[614] $1^{11} 1^{12} 321^1 6 \quad 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 180, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 6 & 4 & 2 & 3 & 14 & 8 \\ 2 & 2 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

[615] $1^{11} 1^{13} 321^1 6 \quad 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 188, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.68 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 7 & 4 & 3 & 3 & 19 & 9 \\ 2 & 2 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[616] $1^{11} 1^{13} 321^1 6 \quad 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 204, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 6 & 9 & 0 & 34 & 12 \\ 2 & 2 & 2 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[617] $1^{11} 1^{13} 321^1 6 \quad 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 248, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 7 & 12 & 0 & 55 & 15 \\ 2 & 2 & 2 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[618] $1^{11} 1^{13} 321^1 6 \quad 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 260, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.66 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 8 & 15 & 0 & 82 & 18 \\ 2 & 2 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

4.451 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{e}_6$

$T = 8 \quad [0] 1^{11} 1^{13} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{e}}_6 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 0 & 4 & -4 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{112} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{e}}_6 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.7$
 $T_{\min} = 14.21 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -2 & 6 & 0 & 0 \\ 6 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 0 & 8 & -4 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{113} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{e}}_6 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 739, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 14.21 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 \\ 9 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 0 & 12 & -4 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \quad 1^{11} 3 2 1^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_2 + \textcolor{red}{c}_6 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 5 & 0 & 3 & -2 & 0 \\ 3 & 5 & -3 & 4 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3 2 1^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_2 + \textcolor{red}{c}_6 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 2 & 3 \\ 4 & 9 & 0 & 12 & 4 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 12 & 0 & 14 & 7 & 8 \\ 2 & 4 & 0 & 7 & -2 & 0 \\ 3 & 5 & -3 & 8 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 3 2 1^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_2 + \textcolor{red}{c}_6 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.34 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 3 \\ 4 & 9 & 0 & 8 & 4 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 8 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & -2 & 0 \\ 3 & 5 & -3 & 4 & 0 & -1 \end{pmatrix}_{b_0}$$

4.452 NHC(-2)(-3) + f₄ + su₂ + f₄

$$T = 8 \quad [0] \quad 1^{111} 3 1^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 100, \Delta = -160, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 4 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{112} 3 1^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 100, \Delta = -160, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.9$$

$$T_{\min} = 13.36 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -2 & 6 & 0 & 0 \\ 6 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 8 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{113} 3 1^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 100, \Delta = -160, \det = 552, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 \\ 9 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 12 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{114} 3 1^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 100, \Delta = -160, \det = 832, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 \\ 12 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 16 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{115} 3 1^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 100, \Delta = -160, \det = 1192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 40 & 0 & 5 \\ 2 & 4 & -2 & 15 & 0 & 0 \\ 15 & 40 & 15 & 148 & 0 & 20 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 20 & -3 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \quad 1^{11} 3 2 1^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_2 + \textcolor{red}{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 93, T \in [16, 93]$$

$$\begin{pmatrix} 5 & 13 & 9 & 1 & 5 \\ 13 & 30 & 20 & 0 & 10 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 10 & 7 & -3 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{111}} 3 21^{\overline{11}} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_2 + \textcolor{red}{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$
 $T_{\min} = 13.92 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$
 $\begin{pmatrix} 5 & 12 & 19 & 1 & 5 \\ 12 & 25 & 40 & 0 & 9 \\ 19 & 40 & 63 & 0 & 15 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 15 & -3 & 1 \end{pmatrix}_{b_0}$

$$[11] \quad 1^1 1^3 21^{113} + [\textcolor{blue}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_2 + \textcolor{red}{f}_4 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -80, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.40 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 1 & 9 & 0 & 3 \\ 1 & 4 & 0 & 1 & 9 & 0 & 2 \\ 1 & 1 & 1 & -3 & 3 & 0 & -1 \\ 3 & 9 & 9 & 3 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & -1 & 9 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11} 3^{21} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{f}_4 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -80, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.40 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 4 & 1 & 12 & 0 & 3 \\ 1 & 4 & 0 & 1 & 12 & 0 & 2 \\ 1 & 1 & 1 & -3 & 4 & 0 & -1 \\ 4 & 12 & 12 & 4 & 46 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & -1 & 12 & -3 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{11} 1^1 3^{21} 1^{15} + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s} \textcolor{orange}{u}_2 + \textcolor{red}{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.5$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 4 & 1 & 15 & 0 & 3 \\ 1 & 4 & 0 & 1 & 15 & 0 & 2 \\ 1 & 1 & 1 & -3 & 5 & 0 & -1 \\ 5 & 15 & 15 & 5 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & -1 & 15 & -3 & -1 \end{pmatrix}_{b_0}$

$$[7] \quad 1^1 \bar{1}^3 2^1 \bar{1}^{17} + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{5}\textcolor{blue}{u}_2 + \textcolor{red}{f}_4 \quad T = 9, \, T_{\max} = 93, \, \Delta = -132, \, \det = -80, \, (n_+, n_-, n_0) = (1, 9, 0), \, c_{\text{ext}} = 15.7$$

$$T_{\min} = 13.40 \rightarrow 14, \, T_{\max} = 93, \, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 1 & 4 & 1 & 21 & 0 & 3 \\ 1 & 4 & 0 & 1 & 21 & 0 & 2 \\ 1 & 1 & 1 & -3 & 7 & 0 & -1 \\ 7 & 21 & 21 & 7 & 145 & 0 & 21 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & -1 & 21 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^1 \bar{1}^3 2^1 \bar{1}^{16} + [\textcolor{red}{\mathfrak{s}}\textcolor{blue}{\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{s}}\textcolor{red}{\mathfrak{u}}_2 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -80, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.6$$

$$T_{\min} = 13.40 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 4 & 1 & 18 & 0 & 3 \\ 1 & 4 & 0 & 1 & 18 & 0 & 2 \\ 1 & 1 & 1 & -3 & 6 & 0 & -1 \\ 6 & 18 & 18 & 6 & 106 & 0 & 18 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & -1 & 18 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^1 \bar{3} 21^{111} + [\textcolor{red}{5} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{5} \textcolor{red}{u}_2 + \textcolor{red}{f}_4 \quad T = 9, \, T_{\max} = 93, \, \Delta = -132, \, \det = -80, \, (n_+, n_-, n_0) = (1, 9, 0), \, c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.40 \rightarrow 14, \, T_{\max} = 93, \, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 3 & 0 & 3 \\ 1 & 4 & 0 & 1 & 3 & 0 & 2 \\ 1 & 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 3 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & -1 & 3 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^1 1^3 21^{112} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_2 + \textcolor{red}{f}_4 \quad T = 9, \, T_{\max} = 93, \, \Delta = -132, \, \det = -80, \, (n_+, n_-, n_0) = (1, 9, 0), \, c_{\text{ext}} = 14.9$$

$$T_{\min} = 13.40 \rightarrow 14, \, T_{\max} = 93, \, T \in [14, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 1 & 6 & 0 & 3 \\ 1 & 4 & 0 & 1 & 6 & 0 & 2 \\ 1 & 1 & 1 & -3 & 2 & 0 & -1 \\ 2 & 6 & 6 & 2 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & -1 & 6 & -3 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{11}1^1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{red}{f}_4 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -165, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.4$
 $T_{\min} = 13.61 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$
 $\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 7 & -3 & 1 \end{pmatrix}_{b_0}$

$$T = 10 \quad [15] \quad 1^{11} 3^{221} 1^{21} + [\textcolor{blue}{s} \textcolor{red}{u} \textcolor{red}{2} \cdot \textcolor{blue}{g} \textcolor{red}{2}] + \textcolor{green}{f}_4 + \textcolor{orange}{s} \textcolor{red}{u} \textcolor{red}{2} + \textcolor{red}{f}_4 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -103, \quad \det = 103, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.63 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 6 & 0 & 17 & 6 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 17 & 0 & 36 & 13 & 14 \\ 2 & 6 & 0 & 13 & 0 & 2 \\ 3 & 6 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

4.453 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 & 3 & 0 \\ 3 & 8 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 6 & 0 & 4 & 4 \\ 3 & 5 & 0 & 4 & -3 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \quad 1^{\overline{1}} 3^{\overline{1}} 2^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 3 & 0 & 3 \\ 1 & 4 & 0 & 1 & 3 & 0 & 3 \\ 1 & 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 3 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -5 & 0 \\ 1 & 3 & 3 & 1 & 3 & 0 & 1 \\ 2 & 3 & 2 & -1 & 3 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1}} 3^{\overline{1}} 2^{\overline{1112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.9$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 3 & 0 & 3 \\ 1 & 4 & 0 & 1 & 3 & 0 & 3 \\ 1 & 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 3 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -5 & 0 \\ 2 & 6 & 6 & 2 & 6 & 0 & 10 \\ 2 & 3 & 2 & -1 & 3 & -3 & 6 \end{pmatrix}_{b_0}$$

4.454 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 2 & 3 \\ 4 & 9 & 4 & 8 & 0 & 4 & 5 \\ 2 & 4 & -2 & 3 & 0 & 2 & 0 \\ 3 & 8 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 2 & 3 & 0 & 0 & 2 \\ 3 & 5 & 0 & 4 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.455 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \quad 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 3 & 5 & 0 & 6 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -3 & 4 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{131}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 2 & 3 \\ 3 & 4 & 0 & 19 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 19 & 0 & 73 & 21 & 12 & 15 \\ 3 & 5 & 0 & 21 & 4 & 4 & 4 \\ 2 & 4 & 0 & 12 & 4 & 0 & 2 \\ 3 & 4 & -3 & 15 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 9 & 6 & 7 \\ 3 & 5 & 0 & 9 & 4 & 4 & 4 \\ 2 & 4 & 0 & 6 & 4 & 0 & 2 \\ 3 & 4 & -3 & 7 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3 1^{\overline{121}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -396, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 & 2 & 3 \\ 3 & 4 & 0 & 14 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 15 & 9 & 11 \\ 3 & 5 & 0 & 15 & 4 & 4 & 4 \\ 2 & 4 & 0 & 9 & 4 & 0 & 2 \\ 3 & 4 & -3 & 11 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{121}} 3 1^{\overline{121}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 & 2 & 3 \\ 3 & 4 & 0 & 18 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 18 & 12 & 14 \\ 3 & 5 & 0 & 18 & 4 & 4 & 4 \\ 2 & 4 & 0 & 12 & 4 & 0 & 2 \\ 3 & 4 & -3 & 14 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{121}} 3 1^{\overline{141}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -936, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 & 2 & 3 \\ 3 & 4 & 0 & 28 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 16 & 28 & 0 & 154 & 30 & 18 & 22 \\ 3 & 5 & 0 & 30 & 4 & 4 & 4 \\ 2 & 4 & 0 & 18 & 4 & 0 & 2 \\ 3 & 4 & -3 & 22 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{121}} 3 1^{\overline{131}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -956, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 & 2 & 3 \\ 3 & 4 & 0 & 23 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 13 & 23 & 0 & 100 & 24 & 15 & 18 \\ 3 & 5 & 0 & 24 & 4 & 4 & 4 \\ 2 & 4 & 0 & 15 & 4 & 0 & 2 \\ 3 & 4 & -3 & 18 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{131}} 3 1^{\overline{131}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -1476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 & 2 & 3 \\ 3 & 4 & 0 & 27 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 15 & 27 & 0 & 133 & 27 & 18 & 21 \\ 3 & 5 & 0 & 27 & 4 & 4 & 4 \\ 2 & 4 & 0 & 18 & 4 & 0 & 2 \\ 3 & 4 & -3 & 21 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{131}} 3 1^{\overline{141}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -1796, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 & 2 & 3 \\ 3 & 4 & 0 & 32 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 18 & 32 & 0 & 193 & 33 & 21 & 25 \\ 3 & 5 & 0 & 33 & 4 & 4 & 4 \\ 2 & 4 & 0 & 21 & 4 & 0 & 2 \\ 3 & 4 & -3 & 25 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{141}} 3 1^{\overline{141}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -2456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 & 2 & 3 \\ 3 & 4 & 0 & 36 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 20 & 36 & 0 & 238 & 36 & 24 & 28 \\ 3 & 5 & 0 & 36 & 4 & 4 & 4 \\ 2 & 4 & 0 & 24 & 4 & 0 & 2 \\ 3 & 4 & -3 & 28 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

4.456 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 3 & 3 & 4 \\ 3 & 5 & 0 & 6 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 2 & 2 \\ 2 & 3 & 0 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 4 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 600, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 9 & 6 & 5 & 7 \\ 3 & 5 & 0 & 9 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 6 & 4 & 0 & 2 & 2 \\ 2 & 3 & 0 & 5 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 7 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.457 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 0 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 44 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 44 & 84 & 0 & 346 & 34 \\ 5 & 8 & -3 & 34 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 \\ 13 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 12 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 17 & 0 & 13 & 7 \\ 5 & 8 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.68 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 4 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 240, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.67 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -2 & 6 & 0 & 0 \\ 6 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 8 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 291, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.61 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 35 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 35 & 67 & 0 & 217 & 27 \\ 5 & 8 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 331, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.61 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 17 & 33 & 0 & 49 & 13 \\ 5 & 8 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{112}} 3^{\overline{121}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 336, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.57 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -2 & 6 & 0 & 0 \\ 8 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 8 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{113}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 365, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.71 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 \\ 9 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 12 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{111}} 3^{\overline{112}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 370, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.60 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 26 & 50 & 0 & 118 & 20 \\ 5 & 8 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 424, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.58 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -2 & 6 & 0 & 0 \\ 7 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 8 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{\overline{114}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 540, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 \\ 12 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 16 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{\overline{113}} 3^{\overline{131}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 581, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.57 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 \\ 12 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 12 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{\overline{113}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 709, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.60 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 \\ 10 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 12 & -2 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{115} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 765, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.3$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 40 & 0 & 5 \\ 2 & 4 & -2 & 15 & 0 & 0 \\ 15 & 40 & 15 & 148 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 20 & -2 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{113} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 781, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 \\ 11 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 12 & -2 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{114} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 924, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 \\ 16 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 16 & -2 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{114} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 1044, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 \\ 13 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 16 & -2 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{114} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 1236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 \\ 15 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 16 & -2 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{114} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 1276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 \\ 14 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 16 & -2 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [23] $1^{11} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 1 & 5 \\ 12 & 25 & 0 & 0 & 9 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 0 & -2 & -1 \end{pmatrix}_{b_0}$$

[24] $1^1 3 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 1 & 6 & 0 & 3 \\ 1 & 4 & 0 & 1 & 6 & 0 & 2 \\ 1 & 1 & 1 & -3 & 2 & 0 & -1 \\ 2 & 6 & 6 & 2 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -1 & 6 & -2 & -3 \end{pmatrix}_{b_0}$$

[25] $1^{11} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 18, T_{\max} = 21, T \in [18, 21]$

$$\begin{pmatrix} 5 & 13 & 9 & 1 & 5 \\ 13 & 30 & 20 & 0 & 10 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 10 & 7 & -2 & -3 \end{pmatrix}_{b_0}$$

[26] $1^{11} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 15.11 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 3 \\ 3 & 3 & 5 & -3 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{111} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 2 & 3 \\ 4 & 9 & 0 & 12 & 4 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 12 & 0 & 14 & 7 & 8 \\ 2 & 4 & 0 & 7 & 0 & 2 \\ 3 & 5 & -3 & 8 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{11} 3 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 3 & 0 \\ 1 & 4 & 0 & 1 & 3 & 0 \\ 1 & 1 & 1 & -3 & 1 & 0 \\ 1 & 3 & 3 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{111} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 12 & 19 & 1 & 5 \\ 12 & 25 & 40 & 0 & 9 \\ 19 & 40 & 63 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{11} 3 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 4 & 4 \\ 2 & 5 & 0 & 4 & 0 & 2 \\ 3 & 5 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{111} 3 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 4 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 12 & 0 & 10 & 6 \\ 3 & 6 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 & 1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 \\ 2 & 2 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{111} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 3 \\ 4 & 9 & 0 & 8 & 4 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 8 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 5 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{14} 3 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 5 & 0 & 30 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 16 & 30 & 0 & 158 & 20 & 24 \\ 2 & 5 & 0 & 20 & 0 & 2 \\ 3 & 5 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{12} 3 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 5 & 0 & 15 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 38 & 10 & 12 \\ 2 & 5 & 0 & 10 & 0 & 2 \\ 3 & 5 & -3 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{11} 3 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -102, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 1 & 6 & 0 & 3 \\ 1 & 4 & 0 & 1 & 6 & 0 & 2 \\ 1 & 1 & 1 & -3 & 2 & 0 & -1 \\ 3 & 6 & 6 & 2 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -1 & 6 & -2 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{11}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 15 & 0 & 13 & 7 \\ 3 & 6 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -137, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 10 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 10 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 5 & 0 & 10 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 10 & 0 & 14 & 7 & 8 \\ 2 & 5 & 0 & 7 & 0 & 2 \\ 3 & 5 & -3 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 4 & 1 & 9 & 0 & 3 \\ 1 & 4 & 0 & 1 & 9 & 0 & 2 \\ 1 & 1 & 1 & -3 & 3 & 0 & -1 \\ 4 & 9 & 9 & 3 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -1 & 9 & -2 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 4 & 1 & 12 & 0 & 3 \\ 1 & 4 & 0 & 1 & 12 & 0 & 2 \\ 1 & 1 & 1 & -3 & 4 & 0 & -1 \\ 5 & 12 & 12 & 4 & 46 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -1 & 12 & -2 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{11}321^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -310, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 5 & 0 & 15 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 15 & 0 & 32 & 11 & 12 \\ 2 & 5 & 0 & 11 & 0 & 2 \\ 3 & 5 & -3 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{11}321^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -310, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 5 & 0 & 25 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 25 & 0 & 92 & 19 & 20 \\ 2 & 5 & 0 & 19 & 0 & 2 \\ 3 & 5 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{11}321^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -350, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 5 & 0 & 20 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 20 & 0 & 58 & 15 & 16 \\ 2 & 5 & 0 & 15 & 0 & 2 \\ 3 & 5 & -3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{13}321^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -390, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 5 & 0 & 25 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 25 & 0 & 104 & 17 & 20 \\ 2 & 5 & 0 & 17 & 0 & 2 \\ 3 & 5 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{111}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 9 & 3 \\ 4 & 10 & 13 & 0 & 24 & 6 \\ 5 & 13 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 24 & 27 & 0 & 49 & 13 \\ 3 & 6 & 6 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{12}321^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -460, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$

$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 5 & 0 & 20 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 20 & 0 & 62 & 14 & 16 \\ 2 & 5 & 0 & 14 & 0 & 2 \\ 3 & 5 & -3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{12}321^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -740, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$

$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 5 & 0 & 25 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 25 & 0 & 94 & 18 & 20 \\ 2 & 5 & 0 & 18 & 0 & 2 \\ 3 & 5 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{14}321^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -860, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 \\ 3 & 5 & 0 & 35 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 18 & 35 & 0 & 202 & 24 & 28 \\ 2 & 5 & 0 & 24 & 0 & 2 \\ 3 & 5 & -3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{13}321^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -910, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 5 & 0 & 30 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 30 & 0 & 142 & 21 & 24 \\ 2 & 5 & 0 & 21 & 0 & 2 \\ 3 & 5 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{12}321^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -940, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$

$T_{\min} = 12.68 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 5 & 0 & 30 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 30 & 0 & 134 & 22 & 24 \\ 2 & 5 & 0 & 22 & 0 & 2 \\ 3 & 5 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

[53] $1^{13}321^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1350, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 5 & 0 & 35 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 35 & 0 & 188 & 25 & 28 \\ 2 & 5 & 0 & 25 & 0 & 2 \\ 3 & 5 & -3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

[54] $1^{14}321^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -1540, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 5 & 0 & 40 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 20 & 40 & 0 & 254 & 28 & 32 \\ 2 & 5 & 0 & 28 & 0 & 2 \\ 3 & 5 & -3 & 32 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [55] $1^{111}2321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 34, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 22 & 5 \\ 12 & 26 & 1 & 48 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 48 & 0 & 86 & 18 \\ 5 & 10 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

- [56] $1^{11} 3^{221} 1^{21} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \mathfrak{so}_8$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.69 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 6 & 0 & 17 & 6 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 17 & 0 & 36 & 13 & 14 \\ 2 & 6 & 0 & 13 & 1 & 3 \\ 3 & 6 & -3 & 14 & 3 & 2 \end{pmatrix}_{b_0}$
- [57] $1^{11} 4^{131} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \mathfrak{so}_8$ $T = 10, T_{\max} = 22, \Delta = -106, \det = 101, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.56 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 5 & 10 & 1 & 9 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 14 & 0 & 13 & 7 \\ 5 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$
- [58] $1^{121} 2^1 3^{21} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \mathfrak{so}_8$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 5 & 12 & 1 & 34 & 5 \\ 12 & 26 & 1 & 76 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 76 & 0 & 210 & 28 \\ 5 & 10 & -1 & 28 & 3 \end{pmatrix}_{b_0}$
- [59] $1^{111} 2^1 3^{21} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \mathfrak{so}_8$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 5 & 12 & 1 & 13 & 5 \\ 12 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 28 & 0 & 26 & 10 \\ 5 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$
- [60] $1^{131} 2^1 3^{21} 1^2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \mathfrak{so}_8$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 146, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 5 & 12 & 1 & 56 & 5 \\ 12 & 26 & 1 & 124 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 56 & 124 & 0 & 570 & 46 \\ 5 & 10 & -1 & 46 & 3 \end{pmatrix}_{b_0}$
- [61] $1^{12} 3^{221} 1^{41} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \mathfrak{so}_8$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 154, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 6 & 0 & 34 & 6 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 34 & 0 & 150 & 26 & 28 \\ 2 & 6 & 0 & 26 & 1 & 3 \\ 3 & 6 & -3 & 28 & 3 & 2 \end{pmatrix}_{b_0}$
- [62] $1^{121} 2^1 3^{21} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \mathfrak{so}_8$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 199, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 5 & 12 & 1 & 25 & 5 \\ 12 & 26 & 1 & 56 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 56 & 0 & 110 & 20 \\ 5 & 10 & -1 & 20 & 3 \end{pmatrix}_{b_0}$
- [63] $1^{141} 2^1 3^{21} 1^2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \mathfrak{so}_8$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 294, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 5 & 12 & 1 & 68 & 5 \\ 12 & 26 & 1 & 152 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 68 & 152 & 0 & 846 & 56 \\ 5 & 10 & -1 & 56 & 3 \end{pmatrix}_{b_0}$
- [64] $1^{121} 2^1 3^{21} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \mathfrak{so}_8$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 299, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 5 & 12 & 1 & 35 & 5 \\ 12 & 26 & 1 & 76 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 35 & 76 & 0 & 210 & 28 \\ 5 & 10 & -1 & 28 & 3 \end{pmatrix}_{b_0}$
- [65] $1^{121} 2^1 3^{21} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \mathfrak{so}_8$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 314, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 5 & 12 & 1 & 26 & 5 \\ 12 & 26 & 1 & 56 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 56 & 0 & 110 & 20 \\ 5 & 10 & -1 & 20 & 3 \end{pmatrix}_{b_0}$
- [66] $1^{121} 2^1 3^{21} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \mathfrak{so}_8$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 314, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 36 & 5 \\ 12 & 26 & 1 & 76 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 36 & 76 & 0 & 210 & 28 \\ 5 & 10 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[67] $1^{1^{131^{11}}}$ $4 \quad 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 317, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.56 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 10 & 1 & 7 & 5 \\ 10 & 13 & 0 & 10 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 10 & 0 & 5 & 5 \\ 5 & 7 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{1312321^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 395, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 57 & 5 \\ 12 & 26 & 1 & 124 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 57 & 124 & 0 & 570 & 46 \\ 5 & 10 & -1 & 46 & 3 \end{pmatrix}_{b_0}$$

[69] $1^{1412321^{13}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 419, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 79 & 5 \\ 12 & 26 & 1 & 172 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 79 & 172 & 0 & 1106 & 64 \\ 5 & 10 & -1 & 64 & 3 \end{pmatrix}_{b_0}$$

[70] $1^{1312321^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 486, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 58 & 5 \\ 12 & 26 & 1 & 124 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 58 & 124 & 0 & 570 & 46 \\ 5 & 10 & -1 & 46 & 3 \end{pmatrix}_{b_0}$$

[71] $1^{11^{131^{11}}}$ $4 \quad 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 486, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 1 & 16 & 5 \\ 10 & 13 & 0 & 24 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 24 & 0 & 38 & 12 \\ 5 & 7 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{1312321^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 486, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 38 & 5 \\ 12 & 26 & 1 & 84 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 38 & 84 & 0 & 250 & 30 \\ 5 & 10 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[73] $1^{1312321^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 495, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 47 & 5 \\ 12 & 26 & 1 & 104 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 47 & 104 & 0 & 390 & 38 \\ 5 & 10 & -1 & 38 & 3 \end{pmatrix}_{b_0}$$

[74] $1^{1412321^{13}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 586, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 80 & 5 \\ 12 & 26 & 1 & 172 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 80 & 172 & 0 & 1106 & 64 \\ 5 & 10 & -1 & 64 & 3 \end{pmatrix}_{b_0}$$

[75] $1^{1412321^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 619, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 59 & 5 \\ 12 & 26 & 1 & 132 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 59 & 132 & 0 & 626 & 48 \\ 5 & 10 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

[76] $1^{1312321^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 686, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$

14.0

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 48 & 5 \\ 12 & 26 & 1 & 104 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 48 & 104 & 0 & 390 & 38 \\ 5 & 10 & -1 & 38 & 3 \end{pmatrix}_{b_0}$$

[77] $1^{141} 2^3 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 719, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 69 & 5 \\ 12 & 26 & 1 & 152 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 69 & 152 & 0 & 846 & 56 \\ 5 & 10 & -1 & 56 & 3 \end{pmatrix}_{b_0}$$

[78] $1^{141} 2^3 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 986, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 70 & 5 \\ 12 & 26 & 1 & 152 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 70 & 152 & 0 & 846 & 56 \\ 5 & 10 & -1 & 56 & 3 \end{pmatrix}_{b_0}$$

[79] $1^{141} 2^3 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 986, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 60 & 5 \\ 12 & 26 & 1 & 132 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 60 & 132 & 0 & 626 & 48 \\ 5 & 10 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

4.458 NHC(-2)(-3) + $\mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$

$T = 9$ [0] $1^{112} 3 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.7$

$$T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 0 & 6 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 7 & 16 & 6 & 0 & 0 & 22 & 8 \\ 3 & 5 & 0 & 0 & -2 & 8 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 12 & 7 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -3 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -3 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 4 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 & 3 & 0 \\ 3 & 8 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 6 & 0 & 4 & 4 \\ 3 & 5 & 0 & 4 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111} 3 1^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9, T_{\max} = 21, \Delta = -106, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 35 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 35 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -3 & 7 & 27 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{1111}3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -380$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 & 3 & 0 \\ 3 & 8 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 6 & 0 & 4 & 4 \\ 3 & 5 & 0 & 4 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{1111}3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -404$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -3 & 7 & 13 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{1111}3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -476$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 4 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 & 3 & 0 \\ 4 & 8 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 6 & 0 & 4 & 4 \\ 3 & 5 & 0 & 4 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{1111}3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -482$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 26 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 26 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -3 & 7 & 20 & 1 \end{pmatrix}_{b_0}$$

4.459 NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{1111}3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -218$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 9 & 7 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -3 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{1111}3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -222$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 2 & 3 \\ 4 & 9 & 4 & 8 & 0 & 4 & 5 \\ 2 & 4 & -2 & 3 & 0 & 2 & 0 \\ 3 & 8 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 2 & 3 & 0 & 0 & 2 \\ 3 & 5 & 0 & 4 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{1111}3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -302$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 2 & 3 \\ 4 & 9 & 4 & 8 & 0 & 4 & 5 \\ 2 & 4 & -2 & 3 & 0 & 2 & 0 \\ 4 & 8 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 2 & 3 & 0 & 0 & 2 \\ 3 & 5 & 0 & 4 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

4.460 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 + \text{su2n3mix}$

$$T = 9 \quad [0] \quad 1^{\overset{1}{11}} 3^{\overset{1}{11}} 1^{\overset{1}{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 15 & 5 \\ 11 & 20 & 0 & 17 & 28 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 24 & 7 \\ 15 & 28 & 0 & 24 & 37 & 11 \\ 5 & 8 & -3 & 7 & 11 & 1 \end{pmatrix}_{b_0}$$

4.461 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{\overset{1}{11}} 3^{\overset{1}{11}} 1^{\overset{1}{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 & 0 \\ 3 & 8 & 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 4 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overset{1}{11}} 3^{\overset{1}{11}} 1^{\overset{1}{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -3 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overset{1}{11}} 3^{\overset{1}{11}} 1^{\overset{1}{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overset{1}{11}} 3^{\overset{1}{11}} 1^{\overset{1}{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -3 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

4.462 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^{\overset{1}{11}} 3^{\overset{1}{11}} 1^{\overset{1}{21}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 & 3 \\ 3 & 4 & 0 & 10 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 10 & 0 & 22 & 12 & 8 \\ 3 & 5 & 0 & 12 & 4 & 4 \\ 3 & 4 & -3 & 8 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overset{1}{11}} 3^{\overset{1}{11}} 1^{\overset{1}{21}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 4 & 3 \\ 4 & 9 & 4 & 8 & 8 & 5 \\ 2 & 4 & -2 & 3 & 3 & 0 \\ 5 & 8 & 3 & 4 & 6 & 4 \\ 4 & 8 & 3 & 6 & 4 & 4 \\ 3 & 5 & 0 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overset{1}{11}} 3^{\overset{1}{11}} 1^{\overset{1}{21}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 14.52 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$$

- [4] $21^{111}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$
- [3] $1^{1111}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 8 & 5 \\ 3 & 4 & -2 & 3 & 3 & 0 \\ 3 & 8 & 3 & 4 & 6 & 4 \\ 3 & 8 & 3 & 6 & 4 & 4 \\ 3 & 5 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$
- [5] $1^{133}31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 15 & 3 \\ 3 & 4 & 0 & 17 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 81 & 13 \\ 15 & 27 & 0 & 81 & 133 & 21 \\ 3 & 4 & -3 & 13 & 21 & 1 \end{pmatrix}_{b_0}$$
- [6] $1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -3 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{111}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.44 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$
- [8] $1^{121}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{124}31^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 17 & 3 \\ 3 & 4 & 0 & 8 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 42 & 6 \\ 17 & 31 & 0 & 42 & 172 & 24 \\ 3 & 4 & -3 & 6 & 24 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{11}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 3 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -3 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{112}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 0 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$
- [12] $1^{11}31^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 10 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 10 & 0 & 22 & 30 & 8 \\ 8 & 14 & 0 & 30 & 37 & 11 \\ 3 & 4 & -3 & 8 & 11 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{13}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 9 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -3 & 4 & 13 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{111}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -3 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{112}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 4 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{113}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & 0 & 12 & -2 \end{pmatrix}_{b_0}$$

[17] $1^{122}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 4 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{113}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 6 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{112}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 13.19 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 0 & 0 & 8 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{111}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.21 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 9 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 42 & 7 \\ 14 & 24 & 0 & 42 & 121 & 19 \\ 3 & 4 & -3 & 7 & 19 & -1 \end{pmatrix}_{b_0}$$

[21] $1^{123}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 6 & 12 & 0 & 18 & 25 & 9 \\ 3 & 4 & -3 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{111}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 8 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & -3 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{133}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 9 & 3 \\ 3 & 4 & 0 & 12 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 36 & 9 \\ 9 & 17 & 0 & 36 & 49 & 13 \\ 3 & 4 & -3 & 9 & 13 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{114}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 8 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -3 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{133}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 0 & 12 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 27 & 9 \\ 6 & 12 & 0 & 27 & 25 & 9 \\ 3 & 4 & -3 & 9 & 9 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{124}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 8 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 24 & 6 \\ 8 & 16 & 0 & 24 & 46 & 12 \\ 3 & 4 & -3 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{134}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 134, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 12 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 36 & 9 \\ 8 & 16 & 0 & 36 & 46 & 12 \\ 3 & 4 & -3 & 9 & 12 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{115}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 10 & 20 & 0 & 15 & 73 & 15 \\ 3 & 4 & -3 & 3 & 15 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{1111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 4 & 3 \\ 4 & 9 & 4 & 8 & 8 & 5 \\ 2 & 4 & -2 & 3 & 3 & 0 \\ 3 & 8 & 3 & 4 & 6 & 4 \\ 4 & 8 & 3 & 6 & 4 & 4 \\ 3 & 5 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{125}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 8 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 30 & 6 \\ 10 & 20 & 0 & 30 & 73 & 15 \\ 3 & 4 & -3 & 6 & 15 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{12}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 7 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -3 & 4 & 10 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{144} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 3 \\ 3 & 4 & 0 & 16 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 46 & 48 & 12 \\ 8 & 16 & 0 & 48 & 46 & 12 \\ 3 & 4 & -3 & 12 & 12 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 14.65 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 16 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & 0 & 16 & -2 \end{pmatrix}_{b_0}$$

[35] $1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -3 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{123} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 8 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 30 & 6 \\ 12 & 22 & 0 & 30 & 85 & 17 \\ 3 & 4 & -3 & 6 & 17 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 7 & 3 \\ 3 & 4 & 0 & 14 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 33 & 11 \\ 7 & 13 & 0 & 33 & 28 & 10 \\ 3 & 4 & -3 & 11 & 10 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{135} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.1$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 10 & 3 \\ 3 & 4 & 0 & 12 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 45 & 9 \\ 10 & 20 & 0 & 45 & 73 & 15 \\ 3 & 4 & -3 & 9 & 15 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{116} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 158, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 12 & 24 & 0 & 18 & 106 & 18 \\ 3 & 4 & -3 & 3 & 18 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.66 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

[41] $1^{122} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -3 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

[42] $1^1 1^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.32 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 8 & 5 \\ 2 & 4 & -2 & 3 & 3 & 0 \\ 3 & 8 & 3 & 4 & 6 & 4 \\ 3 & 8 & 3 & 6 & 4 & 4 \\ 3 & 5 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{126} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.9$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 8 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 36 & 6 \\ 12 & 24 & 0 & 36 & 106 & 18 \\ 3 & 4 & -3 & 6 & 18 & -1 \end{pmatrix}_{b_0}$$

[44] $1^{145} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 13.05 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 10 & 3 \\ 3 & 4 & 0 & 16 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 46 & 60 & 12 \\ 10 & 20 & 0 & 60 & 73 & 15 \\ 3 & 4 & -3 & 12 & 15 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{136} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 12 & 3 \\ 3 & 4 & 0 & 12 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 54 & 9 \\ 12 & 24 & 0 & 54 & 106 & 18 \\ 3 & 4 & -3 & 9 & 18 & -1 \end{pmatrix}_{b_0}$$

[46] $1^{117} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.24 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 14 & 28 & 0 & 21 & 145 & 21 \\ 3 & 4 & -3 & 3 & 21 & -1 \end{pmatrix}_{b_0}$$

[47] $1^{155} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.24 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 10 & 3 \\ 3 & 4 & 0 & 20 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 20 & 0 & 73 & 75 & 15 \\ 10 & 20 & 0 & 75 & 73 & 15 \\ 3 & 4 & -3 & 15 & 15 & -1 \end{pmatrix}_{b_0}$$

[49] $1^{1111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 4 & 3 \\ 4 & 9 & 4 & 8 & 8 & 5 \\ 2 & 4 & -2 & 3 & 3 & 0 \\ 4 & 8 & 3 & 4 & 6 & 4 \\ 4 & 8 & 3 & 6 & 4 & 4 \\ 3 & 5 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[48] $1^{146} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 13.28 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 4 & 0 & 16 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 46 & 72 & 12 \\ 12 & 24 & 0 & 72 & 106 & 18 \\ 3 & 4 & -3 & 12 & 18 & -1 \end{pmatrix}_{b_0}$$

[50] $1^{127} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 190, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.29 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 8 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 42 & 6 \\ 14 & 28 & 0 & 42 & 145 & 21 \\ 3 & 4 & -3 & 6 & 21 & -1 \end{pmatrix}_{b_0}$$

[51] $1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.12 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

[52] $1^{137} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 14 & 3 \\ 3 & 4 & 0 & 12 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 63 & 9 \\ 14 & 28 & 0 & 63 & 145 & 21 \\ 3 & 4 & -3 & 9 & 21 & -1 \end{pmatrix}_{b_0}$$

[54] $1^{156} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 12 & 3 \\ 3 & 4 & 0 & 20 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 20 & 0 & 73 & 90 & 15 \\ 12 & 24 & 0 & 90 & 106 & 18 \\ 3 & 4 & -3 & 15 & 18 & -1 \end{pmatrix}_{b_0}$$

[53] $1^{112} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 16 & 28 & 0 & 18 & 154 & 22 \\ 3 & 4 & -3 & 3 & 22 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{11} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 5 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 8 & 14 & 0 & 15 & 37 & 11 \\ 3 & 4 & -3 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 14.35 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & 0 & 16 & -2 \end{pmatrix}_{b_0}$$

[57] $1^{147} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 14 & 3 \\ 3 & 4 & 0 & 16 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 46 & 84 & 12 \\ 14 & 28 & 0 & 84 & 145 & 21 \\ 3 & 4 & -3 & 12 & 21 & -1 \end{pmatrix}_{b_0}$$

[58] $1^{166} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 13.58 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 12 & 3 \\ 3 & 4 & 0 & 24 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 24 & 0 & 106 & 108 & 18 \\ 12 & 24 & 0 & 108 & 106 & 18 \\ 3 & 4 & -3 & 18 & 18 & -1 \end{pmatrix}_{b_0}$$

[59] $1^{157} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 13.60 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 14 & 3 \\ 3 & 4 & 0 & 20 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 20 & 0 & 73 & 105 & 15 \\ 14 & 28 & 0 & 105 & 145 & 21 \\ 3 & 4 & -3 & 15 & 21 & -1 \end{pmatrix}_{b_0}$$

[60] $1^{134} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 239, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 11 & 3 \\ 3 & 4 & 0 & 12 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 45 & 9 \\ 11 & 21 & 0 & 45 & 76 & 16 \\ 3 & 4 & -3 & 9 & 16 & 1 \end{pmatrix}_{b_0}$$

[61] $1^{132} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 246, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 10 & 3 \\ 3 & 4 & 0 & 17 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 54 & 13 \\ 10 & 18 & 0 & 54 & 58 & 14 \\ 3 & 4 & -3 & 13 & 14 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{167} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.72 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 14 & 3 \\ 3 & 4 & 0 & 24 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 24 & 0 & 106 & 126 & 18 \\ 14 & 28 & 0 & 126 & 145 & 21 \\ 3 & 4 & -3 & 18 & 21 & -1 \end{pmatrix}_{b_0}$$

[63] $1^{11} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 255, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 21 & 4 \\ 11 & 19 & 0 & 21 & 73 & 15 \\ 3 & 4 & -3 & 4 & 15 & 1 \end{pmatrix}_{b_0}$$

[64] $1^{11} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 33 & 7 \\ 11 & 19 & 0 & 33 & 73 & 15 \\ 3 & 4 & -3 & 7 & 15 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 263, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -3 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{143} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 279, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 12 & 3 \\ 3 & 4 & 0 & 21 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 81 & 16 \\ 12 & 22 & 0 & 81 & 85 & 17 \\ 3 & 4 & -3 & 16 & 17 & 1 \end{pmatrix}_{b_0}$$

[67] $1^{11} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 279, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 5 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 27 & 4 \\ 14 & 24 & 0 & 27 & 121 & 19 \\ 3 & 4 & -3 & 4 & 19 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{177} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 13.84 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 14 & 3 \\ 3 & 4 & 0 & 28 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 28 & 0 & 145 & 147 & 21 \\ 14 & 28 & 0 & 147 & 145 & 21 \\ 3 & 4 & -3 & 21 & 21 & -1 \end{pmatrix}_{b_0}$$

[69] $1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -3 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{123} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 282, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 24 & 6 \\ 9 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -3 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{122} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 284, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 13 & 3 \\ 3 & 4 & 0 & 13 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 54 & 10 \\ 13 & 23 & 0 & 54 & 100 & 18 \\ 3 & 4 & -3 & 10 & 18 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{114} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & 0 & 16 & -1 \end{pmatrix}_{b_0}$$

[73] $1^{113} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 4 & 0 & 14 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 57 & 11 \\ 12 & 22 & 0 & 57 & 85 & 17 \\ 3 & 4 & -3 & 11 & 17 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{111} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 317, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 9 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 24 & 7 \\ 8 & 14 & 0 & 24 & 37 & 11 \\ 3 & 4 & -3 & 7 & 11 & 1 \end{pmatrix}_{b_0}$$

[76] $1^{114} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 33 & 7 \\ 11 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -3 & 7 & 16 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{112} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 13 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & -3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{12} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 324, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 5 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 10 & 18 & 0 & 18 & 58 & 14 \\ 3 & 4 & -3 & 4 & 14 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 324, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 13.21 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 14 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & 0 & 16 & -1 \end{pmatrix}_{b_0}$$

[79] $1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 338, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 10 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -3 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

[80] $1^{112} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 359, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 21 & 7 \\ 7 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -3 & 7 & 10 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{14} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 5 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 14 & 26 & 0 & 24 & 118 & 20 \\ 3 & 4 & -3 & 4 & 20 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{113} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 27 & 7 \\ 9 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & -3 & 7 & 13 & 1 \end{pmatrix}_{b_0}$$

[83] $1^{13}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 375, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 5 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 21 & 4 \\ 12 & 22 & 0 & 21 & 85 & 17 \\ 3 & 4 & -3 & 4 & 17 & 1 \end{pmatrix}_{b_0}$$

[84] $1^{124}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 380, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 8 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 36 & 6 \\ 14 & 26 & 0 & 36 & 118 & 20 \\ 3 & 4 & -3 & 6 & 20 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -3 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{124}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 0 & 8 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 30 & 6 \\ 11 & 21 & 0 & 30 & 76 & 16 \\ 3 & 4 & -3 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{111}3^11^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 3 \\ 3 & 4 & 0 & 14 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 39 & 11 \\ 8 & 14 & 0 & 39 & 37 & 11 \\ 3 & 4 & -3 & 11 & 11 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{12}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 13 & 3 \\ 3 & 4 & 0 & 5 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 13 & 23 & 0 & 24 & 100 & 18 \\ 3 & 4 & -3 & 4 & 18 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{114}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 503, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 11 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -3 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{112}3^11^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 506, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 10 & 3 \\ 3 & 4 & 0 & 14 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 48 & 11 \\ 10 & 18 & 0 & 48 & 58 & 14 \\ 3 & 4 & -3 & 11 & 14 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 508, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 30 & 10 \\ 7 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & -3 & 10 & 10 & 1 \end{pmatrix}_{b_0}$$

[92] $1^{123}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 543, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 18 & 3 \\ 3 & 4 & 0 & 13 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 75 & 10 \\ 18 & 32 & 0 & 75 & 193 & 25 \\ 3 & 4 & -3 & 10 & 25 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{122}3^11^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 0 & 13 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 42 & 10 \\ 10 & 18 & 0 & 42 & 58 & 14 \\ 3 & 4 & -3 & 10 & 14 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{113}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 12 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -3 & 3 & 17 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{112}3^11^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 574, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 9 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 30 & 7 \\ 10 & 18 & 0 & 30 & 58 & 14 \\ 3 & 4 & -3 & 7 & 14 & 1 \end{pmatrix}_{b_0}$$

[96] $1^{123}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 591, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 39 & 10 \\ 9 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -3 & 10 & 13 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{124}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 0 & 13 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 48 & 10 \\ 11 & 21 & 0 & 48 & 76 & 16 \\ 3 & 4 & -3 & 10 & 16 & 1 \end{pmatrix}_{b_0}$$

[98] $1^{12}3^11^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 612, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 16 & 3 \\ 3 & 4 & 0 & 5 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 16 & 28 & 0 & 30 & 154 & 22 \\ 3 & 4 & -3 & 4 & 22 & 1 \end{pmatrix}_{b_0}$$

[99] $1^{13}3^11^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 639, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 3 \\ 3 & 4 & 0 & 5 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 27 & 4 \\ 15 & 27 & 0 & 27 & 133 & 21 \\ 3 & 4 & -3 & 4 & 21 & 1 \end{pmatrix}_{b_0}$$

[100] $1^{113}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 648, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 18 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & -3 & 3 & 25 & 1 \end{pmatrix}_{b_0}$$

[101] $1^{113}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 653, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 15 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & -3 & 3 & 21 & 1 \end{pmatrix}_{b_0}$$

[102] $21^{1111}3^11 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 692, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 6 & 4 \\ 8 & 9 & 4 & 8 & 8 & 5 \\ 3 & 4 & -2 & 3 & 3 & 0 \\ 6 & 8 & 3 & 4 & 6 & 4 \\ 6 & 8 & 3 & 6 & 4 & 4 \\ 4 & 5 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[103] $1^{112}3^11^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 695$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 9 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 39 & 7 \\ 13 & 23 & 0 & 39 & 100 & 18 \\ 3 & 4 & -3 & 7 & 18 & 1 \end{pmatrix}_{b_0}$$

[104] $1^{133}3^11^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 717$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 11.51 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 12 & 3 \\ 3 & 4 & 0 & 17 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 66 & 13 \\ 12 & 22 & 0 & 66 & 85 & 17 \\ 3 & 4 & -3 & 13 & 17 & 1 \end{pmatrix}_{b_0}$$

[105] $1^{112}3^11^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 722$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 16 & 3 \\ 3 & 4 & 0 & 9 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 48 & 7 \\ 16 & 28 & 0 & 48 & 154 & 22 \\ 3 & 4 & -3 & 7 & 22 & 1 \end{pmatrix}_{b_0}$$

[106] $1^{143}3^11^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 732$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.53 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 3 \\ 3 & 4 & 0 & 5 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 17 & 31 & 0 & 30 & 172 & 24 \\ 3 & 4 & -3 & 4 & 24 & 1 \end{pmatrix}_{b_0}$$

[107] $1^{133}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 744$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 11.53 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 0 & 17 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 51 & 13 \\ 9 & 17 & 0 & 51 & 49 & 13 \\ 3 & 4 & -3 & 13 & 13 & 1 \end{pmatrix}_{b_0}$$

[108] $1^{134}3^11^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 759$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 17 & 3 \\ 3 & 4 & 0 & 17 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 93 & 13 \\ 17 & 31 & 0 & 93 & 172 & 24 \\ 3 & 4 & -3 & 13 & 24 & 1 \end{pmatrix}_{b_0}$$

[109] $1^{124}3^11^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 764$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 14 & 3 \\ 3 & 4 & 0 & 18 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 84 & 14 \\ 14 & 26 & 0 & 84 & 118 & 20 \\ 3 & 4 & -3 & 14 & 20 & 1 \end{pmatrix}_{b_0}$$

[110] $1^{113}3^11^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 765$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 9 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 36 & 7 \\ 12 & 22 & 0 & 36 & 85 & 17 \\ 3 & 4 & -3 & 7 & 17 & 1 \end{pmatrix}_{b_0}$$

[111] $1^{114}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 794$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.51 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 14 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & -3 & 3 & 20 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{144}3^11^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 824$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.55 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 14 & 3 \\ 3 & 4 & 0 & 21 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 96 & 16 \\ 14 & 26 & 0 & 96 & 118 & 20 \\ 3 & 4 & -3 & 16 & 20 & 1 \end{pmatrix}_{b_0}$$

[113] $1^{134}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 831$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 11 & 3 \\ 3 & 4 & 0 & 17 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 63 & 13 \\ 11 & 21 & 0 & 63 & 76 & 16 \\ 3 & 4 & -3 & 13 & 16 & 1 \end{pmatrix}_{b_0}$$

[114] $1^{123}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 863$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 11.53 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 15 & 3 \\ 3 & 4 & 0 & 13 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 63 & 10 \\ 15 & 27 & 0 & 63 & 133 & 21 \\ 3 & 4 & -3 & 10 & 21 & 1 \end{pmatrix}_{b_0}$$

[115] $1^{122}31^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 868$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.53 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 10 & 3 \\ 3 & 4 & 0 & 18 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 60 & 14 \\ 10 & 18 & 0 & 60 & 58 & 14 \\ 3 & 4 & -3 & 14 & 14 & 1 \end{pmatrix}_{b_0}$$

[116] $1^{13}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 879$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 18 & 3 \\ 3 & 4 & 0 & 5 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 33 & 4 \\ 18 & 32 & 0 & 33 & 193 & 25 \\ 3 & 4 & -3 & 4 & 25 & 1 \end{pmatrix}_{b_0}$$

[117] $1^{123}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 879$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 11.51 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 12 & 3 \\ 3 & 4 & 0 & 13 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 51 & 10 \\ 12 & 22 & 0 & 51 & 85 & 17 \\ 3 & 4 & -3 & 10 & 17 & 1 \end{pmatrix}_{b_0}$$

[118] $1^{114}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 890$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.51 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 9 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 42 & 7 \\ 14 & 26 & 0 & 42 & 118 & 20 \\ 3 & 4 & -3 & 7 & 20 & 1 \end{pmatrix}_{b_0}$$

[119] $1^{123}31^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 954$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 12 & 3 \\ 3 & 4 & 0 & 18 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 72 & 14 \\ 12 & 22 & 0 & 72 & 85 & 17 \\ 3 & 4 & -3 & 14 & 17 & 1 \end{pmatrix}_{b_0}$$

[120] $1^{144}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 988$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 11 & 3 \\ 3 & 4 & 0 & 21 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 78 & 16 \\ 11 & 21 & 0 & 78 & 76 & 16 \\ 3 & 4 & -3 & 16 & 16 & 1 \end{pmatrix}_{b_0}$$

[121] $1^{114}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 991$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 17 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -3 & 3 & 24 & 1 \end{pmatrix}_{b_0}$$

[122] $1^{113}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1064$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 4 & 0 & 9 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 45 & 7 \\ 15 & 27 & 0 & 45 & 135 & 21 \\ 3 & 4 & -3 & 7 & 21 & 1 \end{pmatrix}_{b_0}$$

[123] $1^{14} 3^1 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 1080, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 20 & 3 \\ 3 & 4 & 0 & 5 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 36 & 4 \\ 20 & 36 & 0 & 36 & 238 & 28 \\ 3 & 4 & -3 & 4 & 28 & 1 \end{pmatrix}_{b_0}$$

[124] $1^{14} 3^1 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 1094, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 24 & 3 \\ 20 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & -3 & 3 & 28 & 1 \end{pmatrix}_{b_0}$$

[125] $1^{134} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 1122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 14 & 3 \\ 3 & 4 & 0 & 17 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 78 & 13 \\ 14 & 26 & 0 & 78 & 118 & 20 \\ 3 & 4 & -3 & 13 & 20 & 1 \end{pmatrix}_{b_0}$$

[126] $1^{124} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 1144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 14 & 3 \\ 3 & 4 & 0 & 13 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 60 & 10 \\ 14 & 26 & 0 & 60 & 118 & 20 \\ 3 & 4 & -3 & 10 & 20 & 1 \end{pmatrix}_{b_0}$$

[127] $1^{113} 3^1 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 1269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 18 & 3 \\ 3 & 4 & 0 & 9 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 54 & 7 \\ 18 & 32 & 0 & 54 & 193 & 25 \\ 3 & 4 & -3 & 7 & 25 & 1 \end{pmatrix}_{b_0}$$

[128] $1^{124} 3^1 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 1304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 20 & 3 \\ 3 & 4 & 0 & 13 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 84 & 10 \\ 20 & 36 & 0 & 84 & 238 & 28 \\ 3 & 4 & -3 & 10 & 28 & 1 \end{pmatrix}_{b_0}$$

[129] $1^{133} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 1320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 12 & 3 \\ 3 & 4 & 0 & 22 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 0 & 85 & 87 & 17 \\ 12 & 22 & 0 & 87 & 85 & 17 \\ 3 & 4 & -3 & 17 & 17 & 1 \end{pmatrix}_{b_0}$$

[130] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 1367, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 17 & 3 \\ 3 & 4 & 0 & 9 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 51 & 7 \\ 17 & 31 & 0 & 51 & 172 & 24 \\ 3 & 4 & -3 & 7 & 24 & 1 \end{pmatrix}_{b_0}$$

[131] $1^{124} 3^1 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 1376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 17 & 3 \\ 3 & 4 & 0 & 13 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 72 & 10 \\ 17 & 31 & 0 & 72 & 172 & 24 \\ 3 & 4 & -3 & 10 & 24 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{134} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 1410, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 14 & 3 \\ 3 & 4 & 0 & 22 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 0 & 85 & 102 & 17 \\ 14 & 26 & 0 & 102 & 118 & 20 \\ 3 & 4 & -3 & 17 & 20 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{114} 3^1 1^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1750$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.55 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 20 & 3 \\ 3 & 4 & 0 & 9 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 60 & 7 \\ 20 & 36 & 0 & 60 & 238 & 28 \\ 3 & 4 & -3 & 7 & 28 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{144} 3^1 1^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1780$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.53 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 14 & 3 \\ 3 & 4 & 0 & 26 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 0 & 118 & 120 & 20 \\ 14 & 26 & 0 & 120 & 118 & 20 \\ 3 & 4 & -3 & 20 & 20 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [135] $1^{111} 3^1 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -14$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 & 0 \\ 2 & 1 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

[136] $1^{11} 3^1 21^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -20$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 4 & 4 \\ 2 & 5 & 0 & 4 & 2 & 4 \\ 3 & 5 & -3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

[137] $1^{111} 3^1 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 1 & 0 \\ 2 & 3 & 0 & 4 & 5 & 3 & 5 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & 0 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$

[138] $1^{11} 3^1 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{122} 3^1 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 5 & 0 \\ 2 & 4 & 0 & 2 & 6 & 10 & 4 \\ 3 & 5 & 0 & 5 & 10 & 13 & 7 \\ 2 & 2 & -1 & 0 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{121} 3^1 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 1 & 0 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[141] $1^{124} 3^1 21^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 4 & 0 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 7 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & 0 & 4 & 8 & 1 \end{pmatrix}_{b_0}$$

[146] $1^{111}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 1 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 2 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & 2 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[142] $1^{11}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 4 & 5 & 5 \\ 2 & 2 & 2 & -3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[143] $1^{11}321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 1 & 4 & 2 \\ 1 & 2 & 0 & 0 & 3 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 7 & 3 \\ 3 & 4 & 7 & 0 & 7 & 16 & 8 \\ 2 & 2 & 2 & -3 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

[144] $1^{13}321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 1 & 8 & 2 \\ 1 & 2 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 9 & 3 \\ 5 & 8 & 9 & 0 & 9 & 40 & 12 \\ 2 & 2 & 2 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[145] $1^{111}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{112}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -45$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 9 & 3 \\ 4 & 12 & 9 & 0 & 9 & 28 & 10 \\ 2 & 4 & 2 & -3 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{14}31^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 10 & 2 \\ 2 & 4 & 4 & 0 & 6 & 30 & 4 \\ 1 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 30 & 4 \\ 10 & 30 & 21 & 0 & 30 & 172 & 24 \\ 2 & 4 & 2 & -3 & 4 & 24 & 1 \end{pmatrix}_{b_0}$$

[149] $1^{14}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -54$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 0 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 7 & 0 & 41 & 11 \\ 2 & 2 & -1 & 0 & 0 & 11 & 1 \end{pmatrix}_{b_0}$$

[150] $1^{13}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -59$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 16 & 6 \\ 2 & 2 & -1 & 0 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

- [151] $1^{111}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -60$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 5 & 0 & 5 & 10 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 7 & 4 \\ 5 & 10 & 0 & 7 & 14 & 8 \\ 3 & 5 & -3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$
- [152] $1^{111}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 \\ 2 & 2 & -1 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$
- [153] $1^{121}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 1 & 0 \\ 2 & 4 & 0 & 2 & 6 & 4 & 4 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$
- [154] $1^{112}321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 8 & 0 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 5 & 6 & 0 & 8 & 6 & 26 & 10 \\ 2 & 2 & -1 & 0 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$
- [155] $1^{111}321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 4 & 0 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 4 & 3 & 0 & 4 & 3 & 5 & 5 \\ 2 & 2 & -1 & 0 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$
- [156] $1^{111}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -74$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.51 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 0 & 1 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 5 & 5 \\ 2 & 2 & 2 & -3 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$
- [157] $1^{111}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -79$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 4 & 0 \\ 2 & 2 & 0 & 1 & 0 & 3 & 2 \\ 3 & 3 & 0 & 4 & 3 & 5 & 5 \\ 2 & 2 & -1 & 0 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$
- [158] $1^{111}321^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 5 & 0 & 10 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 10 & 0 & 14 & 30 & 8 \\ 9 & 20 & 0 & 30 & 58 & 16 \\ 3 & 5 & -3 & 8 & 16 & 2 \end{pmatrix}_{b_0}$$
- [160] $1^{113}321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 9 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 9 & 8 & 40 & 12 \\ 2 & 2 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$
- [159] $1^{111}321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 5 & 0 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 3 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[170] $1^{111} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[171] $1^{123} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 6 & 0 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 4 & 7 & 0 & 6 & 14 & 25 & 9 \\ 2 & 2 & -1 & 0 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

[172] $1^{122} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[173] $1^{124} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 7 & 0 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 7 & 9 & 0 & 7 & 18 & 41 & 11 \\ 2 & 2 & -1 & 0 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$

[174] $1^{13} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 0 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 7 & 0 & 6 & 0 & 25 & 9 \\ 2 & 2 & -1 & 0 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[177] $1^{11} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 11 & 4 \\ 7 & 15 & 0 & 11 & 32 & 12 \\ 3 & 5 & -3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

[175] $1^{11} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 19 & 4 \\ 11 & 25 & 0 & 19 & 92 & 20 \\ 3 & 5 & -3 & 4 & 20 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{111} 3^{11} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 3 \\ 3 & 5 & 0 & 15 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 15 & 0 & 32 & 56 & 12 \\ 11 & 25 & 0 & 56 & 92 & 20 \\ 3 & 5 & -3 & 12 & 20 & 2 \end{pmatrix}_{b_0}$$

[179] $1^{112} 3^{11} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 0 & 1 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 6 & 2 \\ 4 & 6 & 8 & 0 & 6 & 26 & 10 \\ 2 & 2 & 2 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{12} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -134$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.55 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 5 & 0 & 5 & 0 & 13 & 7 \\ 2 & 2 & -1 & 0 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

[180] $1^{11} 31^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -139$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 6 & 3 \\ 3 & 9 & 6 & 0 & 6 & 13 & 7 \\ 2 & 4 & 2 & -3 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[181] $1^{11} 3 21^{11} 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 & 5 \\ 3 & 5 & 0 & 5 & 20 & 5 & 3 \\ 1 & 0 & -5 & 0 & 0 & -3 & 5 \\ 2 & 5 & 0 & 2 & 15 & 4 & 9 \\ 9 & 20 & 0 & 15 & 58 & 16 & 2 \\ 3 & 5 & -3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[182] $1^{12} 3 21^{11} 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -144$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 1 & 7 & 2 \\ 1 & 2 & 0 & 0 & 3 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 11 & 3 \\ 5 & 7 & 11 & 0 & 11 & 45 & 13 \\ 2 & 2 & 2 & -3 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

[184] $1^{14} 3 21^{11} 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -144$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 1 & 11 & 2 \\ 1 & 2 & 0 & 0 & 3 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 13 & 3 \\ 7 & 11 & 13 & 0 & 13 & 81 & 17 \\ 2 & 2 & 2 & -3 & 3 & 17 & 1 \end{pmatrix}_{b_0}$$

[183] $1^{11} 3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -144$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 8 & 0 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 6 & 6 & 0 & 8 & 6 & 26 & 10 \\ 2 & 2 & -1 & 0 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[185] $1^{11} 3 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -149$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 12 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 8 & 9 & 0 & 12 & 9 & 61 & 15 \\ 2 & 2 & -1 & 0 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[186] $1^{11} 3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -149$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 8 & 0 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 7 & 6 & 0 & 8 & 6 & 26 & 10 \\ 2 & 2 & -1 & 0 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[187] $1^{11} 3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -154$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

10.2

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 0 & 4 & 3 & 2 \\ 1 & 2 & 0 & 0 & 7 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 16 & 11 & 8 \\ 2 & 3 & 4 & 0 & 11 & 5 & 5 \\ 2 & 2 & 2 & -3 & 8 & 5 & 1 \end{pmatrix}_{b_0}$$

[189] $1^{113} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 \\ 1 & 2 & 0 & 0 & 4 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 13 & 5 \\ 4 & 7 & 6 & 0 & 13 & 25 & 9 \\ 2 & 2 & 2 & -3 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

[188] $1^{112} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 0 & 1 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 5 & 2 \\ 3 & 5 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & 2 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[190] $1^{124} 3^{21} 1^{22} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 10 & 2 \\ 1 & 2 & 0 & 0 & 8 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 26 & 40 & 10 \\ 6 & 10 & 10 & 0 & 40 & 58 & 14 \\ 2 & 2 & 2 & -3 & 10 & 14 & 1 \end{pmatrix}_{b_0}$$

[191] $1^{112} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & 0 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

[192] $1^{113} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 6 & 0 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 4 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & 0 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[193] $1^{124} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 7 & 0 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 5 & 9 & 0 & 7 & 18 & 41 & 11 \\ 2 & 2 & -1 & 0 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$

[194] $1^{123} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 3 & 0 \\ 2 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & 0 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

[195] $1^{111} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 5 & 0 & 10 & 10 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 10 & 0 & 14 & 16 & 8 \\ 5 & 10 & 0 & 16 & 14 & 8 \\ 3 & 5 & -3 & 8 & 8 & 2 \end{pmatrix}_{b_0}$$

- [196] $1^{111} 3 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 4 & 2 \\ 1 & 2 & 0 & 0 & 7 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 16 & 18 & 8 \\ 3 & 4 & 7 & 0 & 18 & 16 & 8 \\ 2 & 2 & 2 & -3 & 8 & 8 & 1 \end{pmatrix}_{b_0}$$
- [197] $1^{111} 3 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 7 & 5 \\ 2 & 3 & 4 & 0 & 7 & 5 & 5 \\ 2 & 2 & 2 & -3 & 5 & 5 & 1 \end{pmatrix}_{b_0}$$
- [198] $1^{13} 3 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 1 & 9 & 2 \\ 1 & 2 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 12 & 3 \\ 6 & 9 & 12 & 0 & 12 & 61 & 15 \\ 2 & 2 & 2 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$
- [199] $1^{111} 3 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 1 & 0 \\ 2 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [200] $1^{111} 3 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -210, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 5 & 0 & 10 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 10 & 0 & 14 & 23 & 8 \\ 7 & 15 & 0 & 23 & 32 & 12 \\ 3 & 5 & -3 & 8 & 12 & 2 \end{pmatrix}_{b_0}$$
- [203] $1^{112} 3 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -214, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 6 & 2 \\ 1 & 2 & 0 & 0 & 7 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 16 & 22 & 8 \\ 4 & 6 & 8 & 0 & 22 & 26 & 10 \\ 2 & 2 & 2 & -3 & 8 & 10 & 1 \end{pmatrix}_{b_0}$$
- [202] $1^{112} 3 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -214, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 0 & 3 & 8 & 2 \\ 1 & 2 & 0 & 0 & 4 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 22 & 5 \\ 6 & 8 & 14 & 0 & 22 & 70 & 16 \\ 2 & 2 & 2 & -3 & 5 & 16 & 1 \end{pmatrix}_{b_0}$$
- [201] $1^{112} 3 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -214, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 5 & 0 \\ 2 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$
- [204] $1^{111} 3 1^{123} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 9 & 3 \\ 5 & 15 & 9 & 0 & 9 & 37 & 11 \\ 2 & 4 & 2 & -3 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

[205] $1^{114} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 4 & 0 \\ 2 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[206] $1^{112} 3^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 5 & 0 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 7 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{112} 3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 5 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & 0 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

[208] $1^{113} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 0 & 1 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 9 & 2 \\ 6 & 9 & 12 & 0 & 9 & 61 & 15 \\ 2 & 2 & 2 & -3 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[209] $1^{113} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 0 & 1 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 7 & 2 \\ 4 & 7 & 6 & 0 & 7 & 25 & 9 \\ 2 & 2 & 2 & -3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[210] $1^{113} 3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 12 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 9 & 9 & 0 & 12 & 9 & 61 & 15 \\ 2 & 2 & -1 & 0 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[211] $1^{112} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 5 & 0 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[212] $1^{112} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 5 & 2 \\ 1 & 2 & 0 & 0 & 4 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 10 & 5 \\ 3 & 5 & 5 & 0 & 10 & 13 & 7 \\ 2 & 2 & 2 & -3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

- [213] $1^{114}3^{11}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 10 & 0 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 6 & 10 & 0 & 10 & 10 & 58 & 14 \\ 2 & 2 & -1 & 0 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$
- [214] $1^{112}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 4 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & 0 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$
- [215] $1^{114}31^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -250$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 8 & 2 \\ 2 & 4 & 4 & 0 & 9 & 24 & 4 \\ 1 & 4 & 0 & 0 & 6 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 42 & 7 \\ 8 & 24 & 18 & 0 & 42 & 118 & 20 \\ 2 & 4 & 2 & -3 & 7 & 20 & 1 \end{pmatrix}_{b_0}$$
- [216] $1^{123}3^{11}21^{144} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -254$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 7 & 2 \\ 1 & 0 & 2 & 0 & 8 & 10 & 2 \\ 1 & 2 & 0 & 0 & 14 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 8 & 14 & 0 & 70 & 80 & 16 \\ 7 & 10 & 15 & 0 & 80 & 88 & 18 \\ 2 & 2 & 2 & -3 & 16 & 18 & 1 \end{pmatrix}_{b_0}$$
- [217] $1^{112}3^{111}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -259$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.68 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 6 & 4 \\ 2 & 2 & -1 & 0 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$
- [218] $1^{12}3^{11}21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -260$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.71 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 18 & 4 \\ 12 & 25 & 0 & 18 & 94 & 20 \\ 3 & 5 & -3 & 4 & 20 & 2 \end{pmatrix}_{b_0}$$
- [219] $1^{113}3^{111}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -264$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 3 & 0 \\ 2 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & 0 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$
- [220] $1^{124}3^{111}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -264$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 7 & 0 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 6 & 9 & 0 & 7 & 18 & 41 & 11 \\ 2 & 2 & -1 & 0 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$
- [222] $1^{124}3^{112}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -264$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 4 & 0 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 6 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & 0 & 4 & 8 & 1 \end{pmatrix}_{b_0}$$

[221] $1^{113}3^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -264$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 15 & 3 \\ 7 & 21 & 15 & 0 & 15 & 85 & 17 \\ 2 & 4 & 2 & -3 & 3 & 17 & 1 \end{pmatrix}_{b_0}$$

[223] $1^{113}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -269$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 3 & 0 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & 0 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[225] $1^{114}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -269$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 7 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 5 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & 0 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[224] $1^{113}3^{114}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -269$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 12 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 10 & 9 & 0 & 12 & 9 & 61 & 15 \\ 2 & 2 & -1 & 0 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[226] $1^{124}3^{111}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -274$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 4 & 0 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & 0 & 4 & 8 & 1 \end{pmatrix}_{b_0}$$

[227] $1^{123}3^{111}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -279$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 6 & 0 \\ 3 & 4 & 0 & 2 & 6 & 14 & 4 \\ 5 & 7 & 0 & 6 & 14 & 25 & 9 \\ 2 & 2 & -1 & 0 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

[228] $1^{13}3^{11}21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -284$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 1 & 10 & 2 \\ 1 & 2 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 15 & 3 \\ 7 & 10 & 15 & 0 & 15 & 88 & 18 \\ 2 & 2 & 2 & -3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

[229] $1^{131}3^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -288$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 6 & 4 \\ 2 & 6 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & 2 & -3 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[230] $1^{111}3^{11}21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -290$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 5 & 0 & 15 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 15 & 0 & 32 & 45 & 12 \\ 9 & 20 & 0 & 45 & 58 & 16 \\ 3 & 5 & -3 & 12 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[231] \ 1^{111} 3 21^{143} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -290, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 9 & 3 \\ 3 & 5 & 0 & 25 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 25 & 0 & 92 & 75 & 20 \\ 9 & 20 & 0 & 75 & 58 & 16 \\ 3 & 5 & -3 & 20 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[232] \ 1^{134} 3 21^{111} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 7 & 0 \\ 4 & 6 & 0 & 3 & 16 & 27 & 6 \\ 6 & 9 & 0 & 7 & 27 & 41 & 11 \\ 2 & 2 & -1 & 0 & 6 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[233] \ 1^{113} 3 21^{111} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 6 & 0 \\ 2 & 2 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & 0 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[234] \ 1^{113} 31^{111} 31^1 + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -301, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 9 & 4 \\ 3 & 9 & 6 & 0 & 9 & 13 & 7 \\ 2 & 4 & 2 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[235] \ 1^{114} 3 21^{13} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 13 & 0 \\ 1 & 2 & 0 & 1 & 0 & 11 & 2 \\ 8 & 11 & 0 & 13 & 11 & 81 & 17 \\ 2 & 2 & -1 & 0 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[236] \ 1^{122} 3 21^{112} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 6 & 2 \\ 1 & 2 & 0 & 0 & 5 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 13 & 20 & 7 \\ 4 & 6 & 8 & 0 & 20 & 26 & 10 \\ 2 & 2 & 2 & -3 & 7 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[237] \ 1^{112} 3 21^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 5 & 0 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 6 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[238] \ 1^{112} 3 21^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -309, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 5 & 0 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 5 & 5 & 0 & 5 & 5 & 13 & 7 \\ 2 & 2 & -1 & 0 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[239] $1^{113} 3^{21} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -309$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 9 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 6 & 8 & 0 & 9 & 8 & 40 & 12 \\ 2 & 2 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[241] $1^{113} 3^{21} 1^{23} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -314$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 9 & 2 \\ 1 & 2 & 0 & 0 & 7 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 16 & 33 & 8 \\ 6 & 9 & 12 & 0 & 33 & 61 & 15 \\ 2 & 2 & 2 & -3 & 8 & 15 & 1 \end{pmatrix}_{b_0}$$

[240] $1^{133} 3^{21} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -314$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 8 & 2 \\ 1 & 2 & 0 & 0 & 6 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 25 & 33 & 9 \\ 5 & 8 & 9 & 0 & 33 & 40 & 12 \\ 2 & 2 & 2 & -3 & 9 & 12 & 1 \end{pmatrix}_{b_0}$$

[242] $1^{111} 3^{21} 1^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -320$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 5 & 0 & 15 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 15 & 0 & 32 & 34 & 12 \\ 7 & 15 & 0 & 34 & 32 & 12 \\ 3 & 5 & -3 & 12 & 12 & 2 \end{pmatrix}_{b_0}$$

[243] $1^{111} 3^{21} 1^{44} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -320$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 11 & 3 \\ 3 & 5 & 0 & 25 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 25 & 0 & 92 & 94 & 20 \\ 11 & 25 & 0 & 94 & 92 & 20 \\ 3 & 5 & -3 & 20 & 20 & 2 \end{pmatrix}_{b_0}$$

[244] $1^{113} 3^{21} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -324$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 0 & 1 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 8 & 2 \\ 5 & 8 & 9 & 0 & 8 & 40 & 12 \\ 2 & 2 & 2 & -3 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[245] $1^{121} 31^{111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -325$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 3 & 2 \\ 2 & 4 & 4 & 0 & 12 & 9 & 4 \\ 1 & 4 & 0 & 0 & 9 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 9 & 0 & 28 & 21 & 10 \\ 3 & 9 & 6 & 0 & 21 & 13 & 7 \\ 2 & 4 & 2 & -3 & 10 & 7 & 1 \end{pmatrix}_{b_0}$$

[246] $1^{114} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -334$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 7 & 0 \\ 2 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & 0 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[247] $1^{122}31^{111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 9 & 0 & 28 & 30 & 10 \\ 4 & 12 & 9 & 0 & 30 & 28 & 10 \\ 2 & 4 & 2 & -3 & 10 & 10 & 1 \end{pmatrix}_{b_0}$$

[248] $1^{14}321^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 1 & 12 & 2 \\ 1 & 2 & 0 & 0 & 3 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 16 & 3 \\ 8 & 12 & 16 & 0 & 16 & 110 & 20 \\ 2 & 2 & 2 & -3 & 3 & 20 & 1 \end{pmatrix}_{b_0}$$

[249] $1^{112}31^{123}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 18 & 4 \\ 1 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 12 & 3 \\ 6 & 18 & 12 & 0 & 12 & 58 & 14 \\ 2 & 4 & 2 & -3 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

[250] $1^{114}321^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -349, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 4 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[251] $1^{111}321^{133} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -360, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 3 \\ 3 & 5 & 0 & 20 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 20 & 0 & 58 & 60 & 16 \\ 9 & 20 & 0 & 60 & 58 & 16 \\ 3 & 5 & -3 & 16 & 16 & 2 \end{pmatrix}_{b_0}$$

[252] $1^{12}321^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -360, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 5 & 0 & 5 & 30 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 22 & 4 \\ 14 & 30 & 0 & 22 & 134 & 24 \\ 3 & 5 & -3 & 4 & 24 & 2 \end{pmatrix}_{b_0}$$

[253] $1^{122}321^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 2 & 0 \\ 3 & 4 & 0 & 2 & 6 & 8 & 4 \\ 3 & 4 & 0 & 2 & 8 & 6 & 4 \\ 2 & 2 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[254] $1^{114}321^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -374, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 0 & 1 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 12 & 2 \\ 8 & 12 & 16 & 0 & 12 & 110 & 20 \\ 2 & 2 & 2 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

[255] $1^{122}321^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 13 & 15 & 7 \\ 3 & 5 & 5 & 0 & 15 & 13 & 7 \\ 2 & 2 & 2 & -3 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

[257] $1^{112} 3 21^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 0 & 7 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 16 & 29 & 8 \\ 5 & 7 & 11 & 0 & 29 & 45 & 13 \\ 2 & 2 & 2 & -3 & 8 & 13 & 1 \end{pmatrix}_{b_0}$$

[258] $1^{113} 3 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 6 & 0 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & 0 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[259] $1^{123} 3 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 7 & 2 \\ 1 & 2 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 13 & 20 & 7 \\ 4 & 7 & 6 & 0 & 20 & 25 & 9 \\ 2 & 2 & 2 & -3 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

[260] $1^{112} 3 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 \\ 1 & 2 & 0 & 0 & 4 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 18 & 5 \\ 5 & 7 & 11 & 0 & 18 & 45 & 13 \\ 2 & 2 & 2 & -3 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

[261] $1^{123} 3 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 9 & 2 \\ 1 & 2 & 0 & 0 & 5 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 13 & 30 & 7 \\ 6 & 9 & 12 & 0 & 30 & 61 & 15 \\ 2 & 2 & 2 & -3 & 7 & 15 & 1 \end{pmatrix}_{b_0}$$

[256] $1^{113} 3 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 3 & 0 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 5 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & 0 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[262] $1^{112} 3 21^{124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -414, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 7 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 16 & 36 & 8 \\ 6 & 8 & 14 & 0 & 36 & 70 & 16 \\ 2 & 2 & 2 & -3 & 8 & 16 & 1 \end{pmatrix}_{b_0}$$

[263] $1^{112} 3 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -414, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 6 & 2 \\ 1 & 2 & 0 & 0 & 4 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 14 & 5 \\ 4 & 6 & 8 & 0 & 14 & 26 & 10 \\ 2 & 2 & 2 & -3 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

[264] $1^{124} 3 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -414, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 7 & 0 \\ 3 & 4 & 0 & 2 & 6 & 18 & 4 \\ 6 & 9 & 0 & 7 & 18 & 41 & 11 \\ 2 & 2 & -1 & 0 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[265] \ 1^{123} 3^{111} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -419, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 3 & 0 \\ 3 & 4 & 0 & 2 & 6 & 12 & 4 \\ 4 & 6 & 0 & 3 & 12 & 16 & 6 \\ 2 & 2 & -1 & 0 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[266] \ 1^{124} 3^{111} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 4 & 0 \\ 3 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 4 & 16 & 30 & 8 \\ 2 & 2 & -1 & 0 & 4 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[267] \ 1^{13} 3^{11} 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 5 & 0 & 5 & 35 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 25 & 4 \\ 17 & 35 & 0 & 25 & 188 & 28 \\ 3 & 5 & -3 & 4 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[268] \ 1^{114} 3^{11} 21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -454, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 8 & 2 \\ 1 & 0 & 2 & 0 & 4 & 12 & 2 \\ 1 & 2 & 0 & 0 & 7 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 16 & 44 & 8 \\ 8 & 12 & 16 & 0 & 44 & 110 & 20 \\ 2 & 2 & 2 & -3 & 8 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[269] \ 1^{113} 3^{112} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 9 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 7 & 8 & 0 & 9 & 8 & 40 & 12 \\ 2 & 2 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[270] \ 1^{113} 3^{114} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 3 & 0 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 7 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & 0 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[271] \ 1^{122} 3^{11} 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 6 & 7 & 2 \\ 1 & 2 & 0 & 0 & 8 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 26 & 36 & 10 \\ 5 & 7 & 11 & 0 & 36 & 45 & 13 \\ 2 & 2 & 2 & -3 & 10 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[272] \ 1^{122} 3^{11} 21^{143} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 7 & 2 \\ 1 & 2 & 0 & 0 & 14 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 8 & 14 & 0 & 70 & 58 & 16 \\ 5 & 7 & 11 & 0 & 58 & 45 & 13 \\ 2 & 2 & 2 & -3 & 16 & 13 & 1 \end{pmatrix}_{b_0}$$

[273] $1^{113} 3^{113} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -469$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 3 & 0 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 6 & 6 & 0 & 3 & 6 & 16 & 6 \\ 2 & 2 & -1 & 0 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[274] $1^{114} 3^{11} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -494$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 0 & 1 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 10 & 2 \\ 6 & 10 & 10 & 0 & 10 & 58 & 14 \\ 2 & 2 & 2 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[275] $1^{12} 31^{112} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -508$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 6 & 2 \\ 2 & 4 & 4 & 0 & 6 & 18 & 4 \\ 1 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 18 & 4 \\ 6 & 18 & 12 & 0 & 18 & 58 & 14 \\ 2 & 4 & 2 & -3 & 4 & 14 & 1 \end{pmatrix}_{b_0}$$

[276] $1^{134} 3^{11} 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -524$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 9 & 2 \\ 1 & 2 & 0 & 0 & 6 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 25 & 34 & 9 \\ 5 & 9 & 7 & 0 & 34 & 41 & 11 \\ 2 & 2 & 2 & -3 & 9 & 11 & 1 \end{pmatrix}_{b_0}$$

[277] $1^{111} 31^{111} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -524$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 15 & 7 \\ 3 & 9 & 6 & 0 & 15 & 13 & 7 \\ 2 & 4 & 2 & -3 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

[278] $1^{114} 3^{11} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -524$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 0 & 1 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 11 & 2 \\ 7 & 11 & 13 & 0 & 11 & 81 & 17 \\ 2 & 2 & 2 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

[279] $1^{133} 3^{111} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -524$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 3 & 0 \\ 4 & 6 & 0 & 3 & 16 & 18 & 6 \\ 4 & 6 & 0 & 3 & 18 & 16 & 6 \\ 2 & 2 & -1 & 0 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[280] $1^{133} 3^{11} 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -524$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 7 & 2 \\ 1 & 2 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 25 & 27 & 9 \\ 4 & 7 & 6 & 0 & 27 & 25 & 9 \\ 2 & 2 & 2 & -3 & 9 & 9 & 1 \end{pmatrix}_{b_0}$$

[281] $1^{122} 31^{112} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -532$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 6 & 2 \\ 2 & 4 & 4 & 0 & 12 & 18 & 4 \\ 1 & 4 & 0 & 0 & 9 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 9 & 0 & 28 & 42 & 10 \\ 6 & 18 & 12 & 0 & 42 & 58 & 14 \\ 2 & 4 & 2 & -3 & 10 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[282] \ 1^{124} 3 21^{114} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 8 & 2 \\ 1 & 0 & 2 & 0 & 5 & 12 & 2 \\ 1 & 2 & 0 & 0 & 5 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 13 & 40 & 7 \\ 8 & 12 & 16 & 0 & 40 & 110 & 20 \\ 2 & 2 & 2 & -3 & 7 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[283] \ 1^{113} 3 21^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -549, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 6 & 0 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 6 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & 0 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[284] \ 1^{112} 31^{13} 31^1 + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -549, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 24 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 15 & 3 \\ 8 & 24 & 15 & 0 & 15 & 100 & 18 \\ 2 & 4 & 2 & -3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[285] \ 1^{113} 3 21^{12} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -549, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 9 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 8 & 8 & 0 & 9 & 8 & 40 & 12 \\ 2 & 2 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[286] \ 1^{114} 3 21^{12} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -549, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 10 & 0 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 7 & 10 & 0 & 10 & 10 & 58 & 14 \\ 2 & 2 & -1 & 0 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[287] \ 1^{114} 3 21^{13} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -549, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 13 & 0 \\ 1 & 2 & 0 & 1 & 0 & 11 & 2 \\ 9 & 11 & 0 & 13 & 11 & 81 & 17 \\ 2 & 2 & -1 & 0 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[288] \ 1^{113} 3 21^{112} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 0 & 3 & 8 & 2 \\ 1 & 2 & 0 & 0 & 4 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 17 & 5 \\ 5 & 8 & 9 & 0 & 17 & 40 & 12 \\ 2 & 2 & 2 & -3 & 5 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[292] \ 1^{114} 3 21^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 7 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & 0 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[289] $1^{123} 3^{21} 1^{33} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 9 & 2 \\ 1 & 2 & 0 & 0 & 11 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 11 & 0 & 45 & 54 & 13 \\ 6 & 9 & 12 & 0 & 54 & 61 & 15 \\ 2 & 2 & 2 & -3 & 13 & 15 & 1 \end{pmatrix}_{b_0}$$

[290] $1^{113} 3^{21} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 9 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 9 & 8 & 0 & 9 & 8 & 40 & 12 \\ 2 & 2 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[289] $1^{114} 3^{21} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 4 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 6 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[293] $1^{114} 3^{21} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -574, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 0 & 4 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 20 & 5 \\ 6 & 10 & 10 & 0 & 20 & 58 & 14 \\ 2 & 2 & 2 & -3 & 5 & 14 & 1 \end{pmatrix}_{b_0}$$

[294] $1^{134} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -579, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 4 & 0 \\ 4 & 6 & 0 & 3 & 16 & 24 & 6 \\ 5 & 8 & 0 & 4 & 24 & 30 & 8 \\ 2 & 2 & -1 & 0 & 6 & 8 & 1 \end{pmatrix}_{b_0}$$

[295] $1^{123} 3^{21} 1^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -614, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 6 & 8 & 2 \\ 1 & 2 & 0 & 0 & 8 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 26 & 34 & 10 \\ 5 & 8 & 9 & 0 & 34 & 40 & 12 \\ 2 & 2 & 2 & -3 & 10 & 12 & 1 \end{pmatrix}_{b_0}$$

[296] $1^{114} 3^{13} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -621, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 30 & 4 \\ 1 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 21 & 3 \\ 10 & 30 & 21 & 0 & 21 & 172 & 24 \\ 2 & 4 & 2 & -3 & 3 & 24 & 1 \end{pmatrix}_{b_0}$$

[297] $1^{122} 3^{21} 1^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 6 & 2 \\ 1 & 2 & 0 & 0 & 8 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 26 & 28 & 10 \\ 4 & 6 & 8 & 0 & 28 & 26 & 10 \\ 2 & 2 & 2 & -3 & 10 & 10 & 1 \end{pmatrix}_{b_0}$$

[298] $1^{113} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 6 & 0 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 7 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & 0 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[299] $1^{122} 3^{11} 21^{144} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 8 & 2 \\ 1 & 2 & 0 & 0 & 14 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 8 & 14 & 0 & 70 & 72 & 16 \\ 6 & 8 & 14 & 0 & 72 & 70 & 16 \\ 2 & 2 & 2 & -3 & 16 & 16 & 1 \end{pmatrix}_{b_0}$$

[300] $1^{113} 3^{11} 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -629, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 6 & 0 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 8 & 7 & 0 & 6 & 7 & 25 & 9 \\ 2 & 2 & -1 & 0 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[301] $1^{113} 3^{11} 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -644, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 0 & 7 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 16 & 40 & 8 \\ 7 & 10 & 15 & 0 & 40 & 88 & 18 \\ 2 & 2 & 2 & -3 & 8 & 18 & 1 \end{pmatrix}_{b_0}$$

[303] $1^{123} 3^{11} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -644, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 8 & 2 \\ 1 & 2 & 0 & 0 & 5 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 13 & 25 & 7 \\ 5 & 8 & 9 & 0 & 25 & 40 & 12 \\ 2 & 2 & 2 & -3 & 7 & 12 & 1 \end{pmatrix}_{b_0}$$

[302] $2^{1111} 2^2 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -644, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 & 2 \\ 2 & 1 & 4 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 1 & 3 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & 1 & -1 \\ 2 & 3 & 3 & 1 & 1 & 3 & 3 \\ 2 & 3 & 3 & 1 & 3 & 1 & 3 \\ 2 & 3 & 2 & -1 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[304] $1^{144} 3^{111} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -684, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 4 & 0 \\ 5 & 8 & 0 & 4 & 30 & 32 & 8 \\ 5 & 8 & 0 & 4 & 32 & 30 & 8 \\ 2 & 2 & -1 & 0 & 8 & 8 & 1 \end{pmatrix}_{b_0}$$

[305] $1^{113} 31^{13} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -691, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 18 & 3 \\ 9 & 27 & 18 & 0 & 18 & 133 & 21 \\ 2 & 4 & 2 & -3 & 3 & 21 & 1 \end{pmatrix}_{b_0}$$

[306] $1^{11}31^{112}31^1 + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -693, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 5 & 2 \\ 2 & 4 & 4 & 0 & 6 & 15 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 15 & 4 \\ 5 & 15 & 9 & 0 & 15 & 37 & 11 \\ 2 & 4 & 2 & -3 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$

[307] $1^{124}321^{134} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 0 & 7 & 12 & 2 \\ 1 & 2 & 0 & 0 & 11 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 11 & 0 & 45 & 72 & 13 \\ 8 & 12 & 16 & 0 & 72 & 110 & 20 \\ 2 & 2 & 2 & -3 & 13 & 20 & 1 \end{pmatrix}_{b_0}$$

[308] $1^{122}321^{133} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 7 & 2 \\ 1 & 2 & 0 & 0 & 11 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 11 & 0 & 45 & 47 & 13 \\ 5 & 7 & 11 & 0 & 47 & 45 & 13 \\ 2 & 2 & 2 & -3 & 13 & 13 & 1 \end{pmatrix}_{b_0}$$

[309] $1^{114}321^{13} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -709, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 4 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 7 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[310] $1^{133}321^{123} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -714, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 9 & 2 \\ 1 & 2 & 0 & 0 & 9 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 9 & 0 & 40 & 51 & 12 \\ 6 & 9 & 12 & 0 & 51 & 61 & 15 \\ 2 & 2 & 2 & -3 & 12 & 15 & 1 \end{pmatrix}_{b_0}$$

[311] $1^{114}321^{13} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -724, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 13 & 0 \\ 1 & 2 & 0 & 1 & 0 & 11 & 2 \\ 10 & 11 & 0 & 13 & 11 & 81 & 17 \\ 2 & 2 & -1 & 0 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

[312] $1^{134}321^{112} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -734, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 10 & 2 \\ 1 & 2 & 0 & 0 & 6 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 25 & 40 & 9 \\ 6 & 10 & 10 & 0 & 40 & 58 & 14 \\ 2 & 2 & 2 & -3 & 9 & 14 & 1 \end{pmatrix}_{b_0}$$

[313] $1^{113}321^{113} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -764, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 0 & 3 & 9 & 2 \\ 1 & 2 & 0 & 0 & 4 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 21 & 5 \\ 6 & 9 & 12 & 0 & 21 & 61 & 15 \\ 2 & 2 & 2 & -3 & 5 & 15 & 1 \end{pmatrix}_{b_0}$$

[314] $1^{114} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -784$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 4 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 8 & 8 & 0 & 4 & 8 & 30 & 8 \\ 2 & 2 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[315] $1^{114} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -784$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 10 & 0 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 8 & 10 & 0 & 10 & 10 & 58 & 14 \\ 2 & 2 & -1 & 0 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[316] $1^{114} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -789$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 7 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 7 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & 0 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[317] $1^{113} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -794$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 0 & 4 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 25 & 5 \\ 7 & 10 & 15 & 0 & 25 & 88 & 18 \\ 2 & 2 & 2 & -3 & 5 & 18 & 1 \end{pmatrix}_{b_0}$$

[318] $1^{111} 3^{112} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -811$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 5 & 2 \\ 2 & 4 & 4 & 0 & 9 & 15 & 4 \\ 1 & 4 & 0 & 0 & 6 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 24 & 7 \\ 5 & 15 & 9 & 0 & 24 & 37 & 11 \\ 2 & 4 & 2 & -3 & 7 & 11 & 1 \end{pmatrix}_{b_0}$$

[319] $1^{144} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -824$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 11 & 2 \\ 1 & 2 & 0 & 0 & 10 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 10 & 10 & 0 & 58 & 70 & 14 \\ 7 & 11 & 13 & 0 & 70 & 81 & 17 \\ 2 & 2 & 2 & -3 & 14 & 17 & 1 \end{pmatrix}_{b_0}$$

[320] $1^{114} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -829$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 13 & 0 \\ 1 & 2 & 0 & 1 & 0 & 11 & 2 \\ 11 & 11 & 0 & 13 & 11 & 81 & 17 \\ 2 & 2 & -1 & 0 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

[321] $1^{13} 3^{113} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -853$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 9 & 2 \\ 2 & 4 & 4 & 0 & 6 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 27 & 4 \\ 9 & 27 & 18 & 0 & 27 & 133 & 21 \\ 2 & 4 & 2 & -3 & 4 & 21 & 1 \end{pmatrix}_{b_0}$$

[323] $1^{134} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -854$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 0 & 8 & 12 & 2 \\ 1 & 2 & 0 & 0 & 9 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 9 & 0 & 40 & 68 & 12 \\ 8 & 12 & 16 & 0 & 68 & 110 & 20 \\ 2 & 2 & 2 & -3 & 12 & 20 & 1 \end{pmatrix}_{b_0}$$

[322] $1^{123}3^{11}21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -854$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$

$T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 10 & 2 \\ 1 & 2 & 0 & 0 & 8 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 26 & 50 & 10 \\ 7 & 10 & 15 & 0 & 50 & 88 & 18 \\ 2 & 2 & 2 & -3 & 10 & 18 & 1 \end{pmatrix}_{b_0}$$

[324] $1^{113}31^{112}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -859$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 7 & 2 \\ 2 & 4 & 4 & 0 & 9 & 21 & 4 \\ 1 & 4 & 0 & 0 & 6 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 36 & 7 \\ 7 & 21 & 15 & 0 & 36 & 85 & 17 \\ 2 & 4 & 2 & -3 & 7 & 17 & 1 \end{pmatrix}_{b_0}$$

[325] $1^{124}3^{11}21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -864$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 10 & 2 \\ 1 & 2 & 0 & 0 & 5 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 13 & 30 & 7 \\ 6 & 10 & 10 & 0 & 30 & 58 & 14 \\ 2 & 2 & 2 & -3 & 7 & 14 & 1 \end{pmatrix}_{b_0}$$

[327] $1^{114}3^{113}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -944$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 7 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 8 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & 0 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[326] $1^{124}3^{11}21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -944$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 11.61 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 0 & 5 & 11 & 2 \\ 1 & 2 & 0 & 0 & 5 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 13 & 35 & 7 \\ 7 & 11 & 13 & 0 & 35 & 81 & 17 \\ 2 & 2 & 2 & -3 & 7 & 17 & 1 \end{pmatrix}_{b_0}$$

[328] $1^{114}3^{113}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -949$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 10 & 0 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 9 & 10 & 0 & 10 & 10 & 58 & 14 \\ 2 & 2 & -1 & 0 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[329] $1^{133}3^{11}21^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -954$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 7 & 2 \\ 1 & 0 & 2 & 0 & 9 & 10 & 2 \\ 1 & 2 & 0 & 0 & 12 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 9 & 12 & 0 & 61 & 75 & 15 \\ 7 & 10 & 15 & 0 & 75 & 88 & 18 \\ 2 & 2 & 2 & -3 & 15 & 18 & 1 \end{pmatrix}_{b_0}$$

[331] $1^{123}3^{11}21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -974$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$

$T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 9 & 2 \\ 1 & 2 & 0 & 0 & 8 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 26 & 42 & 10 \\ 6 & 9 & 12 & 0 & 42 & 61 & 15 \\ 2 & 2 & 2 & -3 & 10 & 15 & 1 \end{pmatrix}_{b_0}$$

[330] $1^{134} 3^{21} 1^{22} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -974$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.66 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 10 & 2 \\ 1 & 2 & 0 & 0 & 9 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 9 & 0 & 40 & 50 & 12 \\ 6 & 10 & 10 & 0 & 50 & 58 & 14 \\ 2 & 2 & 2 & -3 & 12 & 14 & 1 \end{pmatrix}_{b_0}$$

[332] $1^{111} 31^{113} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1000$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 7 & 2 \\ 2 & 4 & 4 & 0 & 9 & 21 & 4 \\ 1 & 4 & 0 & 0 & 6 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 33 & 7 \\ 7 & 21 & 12 & 0 & 33 & 73 & 15 \\ 2 & 4 & 2 & -3 & 7 & 15 & 1 \end{pmatrix}_{b_0}$$

[334] $1^{114} 3^{21} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1004$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 0 & 3 & 11 & 2 \\ 1 & 2 & 0 & 0 & 4 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 24 & 5 \\ 7 & 11 & 13 & 0 & 24 & 81 & 17 \\ 2 & 2 & 2 & -3 & 5 & 17 & 1 \end{pmatrix}_{b_0}$$

[333] $1^{133} 3^{21} 1^{22} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1004$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 11.61 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 8 & 2 \\ 1 & 2 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 9 & 0 & 40 & 42 & 12 \\ 5 & 8 & 9 & 0 & 42 & 40 & 12 \\ 2 & 2 & 2 & -3 & 12 & 12 & 1 \end{pmatrix}_{b_0}$$

[335] $1^{112} 31^{122} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1018$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 6 & 2 \\ 2 & 4 & 4 & 0 & 15 & 18 & 4 \\ 1 & 4 & 0 & 0 & 9 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 15 & 9 & 0 & 37 & 48 & 11 \\ 6 & 18 & 12 & 0 & 48 & 58 & 14 \\ 2 & 4 & 2 & -3 & 11 & 14 & 1 \end{pmatrix}_{b_0}$$

[336] $1^{113} 31^{14} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1020$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 3 & 33 & 4 \\ 1 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 21 & 3 \\ 11 & 33 & 21 & 0 & 21 & 193 & 25 \\ 2 & 4 & 2 & -3 & 3 & 25 & 1 \end{pmatrix}_{b_0}$$

[337] $1^{114} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1029$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 7 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 9 & 9 & 0 & 7 & 9 & 41 & 11 \\ 2 & 2 & -1 & 0 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[338] $2^{1112} 2^{31} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1034$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 4 & 2 \\ 2 & 1 & 4 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 1 & 3 & 6 & 2 \\ 1 & 1 & 1 & -3 & 1 & 2 & -1 \\ 2 & 3 & 3 & 1 & 1 & 6 & 3 \\ 4 & 6 & 6 & 2 & 6 & 10 & 6 \\ 2 & 3 & 2 & -1 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[339] $1^{114} 3^{21^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1044$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 10 & 0 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 10 & 10 & 0 & 10 & 10 & 58 & 14 \\ 2 & 2 & -1 & 0 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[340] $1^{123} 3^{21^{134}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1044$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$

$T_{\min} = 11.61 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 10 & 2 \\ 1 & 2 & 0 & 0 & 11 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 11 & 0 & 45 & 65 & 13 \\ 7 & 10 & 15 & 0 & 65 & 88 & 18 \\ 2 & 2 & 2 & -3 & 13 & 18 & 1 \end{pmatrix}_{b_0}$$

[341] $1^{112} 3^{1^{12} 3^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1046$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 6 & 2 \\ 2 & 4 & 4 & 0 & 9 & 18 & 4 \\ 1 & 4 & 0 & 0 & 6 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 30 & 7 \\ 6 & 18 & 12 & 0 & 30 & 58 & 14 \\ 2 & 4 & 2 & -3 & 7 & 14 & 1 \end{pmatrix}_{b_0}$$

[342] $1^{124} 3^{21^{123}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1064$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 11 & 2 \\ 1 & 2 & 0 & 0 & 8 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 26 & 48 & 10 \\ 7 & 11 & 13 & 0 & 48 & 81 & 17 \\ 2 & 2 & 2 & -3 & 10 & 17 & 1 \end{pmatrix}_{b_0}$$

[343] $1^{134} 3^{21^{144}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1094$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 8 & 2 \\ 1 & 0 & 2 & 0 & 10 & 12 & 2 \\ 1 & 2 & 0 & 0 & 15 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 10 & 15 & 0 & 88 & 100 & 18 \\ 8 & 12 & 16 & 0 & 100 & 110 & 20 \\ 2 & 2 & 2 & -3 & 18 & 20 & 1 \end{pmatrix}_{b_0}$$

[344] $1^{134} 3^{21^{133}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1164$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 7 & 2 \\ 1 & 0 & 2 & 0 & 9 & 11 & 2 \\ 1 & 2 & 0 & 0 & 12 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 9 & 12 & 0 & 61 & 72 & 15 \\ 7 & 11 & 13 & 0 & 72 & 81 & 17 \\ 2 & 2 & 2 & -3 & 15 & 17 & 1 \end{pmatrix}_{b_0}$$

[345] $1^{114} 3^{1^{14} 3^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1174$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 3 & 36 & 4 \\ 1 & 4 & 0 & 0 & 3 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 24 & 3 \\ 12 & 36 & 24 & 0 & 24 & 238 & 28 \\ 2 & 4 & 2 & -3 & 3 & 28 & 1 \end{pmatrix}_{b_0}$$

[346] $1^{113} 3^{1^{11} 3^{13}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1197$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 7 & 2 \\ 2 & 4 & 4 & 0 & 6 & 21 & 4 \\ 1 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 21 & 4 \\ 7 & 21 & 12 & 0 & 21 & 73 & 15 \\ 2 & 4 & 2 & -3 & 4 & 15 & 1 \end{pmatrix}_{b_0}$$

[347] $1^{123} 3^{1^{11} 3^{13}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1236$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 8 & 2 \\ 2 & 4 & 4 & 0 & 6 & 24 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 24 & 4 \\ 8 & 24 & 15 & 0 & 24 & 100 & 18 \\ 2 & 4 & 2 & -3 & 4 & 18 & 1 \end{pmatrix}_{b_0}$$

[348] $1^{114}3^{11}21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1254$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 0 & 3 & 12 & 2 \\ 1 & 2 & 0 & 0 & 4 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 28 & 5 \\ 8 & 12 & 16 & 0 & 28 & 110 & 20 \\ 2 & 2 & 2 & -3 & 5 & 20 & 1 \end{pmatrix}_{b_0}$$

[349] $1^{144}3^{11}21^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1304$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 8 & 2 \\ 1 & 0 & 2 & 0 & 11 & 12 & 2 \\ 1 & 2 & 0 & 0 & 13 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 11 & 13 & 0 & 81 & 96 & 17 \\ 8 & 12 & 16 & 0 & 96 & 110 & 20 \\ 2 & 2 & 2 & -3 & 17 & 20 & 1 \end{pmatrix}_{b_0}$$

[350] $1^{111}31^{122}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1308$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 15 & 4 \\ 1 & 4 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 15 & 9 & 0 & 37 & 39 & 11 \\ 5 & 15 & 9 & 0 & 39 & 37 & 11 \\ 2 & 4 & 2 & -3 & 11 & 11 & 1 \end{pmatrix}_{b_0}$$

[351] $1^{133}3^{11}21^{133} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1324$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 9 & 2 \\ 1 & 2 & 0 & 0 & 12 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 9 & 12 & 0 & 61 & 63 & 15 \\ 6 & 9 & 12 & 0 & 63 & 61 & 15 \\ 2 & 2 & 2 & -3 & 15 & 15 & 1 \end{pmatrix}_{b_0}$$

[352] $1^{143}31^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1336$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 12 & 2 \\ 2 & 4 & 4 & 0 & 6 & 36 & 4 \\ 1 & 4 & 0 & 0 & 3 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 36 & 4 \\ 12 & 36 & 24 & 0 & 36 & 238 & 28 \\ 2 & 4 & 2 & -3 & 4 & 28 & 1 \end{pmatrix}_{b_0}$$

[353] $1^{144}3^{11}21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.68 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 0 & 10 & 10 & 2 \\ 1 & 2 & 0 & 0 & 10 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 10 & 10 & 0 & 58 & 60 & 14 \\ 6 & 10 & 10 & 0 & 60 & 58 & 14 \\ 2 & 2 & 2 & -3 & 14 & 14 & 1 \end{pmatrix}_{b_0}$$

[354] $1^{134}3^{11}21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1404$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.61 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 0 & 8 & 11 & 2 \\ 1 & 2 & 0 & 0 & 9 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 9 & 0 & 40 & 59 & 12 \\ 7 & 11 & 13 & 0 & 59 & 81 & 17 \\ 2 & 2 & 2 & -3 & 12 & 17 & 1 \end{pmatrix}_{b_0}$$

[355] $2^12^131^{1122} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1424$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 4 & 4 & 2 \\ 2 & 1 & 4 & 1 & 6 & 6 & 3 \\ 2 & 4 & 0 & 1 & 6 & 6 & 2 \\ 1 & 1 & 1 & -3 & 2 & 2 & -1 \\ 4 & 6 & 6 & 2 & 10 & 12 & 6 \\ 4 & 6 & 6 & 2 & 12 & 10 & 6 \\ 2 & 3 & 2 & -1 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[356] $1^{124}3^{11}21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1464$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 2 & 0 & 6 & 12 & 2 \\ 1 & 2 & 0 & 0 & 8 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 26 & 56 & 10 \\ 8 & 12 & 16 & 0 & 56 & 110 & 20 \\ 2 & 2 & 2 & -3 & 10 & 20 & 1 \end{pmatrix}_{b_0}$$

[357] $1^{133}3^{11}21^{144} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1484$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.8$

$T_{\min} = 11.61 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 10 & 2 \\ 1 & 2 & 0 & 0 & 15 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 10 & 15 & 0 & 88 & 90 & 18 \\ 7 & 10 & 15 & 0 & 90 & 88 & 18 \\ 2 & 2 & 2 & -3 & 18 & 18 & 1 \end{pmatrix}_{b_0}$$

[358] $1^{122}31^{122}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1568$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 18 & 4 \\ 1 & 4 & 0 & 0 & 12 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 18 & 12 & 0 & 58 & 60 & 14 \\ 6 & 18 & 12 & 0 & 60 & 58 & 14 \\ 2 & 4 & 2 & -3 & 14 & 14 & 1 \end{pmatrix}_{b_0}$$

[359] $1^{112}31^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1669$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 8 & 2 \\ 2 & 4 & 4 & 0 & 9 & 24 & 4 \\ 1 & 4 & 0 & 0 & 6 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 39 & 7 \\ 8 & 24 & 15 & 0 & 39 & 100 & 18 \\ 2 & 4 & 2 & -3 & 7 & 18 & 1 \end{pmatrix}_{b_0}$$

[360] $2^{1113}2^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1684$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 6 & 2 \\ 2 & 1 & 4 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 1 & 3 & 9 & 2 \\ 1 & 1 & 1 & -3 & 1 & 3 & -1 \\ 2 & 3 & 3 & 1 & 1 & 9 & 3 \\ 6 & 9 & 9 & 3 & 9 & 25 & 9 \\ 2 & 3 & 2 & -1 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[361] $1^{114}31^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1741$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 10 & 2 \\ 2 & 4 & 4 & 0 & 9 & 30 & 4 \\ 1 & 4 & 0 & 0 & 6 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 51 & 7 \\ 10 & 30 & 21 & 0 & 51 & 172 & 24 \\ 2 & 4 & 2 & -3 & 7 & 24 & 1 \end{pmatrix}_{b_0}$$

[362] $1^{11}31^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1813$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 9 & 2 \\ 2 & 4 & 4 & 0 & 6 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 27 & 4 \\ 9 & 27 & 15 & 0 & 27 & 121 & 19 \\ 2 & 4 & 2 & -3 & 4 & 19 & 1 \end{pmatrix}_{b_0}$$

[363] $1^{134}3^{11}21^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1814$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 8 & 2 \\ 1 & 0 & 2 & 0 & 9 & 12 & 2 \\ 1 & 2 & 0 & 0 & 12 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 9 & 12 & 0 & 61 & 84 & 15 \\ 8 & 12 & 16 & 0 & 84 & 110 & 20 \\ 2 & 2 & 2 & -3 & 15 & 20 & 1 \end{pmatrix}_{b_0}$$

[364] $1^{144}3^{11}21^{133} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1904$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.61 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 0 & 11 & 11 & 2 \\ 1 & 2 & 0 & 0 & 13 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 11 & 13 & 0 & 81 & 83 & 17 \\ 7 & 11 & 13 & 0 & 83 & 81 & 17 \\ 2 & 2 & 2 & -3 & 17 & 17 & 1 \end{pmatrix}_{b_0}$$

[365] $1^{113}31^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1916$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 9 & 2 \\ 2 & 4 & 4 & 0 & 9 & 27 & 4 \\ 1 & 4 & 0 & 0 & 6 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 45 & 7 \\ 9 & 27 & 18 & 0 & 45 & 133 & 21 \\ 2 & 4 & 2 & -3 & 7 & 21 & 1 \end{pmatrix}_{b_0}$$

[366] $1^{13}31^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1917$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 11 & 2 \\ 2 & 4 & 4 & 0 & 6 & 33 & 4 \\ 1 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 33 & 4 \\ 11 & 33 & 21 & 0 & 33 & 193 & 25 \\ 2 & 4 & 2 & -3 & 4 & 25 & 1 \end{pmatrix}_{b_0}$$

[367] $1^{12}31^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -2076$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 10 & 2 \\ 2 & 4 & 4 & 0 & 6 & 30 & 4 \\ 1 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 30 & 4 \\ 10 & 30 & 18 & 0 & 30 & 154 & 22 \\ 2 & 4 & 2 & -3 & 4 & 22 & 1 \end{pmatrix}_{b_0}$$

[368] $1^{112}31^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -2194$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 10 & 2 \\ 2 & 4 & 4 & 0 & 9 & 30 & 4 \\ 1 & 4 & 0 & 0 & 6 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 48 & 7 \\ 10 & 30 & 18 & 0 & 48 & 154 & 22 \\ 2 & 4 & 2 & -3 & 7 & 22 & 1 \end{pmatrix}_{b_0}$$

[369] $1^{144}321^{144} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -2304$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 8 & 2 \\ 1 & 0 & 2 & 0 & 12 & 12 & 2 \\ 1 & 2 & 0 & 0 & 16 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 12 & 16 & 0 & 110 & 112 & 20 \\ 8 & 12 & 16 & 0 & 112 & 110 & 20 \\ 2 & 2 & 2 & -3 & 20 & 20 & 1 \end{pmatrix}_{b_0}$$

[370] $2^{1114}2^231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -2594$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 8 & 2 \\ 2 & 1 & 4 & 1 & 3 & 12 & 3 \\ 2 & 4 & 0 & 1 & 3 & 12 & 2 \\ 1 & 1 & 1 & -3 & 1 & 4 & -1 \\ 2 & 3 & 3 & 1 & 1 & 12 & 3 \\ 8 & 12 & 12 & 4 & 12 & 46 & 12 \\ 2 & 3 & 2 & -1 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[371] $1^{113}31^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -2875$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 11 & 2 \\ 2 & 4 & 4 & 0 & 9 & 33 & 4 \\ 1 & 4 & 0 & 0 & 6 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 54 & 7 \\ 11 & 33 & 21 & 0 & 54 & 193 & 25 \\ 2 & 4 & 2 & -3 & 7 & 25 & 1 \end{pmatrix}_{b_0}$$

[372] $1^{114}31^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -3134$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 12 & 2 \\ 2 & 4 & 4 & 0 & 9 & 36 & 4 \\ 1 & 4 & 0 & 0 & 6 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 60 & 7 \\ 12 & 36 & 24 & 0 & 60 & 238 & 28 \\ 2 & 4 & 2 & -3 & 7 & 28 & 1 \end{pmatrix}_{b_0}$$

[373] $2^{1115}2^231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -3764$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.3$

$T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 10 & 2 \\ 2 & 1 & 4 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 1 & 3 & 15 & 2 \\ 1 & 1 & 1 & -3 & 1 & 5 & -1 \\ 2 & 3 & 3 & 1 & 1 & 15 & 3 \\ 10 & 15 & 15 & 5 & 15 & 73 & 15 \\ 2 & 3 & 2 & -1 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [374] \quad 1^{11}231^{11}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 1 & 1 & 13 & 3 \\ 5 & 13 & 1 & 0 & 30 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 30 & 0 & 0 & 63 & 15 \\ 3 & 7 & -1 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[375] \quad 1^{11}231^{11}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 1 & 1 & 11 & 3 \\ 5 & 13 & 1 & 0 & 30 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 30 & 0 & 0 & 63 & 15 \\ 3 & 7 & -1 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[376] \quad 1^{11}231^{11}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 1 & 1 & 12 & 3 \\ 5 & 13 & 1 & 0 & 30 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 30 & 0 & 0 & 63 & 15 \\ 3 & 7 & -1 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[377] \quad 1^{12}231^{12}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 208, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 1 & 1 & 23 & 3 \\ 5 & 13 & 1 & 0 & 60 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 23 & 60 & 0 & 0 & 258 & 30 \\ 3 & 7 & -1 & 0 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[378] \quad 1^{13}231^{13}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 268, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 1 & 1 & 34 & 3 \\ 5 & 13 & 1 & 0 & 90 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 34 & 90 & 0 & 0 & 583 & 45 \\ 3 & 7 & -1 & 0 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[379] \quad 1^{12}231^{12}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 288, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 1 & 1 & 24 & 3 \\ 5 & 13 & 1 & 0 & 60 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 24 & 60 & 0 & 0 & 258 & 30 \\ 3 & 7 & -1 & 0 & 30 & 3 \end{pmatrix}_{b_0}$$

4.463 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4$

$$T = 9 \quad [0] \quad 1^{1111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 3 & 0 & 0 \\ 3 & 8 & 3 & 4 & 6 & 0 & 4 \\ 3 & 8 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 4 & 4 & -3 & -1 \end{pmatrix}_{b_0}$$

4.464 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{112}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.7$$

$$T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 1 & 3 \\ 4 & 9 & 4 & 0 & 16 & 0 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 16 & 6 & 0 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 0 & 8 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -106, \ \det = -90, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.07 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 12 & 7 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -3 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -106, \ \det = -104, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -3 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -106, \ \det = -288, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 3 & 0 & 0 \\ 3 & 8 & 3 & 4 & 6 & 0 & 4 \\ 4 & 8 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 4 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -106, \ \det = -324, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.67 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 35 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 35 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -3 & 7 & 27 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -106, \ \det = -380, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 3 & 0 & 0 \\ 3 & 8 & 3 & 4 & 6 & 0 & 4 \\ 3 & 8 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 4 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -106, \ \det = -404, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.65 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -3 & 7 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -106, \ \det = -476, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.57 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 3 & 0 & 0 \\ 4 & 8 & 3 & 4 & 6 & 0 & 4 \\ 4 & 8 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 4 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -106, \ \det = -482, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 26 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 26 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -3 & 7 & 20 & 1 \end{pmatrix}_{b_0}$$

4.465 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{124} 3^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 17 & 2 & 3 \\ 3 & 4 & 0 & 8 & 31 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 42 & 4 & 6 \\ 17 & 31 & 0 & 42 & 172 & 17 & 24 \\ 2 & 3 & 0 & 4 & 17 & 0 & 2 \\ 3 & 4 & -3 & 6 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{122} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 8 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 4 & 6 \\ 4 & 8 & 0 & 12 & 10 & 4 & 6 \\ 2 & 3 & 0 & 4 & 4 & 0 & 2 \\ 3 & 4 & -3 & 6 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 4 & 6 \\ 5 & 9 & 0 & 12 & 13 & 5 & 7 \\ 2 & 3 & 0 & 4 & 5 & 0 & 2 \\ 3 & 4 & -3 & 6 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 2 & 3 \\ 2 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 4 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{13} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 0 & 5 & 17 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 15 & 3 & 4 \\ 9 & 17 & 0 & 15 & 49 & 9 & 13 \\ 2 & 3 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -3 & 4 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 4 & 8 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 2 & 3 \\ 4 & 8 & 0 & 6 & 10 & 4 & 6 \\ 2 & 3 & 0 & 2 & 4 & 0 & 2 \\ 3 & 4 & -3 & 3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{13} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 3 & 5 & 0 & 6 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 2 & 3 \\ 2 & 4 & 0 & 3 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{123}3^11^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -164, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 2 & 3 \\ 3 & 4 & 0 & 8 & 22 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 30 & 4 & 6 \\ 12 & 22 & 0 & 30 & 85 & 12 & 17 \\ 2 & 3 & 0 & 4 & 12 & 0 & 2 \\ 3 & 4 & -3 & 6 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{111}3^11^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 4 & 14 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 2 & 3 \\ 8 & 14 & 0 & 9 & 37 & 8 & 11 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 4 & -3 & 3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{122}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.57 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 2 & 3 \\ 3 & 4 & 0 & 8 & 13 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 18 & 4 & 6 \\ 7 & 13 & 0 & 18 & 28 & 7 & 10 \\ 2 & 3 & 0 & 4 & 7 & 0 & 2 \\ 3 & 4 & -3 & 6 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 0 & 5 & 13 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 12 & 3 & 4 \\ 7 & 13 & 0 & 12 & 28 & 7 & 10 \\ 2 & 3 & 0 & 3 & 7 & 0 & 2 \\ 3 & 4 & -3 & 4 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -234, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 0 & 9 & 13 & 5 & 7 \\ 2 & 3 & 0 & 3 & 5 & 0 & 2 \\ 3 & 4 & -3 & 4 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -254, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.54 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 2 & 3 \\ 5 & 9 & 0 & 6 & 13 & 5 & 7 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -3 & 3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{123}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -284, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 2 & 3 \\ 3 & 4 & 0 & 8 & 17 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 24 & 4 & 6 \\ 9 & 17 & 0 & 24 & 49 & 9 & 13 \\ 2 & 3 & 0 & 4 & 9 & 0 & 2 \\ 3 & 4 & -3 & 6 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{114}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -314, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 2 & 3 \\ 3 & 4 & 0 & 9 & 21 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 33 & 5 & 7 \\ 11 & 21 & 0 & 33 & 76 & 11 & 16 \\ 2 & 3 & 0 & 5 & 11 & 0 & 2 \\ 3 & 4 & -3 & 7 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{124}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 2 & 3 \\ 3 & 4 & 0 & 8 & 21 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 30 & 4 & 6 \\ 11 & 21 & 0 & 30 & 76 & 11 & 16 \\ 2 & 3 & 0 & 4 & 11 & 0 & 2 \\ 3 & 4 & -3 & 6 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{112}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 4 & 13 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 2 & 3 \\ 7 & 13 & 0 & 9 & 28 & 7 & 10 \\ 2 & 3 & 0 & 2 & 7 & 0 & 2 \\ 3 & 4 & -3 & 3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{124}3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 2 & 3 \\ 3 & 4 & 0 & 8 & 26 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 36 & 4 & 6 \\ 14 & 26 & 0 & 36 & 118 & 14 & 20 \\ 2 & 3 & 0 & 4 & 14 & 0 & 2 \\ 3 & 4 & -3 & 6 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{111}3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 15 & 5 & 7 \\ 5 & 9 & 0 & 15 & 13 & 5 & 7 \\ 2 & 3 & 0 & 5 & 5 & 0 & 2 \\ 3 & 4 & -3 & 7 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{112}3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -474, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 2 & 3 \\ 3 & 4 & 0 & 4 & 23 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 2 & 3 \\ 13 & 23 & 0 & 15 & 100 & 13 & 18 \\ 2 & 3 & 0 & 2 & 13 & 0 & 2 \\ 3 & 4 & -3 & 3 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{113}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 0 & 4 & 17 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 2 & 3 \\ 9 & 17 & 0 & 12 & 49 & 9 & 13 \\ 2 & 3 & 0 & 2 & 9 & 0 & 2 \\ 3 & 4 & -3 & 3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{113}3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 2 & 3 \\ 3 & 4 & 0 & 9 & 17 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 27 & 5 & 7 \\ 9 & 17 & 0 & 27 & 49 & 9 & 13 \\ 2 & 3 & 0 & 5 & 9 & 0 & 2 \\ 3 & 4 & -3 & 7 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{112}3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 0 & 4 & 18 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 2 & 3 \\ 10 & 18 & 0 & 12 & 58 & 10 & 14 \\ 2 & 3 & 0 & 2 & 10 & 0 & 2 \\ 3 & 4 & -3 & 3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{114}3^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 4 & 21 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 2 & 3 \\ 11 & 21 & 0 & 15 & 76 & 11 & 16 \\ 2 & 3 & 0 & 2 & 11 & 0 & 2 \\ 3 & 4 & -3 & 3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{112}3^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 2 & 3 \\ 3 & 4 & 0 & 9 & 13 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 21 & 5 & 7 \\ 7 & 13 & 0 & 21 & 28 & 7 & 10 \\ 2 & 3 & 0 & 5 & 7 & 0 & 2 \\ 3 & 4 & -3 & 7 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{124}3^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 2 & 3 \\ 3 & 4 & 0 & 13 & 21 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 48 & 7 & 10 \\ 11 & 21 & 0 & 48 & 76 & 11 & 16 \\ 2 & 3 & 0 & 7 & 11 & 0 & 2 \\ 3 & 4 & -3 & 10 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{122}3^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 13 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 30 & 7 & 10 \\ 7 & 13 & 0 & 30 & 28 & 7 & 10 \\ 2 & 3 & 0 & 7 & 7 & 0 & 2 \\ 3 & 4 & -3 & 10 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{123}3^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -794, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 2 & 3 \\ 3 & 4 & 0 & 13 & 17 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 39 & 7 & 10 \\ 9 & 17 & 0 & 39 & 49 & 9 & 13 \\ 2 & 3 & 0 & 7 & 9 & 0 & 2 \\ 3 & 4 & -3 & 10 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{113}3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -824, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 2 & 3 \\ 3 & 4 & 0 & 4 & 22 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 2 & 3 \\ 12 & 22 & 0 & 15 & 85 & 12 & 17 \\ 2 & 3 & 0 & 2 & 12 & 0 & 2 \\ 3 & 4 & -3 & 3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{113}3^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 2 & 3 \\ 3 & 4 & 0 & 4 & 32 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 21 & 2 & 3 \\ 18 & 32 & 0 & 21 & 193 & 18 & 25 \\ 2 & 3 & 0 & 2 & 18 & 0 & 2 \\ 3 & 4 & -3 & 3 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{133}3^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 2 & 3 \\ 3 & 4 & 0 & 17 & 17 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 51 & 9 & 13 \\ 9 & 17 & 0 & 51 & 49 & 9 & 13 \\ 2 & 3 & 0 & 9 & 9 & 0 & 2 \\ 3 & 4 & -3 & 13 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{134}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -954, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 11 & 2 & 3 \\ 3 & 4 & 0 & 17 & 21 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 63 & 9 & 13 \\ 11 & 21 & 0 & 63 & 76 & 11 & 16 \\ 2 & 3 & 0 & 9 & 11 & 0 & 2 \\ 3 & 4 & -3 & 13 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{113}3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -974, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 2 & 3 \\ 3 & 4 & 0 & 4 & 27 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 18 & 2 & 3 \\ 15 & 27 & 0 & 18 & 133 & 15 & 21 \\ 2 & 3 & 0 & 2 & 15 & 0 & 2 \\ 3 & 4 & -3 & 3 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{144}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1064, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 11 & 2 & 3 \\ 3 & 4 & 0 & 21 & 21 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 78 & 11 & 16 \\ 11 & 21 & 0 & 78 & 76 & 11 & 16 \\ 2 & 3 & 0 & 11 & 11 & 0 & 2 \\ 3 & 4 & -3 & 16 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{114}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1084, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 4 & 0 & 4 & 26 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 18 & 2 & 3 \\ 14 & 26 & 0 & 18 & 118 & 14 & 20 \\ 2 & 3 & 0 & 2 & 14 & 0 & 2 \\ 3 & 4 & -3 & 3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{114}3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1434, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 2 & 3 \\ 3 & 4 & 0 & 4 & 31 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 21 & 2 & 3 \\ 17 & 31 & 0 & 21 & 172 & 17 & 24 \\ 2 & 3 & 0 & 2 & 17 & 0 & 2 \\ 3 & 4 & -3 & 3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{114}3^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1604, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 2 & 3 \\ 3 & 4 & 0 & 4 & 36 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 24 & 2 & 3 \\ 20 & 36 & 0 & 24 & 238 & 20 & 28 \\ 2 & 3 & 0 & 2 & 20 & 0 & 2 \\ 3 & 4 & -3 & 3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [38] \quad 1^{111}3^{111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 247, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 0 & 3 & 9 & 3 & 4 \\ 1 & 4 & 0 & 3 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 6 & 2 \\ 3 & 9 & 6 & 0 & 6 & 13 & 5 \\ 1 & 3 & 2 & 0 & 2 & 5 & 0 \\ 2 & 4 & 2 & -3 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{13}3^{111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 472, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 6 & 3 & 4 \\ 2 & 6 & 3 & 0 & 6 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 3 & 3 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{11}3^{11}1^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 517, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 9 & 3 & 4 \\ 3 & 9 & 6 & 0 & 9 & 13 & 5 & 7 \\ 1 & 3 & 2 & 0 & 3 & 5 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{11}3^{11}1^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 648, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 6 & 3 \\ 4 & 8 & 6 & 0 & 15 & 15 & 6 \\ 2 & 6 & 1 & 0 & 9 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 15 & 9 & 0 & 22 & 24 & 10 \\ 6 & 15 & 9 & 0 & 24 & 22 & 10 \\ 3 & 6 & 3 & -3 & 10 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{11}3^{11}1^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 912, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 15 & 5 & 7 \\ 3 & 9 & 6 & 0 & 15 & 13 & 5 & 7 \\ 1 & 3 & 2 & 0 & 5 & 5 & 0 & 2 \\ 2 & 4 & 2 & -3 & 7 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.466 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 72, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 3 & 3 & 4 \\ 3 & 5 & 0 & 6 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 3 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 4 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 392, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 2 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 2 & 2 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 462, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 9 & 3 & 3 & 4 \\ 5 & 9 & 0 & 9 & 13 & 5 & 6 & 7 \\ 2 & 3 & 0 & 3 & 5 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 6 & 2 & 0 & 2 \\ 3 & 4 & -3 & 4 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 512, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 2 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 2 & 6 & 3 \\ 2 & 3 & 0 & 2 & 2 & 0 & 5 & 2 \\ 5 & 9 & 0 & 6 & 6 & 5 & 12 & 6 \\ 3 & 4 & -3 & 3 & 3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{1111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 2 & 3 & 3 \\ 5 & 9 & 0 & 6 & 13 & 5 & 6 & 7 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 6 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{1111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 9 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 15 & 5 & 6 & 7 \\ 5 & 9 & 0 & 15 & 13 & 5 & 6 & 7 \\ 2 & 3 & 0 & 5 & 5 & 0 & 2 & 2 \\ 2 & 4 & 0 & 6 & 6 & 2 & 0 & 2 \\ 3 & 4 & -3 & 7 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.467 NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_{2n3\text{mix}}$

$T = 9$ [0] $1^{1111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 4 & 0 & 3 & 1 & 5 & 3 \\ 4 & 7 & 0 & 6 & 5 & 7 & 5 \\ 3 & 4 & -3 & 4 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{1211}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 4 & 3 \\ 3 & 4 & 0 & 8 & 9 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 10 & 6 \\ 5 & 9 & 0 & 12 & 13 & 11 & 7 \\ 4 & 7 & 0 & 10 & 11 & 7 & 5 \\ 3 & 4 & -3 & 6 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{1122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 9 & 3 \\ 3 & 4 & 0 & 4 & 8 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 11 & 3 \\ 4 & 8 & 0 & 6 & 10 & 22 & 6 \\ 9 & 16 & 0 & 11 & 22 & 44 & 12 \\ 3 & 4 & -3 & 3 & 6 & 12 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{1131}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 4 & 3 \\ 3 & 4 & 0 & 4 & 12 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 5 & 3 \\ 6 & 12 & 0 & 9 & 25 & 15 & 9 \\ 4 & 7 & 0 & 5 & 15 & 7 & 5 \\ 3 & 4 & -3 & 3 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{1311}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 0 & 5 & 17 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 15 & 6 & 4 \\ 9 & 17 & 0 & 15 & 49 & 21 & 13 \\ 4 & 7 & 0 & 6 & 21 & 7 & 5 \\ 3 & 4 & -3 & 4 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{1111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 0 & 5 & 9 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 9 & 6 & 4 \\ 5 & 9 & 0 & 9 & 13 & 11 & 7 \\ 4 & 7 & 0 & 6 & 11 & 7 & 5 \\ 3 & 4 & -3 & 4 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{1212}3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 9 & 3 \\ 3 & 4 & 0 & 8 & 9 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 22 & 6 \\ 5 & 9 & 0 & 12 & 13 & 26 & 7 \\ 9 & 16 & 0 & 22 & 26 & 44 & 12 \\ 3 & 4 & -3 & 6 & 7 & 12 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{1221}3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 10 & 6 \\ 4 & 8 & 0 & 12 & 10 & 10 & 6 \\ 4 & 7 & 0 & 10 & 10 & 7 & 5 \\ 3 & 4 & -3 & 6 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{1131}3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 7 & 3 \\ 3 & 4 & 0 & 4 & 17 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 8 & 3 \\ 9 & 17 & 0 & 12 & 49 & 36 & 13 \\ 7 & 12 & 0 & 8 & 36 & 25 & 9 \\ 3 & 4 & -3 & 3 & 13 & 9 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{1112}3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 6 & 3 \\ 3 & 4 & 0 & 9 & 9 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 15 & 17 & 7 \\ 5 & 9 & 0 & 15 & 13 & 17 & 7 \\ 6 & 11 & 0 & 17 & 17 & 20 & 8 \\ 3 & 4 & -3 & 7 & 7 & 8 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{1113}3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 0 & 5 & 4 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 18 & 4 \\ 2 & 4 & 0 & 3 & 1 & 14 & 3 \\ 11 & 20 & 0 & 18 & 14 & 69 & 15 \\ 3 & 4 & -3 & 4 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{1113}3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 12 & 4 \\ 2 & 4 & 0 & 3 & 1 & 8 & 3 \\ 7 & 12 & 0 & 12 & 8 & 25 & 9 \\ 3 & 4 & -3 & 4 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{1213}3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 0 & 5 & 13 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 12 & 6 & 4 \\ 7 & 13 & 0 & 12 & 28 & 16 & 10 \\ 4 & 7 & 0 & 6 & 16 & 7 & 5 \\ 3 & 4 & -3 & 4 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{1121}3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 5 & 3 \\ 4 & 8 & 0 & 6 & 10 & 10 & 6 \\ 4 & 7 & 0 & 5 & 10 & 7 & 5 \\ 3 & 4 & -3 & 3 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{1112}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{red}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -246, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 8 & 3 \\ 5 & 9 & 0 & 6 & 13 & 17 & 7 \\ 6 & 11 & 0 & 8 & 17 & 20 & 8 \\ 3 & 4 & -3 & 3 & 7 & 8 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{1112}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{red}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 5 & 4 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 15 & 4 \\ 2 & 4 & 0 & 3 & 1 & 11 & 3 \\ 9 & 16 & 0 & 15 & 11 & 44 & 12 \\ 3 & 4 & -3 & 4 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{1111}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{red}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 8 & 3 \\ 2 & 4 & 0 & 3 & 1 & 8 & 3 \\ 7 & 12 & 0 & 8 & 8 & 25 & 9 \\ 3 & 4 & -3 & 3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{1111}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{red}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 5 & 3 \\ 2 & 4 & 0 & 3 & 1 & 5 & 3 \\ 4 & 7 & 0 & 5 & 5 & 7 & 5 \\ 3 & 4 & -3 & 3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{1111}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{red}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 4 & 3 \\ 3 & 4 & 0 & 4 & 9 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 5 & 3 \\ 5 & 9 & 0 & 6 & 13 & 11 & 7 \\ 4 & 7 & 0 & 5 & 11 & 7 & 5 \\ 3 & 4 & -3 & 3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{1221}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{red}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -326, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 4 & 3 \\ 3 & 4 & 0 & 8 & 13 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 18 & 10 & 6 \\ 7 & 13 & 0 & 18 & 28 & 16 & 10 \\ 4 & 7 & 0 & 10 & 16 & 7 & 5 \\ 3 & 4 & -3 & 6 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{1122}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{red}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 6 & 3 \\ 3 & 4 & 0 & 4 & 8 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 8 & 3 \\ 4 & 8 & 0 & 6 & 10 & 16 & 6 \\ 6 & 11 & 0 & 8 & 16 & 20 & 8 \\ 3 & 4 & -3 & 3 & 6 & 8 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{1121}3^11^{12}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{red}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 4 & 3 \\ 3 & 4 & 0 & 4 & 18 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 5 & 3 \\ 10 & 18 & 0 & 12 & 58 & 22 & 14 \\ 4 & 7 & 0 & 5 & 22 & 7 & 5 \\ 3 & 4 & -3 & 3 & 14 & 5 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{1112}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 8 & 3 \\ 2 & 4 & 0 & 3 & 1 & 8 & 3 \\ 6 & 11 & 0 & 8 & 8 & 20 & 8 \\ 3 & 4 & -3 & 3 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{1122}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 6 & 3 \\ 3 & 4 & 0 & 9 & 13 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 21 & 17 & 7 \\ 7 & 13 & 0 & 21 & 28 & 25 & 10 \\ 6 & 11 & 0 & 17 & 25 & 20 & 8 \\ 3 & 4 & -3 & 7 & 10 & 8 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{1111}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 9 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 15 & 11 & 7 \\ 5 & 9 & 0 & 15 & 13 & 11 & 7 \\ 4 & 7 & 0 & 11 & 11 & 7 & 5 \\ 3 & 4 & -3 & 7 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{1131}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 4 & 3 \\ 3 & 4 & 0 & 4 & 27 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 18 & 5 & 3 \\ 15 & 27 & 0 & 18 & 133 & 33 & 21 \\ 4 & 7 & 0 & 5 & 33 & 7 & 5 \\ 3 & 4 & -3 & 3 & 21 & 5 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{1112}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 14 & 3 \\ 2 & 4 & 0 & 3 & 1 & 14 & 3 \\ 12 & 21 & 0 & 14 & 14 & 80 & 16 \\ 3 & 4 & -3 & 3 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{1121}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -514, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 8 & 3 \\ 7 & 13 & 0 & 9 & 28 & 28 & 10 \\ 7 & 12 & 0 & 8 & 28 & 25 & 9 \\ 3 & 4 & -3 & 3 & 10 & 9 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{1111}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 7 & 3 \\ 3 & 4 & 0 & 4 & 9 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 8 & 3 \\ 5 & 9 & 0 & 6 & 13 & 20 & 7 \\ 7 & 12 & 0 & 8 & 20 & 25 & 9 \\ 3 & 4 & -3 & 3 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{1121}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 4 & 3 \\ 3 & 4 & 0 & 4 & 13 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 5 & 3 \\ 7 & 13 & 0 & 9 & 28 & 16 & 10 \\ 4 & 7 & 0 & 5 & 16 & 7 & 5 \\ 3 & 4 & -3 & 3 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1123}3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -546, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 6 & 3 \\ 3 & 4 & 0 & 4 & 13 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 8 & 3 \\ 7 & 13 & 0 & 9 & 28 & 25 & 10 \\ 6 & 11 & 0 & 8 & 25 & 20 & 8 \\ 3 & 4 & -3 & 3 & 10 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1141}3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -546, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 4 & 3 \\ 3 & 4 & 0 & 9 & 21 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 33 & 11 & 7 \\ 11 & 21 & 0 & 33 & 76 & 26 & 16 \\ 4 & 7 & 0 & 11 & 26 & 7 & 5 \\ 3 & 4 & -3 & 7 & 16 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1113}3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 11 & 3 \\ 2 & 4 & 0 & 3 & 1 & 11 & 3 \\ 8 & 15 & 0 & 11 & 11 & 39 & 11 \\ 3 & 4 & -3 & 3 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1141}3^1 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -616, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 4 & 3 \\ 3 & 4 & 0 & 4 & 36 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 24 & 5 & 3 \\ 20 & 36 & 0 & 24 & 238 & 44 & 28 \\ 4 & 7 & 0 & 5 & 44 & 7 & 5 \\ 3 & 4 & -3 & 3 & 28 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1112}3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -620, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 4 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 11 & 3 \\ 2 & 4 & 0 & 3 & 1 & 11 & 3 \\ 9 & 16 & 0 & 11 & 11 & 44 & 12 \\ 3 & 4 & -3 & 3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1121}3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 4 & 3 \\ 3 & 4 & 0 & 9 & 13 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 21 & 11 & 7 \\ 7 & 13 & 0 & 21 & 28 & 16 & 10 \\ 4 & 7 & 0 & 11 & 16 & 7 & 5 \\ 3 & 4 & -3 & 7 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1131}3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 4 & 3 \\ 3 & 4 & 0 & 9 & 17 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 27 & 11 & 7 \\ 9 & 17 & 0 & 27 & 49 & 21 & 13 \\ 4 & 7 & 0 & 11 & 21 & 7 & 5 \\ 3 & 4 & -3 & 7 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1131}3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -716, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 4 & 3 \\ 3 & 4 & 0 & 4 & 17 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 5 & 3 \\ 9 & 17 & 0 & 12 & 49 & 21 & 13 \\ 4 & 7 & 0 & 5 & 21 & 7 & 5 \\ 3 & 4 & -3 & 3 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{1114}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -720, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 4 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 14 & 3 \\ 2 & 4 & 0 & 3 & 1 & 14 & 3 \\ 10 & 19 & 0 & 14 & 14 & 64 & 14 \\ 3 & 4 & -3 & 3 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

[39] $1^{1112}3^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -810, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 9 & 3 \\ 3 & 4 & 0 & 4 & 9 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 11 & 3 \\ 5 & 9 & 0 & 6 & 13 & 26 & 7 \\ 9 & 16 & 0 & 11 & 26 & 44 & 12 \\ 3 & 4 & -3 & 3 & 7 & 12 & 1 \end{pmatrix}_{b_0}$$

[40] $1^{1113}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -836, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 4 & 3 \\ 3 & 4 & 0 & 4 & 22 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 5 & 3 \\ 12 & 22 & 0 & 15 & 85 & 27 & 17 \\ 4 & 7 & 0 & 5 & 27 & 7 & 5 \\ 3 & 4 & -3 & 3 & 17 & 5 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{1113}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -840, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 4 & 30 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 20 & 3 \\ 2 & 4 & 0 & 3 & 1 & 20 & 3 \\ 17 & 30 & 0 & 20 & 20 & 165 & 23 \\ 3 & 4 & -3 & 3 & 3 & 23 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{1141}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -856, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 4 & 3 \\ 3 & 4 & 0 & 4 & 21 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 5 & 3 \\ 11 & 21 & 0 & 15 & 76 & 26 & 16 \\ 4 & 7 & 0 & 5 & 26 & 7 & 5 \\ 3 & 4 & -3 & 3 & 16 & 5 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{1221}3^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 4 & 3 \\ 3 & 4 & 0 & 13 & 13 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 30 & 16 & 10 \\ 7 & 13 & 0 & 30 & 28 & 16 & 10 \\ 4 & 7 & 0 & 16 & 16 & 7 & 5 \\ 3 & 4 & -3 & 10 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{1113}3^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -964, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 11 & 3 \\ 3 & 4 & 0 & 4 & 9 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 14 & 3 \\ 5 & 9 & 0 & 6 & 13 & 32 & 7 \\ 11 & 20 & 0 & 14 & 32 & 69 & 15 \\ 3 & 4 & -3 & 3 & 7 & 15 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{1113}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -984, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 4 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 14 & 3 \\ 2 & 4 & 0 & 3 & 1 & 14 & 3 \\ 11 & 20 & 0 & 14 & 14 & 69 & 15 \\ 3 & 4 & -3 & 3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{111431^1 1^{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -986, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.67 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 13 & 3 \\ 3 & 4 & 0 & 4 & 9 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 17 & 3 \\ 5 & 9 & 0 & 6 & 13 & 38 & 7 \\ 13 & 24 & 0 & 17 & 38 & 100 & 18 \\ 3 & 4 & -3 & 3 & 7 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{112231^1 1^{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1020, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 9 & 3 \\ 3 & 4 & 0 & 4 & 13 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 11 & 3 \\ 7 & 13 & 0 & 9 & 28 & 37 & 10 \\ 9 & 16 & 0 & 11 & 37 & 44 & 12 \\ 3 & 4 & -3 & 3 & 10 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{111331^1 1^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1076, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 4 & 25 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 17 & 3 \\ 2 & 4 & 0 & 3 & 1 & 17 & 3 \\ 14 & 25 & 0 & 17 & 17 & 111 & 19 \\ 3 & 4 & -3 & 3 & 3 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{111431^1 1^{13}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1176, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 4 & 3 \\ 3 & 4 & 0 & 4 & 31 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 21 & 5 & 3 \\ 17 & 31 & 0 & 21 & 172 & 38 & 24 \\ 4 & 7 & 0 & 5 & 38 & 7 & 5 \\ 3 & 4 & -3 & 3 & 24 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{111431^1 1^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.61 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 4 & 3 \\ 3 & 4 & 0 & 4 & 26 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 18 & 5 & 3 \\ 14 & 26 & 0 & 18 & 118 & 32 & 20 \\ 4 & 7 & 0 & 5 & 32 & 7 & 5 \\ 3 & 4 & -3 & 3 & 20 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{111431^1 1^{14}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1296, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 22 & 3 \\ 3 & 4 & 0 & 4 & 4 & 39 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 26 & 3 \\ 2 & 4 & 0 & 3 & 1 & 26 & 3 \\ 22 & 39 & 0 & 26 & 26 & 280 & 30 \\ 3 & 4 & -3 & 3 & 3 & 30 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{111431^1 1^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1356, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 4 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 17 & 3 \\ 2 & 4 & 0 & 3 & 1 & 17 & 3 \\ 13 & 24 & 0 & 17 & 17 & 100 & 18 \\ 3 & 4 & -3 & 3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{111431^1 1^{13}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1644, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 19 & 3 \\ 3 & 4 & 0 & 4 & 4 & 34 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 23 & 3 \\ 2 & 4 & 0 & 3 & 1 & 23 & 3 \\ 19 & 34 & 0 & 23 & 23 & 208 & 26 \\ 3 & 4 & -3 & 3 & 3 & 26 & 1 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1114}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 4 & 29 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 20 & 3 \\ 2 & 4 & 0 & 3 & 1 & 20 & 3 \\ 16 & 29 & 0 & 20 & 20 & 148 & 22 \\ 3 & 4 & -3 & 3 & 3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [55] \quad 1^{11}31^{111}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 5 \\ 1 & 4 & 0 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 \\ 2 & 6 & 3 & 0 & 4 & 6 & 5 \\ 2 & 6 & 3 & 0 & 6 & 4 & 5 \\ 2 & 5 & 4 & 0 & 5 & 5 & 4 \\ 2 & 4 & 2 & -3 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{111}31^{11}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 172, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 5 \\ 1 & 4 & 0 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 \\ 1 & 3 & 3 & 0 & 1 & 6 & 4 \\ 3 & 9 & 6 & 0 & 6 & 13 & 9 \\ 2 & 5 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 2 & -3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{111}31^{111}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 262, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 5 \\ 1 & 4 & 0 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 \\ 2 & 6 & 3 & 0 & 4 & 9 & 5 \\ 3 & 9 & 6 & 0 & 9 & 13 & 9 \\ 2 & 5 & 4 & 0 & 5 & 9 & 4 \\ 2 & 4 & 2 & -3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{111}31^{111}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 752, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 5 \\ 1 & 4 & 0 & 0 & 6 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 \\ 3 & 9 & 6 & 0 & 13 & 15 & 9 \\ 3 & 9 & 6 & 0 & 15 & 13 & 9 \\ 2 & 5 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 2 & -3 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

4.468 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -44, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.82 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 3 & 5 & 0 & 6 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -69, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 0 & 5 & 19 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 21 & 3 & 4 \\ 11 & 19 & 0 & 21 & 73 & 12 & 15 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -3 & 4 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -72, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} =$$

12.2

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 3 & 3 \\ 8 & 14 & 0 & 9 & 37 & 9 & 11 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -3 & 3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{1111}31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 2 & 3 \\ 3 & 4 & 0 & 9 & 19 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 33 & 6 & 7 \\ 11 & 19 & 0 & 33 & 73 & 12 & 15 \\ 2 & 4 & 0 & 6 & 12 & 0 & 2 \\ 3 & 4 & -3 & 7 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{1121}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 5 & 3 \\ 3 & 4 & 0 & 4 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 6 & 3 \\ 4 & 8 & 0 & 6 & 10 & 12 & 6 \\ 5 & 9 & 0 & 6 & 12 & 12 & 6 \\ 3 & 4 & -3 & 3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{1111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 9 & 6 & 12 & 6 \\ 3 & 4 & -3 & 4 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{1121}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 12 & 4 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 7 & 13 & 0 & 12 & 9 & 27 & 9 \\ 3 & 4 & -3 & 4 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{1111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{1111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 0 & 9 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & -3 & 4 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{1111}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 0 & 5 & 14 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 15 & 3 & 4 \\ 8 & 14 & 0 & 15 & 37 & 9 & 11 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -3 & 4 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{1211}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 12 & 3 & 4 \\ 7 & 13 & 0 & 12 & 28 & 9 & 10 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -3 & 4 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & -3 & 3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1121} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 2 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 3 & 3 \\ 13 & 23 & 0 & 15 & 100 & 15 & 18 \\ 2 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & -3 & 3 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1221} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 5 & 3 \\ 3 & 4 & 0 & 8 & 13 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 18 & 12 & 6 \\ 7 & 13 & 0 & 18 & 28 & 21 & 10 \\ 5 & 9 & 0 & 12 & 21 & 12 & 6 \\ 3 & 4 & -3 & 6 & 10 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & -3 & 3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 15 & 12 & 6 \\ 3 & 4 & -3 & 3 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1111} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -605, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 2 & 3 \\ 3 & 4 & 0 & 9 & 14 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 24 & 6 & 7 \\ 8 & 14 & 0 & 24 & 37 & 9 & 11 \\ 2 & 4 & 0 & 6 & 9 & 0 & 2 \\ 3 & 4 & -3 & 7 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{1211} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 0 & 5 & 18 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 18 & 3 & 4 \\ 10 & 18 & 0 & 18 & 58 & 12 & 14 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -3 & 4 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1211} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -650, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 0 & 5 & 28 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 30 & 3 & 4 \\ 16 & 28 & 0 & 30 & 154 & 18 & 22 \\ 2 & 4 & 0 & 3 & 18 & 0 & 2 \\ 3 & 4 & -3 & 4 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -704, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 15 & 6 & 7 \\ 5 & 9 & 0 & 15 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 6 & 0 & 2 \\ 3 & 4 & -3 & 7 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1111} 3^1 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -716, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.85 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 2 & 3 \\ 3 & 4 & 0 & 14 & 14 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 39 & 9 & 11 \\ 8 & 14 & 0 & 39 & 37 & 9 & 11 \\ 2 & 4 & 0 & 9 & 9 & 0 & 2 \\ 3 & 4 & -3 & 11 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -723, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 3 & 3 \\ 7 & 13 & 0 & 9 & 28 & 9 & 10 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -3 & 3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1121} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -738, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 3 & 3 \\ 10 & 18 & 0 & 12 & 58 & 12 & 14 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -3 & 3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{121} 3^1 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -744, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 0 & 5 & 23 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 24 & 3 & 4 \\ 13 & 23 & 0 & 24 & 100 & 15 & 18 \\ 2 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & -3 & 4 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{131} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -779, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 0 & 5 & 22 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 21 & 3 & 4 \\ 12 & 22 & 0 & 21 & 85 & 15 & 17 \\ 2 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & -3 & 4 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1131} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -828, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 2 & 3 \\ 3 & 4 & 0 & 4 & 32 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 21 & 3 & 3 \\ 18 & 32 & 0 & 21 & 193 & 21 & 25 \\ 2 & 4 & 0 & 3 & 21 & 0 & 2 \\ 3 & 4 & -3 & 3 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1121} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -845, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 5 & 3 \\ 3 & 4 & 0 & 4 & 13 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 6 & 3 \\ 7 & 13 & 0 & 9 & 28 & 21 & 10 \\ 5 & 9 & 0 & 6 & 21 & 12 & 6 \\ 3 & 4 & -3 & 3 & 10 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1121} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -941, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 2 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 21 & 6 & 7 \\ 7 & 13 & 0 & 21 & 28 & 9 & 10 \\ 2 & 4 & 0 & 6 & 9 & 0 & 2 \\ 3 & 4 & -3 & 7 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1121} 31^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -962, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 10 & 2 & 3 \\ 3 & 4 & 0 & 14 & 18 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 48 & 9 & 11 \\ 10 & 18 & 0 & 48 & 58 & 12 & 14 \\ 2 & 4 & 0 & 9 & 12 & 0 & 2 \\ 3 & 4 & -3 & 11 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1131} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -980, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 2 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 27 & 6 & 7 \\ 9 & 17 & 0 & 27 & 49 & 12 & 13 \\ 2 & 4 & 0 & 6 & 12 & 0 & 2 \\ 3 & 4 & -3 & 7 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1121} 31^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -980, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 16 & 2 & 3 \\ 3 & 4 & 0 & 9 & 28 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 48 & 6 & 7 \\ 16 & 28 & 0 & 48 & 154 & 18 & 22 \\ 2 & 4 & 0 & 6 & 18 & 0 & 2 \\ 3 & 4 & -3 & 7 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1131} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -989, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 5 & 3 \\ 3 & 4 & 0 & 4 & 17 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 6 & 3 \\ 9 & 17 & 0 & 12 & 49 & 27 & 13 \\ 5 & 9 & 0 & 6 & 27 & 12 & 6 \\ 3 & 4 & -3 & 3 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1141} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1005, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 5 & 3 \\ 3 & 4 & 0 & 4 & 21 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 6 & 3 \\ 11 & 21 & 0 & 15 & 76 & 33 & 16 \\ 5 & 9 & 0 & 6 & 33 & 12 & 6 \\ 3 & 4 & -3 & 3 & 16 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1131} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1077, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 3 & 3 \\ 9 & 17 & 0 & 12 & 49 & 12 & 13 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -3 & 3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{131} 31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1221, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 0 & 5 & 27 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 27 & 3 & 4 \\ 15 & 27 & 0 & 27 & 133 & 18 & 21 \\ 2 & 4 & 0 & 3 & 18 & 0 & 2 \\ 3 & 4 & -3 & 4 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1221} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1298, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 2 & 3 \\ 3 & 4 & 0 & 13 & 18 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 42 & 9 & 10 \\ 10 & 18 & 0 & 42 & 58 & 12 & 14 \\ 2 & 4 & 0 & 9 & 12 & 0 & 2 \\ 3 & 4 & -3 & 10 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1131} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1333, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 2 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 18 & 3 & 3 \\ 15 & 27 & 0 & 18 & 133 & 18 & 21 \\ 2 & 4 & 0 & 3 & 18 & 0 & 2 \\ 3 & 4 & -3 & 3 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1121} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 2 & 3 \\ 3 & 4 & 0 & 9 & 18 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 30 & 6 & 7 \\ 10 & 18 & 0 & 30 & 58 & 12 & 14 \\ 2 & 4 & 0 & 6 & 12 & 0 & 2 \\ 3 & 4 & -3 & 7 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1221} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1388, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 30 & 9 & 10 \\ 7 & 13 & 0 & 30 & 28 & 9 & 10 \\ 2 & 4 & 0 & 9 & 9 & 0 & 2 \\ 3 & 4 & -3 & 10 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1121} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1389, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 2 & 3 \\ 3 & 4 & 0 & 9 & 23 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 39 & 6 & 7 \\ 13 & 23 & 0 & 39 & 100 & 15 & 18 \\ 2 & 4 & 0 & 6 & 15 & 0 & 2 \\ 3 & 4 & -3 & 7 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1131} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 2 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 3 & 3 \\ 12 & 22 & 0 & 15 & 85 & 15 & 17 \\ 2 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & -3 & 3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{131} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1451, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 0 & 5 & 32 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 33 & 3 & 4 \\ 18 & 32 & 0 & 33 & 193 & 21 & 25 \\ 2 & 4 & 0 & 3 & 21 & 0 & 2 \\ 3 & 4 & -3 & 4 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{141} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1500, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 0 & 5 & 31 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 30 & 3 & 4 \\ 17 & 31 & 0 & 30 & 172 & 21 & 24 \\ 2 & 4 & 0 & 3 & 21 & 0 & 2 \\ 3 & 4 & -3 & 4 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1131} 31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1949, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 2 & 3 \\ 3 & 4 & 0 & 9 & 22 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 36 & 6 & 7 \\ 12 & 22 & 0 & 36 & 85 & 15 & 17 \\ 2 & 4 & 0 & 6 & 15 & 0 & 2 \\ 3 & 4 & -3 & 7 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1221} 31^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2048, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 10 & 2 & 3 \\ 3 & 4 & 0 & 18 & 18 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 60 & 12 & 14 \\ 10 & 18 & 0 & 60 & 58 & 12 & 14 \\ 2 & 4 & 0 & 12 & 12 & 0 & 2 \\ 3 & 4 & -3 & 14 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{141} 31^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2054, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 20 & 2 & 3 \\ 3 & 4 & 0 & 5 & 36 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 36 & 3 & 4 \\ 20 & 36 & 0 & 36 & 238 & 24 & 28 \\ 2 & 4 & 0 & 3 & 24 & 0 & 2 \\ 3 & 4 & -3 & 4 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1141} 31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2106, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 18 & 3 & 3 \\ 14 & 26 & 0 & 18 & 118 & 18 & 20 \\ 2 & 4 & 0 & 3 & 18 & 0 & 2 \\ 3 & 4 & -3 & 3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1141} 31^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 2 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 24 & 3 & 3 \\ 20 & 36 & 0 & 24 & 238 & 24 & 28 \\ 2 & 4 & 0 & 3 & 24 & 0 & 2 \\ 3 & 4 & -3 & 3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1141} 31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2324, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 2 & 3 \\ 3 & 4 & 0 & 9 & 26 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 42 & 6 & 7 \\ 14 & 26 & 0 & 42 & 118 & 18 & 20 \\ 2 & 4 & 0 & 6 & 18 & 0 & 2 \\ 3 & 4 & -3 & 7 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1141} 31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2347, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 2 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 21 & 3 & 3 \\ 17 & 31 & 0 & 21 & 172 & 21 & 24 \\ 2 & 4 & 0 & 3 & 21 & 0 & 2 \\ 3 & 4 & -3 & 3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1131} 31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2496, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 2 & 3 \\ 3 & 4 & 0 & 9 & 27 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 45 & 6 & 7 \\ 15 & 27 & 0 & 45 & 133 & 18 & 21 \\ 2 & 4 & 0 & 6 & 18 & 0 & 2 \\ 3 & 4 & -3 & 7 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1131^1} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2621, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 18 & 2 & 3 \\ 3 & 4 & 0 & 9 & 32 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 54 & 6 & 7 \\ 18 & 32 & 0 & 54 & 193 & 21 & 25 \\ 2 & 4 & 0 & 6 & 21 & 0 & 2 \\ 3 & 4 & -3 & 7 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1141^1} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -3405, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 17 & 2 & 3 \\ 3 & 4 & 0 & 9 & 31 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 51 & 6 & 7 \\ 17 & 31 & 0 & 51 & 172 & 21 & 24 \\ 2 & 4 & 0 & 6 & 21 & 0 & 2 \\ 3 & 4 & -3 & 7 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1141^1} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -4064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 20 & 2 & 3 \\ 3 & 4 & 0 & 9 & 36 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 60 & 6 & 7 \\ 20 & 36 & 0 & 60 & 238 & 24 & 28 \\ 2 & 4 & 0 & 6 & 24 & 0 & 2 \\ 3 & 4 & -3 & 7 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [54] \quad 1^{11^1} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 32, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 1 & 5 \\ 9 & 13 & 1 & 0 & 0 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 7 & -1 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{11^1} 321^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 5 & 0 & 5 & 10 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 7 & 4 & 4 \\ 5 & 10 & 0 & 7 & 14 & 7 & 8 \\ 2 & 5 & 0 & 4 & 7 & 1 & 3 \\ 3 & 5 & -3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{11^1} 321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 0 & 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 2 & 3 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 3 & 4 & 4 \\ 2 & 2 & 2 & -3 & 2 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{11^1} 321^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 160, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 5 & 0 & 5 & 5 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 4 & 15 & 4 \\ 2 & 5 & 0 & 4 & 2 & 15 & 4 \\ 9 & 20 & 0 & 15 & 15 & 57 & 15 \\ 3 & 5 & -3 & 4 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{11^1} 321^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 172, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 & 3 & 2 \\ 1 & 2 & 0 & 0 & 3 & 4 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 4 & 4 & 3 \\ 2 & 3 & 4 & 0 & 4 & 5 & 7 & 5 \\ 2 & 3 & 4 & 0 & 4 & 7 & 4 & 4 \\ 2 & 2 & 2 & -3 & 3 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{111} 3^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 172, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 20 & 5 \\ 9 & 13 & 1 & 0 & 30 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 20 & 30 & 0 & 0 & 63 & 15 \\ 5 & 7 & -1 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{11} 3^{21} 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 220, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 11 & 4 & 4 \\ 7 & 15 & 0 & 11 & 32 & 11 & 12 \\ 2 & 5 & 0 & 4 & 11 & 1 & 3 \\ 3 & 5 & -3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{111} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 262, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 3 & 2 \\ 1 & 2 & 0 & 0 & 1 & 4 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 3 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 5 & 7 & 5 \\ 2 & 3 & 4 & 0 & 3 & 7 & 4 & 4 \\ 2 & 2 & 2 & -3 & 2 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{12} 3^{21} 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 340, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 2 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 18 & 4 & 4 \\ 12 & 25 & 0 & 18 & 94 & 18 & 20 \\ 2 & 5 & 0 & 4 & 18 & 1 & 3 \\ 3 & 5 & -3 & 4 & 20 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{111} 3^{21} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 360, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 2 & 3 \\ 3 & 5 & 0 & 10 & 10 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 10 & 0 & 14 & 16 & 7 & 8 \\ 5 & 10 & 0 & 16 & 14 & 7 & 8 \\ 2 & 5 & 0 & 7 & 7 & 1 & 3 \\ 3 & 5 & -3 & 8 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{13} 3^{21} 1^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 380, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 2 & 3 \\ 3 & 5 & 0 & 5 & 35 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 25 & 4 & 4 \\ 17 & 35 & 0 & 25 & 188 & 25 & 28 \\ 2 & 5 & 0 & 4 & 25 & 1 & 3 \\ 3 & 5 & -3 & 4 & 28 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{111} 3^{21} 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 390, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 2 & 3 \\ 3 & 5 & 0 & 10 & 15 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 10 & 0 & 14 & 23 & 7 & 8 \\ 7 & 15 & 0 & 23 & 32 & 11 & 12 \\ 2 & 5 & 0 & 7 & 11 & 1 & 3 \\ 3 & 5 & -3 & 8 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{111} 3^{21} 1^{231} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 630, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 2 & 3 \\ 3 & 5 & 0 & 15 & 20 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 15 & 0 & 32 & 45 & 11 & 12 \\ 9 & 20 & 0 & 45 & 58 & 15 & 16 \\ 2 & 5 & 0 & 11 & 15 & 1 & 3 \\ 3 & 5 & -3 & 12 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{12} 3^{11} 21^{1141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 640, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 5 & 0 & 5 & 30 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 22 & 4 & 4 \\ 14 & 30 & 0 & 22 & 134 & 22 & 24 \\ 2 & 5 & 0 & 4 & 22 & 1 & 3 \\ 3 & 5 & -3 & 4 & 24 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{111} 3^{11} 21^{1221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 720, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 2 & 3 \\ 3 & 5 & 0 & 15 & 15 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 15 & 0 & 32 & 34 & 11 & 12 \\ 7 & 15 & 0 & 34 & 32 & 11 & 12 \\ 2 & 5 & 0 & 11 & 11 & 1 & 3 \\ 3 & 5 & -3 & 12 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{111} 3^{11} 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 752, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 7 & 5 \\ 2 & 3 & 4 & 0 & 7 & 5 & 5 \\ 2 & 3 & 4 & 0 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{111} 3^{11} 21^{1331} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 840, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.00 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 2 & 3 \\ 3 & 5 & 0 & 20 & 20 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 20 & 0 & 58 & 60 & 15 & 16 \\ 9 & 20 & 0 & 60 & 58 & 15 & 16 \\ 2 & 5 & 0 & 15 & 15 & 1 & 3 \\ 3 & 5 & -3 & 16 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

$$4.469 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 3 & 3 & 4 \\ 3 & 5 & 0 & 6 & 4 & 3 & 3 & 4 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 0 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 4 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 392, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 3 & 2 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 2 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 2 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 462, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 9 & 3 & 3 & 4 \\ 5 & 9 & 0 & 9 & 13 & 6 & 5 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 2 & 2 \\ 2 & 3 & 0 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & -3 & 4 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 + \mathfrak{s}u_3 + \mathfrak{s}u_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 512, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 6 & 2 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 2 & 3 \\ 5 & 9 & 0 & 6 & 6 & 12 & 5 & 6 \\ 2 & 3 & 0 & 2 & 2 & 5 & 0 & 2 \\ 3 & 4 & -3 & 3 & 3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 + \mathfrak{s}u_3 + \mathfrak{s}u_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 622, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 3 & 2 & 3 \\ 5 & 9 & 0 & 6 & 13 & 6 & 5 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 5 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 + \mathfrak{s}u_3 + \mathfrak{s}u_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 1152, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 15 & 6 & 5 & 7 \\ 5 & 9 & 0 & 15 & 13 & 6 & 5 & 7 \\ 2 & 4 & 0 & 6 & 6 & 0 & 2 & 2 \\ 2 & 3 & 0 & 5 & 5 & 2 & 0 & 2 \\ 3 & 4 & -3 & 7 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.470 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{s}u_2$ + $\mathfrak{s}u_2 \rightarrow (-3)$

$$T = 8 \quad [0] \ 1^{\overline{13}} 3^{\overline{11}} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 2, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 15, \ T_{\max} = 20, \ T \in [15, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 6 & 9 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & -3 & 9 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{13}} 3^{\overline{11}} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 12, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 13.33 \rightarrow 14, \ T_{\max} = 20, \ T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 10 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 10 & 0 & 22 & 6 & 8 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & -3 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 21^{\overline{13}} 3^{\overline{11}} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 15, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 2 & 3 \\ 4 & 9 & 4 & 8 & 4 & 5 \\ 3 & 4 & -2 & 3 & 2 & 0 \\ 3 & 8 & 3 & 4 & 3 & 4 \\ 2 & 4 & 2 & 3 & 0 & 2 \\ 3 & 5 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 15, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 2 & 3 \\ 4 & 9 & 4 & 8 & 4 & 5 \\ 3 & 4 & -2 & 3 & 2 & 0 \\ 3 & 8 & 3 & 4 & 3 & 4 \\ 2 & 4 & 2 & 3 & 0 & 2 \\ 3 & 5 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 18, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 15, \ T_{\max} = 20, \ T \in [15, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 2 & 3 \\ 4 & 9 & 4 & 8 & 4 & 5 \\ 2 & 4 & -2 & 3 & 2 & 0 \\ 5 & 8 & 3 & 4 & 3 & 4 \\ 2 & 4 & 2 & 3 & 0 & 2 \\ 3 & 5 & 0 & 4 & 2 & -2 \end{pmatrix}_{b_0}$$

- [5] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & 0 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$
- [6] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 6 \\ 2 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$
- [8] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$
- [9] $13^{11}1^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 2 & 3 \\ 4 & 9 & 4 & 8 & 4 & 5 \\ 2 & 4 & -2 & 3 & 2 & 0 \\ 3 & 8 & 3 & 4 & 3 & 4 \\ 2 & 4 & 2 & 3 & 0 & 2 \\ 3 & 5 & 0 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 2 & 3 \\ 4 & 9 & 4 & 8 & 4 & 5 \\ 2 & 4 & -2 & 3 & 2 & 0 \\ 4 & 8 & 3 & 4 & 3 & 4 \\ 2 & 4 & 2 & 3 & 0 & 2 \\ 3 & 5 & 0 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 5 & 7 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{11}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.14 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 24 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 24 & 0 & 121 & 14 & 19 \\ 2 & 3 & 0 & 14 & 0 & 2 \\ 3 & 4 & -3 & 19 & 2 & 0 \end{pmatrix}_{b_0}$$
- [14] $1^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 14 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 8 & 11 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$
- [13] $1^{11}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} =$

10.2

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 19 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 19 & 0 & 73 & 11 & 15 \\ 2 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & -3 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 7 & 10 \\ 2 & 3 & 0 & 7 & 0 & 2 \\ 3 & 4 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 282, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 17 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 9 & 13 \\ 2 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{14} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 11 & 16 \\ 2 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & -3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 18 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 10 & 14 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 3 & 4 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 21^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 3 & 4 \\ 8 & 9 & 4 & 8 & 4 & 5 \\ 3 & 4 & -2 & 3 & 2 & 0 \\ 6 & 8 & 3 & 4 & 3 & 4 \\ 3 & 4 & 2 & 3 & 0 & 2 \\ 4 & 5 & 0 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{12} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 23 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 23 & 0 & 100 & 13 & 18 \\ 2 & 3 & 0 & 13 & 0 & 2 \\ 3 & 4 & -3 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{13} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 522, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 22 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 0 & 85 & 12 & 17 \\ 2 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{12} 3^{11} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 532, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 28 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 16 & 28 & 0 & 154 & 16 & 22 \\ 2 & 3 & 0 & 16 & 0 & 2 \\ 3 & 4 & -3 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{14}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 652, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 26 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 0 & 118 & 14 & 20 \\ 2 & 3 & 0 & 14 & 0 & 2 \\ 3 & 4 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{13}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 722, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 27 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 0 & 133 & 15 & 21 \\ 2 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & -3 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{13}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 882, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 0 & 32 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 18 & 32 & 0 & 193 & 18 & 25 \\ 2 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & -3 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{14}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 952, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 31 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 31 & 0 & 172 & 17 & 24 \\ 2 & 3 & 0 & 17 & 0 & 2 \\ 3 & 4 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{14}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 0 & 36 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 20 & 36 & 0 & 238 & 20 & 28 \\ 2 & 3 & 0 & 20 & 0 & 2 \\ 3 & 4 & -3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [28] \quad 1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 11 & 5 \\ 6 & 5 & 1 & 10 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 10 & 0 & 18 & 8 \\ 5 & 5 & -1 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{11}31^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 8 & 6 & 0 & 5 & 6 \\ 2 & 6 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 2 & 4 \\ 3 & 6 & 3 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{12}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 12 & 5 \\ 6 & 5 & 1 & 14 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 14 & 0 & 26 & 10 \\ 5 & 5 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 16 & 5 \\ 6 & 5 & 1 & 17 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 17 & 0 & 45 & 13 \\ 5 & 5 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{11}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 1 & 5 \\ 6 & 5 & 1 & 0 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 5 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{11}3^11^131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 2 \\ 1 & 2 & 3 & 0 & 2 & 0 \\ 2 & 3 & 3 & -3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{11}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 6 & 5 \\ 6 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 7 & 0 & 5 & 5 \\ 5 & 5 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{12}31^{11}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 8 & 6 & 0 & 20 & 6 \\ 2 & 6 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 13 & 0 & 44 & 14 \\ 3 & 6 & 3 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{11}3^11^131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 \\ 2 & 4 & 2 & -3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{11}3^11^131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 0 & 6 & 2 & 3 \\ 1 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 6 & 6 & 0 & 13 & 5 & 7 \\ 1 & 2 & 2 & 0 & 5 & 0 & 2 \\ 2 & 3 & 2 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{13}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 38 & 5 \\ 6 & 5 & 1 & 41 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 38 & 41 & 0 & 261 & 31 \\ 5 & 5 & -1 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{11}3^11^131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 2 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 2 & 3 \\ 1 & 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 2 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{12}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 22 & 5 \\ 6 & 5 & 1 & 24 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 24 & 0 & 86 & 18 \\ 5 & 5 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{11}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -119, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.71 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 7 & 5 \\ 6 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 7 & 0 & 5 & 5 \\ 5 & 5 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{13}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -129, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 28 & 5 \\ 6 & 5 & 1 & 31 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 31 & 0 & 141 & 23 \\ 5 & 5 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{11}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -145, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.84 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 28 & 5 \\ 6 & 5 & 1 & 27 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 27 & 0 & 125 & 21 \\ 5 & 5 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{11}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -145, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.84 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 8 & 5 \\ 6 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 7 & 0 & 5 & 5 \\ 5 & 5 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{14}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -146, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.75 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 34 & 5 \\ 6 & 5 & 1 & 38 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 38 & 0 & 210 & 28 \\ 5 & 5 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{11}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -159, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.68 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 17 & 5 \\ 6 & 5 & 1 & 17 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 17 & 0 & 45 & 13 \\ 5 & 5 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{14}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.73 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 44 & 5 \\ 6 & 5 & 1 & 48 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 44 & 48 & 0 & 350 & 36 \\ 5 & 5 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{12}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.73 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 13 & 5 \\ 6 & 5 & 1 & 14 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 14 & 0 & 26 & 10 \\ 5 & 5 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{13}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.04 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 8 & 6 & 0 & 10 & 6 \\ 2 & 6 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 10 & 5 & 0 & 8 & 6 \\ 3 & 6 & 3 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{11} 1^3 3^1 2^1 1^3 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -195, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.78 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 49 & 5 \\ 6 & 5 & 1 & 51 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 49 & 51 & 0 & 421 & 39 \\ 5 & 5 & -1 & 39 & 3 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{11} 1^3 3^1 2^1 1^3 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -195, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.78 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 19 & 5 \\ 6 & 5 & 1 & 21 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 21 & 0 & 61 & 15 \\ 5 & 5 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{13} 1^1 1^1 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -201, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{14} 3^1 2^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -206, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.84 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 25 & 5 \\ 6 & 5 & 1 & 28 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 28 & 0 & 110 & 20 \\ 5 & 5 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{12} 3^1 2^1 1^3 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -206, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.84 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 44 & 5 \\ 6 & 5 & 1 & 44 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 44 & 44 & 0 & 326 & 34 \\ 5 & 5 & -1 & 34 & 3 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{12} 3^1 2^1 1^2 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -206, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.71 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 33 & 5 \\ 6 & 5 & 1 & 34 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 33 & 34 & 0 & 186 & 26 \\ 5 & 5 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{12} 3^1 1^1 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -221, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 9 & 0 & 28 & 7 & 10 \\ 1 & 3 & 2 & 0 & 7 & 0 & 2 \\ 2 & 4 & 2 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{11} 3^1 1^1 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -234, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 8 & 6 & 0 & 15 & 6 \\ 2 & 6 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 15 & 9 & 0 & 22 & 10 \\ 3 & 6 & 3 & -3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{11} 3^1 2^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -245, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.74 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 18 & 5 \\ 6 & 5 & 1 & 17 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 17 & 0 & 45 & 13 \\ 5 & 5 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{12}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -266, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.79 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 14 & 5 \\ 6 & 5 & 1 & 14 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 14 & 0 & 26 & 10 \\ 5 & 5 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{13}31^{12}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -274, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.03 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 8 & 6 & 0 & 35 & 6 \\ 2 & 6 & 1 & 0 & 22 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 35 & 22 & 0 & 134 & 24 \\ 3 & 6 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{12}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -286, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.68 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 23 & 5 \\ 6 & 5 & 1 & 24 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 24 & 0 & 86 & 18 \\ 5 & 5 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{13}1^{12}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -304, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.11 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 8 & 6 & 0 & 20 & 6 \\ 2 & 6 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 10 & 0 & 38 & 12 \\ 3 & 6 & 3 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{13}3^11^{12}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -306, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.03 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 6 & 0 & 22 & 6 & 8 \\ 1 & 3 & 2 & 0 & 6 & 0 & 2 \\ 2 & 4 & 2 & -3 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{11}3^11^{11}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -311, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 5 & 7 \\ 1 & 3 & 2 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{12}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -369, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.79 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 20 & 5 \\ 6 & 5 & 1 & 21 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 21 & 0 & 61 & 15 \\ 5 & 5 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{14}3^11^{12}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -386, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 0 & 24 & 3 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 24 & 18 & 0 & 118 & 14 & 20 \\ 1 & 3 & 2 & 0 & 14 & 0 & 2 \\ 2 & 4 & 2 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{11}3^121^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -395, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.69 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 39 & 5 \\ 6 & 5 & 1 & 41 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 41 & 0 & 261 & 31 \\ 5 & 5 & -1 & 31 & 3 \end{pmatrix}_{b_0}$$

[67] $1^{13}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -395, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$

$T_{\min} = 11.69 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 29 & 5 \\ 6 & 5 & 1 & 31 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 31 & 0 & 141 & 23 \\ 5 & 5 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

[69] $1^{12}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -426, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$

$T_{\min} = 11.72 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 34 & 5 \\ 6 & 5 & 1 & 34 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 34 & 0 & 186 & 26 \\ 5 & 5 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

[70] $1^{12}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -446, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$

$T_{\min} = 11.71 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 24 & 5 \\ 6 & 5 & 1 & 24 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 24 & 0 & 86 & 18 \\ 5 & 5 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[71] $1^{14}3^121^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -446, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$

$T_{\min} = 11.71 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 55 & 5 \\ 6 & 5 & 1 & 58 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 55 & 58 & 0 & 530 & 44 \\ 5 & 5 & -1 & 44 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{14}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -454, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$

$T_{\min} = 11.80 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 26 & 5 \\ 6 & 5 & 1 & 28 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 28 & 0 & 110 & 20 \\ 5 & 5 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[73] $1^{13}3^11^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -481, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$

$T_{\min} = 12.07 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 18 & 3 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 18 & 9 & 0 & 52 & 9 & 12 \\ 1 & 3 & 2 & 0 & 9 & 0 & 2 \\ 2 & 4 & 2 & -3 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{14}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -486, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$

$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 35 & 5 \\ 6 & 5 & 1 & 38 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 35 & 38 & 0 & 210 & 28 \\ 5 & 5 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[76] $1^{13}3^11^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -504, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$

$T_{\min} = 12.14 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 8 & 6 & 0 & 30 & 6 \\ 2 & 6 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 30 & 15 & 0 & 88 & 18 \\ 3 & 6 & 3 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{12}31^{12}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -504, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$

$T_{\min} = 12, \quad T_{\max} = 13, \quad T \in [12, 13]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 8 & 6 & 0 & 30 & 6 \\ 2 & 6 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 30 & 18 & 0 & 94 & 20 \\ 3 & 6 & 3 & -3 & 20 & 2 \end{pmatrix}_{b_0} \\
[77] \quad & 1^{11}3^{11}3^{12}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -514, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7 \\
& T_{\min} = 12.02 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 8 & 6 & 0 & 25 & 6 \\ 2 & 6 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 25 & 14 & 0 & 62 & 16 \\ 3 & 6 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0} \\
[78] \quad & 1^{13}3^{11}21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -549, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.72 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 50 & 5 \\ 6 & 5 & 1 & 51 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 50 & 51 & 0 & 421 & 39 \\ 5 & 5 & -1 & 39 & 3 \end{pmatrix}_{b_0} \\
[79] \quad & 1^{14}3^{11}21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -566, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.69 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 45 & 5 \\ 6 & 5 & 1 & 48 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 45 & 48 & 0 & 350 & 36 \\ 5 & 5 & -1 & 36 & 3 \end{pmatrix}_{b_0} \\
[80] \quad & 1^{11}3^{11}1^{12}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -626, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7 \\
& T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 15 & 9 & 0 & 37 & 8 & 11 \\ 1 & 3 & 2 & 0 & 8 & 0 & 2 \\ 2 & 4 & 2 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0} \\
[81] \quad & 1^{13}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -629, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.71 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 30 & 5 \\ 6 & 5 & 1 & 31 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 31 & 0 & 141 & 23 \\ 5 & 5 & -1 & 23 & 3 \end{pmatrix}_{b_0} \\
[82] \quad & 1^{14}3^{11}1^{12}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0 \\
& T_{\min} = 12.01 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 8 & 6 & 0 & 50 & 6 \\ 2 & 6 & 1 & 0 & 31 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 50 & 31 & 0 & 272 & 34 \\ 3 & 6 & 3 & -3 & 34 & 1 \end{pmatrix}_{b_0} \\
[83] \quad & 1^{13}3^{11}1^{12}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -666, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7 \\
& T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 21 & 3 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 21 & 15 & 0 & 85 & 12 & 17 \\ 1 & 3 & 2 & 0 & 12 & 0 & 2 \\ 2 & 4 & 2 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0} \\
[84] \quad & 1^{13}3^{11}21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -689, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 40 & 5 \\ 6 & 5 & 1 & 41 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 40 & 41 & 0 & 261 & 31 \\ 5 & 5 & -1 & 31 & 3 \end{pmatrix}_{b_0} \\
[85] \quad & 1^{13}1^{14}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -726, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2 \\
& T_{\min} = 12.09 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 0 & 24 & 3 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 24 & 12 & 0 & 94 & 12 & 16 \\ 1 & 3 & 2 & 0 & 12 & 0 & 2 \\ 2 & 4 & 2 & -3 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{12}3^{11}1^{12}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -746, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 18 & 3 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 18 & 12 & 0 & 58 & 10 & 14 \\ 1 & 3 & 2 & 0 & 10 & 0 & 2 \\ 2 & 4 & 2 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{13}31^{14}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -784, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.16 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 8 & 6 & 0 & 40 & 6 \\ 2 & 6 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 40 & 20 & 0 & 158 & 24 \\ 3 & 6 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{14}3^{11}21^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -794, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.71 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 36 & 5 \\ 6 & 5 & 1 & 38 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 36 & 38 & 0 & 210 & 28 \\ 5 & 5 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{14}3^{11}21^{13}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -874, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 56 & 5 \\ 6 & 5 & 1 & 58 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 56 & 58 & 0 & 530 & 44 \\ 5 & 5 & -1 & 44 & 3 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{11}31^{13}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -874, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.04 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 8 & 6 & 0 & 35 & 6 \\ 2 & 6 & 1 & 0 & 19 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 35 & 19 & 0 & 122 & 22 \\ 3 & 6 & 3 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{14}3^{11}21^{12}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -934, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 46 & 5 \\ 6 & 5 & 1 & 48 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 46 & 48 & 0 & 350 & 36 \\ 5 & 5 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{13}31^{13}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -954, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 3 \\ 4 & 8 & 6 & 0 & 45 & 6 \\ 2 & 6 & 1 & 0 & 27 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 45 & 27 & 0 & 214 & 30 \\ 3 & 6 & 3 & -3 & 30 & 2 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{11}31^{13}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -1011, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 21 & 3 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 21 & 12 & 0 & 73 & 11 & 15 \\ 1 & 3 & 2 & 0 & 11 & 0 & 2 \\ 2 & 4 & 2 & -3 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{12}31^{13}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -1024, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.01 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 8 & 6 & 0 & 40 & 6 \\ 2 & 6 & 1 & 0 & 23 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 40 & 23 & 0 & 164 & 26 \\ 3 & 6 & 3 & -3 & 26 & 1 \end{pmatrix}_{b_0}$$

$$[95] \ 1^1 3^1 1^{15} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1041, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.3$$

$$T_{\min} = 12.11 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 0 & 30 & 3 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 30 & 15 & 0 & 148 & 15 & 20 \\ 1 & 3 & 2 & 0 & 15 & 0 & 2 \\ 2 & 4 & 2 & -3 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[96] \ 1^1 3^1 1^{15} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.3$$

$$T_{\min} = 12.17 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 8 & 6 & 0 & 50 & 6 \\ 2 & 6 & 1 & 0 & 25 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 50 & 25 & 0 & 248 & 30 \\ 3 & 6 & 3 & -3 & 30 & 1 \end{pmatrix}_{b_0}$$

$$[97] \ 1^1 3^1 1^{14} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1314, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.05 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 3 \\ 4 & 8 & 6 & 0 & 45 & 6 \\ 2 & 6 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 45 & 24 & 0 & 202 & 28 \\ 3 & 6 & 3 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[98] \ 1^{12} 3^1 1^{13} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1341, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 0 & 24 & 3 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 24 & 15 & 0 & 100 & 13 & 18 \\ 1 & 3 & 2 & 0 & 13 & 0 & 2 \\ 2 & 4 & 2 & -3 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[99] \ 1^{14} 3^1 1^{13} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 0 & 30 & 3 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 30 & 21 & 0 & 172 & 17 & 24 \\ 1 & 3 & 2 & 0 & 17 & 0 & 2 \\ 2 & 4 & 2 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[100] \ 1^{11} 3^1 1^{14} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1466, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.00 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 0 & 27 & 3 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 27 & 15 & 0 & 121 & 14 & 19 \\ 1 & 3 & 2 & 0 & 14 & 0 & 2 \\ 2 & 4 & 2 & -3 & 19 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[101] \ 1^{13} 3^1 1^{13} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1471, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 0 & 27 & 3 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 27 & 18 & 0 & 133 & 15 & 21 \\ 1 & 3 & 2 & 0 & 15 & 0 & 2 \\ 2 & 4 & 2 & -3 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[102] \ 1^{14} 3^1 1^{14} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1584, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 24 & 3 \\ 4 & 8 & 6 & 0 & 60 & 6 \\ 2 & 6 & 1 & 0 & 36 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 24 & 60 & 36 & 0 & 382 & 40 \\ 3 & 6 & 3 & -3 & 40 & 2 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{12} 3^1 1^{14} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1624, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.02 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 8 & 6 & 0 & 50 & 6 \\ 2 & 6 & 1 & 0 & 28 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 50 & 28 & 0 & 254 & 32 \\ 3 & 6 & 3 & -3 & 32 & 1 \end{pmatrix}_{b_0} \\
[104] \quad & 1^{13}31^{14}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1714, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 12.00 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 22 & 3 \\ 4 & 8 & 6 & 0 & 55 & 6 \\ 2 & 6 & 1 & 0 & 32 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 22 & 55 & 32 & 0 & 314 & 36 \\ 3 & 6 & 3 & -3 & 36 & 1 \end{pmatrix}_{b_0} \\
[105] \quad & 1^{12}3^{11}1^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2006, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 0 & 30 & 3 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 30 & 18 & 0 & 154 & 16 & 22 \\ 1 & 3 & 2 & 0 & 16 & 0 & 2 \\ 2 & 4 & 2 & -3 & 22 & 2 & 1 \end{pmatrix}_{b_0} \\
[106] \quad & 1^{13}3^{11}1^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 1 & 2 \\ 2 & 4 & 4 & 0 & 33 & 3 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 33 & 21 & 0 & 193 & 18 & 25 \\ 1 & 3 & 2 & 0 & 18 & 0 & 2 \\ 2 & 4 & 2 & -3 & 25 & 2 & 1 \end{pmatrix}_{b_0} \\
[107] \quad & 1^{14}3^{11}1^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2486, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 1 & 2 \\ 2 & 4 & 4 & 0 & 36 & 3 & 4 \\ 1 & 4 & 0 & 0 & 24 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 36 & 24 & 0 & 238 & 20 & 28 \\ 1 & 3 & 2 & 0 & 20 & 0 & 2 \\ 2 & 4 & 2 & -3 & 28 & 2 & 1 \end{pmatrix}_{b_0} \\
T = 10 \quad [108] \quad & 1^{11}4^{13}1^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 3 & 0 & 2 \end{pmatrix}_{b_0} \\
[110] \quad & 1^{11}4^{13}1^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 5 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 4 & 0 & 2 \end{pmatrix}_{b_0} \\
[109] \quad & 1^{11}4^{13}1^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 3 \\ 3 & 4 & 0 & 9 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 7 & 0 & 2 \end{pmatrix}_{b_0} \\
[111] \quad & 1^{11}4^{13}1^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 6 & 1 & 3 \\ 3 & 4 & 0 & 8 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 0 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 7 & 0 & 2 \end{pmatrix}_{b_0} \\
[112] \quad & 1^{14}4^{13}1^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 66, (n_+, n_-, n_0) =
\end{aligned}$$

$$(1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 1 & 3 \\ 3 & 4 & 0 & 19 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 19 & 0 & 65 & 0 & 15 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 15 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[113] \quad 1^{11} 4^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 3 \\ 3 & 4 & 0 & 7 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[115] \quad 1^{12} 4^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 98, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 1 & 3 \\ 3 & 4 & 0 & 11 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 0 & 21 & 0 & 9 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 9 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[114] \quad 1^{13} 4^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 98, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 1 & 3 \\ 3 & 4 & 0 & 15 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 15 & 0 & 40 & 0 & 12 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 12 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[116] \quad 1^{11} 4^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 1 & 3 \\ 3 & 4 & 0 & 12 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 0 & 26 & 0 & 10 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 10 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[118] \quad 1^{12} 4^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 176, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 1 & 3 \\ 3 & 4 & 0 & 16 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 16 & 0 & 45 & 0 & 13 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 13 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[117] \quad 1^{14} 4^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 176, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 1 & 3 \\ 3 & 4 & 0 & 24 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 24 & 0 & 101 & 0 & 19 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 19 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[119] \quad 1^{13} 4^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 1 & 3 \\ 3 & 4 & 0 & 20 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 20 & 0 & 70 & 0 & 16 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 16 & 0 & 2 \end{pmatrix}_{b_0}$$

$$4.471 \quad \text{NHC}(-2)(-3) + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{f}_4$$

$$T = 9 \quad [0] \quad 1^{11} 3^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 2 & 1 & 3 \\ 4 & 9 & 4 & 8 & 4 & 0 & 5 \\ 2 & 4 & -2 & 3 & 2 & 0 & 0 \\ 3 & 8 & 3 & 4 & 3 & 0 & 4 \\ 2 & 4 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 4 & 2 & -3 & -1 \end{pmatrix}_{b_0}$$

4.472 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 200, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 3 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 3 \\ 2 & 3 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 4 \\ 2 & 4 & 0 & 3 & 2 & 4 & 0 \\ 3 & 4 & -3 & 4 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 600, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 9 & 3 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 5 & 9 & 6 & 7 \\ 2 & 3 & 0 & 5 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 9 & 3 & 4 & 4 & 4 \\ 2 & 4 & 0 & 6 & 2 & 4 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

4.473 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -218, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 9 & 7 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -3 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -222, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 2 & 1 & 3 \\ 4 & 9 & 4 & 8 & 4 & 0 & 5 \\ 2 & 4 & -2 & 3 & 2 & 0 & 0 \\ 3 & 8 & 3 & 4 & 3 & 0 & 4 \\ 2 & 4 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 4 & 2 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -302, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 8 & 4 & 0 & 5 \\ 2 & 4 & -2 & 3 & 2 & 0 & 0 \\ 4 & 8 & 3 & 4 & 3 & 0 & 4 \\ 2 & 4 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 4 & 2 & -2 & 0 \end{pmatrix}_{b_0}$$

4.474 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 9 \quad [1] \quad 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 4 & 3 \\ 3 & 4 & 0 & 8 & 3 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 12 & 6 \\ 2 & 3 & 0 & 4 & 0 & 4 & 2 \\ 4 & 8 & 0 & 12 & 4 & 10 & 6 \\ 3 & 4 & -3 & 6 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{12} 3^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 17 & 3 \\ 3 & 4 & 0 & 8 & 3 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 42 & 6 \\ 2 & 3 & 0 & 4 & 0 & 17 & 2 \\ 17 & 31 & 0 & 42 & 17 & 172 & 24 \\ 3 & 4 & -3 & 6 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 5 & 3 \\ 3 & 4 & 0 & 8 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 12 & 6 \\ 2 & 3 & 0 & 4 & 0 & 5 & 2 \\ 5 & 9 & 0 & 12 & 5 & 13 & 7 \\ 3 & 4 & -3 & 6 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & -3 & 4 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 5 & 3 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 3 & 0 & 9 & 2 \\ 9 & 17 & 0 & 15 & 9 & 49 & 13 \\ 3 & 4 & -3 & 4 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 2 & 3 \\ 3 & 4 & 0 & 8 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 6 & 6 \\ 2 & 3 & 0 & 4 & 0 & 2 & 2 \\ 2 & 4 & 0 & 6 & 2 & 1 & 3 \\ 3 & 4 & -3 & 6 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{131}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 3 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 3 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & -3 & 4 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & -3 & 3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{123}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 12 & 3 \\ 3 & 4 & 0 & 8 & 3 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 30 & 6 \\ 2 & 3 & 0 & 4 & 0 & 12 & 2 \\ 12 & 22 & 0 & 30 & 12 & 85 & 17 \\ 3 & 4 & -3 & 6 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{111}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 3 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 9 & 3 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 8 & 14 & 0 & 9 & 8 & 37 & 11 \\ 3 & 4 & -3 & 3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{12} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 7 & 3 \\ 3 & 4 & 0 & 8 & 3 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 18 & 6 \\ 2 & 3 & 0 & 4 & 0 & 7 & 2 \\ 7 & 13 & 0 & 18 & 7 & 28 & 10 \\ 3 & 4 & -3 & 6 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 3 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 3 & 0 & 7 & 2 \\ 7 & 13 & 0 & 12 & 7 & 28 & 10 \\ 3 & 4 & -3 & 4 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 2 & 3 & 0 & 3 & 0 & 5 & 2 \\ 5 & 9 & 0 & 9 & 5 & 13 & 7 \\ 3 & 4 & -3 & 4 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 6 & 3 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 5 & 9 & 0 & 6 & 5 & 13 & 7 \\ 3 & 4 & -3 & 3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{12} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 9 & 3 \\ 3 & 4 & 0 & 8 & 3 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 24 & 6 \\ 2 & 3 & 0 & 4 & 0 & 9 & 2 \\ 9 & 17 & 0 & 24 & 9 & 49 & 13 \\ 3 & 4 & -3 & 6 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11} 4^1 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 11 & 3 \\ 3 & 4 & 0 & 9 & 3 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 5 & 33 & 7 \\ 2 & 3 & 0 & 5 & 0 & 11 & 2 \\ 11 & 21 & 0 & 33 & 11 & 76 & 16 \\ 3 & 4 & -3 & 7 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{12} 4^1 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 11 & 3 \\ 3 & 4 & 0 & 8 & 3 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 30 & 6 \\ 2 & 3 & 0 & 4 & 0 & 11 & 2 \\ 11 & 21 & 0 & 30 & 11 & 76 & 16 \\ 3 & 4 & -3 & 6 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 4^1 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 3 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 9 & 3 \\ 2 & 3 & 0 & 2 & 0 & 7 & 2 \\ 7 & 13 & 0 & 9 & 7 & 28 & 10 \\ 3 & 4 & -3 & 3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{12} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 14 & 3 \\ 3 & 4 & 0 & 8 & 3 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 36 & 6 \\ 2 & 3 & 0 & 4 & 0 & 14 & 2 \\ 14 & 26 & 0 & 36 & 14 & 118 & 20 \\ 3 & 4 & -3 & 6 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 5 & 15 & 7 \\ 2 & 3 & 0 & 5 & 0 & 5 & 2 \\ 5 & 9 & 0 & 15 & 5 & 13 & 7 \\ 3 & 4 & -3 & 7 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -474, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 3 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 15 & 3 \\ 2 & 3 & 0 & 2 & 0 & 13 & 2 \\ 13 & 23 & 0 & 15 & 13 & 100 & 18 \\ 3 & 4 & -3 & 3 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 3 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 12 & 3 \\ 2 & 3 & 0 & 2 & 0 & 9 & 2 \\ 9 & 17 & 0 & 12 & 9 & 49 & 13 \\ 3 & 4 & -3 & 3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 9 & 3 \\ 3 & 4 & 0 & 9 & 3 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 5 & 27 & 7 \\ 2 & 3 & 0 & 5 & 0 & 9 & 2 \\ 9 & 17 & 0 & 27 & 9 & 49 & 13 \\ 3 & 4 & -3 & 7 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 3 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 12 & 3 \\ 2 & 3 & 0 & 2 & 0 & 10 & 2 \\ 10 & 18 & 0 & 12 & 10 & 58 & 14 \\ 3 & 4 & -3 & 3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{11} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 3 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 15 & 3 \\ 2 & 3 & 0 & 2 & 0 & 11 & 2 \\ 11 & 21 & 0 & 15 & 11 & 76 & 16 \\ 3 & 4 & -3 & 3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{11} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 7 & 3 \\ 3 & 4 & 0 & 9 & 3 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 5 & 21 & 7 \\ 2 & 3 & 0 & 5 & 0 & 7 & 2 \\ 7 & 13 & 0 & 21 & 7 & 28 & 10 \\ 3 & 4 & -3 & 7 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{12431^1 1^{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 11 & 3 \\ 3 & 4 & 0 & 13 & 3 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 7 & 48 & 10 \\ 2 & 3 & 0 & 7 & 0 & 11 & 2 \\ 11 & 21 & 0 & 48 & 11 & 76 & 16 \\ 3 & 4 & -3 & 10 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{12231^1 1^{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 7 & 3 \\ 3 & 4 & 0 & 13 & 3 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 7 & 30 & 10 \\ 2 & 3 & 0 & 7 & 0 & 7 & 2 \\ 7 & 13 & 0 & 30 & 7 & 28 & 10 \\ 3 & 4 & -3 & 10 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{12331^1 1^{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -794, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 9 & 3 \\ 3 & 4 & 0 & 13 & 3 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 7 & 39 & 10 \\ 2 & 3 & 0 & 7 & 0 & 9 & 2 \\ 9 & 17 & 0 & 39 & 9 & 49 & 13 \\ 3 & 4 & -3 & 10 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{11331^1 1^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -824, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 3 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 15 & 3 \\ 2 & 3 & 0 & 2 & 0 & 12 & 2 \\ 12 & 22 & 0 & 15 & 12 & 85 & 17 \\ 3 & 4 & -3 & 3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{11331^1 1^{14}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 3 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 21 & 3 \\ 2 & 3 & 0 & 2 & 0 & 18 & 2 \\ 18 & 32 & 0 & 21 & 18 & 193 & 25 \\ 3 & 4 & -3 & 3 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{11331^1 1^{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 9 & 3 \\ 3 & 4 & 0 & 17 & 3 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 9 & 51 & 13 \\ 2 & 3 & 0 & 9 & 0 & 9 & 2 \\ 9 & 17 & 0 & 51 & 9 & 49 & 13 \\ 3 & 4 & -3 & 13 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{13431^1 1^{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -954, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 11 & 3 \\ 3 & 4 & 0 & 17 & 3 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 9 & 63 & 13 \\ 2 & 3 & 0 & 9 & 0 & 11 & 2 \\ 11 & 21 & 0 & 63 & 11 & 76 & 16 \\ 3 & 4 & -3 & 13 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{11331^1 1^{13}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -974, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 3 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 18 & 3 \\ 2 & 3 & 0 & 2 & 0 & 15 & 2 \\ 15 & 27 & 0 & 18 & 15 & 133 & 21 \\ 3 & 4 & -3 & 3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{14} 4^3 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1064, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 11 & 3 \\ 3 & 4 & 0 & 21 & 3 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 11 & 78 & 16 \\ 2 & 3 & 0 & 11 & 0 & 11 & 2 \\ 11 & 21 & 0 & 78 & 11 & 76 & 16 \\ 3 & 4 & -3 & 16 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{11} 4^3 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1084, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 3 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 18 & 3 \\ 2 & 3 & 0 & 2 & 0 & 14 & 2 \\ 14 & 26 & 0 & 18 & 14 & 118 & 20 \\ 3 & 4 & -3 & 3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{11} 4^3 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1434, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 3 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 21 & 3 \\ 2 & 3 & 0 & 2 & 0 & 17 & 2 \\ 17 & 31 & 0 & 21 & 17 & 172 & 24 \\ 3 & 4 & -3 & 3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{11} 4^3 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1604, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 3 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 24 & 3 \\ 2 & 3 & 0 & 2 & 0 & 20 & 2 \\ 20 & 36 & 0 & 24 & 20 & 238 & 28 \\ 3 & 4 & -3 & 3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [38] \quad 1^{111} 3^1 1^{11} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 247, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 2 & 6 \\ 1 & 3 & 2 & 0 & 2 & 0 & 5 \\ 3 & 9 & 6 & 0 & 6 & 5 & 13 \\ 2 & 4 & 2 & -3 & 3 & 2 & 7 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^1 3^1 1^{11} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 472, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 3 & 6 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 6 & 3 & 4 & 4 \\ 2 & 4 & 2 & -3 & 4 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{11} 3^1 1^{11} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 517, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 3 & 9 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 5 & 2 \\ 3 & 9 & 6 & 0 & 9 & 5 & 13 & 7 \\ 2 & 4 & 2 & -3 & 4 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{111} 3^1 1^{11} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 648, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 6 & 3 \\ 4 & 8 & 6 & 0 & 15 & 15 & 6 \\ 2 & 6 & 1 & 0 & 9 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 15 & 9 & 0 & 22 & 24 & 10 \\ 6 & 15 & 9 & 0 & 24 & 22 & 10 \\ 3 & 6 & 3 & -3 & 10 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{111}3^{11}1^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 912, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 5 & 15 & 7 \\ 1 & 3 & 2 & 0 & 5 & 0 & 5 & 2 \\ 3 & 9 & 6 & 0 & 15 & 5 & 13 & 7 \\ 2 & 4 & 2 & -3 & 7 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$4.475 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3$$

$$T = 10 \quad [0] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 72, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 2 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 392, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 2 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 462, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 9 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 5 & 2 & 2 \\ 5 & 9 & 0 & 9 & 5 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 2 & 6 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 512, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 & 3 \\ 3 & 4 & 0 & 4 & 3 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 3 & 6 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 & 5 & 3 \\ 2 & 4 & 0 & 3 & 2 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 5 & 13 & 6 & 7 \\ 3 & 4 & -3 & 3 & 2 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 622, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 6 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 & 2 \\ 5 & 9 & 0 & 6 & 5 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 2 & 6 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 1152, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 3 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 5 & 15 & 6 & 7 \\ 2 & 3 & 0 & 5 & 0 & 5 & 2 & 2 \\ 5 & 9 & 0 & 15 & 5 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 2 & 6 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.476 NHC(-2)(-3) + f₄ + su₂ + su₂→(-3) + su_{2n3mix}

$$T = 10 \quad [0] \quad 1^{111}3^{111}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -60, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 5 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 2 & 4 & 3 \\ 1 & 3 & 2 & 0 & 2 & 0 & 3 & 2 \\ 2 & 5 & 4 & 0 & 4 & 3 & 4 & 4 \\ 2 & 4 & 2 & -3 & 3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{111}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -60, \quad \det = 175, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 5 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 3 & 5 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 3 & 2 \\ 2 & 5 & 4 & 0 & 5 & 3 & 4 & 4 \\ 2 & 4 & 2 & -3 & 4 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^{111}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -60, \quad \det = 475, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 5 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 5 & 9 & 7 \\ 1 & 3 & 2 & 0 & 5 & 0 & 3 & 2 \\ 2 & 5 & 4 & 0 & 9 & 3 & 4 & 4 \\ 2 & 4 & 2 & -3 & 7 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

4.477 NHC(-2)(-3) + f₄ + su₂ + su₂→(-3) + su₃

$$T = 9 \quad [0] \quad 1^{121}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -96, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 5 & 3 \\ 3 & 4 & 0 & 8 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 12 & 6 \\ 2 & 3 & 0 & 4 & 0 & 5 & 2 \\ 5 & 9 & 0 & 12 & 5 & 12 & 6 \\ 3 & 4 & -3 & 6 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -124, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 2 & 1 & 3 \\ 4 & 9 & 4 & 8 & 4 & 0 & 5 \\ 2 & 4 & -2 & 3 & 2 & 0 & 0 \\ 3 & 8 & 3 & 4 & 3 & 0 & 4 \\ 2 & 4 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 4 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{121}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -136, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 2 & 3 \\ 3 & 4 & 0 & 8 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 4 & 6 & 6 \\ 2 & 3 & 0 & 4 & 0 & 2 & 2 \\ 2 & 4 & 0 & 6 & 2 & 0 & 2 \\ 3 & 4 & -3 & 6 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -136, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 2 & 3 \\ 3 & 4 & 0 & 19 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 19 & 0 & 73 & 11 & 12 & 15 \\ 2 & 3 & 0 & 11 & 0 & 2 & 2 \\ 2 & 4 & 0 & 12 & 2 & 0 & 2 \\ 3 & 4 & -3 & 15 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -146, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -156, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 3 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 4 & -3 & 3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.31 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 3 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 6 & 3 \\ 2 & 3 & 0 & 2 & 0 & 4 & 2 \\ 4 & 8 & 0 & 6 & 4 & 9 & 5 \\ 3 & 4 & -3 & 3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.13 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 8 & 4 & 0 & 5 \\ 2 & 4 & -2 & 3 & 2 & 0 & 0 \\ 4 & 8 & 3 & 4 & 3 & 0 & 4 \\ 2 & 4 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 4 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -206, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 4 & 3 \\ 3 & 4 & 0 & 9 & 3 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 5 & 12 & 7 \\ 2 & 3 & 0 & 5 & 0 & 4 & 2 \\ 4 & 8 & 0 & 12 & 4 & 9 & 5 \\ 3 & 4 & -3 & 7 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -226, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -336, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 6 & 3 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 5 & 9 & 0 & 6 & 5 & 12 & 6 \\ 3 & 4 & -3 & 3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{111}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -386, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 2 & 3 \\ 3 & 4 & 0 & 14 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 8 & 9 & 11 \\ 2 & 3 & 0 & 8 & 0 & 2 & 2 \\ 2 & 4 & 0 & 9 & 2 & 0 & 2 \\ 3 & 4 & -3 & 11 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 5 & 6 & 7 \\ 2 & 3 & 0 & 5 & 0 & 2 & 2 \\ 2 & 4 & 0 & 6 & 2 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 3 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 9 & 3 \\ 2 & 3 & 0 & 2 & 0 & 7 & 2 \\ 7 & 13 & 0 & 9 & 7 & 27 & 9 \\ 3 & 4 & -3 & 3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{12} 1^{13} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 0 & 13 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 7 & 9 & 10 \\ 2 & 3 & 0 & 7 & 0 & 2 & 2 \\ 2 & 4 & 0 & 9 & 2 & 0 & 2 \\ 3 & 4 & -3 & 10 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{13} 1^{13} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 2 & 3 \\ 3 & 4 & 0 & 17 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 9 & 12 & 13 \\ 2 & 3 & 0 & 9 & 0 & 2 & 2 \\ 2 & 4 & 0 & 12 & 2 & 0 & 2 \\ 3 & 4 & -3 & 13 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{12} 1^{13} 3^{11} 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 2 & 3 \\ 3 & 4 & 0 & 28 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 16 & 28 & 0 & 154 & 16 & 18 & 22 \\ 2 & 3 & 0 & 16 & 0 & 2 & 2 \\ 2 & 4 & 0 & 18 & 2 & 0 & 2 \\ 3 & 4 & -3 & 22 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{14} 1^{13} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 2 & 3 \\ 3 & 4 & 0 & 21 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 11 & 15 & 16 \\ 2 & 3 & 0 & 11 & 0 & 2 & 2 \\ 2 & 4 & 0 & 15 & 2 & 0 & 2 \\ 3 & 4 & -3 & 16 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{12} 1^{13} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 2 & 3 \\ 3 & 4 & 0 & 18 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 10 & 12 & 14 \\ 2 & 3 & 0 & 10 & 0 & 2 & 2 \\ 2 & 4 & 0 & 12 & 2 & 0 & 2 \\ 3 & 4 & -3 & 14 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{12} 1^{13} 3^{11} 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 2 & 3 \\ 3 & 4 & 0 & 23 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 13 & 23 & 0 & 100 & 13 & 15 & 18 \\ 2 & 3 & 0 & 13 & 0 & 2 & 2 \\ 2 & 4 & 0 & 15 & 2 & 0 & 2 \\ 3 & 4 & -3 & 18 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{13}1^13^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 2 & 3 \\ 3 & 4 & 0 & 22 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 12 & 22 & 0 & 85 & 12 & 15 & 17 \\ 2 & 3 & 0 & 12 & 0 & 2 & 2 \\ 2 & 4 & 0 & 15 & 2 & 0 & 2 \\ 3 & 4 & -3 & 17 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{13}1^13^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 2 & 3 \\ 3 & 4 & 0 & 27 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 15 & 27 & 0 & 133 & 15 & 18 & 21 \\ 2 & 3 & 0 & 15 & 0 & 2 & 2 \\ 2 & 4 & 0 & 18 & 2 & 0 & 2 \\ 3 & 4 & -3 & 21 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{14}1^13^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 2 & 3 \\ 3 & 4 & 0 & 26 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 14 & 26 & 0 & 118 & 14 & 18 & 20 \\ 2 & 3 & 0 & 14 & 0 & 2 & 2 \\ 2 & 4 & 0 & 18 & 2 & 0 & 2 \\ 3 & 4 & -3 & 20 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{13}1^13^{11}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1826, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 2 & 3 \\ 3 & 4 & 0 & 32 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 18 & 32 & 0 & 193 & 18 & 21 & 25 \\ 2 & 3 & 0 & 18 & 0 & 2 & 2 \\ 2 & 4 & 0 & 21 & 2 & 0 & 2 \\ 3 & 4 & -3 & 25 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{14}1^13^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -2386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 2 & 3 \\ 3 & 4 & 0 & 31 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 17 & 31 & 0 & 172 & 17 & 21 & 24 \\ 2 & 3 & 0 & 17 & 0 & 2 & 2 \\ 2 & 4 & 0 & 21 & 2 & 0 & 2 \\ 3 & 4 & -3 & 24 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{14}1^13^{11}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -2816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 2 & 3 \\ 3 & 4 & 0 & 36 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 20 & 36 & 0 & 238 & 20 & 24 & 28 \\ 2 & 3 & 0 & 20 & 0 & 2 & 2 \\ 2 & 4 & 0 & 24 & 2 & 0 & 2 \\ 3 & 4 & -3 & 28 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 11$ [26] $1^{11}1^14^{13}1^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 5 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 3 & 0 & 0 & 2 \\ 3 & 4 & -2 & 4 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{11}1^11^14^{13}1^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 2 & 3 \\ 3 & 4 & 0 & 7 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 5 & 7 & 0 & 8 & 0 & 5 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 5 & 0 & 0 & 2 \\ 3 & 4 & -2 & 6 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[28] $1^{111}3^{111}4^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 1 & 2 & 3 \\ 3 & 4 & 0 & 12 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 8 & 12 & 0 & 26 & 0 & 8 & 10 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 8 & 0 & 0 & 2 \\ 3 & 4 & -2 & 10 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

4.478 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2$

$T = 10$ [0] $1^{111}3^{111}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 6 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 3 & 2 \\ 2 & 4 & 0 & 3 & 2 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 3 & 4 & 4 \\ 3 & 4 & -3 & 4 & 2 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^{111}3^{111}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & -3 & 3 & 2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^{111}3^{111}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 9 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 5 & 2 \\ 2 & 4 & 0 & 3 & 2 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 5 & 6 & 13 & 7 \\ 3 & 4 & -3 & 4 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^{111}3^{111}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 6 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 & 2 \\ 5 & 9 & 0 & 6 & 5 & 12 & 6 & 6 \\ 2 & 4 & 0 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & -3 & 3 & 2 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^{111}3^{111}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 2 & 6 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 5 & 6 & 13 & 7 \\ 3 & 4 & -3 & 3 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{111}3^{111}3^{111}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 3 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 5 & 6 & 15 & 7 \\ 2 & 3 & 0 & 5 & 0 & 2 & 5 & 2 \\ 2 & 4 & 0 & 6 & 2 & 0 & 6 & 2 \\ 5 & 9 & 0 & 15 & 5 & 6 & 13 & 7 \\ 3 & 4 & -3 & 7 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.479 NHC(-2)(-3) + f₄ + su₂ + su₂n3mix

$$T = 8 \quad [1] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 3 \\ 5 & 8 & 0 & 5 & 12 & 6 \\ 3 & 4 & -3 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.63 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 12 & 5 & 0 & 12 & 6 \\ 3 & 5 & 0 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 19 & 10 \\ 5 & 8 & 0 & 19 & 12 & 6 \\ 3 & 4 & -3 & 10 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 4 & 3 \\ 3 & 4 & 0 & 14 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 17 & 11 \\ 4 & 7 & 0 & 17 & 7 & 5 \\ 3 & 4 & -3 & 11 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{13} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.3$$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 8 & 3 \\ 3 & 4 & 0 & 20 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 20 & 0 & 73 & 55 & 15 \\ 8 & 15 & 0 & 55 & 39 & 11 \\ 3 & 4 & -3 & 15 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 9 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 23 & 7 \\ 8 & 15 & 0 & 23 & 39 & 11 \\ 3 & 4 & -3 & 7 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 12 & 5 & 0 & 12 & 6 \\ 3 & 5 & 0 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 5 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 4 & 7 & 0 & 6 & 7 & 5 \\ 3 & 4 & -3 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{14} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 6 & 3 \\ 3 & 4 & 0 & 16 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 46 & 32 & 12 \\ 6 & 11 & 0 & 32 & 20 & 8 \\ 3 & 4 & -3 & 12 & 8 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{142}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 6 & 3 \\ 3 & 4 & 0 & 31 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 31 & 0 & 172 & 59 & 24 \\ 6 & 11 & 0 & 59 & 20 & 8 \\ 3 & 4 & -3 & 24 & 8 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{131}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 4 & 3 \\ 3 & 4 & 0 & 12 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 15 & 9 \\ 4 & 7 & 0 & 15 & 7 & 5 \\ 3 & 4 & -3 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.64 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 13 & 22 & 0 & 14 & 97 & 17 \\ 3 & 4 & -3 & 3 & 17 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{123}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 8 & 25 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 34 & 6 \\ 14 & 25 & 0 & 34 & 111 & 19 \\ 3 & 4 & -3 & 6 & 19 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{143}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 13.15 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 8 & 3 \\ 3 & 4 & 0 & 26 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 0 & 118 & 68 & 20 \\ 8 & 15 & 0 & 68 & 39 & 11 \\ 3 & 4 & -3 & 20 & 11 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{134}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 13 & 3 \\ 3 & 4 & 0 & 12 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 51 & 9 \\ 13 & 24 & 0 & 51 & 100 & 18 \\ 3 & 4 & -3 & 9 & 18 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{121}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 4 & 3 \\ 3 & 4 & 0 & 23 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 23 & 0 & 100 & 28 & 18 \\ 4 & 7 & 0 & 28 & 7 & 5 \\ 3 & 4 & -3 & 18 & 5 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{131}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 9 & 4 \\ 5 & 8 & 0 & 9 & 12 & 6 \\ 3 & 4 & -3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{143}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 13 & 3 \\ 3 & 4 & 0 & 5 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 21 & 4 \\ 13 & 24 & 0 & 21 & 100 & 18 \\ 3 & 4 & -3 & 4 & 18 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{121}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 10 & 6 \\ 4 & 7 & 0 & 10 & 7 & 5 \\ 3 & 4 & -3 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{124}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 0 & 13 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 43 & 10 \\ 10 & 19 & 0 & 43 & 64 & 14 \\ 3 & 4 & -3 & 10 & 14 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{154}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 10 & 3 \\ 3 & 4 & 0 & 20 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 20 & 0 & 73 & 70 & 15 \\ 10 & 19 & 0 & 70 & 64 & 14 \\ 3 & 4 & -3 & 15 & 14 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{132}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 138, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 0 & 12 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 24 & 9 \\ 6 & 11 & 0 & 24 & 20 & 8 \\ 3 & 4 & -3 & 9 & 8 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{122}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 8 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 22 & 6 \\ 9 & 16 & 0 & 22 & 44 & 12 \\ 3 & 4 & -3 & 6 & 12 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{143}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 3 \\ 3 & 4 & 0 & 16 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 46 & 44 & 12 \\ 8 & 15 & 0 & 44 & 39 & 11 \\ 3 & 4 & -3 & 12 & 11 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{111}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 3 \\ 4 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & -3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{112}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 158, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 9 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 17 & 7 \\ 6 & 11 & 0 & 17 & 20 & 8 \\ 3 & 4 & -3 & 7 & 8 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{113}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 14 & 7 \\ 5 & 8 & 0 & 14 & 12 & 6 \\ 3 & 4 & -3 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{122}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 0 & 8 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 16 & 6 \\ 6 & 11 & 0 & 16 & 20 & 8 \\ 3 & 4 & -3 & 6 & 8 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{111}3^11^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 11 & 3 \\ 10 & 17 & 0 & 11 & 55 & 13 \\ 3 & 4 & -3 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{131}3^11^{111} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 7 & 3 \\ 3 & 4 & 0 & 17 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 36 & 13 \\ 7 & 12 & 0 & 36 & 25 & 9 \\ 3 & 4 & -3 & 13 & 9 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{111}3^11^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 11 & 7 \\ 4 & 7 & 0 & 11 & 7 & 5 \\ 3 & 4 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{113}3^11^{111} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 7 & 12 & 0 & 12 & 25 & 9 \\ 3 & 4 & -3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{133}3^11^{111} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 11 & 20 & 0 & 18 & 69 & 15 \\ 3 & 4 & -3 & 4 & 15 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{133}3^11^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 12 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 33 & 9 \\ 8 & 15 & 0 & 33 & 39 & 11 \\ 3 & 4 & -3 & 9 & 11 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{111}3^11^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 8 & 3 \\ 7 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{112}3^11^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 218, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 8 & 3 \\ 6 & 11 & 0 & 8 & 20 & 8 \\ 3 & 4 & -3 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{144}3^11^{11} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 10 & 3 \\ 3 & 4 & 0 & 16 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 46 & 56 & 12 \\ 10 & 19 & 0 & 56 & 64 & 14 \\ 3 & 4 & -3 & 12 & 14 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{131}3^11^{14} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 4 & 3 \\ 3 & 4 & 0 & 32 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 18 & 32 & 0 & 193 & 39 & 25 \\ 4 & 7 & 0 & 39 & 7 & 5 \\ 3 & 4 & -3 & 25 & 5 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{123}3^11^{111} + [\mathfrak{s}u_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}u_2 + \mathfrak{s}u_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 9 & 16 & 0 & 15 & 44 & 12 \\ 3 & 4 & -3 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$

[39] $1^{123}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 8 & 3 \\ 3 & 4 & 0 & 13 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 34 & 10 \\ 8 & 15 & 0 & 34 & 39 & 11 \\ 3 & 4 & -3 & 10 & 11 & 1 \end{pmatrix}_{b_0}$$

[40] $1^{123}3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 258, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 6 & 3 \\ 3 & 4 & 0 & 22 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 0 & 85 & 42 & 17 \\ 6 & 11 & 0 & 42 & 20 & 8 \\ 3 & 4 & -3 & 17 & 8 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{123}3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 262, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 8 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 22 & 6 \\ 8 & 15 & 0 & 22 & 39 & 11 \\ 3 & 4 & -3 & 6 & 11 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{134}3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 10 & 3 \\ 3 & 4 & 0 & 12 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 42 & 9 \\ 10 & 19 & 0 & 42 & 64 & 14 \\ 3 & 4 & -3 & 9 & 14 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{113}3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 11 & 3 \\ 8 & 15 & 0 & 11 & 39 & 11 \\ 3 & 4 & -3 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{121}3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 298, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 4 & 3 \\ 3 & 4 & 0 & 18 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 22 & 14 \\ 4 & 7 & 0 & 22 & 7 & 5 \\ 3 & 4 & -3 & 14 & 5 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{121}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 303, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 13 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 16 & 10 \\ 4 & 7 & 0 & 16 & 7 & 5 \\ 3 & 4 & -3 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{122}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 308, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 0 & 13 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 25 & 10 \\ 6 & 11 & 0 & 25 & 20 & 8 \\ 3 & 4 & -3 & 10 & 8 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{142}3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 9 & 3 \\ 3 & 4 & 0 & 21 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 59 & 16 \\ 9 & 16 & 0 & 59 & 44 & 12 \\ 3 & 4 & -3 & 16 & 12 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{123}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 322, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 0 & 8 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 28 & 6 \\ 11 & 20 & 0 & 28 & 69 & 15 \\ 3 & 4 & -3 & 6 & 15 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{134}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 330, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.07 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 10 & 3 \\ 3 & 4 & 0 & 17 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 57 & 13 \\ 10 & 19 & 0 & 57 & 64 & 14 \\ 3 & 4 & -3 & 13 & 14 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{124}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 8 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 28 & 6 \\ 10 & 19 & 0 & 28 & 64 & 14 \\ 3 & 4 & -3 & 6 & 14 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{112}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 362, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 20 & 3 \\ 18 & 31 & 0 & 20 & 188 & 24 \\ 3 & 4 & -3 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{121}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 367, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 28 & 10 \\ 7 & 12 & 0 & 28 & 25 & 9 \\ 3 & 4 & -3 & 10 & 9 & 1 \end{pmatrix}_{b_0}$$

[53] $1^{114}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 10 & 19 & 0 & 14 & 64 & 14 \\ 3 & 4 & -3 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

[54] $1^{124}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 16 & 3 \\ 3 & 4 & 0 & 8 & 29 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 40 & 6 \\ 16 & 29 & 0 & 40 & 148 & 22 \\ 3 & 4 & -3 & 6 & 22 & 1 \end{pmatrix}_{b_0}$$

[55] $1^{111}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 20 & 7 \\ 7 & 12 & 0 & 20 & 25 & 9 \\ 3 & 4 & -3 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{131}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 393, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 0 & 17 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 21 & 13 \\ 4 & 7 & 0 & 21 & 7 & 5 \\ 3 & 4 & -3 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

[57] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 395, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 11 & 3 \\ 9 & 16 & 0 & 11 & 44 & 12 \\ 3 & 4 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{133}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 417, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 8 & 3 \\ 3 & 4 & 0 & 17 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 45 & 13 \\ 8 & 15 & 0 & 45 & 39 & 11 \\ 3 & 4 & -3 & 13 & 11 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{132}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 438, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 6 & 3 \\ 3 & 4 & 0 & 17 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 33 & 13 \\ 6 & 11 & 0 & 33 & 20 & 8 \\ 3 & 4 & -3 & 13 & 8 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{141}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 463, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 4 & 3 \\ 3 & 4 & 0 & 21 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 26 & 16 \\ 4 & 7 & 0 & 26 & 7 & 5 \\ 3 & 4 & -3 & 16 & 5 & 1 \end{pmatrix}_{b_0}$$

[61] $1^{112}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 467, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 17 & 3 \\ 15 & 26 & 0 & 17 & 128 & 20 \\ 3 & 4 & -3 & 3 & 20 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{131}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 473, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 4 & 3 \\ 3 & 4 & 0 & 27 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 0 & 133 & 33 & 21 \\ 4 & 7 & 0 & 33 & 7 & 5 \\ 3 & 4 & -3 & 21 & 5 & 1 \end{pmatrix}_{b_0}$$

[63] $1^{112}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 12 & 21 & 0 & 14 & 80 & 16 \\ 3 & 4 & -3 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

[64] $1^{124}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 508, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 13 & 3 \\ 3 & 4 & 0 & 8 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 34 & 6 \\ 13 & 24 & 0 & 34 & 100 & 18 \\ 3 & 4 & -3 & 6 & 18 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{112}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 515, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 26 & 7 \\ 9 & 16 & 0 & 26 & 44 & 12 \\ 3 & 4 & -3 & 7 & 12 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{144}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 520, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 10 & 3 \\ 3 & 4 & 0 & 21 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 71 & 16 \\ 10 & 19 & 0 & 71 & 64 & 14 \\ 3 & 4 & -3 & 16 & 14 & 1 \end{pmatrix}_{b_0}$$

[67] $1^{131}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 4 & 3 \\ 3 & 4 & 0 & 22 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 0 & 85 & 27 & 17 \\ 4 & 7 & 0 & 27 & 7 & 5 \\ 3 & 4 & -3 & 17 & 5 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{142}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 6 & 3 \\ 3 & 4 & 0 & 21 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 41 & 16 \\ 6 & 11 & 0 & 41 & 20 & 8 \\ 3 & 4 & -3 & 16 & 8 & 1 \end{pmatrix}_{b_0}$$

[68] $1^{142}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 6 & 3 \\ 3 & 4 & 0 & 26 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 0 & 118 & 50 & 20 \\ 6 & 11 & 0 & 50 & 20 & 8 \\ 3 & 4 & -3 & 20 & 8 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{132}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 555, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 0 & 17 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 48 & 13 \\ 9 & 16 & 0 & 48 & 44 & 12 \\ 3 & 4 & -3 & 13 & 12 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{143}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 567, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 8 & 3 \\ 3 & 4 & 0 & 21 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 56 & 16 \\ 8 & 15 & 0 & 56 & 39 & 11 \\ 3 & 4 & -3 & 16 & 11 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 577, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 11 & 20 & 0 & 14 & 69 & 15 \\ 3 & 4 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{113}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 592, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 32 & 7 \\ 11 & 20 & 0 & 32 & 69 & 15 \\ 3 & 4 & -3 & 7 & 15 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{114}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 9 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 38 & 7 \\ 13 & 24 & 0 & 38 & 100 & 18 \\ 3 & 4 & -3 & 7 & 18 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{122}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 620, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 13 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 37 & 10 \\ 9 & 16 & 0 & 37 & 44 & 12 \\ 3 & 4 & -3 & 10 & 12 & 1 \end{pmatrix}_{b_0}$$

[76] $1^{141}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 718, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 4 & 3 \\ 3 & 4 & 0 & 36 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 20 & 36 & 0 & 238 & 44 & 28 \\ 4 & 7 & 0 & 44 & 7 & 5 \\ 3 & 4 & -3 & 28 & 5 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{143}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 727, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 11 & 3 \\ 3 & 4 & 0 & 21 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 74 & 16 \\ 11 & 20 & 0 & 74 & 69 & 15 \\ 3 & 4 & -3 & 16 & 15 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{141}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 738, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 4 & 3 \\ 3 & 4 & 0 & 26 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 0 & 118 & 32 & 20 \\ 4 & 7 & 0 & 32 & 7 & 5 \\ 3 & 4 & -3 & 20 & 5 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{114}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 763, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 17 & 3 \\ 13 & 24 & 0 & 17 & 100 & 18 \\ 3 & 4 & -3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

[80] $1^{113}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 25 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 17 & 3 \\ 14 & 25 & 0 & 17 & 111 & 19 \\ 3 & 4 & -3 & 3 & 19 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{123}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 807, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 0 & 13 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 46 & 10 \\ 11 & 20 & 0 & 46 & 69 & 15 \\ 3 & 4 & -3 & 10 & 15 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{141}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 823, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 4 & 3 \\ 3 & 4 & 0 & 31 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 31 & 0 & 172 & 38 & 24 \\ 4 & 7 & 0 & 38 & 7 & 5 \\ 3 & 4 & -3 & 24 & 5 & 1 \end{pmatrix}_{b_0}$$

[83] $1^{133}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 852, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 11 & 3 \\ 3 & 4 & 0 & 17 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 60 & 13 \\ 11 & 20 & 0 & 60 & 69 & 15 \\ 3 & 4 & -3 & 13 & 15 & 1 \end{pmatrix}_{b_0}$$

[84] $1^{113}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 865, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 30 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 20 & 3 \\ 17 & 30 & 0 & 20 & 165 & 23 \\ 3 & 4 & -3 & 3 & 23 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{113}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 868, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 35 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 23 & 3 \\ 20 & 35 & 0 & 23 & 231 & 27 \\ 3 & 4 & -3 & 3 & 27 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{124}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 928, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 13 & 3 \\ 3 & 4 & 0 & 13 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 55 & 10 \\ 13 & 24 & 0 & 55 & 100 & 18 \\ 3 & 4 & -3 & 10 & 18 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{114}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1062, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 29 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 20 & 3 \\ 16 & 29 & 0 & 20 & 148 & 22 \\ 3 & 4 & -3 & 3 & 22 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{144}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1068, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 13 & 3 \\ 3 & 4 & 0 & 21 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 89 & 16 \\ 13 & 24 & 0 & 89 & 100 & 18 \\ 3 & 4 & -3 & 16 & 18 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{134}3^11^{111} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{brown}{\mathfrak{f}_4} + \textcolor{brown}{\mathfrak{su}_2} + \textcolor{brown}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1083, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 13 & 3 \\ 3 & 4 & 0 & 17 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 72 & 13 \\ 13 & 24 & 0 & 72 & 100 & 18 \\ 3 & 4 & -3 & 13 & 18 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{114}3^11^{13} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{brown}{\mathfrak{f}_4} + \textcolor{brown}{\mathfrak{su}_2} + \textcolor{brown}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 19 & 3 \\ 3 & 4 & 0 & 4 & 34 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 23 & 3 \\ 19 & 34 & 0 & 23 & 208 & 26 \\ 3 & 4 & -3 & 3 & 26 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{114}3^11^{14} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{brown}{\mathfrak{f}_4} + \textcolor{brown}{\mathfrak{su}_2} + \textcolor{brown}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 22 & 3 \\ 3 & 4 & 0 & 4 & 39 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 26 & 3 \\ 22 & 39 & 0 & 26 & 280 & 30 \\ 3 & 4 & -3 & 3 & 30 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [92] $1^{124}3^121^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{brown}{\mathfrak{f}_4} + \textcolor{brown}{\mathfrak{su}_2} + \textcolor{brown}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 8 & 0 \\ 2 & 4 & 0 & 2 & 6 & 20 & 4 \\ 8 & 10 & 0 & 8 & 20 & 52 & 12 \\ 2 & 2 & -1 & 0 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{111}3^121^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{brown}{\mathfrak{f}_4} + \textcolor{brown}{\mathfrak{su}_2} + \textcolor{brown}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 2 & 0 \\ 2 & 3 & 0 & 4 & 5 & 5 & 5 \\ 2 & 3 & 0 & 2 & 5 & 3 & 3 \\ 2 & 2 & -1 & 0 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{111}3^11^{131} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{brown}{\mathfrak{f}_4} + \textcolor{brown}{\mathfrak{su}_2} + \textcolor{brown}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 5 & 3 \\ 2 & 4 & 3 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 5 & 3 \\ 2 & 5 & 6 & 0 & 5 & 7 & 5 \\ 2 & 3 & 3 & -3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

[95] $1^{113}3^11^{131} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{brown}{\mathfrak{f}_4} + \textcolor{brown}{\mathfrak{su}_2} + \textcolor{brown}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 17 & 4 \\ 1 & 4 & 0 & 0 & 3 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 13 & 3 \\ 6 & 17 & 13 & 0 & 13 & 60 & 14 \\ 2 & 4 & 2 & -3 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

[96] $1^{111}3^121^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{brown}{\mathfrak{f}_4} + \textcolor{brown}{\mathfrak{su}_2} + \textcolor{brown}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 3 & 2 & 0 & 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & -1 & 0 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{123}3^121^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{brown}{\mathfrak{f}_4} + \textcolor{brown}{\mathfrak{su}_2} + \textcolor{brown}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 7 & 0 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 6 & 8 & 0 & 7 & 16 & 34 & 10 \\ 2 & 2 & -1 & 0 & 4 & 10 & 1 \end{pmatrix}_{b_0}$$

[98] $1^{141}3^{12}21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 10 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 2 & 0 \\ 6 & 10 & 0 & 10 & 58 & 16 & 14 \\ 2 & 3 & 0 & 2 & 16 & 3 & 3 \\ 2 & 2 & -1 & 0 & 14 & 3 & 1 \end{pmatrix}_{b_0}$$

[99] $1^{111}3^{12}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -47$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 3 & 3 \\ 2 & 2 & -1 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[100] $1^{123}3^{12}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 4 & 0 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 6 & 7 & 0 & 4 & 14 & 23 & 7 \\ 2 & 2 & -1 & 0 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

[101] $1^{111}3^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 5 & 4 \\ 1 & 4 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 4 & 3 \\ 2 & 5 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 2 & -3 & 3 & 4 & 1 \end{pmatrix}_{b_0}$$

[102] $1^{111}3^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -59$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.49 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 5 & 3 \\ 3 & 9 & 5 & 0 & 5 & 12 & 6 \\ 2 & 4 & 2 & -3 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[103] $1^{111}3^{12}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 5 & 0 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 5 & 4 & 10 & 6 \\ 2 & 2 & -1 & 0 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[104] $1^{111}3^{12}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -67$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 2 & 0 \\ 3 & 3 & 0 & 4 & 5 & 5 & 5 \\ 2 & 3 & 0 & 2 & 5 & 3 & 3 \\ 2 & 2 & -1 & 0 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[105] $1^{111}3^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 4 & 3 \\ 2 & 4 & 3 & 0 & 3 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 4 & 3 \\ 2 & 4 & 5 & 0 & 4 & 4 & 4 \\ 2 & 3 & 3 & -3 & 3 & 4 & 0 \end{pmatrix}_{b_0}$$

[106] $1^{121}3^{12}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -82$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 \\ 2 & 2 & -1 & 0 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

[107] $1^{11}1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 4 & 3 \\ 1 & 4 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 4 & 3 \\ 2 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 3 & 2 & -3 & 3 & 4 & 0 \end{pmatrix}_{b_0}$$

[108] $1^{12}1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -85$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 12.29 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 11 & 4 \\ 1 & 4 & 0 & 0 & 9 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 9 & 0 & 28 & 25 & 10 \\ 4 & 11 & 7 & 0 & 25 & 20 & 8 \\ 2 & 4 & 2 & -3 & 10 & 8 & 0 \end{pmatrix}_{b_0}$$

[109] $1^{11}1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -89$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 4 & 0 & 6 & 4 & 3 \\ 1 & 4 & 0 & 0 & 6 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 6 & 6 & 0 & 13 & 9 & 7 \\ 2 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 3 & 2 & -3 & 7 & 4 & 0 \end{pmatrix}_{b_0}$$

[110] $1^{11}1^{12}3^{11}2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 3 & 0 \\ 2 & 4 & 0 & 2 & 6 & 10 & 4 \\ 4 & 5 & 0 & 3 & 10 & 11 & 5 \\ 2 & 2 & -1 & 0 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

[111] $1^{11}1^{12}3^{11}2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -97$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 2 & 0 \\ 3 & 5 & 0 & 5 & 13 & 8 & 7 \\ 2 & 3 & 0 & 2 & 8 & 3 & 3 \\ 2 & 2 & -1 & 0 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{11}1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -101$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 5 & 4 \\ 1 & 4 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 5 & 4 \\ 2 & 5 & 4 & 0 & 5 & 4 & 4 \\ 2 & 4 & 2 & -3 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[113] $1^{11}1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -104$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 6 & 4 \\ 1 & 4 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 11 & 7 \\ 2 & 6 & 5 & 0 & 11 & 7 & 5 \\ 2 & 4 & 2 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

[114] $1^{11}1^{11}3^{11}2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -109$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & -1 & 0 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[115] $1^{11}1^{12}3^{11}2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 3 & 0 \\ 2 & 2 & 0 & 1 & 0 & 5 & 2 \\ 3 & 5 & 0 & 3 & 5 & 11 & 5 \\ 2 & 2 & -1 & 0 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[116] $1^{1^{12}} 1^{14} 3^1 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -117$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 & 2 \\ 1 & 0 & 1 & 1 & 11 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 2 & 0 \\ 9 & 11 & 0 & 13 & 81 & 18 & 17 \\ 2 & 3 & 0 & 2 & 18 & 3 & 3 \\ 2 & 2 & -1 & 0 & 17 & 3 & 1 \end{pmatrix}_{b_0}$$

[117] $1^{1^{11}} 1^{11} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -125$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 5 & 0 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 4 & 4 & 0 & 5 & 4 & 10 & 6 \\ 2 & 2 & -1 & 0 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[118] $1^{1^{12}} 1^{23} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -128$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 7 & 0 \\ 2 & 4 & 0 & 2 & 6 & 16 & 4 \\ 5 & 8 & 0 & 7 & 16 & 34 & 10 \\ 2 & 2 & -1 & 0 & 4 & 10 & 1 \end{pmatrix}_{b_0}$$

[119] $1^{1^{11}} 31^{11} 31^{11} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -129$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 8 & 4 \\ 1 & 4 & 0 & 0 & 6 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 12 & 7 \\ 3 & 8 & 4 & 0 & 12 & 9 & 5 \\ 2 & 4 & 2 & -3 & 7 & 5 & 0 \end{pmatrix}_{b_0}$$

[120] $1^{1^{11}} 31^{14} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 & 2 \\ 2 & 4 & 4 & 0 & 27 & 5 & 4 \\ 1 & 4 & 0 & 0 & 15 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 27 & 15 & 0 & 121 & 24 & 19 \\ 2 & 5 & 4 & 0 & 24 & 4 & 4 \\ 2 & 4 & 2 & -3 & 19 & 4 & 1 \end{pmatrix}_{b_0}$$

[121] $1^{1^{11}} 1^{11} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -142$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 2 & 2 & 0 & 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & -1 & 0 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[122] $1^{1^{12}} 1^{32} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -145$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & 1 & 7 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 3 & 0 \\ 4 & 7 & 0 & 6 & 25 & 18 & 9 \\ 3 & 5 & 0 & 3 & 18 & 11 & 5 \\ 2 & 2 & -1 & 0 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

[123] $1^{1^{11}} 1^{12} 3^1 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -148$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 9 & 0 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 6 & 7 & 0 & 9 & 7 & 35 & 11 \\ 2 & 2 & -1 & 0 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[124] $1^{112}3^121^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -157$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 3 & 0 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 3 & 5 & 0 & 3 & 5 & 11 & 5 \\ 2 & 2 & -1 & 0 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[125] $1^{131}3^121^{12}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -158$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 2 & 0 \\ 6 & 8 & 0 & 9 & 40 & 13 & 12 \\ 2 & 3 & 0 & 2 & 13 & 3 & 3 \\ 2 & 2 & -1 & 0 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

[126] $1^{111}3^121^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -160$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 3 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & -1 & 0 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[127] $1^{131}3^121^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -162$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 2 & 0 \\ 4 & 7 & 0 & 6 & 25 & 11 & 9 \\ 2 & 3 & 0 & 2 & 11 & 3 & 3 \\ 2 & 2 & -1 & 0 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[128] $1^{123}1^{111}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -164$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 5 & 2 \\ 2 & 4 & 4 & 0 & 6 & 14 & 4 \\ 1 & 4 & 0 & 0 & 3 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 14 & 4 \\ 5 & 14 & 10 & 0 & 14 & 37 & 11 \\ 2 & 4 & 2 & -3 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$

[129] $1^{111}3^11^{11}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -175$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.03 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 11 & 4 \\ 1 & 4 & 0 & 0 & 3 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 7 & 3 \\ 4 & 11 & 7 & 0 & 7 & 20 & 8 \\ 2 & 4 & 2 & -3 & 3 & 8 & 0 \end{pmatrix}_{b_0}$$

[130] $1^{111}3^11^{11}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -181$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 8 & 3 \\ 4 & 12 & 8 & 0 & 8 & 25 & 9 \\ 2 & 4 & 2 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[131] $1^{132}3^121^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -185$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 3 & 0 \\ 5 & 6 & 0 & 3 & 16 & 15 & 6 \\ 3 & 5 & 0 & 3 & 15 & 11 & 5 \\ 2 & 2 & -1 & 0 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{112}3^121^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{f}_4+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 6 & 0 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 4 & 6 & 0 & 6 & 6 & 20 & 8 \\ 2 & 2 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{111}31^{12}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 17 & 4 \\ 1 & 4 & 0 & 0 & 3 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 10 & 3 \\ 6 & 17 & 10 & 0 & 10 & 48 & 12 \\ 2 & 4 & 2 & -3 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

[134] $1^{11}3^121^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 10 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 10 & 9 & 51 & 13 \\ 2 & 2 & -1 & 0 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[135] $1^{112}31^{11}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -209, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 14 & 4 \\ 1 & 4 & 0 & 0 & 3 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 10 & 3 \\ 5 & 14 & 10 & 0 & 10 & 37 & 11 \\ 2 & 4 & 2 & -3 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

[136] $1^{141}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -209, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 2 & 0 \\ 5 & 9 & 0 & 7 & 41 & 14 & 11 \\ 2 & 3 & 0 & 2 & 14 & 3 & 3 \\ 2 & 2 & -1 & 0 & 11 & 3 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{112}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -214, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 2 & 0 \\ 4 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 \\ 2 & 2 & -1 & 0 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

[138] $1^{111}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 2 & 0 \\ 3 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 3 & 0 & 2 & 6 & 3 & 3 \\ 2 & 2 & -1 & 0 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{121}31^{11}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 & 2 \\ 2 & 4 & 4 & 0 & 12 & 5 & 4 \\ 1 & 4 & 0 & 0 & 9 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 9 & 0 & 28 & 13 & 10 \\ 2 & 5 & 4 & 0 & 13 & 4 & 4 \\ 2 & 4 & 2 & -3 & 10 & 4 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{122}3^121^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 3 & 0 \\ 3 & 4 & 0 & 2 & 6 & 10 & 4 \\ 3 & 5 & 0 & 3 & 10 & 11 & 5 \\ 2 & 2 & -1 & 0 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

[141] $1^{142}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & 1 & 9 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 3 & 0 \\ 5 & 9 & 0 & 7 & 41 & 23 & 11 \\ 3 & 5 & 0 & 3 & 23 & 11 & 5 \\ 2 & 2 & -1 & 0 & 11 & 5 & 1 \end{pmatrix}_{b_0}$$

- [142] $1^{1^{11}} 1^{123} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -224$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 4 & 0 \\ 2 & 4 & 0 & 2 & 6 & 14 & 4 \\ 5 & 7 & 0 & 4 & 14 & 23 & 7 \\ 2 & 2 & -1 & 0 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$
- [143] $1^{1^{12}} 1^{112} 3^1 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -229$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 9 & 0 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 7 & 7 & 0 & 9 & 7 & 35 & 11 \\ 2 & 2 & -1 & 0 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$
- [144] $1^{1^{11}} 1^{124} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -242$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 8 & 0 \\ 2 & 4 & 0 & 2 & 6 & 20 & 4 \\ 7 & 10 & 0 & 8 & 20 & 52 & 12 \\ 2 & 2 & -1 & 0 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$
- [145] $1^{1^{11}} 1^{111} 3^1 1^{14} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 30 & 4 \\ 1 & 4 & 0 & 0 & 3 & 17 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 17 & 3 \\ 10 & 30 & 17 & 0 & 17 & 151 & 21 \\ 2 & 4 & 2 & -3 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$
- [146] $1^{1^{11}} 1^{121} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -254$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 2 & 0 \\ 4 & 5 & 0 & 5 & 13 & 8 & 7 \\ 2 & 3 & 0 & 2 & 8 & 3 & 3 \\ 2 & 2 & -1 & 0 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$
- [147] $1^{1^{11}} 1^{124} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 8 & 0 \\ 2 & 4 & 0 & 2 & 6 & 20 & 4 \\ 6 & 10 & 0 & 8 & 20 & 52 & 12 \\ 2 & 2 & -1 & 0 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$
- [148] $1^{1^{12}} 1^{124} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -258$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 5 & 0 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 7 & 9 & 0 & 5 & 18 & 39 & 9 \\ 2 & 2 & -1 & 0 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$
- [149] $1^{1^{12}} 1^{121} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -269$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 2 & 0 \\ 5 & 5 & 0 & 5 & 13 & 8 & 7 \\ 2 & 3 & 0 & 2 & 8 & 3 & 3 \\ 2 & 2 & -1 & 0 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$
- [150] $1^{1^{11}} 1^{11} 3^1 1^{11} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 5 & 4 \\ 1 & 4 & 0 & 0 & 6 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 9 & 7 \\ 2 & 5 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 2 & -3 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$

[151] $1^{1^{12}} 1^{13} 3^1 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 13 & 0 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 9 & 10 & 0 & 13 & 10 & 74 & 16 \\ 2 & 2 & -1 & 0 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

[152] $1^{1^{13}} 31^{11} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 6 & 2 \\ 2 & 4 & 4 & 0 & 9 & 17 & 4 \\ 1 & 4 & 0 & 0 & 6 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 30 & 7 \\ 6 & 17 & 13 & 0 & 30 & 60 & 14 \\ 2 & 4 & 2 & -3 & 7 & 14 & 1 \end{pmatrix}_{b_0}$$

[153] $1^{1^{11}} 1^{13} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -278, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 2 & 0 \\ 4 & 6 & 0 & 3 & 16 & 9 & 6 \\ 2 & 3 & 0 & 2 & 9 & 3 & 3 \\ 2 & 2 & -1 & 0 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

[154] $1^{1^{11}} 1^{12} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 3 & 0 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 4 & 5 & 0 & 3 & 5 & 11 & 5 \\ 2 & 2 & -1 & 0 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[155] $1^{1^{12}} 1^{13} 3^1 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -297, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 2 & 0 \\ 7 & 8 & 0 & 9 & 40 & 13 & 12 \\ 2 & 3 & 0 & 2 & 13 & 3 & 3 \\ 2 & 2 & -1 & 0 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

[156] $1^{1^{11}} 3^1 1^{12} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 18 & 4 \\ 1 & 4 & 0 & 0 & 3 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 11 & 3 \\ 6 & 18 & 11 & 0 & 11 & 55 & 13 \\ 2 & 4 & 2 & -3 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

[157] $1^{1^{11}} 31^{11} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 8 & 4 \\ 1 & 4 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 8 & 4 \\ 3 & 8 & 4 & 0 & 8 & 9 & 5 \\ 2 & 4 & 2 & -3 & 4 & 5 & 0 \end{pmatrix}_{b_0}$$

[158] $1^{1^{11}} 1^{13} 3^1 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -311, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 3 & 0 \\ 4 & 6 & 0 & 3 & 16 & 15 & 6 \\ 3 & 5 & 0 & 3 & 15 & 11 & 5 \\ 2 & 2 & -1 & 0 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

[159] $1^{1^{11}} 31^{13} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -311, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 & 2 \\ 2 & 4 & 4 & 0 & 21 & 5 & 4 \\ 1 & 4 & 0 & 0 & 12 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 21 & 12 & 0 & 73 & 19 & 15 \\ 2 & 5 & 4 & 0 & 19 & 4 & 4 \\ 2 & 4 & 2 & -3 & 15 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[160] \ 1^{11}3^1 1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 4 & 0 & 6 & 12 & 4 \\ 1 & 4 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 12 & 4 \\ 4 & 12 & 8 & 0 & 12 & 25 & 9 \\ 2 & 4 & 2 & -3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[161] \ 1^{11}3^1 1^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 2 & 0 \\ 5 & 8 & 0 & 4 & 30 & 12 & 8 \\ 2 & 3 & 0 & 2 & 12 & 3 & 3 \\ 2 & 2 & -1 & 0 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[162] \ 1^{11}3^1 1^{11}21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 7 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 5 & 8 & 0 & 7 & 8 & 34 & 10 \\ 2 & 2 & -1 & 0 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[163] \ 1^{11}3^1 1^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 24 & 4 \\ 1 & 4 & 0 & 0 & 3 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 14 & 3 \\ 8 & 24 & 14 & 0 & 14 & 97 & 17 \\ 2 & 4 & 2 & -3 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[164] \ 1^{11}3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 6 & 0 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 5 & 6 & 0 & 6 & 6 & 20 & 8 \\ 2 & 2 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[165] \ 1^{11}4^1 31^{12}3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 26 & 4 \\ 1 & 4 & 0 & 0 & 3 & 19 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 19 & 3 \\ 9 & 26 & 19 & 0 & 19 & 135 & 21 \\ 2 & 4 & 2 & -3 & 3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[166] \ 1^{11}3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -349, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 3 & 0 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 5 & 5 & 0 & 3 & 5 & 11 & 5 \\ 2 & 2 & -1 & 0 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[167] \ 1^{11}4^1 31^{12}3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 5 & 0 \\ 2 & 4 & 0 & 2 & 6 & 18 & 4 \\ 6 & 9 & 0 & 5 & 18 & 39 & 9 \\ 2 & 2 & -1 & 0 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[168] \ 1^{11}3^1 1^{12}3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

10.7

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 & 2 \\ 2 & 4 & 4 & 0 & 15 & 5 & 4 \\ 1 & 4 & 0 & 0 & 9 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 15 & 9 & 0 & 37 & 14 & 11 \\ 2 & 5 & 4 & 0 & 14 & 4 & 4 \\ 2 & 4 & 2 & -3 & 11 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[169] \ 1^1 3^1 1^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -369, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 9 & 4 \\ 3 & 9 & 5 & 0 & 9 & 12 & 6 \\ 2 & 4 & 2 & -3 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[170] \ 1^1 3^1 1^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -374, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 14 & 7 \\ 3 & 9 & 5 & 0 & 14 & 12 & 6 \\ 2 & 4 & 2 & -3 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[171] \ 1^{12} 1^{13} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -377, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 2 & 0 \\ 5 & 6 & 0 & 3 & 16 & 9 & 6 \\ 2 & 3 & 0 & 2 & 9 & 3 & 3 \\ 2 & 2 & -1 & 0 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{11} 1^{14} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -382, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 3 & 0 \\ 5 & 8 & 0 & 4 & 30 & 20 & 8 \\ 3 & 5 & 0 & 3 & 20 & 11 & 5 \\ 2 & 2 & -1 & 0 & 8 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[173] \ 1^{14} 1^{13} 31^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 & 2 \\ 2 & 4 & 4 & 0 & 24 & 5 & 4 \\ 1 & 4 & 0 & 0 & 18 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 24 & 18 & 0 & 118 & 26 & 20 \\ 2 & 5 & 4 & 0 & 26 & 4 & 4 \\ 2 & 4 & 2 & -3 & 20 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[174] \ 1^{12} 1^{14} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -390, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 3 & 0 \\ 6 & 8 & 0 & 4 & 30 & 20 & 8 \\ 3 & 5 & 0 & 3 & 20 & 11 & 5 \\ 2 & 2 & -1 & 0 & 8 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[175] \ 1^{12} 1^{12} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 6 & 0 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 6 & 6 & 0 & 6 & 6 & 20 & 8 \\ 2 & 2 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[176] \ 1^{11} 1^{14} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -413, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 11 & 0 \\ 1 & 2 & 0 & 1 & 0 & 11 & 2 \\ 7 & 11 & 0 & 11 & 11 & 71 & 15 \\ 2 & 2 & -1 & 0 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[177] $1^{113}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 4 & 0 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 5 & 7 & 0 & 4 & 7 & 23 & 7 \\ 2 & 2 & -1 & 0 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{141}3^121^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -419$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 1 & 1 & 10 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 2 & 0 \\ 7 & 10 & 0 & 10 & 58 & 16 & 14 \\ 2 & 3 & 0 & 2 & 16 & 3 & 3 \\ 2 & 2 & -1 & 0 & 14 & 3 & 1 \end{pmatrix}_{b_0}$$

[179] $1^{131}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -423$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 2 & 0 \\ 5 & 7 & 0 & 6 & 25 & 11 & 9 \\ 2 & 3 & 0 & 2 & 11 & 3 & 3 \\ 2 & 2 & -1 & 0 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[180] $1^{113}3^121^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -436$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.71 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 10 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 7 & 9 & 0 & 10 & 9 & 51 & 13 \\ 2 & 2 & -1 & 0 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[181] $1^{113}31^{111}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -445$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 4 & 0 & 6 & 11 & 4 \\ 1 & 4 & 0 & 0 & 3 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 11 & 4 \\ 4 & 11 & 7 & 0 & 11 & 20 & 8 \\ 2 & 4 & 2 & -3 & 4 & 8 & 1 \end{pmatrix}_{b_0}$$

[182] $1^{114}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -448$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 8 & 0 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 6 & 10 & 0 & 8 & 10 & 52 & 12 \\ 2 & 2 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[183] $1^{112}31^{12}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -454$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 20 & 4 \\ 1 & 4 & 0 & 0 & 3 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 13 & 3 \\ 7 & 20 & 13 & 0 & 13 & 71 & 15 \\ 2 & 4 & 2 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

[184] $1^{142}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -462$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 1 & 1 & 9 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 3 & 0 \\ 6 & 9 & 0 & 7 & 41 & 23 & 11 \\ 3 & 5 & 0 & 3 & 23 & 11 & 5 \\ 2 & 2 & -1 & 0 & 11 & 5 & 1 \end{pmatrix}_{b_0}$$

[185] $1^{122}31^{111}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -464$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 5 & 2 \\ 2 & 4 & 4 & 0 & 12 & 14 & 4 \\ 1 & 4 & 0 & 0 & 9 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 9 & 0 & 28 & 34 & 10 \\ 5 & 14 & 10 & 0 & 34 & 37 & 11 \\ 2 & 4 & 2 & -3 & 10 & 11 & 1 \end{pmatrix}_{b_0}$$

[186] $1^{114}3^{11}21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -469, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 14 & 0 \\ 1 & 2 & 0 & 1 & 0 & 12 & 2 \\ 9 & 12 & 0 & 14 & 12 & 96 & 18 \\ 2 & 2 & -1 & 0 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[187] $1^{113}3^{11}2^{13}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 23 & 4 \\ 1 & 4 & 0 & 0 & 3 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 16 & 3 \\ 8 & 23 & 16 & 0 & 16 & 100 & 18 \\ 2 & 4 & 2 & -3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

[188] $1^{141}3^{11}21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -522, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 2 & 0 \\ 6 & 8 & 0 & 4 & 30 & 12 & 8 \\ 2 & 3 & 0 & 2 & 12 & 3 & 3 \\ 2 & 2 & -1 & 0 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

[189] $1^{113}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -541, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 7 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 6 & 8 & 0 & 7 & 8 & 34 & 10 \\ 2 & 2 & -1 & 0 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[190] $1^{131}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -542, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 2 & 0 \\ 6 & 7 & 0 & 6 & 25 & 11 & 9 \\ 2 & 3 & 0 & 2 & 11 & 3 & 3 \\ 2 & 2 & -1 & 0 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[191] $1^{114}3^{11}21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 5 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 6 & 9 & 0 & 5 & 9 & 39 & 9 \\ 2 & 2 & -1 & 0 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[192] $1^{113}3^{11}21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -557, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 4 & 0 \\ 1 & 2 & 0 & 1 & 0 & 7 & 2 \\ 6 & 7 & 0 & 4 & 7 & 23 & 7 \\ 2 & 2 & -1 & 0 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[193] $1^{141}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -574, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 2 & 0 \\ 6 & 9 & 0 & 7 & 41 & 14 & 11 \\ 2 & 3 & 0 & 2 & 14 & 3 & 3 \\ 2 & 2 & -1 & 0 & 11 & 3 & 1 \end{pmatrix}_{b_0}$$

[194] $1^{12}31^{12}3^11^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -586, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 & 2 \\ 2 & 4 & 4 & 0 & 18 & 5 & 4 \\ 1 & 4 & 0 & 0 & 12 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 18 & 12 & 0 & 58 & 18 & 14 \\ 2 & 5 & 4 & 0 & 18 & 4 & 4 \\ 2 & 4 & 2 & -3 & 14 & 4 & 1 \end{pmatrix}_{b_0}$$

[195] $1^{13}31^{12}3^11^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -596, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 & 2 \\ 2 & 4 & 4 & 0 & 21 & 5 & 4 \\ 1 & 4 & 0 & 0 & 15 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 21 & 15 & 0 & 85 & 22 & 17 \\ 2 & 5 & 4 & 0 & 22 & 4 & 4 \\ 2 & 4 & 2 & -3 & 17 & 4 & 1 \end{pmatrix}_{b_0}$$

[196] $1^{112}3^121^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -597, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 10 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 8 & 9 & 0 & 10 & 9 & 51 & 13 \\ 2 & 2 & -1 & 0 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[197] $1^{111}31^{111}3^11^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -600, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 4 & 2 \\ 2 & 4 & 4 & 0 & 9 & 11 & 4 \\ 1 & 4 & 0 & 0 & 6 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 18 & 7 \\ 4 & 11 & 7 & 0 & 18 & 20 & 8 \\ 2 & 4 & 2 & -3 & 7 & 8 & 1 \end{pmatrix}_{b_0}$$

[198] $1^{111}3^11^{111}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -601, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 4 & 2 \\ 2 & 4 & 4 & 0 & 9 & 12 & 4 \\ 1 & 4 & 0 & 0 & 6 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 20 & 7 \\ 4 & 12 & 8 & 0 & 20 & 25 & 9 \\ 2 & 4 & 2 & -3 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

[199] $1^{112}31^{13}3^11^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -601, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 26 & 4 \\ 1 & 4 & 0 & 0 & 3 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 16 & 3 \\ 9 & 26 & 16 & 0 & 16 & 117 & 19 \\ 2 & 4 & 2 & -3 & 3 & 19 & 0 \end{pmatrix}_{b_0}$$

[200] $1^{112}31^{111}3^11^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -649, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 5 & 2 \\ 2 & 4 & 4 & 0 & 9 & 14 & 4 \\ 1 & 4 & 0 & 0 & 6 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 24 & 7 \\ 5 & 14 & 10 & 0 & 24 & 37 & 11 \\ 2 & 4 & 2 & -3 & 7 & 11 & 1 \end{pmatrix}_{b_0}$$

[201] $1^{112}31^{14}3^11^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -650, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 3 & 32 & 4 \\ 1 & 4 & 0 & 0 & 3 & 19 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 19 & 3 \\ 11 & 32 & 19 & 0 & 19 & 175 & 23 \\ 2 & 4 & 2 & -3 & 3 & 23 & 0 \end{pmatrix}_{b_0}$$

[202] $1^{141}3^121^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -662, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 1 & 1 & 10 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 2 & 0 \\ 8 & 10 & 0 & 10 & 58 & 16 & 14 \\ 2 & 3 & 0 & 2 & 16 & 3 & 3 \\ 2 & 2 & -1 & 0 & 14 & 3 & 1 \end{pmatrix}_{b_0}$$

[203] $1^{1^{12}} 1^{113} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -692$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 7 & 0 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 7 & 8 & 0 & 7 & 8 & 34 & 10 \\ 2 & 2 & -1 & 0 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[204] $1^{1^{12}} 1^{114} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -720$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.71 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 14 & 0 \\ 1 & 2 & 0 & 1 & 0 & 12 & 2 \\ 10 & 12 & 0 & 14 & 12 & 96 & 18 \\ 2 & 2 & -1 & 0 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[205] $1^{1^{11}} 1^{114} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -724$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.68 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 11 & 0 \\ 1 & 2 & 0 & 1 & 0 & 11 & 2 \\ 8 & 11 & 0 & 11 & 11 & 71 & 15 \\ 2 & 2 & -1 & 0 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[206] $1^{1^{11}} 1^{114} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -749$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.71 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 8 & 0 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 7 & 10 & 0 & 8 & 10 & 52 & 12 \\ 2 & 2 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{1^{12}} 1^{114} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -765$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 5 & 0 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 7 & 9 & 0 & 5 & 9 & 39 & 9 \\ 2 & 2 & -1 & 0 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[208] $1^{1^{12}} 1^{141} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -797$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 2 & 0 \\ 7 & 9 & 0 & 7 & 41 & 14 & 11 \\ 2 & 3 & 0 & 2 & 14 & 3 & 3 \\ 2 & 2 & -1 & 0 & 11 & 3 & 1 \end{pmatrix}_{b_0}$$

[209] $1^{1^{21}} 3^1 1^{13} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -821$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 & 2 \\ 2 & 4 & 4 & 0 & 24 & 5 & 4 \\ 1 & 4 & 0 & 0 & 15 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 24 & 15 & 0 & 100 & 23 & 18 \\ 2 & 5 & 4 & 0 & 23 & 4 & 4 \\ 2 & 4 & 2 & -3 & 18 & 4 & 1 \end{pmatrix}_{b_0}$$

[210] $1^{1^{13}} 3^1 1^{13} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -871$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 29 & 4 \\ 1 & 4 & 0 & 0 & 3 & 19 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 19 & 3 \\ 10 & 29 & 19 & 0 & 19 & 152 & 22 \\ 2 & 4 & 2 & -3 & 3 & 22 & 1 \end{pmatrix}_{b_0}$$

[211] $1^{12}31^{14}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -926, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 & 2 \\ 2 & 4 & 4 & 0 & 30 & 5 & 4 \\ 1 & 4 & 0 & 0 & 18 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 30 & 18 & 0 & 154 & 28 & 22 \\ 2 & 5 & 4 & 0 & 28 & 4 & 4 \\ 2 & 4 & 2 & -3 & 22 & 4 & 1 \end{pmatrix}_{b_0}$$

[212] $1^{114}31^{13}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -929, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 3 & 32 & 4 \\ 1 & 4 & 0 & 0 & 3 & 22 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 22 & 3 \\ 11 & 32 & 22 & 0 & 22 & 193 & 25 \\ 2 & 4 & 2 & -3 & 3 & 25 & 1 \end{pmatrix}_{b_0}$$

[213] $1^{112}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -965, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 11 & 0 \\ 1 & 2 & 0 & 1 & 0 & 11 & 2 \\ 9 & 11 & 0 & 11 & 11 & 71 & 15 \\ 2 & 2 & -1 & 0 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[214] $1^{112}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -980, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 8 & 0 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 8 & 10 & 0 & 8 & 10 & 52 & 12 \\ 2 & 2 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[215] $1^{131}31^{13}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 & 2 \\ 2 & 4 & 4 & 0 & 27 & 5 & 4 \\ 1 & 4 & 0 & 0 & 18 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 27 & 18 & 0 & 133 & 27 & 21 \\ 2 & 5 & 4 & 0 & 27 & 4 & 4 \\ 2 & 4 & 2 & -3 & 21 & 4 & 1 \end{pmatrix}_{b_0}$$

[216] $1^{113}31^{14}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 3 & 35 & 4 \\ 1 & 4 & 0 & 0 & 3 & 22 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 22 & 3 \\ 12 & 35 & 22 & 0 & 22 & 216 & 26 \\ 2 & 4 & 2 & -3 & 3 & 26 & 1 \end{pmatrix}_{b_0}$$

[217] $1^{141}31^{13}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 & 2 \\ 2 & 4 & 4 & 0 & 30 & 5 & 4 \\ 1 & 4 & 0 & 0 & 21 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 30 & 21 & 0 & 172 & 31 & 24 \\ 2 & 5 & 4 & 0 & 31 & 4 & 4 \\ 2 & 4 & 2 & -3 & 24 & 4 & 1 \end{pmatrix}_{b_0}$$

[218] $1^{114}31^{14}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 3 & 38 & 4 \\ 1 & 4 & 0 & 0 & 3 & 25 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 25 & 3 \\ 13 & 38 & 25 & 0 & 25 & 263 & 29 \\ 2 & 4 & 2 & -3 & 3 & 29 & 1 \end{pmatrix}_{b_0}$$

[219] $1^{131}31^{14}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

11.2

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 & 2 \\ 2 & 4 & 4 & 0 & 33 & 5 & 4 \\ 1 & 4 & 0 & 0 & 21 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 33 & 21 & 0 & 193 & 32 & 25 \\ 2 & 5 & 4 & 0 & 32 & 4 & 4 \\ 2 & 4 & 2 & -3 & 25 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[220] 1^{141} 31^{143} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1846, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 & 2 \\ 2 & 4 & 4 & 0 & 36 & 5 & 4 \\ 1 & 4 & 0 & 0 & 24 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 36 & 24 & 0 & 238 & 36 & 28 \\ 2 & 5 & 4 & 0 & 36 & 4 & 4 \\ 2 & 4 & 2 & -3 & 28 & 4 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [221] 1^{12} 23^1 1^{11} 31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -61, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 1 & 1 & 21 & 3 \\ 5 & 13 & 1 & 0 & 54 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 21 & 54 & 0 & 0 & 217 & 27 \\ 3 & 7 & -1 & 0 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[222] 1^{13} 23^1 1^{12} 31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -61, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 1 & 1 & 32 & 3 \\ 5 & 13 & 1 & 0 & 84 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 32 & 84 & 0 & 0 & 520 & 42 \\ 3 & 7 & -1 & 0 & 42 & 3 \end{pmatrix}_{b_0}$$

$$[223] 1^{11} 4^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 17, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 3 & 0 & 1 & 2 & 3 \\ 3 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -2 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[224] 1^{11} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 17, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[225] 1^{11} 4^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 3 & 0 & 1 & 3 & 3 \\ 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[226] 1^{12} 23^1 1^{11} 31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -61, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 1 & 1 & 22 & 3 \\ 5 & 13 & 1 & 0 & 54 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 22 & 54 & 0 & 0 & 217 & 27 \\ 3 & 7 & -1 & 0 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[227] 1^{121} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 6 & 6 \\ 3 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -2 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[228] 1^{122} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 34, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$$

10.7

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 8 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -2 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[229] \ 1^{131} 4 \ 13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 12 & 13 \\ 3 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -2 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[230] \ 1^{112} 4 \ 13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 46, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 12 & 7 \\ 5 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -2 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[231] \ 1^{111} 4 \ 13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 57, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 6 & 7 \\ 3 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -2 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[232] \ 1^{111} 4 \ 1^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 8 & 3 \\ 9 & 12 & 0 & 8 & 26 & 10 \\ 3 & 4 & -2 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[233] \ 1^{121} 4 \ 13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 65, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 9 & 10 \\ 3 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -2 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[235] \ 1^{121} 4 \ 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 0 & 11 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 0 & 21 & 17 & 9 \\ 6 & 9 & 0 & 17 & 13 & 7 \\ 3 & 4 & -2 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[234] \ 1^{111} 4 \ 13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[236] \ 1^{122} 4 \ 13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 18 & 10 \\ 5 & 8 & 0 & 18 & 10 & 6 \\ 3 & 4 & -2 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[237] $1^{122}4^{13^1}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 5 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -2 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[238] $1^{112}4^{1^1 3^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 110, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 7 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 8 & 10 & 6 \\ 5 & 8 & 0 & 10 & 10 & 6 \\ 3 & 4 & -2 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[239] $1^{112}4^{1^1 3^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 7 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 8 & 16 & 6 \\ 8 & 13 & 0 & 16 & 28 & 10 \\ 3 & 4 & -2 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[240] $1^{111}4^{1^1 3^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 115, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 8 & 11 & 6 \\ 6 & 9 & 0 & 11 & 13 & 7 \\ 3 & 4 & -2 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[241] $1^{111}4^{1^1 3^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 133, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[242] $1^{111}4^{1^1 3^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 137, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 8 & 5 & 6 \\ 3 & 4 & 0 & 5 & 1 & 3 \\ 3 & 4 & -2 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[243] $1^{112}4^{1^1 3^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 5 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -2 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[244] $1^{112}4^{1^1 3^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 8 & 3 \\ 8 & 11 & 0 & 8 & 21 & 9 \\ 3 & 4 & -2 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[245] $1^{111}4^{1^1 3^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 176, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 0 & 26 & 8 & 10 \\ 3 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -2 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[246] $1^{122}1^{13^11^1}4^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 176, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 5 & 3 \\ 3 & 4 & 0 & 16 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 16 & 0 & 45 & 22 & 13 \\ 5 & 8 & 0 & 22 & 10 & 6 \\ 3 & 4 & -2 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[247] $1^{122}1^{13^11^1}4^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 203, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 8 & 3 \\ 3 & 4 & 0 & 11 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 0 & 21 & 25 & 9 \\ 8 & 13 & 0 & 25 & 28 & 10 \\ 3 & 4 & -2 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[248] $1^{113}4^{13^11^1}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 208, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 7 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -2 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[249] $1^{121}1^{13^11^1}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 225, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 11 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 0 & 21 & 8 & 9 \\ 3 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -2 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[250] $1^{122}1^{13^11^1}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 254, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 11 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 0 & 21 & 16 & 9 \\ 5 & 8 & 0 & 16 & 10 & 6 \\ 3 & 4 & -2 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[251] $1^{114}4^{13^11^1}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 272, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 9 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[252] $1^{112}4^113^{11^1}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 277, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 8 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[253] $1^{131}1^{13^11^1}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 281, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 15 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 15 & 0 & 40 & 11 & 12 \\ 3 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & -2 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[254] $1^{141}1^{13^11^1}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 305, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 19 & 0 & 65 & 14 & 15 \\ 3 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -2 & 15 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[255] 1^{112} 1^{13^1 1^1} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 320, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 11 & 3 \\ 11 & 16 & 0 & 11 & 45 & 13 \\ 3 & 4 & -2 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[256] 1^{121} 1^{13^1 1^1} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 320, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 16 & 0 & 45 & 11 & 13 \\ 3 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & -2 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[257] 1^{113} 1^{13^1 1^1} 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 349, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 11 & 3 \\ 10 & 15 & 0 & 11 & 40 & 12 \\ 3 & 4 & -2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[258] 1^{113} 1^{11} 4^1 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 421, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 10 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[259] 1^{131} 1^{13^1 1^1} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 432, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 & 3 \\ 3 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 20 & 0 & 70 & 14 & 16 \\ 3 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -2 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[260] 1^{141} 1^{13^1 1^1} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 24 & 0 & 101 & 17 & 19 \\ 3 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & -2 & 19 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[261] 1^{114} 1^{13^1 1^1} 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 525, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 12 & 19 & 0 & 14 & 65 & 15 \\ 3 & 4 & -2 & 3 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[262] 1^{114} 1^{11} 4^1 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 565, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 12 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[263] \ 1^{113} 1^{131} 1^{11} 4 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -79, \ \det = 576, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.60 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 13 & 20 & 0 & 14 & 70 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[264] \ 1^{114} 1^{131} 1^{11} 4 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -79, \ \det = 832, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 17 & 3 \\ 15 & 24 & 0 & 17 & 101 & 19 \\ 3 & 4 & -2 & 3 & 19 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [265] \ 1^{111} 31^{14} 13^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 13, \ \Delta = -58, \ \det = -16, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 33 & 4 \\ 1 & 4 & 0 & 0 & 33 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 \\ 1 & 3 & 3 & 0 & 13 & 3 \\ 2 & 3 & 3 & 0 & 31 & 3 \\ 2 & 4 & 2 & -2 & 33 & 2 \end{pmatrix}_{b_0}$$

$$[266] \ 1^{111} 31^{14} 13^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 13, \ \Delta = -58, \ \det = -28, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 14 & 2 \\ 2 & 4 & 4 & 0 & 38 & 4 \\ 1 & 4 & 0 & 0 & 36 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 \\ 1 & 3 & 3 & 0 & 16 & 3 \\ 4 & 8 & 6 & 0 & 613 & 7 \\ 2 & 4 & 2 & -2 & 37 & 2 \end{pmatrix}_{b_0}$$

$$[267] \ 1^{112} 31^{11} 41^{13} 1^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 13, \ \Delta = -58, \ \det = -48, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 17 & 2 \\ 2 & 4 & 4 & 0 & 314 & 4 \\ 1 & 4 & 0 & 0 & 310 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 \\ 1 & 3 & 3 & 0 & 110 & 3 \\ 7 & 14 & 10 & 0 & 1038 & 12 \\ 2 & 4 & 2 & -2 & 312 & 2 \end{pmatrix}_{b_0}$$

$$[268] \ 1^{141} 31^{11} 41^{13} 1^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 13, \ \Delta = -58, \ \det = -96, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 20 & 3 \\ 1 & 4 & 0 & 0 & 16 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 \\ 8 & 20 & 16 & 0 & 90 & 18 \\ 2 & 3 & 3 & 0 & 16 & 1 \\ 2 & 4 & 2 & -2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[269] \ 1^{111} 31^{11} 41^{13} 1^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 13, \ \Delta = -58, \ \det = -96, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 113 & 4 \\ 1 & 4 & 0 & 0 & 73 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 \\ 5 & 11 & 7 & 0 & 217 & 9 \\ 2 & 3 & 3 & 0 & 71 & 3 \\ 2 & 4 & 2 & -2 & 93 & 2 \end{pmatrix}_{b_0}$$

$$[270] \ 1^{114} 31^{11} 41^{13} 1^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 13, \ \Delta = -58, \ \det = -112, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 320 & 4 \\ 1 & 4 & 0 & 0 & 316 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 \\ 1 & 3 & 3 & 0 & 116 & 3 \\ 9 & 20 & 16 & 0 & 1690 & 18 \\ 2 & 4 & 2 & -2 & 318 & 2 \end{pmatrix}_{b_0}$$

$$[271] \ 1^{113} 31^{11} 41^{13} 1^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 13, \ \Delta = -58, \ \det = -160, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 17 & 4 \\ 1 & 4 & 0 & 0 & 3 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 3 & 3 & 0 & 1 & 13 & 3 \\ 8 & 17 & 13 & 0 & 13 & 61 & 15 \\ 2 & 4 & 2 & -2 & 3 & 15 & 2 \end{pmatrix}_{b_0}$$

[272] $1^{121}3^{11}4^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -160, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 & 2 \\ 2 & 4 & 4 & 0 & 14 & 3 & 4 \\ 1 & 4 & 0 & 0 & 10 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 14 & 10 & 0 & 38 & 10 & 12 \\ 2 & 3 & 3 & 0 & 10 & 1 & 3 \\ 2 & 4 & 2 & -2 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[273] $1^{131}3^{11}4^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -160, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 & 2 \\ 2 & 4 & 4 & 0 & 17 & 3 & 4 \\ 1 & 4 & 0 & 0 & 13 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 17 & 13 & 0 & 61 & 13 & 15 \\ 2 & 3 & 3 & 0 & 13 & 1 & 3 \\ 2 & 4 & 2 & -2 & 15 & 3 & 2 \end{pmatrix}_{b_0}$$

4.480 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}2\mathfrak{n}3\mathfrak{mix} + \mathfrak{so}_8$

$T = 9$ [0] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9, T_{\max} = 21, \Delta = -108, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 15 & 5 \\ 11 & 20 & 0 & 17 & 28 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 24 & 7 \\ 15 & 28 & 0 & 24 & 37 & 11 \\ 5 & 8 & -3 & 7 & 11 & 1 \end{pmatrix}_{b_0}$$

4.481 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}2\mathfrak{n}3\mathfrak{mix} + \mathfrak{su}_2$

$T = 9$ [0] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 0 & 5 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 4 & 7 & 0 & 6 & 7 & 5 & 5 \\ 2 & 4 & 0 & 3 & 5 & 1 & 3 \\ 3 & 4 & -3 & 4 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 7 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 10 & 12 & 6 \\ 4 & 7 & 0 & 10 & 7 & 11 & 5 \\ 5 & 9 & 0 & 12 & 11 & 13 & 7 \\ 3 & 4 & -3 & 6 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{122}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 2 & 3 \\ 3 & 4 & 0 & 8 & 16 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 22 & 6 & 6 \\ 9 & 16 & 0 & 22 & 44 & 11 & 12 \\ 2 & 4 & 0 & 6 & 11 & 1 & 3 \\ 3 & 4 & -3 & 6 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{131}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 4 & 2 & 3 \\ 3 & 4 & 0 & 12 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 15 & 9 & 9 \\ 4 & 7 & 0 & 15 & 7 & 5 & 5 \\ 2 & 4 & 0 & 9 & 5 & 1 & 3 \\ 3 & 4 & -3 & 9 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overset{1^1}{111}3}1^{\overset{1^1}{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -146, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 0 & 5 & 7 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 9 & 4 \\ 4 & 7 & 0 & 6 & 7 & 11 & 5 \\ 5 & 9 & 0 & 9 & 11 & 13 & 7 \\ 3 & 4 & -3 & 4 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overset{1^1}{113}3}1^{\overset{1^1}{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -146, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.70 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 0 & 5 & 7 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 15 & 4 \\ 4 & 7 & 0 & 6 & 7 & 21 & 5 \\ 9 & 17 & 0 & 15 & 21 & 49 & 13 \\ 3 & 4 & -3 & 4 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overset{1^1}{122}3}1^{\overset{1^1}{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -150, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 9 & 3 \\ 3 & 4 & 0 & 8 & 16 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 22 & 12 & 6 \\ 9 & 16 & 0 & 22 & 44 & 26 & 12 \\ 5 & 9 & 0 & 12 & 26 & 13 & 7 \\ 3 & 4 & -3 & 6 & 12 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overset{1^1}{1212}3}1^{\overset{1^1}{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -156, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 7 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 10 & 12 & 6 \\ 4 & 7 & 0 & 10 & 7 & 10 & 5 \\ 4 & 8 & 0 & 12 & 10 & 10 & 6 \\ 3 & 4 & -3 & 6 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overset{1^1}{1311}3}1^{\overset{1^1}{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -164, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 7 & 2 & 3 \\ 3 & 4 & 0 & 17 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 36 & 12 & 13 \\ 7 & 12 & 0 & 36 & 25 & 8 & 9 \\ 2 & 4 & 0 & 12 & 8 & 1 & 3 \\ 3 & 4 & -3 & 13 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overset{1^1}{1121}3}1^{\overset{1^1}{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 5 & 3 \\ 3 & 4 & 0 & 9 & 11 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 17 & 15 & 7 \\ 6 & 11 & 0 & 17 & 20 & 17 & 8 \\ 5 & 9 & 0 & 15 & 17 & 13 & 7 \\ 3 & 4 & -3 & 7 & 8 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overset{1^1}{111}3}1^{\overset{1^1}{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -194, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 0 & 5 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 12 & 3 & 4 \\ 7 & 12 & 0 & 12 & 25 & 8 & 9 \\ 2 & 4 & 0 & 3 & 8 & 1 & 3 \\ 3 & 4 & -3 & 4 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overset{1^1}{113}3}1^{\overset{1^1}{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -194, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 0 & 5 & 20 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 18 & 3 & 4 \\ 11 & 20 & 0 & 18 & 69 & 14 & 15 \\ 2 & 4 & 0 & 3 & 14 & 1 & 3 \\ 3 & 4 & -3 & 4 & 15 & 3 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{1211}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{red}{f}_4+\textcolor{red}{su}_2+\textcolor{red}{su}_2+\textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 10 & 6 & 6 \\ 4 & 7 & 0 & 10 & 7 & 5 & 5 \\ 2 & 4 & 0 & 6 & 5 & 1 & 3 \\ 3 & 4 & -3 & 6 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{1121}3^11^{111}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{red}{f}_4+\textcolor{red}{su}_2+\textcolor{red}{su}_2+\textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 0 & 5 & 7 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 12 & 4 \\ 4 & 7 & 0 & 6 & 7 & 16 & 5 \\ 7 & 13 & 0 & 12 & 16 & 28 & 10 \\ 3 & 4 & -3 & 4 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{1121}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{red}{f}_4+\textcolor{red}{su}_2+\textcolor{red}{su}_2+\textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -246, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 2 & 3 \\ 3 & 4 & 0 & 9 & 11 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 17 & 6 & 7 \\ 6 & 11 & 0 & 17 & 20 & 8 & 8 \\ 2 & 4 & 0 & 6 & 8 & 1 & 3 \\ 3 & 4 & -3 & 7 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{1211}3^11^{111}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{red}{f}_4+\textcolor{red}{su}_2+\textcolor{red}{su}_2+\textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 0 & 5 & 16 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 15 & 3 & 4 \\ 9 & 16 & 0 & 15 & 44 & 11 & 12 \\ 2 & 4 & 0 & 3 & 11 & 1 & 3 \\ 3 & 4 & -3 & 4 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{1111}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{red}{f}_4+\textcolor{red}{su}_2+\textcolor{red}{su}_2+\textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 8 & 3 & 3 \\ 7 & 12 & 0 & 8 & 25 & 8 & 9 \\ 2 & 4 & 0 & 3 & 8 & 1 & 3 \\ 3 & 4 & -3 & 3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{1111}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{red}{f}_4+\textcolor{red}{su}_2+\textcolor{red}{su}_2+\textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 3 & 3 \\ 4 & 7 & 0 & 5 & 7 & 5 & 5 \\ 2 & 4 & 0 & 3 & 5 & 1 & 3 \\ 3 & 4 & -3 & 3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{1111}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{red}{f}_4+\textcolor{red}{su}_2+\textcolor{red}{su}_2+\textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 5 & 3 \\ 3 & 4 & 0 & 4 & 7 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 6 & 3 \\ 4 & 7 & 0 & 5 & 7 & 11 & 5 \\ 5 & 9 & 0 & 6 & 11 & 13 & 7 \\ 3 & 4 & -3 & 3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{1212}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{red}{f}_4+\textcolor{red}{su}_2+\textcolor{red}{su}_2+\textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -326, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 7 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 10 & 18 & 6 \\ 4 & 7 & 0 & 10 & 7 & 16 & 5 \\ 7 & 13 & 0 & 18 & 16 & 28 & 10 \\ 3 & 4 & -3 & 6 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{11221}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 6 & 2 & 3 \\ 3 & 4 & 0 & 8 & 11 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 16 & 6 & 6 \\ 6 & 11 & 0 & 16 & 20 & 8 & 8 \\ 2 & 4 & 0 & 6 & 8 & 1 & 3 \\ 3 & 4 & -3 & 6 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{1112}3^11^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 10 & 3 \\ 3 & 4 & 0 & 4 & 7 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 12 & 3 \\ 4 & 7 & 0 & 5 & 7 & 22 & 5 \\ 10 & 18 & 0 & 12 & 22 & 58 & 14 \\ 3 & 4 & -3 & 3 & 5 & 14 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{1121}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 2 & 3 \\ 3 & 4 & 0 & 4 & 11 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 8 & 3 & 3 \\ 6 & 11 & 0 & 8 & 20 & 8 & 8 \\ 2 & 4 & 0 & 3 & 8 & 1 & 3 \\ 3 & 4 & -3 & 3 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{1122}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 7 & 3 \\ 3 & 4 & 0 & 9 & 11 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 17 & 21 & 7 \\ 6 & 11 & 0 & 17 & 20 & 25 & 8 \\ 7 & 13 & 0 & 21 & 25 & 28 & 10 \\ 3 & 4 & -3 & 7 & 8 & 10 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{1111}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 5 & 3 \\ 3 & 4 & 0 & 9 & 7 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 11 & 15 & 7 \\ 4 & 7 & 0 & 11 & 7 & 11 & 5 \\ 5 & 9 & 0 & 15 & 11 & 13 & 7 \\ 3 & 4 & -3 & 7 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{1113}3^11^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 15 & 3 \\ 3 & 4 & 0 & 4 & 7 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 18 & 3 \\ 4 & 7 & 0 & 5 & 7 & 33 & 5 \\ 15 & 27 & 0 & 18 & 33 & 133 & 21 \\ 3 & 4 & -3 & 3 & 5 & 21 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{1121}3^11^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 2 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 14 & 3 & 3 \\ 12 & 21 & 0 & 14 & 80 & 14 & 16 \\ 2 & 4 & 0 & 3 & 14 & 1 & 3 \\ 3 & 4 & -3 & 3 & 16 & 3 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{1112}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -514, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 7 & 3 \\ 3 & 4 & 0 & 4 & 12 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 8 & 9 & 3 \\ 7 & 12 & 0 & 8 & 25 & 28 & 9 \\ 7 & 13 & 0 & 9 & 28 & 28 & 10 \\ 3 & 4 & -3 & 3 & 9 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{\overline{11113}} 3^{\overline{1}} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -524, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 5 & 3 \\ 3 & 4 & 0 & 4 & 12 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 8 & 6 & 3 \\ 7 & 12 & 0 & 8 & 25 & 20 & 9 \\ 5 & 9 & 0 & 6 & 20 & 13 & 7 \\ 3 & 4 & -3 & 3 & 9 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{\overline{11123}} 3^{\overline{1}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -536, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 7 & 3 \\ 3 & 4 & 0 & 4 & 7 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 9 & 3 \\ 4 & 7 & 0 & 5 & 7 & 16 & 5 \\ 7 & 13 & 0 & 9 & 16 & 28 & 10 \\ 3 & 4 & -3 & 3 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{\overline{11143}} 3^{\overline{1}} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -546, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.66 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 11 & 3 \\ 3 & 4 & 0 & 9 & 7 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 11 & 33 & 7 \\ 4 & 7 & 0 & 11 & 7 & 26 & 5 \\ 11 & 21 & 0 & 33 & 26 & 76 & 16 \\ 3 & 4 & -3 & 7 & 5 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{\overline{11223}} 3^{\overline{1}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -546, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 7 & 3 \\ 3 & 4 & 0 & 4 & 11 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 8 & 9 & 3 \\ 6 & 11 & 0 & 8 & 20 & 25 & 8 \\ 7 & 13 & 0 & 9 & 25 & 28 & 10 \\ 3 & 4 & -3 & 3 & 8 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{\overline{11313}} 3^{\overline{1}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -564, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.81 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 4 & 15 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 11 & 3 & 3 \\ 8 & 15 & 0 & 11 & 39 & 11 & 11 \\ 2 & 4 & 0 & 3 & 11 & 1 & 3 \\ 3 & 4 & -3 & 3 & 11 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{\overline{11143}} 3^{\overline{1}} 1^{\overline{14}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -616, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 20 & 3 \\ 3 & 4 & 0 & 4 & 7 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 24 & 3 \\ 4 & 7 & 0 & 5 & 7 & 44 & 5 \\ 20 & 36 & 0 & 24 & 44 & 238 & 28 \\ 3 & 4 & -3 & 3 & 5 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{\overline{11121}} 3^{\overline{1}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -620, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 11 & 3 & 3 \\ 9 & 16 & 0 & 11 & 44 & 11 & 12 \\ 2 & 4 & 0 & 3 & 11 & 1 & 3 \\ 3 & 4 & -3 & 3 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{\overline{11123}} 3^{\overline{1}} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -626, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 7 & 3 \\ 3 & 4 & 0 & 9 & 7 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 11 & 21 & 7 \\ 4 & 7 & 0 & 11 & 7 & 16 & 5 \\ 7 & 13 & 0 & 21 & 16 & 28 & 10 \\ 3 & 4 & -3 & 7 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{\overline{1113}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -656, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 9 & 3 \\ 3 & 4 & 0 & 9 & 7 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 11 & 27 & 7 \\ 4 & 7 & 0 & 11 & 7 & 21 & 5 \\ 9 & 17 & 0 & 27 & 21 & 49 & 13 \\ 3 & 4 & -3 & 7 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{\overline{1113}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -716, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 9 & 3 \\ 3 & 4 & 0 & 4 & 7 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 12 & 3 \\ 4 & 7 & 0 & 5 & 7 & 21 & 5 \\ 9 & 17 & 0 & 12 & 21 & 49 & 13 \\ 3 & 4 & -3 & 3 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{\overline{1141}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -720, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 14 & 3 & 3 \\ 10 & 19 & 0 & 14 & 64 & 14 & 14 \\ 2 & 4 & 0 & 3 & 14 & 1 & 3 \\ 3 & 4 & -3 & 3 & 14 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{\overline{1121}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -810, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 5 & 3 \\ 3 & 4 & 0 & 4 & 16 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 11 & 6 & 3 \\ 9 & 16 & 0 & 11 & 44 & 26 & 12 \\ 5 & 9 & 0 & 6 & 26 & 13 & 7 \\ 3 & 4 & -3 & 3 & 12 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{\overline{1113}} 3^{\overline{1}} 1^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -836, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 12 & 3 \\ 3 & 4 & 0 & 4 & 7 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 15 & 3 \\ 4 & 7 & 0 & 5 & 7 & 27 & 5 \\ 12 & 22 & 0 & 15 & 27 & 85 & 17 \\ 3 & 4 & -3 & 3 & 5 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{\overline{1131}} 3^{\overline{1}} 1^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -840, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 2 & 3 \\ 3 & 4 & 0 & 4 & 30 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 20 & 3 & 3 \\ 17 & 30 & 0 & 20 & 165 & 20 & 23 \\ 2 & 4 & 0 & 3 & 20 & 1 & 3 \\ 3 & 4 & -3 & 3 & 23 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{\overline{1114}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -856, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 11 & 3 \\ 3 & 4 & 0 & 4 & 7 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 15 & 3 \\ 4 & 7 & 0 & 5 & 7 & 26 & 5 \\ 11 & 21 & 0 & 15 & 26 & 76 & 16 \\ 3 & 4 & -3 & 3 & 5 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{\overline{1212}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -896, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 7 & 3 \\ 3 & 4 & 0 & 13 & 7 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 16 & 30 & 10 \\ 4 & 7 & 0 & 16 & 7 & 16 & 5 \\ 7 & 13 & 0 & 30 & 16 & 28 & 10 \\ 3 & 4 & -3 & 10 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{11313}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -964, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 5 & 3 \\ 3 & 4 & 0 & 4 & 20 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 14 & 6 & 3 \\ 11 & 20 & 0 & 14 & 69 & 32 & 15 \\ 5 & 9 & 0 & 6 & 32 & 13 & 7 \\ 3 & 4 & -3 & 3 & 15 & 7 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{11313}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -984, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 14 & 3 & 3 \\ 11 & 20 & 0 & 14 & 69 & 14 & 15 \\ 2 & 4 & 0 & 3 & 14 & 1 & 3 \\ 3 & 4 & -3 & 3 & 15 & 3 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{11413}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -986, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 2 & 3 \\ 3 & 4 & 0 & 9 & 24 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 38 & 6 & 7 \\ 13 & 24 & 0 & 38 & 100 & 17 & 18 \\ 2 & 4 & 0 & 6 & 17 & 1 & 3 \\ 3 & 4 & -3 & 7 & 18 & 3 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{11223}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -1020, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 7 & 3 \\ 3 & 4 & 0 & 4 & 16 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 11 & 9 & 3 \\ 9 & 16 & 0 & 11 & 44 & 37 & 12 \\ 7 & 13 & 0 & 9 & 37 & 28 & 10 \\ 3 & 4 & -3 & 3 & 12 & 10 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{11313}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -1076, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 4 & 0 & 4 & 25 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 17 & 3 & 3 \\ 14 & 25 & 0 & 17 & 111 & 17 & 19 \\ 2 & 4 & 0 & 3 & 17 & 1 & 3 \\ 3 & 4 & -3 & 3 & 19 & 3 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{11143}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -1176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 17 & 3 \\ 3 & 4 & 0 & 4 & 7 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 21 & 3 \\ 4 & 7 & 0 & 5 & 7 & 38 & 5 \\ 17 & 31 & 0 & 21 & 38 & 172 & 24 \\ 3 & 4 & -3 & 3 & 5 & 24 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{11143}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -1256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 14 & 3 \\ 3 & 4 & 0 & 4 & 7 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 18 & 3 \\ 4 & 7 & 0 & 5 & 7 & 32 & 5 \\ 14 & 26 & 0 & 18 & 32 & 118 & 20 \\ 3 & 4 & -3 & 3 & 5 & 20 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{11143}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -1296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 22 & 2 & 3 \\ 3 & 4 & 0 & 4 & 39 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 26 & 3 & 3 \\ 22 & 39 & 0 & 26 & 280 & 26 & 30 \\ 2 & 4 & 0 & 3 & 26 & 1 & 3 \\ 3 & 4 & -3 & 3 & 30 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{11413^1 1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1356, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 2 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 17 & 3 & 3 \\ 13 & 24 & 0 & 17 & 100 & 17 & 18 \\ 2 & 4 & 0 & 3 & 17 & 1 & 3 \\ 3 & 4 & -3 & 3 & 18 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{11413^1 1^{13}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1644, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 19 & 2 & 3 \\ 3 & 4 & 0 & 4 & 34 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 23 & 3 & 3 \\ 19 & 34 & 0 & 23 & 208 & 23 & 26 \\ 2 & 4 & 0 & 3 & 23 & 1 & 3 \\ 3 & 4 & -3 & 3 & 26 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{11413^1 1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 2 & 3 \\ 3 & 4 & 0 & 4 & 29 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 20 & 3 & 3 \\ 16 & 29 & 0 & 20 & 148 & 20 & 22 \\ 2 & 4 & 0 & 3 & 20 & 1 & 3 \\ 3 & 4 & -3 & 3 & 22 & 3 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [55] \quad 1^{11131^{11} 3^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 5 & 6 \\ 1 & 4 & 0 & 0 & 3 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 \\ 2 & 6 & 3 & 0 & 4 & 5 & 6 \\ 2 & 5 & 4 & 0 & 5 & 4 & 5 \\ 2 & 6 & 3 & 0 & 6 & 5 & 4 \\ 2 & 4 & 2 & -3 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{111131^{11} 3^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 172, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 \\ 2 & 4 & 4 & 0 & 3 & 5 & 9 \\ 1 & 4 & 0 & 0 & 3 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 \\ 1 & 3 & 3 & 0 & 1 & 4 & 6 \\ 2 & 5 & 4 & 0 & 4 & 4 & 9 \\ 3 & 9 & 6 & 0 & 6 & 9 & 13 \\ 2 & 4 & 2 & -3 & 3 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{111131^{11} 3^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 262, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 3 \\ 2 & 4 & 4 & 0 & 6 & 5 & 9 \\ 1 & 4 & 0 & 0 & 3 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 \\ 2 & 6 & 3 & 0 & 4 & 5 & 9 \\ 2 & 5 & 4 & 0 & 5 & 4 & 9 \\ 3 & 9 & 6 & 0 & 9 & 9 & 13 \\ 2 & 4 & 2 & -3 & 4 & 4 & 7 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{111131^{11} 3^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 752, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 3 \\ 2 & 4 & 4 & 0 & 9 & 5 & 9 \\ 1 & 4 & 0 & 0 & 6 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 \\ 3 & 9 & 6 & 0 & 13 & 9 & 15 \\ 2 & 5 & 4 & 0 & 9 & 4 & 9 \\ 3 & 9 & 6 & 0 & 15 & 9 & 13 \\ 2 & 4 & 2 & -3 & 7 & 4 & 7 \end{pmatrix}_{b_0}$$

4.482 NHC(-2)(-3) + f₄ + su₂ + su₂n₃mix + su₂→(-3)

$$T = 10 \quad [0] \quad 1^{11113^1 1^1 3^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -60, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 5 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 4 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 4 & 2 & 3 \\ 2 & 5 & 4 & 0 & 4 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & -3 & 3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 175, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 5 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 4 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 5 & 3 & 4 \\ 2 & 5 & 4 & 0 & 5 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 3 & 3 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 475, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 5 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 4 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 9 & 5 & 7 \\ 2 & 5 & 4 & 0 & 9 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 5 & 3 & 0 & 2 \\ 2 & 4 & 2 & -3 & 7 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

4.483 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -104, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 8 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 9 & 3 & 4 & 4 \\ 5 & 8 & 0 & 9 & 12 & 5 & 6 & 6 \\ 2 & 4 & 0 & 3 & 5 & 0 & 2 & 2 \\ 3 & 4 & -3 & 4 & 6 & 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 11 & 3 & 3 & 3 \\ 10 & 17 & 0 & 11 & 55 & 11 & 13 & 13 \\ 2 & 4 & 0 & 3 & 11 & 0 & 2 & 2 \\ 3 & 4 & -3 & 3 & 13 & 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -119, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 7 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 & 4 \\ 4 & 7 & 0 & 6 & 7 & 5 & 5 & 5 \\ 2 & 4 & 0 & 3 & 5 & 0 & 2 & 2 \\ 3 & 4 & -3 & 4 & 5 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 11 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 8 & 3 & 3 & 3 \\ 6 & 11 & 0 & 8 & 20 & 9 & 8 & 8 \\ 3 & 5 & 0 & 3 & 9 & 3 & 3 & 3 \\ 3 & 4 & -3 & 3 & 8 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -204, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.14 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 7 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 3 & 3 & 3 \\ 4 & 7 & 0 & 5 & 7 & 6 & 5 & 5 \\ 3 & 5 & 0 & 3 & 6 & 3 & 3 & 3 \\ 3 & 4 & -3 & 3 & 5 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -269, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} =$$

12.2

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 14 & 6 & 7 \\ 5 & 8 & 0 & 14 & 12 & 5 & 6 \\ 2 & 4 & 0 & 6 & 5 & 0 & 2 \\ 3 & 4 & -3 & 7 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 0 & 5 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 12 & 3 & 4 \\ 7 & 12 & 0 & 12 & 25 & 8 & 9 \\ 2 & 4 & 0 & 3 & 8 & 0 & 2 \\ 3 & 4 & -3 & 4 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1311}} 3^{\overline{11}} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 0 & 5 & 20 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 18 & 3 & 4 \\ 11 & 20 & 0 & 18 & 69 & 14 & 15 \\ 2 & 4 & 0 & 3 & 14 & 0 & 2 \\ 3 & 4 & -3 & 4 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 3 & 3 \\ 4 & 7 & 0 & 5 & 7 & 5 & 5 \\ 2 & 4 & 0 & 3 & 5 & 0 & 2 \\ 3 & 4 & -3 & 3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 5 & 3 \\ 3 & 4 & 0 & 4 & 7 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 5 & 6 & 3 \\ 4 & 7 & 0 & 5 & 7 & 11 & 5 \\ 5 & 9 & 0 & 6 & 11 & 12 & 6 \\ 3 & 4 & -3 & 3 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -481, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 8 & 3 & 3 \\ 7 & 12 & 0 & 8 & 25 & 8 & 9 \\ 2 & 4 & 0 & 3 & 8 & 0 & 2 \\ 3 & 4 & -3 & 3 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1211}} 3^{\overline{11}} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 0 & 5 & 16 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 15 & 3 & 4 \\ 9 & 16 & 0 & 15 & 44 & 11 & 12 \\ 2 & 4 & 0 & 3 & 11 & 0 & 2 \\ 3 & 4 & -3 & 4 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 2 & 3 \\ 3 & 4 & 0 & 9 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 11 & 6 & 7 \\ 4 & 7 & 0 & 11 & 7 & 5 & 5 \\ 2 & 4 & 0 & 6 & 5 & 0 & 2 \\ 3 & 4 & -3 & 7 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 2 & 3 \\ 3 & 4 & 0 & 9 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 20 & 6 & 7 \\ 7 & 12 & 0 & 20 & 25 & 8 & 9 \\ 2 & 4 & 0 & 6 & 8 & 0 & 2 \\ 3 & 4 & -3 & 7 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$4.484 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$$

$$T = 8 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 15.51 \rightarrow 16, T_{\max} = 28, T \in [16, 28]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 0 & 0 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{132}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 12 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 27 & 9 \\ 7 & 13 & 0 & 27 & 27 & 9 \\ 3 & 4 & -3 & 9 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.11 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 0 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & -3 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 3 & 5 & 0 & 6 & 3 & 3 \\ 3 & 4 & -3 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{121}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 0 & 12 & 12 & 6 \\ 3 & 4 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 4 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{13}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.15 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 9 & 17 & 0 & 15 & 48 & 12 \\ 3 & 4 & -3 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -2 & 6 & 0 & 0 \\ 6 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 137, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 4 & 8 & 0 & 6 & 9 & 5 \\ 3 & 4 & -3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 19 & 0 & 73 & 12 & 15 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -3 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 6 & 6 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 12 & 7 \\ 4 & 8 & 0 & 12 & 9 & 5 \\ 3 & 4 & -3 & 7 & 5 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{122}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 12.48 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 4 & 8 & 0 & 12 & 9 & 5 \\ 3 & 4 & -3 & 6 & 5 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 6 & 12 & 0 & 9 & 24 & 8 \\ 3 & 4 & -3 & 3 & 8 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{131}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 9 & 9 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -3 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 159, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.34 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 8 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -3 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{132}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 4 & 3 \\ 3 & 4 & 0 & 12 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 25 & 18 & 9 \\ 4 & 8 & 0 & 18 & 9 & 5 \\ 3 & 4 & -3 & 9 & 5 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{122}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 4 & 3 \\ 3 & 4 & 0 & 18 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 24 & 14 \\ 4 & 8 & 0 & 24 & 9 & 5 \\ 3 & 4 & -3 & 14 & 5 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{11}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 12 & 6 \\ 3 & 4 & -3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{141}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 46 & 12 & 12 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -3 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{113}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 \\ 9 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{12}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 7 & 13 & 0 & 12 & 27 & 9 \\ 3 & 4 & -3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & -3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{132}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 197, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.16 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 4 & 3 \\ 3 & 4 & 0 & 27 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 0 & 133 & 36 & 21 \\ 4 & 8 & 0 & 36 & 9 & 5 \\ 3 & 4 & -3 & 21 & 5 & -1 \end{pmatrix}_{b_0}$$

[28] $1^{112}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -2 & 6 & 0 & 0 \\ 8 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{151}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 20 & 0 & 73 & 15 & 15 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -3 & 15 & 2 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{122}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 0 & 18 & 27 & 9 \\ 3 & 4 & -3 & 6 & 9 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 9 & 7 \\ 3 & 5 & 0 & 9 & 3 & 3 \\ 3 & 4 & -3 & 7 & 3 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{14}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 21 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 18 & 16 \\ 3 & 5 & 0 & 18 & 3 & 3 \\ 3 & 4 & -3 & 16 & 3 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{161}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 13.42 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 24 & 0 & 106 & 18 & 18 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -3 & 18 & 2 & -1 \end{pmatrix}_{b_0}$$

[34] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 9 & 11 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 \\ 12 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & -2 & 6 & 0 & 0 \\ 7 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{171}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 275, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.69 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 28 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 28 & 0 & 145 & 21 & 21 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & -3 & 21 & 2 & -1 \end{pmatrix}_{b_0}$$

[39] $1^{12}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 12 & 10 \\ 3 & 5 & 0 & 12 & 3 & 3 \\ 3 & 4 & -3 & 10 & 3 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{13}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 294, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 15 & 13 \\ 3 & 5 & 0 & 15 & 3 & 3 \\ 3 & 4 & -3 & 13 & 3 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 306, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 0 & 9 & 27 & 9 \\ 3 & 4 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 18 & 10 \\ 4 & 8 & 0 & 18 & 9 & 5 \\ 3 & 4 & -3 & 10 & 5 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{115}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 338, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 40 & 0 & 5 \\ 2 & 4 & -2 & 15 & 0 & 0 \\ 15 & 40 & 15 & 148 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 20 & -1 & -1 \end{pmatrix}_{b_0}$$

[44] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 \\ 12 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 371, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 0 & 15 & 12 & 6 \\ 3 & 4 & -3 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{112}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 10 & 18 & 0 & 12 & 57 & 13 \\ 3 & 4 & -3 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 411, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 9 & 10 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 48 & 12 \\ 3 & 4 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{113} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 \\ 10 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{112} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 21 & 7 \\ 7 & 13 & 0 & 21 & 27 & 9 \\ 3 & 4 & -3 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{113} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 463, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 27 & 7 \\ 9 & 17 & 0 & 27 & 48 & 12 \\ 3 & 4 & -3 & 7 & 12 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{132} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 489, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 0 & 17 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 24 & 13 \\ 4 & 8 & 0 & 24 & 9 & 5 \\ 3 & 4 & -3 & 13 & 5 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{113} 3^1 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 490, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 \\ 11 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{132} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 501, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 4 & 3 \\ 3 & 4 & 0 & 22 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 0 & 85 & 30 & 17 \\ 4 & 8 & 0 & 30 & 9 & 5 \\ 3 & 4 & -3 & 17 & 5 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{121} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 507, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 21 & 10 \\ 5 & 9 & 0 & 21 & 12 & 6 \\ 3 & 4 & -3 & 10 & 6 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{114} 3^1 1^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 \\ 16 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{121} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 576, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{131} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 579, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 27 & 13 \\ 5 & 9 & 0 & 27 & 12 & 6 \\ 3 & 4 & -3 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{141} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 587, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 5 & 3 \\ 3 & 4 & 0 & 21 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 33 & 16 \\ 5 & 9 & 0 & 33 & 12 & 6 \\ 3 & 4 & -3 & 16 & 6 & 1 \end{pmatrix}_{b_0}$$

[61] $1^{121} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 28 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 16 & 28 & 0 & 154 & 18 & 22 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -3 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{131} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 12 & 13 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 607, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 12 & 22 & 0 & 15 & 84 & 16 \\ 3 & 4 & -3 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

[63] $1^{114} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 626, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 \\ 13 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{121} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 635, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 23 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 23 & 0 & 100 & 15 & 18 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -3 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{122} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 30 & 10 \\ 7 & 13 & 0 & 30 & 27 & 9 \\ 3 & 4 & -3 & 10 & 9 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{114} 3^1 1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 770, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 \\ 15 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{141} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 771, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 15 & 16 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -3 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{132} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 798, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 7 & 3 \\ 3 & 4 & 0 & 17 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 39 & 13 \\ 7 & 13 & 0 & 39 & 27 & 9 \\ 3 & 4 & -3 & 13 & 9 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{114} 3^1 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 \\ 14 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{131} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 915, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 0 & 85 & 15 & 17 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{131} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 27 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 0 & 133 & 18 & 21 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -3 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{131} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1251, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 0 & 32 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 18 & 32 & 0 & 193 & 21 & 25 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & -3 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{141} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 26 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 0 & 118 & 18 & 20 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{141} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1643, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 31 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 31 & 0 & 172 & 21 & 24 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{141} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1920, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 0 & 36 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 20 & 36 & 0 & 238 & 24 & 28 \\ 2 & 4 & 0 & 24 & 0 & 2 \\ 3 & 4 & -3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [76] $1^{12} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 5 & 0 & 15 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 38 & 10 & 12 \\ 2 & 5 & 0 & 10 & 1 & 3 \\ 3 & 5 & -3 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{111} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 19 & 1 & 5 \\ 12 & 25 & 40 & 0 & 9 \\ 19 & 40 & 63 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 15 & -1 & 2 \end{pmatrix}_{b_0}$$

[78] $1^{111} 3^1 1^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 1 & 3 & 3 & 0 & 3 & 0 \\ 2 & 3 & 3 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{\overline{11}} 3 2 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -40$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 4 & 4 \\ 2 & 5 & 0 & 4 & 1 & 3 \\ 3 & 5 & -3 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{\overline{11}} 3 1^{\overline{11}} 3 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 6 & 0 & 4 & 3 & 4 \\ 1 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[86] $1^{\overline{111}} 3 2 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 23 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[87] $1^{\overline{11}} 3 2 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 4 & 4 & 4 \\ 2 & 2 & 2 & -3 & 3 & 4 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{\overline{11}} 3 1^{\overline{11}} 3 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 2 & -3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[82] $1^{\overline{13}} 3 2 1^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 1 & 8 & 2 \\ 1 & 2 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 9 & 3 \\ 5 & 8 & 9 & 0 & 9 & 39 & 11 \\ 2 & 2 & 2 & -3 & 3 & 11 & 0 \end{pmatrix}_{b_0}$$

[83] $1^{\overline{11}} 3 2 1^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 1 & 4 & 2 \\ 1 & 2 & 0 & 0 & 3 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 7 & 3 \\ 3 & 4 & 7 & 0 & 7 & 15 & 7 \\ 2 & 2 & 2 & -3 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[84] $1^{\overline{11}} 3 2 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 5 \\ 9 & 13 & 1 & 0 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 7 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

[85] $1^{\overline{111}} 3 2 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 19 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[88] $1^{\overline{11}} 3 2 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -61$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 & 0 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & 0 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

[89] $1^{121} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 13 & 14 & 1 & 28 & 7 \\ 14 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 30 & 0 & 58 & 14 \\ 7 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[90] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

[91] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 6 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 3 & 0 & 4 & 4 \\ 3 & 4 & 3 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[92] $1^{11} 31^{12} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 6 & 0 & 22 & 6 & 8 \\ 1 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 4 & 2 & -3 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

[93] $1^{11} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 5 & 0 & 5 & 10 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 7 & 4 \\ 5 & 10 & 0 & 7 & 13 & 7 \\ 3 & 5 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 10 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 10 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

[95] $1^{11} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 3 & 3 & 3 \\ 1 & 4 & 0 & 1 & 3 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & 1 & -1 \\ 1 & 3 & 3 & 1 & 1 & 3 & 3 \\ 2 & 3 & 3 & 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[96] $1^{11} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -89, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 4 & 1 & 3 & 3 & 3 \\ 1 & 4 & 0 & 1 & 3 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & 1 & -1 \\ 2 & 3 & 3 & 1 & 1 & 3 & 3 \\ 2 & 3 & 3 & 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & -1 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[103] $1^{111} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 12 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[97] $1^{121} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 18 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[98] $1^{12} 3^{11} 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.17 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 0 & 5 & 1 & 2 \\ 1 & 2 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 13 & 5 & 7 \\ 1 & 1 & 3 & 0 & 5 & 0 & 2 \\ 2 & 2 & 2 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[99] $1^{141} 3^{21} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 66 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 66 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[100] $1^{111} 3^{21} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 9 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[101] $1^{121} 3^{21} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 51 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 51 & 75 & 0 & 418 & 38 \\ 5 & 7 & -1 & 38 & 2 \end{pmatrix}_{b_0}$$

[102] $1^{112} 3^{21} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 12.71 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & 0 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[105] $1^{111} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -101$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 0 & 1 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 4 & 4 \\ 2 & 2 & 2 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

[104] $1^{112} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -101$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & 0 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[107] $1^{11} 3^{21} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -111$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.19 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 1 & 2 \\ 1 & 2 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 4 & 5 \\ 1 & 1 & 3 & 0 & 4 & 0 & 2 \\ 2 & 2 & 2 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

[106] $1^{11} 3^{21} 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -111$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 0 & 4 & 1 & 2 \\ 1 & 2 & 0 & 0 & 7 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 16 & 7 & 8 \\ 1 & 1 & 3 & 0 & 7 & 0 & 2 \\ 2 & 2 & 2 & -3 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

[109] $1^{121} 3^{21} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 21 & 18 & 0 & 94 & 20 \\ 3 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[108] $1^{11} 3 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 1 & 6 \\ 1 & 2 & 0 & 0 & 3 & 8 \\ 1 & 0 & 0 & -5 & 0 & 0 \\ 1 & 1 & 3 & 0 & 1 & 8 \\ 4 & 6 & 8 & 0 & 8 & 25 \\ 2 & 2 & 2 & -3 & 3 & 9 \end{pmatrix}_{b_0}$$

[110] $1^{11} 3 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -125$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 5 & 0 & 10 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 10 & 0 & 14 & 7 & 8 \\ 2 & 5 & 0 & 7 & 1 & 3 \\ 3 & 5 & -3 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[111] $1^{141} 3 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 2 & 0 & 9 & 3 & 2 \\ 1 & 2 & 0 & 0 & 7 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 7 & 0 & 41 & 16 & 11 \\ 2 & 3 & 4 & 0 & 16 & 4 & 4 \\ 2 & 2 & 2 & -3 & 11 & 4 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{11} 3 1^{11} 3 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -137$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.30 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[114] $1^{11} 3 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 & 5 \\ 3 & 5 & 0 & 5 & 15 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 11 & 4 & 4 \\ 7 & 15 & 0 & 11 & 31 & 11 & 11 \\ 3 & 5 & -3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{11} 3 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 19 & 4 & 4 \\ 11 & 25 & 0 & 19 & 91 & 19 & 19 \\ 3 & 5 & -3 & 4 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

[116] $1^{111} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -141$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 & 4 \\ 3 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 1 & 0 & 7 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 7 & 0 & 14 & 8 & 8 \\ 3 & 4 & 3 & -3 & 8 & 1 & 1 \end{pmatrix}_{b_0}$$

[115] $1^{111} 3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -141$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 13 & 14 & 1 & 14 & 7 & 7 \\ 14 & 13 & 1 & 15 & 7 & 7 \\ 1 & 1 & -3 & 0 & -1 & -1 \\ 14 & 15 & 0 & 13 & 7 & 7 \\ 7 & 7 & -1 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[117] $1^{11} 3 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -160$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 & 5 \\ 3 & 5 & 0 & 5 & 20 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 15 & 4 & 4 \\ 9 & 20 & 0 & 15 & 57 & 15 & 15 \\ 3 & 5 & -3 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[118] $1^{112} 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -161$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 0 & 1 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 6 & 2 \\ 4 & 6 & 8 & 0 & 6 & 25 & 9 \\ 2 & 2 & 2 & -3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{111} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 1 & 6 & 3 \\ 3 & 9 & 6 & 0 & 6 & 12 & 6 \\ 2 & 4 & 2 & -3 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{13} 3 21^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 5 & 0 & 25 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 25 & 0 & 104 & 17 & 20 \\ 2 & 5 & 0 & 17 & 1 & 3 \\ 3 & 5 & -3 & 20 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{121} 3 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 52 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 52 & 75 & 0 & 418 & 38 \\ 5 & 7 & -1 & 38 & 2 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{111} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 20 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{131} 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 47 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 47 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{131} 3 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 58 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 58 & 90 & 0 & 583 & 45 \\ 5 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{111} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 22 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{111} 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 11 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{121} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 28 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[127] $1^{111} 3^{11} 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -176$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 10 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[129] $1^{131} 31^{11} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -188$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.30 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 5 & 1 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 12 & 15 & 0 & 49 & 12 & 13 \\ 1 & 3 & 3 & 0 & 12 & 0 & 2 \\ 2 & 3 & 3 & -3 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

[130] $1^{112} 321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -191$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 0 & 1 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 5 & 2 \\ 3 & 5 & 5 & 0 & 5 & 12 & 6 \\ 2 & 2 & 2 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[132] $1^{11} 321^{141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -215$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.10 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 5 & 0 & 25 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 25 & 0 & 92 & 19 & 20 \\ 2 & 5 & 0 & 19 & 1 & 3 \\ 3 & 5 & -3 & 20 & 3 & 1 \end{pmatrix}_{b_0}$$

[131] $1^{11} 321^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -215$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 5 & 0 & 15 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 15 & 0 & 32 & 11 & 12 \\ 2 & 5 & 0 & 11 & 1 & 3 \\ 3 & 5 & -3 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[134] $1^{111} 321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -216$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 21 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[135] $1^{14} 321^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -216$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.19 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 0 & 10 & 1 & 2 \\ 1 & 2 & 0 & 0 & 10 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 10 & 10 & 0 & 58 & 10 & 14 \\ 1 & 1 & 3 & 0 & 10 & 0 & 2 \\ 2 & 2 & 2 & -3 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

[133] $1^{121} 321^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -216$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 39 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 60 & 0 & 258 & 30 \\ 5 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[137] $1^{112} 321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -221$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 3 & 4 & 0 & 2 & 4 & 5 & 3 \\ 2 & 2 & -1 & 0 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[138] $1^{111} 321^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -221$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 0 & 4 & 3 & 2 \\ 1 & 2 & 0 & 0 & 7 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 16 & 11 & 8 \\ 2 & 3 & 4 & 0 & 11 & 4 & 4 \\ 2 & 2 & 2 & -3 & 8 & 4 & 1 \end{pmatrix}_{b_0}$$

[136] $1^{111} 3^{21} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -221$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 4 & 2 \\ 1 & 2 & 0 & 0 & 4 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 11 & 5 \\ 3 & 4 & 7 & 0 & 11 & 15 & 7 \\ 2 & 2 & 2 & -3 & 5 & 7 & 0 \end{pmatrix}_{b_0}$$

[139] $1^{11} 3^{21} 1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -245$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 5 & 0 & 20 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 20 & 0 & 58 & 15 & 16 \\ 2 & 5 & 0 & 15 & 1 & 3 \\ 3 & 5 & -3 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[140] $1^{111} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 7 & 5 \\ 2 & 3 & 4 & 0 & 7 & 4 & 4 \\ 2 & 2 & 2 & -3 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

[141] $1^{12} 3^{21} 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -276$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 12.22 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 0 & 6 & 1 & 2 \\ 1 & 2 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 26 & 8 & 10 \\ 1 & 1 & 3 & 0 & 8 & 0 & 2 \\ 2 & 2 & 2 & -3 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

[142] $1^{121} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -276$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 12.32 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 15 & 4 \\ 2 & 4 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 10 & 0 & 38 & 12 \\ 3 & 4 & 3 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[143] $1^{111} 3^1 1^{31} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -284$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.25 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 & 3 \\ 2 & 4 & 3 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 6 & 9 & 0 & 13 & 6 & 7 \\ 1 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 3 & 3 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[144] $1^{12} 3^{21} 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -290$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 5 & 0 & 20 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 20 & 0 & 62 & 14 & 16 \\ 2 & 5 & 0 & 14 & 1 & 3 \\ 3 & 5 & -3 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[145] $1^{131} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -296$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 37 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 37 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[146] $1^{121} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -296$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 19 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[147] $1^{121} 321^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -306, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 14 & 0 & 62 & 16 \\ 3 & 4 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{112} 321^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -311, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 5 & 2 \\ 1 & 2 & 0 & 0 & 4 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 10 & 5 \\ 3 & 5 & 5 & 0 & 10 & 12 & 6 \\ 2 & 2 & 2 & -3 & 5 & 6 & 0 \end{pmatrix}_{b_0}$$

[149] $1^{131} 321^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -311, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 2 & 0 & 7 & 3 & 2 \\ 1 & 2 & 0 & 0 & 6 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 25 & 13 & 9 \\ 2 & 3 & 4 & 0 & 13 & 4 & 4 \\ 2 & 2 & 2 & -3 & 9 & 4 & 1 \end{pmatrix}_{b_0}$$

[150] $1^{13} 321^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -311, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 0 & 8 & 1 & 2 \\ 1 & 2 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 9 & 0 & 40 & 9 & 12 \\ 1 & 1 & 3 & 0 & 9 & 0 & 2 \\ 2 & 2 & 2 & -3 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

[151] $1^{111} 31^{11} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.31 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 6 & 7 \\ 1 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 4 & 2 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[152] $1^{121} 31^{11} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -332, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 4 & 1 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 9 & 12 & 0 & 28 & 9 & 10 \\ 1 & 3 & 3 & 0 & 9 & 0 & 2 \\ 2 & 3 & 3 & -3 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

[153] $1^{12} 321^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.48 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 0 & 7 & 1 & 2 \\ 1 & 2 & 0 & 0 & 11 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 11 & 0 & 45 & 11 & 13 \\ 1 & 1 & 3 & 0 & 11 & 0 & 2 \\ 2 & 2 & 2 & -3 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

[154] $1^{131} 321^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -351, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 30 & 4 \\ 2 & 4 & 1 & 0 & 25 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 30 & 25 & 0 & 188 & 28 \\ 3 & 4 & 3 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[155] $1^{121} 321^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 0 & 5 & 3 & 2 \\ 1 & 2 & 0 & 0 & 5 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 13 & 10 & 7 \\ 2 & 3 & 4 & 0 & 10 & 4 & 4 \\ 2 & 2 & 2 & -3 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$

[156] $1^{131} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 13 & 0 & 74 & 16 \\ 3 & 4 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[157] $1^{121} 3 21^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 2 & 0 & 8 & 3 & 2 \\ 1 & 2 & 0 & 0 & 14 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 8 & 14 & 0 & 70 & 22 & 16 \\ 2 & 3 & 4 & 0 & 22 & 4 & 4 \\ 2 & 2 & 2 & -3 & 16 & 4 & 1 \end{pmatrix}_{b_0}$$

[158] $1^{131} 1^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 6 & 4 \\ 2 & 6 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 2 & -3 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

[159] $1^{121} 3 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 22 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[160] $1^{131} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 28 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[161] $1^{121} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 29 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[162] $1^{121} 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 40 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 40 & 60 & 0 & 258 & 30 \\ 5 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[163] $1^{141} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 46 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 46 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

[164] $1^{141} 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 56 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 56 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

[165] $1^{121} 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 20 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[166] $1^{11}31^{111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -417, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 4 & 9 & 4 \\ 3 & 9 & 6 & 0 & 9 & 12 & 6 \\ 2 & 4 & 2 & -3 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

[167] $1^{14}3^{11}21^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 & 5 \\ 3 & 5 & 0 & 35 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 & 2 \\ 18 & 35 & 0 & 202 & 24 & 28 & 2 \\ 2 & 5 & 0 & 24 & 1 & 3 & 3 \\ 3 & 5 & -3 & 28 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[168] $1^{141}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 & 4 \\ 2 & 4 & 1 & 0 & 16 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 & 2 \\ 14 & 27 & 16 & 0 & 122 & 20 & 2 \\ 3 & 4 & 3 & -3 & 20 & 1 & 1 \end{pmatrix}_{b_0}$$

[169] $1^{121}3^{113}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 12.11 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 21 & 5 & 7 \\ 9 & 13 & 1 & 30 & 7 & 7 \\ 1 & 1 & -3 & 0 & -1 & 2 \\ 21 & 30 & 0 & 58 & 14 & 14 \\ 5 & 7 & -1 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[170] $1^{11}31^{111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 6 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 9 & 7 \\ 2 & 6 & 3 & 0 & 9 & 3 & 3 \\ 2 & 4 & 2 & -3 & 7 & 3 & 0 \end{pmatrix}_{b_0}$$

[171] $1^{112}3^{11}21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -481, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 6 & 2 \\ 1 & 2 & 0 & 0 & 4 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 14 & 5 \\ 4 & 6 & 8 & 0 & 14 & 25 & 9 \\ 2 & 2 & 2 & -3 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

[172] $1^{12}3^{11}21^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 & 3 \\ 3 & 5 & 0 & 25 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 & 2 \\ 12 & 25 & 0 & 94 & 18 & 20 & 2 \\ 2 & 5 & 0 & 18 & 1 & 3 & 3 \\ 3 & 5 & -3 & 20 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[173] $1^{131}3^{112}21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 59 & 5 & 7 \\ 9 & 13 & 1 & 90 & 7 & 7 \\ 1 & 1 & -3 & 0 & -1 & 2 \\ 59 & 90 & 0 & 583 & 45 & 45 \\ 5 & 7 & -1 & 45 & 3 & 3 \end{pmatrix}_{b_0}$$

[174] $1^{121}3^{113}21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 41 & 5 & 7 \\ 9 & 13 & 1 & 60 & 7 & 7 \\ 1 & 1 & -3 & 0 & -1 & 2 \\ 41 & 60 & 0 & 258 & 30 & 30 \\ 5 & 7 & -1 & 30 & 3 & 3 \end{pmatrix}_{b_0}$$

[175] $1^{141}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 37 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 37 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{13} 3^{21} 1^{31} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -551$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 0 & 9 & 1 & 2 \\ 1 & 2 & 0 & 0 & 12 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 9 & 12 & 0 & 61 & 12 & 15 \\ 1 & 1 & 3 & 0 & 12 & 0 & 2 \\ 2 & 2 & 2 & -3 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

[177] $1^{13} 3^{21} 1^{31} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -565$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 5 & 0 & 30 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 30 & 0 & 142 & 21 & 24 \\ 2 & 5 & 0 & 21 & 1 & 3 \\ 3 & 5 & -3 & 24 & 3 & 2 \end{pmatrix}_{b_0}$$

[180] $1^{121} 3^{21} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -576$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.02 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 30 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[179] $1^{131} 3^{21} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -576$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.02 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 48 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 48 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[178] $1^{121} 3^{21} 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -576$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 42 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 42 & 60 & 0 & 258 & 30 \\ 5 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[181] $1^{131} 3^{21} 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -581$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 21 & 0 & 142 & 24 \\ 3 & 4 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[182] $1^{121} 3^{21} 1^{21} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -586$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 2 & 0 & 6 & 3 & 2 \\ 1 & 2 & 0 & 0 & 8 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 26 & 14 & 10 \\ 2 & 3 & 4 & 0 & 14 & 4 & 4 \\ 2 & 2 & 2 & -3 & 10 & 4 & 1 \end{pmatrix}_{b_0}$$

[183] $1^{111} 3^1 1^{23} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -587$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 15 & 9 & 0 & 37 & 9 & 11 \\ 1 & 3 & 3 & 0 & 9 & 0 & 2 \\ 2 & 4 & 2 & -3 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

[184] $1^{131} 3^{21} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -591$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.27 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 17 & 0 & 104 & 20 \\ 3 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[185] $1^{121} 3 21^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -596, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 2 & 0 & 7 & 3 & 2 \\ 1 & 2 & 0 & 0 & 11 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 11 & 0 & 45 & 18 & 13 \\ 2 & 3 & 4 & 0 & 18 & 4 & 4 \\ 2 & 2 & 2 & -3 & 13 & 4 & 1 \end{pmatrix}_{b_0}$$

[186] $1^{14} 3 21^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -636, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 0 & 11 & 1 & 2 \\ 1 & 2 & 0 & 0 & 13 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 11 & 13 & 0 & 81 & 13 & 17 \\ 1 & 1 & 3 & 0 & 13 & 0 & 2 \\ 2 & 2 & 2 & -3 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

[187] $1^{12} 3 21^{141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -650, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 5 & 0 & 30 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 30 & 0 & 134 & 22 & 24 \\ 2 & 5 & 0 & 22 & 1 & 3 \\ 3 & 5 & -3 & 24 & 3 & 2 \end{pmatrix}_{b_0}$$

[192] $1^{131} 3 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 29 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[191] $1^{131} 3 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 38 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 38 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[190] $1^{121} 3 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 32 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 32 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[189] $1^{141} 3 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 67 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 67 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[188] $1^{121} 3 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 31 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[193] $1^{13} 3 21^{141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -671, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 0 & 10 & 1 & 2 \\ 1 & 2 & 0 & 0 & 15 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 10 & 15 & 0 & 88 & 15 & 18 \\ 1 & 1 & 3 & 0 & 15 & 0 & 2 \\ 2 & 2 & 2 & -3 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

[194] $1^{111} 31^{111} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -699, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.20 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 13 & 15 & 7 \\ 3 & 9 & 6 & 0 & 15 & 12 & 6 \\ 2 & 4 & 2 & -3 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

[196] $1^{131} 3^{21} 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 60 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 60 & 90 & 0 & 583 & 45 \\ 5 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

[195] $1^{131} 3^{21} 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 30 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[197] $1^{111} 31^{13} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -818, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 21 & 3 & 4 \\ 1 & 4 & 0 & 0 & 12 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 21 & 12 & 0 & 73 & 12 & 15 \\ 1 & 3 & 3 & 0 & 12 & 0 & 2 \\ 2 & 4 & 2 & -3 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

[198] $1^{131} 3^{21} 1^{21} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -821, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 2 & 0 & 8 & 3 & 2 \\ 1 & 2 & 0 & 0 & 9 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 9 & 0 & 40 & 17 & 12 \\ 2 & 3 & 4 & 0 & 17 & 4 & 4 \\ 2 & 2 & 2 & -3 & 12 & 4 & 1 \end{pmatrix}_{b_0}$$

[199] $1^{141} 3^{21} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -846, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 30 & 4 \\ 2 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 30 & 20 & 0 & 158 & 24 \\ 3 & 4 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[200] $1^{13} 3^{21} 1^{41} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -895, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 5 & 0 & 35 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 35 & 0 & 188 & 25 & 28 \\ 2 & 5 & 0 & 25 & 1 & 3 \\ 3 & 5 & -3 & 28 & 3 & 2 \end{pmatrix}_{b_0}$$

[201] $1^{141} 3^{21} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 38 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 38 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[202] $1^{131} 3^{21} 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 31 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{131} 3^{21} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 49 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 49 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[204] $1^{141} 3^{21} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

12.2

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 47 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 47 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

$$[205] \ 2 \overset{1^{111}}{2} \ 31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -901, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 \\ 2 & 1 & 4 & 1 & 3 & 3 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 1 & 1 & 1 & -3 & 1 & 1 \\ 2 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 3 & 1 & 3 & 0 \\ 2 & 3 & 2 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[206] \ 1 \overset{1^1}{141} \ 321^{\overline{121}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -926, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 2 & 0 & 10 & 3 & 2 \\ 1 & 2 & 0 & 0 & 10 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 10 & 10 & 0 & 58 & 20 & 14 \\ 2 & 3 & 4 & 0 & 20 & 4 & 4 \\ 2 & 2 & 2 & -3 & 14 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[207] \ 1 \overset{1^{12}}{131} \ 3 \ 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -936, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 39 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

$$[208] \ 1 \overset{1^{11}}{141} \ 3 \ 21^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -936, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 57 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 57 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

$$[209] \ 1 \overset{1^1}{14} \ 321^{\overline{141}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -936, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 0 & 12 & 1 & 2 \\ 1 & 2 & 0 & 0 & 16 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 12 & 16 & 0 & 110 & 16 & 20 \\ 1 & 1 & 3 & 0 & 16 & 0 & 2 \\ 2 & 2 & 2 & -3 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[210] \ 1 \overset{1^1}{14} \ 321^{\overline{141}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -950, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 5 & 0 & 40 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 20 & 40 & 0 & 254 & 28 & 32 \\ 2 & 5 & 0 & 28 & 1 & 3 \\ 3 & 5 & -3 & 32 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[211] \ 1 \overset{1^1}{141} \ 321^{\overline{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -966, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 1 & 0 & 28 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 28 & 0 & 254 & 32 \\ 3 & 4 & 3 & -3 & 32 & 1 \end{pmatrix}_{b_0}$$

$$[212] \ 1 \overset{1^{11}}{141} \ 31^{\overline{1431}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -1007, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 0 & 27 & 3 & 4 \\ 1 & 4 & 0 & 0 & 15 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 27 & 15 & 0 & 121 & 15 & 19 \\ 1 & 3 & 3 & 0 & 15 & 0 & 2 \\ 2 & 4 & 2 & -3 & 19 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[213] \ 1 \overset{1^1}{141} \ 321^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1016, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 33 & 4 \\ 2 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 33 & 24 & 0 & 202 & 28 \\ 3 & 4 & 3 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[214] $1^{131} 3^{21} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1016$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 61 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 61 & 90 & 0 & 583 & 45 \\ 5 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

[215] $1^{131} 3^{21} 1^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1111$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 2 & 0 & 9 & 3 & 2 \\ 1 & 2 & 0 & 0 & 12 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 9 & 12 & 0 & 61 & 21 & 15 \\ 2 & 3 & 4 & 0 & 21 & 4 & 4 \\ 2 & 2 & 2 & -3 & 15 & 4 & 1 \end{pmatrix}_{b_0}$$

[216] $1^{141} 3^{21} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 68 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 68 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{131} 3^{21} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 40 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 40 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[217] $1^{131} 3^{21} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 50 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 50 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[219] $1^{141} 3^{21} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1176$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 39 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[220] $1^{131} 3^{21} 1^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1181$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 2 & 0 & 10 & 3 & 2 \\ 1 & 2 & 0 & 0 & 15 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 10 & 15 & 0 & 88 & 25 & 18 \\ 2 & 3 & 4 & 0 & 25 & 4 & 4 \\ 2 & 2 & 2 & -3 & 18 & 4 & 1 \end{pmatrix}_{b_0}$$

[221] $1^{131} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 41 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 41 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[222] $1^{131} 3^{21} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1296$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 51 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 51 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[223] $1^{141} 3^{12} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 48 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 48 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

[224] $1^{141} 3^{12} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 58 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 58 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

[225] $1^{141} 3^{14} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 40 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 40 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[226] $1^{141} 3^{11} 21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 2 & 0 & 11 & 3 & 2 \\ 1 & 2 & 0 & 0 & 13 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 11 & 13 & 0 & 81 & 24 & 17 \\ 2 & 3 & 4 & 0 & 24 & 4 & 4 \\ 2 & 2 & 2 & -3 & 17 & 4 & 1 \end{pmatrix}_{b_0}$$

[227] $1^{141} 3^{13} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 69 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 69 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[228] $1^{141} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1616, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 49 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 49 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

[229] $1^{141} 3^{13} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 59 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 59 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

[230] $1^{141} 3^{11} 21^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1846, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 2 & 0 & 12 & 3 & 2 \\ 1 & 2 & 0 & 0 & 16 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 12 & 16 & 0 & 110 & 28 & 20 \\ 2 & 3 & 4 & 0 & 28 & 4 & 4 \\ 2 & 2 & 2 & -3 & 20 & 4 & 1 \end{pmatrix}_{b_0}$$

[232] $1^{141} 3^{13} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1856, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 70 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 70 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[231] $1^{141} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1856$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.06 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 50 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 50 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

[233] $1^{141} 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -2016$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.02 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 60 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 60 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

$T = 10$ [234] $1^{111} 2 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -61$, $\det = 16$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 1 & 0 \\ 2 & 4 & 0 & 4 & 6 & 4 & 6 \\ 1 & 3 & 0 & 1 & 4 & 0 & 2 \\ 2 & 3 & -1 & 0 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{11} 3 221^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 19$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 6 & 0 & 17 & 6 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 17 & 0 & 36 & 13 & 14 \\ 2 & 6 & 0 & 13 & 2 & 4 \\ 3 & 6 & -3 & 14 & 4 & 3 \end{pmatrix}_{b_0}$$

[236] $1^{12} 3 221^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 40$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 6 & 0 & 34 & 6 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 34 & 0 & 150 & 26 & 28 \\ 2 & 6 & 0 & 26 & 2 & 4 \\ 3 & 6 & -3 & 28 & 4 & 3 \end{pmatrix}_{b_0}$$

[237] $1^{11} 4 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -86$, $\det = 56$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

[238] $1^{11} 4 1^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -86$, $\det = 66$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 3 \\ 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

[239] $1^{111} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -86$, $\det = 117$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 5 & 3 \\ 5 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & -2 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[240] $1^{112} 3 221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 136$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.4$

$T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 3 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 11 & 3 \\ 10 & 11 & 0 & 11 & 49 & 13 \\ 3 & 3 & -1 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[241] $1^{\overline{111}} 1^{\overline{131}} 4^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 8 & 5 & 6 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[242] $1^{\overline{111}} 3^{\overline{11}} 221^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 171, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 1 & 0 & 5 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 6 & 8 & 6 \\ 2 & 3 & 5 & 0 & 8 & 5 & 5 \\ 2 & 2 & 3 & -3 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[243] $1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 201, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 3 \\ 4 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[244] $1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 222, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 2 & 3 \\ 4 & 4 & 0 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 8 & 5 & 6 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[245] $1^{\overline{111}} 4^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 288, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 0 & 26 & 8 & 10 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[246] $1^{\overline{112}} 3^{\overline{11}} 221^{\overline{112}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 297, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.4$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 6 & 2 \\ 1 & 2 & 1 & 0 & 5 & 10 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 6 & 16 & 6 \\ 4 & 6 & 10 & 0 & 16 & 29 & 11 \\ 2 & 2 & 3 & -3 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$

[247] $1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 311, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 6 & 2 & 3 \\ 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 5 & 7 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[248] $1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 338, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 8 & 2 & 3 \\ 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 0 & 26 & 8 & 10 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[249] $1^{\overline{121}} 3^{\overline{11}} 221^{\overline{121}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 2 & 0 & 6 & 3 & 2 \\ 1 & 2 & 1 & 0 & 10 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 10 & 0 & 30 & 16 & 12 \\ 2 & 3 & 5 & 0 & 16 & 5 & 5 \\ 2 & 2 & 3 & -3 & 12 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[250] \ 1^{131} 3^{221} 1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 675, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 2 & 0 & 9 & 3 & 2 \\ 1 & 2 & 1 & 0 & 15 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 9 & 15 & 0 & 70 & 24 & 18 \\ 2 & 3 & 5 & 0 & 24 & 5 & 5 \\ 2 & 2 & 3 & -3 & 18 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[251] \ 1^{141} 3^{221} 1^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 1116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 2 & 0 & 12 & 3 & 2 \\ 1 & 2 & 1 & 0 & 20 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 12 & 20 & 0 & 126 & 32 & 24 \\ 2 & 3 & 5 & 0 & 32 & 5 & 5 \\ 2 & 2 & 3 & -3 & 24 & 5 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [252] \ 1^{125} 1^{31} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -34, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 12 & 1 & 5 \\ 9 & 13 & 20 & 0 & 7 \\ 12 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[253] \ 1^{115} 1^{31} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -61, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 5 \\ 9 & 13 & 10 & 0 & 7 \\ 6 & 10 & 5 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

4.485 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 & 0 \\ 3 & 8 & 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 4 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -3 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -3 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

4.486 NHC(-2)(-3) + f₄ + su₂ + su₃ + su₂

$$T = 9 \quad [0] \quad 1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & -3 & 4 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 0 & 5 & 4 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 21 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 11 & 19 & 0 & 21 & 12 & 73 & 15 \\ 3 & 4 & -3 & 4 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 9 & 37 & 11 \\ 3 & 4 & -3 & 3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3 1^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 5 & 3 \\ 3 & 4 & 0 & 19 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 19 & 0 & 73 & 12 & 33 & 15 \\ 2 & 4 & 0 & 12 & 0 & 6 & 2 \\ 5 & 9 & 0 & 33 & 6 & 13 & 7 \\ 3 & 4 & -3 & 15 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{121} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 6 & 6 \\ 5 & 9 & 0 & 12 & 12 & 6 & 6 \\ 2 & 4 & 0 & 6 & 6 & 1 & 3 \\ 3 & 4 & -3 & 6 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 0 & 9 & 12 & 6 & 6 \\ 2 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & -3 & 4 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{121} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 12 & 3 & 4 \\ 7 & 13 & 0 & 12 & 27 & 9 & 9 \\ 2 & 4 & 0 & 3 & 9 & 1 & 3 \\ 3 & 4 & -3 & 4 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

12.2

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 \\ 3 & 4 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & -3 & 4 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{111}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 0 & 5 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 15 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 8 & 14 & 0 & 15 & 9 & 37 & 11 \\ 3 & 4 & -3 & 4 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{112}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 12 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 7 & 13 & 0 & 12 & 9 & 28 & 10 \\ 3 & 4 & -3 & 4 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 6 & 12 & 6 & 6 \\ 2 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & -3 & 3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{112}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 4 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 13 & 23 & 0 & 15 & 15 & 100 & 18 \\ 3 & 4 & -3 & 3 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{1212}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 7 & 3 \\ 3 & 4 & 0 & 8 & 9 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 18 & 6 \\ 5 & 9 & 0 & 12 & 12 & 21 & 6 \\ 7 & 13 & 0 & 18 & 21 & 28 & 10 \\ 3 & 4 & -3 & 6 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & -3 & 3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 6 & 3 \\ 5 & 9 & 0 & 6 & 12 & 15 & 6 \\ 5 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & -3 & 3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1111}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -605, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 8 & 3 \\ 3 & 4 & 0 & 9 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 24 & 7 \\ 2 & 4 & 0 & 6 & 0 & 9 & 2 \\ 8 & 14 & 0 & 24 & 9 & 37 & 11 \\ 3 & 4 & -3 & 7 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{112}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 0 & 5 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 18 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 10 & 18 & 0 & 18 & 12 & 58 & 14 \\ 3 & 4 & -3 & 4 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{112}31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -650, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 0 & 5 & 4 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 30 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 16 & 28 & 0 & 30 & 18 & 154 & 22 \\ 3 & 4 & -3 & 4 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{1111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 15 & 7 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 0 & 15 & 6 & 13 & 7 \\ 3 & 4 & -3 & 7 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1111}31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -716, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 8 & 3 \\ 3 & 4 & 0 & 14 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 9 & 39 & 11 \\ 2 & 4 & 0 & 9 & 0 & 9 & 2 \\ 8 & 14 & 0 & 39 & 9 & 37 & 11 \\ 3 & 4 & -3 & 11 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1112}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -723, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 7 & 13 & 0 & 9 & 9 & 28 & 10 \\ 3 & 4 & -3 & 3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{1112}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -738, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 10 & 18 & 0 & 12 & 12 & 58 & 14 \\ 3 & 4 & -3 & 3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{112}31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 0 & 5 & 4 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 24 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 13 & 23 & 0 & 24 & 15 & 100 & 18 \\ 3 & 4 & -3 & 4 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1113} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -779, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 0 & 5 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 21 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 12 & 22 & 0 & 21 & 15 & 85 & 17 \\ 3 & 4 & -3 & 4 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1113} 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -828, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 4 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 21 & 3 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 18 & 32 & 0 & 21 & 21 & 193 & 25 \\ 3 & 4 & -3 & 3 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1112} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -845, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 7 & 3 \\ 3 & 4 & 0 & 4 & 9 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 9 & 3 \\ 5 & 9 & 0 & 6 & 12 & 21 & 6 \\ 7 & 13 & 0 & 9 & 21 & 28 & 10 \\ 3 & 4 & -3 & 3 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1112} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -941, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 7 & 3 \\ 3 & 4 & 0 & 9 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 21 & 7 \\ 2 & 4 & 0 & 6 & 0 & 9 & 2 \\ 7 & 13 & 0 & 21 & 9 & 28 & 10 \\ 3 & 4 & -3 & 7 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1112} 31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -962, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 10 & 3 \\ 3 & 4 & 0 & 14 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 9 & 48 & 11 \\ 2 & 4 & 0 & 9 & 0 & 12 & 2 \\ 10 & 18 & 0 & 48 & 12 & 58 & 14 \\ 3 & 4 & -3 & 11 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1113} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -980, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 9 & 3 \\ 3 & 4 & 0 & 9 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 27 & 7 \\ 2 & 4 & 0 & 6 & 0 & 12 & 2 \\ 9 & 17 & 0 & 27 & 12 & 49 & 13 \\ 3 & 4 & -3 & 7 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1112} 31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -980, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 16 & 3 \\ 3 & 4 & 0 & 9 & 4 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 48 & 7 \\ 2 & 4 & 0 & 6 & 0 & 18 & 2 \\ 16 & 28 & 0 & 48 & 18 & 154 & 22 \\ 3 & 4 & -3 & 7 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1113} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -989, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 9 & 3 \\ 3 & 4 & 0 & 4 & 9 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 12 & 3 \\ 5 & 9 & 0 & 6 & 12 & 27 & 6 \\ 9 & 17 & 0 & 12 & 27 & 49 & 13 \\ 3 & 4 & -3 & 3 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1114} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1005, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 11 & 3 \\ 3 & 4 & 0 & 4 & 9 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 15 & 3 \\ 5 & 9 & 0 & 6 & 12 & 33 & 6 \\ 11 & 21 & 0 & 15 & 33 & 76 & 16 \\ 3 & 4 & -3 & 3 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1113} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1077, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 9 & 17 & 0 & 12 & 12 & 49 & 13 \\ 3 & 4 & -3 & 3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{113} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1221, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 0 & 5 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 27 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 15 & 27 & 0 & 27 & 18 & 133 & 21 \\ 3 & 4 & -3 & 4 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1212} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1298, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 10 & 3 \\ 3 & 4 & 0 & 13 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 9 & 42 & 10 \\ 2 & 4 & 0 & 9 & 0 & 12 & 2 \\ 10 & 18 & 0 & 42 & 12 & 58 & 14 \\ 3 & 4 & -3 & 10 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1113} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1333, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 15 & 27 & 0 & 18 & 18 & 133 & 21 \\ 3 & 4 & -3 & 3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1112} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 10 & 3 \\ 3 & 4 & 0 & 9 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 30 & 7 \\ 2 & 4 & 0 & 6 & 0 & 12 & 2 \\ 10 & 18 & 0 & 30 & 12 & 58 & 14 \\ 3 & 4 & -3 & 7 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1212} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1388, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 9 & 30 & 10 \\ 2 & 4 & 0 & 9 & 0 & 9 & 2 \\ 7 & 13 & 0 & 30 & 9 & 28 & 10 \\ 3 & 4 & -3 & 10 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1112} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1389, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 13 & 3 \\ 3 & 4 & 0 & 9 & 4 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 39 & 7 \\ 2 & 4 & 0 & 6 & 0 & 15 & 2 \\ 13 & 23 & 0 & 39 & 15 & 100 & 18 \\ 3 & 4 & -3 & 7 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1113}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 12 & 22 & 0 & 15 & 15 & 85 & 17 \\ 3 & 4 & -3 & 3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{1113}31^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1451, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 0 & 5 & 4 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 33 & 4 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 18 & 32 & 0 & 33 & 21 & 193 & 25 \\ 3 & 4 & -3 & 4 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1113}31^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1500, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 0 & 5 & 4 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 30 & 4 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 17 & 31 & 0 & 30 & 21 & 172 & 24 \\ 3 & 4 & -3 & 4 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1113}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1949, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 12 & 3 \\ 3 & 4 & 0 & 9 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 36 & 7 \\ 2 & 4 & 0 & 6 & 0 & 15 & 2 \\ 12 & 22 & 0 & 36 & 15 & 85 & 17 \\ 3 & 4 & -3 & 7 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1212}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2048, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 10 & 3 \\ 3 & 4 & 0 & 18 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 12 & 60 & 14 \\ 2 & 4 & 0 & 12 & 0 & 12 & 2 \\ 10 & 18 & 0 & 60 & 12 & 58 & 14 \\ 3 & 4 & -3 & 14 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1114}31^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2054, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 0 & 5 & 4 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 36 & 4 \\ 2 & 4 & 0 & 3 & 0 & 24 & 2 \\ 20 & 36 & 0 & 36 & 24 & 238 & 28 \\ 3 & 4 & -3 & 4 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1114}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2106, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 4 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 14 & 26 & 0 & 18 & 18 & 118 & 20 \\ 3 & 4 & -3 & 3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1114}31^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 4 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 24 & 3 \\ 2 & 4 & 0 & 3 & 0 & 24 & 2 \\ 20 & 36 & 0 & 24 & 24 & 238 & 28 \\ 3 & 4 & -3 & 3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1114} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2324, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 14 & 3 \\ 3 & 4 & 0 & 9 & 4 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 42 & 7 \\ 2 & 4 & 0 & 6 & 0 & 18 & 2 \\ 14 & 26 & 0 & 42 & 18 & 118 & 20 \\ 3 & 4 & -3 & 7 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1114} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2347, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 4 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 21 & 3 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 17 & 31 & 0 & 21 & 21 & 172 & 24 \\ 3 & 4 & -3 & 3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1113} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2496, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 15 & 3 \\ 3 & 4 & 0 & 9 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 45 & 7 \\ 2 & 4 & 0 & 6 & 0 & 18 & 2 \\ 15 & 27 & 0 & 45 & 18 & 133 & 21 \\ 3 & 4 & -3 & 7 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1113} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2621, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 18 & 3 \\ 3 & 4 & 0 & 9 & 4 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 54 & 7 \\ 2 & 4 & 0 & 6 & 0 & 21 & 2 \\ 18 & 32 & 0 & 54 & 21 & 193 & 25 \\ 3 & 4 & -3 & 7 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1114} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -3405, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 17 & 3 \\ 3 & 4 & 0 & 9 & 4 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 51 & 7 \\ 2 & 4 & 0 & 6 & 0 & 21 & 2 \\ 17 & 31 & 0 & 51 & 21 & 172 & 24 \\ 3 & 4 & -3 & 7 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1114} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -4064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 20 & 3 \\ 3 & 4 & 0 & 9 & 4 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 60 & 7 \\ 2 & 4 & 0 & 6 & 0 & 24 & 2 \\ 20 & 36 & 0 & 60 & 24 & 238 & 28 \\ 3 & 4 & -3 & 7 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [54] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 32, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 1 & 5 \\ 9 & 13 & 1 & 0 & 0 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 7 & -1 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{11} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 5 & 0 & 5 & 5 & 10 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 4 & 7 & 4 \\ 2 & 5 & 0 & 4 & 1 & 7 & 3 \\ 5 & 10 & 0 & 7 & 7 & 14 & 8 \\ 3 & 5 & -3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{111111} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 4 & 1 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 3 & 2 & 2 \\ 2 & 3 & 4 & 0 & 3 & 4 & 3 & 4 \\ 1 & 2 & 1 & 0 & 2 & 3 & 0 & 2 \\ 2 & 2 & 2 & -3 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{11} 3 21^{1131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 5 & 0 & 5 & 20 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 15 & 4 & 4 \\ 9 & 20 & 0 & 15 & 57 & 15 & 15 \\ 2 & 5 & 0 & 4 & 15 & 2 & 4 \\ 3 & 5 & -3 & 4 & 15 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1111} 3 21^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 & 3 & 2 \\ 1 & 2 & 0 & 0 & 3 & 4 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 1 & 4 & 4 & 3 \\ 2 & 3 & 4 & 0 & 4 & 4 & 7 & 4 \\ 2 & 3 & 4 & 0 & 4 & 7 & 5 & 5 \\ 2 & 2 & 2 & -3 & 3 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{1111} 3 21^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 20 & 5 \\ 9 & 13 & 1 & 0 & 30 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 20 & 30 & 0 & 0 & 63 & 15 \\ 5 & 7 & -1 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{11} 3 21^{1112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 5 & 0 & 5 & 5 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 4 & 11 & 4 \\ 2 & 5 & 0 & 4 & 1 & 11 & 3 \\ 7 & 15 & 0 & 11 & 11 & 32 & 12 \\ 3 & 5 & -3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{111111} 3 21^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 262, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 3 & 2 \\ 1 & 2 & 0 & 0 & 1 & 4 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 2 & 1 & 0 & 0 & 3 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 4 & 7 & 4 \\ 2 & 3 & 4 & 0 & 3 & 7 & 5 & 5 \\ 2 & 2 & 2 & -3 & 2 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{12} 3 21^{1113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 340, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 5 & 0 & 5 & 5 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 4 & 18 & 4 \\ 2 & 5 & 0 & 4 & 1 & 18 & 3 \\ 12 & 25 & 0 & 18 & 18 & 94 & 20 \\ 3 & 5 & -3 & 4 & 3 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{1111} 3 21^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 5 & 0 & 10 & 5 & 10 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 10 & 0 & 14 & 7 & 16 & 8 \\ 2 & 5 & 0 & 7 & 1 & 7 & 3 \\ 5 & 10 & 0 & 16 & 7 & 14 & 8 \\ 3 & 5 & -3 & 8 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{13} 321^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 380, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 5 & 0 & 5 & 5 & 35 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 4 & 25 & 4 \\ 2 & 5 & 0 & 4 & 1 & 25 & 3 \\ 17 & 35 & 0 & 25 & 25 & 188 & 28 \\ 3 & 5 & -3 & 4 & 3 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{111} 321^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 390, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 7 & 3 \\ 3 & 5 & 0 & 10 & 5 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 10 & 0 & 14 & 7 & 23 & 8 \\ 2 & 5 & 0 & 7 & 1 & 11 & 3 \\ 7 & 15 & 0 & 23 & 11 & 32 & 12 \\ 3 & 5 & -3 & 8 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{111} 321^{1213} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 630, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 9 & 3 \\ 3 & 5 & 0 & 15 & 5 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 15 & 0 & 32 & 11 & 45 & 12 \\ 2 & 5 & 0 & 11 & 1 & 15 & 3 \\ 9 & 20 & 0 & 45 & 15 & 58 & 16 \\ 3 & 5 & -3 & 12 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{12} 321^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 640, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 5 & 0 & 5 & 5 & 30 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 2 & 4 & 22 & 4 \\ 2 & 5 & 0 & 4 & 1 & 22 & 3 \\ 14 & 30 & 0 & 22 & 22 & 134 & 24 \\ 3 & 5 & -3 & 4 & 3 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{111} 321^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 720, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 7 & 3 \\ 3 & 5 & 0 & 15 & 5 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 15 & 0 & 32 & 11 & 34 & 12 \\ 2 & 5 & 0 & 11 & 1 & 11 & 3 \\ 7 & 15 & 0 & 34 & 11 & 32 & 12 \\ 3 & 5 & -3 & 12 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{1111} 321^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 752, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 5 & 7 & 5 \\ 2 & 3 & 4 & 0 & 7 & 4 & 7 \\ 2 & 3 & 4 & 0 & 7 & 7 & 5 \\ 2 & 2 & 2 & -3 & 5 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{111} 321^{1313} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 840, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.00 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 9 & 3 \\ 3 & 5 & 0 & 20 & 5 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 20 & 0 & 58 & 15 & 60 & 16 \\ 2 & 5 & 0 & 15 & 1 & 15 & 3 \\ 9 & 20 & 0 & 60 & 15 & 58 & 16 \\ 3 & 5 & -3 & 16 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [71] \quad 1^{111} 3221^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -52, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 7 & 3 \\ 3 & 6 & 0 & 17 & 6 & 17 & 6 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 17 & 0 & 36 & 13 & 38 & 14 \\ 2 & 6 & 0 & 13 & 2 & 13 & 4 \\ 7 & 17 & 0 & 38 & 13 & 36 & 14 \\ 3 & 6 & -3 & 14 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{1111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -468, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 3 \\ 1 & 2 & 1 & 0 & 5 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 6 & 8 & 8 \\ 2 & 3 & 5 & 0 & 8 & 5 & 8 \\ 2 & 3 & 5 & 0 & 8 & 5 & 8 \\ 2 & 2 & 3 & -3 & 6 & 5 & 6 \end{pmatrix}_{b_0}$$

4.487 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 10 \quad [0] \quad 1^{11}3^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 2 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{1111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 3 & 2 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 2 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{1111}3^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 9 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 5 & 7 \\ 2 & 3 & 0 & 3 & 2 & 5 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{1111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 3 & 2 & 3 \\ 5 & 9 & 0 & 6 & 12 & 6 & 5 & 6 \\ 2 & 4 & 0 & 3 & 6 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 5 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 6 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{1111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 3 & 2 & 3 \\ 5 & 9 & 0 & 6 & 12 & 6 & 5 & 6 \\ 2 & 4 & 0 & 3 & 6 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 5 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{1111}3^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 15 & 5 & 7 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 & 2 \\ 5 & 9 & 0 & 15 & 6 & 13 & 5 & 7 \\ 2 & 3 & 0 & 5 & 2 & 5 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.488 NHC(-2)(-3) + f₄ + su₂ + su₃ + su₂→(-3)

$$T = 9 \quad [0] \quad 1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 12 & 4 & 6 \\ 5 & 9 & 0 & 12 & 12 & 5 & 6 \\ 2 & 3 & 0 & 4 & 5 & 0 & 2 \\ 3 & 4 & -3 & 6 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 2 & 3 \\ 4 & 9 & 4 & 3 & 0 & 4 & 5 \\ 2 & 4 & -2 & 3 & 0 & 2 & 0 \\ 3 & 8 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 3 & 0 & 0 & 2 \\ 3 & 5 & 0 & 4 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 10 & 6 & 4 & 6 \\ 2 & 4 & 0 & 6 & 0 & 2 & 2 \\ 2 & 3 & 0 & 4 & 2 & 0 & 2 \\ 3 & 4 & -3 & 6 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 19 & 0 & 73 & 12 & 11 & 15 \\ 2 & 4 & 0 & 12 & 0 & 2 & 2 \\ 2 & 3 & 0 & 11 & 2 & 0 & 2 \\ 3 & 4 & -3 & 15 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 2 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.31 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 4 & 8 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 2 & 3 \\ 4 & 8 & 0 & 6 & 9 & 4 & 5 \\ 2 & 3 & 0 & 2 & 4 & 0 & 2 \\ 3 & 4 & -3 & 3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[7] \quad & 1^{11} 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -184, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 2 & 3 \\ 4 & 9 & 4 & 8 & 0 & 4 & 5 \\ 2 & 4 & -2 & 3 & 0 & 2 & 0 \\ 4 & 8 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 3 & 0 & 0 & 2 \\ 3 & 5 & 0 & 4 & -1 & 2 & 0 \end{pmatrix}_{b_0} \\
[8] \quad & 1^{11} 1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -206, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 13.4 \\
& T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 2 & 3 \\ 3 & 4 & 0 & 9 & 8 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 12 & 5 & 7 \\ 4 & 8 & 0 & 12 & 9 & 4 & 5 \\ 2 & 3 & 0 & 5 & 4 & 0 & 2 \\ 3 & 4 & -3 & 7 & 5 & 2 & 0 \end{pmatrix}_{b_0} \\
[9] \quad & 1^{11} 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -226, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 2 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[10] \quad & 1^{11} 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -336, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 2 & 3 \\ 5 & 9 & 0 & 6 & 12 & 5 & 6 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -3 & 3 & 6 & 2 & 1 \end{pmatrix}_{b_0} \\
[11] \quad & 1^{11} 1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -386, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 9 & 8 & 11 \\ 2 & 4 & 0 & 9 & 0 & 2 & 2 \\ 2 & 3 & 0 & 8 & 2 & 0 & 2 \\ 3 & 4 & -3 & 11 & 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[12] \quad & 1^{11} 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 5 & 7 \\ 2 & 4 & 0 & 6 & 0 & 2 & 2 \\ 2 & 3 & 0 & 5 & 2 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[13] \quad & 1^{11} 1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -476, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 13.4 \\
& T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 4 & 13 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 9 & 2 & 3 \\ 7 & 13 & 0 & 9 & 27 & 7 & 9 \\ 2 & 3 & 0 & 2 & 7 & 0 & 2 \\ 3 & 4 & -3 & 3 & 9 & 2 & 1 \end{pmatrix}_{b_0} \\
[14] \quad & 1^{12} 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -626, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.7 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 9 & 7 & 10 \\ 2 & 4 & 0 & 9 & 0 & 2 & 2 \\ 2 & 3 & 0 & 7 & 2 & 0 & 2 \\ 3 & 4 & -3 & 10 & 2 & 2 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

[15] $1^{131}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 12 & 9 & 13 \\ 2 & 4 & 0 & 12 & 0 & 2 & 2 \\ 2 & 3 & 0 & 9 & 2 & 0 & 2 \\ 3 & 4 & -3 & 13 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{121}3^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 2 & 3 \\ 3 & 4 & 0 & 28 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 16 & 28 & 0 & 154 & 18 & 16 & 22 \\ 2 & 4 & 0 & 18 & 0 & 2 & 2 \\ 2 & 3 & 0 & 16 & 2 & 0 & 2 \\ 3 & 4 & -3 & 22 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{141}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 15 & 11 & 16 \\ 2 & 4 & 0 & 15 & 0 & 2 & 2 \\ 2 & 3 & 0 & 11 & 2 & 0 & 2 \\ 3 & 4 & -3 & 16 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{121}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 10 & 18 & 0 & 58 & 12 & 10 & 14 \\ 2 & 4 & 0 & 12 & 0 & 2 & 2 \\ 2 & 3 & 0 & 10 & 2 & 0 & 2 \\ 3 & 4 & -3 & 14 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{121}3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 2 & 3 \\ 3 & 4 & 0 & 23 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 13 & 23 & 0 & 100 & 15 & 13 & 18 \\ 2 & 4 & 0 & 15 & 0 & 2 & 2 \\ 2 & 3 & 0 & 13 & 2 & 0 & 2 \\ 3 & 4 & -3 & 18 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{131}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 12 & 22 & 0 & 85 & 15 & 12 & 17 \\ 2 & 4 & 0 & 15 & 0 & 2 & 2 \\ 2 & 3 & 0 & 12 & 2 & 0 & 2 \\ 3 & 4 & -3 & 17 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{131}3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 2 & 3 \\ 3 & 4 & 0 & 27 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 15 & 27 & 0 & 133 & 18 & 15 & 21 \\ 2 & 4 & 0 & 18 & 0 & 2 & 2 \\ 2 & 3 & 0 & 15 & 2 & 0 & 2 \\ 3 & 4 & -3 & 21 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{141}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 2 & 3 \\ 3 & 4 & 0 & 26 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 14 & 26 & 0 & 118 & 18 & 14 & 20 \\ 2 & 4 & 0 & 18 & 0 & 2 & 2 \\ 2 & 3 & 0 & 14 & 2 & 0 & 2 \\ 3 & 4 & -3 & 20 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{131}3^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1826, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 2 & 3 \\ 3 & 4 & 0 & 32 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 18 & 32 & 0 & 193 & 21 & 18 & 25 \\ 2 & 4 & 0 & 21 & 0 & 2 & 2 \\ 2 & 3 & 0 & 18 & 2 & 0 & 2 \\ 3 & 4 & -3 & 25 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{141}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 2 & 3 \\ 3 & 4 & 0 & 31 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 17 & 31 & 0 & 172 & 21 & 17 & 24 \\ 2 & 4 & 0 & 21 & 0 & 2 & 2 \\ 2 & 3 & 0 & 17 & 2 & 0 & 2 \\ 3 & 4 & -3 & 24 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{141}3^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 2 & 3 \\ 3 & 4 & 0 & 36 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 20 & 36 & 0 & 238 & 24 & 20 & 28 \\ 2 & 4 & 0 & 24 & 0 & 2 & 2 \\ 2 & 3 & 0 & 20 & 2 & 0 & 2 \\ 3 & 4 & -3 & 28 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [26] \quad 1^{11}4^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 0 & 5 & 4 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 4 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{111}4^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 1 & 3 \\ 3 & 4 & 0 & 7 & 4 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 5 & 7 & 0 & 8 & 5 & 0 & 6 \\ 2 & 4 & 0 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 6 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{111}4^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 1 & 3 \\ 3 & 4 & 0 & 12 & 4 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 8 & 12 & 0 & 26 & 8 & 0 & 10 \\ 2 & 4 & 0 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 10 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$4.489 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$$

$$T = 10 \quad [0] \quad 1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 3 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 & 3 \\ 2 & 3 & 0 & 3 & 2 & 0 & 3 \\ 3 & 5 & 0 & 6 & 3 & 3 & 4 \\ 3 & 4 & -3 & 4 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111}3^{11}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 2 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 & 3 \\ 2 & 3 & 0 & 2 & 2 & 0 & 2 \\ 2 & 4 & 0 & 3 & 3 & 2 & 1 \\ 3 & 4 & -3 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 & 6 & 2 \\ 2 & 3 & 0 & 3 & 2 & 0 & 5 & 2 \\ 5 & 9 & 0 & 9 & 6 & 5 & 13 & 7 \\ 3 & 4 & -3 & 4 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 2 & 3 & 3 \\ 5 & 9 & 0 & 6 & 12 & 5 & 6 & 6 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 6 & 2 & 1 & 3 \\ 3 & 4 & -3 & 3 & 6 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 2 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 & 6 & 2 \\ 2 & 3 & 0 & 2 & 2 & 0 & 5 & 2 \\ 5 & 9 & 0 & 6 & 6 & 5 & 13 & 7 \\ 3 & 4 & -3 & 3 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 5 & 15 & 7 \\ 2 & 4 & 0 & 6 & 0 & 2 & 6 & 2 \\ 2 & 3 & 0 & 5 & 2 & 0 & 5 & 2 \\ 5 & 9 & 0 & 15 & 6 & 5 & 13 & 7 \\ 3 & 4 & -3 & 7 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.490 NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_{2n3\text{mix}}$

$$T = 9 \quad [0] \quad 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 5 & 2 \\ 5 & 8 & 0 & 9 & 5 & 12 & 6 \\ 3 & 4 & -3 & 4 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 11 & 3 \\ 2 & 4 & 0 & 3 & 0 & 11 & 2 \\ 10 & 17 & 0 & 11 & 11 & 55 & 13 \\ 3 & 4 & -3 & 3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 4 & 0 & 3 & 0 & 5 & 2 \\ 4 & 7 & 0 & 6 & 5 & 7 & 5 \\ 3 & 4 & -3 & 4 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1123}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 6 & 3 \\ 3 & 4 & 0 & 4 & 5 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 8 & 3 \\ 3 & 5 & 0 & 3 & 3 & 9 & 3 \\ 6 & 11 & 0 & 8 & 9 & 20 & 8 \\ 3 & 4 & -3 & 3 & 3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1113}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -204, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.14 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 0 & 4 & 5 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 5 & 3 \\ 3 & 5 & 0 & 3 & 3 & 6 & 3 \\ 4 & 7 & 0 & 5 & 6 & 7 & 5 \\ 3 & 4 & -3 & 3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1113}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -269, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 14 & 7 \\ 2 & 4 & 0 & 6 & 0 & 5 & 2 \\ 5 & 8 & 0 & 14 & 5 & 12 & 6 \\ 3 & 4 & -3 & 7 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1113}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 12 & 4 \\ 2 & 4 & 0 & 3 & 0 & 8 & 2 \\ 7 & 12 & 0 & 12 & 8 & 25 & 9 \\ 3 & 4 & -3 & 4 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1113}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 0 & 5 & 4 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 18 & 4 \\ 2 & 4 & 0 & 3 & 0 & 14 & 2 \\ 11 & 20 & 0 & 18 & 14 & 69 & 15 \\ 3 & 4 & -3 & 4 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1113}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -424, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.07 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 5 & 3 \\ 2 & 4 & 0 & 3 & 0 & 5 & 2 \\ 4 & 7 & 0 & 5 & 5 & 7 & 5 \\ 3 & 4 & -3 & 3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1113}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 4 & 3 \\ 3 & 4 & 0 & 4 & 9 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 6 & 5 & 3 \\ 5 & 9 & 0 & 6 & 12 & 11 & 6 \\ 4 & 7 & 0 & 5 & 11 & 7 & 5 \\ 3 & 4 & -3 & 3 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1113}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -481, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 8 & 3 \\ 2 & 4 & 0 & 3 & 0 & 8 & 2 \\ 7 & 12 & 0 & 8 & 8 & 25 & 9 \\ 3 & 4 & -3 & 3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{112}3^{\overline{1}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2} \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -500, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 5 & 4 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 15 & 4 \\ 2 & 4 & 0 & 3 & 0 & 11 & 2 \\ 9 & 16 & 0 & 15 & 11 & 44 & 12 \\ 3 & 4 & -3 & 4 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{111}3^{\overline{1}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2} \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -509, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.06 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 4 & 3 \\ 3 & 4 & 0 & 9 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 11 & 7 \\ 2 & 4 & 0 & 6 & 0 & 5 & 2 \\ 4 & 7 & 0 & 11 & 5 & 7 & 5 \\ 3 & 4 & -3 & 7 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{111}3^{\overline{1}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2} \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -896, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 7 & 3 \\ 3 & 4 & 0 & 9 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 20 & 7 \\ 2 & 4 & 0 & 6 & 0 & 8 & 2 \\ 7 & 12 & 0 & 20 & 8 & 25 & 9 \\ 3 & 4 & -3 & 7 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

4.491 NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{111}3^{\overline{1}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3} \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -39, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 & 0 \\ 4 & 8 & 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}3^{\overline{1}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3} \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -119, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 3 & 3 \\ 3 & 4 & -3 & 4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}3^{\overline{1}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3} \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -126, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.05 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{113}3^{\overline{1}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3} \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -159, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 1 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 & 0 \\ 10 & 24 & 9 & 52 & 0 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 12 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{112}3^{\overline{1}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3} \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -174, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 1 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 0 & 5 \\ 2 & 4 & -2 & 6 & 0 & 0 & 0 \\ 7 & 16 & 6 & 22 & 0 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 8 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -182$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.26 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{122}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -200$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.9$

$T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 4 & 3 \\ 3 & 4 & 0 & 13 & 5 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 12 & 18 & 10 \\ 3 & 5 & 0 & 12 & 3 & 6 & 3 \\ 4 & 8 & 0 & 18 & 6 & 9 & 5 \\ 3 & 4 & -3 & 10 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -203$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.34 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 9 & 36 & 10 \\ 3 & 4 & -3 & 3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{132}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -206$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 4 & 3 \\ 3 & 4 & 0 & 17 & 5 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 15 & 24 & 13 \\ 3 & 5 & 0 & 15 & 3 & 6 & 3 \\ 4 & 8 & 0 & 24 & 6 & 9 & 5 \\ 3 & 4 & -3 & 13 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{111}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -315$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 8 & 14 & 0 & 37 & 9 & 9 & 11 \\ 2 & 4 & 0 & 9 & 0 & 3 & 2 \\ 2 & 4 & 0 & 9 & 3 & 0 & 2 \\ 3 & 4 & -3 & 11 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{113}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -321$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.21 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 0 & 5 \\ 2 & 4 & -2 & 9 & 0 & 0 & 0 \\ 11 & 24 & 9 & 52 & 0 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 12 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 12 & 6 \\ 3 & 4 & -3 & 4 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{111}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -399$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.53 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 \\ 3 & 4 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{114} 3^{1^{311}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 & 0 \\ 15 & 32 & 12 & 94 & 0 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 16 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3^{1^{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 9 & 6 & 7 \\ 3 & 5 & 0 & 9 & 3 & 3 & 3 \\ 2 & 4 & 0 & 6 & 3 & 0 & 2 \\ 3 & 4 & -3 & 7 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3^{1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & -3 & 3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{114} 3^{1^{211}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 0 & 5 \\ 2 & 4 & -2 & 12 & 0 & 0 & 0 \\ 14 & 32 & 12 & 94 & 0 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 16 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{141} 3^{1^{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -527, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 2 & 3 \\ 3 & 4 & 0 & 21 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 21 & 0 & 76 & 18 & 15 & 16 \\ 3 & 5 & 0 & 18 & 3 & 3 & 3 \\ 2 & 4 & 0 & 15 & 3 & 0 & 2 \\ 3 & 4 & -3 & 16 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111} 3^{1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -621, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 6 & 7 \\ 2 & 4 & 0 & 6 & 0 & 3 & 2 \\ 2 & 4 & 0 & 6 & 3 & 0 & 2 \\ 3 & 4 & -3 & 7 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{121} 3^{1^{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -707, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 2 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 28 & 12 & 9 & 10 \\ 3 & 5 & 0 & 12 & 3 & 3 & 3 \\ 2 & 4 & 0 & 9 & 3 & 0 & 2 \\ 3 & 4 & -3 & 10 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{131} 3^{1^{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -713, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 2 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 17 & 0 & 49 & 15 & 12 & 13 \\ 3 & 5 & 0 & 15 & 3 & 3 & 3 \\ 2 & 4 & 0 & 12 & 3 & 0 & 2 \\ 3 & 4 & -3 & 13 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1111} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 14, \quad \Delta = -95, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.22 \rightarrow 13, \quad T_{\max} = 14, \quad T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 13 & 6 & 15 & 7 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 0 & 15 & 6 & 12 & 6 \\ 3 & 4 & -3 & 7 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

4.492 $\text{NHC}(-2)(-3) + \mathbf{f}_4 + \mathbf{su}_2 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \quad 1^1 3^1 1^{11} 2^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + (-3) \rightarrow (-2) \quad T = 8, \quad T_{\max} = 31, \quad \Delta = -117, \quad \det = 725, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 31, \quad T \in [12, 31]$$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 1 & 4 \\ 8 & 9 & 4 & 8 & 0 & 5 \\ 3 & 4 & -2 & 3 & 0 & 0 \\ 6 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

4.493 $\text{NHC}(-2)(-3) + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3)$

$$T = 7 \quad [0] \quad 2^1 1^3 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -139, \quad \det = -20, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 97, \quad T \in [12, 97]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 3 & 4 & -2 & 2 & 0 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 5 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -139, \quad \det = -20, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 97, \quad T \in [12, 97]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 3 & 4 & -2 & 2 & 0 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 5 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 1^3 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -139, \quad \det = -44, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.09 \rightarrow 13, \quad T_{\max} = 97, \quad T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 2 & 4 & -2 & 2 & 0 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 5 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^1 1^3 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -139, \quad \det = -46, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.48 \rightarrow 12, \quad T_{\max} = 97, \quad T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 2^1 1^3 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -139, \quad \det = -170, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 97, \quad T \in [12, 97]$$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 4 \\ 8 & 9 & 4 & 4 & 5 \\ 3 & 4 & -2 & 2 & 0 \\ 3 & 4 & 2 & 0 & 2 \\ 4 & 5 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [5] \quad 1^{11} 3^1 1^1 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 15, \quad T_{\max} = 35, \quad T \in [15, 35]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 2 & 5 & 0 & 4 \\ 2 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 2^1 1^3 3^1 2^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 22, \quad \Delta = -112, \quad \det = 13, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 12, \quad T_{\max} = 22, \quad T \in [12, 22]$$

$$\begin{pmatrix} 13 & 20 & 9 & 7 \\ 20 & 30 & 13 & 10 \\ 9 & 13 & 4 & 4 \\ 7 & 10 & 4 & 3 \end{pmatrix}_{b_0}$$

[7] $13^1 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 5 & 13 & 6 & 5 \\ 13 & 30 & 13 & 10 \\ 6 & 13 & 4 & 4 \\ 5 & 10 & 4 & 2 \end{pmatrix}_{b_0}$

[9] $13^{11} 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 5 & 12 & 6 & 5 \\ 12 & 25 & 12 & 9 \\ 6 & 12 & 4 & 4 \\ 5 & 9 & 4 & 2 \end{pmatrix}_{b_0}$

[8] $1^{11} 3^1 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 2 & 0 & 0 & 2 \\ 2 & 3 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$

[10] $1^1 3^1 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 6 & 1 & 5 \\ 6 & 5 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 5 & 5 & -1 & 3 \end{pmatrix}_{b_0}$

[11] $1^1 31^{13} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$
 $\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 6 & 0 & 4 \\ 4 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & -3 & 1 \end{pmatrix}_{b_0}$

[12] $1^1 3^1 1^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$

[13] $1^1 31^{13} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12, T_{\max} = 35, T \in [12, 35]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 8 & 6 & 0 & 6 \\ 2 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & -3 & 2 \end{pmatrix}_{b_0}$

[14] $1^1 3^1 1^1 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 2 & 4 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$

[15] $21^{11} 3^1 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 13 & 20 & 9 & 7 \\ 20 & 25 & 12 & 9 \\ 9 & 12 & 4 & 4 \\ 7 & 9 & 4 & 3 \end{pmatrix}_{b_0}$

[16] $1^1 321^{11} 3^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 45, \Delta = -119, \det = 375, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 45, T \in [12, 45]$
 $\begin{pmatrix} 0 & 3 & 3 & 1 & 1 & 2 \\ 3 & 1 & 4 & 1 & 0 & 3 \\ 3 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$

$T = 9$ [17] $1^1 4^1 13^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$
 $\begin{pmatrix} 1 & 4 & 1 & 1 & 3 \\ 4 & 9 & 0 & 0 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & -2 & 0 & -1 \end{pmatrix}_{b_0}$

- [18] $21^{\overline{1}1}413^{\overline{1}1}1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 21$, $\Delta = -109$, $\det = -2$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 16$, $T_{\max} = 21$, $T \in [16, 21]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 0 & 0 & 4 \\ 2 & 7 & 3 & 0 & 0 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 3 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$
- [19] $1^{\overline{1}1}413^{\overline{1}1}1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 21$, $\Delta = -109$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$
- [20] $1^{\overline{1}1}23^{\overline{1}1}131^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -17$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 3 & 3 \\ 5 & 18 & 1 & 10 & 8 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 10 & 0 & 4 & 4 \\ 3 & 8 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$
- [21] $1^{\overline{1}1}2313^{\overline{1}1}1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -20$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 0 & 0 & 0 & 2 \\ 3 & 0 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$
- [22] $1^{\overline{1}1}231^{\overline{1}1}3^{\overline{1}1}1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -40$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.80 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 8 & 6 & 0 & 6 \\ 2 & 6 & 2 & 0 & 4 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 6 & 4 & -3 & 2 \end{pmatrix}_{b_0}$$
- [23] $21^{\overline{1}1}41^{\overline{1}1}3^{\overline{1}1}1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 21$, $\Delta = -109$, $\det = -42$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 0 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & -2 & 0 & 1 \end{pmatrix}_{b_0}$$
- [24] $1^{\overline{1}1}23^{\overline{1}1}1^{\overline{1}}31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -52$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 3 & 3 \\ 5 & 13 & 1 & 9 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 9 & 0 & 4 & 4 \\ 3 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$
- [25] $1^{\overline{1}1}23^{\overline{1}1}1^{\overline{1}}31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -52$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.46 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 2 & 4 & 3 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$
- [26] $1^{\overline{1}1}231^{\overline{1}1}3^{\overline{1}1}1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -60$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 2 & 3 \\ 5 & 13 & 1 & 5 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 0 & 2 \\ 3 & 7 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$
- [27] $1^{\overline{1}1}4^{\overline{1}1}1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 21$, $\Delta = -109$, $\det = -72$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 4 & 1 & 1 & 3 \\ 4 & 4 & 0 & 0 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$
- [28] $1^{\overline{1}1}2321^{\overline{1}1}3^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -265$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.22 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 1 & 2 \\ 3 & 1 & 4 & 1 & 0 & 3 \\ 3 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
T = 10 \quad [29] \quad & 1^1 5 13^1 1^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 16, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 5 & 9 & 8 & 5 \\ 9 & 13 & 12 & 7 \\ 8 & 12 & 10 & 6 \\ 5 & 7 & 6 & 3 \end{pmatrix}_{b_0} \\
[30] \quad & 21^1 5 13^1 1^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = \\
& 9.2 \\
& T_{\min} = 13.44 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 5 & 10 & 8 & 5 \\ 10 & 13 & 12 & 7 \\ 8 & 12 & 10 & 6 \\ 5 & 7 & 6 & 3 \end{pmatrix}_{b_0} \\
[31] \quad & 21^1 5 13^1 1^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = \\
& 9.2 \\
& T_{\min} = 13.44 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 13 & 14 & 12 & 7 \\ 14 & 13 & 12 & 7 \\ 12 & 12 & 10 & 6 \\ 7 & 7 & 6 & 3 \end{pmatrix}_{b_0} \\
[32] \quad & 1^1 31^1 413^1 1^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = \\
& 9.2 \\
& T_{\min} = 13.40 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0} \\
[33] \quad & 1^1 3^1 1^1 4131^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = \\
& 9.2 \\
& T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & -2 & 2 & 0 \\ 1 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0} \\
[34] \quad & 1^1 3^1 1^1 4131^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = \\
& 9.2 \\
& T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 2 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0} \\
T = 11 \quad [35] \quad & 1^1 31^1 5 13^1 1^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -8, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -2 & 2 & 0 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 2 & 0 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0} \\
[36] \quad & 1^1 3^1 151^1 231^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -10, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 8 & 1 & 4 & 3 \\ 8 & 38 & 1 & 20 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 20 & 0 & 10 & 6 \\ 3 & 12 & -1 & 6 & 3 \end{pmatrix}_{b_0} \\
[37] \quad & 1^1 3^1 15 131^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = \\
& 9.2 \\
& T_{\min} = 13.60 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 2 & 0 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0} \\
[38] \quad & 1^1 3^1 15 131^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{s}u_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -21, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = \\
& 9.2
\end{aligned}$$

$$T_{\min} = 13.67 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 3 & -1 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[39] \ 1^1 3^1 1^1 2^1 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -85, \det = -84, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 8 & 1 & 2 & 3 \\ 8 & 38 & 1 & 8 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 8 & 0 & 0 & 2 \\ 3 & 12 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [40] \ 1^1 3^1 1^1 2^1 6^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[41] \ 1^1 3^1 1^1 3^1 6^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 52, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 15.15 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 3 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[42] \ 1^1 3^1 2^1 1^1 6^1 2^1 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 15, \Delta = -83, \det = 84, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 1 & 2 \\ 3 & 10 & 5 & 0 & 4 & 6 \\ 1 & 5 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 4 & 2 & 0 & 0 & 2 \\ 2 & 6 & 2 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [43] \ 1^1 7^1 1^1 1^1 1^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 16, \Delta = -81, \det = -20, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 15.60 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 2 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [45] \ 1^1 8^1 1^1 1^1 1^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 14, T_{\max} = 17, \Delta = -80, \det = 20, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 16.60 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 2 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[44] \ 1^1 8^1 1^1 1^1 1^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 14, T_{\max} = 17, \Delta = -80, \det = 20, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 16.60 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [46] \ 1^1 8^1 1^1 1^1 1^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 15, T_{\max} = 18, \Delta = -79, \det = -20, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 17.60 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 5 & 1 & 9 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 9 & 0 & 13 & 7 \\ 3 & 5 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

4.494 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -162, \quad \det = 160, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.40 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -2 & 2 & 0 & 0 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 2 & -3 & -1 \end{pmatrix}_{b_0}$$

4.495 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -320, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.40 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 2 & 0 & 3 & 0 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & 2 & -3 & 4 & -1 \end{pmatrix}_{b_0}$$

4.496 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -89, \quad \det = -160, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.25 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 2 & 3 & 0 & 0 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 3 & 8 & 3 & 3 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & 0 & 2 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -89, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 & 4 \\ 2 & 4 & 0 & 2 & 4 & 0 & 2 \\ 3 & 4 & -3 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

4.497 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \text{NHC}(-2)(-3) + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 200, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 2 & 4 & 0 & 3 & 2 \\ 2 & 4 & 0 & 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & -3 & 2 & 4 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 600, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 & 5 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 2 & 4 & 0 & 6 & 2 \\ 5 & 9 & 0 & 5 & 9 & 6 & 13 & 7 \\ 3 & 4 & -3 & 2 & 4 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.498 NHC(-2)(-3) + f₄ + su₂→(-3) + so₈

$$T = 8 \quad [0] \quad 1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -2 & 2 & 0 & 0 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 2 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.68 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 5 \\ 11 & 20 & 0 & 11 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 11 & 0 & 4 & 4 \\ 5 & 8 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \quad 1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{so}_8 \quad T = 9, T_{\max} = 22, \Delta = -111, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 5 & 13 & 6 & 1 & 5 \\ 13 & 30 & 13 & 0 & 10 \\ 6 & 13 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 10 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{so}_8 \quad T = 9, T_{\max} = 22, \Delta = -111, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11}3^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 2 & 6 & 0 & 4 & 4 \\ 4 & 6 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 0 & 0 & 2 \\ 3 & 4 & 5 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11}3^{11}1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 & 3 \\ 4 & 10 & 13 & 0 & 8 & 6 \\ 5 & 13 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 9 & 0 & 4 & 4 \\ 3 & 6 & 6 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [6] \quad 1^{11}4^{11}13^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 5 & 12 & 1 & 1 & 5 \\ 12 & 25 & 0 & 0 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 9 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11}23^{11}1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{so}_8 \quad T = 10, T_{\max} = 14, \Delta = -90, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 \\ 1 & 4 & 1 & 0 & 2 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 1 & 3 & 4 & 0 & 2 & 0 \\ 2 & 4 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11}3^{11}4^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 202, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.63 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 1 & 1 & 5 \\ 10 & 13 & 0 & 0 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 7 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [9] \quad 1^{11}3^{11}5^{11}131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 66, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 14.94 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 26 & 1 & 20 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 10 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$4.499 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2$$

$$T = 9 \quad [0] \quad 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 13 & 7 \\ 5 & 8 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 2 & 0 & 3 & 0 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & 2 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 2 & 0 & 3 & 0 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & 2 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$4.500 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_3$$

$$T = 9 \quad [0] \quad 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 12 & 6 \\ 5 & 8 & -3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

$$4.501 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$$

$$T = 8 \quad [0] \quad 1^{13}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 25 & 9 \\ 3 & 4 & -3 & 2 & 9 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{13}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 10 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 6 & 10 & 0 & 6 & 22 & 8 \\ 3 & 4 & -3 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

- [2] $21^1 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 3 \\ 4 & 9 & 4 & 4 & 8 & 5 \\ 3 & 4 & -2 & 2 & 3 & 0 \\ 2 & 4 & 2 & 0 & 3 & 2 \\ 3 & 8 & 3 & 3 & 4 & 4 \\ 3 & 5 & 0 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$
- [3] $1^{111} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 3 \\ 4 & 9 & 4 & 4 & 8 & 5 \\ 3 & 4 & -2 & 2 & 3 & 0 \\ 2 & 4 & 2 & 0 & 3 & 2 \\ 3 & 8 & 3 & 3 & 4 & 4 \\ 3 & 5 & 0 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$
- [4] $1^{12} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 5 & 3 \\ 4 & 9 & 4 & 4 & 8 & 5 \\ 2 & 4 & -2 & 2 & 3 & 0 \\ 2 & 4 & 2 & 0 & 3 & 2 \\ 5 & 8 & 3 & 3 & 4 & 4 \\ 3 & 5 & 0 & 2 & 4 & -2 \end{pmatrix}_{b_0}$$
- [5] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -2 & 2 & 0 & 0 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & 2 & 0 & -1 \end{pmatrix}_{b_0}$$
- [6] $1^{12} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 4 & 8 & 0 & 4 & 10 & 6 \\ 3 & 4 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$
- [8] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [9] $13^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 & 3 \\ 4 & 9 & 4 & 4 & 8 & 5 \\ 2 & 4 & -2 & 2 & 3 & 0 \\ 2 & 4 & 2 & 0 & 3 & 2 \\ 3 & 8 & 3 & 3 & 4 & 4 \\ 3 & 5 & 0 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{111} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 4 & 3 \\ 4 & 9 & 4 & 4 & 8 & 5 \\ 2 & 4 & -2 & 2 & 3 & 0 \\ 2 & 4 & 2 & 0 & 3 & 2 \\ 4 & 8 & 3 & 3 & 4 & 4 \\ 3 & 5 & 0 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{11} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 14 & 24 & 0 & 14 & 121 & 19 \\ 3 & 4 & -3 & 2 & 19 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 5 & 9 & 0 & 5 & 13 & 7 \\ 3 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 8 & 14 & 0 & 8 & 37 & 11 \\ 3 & 4 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 11 & 19 & 0 & 11 & 73 & 15 \\ 3 & 4 & -3 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 7 & 13 & 0 & 7 & 28 & 10 \\ 3 & 4 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 282, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 9 & 17 & 0 & 9 & 49 & 13 \\ 3 & 4 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{14}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 11 & 21 & 0 & 11 & 76 & 16 \\ 3 & 4 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{12}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 10 & 18 & 0 & 10 & 58 & 14 \\ 3 & 4 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 21^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 6 & 4 \\ 8 & 9 & 4 & 4 & 8 & 5 \\ 3 & 4 & -2 & 2 & 3 & 0 \\ 3 & 4 & 2 & 0 & 3 & 2 \\ 6 & 8 & 3 & 3 & 4 & 4 \\ 4 & 5 & 0 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{12}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 13 & 2 \\ 13 & 23 & 0 & 13 & 100 & 18 \\ 3 & 4 & -3 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 522, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 12 & 22 & 0 & 12 & 85 & 17 \\ 3 & 4 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{12}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 532, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 3 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 16 & 2 \\ 16 & 28 & 0 & 16 & 154 & 22 \\ 3 & 4 & -3 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{14}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 652, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 14 & 26 & 0 & 14 & 118 & 20 \\ 3 & 4 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{13}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 722, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 3 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 15 & 2 \\ 15 & 27 & 0 & 15 & 133 & 21 \\ 3 & 4 & -3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{13}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 882, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 3 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 18 & 2 \\ 18 & 32 & 0 & 18 & 193 & 25 \\ 3 & 4 & -3 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{14}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 952, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 3 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 17 & 2 \\ 17 & 31 & 0 & 17 & 172 & 24 \\ 3 & 4 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{14}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 3 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 20 & 2 \\ 20 & 36 & 0 & 20 & 238 & 28 \\ 3 & 4 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [28] \quad 1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 11 & 5 \\ 6 & 5 & 1 & 10 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 10 & 0 & 18 & 8 \\ 5 & 5 & -1 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{11}3^{11}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 8 & 6 & 0 & 5 & 6 \\ 2 & 6 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 2 & 4 \\ 3 & 6 & 3 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{12}3^{12}2^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 12 & 5 \\ 6 & 5 & 1 & 14 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 14 & 0 & 26 & 10 \\ 5 & 5 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{11}3^{12}2^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 16 & 5 \\ 6 & 5 & 1 & 17 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 17 & 0 & 45 & 13 \\ 5 & 5 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{11}3^{12}2^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 1 & 5 \\ 6 & 5 & 1 & 0 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 5 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{11}3^{12}2^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 6 & 5 \\ 6 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 7 & 0 & 5 & 5 \\ 5 & 5 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{11}3^{11}1^13^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 3 & 0 & 0 & 2 \\ 1 & 3 & 3 & 0 & 2 & 1 \\ 2 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{12}3^{11}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 8 & 6 & 0 & 20 & 6 \\ 2 & 6 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 13 & 0 & 44 & 14 \\ 3 & 6 & 3 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{11}3^{11}1^13^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 1 & 3 & 3 & 0 & 2 & 1 \\ 2 & 4 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{11}3^{11}1^13^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 & 3 \\ 1 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 2 & 0 & 0 & 2 & 2 \\ 1 & 3 & 3 & 0 & 2 & 1 & 3 \\ 2 & 3 & 2 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{11}3^1 1^1 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 2 & 6 & 3 \\ 1 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 2 & 0 & 0 & 5 & 2 \\ 3 & 6 & 6 & 0 & 5 & 13 & 7 \\ 2 & 3 & 2 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{13}3^1 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 38 & 5 \\ 6 & 5 & 1 & 41 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 38 & 41 & 0 & 261 & 31 \\ 5 & 5 & -1 & 31 & 3 \end{pmatrix}_{b_0}$$

[40] $1^{12}3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 22 & 5 \\ 6 & 5 & 1 & 24 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 24 & 0 & 86 & 18 \\ 5 & 5 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[41] $1^{11}3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 7 & 5 \\ 6 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 7 & 0 & 5 & 5 \\ 5 & 5 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

[42] $1^{13}3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 28 & 5 \\ 6 & 5 & 1 & 31 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 31 & 0 & 141 & 23 \\ 5 & 5 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

[43] $1^{11}3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 8 & 5 \\ 6 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 7 & 0 & 5 & 5 \\ 5 & 5 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

[44] $1^{11}3^1 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 28 & 5 \\ 6 & 5 & 1 & 27 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 27 & 0 & 125 & 21 \\ 5 & 5 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[45] $1^{14}3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 34 & 5 \\ 6 & 5 & 1 & 38 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 38 & 0 & 210 & 28 \\ 5 & 5 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[46] $1^{11}3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 17 & 5 \\ 6 & 5 & 1 & 17 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 17 & 0 & 45 & 13 \\ 5 & 5 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{14}3^{11}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.73 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 44 & 5 \\ 6 & 5 & 1 & 48 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 44 & 48 & 0 & 350 & 36 \\ 5 & 5 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{11}3^{11}21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.73 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 13 & 5 \\ 6 & 5 & 1 & 14 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 14 & 0 & 26 & 10 \\ 5 & 5 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{13}3^{11}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.04 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 8 & 6 & 0 & 10 & 6 \\ 2 & 6 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 10 & 5 & 0 & 8 & 6 \\ 3 & 6 & 3 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{13}3^{11}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -195, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.78 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 49 & 5 \\ 6 & 5 & 1 & 51 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 49 & 51 & 0 & 421 & 39 \\ 5 & 5 & -1 & 39 & 3 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{13}3^{11}21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -195, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.78 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 19 & 5 \\ 6 & 5 & 1 & 21 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 21 & 0 & 61 & 15 \\ 5 & 5 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{13}3^{11}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -201, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{14}3^{11}21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -206, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.84 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 25 & 5 \\ 6 & 5 & 1 & 28 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 28 & 0 & 110 & 20 \\ 5 & 5 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{12}3^{11}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -206, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.71 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 33 & 5 \\ 6 & 5 & 1 & 34 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 33 & 34 & 0 & 186 & 26 \\ 5 & 5 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{12}3^{11}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -206, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.84 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 5 & 6 & 1 & 44 & 5 \\ 6 & 5 & 1 & 44 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 44 & 44 & 0 & 326 & 34 \\ 5 & 5 & -1 & 34 & 3 \end{pmatrix}_{b_0} \\
[56] \quad & 1^{12}3^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -221, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 2 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 7 & 2 \\ 4 & 12 & 9 & 0 & 7 & 28 & 10 \\ 2 & 4 & 2 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0} \\
[57] \quad & 1^{11}31^{11}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -234, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2 \\
& T_{\min} = 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 8 & 6 & 0 & 15 & 6 \\ 2 & 6 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 15 & 9 & 0 & 22 & 10 \\ 3 & 6 & 3 & -3 & 10 & 2 \end{pmatrix}_{b_0} \\
[58] \quad & 1^{12}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -245, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.74 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 18 & 5 \\ 6 & 5 & 1 & 17 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 17 & 0 & 45 & 13 \\ 5 & 5 & -1 & 13 & 3 \end{pmatrix}_{b_0} \\
[59] \quad & 1^{12}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -266, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7 \\
& T_{\min} = 11.79 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 14 & 5 \\ 6 & 5 & 1 & 14 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 14 & 0 & 26 & 10 \\ 5 & 5 & -1 & 10 & 3 \end{pmatrix}_{b_0} \\
[60] \quad & 1^{13}31^{12}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -274, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7 \\
& T_{\min} = 12.03 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 8 & 6 & 0 & 35 & 6 \\ 2 & 6 & 1 & 0 & 22 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 35 & 22 & 0 & 134 & 24 \\ 3 & 6 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0} \\
[61] \quad & 1^{12}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -286, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7 \\
& T_{\min} = 11.68 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 23 & 5 \\ 6 & 5 & 1 & 24 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 24 & 0 & 86 & 18 \\ 5 & 5 & -1 & 18 & 3 \end{pmatrix}_{b_0} \\
[62] \quad & 1^{13}1^{12}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -304, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7 \\
& T_{\min} = 12.11 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 8 & 6 & 0 & 20 & 6 \\ 2 & 6 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 10 & 0 & 38 & 12 \\ 3 & 6 & 3 & -3 & 12 & 1 \end{pmatrix}_{b_0} \\
[63] \quad & 1^{13}1^{12}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -306, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7 \\
& T_{\min} = 12.03 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 6 & 2 \\ 4 & 12 & 6 & 0 & 6 & 22 & 8 \\ 2 & 4 & 2 & -3 & 2 & 8 & 0 \end{pmatrix}_{b_0} \\
[64] \quad & 1^{11}3^{11}1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -311, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 5 & 2 \\ 3 & 9 & 6 & 0 & 5 & 13 & 7 \\ 2 & 4 & 2 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{13}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -369, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.79 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 20 & 5 \\ 6 & 5 & 1 & 21 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 21 & 0 & 61 & 15 \\ 5 & 5 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{14}3^11^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -386, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 24 & 4 \\ 1 & 4 & 0 & 0 & 2 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 14 & 2 \\ 8 & 24 & 18 & 0 & 14 & 118 & 20 \\ 2 & 4 & 2 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{13}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -395, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.69 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 29 & 5 \\ 6 & 5 & 1 & 31 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 31 & 0 & 141 & 23 \\ 5 & 5 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{13}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -395, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.69 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 39 & 5 \\ 6 & 5 & 1 & 41 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 41 & 0 & 261 & 31 \\ 5 & 5 & -1 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{12}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -426, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.72 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 34 & 5 \\ 6 & 5 & 1 & 34 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 34 & 0 & 186 & 26 \\ 5 & 5 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{12}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -446, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.71 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 24 & 5 \\ 6 & 5 & 1 & 24 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 24 & 0 & 86 & 18 \\ 5 & 5 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{14}3^121^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -446, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.71 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 55 & 5 \\ 6 & 5 & 1 & 58 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 55 & 58 & 0 & 530 & 44 \\ 5 & 5 & -1 & 44 & 3 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{14}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -454, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.80 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 26 & 5 \\ 6 & 5 & 1 & 28 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 28 & 0 & 110 & 20 \\ 5 & 5 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{13}1^{13}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -481, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.07 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 18 & 4 \\ 1 & 4 & 0 & 0 & 2 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 9 & 2 \\ 6 & 18 & 9 & 0 & 9 & 52 & 12 \\ 2 & 4 & 2 & -3 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{14}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -486, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 35 & 5 \\ 6 & 5 & 1 & 38 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 35 & 38 & 0 & 210 & 28 \\ 5 & 5 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{13}3^{13}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -504, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.14 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 8 & 6 & 0 & 30 & 6 \\ 2 & 6 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 30 & 15 & 0 & 88 & 18 \\ 3 & 6 & 3 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{12}3^{12}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -504, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 8 & 6 & 0 & 30 & 6 \\ 2 & 6 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 30 & 18 & 0 & 94 & 20 \\ 3 & 6 & 3 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{11}3^{12}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -514, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.02 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 8 & 6 & 0 & 25 & 6 \\ 2 & 6 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 25 & 14 & 0 & 62 & 16 \\ 3 & 6 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{13}3^{12}21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -549, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.72 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 50 & 5 \\ 6 & 5 & 1 & 51 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 50 & 51 & 0 & 421 & 39 \\ 5 & 5 & -1 & 39 & 3 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{14}3^{11}21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -566, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 45 & 5 \\ 6 & 5 & 1 & 48 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 45 & 48 & 0 & 350 & 36 \\ 5 & 5 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{11}3^{11}123^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -626, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 2 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 8 & 2 \\ 5 & 15 & 9 & 0 & 8 & 37 & 11 \\ 2 & 4 & 2 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{13}3^{12}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -629, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.71 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 30 & 5 \\ 6 & 5 & 1 & 31 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 31 & 0 & 141 & 23 \\ 5 & 5 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{14}3^{13}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.01 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 8 & 6 & 0 & 50 & 6 \\ 2 & 6 & 1 & 0 & 31 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 50 & 31 & 0 & 272 & 34 \\ 3 & 6 & 3 & -3 & 34 & 1 \end{pmatrix}_{b_0} \\
[83] \quad & 1^{13}3^{11}2^{12}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -666, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7 \\
& T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 2 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 12 & 2 \\ 7 & 21 & 15 & 0 & 12 & 85 & 17 \\ 2 & 4 & 2 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0} \\
[84] \quad & 1^{12}3^{11}2^{12}21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -689, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 40 & 5 \\ 6 & 5 & 1 & 41 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 40 & 41 & 0 & 261 & 31 \\ 5 & 5 & -1 & 31 & 3 \end{pmatrix}_{b_0} \\
[85] \quad & 1^{13}3^{11}4^{14}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -726, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2 \\
& T_{\min} = 12.09 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 24 & 4 \\ 1 & 4 & 0 & 0 & 2 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 12 & 2 \\ 8 & 24 & 12 & 0 & 12 & 94 & 16 \\ 2 & 4 & 2 & -3 & 2 & 16 & 0 \end{pmatrix}_{b_0} \\
[86] \quad & 1^{12}3^{11}2^{12}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -746, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7 \\
& T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 18 & 4 \\ 1 & 4 & 0 & 0 & 2 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 10 & 2 \\ 6 & 18 & 12 & 0 & 10 & 58 & 14 \\ 2 & 4 & 2 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0} \\
[87] \quad & 1^{13}3^{11}4^{14}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -784, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2 \\
& T_{\min} = 12.16 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 8 & 6 & 0 & 40 & 6 \\ 2 & 6 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 40 & 20 & 0 & 158 & 24 \\ 3 & 6 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0} \\
[88] \quad & 1^{14}3^{11}2^{12}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -794, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.71 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 36 & 5 \\ 6 & 5 & 1 & 38 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 36 & 38 & 0 & 210 & 28 \\ 5 & 5 & -1 & 28 & 3 \end{pmatrix}_{b_0} \\
[89] \quad & 1^{14}3^{11}2^{12}21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -874, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 5 & 6 & 1 & 56 & 5 \\ 6 & 5 & 1 & 58 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 56 & 58 & 0 & 530 & 44 \\ 5 & 5 & -1 & 44 & 3 \end{pmatrix}_{b_0} \\
[90] \quad & 1^{11}3^{13}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -874, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0 \\
& T_{\min} = 12.04 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 8 & 6 & 0 & 35 & 6 \\ 2 & 6 & 1 & 0 & 19 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 35 & 19 & 0 & 122 & 22 \\ 3 & 6 & 3 & -3 & 22 & 1 \end{pmatrix}_{b_0} \\
[91] \quad & 1^{14}3^{11}2^{12}21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -934, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 5 & 6 & 1 & 46 & 5 \\ 6 & 5 & 1 & 48 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 46 & 48 & 0 & 350 & 36 \\ 5 & 5 & -1 & 36 & 3 \end{pmatrix}_{b_0} \\
[92] \quad & 1^{13}31^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -954, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 3 \\ 4 & 8 & 6 & 0 & 45 & 6 \\ 2 & 6 & 1 & 0 & 27 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 45 & 27 & 0 & 214 & 30 \\ 3 & 6 & 3 & -3 & 30 & 2 \end{pmatrix}_{b_0} \\
[93] \quad & 1^{11}3^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1011, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 2 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 11 & 2 \\ 7 & 21 & 12 & 0 & 11 & 73 & 15 \\ 2 & 4 & 2 & -3 & 2 & 15 & 1 \end{pmatrix}_{b_0} \\
[94] \quad & 1^{12}31^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1024, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 8 & 6 & 0 & 40 & 6 \\ 2 & 6 & 1 & 0 & 23 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 40 & 23 & 0 & 164 & 26 \\ 3 & 6 & 3 & -3 & 26 & 1 \end{pmatrix}_{b_0} \\
[95] \quad & 1^{13}1^{15}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1041, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.3 \\
& T_{\min} = 12.11 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 30 & 4 \\ 1 & 4 & 0 & 0 & 2 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 15 & 2 \\ 10 & 30 & 15 & 0 & 15 & 148 & 20 \\ 2 & 4 & 2 & -3 & 2 & 20 & 0 \end{pmatrix}_{b_0} \\
[96] \quad & 1^{13}1^{15}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.3 \\
& T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 8 & 6 & 0 & 50 & 6 \\ 2 & 6 & 1 & 0 & 25 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 50 & 25 & 0 & 248 & 30 \\ 3 & 6 & 3 & -3 & 30 & 1 \end{pmatrix}_{b_0} \\
[97] \quad & 1^{11}31^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 3 \\ 4 & 8 & 6 & 0 & 45 & 6 \\ 2 & 6 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 45 & 24 & 0 & 202 & 28 \\ 3 & 6 & 3 & -3 & 28 & 1 \end{pmatrix}_{b_0} \\
[98] \quad & 1^{12}3^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1341, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 24 & 4 \\ 1 & 4 & 0 & 0 & 2 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 13 & 2 \\ 8 & 24 & 15 & 0 & 13 & 100 & 18 \\ 2 & 4 & 2 & -3 & 2 & 18 & 1 \end{pmatrix}_{b_0} \\
[99] \quad & 1^{14}3^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 30 & 4 \\ 1 & 4 & 0 & 0 & 2 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 17 & 2 \\ 10 & 30 & 21 & 0 & 17 & 172 & 24 \\ 2 & 4 & 2 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0} \\
[100] \quad & 1^{11}3^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1466, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 12.00 \rightarrow 12, T_{\max} = 13, T \in [12, 13]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 2 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 14 & 2 \\ 9 & 27 & 15 & 0 & 14 & 121 & 19 \\ 2 & 4 & 2 & -3 & 2 & 19 & 1 \end{pmatrix}_{b_0}$$

[101] $1^{13}3^{11}1^{13}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1471, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 2 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 15 & 2 \\ 9 & 27 & 18 & 0 & 15 & 133 & 21 \\ 2 & 4 & 2 & -3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

[102] $1^{14}3^{11}1^{14}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 24 & 3 \\ 4 & 8 & 6 & 0 & 60 & 6 \\ 2 & 6 & 1 & 0 & 36 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 24 & 60 & 36 & 0 & 382 & 40 \\ 3 & 6 & 3 & -3 & 40 & 2 \end{pmatrix}_{b_0}$$

[103] $1^{12}3^{11}1^{14}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 8 & 6 & 0 & 50 & 6 \\ 2 & 6 & 1 & 0 & 28 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 50 & 28 & 0 & 254 & 32 \\ 3 & 6 & 3 & -3 & 32 & 1 \end{pmatrix}_{b_0}$$

[104] $1^{13}3^{11}1^{14}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1714, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.00 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 22 & 3 \\ 4 & 8 & 6 & 0 & 55 & 6 \\ 2 & 6 & 1 & 0 & 32 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 22 & 55 & 32 & 0 & 314 & 36 \\ 3 & 6 & 3 & -3 & 36 & 1 \end{pmatrix}_{b_0}$$

[105] $1^{12}3^{11}1^{14}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -2006, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 30 & 4 \\ 1 & 4 & 0 & 0 & 2 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 16 & 2 \\ 10 & 30 & 18 & 0 & 16 & 154 & 22 \\ 2 & 4 & 2 & -3 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

[106] $1^{13}3^{11}1^{14}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -2346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 3 & 33 & 4 \\ 1 & 4 & 0 & 0 & 2 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 18 & 2 \\ 11 & 33 & 21 & 0 & 18 & 193 & 25 \\ 2 & 4 & 2 & -3 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

[107] $1^{14}3^{11}1^{14}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -2486, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 3 & 36 & 4 \\ 1 & 4 & 0 & 0 & 2 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 20 & 2 \\ 12 & 36 & 24 & 0 & 20 & 238 & 28 \\ 2 & 4 & 2 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [108] $1^{11}4^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & -2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[109] $1^{11}4^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 0 & 13 & 7 \\ 3 & 4 & -2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{\overline{11}} 4 \ 13^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 0 & 4 & 4 \\ 3 & 4 & -2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{\overline{11}} 4 \ 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 6 & 3 \\ 3 & 4 & 0 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 8 & 0 & 0 & 13 & 7 \\ 3 & 4 & -2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{\overline{14}} 4 \ 1^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 11 & 3 \\ 3 & 4 & 0 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 19 & 0 & 0 & 65 & 15 \\ 3 & 4 & -2 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{\overline{11}} 4 \ 1^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 7 & 0 & 0 & 8 & 6 \\ 3 & 4 & -2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{\overline{12}} 4 \ 1^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 98, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 7 & 3 \\ 3 & 4 & 0 & 0 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 11 & 0 & 0 & 21 & 9 \\ 3 & 4 & -2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[114] \ 1^{\overline{13}} 4 \ 1^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 98, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 9 & 3 \\ 3 & 4 & 0 & 0 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 0 & 0 & 40 & 12 \\ 3 & 4 & -2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{\overline{11}} 4 \ 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 3 \\ 3 & 4 & 0 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 12 & 0 & 0 & 26 & 10 \\ 3 & 4 & -2 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{\overline{12}} 4 \ 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 176, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 10 & 3 \\ 3 & 4 & 0 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 16 & 0 & 0 & 45 & 13 \\ 3 & 4 & -2 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{14} \ 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -79, \ \det = 176, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.59 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 14 & 3 \\ 3 & 4 & 0 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 14 & 24 & 0 & 0 & 101 & 19 \\ 3 & 4 & -2 & 0 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{13} \ 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -79, \ \det = 192, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.58 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 12 & 3 \\ 3 & 4 & 0 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 20 & 0 & 0 & 70 & 16 \\ 3 & 4 & -2 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

4.502 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{f}_4$

$$T = 9 \quad [0] \ 1^{11} \ 3^{11} \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -320, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.40 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 2 & 3 & 0 & 0 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 3 & 8 & 3 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 2 & 4 & -3 & -1 \end{pmatrix}_{b_0}$$

4.503 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \ 1^{11} \ 3^{11} \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 200, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 3 & 4 \\ 3 & 5 & 0 & 3 & 6 & 4 & 4 & 4 \\ 2 & 4 & 0 & 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & -3 & 2 & 4 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} \ 3^{11} \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 600, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 9 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 3 & 2 & 2 \\ 5 & 9 & 0 & 5 & 13 & 9 & 6 & 7 \\ 3 & 5 & 0 & 3 & 9 & 4 & 4 & 4 \\ 2 & 4 & 0 & 2 & 6 & 4 & 0 & 2 \\ 3 & 4 & -3 & 2 & 7 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

4.504 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{11} \ 3^{11} \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -218, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 13 & 7 \\ 5 & 8 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} \ 3^{11} \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -222, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 2 & 3 & 0 & 0 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 3 & 8 & 3 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 2 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^{11}1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -302$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 2 & 3 & 0 & 0 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 4 & 8 & 3 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 2 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$

4.505 NHC(-2)(-3) + f₄ + su₂→(-3) + su₂ + su₂

$T = 9$ [1] $1^{122}3^{11}1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 0 & 3 & 8 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 4 & 2 \\ 4 & 8 & 0 & 4 & 10 & 12 & 6 \\ 4 & 8 & 0 & 4 & 12 & 10 & 6 \\ 3 & 4 & -3 & 2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[0] $1^{124}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.7$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 17 & 3 \\ 3 & 4 & 0 & 3 & 8 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 17 & 2 \\ 4 & 8 & 0 & 4 & 10 & 42 & 6 \\ 17 & 31 & 0 & 17 & 42 & 172 & 24 \\ 3 & 4 & -3 & 2 & 6 & 24 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 11.55 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 5 & 3 \\ 3 & 4 & 0 & 3 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 5 & 2 \\ 4 & 8 & 0 & 4 & 10 & 12 & 6 \\ 5 & 9 & 0 & 5 & 12 & 13 & 7 \\ 3 & 4 & -3 & 2 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{113}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -74$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 4 \\ 2 & 4 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & -3 & 2 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{133}3^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -74$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 9 & 3 \\ 3 & 4 & 0 & 3 & 5 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 9 & 2 \\ 3 & 5 & 0 & 3 & 4 & 15 & 4 \\ 9 & 17 & 0 & 9 & 15 & 49 & 13 \\ 3 & 4 & -3 & 2 & 4 & 13 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 2 & 2 \\ 4 & 8 & 0 & 4 & 10 & 6 & 6 \\ 2 & 4 & 0 & 2 & 6 & 1 & 3 \\ 3 & 4 & -3 & 2 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{131}3^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -104$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & 0 & 3 & 6 & 4 \\ 3 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -144, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.50 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 \\ 2 & 4 & 0 & 2 & 3 & 1 \\ 3 & 4 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{12}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -164, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 12 & 3 \\ 3 & 4 & 0 & 3 & 8 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 12 & 2 \\ 4 & 8 & 0 & 4 & 10 & 30 & 6 \\ 12 & 22 & 0 & 12 & 30 & 85 & 17 \\ 3 & 4 & -3 & 2 & 6 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{12}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.57 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 7 & 3 \\ 3 & 4 & 0 & 3 & 8 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 7 & 2 \\ 4 & 8 & 0 & 4 & 10 & 18 & 6 \\ 7 & 13 & 0 & 7 & 18 & 28 & 10 \\ 3 & 4 & -3 & 2 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 8 & 2 \\ 2 & 4 & 0 & 2 & 1 & 9 & 3 \\ 8 & 14 & 0 & 8 & 9 & 37 & 11 \\ 3 & 4 & -3 & 2 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 7 & 3 \\ 3 & 4 & 0 & 3 & 5 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 7 & 2 \\ 3 & 5 & 0 & 3 & 4 & 12 & 4 \\ 7 & 13 & 0 & 7 & 12 & 28 & 10 \\ 3 & 4 & -3 & 2 & 4 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -234, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 3 & 5 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 5 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 4 \\ 5 & 9 & 0 & 5 & 9 & 13 & 7 \\ 3 & 4 & -3 & 2 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -254, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.54 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 \\ 2 & 4 & 0 & 2 & 1 & 6 & 3 \\ 5 & 9 & 0 & 5 & 6 & 13 & 7 \\ 3 & 4 & -3 & 2 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{12}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -284, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 9 & 3 \\ 3 & 4 & 0 & 3 & 8 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 9 & 2 \\ 4 & 8 & 0 & 4 & 10 & 24 & 6 \\ 9 & 17 & 0 & 9 & 24 & 49 & 13 \\ 3 & 4 & -3 & 2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11} 1^{12} 3^{11} 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 11 & 3 \\ 3 & 4 & 0 & 3 & 9 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 11 & 2 \\ 5 & 9 & 0 & 5 & 13 & 33 & 7 \\ 11 & 21 & 0 & 11 & 33 & 76 & 16 \\ 3 & 4 & -3 & 2 & 7 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{12} 4^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 11 & 3 \\ 3 & 4 & 0 & 3 & 8 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 11 & 2 \\ 4 & 8 & 0 & 4 & 10 & 30 & 6 \\ 11 & 21 & 0 & 11 & 30 & 76 & 16 \\ 3 & 4 & -3 & 2 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 7 & 2 \\ 2 & 4 & 0 & 2 & 1 & 9 & 3 \\ 7 & 13 & 0 & 7 & 9 & 28 & 10 \\ 3 & 4 & -3 & 2 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{12} 4^{11} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 14 & 3 \\ 3 & 4 & 0 & 3 & 8 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 14 & 2 \\ 4 & 8 & 0 & 4 & 10 & 36 & 6 \\ 14 & 26 & 0 & 14 & 36 & 118 & 20 \\ 3 & 4 & -3 & 2 & 6 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11} 1^{12} 3^{11} 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 3 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 5 & 2 \\ 5 & 9 & 0 & 5 & 13 & 15 & 7 \\ 5 & 9 & 0 & 5 & 15 & 13 & 7 \\ 3 & 4 & -3 & 2 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11} 1^{12} 3^{11} 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -474, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 4 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 13 & 2 \\ 2 & 4 & 0 & 2 & 1 & 15 & 3 \\ 13 & 23 & 0 & 13 & 15 & 100 & 18 \\ 3 & 4 & -3 & 2 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 9 & 2 \\ 2 & 4 & 0 & 2 & 1 & 12 & 3 \\ 9 & 17 & 0 & 9 & 12 & 49 & 13 \\ 3 & 4 & -3 & 2 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11} 1^{12} 3^{11} 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 9 & 3 \\ 3 & 4 & 0 & 3 & 9 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 9 & 2 \\ 5 & 9 & 0 & 5 & 13 & 27 & 7 \\ 9 & 17 & 0 & 9 & 27 & 49 & 13 \\ 3 & 4 & -3 & 2 & 7 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{112}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 10 & 2 \\ 2 & 4 & 0 & 2 & 1 & 12 & 3 \\ 10 & 18 & 0 & 10 & 12 & 58 & 14 \\ 3 & 4 & -3 & 2 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{112}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 7 & 3 \\ 3 & 4 & 0 & 3 & 9 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 7 & 2 \\ 5 & 9 & 0 & 5 & 13 & 21 & 7 \\ 7 & 13 & 0 & 7 & 21 & 28 & 10 \\ 3 & 4 & -3 & 2 & 7 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{114}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 11 & 2 \\ 2 & 4 & 0 & 2 & 1 & 15 & 3 \\ 11 & 21 & 0 & 11 & 15 & 76 & 16 \\ 3 & 4 & -3 & 2 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{123}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 11 & 3 \\ 3 & 4 & 0 & 3 & 13 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 7 & 11 & 2 \\ 7 & 13 & 0 & 7 & 28 & 48 & 10 \\ 11 & 21 & 0 & 11 & 48 & 76 & 16 \\ 3 & 4 & -3 & 2 & 10 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{123}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 7 & 3 \\ 3 & 4 & 0 & 3 & 13 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 7 & 7 & 2 \\ 7 & 13 & 0 & 7 & 28 & 30 & 10 \\ 7 & 13 & 0 & 7 & 30 & 28 & 10 \\ 3 & 4 & -3 & 2 & 10 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{123}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -794, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 9 & 3 \\ 3 & 4 & 0 & 3 & 13 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 7 & 9 & 2 \\ 7 & 13 & 0 & 7 & 28 & 39 & 10 \\ 9 & 17 & 0 & 9 & 39 & 49 & 13 \\ 3 & 4 & -3 & 2 & 10 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{113}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -824, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 12 & 2 \\ 2 & 4 & 0 & 2 & 1 & 15 & 3 \\ 12 & 22 & 0 & 12 & 15 & 85 & 17 \\ 3 & 4 & -3 & 2 & 3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{133}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 9 & 3 \\ 3 & 4 & 0 & 3 & 17 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 9 & 9 & 2 \\ 9 & 17 & 0 & 9 & 49 & 51 & 13 \\ 9 & 17 & 0 & 9 & 51 & 49 & 13 \\ 3 & 4 & -3 & 2 & 13 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{113}3^{11}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 18 & 3 \\ 3 & 4 & 0 & 3 & 4 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 18 & 2 \\ 2 & 4 & 0 & 2 & 1 & 21 & 3 \\ 18 & 32 & 0 & 18 & 21 & 193 & 25 \\ 3 & 4 & -3 & 2 & 3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{134}3^{11}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -954, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 11 & 3 \\ 3 & 4 & 0 & 3 & 17 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 9 & 11 & 2 \\ 9 & 17 & 0 & 9 & 49 & 63 & 13 \\ 11 & 21 & 0 & 11 & 63 & 76 & 16 \\ 3 & 4 & -3 & 2 & 13 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{113}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -974, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 3 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 15 & 2 \\ 2 & 4 & 0 & 2 & 1 & 18 & 3 \\ 15 & 27 & 0 & 15 & 18 & 133 & 21 \\ 3 & 4 & -3 & 2 & 3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{144}3^{11}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1064, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 11 & 3 \\ 3 & 4 & 0 & 3 & 21 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 11 & 11 & 2 \\ 11 & 21 & 0 & 11 & 76 & 78 & 16 \\ 11 & 21 & 0 & 11 & 78 & 76 & 16 \\ 3 & 4 & -3 & 2 & 16 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{114}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1084, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 4 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 14 & 2 \\ 2 & 4 & 0 & 2 & 1 & 18 & 3 \\ 14 & 26 & 0 & 14 & 18 & 118 & 20 \\ 3 & 4 & -3 & 2 & 3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{114}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1434, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 3 & 4 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 17 & 2 \\ 2 & 4 & 0 & 2 & 1 & 21 & 3 \\ 17 & 31 & 0 & 17 & 21 & 172 & 24 \\ 3 & 4 & -3 & 2 & 3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{114}3^{11}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1604, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 0 & 3 & 4 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 20 & 2 \\ 2 & 4 & 0 & 2 & 1 & 24 & 3 \\ 20 & 36 & 0 & 20 & 24 & 238 & 28 \\ 3 & 4 & -3 & 2 & 3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [38] \quad 1^{111}3^{11}1^{11}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 247, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 9 \\ 1 & 4 & 0 & 0 & 2 & 3 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 2 & 5 \\ 1 & 3 & 3 & 0 & 2 & 1 & 6 \\ 3 & 9 & 6 & 0 & 5 & 6 & 13 \\ 2 & 4 & 2 & -3 & 2 & 3 & 7 \end{pmatrix}_{b_0}$$

[39] $1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 472, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 6 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 3 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 6 & 4 \\ 2 & 6 & 3 & 0 & 3 & 6 & 4 & 4 \\ 2 & 4 & 2 & -3 & 2 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[40] $1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 517, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 3 & 5 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 9 & 4 \\ 3 & 9 & 6 & 0 & 5 & 9 & 13 & 7 \\ 2 & 4 & 2 & -3 & 2 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{11}31^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 648, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 6 & 3 \\ 4 & 8 & 6 & 0 & 15 & 15 & 6 \\ 2 & 6 & 1 & 0 & 9 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 15 & 9 & 0 & 22 & 24 & 10 \\ 6 & 15 & 9 & 0 & 24 & 22 & 10 \\ 3 & 6 & 3 & -3 & 10 & 10 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 912, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 2 & 6 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 5 & 5 & 2 \\ 3 & 9 & 6 & 0 & 5 & 13 & 15 & 7 \\ 3 & 9 & 6 & 0 & 5 & 15 & 13 & 7 \\ 2 & 4 & 2 & -3 & 2 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

4.506 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$T = 10$ [0] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 3 & 4 \\ 3 & 5 & 0 & 3 & 6 & 4 & 3 & 4 \\ 2 & 4 & 0 & 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 4 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 2 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 5 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 3 & 4 \\ 5 & 9 & 0 & 5 & 9 & 13 & 6 & 7 \\ 2 & 4 & 0 & 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & -3 & 2 & 4 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{s}\mathfrak{u}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}\mathfrak{u}_2 \rightarrow (-3) + \mathfrak{s}\mathfrak{u}_2 + \mathfrak{s}\mathfrak{u}_2 + \mathfrak{s}\mathfrak{u}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 512, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 2 & 5 \\ 2 & 4 & 0 & 2 & 1 & 3 & 6 \\ 2 & 4 & 0 & 2 & 3 & 1 & 6 \\ 5 & 9 & 0 & 5 & 6 & 6 & 12 \\ 3 & 4 & -3 & 2 & 3 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{s}\mathfrak{u}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}\mathfrak{u}_2 \rightarrow (-3) + \mathfrak{s}\mathfrak{u}_2 + \mathfrak{s}\mathfrak{u}_2 + \mathfrak{s}\mathfrak{u}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 622, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 5 & 6 & 13 & 6 & 7 \\ 2 & 4 & 0 & 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & -3 & 2 & 3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{s}\mathfrak{u}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}\mathfrak{u}_2 \rightarrow (-3) + \mathfrak{s}\mathfrak{u}_2 + \mathfrak{s}\mathfrak{u}_2 + \mathfrak{s}\mathfrak{u}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 1152, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 2 & 3 \\ 3 & 4 & 0 & 3 & 9 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 5 & 2 & 2 \\ 5 & 9 & 0 & 5 & 13 & 15 & 6 & 7 \\ 5 & 9 & 0 & 5 & 15 & 13 & 6 & 7 \\ 2 & 4 & 0 & 2 & 6 & 6 & 0 & 2 \\ 3 & 4 & -3 & 2 & 7 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.507 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{s}\mathfrak{u}_2 \rightarrow (-3)$ + $\mathfrak{s}\mathfrak{u}_2$ + su2n3mix

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{s}\mathfrak{u}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}\mathfrak{u}_2 \rightarrow (-3) + \mathfrak{s}\mathfrak{u}_2 + \mathfrak{s}\mathfrak{u}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 100, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 5 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 2 & 3 & 2 \\ 1 & 3 & 3 & 0 & 2 & 1 & 4 & 3 \\ 2 & 5 & 4 & 0 & 3 & 4 & 4 & 4 \\ 2 & 4 & 2 & -3 & 2 & 3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{s}\mathfrak{u}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}\mathfrak{u}_2 \rightarrow (-3) + \mathfrak{s}\mathfrak{u}_2 + \mathfrak{s}\mathfrak{u}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 175, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 5 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 3 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 5 & 4 \\ 2 & 5 & 4 & 0 & 3 & 5 & 4 & 4 \\ 2 & 4 & 2 & -3 & 2 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{s}\mathfrak{u}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}\mathfrak{u}_2 \rightarrow (-3) + \mathfrak{s}\mathfrak{u}_2 + \mathfrak{s}\mathfrak{u}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -60, \ \det = 475, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 5 & 4 \\ 1 & 4 & 0 & 0 & 2 & 6 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 5 & 3 & 2 \\ 3 & 9 & 6 & 0 & 5 & 13 & 9 & 7 \\ 2 & 5 & 4 & 0 & 3 & 9 & 4 & 4 \\ 2 & 4 & 2 & -3 & 2 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$

4.508 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{s}\mathfrak{u}_2 \rightarrow (-3)$ + $\mathfrak{s}\mathfrak{u}_2$ + $\mathfrak{s}\mathfrak{u}_3$

$$T = 9 \quad [0] \ 1^{\overline{1211}} 3^{\overline{1111}} + [\mathfrak{s}\mathfrak{u}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}\mathfrak{u}_2 \rightarrow (-3) + \mathfrak{s}\mathfrak{u}_2 + \mathfrak{s}\mathfrak{u}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -96, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 5 & 3 \\ 3 & 4 & 0 & 3 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 5 & 2 \\ 4 & 8 & 0 & 4 & 10 & 12 & 6 \\ 5 & 9 & 0 & 5 & 12 & 12 & 6 \\ 3 & 4 & -3 & 2 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{s}\mathfrak{u}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{s}\mathfrak{u}_2 \rightarrow (-3) + \mathfrak{s}\mathfrak{u}_2 + \mathfrak{s}\mathfrak{u}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -124, \ (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 2 & 3 & 0 & 0 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 3 & 8 & 3 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 2 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{12}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 2 & 2 \\ 4 & 8 & 0 & 4 & 10 & 6 & 6 \\ 2 & 4 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -3 & 2 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 3 & 19 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 11 & 2 & 2 \\ 11 & 19 & 0 & 11 & 73 & 12 & 15 \\ 2 & 4 & 0 & 2 & 12 & 0 & 2 \\ 3 & 4 & -3 & 2 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 4 \\ 2 & 4 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 3 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 4 & -3 & 2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.31 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 4 & 2 \\ 2 & 4 & 0 & 2 & 1 & 6 & 3 \\ 4 & 8 & 0 & 4 & 6 & 9 & 5 \\ 3 & 4 & -3 & 2 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 2 & 3 & 0 & 0 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 4 & 8 & 3 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 2 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 4 & 3 \\ 3 & 4 & 0 & 3 & 9 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 4 & 2 \\ 5 & 9 & 0 & 5 & 13 & 12 & 7 \\ 4 & 8 & 0 & 4 & 12 & 9 & 5 \\ 3 & 4 & -3 & 2 & 7 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -226, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 & 3 \\ 2 & 4 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 \\ 2 & 4 & 0 & 2 & 1 & 6 & 3 \\ 5 & 9 & 0 & 5 & 6 & 12 & 6 \\ 3 & 4 & -3 & 2 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 3 & 14 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 8 & 2 & 2 \\ 8 & 14 & 0 & 8 & 37 & 9 & 11 \\ 2 & 4 & 0 & 2 & 9 & 0 & 2 \\ 3 & 4 & -3 & 2 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 3 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 2 & 2 \\ 5 & 9 & 0 & 5 & 13 & 6 & 7 \\ 2 & 4 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{112}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 7 & 2 \\ 2 & 4 & 0 & 2 & 1 & 9 & 3 \\ 7 & 13 & 0 & 7 & 9 & 27 & 9 \\ 3 & 4 & -3 & 2 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{121}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 3 & 13 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 7 & 2 & 2 \\ 7 & 13 & 0 & 7 & 28 & 9 & 10 \\ 2 & 4 & 0 & 2 & 9 & 0 & 2 \\ 3 & 4 & -3 & 2 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{121}3^{11}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 2 & 3 \\ 3 & 4 & 0 & 3 & 28 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 16 & 2 & 2 \\ 16 & 28 & 0 & 16 & 154 & 18 & 22 \\ 2 & 4 & 0 & 2 & 18 & 0 & 2 \\ 3 & 4 & -3 & 2 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{131}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 0 & 3 & 17 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 9 & 2 & 2 \\ 9 & 17 & 0 & 9 & 49 & 12 & 13 \\ 2 & 4 & 0 & 2 & 12 & 0 & 2 \\ 3 & 4 & -3 & 2 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{141}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 11 & 2 & 2 \\ 11 & 21 & 0 & 11 & 76 & 15 & 16 \\ 2 & 4 & 0 & 2 & 15 & 0 & 2 \\ 3 & 4 & -3 & 2 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{12} 1^3 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 0 & 3 & 18 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 10 & 2 & 2 \\ 10 & 18 & 0 & 10 & 58 & 12 & 14 \\ 2 & 4 & 0 & 2 & 12 & 0 & 2 \\ 3 & 4 & -3 & 2 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{12} 1^3 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 2 & 3 \\ 3 & 4 & 0 & 3 & 23 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 13 & 2 & 2 \\ 13 & 23 & 0 & 13 & 100 & 15 & 18 \\ 2 & 4 & 0 & 2 & 15 & 0 & 2 \\ 3 & 4 & -3 & 2 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{13} 1^3 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 2 & 3 \\ 3 & 4 & 0 & 3 & 22 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 12 & 2 & 2 \\ 12 & 22 & 0 & 12 & 85 & 15 & 17 \\ 2 & 4 & 0 & 2 & 15 & 0 & 2 \\ 3 & 4 & -3 & 2 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{13} 1^3 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 2 & 3 \\ 3 & 4 & 0 & 3 & 27 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 15 & 2 & 2 \\ 15 & 27 & 0 & 15 & 133 & 18 & 21 \\ 2 & 4 & 0 & 2 & 18 & 0 & 2 \\ 3 & 4 & -3 & 2 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{14} 1^3 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 4 & 0 & 3 & 26 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 14 & 2 & 2 \\ 14 & 26 & 0 & 14 & 118 & 18 & 20 \\ 2 & 4 & 0 & 2 & 18 & 0 & 2 \\ 3 & 4 & -3 & 2 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{13} 1^3 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1826, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 2 & 3 \\ 3 & 4 & 0 & 3 & 32 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 18 & 2 & 2 \\ 18 & 32 & 0 & 18 & 193 & 21 & 25 \\ 2 & 4 & 0 & 2 & 21 & 0 & 2 \\ 3 & 4 & -3 & 2 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{14} 1^3 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 2 & 3 \\ 3 & 4 & 0 & 3 & 31 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 17 & 2 & 2 \\ 17 & 31 & 0 & 17 & 172 & 21 & 24 \\ 2 & 4 & 0 & 2 & 21 & 0 & 2 \\ 3 & 4 & -3 & 2 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{14} 1^3 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2816, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 2 & 3 \\ 3 & 4 & 0 & 3 & 36 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 20 & 2 & 2 \\ 20 & 36 & 0 & 20 & 238 & 24 & 28 \\ 2 & 4 & 0 & 2 & 24 & 0 & 2 \\ 3 & 4 & -3 & 2 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [26] \quad 1^{11} 4^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 5 & 0 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 0 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{11} 1^{13} 1^{11} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 0 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 0 & 8 & 5 & 6 \\ 2 & 4 & 0 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 0 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11} 1^{13} 1^{11} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 8 & 12 & 0 & 0 & 26 & 8 & 10 \\ 2 & 4 & 0 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 0 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

4.509 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 & 3 \\ 3 & 5 & 0 & 3 & 4 & 3 & 6 \\ 2 & 4 & 0 & 2 & 3 & 0 & 3 \\ 3 & 5 & 0 & 3 & 6 & 3 & 4 \\ 3 & 4 & -3 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 1^{13} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 & 3 \\ 2 & 4 & 0 & 2 & 3 & 0 & 3 \\ 2 & 4 & 0 & 2 & 3 & 3 & 1 \\ 3 & 4 & -3 & 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 1^{13} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 & 5 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 2 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 5 & 9 & 6 & 13 & 7 \\ 3 & 4 & -3 & 2 & 4 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 1^{13} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 5 & 6 & 12 & 6 & 6 \\ 2 & 4 & 0 & 2 & 3 & 6 & 1 & 3 \\ 3 & 4 & -3 & 2 & 3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1111}3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 5 & 6 & 12 & 6 & 6 \\ 2 & 4 & 0 & 2 & 3 & 6 & 1 & 3 \\ 3 & 4 & -3 & 2 & 3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1111}3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 9 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 2 & 5 & 2 \\ 5 & 9 & 0 & 5 & 13 & 6 & 15 & 7 \\ 2 & 4 & 0 & 2 & 6 & 0 & 6 & 2 \\ 5 & 9 & 0 & 5 & 15 & 6 & 13 & 7 \\ 3 & 4 & -3 & 2 & 7 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.510 NHC(-2)(-3) + $\mathbf{f}_4$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{11}3^{11}131^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & 4 & 0 & 2 & 2 & 3 & 3 \\ 1 & 4 & 0 & 0 & 2 & 2 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 & 3 \\ 1 & 2 & 2 & 0 & 0 & 2 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & -3 & 2 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^3 1^1 1^3 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 & 3 \\ 4 & 8 & 6 & 0 & 5 & 6 & 3 \\ 2 & 6 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 & 3 \\ 2 & 5 & 3 & 0 & 1 & 3 & 3 \\ 3 & 6 & 3 & -3 & 3 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^3 1^1 1^3 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 & 3 \\ 1 & 3 & 2 & 0 & 0 & 2 & 2 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 2 & -3 & 2 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

4.511 NHC(-2)(-3) + $\mathbf{f}_4$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{11}3^{11}131^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 & 4 \\ 1 & 4 & 0 & 0 & 2 & 5 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 & 3 \\ 1 & 3 & 2 & 0 & 0 & 4 & 2 & 2 \\ 2 & 6 & 5 & 0 & 4 & 7 & 5 & 5 \\ 2 & 4 & 2 & -3 & 2 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^1 1^3 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 4 & 0 & 2 & 4 & 3 & 3 \\ 1 & 4 & 0 & 0 & 2 & 4 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 & 3 \\ 1 & 2 & 2 & 0 & 0 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 & 4 & 4 \\ 2 & 3 & 2 & -3 & 2 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12}3^1 1^{11}3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 8 & 6 & 0 & 25 & 6 \\ 2 & 6 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 25 & 16 & 0 & 69 & 17 \\ 3 & 6 & 3 & -3 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 5 & 4 \\ 1 & 4 & 0 & 0 & 2 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 3 & 2 \\ 2 & 5 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{13}1^{11}1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 8 & 4 \\ 1 & 4 & 0 & 0 & 2 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 4 & 2 \\ 3 & 8 & 4 & 0 & 4 & 9 & 5 \\ 2 & 4 & 2 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{13}1^{11}1^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -250, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 8 & 6 & 0 & 15 & 6 \\ 2 & 6 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 15 & 8 & 0 & 21 & 9 \\ 3 & 6 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11}3^{11}1^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 8 & 6 & 0 & 20 & 6 \\ 2 & 6 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 12 & 0 & 41 & 13 \\ 3 & 6 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{13}3^{11}1^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 17 & 4 \\ 1 & 4 & 0 & 0 & 2 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 10 & 2 \\ 6 & 17 & 13 & 0 & 10 & 60 & 14 \\ 2 & 4 & 2 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{13}1^{11}1^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -303, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 5 & 2 \\ 3 & 9 & 5 & 0 & 5 & 12 & 6 \\ 2 & 4 & 2 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{12}3^{11}1^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 2 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 9 & 2 \\ 5 & 15 & 11 & 0 & 9 & 44 & 12 \\ 2 & 4 & 2 & -3 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11}3^{11}1^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -417, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 0 & 2 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 7 & 2 \\ 4 & 12 & 8 & 0 & 7 & 25 & 9 \\ 2 & 4 & 2 & -3 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11}3^{11}1^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 11 & 4 \\ 1 & 4 & 0 & 0 & 2 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 6 & 2 \\ 4 & 11 & 7 & 0 & 6 & 20 & 8 \\ 2 & 4 & 2 & -3 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{12}3^{11}1^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -465, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 14 & 4 \\ 1 & 4 & 0 & 0 & 2 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 8 & 2 \\ 5 & 14 & 10 & 0 & 8 & 37 & 11 \\ 2 & 4 & 2 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{13}3^{11}1^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -520, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 2 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 8 & 2 \\ 5 & 15 & 8 & 0 & 8 & 36 & 10 \\ 2 & 4 & 2 & -3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11}3^{11}1^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -844, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 18 & 4 \\ 1 & 4 & 0 & 0 & 2 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 10 & 2 \\ 6 & 18 & 11 & 0 & 10 & 55 & 13 \\ 2 & 4 & 2 & -3 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{12}3^{11}1^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -968, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 2 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 12 & 2 \\ 7 & 21 & 14 & 0 & 12 & 80 & 16 \\ 2 & 4 & 2 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

4.512 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \text{su2n3mix} + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11}3^{11}1^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 5 & 4 \\ 1 & 4 & 0 & 0 & 2 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 3 & 2 \\ 2 & 5 & 4 & 0 & 3 & 4 & 4 \\ 1 & 3 & 3 & 0 & 2 & 4 & 3 \\ 2 & 4 & 2 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 175, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 5 & 4 \\ 1 & 4 & 0 & 0 & 2 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 3 & 2 \\ 2 & 5 & 4 & 0 & 3 & 4 & 4 \\ 2 & 6 & 3 & 0 & 3 & 5 & 4 \\ 2 & 4 & 2 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}1^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 475, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 \\ 2 & 4 & 4 & 0 & 3 & 5 & 9 \\ 1 & 4 & 0 & 0 & 2 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 3 & 5 \\ 2 & 5 & 4 & 0 & 3 & 4 & 9 \\ 3 & 9 & 6 & 0 & 5 & 9 & 13 \\ 2 & 4 & 2 & -3 & 2 & 4 & 7 \end{pmatrix}_{b_0}$$

4.513 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$

$$T = 8 \quad [0] \quad 1^{13}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 35, T \in [14, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 24 & 8 \\ 3 & 4 & -3 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.11 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & -2 & 2 & 0 & 0 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 4 & 8 & 0 & 4 & 9 & 5 \\ 3 & 4 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{13}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 4 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 278, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 7 & 13 & 0 & 7 & 27 & 9 \\ 3 & 4 & -3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{13}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 9 & 17 & 0 & 9 & 48 & 12 \\ 3 & 4 & -3 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \ 1^{13}3^{12}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 13 & 6 & 1 & 5 \\ 13 & 30 & 13 & 0 & 10 \\ 6 & 13 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 10 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

- [9] $1^{11}3^{12}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -52$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 4 & -1 & 2 \end{pmatrix}_{b_0}$
- [10] $1^{11}31^{13}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$
 $\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 2 & 6 & 0 & 4 & 4 \\ 4 & 6 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 0 & 1 & 3 \\ 3 & 4 & 5 & -3 & 3 & 1 \end{pmatrix}_{b_0}$
- [11] $1^{12}3^{12}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 5 & 6 & 1 & 12 & 5 \\ 6 & 5 & 1 & 14 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 14 & 0 & 25 & 9 \\ 5 & 5 & -1 & 9 & 2 \end{pmatrix}_{b_0}$
- [12] $1^{11}3^{12}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 5 & 6 & 1 & 6 & 5 \\ 6 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 7 & 0 & 4 & 4 \\ 5 & 5 & -1 & 4 & 2 \end{pmatrix}_{b_0}$
- [13] $1^{11}31^{13}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -106$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.30 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 8 & 6 & 0 & 5 & 6 \\ 2 & 6 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 1 & 3 \\ 3 & 6 & 3 & -3 & 3 & 1 \end{pmatrix}_{b_0}$
- [14] $1^{11}3^{11}1^{13}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -124$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.32 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 3 & 0 & 0 & 2 \\ 1 & 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$
- [15] $1^{12}31^{11}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 8 & 6 & 0 & 20 & 6 \\ 2 & 6 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 13 & 0 & 43 & 13 \\ 3 & 6 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$
- [16] $1^{11}3^{11}1^{13}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -149$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 1 & 3 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$
- [17] $1^{13}1^{11}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 8 & 6 & 0 & 10 & 6 \\ 2 & 6 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 10 & 5 & 0 & 7 & 5 \\ 3 & 6 & 3 & -3 & 5 & 1 \end{pmatrix}_{b_0}$
- [18] $1^{13}1^{11}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -284$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 2 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{11}3^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -306, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 & 3 \\ 4 & 8 & 6 & 0 & 15 & 6 & 6 \\ 2 & 6 & 1 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 15 & 9 & 0 & 21 & 9 & 9 \\ 3 & 6 & 3 & -3 & 9 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{11}3^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 0 & 0 & 5 & 2 \\ 3 & 9 & 6 & 0 & 5 & 12 & 6 \\ 2 & 4 & 2 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [21] \ 1^{11}4^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 0 & 2 & 2 \\ 3 & 4 & -2 & 0 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11}3^{11}4^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 130, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 1 & 2 & 3 & 3 \\ 4 & 4 & 0 & 0 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 2 & 2 \\ 3 & 4 & -2 & 0 & 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [23] \ 1^{11}5^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 8 & 1 & 5 & 5 \\ 9 & 13 & 12 & 0 & 7 & 7 \\ 8 & 12 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 5 & 7 & 6 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

4.514 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 12 & 6 \\ 5 & 8 & -3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

4.515 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{11}2^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 4 & 3 \\ 3 & 4 & 0 & 3 & 9 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 4 & 2 \\ 5 & 9 & 0 & 5 & 12 & 12 & 6 \\ 4 & 8 & 0 & 4 & 12 & 10 & 6 \\ 3 & 4 & -3 & 2 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[1] \quad & 1^{111}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 2 & 0 & 3 & 0 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 3 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & 2 & -1 & 4 & 0 \end{pmatrix}_{b_0} \\
[2] \quad & 1^{111}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 4 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 11 & 2 \\ 2 & 4 & 0 & 2 & 0 & 12 & 2 \\ 11 & 19 & 0 & 11 & 12 & 73 & 15 \\ 3 & 4 & -3 & 2 & 2 & 15 & 1 \end{pmatrix}_{b_0} \\
[3] \quad & 1^{112}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7 \\
& T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 4 & 2 \\ 2 & 4 & 0 & 2 & 0 & 6 & 2 \\ 4 & 8 & 0 & 4 & 6 & 10 & 6 \\ 3 & 4 & -3 & 2 & 2 & 6 & 0 \end{pmatrix}_{b_0} \\
[4] \quad & 1^{113}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 3 & 2 \\ 2 & 4 & 0 & 2 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -3 & 2 & 2 & 4 & 1 \end{pmatrix}_{b_0} \\
[5] \quad & 1^{113}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & -3 & 2 & 3 & 3 & 0 \end{pmatrix}_{b_0} \\
[6] \quad & 1^{121}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4 \\
& T_{\min} = 12.31 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 2 & 2 \\ 4 & 8 & 0 & 4 & 9 & 6 & 5 \\ 2 & 4 & 0 & 2 & 6 & 1 & 3 \\ 3 & 4 & -3 & 2 & 5 & 3 & 0 \end{pmatrix}_{b_0} \\
[7] \quad & 1^{111}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 2 & 0 & 3 & 0 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & 2 & -1 & 4 & 0 \end{pmatrix}_{b_0} \\
[8] \quad & 1^{121}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4 \\
& T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 5 & 3 \\ 3 & 4 & 0 & 3 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 5 & 2 \\ 4 & 8 & 0 & 4 & 9 & 12 & 5 \\ 5 & 9 & 0 & 5 & 12 & 13 & 7 \\ 3 & 4 & -3 & 2 & 5 & 7 & 0 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[9] \quad & 1^{111}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -226, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 2 & 4 & 0 & 2 & 0 & 3 & 2 \\ 2 & 4 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & -3 & 2 & 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[10] \quad & 1^{111}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -336, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 3 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 5 & 2 & 2 \\ 5 & 9 & 0 & 5 & 12 & 6 & 6 \\ 2 & 4 & 0 & 2 & 6 & 1 & 3 \\ 3 & 4 & -3 & 2 & 6 & 3 & 1 \end{pmatrix}_{b_0} \\
[11] \quad & 1^{111}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -386, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 8 & 2 \\ 2 & 4 & 0 & 2 & 0 & 9 & 2 \\ 8 & 14 & 0 & 8 & 9 & 37 & 11 \\ 3 & 4 & -3 & 2 & 2 & 11 & 1 \end{pmatrix}_{b_0} \\
[12] \quad & 1^{111}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.2 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 \\ 2 & 4 & 0 & 2 & 0 & 6 & 2 \\ 5 & 9 & 0 & 5 & 6 & 13 & 7 \\ 3 & 4 & -3 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0} \\
[13] \quad & 1^{121}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -476, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 13.4 \\
& T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 3 & 13 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 7 & 2 & 2 \\ 7 & 13 & 0 & 7 & 27 & 9 & 9 \\ 2 & 4 & 0 & 2 & 9 & 1 & 3 \\ 3 & 4 & -3 & 2 & 9 & 3 & 1 \end{pmatrix}_{b_0} \\
[14] \quad & 1^{112}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -626, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.7 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 7 & 2 \\ 2 & 4 & 0 & 2 & 0 & 9 & 2 \\ 7 & 13 & 0 & 7 & 9 & 28 & 10 \\ 3 & 4 & -3 & 2 & 2 & 10 & 1 \end{pmatrix}_{b_0} \\
[15] \quad & 1^{112}3^{11}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 12.7 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 16 & 3 \\ 3 & 4 & 0 & 3 & 4 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 16 & 2 \\ 2 & 4 & 0 & 2 & 0 & 18 & 2 \\ 16 & 28 & 0 & 16 & 18 & 154 & 22 \\ 3 & 4 & -3 & 2 & 2 & 22 & 1 \end{pmatrix}_{b_0} \\
[16] \quad & 1^{113}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 13.0 \\
& T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 9 & 2 \\ 2 & 4 & 0 & 2 & 0 & 12 & 2 \\ 9 & 17 & 0 & 9 & 12 & 49 & 13 \\ 3 & 4 & -3 & 2 & 2 & 13 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[17] \quad 1^{1143^1 1^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 11 & 2 \\ 2 & 4 & 0 & 2 & 0 & 15 & 2 \\ 11 & 21 & 0 & 11 & 15 & 76 & 16 \\ 3 & 4 & -3 & 2 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1123^1 1^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 10 & 2 \\ 2 & 4 & 0 & 2 & 0 & 12 & 2 \\ 10 & 18 & 0 & 10 & 12 & 58 & 14 \\ 3 & 4 & -3 & 2 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1123^1 1^{13}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 4 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 13 & 2 \\ 2 & 4 & 0 & 2 & 0 & 15 & 2 \\ 13 & 23 & 0 & 13 & 15 & 100 & 18 \\ 3 & 4 & -3 & 2 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1133^1 1^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 12 & 2 \\ 2 & 4 & 0 & 2 & 0 & 15 & 2 \\ 12 & 22 & 0 & 12 & 15 & 85 & 17 \\ 3 & 4 & -3 & 2 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1133^1 1^{13}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 3 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 15 & 2 \\ 2 & 4 & 0 & 2 & 0 & 18 & 2 \\ 15 & 27 & 0 & 15 & 18 & 133 & 21 \\ 3 & 4 & -3 & 2 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1143^1 1^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 4 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 14 & 2 \\ 2 & 4 & 0 & 2 & 0 & 18 & 2 \\ 14 & 26 & 0 & 14 & 18 & 118 & 20 \\ 3 & 4 & -3 & 2 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1133^1 1^{14}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1826, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 18 & 3 \\ 3 & 4 & 0 & 3 & 4 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 18 & 2 \\ 2 & 4 & 0 & 2 & 0 & 21 & 2 \\ 18 & 32 & 0 & 18 & 21 & 193 & 25 \\ 3 & 4 & -3 & 2 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1143^1 1^{13}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 3 & 4 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 17 & 2 \\ 2 & 4 & 0 & 2 & 0 & 21 & 2 \\ 17 & 31 & 0 & 17 & 21 & 172 & 24 \\ 3 & 4 & -3 & 2 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{114}3^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -2816, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 0 & 3 & 4 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 20 & 2 \\ 2 & 4 & 0 & 2 & 0 & 24 & 2 \\ 20 & 36 & 0 & 20 & 24 & 238 & 28 \\ 3 & 4 & -3 & 2 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [26] \ 1^{111}4^{131}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 13, \ \Delta = -58, \ \det = -32, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 0 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{111}4^{131}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 13, \ \Delta = -58, \ \det = -96, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 0 & 4 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 0 & 5 & 2 \\ 5 & 7 & 0 & 0 & 5 & 8 & 6 \\ 3 & 4 & -2 & 0 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{111}4^{131}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 13, \ \Delta = -58, \ \det = -192, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 0 & 8 & 2 \\ 8 & 12 & 0 & 0 & 8 & 26 & 10 \\ 3 & 4 & -2 & 0 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

4.516 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{113}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 72, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 & 5 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 3 & 3 & 2 \\ 2 & 4 & 0 & 2 & 0 & 3 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 4 & 6 & 4 \\ 3 & 5 & 0 & 3 & 3 & 6 & 4 & 4 \\ 3 & 4 & -3 & 2 & 2 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 392, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 0 & 2 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 2 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & -3 & 2 & 2 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 462, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 3 & 4 & 5 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 3 & 5 & 2 \\ 2 & 4 & 0 & 2 & 0 & 3 & 6 & 2 \\ 3 & 5 & 0 & 3 & 3 & 4 & 9 & 4 \\ 5 & 9 & 0 & 5 & 6 & 9 & 13 & 7 \\ 3 & 4 & -3 & 2 & 2 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 512, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 9 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 0 & 5 & 2 & 2 \\ 5 & 9 & 0 & 0 & 5 & 12 & 6 & 6 \\ 2 & 4 & 0 & 2 & 6 & 1 & 3 & 3 \\ 2 & 4 & 0 & 2 & 6 & 3 & 1 & 3 \\ 3 & 4 & -3 & 2 & 6 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 \\ 2 & 4 & 0 & 2 & 0 & 3 & 6 \\ 2 & 4 & 0 & 2 & 3 & 1 & 6 \\ 5 & 9 & 0 & 5 & 6 & 6 & 13 \\ 3 & 4 & -3 & 2 & 2 & 3 & 7 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 3 & 4 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 5 & 5 & 2 \\ 2 & 4 & 0 & 2 & 0 & 6 & 6 & 2 \\ 5 & 9 & 0 & 5 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 5 & 6 & 15 & 13 & 7 \\ 3 & 4 & -3 & 2 & 2 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

4.517 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{123}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 & 3 \\ 3 & 4 & 0 & 3 & 8 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 4 & 3 & 2 \\ 4 & 8 & 0 & 4 & 9 & 6 & 5 \\ 3 & 5 & 0 & 3 & 6 & 3 & 3 \\ 3 & 4 & -3 & 2 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 0 & 5 \\ 2 & 4 & -2 & 2 & 0 & 0 & 0 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 2 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.47 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 2 & 4 & 0 & 2 & 0 & 3 & 2 \\ 2 & 4 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 \\ 2 & 4 & 0 & 2 & 0 & 6 & 2 \\ 5 & 9 & 0 & 5 & 6 & 12 & 6 \\ 3 & 4 & -3 & 2 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

4.518 NHC(-2)(-3) + f₄ + su₂→(-3) + (-3)→(-2)

$$T = 8 \quad [0] \quad 1^1 3^1 1^1 2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 45, \Delta = -119, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 45, T \in [12, 45]$$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 1 & 4 \\ 8 & 9 & 4 & 4 & 0 & 5 \\ 3 & 4 & -2 & 2 & 0 & 0 \\ 3 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [1] \quad 2^1 1^1 4^1 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 1 & 3 \\ 3 & 1 & 6 & 0 & 0 & 3 \\ 5 & 6 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 6 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \quad 1^1 5^1 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 10 & 8 & 1 & 5 \\ 10 & 13 & 12 & 0 & 7 \\ 8 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

4.519 NHC(-2)(-3) + f₄ + su₂n3mix

$$T = 7 \quad [0] \quad 1^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 14.09 \rightarrow 15, T_{\max} = 97, T \in [15, 97]$$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 2 & 4 & -2 & 5 & 0 \\ 7 & 12 & 5 & 12 & 6 \\ 3 & 5 & 0 & 6 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 2^1 3^1 1^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 3 & 4 & -2 & 5 & 0 \\ 5 & 12 & 5 & 12 & 6 \\ 3 & 5 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 3 & 4 & -2 & 5 & 0 \\ 5 & 12 & 5 & 12 & 6 \\ 3 & 5 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^1 3^1 1^1 1^4 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 15.65 \rightarrow 16, T_{\max} = 97, T \in [16, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 14 & 23 & 0 & 120 & 18 \\ 3 & 4 & -3 & 18 & -3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^1 3^1 1^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 5 & 8 & 0 & 12 & 6 \\ 3 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^1 3^1 1^1 1^3 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.31 \rightarrow 14, T_{\max} = 97, T \in [14, 97]$$

- [6] $1^{11}3^{11}1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.34 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$
 $\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 11 & 18 & 0 & 72 & 14 \\ 3 & 4 & -3 & 14 & -1 \end{pmatrix}_{b_0}$
- [7] $1^{11}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 13 & 0 & 36 & 10 \\ 3 & 4 & -3 & 10 & 0 \end{pmatrix}_{b_0}$
- [8] $1^{111}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$
 $\begin{pmatrix} 1 & 4 & 2 & 6 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 2 & 4 & -2 & 5 & 0 \\ 6 & 12 & 5 & 12 & 6 \\ 3 & 5 & 0 & 6 & 0 \end{pmatrix}_{b_0}$
- [9] $13^{11}1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$
 $\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 2 & 4 & -2 & 5 & 0 \\ 5 & 12 & 5 & 12 & 6 \\ 3 & 5 & 0 & 6 & 0 \end{pmatrix}_{b_0}$
- [10] $1^{12}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 11 & 0 & 20 & 8 \\ 3 & 4 & -3 & 8 & 1 \end{pmatrix}_{b_0}$
- [11] $1^{13}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 15 & 0 & 39 & 11 \\ 3 & 4 & -3 & 11 & 1 \end{pmatrix}_{b_0}$
- [12] $1^{113}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 12 & 0 & 25 & 9 \\ 3 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$
- [13] $1^{14}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -190, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 10 & 19 & 0 & 64 & 14 \\ 3 & 4 & -3 & 14 & 1 \end{pmatrix}_{b_0}$
- [14] $1^{113}3^{11}1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 10 & 17 & 0 & 55 & 13 \\ 3 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$
- [15] $1^{123}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -240, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 16 & 0 & 44 & 12 \\ 3 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$
- [16] $1^{113}3^{11}1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -259, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 13 & 22 & 0 & 97 & 17 \\ 3 & 4 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{13}3^{11}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -295, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 16 & 27 & 0 & 151 & 21 \\ 3 & 4 & -3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 21^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -320, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 4 & 8 & 3 & 9 & 4 \\ 8 & 9 & 4 & 12 & 5 \\ 3 & 4 & -2 & 5 & 0 \\ 9 & 12 & 5 & 12 & 6 \\ 4 & 5 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{13}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 11 & 20 & 0 & 69 & 15 \\ 3 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{12}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -354, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 12 & 21 & 0 & 80 & 16 \\ 3 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{14}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 13 & 24 & 0 & 100 & 18 \\ 3 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{12}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -456, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 15 & 26 & 0 & 128 & 20 \\ 3 & 4 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{13}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -499, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 14 & 25 & 0 & 111 & 19 \\ 3 & 4 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{12}3^{11}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -546, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 18 & 31 & 0 & 188 & 24 \\ 3 & 4 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{14}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -646, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 16 & 29 & 0 & 148 & 22 \\ 3 & 4 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{13}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -655, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 17 & 30 & 0 & 165 & 23 \\ 3 & 4 & -3 & 23 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{13}3^{11}1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -799, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 35 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 20 & 35 & 0 & 231 & 27 \\ 3 & 4 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{14}3^{11}1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -856, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 34 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 19 & 34 & 0 & 208 & 26 \\ 3 & 4 & -3 & 26 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{14}3^{11}1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -1054, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 39 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 22 & 39 & 0 & 280 & 30 \\ 3 & 4 & -3 & 30 & 1 \end{pmatrix}_{b_0}$$

$T = 8$ [30] $1^{12}3^{12}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 19.45 \rightarrow 20, T_{\max} = 22, T \in [20, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 5 & 3 & 0 & 7 & 15 & 7 \\ 2 & 2 & -1 & 0 & 7 & -7 \end{pmatrix}_{b_0}$$

[32] $1^{11}3^{11}1^{13}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 5 & 0 & 7 & 5 \\ 2 & 4 & 2 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{13}3^{11}1^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 14 & 0 & 69 & 15 \\ 2 & 4 & 2 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{11}3^{11}1^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.54 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 8 & 3 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 8 & 0 & 25 & 9 \\ 2 & 3 & 2 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 3 & 2 & 0 & 4 & 4 & 4 \\ 2 & 2 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{11}3^{11}1^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 5 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 6 & 0 & 7 & 5 \\ 2 & 3 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 13 & 9 & 1 & 7 \\ 9 & 5 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

[37] $1^{12}31^{13}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 7 & 3 \\ 2 & 4 & 3 & 0 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 7 & 8 & 0 & 15 & 7 \\ 2 & 3 & 3 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{12}3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 25 & 12 & 1 & 9 \\ 12 & 5 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 9 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{11}3^1 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 4 & 3 \\ 1 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{11}3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 3 & 3 \\ 2 & 2 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{11}3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 3 & 4 & 0 & 5 & 10 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{11}3^1 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 5 & 3 \\ 1 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 5 & 0 & 7 & 5 \\ 2 & 3 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{12}3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.32 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 4 & 2 & 0 & 4 & 4 & 4 \\ 2 & 2 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{12}3^1 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.05 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 7 & 6 & 0 & 11 & 42 & 12 \\ 2 & 2 & -1 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

[44] $1^{11}3^1 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.15 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 4 & 3 \\ 2 & 4 & 3 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 0 & 4 & 4 \\ 2 & 3 & 3 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{12}3^1 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 5 & 7 & 0 & 9 & 35 & 11 \\ 2 & 2 & -1 & 0 & 11 & 0 \end{pmatrix}_{b_0}$$

[47] $1^{11}3^1 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 5 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 4 \\ 2 & 4 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

- [48] $1^{12}3^{12}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 3 & 5 & 0 & 3 & 11 & 5 \\ 2 & 2 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$
- [50] $1^{11}3^{11}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 5 & 5 & 0 & 8 & 23 & 9 \\ 2 & 2 & -1 & 0 & 9 & 0 \end{pmatrix}_{b_0}$$
- [49] $1^{13}3^{12}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 0 \\ 7 & 10 & 0 & 13 & 74 & 16 \\ 2 & 2 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$
- [51] $1^{112}3^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 10 & 3 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 10 & 10 & 0 & 37 & 11 \\ 2 & 3 & 2 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$
- [52] $1^{13}3^{12}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 4 & 7 & 0 & 4 & 23 & 7 \\ 2 & 2 & -1 & 0 & 7 & 0 \end{pmatrix}_{b_0}$$
- [53] $1^{11}3^{11}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 3 & 3 & 0 & 2 & 3 & 3 \\ 2 & 2 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$
- [54] $1^{112}3^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.51 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 7 & 3 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 7 & 7 & 0 & 15 & 7 \\ 2 & 3 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [55] $1^{12}3^{12}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 4 & 6 & 0 & 6 & 20 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [56] $1^{14}3^{12}21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 5 & 9 & 0 & 5 & 39 & 9 \\ 2 & 2 & -1 & 0 & 9 & 0 \end{pmatrix}_{b_0}$$
- [57] $1^{12}3^{12}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 12 & 0 \\ 7 & 8 & 0 & 12 & 56 & 14 \\ 2 & 2 & -1 & 0 & 14 & 0 \end{pmatrix}_{b_0}$$
- [58] $1^13^11^13^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 8 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 4 & 0 & 9 & 5 \\ 2 & 4 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{11} 3^{11} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 4 & 4 & 0 & 5 & 10 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{11} 3^{11} 2^3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 0 & 16 & 3 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 16 & 0 & 100 & 18 \\ 2 & 3 & 2 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{12} 3^{11} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 4 & 3 & 0 & 2 & 3 & 3 \\ 2 & 2 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{13} 3^{11} 1^{11} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 167, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 5 & 0 & 12 & 6 \\ 2 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{12} 3^{11} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 4 & 5 & 0 & 3 & 11 & 5 \\ 2 & 2 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{13} 3^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 17 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 17 & 13 & 0 & 60 & 14 \\ 2 & 4 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{12} 3^{11} 1^{11} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 11 & 0 & 44 & 12 \\ 2 & 4 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{13} 3^{11} 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 6 & 9 & 0 & 10 & 51 & 13 \\ 2 & 2 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{13} 3^{11} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 5 & 8 & 0 & 7 & 34 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{11} 3^{11} 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 6 & 5 & 0 & 8 & 23 & 9 \\ 2 & 2 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{114}3^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 205, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.68 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 1 & 4 & 0 & 22 & 3 \\ 1 & 4 & 0 & 0 & 22 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 22 & 22 & 0 & 193 & 25 \\ 2 & 3 & 2 & -3 & 25 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{113}3^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 10 & 3 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 10 & 10 & 0 & 32 & 10 \\ 2 & 3 & 2 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{12}3^{11}2^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 5 & 4 & 0 & 5 & 10 & 6 \\ 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{113}3^{11}1^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 8 & 0 & 25 & 9 \\ 2 & 4 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{12}3^{11}2^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 6 & 7 & 0 & 9 & 35 & 11 \\ 2 & 2 & -1 & 0 & 11 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{14}3^{11}2^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 14 & 0 \\ 8 & 12 & 0 & 14 & 96 & 18 \\ 2 & 2 & -1 & 0 & 18 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{13}3^{11}2^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 5 & 7 & 0 & 4 & 23 & 7 \\ 2 & 2 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

[76] $1^{13}3^{12}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.19 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 14 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 14 & 7 & 0 & 31 & 9 \\ 2 & 4 & 2 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{12}3^{11}2^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 251, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 5 & 6 & 0 & 6 & 20 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{11}3^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 255, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 11 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 11 & 7 & 0 & 20 & 8 \\ 2 & 4 & 2 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

- [79] $1^{14}3^{12}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 256$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 6 & 10 & 0 & 8 & 52 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$
- [80] $1^{11}3^{13}1^{13}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 257$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 13 & 3 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 13 & 13 & 0 & 60 & 14 \\ 2 & 3 & 2 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$
- [81] $1^{12}3^{12}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 259$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 5 & 0 & 3 & 11 & 5 \\ 2 & 2 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$
- [82] $1^{12}3^{11}1^{13}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 272$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 14 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 14 & 10 & 0 & 37 & 11 \\ 2 & 4 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$
- [83] $1^{14}3^{12}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 291$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 7 & 11 & 0 & 11 & 71 & 15 \\ 2 & 2 & -1 & 0 & 15 & 1 \end{pmatrix}_{b_0}$$
- [84] $1^{14}3^{12}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 304$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 6 & 9 & 0 & 5 & 39 & 9 \\ 2 & 2 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$
- [85] $1^{12}3^{12}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 304$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 12 & 0 \\ 8 & 8 & 0 & 12 & 56 & 14 \\ 2 & 2 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$
- [86] $1^{13}3^{12}1^{13}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 307$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.08 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 8 & 0 & 36 & 10 \\ 2 & 4 & 2 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$
- [87] $1^{13}3^{12}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 315$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 0 \\ 8 & 10 & 0 & 13 & 74 & 16 \\ 2 & 2 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$
- [88] $1^{11}4^{13}1^{13}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 322$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.71 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 13 & 3 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 13 & 13 & 0 & 55 & 13 \\ 2 & 3 & 2 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$
- [89] $1^{13}3^{12}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 355$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$

- $T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
- $$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 6 & 8 & 0 & 7 & 34 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$
- [90] $1^{14}3^{11}2^{12}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 363$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
- $$T_{\min} = 11.94 \rightarrow 12$$
- ,
- $T_{\max} = 35$
- ,
- $T \in [12, 35]$
- $$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 20 & 0 & 148 & 22 \\ 2 & 4 & 2 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$
- [91] $1^{13}3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 363$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
- $$T_{\min} = 11.83 \rightarrow 12$$
- ,
- $T_{\max} = 22$
- ,
- $T \in [12, 22]$
- $$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 6 & 7 & 0 & 4 & 23 & 7 \\ 2 & 2 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$
- [92] $1^{12}3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 364$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
- $$T_{\min} = 11.71 \rightarrow 12$$
- ,
- $T_{\max} = 22$
- ,
- $T \in [12, 22]$
- $$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 6 & 6 & 0 & 6 & 20 & 8 \\ 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [93] $1^{12}3^{11}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 379$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
- $$T_{\min} = 11.73 \rightarrow 12$$
- ,
- $T_{\max} = 22$
- ,
- $T \in [12, 22]$
- $$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 0 \\ 7 & 7 & 0 & 9 & 35 & 11 \\ 2 & 2 & -1 & 0 & 11 & 1 \end{pmatrix}_{b_0}$$
- [94] $1^{13}3^{11}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 380$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
- $$T_{\min} = 11.69 \rightarrow 12$$
- ,
- $T_{\max} = 22$
- ,
- $T \in [12, 22]$
- $$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 7 & 9 & 0 & 10 & 51 & 13 \\ 2 & 2 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$
- [95] $1^{114}31^{23}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 398$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
- $$T_{\min} = 12.38 \rightarrow 13$$
- ,
- $T_{\max} = 35$
- ,
- $T \in [13, 35]$
- $$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 4 & 0 & 19 & 3 \\ 1 & 4 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 19 & 19 & 0 & 135 & 21 \\ 2 & 3 & 2 & -3 & 21 & 0 \end{pmatrix}_{b_0}$$
- [96] $1^{131}3^{13}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 420$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
- $$T_{\min} = 12.20 \rightarrow 13$$
- ,
- $T_{\max} = 35$
- ,
- $T \in [13, 35]$
- $$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 20 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 20 & 10 & 0 & 65 & 13 \\ 2 & 4 & 2 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$
- [97] $1^{114}31^{13}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 437$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
- $$T_{\min} = 12.41 \rightarrow 13$$
- ,
- $T_{\max} = 35$
- ,
- $T \in [13, 35]$
- $$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 0 & 16 & 3 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 16 & 16 & 0 & 89 & 17 \\ 2 & 3 & 2 & -3 & 17 & 0 \end{pmatrix}_{b_0}$$
- [98] $1^{14}3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 459$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
- $$T_{\min} = 11.71 \rightarrow 12$$
- ,
- $T_{\max} = 22$
- ,
- $T \in [12, 22]$
- $$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 7 & 10 & 0 & 8 & 52 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$
- [99] $1^{14}3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 467$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
- $$T_{\min} = 11.83 \rightarrow 12$$
- ,
- $T_{\max} = 22$
- ,
- $T \in [12, 22]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 7 & 9 & 0 & 5 & 39 & 9 \\ 2 & 2 & -1 & 0 & 9 & 1 \end{pmatrix}_{b_0} \\
[100] & 1^{11}3^{11}1^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 11.97 \rightarrow 12, T_{\max} = 35, T \in [12, 35] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 11 & 0 & 55 & 13 \\ 2 & 4 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0} \\
[101] & 1^{11}3^{11}1^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 483, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 11.99 \rightarrow 12, T_{\max} = 35, T \in [12, 35] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 17 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 17 & 10 & 0 & 48 & 12 \\ 2 & 4 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0} \\
& 1^{11} \\
[102] & 1^{14}3^{11}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 499, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 11.73 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 14 & 0 \\ 9 & 12 & 0 & 14 & 96 & 18 \\ 2 & 2 & -1 & 0 & 18 & 1 \end{pmatrix}_{b_0} \\
[103] & 1^31^{13}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 503, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.11 \rightarrow 13, T_{\max} = 35, T \in [13, 35] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 11 & 0 & 72 & 14 \\ 2 & 4 & 2 & -3 & 14 & 0 \end{pmatrix}_{b_0} \\
[104] & 1^{13}3^{11}1^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 508, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 35, T \in [12, 35] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 17 & 0 & 111 & 19 \\ 2 & 4 & 2 & -3 & 19 & 1 \end{pmatrix}_{b_0} \\
& 1^{12} \\
[105] & 1^{13}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 508, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 11.69 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 7 & 8 & 0 & 7 & 34 & 10 \\ 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0} \\
& 1^{11} \\
[106] & 1^{14}3^{11}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 524, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 11.68 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 8 & 11 & 0 & 11 & 71 & 15 \\ 2 & 2 & -1 & 0 & 15 & 1 \end{pmatrix}_{b_0} \\
& 1^{12} \\
[107] & 1^{13}3^{11}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 11.73 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 0 \\ 9 & 10 & 0 & 13 & 74 & 16 \\ 2 & 2 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0} \\
[108] & 1^{12}3^{11}1^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 547, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 14 & 0 & 80 & 16 \\ 2 & 4 & 2 & -3 & 16 & 1 \end{pmatrix}_{b_0} \\
[109] & 1^{14}31^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 552, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 26 & 4 \\ 1 & 4 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 26 & 19 & 0 & 135 & 21 \\ 2 & 4 & 2 & -3 & 21 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

[110] $1^{13}3^{12}21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 563$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.68 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 0 \\ 8 & 9 & 0 & 10 & 51 & 13 \\ 2 & 2 & -1 & 0 & 13 & 1 \end{pmatrix}_{b_0}$$

[111] $1^{12}31^{12}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 612$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 20 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 20 & 13 & 0 & 71 & 15 \\ 2 & 4 & 2 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{13}31^{12}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 635$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 23 & 4 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 23 & 16 & 0 & 100 & 18 \\ 2 & 4 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[113] $1^{13}31^{14}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 648$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.20 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 26 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 26 & 13 & 0 & 111 & 17 \\ 2 & 4 & 2 & -3 & 17 & 0 \end{pmatrix}_{b_0}$$

[114] $1^{14}3^{12}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 652$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 8 & 10 & 0 & 8 & 52 & 12 \\ 2 & 2 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[115] $1^{14}3^{12}21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 747$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.66 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 0 \\ 9 & 11 & 0 & 11 & 71 & 15 \\ 2 & 2 & -1 & 0 & 15 & 1 \end{pmatrix}_{b_0}$$

[116] $1^{14}3^{12}21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 752$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.68 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 14 & 0 \\ 10 & 12 & 0 & 14 & 96 & 18 \\ 2 & 2 & -1 & 0 & 18 & 1 \end{pmatrix}_{b_0}$$

[117] $1^{13}1^{14}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 755$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 14 & 0 & 120 & 18 \\ 2 & 4 & 2 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

[118] $1^{11}31^{13}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 767$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.01 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 23 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 23 & 13 & 0 & 88 & 16 \\ 2 & 4 & 2 & -3 & 16 & 0 \end{pmatrix}_{b_0}$$

[119] $1^{11}3^{11}1^{13}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 788$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 14 & 0 & 97 & 17 \\ 2 & 4 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[120] $1^{12}3^{11}1^{13}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 967$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 39 & 4 \\ 1 & 4 & 0 & 0 & 26 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 39 & 26 & 0 & 280 & 30 \\ 2 & 4 & 2 & -3 & 30 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{13}31^{14}3^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 1707$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 35 & 4 \\ 1 & 4 & 0 & 0 & 22 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 35 & 22 & 0 & 216 & 26 \\ 2 & 4 & 2 & -3 & 26 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{14}31^{14}3^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 1848$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 38 & 4 \\ 1 & 4 & 0 & 0 & 25 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 38 & 25 & 0 & 263 & 29 \\ 2 & 4 & 2 & -3 & 29 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [134] $1^1 231^1 3^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -20$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$

$T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 3 & 3 \\ 5 & 13 & 1 & 5 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 5 & 0 & 0 & 2 \\ 3 & 7 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[135] $1^{11} 231^1 3^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -20$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 5 & 4 \\ 1 & 4 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 5 & 0 & 5 & 5 \\ 2 & 4 & 3 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[136] $1^{13} 23^1 1^{12} 31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 31 & 3 \\ 5 & 13 & 1 & 84 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 84 & 0 & 520 & 42 \\ 3 & 7 & -1 & 42 & 3 \end{pmatrix}_{b_0}$$

[137] $1^{12} 23^1 1^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -28$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.29 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 24 & 3 \\ 5 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 69 & 0 & 340 & 34 \\ 3 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[138] $1^{12} 3^{11} 131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 1 & 4 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 0 & 0 & 4 & 4 \\ 3 & 0 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

[139] $1^{11} 23^1 1^{13} 131^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 5 & 0 & 6 & 4 \\ 1 & 5 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 6 & 0 & 8 & 6 \\ 2 & 4 & 3 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{13} 4^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -109$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 8 & 0 & 13 & 7 \\ 3 & 4 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[141] $1^{14} 4^1 13^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -109$, $\det = -35$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 4 & 5 & 0 & 4 & 4 \\ 3 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[143] $1^{12} 231^{11} 3^{11} 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -40$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 5 & 1 & 21 & 3 \\ 5 & 13 & 1 & 50 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 50 & 0 & 185 & 25 \\ 3 & 7 & -1 & 25 & 3 \end{pmatrix}_{b_0} \\
[144] & 1^{12}231^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 19 & 3 \\ 5 & 13 & 1 & 50 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 50 & 0 & 185 & 25 \\ 3 & 7 & -1 & 25 & 3 \end{pmatrix}_{b_0} \\
[142] & 1^{12}231^{11}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 21 & 3 \\ 5 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 54 & 0 & 217 & 27 \\ 3 & 7 & -1 & 27 & 3 \end{pmatrix}_{b_0} \\
[145] & 1^{12}231^{11}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 22 & 3 \\ 5 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 54 & 0 & 217 & 27 \\ 3 & 7 & -1 & 27 & 3 \end{pmatrix}_{b_0} \\
[146] & 1^{11}413^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.58 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 4 & 0 & 1 & 3 \\ 3 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0} \\
[147] & 1^{11}3^{11}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.82 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 4 & 1 & 3 & 3 \\ 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0} \\
[148] & 1^{11}3^{11}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.76 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 7 & 0 & 8 & 6 \\ 3 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0} \\
[149] & 1^{12}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 4 & 3 \\ 5 & 13 & 1 & 9 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 9 & 0 & 4 & 4 \\ 3 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0} \\
[150] & 1^{11}231^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 12 & 3 \\ 5 & 13 & 1 & 35 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 35 & 0 & 85 & 17 \\ 3 & 7 & -1 & 17 & 2 \end{pmatrix}_{b_0} \\
[151] & 1^{11}231^{11}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 16 & 3 \\ 5 & 13 & 1 & 35 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 35 & 0 & 85 & 17 \\ 3 & 7 & -1 & 17 & 2 \end{pmatrix}_{b_0} \\
[152] & 1^{112}2313^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 2 & 5 & 0 & 9 & 4 \\ 1 & 5 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 9 & 0 & 19 & 9 \\ 2 & 4 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0} \\
[153] & 1^{11}3^{11}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.82 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 4 & 1 & 6 & 3 \\ 4 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 8 & 0 & 13 & 7 \\ 3 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0} \\
[154] & 21^{11}41^{13}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 2 & 2 & 1 & 3 & 2 \\ 2 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 2 & 6 & 0 & 4 & 4 \\ 2 & 2 & 3 & -2 & 4 & 1 \end{pmatrix}_{b_0} \\
[155] & 1^{12}231^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 20 & 3 \\ 5 & 13 & 1 & 50 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 50 & 0 & 185 & 25 \\ 3 & 7 & -1 & 25 & 3 \end{pmatrix}_{b_0} \\
[156] & 1^{12}231^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 14 & 4 \\ 1 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 14 & 12 & 0 & 41 & 13 \\ 2 & 4 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0} \\
[157] & 1^{12}413^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.70 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 8 & 0 & 10 & 6 \\ 3 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0} \\
[158] & 1^{11}231^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 18 & 3 \\ 5 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 54 & 0 & 200 & 26 \\ 3 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0} \\
[159] & 1^{13}4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.93 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 4 & 1 & 4 & 3 \\ 4 & 4 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 4 & 5 & 0 & 4 & 4 \\ 3 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0} \\
[160] & 1^{13}231^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 20 & 0 & 120 & 22 \\ 2 & 4 & 3 & -3 & 22 & 1 \end{pmatrix}_{b_0} \\
[161] & 1^{12}3^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 8 & 3 \\ 5 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 24 & 0 & 37 & 11 \\ 3 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0} \\
[162] & 1^{11}413^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 9 & 0 & 13 & 7 \\ 3 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0} \\
[163] & 1^{13}413^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.82 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 12 & 0 & 25 & 9 \\ 3 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0} \\
[164] & 1^{12}3^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 11 & 3 \\ 5 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 24 & 0 & 37 & 11 \\ 3 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0} \\
[165] & 1^{12}31^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13]
\end{aligned}$$

- [166] $1^{11}231^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 7 & 3 \\ 5 & 13 & 1 & 20 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 20 & 0 & 25 & 9 \\ 3 & 7 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$
- [167] $1^{11}231^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 14 & 3 \\ 5 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 39 & 0 & 107 & 19 \\ 3 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$
- [168] $1^{11}231^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 9 & 0 & 26 & 10 \\ 2 & 4 & 3 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$
- [169] $1^{14}413^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 16 & 0 & 46 & 12 \\ 3 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$
- [170] $1^{12}4^{13}1^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 11 & 0 & 21 & 9 \\ 3 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$
- [171] $1^{11}231^{11}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 17 & 3 \\ 5 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 39 & 0 & 107 & 19 \\ 3 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$
- [174] $1^{11}231^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 11 & 4 \\ 1 & 4 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 11 & 8 & 0 & 21 & 9 \\ 2 & 4 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$
- [172] $1^{11}4^{13}1^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 12 & 0 & 26 & 10 \\ 3 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$
- [173] $1^{11}231^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.20 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 8 & 3 \\ 5 & 13 & 1 & 20 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 20 & 0 & 25 & 9 \\ 3 & 7 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$
- [175] $1^{12}23^{11}1^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 1 & 11 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 0 & 0 & 37 & 11 \\ 3 & 0 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$
- [176] $1^{12}4^{13}1^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 13 & 0 & 28 & 10 \\ 3 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$
- [179] $1^{11}231^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$

$$\begin{aligned}
& T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 13 & 3 \\ 5 & 13 & 1 & 35 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 35 & 0 & 85 & 17 \\ 3 & 7 & -1 & 17 & 2 \end{pmatrix}_{b_0} \\
[177] & 1^{11}231^{11}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 15 & 3 \\ 5 & 13 & 1 & 35 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 35 & 0 & 85 & 17 \\ 3 & 7 & -1 & 17 & 2 \end{pmatrix}_{b_0} \\
[178] & 1^{12}231^{12}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 13 & 3 \\ 5 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 39 & 0 & 100 & 18 \\ 3 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0} \\
[180] & 1^{12}231^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 9 & 3 \\ 5 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 24 & 0 & 37 & 11 \\ 3 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0} \\
[182] & 1^{13}231^{12}31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 32 & 3 \\ 5 & 13 & 1 & 84 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 32 & 84 & 0 & 520 & 42 \\ 3 & 7 & -1 & 42 & 3 \end{pmatrix}_{b_0} \\
[183] & 1^{11}231^{13}1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 1 & 1 & 12 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 0 & 0 & 44 & 12 \\ 3 & 0 & -1 & 12 & 2 \end{pmatrix}_{b_0} \\
[181] & 1^{12}231^{11}31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 10 & 3 \\ 5 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 24 & 0 & 37 & 11 \\ 3 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0} \\
[184] & 1^{11}231^{11}31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 15 & 3 \\ 5 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 39 & 0 & 107 & 19 \\ 3 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0} \\
[186] & 1^{11}231^{11}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 16 & 3 \\ 5 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 39 & 0 & 107 & 19 \\ 3 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0} \\
[185] & 1^{11}231^{11}31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 14 & 3 \\ 5 & 13 & 1 & 35 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 35 & 0 & 85 & 17 \\ 3 & 7 & -1 & 17 & 2 \end{pmatrix}_{b_0} \\
[187] & 1^{13}4^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 15 & 0 & 40 & 12 \\ 3 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0} \\
[188] & 1^{13}4^{11}131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 17 & 0 & 49 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0} \\
[189] & 1^{12}231^{12}31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2
\end{aligned}$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 1 & 25 & 3 \\ 5 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 69 & 0 & 340 & 34 \\ 3 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[190] \ 1^{12} 4 \ 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 16 & 0 & 45 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[191] \ 1^{12} 23^{11} 1^{12} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 16 & 0 & 84 & 18 \\ 2 & 4 & 3 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[192] \ 1^{11} 23^{11} 1^{12} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.47 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 12 & 0 & 56 & 14 \\ 2 & 4 & 3 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[193] \ 1^{14} 4 \ 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -323, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 19 & 0 & 65 & 15 \\ 3 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[194] \ 1^{14} 4 \ 13^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -323, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 21 & 0 & 76 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[195] \ 1^{11} 23^{11} 1^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -328, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 1 & 19 & 3 \\ 5 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 54 & 0 & 200 & 26 \\ 3 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[196] \ 1^{12} 23^{11} 1^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -348, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 1 & 14 & 3 \\ 5 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 39 & 0 & 100 & 18 \\ 3 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[197] \ 1^{12} 23^{11} 1^{12} 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 1 & 17 & 3 \\ 5 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 39 & 0 & 100 & 18 \\ 3 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[198] \ 1^{13} 4 \ 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 20 & 0 & 70 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[199] \ 1^{12} 23^{11} 1^{12} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 1 & 15 & 3 \\ 5 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 39 & 0 & 100 & 18 \\ 3 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[201] \ 1^{12} 23^{11} 1^{12} 31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 1 & 16 & 3 \\ 5 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 39 & 0 & 100 & 18 \\ 3 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

- [200] $1^{12}23^11^{12}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -444$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.02 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 26 & 3 \\ 5 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 69 & 0 & 340 & 34 \\ 3 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$
- [202] $1^{11}23^11^{12}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -496$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.06 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 20 & 3 \\ 5 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 54 & 0 & 200 & 26 \\ 3 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$
- [203] $1^{12}23^11^{12}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -532$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.02 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 27 & 3 \\ 5 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 69 & 0 & 340 & 34 \\ 3 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$
- [204] $1^{14}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -109$, $\det = -544$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 24 & 0 & 101 & 19 \\ 3 & 4 & -2 & 19 & 2 \end{pmatrix}_{b_0}$
- [205] $1^{11}23^11^{12}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -584$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.05 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 21 & 3 \\ 5 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 54 & 0 & 200 & 26 \\ 3 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$
- [206] $1^{11}23^11^{12}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -592$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.05 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 1 & 22 & 3 \\ 5 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 54 & 0 & 200 & 26 \\ 3 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$
- $T = 10$ [207] $1^{15}513^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -104$, $\det = 11$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$
 $\begin{pmatrix} 5 & 9 & 9 & 5 \\ 9 & 13 & 12 & 7 \\ 9 & 12 & 10 & 6 \\ 5 & 7 & 6 & 3 \end{pmatrix}_{b_0}$
- [208] $1^{13}1^14131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -88$, $\det = 20$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 3 & 0 & -2 & 2 & 0 \\ 2 & 3 & 1 & 2 & 1 & 3 \\ 2 & 4 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0}$
- [209] $1^{11}3^11^14131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -88$, $\det = 20$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 6 & 5 & 0 & 7 & 5 \\ 2 & 4 & 2 & -2 & 5 & 1 \end{pmatrix}_{b_0}$
- [210] $1^{11}31^{11}41^13^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -88$, $\det = 24$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13$, $T_{\max} = 14$, $T \in [13, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 11 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 7 & 0 & 21 & 9 \\ 2 & 4 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$
- [211] $1^{11}31^1413^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -88$, $\det = 24$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 13$, $T_{\max} = 14$, $T \in [13, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$
- [212] $1^{11}513^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -104$, $\det = 27$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.2$

$$\begin{aligned}
& T_{\min} = 13.48 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 5 & 9 & 17 & 5 \\ 9 & 13 & 22 & 7 \\ 17 & 22 & 33 & 11 \\ 5 & 7 & 11 & 3 \end{pmatrix}_{b_0} \\
[213] & 1^{12}3^{11}41^{13}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13, T_{\max} = 14, T \in [13, 14] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 11 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 11 & 9 & 0 & 28 & 10 \\ 2 & 4 & 2 & -2 & 10 & 2 \end{pmatrix}_{b_0} \\
[214] & 1^3 3^{11} 41^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13.30 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 5 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & -2 & 3 & 0 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 2 & 4 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0} \\
[216] & 1^{11}3^{11}41^{13}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13, T_{\max} = 14, T \in [13, 14] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 8 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & -2 & 7 & 2 \end{pmatrix}_{b_0} \\
[215] & 1^{11}5^{11}13^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13.58 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 5 & 9 & 14 & 5 \\ 9 & 13 & 22 & 7 \\ 14 & 22 & 33 & 11 \\ 5 & 7 & 11 & 3 \end{pmatrix}_{b_0} \\
[217] & 1^3 3^{11} 41^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13.38 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 3 & 0 & -2 & 3 & 0 \\ 4 & 6 & 1 & 3 & 5 & 5 \\ 2 & 4 & 0 & 0 & 5 & 1 \end{pmatrix}_{b_0} \\
[218] & 1^3 3^{11} 41^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13.15 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 3 & 0 & 4 & 4 \\ 2 & 4 & 2 & -2 & 4 & 1 \end{pmatrix}_{b_0} \\
[219] & 1^{11}5^{11}13^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 5 & 9 & 16 & 5 \\ 9 & 13 & 22 & 7 \\ 16 & 22 & 33 & 11 \\ 5 & 7 & 11 & 3 \end{pmatrix}_{b_0} \\
[220] & 1^{11}5^{11}13^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 67, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13.43 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 5 & 9 & 15 & 5 \\ 9 & 13 & 22 & 7 \\ 15 & 22 & 33 & 11 \\ 5 & 7 & 11 & 3 \end{pmatrix}_{b_0} \\
[221] & 1^{11}5^{11}13^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13.70 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 5 & 9 & 20 & 5 \\ 9 & 13 & 32 & 7 \\ 20 & 32 & 70 & 16 \\ 5 & 7 & 16 & 3 \end{pmatrix}_{b_0} \\
[222] & 1^3 3^{11} 41^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13.29 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 1 & 3 & 8 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & -2 & 4 & 0 \\ 5 & 8 & 0 & 4 & 10 & 6 \\ 2 & 4 & 0 & 0 & 6 & 1 \end{pmatrix}_{b_0} \\
[223] & 1^{12}5^{11}13^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]
\end{aligned}$$

$$\begin{pmatrix} 5 & 9 & 24 & 5 \\ 9 & 13 & 32 & 7 \\ 24 & 32 & 70 & 16 \\ 5 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[224] \ 21^{11} 5^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -104, \ \det = 100, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.40 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 15 & 5 \\ 10 & 13 & 22 & 7 \\ 15 & 22 & 33 & 11 \\ 5 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[225] \ 1^{13} 31^{11} 413^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -88, \ \det = 104, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.42 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[226] \ 1^{12} 5^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -104, \ \det = 112, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.82 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 26 & 5 \\ 9 & 13 & 42 & 7 \\ 26 & 42 & 121 & 21 \\ 5 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[227] \ 1^{12} 5^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -104, \ \det = 123, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.50 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 21 & 5 \\ 9 & 13 & 32 & 7 \\ 21 & 32 & 70 & 16 \\ 5 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[228] \ 1^{13} 31^{11} 413^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -88, \ \det = 124, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.06 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 15 & 12 & 0 & 49 & 13 \\ 2 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[229] \ 1^{12} 5^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -104, \ \det = 131, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.40 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 23 & 5 \\ 9 & 13 & 32 & 7 \\ 23 & 32 & 70 & 16 \\ 5 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[230] \ 1^{12} 5^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -104, \ \det = 140, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.43 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 22 & 5 \\ 9 & 13 & 32 & 7 \\ 22 & 32 & 70 & 16 \\ 5 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[231] \ 1^{11} 31^{11} 413^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -88, \ \det = 156, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.05 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 9 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[232] \ 1^{12} 31^{11} 413^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -88, \ \det = 160, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 14 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 14 & 10 & 0 & 38 & 12 \\ 2 & 4 & 2 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[233] \ 1^{13} 5^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -104, \ \det = 179, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.56 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 27 & 5 \\ 9 & 13 & 42 & 7 \\ 27 & 42 & 121 & 21 \\ 5 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[234] \ 1^{12} 31^{11} 413^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -88, \ \det = 180, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.04 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 9 & 0 & 28 & 10 \\ 2 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[235] $1^{14}3^{11}4^{11}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 20 & 4 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 20 & 16 & 0 & 90 & 18 \\ 2 & 4 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[236] $1^{13}3^{11}4^{11}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 216, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 17 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 17 & 13 & 0 & 61 & 15 \\ 2 & 4 & 2 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[237] $21^{12}5^{13}1^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -104, \det = 218, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 10 & 22 & 5 \\ 10 & 13 & 32 & 7 \\ 22 & 32 & 70 & 16 \\ 5 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[238] $1^{13}5^{12}1^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -104, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 9 & 28 & 5 \\ 9 & 13 & 42 & 7 \\ 28 & 42 & 121 & 21 \\ 5 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[239] $1^{13}5^{14}1^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -104, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 9 & 30 & 5 \\ 9 & 13 & 42 & 7 \\ 30 & 42 & 121 & 21 \\ 5 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[240] $1^{13}5^{13}1^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -104, \det = 235, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 9 & 29 & 5 \\ 9 & 13 & 42 & 7 \\ 29 & 42 & 121 & 21 \\ 5 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[241] $21^{13}5^{13}1^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -104, \det = 372, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 10 & 29 & 5 \\ 10 & 13 & 42 & 7 \\ 29 & 42 & 121 & 21 \\ 5 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

$T = 11$ [242] $1^{11}3^{11}1^15^{13}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 11, T_{\max} = 14, \Delta = -83, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.78 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 2 & 5 & 0 & 2 & 7 & 5 \\ 2 & 3 & -1 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[243] $1^{11}3^{11}5^{13}1^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 11, T_{\max} = 14, \Delta = -83, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 3 & 0 \\ 2 & 0 & 4 & 3 & 5 & 5 \\ 2 & 0 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[244] $1^{11}3^{11}5^{13}1^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 11, T_{\max} = 14, \Delta = -83, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.79 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 2 & 4 & 0 & 3 & 5 & 5 \\ 2 & 3 & -1 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[245] $1^{13}3^{11}5^{13}1^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2$ $T = 11, T_{\max} = 14, \Delta = -83, \det = -41, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 5 & 0 \\ 7 & 0 & 10 & 5 & 33 & 11 \\ 2 & 0 & 2 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[266] $1^{14}31^1513^{11}1^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -420, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.79 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 6 & 0 \\ 8 & 0 & 13 & 6 & 56 & 14 \\ 2 & 0 & 2 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

$T = 12$ [267] $1^{11}3^{121}6^{131}1^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$

$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 4 & 2 & 0 & 2 & 4 & 4 \\ 2 & 2 & 0 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

[268] $1^{11}3^{1131}6^{131}1^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 40, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 15, T_{\max} = 16, T \in [15, 16]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 3 & 2 & 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 2 & 2 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[269] $1^{11}3^{121}6^{131}1^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 44, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.82 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 2 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[270] $1^{11}3^{131}6^{131}1^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 56, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 5 & 2 & 1 & 2 & 5 & 5 \\ 2 & 2 & 0 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[271] $1^{11}3^{131}6^{131}1^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 68, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.82 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 3 & 2 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[272] $1^{11}3^{131}6^{131}1^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.78 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 4 & 0 & 3 & 10 & 6 \\ 2 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[273] $1^{11}3^{131}6^{131}1^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 176, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.70 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 6 & 4 & 1 & 3 & 11 & 7 \\ 2 & 2 & 0 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[274] $1^{11}3^{121}6^{131}1^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 212, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.74 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 7 & 4 & 2 & 3 & 14 & 8 \\ 2 & 2 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [275] \quad 1^{\overline{1}} \quad 8 \quad 123^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 14, \quad T_{\max} = 17, \quad \Delta = -80, \quad \det = 20, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 16.60 \rightarrow 17, \quad T_{\max} = 17, \quad T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 2 & 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[276] \quad 1^{\overline{1}} \quad 8 \quad 123^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 14, \quad T_{\max} = 17, \quad \Delta = -80, \quad \det = 24, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 16.83 \rightarrow 17, \quad T_{\max} = 17, \quad T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 1 & 1 & -2 & 2 & 0 \\ 3 & 2 & 5 & 2 & 7 & 5 \\ 2 & 2 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[277] \quad 1^{\overline{1}} \quad 8 \quad 123^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 14, \quad T_{\max} = 17, \quad \Delta = -80, \quad \det = 104, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 16.65 \rightarrow 17, \quad T_{\max} = 17, \quad T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 5 & 8 & 0 & 31 & 11 \\ 2 & 2 & 2 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[278] \quad 1^{\overline{1}} \quad 8 \quad 123^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 14, \quad T_{\max} = 17, \quad \Delta = -80, \quad \det = 156, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 16.64 \rightarrow 17, \quad T_{\max} = 17, \quad T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 6 & 11 & 0 & 50 & 14 \\ 2 & 2 & 2 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [279] \quad 1^{\overline{1}} \quad 8 \quad 123^{\overline{1}} 141^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 15, \quad T_{\max} = 18, \quad \Delta = -79, \quad \det = -24, \quad (n_+, n_-, n_0) = (1, 15, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 17.83 \rightarrow 18, \quad T_{\max} = 18, \quad T \in [18, 18]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 5 & 1 & 9 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 13 & 7 \\ 3 & 5 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

4.520 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{f}_4$

$$T = 8 \quad [0] \quad 1^{\overline{1}} \quad 1^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{green}{\mathfrak{f}}_4 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -162, \quad \det = 360, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.40 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & -2 & 5 & 0 & 0 \\ 5 & 12 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 6 & -3 & -1 \end{pmatrix}_{b_0}$$

4.521 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \text{su2n3mix} + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \quad 1^{\overline{1}} \quad 1^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -89, \quad \det = -180, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.04 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 8 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 8 & 0 & 12 & 9 & 5 & 6 \\ 3 & 5 & 0 & 9 & 4 & 4 & 4 \\ 2 & 4 & 0 & 5 & 4 & 0 & 2 \\ 3 & 4 & -3 & 6 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1}} \quad 1^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -89, \quad \det = -500, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 2 & 3 \\ 3 & 4 & 0 & 12 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 12 & 8 & 9 \\ 3 & 5 & 0 & 12 & 4 & 4 & 4 \\ 2 & 4 & 0 & 8 & 4 & 0 & 2 \\ 3 & 4 & -3 & 9 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \quad 1^{11} 3^{11} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + [\text{su}_2 \cdot \text{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 3 & 3 & 4 \\ 3 & 3 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

4.522 NHC(-2)(-3) + f₄ + su2n3mix + so₈

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 24 & 5 \\ 11 & 20 & 0 & 45 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 24 & 45 & 0 & 100 & 18 \\ 5 & 8 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 5 \\ 11 & 20 & 0 & 28 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 15 & 28 & 0 & 37 & 11 \\ 5 & 8 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & -2 & 5 & 0 & 0 \\ 6 & 12 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 6 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & -2 & 5 & 0 & 0 \\ 5 & 12 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 6 & -2 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 6 & 3 \\ 4 & 10 & 13 & 0 & 16 & 6 \\ 5 & 13 & 12 & 0 & 17 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 16 & 17 & 0 & 20 & 8 \\ 3 & 6 & 6 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [5] \quad 1^{11} 3^{11} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 233, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.59 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 1 & 10 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 14 & 0 & 13 & 7 \\ 5 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [6] \quad 1^{11} 3^{11} 5^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 94, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 5 & 12 & 1 & 20 & 5 \\ 12 & 26 & 1 & 44 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 44 & 0 & 70 & 16 \\ 5 & 10 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

4.523 NHC(-2)(-3) + f₄ + su2n3mix + s₀₈ + su₂

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{s}_{08} + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 9 & 5 \\ 11 & 20 & 0 & 28 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 28 & 0 & 37 & 24 & 11 \\ 9 & 17 & 0 & 24 & 13 & 7 \\ 5 & 8 & -3 & 11 & 7 & 1 \end{pmatrix}_{b_0}$$

4.524 NHC(-2)(-3) + f₄ + su2n3mix + su₂

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.63 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & -2 & 5 & 0 & 0 \\ 6 & 12 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & 6 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 0 & 12 & 5 & 6 \\ 2 & 4 & 0 & 5 & 1 & 3 \\ 3 & 4 & -3 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 7 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 17 & 5 \\ 8 & 14 & 0 & 17 & 37 & 11 \\ 3 & 4 & -3 & 5 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 0 & 12 & 19 & 6 \\ 7 & 13 & 0 & 19 & 28 & 10 \\ 3 & 4 & -3 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.3$$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 10 & 3 \\ 3 & 4 & 0 & 15 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 55 & 11 \\ 10 & 20 & 0 & 55 & 73 & 15 \\ 3 & 4 & -3 & 11 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 5 & 3 \\ 3 & 4 & 0 & 15 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 23 & 11 \\ 5 & 9 & 0 & 23 & 13 & 7 \\ 3 & 4 & -3 & 11 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 7 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 6 & 5 \\ 3 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -3 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & -2 & 5 & 0 & 0 \\ 5 & 12 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & 6 & 0 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{124}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 11 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 32 & 8 \\ 8 & 16 & 0 & 32 & 46 & 12 \\ 3 & 4 & -3 & 8 & 12 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{113}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 15 & 5 \\ 6 & 12 & 0 & 15 & 25 & 9 \\ 3 & 4 & -3 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{124}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 17 & 3 \\ 3 & 4 & 0 & 11 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 59 & 8 \\ 17 & 31 & 0 & 59 & 172 & 24 \\ 3 & 4 & -3 & 8 & 24 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.64 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 22 & 0 & 97 & 14 & 17 \\ 2 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -3 & 17 & 3 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{132}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 4 & 3 \\ 3 & 4 & 0 & 25 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 25 & 0 & 111 & 34 & 19 \\ 4 & 8 & 0 & 34 & 10 & 6 \\ 3 & 4 & -3 & 19 & 6 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{134}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 13.15 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 14 & 3 \\ 3 & 4 & 0 & 15 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 68 & 11 \\ 14 & 26 & 0 & 68 & 118 & 20 \\ 3 & 4 & -3 & 11 & 20 & -1 \end{pmatrix}_{b_0}$$

[14] $1^{143}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 6 & 3 \\ 3 & 4 & 0 & 24 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 0 & 100 & 51 & 18 \\ 6 & 12 & 0 & 51 & 25 & 9 \\ 3 & 4 & -3 & 18 & 9 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{112}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 13 & 3 \\ 3 & 4 & 0 & 7 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 28 & 5 \\ 13 & 23 & 0 & 28 & 100 & 18 \\ 3 & 4 & -3 & 5 & 18 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{14}3^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 & 3 \\ 3 & 4 & 0 & 24 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 0 & 100 & 21 & 18 \\ 3 & 5 & 0 & 21 & 4 & 4 \\ 3 & 4 & -3 & 18 & 4 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{13}3^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 8 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 0 & 12 & 9 & 6 \\ 3 & 5 & 0 & 9 & 4 & 4 \\ 3 & 4 & -3 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{112}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 7 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 10 & 5 \\ 4 & 8 & 0 & 10 & 10 & 6 \\ 3 & 4 & -3 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{142}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 7 & 3 \\ 3 & 4 & 0 & 19 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 0 & 64 & 43 & 14 \\ 7 & 13 & 0 & 43 & 28 & 10 \\ 3 & 4 & -3 & 14 & 10 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{145}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 10 & 3 \\ 3 & 4 & 0 & 19 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 0 & 64 & 70 & 14 \\ 10 & 20 & 0 & 70 & 73 & 15 \\ 3 & 4 & -3 & 14 & 15 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{123}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 138, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 0 & 11 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 24 & 8 \\ 6 & 12 & 0 & 24 & 25 & 9 \\ 3 & 4 & -3 & 8 & 9 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{122}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 0 & 16 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 22 & 12 \\ 4 & 8 & 0 & 22 & 10 & 6 \\ 3 & 4 & -3 & 12 & 6 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{134}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 3 \\ 3 & 4 & 0 & 15 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 44 & 11 \\ 8 & 16 & 0 & 44 & 46 & 12 \\ 3 & 4 & -3 & 11 & 12 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 5 \\ 2 & 4 & 0 & 5 & 1 & 3 \\ 3 & 4 & -3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 158, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 11 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 17 & 8 \\ 5 & 9 & 0 & 17 & 13 & 7 \\ 3 & 4 & -3 & 8 & 7 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 0 & 12 & 14 & 6 \\ 5 & 9 & 0 & 14 & 13 & 7 \\ 3 & 4 & -3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{122}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 4 & 3 \\ 3 & 4 & 0 & 11 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 16 & 8 \\ 4 & 8 & 0 & 16 & 10 & 6 \\ 3 & 4 & -3 & 8 & 6 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{111}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 17 & 0 & 55 & 11 & 13 \\ 2 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & -3 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{113}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 12 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 36 & 9 \\ 9 & 17 & 0 & 36 & 49 & 13 \\ 3 & 4 & -3 & 9 & 13 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 11 & 5 \\ 5 & 9 & 0 & 11 & 13 & 7 \\ 3 & 4 & -3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{13}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 20 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 18 & 15 \\ 3 & 5 & 0 & 18 & 4 & 4 \\ 3 & 4 & -3 & 15 & 4 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{113}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 12 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 12 & 9 \\ 3 & 5 & 0 & 12 & 4 & 4 \\ 3 & 4 & -3 & 9 & 4 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{133}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 6 & 3 \\ 3 & 4 & 0 & 15 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 33 & 11 \\ 6 & 12 & 0 & 33 & 25 & 9 \\ 3 & 4 & -3 & 11 & 9 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 8 & 9 \\ 2 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{121}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 218, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 11 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 8 & 8 \\ 2 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -3 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{144}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 8 & 3 \\ 3 & 4 & 0 & 19 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 0 & 64 & 56 & 14 \\ 8 & 16 & 0 & 56 & 46 & 12 \\ 3 & 4 & -3 & 14 & 12 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{113}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 18 & 3 \\ 3 & 4 & 0 & 7 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 39 & 5 \\ 18 & 32 & 0 & 39 & 193 & 25 \\ 3 & 4 & -3 & 5 & 25 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{12}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 16 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 15 & 12 \\ 3 & 5 & 0 & 15 & 4 & 4 \\ 3 & 4 & -3 & 12 & 4 & 1 \end{pmatrix}_{b_0}$$

[39] $1^{12}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 7 & 3 \\ 3 & 4 & 0 & 15 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 34 & 11 \\ 7 & 13 & 0 & 34 & 28 & 10 \\ 3 & 4 & -3 & 11 & 10 & 1 \end{pmatrix}_{b_0}$$

[40] $1^{12}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 258, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 12 & 3 \\ 3 & 4 & 0 & 11 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 42 & 8 \\ 12 & 22 & 0 & 42 & 85 & 17 \\ 3 & 4 & -3 & 8 & 17 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{12}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 262, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 4 & 3 \\ 3 & 4 & 0 & 15 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 22 & 11 \\ 4 & 8 & 0 & 22 & 10 & 6 \\ 3 & 4 & -3 & 11 & 6 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{14}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 6 & 3 \\ 3 & 4 & 0 & 19 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 0 & 64 & 42 & 14 \\ 6 & 12 & 0 & 42 & 25 & 9 \\ 3 & 4 & -3 & 14 & 9 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{13}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 15 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 11 & 11 \\ 2 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & -3 & 11 & 3 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{11}2^31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 298, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 7 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 22 & 5 \\ 10 & 18 & 0 & 22 & 58 & 14 \\ 3 & 4 & -3 & 5 & 14 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{11}2^31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 303, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 7 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 16 & 5 \\ 7 & 13 & 0 & 16 & 28 & 10 \\ 3 & 4 & -3 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{12}2^31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 308, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 11 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 25 & 8 \\ 7 & 13 & 0 & 25 & 28 & 10 \\ 3 & 4 & -3 & 8 & 10 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{12}4^31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 11 & 3 \\ 3 & 4 & 0 & 16 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 59 & 12 \\ 11 & 21 & 0 & 59 & 76 & 16 \\ 3 & 4 & -3 & 12 & 16 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{132}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 322, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 4 & 3 \\ 3 & 4 & 0 & 20 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 28 & 15 \\ 4 & 8 & 0 & 28 & 10 & 6 \\ 3 & 4 & -3 & 15 & 6 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{143}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 330, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.07 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 9 & 3 \\ 3 & 4 & 0 & 19 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 0 & 64 & 57 & 14 \\ 9 & 17 & 0 & 57 & 49 & 13 \\ 3 & 4 & -3 & 14 & 13 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{142}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 4 & 3 \\ 3 & 4 & 0 & 19 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 0 & 64 & 28 & 14 \\ 4 & 8 & 0 & 28 & 10 & 6 \\ 3 & 4 & -3 & 14 & 6 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{121}3^11^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 362, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 0 & 31 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 18 & 31 & 0 & 188 & 20 & 24 \\ 2 & 4 & 0 & 20 & 1 & 3 \\ 3 & 4 & -3 & 24 & 3 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{112}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 367, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 12 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 28 & 9 \\ 7 & 13 & 0 & 28 & 28 & 10 \\ 3 & 4 & -3 & 9 & 10 & 1 \end{pmatrix}_{b_0}$$

[53] $1^{141}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 0 & 64 & 14 & 14 \\ 2 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -3 & 14 & 3 & 1 \end{pmatrix}_{b_0}$$

[55] $1^{142}3^11^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 4 & 3 \\ 3 & 4 & 0 & 29 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 16 & 29 & 0 & 148 & 40 & 22 \\ 4 & 8 & 0 & 40 & 10 & 6 \\ 3 & 4 & -3 & 22 & 6 & 1 \end{pmatrix}_{b_0}$$

[54] $1^{111}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 12 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 20 & 9 \\ 5 & 9 & 0 & 20 & 13 & 7 \\ 3 & 4 & -3 & 9 & 7 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{113}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 393, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 7 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 21 & 5 \\ 9 & 17 & 0 & 21 & 49 & 13 \\ 3 & 4 & -3 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

[57] $1^{121}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 395, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 11 & 12 \\ 2 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & -3 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{133}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 417, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 9 & 3 \\ 3 & 4 & 0 & 15 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 45 & 11 \\ 9 & 17 & 0 & 45 & 49 & 13 \\ 3 & 4 & -3 & 11 & 13 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{123}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 438, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 9 & 3 \\ 3 & 4 & 0 & 11 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 33 & 8 \\ 9 & 17 & 0 & 33 & 49 & 13 \\ 3 & 4 & -3 & 8 & 13 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{114}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 463, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 0 & 7 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 26 & 5 \\ 11 & 21 & 0 & 26 & 76 & 16 \\ 3 & 4 & -3 & 5 & 16 & 1 \end{pmatrix}_{b_0}$$

[61] $1^{121}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 467, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 26 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 26 & 0 & 128 & 17 & 20 \\ 2 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & -3 & 20 & 3 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{113}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 473, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 15 & 3 \\ 3 & 4 & 0 & 7 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 33 & 5 \\ 15 & 27 & 0 & 33 & 133 & 21 \\ 3 & 4 & -3 & 5 & 21 & 1 \end{pmatrix}_{b_0}$$

[63] $1^{121}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 21 & 0 & 80 & 14 & 16 \\ 2 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -3 & 16 & 3 & 1 \end{pmatrix}_{b_0}$$

[64] $1^{142}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 508, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 4 & 3 \\ 3 & 4 & 0 & 24 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 0 & 100 & 34 & 18 \\ 4 & 8 & 0 & 34 & 10 & 6 \\ 3 & 4 & -3 & 18 & 6 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{121}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 515, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 16 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 26 & 12 \\ 5 & 9 & 0 & 26 & 13 & 7 \\ 3 & 4 & -3 & 12 & 7 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{144}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 520, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 11 & 3 \\ 3 & 4 & 0 & 19 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 0 & 64 & 71 & 14 \\ 11 & 21 & 0 & 71 & 76 & 16 \\ 3 & 4 & -3 & 14 & 16 & 1 \end{pmatrix}_{b_0}$$

[67] $1^{113}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 7 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 27 & 5 \\ 12 & 22 & 0 & 27 & 85 & 17 \\ 3 & 4 & -3 & 5 & 17 & 1 \end{pmatrix}_{b_0}$$

[68] $1^{124}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 11 & 3 \\ 3 & 4 & 0 & 11 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 41 & 8 \\ 11 & 21 & 0 & 41 & 76 & 16 \\ 3 & 4 & -3 & 8 & 16 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{124}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 14 & 3 \\ 3 & 4 & 0 & 11 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 50 & 8 \\ 14 & 26 & 0 & 50 & 118 & 20 \\ 3 & 4 & -3 & 8 & 20 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{123}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 555, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 0 & 16 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 48 & 12 \\ 9 & 17 & 0 & 48 & 49 & 13 \\ 3 & 4 & -3 & 12 & 13 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{134}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 567, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 11 & 3 \\ 3 & 4 & 0 & 15 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 56 & 11 \\ 11 & 21 & 0 & 56 & 76 & 16 \\ 3 & 4 & -3 & 11 & 16 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{131}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 577, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 14 & 15 \\ 2 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -3 & 15 & 3 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{131}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 592, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 5 & 3 \\ 3 & 4 & 0 & 20 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 32 & 15 \\ 5 & 9 & 0 & 32 & 13 & 7 \\ 3 & 4 & -3 & 15 & 7 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{141}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 5 & 3 \\ 3 & 4 & 0 & 24 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 0 & 100 & 38 & 18 \\ 5 & 9 & 0 & 38 & 13 & 7 \\ 3 & 4 & -3 & 18 & 7 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{122}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 620, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 7 & 3 \\ 3 & 4 & 0 & 16 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 37 & 12 \\ 7 & 13 & 0 & 37 & 28 & 10 \\ 3 & 4 & -3 & 12 & 10 & 1 \end{pmatrix}_{b_0}$$

[76] $1^{114}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 718, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 20 & 3 \\ 3 & 4 & 0 & 7 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 44 & 5 \\ 20 & 36 & 0 & 44 & 238 & 28 \\ 3 & 4 & -3 & 5 & 28 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{134}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 727, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 11 & 3 \\ 3 & 4 & 0 & 20 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 74 & 15 \\ 11 & 21 & 0 & 74 & 76 & 16 \\ 3 & 4 & -3 & 15 & 16 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{114}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 738, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 7 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 32 & 5 \\ 14 & 26 & 0 & 32 & 118 & 20 \\ 3 & 4 & -3 & 5 & 20 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{141}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 763, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 0 & 100 & 17 & 18 \\ 2 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & -3 & 18 & 3 & 1 \end{pmatrix}_{b_0}$$

[80] $1^{131}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 25 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 25 & 0 & 111 & 17 & 19 \\ 2 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & -3 & 19 & 3 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{132}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 807, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 7 & 3 \\ 3 & 4 & 0 & 20 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 46 & 15 \\ 7 & 13 & 0 & 46 & 28 & 10 \\ 3 & 4 & -3 & 15 & 10 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{114}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 823, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 17 & 3 \\ 3 & 4 & 0 & 7 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 38 & 5 \\ 17 & 31 & 0 & 38 & 172 & 24 \\ 3 & 4 & -3 & 5 & 24 & 1 \end{pmatrix}_{b_0}$$

[83] $1^{133}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 852, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 9 & 3 \\ 3 & 4 & 0 & 20 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 60 & 15 \\ 9 & 17 & 0 & 60 & 49 & 13 \\ 3 & 4 & -3 & 15 & 13 & 1 \end{pmatrix}_{b_0}$$

[84] $1^{131}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 865, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 30 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 30 & 0 & 165 & 20 & 23 \\ 2 & 4 & 0 & 20 & 1 & 3 \\ 3 & 4 & -3 & 23 & 3 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{131}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 868, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 0 & 35 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 20 & 35 & 0 & 231 & 23 & 27 \\ 2 & 4 & 0 & 23 & 1 & 3 \\ 3 & 4 & -3 & 27 & 3 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{142}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 928, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 7 & 3 \\ 3 & 4 & 0 & 24 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 0 & 100 & 55 & 18 \\ 7 & 13 & 0 & 55 & 28 & 10 \\ 3 & 4 & -3 & 18 & 10 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{141}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1062, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 29 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 16 & 29 & 0 & 148 & 20 & 22 \\ 2 & 4 & 0 & 20 & 1 & 3 \\ 3 & 4 & -3 & 22 & 3 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{144}3^11^{111}+[\textcolor{blue}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{f}}_4+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 1068$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 11 & 3 \\ 3 & 4 & 0 & 24 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 0 & 100 & 89 & 18 \\ 11 & 21 & 0 & 89 & 76 & 16 \\ 3 & 4 & -3 & 18 & 16 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{143}3^11^{111}+[\textcolor{blue}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{f}}_4+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 1083$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 9 & 3 \\ 3 & 4 & 0 & 24 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 0 & 100 & 72 & 18 \\ 9 & 17 & 0 & 72 & 49 & 13 \\ 3 & 4 & -3 & 18 & 13 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{141}3^11^{13}+[\textcolor{blue}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{f}}_4+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 1267$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 2 & 3 \\ 3 & 4 & 0 & 34 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 19 & 34 & 0 & 208 & 23 & 26 \\ 2 & 4 & 0 & 23 & 1 & 3 \\ 3 & 4 & -3 & 26 & 3 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{141}3^11^{14}+[\textcolor{blue}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{f}}_4+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 1378$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 2 & 3 \\ 3 & 4 & 0 & 39 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 22 & 39 & 0 & 280 & 26 & 30 \\ 2 & 4 & 0 & 26 & 1 & 3 \\ 3 & 4 & -3 & 30 & 3 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [92] $1^{142}3^121^{11}+[\textcolor{blue}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{f}}_4+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 2 & 0 \\ 8 & 10 & 0 & 8 & 52 & 20 & 12 \\ 2 & 4 & 0 & 2 & 20 & 6 & 4 \\ 2 & 2 & -1 & 0 & 12 & 4 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{111}3^121^{11}+[\textcolor{blue}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{f}}_4+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -14$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 4 & 0 \\ 2 & 3 & 0 & 2 & 3 & 5 & 3 \\ 2 & 3 & 0 & 4 & 5 & 5 & 5 \\ 2 & 2 & -1 & 0 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{111}3^11^{13}1^1+[\textcolor{blue}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{f}}_4+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 5 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 6 & 0 & 7 & 5 & 5 \\ 1 & 3 & 3 & 0 & 5 & 1 & 3 \\ 2 & 3 & 3 & -3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

[95] $1^{131}3^11^{13}1^1+[\textcolor{blue}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{f}}_4+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 17 & 3 & 4 \\ 1 & 4 & 0 & 0 & 13 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 17 & 13 & 0 & 60 & 13 & 14 \\ 1 & 3 & 3 & 0 & 13 & 1 & 3 \\ 2 & 4 & 2 & -3 & 14 & 3 & 0 \end{pmatrix}_{b_0}$$

[96] $1^{111}3^121^1+[\textcolor{blue}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{f}}_4+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -33$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 1 & 0 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 3 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & 0 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[98] $1^{132}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 2 & 0 \\ 6 & 8 & 0 & 7 & 34 & 16 & 10 \\ 2 & 4 & 0 & 2 & 16 & 6 & 4 \\ 2 & 2 & -1 & 0 & 10 & 4 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{114}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 10 & 0 \\ 2 & 3 & 0 & 2 & 3 & 16 & 3 \\ 6 & 10 & 0 & 10 & 16 & 58 & 14 \\ 2 & 2 & -1 & 0 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

[99] $1^{113}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

[100] $1^{132}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 2 & 0 \\ 6 & 7 & 0 & 4 & 23 & 14 & 7 \\ 2 & 4 & 0 & 2 & 14 & 6 & 4 \\ 2 & 2 & -1 & 0 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$

[101] $1^{111}3^11^31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 4 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 4 & 1 & 3 \\ 2 & 4 & 2 & -3 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

[102] $1^{113}3^11^131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 5 & 0 & 12 & 5 & 6 \\ 1 & 3 & 3 & 0 & 5 & 1 & 3 \\ 2 & 4 & 2 & -3 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

[103] $1^{111}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 1 & 0 \\ 3 & 4 & 0 & 5 & 10 & 4 & 6 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & 0 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[104] $1^{111}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 4 & 0 \\ 2 & 3 & 0 & 2 & 3 & 5 & 3 \\ 3 & 3 & 0 & 4 & 5 & 5 & 5 \\ 2 & 2 & -1 & 0 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[105] $1^{111}3^11^31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 4 & 3 & 3 \\ 2 & 4 & 3 & 0 & 5 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 5 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 4 & 1 & 3 \\ 2 & 3 & 3 & -3 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

[106] $1^{112}3^121^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 \\ 2 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 2 & -1 & 0 & 3 & 4 & 1 \end{pmatrix}_{b_0}$$

[107] $1^{1111}313^11^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 4 & 3 & 3 \\ 1 & 4 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 4 & 1 & 3 \\ 2 & 3 & 2 & -3 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

[108] $1^{112}31^{111}3^11^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 4 & 2 \\ 2 & 4 & 4 & 0 & 11 & 12 & 4 \\ 1 & 4 & 0 & 0 & 7 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 11 & 7 & 0 & 20 & 25 & 8 \\ 4 & 12 & 9 & 0 & 25 & 28 & 10 \\ 2 & 4 & 2 & -3 & 8 & 10 & 0 \end{pmatrix}_{b_0}$$

[109] $1^{1111}31^13^11^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -89, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 4 & 0 & 4 & 6 & 3 \\ 1 & 4 & 0 & 0 & 4 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 4 & 9 & 4 \\ 3 & 6 & 6 & 0 & 9 & 13 & 7 \\ 2 & 3 & 2 & -3 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

[110] $1^{122}3^121^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 2 & 0 \\ 4 & 5 & 0 & 3 & 11 & 10 & 5 \\ 2 & 4 & 0 & 2 & 10 & 6 & 4 \\ 2 & 2 & -1 & 0 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

[111] $1^{112}3^121^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -97, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 5 & 0 \\ 2 & 3 & 0 & 2 & 3 & 8 & 3 \\ 3 & 5 & 0 & 5 & 8 & 13 & 7 \\ 2 & 2 & -1 & 0 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{11}31^{11}3^11^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 5 & 6 & 4 \\ 1 & 4 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 5 & 4 \\ 2 & 6 & 3 & 0 & 5 & 4 & 4 \\ 2 & 4 & 2 & -3 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[113] $1^{111}3^11^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 5 & 0 & 7 & 11 & 5 \\ 3 & 9 & 6 & 0 & 11 & 13 & 7 \\ 2 & 4 & 2 & -3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[114] $1^{111}3^121^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 1 & 0 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & 0 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

- [115] $1^{1^{11}} 1^{121} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 1 & 0 \\ 3 & 5 & 0 & 3 & 11 & 5 & 5 \\ 2 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & 0 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$
- [116] $1^{1^{12}} 1^{143} 3^1 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -117$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 9 & 2 \\ 1 & 0 & 1 & 1 & 3 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 13 & 0 \\ 2 & 3 & 0 & 2 & 3 & 18 & 3 \\ 9 & 11 & 0 & 13 & 18 & 81 & 17 \\ 2 & 2 & -1 & 0 & 3 & 17 & 1 \end{pmatrix}_{b_0}$$
- [117] $1^{1^{11}} 1^{111} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -125$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 1 & 0 \\ 4 & 4 & 0 & 5 & 10 & 4 & 6 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & 0 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$
- [118] $1^{1^{132}} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -128$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 2 & 0 \\ 5 & 8 & 0 & 7 & 34 & 16 & 10 \\ 2 & 4 & 0 & 2 & 16 & 6 & 4 \\ 2 & 2 & -1 & 0 & 10 & 4 & 1 \end{pmatrix}_{b_0}$$
- [119] $1^{1^{11}} 1^{11} 31^{111} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -129$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 8 & 9 & 4 \\ 1 & 4 & 0 & 0 & 4 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 8 & 4 & 0 & 9 & 12 & 5 \\ 3 & 9 & 6 & 0 & 12 & 13 & 7 \\ 2 & 4 & 2 & -3 & 5 & 7 & 0 \end{pmatrix}_{b_0}$$
- [120] $1^{1^{11}} 1^{11} 31^{143} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 9 & 2 \\ 2 & 4 & 4 & 0 & 5 & 27 & 4 \\ 1 & 4 & 0 & 0 & 4 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 24 & 4 \\ 9 & 27 & 15 & 0 & 24 & 121 & 19 \\ 2 & 4 & 2 & -3 & 4 & 19 & 1 \end{pmatrix}_{b_0}$$
- [121] $1^{1^{11}} 1^{111} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -142$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 1 & 0 \\ 2 & 3 & 0 & 2 & 3 & 3 & 3 \\ 2 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & 0 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$
- [122] $1^{1^{13}} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -145$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 6 & 0 \\ 3 & 5 & 0 & 3 & 11 & 18 & 5 \\ 4 & 7 & 0 & 6 & 18 & 25 & 9 \\ 2 & 2 & -1 & 0 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$
- [123] $1^{1^{11}} 1^{121} 3^1 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -148$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 1 & 0 \\ 6 & 7 & 0 & 9 & 35 & 7 & 11 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & 0 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

[124] $1^{1^{12}3^1}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 1 & 0 \\ 3 & 5 & 0 & 3 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & 0 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$

[125] $1^{1^{11}3^1}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -158, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 9 & 0 \\ 2 & 3 & 0 & 2 & 3 & 13 & 3 \\ 6 & 8 & 0 & 9 & 13 & 40 & 12 \\ 2 & 2 & -1 & 0 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[126] $1^{1^{11}3^1}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 1 & 0 \\ 3 & 3 & 0 & 2 & 3 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 2 & 2 & -1 & 0 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[127] $1^{1^{11}3^1}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 6 & 0 \\ 2 & 3 & 0 & 2 & 3 & 11 & 3 \\ 4 & 7 & 0 & 6 & 11 & 25 & 9 \\ 2 & 2 & -1 & 0 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[128] $1^{1^{12}3^1}1^{11}3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 & 2 \\ 2 & 4 & 4 & 0 & 14 & 6 & 4 \\ 1 & 4 & 0 & 0 & 10 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 14 & 10 & 0 & 37 & 14 & 11 \\ 2 & 6 & 3 & 0 & 14 & 4 & 4 \\ 2 & 4 & 2 & -3 & 11 & 4 & 1 \end{pmatrix}_{b_0}$$

[129] $1^{1^{11}3^1}1^{11}3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 11 & 3 & 4 \\ 1 & 4 & 0 & 0 & 7 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 11 & 7 & 0 & 20 & 7 & 8 \\ 1 & 3 & 3 & 0 & 7 & 1 & 3 \\ 2 & 4 & 2 & -3 & 8 & 3 & 0 \end{pmatrix}_{b_0}$$

[130] $1^{1^{11}3^1}1^{11}3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 8 & 0 & 25 & 8 & 9 \\ 1 & 3 & 3 & 0 & 8 & 1 & 3 \\ 2 & 4 & 2 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[131] $1^{1^{12}3^1}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -185, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 3 & 0 \\ 3 & 5 & 0 & 3 & 11 & 15 & 5 \\ 5 & 6 & 0 & 3 & 15 & 16 & 6 \\ 2 & 2 & -1 & 0 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{1^{12}3^1}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 1 & 0 \\ 4 & 6 & 0 & 6 & 20 & 6 & 8 \\ 1 & 2 & 0 & 1 & 6 & 0 & 2 \\ 2 & 2 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{111}3^1 1^{12}3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 17 & 3 & 4 \\ 1 & 4 & 0 & 0 & 10 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 17 & 10 & 0 & 48 & 10 & 12 \\ 1 & 3 & 3 & 0 & 10 & 1 & 3 \\ 2 & 4 & 2 & -3 & 12 & 3 & 0 \end{pmatrix}_{b_0}$$

[134] $1^{131}3^1 2^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 1 & 0 \\ 6 & 9 & 0 & 10 & 51 & 9 & 13 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & 0 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

[136] $1^{121}3^1 1^{11}3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -209, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 14 & 3 & 4 \\ 1 & 4 & 0 & 0 & 10 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 14 & 10 & 0 & 37 & 10 & 11 \\ 1 & 3 & 3 & 0 & 10 & 1 & 3 \\ 2 & 4 & 2 & -3 & 11 & 3 & 1 \end{pmatrix}_{b_0}$$

[135] $1^{114}3^1 2^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -209, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 7 & 0 \\ 2 & 3 & 0 & 2 & 3 & 14 & 3 \\ 5 & 9 & 0 & 7 & 14 & 41 & 11 \\ 2 & 2 & -1 & 0 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{112}3^1 2^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -214, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 \\ 4 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 2 & -1 & 0 & 3 & 4 & 1 \end{pmatrix}_{b_0}$$

[138] $1^{111}3^1 2^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 3 & 6 & 3 \\ 3 & 4 & 0 & 2 & 6 & 6 & 4 \\ 2 & 2 & -1 & 0 & 3 & 4 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{112}3^1 1^{11}3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 4 & 0 & 5 & 12 & 4 \\ 1 & 4 & 0 & 0 & 4 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 13 & 4 \\ 4 & 12 & 9 & 0 & 13 & 28 & 10 \\ 2 & 4 & 2 & -3 & 4 & 10 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{122}3^1 2^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 2 & 0 \\ 3 & 5 & 0 & 3 & 11 & 10 & 5 \\ 3 & 4 & 0 & 2 & 10 & 6 & 4 \\ 2 & 2 & -1 & 0 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

[141] $1^{124}3^1 2^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 7 & 0 \\ 3 & 5 & 0 & 3 & 11 & 23 & 5 \\ 5 & 9 & 0 & 7 & 23 & 41 & 11 \\ 2 & 2 & -1 & 0 & 5 & 11 & 1 \end{pmatrix}_{b_0}$$

[142] $1^{132}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 2 & 0 \\ 5 & 7 & 0 & 4 & 23 & 14 & 7 \\ 2 & 4 & 0 & 2 & 14 & 6 & 4 \\ 2 & 2 & -1 & 0 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$

[143] $1^{121}3^121^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 9 & 1 & 0 \\ 7 & 7 & 0 & 9 & 35 & 7 & 11 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & 0 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

[144] $1^{142}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 2 & 0 \\ 7 & 10 & 0 & 8 & 52 & 20 & 12 \\ 2 & 4 & 0 & 2 & 20 & 6 & 4 \\ 2 & 2 & -1 & 0 & 12 & 4 & 1 \end{pmatrix}_{b_0}$$

[145] $1^{111}3^11^{14}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 0 & 30 & 3 & 4 \\ 1 & 4 & 0 & 0 & 17 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 30 & 17 & 0 & 151 & 17 & 21 \\ 1 & 3 & 3 & 0 & 17 & 1 & 3 \\ 2 & 4 & 2 & -3 & 21 & 3 & 0 \end{pmatrix}_{b_0}$$

[146] $1^{112}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 5 & 0 \\ 2 & 3 & 0 & 2 & 3 & 8 & 3 \\ 4 & 5 & 0 & 5 & 8 & 13 & 7 \\ 2 & 2 & -1 & 0 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{142}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 2 & 0 \\ 6 & 10 & 0 & 8 & 52 & 20 & 12 \\ 2 & 4 & 0 & 2 & 20 & 6 & 4 \\ 2 & 2 & -1 & 0 & 12 & 4 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{142}3^121^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 2 & 0 \\ 7 & 9 & 0 & 5 & 39 & 18 & 9 \\ 2 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & 0 & 9 & 4 & 1 \end{pmatrix}_{b_0}$$

[149] $1^{112}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 5 & 0 \\ 2 & 3 & 0 & 2 & 3 & 8 & 3 \\ 5 & 5 & 0 & 5 & 8 & 13 & 7 \\ 2 & 2 & -1 & 0 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[150] $1^{111}31^{11}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 5 & 9 & 4 \\ 1 & 4 & 0 & 0 & 4 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 9 & 6 & 0 & 9 & 13 & 7 \\ 2 & 4 & 2 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

[151] $1^{112}3^121^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 13 & 1 & 0 \\ 9 & 10 & 0 & 13 & 74 & 10 & 16 \\ 1 & 2 & 0 & 1 & 10 & 0 & 2 \\ 2 & 2 & -1 & 0 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

[152] $1^{131}31^{111}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 3 & 2 \\ 2 & 4 & 4 & 0 & 17 & 9 & 4 \\ 1 & 4 & 0 & 0 & 13 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 17 & 13 & 0 & 60 & 30 & 14 \\ 3 & 9 & 6 & 0 & 30 & 13 & 7 \\ 2 & 4 & 2 & -3 & 14 & 7 & 1 \end{pmatrix}_{b_0}$$

[153] $1^{111}3^121^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -278, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 3 & 0 \\ 2 & 3 & 0 & 2 & 3 & 9 & 3 \\ 4 & 6 & 0 & 3 & 9 & 16 & 6 \\ 2 & 2 & -1 & 0 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[154] $1^{111}3^121^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 1 & 0 \\ 4 & 5 & 0 & 3 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & 0 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$

[155] $1^{112}3^121^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -297, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 9 & 0 \\ 2 & 3 & 0 & 2 & 3 & 13 & 3 \\ 7 & 8 & 0 & 9 & 13 & 40 & 12 \\ 2 & 2 & -1 & 0 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[156] $1^{111}3^11^{12}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 18 & 3 & 4 \\ 1 & 4 & 0 & 0 & 11 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 18 & 11 & 0 & 55 & 11 & 13 \\ 1 & 3 & 3 & 0 & 11 & 1 & 3 \\ 2 & 4 & 2 & -3 & 13 & 3 & 0 \end{pmatrix}_{b_0}$$

[157] $1^{131}1^{111}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 8 & 6 & 4 \\ 1 & 4 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 8 & 4 & 0 & 9 & 8 & 5 \\ 2 & 6 & 3 & 0 & 8 & 4 & 4 \\ 2 & 4 & 2 & -3 & 5 & 4 & 0 \end{pmatrix}_{b_0}$$

[159] $1^{123}3^121^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -311, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 3 & 0 \\ 3 & 5 & 0 & 3 & 11 & 15 & 5 \\ 4 & 6 & 0 & 3 & 15 & 16 & 6 \\ 2 & 2 & -1 & 0 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[158] $1^{11}3^{11}3^{13}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -311$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 7 & 2 \\ 2 & 4 & 4 & 0 & 5 & 21 & 4 \\ 1 & 4 & 0 & 0 & 4 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 19 & 4 \\ 7 & 21 & 12 & 0 & 19 & 73 & 15 \\ 2 & 4 & 2 & -3 & 4 & 15 & 1 \end{pmatrix}_{b_0}$$

[160] $1^{11}3^{11}1^{11}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -316$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 & 2 \\ 2 & 4 & 4 & 0 & 12 & 6 & 4 \\ 1 & 4 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 8 & 0 & 25 & 12 & 9 \\ 2 & 6 & 3 & 0 & 12 & 4 & 4 \\ 2 & 4 & 2 & -3 & 9 & 4 & 1 \end{pmatrix}_{b_0}$$

[161] $1^{11}1^{14}3^{11}2^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -319$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 4 & 0 \\ 2 & 3 & 0 & 2 & 3 & 12 & 3 \\ 5 & 8 & 0 & 4 & 12 & 30 & 8 \\ 2 & 2 & -1 & 0 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

[162] $1^{11}1^{13}3^{11}2^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -320$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 1 & 0 \\ 5 & 8 & 0 & 7 & 34 & 8 & 10 \\ 1 & 2 & 0 & 1 & 8 & 0 & 2 \\ 2 & 2 & -1 & 0 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

[163] $1^{11}1^{11}3^{11}1^{13}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -321$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 0 & 24 & 3 & 4 \\ 1 & 4 & 0 & 0 & 14 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 24 & 14 & 0 & 97 & 14 & 17 \\ 1 & 3 & 3 & 0 & 14 & 1 & 3 \\ 2 & 4 & 2 & -3 & 17 & 3 & 0 \end{pmatrix}_{b_0}$$

[164] $1^{11}1^{12}3^{11}2^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -333$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 1 & 0 \\ 5 & 6 & 0 & 6 & 20 & 6 & 8 \\ 1 & 2 & 0 & 1 & 6 & 0 & 2 \\ 2 & 2 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

[165] $1^{14}1^{11}3^{11}1^{12}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -334$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 0 & 26 & 3 & 4 \\ 1 & 4 & 0 & 0 & 19 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 26 & 19 & 0 & 135 & 19 & 21 \\ 1 & 3 & 3 & 0 & 19 & 1 & 3 \\ 2 & 4 & 2 & -3 & 21 & 3 & 1 \end{pmatrix}_{b_0}$$

[166] $1^{12}1^{12}3^{11}2^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -349$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 1 & 0 \\ 5 & 5 & 0 & 3 & 11 & 5 & 5 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & -1 & 0 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$

[167] $1^{11}1^{14}2^{11}3^{11}2^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -352$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 2 & 0 \\ 6 & 9 & 0 & 5 & 39 & 18 & 9 \\ 2 & 4 & 0 & 2 & 18 & 6 & 4 \\ 2 & 2 & -1 & 0 & 9 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[168] \ 1^{111}31^{123}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 5 & 2 \\ 2 & 4 & 4 & 0 & 5 & 15 & 4 \\ 1 & 4 & 0 & 0 & 4 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 14 & 4 \\ 5 & 15 & 9 & 0 & 14 & 37 & 11 \\ 2 & 4 & 2 & -3 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[169] \ 1^13^11^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -369, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 6 & 4 \\ 1 & 4 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 5 & 0 & 12 & 9 & 6 \\ 2 & 6 & 3 & 0 & 9 & 4 & 4 \\ 2 & 4 & 2 & -3 & 6 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[170] \ 1^{11}3^11^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -374, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 5 & 0 & 12 & 14 & 6 \\ 3 & 9 & 6 & 0 & 14 & 13 & 7 \\ 2 & 4 & 2 & -3 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[171] \ 1^{112}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -377, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 3 & 0 \\ 2 & 3 & 0 & 2 & 3 & 9 & 3 \\ 5 & 6 & 0 & 3 & 9 & 16 & 6 \\ 2 & 2 & -1 & 0 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{124}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -382, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 4 & 0 \\ 3 & 5 & 0 & 3 & 11 & 20 & 5 \\ 5 & 8 & 0 & 4 & 20 & 30 & 8 \\ 2 & 2 & -1 & 0 & 5 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[173] \ 1^{114}31^{123}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 8 & 2 \\ 2 & 4 & 4 & 0 & 5 & 24 & 4 \\ 1 & 4 & 0 & 0 & 4 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 26 & 4 \\ 8 & 24 & 18 & 0 & 26 & 118 & 20 \\ 2 & 4 & 2 & -3 & 4 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[174] \ 1^{124}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -390, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 4 & 0 \\ 3 & 5 & 0 & 3 & 11 & 20 & 5 \\ 6 & 8 & 0 & 4 & 20 & 30 & 8 \\ 2 & 2 & -1 & 0 & 5 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[175] \ 1^{121}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 6 & 1 & 0 \\ 6 & 6 & 0 & 6 & 20 & 6 & 8 \\ 1 & 2 & 0 & 1 & 6 & 0 & 2 \\ 2 & 2 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

[176] $1^{141}3^121^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -413$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 1 & 0 \\ 7 & 11 & 0 & 11 & 71 & 11 & 15 \\ 1 & 2 & 0 & 1 & 11 & 0 & 2 \\ 2 & 2 & -1 & 0 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[177] $1^{131}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 1 & 0 \\ 5 & 7 & 0 & 4 & 23 & 7 & 7 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & 0 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{114}3^121^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -419$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 10 & 0 \\ 2 & 3 & 0 & 2 & 3 & 16 & 3 \\ 7 & 10 & 0 & 10 & 16 & 58 & 14 \\ 2 & 2 & -1 & 0 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

[179] $1^{113}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -423$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 6 & 0 \\ 2 & 3 & 0 & 2 & 3 & 11 & 3 \\ 5 & 7 & 0 & 6 & 11 & 25 & 9 \\ 2 & 2 & -1 & 0 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[180] $1^{131}3^121^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -436$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.71 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 1 & 0 \\ 7 & 9 & 0 & 10 & 51 & 9 & 13 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & 0 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

[181] $1^{11}31^{111}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -445$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 & 2 \\ 2 & 4 & 4 & 0 & 11 & 6 & 4 \\ 1 & 4 & 0 & 0 & 7 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 11 & 7 & 0 & 20 & 11 & 8 \\ 2 & 6 & 3 & 0 & 11 & 4 & 4 \\ 2 & 4 & 2 & -3 & 8 & 4 & 1 \end{pmatrix}_{b_0}$$

[182] $1^{141}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -448$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 1 & 0 \\ 6 & 10 & 0 & 8 & 52 & 10 & 12 \\ 1 & 2 & 0 & 1 & 10 & 0 & 2 \\ 2 & 2 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[183] $1^{121}31^{12}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -454$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 20 & 3 & 4 \\ 1 & 4 & 0 & 0 & 13 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 20 & 13 & 0 & 71 & 13 & 15 \\ 1 & 3 & 3 & 0 & 13 & 1 & 3 \\ 2 & 4 & 2 & -3 & 15 & 3 & 1 \end{pmatrix}_{b_0}$$

[184] $1^{124}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -462$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 7 & 0 \\ 3 & 5 & 0 & 3 & 11 & 23 & 5 \\ 6 & 9 & 0 & 7 & 23 & 41 & 11 \\ 2 & 2 & -1 & 0 & 5 & 11 & 1 \end{pmatrix}_{b_0}$$

[185] $1^{122}3^11^{11}3^11^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 4 & 2 \\ 2 & 4 & 4 & 0 & 14 & 12 & 4 \\ 1 & 4 & 0 & 0 & 10 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 14 & 10 & 0 & 37 & 34 & 11 \\ 4 & 12 & 9 & 0 & 34 & 28 & 10 \\ 2 & 4 & 2 & -3 & 11 & 10 & 1 \end{pmatrix}_{b_0}$$

[186] $1^{141}3^12^11^{13} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -469, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 14 & 1 & 0 \\ 9 & 12 & 0 & 14 & 96 & 12 & 18 \\ 1 & 2 & 0 & 1 & 12 & 0 & 2 \\ 2 & 2 & -1 & 0 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

[187] $1^{131}3^11^{12}3^11^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 0 & 23 & 3 & 4 \\ 1 & 4 & 0 & 0 & 16 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 23 & 16 & 0 & 100 & 16 & 18 \\ 1 & 3 & 3 & 0 & 16 & 1 & 3 \\ 2 & 4 & 2 & -3 & 18 & 3 & 1 \end{pmatrix}_{b_0}$$

[188] $1^{112}3^12^12^11^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -522, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 4 & 0 \\ 2 & 3 & 0 & 2 & 3 & 12 & 3 \\ 6 & 8 & 0 & 4 & 12 & 30 & 8 \\ 2 & 2 & -1 & 0 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

[189] $1^{111}3^12^12^11^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -541, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 1 & 0 \\ 6 & 8 & 0 & 7 & 34 & 8 & 10 \\ 1 & 2 & 0 & 1 & 8 & 0 & 2 \\ 2 & 2 & -1 & 0 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

[190] $1^{112}3^12^12^11^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -542, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 6 & 0 \\ 2 & 3 & 0 & 2 & 3 & 11 & 3 \\ 6 & 7 & 0 & 6 & 11 & 25 & 9 \\ 2 & 2 & -1 & 0 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[191] $1^{111}3^12^12^11^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 1 & 0 \\ 6 & 9 & 0 & 5 & 39 & 9 & 9 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & 0 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[192] $1^{112}3^12^12^11^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -557, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 4 & 1 & 0 \\ 6 & 7 & 0 & 4 & 23 & 7 & 7 \\ 1 & 2 & 0 & 1 & 7 & 0 & 2 \\ 2 & 2 & -1 & 0 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[193] $1^{114}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -574, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 7 & 0 \\ 2 & 3 & 0 & 2 & 3 & 14 & 3 \\ 6 & 9 & 0 & 7 & 14 & 41 & 11 \\ 2 & 2 & -1 & 0 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

[194] $1^{112}3^{12}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -586, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 6 & 2 \\ 2 & 4 & 4 & 0 & 5 & 18 & 4 \\ 1 & 4 & 0 & 0 & 4 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 18 & 4 \\ 6 & 18 & 12 & 0 & 18 & 58 & 14 \\ 2 & 4 & 2 & -3 & 4 & 14 & 1 \end{pmatrix}_{b_0}$$

[195] $1^{113}3^{12}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -596, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 7 & 2 \\ 2 & 4 & 4 & 0 & 5 & 21 & 4 \\ 1 & 4 & 0 & 0 & 4 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 22 & 4 \\ 7 & 21 & 15 & 0 & 22 & 85 & 17 \\ 2 & 4 & 2 & -3 & 4 & 17 & 1 \end{pmatrix}_{b_0}$$

[196] $1^{131}3^{11}21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -597, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 10 & 1 & 0 \\ 8 & 9 & 0 & 10 & 51 & 9 & 13 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & 0 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

[197] $1^{111}3^{11}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -600, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 3 & 2 \\ 2 & 4 & 4 & 0 & 11 & 9 & 4 \\ 1 & 4 & 0 & 0 & 7 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 11 & 7 & 0 & 20 & 18 & 8 \\ 3 & 9 & 6 & 0 & 18 & 13 & 7 \\ 2 & 4 & 2 & -3 & 8 & 7 & 1 \end{pmatrix}_{b_0}$$

[198] $1^{111}3^{11}1^{111}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -601, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 3 & 2 \\ 2 & 4 & 4 & 0 & 12 & 9 & 4 \\ 1 & 4 & 0 & 0 & 8 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 8 & 0 & 25 & 20 & 9 \\ 3 & 9 & 6 & 0 & 20 & 13 & 7 \\ 2 & 4 & 2 & -3 & 9 & 7 & 1 \end{pmatrix}_{b_0}$$

[199] $1^{121}3^{13}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -601, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 0 & 26 & 3 & 4 \\ 1 & 4 & 0 & 0 & 16 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 26 & 16 & 0 & 117 & 16 & 19 \\ 1 & 3 & 3 & 0 & 16 & 1 & 3 \\ 2 & 4 & 2 & -3 & 19 & 3 & 0 \end{pmatrix}_{b_0}$$

[200] $1^{121}3^{11}1^{111}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -649, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 3 & 2 \\ 2 & 4 & 4 & 0 & 14 & 9 & 4 \\ 1 & 4 & 0 & 0 & 10 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 14 & 10 & 0 & 37 & 24 & 11 \\ 3 & 9 & 6 & 0 & 24 & 13 & 7 \\ 2 & 4 & 2 & -3 & 11 & 7 & 1 \end{pmatrix}_{b_0}$$

[201] $1^{121}3^{14}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -650, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 1 & 2 \\ 2 & 4 & 4 & 0 & 32 & 3 & 4 \\ 1 & 4 & 0 & 0 & 19 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 32 & 19 & 0 & 175 & 19 & 23 \\ 1 & 3 & 3 & 0 & 19 & 1 & 3 \\ 2 & 4 & 2 & -3 & 23 & 3 & 0 \end{pmatrix}_{b_0}$$

[202] $1^{114}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -662, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 10 & 0 \\ 2 & 3 & 0 & 2 & 3 & 16 & 3 \\ 8 & 10 & 0 & 10 & 16 & 58 & 14 \\ 2 & 2 & -1 & 0 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

[203] $1^{131}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -692, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 7 & 1 & 0 \\ 7 & 8 & 0 & 7 & 34 & 8 & 10 \\ 1 & 2 & 0 & 1 & 8 & 0 & 2 \\ 2 & 2 & -1 & 0 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

[204] $1^{141}3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -720, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 14 & 1 & 0 \\ 10 & 12 & 0 & 14 & 96 & 12 & 18 \\ 1 & 2 & 0 & 1 & 12 & 0 & 2 \\ 2 & 2 & -1 & 0 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

[205] $1^{141}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -724, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 1 & 0 \\ 8 & 11 & 0 & 11 & 71 & 11 & 15 \\ 1 & 2 & 0 & 1 & 11 & 0 & 2 \\ 2 & 2 & -1 & 0 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[206] $1^{141}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -749, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 1 & 0 \\ 7 & 10 & 0 & 8 & 52 & 10 & 12 \\ 1 & 2 & 0 & 1 & 10 & 0 & 2 \\ 2 & 2 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{141}3^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -765, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 5 & 1 & 0 \\ 7 & 9 & 0 & 5 & 39 & 9 & 9 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & -1 & 0 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[208] $1^{114}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -797, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 7 & 0 \\ 2 & 3 & 0 & 2 & 3 & 14 & 3 \\ 7 & 9 & 0 & 7 & 14 & 41 & 11 \\ 2 & 2 & -1 & 0 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

[209] $1^{112}3^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -821, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 8 & 2 \\ 2 & 4 & 4 & 0 & 5 & 24 & 4 \\ 1 & 4 & 0 & 0 & 4 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 23 & 4 \\ 8 & 24 & 15 & 0 & 23 & 100 & 18 \\ 2 & 4 & 2 & -3 & 4 & 18 & 1 \end{pmatrix}_{b_0}$$

[210] $1^{131}31^{133}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -871, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 0 & 29 & 3 & 4 \\ 1 & 4 & 0 & 0 & 19 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 29 & 19 & 0 & 152 & 19 & 22 \\ 1 & 3 & 3 & 0 & 19 & 1 & 3 \\ 2 & 4 & 2 & -3 & 22 & 3 & 1 \end{pmatrix}_{b_0}$$

[211] $1^{112}31^{143}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -926, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 10 & 2 \\ 2 & 4 & 4 & 0 & 5 & 30 & 4 \\ 1 & 4 & 0 & 0 & 4 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 28 & 4 \\ 10 & 30 & 18 & 0 & 28 & 154 & 22 \\ 2 & 4 & 2 & -3 & 4 & 22 & 1 \end{pmatrix}_{b_0}$$

[212] $1^{141}31^{133}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -929, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 1 & 2 \\ 2 & 4 & 4 & 0 & 32 & 3 & 4 \\ 1 & 4 & 0 & 0 & 22 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 32 & 22 & 0 & 193 & 22 & 25 \\ 1 & 3 & 3 & 0 & 22 & 1 & 3 \\ 2 & 4 & 2 & -3 & 25 & 3 & 1 \end{pmatrix}_{b_0}$$

[213] $1^{112}1^{141}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -965, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 11 & 1 & 0 \\ 9 & 11 & 0 & 11 & 71 & 11 & 15 \\ 1 & 2 & 0 & 1 & 11 & 0 & 2 \\ 2 & 2 & -1 & 0 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[214] $1^{112}1^{141}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -980, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 8 & 1 & 0 \\ 8 & 10 & 0 & 8 & 52 & 10 & 12 \\ 1 & 2 & 0 & 1 & 10 & 0 & 2 \\ 2 & 2 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[215] $1^{113}31^{133}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 9 & 2 \\ 2 & 4 & 4 & 0 & 5 & 27 & 4 \\ 1 & 4 & 0 & 0 & 4 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 27 & 4 \\ 9 & 27 & 18 & 0 & 27 & 133 & 21 \\ 2 & 4 & 2 & -3 & 4 & 21 & 1 \end{pmatrix}_{b_0}$$

[216] $1^{131}31^{143}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 1 & 2 \\ 2 & 4 & 4 & 0 & 35 & 3 & 4 \\ 1 & 4 & 0 & 0 & 22 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 35 & 22 & 0 & 216 & 22 & 26 \\ 1 & 3 & 3 & 0 & 22 & 1 & 3 \\ 2 & 4 & 2 & -3 & 26 & 3 & 1 \end{pmatrix}_{b_0}$$

[217] $1^{114}31^{133}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 10 & 2 \\ 2 & 4 & 4 & 0 & 5 & 30 & 4 \\ 1 & 4 & 0 & 0 & 4 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 31 & 4 \\ 10 & 30 & 21 & 0 & 31 & 172 & 24 \\ 2 & 4 & 2 & -3 & 4 & 24 & 1 \end{pmatrix}_{b_0}$$

[218] $1^{141}31^{143}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

10.2

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 1 & 2 \\ 2 & 4 & 4 & 0 & 38 & 3 & 4 \\ 1 & 4 & 0 & 0 & 25 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 38 & 25 & 0 & 263 & 25 & 29 \\ 1 & 3 & 3 & 0 & 25 & 1 & 3 \\ 2 & 4 & 2 & -3 & 29 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[219] 1^{113} 31^{143} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 11 & 2 \\ 2 & 4 & 4 & 0 & 5 & 33 & 4 \\ 1 & 4 & 0 & 0 & 4 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 32 & 4 \\ 11 & 33 & 21 & 0 & 32 & 193 & 25 \\ 2 & 4 & 2 & -3 & 4 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[220] 1^{114} 31^{143} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1846, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 12 & 2 \\ 2 & 4 & 4 & 0 & 5 & 36 & 4 \\ 1 & 4 & 0 & 0 & 4 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 36 & 4 \\ 12 & 36 & 24 & 0 & 36 & 238 & 28 \\ 2 & 4 & 2 & -3 & 4 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [221] 1^{12} 23^1 1^{11} 31^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -61, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 1 & 21 & 1 & 3 \\ 5 & 13 & 1 & 54 & 0 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 21 & 54 & 0 & 217 & 0 & 27 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 7 & -1 & 27 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[222] 1^{13} 23^1 1^{12} 31^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -61, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 1 & 32 & 1 & 3 \\ 5 & 13 & 1 & 84 & 0 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 32 & 84 & 0 & 520 & 0 & 42 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 7 & -1 & 42 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[224] 1^{11} 3^1 1^1 4 \quad 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 17, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 2 & 3 \\ 3 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & -2 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[223] 1^{11} 4 \quad 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 17, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 3 & 3 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[225] 1^{11} 3^1 1^1 4 \quad 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 3 & 4 \\ 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[226] 1^{12} 23^1 1^{11} 31^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -61, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 1 & 22 & 1 & 3 \\ 5 & 13 & 1 & 54 & 0 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 22 & 54 & 0 & 217 & 0 & 27 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 7 & -1 & 27 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[227] 1^{112} 4 \quad 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$$

10.7

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 6 & 3 \\ 4 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -2 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[228] \ 1^{122} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 34, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 4 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 18 & 10 \\ 4 & 8 & 0 & 18 & 10 & 6 \\ 3 & 4 & -2 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[229] \ 1^{113} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[230] \ 1^{121} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 46, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -2 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[231] \ 1^{111} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 57, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[232] \ 1^{111} 4^{13} 1^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 12 & 0 & 26 & 8 & 10 \\ 2 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -2 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[233] \ 1^{112} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 65, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[235] \ 1^{111} 4^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[234] \ 1^{112} 4^{13} 1^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 9 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 17 & 7 \\ 7 & 11 & 0 & 17 & 21 & 9 \\ 3 & 4 & -2 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[236] $1^{122} 4^{13^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -2 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[237] $1^{122} 4^{13^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 12 & 6 \\ 4 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -2 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[238] $1^{121} 4^{1^3 1^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 110, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 10 & 6 \\ 5 & 7 & 0 & 10 & 8 & 6 \\ 3 & 4 & -2 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[239] $1^{121} 4^{1^3 1^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 5 & 3 \\ 3 & 4 & 0 & 13 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 16 & 10 \\ 5 & 7 & 0 & 16 & 8 & 6 \\ 3 & 4 & -2 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[240] $1^{111} 4^{1^3 1^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 115, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 9 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 11 & 7 \\ 5 & 7 & 0 & 11 & 8 & 6 \\ 3 & 4 & -2 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[241] $1^{111} 4^{13^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 133, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -2 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[242] $1^{111} 4^{1^3 1^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 137, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 5 & 3 \\ 5 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & -2 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[243] $1^{121} 4^{13^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 6 & 6 \\ 2 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -2 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[244] $1^{121} 4^{1^3 1^1 1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 11 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 11 & 0 & 21 & 8 & 9 \\ 2 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -2 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[246] $1^{111} 1^{13^1 1^1} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 176, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 8 & 3 \\ 8 & 12 & 0 & 8 & 26 & 10 \\ 3 & 4 & -2 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[245] $1^{122} 1^{13^1 1^1} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 176, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 8 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 22 & 6 \\ 10 & 16 & 0 & 22 & 45 & 13 \\ 3 & 4 & -2 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[247] $1^{122} 1^{13^1 1^1} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 203, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 7 & 3 \\ 3 & 4 & 0 & 13 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 25 & 10 \\ 7 & 11 & 0 & 25 & 21 & 9 \\ 3 & 4 & -2 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[248] $1^{131} 4^{13^1 1^1} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 208, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 9 & 9 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -2 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[249] $1^{112} 1^{13^1 1^1} 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 225, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 4 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 8 & 3 \\ 7 & 11 & 0 & 8 & 21 & 9 \\ 3 & 4 & -2 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[250] $1^{122} 1^{13^1 1^1} 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 254, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 8 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 16 & 6 \\ 7 & 11 & 0 & 16 & 21 & 9 \\ 3 & 4 & -2 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[251] $1^{141} 4^{13^1 1^1} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 272, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 46 & 12 & 12 \\ 2 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -2 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[252] $1^{121} 1^{11} 4^1 13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 277, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 9 & 10 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -2 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[253] $1^{113} 1^{13^1 1^1} 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 281, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 11 & 3 \\ 9 & 15 & 0 & 11 & 40 & 12 \\ 3 & 4 & -2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[254] \ 1^{114} 1^{13^1 1^1} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 305, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 14 & 3 \\ 11 & 19 & 0 & 14 & 65 & 15 \\ 3 & 4 & -2 & 3 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[255] \ 1^{121} 1^{13^1 1^1} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 320, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 16 & 0 & 45 & 11 & 13 \\ 2 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & -2 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[256] \ 1^{112} 1^{13^1 1^1} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 320, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 11 & 3 \\ 10 & 16 & 0 & 11 & 45 & 13 \\ 3 & 4 & -2 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[257] \ 1^{131} 1^{13^1 1^1} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 349, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 15 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 15 & 0 & 40 & 11 & 12 \\ 2 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & -2 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[258] \ 1^{131} 1^{11} 4 \ 13^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 421, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 12 & 13 \\ 2 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -2 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[259] \ 1^{113} 1^{13^1 1^1} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 432, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 14 & 3 \\ 12 & 20 & 0 & 14 & 70 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[260] \ 1^{114} 1^{13^1 1^1} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 17 & 3 \\ 14 & 24 & 0 & 17 & 101 & 19 \\ 3 & 4 & -2 & 3 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[261] \ 1^{141} 1^{13^1 1^1} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 525, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 19 & 0 & 65 & 14 & 15 \\ 2 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -2 & 15 & 3 & 2 \end{pmatrix}_{b_0}$$

[262] $1^{141} 4^{13} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 565, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 15 & 16 \\ 2 & 4 & 0 & 15 & 1 & 3 \\ 3 & 4 & -2 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[263] $1^{131} 4^{13} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 576, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 20 & 0 & 70 & 14 & 16 \\ 2 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -2 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[264] $1^{141} 4^{13} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 832, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 24 & 0 & 101 & 17 & 19 \\ 2 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & -2 & 19 & 3 & 2 \end{pmatrix}_{b_0}$$

$T = 11$ [265] $1^{111} 31^{14} 13^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 1 & 3 & 3 & 0 & 3 & 1 \\ 2 & 4 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[266] $1^{111} 31^{14} 13^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 8 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 6 & 0 & 13 & 6 & 7 \\ 1 & 3 & 3 & 0 & 6 & 1 & 3 \\ 2 & 4 & 2 & -2 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[267] $1^{121} 31^{11} 41^{13} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -48, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 14 & 3 & 4 \\ 1 & 4 & 0 & 0 & 10 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 14 & 10 & 0 & 38 & 10 & 12 \\ 1 & 3 & 3 & 0 & 10 & 1 & 3 \\ 2 & 4 & 2 & -2 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[268] $1^{114} 31^{11} 41^{13} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 20 & 4 \\ 1 & 4 & 0 & 0 & 3 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 1 & 16 & 3 \\ 8 & 20 & 16 & 0 & 16 & 90 & 18 \\ 2 & 4 & 2 & -2 & 3 & 18 & 2 \end{pmatrix}_{b_0}$$

[269] $1^{111} 31^{11} 41^{13} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 11 & 4 \\ 1 & 4 & 0 & 0 & 3 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 1 & 7 & 3 \\ 5 & 11 & 7 & 0 & 7 & 21 & 9 \\ 2 & 4 & 2 & -2 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[270] $1^{141} 31^{11} 41^{13} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -112, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 0 & 20 & 3 & 4 \\ 1 & 4 & 0 & 0 & 16 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 20 & 16 & 0 & 90 & 16 & 18 \\ 1 & 3 & 3 & 0 & 16 & 1 & 3 \\ 2 & 4 & 2 & -2 & 18 & 3 & 2 \end{pmatrix}_{b_0}$$

[273] $1^{13}31^{11}41^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -160, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 0 & 17 & 3 & 4 \\ 1 & 4 & 0 & 0 & 13 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 17 & 13 & 0 & 61 & 13 & 15 \\ 1 & 3 & 3 & 0 & 13 & 1 & 3 \\ 2 & 4 & 2 & -2 & 15 & 3 & 2 \end{pmatrix}_{b_0}$$

[272] $1^{113}31^{11}41^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -160, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 17 & 4 \\ 1 & 4 & 0 & 0 & 3 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 1 & 13 & 3 \\ 7 & 17 & 13 & 0 & 13 & 61 & 15 \\ 2 & 4 & 2 & -2 & 3 & 15 & 2 \end{pmatrix}_{b_0}$$

[271] $1^{112}31^{11}41^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -160, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 14 & 4 \\ 1 & 4 & 0 & 0 & 3 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 1 & 10 & 3 \\ 6 & 14 & 10 & 0 & 10 & 38 & 12 \\ 2 & 4 & 2 & -2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

4.525 $\text{NHC}(-2)(-3) + \text{f}_4 + \text{su2n3mix} + \text{su}_2 + \text{so}_8$

$T = 9 \quad [0] \quad 1^{113}31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 9 & 5 \\ 11 & 20 & 0 & 28 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 28 & 0 & 37 & 24 & 11 \\ 9 & 17 & 0 & 24 & 13 & 7 \\ 5 & 8 & -3 & 11 & 7 & 1 \end{pmatrix}_{b_0}$$

4.526 $\text{NHC}(-2)(-3) + \text{f}_4 + \text{su2n3mix} + \text{su}_2 + \text{su}_2$

$T = 9 \quad [0] \quad 1^{1113}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 2 & 3 \\ 3 & 4 & 0 & 7 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 6 & 5 & 5 \\ 3 & 5 & 0 & 6 & 4 & 3 & 4 \\ 2 & 4 & 0 & 5 & 3 & 1 & 3 \\ 3 & 4 & -3 & 5 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{11213}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 5 & 3 \\ 3 & 4 & 0 & 7 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 10 & 11 & 5 \\ 4 & 8 & 0 & 10 & 10 & 12 & 6 \\ 5 & 9 & 0 & 11 & 12 & 13 & 7 \\ 3 & 4 & -3 & 5 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{12213}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 2 & 3 \\ 3 & 4 & 0 & 16 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 22 & 11 & 12 \\ 4 & 8 & 0 & 22 & 10 & 6 & 6 \\ 2 & 4 & 0 & 11 & 6 & 1 & 3 \\ 3 & 4 & -3 & 12 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{11313}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

12.0

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 6 & 2 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 15 & 5 & 5 \\ 6 & 12 & 0 & 15 & 25 & 9 & 9 \\ 2 & 4 & 0 & 5 & 9 & 1 & 3 \\ 3 & 4 & -3 & 5 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}1} 3^{\overline{111}} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 5 & 3 \\ 3 & 4 & 0 & 7 & 5 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 6 & 11 & 5 \\ 3 & 5 & 0 & 6 & 4 & 9 & 4 \\ 5 & 9 & 0 & 11 & 9 & 13 & 7 \\ 3 & 4 & -3 & 5 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{113}3} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 9 & 3 \\ 3 & 4 & 0 & 7 & 5 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 6 & 21 & 5 \\ 3 & 5 & 0 & 6 & 4 & 15 & 4 \\ 9 & 17 & 0 & 21 & 15 & 49 & 13 \\ 3 & 4 & -3 & 5 & 4 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{122}1} 3^{\overline{111}} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 5 & 3 \\ 3 & 4 & 0 & 16 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 22 & 26 & 12 \\ 4 & 8 & 0 & 22 & 10 & 12 & 6 \\ 5 & 9 & 0 & 26 & 12 & 13 & 7 \\ 3 & 4 & -3 & 12 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{112}2} 3^{\overline{11}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 4 & 3 \\ 3 & 4 & 0 & 7 & 8 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 10 & 10 & 5 \\ 4 & 8 & 0 & 10 & 10 & 12 & 6 \\ 4 & 8 & 0 & 10 & 12 & 10 & 6 \\ 3 & 4 & -3 & 5 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{113}1} 3^{\overline{111}} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 2 & 3 \\ 3 & 4 & 0 & 12 & 17 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 36 & 8 & 9 \\ 9 & 17 & 0 & 36 & 49 & 12 & 13 \\ 2 & 4 & 0 & 8 & 12 & 1 & 3 \\ 3 & 4 & -3 & 9 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{121}1} 3^{\overline{111}} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 5 & 3 \\ 3 & 4 & 0 & 11 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 17 & 17 & 8 \\ 5 & 9 & 0 & 17 & 13 & 15 & 7 \\ 5 & 9 & 0 & 17 & 15 & 13 & 7 \\ 3 & 4 & -3 & 8 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{111}1} 3^{\overline{111}} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 2 & 3 \\ 3 & 4 & 0 & 12 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 12 & 8 & 9 \\ 3 & 5 & 0 & 12 & 4 & 3 & 4 \\ 2 & 4 & 0 & 8 & 3 & 1 & 3 \\ 3 & 4 & -3 & 9 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{131}1} 3^{\overline{111}} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

11.2

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 2 & 3 \\ 3 & 4 & 0 & 20 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 18 & 14 & 15 \\ 3 & 5 & 0 & 18 & 4 & 3 & 4 \\ 2 & 4 & 0 & 14 & 3 & 1 & 3 \\ 3 & 4 & -3 & 15 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{1121}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 7 & 3 \\ 3 & 4 & 0 & 7 & 5 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 6 & 16 & 5 \\ 3 & 5 & 0 & 6 & 4 & 12 & 4 \\ 7 & 13 & 0 & 16 & 12 & 28 & 10 \\ 3 & 4 & -3 & 5 & 4 & 10 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{1121}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 2 & 3 \\ 3 & 4 & 0 & 7 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 10 & 5 & 5 \\ 4 & 8 & 0 & 10 & 10 & 6 & 6 \\ 2 & 4 & 0 & 5 & 6 & 1 & 3 \\ 3 & 4 & -3 & 5 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{1211}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -246, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 2 & 3 \\ 3 & 4 & 0 & 11 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 17 & 8 & 8 \\ 5 & 9 & 0 & 17 & 13 & 6 & 7 \\ 2 & 4 & 0 & 8 & 6 & 1 & 3 \\ 3 & 4 & -3 & 8 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{1211}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 2 & 3 \\ 3 & 4 & 0 & 16 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 15 & 11 & 12 \\ 3 & 5 & 0 & 15 & 4 & 3 & 4 \\ 2 & 4 & 0 & 11 & 3 & 1 & 3 \\ 3 & 4 & -3 & 12 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{1111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 8 & 8 & 9 \\ 2 & 4 & 0 & 8 & 1 & 3 & 3 \\ 2 & 4 & 0 & 8 & 3 & 1 & 3 \\ 3 & 4 & -3 & 9 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{1111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 5 & 5 \\ 2 & 4 & 0 & 5 & 1 & 3 & 3 \\ 2 & 4 & 0 & 5 & 3 & 1 & 3 \\ 3 & 4 & -3 & 5 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{1111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 11 & 5 \\ 2 & 4 & 0 & 5 & 1 & 6 & 3 \\ 5 & 9 & 0 & 11 & 6 & 13 & 7 \\ 3 & 4 & -3 & 5 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{1122}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -326, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

12.2

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 7 & 3 \\ 3 & 4 & 0 & 7 & 8 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 10 & 16 & 5 \\ 4 & 8 & 0 & 10 & 10 & 18 & 6 \\ 7 & 13 & 0 & 16 & 18 & 28 & 10 \\ 3 & 4 & -3 & 5 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1221} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 4 & 2 & 3 \\ 3 & 4 & 0 & 11 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 16 & 8 & 8 \\ 4 & 8 & 0 & 16 & 10 & 6 & 6 \\ 2 & 4 & 0 & 8 & 6 & 1 & 3 \\ 3 & 4 & -3 & 8 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 10 & 3 \\ 3 & 4 & 0 & 7 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 22 & 5 \\ 2 & 4 & 0 & 5 & 1 & 12 & 3 \\ 10 & 18 & 0 & 22 & 12 & 58 & 14 \\ 3 & 4 & -3 & 5 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{1211} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 2 & 3 \\ 3 & 4 & 0 & 11 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 8 & 8 & 8 \\ 2 & 4 & 0 & 8 & 1 & 3 & 3 \\ 2 & 4 & 0 & 8 & 3 & 1 & 3 \\ 3 & 4 & -3 & 8 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{1212} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 7 & 3 \\ 3 & 4 & 0 & 11 & 9 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 17 & 25 & 8 \\ 5 & 9 & 0 & 17 & 13 & 21 & 7 \\ 7 & 13 & 0 & 25 & 21 & 28 & 10 \\ 3 & 4 & -3 & 8 & 7 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{1111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 5 & 3 \\ 3 & 4 & 0 & 7 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 11 & 11 & 5 \\ 5 & 9 & 0 & 11 & 13 & 15 & 7 \\ 5 & 9 & 0 & 11 & 15 & 13 & 7 \\ 3 & 4 & -3 & 5 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{1112} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 15 & 3 \\ 3 & 4 & 0 & 7 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 33 & 5 \\ 2 & 4 & 0 & 5 & 1 & 18 & 3 \\ 15 & 27 & 0 & 33 & 18 & 133 & 21 \\ 3 & 4 & -3 & 5 & 3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{1211} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 12 & 21 & 0 & 80 & 14 & 14 & 16 \\ 2 & 4 & 0 & 14 & 1 & 3 & 3 \\ 2 & 4 & 0 & 14 & 3 & 1 & 3 \\ 3 & 4 & -3 & 16 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{1112} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -514, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 7 & 3 \\ 3 & 4 & 0 & 12 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 8 & 28 & 9 \\ 2 & 4 & 0 & 8 & 1 & 9 & 3 \\ 7 & 13 & 0 & 28 & 9 & 28 & 10 \\ 3 & 4 & -3 & 9 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{111123} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 5 & 3 \\ 3 & 4 & 0 & 12 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 8 & 20 & 9 \\ 2 & 4 & 0 & 8 & 1 & 6 & 3 \\ 5 & 9 & 0 & 20 & 6 & 13 & 7 \\ 3 & 4 & -3 & 9 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{11123} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 7 & 3 \\ 3 & 4 & 0 & 7 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 16 & 5 \\ 2 & 4 & 0 & 5 & 1 & 9 & 3 \\ 7 & 13 & 0 & 16 & 9 & 28 & 10 \\ 3 & 4 & -3 & 5 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{12123} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -546, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 7 & 3 \\ 3 & 4 & 0 & 11 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 8 & 25 & 8 \\ 2 & 4 & 0 & 8 & 1 & 9 & 3 \\ 7 & 13 & 0 & 25 & 9 & 28 & 10 \\ 3 & 4 & -3 & 8 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{11143} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -546, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 11 & 3 \\ 3 & 4 & 0 & 7 & 9 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 11 & 26 & 5 \\ 5 & 9 & 0 & 11 & 13 & 33 & 7 \\ 11 & 21 & 0 & 26 & 33 & 76 & 16 \\ 3 & 4 & -3 & 5 & 7 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{13113} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 2 & 3 \\ 3 & 4 & 0 & 15 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 11 & 11 & 11 \\ 2 & 4 & 0 & 11 & 1 & 3 & 3 \\ 2 & 4 & 0 & 11 & 3 & 1 & 3 \\ 3 & 4 & -3 & 11 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{11143} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -616, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 20 & 3 \\ 3 & 4 & 0 & 7 & 4 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 44 & 5 \\ 2 & 4 & 0 & 5 & 1 & 24 & 3 \\ 20 & 36 & 0 & 44 & 24 & 238 & 28 \\ 3 & 4 & -3 & 5 & 3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{12113} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -620, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 11 & 11 & 12 \\ 2 & 4 & 0 & 11 & 1 & 3 & 3 \\ 2 & 4 & 0 & 11 & 3 & 1 & 3 \\ 3 & 4 & -3 & 12 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{11123} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -626, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 7 & 3 \\ 3 & 4 & 0 & 7 & 9 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 11 & 16 & 5 \\ 5 & 9 & 0 & 11 & 13 & 21 & 7 \\ 7 & 13 & 0 & 16 & 21 & 28 & 10 \\ 3 & 4 & -3 & 5 & 7 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1113}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 9 & 3 \\ 3 & 4 & 0 & 7 & 9 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 11 & 21 & 5 \\ 5 & 9 & 0 & 11 & 13 & 27 & 7 \\ 9 & 17 & 0 & 21 & 27 & 49 & 13 \\ 3 & 4 & -3 & 5 & 7 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1113}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -716, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 9 & 3 \\ 3 & 4 & 0 & 7 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 21 & 5 \\ 2 & 4 & 0 & 5 & 1 & 12 & 3 \\ 9 & 17 & 0 & 21 & 12 & 49 & 13 \\ 3 & 4 & -3 & 5 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1411}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -720, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 10 & 19 & 0 & 64 & 14 & 14 & 14 \\ 2 & 4 & 0 & 14 & 1 & 3 & 3 \\ 2 & 4 & 0 & 14 & 3 & 1 & 3 \\ 3 & 4 & -3 & 14 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1211}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -810, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 5 & 3 \\ 3 & 4 & 0 & 16 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 11 & 26 & 12 \\ 2 & 4 & 0 & 11 & 1 & 6 & 3 \\ 5 & 9 & 0 & 26 & 6 & 13 & 7 \\ 3 & 4 & -3 & 12 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1113}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -836, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 12 & 3 \\ 3 & 4 & 0 & 7 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 27 & 5 \\ 2 & 4 & 0 & 5 & 1 & 15 & 3 \\ 12 & 22 & 0 & 27 & 15 & 85 & 17 \\ 3 & 4 & -3 & 5 & 3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{1311}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -840, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 2 & 3 \\ 3 & 4 & 0 & 30 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 17 & 30 & 0 & 165 & 20 & 20 & 23 \\ 2 & 4 & 0 & 20 & 1 & 3 & 3 \\ 2 & 4 & 0 & 20 & 3 & 1 & 3 \\ 3 & 4 & -3 & 23 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1114}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -856, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 11 & 3 \\ 3 & 4 & 0 & 7 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 26 & 5 \\ 2 & 4 & 0 & 5 & 1 & 15 & 3 \\ 11 & 21 & 0 & 26 & 15 & 76 & 16 \\ 3 & 4 & -3 & 5 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1122}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -896, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 7 & 3 \\ 3 & 4 & 0 & 7 & 13 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 16 & 16 & 5 \\ 7 & 13 & 0 & 16 & 28 & 30 & 10 \\ 7 & 13 & 0 & 16 & 30 & 28 & 10 \\ 3 & 4 & -3 & 5 & 10 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1311}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -964, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 5 & 3 \\ 3 & 4 & 0 & 20 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 14 & 32 & 15 \\ 2 & 4 & 0 & 14 & 1 & 6 & 3 \\ 5 & 9 & 0 & 32 & 6 & 13 & 7 \\ 3 & 4 & -3 & 15 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1311}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -984, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 2 & 3 \\ 3 & 4 & 0 & 20 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 14 & 14 & 15 \\ 2 & 4 & 0 & 14 & 1 & 3 & 3 \\ 2 & 4 & 0 & 14 & 3 & 1 & 3 \\ 3 & 4 & -3 & 15 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1411}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -986, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 5 & 2 & 3 \\ 3 & 4 & 0 & 24 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 13 & 24 & 0 & 100 & 38 & 17 & 18 \\ 5 & 9 & 0 & 38 & 13 & 6 & 7 \\ 2 & 4 & 0 & 17 & 6 & 1 & 3 \\ 3 & 4 & -3 & 18 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1212}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1020, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 7 & 3 \\ 3 & 4 & 0 & 16 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 11 & 37 & 12 \\ 2 & 4 & 0 & 11 & 1 & 9 & 3 \\ 7 & 13 & 0 & 37 & 9 & 28 & 10 \\ 3 & 4 & -3 & 12 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1311}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1076, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 2 & 3 \\ 3 & 4 & 0 & 25 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 14 & 25 & 0 & 111 & 17 & 17 & 19 \\ 2 & 4 & 0 & 17 & 1 & 3 & 3 \\ 2 & 4 & 0 & 17 & 3 & 1 & 3 \\ 3 & 4 & -3 & 19 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1114}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 17 & 3 \\ 3 & 4 & 0 & 7 & 4 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 38 & 5 \\ 2 & 4 & 0 & 5 & 1 & 21 & 3 \\ 17 & 31 & 0 & 38 & 21 & 172 & 24 \\ 3 & 4 & -3 & 5 & 3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1114}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 14 & 3 \\ 3 & 4 & 0 & 7 & 4 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 32 & 5 \\ 2 & 4 & 0 & 5 & 1 & 18 & 3 \\ 14 & 26 & 0 & 32 & 18 & 118 & 20 \\ 3 & 4 & -3 & 5 & 3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1411}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1296, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 2 & 2 & 3 \\ 3 & 4 & 0 & 39 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 22 & 39 & 0 & 280 & 26 & 26 & 30 \\ 2 & 4 & 0 & 26 & 1 & 3 & 3 \\ 2 & 4 & 0 & 26 & 3 & 1 & 3 \\ 3 & 4 & -3 & 30 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1411311^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 13 & 24 & 0 & 100 & 17 & 17 & 18 \\ 2 & 4 & 0 & 17 & 1 & 3 & 3 \\ 2 & 4 & 0 & 17 & 3 & 1 & 3 \\ 3 & 4 & -3 & 18 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1411311^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1644, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 2 & 2 & 3 \\ 3 & 4 & 0 & 34 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 19 & 34 & 0 & 208 & 23 & 23 & 26 \\ 2 & 4 & 0 & 23 & 1 & 3 & 3 \\ 2 & 4 & 0 & 23 & 3 & 1 & 3 \\ 3 & 4 & -3 & 26 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1411311^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1664, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 2 & 3 \\ 3 & 4 & 0 & 29 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 16 & 29 & 0 & 148 & 20 & 20 & 22 \\ 2 & 4 & 0 & 20 & 1 & 3 & 3 \\ 2 & 4 & 0 & 20 & 3 & 1 & 3 \\ 3 & 4 & -3 & 22 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [55] \quad 1^{1131111311^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 5 & 6 & 4 \\ 1 & 4 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 5 & 4 \\ 2 & 6 & 3 & 0 & 5 & 4 & 4 \\ 2 & 6 & 3 & 0 & 5 & 6 & 4 \\ 2 & 4 & 2 & -3 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{11113111311^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 5 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 4 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 3 & 3 & 0 & 4 & 1 & 6 & 3 \\ 3 & 9 & 6 & 0 & 9 & 6 & 13 & 7 \\ 2 & 4 & 2 & -3 & 4 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{111131111311^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 262, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 5 & 6 & 9 & 4 \\ 1 & 4 & 0 & 0 & 4 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 5 & 9 & 4 \\ 2 & 6 & 3 & 0 & 5 & 4 & 9 & 4 \\ 3 & 9 & 6 & 0 & 9 & 9 & 13 & 7 \\ 2 & 4 & 2 & -3 & 4 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{111131111311^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 752, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 5 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 4 & 6 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 9 & 9 & 4 \\ 3 & 9 & 6 & 0 & 9 & 13 & 15 & 7 \\ 3 & 9 & 6 & 0 & 9 & 15 & 13 & 7 \\ 2 & 4 & 2 & -3 & 4 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

4.527 NHC(-2)(-3) + f₄ + su₂n₃mix + su₂ + su₂→(-3)

$$T = 10 \quad [0] \quad 1^{111}3^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -60, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 4 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 4 & 3 \\ 1 & 3 & 3 & 0 & 4 & 1 & 2 \\ 1 & 3 & 2 & 0 & 3 & 2 & 0 \\ 2 & 4 & 2 & -3 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -60, \quad \det = 175, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 4 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 5 & 3 \\ 2 & 6 & 3 & 0 & 5 & 4 & 3 \\ 1 & 3 & 2 & 0 & 3 & 3 & 0 \\ 2 & 4 & 2 & -3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -60, \quad \det = 475, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 5 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 4 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 9 & 3 & 4 \\ 3 & 9 & 6 & 0 & 9 & 13 & 5 & 7 \\ 1 & 3 & 2 & 0 & 3 & 5 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.528 NHC(-2)(-3) + f₄ + su₂n₃mix + su₂ + su₃

$$T = 9 \quad [0] \quad 1^{111}3^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -104, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 8 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 8 & 0 & 12 & 9 & 5 & 6 \\ 3 & 5 & 0 & 9 & 4 & 3 & 4 \\ 2 & 4 & 0 & 5 & 3 & 0 & 2 \\ 3 & 4 & -3 & 6 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -116, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 10 & 17 & 0 & 55 & 11 & 11 & 13 \\ 2 & 4 & 0 & 11 & 1 & 3 & 3 \\ 2 & 4 & 0 & 11 & 3 & 0 & 2 \\ 3 & 4 & -3 & 13 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -119, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 2 & 3 \\ 3 & 4 & 0 & 7 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 6 & 5 & 5 \\ 3 & 5 & 0 & 6 & 4 & 3 & 4 \\ 2 & 4 & 0 & 5 & 3 & 0 & 2 \\ 3 & 4 & -3 & 5 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{121}3^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -124, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 & 3 \\ 3 & 4 & 0 & 11 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 8 & 9 & 8 \\ 2 & 4 & 0 & 8 & 1 & 3 & 3 \\ 3 & 5 & 0 & 9 & 3 & 3 & 3 \\ 3 & 4 & -3 & 8 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -204, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.14 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 & 3 \\ 3 & 4 & 0 & 7 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 6 & 5 \\ 2 & 4 & 0 & 5 & 1 & 3 & 3 \\ 3 & 5 & 0 & 6 & 3 & 3 & 3 \\ 3 & 4 & -3 & 5 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{1111}1}3^{\overline{111}}1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -269$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 8 & 0 & 12 & 14 & 5 & 6 \\ 5 & 9 & 0 & 14 & 13 & 6 & 7 \\ 2 & 4 & 0 & 5 & 6 & 0 & 2 \\ 3 & 4 & -3 & 6 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{131}1}3^{\overline{111}}1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -401$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 2 & 3 \\ 3 & 4 & 0 & 20 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 18 & 14 & 15 \\ 3 & 5 & 0 & 18 & 4 & 3 & 4 \\ 2 & 4 & 0 & 14 & 3 & 0 & 2 \\ 3 & 4 & -3 & 15 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{\overline{111}1}3^{\overline{111}}1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -401$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 2 & 3 \\ 3 & 4 & 0 & 12 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 12 & 8 & 9 \\ 3 & 5 & 0 & 12 & 4 & 3 & 4 \\ 2 & 4 & 0 & 8 & 3 & 0 & 2 \\ 3 & 4 & -3 & 9 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{\overline{111}1}3^{\overline{111}}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -424$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 5 & 5 \\ 2 & 4 & 0 & 5 & 1 & 3 & 3 \\ 2 & 4 & 0 & 5 & 3 & 0 & 2 \\ 3 & 4 & -3 & 5 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{\overline{111}1}3^{\overline{111}}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -464$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 11 & 5 \\ 2 & 4 & 0 & 5 & 1 & 6 & 3 \\ 5 & 9 & 0 & 11 & 6 & 12 & 6 \\ 3 & 4 & -3 & 5 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{\overline{111}1}3^{\overline{111}}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -481$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 8 & 8 & 9 \\ 2 & 4 & 0 & 8 & 1 & 3 & 3 \\ 2 & 4 & 0 & 8 & 3 & 0 & 2 \\ 3 & 4 & -3 & 9 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{\overline{121}1}3^{\overline{111}}1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -500$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 2 & 3 \\ 3 & 4 & 0 & 16 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 15 & 11 & 12 \\ 3 & 5 & 0 & 15 & 4 & 3 & 4 \\ 2 & 4 & 0 & 11 & 3 & 0 & 2 \\ 3 & 4 & -3 & 12 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{\overline{111}1}3^{\overline{111}}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -509$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 2 & 3 \\ 3 & 4 & 0 & 7 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 11 & 5 & 5 \\ 5 & 9 & 0 & 11 & 13 & 6 & 7 \\ 2 & 4 & 0 & 5 & 6 & 0 & 2 \\ 3 & 4 & -3 & 5 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{\overline{111}13^{\overline{11}1}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3} \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 2 & 3 \\ 3 & 4 & 0 & 12 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 20 & 8 & 9 \\ 5 & 9 & 0 & 20 & 13 & 6 & 7 \\ 2 & 4 & 0 & 8 & 6 & 0 & 2 \\ 3 & 4 & -3 & 9 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.529 NHC(-2)(-3) + $\mathfrak{f}_4$ + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{\overline{11}3^{\overline{11}1}1^{\overline{11}1}1^{\overline{11}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)} \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -91, \quad \det = -60, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.13 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 5 & 0 & 7 & 4 & 5 \\ 1 & 3 & 2 & 0 & 4 & 0 & 2 \\ 2 & 4 & 2 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}1}3^{\overline{11}1}13^{\overline{11}1}1^{\overline{11}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)} \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -90, \quad \det = -85, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 4 & 2 & 3 \\ 1 & 4 & 0 & 0 & 4 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 4 & 3 & 4 \\ 1 & 2 & 2 & 0 & 3 & 0 & 2 \\ 2 & 3 & 2 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{12}3^{\overline{11}1}1^{\overline{11}3^{\overline{11}1}1^{\overline{11}1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)} \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -90, \quad \det = -90, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 8 & 6 & 0 & 25 & 6 \\ 2 & 6 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 25 & 16 & 0 & 69 & 17 \\ 3 & 6 & 3 & -3 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{11}3^{\overline{11}1}1^{\overline{11}3^{\overline{11}1}1^{\overline{11}1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)} \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -90, \quad \det = -150, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{13}1}1^{\overline{11}3^{\overline{11}1}1^{\overline{11}1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)} \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -90, \quad \det = -205, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.16 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 8 & 3 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 8 & 4 & 0 & 9 & 4 & 5 \\ 1 & 3 & 2 & 0 & 4 & 0 & 2 \\ 2 & 4 & 2 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{13}1}1^{\overline{11}3^{\overline{11}1}1^{\overline{11}1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)} \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -90, \quad \det = -250, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.13 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 8 & 6 & 0 & 15 & 6 \\ 2 & 6 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 15 & 8 & 0 & 21 & 9 \\ 3 & 6 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{11}3^{\overline{11}1}1^{\overline{11}3^{\overline{11}1}1^{\overline{11}1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)} \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -90, \quad \det = -280, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.03 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 8 & 6 & 0 & 20 & 6 \\ 2 & 6 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 12 & 0 & 41 & 13 \\ 3 & 6 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\overline{13}3^{\overline{11}1}1^{\overline{11}3^{\overline{11}1}1^{\overline{11}1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)} \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -90, \quad \det = -295, \quad (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 17 & 3 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 17 & 13 & 0 & 60 & 10 & 14 \\ 1 & 3 & 2 & 0 & 10 & 0 & 2 \\ 2 & 4 & 2 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{13}3^{11}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -303, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 5 & 0 & 12 & 5 & 6 \\ 1 & 3 & 2 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & -3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{12}3^{11}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 15 & 11 & 0 & 44 & 9 & 12 \\ 1 & 3 & 2 & 0 & 9 & 0 & 2 \\ 2 & 4 & 2 & -3 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11}3^{11}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -417, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 12 & 8 & 0 & 25 & 7 & 9 \\ 1 & 3 & 2 & 0 & 7 & 0 & 2 \\ 2 & 4 & 2 & -3 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11}3^{11}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 11 & 3 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 11 & 7 & 0 & 20 & 6 & 8 \\ 1 & 3 & 2 & 0 & 6 & 0 & 2 \\ 2 & 4 & 2 & -3 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{12}3^{11}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -465, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 14 & 3 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 14 & 10 & 0 & 37 & 8 & 11 \\ 1 & 3 & 2 & 0 & 8 & 0 & 2 \\ 2 & 4 & 2 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{13}3^{11}1^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -520, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 15 & 8 & 0 & 36 & 8 & 10 \\ 1 & 3 & 2 & 0 & 8 & 0 & 2 \\ 2 & 4 & 2 & -3 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11}3^{11}1^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -844, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 18 & 3 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 18 & 11 & 0 & 55 & 10 & 13 \\ 1 & 3 & 2 & 0 & 10 & 0 & 2 \\ 2 & 4 & 2 & -3 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{12}3^{11}1^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -968, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 0 & 21 & 3 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 21 & 14 & 0 & 80 & 12 & 16 \\ 1 & 3 & 2 & 0 & 12 & 0 & 2 \\ 2 & 4 & 2 & -3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

4.530 NHC(-2)(-3) + f₄ + su2n3mix + su₂→(-3) + su₂

$$T = 10 \quad [0] \quad 1^{11}3^{11}1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -60, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 2 \\ 1 & 3 & 3 & 0 & 4 & 2 & 1 \\ 2 & 4 & 2 & -3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -60, \quad \det = 175, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 3 & 5 \\ 1 & 3 & 2 & 0 & 3 & 0 & 3 \\ 2 & 6 & 3 & 0 & 5 & 3 & 4 \\ 2 & 4 & 2 & -3 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -60, \quad \det = 475, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 5 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 3 & 9 & 4 \\ 1 & 3 & 2 & 0 & 3 & 0 & 5 & 2 \\ 3 & 9 & 6 & 0 & 9 & 5 & 13 & 7 \\ 2 & 4 & 2 & -3 & 4 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.531 NHC(-2)(-3) + f₄ + su2n3mix + su2n3mix

$$T = 9 \quad [0] \quad 1^{11}3^{11}1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -90, \quad \det = -40, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 4 & 0 & 6 & 11 & 4 \\ 1 & 4 & 0 & 0 & 5 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 5 & 0 & 7 & 13 & 5 \\ 4 & 11 & 7 & 0 & 13 & 20 & 8 \\ 2 & 4 & 2 & -3 & 5 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -90, \quad \det = -60, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.13 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 5 & 4 \\ 1 & 4 & 0 & 0 & 5 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 5 & 0 & 7 & 7 & 5 \\ 2 & 5 & 4 & 0 & 7 & 4 & 4 \\ 2 & 4 & 2 & -3 & 5 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -90, \quad \det = -135, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.04 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 5 & 9 & 4 \\ 1 & 4 & 0 & 0 & 4 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 8 & 4 \\ 3 & 9 & 5 & 0 & 8 & 12 & 6 \\ 2 & 4 & 2 & -3 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}3^{11}1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -90, \quad \det = -325, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 4 & 0 & 5 & 12 & 4 \\ 1 & 4 & 0 & 0 & 4 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 12 & 4 \\ 4 & 12 & 8 & 0 & 12 & 25 & 9 \\ 2 & 4 & 2 & -3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [4] \quad 1^{11}3^{11}1^{11}41^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 11, \quad T_{\max} = 13, \quad \Delta = -60, \quad \det = -28, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 8 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 1 & 6 & 3 \\ 4 & 8 & 6 & 0 & 6 & 13 & 7 \\ 2 & 4 & 2 & -2 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

4.532 NHC(-2)(-3) + f₄ + su₂n₃mix + su₃

$$T = 8 \quad [0] \ 1^{141} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 5 & 3 \\ 3 & 4 & 0 & 19 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 0 & 64 & 29 & 14 \\ 5 & 9 & 0 & 29 & 12 & 6 \\ 3 & 4 & -3 & 14 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 102, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 & 3 \\ 3 & 4 & 0 & 11 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 9 & 8 \\ 3 & 5 & 0 & 9 & 3 & 3 \\ 3 & 4 & -3 & 8 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 0 & 12 & 5 & 6 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 12 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 16 & 9 \\ 4 & 8 & 0 & 16 & 9 & 5 \\ 3 & 4 & -3 & 9 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & -2 & 5 & 0 & 0 \\ 6 & 12 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 7 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 6 & 5 \\ 3 & 5 & 0 & 6 & 3 & 3 \\ 3 & 4 & -3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 27 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 16 & 27 & 0 & 151 & 17 & 21 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 3 & 4 & -3 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & -2 & 5 & 0 & 0 \\ 5 & 12 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{112} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 197, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 7 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 10 & 5 \\ 4 & 8 & 0 & 10 & 9 & 5 \\ 3 & 4 & -3 & 5 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{131} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 5 & 3 \\ 3 & 4 & 0 & 15 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 23 & 11 \\ 5 & 9 & 0 & 23 & 12 & 6 \\ 3 & 4 & -3 & 11 & 6 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 227, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.04 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 5 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 11 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 17 & 8 \\ 5 & 9 & 0 & 17 & 12 & 6 \\ 3 & 4 & -3 & 8 & 6 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 11 & 5 \\ 5 & 9 & 0 & 11 & 12 & 6 \\ 3 & 4 & -3 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{111}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 22 & 0 & 97 & 14 & 17 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 3 & 4 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.14 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 11 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 8 & 8 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 3 & 4 & -3 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 8 & 9 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 3 & 4 & -3 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 7 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 16 & 5 \\ 7 & 13 & 0 & 16 & 27 & 9 \\ 3 & 4 & -3 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{111}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 403, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 17 & 0 & 55 & 11 & 13 \\ 2 & 4 & 0 & 11 & 0 & 2 \\ 3 & 4 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{131}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 443, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 15 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 0 & 39 & 11 & 11 \\ 2 & 4 & 0 & 11 & 0 & 2 \\ 3 & 4 & -3 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{141}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 560, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 0 & 64 & 14 & 14 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 3 & 4 & -3 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 635, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 11 & 12 \\ 2 & 4 & 0 & 11 & 0 & 2 \\ 3 & 4 & -3 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{121}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 832, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 21 & 0 & 80 & 14 & 16 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 3 & 4 & -3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{121}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 908, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 0 & 31 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 18 & 31 & 0 & 188 & 20 & 24 \\ 2 & 4 & 0 & 20 & 0 & 2 \\ 3 & 4 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{131}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 908, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.01 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 14 & 15 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 3 & 4 & -3 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{121}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 923, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 26 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 26 & 0 & 128 & 17 & 20 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 3 & 4 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{141}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 0 & 100 & 17 & 18 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 3 & 4 & -3 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{131}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 25 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 14 & 25 & 0 & 111 & 17 & 19 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 3 & 4 & -3 & 19 & 2 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{131}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1520, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 30 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 30 & 0 & 165 & 20 & 23 \\ 2 & 4 & 0 & 20 & 0 & 2 \\ 3 & 4 & -3 & 23 & 2 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{131}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1667, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 0 & 35 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 20 & 35 & 0 & 231 & 23 & 27 \\ 2 & 4 & 0 & 23 & 0 & 2 \\ 3 & 4 & -3 & 27 & 2 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{141}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1708, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 29 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 16 & 29 & 0 & 148 & 20 & 22 \\ 2 & 4 & 0 & 20 & 0 & 2 \\ 3 & 4 & -3 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{141}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 2123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 2 & 3 \\ 3 & 4 & 0 & 34 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 19 & 34 & 0 & 208 & 23 & 26 \\ 2 & 4 & 0 & 23 & 0 & 2 \\ 3 & 4 & -3 & 26 & 2 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{141}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 2432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 2 & 3 \\ 3 & 4 & 0 & 39 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 22 & 39 & 0 & 280 & 26 & 30 \\ 2 & 4 & 0 & 26 & 0 & 2 \\ 3 & 4 & -3 & 30 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [32] $1^{111}3^121^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 37 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 37 & 54 & 0 & 217 & 27 \\ 5 & 7 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

[33] $1^{11}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 8 & 3 & 4 \\ 1 & 4 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 8 & 4 & 0 & 9 & 4 & 5 \\ 1 & 3 & 3 & 0 & 4 & 0 & 2 \\ 2 & 4 & 2 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{121}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 0 & 7 & 3 & 3 \\ 2 & 4 & 3 & 0 & 8 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 7 & 8 & 0 & 15 & 7 & 7 \\ 1 & 3 & 3 & 0 & 7 & 0 & 2 \\ 2 & 3 & 3 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{111}3^11^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 5 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 6 & 0 & 7 & 5 & 5 \\ 1 & 3 & 3 & 0 & 5 & 0 & 2 \\ 2 & 3 & 3 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{121}3^121^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 44 & 5 \\ 9 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 44 & 69 & 0 & 340 & 34 \\ 5 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{111}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 13 & 9 & 1 & 9 & 7 \\ 9 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 7 & 0 & 4 & 4 \\ 7 & 5 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{121}3^121^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 56 & 5 \\ 9 & 13 & 1 & 84 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 56 & 84 & 0 & 520 & 42 \\ 5 & 7 & -1 & 42 & 3 \end{pmatrix}_{b_0}$$

[38] $1^{111}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 25 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 39 & 0 & 107 & 19 \\ 5 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{111}3^{11}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 5 & 3 \\ 1 & 4 & 0 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 \\ 2 & 5 & 4 & 0 & 4 & 4 \\ 1 & 3 & 3 & 0 & 4 & 0 \\ 2 & 4 & 2 & -3 & 4 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{111}3^{11}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 1 \\ 1 & 1 & 4 & 0 & 4 & 3 \\ 2 & 4 & 3 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 \\ 2 & 4 & 5 & 0 & 4 & 4 \\ 1 & 3 & 3 & 0 & 4 & 0 \\ 2 & 3 & 3 & -3 & 4 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{111}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 15 & 5 \\ 9 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 24 & 0 & 37 & 11 \\ 5 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{11}3^{11}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 5 & 6 \\ 1 & 4 & 0 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 \\ 2 & 5 & 4 & 0 & 4 & 5 \\ 2 & 6 & 3 & 0 & 5 & 3 \\ 2 & 4 & 2 & -3 & 4 & 3 \end{pmatrix}_{b_0}$$

[44] $1^{11}3^{11}43^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 1 \\ 2 & 4 & 4 & 0 & 27 & 3 \\ 1 & 4 & 0 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 \\ 9 & 27 & 14 & 0 & 120 & 14 \\ 1 & 3 & 3 & 0 & 14 & 0 \\ 2 & 4 & 2 & -3 & 18 & 2 \end{pmatrix}_{b_0}$$

[45] $1^{11}3^{11}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 \\ 2 & 4 & 4 & 0 & 9 & 3 \\ 1 & 4 & 0 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 \\ 3 & 9 & 5 & 0 & 12 & 5 \\ 1 & 3 & 3 & 0 & 5 & 0 \\ 2 & 4 & 2 & -3 & 6 & 2 \end{pmatrix}_{b_0}$$

[46] $1^{121}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 24 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 39 & 0 & 100 & 18 \\ 5 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[47] $1^{11}3^{11}3^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 1 \\ 2 & 4 & 4 & 0 & 21 & 3 \\ 1 & 4 & 0 & 0 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 \\ 7 & 21 & 11 & 0 & 72 & 11 \\ 1 & 3 & 3 & 0 & 11 & 0 \\ 2 & 4 & 2 & -3 & 14 & 2 \end{pmatrix}_{b_0}$$

[48] $1^{11}3^{11}23^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -261, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 15 & 8 & 0 & 36 & 8 & 10 \\ 1 & 3 & 3 & 0 & 8 & 0 & 2 \\ 2 & 4 & 2 & -3 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{111} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -272, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.04 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 26 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 39 & 0 & 107 & 19 \\ 5 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{111} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -272, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.18 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 16 & 5 \\ 9 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 24 & 0 & 37 & 11 \\ 5 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{111} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -296, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.16 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 17 & 5 \\ 9 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 24 & 0 & 37 & 11 \\ 5 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{121} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -320, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.15 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 34 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 54 & 0 & 200 & 26 \\ 5 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{111} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -336, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.04 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 27 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 39 & 0 & 107 & 19 \\ 5 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{111} 31^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -354, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.20 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 0 & 5 & 9 & 4 \\ 1 & 4 & 0 & 0 & 4 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 9 & 6 & 0 & 9 & 12 & 6 \\ 2 & 4 & 2 & -3 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[55] \ 1^{121} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -384, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.03 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 45 & 5 \\ 9 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 45 & 69 & 0 & 340 & 34 \\ 5 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{111} 31^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -430, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.33 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 0 & 11 & 3 & 4 \\ 1 & 4 & 0 & 0 & 7 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 11 & 7 & 0 & 20 & 7 & 8 \\ 1 & 3 & 3 & 0 & 7 & 0 & 2 \\ 2 & 4 & 2 & -3 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{121} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 25 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 39 & 0 & 100 & 18 \\ 5 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{121} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -584, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 35 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 35 & 54 & 0 & 200 & 26 \\ 5 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{121} 3^{121} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -608, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.18 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 26 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 39 & 0 & 100 & 18 \\ 5 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{121} 3^{121} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -608, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.02 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 46 & 5 \\ 9 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 46 & 69 & 0 & 340 & 34 \\ 5 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{111} 31^{123} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -679, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.29 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 0 & 17 & 3 & 4 \\ 1 & 4 & 0 & 0 & 10 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 17 & 10 & 0 & 48 & 10 & 12 \\ 1 & 3 & 3 & 0 & 10 & 0 & 2 \\ 2 & 4 & 2 & -3 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{121} 3^{121} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -768, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.06 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 36 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 36 & 54 & 0 & 200 & 26 \\ 5 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{111} 31^{133} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -886, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.28 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 0 & 23 & 3 & 4 \\ 1 & 4 & 0 & 0 & 13 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 23 & 13 & 0 & 88 & 13 & 16 \\ 1 & 3 & 3 & 0 & 13 & 0 & 2 \\ 2 & 4 & 2 & -3 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{111} 31^{143} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -1051, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.28 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 0 & 29 & 3 & 4 \\ 1 & 4 & 0 & 0 & 16 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 29 & 16 & 0 & 140 & 16 & 20 \\ 1 & 3 & 3 & 0 & 16 & 0 & 2 \\ 2 & 4 & 2 & -3 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [65] \ 1^{111} 4^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -88, \ \det = 24, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{111} 4^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -88, \ \det = 56, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 7 & 0 & 8 & 5 & 6 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[67] \ 1^{13} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -88, \ \det = 79, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 2 & 3 \\ 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{11} 1^{13} 1^1 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 105, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 1 & 6 & 2 & 3 \\ 4 & 4 & 0 & 8 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 0 & 13 & 6 & 7 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{111} 1^1 4^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{11} 1^{13} 1^1 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 1 & 4 & 2 & 3 \\ 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

[71] $1^{111} 1^{13} 1^1 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 12 & 0 & 26 & 8 & 10 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{111} 1^1 4^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 11$ [73] $1^{11} 1^{13} 1^1 5^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 14, \Delta = -83, \det = -89, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 5 & 9 & 14 & 1 & 5 \\ 9 & 13 & 22 & 0 & 7 \\ 14 & 22 & 33 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 11 & -1 & 3 \end{pmatrix}_{b_0}$$

[74] $1^{12} 1^{13} 1^1 5^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 14, \Delta = -83, \det = -126, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 5 & 9 & 20 & 1 & 5 \\ 9 & 13 & 32 & 0 & 7 \\ 20 & 32 & 70 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

[75] $1^{13} 1^{13} 1^1 5^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 14, \Delta = -83, \det = -145, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 5 & 9 & 26 & 1 & 5 \\ 9 & 13 & 42 & 0 & 7 \\ 26 & 42 & 121 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

4.533 NHC(-2)(-3) + f₄ + su₂n3mix + su₃ + su₂

$T = 9$ [0] $1^{11} 1^{13} 1^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 & 3 \\ 3 & 4 & 0 & 8 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 8 & 0 & 12 & 5 & 9 & 6 \\ 2 & 4 & 0 & 5 & 0 & 3 & 2 \\ 3 & 5 & 0 & 9 & 3 & 4 & 4 \\ 3 & 4 & -3 & 6 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}1} 3^{\overline{1}} 1^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 10 & 17 & 0 & 55 & 11 & 11 & 13 \\ 2 & 4 & 0 & 11 & 0 & 3 & 2 \\ 2 & 4 & 0 & 11 & 3 & 1 & 3 \\ 3 & 4 & -3 & 13 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}1} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 & 3 \\ 3 & 4 & 0 & 7 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 6 & 5 \\ 2 & 4 & 0 & 5 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & -3 & 5 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{121}1} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 & 2 & 3 \\ 3 & 4 & 0 & 11 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 9 & 8 & 8 \\ 3 & 5 & 0 & 9 & 3 & 3 & 3 \\ 2 & 4 & 0 & 8 & 3 & 1 & 3 \\ 3 & 4 & -3 & 8 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}1} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 2 & 3 \\ 3 & 4 & 0 & 7 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 6 & 5 & 5 \\ 3 & 5 & 0 & 6 & 3 & 3 & 3 \\ 2 & 4 & 0 & 5 & 3 & 1 & 3 \\ 3 & 4 & -3 & 5 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}1} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 8 & 0 & 12 & 5 & 14 & 6 \\ 2 & 4 & 0 & 5 & 0 & 6 & 2 \\ 5 & 9 & 0 & 14 & 6 & 13 & 7 \\ 3 & 4 & -3 & 6 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{131}1} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 & 3 \\ 3 & 4 & 0 & 20 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 11 & 20 & 0 & 69 & 14 & 18 & 15 \\ 2 & 4 & 0 & 14 & 0 & 3 & 2 \\ 3 & 5 & 0 & 18 & 3 & 4 & 4 \\ 3 & 4 & -3 & 15 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}1} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 & 3 \\ 3 & 4 & 0 & 12 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 8 & 12 & 9 \\ 2 & 4 & 0 & 8 & 0 & 3 & 2 \\ 3 & 5 & 0 & 12 & 3 & 4 & 4 \\ 3 & 4 & -3 & 9 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}1} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 5 & 5 \\ 2 & 4 & 0 & 5 & 0 & 3 & 2 \\ 2 & 4 & 0 & 5 & 3 & 1 & 3 \\ 3 & 4 & -3 & 5 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 2 & 3 \\ 3 & 4 & 0 & 7 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 11 & 5 & 5 \\ 5 & 9 & 0 & 11 & 12 & 6 & 6 \\ 2 & 4 & 0 & 5 & 6 & 1 & 3 \\ 3 & 4 & -3 & 5 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -481, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 8 & 8 & 9 \\ 2 & 4 & 0 & 8 & 0 & 3 & 2 \\ 2 & 4 & 0 & 8 & 3 & 1 & 3 \\ 3 & 4 & -3 & 9 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1211}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 & 3 \\ 3 & 4 & 0 & 16 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 16 & 0 & 44 & 11 & 15 & 12 \\ 2 & 4 & 0 & 11 & 0 & 3 & 2 \\ 3 & 5 & 0 & 15 & 3 & 4 & 4 \\ 3 & 4 & -3 & 12 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 5 & 11 & 5 \\ 2 & 4 & 0 & 5 & 0 & 6 & 2 \\ 5 & 9 & 0 & 11 & 6 & 13 & 7 \\ 3 & 4 & -3 & 5 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 5 & 3 \\ 3 & 4 & 0 & 12 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 0 & 25 & 8 & 20 & 9 \\ 2 & 4 & 0 & 8 & 0 & 6 & 2 \\ 5 & 9 & 0 & 20 & 6 & 13 & 7 \\ 3 & 4 & -3 & 9 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.534 NHC(-2)(-3) + $\mathfrak{f}_4$ + su2n3mix + $\mathfrak{su}_3$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 8 & 0 & 12 & 5 & 5 & 6 \\ 2 & 4 & 0 & 5 & 0 & 3 & 2 \\ 2 & 4 & 0 & 5 & 3 & 0 & 2 \\ 3 & 4 & -3 & 6 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -81, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 0 & 5 \\ 2 & 4 & -2 & 5 & 0 & 0 & 0 \\ 5 & 12 & 5 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & 6 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] 1^{11231^1} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 4 & 3 \\ 3 & 4 & 0 & 7 & 5 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 6 & 10 & 5 \\ 3 & 5 & 0 & 6 & 3 & 6 & 3 \\ 4 & 8 & 0 & 10 & 6 & 9 & 5 \\ 3 & 4 & -3 & 5 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] 1^{12131^1} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 & 2 & 3 \\ 3 & 4 & 0 & 11 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 6 & 11 & 0 & 20 & 9 & 8 & 8 \\ 3 & 5 & 0 & 9 & 3 & 3 & 3 \\ 2 & 4 & 0 & 8 & 3 & 0 & 2 \\ 3 & 4 & -3 & 8 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] 1^{11131^1} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 2 & 3 \\ 3 & 4 & 0 & 7 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 7 & 0 & 7 & 6 & 5 & 5 \\ 3 & 5 & 0 & 6 & 3 & 3 & 3 \\ 2 & 4 & 0 & 5 & 3 & 0 & 2 \\ 3 & 4 & -3 & 5 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

4.535 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_3$

$$T = 7 \quad [0] 1^{11131^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 3 & 4 & -2 & 3 & 0 \\ 3 & 8 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] 21^{131^{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 3 & 4 & -2 & 3 & 0 \\ 3 & 8 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] 1^{11131^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 \\ 5 & 8 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[3] 1^{131^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 97, T \in [14, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 10 & 0 & 21 & 7 \\ 3 & 4 & -3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[4] 1^{1131^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[5] 1^{131^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 \\ 3 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] 1^{1131^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{12}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 4 & 8 & 0 & 9 & 5 \\ 3 & 4 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{13}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 12 & 0 & 24 & 8 \\ 3 & 4 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{14}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.10 \rightarrow 14, T_{\max} = 97, T \in [14, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 16 & 0 & 45 & 11 \\ 3 & 4 & -3 & 11 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{15}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 97, T \in [14, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 10 & 20 & 0 & 72 & 14 \\ 3 & 4 & -3 & 14 & -1 \end{pmatrix}_{b_0}$$

[11] $1^113^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -93, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 \\ 3 & 8 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{111}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 \\ 4 & 8 & 3 & 3 & 3 \\ 3 & 5 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{11}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 5 & 9 & 0 & 12 & 6 \\ 3 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{11}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 14 & 0 & 36 & 10 \\ 3 & 4 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{12}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -169, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 13 & 0 & 27 & 9 \\ 3 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{13}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -228, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 17 & 0 & 48 & 12 \\ 3 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{12}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -259, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 10 & 18 & 0 & 57 & 13 \\ 3 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{14}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -289, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 11 & 21 & 0 & 75 & 15 \\ 3 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{13}3^{11}2^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -372, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 12 & 22 & 0 & 84 & 16 \\ 3 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[20] $21^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -379, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 4 \\ 8 & 9 & 4 & 8 & 5 \\ 3 & 4 & -2 & 3 & 0 \\ 6 & 8 & 3 & 3 & 3 \\ 4 & 5 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{14}3^{11}2^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -487, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -5 & 0 & -3 \\ 14 & 26 & 0 & 117 & 19 \\ 3 & 4 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

$T = 8$ [22] $1^{11}3^{11}2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 15, T_{\max} = 35, T \in [15, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 \\ 1 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

[23] $1^{11}3^{11}2^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 35, T \in [15, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 5 & 0 & 10 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 4 & 10 & 0 & 13 & 7 \\ 3 & 5 & -3 & 7 & -1 \end{pmatrix}_{b_0}$$

[24] $1^{11}3^{11}2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 14.11 \rightarrow 15, T_{\max} = 35, T \in [15, 35]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 5 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

[25] $1^{11}3^{11}2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 5 & 13 & 9 & 5 \\ 13 & 30 & 20 & 10 \\ 9 & 20 & 12 & 6 \\ 5 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{11}3^{11}2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 5 & 13 & 1 & 5 \\ 13 & 30 & 0 & 10 \\ 1 & 0 & -3 & -1 \\ 5 & 10 & -1 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{11}3^{11}2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.14 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 2 \\ 2 & 2 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{12}3^{11}2^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 14.14 \rightarrow 15, T_{\max} = 35, T \in [15, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 2 & 0 & 5 & 3 \\ 2 & 2 & 2 & -3 & 3 & -2 \end{pmatrix}_{b_0}$$

[29] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 35, T \in [14, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 \\ 3 & 5 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 10 & 18 & 12 & 6 \\ 18 & 30 & 20 & 10 \\ 12 & 20 & 12 & 6 \\ 6 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

[31] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -3 & -1 \\ 5 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 5 & 0 & 1 & 3 \\ 3 & 5 & -3 & 3 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{12}321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 12.54 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 1 & 6 & 3 \\ 1 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 4 & 6 & 6 & 2 & 9 & 5 \\ 2 & 3 & 2 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{11}321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.35 \rightarrow 14, T_{\max} = 35, T \in [14, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 5 & 10 & 0 & 32 & 10 \\ 2 & 2 & 2 & -3 & 10 & -1 \end{pmatrix}_{b_0}$$

[37] $1^{12}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[39] $1^{13}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 35, T \in [14, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & 0 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{11} 3 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 3 & 3 \\ 1 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 3 & 3 & 1 & 0 & 2 \\ 2 & 3 & 2 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{11} 3 1^1 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 6 & 0 & 12 & 6 \\ 2 & 3 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{11} 3 1^1 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{11} 3 1 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 83, T \in [14, 83]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 2 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1 3 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 35, T \in [14, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 3 & 3 \\ 1 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 1 & 3 & 3 & 1 & 0 & 2 \\ 2 & 3 & 2 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[45] \quad 2 2 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 10 & 19 & 12 & 6 \\ 19 & 30 & 20 & 10 \\ 12 & 20 & 12 & 6 \\ 6 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{11} 3 1^1 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{11} 3 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 5 & 0 & 10 & 5 \\ 1 & 0 & -5 & 0 & -3 \\ 5 & 10 & 0 & 13 & 7 \\ 3 & 5 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{11} 3 2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 13 & 14 & 1 & 7 \\ 14 & 13 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 7 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{11} 3 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 4 \\ 2 & 2 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{11} 3 2 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 4 & 7 & 0 & 15 & 7 \\ 2 & 2 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{11} 13 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 13.57 \rightarrow 14, T_{\max} = 35, T \in [14, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 6 & 3 \\ 1 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 2 & 6 & 6 & 2 & 9 & 5 \\ 2 & 3 & 2 & -1 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{11} 13 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 5 \\ 3 & 5 & 0 & 15 & 5 & 3 \\ 1 & 0 & -5 & 0 & -3 & 3 \\ 7 & 15 & 0 & 31 & 11 & 11 \\ 3 & 5 & -3 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{11} 13 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 5 & 0 & 25 & 5 & 5 \\ 1 & 0 & -5 & 0 & -3 & 3 \\ 11 & 25 & 0 & 91 & 19 & 19 \\ 3 & 5 & -3 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{11} 13 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 5 \\ 3 & 5 & 0 & 20 & 5 & 5 \\ 1 & 0 & -5 & 0 & -3 & 3 \\ 9 & 20 & 0 & 57 & 15 & 15 \\ 3 & 5 & -3 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{11} 13 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 1 & 6 & 3 \\ 1 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 3 & 6 & 6 & 2 & 9 & 5 \\ 2 & 3 & 2 & -1 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{12} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 3 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{12} 13 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 0 & 12 & 6 \\ 2 & 2 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{13} 11 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{13} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 15 & 5 \\ 2 & 2 & -1 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{13} 13 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 0 & 24 & 8 \\ 2 & 2 & 2 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{11}31^{11}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{12}31^{11}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 9 & 0 & 27 & 9 \\ 2 & 4 & 2 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{12}321^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 0 & 25 & 9 \\ 2 & 2 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{12}321^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 7 & 11 & 0 & 44 & 12 \\ 2 & 2 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{11}31^{11}31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 219, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 0 & 12 & 6 \\ 2 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{13}321^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 283, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 9 & 0 & 39 & 11 \\ 2 & 2 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[67] \quad 2^2 2^1 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 353, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 \\ 2 & 1 & 4 & 1 & 3 & 3 \\ 2 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 3 & 3 & 1 & 0 & 2 \\ 2 & 3 & 2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{13}321^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 363, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 12 & 0 & 60 & 14 \\ 2 & 2 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[69] \quad 2^2 2^1 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 4 & 2 \\ 2 & 1 & 4 & 1 & 6 & 3 \\ 2 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -3 & 2 & -1 \\ 4 & 6 & 6 & 2 & 9 & 5 \\ 2 & 3 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [70] \quad 1^{13}3221^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 6 & 6 \\ 1 & 0 & -5 & 0 & -3 \\ 2 & 6 & 0 & 2 & 4 \\ 3 & 6 & -3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{11} 3^{11} 221^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -89, \ \det = -12, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 6 & 0 & 23 & 6 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 23 & 0 & 66 & 18 \\ 3 & 6 & -3 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{11} 2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -17, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 3 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{11} 3^{11} 221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -89, \ \det = -18, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.44 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 4 & 9 & 0 & 19 & 9 \\ 2 & 2 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{11} 2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -20, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 1 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{112} 231^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -21, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.57 \rightarrow 14, \ T_{\max} = 15, \ T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 2 & 5 & 0 & 11 & 4 \\ 1 & 5 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 11 & 11 & 0 & 31 & 11 \\ 2 & 4 & 3 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{11} 2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -24, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 3 & 0 & 1 & 0 & 2 \\ 2 & 3 & -1 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[77] \ 1^{11} 4^{131} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 9, \ T_{\max} = 29, \ \Delta = -116, \ \det = -27, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.37 \rightarrow 14, \ T_{\max} = 29, \ T \in [14, 29]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[78] \ 1^{11} 3^{11} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -89, \ \det = -29, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.59 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{11} 4^{131} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 9, \ T_{\max} = 29, \ \Delta = -116, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.38 \rightarrow 14, \ T_{\max} = 29, \ T \in [14, 29]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 5 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[80] \ 1^{12} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 0 & 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

- [79] $1^{11}3221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -89$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$
- [82] $1^{11}231^{131}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -36$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 13$, $T_{\max} = 15$, $T \in [13, 15]$
 $\begin{pmatrix} 1 & 1 & 1 & 6 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 0 & 0 & 12 & 6 \\ 3 & 0 & -1 & 6 & 2 \end{pmatrix}_{b_0}$
- [83] $1^{12}3221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -89$, $\det = -39$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.4$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 11 & 0 & 49 & 13 \\ 3 & 3 & -1 & 13 & 2 \end{pmatrix}_{b_0}$
- [85] $1^{11}23131^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -41$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 14.17 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 1 & 1 & 1 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 0 & -1 & -1 & 0 \end{pmatrix}_{b_0}$
- [84] $1^{111}23131^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -41$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 14.17 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 5 & 0 & 4 & 4 \\ 1 & 5 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 4 & 4 & 0 & 1 & 3 \\ 2 & 4 & 3 & -3 & 3 & -1 \end{pmatrix}_{b_0}$
- [86] $1^{11}231^{131}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$
 $\begin{pmatrix} 1 & 1 & 5 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 5 & 1 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 0 & 6 & -1 & 2 \end{pmatrix}_{b_0}$
- [88] $1^{11}4131^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 29$, $\Delta = -116$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 13$, $T_{\max} = 29$, $T \in [13, 29]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$
- [87] $1^{11}3221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -89$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 6 & 0 & 11 & 6 \\ 1 & 0 & -5 & 0 & -3 \\ 5 & 11 & 0 & 14 & 8 \\ 3 & 6 & -3 & 8 & 2 \end{pmatrix}_{b_0}$
- [89] $1^{11}3221^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -89$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.18 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 6 & 0 & 17 & 6 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 17 & 0 & 35 & 13 \\ 3 & 6 & -3 & 13 & 2 \end{pmatrix}_{b_0}$
- [92] $1^{11}3221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -89$, $\det = -54$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 5 & 0 & 5 & 5 \\ 2 & 2 & 3 & -3 & 5 & 2 \end{pmatrix}_{b_0}$
- [91] $1^{11}3221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -89$, $\det = -54$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 8 & 0 & 30 & 10 \\ 3 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$
- [90] $1^{11}3221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -89$, $\det = -54$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

- $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & 0 & 2 \\ 3 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$
- [93] $1^{1231}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$
 $\begin{pmatrix} 1 & 5 & 1 & 1 & 3 \\ 5 & 13 & 1 & 0 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & -1 & -1 & 2 \end{pmatrix}_{b_0}$
- [94] $1^{12}2^{11}3^{21}1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.4$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 2 & 0 \\ 3 & 0 & 6 & 2 & 9 & 5 \\ 2 & 0 & 2 & 0 & 5 & 1 \end{pmatrix}_{b_0}$
- [95] $1^{112}23131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.4$
 $T_{\min} = 14.93 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 5 & 0 & 8 & 4 \\ 1 & 5 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 8 & 8 & 0 & 13 & 7 \\ 2 & 4 & 3 & -3 & 7 & -1 \end{pmatrix}_{b_0}$
- [96] $1^{11}2^{11}3^{21}1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -60$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 2 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & 0 & 2 & 1 \end{pmatrix}_{b_0}$
- [97] $1^{13}3^{221}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -89$, $\det = -74$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 14 & 0 & 74 & 16 \\ 3 & 3 & -1 & 16 & 2 \end{pmatrix}_{b_0}$
- [98] $1^{12}3^{221}1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -89$, $\det = -77$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.4$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 6 & 0 & 9 & 5 \\ 3 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$
- [99] $1^{12}3^{221}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -89$, $\det = -79$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.4$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 7 & 14 & 0 & 53 & 15 \\ 2 & 2 & 3 & -3 & 15 & 2 \end{pmatrix}_{b_0}$
- [102] $1^{11}41^{1231}1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & -2 & 2 \end{pmatrix}_{b_0}$
- [101] $1^{11}4^{131}1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 29$, $\Delta = -116$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 29$, $T \in [13, 29]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 7 & 0 & 7 & 5 \\ 3 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$
- [100] $1^{131}4^{11}1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 29$, $\Delta = -116$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 29$, $T \in [13, 29]$
 $\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$
- [103] $1^{1231}1^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{12} 3^{11} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 6 & 0 & 13 & 7 \\ 2 & 2 & 3 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{11} 2^3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 3 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[107] \ 1^{13} 3^{11} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 24 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{13} 3^{11} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 7 & 0 & 25 & 9 \\ 2 & 2 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{12} 31^{11} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 5 & 1 & 5 & 3 \\ 5 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 15 & 0 & 12 & 6 \\ 3 & 7 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[108] \ 1^{12} 31^{11} 31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 5 & 1 & 7 & 3 \\ 5 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 15 & 0 & 12 & 6 \\ 3 & 7 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{12} 3^{11} 221^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 10 & 0 & 29 & 11 \\ 2 & 2 & 3 & -3 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{12} 3^{11} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 11 & 0 & 49 & 13 \\ 3 & 3 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{11} 231^{11} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 7 & 0 & 13 & 7 \\ 2 & 4 & 3 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{12} 2^3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 2 & 0 \\ 4 & 0 & 6 & 2 & 9 & 5 \\ 2 & 0 & 2 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

[114] $1^{13}2^13^121^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 3 & 0 \\ 5 & 0 & 9 & 3 & 24 & 8 \\ 2 & 0 & 2 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[115] $1^{1231}1^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 5 & 1 & 6 & 3 \\ 5 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 15 & 0 & 12 & 6 \\ 3 & 7 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[116] $1^{13}3^1221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -178, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 14 & 0 & 74 & 16 \\ 3 & 3 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[117] $1^{13}3^1221^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 1 & 0 & 19 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 10 & 19 & 0 & 103 & 21 \\ 2 & 2 & 3 & -3 & 21 & 2 \end{pmatrix}_{b_0}$$

[118] $1^{13}3^1221^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 11 & 0 & 43 & 13 \\ 2 & 2 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[119] $1^{13}3^1221^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 15 & 0 & 69 & 17 \\ 2 & 2 & 3 & -3 & 17 & 2 \end{pmatrix}_{b_0}$$

[120] $1^{141}1^{11}231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 1 & 6 & 1 & 5 & 3 \\ 6 & 13 & 0 & 12 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 12 & 0 & 7 & 5 \\ 3 & 7 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$T = 10$ [122] $1^{1231}1^{41}1^{31}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 8 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 7 & 0 & 15 & 7 \\ 2 & 4 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[121] $1^{131}1^{41}1^{31}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 13 & 21 & 1 & 7 \\ 21 & 33 & 1 & 11 \\ 1 & 1 & -2 & 0 \\ 7 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

[123] $1^{1131}1^{41}1^{31}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.29 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 5 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 4 & 0 & 4 & 4 \\ 2 & 4 & 2 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

[124] $1^{115}1^{131}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 13.60 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{aligned}
& \begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 0 & 7 \\ 1 & 0 & -3 & -1 \\ 5 & 7 & -1 & 3 \end{pmatrix}_{b_0} \\
[125] & 1^{\overline{11}} 31^{\overline{1}} 4131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 13.57 \rightarrow 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0} \\
[126] & 1^{\overline{1}} 5^{\overline{1}} 1^{\overline{1}} 231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 147, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 13.74 \rightarrow 14, T_{\max} = 24, T \in [14, 24] \\
& \begin{pmatrix} 5 & 23 & 1 & 5 \\ 23 & 90 & 0 & 18 \\ 1 & 0 & -3 & -1 \\ 5 & 18 & -1 & 3 \end{pmatrix}_{b_0} \\
T = 11 & [127] 1^{\overline{11}} 31^{\overline{1}} 5^{\overline{1}} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -90, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 14.40 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 1 & 0 & 3 & 1 & 0 & 2 \\ 2 & 0 & 2 & 0 & 2 & 1 \end{pmatrix}_{b_0} \\
[128] & 1^{\overline{11}} 31^{\overline{1}} 5^{\overline{1}} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -90, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 14, T_{\max} = 15, T \in [14, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 3 & 0 & 1 & 0 & 2 \\ 2 & 3 & -1 & 0 & 2 & 2 \end{pmatrix}_{b_0} \\
[129] & 1^{\overline{1}} 3151^{\overline{1}} 231^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 14.40 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 8 & 1 & 1 & 3 \\ 8 & 38 & 1 & 0 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 12 & -1 & -1 & 2 \end{pmatrix}_{b_0} \\
T = 12 & [131] 1^{\overline{1}} 6^{\overline{1321}} 131^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 15.40 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 1 & 3 & 0 & 0 & 2 \\ 2 & 2 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0} \\
[130] & 1^{\overline{1}} 6^{\overline{1321}} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 15.40 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0} \\
[132] & 1^{\overline{1}} 6^{\overline{131}} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -87, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 14.83 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 0 & -1 & 2 \end{pmatrix}_{b_0} \\
[133] & 1^{\overline{1}} 6^{\overline{131}} 131^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -87, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 2 & 1 & 3 & 1 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0} \\
[134] & 1^{\overline{11}} 321^{\overline{1}} 61231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 \quad T = 12, T_{\max} = 16, \Delta = -89, \det = 50, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 15.40 \rightarrow 16, T_{\max} = 16, T \in [16, 16]
\end{aligned}$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 1 & 2 \\ 3 & 10 & 5 & 0 & 4 & 6 \\ 1 & 5 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 4 & 3 & 0 & 0 & 2 \\ 2 & 6 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[135] $31^{\overset{1}{11}}316131^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 17, \Delta = -95, \det = 193, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 15.21 \rightarrow 16, T_{\max} = 17, T \in [16, 17]$
 $\begin{pmatrix} 5 & 8 & 8 & 5 & 5 \\ 8 & 6 & 9 & 6 & 5 \\ 8 & 9 & 5 & 5 & 5 \\ 5 & 6 & 5 & 5 & 3 \end{pmatrix}_{b_0}$

$T = 14$ [136] $1^{\overset{1}{1}}6^{\overset{1}{15}131}131^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 14, T_{\max} = 17, \Delta = -81, \det = 20, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 16.60 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 2 \end{pmatrix}_{b_0}$

$T = 16$ [137] $1^{\overset{1}{1}}8^{\overset{1}{15}1321}1231^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 16, T_{\max} = 19, \Delta = -80, \det = 20, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 18.60 \rightarrow 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 2 \end{pmatrix}_{b_0}$

4.536 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{f}_4$

$T = 8$ [0] $1^{\overset{1}{11}}31^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 101, \Delta = -169, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.4$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$
 $\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 0 & 3 & -3 & -1 \end{pmatrix}_{b_0}$

$T = 9$ [1] $1^{\overset{1}{11}}321^{\overset{1}{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$
 $T_{\min} = 13.83 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$
 $\begin{pmatrix} 5 & 13 & 9 & 1 & 5 \\ 13 & 30 & 20 & 0 & 10 \\ 9 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 10 & 6 & -3 & 1 \end{pmatrix}_{b_0}$

[3] $1^{\overset{1}{1}}321^{\overset{1}{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 3 & 0 & 3 \\ 1 & 4 & 0 & 1 & 3 & 0 & 2 \\ 1 & 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 3 & 3 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & -1 & 2 & -3 & -1 \end{pmatrix}_{b_0}$

4.537 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$

$T = 9$ [0] $1^{\overset{1}{11}}31^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & 0 & -1 & 4 & -1 & 0 \end{pmatrix}_{b_0}$

[1] $1^{\overset{1}{121}}31^{\overset{1}{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 & 3 \\ 3 & 4 & 0 & 8 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 6 & 6 & 5 \\ 2 & 4 & 0 & 6 & 1 & 4 & 3 \\ 3 & 5 & 0 & 6 & 4 & 3 & 3 \\ 3 & 4 & -3 & 5 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 6 & 3 & 3 \\ 3 & 5 & 0 & 6 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -3 & 3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 4 & 3 \\ 3 & 5 & 0 & 3 & 4 & 3 & 3 \\ 3 & 4 & -3 & 2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 9 & 6 & 6 \\ 3 & 5 & 0 & 9 & 4 & 4 & 4 \\ 2 & 4 & 0 & 6 & 4 & 0 & 2 \\ 3 & 4 & -3 & 6 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

4.538 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 98, T \in [15, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 18 & 5 \\ 11 & 20 & 0 & 34 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 18 & 34 & 0 & 57 & 13 \\ 5 & 8 & -3 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 17 & 0 & 12 & 6 \\ 5 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 235, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 3 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 259, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 4 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & 3 & -2 & 0 \end{pmatrix}_{b_0}$$

- $T = 9$ $[5] 1^1 3 2 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{so}_8$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -23$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.4$
 $T_{\min} = 14.83 \rightarrow 15$, $T_{\max} = 32$, $T \in [15, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 & 3 \\ 1 & 1 & 4 & 1 & 6 & 0 & 0 & 3 \\ 1 & 4 & 0 & 1 & 6 & 0 & 0 & 2 \\ 1 & 1 & 1 & -3 & 2 & 0 & 0 & -1 \\ 2 & 6 & 6 & 2 & 9 & 0 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 & 0 \\ 2 & 3 & 2 & -1 & 5 & -2 & -2 & 0 \end{pmatrix}_{b_0}$$
- [6] $1^{11} 3 2 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{so}_8$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -40$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 & 0 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 5 & 0 & 3 & 0 & 2 & 2 \\ 3 & 5 & -3 & 3 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$
- [7] $1^{11} 3 2 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{so}_8$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -47$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.13 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 & 2 & 2 \\ 1 & 2 & 0 & 0 & 3 & 1 & 1 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 & 0 \\ 1 & 1 & 3 & 0 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 2 & 2 & -3 & 2 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{11} 3 2 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{so}_8$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 5 & 13 & 9 & 1 & 5 & 5 \\ 13 & 30 & 20 & 0 & 10 & 10 \\ 9 & 20 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & -4 & -2 & 0 \\ 5 & 10 & 6 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [9] $1^{11} 3 2 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{so}_8$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -53$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.23 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 3 & 0 & 3 & 3 \\ 1 & 4 & 0 & 1 & 3 & 0 & 0 & 2 \\ 1 & 1 & 1 & -3 & 1 & 0 & -1 & 0 \\ 1 & 3 & 3 & 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 & 0 \\ 2 & 3 & 2 & -1 & 2 & -2 & -1 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{11} 3 2 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{so}_8$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -62$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.10 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 1 & 0 & 4 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 & 0 \\ 2 & 3 & 4 & 0 & 0 & 2 & 2 \\ 3 & 4 & 3 & -3 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{11} 3 2 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{so}_8$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -110$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.49 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 & 0 \\ 2 & 5 & 0 & 1 & 4 & 3 & 3 \\ 2 & 5 & 0 & 4 & 0 & 2 & 2 \\ 3 & 5 & -3 & 3 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{11} 3 2 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{so}_8$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -147$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.12 \rightarrow 14$, $T_{\max} = 32$, $T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 & 0 \\ 1 & 1 & 0 & -2 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 & 0 \\ 1 & 2 & 0 & 1 & 0 & -2 & 0 & 0 \\ 2 & 2 & -1 & 0 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$
- [13] $1^{11} 3 2 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{so}_8$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -240$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 & 3 \\ 3 & 5 & 0 & 10 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 & 0 \\ 5 & 10 & 0 & 13 & 7 & 7 & 7 \\ 2 & 5 & 0 & 7 & 0 & 2 & 2 \\ 3 & 5 & -3 & 7 & 2 & 1 & 0 \end{pmatrix}_{b_0}$$
- [14] $1^1 3 1^1 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{so}_8$ $T = 9$, $T_{\max} = 82$, $\Delta = -124$, $\det = -277$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 82$, $T \in [14, 82]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 0 & 3 \\ 1 & 3 & 3 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 3 & 0 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [15] \quad 1^{11} 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -90, \quad \det = 45, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.38 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 5 & 1 & 12 & 1 & 5 \\ 1 & -2 & 1 & 0 & 0 \\ 12 & 1 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 0 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11} 3^{11} 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -88, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 14, \quad T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 6 & 0 & 11 & 6 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 11 & 0 & 14 & 8 & 8 \\ 2 & 6 & 0 & 8 & 1 & 3 \\ 3 & 6 & -3 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 3^{11} 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -88, \quad \det = 98, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.61 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 5 & 8 & 1 & 1 & 5 \\ 8 & 10 & 1 & 0 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 6 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

4.539 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 26, \quad \Delta = -115, \quad \det = -83, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, \quad T_{\max} = 26, \quad T \in [14, 26]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 0 & 0 & 4 & 4 \\ 3 & 5 & 0 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 26, \quad \Delta = -115, \quad \det = -215, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, \quad T_{\max} = 26, \quad T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 15 & 6 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 26, \quad \Delta = -115, \quad \det = -264, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, \quad T_{\max} = 26, \quad T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 26, \quad \Delta = -115, \quad \det = -735, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, \quad T_{\max} = 26, \quad T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -3 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

4.540 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 32, \quad \Delta = -117, \quad \det = -392, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, \quad T_{\max} = 32, \quad T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 9 & 6 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -3 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

4.541 NHC(-2)(-3) + f₄ + su₃ + su₂

$$T = 8 \quad [0] \ 1^{111} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 15.51 \rightarrow 16, T_{\max} = 28, T \in [16, 28]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & 3 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 0 & 13 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 27 & 9 \\ 6 & 12 & 0 & 27 & 25 & 9 \\ 3 & 4 & -3 & 9 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.11 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & -2 & 3 & 0 & 0 \\ 4 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 0 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -3 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 6 & 3 \\ 3 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -3 & 3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 12 & 6 \\ 4 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

- [9] $1^{13}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 123$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 48 & 15 & 12 \\ 3 & 5 & 0 & 15 & 4 & 4 \\ 3 & 4 & -3 & 12 & 4 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 128$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 0 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 131$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 137$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 12.29 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 6 & 5 \\ 2 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$
- [13] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 140$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 4 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$
- [14] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 140$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 11 & 19 & 0 & 12 & 73 & 15 \\ 3 & 4 & -3 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$
- [15] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 141$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 12 & 5 \\ 5 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -3 & 5 & 7 & 0 \end{pmatrix}_{b_0}$$
- [16] $1^{122}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 146$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.9$
 $T_{\min} = 12.48 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 12 & 5 \\ 4 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -3 & 5 & 6 & 0 \end{pmatrix}_{b_0}$$
- [17] $1^{131}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 147$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 24 & 9 & 8 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -3 & 8 & 3 & 0 \end{pmatrix}_{b_0}$$
- [18] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 155$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 6 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -3 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

- [19] $1^{111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 159$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.34 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 0 & 36 & 9 & 10 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -3 & 10 & 3 & 0 \end{pmatrix}_{b_0}$$
- [20] $1^{123}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 161$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 18 & 5 \\ 6 & 12 & 0 & 18 & 25 & 9 \\ 3 & 4 & -3 & 5 & 9 & 0 \end{pmatrix}_{b_0}$$
- [21] $1^{122}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 162$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.9$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 8 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 24 & 5 \\ 10 & 18 & 0 & 24 & 58 & 14 \\ 3 & 4 & -3 & 5 & 14 & 0 \end{pmatrix}_{b_0}$$
- [22] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 171$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 9 & 6 \\ 3 & 5 & 0 & 9 & 4 & 4 \\ 3 & 4 & -3 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$
- [23] $1^{114}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 8 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -3 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$
- [24] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 178$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 9 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$
- [25] $1^{123}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 180$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 12 & 9 \\ 3 & 5 & 0 & 12 & 4 & 4 \\ 3 & 4 & -3 & 9 & 4 & 1 \end{pmatrix}_{b_0}$$
- [26] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 192$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 6 & 6 \\ 2 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$
- [27] $1^{123}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 197$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.16 \rightarrow 14$, $T_{\max} = 28$, $T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 15 & 3 \\ 3 & 4 & 0 & 8 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 36 & 5 \\ 15 & 27 & 0 & 36 & 133 & 21 \\ 3 & 4 & -3 & 5 & 21 & -1 \end{pmatrix}_{b_0}$$
- [28] $1^{112}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 200$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 0 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

- [29] $1^{115}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 203$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.3$
 $T_{\min} = 13.11 \rightarrow 14$, $T_{\max} = 28$, $T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 10 & 20 & 0 & 15 & 73 & 15 \\ 3 & 4 & -3 & 2 & 15 & -1 \end{pmatrix}_{b_0}$$
- [30] $1^{122}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 210$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 18 & 9 \\ 4 & 8 & 0 & 18 & 10 & 6 \\ 3 & 4 & -3 & 9 & 6 & 1 \end{pmatrix}_{b_0}$$
- [31] $1^{11}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 226$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.10 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 9 & 3 \\ 5 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -3 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [32] $1^{14}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 232$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 18 & 3 \\ 11 & 21 & 0 & 18 & 76 & 16 \\ 3 & 4 & -3 & 3 & 16 & 0 \end{pmatrix}_{b_0}$$
- [33] $1^{116}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 236$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 13.42 \rightarrow 14$, $T_{\max} = 28$, $T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 12 & 24 & 0 & 18 & 106 & 18 \\ 3 & 4 & -3 & 2 & 18 & -1 \end{pmatrix}_{b_0}$$
- [34] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 240$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$
- [35] $1^{111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 243$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$
- [36] $1^{114}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 248$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$
- [37] $1^{112}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 266$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 7 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 0 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$
- [38] $1^{117}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 275$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 13.69 \rightarrow 14$, $T_{\max} = 28$, $T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 14 & 28 & 0 & 21 & 145 & 21 \\ 3 & 4 & -3 & 2 & 21 & -1 \end{pmatrix}_{b_0}$$

[39] $1^{12}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 292$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.05 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 12 & 3 \\ 7 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -3 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{13}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 294$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.06 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 15 & 3 \\ 9 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -3 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 306$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 9 & 9 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 312$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.9$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 18 & 5 \\ 7 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -3 & 5 & 10 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{115}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 338$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.3$
 $T_{\min} = 13.06 \rightarrow 14$, $T_{\max} = 28$, $T \in [14, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & -2 & 0 & 15 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 40 & 15 & 0 & 148 & 20 \\ 3 & 5 & 0 & -1 & 20 & -1 \end{pmatrix}_{b_0}$$

[44] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 340$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 371$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 15 & 6 \\ 5 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{121}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 381$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 0 & 57 & 12 & 13 \\ 2 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -3 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 411$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 7 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{131}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 424$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 48 & 12 & 12 \\ 2 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -3 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

- [49] $1^{113} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 436$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.17 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & -1 & 12 & 0 \end{pmatrix}_{b_0}$
- [50] $1^{121} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 450$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 21 & 9 \\ 5 & 9 & 0 & 21 & 13 & 7 \\ 3 & 4 & -3 & 9 & 7 & 1 \end{pmatrix}_{b_0}$
- [51] $1^{131} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 463$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 48 & 27 & 12 \\ 5 & 9 & 0 & 27 & 13 & 7 \\ 3 & 4 & -3 & 12 & 7 & 1 \end{pmatrix}_{b_0}$
- [52] $1^{123} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 489$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 24 & 5 \\ 9 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -3 & 5 & 13 & 0 \end{pmatrix}_{b_0}$
- [53] $1^{113} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 490$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.09 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 11 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 0 & -1 & 12 & 0 \end{pmatrix}_{b_0}$
- [54] $1^{123} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 501$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.30 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 8 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 30 & 5 \\ 12 & 22 & 0 & 30 & 85 & 17 \\ 3 & 4 & -3 & 5 & 17 & 0 \end{pmatrix}_{b_0}$
- [55] $1^{112} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 507$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 21 & 6 \\ 7 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -3 & 6 & 10 & 1 \end{pmatrix}_{b_0}$
- [56] $1^{114} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 536$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 16 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & -1 & 16 & 0 \end{pmatrix}_{b_0}$
- [57] $1^{112} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 576$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 10 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$
- [58] $1^{113} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 579$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 27 & 6 \\ 9 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & -3 & 6 & 13 & 1 \end{pmatrix}_{b_0}$

[59] $1^{114}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 587, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 33 & 6 \\ 11 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -3 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

[61] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 9 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{112}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 16 & 28 & 0 & 18 & 154 & 22 \\ 3 & 4 & -3 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{131}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 607, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 0 & 84 & 15 & 16 \\ 2 & 4 & 0 & 15 & 1 & 3 \\ 3 & 4 & -3 & 16 & 3 & 1 \end{pmatrix}_{b_0}$$

[63] $1^{114}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 626, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 13 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{112}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 635, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 13 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & -3 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{122}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 30 & 9 \\ 7 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & -3 & 9 & 10 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{114}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 770, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{114}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 771, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 11 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -3 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{123}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 798, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 39 & 9 \\ 9 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -3 & 9 & 13 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{114} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 800$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 0 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{113} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 915$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 12 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{113} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1136$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 15 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & -3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{113} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1251$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 18 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & -3 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1260$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 14 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{114} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1643$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 17 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{114} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1920$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 20 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [76] $1^{12} 3 2 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -20$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 10 & 3 \\ 8 & 15 & 0 & 10 & 38 & 12 \\ 3 & 5 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{111} 3 2 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 19 & 5 \\ 12 & 25 & 0 & 40 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 40 & 0 & 63 & 15 \\ 5 & 9 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[78] $1^{111} 3 1^{13} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 3 & 3 & 0 & 3 & 1 \\ 2 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

- [79] $1^{\overline{1}1} 3 2 1^{\overline{1}11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -40$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 4 & 3 \\ 2 & 5 & 0 & 4 & 2 & 4 \\ 3 & 5 & -3 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$
- [80] $1^{\overline{1}1} 3 1^{\overline{1}1} 3 1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 6 & 0 & 3 & 4 & 4 \\ 2 & 3 & 3 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$
- [86] $1^{\overline{1}1} 3 2 1^{\overline{1}21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 0 & 4 & 1 & 2 \\ 1 & 2 & 0 & 0 & 7 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 15 & 7 & 7 \\ 1 & 1 & 3 & 0 & 7 & 1 & 3 \\ 2 & 2 & 2 & -3 & 7 & 3 & 0 \end{pmatrix}_{b_0}$$
- [85] $1^{\overline{1}1} 3 2 1^{\overline{1}21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 0 & 8 & 1 & 2 \\ 1 & 2 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 9 & 0 & 39 & 9 & 11 \\ 1 & 1 & 3 & 0 & 9 & 1 & 3 \\ 2 & 2 & 2 & -3 & 11 & 3 & 0 \end{pmatrix}_{b_0}$$
- [84] $1^{\overline{1}1} 3 2 1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 19 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$
- [83] $1^{\overline{1}1} 3 2 1^{\overline{1}14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 23 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$
- [82] $1^{\overline{1}1} 3 2 1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 5 \\ 9 & 13 & 1 & 0 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 7 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$
- [81] $1^{\overline{1}1} 3 1^{\overline{1}1} 3 1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 \\ 1 & 3 & 3 & 0 & 3 & 1 & 3 \\ 2 & 4 & 2 & -3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$
- [87] $1^{\overline{1}1} 3 2 1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 1 & 2 \\ 1 & 2 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 1 & 3 & 0 & 4 & 1 & 3 \\ 2 & 2 & 2 & -3 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$
- [88] $1^{\overline{1}1} 3 2 1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -61$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & 0 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

- [89] $1^{112} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 13 & 14 & 1 & 28 & 7 \\ 14 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 30 & 0 & 58 & 14 \\ 7 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$
- [90] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 9 & -1 & 7 & 2 \end{pmatrix}_{b_0}$
- [91] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -71$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 6 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 3 & 0 & 4 & 4 \\ 3 & 4 & 3 & -3 & 4 & 1 \end{pmatrix}_{b_0}$
- [92] $1^{11} 31^{12} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -74$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 12 \\ 1 & 4 & 0 & 0 & 3 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 6 \\ 4 & 12 & 6 & 0 & 6 & 22 \end{pmatrix}_{b_0}$
- [93] $1^{11} 3 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 5 & 0 & 10 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 10 & 0 & 13 & 7 & 7 \\ 2 & 5 & 0 & 7 & 2 & 4 \\ 3 & 5 & -3 & 7 & 4 & 1 \end{pmatrix}_{b_0}$
- [94] $1^{11} 3 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 1 & 3 & 3 & 3 \\ 1 & 4 & 0 & 1 & 3 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & 1 & -1 \\ 2 & 3 & 3 & 1 & 0 & 3 & 2 \\ 1 & 3 & 3 & 1 & 3 & 1 & 3 \\ 2 & 3 & 2 & -1 & 2 & 3 & 0 \end{pmatrix}_{b_0}$
- [95] $1^{111} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 5 & 12 & 1 & 10 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 20 & 0 & 13 & 7 \\ 5 & 9 & -1 & 7 & 2 \end{pmatrix}_{b_0}$
- [96] $1^{11} 3 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -89$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 4 & 1 & 3 & 3 & 3 \\ 1 & 4 & 0 & 1 & 3 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & 1 & -1 \\ 2 & 3 & 3 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 3 & 1 & 3 \\ 2 & 3 & 2 & -1 & 2 & 3 & 0 \end{pmatrix}_{b_0}$
- [97] $1^{121} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 12.71 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 1 & 0 \\ 2 & 4 & 0 & 2 & 5 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & 0 & 3 & 2 & 0 \end{pmatrix}_{b_0}$
- [98] $1^{12} 3 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.17 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 1 & 5 & 2 \\ 1 & 2 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 5 & 2 \\ 3 & 5 & 5 & 0 & 5 & 13 & 7 \\ 2 & 2 & 2 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$

- [99] $1^{112} 3^{13} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 51 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 51 & 75 & 0 & 418 & 38 \\ 5 & 7 & -1 & 38 & 2 \end{pmatrix}_{b_0}$$
- [100] $1^{111} 3^{21} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 9 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$
- [101] $1^{114} 3^{21} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 66 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 66 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$
- [102] $1^{112} 3^{21} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 18 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$
- [103] $1^{111} 3^{21} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 12 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$
- [105] $1^{121} 3^{21} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -101$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 1 & 0 \\ 2 & 4 & 0 & 2 & 5 & 4 & 3 \\ 2 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & 0 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$
- [104] $1^{111} 3^{21} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -101$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 & 2 \\ 1 & 2 & 0 & 0 & 4 & 1 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 2 & 1 & 0 & 3 & 0 & 2 \\ 2 & 2 & 2 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$
- [106] $1^{11} 3^{21} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -111$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.19 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 4 & 2 \\ 2 & 3 & 4 & 0 & 4 & 5 & 5 \\ 2 & 2 & 2 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$
- [107] $1^{11} 3^{21} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -111$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 1 & 4 & 2 \\ 1 & 2 & 0 & 0 & 3 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 7 & 2 \\ 3 & 4 & 7 & 0 & 7 & 16 & 8 \\ 2 & 2 & 2 & -3 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$
- [108] $1^{112} 3^{21} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 21 & 18 & 0 & 94 & 20 \\ 3 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$
- [109] $1^{12} 3^{21} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 0 & 6 & 1 & 2 \\ 1 & 2 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 25 & 8 & 9 \\ 1 & 1 & 3 & 0 & 8 & 1 & 3 \\ 2 & 2 & 2 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{11} 3^{11} 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 5 & 0 & 5 & 10 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 7 & 3 \\ 5 & 10 & 0 & 7 & 14 & 8 \\ 3 & 5 & -3 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{114} 3^{11} 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 0 & 3 & 9 & 2 \\ 1 & 2 & 0 & 0 & 4 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 16 & 4 \\ 5 & 9 & 7 & 0 & 16 & 41 & 11 \\ 2 & 2 & 2 & -3 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{11} 31^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -137, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{11} 3^{11} 21^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 5 & 0 & 25 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 11 & 25 & 0 & 91 & 19 & 19 \\ 2 & 5 & 0 & 19 & 2 & 4 \\ 3 & 5 & -3 & 19 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[114] \ 1^{11} 3^{11} 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 5 & 0 & 15 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 15 & 0 & 31 & 11 & 11 \\ 2 & 5 & 0 & 11 & 2 & 4 \\ 3 & 5 & -3 & 11 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{111} 3^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -141, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 13 & 14 & 1 & 14 & 7 \\ 14 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 15 & 0 & 13 & 7 \\ 7 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{111} 3^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -141, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 7 & 0 & 14 & 8 \\ 3 & 4 & 3 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{11} 3^{11} 21^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 5 & 0 & 20 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 20 & 0 & 57 & 15 & 15 \\ 2 & 5 & 0 & 15 & 2 & 4 \\ 3 & 5 & -3 & 15 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{121} 3^{11} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 & 2 \\ 1 & 2 & 0 & 0 & 8 & 1 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 25 & 6 & 9 \\ 1 & 2 & 1 & 0 & 6 & 0 & 2 \\ 2 & 2 & 2 & -3 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{\overline{111}} 3 21^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 12 & 6 & 6 \\ 1 & 3 & 3 & 0 & 6 & 1 & 3 \\ 2 & 4 & 2 & -3 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{\overline{13}} 3 21^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 17 & 3 \\ 13 & 25 & 0 & 17 & 104 & 20 \\ 3 & 5 & -3 & 3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{\overline{113}} 3 21^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 58 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 58 & 90 & 0 & 583 & 45 \\ 5 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{\overline{111}} 3 21^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 20 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{\overline{111}} 3 21^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 11 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{\overline{111}} 3 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 10 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{\overline{112}} 3 21^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 52 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 52 & 75 & 0 & 418 & 38 \\ 5 & 7 & -1 & 38 & 2 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{\overline{113}} 3 21^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 47 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 47 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{\overline{112}} 3 21^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 28 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{\overline{111}} 3 21^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 22 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[129] $1^{113}31^{11}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 3 & 12 & 3 \\ 2 & 4 & 3 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 12 & 2 \\ 5 & 12 & 15 & 0 & 12 & 49 & 13 \\ 2 & 3 & 3 & -3 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

[130] $1^{121}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 & 2 \\ 1 & 2 & 0 & 0 & 5 & 1 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 12 & 5 & 6 \\ 1 & 2 & 1 & 0 & 5 & 0 & 2 \\ 2 & 2 & 2 & -3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

[131] $1^{11}321^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 19 & 3 \\ 11 & 25 & 0 & 19 & 92 & 20 \\ 3 & 5 & -3 & 3 & 20 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{11}321^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 11 & 3 \\ 7 & 15 & 0 & 11 & 32 & 12 \\ 3 & 5 & -3 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[133] $1^{14}321^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 1 & 10 & 2 \\ 1 & 2 & 0 & 0 & 3 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 10 & 2 \\ 6 & 10 & 10 & 0 & 10 & 58 & 14 \\ 2 & 2 & 2 & -3 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

[134] $1^{112}321^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 39 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 60 & 0 & 258 & 30 \\ 5 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[135] $1^{111}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 21 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[136] $1^{111}321^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 4 & 2 \\ 1 & 2 & 0 & 0 & 4 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 11 & 4 \\ 3 & 4 & 7 & 0 & 11 & 16 & 8 \\ 2 & 2 & 2 & -3 & 4 & 8 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{111}321^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 0 & 4 & 3 & 2 \\ 1 & 2 & 0 & 0 & 7 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 7 & 0 & 15 & 11 & 7 \\ 2 & 3 & 4 & 0 & 11 & 5 & 5 \\ 2 & 2 & 2 & -3 & 7 & 5 & 0 \end{pmatrix}_{b_0}$$

[138] $1^{121}321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 2 & 1 & 0 \\ 3 & 4 & 0 & 2 & 5 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 2 & 2 & -1 & 0 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{11} 3^{21} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 5 & 0 & 5 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 15 & 3 \\ 9 & 20 & 0 & 15 & 58 & 16 \\ 3 & 5 & -3 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{11} 3^{21} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 7 & 4 \\ 2 & 3 & 4 & 0 & 7 & 5 & 5 \\ 2 & 2 & 2 & -3 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{11} 3^{21} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 15 & 4 \\ 2 & 4 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 10 & 0 & 38 & 12 \\ 3 & 4 & 3 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[141] \ 1^{12} 3^{21} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 1 & 6 & 2 \\ 1 & 2 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 8 & 2 \\ 4 & 6 & 8 & 0 & 8 & 26 & 10 \\ 2 & 2 & 2 & -3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[143] \ 1^{11} 3^1 1^{13} 1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 3 & 6 & 9 & 0 & 6 & 13 & 7 \\ 2 & 3 & 3 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[144] \ 1^{12} 3^{21} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -290, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 5 & 0 & 5 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 14 & 3 \\ 10 & 20 & 0 & 14 & 62 & 16 \\ 3 & 5 & -3 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[146] \ 1^{11} 3^{21} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 19 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[145] \ 1^{11} 3^{21} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 37 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 37 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

$$[147] \ 1^{11} 3^{21} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -306, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 14 & 0 & 62 & 16 \\ 3 & 4 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{121}321^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -311, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$

$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 0 & 5 & 3 & 2 \\ 1 & 2 & 0 & 0 & 5 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 12 & 10 & 6 \\ 2 & 3 & 4 & 0 & 10 & 5 & 5 \\ 2 & 2 & 2 & -3 & 6 & 5 & 0 \end{pmatrix}_{b_0}$$

[149] $1^{113}321^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -311, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 \\ 1 & 2 & 0 & 0 & 4 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 13 & 4 \\ 4 & 7 & 6 & 0 & 13 & 25 & 9 \\ 2 & 2 & 2 & -3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

[150] $1^{13}321^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -311, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 1 & 8 & 2 \\ 1 & 2 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 9 & 2 \\ 5 & 8 & 9 & 0 & 9 & 40 & 12 \\ 2 & 2 & 2 & -3 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

[151] $1^{111}31^{11}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.31 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 3 & 9 & 6 & 0 & 6 & 13 & 7 \\ 2 & 4 & 2 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[152] $1^{112}31^{11}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -332, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 3 & 9 & 3 \\ 2 & 4 & 3 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 9 & 2 \\ 4 & 9 & 12 & 0 & 9 & 28 & 10 \\ 2 & 3 & 3 & -3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

[153] $1^{12}321^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 1 & 7 & 2 \\ 1 & 2 & 0 & 0 & 3 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 11 & 2 \\ 5 & 7 & 11 & 0 & 11 & 45 & 13 \\ 2 & 2 & 2 & -3 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

[154] $1^{113}321^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -351, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 30 & 4 \\ 2 & 4 & 1 & 0 & 25 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 30 & 25 & 0 & 188 & 28 \\ 3 & 4 & 3 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[155] $1^{112}321^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 5 & 2 \\ 1 & 2 & 0 & 0 & 4 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 10 & 4 \\ 3 & 5 & 5 & 0 & 10 & 13 & 7 \\ 2 & 2 & 2 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

[156] $1^{113}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 13 & 0 & 74 & 16 \\ 3 & 4 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[157] $1^{112}321^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -386$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 0 & 3 & 8 & 2 \\ 1 & 2 & 0 & 0 & 4 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 22 & 4 \\ 6 & 8 & 14 & 0 & 22 & 70 & 16 \\ 2 & 2 & 2 & -3 & 4 & 16 & 1 \end{pmatrix}_{b_0}$$

[158] $1^{131}1^{111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -404$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.42 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 3 & 6 & 3 \\ 2 & 6 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & 2 & -3 & 3 & 4 & 0 \end{pmatrix}_{b_0}$$

[159] $1^{112}321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 22 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[160] $1^{114}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.26 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 46 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 46 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

[161] $1^{114}321^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 56 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 56 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

[162] $1^{112}321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 20 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[163] $1^{112}321^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 40 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 40 & 60 & 0 & 258 & 30 \\ 5 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[164] $1^{113}321^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.26 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 28 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[165] $1^{112}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.03 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 29 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[166] $1^{111}31^{111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -417$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.19 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 6 & 4 \\ 1 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 12 & 9 & 6 \\ 2 & 6 & 3 & 0 & 9 & 4 & 4 \\ 2 & 4 & 2 & -3 & 6 & 4 & 0 \end{pmatrix}_{b_0}$$

[167] $1^{14} 3^{21} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -440$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.99 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 5 & 0 & 5 & 35 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 24 & 3 \\ 18 & 35 & 0 & 24 & 202 & 28 \\ 3 & 5 & -3 & 3 & 28 & 2 \end{pmatrix}_{b_0}$$

[168] $1^{114} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -456$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 27 & 16 & 0 & 122 & 20 \\ 3 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[169] $1^{112} 3^{21} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -456$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 21 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[170] $1^{11} 3^{111} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -476$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 9 \\ 1 & 4 & 0 & 0 & 3 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 \\ 2 & 6 & 3 & 0 & 3 & 9 \\ 3 & 9 & 6 & 0 & 9 & 13 \\ 2 & 4 & 2 & -3 & 3 & 7 \end{pmatrix}_{b_0}$$

[171] $1^{121} 3^{21} 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -481$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 2 & 0 & 6 & 3 & 2 \\ 1 & 2 & 0 & 0 & 8 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 25 & 14 & 9 \\ 2 & 3 & 4 & 0 & 14 & 5 & 5 \\ 2 & 2 & 2 & -3 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

[172] $1^{12} 3^{21} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -500$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 18 & 3 \\ 12 & 25 & 0 & 18 & 94 & 20 \\ 3 & 5 & -3 & 3 & 20 & 2 \end{pmatrix}_{b_0}$$

[173] $1^{113} 3^{21} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -536$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 59 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 59 & 90 & 0 & 583 & 45 \\ 5 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

[174] $1^{112} 3^{21} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -536$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 41 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 41 & 60 & 0 & 258 & 30 \\ 5 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[175] $1^{114} 3^{21} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -536$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.36 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 37 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 37 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{13} 3^{21} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -551$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 1 & 9 & 2 \\ 1 & 2 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 12 & 2 \\ 6 & 9 & 12 & 0 & 12 & 61 & 15 \\ 2 & 2 & 2 & -3 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$

[177] $1^{13} 3^{11} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -565$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 5 & 0 & 5 & 30 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 21 & 3 \\ 15 & 30 & 0 & 21 & 142 & 24 \\ 3 & 5 & -3 & 3 & 24 & 2 \end{pmatrix}_{b_0}$$

[180] $1^{112} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -576$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 12.02 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 30 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[179] $1^{112} 3^{11} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -576$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 42 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 42 & 60 & 0 & 258 & 30 \\ 5 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[178] $1^{113} 3^{11} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -576$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.02 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 48 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 48 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[181] $1^{113} 3^{11} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -581$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 21 & 0 & 142 & 24 \\ 3 & 4 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[182] $1^{112} 3^{11} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -586$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 6 & 2 \\ 1 & 2 & 0 & 0 & 4 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 14 & 4 \\ 4 & 6 & 8 & 0 & 14 & 26 & 10 \\ 2 & 2 & 2 & -3 & 4 & 10 & 1 \end{pmatrix}_{b_0}$$

[183] $1^{111} 31^{12} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -587$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 9 & 2 \\ 5 & 15 & 9 & 0 & 9 & 37 & 11 \\ 2 & 4 & 2 & -3 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

[184] $1^{113} 3^{11} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -591$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.27 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 17 & 0 & 104 & 20 \\ 3 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[185] $1^{112} 3^{11} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -596$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 \\ 1 & 2 & 0 & 0 & 4 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 18 & 4 \\ 5 & 7 & 11 & 0 & 18 & 45 & 13 \\ 2 & 2 & 2 & -3 & 4 & 13 & 1 \end{pmatrix}_{b_0}$$

[186] $1^{14} 3^{11} 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -636$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 1 & 11 & 2 \\ 1 & 2 & 0 & 0 & 3 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 13 & 2 \\ 7 & 11 & 13 & 0 & 13 & 81 & 17 \\ 2 & 2 & 2 & -3 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

[187] $1^{12} 3^{11} 21^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -650$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 5 & 0 & 5 & 30 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 22 & 3 \\ 14 & 30 & 0 & 22 & 134 & 24 \\ 3 & 5 & -3 & 3 & 24 & 2 \end{pmatrix}_{b_0}$$

[192] $1^{114} 3^{11} 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -656$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.02 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 67 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 67 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[191] $1^{113} 3^{11} 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -656$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 38 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 38 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[190] $1^{113} 3^{12} 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -656$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 29 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[189] $1^{112} 3^{13} 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -656$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.02 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 31 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[188] $1^{112} 3^{14} 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -656$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.02 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 32 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 32 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[193] $1^{13} 3^{11} 21^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -671$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.39 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 1 & 10 & 2 \\ 1 & 2 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 15 & 2 \\ 7 & 10 & 15 & 0 & 15 & 88 & 18 \\ 2 & 2 & 2 & -3 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

[194] $1^{111} 31^{11} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -699$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.20 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 12 & 15 & 6 \\ 3 & 9 & 6 & 0 & 15 & 13 & 7 \\ 2 & 4 & 2 & -3 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

- [195] $1^{113} 3 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 60 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 60 & 90 & 0 & 583 & 45 \\ 5 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$
- [196] $1^{113} 3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 30 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$
- [197] $1^{111} 31^{13} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -818, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 12 & 2 \\ 7 & 21 & 12 & 0 & 12 & 73 & 15 \\ 2 & 4 & 2 & -3 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$
- [198] $1^{113} 3 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -821, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 0 & 3 & 8 & 2 \\ 1 & 2 & 0 & 0 & 4 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 17 & 4 \\ 5 & 8 & 9 & 0 & 17 & 40 & 12 \\ 2 & 2 & 2 & -3 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$
- [199] $1^{114} 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -846, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 30 & 4 \\ 2 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 30 & 20 & 0 & 158 & 24 \\ 3 & 4 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$
- [200] $1^{13} 3 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -895, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 5 & 0 & 5 & 35 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 25 & 3 \\ 17 & 35 & 0 & 25 & 188 & 28 \\ 3 & 5 & -3 & 3 & 28 & 2 \end{pmatrix}_{b_0}$$
- [202] $1^{114} 3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 38 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 38 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$
- [204] $1^{113} 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 49 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 49 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$
- [203] $1^{113} 3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 31 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$
- [201] $1^{114} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 47 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 47 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

[205] $2 \overset{1^{1111}}{2} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -901, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 & 2 \\ 2 & 1 & 4 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 1 & 3 & 3 & 2 \\ 1 & 1 & 1 & -3 & 1 & 1 & -1 \\ 2 & 3 & 3 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 3 & 1 & 3 \\ 2 & 3 & 2 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[206] $1 \overset{1^1}{1^{114}} 3 21^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -926, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 0 & 4 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 20 & 4 \\ 6 & 10 & 10 & 0 & 20 & 58 & 14 \\ 2 & 2 & 2 & -3 & 4 & 14 & 1 \end{pmatrix}_{b_0}$$

[209] $1 \overset{1^{11}}{1^{114}} 3 21^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -936, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 57 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 57 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

[208] $1 \overset{1^1}{1^{14}} 3 21^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -936, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 1 & 12 & 2 \\ 1 & 2 & 0 & 0 & 3 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 16 & 2 \\ 8 & 12 & 16 & 0 & 16 & 110 & 20 \\ 2 & 2 & 2 & -3 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

[207] $1 \overset{1^{12}}{1^{113}} 3 21^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -936, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 39 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[210] $1 \overset{1^1}{1^{14}} 3 21^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -950, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 5 & 0 & 5 & 40 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 28 & 3 \\ 20 & 40 & 0 & 28 & 254 & 32 \\ 3 & 5 & -3 & 3 & 32 & 2 \end{pmatrix}_{b_0}$$

[211] $1 \overset{1^1}{1^{114}} 3 21^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -966, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 1 & 0 & 28 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 28 & 0 & 254 & 32 \\ 3 & 4 & 3 & -3 & 32 & 1 \end{pmatrix}_{b_0}$$

[212] $1 \overset{1^{11}}{1^{111}} 31^{\overline{14}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -1007, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 15 & 2 \\ 9 & 27 & 15 & 0 & 15 & 121 & 19 \\ 2 & 4 & 2 & -3 & 2 & 19 & 0 \end{pmatrix}_{b_0}$$

[213] $1 \overset{1^1}{1^{114}} 3 21^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1016, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 33 & 4 \\ 2 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 33 & 24 & 0 & 202 & 28 \\ 3 & 4 & 3 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[214] $1^{113} 3^{14} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1016$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 61 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 61 & 90 & 0 & 583 & 45 \\ 5 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

[215] $1^{113} 3^{11} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1111$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 0 & 3 & 9 \\ 1 & 2 & 0 & 0 & 4 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 21 \\ 6 & 9 & 12 & 0 & 21 & 61 \\ 2 & 2 & 2 & -3 & 4 & 15 \end{pmatrix}_{b_0}$$

[216] $1^{114} 3^{12} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 68 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 68 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[217] $1^{113} 3^{11} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 40 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 40 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{113} 3^{12} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 50 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 50 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[219] $1^{114} 3^{11} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1176$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 39 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[220] $1^{113} 3^{11} 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1181$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 0 & 4 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 25 & 4 \\ 7 & 10 & 15 & 0 & 25 & 88 & 18 \\ 2 & 2 & 2 & -3 & 4 & 18 & 1 \end{pmatrix}_{b_0}$$

[221] $1^{113} 3^{11} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 41 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 41 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[222] $1^{113} 3^{12} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1296$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 51 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 51 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[223] $1^{114} 3^{12} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1296$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.08 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 48 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 48 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

[224] $1^{114} 3^{12} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1376$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.03 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 58 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 58 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

[225] $1^{114} 3^{14} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1376$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.14 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 40 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 40 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[226] $1^{114} 3^{11} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1496$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 0 & 3 & 11 & 2 \\ 1 & 2 & 0 & 0 & 4 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 24 & 4 \\ 7 & 11 & 13 & 0 & 24 & 81 & 17 \\ 2 & 2 & 2 & -3 & 4 & 17 & 1 \end{pmatrix}_{b_0}$$

[227] $1^{114} 3^{13} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1536$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.01 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 69 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 69 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[228] $1^{114} 3^{11} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1616$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 49 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 49 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

[229] $1^{114} 3^{13} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1736$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.03 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 59 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 59 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

[230] $1^{114} 3^{11} 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1846$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 0 & 3 & 12 & 2 \\ 1 & 2 & 0 & 0 & 4 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 28 & 4 \\ 8 & 12 & 16 & 0 & 28 & 110 & 20 \\ 2 & 2 & 2 & -3 & 4 & 20 & 1 \end{pmatrix}_{b_0}$$

[232] $1^{114} 3^{13} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1856$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.01 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 70 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 70 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[231] $1^{114} 3^{114} 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1856$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.06 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 50 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 50 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

[233] $1^{114} 3^{114} 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -2016$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.02 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 60 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 60 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

$T = 10$ [234] $1^{111} 2^3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -61$, $\det = 16$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 3 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 1 & 4 & 0 \\ 1 & 3 & 0 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 4 & 4 & 6 & 6 \\ 2 & 3 & -1 & 0 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{11} 3^{11} 221^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 19$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 6 & 0 & 6 & 17 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 2 & 13 & 4 \\ 7 & 17 & 0 & 13 & 36 & 14 \\ 3 & 6 & -3 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[236] $1^{12} 3^{12} 221^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 40$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 6 & 0 & 6 & 34 & 6 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 0 & 2 & 26 & 4 \\ 14 & 34 & 0 & 26 & 150 & 28 \\ 3 & 6 & -3 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

[237] $1^{11} 4^{11} 131^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -86$, $\det = 56$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[238] $1^{11} 4^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -86$, $\det = 66$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[239] $1^{111} 4^{111} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -86$, $\det = 117$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 7 & 5 & 5 \\ 2 & 4 & 0 & 5 & 1 & 3 \\ 3 & 4 & -2 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

[240] $1^{121} 3^{121} 221^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 136$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.4$

$T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 2 & 3 \\ 2 & 1 & 1 & 11 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 11 & 0 & 49 & 11 & 13 \\ 2 & 3 & 0 & 11 & 1 & 3 \\ 3 & 3 & -1 & 13 & 3 & 3 \end{pmatrix}_{b_0}$$

[241] $1^{\overline{111}} 1^{\overline{131}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 5 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & -2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[242] $1^{\overline{111}} 1^{\overline{11}} 3^{\overline{221}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 171, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 1 & 0 & 5 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 5 & 8 & 5 \\ 2 & 3 & 5 & 0 & 8 & 6 & 6 \\ 2 & 2 & 3 & -3 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[243] $1^{\overline{11}} 1^{\overline{31}} 1^{\overline{11}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 201, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 3 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[244] $1^{\overline{11}} 1^{\overline{31}} 1^{\overline{11}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 222, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 5 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & -2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[245] $1^{\overline{111}} 1^{\overline{131}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 288, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 8 & 12 & 0 & 8 & 26 & 10 \\ 3 & 4 & -2 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[246] $1^{\overline{121}} 1^{\overline{11}} 3^{\overline{221}} 1^{\overline{121}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 297, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.4$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 2 & 0 & 6 & 3 & 2 \\ 1 & 2 & 1 & 0 & 10 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 10 & 0 & 29 & 16 & 11 \\ 2 & 3 & 5 & 0 & 16 & 6 & 6 \\ 2 & 2 & 3 & -3 & 11 & 6 & 2 \end{pmatrix}_{b_0}$$

[247] $1^{\overline{11}} 1^{\overline{31}} 1^{\overline{11}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 311, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 6 & 9 & 0 & 5 & 13 & 7 \\ 3 & 4 & -2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[248] $1^{\overline{11}} 1^{\overline{31}} 1^{\overline{11}} 4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 338, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 8 & 3 \\ 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 8 & 12 & 0 & 8 & 26 & 10 \\ 3 & 4 & -2 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[249] $1^{\overline{112}} 1^{\overline{11}} 3^{\overline{221}} 1^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 6 & 2 \\ 1 & 2 & 1 & 0 & 5 & 10 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 5 & 16 & 5 \\ 4 & 6 & 10 & 0 & 16 & 30 & 12 \\ 2 & 2 & 3 & -3 & 5 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[250] \ 1^{113} 3^{11} 221^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 675, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 0 & 3 & 9 & 2 \\ 1 & 2 & 1 & 0 & 5 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 5 & 24 & 5 \\ 6 & 9 & 15 & 0 & 24 & 70 & 18 \\ 2 & 2 & 3 & -3 & 5 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[251] \ 1^{114} 3^{11} 221^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 1116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 0 & 3 & 12 & 2 \\ 1 & 2 & 1 & 0 & 5 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 5 & 32 & 5 \\ 8 & 12 & 20 & 0 & 32 & 126 & 24 \\ 2 & 2 & 3 & -3 & 5 & 24 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [252] \ 1^{12} 5^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -34, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 1 & 12 & 5 \\ 9 & 13 & 0 & 20 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 20 & 0 & 26 & 10 \\ 5 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[253] \ 1^{11} 5^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -61, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 1 & 6 & 5 \\ 9 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 10 & 0 & 5 & 5 \\ 5 & 7 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

4.542 NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 0 & -1 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 15 & 6 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -3 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

4.543 NHC(-2)(-3) + f₄ + su₃ + su₂ + su₂

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & 0 & 3 & 6 & 4 \\ 3 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 11 & 3 \\ 3 & 4 & 0 & 4 & 5 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 3 & 5 & 0 & 3 & 4 & 21 & 4 \\ 11 & 19 & 0 & 12 & 21 & 73 & 15 \\ 3 & 4 & -3 & 2 & 4 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 8 & 14 & 0 & 9 & 9 & 37 & 11 \\ 3 & 4 & -3 & 2 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 5 & 3 \\ 3 & 4 & 0 & 4 & 19 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 12 & 6 & 2 \\ 11 & 19 & 0 & 12 & 73 & 33 & 15 \\ 5 & 9 & 0 & 6 & 33 & 13 & 7 \\ 3 & 4 & -3 & 2 & 15 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 2 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 12 & 6 & 6 \\ 4 & 8 & 0 & 12 & 10 & 6 & 6 \\ 2 & 4 & 0 & 6 & 6 & 1 & 3 \\ 3 & 4 & -3 & 6 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 9 & 6 & 6 \\ 3 & 5 & 0 & 9 & 4 & 3 & 4 \\ 2 & 4 & 0 & 6 & 3 & 1 & 3 \\ 3 & 4 & -3 & 6 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{121} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 2 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 12 & 9 & 9 \\ 3 & 5 & 0 & 12 & 4 & 3 & 4 \\ 2 & 4 & 0 & 9 & 3 & 1 & 3 \\ 3 & 4 & -3 & 9 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

12.2

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 \\ 3 & 4 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 4 \\ 5 & 9 & 0 & 6 & 9 & 13 & 7 \\ 3 & 4 & -3 & 2 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{112}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 8 & 3 \\ 3 & 4 & 0 & 4 & 5 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 3 & 5 & 0 & 3 & 4 & 15 & 4 \\ 8 & 14 & 0 & 9 & 15 & 37 & 11 \\ 3 & 4 & -3 & 2 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 7 & 3 \\ 3 & 4 & 0 & 4 & 5 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 3 & 5 & 0 & 3 & 4 & 12 & 4 \\ 7 & 13 & 0 & 9 & 12 & 28 & 10 \\ 3 & 4 & -3 & 2 & 4 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 6 & 6 & 6 \\ 2 & 4 & 0 & 6 & 1 & 3 & 3 \\ 2 & 4 & 0 & 6 & 3 & 1 & 3 \\ 3 & 4 & -3 & 6 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 4 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 2 & 4 & 0 & 3 & 1 & 15 & 3 \\ 13 & 23 & 0 & 15 & 15 & 100 & 18 \\ 3 & 4 & -3 & 2 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{1122}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 7 & 3 \\ 3 & 4 & 0 & 9 & 8 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 12 & 21 & 6 \\ 4 & 8 & 0 & 12 & 10 & 18 & 6 \\ 7 & 13 & 0 & 21 & 18 & 28 & 10 \\ 3 & 4 & -3 & 6 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & -3 & 2 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 6 & 15 & 6 \\ 2 & 4 & 0 & 6 & 1 & 6 & 3 \\ 5 & 9 & 0 & 15 & 6 & 13 & 7 \\ 3 & 4 & -3 & 6 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1111}31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -605, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 8 & 3 \\ 3 & 4 & 0 & 4 & 9 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 9 & 2 \\ 5 & 9 & 0 & 6 & 13 & 24 & 7 \\ 8 & 14 & 0 & 9 & 24 & 37 & 11 \\ 3 & 4 & -3 & 2 & 7 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{112}31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 10 & 3 \\ 3 & 4 & 0 & 4 & 5 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 3 & 5 & 0 & 3 & 4 & 18 & 4 \\ 10 & 18 & 0 & 12 & 18 & 58 & 14 \\ 3 & 4 & -3 & 2 & 4 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{112}31^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -650, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 16 & 3 \\ 3 & 4 & 0 & 4 & 5 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 3 & 5 & 0 & 3 & 4 & 30 & 4 \\ 16 & 28 & 0 & 18 & 30 & 154 & 22 \\ 3 & 4 & -3 & 2 & 4 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{1111}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 5 & 9 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & -3 & 2 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1111}31^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -716, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 9 & 2 \\ 8 & 14 & 0 & 9 & 37 & 39 & 11 \\ 8 & 14 & 0 & 9 & 39 & 37 & 11 \\ 3 & 4 & -3 & 2 & 11 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1112}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -723, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 7 & 13 & 0 & 9 & 9 & 28 & 10 \\ 3 & 4 & -3 & 2 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{1112}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -738, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 4 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 \\ 10 & 18 & 0 & 12 & 12 & 58 & 14 \\ 3 & 4 & -3 & 2 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{112}31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 13 & 3 \\ 3 & 4 & 0 & 4 & 5 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 3 & 5 & 0 & 3 & 4 & 24 & 4 \\ 13 & 23 & 0 & 15 & 24 & 100 & 18 \\ 3 & 4 & -3 & 2 & 4 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1113} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -779, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 12 & 3 \\ 3 & 4 & 0 & 4 & 5 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 3 & 5 & 0 & 3 & 4 & 21 & 4 \\ 12 & 22 & 0 & 15 & 21 & 85 & 17 \\ 3 & 4 & -3 & 2 & 4 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1113} 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -828, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 4 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 2 & 4 & 0 & 3 & 1 & 21 & 3 \\ 18 & 32 & 0 & 21 & 21 & 193 & 25 \\ 3 & 4 & -3 & 2 & 3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1112} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -845, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 7 & 3 \\ 3 & 4 & 0 & 9 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 6 & 21 & 6 \\ 2 & 4 & 0 & 6 & 1 & 9 & 3 \\ 7 & 13 & 0 & 21 & 9 & 28 & 10 \\ 3 & 4 & -3 & 6 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1112} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -941, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 7 & 3 \\ 3 & 4 & 0 & 4 & 9 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 9 & 2 \\ 5 & 9 & 0 & 6 & 13 & 21 & 7 \\ 7 & 13 & 0 & 9 & 21 & 28 & 10 \\ 3 & 4 & -3 & 2 & 7 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1112} 31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -962, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 10 & 3 \\ 3 & 4 & 0 & 4 & 14 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 12 & 2 \\ 8 & 14 & 0 & 9 & 37 & 48 & 11 \\ 10 & 18 & 0 & 12 & 48 & 58 & 14 \\ 3 & 4 & -3 & 2 & 11 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1113} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -980, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 9 & 3 \\ 3 & 4 & 0 & 4 & 9 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 12 & 2 \\ 5 & 9 & 0 & 6 & 13 & 27 & 7 \\ 9 & 17 & 0 & 12 & 27 & 49 & 13 \\ 3 & 4 & -3 & 2 & 7 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1112} 31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -980, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 16 & 3 \\ 3 & 4 & 0 & 4 & 9 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 18 & 2 \\ 5 & 9 & 0 & 6 & 13 & 48 & 7 \\ 16 & 28 & 0 & 18 & 48 & 154 & 22 \\ 3 & 4 & -3 & 2 & 7 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1113} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{f}_4 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -989, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 9 & 3 \\ 3 & 4 & 0 & 9 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 6 & 27 & 6 \\ 2 & 4 & 0 & 6 & 1 & 12 & 3 \\ 9 & 17 & 0 & 27 & 12 & 49 & 13 \\ 3 & 4 & -3 & 6 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1114} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1005, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 11 & 3 \\ 3 & 4 & 0 & 9 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 6 & 33 & 6 \\ 2 & 4 & 0 & 6 & 1 & 15 & 3 \\ 11 & 21 & 0 & 33 & 15 & 76 & 16 \\ 3 & 4 & -3 & 6 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1113} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1077, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 2 & 4 & 0 & 3 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 12 & 49 & 13 \\ 3 & 4 & -3 & 2 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1113} 3^1 1^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1221, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 15 & 3 \\ 3 & 4 & 0 & 4 & 5 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 3 & 5 & 0 & 3 & 4 & 27 & 4 \\ 15 & 27 & 0 & 18 & 27 & 133 & 21 \\ 3 & 4 & -3 & 2 & 4 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1122} 3^1 1^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1298, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 10 & 3 \\ 3 & 4 & 0 & 4 & 13 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 12 & 2 \\ 7 & 13 & 0 & 9 & 28 & 42 & 10 \\ 10 & 18 & 0 & 12 & 42 & 58 & 14 \\ 3 & 4 & -3 & 2 & 10 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1113} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1333, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 2 & 4 & 0 & 3 & 1 & 18 & 3 \\ 15 & 27 & 0 & 18 & 18 & 133 & 21 \\ 3 & 4 & -3 & 2 & 3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1112} 3^1 1^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 10 & 3 \\ 3 & 4 & 0 & 4 & 9 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 12 & 2 \\ 5 & 9 & 0 & 6 & 13 & 30 & 7 \\ 10 & 18 & 0 & 12 & 30 & 58 & 14 \\ 3 & 4 & -3 & 2 & 7 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1122} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1388, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 9 & 2 \\ 7 & 13 & 0 & 9 & 28 & 30 & 10 \\ 7 & 13 & 0 & 9 & 30 & 28 & 10 \\ 3 & 4 & -3 & 2 & 10 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1112} 3^1 1^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1389, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 13 & 3 \\ 3 & 4 & 0 & 4 & 9 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 15 & 2 \\ 5 & 9 & 0 & 6 & 13 & 39 & 7 \\ 13 & 23 & 0 & 15 & 39 & 100 & 18 \\ 3 & 4 & -3 & 2 & 7 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1113}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 15 \\ 12 & 22 & 0 & 15 & 15 & 85 & 17 \\ 3 & 4 & -3 & 2 & 3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{1113}31^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1451, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 18 & 3 \\ 3 & 4 & 0 & 4 & 5 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 3 & 5 & 0 & 3 & 4 & 33 & 4 \\ 18 & 32 & 0 & 21 & 33 & 193 & 25 \\ 3 & 4 & -3 & 2 & 4 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1113}31^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1500, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 17 & 3 \\ 3 & 4 & 0 & 4 & 5 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 3 & 5 & 0 & 3 & 4 & 30 & 4 \\ 17 & 31 & 0 & 21 & 30 & 172 & 24 \\ 3 & 4 & -3 & 2 & 4 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1113}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1949, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 12 & 3 \\ 3 & 4 & 0 & 4 & 9 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 15 & 2 \\ 5 & 9 & 0 & 6 & 13 & 36 & 7 \\ 12 & 22 & 0 & 15 & 36 & 85 & 17 \\ 3 & 4 & -3 & 2 & 7 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1122}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2048, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 12 & 12 & 2 \\ 10 & 18 & 0 & 12 & 58 & 60 & 14 \\ 10 & 18 & 0 & 12 & 60 & 58 & 14 \\ 3 & 4 & -3 & 2 & 14 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1114}31^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2054, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 20 & 3 \\ 3 & 4 & 0 & 4 & 5 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 24 & 2 \\ 3 & 5 & 0 & 3 & 4 & 36 & 4 \\ 20 & 36 & 0 & 24 & 36 & 238 & 28 \\ 3 & 4 & -3 & 2 & 4 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1114}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2106, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 4 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 2 & 4 & 0 & 0 & 3 & 1 & 18 \\ 14 & 26 & 0 & 18 & 18 & 118 & 20 \\ 3 & 4 & -3 & 2 & 3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1114}31^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 4 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 24 & 2 \\ 2 & 4 & 0 & 3 & 1 & 24 & 3 \\ 20 & 36 & 0 & 24 & 24 & 238 & 28 \\ 3 & 4 & -3 & 2 & 3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1114} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2324, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 14 & 3 \\ 3 & 4 & 0 & 4 & 9 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 18 & 2 \\ 5 & 9 & 0 & 6 & 13 & 42 & 7 \\ 14 & 26 & 0 & 18 & 42 & 118 & 20 \\ 3 & 4 & -3 & 2 & 7 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1114} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2347, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 4 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 2 & 4 & 0 & 3 & 1 & 21 & 3 \\ 17 & 31 & 0 & 21 & 21 & 172 & 24 \\ 3 & 4 & -3 & 2 & 3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1113} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2496, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 15 & 3 \\ 3 & 4 & 0 & 4 & 9 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 18 & 2 \\ 5 & 9 & 0 & 6 & 13 & 45 & 7 \\ 15 & 27 & 0 & 18 & 45 & 133 & 21 \\ 3 & 4 & -3 & 2 & 7 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1113} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2621, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 18 & 3 \\ 3 & 4 & 0 & 4 & 9 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 21 & 2 \\ 5 & 9 & 0 & 6 & 13 & 54 & 7 \\ 18 & 32 & 0 & 21 & 54 & 193 & 25 \\ 3 & 4 & -3 & 2 & 7 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1114} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -3405, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 17 & 3 \\ 3 & 4 & 0 & 4 & 9 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 21 & 2 \\ 5 & 9 & 0 & 6 & 13 & 51 & 7 \\ 17 & 31 & 0 & 21 & 51 & 172 & 24 \\ 3 & 4 & -3 & 2 & 7 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1114} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -4064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 20 & 3 \\ 3 & 4 & 0 & 4 & 9 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 24 & 2 \\ 5 & 9 & 0 & 6 & 13 & 60 & 7 \\ 20 & 36 & 0 & 24 & 60 & 238 & 28 \\ 3 & 4 & -3 & 2 & 7 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [54] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 32, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 1 & 5 \\ 9 & 13 & 1 & 0 & 0 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 7 & -1 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{11} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 5 & 0 & 5 & 5 & 10 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 4 & 7 & 3 \\ 2 & 5 & 0 & 4 & 2 & 7 & 4 \\ 5 & 10 & 0 & 7 & 7 & 14 & 8 \\ 3 & 5 & -3 & 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{111111} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 & 2 \\ 1 & 2 & 0 & 0 & 4 & 1 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 3 & 3 \\ 1 & 2 & 1 & 0 & 3 & 0 & 2 \\ 2 & 2 & 2 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{11} 3 21^{1311} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 2 & 3 \\ 3 & 5 & 0 & 20 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 9 & 20 & 0 & 57 & 15 & 15 & 15 \\ 2 & 5 & 0 & 15 & 2 & 4 & 4 \\ 2 & 5 & 0 & 15 & 4 & 2 & 4 \\ 3 & 5 & -3 & 15 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1111} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 20 & 5 \\ 9 & 13 & 1 & 0 & 30 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 20 & 30 & 0 & 0 & 63 & 15 \\ 5 & 7 & -1 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{1111} 3 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 1 & 3 \\ 1 & 2 & 0 & 0 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 4 & 7 \\ 1 & 1 & 3 & 0 & 4 & 1 & 4 \\ 2 & 3 & 4 & 0 & 7 & 4 & 5 \\ 2 & 2 & 2 & -3 & 4 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{11} 3 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 5 & 0 & 5 & 5 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 4 & 11 & 3 \\ 2 & 5 & 0 & 4 & 2 & 11 & 4 \\ 7 & 15 & 0 & 11 & 11 & 32 & 12 \\ 3 & 5 & -3 & 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{111111} 3 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 262, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 & 3 \\ 1 & 2 & 0 & 0 & 4 & 1 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 3 & 7 \\ 1 & 2 & 1 & 0 & 3 & 0 & 3 \\ 2 & 3 & 4 & 0 & 7 & 3 & 5 \\ 2 & 2 & 2 & -3 & 4 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{12} 3 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 340, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 5 & 0 & 5 & 5 & 25 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 4 & 18 & 3 \\ 2 & 5 & 0 & 4 & 2 & 18 & 4 \\ 12 & 25 & 0 & 18 & 18 & 94 & 20 \\ 3 & 5 & -3 & 3 & 4 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{1111} 3 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 5 & 0 & 5 & 10 & 10 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 7 & 7 & 3 \\ 5 & 10 & 0 & 7 & 14 & 16 & 8 \\ 5 & 10 & 0 & 7 & 16 & 14 & 8 \\ 3 & 5 & -3 & 3 & 8 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{13} 321^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 380, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 5 & 0 & 5 & 5 & 35 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 4 & 25 & 3 \\ 2 & 5 & 0 & 4 & 2 & 25 & 4 \\ 17 & 35 & 0 & 25 & 25 & 188 & 28 \\ 3 & 5 & -3 & 3 & 4 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{111} 321^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 390, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 7 & 3 \\ 3 & 5 & 0 & 5 & 10 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 7 & 11 & 3 \\ 5 & 10 & 0 & 7 & 14 & 23 & 8 \\ 7 & 15 & 0 & 11 & 23 & 32 & 12 \\ 3 & 5 & -3 & 3 & 8 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{111} 321^{1123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 630, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 9 & 3 \\ 3 & 5 & 0 & 5 & 15 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 11 & 15 & 3 \\ 7 & 15 & 0 & 11 & 32 & 45 & 12 \\ 9 & 20 & 0 & 15 & 45 & 58 & 16 \\ 3 & 5 & -3 & 3 & 12 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{12} 321^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 640, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 5 & 0 & 5 & 5 & 30 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 4 & 22 & 3 \\ 2 & 5 & 0 & 4 & 2 & 22 & 4 \\ 14 & 30 & 0 & 22 & 22 & 134 & 24 \\ 3 & 5 & -3 & 3 & 4 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{111} 321^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 720, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 7 & 3 \\ 3 & 5 & 0 & 5 & 15 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 11 & 11 & 3 \\ 7 & 15 & 0 & 11 & 32 & 34 & 12 \\ 7 & 15 & 0 & 11 & 34 & 32 & 12 \\ 3 & 5 & -3 & 3 & 12 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{1111} 321^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 752, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 7 & 4 \\ 2 & 3 & 4 & 0 & 7 & 5 & 5 \\ 2 & 3 & 4 & 0 & 7 & 7 & 5 \\ 2 & 2 & 2 & -3 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{111} 321^{1133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 840, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.00 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 9 & 3 \\ 3 & 5 & 0 & 5 & 20 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 15 & 15 & 3 \\ 9 & 20 & 0 & 15 & 58 & 60 & 16 \\ 9 & 20 & 0 & 15 & 60 & 58 & 16 \\ 3 & 5 & -3 & 3 & 16 & 16 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [71] \quad 1^{111} 3221^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -52, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 7 & 3 \\ 3 & 6 & 0 & 6 & 17 & 17 & 6 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 6 & 0 & 2 & 13 & 13 & 4 \\ 7 & 17 & 0 & 13 & 36 & 38 & 14 \\ 7 & 17 & 0 & 13 & 38 & 36 & 14 \\ 3 & 6 & -3 & 4 & 14 & 14 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{1111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -468, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 1 & 0 & 5 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 5 & 8 & 5 \\ 2 & 3 & 5 & 0 & 8 & 6 & 6 \\ 2 & 2 & 3 & -3 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

4.544 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 10 \quad [0] \quad 1^{111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 5 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 3 & 4 \\ 3 & 5 & 0 & 3 & 6 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 4 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{1111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 & 2 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 2 & 2 & 0 & 2 \\ 3 & 4 & -3 & 2 & 3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 3 & 4 \\ 5 & 9 & 0 & 6 & 9 & 13 & 5 & 7 \\ 2 & 3 & 0 & 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & -3 & 2 & 4 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{1111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 6 & 6 & 5 & 6 \\ 2 & 4 & 0 & 6 & 1 & 3 & 2 & 3 \\ 2 & 4 & 0 & 6 & 3 & 1 & 2 & 3 \\ 2 & 3 & 0 & 5 & 2 & 2 & 0 & 2 \\ 3 & 4 & -3 & 6 & 3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{1111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 & 2 \\ 2 & 4 & 0 & 3 & 1 & 6 & 2 & 3 \\ 5 & 9 & 0 & 6 & 6 & 13 & 5 & 7 \\ 2 & 3 & 0 & 2 & 2 & 5 & 0 & 2 \\ 3 & 4 & -3 & 2 & 3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{1111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 & 2 \\ 5 & 9 & 0 & 6 & 13 & 15 & 5 & 7 \\ 5 & 9 & 0 & 6 & 15 & 13 & 5 & 7 \\ 2 & 3 & 0 & 2 & 5 & 5 & 0 & 2 \\ 3 & 4 & -3 & 2 & 7 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.545 NHC(-2)(-3) + f₄ + su₃ + su₂ + su₂→(-3)

$$T = 9 \quad [0] \quad 1^{112} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 2 & 3 \\ 3 & 4 & 0 & 9 & 8 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 12 & 5 & 6 \\ 4 & 8 & 0 & 12 & 10 & 4 & 6 \\ 2 & 3 & 0 & 5 & 4 & 0 & 2 \\ 3 & 4 & -3 & 6 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 2 & 3 \\ 4 & 9 & 4 & 0 & 8 & 4 & 5 \\ 2 & 4 & -2 & 0 & 3 & 2 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 3 & 4 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 3 & 5 & 0 & -1 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{112} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 4 & 8 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 2 & 2 \\ 4 & 8 & 0 & 6 & 10 & 4 & 6 \\ 2 & 3 & 0 & 2 & 4 & 0 & 2 \\ 3 & 4 & -3 & 2 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 4 & 19 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 12 & 2 & 2 \\ 11 & 19 & 0 & 12 & 73 & 11 & 15 \\ 2 & 3 & 0 & 2 & 11 & 0 & 2 \\ 3 & 4 & -3 & 2 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 2 & 3 \\ 2 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{121} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.31 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 6 & 4 & 5 \\ 2 & 4 & 0 & 6 & 1 & 2 & 3 \\ 2 & 3 & 0 & 4 & 2 & 0 & 2 \\ 3 & 4 & -3 & 5 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 0 & 8 & 4 & 5 \\ 2 & 4 & -2 & 0 & 3 & 2 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 3 & 4 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 3 & 5 & 0 & -1 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{121}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 12 & 4 & 5 \\ 5 & 9 & 0 & 12 & 13 & 5 & 7 \\ 2 & 3 & 0 & 4 & 5 & 0 & 2 \\ 3 & 4 & -3 & 5 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{111}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 & 2 \\ 2 & 4 & 0 & 3 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & -3 & 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{111}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 6 & 5 & 6 \\ 2 & 4 & 0 & 6 & 1 & 2 & 3 \\ 2 & 3 & 0 & 5 & 2 & 0 & 2 \\ 3 & 4 & -3 & 6 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 4 & 14 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 2 & 2 \\ 8 & 14 & 0 & 9 & 37 & 8 & 11 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 4 & -3 & 2 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{111}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 2 & 2 \\ 5 & 9 & 0 & 6 & 13 & 5 & 7 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{121}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 9 & 7 & 9 \\ 2 & 4 & 0 & 9 & 1 & 2 & 3 \\ 2 & 3 & 0 & 7 & 2 & 0 & 2 \\ 3 & 4 & -3 & 9 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{112}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 4 & 13 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 2 & 2 \\ 7 & 13 & 0 & 9 & 28 & 7 & 10 \\ 2 & 3 & 0 & 2 & 7 & 0 & 2 \\ 3 & 4 & -3 & 2 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{112}3^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 2 & 3 \\ 3 & 4 & 0 & 4 & 28 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 18 & 2 & 2 \\ 16 & 28 & 0 & 18 & 154 & 16 & 22 \\ 2 & 3 & 0 & 2 & 16 & 0 & 2 \\ 3 & 4 & -3 & 2 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{113}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 0 & 4 & 17 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 12 & 2 & 2 \\ 9 & 17 & 0 & 12 & 49 & 9 & 13 \\ 2 & 3 & 0 & 2 & 9 & 0 & 2 \\ 3 & 4 & -3 & 2 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{114}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 4 & 21 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 15 & 2 & 2 \\ 11 & 21 & 0 & 15 & 76 & 11 & 16 \\ 2 & 3 & 0 & 2 & 11 & 0 & 2 \\ 3 & 4 & -3 & 2 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{112}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 0 & 4 & 18 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 12 & 2 & 2 \\ 10 & 18 & 0 & 12 & 58 & 10 & 14 \\ 2 & 3 & 0 & 2 & 10 & 0 & 2 \\ 3 & 4 & -3 & 2 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{112}3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 2 & 3 \\ 3 & 4 & 0 & 4 & 23 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 15 & 2 & 2 \\ 13 & 23 & 0 & 15 & 100 & 13 & 18 \\ 2 & 3 & 0 & 2 & 13 & 0 & 2 \\ 3 & 4 & -3 & 2 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{113}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 2 & 3 \\ 3 & 4 & 0 & 4 & 22 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 15 & 2 & 2 \\ 12 & 22 & 0 & 15 & 85 & 12 & 17 \\ 2 & 3 & 0 & 2 & 12 & 0 & 2 \\ 3 & 4 & -3 & 2 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{113}3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 2 & 3 \\ 3 & 4 & 0 & 4 & 27 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 18 & 2 & 2 \\ 15 & 27 & 0 & 18 & 133 & 15 & 21 \\ 2 & 3 & 0 & 2 & 15 & 0 & 2 \\ 3 & 4 & -3 & 2 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{114}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 4 & 0 & 4 & 26 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 18 & 2 & 2 \\ 14 & 26 & 0 & 18 & 118 & 14 & 20 \\ 2 & 3 & 0 & 2 & 14 & 0 & 2 \\ 3 & 4 & -3 & 2 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11} 3^{11} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1826, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 2 & 3 \\ 3 & 4 & 0 & 4 & 32 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 21 & 2 & 2 \\ 18 & 32 & 0 & 21 & 193 & 18 & 25 \\ 2 & 3 & 0 & 2 & 18 & 0 & 2 \\ 3 & 4 & -3 & 2 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{11} 3^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 2 & 3 \\ 3 & 4 & 0 & 4 & 31 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 21 & 2 & 2 \\ 17 & 31 & 0 & 21 & 172 & 17 & 24 \\ 2 & 3 & 0 & 2 & 17 & 0 & 2 \\ 3 & 4 & -3 & 2 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{11} 3^{11} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 2 & 3 \\ 3 & 4 & 0 & 4 & 36 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 24 & 2 & 2 \\ 20 & 36 & 0 & 24 & 238 & 20 & 28 \\ 2 & 3 & 0 & 2 & 20 & 0 & 2 \\ 3 & 4 & -3 & 2 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [26] \quad 1^{11} 4^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 & 5 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 0 & 2 \\ 3 & 5 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 2 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{11} 1^{13} 1^{11} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 1 & 3 \\ 3 & 4 & 0 & 4 & 7 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 5 & 0 & 2 \\ 5 & 7 & 0 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 2 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11} 1^{13} 1^{11} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 1 & 3 \\ 3 & 4 & 0 & 4 & 12 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 8 & 0 & 2 \\ 8 & 12 & 0 & 8 & 26 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 2 & 10 & 0 & 2 \end{pmatrix}_{b_0}$$

$$4.546 \quad \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 6 \\ 2 & 3 & 0 & 2 & 3 & 0 & 3 \\ 3 & 5 & 0 & 3 & 6 & 3 & 4 \\ 3 & 4 & -3 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 1^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 & 3 \\ 2 & 4 & 0 & 3 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 2 & 0 & 2 \\ 2 & 4 & 0 & 3 & 3 & 2 & 1 \\ 3 & 4 & -3 & 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 & 6 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 9 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 5 & 2 \\ 5 & 9 & 0 & 6 & 9 & 5 & 13 & 7 \\ 3 & 4 & -3 & 2 & 4 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 6 & 5 & 6 & 6 \\ 2 & 4 & 0 & 6 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 5 & 2 & 0 & 2 & 2 \\ 2 & 4 & 0 & 6 & 3 & 2 & 1 & 3 \\ 3 & 4 & -3 & 6 & 3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 & 6 & 2 \\ 2 & 4 & 0 & 3 & 1 & 2 & 6 & 3 \\ 2 & 3 & 0 & 2 & 2 & 0 & 5 & 2 \\ 5 & 9 & 0 & 6 & 6 & 5 & 13 & 7 \\ 3 & 4 & -3 & 2 & 3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 2 & 6 & 2 \\ 5 & 9 & 0 & 6 & 13 & 5 & 15 & 7 \\ 2 & 3 & 0 & 2 & 5 & 0 & 5 & 2 \\ 5 & 9 & 0 & 6 & 15 & 5 & 13 & 7 \\ 3 & 4 & -3 & 2 & 7 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.547 NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_{2n3\text{mix}}$

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 5 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 4 \\ 5 & 8 & 0 & 5 & 9 & 12 & 6 \\ 3 & 4 & -3 & 2 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 11 & 2 \\ 2 & 4 & 0 & 3 & 1 & 11 & 3 \\ 10 & 17 & 0 & 11 & 11 & 55 & 13 \\ 3 & 4 & -3 & 2 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 0 & 4 & 5 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 5 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 4 \\ 4 & 7 & 0 & 5 & 6 & 7 & 5 \\ 3 & 4 & -3 & 2 & 4 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1123}} 3^{\overline{1}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 0 & 5 & 4 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 1 & 8 & 3 \\ 6 & 11 & 0 & 9 & 8 & 20 & 8 \\ 3 & 4 & -3 & 3 & 3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1113}} 3^{\overline{1}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -204, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.14 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 5 & 3 \\ 4 & 7 & 0 & 6 & 5 & 7 & 5 \\ 3 & 4 & -3 & 3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1113}} 3^{\overline{1}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -269, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 5 & 2 \\ 5 & 9 & 0 & 6 & 13 & 14 & 7 \\ 5 & 8 & 0 & 5 & 14 & 12 & 6 \\ 3 & 4 & -3 & 2 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1113}} 3^{\overline{1}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 11 & 3 \\ 3 & 4 & 0 & 4 & 5 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 14 & 2 \\ 3 & 5 & 0 & 3 & 4 & 18 & 4 \\ 11 & 20 & 0 & 14 & 18 & 69 & 15 \\ 3 & 4 & -3 & 2 & 4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1113}} 3^{\overline{1}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 7 & 3 \\ 3 & 4 & 0 & 4 & 5 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 8 & 2 \\ 3 & 5 & 0 & 3 & 4 & 12 & 4 \\ 7 & 12 & 0 & 8 & 12 & 25 & 9 \\ 3 & 4 & -3 & 2 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1113}} 3^{\overline{1}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -424, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.07 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 5 & 2 \\ 2 & 4 & 0 & 3 & 1 & 5 & 3 \\ 4 & 7 & 0 & 5 & 5 & 7 & 5 \\ 3 & 4 & -3 & 2 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1113}} 3^{\overline{1}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 4 & 3 \\ 3 & 4 & 0 & 9 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 6 & 11 & 6 \\ 2 & 4 & 0 & 6 & 1 & 5 & 3 \\ 4 & 7 & 0 & 11 & 5 & 7 & 5 \\ 3 & 4 & -3 & 6 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1113}} 3^{\overline{1}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -481, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 8 & 2 \\ 2 & 4 & 0 & 3 & 1 & 8 & 3 \\ 7 & 12 & 0 & 8 & 8 & 25 & 9 \\ 3 & 4 & -3 & 2 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{112}3^{\overline{111}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2} \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -500, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 9 & 3 \\ 3 & 4 & 0 & 4 & 5 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 11 & 2 \\ 3 & 5 & 0 & 3 & 4 & 15 & 4 \\ 9 & 16 & 0 & 11 & 15 & 44 & 12 \\ 3 & 4 & -3 & 2 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{111}1^{\overline{111}}3^{\overline{111}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2} \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -509, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.06 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 4 & 3 \\ 3 & 4 & 0 & 4 & 9 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 5 & 2 \\ 5 & 9 & 0 & 6 & 13 & 11 & 7 \\ 4 & 7 & 0 & 5 & 11 & 7 & 5 \\ 3 & 4 & -3 & 2 & 7 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{111}1^{\overline{111}}3^{\overline{111}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2} \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -896, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 7 & 3 \\ 3 & 4 & 0 & 4 & 9 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 8 & 2 \\ 5 & 9 & 0 & 6 & 13 & 20 & 7 \\ 7 & 12 & 0 & 8 & 20 & 25 & 9 \\ 3 & 4 & -3 & 2 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

4.548 NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{111}1^{\overline{111}}3^{\overline{111}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3} \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -39, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 1 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 8 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}1^{\overline{111}}3^{\overline{111}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3} \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -119, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 4 \\ 3 & 5 & 0 & 3 & 6 & 3 & 3 \\ 3 & 4 & -3 & 2 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}1^{\overline{111}}3^{\overline{111}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3} \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -126, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.05 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{113}1^{\overline{111}}3^{\overline{111}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3} \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -159, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 1 & 3 \\ 4 & 9 & 4 & 0 & 24 & 0 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 10 & 24 & 9 & 0 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{112}1^{\overline{111}}3^{\overline{111}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3} \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -174, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 1 & 3 \\ 4 & 9 & 4 & 0 & 16 & 0 & 5 \\ 2 & 4 & -2 & 0 & 6 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 7 & 16 & 6 & 0 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{122}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.9$

$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 0 & 5 & 13 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 12 & 6 & 3 \\ 7 & 13 & 0 & 12 & 28 & 18 & 10 \\ 4 & 8 & 0 & 6 & 18 & 9 & 5 \\ 3 & 4 & -3 & 3 & 10 & 5 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.34 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 2 & 4 & 0 & 3 & 1 & 9 & 3 \\ 8 & 14 & 0 & 9 & 9 & 36 & 10 \\ 3 & 4 & -3 & 2 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{132}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 0 & 5 & 17 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 15 & 6 & 3 \\ 9 & 17 & 0 & 15 & 49 & 24 & 13 \\ 4 & 8 & 0 & 6 & 24 & 9 & 5 \\ 3 & 4 & -3 & 3 & 13 & 5 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{111}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 3 & 2 \\ 8 & 14 & 0 & 9 & 37 & 9 & 11 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -3 & 2 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{113}31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 1 & 3 \\ 4 & 9 & 4 & 0 & 24 & 0 & 5 \\ 2 & 4 & -2 & 0 & 9 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 11 & 24 & 9 & 0 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 3 & 5 & 0 & 3 & 4 & 9 & 4 \\ 5 & 9 & 0 & 6 & 9 & 12 & 6 \\ 3 & 4 & -3 & 2 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -399, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{114} 3^{1^{131}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 1 & 3 \\ 4 & 9 & 4 & 0 & 32 & 0 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 15 & 32 & 12 & 0 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{111} 3^{1^{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 9 & 3 & 3 \\ 5 & 9 & 0 & 9 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & -3 & 3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111} 3^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & -3 & 2 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{114} 3^{1^{121}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 1 & 3 \\ 4 & 9 & 4 & 0 & 32 & 0 & 5 \\ 2 & 4 & -2 & 0 & 12 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 14 & 32 & 12 & 0 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{141} 3^{1^{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -527, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 18 & 3 & 3 \\ 11 & 21 & 0 & 18 & 76 & 15 & 16 \\ 2 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & -3 & 3 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{111} 3^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -621, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 5 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & -3 & 2 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{121} 3^{1^{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -707, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$

$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 12 & 3 & 3 \\ 7 & 13 & 0 & 12 & 28 & 9 & 10 \\ 2 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & -3 & 3 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{131} 3^{1^{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -713, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 15 & 3 & 3 \\ 9 & 17 & 0 & 15 & 49 & 12 & 13 \\ 2 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & -3 & 3 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 14, \quad \Delta = -95, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.22 \rightarrow 13, \quad T_{\max} = 14, \quad T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 5 & 9 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 15 & 12 & 6 \\ 3 & 4 & -3 & 2 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

4.549 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \quad 1^{13} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 48, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.33 \rightarrow 14, \quad T_{\max} = 35, \quad T \in [14, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 24 & 6 & 8 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & -3 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.11 \rightarrow 13, \quad T_{\max} = 35, \quad T \in [13, 35]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & 0 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 98, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.22 \rightarrow 13, \quad T_{\max} = 35, \quad T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 4 & 5 \\ 2 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{13} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 118, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.19 \rightarrow 13, \quad T_{\max} = 35, \quad T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 128, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 35, \quad T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 208, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 35, \quad T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 5 & 6 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 278, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 35, \quad T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 7 & 9 \\ 2 & 3 & 0 & 7 & 0 & 2 \\ 3 & 4 & -3 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{13}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 35, \ \Delta = -118, \ \det = 328, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 35, \ T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 17 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 48 & 9 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & -3 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \ 1^{13}3^{121}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 13 & 1 & 6 & 5 \\ 13 & 30 & 0 & 13 & 10 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 13 & 0 & 4 & 4 \\ 5 & 10 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11}3^{121}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -52, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 5 \\ 12 & 25 & 0 & 12 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 12 & 0 & 4 & 4 \\ 5 & 9 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11}3^{131}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -64, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.50 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 2 & 6 & 0 & 4 & 4 \\ 4 & 6 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 0 & 1 & 3 \\ 3 & 4 & 5 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{12}3^{121}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -69, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.59 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 6 & 1 & 12 & 5 \\ 6 & 5 & 1 & 14 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 14 & 0 & 25 & 9 \\ 5 & 5 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11}3^{121}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -96, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 6 & 5 \\ 9 & 13 & 1 & 9 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 4 & 4 \\ 5 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11}3^{131}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -106, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.30 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 8 & 6 & 0 & 5 & 6 \\ 2 & 6 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 1 & 3 \\ 3 & 6 & 3 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11}3^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.32 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 2 & 3 & 0 & 2 & 0 \\ 2 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{12}3^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -136, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 8 & 6 & 0 & 20 & 6 \\ 2 & 6 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 13 & 0 & 43 & 13 \\ 3 & 6 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11}3^{11}1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -149, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 3 & 2 & 0 & 2 & 0 \\ 2 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{13}1^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 8 & 6 & 0 & 10 & 6 \\ 2 & 6 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 10 & 5 & 0 & 7 & 5 \\ 3 & 6 & 3 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{13}1^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 \\ 1 & 3 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11}3^{11}1^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -306, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 8 & 6 & 0 & 15 & 6 \\ 2 & 6 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 15 & 9 & 0 & 21 & 9 \\ 3 & 6 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11}3^{11}1^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 12 & 5 & 6 \\ 1 & 3 & 2 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & -3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [21] \quad 1^{11}4^{11}13^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11}3^{11}1^{11}4^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 130, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 4 & 0 & 3 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [23] \quad 1^{11}5^{11}13^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 1 & 8 & 5 \\ 9 & 13 & 0 & 12 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 12 & 0 & 10 & 6 \\ 5 & 7 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

4.550 NHC(-2)(-3) + f₄ + su₃ + su₂→(-3) + so₈

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 9 & 6 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -3 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

4.551 NHC(-2)(-3) + f₄ + su₃ + su₂→(-3) + su₂

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 4 & 3 \\ 3 & 4 & 0 & 9 & 3 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 5 & 12 & 6 \\ 2 & 3 & 0 & 5 & 0 & 4 & 2 \\ 4 & 8 & 0 & 12 & 4 & 10 & 6 \\ 3 & 4 & -3 & 6 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 & 3 \\ 4 & 9 & 4 & 0 & 4 & 8 & 5 \\ 2 & 4 & -2 & 0 & 2 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 3 & 8 & 3 & 0 & 3 & 4 & 4 \\ 3 & 5 & 0 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 3 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 12 & 2 \\ 2 & 3 & 0 & 2 & 0 & 11 & 2 \\ 11 & 19 & 0 & 12 & 11 & 73 & 15 \\ 3 & 4 & -3 & 2 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 3 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 6 & 2 \\ 2 & 3 & 0 & 2 & 0 & 4 & 2 \\ 4 & 8 & 0 & 6 & 4 & 10 & 6 \\ 3 & 4 & -3 & 2 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 3 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -3 & 2 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & -3 & 3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.31 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 2 & 3 \\ 3 & 4 & 0 & 8 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 4 & 6 & 5 \\ 2 & 3 & 0 & 4 & 0 & 2 & 2 \\ 2 & 4 & 0 & 6 & 2 & 1 & 3 \\ 3 & 4 & -3 & 5 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.13 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 4 & 3 \\ 4 & 9 & 4 & 0 & 4 & 8 & 5 \\ 2 & 4 & -2 & 0 & 2 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 4 & 8 & 3 & 0 & 3 & 4 & 4 \\ 3 & 5 & 0 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -206, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 5 & 3 \\ 3 & 4 & 0 & 8 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 4 & 12 & 5 \\ 2 & 3 & 0 & 4 & 0 & 5 & 2 \\ 5 & 9 & 0 & 12 & 5 & 13 & 7 \\ 3 & 4 & -3 & 5 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -226, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & -3 & 2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -336, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 5 & 6 & 6 \\ 2 & 3 & 0 & 5 & 0 & 2 & 2 \\ 2 & 4 & 0 & 6 & 2 & 1 & 3 \\ 3 & 4 & -3 & 6 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{111}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -386, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 3 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 9 & 2 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 8 & 14 & 0 & 9 & 8 & 37 & 11 \\ 3 & 4 & -3 & 2 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 6 & 2 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 5 & 9 & 0 & 6 & 5 & 13 & 7 \\ 3 & 4 & -3 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -476, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 0 & 13 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 7 & 9 & 9 \\ 2 & 3 & 0 & 7 & 0 & 2 & 2 \\ 2 & 4 & 0 & 9 & 2 & 1 & 3 \\ 3 & 4 & -3 & 9 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -626, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 3 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 9 & 2 \\ 2 & 3 & 0 & 2 & 0 & 7 & 2 \\ 7 & 13 & 0 & 9 & 7 & 28 & 10 \\ 3 & 4 & -3 & 2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11} 1^4 3^1 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 3 & 28 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 18 & 2 \\ 2 & 3 & 0 & 2 & 0 & 16 & 2 \\ 16 & 28 & 0 & 18 & 16 & 154 & 22 \\ 3 & 4 & -3 & 2 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11} 1^4 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 3 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 12 & 2 \\ 2 & 3 & 0 & 2 & 0 & 9 & 2 \\ 9 & 17 & 0 & 12 & 9 & 49 & 13 \\ 3 & 4 & -3 & 2 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 1^4 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 3 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 15 & 2 \\ 2 & 3 & 0 & 2 & 0 & 11 & 2 \\ 11 & 21 & 0 & 15 & 11 & 76 & 16 \\ 3 & 4 & -3 & 2 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11} 1^4 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 3 & 18 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 12 & 2 \\ 2 & 3 & 0 & 2 & 0 & 10 & 2 \\ 10 & 18 & 0 & 12 & 10 & 58 & 14 \\ 3 & 4 & -3 & 2 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11} 1^4 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 3 & 23 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 15 & 2 \\ 2 & 3 & 0 & 2 & 0 & 13 & 2 \\ 13 & 23 & 0 & 15 & 13 & 100 & 18 \\ 3 & 4 & -3 & 2 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11} 1^4 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 3 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 15 & 2 \\ 2 & 3 & 0 & 2 & 0 & 12 & 2 \\ 12 & 22 & 0 & 15 & 12 & 85 & 17 \\ 3 & 4 & -3 & 2 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11} 1^4 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 3 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 18 & 2 \\ 2 & 3 & 0 & 2 & 0 & 15 & 2 \\ 15 & 27 & 0 & 18 & 15 & 133 & 21 \\ 3 & 4 & -3 & 2 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11} 1^4 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 3 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 18 & 2 \\ 2 & 3 & 0 & 2 & 0 & 14 & 2 \\ 14 & 26 & 0 & 18 & 14 & 118 & 20 \\ 3 & 4 & -3 & 2 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{114} 3^{11} 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -1826, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 3 & 32 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 21 & 2 \\ 2 & 3 & 0 & 2 & 0 & 18 & 2 \\ 18 & 32 & 0 & 21 & 18 & 193 & 25 \\ 3 & 4 & -3 & 2 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{114} 3^{11} 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -2386, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 3 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 21 & 2 \\ 2 & 3 & 0 & 2 & 0 & 17 & 2 \\ 17 & 31 & 0 & 21 & 17 & 172 & 24 \\ 3 & 4 & -3 & 2 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{114} 3^{11} 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -2816, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 3 & 36 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 24 & 2 \\ 2 & 3 & 0 & 2 & 0 & 20 & 2 \\ 20 & 36 & 0 & 24 & 20 & 238 & 28 \\ 3 & 4 & -3 & 2 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [26] \quad 1^{11} 4^{13} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, \quad T_{\max} = 13, \quad \Delta = -58, \quad \det = -32, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -2 & 2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{111} 1^3 3^{11} 4^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, \quad T_{\max} = 13, \quad \Delta = -58, \quad \det = -96, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 0 & 4 & 0 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 7 & 0 & 5 & 0 & 8 & 6 \\ 3 & 4 & -2 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{111} 1^3 3^{11} 4^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, \quad T_{\max} = 13, \quad \Delta = -58, \quad \det = -192, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 8 & 3 \\ 3 & 4 & 0 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 12 & 0 & 8 & 0 & 26 & 10 \\ 3 & 4 & -2 & 2 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

4.552 $\text{NHC}(-2)(-3) + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$

$$T = 10 \quad [0] \quad 1^{113} 3^{11} 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 3 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 4 \\ 3 & 5 & 0 & 3 & 3 & 6 & 4 \\ 3 & 4 & -3 & 2 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}3^11^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 392, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 1 & 3 \\ 2 & 4 & 0 & 3 & 2 & 3 & 1 \\ 3 & 4 & -3 & 2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}3^11^{\overline{111}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 462, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 3 & 5 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 3 & 6 & 2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 5 & 2 \\ 3 & 5 & 0 & 3 & 3 & 4 & 9 & 4 \\ 5 & 9 & 0 & 6 & 5 & 9 & 13 & 7 \\ 3 & 4 & -3 & 2 & 2 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}3^11^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 512, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 3 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 5 & 6 & 6 & 6 \\ 2 & 3 & 0 & 5 & 0 & 2 & 2 & 2 \\ 2 & 4 & 0 & 6 & 2 & 1 & 3 & 3 \\ 2 & 4 & 0 & 6 & 2 & 3 & 1 & 3 \\ 3 & 4 & -3 & 6 & 2 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}3^11^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 622, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 3 & 6 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 & 5 & 2 \\ 2 & 4 & 0 & 3 & 2 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 5 & 6 & 13 & 7 \\ 3 & 4 & -3 & 2 & 2 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}3^11^{\overline{111}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 1152, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 3 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 6 & 6 & 2 \\ 2 & 3 & 0 & 2 & 0 & 5 & 5 & 2 \\ 5 & 9 & 0 & 6 & 5 & 13 & 15 & 7 \\ 5 & 9 & 0 & 6 & 5 & 15 & 13 & 7 \\ 3 & 4 & -3 & 2 & 2 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

4.553 NHC(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{12}3^11^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -34, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 & 3 \\ 3 & 4 & 0 & 8 & 3 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 4 & 6 & 5 \\ 2 & 3 & 0 & 4 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 3 & 3 \\ 3 & 4 & -3 & 5 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}3^11^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -36, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.33 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 4 & 0 & 5 \\ 2 & 4 & -2 & 0 & 2 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}3^11^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -274, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.47 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 2 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 3 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 2 & 6 & 2 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 5 & 9 & 0 & 6 & 5 & 12 & 6 \\ 3 & 4 & -3 & 2 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

4.554 NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_{2n3\text{mix}}$

$$T = 8 \quad [0] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 9 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 29 & 6 \\ 10 & 19 & 0 & 29 & 64 & 14 \\ 3 & 4 & -3 & 6 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 102, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 5 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 9 & 3 \\ 6 & 11 & 0 & 9 & 20 & 8 \\ 3 & 4 & -3 & 3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 5 & 8 & 0 & 5 & 12 & 6 \\ 3 & 4 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 16 & 5 \\ 7 & 12 & 0 & 16 & 25 & 9 \\ 3 & 4 & -3 & 5 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 5 & 0 & 12 & 6 \\ 3 & 5 & 0 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 5 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 6 & 3 \\ 4 & 7 & 0 & 6 & 7 & 5 \\ 3 & 4 & -3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

- [6] $1^{\overline{111}3^11^{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 155$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.06 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 16 & 27 & 0 & 17 & 151 & 21 \\ 3 & 4 & -3 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$
- [7] $1^{\overline{111}3^11^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 162$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 5 & 0 & 12 & 6 \\ 3 & 5 & 0 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{\overline{121}3^11^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 197$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 12.26 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 0 & 9 & 10 & 5 \\ 4 & 7 & 0 & 10 & 7 & 5 \\ 3 & 4 & -3 & 5 & 5 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{\overline{113}3^11^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 208$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 9 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 23 & 6 \\ 8 & 15 & 0 & 23 & 39 & 11 \\ 3 & 4 & -3 & 6 & 11 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{\overline{111}3^11^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 227$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.04 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 4 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{\overline{111}3^11^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 272$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 11 & 6 \\ 4 & 7 & 0 & 11 & 7 & 5 \\ 3 & 4 & -3 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{\overline{112}3^11^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 272$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 9 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 17 & 6 \\ 6 & 11 & 0 & 17 & 20 & 8 \\ 3 & 4 & -3 & 6 & 8 & 1 \end{pmatrix}_{b_0}$$
- [14] $1^{\overline{111}3^11^{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 332$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 13 & 22 & 0 & 14 & 97 & 17 \\ 3 & 4 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$
- [13] $1^{\overline{112}3^11^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 332$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.14 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 6 & 11 & 0 & 8 & 20 & 8 \\ 3 & 4 & -3 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$
- [15] $1^{\overline{111}3^11^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 368$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 7 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & -3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

- [16] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 382$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 13 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 0 & 27 & 16 & 9 \\ 4 & 7 & 0 & 16 & 7 & 5 \\ 3 & 4 & -3 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$
- [17] $1^{111}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 403$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 10 & 17 & 0 & 11 & 55 & 13 \\ 3 & 4 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$
- [18] $1^{113}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 443$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 8 & 15 & 0 & 11 & 39 & 11 \\ 3 & 4 & -3 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$
- [19] $1^{114}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 560$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.33 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 10 & 19 & 0 & 14 & 64 & 14 \\ 3 & 4 & -3 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$
- [20] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 635$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 9 & 16 & 0 & 11 & 44 & 12 \\ 3 & 4 & -3 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$
- [21] $1^{112}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 832$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 12 & 21 & 0 & 14 & 80 & 16 \\ 3 & 4 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$
- [22] $1^{112}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 908$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 20 & 2 \\ 18 & 31 & 0 & 20 & 188 & 24 \\ 3 & 4 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$
- [23] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 908$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.01 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 11 & 20 & 0 & 14 & 69 & 15 \\ 3 & 4 & -3 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$
- [24] $1^{112}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 923$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 15 & 26 & 0 & 17 & 128 & 20 \\ 3 & 4 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$
- [25] $1^{114}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 1187$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.05 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 13 & 24 & 0 & 17 & 100 & 18 \\ 3 & 4 & -3 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{113}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 1267$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 25 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 14 & 25 & 0 & 17 & 111 & 19 \\ 3 & 4 & -3 & 2 & 19 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{113}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 1520$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 30 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 20 & 2 \\ 17 & 30 & 0 & 20 & 165 & 23 \\ 3 & 4 & -3 & 2 & 23 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{113}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 1667$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 35 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 23 & 2 \\ 20 & 35 & 0 & 23 & 231 & 27 \\ 3 & 4 & -3 & 2 & 27 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{114}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 1708$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 29 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 20 & 2 \\ 16 & 29 & 0 & 20 & 148 & 22 \\ 3 & 4 & -3 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{114}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 2123$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 19 & 3 \\ 3 & 4 & 0 & 4 & 34 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 23 & 2 \\ 19 & 34 & 0 & 23 & 208 & 26 \\ 3 & 4 & -3 & 2 & 26 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{114}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 2432$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 22 & 3 \\ 3 & 4 & 0 & 4 & 39 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 26 & 2 \\ 22 & 39 & 0 & 26 & 280 & 30 \\ 3 & 4 & -3 & 2 & 30 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [32] $1^{111}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 37 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 37 & 54 & 0 & 217 & 27 \\ 5 & 7 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

[33] $1^{11}31^{113}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -61$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 38 & 4 \\ 1 & 4 & 0 & 0 & 34 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 4 \\ 3 & 8 & 4 & 0 & 49 & 5 \\ 2 & 4 & 2 & -3 & 2 & 5 \end{pmatrix}_{b_0}$$

[34] $1^{112}31^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.25 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 13 & 2 \\ 1 & 1 & 4 & 0 & 37 & 3 \\ 2 & 4 & 3 & 0 & 38 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 7 \\ 3 & 7 & 8 & 0 & 7 & 15 \\ 2 & 3 & 3 & -3 & 2 & 7 \end{pmatrix}_{b_0}$$

[35] $1^{111}3^11^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -76$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.26 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 5 & 3 \\ 2 & 4 & 3 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 5 & 2 \\ 2 & 5 & 6 & 0 & 5 & 7 & 5 \\ 2 & 3 & 3 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{112}3^121^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 44 & 5 \\ 9 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 44 & 69 & 0 & 340 & 34 \\ 5 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{111}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 7 & 5 \\ 9 & 13 & 1 & 9 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 4 & 4 \\ 5 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{111}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 25 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 39 & 0 & 107 & 19 \\ 5 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{112}3^121^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 56 & 5 \\ 9 & 13 & 1 & 84 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 56 & 84 & 0 & 520 & 42 \\ 5 & 7 & -1 & 42 & 3 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{111}3^11^31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 5 & 4 \\ 1 & 4 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 4 & 2 \\ 2 & 5 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 2 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{111}3^11^31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 4 & 3 \\ 2 & 4 & 3 & 0 & 3 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 4 & 2 \\ 2 & 4 & 5 & 0 & 4 & 4 & 4 \\ 2 & 3 & 3 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{111}3^121^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 15 & 5 \\ 9 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 24 & 0 & 37 & 11 \\ 5 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{11}3^11^31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 5 & 4 \\ 1 & 4 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 0 & 3 & 5 & 3 \\ 2 & 5 & 4 & 0 & 5 & 4 & 4 \\ 2 & 4 & 2 & -3 & 3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{11}3^11^43^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 0 & 3 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 14 & 2 \\ 9 & 27 & 14 & 0 & 14 & 120 & 18 \\ 2 & 4 & 2 & -3 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{11}3^11^43^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 3 & 9 & 5 & 0 & 5 & 12 \\ 2 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{11} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 24 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 39 & 0 & 100 & 18 \\ 5 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{11} 3^1 1^{13} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 0 & 3 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 11 & 2 \\ 7 & 21 & 11 & 0 & 11 & 72 & 14 \\ 2 & 4 & 2 & -3 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{11} 3^1 1^{12} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -261, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 8 & 2 \\ 5 & 15 & 8 & 0 & 8 & 36 & 10 \\ 2 & 4 & 2 & -3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{11} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 26 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 39 & 0 & 107 & 19 \\ 5 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{11} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 16 & 5 \\ 9 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 24 & 0 & 37 & 11 \\ 5 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{11} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 17 & 5 \\ 9 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 24 & 0 & 37 & 11 \\ 5 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{11} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 34 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 54 & 0 & 200 & 26 \\ 5 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{11} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 27 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 39 & 0 & 107 & 19 \\ 5 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{11} 31^{11} 3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 0 & 9 & 5 & 4 \\ 1 & 4 & 0 & 0 & 6 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 0 & 12 & 9 & 6 \\ 2 & 5 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 2 & -3 & 6 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[55] \ 1^{11} 3^1 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 45 & 5 \\ 9 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 45 & 69 & 0 & 340 & 34 \\ 5 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[56] 1^{11} 31^{11} 31^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -430, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 3 & 11 & 4 \\ 1 & 4 & 0 & 0 & 3 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 7 & 2 \\ 4 & 11 & 7 & 0 & 7 & 20 & 8 \\ 2 & 4 & 2 & -3 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[57] 1^{11} 31^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 25 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 39 & 0 & 100 & 18 \\ 5 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[58] 1^{11} 31^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 35 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 35 & 54 & 0 & 200 & 26 \\ 5 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[60] 1^{12} 31^{12} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 26 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 39 & 0 & 100 & 18 \\ 5 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[59] 1^{12} 31^{12} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 46 & 5 \\ 9 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 46 & 69 & 0 & 340 & 34 \\ 5 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[61] 1^{11} 31^{12} 31^{12} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -679, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 17 & 4 \\ 1 & 4 & 0 & 0 & 3 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 10 & 2 \\ 6 & 17 & 10 & 0 & 10 & 48 & 12 \\ 2 & 4 & 2 & -3 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[62] 1^{12} 31^{12} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -768, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 36 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 36 & 54 & 0 & 200 & 26 \\ 5 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[63] 1^{11} 31^{13} 31^{13} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -886, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 3 & 23 & 4 \\ 1 & 4 & 0 & 0 & 3 & 13 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 13 & 2 \\ 8 & 23 & 13 & 0 & 13 & 88 & 16 \\ 2 & 4 & 2 & -3 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[64] 1^{11} 31^{14} 31^{14} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -1051, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 3 & 29 & 4 \\ 1 & 4 & 0 & 0 & 3 & 16 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 16 & 2 \\ 10 & 29 & 16 & 0 & 16 & 140 & 20 \\ 2 & 4 & 2 & -3 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [65] \quad 1^{11} 4^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{111} 4^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 6 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & -2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{11} 4^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 79, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{11} 4^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 105, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 6 & 8 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{111} 4^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{11} 4^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{111} 4^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 9 & 12 & 0 & 8 & 26 & 10 \\ 3 & 4 & -2 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{111} 4^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [73] \quad 1^{11} 5^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -89, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 1 & 14 & 5 \\ 9 & 13 & 0 & 22 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 22 & 0 & 33 & 11 \\ 5 & 7 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[74] $1^{12} \overset{1}{5} 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 11, T_{\max} = 14, \Delta = -83, \det = -126, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 5 & 9 & 1 & 20 & 5 \\ 9 & 13 & 0 & 32 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 32 & 0 & 70 & 16 \\ 5 & 7 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

4.555 NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_{2n3\text{mix}} + \mathfrak{su}_2$

$T = 9$ [0] $1^{11} \overset{1}{3} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 3 \\ 3 & 4 & 0 & 4 & 8 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 5 & 3 & 2 \\ 5 & 8 & 0 & 5 & 12 & 9 & 6 \\ 3 & 5 & 0 & 3 & 9 & 4 & 4 \\ 3 & 4 & -3 & 2 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111} \overset{1}{3} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 11 & 3 & 2 \\ 10 & 17 & 0 & 11 & 55 & 11 & 13 \\ 2 & 4 & 0 & 3 & 11 & 1 & 3 \\ 3 & 4 & -3 & 2 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111} \overset{1}{3} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 & 3 \\ 3 & 4 & 0 & 4 & 7 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 5 & 3 & 2 \\ 4 & 7 & 0 & 5 & 7 & 6 & 5 \\ 3 & 5 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -3 & 2 & 5 & 4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{121} \overset{1}{3} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 0 & 5 & 11 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 9 & 3 & 3 \\ 6 & 11 & 0 & 9 & 20 & 8 & 8 \\ 2 & 4 & 0 & 3 & 8 & 1 & 3 \\ 3 & 4 & -3 & 3 & 8 & 3 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111} \overset{1}{3} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 0 & 5 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 6 & 3 & 3 \\ 4 & 7 & 0 & 6 & 7 & 5 & 5 \\ 2 & 4 & 0 & 3 & 5 & 1 & 3 \\ 3 & 4 & -3 & 3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111} \overset{1}{3} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 8 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 5 & 6 & 2 \\ 5 & 8 & 0 & 5 & 12 & 14 & 6 \\ 5 & 9 & 0 & 6 & 14 & 13 & 7 \\ 3 & 4 & -3 & 2 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{113} \overset{1}{3} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 & 3 \\ 3 & 4 & 0 & 4 & 20 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 14 & 3 & 2 \\ 11 & 20 & 0 & 14 & 69 & 18 & 15 \\ 3 & 5 & 0 & 3 & 18 & 4 & 4 \\ 3 & 4 & -3 & 2 & 15 & 4 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{1111}1}3^{\overline{111}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -401$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 & 3 \\ 3 & 4 & 0 & 4 & 12 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 8 & 3 & 2 \\ 7 & 12 & 0 & 8 & 25 & 12 & 9 \\ 3 & 5 & 0 & 3 & 12 & 4 & 4 \\ 3 & 4 & -3 & 2 & 9 & 4 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{\overline{1111}1}3^{\overline{111}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -424$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 5 & 3 & 2 \\ 4 & 7 & 0 & 5 & 7 & 5 & 5 \\ 2 & 4 & 0 & 3 & 5 & 1 & 3 \\ 3 & 4 & -3 & 2 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{\overline{1111}1}3^{\overline{111}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -464$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 2 & 3 \\ 3 & 4 & 0 & 9 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 0 & 12 & 11 & 6 & 6 \\ 4 & 7 & 0 & 11 & 7 & 5 & 5 \\ 2 & 4 & 0 & 6 & 5 & 1 & 3 \\ 3 & 4 & -3 & 6 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{\overline{1111}1}3^{\overline{111}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -481$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 8 & 3 & 2 \\ 7 & 12 & 0 & 8 & 25 & 8 & 9 \\ 2 & 4 & 0 & 3 & 8 & 1 & 3 \\ 3 & 4 & -3 & 2 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{\overline{1112}1}3^{\overline{111}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -500$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.08 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 & 3 \\ 3 & 4 & 0 & 4 & 16 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 11 & 3 & 2 \\ 9 & 16 & 0 & 11 & 44 & 15 & 12 \\ 3 & 5 & 0 & 3 & 15 & 4 & 4 \\ 3 & 4 & -3 & 2 & 12 & 4 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{\overline{1111}1}3^{\overline{111}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -509$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.06 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 5 & 3 \\ 3 & 4 & 0 & 4 & 7 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 5 & 6 & 2 \\ 4 & 7 & 0 & 5 & 7 & 11 & 5 \\ 5 & 9 & 0 & 6 & 11 & 13 & 7 \\ 3 & 4 & -3 & 2 & 5 & 7 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{\overline{1111}1}3^{\overline{111}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -896$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 5 & 3 \\ 3 & 4 & 0 & 4 & 12 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 8 & 6 & 2 \\ 7 & 12 & 0 & 8 & 25 & 20 & 9 \\ 5 & 9 & 0 & 6 & 20 & 13 & 7 \\ 3 & 4 & -3 & 2 & 9 & 7 & 1 \end{pmatrix}_{b_0}$$

4.556 NHC(-2)(-3) + f₄ + su₃ + su_{2n3mix} + su₃

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 5 & 3 & 2 \\ 5 & 8 & 0 & 5 & 12 & 5 & 6 \\ 2 & 4 & 0 & 3 & 5 & 0 & 2 \\ 3 & 4 & -3 & 2 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -81, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 1 & 3 \\ 4 & 9 & 4 & 0 & 12 & 0 & 5 \\ 2 & 4 & -2 & 0 & 5 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 5 & 12 & 5 & 0 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & 6 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 0 & 5 & 7 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 6 & 6 & 3 \\ 4 & 7 & 0 & 6 & 7 & 10 & 5 \\ 4 & 8 & 0 & 6 & 10 & 9 & 5 \\ 3 & 4 & -3 & 3 & 5 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 0 & 5 & 11 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 9 & 3 & 3 \\ 6 & 11 & 0 & 9 & 20 & 8 & 8 \\ 2 & 4 & 0 & 3 & 8 & 0 & 2 \\ 3 & 4 & -3 & 3 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 0 & 5 & 7 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 6 & 3 & 3 \\ 4 & 7 & 0 & 6 & 7 & 5 & 5 \\ 2 & 4 & 0 & 3 & 5 & 0 & 2 \\ 3 & 4 & -3 & 3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

4.557 NHC(-2)(-3) + f₄ + su₃ + su₃

$$T = 8 \quad [0] \quad 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 9 & 4 & 0 & 0 & 5 \\ 2 & 4 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 0 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 5 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 6 & 3 \\ 4 & 8 & 0 & 6 & 9 & 5 \\ 3 & 4 & -3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

- [3] $1^{111}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 0 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$
- [4] $1^{111}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.28 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 0 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$
- [5] $1^{111}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$
- [6] $1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$
- [7] $1^{111}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.19 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$
- $T = 9$ [8] $1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 3 \\ 3 & 4 & 3 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{11}321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 5 & 12 & 1 & 1 & 5 & 5 \\ 12 & 25 & 0 & 0 & 9 & 9 \\ 1 & 0 & -3 & 0 & -1 & -1 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 5 & 9 & -1 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$
- [10] $1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 5 & 13 & 9 & 1 & 5 & 5 \\ 13 & 30 & 20 & 0 & 10 & 10 \\ 9 & 20 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 5 & 10 & 6 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$
- [11] $1^{12}321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 5 & 3 \\ 2 & 2 & -1 & 0 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{111}321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{f}}_4 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 11 & 0 & 31 & 11 \\ 3 & 4 & 3 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{11}321^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 4 & 3 \\ 2 & 5 & 0 & 4 & 1 & 3 \\ 3 & 5 & -3 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{111}31^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 3 & 3 & 0 & 3 & 0 \\ 2 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{11}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 5 \\ 9 & 13 & 1 & 0 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -105, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[17] $1^{111}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 6 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 3 & 0 & 3 & 3 \\ 3 & 4 & 3 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{12}321^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 1 & 5 & 2 \\ 1 & 2 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 5 & 2 \\ 3 & 5 & 5 & 0 & 5 & 12 & 6 \\ 2 & 2 & 2 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{11}321^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 0 & 0 & 4 & 2 \\ 2 & 3 & 4 & 0 & 4 & 4 & 4 \\ 2 & 2 & 2 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{111}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 5 & 9 & 1 & 9 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 12 & 6 \\ 5 & 7 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{11}321^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 5 & 0 & 5 & 10 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 7 & 3 \\ 5 & 10 & 0 & 7 & 13 & 7 \\ 3 & 5 & -3 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{111}321^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 7 & 0 & 13 & 7 \\ 3 & 4 & 3 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{11}31^{11}31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.52 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 2 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{111}321^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 5 & 9 & 1 & 10 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 12 & 6 \\ 5 & 7 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{11}321^{112} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -250, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 11 & 3 \\ 7 & 15 & 0 & 11 & 31 & 11 \\ 3 & 5 & -3 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{11}321^{113} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 5 & 0 & 5 & 20 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 0 & 1 & 15 & 3 \\ 9 & 20 & 0 & 15 & 57 & 15 \\ 3 & 5 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{111}321^{111} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 4 & 7 & 4 \\ 2 & 3 & 4 & 0 & 7 & 4 & 4 \\ 2 & 2 & 2 & -3 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [28] $1^{11}4131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_3} \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 9 & 0 & 4 & 4 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 3 & 5 & -2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{11}41^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_3} \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 4 & -2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{11}41^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_3} \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 102, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 3 & 4 & -2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.558 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$

$T = 9$ [0] $1^{111}31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 0 & 4 & 4 \\ 3 & 5 & 0 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -119$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & -3 & 3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -126$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 13.05 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -3 & 2 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{113}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -159$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.21 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 0 & 9 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 10 & 24 & 9 & 0 & 0 & 52 & 12 \\ 3 & 5 & 0 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{112}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -174$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.7$

$T_{\min} = 12.21 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 0 & 16 & 5 \\ 2 & 4 & -2 & 0 & 0 & 6 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 16 & 6 & 0 & 0 & 22 & 8 \\ 3 & 5 & 0 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -182$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.26 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -3 & 3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{122}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -200$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.9$

$T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 0 & 5 & 8 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 6 & 12 & 3 \\ 4 & 8 & 0 & 6 & 9 & 18 & 5 \\ 7 & 13 & 0 & 12 & 18 & 28 & 10 \\ 3 & 4 & -3 & 3 & 5 & 10 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{1111}31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -203$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.34 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 9 & 3 & 2 \\ 8 & 14 & 0 & 9 & 36 & 9 & 10 \\ 2 & 4 & 0 & 3 & 9 & 1 & 3 \\ 3 & 4 & -3 & 2 & 10 & 3 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{123}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -206$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 0 & 5 & 8 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 6 & 15 & 3 \\ 4 & 8 & 0 & 6 & 9 & 24 & 5 \\ 9 & 17 & 0 & 15 & 24 & 49 & 13 \\ 3 & 4 & -3 & 3 & 5 & 13 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{1111}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 4 & 14 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 9 & 37 & 11 \\ 3 & 4 & -3 & 2 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{113}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 0 & 24 & 5 \\ 2 & 4 & -2 & 0 & 0 & 9 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 11 & 24 & 9 & 0 & 0 & 52 & 12 \\ 3 & 5 & 0 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 3 \\ 3 & 4 & 0 & 4 & 9 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 5 & 9 & 0 & 6 & 12 & 9 & 6 \\ 3 & 5 & 0 & 3 & 9 & 4 & 4 \\ 3 & 4 & -3 & 2 & 6 & 4 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -399, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -3 & 2 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{114}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 0 & 12 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 15 & 32 & 12 & 0 & 0 & 94 & 16 \\ 3 & 5 & 0 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & -3 & 3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 5 & 9 & 0 & 6 & 12 & 6 & 6 \\ 2 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & -3 & 2 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{114}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 0 & 32 & 5 \\ 2 & 4 & -2 & 0 & 0 & 12 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 14 & 32 & 12 & 0 & 0 & 94 & 16 \\ 3 & 5 & 0 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{114} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -527, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 0 & 5 & 4 & 21 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 18 & 3 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 11 & 21 & 0 & 18 & 15 & 76 & 16 \\ 3 & 4 & -3 & 3 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{114} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -621, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & -3 & 2 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{114} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -707, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 12 & 3 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 7 & 13 & 0 & 12 & 9 & 28 & 10 \\ 3 & 4 & -3 & 3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{113} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -713, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 5 & 4 & 17 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 15 & 3 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 9 & 17 & 0 & 15 & 12 & 49 & 13 \\ 3 & 4 & -3 & 3 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 5 & 9 & 0 & 6 & 12 & 15 & 6 \\ 5 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & -3 & 2 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

4.559 NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{123} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 0 & 5 & 8 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 6 & 3 & 3 \\ 4 & 8 & 0 & 6 & 9 & 4 & 5 \\ 2 & 3 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -3 & 3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 0 & 4 & 5 \\ 2 & 4 & -2 & 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 0 & 2 \\ 3 & 5 & 0 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -274, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.47 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 2 & 2 \\ 2 & 4 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & -3 & 2 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 6 & 2 & 2 \\ 5 & 9 & 0 & 6 & 12 & 5 & 6 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -3 & 2 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

4.560 NHC(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_{2n3\text{mix}}$

$$T = 9 \quad [0] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 4 & 8 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 0 & 3 & 5 & 2 \\ 2 & 4 & 0 & 3 & 0 & 5 & 2 \\ 5 & 8 & 0 & 5 & 5 & 12 & 6 \\ 3 & 4 & -3 & 2 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -81, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 0 & 12 & 5 \\ 2 & 4 & -2 & 0 & 0 & 5 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 5 & 0 & 0 & 12 & 6 \\ 3 & 5 & 0 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 0 & 5 & 8 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 6 & 6 & 3 \\ 4 & 8 & 0 & 6 & 9 & 10 & 5 \\ 4 & 7 & 0 & 6 & 10 & 7 & 5 \\ 3 & 4 & -3 & 3 & 5 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 0 & 5 & 4 & 11 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 0 & 8 & 2 \\ 6 & 11 & 0 & 9 & 8 & 20 & 8 \\ 3 & 4 & -3 & 3 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 & 7 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 0 & 5 & 2 \\ 4 & 7 & 0 & 6 & 5 & 7 & 5 \\ 3 & 4 & -3 & 3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

4.561 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 3 & 0 & 0 & 3 & 3 \\ 3 & 5 & 0 & -1 & -1 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 0 & 8 & 5 \\ 2 & 4 & -2 & 0 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 3 & 0 & 0 & 3 & 3 \\ 3 & 5 & 0 & -1 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

4.562 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + (-3) \rightarrow (-2)$

$$T = 7 \quad [0] \quad 1^{\overline{1}} 31^{\overline{11}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 97, \Delta = -147, \det = -275, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [1] \quad 1^{\overline{1}} 231^{\overline{11}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 45, \Delta = -119, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 45, T \in [13, 45]$$

$$\begin{pmatrix} 13 & 20 & 1 & 7 \\ 20 & 25 & 0 & 9 \\ 1 & 0 & -3 & -1 \\ 7 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \quad 2^{\overline{1}} 1^{\overline{1}} 41^{\overline{11}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 1 & 6 & 0 & 3 \\ 5 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 3 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [3] \quad 1^{\overline{1}} 5^{\overline{21}} 131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 13 & 0 & 7 \\ 1 & 0 & -3 & -1 \\ 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

4.563 $\text{NHC}(-2)(-3) + \mathfrak{f}_4 + (-3) \rightarrow (-2) + \mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^{\overline{1}} 31^{\overline{11}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 725, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 31, T \in [12, 31]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 6 & 4 \\ 8 & 9 & 4 & 0 & 8 & 5 \\ 3 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 8 & 3 & 0 & 4 & 4 \\ 4 & 5 & 0 & -1 & 4 & 1 \end{pmatrix}_{b_0}$$

4.564 NHC(-2)(-3) + f₄ + (-3)→(-2) + su₂→(-3)

$$T = 8 \quad [0] \quad 1^1 3^1 1^1 2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + (-3) \rightarrow (-2) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 45, \Delta = -119, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 45, T \in [12, 45]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 3 & 4 \\ 8 & 9 & 4 & 0 & 4 & 5 \\ 3 & 4 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 2 & 0 & 0 & 2 \\ 4 & 5 & 0 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [1] \quad 2^1 1^1 4^1 1^3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + (-3) \rightarrow (-2) + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 14, \Delta = -89, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 1 & 3 \\ 3 & 1 & 6 & 0 & 0 & 3 \\ 5 & 6 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 6 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \quad 1^1 5^1 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + (-3) \rightarrow (-2) + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 10 & 1 & 8 & 5 \\ 10 & 13 & 0 & 12 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 12 & 0 & 10 & 6 \\ 5 & 7 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

4.565 NHC(-2)(-3) + NHC(-2)(-3)

$$T = 7 \quad [0] \quad 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -2 & 1 & 0 \\ 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 4 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 4 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 3 & 5 & 4 & 4 & 4 \\ 2 & 4 & 4 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 2^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 4 & 5 & 4 \\ 3 & 4 & 1 & 4 & 3 \\ 6 & 5 & 4 & 3 & 3 \\ 4 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 2 & 4 & 1 & 4 & 3 \\ 3 & 5 & 4 & 3 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0} \\
[6] & 131^{\overline{11}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 1 & 4 & 3 \\ 2 & 4 & 4 & 0 & 2 \\ 3 & 4 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[7] & 21^{\overline{11}}31^{\overline{11}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 6 & 3 & 4 \\ 5 & 4 & 5 & 4 & 4 \\ 6 & 5 & 4 & 4 & 4 \\ 3 & 4 & 4 & 0 & 2 \\ 4 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0} \\
[8] & 21^{\overline{11}}31^{\overline{11}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 4 & 5 & 4 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 5 & 4 & 3 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0} \\
[9] & 21^{\overline{11}}31^{\overline{11}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
\\
T = 8 & [10] 1^{\overline{11}}41^{\overline{11}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0} \\
[11] & 1^{\overline{11}}321^{\overline{11}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 10, \Delta = -66, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 2 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 5 & 4 & -1 & 4 \end{pmatrix}_{b_0} \\
[12] & 21^{\overline{11}}41^{\overline{11}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 3 & 5 & 4 & 4 & 4 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[13] & 21^{\overline{11}}41^{\overline{11}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[14] & 141^{\overline{11}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[15] & 221^{\overline{11}}41^{\overline{11}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 58, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 7 & 8 & 4 \\ 5 & 7 & 7 & 12 & 5 \\ 6 & 8 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0} \\
[16] & 21^{\overline{11}}41^{\overline{11}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[17] \quad & 31^{11}31^{11}3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{orange}{\mathfrak{g}}_2] \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 3 & 6 \\ 3 & 4 & 0 & 3 & 3 \\ 3 & 3 & 3 & 1 & 4 \\ 3 & 6 & 3 & 4 & 3 \\ 2 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
T = 9 \quad & [18] \quad 21^1 \overset{1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{orange}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 3 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[19] \quad & 21^1 \overset{1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{orange}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 3 & 3 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[21] \quad & 221^1 \overset{1}{5} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{orange}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 6 & 4 \\ 2 & 4 & 1 & 5 & 3 \\ 5 & 6 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0} \\
[20] \quad & 1 \overset{1}{1^1} \overset{1}{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{orange}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[22] \quad & 21^1 \overset{1}{5} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{orange}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 4 & 6 & 4 \\ 3 & 4 & 1 & 5 & 3 \\ 6 & 6 & 5 & 7 & 5 \\ 4 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0} \\
[23] \quad & 1^1 4^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{orange}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 11, \Delta = -64, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 2 & 3 & 3 & 4 \\ 3 & 1 & 2 & 3 \\ 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 5 \end{pmatrix}_{b_0} \\
[24] \quad & 1^1 \overset{1}{5} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{orange}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[25] \quad & 21^1 \overset{1}{5} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{orange}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 3 & 4 & 3 \\ 4 & 4 & 4 & 6 & 4 \\ 3 & 4 & 1 & 5 & 3 \\ 4 & 6 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0} \\
[26] \quad & 21^1 \overset{1}{5} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{orange}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[27] \quad 1 \overset{1}{\underset{1^1, 1^{11}}{5}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[28] \quad 1 \overset{1}{\underset{1^1}{5}} 1^{11} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[29] \quad 221 \overset{1}{\underset{1^1}{5}} 1^{11} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 5 & 2 & 3 \\ 6 & 13 & 12 & 5 & 7 \\ 5 & 12 & 8 & 5 & 6 \\ 2 & 5 & 5 & 0 & 2 \\ 3 & 7 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [30] \quad 1 \overset{1}{\underset{1^1}{5}} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 2 & 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[31] \quad 1 \overset{1}{\underset{1^1}{5}} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 5 & 5 & 3 & 5 \\ 2 & 5 & 2 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 5 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[32] \quad 1 \overset{1}{\underset{1^1}{5}} 1231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 2 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 5 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[33] \quad 2 \overset{1}{\underset{1^1}{3}} 1 \overset{1}{\underset{1^1}{5}} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 5 & 5 & 3 & 5 \\ 3 & 5 & 2 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 5 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[34] \quad 22 \overset{1}{\underset{1^1}{3}} 151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 243, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 5 & 5 & 3 \\ 6 & 14 & 13 & 13 & 8 \\ 5 & 13 & 9 & 12 & 7 \\ 5 & 13 & 12 & 8 & 6 \\ 3 & 8 & 7 & 6 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [35] \quad 1^{11} 4131 \overset{1}{\underset{1^1}{5}} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[36] \quad 41 \overset{1}{\underset{1^1}{11}} 231 \overset{1}{\underset{1^1}{5}} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -108, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 9 & 8 & 1 & 3 \\ 9 & 51 & 48 & 0 & 15 \\ 8 & 48 & 42 & 1 & 14 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 15 & 14 & -1 & 4 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [37] \quad 1^{\overline{1}^1 \overline{5}^1 131} 6^{11} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 3 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

4.566 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{e}_6$

$$T = 8 \quad [0] \quad 1^{\overline{1}^1 \overline{1}^1 \overline{3}^1} 31^{\overline{1}^1 \overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 4 & 4 & 0 & 4 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1}^1 \overline{1}^1 \overline{1}^1} 31^{\overline{1}^1 \overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 2 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1}^1 \overline{1}^1 \overline{1}^1} 31^{\overline{1}^1 \overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 \\ 3 & 5 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$

4.567 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \quad 1^{\overline{1}^1 \overline{1}^1 \overline{1}^1 \overline{1}^1} 31^{\overline{1}^1 \overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 4 & 3 \\ 3 & 5 & 3 & 3 & 0 & 4 & 3 & 3 \\ 3 & 4 & 3 & 2 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

4.568 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\overline{1}^1 \overline{1}^1 \overline{1}^1} 31^{\overline{1}^1 \overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 13 & 0 & 6 \\ 9 & 17 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

4.569 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\overline{1}^1 \overline{1}^1 \overline{1}^1} 31^{\overline{1}^1 \overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 2 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

13.5

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 5 & 4 & 0 & 23 & 4 \\ 3 & 5 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 24 & 15 & 0 & 100 & 18 \\ 3 & 4 & 4 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{113} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -1892, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 5 & 4 & 0 & 27 & 4 \\ 3 & 5 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 4 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{113} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -2307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 5 & 4 & 0 & 32 & 4 \\ 3 & 5 & 4 & 4 & 0 & 33 & 4 \\ 2 & 4 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 33 & 21 & 0 & 193 & 25 \\ 3 & 4 & 4 & 2 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{114} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -3152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 5 & 4 & 0 & 36 & 4 \\ 3 & 5 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 4 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

4.570 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{1111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 & 4 \\ 3 & 5 & 4 & 4 & 0 & 3 & 3 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & 2 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 429, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 4 & 0 & 3 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 1264, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 9 & 4 \\ 3 & 5 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

4.571 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{1111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.572 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 3 & 4 & 3 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 5 & 0 & 8 & 4 \\ 2 & 4 & 1 & 4 & 0 & 6 & 3 \\ 3 & 5 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 6 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & 3 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 & 3 \\ 3 & 5 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -588, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 4 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

4.573 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

4.574 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 5 & 4 \\ 3 & 5 & 4 & 4 & 0 & 3 & 6 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 3 & 0 & 3 & 3 & 3 \\ 3 & 4 & 4 & 2 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -101, \ \det = 825, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.42 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.575 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{f}_4$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 \quad T = 8, \ T_{\max} = 31, \ \Delta = -117, \ \det = 108, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 31, \ T \in [12, 31]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 & -3 \\ 3 & 4 & 4 & 2 & -3 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 \quad T = 8, \ T_{\max} = 31, \ \Delta = -117, \ \det = 112, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.25 \rightarrow 13, \ T_{\max} = 31, \ T \in [13, 31]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 & -3 \\ 3 & 4 & 3 & 2 & -3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 \quad T = 8, \ T_{\max} = 31, \ \Delta = -117, \ \det = 115, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.09 \rightarrow 13, \ T_{\max} = 31, \ T \in [13, 31]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 & 3 \\ 3 & 5 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 & -3 \\ 3 & 4 & 3 & 3 & -3 & 0 & 0 \end{pmatrix}_{b_0}$$

4.576 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 29, \ \Delta = -116, \ \det = -395, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, \ T_{\max} = 29, \ T \in [13, 29]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 & 8 \\ 8 & 16 & 10 & 13 & 0 & 6 & 6 \\ 9 & 17 & 13 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 & -3 \\ 5 & 8 & 6 & 6 & -3 & 1 & 1 \end{pmatrix}_{b_0}$$

4.577 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -87, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -87, \ \det = -236, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 5 & 4 & 0 & 19 & 4 \\ 3 & 5 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 19 & 21 & 12 & 0 & 73 & 15 \\ 3 & 4 & 4 & 2 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

4.578 NHC(-2)(-3) + NHC(-2)(-3) + f₄ + su₂ + su₂→(-3)

$$T = 10 \quad [0] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 3 & 4 \\ 3 & 5 & 4 & 4 & 0 & 3 & 3 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 3 & 2 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 600, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 3 & 4 \\ 3 & 5 & 4 & 4 & 0 & 9 & 3 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 6 & 0 & 13 & 5 & 7 \\ 2 & 3 & 3 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.579 NHC(-2)(-3) + NHC(-2)(-3) + f₄ + su₂→(-3)

$$T = 9 \quad [0] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 4 & 0 & 2 & 3 \\ 2 & 4 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 3 & 4 \\ 3 & 5 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

4.580 NHC(-2)(-3) + NHC(-2)(-3) + f₄ + su₂→(-3) + su₂

$$T = 10 \quad [0] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 3 & 4 \\ 3 & 5 & 4 & 4 & 0 & 3 & 3 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 4 & 2 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 600, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 3 & 9 & 4 \\ 3 & 5 & 4 & 4 & 0 & 3 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 5 & 2 \\ 5 & 9 & 9 & 6 & 0 & 5 & 13 & 7 \\ 3 & 4 & 4 & 2 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.581 NHC(-2)(-3) + NHC(-2)(-3) + f₄ + su₂n3mix

$$T = 9 \quad [0] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + [\text{su}_2 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 5 & 4 & 0 & 7 & 4 \\ 3 & 5 & 4 & 4 & 0 & 6 & 4 \\ 2 & 4 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 5 & 0 & 7 & 5 \\ 3 & 4 & 4 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -89, \ \det = -500, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 12 & 4 \\ 3 & 5 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 12 & 8 & 0 & 25 & 9 \\ 3 & 4 & 4 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

4.582 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -96, \ \det = -61, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.28 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 3 & 4 & 3 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -96, \ \det = -85, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.29 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 5 & 0 & 8 & 4 \\ 2 & 4 & 1 & 4 & 0 & 6 & 3 \\ 3 & 5 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -96, \ \det = -224, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.48 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -96, \ \det = -280, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.29 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 & 3 \\ 3 & 5 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -96, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

4.583 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{so}_8$

$$T = 8 \quad [0] \ 21^{\overline{11}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 25, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 12 & 15 & 7 \\ 17 & 20 & 16 & 17 & 8 \\ 12 & 16 & 10 & 13 & 6 \\ 15 & 17 & 13 & 12 & 6 \\ 7 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 42, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 1 & 0 \\ 9 & 17 & 1 & 12 & 6 \\ 5 & 8 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 13 & 1 & 7 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 1 & 0 \\ 12 & 17 & 1 & 12 & 6 \\ 6 & 8 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 4 & 4 & 0 & 4 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 \\ 3 & 5 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 8 & 16 & 10 & 1 & 6 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 11 & 12 & 6 \\ 15 & 20 & 16 & 17 & 8 \\ 11 & 16 & 10 & 13 & 6 \\ 12 & 17 & 13 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 8 & 16 & 10 & 13 & 6 \\ 9 & 17 & 13 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 2 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 16 & 12 & 12 & 6 \\ 16 & 20 & 16 & 17 & 8 \\ 12 & 16 & 10 & 13 & 6 \\ 12 & 17 & 13 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [11] \ 1^{11} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{orange}{g}_2] + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

4.584 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}_6$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 13 & 0 & 6 \\ 9 & 17 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

4.585 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 13 & 0 & 6 \\ 9 & 17 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

4.586 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 1 & 5 \\ 11 & 20 & 0 & 17 & 0 & 8 \\ 1 & 0 & -2 & 1 & 0 & 0 \\ 9 & 17 & 1 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 0 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -290, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 13 & 0 & 6 \\ 9 & 17 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

4.587 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -64, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \quad 1^{\overline{111}} 5^{\overline{1}} 13^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 0 & 13 & 2 \\ 1 & 1 & 1 & 0 & 1 & 10 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 7 & 8 & 13 & 10 & 0 & 75 & 17 \\ 2 & 2 & 2 & 2 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 5^{\overline{1}} 13^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 5 & 5 & 0 & 30 & 5 \\ 2 & 5 & 2 & 1 & 22 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 16 & 30 & 22 & 0 & 154 & 26 \\ 3 & 5 & 4 & -1 & 26 & 4 \end{pmatrix}_{b_0}$$

$$[10] \ 1 \overset{1}{\overset{1}{5}} 1^3 231^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 24 & 3 \\ 3 & 5 & 5 & 0 & 45 & 5 \\ 2 & 5 & 2 & 1 & 33 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 24 & 45 & 33 & 0 & 349 & 39 \\ 3 & 5 & 4 & -1 & 39 & 4 \end{pmatrix}_{b_0}$$

$$[11] \ 1 \overset{1}{\overset{1}{5}} 131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 5 & 5 & 3 & 5 & 5 \\ 2 & 5 & 2 & 3 & 4 & 4 \\ 2 & 3 & 3 & 0 & 2 & 2 \\ 3 & 5 & 4 & 2 & 2 & 4 \\ 3 & 5 & 4 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[12] \ 1 \overset{1}{\overset{1}{5}} 131^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 3 \\ 3 & 5 & 5 & 3 & 10 & 5 \\ 2 & 5 & 2 & 3 & 8 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 \\ 6 & 10 & 8 & 4 & 14 & 8 \\ 3 & 5 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[13] \ 1 \overset{1}{\overset{1}{5}} 131^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 9 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 2 & 5 & 2 & 3 & 12 & 4 \\ 2 & 3 & 3 & 0 & 6 & 2 \\ 9 & 15 & 12 & 6 & 34 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[14] \ 1 \overset{1}{\overset{1}{5}} 131^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 12 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 2 & 5 & 2 & 3 & 16 & 4 \\ 2 & 3 & 3 & 0 & 8 & 2 \\ 12 & 20 & 16 & 8 & 62 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[17] \ 1 \overset{1}{\overset{1}{5}} 131^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -26, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 0 & 15 & 2 \\ 1 & 1 & 1 & 0 & 1 & 12 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 9 & 12 & 15 & 12 & 0 & 115 & 21 \\ 2 & 2 & 2 & 2 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[21] \ 2 \ 3 \ 151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 18 & 3 \\ 4 & 5 & 5 & 3 & 30 & 5 \\ 3 & 5 & 2 & 3 & 24 & 4 \\ 2 & 3 & 3 & 0 & 12 & 2 \\ 18 & 30 & 24 & 12 & 142 & 24 \\ 3 & 5 & 4 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[20] \ 2 \ 3 \ 151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 6 & 3 \\ 4 & 5 & 5 & 3 & 10 & 5 \\ 3 & 5 & 2 & 3 & 8 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 \\ 6 & 10 & 8 & 4 & 14 & 8 \\ 3 & 5 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[19] \ 2 \ 3 \ 151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 12 & 3 \\ 4 & 5 & 5 & 3 & 20 & 5 \\ 3 & 5 & 2 & 3 & 16 & 4 \\ 2 & 3 & 3 & 0 & 8 & 2 \\ 12 & 20 & 16 & 8 & 62 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[18] \quad 2 \quad 3 \quad 15 \quad 1^{\overline{113}} \quad 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 9 & 3 \\ 4 & 5 & 5 & 3 & 15 & 5 \\ 3 & 5 & 2 & 3 & 12 & 4 \\ 2 & 3 & 3 & 0 & 6 & 2 \\ 9 & 15 & 12 & 6 & 34 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[24] \quad 2 \quad 3 \quad 15 \quad 1^{\overline{115}} \quad 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.1$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 15 & 3 \\ 4 & 5 & 5 & 3 & 25 & 5 \\ 3 & 5 & 2 & 3 & 20 & 4 \\ 2 & 3 & 3 & 0 & 10 & 2 \\ 15 & 25 & 20 & 10 & 98 & 20 \\ 3 & 5 & 4 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[22] \quad 2 \quad 3 \quad 15 \quad 1^{\overline{117}} \quad 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 21 & 3 \\ 4 & 5 & 5 & 3 & 35 & 5 \\ 3 & 5 & 2 & 3 & 28 & 4 \\ 2 & 3 & 3 & 0 & 14 & 2 \\ 21 & 35 & 28 & 14 & 194 & 28 \\ 3 & 5 & 4 & 2 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 2 \quad 3 \quad 15 \quad 1^{\overline{111}} \quad 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 3 \\ 4 & 5 & 5 & 3 & 5 & 5 \\ 3 & 5 & 2 & 3 & 4 & 4 \\ 2 & 3 & 3 & 0 & 2 & 2 \\ 3 & 5 & 4 & 2 & 4 & 4 \\ 3 & 5 & 4 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [25] \quad 41 \quad 1^{\overline{111}} \quad 231 \quad 5 \quad 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 216, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 9 & 8 & 1 & 8 & 3 \\ 9 & 51 & 48 & 0 & 48 & 15 \\ 8 & 48 & 42 & 1 & 44 & 14 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 48 & 44 & 0 & 42 & 14 \\ 3 & 15 & 14 & -1 & 14 & 4 \end{pmatrix}_{b_0}$$

4.588 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$

$$T = 9 \quad [0] \quad 1^{\overline{11}} \quad 3 \quad 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -34, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 1 & 3 \\ 3 & 4 & 4 & 0 & 5 & 0 & 4 \\ 2 & 4 & 1 & 1 & 3 & 0 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 3 & 5 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & -1 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [6] \quad 2 \quad 3 \quad 15 \quad 1^{\overline{1121}} \quad 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 6 & 3 & 3 \\ 4 & 5 & 5 & 3 & 10 & 5 & 5 \\ 3 & 5 & 2 & 3 & 8 & 4 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 & 2 \\ 6 & 10 & 8 & 4 & 14 & 8 & 8 \\ 3 & 5 & 4 & 2 & 8 & 2 & 4 \\ 3 & 5 & 4 & 2 & 8 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 2 \quad 3 \quad 15 \quad 1^{\overline{1131}} \quad 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 9 & 3 & 3 \\ 4 & 5 & 5 & 3 & 15 & 5 & 5 \\ 3 & 5 & 2 & 3 & 12 & 4 & 4 \\ 2 & 3 & 3 & 0 & 6 & 2 & 2 \\ 9 & 15 & 12 & 6 & 34 & 12 & 12 \\ 3 & 5 & 4 & 2 & 12 & 2 & 4 \\ 3 & 5 & 4 & 2 & 12 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[4] \quad 2 \quad 3 \quad 15 \quad 1^{\overline{1141}} \quad 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{green}{su}_2 \cdot \textcolor{green}{g}_2] + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 12 & 3 & 3 \\ 4 & 5 & 5 & 3 & 20 & 5 & 5 \\ 3 & 5 & 2 & 3 & 16 & 4 & 4 \\ 2 & 3 & 3 & 0 & 8 & 2 & 2 \\ 12 & 20 & 16 & 8 & 62 & 16 & 16 \\ 3 & 5 & 4 & 2 & 16 & 2 & 4 \\ 3 & 5 & 4 & 2 & 16 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[3] \ 2 \ 3 \ 15 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{red}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.1$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 15 & 3 \\ 4 & 5 & 5 & 3 & 5 & 25 & 5 \\ 3 & 5 & 2 & 3 & 4 & 20 & 4 \\ 2 & 3 & 3 & 0 & 2 & 10 & 2 \\ 3 & 5 & 4 & 2 & 2 & 20 & 4 \\ 15 & 25 & 20 & 10 & 20 & 98 & 20 \\ 3 & 5 & 4 & 2 & 4 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[2] \ 2 \ 3 \ 15 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{red}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 6 & 6 & 3 \\ 4 & 5 & 5 & 3 & 10 & 10 & 5 \\ 3 & 5 & 2 & 3 & 8 & 8 & 4 \\ 2 & 3 & 3 & 0 & 4 & 4 & 2 \\ 6 & 10 & 8 & 4 & 14 & 16 & 8 \\ 6 & 10 & 8 & 4 & 16 & 14 & 8 \\ 3 & 5 & 4 & 2 & 8 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 2 \ 3 \ 15 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{red}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 3 & 3 \\ 4 & 5 & 5 & 3 & 5 & 5 & 5 \\ 3 & 5 & 2 & 3 & 4 & 4 & 4 \\ 2 & 3 & 3 & 0 & 2 & 2 & 2 \\ 3 & 5 & 4 & 2 & 2 & 4 & 4 \\ 3 & 5 & 4 & 2 & 4 & 2 & 4 \\ 3 & 5 & 4 & 2 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

4.589 NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 12 \quad [8] \ 1^{11} \ 5 \ 13^1 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{red}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 4 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 3 & 4 & 5 & 4 & 0 & 11 & 4 & 7 \\ 1 & 1 & 2 & 1 & 0 & 4 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{13} \ 5 \ 13^1 \ 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{red}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 12 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 15 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 12 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 9 & 12 & 15 & 12 & 0 & 115 & 12 & 21 \\ 1 & 1 & 2 & 1 & 0 & 12 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{12} \ 5 \ 13^1 \ 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{red}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 8 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 6 & 8 & 10 & 8 & 0 & 50 & 8 & 14 \\ 1 & 1 & 2 & 1 & 0 & 8 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

4.590 NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \ 1^{11} \ 5 \ 13^1 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + [\textcolor{green}{\mathfrak{su}}_2 \cdot \textcolor{red}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 1 & 1 & 0 & 1 & 1 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

4.591 NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 12 \quad [8] \quad 1^{11} \overset{1}{5} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 4 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 5 & 2 \\ 1 & 1 & 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 4 & 2 \\ 3 & 4 & 5 & 4 & 0 & 4 & 11 & 7 \\ 2 & 2 & 2 & 2 & -1 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{12} \overset{1}{5} 13^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 8 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 10 & 2 \\ 1 & 1 & 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 8 & 2 \\ 6 & 8 & 10 & 8 & 0 & 8 & 50 & 14 \\ 2 & 2 & 2 & 2 & -1 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{13} \overset{1}{5} 13^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 12 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 15 & 2 \\ 1 & 1 & 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 12 & 2 \\ 9 & 12 & 15 & 12 & 0 & 12 & 115 & 21 \\ 2 & 2 & 2 & 2 & -1 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

4.592 NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{su}_2 \mathfrak{n}_3 \text{mix}$

$$T = 11 \quad [0] \quad 1^{12} \overset{1}{5} 13^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 11 & 2 \\ 1 & 2 & 0 & 1 & 0 & 1 & 16 & 2 \\ 1 & 1 & 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 9 & 11 & 16 & 12 & 0 & 11 & 17 & 21 \\ 2 & 2 & 2 & 2 & -1 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{12} \overset{1}{5} 13^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 9 & 2 \\ 1 & 2 & 0 & 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 7 & 9 & 12 & 9 & 0 & 6 & 8 & 16 \\ 2 & 2 & 2 & 2 & -1 & 1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{13} \overset{1}{5} 13^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 13 & 2 \\ 1 & 2 & 0 & 1 & 0 & 1 & 17 & 2 \\ 1 & 1 & 1 & 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 0 & 1 & -3 & 0 & 0 & -1 \\ 10 & 13 & 17 & 13 & 0 & 14 & 1 & 23 \\ 2 & 2 & 2 & 2 & -1 & 2 & 23 & 3 \end{pmatrix}_{b_0}$$

4.593 NHC(-2)(-3) + NHC(-2)(-3) + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{11} \overset{1}{5} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 \\ 2 & 4 & 1 & 3 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 2 \\ 3 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 21 \overset{1}{1} \overset{1}{5} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 12, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 \\ 2 & 4 & 1 & 3 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 2 \\ 3 & 3 & 2 & 2 & 0 & 3 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1 \overset{1}{1} \overset{1}{5} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 21, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 1 & 3 & 0 & 3 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 21 \overset{1}{1} \overset{1}{5} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 34, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 \\ 2 & 4 & 1 & 3 & 2 & 3 \\ 3 & 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 21 \overset{1}{1} \overset{1}{5} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 105, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 & 3 \\ 4 & 4 & 4 & 3 & 3 & 4 \\ 3 & 4 & 1 & 3 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

4.594 $\text{NHC}(-2)(-3) + \text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 11 \quad [0] \ 1 \overset{1}{1} \overset{1}{5} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -11, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 & 4 \\ 2 & 3 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 2 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1 \overset{1}{1} \overset{1}{5} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -15, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 3 \\ 3 & 4 & 3 & 3 & 0 & 10 & 4 \\ 2 & 3 & 1 & 3 & 0 & 8 & 3 \\ 2 & 3 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 10 & 8 & 6 & 0 & 21 & 9 \\ 3 & 4 & 3 & 2 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1 \overset{1}{1} \overset{1}{5} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -23, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 3 & 3 & 0 & 17 & 4 \\ 2 & 3 & 1 & 3 & 0 & 13 & 3 \\ 2 & 3 & 3 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 10 & 17 & 13 & 10 & 0 & 61 & 15 \\ 3 & 4 & 3 & 2 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1 \overset{1}{1} \overset{1}{5} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -44, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 3 & 3 & 0 & 7 & 4 \\ 2 & 3 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 3 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 7 & 5 & 4 & 0 & 8 & 6 \\ 3 & 4 & 3 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{12} \overset{1}{5} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 3 & 3 & 0 & 14 & 4 \\ 2 & 3 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 14 & 10 & 8 & 0 & 38 & 12 \\ 3 & 4 & 3 & 2 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{13} \overset{1}{5} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 12 & 3 \\ 3 & 4 & 3 & 3 & 0 & 21 & 4 \\ 2 & 3 & 1 & 3 & 0 & 15 & 3 \\ 2 & 3 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 12 & 21 & 15 & 12 & 0 & 88 & 18 \\ 3 & 4 & 3 & 2 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

4.595 NHC(-2)(-3) + $\mathfrak{so}_8$

$$T = 6 \quad [0] \quad 21^1 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{so}_8 \quad T = 6, T_{\max} = 96, \Delta = -143, \det = 6, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 96, T \in [11, 96]$$

$$\begin{pmatrix} 10 & 16 & 6 \\ 16 & 25 & 9 \\ 6 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{so}_8 \quad T = 6, T_{\max} = 96, \Delta = -143, \det = 6, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 96, T \in [11, 96]$$

$$\begin{pmatrix} 10 & 16 & 6 \\ 16 & 25 & 9 \\ 6 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{so}_8 \quad T = 6, T_{\max} = 96, \Delta = -143, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -4 & -2 \\ 3 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{so}_8 \quad T = 6, T_{\max} = 96, \Delta = -143, \det = 19, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 96, T \in [12, 96]$$

$$\begin{pmatrix} 5 & 12 & 5 \\ 12 & 25 & 9 \\ 5 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{so}_8 \quad T = 6, T_{\max} = 96, \Delta = -143, \det = 21, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 96, T \in [12, 96]$$

$$\begin{pmatrix} 5 & 11 & 5 \\ 11 & 20 & 8 \\ 5 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{so}_8 \quad T = 6, T_{\max} = 96, \Delta = -143, \det = 25, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$$

$$\begin{pmatrix} 10 & 15 & 6 \\ 15 & 20 & 8 \\ 6 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 21^1 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{so}_8 \quad T = 6, T_{\max} = 96, \Delta = -143, \det = 29, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$$

$$\begin{pmatrix} 13 & 17 & 7 \\ 17 & 20 & 8 \\ 7 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 21^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{so}_8 \quad T = 6, T_{\max} = 96, \Delta = -143, \det = 56, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$$

$$\begin{pmatrix} 10 & 16 & 6 \\ 16 & 20 & 8 \\ 6 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 21^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{so}_8 \quad T = 6, T_{\max} = 96, \Delta = -143, \det = 75, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 96, T \in [11, 96]$$

$$\begin{pmatrix} 13 & 20 & 7 \\ 20 & 25 & 9 \\ 7 & 9 & 3 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [10] \quad 21^1 321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 24, T \in [11, 24]$$

- [11] $\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 5 & 5 & 5 \\ 3 & 5 & 0 & 2 \\ 4 & 5 & 2 & 3 \end{pmatrix}_{b_0}$ $131^1 31^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -6$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 1 & 4 & 0 & 3 & 2 \\ 2 & 3 & 3 & 2 & 2 \\ 2 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [12] $\begin{matrix} 1^1 \\ 1^1 \end{matrix} 141^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8$ $T = 7$, $T_{\max} = 92$, $\Delta = -134$, $\det = -6$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 92$, $T \in [12, 92]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 9 & 8 & 5 \\ 4 & 8 & 6 & 4 \\ 3 & 5 & 4 & 2 \end{pmatrix}_{b_0}$
- [13] $\begin{matrix} 21^1 \\ 1^1 \end{matrix} 41^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8$ $T = 7$, $T_{\max} = 92$, $\Delta = -134$, $\det = -6$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 92$, $T \in [12, 92]$
 $\begin{pmatrix} 10 & 16 & 6 \\ 16 & 25 & 9 \\ 6 & 9 & 3 \end{pmatrix}_{b_0}$
- [14] $21^1 321^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -7$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 13$, $T_{\max} = 24$, $T \in [13, 24]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 1 & 5 & 3 \\ 4 & 5 & 9 & 5 \\ 3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$
- [15] $\begin{matrix} 1^1 \\ 1^1 \end{matrix} 321^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -11$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -4 & -2 \\ 3 & 5 & -2 & 3 \end{pmatrix}_{b_0}$
- [16] $1^1 321^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -11$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 3 & 3 & -1 & 3 \end{pmatrix}_{b_0}$
- [17] $1^1 321^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -12$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & -2 & 2 \end{pmatrix}_{b_0}$
- [18] $\begin{matrix} 1^1 \\ 1^1 \end{matrix} 1321^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -12$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 3 \\ 1 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 1 \end{pmatrix}_{b_0}$
- [19] $1^1 13131^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -12$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$
- [20] $\begin{matrix} 1^1 \\ 1^1 \end{matrix} 1321^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -12$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & 0 & 2 \\ 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$
- [21] $1^1 1321^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -12$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 5 & 3 \\ 4 & 5 & 9 & 5 \\ 3 & 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [22] $\begin{matrix} 1^1 \\ 1^1 \end{matrix} 21^1 41^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8$ $T = 7$, $T_{\max} = 92$, $\Delta = -134$, $\det = -14$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 92$, $T \in [12, 92]$

- [23] $\begin{pmatrix} 10 & 12 & 6 \\ 12 & 13 & 7 \\ 6 & 7 & 3 \end{pmatrix}_{b_0}$
 $1^{11} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 92, \Delta = -134, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 92, T \in [12, 92]$
- [28] $\begin{pmatrix} 10 & 12 & 6 \\ 12 & 13 & 7 \\ 6 & 7 & 3 \end{pmatrix}_{b_0}$
 $1^{11} 3 1^1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$
- [24] $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 3 & -1 & 2 & 1 \end{pmatrix}_{b_0}$
 $1^{11} 3 1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$
- [25] $\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 3 & 6 & -1 & 2 \end{pmatrix}_{b_0}$
 $1^{11} 3 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$
- [26] $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 2 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$
 $1^{11} 3 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$
- [27] $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$
 $1^1 1^1 3 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$
- [30] $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 0 & 2 \\ 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$
 $1^1 1^1 3 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$
- [29] $\begin{pmatrix} 4 & 3 & 7 & 4 \\ 3 & 1 & 5 & 3 \\ 7 & 5 & 9 & 5 \\ 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$
 $1^{11} 1 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 92, \Delta = -134, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 92, T \in [12, 92]$
- [33] $\begin{pmatrix} 5 & 9 & 5 \\ 9 & 13 & 7 \\ 5 & 7 & 3 \end{pmatrix}_{b_0}$
 $1^{11} 12 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$
- [32] $\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 9 & 0 & 5 \\ 1 & 0 & -4 & -2 \\ 3 & 5 & -2 & 1 \end{pmatrix}_{b_0}$
 $1^1 2 1 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 92, \Delta = -134, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$
- [31] $\begin{pmatrix} 5 & 12 & 5 \\ 12 & 25 & 9 \\ 5 & 9 & 2 \end{pmatrix}_{b_0}$
 $1^{11} 1 4 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 92, \Delta = -134, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$
- [37] $\begin{pmatrix} 5 & 12 & 5 \\ 12 & 25 & 9 \\ 5 & 9 & 2 \end{pmatrix}_{b_0}$
 $2 1^{11} 1 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 92, \Delta = -134, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 92, T \in [12, 92]$
- [37] $\begin{pmatrix} 4 & 8 & 6 & 4 \\ 8 & 9 & 8 & 5 \\ 6 & 8 & 6 & 4 \\ 4 & 5 & 4 & 2 \end{pmatrix}_{b_0}$

- [36] $1^1 1^1 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 \\ 3 & 5 & 0 & 2 \\ 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$
- [34] $1^{11} 1^1 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 3 & 3 \\ 2 & 2 & -1 & 3 & 2 \end{pmatrix}_{b_0}$
- [35] $2 1^{11} 4 1^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 92$, $\Delta = -134$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 92$, $T \in [12, 92]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 7 & 2 & 4 \\ 2 & 7 & 3 & 2 & 3 \\ 2 & 2 & 2 & -2 & 0 \\ 2 & 4 & 3 & 0 & 1 \end{pmatrix}_{b_0}$
- [38] $12 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12.14 \rightarrow 13$, $T_{\max} = 24$, $T \in [13, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 \end{pmatrix}_{b_0}$
- [41] $2 2 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -23$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$
- [40] $1^1 3 1^1 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -23$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 \\ 2 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & -2 & 1 \end{pmatrix}_{b_0}$
- [39] $3 1^1 2 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 77$, $\Delta = -123$, $\det = -23$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 77$, $T \in [12, 77]$
 $\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$
- [42] $1^1 1^1 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -27$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 3 & 4 & 3 & 3 \\ 2 & 2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [43] $2 1^{11} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 92$, $\Delta = -134$, $\det = -27$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 92$, $T \in [12, 92]$
 $\begin{pmatrix} 13 & 14 & 7 \\ 14 & 13 & 7 \\ 7 & 7 & 3 \end{pmatrix}_{b_0}$
- [44] $1^{11} 1^1 3 1^1 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 6 & 0 & 11 & 5 \\ 2 & 3 & -1 & 5 & 1 \end{pmatrix}_{b_0}$
- [45] $1 3 1^1 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -29$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 10 & 13 & 6 \\ 5 & 13 & 12 & 6 \\ 3 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [46] $2 2 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -31$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 4 & 3 & 8 & 4 \\ 3 & 1 & 5 & 3 \\ 8 & 5 & 9 & 5 \\ 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$

- [48] $21^{11}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 2 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$
- [47] $1^{11}31^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 4 & 7 & 8 & 4 \\ 7 & 10 & 13 & 6 \\ 8 & 13 & 12 & 6 \\ 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [49] $1^131^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11$, $T_{\max} = 71$, $T \in [11, 71]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & -2 & 2 \end{pmatrix}_{b_0}$
- [51] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -35$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 3 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [50] $21^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 92$, $\Delta = -134$, $\det = -35$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 92$, $T \in [12, 92]$
 $\begin{pmatrix} 5 & 10 & 5 \\ 10 & 13 & 7 \\ 5 & 7 & 3 \end{pmatrix}_{b_0}$
- [52] $21^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -47$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 3 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [53] $21^{11}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -47$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 4 & 8 & 3 & 4 \\ 8 & 9 & 4 & 5 \\ 3 & 4 & 0 & 2 \\ 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$
- [54] $1^131^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 4 & 8 & 8 & 4 \\ 8 & 13 & 16 & 7 \\ 8 & 16 & 12 & 6 \\ 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$
- [55] $2^231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -50$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 0 & 2 \\ 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$
- [56] $2^231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -56$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 3 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [57] $2^231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -68$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 1 & 3 \\ 2 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 2 \end{pmatrix}_{b_0}$
- [59] $21^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 92$, $\Delta = -134$, $\det = -75$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 92$, $T \in [12, 92]$
 $\begin{pmatrix} 13 & 20 & 7 \\ 20 & 25 & 9 \\ 7 & 9 & 3 \end{pmatrix}_{b_0}$
- [58] $131^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -75$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$

$$\begin{pmatrix} 1 & 5 & 5 & 3 \\ 5 & 13 & 16 & 7 \\ 5 & 16 & 12 & 6 \\ 3 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[60] \quad 2 \overset{1}{3} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 30, \Delta = -117, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 30, T \in [11, 30]$$

$$\begin{pmatrix} 4 & 6 & 6 & 4 \\ 6 & 5 & 8 & 5 \\ 6 & 8 & 4 & 4 \\ 4 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 31 \overset{1}{1} 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -95, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$$

$$\begin{pmatrix} 1 & 5 & 4 & 3 \\ 5 & 5 & 5 & 5 \\ 4 & 5 & 0 & 2 \\ 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[62] \quad 2 \overset{1}{3} 1^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -98, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$$

$$\begin{pmatrix} 4 & 8 & 7 & 4 \\ 8 & 10 & 9 & 6 \\ 7 & 9 & 4 & 4 \\ 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[63] \quad 31 \overset{1}{1} 31^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 90, \Delta = -130, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 90, T \in [12, 90]$$

$$\begin{pmatrix} 9 & 12 & 12 & 5 \\ 12 & 13 & 16 & 7 \\ 12 & 16 & 12 & 6 \\ 5 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[64] \quad 31 \overset{1}{1} 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[65] \quad 221 \overset{1}{1} 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 7, T_{\max} = 92, \Delta = -134, \det = -173, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 92, T \in [12, 92]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 4 & 7 & 2 & 4 \\ 3 & 7 & 3 & 2 & 3 \\ 1 & 2 & 2 & -2 & 0 \\ 2 & 4 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [67] \quad 21^1 4131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 23, T \in [12, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 1 & 4 \\ 2 & 1 & -2 & 0 \\ 3 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[66] \quad 131^1 321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 2 & 4 & 4 & 4 \\ 2 & 4 & 3 & 3 & 3 \\ 1 & 4 & 3 & 0 & 2 \\ 2 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^1 3 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 6 & 0 & 6 \\ 1 & 0 & -4 & -2 \\ 3 & 6 & -2 & 4 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^1 1^1 3 221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^1 2 3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 3 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[72] \quad 21^1 41^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 23, T \in [12, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[71] $1^1 1^1 3 2 2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -86$, $\det = 8$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.50 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 2 & 3 & -2 & 3 \end{pmatrix}_{b_0}$$

[74] $1^1 1^1 2 3 2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 8$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.50 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & -2 & 3 \end{pmatrix}_{b_0}$$

[75] $1^1 1^1 3 1^1 4 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 23$, $\Delta = -113$, $\det = 8$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 23$, $T \in [12, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[73] $1^1 1^1 2 3 2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 8$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.50 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 3 \\ 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

[77] $1^1 1^1 2 3 1 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 9$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 4 & 0 & 0 & 2 \\ 2 & 4 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[76] $1^1 2 3 1 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 9$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 0 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

[79] $1^1 1^2 3 2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 9$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 10 & 5 & 6 \\ 2 & 5 & 1 & 3 \\ 3 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

[78] $4 1^1 2 2 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 24$, $\Delta = -114$, $\det = 9$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 24$, $T \in [12, 24]$

$$\begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 0 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

[80] $1^1 2 3 1^1 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 10$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 2 & 4 & 0 & 4 \\ 2 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 4 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[82] $1^1 1^1 4 1 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 23$, $\Delta = -113$, $\det = 11$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 23$, $T \in [12, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -3 & -1 \\ 3 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

[83] $1^1 1^1 2 3 2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 11$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 4 & 7 & 3 & 4 \\ 7 & 10 & 5 & 6 \\ 3 & 5 & 1 & 3 \\ 4 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

[81] $1^1 1^1 4 1 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 23$, $\Delta = -113$, $\det = 11$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$

- $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 23$, $T \in [12, 23]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 3 & 2 & 0 & 3 \end{pmatrix}_{b_0}$
- [89] $1^{\overline{1}} 4 1231^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 24$, $\Delta = -114$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 13$, $T_{\max} = 24$, $T \in [13, 24]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 2 & 0 & 2 \\ 1 & 0 & -2 & 0 \\ 3 & 2 & 0 & 2 \end{pmatrix}_{b_0}$
- [88] $1^{\overline{11}} 4 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 23$, $\Delta = -113$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 13$, $T_{\max} = 23$, $T \in [13, 23]$
 $\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 9 & 0 & 5 \\ 1 & 0 & -3 & -1 \\ 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$
- [87] $1^{\overline{11}} 2^{\overline{1}} 321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 4 & 2 \\ 2 & 0 & 4 & 4 & 4 \\ 2 & 0 & 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [85] $1^{\overline{1}} 4 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 23$, $\Delta = -113$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 23$, $T \in [12, 23]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$
- [84] $1^{\overline{11}} 4 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 24$, $\Delta = -114$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 13$, $T_{\max} = 24$, $T \in [13, 24]$
 $\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 9 & 0 & 5 \\ 1 & 0 & -3 & -1 \\ 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$
- [86] $1^{\overline{1}} 4 131^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 23$, $\Delta = -113$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 13$, $T_{\max} = 23$, $T \in [13, 23]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 2 & 0 & 2 \\ 1 & 0 & -2 & 0 \\ 3 & 2 & 0 & 2 \end{pmatrix}_{b_0}$
- [90] $1^{\overline{1}} 231^{\overline{1}} 31^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 14$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 3 & 3 \\ 1 & 0 & 3 & 0 & 2 \\ 2 & 0 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [92] $1^{\overline{11}} 231^{\overline{1}} 31^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 15$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 2 & 4 & 4 & 4 \\ 2 & 4 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 2 \\ 2 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [91] $1^{\overline{1}} 4 131^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 23$, $\Delta = -113$, $\det = 15$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 0 & 1 & 2 \\ 2 & 1 & -2 & 0 \\ 3 & 2 & 0 & 2 \end{pmatrix}_{b_0}$
- [93] $1^{\overline{1}} 3 221^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -86$, $\det = 15$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 6 & 6 & 6 \\ 3 & 6 & 1 & 3 \\ 3 & 6 & 3 & 4 \end{pmatrix}_{b_0}$
- [94] $1^{\overline{1}} 4 1231^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 24$, $\Delta = -114$, $\det = 15$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 24$, $T \in [13, 24]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 0 & 1 & 2 \\ 2 & 1 & -2 & 0 \\ 3 & 2 & 0 & 2 \end{pmatrix}_{b_0}$
- [95] $1^{\overline{11}} 231^{\overline{1}} 31^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 15$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 4 \\ 1 & 4 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 2 \\ 2 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{11}2321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 4 & 4 \\ 2 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[97] $21^{11}3221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 3 & 5 & 4 & 4 \\ 2 & 2 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

[98] $13141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 5 & 12 & 5 \\ 12 & 25 & 9 \\ 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[100] $1^{11}231^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 4 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

[99] $1^{11}231^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 0 & 2 \\ 2 & 4 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[101] $1^{11}23131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 0 & 4 & 0 & 2 \\ 2 & 0 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[102] $1^{11}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 4 & 4 & 4 & 4 \\ 2 & 3 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[103] $1^{11}4131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 0 & 2 & 2 \\ 3 & 2 & 0 & 2 \\ 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[105] $1^{11}231^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 0 & 3 & 0 & 2 \\ 2 & 0 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[104] $141^{11}231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 4 & 1 & 4 \\ 2 & 1 & -2 & 0 \\ 3 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

[106] $1^{11}2321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 3 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[107] $1^{11}3^{11}221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 3 & 5 & 4 & 4 \\ 2 & 2 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

[108] $1^{\overline{1}} 4^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 4 & 3 & 4 \\ 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[109] $1^{\overline{1}} 4^{\overline{1}} 231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 4 & 3 & 4 \\ 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[110] $21^{\overline{1}} 4^{\overline{1}} 31^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

[111] $1231^{\overline{1}} 31^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 0 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{\overline{1}} 4^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[114] $1^{\overline{1}} 4^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 4 & 0 & 4 \\ 1 & 0 & -3 & -1 \\ 3 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

[113] $1^{\overline{1}} 4^{\overline{1}} 231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 4 & 2 & 4 \\ 1 & 2 & -2 & 0 \\ 3 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

[115] $131^{\overline{1}} 4^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 5 & 10 & 5 \\ 10 & 13 & 7 \\ 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[116] $1^{\overline{1}} 4^{\overline{1}} 231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 3 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

[117] $2^{\overline{31}} 3^{\overline{1}} 21^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 14, \Delta = -96, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$

$$\begin{pmatrix} 0 & 3 & 2 & 2 & 2 \\ 3 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 \\ 2 & 2 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[118] $31^{\overline{1}} 2321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 14, \Delta = -96, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 9 & 12 & 4 & 5 \\ 12 & 10 & 5 & 6 \\ 4 & 5 & 1 & 3 \\ 5 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

[119] $31^{\overline{1}} 31^{\overline{1}} 41^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$

$$\begin{pmatrix} 1 & 6 & 4 & 3 \\ 6 & 13 & 6 & 7 \\ 4 & 6 & 0 & 2 \\ 3 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[120] $21^{1\bar{1}1}4131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 4 & 7 & 0 & 4 \\ 2 & 7 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[121] $2\bar{3}141^{1\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 0 & 2 \\ 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[122] $31^{1\bar{1}1}2321^{1\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 14, \Delta = -96, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$

$$\begin{pmatrix} 0 & 3 & 1 & 4 & 2 \\ 3 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 4 & 4 & 4 & 4 \\ 2 & 3 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[123] $1^141^{1\bar{1}1}231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 4 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[124] $1^141^{1\bar{1}1}231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 13 & 0 & 7 \\ 1 & 0 & -3 & -1 \\ 3 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[125] $2\bar{3}1^{1\bar{1}1}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 1 & 6 & 3 & 3 \\ 6 & 13 & 6 & 7 \\ 3 & 6 & 0 & 2 \\ 3 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

[126] $2\bar{3}141^{1\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 1 & 3 \\ 2 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

[127] $2\bar{3}1^{1\bar{1}1}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 0 & 3 & 2 & 1 & 2 \\ 3 & 4 & 4 & 2 & 4 \\ 2 & 4 & 0 & 1 & 2 \\ 1 & 2 & 1 & -2 & 0 \\ 2 & 4 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

[128] $31^{1\bar{1}1}2321^{1\bar{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 14, \Delta = -96, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 2 & 5 & 4 \\ 4 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

[129] $31^{1\bar{1}1}31^{1\bar{1}1}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 4 & 4 & 2 & 4 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 2 & 1 & -2 & 0 \\ 2 & 4 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

[130] $31^{1\bar{1}1}3141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

[131] $41^{1\bar{1}1}2231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 26 & 29 & 0 & 10 \\ 4 & 29 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 10 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& T = 9 \quad [133] \ 21^1 51231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 5 & 5 & 5 \\ 3 & 5 & 0 & 2 \\ 4 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
& [132] \ 1^1 5131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 5 & 5 & 5 \\ 3 & 5 & 0 & 2 \\ 4 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
& [134] \ 1^1 41321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 4 & 5 \\ 4 & 2 & 4 \\ 5 & 4 & 5 \end{pmatrix}_{b_0} \\
& [135] \ 1^1 41321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0} \\
& [136] \ 231^1 51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 \\ 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
& [138] \ 151^1 2231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 10 & 21 & 6 \\ 9 & 21 & 42 & 12 \\ 3 & 6 & 12 & 3 \end{pmatrix}_{b_0} \\
& [137] \ 1^{11} 314131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 3 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
& [144] \ 1^1 4141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.45 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -4 & -2 \\ 3 & 5 & -2 & 3 \end{pmatrix}_{b_0} \\
& [143] \ 21^1 4141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.45 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 5 & 4 & 5 \\ 3 & 4 & 0 & 2 \\ 4 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
& [142] \ 1^1 51231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.45 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -3 & -1 \\ 3 & 3 & -1 & 3 \end{pmatrix}_{b_0} \\
& [141] \ 1^1 4141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.45 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 4 & 5 \\ 3 & 4 & 0 & 2 \\ 3 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
& [140] \ 1^1 4141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.45 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 1 & 3 \\ 1 & 3 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 2 \end{pmatrix}_{b_0} \\
& [139] \ 1^1 5131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0
\end{aligned}$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[148] \quad 1 \overset{1^{11}}{4} 1 \overset{1^1}{4} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 \\ 1 & 4 & 0 & 1 \\ 1 & 1 & 1 & -2 \\ 2 & 3 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[147] \quad 1 \overset{1^1}{4} 1 \overset{1^1}{4} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 4 & 5 \\ 2 & 4 & 0 & 2 \\ 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[146] \quad 21 \overset{1^{11}}{1} 51 \overset{1^{11}}{3} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 4 & 8 & 6 & 4 \\ 8 & 10 & 9 & 6 \\ 6 & 9 & 7 & 5 \\ 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[145] \quad 1 \overset{1^{11}}{1} 413141 \overset{1^1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 \\ 1 & 4 & 0 & 0 \\ 1 & 0 & 0 & -4 \\ 2 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[154] \quad 1 \overset{1^1}{5} 1231 \overset{1^{11}}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 0 & 2 \\ 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[153] \quad 1 \overset{1^1}{5} 131 \overset{1^{11}}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 0 & 2 \\ 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[151] \quad 1 \overset{1^{11}}{1} 322151 \overset{1^1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 3 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[150] \quad 1 \overset{1^1}{4} 1321 \overset{1^{11}}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[149] \quad 21 \overset{1^1}{5} 131 \overset{1^{11}}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 0 & 2 \\ 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[152] \quad 21 \overset{1^1}{1} 4141 \overset{1^1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[155] \quad 21 \overset{1^1}{1} 5131 \overset{1^1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 3 & 4 \\ 3 & 3 & 0 & 2 \\ 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[157] \quad 214141 \overset{1^1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$\begin{aligned}
& T_{\min} = 14.37 \rightarrow 15, \quad T_{\max} = 21, \quad T \in [15, 21] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 8 & 5 \\ 3 & 8 & 4 & 4 \\ 3 & 5 & 4 & 1 \end{pmatrix}_{b_0} \\
[156] & 14141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -104, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 14.37 \rightarrow 15, \quad T_{\max} = 21, \quad T \in [15, 21] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 8 & 5 \\ 3 & 8 & 4 & 4 \\ 3 & 5 & 4 & 1 \end{pmatrix}_{b_0} \\
[159] & 1^1 31^1 4131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -92, \quad \det = -20, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.40 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 1 & 3 \\ 5 & 13 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 3 & 7 & 0 & 3 \end{pmatrix}_{b_0} \\
[158] & 1^1 31^1 4131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -92, \quad \det = -20, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.40 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0} \\
[160] & 21^1 5131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -23, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.57 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21] \\
& \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[161] & 21^1 51^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -109, \quad \det = -38, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.63 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21] \\
& \begin{pmatrix} 4 & 11 & 3 & 4 \\ 11 & 26 & 8 & 10 \\ 3 & 8 & 0 & 2 \\ 4 & 10 & 2 & 3 \end{pmatrix}_{b_0} \\
[162] & 2^1 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -87, \quad \det = -44, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 6 & 6 & 6 \\ 3 & 6 & 1 & 3 \\ 3 & 6 & 3 & 4 \end{pmatrix}_{b_0} \\
[163] & 14141^{11} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -104, \quad \det = -47, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 13.30 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21] \\
& \begin{pmatrix} 4 & 8 & 3 & 4 \\ 8 & 9 & 4 & 5 \\ 3 & 4 & 0 & 2 \\ 4 & 5 & 2 & 2 \end{pmatrix}_{b_0} \\
[164] & 1^1 51^1 32 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -109, \quad \det = -50, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.88 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21] \\
& \begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 0 & 2 \\ 3 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[166] & 1^1 3221^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -109, \quad \det = -53, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21] \\
& \begin{pmatrix} 1 & 1 & 9 & 3 \\ 1 & -2 & 1 & 0 \\ 9 & 1 & 42 & 12 \\ 3 & 0 & 12 & 3 \end{pmatrix}_{b_0} \\
[165] & 41^1 231^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, \quad T_{\max} = 24, \quad \Delta = -113, \quad \det = -53, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 24, \quad T \in [13, 24] \\
& \begin{pmatrix} 0 & 3 & 4 & 1 & 2 \\ 3 & 10 & 17 & 0 & 6 \\ 4 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 6 & 9 & -2 & 2 \end{pmatrix}_{b_0} \\
[167] & 1^1 41^1 3141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -55, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23] \\
& \begin{pmatrix} 1 & 8 & 1 & 3 \\ 8 & 38 & 1 & 12 \\ 1 & 1 & -3 & -1 \\ 3 & 12 & -1 & 3 \end{pmatrix}_{b_0} \\
[169] & 1^1 5^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -109, \quad \det = -56, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21] \\
& \begin{pmatrix} 1 & 7 & 2 & 3 \\ 7 & 26 & 8 & 10 \\ 2 & 8 & 0 & 2 \\ 3 & 10 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[168] $1^1 4^1 1^1 3 1 4 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 13 & 6 & 0 & 7 \\ 1 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 7 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

[170] $3 1^1 4^1 1 4 1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 9, T_{\max} = 23, \Delta = -112, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 3 & 1 & 2 & 2 \\ 3 & 13 & 1 & 9 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 9 & 0 & 3 & 3 \\ 2 & 7 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$

[171] $2^1 3^1 2 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 9, T_{\max} = 12, \Delta = -87, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 2 & 5 & 1 & 4 \\ 2 & 5 & 1 & 1 & 3 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 4 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[172] $3 1^1 4^1 1 4 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 9, T_{\max} = 23, \Delta = -112, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 5 & 8 & 0 & 5 \\ 2 & 8 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 5 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

[173] $2^1 3^1 2^1 2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 9, T_{\max} = 12, \Delta = -87, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 5 & 3 & 4 \\ 2 & 5 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[174] $3 1^1 4^1 1 4 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 9, T_{\max} = 23, \Delta = -112, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 3 & 2 & 1 & 2 \\ 3 & 13 & 10 & 0 & 7 \\ 2 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 7 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

[175] $2 3 2 1^1 5 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 9, T_{\max} = 22, \Delta = -111, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 1 & 8 & 3 & 3 \\ 8 & 26 & 8 & 10 \\ 3 & 8 & 0 & 2 \\ 3 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[176] $1^1 3 2 2 1^1 5 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 9, T_{\max} = 21, \Delta = -109, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 9 & 1 & 3 \\ 9 & 43 & 1 & 13 \\ 1 & 1 & -3 & -1 \\ 3 & 13 & -1 & 3 \end{pmatrix}_{b_0}$$

[177] $2 2 3^1 1 5 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 9, T_{\max} = 21, \Delta = -109, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 6 & 5 & 3 \\ 6 & 14 & 13 & 8 \\ 5 & 13 & 7 & 5 \\ 3 & 8 & 5 & 3 \end{pmatrix}_{b_0}$$

$T = 10$ [178] $1^1 4^1 1 4 1 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 10, T_{\max} = 13, \Delta = -83, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 2 & 4 & 4 \\ 4 & 5 & 5 \\ 4 & 5 & 5 \end{pmatrix}_{b_0}$$

[179] $1^1 3 1 5 1 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 10, T_{\max} = 13, \Delta = -87, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 3 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

[181] $1^1 4^1 1 4 1 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 10, T_{\max} = 13, \Delta = -83, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 3 & 5 & 0 & 4 \end{pmatrix}_{b_0}$$

[180] $1^{11}315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 3 \\ 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

[184] $1^{11}3151231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 4 & 0 & 0 & 2 \\ 2 & 4 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[183] $1^{11}3151231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 0 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

[182] $21^{11}5131^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.56 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 4 & 5 & 1 & 4 \\ 5 & 5 & 2 & 5 \\ 1 & 2 & -2 & 0 \\ 4 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

[186] $1^{11}413151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 1 & 5 \\ 1 & -2 & 0 \\ 5 & 0 & 4 \end{pmatrix}_{b_0}$$

[185] $1^{11}414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 4 & 3 & 1 & 4 \\ 3 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 4 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[187] $1^{11}414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 15, T_{\max} = 22, T \in [15, 22]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 9 & 0 & 5 \\ 1 & 0 & -3 & -1 \\ 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[188] $1^{11}414131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[189] $1^{11}315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 4 & 4 \\ 2 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[190] $1^{11}414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[191] $1^{11}414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 4 & 4 & 4 & 4 \\ 4 & 2 & 5 & 4 \\ 4 & 5 & 1 & 3 \\ 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

[192] $2^13^15131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.77 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 4 & 8 & 3 & 4 \\ 8 & 10 & 5 & 6 \\ 3 & 5 & 1 & 3 \\ 4 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

[193] $1^{11}414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.29 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 5 & 0 & 5 \\ 1 & 0 & -3 & -1 \\ 3 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

[194] $1^{11}3151^{1231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.27 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 13 & 6 & 0 & 7 \\ 1 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 7 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

[195] $1^{11}3151^{1231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 13 & 6 & 0 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 5 & 0 & 0 & 2 \\ 2 & 7 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[197] $1^{11}41^{132151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.82 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 4 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[196] $1^{11}5131^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.82 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 4 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[198] $1^{11}414141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 68, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.82 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 1 & 3 \\ 2 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

[199] $1^{11}41321^{151} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 88, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.64 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 2 & 1 & 1 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[200] $31^{11}315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 89, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 2 & 5 & 4 \\ 4 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$T = 11$ [201] $1^{11}3151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 3 & 1 & 5 \\ 1 & 1 & -3 & -1 \\ 3 & 5 & -1 & 5 \end{pmatrix}_{b_0}$$

[202] $1^{11}3151321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 2 & 4 \\ 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

[203] $1^{11}6131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 14, \Delta = -85, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[204] $1^{\overline{1}1} 6^{\overline{1}} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 14, \Delta = -85, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$

[205] $1^{\overline{1}1} 6^{\overline{1}} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$

[206] $2^{\overline{1}1} 3^{\overline{1}} 151321^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 13, \Delta = -60, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 5 & 5 \\ 2 & 3 & 3 & 5 & 4 \end{pmatrix}_{b_0}$

[207] $1^{\overline{1}1} 31^{\overline{1}} 51321^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 5 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 5 & 5 \\ 2 & 4 & -1 & 5 & 4 \end{pmatrix}_{b_0}$

[208] $31^{\overline{1}1} 1316131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 15, \Delta = -92, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 3 \\ 4 & 3 & 1 & 3 \\ 3 & 3 & 3 & 3 \end{pmatrix}_{b_0}$

[209] $31^{\overline{1}1} 321^{\overline{1}} 6131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 14.40 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 1 & 7 & 4 & 3 \\ 7 & 19 & 8 & 9 \\ 4 & 8 & 1 & 3 \\ 3 & 9 & 3 & 3 \end{pmatrix}_{b_0}$

[210] $31^{\overline{1}1} 13161231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -89, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 14.73 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 2 & 5 & 4 \\ 4 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$

[211] $1^{\overline{1}1} 6^{\overline{1}} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 15, \Delta = -92, \det = -89, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 14.73 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 2 & 5 & 4 \\ 4 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$

[212] $1^{\overline{1}} 4141^{\overline{1}} 3151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 23, \Delta = -106, \det = -127, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 14.54 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 5 & 1 & 1 & 2 \\ 5 & 8 & 1 & 2 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & -2 & 0 \\ 2 & 6 & 0 & 0 & 2 \end{pmatrix}_{b_0}$

[213] $41^{\overline{1}1} 4^{\overline{1}} 14141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -145, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 14.39 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$
 $\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 2 & 5 & 1 & 4 \\ 3 & 5 & 1 & 1 & 3 \\ 1 & 1 & 1 & -2 & 0 \\ 2 & 4 & 3 & 0 & 2 \end{pmatrix}_{b_0}$

$T = 12$ [214] $1^{\overline{1}1} 31^{\overline{1}} 513221^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & -2 & 1 & 0 \\ 2 & 1 & 3 & 5 \\ 3 & 0 & 5 & 6 \end{pmatrix}_{b_0}$

[215] $1^{\overline{1}1} 6^{\overline{1}} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.60 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 2 & 3 & 4 \\ 3 & 2 & 4 \\ 4 & 4 & 6 \end{pmatrix}_{b_0}$$

[216] $1^{1\bar{1}31} 6^{131} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 \quad T = 12, T_{\max} = 14, \Delta = -63, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.60 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & -1 & 4 \end{pmatrix}_{b_0}$$

[220] $1^{1\bar{1}31} 5^{13221} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 6 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 6 & 6 \\ 2 & 5 & -1 & 6 & 5 \end{pmatrix}_{b_0}$$

[218] $1^{1\bar{1}31} 6^{13141} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 \quad T = 12, T_{\max} = 15, \Delta = -83, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 14.50 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[217] $1^{1\bar{1}31} 6^{13141} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 \quad T = 12, T_{\max} = 15, \Delta = -83, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 14.50 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 0 & 3 \\ 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & -2 & 3 \end{pmatrix}_{b_0}$$

[219] $1^{1\bar{1}321} 7^{1231} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 \quad T = 12, T_{\max} = 15, \Delta = -84, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 14.50 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[221] $31^{1\bar{1}31613141} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 \quad T = 12, T_{\max} = 16, \Delta = -91, \det = 89, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 15.73 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 2 & 5 & 4 \\ 4 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$T = 13$ [222] $2^{1\bar{1}} 3^{1\bar{1}} 21^1 8^{1231} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 \quad T = 13, T_{\max} = 16, \Delta = -85, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 4 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[225] $1^{1\bar{1}3221} 8^{1231} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 15.50 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[223] $1^{1\bar{1}3221} 8^{1231} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 15.50 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[224] $1^{1\bar{1}321} 8^{1231} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 \quad T = 13, T_{\max} = 16, \Delta = -83, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 15.50 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[226] $31^{1\bar{1}321} 8^{1231} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 \quad T = 13, T_{\max} = 17, \Delta = -91, \det = -20, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 16.40 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 3 \\ 4 & 3 & 1 & 3 \\ 3 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

[227] $31^1 3221^1 81231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 13, T_{\max} = 17, \Delta = -91, \det = -29, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 16.59 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 1 & 9 & 4 & 3 \\ 9 & 34 & 10 & 12 \\ 4 & 10 & 1 & 3 \\ 3 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

[228] $31^1 6^{15131} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 13, T_{\max} = 16, \Delta = -86, \det = -44, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 15.91 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 0 & 2 \\ 2 & 4 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$T = 14$ [229] $1^1 8^{11321} 123141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 14, T_{\max} = 17, \Delta = -82, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 16.50 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[230] $1^1 8^{11321} 123141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 14, T_{\max} = 17, \Delta = -82, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 16.50 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 0 & 3 \\ 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & -2 & 3 \end{pmatrix}_{b_0}$$

[231] $23218^{11} 12321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 2 & 5 & 5 & 5 \\ 2 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

$T = 15$ [232] $1^1 6^{12315131} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 15, T_{\max} = 16, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

[233] $1^1 8^{1512321} 1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 15, T_{\max} = 18, \Delta = -79, \det = -8, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 17.50 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$T = 18$ [234] $1^1 3221^{11} (12) 12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 18, T_{\max} = 20, \Delta = -62, \det = 5, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 19.60 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & -1 & 4 \end{pmatrix}_{b_0}$$

$T = 19$ [235] $3221^{11} (12) 12231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 \quad T = 19, T_{\max} = 22, \Delta = -85, \det = -44, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 21.91 \rightarrow 22, T_{\max} = 22, T \in [22, 22]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 0 & 2 \\ 2 & 4 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

4.596 NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{e}_6$

- $T = 7$ [0] $1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6$ $T = 7, T_{\max} = 101, \Delta = -192, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -6 & -4 \\ 5 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$
- [1] $1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6$ $T = 7, T_{\max} = 101, \Delta = -192, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 1 & 5 \\ 11 & 20 & 0 & 8 \\ 1 & 0 & -6 & -4 \\ 5 & 8 & -4 & 0 \end{pmatrix}_{b_0}$$
- $T = 8$ [2] $1^{11}31^131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6$ $T = 8, T_{\max} = 101, \Delta = -171, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 16.80 \rightarrow 17, T_{\max} = 101, T \in [17, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 2 & -4 & -4 & 0 \end{pmatrix}_{b_0}$$
- [3] $1^{11}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6$ $T = 8, T_{\max} = 101, \Delta = -164, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.68 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 5 & 0 & -2 & 0 \\ 3 & 5 & -2 & 0 & 0 \end{pmatrix}_{b_0}$$
- [4] $1^{11}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6$ $T = 8, T_{\max} = 101, \Delta = -164, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & -2 & 0 \\ 3 & 5 & -2 & 0 & 0 \end{pmatrix}_{b_0}$$
- [5] $1^{11}321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6$ $T = 8, T_{\max} = 101, \Delta = -164, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.60 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & -3 & -1 \\ 2 & 2 & -1 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$
- [6] $1^{11}31^131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6$ $T = 8, T_{\max} = 101, \Delta = -171, \det = 125, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 10 & 13 & 0 & 6 \\ 5 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$
- [7] $1^{11}31^131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6$ $T = 8, T_{\max} = 101, \Delta = -171, \det = 350, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.79 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 3 \\ 5 & 13 & 16 & 0 & 7 \\ 5 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 7 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$
- $T = 9$ [8] $1^{11}231^131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6$ $T = 9, T_{\max} = 99, \Delta = -143, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 2 & 4 & 0 & 4 & 4 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 4 & 3 & 0 & -2 & 0 \\ 2 & 4 & 3 & -2 & 0 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{11}3221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6$ $T = 9, T_{\max} = 95, \Delta = -135, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14, T_{\max} = 95, T \in [14, 95]$

$$\begin{pmatrix} 5 & 15 & 1 & 5 \\ 15 & 42 & 0 & 12 \\ 1 & 0 & -4 & -2 \\ 5 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$
- [10] $1^{11}2321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6$ $T = 9, T_{\max} = 99, \Delta = -137, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.35 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{11}2321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 99, \Delta = -137, \det = -37, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.41 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 1 & 0 & -3 & -1 \\ 2 & 3 & 2 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{11}4131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 102, \Delta = -162, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -5 & -3 \\ 5 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[13] $1^1231^131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.87 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 3 & 0 & -2 & 0 \\ 2 & 0 & 3 & -2 & 0 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{11}4^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 102, \Delta = -162, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.42 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$

$$\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 13 & 0 & 7 \\ 1 & 0 & -5 & -3 \\ 5 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [15] $1^14^11321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{e}}_6 \quad T = 10, T_{\max} = 95, \Delta = -134, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 15, T_{\max} = 95, T \in [15, 95]$

$$\begin{pmatrix} 5 & 4 & 1 & 5 \\ 4 & 2 & 2 & 4 \\ 1 & 2 & -4 & -2 \\ 5 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$T = 12$ [16] $1^{11}31^151321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{e}}_6 \quad T = 12, T_{\max} = 24, \Delta = -108, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 22, T_{\max} = 24, T \in [22, 24]$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 28 & 1 & 10 \\ 1 & 1 & -3 & -1 \\ 5 & 10 & -1 & -3 \end{pmatrix}_{b_0}$$

4.597 $\text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{e}}_6 + \text{NHC}(-2)(-3)$

$T = 9$ [0] $1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{e}}_6 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 10 & 13 & 6 \\ 9 & 17 & 0 & 13 & 12 & 6 \\ 5 & 8 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

4.598 $\text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2$

$T = 8$ [0] $1^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 5 \\ 12 & 25 & 0 & 16 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 9 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.51 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 9 & 17 & 0 & 13 & 7 \\ 5 & 8 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.71 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 1 & 44 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 44 & 84 & 0 & 346 & 34 \\ 5 & 8 & -4 & 34 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{113}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 59 & 5 \\ 12 & 25 & 0 & 124 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 59 & 124 & 0 & 613 & 45 \\ 5 & 9 & -4 & 45 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 9 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.53 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 17 & 36 & 0 & 49 & 13 \\ 5 & 9 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 318, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 18 & 5 \\ 12 & 25 & 0 & 40 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 18 & 40 & 0 & 58 & 14 \\ 5 & 9 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 379, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 1 & 35 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 35 & 67 & 0 & 217 & 27 \\ 5 & 8 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 17 & 33 & 0 & 49 & 13 \\ 5 & 8 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 430, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 34 & 5 \\ 12 & 25 & 0 & 72 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 34 & 72 & 0 & 202 & 26 \\ 5 & 9 & -4 & 26 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 26 & 50 & 0 & 118 & 20 \\ 5 & 8 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 493, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 27 & 5 \\ 12 & 25 & 0 & 60 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 27 & 60 & 0 & 133 & 21 \\ 5 & 9 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{112}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 546$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 26 & 5 \\ 12 & 25 & 0 & 56 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 26 & 56 & 0 & 118 & 20 \\ 5 & 9 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{114}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 738$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 36 & 5 \\ 12 & 25 & 0 & 80 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 36 & 80 & 0 & 238 & 28 \\ 5 & 9 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{113}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 745$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 51 & 5 \\ 12 & 25 & 0 & 108 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 51 & 108 & 0 & 457 & 39 \\ 5 & 9 & -4 & 39 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{113}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 921$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 35 & 5 \\ 12 & 25 & 0 & 76 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 35 & 76 & 0 & 217 & 27 \\ 5 & 9 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{113}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1005$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 43 & 5 \\ 12 & 25 & 0 & 92 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 43 & 92 & 0 & 325 & 33 \\ 5 & 9 & -4 & 33 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{115}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1053$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.1$
 $T_{\min} = 13.52 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 45 & 5 \\ 12 & 25 & 0 & 100 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 45 & 100 & 0 & 373 & 35 \\ 5 & 9 & -4 & 35 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{114}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1186$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 68 & 5 \\ 12 & 25 & 0 & 144 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 68 & 144 & 0 & 814 & 52 \\ 5 & 9 & -4 & 52 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{114}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1366$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 44 & 5 \\ 12 & 25 & 0 & 96 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 44 & 96 & 0 & 346 & 34 \\ 5 & 9 & -4 & 34 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{114}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1590$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 60 & 5 \\ 12 & 25 & 0 & 128 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 60 & 128 & 0 & 634 & 46 \\ 5 & 9 & -4 & 46 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{114}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1650$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 1 & 52 & 5 \\ 12 & 25 & 0 & 112 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 52 & 112 & 0 & 478 & 40 \\ 5 & 9 & -4 & 40 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [23] $1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 33, T_{\max} = 99, T \in [33, 99]$$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 4 & 3 \\ 5 & 13 & 16 & 0 & 12 & 7 \\ 5 & 16 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 12 & 0 & 10 & 6 \\ 3 & 7 & 6 & -4 & 6 & -19 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{111} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 9 & 0 & 4 & 0 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & -2 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{111} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 16.11 \rightarrow 17, T_{\max} = 94, T \in [17, 94]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 6 & 3 \\ 4 & 9 & 0 & 4 & 12 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -2 & 7 & 0 \\ 6 & 12 & 0 & 7 & 14 & 8 \\ 3 & 5 & -2 & 0 & 8 & -3 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{11} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.15 \rightarrow 16, T_{\max} = 94, T \in [16, 94]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & -2 & 3 & 0 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & -2 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{111} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -81, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 9 & 0 & 4 & 12 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -2 & 7 & 0 \\ 5 & 12 & 0 & 7 & 14 & 8 \\ 3 & 5 & -2 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{111} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 4 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 12 & 0 & 10 & 6 \\ 3 & 6 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{111} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.63 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{11} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 15 & 0 & 13 & 7 \\ 3 & 6 & 6 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{111} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -165, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.61 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -2 & 3 & 0 \\ 3 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -2 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{111} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.57 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -2 & 3 & 0 \\ 4 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -2 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{111} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -571, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 9 & 3 \\ 4 & 10 & 13 & 0 & 24 & 6 \\ 5 & 13 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 24 & 27 & 0 & 49 & 13 \\ 3 & 6 & 6 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [34] \quad 1^{11} 2^3 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 17.67 \rightarrow 18, T_{\max} = 22, T \in [18, 22]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{11} 3 \, 221^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 15 & 1 & 13 & 5 \\ 15 & 42 & 0 & 36 & 12 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 36 & 0 & 28 & 10 \\ 5 & 12 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{11} 231^{131} 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & -2 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{11} 3 \, 221^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 15 & 1 & 21 & 5 \\ 15 & 42 & 0 & 59 & 12 \\ 1 & 0 & -4 & 0 & -2 \\ 21 & 59 & 0 & 79 & 17 \\ 5 & 12 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{11} 1^{31} 4 \quad 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 94, \Delta = -132, \det = 133, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 14 & 0 & 13 & 7 \\ 5 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{11} 1^{31} 4 \quad 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 94, \Delta = -132, \det = 405, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.42 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 5 & 10 & 1 & 7 & 5 \\ 10 & 13 & 0 & 10 & 7 \\ 1 & 0 & -5 & 0 & -3 \\ 7 & 10 & 0 & 5 & 5 \\ 5 & 7 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{11} 1^{31} 4 \quad 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 94, \Delta = -132, \det = 628, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.42 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 5 & 10 & 1 & 16 & 5 \\ 10 & 13 & 0 & 24 & 7 \\ 1 & 0 & -5 & 0 & -3 \\ 16 & 24 & 0 & 38 & 12 \\ 5 & 7 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

4.599 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{112} 31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 26 & 5 \\ 12 & 25 & 0 & 16 & 56 & 9 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 10 & 36 & 6 \\ 26 & 56 & 0 & 36 & 118 & 20 \\ 5 & 9 & -4 & 6 & 20 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

15.0

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 12 & 7 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{111}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{111}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 17 & 5 \\ 12 & 25 & 0 & 20 & 36 & 9 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 20 & 0 & 13 & 27 & 7 \\ 17 & 36 & 0 & 27 & 49 & 13 \\ 5 & 9 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{111}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 35 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 35 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -4 & 7 & 27 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3^{\overline{111}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 9 & 5 \\ 12 & 25 & 0 & 20 & 20 & 9 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 20 & 0 & 13 & 15 & 7 \\ 9 & 20 & 0 & 15 & 13 & 7 \\ 5 & 9 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{111}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3^{\overline{111}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 17 & 5 \\ 12 & 25 & 0 & 36 & 36 & 9 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 17 & 36 & 0 & 49 & 51 & 13 \\ 17 & 36 & 0 & 51 & 49 & 13 \\ 5 & 9 & -4 & 13 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3^{\overline{111}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -618, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 26 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 26 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -4 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [9] \ 1^{\overline{11}} 3^{\overline{111}} 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [10] \quad 1^{11} 1^{11} 31^{11} 4 \quad 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -68, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 7 & 5 \\ 10 & 13 & 0 & 14 & 10 & 7 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 14 & 0 & 13 & 9 & 7 \\ 7 & 10 & 0 & 9 & 5 & 5 \\ 5 & 7 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

4.600 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$

$$T = 10 \quad [0] \quad 1^{11} 1^{11} 31^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 372, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 27 & 7 \\ 9 & 17 & 0 & 15 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 27 & 49 & 13 \\ 5 & 8 & -4 & 7 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

4.601 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 1^{11} 31^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 8 & 5 \\ 12 & 25 & 0 & 20 & 16 & 9 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 20 & 0 & 13 & 12 & 7 \\ 8 & 16 & 0 & 12 & 9 & 5 \\ 5 & 9 & -4 & 7 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 1^{11} 31^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 1^{11} 31^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 1^{11} 31^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -4 & 13 & 6 & 0 \end{pmatrix}_{b_0}$$

4.602 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{11} 1^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 99, \Delta = -137, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.69 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 3 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & -2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 5 & -2 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 99, \Delta = -137, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 9 & 0 & 4 & 4 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 5 & -2 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^{11}1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 & 3 \\ 4 & 10 & 13 & 0 & 8 & 6 \\ 5 & 13 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 9 & 0 & 4 & 4 \\ 3 & 6 & 6 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$T = 11$ [3] $1^{11}4^{11}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 5 & 4 & 1 & 1 & 5 \\ 4 & 2 & 2 & 0 & 4 \\ 1 & 2 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 4 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

4.603 NHC(-2)(-3) + so₈ + e₆ + su₃

$T = 8$ [1] $1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 5 \\ 12 & 25 & 0 & 16 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 8 & 16 & 0 & 9 & 5 \\ 5 & 9 & -4 & 5 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -6 & 0 & -4 \\ 9 & 17 & 0 & 12 & 6 \\ 5 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 302, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 9 & 20 & 0 & 12 & 6 \\ 5 & 9 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 330, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 13.68 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -6 & 0 & -4 \\ 17 & 36 & 0 & 48 & 12 \\ 5 & 9 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [5] $1^{11}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & -2 & 3 & 0 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & -2 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{11}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & 0 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{11}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 100$, $\Delta = -150$, $\det = -354$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 14.07 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 4 & 3 \\ 5 & 13 & 16 & 0 & 12 & 7 \\ 5 & 16 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 12 & 0 & 9 & 5 \\ 3 & 7 & 6 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

4.604 $\text{NHC}(-2)(-3) + \textcolor{blue}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$

$T = 9$ [0] $1^{11}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 14.23 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 9 & 5 \\ 12 & 25 & 0 & 16 & 20 & 9 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 0 & 9 & 12 & 5 \\ 9 & 20 & 0 & 12 & 13 & 7 \\ 5 & 9 & -4 & 5 & 7 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 15 & 6 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -4 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -334$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -931$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

4.605 $\text{NHC}(-2)(-3) + \textcolor{blue}{so}_8 + \textcolor{red}{e}_7$

$T = 8$ [0] $1^{11}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_7$ $T = 8$, $T_{\max} = 103$, $\Delta = -219$, $\det = 108$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.33 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & -4 & -2 \\ 3 & 5 & -2 & -2 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{11}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_7$ $T = 8$, $T_{\max} = 103$, $\Delta = -219$, $\det = 127$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 15.39 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & -5 & -3 \\ 2 & 2 & -1 & 0 & -3 & -3 \end{pmatrix}_{b_0}$$

$T = 9$ [2] $1^13221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{e}_7$ $T = 9$, $T_{\max} = 103$, $\Delta = -190$, $\det = -30$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 6 & 6 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 6 & 0 & -3 & -1 \\ 3 & 6 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{1231}1^{31}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_7$ $T = 9$, $T_{\max} = 103$, $\Delta = -198$, $\det = -154$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 15.64 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 3 & 0 & -4 & -2 \\ 2 & 0 & 3 & -2 & -2 & -2 \end{pmatrix}_{b_0}$$

4.606 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_2$

$T = 9$ [0] $1^{11}1^{11}3^{21}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 103$, $\Delta = -189$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 9 & 0 & 4 & 0 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & -2 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{11}1^{11}3^{21}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 103$, $\Delta = -189$, $\det = -167$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.43 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & -5 & 0 & -3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & -3 & 0 & -3 \end{pmatrix}_{b_0}$$

[2] $1^{11}1^{11}3^{21}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 103$, $\Delta = -189$, $\det = -261$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.34 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -4 & 3 & -2 \\ 3 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -2 & -2 & 4 & -2 \end{pmatrix}_{b_0}$$

[3] $1^{11}1^{11}3^{21}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 103$, $\Delta = -189$, $\det = -293$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.33 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -4 & 3 & -2 \\ 4 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -2 & -2 & 4 & -2 \end{pmatrix}_{b_0}$$

$T = 10$ [4] $1^{1231}1^{31}1^{11}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 103$, $\Delta = -168$, $\det = 124$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 15.68 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 3 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & -2 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

4.607 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$

$T = 10$ [0] $1^{11}1^{11}3^{21}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 102$, $\Delta = -159$, $\det = 160$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 15.57 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & -5 & 0 & 0 & -3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & -3 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

4.608 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{11}1^{11}3^{21}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 103$, $\Delta = -192$, $\det = -180$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 9 & 0 & 4 & 4 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -4 & 2 & -2 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 5 & -2 & -2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$4.609 \quad \text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_3$$

$$T = 9 \quad [0] \quad 1^{11} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 103, \Delta = -198, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.63 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & -5 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$$4.610 \quad \text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}'_7$$

$$T = 8 \quad [0] \quad 1^{11} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}'_7 \quad T = 8, T_{\max} = 102, \Delta = -191, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & -3 & -1 \\ 3 & 5 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}'_7 \quad T = 8, T_{\max} = 102, \Delta = -191, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.47 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & -4 & -2 \\ 2 & 2 & -1 & 0 & -2 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \quad 1^{11} 3 221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}'_7 \quad T = 9, T_{\max} = 102, \Delta = -162, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 6 & 6 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 6 & 0 & -2 & 0 \\ 3 & 6 & -2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1231} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}'_7 \quad T = 9, T_{\max} = 102, \Delta = -170, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 3 & 0 & -3 & -1 \\ 2 & 0 & 3 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$4.611 \quad \text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_2$$

$$T = 9 \quad [0] \quad 1^{11} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 9 & 0 & 4 & 0 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & -2 & -1 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.51 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 0 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & -4 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -213, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.45 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -3 & 3 & -1 \\ 3 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -2 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 101, \Delta = -161, \det = -241, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -3 & 3 & -1 \\ 4 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -2 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] 1^1 231^1 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -140, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 3 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 3 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & -2 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$4.612 \quad \text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$$

$$T = 10 \quad [0] 1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 93, \Delta = -131, \det = 132, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 0 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & -2 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

$$4.613 \quad \text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 \rightarrow (-3)$$

$$T = 9 \quad [0] 1^{11} 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 102, \Delta = -164, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.49 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 9 & 0 & 4 & 4 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & -3 & 2 & -1 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 5 & -2 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

$$4.614 \quad \text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_3$$

$$T = 9 \quad [0] 1^{11} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -170, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & 0 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$4.615 \quad \text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{red}{f}_4$$

$$T = 7 \quad [0] 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 100, \Delta = -166, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -5 & -3 \\ 5 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[1] 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 100, \Delta = -166, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 5 & 11 & 1 & 5 \\ 11 & 20 & 0 & 8 \\ 1 & 0 & -5 & -3 \\ 5 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [2] \ 1^{\overline{1}} 3 2 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 \\ 1 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & 0 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1}} 3 2 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 4 & 5 & 9 & 0 & 5 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 3 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1}} 3 2 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1}} 3 2 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 13 & 1 & 5 \\ 13 & 30 & 0 & 10 \\ 1 & 0 & -4 & -2 \\ 5 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1}} 3 2 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1}} 3 2 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -4 & -2 \\ 5 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1}} 3 2 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1}} 3 2 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1}} 3 1^{\overline{1}} 3 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 10 & 13 & 0 & 6 \\ 5 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1}} 3 2 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 3 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1}} 3 1^{\overline{1}} 3 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 275, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 3 \\ 5 & 13 & 16 & 0 & 7 \\ 5 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 7 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [13] \quad 1^1 3^1 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 6 & 6 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 6 & 0 & 0 & 2 \\ 3 & 6 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^1 3^1 221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 \quad T = 9, T_{\max} = 21, \Delta = -109, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 15.15 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 5 & 1 & 3 \\ 2 & 1 & 6 & 0 & 3 \\ 5 & 6 & 17 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 3 & 7 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11} 2321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 \quad T = 9, T_{\max} = 22, \Delta = -111, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 18 & 1 & 8 \\ 1 & 1 & -3 & -1 \\ 5 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11} 2321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 \quad T = 9, T_{\max} = 22, \Delta = -111, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 1 & 0 & -2 & 0 \\ 2 & 3 & 2 & -2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 4131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 \quad T = 9, T_{\max} = 99, \Delta = -136, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -4 & -2 \\ 5 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^1 231^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 1 & 1 & 5 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 5 & 1 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 0 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^1 1^{31} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 \quad T = 9, T_{\max} = 99, \Delta = -136, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 13 & 0 & 7 \\ 1 & 0 & -4 & -2 \\ 5 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [20] \quad 1^1 4^1 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 4 & 1 & 5 \\ 4 & 2 & 2 & 4 \\ 1 & 2 & -3 & -1 \\ 5 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

4.616 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 10 & 13 & 6 \\ 9 & 17 & 0 & 13 & 12 & 6 \\ 5 & 8 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

4.617 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 5 \\ 12 & 25 & 0 & 16 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 9 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 44 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 44 & 84 & 0 & 346 & 34 \\ 5 & 8 & -3 & 34 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 59 & 5 \\ 12 & 25 & 0 & 124 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 59 & 124 & 0 & 613 & 45 \\ 5 & 9 & -3 & 45 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 17 & 0 & 13 & 7 \\ 5 & 8 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 9 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.68 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 17 & 36 & 0 & 49 & 13 \\ 5 & 9 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 18 & 5 \\ 12 & 25 & 0 & 40 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 18 & 40 & 0 & 58 & 14 \\ 5 & 9 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 291, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 35 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 35 & 67 & 0 & 217 & 27 \\ 5 & 8 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 17 & 33 & 0 & 49 & 13 \\ 5 & 8 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 34 & 5 \\ 12 & 25 & 0 & 72 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 34 & 72 & 0 & 202 & 26 \\ 5 & 9 & -3 & 26 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 365, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.71 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 1 & 27 & 5 \\ 12 & 25 & 0 & 60 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 27 & 60 & 0 & 133 & 21 \\ 5 & 9 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 26 & 50 & 0 & 118 & 20 \\ 5 & 8 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 1 & 26 & 5 \\ 12 & 25 & 0 & 56 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 26 & 56 & 0 & 118 & 20 \\ 5 & 9 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 540, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 1 & 36 & 5 \\ 12 & 25 & 0 & 80 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 36 & 80 & 0 & 238 & 28 \\ 5 & 9 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{113}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 581, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 1 & 51 & 5 \\ 12 & 25 & 0 & 108 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 51 & 108 & 0 & 457 & 39 \\ 5 & 9 & -3 & 39 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 709, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 1 & 35 & 5 \\ 12 & 25 & 0 & 76 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 35 & 76 & 0 & 217 & 27 \\ 5 & 9 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{115}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 765, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.3$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 1 & 45 & 5 \\ 12 & 25 & 0 & 100 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 45 & 100 & 0 & 373 & 35 \\ 5 & 9 & -3 & 35 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 781, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 1 & 43 & 5 \\ 12 & 25 & 0 & 92 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 43 & 92 & 0 & 325 & 33 \\ 5 & 9 & -3 & 33 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{114}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 924, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 1 & 68 & 5 \\ 12 & 25 & 0 & 144 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 68 & 144 & 0 & 814 & 52 \\ 5 & 9 & -3 & 52 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 1044, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 1 & 44 & 5 \\ 12 & 25 & 0 & 96 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 44 & 96 & 0 & 346 & 34 \\ 5 & 9 & -3 & 34 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{114}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 1236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 1 & 60 & 5 \\ 12 & 25 & 0 & 128 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 60 & 128 & 0 & 634 & 46 \\ 5 & 9 & -3 & 46 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{114} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 1276, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.59 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 52 & 5 \\ 12 & 25 & 0 & 112 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 52 & 112 & 0 & 478 & 40 \\ 5 & 9 & -3 & 40 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [23] \ 1^{13} 3 2 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.7$$

$$T_{\min} = 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 0 & 6 & 3 \\ 1 & 4 & 0 & 1 & 0 & 6 & 2 \\ 1 & 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 6 & 2 & 0 & 10 & 6 \\ 2 & 3 & 2 & 0 & -3 & 6 & -3 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11} 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 18, \ T_{\max} = 21, \ T \in [18, 21]$$

$$\begin{pmatrix} 5 & 13 & 1 & 9 & 5 \\ 13 & 30 & 0 & 20 & 10 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 10 & -2 & 7 & -3 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{11} 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 1 & 5 \\ 12 & 25 & 0 & 0 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 9 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{111} 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -18, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 7 & 0 & 14 & 8 \\ 3 & 5 & 2 & -3 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{11} 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -18, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 15.11 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 3 \\ 3 & 3 & 5 & -3 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{13} 3 2 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 & 3 \\ 1 & 4 & 0 & 1 & 0 & 3 & 2 \\ 1 & 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & 0 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{111} 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -47, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.13 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 19 & 5 \\ 12 & 25 & 0 & 40 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 19 & 40 & 0 & 63 & 15 \\ 5 & 9 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{13} 3 2 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 2 & 4 \\ 3 & 5 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{111} 3 1^3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 26, \ \Delta = -115, \ \det = -64, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.87 \rightarrow 13, \ T_{\max} = 26, \ T \in [13, 26]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 4 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 12 & 0 & 10 & 6 \\ 3 & 6 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{11} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{11} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.22 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{14} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 5 & 5 & 0 & 30 & 5 \\ 2 & 5 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 30 & 20 & 0 & 158 & 24 \\ 3 & 5 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{12} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 5 & 5 & 0 & 15 & 5 \\ 2 & 5 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 10 & 0 & 38 & 12 \\ 3 & 5 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{11} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -102, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 1 & 0 & 7 & 3 \\ 1 & 4 & 0 & 1 & 0 & 7 & 2 \\ 1 & 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 7 & 7 & 4 & 0 & 16 & 8 \\ 2 & 3 & 2 & 0 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{11} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 15 & 0 & 13 & 7 \\ 3 & 6 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{11} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 9 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{11} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -137, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 12 & 1 & 10 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 20 & 0 & 13 & 7 \\ 5 & 9 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{11} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 5 & 5 & 0 & 10 & 5 \\ 2 & 5 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 10 & 7 & 0 & 14 & 8 \\ 3 & 5 & 2 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{11} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 1 & 0 & 10 & 3 \\ 1 & 4 & 0 & 1 & 0 & 10 & 2 \\ 1 & 1 & 1 & -2 & 0 & 5 & 0 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 10 & 10 & 5 & 0 & 33 & 11 \\ 2 & 3 & 2 & 0 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{11}321^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -302$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 1 & 0 & 13 & 3 \\ 1 & 4 & 0 & 1 & 0 & 13 & 2 \\ 1 & 1 & 1 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 13 & 13 & 6 & 0 & 56 & 14 \\ 2 & 3 & 2 & 0 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{11}321^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -310$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 5 & 5 & 0 & 25 & 5 \\ 2 & 5 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 25 & 19 & 0 & 92 & 20 \\ 3 & 5 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{11}321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -310$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.66 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 5 & 5 & 0 & 15 & 5 \\ 2 & 5 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 15 & 11 & 0 & 32 & 12 \\ 3 & 5 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{11}321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -350$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 5 & 5 & 0 & 20 & 5 \\ 2 & 5 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 15 & 0 & 58 & 16 \\ 3 & 5 & 2 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{13}321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -390$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 5 & 5 & 0 & 25 & 5 \\ 2 & 5 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 25 & 17 & 0 & 104 & 20 \\ 3 & 5 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{111}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -435$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 9 & 3 \\ 4 & 10 & 13 & 0 & 24 & 6 \\ 5 & 13 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 24 & 27 & 0 & 49 & 13 \\ 3 & 6 & 6 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{12}321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -460$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 5 & 5 & 0 & 20 & 5 \\ 2 & 5 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 20 & 14 & 0 & 62 & 16 \\ 3 & 5 & 2 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{12}321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -740$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.64 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 5 & 5 & 0 & 25 & 5 \\ 2 & 5 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 25 & 18 & 0 & 94 & 20 \\ 3 & 5 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{14}321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -860$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 5 & 5 & 0 & 35 & 5 \\ 2 & 5 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 35 & 24 & 0 & 202 & 28 \\ 3 & 5 & 2 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{13}321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -910$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 5 & 5 & 0 & 30 & 5 \\ 2 & 5 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 30 & 21 & 0 & 142 & 24 \\ 3 & 5 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{12}3^{11}21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -940$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.68 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 5 & 5 & 0 & 30 & 5 \\ 2 & 5 & 0 & 0 & 22 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 30 & 22 & 0 & 134 & 24 \\ 3 & 5 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[53] $1^{13}3^{11}21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -1350$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.64 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 5 & 5 & 0 & 35 & 5 \\ 2 & 5 & 0 & 0 & 25 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 35 & 25 & 0 & 188 & 28 \\ 3 & 5 & 2 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[54] $1^{14}3^{11}21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -1540$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 5 & 5 & 0 & 40 & 5 \\ 2 & 5 & 0 & 0 & 28 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 40 & 28 & 0 & 254 & 32 \\ 3 & 5 & 2 & -3 & 32 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [55] $1^{11}3^{11}221^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 30$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 6 & 0 & 6 & 6 & 6 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 6 & 0 & 0 & 5 & 2 \\ 3 & 6 & 0 & 5 & 3 & 5 \\ 3 & 6 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[56] $1^{11}3^{11}221^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 44$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 6 & 0 & 6 & 11 & 6 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 6 & 0 & 0 & 8 & 2 \\ 5 & 11 & 0 & 8 & 15 & 9 \\ 3 & 6 & -2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[57] $1^{11}3^{11}4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -106$, $\det = 101$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 13.56 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 5 & 5 \\ 10 & 13 & 0 & 14 & 7 & 7 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 14 & 0 & 13 & 7 & 7 \\ 5 & 7 & -2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[58] $1^{12}3^{11}221^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 104$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.7$

$T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 6 & 0 & 6 & 22 & 6 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 6 & 0 & 0 & 16 & 2 \\ 10 & 22 & 0 & 16 & 66 & 18 \\ 3 & 6 & -2 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[59] $1^{11}3^{11}221^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 110$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 6 & 0 & 6 & 17 & 6 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 6 & 0 & 0 & 13 & 2 \\ 8 & 17 & 0 & 13 & 36 & 14 \\ 3 & 6 & -2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[60] $1^{13}3^{11}221^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 204$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 6 & 0 & 6 & 33 & 6 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 6 & 0 & 0 & 24 & 2 \\ 15 & 33 & 0 & 24 & 151 & 27 \\ 3 & 6 & -2 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

[61] $1^{12} 3^{11} 221^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 230, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.7$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 6 & 0 & 6 & 28 & 6 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 6 & 0 & 0 & 21 & 2 \\ 13 & 28 & 0 & 21 & 103 & 23 \\ 3 & 6 & -2 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{131} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 317, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 10 & 1 & 7 & 5 \\ 10 & 13 & 0 & 10 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 10 & 0 & 5 & 5 \\ 5 & 7 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{11} 3^{11} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -106, \det = 486, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 10 & 1 & 16 & 5 \\ 10 & 13 & 0 & 24 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 24 & 0 & 38 & 12 \\ 5 & 7 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

4.618 NHC(-2)(-3) + so₈ + f₄ + su₂ + su₂

$T = 9 \quad [0] \quad 1^{112} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.7$

$T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 26 & 5 \\ 12 & 25 & 0 & 16 & 56 & 9 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 10 & 36 & 6 \\ 26 & 56 & 0 & 36 & 118 & 20 \\ 5 & 9 & -3 & 6 & 20 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 12 & 7 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -3 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -3 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 17 & 5 \\ 12 & 25 & 0 & 20 & 36 & 9 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 20 & 0 & 13 & 27 & 7 \\ 17 & 36 & 0 & 27 & 49 & 13 \\ 5 & 9 & -3 & 7 & 13 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{111} 3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 35 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 35 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -3 & 7 & 27 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{111} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 9 & 5 \\ 12 & 25 & 0 & 20 & 20 & 9 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 20 & 0 & 13 & 15 & 7 \\ 9 & 20 & 0 & 15 & 13 & 7 \\ 5 & 9 & -3 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{111}3^{111} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{o}}_8 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{blue}{\mathfrak{s}\mathfrak{u}}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -3 & 7 & 13 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^{111} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{o}}_8 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{blue}{\mathfrak{s}\mathfrak{u}}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 17 & 5 \\ 12 & 25 & 0 & 36 & 36 & 9 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 36 & 0 & 49 & 51 & 13 \\ 17 & 36 & 0 & 51 & 49 & 13 \\ 5 & 9 & -3 & 13 & 13 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{111}3^{112} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{o}}_8 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{blue}{\mathfrak{s}\mathfrak{u}}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -482, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 26 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 26 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -3 & 7 & 20 & 1 \end{pmatrix}_{b_0}$$

4.619 $\text{NHC}(-2)(-3) + \mathfrak{s}\mathfrak{o}_8 + \mathfrak{f}_4 + \mathfrak{s}\mathfrak{u}_2 + \mathfrak{s}\mathfrak{u}_2 \rightarrow (-3)$

$T = 9 \quad [0] 1^{111}3^{111} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{o}}_8 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{blue}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 9 & 7 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -3 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^{111} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{o}}_8 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{blue}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 6 & 5 \\ 12 & 25 & 0 & 20 & 12 & 9 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 20 & 0 & 13 & 9 & 7 \\ 6 & 12 & 0 & 9 & 4 & 4 \\ 5 & 9 & -3 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^{111} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{o}}_8 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{blue}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 6 & 5 \\ 12 & 25 & 0 & 36 & 12 & 9 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 36 & 0 & 49 & 17 & 13 \\ 6 & 12 & 0 & 17 & 4 & 4 \\ 5 & 9 & -3 & 13 & 4 & 1 \end{pmatrix}_{b_0}$$

4.620 $\text{NHC}(-2)(-3) + \mathfrak{s}\mathfrak{o}_8 + \mathfrak{f}_4 + \mathfrak{s}\mathfrak{u}_2 + \mathfrak{s}\mathfrak{u}_2 \text{n3mix}$

$T = 9 \quad [0] 1^{111}3^{111} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{o}}_8 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{blue}{\mathfrak{s}\mathfrak{u}}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 15 & 5 \\ 11 & 20 & 0 & 17 & 28 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 24 & 7 \\ 15 & 28 & 0 & 24 & 37 & 11 \\ 5 & 8 & -3 & 7 & 11 & 1 \end{pmatrix}_{b_0}$$

4.621 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 8 & 5 \\ 12 & 25 & 0 & 20 & 16 & 9 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 20 & 0 & 13 & 12 & 7 \\ 8 & 16 & 0 & 12 & 9 & 5 \\ 5 & 9 & -3 & 7 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -3 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -3 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

4.622 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 5 \\ 12 & 25 & 0 & 12 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 12 & 0 & 4 & 4 \\ 5 & 9 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.68 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 5 \\ 11 & 20 & 0 & 11 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 6 & 11 & 0 & 4 & 4 \\ 5 & 8 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \quad 1^{\overline{13}} 3^{\overline{1}} 2^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 22, \Delta = -111, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 5 & 13 & 1 & 6 & 5 \\ 13 & 30 & 0 & 13 & 10 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 13 & 0 & 4 & 4 \\ 5 & 10 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{11}} 3^{\overline{1}} 2^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 22, \Delta = -111, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 5 \\ 12 & 25 & 0 & 12 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 12 & 0 & 4 & 4 \\ 5 & 9 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -112, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 3 \\ 4 & 10 & 13 & 0 & 4 & 6 \\ 5 & 13 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 0 & 0 & 2 \\ 3 & 6 & 6 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 & 3 \\ 4 & 10 & 13 & 0 & 8 & 6 \\ 5 & 13 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 9 & 0 & 4 & 4 \\ 3 & 6 & 6 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [6] \ 1^{11}4^{11}13^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 5 & 12 & 1 & 1 & 5 \\ 12 & 25 & 0 & 0 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 9 & -2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11}3^{11}1^{11}4^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 202, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.63 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 1 & 1 & 5 \\ 10 & 13 & 0 & 0 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 7 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [8] \ 1^{11}4^{11}13^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 4 & 1 & 1 & 5 \\ 4 & 2 & 2 & 0 & 4 \\ 1 & 2 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 4 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

4.623 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 13 & 7 \\ 5 & 8 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 9 & 5 \\ 12 & 25 & 0 & 12 & 20 & 9 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 4 & 9 & 4 \\ 9 & 20 & 0 & 9 & 13 & 7 \\ 5 & 9 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 17 & 5 \\ 12 & 25 & 0 & 12 & 36 & 9 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 0 & 4 & 17 & 4 \\ 17 & 36 & 0 & 17 & 49 & 13 \\ 5 & 9 & -3 & 4 & 13 & 1 \end{pmatrix}_{b_0}$$

4.624 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 12 & 6 \\ 5 & 8 & -3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

4.625 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \mathfrak{n}3\text{mix}$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 24 & 5 \\ 11 & 20 & 0 & 45 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 24 & 45 & 0 & 100 & 18 \\ 5 & 8 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 5 \\ 11 & 20 & 0 & 28 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 15 & 28 & 0 & 37 & 11 \\ 5 & 8 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 23 & 5 \\ 12 & 25 & 0 & 48 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 23 & 48 & 0 & 89 & 17 \\ 5 & 9 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 15 & 5 \\ 12 & 25 & 0 & 32 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 15 & 32 & 0 & 37 & 11 \\ 5 & 9 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 6 & 3 \\ 4 & 10 & 13 & 0 & 16 & 6 \\ 5 & 13 & 12 & 0 & 17 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 16 & 17 & 0 & 20 & 8 \\ 3 & 6 & 6 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [5] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} 1^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -108, \det = 233, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.59 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 1 & 10 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 14 & 0 & 13 & 7 \\ 5 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

4.626 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \mathfrak{n}3\text{mix} + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 9 & 5 \\ 11 & 20 & 0 & 28 & 17 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 15 & 28 & 0 & 37 & 24 & 11 \\ 9 & 17 & 0 & 24 & 13 & 7 \\ 5 & 8 & -3 & 11 & 7 & 1 \end{pmatrix}_{b_0}$$

4.627 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{f}_4$ + $\mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{11} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 98, T \in [15, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 18 & 5 \\ 11 & 20 & 0 & 34 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 18 & 34 & 0 & 57 & 13 \\ 5 & 8 & -3 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 5 \\ 12 & 25 & 0 & 16 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 8 & 16 & 0 & 9 & 5 \\ 5 & 9 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 17 & 0 & 12 & 6 \\ 5 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 235, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 9 & 20 & 0 & 12 & 6 \\ 5 & 9 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 259, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -5 & 0 & -3 \\ 17 & 36 & 0 & 48 & 12 \\ 5 & 9 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \ 1^{11} 3^{11} 2^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 14.83 \rightarrow 15, T_{\max} = 32, T \in [15, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 0 & 6 & 3 \\ 1 & 4 & 0 & 1 & 0 & 6 & 2 \\ 1 & 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 6 & 2 & 0 & 9 & 5 \\ 2 & 3 & 2 & 0 & -3 & 5 & -2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^{11} 2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 3 & 5 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^{11} 2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 0 & 1 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 1 & 2 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 1 & 3 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 0 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^{11} 2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 13 & 1 & 9 & 5 \\ 13 & 30 & 0 & 20 & 10 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 20 & 0 & 12 & 6 \\ 5 & 10 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^{11} 2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 & 3 \\ 1 & 4 & 0 & 1 & 0 & 3 & 2 \\ 1 & 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 1 & 0 & 0 & 2 \\ 2 & 3 & 2 & 0 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{11}3^{11}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.10 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 5 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 & 3 \\ 2 & 3 & 4 & 0 & 0 & 2 & 3 \\ 3 & 3 & 5 & -3 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[11] $1^{11}3^{11}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 & 3 \\ 2 & 5 & 0 & 0 & 4 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 & 3 \\ 2 & 5 & 4 & 0 & 1 & 3 & 3 \\ 3 & 5 & 2 & -3 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[12] $1^{11}3^{11}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.12 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & 0 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[13] $1^{11}3^{11}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 & 3 \\ 3 & 5 & 5 & 0 & 10 & 5 & 3 \\ 2 & 5 & 0 & 0 & 7 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 & 3 \\ 5 & 10 & 7 & 0 & 13 & 7 & 3 \\ 3 & 5 & 2 & -3 & 7 & 1 & 3 \end{pmatrix}_{b_0}$$

[14] $1^{11}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 82, \Delta = -124, \det = -277, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 82, T \in [14, 82]$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 4 & 3 & 3 \\ 5 & 13 & 16 & 0 & 12 & 7 & 3 \\ 5 & 16 & 12 & 0 & 12 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 & 3 \\ 4 & 12 & 12 & 0 & 9 & 5 & 3 \\ 3 & 7 & 6 & -3 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

4.628 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$

$T = 9 \quad [0] \quad 1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 9 & 5 & 5 \\ 12 & 25 & 0 & 16 & 20 & 9 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 & 5 \\ 8 & 16 & 0 & 9 & 12 & 5 & 5 \\ 9 & 20 & 0 & 12 & 13 & 7 & 5 \\ 5 & 9 & -3 & 5 & 7 & 0 & 5 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 & 5 \\ 9 & 17 & 0 & 12 & 15 & 6 & 5 \\ 9 & 17 & 0 & 15 & 13 & 7 & 5 \\ 5 & 8 & -3 & 6 & 7 & 1 & 5 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 & 5 \\ 1 & 0 & -5 & 0 & 0 & -3 & 5 \\ 9 & 17 & 0 & 12 & 12 & 6 & 5 \\ 8 & 16 & 0 & 12 & 10 & 6 & 5 \\ 5 & 8 & -3 & 6 & 6 & 1 & 5 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -3 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

4.629 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 0 & 12 & 9 & 6 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -3 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

4.630 NHC(-2)(-3) + $\mathfrak{so}_8$ + NHC(-2)(-3)

$$T = 8 \quad [0] \ 21^{\overline{11}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 12 & 15 & 7 \\ 17 & 20 & 16 & 17 & 8 \\ 12 & 16 & 10 & 13 & 6 \\ 15 & 17 & 13 & 12 & 6 \\ 7 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 42, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 1 & 0 \\ 3 & 5 & 0 & 1 & 3 & 3 \\ 3 & 4 & -2 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 1 & 0 \\ 2 & 4 & 0 & 1 & 0 & 2 \\ 3 & 4 & -2 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 1 & 0 \\ 12 & 17 & 1 & 12 & 6 \\ 6 & 8 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 4 & 3 \\ 2 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & -2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 4 & 3 \\ 3 & 5 & 0 & 4 & 3 & 3 \\ 3 & 4 & -2 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 8 & 16 & 10 & 1 & 6 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 10 & 15 & 11 & 12 & 6 \\ 15 & 20 & 16 & 17 & 8 \\ 11 & 16 & 10 & 13 & 6 \\ 12 & 17 & 13 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{11}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 8 & 16 & 10 & 13 & 6 \\ 9 & 17 & 13 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[10] $21^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 10 & 16 & 12 & 12 & 6 \\ 16 & 20 & 16 & 17 & 8 \\ 12 & 16 & 10 & 13 & 6 \\ 12 & 17 & 13 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [12] $1^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 12 & 8 & 6 \\ 12 & 13 & 14 & 10 & 7 \\ 12 & 14 & 13 & 10 & 7 \\ 8 & 10 & 10 & 4 & 4 \\ 6 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[11] $21^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 12 & 8 & 6 \\ 12 & 13 & 14 & 10 & 7 \\ 12 & 14 & 13 & 10 & 7 \\ 8 & 10 & 10 & 4 & 4 \\ 6 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[13] $1^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 5 \\ 9 & 13 & 14 & 0 & 7 \\ 9 & 14 & 13 & 1 & 7 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[14] $1^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 5 \\ 9 & 13 & 10 & 0 & 7 \\ 6 & 10 & 5 & 1 & 5 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

[15] $21^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 14 & 15 & 9 & 7 \\ 14 & 13 & 14 & 10 & 7 \\ 15 & 14 & 13 & 10 & 7 \\ 9 & 10 & 10 & 4 & 4 \\ 7 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

4.631 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \text{NHC}(-2)(-3) + \textcolor{blue}{e}_6$

$T = 9$ [0] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 13 & 0 & 6 \\ 9 & 17 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

4.632 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{NHC}(-2)(-3) + \mathfrak{f}_4$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 13 & 0 & 6 \\ 9 & 17 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

4.633 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{NHC}(-2)(-3) + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 1 & 0 \\ 9 & 17 & 0 & 1 & 12 & 6 \\ 5 & 8 & -2 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -290, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 10 & 13 & 6 \\ 9 & 17 & 0 & 13 & 12 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

4.634 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8$

$$T = 7 \quad [0] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 97, \Delta = -142, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -4 & -2 \\ 5 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 97, \Delta = -142, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 5 & 11 & 1 & 5 \\ 11 & 20 & 0 & 8 \\ 1 & 0 & -4 & -2 \\ 5 & 8 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [2] \quad 1^{11} 3^1 2^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 \\ 1 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 0 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 2^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 10 & 11 & 16 & 6 \\ 11 & 10 & 17 & 6 \\ 16 & 17 & 25 & 9 \\ 6 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 2^1 1^1 4^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 10 & 16 & 8 & 6 \\ 16 & 25 & 12 & 9 \\ 8 & 12 & 3 & 3 \\ 6 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 4^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 4 & 4 & 1 & 3 \\ 4 & 9 & 8 & 0 & 5 \\ 4 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

- [6] $1^{11} 1^1 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 24$, $\Delta = -114$, $\det = 29$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.41 \rightarrow 13$, $T_{\max} = 24$, $T \in [13, 24]$
 $\begin{pmatrix} 5 & 8 & 12 & 5 \\ 8 & 10 & 17 & 6 \\ 12 & 17 & 25 & 9 \\ 5 & 6 & 9 & 2 \end{pmatrix}_{b_0}$
- [7] $1^{11} 1^1 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 24$, $\Delta = -114$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 24$, $T \in [13, 24]$
 $\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 4 & 5 & 9 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 3 & 5 & -2 & 1 \end{pmatrix}_{b_0}$
- [8] $1^{11} 1^1 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 24$, $\Delta = -114$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 24$, $T \in [13, 24]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & -2 & 1 \end{pmatrix}_{b_0}$
- [9] $1^{11} 1^1 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 24$, $\Delta = -114$, $\det = 35$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 5 & 0 & 0 & 2 \\ 3 & 5 & -2 & 2 & 2 \end{pmatrix}_{b_0}$
- [10] $1^{11} 1^1 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 24$, $\Delta = -114$, $\det = 35$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 13 & 12 & 19 & 7 \\ 12 & 10 & 17 & 6 \\ 19 & 17 & 25 & 9 \\ 7 & 6 & 9 & 3 \end{pmatrix}_{b_0}$
- [11] $1^{11} 1^1 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 24$, $\Delta = -114$, $\det = 43$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 24$, $T \in [13, 24]$
 $\begin{pmatrix} 5 & 8 & 1 & 5 \\ 8 & 10 & 1 & 6 \\ 1 & 1 & -3 & -1 \\ 5 & 6 & -1 & 2 \end{pmatrix}_{b_0}$
- [12] $1^{11} 1^1 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 24$, $\Delta = -114$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.27 \rightarrow 13$, $T_{\max} = 24$, $T \in [13, 24]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 2 & 1 & 0 & -2 & 0 \\ 2 & 2 & 2 & -2 & 0 & 0 \end{pmatrix}_{b_0}$
- [13] $1^{11} 1^1 3 1^1 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 67$, $\Delta = -121$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.27 \rightarrow 13$, $T_{\max} = 67$, $T \in [13, 67]$
 $\begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 12 & 0 & 11 & 5 \\ 3 & 6 & -1 & 5 & 1 \end{pmatrix}_{b_0}$
- [15] $1^{11} 1^1 4 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 51$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$
 $\begin{pmatrix} 5 & 12 & 6 & 5 \\ 12 & 25 & 12 & 9 \\ 6 & 12 & 3 & 3 \\ 5 & 9 & 3 & 2 \end{pmatrix}_{b_0}$
- [14] $1^{11} 1^1 4 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 51$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.35 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$
 $\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 0 & 7 \\ 1 & 0 & -4 & -2 \\ 5 & 7 & -2 & 2 \end{pmatrix}_{b_0}$
- [16] $1^{11} 2^2 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 24$, $\Delta = -114$, $\det = 53$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 24$, $T \in [12, 24]$
 $\begin{pmatrix} 13 & 12 & 20 & 7 \\ 12 & 10 & 17 & 6 \\ 20 & 17 & 25 & 9 \\ 7 & 6 & 9 & 3 \end{pmatrix}_{b_0}$
- [17] $1^{11} 1^1 3 1^1 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 67$, $\Delta = -121$, $\det = 67$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 67$, $T \in [13, 67]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 10 & 13 & 0 & 6 \\ 5 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 0 & 3 & 3 \\ 2 & 2 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[19] $21^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 13 & 20 & 9 & 7 \\ 20 & 25 & 12 & 9 \\ 9 & 12 & 3 & 3 \\ 7 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{13}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.20 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 3 \\ 5 & 13 & 16 & 0 & 7 \\ 5 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [21] $1^{11}3221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 6 & 6 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 6 & 0 & 1 & 3 \\ 3 & 6 & -2 & 3 & 3 \end{pmatrix}_{b_0}$$

[22] $1^{11}4131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & -2 & 1 & 0 \\ 1 & 0 & 1 & -2 & 0 \\ 3 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{11}4131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -3 & -1 \\ 5 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

[24] $1^{11}4131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & -2 & 0 \\ 3 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{11}231^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 1 & 0 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 0 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{11}3221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 5 & 0 & 4 & 4 \\ 2 & 2 & 3 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{11}2321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[28] $1^{11}41^{11}231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 1 & 2 & 4 \\ 2 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & -2 & 0 \\ 3 & 4 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{\overline{1}131^{\overline{1}}} 4^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -84, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0 \\ T_{\min} = 12.86 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{\overline{1}131^{\overline{1}}} 4^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -92, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0 \\ T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 13 & 0 & 7 \\ 1 & 0 & -3 & -1 \\ 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [31] \ 1^{\overline{1}1} 4^{\overline{1}} 1321^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 6, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 4 & 1 & 5 \\ 4 & 2 & 2 & 4 \\ 1 & 2 & -2 & 0 \\ 5 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{\overline{1}1} 4^{\overline{1}} 141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 10, \ T_{\max} = 22, \ \Delta = -103, \ \det = 9, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.0 \\ T_{\min} = 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 3 \\ 1 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 0 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{\overline{1}1} 4^{\overline{1}} 141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 10, \ T_{\max} = 22, \ \Delta = -103, \ \det = 21, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.0 \\ T_{\min} = 13.43 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 1 & 0 & 3 \\ 1 & 3 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 0 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[34] \ 21^{\overline{1}1} 4^{\overline{1}} 141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 10, \ T_{\max} = 22, \ \Delta = -103, \ \det = 30, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.0 \\ T_{\min} = 13.40 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 4 \\ 8 & 9 & 4 & 4 & 5 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 0 & 2 \\ 4 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{\overline{1}1} 4^{\overline{1}} 141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 10, \ T_{\max} = 22, \ \Delta = -103, \ \det = 32, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.0 \\ T_{\min} = 13.38 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 5 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{\overline{1}1} 4^{\overline{1}} 141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 10, \ T_{\max} = 22, \ \Delta = -103, \ \det = 32, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.0 \\ T_{\min} = 14.38 \rightarrow 15, \ T_{\max} = 22, \ T \in [15, 22]$$

$$\begin{pmatrix} 5 & 9 & 9 & 5 \\ 9 & 14 & 17 & 8 \\ 9 & 17 & 13 & 7 \\ 5 & 8 & 7 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [37] \ 1^{\overline{1}1} 4^{\overline{1}} 14131^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 11, \ T_{\max} = 14, \ \Delta = -82, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 2 & 4 & 2 & 4 \\ 4 & 5 & 1 & 5 \\ 2 & 1 & -2 & 0 \\ 4 & 5 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{\overline{1}1} 4^{\overline{1}} 141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 \quad T = 11, \ T_{\max} = 23, \ \Delta = -102, \ \det = -165, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 11.0 \\ T_{\min} = 14.39 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 1 & 4 & 1 & 0 & 3 \\ 2 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

4.635 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8 + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 1 & 0 \\ 9 & 17 & 0 & 1 & 12 & 6 \\ 5 & 8 & -2 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -290, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 10 & 13 & 6 \\ 9 & 17 & 0 & 13 & 12 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [2] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 9 & 6 & 5 \\ 9 & 13 & 0 & 14 & 10 & 7 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 14 & 0 & 13 & 10 & 7 \\ 6 & 10 & 0 & 10 & 4 & 4 \\ 5 & 7 & -2 & 7 & 4 & 2 \end{pmatrix}_{b_0}$$

4.636 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 2^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 12 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 6 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 11, T_{\max} = 23, \Delta = -102, \det = -21, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 5 \\ 9 & 14 & 17 & 0 & 8 \\ 9 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

4.637 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 5 \\ 12 & 25 & 0 & 16 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 9 & -2 & 6 & -1 \end{pmatrix}_{b_0}$$

- [1] $1^{113} 3 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.71 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 12 & 1 & 54 & 5 \\ 12 & 25 & 0 & 108 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 54 & 108 & 0 & 457 & 39 \\ 5 & 9 & -2 & 39 & 0 \end{pmatrix}_{b_0}$$
- [3] $1^{11} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 8 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$
- [2] $1^{111} 3 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 1 & 44 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 44 & 84 & 0 & 346 & 34 \\ 5 & 8 & -2 & 34 & 2 \end{pmatrix}_{b_0}$$
- [4] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 17 & 0 & 13 & 7 \\ 5 & 8 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$
- [5] $1^{111} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 16 & 0 & 10 & 6 \\ 5 & 8 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$
- [6] $1^{112} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 5 & 12 & 1 & 36 & 5 \\ 12 & 25 & 0 & 72 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 36 & 72 & 0 & 202 & 26 \\ 5 & 9 & -2 & 26 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 1 & 20 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 20 & 33 & 0 & 49 & 13 \\ 5 & 8 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$
- [8] $1^{113} 3 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 1 & 59 & 5 \\ 12 & 25 & 0 & 124 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 59 & 124 & 0 & 613 & 45 \\ 5 & 9 & -2 & 45 & 2 \end{pmatrix}_{b_0}$$
- [9] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 1 & 39 & 5 \\ 12 & 25 & 0 & 76 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 39 & 76 & 0 & 217 & 27 \\ 5 & 9 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$
- [10] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 13 & 7 \\ 5 & 8 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$
- [11] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 1 & 18 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 36 & 0 & 49 & 13 \\ 5 & 9 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

- [12] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 107$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 1 & 39 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 39 & 67 & 0 & 217 & 27 \\ 5 & 8 & -2 & 27 & 2 \end{pmatrix}_{b_0}$
- [13] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 117$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 9 & -2 & 7 & 2 \end{pmatrix}_{b_0}$
- [14] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 128$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 1 & 8 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 8 & -2 & 6 & 2 \end{pmatrix}_{b_0}$
- [15] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 137$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 1 & 17 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 36 & 0 & 49 & 13 \\ 5 & 9 & -2 & 13 & 2 \end{pmatrix}_{b_0}$
- [16] $1^{112} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 162$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.04 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 5 & 12 & 1 & 18 & 5 \\ 12 & 25 & 0 & 40 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 40 & 0 & 58 & 14 \\ 5 & 9 & -2 & 14 & 1 \end{pmatrix}_{b_0}$
- [18] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 16 & 0 & 10 & 6 \\ 5 & 8 & -2 & 6 & 2 \end{pmatrix}_{b_0}$
- [17] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 1 & 48 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 48 & 84 & 0 & 346 & 34 \\ 5 & 8 & -2 & 34 & 2 \end{pmatrix}_{b_0}$
- [19] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 203$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 1 & 35 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 35 & 67 & 0 & 217 & 27 \\ 5 & 8 & -2 & 27 & 2 \end{pmatrix}_{b_0}$
- [20] $1^{113} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 237$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 5 & 12 & 1 & 27 & 5 \\ 12 & 25 & 0 & 60 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 27 & 60 & 0 & 133 & 21 \\ 5 & 9 & -2 & 21 & 1 \end{pmatrix}_{b_0}$
- [21] $1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 242$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 1 & 34 & 5 \\ 12 & 25 & 0 & 72 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 34 & 72 & 0 & 202 & 26 \\ 5 & 9 & -2 & 26 & 2 \end{pmatrix}_{b_0}$
- [22] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 243$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 1 & 17 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 33 & 0 & 49 & 13 \\ 5 & 8 & -2 & 13 & 2 \end{pmatrix}_{b_0}$

[illegible]

- [56] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 1038$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 1 & 45 & 5 \\ 12 & 25 & 0 & 96 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 45 & 96 & 0 & 346 & 34 \\ 5 & 9 & -2 & 34 & 2 \end{pmatrix}_{b_0}$
- [57] $1^{114} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 1058$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 1 & 55 & 5 \\ 12 & 25 & 0 & 112 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 55 & 112 & 0 & 478 & 40 \\ 5 & 9 & -2 & 40 & 2 \end{pmatrix}_{b_0}$
- [58] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 1070$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 1 & 47 & 5 \\ 12 & 25 & 0 & 96 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 47 & 96 & 0 & 346 & 34 \\ 5 & 9 & -2 & 34 & 2 \end{pmatrix}_{b_0}$
- [59] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 1154$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 1 & 46 & 5 \\ 12 & 25 & 0 & 96 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 46 & 96 & 0 & 346 & 34 \\ 5 & 9 & -2 & 34 & 2 \end{pmatrix}_{b_0}$
- [60] $1^{114} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 1206$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 1 & 54 & 5 \\ 12 & 25 & 0 & 112 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 54 & 112 & 0 & 478 & 40 \\ 5 & 9 & -2 & 40 & 2 \end{pmatrix}_{b_0}$
- $T = 9$ [61] $1^{13} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -21$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 5 & 9 & 1 & 18 & 5 \\ 9 & 13 & 0 & 30 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 30 & 0 & 61 & 15 \\ 5 & 7 & -2 & 15 & 2 \end{pmatrix}_{b_0}$
- [62] $1^{111} 3 1^{13} 1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.33 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 1 & 5 & 4 & 3 \\ 4 & 10 & 1 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 12 & 0 & 11 & 12 & 5 \\ 4 & 12 & 0 & 12 & 10 & 6 \\ 3 & 6 & -1 & 5 & 6 & 1 \end{pmatrix}_{b_0}$
- [63] $1^{111} 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 13 & 12 & 19 & 1 & 7 \\ 12 & 10 & 17 & 0 & 6 \\ 19 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -2 & 0 \\ 7 & 6 & 9 & 0 & 3 \end{pmatrix}_{b_0}$
- [64] $1^{111} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -33$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$
 $\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 3 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 9 & 3 & 0 & 1 \end{pmatrix}_{b_0}$
- [65] $1^{111} 3 1^1 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 5 & 1 & 4 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 12 & 12 & 0 & 10 & 6 \\ 3 & 6 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$
- [66] $1^{111} 3 2 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -42$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 1 & 0 & 7 & 3 \\ 1 & 4 & 0 & 1 & 0 & 7 & 2 \\ 1 & 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 7 & 7 & 4 & 0 & 16 & 8 \\ 2 & 3 & 2 & 0 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[67] \ 1^{12} \overset{1}{4} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -66, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.36 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 1 & 12 & 5 \\ 9 & 13 & 0 & 20 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 20 & 0 & 26 & 10 \\ 5 & 7 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{12} \overset{1}{3} 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -70, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 10 & 2 \\ 8 & 15 & 0 & 10 & 38 & 12 \\ 3 & 5 & -2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{14} \overset{1}{3} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -70, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 5 & 0 & 5 & 30 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 20 & 2 \\ 16 & 30 & 0 & 20 & 158 & 24 \\ 3 & 5 & -2 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{11} \overset{1}{3} 1^1 \overset{1}{3} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -79, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.16 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 15 & 0 & 13 & 7 \\ 3 & 6 & 6 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{11} \overset{1}{4} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -81, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.41 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 1 & 9 & 5 \\ 9 & 13 & 0 & 14 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 14 & 0 & 13 & 7 \\ 5 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{11} \overset{1}{4} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -93, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.35 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 1 & 6 & 5 \\ 9 & 13 & 0 & 10 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 10 & 0 & 5 & 5 \\ 5 & 7 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{111} \overset{1}{3} 1^1 \overset{1}{3} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -94, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 12 & 0 & 10 & 6 \\ 3 & 6 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{11} \overset{1}{3} 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -94, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.87 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 4 & 2 \\ 3 & 5 & 0 & 4 & 2 & 4 \\ 3 & 5 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{11} \overset{1}{3} 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -111, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 5 & 0 & 5 & 10 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 7 & 2 \\ 6 & 10 & 0 & 7 & 14 & 8 \\ 3 & 5 & -2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{111} \overset{1}{3} 1^1 \overset{1}{3} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -117, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.35 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 5 & 3 \\ 4 & 10 & 1 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 12 & 0 & 11 & 12 & 5 \\ 5 & 12 & 0 & 12 & 10 & 6 \\ 3 & 6 & -1 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{11} 3^{21} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -131$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 5 & 0 & 5 & 10 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 7 & 2 \\ 5 & 10 & 0 & 7 & 14 & 8 \\ 3 & 5 & -2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[78] $1^{11} 3^{21} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -138$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 1 & 0 & 13 & 3 \\ 1 & 4 & 0 & 1 & 0 & 13 & 2 \\ 1 & 1 & 1 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 6 & 13 & 13 & 6 & 0 & 56 & 14 \\ 2 & 3 & 2 & 0 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{11} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -186$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.39 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 9 & 1 & 15 & 5 \\ 9 & 13 & 0 & 24 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 24 & 0 & 38 & 12 \\ 5 & 7 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{11} 3^{21} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -199$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 11 & 2 \\ 7 & 15 & 0 & 11 & 32 & 12 \\ 3 & 5 & -2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[81] $1^{13} 3^{21} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -251$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 17 & 2 \\ 14 & 25 & 0 & 17 & 104 & 20 \\ 3 & 5 & -2 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[82] $1^{13} 3^{21} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 17 & 2 \\ 13 & 25 & 0 & 17 & 104 & 20 \\ 3 & 5 & -2 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{11} 3^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.20 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 10 & 3 \\ 4 & 10 & 13 & 0 & 24 & 6 \\ 5 & 13 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 24 & 27 & 0 & 49 & 13 \\ 3 & 6 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[84] $1^{12} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -273$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 9 & 1 & 21 & 5 \\ 9 & 13 & 0 & 34 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 21 & 34 & 0 & 77 & 17 \\ 5 & 7 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[85] $1^{11} 3^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -299$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 9 & 3 \\ 4 & 10 & 13 & 0 & 24 & 6 \\ 5 & 13 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 24 & 27 & 0 & 49 & 13 \\ 3 & 6 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

- [86] $1^{12} 3^{11} 21^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 5 & 0 & 5 & 20 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 14 & 2 \\ 10 & 20 & 0 & 14 & 62 & 16 \\ 3 & 5 & -2 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$
- [87] $1^{11} 3^{11} 21^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 11 & 2 \\ 8 & 15 & 0 & 11 & 32 & 12 \\ 3 & 5 & -2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$
- [88] $1^{13} 4^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -342, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 9 & 1 & 27 & 5 \\ 9 & 13 & 0 & 44 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 27 & 44 & 0 & 130 & 22 \\ 5 & 7 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$
- [89] $1^{12} 3^{11} 21^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -374, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 5 & 0 & 5 & 20 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 14 & 2 \\ 11 & 20 & 0 & 14 & 62 & 16 \\ 3 & 5 & -2 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$
- [90] $1^{14} 4^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -393, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 9 & 1 & 33 & 5 \\ 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 33 & 54 & 0 & 197 & 27 \\ 5 & 7 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$
- [91] $1^{12} 3^{11} 21^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -486, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 18 & 2 \\ 12 & 25 & 0 & 18 & 94 & 20 \\ 3 & 5 & -2 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$
- [92] $1^{11} 3^{11} 21^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -495, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 5 & 0 & 5 & 20 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 15 & 2 \\ 10 & 20 & 0 & 15 & 58 & 16 \\ 3 & 5 & -2 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$
- [93] $1^{14} 3^{11} 21^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -594, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 5 & 0 & 5 & 35 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 24 & 2 \\ 18 & 35 & 0 & 24 & 202 & 28 \\ 3 & 5 & -2 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$
- [94] $1^{13} 3^{11} 21^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -619, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 5 & 0 & 5 & 30 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 21 & 2 \\ 15 & 30 & 0 & 21 & 142 & 24 \\ 3 & 5 & -2 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$
- [95] $1^{14} 3^{11} 21^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -654, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 19 & 3 \\ 3 & 5 & 0 & 5 & 35 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 24 & 2 \\ 19 & 35 & 0 & 24 & 202 & 28 \\ 3 & 5 & -2 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{12} 3^{11} 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -706, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 5 & 0 & 5 & 25 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 18 & 2 \\ 13 & 25 & 0 & 18 & 94 & 20 \\ 3 & 5 & -2 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[97] $1^{13} 3^{11} 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -759, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 5 & 0 & 5 & 30 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 21 & 2 \\ 16 & 30 & 0 & 21 & 142 & 24 \\ 3 & 5 & -2 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[98] $1^{13} 3^{11} 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -895, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 5 & 0 & 5 & 35 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 25 & 2 \\ 17 & 35 & 0 & 25 & 188 & 28 \\ 3 & 5 & -2 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

[99] $1^{12} 3^{11} 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -966, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 5 & 0 & 5 & 30 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 22 & 2 \\ 15 & 30 & 0 & 22 & 134 & 24 \\ 3 & 5 & -2 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[100] $1^{14} 3^{11} 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -1046, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 5 & 0 & 5 & 40 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 28 & 2 \\ 20 & 40 & 0 & 28 & 254 & 32 \\ 3 & 5 & -2 & 2 & 32 & 2 \end{pmatrix}_{b_0}$$

[101] $1^{13} 3^{11} 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -1195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 5 & 0 & 5 & 35 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 25 & 2 \\ 18 & 35 & 0 & 25 & 188 & 28 \\ 3 & 5 & -2 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

[102] $1^{14} 3^{11} 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -1266, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 21 & 3 \\ 3 & 5 & 0 & 5 & 40 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 28 & 2 \\ 21 & 40 & 0 & 28 & 254 & 32 \\ 3 & 5 & -2 & 2 & 32 & 2 \end{pmatrix}_{b_0}$$

$T = 10$ [103] $1^{11} 4^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 1 & 0 & -2 & 0 & 0 \\ 2 & 1 & 1 & 0 & 0 & 2 \\ 3 & 2 & 0 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[104] $1^{11} 4^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$

12.0

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 2 & 1 & 0 & -2 & 2 & 0 \\ 9 & 6 & 4 & 2 & 26 & 10 \\ 3 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[114] \ 1^{\overset{1}{\textcolor{blue}{1}} \overset{11}{\textcolor{red}{4}} \overset{31}{\textcolor{red}{2}}} 4 \ 1^{\overset{1}{\textcolor{red}{1}}} + [\textcolor{red}{\text{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\text{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 210, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 12 & 3 \\ 3 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & -2 & 1 & 2 & 0 \\ 1 & 0 & 1 & -2 & 2 & 0 \\ 12 & 12 & 2 & 2 & 38 & 12 \\ 3 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{\overset{1}{\textcolor{blue}{1}} \overset{11}{\textcolor{red}{3}} \overset{31}{\textcolor{red}{1}}} 4 \ 1^{\overset{1}{\textcolor{red}{1}}} + [\textcolor{red}{\text{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\text{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 216, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & -2 & 1 & 0 \\ 2 & 2 & 0 & 1 & 1 & 0 & 2 \\ 2 & 2 & 0 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{\overset{1}{\textcolor{blue}{1}} \overset{11}{\textcolor{red}{3}} \overset{31}{\textcolor{red}{1}}} 4 \ 1^{\overset{1}{\textcolor{red}{1}}} + [\textcolor{red}{\text{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\text{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 229, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 10 & 1 & 7 & 5 \\ 10 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 10 & 0 & 5 & 5 \\ 5 & 7 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{\overset{1}{\textcolor{blue}{1}} \overset{11}{\textcolor{red}{3}} \overset{31}{\textcolor{red}{1}}} 4 \ 1^{\overset{1}{\textcolor{red}{1}}} + [\textcolor{red}{\text{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\text{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 234, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 2 & 1 & 0 & -2 & 2 & 0 \\ 5 & 4 & 2 & 2 & 6 & 6 \\ 3 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{\overset{1}{\textcolor{blue}{1}} \overset{11}{\textcolor{red}{3}} \overset{31}{\textcolor{red}{2}}} 4 \ 1^{\overset{1}{\textcolor{red}{1}}} + [\textcolor{red}{\text{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\text{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 259, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 3 \\ 2 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 2 & 1 & 0 & -2 & 2 & 0 \\ 7 & 5 & 3 & 2 & 14 & 8 \\ 3 & 2 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{\overset{1}{\textcolor{blue}{1}} \overset{11}{\textcolor{red}{3}} \overset{31}{\textcolor{red}{1}}} 4 \ 1^{\overset{1}{\textcolor{red}{1}}} + [\textcolor{red}{\text{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\text{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 285, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 & -2 & 1 & 0 \\ 3 & 2 & 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{\overset{1}{\textcolor{blue}{1}} \overset{11}{\textcolor{red}{3}} \overset{31}{\textcolor{red}{1}}} 4 \ 1^{\overset{1}{\textcolor{red}{1}}} + [\textcolor{red}{\text{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\text{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 344, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 10 & 1 & 16 & 5 \\ 10 & 13 & 0 & 24 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 24 & 0 & 38 & 12 \\ 5 & 7 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{\overset{1}{\textcolor{blue}{1}} \overset{11}{\textcolor{red}{3}} \overset{31}{\textcolor{red}{1}}} 4 \ 1^{\overset{1}{\textcolor{red}{1}}} + [\textcolor{red}{\text{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\text{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 408, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 10 & 1 & 17 & 5 \\ 10 & 13 & 0 & 24 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 17 & 24 & 1 & 39 & 13 \\ 5 & 7 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

4.638 NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -50, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 17 & 0 & 0 & 13 & 7 \\ 5 & 8 & -2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -72, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -2 & 7 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -76, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.89 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -2 & 7 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -88, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 39 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 39 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -2 & 7 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -96, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 45 & 5 \\ 11 & 20 & 0 & 0 & 84 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 45 & 84 & 0 & 0 & 346 & 34 \\ 5 & 8 & -2 & 0 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 16 & 0 & 0 & 10 & 6 \\ 5 & 8 & -2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -160, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 10 & 5 \\ 12 & 25 & 0 & 36 & 20 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 36 & 0 & 49 & 27 & 13 \\ 10 & 20 & 0 & 27 & 13 & 7 \\ 5 & 9 & -2 & 13 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -168, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 18 & 18 & 5 \\ 12 & 25 & 0 & 36 & 36 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 36 & 0 & 49 & 51 & 13 \\ 18 & 36 & 0 & 51 & 49 & 13 \\ 5 & 9 & -2 & 13 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -172, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 18 & 17 & 5 \\ 12 & 25 & 0 & 36 & 36 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 36 & 0 & 49 & 51 & 13 \\ 17 & 36 & 0 & 51 & 49 & 13 \\ 5 & 9 & -2 & 13 & 13 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{1111}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 17 & 5 \\ 12 & 25 & 0 & 20 & 36 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 20 & 0 & 13 & 27 & 7 \\ 17 & 36 & 0 & 27 & 49 & 13 \\ 5 & 9 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{111}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 35 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 35 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -2 & 7 & 27 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 48 & 5 \\ 11 & 20 & 0 & 17 & 84 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 72 & 7 \\ 48 & 84 & 0 & 72 & 346 & 34 \\ 5 & 8 & -2 & 7 & 34 & 2 \end{pmatrix}_{b_0}$$

[12] $1^{111}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 18 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 13 & 27 & 7 \\ 18 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{111}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 36 & 5 \\ 11 & 20 & 0 & 0 & 67 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 36 & 67 & 0 & 0 & 217 & 27 \\ 5 & 8 & -2 & 0 & 27 & 2 \end{pmatrix}_{b_0}$$

[14] $1^{111}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 19 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 19 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{111}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 18 & 5 \\ 11 & 20 & 0 & 0 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 18 & 33 & 0 & 0 & 49 & 13 \\ 5 & 8 & -2 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{111}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -262, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 29 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 29 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{1111}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

13.0

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 9 & 5 \\ 12 & 25 & 0 & 20 & 20 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 20 & 0 & 13 & 15 & 7 \\ 9 & 20 & 0 & 15 & 13 & 7 \\ 5 & 9 & -2 & 7 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{\overline{111}} \overline{3} \ 1^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -286, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 45 & 5 \\ 11 & 20 & 0 & 17 & 84 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 13 & 72 & 7 \\ 45 & 84 & 0 & 72 & 346 & 34 \\ 5 & 8 & -2 & 7 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{\overline{111}} \overline{3} \ 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 27 & 5 \\ 11 & 20 & 0 & 0 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 27 & 50 & 0 & 0 & 118 & 20 \\ 5 & 8 & -2 & 0 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{\overline{111}} \overline{3} \ 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{\overline{111}} \overline{3} \ 1^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 45 & 5 \\ 11 & 20 & 0 & 17 & 84 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 72 & 7 \\ 45 & 84 & 0 & 72 & 346 & 34 \\ 5 & 8 & -2 & 7 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{\overline{111}} \overline{3} \ 1^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -342, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 27 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 13 & 42 & 7 \\ 27 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{\overline{1111}} \overline{3} \ 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 17 & 5 \\ 12 & 25 & 0 & 36 & 36 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 36 & 0 & 49 & 51 & 13 \\ 17 & 36 & 0 & 51 & 49 & 13 \\ 5 & 9 & -2 & 13 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{\overline{111}} \overline{3} \ 1^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 26 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 26 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{\overline{111}} \overline{3} \ 1^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -360, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 36 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 13 & 57 & 7 \\ 36 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -2 & 7 & 27 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{111} 3 1^{111} + [\textcolor{red}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s} \textcolor{green}{o}_8 + \textcolor{orange}{s} \textcolor{orange}{o}_8 + \textcolor{red}{s} \textcolor{red}{u}_2 + \textcolor{blue}{s} \textcolor{blue}{u}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 18 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 18 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{111}31^{113} + [\textcolor{red}{s}2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}08 + \textcolor{brown}{s}08 + \textcolor{red}{s}2 + \textcolor{blue}{s}2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -468, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 38 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 38 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -2 & 7 & 27 & 2 \end{pmatrix}_b$$

[28] $1^{\text{111}} 3^{\text{114}} 1^{\text{13}} + [\text{5u}_2 \cdot \text{g}_2] + \text{5o}_8 + \text{5o}_8 + \text{5u}_2 + \text{5u}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 47 & 5 \\ 11 & 20 & 0 & 17 & 84 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 72 & 7 \\ 47 & 84 & 0 & 72 & 346 & 34 \\ 5 & 8 & -2 & 7 & 34 & 2 \end{pmatrix}_b$$

[29] $1^{\text{111}} 3^{\text{112}} + [\text{5u}_2 \cdot \text{g}_2] + \text{5o}_8 + \text{5o}_8 + \text{5u}_2 + \text{5u}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -522, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 28 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 28 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -2 & 7 & 20 & 2 \end{pmatrix}_b$$

[30] $1^{111}3^{112}1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -530, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 46 & 5 \\ 11 & 20 & 0 & 17 & 84 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 72 & 7 \\ 46 & 84 & 0 & 72 & 346 & 34 \\ 5 & 8 & -2 & 7 & 34 & 2 \end{pmatrix}_b$$

[31] $1^{111}3^11^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -532, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 36 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 36 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -2 & 7 & 27 & 2 \end{pmatrix}_b$$

[32] $1^{111}3^11^{12} + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{5}\textcolor{brown}{o}_8 + \textcolor{brown}{5}\textcolor{brown}{o}_8 + \textcolor{red}{5}\textcolor{red}{u}_2 + \textcolor{blue}{5}\textcolor{blue}{u}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -550, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 27 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 27 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -2 & 7 & 20 & 2 \end{pmatrix}_b$$

[33] $1^{\text{red}1\text{red}1\text{red}}3^{\text{red}1\text{red}1\text{red}} + [\text{red}5\text{red}2 \cdot \text{blue}2] + \text{green}5\text{green}8 + \text{orange}5\text{orange}8 + \text{red}5\text{red}2 + \text{blue}5\text{blue}2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -616, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 37 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 37 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -2 & 7 & 27 & 2 \end{pmatrix}_b$$

4.639 $\text{NHC}(-2)(-3) + \mathfrak{s}\mathfrak{o}_8 + \mathfrak{s}\mathfrak{o}_8 + \mathfrak{s}\mathfrak{u}_2 + \mathfrak{s}\mathfrak{u}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{\textcolor{brown}{11}} 3^{\textcolor{blue}{1}} 1^{\textcolor{red}{1}} + [\textcolor{red}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s} \textcolor{green}{o}_8 + \textcolor{orange}{s} \textcolor{orange}{o}_8 + \textcolor{red}{s} \textcolor{red}{u}_2 + \textcolor{blue}{s} \textcolor{blue}{u}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 6 & 5 \\ 11 & 20 & 0 & 0 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 11 & 0 & 0 & 4 & 4 \\ 5 & 8 & -2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 18 & 6 & 5 \\ 12 & 25 & 0 & 36 & 12 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 36 & 0 & 49 & 17 & 13 \\ 6 & 12 & 0 & 17 & 4 & 4 \\ 5 & 9 & -2 & 13 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 6 & 5 \\ 12 & 25 & 0 & 20 & 12 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 20 & 0 & 13 & 9 & 7 \\ 6 & 12 & 0 & 9 & 4 & 4 \\ 5 & 9 & -2 & 7 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 9 & 7 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -2 & 7 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -217, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 6 & 5 \\ 12 & 25 & 0 & 36 & 12 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 36 & 0 & 49 & 17 & 13 \\ 6 & 12 & 0 & 17 & 4 & 4 \\ 5 & 9 & -2 & 13 & 4 & 2 \end{pmatrix}_{b_0}$$

4.640 NHC(-2)(-3) + so₈ + so₈ + su₂ + su₂n3mix

$$T = 9 \quad [0] \ 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 15 & 5 \\ 11 & 20 & 0 & 0 & 28 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 28 & 0 & 0 & 37 & 11 \\ 5 & 8 & -2 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 15 & 5 \\ 11 & 20 & 0 & 17 & 28 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 13 & 24 & 7 \\ 15 & 28 & 0 & 24 & 37 & 11 \\ 5 & 8 & -2 & 7 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 16 & 5 \\ 11 & 20 & 0 & 17 & 28 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 24 & 7 \\ 16 & 28 & 0 & 24 & 37 & 11 \\ 5 & 8 & -2 & 7 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 15 & 5 \\ 11 & 20 & 0 & 17 & 28 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 24 & 7 \\ 15 & 28 & 0 & 24 & 37 & 11 \\ 5 & 8 & -2 & 7 & 11 & 2 \end{pmatrix}_{b_0}$$

4.641 NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 8 & 5 \\ 12 & 25 & 0 & 20 & 16 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 20 & 0 & 13 & 12 & 7 \\ 8 & 16 & 0 & 12 & 9 & 5 \\ 5 & 9 & -2 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{113} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 35 & 1 & 5 \\ 12 & 25 & 0 & 76 & 0 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 35 & 76 & 0 & 217 & 0 & 27 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & -2 & 27 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -2 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 1 & 5 \\ 12 & 25 & 0 & 36 & 0 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 36 & 0 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & -2 & 13 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -2 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -133, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 1 & 5 \\ 11 & 20 & 0 & 17 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 7 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 10 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 10 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -2 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -2 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 1 & 5 \\ 11 & 20 & 0 & 16 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 26 & 1 & 5 \\ 12 & 25 & 0 & 56 & 0 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 26 & 56 & 0 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & -2 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 34 & 1 & 5 \\ 12 & 25 & 0 & 72 & 0 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 34 & 72 & 0 & 202 & 0 & 26 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & -2 & 26 & -1 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 10 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 10 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 1 & 5 \\ 11 & 20 & 0 & 50 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 26 & 50 & 0 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 1 & 5 \\ 11 & 20 & 0 & 33 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 33 & 0 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 13 & -1 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{113} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -295, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 51 & 1 & 5 \\ 12 & 25 & 0 & 108 & 0 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 51 & 108 & 0 & 457 & 0 & 39 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & -2 & 39 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{113} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 43 & 1 & 5 \\ 12 & 25 & 0 & 92 & 0 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 43 & 92 & 0 & 325 & 0 & 33 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & -2 & 33 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{114} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 52 & 1 & 5 \\ 12 & 25 & 0 & 112 & 0 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 52 & 112 & 0 & 478 & 0 & 40 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & -2 & 40 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{114} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 68 & 1 & 5 \\ 12 & 25 & 0 & 144 & 0 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 68 & 144 & 0 & 814 & 0 & 52 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & -2 & 52 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{114} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 60 & 1 & 5 \\ 12 & 25 & 0 & 128 & 0 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 60 & 128 & 0 & 634 & 0 & 46 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & -2 & 46 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 19 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 19 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 18 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \quad 1^{11} 4 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 9 & 6 & 5 \\ 9 & 13 & 0 & 14 & 10 & 7 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 14 & 0 & 13 & 9 & 7 \\ 6 & 10 & 0 & 9 & 4 & 4 \\ 5 & 7 & -2 & 7 & 4 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{111}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 261, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 6 & 6 & 5 \\ 9 & 13 & 0 & 10 & 10 & 7 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 10 & 0 & 5 & 7 & 5 \\ 6 & 10 & 0 & 7 & 4 & 4 \\ 5 & 7 & -2 & 5 & 4 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{111}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 480, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 15 & 6 & 5 \\ 9 & 13 & 0 & 24 & 10 & 7 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 24 & 0 & 38 & 16 & 12 \\ 6 & 10 & 0 & 16 & 4 & 4 \\ 5 & 7 & -2 & 12 & 4 & 2 \end{pmatrix}_{b_0}$$

4.642 NHC(-2)(-3) + so₈ + so₈ + su₂ → (-3)

$T = 8 \quad [0] \quad 1^{111}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 5 \\ 12 & 25 & 0 & 12 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 12 & 0 & 4 & 4 \\ 5 & 9 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 5 \\ 11 & 20 & 0 & 11 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 11 & 0 & 4 & 4 \\ 5 & 8 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$T = 9 \quad [2] \quad 1^{111}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 3 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 5 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 3 \\ 4 & 10 & 13 & 0 & 4 & 6 \\ 5 & 13 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 5 & 0 & 0 & 2 \\ 3 & 6 & 6 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 & 3 \\ 4 & 10 & 13 & 0 & 8 & 6 \\ 5 & 13 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 9 & 0 & 4 & 4 \\ 3 & 6 & 6 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

$T = 10 \quad [5] \quad 1^{111}413^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 1 & 3 \\ 3 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & -2 & 1 & 0 & 0 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 2 & 0 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[6] $1^1 4 13^1 1^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 2 & 0 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[7] $1^1 3^1 1^1 4^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 145, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 10 & 1 & 1 & 5 \\ 10 & 13 & 0 & 0 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 7 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

[8] $1^1 3^1 1^1 4^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 153, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 \\ 2 & 2 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

4.643 NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$T = 9 \quad [0] 1^{11} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 1 & 5 \\ 11 & 20 & 0 & 11 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 8 & -2 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 9 & 5 \\ 12 & 25 & 0 & 12 & 20 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 12 & 0 & 4 & 9 & 4 \\ 9 & 20 & 0 & 9 & 13 & 7 \\ 5 & 9 & -2 & 4 & 7 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 18 & 5 \\ 12 & 25 & 0 & 12 & 36 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 12 & 0 & 4 & 17 & 4 \\ 18 & 36 & 0 & 17 & 49 & 13 \\ 5 & 9 & -2 & 4 & 13 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 13 & 7 \\ 5 & 8 & -2 & 4 & 7 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -217, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 17 & 5 \\ 12 & 25 & 0 & 12 & 36 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 12 & 0 & 4 & 17 & 4 \\ 17 & 36 & 0 & 17 & 49 & 13 \\ 5 & 9 & -2 & 4 & 13 & 2 \end{pmatrix}_{b_0}$$

4.644 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -87, \quad \det = -132, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 6 & 5 \\ 12 & 25 & 0 & 12 & 12 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 12 & 0 & 4 & 6 & 4 \\ 6 & 12 & 0 & 6 & 4 & 4 \\ 5 & 9 & -2 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

4.645 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 14, \quad \Delta = -93, \quad \det = -294, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, \quad T_{\max} = 14, \quad T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 12 & 6 \\ 5 & 8 & -2 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

4.646 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \mathfrak{n} 3 \text{mix}$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 24, \quad \Delta = -114, \quad \det = 35, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.83 \rightarrow 12, \quad T_{\max} = 24, \quad T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 1 & 24 & 5 \\ 11 & 20 & 0 & 45 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 24 & 45 & 0 & 100 & 18 \\ 5 & 8 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 24, \quad \Delta = -114, \quad \det = 59, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 24, \quad T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 1 & 7 & 5 \\ 11 & 20 & 0 & 11 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 11 & 0 & 4 & 4 \\ 5 & 8 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 24, \quad \Delta = -114, \quad \det = 69, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.83 \rightarrow 12, \quad T_{\max} = 24, \quad T \in [12, 24]$$

$$\begin{pmatrix} 5 & 12 & 1 & 24 & 5 \\ 12 & 25 & 0 & 48 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 24 & 48 & 0 & 89 & 17 \\ 5 & 9 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 24, \quad \Delta = -114, \quad \det = 75, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.81 \rightarrow 12, \quad T_{\max} = 24, \quad T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 1 & 25 & 5 \\ 11 & 20 & 0 & 45 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 25 & 45 & 0 & 100 & 18 \\ 5 & 8 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 24, \quad \Delta = -114, \quad \det = 112, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.82 \rightarrow 12, \quad T_{\max} = 24, \quad T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 1 & 16 & 5 \\ 11 & 20 & 0 & 28 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 28 & 0 & 37 & 11 \\ 5 & 8 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 24, \quad \Delta = -114, \quad \det = 128, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.84 \rightarrow 12, \quad T_{\max} = 24, \quad T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 5 \\ 11 & 20 & 0 & 28 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 28 & 0 & 37 & 11 \\ 5 & 8 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 161, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.85 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 12 & 1 & 23 & 5 \\ 12 & 25 & 0 & 48 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 23 & 48 & 0 & 89 & 17 \\ 5 & 9 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 189, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 12 & 1 & 15 & 5 \\ 12 & 25 & 0 & 32 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 32 & 0 & 37 & 11 \\ 5 & 9 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \ 1^{\overline{12}} 3^{\overline{1}} 21^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -87, \ \det = -12, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 0 & 4 & 3 \\ 1 & 4 & 0 & 1 & 0 & 4 & 2 \\ 1 & 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 4 & 2 & 0 & 4 & 4 \\ 2 & 3 & 2 & 0 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{12}} 3^{\overline{1}} 21^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -87, \ \det = -40, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 5 & 0 & 5 & 18 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 12 & 2 \\ 10 & 18 & 0 & 12 & 56 & 14 \\ 3 & 5 & -2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{111}} 3^{\overline{1}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 14, \ \Delta = -93, \ \det = -66, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.18 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 7 & 3 \\ 4 & 10 & 13 & 0 & 16 & 6 \\ 5 & 13 & 12 & 0 & 17 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 16 & 17 & 0 & 20 & 8 \\ 3 & 6 & 6 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -87, \ \det = -161, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 5 & 0 & 5 & 13 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 9 & 2 \\ 7 & 13 & 0 & 9 & 26 & 10 \\ 3 & 5 & -2 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{111}} 3^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 14, \ \Delta = -93, \ \det = -166, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.19 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 6 & 3 \\ 4 & 10 & 13 & 0 & 16 & 6 \\ 5 & 13 & 12 & 0 & 17 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 16 & 17 & 0 & 20 & 8 \\ 3 & 6 & 6 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -87, \ \det = -349, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 5 & 0 & 5 & 18 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 13 & 2 \\ 9 & 18 & 0 & 13 & 48 & 14 \\ 3 & 5 & -2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{\overline{12}} 3^{\overline{1}} 21^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -87, \ \det = -404, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 5 & 0 & 5 & 23 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 16 & 2 \\ 12 & 23 & 0 & 16 & 84 & 18 \\ 3 & 5 & -2 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{12} 3^{11} 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -87, \ \det = -696, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 5 & 0 & 5 & 28 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 20 & 2 \\ 14 & 28 & 0 & 20 & 120 & 22 \\ 3 & 5 & -2 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [16] \ 1^{111} 3^{11} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 12, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 13 & 5 & 1 & 1 & 7 \\ 5 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & -2 & 0 \\ 7 & 3 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{111} 3^{11} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 26, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.85 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 4 & 3 \\ 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 2 & 1 & 0 & -2 & 0 & 0 \\ 4 & 3 & 2 & 0 & 4 & 4 \\ 3 & 2 & 0 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111} 4^{11} 13^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 45, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.84 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 1 & 0 & -2 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 2 \\ 3 & 2 & 0 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{111} 3^{11} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 49, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.86 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 1 & 0 & -2 & 0 & 0 \\ 3 & 1 & 1 & 0 & 0 & 2 \\ 3 & 2 & 0 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111} 3^{11} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 56, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 10 & 1 & 11 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 11 & 14 & 1 & 14 & 8 \\ 5 & 7 & -1 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111} 3^{11} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 168, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 10 & 1 & 10 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 14 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

4.647 NHC(-2)(-3) + so₈ + so₈ + su₂n3mix + su₂

$$T = 9 \quad [0] \ 1^{111} 3^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -80, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 1 & 5 \\ 11 & 20 & 0 & 28 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 28 & 0 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 8 & -2 & 11 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^{111} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -117, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 10 & 5 \\ 11 & 20 & 0 & 28 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 28 & 0 & 37 & 24 & 11 \\ 10 & 17 & 0 & 24 & 13 & 7 \\ 5 & 8 & -2 & 11 & 7 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 16 & 9 & 5 \\ 11 & 20 & 0 & 28 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 28 & 0 & 37 & 24 & 11 \\ 9 & 17 & 0 & 24 & 13 & 7 \\ 5 & 8 & -2 & 11 & 7 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 9 & 5 \\ 11 & 20 & 0 & 28 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 28 & 0 & 37 & 24 & 11 \\ 9 & 17 & 0 & 24 & 13 & 7 \\ 5 & 8 & -2 & 11 & 7 & 2 \end{pmatrix}_{b_0}$$

4.648 NHC(-2)(-3) + so₈ + so₈ + su_{2n3mix} + su₃

$T = 9$ [0] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -93, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 5 & 11 & 1 & 7 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 12 & 6 \\ 5 & 8 & -2 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

4.649 NHC(-2)(-3) + so₈ + so₈ + su₃

$T = 8$ [0] $1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 67, \Delta = -121, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 67, T \in [15, 67]$

$$\begin{pmatrix} 5 & 11 & 1 & 18 & 5 \\ 11 & 20 & 0 & 34 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 34 & 0 & 57 & 13 \\ 5 & 8 & -2 & 13 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 67, \Delta = -121, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$

$$\begin{pmatrix} 5 & 11 & 1 & 19 & 5 \\ 11 & 20 & 0 & 34 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 19 & 34 & 0 & 57 & 13 \\ 5 & 8 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 67, \Delta = -121, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 5 \\ 12 & 25 & 0 & 16 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 16 & 0 & 9 & 5 \\ 5 & 9 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 67, \Delta = -121, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 67, \Delta = -121, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 17 & 0 & 12 & 6 \\ 5 & 8 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 67, \Delta = -121, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12, T_{\max} = 67, T \in [12, 67]$

$$\begin{pmatrix} 5 & 12 & 1 & 18 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 36 & 0 & 48 & 12 \\ 5 & 9 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 67$, $\Delta = -121$, $\det = 147$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 67$, $T \in [13, 67]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 12 & 6 \\ 5 & 8 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 67$, $\Delta = -121$, $\det = 168$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 67$, $T \in [13, 67]$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 20 & 0 & 12 & 6 \\ 5 & 9 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 67$, $\Delta = -121$, $\det = 188$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.13 \rightarrow 13$, $T_{\max} = 67$, $T \in [13, 67]$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 36 & 0 & 48 & 12 \\ 5 & 9 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [9] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 21$, $\Delta = -100$, $\det = -5$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 15.40 \rightarrow 16$, $T_{\max} = 21$, $T \in [16, 21]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 1 & 3 \\ 4 & 10 & 13 & 0 & 0 & 6 \\ 5 & 13 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & 6 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

[10] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -26$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.46 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & -2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -33$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.27 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 \\ 1 & 2 & 0 & 0 & 1 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 \\ 1 & 2 & 1 & 0 & -2 & 1 \\ 1 & 1 & 3 & 0 & 1 & 0 \\ 2 & 2 & 2 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

[12] $1^{11}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 21$, $\Delta = -100$, $\det = -41$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.41 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 1 & 3 \\ 4 & 10 & 1 & 12 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 12 & 0 & 11 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & 5 & -1 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -45$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 3 & 5 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.42 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 12 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 6 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.17 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 4 & 2 \\ 2 & 5 & 0 & 4 & 1 & 3 \\ 3 & 5 & -2 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 8 & 1 & 1 & 5 \\ 8 & 10 & 1 & 0 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 6 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 5 & 9 & 1 & 6 & 5 \\ 9 & 13 & 0 & 10 & 7 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 10 & 0 & 4 & 4 \\ 5 & 7 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11} 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 5 & 0 & 5 & 10 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 7 & 2 \\ 5 & 10 & 0 & 7 & 13 & 7 \\ 3 & 5 & -2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11} 31^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 4 & 3 \\ 5 & 13 & 16 & 0 & 12 & 7 \\ 5 & 16 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 12 & 12 & 0 & 9 & 5 \\ 3 & 7 & 6 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [21] \quad 1^{11} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 14 & 10 & 2 & 8 \\ 10 & 5 & 1 & 5 \\ 2 & 1 & -2 & 0 \\ 8 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 10 & 1 & 1 & 5 \\ 10 & 13 & 0 & 0 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 1 & 0 & 1 & -2 & 0 \\ 5 & 7 & -1 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 129, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 0 & 1 & 1 & 1 & 0 \\ 1 & 1 & -2 & 0 & 0 & 1 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 1 & 0 & 1 & 0 & 0 & -2 \\ 2 & 2 & 0 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [24] \quad 1^{11} 4 1 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 1 & 0 & 2 & 3 \\ 1 & 4 & 0 & 1 & 0 & 2 & 2 \\ 1 & 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 2 & 2 & 1 & 0 & 0 & 2 \\ 2 & 3 & 2 & 0 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{11} 1^{11} 4 1 4 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3$ $T = 11, T_{\max} = 14, \Delta = -82, \det = -48, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 1 & 0 & 3 & 3 \\ 1 & 3 & 0 & 1 & 0 & 2 & 2 \\ 1 & 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 2 & 1 & 0 & 0 & 2 \\ 2 & 3 & 2 & 0 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{11} 1^{11} 4 1 4 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3$ $T = 11, T_{\max} = 14, \Delta = -82, \det = -69, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 2 & 5 & 0 & 3 & 1 & 3 \\ 3 & 5 & -2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

4.650 NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$

$T = 9$ [0] $1^{11} 1^{11} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 9 & 5 \\ 12 & 25 & 0 & 16 & 20 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 12 & 5 \\ 9 & 20 & 0 & 12 & 13 & 7 \\ 5 & 9 & -2 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{11} 1^{11} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$

$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 28 & 5 \\ 12 & 25 & 0 & 16 & 60 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 36 & 5 \\ 28 & 60 & 0 & 36 & 133 & 21 \\ 5 & 9 & -2 & 5 & 21 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{11} 1^{11} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 1 & 5 \\ 11 & 20 & 0 & 17 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 8 & -2 & 6 & 0 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{11} 1^{11} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 10 & 5 \\ 12 & 25 & 0 & 16 & 20 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 12 & 5 \\ 10 & 20 & 0 & 12 & 13 & 7 \\ 5 & 9 & -2 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{11} 1^{11} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 10 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 15 & 6 \\ 10 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -2 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{11} 1^{11} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -133, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 13 & 7 \\ 5 & 8 & -2 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{11} 1^{11} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

14.0

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 12 & 15 & 6 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -2 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 15 & 6 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -2 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 8 & 5 \\ 11 & 20 & 0 & 0 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 16 & 0 & 0 & 10 & 6 \\ 5 & 8 & -2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 19 & 5 \\ 12 & 25 & 0 & 16 & 40 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 24 & 5 \\ 19 & 40 & 0 & 24 & 58 & 14 \\ 5 & 9 & -2 & 5 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 20 & 5 \\ 12 & 25 & 0 & 16 & 40 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 24 & 5 \\ 20 & 40 & 0 & 24 & 58 & 14 \\ 5 & 9 & -2 & 5 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 9 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 26 & 5 \\ 11 & 20 & 0 & 0 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 26 & 50 & 0 & 0 & 118 & 20 \\ 5 & 8 & -2 & -1 & 20 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 17 & 5 \\ 11 & 20 & 0 & 0 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 17 & 33 & 0 & 0 & 49 & 13 \\ 5 & 8 & -2 & -1 & 13 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 30 & 5 \\ 12 & 25 & 0 & 16 & 60 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 36 & 5 \\ 30 & 60 & 0 & 36 & 133 & 21 \\ 5 & 9 & -2 & 5 & 21 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 29 & 5 \\ 12 & 25 & 0 & 16 & 60 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 36 & 5 \\ 29 & 60 & 0 & 36 & 133 & 21 \\ 5 & 9 & -2 & 5 & 21 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 38 & 5 \\ 12 & 25 & 0 & 16 & 80 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 48 & 5 \\ 38 & 80 & 0 & 48 & 238 & 28 \\ 5 & 9 & -2 & 5 & 28 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 40 & 5 \\ 12 & 25 & 0 & 16 & 80 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 48 & 5 \\ 40 & 80 & 0 & 48 & 238 & 28 \\ 5 & 9 & -2 & 5 & 28 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 39 & 5 \\ 12 & 25 & 0 & 16 & 80 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 48 & 5 \\ 39 & 80 & 0 & 48 & 238 & 28 \\ 5 & 9 & -2 & 5 & 28 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{112} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 19 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 19 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 18 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 18 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \quad 1^{11} 4^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 6 & 9 & 5 \\ 9 & 13 & 0 & 10 & 14 & 7 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 10 & 0 & 4 & 9 & 4 \\ 9 & 14 & 0 & 9 & 13 & 7 \\ 5 & 7 & -2 & 4 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{11} 4^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 261, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 6 & 6 & 5 \\ 9 & 13 & 0 & 10 & 10 & 7 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 10 & 0 & 4 & 7 & 4 \\ 6 & 10 & 0 & 7 & 5 & 5 \\ 5 & 7 & -2 & 4 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{11} 4^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 480, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 6 & 15 & 5 \\ 9 & 13 & 0 & 10 & 24 & 7 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 10 & 0 & 4 & 16 & 4 \\ 15 & 24 & 0 & 16 & 38 & 12 \\ 5 & 7 & -2 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

4.651 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 9 & 6 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -2 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

4.652 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \text{n3mix}$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 7 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 9 & 6 \\ 7 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -2 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

4.653 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 1 & 5 \\ 12 & 25 & 0 & 16 & 0 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & -2 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 9 & 5 \\ 12 & 25 & 0 & 16 & 20 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 12 & 5 \\ 9 & 20 & 0 & 12 & 12 & 6 \\ 5 & 9 & -2 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 21$, $\Delta = -100$, $\det = -180$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 1 & 8 & 10 & 5 \\ 12 & 25 & 0 & 16 & 20 & 9 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 9 & 12 & 5 \\ 10 & 20 & 0 & 12 & 12 & 6 \\ 5 & 9 & -2 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 21$, $\Delta = -100$, $\det = -245$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [4] $1^{11} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 45$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.20 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 1 & 5 \\ 12 & 25 & 12 & 0 & 0 & 9 \\ 6 & 12 & 3 & 0 & 0 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 3 & -1 & -1 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{11} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 168$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 4 & 1 & 3 \\ 5 & 13 & 16 & 0 & 12 & 0 & 7 \\ 5 & 16 & 12 & 0 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 12 & 12 & 0 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 6 & -2 & 5 & -1 & 1 \end{pmatrix}_{b_0}$$

4.654 NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_2$

$T = 7$ [0] $21^1 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -2$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 25 & 20 & 9 \\ 12 & 20 & 13 & 7 \\ 6 & 9 & 7 & 0 \end{pmatrix}_{b_0}$$

[1] $1 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -2$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 12 & 8 & 5 \\ 12 & 25 & 16 & 9 \\ 8 & 16 & 10 & 6 \\ 5 & 9 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -2$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 25 & 20 & 9 \\ 12 & 20 & 13 & 7 \\ 6 & 9 & 7 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{113} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -2$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 10 & 16 & 80 & 6 \\ 16 & 25 & 124 & 9 \\ 80 & 124 & 613 & 45 \\ 6 & 9 & 45 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -5$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 14 & 6 \\ 15 & 20 & 17 & 8 \\ 14 & 17 & 13 & 7 \\ 6 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

[5] $1^{111} 3 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 12 & 11 & 5 \\ 12 & 25 & 20 & 9 \\ 11 & 20 & 13 & 7 \\ 5 & 9 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{114} 3 \ 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 10 & 16 & 74 & 6 \\ 16 & 25 & 112 & 9 \\ 74 & 112 & 478 & 40 \\ 6 & 9 & 40 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{112} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 10 & 16 & 37 & 6 \\ 16 & 25 & 56 & 9 \\ 37 & 56 & 118 & 20 \\ 6 & 9 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14.09 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 5 & 0 & 4 & 4 \\ 3 & 4 & -2 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{113} 3 \ 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 20.33 \rightarrow 21, T_{\max} = 22, T \in [21, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 10 & 0 & 22 & 8 \\ 3 & 4 & -2 & 8 & -8 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3 \ 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 10 & 0 & 22 & 8 \\ 3 & 4 & -2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 3 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -13, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -2 & 0 \\ 5 & 9 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 3 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 16 & 23 & 6 \\ 16 & 25 & 36 & 9 \\ 23 & 36 & 49 & 13 \\ 6 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{113} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12.41 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 10 & 16 & 47 & 6 \\ 16 & 25 & 76 & 9 \\ 47 & 76 & 217 & 27 \\ 6 & 9 & 27 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 21^{13} 3 \ 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 10 & 16 & 84 & 6 \\ 16 & 25 & 128 & 9 \\ 84 & 128 & 634 & 46 \\ 6 & 9 & 46 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{114} 3 \ 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 10 & 16 & 84 & 6 \\ 16 & 25 & 128 & 9 \\ 84 & 128 & 634 & 46 \\ 6 & 9 & 46 & 1 \end{pmatrix}_{b_0}$$

- [17] $1^{111}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -22$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 13.09 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$
 $\begin{pmatrix} 5 & 11 & 30 & 5 \\ 11 & 20 & 50 & 8 \\ 30 & 50 & 118 & 20 \\ 5 & 8 & 20 & 0 \end{pmatrix}_{b_0}$
- [18] $1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -22$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 10 & 5 \\ 11 & 20 & 17 & 8 \\ 10 & 17 & 13 & 7 \\ 5 & 8 & 7 & 2 \end{pmatrix}_{b_0}$
- [19] $1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -22$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 1 & 5 \\ 11 & 20 & 0 & 8 \\ 1 & 0 & -2 & 0 \\ 5 & 8 & 0 & 2 \end{pmatrix}_{b_0}$
- [20] $1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -25$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 12 & 6 \\ 15 & 20 & 17 & 8 \\ 12 & 17 & 13 & 7 \\ 6 & 8 & 7 & 3 \end{pmatrix}_{b_0}$
- [21] $1^{112}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -27$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.22 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 10 & 16 & 35 & 6 \\ 16 & 25 & 56 & 9 \\ 35 & 56 & 118 & 20 \\ 6 & 9 & 20 & 2 \end{pmatrix}_{b_0}$
- [22] $1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -27$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.48 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 5 & 0 & 4 & 4 \\ 3 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$
- [23] $1^131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 9 & 5 \\ 11 & 20 & 17 & 8 \\ 9 & 17 & 13 & 7 \\ 5 & 8 & 7 & 2 \end{pmatrix}_{b_0}$
- [24] $1^{112}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -29$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 13.34 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$
 $\begin{pmatrix} 5 & 12 & 29 & 5 \\ 12 & 25 & 56 & 9 \\ 29 & 56 & 118 & 20 \\ 5 & 9 & 20 & 0 \end{pmatrix}_{b_0}$
- [26] $1^{111}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -30$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.20 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 10 & 5 \\ 11 & 20 & 16 & 8 \\ 10 & 16 & 10 & 6 \\ 5 & 8 & 6 & 2 \end{pmatrix}_{b_0}$
- [27] $1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -30$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 1 & 6 \\ 15 & 20 & 0 & 8 \\ 1 & 0 & -2 & 0 \\ 6 & 8 & 0 & 3 \end{pmatrix}_{b_0}$
- [25] $21^131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -30$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.20 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 13 & 17 & 12 & 7 \\ 17 & 20 & 16 & 8 \\ 12 & 16 & 10 & 6 \\ 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$
- [28] $1^{14}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.25 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 16 & 0 & 46 & 12 \\ 3 & 4 & -2 & 12 & 0 \end{pmatrix}_{b_0}$

- [31] $1^1 3^1 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 14.75 \rightarrow 15$, $T_{\max} = 22$, $T \in [15, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 20 & 0 & 94 & 16 \\ 3 & 4 & -2 & 16 & -2 \end{pmatrix}_{b_0}$$
- [32] $1^{12} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 14.75 \rightarrow 15$, $T_{\max} = 22$, $T \in [15, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 8 & 0 & 10 & 6 \\ 3 & 4 & -2 & 6 & -2 \end{pmatrix}_{b_0}$$
- [33] $1^{12} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 4 & 8 & 0 & 10 & 6 \\ 3 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$
- [34] $1^{14} 1^1 3^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 59 & 5 \\ 12 & 25 & 124 & 9 \\ 59 & 124 & 613 & 45 \\ 5 & 9 & 45 & 2 \end{pmatrix}_{b_0}$$
- [35] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 4 & 4 & 0 & 1 & 3 \\ 3 & 4 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [36] $1^{13} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 12 & 0 & 25 & 9 \\ 3 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$
- [37] $1^{17} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.3$
 $T_{\min} = 16.37 \rightarrow 17$, $T_{\max} = 22$, $T \in [17, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 28 & 0 & 145 & 21 \\ 3 & 4 & -2 & 21 & -4 \end{pmatrix}_{b_0}$$
- [38] $1^{16} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 14.75 \rightarrow 15$, $T_{\max} = 22$, $T \in [15, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 24 & 0 & 106 & 18 \\ 3 & 4 & -2 & 18 & -2 \end{pmatrix}_{b_0}$$
- [39] $1^{15} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.1$
 $T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 20 & 0 & 73 & 15 \\ 3 & 4 & -2 & 15 & -1 \end{pmatrix}_{b_0}$$
- [30] $1^{112} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 10 & 16 & 46 & 6 \\ 16 & 25 & 72 & 9 \\ 46 & 72 & 202 & 26 \\ 6 & 9 & 26 & 3 \end{pmatrix}_{b_0}$$
- [29] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.38 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$
- [41] $1^{112} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -33$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 5 & 12 & 21 & 5 \\ 12 & 25 & 40 & 9 \\ 21 & 40 & 58 & 14 \\ 5 & 9 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{113} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 60 & 5 \\ 12 & 25 & 124 & 9 \\ 60 & 124 & 613 & 45 \\ 5 & 9 & 45 & 2 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 27 & 6 \\ 15 & 20 & 33 & 8 \\ 27 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & -2 & 7 & -2 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 13 & 6 \\ 15 & 20 & 17 & 8 \\ 13 & 17 & 13 & 7 \\ 6 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 4 & 5 & 0 & 4 & 4 \\ 3 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[46] \ 21^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 13 & 17 & 15 & 7 \\ 17 & 20 & 17 & 8 \\ 15 & 17 & 13 & 7 \\ 7 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 10 & 16 & 50 & 6 \\ 16 & 25 & 76 & 9 \\ 50 & 76 & 217 & 27 \\ 6 & 9 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 18 & 5 \\ 12 & 25 & 36 & 9 \\ 18 & 36 & 49 & 13 \\ 5 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{11} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 15 & 0 & 52 & 12 \\ 3 & 4 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[50] \ 21^{11} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 16 & 36 & 6 \\ 16 & 25 & 56 & 9 \\ 36 & 56 & 118 & 20 \\ 6 & 9 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 16 & 36 & 6 \\ 16 & 25 & 56 & 9 \\ 36 & 56 & 118 & 20 \\ 6 & 9 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{11} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 10 & 0 & 22 & 8 \\ 3 & 4 & -2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[53] \ 1^1 1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.36 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 10 & 0 & 22 & 8 \\ 3 & 4 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[54] \ 1^1 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.48 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 8 & 5 \\ 11 & 20 & 16 & 8 \\ 8 & 16 & 10 & 6 \\ 5 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[55] \ 1^1 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.48 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 20 & 5 \\ 11 & 20 & 33 & 8 \\ 20 & 33 & 49 & 13 \\ 5 & 8 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[56] \ 1^1 1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.48 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 44 & 5 \\ 11 & 20 & 84 & 8 \\ 44 & 84 & 346 & 34 \\ 5 & 8 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[57] \ 1^1 1^{11} 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 9 & 5 \\ 12 & 25 & 20 & 9 \\ 9 & 20 & 13 & 7 \\ 5 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[58] \ 1^1 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.58 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 4 & 0 & 1 & 3 \\ 3 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[59] \ 21^{12} 3^1 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 10 & 16 & 60 & 6 \\ 16 & 25 & 92 & 9 \\ 60 & 92 & 325 & 33 \\ 6 & 9 & 33 & 2 \end{pmatrix}_{b_0}$$

$$[60] \ 1^1 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 11 & 6 \\ 15 & 20 & 16 & 8 \\ 11 & 16 & 10 & 6 \\ 6 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[61] \ 1^1 1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 5 & 12 & 36 & 5 \\ 12 & 25 & 72 & 9 \\ 36 & 72 & 202 & 26 \\ 5 & 9 & 26 & 1 \end{pmatrix}_{b_0}$$

$$[62] \ 1^1 1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 35 & 6 \\ 15 & 20 & 50 & 8 \\ 35 & 50 & 118 & 20 \\ 6 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[63] \ 1^1 1^{13} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 10 & 16 & 60 & 6 \\ 16 & 25 & 92 & 9 \\ 60 & 92 & 325 & 33 \\ 6 & 9 & 33 & 2 \end{pmatrix}_{b_0}$$

$$[64] \ 1^1 1^{11} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 13 & 6 \\ 15 & 20 & 16 & 8 \\ 13 & 16 & 10 & 6 \\ 6 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{11} 1^3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 17 & 5 \\ 12 & 25 & 36 & 9 \\ 17 & 36 & 49 & 13 \\ 5 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{11} 1^3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -53, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 10 & 5 \\ 12 & 25 & 20 & 9 \\ 10 & 20 & 13 & 7 \\ 5 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{11} 1^3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 13.36 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 12 & 47 & 5 \\ 12 & 25 & 92 & 9 \\ 47 & 92 & 325 & 33 \\ 5 & 9 & 33 & 0 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{11} 1^3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 16 & 69 & 6 \\ 16 & 25 & 108 & 9 \\ 69 & 108 & 457 & 39 \\ 6 & 9 & 39 & 3 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{11} 1^3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 9 & 5 \\ 11 & 20 & 16 & 8 \\ 9 & 16 & 10 & 6 \\ 5 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{11} 1^3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 15 & 0 & 52 & 12 \\ 3 & 4 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{11} 1^3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -62, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 10 & 16 & 58 & 6 \\ 16 & 25 & 92 & 9 \\ 58 & 92 & 325 & 33 \\ 6 & 9 & 33 & 2 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{11} 1^3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 39 & 5 \\ 11 & 20 & 67 & 8 \\ 39 & 67 & 217 & 27 \\ 5 & 8 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{11} 1^3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 5 & 12 & 54 & 5 \\ 12 & 25 & 108 & 9 \\ 54 & 108 & 457 & 39 \\ 5 & 9 & 39 & 1 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{11} 1^3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 15 & 0 & 52 & 12 \\ 3 & 4 & -2 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{11} 1^3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.55 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 9 & 0 & 13 & 7 \\ 3 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

- [76] $1^{111} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -70$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.86 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 12 & 6 \\ 15 & 20 & 16 & 8 \\ 12 & 16 & 10 & 6 \\ 6 & 8 & 6 & 3 \end{pmatrix}_{b_0}$
- [77] $21^{111} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -72$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.11 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 10 & 16 & 36 & 6 \\ 16 & 20 & 50 & 8 \\ 36 & 50 & 118 & 20 \\ 6 & 8 & 20 & 2 \end{pmatrix}_{b_0}$
- [78] $1^{114} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -72$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 10 & 16 & 70 & 6 \\ 16 & 25 & 112 & 9 \\ 70 & 112 & 478 & 40 \\ 6 & 9 & 40 & 2 \end{pmatrix}_{b_0}$
- [79] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -74$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 10 & 16 & 48 & 6 \\ 16 & 25 & 76 & 9 \\ 48 & 76 & 217 & 27 \\ 6 & 9 & 27 & 2 \end{pmatrix}_{b_0}$
- [80] $21^{11} 31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -74$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 10 & 16 & 48 & 6 \\ 16 & 25 & 76 & 9 \\ 48 & 76 & 217 & 27 \\ 6 & 9 & 27 & 2 \end{pmatrix}_{b_0}$
- [81] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -75$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 10 & 16 & 63 & 6 \\ 16 & 25 & 96 & 9 \\ 63 & 96 & 346 & 34 \\ 6 & 9 & 34 & 2 \end{pmatrix}_{b_0}$
- [82] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -75$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 23 & 6 \\ 15 & 20 & 33 & 8 \\ 23 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0}$
- [83] $1^{111} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -78$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 48 & 5 \\ 11 & 20 & 84 & 8 \\ 48 & 84 & 346 & 34 \\ 5 & 8 & 34 & 2 \end{pmatrix}_{b_0}$
- [84] $21^{112} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -78$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.87 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 13 & 20 & 54 & 7 \\ 20 & 25 & 72 & 9 \\ 54 & 72 & 202 & 26 \\ 7 & 9 & 26 & 3 \end{pmatrix}_{b_0}$
- [85] $131^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -78$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 18 & 5 \\ 12 & 25 & 40 & 9 \\ 18 & 40 & 58 & 14 \\ 5 & 9 & 14 & 2 \end{pmatrix}_{b_0}$
- [86] $1^{12} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -80$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.70 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 8 & 0 & 10 & 6 \\ 3 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$
- [87] $1^{12} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -80$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 8 & 0 & 10 & 6 \\ 3 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$
- [88] $1^1 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -80$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 14.30 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 20 & 0 & 94 & 16 \\ 3 & 4 & -2 & 16 & -2 \end{pmatrix}_{b_0}$$

$$[89] \ 1^{12} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 20 & 0 & 94 & 16 \\ 3 & 4 & -2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[90] \ 1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 16 & 49 & 6 \\ 16 & 25 & 76 & 9 \\ 49 & 76 & 217 & 27 \\ 6 & 9 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[92] \ 1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 10 & 16 & 60 & 6 \\ 16 & 25 & 96 & 9 \\ 60 & 96 & 346 & 34 \\ 6 & 9 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[91] \ 21^{11} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 10 & 16 & 60 & 6 \\ 16 & 25 & 96 & 9 \\ 60 & 96 & 346 & 34 \\ 6 & 9 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[93] \ 1^{111} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 48 & 6 \\ 15 & 20 & 67 & 8 \\ 48 & 67 & 217 & 27 \\ 6 & 8 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[94] \ 1^{114} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13.02 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 12 & 72 & 5 \\ 12 & 25 & 144 & 9 \\ 72 & 144 & 814 & 52 \\ 5 & 9 & 52 & 0 \end{pmatrix}_{b_0}$$

$$[96] \ 1^{13} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 35 & 5 \\ 11 & 20 & 67 & 8 \\ 35 & 67 & 217 & 27 \\ 5 & 8 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[97] \ 1 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 17 & 5 \\ 11 & 20 & 33 & 8 \\ 17 & 33 & 49 & 13 \\ 5 & 8 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[95] \ 1^{113} 3 1^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 31 & 5 \\ 12 & 25 & 60 & 9 \\ 31 & 60 & 133 & 21 \\ 5 & 9 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[98] \ 1^{111} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 61 & 6 \\ 15 & 20 & 84 & 8 \\ 61 & 84 & 346 & 34 \\ 6 & 8 & 34 & 3 \end{pmatrix}_{b_0}$$

$$[99] \ 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 9 & 0 & 13 & 7 \\ 3 & 4 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[100] \ 1^{11} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 14 & 0 & 37 & 11 \\ 3 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[101] \ 1^{1^4} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 16 & 92 & 6 \\ 16 & 25 & 144 & 9 \\ 92 & 144 & 814 & 52 \\ 6 & 9 & 52 & 3 \end{pmatrix}_{b_0}$$

$$[102] \ 1^{1^2} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 34 & 5 \\ 12 & 25 & 72 & 9 \\ 34 & 72 & 202 & 26 \\ 5 & 9 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{1^3} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 20 & 0 & 94 & 16 \\ 3 & 4 & -2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{1^2} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 8 & 0 & 10 & 6 \\ 3 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{1^2} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 35 & 5 \\ 12 & 25 & 72 & 9 \\ 35 & 72 & 202 & 26 \\ 5 & 9 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{1^2} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -98, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 20 & 5 \\ 12 & 25 & 40 & 9 \\ 20 & 40 & 58 & 14 \\ 5 & 9 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[107] \ 1^{1^4} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.06 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 10 & 16 & 73 & 6 \\ 16 & 25 & 112 & 9 \\ 73 & 112 & 478 & 40 \\ 6 & 9 & 40 & 2 \end{pmatrix}_{b_0}$$

$$[108] \ 1^{1^2} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 19 & 5 \\ 11 & 20 & 33 & 8 \\ 19 & 33 & 49 & 13 \\ 5 & 8 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{1^3} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 29 & 5 \\ 11 & 20 & 50 & 8 \\ 29 & 50 & 118 & 20 \\ 5 & 8 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{1^2} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 26 & 5 \\ 11 & 20 & 50 & 8 \\ 26 & 50 & 118 & 20 \\ 5 & 8 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{1^3} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 26 & 6 \\ 15 & 20 & 33 & 8 \\ 26 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

- [112] $1^{114}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -107$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 10 & 16 & 81 & 6 \\ 16 & 25 & 128 & 9 \\ 81 & 128 & 634 & 46 \\ 6 & 9 & 46 & 2 \end{pmatrix}_{b_0}$
- [113] $1^{11}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -107$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.09 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 19 & 0 & 73 & 15 \\ 3 & 4 & -2 & 15 & 1 \end{pmatrix}_{b_0}$
- [114] $1^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -107$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.64 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 13 & 0 & 28 & 10 \\ 3 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$
- [115] $1^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -107$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.64 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 9 & 0 & 13 & 7 \\ 3 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$
- [116] $1^{113}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -108$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 39 & 5 \\ 12 & 25 & 76 & 9 \\ 39 & 76 & 217 & 27 \\ 5 & 9 & 27 & 2 \end{pmatrix}_{b_0}$
- [117] $1^{112}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -110$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.49 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 28 & 5 \\ 12 & 25 & 56 & 9 \\ 28 & 56 & 118 & 20 \\ 5 & 9 & 20 & 2 \end{pmatrix}_{b_0}$
- [118] $1^{11}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -112$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 24 & 0 & 121 & 19 \\ 3 & 4 & -2 & 19 & 1 \end{pmatrix}_{b_0}$
- [119] $1^{113}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -112$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 12 & 0 & 25 & 9 \\ 3 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$
- [120] $1^{112}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -113$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 19 & 5 \\ 12 & 25 & 40 & 9 \\ 19 & 40 & 58 & 14 \\ 5 & 9 & 14 & 2 \end{pmatrix}_{b_0}$
- [121] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -114$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 18 & 5 \\ 11 & 20 & 33 & 8 \\ 18 & 33 & 49 & 13 \\ 5 & 8 & 13 & 2 \end{pmatrix}_{b_0}$
- [122] $1^{111}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -114$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 45 & 5 \\ 11 & 20 & 84 & 8 \\ 45 & 84 & 346 & 34 \\ 5 & 8 & 34 & 2 \end{pmatrix}_{b_0}$
- [123] $1^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -115$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 9 & 0 & 13 & 7 \\ 3 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[124] \quad 1^{11} 3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -122, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.51 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 26 & 5 \\ 12 & 25 & 56 & 9 \\ 26 & 56 & 118 & 20 \\ 5 & 9 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[125] \quad 1^{11} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -125, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 24 & 6 \\ 15 & 20 & 33 & 8 \\ 24 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[126] \quad 1^1 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 5 & 12 & 27 & 5 \\ 12 & 25 & 60 & 9 \\ 27 & 60 & 133 & 21 \\ 5 & 9 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[127] \quad 1^{114} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 16 & 62 & 6 \\ 16 & 25 & 96 & 9 \\ 62 & 96 & 346 & 34 \\ 6 & 9 & 34 & 3 \end{pmatrix}_{b_0}$$

$$[130] \quad 1^{111} 3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 39 & 6 \\ 15 & 20 & 50 & 8 \\ 39 & 50 & 118 & 20 \\ 6 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[129] \quad 1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 38 & 5 \\ 11 & 20 & 67 & 8 \\ 38 & 67 & 217 & 27 \\ 5 & 8 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[128] \quad 1^{111} 3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 36 & 6 \\ 15 & 20 & 50 & 8 \\ 36 & 50 & 118 & 20 \\ 6 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[131] \quad 1^{111} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 47 & 5 \\ 11 & 20 & 84 & 8 \\ 47 & 84 & 346 & 34 \\ 5 & 8 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[132] \quad 1^{114} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.05 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 10 & 16 & 61 & 6 \\ 16 & 25 & 96 & 9 \\ 61 & 96 & 346 & 34 \\ 6 & 9 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[133] \quad 21^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 13 & 20 & 27 & 7 \\ 20 & 25 & 36 & 9 \\ 27 & 36 & 49 & 13 \\ 7 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[134] \quad 1^{111} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 25 & 6 \\ 15 & 20 & 33 & 8 \\ 25 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[135] \quad 21^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

- [136] $21^{12}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7, T_{\max} = 22, \Delta = -113, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 13 & 17 & 27 & 7 \\ 17 & 20 & 33 & 8 \\ 27 & 33 & 49 & 13 \\ 7 & 8 & 13 & 3 \end{pmatrix}_{b_0}$
- [137] $1^{14}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7, T_{\max} = 22, \Delta = -113, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 16 & 72 & 6 \\ 16 & 25 & 112 & 9 \\ 72 & 112 & 478 & 40 \\ 6 & 9 & 40 & 3 \end{pmatrix}_{b_0}$
- [138] $1^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7, T_{\max} = 22, \Delta = -113, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 27 & 5 \\ 12 & 25 & 56 & 9 \\ 27 & 56 & 118 & 20 \\ 5 & 9 & 20 & 2 \end{pmatrix}_{b_0}$
- [139] $1^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7, T_{\max} = 22, \Delta = -113, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 27 & 5 \\ 11 & 20 & 50 & 8 \\ 27 & 50 & 118 & 20 \\ 5 & 8 & 20 & 2 \end{pmatrix}_{b_0}$
- [140] $1^{11}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7, T_{\max} = 22, \Delta = -113, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 36 & 5 \\ 11 & 20 & 67 & 8 \\ 36 & 67 & 217 & 27 \\ 5 & 8 & 27 & 2 \end{pmatrix}_{b_0}$
- [141] $1^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7, T_{\max} = 22, \Delta = -113, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 28 & 5 \\ 11 & 20 & 50 & 8 \\ 28 & 50 & 118 & 20 \\ 5 & 8 & 20 & 2 \end{pmatrix}_{b_0}$
- [142] $1^{11}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7, T_{\max} = 22, \Delta = -113, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 46 & 5 \\ 11 & 20 & 84 & 8 \\ 46 & 84 & 346 & 34 \\ 5 & 8 & 34 & 2 \end{pmatrix}_{b_0}$
- [143] $21^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7, T_{\max} = 22, \Delta = -113, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 20 & 16 & 8 \\ 12 & 16 & 10 & 6 \\ 6 & 8 & 6 & 3 \end{pmatrix}_{b_0}$
- [144] $1^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7, T_{\max} = 22, \Delta = -113, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 16 & 0 & 46 & 12 \\ 3 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$
- [145] $1^{13}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7, T_{\max} = 22, \Delta = -113, \det = -147, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 49 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$
- [146] $1^{13}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7, T_{\max} = 22, \Delta = -113, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 53 & 5 \\ 12 & 25 & 108 & 9 \\ 53 & 108 & 457 & 39 \\ 5 & 9 & 39 & 2 \end{pmatrix}_{b_0}$
- [147] $1^{11}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7, T_{\max} = 22, \Delta = -113, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 62 & 6 \\ 15 & 20 & 84 & 8 \\ 62 & 84 & 346 & 34 \\ 6 & 8 & 34 & 3 \end{pmatrix}_{b_0}$$

[148] $1^{111} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -155, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 49 & 6 \\ 15 & 20 & 67 & 8 \\ 49 & 67 & 217 & 27 \\ 6 & 8 & 27 & 3 \end{pmatrix}_{b_0}$

[149] $1^{111} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 37 & 5 \\ 11 & 20 & 67 & 8 \\ 37 & 67 & 217 & 27 \\ 5 & 8 & 27 & 2 \end{pmatrix}_{b_0}$

[150] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 14 & 0 & 37 & 11 \\ 3 & 4 & -2 & 11 & 1 \end{pmatrix}_{b_0}$

[151] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 14 & 0 & 37 & 11 \\ 3 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$

[153] $1^{13} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 12 & 0 & 25 & 9 \\ 3 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$

[152] $1^{13} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 12 & 0 & 25 & 9 \\ 3 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$

[154] $1^{13} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 51 & 5 \\ 12 & 25 & 108 & 9 \\ 51 & 108 & 457 & 39 \\ 5 & 9 & 39 & 2 \end{pmatrix}_{b_0}$

[155] $1^{111} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -170, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 63 & 6 \\ 15 & 20 & 84 & 8 \\ 63 & 84 & 346 & 34 \\ 6 & 8 & 34 & 3 \end{pmatrix}_{b_0}$

[156] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -170, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 38 & 6 \\ 15 & 20 & 50 & 8 \\ 38 & 50 & 118 & 20 \\ 6 & 8 & 20 & 3 \end{pmatrix}_{b_0}$

[157] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -170, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 37 & 6 \\ 15 & 20 & 50 & 8 \\ 37 & 50 & 118 & 20 \\ 6 & 8 & 20 & 3 \end{pmatrix}_{b_0}$

[158] $1^{12} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 18 & 0 & 58 & 14 \\ 3 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$

- [159] $1^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -173$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 30 & 5 \\ 12 & 25 & 60 & 9 \\ 30 & 60 & 133 & 21 \\ 5 & 9 & 21 & 2 \end{pmatrix}_{b_0}$
- [160] $1^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -173$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 46 & 5 \\ 12 & 25 & 92 & 9 \\ 46 & 92 & 325 & 33 \\ 5 & 9 & 33 & 2 \end{pmatrix}_{b_0}$
- [161] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -175$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 51 & 6 \\ 15 & 20 & 67 & 8 \\ 51 & 67 & 217 & 27 \\ 6 & 8 & 27 & 3 \end{pmatrix}_{b_0}$
- [162] $1^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -176$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 12 & 0 & 25 & 9 \\ 3 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$
- [163] $1^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -179$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.65 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 13 & 0 & 28 & 10 \\ 3 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$
- [164] $1^{113}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -181$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 52 & 5 \\ 12 & 25 & 108 & 9 \\ 52 & 108 & 457 & 39 \\ 5 & 9 & 39 & 2 \end{pmatrix}_{b_0}$
- [165] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -185$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 50 & 6 \\ 15 & 20 & 67 & 8 \\ 50 & 67 & 217 & 27 \\ 6 & 8 & 27 & 3 \end{pmatrix}_{b_0}$
- [166] $1^{14}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -187$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 21 & 0 & 76 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$
- [167] $1^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -188$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 14 & 0 & 37 & 11 \\ 3 & 4 & -2 & 11 & 1 \end{pmatrix}_{b_0}$
- [168] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -188$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 14 & 0 & 37 & 11 \\ 3 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$
- [169] $21^{14}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -190$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 13 & 17 & 72 & 7 \\ 17 & 20 & 84 & 8 \\ 72 & 84 & 346 & 34 \\ 7 & 8 & 34 & 3 \end{pmatrix}_{b_0}$
- [170] $21^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -190$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.86 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 16 & 24 & 6 \\ 16 & 20 & 33 & 8 \\ 24 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0}$

- [171] $1^{113}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -193$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 28 & 5 \\ 12 & 25 & 60 & 9 \\ 28 & 60 & 133 & 21 \\ 5 & 9 & 21 & 2 \end{pmatrix}_{b_0}$
- [172] $1^{11}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -195$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 19 & 0 & 73 & 15 \\ 3 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$
- [173] $21^{12}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -198$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 13 & 17 & 42 & 7 \\ 17 & 20 & 50 & 8 \\ 42 & 50 & 118 & 20 \\ 7 & 8 & 20 & 3 \end{pmatrix}_{b_0}$
- [174] $1^{11}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -198$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.20 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 5 & 12 & 36 & 5 \\ 12 & 25 & 80 & 9 \\ 36 & 80 & 238 & 28 \\ 5 & 9 & 28 & 1 \end{pmatrix}_{b_0}$
- [175] $21^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -200$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 13 & 20 & 15 & 7 \\ 20 & 25 & 20 & 9 \\ 15 & 20 & 13 & 7 \\ 7 & 9 & 7 & 3 \end{pmatrix}_{b_0}$
- [176] $1^{12}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -203$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 13 & 0 & 28 & 10 \\ 3 & 4 & -2 & 10 & 1 \end{pmatrix}_{b_0}$
- [177] $1^{114}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -205$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.68 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 71 & 5 \\ 12 & 25 & 144 & 9 \\ 71 & 144 & 814 & 52 \\ 5 & 9 & 52 & 2 \end{pmatrix}_{b_0}$
- [178] $1^{113}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -208$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 29 & 5 \\ 12 & 25 & 60 & 9 \\ 29 & 60 & 133 & 21 \\ 5 & 9 & 21 & 2 \end{pmatrix}_{b_0}$
- [179] $1^{113}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -209$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 38 & 5 \\ 12 & 25 & 76 & 9 \\ 38 & 76 & 217 & 27 \\ 5 & 9 & 27 & 2 \end{pmatrix}_{b_0}$
- [180] $1^{11}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -212$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 35 & 5 \\ 12 & 25 & 76 & 9 \\ 35 & 76 & 217 & 27 \\ 5 & 9 & 27 & 2 \end{pmatrix}_{b_0}$
- [181] $21^{13}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -216$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 13 & 17 & 57 & 7 \\ 17 & 20 & 67 & 8 \\ 57 & 67 & 217 & 27 \\ 7 & 8 & 27 & 3 \end{pmatrix}_{b_0}$
- [182] $1^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -219$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 13 & 0 & 28 & 10 \\ 3 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$

- [183] $1^{14} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -224$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 16 & 0 & 46 & 12 \\ 3 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$
- [184] $1^{12} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -224$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 43 & 5 \\ 12 & 25 & 92 & 9 \\ 43 & 92 & 325 & 33 \\ 5 & 9 & 33 & 2 \end{pmatrix}_{b_0}$
- [185] $1^{11} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -224$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 24 & 0 & 121 & 19 \\ 3 & 4 & -2 & 19 & 1 \end{pmatrix}_{b_0}$
- [186] $1^{12} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -227$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 23 & 0 & 100 & 18 \\ 3 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$
- [187] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -227$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 13 & 0 & 28 & 10 \\ 3 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$
- [188] $1^{11} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -240$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 45 & 5 \\ 12 & 25 & 92 & 9 \\ 45 & 92 & 325 & 33 \\ 5 & 9 & 33 & 2 \end{pmatrix}_{b_0}$
- [189] $1^{14} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -242$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 64 & 5 \\ 12 & 25 & 128 & 9 \\ 64 & 128 & 634 & 46 \\ 5 & 9 & 46 & 2 \end{pmatrix}_{b_0}$
- [191] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -251$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.69 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 17 & 0 & 49 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$
- [190] $1^{12} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -251$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 19 & 0 & 73 & 15 \\ 3 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$
- [192] $1^{13} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -252$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.73 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 22 & 0 & 85 & 17 \\ 3 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$
- [193] $1^{11} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -257$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 44 & 5 \\ 12 & 25 & 92 & 9 \\ 44 & 92 & 325 & 33 \\ 5 & 9 & 33 & 2 \end{pmatrix}_{b_0}$
- [194] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -260$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 37 & 5 \\ 12 & 25 & 76 & 9 \\ 37 & 76 & 217 & 27 \\ 5 & 9 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[195] \ 1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 36 & 5 \\ 12 & 25 & 76 & 9 \\ 36 & 76 & 217 & 27 \\ 5 & 9 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[196] \ 1^{14} 3 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -262, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.42 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 68 & 5 \\ 12 & 25 & 144 & 9 \\ 68 & 144 & 814 & 52 \\ 5 & 9 & 52 & 2 \end{pmatrix}_{b_0}$$

$$[197] \ 1^{114} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -267, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 19 & 0 & 73 & 15 \\ 3 & 4 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[198] \ 1^{12} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 18 & 0 & 58 & 14 \\ 3 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[199] \ 1^{13} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -272, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 28 & 0 & 154 & 22 \\ 3 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[200] \ 1^{13} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -272, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 16 & 0 & 46 & 12 \\ 3 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[201] \ 1^{114} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -274, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 70 & 5 \\ 12 & 25 & 144 & 9 \\ 70 & 144 & 814 & 52 \\ 5 & 9 & 52 & 2 \end{pmatrix}_{b_0}$$

$$[202] \ 1^{113} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -275, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 19 & 0 & 73 & 15 \\ 3 & 4 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[203] \ 1^{114} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -278, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 40 & 5 \\ 12 & 25 & 80 & 9 \\ 40 & 80 & 238 & 28 \\ 5 & 9 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[204] \ 1^{14} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -288, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 16 & 0 & 46 & 12 \\ 3 & 4 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[205] \ 1^1 3 1^{115} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -288, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.1$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 5 & 12 & 45 & 5 \\ 12 & 25 & 100 & 9 \\ 45 & 100 & 373 & 35 \\ 5 & 9 & 35 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{14}3^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -293, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 69 & 5 \\ 12 & 25 & 144 & 9 \\ 69 & 144 & 814 & 52 \\ 5 & 9 & 52 & 2 \end{pmatrix}_{b_0}$

[206] $1^{14}3^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -293, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 37 & 5 \\ 12 & 25 & 80 & 9 \\ 37 & 80 & 238 & 28 \\ 5 & 9 & 28 & 2 \end{pmatrix}_{b_0}$

[208] $1^{11}3^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 24 & 0 & 121 & 19 \\ 3 & 4 & -2 & 19 & 1 \end{pmatrix}_{b_0}$

[209] $1^{11}3^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -322, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 44 & 5 \\ 12 & 25 & 96 & 9 \\ 44 & 96 & 346 & 34 \\ 5 & 9 & 34 & 2 \end{pmatrix}_{b_0}$

[211] $1^{14}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -323, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 21 & 0 & 76 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$

[210] $1^{13}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -323, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 17 & 0 & 49 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$

[212] $1^{14}3^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -326, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 56 & 5 \\ 12 & 25 & 112 & 9 \\ 56 & 112 & 478 & 40 \\ 5 & 9 & 40 & 2 \end{pmatrix}_{b_0}$

[214] $1^{12}3^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -332, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 18 & 0 & 58 & 14 \\ 3 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$

[213] $1^{14}3^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -332, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 26 & 0 & 118 & 20 \\ 3 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$

[215] $1^{14}3^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -333, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 39 & 5 \\ 12 & 25 & 80 & 9 \\ 39 & 80 & 238 & 28 \\ 5 & 9 & 28 & 2 \end{pmatrix}_{b_0}$

[217] $1^{14}3^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -338, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 38 & 5 \\ 12 & 25 & 80 & 9 \\ 38 & 80 & 238 & 28 \\ 5 & 9 & 28 & 2 \end{pmatrix}_{b_0}$

- [216] $1^{114}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -338, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 48 & 5 \\ 12 & 25 & 96 & 9 \\ 48 & 96 & 346 & 34 \\ 5 & 9 & 34 & 2 \end{pmatrix}_{b_0}$
- [218] $1^{114}31^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -345, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 63 & 5 \\ 12 & 25 & 128 & 9 \\ 63 & 128 & 634 & 46 \\ 5 & 9 & 46 & 2 \end{pmatrix}_{b_0}$
- [220] $1^{12}31^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -347, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 23 & 0 & 100 & 18 \\ 3 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$
- [219] $1^{13}31^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -347, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 27 & 0 & 133 & 21 \\ 3 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$
- [221] $21^{112}31^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -350, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 13 & 20 & 30 & 7 \\ 20 & 25 & 40 & 9 \\ 30 & 40 & 58 & 14 \\ 7 & 9 & 14 & 3 \end{pmatrix}_{b_0}$
- [222] $1^{11}31^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -352, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 24 & 0 & 121 & 19 \\ 3 & 4 & -2 & 19 & 1 \end{pmatrix}_{b_0}$
- [223] $1^331^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -354, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.47 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 60 & 5 \\ 12 & 25 & 128 & 9 \\ 60 & 128 & 634 & 46 \\ 5 & 9 & 46 & 2 \end{pmatrix}_{b_0}$
- [224] $21^{112}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -362, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 13 & 20 & 42 & 7 \\ 20 & 25 & 56 & 9 \\ 42 & 56 & 118 & 20 \\ 7 & 9 & 20 & 3 \end{pmatrix}_{b_0}$
- [225] $1^{13}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -363, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 17 & 0 & 49 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$
- [227] $1^{12}31^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 18 & 0 & 58 & 14 \\ 3 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$
- [226] $1^{12}31^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 18 & 0 & 58 & 14 \\ 3 & 4 & -2 & 14 & 1 \end{pmatrix}_{b_0}$
- [228] $1^{11}31^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -368, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 24 & 0 & 121 & 19 \\ 3 & 4 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[229] $1^{13} 3^{14} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -371, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 17 & 0 & 49 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[230] $1^{12} 1^3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -374, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 52 & 5 \\ 12 & 25 & 112 & 9 \\ 52 & 112 & 478 & 40 \\ 5 & 9 & 40 & 2 \end{pmatrix}_{b_0}$$

[231] $1^{13} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -380, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 22 & 0 & 85 & 17 \\ 3 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[232] $1^{14} 3^{13} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -398, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 62 & 5 \\ 12 & 25 & 128 & 9 \\ 62 & 128 & 634 & 46 \\ 5 & 9 & 46 & 2 \end{pmatrix}_{b_0}$$

[233] $1^{14} 3^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -401, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 61 & 5 \\ 12 & 25 & 128 & 9 \\ 61 & 128 & 634 & 46 \\ 5 & 9 & 46 & 2 \end{pmatrix}_{b_0}$$

[234] $1^{14} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -401, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 45 & 5 \\ 12 & 25 & 96 & 9 \\ 45 & 96 & 346 & 34 \\ 5 & 9 & 34 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{14} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -409, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 47 & 5 \\ 12 & 25 & 96 & 9 \\ 47 & 96 & 346 & 34 \\ 5 & 9 & 34 & 2 \end{pmatrix}_{b_0}$$

[236] $1^{14} 3^{12} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -413, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 55 & 5 \\ 12 & 25 & 112 & 9 \\ 55 & 112 & 478 & 40 \\ 5 & 9 & 40 & 2 \end{pmatrix}_{b_0}$$

[237] $1^{12} 3^{14} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -416, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 28 & 0 & 154 & 22 \\ 3 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[238] $1^{14} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -427, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 21 & 0 & 76 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[239] $1^{14} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -430, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 46 & 5 \\ 12 & 25 & 96 & 9 \\ 46 & 96 & 346 & 34 \\ 5 & 9 & 34 & 2 \end{pmatrix}_{b_0}$$

- [240] $1^{13}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -432$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 32 & 0 & 193 & 25 \\ 3 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$
- [241] $1^{12}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -435$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 23 & 0 & 100 & 18 \\ 3 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$
- [242] $1^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -437$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 53 & 5 \\ 12 & 25 & 112 & 9 \\ 53 & 112 & 478 & 40 \\ 5 & 9 & 40 & 2 \end{pmatrix}_{b_0}$
- [243] $1^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -450$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 54 & 5 \\ 12 & 25 & 112 & 9 \\ 54 & 112 & 478 & 40 \\ 5 & 9 & 40 & 2 \end{pmatrix}_{b_0}$
- [244] $1^{14}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -467$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.76 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 31 & 0 & 172 & 24 \\ 3 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$
- [245] $1^{12}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -476$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.72 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 22 & 0 & 85 & 17 \\ 3 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$
- [246] $21^{13}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -488$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 13 & 20 & 69 & 7 \\ 20 & 25 & 92 & 9 \\ 69 & 92 & 325 & 33 \\ 7 & 9 & 33 & 3 \end{pmatrix}_{b_0}$
- [247] $1^{12}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -491$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.87 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 23 & 0 & 100 & 18 \\ 3 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$
- [248] $1^{14}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -492$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.71 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 26 & 0 & 118 & 20 \\ 3 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$
- [249] $1^{13}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -499$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 21 & 0 & 76 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$
- [250] $1^{13}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -499$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.73 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 27 & 0 & 133 & 21 \\ 3 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$
- [251] $1^{12}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -515$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.99 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 23 & 0 & 100 & 18 \\ 3 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[252] $1^{12} 3^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -528, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 28 & 0 & 154 & 22 \\ 3 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[253] $1^{14} 3^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -539, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 21 & 0 & 76 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[254] $1^{13} 3^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -540, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 22 & 0 & 85 & 17 \\ 3 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[255] $1^{13} 3^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -572, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 22 & 0 & 85 & 17 \\ 3 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[256] $21^{14} 31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -578, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 20 & 96 & 7 \\ 20 & 25 & 128 & 9 \\ 96 & 128 & 634 & 46 \\ 7 & 9 & 46 & 3 \end{pmatrix}_{b_0}$$

[257] $1^{14} 3^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -592, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 20 & 36 & 0 & 238 & 28 \\ 3 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[258] $21^{13} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 20 & 45 & 7 \\ 20 & 25 & 60 & 9 \\ 45 & 60 & 133 & 21 \\ 7 & 9 & 21 & 3 \end{pmatrix}_{b_0}$$

[260] $1^{12} 3^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -608, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 19 & 28 & 0 & 154 & 22 \\ 3 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[259] $1^{13} 3^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -608, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 19 & 32 & 0 & 193 & 25 \\ 3 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[261] $1^{12} 3^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -619, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 27 & 0 & 133 & 21 \\ 3 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[262] $1^{14} 3^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -620, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 26 & 0 & 118 & 20 \\ 3 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[263] $1^{14} 3^1 1^{13} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{s}o_8 + \textcolor{orange}{s}u_2 \quad T = 7, \, T_{\max} = 22, \, \Delta = -113, \, \det = -651, \, (n_+, n_-, n_0) = (1, 7, 0), \, c_{\text{ext}} = 9.0$
 $T_{\min} = 10.72 \rightarrow 11, \, T_{\max} = 22, \, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 31 & 0 & 172 & 24 \\ 3 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$

[264] $1^{12} 3^1 1^{14} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}o_8 + \textcolor{orange}{s}u_2 \quad T = 7, \quad T_{\max} = 22, \quad \Delta = -113, \quad \det = -656, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.5$
 $T_{\min} = 10.99 \rightarrow 11, \quad T_{\max} = 22, \quad T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 20 & 28 & 0 & 154 & 22 \\ 3 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$

[265] $21^{113}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -692, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\left(\begin{array}{cccc} 13 & 20 & 57 & 7 \\ 20 & 25 & 76 & 9 \\ 57 & 76 & 217 & 27 \\ 7 & 9 & 27 & 3 \end{array} \right)_{b_0}$

[266] $1^{13} 3 1^{13} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}o_8 + \textcolor{orange}{s}u_2 \quad T = 7, \, T_{\max} = 22, \, \Delta = -113, \, \det = -707, \, (n_+, n_-, n_0) = (1, 7, 0), \, c_{\text{ext}} = 8.8$
 $T_{\min} = 10.79 \rightarrow 11, \, T_{\max} = 22, \, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 27 & 0 & 133 & 21 \\ 3 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$

[267] $1^{14} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -716, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 26 & 0 & 118 & 20 \\ 3 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$

[268] $1^{13} 3^{12} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -752, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 20 & 32 & 0 & 193 & 25 \\ 3 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$

[269] $1^{13} 3^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -763, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 19 & 27 & 0 & 133 & 21 \\ 3 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$

$$[270] \ 1^{\textcolor{brown}{14}} 3 \ 1^{\textcolor{brown}{12}} + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{5}\textcolor{blue}{o}_8 + \textcolor{brown}{5}\textcolor{blue}{u}_2 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -780, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, \ T_{\max} = 22, \ T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 26 & 0 & 118 & 20 \\ 3 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[271] \ 1^{\textcolor{brown}{14}} 3^{\textcolor{brown}{14}} + [\textcolor{red}{5}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{5}\textcolor{green}{o}_8 + \textcolor{orange}{5}\textcolor{orange}{u}_2 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -800, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.75 \rightarrow 11, \ T_{\max} = 22, \ T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 21 & 36 & 0 & 238 & 28 \\ 3 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[272] \ 1^{14} 3^{12} 1^{13} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}o_8 + \textcolor{orange}{s}u_2 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -803, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.72 \rightarrow 11, \ T_{\max} = 22, \ T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 19 & 31 & 0 & 172 & 24 \\ 3 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[273] $1^{13} 3^1 1^{14} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}o_8 + \textcolor{orange}{s}u_2 \quad T = 7, \, T_{\max} = 22, \, \Delta = -113, \, \det = -864, \, (n_+, n_-, n_0) = (1, 7, 0), \, c_{\text{ext}} = 8.8$
 $T_{\min} = 10.81 \rightarrow 11, \, T_{\max} = 22, \, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 21 & 32 & 0 & 193 & 25 \\ 3 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$

$$[274] \ 1^{14} 3^1 1^{13} + [\textcolor{red}{5u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{5o}_8 + \textcolor{red}{5u}_2 \quad T = 7, \ T_{\max} = 22, \ \Delta = -113, \ \det = -923, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 20 & 31 & 0 & 172 & 24 \\ 3 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[275] \ 1^{13} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -944, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 22 & 32 & 0 & 193 & 25 \\ 3 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[276] \ 21^{11} 4^1 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -950, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 13 & 20 & 60 & 7 \\ 20 & 25 & 80 & 9 \\ 60 & 80 & 238 & 28 \\ 7 & 9 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[277] \ 1^{14} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -976, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 22 & 36 & 0 & 238 & 28 \\ 3 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[278] \ 21^{11} 4^1 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -998, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 13 & 20 & 84 & 7 \\ 20 & 25 & 112 & 9 \\ 84 & 112 & 478 & 40 \\ 7 & 9 & 40 & 3 \end{pmatrix}_{b_0}$$

$$[279] \ 1^{14} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -1011, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 21 & 31 & 0 & 172 & 24 \\ 3 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[280] \ 1^{14} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -1120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 23 & 36 & 0 & 238 & 28 \\ 3 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[281] \ 21^{11} 4^1 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -1122, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 13 & 20 & 72 & 7 \\ 20 & 25 & 96 & 9 \\ 72 & 96 & 346 & 34 \\ 7 & 9 & 34 & 3 \end{pmatrix}_{b_0}$$

$$[282] \ 1^{14} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -1232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 24 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 24 & 36 & 0 & 238 & 28 \\ 3 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[283] \ 21^{11} 5^1 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -1400, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.1$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 13 & 20 & 75 & 7 \\ 20 & 25 & 100 & 9 \\ 75 & 100 & 373 & 35 \\ 7 & 9 & 35 & 3 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [284] \ 1^{11} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 4 & 4 & 1 & 3 \\ 4 & 9 & 8 & 0 & 5 \\ 4 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & 4 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[285] \ 21^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 10 & 16 & 8 & 6 \\ 16 & 25 & 12 & 9 \\ 8 & 12 & 5 & 5 \\ 6 & 9 & 5 & 0 \end{pmatrix}_{b_0}$$

[286] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 3 & 5 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 4 \\ 3 & 5 & 2 & 4 & 3 \end{pmatrix}_{b_0}$

[287] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 4 & 0 & 10 & 6 \\ 3 & 3 & -1 & 6 & 3 \end{pmatrix}_{b_0}$

[288] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 1 & 3 & 0 & 1 & 3 \\ 2 & 2 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$

[289] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & -2 & 0 & 2 \end{pmatrix}_{b_0}$

[290] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & 2 & 0 & 1 \end{pmatrix}_{b_0}$

[291] $1^{12}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -104, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 11 & 5 \\ 9 & 13 & 14 & 7 \\ 11 & 14 & 13 & 7 \\ 5 & 7 & 7 & 3 \end{pmatrix}_{b_0}$

[292] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 5 & 0 & 2 & 4 \\ 3 & 5 & -2 & 4 & 3 \end{pmatrix}_{b_0}$

[293] $1^{11}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -104, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13.54 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$
 $\begin{pmatrix} 5 & 12 & 6 & 5 \\ 12 & 25 & 12 & 9 \\ 6 & 12 & 5 & 5 \\ 5 & 9 & 5 & 1 \end{pmatrix}_{b_0}$

[294] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 3 & 2 \\ 1 & 3 & 0 & 3 & 1 & 3 \\ 2 & 3 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$

[295] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 12 & 0 & 13 & 7 \\ 3 & 6 & -1 & 7 & 2 \end{pmatrix}_{b_0}$

[296] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 4 & 7 & 8 & 1 & 4 \\ 7 & 10 & 13 & 0 & 6 \\ 8 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 4 & 6 & 6 & 0 & 2 \end{pmatrix}_{b_0}$

- [297] $1^{12} 3^{11} 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 5 & 5 & 15 & 5 \\ 3 & 5 & 0 & 10 & 2 \\ 8 & 15 & 10 & 38 & 12 \\ 3 & 5 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$
- [298] $1^{14} 3^{11} 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 5 & 5 & 30 & 5 \\ 3 & 5 & 0 & 20 & 2 \\ 17 & 30 & 20 & 158 & 24 \\ 3 & 5 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$
- [299] $1^{13} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 3 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$
- [300] $1^{11} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$
- [301] $1^{12} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -104, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 16 & 5 \\ 9 & 13 & 20 & 7 \\ 16 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$
- [302] $1^{11} 31^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 7 & 8 & 9 & 4 \\ 7 & 10 & 13 & 12 & 6 \\ 8 & 13 & 12 & 15 & 6 \\ 9 & 12 & 15 & 13 & 7 \\ 4 & 6 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$
- [303] $1^{11} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 7 & 0 & 21 & 9 \\ 3 & 3 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$
- [310] $21^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -104, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 12 & 8 & 6 \\ 12 & 13 & 10 & 7 \\ 8 & 10 & 5 & 5 \\ 6 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$
- [309] $1^{12} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -104, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 12 & 28 & 6 \\ 12 & 13 & 34 & 7 \\ 28 & 34 & 77 & 17 \\ 6 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$
- [308] $1^{14} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -104, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 12 & 12 & 6 \\ 12 & 13 & 14 & 7 \\ 12 & 14 & 13 & 7 \\ 6 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$
- [307] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 20, \Delta = -104, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 12 & 8 & 6 \\ 12 & 13 & 10 & 7 \\ 8 & 10 & 5 & 5 \\ 6 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$
- [306] $1^{11} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[305] \ 21^{11} 4 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 12 & 28 & 6 \\ 12 & 13 & 34 & 7 \\ 28 & 34 & 77 & 17 \\ 6 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[304] \ 21^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 12 & 12 & 6 \\ 12 & 13 & 14 & 7 \\ 12 & 14 & 13 & 7 \\ 6 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[311] \ 1^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 0 & 7 \\ 1 & 0 & -2 & 0 \\ 5 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[313] \ 21^{11} 3 2 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 9 & 4 \\ 5 & 5 & 5 & 10 & 5 \\ 3 & 5 & 0 & 7 & 2 \\ 9 & 10 & 7 & 14 & 8 \\ 4 & 5 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[312] \ 1^{11} 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[315] \ 1^{11} 3 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 4 & 3 & 6 & 3 \\ 1 & 4 & 0 & 3 & 6 & 2 \\ 2 & 3 & 3 & 2 & 6 & 2 \\ 3 & 6 & 6 & 6 & 10 & 6 \\ 2 & 3 & 2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[314] \ 1^{11} 3 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 3 & 3 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[316] \ 1^{11} 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 1 & 3 \\ 3 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[317] \ 1^{12} 3 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 6 & 0 & 10 & 6 \\ 2 & 3 & 3 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[318] \ 1^{11} 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 1 & 3 & 0 & 1 & 3 \\ 2 & 2 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[319] \ 1^{11} 3 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

- [320] $1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 24$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$
- [321] $1^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 19 & 5 \\ 9 & 13 & 24 & 7 \\ 19 & 24 & 38 & 12 \\ 5 & 7 & 12 & 3 \end{pmatrix}_{b_0}$
- [322] $1^141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 9 & 5 \\ 9 & 13 & 14 & 7 \\ 9 & 14 & 13 & 7 \\ 5 & 7 & 7 & 3 \end{pmatrix}_{b_0}$
- [325] $1^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 28$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 8 & 5 \\ 9 & 13 & 10 & 7 \\ 8 & 10 & 5 & 5 \\ 5 & 7 & 5 & 3 \end{pmatrix}_{b_0}$
- [323] $1^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 28$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 10 & 12 & 20 & 6 \\ 12 & 13 & 24 & 7 \\ 20 & 24 & 38 & 12 \\ 6 & 7 & 12 & 3 \end{pmatrix}_{b_0}$
- [324] $21^141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 28$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 10 & 12 & 20 & 6 \\ 12 & 13 & 24 & 7 \\ 20 & 24 & 38 & 12 \\ 6 & 7 & 12 & 3 \end{pmatrix}_{b_0}$
- [327] $1^{12}31^{11}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 30$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 6 & 6 & 0 & 6 & 10 & 6 \\ 2 & 3 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [326] $1^{11}31^{11}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 30$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 3 & 2 \\ 3 & 3 & 0 & 3 & 1 & 3 \\ 2 & 3 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [328] $1^{11}3131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 30$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 6 & 3 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 12 & 0 & 10 & 6 \\ 3 & 6 & -1 & 6 & 2 \end{pmatrix}_{b_0}$
- [329] $1^1321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 31$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 3 & 5 & 0 & 4 & 2 \\ 2 & 5 & 4 & 2 & 4 \\ 3 & 5 & 2 & 4 & 3 \end{pmatrix}_{b_0}$
- [336] $1^141^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$

- $$\begin{pmatrix} 5 & 9 & 18 & 5 \\ 9 & 13 & 30 & 7 \\ 18 & 30 & 61 & 15 \\ 5 & 7 & 15 & 2 \end{pmatrix}_{b_0}$$
- [335] $141^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 5 & 9 & 24 & 5 \\ 9 & 13 & 40 & 7 \\ 24 & 40 & 110 & 20 \\ 5 & 7 & 20 & 1 \end{pmatrix}_{b_0}$$
- [334] $141^{115} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.1$
 $T_{\min} = 14.37 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 5 & 9 & 30 & 5 \\ 9 & 13 & 50 & 7 \\ 30 & 50 & 173 & 25 \\ 5 & 7 & 25 & 0 \end{pmatrix}_{b_0}$$
- [337] $141^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 10 & 5 \\ 9 & 13 & 14 & 7 \\ 10 & 14 & 13 & 7 \\ 5 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$
- [338] $141^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 12 & 5 \\ 9 & 13 & 20 & 7 \\ 12 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$
- [333] $141^{116} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 15.75 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 5 & 9 & 36 & 5 \\ 9 & 13 & 60 & 7 \\ 36 & 60 & 250 & 30 \\ 5 & 7 & 30 & -1 \end{pmatrix}_{b_0}$$
- [332] $141^{117} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.3$
 $T_{\min} = 17.38 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$

$$\begin{pmatrix} 5 & 9 & 42 & 5 \\ 9 & 13 & 70 & 7 \\ 42 & 70 & 341 & 35 \\ 5 & 7 & 35 & -3 \end{pmatrix}_{b_0}$$
- [331] $141^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 6 & 5 \\ 9 & 13 & 10 & 7 \\ 6 & 10 & 5 & 5 \\ 5 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$
- [330] $21^1 41^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 13 & 14 & 9 & 7 \\ 14 & 13 & 10 & 7 \\ 9 & 10 & 5 & 5 \\ 7 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$
- [339] $1^{11} 31^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 2 & 3 & 3 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [340] $1^1 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 5 & 0 & 2 & 4 \\ 3 & 5 & -2 & 4 & 3 \end{pmatrix}_{b_0}$$
- [341] $1^{11} 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 3 & 3 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 0 & 2 \end{pmatrix}_{b_0} \\
[342] & 1^{13}31^{11}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 9 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 12 & 15 & 0 & 49 & 13 \\ 2 & 3 & 3 & -2 & 13 & 1 \end{pmatrix}_{b_0} \\
[343] & 1^{11}31^{14}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 1 & 25 & 3 \\ 4 & 10 & 1 & 60 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 60 & 0 & 346 & 34 \\ 3 & 6 & -1 & 34 & 2 \end{pmatrix}_{b_0} \\
[344] & 131^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 5 & 5 & 3 \\ 4 & 10 & 13 & 12 & 6 \\ 5 & 13 & 12 & 15 & 6 \\ 5 & 12 & 15 & 13 & 7 \\ 3 & 6 & 6 & 7 & 2 \end{pmatrix}_{b_0} \\
[347] & 1^{11}31^{14}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 7 & 8 & 7 & 4 \\ 7 & 10 & 13 & 12 & 6 \\ 8 & 13 & 12 & 12 & 6 \\ 7 & 12 & 12 & 10 & 6 \\ 4 & 6 & 6 & 6 & 2 \end{pmatrix}_{b_0} \\
[345] & 1^{12}31^{14}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 3 & 6 & 0 & 6 & 10 & 6 \\ 2 & 3 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[346] & 1^{11}31^{14}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 7 & 8 & 8 & 4 \\ 7 & 10 & 13 & 12 & 6 \\ 8 & 13 & 12 & 15 & 6 \\ 8 & 12 & 15 & 13 & 7 \\ 4 & 6 & 6 & 7 & 2 \end{pmatrix}_{b_0} \\
[349] & 1^{11}31^{14}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 11 & 6 & 5 \\ 1 & 3 & 0 & 6 & 1 & 3 \\ 2 & 3 & -1 & 5 & 3 & 1 \end{pmatrix}_{b_0} \\
[348] & 1^{12}321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 5 & 5 & 15 & 5 \\ 3 & 5 & 0 & 10 & 2 \\ 9 & 15 & 10 & 38 & 12 \\ 3 & 5 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[350] & 1^{11}321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 5 & 0 & 15 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 15 & 0 & 34 & 12 \\ 3 & 5 & -2 & 12 & 3 \end{pmatrix}_{b_0} \\
[351] & 1^{11}31^{14}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 3 & 2 \\ 2 & 3 & 0 & 3 & 1 & 3 \\ 2 & 3 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[352] & 1^{11}31^{14}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 42, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 6 & 9 & 0 & 13 & 7 \\ 2 & 3 & 3 & -2 & 7 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [353] $1^{11}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 43$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 7 & 5 \\ 9 & 13 & 10 & 7 \\ 7 & 10 & 5 & 5 \\ 5 & 7 & 5 & 3 \end{pmatrix}_{b_0}$
- [354] $1^{11}321^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 43$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 5 & 0 & 25 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 25 & 0 & 92 & 20 \\ 3 & 5 & -2 & 20 & 3 \end{pmatrix}_{b_0}$
- [355] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 43$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & 1 & 3 \\ 3 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$
- [356] $1^{11}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 7 & 0 & 16 & 8 \\ 2 & 2 & 2 & -2 & 8 & 2 \end{pmatrix}_{b_0}$
- [357] $1^{11}31^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 5 & 5 & 3 \\ 4 & 10 & 13 & 12 & 6 \\ 5 & 13 & 12 & 12 & 6 \\ 5 & 12 & 12 & 10 & 6 \\ 3 & 6 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [358] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.74 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 5 & 0 & 10 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 10 & 0 & 14 & 8 \\ 3 & 5 & -2 & 8 & 3 \end{pmatrix}_{b_0}$
- [359] $1^{11}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.74 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 5 & 0 & 10 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 10 & 0 & 14 & 8 \\ 3 & 5 & -2 & 8 & 3 \end{pmatrix}_{b_0}$
- [360] $1^{12}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.74 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 10 & 0 & 38 & 12 \\ 3 & 3 & -1 & 12 & 3 \end{pmatrix}_{b_0}$
- [361] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.58 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & -2 & 5 & 2 \end{pmatrix}_{b_0}$
- [362] $1^{11}31^131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.17 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 4 & 7 & 8 & 8 & 4 \\ 7 & 10 & 13 & 12 & 6 \\ 8 & 13 & 12 & 12 & 6 \\ 8 & 12 & 12 & 10 & 6 \\ 4 & 6 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [363] $1^{14}31^131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 4 & 3 & 12 & 3 \\ 1 & 4 & 0 & 3 & 12 & 2 \\ 2 & 3 & 3 & 2 & 12 & 2 \\ 6 & 12 & 12 & 12 & 46 & 12 \\ 2 & 3 & 2 & 2 & 12 & 1 \end{pmatrix}_{b_0}$
- [364] $1^{12}31^{12}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 12 & 0 & 58 & 14 \\ 2 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[365] \ 21^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 13 & 20 & 9 & 7 \\ 20 & 25 & 12 & 9 \\ 9 & 12 & 5 & 5 \\ 7 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[366] \ 21^{12}321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 18 & 4 \\ 5 & 5 & 5 & 20 & 5 \\ 3 & 5 & 0 & 14 & 2 \\ 18 & 20 & 14 & 62 & 16 \\ 4 & 5 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[367] \ 1^{11}31^331^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 1 & 19 & 3 \\ 4 & 10 & 1 & 48 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 48 & 0 & 217 & 27 \\ 3 & 6 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[369] \ 1^{12}31^{11}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 7 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 12 & 0 & 28 & 10 \\ 2 & 3 & 3 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[368] \ 1^131^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 6 & 0 & 4 & 4 \\ 2 & 3 & 3 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[371] \ 1^{14}31^{11}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 18 & 15 & 0 & 76 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[372] \ 1^131^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 3 & 0 & 4 & 4 \\ 2 & 4 & 2 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[370] \ 1^{11}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 11 & 6 & 5 \\ 2 & 3 & 0 & 6 & 1 & 3 \\ 2 & 3 & -1 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[373] \ 1^1321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 3 & 5 & 0 & 4 & 2 \\ 3 & 5 & 4 & 2 & 4 \\ 3 & 5 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[374] \ 21^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 14 & 15 & 7 \\ 14 & 13 & 14 & 7 \\ 15 & 14 & 13 & 7 \\ 7 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[377] \ 1^{11}3131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 12 & 0 & 10 & 6 \\ 3 & 6 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[375] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 59$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 5 & 5 & 10 & 5 \\ 3 & 5 & 0 & 7 & 2 \\ 5 & 10 & 7 & 14 & 8 \\ 3 & 5 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[376] $1^{12}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 59$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.36 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 15 & 5 \\ 9 & 13 & 20 & 7 \\ 15 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[378] $1^{11}23131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 60$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 0 & 10 & 6 \\ 2 & 3 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[379] $1^{11}31^{11}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 60$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[380] $1^{13}31^{11}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 60$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 15 & 12 & 0 & 49 & 13 \\ 2 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[381] $1^{13}321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 61$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 17 & 0 & 121 & 21 \\ 3 & 3 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[382] $1^{11}321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 62$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.74 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 5 & 0 & 20 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 20 & 0 & 58 & 16 \\ 3 & 5 & -2 & 16 & 3 \end{pmatrix}_{b_0}$$

[383] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 63$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.70 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 7 & 0 & 21 & 9 \\ 3 & 3 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

[384] $1^{11}321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 63$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.70 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 5 & 0 & 15 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 15 & 0 & 32 & 12 \\ 3 & 5 & -2 & 12 & 3 \end{pmatrix}_{b_0}$$

[385] $1^{13}321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 64$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 5 & 5 & 25 & 5 \\ 3 & 5 & 0 & 17 & 2 \\ 13 & 25 & 17 & 104 & 20 \\ 3 & 5 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[386] $1^{14}31^{11}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 66$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 12 & 2 \\ 6 & 12 & 0 & 12 & 46 & 12 \\ 2 & 3 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[389] $1^{13}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 15 & 0 & 49 & 13 \\ 2 & 3 & 3 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[388] $1^{11}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 1 & 9 & 3 \\ 4 & 10 & 1 & 24 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 24 & 0 & 49 & 13 \\ 3 & 6 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[387] $1^{11}31^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 1 & 21 & 3 \\ 4 & 10 & 1 & 48 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 48 & 0 & 217 & 27 \\ 3 & 6 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[391] $1^1 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 15 & 5 \\ 9 & 13 & 24 & 7 \\ 15 & 24 & 38 & 12 \\ 5 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

[390] $1^{12} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 13 & 5 \\ 9 & 13 & 20 & 7 \\ 13 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[392] $1^{12}31^{11}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 5 & 6 & 0 & 6 & 10 & 6 \\ 2 & 3 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[393] $21^{11} 4^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 13 & 14 & 51 & 7 \\ 14 & 13 & 54 & 7 \\ 51 & 54 & 197 & 27 \\ 7 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

[394] $1^{11}31^{12}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 15 & 9 & 0 & 37 & 11 \\ 2 & 4 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

[395] $1^{13}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 13 & 0 & 61 & 15 \\ 3 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[396] $1^{11} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 11 & 0 & 65 & 15 \\ 3 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[399] $1^{12}31^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 4 & 6 & 0 & 6 & 10 & 6 \\ 2 & 3 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

- [398] $1^{11}31^{13}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 74$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 1 & 11 & 3 \\ 4 & 10 & 1 & 24 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 24 & 0 & 49 & 13 \\ 3 & 6 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$
- [397] $1^{11}31^{23}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 74$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 1 & 14 & 3 \\ 4 & 10 & 1 & 36 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 36 & 0 & 118 & 20 \\ 3 & 6 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$
- [401] $1^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 76$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 18 & 5 \\ 9 & 13 & 24 & 7 \\ 18 & 24 & 38 & 12 \\ 5 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$
- [400] $1^{12}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 76$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 14 & 5 \\ 9 & 13 & 20 & 7 \\ 14 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$
- [402] $1^{11}321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 79$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 5 & 5 & 10 & 5 \\ 3 & 5 & 0 & 8 & 2 \\ 5 & 10 & 8 & 14 & 8 \\ 3 & 5 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$
- [403] $1^{12}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 80$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 5 & 0 & 13 & 7 \\ 2 & 2 & 2 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$
- [404] $1^{11}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 80$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$
- [405] $1^{13}1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 84$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 6 & 3 & 0 & 4 & 4 \\ 2 & 4 & 2 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$
- [406] $1^{12}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 84$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 12 & 9 & 0 & 28 & 10 \\ 2 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$
- [407] $1^{14}31^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 84$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 18 & 15 & 0 & 76 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$
- [408] $1^{11}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 84$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 5 & 5 & 10 & 5 \\ 3 & 5 & 0 & 7 & 2 \\ 6 & 10 & 7 & 14 & 8 \\ 3 & 5 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

- [409] $2 \overset{1^{11}}{2} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 3 & 2 \\ 2 & 1 & 4 & 1 & 4 & 3 \\ 2 & 4 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & -2 & 3 & 0 \\ 3 & 4 & 4 & 3 & 5 & 5 \\ 2 & 3 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$
- [410] $1 \overset{1^{11}}{1} 31^{12} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 16 & 3 \\ 4 & 10 & 1 & 36 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 36 & 0 & 118 & 20 \\ 3 & 6 & -1 & 20 & 2 \end{pmatrix}_{b_0}$
- [411] $1 \overset{1^3}{1} 31^{12} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 9 & 2 \\ 5 & 9 & 0 & 9 & 25 & 9 \\ 2 & 3 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0}$
- [412] $21 \overset{1^1}{1} 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 20, \Delta = -104, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 5 & 10 & 7 & 5 \\ 10 & 13 & 10 & 7 \\ 7 & 10 & 5 & 5 \\ 5 & 7 & 5 & 3 \end{pmatrix}_{b_0}$
- [414] $1 \overset{1^{12}}{1} 31^{12} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 9 & 9 & 0 & 28 & 10 \\ 2 & 3 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$
- [413] $1 \overset{1^{11}}{1} 31^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 20 & 3 \\ 4 & 10 & 1 & 48 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 48 & 0 & 217 & 27 \\ 3 & 6 & -1 & 27 & 2 \end{pmatrix}_{b_0}$
- [415] $1 \overset{1^3}{1} 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 20, \Delta = -104, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 19 & 5 \\ 9 & 13 & 30 & 7 \\ 19 & 30 & 61 & 15 \\ 5 & 7 & 15 & 3 \end{pmatrix}_{b_0}$
- [417] $1 \overset{1^{11}}{1} 31^{12} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 4 & 7 & 8 & 17 & 4 \\ 7 & 10 & 13 & 24 & 6 \\ 8 & 13 & 12 & 27 & 6 \\ 17 & 24 & 27 & 49 & 13 \\ 4 & 6 & 6 & 13 & 2 \end{pmatrix}_{b_0}$
- [418] $1 \overset{1^1}{1} 321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 5 & 5 & 15 & 5 \\ 3 & 5 & 0 & 11 & 2 \\ 7 & 15 & 11 & 32 & 12 \\ 3 & 5 & 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [416] $1 \overset{1^1}{1} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 20, \Delta = -104, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 16 & 5 \\ 9 & 13 & 24 & 7 \\ 16 & 24 & 38 & 12 \\ 5 & 7 & 12 & 3 \end{pmatrix}_{b_0}$
- [419] $1 \overset{1^{12}}{1} 321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 14 & 0 & 90 & 18 \\ 3 & 3 & -1 & 18 & 3 \end{pmatrix}_{b_0}$
- [420] $1 \overset{1^2}{1} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 20, \Delta = -104, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$

$$\begin{aligned}
& T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 9 & 17 & 5 \\ 9 & 13 & 24 & 7 \\ 17 & 24 & 38 & 12 \\ 5 & 7 & 12 & 3 \end{pmatrix}_{b_0} \\
[421] & 21^{13} 321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 4 & 5 & 3 & 27 & 4 \\ 5 & 5 & 5 & 30 & 5 \\ 3 & 5 & 0 & 21 & 2 \\ 27 & 30 & 21 & 142 & 24 \\ 4 & 5 & 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[422] & 1^{112} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 7 & 0 & 21 & 9 \\ 3 & 3 & -1 & 9 & 3 \end{pmatrix}_{b_0} \\
[423] & 1^{111} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 1 & 10 & 3 \\ 4 & 10 & 1 & 24 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 24 & 0 & 49 & 13 \\ 3 & 6 & -1 & 13 & 2 \end{pmatrix}_{b_0} \\
[425] & 1^{111} 3 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 5 & 10 & 0 & 33 & 11 \\ 2 & 2 & 2 & -2 & 11 & 2 \end{pmatrix}_{b_0} \\
[424] & 1^{13} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 6 & 0 & 25 & 9 \\ 2 & 2 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0} \\
[426] & 1^{11} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 9 & 0 & 13 & 7 \\ 2 & 3 & 3 & -2 & 7 & 1 \end{pmatrix}_{b_0} \\
[427] & 1^{11} 4 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 9 & 21 & 5 \\ 9 & 13 & 34 & 7 \\ 21 & 34 & 77 & 17 \\ 5 & 7 & 17 & 3 \end{pmatrix}_{b_0} \\
[428] & 1^{111} 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 4 & 7 & 0 & 16 & 8 \\ 2 & 2 & 2 & -2 & 8 & 2 \end{pmatrix}_{b_0} \\
[429] & 1^{12} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 9 & 12 & 0 & 28 & 10 \\ 2 & 3 & 3 & -2 & 10 & 1 \end{pmatrix}_{b_0} \\
[430] & 1^{111} 31^{21} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 1 & 15 & 3 \\ 4 & 10 & 1 & 36 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 36 & 0 & 118 & 20 \\ 3 & 6 & -1 & 20 & 2 \end{pmatrix}_{b_0} \\
[431] & 1^{13} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20]
\end{aligned}$$

$$\begin{pmatrix} 5 & 9 & 22 & 5 \\ 9 & 13 & 30 & 7 \\ 22 & 30 & 61 & 15 \\ 5 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[432] \ 1^{11} 3^{11} 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 115, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.97 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 5 & 5 & 20 & 5 \\ 3 & 5 & 0 & 15 & 2 \\ 9 & 20 & 15 & 58 & 16 \\ 3 & 5 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[433] \ 1^{14} 4^{11} 1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -104, \ \det = 115, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.74 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 5 & 9 & 25 & 5 \\ 9 & 13 & 40 & 7 \\ 25 & 40 & 110 & 20 \\ 5 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[434] \ 1^{12} 3^{11} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 116, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.83 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 5 & 5 & 20 & 5 \\ 3 & 5 & 0 & 14 & 2 \\ 10 & 20 & 14 & 62 & 16 \\ 3 & 5 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[435] \ 1^{12} 3^{11} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 118, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 5 & 0 & 20 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 20 & 0 & 62 & 16 \\ 3 & 5 & -2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[437] \ 1^{12} 3^{11} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 118, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 18 & 0 & 166 & 24 \\ 3 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[436] \ 1^{12} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 118, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 10 & 0 & 38 & 12 \\ 3 & 3 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[438] \ 1^{14} 3^{11} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 118, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 20 & 0 & 158 & 24 \\ 3 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[439] \ 1^{12} 3^{11} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 120, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.73 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 6 & 8 & 0 & 26 & 10 \\ 2 & 2 & 2 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[440] \ 2^{11} 2^{11} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 121, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 5 & 5 & 25 & 5 \\ 3 & 5 & 0 & 17 & 2 \\ 15 & 25 & 17 & 104 & 20 \\ 3 & 5 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[441] \ 1^{13} 3^{11} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 123, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.85 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 19 & 3 \\ 2 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 21 & 0 & 205 & 27 \\ 3 & 3 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[442] \ 1^{11} 3^{11} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 123, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.85 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 5 & 0 & 15 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 15 & 0 & 32 & 12 \\ 3 & 5 & -2 & 12 & 3 \end{pmatrix}_{b_0}$$

[443] $1^{11}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 124$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 4 & 7 & 8 & 15 & 4 \\ 7 & 10 & 13 & 24 & 6 \\ 8 & 13 & 12 & 27 & 6 \\ 15 & 24 & 27 & 49 & 13 \\ 4 & 6 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[444] $1^{12}41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 124$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 20 & 5 \\ 9 & 13 & 30 & 7 \\ 20 & 30 & 61 & 15 \\ 5 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[445] $1^{11}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 124$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 9 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[446] $2^1 2^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 124$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 5 & 5 & 5 & 5 \\ 3 & 5 & 0 & 4 & 2 \\ 3 & 5 & 4 & 2 & 4 \\ 3 & 5 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[447] $1^{13}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 124$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 15 & 12 & 0 & 49 & 13 \\ 2 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[448] $21^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 126$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 13 & 14 & 24 & 7 \\ 14 & 13 & 24 & 7 \\ 24 & 24 & 38 & 12 \\ 7 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

[449] $1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 126$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.35 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 6 & 9 & 0 & 13 & 7 \\ 2 & 3 & 3 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[450] $1^{13}321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 127$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 11 & 0 & 65 & 15 \\ 3 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[452] $1^{12}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 128$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 5 & 0 & 13 & 7 \\ 2 & 2 & 2 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[451] $1^{12}321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 128$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 11 & 0 & 45 & 13 \\ 2 & 2 & 2 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[453] $1^{11}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 129$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 5 & 10 & 3 \\ 4 & 10 & 13 & 24 & 6 \\ 5 & 13 & 12 & 27 & 6 \\ 10 & 24 & 27 & 49 & 13 \\ 3 & 6 & 6 & 13 & 2 \end{pmatrix}_{b_0} \\
[454] \quad & 31^{11}231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.32 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 3 & 3 & 1 & 4 & 2 \\ 3 & 1 & 4 & 1 & 4 & 3 \\ 3 & 4 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & -2 & 3 & 0 \\ 4 & 4 & 4 & 3 & 5 & 5 \\ 2 & 3 & 2 & 0 & 5 & 1 \end{pmatrix}_{b_0} \\
[455] \quad & 1^{13}31^{13}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 9 & 2 \\ 6 & 9 & 0 & 9 & 25 & 9 \\ 2 & 3 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0} \\
[456] \quad & 1^{13}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 9 & 21 & 5 \\ 9 & 13 & 30 & 7 \\ 21 & 30 & 61 & 15 \\ 5 & 7 & 15 & 3 \end{pmatrix}_{b_0} \\
[457] \quad & 1^{13}31^{13}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 134, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 9 & 2 \\ 7 & 9 & 0 & 9 & 25 & 9 \\ 2 & 3 & -1 & 2 & 9 & 1 \end{pmatrix}_{b_0} \\
[458] \quad & 21^{11}41^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 134, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 13 & 14 & 42 & 7 \\ 14 & 13 & 44 & 7 \\ 42 & 44 & 130 & 22 \\ 7 & 7 & 22 & 3 \end{pmatrix}_{b_0} \\
[459] \quad & 131^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 5 & 9 & 3 \\ 4 & 10 & 13 & 24 & 6 \\ 5 & 13 & 12 & 27 & 6 \\ 9 & 24 & 27 & 49 & 13 \\ 3 & 6 & 6 & 13 & 2 \end{pmatrix}_{b_0} \\
[460] \quad & 1^{14}321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 138, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 25 & 3 \\ 2 & 1 & 1 & 28 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 28 & 0 & 366 & 36 \\ 3 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0} \\
[461] \quad & 1^{13}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 5 & 5 & 25 & 5 \\ 3 & 5 & 0 & 17 & 2 \\ 14 & 25 & 17 & 104 & 20 \\ 3 & 5 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[462] \quad & 21^{12}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 10 & 14 & 5 \\ 10 & 13 & 20 & 7 \\ 14 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0} \\
[463] \quad & 21^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 10 & 16 & 5 \\ 10 & 13 & 24 & 7 \\ 16 & 24 & 38 & 12 \\ 5 & 7 & 12 & 3 \end{pmatrix}_{b_0} \\
[466] \quad & 1^{14}31^{12}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 24 & 18 & 0 & 118 & 20 \\ 2 & 4 & 2 & -2 & 20 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

- [465] $1^1 31^{12} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.22 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 6 & 0 & 22 & 8 \\ 2 & 4 & 2 & -2 & 8 & 1 \end{pmatrix}_{b_0}$
- [464] $1^1 31^{12} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.22 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 12 & 6 & 0 & 22 & 8 \\ 2 & 4 & 2 & -2 & 8 & 1 \end{pmatrix}_{b_0}$
- [467] $1^1 4^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 147$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 27 & 5 \\ 9 & 13 & 44 & 7 \\ 27 & 44 & 130 & 22 \\ 5 & 7 & 22 & 3 \end{pmatrix}_{b_0}$
- [468] $1^1 231^{11} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.05 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 9 & 0 & 28 & 10 \\ 2 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$
- [469] $1^1 4^1 321^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 151$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 5 & 5 & 35 & 5 \\ 3 & 5 & 0 & 24 & 2 \\ 18 & 35 & 24 & 202 & 28 \\ 3 & 5 & 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [470] $21^{11} 41^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 152$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 13 & 14 & 33 & 7 \\ 14 & 13 & 34 & 7 \\ 33 & 34 & 77 & 17 \\ 7 & 7 & 17 & 3 \end{pmatrix}_{b_0}$
- [471] $1^{12} 3^1 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 153$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 24 & 3 \\ 2 & 1 & 1 & 25 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 25 & 0 & 313 & 33 \\ 3 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$
- [472] $1^{14} 31^1 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 155$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 12 & 2 \\ 7 & 12 & 0 & 12 & 46 & 12 \\ 2 & 3 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$
- [473] $1^{13} 31^{11} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 156$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.05 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 15 & 12 & 0 & 49 & 13 \\ 2 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$
- [474] $1^{11} 31^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 156$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.05 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 9 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$
- [475] $1^{11} 31^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 159$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 4 & 7 & 8 & 16 & 4 \\ 7 & 10 & 13 & 24 & 6 \\ 8 & 13 & 12 & 27 & 6 \\ 16 & 24 & 27 & 49 & 13 \\ 4 & 6 & 6 & 13 & 2 \end{pmatrix}_{b_0}$
- [476] $1^{12} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 160$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$

$$T_{\min} = 11.47 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 9 & 22 & 5 \\ 9 & 13 & 34 & 7 \\ 22 & 34 & 77 & 17 \\ 5 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[477] \ 1^{12} 3^{21} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 5 & 0 & 25 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 25 & 0 & 94 & 20 \\ 3 & 5 & -2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[478] \ 2 \ 2 \ 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 5 & 5 & 10 & 5 \\ 3 & 5 & 0 & 7 & 2 \\ 6 & 10 & 7 & 14 & 8 \\ 3 & 5 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[479] \ 1^{11} 3 \ 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 5 & 0 & 20 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 20 & 0 & 58 & 16 \\ 3 & 5 & -2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[480] \ 1^{12} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 9 & 25 & 5 \\ 9 & 13 & 34 & 7 \\ 25 & 34 & 77 & 17 \\ 5 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[482] \ 1^{13} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 6 & 0 & 25 & 9 \\ 2 & 2 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[481] \ 1^{14} 4^{13} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 12 & 0 & 46 & 12 \\ 2 & 3 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[483] \ 21^{14} 321^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 170, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 36 & 4 \\ 5 & 5 & 5 & 40 & 5 \\ 3 & 5 & 0 & 28 & 2 \\ 36 & 40 & 28 & 254 & 32 \\ 4 & 5 & 2 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[485] \ 1^{11} 3 \ 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 5 & 5 & 15 & 5 \\ 3 & 5 & 0 & 11 & 2 \\ 8 & 15 & 11 & 32 & 12 \\ 3 & 5 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[484] \ 1^{13} 3 \ 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 13 & 0 & 61 & 15 \\ 3 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[486] \ 1^{14} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 9 & 26 & 5 \\ 9 & 13 & 40 & 7 \\ 26 & 40 & 110 & 20 \\ 5 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[487] \ 1^{12} 31^{11} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 12 & 0 & 28 & 10 \\ 2 & 3 & 3 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[489] $1^{13}31^{13}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 180$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 18 & 9 & 0 & 52 & 12 \\ 2 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[488] $1^{12}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 180$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.04 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 9 & 0 & 28 & 10 \\ 2 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[490] $2^{111}2^231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 181$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 \\ 2 & 1 & 4 & 1 & 3 & 3 \\ 2 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 2 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[491] $1^{12}3^221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 182$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 10 & 0 & 38 & 12 \\ 3 & 3 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

[493] $1^{111}3^221^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 183$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 5 & 0 & 25 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 25 & 0 & 92 & 20 \\ 3 & 5 & -2 & 20 & 3 \end{pmatrix}_{b_0}$$

[492] $1^{13}3^221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 183$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 17 & 0 & 121 & 21 \\ 3 & 3 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[494] $1^{14}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 183$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 7 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 15 & 18 & 0 & 76 & 16 \\ 2 & 3 & 3 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[495] $1^{12}3^221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 186$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 14 & 0 & 90 & 18 \\ 3 & 3 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[496] $1^{12}3^221^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 186$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 5 & 0 & 30 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 30 & 0 & 134 & 24 \\ 3 & 5 & -2 & 24 & 3 \end{pmatrix}_{b_0}$$

[497] $1^{11}4^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 187$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 33 & 5 \\ 9 & 13 & 54 & 7 \\ 33 & 54 & 197 & 27 \\ 5 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

[498] $1^{12}4^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 187$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 9 & 23 & 5 \\ 9 & 13 & 34 & 7 \\ 23 & 34 & 77 & 17 \\ 5 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[500] \ 1^{13} 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 9 & 0 & 40 & 12 \\ 2 & 2 & 2 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[499] \ 1^{12} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 9 & 24 & 5 \\ 9 & 13 & 34 & 7 \\ 24 & 34 & 77 & 17 \\ 5 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[501] \ 1^{12} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 5 & 5 & 20 & 5 \\ 3 & 5 & 0 & 14 & 2 \\ 11 & 20 & 14 & 62 & 16 \\ 3 & 5 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[502] \ 1^{13} 31^{14} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 24 & 12 & 0 & 94 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[504] \ 1^{14} 31^{11} 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 10 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 15 & 18 & 0 & 76 & 16 \\ 2 & 3 & 3 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[503] \ 1^{12} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 12 & 0 & 28 & 10 \\ 2 & 3 & 3 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[505] \ 2 \ 2 \ 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 5 & 5 & 10 & 5 \\ 3 & 5 & 0 & 8 & 2 \\ 6 & 10 & 8 & 14 & 8 \\ 3 & 5 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[506] \ 1^{13} 31^{11} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 8 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 15 & 0 & 49 & 13 \\ 2 & 3 & 3 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[507] \ 1^{12} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 5 & 5 & 25 & 5 \\ 3 & 5 & 0 & 18 & 2 \\ 12 & 25 & 18 & 94 & 20 \\ 3 & 5 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[508] \ 1^{12} 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 6 & 8 & 0 & 26 & 10 \\ 2 & 2 & 2 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[509] \ 1^{14} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 9 & 27 & 5 \\ 9 & 13 & 40 & 7 \\ 27 & 40 & 110 & 20 \\ 5 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[510] \ 1^{14} 3^1 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 12 & 2 \\ 8 & 12 & 0 & 12 & 46 & 12 \\ 2 & 3 & -1 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[511] \ 1^{14} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.42 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 9 & 28 & 5 \\ 9 & 13 & 40 & 7 \\ 28 & 40 & 110 & 20 \\ 5 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[512] \ 1^{13} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 5 & 5 & 30 & 5 \\ 3 & 5 & 0 & 21 & 2 \\ 15 & 30 & 21 & 142 & 24 \\ 3 & 5 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[513] \ 1^{13} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 15 & 0 & 49 & 13 \\ 2 & 3 & 3 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[515] \ 1^{14} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 222, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 16 & 0 & 90 & 18 \\ 3 & 3 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[514] \ 1^{12} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 222, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 19 & 3 \\ 2 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 18 & 0 & 166 & 24 \\ 3 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[516] \ 1^{13} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 9 & 28 & 5 \\ 9 & 13 & 44 & 7 \\ 28 & 44 & 130 & 22 \\ 5 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[517] \ 21^{13} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 10 & 21 & 5 \\ 10 & 13 & 30 & 7 \\ 21 & 30 & 61 & 15 \\ 5 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[520] \ 1^{14} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 21 & 3 \\ 2 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 24 & 0 & 250 & 30 \\ 3 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[519] \ 1^{13} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 5 & 0 & 30 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 30 & 0 & 142 & 24 \\ 3 & 5 & -2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[518] \ 1^{12} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 10 & 0 & 38 & 12 \\ 3 & 3 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

[521] $1^{13}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 12 & 0 & 61 & 15 \\ 2 & 2 & 2 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[522] $1^{12}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 7 & 11 & 0 & 45 & 13 \\ 2 & 2 & 2 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[523] $1^{11}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 5 & 5 & 20 & 5 \\ 3 & 5 & 0 & 15 & 2 \\ 10 & 20 & 15 & 58 & 16 \\ 3 & 5 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[524] $1^{13}31^{13}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 18 & 9 & 0 & 52 & 12 \\ 2 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[525] $1^{14}3^{11}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 10 & 0 & 58 & 14 \\ 2 & 2 & 2 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[526] $1^{12}3^{11}21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 8 & 14 & 0 & 70 & 16 \\ 2 & 2 & 2 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[527] $1^{11}4^{13}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 15 & 15 & 0 & 76 & 16 \\ 2 & 3 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[528] $1^{13}31^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 258, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 7 & 2 \\ 1 & 1 & 4 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 15 & 0 & 49 & 13 \\ 2 & 3 & 3 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[529] $1^{13}3^{11}21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 10 & 15 & 0 & 88 & 18 \\ 2 & 2 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[530] $1^{13}3^{12}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 21 & 0 & 205 & 27 \\ 3 & 3 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

- [531] $1^{12}3^{11}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 261$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 5 & 0 & 25 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 25 & 0 & 94 & 20 \\ 3 & 5 & -2 & 20 & 3 \end{pmatrix}_{b_0}$$
- [532] $1^{12}3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 261$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 13 & 0 & 61 & 15 \\ 3 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$
- [533] $1^{13}31^{12}31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 264$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 15 & 0 & 85 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$
- [534] $1^{11}31^{12}31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 264$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 15 & 9 & 0 & 37 & 11 \\ 2 & 4 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$
- [535] $1^{12}3^{11}21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 264$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.91 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 5 & 5 & 30 & 5 \\ 3 & 5 & 0 & 22 & 2 \\ 14 & 30 & 22 & 134 & 24 \\ 3 & 5 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$
- [536] $1^{11}31^{12}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 264$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 15 & 9 & 0 & 37 & 11 \\ 2 & 4 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$
- [537] $1^{13}31^{12}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 264$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 21 & 15 & 0 & 85 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$
- [538] $1^{14}3^{11}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 266$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 20 & 0 & 158 & 24 \\ 3 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$
- [539] $1^{12}3^{13}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 266$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 14 & 0 & 90 & 18 \\ 3 & 3 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$
- [540] $1^{13}41^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 275$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 29 & 5 \\ 9 & 13 & 44 & 7 \\ 29 & 44 & 130 & 22 \\ 5 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$
- [541] $1^{14}3^{11}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 276$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 5 & 5 & 35 & 5 \\ 3 & 5 & 0 & 24 & 2 \\ 19 & 35 & 24 & 202 & 28 \\ 3 & 5 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$
- [542] $1^{13}31^{13}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 276$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$

$$\begin{aligned}
& T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 18 & 9 & 0 & 52 & 12 \\ 2 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0} \\
[544] & 1^{114} 31^{231} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 284, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 0 & 18 & 3 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 18 & 18 & 0 & 118 & 20 \\ 2 & 3 & 2 & -2 & 20 & 1 \end{pmatrix}_{b_0} \\
[543] & 1^{113} 3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 284, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 9 & 0 & 40 & 12 \\ 2 & 2 & 2 & -2 & 12 & 2 \end{pmatrix}_{b_0} \\
[545] & 21^{12} 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 10 & 23 & 5 \\ 10 & 13 & 34 & 7 \\ 23 & 34 & 77 & 17 \\ 5 & 7 & 17 & 3 \end{pmatrix}_{b_0} \\
[546] & 1^{14} 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 9 & 34 & 5 \\ 9 & 13 & 54 & 7 \\ 34 & 54 & 197 & 27 \\ 5 & 7 & 27 & 3 \end{pmatrix}_{b_0} \\
[547] & 1^{14} 31^{11} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 291, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 8 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 13 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 15 & 18 & 0 & 76 & 16 \\ 2 & 3 & 3 & -2 & 16 & 1 \end{pmatrix}_{b_0} \\
[548] & 1^{111} 3 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 291, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 5 & 5 & 25 & 5 \\ 3 & 5 & 0 & 19 & 2 \\ 12 & 25 & 19 & 92 & 20 \\ 3 & 5 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[549] & 1^{14} 31^{11} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 295, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 9 & 2 \\ 1 & 1 & 4 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 18 & 0 & 76 & 16 \\ 2 & 3 & 3 & -2 & 16 & 1 \end{pmatrix}_{b_0} \\
[550] & 1^{13} 3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 297, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 17 & 0 & 121 & 21 \\ 3 & 3 & -1 & 21 & 3 \end{pmatrix}_{b_0} \\
[551] & 1^{13} 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 9 & 31 & 5 \\ 9 & 13 & 44 & 7 \\ 31 & 44 & 130 & 22 \\ 5 & 7 & 22 & 3 \end{pmatrix}_{b_0} \\
[552] & 1^{13} 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 300, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 9 & 30 & 5 \\ 9 & 13 & 44 & 7 \\ 30 & 44 & 130 & 22 \\ 5 & 7 & 22 & 3 \end{pmatrix}_{b_0} \\
[553] & 2 2 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 300, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0
\end{aligned}$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 5 & 2 \\ 2 & 1 & 4 & 1 & 7 & 3 \\ 2 & 4 & 0 & 1 & 7 & 2 \\ 1 & 1 & 1 & -2 & 4 & 0 \\ 5 & 7 & 7 & 4 & 16 & 8 \\ 2 & 3 & 2 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[554] \ 1^{12} 31^{12} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 18 & 12 & 0 & 58 & 14 \\ 2 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[555] \ 1^{12} 31^{12} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 18 & 12 & 0 & 58 & 14 \\ 2 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[556] \ 1^{13} 321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 5 & 0 & 35 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 35 & 0 & 188 & 28 \\ 3 & 5 & -2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[557] \ 1^{13} 321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 25 & 3 \\ 2 & 1 & 1 & 25 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 25 & 0 & 313 & 33 \\ 3 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[558] \ 2^2 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 4 & 2 \\ 2 & 1 & 4 & 1 & 6 & 3 \\ 2 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -2 & 2 & 0 \\ 4 & 6 & 2 & 10 & 6 \\ 2 & 3 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[559] \ 2^2 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 5 & 5 & 15 & 5 \\ 3 & 5 & 0 & 12 & 2 \\ 9 & 15 & 12 & 34 & 12 \\ 3 & 5 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[560] \ 1^{12} 321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 318, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 18 & 0 & 166 & 24 \\ 3 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[561] \ 1^{12} 321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 324, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 5 & 5 & 25 & 5 \\ 3 & 5 & 0 & 18 & 2 \\ 13 & 25 & 18 & 94 & 20 \\ 3 & 5 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[562] \ 1^{14} 321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 326, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 26 & 3 \\ 2 & 1 & 1 & 28 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 28 & 0 & 366 & 36 \\ 3 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[563] \ 1^{12} 321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 326, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 5 & 0 & 30 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 30 & 0 & 134 & 24 \\ 3 & 5 & -2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[564] \ 1^{11} 31^{12} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 15 & 9 & 0 & 37 & 11 \\ 2 & 4 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[565] \ 1^{13} 3^{11} 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 5 & 5 & 30 & 5 \\ 3 & 5 & 0 & 21 & 2 \\ 16 & 30 & 21 & 142 & 24 \\ 3 & 5 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[566] \ 1^{14} 3^{112} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 338, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 16 & 0 & 90 & 18 \\ 3 & 3 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[567] \ 1^{12} 3^{114} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 338, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 14 & 0 & 90 & 18 \\ 3 & 3 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[568] \ 1^{13} 3^{114} 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 5 & 5 & 35 & 5 \\ 3 & 5 & 0 & 25 & 2 \\ 17 & 35 & 25 & 188 & 28 \\ 3 & 5 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[569] \ 1^{13} 3^{113} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 13 & 0 & 61 & 15 \\ 3 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[570] \ 1^{14} 3^{114} 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 344, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 5 & 5 & 40 & 5 \\ 3 & 5 & 0 & 28 & 2 \\ 20 & 40 & 28 & 254 & 32 \\ 3 & 5 & 2 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[571] \ 1^{14} 3^{113} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 344, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 11 & 13 & 0 & 81 & 17 \\ 2 & 2 & 2 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[572] \ 2^2 2^2 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 344, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 5 & 5 & 20 & 5 \\ 3 & 5 & 0 & 14 & 2 \\ 12 & 20 & 14 & 62 & 16 \\ 3 & 5 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[573] \ 1^{11} 31^{13} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 348, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 21 & 12 & 0 & 73 & 15 \\ 2 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[574] \ 21^{14} 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 358, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 10 & 28 & 5 \\ 10 & 13 & 40 & 7 \\ 28 & 40 & 110 & 20 \\ 5 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[575] \ 1^{14} 3^{11} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 10 & 10 & 0 & 58 & 14 \\ 2 & 2 & 2 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[576] \ 1^{14} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 363, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 9 & 35 & 5 \\ 9 & 13 & 54 & 7 \\ 35 & 54 & 197 & 27 \\ 5 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[577] \ 2 \ 2 \ 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 21 & 3 \\ 4 & 5 & 5 & 35 & 5 \\ 3 & 5 & 0 & 24 & 2 \\ 21 & 35 & 24 & 202 & 28 \\ 3 & 5 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[578] \ 1^{13} 3 \ 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 12 & 0 & 61 & 15 \\ 2 & 2 & 2 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[579] \ 2 \ 2 \ 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 5 & 5 & 15 & 5 \\ 3 & 5 & 0 & 11 & 2 \\ 9 & 15 & 11 & 32 & 12 \\ 3 & 5 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[581] \ 1^{13} 1^{14} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 24 & 12 & 0 & 94 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[580] \ 1^{13} 1^{14} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 24 & 12 & 0 & 94 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[582] \ 1^{13} 3 \ 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 391, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 21 & 3 \\ 2 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 21 & 0 & 205 & 27 \\ 3 & 3 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[584] \ 1^{14} 31^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 24 & 18 & 0 & 118 & 20 \\ 2 & 4 & 2 & -2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[583] \ 1^{14} 31^{12} 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 24 & 18 & 0 & 118 & 20 \\ 2 & 4 & 2 & -2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[585] \ 1^{14} 3 \ 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 402, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 22 & 3 \\ 2 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 24 & 0 & 250 & 30 \\ 3 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[586] \ 1^{13} 3 \ 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 403, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 17 & 0 & 121 & 21 \\ 3 & 3 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[587] \ 1^{14} 3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 20 & 0 & 158 & 24 \\ 3 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[588] \ 1^{14} 3 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 5 & 0 & 40 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 20 & 40 & 0 & 254 & 32 \\ 3 & 5 & -2 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[589] \ 1^{11} 31^{14} 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 408, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 27 & 15 & 0 & 121 & 19 \\ 2 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[590] \ 1^{14} 3 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 408, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 16 & 0 & 110 & 20 \\ 2 & 2 & 2 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[591] \ 31^{11} 231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 409, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 3 & 2 \\ 3 & 1 & 4 & 1 & 3 & 3 \\ 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 3 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[592] \ 1^{14} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 412, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 9 & 36 & 5 \\ 9 & 13 & 54 & 7 \\ 36 & 54 & 197 & 27 \\ 5 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[593] \ 1^{13} 3 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 420, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 10 & 15 & 0 & 88 & 18 \\ 2 & 2 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[594] \ 1^{12} 31^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 18 & 12 & 0 & 58 & 14 \\ 2 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[595] \ 1^{12} 31^{12} 31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 18 & 12 & 0 & 58 & 14 \\ 2 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[596] \ 1^{14} 4 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 435, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.42 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 9 & 37 & 5 \\ 9 & 13 & 54 & 7 \\ 37 & 54 & 197 & 27 \\ 5 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[597] \ 1^{12} 31^{13} 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 24 & 15 & 0 & 100 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[598] \ 1^{12} 3^{11} 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 439, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 5 & 5 & 30 & 5 \\ 3 & 5 & 0 & 22 & 2 \\ 15 & 30 & 22 & 134 & 24 \\ 3 & 5 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[599] \ 1^{11} 31^{13} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 444, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 21 & 12 & 0 & 73 & 15 \\ 2 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[600] \ 1^{14} 3^{13} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 446, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 16 & 0 & 90 & 18 \\ 3 & 3 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[601] \ 1^{13} 3^{11} 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 447, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 5 & 0 & 35 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 35 & 0 & 188 & 28 \\ 3 & 5 & -2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[602] \ 1^{13} 31^{14} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 24 & 12 & 0 & 94 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[603] \ 1^{13} 3^{14} 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 453, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 26 & 3 \\ 2 & 1 & 1 & 25 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 25 & 0 & 313 & 33 \\ 3 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[604] \ 1^{13} 31^{12} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 21 & 15 & 0 & 85 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[605] \ 1^{13} 31^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 21 & 15 & 0 & 85 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[606] \ 21^{13} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 470, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 10 & 30 & 5 \\ 10 & 13 & 44 & 7 \\ 30 & 44 & 130 & 22 \\ 5 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[607] \ 2^{114} 2^2 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 484, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 5 & 5 & 20 & 5 \\ 3 & 5 & 0 & 16 & 2 \\ 12 & 20 & 16 & 62 & 16 \\ 3 & 5 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[608] \ 1^{14} 3^{11} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 488, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 11 & 13 & 0 & 81 & 17 \\ 2 & 2 & 2 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[609] $1^{13} 3^{14} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 501$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 17 & 0 & 121 & 21 \\ 3 & 3 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[610] $1^{14} 3^{13} 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 506$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 27 & 3 \\ 2 & 1 & 1 & 28 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 28 & 0 & 366 & 36 \\ 3 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

[611] $1^{11} 31^{13} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 508$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 21 & 12 & 0 & 73 & 15 \\ 2 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[612] $1^{13} 3^{14} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 513$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 22 & 3 \\ 2 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 21 & 0 & 205 & 27 \\ 3 & 3 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

[613] $1^{13} 3^{11} 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 515$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 5 & 5 & 35 & 5 \\ 3 & 5 & 0 & 25 & 2 \\ 18 & 35 & 25 & 188 & 28 \\ 3 & 5 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[614] $1^{14} 3^{11} 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 519$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 5 & 5 & 40 & 5 \\ 3 & 5 & 0 & 28 & 2 \\ 21 & 40 & 28 & 254 & 32 \\ 3 & 5 & 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[616] $1^{13} 31^{12} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 520$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 21 & 15 & 0 & 85 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

[615] $21^{15} 41^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 520$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.1$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 5 & 10 & 35 & 5 \\ 10 & 13 & 50 & 7 \\ 35 & 50 & 173 & 25 \\ 5 & 7 & 25 & 3 \end{pmatrix}_{b_0}$$

[617] $1^{14} 31^{12} 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 528$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 24 & 18 & 0 & 118 & 20 \\ 2 & 4 & 2 & -2 & 20 & 1 \end{pmatrix}_{b_0}$$

[618] $1^{14} 31^{12} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 528$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 24 & 18 & 0 & 118 & 20 \\ 2 & 4 & 2 & -2 & 20 & 1 \end{pmatrix}_{b_0}$$

[619] $1^{14} 3^{13} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 538$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

[620] $1^{11}31^{13}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 540, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 21 & 12 & 0 & 73 & 15 \\ 2 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[621] $2^1 2^1 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 541, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 1 & 9 & 2 \\ 2 & 4 & 0 & 1 & 9 & 2 \\ 1 & 1 & 1 & -2 & 3 & 0 \\ 6 & 9 & 9 & 3 & 25 & 9 \\ 2 & 3 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[622] $1^{14} 3^1 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 558, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 23 & 3 \\ 2 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 24 & 0 & 250 & 30 \\ 3 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[623] $1^{13} 31^{15} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.1$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 30 & 15 & 0 & 148 & 20 \\ 2 & 4 & 2 & -2 & 20 & 1 \end{pmatrix}_{b_0}$$

[624] $1^{12} 31^{13} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 24 & 15 & 0 & 100 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

[625] $1^{14} 31^{13} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 30 & 21 & 0 & 172 & 24 \\ 2 & 4 & 2 & -2 & 24 & 1 \end{pmatrix}_{b_0}$$

[626] $1^{14} 3^1 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 584, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 12 & 16 & 0 & 110 & 20 \\ 2 & 2 & 2 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[627] $1^{13} 31^{13} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 604, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 27 & 18 & 0 & 133 & 21 \\ 2 & 4 & 2 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

[628] $2^1 2^1 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 605, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 7 & 2 \\ 2 & 1 & 4 & 1 & 10 & 3 \\ 2 & 4 & 0 & 1 & 10 & 2 \\ 1 & 1 & 1 & -2 & 5 & 0 \\ 7 & 10 & 10 & 5 & 33 & 11 \\ 2 & 3 & 2 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[629] $2^1 2^1 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 625, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 5 & 5 & 20 & 5 \\ 3 & 5 & 0 & 15 & 2 \\ 12 & 20 & 15 & 58 & 16 \\ 3 & 5 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[630] $2^1 2^1 31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 649, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$

$$\begin{aligned}
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 3 & 18 & 3 \\ 4 & 5 & 5 & 30 & 5 \\ 3 & 5 & 0 & 21 & 2 \\ 18 & 30 & 21 & 142 & 24 \\ 3 & 5 & 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[631] & 1^{12}31^{13}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 660, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 24 & 15 & 0 & 100 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0} \\
[632] & 1^{14}3^{14}21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 662, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 20 & 0 & 158 & 24 \\ 3 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0} \\
[633] & 1^{11}31^{14}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 664, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.19 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 27 & 15 & 0 & 121 & 19 \\ 2 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0} \\
[634] & 1^{11}31^{14}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 664, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.19 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 27 & 15 & 0 & 121 & 19 \\ 2 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0} \\
[635] & 1^{14}3^{14}21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 678, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 28 & 3 \\ 2 & 1 & 1 & 28 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 28 & 0 & 366 & 36 \\ 3 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0} \\
[636] & 21^{14}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 688, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 10 & 37 & 5 \\ 10 & 13 & 54 & 7 \\ 37 & 54 & 197 & 27 \\ 5 & 7 & 27 & 3 \end{pmatrix}_{b_0} \\
[637] & 1^{14}3^{14}21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 706, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 24 & 3 \\ 2 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 24 & 0 & 250 & 30 \\ 3 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0} \\
[638] & 21^{16}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 718, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2 \\
& T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 10 & 42 & 5 \\ 10 & 13 & 60 & 7 \\ 42 & 60 & 250 & 30 \\ 5 & 7 & 30 & 3 \end{pmatrix}_{b_0} \\
[639] & 1^{12}31^{13}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 24 & 15 & 0 & 100 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0} \\
[640] & 1^{13}31^{13}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 732, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 27 & 18 & 0 & 133 & 21 \\ 2 & 4 & 2 & -2 & 21 & 1 \end{pmatrix}_{b_0} \\
[641] & 2^{113}2^{12}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 736, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 5 & 5 & 25 & 5 \\ 3 & 5 & 0 & 18 & 2 \\ 15 & 25 & 18 & 94 & 20 \\ 3 & 5 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[642] $1^{12}31^{13}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 756, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 24 & 15 & 0 & 100 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

[643] $1^{13}31^{16}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 784, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 36 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 36 & 18 & 0 & 214 & 24 \\ 2 & 4 & 2 & -2 & 24 & 1 \end{pmatrix}_{b_0}$$

[644] $1^{11}31^{14}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 792, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 27 & 15 & 0 & 121 & 19 \\ 2 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[645] $1^{11}31^{14}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 792, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 27 & 15 & 0 & 121 & 19 \\ 2 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[646] $1^{13}31^{13}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 828, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 27 & 18 & 0 & 133 & 21 \\ 2 & 4 & 2 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

[647] $1^{14}31^{13}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 852, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 30 & 21 & 0 & 172 & 24 \\ 2 & 4 & 2 & -2 & 24 & 1 \end{pmatrix}_{b_0}$$

[648] $2^{114}2^231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 856, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 8 & 2 \\ 2 & 1 & 4 & 1 & 12 & 3 \\ 2 & 4 & 0 & 1 & 12 & 2 \\ 1 & 1 & 1 & -2 & 4 & 0 \\ 8 & 12 & 12 & 4 & 46 & 12 \\ 2 & 3 & 2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[649] $1^{12}31^{14}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 864, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 30 & 18 & 0 & 154 & 22 \\ 2 & 4 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

[650] $1^{12}31^{14}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 864, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 14 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 30 & 18 & 0 & 154 & 22 \\ 2 & 4 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

[651] $1^{13}31^{13}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 892, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 27 & 18 & 0 & 133 & 21 \\ 2 & 4 & 2 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

[652] $1^{13}31^{13}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 924, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 27 & 18 & 0 & 133 & 21 \\ 2 & 4 & 2 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

[653] $2^1 2^{114} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 929, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 5 & 5 & 25 & 5 \\ 3 & 5 & 0 & 19 & 2 \\ 15 & 25 & 19 & 92 & 20 \\ 3 & 5 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[654] $1^{14} 31^{13} 31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 948, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 14 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 30 & 21 & 0 & 172 & 24 \\ 2 & 4 & 2 & -2 & 24 & 1 \end{pmatrix}_{b_0}$$

[655] $21^{17} 41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 952, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.3$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 10 & 49 & 5 \\ 10 & 13 & 70 & 7 \\ 49 & 70 & 341 & 35 \\ 5 & 7 & 35 & 3 \end{pmatrix}_{b_0}$$

[656] $1^{13} 31^{14} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 984, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 33 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 33 & 21 & 0 & 193 & 25 \\ 2 & 4 & 2 & -2 & 25 & 1 \end{pmatrix}_{b_0}$$

[657] $2^1 2^{114} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1000, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 9 & 2 \\ 2 & 1 & 4 & 1 & 13 & 3 \\ 2 & 4 & 0 & 1 & 13 & 2 \\ 1 & 1 & 1 & -2 & 6 & 0 \\ 9 & 13 & 13 & 6 & 56 & 14 \\ 2 & 3 & 2 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

[658] $1^{14} 31^{13} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1012, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 30 & 21 & 0 & 172 & 24 \\ 2 & 4 & 2 & -2 & 24 & 1 \end{pmatrix}_{b_0}$$

[659] $1^{14} 31^{14} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1024, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 36 & 4 \\ 1 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 36 & 24 & 0 & 238 & 28 \\ 2 & 4 & 2 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

[660] $1^1 31^{17} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1044, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.3$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 14 & 2 \\ 2 & 4 & 4 & 0 & 42 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 42 & 21 & 0 & 292 & 28 \\ 2 & 4 & 2 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

[661] $1^{14} 31^{13} 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1044, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 30 & 21 & 0 & 172 & 24 \\ 2 & 4 & 2 & -2 & 24 & 1 \end{pmatrix}_{b_0}$$

[662] $1^{12} 31^{14} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1056, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 30 & 18 & 0 & 154 & 22 \\ 2 & 4 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

[663] $1^{12} 31^{14} 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1056, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 30 & 18 & 0 & 154 & 22 \\ 2 & 4 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

[664] $2^1 2^1 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1076, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 24 & 3 \\ 4 & 5 & 5 & 40 & 5 \\ 3 & 5 & 0 & 28 & 2 \\ 24 & 40 & 28 & 254 & 32 \\ 3 & 5 & 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[665] $1^{12} 31^{14} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 30 & 4 \\ 1 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 30 & 18 & 0 & 154 & 22 \\ 2 & 4 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

[666] $2^1 2^1 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 18 & 3 \\ 4 & 5 & 5 & 30 & 5 \\ 3 & 5 & 0 & 22 & 2 \\ 18 & 30 & 22 & 134 & 24 \\ 3 & 5 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[667] $2^1 2^1 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1225, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 21 & 3 \\ 4 & 5 & 5 & 35 & 5 \\ 3 & 5 & 0 & 25 & 2 \\ 21 & 35 & 25 & 188 & 28 \\ 3 & 5 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[669] $1^{13} 31^{14} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 33 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 33 & 21 & 0 & 193 & 25 \\ 2 & 4 & 2 & -2 & 25 & 1 \end{pmatrix}_{b_0}$$

[668] $1^{13} 31^{14} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 15 & 2 \\ 2 & 4 & 4 & 0 & 33 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 33 & 21 & 0 & 193 & 25 \\ 2 & 4 & 2 & -2 & 25 & 1 \end{pmatrix}_{b_0}$$

[670] $2^1 2^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.1$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 10 & 2 \\ 2 & 1 & 4 & 1 & 15 & 3 \\ 2 & 4 & 0 & 1 & 15 & 2 \\ 1 & 1 & 1 & -2 & 5 & 0 \\ 10 & 15 & 15 & 5 & 73 & 15 \\ 2 & 3 & 2 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[671] $1^{14} 31^{14} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1344, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 36 & 4 \\ 1 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 36 & 24 & 0 & 238 & 28 \\ 2 & 4 & 2 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

[672] $1^{13} 31^{14} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 14 & 2 \\ 2 & 4 & 4 & 0 & 33 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 33 & 21 & 0 & 193 & 25 \\ 2 & 4 & 2 & -2 & 25 & 1 \end{pmatrix}_{b_0}$$

[673] $1^{13} 31^{14} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 33 & 4 \\ 1 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 33 & 21 & 0 & 193 & 25 \\ 2 & 4 & 2 & -2 & 25 & 1 \end{pmatrix}_{b_0}$$

[674] $1^{14} 31^{14} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 16 & 2 \\ 2 & 4 & 4 & 0 & 36 & 4 \\ 1 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 36 & 24 & 0 & 238 & 28 \\ 2 & 4 & 2 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

[675] $1^{14}31^{14}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 1536$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 14 & 2 \\ 2 & 4 & 4 & 0 & 36 & 4 \\ 1 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 36 & 24 & 0 & 238 & 28 \\ 2 & 4 & 2 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

[676] $1^{14}31^{14}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 1600$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 15 & 2 \\ 2 & 4 & 4 & 0 & 36 & 4 \\ 1 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 36 & 24 & 0 & 238 & 28 \\ 2 & 4 & 2 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

[677] $2^1 2^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 1756$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 12 & 2 \\ 2 & 1 & 4 & 1 & 18 & 3 \\ 2 & 4 & 0 & 1 & 18 & 2 \\ 1 & 1 & 1 & -2 & 6 & 0 \\ 12 & 18 & 18 & 6 & 106 & 18 \\ 2 & 3 & 2 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

[678] $2^1 2^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 2341$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.3$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 14 & 2 \\ 2 & 1 & 4 & 1 & 21 & 3 \\ 2 & 4 & 0 & 1 & 21 & 2 \\ 1 & 1 & 1 & -2 & 7 & 0 \\ 14 & 21 & 21 & 7 & 145 & 21 \\ 2 & 3 & 2 & 0 & 21 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [679] $1^{111} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 8 & 3 \\ 2 & 0 & 2 & 4 & 2 \\ 3 & 2 & 0 & 5 & 2 \\ 8 & 4 & 5 & 18 & 8 \\ 3 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[680] $1^{11} 4^1 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 0 & 2 & 2 & 2 \\ 3 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 3 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[681] $1^{11} 2^1 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 6 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 8 & 3 \\ 6 & 0 & 8 & 22 & 10 \\ 3 & 0 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[682] $21^1 41^1 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 1 & 2 & 4 \\ 2 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 0 & 2 \\ 3 & 4 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[683] $21^{11} 41^1 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 1 & 15 & 4 \\ 2 & 1 & -2 & 2 & 0 \\ 12 & 15 & 2 & 54 & 14 \\ 3 & 4 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[684] $1^{11} 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 4 & 3 \\ 2 & 0 & 4 & 2 & 4 \\ 3 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[685] $1^{11} 2^1 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -5$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 1 \\ 1 & 0 & 0 & -4 & 0 \\ 1 & 3 & 1 & 0 & 1 \\ 2 & 3 & 2 & -2 & 3 \end{pmatrix}_{b_0}$$

$$[688] \ 1^{11} 4^{11} 23^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 4 \\ 3 & 4 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[686] \ 1^{11} 4^{11} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 4 & 5 & 1 & 5 & 5 \\ 3 & 4 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[687] \ 1^{11} 4^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 2 & 4 & 4 & 4 \\ 4 & 4 & 3 & 4 \\ 4 & 3 & 1 & 3 \\ 4 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[689] \ 1^{11} 4^{11} 1^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[690] \ 1^{11} 4^{11} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 3 & 1 & 5 \\ 3 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 5 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[692] \ 1^{11} 41^{11} 231^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 5 & 2 \\ 3 & 4 & 1 & 2 & 6 & 4 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & -2 & 2 & 0 \\ 5 & 6 & 2 & 2 & 8 & 6 \\ 2 & 4 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[691] \ 2^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 3 & 3 & 3 \\ 6 & 13 & 6 & 6 & 7 \\ 3 & 6 & 0 & 4 & 2 \\ 3 & 6 & 4 & 2 & 4 \\ 3 & 7 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[693] \ 1^{11} 23^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 2 & 1 & 8 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 8 & 0 & 22 & 10 \\ 3 & 4 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[695] \ 1^{11} 2^1 3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 0 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[694] \ 1^{11} 4^{11} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 0 & 2 \\ 3 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[696] \ 21^1 41^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[714] \ 1^{\overline{11}31^{\overline{1}}} 4^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 1 & 4 & 3 \\ 4 & 4 & 0 & 5 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 4 & 5 & 1 & 5 & 5 \\ 3 & 4 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[712] \ 1^{\overline{11}31^{\overline{1}}} 4^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 4 & 3 \\ 4 & 4 & 3 & 5 & 4 \\ 3 & 3 & 0 & 4 & 2 \\ 4 & 5 & 4 & 5 & 5 \\ 3 & 4 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[713] \ 1^{\overline{11}31^{\overline{1}}} 4^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 \\ 4 & 4 & 0 & 5 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 5 & 0 & 4 & 4 \\ 3 & 4 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[711] \ 21^{\overline{11}41^{\overline{11}31^{\overline{1}}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 \\ 2 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 2 & 0 & 0 & 2 \\ 2 & 2 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[710] \ 1^{\overline{11}31^{\overline{1}}} 4^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 3 & 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[709] \ 21^{\overline{11}41^{\overline{11}31^{\overline{1}}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 3 & 2 \\ 2 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -3 & 1 & -1 \\ 3 & 2 & 6 & 1 & 5 & 5 \\ 2 & 2 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[708] \ 1^{\overline{11}31^{\overline{1}}} 4^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 4 & 1 & 5 \\ 4 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 5 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[715] \ 21^{\overline{11}41^{\overline{11}31^{\overline{11}}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 1 & 13 & 4 \\ 2 & 1 & -2 & 2 & 0 \\ 10 & 13 & 2 & 36 & 12 \\ 3 & 4 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[716] \ 1^{\overline{13}2321^{\overline{11}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 2 & 1 & 16 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 16 & 0 & 70 & 18 \\ 3 & 4 & -1 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[717] \ 1^{\overline{12}2321^{\overline{11}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 2 & 1 & 12 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 12 & 0 & 42 & 14 \\ 3 & 4 & -1 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[718] \ 2^{\overline{11}41^{\overline{1}}} 3^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 3 & 3 & 3 \\ 6 & 13 & 6 & 8 & 7 \\ 3 & 6 & 0 & 3 & 2 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 7 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[719] $1^{11}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -27$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 4 & 0 & 2 & 4 \\ 3 & 4 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

[720] $1^{11}2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 4 & 3 \\ 3 & 0 & 4 & 2 & 4 \\ 3 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[721] $1^1 4 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 3 \\ 2 & 0 & 2 & 1 & 2 \\ 3 & 2 & 0 & 2 & 2 \\ 2 & 1 & 2 & 0 & 2 \\ 3 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[722] $1^{11}2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 4 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 8 & 4 \\ 2 & 4 & 0 & 8 & 6 & 6 \\ 2 & 3 & -1 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

[723] $1^{11}2^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & -2 & 6 & 3 \end{pmatrix}_{b_0}$$

[724] $1^{11}2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 4 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 16 & 4 \\ 4 & 8 & 0 & 16 & 30 & 12 \\ 2 & 3 & -1 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[725] $1^{11}4^2 3 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 1 & 4 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 32 & 4 \\ 8 & 16 & 0 & 32 & 126 & 24 \\ 2 & 3 & -1 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[726] $1^{11}2^3 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 4 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 8 & 4 \\ 4 & 4 & 0 & 8 & 6 & 6 \\ 2 & 3 & -1 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

[727] $1^{11}3^2 3 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 4 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 24 & 4 \\ 6 & 12 & 0 & 24 & 70 & 18 \\ 2 & 3 & -1 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[728] $1^1 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -33$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 8 & 0 & 13 & 7 \\ 3 & 4 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[729] $1^{11}41^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -33$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 1 & 3 & 3 \\ 6 & 13 & 0 & 8 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 8 & 0 & 4 & 4 \\ 3 & 7 & -1 & 4 & 3 \end{pmatrix}_{b_0}$

[730] $21^{11}41^{13}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 1 & 9 & 4 \\ 2 & 1 & -2 & 2 & 0 \\ 6 & 9 & 2 & 12 & 8 \\ 3 & 4 & 0 & 8 & 3 \end{pmatrix}_{b_0}$

[731] $21^{11}41^{23}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -36$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 1 & 11 & 4 \\ 2 & 1 & -2 & 2 & 0 \\ 8 & 11 & 2 & 22 & 10 \\ 3 & 4 & 0 & 10 & 3 \end{pmatrix}_{b_0}$

[732] $1^{13}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -36$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 10 & 6 & 0 & 34 & 12 \\ 2 & 3 & 2 & -2 & 12 & 3 \end{pmatrix}_{b_0}$

[733] $1^{13}2321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -38$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 2 & 1 & 20 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 20 & 0 & 130 & 24 \\ 3 & 4 & -1 & 24 & 4 \end{pmatrix}_{b_0}$

[734] $1^{11}41^{31}2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -41$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 10 & 0 & 21 & 9 \\ 3 & 4 & -1 & 9 & 3 \end{pmatrix}_{b_0}$

[735] $1^{11}41^{31}4 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -43$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 6 & 1 & 32 & 10 \\ 3 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$

[736] $1^{11}41^{31}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -43$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 9 & 0 & 13 & 7 \\ 3 & 4 & -1 & 7 & 3 \end{pmatrix}_{b_0}$

[737] $1^{11}41^{31}2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 3 & 2 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & 0 & 4 & 2 \end{pmatrix}_{b_0}$

[738] $1^{11}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -45$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 7 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 8 & 3 \\ 7 & 0 & 8 & 22 & 10 \\ 3 & 0 & 3 & 10 & 4 \end{pmatrix}_{b_0}$

[739] $1^{12}2321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -45$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 7 & 5 & 0 & 17 & 9 \\ 2 & 3 & 2 & -2 & 9 & 3 \end{pmatrix}_{b_0}$

[740] $1^{12}4^{11}131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 5 & 8 & 1 & 11 & 7 \\ 3 & 4 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[741] $1^{11}4^{11}1^{31}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 2 & 1 & 2 & 4 \\ 3 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

[742] $1^{11}4^{11}1^{31}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 8 & 4 & 1 & 18 & 8 \\ 3 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[743] $1^{11}4^{11}1^{31}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 7 & 0 & 8 & 6 \\ 3 & 4 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[744] $1^{11}4^{11}131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 3 & 1 & 2 & 4 \\ 3 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

[745] $1^{11}4^{11}131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 2 & 2 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[746] $2^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -49, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 3 \\ 4 & 5 & 5 & 7 & 5 \\ 3 & 5 & 0 & 5 & 2 \\ 5 & 7 & 5 & 8 & 6 \\ 3 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[747] $1^{12}2^3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 16 & 3 \\ 13 & 0 & 16 & 94 & 20 \\ 3 & 0 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

[748] $1^{11}4^{11}1^{31}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 4 & 3 \\ 4 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 3 & 2 \\ 4 & 4 & 3 & 2 & 4 \\ 3 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[751] $1^{11}4^{11}1^{31}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 6 & 3 \\ 2 & 0 & 2 & 3 & 2 \\ 3 & 2 & 0 & 4 & 2 \\ 6 & 3 & 4 & 8 & 6 \\ 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[750] $2^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 3 & 6 & 3 \\ 6 & 13 & 6 & 14 & 7 \\ 3 & 6 & 0 & 7 & 2 \\ 6 & 14 & 7 & 14 & 8 \\ 3 & 7 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[749] $1^{\overline{1}1231^{\overline{1}}}4^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 3 & 1 & 8 & 6 \\ 3 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[752] $1^{\overline{1}11}4^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 3 & 3 & 0 & 2 & 4 & 4 \\ 2 & 2 & 0 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

[753] $1^{\overline{1}11}4^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 9 & 3 \\ 2 & 0 & 2 & 6 & 2 \\ 3 & 2 & 0 & 6 & 2 \\ 9 & 6 & 6 & 26 & 10 \\ 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[754] $1^{\overline{1}11}4^{\overline{1}1}131^{\overline{1}11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 0 & 2 & 3 & 2 \\ 3 & 2 & 0 & 3 & 2 \\ 3 & 3 & 3 & 2 & 4 \\ 3 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[755] $1^{\overline{1}1}4^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[756] $1^{\overline{1}3}4^{\overline{1}1}131^{\overline{1}2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 8 & 12 & 2 & 29 & 11 \\ 3 & 4 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[758] $1^{\overline{1}4}2321^{\overline{1}2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 2 & 1 & 24 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 24 & 0 & 174 & 28 \\ 3 & 4 & -1 & 28 & 4 \end{pmatrix}_{b_0}$$

[757] $1^{\overline{1}1}4^{\overline{1}1}131^{\overline{1}3} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 5 & 1 & 18 & 8 \\ 3 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[760] $1^{\overline{1}11}4^{\overline{1}1}131^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 6 & 9 & 1 & 14 & 8 \\ 3 & 4 & -1 & 8 & 3 \end{pmatrix}_{b_0}$$

[759] $1^{\overline{1}11}4^{\overline{1}1}131^{\overline{1}2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 4 & 1 & 8 & 6 \\ 3 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[763] $1^{\overline{1}1}4^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 3 & 3 & 0 & 2 & 4 & 4 \\ 2 & 2 & 0 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

- [762] $1^{11}2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 4 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 8 & 4 \\ 3 & 4 & 0 & 8 & 6 & 6 \\ 2 & 3 & -1 & 4 & 6 & 3 \end{pmatrix}_{b_0}$
- [761] $1^{11}4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 4 & 3 \\ 2 & 0 & 2 & 2 & 2 \\ 3 & 2 & 0 & 3 & 2 \\ 4 & 2 & 3 & 2 & 4 \\ 3 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$
- [764] $1^{12}2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.86 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 12 & 3 \\ 9 & 0 & 12 & 42 & 14 \\ 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$
- [765] $31^{11}31^{11}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -72$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 6 & 4 & 8 & 3 \\ 6 & 13 & 6 & 20 & 7 \\ 4 & 6 & 0 & 8 & 2 \\ 8 & 20 & 8 & 26 & 10 \\ 3 & 7 & 2 & 10 & 2 \end{pmatrix}_{b_0}$
- [768] $1^{11}4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -73$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 3 & 2 & 8 & 6 \\ 3 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$
- [767] $1^{14}1^{11}231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -73$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 1 & 6 & 3 \\ 6 & 13 & 0 & 12 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 6 & 12 & 1 & 9 & 7 \\ 3 & 7 & -1 & 7 & 3 \end{pmatrix}_{b_0}$
- [766] $1^{13}4^{11}131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -73$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 7 & 12 & 1 & 26 & 10 \\ 3 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$
- [769] $1^{12}2^3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 2 & 1 & 12 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 12 & 0 & 42 & 14 \\ 3 & 4 & -1 & 14 & 4 \end{pmatrix}_{b_0}$
- [771] $1^{11}4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -78$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 3 & 8 & 3 \\ 4 & 4 & 3 & 9 & 4 \\ 3 & 3 & 0 & 7 & 2 \\ 8 & 9 & 7 & 17 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [770] $1^{11}4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -78$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 4 & 3 & 9 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 6 & 9 & 5 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$
- [772] $31^{11}31^{11}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -79$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.06 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 3 & 3 & 1 & 3 & 2 \\ 3 & 4 & 4 & 2 & 3 & 4 \\ 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & -2 & 1 & 0 \\ 3 & 3 & 3 & 1 & 1 & 3 \\ 2 & 4 & 2 & 0 & 3 & 1 \end{pmatrix}_{b_0}$
- [773] $1^{12}4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -79$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 14 & 17 & 1 & 62 & 16 \\ 3 & 4 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[774] \ 1^{12} 4^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 3 & 4 & 1 & 2 & 7 & 5 \\ 2 & 2 & 0 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[775] \ 1^{11} 2^3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 8 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 8 & 3 \\ 8 & 0 & 8 & 22 & 10 \\ 3 & 0 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[776] \ 1^{12} 2321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 8 & 3 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 8 & 0 & 30 & 12 \\ 2 & 3 & 2 & -2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[777] \ 1^{12} 4^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 11 & 0 & 21 & 9 \\ 3 & 4 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[778] \ 1^{12} 4^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 4 & 5 & 0 & 3 & 12 & 6 \\ 2 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[779] \ 1^{11} 4^{11} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 4 & 0 & 3 & 12 & 6 \\ 2 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[780] \ 1^{13} 2^3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 15 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 20 & 3 \\ 15 & 0 & 20 & 130 & 24 \\ 3 & 0 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[781] \ 1^{12} 4^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 8 & 11 & 1 & 22 & 10 \\ 3 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[783] \ 1^{13} 31^{11} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 6 & 3 & 3 & 2 & 13 & 7 \\ 2 & 2 & 0 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[784] \ 1^{13} 31^{11} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 4 & 0 & 7 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 7 & 0 & 8 & 6 \\ 3 & 4 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[782] $1^{\overline{11}31^{\overline{11}}}4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 6 & 3 \\ 4 & 4 & 0 & 7 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 6 & 7 & 1 & 9 & 7 \\ 3 & 4 & -1 & 7 & 3 \end{pmatrix}_{b_0}$

[785] $1^{\overline{11}31^{\overline{11}}}4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 9 & 12 & 1 & 27 & 11 \\ 3 & 4 & -1 & 11 & 3 \end{pmatrix}_{b_0}$

[786] $1^{\overline{11}31^{\overline{2}}}4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 12 & 15 & 1 & 46 & 14 \\ 3 & 4 & -1 & 14 & 3 \end{pmatrix}_{b_0}$

[787] $1^{\overline{12}31^{\overline{2}}}4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 11 & 14 & 1 & 39 & 13 \\ 3 & 4 & -1 & 13 & 3 \end{pmatrix}_{b_0}$

[788] $1^{\overline{11}31^{\overline{1}}}4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 12 & 0 & 26 & 10 \\ 3 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$

[789] $1^{\overline{12}4131^{\overline{12}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 9 & 13 & 2 & 32 & 12 \\ 3 & 4 & -1 & 12 & 3 \end{pmatrix}_{b_0}$

[790] $1^{\overline{11}31^{\overline{2}}}4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 7 & 3 \\ 2 & 0 & 2 & 5 & 2 \\ 3 & 2 & 0 & 5 & 2 \\ 7 & 5 & 5 & 14 & 8 \\ 3 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[791] $2^{\overline{14}1^{\overline{1}}}3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 2 & 1 & 2 & 2 \\ 3 & 4 & 4 & 2 & 3 & 4 \\ 2 & 4 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & -2 & 1 & 0 \\ 2 & 3 & 3 & 1 & 1 & 3 \\ 2 & 4 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$

[792] $1^{\overline{11}31^{\overline{1}}}4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 5 & 3 \\ 2 & 0 & 2 & 4 & 2 \\ 3 & 2 & 0 & 4 & 2 \\ 5 & 4 & 4 & 6 & 6 \\ 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[793] $1^{\overline{11}31^{\overline{1}}}4^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 5 & 4 & 2 & 6 & 6 \\ 3 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$

[794] $1^{\overline{13}2321^{\overline{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 2 & 1 & 16 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 16 & 0 & 70 & 18 \\ 3 & 4 & -1 & 18 & 4 \end{pmatrix}_{b_0}$

[797] $1^{13}4^{11}131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -104$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 4 & 6 & 1 & 3 & 17 & 7 \\ 2 & 2 & 0 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[796] $1^{11}4^{13}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -104$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 6 & 3 & 3 & 2 & 13 & 7 \\ 2 & 2 & 0 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[795] $1^{11}4^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -104$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 4 & 3 & 1 & 2 & 5 & 5 \\ 2 & 2 & 0 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[798] $31^{11}31^{11}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -109$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.30 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 6 & 4 & 5 & 3 \\ 6 & 13 & 6 & 12 & 7 \\ 4 & 6 & 0 & 5 & 2 \\ 5 & 12 & 5 & 8 & 6 \\ 3 & 7 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[799] $21^{11}41^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -114$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 4 & 2 \\ 2 & 0 & 3 & 0 & 4 & 2 \\ 2 & 3 & 3 & 0 & 8 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 4 & 8 & 0 & 10 & 6 \\ 2 & 2 & 3 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[800] $1^{14}2^3 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 2 & 1 & 20 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 20 & 0 & 106 & 22 \\ 3 & 4 & -1 & 22 & 4 \end{pmatrix}_{b_0}$$

[801] $1^{13}2^3 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 2 & 1 & 20 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 20 & 0 & 130 & 24 \\ 3 & 4 & -1 & 24 & 4 \end{pmatrix}_{b_0}$$

[802] $1^{13}2^3 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -117$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 11 & 9 & 0 & 49 & 15 \\ 2 & 3 & 2 & -2 & 15 & 3 \end{pmatrix}_{b_0}$$

[803] $1^{12}4^{11}131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -118$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 8 & 13 & 1 & 29 & 11 \\ 3 & 4 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[804] $1^{14}4^{11}131^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -118$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 10 & 16 & 2 & 50 & 14 \\ 3 & 4 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

[805] $1^{13}4^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -118$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 15 & 0 & 40 & 12 \\ 3 & 4 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

[806] $1^{\overline{1}^{12}31^2} 4^{\overline{1}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 4 & 2 & 14 & 8 \\ 3 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[807] $1^{\overline{1}^{12}31^2} 4^{\overline{1}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 14 & 0 & 38 & 12 \\ 3 & 4 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

[808] $1^{\overline{1}^{11}31^3} 4^{\overline{1}^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 18 & 0 & 70 & 16 \\ 3 & 4 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[809] $1^{\overline{1}^{12}31^1} 4^{\overline{1}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 11 & 15 & 2 & 44 & 14 \\ 3 & 4 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

[810] $1^{\overline{1}^{12}4131^2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 4 & 4 & 2 & 2 & 10 & 6 \\ 2 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[811] $1^{\overline{1}^{12}31^1} 4^{\overline{1}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 5 & 3 & 2 & 2 & 8 & 6 \\ 2 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[812] $1^{\overline{1}^{12}31^3} 4^{\overline{1}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 17 & 0 & 61 & 15 \\ 3 & 4 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[813] $1^{\overline{1}^{11}31^2} 4^{\overline{1}^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 15 & 0 & 45 & 13 \\ 3 & 4 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

[814] $1^{\overline{1}^{11}4131^3} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 12 & 17 & 3 & 58 & 16 \\ 3 & 4 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[815] $21^{\overline{1}^{11}41^{11}31^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 5 & 2 \\ 2 & 0 & 3 & 0 & 4 & 2 \\ 2 & 3 & 3 & 0 & 8 & 3 \\ 1 & 0 & 0 & -3 & 1 & -1 \\ 5 & 4 & 8 & 1 & 11 & 7 \\ 2 & 2 & 3 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[816] $1^{\overline{1}^{11}31^{11}} 4^{\overline{1}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 7 & 3 \\ 4 & 4 & 3 & 9 & 4 \\ 3 & 3 & 0 & 6 & 2 \\ 7 & 9 & 6 & 14 & 8 \\ 3 & 4 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[819] \ 1^{11} 4 \ 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 1 & 8 & 3 \\ 4 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 12 & 0 & 26 & 10 \\ 3 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[820] \ 1^{11} 4 \ 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 1 & 10 & 3 \\ 4 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 10 & 12 & 2 & 30 & 12 \\ 3 & 4 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[817] \ 1^{11} 2 \ 3 \ 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 4 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 16 & 4 \\ 5 & 8 & 0 & 16 & 30 & 12 \\ 2 & 3 & -1 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[818] \ 1^{11} 4 \ 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 5 & 4 & 3 & 2 & 15 & 7 \\ 2 & 2 & 0 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[821] \ 1^{14} 2 \ 3 \ 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 14 & 3 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 14 & 10 & 0 & 74 & 18 \\ 2 & 3 & 2 & -2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[823] \ 1^{12} 2 \ 3 \ 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 14 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 16 & 3 \\ 14 & 0 & 16 & 94 & 20 \\ 3 & 0 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[822] \ 1^{12} 2 \ 3 \ 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 12 & 3 \\ 10 & 0 & 12 & 42 & 14 \\ 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[824] \ 1^{14} 4 \ 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -138, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 4 & -1 \\ 15 & 21 & 4 & 92 & 20 \\ 3 & 4 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[825] \ 1^{14} 4 \ 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 14 & 19 & 3 & 74 & 18 \\ 3 & 4 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[826] \ 1^{13} 4 \ 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -142, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 17 & 21 & 2 & 92 & 20 \\ 3 & 4 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[829] $1^{13} 4^{12} 31^2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -145$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 14 & 18 & 2 & 65 & 17 \\ 3 & 4 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[828] $1^{12} 4^{11} 31^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -145$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 12 & 16 & 2 & 49 & 15 \\ 3 & 4 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[827] $1^{13} 4^{11} 31^2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -145$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 5 & 2 & 24 & 10 \\ 3 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[830] $1^{14} 2^3 21^{13} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -147$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 21 & 3 \\ 2 & 2 & 1 & 28 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 28 & 0 & 266 & 34 \\ 3 & 4 & -1 & 34 & 4 \end{pmatrix}_{b_0}$$

[831] $1^{12} 4^{11} 31^3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -149$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 18 & 22 & 2 & 101 & 21 \\ 3 & 4 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[832] $1^{11} 4^{11} 31^2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -150$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 7 & 5 & 3 & 14 & 8 \\ 3 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[833] $1^{14} 1^{11} 231^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -150$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 3 & 2 \\ 3 & 4 & 1 & 2 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 2 & 0 \\ 3 & 6 & 0 & 2 & 4 & 4 \\ 2 & 4 & 0 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

[836] $1^{12} 4^{11} 31^2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -152$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 6 & 6 & 0 & 4 & 22 & 8 \\ 2 & 2 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

[835] $1^{11} 4^{11} 31^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -152$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 4 & 3 & 1 & 2 & 5 & 5 \\ 2 & 2 & 0 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[834] $1^{11} 4^{11} 31^2 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -152$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 6 & 4 & 1 & 3 & 13 & 7 \\ 2 & 2 & 0 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[839] $1^{13} 4^{11} 31^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -154$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 10 & 15 & 1 & 41 & 13 \\ 3 & 4 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

[837] $1^{11} 4^{1231^1} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 7 & 5 & 2 & 12 & 8 \\ 3 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[838] $1^1 4^{1431^2} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 6 & 2 & 38 & 12 \\ 3 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[840] $1^{13} 2^1 3^{12} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 20 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 24 & 3 \\ 20 & 0 & 24 & 214 & 30 \\ 3 & 0 & 3 & 30 & 4 \end{pmatrix}_{b_0}$$

[841] $1^{12} 3^{11} 4^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 5 & 3 & 2 & 2 & 8 & 6 \\ 2 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[842] $1^{13} 2^1 3^{12} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 12 & 3 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 12 & 0 & 70 & 18 \\ 2 & 3 & 2 & -2 & 18 & 3 \end{pmatrix}_{b_0}$$

[844] $1^{12} 2^1 3^{12} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 4 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 16 & 4 \\ 6 & 8 & 0 & 16 & 30 & 12 \\ 2 & 3 & -1 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[843] $1^{13} 2^1 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 12 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 16 & 3 \\ 12 & 0 & 16 & 70 & 18 \\ 3 & 0 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[845] $1^{12} 3^{11} 4^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 16 & 0 & 45 & 13 \\ 3 & 4 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

[846] $1^{14} 1^{11} 2^{31} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 5 & 3 \\ 6 & 13 & 0 & 12 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 12 & 0 & 8 & 6 \\ 3 & 7 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[847] $1^{12} 3^{12} 4^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 15 & 19 & 2 & 72 & 18 \\ 3 & 4 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[849] $1^{13} 2^1 3^{12} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 2 & 1 & 16 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 16 & 0 & 70 & 18 \\ 3 & 4 & -1 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[848] \ 31^1 31^{11} 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 3 & 2 \\ 3 & 4 & 4 & 2 & 6 & 4 \\ 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & -2 & 2 & 0 \\ 3 & 6 & 3 & 2 & 4 & 4 \\ 2 & 4 & 2 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[850] \ 1^{14} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 2 & 1 & 24 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 24 & 0 & 174 & 28 \\ 3 & 4 & -1 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[851] \ 1^{13} 4^{11} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 10 & 17 & 1 & 50 & 14 \\ 3 & 4 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[852] \ 1^{12} 31^3 4^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 10 & 5 & 3 & 24 & 10 \\ 3 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[853] \ 1^{12} 1^{11} 31^1 4^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 5 & 1 & 3 & 13 & 7 \\ 2 & 2 & 0 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[854] \ 2^1 3^{14} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 5 & 2 \\ 2 & 1 & 4 & 1 & 6 & 3 \\ 2 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -2 & 3 & 0 \\ 5 & 6 & 6 & 3 & 13 & 7 \\ 2 & 3 & 2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[855] \ 2^1 3^{14} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 \\ 2 & 1 & 4 & 1 & 3 & 3 \\ 2 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 2 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[856] \ 1^{14} 4^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 17 & 22 & 3 & 99 & 21 \\ 3 & 4 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[857] \ 1^{13} 4^{11} 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -183, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 11 & 17 & 2 & 53 & 15 \\ 3 & 4 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[858] \ 1^{13} 4^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 15 & 20 & 3 & 79 & 19 \\ 3 & 4 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

[860] $1^{13} 4^{11} 13^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -188$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 6 & 2 & 3 & 20 & 8 \\ 2 & 2 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

[859] $1^{11} 4^{12} 31^{21} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -188$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 7 & 4 & 2 & 3 & 16 & 8 \\ 2 & 2 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

[862] $1^{11} 4^{13} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -189$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.71 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 9 & 6 & 2 & 22 & 10 \\ 3 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[861] $1^{12} 4^{11} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -189$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.71 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 11 & 16 & 1 & 46 & 14 \\ 3 & 4 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

[863] $1^{12} 2^3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -189$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 11 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 12 & 3 \\ 11 & 0 & 12 & 42 & 14 \\ 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[866] $1^{11} 4^{11} 231^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 9 & 3 \\ 6 & 13 & 0 & 20 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 9 & 20 & 1 & 27 & 11 \\ 3 & 7 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[867] $1^{11} 4^{13} 31^{21} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 8 & 4 & 3 & 3 & 21 & 9 \\ 2 & 2 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[868] $1^{11} 4^{11} 31^{31} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 8 & 5 & 1 & 4 & 25 & 9 \\ 2 & 2 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[869] $1^{13} 2^3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 1 & 4 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 24 & 4 \\ 7 & 12 & 0 & 24 & 70 & 18 \\ 2 & 3 & -1 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[865] $1^{12} 2^3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 16 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 20 & 3 \\ 16 & 0 & 20 & 130 & 24 \\ 3 & 0 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[864] $1^{11} 4^{11} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 1 & 9 & 3 \\ 4 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 9 & 12 & 1 & 27 & 11 \\ 3 & 4 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[870] $1^{13} 4^1 1^{131^2} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -193, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 18 & 0 & 61 & 15 \\ 3 & 4 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[872] $1^{14} 4^1 1^{1331^3} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 20 & 25 & 3 & 130 & 24 \\ 3 & 4 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[871] $1^{141} 1^{11} 231^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 4 & 2 \\ 3 & 4 & 1 & 2 & 6 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & -2 & 2 & 0 \\ 4 & 6 & 1 & 2 & 5 & 5 \\ 2 & 4 & 0 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[873] $1^{141} 1^{12} 231^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 9 & 2 \\ 3 & 4 & 1 & 2 & 12 & 4 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 2 & 0 & -2 & 4 & 0 \\ 9 & 12 & 3 & 4 & 31 & 11 \\ 2 & 4 & 0 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[874] $1^{12} 2^3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 15 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 16 & 3 \\ 15 & 0 & 16 & 94 & 20 \\ 3 & 0 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

[875] $1^{13} 4^1 1^{1131^2} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 13 & 18 & 1 & 62 & 16 \\ 3 & 4 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[876] $1^{11} 4^1 1^{1431^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 11 & 7 & 2 & 36 & 12 \\ 3 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[877] $1^{142} 3^1 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 2 & 1 & 20 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 20 & 0 & 106 & 22 \\ 3 & 4 & -1 & 22 & 4 \end{pmatrix}_{b_0}$$

[878] $1^{11} 41^{11} 231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 8 & 3 \\ 6 & 13 & 0 & 20 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 20 & 0 & 26 & 10 \\ 3 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[880] $1^{13} 4^1 1^{131^2} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 7 & 8 & 0 & 5 & 36 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[879] $1^{12} 4^1 1^{131^3} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 8 & 7 & 0 & 5 & 36 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[881] $1^{14} 1^{12} 31^1 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 13 & 19 & 2 & 69 & 17 \\ 3 & 4 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[882] $1^{14} 1^{14} 31^1 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -217, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 4 & -1 \\ 18 & 24 & 4 & 117 & 23 \\ 3 & 4 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

[883] $1^{13} 1^{11} 31^4 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -217, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 19 & 24 & 1 & 122 & 22 \\ 3 & 4 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[884] $1^{14} 1^{11} 31^1 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 12 & 19 & 1 & 66 & 16 \\ 3 & 4 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[885] $1^{13} 1^{13} 31^2 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 18 & 23 & 3 & 106 & 22 \\ 3 & 4 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[886] $1^{14} 2321^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 15 & 3 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 15 & 13 & 0 & 97 & 21 \\ 2 & 3 & 2 & -2 & 21 & 3 \end{pmatrix}_{b_0}$$

[888] $1^{12} 1^{11} 31^4 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 20 & 25 & 1 & 133 & 23 \\ 3 & 4 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

[887] $1^{13} 1^{13} 31^3 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 21 & 26 & 3 & 139 & 25 \\ 3 & 4 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

[889] $1^{13} 1^{11} 31^1 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 20 & 0 & 70 & 16 \\ 3 & 4 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[890] $1^{14} 4 \ 131^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 12 & 21 & 1 & 77 & 17 \\ 3 & 4 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[891] $1^{13} 1^{11} 31^3 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 16 & 21 & 1 & 89 & 19 \\ 3 & 4 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[892] \ 1^{12} 4 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 19 & 0 & 68 & 16 \\ 3 & 4 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[893] \ 1^{14} 2 \ 3 \ 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 22 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 28 & 3 \\ 22 & 0 & 28 & 266 & 34 \\ 3 & 0 & 3 & 34 & 4 \end{pmatrix}_{b_0}$$

$$[894] \ 1^{14} 2 \ 3 \ 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 18 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 24 & 3 \\ 18 & 0 & 24 & 174 & 28 \\ 3 & 0 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[895] \ 1^{12} 4 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -230, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 14 & 19 & 1 & 69 & 17 \\ 3 & 4 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[896] \ 1^{12} 4 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 6 & 5 & 2 & 3 & 16 & 8 \\ 2 & 2 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[897] \ 1^{13} 4 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 12 & 6 & 3 & 34 & 12 \\ 3 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[898] \ 1^{14} 4 \ 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 14 & 21 & 3 & 85 & 19 \\ 3 & 4 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[899] \ 1^{13} 4 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 21 & 0 & 88 & 18 \\ 3 & 4 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[900] \ 1^{11} 4 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -239, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 9 & 6 & 3 & 20 & 10 \\ 3 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[901] \ 1^{13} 4 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 6 & 7 & 1 & 4 & 25 & 9 \\ 2 & 2 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

- [902] $1^{13}4^{11}13^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -240$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 6 & 6 & 3 & 3 & 25 & 9 \\ 2 & 2 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$
- [903] $1^{12}4^{11}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -241$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 17 & 22 & 1 & 98 & 20 \\ 3 & 4 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$
- [904] $1^{13}2^3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -245$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 16 & 3 \\ 13 & 0 & 16 & 70 & 18 \\ 3 & 0 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$
- [905] $1^{11}1^{12}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -248$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 9 & 5 & 2 & 4 & 28 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$
- [906] $1^{14}4^{11}13^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -248$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 6 & 8 & 2 & 4 & 34 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$
- [907] $1^{13}4^{11}31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -253$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 24 & 0 & 121 & 21 \\ 3 & 4 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$
- [908] $1^{11}4^2 3 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 1 & 4 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 32 & 4 \\ 9 & 16 & 0 & 32 & 126 & 24 \\ 2 & 3 & -1 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$
- [909] $1^{14}2^3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -258$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 2 & 1 & 24 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 24 & 0 & 174 & 28 \\ 3 & 4 & -1 & 28 & 4 \end{pmatrix}_{b_0}$$
- [911] $1^{13}4^{11}13^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -260$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 7 & 6 & 4 & 3 & 32 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$
- [910] $1^{12}4^{11}31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -260$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 8 & 5 & 4 & 3 & 28 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$
- [912] $1^{12}4^{11}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -261$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 22 & 0 & 97 & 19 \\ 3 & 4 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

[914] $1^{12} 4^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -264$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 7 & 6 & 1 & 4 & 23 & 9 \\ 2 & 2 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[913] $1^{12} 4^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -264$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 7 & 5 & 3 & 3 & 21 & 9 \\ 2 & 2 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[917] $1^{12} 4^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -266$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 25 & 0 & 132 & 22 \\ 3 & 4 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[918] $1^{14} 4^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -266$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 4 & -1 \\ 21 & 27 & 4 & 148 & 26 \\ 3 & 4 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

[915] $1^{14} 4^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -266$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 22 & 0 & 90 & 18 \\ 3 & 4 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[916] $1^{14} 4^{11} 13^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -266$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 13 & 21 & 2 & 80 & 18 \\ 3 & 4 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[919] $1^{14} 12^{23} 1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -270$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 6 & 2 \\ 3 & 4 & 1 & 2 & 12 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 4 & 0 \\ 6 & 12 & 0 & 4 & 22 & 8 \\ 2 & 4 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

[920] $1^{13} 4^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -270$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 14 & 20 & 2 & 74 & 18 \\ 3 & 4 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[921] $1^{11} 4^{13} 3^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -272$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 10 & 5 & 3 & 4 & 33 & 11 \\ 2 & 2 & 0 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[922] $1^{12} 3^{14} 2^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -272$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 16 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 16 & 3 \\ 16 & 0 & 16 & 94 & 20 \\ 3 & 0 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

[923] $1^{14} 2^{32} 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -272$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 16 & 3 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 16 & 0 & 126 & 24 \\ 2 & 3 & 2 & -2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[924] \ 1^{14} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 27 & 3 \\ 3 & 4 & 0 & 33 & 4 \\ 1 & 0 & -3 & 4 & -1 \\ 27 & 33 & 4 & 228 & 32 \\ 3 & 4 & -1 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[925] \ 1^{14} 41^{12} 231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 1 & 11 & 3 \\ 6 & 13 & 0 & 24 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 11 & 24 & 1 & 39 & 13 \\ 3 & 7 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[926] \ 1^{13} 2 \ 3 \ 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -277, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 21 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 24 & 3 \\ 21 & 0 & 24 & 214 & 30 \\ 3 & 0 & 3 & 30 & 4 \end{pmatrix}_{b_0}$$

$$[927] \ 1^{14} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -281, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 24 & 0 & 101 & 19 \\ 3 & 4 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[928] \ 1^{13} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -282, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 13 & 20 & 1 & 71 & 17 \\ 3 & 4 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[929] \ 1^{14} 4 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -282, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 16 & 22 & 2 & 94 & 20 \\ 3 & 4 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[930] \ 1^{14} 4 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 14 & 7 & 3 & 48 & 14 \\ 3 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[931] \ 1^{14} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -285, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 24 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -3 & 4 & -1 \\ 24 & 30 & 4 & 185 & 29 \\ 3 & 4 & -1 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[932] \ 1^{13} 2 \ 3 \ 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 1 & 4 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 24 & 4 \\ 8 & 12 & 0 & 24 & 70 & 18 \\ 2 & 3 & -1 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[933] \ 1^{13} 2 \ 3 \ 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 17 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 20 & 3 \\ 17 & 0 & 20 & 130 & 24 \\ 3 & 0 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[934] \ 1^{14} 4 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 7 & 9 & 1 & 5 & 41 & 11 \\ 2 & 2 & 0 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[935] \ 1^{11} 4^{1231^4} 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 11 & 6 & 2 & 5 & 44 & 12 \\ 2 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[936] \ 1^{41} 1^{12} 231^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 1 & 10 & 3 \\ 6 & 13 & 0 & 24 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 24 & 0 & 38 & 12 \\ 3 & 7 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[938] \ 1^{11} 4^{1331^2} 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -310, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 7 & 3 & 30 & 12 \\ 3 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[937] \ 1^{14} 4^{1131^2} 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -310, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 15 & 22 & 1 & 91 & 19 \\ 3 & 4 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[939] \ 1^{13} 4^{1131^3} 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -312, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 9 & 9 & 0 & 6 & 52 & 12 \\ 2 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[940] \ 1^{11} 4^{1231^3} 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -318, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 11 & 7 & 4 & 32 & 12 \\ 3 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[941] \ 1^{14} 4^{1231^3} 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 19 & 25 & 2 & 125 & 23 \\ 3 & 4 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[943] \ 1^{13} 2^3 21^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 14 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 16 & 3 \\ 14 & 0 & 16 & 70 & 18 \\ 3 & 0 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[942] \ 1^{13} 2^3 21^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 26 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 28 & 3 \\ 26 & 0 & 28 & 322 & 36 \\ 3 & 0 & 3 & 36 & 4 \end{pmatrix}_{b_0}$$

$$[945] \ 1^{13} 4^{131^4} 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -325, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 14 & 7 & 4 & 48 & 14 \\ 3 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

- [944] $1^{14}2^321^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -325$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 1 & 2 & 27 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 32 & 3 \\ 27 & 0 & 32 & 382 & 40 \\ 3 & 0 & 3 & 40 & 4 \end{pmatrix}_{b_0}$
- [946] $1^{14}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -326$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 22 & 28 & 2 & 162 & 26 \\ 3 & 4 & -1 & 26 & 3 \end{pmatrix}_{b_0}$
- [948] $1^{13}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -327$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 17 & 23 & 2 & 101 & 21 \\ 3 & 4 & -1 & 21 & 3 \end{pmatrix}_{b_0}$
- [947] $1^{13}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -327$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 23 & 0 & 97 & 19 \\ 3 & 4 & -1 & 19 & 3 \end{pmatrix}_{b_0}$
- [949] $1^{14}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -341$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 17 & 24 & 3 & 110 & 22 \\ 3 & 4 & -1 & 22 & 3 \end{pmatrix}_{b_0}$
- [950] $1^{12}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 8 & 6 & 2 & 4 & 26 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$
- [953] $1^{13}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 7 & 7 & 2 & 4 & 28 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$
- [954] $1^{14}4^113^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 7 & 8 & 3 & 4 & 39 & 11 \\ 2 & 2 & 0 & 0 & 11 & 2 \end{pmatrix}_{b_0}$
- [951] $1^{12}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 9 & 7 & 1 & 5 & 37 & 11 \\ 2 & 2 & 0 & 0 & 11 & 2 \end{pmatrix}_{b_0}$
- [952] $1^{11}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.99 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 12 & 6 & 3 & 5 & 49 & 13 \\ 2 & 2 & 0 & 0 & 13 & 2 \end{pmatrix}_{b_0}$
- [955] $1^{13}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -351$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 23 & 29 & 2 & 173 & 27 \\ 3 & 4 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

[956] $1^{14} 4^{11} 31^3 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -351$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 25 & 0 & 121 & 21 \\ 3 & 4 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[957] $1^{13} 4^{11} 31^3 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -354$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 20 & 26 & 2 & 134 & 24 \\ 3 & 4 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[958] $1^{14} 2^3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -357$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 19 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 24 & 3 \\ 19 & 0 & 24 & 174 & 28 \\ 3 & 0 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

[959] $1^{13} 4^{11} 31^2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -363$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 16 & 23 & 1 & 98 & 20 \\ 3 & 4 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[960] $1^{11} 4^{14} 31^2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -363$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 8 & 3 & 44 & 14 \\ 3 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[961] $1^{13} 4^{11} 31^2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -368$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 8 & 8 & 1 & 5 & 37 & 11 \\ 2 & 2 & 0 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[962] $1^{14} 4^{11} 31^3 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -371$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 18 & 25 & 1 & 122 & 22 \\ 3 & 4 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[963] $1^{14} 2^3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -373$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 23 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 28 & 3 \\ 23 & 0 & 28 & 266 & 34 \\ 3 & 0 & 3 & 34 & 4 \end{pmatrix}_{b_0}$$

[964] $1^{14} 4^{11} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -373$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 15 & 24 & 1 & 102 & 20 \\ 3 & 4 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[965] $1^{13} 2^3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -388$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 22 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 24 & 3 \\ 22 & 0 & 24 & 214 & 30 \\ 3 & 0 & 3 & 30 & 4 \end{pmatrix}_{b_0}$$

[966] $1^{13} 2^3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -388$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 18 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 20 & 3 \\ 18 & 0 & 20 & 130 & 24 \\ 3 & 0 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[967] $1^{12} 4^{13} 31^2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 9 & 6 & 3 & 4 & 31 & 11 \\ 2 & 2 & 0 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[968] $1^{14} 4^{12} 31^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -393, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 16 & 24 & 2 & 105 & 21 \\ 3 & 4 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[969] $1^{13} 4^{11} 31^3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -402, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 26 & 0 & 130 & 22 \\ 3 & 4 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[970] $1^{14} 4^{11} 31^4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -402, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 21 & 28 & 1 & 159 & 25 \\ 3 & 4 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

[971] $1^{13} 4^{11} 31^4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 11 & 10 & 0 & 7 & 72 & 14 \\ 2 & 2 & 0 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

[972] $1^{14} 4^{11} 31^3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 10 & 11 & 0 & 7 & 72 & 14 \\ 2 & 2 & 0 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

[974] $1^{14} 4^{11} 31^4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -406, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 16 & 8 & 4 & 62 & 16 \\ 3 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[973] $1^{14} 4^{11} 31^4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -406, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 28 & 0 & 158 & 24 \\ 3 & 4 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[976] $1^{12} 4^{14} 31^2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -408, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 10 & 6 & 4 & 4 & 38 & 12 \\ 2 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[975] $1^{14} 4^{13} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -408, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 8 & 8 & 4 & 4 & 46 & 12 \\ 2 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

- [977] $1^{14} 4^{1^{13}31^2} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -414$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 20 & 27 & 3 & 141 & 25 \\ 3 & 4 & -1 & 25 & 3 \end{pmatrix}_{b_0}$
- [978] $1^{13} 4^{1^{11}31^3} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -414$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 19 & 26 & 1 & 131 & 23 \\ 3 & 4 & -1 & 23 & 3 \end{pmatrix}_{b_0}$
- [979] $1^{13} 4^{1^{13}31^1} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 8 & 7 & 3 & 4 & 33 & 11 \\ 2 & 2 & 0 & 0 & 11 & 2 \end{pmatrix}_{b_0}$
- [980] $1^{12} 4^{1^{11}31^4} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 11 & 8 & 1 & 6 & 55 & 13 \\ 2 & 2 & 0 & 0 & 13 & 2 \end{pmatrix}_{b_0}$
- [981] $1^{114} 2^3 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 1 & 4 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 32 & 4 \\ 10 & 16 & 0 & 32 & 126 & 24 \\ 2 & 3 & -1 & 4 & 24 & 3 \end{pmatrix}_{b_0}$
- [982] $1^{11} 4^{1^{13}31^3} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -425$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 13 & 8 & 4 & 42 & 14 \\ 3 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$
- [983] $1^{14} 4^{1^{13}31^2} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -426$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 27 & 0 & 132 & 22 \\ 3 & 4 & -1 & 22 & 3 \end{pmatrix}_{b_0}$
- [984] $1^{13} 4^{1^{11}31^4} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -435$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 22 & 29 & 1 & 170 & 26 \\ 3 & 4 & -1 & 26 & 3 \end{pmatrix}_{b_0}$
- [985] $31^{1131} 11^{41} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -439$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 3 & 3 & 1 & 6 & 2 \\ 3 & 4 & 4 & 2 & 9 & 4 \\ 3 & 4 & 0 & 1 & 6 & 2 \\ 1 & 2 & 1 & -2 & 3 & 0 \\ 6 & 9 & 6 & 3 & 13 & 7 \\ 2 & 4 & 2 & 0 & 7 & 1 \end{pmatrix}_{b_0}$
- [986] $1^{12} 4^{1^{12}31^3} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -444$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 10 & 7 & 2 & 5 & 40 & 12 \\ 2 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$
- [987] $1^{14} 4^{1^{12}31^1} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -444$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 8 & 9 & 2 & 5 & 44 & 12 \\ 2 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[988] $1^{13} 4^{131^4} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -447$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 29 & 0 & 169 & 25 \\ 3 & 4 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

[989] $1^{41} 12^{231^{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -450$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 8 & 2 \\ 3 & 4 & 1 & 2 & 12 & 4 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & -2 & 4 & 0 \\ 8 & 12 & 2 & 4 & 26 & 10 \\ 2 & 4 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[990] $1^{13} 4^{1431^1} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -456$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 9 & 7 & 4 & 4 & 40 & 12 \\ 2 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[991] $1^{14} 4^{1331^3} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -457$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 23 & 30 & 3 & 178 & 28 \\ 3 & 4 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[992] $1^{14} 4^{1131^2} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -464$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 9 & 10 & 1 & 6 & 55 & 13 \\ 2 & 2 & 0 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

[993] $1^{41} 12^{231^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -470$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 9 & 2 \\ 3 & 4 & 1 & 2 & 18 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 6 & 0 \\ 9 & 18 & 0 & 6 & 52 & 12 \\ 2 & 4 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[994] $1^{14} 4^{1331^4} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -470$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 26 & 3 \\ 3 & 4 & 0 & 33 & 4 \\ 1 & 0 & -3 & 3 & -1 \\ 26 & 33 & 3 & 221 & 31 \\ 3 & 4 & -1 & 31 & 3 \end{pmatrix}_{b_0}$$

[995] $1^{41} 12^{231^{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -475$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 7 & 2 \\ 3 & 4 & 1 & 2 & 12 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & -2 & 4 & 0 \\ 7 & 12 & 1 & 4 & 23 & 9 \\ 2 & 4 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[997] $1^{14} 2^3 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -480$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 20 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 24 & 3 \\ 20 & 0 & 24 & 174 & 28 \\ 3 & 0 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

[996] $1^{14} 2^3 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -480$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 28 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 32 & 3 \\ 28 & 0 & 32 & 382 & 40 \\ 3 & 0 & 3 & 40 & 4 \end{pmatrix}_{b_0}$$

[998] $1^{11} 41^{12} 231^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -481$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 13 & 3 \\ 6 & 13 & 0 & 32 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 32 & 0 & 68 & 16 \\ 3 & 7 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[999] $1^{13} 4^{11} 31^3 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -488$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 10 & 9 & 1 & 6 & 53 & 13 \\ 2 & 2 & 0 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

[1000] $1^{14} 4^{11} 31^2 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -490$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 19 & 27 & 2 & 136 & 24 \\ 3 & 4 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[1001] $1^{13} 4^{11} 31^2 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -492$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 9 & 8 & 2 & 5 & 40 & 12 \\ 2 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[1002] $1^{14} 4^{11} 31^2 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -494$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 18 & 27 & 1 & 133 & 23 \\ 3 & 4 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

[1004] $1^{14} 2^3 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -512$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 1 & 2 & 24 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 28 & 3 \\ 24 & 0 & 28 & 266 & 34 \\ 3 & 0 & 3 & 34 & 4 \end{pmatrix}_{b_0}$$

[1003] $1^{12} 4^{11} 31^3 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -512$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 11 & 7 & 3 & 5 & 45 & 13 \\ 2 & 2 & 0 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

[1005] $1^{11} 4^{14} 31^3 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -514$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 15 & 9 & 4 & 56 & 16 \\ 3 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1006] $1^{11} 41^{12} 231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -521$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 14 & 3 \\ 6 & 13 & 0 & 32 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 14 & 32 & 1 & 69 & 17 \\ 3 & 7 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[1007] $1^{11} 4^{13} 31^4 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -534$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 5 & 0 \\ 15 & 9 & 5 & 58 & 16 \\ 3 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1008] $1^{12} 4^{11} 31^4 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -536$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 12 & 8 & 2 & 6 & 58 & 14 \\ 2 & 2 & 0 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

[1009] $1^{14} 4^{11} 31^4 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -536$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 12 & 12 & 0 & 8 & 94 & 16 \\ 2 & 2 & 0 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[1010] \ 1^{14} 4 \ 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -541, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 30 & 0 & 169 & 25 \\ 3 & 4 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[1011] \ 1^{12} 4 \ 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -548, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 12 & 7 & 4 & 5 & 52 & 14 \\ 2 & 2 & 0 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[1012] \ 1^{14} 1^{13} 231^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -550, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 13 & 2 \\ 3 & 4 & 1 & 2 & 18 & 4 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 2 & 0 & -2 & 6 & 0 \\ 13 & 18 & 4 & 6 & 68 & 16 \\ 2 & 4 & 0 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[1013] \ 1^{14} 1^{13} 231^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -553, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 1 & 15 & 3 \\ 6 & 13 & 0 & 36 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 36 & 0 & 88 & 18 \\ 3 & 7 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1014] \ 1^{14} 4 \ 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -557, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 22 & 30 & 2 & 173 & 27 \\ 3 & 4 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1015] \ 1^{14} 4 \ 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 9 & 9 & 3 & 5 & 49 & 13 \\ 2 & 2 & 0 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[1016] \ 1^{14} 1^{13} 231^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -577, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 1 & 16 & 3 \\ 6 & 13 & 0 & 36 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 16 & 36 & 1 & 89 & 19 \\ 3 & 7 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1017] \ 1^{13} 4 \ 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 10 & 8 & 3 & 5 & 45 & 13 \\ 2 & 2 & 0 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[1018] \ 1^{14} 4 \ 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -585, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 21 & 30 & 1 & 170 & 26 \\ 3 & 4 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[1019] \ 1^{14} 4 \ 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -594, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 25 & 3 \\ 3 & 4 & 0 & 33 & 4 \\ 1 & 0 & -3 & 2 & -1 \\ 25 & 33 & 2 & 216 & 30 \\ 3 & 4 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[1020] \ 1^{13} 4 \ 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -600, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 12 & 10 & 1 & 7 & 73 & 15 \\ 2 & 2 & 0 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[1022] \ 1^{14} 4 \ 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 11 & 11 & 1 & 7 & 73 & 15 \\ 2 & 2 & 0 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[1021] \ 1^{12} 4 \ 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 13 & 8 & 3 & 6 & 63 & 15 \\ 2 & 2 & 0 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[1023] \ 1^{14} 4 \ 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 33 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 33 & 0 & 212 & 28 \\ 3 & 4 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[1025] \ 1^{13} 4 \ 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -632, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 11 & 9 & 2 & 6 & 56 & 14 \\ 2 & 2 & 0 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[1024] \ 1^{14} 4 \ 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -632, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 10 & 10 & 2 & 6 & 58 & 14 \\ 2 & 2 & 0 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[1027] \ 1^{13} 4 \ 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -644, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 11 & 8 & 4 & 5 & 52 & 14 \\ 2 & 2 & 0 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[1026] \ 1^{14} 4 \ 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -644, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 10 & 9 & 4 & 5 & 56 & 14 \\ 2 & 2 & 0 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[1028] \ 1^{14} 4 \ 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -646, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 24 & 3 \\ 3 & 4 & 0 & 33 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 24 & 33 & 1 & 213 & 29 \\ 3 & 4 & -1 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[1029] \ 1^{11} 4 \ 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -659, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 5 & 0 \\ 17 & 10 & 5 & 72 & 18 \\ 3 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1030] \ 1^{12} 4 \ 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -680, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 14 & 8 & 4 & 6 & 70 & 16 \\ 2 & 2 & 0 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

[1031] $1^{13} 4^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -744$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 12 & 9 & 3 & 6 & 61 & 15 \\ 2 & 2 & 0 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[1032] $1^{14} 1^{14} 231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -750$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 12 & 2 \\ 3 & 4 & 1 & 2 & 24 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 8 & 0 \\ 12 & 24 & 0 & 8 & 94 & 16 \\ 2 & 4 & 0 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

[1033] $1^{13} 4^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -764$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 13 & 10 & 2 & 7 & 76 & 16 \\ 2 & 2 & 0 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

[1034] $1^{14} 4^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -768$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 11 & 10 & 3 & 6 & 63 & 15 \\ 2 & 2 & 0 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[1035] $1^{14} 4^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -776$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 13 & 12 & 1 & 8 & 95 & 17 \\ 2 & 2 & 0 & 0 & 17 & 2 \end{pmatrix}_{b_0}$$

[1036] $1^{14} 4^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -812$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 12 & 11 & 2 & 7 & 76 & 16 \\ 2 & 2 & 0 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

[1037] $1^{13} 4^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -824$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 13 & 9 & 4 & 6 & 68 & 16 \\ 2 & 2 & 0 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

[1038] $1^{14} 1^{13} 231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -835$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 10 & 2 \\ 3 & 4 & 1 & 2 & 18 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & -2 & 6 & 0 \\ 10 & 18 & 1 & 6 & 53 & 13 \\ 2 & 4 & 0 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

[1039] $1^{11} 41^{13} 231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -852$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 18 & 3 \\ 6 & 13 & 0 & 44 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 44 & 0 & 130 & 22 \\ 3 & 7 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[1040] $1^{14} 4^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -872$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 6 & 0 \\ 12 & 10 & 4 & 6 & 70 & 16 \\ 2 & 2 & 0 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

[1041] $1^{141}13^{231}13 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -875$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 12 & 2 \\ 3 & 4 & 1 & 2 & 18 & 4 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 12 & 18 & 3 & 6 & 61 & 15 \\ 2 & 4 & 0 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[1043] $1^{13}1^{13}31^4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -896$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 14 & 10 & 3 & 7 & 81 & 17 \\ 2 & 2 & 0 & 0 & 17 & 2 \end{pmatrix}_{b_0}$$

[1042] $1^{141}14^{231}1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -896$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 20 & 3 \\ 6 & 13 & 0 & 48 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 48 & 0 & 158 & 24 \\ 3 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[1044] $1^{141}1^{13}231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -948$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 19 & 3 \\ 6 & 13 & 0 & 44 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 19 & 44 & 1 & 131 & 23 \\ 3 & 7 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

[1045] $1^{14}1^{13}31^3 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -968$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 13 & 11 & 3 & 7 & 81 & 17 \\ 2 & 2 & 0 & 0 & 17 & 2 \end{pmatrix}_{b_0}$$

[1046] $1^{141}1^{13}231^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -970$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 11 & 2 \\ 3 & 4 & 1 & 2 & 18 & 4 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & -2 & 6 & 0 \\ 11 & 18 & 2 & 6 & 56 & 14 \\ 2 & 4 & 0 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

[1047] $1^{141}14^{231}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -976$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 21 & 3 \\ 6 & 13 & 0 & 48 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 21 & 48 & 1 & 159 & 25 \\ 3 & 7 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

[1048] $1^{14}1^{12}31^4 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -984$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 14 & 12 & 2 & 8 & 98 & 18 \\ 2 & 2 & 0 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

[1049] $1^{13}1^{14}31^4 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -996$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 15 & 10 & 4 & 7 & 88 & 18 \\ 2 & 2 & 0 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

[1050] $1^{14}1^{14}31^3 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -1092$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 7 & 0 \\ 14 & 11 & 4 & 7 & 88 & 18 \\ 2 & 2 & 0 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

[1051] $1^{14}1^{13}31^4 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -1160$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 15 & 12 & 3 & 8 & 103 & 19 \\ 2 & 2 & 0 & 0 & 19 & 2 \end{pmatrix}_{b_0} \\
[1052] \quad & 1^{14} 41^{14} 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -1275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 1 & 1 & 13 & 2 \\ 3 & 4 & 1 & 2 & 24 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & -2 & 8 & 0 \\ 13 & 24 & 1 & 8 & 95 & 17 \\ 2 & 4 & 0 & 0 & 17 & 2 \end{pmatrix}_{b_0} \\
[1053] \quad & 1^{14} 4^{14} 1^{14} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -1304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 8 & 0 \\ 16 & 12 & 4 & 8 & 110 & 20 \\ 2 & 2 & 0 & 0 & 20 & 2 \end{pmatrix}_{b_0} \\
[1054] \quad & 1^{11} 41^{14} 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -1321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 6 & 1 & 23 & 3 \\ 6 & 13 & 0 & 56 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 56 & 0 & 212 & 28 \\ 3 & 7 & -1 & 28 & 3 \end{pmatrix}_{b_0} \\
[1055] \quad & 1^{14} 41^{14} 231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -1470, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 1 & 1 & 16 & 2 \\ 3 & 4 & 1 & 2 & 24 & 4 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 2 & 0 & -2 & 8 & 0 \\ 16 & 24 & 4 & 8 & 110 & 20 \\ 2 & 4 & 0 & 0 & 20 & 2 \end{pmatrix}_{b_0} \\
[1056] \quad & 1^{11} 41^{14} 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -1473, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 6 & 1 & 24 & 3 \\ 6 & 13 & 0 & 56 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 24 & 56 & 1 & 213 & 29 \\ 3 & 7 & -1 & 29 & 3 \end{pmatrix}_{b_0} \\
[1057] \quad & 1^{14} 41^{14} 231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -1570, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 1 & 1 & 14 & 2 \\ 3 & 4 & 1 & 2 & 24 & 4 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & -2 & 8 & 0 \\ 14 & 24 & 2 & 8 & 98 & 18 \\ 2 & 4 & 0 & 0 & 18 & 2 \end{pmatrix}_{b_0} \\
[1058] \quad & 1^{14} 41^{14} 231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -1635, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 1 & 1 & 15 & 2 \\ 3 & 4 & 1 & 2 & 24 & 4 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 2 & 0 & -2 & 8 & 0 \\ 15 & 24 & 3 & 8 & 103 & 19 \\ 2 & 4 & 0 & 0 & 19 & 2 \end{pmatrix}_{b_0} \\
T = 10 \quad [1059] \quad & 231^{11} 51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
[1060] \quad & 1^{11} 51^{11} 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 7 & 0 & 26 & 10 \\ 3 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0} \\
[1061] \quad & 1^{11} 51^{11} 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 17, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 7 & 0 & 21 & 9 \\ 3 & 3 & -1 & 9 & 3 \end{pmatrix}_{b_0} \\
[1062] \quad & 1^{11} 51^{11} 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0
\end{aligned}$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 1 & 3 \\ 3 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1063] \quad 1^{11} 2^3 1^1 5^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 3 & 0 & 6 & 2 \\ 5 & 7 & 6 & 9 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1064] \quad 1^{11} 4^1 1^1 3^1 1^1 4^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 1 & 2 \\ 3 & 13 & 6 & 0 & 5 & 7 \\ 1 & 6 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 5 & 3 & 0 & 1 & 3 \\ 2 & 7 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1065] \quad 1^{11} 4^1 1^1 3^1 1^1 4^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 8 & 1 & 7 & 3 \\ 8 & 38 & 1 & 32 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 32 & 0 & 26 & 10 \\ 3 & 12 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1066] \quad 1^{11} 5^1 1^{23} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 21 & 0 & 225 & 29 \\ 3 & 3 & -1 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[1067] \quad 1^{11} 5^1 1^{23} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 43, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 14 & 0 & 95 & 19 \\ 3 & 3 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1068] \quad 1^{12} 5^1 1^{23} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 46, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 10 & 0 & 38 & 12 \\ 3 & 3 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1069] \quad 1^{11} 5^1 1^{23} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 7 & 2 & 1 & 3 \\ 7 & 26 & 8 & 0 & 10 \\ 2 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 10 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[1070] \quad 1^{11} 4^1 1^1 3^1 1^1 4^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 49, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 4 & 2 \\ 3 & 13 & 6 & 0 & 17 & 7 \\ 1 & 6 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 17 & 7 & 0 & 21 & 9 \\ 2 & 7 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1071] \quad 2^3 1^1 5^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 58, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 3 & 3 & 0 & 8 & 2 \\ 7 & 11 & 8 & 22 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1072] \quad 1^{11} 5^1 1^{32} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 23 & 3 \\ 2 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 24 & 0 & 295 & 33 \\ 3 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[1073] \quad 1^{11} 5^1 1^{23} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 10 & 0 & 43 & 13 \\ 3 & 3 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

[1074] $2^{\overline{11}} 3^{\overline{11}} 1^{\overline{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 67, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 9 & 15 & 10 & 41 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1075] $1^{\overline{13}} 5^{\overline{11}} 1231^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 73, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 13 & 0 & 61 & 15 \\ 3 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[1076] $1^{\overline{11}} 5^{\overline{11}} 1^{\overline{2}} 231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 77, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 17 & 0 & 141 & 23 \\ 3 & 3 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

[1077] $1^{\overline{11}} 3221^{\overline{11}} 51^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 90, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 1 & 13 & 3 \\ 9 & 43 & 1 & 65 & 13 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 65 & 0 & 95 & 19 \\ 3 & 13 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

[1078] $31^{\overline{11}} 4^{\overline{11}} 141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 97, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 2 & 1 & 2 & 2 \\ 3 & 13 & 10 & 0 & 9 & 7 \\ 2 & 10 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 9 & 7 & 0 & 5 & 5 \\ 2 & 7 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[1079] $1^{\overline{12}} 5^{\overline{11}} 1^{\overline{1}} 231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.66 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 13 & 0 & 66 & 16 \\ 3 & 3 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[1080] $1^{\overline{12}} 5^{\overline{11}} 1^{\overline{1}} 231^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 17 & 0 & 126 & 22 \\ 3 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[1081] $1^{\overline{14}} 1^{\overline{11}} 31^{\overline{1}} 41^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 145, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 6 & 2 \\ 3 & 13 & 6 & 0 & 27 & 7 \\ 1 & 6 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 27 & 9 & 0 & 49 & 13 \\ 2 & 7 & 2 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[1082] $1^{\overline{14}} 1^{\overline{11}} 3141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 8 & 1 & 8 & 3 \\ 8 & 38 & 1 & 40 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 40 & 0 & 38 & 12 \\ 3 & 12 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

[1083] $31^{\overline{11}} 4^{\overline{11}} 1414^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 151, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 2 & 3 & 2 \\ 3 & 13 & 1 & 9 & 15 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 9 & 0 & 3 & 9 & 3 \\ 3 & 15 & 0 & 9 & 13 & 7 \\ 2 & 7 & -1 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[1084] $1^{\overline{14}} 1^{\overline{11}} 3141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 161, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 3 & 2 \\ 3 & 13 & 6 & 0 & 15 & 7 \\ 1 & 6 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 15 & 5 & 0 & 13 & 7 \\ 2 & 7 & 2 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1085] $1^{13}5^11^{231}1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 167, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 16 & 0 & 95 & 19 \\ 3 & 3 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

[1086] $1^{12}5^11^{231}1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 182, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 22 & 3 \\ 2 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 24 & 0 & 270 & 32 \\ 3 & 3 & -1 & 32 & 3 \end{pmatrix}_{b_0}$$

[1088] $1^{141}1^{11}31^{241}1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 186, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 8 & 1 & 22 & 3 \\ 8 & 38 & 1 & 104 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 104 & 0 & 278 & 32 \\ 3 & 12 & -1 & 32 & 3 \end{pmatrix}_{b_0}$$

[1087] $1^{12}5^11^{231}1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 186, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 20 & 0 & 178 & 26 \\ 3 & 3 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

[1089] $31^{11}4^{141}1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 190, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 2 & 4 & 2 \\ 3 & 13 & 1 & 9 & 15 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 9 & 0 & 3 & 9 & 3 \\ 4 & 15 & 0 & 9 & 13 & 7 \\ 2 & 7 & -1 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[1090] $31^{11}4^{141}1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 193, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 2 & 1 & 3 & 2 \\ 3 & 13 & 10 & 0 & 15 & 7 \\ 2 & 10 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 15 & 9 & 0 & 13 & 7 \\ 2 & 7 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1091] $1^{141}1^{11}31^{141}1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 195, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 8 & 1 & 9 & 3 \\ 8 & 38 & 1 & 40 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 40 & 0 & 38 & 12 \\ 3 & 12 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

[1094] $1^{12}5^11^{3231}1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 208, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 29 & 3 \\ 2 & 1 & 1 & 31 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 31 & 0 & 470 & 42 \\ 3 & 3 & -1 & 42 & 3 \end{pmatrix}_{b_0}$$

[1093] $1^{11}4^{11}31^{141}1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 208, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 4 & 2 \\ 3 & 13 & 6 & 0 & 20 & 7 \\ 1 & 6 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 20 & 8 & 0 & 26 & 10 \\ 2 & 7 & 2 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1092] $1^{11}4^{11}31^{241}1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 208, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 10 & 2 \\ 3 & 13 & 6 & 0 & 44 & 7 \\ 1 & 6 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 44 & 16 & 0 & 138 & 22 \\ 2 & 7 & 2 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[1095] $1^{13}5^11^{231}1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 211, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$

$$\begin{aligned}
& T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 20 & 0 & 163 & 25 \\ 3 & 3 & -1 & 25 & 3 \end{pmatrix}_{b_0} \\
[1096] & 1^1 41^{11} 31^2 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 211, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 8 & 1 & 23 & 3 \\ 8 & 38 & 1 & 104 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 104 & 0 & 278 & 32 \\ 3 & 12 & -1 & 32 & 3 \end{pmatrix}_{b_0} \\
[1097] & 1^1 5^1 1^3 231^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 12.61 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 25 & 3 \\ 2 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 27 & 0 & 346 & 36 \\ 3 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0} \\
[1098] & 1^1 41^{11} 31^1 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 239, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 8 & 1 & 15 & 3 \\ 8 & 38 & 1 & 72 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 72 & 0 & 130 & 22 \\ 3 & 12 & -1 & 22 & 3 \end{pmatrix}_{b_0} \\
[1099] & 1^1 3221^{11} 51^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 266, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 9 & 1 & 9 & 3 \\ 9 & 43 & 1 & 45 & 13 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 45 & 0 & 43 & 13 \\ 3 & 13 & -1 & 13 & 3 \end{pmatrix}_{b_0} \\
[1100] & 1^1 41^{11} 31^1 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 274, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 8 & 1 & 16 & 3 \\ 8 & 38 & 1 & 72 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 72 & 0 & 130 & 22 \\ 3 & 12 & -1 & 22 & 3 \end{pmatrix}_{b_0} \\
[1101] & 1^1 5^1 1^2 231^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 293, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 12.65 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 23 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 23 & 0 & 221 & 29 \\ 3 & 3 & -1 & 29 & 3 \end{pmatrix}_{b_0} \\
[1102] & 1^1 5^1 1^2 231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 321, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 24 & 3 \\ 2 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 27 & 0 & 321 & 35 \\ 3 & 3 & -1 & 35 & 3 \end{pmatrix}_{b_0} \\
[1103] & 1^1 41^{11} 31^1 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 336, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 3 & 1 & 1 & 7 & 2 \\ 3 & 13 & 6 & 0 & 32 & 7 \\ 1 & 6 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 32 & 12 & 0 & 70 & 16 \\ 2 & 7 & 2 & -2 & 16 & 2 \end{pmatrix}_{b_0} \\
[1104] & 31^{11} 4^1 141^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 352, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 3 & 2 & 1 & 5 & 2 \\ 3 & 13 & 10 & 0 & 24 & 7 \\ 2 & 10 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 24 & 16 & 0 & 38 & 12 \\ 2 & 7 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0} \\
[1105] & 1^1 5^1 1^3 231^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 391, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 27 & 3 \\ 2 & 1 & 1 & 30 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 30 & 0 & 403 & 39 \\ 3 & 3 & -1 & 39 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1106] $1^{13}51^3231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 403, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 31 & 3 \\ 2 & 1 & 1 & 34 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 34 & 0 & 535 & 45 \\ 3 & 3 & -1 & 45 & 3 \end{pmatrix}_{b_0}$

[1107] $1^{11}3221^{12}51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 440, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 9 & 1 & 22 & 3 \\ 9 & 43 & 1 & 110 & 13 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 110 & 0 & 270 & 32 \\ 3 & 13 & -1 & 32 & 3 \end{pmatrix}_{b_0}$

[1108] $1^{13}221^{12}51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 536, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 9 & 1 & 18 & 3 \\ 9 & 43 & 1 & 90 & 13 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 90 & 0 & 178 & 26 \\ 3 & 13 & -1 & 26 & 3 \end{pmatrix}_{b_0}$

[1109] $1^{11}3221^{13}51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 970, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 9 & 1 & 31 & 3 \\ 9 & 43 & 1 & 155 & 13 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 155 & 0 & 535 & 45 \\ 3 & 13 & -1 & 45 & 3 \end{pmatrix}_{b_0}$

[1110] $1^{13}221^{13}51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 986, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 9 & 1 & 27 & 3 \\ 9 & 43 & 1 & 135 & 13 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 135 & 0 & 403 & 39 \\ 3 & 13 & -1 & 39 & 3 \end{pmatrix}_{b_0}$

$T = 12$ [1111] $1^{11}3151321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 1 & 2 & 6 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 9 & 4 \\ 6 & 0 & 9 & 23 & 11 \\ 3 & 0 & 4 & 11 & 5 \end{pmatrix}_{b_0}$

[1112] $1^{11}3151321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 3 & 1 & 14 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 14 & 0 & 46 & 16 \\ 3 & 5 & -1 & 16 & 5 \end{pmatrix}_{b_0}$

[1113] $1^{12}3151321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 3 & 1 & 28 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 28 & 0 & 190 & 32 \\ 3 & 5 & -1 & 32 & 5 \end{pmatrix}_{b_0}$

[1114] $1^{13}151321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 5 & 4 \\ 3 & 0 & 5 & 3 & 5 \\ 3 & 0 & 4 & 5 & 5 \end{pmatrix}_{b_0}$

[1116] $1^{12}3151321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 2 & 1 & 5 & 10 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 5 & 18 & 5 \\ 4 & 10 & 0 & 18 & 34 & 14 \\ 2 & 4 & -1 & 5 & 14 & 4 \end{pmatrix}_{b_0}$

[1117] $1^{11}3151321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 5 & 9 & 5 \\ 2 & 5 & 0 & 9 & 7 & 7 \\ 2 & 4 & -1 & 5 & 7 & 4 \end{pmatrix}_{b_0}$

[1115] $1^{13}31\bar{5}1321^{113}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 24$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 2 & 1 & 5 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 5 & 27 & 5 \\ 6 & 15 & 0 & 27 & 79 & 21 \\ 2 & 4 & -1 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

[1118] $1^{14}31\bar{5}1321^{114}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 24$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 2 & 1 & 5 & 20 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 5 & 36 & 5 \\ 8 & 20 & 0 & 36 & 142 & 28 \\ 2 & 4 & -1 & 5 & 28 & 4 \end{pmatrix}_{b_0}$$

[1119] $1^{11}31\bar{5}1321^{11}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 27$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.96 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 7 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 9 & 4 \\ 7 & 0 & 9 & 23 & 11 \\ 3 & 0 & 4 & 11 & 5 \end{pmatrix}_{b_0}$$

[1120] $1^{11}31\bar{5}1321^{12}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 37$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 14 & 4 \\ 9 & 0 & 14 & 46 & 16 \\ 3 & 0 & 4 & 16 & 5 \end{pmatrix}_{b_0}$$

[1121] $1^{12}31\bar{5}1321^{13}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 43$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 12.98 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 15 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 23 & 4 \\ 15 & 0 & 23 & 139 & 27 \\ 3 & 0 & 4 & 27 & 5 \end{pmatrix}_{b_0}$$

[1123] $1^{11}31\bar{5}1321^{13}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 82$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 12 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 19 & 4 \\ 12 & 0 & 19 & 79 & 21 \\ 3 & 0 & 4 & 21 & 5 \end{pmatrix}_{b_0}$$

[1122] $1^{11}31\bar{5}1321^{12}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 82$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 14 & 4 \\ 10 & 0 & 14 & 46 & 16 \\ 3 & 0 & 4 & 16 & 5 \end{pmatrix}_{b_0}$$

[1125] $1^{12}31\bar{5}1321^{13}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 118$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 16 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 23 & 4 \\ 16 & 0 & 23 & 139 & 27 \\ 3 & 0 & 4 & 27 & 5 \end{pmatrix}_{b_0}$$

[1124] $1^{12}31\bar{5}1321^{14}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 118$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 18 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 28 & 4 \\ 18 & 0 & 28 & 190 & 32 \\ 3 & 0 & 4 & 32 & 5 \end{pmatrix}_{b_0}$$

[1126] $1^{13}31\bar{5}1321^{14}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 138$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 22 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 32 & 4 \\ 22 & 0 & 32 & 282 & 38 \\ 3 & 0 & 4 & 38 & 5 \end{pmatrix}_{b_0}$$

[1127] $1^{11}31\bar{5}1321^{13}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 147$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} =$

8.0

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 19 & 4 \\ 13 & 0 & 19 & 79 & 21 \\ 3 & 0 & 4 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[1128] \ 1^{13} 31 \ 5 \ 1321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 183, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 17 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 23 & 4 \\ 17 & 0 & 23 & 139 & 27 \\ 3 & 0 & 4 & 27 & 5 \end{pmatrix}_{b_0}$$

$$[1129] \ 1^{13} 31 \ 5 \ 1321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 213, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 19 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 28 & 4 \\ 19 & 0 & 28 & 190 & 32 \\ 3 & 0 & 4 & 32 & 5 \end{pmatrix}_{b_0}$$

$$[1130] \ 1^{13} 31 \ 5 \ 1321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 233, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 23 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 32 & 4 \\ 23 & 0 & 32 & 282 & 38 \\ 3 & 0 & 4 & 38 & 5 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [1133] \ 1^{11} 31 \ 6 \ 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 1 \\ 1 & 0 & 0 & -3 & 0 \\ 1 & 3 & 1 & 0 & 1 \\ 2 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1134] \ 1^{11} 31 \ 5 \ 13221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 12 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 22 & 5 \\ 12 & 0 & 22 & 88 & 24 \\ 3 & 0 & 5 & 24 & 6 \end{pmatrix}_{b_0}$$

$$[1135] \ 1^{11} 31 \ 5 \ 13221^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 16 & 5 \\ 9 & 0 & 16 & 50 & 18 \\ 3 & 0 & 5 & 18 & 6 \end{pmatrix}_{b_0}$$

$$[1136] \ 1^{12} 31 \ 5 \ 13221^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 18 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 3 & 32 & 5 \\ 18 & 0 & 32 & 206 & 36 \\ 3 & 0 & 5 & 36 & 6 \end{pmatrix}_{b_0}$$

$$[1137] \ 1^{12} 31 \ 5 \ 13221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 3 & 1 & 6 & 12 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 6 & 20 & 6 \\ 4 & 12 & 0 & 20 & 38 & 16 \\ 2 & 5 & -1 & 6 & 16 & 5 \end{pmatrix}_{b_0}$$

$$[1138] \ 1^{11} 31 \ 5 \ 13221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 3 & 1 & 6 & 6 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 6 & 10 & 6 \\ 2 & 6 & 0 & 10 & 8 & 8 \\ 2 & 5 & -1 & 6 & 8 & 5 \end{pmatrix}_{b_0}$$

$$[1140] \ 1^{14}31\bar{5}13221^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 13, \ T_{\max} = 13, \ \Delta = 0, \ \det = -16, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 3 & 1 & 6 & 24 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 6 & 40 & 6 \\ 8 & 24 & 0 & 40 & 158 & 32 \\ 2 & 5 & -1 & 6 & 32 & 5 \end{pmatrix}_{b_0}$$

$$[1139] \ 1^{13}31\bar{5}13221^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 13, \ T_{\max} = 13, \ \Delta = 0, \ \det = -16, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 3 & 1 & 6 & 18 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 6 & 30 & 6 \\ 6 & 18 & 0 & 30 & 88 & 24 \\ 2 & 5 & -1 & 6 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[1141] \ 1^{11}31\bar{6}131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 13, \ T_{\max} = 14, \ \Delta = -33, \ \det = -18, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1142] \ 1^{12}31\bar{6}131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 13, \ T_{\max} = 14, \ \Delta = -33, \ \det = -50, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -3 & 1 & -1 \\ 4 & 7 & 5 & 1 & 18 & 10 \\ 2 & 3 & 2 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

$$T = 19 \quad [1143] \ 1^{11}3221(12)12231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 19, \ T_{\max} = 20, \ \Delta = -32, \ \det = -2, \ (n_+, n_-, n_0) = (1, 19, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 20, \ T_{\max} = 20, \ T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1144] \ 1^{11}3221(12)12231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 19, \ T_{\max} = 20, \ \Delta = -32, \ \det = -18, \ (n_+, n_-, n_0) = (1, 19, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 20, \ T_{\max} = 20, \ T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1145] \ 1^{12}3221(12)12231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 19, \ T_{\max} = 20, \ \Delta = -32, \ \det = -50, \ (n_+, n_-, n_0) = (1, 19, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 20, \ T_{\max} = 20, \ T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -3 & 1 & -1 \\ 4 & 7 & 5 & 1 & 18 & 10 \\ 2 & 3 & 2 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

4.655 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{violet}{e}_6$

$$T = 8 \quad [0] \ 1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{violet}{e}_6 \quad T = 8, \ T_{\max} = 101, \ \Delta = -162, \ \det = 6, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.33 \rightarrow 15, \ T_{\max} = 101, \ T \in [15, 101]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 6 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{violet}{e}_6 \quad T = 8, \ T_{\max} = 101, \ \Delta = -162, \ \det = 139, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.51 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.71 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 44 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 44 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 34 & -4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{113}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 59 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 59 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 45 & -4 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.53 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 318, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 25 & 40 & 0 & 9 \\ 18 & 40 & 58 & 0 & 14 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 14 & -4 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 379, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 35 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 35 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 27 & -4 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 17 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 430, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 34 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 26 & -4 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 26 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 20 & -4 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 493, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 27 & 1 & 5 \\ 12 & 25 & 60 & 0 & 9 \\ 27 & 60 & 133 & 0 & 21 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 21 & -4 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 546, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 26 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 20 & -4 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 738, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.51 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 36 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 36 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 28 & -4 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{113}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 745, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 51 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 39 & -4 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 921, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13.44 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 35 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 27 & -4 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1005, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 43 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 43 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 33 & -4 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{115}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1053, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.1$
 $T_{\min} = 13.52 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 100 & 0 & 9 \\ 45 & 100 & 373 & 0 & 35 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 35 & -4 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{114}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 68 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 52 & -4 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 44 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 44 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 34 & -4 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{114}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1590, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 60 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 46 & -4 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{114}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1650, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 52 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 40 & -4 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [23] $1^{11}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 15 & 0 & 6 \\ 5 & 12 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[24] 1^{111} 31^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -571, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 9 & 1 & 3 \\ 4 & 10 & 13 & 24 & 0 & 6 \\ 5 & 13 & 12 & 27 & 0 & 6 \\ 9 & 24 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

4.656 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2$ + $\mathfrak{e}_6$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] 1^{112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 26 & 5 \\ 12 & 25 & 16 & 0 & 56 & 9 \\ 8 & 16 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 56 & 36 & 0 & 118 & 20 \\ 5 & 9 & 6 & -4 & 20 & -1 \end{pmatrix}_{b_0}$$

$$[1] 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 7 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] 1^{11} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 17 & 5 \\ 12 & 25 & 20 & 0 & 36 & 9 \\ 9 & 20 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 36 & 27 & 0 & 49 & 13 \\ 5 & 9 & 7 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[4] 1^{111} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 35 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 35 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

$$[5] 1^{111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 9 & 5 \\ 12 & 25 & 20 & 0 & 20 & 9 \\ 9 & 20 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 20 & 15 & 0 & 13 & 7 \\ 5 & 9 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[6] 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -608, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 17 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -618, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.51 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 26 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$4.657 \quad \text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 372, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 17 & 5 \\ 11 & 20 & 17 & 0 & 17 & 33 & 8 \\ 9 & 17 & 13 & 0 & 15 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 27 & 7 \\ 17 & 33 & 27 & 0 & 27 & 49 & 13 \\ 5 & 8 & 7 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$4.658 \quad \text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3$$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -115, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, \ T_{\max} = 99, \ T \in [15, 99]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 8 & 5 \\ 12 & 25 & 20 & 0 & 16 & 9 \\ 9 & 20 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 9 & 5 \\ 5 & 9 & 7 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -271, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -334, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 8 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -931, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

4.659 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2$ + $\mathfrak{f}_4$

$$T = 8 \quad [0] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 44 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 44 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 34 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 59 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 59 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 45 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.68 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 25 & 40 & 0 & 9 \\ 18 & 40 & 58 & 0 & 14 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 14 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 291, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 35 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 35 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 27 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 17 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{112} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 34 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 26 & -3 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 365, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.71 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 27 & 1 & 5 \\ 12 & 25 & 60 & 0 & 9 \\ 27 & 60 & 133 & 0 & 21 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 21 & -3 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 26 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 20 & -3 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 26 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 20 & -3 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 540, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 36 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 36 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 28 & -3 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{113}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 581, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 51 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 39 & -3 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 709, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 35 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 27 & -3 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{115}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 765, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.3$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 100 & 0 & 9 \\ 45 & 100 & 373 & 0 & 35 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 35 & -3 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 781, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 43 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 43 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 33 & -3 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{114}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 924, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 68 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 52 & -3 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 1044, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 44 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 44 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 34 & -3 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{114}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 1236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 60 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 46 & -3 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{114}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 1276$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 52 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 40 & -3 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [23] $1^{11}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 15 & 0 & 6 \\ 5 & 12 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{111}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -435$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 1 & 4 & 5 & 9 & 1 & 3 \\ 4 & 10 & 13 & 24 & 0 & 6 \\ 5 & 13 & 12 & 27 & 0 & 6 \\ 9 & 24 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

4.660 NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$

$T = 9$ [0] $1^{112}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.7$

$T_{\min} = 15$, $T_{\max} = 21$, $T \in [15, 21]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 26 & 5 \\ 12 & 25 & 16 & 0 & 56 & 9 \\ 8 & 16 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 56 & 36 & 0 & 118 & 20 \\ 5 & 9 & 6 & -3 & 20 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -90$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 13.07 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 7 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -104$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -288$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 17 & 5 \\ 12 & 25 & 20 & 0 & 36 & 9 \\ 9 & 20 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 36 & 27 & 0 & 49 & 13 \\ 5 & 9 & 7 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -324$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 12.67 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 35 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 35 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -380$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

14.2

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 9 & 5 \\ 12 & 25 & 20 & 0 & 20 & 9 \\ 9 & 20 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 15 & 0 & 13 & 7 \\ 5 & 9 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 17 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -482, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 26 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

4.661 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 7 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 6 & 5 \\ 12 & 25 & 20 & 0 & 12 & 9 \\ 9 & 20 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 9 & 0 & 4 & 4 \\ 5 & 9 & 7 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 6 & 5 \\ 12 & 25 & 36 & 0 & 12 & 9 \\ 17 & 36 & 49 & 0 & 17 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 17 & 0 & 4 & 4 \\ 5 & 9 & 13 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

4.662 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \text{n3mix}$

$$T = 9 \quad [0] \ 1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 15 & 5 \\ 11 & 20 & 17 & 0 & 28 & 8 \\ 9 & 17 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 28 & 24 & 0 & 37 & 11 \\ 5 & 8 & 7 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

4.663 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 8 & 5 \\ 12 & 25 & 20 & 0 & 16 & 9 \\ 9 & 20 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 9 & 5 \\ 5 & 9 & 7 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 8 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

4.664 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 8 \quad [0] \quad 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 6 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{113} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.71 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 12 & 54 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 54 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 39 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 8 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 44 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 44 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 34 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 10 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

- [5] $1^{111} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 10 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$
- [6] $1^{112} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 5 & 12 & 36 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 36 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 26 & -2 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{111} 3 1^{31} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 20 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 20 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$
- [8] $1^{113} 3 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 59 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 59 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 45 & -2 & 2 \end{pmatrix}_{b_0}$$
- [9] $1^{113} 3 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 60 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 45 & -2 & 2 \end{pmatrix}_{b_0}$$
- [10] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 13 & 7 \\ 5 & 8 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$
- [11] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 18 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$
- [12] $1^{111} 3 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 39 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 39 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 27 & -2 & 2 \end{pmatrix}_{b_0}$$
- [13] $1^{111} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$
- [14] $1^{111} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 8 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$
- [15] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 22, \Delta = -112, \det = 137, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

- [16] $1^{112} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 162$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.04 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 25 & 40 & 0 & 9 \\ 18 & 40 & 58 & 0 & 14 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 14 & -2 & 1 \end{pmatrix}_{b_0}$
- [17] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 48 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 48 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 34 & -2 & 2 \end{pmatrix}_{b_0}$
- [18] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 16 & 0 & 10 & 6 \\ 5 & 8 & -2 & 6 & 2 \end{pmatrix}_{b_0}$
- [19] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 203$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 35 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 35 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 27 & -2 & 2 \end{pmatrix}_{b_0}$
- [20] $1^{113} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 237$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 5 & 12 & 27 & 1 & 5 \\ 12 & 25 & 60 & 0 & 9 \\ 27 & 60 & 133 & 0 & 21 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 21 & -2 & 1 \end{pmatrix}_{b_0}$
- [21] $1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 242$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 34 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 34 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 26 & -2 & 2 \end{pmatrix}_{b_0}$
- [22] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 243$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 1 & 17 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 33 & 0 & 49 & 13 \\ 5 & 8 & -2 & 13 & 2 \end{pmatrix}_{b_0}$
- [23] $1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 254$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 35 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 35 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 26 & -2 & 2 \end{pmatrix}_{b_0}$
- [24] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 268$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 29 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 29 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 20 & -2 & 2 \end{pmatrix}_{b_0}$
- [25] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 268$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 1 & 26 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 26 & 50 & 0 & 118 & 20 \\ 5 & 8 & -2 & 20 & 2 \end{pmatrix}_{b_0}$
- [26] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 291$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 19 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 19 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & -2 & 2 \end{pmatrix}_{b_0}$

- [27] $1^{112} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 302$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 19 & 1 & 5 \\ 12 & 25 & 40 & 0 & 9 \\ 19 & 40 & 58 & 0 & 14 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 14 & -2 & 2 \end{pmatrix}_{b_0}$
- [28] $1^{111} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 320$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 45 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 45 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 34 & -2 & 2 \end{pmatrix}_{b_0}$
- [29] $1^{114} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 342$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 5 & 12 & 36 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 36 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 28 & -2 & 1 \end{pmatrix}_{b_0}$
- [30] $1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 347$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 1 & 18 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 33 & 0 & 49 & 13 \\ 5 & 8 & -2 & 13 & 2 \end{pmatrix}_{b_0}$
- [31] $1^{113} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 353$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 53 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 53 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 39 & -2 & 2 \end{pmatrix}_{b_0}$
- [32] $1^{111} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 371$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 38 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 38 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 27 & -2 & 2 \end{pmatrix}_{b_0}$
- [33] $1^{112} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 378$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 27 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 27 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 20 & -2 & 2 \end{pmatrix}_{b_0}$
- [34] $1^{111} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 384$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 47 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 47 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 34 & -2 & 2 \end{pmatrix}_{b_0}$
- [35] $1^{113} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 417$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 51 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 51 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 39 & -2 & 2 \end{pmatrix}_{b_0}$
- [36] $1^{111} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 419$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 36 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 36 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 27 & -2 & 2 \end{pmatrix}_{b_0}$
- [38] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 428$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 28 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 28 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 20 & -2 & 2 \end{pmatrix}_{b_0}$

- [37] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 428$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 27 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 27 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 20 & -2 & 2 \end{pmatrix}_{b_0}$
- [39] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 432$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 46 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 46 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 34 & -2 & 2 \end{pmatrix}_{b_0}$
- [40] $1^{114} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 434$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 71 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 71 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 52 & -2 & 2 \end{pmatrix}_{b_0}$
- [41] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 475$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 37 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 37 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 27 & -2 & 2 \end{pmatrix}_{b_0}$
- [42] $1^{115} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 477$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.1$
 $T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$
 $\begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 100 & 0 & 9 \\ 45 & 100 & 373 & 0 & 35 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 35 & -2 & 1 \end{pmatrix}_{b_0}$
- [43] $1^{113} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 485$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 52 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 52 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 39 & -2 & 2 \end{pmatrix}_{b_0}$
- [44] $1^{113} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 497$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 28 & 1 & 5 \\ 12 & 25 & 60 & 0 & 9 \\ 28 & 60 & 133 & 0 & 21 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 21 & -2 & 2 \end{pmatrix}_{b_0}$
- [45] $1^{113} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 557$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 29 & 1 & 5 \\ 12 & 25 & 60 & 0 & 9 \\ 29 & 60 & 133 & 0 & 21 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 21 & -2 & 2 \end{pmatrix}_{b_0}$
- [46] $1^{113} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 621$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 45 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 33 & -2 & 2 \end{pmatrix}_{b_0}$
- [47] $1^{114} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 662$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 68 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 68 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 52 & -2 & 2 \end{pmatrix}_{b_0}$
- [48] $1^{113} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 689$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 44 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 44 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 33 & -2 & 2 \end{pmatrix}_{b_0}$

[49] $1^{11} 1^{13} 3^1 1^{11} + [\textcolor{red}{5}\textcolor{blue}{2}\textcolor{blue}{2} \cdot \textcolor{blue}{9}\textcolor{blue}{2}] + \textcolor{teal}{5}\textcolor{teal}{0}\textcolor{teal}{8} + \textcolor{orange}{5}\textcolor{orange}{2}\textcolor{orange}{2} + \textcolor{red}{5}\textcolor{red}{0}\textcolor{red}{8} \quad T = 8, \quad T_{\max} = 22, \quad \Delta = -112, \quad \det = 693, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 12.8$
 $T_{\min} = 11.84 \rightarrow 12, \quad T_{\max} = 22, \quad T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 36 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 36 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 27 & -2 & 2 \end{pmatrix}_{b_0}$$

[50] $1^{14} 3^1 1^{14} + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{5}\textcolor{blue}{o}_8 + \textcolor{brown}{5}\textcolor{blue}{u}_2 + \textcolor{red}{5}\textcolor{blue}{o}_8 \quad T = 8, \quad T_{\max} = 22, \quad \Delta = -112, \quad \det = 710, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 13.0$
 $T_{\min} = 11.84 \rightarrow 12, \quad T_{\max} = 22, \quad T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 70 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 70 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 52 & -2 & 2 \end{pmatrix}_{b_0}$

$$[51] \ 1^{11} 1^{14} 3 \ 1^{\textcolor{teal}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{s}\textcolor{teal}{o}\textcolor{teal}{g}_8 + \textcolor{brown}{s}\textcolor{brown}{u}_2 + \textcolor{brown}{s}\textcolor{brown}{o}_8 \quad T = 8, \ T_{\max} = 22, \ \Delta = -112, \ \det = 722, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 37 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 37 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 28 & -2 & 2 \end{pmatrix}_{b_0}$$

[52] $1^{11} 1^{14} 3 1^{14} + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{5}\textcolor{brown}{o}_8 + \textcolor{brown}{5}\textcolor{blue}{u}_2 + \textcolor{red}{5}\textcolor{brown}{o}_8 \quad T = 8, \quad T_{\max} = 22, \quad \Delta = -112, \quad \det = 786, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 13.0$
 $T_{\min} = 11.82 \rightarrow 12, \quad T_{\max} = 22, \quad T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 69 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 69 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 52 & -2 & 2 \end{pmatrix}_{b_0}$

[53] $1^{114} 3^{131} 1^{13} + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{5}\textcolor{blue}{o}_8 + \textcolor{orange}{5}\textcolor{blue}{u}_2 + \textcolor{red}{5}\textcolor{blue}{o}_8 \quad T = 8, \quad T_{\max} = 22, \quad \Delta = -112, \quad \det = 846, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 13.0$
 $T_{\min} = 11.84 \rightarrow 12, \quad T_{\max} = 22, \quad T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 63 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 63 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 46 & -2 & 2 \end{pmatrix}_{b_0}$

[54] $1^{11} 1^4 3 1^{31} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{red}{\mathbf{g}}_2] + \textcolor{teal}{\mathbf{so}}_8 + \textcolor{blue}{\mathbf{so}}_2 + \textcolor{violet}{\mathbf{so}}_8 \quad T = 8, \quad T_{\max} = 22, \quad \Delta = -112, \quad \det = 882, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 13.0$
 $T_{\min} = 11.84 \rightarrow 12, \quad T_{\max} = 22, \quad T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 39 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 39 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 28 & -2 & 2 \end{pmatrix}_{b_0}$

$$[55] \text{ } 1^{11} 1^4 3^1 1^{21} + [\textcolor{red}{\mathfrak{s}}\textcolor{blue}{\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}}\textcolor{green}{\mathfrak{o}}_8 + \textcolor{orange}{\mathfrak{u}}_2 + \textcolor{red}{\mathfrak{s}}\textcolor{red}{\mathfrak{o}}_8 \quad T = 8, \quad T_{\max} = 22, \quad \Delta = -112, \quad \det = 902, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 22, \quad T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 38 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 38 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 28 & -2 & 2 \end{pmatrix}_{b_0}$$

[56] $1^{11} 1^{14} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{so}_2 + \text{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 1038, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 45 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 34 & -2 & 2 \end{pmatrix}_{b_0}$

[57] $1^{114} 3^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 1058, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 62 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 62 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 46 & -2 & 2 \end{pmatrix}_{b_0}$

[58] $1^{11} 4^{11} 3^1 1^{13} + [\textcolor{red}{su}_2, \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 22, \ \Delta = -112, \ \det = 1070, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.0$
 $T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 22, \ T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 61 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 61 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 46 & -2 & 2 \end{pmatrix}_{b_0}$

[59] $1^{114} 3^{121} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 1154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 46 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 46 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 34 & -2 & 2 \end{pmatrix}_{b_0}$

- [60] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 1206$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 54 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 54 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 40 & -2 & 2 \end{pmatrix}_{b_0}$
- $T = 9$ [61] $1^{113} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -21$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 5 & 9 & 18 & 1 & 5 \\ 9 & 13 & 30 & 0 & 7 \\ 18 & 30 & 61 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 15 & -2 & 2 \end{pmatrix}_{b_0}$
- [62] $1^{111} 3 1^1 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.33 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 1 & 5 & 4 & 3 \\ 4 & 10 & 1 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 12 & 0 & 11 & 12 & 5 \\ 4 & 12 & 0 & 12 & 10 & 6 \\ 3 & 6 & -1 & 5 & 6 & 1 \end{pmatrix}_{b_0}$
- [63] $1^{11} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -33$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$
 $\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 5 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & -2 & 1 \end{pmatrix}_{b_0}$
- [64] $1^{111} 3 1^1 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 5 & 1 & 4 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 12 & 12 & 0 & 10 & 6 \\ 3 & 6 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$
- [65] $1^{12} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.36 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 5 & 9 & 12 & 1 & 5 \\ 9 & 13 & 20 & 0 & 7 \\ 12 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 10 & -2 & 2 \end{pmatrix}_{b_0}$
- [66] $1^{11} 3 1^1 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -79$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 15 & 0 & 6 \\ 5 & 12 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 6 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$
- [67] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -81$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.41 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 5 & 9 & 9 & 1 & 5 \\ 9 & 13 & 14 & 0 & 7 \\ 9 & 14 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 7 & -2 & 2 \end{pmatrix}_{b_0}$
- [68] $1^{11} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -93$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.35 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 5 & 9 & 6 & 1 & 5 \\ 9 & 13 & 10 & 0 & 7 \\ 6 & 10 & 5 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 5 & -2 & 2 \end{pmatrix}_{b_0}$
- [69] $1^{11} 3 2 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 2 & 4 & 4 \\ 2 & 5 & 0 & 4 & 0 & 2 \\ 3 & 5 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [70] $1^{111} 3 1^1 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$\begin{aligned}
& T_{\min} = 12.21 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 3 \\ 4 & 10 & 13 & 0 & 12 & 6 \\ 5 & 13 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 12 & 0 & 10 & 6 \\ 3 & 6 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0} \\
[71] & 1^{11} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 1 & 5 & 5 & 3 \\ 4 & 10 & 1 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 12 & 0 & 11 & 12 & 5 \\ 5 & 12 & 0 & 12 & 10 & 6 \\ 3 & 6 & -1 & 5 & 6 & 1 \end{pmatrix}_{b_0} \\
[72] & 1^{11} 321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 5 & 0 & 10 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 10 & 0 & 14 & 7 & 8 \\ 2 & 5 & 0 & 7 & 0 & 2 \\ 3 & 5 & -2 & 8 & 2 & 2 \end{pmatrix}_{b_0} \\
[73] & 1^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 12.39 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 5 & 9 & 15 & 1 & 5 \\ 9 & 13 & 24 & 0 & 7 \\ 15 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 12 & -2 & 2 \end{pmatrix}_{b_0} \\
[74] & 1^{11} 321^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 5 & 0 & 15 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 15 & 0 & 32 & 11 & 12 \\ 2 & 5 & 0 & 11 & 0 & 2 \\ 3 & 5 & -2 & 12 & 2 & 2 \end{pmatrix}_{b_0} \\
[75] & 1^{11} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 12.20 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 5 & 1 & 10 & 3 \\ 4 & 10 & 13 & 0 & 24 & 6 \\ 5 & 13 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 24 & 27 & 0 & 49 & 13 \\ 3 & 6 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0} \\
[76] & 1^{12} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -273, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5 \\
& T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21] \\
& \begin{pmatrix} 5 & 9 & 21 & 1 & 5 \\ 9 & 13 & 34 & 0 & 7 \\ 21 & 34 & 77 & 0 & 17 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 17 & -2 & 2 \end{pmatrix}_{b_0} \\
[77] & 1^{11} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 5 & 1 & 9 & 3 \\ 4 & 10 & 13 & 0 & 24 & 6 \\ 5 & 13 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 24 & 27 & 0 & 49 & 13 \\ 3 & 6 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0} \\
[78] & 1^{12} 321^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5 \\
& T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 5 & 0 & 20 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 20 & 0 & 62 & 14 & 16 \\ 2 & 5 & 0 & 14 & 0 & 2 \\ 3 & 5 & -2 & 16 & 2 & 2 \end{pmatrix}_{b_0} \\
[79] & 1^{11} 321^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 5 & 0 & 15 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 15 & 0 & 32 & 11 & 12 \\ 2 & 5 & 0 & 11 & 0 & 2 \\ 3 & 5 & -2 & 12 & 2 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [80] $1^{13} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -342$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.39 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 5 & 9 & 27 & 1 & 5 \\ 9 & 13 & 44 & 0 & 7 \\ 27 & 44 & 130 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 22 & -2 & 2 \end{pmatrix}_{b_0}$
- [81] $1^{14} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -393$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.39 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 5 & 9 & 33 & 1 & 5 \\ 9 & 13 & 54 & 0 & 7 \\ 33 & 54 & 197 & 0 & 27 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 27 & -2 & 2 \end{pmatrix}_{b_0}$
- [82] $1^{12} 3^{21} 1^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -486$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 5 & 0 & 25 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 25 & 0 & 94 & 18 & 20 \\ 2 & 5 & 0 & 18 & 0 & 2 \\ 3 & 5 & -2 & 20 & 2 & 2 \end{pmatrix}_{b_0}$
- [83] $1^{11} 3^{21} 1^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -495$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 5 & 0 & 20 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 20 & 0 & 58 & 15 & 16 \\ 2 & 5 & 0 & 15 & 0 & 2 \\ 3 & 5 & -2 & 16 & 2 & 2 \end{pmatrix}_{b_0}$
- [84] $1^{13} 3^{21} 1^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -619$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 5 & 0 & 30 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 30 & 0 & 142 & 21 & 24 \\ 2 & 5 & 0 & 21 & 0 & 2 \\ 3 & 5 & -2 & 24 & 2 & 2 \end{pmatrix}_{b_0}$
- [85] $1^{12} 3^{21} 1^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -706$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 5 & 0 & 25 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 25 & 0 & 94 & 18 & 20 \\ 2 & 5 & 0 & 18 & 0 & 2 \\ 3 & 5 & -2 & 20 & 2 & 2 \end{pmatrix}_{b_0}$
- [86] $1^{13} 3^{21} 1^{41} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -895$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 5 & 0 & 35 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 35 & 0 & 188 & 25 & 28 \\ 2 & 5 & 0 & 25 & 0 & 2 \\ 3 & 5 & -2 & 28 & 2 & 2 \end{pmatrix}_{b_0}$
- [87] $1^{12} 3^{21} 1^{41} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -966$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 5 & 0 & 30 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 30 & 0 & 134 & 22 & 24 \\ 2 & 5 & 0 & 22 & 0 & 2 \\ 3 & 5 & -2 & 24 & 2 & 2 \end{pmatrix}_{b_0}$
- [88] $1^{14} 3^{21} 1^{41} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -1046$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 5 & 0 & 40 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 20 & 40 & 0 & 254 & 28 & 32 \\ 2 & 5 & 0 & 28 & 0 & 2 \\ 3 & 5 & -2 & 32 & 2 & 2 \end{pmatrix}_{b_0}$
- [89] $1^{13} 3^{21} 1^{41} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -1195$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 \\ 3 & 5 & 0 & 35 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 35 & 0 & 188 & 25 & 28 \\ 2 & 5 & 0 & 25 & 0 & 2 \\ 3 & 5 & -2 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [90] \quad 1^{\underline{1}} 1^{\underline{11}31^{\underline{1}}} 4 \quad 1^{\underline{1}} + [\underline{\text{su}}_2 \cdot \underline{\text{g}}_2] + \underline{\text{so}}_8 + \underline{\text{su}}_2 + \underline{\text{so}}_8 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 & 0 & 2 \\ 2 & 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 0 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{\underline{1}} 1^{\underline{11}31^{\underline{1}}} 4 \quad 1^{\underline{1}} + [\underline{\text{su}}_2 \cdot \underline{\text{g}}_2] + \underline{\text{so}}_8 + \underline{\text{su}}_2 + \underline{\text{so}}_8 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 10 & 1 & 10 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 10 & 14 & 1 & 14 & 8 \\ 5 & 7 & -1 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{\underline{1}} 1^{\underline{11}31^{\underline{1}}} 4 \quad 1^{\underline{1}} + [\underline{\text{su}}_2 \cdot \underline{\text{g}}_2] + \underline{\text{so}}_8 + \underline{\text{su}}_2 + \underline{\text{so}}_8 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 14 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{\underline{1}} 1^{\underline{11}31^{\underline{1}}} 4 \quad 1^{\underline{1}} 1^{\underline{11}31^{\underline{1}}} + [\underline{\text{su}}_2 \cdot \underline{\text{g}}_2] + \underline{\text{so}}_8 + \underline{\text{su}}_2 + \underline{\text{so}}_8 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 2 & 3 \\ 2 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 3 & 3 & 1 & 2 & 2 & 4 \\ 2 & 1 & 0 & 2 & -2 & 0 \\ 3 & 2 & 0 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{\underline{1}} 1^{\underline{11}31^{\underline{1}}} 4 \quad 1^{\underline{1}} + [\underline{\text{su}}_2 \cdot \underline{\text{g}}_2] + \underline{\text{so}}_8 + \underline{\text{su}}_2 + \underline{\text{so}}_8 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 2 & 3 \\ 2 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 4 & 2 & 1 & 2 & 2 & 4 \\ 2 & 1 & 0 & 2 & -2 & 0 \\ 3 & 2 & 0 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{\underline{1}} 1^{\underline{11}31^{\underline{1}}} 4 \quad 1^{\underline{1}} + [\underline{\text{su}}_2 \cdot \underline{\text{g}}_2] + \underline{\text{so}}_8 + \underline{\text{su}}_2 + \underline{\text{so}}_8 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 2 & 3 \\ 2 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 6 & 3 & 2 & 8 & 2 & 6 \\ 2 & 1 & 0 & 2 & -2 & 0 \\ 3 & 2 & 0 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[96] \quad 1^{\underline{1}} 1^{\underline{11}31^{\underline{1}}} 4 \quad 1^{\underline{1}} + [\underline{\text{su}}_2 \cdot \underline{\text{g}}_2] + \underline{\text{so}}_8 + \underline{\text{su}}_2 + \underline{\text{so}}_8 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 234, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 3 \\ 2 & 0 & 1 & 4 & 1 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 5 & 4 & 2 & 6 & 2 & 6 \\ 2 & 1 & 0 & 2 & -2 & 0 \\ 3 & 2 & 0 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[97] \quad 1^{\underline{1}} 1^{\underline{11}31^{\underline{1}}} 4 \quad 1^{\underline{1}} + [\underline{\text{su}}_2 \cdot \underline{\text{g}}_2] + \underline{\text{so}}_8 + \underline{\text{su}}_2 + \underline{\text{so}}_8 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 259, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 2 & 3 \\ 2 & 0 & 1 & 5 & 1 & 2 \\ 1 & 1 & -2 & 3 & 0 & 0 \\ 7 & 5 & 3 & 14 & 2 & 8 \\ 2 & 1 & 0 & 2 & -2 & 0 \\ 3 & 2 & 0 & 8 & 0 & 2 \end{pmatrix}_{b_0}$$

4.665 NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 17 & 0 & 0 & 13 & 7 \\ 5 & 8 & -2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 10 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -2 & 7 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 39 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 39 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -2 & 7 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 45 & 5 \\ 11 & 20 & 0 & 0 & 84 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 45 & 84 & 0 & 0 & 346 & 34 \\ 5 & 8 & -2 & 0 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 16 & 0 & 0 & 10 & 6 \\ 5 & 8 & -2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 10 & 5 \\ 12 & 25 & 36 & 0 & 20 & 9 \\ 17 & 36 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 20 & 27 & 0 & 13 & 7 \\ 5 & 9 & 13 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 18 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 18 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 18 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{1111}3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 17 & 5 \\ 12 & 25 & 20 & 0 & 36 & 9 \\ 9 & 20 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 36 & 27 & 0 & 49 & 13 \\ 5 & 9 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{111}3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 35 & 5 \\ 11 & 20 & 0 & 17 & 67 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 57 & 7 \\ 35 & 67 & 0 & 57 & 217 & 27 \\ 5 & 8 & -2 & 7 & 27 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 48 & 5 \\ 11 & 20 & 0 & 17 & 84 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 72 & 7 \\ 48 & 84 & 0 & 72 & 346 & 34 \\ 5 & 8 & -2 & 7 & 34 & 2 \end{pmatrix}_{b_0}$$

[12] $1^{111}3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 18 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 10 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{111}3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 36 & 5 \\ 11 & 20 & 0 & 0 & 67 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 36 & 67 & 0 & 0 & 217 & 27 \\ 5 & 8 & -2 & 0 & 27 & 2 \end{pmatrix}_{b_0}$$

[14] $1^{111}3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 19 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 19 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{111}3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 18 & 5 \\ 11 & 20 & 0 & 0 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 18 & 33 & 0 & 0 & 49 & 13 \\ 5 & 8 & -2 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{111}3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -262, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 29 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 29 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{1111}3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

13.0

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 9 & 5 \\ 12 & 25 & 20 & 0 & 20 & 9 \\ 9 & 20 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 20 & 15 & 0 & 13 & 7 \\ 5 & 9 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111} 3 \ 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -286, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 45 & 5 \\ 11 & 20 & 17 & 0 & 84 & 8 \\ 10 & 17 & 13 & 0 & 72 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 45 & 84 & 72 & 0 & 346 & 34 \\ 5 & 8 & 7 & -2 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{111} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 27 & 5 \\ 11 & 20 & 0 & 0 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 27 & 50 & 0 & 0 & 118 & 20 \\ 5 & 8 & -2 & 0 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111} 3 \ 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 45 & 5 \\ 11 & 20 & 0 & 17 & 84 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 72 & 7 \\ 45 & 84 & 0 & 72 & 346 & 34 \\ 5 & 8 & -2 & 7 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111} 3 \ 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -342, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 27 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 10 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 27 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 17 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{111} 3 \ 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 26 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 26 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{111} 3 \ 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -360, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 36 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 10 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 36 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 6 & 5 \\ 11 & 20 & 0 & 0 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 11 & 0 & 0 & 4 & 4 \\ 5 & 8 & -2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 6 & 5 \\ 12 & 25 & 20 & 0 & 12 & 9 \\ 9 & 20 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 9 & 0 & 4 & 4 \\ 5 & 9 & 7 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 6 & 5 \\ 12 & 25 & 36 & 0 & 12 & 9 \\ 18 & 36 & 49 & 0 & 17 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 17 & 0 & 4 & 4 \\ 5 & 9 & 13 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 9 & 7 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -2 & 7 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -217, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 6 & 5 \\ 12 & 25 & 36 & 0 & 12 & 9 \\ 17 & 36 & 49 & 0 & 17 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 17 & 0 & 4 & 4 \\ 5 & 9 & 13 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

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$$T = 9 \quad [0] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 15 & 5 \\ 11 & 20 & 0 & 0 & 28 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 28 & 0 & 0 & 37 & 11 \\ 5 & 8 & -2 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 15 & 5 \\ 11 & 20 & 17 & 0 & 28 & 8 \\ 10 & 17 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 28 & 24 & 0 & 37 & 11 \\ 5 & 8 & 7 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 16 & 5 \\ 11 & 20 & 0 & 17 & 28 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 24 & 7 \\ 16 & 28 & 0 & 24 & 37 & 11 \\ 5 & 8 & -2 & 7 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 15 & 5 \\ 11 & 20 & 0 & 17 & 28 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 24 & 7 \\ 15 & 28 & 0 & 24 & 37 & 11 \\ 5 & 8 & -2 & 7 & 11 & 2 \end{pmatrix}_{b_0}$$

4.668 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 8 & 5 \\ 12 & 25 & 20 & 0 & 16 & 9 \\ 9 & 20 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 12 & 0 & 9 & 5 \\ 5 & 9 & 7 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 28 & 1 & 8 & 5 \\ 12 & 25 & 60 & 0 & 16 & 9 \\ 28 & 60 & 133 & 0 & 36 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 36 & 0 & 9 & 5 \\ 5 & 9 & 21 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -2 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 1 & 5 \\ 12 & 25 & 36 & 0 & 0 & 9 \\ 17 & 36 & 49 & 0 & 0 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 13 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 10 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -133, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 1 & 5 \\ 11 & 20 & 0 & 17 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 7 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 10 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 10 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -2 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 12 & 6 \\ 5 & 8 & -2 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -160$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 1 & 5 \\ 11 & 20 & 0 & 16 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -162$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.5$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 19 & 1 & 8 & 5 \\ 12 & 25 & 40 & 0 & 16 & 9 \\ 19 & 40 & 58 & 0 & 24 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 24 & 0 & 9 & 5 \\ 5 & 9 & 14 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{112} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -190$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.5$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 1 & 5 \\ 12 & 25 & 72 & 0 & 0 & 9 \\ 34 & 72 & 202 & 0 & 0 & 26 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 26 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -194$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -215$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 10 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 10 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -222$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -244$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 1 & 5 \\ 11 & 20 & 0 & 50 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 26 & 50 & 0 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -293$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 1 & 5 \\ 11 & 20 & 0 & 33 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 33 & 0 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 13 & -1 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{113} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -295$, $(n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 1 & 5 \\ 12 & 25 & 108 & 0 & 0 & 9 \\ 51 & 108 & 457 & 0 & 0 & 39 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 39 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 29 & 1 & 8 & 5 \\ 12 & 25 & 60 & 0 & 16 & 9 \\ 29 & 60 & 133 & 0 & 36 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 36 & 0 & 9 & 5 \\ 5 & 9 & 21 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 38 & 1 & 8 & 5 \\ 12 & 25 & 80 & 0 & 16 & 9 \\ 38 & 80 & 238 & 0 & 48 & 28 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 48 & 0 & 9 & 5 \\ 5 & 9 & 28 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{114} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 1 & 5 \\ 12 & 25 & 144 & 0 & 0 & 9 \\ 68 & 144 & 814 & 0 & 0 & 52 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 52 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 39 & 1 & 8 & 5 \\ 12 & 25 & 80 & 0 & 16 & 9 \\ 39 & 80 & 238 & 0 & 48 & 28 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 48 & 0 & 9 & 5 \\ 5 & 9 & 28 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 19 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 18 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \quad 1^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 6 & 5 \\ 9 & 13 & 14 & 0 & 10 & 7 \\ 9 & 14 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 9 & 0 & 4 & 4 \\ 5 & 7 & 7 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{111} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 261, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 6 & 5 \\ 9 & 13 & 10 & 0 & 10 & 7 \\ 6 & 10 & 5 & 0 & 7 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 7 & 0 & 4 & 4 \\ 5 & 7 & 5 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{111} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 480, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 15 & 1 & 6 & 5 \\ 9 & 13 & 24 & 0 & 10 & 7 \\ 15 & 24 & 38 & 0 & 16 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 16 & 0 & 4 & 4 \\ 5 & 7 & 12 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

4.669 NHC(-2)(-3) + so₈ + su₂ + su₂

$T = 8$ [0] $1^{111} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 14 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 14 & 17 & 13 & 15 & 7 \\ 12 & 17 & 15 & 13 & 7 \\ 6 & 8 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & -2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 0 & 4 & 4 \\ 3 & 4 & -2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 0 & 13 & 7 \\ 3 & 4 & -2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 1 & 1 & 6 \\ 15 & 20 & 0 & 0 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 6 & 8 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 1 & 6 \\ 15 & 20 & 17 & 0 & 8 \\ 12 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -2 & 0 \\ 6 & 8 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

[6] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 48 & 6 \\ 15 & 20 & 0 & 67 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 48 & 67 & 0 & 217 & 27 \\ 6 & 8 & 0 & 27 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{133} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 15 & 3 \\ 3 & 4 & 0 & 17 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 81 & 13 \\ 15 & 27 & 0 & 81 & 133 & 21 \\ 3 & 4 & -2 & 13 & 21 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{111} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 35 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 35 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[10] $1^{1111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 16 & 23 & 13 & 6 \\ 16 & 25 & 36 & 20 & 9 \\ 23 & 36 & 49 & 27 & 13 \\ 13 & 20 & 27 & 13 & 7 \\ 6 & 9 & 13 & 7 & 3 \end{pmatrix}_{b_0}$$

[11] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -2 & 4 & 7 & 2 \end{pmatrix}_{b_0}$$

[12] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 13 & 15 & 7 \\ 13 & 17 & 15 & 13 & 7 \\ 6 & 8 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[14] $1^{121} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 6 & 6 \\ 3 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -2 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -2 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 8 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -2 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

- [17] $1^{124} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 17 & 3 \\ 3 & 4 & 0 & 8 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 42 & 6 \\ 17 & 31 & 0 & 42 & 172 & 24 \\ 3 & 4 & -2 & 6 & 24 & 2 \end{pmatrix}_{b_0}$$
- [18] $1^{111} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 48 & 6 \\ 15 & 20 & 17 & 67 & 8 \\ 12 & 17 & 13 & 57 & 7 \\ 48 & 67 & 57 & 217 & 27 \\ 6 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$
- [19] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 13 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 13 & 17 & 13 & 15 & 7 \\ 13 & 17 & 15 & 13 & 7 \\ 6 & 8 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$
- [20] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 12 & 4 \\ 8 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -2 & 4 & 10 & 2 \end{pmatrix}_{b_0}$$
- [21] $1^{133} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 16 & 3 \\ 3 & 4 & 0 & 17 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 81 & 13 \\ 16 & 27 & 0 & 81 & 133 & 21 \\ 3 & 4 & -2 & 13 & 21 & 2 \end{pmatrix}_{b_0}$$
- [22] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 23 & 6 \\ 15 & 20 & 0 & 33 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 23 & 33 & 0 & 49 & 13 \\ 6 & 8 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$
- [23] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 13 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 17 & 0 & 13 & 7 \\ 6 & 8 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$
- [24] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$
- [25] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$
- [26] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 3 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -2 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$
- [27] $1^{111} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 11 & 6 \\ 15 & 20 & 0 & 16 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 16 & 0 & 10 & 6 \\ 6 & 8 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 16 & 23 & 23 & 6 \\ 16 & 25 & 36 & 36 & 9 \\ 23 & 36 & 49 & 51 & 13 \\ 23 & 36 & 51 & 49 & 13 \\ 6 & 9 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{11} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 5 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 15 & 4 \\ 8 & 14 & 0 & 15 & 37 & 11 \\ 3 & 4 & -2 & 4 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{111} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 13 & 35 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 13 & 17 & 13 & 42 & 7 \\ 35 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{113} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 12 & 3 \\ 3 & 4 & 0 & 9 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 36 & 7 \\ 12 & 22 & 0 & 36 & 85 & 17 \\ 3 & 4 & -2 & 7 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{123} 3^1 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 10 & 3 \\ 3 & 4 & 0 & 18 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 54 & 14 \\ 10 & 17 & 0 & 54 & 49 & 13 \\ 3 & 4 & -2 & 14 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 12 & 7 \\ 5 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -2 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & -2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{13} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 7 & 3 \\ 3 & 4 & 0 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 0 & 25 & 9 \\ 3 & 4 & -2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{11} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 4 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -2 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 13 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 13 & 17 & 13 & 27 & 7 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

- [38] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$
- [39] $1^{13} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 9 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -2 & 4 & 13 & 2 \end{pmatrix}_{b_0}$$
- [42] $1^{123} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 8 & 12 & 0 & 18 & 25 & 9 \\ 3 & 4 & -2 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$
- [40] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 4 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -2 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$
- [43] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 8 & 0 & 0 & 10 & 6 \\ 3 & 4 & -2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$
- [41] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$
- [44] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$
- [46] $1^{123} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 39 & 10 \\ 11 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -2 & 10 & 13 & 2 \end{pmatrix}_{b_0}$$
- [45] $1^{114} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 12 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 33 & 7 \\ 12 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -2 & 7 & 16 & 2 \end{pmatrix}_{b_0}$$
- [47] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 9 & 3 \\ 9 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & -2 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

- [48] $1^{14} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 12 & 21 & 0 & 18 & 76 & 16 \\ 3 & 4 & -2 & 4 & 16 & 2 \end{pmatrix}_{b_0}$$
- [49] $1^{123} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 27 & 10 \\ 7 & 12 & 0 & 27 & 25 & 9 \\ 3 & 4 & -2 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$
- [50] $1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 13 & 27 & 7 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$
- [51] $1^{111} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$
- [52] $1^{111} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 39 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 39 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$
- [54] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 8 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & -2 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$
- [53] $1^{111} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 16 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 16 & 0 & 10 & 6 \\ 6 & 8 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$
- [56] $1^{122} 3 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 14 & 3 \\ 3 & 4 & 0 & 13 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 54 & 10 \\ 14 & 23 & 0 & 54 & 100 & 18 \\ 3 & 4 & -2 & 10 & 18 & 2 \end{pmatrix}_{b_0}$$
- [55] $1^{113} 3 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 15 & 3 \\ 3 & 4 & 0 & 9 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 45 & 7 \\ 15 & 27 & 0 & 45 & 133 & 21 \\ 3 & 4 & -2 & 7 & 21 & 2 \end{pmatrix}_{b_0}$$
- [59] $1^{124} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 8 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 30 & 6 \\ 11 & 21 & 0 & 30 & 76 & 16 \\ 3 & 4 & -2 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$
- [58] $1^{122} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -2 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[57] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 12 & 6 \\ 6 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -2 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[60] $1^{122} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 5 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -2 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[61] $1^{132} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 8 & 3 \\ 3 & 4 & 0 & 12 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 27 & 9 \\ 8 & 13 & 0 & 27 & 28 & 10 \\ 3 & 4 & -2 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 6 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 0 & 13 & 7 \\ 3 & 4 & -2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{144} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 12 & 3 \\ 3 & 4 & 0 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 21 & 0 & 0 & 76 & 16 \\ 3 & 4 & -2 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

[64] $1^{112} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 9 & 4 \\ 7 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -2 & 4 & 7 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{113} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -2 & 4 & 7 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 25 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 13 & 27 & 7 \\ 25 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[67] $1^{121} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.66 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 21 & 10 \\ 6 & 9 & 0 & 21 & 13 & 7 \\ 3 & 4 & -2 & 10 & 7 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{123} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 8 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 30 & 6 \\ 12 & 22 & 0 & 30 & 85 & 17 \\ 3 & 4 & -2 & 6 & 17 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 7 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -2 & 4 & 10 & 2 \end{pmatrix}_{b_0}$$

[70] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 13 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 13 & 21 & 0 & 18 & 76 & 16 \\ 3 & 4 & -2 & 4 & 16 & 2 \end{pmatrix}_{b_0}$$

[71] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{111} 3^1 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 6 & 3 \\ 3 & 4 & 0 & 14 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 24 & 11 \\ 6 & 9 & 0 & 24 & 13 & 7 \\ 3 & 4 & -2 & 11 & 7 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{112} 3^1 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 7 & 3 \\ 3 & 4 & 0 & 14 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 33 & 11 \\ 7 & 13 & 0 & 33 & 28 & 10 \\ 3 & 4 & -2 & 11 & 10 & 2 \end{pmatrix}_{b_0}$$

[74] $1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 24 & 6 \\ 9 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -2 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[76] $1^{11} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 12 & 3 \\ 3 & 4 & 0 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 19 & 0 & 0 & 73 & 15 \\ 3 & 4 & -2 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[75] $1^{111} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 62 & 6 \\ 15 & 20 & 17 & 84 & 8 \\ 12 & 17 & 13 & 72 & 7 \\ 62 & 84 & 72 & 346 & 34 \\ 6 & 8 & 7 & 34 & 3 \end{pmatrix}_{b_0}$$

[77] $1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -2 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[78] $1^{13} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 16 & 3 \\ 3 & 4 & 0 & 5 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 27 & 4 \\ 16 & 27 & 0 & 27 & 133 & 21 \\ 3 & 4 & -2 & 4 & 21 & 2 \end{pmatrix}_{b_0}$$

[79] $1^{111} 3 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 13 & 48 & 6 \\ 15 & 20 & 17 & 67 & 8 \\ 13 & 17 & 13 & 57 & 7 \\ 48 & 67 & 57 & 217 & 27 \\ 6 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

[81] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 9 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & -2 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 24 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 13 & 27 & 7 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[82] $1^{134} 3 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 12 & 3 \\ 3 & 4 & 0 & 22 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 81 & 17 \\ 12 & 21 & 0 & 81 & 76 & 16 \\ 3 & 4 & -2 & 17 & 16 & 2 \end{pmatrix}_{b_0}$$

[84] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 36 & 6 \\ 15 & 20 & 0 & 50 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 36 & 50 & 0 & 118 & 20 \\ 6 & 8 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[83] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 9 & 3 \\ 3 & 4 & 0 & 0 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 14 & 0 & 0 & 37 & 11 \\ 3 & 4 & -2 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[85] $1^{111} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 36 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 36 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[86] $1^{112} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 6 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -2 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[87] $1^{111} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[88] $1^{133} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 12 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 36 & 9 \\ 9 & 17 & 0 & 36 & 49 & 13 \\ 3 & 4 & -2 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

- [90] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 3 \\ 3 & 4 & 0 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 13 & 0 & 0 & 28 & 10 \\ 3 & 4 & -2 & 0 & 10 & 2 \end{pmatrix}_{b_0}$
- [89] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 10 & 3 \\ 3 & 4 & 0 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 17 & 0 & 0 & 49 & 13 \\ 3 & 4 & -2 & 0 & 13 & 2 \end{pmatrix}_{b_0}$
- [91] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 133, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 3 & 7 & 2 \end{pmatrix}_{b_0}$
- [92] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.68 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 9 & 4 \\ 6 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -2 & 4 & 7 & 2 \end{pmatrix}_{b_0}$
- [93] $1^{123} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 8 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 30 & 6 \\ 13 & 22 & 0 & 30 & 85 & 17 \\ 3 & 4 & -2 & 6 & 17 & 2 \end{pmatrix}_{b_0}$
- [94] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 10 & 15 & 13 & 24 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 13 & 17 & 13 & 27 & 7 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$
- [95] $1^{111} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 10 & 15 & 12 & 49 & 6 \\ 15 & 20 & 17 & 67 & 8 \\ 12 & 17 & 13 & 57 & 7 \\ 49 & 67 & 57 & 217 & 27 \\ 6 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$
- [96] $1^{123} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 17 & 3 \\ 3 & 4 & 0 & 13 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 63 & 10 \\ 17 & 27 & 0 & 63 & 133 & 21 \\ 3 & 4 & -2 & 10 & 21 & 2 \end{pmatrix}_{b_0}$
- [97] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 9 & 4 \\ 6 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -2 & 4 & 7 & 2 \end{pmatrix}_{b_0}$
- [98] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 10 & 15 & 1 & 24 & 6 \\ 15 & 20 & 0 & 33 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 24 & 33 & 0 & 49 & 13 \\ 6 & 8 & 0 & 13 & 3 \end{pmatrix}_{b_0}$
- [100] $1^{13} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 143, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 10 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -2 & 4 & 13 & 2 \end{pmatrix}_{b_0}$$

[99] $1^{132} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 143, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 8 & 3 \\ 3 & 4 & 0 & 17 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 39 & 13 \\ 8 & 13 & 0 & 39 & 28 & 10 \\ 3 & 4 & -2 & 13 & 10 & 2 \end{pmatrix}_{b_0}$$

[101] $1^{112} 3 1^{\textcolor{blue}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 5 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -2 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[102] $1^{112} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 21 & 7 \\ 9 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -2 & 7 & 10 & 2 \end{pmatrix}_{b_0}$$

[103] $1^{124} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 17 & 3 \\ 3 & 4 & 0 & 13 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 72 & 10 \\ 17 & 31 & 0 & 72 & 172 & 24 \\ 3 & 4 & -2 & 10 & 24 & 2 \end{pmatrix}_{b_0}$$

[104] $1^{11} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 5 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 8 & 14 & 0 & 15 & 37 & 11 \\ 3 & 4 & -2 & 4 & 11 & 2 \end{pmatrix}_{b_0}$$

[105] $1^{143} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 10 & 3 \\ 3 & 4 & 0 & 21 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 63 & 16 \\ 10 & 17 & 0 & 63 & 49 & 13 \\ 3 & 4 & -2 & 16 & 13 & 2 \end{pmatrix}_{b_0}$$

[106] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 9 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -2 & 4 & 10 & 2 \end{pmatrix}_{b_0}$$

[107] $1^{134} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 0 & 12 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 45 & 9 \\ 11 & 21 & 0 & 45 & 76 & 16 \\ 3 & 4 & -2 & 9 & 16 & 2 \end{pmatrix}_{b_0}$$

[108] $1^{111} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 38 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 38 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[109] $1^{13} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 11 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -2 & 4 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{112} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 156, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 10 & 3 \\ 3 & 4 & 0 & 14 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 14 & 0 & 37 & 48 & 11 \\ 10 & 18 & 0 & 48 & 58 & 14 \\ 3 & 4 & -2 & 11 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{112} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 158, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.72 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 10 & 3 \\ 3 & 4 & 0 & 9 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 30 & 7 \\ 10 & 18 & 0 & 30 & 58 & 14 \\ 3 & 4 & -2 & 7 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{12} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 160, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 17 & 3 \\ 3 & 4 & 0 & 0 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 17 & 28 & 0 & 0 & 154 & 22 \\ 3 & 4 & -2 & 0 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[114] \ 1^{113} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 160, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.80 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 13 & 3 \\ 3 & 4 & 0 & 14 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 57 & 11 \\ 13 & 22 & 0 & 57 & 85 & 17 \\ 3 & 4 & -2 & 11 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{123} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 160, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.95 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 7 & 12 & 0 & 18 & 25 & 9 \\ 3 & 4 & -2 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{111} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 161, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 63 & 6 \\ 15 & 20 & 17 & 84 & 8 \\ 12 & 17 & 13 & 72 & 7 \\ 63 & 84 & 72 & 346 & 34 \\ 6 & 8 & 7 & 34 & 3 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{144} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 162, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.95 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 11 & 3 \\ 3 & 4 & 0 & 16 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 46 & 60 & 12 \\ 11 & 21 & 0 & 60 & 76 & 16 \\ 3 & 4 & -2 & 12 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{12} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 164, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.66 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 8 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -2 & 4 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{111} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 166, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 13 & 36 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 13 & 17 & 13 & 42 & 7 \\ 36 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{12} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 166, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.83 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 0 & 5 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 18 & 4 \\ 11 & 18 & 0 & 18 & 58 & 14 \\ 3 & 4 & -2 & 4 & 14 & 2 \end{pmatrix}_{b_0}$$

[120] $1^{134} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 167, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 18 & 3 \\ 3 & 4 & 0 & 17 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 93 & 13 \\ 18 & 31 & 0 & 93 & 172 & 24 \\ 3 & 4 & -2 & 13 & 24 & 2 \end{pmatrix}_{b_0}$$

[121] $1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 11 & 3 \\ 3 & 4 & 0 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 18 & 0 & 0 & 58 & 14 \\ 3 & 4 & -2 & 0 & 14 & 2 \end{pmatrix}_{b_0}$$

[122] $1^{14} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 15 & 3 \\ 3 & 4 & 0 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 26 & 0 & 0 & 118 & 20 \\ 3 & 4 & -2 & 0 & 20 & 2 \end{pmatrix}_{b_0}$$

[123] $1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 37 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 37 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[124] $1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 51 & 6 \\ 15 & 20 & 17 & 67 & 8 \\ 12 & 17 & 13 & 57 & 7 \\ 51 & 67 & 57 & 217 & 27 \\ 6 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

[125] $1^{132} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 10 & 3 \\ 3 & 4 & 0 & 17 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 54 & 13 \\ 10 & 18 & 0 & 54 & 58 & 14 \\ 3 & 4 & -2 & 13 & 14 & 2 \end{pmatrix}_{b_0}$$

[126] $1^{124} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 14 & 3 \\ 3 & 4 & 0 & 13 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 60 & 10 \\ 14 & 26 & 0 & 60 & 118 & 20 \\ 3 & 4 & -2 & 10 & 20 & 2 \end{pmatrix}_{b_0}$$

[127] $1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 6 & 3 \\ 5 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -2 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[128] $1^{114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 12 & 3 \\ 9 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[129] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 177, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[130] $1^{12} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 14 & 3 \\ 3 & 4 & 0 & 0 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 14 & 23 & 0 & 0 & 100 & 18 \\ 3 & 4 & -2 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

[131] $1^{123} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 24 & 6 \\ 10 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -2 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[132] $1^{124} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 13 & 3 \\ 3 & 4 & 0 & 8 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 30 & 6 \\ 13 & 21 & 0 & 30 & 76 & 16 \\ 3 & 4 & -2 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[133] $1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 50 & 6 \\ 15 & 20 & 17 & 67 & 8 \\ 12 & 17 & 13 & 57 & 7 \\ 50 & 67 & 57 & 217 & 27 \\ 6 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

[134] $1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 13 & 3 \\ 3 & 4 & 0 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 22 & 0 & 0 & 85 & 17 \\ 3 & 4 & -2 & 0 & 17 & 2 \end{pmatrix}_{b_0}$$

[136] $1^{112} 3^1 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 8 & 3 \\ 3 & 4 & 0 & 14 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 14 & 0 & 37 & 33 & 11 \\ 8 & 13 & 0 & 33 & 28 & 10 \\ 3 & 4 & -2 & 11 & 10 & 2 \end{pmatrix}_{b_0}$$

[135] $1^{133} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 13 & 3 \\ 3 & 4 & 0 & 17 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 66 & 13 \\ 13 & 22 & 0 & 66 & 85 & 17 \\ 3 & 4 & -2 & 13 & 17 & 2 \end{pmatrix}_{b_0}$$

[137] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[138] $1^{143} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 12 & 3 \\ 3 & 4 & 0 & 21 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 81 & 16 \\ 12 & 22 & 0 & 81 & 85 & 17 \\ 3 & 4 & -2 & 16 & 17 & 2 \end{pmatrix}_{b_0}$$

[139] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 9 & 3 \\ 7 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -2 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[140] $1^{124} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 192$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 8 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 24 & 6 \\ 10 & 16 & 0 & 24 & 46 & 12 \\ 3 & 4 & -2 & 6 & 12 & 2 \end{pmatrix}_{b_0}$$

[141] $1^{123} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 194$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.3$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 24 & 6 \\ 9 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -2 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[142] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 200$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.66 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 15 & 7 \\ 6 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -2 & 7 & 7 & 2 \end{pmatrix}_{b_0}$$

[143] $1^{124} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 200$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 10.76 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 15 & 3 \\ 3 & 4 & 0 & 8 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 36 & 6 \\ 15 & 26 & 0 & 36 & 118 & 20 \\ 3 & 4 & -2 & 6 & 20 & 2 \end{pmatrix}_{b_0}$$

[144] $1^{113} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 201$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 16 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 18 & 3 \\ 16 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & -2 & 3 & 21 & 2 \end{pmatrix}_{b_0}$$

[145] $1^{122} 3^1 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 202$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.72 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 8 & 3 \\ 3 & 4 & 0 & 18 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 42 & 14 \\ 8 & 13 & 0 & 42 & 28 & 10 \\ 3 & 4 & -2 & 14 & 10 & 2 \end{pmatrix}_{b_0}$$

[146] $1^{122} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 204$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 13 & 3 \\ 3 & 4 & 0 & 13 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 54 & 10 \\ 13 & 23 & 0 & 54 & 100 & 18 \\ 3 & 4 & -2 & 10 & 18 & 2 \end{pmatrix}_{b_0}$$

[147] $1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 204$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.65 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -2 & 7 & 7 & 2 \end{pmatrix}_{b_0}$$

[148] $1^{13} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 208$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 19 & 3 \\ 3 & 4 & 0 & 0 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 19 & 32 & 0 & 0 & 193 & 25 \\ 3 & 4 & -2 & 0 & 25 & 2 \end{pmatrix}_{b_0}$$

[149] $1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 208$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 7 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -2 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[150] $1^{11} 1^{111} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 5 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 15 & 4 \\ 9 & 14 & 0 & 15 & 37 & 11 \\ 3 & 4 & -2 & 4 & 11 & 2 \end{pmatrix}_{b_0}$$

[151] $1^{14} 1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 18 & 3 \\ 3 & 4 & 0 & 0 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 18 & 31 & 0 & 0 & 172 & 24 \\ 3 & 4 & -2 & 0 & 24 & 2 \end{pmatrix}_{b_0}$$

[152] $1^{13} 1^{12} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 16 & 3 \\ 3 & 4 & 0 & 0 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 16 & 27 & 0 & 0 & 133 & 21 \\ 3 & 4 & -2 & 0 & 21 & 2 \end{pmatrix}_{b_0}$$

[153] $1^{123} 1^{12} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 14 & 3 \\ 3 & 4 & 0 & 13 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 51 & 10 \\ 14 & 22 & 0 & 51 & 85 & 17 \\ 3 & 4 & -2 & 10 & 17 & 2 \end{pmatrix}_{b_0}$$

[154] $21^{14} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 17 & 15 & 72 & 7 \\ 17 & 20 & 17 & 84 & 8 \\ 15 & 17 & 13 & 72 & 7 \\ 72 & 84 & 72 & 346 & 34 \\ 7 & 8 & 7 & 34 & 3 \end{pmatrix}_{b_0}$$

[155] $1^{114} 1^1 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 219, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 33 & 7 \\ 11 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -2 & 7 & 16 & 2 \end{pmatrix}_{b_0}$$

[158] $1^{113} 1^1 31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 4 & 0 & 14 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 57 & 11 \\ 12 & 22 & 0 & 57 & 85 & 17 \\ 3 & 4 & -2 & 11 & 17 & 2 \end{pmatrix}_{b_0}$$

[157] $1^{14} 1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 21 & 3 \\ 3 & 4 & 0 & 0 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 21 & 36 & 0 & 0 & 238 & 28 \\ 3 & 4 & -2 & 0 & 28 & 2 \end{pmatrix}_{b_0}$$

[156] $1^{113} 1^{111} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 13 & 3 \\ 3 & 4 & 0 & 14 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 14 & 0 & 37 & 57 & 11 \\ 13 & 22 & 0 & 57 & 85 & 17 \\ 3 & 4 & -2 & 11 & 17 & 2 \end{pmatrix}_{b_0}$$

[160] $1^{112} 1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 12 & 3 \\ 11 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -2 & 3 & 14 & 2 \end{pmatrix}_{b_0}$$

[159] $21^{1111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 228$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 13 & 20 & 27 & 27 & 7 \\ 20 & 25 & 36 & 36 & 9 \\ 27 & 36 & 49 & 51 & 13 \\ 27 & 36 & 51 & 49 & 13 \\ 7 & 9 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$

[162] $1^{112}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 229$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 13 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & -2 & 3 & 18 & 2 \end{pmatrix}_{b_0}$$

[161] $1^{111}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 229$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 9 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 24 & 7 \\ 8 & 14 & 0 & 24 & 37 & 11 \\ 3 & 4 & -2 & 7 & 11 & 2 \end{pmatrix}_{b_0}$$

[163] $21^{112}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 232$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 13 & 17 & 15 & 42 & 7 \\ 17 & 20 & 17 & 50 & 8 \\ 15 & 17 & 13 & 42 & 7 \\ 42 & 50 & 42 & 118 & 20 \\ 7 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[164] $1^{11}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 233$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 9 & 14 & 0 & 15 & 37 & 11 \\ 3 & 4 & -2 & 4 & 11 & 2 \end{pmatrix}_{b_0}$$

[165] $1^{12}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 234$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.76 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 5 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 10 & 18 & 0 & 18 & 58 & 14 \\ 3 & 4 & -2 & 4 & 14 & 2 \end{pmatrix}_{b_0}$$

[167] $1^{13}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 241$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 19 & 3 \\ 3 & 4 & 0 & 5 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 33 & 4 \\ 19 & 32 & 0 & 33 & 193 & 25 \\ 3 & 4 & -2 & 4 & 25 & 2 \end{pmatrix}_{b_0}$$

[166] $1^{123}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 241$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.3$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 19 & 3 \\ 3 & 4 & 0 & 13 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 75 & 10 \\ 19 & 32 & 0 & 75 & 193 & 25 \\ 3 & 4 & -2 & 10 & 25 & 2 \end{pmatrix}_{b_0}$$

[169] $1^{112}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 244$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.72 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 10 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -2 & 3 & 14 & 2 \end{pmatrix}_{b_0}$$

[168] $21^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 244$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 13 & 20 & 27 & 15 & 7 \\ 20 & 25 & 36 & 20 & 9 \\ 27 & 36 & 49 & 27 & 13 \\ 15 & 20 & 27 & 13 & 7 \\ 7 & 9 & 13 & 7 & 3 \end{pmatrix}_{b_0}$$

[170] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 5 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 10 & 14 & 0 & 15 & 37 & 11 \\ 3 & 4 & -2 & 4 & 11 & 2 \end{pmatrix}_{b_0}$$

[171] $1^{132} 3 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 11 & 3 \\ 3 & 4 & 0 & 22 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 72 & 17 \\ 11 & 18 & 0 & 72 & 58 & 14 \\ 3 & 4 & -2 & 17 & 14 & 2 \end{pmatrix}_{b_0}$$

[172] $1^{122} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 14 & 3 \\ 3 & 4 & 0 & 13 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 54 & 10 \\ 14 & 23 & 0 & 54 & 100 & 18 \\ 3 & 4 & -2 & 10 & 18 & 2 \end{pmatrix}_{b_0}$$

[173] $1^{14} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 5 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 14 & 26 & 0 & 24 & 118 & 20 \\ 3 & 4 & -2 & 4 & 20 & 2 \end{pmatrix}_{b_0}$$

[174] $1^{122} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 12 & 6 \\ 5 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -2 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[175] $1^{113} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 8 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -2 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{112} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 259, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.68 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 21 & 7 \\ 7 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -2 & 7 & 10 & 2 \end{pmatrix}_{b_0}$$

[178] $1^{113} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 27 & 7 \\ 9 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

[179] $1^{124} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 8 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 36 & 6 \\ 14 & 26 & 0 & 36 & 118 & 20 \\ 3 & 4 & -2 & 6 & 20 & 2 \end{pmatrix}_{b_0}$$

[177] $1^{12} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 5 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 24 & 4 \\ 14 & 23 & 0 & 24 & 100 & 18 \\ 3 & 4 & -2 & 4 & 18 & 2 \end{pmatrix}_{b_0}$$

- [180] $1^{111} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 12 & 3 \\ 3 & 4 & 0 & 9 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 33 & 7 \\ 12 & 19 & 0 & 33 & 73 & 15 \\ 3 & 4 & -2 & 7 & 15 & 2 \end{pmatrix}_{b_0}$
- [181] $21^{113} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 13 & 17 & 15 & 57 & 7 \\ 17 & 20 & 17 & 67 & 8 \\ 15 & 17 & 13 & 57 & 7 \\ 57 & 67 & 57 & 217 & 27 \\ 7 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$
- [184] $1^{13} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 5 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 21 & 4 \\ 12 & 22 & 0 & 21 & 85 & 17 \\ 3 & 4 & -2 & 4 & 17 & 2 \end{pmatrix}_{b_0}$
- [185] $1^{113} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 2 \end{pmatrix}_{b_0}$
- [183] $1^{123} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 4 & 0 & 13 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 51 & 10 \\ 12 & 22 & 0 & 51 & 85 & 17 \\ 3 & 4 & -2 & 10 & 17 & 2 \end{pmatrix}_{b_0}$
- [182] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 9 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 3 & 10 & 2 \end{pmatrix}_{b_0}$
- [186] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 270, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 18 & 6 \\ 8 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -2 & 6 & 10 & 2 \end{pmatrix}_{b_0}$
- [187] $1^{111} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 9 & 3 \\ 3 & 4 & 0 & 14 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 39 & 11 \\ 9 & 14 & 0 & 39 & 37 & 11 \\ 3 & 4 & -2 & 11 & 11 & 2 \end{pmatrix}_{b_0}$
- [188] $1^{114} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 9 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$
- [189] $1^{111} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 273, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 24 & 7 \\ 9 & 14 & 0 & 24 & 37 & 11 \\ 3 & 4 & -2 & 7 & 11 & 2 \end{pmatrix}_{b_0}$
- [190] $1^{124} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 0 & 8 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 30 & 6 \\ 11 & 21 & 0 & 30 & 76 & 16 \\ 3 & 4 & -2 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[191] \ 1^{111} 3 \ 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 9 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 33 & 7 \\ 12 & 19 & 0 & 33 & 73 & 15 \\ 3 & 4 & -2 & 7 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[192] \ 1^{112} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 277, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.66 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 8 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[193] \ 1^{113} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 27 & 7 \\ 11 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[194] \ 1^{124} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 13 & 3 \\ 3 & 4 & 0 & 13 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 48 & 10 \\ 13 & 21 & 0 & 48 & 76 & 16 \\ 3 & 4 & -2 & 10 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[195] \ 1^{112} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 281, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 9 & 3 \\ 8 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[196] \ 1^{133} 3 \ 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 293, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 12 & 3 \\ 3 & 4 & 0 & 17 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 66 & 13 \\ 12 & 22 & 0 & 66 & 85 & 17 \\ 3 & 4 & -2 & 13 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[197] \ 1^{132} 3 \ 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 298, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 11 & 3 \\ 3 & 4 & 0 & 17 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 54 & 13 \\ 11 & 18 & 0 & 54 & 58 & 14 \\ 3 & 4 & -2 & 13 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[199] \ 1^{123} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 18 & 6 \\ 7 & 12 & 0 & 18 & 25 & 9 \\ 3 & 4 & -2 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[198] \ 1^{111} 3 \ 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 3 \\ 3 & 4 & 0 & 14 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 39 & 11 \\ 8 & 14 & 0 & 39 & 37 & 11 \\ 3 & 4 & -2 & 11 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[200] \ 1^{122} 3 \ 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 0 & 13 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 42 & 10 \\ 11 & 18 & 0 & 42 & 58 & 14 \\ 3 & 4 & -2 & 10 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[201] 1^{122} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.68 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 8 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 30 & 10 \\ 8 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & -2 & 10 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[202] 1^{133} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 313, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 0 & 12 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 36 & 9 \\ 10 & 17 & 0 & 36 & 49 & 13 \\ 3 & 4 & -2 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[204] 1^{143} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 13 & 3 \\ 3 & 4 & 0 & 26 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 102 & 20 \\ 13 & 22 & 0 & 102 & 85 & 17 \\ 3 & 4 & -2 & 20 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[203] 1^{112} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 11 & 3 \\ 3 & 4 & 0 & 14 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 48 & 11 \\ 11 & 18 & 0 & 48 & 58 & 14 \\ 3 & 4 & -2 & 11 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[205] 1^{112} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.66 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 21 & 7 \\ 8 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -2 & 7 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[206] 1^{124} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 8 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 24 & 6 \\ 9 & 16 & 0 & 24 & 46 & 12 \\ 3 & 4 & -2 & 6 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[207] 1^{124} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 322, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 8 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 30 & 6 \\ 12 & 21 & 0 & 30 & 76 & 16 \\ 3 & 4 & -2 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[208] 1^{113} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 10 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 27 & 7 \\ 10 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[209] 1^{112} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 333, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 14 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & -2 & 3 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[210] 1^{12} 3 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 17 & 3 \\ 3 & 4 & 0 & 5 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 30 & 4 \\ 17 & 28 & 0 & 30 & 154 & 22 \\ 3 & 4 & -2 & 4 & 22 & 2 \end{pmatrix}_{b_0}$$

[211] $1^{112} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 340$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 12 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -2 & 3 & 14 & 2 \end{pmatrix}_{b_0}$$

[212] $1^{112} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 340$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.71 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 11 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -2 & 3 & 14 & 2 \end{pmatrix}_{b_0}$$

[213] $1^{112} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 341$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 15 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & -2 & 3 & 18 & 2 \end{pmatrix}_{b_0}$$

[214] $1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 342$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.73 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 11 & 18 & 0 & 18 & 58 & 14 \\ 3 & 4 & -2 & 4 & 14 & 2 \end{pmatrix}_{b_0}$$

[215] $1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 342$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.87 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 13 & 3 \\ 3 & 4 & 0 & 5 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 13 & 18 & 0 & 18 & 58 & 14 \\ 3 & 4 & -2 & 4 & 14 & 2 \end{pmatrix}_{b_0}$$

[217] $1^{12} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 344$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 13 & 3 \\ 3 & 4 & 0 & 5 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 13 & 23 & 0 & 24 & 100 & 18 \\ 3 & 4 & -2 & 4 & 18 & 2 \end{pmatrix}_{b_0}$$

[216] $1^{124} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 344$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 10.72 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 4 & 0 & 8 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 36 & 6 \\ 15 & 26 & 0 & 36 & 118 & 20 \\ 3 & 4 & -2 & 6 & 20 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{123} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 345$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.3$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 19 & 3 \\ 3 & 4 & 0 & 13 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 75 & 10 \\ 19 & 32 & 0 & 75 & 193 & 25 \\ 3 & 4 & -2 & 10 & 25 & 2 \end{pmatrix}_{b_0}$$

[219] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 349$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.87 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 11 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[221] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 353$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 18 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 21 & 3 \\ 18 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -2 & 3 & 24 & 2 \end{pmatrix}_{b_0}$$

[220] $1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 353$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.71 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 12 & 3 \\ 10 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[222] $1^{113} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 356$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 13 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 15 & 3 \\ 13 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -2 & 3 & 17 & 2 \end{pmatrix}_{b_0}$$

[223] $1^{143} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 361$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 10.74 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 13 & 3 \\ 3 & 4 & 0 & 21 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 81 & 16 \\ 13 & 22 & 0 & 81 & 85 & 17 \\ 3 & 4 & -2 & 16 & 17 & 2 \end{pmatrix}_{b_0}$$

[224] $1^{114} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 363$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 33 & 7 \\ 13 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -2 & 7 & 16 & 2 \end{pmatrix}_{b_0}$$

[225] $1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 364$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.71 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 15 & 7 \\ 6 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -2 & 7 & 7 & 2 \end{pmatrix}_{b_0}$$

[227] $1^{112} 3 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 364$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 10 & 3 \\ 3 & 4 & 0 & 14 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 48 & 11 \\ 10 & 18 & 0 & 48 & 58 & 14 \\ 3 & 4 & -2 & 11 & 14 & 2 \end{pmatrix}_{b_0}$$

[226] $1^{122} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 364$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 30 & 10 \\ 7 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & -2 & 10 & 10 & 2 \end{pmatrix}_{b_0}$$

[228] $1^{114} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 365$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 16 & 3 \\ 3 & 4 & 0 & 5 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 21 & 4 \\ 16 & 22 & 0 & 21 & 85 & 17 \\ 3 & 4 & -2 & 4 & 17 & 2 \end{pmatrix}_{b_0}$$

[229] $1^{113} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 369$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.87 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 4 & 0 & 9 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 36 & 7 \\ 15 & 22 & 0 & 36 & 85 & 17 \\ 3 & 4 & -2 & 7 & 17 & 2 \end{pmatrix}_{b_0}$$

[230] $1^{134} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 378$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 14 & 3 \\ 3 & 4 & 0 & 17 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 78 & 13 \\ 14 & 26 & 0 & 78 & 118 & 20 \\ 3 & 4 & -2 & 13 & 20 & 2 \end{pmatrix}_{b_0}$$

[231] $1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 5 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 12 & 18 & 0 & 18 & 58 & 14 \\ 3 & 4 & -2 & 4 & 14 & 2 \end{pmatrix}_{b_0}$$

[232] $1^{14} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 14 & 3 \\ 3 & 4 & 0 & 21 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 96 & 16 \\ 14 & 26 & 0 & 96 & 118 & 20 \\ 3 & 4 & -2 & 16 & 20 & 2 \end{pmatrix}_{b_0}$$

[234] $1^{14} 3^1 1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 10 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[233] $1^{13} 3^1 1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 12 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 27 & 9 \\ 7 & 12 & 0 & 27 & 25 & 9 \\ 3 & 4 & -2 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 388, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 27 & 7 \\ 10 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

[236] $1^{12} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 389, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 18 & 3 \\ 3 & 4 & 0 & 13 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 75 & 10 \\ 18 & 32 & 0 & 75 & 193 & 25 \\ 3 & 4 & -2 & 10 & 25 & 2 \end{pmatrix}_{b_0}$$

[237] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 10 & 3 \\ 3 & 4 & 0 & 17 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 51 & 13 \\ 10 & 17 & 0 & 51 & 49 & 13 \\ 3 & 4 & -2 & 13 & 13 & 2 \end{pmatrix}_{b_0}$$

[239] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 393, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 15 & 3 \\ 12 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[238] $1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 393, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 9 & 3 \\ 3 & 4 & 0 & 9 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 24 & 7 \\ 9 & 14 & 0 & 24 & 37 & 11 \\ 3 & 4 & -2 & 7 & 11 & 2 \end{pmatrix}_{b_0}$$

[240] $1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 394, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 0 & 13 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 42 & 10 \\ 10 & 18 & 0 & 42 & 58 & 14 \\ 3 & 4 & -2 & 10 & 14 & 2 \end{pmatrix}_{b_0}$$

[241] $1^{112} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 398, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 9 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 30 & 7 \\ 12 & 18 & 0 & 30 & 58 & 14 \\ 3 & 4 & -2 & 7 & 14 & 2 \end{pmatrix}_{b_0}$$

[242] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 11 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[243] $1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 401, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 13 & 3 \\ 3 & 4 & 0 & 5 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 21 & 4 \\ 13 & 22 & 0 & 21 & 85 & 17 \\ 3 & 4 & -2 & 4 & 17 & 2 \end{pmatrix}_{b_0}$$

[244] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 12 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -2 & 3 & 17 & 2 \end{pmatrix}_{b_0}$$

[245] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 407, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 33 & 7 \\ 12 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -2 & 7 & 16 & 2 \end{pmatrix}_{b_0}$$

[246] $1^{14} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 410, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 3 \\ 3 & 4 & 0 & 5 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 15 & 26 & 0 & 24 & 118 & 20 \\ 3 & 4 & -2 & 4 & 20 & 2 \end{pmatrix}_{b_0}$$

[248] $1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 414, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 24 & 6 \\ 10 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -2 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[247] $1^{112} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 414, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 9 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 30 & 7 \\ 10 & 18 & 0 & 30 & 58 & 14 \\ 3 & 4 & -2 & 7 & 14 & 2 \end{pmatrix}_{b_0}$$

[249] $1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 39 & 10 \\ 9 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -2 & 10 & 13 & 2 \end{pmatrix}_{b_0}$$

[250] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 421, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 10 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[251] $1^{124} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 424$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 0 & 13 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 48 & 10 \\ 11 & 21 & 0 & 48 & 76 & 16 \\ 3 & 4 & -2 & 10 & 16 & 2 \end{pmatrix}_{b_0}$$

[252] $1^{112} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 427$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.69 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 21 & 7 \\ 8 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -2 & 7 & 10 & 2 \end{pmatrix}_{b_0}$$

[253] $1^{134} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 432$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 12 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 36 & 9 \\ 9 & 16 & 0 & 36 & 46 & 12 \\ 3 & 4 & -2 & 9 & 12 & 2 \end{pmatrix}_{b_0}$$

[254] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 437$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 12 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[255] $1^{144} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 440$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 12 & 3 \\ 3 & 4 & 0 & 21 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 78 & 16 \\ 12 & 21 & 0 & 78 & 76 & 16 \\ 3 & 4 & -2 & 16 & 16 & 2 \end{pmatrix}_{b_0}$$

[256] $1^{14} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 446$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.87 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 18 & 3 \\ 3 & 4 & 0 & 5 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 18 & 26 & 0 & 24 & 118 & 20 \\ 3 & 4 & -2 & 4 & 20 & 2 \end{pmatrix}_{b_0}$$

[257] $1^{13} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 449$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 3 \\ 3 & 4 & 0 & 5 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 21 & 4 \\ 15 & 22 & 0 & 21 & 85 & 17 \\ 3 & 4 & -2 & 4 & 17 & 2 \end{pmatrix}_{b_0}$$

[258] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 452$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 18 & 3 \\ 15 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & -2 & 3 & 20 & 2 \end{pmatrix}_{b_0}$$

[259] $1^{13} 3^1 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 459$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 3 \\ 3 & 4 & 0 & 5 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 27 & 4 \\ 15 & 27 & 0 & 27 & 133 & 21 \\ 3 & 4 & -2 & 4 & 21 & 2 \end{pmatrix}_{b_0}$$

[260] $1^{13} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 461$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.72 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 5 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 21 & 4 \\ 14 & 22 & 0 & 21 & 85 & 17 \\ 3 & 4 & -2 & 4 & 17 & 2 \end{pmatrix}_{b_0}$$

[261] $1^{\overline{113}} 3^{\overline{114}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 18 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & -2 & 3 & 25 & 2 \end{pmatrix}_{b_0}$$

[262] $1^{\overline{113}} 3^{\overline{113}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 469, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 15 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & -2 & 3 & 21 & 2 \end{pmatrix}_{b_0}$$

[263] $1^{\overline{123}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 39 & 10 \\ 10 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -2 & 10 & 13 & 2 \end{pmatrix}_{b_0}$$

[264] $1^{\overline{113}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 477, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 11 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[265] $1^{\overline{112}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 17 & 3 \\ 3 & 4 & 0 & 9 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 48 & 7 \\ 17 & 28 & 0 & 48 & 154 & 22 \\ 3 & 4 & -2 & 7 & 22 & 2 \end{pmatrix}_{b_0}$$

[266] $1^{\overline{122}} 3^{\overline{122}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 11 & 3 \\ 3 & 4 & 0 & 18 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 60 & 14 \\ 11 & 18 & 0 & 60 & 58 & 14 \\ 3 & 4 & -2 & 14 & 14 & 2 \end{pmatrix}_{b_0}$$

[267] $1^{\overline{14}} 3^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 494, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 16 & 3 \\ 3 & 4 & 0 & 5 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 16 & 26 & 0 & 24 & 118 & 20 \\ 3 & 4 & -2 & 4 & 20 & 2 \end{pmatrix}_{b_0}$$

[268] $1^{\overline{134}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 497, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 12 & 3 \\ 3 & 4 & 0 & 12 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 45 & 9 \\ 12 & 21 & 0 & 45 & 76 & 16 \\ 3 & 4 & -2 & 9 & 16 & 2 \end{pmatrix}_{b_0}$$

[269] $1^{\overline{112}} 3^{\overline{113}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 499, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 9 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 39 & 7 \\ 13 & 23 & 0 & 39 & 100 & 18 \\ 3 & 4 & -2 & 7 & 18 & 2 \end{pmatrix}_{b_0}$$

[271] $21^{\overline{1111}} 31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 500, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 20 & 15 & 15 & 7 \\ 20 & 25 & 20 & 20 & 9 \\ 15 & 20 & 13 & 15 & 7 \\ 15 & 20 & 15 & 13 & 7 \\ 7 & 9 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[270] $1^{14}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 500, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 5 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 14 & 23 & 0 & 24 & 100 & 18 \\ 3 & 4 & -2 & 4 & 18 & 2 \end{pmatrix}_{b_0}$$

[272] $1^{14}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 506, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 3 \\ 3 & 4 & 0 & 5 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 17 & 26 & 0 & 24 & 118 & 20 \\ 3 & 4 & -2 & 4 & 20 & 2 \end{pmatrix}_{b_0}$$

[273] $1^{13}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 509, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 12 & 3 \\ 3 & 4 & 0 & 17 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 66 & 13 \\ 12 & 22 & 0 & 66 & 85 & 17 \\ 3 & 4 & -2 & 13 & 17 & 2 \end{pmatrix}_{b_0}$$

[274] $1^{11}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 514, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 16 & 3 \\ 3 & 4 & 0 & 9 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 48 & 7 \\ 16 & 28 & 0 & 48 & 154 & 22 \\ 3 & 4 & -2 & 7 & 22 & 2 \end{pmatrix}_{b_0}$$

[275] $1^{11}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 522, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 30 & 7 \\ 11 & 18 & 0 & 30 & 58 & 14 \\ 3 & 4 & -2 & 7 & 14 & 2 \end{pmatrix}_{b_0}$$

[277] $1^{13}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 524, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 0 & 17 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 51 & 13 \\ 9 & 17 & 0 & 51 & 49 & 13 \\ 3 & 4 & -2 & 13 & 13 & 2 \end{pmatrix}_{b_0}$$

[276] $1^{14}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 524, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 3 \\ 3 & 4 & 0 & 5 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 17 & 31 & 0 & 30 & 172 & 24 \\ 3 & 4 & -2 & 4 & 24 & 2 \end{pmatrix}_{b_0}$$

[278] $1^{12}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 526, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 8 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 30 & 6 \\ 12 & 21 & 0 & 30 & 76 & 16 \\ 3 & 4 & -2 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[279] $1^{12}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 15 & 3 \\ 3 & 4 & 0 & 18 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 84 & 14 \\ 15 & 26 & 0 & 84 & 118 & 20 \\ 3 & 4 & -2 & 14 & 20 & 2 \end{pmatrix}_{b_0}$$

[280] $1^{11}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 534, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 15 & 3 \\ 3 & 4 & 0 & 9 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 42 & 7 \\ 15 & 26 & 0 & 42 & 118 & 20 \\ 3 & 4 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

[281] $1^{12} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 3 \\ 3 & 4 & 0 & 5 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 17 & 23 & 0 & 24 & 100 & 18 \\ 3 & 4 & -2 & 4 & 18 & 2 \end{pmatrix}_{b_0}$$

[282] $1^{134} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 539, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 17 & 3 \\ 3 & 4 & 0 & 17 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 93 & 13 \\ 17 & 31 & 0 & 93 & 172 & 24 \\ 3 & 4 & -2 & 13 & 24 & 2 \end{pmatrix}_{b_0}$$

[283] $1^{124} 3^1 1^{22} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 14 & 3 \\ 3 & 4 & 0 & 18 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 84 & 14 \\ 14 & 26 & 0 & 84 & 118 & 20 \\ 3 & 4 & -2 & 14 & 20 & 2 \end{pmatrix}_{b_0}$$

[284] $1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 549, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 9 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 36 & 7 \\ 12 & 22 & 0 & 36 & 85 & 17 \\ 3 & 4 & -2 & 7 & 17 & 2 \end{pmatrix}_{b_0}$$

[285] $1^{111} 3^1 1^{22} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 560, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 0 & 14 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 14 & 0 & 37 & 39 & 11 \\ 9 & 14 & 0 & 39 & 37 & 11 \\ 3 & 4 & -2 & 11 & 11 & 2 \end{pmatrix}_{b_0}$$

[286] $1^{114} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 14 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & -2 & 3 & 20 & 2 \end{pmatrix}_{b_0}$$

[287] $1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 13 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -2 & 3 & 17 & 2 \end{pmatrix}_{b_0}$$

[288] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 565, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 12 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[289] $1^{144} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 574, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 14 & 3 \\ 3 & 4 & 0 & 21 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 96 & 16 \\ 14 & 26 & 0 & 96 & 118 & 20 \\ 3 & 4 & -2 & 16 & 20 & 2 \end{pmatrix}_{b_0}$$

[291] $1^{134} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 579, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 11 & 3 \\ 3 & 4 & 0 & 17 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 63 & 13 \\ 11 & 21 & 0 & 63 & 76 & 16 \\ 3 & 4 & -2 & 13 & 16 & 2 \end{pmatrix}_{b_0}$$

[290] $1^{112} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 579, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 4 & 0 & 9 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 39 & 7 \\ 15 & 23 & 0 & 39 & 100 & 18 \\ 3 & 4 & -2 & 7 & 18 & 2 \end{pmatrix}_{b_0}$$

[292] $1^{124} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 12 & 3 \\ 3 & 4 & 0 & 13 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 48 & 10 \\ 12 & 21 & 0 & 48 & 76 & 16 \\ 3 & 4 & -2 & 10 & 16 & 2 \end{pmatrix}_{b_0}$$

[293] $1^{112} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 584, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 3 \\ 3 & 4 & 0 & 5 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 15 & 23 & 0 & 24 & 100 & 18 \\ 3 & 4 & -2 & 4 & 18 & 2 \end{pmatrix}_{b_0}$$

[294] $1^{112} 3 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 11 & 3 \\ 3 & 4 & 0 & 14 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 14 & 0 & 37 & 48 & 11 \\ 11 & 18 & 0 & 48 & 58 & 14 \\ 3 & 4 & -2 & 11 & 14 & 2 \end{pmatrix}_{b_0}$$

[295] $1^{134} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 591, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 12 & 3 \\ 3 & 4 & 0 & 17 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 63 & 13 \\ 12 & 21 & 0 & 63 & 76 & 16 \\ 3 & 4 & -2 & 13 & 16 & 2 \end{pmatrix}_{b_0}$$

[296] $1^{113} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 15 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -2 & 3 & 17 & 2 \end{pmatrix}_{b_0}$$

[297] $1^{12} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 16 & 3 \\ 3 & 4 & 0 & 5 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 24 & 4 \\ 16 & 23 & 0 & 24 & 100 & 18 \\ 3 & 4 & -2 & 4 & 18 & 2 \end{pmatrix}_{b_0}$$

[298] $1^{124} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 598, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 22 & 3 \\ 3 & 4 & 0 & 13 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 84 & 10 \\ 22 & 36 & 0 & 84 & 238 & 28 \\ 3 & 4 & -2 & 10 & 28 & 2 \end{pmatrix}_{b_0}$$

[299] $1^{112} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 602, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 11 & 3 \\ 3 & 4 & 0 & 9 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 30 & 7 \\ 11 & 18 & 0 & 30 & 58 & 14 \\ 3 & 4 & -2 & 7 & 14 & 2 \end{pmatrix}_{b_0}$$

[300] $1^{123} 3 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 604, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 13 & 3 \\ 3 & 4 & 0 & 18 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 72 & 14 \\ 13 & 22 & 0 & 72 & 85 & 17 \\ 3 & 4 & -2 & 14 & 17 & 2 \end{pmatrix}_{b_0}$$

[301] $1^{124} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 606, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 16 & 3 \\ 3 & 4 & 0 & 13 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 60 & 10 \\ 16 & 26 & 0 & 60 & 118 & 20 \\ 3 & 4 & -2 & 10 & 20 & 2 \end{pmatrix}_{b_0}$$

[302] $1^{123} 3^1 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 607, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 16 & 3 \\ 3 & 4 & 0 & 13 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 63 & 10 \\ 16 & 27 & 0 & 63 & 133 & 21 \\ 3 & 4 & -2 & 10 & 21 & 2 \end{pmatrix}_{b_0}$$

[303] $1^{134} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 614, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 15 & 3 \\ 3 & 4 & 0 & 17 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 78 & 13 \\ 15 & 26 & 0 & 78 & 118 & 20 \\ 3 & 4 & -2 & 13 & 20 & 2 \end{pmatrix}_{b_0}$$

[304] $1^{124} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 616, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 4 & 0 & 13 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 48 & 10 \\ 12 & 21 & 0 & 48 & 76 & 16 \\ 3 & 4 & -2 & 10 & 16 & 2 \end{pmatrix}_{b_0}$$

[305] $1^{112} 3^1 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 619, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 14 & 3 \\ 3 & 4 & 0 & 9 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 39 & 7 \\ 14 & 23 & 0 & 39 & 100 & 18 \\ 3 & 4 & -2 & 7 & 18 & 2 \end{pmatrix}_{b_0}$$

[306] $1^{123} 3^1 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 619, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 15 & 3 \\ 3 & 4 & 0 & 13 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 63 & 10 \\ 15 & 27 & 0 & 63 & 133 & 21 \\ 3 & 4 & -2 & 10 & 21 & 2 \end{pmatrix}_{b_0}$$

[307] $1^{122} 3^1 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 624, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 10 & 3 \\ 3 & 4 & 0 & 18 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 60 & 14 \\ 10 & 18 & 0 & 60 & 58 & 14 \\ 3 & 4 & -2 & 14 & 14 & 2 \end{pmatrix}_{b_0}$$

[308] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 628, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 14 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -2 & 3 & 17 & 2 \end{pmatrix}_{b_0}$$

[310] $1^{123} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 12 & 3 \\ 3 & 4 & 0 & 13 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 51 & 10 \\ 12 & 22 & 0 & 51 & 85 & 17 \\ 3 & 4 & -2 & 10 & 17 & 2 \end{pmatrix}_{b_0}$$

[309] $1^{13} 3^1 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 18 & 3 \\ 3 & 4 & 0 & 5 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 33 & 4 \\ 18 & 32 & 0 & 33 & 193 & 25 \\ 3 & 4 & -2 & 4 & 25 & 2 \end{pmatrix}_{b_0}$$

[311] $1^{114} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 634, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 9 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 42 & 7 \\ 14 & 26 & 0 & 42 & 118 & 20 \\ 3 & 4 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

[312] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 15 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[313] $1^{113} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 16 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & -2 & 3 & 21 & 2 \end{pmatrix}_{b_0}$$

[314] $1^{12} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 638, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 3 \\ 3 & 4 & 0 & 5 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 17 & 28 & 0 & 30 & 154 & 22 \\ 3 & 4 & -2 & 4 & 22 & 2 \end{pmatrix}_{b_0}$$

[315] $1^{13} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 639, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 16 & 3 \\ 3 & 4 & 0 & 5 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 27 & 4 \\ 16 & 27 & 0 & 27 & 133 & 21 \\ 3 & 4 & -2 & 4 & 21 & 2 \end{pmatrix}_{b_0}$$

[316] $1^{113} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 640, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 19 & 3 \\ 3 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 19 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & -2 & 3 & 25 & 2 \end{pmatrix}_{b_0}$$

[317] $1^{122} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 11 & 3 \\ 3 & 4 & 0 & 13 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 42 & 10 \\ 11 & 18 & 0 & 42 & 58 & 14 \\ 3 & 4 & -2 & 10 & 14 & 2 \end{pmatrix}_{b_0}$$

[318] $1^{114} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 17 & 3 \\ 3 & 4 & 0 & 9 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 42 & 7 \\ 17 & 26 & 0 & 42 & 118 & 20 \\ 3 & 4 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

[319] $1^{123} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 649, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 13 & 3 \\ 3 & 4 & 0 & 13 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 51 & 10 \\ 13 & 22 & 0 & 51 & 85 & 17 \\ 3 & 4 & -2 & 10 & 17 & 2 \end{pmatrix}_{b_0}$$

[320] $1^{113} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 649, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 13 & 3 \\ 3 & 4 & 0 & 9 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 36 & 7 \\ 13 & 22 & 0 & 36 & 85 & 17 \\ 3 & 4 & -2 & 7 & 17 & 2 \end{pmatrix}_{b_0}$$

[321] $1^{122} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 652, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 30 & 10 \\ 8 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & -2 & 10 & 10 & 2 \end{pmatrix}_{b_0}$$

[322] $1^{112} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 655, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 9 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 39 & 7 \\ 14 & 23 & 0 & 39 & 100 & 18 \\ 3 & 4 & -2 & 7 & 18 & 2 \end{pmatrix}_{b_0}$$

[323] $1^{133} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 13 & 3 \\ 3 & 4 & 0 & 22 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 87 & 17 \\ 13 & 22 & 0 & 87 & 85 & 17 \\ 3 & 4 & -2 & 17 & 17 & 2 \end{pmatrix}_{b_0}$$

[324] $1^{113} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 661, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 9 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 36 & 7 \\ 14 & 22 & 0 & 36 & 85 & 17 \\ 3 & 4 & -2 & 7 & 17 & 2 \end{pmatrix}_{b_0}$$

[325] $1^{124} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 15 & 3 \\ 3 & 4 & 0 & 18 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 18 & 0 & 58 & 84 & 14 \\ 15 & 26 & 0 & 84 & 118 & 20 \\ 3 & 4 & -2 & 14 & 20 & 2 \end{pmatrix}_{b_0}$$

[327] $1^{144} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 684, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 11 & 3 \\ 3 & 4 & 0 & 21 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 78 & 16 \\ 11 & 21 & 0 & 78 & 76 & 16 \\ 3 & 4 & -2 & 16 & 16 & 2 \end{pmatrix}_{b_0}$$

[326] $1^{123} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 684, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 12 & 3 \\ 3 & 4 & 0 & 18 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 72 & 14 \\ 12 & 22 & 0 & 72 & 85 & 17 \\ 3 & 4 & -2 & 14 & 17 & 2 \end{pmatrix}_{b_0}$$

[329] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 685, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 13 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[328] $1^{113} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 685, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 18 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & -2 & 3 & 21 & 2 \end{pmatrix}_{b_0}$$

[330] $1^{112} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 690, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 18 & 3 \\ 3 & 4 & 0 & 9 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 48 & 7 \\ 18 & 28 & 0 & 48 & 154 & 22 \\ 3 & 4 & -2 & 7 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[331] 1^{113} 3^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 704, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 21 & 3 \\ 3 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 21 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & -2 & 3 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[334] 1^{113} 3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 709, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 17 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & -2 & 3 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[332] 1^{114} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 709, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 14 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[333] 1^{114} 3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 709, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 17 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -2 & 3 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[335] 1^{124} 3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 712, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 19 & 3 \\ 3 & 4 & 0 & 13 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 72 & 10 \\ 19 & 31 & 0 & 72 & 172 & 24 \\ 3 & 4 & -2 & 10 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[336] 1^{123} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 715, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 10 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 39 & 10 \\ 10 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -2 & 10 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[337] 1^{112} 3^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 718, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 17 & 3 \\ 3 & 4 & 0 & 9 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 48 & 7 \\ 17 & 28 & 0 & 48 & 154 & 22 \\ 3 & 4 & -2 & 7 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[338] 1^{113} 3^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 720, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 20 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & -2 & 3 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[339] 1^{113} 3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 721, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 9 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 36 & 7 \\ 13 & 22 & 0 & 36 & 85 & 17 \\ 3 & 4 & -2 & 7 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[340] 1^{114} 3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 723, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 21 & 3 \\ 3 & 4 & 0 & 9 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 51 & 7 \\ 21 & 31 & 0 & 51 & 172 & 24 \\ 3 & 4 & -2 & 7 & 24 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 18 & 3 \\ 3 & 4 & 0 & 5 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 18 & 31 & 0 & 30 & 172 & 24 \\ 3 & 4 & -2 & 4 & 24 & 2 \end{pmatrix} b_0$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 18 & 3 \\ 3 & 4 & 0 & 9 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 45 & 7 \\ 18 & 27 & 0 & 45 & 133 & 21 \\ 3 & 4 & -2 & 7 & 21 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 19 & 3 \\ 3 & 4 & 0 & 5 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 27 & 4 \\ 19 & 27 & 0 & 27 & 133 & 21 \\ 3 & 4 & -2 & 4 & 21 & 2 \end{pmatrix}_{b_1}$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 3 \\ 3 & 4 & 0 & 5 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 27 & 4 \\ 17 & 27 & 0 & 27 & 133 & 21 \\ 3 & 4 & -2 & 4 & 21 & 2 \end{pmatrix}_L$$

$$n_{\min} = 10.72 \rightarrow 11, T_{\max} \begin{pmatrix} 1 & 3 & 1 & 10 & 18 & 3 \\ 3 & 4 & 0 & 17 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 93 & 13 \\ 18 & 31 & 0 & 93 & 172 & 24 \\ 3 & 4 & -2 & 13 & 24 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 4 & 0 & 9 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 45 & 7 \\ 15 & 27 & 0 & 45 & 133 & 21 \\ 3 & 4 & -2 & 7 & 21 & 2 \end{pmatrix} b_0$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 18 & 3 \\ 3 & 4 & 0 & 5 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 18 & 28 & 0 & 30 & 154 & 22 \\ 3 & 4 & -2 & 4 & 22 & 2 \end{pmatrix}_b$$

$$\min = 10.02 \quad / \quad 11, \quad 1_{\max} = \begin{pmatrix} 1 & 3 & 1 & 3 & 20 & 3 \\ 3 & 4 & 0 & 5 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 36 & 4 \\ 20 & 36 & 0 & 36 & 238 & 28 \\ 3 & 4 & -2 & 4 & 28 & 2 \end{pmatrix},$$

$$b] \ 1^{\textcolor{red}{13}} 3 \ 1^{\textcolor{blue}{11} \textcolor{red}{3}} + [\textcolor{red}{5} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] \cdot$$

$$T_{\min} = 10.79 \rightarrow 11, \quad T_{\max} = \begin{pmatrix} 1 & 3 & 1 & 3 & 18 & 3 \\ 3 & 4 & 0 & 5 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 27 & 4 \\ 18 & 27 & 0 & 27 & 133 & 21 \end{pmatrix}$$

$$\left(\begin{array}{cccccc} 18 & 27 & 0 & 27 & 133 & 21 \\ 3 & 4 & -2 & 4 & 21 & 2 \end{array} \right)_{b_0}$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 24 & 3 \\ 20 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & -2 & 3 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[351] \ 1^{123} 1^{111} 3 \ 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 787, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 16 & 3 \\ 3 & 4 & 0 & 13 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 63 & 10 \\ 16 & 27 & 0 & 63 & 133 & 21 \\ 3 & 4 & -2 & 10 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[352] \ 1^{114} 1^{11} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 788, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 15 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & -2 & 3 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[353] \ 1^{134} 1^{11} 3 \ 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 794, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 14 & 3 \\ 3 & 4 & 0 & 17 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 78 & 13 \\ 14 & 26 & 0 & 78 & 118 & 20 \\ 3 & 4 & -2 & 13 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[354] \ 1^{144} 1^{11} 3 \ 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 15 & 3 \\ 3 & 4 & 0 & 26 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 120 & 20 \\ 15 & 26 & 0 & 120 & 118 & 20 \\ 3 & 4 & -2 & 20 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[355] \ 1^{113} 1^{111} 3 \ 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 812, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 16 & 3 \\ 3 & 4 & 0 & 9 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 45 & 7 \\ 16 & 27 & 0 & 45 & 133 & 21 \\ 3 & 4 & -2 & 7 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[356] \ 1^{124} 1^{11} 3 \ 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 814, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 14 & 3 \\ 3 & 4 & 0 & 13 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 60 & 10 \\ 14 & 26 & 0 & 60 & 118 & 20 \\ 3 & 4 & -2 & 10 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[357] \ 1^{113} 1^{111} 3 \ 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 817, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 19 & 3 \\ 3 & 4 & 0 & 9 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 54 & 7 \\ 19 & 32 & 0 & 54 & 193 & 25 \\ 3 & 4 & -2 & 7 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[358] \ 1^{12} 1^{14} 3 \ 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 818, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 20 & 3 \\ 3 & 4 & 0 & 5 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 20 & 28 & 0 & 30 & 154 & 22 \\ 3 & 4 & -2 & 4 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[359] \ 1^{12} 1^{13} 3 \ 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 830, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 19 & 3 \\ 3 & 4 & 0 & 5 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 19 & 28 & 0 & 30 & 154 & 22 \\ 3 & 4 & -2 & 4 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[360] \ 1^{114} 1^{111} 3 \ 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 843, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} =$$

10.0

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 18 & 3 \\ 3 & 4 & 0 & 9 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 51 & 7 \\ 18 & 31 & 0 & 51 & 172 & 24 \\ 3 & 4 & -2 & 7 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[361] \ 1^{133} 3 \ 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 849, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 13 & 3 \\ 3 & 4 & 0 & 17 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 66 & 13 \\ 13 & 22 & 0 & 66 & 85 & 17 \\ 3 & 4 & -2 & 13 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[362] \ 1^{124} 3 \ 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 850, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 21 & 3 \\ 3 & 4 & 0 & 13 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 84 & 10 \\ 21 & 36 & 0 & 84 & 238 & 28 \\ 3 & 4 & -2 & 10 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[363] \ 1^{13} 3 \ 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 857, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 19 & 3 \\ 3 & 4 & 0 & 5 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 33 & 4 \\ 19 & 32 & 0 & 33 & 193 & 25 \\ 3 & 4 & -2 & 4 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[364] \ 1^{134} 3 \ 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 860, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 15 & 3 \\ 3 & 4 & 0 & 22 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 102 & 17 \\ 15 & 26 & 0 & 102 & 118 & 20 \\ 3 & 4 & -2 & 17 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[365] \ 1^{14} 3 \ 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 860, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 19 & 3 \\ 3 & 4 & 0 & 5 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 19 & 31 & 0 & 30 & 172 & 24 \\ 3 & 4 & -2 & 4 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[366] \ 1^{114} 3 \ 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 870, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 4 & 0 & 9 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 42 & 7 \\ 15 & 26 & 0 & 42 & 118 & 20 \\ 3 & 4 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[367] \ 1^{114} 3 \ 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 874, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 16 & 3 \\ 3 & 4 & 0 & 9 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 42 & 7 \\ 16 & 26 & 0 & 42 & 118 & 20 \\ 3 & 4 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[368] \ 1^{114} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 884, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 18 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & -2 & 3 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[369] \ 1^{14} 3 \ 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 908, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 21 & 3 \\ 3 & 4 & 0 & 5 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 21 & 31 & 0 & 30 & 172 & 24 \\ 3 & 4 & -2 & 4 & 24 & 2 \end{pmatrix}_{b_0}$$

[370] $1^{113} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 909, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 18 & 3 \\ 3 & 4 & 0 & 9 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 54 & 7 \\ 18 & 32 & 0 & 54 & 193 & 25 \\ 3 & 4 & -2 & 7 & 25 & 2 \end{pmatrix}_{b_0}$$

[371] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 916, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 16 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & -2 & 3 & 20 & 2 \end{pmatrix}_{b_0}$$

[372] $1^{14} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 920, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 20 & 3 \\ 3 & 4 & 0 & 5 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 30 & 4 \\ 20 & 31 & 0 & 30 & 172 & 24 \\ 3 & 4 & -2 & 4 & 24 & 2 \end{pmatrix}_{b_0}$$

[373] $1^{124} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 934, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 20 & 3 \\ 3 & 4 & 0 & 13 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 84 & 10 \\ 20 & 36 & 0 & 84 & 238 & 28 \\ 3 & 4 & -2 & 10 & 28 & 2 \end{pmatrix}_{b_0}$$

[374] $1^{124} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 938, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 15 & 3 \\ 3 & 4 & 0 & 13 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 60 & 10 \\ 15 & 26 & 0 & 60 & 118 & 20 \\ 3 & 4 & -2 & 10 & 20 & 2 \end{pmatrix}_{b_0}$$

[375] $1^{133} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 940, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 10 & 3 \\ 3 & 4 & 0 & 17 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 51 & 13 \\ 10 & 17 & 0 & 51 & 49 & 13 \\ 3 & 4 & -2 & 13 & 13 & 2 \end{pmatrix}_{b_0}$$

[376] $1^{114} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 941, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 18 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -2 & 3 & 24 & 2 \end{pmatrix}_{b_0}$$

[377] $1^{133} 3 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 944, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 12 & 3 \\ 3 & 4 & 0 & 22 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 87 & 17 \\ 12 & 22 & 0 & 87 & 85 & 17 \\ 3 & 4 & -2 & 17 & 17 & 2 \end{pmatrix}_{b_0}$$

[378] $1^{123} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 945, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 13 & 3 \\ 3 & 4 & 0 & 13 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 51 & 10 \\ 13 & 22 & 0 & 51 & 85 & 17 \\ 3 & 4 & -2 & 10 & 17 & 2 \end{pmatrix}_{b_0}$$

[379] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 948, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 18 & 3 \\ 17 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & -2 & 3 & 20 & 2 \end{pmatrix}_{b_0}$$

[380] $1^{113} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 972, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 17 & 3 \\ 3 & 4 & 0 & 9 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 45 & 7 \\ 17 & 27 & 0 & 45 & 133 & 21 \\ 3 & 4 & -2 & 7 & 21 & 2 \end{pmatrix}_{b_0}$$

[381] $1^{114} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 979, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 17 & 3 \\ 3 & 4 & 0 & 9 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 51 & 7 \\ 17 & 31 & 0 & 51 & 172 & 24 \\ 3 & 4 & -2 & 7 & 24 & 2 \end{pmatrix}_{b_0}$$

[382] $1^{113} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 984, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 16 & 3 \\ 3 & 4 & 0 & 9 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 45 & 7 \\ 16 & 27 & 0 & 45 & 133 & 21 \\ 3 & 4 & -2 & 7 & 21 & 2 \end{pmatrix}_{b_0}$$

[383] $1^{124} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 984, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 17 & 3 \\ 3 & 4 & 0 & 13 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 72 & 10 \\ 17 & 31 & 0 & 72 & 172 & 24 \\ 3 & 4 & -2 & 10 & 24 & 2 \end{pmatrix}_{b_0}$$

[384] $1^{124} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 994, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 21 & 3 \\ 3 & 4 & 0 & 13 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 84 & 10 \\ 21 & 36 & 0 & 84 & 238 & 28 \\ 3 & 4 & -2 & 10 & 28 & 2 \end{pmatrix}_{b_0}$$

[385] $1^{114} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 994, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 21 & 3 \\ 3 & 4 & 0 & 9 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 60 & 7 \\ 21 & 36 & 0 & 60 & 238 & 28 \\ 3 & 4 & -2 & 7 & 28 & 2 \end{pmatrix}_{b_0}$$

[386] $1^{144} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1002, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 15 & 3 \\ 3 & 4 & 0 & 21 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 96 & 16 \\ 15 & 26 & 0 & 96 & 118 & 20 \\ 3 & 4 & -2 & 16 & 20 & 2 \end{pmatrix}_{b_0}$$

[387] $1^{134} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1003, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 12 & 3 \\ 3 & 4 & 0 & 17 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 63 & 13 \\ 12 & 21 & 0 & 63 & 76 & 16 \\ 3 & 4 & -2 & 13 & 16 & 2 \end{pmatrix}_{b_0}$$

[388] $1^{134} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1004, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 14 & 3 \\ 3 & 4 & 0 & 22 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 102 & 17 \\ 14 & 26 & 0 & 102 & 118 & 20 \\ 3 & 4 & -2 & 17 & 20 & 2 \end{pmatrix}_{b_0}$$

[389] $1^{122} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1008, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 11 & 3 \\ 3 & 4 & 0 & 18 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 18 & 0 & 58 & 60 & 14 \\ 11 & 18 & 0 & 60 & 58 & 14 \\ 3 & 4 & -2 & 14 & 14 & 2 \end{pmatrix}_{b_0}$$

[390] $1^{12} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1013, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 20 & 3 \\ 3 & 4 & 0 & 5 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 33 & 4 \\ 20 & 32 & 0 & 33 & 193 & 25 \\ 3 & 4 & -2 & 4 & 25 & 2 \end{pmatrix}_{b_0}$$

[391] $1^{113} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1017, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 21 & 3 \\ 3 & 4 & 0 & 9 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 54 & 7 \\ 21 & 32 & 0 & 54 & 193 & 25 \\ 3 & 4 & -2 & 7 & 25 & 2 \end{pmatrix}_{b_0}$$

[392] $1^{114} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1024, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 21 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 24 & 3 \\ 21 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & -2 & 3 & 28 & 2 \end{pmatrix}_{b_0}$$

[393] $1^{14} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1026, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 21 & 3 \\ 3 & 4 & 0 & 5 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 36 & 4 \\ 21 & 36 & 0 & 36 & 238 & 28 \\ 3 & 4 & -2 & 4 & 28 & 2 \end{pmatrix}_{b_0}$$

[394] $1^{123} 3 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1036, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 13 & 3 \\ 3 & 4 & 0 & 18 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 18 & 0 & 58 & 72 & 14 \\ 13 & 22 & 0 & 72 & 85 & 17 \\ 3 & 4 & -2 & 14 & 17 & 2 \end{pmatrix}_{b_0}$$

[395] $1^{114} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1061, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 21 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 21 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -2 & 3 & 24 & 2 \end{pmatrix}_{b_0}$$

[396] $1^{124} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1076, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 18 & 3 \\ 3 & 4 & 0 & 13 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 72 & 10 \\ 18 & 31 & 0 & 72 & 172 & 24 \\ 3 & 4 & -2 & 10 & 24 & 2 \end{pmatrix}_{b_0}$$

[397] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1077, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 19 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 19 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -2 & 3 & 24 & 2 \end{pmatrix}_{b_0}$$

[398] $1^{124} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1082, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 15 & 3 \\ 3 & 4 & 0 & 13 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 60 & 10 \\ 15 & 26 & 0 & 60 & 118 & 20 \\ 3 & 4 & -2 & 10 & 20 & 2 \end{pmatrix}_{b_0}$$

[399] $1^{13} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1097, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 21 & 3 \\ 3 & 4 & 0 & 5 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 33 & 4 \\ 21 & 32 & 0 & 33 & 193 & 25 \\ 3 & 4 & -2 & 4 & 25 & 2 \end{pmatrix}_{b_0}$$

[400] $1^{13} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 22 & 3 \\ 3 & 4 & 0 & 5 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 33 & 4 \\ 22 & 32 & 0 & 33 & 193 & 25 \\ 3 & 4 & -2 & 4 & 25 & 2 \end{pmatrix}_{b_0}$$

[401] $1^{14} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 21 & 3 \\ 20 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -2 & 3 & 24 & 2 \end{pmatrix}_{b_0}$$

[402] $1^{14} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 20 & 3 \\ 3 & 4 & 0 & 9 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 51 & 7 \\ 20 & 31 & 0 & 51 & 172 & 24 \\ 3 & 4 & -2 & 7 & 24 & 2 \end{pmatrix}_{b_0}$$

[403] $1^{124} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 18 & 3 \\ 3 & 4 & 0 & 13 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 72 & 10 \\ 18 & 31 & 0 & 72 & 172 & 24 \\ 3 & 4 & -2 & 10 & 24 & 2 \end{pmatrix}_{b_0}$$

[405] $1^{14} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 24 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 24 & 3 \\ 24 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & -2 & 3 & 28 & 2 \end{pmatrix}_{b_0}$$

[404] $1^{14} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 22 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 24 & 3 \\ 22 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & -2 & 3 & 28 & 2 \end{pmatrix}_{b_0}$$

[406] $1^{113} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1177, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 19 & 3 \\ 3 & 4 & 0 & 9 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 54 & 7 \\ 19 & 32 & 0 & 54 & 193 & 25 \\ 3 & 4 & -2 & 7 & 25 & 2 \end{pmatrix}_{b_0}$$

[407] $1^{14} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1190, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 24 & 3 \\ 3 & 4 & 0 & 9 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 60 & 7 \\ 24 & 36 & 0 & 60 & 238 & 28 \\ 3 & 4 & -2 & 7 & 28 & 2 \end{pmatrix}_{b_0}$$

[408] $1^{1^{12}} 3 1^{1^{14}} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 1206$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 22 & 3 \\ 3 & 4 & 0 & 5 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 36 & 4 \\ 22 & 36 & 0 & 36 & 238 & 28 \\ 3 & 4 & -2 & 4 & 28 & 2 \end{pmatrix}_{b_0}$$

[409] $1^{1^{12}} 3 1^{1^{14}} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 1213$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 20 & 3 \\ 3 & 4 & 0 & 9 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 54 & 7 \\ 20 & 32 & 0 & 54 & 193 & 25 \\ 3 & 4 & -2 & 7 & 25 & 2 \end{pmatrix}_{b_0}$$

[410] $1^{1^{13}} 3 1^{1^{14}} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 1216$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 23 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 24 & 3 \\ 23 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & -2 & 3 & 28 & 2 \end{pmatrix}_{b_0}$$

[411] $1^{1^{14}} 3 1^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 1228$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 10.76 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 12 & 3 \\ 3 & 4 & 0 & 21 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 78 & 16 \\ 12 & 21 & 0 & 78 & 76 & 16 \\ 3 & 4 & -2 & 16 & 16 & 2 \end{pmatrix}_{b_0}$$

[412] $1^{1^{14}} 3 1^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 1238$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 10.72 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 15 & 3 \\ 3 & 4 & 0 & 17 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 78 & 13 \\ 15 & 26 & 0 & 78 & 118 & 20 \\ 3 & 4 & -2 & 13 & 20 & 2 \end{pmatrix}_{b_0}$$

[413] $1^{1^{14}} 3 1^{1^{14}} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 1254$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 20 & 3 \\ 3 & 4 & 0 & 9 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 60 & 7 \\ 20 & 36 & 0 & 60 & 238 & 28 \\ 3 & 4 & -2 & 7 & 28 & 2 \end{pmatrix}_{b_0}$$

[414] $1^{1^{14}} 3 1^{1^{13}} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 1263$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 18 & 3 \\ 3 & 4 & 0 & 9 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 51 & 7 \\ 18 & 31 & 0 & 51 & 172 & 24 \\ 3 & 4 & -2 & 7 & 24 & 2 \end{pmatrix}_{b_0}$$

[415] $1^{1^{14}} 3 1^{1^{22}} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 1264$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 14 & 3 \\ 3 & 4 & 0 & 26 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 120 & 20 \\ 14 & 26 & 0 & 120 & 118 & 20 \\ 3 & 4 & -2 & 20 & 20 & 2 \end{pmatrix}_{b_0}$$

[416] $1^{1^{13}} 3 1^{1^{14}} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 1314$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 23 & 3 \\ 3 & 4 & 0 & 5 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 36 & 4 \\ 23 & 36 & 0 & 36 & 238 & 28 \\ 3 & 4 & -2 & 4 & 28 & 2 \end{pmatrix}_{b_0}$$

[417] $1^{1^{12}} 3 1^{1^{13}} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 1315$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 19 & 3 \\ 3 & 4 & 0 & 9 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 51 & 7 \\ 19 & 31 & 0 & 51 & 172 & 24 \\ 3 & 4 & -2 & 7 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[418] \ 1^{14} 3 \ 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1350, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 24 & 3 \\ 3 & 4 & 0 & 5 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 36 & 4 \\ 24 & 36 & 0 & 36 & 238 & 28 \\ 3 & 4 & -2 & 4 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[419] \ 1^{133} 3 \ 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 13 & 3 \\ 3 & 4 & 0 & 22 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 22 & 0 & 85 & 87 & 17 \\ 13 & 22 & 0 & 87 & 85 & 17 \\ 3 & 4 & -2 & 17 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[420] \ 1^{134} 3 \ 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1484, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 15 & 3 \\ 3 & 4 & 0 & 22 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 22 & 0 & 85 & 102 & 17 \\ 15 & 26 & 0 & 102 & 118 & 20 \\ 3 & 4 & -2 & 17 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[421] \ 1^{114} 3 \ 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1554, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 23 & 3 \\ 3 & 4 & 0 & 9 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 60 & 7 \\ 23 & 36 & 0 & 60 & 238 & 28 \\ 3 & 4 & -2 & 7 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[422] \ 1^{114} 3 \ 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1586, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 21 & 3 \\ 3 & 4 & 0 & 9 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 60 & 7 \\ 21 & 36 & 0 & 60 & 238 & 28 \\ 3 & 4 & -2 & 7 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[423] \ 1^{114} 3 \ 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1686, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 22 & 3 \\ 3 & 4 & 0 & 9 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 60 & 7 \\ 22 & 36 & 0 & 60 & 238 & 28 \\ 3 & 4 & -2 & 7 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[424] \ 1^{144} 3 \ 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1904, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 15 & 3 \\ 3 & 4 & 0 & 26 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 26 & 0 & 118 & 120 & 20 \\ 15 & 26 & 0 & 120 & 118 & 20 \\ 3 & 4 & -2 & 20 & 20 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [425] \ 1^{111} 31^1 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 1 & 3 & 3 & 0 & 3 & 1 \\ 2 & 4 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

4.670 NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -50, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 17 & 0 & 0 & 13 & 7 \\ 5 & 8 & -2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -72, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 10 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 10 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -2 & 7 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -76, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.89 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 15 & 7 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -2 & 7 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 16 & 0 & 0 & 10 & 6 \\ 5 & 8 & -2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -232, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 18 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 13 & 27 & 7 \\ 18 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 \ 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -242, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 36 & 5 \\ 11 & 20 & 0 & 0 & 67 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 36 & 67 & 0 & 0 & 217 & 27 \\ 5 & 8 & -2 & 0 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -248, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.85 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 19 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 19 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -258, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 18 & 5 \\ 11 & 20 & 0 & 0 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 18 & 33 & 0 & 0 & 49 & 13 \\ 5 & 8 & -2 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3 \ 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -296, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 27 & 5 \\ 11 & 20 & 0 & 0 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 27 & 50 & 0 & 0 & 118 & 20 \\ 5 & 8 & -2 & 0 & 20 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{111} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -342, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 27 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 13 & 42 & 7 \\ 27 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{111} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 26 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 26 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

[12] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 18 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 27 & 7 \\ 18 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{111} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -522, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 28 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 28 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

[14] $1^{111} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -550, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 27 & 5 \\ 11 & 20 & 0 & 17 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 42 & 7 \\ 27 & 50 & 0 & 42 & 118 & 20 \\ 5 & 8 & -2 & 7 & 20 & 2 \end{pmatrix}_{b_0}$$

4.671 NHC(-2)(-3) + so₈ + su₂ + su₂ + su₃

$T = 9$ [0] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 11, \Delta = -62, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 4 & 5 & 0 & 6 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 11, \Delta = -62, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 5 & 0 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 0 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 3 & 5 & 0 & 6 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

4.672 NHC(-2)(-3) + so₈ + su₂ + su₂ → (-3)

$$T = 8 \quad [0] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 22, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 16 & 23 & 8 & 6 \\ 16 & 25 & 36 & 12 & 9 \\ 23 & 36 & 49 & 17 & 13 \\ 8 & 12 & 17 & 4 & 4 \\ 6 & 9 & 13 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 2 \\ 3 & 4 & -2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 4 & 6 \\ 2 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 49, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 8 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 12 & 17 & 13 & 9 & 7 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 54, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.70 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 59, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 1 & 8 & 6 \\ 15 & 20 & 0 & 11 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 11 & 0 & 4 & 4 \\ 6 & 8 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 61, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.62 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 66, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 13 & 8 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 13 & 17 & 13 & 9 & 7 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[9] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 5 & 7 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 4 & 6 \\ 2 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 6 & 9 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[12] $1^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 14 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 8 & 11 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[14] $1^{14}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 16 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 46 & 8 & 12 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 7 & 10 \\ 2 & 3 & 0 & 7 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$\begin{aligned}
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 17 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 9 & 13 \\ 2 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 13 & 2 & 2 \end{pmatrix}_{b_0} \\
[16] & 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 5 & 7 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 7 & 2 & 2 \end{pmatrix}_{b_0} \\
[18] & 1^{14} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 11 & 16 \\ 2 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & -2 & 16 & 2 & 2 \end{pmatrix}_{b_0} \\
[19] & 21^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 13 & 20 & 27 & 9 & 7 \\ 20 & 25 & 36 & 12 & 9 \\ 27 & 36 & 49 & 17 & 13 \\ 9 & 12 & 17 & 4 & 4 \\ 7 & 9 & 13 & 4 & 3 \end{pmatrix}_{b_0} \\
[20] & 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 18 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 10 & 14 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 3 & 4 & -2 & 14 & 2 & 2 \end{pmatrix}_{b_0} \\
[21] & 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 14 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 14 & 0 & 37 & 8 & 11 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 11 & 2 & 2 \end{pmatrix}_{b_0} \\
[22] & 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 306, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 13 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 7 & 10 \\ 2 & 3 & 0 & 7 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 2 \end{pmatrix}_{b_0} \\
[23] & 1^{11} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 311, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 19 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 19 & 0 & 73 & 11 & 15 \\ 2 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & -2 & 15 & 2 & 2 \end{pmatrix}_{b_0} \\
[24] & 21^{11} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 13 & 20 & 15 & 9 & 7 \\ 20 & 25 & 20 & 12 & 9 \\ 15 & 20 & 13 & 9 & 7 \\ 9 & 12 & 9 & 4 & 4 \\ 7 & 9 & 7 & 4 & 3 \end{pmatrix}_{b_0} \\
[25] & 1^{12} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 23 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 23 & 0 & 100 & 13 & 18 \\ 2 & 3 & 0 & 13 & 0 & 2 \\ 3 & 4 & -2 & 18 & 2 & 2 \end{pmatrix}_{b_0} \\
[26] & 1^{13} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 22 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 12 & 17 \\ 2 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 17 & 2 & 2 \end{pmatrix}_{b_0} \\
[27] & 1^{12} 3^{11} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 28 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 28 & 0 & 154 & 16 & 22 \\ 2 & 3 & 0 & 16 & 0 & 2 \\ 3 & 4 & -2 & 22 & 2 & 2 \end{pmatrix}_{b_0} \\
[28] & 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 399, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 17 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 9 & 13 \\ 2 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 13 & 2 & 2 \end{pmatrix}_{b_0} \\
[29] & 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 18 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 18 & 0 & 58 & 10 & 14 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 3 & 4 & -2 & 14 & 2 & 2 \end{pmatrix}_{b_0} \\
[30] & 1^{14} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 26 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 14 & 20 \\ 2 & 3 & 0 & 14 & 0 & 2 \\ 3 & 4 & -2 & 20 & 2 & 2 \end{pmatrix}_{b_0} \\
[31] & 1^{14} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 474, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 11 & 16 \\ 2 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & -2 & 16 & 2 & 2 \end{pmatrix}_{b_0} \\
[32] & 1^{13} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 511, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8 \\
& T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 27 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 27 & 0 & 133 & 15 & 21 \\ 2 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 21 & 2 & 2 \end{pmatrix}_{b_0} \\
[33] & 1^{12} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 570, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 23 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 23 & 0 & 100 & 13 & 18 \\ 2 & 3 & 0 & 13 & 0 & 2 \\ 3 & 4 & -2 & 18 & 2 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[34] $1^{13}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 621, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 0 & 32 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 32 & 0 & 193 & 18 & 25 \\ 2 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & -2 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 621, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 22 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 22 & 0 & 85 & 12 & 17 \\ 2 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{12}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 654, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 28 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 28 & 0 & 154 & 16 & 22 \\ 2 & 3 & 0 & 16 & 0 & 2 \\ 3 & 4 & -2 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{14}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 670, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 31 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 31 & 0 & 172 & 17 & 24 \\ 2 & 3 & 0 & 17 & 0 & 2 \\ 3 & 4 & -2 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

[38] $1^{14}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 770, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 26 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 26 & 0 & 118 & 14 & 20 \\ 2 & 3 & 0 & 14 & 0 & 2 \\ 3 & 4 & -2 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{13}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 811, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 27 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 27 & 0 & 133 & 15 & 21 \\ 2 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{14}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 854, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 0 & 36 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 20 & 36 & 0 & 238 & 20 & 28 \\ 2 & 3 & 0 & 20 & 0 & 2 \\ 3 & 4 & -2 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{13}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 969, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 2 & 3 \\ 3 & 4 & 0 & 32 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 19 & 32 & 0 & 193 & 18 & 25 \\ 2 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & -2 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{14}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 1034, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 0 & 31 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 31 & 0 & 172 & 17 & 24 \\ 2 & 3 & 0 & 17 & 0 & 2 \\ 3 & 4 & -2 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{14}3^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 2 & 3 \\ 3 & 4 & 0 & 36 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 21 & 36 & 0 & 238 & 20 & 28 \\ 2 & 3 & 0 & 20 & 0 & 2 \\ 3 & 4 & -2 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [44] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 & 3 & 5 \\ 3 & 5 & 0 & 3 & 2 & 2 \\ 3 & 5 & 3 & 4 & 3 & 4 \\ 2 & 3 & 2 & 3 & 0 & 2 \\ 3 & 5 & 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[45] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 2 & 4 & 0 & 2 & 4 \\ 3 & 3 & 3 & -2 & 4 & 3 \end{pmatrix}_{b_0}$$

[46] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 3 \\ 2 & 1 & 1 & 4 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 4 & 0 & 10 & 4 & 6 \\ 2 & 2 & 0 & 4 & 0 & 2 \\ 3 & 3 & -1 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[47] $1^{11}3^{11}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 3 \\ 2 & 1 & 3 & 0 & 7 & 3 \\ 2 & 3 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 7 & 10 & 0 & 25 & 11 \\ 3 & 3 & 3 & -2 & 11 & 3 \end{pmatrix}_{b_0}$$

[48] $21^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 9 & 3 & 4 \\ 5 & 5 & 5 & 10 & 3 & 5 \\ 3 & 5 & 0 & 7 & 2 & 2 \\ 9 & 10 & 7 & 14 & 5 & 8 \\ 3 & 3 & 2 & 5 & 0 & 2 \\ 4 & 5 & 2 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[49] $1^{12}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -22, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 2 & 3 \\ 3 & 5 & 5 & 15 & 3 & 5 \\ 3 & 5 & 0 & 10 & 2 & 2 \\ 8 & 15 & 10 & 38 & 8 & 12 \\ 2 & 3 & 2 & 8 & 0 & 2 \\ 3 & 5 & 2 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[50] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 2 & 3 \\ 2 & 1 & 1 & 7 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 6 & 9 \\ 2 & 2 & 0 & 6 & 0 & 2 \\ 3 & 3 & -1 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[51] $1^{11} 1^3 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 5 & 5 & 5 & 3 & 5 \\ 3 & 5 & 0 & 4 & 2 & 2 \\ 2 & 5 & 4 & 2 & 2 & 4 \\ 2 & 3 & 2 & 2 & 0 & 2 \\ 3 & 5 & 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[52] $1^{11} 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 2 & 3 \\ 3 & 5 & 5 & 20 & 3 & 5 \\ 3 & 5 & 0 & 15 & 2 & 2 \\ 9 & 20 & 15 & 58 & 9 & 16 \\ 2 & 3 & 2 & 9 & 0 & 2 \\ 3 & 5 & 2 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[53] $1^{11} 1^3 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 2 & 3 \\ 2 & 1 & 1 & 0 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 2 & 0 & 0 & 0 & 2 \\ 3 & 3 & -1 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[54] $1^{11} 1^{14} 3^1 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 25 & 2 & 3 \\ 2 & 1 & 1 & 28 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 25 & 28 & 0 & 366 & 24 & 36 \\ 2 & 2 & 0 & 24 & 0 & 2 \\ 3 & 3 & -1 & 36 & 2 & 3 \end{pmatrix}_{b_0}$$

[55] $1^{11} 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -37, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 3 & 3 \\ 2 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 2 & 0 & 1 & 3 \\ 3 & 3 & 3 & -2 & 3 & 3 \end{pmatrix}_{b_0}$$

[56] $1^{11} 1^{11} 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 & 0 & 2 \\ 3 & 3 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[57] $1^{11} 1^{13} 3^1 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 2 & 3 \\ 2 & 1 & 1 & 17 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 14 & 17 & 0 & 121 & 14 & 21 \\ 2 & 2 & 0 & 14 & 0 & 2 \\ 3 & 3 & -1 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[58] $21^{12} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 18 & 3 & 4 \\ 5 & 5 & 5 & 20 & 3 & 5 \\ 3 & 5 & 0 & 14 & 2 & 2 \\ 18 & 20 & 14 & 62 & 10 & 16 \\ 3 & 3 & 2 & 10 & 0 & 2 \\ 4 & 5 & 2 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[59] $1^{12} 3^1 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 3 \\ 2 & 1 & 3 & 0 & 12 & 3 \\ 2 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 12 & 16 & 0 & 70 & 18 \\ 3 & 3 & 3 & -2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{12} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.88 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 2 & 3 \\ 2 & 1 & 1 & 10 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 8 & 12 \\ 2 & 2 & 0 & 8 & 0 & 2 \\ 3 & 3 & -1 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{11} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -69, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 3 \\ 2 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 6 & 0 & 9 & 7 \\ 3 & 3 & 3 & -2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{11} 3^1 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -83, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 3 \\ 3 & 5 & 5 & 10 & 3 & 3 \\ 3 & 5 & 0 & 7 & 2 & 2 \\ 5 & 10 & 7 & 14 & 5 & 8 \\ 2 & 3 & 2 & 5 & 0 & 2 \\ 3 & 5 & 2 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{12} 3^1 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -88, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.95 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 2 & 3 \\ 2 & 1 & 1 & 11 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 12 & 11 & 0 & 65 & 10 & 15 \\ 2 & 2 & 0 & 10 & 0 & 2 \\ 3 & 3 & -1 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{13} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -88, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.95 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 2 & 3 \\ 2 & 1 & 1 & 13 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 13 & 0 & 61 & 10 & 15 \\ 2 & 2 & 0 & 10 & 0 & 2 \\ 3 & 3 & -1 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{12} 3^1 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -88, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.95 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 24 & 2 & 3 \\ 2 & 1 & 1 & 25 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 24 & 25 & 0 & 313 & 22 & 33 \\ 2 & 2 & 0 & 22 & 0 & 2 \\ 3 & 3 & -1 & 33 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{11} 3^1 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -91, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.91 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 2 & 3 \\ 3 & 5 & 5 & 15 & 3 & 5 \\ 3 & 5 & 0 & 11 & 2 & 2 \\ 7 & 15 & 11 & 32 & 7 & 12 \\ 2 & 3 & 2 & 7 & 0 & 2 \\ 3 & 5 & 2 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{11} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -108, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 2 & 3 \\ 2 & 1 & 1 & 7 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 7 & 7 & 0 & 21 & 6 & 9 \\ 2 & 2 & 0 & 6 & 0 & 2 \\ 3 & 3 & -1 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[67] $1^{11} 1^{13} 3^1 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 19 & 2 & 3 \\ 2 & 1 & 1 & 21 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 19 & 21 & 0 & 205 & 18 & 27 \\ 2 & 2 & 0 & 18 & 0 & 2 \\ 3 & 3 & -1 & 27 & 2 & 3 \end{pmatrix}_{b_0}$$

[71] $1^{12} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 3 \\ 2 & 1 & 3 & 0 & 8 & 3 \\ 2 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 8 & 0 & 22 & 10 \\ 3 & 3 & 3 & -2 & 10 & 3 \end{pmatrix}_{b_0}$$

[70] $1^{14} 3^1 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 2 & 3 \\ 2 & 1 & 1 & 20 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 16 & 20 & 0 & 158 & 16 & 24 \\ 2 & 2 & 0 & 16 & 0 & 2 \\ 3 & 3 & -1 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

[69] $1^{12} 3^1 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 2 & 3 \\ 2 & 1 & 1 & 18 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 18 & 18 & 0 & 166 & 16 & 24 \\ 2 & 2 & 0 & 16 & 0 & 2 \\ 3 & 3 & -1 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

[73] $1^{12} 3^1 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 2 & 3 \\ 2 & 1 & 1 & 14 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 13 & 14 & 0 & 90 & 12 & 18 \\ 2 & 2 & 0 & 12 & 0 & 2 \\ 3 & 3 & -1 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{12} 3^1 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 3 \\ 2 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 10 & 12 & 0 & 42 & 14 \\ 3 & 3 & 3 & -2 & 14 & 3 \end{pmatrix}_{b_0}$$

[74] $1^{12} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 2 & 3 \\ 3 & 5 & 5 & 20 & 3 & 5 \\ 3 & 5 & 0 & 14 & 2 & 2 \\ 10 & 20 & 14 & 62 & 10 & 16 \\ 2 & 3 & 2 & 10 & 0 & 2 \\ 3 & 5 & 2 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[75] $1^{13} 3^1 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 14 & 3 \\ 2 & 1 & 3 & 0 & 17 & 3 \\ 2 & 3 & 1 & 0 & 22 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 17 & 22 & 0 & 137 & 25 \\ 3 & 3 & 3 & -2 & 25 & 3 \end{pmatrix}_{b_0}$$

[76] $1^{12} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 2 & 3 \\ 2 & 1 & 1 & 7 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 7 & 0 & 21 & 6 & 9 \\ 2 & 2 & 0 & 6 & 0 & 2 \\ 3 & 3 & -1 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{12} 3^1 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -178, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.92 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 2 & 3 \\ 3 & 5 & 5 & 30 & 3 & 5 \\ 3 & 5 & 0 & 22 & 2 & 2 \\ 14 & 30 & 22 & 134 & 14 & 24 \\ 2 & 3 & 2 & 14 & 0 & 2 \\ 3 & 5 & 2 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{11} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -196, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.92 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 2 & 3 \\ 2 & 1 & 1 & 10 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 9 & 10 & 0 & 38 & 8 & 12 \\ 2 & 2 & 0 & 8 & 0 & 2 \\ 3 & 3 & -1 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{12} 3^1 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -198, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.91 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 2 & 3 \\ 3 & 5 & 5 & 25 & 3 & 5 \\ 3 & 5 & 0 & 18 & 2 & 2 \\ 12 & 25 & 18 & 94 & 12 & 20 \\ 2 & 3 & 2 & 12 & 0 & 2 \\ 3 & 5 & 2 & 20 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{13} 3^1 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -229, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.92 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 3 \\ 2 & 1 & 3 & 0 & 13 & 3 \\ 2 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 13 & 14 & 0 & 65 & 17 \\ 3 & 3 & 3 & -2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{13} 3^1 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -237, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.89 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 12 & 3 \\ 2 & 1 & 3 & 0 & 15 & 3 \\ 2 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 15 & 18 & 0 & 97 & 21 \\ 3 & 3 & 3 & -2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{11} 3^1 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 2 & 3 \\ 2 & 1 & 1 & 17 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 15 & 17 & 0 & 121 & 14 & 21 \\ 2 & 2 & 0 & 14 & 0 & 2 \\ 3 & 3 & -1 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{11} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.96 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 2 & 3 \\ 2 & 1 & 1 & 13 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 11 & 13 & 0 & 61 & 10 & 15 \\ 2 & 2 & 0 & 10 & 0 & 2 \\ 3 & 3 & -1 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{12} 3^1 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -288, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.92 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 2 & 3 \\ 2 & 1 & 1 & 14 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 14 & 14 & 0 & 90 & 12 & 18 \\ 2 & 2 & 0 & 12 & 0 & 2 \\ 3 & 3 & -1 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[85] $1^{11} 1^{14} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 21 & 2 & 3 \\ 2 & 1 & 1 & 24 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 21 & 24 & 0 & 250 & 20 & 30 \\ 2 & 2 & 0 & 20 & 0 & 2 \\ 3 & 3 & -1 & 30 & 2 & 3 \end{pmatrix}_{b_0}$$

[87] $1^{12} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 2 & 3 \\ 2 & 1 & 1 & 10 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 10 & 0 & 38 & 8 & 12 \\ 2 & 2 & 0 & 8 & 0 & 2 \\ 3 & 3 & -1 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[86] $1^{11} 1^{14} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 12 & 3 \\ 2 & 1 & 3 & 0 & 16 & 3 \\ 2 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 16 & 16 & 0 & 94 & 20 \\ 3 & 3 & 3 & -2 & 20 & 3 \end{pmatrix}_{b_0}$$

[88] $1^{11} 1^{14} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 2 & 3 \\ 2 & 1 & 1 & 16 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 13 & 16 & 0 & 90 & 12 & 18 \\ 2 & 2 & 0 & 12 & 0 & 2 \\ 3 & 3 & -1 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[89] $1^{12} 1^{13} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -360, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 2 & 3 \\ 2 & 1 & 1 & 21 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 20 & 21 & 0 & 205 & 18 & 27 \\ 2 & 2 & 0 & 18 & 0 & 2 \\ 3 & 3 & -1 & 27 & 2 & 3 \end{pmatrix}_{b_0}$$

[91] $1^{12} 1^{14} 3^1 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 26 & 2 & 3 \\ 2 & 1 & 1 & 28 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 26 & 28 & 0 & 366 & 24 & 36 \\ 2 & 2 & 0 & 24 & 0 & 2 \\ 3 & 3 & -1 & 36 & 2 & 3 \end{pmatrix}_{b_0}$$

[90] $1^{11} 1^{14} 3^1 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 16 & 3 \\ 2 & 1 & 3 & 0 & 20 & 3 \\ 2 & 3 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 20 & 24 & 0 & 174 & 28 \\ 3 & 3 & 3 & -2 & 28 & 3 \end{pmatrix}_{b_0}$$

[93] $1^{11} 1^{14} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 2 & 3 \\ 2 & 1 & 1 & 20 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 17 & 20 & 0 & 158 & 16 & 24 \\ 2 & 2 & 0 & 16 & 0 & 2 \\ 3 & 3 & -1 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

[92] $1^{11} 1^{14} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 14 & 3 \\ 2 & 1 & 3 & 0 & 18 & 3 \\ 2 & 3 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 18 & 20 & 0 & 130 & 24 \\ 3 & 3 & 3 & -2 & 24 & 3 \end{pmatrix}_{b_0}$$

[94] $1^{13}3^{12}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 2 & 3 \\ 2 & 1 & 1 & 13 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 12 & 13 & 0 & 61 & 10 & 15 \\ 2 & 2 & 0 & 10 & 0 & 2 \\ 3 & 3 & -1 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[95] $1^{13}3^{12}21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -472, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 2 & 3 \\ 2 & 1 & 1 & 17 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 16 & 17 & 0 & 121 & 14 & 21 \\ 2 & 2 & 0 & 14 & 0 & 2 \\ 3 & 3 & -1 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[96] $1^{14}3^{12}21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -592, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 22 & 2 & 3 \\ 2 & 1 & 1 & 24 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 22 & 24 & 0 & 250 & 20 & 30 \\ 2 & 2 & 0 & 20 & 0 & 2 \\ 3 & 3 & -1 & 30 & 2 & 3 \end{pmatrix}_{b_0}$$

[97] $1^{14}3^{12}21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -640, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 2 & 3 \\ 2 & 1 & 1 & 20 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 18 & 20 & 0 & 158 & 16 & 24 \\ 2 & 2 & 0 & 16 & 0 & 2 \\ 3 & 3 & -1 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

4.673 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$

$T = 9 \quad [0] \quad 1^{11}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 6 & 5 \\ 11 & 20 & 0 & 0 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 11 & 0 & 0 & 4 & 4 \\ 5 & 8 & -2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 9 & 7 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -2 & 7 & 4 & 2 \end{pmatrix}_{b_0}$$

4.674 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9 \quad [0] \quad 1^{12}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 2 & 3 \\ 3 & 4 & 0 & 8 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 4 & 4 & 6 \\ 2 & 3 & 0 & 4 & 0 & 2 & 2 \\ 2 & 3 & 0 & 4 & 2 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{14}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -76, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 2 & 3 \\ 3 & 4 & 0 & 21 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 11 & 11 & 16 \\ 2 & 3 & 0 & 11 & 0 & 2 & 2 \\ 2 & 3 & 0 & 11 & 2 & 0 & 2 \\ 3 & 4 & -2 & 16 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^1 1^3 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & -2 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^1 1^3 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 2 & 2 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 3 & 4 & -2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^1 1^3 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 2 & 3 \\ 3 & 4 & 0 & 17 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 9 & 9 & 13 \\ 2 & 3 & 0 & 9 & 0 & 2 & 2 \\ 2 & 3 & 0 & 9 & 2 & 0 & 2 \\ 3 & 4 & -2 & 13 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^1 1^3 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 9 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 5 & 5 & 7 \\ 2 & 3 & 0 & 5 & 0 & 2 & 2 \\ 2 & 3 & 0 & 5 & 2 & 0 & 2 \\ 3 & 4 & -2 & 7 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^1 1^3 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 2 & 3 \\ 3 & 4 & 0 & 14 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 8 & 8 & 11 \\ 2 & 3 & 0 & 8 & 0 & 2 & 2 \\ 2 & 3 & 0 & 8 & 2 & 0 & 2 \\ 3 & 4 & -2 & 11 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 0 & 13 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 7 & 7 & 10 \\ 2 & 3 & 0 & 7 & 0 & 2 & 2 \\ 2 & 3 & 0 & 7 & 2 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^1 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 2 & 3 \\ 3 & 4 & 0 & 18 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 10 & 10 & 14 \\ 2 & 3 & 0 & 10 & 0 & 2 & 2 \\ 2 & 3 & 0 & 10 & 2 & 0 & 2 \\ 3 & 4 & -2 & 14 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^1 1^{13} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -468, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 2 & 3 \\ 3 & 4 & 0 & 22 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 12 & 12 & 17 \\ 2 & 3 & 0 & 12 & 0 & 2 & 2 \\ 2 & 3 & 0 & 12 & 2 & 0 & 2 \\ 3 & 4 & -2 & 17 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{14} 3^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2 \rightarrow (-3)} \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -488, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 2 & 3 \\ 3 & 4 & 0 & 26 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 14 & 14 & 20 \\ 2 & 3 & 0 & 14 & 0 & 2 & 2 \\ 2 & 3 & 0 & 14 & 2 & 0 & 2 \\ 3 & 4 & -2 & 20 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

4.675 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + su2n3mix

$$T = 9 \quad [0] \ 1^{11} 3^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 4 & 3 \\ 3 & 4 & 0 & 14 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 8 & 17 & 11 \\ 2 & 3 & 0 & 8 & 0 & 4 & 2 \\ 4 & 7 & 0 & 17 & 4 & 7 & 5 \\ 3 & 4 & -2 & 11 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 9 & 3 \\ 3 & 4 & 0 & 8 & 3 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 4 & 22 & 6 \\ 2 & 3 & 0 & 4 & 0 & 9 & 2 \\ 9 & 16 & 0 & 22 & 9 & 44 & 12 \\ 3 & 4 & -2 & 6 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 4 & 3 \\ 3 & 4 & 0 & 8 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 4 & 10 & 6 \\ 2 & 3 & 0 & 4 & 0 & 4 & 2 \\ 4 & 7 & 0 & 10 & 4 & 7 & 5 \\ 3 & 4 & -2 & 6 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 0 & 5 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 3 & 0 & 3 & 0 & 4 & 2 \\ 4 & 7 & 0 & 6 & 4 & 7 & 5 \\ 3 & 4 & -2 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 2 & 5 & 3 \\ 2 & 3 & 0 & 2 & 0 & 4 & 2 \\ 4 & 7 & 0 & 5 & 4 & 7 & 5 \\ 3 & 4 & -2 & 3 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 3 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 2 & 8 & 3 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 6 & 11 & 0 & 8 & 6 & 20 & 8 \\ 3 & 4 & -2 & 3 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[6] \quad & 1^{123}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -238, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 11 & 3 \\ 3 & 4 & 0 & 8 & 3 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 4 & 28 & 6 \\ 2 & 3 & 0 & 4 & 0 & 11 & 2 \\ 11 & 20 & 0 & 28 & 11 & 69 & 15 \\ 3 & 4 & -2 & 6 & 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[7] \quad & 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -251, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 4 & 3 \\ 3 & 4 & 0 & 9 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 5 & 11 & 7 \\ 2 & 3 & 0 & 5 & 0 & 4 & 2 \\ 4 & 7 & 0 & 11 & 4 & 7 & 5 \\ 3 & 4 & -2 & 7 & 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[8] \quad & 1^{124}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -264, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 16 & 3 \\ 3 & 4 & 0 & 8 & 3 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 4 & 40 & 6 \\ 2 & 3 & 0 & 4 & 0 & 16 & 2 \\ 16 & 29 & 0 & 40 & 16 & 148 & 22 \\ 3 & 4 & -2 & 6 & 2 & 22 & 2 \end{pmatrix}_{b_0} \\
[9] \quad & 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -287, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 3 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 2 & 8 & 3 \\ 2 & 3 & 0 & 2 & 0 & 7 & 2 \\ 7 & 12 & 0 & 8 & 7 & 25 & 9 \\ 3 & 4 & -2 & 3 & 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[10] \quad & 1^{141}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -299, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 4 & 3 \\ 3 & 4 & 0 & 21 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 11 & 26 & 16 \\ 2 & 3 & 0 & 11 & 0 & 4 & 2 \\ 4 & 7 & 0 & 26 & 4 & 7 & 5 \\ 3 & 4 & -2 & 16 & 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[11] \quad & 1^{121}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -314, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 4 & 3 \\ 3 & 4 & 0 & 18 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 10 & 22 & 14 \\ 2 & 3 & 0 & 10 & 0 & 4 & 2 \\ 4 & 7 & 0 & 22 & 4 & 7 & 5 \\ 3 & 4 & -2 & 14 & 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[12] \quad & 1^{122}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -336, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 6 & 3 \\ 3 & 4 & 0 & 13 & 3 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 7 & 25 & 10 \\ 2 & 3 & 0 & 7 & 0 & 6 & 2 \\ 6 & 11 & 0 & 25 & 6 & 20 & 8 \\ 3 & 4 & -2 & 10 & 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[13] \quad & 1^{121}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -339, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 4 & 3 \\ 3 & 4 & 0 & 13 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 7 & 16 & 10 \\ 2 & 3 & 0 & 7 & 0 & 4 & 2 \\ 4 & 7 & 0 & 16 & 4 & 7 & 5 \\ 3 & 4 & -2 & 10 & 2 & 5 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[14] $1^{13}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 4 & 3 \\ 3 & 4 & 0 & 17 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 9 & 21 & 13 \\ 2 & 3 & 0 & 9 & 0 & 4 & 2 \\ 4 & 7 & 0 & 21 & 4 & 7 & 5 \\ 3 & 4 & -2 & 13 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{12}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 13 & 3 \\ 3 & 4 & 0 & 8 & 3 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 4 & 34 & 6 \\ 2 & 3 & 0 & 4 & 0 & 13 & 2 \\ 13 & 24 & 0 & 34 & 13 & 100 & 18 \\ 3 & 4 & -2 & 6 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{13}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 6 & 3 \\ 3 & 4 & 0 & 17 & 3 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 9 & 33 & 13 \\ 2 & 3 & 0 & 9 & 0 & 6 & 2 \\ 6 & 11 & 0 & 33 & 6 & 20 & 8 \\ 3 & 4 & -2 & 13 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{14}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 6 & 3 \\ 3 & 4 & 0 & 21 & 3 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 11 & 41 & 16 \\ 2 & 3 & 0 & 11 & 0 & 6 & 2 \\ 6 & 11 & 0 & 41 & 6 & 20 & 8 \\ 3 & 4 & -2 & 16 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -477, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 3 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 2 & 11 & 3 \\ 2 & 3 & 0 & 2 & 0 & 9 & 2 \\ 9 & 16 & 0 & 11 & 9 & 44 & 12 \\ 3 & 4 & -2 & 3 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{13}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -526, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 4 & 3 \\ 3 & 4 & 0 & 22 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 12 & 27 & 17 \\ 2 & 3 & 0 & 12 & 0 & 4 & 2 \\ 4 & 7 & 0 & 27 & 4 & 7 & 5 \\ 3 & 4 & -2 & 17 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{14}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 6 & 3 \\ 3 & 4 & 0 & 26 & 3 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 14 & 50 & 20 \\ 2 & 3 & 0 & 14 & 0 & 6 & 2 \\ 6 & 11 & 0 & 50 & 6 & 20 & 8 \\ 3 & 4 & -2 & 20 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{11}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 3 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 2 & 14 & 3 \\ 2 & 3 & 0 & 2 & 0 & 12 & 2 \\ 12 & 21 & 0 & 14 & 12 & 80 & 16 \\ 3 & 4 & -2 & 3 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -631, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 3 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 2 & 14 & 3 \\ 2 & 3 & 0 & 2 & 0 & 11 & 2 \\ 11 & 20 & 0 & 14 & 11 & 69 & 15 \\ 3 & 4 & -2 & 3 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{14}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -666, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 4 & 3 \\ 3 & 4 & 0 & 26 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 14 & 32 & 20 \\ 2 & 3 & 0 & 14 & 0 & 4 & 2 \\ 4 & 7 & 0 & 32 & 4 & 7 & 5 \\ 3 & 4 & -2 & 20 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{114}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -749, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 3 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 2 & 17 & 3 \\ 2 & 3 & 0 & 2 & 0 & 13 & 2 \\ 13 & 24 & 0 & 17 & 13 & 100 & 18 \\ 3 & 4 & -2 & 3 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{113}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -886, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 3 & 25 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 2 & 17 & 3 \\ 2 & 3 & 0 & 2 & 0 & 14 & 2 \\ 14 & 25 & 0 & 17 & 14 & 111 & 19 \\ 3 & 4 & -2 & 3 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{114}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -1152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 3 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 2 & 20 & 3 \\ 2 & 3 & 0 & 2 & 0 & 16 & 2 \\ 16 & 29 & 0 & 20 & 16 & 148 & 22 \\ 3 & 4 & -2 & 3 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

4.676 NHC(-2)(-3) + so₈ + su₂ + su₂n3mix

$$T = 8 \quad [0] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 20 & 6 \\ 15 & 20 & 17 & 28 & 8 \\ 12 & 17 & 13 & 24 & 7 \\ 20 & 28 & 24 & 37 & 11 \\ 6 & 8 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 1 & 20 & 6 \\ 15 & 20 & 0 & 28 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 20 & 28 & 0 & 37 & 11 \\ 6 & 8 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 8 & 0 & 0 & 12 & 6 \\ 3 & 4 & -2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 9 & 3 \\ 3 & 4 & 0 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 16 & 0 & 0 & 44 & 12 \\ 3 & 4 & -2 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{144} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 13 & 3 \\ 3 & 4 & 0 & 16 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 46 & 68 & 12 \\ 13 & 24 & 0 & 68 & 100 & 18 \\ 3 & 4 & -2 & 12 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 4 & 3 \\ 3 & 4 & 0 & 14 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 17 & 11 \\ 4 & 7 & 0 & 17 & 7 & 5 \\ 3 & 4 & -2 & 11 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 4 & 3 \\ 3 & 4 & 0 & 0 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 7 & 0 & 0 & 7 & 5 \\ 3 & 4 & -2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 5 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 4 & 7 & 0 & 6 & 7 & 5 \\ 3 & 4 & -2 & 4 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 7 & 3 \\ 3 & 4 & 0 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 0 & 25 & 9 \\ 3 & 4 & -2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 13 & 20 & 6 \\ 15 & 20 & 17 & 28 & 8 \\ 13 & 17 & 13 & 24 & 7 \\ 20 & 28 & 24 & 37 & 11 \\ 6 & 8 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 5 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 12 & 4 \\ 7 & 12 & 0 & 12 & 25 & 9 \\ 3 & 4 & -2 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{123} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 8 & 25 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 34 & 6 \\ 14 & 25 & 0 & 34 & 111 & 19 \\ 3 & 4 & -2 & 6 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{144} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 14 & 3 \\ 3 & 4 & 0 & 21 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 89 & 16 \\ 14 & 24 & 0 & 89 & 100 & 18 \\ 3 & 4 & -2 & 16 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 21 & 6 \\ 15 & 20 & 17 & 28 & 8 \\ 12 & 17 & 13 & 24 & 7 \\ 21 & 28 & 24 & 37 & 11 \\ 6 & 8 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{121} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 4 & 3 \\ 3 & 4 & 0 & 23 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 23 & 0 & 100 & 28 & 18 \\ 4 & 7 & 0 & 28 & 7 & 5 \\ 3 & 4 & -2 & 18 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 0 & 14 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 14 & 0 & 37 & 17 & 11 \\ 4 & 7 & 0 & 17 & 7 & 5 \\ 3 & 4 & -2 & 11 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[16] \quad 21^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 17 & 15 & 24 & 7 \\ 17 & 20 & 17 & 28 & 8 \\ 15 & 17 & 13 & 24 & 7 \\ 24 & 28 & 24 & 37 & 11 \\ 7 & 8 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{133} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 0 & 12 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 42 & 9 \\ 11 & 20 & 0 & 42 & 69 & 15 \\ 3 & 4 & -2 & 9 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 8 & 3 \\ 7 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & -2 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{122} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 16 & 6 \\ 7 & 11 & 0 & 16 & 20 & 8 \\ 3 & 4 & -2 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{121} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 10 & 6 \\ 4 & 7 & 0 & 10 & 7 & 5 \\ 3 & 4 & -2 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 8 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 22 & 6 \\ 9 & 16 & 0 & 22 & 44 & 12 \\ 3 & 4 & -2 & 6 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 8 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 22 & 6 \\ 9 & 16 & 0 & 22 & 44 & 12 \\ 3 & 4 & -2 & 6 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 4 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 8 & 3 \\ 6 & 11 & 0 & 8 & 20 & 8 \\ 3 & 4 & -2 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[24] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 109$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 5 & 3 \\ 4 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & -2 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 113$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 5 & 3 \\ 4 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & -2 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{124}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 114$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 17 & 3 \\ 3 & 4 & 0 & 8 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 40 & 6 \\ 17 & 29 & 0 & 40 & 148 & 22 \\ 3 & 4 & -2 & 6 & 22 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 115$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 9 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 17 & 7 \\ 6 & 11 & 0 & 17 & 20 & 8 \\ 3 & 4 & -2 & 7 & 8 & 2 \end{pmatrix}_{b_0}$$

[28] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 117$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 5 & 3 \\ 5 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & -2 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 120$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 14 & 7 \\ 5 & 8 & 0 & 14 & 12 & 6 \\ 3 & 4 & -2 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 128$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 11 & 3 \\ 9 & 16 & 0 & 11 & 44 & 12 \\ 3 & 4 & -2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{123}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 128$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 8 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 28 & 6 \\ 12 & 20 & 0 & 28 & 69 & 15 \\ 3 & 4 & -2 & 6 & 15 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{131}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 129$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 7 & 3 \\ 3 & 4 & 0 & 17 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 36 & 13 \\ 7 & 12 & 0 & 36 & 25 & 9 \\ 3 & 4 & -2 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 142$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 11 & 7 \\ 4 & 7 & 0 & 11 & 7 & 5 \\ 3 & 4 & -2 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{11} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 14 & 3 \\ 11 & 20 & 0 & 14 & 69 & 15 \\ 3 & 4 & -2 & 3 & 15 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{11} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 7 & 3 \\ 3 & 4 & 0 & 21 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 44 & 16 \\ 7 & 12 & 0 & 44 & 25 & 9 \\ 3 & 4 & -2 & 16 & 9 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{11} 1^{11} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 4 & 3 \\ 3 & 4 & 0 & 32 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 32 & 0 & 193 & 39 & 25 \\ 4 & 7 & 0 & 39 & 7 & 5 \\ 3 & 4 & -2 & 25 & 5 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{11} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 11 & 20 & 0 & 18 & 69 & 15 \\ 3 & 4 & -2 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$

[38] $1^{11} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 14 & 7 \\ 5 & 8 & 0 & 14 & 12 & 6 \\ 3 & 4 & -2 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{11} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 7 & 12 & 0 & 12 & 25 & 9 \\ 3 & 4 & -2 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{11} 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 157, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 8 & 3 \\ 6 & 11 & 0 & 8 & 20 & 8 \\ 3 & 4 & -2 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{11} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 8 & 3 \\ 7 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & -2 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{11} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 4 & 3 \\ 3 & 4 & 0 & 9 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 11 & 7 \\ 4 & 7 & 0 & 11 & 7 & 5 \\ 3 & 4 & -2 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{11} 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 10 & 6 \\ 4 & 7 & 0 & 10 & 7 & 5 \\ 3 & 4 & -2 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[46] $1^{123}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 8 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 28 & 6 \\ 11 & 20 & 0 & 28 & 69 & 15 \\ 3 & 4 & -2 & 6 & 15 & 2 \end{pmatrix}_{b_0}$$

[44] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 8 & 3 \\ 8 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & -2 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[45] $1^{123}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 9 & 16 & 0 & 15 & 44 & 12 \\ 3 & 4 & -2 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

[47] $1^{14}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 5 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 21 & 4 \\ 14 & 24 & 0 & 21 & 100 & 18 \\ 3 & 4 & -2 & 4 & 18 & 2 \end{pmatrix}_{b_0}$$

[48] $1^{132}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 6 & 3 \\ 3 & 4 & 0 & 22 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 42 & 17 \\ 6 & 11 & 0 & 42 & 20 & 8 \\ 3 & 4 & -2 & 17 & 8 & 2 \end{pmatrix}_{b_0}$$

[49] $1^{122}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 6 & 3 \\ 3 & 4 & 0 & 13 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 25 & 10 \\ 6 & 11 & 0 & 25 & 20 & 8 \\ 3 & 4 & -2 & 10 & 8 & 2 \end{pmatrix}_{b_0}$$

[50] $1^{113}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 5 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 8 & 12 & 0 & 12 & 25 & 9 \\ 3 & 4 & -2 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{122}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 190, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 8 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 16 & 6 \\ 6 & 11 & 0 & 16 & 20 & 8 \\ 3 & 4 & -2 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

[52] $1^{134}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 13 & 3 \\ 3 & 4 & 0 & 12 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 51 & 9 \\ 13 & 24 & 0 & 51 & 100 & 18 \\ 3 & 4 & -2 & 9 & 18 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{123}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 8 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 22 & 6 \\ 9 & 15 & 0 & 22 & 39 & 11 \\ 3 & 4 & -2 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$

[54] $1^{121}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 4 & 3 \\ 3 & 4 & 0 & 23 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 23 & 0 & 100 & 28 & 18 \\ 4 & 7 & 0 & 28 & 7 & 5 \\ 3 & 4 & -2 & 18 & 5 & 2 \end{pmatrix}_{b_0}$$

[55] $1^{142}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 9 & 3 \\ 3 & 4 & 0 & 21 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 59 & 16 \\ 9 & 16 & 0 & 59 & 44 & 12 \\ 3 & 4 & -2 & 16 & 12 & 2 \end{pmatrix}_{b_0}$$

[56] $1^{133}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 12 & 3 \\ 3 & 4 & 0 & 17 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 60 & 13 \\ 12 & 20 & 0 & 60 & 69 & 15 \\ 3 & 4 & -2 & 13 & 15 & 2 \end{pmatrix}_{b_0}$$

[57] $1^{121}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 4 & 3 \\ 3 & 4 & 0 & 18 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 22 & 14 \\ 4 & 7 & 0 & 22 & 7 & 5 \\ 3 & 4 & -2 & 14 & 5 & 2 \end{pmatrix}_{b_0}$$

[58] $1^{112}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 9 & 3 \\ 3 & 4 & 0 & 9 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 26 & 7 \\ 9 & 16 & 0 & 26 & 44 & 12 \\ 3 & 4 & -2 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[59] $1^{131}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 12 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 15 & 9 \\ 4 & 7 & 0 & 15 & 7 & 5 \\ 3 & 4 & -2 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[60] $1^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 219, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 13 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 16 & 10 \\ 4 & 7 & 0 & 16 & 7 & 5 \\ 3 & 4 & -2 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[61] $1^{122}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 222, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 0 & 13 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 25 & 10 \\ 6 & 11 & 0 & 25 & 20 & 8 \\ 3 & 4 & -2 & 10 & 8 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{123}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 0 & 8 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 28 & 6 \\ 11 & 20 & 0 & 28 & 69 & 15 \\ 3 & 4 & -2 & 6 & 15 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{124}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 8 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 34 & 6 \\ 13 & 24 & 0 & 34 & 100 & 18 \\ 3 & 4 & -2 & 6 & 18 & 2 \end{pmatrix}_{b_0}$$

[64] $1^{11} 1^{12} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 8 & 3 \\ 7 & 11 & 0 & 8 & 20 & 8 \\ 3 & 4 & -2 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{11} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 9 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 20 & 7 \\ 8 & 12 & 0 & 20 & 25 & 9 \\ 3 & 4 & -2 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 233, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 5 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 12 & 20 & 0 & 18 & 69 & 15 \\ 3 & 4 & -2 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 5 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 10 & 16 & 0 & 15 & 44 & 12 \\ 3 & 4 & -2 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{14} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 242, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 0 & 16 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 46 & 20 & 12 \\ 4 & 7 & 0 & 20 & 7 & 5 \\ 3 & 4 & -2 & 12 & 5 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{13} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 0 & 12 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 24 & 9 \\ 6 & 11 & 0 & 24 & 20 & 8 \\ 3 & 4 & -2 & 9 & 8 & 2 \end{pmatrix}_{b_0}$$

[70] $1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 258, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 16 & 3 \\ 3 & 4 & 0 & 8 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 40 & 6 \\ 16 & 29 & 0 & 40 & 148 & 22 \\ 3 & 4 & -2 & 6 & 22 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 28 & 10 \\ 7 & 12 & 0 & 28 & 25 & 9 \\ 3 & 4 & -2 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[71] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 9 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 20 & 7 \\ 7 & 12 & 0 & 20 & 25 & 9 \\ 3 & 4 & -2 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 0 & 13 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 37 & 10 \\ 10 & 16 & 0 & 37 & 44 & 12 \\ 3 & 4 & -2 & 10 & 12 & 2 \end{pmatrix}_{b_0}$$

[74] $1^{111}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 273, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 20 & 7 \\ 7 & 12 & 0 & 20 & 25 & 9 \\ 3 & 4 & -2 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[75] $1^{131}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 278, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 0 & 17 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 21 & 13 \\ 4 & 7 & 0 & 21 & 7 & 5 \\ 3 & 4 & -2 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[76] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 11 & 3 \\ 9 & 16 & 0 & 11 & 44 & 12 \\ 3 & 4 & -2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[77] $1^{132}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 311, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 6 & 3 \\ 3 & 4 & 0 & 17 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 33 & 13 \\ 6 & 11 & 0 & 33 & 20 & 8 \\ 3 & 4 & -2 & 13 & 8 & 2 \end{pmatrix}_{b_0}$$

[78] $1^{131}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 313, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 7 & 3 \\ 3 & 4 & 0 & 17 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 36 & 13 \\ 7 & 12 & 0 & 36 & 25 & 9 \\ 3 & 4 & -2 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[79] $1^{141}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 4 & 3 \\ 3 & 4 & 0 & 21 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 26 & 16 \\ 4 & 7 & 0 & 26 & 7 & 5 \\ 3 & 4 & -2 & 16 & 5 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{124}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 8 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 34 & 6 \\ 14 & 24 & 0 & 34 & 100 & 18 \\ 3 & 4 & -2 & 6 & 18 & 2 \end{pmatrix}_{b_0}$$

[81] $1^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 327, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 4 & 3 \\ 3 & 4 & 0 & 13 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 16 & 10 \\ 4 & 7 & 0 & 16 & 7 & 5 \\ 3 & 4 & -2 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[82] $1^{131}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 4 & 3 \\ 3 & 4 & 0 & 27 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 27 & 0 & 133 & 33 & 21 \\ 4 & 7 & 0 & 33 & 7 & 5 \\ 3 & 4 & -2 & 21 & 5 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{121}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 338, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 4 & 3 \\ 3 & 4 & 0 & 18 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 18 & 0 & 58 & 22 & 14 \\ 4 & 7 & 0 & 22 & 7 & 5 \\ 3 & 4 & -2 & 14 & 5 & 2 \end{pmatrix}_{b_0}$$

[84] $1^{11} 1^{12} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 341, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 11 & 3 \\ 9 & 15 & 0 & 11 & 39 & 11 \\ 3 & 4 & -2 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

[85] $1^{11} 1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 349, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 12 & 21 & 0 & 14 & 80 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[86] $1^{124} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 13 & 3 \\ 3 & 4 & 0 & 8 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 34 & 6 \\ 13 & 24 & 0 & 34 & 100 & 18 \\ 3 & 4 & -2 & 6 & 18 & 2 \end{pmatrix}_{b_0}$$

[87] $1^{121} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 353, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 7 & 3 \\ 3 & 4 & 0 & 13 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 28 & 10 \\ 7 & 12 & 0 & 28 & 25 & 9 \\ 3 & 4 & -2 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[88] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 11 & 3 \\ 10 & 16 & 0 & 11 & 44 & 12 \\ 3 & 4 & -2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[89] $1^{112} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 26 & 7 \\ 9 & 16 & 0 & 26 & 44 & 12 \\ 3 & 4 & -2 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[90] $1^{131} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 377, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 4 & 3 \\ 3 & 4 & 0 & 22 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 27 & 17 \\ 4 & 7 & 0 & 27 & 7 & 5 \\ 3 & 4 & -2 & 17 & 5 & 2 \end{pmatrix}_{b_0}$$

[91] $1^{123} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 11 & 3 \\ 3 & 4 & 0 & 13 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 46 & 10 \\ 11 & 20 & 0 & 46 & 69 & 15 \\ 3 & 4 & -2 & 10 & 15 & 2 \end{pmatrix}_{b_0}$$

[92] $1^{142} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 6 & 3 \\ 3 & 4 & 0 & 21 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 41 & 16 \\ 6 & 11 & 0 & 41 & 20 & 8 \\ 3 & 4 & -2 & 16 & 8 & 2 \end{pmatrix}_{b_0}$$

[93] $1^{132} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 387, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 6 & 3 \\ 3 & 4 & 0 & 17 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 33 & 13 \\ 6 & 11 & 0 & 33 & 20 & 8 \\ 3 & 4 & -2 & 13 & 8 & 2 \end{pmatrix}_{b_0}$$

[94] $1^{142}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 6 & 3 \\ 3 & 4 & 0 & 26 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 50 & 20 \\ 6 & 11 & 0 & 50 & 20 & 8 \\ 3 & 4 & -2 & 20 & 8 & 2 \end{pmatrix}_{b_0}$$

[95] $1^{132}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 0 & 17 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 48 & 13 \\ 9 & 16 & 0 & 48 & 44 & 12 \\ 3 & 4 & -2 & 13 & 12 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{131}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 393, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 4 & 3 \\ 3 & 4 & 0 & 32 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 19 & 32 & 0 & 193 & 39 & 25 \\ 4 & 7 & 0 & 39 & 7 & 5 \\ 3 & 4 & -2 & 25 & 5 & 2 \end{pmatrix}_{b_0}$$

[97] $1^{112}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 396, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 9 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 26 & 7 \\ 10 & 16 & 0 & 26 & 44 & 12 \\ 3 & 4 & -2 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[98] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 11 & 20 & 0 & 14 & 69 & 15 \\ 3 & 4 & -2 & 3 & 15 & 2 \end{pmatrix}_{b_0}$$

[99] $1^{113}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 32 & 7 \\ 11 & 20 & 0 & 32 & 69 & 15 \\ 3 & 4 & -2 & 7 & 15 & 2 \end{pmatrix}_{b_0}$$

[100] $1^{114}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 9 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 38 & 7 \\ 13 & 24 & 0 & 38 & 100 & 18 \\ 3 & 4 & -2 & 7 & 18 & 2 \end{pmatrix}_{b_0}$$

[101] $1^{112}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 17 & 3 \\ 16 & 26 & 0 & 17 & 128 & 20 \\ 3 & 4 & -2 & 3 & 20 & 2 \end{pmatrix}_{b_0}$$

[102] $1^{112}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 437, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 13 & 21 & 0 & 14 & 80 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[104] $1^{122}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 9 & 3 \\ 3 & 4 & 0 & 13 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 37 & 10 \\ 9 & 16 & 0 & 37 & 44 & 12 \\ 3 & 4 & -2 & 10 & 12 & 2 \end{pmatrix}_{b_0}$$

[103] $1^{122}3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 13 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 37 & 10 \\ 9 & 16 & 0 & 37 & 44 & 12 \\ 3 & 4 & -2 & 10 & 12 & 2 \end{pmatrix}_{b_0}$$

[105] $1^{114}3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 453, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 11 & 19 & 0 & 14 & 64 & 14 \\ 3 & 4 & -2 & 3 & 14 & 2 \end{pmatrix}_{b_0}$$

[106] $1^{131}3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 474, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 4 & 3 \\ 3 & 4 & 0 & 17 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 21 & 13 \\ 4 & 7 & 0 & 21 & 7 & 5 \\ 3 & 4 & -2 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[107] $1^{142}3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 482, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 6 & 3 \\ 3 & 4 & 0 & 26 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 26 & 0 & 118 & 50 & 20 \\ 6 & 11 & 0 & 50 & 20 & 8 \\ 3 & 4 & -2 & 20 & 8 & 2 \end{pmatrix}_{b_0}$$

[108] $1^{143}3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 501, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 11 & 3 \\ 3 & 4 & 0 & 21 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 74 & 16 \\ 11 & 20 & 0 & 74 & 69 & 15 \\ 3 & 4 & -2 & 16 & 15 & 2 \end{pmatrix}_{b_0}$$

[109] $1^{141}3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 502, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 4 & 3 \\ 3 & 4 & 0 & 36 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 20 & 36 & 0 & 238 & 44 & 28 \\ 4 & 7 & 0 & 44 & 7 & 5 \\ 3 & 4 & -2 & 28 & 5 & 2 \end{pmatrix}_{b_0}$$

[110] $1^{123}3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 505, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 12 & 3 \\ 3 & 4 & 0 & 13 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 46 & 10 \\ 12 & 20 & 0 & 46 & 69 & 15 \\ 3 & 4 & -2 & 10 & 15 & 2 \end{pmatrix}_{b_0}$$

[111] $1^{113}3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 513, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 9 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 32 & 7 \\ 12 & 20 & 0 & 32 & 69 & 15 \\ 3 & 4 & -2 & 7 & 15 & 2 \end{pmatrix}_{b_0}$$

[112] $1^{134}3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 516, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 14 & 3 \\ 3 & 4 & 0 & 17 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 72 & 13 \\ 14 & 24 & 0 & 72 & 100 & 18 \\ 3 & 4 & -2 & 13 & 18 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{141}3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 522, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 4 & 3 \\ 3 & 4 & 0 & 26 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 32 & 20 \\ 4 & 7 & 0 & 32 & 7 & 5 \\ 3 & 4 & -2 & 20 & 5 & 2 \end{pmatrix}_{b_0}$$

[114] $1^{11} 1^{142} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 9 & 3 \\ 3 & 4 & 0 & 21 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 59 & 16 \\ 9 & 16 & 0 & 59 & 44 & 12 \\ 3 & 4 & -2 & 16 & 12 & 2 \end{pmatrix}_{b_0}$$

[116] $1^{11} 1^{134} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 13 & 3 \\ 3 & 4 & 0 & 17 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 72 & 13 \\ 13 & 24 & 0 & 72 & 100 & 18 \\ 3 & 4 & -2 & 13 & 18 & 2 \end{pmatrix}_{b_0}$$

[115] $1^{11} 1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 17 & 3 \\ 13 & 24 & 0 & 17 & 100 & 18 \\ 3 & 4 & -2 & 3 & 18 & 2 \end{pmatrix}_{b_0}$$

[117] $1^{11} 1^{132} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 552, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 9 & 3 \\ 3 & 4 & 0 & 17 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 48 & 13 \\ 9 & 16 & 0 & 48 & 44 & 12 \\ 3 & 4 & -2 & 13 & 12 & 2 \end{pmatrix}_{b_0}$$

[118] $1^{11} 1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 557, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 25 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 17 & 3 \\ 14 & 25 & 0 & 17 & 111 & 19 \\ 3 & 4 & -2 & 3 & 19 & 2 \end{pmatrix}_{b_0}$$

[119] $1^{11} 1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 560, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 14 & 3 \\ 12 & 20 & 0 & 14 & 69 & 15 \\ 3 & 4 & -2 & 3 & 15 & 2 \end{pmatrix}_{b_0}$$

[120] $1^{11} 1^{131} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 562, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 4 & 3 \\ 3 & 4 & 0 & 27 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 27 & 0 & 133 & 33 & 21 \\ 4 & 7 & 0 & 33 & 7 & 5 \\ 3 & 4 & -2 & 21 & 5 & 2 \end{pmatrix}_{b_0}$$

[121] $1^{11} 1^{142} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 570, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 6 & 3 \\ 3 & 4 & 0 & 21 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 41 & 16 \\ 6 & 11 & 0 & 41 & 20 & 8 \\ 3 & 4 & -2 & 16 & 8 & 2 \end{pmatrix}_{b_0}$$

[122] $1^{11} 1^{114} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 9 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 38 & 7 \\ 14 & 24 & 0 & 38 & 100 & 18 \\ 3 & 4 & -2 & 7 & 18 & 2 \end{pmatrix}_{b_0}$$

[123] $1^{11} 1^{123} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 581, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 0 & 13 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 46 & 10 \\ 11 & 20 & 0 & 46 & 69 & 15 \\ 3 & 4 & -2 & 10 & 15 & 2 \end{pmatrix}_{b_0}$$

[124] $1^{141}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 583, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 4 & 3 \\ 3 & 4 & 0 & 31 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 31 & 0 & 172 & 38 & 24 \\ 4 & 7 & 0 & 38 & 7 & 5 \\ 3 & 4 & -2 & 24 & 5 & 2 \end{pmatrix}_{b_0}$$

[125] $1^{131}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 589, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 4 & 3 \\ 3 & 4 & 0 & 22 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 22 & 0 & 85 & 27 & 17 \\ 4 & 7 & 0 & 27 & 7 & 5 \\ 3 & 4 & -2 & 17 & 5 & 2 \end{pmatrix}_{b_0}$$

[126] $1^{141}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 4 & 3 \\ 3 & 4 & 0 & 21 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 26 & 16 \\ 4 & 7 & 0 & 26 & 7 & 5 \\ 3 & 4 & -2 & 16 & 5 & 2 \end{pmatrix}_{b_0}$$

[127] $1^{133}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 605, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 11 & 3 \\ 3 & 4 & 0 & 17 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 60 & 13 \\ 11 & 20 & 0 & 60 & 69 & 15 \\ 3 & 4 & -2 & 13 & 15 & 2 \end{pmatrix}_{b_0}$$

[128] $1^{113}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 628, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 30 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 20 & 3 \\ 17 & 30 & 0 & 20 & 165 & 23 \\ 3 & 4 & -2 & 3 & 23 & 2 \end{pmatrix}_{b_0}$$

[129] $1^{133}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 11 & 3 \\ 3 & 4 & 0 & 17 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 60 & 13 \\ 11 & 20 & 0 & 60 & 69 & 15 \\ 3 & 4 & -2 & 13 & 15 & 2 \end{pmatrix}_{b_0}$$

[130] $1^{113}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 35 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 23 & 3 \\ 20 & 35 & 0 & 23 & 231 & 27 \\ 3 & 4 & -2 & 3 & 27 & 2 \end{pmatrix}_{b_0}$$

[131] $1^{124}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 664, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 13 & 3 \\ 3 & 4 & 0 & 13 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 55 & 10 \\ 13 & 24 & 0 & 55 & 100 & 18 \\ 3 & 4 & -2 & 10 & 18 & 2 \end{pmatrix}_{b_0}$$

[132] $1^{124}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 692, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 14 & 3 \\ 3 & 4 & 0 & 13 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 55 & 10 \\ 14 & 24 & 0 & 55 & 100 & 18 \\ 3 & 4 & -2 & 10 & 18 & 2 \end{pmatrix}_{b_0}$$

[133] $1^{113}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 709, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 25 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 17 & 3 \\ 15 & 25 & 0 & 17 & 111 & 19 \\ 3 & 4 & -2 & 3 & 19 & 2 \end{pmatrix}_{b_0}$$

[134] $1^{144}3^11^{111}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 744, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 13 & 3 \\ 3 & 4 & 0 & 21 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 89 & 16 \\ 13 & 24 & 0 & 89 & 100 & 18 \\ 3 & 4 & -2 & 16 & 18 & 2 \end{pmatrix}_{b_0}$$

[135] $1^{143}3^11^{111}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 749, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 11 & 3 \\ 3 & 4 & 0 & 21 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 74 & 16 \\ 11 & 20 & 0 & 74 & 69 & 15 \\ 3 & 4 & -2 & 16 & 15 & 2 \end{pmatrix}_{b_0}$$

[136] $1^{114}3^11^{11}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 752, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 17 & 3 \\ 14 & 24 & 0 & 17 & 100 & 18 \\ 3 & 4 & -2 & 3 & 18 & 2 \end{pmatrix}_{b_0}$$

[137] $1^{114}3^11^{12}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 765, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 20 & 3 \\ 16 & 29 & 0 & 20 & 148 & 22 \\ 3 & 4 & -2 & 3 & 22 & 2 \end{pmatrix}_{b_0}$$

[138] $1^{134}3^11^{111}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 13 & 3 \\ 3 & 4 & 0 & 17 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 72 & 13 \\ 13 & 24 & 0 & 72 & 100 & 18 \\ 3 & 4 & -2 & 13 & 18 & 2 \end{pmatrix}_{b_0}$$

[139] $1^{113}3^11^{13}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 788, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 30 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 20 & 3 \\ 18 & 30 & 0 & 20 & 165 & 23 \\ 3 & 4 & -2 & 3 & 23 & 2 \end{pmatrix}_{b_0}$$

[140] $1^{113}3^11^{14}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 797, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 21 & 3 \\ 3 & 4 & 0 & 4 & 35 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 23 & 3 \\ 21 & 35 & 0 & 23 & 231 & 27 \\ 3 & 4 & -2 & 3 & 27 & 2 \end{pmatrix}_{b_0}$$

[141] $1^{144}3^11^{111}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 808, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 13 & 3 \\ 3 & 4 & 0 & 21 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 89 & 16 \\ 13 & 24 & 0 & 89 & 100 & 18 \\ 3 & 4 & -2 & 16 & 18 & 2 \end{pmatrix}_{b_0}$$

[142] $1^{141}3^11^{12}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 822, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 4 & 3 \\ 3 & 4 & 0 & 26 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 26 & 0 & 118 & 32 & 20 \\ 4 & 7 & 0 & 32 & 7 & 5 \\ 3 & 4 & -2 & 20 & 5 & 2 \end{pmatrix}_{b_0}$$

[143] $1^{141}3^11^{14}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{so}_8+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 834, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 4 & 3 \\ 3 & 4 & 0 & 36 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 21 & 36 & 0 & 238 & 44 & 28 \\ 4 & 7 & 0 & 44 & 7 & 5 \\ 3 & 4 & -2 & 28 & 5 & 2 \end{pmatrix}_{b_0}$$

[144] $1^{\overline{114}} 3^1 1^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 899, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 4 & 3 \\ 3 & 4 & 0 & 31 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 31 & 0 & 172 & 38 & 24 \\ 4 & 7 & 0 & 38 & 7 & 5 \\ 3 & 4 & -2 & 24 & 5 & 2 \end{pmatrix}_{b_0}$$

[145] $1^{\overline{114}} 3^1 1^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 916, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 19 & 3 \\ 3 & 4 & 0 & 4 & 34 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 23 & 3 \\ 19 & 34 & 0 & 23 & 208 & 26 \\ 3 & 4 & -2 & 3 & 26 & 2 \end{pmatrix}_{b_0}$$

[146] $1^{\overline{114}} 3^1 1^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 981, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 20 & 3 \\ 17 & 29 & 0 & 20 & 148 & 22 \\ 3 & 4 & -2 & 3 & 22 & 2 \end{pmatrix}_{b_0}$$

[147] $1^{\overline{114}} 3^1 1^{\overline{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 997, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 22 & 3 \\ 3 & 4 & 0 & 4 & 39 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 26 & 3 \\ 22 & 39 & 0 & 26 & 280 & 30 \\ 3 & 4 & -2 & 3 & 30 & 2 \end{pmatrix}_{b_0}$$

[148] $1^{\overline{114}} 3^1 1^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 1140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 34 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 23 & 3 \\ 20 & 34 & 0 & 23 & 208 & 26 \\ 3 & 4 & -2 & 3 & 26 & 2 \end{pmatrix}_{b_0}$$

[149] $1^{\overline{114}} 3^1 1^{\overline{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 1229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 23 & 3 \\ 3 & 4 & 0 & 4 & 39 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 26 & 3 \\ 23 & 39 & 0 & 26 & 280 & 30 \\ 3 & 4 & -2 & 3 & 30 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [150] $1^{\overline{11}} 3^1 21^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 8 & 3 \\ 2 & 1 & 1 & 4 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 4 & 0 & 10 & 20 & 6 \\ 8 & 9 & 0 & 20 & 35 & 11 \\ 3 & 3 & -1 & 6 & 11 & 3 \end{pmatrix}_{b_0}$$

[151] $1^{\overline{111}} 3^1 21^{\overline{121}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 4 & 2 \\ 1 & 2 & 0 & 0 & 7 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 7 & 0 & 16 & 14 & 8 \\ 3 & 4 & 5 & 0 & 14 & 10 & 6 \\ 2 & 2 & 2 & -2 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[152] $1^{\overline{122}} 3^1 21^{\overline{132}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 7 & 2 \\ 1 & 2 & 0 & 0 & 11 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 11 & 0 & 45 & 41 & 13 \\ 5 & 7 & 9 & 0 & 41 & 35 & 11 \\ 2 & 2 & 2 & -2 & 13 & 11 & 2 \end{pmatrix}_{b_0}$$

- [153] $1^{11}3^121^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 7 & 3 \\ 3 & 5 & 5 & 5 & 13 & 5 \\ 3 & 5 & 0 & 4 & 9 & 2 \\ 2 & 5 & 4 & 2 & 9 & 4 \\ 7 & 13 & 9 & 9 & 26 & 10 \\ 3 & 5 & 2 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$
- [154] $1^{11}3^121^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 8 & 3 \\ 2 & 1 & 1 & 13 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 11 & 13 & 0 & 61 & 47 & 15 \\ 8 & 9 & 0 & 47 & 35 & 11 \\ 3 & 3 & -1 & 15 & 11 & 3 \end{pmatrix}_{b_0}$$
- [155] $1^{11}3^121^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 5 & 0 & 10 & 13 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 10 & 0 & 14 & 19 & 8 \\ 6 & 13 & 0 & 19 & 24 & 10 \\ 3 & 5 & -2 & 8 & 10 & 3 \end{pmatrix}_{b_0}$$
- [156] $1^{11}3^121^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 1 & 4 \\ 1 & 2 & 0 & 0 & 3 & 5 \\ 1 & 0 & 0 & -4 & 0 & 0 \\ 1 & 1 & 3 & 0 & 1 & 5 \\ 3 & 4 & 5 & 0 & 5 & 10 \\ 2 & 2 & 2 & -2 & 3 & 6 \end{pmatrix}_{b_0}$$
- [157] $1^{121}3^121^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 2 & 0 & 7 & 4 & 2 \\ 1 & 2 & 0 & 0 & 11 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 11 & 0 & 45 & 23 & 13 \\ 3 & 4 & 5 & 0 & 23 & 10 & 6 \\ 2 & 2 & 2 & -2 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$
- [158] $1^{112}3^121^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 0 & 7 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 7 & 0 & 16 & 25 & 8 \\ 5 & 7 & 9 & 0 & 25 & 35 & 11 \\ 2 & 2 & 2 & -2 & 8 & 11 & 2 \end{pmatrix}_{b_0}$$
- [159] $1^{131}3^121^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 8 & 3 \\ 2 & 1 & 1 & 13 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 13 & 0 & 61 & 47 & 15 \\ 8 & 9 & 0 & 47 & 35 & 11 \\ 3 & 3 & -1 & 15 & 11 & 3 \end{pmatrix}_{b_0}$$
- [160] $1^{111}3^121^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 8 & 3 \\ 2 & 1 & 1 & 11 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 12 & 11 & 0 & 65 & 49 & 15 \\ 8 & 9 & 0 & 49 & 35 & 11 \\ 3 & 3 & -1 & 15 & 11 & 3 \end{pmatrix}_{b_0}$$
- [161] $1^{111}3^121^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 9 & 3 \\ 8 & 9 & 0 & 9 & 35 & 11 \\ 3 & 3 & -1 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$
- [162] $1^{123}3^121^{133} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -37, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 10 & 2 \\ 1 & 2 & 0 & 0 & 11 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 11 & 0 & 45 & 59 & 13 \\ 7 & 10 & 13 & 0 & 59 & 74 & 16 \\ 2 & 2 & 2 & -2 & 13 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[163] \ 1^{\overline{13}} 3^{\overline{1}} 21^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -44, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 5 & 0 & 5 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 2 & 6 & 4 \\ 4 & 8 & 0 & 6 & 8 & 6 \\ 3 & 5 & -2 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[164] \ 1^{\overline{13}} 3^{\overline{1}} 21^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -50, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.96 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 5 & 0 & 5 & 13 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 2 & 10 & 4 \\ 6 & 13 & 0 & 10 & 24 & 10 \\ 3 & 5 & -2 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[165] \ 1^{\overline{132}} 3^{\overline{1}} 21^{\overline{142}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -53, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.96 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 5 & 2 \\ 1 & 0 & 2 & 0 & 10 & 7 & 2 \\ 1 & 2 & 0 & 0 & 15 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 10 & 15 & 0 & 88 & 57 & 18 \\ 5 & 7 & 9 & 0 & 57 & 35 & 11 \\ 2 & 2 & 2 & -2 & 18 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[166] \ 1^{\overline{131}} 3^{\overline{1}} 21^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -53, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.96 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 2 & 0 & 7 & 4 & 2 \\ 1 & 2 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 6 & 0 & 25 & 17 & 9 \\ 3 & 4 & 5 & 0 & 17 & 10 & 6 \\ 2 & 2 & 2 & -2 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[167] \ 1^{\overline{123}} 3^{\overline{1}} 21^{\overline{132}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -62, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.97 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 6 & 3 \\ 3 & 5 & 0 & 25 & 13 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 25 & 0 & 94 & 48 & 20 \\ 6 & 13 & 0 & 48 & 24 & 10 \\ 3 & 5 & -2 & 20 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[168] \ 1^{\overline{111}} 3^{\overline{1}} 21^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -62, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 8 & 3 \\ 2 & 1 & 1 & 7 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 29 & 9 \\ 8 & 9 & 0 & 29 & 35 & 11 \\ 3 & 3 & -1 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[169] \ 1^{\overline{111}} 3^{\overline{1}} 21^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -63, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 9 & 3 \\ 2 & 1 & 1 & 7 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 29 & 9 \\ 9 & 9 & 0 & 29 & 35 & 11 \\ 3 & 3 & -1 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[170] \ 1^{\overline{112}} 3^{\overline{1}} 21^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -69, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 15 & 3 \\ 2 & 1 & 1 & 3 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 16 & 3 \\ 15 & 16 & 0 & 16 & 116 & 20 \\ 3 & 3 & -1 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[171] \ 1^{\overline{111}} 3^{\overline{1}} 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -75, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.95 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 9 & 3 \\ 9 & 9 & 0 & 9 & 35 & 11 \\ 3 & 3 & -1 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

[172] $1^{11}3^{121}21^{142} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 6 & 3 \\ 3 & 5 & 0 & 25 & 13 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 25 & 0 & 92 & 49 & 20 \\ 6 & 13 & 0 & 49 & 24 & 10 \\ 3 & 5 & -2 & 20 & 10 & 3 \end{pmatrix}_{b_0}$$

[173] $1^{11}3^{121}21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 0 & 7 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 7 & 0 & 16 & 36 & 8 \\ 7 & 10 & 13 & 0 & 36 & 74 & 16 \\ 2 & 2 & 2 & -2 & 8 & 16 & 2 \end{pmatrix}_{b_0}$$

[174] $1^{124}3^{121}21^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 0 & 7 & 13 & 2 \\ 1 & 2 & 0 & 0 & 11 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 11 & 0 & 45 & 77 & 13 \\ 9 & 13 & 17 & 0 & 77 & 127 & 21 \\ 2 & 2 & 2 & -2 & 13 & 21 & 2 \end{pmatrix}_{b_0}$$

[175] $1^{11}3^{121}21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 6 & 5 \\ 2 & 2 & 4 & 0 & 6 & 4 & 4 \\ 2 & 2 & 2 & -2 & 5 & 4 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{123}3^{121}21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 1 & 7 & 2 \\ 1 & 2 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 1 & 3 & 0 & 1 & 9 & 3 \\ 5 & 7 & 9 & 0 & 9 & 35 & 11 \\ 2 & 2 & 2 & -2 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

[177] $1^{142}3^{121}21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 5 & 2 \\ 1 & 0 & 2 & 0 & 10 & 7 & 2 \\ 1 & 2 & 0 & 0 & 10 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 10 & 10 & 0 & 58 & 46 & 14 \\ 5 & 7 & 9 & 0 & 46 & 35 & 11 \\ 2 & 2 & 2 & -2 & 14 & 11 & 2 \end{pmatrix}_{b_0}$$

[178] $1^{11}3^{121}21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 3 \\ 3 & 5 & 0 & 20 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 20 & 0 & 58 & 24 & 16 \\ 4 & 8 & 0 & 24 & 8 & 6 \\ 3 & 5 & -2 & 16 & 6 & 3 \end{pmatrix}_{b_0}$$

[179] $1^{11}3^{121}21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 0 & 1 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 2 & 1 & 0 & 0 & 4 & 2 \\ 3 & 4 & 5 & 0 & 4 & 10 & 6 \\ 2 & 2 & 2 & -2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[180] $1^{11}3^{121}21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 10 & 3 \\ 2 & 1 & 1 & 7 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 36 & 9 \\ 10 & 12 & 0 & 36 & 56 & 14 \\ 3 & 3 & -1 & 9 & 14 & 3 \end{pmatrix}_{b_0}$$

[181] $1^{132}3^{121}21^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$\begin{aligned}
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 14 & 10 & 3 \\ 2 & 1 & 1 & 17 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 14 & 17 & 0 & 121 & 84 & 21 \\ 10 & 12 & 0 & 84 & 56 & 14 \\ 3 & 3 & -1 & 21 & 14 & 3 \end{pmatrix}_{b_0} \\
[182] & 1^{\overline{131}1} 3^{\overline{1}21} 1^{\overline{141}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8 \\
& T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 2 & 0 & 10 & 4 & 2 \\ 1 & 2 & 0 & 0 & 15 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 10 & 15 & 0 & 88 & 32 & 18 \\ 3 & 4 & 5 & 0 & 32 & 10 & 6 \\ 2 & 2 & 2 & -2 & 18 & 6 & 2 \end{pmatrix}_{b_0} \\
[183] & 1^{\overline{11}1} 3^{\overline{1}21} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 5 & 0 & 10 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 10 & 0 & 14 & 12 & 8 \\ 4 & 8 & 0 & 12 & 8 & 6 \\ 3 & 5 & -2 & 8 & 6 & 3 \end{pmatrix}_{b_0} \\
[184] & 1^{\overline{121}1} 3^{\overline{1}21} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 8 & 8 & 3 \\ 2 & 1 & 1 & 10 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 38 & 12 \\ 8 & 9 & 0 & 38 & 35 & 11 \\ 3 & 3 & -1 & 12 & 11 & 3 \end{pmatrix}_{b_0} \\
[185] & 1^{\overline{122}1} 3^{\overline{1}21} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 8 & 11 & 3 \\ 2 & 1 & 1 & 10 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 48 & 12 \\ 11 & 12 & 0 & 48 & 56 & 14 \\ 3 & 3 & -1 & 12 & 14 & 3 \end{pmatrix}_{b_0} \\
[186] & 1^{\overline{131}1} 3^{\overline{1}21} 1^{\overline{121}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 14 & 8 & 3 \\ 2 & 1 & 1 & 17 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 14 & 17 & 0 & 121 & 67 & 21 \\ 8 & 9 & 0 & 67 & 35 & 11 \\ 3 & 3 & -1 & 21 & 11 & 3 \end{pmatrix}_{b_0} \\
[187] & 1^{\overline{111}1} 3^{\overline{1}21} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 7 & 8 & 3 \\ 2 & 1 & 1 & 7 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 7 & 7 & 0 & 21 & 29 & 9 \\ 8 & 9 & 0 & 29 & 35 & 11 \\ 3 & 3 & -1 & 9 & 11 & 3 \end{pmatrix}_{b_0} \\
[188] & 1^{\overline{11}1} 3^{\overline{1}21} 1^{\overline{122}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 5 & 0 & 15 & 13 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 15 & 0 & 32 & 29 & 12 \\ 6 & 13 & 0 & 29 & 24 & 10 \\ 3 & 5 & -2 & 12 & 10 & 3 \end{pmatrix}_{b_0} \\
[189] & 1^{\overline{122}1} 3^{\overline{1}21} 1^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 8 & 15 & 3 \\ 2 & 1 & 1 & 10 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 68 & 12 \\ 15 & 16 & 0 & 68 & 116 & 20 \\ 3 & 3 & -1 & 12 & 20 & 3 \end{pmatrix}_{b_0} \\
[190] & 1^{\overline{113}1} 3^{\overline{1}21} 1^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0
\end{aligned}$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 16 & 3 \\ 2 & 1 & 1 & 3 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 19 & 3 \\ 16 & 19 & 0 & 19 & 151 & 23 \\ 3 & 3 & -1 & 3 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[191] \ 1^{112} 3^{121} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 22 & 3 \\ 2 & 1 & 1 & 3 & 23 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 23 & 3 \\ 22 & 23 & 0 & 23 & 243 & 29 \\ 3 & 3 & -1 & 3 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[192] \ 1^{111} 3^{121} 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 8 & 3 \\ 2 & 1 & 1 & 10 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 9 & 10 & 0 & 38 & 38 & 12 \\ 8 & 9 & 0 & 38 & 35 & 11 \\ 3 & 3 & -1 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[194] \ 1^{112} 3^{121} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 12 & 3 \\ 10 & 12 & 0 & 12 & 56 & 14 \\ 3 & 3 & -1 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[195] \ 1^{11} 3^{121} 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 5 & 0 & 15 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 15 & 0 & 32 & 18 & 12 \\ 4 & 8 & 0 & 18 & 8 & 6 \\ 3 & 5 & -2 & 12 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[193] \ 1^{112} 3^{121} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 16 & 3 \\ 2 & 1 & 1 & 3 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 16 & 3 \\ 16 & 16 & 0 & 16 & 116 & 20 \\ 3 & 3 & -1 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[196] \ 1^{11} 3^{121} 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 8 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 6 & 8 & 0 & 26 & 12 \\ 2 & 2 & 4 & 0 & 12 & 4 \\ 2 & 2 & 2 & -2 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[197] \ 1^{11} 3^{121} 21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 6 & 3 \\ 3 & 5 & 0 & 20 & 13 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 20 & 0 & 58 & 39 & 16 \\ 6 & 13 & 0 & 39 & 24 & 10 \\ 3 & 5 & -2 & 16 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[198] \ 1^{112} 3^{121} 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 8 & 3 \\ 2 & 1 & 1 & 18 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 18 & 18 & 0 & 166 & 78 & 24 \\ 8 & 9 & 0 & 78 & 35 & 11 \\ 3 & 3 & -1 & 24 & 11 & 3 \end{pmatrix}_{b_0}$$

[199] $1^{141}3^{121}21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 8 & 3 \\ 2 & 1 & 1 & 20 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 16 & 20 & 0 & 158 & 76 & 24 \\ 8 & 9 & 0 & 76 & 35 & 11 \\ 3 & 3 & -1 & 24 & 11 & 3 \end{pmatrix}_{b_0}$$

[200] $1^{123}3^{121}1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 1 & 8 & 2 \\ 1 & 2 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 1 & 3 & 0 & 1 & 12 & 3 \\ 6 & 8 & 12 & 0 & 12 & 56 & 14 \\ 2 & 2 & 2 & -2 & 3 & 14 & 2 \end{pmatrix}_{b_0}$$

[201] $1^{123}3^{121}1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 4 & 3 \\ 3 & 5 & 0 & 20 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 20 & 0 & 62 & 24 & 16 \\ 4 & 8 & 0 & 24 & 8 & 6 \\ 3 & 5 & -2 & 16 & 6 & 3 \end{pmatrix}_{b_0}$$

[202] $1^{114}3^{121}1^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -149, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 13 & 2 \\ 1 & 2 & 0 & 0 & 7 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 7 & 0 & 16 & 47 & 8 \\ 9 & 13 & 17 & 0 & 47 & 127 & 21 \\ 2 & 2 & 2 & -2 & 8 & 21 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{112}3^{121}1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 16 & 3 \\ 2 & 1 & 1 & 7 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 52 & 9 \\ 16 & 16 & 0 & 52 & 116 & 20 \\ 3 & 3 & -1 & 9 & 20 & 3 \end{pmatrix}_{b_0}$$

[204] $1^{133}3^{121}1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 1 & 10 & 2 \\ 1 & 2 & 0 & 0 & 3 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 1 & 3 & 0 & 1 & 13 & 3 \\ 7 & 10 & 13 & 0 & 13 & 74 & 16 \\ 2 & 2 & 2 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[205] $1^{113}3^{121}1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 5 & 0 & 5 & 23 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 2 & 17 & 4 \\ 11 & 23 & 0 & 17 & 78 & 18 \\ 3 & 5 & -2 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[206] $1^{123}3^{121}1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -170, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 21 & 3 \\ 2 & 1 & 1 & 10 & 23 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 98 & 12 \\ 21 & 23 & 0 & 98 & 243 & 29 \\ 3 & 3 & -1 & 12 & 29 & 3 \end{pmatrix}_{b_0}$$

[207] $1^{113}3^{121}1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -170, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$\begin{aligned}
& T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 6 & 16 & 3 \\ 2 & 1 & 1 & 7 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 59 & 9 \\ 16 & 19 & 0 & 59 & 151 & 23 \\ 3 & 3 & -1 & 9 & 23 & 3 \end{pmatrix}_{b_0} \\
[208] & 1^{\overline{111}3^1 21^{\overline{122}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 5 & 2 \\ 1 & 2 & 0 & 0 & 7 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 7 & 0 & 16 & 21 & 8 \\ 4 & 5 & 8 & 0 & 21 & 23 & 9 \\ 2 & 2 & 2 & -2 & 8 & 9 & 2 \end{pmatrix}_{b_0} \\
[209] & 1^{\overline{111}3^1 21^{\overline{12}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 0 & 1 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 2 & 1 & 0 & 0 & 7 & 2 \\ 5 & 7 & 9 & 0 & 7 & 35 & 11 \\ 2 & 2 & 2 & -2 & 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[210] & 1^{\overline{111}3^1 21^{\overline{112}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 6 & 15 & 3 \\ 2 & 1 & 1 & 7 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 52 & 9 \\ 15 & 16 & 0 & 52 & 116 & 20 \\ 3 & 3 & -1 & 9 & 20 & 3 \end{pmatrix}_{b_0} \\
[211] & 1^{\overline{132}3^1 21^{\overline{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 10 & 10 & 3 \\ 2 & 1 & 1 & 13 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 13 & 0 & 61 & 60 & 15 \\ 10 & 12 & 0 & 60 & 56 & 14 \\ 3 & 3 & -1 & 15 & 14 & 3 \end{pmatrix}_{b_0} \\
[212] & 1^{\overline{111}3^1 21^{\overline{114}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 5 & 0 & 5 & 28 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 2 & 21 & 4 \\ 13 & 28 & 0 & 21 & 116 & 22 \\ 3 & 5 & -2 & 4 & 22 & 3 \end{pmatrix}_{b_0} \\
[213] & 1^{\overline{111}3^1 21^{\overline{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 4 & 2 \\ 1 & 2 & 0 & 0 & 4 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 9 & 5 \\ 3 & 4 & 5 & 0 & 9 & 10 & 6 \\ 2 & 2 & 2 & -2 & 5 & 6 & 2 \end{pmatrix}_{b_0} \\
[214] & 1^{\overline{111}3^1 21^{\overline{112}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 6 & 18 & 3 \\ 2 & 1 & 1 & 7 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 59 & 9 \\ 18 & 19 & 0 & 59 & 151 & 23 \\ 3 & 3 & -1 & 9 & 23 & 3 \end{pmatrix}_{b_0} \\
[215] & 1^{\overline{131}3^1 21^{\overline{141}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8 \\
& T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 1 & 24 & 8 & 3 \\ 2 & 1 & 1 & 25 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 24 & 25 & 0 & 313 & 107 & 33 \\ 8 & 9 & 0 & 107 & 35 & 11 \\ 3 & 3 & -1 & 33 & 11 & 3 \end{pmatrix}_{b_0}$$

[216] $1^{11} 1^{142} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 10 & 3 \\ 2 & 1 & 1 & 16 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 13 & 16 & 0 & 90 & 72 & 18 \\ 10 & 12 & 0 & 72 & 56 & 14 \\ 3 & 3 & -1 & 18 & 14 & 3 \end{pmatrix}_{b_0}$$

[217] $1^{12} 1^{124} 3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 28 & 3 \\ 2 & 1 & 1 & 10 & 30 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 128 & 12 \\ 28 & 30 & 0 & 128 & 416 & 38 \\ 3 & 3 & -1 & 12 & 38 & 3 \end{pmatrix}_{b_0}$$

[218] $1^{11} 1^{122} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 10 & 3 \\ 2 & 1 & 1 & 10 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 48 & 12 \\ 10 & 12 & 0 & 48 & 56 & 14 \\ 3 & 3 & -1 & 12 & 14 & 3 \end{pmatrix}_{b_0}$$

[219] $1^{11} 1^{121} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -197, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 4 & 2 \\ 1 & 2 & 0 & 0 & 5 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 5 & 0 & 13 & 13 & 7 \\ 3 & 4 & 5 & 0 & 13 & 10 & 6 \\ 2 & 2 & 2 & -2 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[220] $1^{11} 1^{121} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 8 & 3 \\ 2 & 1 & 1 & 14 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 13 & 14 & 0 & 90 & 58 & 18 \\ 8 & 9 & 0 & 58 & 35 & 11 \\ 3 & 3 & -1 & 18 & 11 & 3 \end{pmatrix}_{b_0}$$

[221] $1^{12} 1^{121} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 8 & 3 \\ 2 & 1 & 1 & 14 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 14 & 14 & 0 & 90 & 58 & 18 \\ 8 & 9 & 0 & 58 & 35 & 11 \\ 3 & 3 & -1 & 18 & 11 & 3 \end{pmatrix}_{b_0}$$

[222] $1^{11} 1^{12} 3^1 21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 6 & 3 \\ 3 & 5 & 0 & 30 & 13 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 30 & 0 & 134 & 58 & 24 \\ 6 & 13 & 0 & 58 & 24 & 10 \\ 3 & 5 & -2 & 24 & 10 & 3 \end{pmatrix}_{b_0}$$

[223] $1^{11} 1^{13} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 & 2 \\ 1 & 2 & 0 & 0 & 12 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 12 & 0 & 61 & 18 & 15 \\ 2 & 2 & 4 & 0 & 18 & 4 & 4 \\ 2 & 2 & 2 & -2 & 15 & 4 & 2 \end{pmatrix}_{b_0}$$

[224] $1^{114}3^121^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 6 & 18 & 3 \\ 2 & 1 & 1 & 7 & 22 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 66 & 9 \\ 18 & 22 & 0 & 66 & 192 & 26 \\ 3 & 3 & -1 & 9 & 26 & 3 \end{pmatrix}_{b_0}$

[225] $1^{112}3^121^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 6 & 20 & 3 \\ 2 & 1 & 1 & 7 & 20 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 68 & 9 \\ 20 & 20 & 0 & 68 & 200 & 26 \\ 3 & 3 & -1 & 9 & 26 & 3 \end{pmatrix}_{b_0}$

[226] $1^{112}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -217, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 11 & 3 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 12 & 3 \\ 11 & 12 & 0 & 12 & 56 & 14 \\ 3 & 3 & -1 & 3 & 14 & 3 \end{pmatrix}_{b_0}$

[227] $1^{111}3^121^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 5 & 2 \\ 1 & 2 & 0 & 0 & 4 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 13 & 5 \\ 4 & 5 & 8 & 0 & 13 & 23 & 9 \\ 2 & 2 & 2 & -2 & 5 & 9 & 2 \end{pmatrix}_{b_0}$

[228] $1^{141}3^121^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 18 & 8 & 3 \\ 2 & 1 & 1 & 20 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 18 & 20 & 0 & 158 & 76 & 24 \\ 8 & 9 & 0 & 76 & 35 & 11 \\ 3 & 3 & -1 & 24 & 11 & 3 \end{pmatrix}_{b_0}$

[229] $1^{131}3^121^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 19 & 8 & 3 \\ 2 & 1 & 1 & 21 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 19 & 21 & 0 & 205 & 87 & 27 \\ 8 & 9 & 0 & 87 & 35 & 11 \\ 3 & 3 & -1 & 27 & 11 & 3 \end{pmatrix}_{b_0}$

[230] $1^{123}3^121^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 12 & 4 & 3 \\ 3 & 5 & 0 & 25 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 25 & 0 & 94 & 30 & 20 \\ 4 & 8 & 0 & 30 & 8 & 6 \\ 3 & 5 & -2 & 20 & 6 & 3 \end{pmatrix}_{b_0}$

[231] $1^{122}3^121^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 9 & 10 & 3 \\ 2 & 1 & 1 & 10 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 9 & 10 & 0 & 38 & 48 & 12 \\ 10 & 12 & 0 & 48 & 56 & 14 \\ 3 & 3 & -1 & 12 & 14 & 3 \end{pmatrix}_{b_0}$

[232] $1^{142}3^121^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 10 & 3 \\ 2 & 1 & 1 & 20 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 16 & 20 & 0 & 158 & 96 & 24 \\ 10 & 12 & 0 & 96 & 56 & 14 \\ 3 & 3 & -1 & 24 & 14 & 3 \end{pmatrix}_{b_0}$$

[233] $1^{11} 1^{13} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 17 & 3 \\ 2 & 1 & 1 & 7 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 59 & 9 \\ 17 & 19 & 0 & 59 & 151 & 23 \\ 3 & 3 & -1 & 9 & 23 & 3 \end{pmatrix}_{b_0}$$

[234] $1^{11} 1^{13} 3^1 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 1 & 11 & 2 \\ 1 & 2 & 0 & 0 & 3 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 1 & 3 & 0 & 1 & 16 & 3 \\ 8 & 11 & 16 & 0 & 16 & 103 & 19 \\ 2 & 2 & 2 & -2 & 3 & 19 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{12} 1^{12} 3^1 21^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -252, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 2 & 0 & 6 & 4 & 2 \\ 1 & 2 & 0 & 0 & 8 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 8 & 0 & 26 & 18 & 10 \\ 3 & 4 & 5 & 0 & 18 & 10 & 6 \\ 2 & 2 & 2 & -2 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[236] $1^{12} 1^{12} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 20 & 3 \\ 2 & 1 & 1 & 10 & 22 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 88 & 12 \\ 20 & 22 & 0 & 88 & 192 & 26 \\ 3 & 3 & -1 & 12 & 26 & 3 \end{pmatrix}_{b_0}$$

[237] $1^{11} 1^{12} 3^1 21^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 4 & 3 \\ 3 & 5 & 0 & 30 & 8 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 30 & 0 & 134 & 36 & 24 \\ 4 & 8 & 0 & 36 & 8 & 6 \\ 3 & 5 & -2 & 24 & 6 & 3 \end{pmatrix}_{b_0}$$

[238] $1^{11} 1^{14} 3^1 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 1 & 13 & 2 \\ 1 & 2 & 0 & 0 & 3 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 1 & 3 & 0 & 1 & 17 & 3 \\ 9 & 13 & 17 & 0 & 17 & 127 & 21 \\ 2 & 2 & 2 & -2 & 3 & 21 & 2 \end{pmatrix}_{b_0}$$

[239] $1^{11} 1^{14} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -261, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 19 & 3 \\ 2 & 1 & 1 & 7 & 22 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 66 & 9 \\ 19 & 22 & 0 & 66 & 192 & 26 \\ 3 & 3 & -1 & 9 & 26 & 3 \end{pmatrix}_{b_0}$$

[241] $1^{11} 1^{14} 3^1 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 18 & 3 \\ 2 & 1 & 1 & 3 & 22 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 22 & 3 \\ 18 & 22 & 0 & 22 & 192 & 26 \\ 3 & 3 & -1 & 3 & 26 & 3 \end{pmatrix}_{b_0}$$

[240] $1^{1^{12}}3^{1^{11}}21^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 12 & 3 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 12 & 3 \\ 12 & 12 & 0 & 12 & 56 & 14 \\ 3 & 3 & -1 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[242] $1^{1^{11}}3^{1^{14}}21^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -265, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 23 & 3 \\ 2 & 1 & 1 & 3 & 26 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 26 & 3 \\ 23 & 26 & 0 & 26 & 292 & 32 \\ 3 & 3 & -1 & 3 & 32 & 3 \end{pmatrix}_{b_0}$$

[243] $1^{1^{12}}3^{1^{13}}21^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -266, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 16 & 3 \\ 2 & 1 & 1 & 10 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 78 & 12 \\ 16 & 19 & 0 & 78 & 151 & 23 \\ 3 & 3 & -1 & 12 & 23 & 3 \end{pmatrix}_{b_0}$$

[244] $1^{1^{11}}3^{1^{13}}21^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 10 & 3 \\ 2 & 1 & 1 & 13 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 11 & 13 & 0 & 61 & 60 & 15 \\ 10 & 12 & 0 & 60 & 56 & 14 \\ 3 & 3 & -1 & 15 & 14 & 3 \end{pmatrix}_{b_0}$$

[245] $1^{1^{12}}3^{1^{13}}21^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -270, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 8 & 3 \\ 2 & 1 & 1 & 17 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 16 & 17 & 0 & 121 & 67 & 21 \\ 8 & 9 & 0 & 67 & 35 & 11 \\ 3 & 3 & -1 & 21 & 11 & 3 \end{pmatrix}_{b_0}$$

[246] $1^{1^{11}}3^{1^{13}}21^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 17 & 3 \\ 2 & 1 & 1 & 3 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 19 & 3 \\ 17 & 19 & 0 & 19 & 151 & 23 \\ 3 & 3 & -1 & 3 & 23 & 3 \end{pmatrix}_{b_0}$$

[247] $1^{1^{12}}3^{1^{14}}21^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 5 & 2 \\ 1 & 2 & 0 & 0 & 8 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 8 & 0 & 26 & 26 & 10 \\ 4 & 5 & 8 & 0 & 26 & 23 & 9 \\ 2 & 2 & 2 & -2 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[248] $1^{1^{11}}3^{1^{13}}21^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -279, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 21 & 3 \\ 2 & 1 & 1 & 7 & 23 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 75 & 9 \\ 21 & 23 & 0 & 75 & 243 & 29 \\ 3 & 3 & -1 & 9 & 29 & 3 \end{pmatrix}_{b_0}$$

[249] $1^{1^{13}}3^{1^{14}}21^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 0 & 1 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 2 & 1 & 0 & 0 & 10 & 2 \\ 7 & 10 & 13 & 0 & 10 & 74 & 16 \\ 2 & 2 & 2 & -2 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[250] $1^{141}3^121^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 2 & 0 & 10 & 4 & 2 \\ 1 & 2 & 0 & 0 & 10 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 10 & 10 & 0 & 58 & 26 & 14 \\ 3 & 4 & 5 & 0 & 26 & 10 & 6 \\ 2 & 2 & 2 & -2 & 14 & 6 & 2 \end{pmatrix}_{b_0}$$

[251] $1^{141}3^121^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 8 & 3 \\ 2 & 1 & 1 & 20 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 17 & 20 & 0 & 158 & 76 & 24 \\ 8 & 9 & 0 & 76 & 35 & 11 \\ 3 & 3 & -1 & 24 & 11 & 3 \end{pmatrix}_{b_0}$$

[252] $1^{112}3^121^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 7 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 7 & 0 & 16 & 32 & 8 \\ 6 & 8 & 12 & 0 & 32 & 56 & 14 \\ 2 & 2 & 2 & -2 & 8 & 14 & 2 \end{pmatrix}_{b_0}$$

[253] $1^{131}3^121^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 8 & 3 \\ 2 & 1 & 1 & 17 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 15 & 17 & 0 & 121 & 67 & 21 \\ 8 & 9 & 0 & 67 & 35 & 11 \\ 3 & 3 & -1 & 21 & 11 & 3 \end{pmatrix}_{b_0}$$

[254] $1^{122}3^121^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 7 & 2 \\ 1 & 2 & 0 & 0 & 5 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 5 & 0 & 13 & 23 & 7 \\ 5 & 7 & 9 & 0 & 23 & 35 & 11 \\ 2 & 2 & 2 & -2 & 7 & 11 & 2 \end{pmatrix}_{b_0}$$

[255] $1^{121}3^121^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -309, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 5 & 2 \\ 1 & 2 & 0 & 0 & 11 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 11 & 0 & 45 & 34 & 13 \\ 4 & 5 & 8 & 0 & 34 & 23 & 9 \\ 2 & 2 & 2 & -2 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[256] $1^{123}3^121^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -310, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 5 & 0 & 5 & 33 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 2 & 24 & 4 \\ 16 & 33 & 0 & 24 & 164 & 26 \\ 3 & 5 & -2 & 4 & 26 & 3 \end{pmatrix}_{b_0}$$

[257] $1^{122}3^121^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 6 & 2 \\ 1 & 2 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 5 & 0 & 13 & 18 & 7 \\ 4 & 6 & 6 & 0 & 18 & 20 & 8 \\ 2 & 2 & 2 & -2 & 7 & 8 & 2 \end{pmatrix}_{b_0}$$

[258] $1^{11} 1^{142} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 21 & 10 & 3 \\ 2 & 1 & 1 & 24 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 21 & 24 & 0 & 250 & 120 & 30 \\ 10 & 12 & 0 & 120 & 56 & 14 \\ 3 & 3 & -1 & 30 & 14 & 3 \end{pmatrix}_{b_0}$$

[259] $1^{11} 1^{14} 3^1 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 & 2 \\ 1 & 2 & 0 & 0 & 16 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 16 & 0 & 110 & 24 & 20 \\ 2 & 2 & 4 & 0 & 24 & 4 & 4 \\ 2 & 2 & 2 & -2 & 20 & 4 & 2 \end{pmatrix}_{b_0}$$

[260] $1^{11} 1^{123} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 17 & 3 \\ 2 & 1 & 1 & 10 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 78 & 12 \\ 17 & 19 & 0 & 78 & 151 & 23 \\ 3 & 3 & -1 & 12 & 23 & 3 \end{pmatrix}_{b_0}$$

[261] $1^{11} 1^{113} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 0 & 1 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 2 & 1 & 0 & 0 & 9 & 2 \\ 6 & 9 & 10 & 0 & 9 & 51 & 13 \\ 2 & 2 & 2 & -2 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[262] $1^{12} 1^{113} 3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 26 & 3 \\ 2 & 1 & 1 & 7 & 27 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 91 & 9 \\ 26 & 27 & 0 & 91 & 359 & 35 \\ 3 & 3 & -1 & 9 & 35 & 3 \end{pmatrix}_{b_0}$$

[263] $1^{11} 1^{131} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 2 & 0 & 8 & 4 & 2 \\ 1 & 2 & 0 & 0 & 9 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 9 & 0 & 40 & 22 & 12 \\ 3 & 4 & 5 & 0 & 22 & 10 & 6 \\ 2 & 2 & 2 & -2 & 12 & 6 & 2 \end{pmatrix}_{b_0}$$

[264] $1^{11} 1^{131} 3^1 21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 0 & 9 & 5 & 2 \\ 1 & 2 & 0 & 0 & 12 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 12 & 0 & 61 & 39 & 15 \\ 4 & 5 & 8 & 0 & 39 & 23 & 9 \\ 2 & 2 & 2 & -2 & 15 & 9 & 2 \end{pmatrix}_{b_0}$$

[265] $1^{12} 1^{113} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 22 & 3 \\ 2 & 1 & 1 & 7 & 23 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 75 & 9 \\ 22 & 23 & 0 & 75 & 243 & 29 \\ 3 & 3 & -1 & 9 & 29 & 3 \end{pmatrix}_{b_0}$$

[266] $1^{11} 1^{13} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 13 & 3 \\ 2 & 1 & 1 & 3 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 15 & 3 \\ 13 & 15 & 0 & 15 & 83 & 17 \\ 3 & 3 & -1 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

[267] $1^{11} 1^{13} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -357, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 2 & 0 & 9 & 4 & 2 \\ 1 & 2 & 0 & 0 & 12 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 12 & 0 & 61 & 27 & 15 \\ 3 & 4 & 5 & 0 & 27 & 10 & 6 \\ 2 & 2 & 2 & -2 & 15 & 6 & 2 \end{pmatrix}_{b_0}$$

[268] $1^{11} 1^{13} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 10 & 3 \\ 2 & 1 & 1 & 17 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 15 & 17 & 0 & 121 & 84 & 21 \\ 10 & 12 & 0 & 84 & 56 & 14 \\ 3 & 3 & -1 & 21 & 14 & 3 \end{pmatrix}_{b_0}$$

[269] $1^{11} 1^{123} 3^1 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 9 & 2 \\ 1 & 2 & 0 & 0 & 8 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 8 & 0 & 26 & 38 & 10 \\ 6 & 9 & 10 & 0 & 38 & 51 & 13 \\ 2 & 2 & 2 & -2 & 10 & 13 & 2 \end{pmatrix}_{b_0}$$

[270] $1^{12} 1^{124} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 24 & 3 \\ 2 & 1 & 1 & 10 & 26 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 108 & 12 \\ 24 & 26 & 0 & 108 & 292 & 32 \\ 3 & 3 & -1 & 12 & 32 & 3 \end{pmatrix}_{b_0}$$

[271] $1^{12} 1^{113} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -378, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 18 & 3 \\ 2 & 1 & 1 & 3 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 19 & 3 \\ 18 & 19 & 0 & 19 & 151 & 23 \\ 3 & 3 & -1 & 3 & 23 & 3 \end{pmatrix}_{b_0}$$

[272] $1^{12} 1^{131} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 8 & 3 \\ 2 & 1 & 1 & 21 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 20 & 21 & 0 & 205 & 87 & 27 \\ 8 & 9 & 0 & 87 & 35 & 11 \\ 3 & 3 & -1 & 27 & 11 & 3 \end{pmatrix}_{b_0}$$

[273] $1^{11} 1^{114} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 23 & 3 \\ 2 & 1 & 1 & 7 & 26 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 82 & 9 \\ 23 & 26 & 0 & 82 & 292 & 32 \\ 3 & 3 & -1 & 9 & 32 & 3 \end{pmatrix}_{b_0}$$

[274] $1^{112}3^121^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -405, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 \\ 1 & 2 & 0 & 0 & 4 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 16 & 5 \\ 5 & 7 & 9 & 0 & 16 & 35 & 11 \\ 2 & 2 & 2 & -2 & 5 & 11 & 2 \end{pmatrix}_{b_0}$$

[275] $1^{114}3^121^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -413, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 0 & 1 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 2 & 1 & 0 & 0 & 13 & 2 \\ 9 & 13 & 17 & 0 & 13 & 127 & 21 \\ 2 & 2 & 2 & -2 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[276] $1^{124}3^121^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 18 & 3 \\ 2 & 1 & 1 & 10 & 22 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 88 & 12 \\ 18 & 22 & 0 & 88 & 192 & 26 \\ 3 & 3 & -1 & 12 & 26 & 3 \end{pmatrix}_{b_0}$$

[277] $1^{141}3^121^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -419, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 21 & 8 & 3 \\ 2 & 1 & 1 & 24 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 21 & 24 & 0 & 250 & 96 & 30 \\ 8 & 9 & 0 & 96 & 35 & 11 \\ 3 & 3 & -1 & 30 & 11 & 3 \end{pmatrix}_{b_0}$$

[278] $1^{113}3^121^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -429, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 8 & 2 \\ 1 & 0 & 2 & 0 & 4 & 11 & 2 \\ 1 & 2 & 0 & 0 & 7 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 7 & 0 & 16 & 43 & 8 \\ 8 & 11 & 16 & 0 & 43 & 103 & 19 \\ 2 & 2 & 2 & -2 & 8 & 19 & 2 \end{pmatrix}_{b_0}$$

[280] $1^{114}3^121^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 24 & 3 \\ 2 & 1 & 1 & 3 & 26 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 26 & 3 \\ 24 & 26 & 0 & 26 & 292 & 32 \\ 3 & 3 & -1 & 3 & 32 & 3 \end{pmatrix}_{b_0}$$

[279] $1^{142}3^121^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 10 & 3 \\ 2 & 1 & 1 & 20 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 18 & 20 & 0 & 158 & 96 & 24 \\ 10 & 12 & 0 & 96 & 56 & 14 \\ 3 & 3 & -1 & 24 & 14 & 3 \end{pmatrix}_{b_0}$$

[281] $1^{123}3^121^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -445, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 0 & 5 & 10 & 2 \\ 1 & 2 & 0 & 0 & 5 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 5 & 0 & 13 & 33 & 7 \\ 7 & 10 & 13 & 0 & 33 & 74 & 16 \\ 2 & 2 & 2 & -2 & 7 & 16 & 2 \end{pmatrix}_{b_0}$$

[282] $1^{114}3^121^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -448, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 24 & 3 \\ 2 & 1 & 1 & 7 & 26 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 82 & 9 \\ 24 & 26 & 0 & 82 & 292 & 32 \\ 3 & 3 & -1 & 9 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[283] \ 1^{112} 3^{121} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -454, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.00 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 14 & 3 \\ 2 & 1 & 1 & 3 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 15 & 3 \\ 14 & 15 & 0 & 15 & 83 & 17 \\ 3 & 3 & -1 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[284] \ 1^{112} 3^{121} 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 0 & 3 & 8 & 2 \\ 1 & 2 & 0 & 0 & 4 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 20 & 5 \\ 6 & 8 & 12 & 0 & 20 & 56 & 14 \\ 2 & 2 & 2 & -2 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[285] \ 1^{141} 3^{121} 21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 0 & 12 & 5 & 2 \\ 1 & 2 & 0 & 0 & 16 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 16 & 0 & 110 & 52 & 20 \\ 4 & 5 & 8 & 0 & 52 & 23 & 9 \\ 2 & 2 & 2 & -2 & 20 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[286] \ 1^{122} 3^{121} 21^{133} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 8 & 2 \\ 1 & 2 & 0 & 0 & 11 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 11 & 0 & 45 & 52 & 13 \\ 6 & 8 & 12 & 0 & 52 & 56 & 14 \\ 2 & 2 & 2 & -2 & 13 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[287] \ 1^{122} 3^{121} 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -468, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 6 & 7 & 2 \\ 1 & 2 & 0 & 0 & 8 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 8 & 0 & 26 & 32 & 10 \\ 5 & 7 & 9 & 0 & 32 & 35 & 11 \\ 2 & 2 & 2 & -2 & 10 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[288] \ 1^{114} 3^{121} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -469, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 19 & 3 \\ 2 & 1 & 1 & 3 & 22 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 22 & 3 \\ 19 & 22 & 0 & 22 & 192 & 26 \\ 3 & 3 & -1 & 3 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[289] \ 1^{124} 3^{121} 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 23 & 3 \\ 2 & 1 & 1 & 10 & 26 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 108 & 12 \\ 23 & 26 & 0 & 108 & 292 & 32 \\ 3 & 3 & -1 & 12 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[290] \ 1^{131} 3^{121} 21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -477, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 0 & 10 & 5 & 2 \\ 1 & 2 & 0 & 0 & 15 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 10 & 15 & 0 & 88 & 47 & 18 \\ 4 & 5 & 8 & 0 & 47 & 23 & 9 \\ 2 & 2 & 2 & -2 & 18 & 9 & 2 \end{pmatrix}_{b_0}$$

[291] $1^{124}3^{11}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -484, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 19 & 3 \\ 2 & 1 & 1 & 10 & 22 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 38 & 88 & 12 \\ 19 & 22 & 0 & 88 & 192 & 26 \\ 3 & 3 & -1 & 12 & 26 & 3 \end{pmatrix}_{b_0}$$

[292] $1^{132}3^{11}21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -485, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 7 & 2 \\ 1 & 2 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 9 & 0 & 40 & 39 & 12 \\ 5 & 7 & 9 & 0 & 39 & 35 & 11 \\ 2 & 2 & 2 & -2 & 12 & 11 & 2 \end{pmatrix}_{b_0}$$

[293] $1^{113}3^{11}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 0 & 3 & 9 & 2 \\ 1 & 2 & 0 & 0 & 4 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 19 & 5 \\ 6 & 9 & 10 & 0 & 19 & 51 & 13 \\ 2 & 2 & 2 & -2 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[294] $1^{141}3^{11}21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 3 & 2 \\ 1 & 0 & 2 & 0 & 12 & 4 & 2 \\ 1 & 2 & 0 & 0 & 16 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 16 & 0 & 110 & 36 & 20 \\ 3 & 4 & 5 & 0 & 36 & 10 & 6 \\ 2 & 2 & 2 & -2 & 20 & 6 & 2 \end{pmatrix}_{b_0}$$

[295] $1^{114}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 0 & 1 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 2 & 1 & 0 & 0 & 12 & 2 \\ 8 & 12 & 14 & 0 & 12 & 96 & 18 \\ 2 & 2 & 2 & -2 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[296] $1^{142}3^{11}21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 0 & 10 & 6 & 2 \\ 1 & 2 & 0 & 0 & 10 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 10 & 10 & 0 & 58 & 36 & 14 \\ 4 & 6 & 6 & 0 & 36 & 20 & 8 \\ 2 & 2 & 2 & -2 & 14 & 8 & 2 \end{pmatrix}_{b_0}$$

[297] $1^{122}3^{11}21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 8 & 2 \\ 1 & 2 & 0 & 0 & 8 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 8 & 0 & 26 & 40 & 10 \\ 6 & 8 & 12 & 0 & 40 & 56 & 14 \\ 2 & 2 & 2 & -2 & 10 & 14 & 2 \end{pmatrix}_{b_0}$$

[298] $1^{114}3^{11}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -532, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 28 & 3 \\ 2 & 1 & 1 & 7 & 30 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 98 & 9 \\ 28 & 30 & 0 & 98 & 416 & 38 \\ 3 & 3 & -1 & 9 & 38 & 3 \end{pmatrix}_{b_0}$$

[299] $1^{141}3^121^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -533, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 2 & 0 & 11 & 4 & 2 \\ 1 & 2 & 0 & 0 & 13 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 13 & 0 & 81 & 31 & 17 \\ 3 & 4 & 5 & 0 & 31 & 10 & 6 \\ 2 & 2 & 2 & -2 & 17 & 6 & 2 \end{pmatrix}_{b_0}$$

[300] $1^{142}3^121^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -538, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 26 & 8 & 3 \\ 2 & 1 & 1 & 28 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 26 & 28 & 0 & 366 & 116 & 36 \\ 8 & 9 & 0 & 116 & 35 & 11 \\ 3 & 3 & -1 & 36 & 11 & 3 \end{pmatrix}_{b_0}$$

[301] $1^{142}3^121^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 10 & 3 \\ 2 & 1 & 1 & 20 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 17 & 20 & 0 & 158 & 96 & 24 \\ 10 & 12 & 0 & 96 & 56 & 14 \\ 3 & 3 & -1 & 24 & 14 & 3 \end{pmatrix}_{b_0}$$

[302] $1^{141}3^121^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -550, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 22 & 8 & 3 \\ 2 & 1 & 1 & 24 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 22 & 24 & 0 & 250 & 96 & 30 \\ 8 & 9 & 0 & 96 & 35 & 11 \\ 3 & 3 & -1 & 30 & 11 & 3 \end{pmatrix}_{b_0}$$

[304] $1^{123}3^121^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 9 & 2 \\ 1 & 2 & 0 & 0 & 5 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 5 & 0 & 13 & 28 & 7 \\ 6 & 9 & 10 & 0 & 28 & 51 & 13 \\ 2 & 2 & 2 & -2 & 7 & 13 & 2 \end{pmatrix}_{b_0}$$

[303] $1^{132}3^121^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 5 & 2 \\ 1 & 0 & 2 & 0 & 9 & 7 & 2 \\ 1 & 2 & 0 & 0 & 12 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 12 & 0 & 61 & 48 & 15 \\ 5 & 7 & 9 & 0 & 48 & 35 & 11 \\ 2 & 2 & 2 & -2 & 15 & 11 & 2 \end{pmatrix}_{b_0}$$

[305] $1^{132}3^121^{133} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -592, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 8 & 2 \\ 1 & 2 & 0 & 0 & 12 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 12 & 0 & 61 & 60 & 15 \\ 6 & 8 & 12 & 0 & 60 & 56 & 14 \\ 2 & 2 & 2 & -2 & 15 & 14 & 2 \end{pmatrix}_{b_0}$$

[306] $1^{124}3^121^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 0 & 5 & 13 & 2 \\ 1 & 2 & 0 & 0 & 5 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 5 & 0 & 13 & 43 & 7 \\ 9 & 13 & 17 & 0 & 43 & 127 & 21 \\ 2 & 2 & 2 & -2 & 7 & 21 & 2 \end{pmatrix}_{b_0}$$

[307] $1^{114}3^{112}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -628, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 20 & 3 \\ 2 & 1 & 1 & 3 & 22 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 22 & 3 \\ 20 & 22 & 0 & 22 & 192 & 26 \\ 3 & 3 & -1 & 3 & 26 & 3 \end{pmatrix}_{b_0}$$

[308] $1^{123}3^{11}21^{134} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -645, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 0 & 7 & 11 & 2 \\ 1 & 2 & 0 & 0 & 11 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 11 & 0 & 45 & 70 & 13 \\ 8 & 11 & 16 & 0 & 70 & 103 & 19 \\ 2 & 2 & 2 & -2 & 13 & 19 & 2 \end{pmatrix}_{b_0}$$

[309] $1^{132}3^{11}21^{143} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -672, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 6 & 2 \\ 1 & 0 & 2 & 0 & 10 & 8 & 2 \\ 1 & 2 & 0 & 0 & 15 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 10 & 15 & 0 & 88 & 72 & 18 \\ 6 & 8 & 12 & 0 & 72 & 56 & 14 \\ 2 & 2 & 2 & -2 & 18 & 14 & 2 \end{pmatrix}_{b_0}$$

[311] $1^{142}3^{11}21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 5 & 2 \\ 1 & 0 & 2 & 0 & 11 & 7 & 2 \\ 1 & 2 & 0 & 0 & 13 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 13 & 0 & 81 & 55 & 17 \\ 5 & 7 & 9 & 0 & 55 & 35 & 11 \\ 2 & 2 & 2 & -2 & 17 & 11 & 2 \end{pmatrix}_{b_0}$$

[310] $1^{142}3^{11}21^{143} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 6 & 2 \\ 1 & 0 & 2 & 0 & 12 & 8 & 2 \\ 1 & 2 & 0 & 0 & 16 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 16 & 0 & 110 & 80 & 20 \\ 6 & 8 & 12 & 0 & 80 & 56 & 14 \\ 2 & 2 & 2 & -2 & 20 & 14 & 2 \end{pmatrix}_{b_0}$$

[312] $1^{113}3^{11}21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -717, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 0 & 4 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 23 & 5 \\ 7 & 10 & 13 & 0 & 23 & 74 & 16 \\ 2 & 2 & 2 & -2 & 5 & 16 & 2 \end{pmatrix}_{b_0}$$

[313] $1^{142}3^{11}21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -720, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 5 & 2 \\ 1 & 0 & 2 & 0 & 12 & 7 & 2 \\ 1 & 2 & 0 & 0 & 16 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 16 & 0 & 110 & 64 & 20 \\ 5 & 7 & 9 & 0 & 64 & 35 & 11 \\ 2 & 2 & 2 & -2 & 20 & 11 & 2 \end{pmatrix}_{b_0}$$

[315] $1^{124}3^{11}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -752, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 0 & 5 & 11 & 2 \\ 1 & 2 & 0 & 0 & 5 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 5 & 0 & 13 & 33 & 7 \\ 7 & 11 & 11 & 0 & 33 & 71 & 15 \\ 2 & 2 & 2 & -2 & 7 & 15 & 2 \end{pmatrix}_{b_0}$$

[314] $1^{124}3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -752, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 2 & 0 & 6 & 12 & 2 \\ 1 & 2 & 0 & 0 & 8 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 8 & 0 & 26 & 52 & 10 \\ 8 & 12 & 14 & 0 & 52 & 96 & 18 \\ 2 & 2 & 2 & -2 & 10 & 18 & 2 \end{pmatrix}_{b_0}$$

[316] $1^{123}3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -780, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 10 & 2 \\ 1 & 2 & 0 & 0 & 8 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 8 & 0 & 26 & 46 & 10 \\ 7 & 10 & 13 & 0 & 46 & 74 & 16 \\ 2 & 2 & 2 & -2 & 10 & 16 & 2 \end{pmatrix}_{b_0}$$

[317] $1^{113}3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -800, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 0 & 3 & 11 & 2 \\ 1 & 2 & 0 & 0 & 4 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 27 & 5 \\ 8 & 11 & 16 & 0 & 27 & 103 & 19 \\ 2 & 2 & 2 & -2 & 5 & 19 & 2 \end{pmatrix}_{b_0}$$

[318] $1^{123}3^1 21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -848, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 2 & 0 & 6 & 11 & 2 \\ 1 & 2 & 0 & 0 & 8 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 8 & 0 & 26 & 54 & 10 \\ 8 & 11 & 16 & 0 & 54 & 103 & 19 \\ 2 & 2 & 2 & -2 & 10 & 19 & 2 \end{pmatrix}_{b_0}$$

[319] $1^{124}3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -852, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 8 & 2 \\ 1 & 0 & 2 & 0 & 5 & 12 & 2 \\ 1 & 2 & 0 & 0 & 5 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 5 & 0 & 13 & 38 & 7 \\ 8 & 12 & 14 & 0 & 38 & 96 & 18 \\ 2 & 2 & 2 & -2 & 7 & 18 & 2 \end{pmatrix}_{b_0}$$

[320] $1^{114}3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -884, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 0 & 3 & 12 & 2 \\ 1 & 2 & 0 & 0 & 4 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 26 & 5 \\ 8 & 12 & 14 & 0 & 26 & 96 & 18 \\ 2 & 2 & 2 & -2 & 5 & 18 & 2 \end{pmatrix}_{b_0}$$

[321] $1^{114}3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -1125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 0 & 3 & 13 & 2 \\ 1 & 2 & 0 & 0 & 4 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 5 & 30 & 5 \\ 9 & 13 & 17 & 0 & 30 & 127 & 21 \\ 2 & 2 & 2 & -2 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$

[322] $1^{124}3^1 21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -1188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 0 & 6 & 13 & 2 \\ 1 & 2 & 0 & 0 & 8 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 8 & 0 & 26 & 60 & 10 \\ 9 & 13 & 17 & 0 & 60 & 127 & 21 \\ 2 & 2 & 2 & -2 & 10 & 21 & 2 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [323] \quad 1^1 3^1 1^1 5 \quad 13221^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 9.0$$

$$[324] \quad 1^1 3^1 1^1 5 \quad 1321^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 5 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 5 & 8 & 4 \\ 3 & 0 & 5 & 3 & 8 & 5 \\ 5 & 0 & 8 & 8 & 15 & 9 \\ 3 & 0 & 4 & 5 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[325] \quad 1^1 3^1 1^1 5 \quad 1321^{131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 13 & 5 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 19 & 8 & 4 \\ 13 & 0 & 19 & 79 & 36 & 21 \\ 5 & 0 & 8 & 36 & 15 & 9 \\ 3 & 0 & 4 & 21 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[326] \quad 1^1 3^1 1^1 5 \quad 1321^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 7 & 5 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 9 & 8 & 4 \\ 7 & 0 & 9 & 23 & 20 & 11 \\ 5 & 0 & 8 & 20 & 15 & 9 \\ 3 & 0 & 4 & 11 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[327] \quad 1^1 3^1 1^1 5 \quad 1321^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 10 & 5 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 14 & 8 & 4 \\ 10 & 0 & 14 & 46 & 28 & 16 \\ 5 & 0 & 8 & 28 & 15 & 9 \\ 3 & 0 & 4 & 16 & 9 & 5 \end{pmatrix}_{b_0}$$

4.677 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^1 3^1 1^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 15 & 5 \\ 11 & 20 & 0 & 0 & 28 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 28 & 0 & 0 & 37 & 11 \\ 5 & 8 & -2 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

4.678 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \text{su2n3mix} + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{111} 3^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 4 & 2 & 3 \\ 3 & 4 & 0 & 14 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 17 & 8 & 11 \\ 4 & 7 & 0 & 17 & 7 & 4 & 5 \\ 2 & 3 & 0 & 8 & 4 & 0 & 2 \\ 3 & 4 & -2 & 11 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{122} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 2 & 3 \\ 3 & 4 & 0 & 8 & 16 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 22 & 4 & 6 \\ 9 & 16 & 0 & 22 & 44 & 9 & 12 \\ 2 & 3 & 0 & 4 & 9 & 0 & 2 \\ 3 & 4 & -2 & 6 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{121}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -86, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 10 & 4 & 6 \\ 4 & 7 & 0 & 10 & 7 & 4 & 5 \\ 2 & 3 & 0 & 4 & 4 & 0 & 2 \\ 3 & 4 & -2 & 6 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -91, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 0 & 5 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 6 & 3 & 4 \\ 4 & 7 & 0 & 6 & 7 & 4 & 5 \\ 2 & 3 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -2 & 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -194, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 4 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 5 & 2 & 3 \\ 4 & 7 & 0 & 5 & 7 & 4 & 5 \\ 2 & 3 & 0 & 2 & 4 & 0 & 2 \\ 3 & 4 & -2 & 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{112}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -236, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 2 & 3 \\ 3 & 4 & 0 & 4 & 11 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 8 & 2 & 3 \\ 6 & 11 & 0 & 8 & 20 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -2 & 3 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{123}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -238, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 2 & 3 \\ 3 & 4 & 0 & 8 & 20 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 28 & 4 & 6 \\ 11 & 20 & 0 & 28 & 69 & 11 & 15 \\ 2 & 3 & 0 & 4 & 11 & 0 & 2 \\ 3 & 4 & -2 & 6 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -251, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 2 & 3 \\ 3 & 4 & 0 & 9 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 11 & 5 & 7 \\ 4 & 7 & 0 & 11 & 7 & 4 & 5 \\ 2 & 3 & 0 & 5 & 4 & 0 & 2 \\ 3 & 4 & -2 & 7 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{124}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -264, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 16 & 2 & 3 \\ 3 & 4 & 0 & 8 & 29 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 40 & 4 & 6 \\ 16 & 29 & 0 & 40 & 148 & 16 & 22 \\ 2 & 3 & 0 & 4 & 16 & 0 & 2 \\ 3 & 4 & -2 & 6 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -287, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 4 & 12 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 8 & 2 & 3 \\ 7 & 12 & 0 & 8 & 25 & 7 & 9 \\ 2 & 3 & 0 & 2 & 7 & 0 & 2 \\ 3 & 4 & -2 & 3 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{141}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 4 & 2 & 3 \\ 3 & 4 & 0 & 21 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 26 & 11 & 16 \\ 4 & 7 & 0 & 26 & 7 & 4 & 5 \\ 2 & 3 & 0 & 11 & 4 & 0 & 2 \\ 3 & 4 & -2 & 16 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{121}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 4 & 2 & 3 \\ 3 & 4 & 0 & 18 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 22 & 10 & 14 \\ 4 & 7 & 0 & 22 & 7 & 4 & 5 \\ 2 & 3 & 0 & 10 & 4 & 0 & 2 \\ 3 & 4 & -2 & 14 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{122}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 2 & 3 \\ 3 & 4 & 0 & 13 & 11 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 25 & 7 & 10 \\ 6 & 11 & 0 & 25 & 20 & 6 & 8 \\ 2 & 3 & 0 & 7 & 6 & 0 & 2 \\ 3 & 4 & -2 & 10 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{121}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 2 & 3 \\ 3 & 4 & 0 & 13 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 16 & 7 & 10 \\ 4 & 7 & 0 & 16 & 7 & 4 & 5 \\ 2 & 3 & 0 & 7 & 4 & 0 & 2 \\ 3 & 4 & -2 & 10 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{131}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 2 & 3 \\ 3 & 4 & 0 & 17 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 21 & 9 & 13 \\ 4 & 7 & 0 & 21 & 7 & 4 & 5 \\ 2 & 3 & 0 & 9 & 4 & 0 & 2 \\ 3 & 4 & -2 & 13 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{124}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 13 & 2 & 3 \\ 3 & 4 & 0 & 8 & 24 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 34 & 4 & 6 \\ 13 & 24 & 0 & 34 & 100 & 13 & 18 \\ 2 & 3 & 0 & 4 & 13 & 0 & 2 \\ 3 & 4 & -2 & 6 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{132}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 6 & 2 & 3 \\ 3 & 4 & 0 & 17 & 11 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 33 & 9 & 13 \\ 6 & 11 & 0 & 33 & 20 & 6 & 8 \\ 2 & 3 & 0 & 9 & 6 & 0 & 2 \\ 3 & 4 & -2 & 13 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{142}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 6 & 2 & 3 \\ 3 & 4 & 0 & 21 & 11 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 41 & 11 & 16 \\ 6 & 11 & 0 & 41 & 20 & 6 & 8 \\ 2 & 3 & 0 & 11 & 6 & 0 & 2 \\ 3 & 4 & -2 & 16 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -477, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 0 & 4 & 16 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 11 & 2 & 3 \\ 9 & 16 & 0 & 11 & 44 & 9 & 12 \\ 2 & 3 & 0 & 2 & 9 & 0 & 2 \\ 3 & 4 & -2 & 3 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{13} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -526, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 4 & 2 & 3 \\ 3 & 4 & 0 & 22 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 27 & 12 & 17 \\ 4 & 7 & 0 & 27 & 7 & 4 & 5 \\ 2 & 3 & 0 & 12 & 4 & 0 & 2 \\ 3 & 4 & -2 & 17 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{14} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -528, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 6 & 2 & 3 \\ 3 & 4 & 0 & 26 & 11 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 50 & 14 & 20 \\ 6 & 11 & 0 & 50 & 20 & 6 & 8 \\ 2 & 3 & 0 & 14 & 6 & 0 & 2 \\ 3 & 4 & -2 & 20 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -584, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 2 & 3 \\ 3 & 4 & 0 & 4 & 21 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 14 & 2 & 3 \\ 12 & 21 & 0 & 14 & 80 & 12 & 16 \\ 2 & 3 & 0 & 2 & 12 & 0 & 2 \\ 3 & 4 & -2 & 3 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -631, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 4 & 20 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 14 & 2 & 3 \\ 11 & 20 & 0 & 14 & 69 & 11 & 15 \\ 2 & 3 & 0 & 2 & 11 & 0 & 2 \\ 3 & 4 & -2 & 3 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{14} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -666, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 4 & 2 & 3 \\ 3 & 4 & 0 & 26 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 32 & 14 & 20 \\ 4 & 7 & 0 & 32 & 7 & 4 & 5 \\ 2 & 3 & 0 & 14 & 4 & 0 & 2 \\ 3 & 4 & -2 & 20 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -749, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.92 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 2 & 3 \\ 3 & 4 & 0 & 4 & 24 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 17 & 2 & 3 \\ 13 & 24 & 0 & 17 & 100 & 13 & 18 \\ 2 & 3 & 0 & 2 & 13 & 0 & 2 \\ 3 & 4 & -2 & 3 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -886, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 4 & 0 & 4 & 25 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 17 & 2 & 3 \\ 14 & 25 & 0 & 17 & 111 & 14 & 19 \\ 2 & 3 & 0 & 2 & 14 & 0 & 2 \\ 3 & 4 & -2 & 3 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{114}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -1152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 2 & 3 \\ 3 & 4 & 0 & 4 & 29 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 20 & 2 & 3 \\ 16 & 29 & 0 & 20 & 148 & 16 & 22 \\ 2 & 3 & 0 & 2 & 16 & 0 & 2 \\ 3 & 4 & -2 & 3 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

4.679 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2$ + su2n3mix + su2n3mix

$T = 9 \quad [0] \quad 1^{1121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 7 & 3 \\ 3 & 4 & 0 & 4 & 11 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 8 & 8 & 3 \\ 6 & 11 & 0 & 8 & 20 & 23 & 8 \\ 7 & 12 & 0 & 8 & 23 & 25 & 9 \\ 3 & 4 & -2 & 3 & 8 & 9 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{1211}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 7 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 10 & 10 & 6 \\ 4 & 7 & 0 & 10 & 7 & 9 & 5 \\ 4 & 7 & 0 & 10 & 9 & 7 & 5 \\ 3 & 4 & -2 & 6 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{1111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 4 & 3 \\ 3 & 4 & 0 & 9 & 7 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 11 & 11 & 7 \\ 4 & 7 & 0 & 11 & 7 & 9 & 5 \\ 4 & 7 & 0 & 11 & 9 & 7 & 5 \\ 3 & 4 & -2 & 7 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{1111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -303, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 7 & 3 \\ 3 & 4 & 0 & 4 & 7 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 5 & 8 & 3 \\ 4 & 7 & 0 & 5 & 7 & 15 & 5 \\ 7 & 12 & 0 & 8 & 15 & 25 & 9 \\ 3 & 4 & -2 & 3 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{1111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 0 & 4 & 7 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 5 & 5 & 3 \\ 4 & 7 & 0 & 5 & 7 & 9 & 5 \\ 4 & 7 & 0 & 5 & 9 & 7 & 5 \\ 3 & 4 & -2 & 3 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{1112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -438, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 12 & 3 \\ 3 & 4 & 0 & 4 & 7 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 5 & 14 & 3 \\ 4 & 7 & 0 & 5 & 7 & 26 & 5 \\ 12 & 21 & 0 & 14 & 26 & 80 & 16 \\ 3 & 4 & -2 & 3 & 5 & 16 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{1211}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -448, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 4 & 3 \\ 3 & 4 & 0 & 13 & 7 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 16 & 16 & 10 \\ 4 & 7 & 0 & 16 & 7 & 9 & 5 \\ 4 & 7 & 0 & 16 & 9 & 7 & 5 \\ 3 & 4 & -2 & 10 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1112} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -589, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 9 & 3 \\ 3 & 4 & 0 & 4 & 7 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 5 & 11 & 3 \\ 4 & 7 & 0 & 5 & 7 & 20 & 5 \\ 9 & 16 & 0 & 11 & 20 & 44 & 12 \\ 3 & 4 & -2 & 3 & 5 & 12 & 2 \end{pmatrix}_{b_0}$$

4.680 NHC(-2)(-3) + so₈ + su₂ + su₃

$$T = 8 \quad [0] \ 1^{11} 3^{112} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 14 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 14 & 17 & 13 & 15 & 7 \\ 12 & 17 & 15 & 12 & 6 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{112} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 19 & 5 \\ 11 & 20 & 0 & 34 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 19 & 34 & 0 & 57 & 13 \\ 5 & 8 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{141} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 16 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 46 & 24 & 12 \\ 5 & 9 & 0 & 24 & 12 & 6 \\ 3 & 4 & -2 & 12 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{132} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 12 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 12 & 0 & 25 & 27 & 9 \\ 7 & 13 & 0 & 27 & 27 & 9 \\ 3 & 4 & -2 & 9 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 13 & 15 & 7 \\ 12 & 17 & 15 & 12 & 6 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 0 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 10 & 17 & 13 & 15 & 7 \\ 10 & 17 & 15 & 12 & 6 \\ 5 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 29, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 20 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 20 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3 \ 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 24 & 0 & 121 & 15 & 19 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 19 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 3 & 4 & -2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.62 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 5 \\ 12 & 25 & 20 & 16 & 9 \\ 9 & 20 & 13 & 12 & 7 \\ 8 & 16 & 12 & 9 & 5 \\ 5 & 9 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 0 & 12 & 6 \\ 3 & 4 & -2 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 10 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 0 & 3 & 3 \\ 3 & 4 & -2 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 37, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 10 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 37, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 17 & 0 & 12 & 6 \\ 5 & 8 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

- [18] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 39$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 11 & 13 & 6 \\ 15 & 20 & 16 & 17 & 8 \\ 11 & 16 & 10 & 12 & 6 \\ 13 & 17 & 12 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [19] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$
- [20] $1^{11} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 11 & 1 & 3 \\ 3 & 4 & 0 & 19 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 19 & 0 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 15 & -1 & 1 \end{pmatrix}_{b_0}$
- [21] $1^{11} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 59 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 59 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 45 & -1 & 2 \end{pmatrix}_{b_0}$
- [22] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 5 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 4 & -1 & 1 \end{pmatrix}_{b_0}$
- [23] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 12 & 1 & 6 \\ 15 & 20 & 17 & 0 & 8 \\ 12 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$
- [24] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 10 & 10 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 10 & 16 & 10 & 12 & 6 \\ 10 & 17 & 12 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [25] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 8 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 6 & 6 \\ 3 & 5 & 0 & 6 & 3 & 3 \\ 3 & 4 & -2 & 6 & 3 & 1 \end{pmatrix}_{b_0}$
- [27] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 12 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 12 & 0 & 25 & 18 & 9 \\ 5 & 8 & 0 & 18 & 9 & 5 \\ 3 & 4 & -2 & 9 & 5 & 1 \end{pmatrix}_{b_0}$
- [28] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 3 & -1 & 1 \end{pmatrix}_{b_0}$
- [26] $1^{11} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 60 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 45 & -1 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 10 & 17 & 13 & 15 & 7 \\ 9 & 17 & 15 & 12 & 6 \\ 5 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 17 & 0 & 12 & 6 \\ 6 & 8 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{114}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 37 & 8 & 5 \\ 12 & 25 & 80 & 16 & 9 \\ 37 & 80 & 238 & 48 & 28 \\ 8 & 16 & 48 & 9 & 5 \\ 5 & 9 & 28 & 5 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 10 & 8 & 5 \\ 12 & 25 & 20 & 16 & 9 \\ 10 & 20 & 13 & 12 & 7 \\ 8 & 16 & 12 & 9 & 5 \\ 5 & 9 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 18 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 13 & 15 & 7 \\ 10 & 17 & 15 & 12 & 6 \\ 5 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 13 & 15 & 7 \\ 9 & 17 & 15 & 12 & 6 \\ 5 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[38] $1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 13 & 15 & 7 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & -2 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[41] $21^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 17 & 15 & 15 & 7 \\ 17 & 20 & 17 & 17 & 8 \\ 15 & 17 & 13 & 15 & 7 \\ 15 & 17 & 15 & 12 & 6 \\ 7 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 6 & 6 \\ 3 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 3 & 3 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & -2 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 12 & 6 \\ 3 & 4 & -2 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{122}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 8 & 13 & 0 & 18 & 27 & 9 \\ 3 & 4 & -2 & 6 & 9 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 0 & 12 & 12 & 6 \\ 3 & 4 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 13 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 13 & 17 & 13 & 15 & 7 \\ 12 & 17 & 15 & 12 & 6 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[48] $1^{13}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 1 & 3 \\ 3 & 4 & 0 & 17 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 13 & -1 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

[49] $1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 8 & 16 & 10 & 12 & 6 \\ 9 & 17 & 12 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[52] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 28 & 8 & 5 \\ 12 & 25 & 60 & 16 & 9 \\ 28 & 60 & 133 & 36 & 21 \\ 8 & 16 & 36 & 9 & 5 \\ 5 & 9 & 21 & 5 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 3 & 5 & 0 & 6 & 3 & 3 \\ 3 & 4 & -2 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 19 & 8 & 5 \\ 12 & 25 & 40 & 16 & 9 \\ 19 & 40 & 58 & 24 & 14 \\ 8 & 16 & 24 & 9 & 5 \\ 5 & 9 & 14 & 5 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 9 & 16 & 10 & 12 & 6 \\ 10 & 17 & 12 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[54] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 11 & 12 & 6 \\ 15 & 20 & 16 & 17 & 8 \\ 11 & 16 & 10 & 12 & 6 \\ 12 & 17 & 12 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[55] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 16 & 17 & 8 \\ 12 & 16 & 10 & 12 & 6 \\ 12 & 17 & 12 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[56] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 9 & 16 & 10 & 12 & 6 \\ 9 & 17 & 12 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[57] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 20 & 8 & 5 \\ 12 & 25 & 40 & 16 & 9 \\ 20 & 40 & 58 & 24 & 14 \\ 8 & 16 & 24 & 9 & 5 \\ 5 & 9 & 14 & 5 & 2 \end{pmatrix}_{b_0}$$

[58] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 0 & 15 & 12 & 6 \\ 3 & 4 & -2 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 6 & 3 \\ 4 & 8 & 0 & 6 & 9 & 5 \\ 3 & 4 & -2 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 14 & 0 & 37 & 9 & 11 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 9 & 0 & 13 & 9 & 7 \\ 3 & 5 & 0 & 9 & 3 & 3 \\ 3 & 4 & -2 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$

[61] $1^{11} 3^{11} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 6 & 4 \\ 4 & 5 & 0 & 6 & 3 & 3 \\ 3 & 4 & -2 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

[64] $1^{11} 3^{11} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 10 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 10 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 1 & 3 \\ 3 & 4 & 0 & 14 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 11 & -1 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 3 \\ 3 & 4 & 0 & 9 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 7 & -1 & 1 \end{pmatrix}_{b_0}$$

[67] $1^{11} 3^{11} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 9 & 17 & 0 & 15 & 48 & 12 \\ 3 & 4 & -2 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 48 & 12 \\ 3 & 4 & -2 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 6 & 4 \\ 3 & 5 & 0 & 6 & 3 & 3 \\ 3 & 4 & -2 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 1 & 3 \\ 3 & 4 & 0 & 13 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 10 & -1 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 11 & 1 & 6 \\ 15 & 20 & 16 & 0 & 8 \\ 11 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 20 & 10 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 20 & 33 & 49 & 27 & 13 \\ 10 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[79] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[80] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 4 & 8 & 0 & 6 & 9 & 5 \\ 3 & 4 & -2 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 6 & 6 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 19 & 0 & 73 & 12 & 15 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[76] $1^{111} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 24 & 0 & 121 & 15 & 19 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 19 & 2 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 19 & 0 & 73 & 12 & 15 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 8 & 0 & 10 & 6 & 6 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{131} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 12 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 18 & 9 \\ 5 & 9 & 0 & 18 & 12 & 6 \\ 3 & 4 & -2 & 9 & 6 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 48 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 48 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 9 & 7 \\ 4 & 5 & 0 & 9 & 3 & 3 \\ 3 & 4 & -2 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$

[84] $1^{14} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 18 & 4 \\ 12 & 21 & 0 & 18 & 75 & 15 \\ 3 & 4 & -2 & 4 & 15 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 12 & 7 \\ 4 & 8 & 0 & 12 & 9 & 5 \\ 3 & 4 & -2 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & -2 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{121} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 12 & 14 \\ 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 12 & 7 \\ 5 & 8 & 0 & 12 & 9 & 5 \\ 3 & 4 & -2 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 0 & 9 & 27 & 9 \\ 3 & 4 & -2 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 35 & 1 & 5 & 8 \\ 11 & 20 & 67 & 0 & 8 & 8 \\ 35 & 67 & 217 & 0 & 27 & 8 \\ 1 & 0 & 0 & -3 & -1 & 1 \\ 5 & 8 & 27 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[92] $1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 17 & 10 & 5 & 8 \\ 11 & 20 & 33 & 17 & 8 & 8 \\ 17 & 33 & 49 & 27 & 13 & 8 \\ 10 & 17 & 27 & 12 & 6 & 6 \\ 5 & 8 & 13 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[93] $1^{12} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 13 & 0 & 28 & 12 & 10 \\ 3 & 5 & 0 & 12 & 3 & 3 \\ 3 & 4 & -2 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 23 & 1 & 6 & 6 \\ 15 & 20 & 33 & 0 & 8 & 8 \\ 23 & 33 & 49 & 0 & 13 & 13 \\ 1 & 0 & 0 & -3 & -1 & 1 \\ 6 & 8 & 13 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[95] $1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 8 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -2 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[96] $1^{141} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 5 & 3 \\ 3 & 4 & 0 & 21 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 21 & 0 & 76 & 33 & 16 \\ 5 & 9 & 0 & 33 & 12 & 6 \\ 3 & 4 & -2 & 16 & 6 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 & 8 \\ 11 & 20 & 16 & 0 & 8 & 8 \\ 9 & 16 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & -3 & -1 & 1 \\ 5 & 8 & 6 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[98] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 30 & 8 & 5 & 9 \\ 12 & 25 & 60 & 16 & 9 & 9 \\ 30 & 60 & 133 & 36 & 21 & 21 \\ 8 & 16 & 36 & 9 & 5 & 5 \\ 5 & 9 & 21 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

[99] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 6 & 7 \\ 3 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[100] $1^{11} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 12 & 6 \\ 3 & 4 & -2 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

[101] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 0 & 12 & 12 & 6 \\ 3 & 4 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[105] $1^{111} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 24 & 0 & 121 & 15 & 19 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 19 & 2 & 1 \end{pmatrix}_{b_0}$$

[103] $1^{111} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[104] $1^{122} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 12 & 6 \\ 5 & 8 & 0 & 12 & 9 & 5 \\ 3 & 4 & -2 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

[102] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 29 & 8 & 5 \\ 12 & 25 & 60 & 16 & 9 \\ 29 & 60 & 133 & 36 & 21 \\ 8 & 16 & 36 & 9 & 5 \\ 5 & 9 & 21 & 5 & 2 \end{pmatrix}_{b_0}$$

[106] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 26 & 12 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 26 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[107] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 7 & 13 & 0 & 12 & 27 & 9 \\ 3 & 4 & -2 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

[108] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 23 & 13 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 23 & 33 & 49 & 27 & 13 \\ 13 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[109] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & -2 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{112} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 20 & 1 & 5 \\ 12 & 25 & 40 & 0 & 9 \\ 20 & 40 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 14 & -1 & 2 \end{pmatrix}_{b_0}$$

[111] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 28 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 28 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{12} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 144, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.39 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 1 & 3 \\ 3 & 4 & 0 & 28 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 28 & 0 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 22 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 146, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.16 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 0 & 18 & 27 & 9 \\ 3 & 4 & -2 & 6 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[114] \ 21^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 148, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.08 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 16 & 12 & 12 & 6 \\ 16 & 20 & 16 & 17 & 8 \\ 12 & 16 & 10 & 12 & 6 \\ 12 & 17 & 12 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 148, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.41 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 34 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 26 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 148, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.41 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 1 & 3 \\ 3 & 4 & 0 & 18 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 14 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 148, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 9 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -2 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 150, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.13 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 23 & 12 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 23 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 151, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.25 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 21 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 18 & 16 \\ 3 & 5 & 0 & 18 & 3 & 3 \\ 3 & 4 & -2 & 16 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 155, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.34 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 17 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 157, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.41 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 35 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 26 & -1 & 2 \end{pmatrix}_{b_0}$$

[122] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 158, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 38 & 8 & 5 \\ 12 & 25 & 80 & 16 & 9 \\ 38 & 80 & 238 & 48 & 28 \\ 8 & 16 & 48 & 9 & 5 \\ 5 & 9 & 28 & 5 & 2 \end{pmatrix}_{b_0}$$

[123] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 9 & 4 \\ 6 & 9 & 0 & 9 & 12 & 6 \\ 3 & 4 & -2 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

[125] $1^{12} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 1 & 3 \\ 3 & 4 & 0 & 23 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 23 & 0 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 18 & -1 & 1 \end{pmatrix}_{b_0}$$

[124] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & -2 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[127] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 9 & 7 \\ 3 & 5 & 0 & 9 & 3 & 3 \\ 3 & 4 & -2 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$

[128] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 26 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[126] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 29 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 29 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[129] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 13 & 0 & 28 & 9 & 10 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

[130] $1^{13} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 170, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 15 & 4 \\ 10 & 17 & 0 & 15 & 48 & 12 \\ 3 & 4 & -2 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$

[131] $1^{141} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 15 & 16 \\ 3 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 9 & 10 \\ 3 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 0 & 18 & 27 & 9 \\ 3 & 4 & -2 & 6 & 9 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 12 & 6 \\ 4 & 8 & 0 & 12 & 9 & 5 \\ 3 & 4 & -2 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

[135] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 5 & 8 & 0 & 6 & 9 & 5 \\ 3 & 4 & -2 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[136] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[138] $1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 37 & 9 & 11 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

[142] $1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 46 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 46 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 33 & -1 & 2 \end{pmatrix}_{b_0}$$

[140] $1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 1 & 3 \\ 3 & 4 & 0 & 22 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 17 & -1 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 26 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[141] $1^{14} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 1 & 14 & 1 & 3 \\ 3 & 4 & 0 & 26 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 20 & -1 & 1 \end{pmatrix}_{b_0} \\
[143] & 21^{111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 13 & 20 & 15 & 12 & 7 \\ 20 & 25 & 20 & 16 & 9 \\ 15 & 20 & 13 & 12 & 7 \\ 12 & 16 & 12 & 9 & 5 \\ 7 & 9 & 7 & 5 & 2 \end{pmatrix}_{b_0} \\
[144] & 1^{132} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0 \\
& T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 12 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 27 & 9 \\ 7 & 13 & 0 & 27 & 27 & 9 \\ 3 & 4 & -2 & 9 & 9 & 1 \end{pmatrix}_{b_0} \\
[145] & 1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 12 & 40 & 8 & 5 \\ 12 & 25 & 80 & 16 & 9 \\ 40 & 80 & 238 & 48 & 28 \\ 8 & 16 & 48 & 9 & 5 \\ 5 & 9 & 28 & 5 & 2 \end{pmatrix}_{b_0} \\
[146] & 1^{131} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 12 & 13 \\ 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 13 & 2 & 1 \end{pmatrix}_{b_0} \\
[147] & 1^{112} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 12 & 19 & 1 & 5 \\ 12 & 25 & 40 & 0 & 9 \\ 19 & 40 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 14 & -1 & 2 \end{pmatrix}_{b_0} \\
[149] & 1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 8 & 13 & 0 & 12 & 27 & 9 \\ 3 & 4 & -2 & 4 & 9 & 1 \end{pmatrix}_{b_0} \\
[148] & 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 11 & 19 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 19 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0} \\
[151] & 1^{111} 1 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 11 & 17 & 9 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 17 & 33 & 49 & 27 & 13 \\ 9 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0} \\
[150] & 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 11 & 19 & 9 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 19 & 33 & 49 & 27 & 13 \\ 9 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0} \\
[152] & 1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 11 & 19 & 10 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 19 & 33 & 49 & 27 & 13 \\ 10 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0} \\
[153] & 1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7
\end{aligned}$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 18 & 10 \\ 5 & 8 & 0 & 18 & 9 & 5 \\ 3 & 4 & -2 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[154] \ 1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 39 & 8 & 5 \\ 12 & 25 & 80 & 16 & 9 \\ 39 & 80 & 238 & 48 & 28 \\ 8 & 16 & 48 & 9 & 5 \\ 5 & 9 & 28 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[155] \ 1^{113} 3^1 1^{21} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 205, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 53 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 53 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 39 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[156] \ 1^{13} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 15 & 13 \\ 3 & 5 & 0 & 15 & 3 & 3 \\ 3 & 4 & -2 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[157] \ 1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 18 & 10 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 18 & 33 & 49 & 27 & 13 \\ 10 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[158] \ 1^{111} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 45 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 45 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[159] \ 1^{121} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 21 & 10 \\ 6 & 9 & 0 & 21 & 12 & 6 \\ 3 & 4 & -2 & 10 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[160] \ 1^{112} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 21 & 7 \\ 7 & 13 & 0 & 21 & 27 & 9 \\ 3 & 4 & -2 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[161] \ 1^{12} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 12 & 10 \\ 3 & 5 & 0 & 12 & 3 & 3 \\ 3 & 4 & -2 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[162] \ 1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 4 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 18 & 10 \\ 4 & 8 & 0 & 18 & 9 & 5 \\ 3 & 4 & -2 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[163] \ 1^{11} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 9 & 0 & 13 & 9 & 7 \\ 3 & 5 & 0 & 9 & 3 & 3 \\ 3 & 4 & -2 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$

[164] $1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 18 & 10 \\ 4 & 8 & 0 & 18 & 9 & 5 \\ 3 & 4 & -2 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

[165] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 0 & 9 & 27 & 9 \\ 3 & 4 & -2 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[166] $1^{131} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 15 & 17 \\ 3 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

[167] $1^{121} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 0 & 10 & 6 & 6 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[168] $1^{132} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 4 & 3 \\ 3 & 4 & 0 & 12 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 0 & 25 & 18 & 9 \\ 4 & 8 & 0 & 18 & 9 & 5 \\ 3 & 4 & -2 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

[169] $1^{131} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 12 & 0 & 25 & 9 & 9 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[170] $1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 10 & 6 & 6 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[171] $1^{111} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 19 & 0 & 73 & 12 & 15 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[172] $1^{111} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 19 & 0 & 73 & 12 & 15 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[173] $1^{114} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 71 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 71 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 52 & -1 & 2 \end{pmatrix}_{b_0}$$

[174] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 233$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 18 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[175] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 234$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 25 & 12 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 25 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 237$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 27 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 27 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[177] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 240$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 240$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 7 & 12 & 0 & 9 & 24 & 8 \\ 3 & 4 & -2 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

[179] $1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 240$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 14 & 0 & 37 & 9 & 11 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

[180] $1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 241$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 38 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 38 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[181] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 241$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 24 & 12 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 24 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[182] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 242$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 9 & 7 \\ 3 & 5 & 0 & 9 & 3 & 3 \\ 3 & 4 & -2 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$

[183] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 244$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 30 & 1 & 5 \\ 12 & 25 & 60 & 0 & 9 \\ 30 & 60 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 21 & -1 & 2 \end{pmatrix}_{b_0}$$

[184] $21^{111} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 244$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 17 & 27 & 15 & 7 \\ 17 & 20 & 33 & 17 & 8 \\ 27 & 33 & 49 & 27 & 13 \\ 15 & 17 & 27 & 12 & 6 \\ 7 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[185] \ 1^{\overline{111}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 18 & 9 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 18 & 33 & 49 & 27 & 13 \\ 9 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[187] \ 1^{\overline{13}} 3^{\overline{113}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 1 & 3 \\ 3 & 4 & 0 & 27 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 27 & 0 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 21 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[186] \ 1^{\overline{113}} 3^{\overline{113}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 51 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 39 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[188] \ 1^{\overline{111}} 3^{\overline{114}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 47 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 47 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[189] \ 1^{\overline{132}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 17 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 24 & 13 \\ 5 & 8 & 0 & 24 & 9 & 5 \\ 3 & 4 & -2 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[190] \ 1^{\overline{111}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 9 & 0 & 13 & 15 & 7 \\ 6 & 9 & 0 & 15 & 12 & 6 \\ 3 & 4 & -2 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[191] \ 1^{\overline{141}} 3^{\overline{113}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 & 3 \\ 3 & 4 & 0 & 31 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 31 & 0 & 172 & 21 & 24 \\ 3 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & -2 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[192] \ 1^{\overline{111}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 0 & 15 & 12 & 6 \\ 3 & 4 & -2 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[193] \ 1^{\overline{111}} 3^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 14 & 0 & 37 & 9 & 11 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[194] \ 1^{\overline{132}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

- [195] $21^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 274$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.32 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 12 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 18 & 9 \\ 4 & 8 & 0 & 18 & 9 & 5 \\ 3 & 4 & -2 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$
- [197] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 276$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.17 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 30 & 12 & 7 \\ 20 & 25 & 40 & 16 & 9 \\ 30 & 40 & 58 & 24 & 14 \\ 12 & 16 & 24 & 9 & 5 \\ 7 & 9 & 14 & 5 & 2 \end{pmatrix}_{b_0}$$
- [196] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 276$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 38 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 38 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$
- [198] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 277$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 36 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 36 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$
- [199] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 285$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 35 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$
- [200] $21^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 286$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 16 & 24 & 12 & 6 \\ 16 & 20 & 33 & 17 & 8 \\ 24 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$
- [201] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 286$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 27 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 27 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$
- [202] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 286$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 28 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 28 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$
- [203] $1^{13}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 288$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 1 & 3 \\ 3 & 4 & 0 & 32 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 32 & 0 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 25 & -1 & 1 \end{pmatrix}_{b_0}$$
- [205] $1^{111}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 290$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 46 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 46 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$
- [204] $1^{132}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 290$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 8 & 3 \\ 3 & 4 & 0 & 17 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 39 & 13 \\ 8 & 13 & 0 & 39 & 27 & 9 \\ 3 & 4 & -2 & 13 & 9 & 1 \end{pmatrix}_{b_0}$$

[206] $1^{132} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 293$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 4 & 3 \\ 3 & 4 & 0 & 17 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 17 & 0 & 49 & 24 & 13 \\ 4 & 8 & 0 & 24 & 9 & 5 \\ 3 & 4 & -2 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 295$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 13 & 0 & 28 & 12 & 10 \\ 3 & 5 & 0 & 12 & 3 & 3 \\ 3 & 4 & -2 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

[208] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 296$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 9 & 10 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

[209] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 301$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.22 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 0 & 12 & 48 & 12 \\ 3 & 4 & -2 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[210] $1^{131} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 303$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.17 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 5 & 3 \\ 3 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 17 & 0 & 49 & 27 & 13 \\ 5 & 9 & 0 & 27 & 12 & 6 \\ 3 & 4 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

[211] $1^{141} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 304$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 & 3 \\ 3 & 4 & 0 & 26 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 18 & 20 \\ 3 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -2 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

[212] $1^{113} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 304$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 52 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 39 & -1 & 2 \end{pmatrix}_{b_0}$$

[213] $1^{113} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 304$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 28 & 1 & 5 \\ 12 & 25 & 60 & 0 & 9 \\ 28 & 60 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 21 & -1 & 2 \end{pmatrix}_{b_0}$$

[214] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 307$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 0 & 15 & 12 & 6 \\ 3 & 4 & -2 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

[215] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 309$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 8 & 13 & 0 & 9 & 27 & 9 \\ 3 & 4 & -2 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[216] $1^{14} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 309, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 1 & 3 \\ 3 & 4 & 0 & 31 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 31 & 0 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 24 & -1 & 1 \end{pmatrix}_{b_0}$$

[217] $1^{11} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 37 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 37 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 9 & 9 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[219] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 326, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 21 & 7 \\ 7 & 13 & 0 & 21 & 27 & 9 \\ 3 & 4 & -2 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

[220] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 27 & 7 \\ 9 & 17 & 0 & 27 & 48 & 12 \\ 3 & 4 & -2 & 7 & 12 & 1 \end{pmatrix}_{b_0}$$

[221] $1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 333, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 43 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 43 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 33 & -1 & 2 \end{pmatrix}_{b_0}$$

[222] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 17 & 0 & 49 & 15 & 13 \\ 3 & 5 & 0 & 15 & 3 & 3 \\ 3 & 4 & -2 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$

[223] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 341, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 0 & 17 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 24 & 13 \\ 4 & 8 & 0 & 24 & 9 & 5 \\ 3 & 4 & -2 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

[224] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 12 & 10 \\ 3 & 5 & 0 & 12 & 3 & 3 \\ 3 & 4 & -2 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

[225] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 349, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 29 & 1 & 5 \\ 12 & 25 & 60 & 0 & 9 \\ 29 & 60 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 21 & -1 & 2 \end{pmatrix}_{b_0}$$

[226] $1^{132} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 4 & 3 \\ 3 & 4 & 0 & 22 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 30 & 17 \\ 4 & 8 & 0 & 30 & 9 & 5 \\ 3 & 4 & -2 & 17 & 5 & 1 \end{pmatrix}_{b_0}$$

[227] $1^{132} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 4 & 3 \\ 3 & 4 & 0 & 22 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 22 & 0 & 85 & 30 & 17 \\ 4 & 8 & 0 & 30 & 9 & 5 \\ 3 & 4 & -2 & 17 & 5 & 1 \end{pmatrix}_{b_0}$$

[228] $1^{121} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 21 & 10 \\ 5 & 9 & 0 & 21 & 12 & 6 \\ 3 & 4 & -2 & 10 & 6 & 1 \end{pmatrix}_{b_0}$$

[229] $1^{121} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 13 & 0 & 28 & 21 & 10 \\ 5 & 9 & 0 & 21 & 12 & 6 \\ 3 & 4 & -2 & 10 & 6 & 1 \end{pmatrix}_{b_0}$$

[230] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 371, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 13 & 0 & 28 & 12 & 10 \\ 3 & 5 & 0 & 12 & 3 & 3 \\ 3 & 4 & -2 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

[231] $1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 11 & 18 & 0 & 12 & 57 & 13 \\ 3 & 4 & -2 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

[232] $1^{113} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 45 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 33 & -1 & 2 \end{pmatrix}_{b_0}$$

[233] $1^{131} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 12 & 0 & 25 & 9 & 9 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[234] $1^{114} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 40 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 40 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 28 & -1 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{114} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 56 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 56 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -1 & 2 \end{pmatrix}_{b_0}$$

- [236] $1^{112} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 386, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 21 & 7 \\ 8 & 13 & 0 & 21 & 27 & 9 \\ 3 & 4 & -2 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$
- [237] $1^{132} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 7 & 3 \\ 3 & 4 & 0 & 17 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 17 & 0 & 49 & 39 & 13 \\ 7 & 13 & 0 & 39 & 27 & 9 \\ 3 & 4 & -2 & 13 & 9 & 1 \end{pmatrix}_{b_0}$$
- [238] $1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 13 & 0 & 28 & 9 & 10 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$
- [239] $1^{13} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 394, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 15 & 13 \\ 3 & 5 & 0 & 15 & 3 & 3 \\ 3 & 4 & -2 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$
- [240] $1^{14} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 395, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 & 3 \\ 3 & 4 & 0 & 21 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 18 & 16 \\ 3 & 5 & 0 & 18 & 3 & 3 \\ 3 & 4 & -2 & 16 & 3 & 1 \end{pmatrix}_{b_0}$$
- [243] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 44 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 44 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$
- [241] $1^{14} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 1 & 3 \\ 3 & 4 & 0 & 36 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 20 & 36 & 0 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 28 & -1 & 1 \end{pmatrix}_{b_0}$$
- [242] $1^{114} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 68 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 52 & -1 & 2 \end{pmatrix}_{b_0}$$
- [244] $1^{141} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 5 & 3 \\ 3 & 4 & 0 & 21 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 33 & 16 \\ 5 & 9 & 0 & 33 & 12 & 6 \\ 3 & 4 & -2 & 16 & 6 & 1 \end{pmatrix}_{b_0}$$
- [245] $1^{131} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 411, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 27 & 13 \\ 5 & 9 & 0 & 27 & 12 & 6 \\ 3 & 4 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$
- [251] $1^{131} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 0 & 25 & 9 & 9 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[246] \ 1^{141} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 46 & 12 & 12 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[247] \ 1^{131} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 12 & 13 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[248] \ 1^{121} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[249] \ 1^{121} 3 \ 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 28 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 28 & 0 & 154 & 18 & 22 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -2 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[250] \ 1^{121} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[252] \ 1^{122} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 8 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 30 & 10 \\ 8 & 13 & 0 & 30 & 27 & 9 \\ 3 & 4 & -2 & 10 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[253] \ 21^{113} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.35 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 20 & 45 & 12 & 7 \\ 20 & 25 & 60 & 16 & 9 \\ 45 & 60 & 133 & 36 & 21 \\ 12 & 16 & 36 & 9 & 5 \\ 7 & 9 & 21 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[256] \ 1^{113} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 37 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 37 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[254] \ 1^{114} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 37 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 37 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 28 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[255] \ 1^{132} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 4 & 3 \\ 3 & 4 & 0 & 17 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 24 & 13 \\ 4 & 8 & 0 & 24 & 9 & 5 \\ 3 & 4 & -2 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

[258] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 36 & 1 & 5 & 9 \\ 12 & 25 & 76 & 0 & 0 & 9 \\ 36 & 76 & 217 & 0 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 & 1 \\ 5 & 9 & 27 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[257] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 44 & 1 & 5 & 9 \\ 12 & 25 & 92 & 0 & 0 & 9 \\ 44 & 92 & 325 & 0 & 0 & 33 \\ 1 & 0 & 0 & -3 & -1 & 1 \\ 5 & 9 & 33 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[259] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 12 & 22 & 0 & 15 & 84 & 16 \\ 3 & 4 & -2 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

[260] $1^{114} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 70 & 1 & 5 & 9 \\ 12 & 25 & 144 & 0 & 0 & 9 \\ 70 & 144 & 814 & 0 & 0 & 52 \\ 1 & 0 & 0 & -3 & -1 & 1 \\ 5 & 9 & 52 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[261] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 48 & 1 & 5 & 9 \\ 12 & 25 & 96 & 0 & 0 & 9 \\ 48 & 96 & 346 & 0 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 & 1 \\ 5 & 9 & 34 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[262] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 453, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 12 & 3 \\ 10 & 17 & 0 & 12 & 48 & 12 \\ 3 & 4 & -2 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[263] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 455, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 13 & 27 & 7 \\ 10 & 17 & 0 & 27 & 48 & 12 \\ 3 & 4 & -2 & 7 & 12 & 1 \end{pmatrix}_{b_0}$$

[264] $1^{121} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 23 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 23 & 0 & 100 & 15 & 18 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

[265] $1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 9 & 10 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

[266] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 458, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 17 & 0 & 49 & 15 & 13 \\ 3 & 5 & 0 & 15 & 3 & 3 \\ 3 & 4 & -2 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$

[267] $1^{122} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 28 & 30 & 10 \\ 7 & 13 & 0 & 30 & 27 & 9 \\ 3 & 4 & -2 & 10 & 9 & 1 \end{pmatrix}_{b_0}$$

[268] $1^{13} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 17 & 0 & 49 & 15 & 13 \\ 3 & 5 & 0 & 15 & 3 & 3 \\ 3 & 4 & -2 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$

[269] $1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 488, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 13 & 0 & 28 & 9 & 10 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

[270] $1^{122} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 491, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.19 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 30 & 10 \\ 7 & 13 & 0 & 30 & 27 & 9 \\ 3 & 4 & -2 & 10 & 9 & 1 \end{pmatrix}_{b_0}$$

[271] $1^{114} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 493, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 69 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 69 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 52 & -1 & 2 \end{pmatrix}_{b_0}$$

[272] $1^{121} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 498, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 5 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 28 & 21 & 10 \\ 5 & 9 & 0 & 21 & 12 & 6 \\ 3 & 4 & -2 & 10 & 6 & 1 \end{pmatrix}_{b_0}$$

[273] $1^{114} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 501, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.47 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 63 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 63 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 46 & -1 & 2 \end{pmatrix}_{b_0}$$

[274] $1^{14} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 503, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 & 3 \\ 3 & 4 & 0 & 21 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 21 & 0 & 76 & 18 & 16 \\ 3 & 5 & 0 & 18 & 3 & 3 \\ 3 & 4 & -2 & 16 & 3 & 1 \end{pmatrix}_{b_0}$$

[275] $1^{114} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 52 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -1 & 2 \end{pmatrix}_{b_0}$$

[276] $1^{114} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.47 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 60 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 46 & -1 & 2 \end{pmatrix}_{b_0}$$

[277] $1^{14} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 535, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 & 3 \\ 3 & 4 & 0 & 21 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 21 & 0 & 76 & 18 & 16 \\ 3 & 5 & 0 & 18 & 3 & 3 \\ 3 & 4 & -2 & 16 & 3 & 1 \end{pmatrix}_{b_0}$$

[278] $1^{141} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 21 & 0 & 76 & 15 & 16 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

[279] $1^{114} 3^1 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 549, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 39 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 39 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 28 & -1 & 2 \end{pmatrix}_{b_0}$$

[280] $1^{114} 3^1 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 38 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 38 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 28 & -1 & 2 \end{pmatrix}_{b_0}$$

[281] $1^{132} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 566, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 7 & 3 \\ 3 & 4 & 0 & 17 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 49 & 39 & 13 \\ 7 & 13 & 0 & 39 & 27 & 9 \\ 3 & 4 & -2 & 13 & 9 & 1 \end{pmatrix}_{b_0}$$

[282] $1^{14} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 571, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 & 3 \\ 3 & 4 & 0 & 21 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 21 & 0 & 76 & 18 & 16 \\ 3 & 5 & 0 & 18 & 3 & 3 \\ 3 & 4 & -2 & 16 & 3 & 1 \end{pmatrix}_{b_0}$$

[283] $1^{121} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 584, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 23 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 23 & 0 & 100 & 15 & 18 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

[284] $1^{141} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 586, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 5 & 3 \\ 3 & 4 & 0 & 21 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 21 & 0 & 76 & 33 & 16 \\ 5 & 9 & 0 & 33 & 12 & 6 \\ 3 & 4 & -2 & 16 & 6 & 1 \end{pmatrix}_{b_0}$$

[285] $1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 1 & 15 & 3 \\ 13 & 22 & 0 & 15 & 84 & 16 \\ 3 & 4 & -2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[286] $1^{131} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 5 & 3 \\ 3 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 17 & 0 & 49 & 27 & 13 \\ 5 & 9 & 0 & 27 & 12 & 6 \\ 3 & 4 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

[290] $1^{141} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 16 & 0 & 46 & 12 & 12 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[287] $1^{121} 3^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

[288] $1^{141} 3^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 16 & 0 & 46 & 12 & 12 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[289] $1^{121} 3^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

[291] $21^{114} 31^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 634, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 60 & 12 & 7 \\ 20 & 25 & 80 & 16 & 9 \\ 60 & 80 & 238 & 48 & 28 \\ 12 & 16 & 48 & 9 & 5 \\ 7 & 9 & 28 & 5 & 2 \end{pmatrix}_{b_0}$$

[292] $1^{114} 3^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 45 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[293] $1^{131} 3^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 639, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 5 & 3 \\ 3 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 27 & 13 \\ 5 & 9 & 0 & 27 & 12 & 6 \\ 3 & 4 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

[294] $1^{114} 3^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 645, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 55 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 55 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -1 & 2 \end{pmatrix}_{b_0}$$

[295] $1^{131} 3^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 85 & 15 & 17 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

[296] $1^{114} 3^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 660, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 62 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 62 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 46 & -1 & 2 \end{pmatrix}_{b_0}$$

[297] $1^{114} 3^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 661, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 47 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 47 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[298] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 669$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 61 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 61 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 46 & -1 & 2 \end{pmatrix}_{b_0}$

$$[302] \ 1^{121} 3 \ 1^{12} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{blue}{s}\textcolor{red}{u}_2 + \textcolor{red}{s}\textcolor{red}{u}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 672, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.07 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[300] \ 1^{131} 1^{14} 3 \ 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 672, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.16 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 17 & 0 & 49 & 12 & 13 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

[299] $1^{131} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, \quad T_{\max} = 12, \quad \Delta = -92, \quad \det = 672, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 12 & 13 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 13 & 2 & 1 \end{pmatrix}_{b_0}$

$$[304] \ 1^{121} 3 \ 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 672, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.07 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 28 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 28 & 0 & 154 & 18 & 22 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -2 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[301] \ 1^{121} 3 \ 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 672, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.07 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 0 & 28 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 20 & 28 & 0 & 154 & 18 & 22 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -2 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

[303] $1^{141} 3^1 1^{13} [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, \quad T_{\max} = 12, \quad \Delta = -92, \quad \det = 672, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.0$
 $T_{\min} = 11.29 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 16 & 0 & 46 & 12 & 12 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[305] $1^{121} 3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 680$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 23 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 23 & 0 & 100 & 15 & 18 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 18 & 2 & 1 \end{pmatrix}_{b_0}$

[306] $1^{114} 3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, \quad T_{\max} = 12, \quad \Delta = -92, \quad \det = 717, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.0$
 $T_{\min} = 11.41 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 53 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 53 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -1 & 2 \end{pmatrix}_{b_0}$

[307] $1^{132} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 722$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.17 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 7 & 3 \\ 3 & 4 & 0 & 17 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 49 & 39 & 13 \\ 7 & 13 & 0 & 39 & 27 & 9 \\ 3 & 4 & -2 & 13 & 9 & 1 \end{pmatrix}_{b_0}$

[308] $1^{114} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 724$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 27 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 27 & 0 & 133 & 18 & 21 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -2 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

[319] $1^{141} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 888, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 76 & 15 & 16 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

[321] $1^{141} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 896, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 26 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 26 & 0 & 118 & 18 & 20 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & -2 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

[320] $1^{131} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 896, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 0 & 32 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 32 & 0 & 193 & 21 & 25 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & -2 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

[322] $21^{115} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 904, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.1$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 75 & 12 & 7 \\ 20 & 25 & 100 & 16 & 9 \\ 75 & 100 & 373 & 60 & 35 \\ 12 & 16 & 60 & 9 & 5 \\ 7 & 9 & 35 & 5 & 2 \end{pmatrix}_{b_0}$$

[323] $1^{131} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 944, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 22 & 0 & 85 & 15 & 17 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

[324] $1^{131} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1040, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 22 & 0 & 85 & 15 & 17 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

[325] $1^{131} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 22 & 0 & 85 & 15 & 17 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

[326] $1^{141} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 21 & 0 & 76 & 15 & 16 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

[327] $1^{131} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 22 & 0 & 85 & 15 & 17 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & -2 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

[328] $1^{131} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 2 & 3 \\ 3 & 4 & 0 & 31 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 21 & 31 & 0 & 172 & 21 & 24 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & -2 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

[349] $1^{141} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 2144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 2 & 3 \\ 3 & 4 & 0 & 36 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 22 & 36 & 0 & 238 & 24 & 28 \\ 2 & 4 & 0 & 24 & 0 & 2 \\ 3 & 4 & -2 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

[350] $1^{141} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 2336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 2 & 3 \\ 3 & 4 & 0 & 36 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 23 & 36 & 0 & 238 & 24 & 28 \\ 2 & 4 & 0 & 24 & 0 & 2 \\ 3 & 4 & -2 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

[351] $1^{141} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 2400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 24 & 2 & 3 \\ 3 & 4 & 0 & 36 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 24 & 36 & 0 & 238 & 24 & 28 \\ 2 & 4 & 0 & 24 & 0 & 2 \\ 3 & 4 & -2 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [352] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 19 & 7 & 5 \\ 9 & 13 & 24 & 10 & 7 \\ 19 & 24 & 38 & 16 & 12 \\ 7 & 10 & 16 & 4 & 4 \\ 5 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

[353] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 5 \\ 9 & 13 & 0 & 0 & 7 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 0 & -1 & 3 \end{pmatrix}_{b_0}$$

[354] $21^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 12 & 8 & 6 \\ 12 & 13 & 14 & 10 & 7 \\ 12 & 14 & 13 & 9 & 7 \\ 8 & 10 & 9 & 4 & 4 \\ 6 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[355] $1^{12} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 12 & 1 & 5 \\ 9 & 13 & 20 & 0 & 7 \\ 12 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[356] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 8 & 8 & 6 \\ 12 & 13 & 10 & 10 & 7 \\ 8 & 10 & 5 & 7 & 5 \\ 8 & 10 & 7 & 4 & 4 \\ 6 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[357] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 12 & 8 & 6 \\ 12 & 13 & 14 & 10 & 7 \\ 12 & 14 & 13 & 9 & 7 \\ 8 & 10 & 9 & 4 & 4 \\ 6 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[358] $21^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 8 & 8 & 6 \\ 12 & 13 & 10 & 10 & 7 \\ 8 & 10 & 5 & 7 & 5 \\ 8 & 10 & 7 & 4 & 4 \\ 6 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[359] $21 \overset{1^1}{1^1} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 14 & 15 & 9 & 7 \\ 14 & 13 & 14 & 10 & 7 \\ 15 & 14 & 13 & 9 & 7 \\ 9 & 10 & 9 & 4 & 4 \\ 7 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[360] $1 \overset{1^1}{1^1} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 9 & 7 & 5 \\ 9 & 13 & 14 & 10 & 7 \\ 9 & 14 & 13 & 9 & 7 \\ 7 & 10 & 9 & 4 & 4 \\ 5 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[361] $1 \overset{1^{21}}{1^1} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -49, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 8 & 1 & 5 \\ 9 & 13 & 10 & 0 & 7 \\ 8 & 10 & 5 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

[362] $1 \overset{1^2}{1^1} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 6 & 8 & 5 \\ 9 & 13 & 10 & 10 & 7 \\ 6 & 10 & 5 & 7 & 5 \\ 8 & 10 & 7 & 4 & 4 \\ 5 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[363] $21 \overset{1^1}{1^1} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 14 & 9 & 9 & 7 \\ 14 & 13 & 10 & 10 & 7 \\ 9 & 10 & 5 & 7 & 5 \\ 9 & 10 & 7 & 4 & 4 \\ 7 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[364] $1 \overset{1^1}{1^1} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 5 \\ 9 & 13 & 14 & 0 & 7 \\ 9 & 14 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[365] $1 \overset{1^1}{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 9 & 6 & 5 \\ 9 & 13 & 14 & 10 & 7 \\ 9 & 14 & 13 & 9 & 7 \\ 6 & 10 & 9 & 4 & 4 \\ 5 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[366] $1 \overset{1^1}{1^1} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 5 \\ 9 & 13 & 10 & 0 & 7 \\ 6 & 10 & 5 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

[367] $21 \overset{1^{11}}{1^1} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 20 & 8 & 6 \\ 12 & 13 & 24 & 10 & 7 \\ 20 & 24 & 38 & 16 & 12 \\ 8 & 10 & 16 & 4 & 4 \\ 6 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

[368] $1 \overset{1^1}{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 20 & 8 & 6 \\ 12 & 13 & 24 & 10 & 7 \\ 20 & 24 & 38 & 16 & 12 \\ 8 & 10 & 16 & 4 & 4 \\ 6 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

[369] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 10 & 7 & 5 \\ 9 & 13 & 14 & 10 & 7 \\ 10 & 14 & 13 & 9 & 7 \\ 7 & 10 & 9 & 4 & 4 \\ 5 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$

[370] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 10 & 1 & 5 \\ 9 & 13 & 14 & 0 & 7 \\ 10 & 14 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$

[371] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 1 & 7 & 5 \\ 9 & 13 & 0 & 10 & 7 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 10 & 0 & 4 & 4 \\ 5 & 7 & 0 & 4 & 3 \end{pmatrix}_{b_0}$

[372] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 7 & 1 & 5 \\ 9 & 13 & 10 & 0 & 7 \\ 7 & 10 & 5 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$

[373] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -97, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 8 & 7 & 5 \\ 9 & 13 & 10 & 10 & 7 \\ 8 & 10 & 5 & 7 & 5 \\ 7 & 10 & 7 & 4 & 4 \\ 5 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$

[374] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 6 & 7 & 5 \\ 9 & 13 & 10 & 10 & 7 \\ 6 & 10 & 5 & 7 & 5 \\ 7 & 10 & 7 & 4 & 4 \\ 5 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$

[375] $1^{12} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 15 & 1 & 5 \\ 9 & 13 & 20 & 0 & 7 \\ 15 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 10 & -1 & 3 \end{pmatrix}_{b_0}$

[376] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 15 & 1 & 5 \\ 9 & 13 & 24 & 0 & 7 \\ 15 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 12 & -1 & 3 \end{pmatrix}_{b_0}$

[377] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 7 & 6 & 5 \\ 9 & 13 & 10 & 10 & 7 \\ 7 & 10 & 5 & 7 & 5 \\ 6 & 10 & 7 & 4 & 4 \\ 5 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$

[378] $1^{12} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 13 & 1 & 5 \\ 9 & 13 & 20 & 0 & 7 \\ 13 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 10 & -1 & 3 \end{pmatrix}_{b_0}$

[379] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -142, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$\begin{aligned}
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 9 & 15 & 7 & 5 \\ 9 & 13 & 24 & 10 & 7 \\ 15 & 24 & 38 & 16 & 12 \\ 7 & 10 & 16 & 4 & 4 \\ 5 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0} \\
[380] & 1^{\frac{11}{1}} 4^{\frac{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 9 & 18 & 1 & 5 \\ 9 & 13 & 24 & 0 & 7 \\ 18 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 12 & -1 & 3 \end{pmatrix}_{b_0} \\
[381] & 1^{\frac{11}{1}} 4^{\frac{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 9 & 7 & 7 & 5 \\ 9 & 13 & 10 & 10 & 7 \\ 7 & 10 & 5 & 7 & 5 \\ 7 & 10 & 7 & 4 & 4 \\ 5 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0} \\
[384] & 1^{\frac{13}{1}} 4^{\frac{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 9 & 19 & 1 & 5 \\ 9 & 13 & 30 & 0 & 7 \\ 19 & 30 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 15 & -1 & 3 \end{pmatrix}_{b_0} \\
[382] & 1^{\frac{12}{1}} 4^{\frac{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 9 & 21 & 1 & 5 \\ 9 & 13 & 34 & 0 & 7 \\ 21 & 34 & 77 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 17 & -1 & 3 \end{pmatrix}_{b_0} \\
[383] & 1^{\frac{12}{1}} 4^{\frac{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 9 & 14 & 1 & 5 \\ 9 & 13 & 20 & 0 & 7 \\ 14 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[385] & 1^{\frac{14}{1}} 4^{\frac{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 9 & 25 & 1 & 5 \\ 9 & 13 & 40 & 0 & 7 \\ 25 & 40 & 110 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 20 & -1 & 3 \end{pmatrix}_{b_0} \\
[386] & 1^{\frac{11}{1}} 4^{\frac{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 9 & 18 & 7 & 5 \\ 9 & 13 & 24 & 10 & 7 \\ 18 & 24 & 38 & 16 & 12 \\ 7 & 10 & 16 & 4 & 4 \\ 5 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0} \\
[387] & 1^{\frac{13}{1}} 4^{\frac{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 9 & 27 & 1 & 5 \\ 9 & 13 & 44 & 0 & 7 \\ 27 & 44 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 22 & -1 & 3 \end{pmatrix}_{b_0} \\
[388] & 1^{\frac{14}{1}} 4^{\frac{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 9 & 33 & 1 & 5 \\ 9 & 13 & 54 & 0 & 7 \\ 33 & 54 & 197 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 27 & -1 & 3 \end{pmatrix}_{b_0} \\
[389] & 1^{\frac{11}{1}} 4^{\frac{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 9 & 22 & 1 & 5 \\ 9 & 13 & 34 & 0 & 7 \\ 22 & 34 & 77 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 17 & -1 & 3 \end{pmatrix}_{b_0} \\
[400] & 1^{12} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 9 & 25 & 1 & 5 \\ 9 & 13 & 34 & 0 & 7 \\ 25 & 34 & 77 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 17 & -1 & 3 \end{pmatrix}_{b_0} \\
[401] & 1^{14} 4^{11} 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 9 & 26 & 1 & 5 \\ 9 & 13 & 40 & 0 & 7 \\ 26 & 40 & 110 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 20 & -1 & 3 \end{pmatrix}_{b_0} \\
[402] & 21^{111} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -382, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 10 & 16 & 7 & 5 \\ 10 & 13 & 24 & 10 & 7 \\ 16 & 24 & 38 & 16 & 12 \\ 7 & 10 & 16 & 4 & 4 \\ 5 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0} \\
[403] & 1^{12} 4^{11} 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -406, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 9 & 23 & 1 & 5 \\ 9 & 13 & 34 & 0 & 7 \\ 23 & 34 & 77 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 17 & -1 & 3 \end{pmatrix}_{b_0} \\
[404] & 1^{12} 4^{11} 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -409, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 9 & 24 & 1 & 5 \\ 9 & 13 & 34 & 0 & 7 \\ 24 & 34 & 77 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 17 & -1 & 3 \end{pmatrix}_{b_0} \\
[405] & 1^{13} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 9 & 28 & 1 & 5 \\ 9 & 13 & 44 & 0 & 7 \\ 28 & 44 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 22 & -1 & 3 \end{pmatrix}_{b_0} \\
[406] & 1^{14} 4^{11} 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -439, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 9 & 27 & 1 & 5 \\ 9 & 13 & 40 & 0 & 7 \\ 27 & 40 & 110 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 20 & -1 & 3 \end{pmatrix}_{b_0} \\
[407] & 1^{14} 4^{11} 1^{41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -454, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 9 & 28 & 1 & 5 \\ 9 & 13 & 40 & 0 & 7 \\ 28 & 40 & 110 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 20 & -1 & 3 \end{pmatrix}_{b_0} \\
[408] & 1^{14} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 9 & 34 & 1 & 5 \\ 9 & 13 & 54 & 0 & 7 \\ 34 & 54 & 197 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 27 & -1 & 3 \end{pmatrix}_{b_0} \\
[409] & 1^{13} 4^{11} 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s0}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -579, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 5 & 9 & 29 & 1 & 5 \\ 9 & 13 & 44 & 0 & 7 \\ 29 & 44 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[410] \ 1^{13} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -651, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 31 & 1 & 5 \\ 9 & 13 & 44 & 0 & 7 \\ 31 & 44 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[411] \ 1^{13} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -654, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.74 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 30 & 1 & 5 \\ 9 & 13 & 44 & 0 & 7 \\ 30 & 44 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[412] \ 1^{14} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -734, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.72 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 35 & 1 & 5 \\ 9 & 13 & 54 & 0 & 7 \\ 35 & 54 & 197 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[413] \ 1^{14} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -881, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.73 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 36 & 1 & 5 \\ 9 & 13 & 54 & 0 & 7 \\ 36 & 54 & 197 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[414] \ 1^{14} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -950, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.75 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 37 & 1 & 5 \\ 9 & 13 & 54 & 0 & 7 \\ 37 & 54 & 197 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [415] \ 1^1 4 \ 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -62, \ \det = 12, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 4 & 1 & 2 & 5 \\ 4 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 0 & 2 \\ 5 & 3 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

4.681 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6$

$$T = 9 \quad [0] \ 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -115, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, \ T_{\max} = 99, \ T \in [15, 99]$$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 9 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 9 & 7 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -271, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 7 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -334, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 12 & 0 & 6 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{e}}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -931$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 17 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 17 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 13 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

4.682 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4$

$T = 9$ [0] $1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{f}}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 9 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 9 & 7 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{f}}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -215$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 7 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{f}}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -264$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 12 & 0 & 6 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{f}}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -735$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 17 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 17 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 13 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

4.683 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8$

$T = 9$ [0] $1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 9 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 7 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{113} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -87$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.8$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 28 & 8 & 1 & 5 \\ 12 & 25 & 60 & 16 & 0 & 9 \\ 28 & 60 & 133 & 36 & 0 & 21 \\ 8 & 16 & 36 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 21 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -2 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}0_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{s}0_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 10 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 10 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 7 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}0_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{s}0_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 10 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}0_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{s}0_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -133, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 1 & 5 \\ 11 & 20 & 0 & 17 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 7 & -1 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}0_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{s}0_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 10 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}0_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{s}0_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{111} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}0_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{s}0_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 1 & 5 \\ 11 & 20 & 0 & 16 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{112} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}0_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{s}0_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 19 & 8 & 1 & 5 \\ 12 & 25 & 40 & 16 & 0 & 9 \\ 19 & 40 & 58 & 24 & 0 & 14 \\ 8 & 16 & 24 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 14 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{112} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}0_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{s}0_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 20 & 8 & 1 & 5 \\ 12 & 25 & 40 & 16 & 0 & 9 \\ 20 & 40 & 58 & 24 & 0 & 14 \\ 8 & 16 & 24 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 14 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}0_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{s}0_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

14.0

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 10 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 10 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 16 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 10 & 12 & 6 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 1 & 5 \\ 11 & 20 & 0 & 50 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 26 & 50 & 0 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 20 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 1 & 5 \\ 11 & 20 & 0 & 33 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 33 & 0 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & 13 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 30 & 8 & 1 & 5 \\ 12 & 25 & 60 & 16 & 0 & 9 \\ 30 & 60 & 133 & 36 & 0 & 21 \\ 8 & 16 & 36 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 21 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 29 & 8 & 1 & 5 \\ 12 & 25 & 60 & 16 & 0 & 9 \\ 29 & 60 & 133 & 36 & 0 & 21 \\ 8 & 16 & 36 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 21 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 38 & 8 & 1 & 5 \\ 12 & 25 & 80 & 16 & 0 & 9 \\ 38 & 80 & 238 & 48 & 0 & 28 \\ 8 & 16 & 48 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 28 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 40 & 8 & 1 & 5 \\ 12 & 25 & 80 & 16 & 0 & 9 \\ 40 & 80 & 238 & 48 & 0 & 28 \\ 8 & 16 & 48 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 28 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{\overline{11}4} 3 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -510, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.37 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 39 & 8 & 1 & 5 \\ 12 & 25 & 80 & 16 & 0 & 9 \\ 39 & 80 & 238 & 48 & 0 & 28 \\ 8 & 16 & 48 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 28 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{\overline{11}1} 3 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -539, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{\overline{11}1} 3 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -539, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 19 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 19 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{\overline{11}1} 3 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -735, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 18 & 9 & 5 \\ 11 & 20 & 0 & 33 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 33 & 0 & 49 & 27 & 13 \\ 9 & 17 & 0 & 27 & 12 & 6 \\ 5 & 8 & -2 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \ 1^{\overline{11}1} 4 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 165, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 9 & 6 & 1 & 5 \\ 9 & 13 & 14 & 10 & 0 & 7 \\ 9 & 14 & 13 & 9 & 0 & 7 \\ 6 & 10 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 7 & 7 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

4.684 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{11}1} 3 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -16, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & -2 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}1} 3 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 0 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}1} 3 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & -2 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

4.685 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2$ + (-3)→(-2)

$$T = 8 \quad [0] \quad 1^{11} 31^{12} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 12 & 1 & 7 \\ 17 & 20 & 16 & 0 & 8 \\ 12 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 27 & 1 & 7 \\ 17 & 20 & 33 & 0 & 8 \\ 27 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 325, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 27 & 1 & 7 \\ 20 & 25 & 36 & 0 & 9 \\ 27 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 31^{14} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 434, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 72 & 1 & 7 \\ 17 & 20 & 84 & 0 & 8 \\ 72 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 31^{12} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 42 & 1 & 7 \\ 17 & 20 & 50 & 0 & 8 \\ 42 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 31^{13} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 499, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 57 & 1 & 7 \\ 17 & 20 & 67 & 0 & 8 \\ 57 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{13} 1^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 525, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 15 & 1 & 7 \\ 20 & 25 & 20 & 0 & 9 \\ 15 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \quad 1^{2^1 1^{11}} 4^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 7 & 1 & 5 \\ 10 & 13 & 10 & 0 & 7 \\ 7 & 10 & 5 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{2^1 1^{11}} 4^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$[3] \ 1^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -35, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.60 \rightarrow 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -43, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.23 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 6 & 5 \\ 11 & 20 & 11 & 8 \\ 6 & 11 & 4 & 4 \\ 5 & 8 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -50, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 10 & 15 & 8 & 6 \\ 15 & 20 & 11 & 8 \\ 8 & 11 & 4 & 4 \\ 6 & 8 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 21^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -57, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 13 & 17 & 9 & 7 \\ 17 & 20 & 11 & 8 \\ 9 & 11 & 4 & 4 \\ 7 & 8 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 21^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -102, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 10 & 16 & 8 & 6 \\ 16 & 20 & 11 & 8 \\ 8 & 11 & 4 & 4 \\ 6 & 8 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 21^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -123, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.82 \rightarrow 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 13 & 20 & 9 & 7 \\ 20 & 25 & 12 & 9 \\ 9 & 12 & 4 & 4 \\ 7 & 9 & 4 & 3 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [9] \ 21^1 3^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 12, \ \Delta = -88, \ \det = 4, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 5 & 5 & 3 & 5 \\ 3 & 5 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 4 & 5 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[10] \ 1^1 3^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 12, \ \Delta = -88, \ \det = 19, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.26 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 3 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 5 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^1 3^1 313^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 20, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.40 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 10 & 1 & 4 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 6 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^1 3^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 12, \ \Delta = -88, \ \det = 20, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 5 & 2 & 3 \\ 4 & 5 & 9 & 4 & 5 \\ 2 & 2 & 4 & 0 & 2 \\ 3 & 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^1 3^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 12, \ \Delta = -88, \ \det = 20, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 2 & 4 & 0 & 2 & 2 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^1 3^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 12, \ \Delta = -88, \ \det = 20, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.40 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 5 & 5 & 3 & 5 \\ 2 & 5 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[11] \quad & 1^1 1^1 3^1 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 2 & 2 & 0 & 0 & 2 \\ 2 & 3 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0} \\
[17] \quad & 1^1 1^1 3^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 3 & 3 & -2 & 3 \end{pmatrix}_{b_0} \\
[16] \quad & 1^1 1^1 3^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 2 & 0 & 0 & 2 \\ 3 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0} \\
[18] \quad & 1^1 1^1 3^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 3 & 7 & 3 & 4 \\ 3 & 1 & 5 & 2 & 3 \\ 7 & 5 & 9 & 4 & 5 \\ 3 & 2 & 4 & 0 & 2 \\ 4 & 3 & 5 & 2 & 2 \end{pmatrix}_{b_0} \\
[19] \quad & 1^1 1^1 3^1 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 1 & 3 & 3 \\ 4 & 10 & 1 & 8 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 8 & 0 & 4 & 4 \\ 3 & 6 & -1 & 4 & 2 \end{pmatrix}_{b_0} \\
[20] \quad & 1^1 1^1 3^1 21^1 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 3 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 0 & 1 \end{pmatrix}_{b_0} \\
[21] \quad & 1^1 1^1 3^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 5 & 3 & 5 \\ 3 & 5 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 5 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[22] \quad & 1^1 1^1 3^1 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 6 & 0 & 4 \\ 4 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & -2 & 2 \end{pmatrix}_{b_0} \\
[23] \quad & 13^1 1^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 5 & 2 & 3 \\ 4 & 10 & 13 & 4 & 6 \\ 5 & 13 & 12 & 5 & 6 \\ 2 & 4 & 5 & 0 & 2 \\ 3 & 6 & 6 & 2 & 2 \end{pmatrix}_{b_0} \\
[24] \quad & 1^1 3^1 1^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 2 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & -2 & 2 & 1 \end{pmatrix}_{b_0} \\
[25] \quad & 1^1 1^1 31^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 4 & 7 & 8 & 3 & 4 \\ 7 & 10 & 13 & 4 & 6 \\ 8 & 13 & 12 & 5 & 6 \\ 3 & 4 & 5 & 0 & 2 \\ 4 & 6 & 6 & 2 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [26] $1^1 3^1 1^1 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 8 & 6 & 0 & 6 \\ 2 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 6 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$
- [27] $13^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 3 & 3 \\ 4 & 10 & 13 & 8 & 6 \\ 5 & 13 & 12 & 9 & 6 \\ 3 & 8 & 9 & 4 & 4 \\ 3 & 6 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$
- [29] $1^1 1^1 3^1 1^1 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 4 & 7 & 8 & 5 & 4 \\ 7 & 10 & 13 & 8 & 6 \\ 8 & 13 & 12 & 9 & 6 \\ 5 & 8 & 9 & 4 & 4 \\ 4 & 6 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$
- [28] $1^1 3^1 1^1 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 2 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [30] $21^1 1^1 3^1 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 4 \\ 8 & 9 & 4 & 4 & 5 \\ 3 & 4 & 0 & 2 & 2 \\ 3 & 4 & 2 & 0 & 2 \\ 4 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [31] $31^1 1^1 23^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 14, \Delta = -96, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$

$$\begin{pmatrix} 1 & 5 & 4 & 2 & 3 \\ 5 & 5 & 5 & 3 & 5 \\ 4 & 5 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [32] $31^1 1^1 23^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 14, \Delta = -96, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$

$$\begin{pmatrix} 5 & 8 & 8 & 5 \\ 8 & 6 & 9 & 6 \\ 8 & 9 & 5 & 5 \\ 5 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$
- [33] $1^1 32^1 1^1 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 13, \Delta = -95, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 1 & 2 \\ 3 & 1 & 4 & 1 & 0 & 3 \\ 3 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$
- $T = 9$ [36] $21^1 41^1 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$
- [35] $1^1 3^1 4^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$
- [34] $1^1 4^1 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$
- [37] $1^1 4^1 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 12, \Delta = -85, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{\overline{1}} 4^{\overline{1}} 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[39] \ 21^{\overline{1}} 41^{\overline{1}} 31^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 0 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & -1 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{\overline{1}} 4^{\overline{1}} 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 0 & 2 & 0 & 2 \\ 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{\overline{1}} 1^{\overline{3}} 4^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{\overline{1}} 41^{\overline{1}} 23^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -73, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 1 & 1 & 3 \\ 6 & 1 & 3 & 0 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 7 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{\overline{1}} 41^{\overline{1}} 23^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 1 & 2 \\ 3 & 4 & 1 & 2 & 0 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[44] \ 23^{\overline{1}} 141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 \\ 2 & 1 & 4 & 1 & 2 & 3 \\ 2 & 4 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 2 & 2 & 2 & 1 & 0 & 2 \\ 2 & 3 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{\overline{1}} 41^{\overline{1}} 31^{\overline{1}} 3^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 1 & 2 \\ 3 & 4 & 4 & 2 & 0 & 4 \\ 3 & 4 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 2 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[46] \ 31^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} 41^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.11 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 & 2 \\ 3 & 4 & 4 & 2 & 3 & 4 \\ 3 & 4 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & -2 & 1 & 0 \\ 2 & 3 & 2 & 1 & 0 & 2 \\ 2 & 4 & 2 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [47] $21^1 5 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[48] $21^1 5 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 3 & 3 & 4 \\ 3 & 3 & 0 & 3 & 2 \\ 3 & 3 & 3 & 1 & 3 \\ 4 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[49] $15^1 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 10, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[50] $1^1 5 123^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 10, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & 4 & 4 \\ 3 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0}$

[51] $1^1 4 13^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0}$

[52] $1^1 4 13^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 22, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$

[53] $21^1 5 1^1 23^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 37, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 4 & 11 & 3 & 3 & 4 \\ 11 & 26 & 8 & 8 & 10 \\ 3 & 8 & 0 & 3 & 2 \\ 3 & 8 & 3 & 1 & 3 \\ 4 & 10 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[54] $1^1 5^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 73, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 7 & 2 & 2 & 3 \\ 7 & 26 & 8 & 8 & 10 \\ 2 & 8 & 0 & 3 & 2 \\ 2 & 8 & 3 & 1 & 3 \\ 3 & 10 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[56] $1^1 4 1^1 3^1 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 97, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 3 & 1 & 1 & 2 & 2 \\ 3 & 13 & 6 & 0 & 9 & 7 \\ 1 & 6 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 9 & 3 & 0 & 4 & 4 \\ 2 & 7 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$

[55] $1^1 4^1 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 97, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 5 & 8 & 0 & 3 & 5 \\ 2 & 8 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 5 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[57] $1^{\overline{1}^3 \overline{1}^1} 4 \ 141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 102, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 & 1 & 2 \\ 3 & 13 & 1 & 9 & 5 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 9 & 0 & 3 & 3 & 3 \\ 1 & 5 & 0 & 3 & 0 & 2 \\ 2 & 7 & -1 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[58] $1^{\overline{1}^3 \overline{1}^1} 41^{\overline{1}} 3^{\overline{1}} 141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 106, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 8 & 1 & 5 & 3 & 3 \\ 8 & 38 & 1 & 24 & 12 & 12 \\ 1 & 1 & -3 & 0 & -1 & -1 \\ 5 & 24 & 0 & 13 & 7 & 7 \\ 3 & 12 & -1 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

[59] $1^{\overline{1}^3 \overline{1}^1} 4 \ 141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 137, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 2 & 1 & 1 & 2 \\ 3 & 13 & 10 & 0 & 5 & 7 \\ 2 & 10 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 5 & 3 & 0 & 0 & 2 \\ 2 & 7 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$T = 11 \quad [61] \ 21^{\overline{1}} 513^{\overline{1}^1} 1^{\overline{1}} 41^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 4 & 5 & 1 & 2 & 4 & 4 \\ 5 & 5 & 2 & 3 & 5 & 5 \\ 1 & 2 & -2 & 1 & 0 & 0 \\ 2 & 3 & 1 & 0 & 2 & 2 \\ 4 & 5 & 0 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[60] $1^{\overline{1}^3} 31^{\overline{1}} 513^{\overline{1}^1} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 3 & 3 \\ 3 & 0 & 3 & 4 & 4 & 4 \\ 3 & 0 & 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

[62] $1^{\overline{1}^3} 413^{\overline{1}^1} 1^{\overline{1}} 51^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 1 & 2 & 5 & 5 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 0 & 2 & 2 \\ 5 & 0 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

[63] $1^{\overline{1}^3} 31^{\overline{1}} 513^{\overline{1}^1} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 3 & 4 \\ 1 & 1 & 0 & 3 & 0 & 2 \\ 2 & 3 & -1 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[65] $1^{\overline{1}^3} 3^{\overline{1}^1} 1^{\overline{1}} 5131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -13, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 2 & 1 & 3 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 & -1 \\ 2 & 3 & 0 & 1 & 3 & 3 \\ 3 & 4 & -1 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[64] $1^{\overline{1}^3} 3^{\overline{1}^1} 1^{\overline{1}} 5131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -13, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 1 & 3 \\ 3 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^1 413^1 21^1 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -81, \quad \det = -106, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.70 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 2 & 2 \\ 4 & 2 & 1 & 1 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 2 & 2 & 0 & 1 & 0 & 2 \\ 2 & 4 & 0 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^1 1513^1 1^1 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -79, \quad \det = -106, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.89 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 & 2 \\ 3 & 4 & 1 & 2 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 1 & 0 \\ 2 & 3 & 0 & 1 & 0 & 2 \\ 2 & 4 & 0 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [68] \quad 1^1 1^1 3^1 21^1 6 \quad 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 14, \quad \Delta = -58, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^1 1^1 3^1 21^1 6 \quad 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 14, \quad \Delta = -58, \quad \det = 13, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.85 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [70] \quad 23218123^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 14, \quad T_{\max} = 16, \quad \Delta = -58, \quad \det = 3, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 16, \quad T_{\max} = 16, \quad T \in [16, 16]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 2 & 3 \\ 3 & 3 & 1 & 3 & 3 \\ 3 & 2 & 3 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

4.687 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_6$

$$T = 9 \quad [0] \quad 1^1 1^1 3^1 21^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_6 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -137, \quad \det = -78, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.69 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 3 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 5 & 0 & 2 & -2 & 0 \\ 3 & 5 & -2 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 1^1 3^1 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_6 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -143, \quad \det = -244, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.77 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 3 & 1 & 3 \\ 4 & 10 & 13 & 8 & 0 & 6 \\ 5 & 13 & 12 & 9 & 0 & 6 \\ 3 & 8 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

4.688 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4$

$$T = 8 \quad [0] \quad 1^1 1^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -138, \quad \det = 116, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.62 \rightarrow 13, \quad T_{\max} = 98, \quad T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, \ T_{\max} = 98, \ \Delta = -138, \ \det = 174, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.68 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 6 & 1 & 5 \\ 11 & 20 & 11 & 0 & 8 \\ 6 & 11 & 4 & 0 & 4 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \ 1^{11}3^{11}21^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, \ T_{\max} = 22, \ \Delta = -111, \ \det = -59, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 22, \ T \in [13, 22]$$

$$\begin{pmatrix} 5 & 13 & 1 & 6 & 5 \\ 13 & 30 & 0 & 13 & 10 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 13 & 0 & 4 & 4 \\ 5 & 10 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}31^{11}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -112, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 32, \ T \in [13, 32]$$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 1 & 3 \\ 4 & 10 & 13 & 4 & 0 & 6 \\ 5 & 13 & 12 & 5 & 0 & 6 \\ 2 & 4 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^{11}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -187, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 32, \ T \in [13, 32]$$

$$\begin{pmatrix} 1 & 4 & 5 & 3 & 1 & 3 \\ 4 & 10 & 13 & 8 & 0 & 6 \\ 5 & 13 & 12 & 9 & 0 & 6 \\ 3 & 8 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

4.689 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -218, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 6 & 1 & 9 & 5 \\ 11 & 20 & 11 & 0 & 17 & 8 \\ 6 & 11 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 9 & 0 & 13 & 7 \\ 5 & 8 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -222, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 9 & 5 \\ 12 & 25 & 12 & 0 & 20 & 9 \\ 6 & 12 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 9 & 0 & 13 & 7 \\ 5 & 9 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -302, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 17 & 5 \\ 12 & 25 & 12 & 0 & 36 & 9 \\ 6 & 12 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 36 & 17 & 0 & 49 & 13 \\ 5 & 9 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

4.690 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -392, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, \ T_{\max} = 32, \ T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 6 & 1 & 9 & 5 \\ 11 & 20 & 11 & 0 & 17 & 8 \\ 6 & 11 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 9 & 0 & 12 & 6 \\ 5 & 8 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

4.691 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 5 \\ 11 & 20 & 0 & 11 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 11 & 0 & 4 & 4 \\ 5 & 8 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \quad 1^{13} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 3 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 5 & 0 & 2 & 0 & 2 \\ 3 & 5 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{13} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 2 & 6 & 0 & 4 & 4 \\ 4 & 6 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 5 & 0 & 0 & 2 \\ 3 & 4 & 5 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 1^{13} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 & 3 \\ 4 & 10 & 13 & 0 & 8 & 6 \\ 5 & 13 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 9 & 0 & 4 & 4 \\ 3 & 6 & 6 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [5] \quad 1^{14} 4^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 2 & 3 \\ 2 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 0 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

4.692 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 1 & 5 \\ 11 & 20 & 0 & 11 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 8 & -2 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 9 & 5 \\ 12 & 25 & 12 & 0 & 20 & 9 \\ 6 & 12 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 20 & 9 & 0 & 13 & 7 \\ 5 & 9 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 18 & 5 \\ 12 & 25 & 12 & 0 & 36 & 9 \\ 6 & 12 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 36 & 17 & 0 & 49 & 13 \\ 5 & 9 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 13 & 7 \\ 5 & 8 & -2 & 4 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -217, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 17 & 5 \\ 12 & 25 & 12 & 0 & 36 & 9 \\ 6 & 12 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 36 & 17 & 0 & 49 & 13 \\ 5 & 9 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

4.693 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 6 & 5 \\ 12 & 25 & 12 & 0 & 12 & 9 \\ 6 & 12 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 6 & 0 & 4 & 4 \\ 5 & 9 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

4.694 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 12 & 6 \\ 5 & 8 & -2 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

4.695 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 16 & 8 & 23 & 6 \\ 16 & 25 & 12 & 36 & 9 \\ 8 & 12 & 4 & 17 & 4 \\ 23 & 36 & 17 & 49 & 13 \\ 6 & 9 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 4 & 8 & 0 & 4 & 10 & 6 \\ 3 & 4 & -2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 1^{\overline{3}} 1^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 3 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

[3] $1^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 10 & 15 & 8 & 12 & 6 \\ 15 & 20 & 11 & 17 & 8 \\ 8 & 11 & 4 & 9 & 4 \\ 12 & 17 & 9 & 13 & 7 \\ 6 & 8 & 4 & 7 & 3 \end{pmatrix}_{b_0}$

[4] $1^1 1^3 1^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$

$$[5] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_8 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 59, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 8 & 1 & 6 \\ 15 & 20 & 11 & 0 & 8 \\ 8 & 11 & 4 & 0 & 4 \\ 1 & 0 & 0 & -2 & 0 \\ 6 & 8 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{brown}{s}\textcolor{brown}{u}_2 \rightarrow (-3) + \textcolor{red}{s}\textcolor{red}{u}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.62 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{\overline{11}} 1^{\overline{3}} 1^{\overline{1}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \, T_{\max} = 11, \, \Delta = -85, \, \det = 66, \, (n_+, n_-, n_0) = (1, 8, 0), \, c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, \, T_{\max} = 11, \, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{brown}{s}\textcolor{blue}{u}_2 \rightarrow (-3) + \textcolor{red}{s}\textcolor{blue}{u}_2 \quad T = 8, \, T_{\max} = 11, \, \Delta = -85, \, \det = 66, \, (n_+, n_-, n_0) = (1, 8, 0), \, c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, \, T_{\max} = 11, \, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 8 & 13 & 6 \\ 15 & 20 & 11 & 17 & 8 \\ 8 & 11 & 4 & 9 & 4 \\ 13 & 17 & 9 & 13 & 7 \\ 6 & 8 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[9] $1^{11}3^11^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 3 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$

$$[10] \quad 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 119, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.71 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 5 & 9 & 0 & 5 & 13 & 7 \\ 3 & 4 & -2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 5 & 8 & 0 & 4 & 10 & 6 \\ 3 & 4 & -2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{11} 1^{13} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 7 & 12 & 0 & 6 & 25 & 9 \\ 3 & 4 & -2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[12] $1^{11} 1^{13} 3^1 1^2 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 8 & 14 & 0 & 8 & 37 & 11 \\ 3 & 4 & -2 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[14] $1^{11} 1^{14} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 9 & 16 & 0 & 8 & 46 & 12 \\ 3 & 4 & -2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{12} 1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 7 & 13 & 0 & 7 & 28 & 10 \\ 3 & 4 & -2 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{11} 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 6 & 9 & 0 & 5 & 13 & 7 \\ 3 & 4 & -2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{13} 1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 9 & 17 & 0 & 9 & 49 & 13 \\ 3 & 4 & -2 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{14} 1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 11 & 21 & 0 & 11 & 76 & 16 \\ 3 & 4 & -2 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[19] $21^{11} 1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 20 & 9 & 27 & 7 \\ 20 & 25 & 12 & 36 & 9 \\ 9 & 12 & 4 & 17 & 4 \\ 27 & 36 & 17 & 49 & 13 \\ 7 & 9 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[20] $1^{12} 1^{13} 3^1 1^2 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$\begin{aligned}
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 10 & 18 & 0 & 10 & 58 & 14 \\ 3 & 4 & -2 & 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[21] & 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 9 & 14 & 0 & 8 & 37 & 11 \\ 3 & 4 & -2 & 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[22] & 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 306, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 8 & 13 & 0 & 7 & 28 & 10 \\ 3 & 4 & -2 & 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[23] & 1^{11} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 311, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 12 & 19 & 0 & 11 & 73 & 15 \\ 3 & 4 & -2 & 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[24] & 21^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 13 & 20 & 9 & 15 & 7 \\ 20 & 25 & 12 & 20 & 9 \\ 9 & 12 & 4 & 9 & 4 \\ 15 & 20 & 9 & 13 & 7 \\ 7 & 9 & 4 & 7 & 3 \end{pmatrix}_{b_0} \\
[25] & 1^{12} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 13 & 2 \\ 13 & 23 & 0 & 13 & 100 & 18 \\ 3 & 4 & -2 & 2 & 18 & 2 \end{pmatrix}_{b_0} \\
[26] & 1^{13} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 12 & 22 & 0 & 12 & 85 & 17 \\ 3 & 4 & -2 & 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[27] & 1^{12} 3^{11} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 3 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 16 & 2 \\ 16 & 28 & 0 & 16 & 154 & 22 \\ 3 & 4 & -2 & 2 & 22 & 2 \end{pmatrix}_{b_0} \\
[28] & 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 399, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 10 & 17 & 0 & 9 & 49 & 13 \\ 3 & 4 & -2 & 2 & 13 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[30] $1^{14}3^11^{12}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{s0}_8+\textcolor{orange}{su}_2\rightarrow(-3)+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 14 & 26 & 0 & 14 & 118 & 20 \\ 3 & 4 & -2 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{12}3^11^{12}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{s0}_8+\textcolor{orange}{su}_2\rightarrow(-3)+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 11 & 18 & 0 & 10 & 58 & 14 \\ 3 & 4 & -2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{14}3^11^{11}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{s0}_8+\textcolor{orange}{su}_2\rightarrow(-3)+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 474, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 12 & 21 & 0 & 11 & 76 & 16 \\ 3 & 4 & -2 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{13}3^11^{13}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{s0}_8+\textcolor{orange}{su}_2\rightarrow(-3)+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 511, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 3 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 15 & 2 \\ 15 & 27 & 0 & 15 & 133 & 21 \\ 3 & 4 & -2 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{12}3^11^{13}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{s0}_8+\textcolor{orange}{su}_2\rightarrow(-3)+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 570, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 13 & 2 \\ 14 & 23 & 0 & 13 & 100 & 18 \\ 3 & 4 & -2 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{13}3^11^{14}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{s0}_8+\textcolor{orange}{su}_2\rightarrow(-3)+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 621, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 3 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 18 & 2 \\ 18 & 32 & 0 & 18 & 193 & 25 \\ 3 & 4 & -2 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{13}3^11^{12}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{s0}_8+\textcolor{orange}{su}_2\rightarrow(-3)+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 621, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 13 & 22 & 0 & 12 & 85 & 17 \\ 3 & 4 & -2 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{12}3^11^{14}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{s0}_8+\textcolor{orange}{su}_2\rightarrow(-3)+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 654, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 3 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 16 & 2 \\ 17 & 28 & 0 & 16 & 154 & 22 \\ 3 & 4 & -2 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{14}3^11^{13}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{s0}_8+\textcolor{orange}{su}_2\rightarrow(-3)+\textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 670, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 3 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 17 & 2 \\ 17 & 31 & 0 & 17 & 172 & 24 \\ 3 & 4 & -2 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[38] $1^{14}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 770, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 3 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 15 & 26 & 0 & 14 & 118 & 20 \\ 3 & 4 & -2 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{13}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 811, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 3 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 15 & 2 \\ 16 & 27 & 0 & 15 & 133 & 21 \\ 3 & 4 & -2 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{14}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 854, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 3 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 20 & 2 \\ 20 & 36 & 0 & 20 & 238 & 28 \\ 3 & 4 & -2 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{13}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 969, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 19 & 3 \\ 3 & 4 & 0 & 3 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 18 & 2 \\ 19 & 32 & 0 & 18 & 193 & 25 \\ 3 & 4 & -2 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{14}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 1034, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 3 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 17 & 2 \\ 18 & 31 & 0 & 17 & 172 & 24 \\ 3 & 4 & -2 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{14}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 1266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 21 & 3 \\ 3 & 4 & 0 & 3 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 20 & 2 \\ 21 & 36 & 0 & 20 & 238 & 28 \\ 3 & 4 & -2 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [44] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 3 \\ 3 & 5 & 5 & 3 & 5 & 5 \\ 3 & 5 & 0 & 2 & 3 & 2 \\ 2 & 3 & 2 & 0 & 3 & 2 \\ 3 & 5 & 3 & 3 & 4 & 4 \\ 3 & 5 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[46] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 1 & 1 & 2 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 4 & 2 \\ 5 & 4 & 0 & 4 & 10 & 6 \\ 3 & 3 & -1 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[45] $1^1 1^3 1^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 2 & 4 & 0 & 2 & 4 \\ 3 & 3 & 3 & -2 & 4 & 3 \end{pmatrix}_{b_0}$$

[47] $1^1 1^1 3^1 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 3 \\ 2 & 1 & 3 & 0 & 7 & 3 \\ 2 & 3 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 7 & 10 & 0 & 25 & 11 \\ 3 & 3 & 3 & -2 & 11 & 3 \end{pmatrix}_{b_0}$$

[48] $21^{11} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 9 & 4 \\ 5 & 5 & 5 & 3 & 10 & 5 \\ 3 & 5 & 0 & 2 & 7 & 2 \\ 3 & 3 & 2 & 0 & 5 & 2 \\ 9 & 10 & 7 & 5 & 14 & 8 \\ 4 & 5 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[49] $1^1 1^2 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -22, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 8 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 3 & 5 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 8 & 15 & 10 & 8 & 38 & 12 \\ 3 & 5 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[50] $1^1 1^1 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 2 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 6 & 2 \\ 6 & 7 & 0 & 6 & 21 & 9 \\ 3 & 3 & -1 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[51] $1^1 1^3 1^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 5 & 5 & 3 & 5 & 5 \\ 3 & 5 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 2 & 5 & 4 & 2 & 2 & 4 \\ 3 & 5 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[52] $1^1 1^1 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 9 & 3 \\ 3 & 5 & 5 & 3 & 20 & 3 \\ 3 & 5 & 0 & 2 & 15 & 2 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 9 & 20 & 15 & 9 & 58 & 16 \\ 3 & 5 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[54] $1^1 1^1 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & -1 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[53] $1^1 1^4 3^1 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 25 & 3 \\ 2 & 1 & 1 & 2 & 28 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 24 & 2 \\ 25 & 28 & 0 & 24 & 366 & 36 \\ 3 & 3 & -1 & 2 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -37, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 3 & 3 \\ 2 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 2 & 0 & 1 & 3 \\ 3 & 3 & 3 & -2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{13}3^{11}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 14 & 3 \\ 2 & 1 & 1 & 2 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 14 & 2 \\ 14 & 17 & 0 & 14 & 121 & 21 \\ 3 & 3 & -1 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[57] \quad 21^{12}3^{11}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 18 & 4 \\ 5 & 5 & 5 & 3 & 20 & 5 \\ 3 & 5 & 0 & 2 & 14 & 2 \\ 3 & 3 & 2 & 0 & 10 & 2 \\ 18 & 20 & 14 & 10 & 62 & 16 \\ 4 & 5 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{12}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 8 & 2 \\ 8 & 10 & 0 & 8 & 38 & 12 \\ 3 & 3 & -1 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{12}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 3 \\ 2 & 1 & 3 & 0 & 12 & 3 \\ 2 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 12 & 16 & 0 & 70 & 18 \\ 3 & 3 & 3 & -2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 3 \\ 2 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 6 & 0 & 9 & 7 \\ 3 & 3 & 3 & -2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 3 \\ 3 & 5 & 5 & 3 & 10 & 5 \\ 3 & 5 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 0 & 5 & 2 \\ 5 & 10 & 7 & 5 & 14 & 8 \\ 3 & 5 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{11}3^{11}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -88, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 12 & 3 \\ 2 & 1 & 1 & 2 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 10 & 2 \\ 12 & 11 & 0 & 10 & 65 & 15 \\ 3 & 3 & -1 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{13} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 10 & 2 \\ 10 & 13 & 0 & 10 & 61 & 15 \\ 3 & 3 & -1 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{12} 3^1 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 24 & 3 \\ 2 & 1 & 1 & 2 & 25 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 22 & 2 \\ 24 & 25 & 0 & 22 & 313 & 33 \\ 3 & 3 & -1 & 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{11} 3^1 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 7 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 3 & 5 & 0 & 2 & 11 & 2 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 7 & 15 & 11 & 7 & 32 & 12 \\ 3 & 5 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{11} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 3 \\ 2 & 1 & 1 & 2 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 6 & 2 \\ 7 & 7 & 0 & 6 & 21 & 9 \\ 3 & 3 & -1 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{11} 3^1 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 19 & 3 \\ 2 & 1 & 1 & 2 & 21 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 18 & 2 \\ 19 & 21 & 0 & 18 & 205 & 27 \\ 3 & 3 & -1 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{12} 3^1 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 18 & 3 \\ 2 & 1 & 1 & 2 & 18 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 16 & 2 \\ 18 & 18 & 0 & 16 & 166 & 24 \\ 3 & 3 & -1 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{14} 3^1 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 16 & 3 \\ 2 & 1 & 1 & 2 & 20 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 16 & 2 \\ 16 & 20 & 0 & 16 & 158 & 24 \\ 3 & 3 & -1 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{12} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 3 \\ 2 & 1 & 3 & 0 & 8 & 3 \\ 2 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 8 & 0 & 22 & 10 \\ 3 & 3 & 3 & -2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{12} 3^1 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -132, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.88 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 3 \\ 2 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 10 & 12 & 0 & 42 & 14 \\ 3 & 3 & 3 & -2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{11} 3^1 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -132, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.88 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 13 & 3 \\ 2 & 1 & 1 & 2 & 14 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 12 & 2 \\ 13 & 14 & 0 & 12 & 90 & 18 \\ 3 & 3 & -1 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{12} 3^1 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -146, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 10 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 3 & 5 & 0 & 2 & 14 & 2 \\ 2 & 3 & 2 & 0 & 10 & 2 \\ 10 & 20 & 14 & 10 & 62 & 16 \\ 3 & 5 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{13} 3^1 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -157, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.89 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 14 & 3 \\ 2 & 1 & 3 & 0 & 17 & 3 \\ 2 & 3 & 1 & 0 & 22 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 17 & 22 & 0 & 137 & 25 \\ 3 & 3 & 3 & -2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{11} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -168, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.98 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 1 & 2 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 6 & 2 \\ 8 & 7 & 0 & 6 & 21 & 9 \\ 3 & 3 & -1 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{12} 3^1 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -178, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.92 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 14 & 3 \\ 3 & 5 & 5 & 3 & 30 & 5 \\ 3 & 5 & 0 & 2 & 22 & 2 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 14 & 30 & 22 & 14 & 134 & 24 \\ 3 & 5 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{11} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -196, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.92 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 2 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 8 & 2 \\ 9 & 10 & 0 & 8 & 38 & 12 \\ 3 & 3 & -1 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{12} 3^1 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -198, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.91 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 12 & 3 \\ 3 & 5 & 5 & 3 & 25 & 5 \\ 3 & 5 & 0 & 2 & 18 & 2 \\ 2 & 3 & 2 & 0 & 12 & 2 \\ 12 & 25 & 18 & 12 & 94 & 20 \\ 3 & 5 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[80] $1^{13}3^121^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 3 \\ 2 & 1 & 3 & 0 & 13 & 3 \\ 2 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 13 & 14 & 0 & 65 & 17 \\ 3 & 3 & 3 & -2 & 17 & 3 \end{pmatrix}_{b_0}$$

[81] $1^{13}3^121^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -237, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 12 & 3 \\ 2 & 1 & 3 & 0 & 15 & 3 \\ 2 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 15 & 18 & 0 & 97 & 21 \\ 3 & 3 & 3 & -2 & 21 & 3 \end{pmatrix}_{b_0}$$

[82] $1^{13}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 11 & 3 \\ 2 & 1 & 1 & 2 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 10 & 2 \\ 11 & 13 & 0 & 10 & 61 & 15 \\ 3 & 3 & -1 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[83] $1^{13}3^121^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 15 & 3 \\ 2 & 1 & 1 & 2 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 14 & 2 \\ 15 & 17 & 0 & 14 & 121 & 21 \\ 3 & 3 & -1 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[84] $1^{12}3^121^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 14 & 3 \\ 2 & 1 & 1 & 2 & 14 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 12 & 2 \\ 14 & 14 & 0 & 12 & 90 & 18 \\ 3 & 3 & -1 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[85] $1^{14}3^121^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 21 & 3 \\ 2 & 1 & 1 & 2 & 24 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 20 & 2 \\ 21 & 24 & 0 & 20 & 250 & 30 \\ 3 & 3 & -1 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

[87] $1^{14}3^121^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 12 & 3 \\ 2 & 1 & 3 & 0 & 16 & 3 \\ 2 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 16 & 16 & 0 & 94 & 20 \\ 3 & 3 & 3 & -2 & 20 & 3 \end{pmatrix}_{b_0}$$

[86] $1^{12}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 1 & 2 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 8 & 2 \\ 10 & 10 & 0 & 8 & 38 & 12 \\ 3 & 3 & -1 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[88] $1^{14}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 13 & 3 \\ 2 & 1 & 1 & 2 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 12 & 2 \\ 13 & 16 & 0 & 12 & 90 & 18 \\ 3 & 3 & -1 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[89] $1^{12} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -360, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 20 & 3 \\ 2 & 1 & 1 & 2 & 21 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 18 & 2 \\ 20 & 21 & 0 & 18 & 205 & 27 \\ 3 & 3 & -1 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[91] $1^{12} 3^1 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 26 & 3 \\ 2 & 1 & 1 & 2 & 28 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 24 & 2 \\ 26 & 28 & 0 & 24 & 366 & 36 \\ 3 & 3 & -1 & 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[90] $1^{11} 3^1 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 16 & 3 \\ 2 & 1 & 3 & 0 & 20 & 3 \\ 2 & 3 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 20 & 24 & 0 & 174 & 28 \\ 3 & 3 & 3 & -2 & 28 & 3 \end{pmatrix}_{b_0}$$

[93] $1^{11} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 17 & 3 \\ 2 & 1 & 1 & 2 & 20 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 16 & 2 \\ 17 & 20 & 0 & 16 & 158 & 24 \\ 3 & 3 & -1 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[92] $1^{11} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 14 & 3 \\ 2 & 1 & 3 & 0 & 18 & 3 \\ 2 & 3 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 18 & 20 & 0 & 130 & 24 \\ 3 & 3 & 3 & -2 & 24 & 3 \end{pmatrix}_{b_0}$$

[94] $1^{12} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 12 & 3 \\ 2 & 1 & 1 & 2 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 10 & 2 \\ 12 & 13 & 0 & 10 & 61 & 15 \\ 3 & 3 & -1 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[95] $1^{12} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -472, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 16 & 3 \\ 2 & 1 & 1 & 2 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 14 & 2 \\ 16 & 17 & 0 & 14 & 121 & 21 \\ 3 & 3 & -1 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[96] $1^{12} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -592, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 22 & 3 \\ 2 & 1 & 1 & 2 & 24 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 20 & 2 \\ 22 & 24 & 0 & 20 & 250 & 30 \\ 3 & 3 & -1 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[97] \quad 1^{14}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -58, \quad \det = -640, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 18 & 3 \\ 2 & 1 & 1 & 2 & 20 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 2 & 0 & 0 & 16 & 2 \\ 18 & 20 & 0 & 16 & 158 & 24 \\ 3 & 3 & -1 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [99] \quad 1^{11}3^15 \quad 13^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 0 & 3 \\ 3 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 0 & 3 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[98] \quad 1^{11}3^115 \quad 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 6 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 8 & 3 \\ 2 & 0 & 3 & 1 & 7 & 3 \\ 6 & 0 & 8 & 7 & 22 & 10 \\ 3 & 0 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[100] \quad 1^{11}3^115 \quad 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 7, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.86 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 2 & 1 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 7 & 3 \\ 6 & 8 & 0 & 7 & 22 & 10 \\ 3 & 4 & -1 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[102] \quad 1^{12}3^15 \quad 13^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 10, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 3 & 0 & 3 & 4 & 14 & 4 \\ 9 & 0 & 12 & 14 & 42 & 14 \\ 3 & 0 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[101] \quad 1^{12}3^15 \quad 13^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 10, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 3 & 0 & 3 & 4 & 14 & 4 \\ 11 & 0 & 12 & 14 & 42 & 14 \\ 3 & 0 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[103] \quad 1^{13}3^115 \quad 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 11, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.91 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 1 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 0 & 3 \\ 2 & 0 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 0 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[104] \quad 1^{11}3^115 \quad 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 11, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.91 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 2 & 1 & 3 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 2 & 4 \\ 3 & 4 & -1 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[105] \quad 1^{12}3^15 \quad 13^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 3 & 0 & 3 & 4 & 14 & 4 \\ 10 & 0 & 12 & 14 & 42 & 14 \\ 3 & 0 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{13}3^{11}5131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 19, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 14 & 3 \\ 2 & 2 & 1 & 3 & 20 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 17 & 3 \\ 14 & 20 & 0 & 17 & 130 & 24 \\ 3 & 4 & -1 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[107] \ 1^{11}3^{11}5131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 24, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 1 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 3 & 8 \\ 1 & 1 & 0 & 3 & 0 & 3 \\ 2 & 4 & 0 & 8 & 3 & 6 \\ 2 & 3 & -1 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[108] \ 1^{12}3^{11}5131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 31, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 12.97 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 16 & 3 \\ 2 & 0 & 3 & 1 & 14 & 3 \\ 13 & 0 & 16 & 14 & 94 & 20 \\ 3 & 0 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{13}3^{11}5131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 31, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 12.97 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 2 & 1 & 3 & 16 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 13 & 3 \\ 10 & 16 & 0 & 13 & 70 & 18 \\ 3 & 4 & -1 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{12}3^{11}5131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 34, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 3 \\ 2 & 2 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 10 & 3 \\ 8 & 12 & 0 & 10 & 42 & 14 \\ 3 & 4 & -1 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{11}3^{11}5131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 45, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.89 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 7 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 8 & 3 \\ 2 & 0 & 3 & 1 & 7 & 3 \\ 7 & 0 & 8 & 7 & 22 & 10 \\ 3 & 0 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{11}3^{11}5131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 58, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 8 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 8 & 3 \\ 2 & 0 & 3 & 1 & 7 & 3 \\ 8 & 0 & 8 & 7 & 22 & 10 \\ 3 & 0 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{14}3^{11}5131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 58, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 16 & 3 \\ 2 & 2 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 20 & 3 \\ 16 & 24 & 0 & 20 & 174 & 28 \\ 3 & 4 & -1 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[114] \ 1^{12}3^{11}5131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 91, \ (n_+, n_-, n_0) =$$

$(1, 12, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 2 & 0 & 3 & 1 & 10 & 3 \\ 9 & 0 & 12 & 10 & 42 & 14 \\ 3 & 0 & 3 & 3 & 14 & 4 \end{pmatrix} b_0$$

$$[115] \ 1^{13} 3^1 1^5 131^{13} + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s} \textcolor{blue}{o}_8 + \textcolor{brown}{s} \textcolor{blue}{u}_2 \rightarrow (-3) + \textcolor{red}{s} \textcolor{blue}{u}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 98, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 20 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 24 & 3 \\ 2 & 0 & 3 & 1 & 21 & 3 \\ 20 & 0 & 24 & 21 & 214 & 30 \\ 3 & 0 & 3 & 3 & 30 & 4 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{12}3^11^5131^{12} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}o_8 + \textcolor{brown}{s}u_2 \rightarrow (-3) + \textcolor{red}{s}u_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 102, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 14 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 16 & 3 \\ 2 & 0 & 3 & 1 & 14 & 3 \\ 14 & 0 & 16 & 14 & 94 & 20 \\ 3 & 0 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{12} 3^1 1^5 13^1 1^2 + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s} \textcolor{blue}{o}_8 + \textcolor{red}{s} \textcolor{blue}{u}_2 \rightarrow (-3) + \textcolor{red}{s} \textcolor{blue}{u}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 143, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 15 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 16 & 3 \\ 2 & 0 & 3 & 1 & 14 & 3 \\ 15 & 0 & 16 & 14 & 94 & 20 \\ 3 & 0 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{12}3^11^5131^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{brown}{s}\textcolor{blue}{u}_2 \rightarrow (-3) + \textcolor{red}{s}\textcolor{blue}{u}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 154, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 2 & 0 & 3 & 1 & 10 & 3 \\ 10 & 0 & 12 & 10 & 42 & 14 \\ 3 & 0 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{12}3^11^5131^{11} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{green}{o}_8 + \textcolor{brown}{s}\textcolor{brown}{u}_2 \rightarrow (-3) + \textcolor{red}{s}\textcolor{red}{u}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 187, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 11 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 \\ 2 & 0 & 3 & 1 & 10 & 3 \\ 11 & 0 & 12 & 10 & 42 & 14 \\ 3 & 0 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{13} 3^1 1^5 131^{13} + [\textcolor{red}{\mathfrak{s}}\textcolor{red}{\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}}\textcolor{green}{\mathfrak{o}}_8 + \textcolor{orange}{\mathfrak{s}}\textcolor{orange}{\mathfrak{u}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{s}}\textcolor{red}{\mathfrak{u}}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 197, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 21 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 24 & 3 \\ 2 & 0 & 3 & 1 & 21 & 3 \\ 21 & 0 & 24 & 21 & 214 & 30 \\ 3 & 0 & 3 & 3 & 30 & 4 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{14} 3^1 1^5 131^{14} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}o_8 + \textcolor{brown}{s}u_2 \rightarrow (-3) + \textcolor{red}{s}u_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 203, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 27 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 32 & 3 \\ 2 & 0 & 3 & 1 & 28 & 3 \\ 27 & 0 & 32 & 28 & 382 & 40 \\ 3 & 0 & 3 & 3 & 40 & 4 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{13}3^11^5131^{12} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}o_8 + \textcolor{brown}{s}u_2 \rightarrow (-3) + \textcolor{red}{s}u_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 214, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 16 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 20 & 3 \\ 2 & 0 & 3 & 1 & 17 & 3 \\ 16 & 0 & 20 & 17 & 130 & 24 \\ 3 & 0 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{14}3^{11}5 \ 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 226, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.98 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 22 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 28 & 3 \\ 2 & 0 & 3 & 1 & 24 & 3 \\ 22 & 0 & 28 & 24 & 266 & 34 \\ 3 & 0 & 3 & 3 & 34 & 4 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{13}3^{11}5 \ 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 305, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 17 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 20 & 3 \\ 2 & 0 & 3 & 1 & 17 & 3 \\ 17 & 0 & 20 & 17 & 130 & 24 \\ 3 & 0 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{14}3^{11}5 \ 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 375, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 23 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 28 & 3 \\ 2 & 0 & 3 & 1 & 24 & 3 \\ 23 & 0 & 28 & 24 & 266 & 34 \\ 3 & 0 & 3 & 3 & 34 & 4 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{14}3^{11}5 \ 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 427, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.00 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 19 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 24 & 3 \\ 2 & 0 & 3 & 1 & 20 & 3 \\ 19 & 0 & 24 & 20 & 174 & 28 \\ 3 & 0 & 3 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

4.696 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -98, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 1 & 5 \\ 11 & 20 & 0 & 11 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 8 & -2 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -163, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 13 & 7 \\ 5 & 8 & -2 & 4 & 7 & 2 \end{pmatrix}_{b_0}$$

4.697 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{12}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -8, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 3 & 8 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 2 & 2 \\ 4 & 8 & 0 & 4 & 10 & 4 & 6 \\ 2 & 3 & 0 & 2 & 4 & 0 & 2 \\ 3 & 4 & -2 & 2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{14}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -76, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.89 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 2 & 3 \\ 3 & 4 & 0 & 3 & 22 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 12 & 2 & 2 \\ 12 & 22 & 0 & 12 & 85 & 12 & 17 \\ 2 & 3 & 0 & 2 & 12 & 0 & 2 \\ 3 & 4 & -2 & 2 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{14}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -488, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 4 & 0 & 3 & 26 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 14 & 2 & 2 \\ 14 & 26 & 0 & 14 & 118 & 14 & 20 \\ 2 & 3 & 0 & 2 & 14 & 0 & 2 \\ 3 & 4 & -2 & 2 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [14] \ 1^{12}3^{11}5^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 13, \ T_{\max} = 13, \ \Delta = -1, \ \det = -4, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 10 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 12 & 3 & 3 \\ 3 & 0 & 3 & 4 & 14 & 3 & 4 \\ 10 & 0 & 12 & 14 & 42 & 10 & 14 \\ 2 & 0 & 3 & 3 & 10 & 1 & 3 \\ 3 & 0 & 3 & 4 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

4.698 NHC(-2)(-3) + so₈ + su₂→(-3) + su₂ + su₂n3mix

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -30, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 4 & 3 \\ 3 & 4 & 0 & 3 & 14 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 8 & 4 & 2 \\ 8 & 14 & 0 & 8 & 37 & 17 & 11 \\ 4 & 7 & 0 & 4 & 17 & 7 & 5 \\ 3 & 4 & -2 & 2 & 11 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -84, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 9 & 3 \\ 3 & 4 & 0 & 3 & 8 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 9 & 2 \\ 4 & 8 & 0 & 4 & 10 & 22 & 6 \\ 9 & 16 & 0 & 9 & 22 & 44 & 12 \\ 3 & 4 & -2 & 2 & 6 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -86, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 0 & 3 & 8 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 4 & 2 \\ 4 & 8 & 0 & 4 & 10 & 10 & 6 \\ 4 & 7 & 0 & 4 & 10 & 7 & 5 \\ 3 & 4 & -2 & 2 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -91, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 0 & 3 & 5 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 3 & 4 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 4 \\ 4 & 7 & 0 & 4 & 6 & 7 & 5 \\ 3 & 4 & -2 & 2 & 4 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -194, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 4 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 4 & 2 \\ 2 & 4 & 0 & 2 & 1 & 5 & 3 \\ 4 & 7 & 0 & 4 & 5 & 7 & 5 \\ 3 & 4 & -2 & 2 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -236, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 4 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 6 & 2 \\ 2 & 4 & 0 & 2 & 1 & 8 & 3 \\ 6 & 11 & 0 & 6 & 8 & 20 & 8 \\ 3 & 4 & -2 & 2 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{123}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -238, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 11 & 3 \\ 3 & 4 & 0 & 3 & 8 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 11 & 2 \\ 4 & 8 & 0 & 4 & 10 & 28 & 6 \\ 11 & 20 & 0 & 11 & 28 & 69 & 15 \\ 3 & 4 & -2 & 2 & 6 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -251, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 4 & 3 \\ 3 & 4 & 0 & 3 & 9 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 5 & 4 & 2 \\ 5 & 9 & 0 & 5 & 13 & 11 & 7 \\ 4 & 7 & 0 & 4 & 11 & 7 & 5 \\ 3 & 4 & -2 & 2 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{124}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -264, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 16 & 3 \\ 3 & 4 & 0 & 3 & 8 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 16 & 2 \\ 4 & 8 & 0 & 4 & 10 & 40 & 6 \\ 16 & 29 & 0 & 16 & 40 & 148 & 22 \\ 3 & 4 & -2 & 2 & 6 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -287, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 7 & 2 \\ 2 & 4 & 0 & 2 & 1 & 8 & 3 \\ 7 & 12 & 0 & 7 & 8 & 25 & 9 \\ 3 & 4 & -2 & 2 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{141}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -299, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.92 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 4 & 3 \\ 3 & 4 & 0 & 3 & 21 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 11 & 4 & 2 \\ 11 & 21 & 0 & 11 & 76 & 26 & 16 \\ 4 & 7 & 0 & 4 & 26 & 7 & 5 \\ 3 & 4 & -2 & 2 & 16 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{121}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -314, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 4 & 3 \\ 3 & 4 & 0 & 3 & 18 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 10 & 4 & 2 \\ 10 & 18 & 0 & 10 & 58 & 22 & 14 \\ 4 & 7 & 0 & 4 & 22 & 7 & 5 \\ 3 & 4 & -2 & 2 & 14 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{122}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -336, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 6 & 3 \\ 3 & 4 & 0 & 3 & 13 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 7 & 6 & 2 \\ 7 & 13 & 0 & 7 & 28 & 25 & 10 \\ 6 & 11 & 0 & 6 & 25 & 20 & 8 \\ 3 & 4 & -2 & 2 & 10 & 8 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 4 & 3 \\ 3 & 4 & 0 & 3 & 13 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 7 & 4 & 2 \\ 7 & 13 & 0 & 7 & 28 & 16 & 10 \\ 4 & 7 & 0 & 4 & 16 & 7 & 5 \\ 3 & 4 & -2 & 2 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[14] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 4 & 3 \\ 3 & 4 & 0 & 3 & 17 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 9 & 4 & 2 \\ 9 & 17 & 0 & 9 & 49 & 21 & 13 \\ 4 & 7 & 0 & 4 & 21 & 7 & 5 \\ 3 & 4 & -2 & 2 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 13 & 3 \\ 3 & 4 & 0 & 3 & 8 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 13 & 2 \\ 4 & 8 & 0 & 4 & 10 & 34 & 6 \\ 13 & 24 & 0 & 13 & 34 & 100 & 18 \\ 3 & 4 & -2 & 2 & 6 & 18 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 6 & 3 \\ 3 & 4 & 0 & 3 & 17 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 9 & 6 & 2 \\ 9 & 17 & 0 & 9 & 49 & 33 & 13 \\ 6 & 11 & 0 & 6 & 33 & 20 & 8 \\ 3 & 4 & -2 & 2 & 13 & 8 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{14}23^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 6 & 3 \\ 3 & 4 & 0 & 3 & 21 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 11 & 6 & 2 \\ 11 & 21 & 0 & 11 & 76 & 41 & 16 \\ 6 & 11 & 0 & 6 & 41 & 20 & 8 \\ 3 & 4 & -2 & 2 & 16 & 8 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{11}23^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -477, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 9 & 2 \\ 2 & 4 & 0 & 2 & 1 & 11 & 3 \\ 9 & 16 & 0 & 9 & 11 & 44 & 12 \\ 3 & 4 & -2 & 2 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -526, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 4 & 3 \\ 3 & 4 & 0 & 3 & 22 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 12 & 4 & 2 \\ 12 & 22 & 0 & 12 & 85 & 27 & 17 \\ 4 & 7 & 0 & 4 & 27 & 7 & 5 \\ 3 & 4 & -2 & 2 & 17 & 5 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{14}23^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 6 & 3 \\ 3 & 4 & 0 & 3 & 26 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 14 & 6 & 2 \\ 14 & 26 & 0 & 14 & 118 & 50 & 20 \\ 6 & 11 & 0 & 6 & 50 & 20 & 8 \\ 3 & 4 & -2 & 2 & 20 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11} 1^{12} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 12 & 2 \\ 2 & 4 & 0 & 2 & 1 & 14 & 3 \\ 12 & 21 & 0 & 12 & 14 & 80 & 16 \\ 3 & 4 & -2 & 2 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -631, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 4 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 11 & 2 \\ 2 & 4 & 0 & 2 & 1 & 14 & 3 \\ 11 & 20 & 0 & 11 & 14 & 69 & 15 \\ 3 & 4 & -2 & 2 & 3 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{14} 1^{12} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -666, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 4 & 3 \\ 3 & 4 & 0 & 3 & 26 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 14 & 4 & 2 \\ 14 & 26 & 0 & 14 & 118 & 32 & 20 \\ 4 & 7 & 0 & 4 & 32 & 7 & 5 \\ 3 & 4 & -2 & 2 & 20 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -749, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 4 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 13 & 2 \\ 2 & 4 & 0 & 2 & 1 & 17 & 3 \\ 13 & 24 & 0 & 13 & 17 & 100 & 18 \\ 3 & 4 & -2 & 2 & 3 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{11} 1^{12} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -886, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 4 & 25 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 14 & 2 \\ 2 & 4 & 0 & 2 & 1 & 17 & 3 \\ 14 & 25 & 0 & 14 & 17 & 111 & 19 \\ 3 & 4 & -2 & 2 & 3 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{11} 1^{12} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -1152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 16 & 3 \\ 3 & 4 & 0 & 3 & 4 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 16 & 2 \\ 2 & 4 & 0 & 2 & 1 & 20 & 3 \\ 16 & 29 & 0 & 16 & 20 & 148 & 22 \\ 3 & 4 & -2 & 2 & 3 & 22 & 2 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [29] \quad 1^{11} 1^{12} 3^{11} 15 13^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 5 & 3 \\ 2 & 2 & 1 & 3 & 8 & 7 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 7 & 6 & 3 \\ 6 & 8 & 0 & 7 & 22 & 19 & 10 \\ 5 & 7 & 0 & 6 & 19 & 14 & 8 \\ 3 & 4 & -1 & 3 & 10 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{11} 1^{12} 3^{11} 15 13^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 5 & 3 \\ 2 & 2 & 1 & 3 & 4 & 7 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 2 & 7 & 4 \\ 5 & 7 & 0 & 6 & 7 & 14 & 8 \\ 3 & 4 & -1 & 3 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{13}3^{11}5^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 5 & 3 \\ 2 & 2 & 1 & 3 & 16 & 7 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 13 & 6 & 3 \\ 10 & 16 & 0 & 13 & 70 & 33 & 18 \\ 5 & 7 & 0 & 6 & 33 & 14 & 8 \\ 3 & 4 & -1 & 3 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{11}3^{11}5^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 13 & 3 \\ 2 & 2 & 1 & 3 & 4 & 19 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 3 & 16 & 3 \\ 2 & 4 & 0 & 3 & 2 & 19 & 4 \\ 13 & 19 & 0 & 16 & 19 & 110 & 22 \\ 3 & 4 & -1 & 3 & 4 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{12}3^{11}5^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 8 & 5 & 3 \\ 2 & 2 & 1 & 3 & 12 & 7 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 10 & 6 & 3 \\ 8 & 12 & 0 & 10 & 42 & 26 & 14 \\ 5 & 7 & 0 & 6 & 26 & 14 & 8 \\ 3 & 4 & -1 & 3 & 14 & 8 & 4 \end{pmatrix}_{b_0}$$

4.699 NHC(-2)(-3) + so₈ + su₂→(-3) + su₂→(-3)

$$T = 8 \quad [0] \ 21^{13}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 10 & 16 & 8 & 8 & 6 \\ 16 & 25 & 12 & 12 & 9 \\ 8 & 12 & 4 & 6 & 4 \\ 8 & 12 & 6 & 4 & 4 \\ 6 & 9 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 10 & 16 & 8 & 8 & 6 \\ 16 & 25 & 12 & 12 & 9 \\ 8 & 12 & 4 & 6 & 4 \\ 8 & 12 & 6 & 4 & 4 \\ 6 & 9 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 13^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 6 & 6 & 5 \\ 12 & 25 & 12 & 12 & 9 \\ 6 & 12 & 4 & 6 & 4 \\ 6 & 12 & 6 & 4 & 4 \\ 5 & 9 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{13}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 21^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 13 & 20 & 9 & 9 & 7 \\ 20 & 25 & 12 & 12 & 9 \\ 9 & 12 & 4 & 6 & 4 \\ 9 & 12 & 6 & 4 & 4 \\ 7 & 9 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [5] \ 1^{14}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^1 1^4 1^3 1^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 1 & 3 \\ 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 2 & 0 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [7] \quad 1^1 3^1 1^5 1^3 1^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 3 & 3 \\ 3 & 0 & 3 & 4 & 3 & 4 \\ 2 & 0 & 3 & 3 & 1 & 3 \\ 3 & 0 & 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

4.700 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^1 1^3 3^1 1^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 6 & 6 & 1 & 5 \\ 12 & 25 & 12 & 12 & 0 & 9 \\ 6 & 12 & 4 & 6 & 0 & 4 \\ 6 & 12 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 4 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

4.701 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^1 1^2 3^1 1^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 3 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 4 & 2 \\ 2 & 3 & 0 & 2 & 0 & 4 & 2 \\ 4 & 8 & 0 & 4 & 4 & 10 & 6 \\ 3 & 4 & -2 & 2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 1^4 3^1 1^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 3 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 11 & 2 \\ 2 & 3 & 0 & 2 & 0 & 11 & 2 \\ 11 & 21 & 0 & 11 & 11 & 76 & 16 \\ 3 & 4 & -2 & 2 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 1^3 3^1 1^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 3 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^1 1^1 3^1 1^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 2 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 3 & 12 & 3 \\ 3 & 0 & 3 & 4 & 3 & 14 & 4 \\ 2 & 0 & 3 & 3 & 1 & 10 & 3 \\ 10 & 0 & 12 & 14 & 10 & 42 & 14 \\ 3 & 0 & 3 & 4 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

4.702 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + su2n3mix

$$T = 8 \quad [0] \quad 1^{1^1} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 8 & 13 & 0 & 8 & 36 & 10 \\ 3 & 4 & -2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1^1} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 5 & 8 & 0 & 5 & 12 & 6 \\ 3 & 4 & -2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1^1} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 4 & 7 & 0 & 4 & 7 & 5 \\ 3 & 4 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1^1} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 10 & 19 & 0 & 10 & 64 & 14 \\ 3 & 4 & -2 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1^1} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 6 & 11 & 0 & 6 & 20 & 8 \\ 3 & 4 & -2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1^1} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 8 & 15 & 0 & 8 & 39 & 11 \\ 3 & 4 & -2 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{1^1} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 7 & 12 & 0 & 7 & 25 & 9 \\ 3 & 4 & -2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{1^1} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.02 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 10 & 17 & 0 & 10 & 55 & 13 \\ 3 & 4 & -2 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{13}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 9 & 16 & 0 & 9 & 44 & 12 \\ 3 & 4 & -2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{13}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 11 & 20 & 0 & 11 & 69 & 15 \\ 3 & 4 & -2 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{14}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 13 & 2 \\ 13 & 24 & 0 & 13 & 100 & 18 \\ 3 & 4 & -2 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{12}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 12 & 21 & 0 & 12 & 80 & 16 \\ 3 & 4 & -2 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[12] $1^{13}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 591, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 25 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 14 & 25 & 0 & 14 & 111 & 19 \\ 3 & 4 & -2 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{14}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 3 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 16 & 2 \\ 16 & 29 & 0 & 16 & 148 & 22 \\ 3 & 4 & -2 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

$T = 10$ [14] $1^{11}4^{13}1^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & -1 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

[15] $1^14^{13}1^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 34, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 & 3 \\ 2 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 1 & 0 & 0 & 0 & 2 \\ 3 & 2 & 0 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

- [16] $1^{12}4^{11}13^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 8 & 0 & 0 & 10 & 6 \\ 3 & 4 & -1 & 0 & 6 & 3 \end{pmatrix}_{b_0}$
- [17] $1^{11}4^{11}13^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 59, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 6 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 0 & 13 & 7 \\ 3 & 4 & -1 & 0 & 7 & 3 \end{pmatrix}_{b_0}$
- [18] $1^{11}4^{11}13^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 65, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 2 & 0 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 0 & 0 & 2 & 2 \end{pmatrix}_{b_0}$
- [19] $1^{12}4^{11}13^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 3 \\ 3 & 4 & 0 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 13 & 0 & 0 & 28 & 10 \\ 3 & 4 & -1 & 0 & 10 & 3 \end{pmatrix}_{b_0}$
- [20] $1^{11}4^{11}13^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 2 & 1 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 0 & 0 & 3 & 2 \end{pmatrix}_{b_0}$
- [22] $1^{11}4^{11}13^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 1 & 10 & 3 \\ 2 & 0 & 1 & 0 & 6 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 6 & 1 & 0 & 32 & 10 \\ 3 & 2 & 0 & 0 & 10 & 3 \end{pmatrix}_{b_0}$
- [21] $1^{11}4^{11}13^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 1 & 4 & 3 \\ 2 & 0 & 1 & 0 & 3 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 3 & 1 & 0 & 2 & 4 \\ 3 & 2 & 0 & 0 & 4 & 3 \end{pmatrix}_{b_0}$
- [23] $1^{11}4^{11}13^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 1 & 6 & 3 \\ 2 & 0 & 1 & 0 & 4 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 4 & 1 & 0 & 8 & 6 \\ 3 & 2 & 0 & 0 & 6 & 3 \end{pmatrix}_{b_0}$
- [24] $1^{11}4^{11}13^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 8 & 3 \\ 2 & 0 & 1 & 0 & 5 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 5 & 1 & 0 & 18 & 8 \\ 3 & 2 & 0 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{12}4^{11}13^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 4 & 1 & 2 & 0 & 7 & 5 \\ 2 & 2 & 0 & 0 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{13}4^{11}13^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 0 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 6 & 1 & 3 & 0 & 17 & 7 \\ 2 & 2 & 0 & 0 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{12}4^{11}13^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 0 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 4 & 2 & 2 & 0 & 10 & 6 \\ 2 & 2 & 0 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{13}4^{11}13^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 337, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 0 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 6 & 2 & 3 & 0 & 20 & 8 \\ 2 & 2 & 0 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{14}4^{11}13^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 388, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 0 & 8 & 2 \\ 1 & 1 & -2 & 0 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 0 & 4 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 8 & 2 & 4 & 0 & 34 & 10 \\ 2 & 2 & 0 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [30] \quad 1^{12}3^{11}5^{13}1^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 12 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 15 & 3 \\ 3 & 0 & 3 & 4 & 17 & 4 \\ 12 & 0 & 15 & 17 & 65 & 17 \\ 3 & 0 & 3 & 4 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{11}3^{11}5^{13}1^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 2 & 1 & 3 & 7 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 3 \\ 5 & 7 & 0 & 6 & 14 & 8 \\ 3 & 4 & -1 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{13}3^{11}5^{13}1^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 13 & 3 \\ 2 & 2 & 1 & 3 & 19 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 16 & 3 \\ 13 & 19 & 0 & 16 & 110 & 22 \\ 3 & 4 & -1 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{12}3^{11}5^{13}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -31, \ \det = 162, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 2 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 3 & 15 & 3 \\ 2 & 0 & 3 & 1 & 13 & 3 \\ 13 & 0 & 15 & 13 & 76 & 18 \\ 3 & 0 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

4.703 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + su2n3mix + $\mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -30, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 8 & 3 \\ 3 & 4 & 0 & 3 & 7 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 8 & 2 \\ 4 & 7 & 0 & 4 & 7 & 17 & 5 \\ 8 & 14 & 0 & 8 & 17 & 37 & 11 \\ 3 & 4 & -2 & 2 & 5 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -84, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 4 & 3 \\ 3 & 4 & 0 & 3 & 16 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 9 & 4 & 2 \\ 9 & 16 & 0 & 9 & 44 & 22 & 12 \\ 4 & 8 & 0 & 4 & 22 & 10 & 6 \\ 3 & 4 & -2 & 2 & 12 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -86, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 0 & 3 & 7 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 4 & 2 \\ 4 & 7 & 0 & 4 & 7 & 10 & 5 \\ 4 & 8 & 0 & 4 & 10 & 10 & 6 \\ 3 & 4 & -2 & 2 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -91, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 & 3 \\ 3 & 4 & 0 & 3 & 7 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 3 & 2 \\ 4 & 7 & 0 & 4 & 7 & 6 & 5 \\ 3 & 5 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -2 & 2 & 5 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -194, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 0 & 3 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 2 & 2 \\ 4 & 7 & 0 & 4 & 7 & 5 & 5 \\ 2 & 4 & 0 & 2 & 5 & 1 & 3 \\ 3 & 4 & -2 & 2 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{12}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -236, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 2 & 3 \\ 3 & 4 & 0 & 3 & 11 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 6 & 2 & 2 \\ 6 & 11 & 0 & 6 & 20 & 8 & 8 \\ 2 & 4 & 0 & 2 & 8 & 1 & 3 \\ 3 & 4 & -2 & 2 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{13}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -238, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 4 & 3 \\ 3 & 4 & 0 & 3 & 20 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 11 & 4 & 2 \\ 11 & 20 & 0 & 11 & 69 & 28 & 15 \\ 4 & 8 & 0 & 4 & 28 & 10 & 6 \\ 3 & 4 & -2 & 2 & 15 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 5 & 3 \\ 3 & 4 & 0 & 3 & 7 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 5 & 2 \\ 4 & 7 & 0 & 4 & 7 & 11 & 5 \\ 5 & 9 & 0 & 5 & 11 & 13 & 7 \\ 3 & 4 & -2 & 2 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{142}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 4 & 3 \\ 3 & 4 & 0 & 3 & 29 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 16 & 4 & 2 \\ 16 & 29 & 0 & 16 & 148 & 40 & 22 \\ 4 & 8 & 0 & 4 & 40 & 10 & 6 \\ 3 & 4 & -2 & 2 & 22 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 7 & 2 & 2 \\ 7 & 12 & 0 & 7 & 25 & 8 & 9 \\ 2 & 4 & 0 & 2 & 8 & 1 & 3 \\ 3 & 4 & -2 & 2 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{114}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 11 & 3 \\ 3 & 4 & 0 & 3 & 7 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 11 & 2 \\ 4 & 7 & 0 & 4 & 7 & 26 & 5 \\ 11 & 21 & 0 & 11 & 26 & 76 & 16 \\ 3 & 4 & -2 & 2 & 5 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{112}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 10 & 3 \\ 3 & 4 & 0 & 3 & 7 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 10 & 2 \\ 4 & 7 & 0 & 4 & 7 & 22 & 5 \\ 10 & 18 & 0 & 10 & 22 & 58 & 14 \\ 3 & 4 & -2 & 2 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{122}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 7 & 3 \\ 3 & 4 & 0 & 3 & 11 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 6 & 7 & 2 \\ 6 & 11 & 0 & 6 & 20 & 25 & 8 \\ 7 & 13 & 0 & 7 & 25 & 28 & 10 \\ 3 & 4 & -2 & 2 & 8 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{112}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 7 & 3 \\ 3 & 4 & 0 & 3 & 7 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 7 & 2 \\ 4 & 7 & 0 & 4 & 7 & 16 & 5 \\ 7 & 13 & 0 & 7 & 16 & 28 & 10 \\ 3 & 4 & -2 & 2 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{113}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 9 & 3 \\ 3 & 4 & 0 & 3 & 7 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 9 & 2 \\ 4 & 7 & 0 & 4 & 7 & 21 & 5 \\ 9 & 17 & 0 & 9 & 21 & 49 & 13 \\ 3 & 4 & -2 & 2 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 4 & 3 \\ 3 & 4 & 0 & 3 & 24 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 13 & 4 & 2 \\ 13 & 24 & 0 & 13 & 100 & 34 & 18 \\ 4 & 8 & 0 & 4 & 34 & 10 & 6 \\ 3 & 4 & -2 & 2 & 18 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 9 & 3 \\ 3 & 4 & 0 & 3 & 11 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 6 & 9 & 2 \\ 6 & 11 & 0 & 6 & 20 & 33 & 8 \\ 9 & 17 & 0 & 9 & 33 & 49 & 13 \\ 3 & 4 & -2 & 2 & 8 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 11 & 3 \\ 3 & 4 & 0 & 3 & 11 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 6 & 11 & 2 \\ 6 & 11 & 0 & 6 & 20 & 41 & 8 \\ 11 & 21 & 0 & 11 & 41 & 76 & 16 \\ 3 & 4 & -2 & 2 & 8 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -477, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 9 & 2 & 2 \\ 9 & 16 & 0 & 9 & 44 & 11 & 12 \\ 2 & 4 & 0 & 2 & 11 & 1 & 3 \\ 3 & 4 & -2 & 2 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11} 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -526, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 12 & 3 \\ 3 & 4 & 0 & 3 & 7 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 12 & 2 \\ 4 & 7 & 0 & 4 & 7 & 27 & 5 \\ 12 & 22 & 0 & 12 & 27 & 85 & 17 \\ 3 & 4 & -2 & 2 & 5 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11} 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 14 & 3 \\ 3 & 4 & 0 & 3 & 11 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 6 & 14 & 2 \\ 6 & 11 & 0 & 6 & 20 & 50 & 8 \\ 14 & 26 & 0 & 14 & 50 & 118 & 20 \\ 3 & 4 & -2 & 2 & 8 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11} 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 2 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 12 & 2 & 2 \\ 12 & 21 & 0 & 12 & 80 & 14 & 16 \\ 2 & 4 & 0 & 2 & 14 & 1 & 3 \\ 3 & 4 & -2 & 2 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -631, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 3 & 20 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 11 & 2 & 2 \\ 11 & 20 & 0 & 11 & 69 & 14 & 15 \\ 2 & 3 & 4 & 0 & 2 & 14 & 1 & 3 \\ 3 & 4 & -2 & 2 & 15 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11} 1^{14} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -666, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 14 & 3 \\ 3 & 4 & 0 & 3 & 7 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 14 & 2 \\ 4 & 7 & 0 & 4 & 7 & 32 & 5 \\ 14 & 26 & 0 & 14 & 32 & 118 & 20 \\ 3 & 4 & -2 & 2 & 5 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{14} 1^{13} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -749, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 2 & 3 \\ 3 & 4 & 0 & 3 & 24 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 13 & 2 & 2 \\ 13 & 24 & 0 & 13 & 100 & 17 & 18 \\ 2 & 4 & 0 & 2 & 17 & 1 & 3 \\ 3 & 4 & -2 & 2 & 18 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{13} 1^{14} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -886, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 4 & 0 & 3 & 25 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 14 & 2 & 2 \\ 14 & 25 & 0 & 14 & 111 & 17 & 19 \\ 2 & 4 & 0 & 2 & 17 & 1 & 3 \\ 3 & 4 & -2 & 2 & 19 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{14} 1^{13} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -1152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 2 & 3 \\ 3 & 4 & 0 & 3 & 29 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 16 & 2 & 2 \\ 16 & 29 & 0 & 16 & 148 & 20 & 22 \\ 2 & 4 & 0 & 2 & 20 & 1 & 3 \\ 3 & 4 & -2 & 2 & 22 & 3 & 2 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [27] \quad 1^{11} 1^{13} 3^{11} 1^{15} 13^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 10 & 3 \\ 2 & 2 & 1 & 3 & 7 & 16 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 13 & 3 \\ 5 & 7 & 0 & 6 & 14 & 33 & 8 \\ 10 & 16 & 0 & 13 & 33 & 70 & 18 \\ 3 & 4 & -1 & 3 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{11} 1^{13} 3^{11} 1^{15} 13^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 2 & 3 \\ 2 & 2 & 1 & 3 & 7 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 3 & 3 \\ 5 & 7 & 0 & 6 & 14 & 7 & 8 \\ 2 & 4 & 0 & 3 & 7 & 2 & 4 \\ 3 & 4 & -1 & 3 & 8 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11} 1^{13} 3^{11} 1^{15} 13^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 6 & 3 \\ 2 & 2 & 1 & 3 & 7 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 7 & 3 \\ 5 & 7 & 0 & 6 & 14 & 19 & 8 \\ 6 & 8 & 0 & 7 & 19 & 22 & 10 \\ 3 & 4 & -1 & 3 & 8 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{13} 1^{13} 3^{11} 1^{15} 13^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 13 & 2 & 3 \\ 2 & 2 & 1 & 3 & 19 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 16 & 3 & 3 \\ 13 & 19 & 0 & 16 & 110 & 19 & 22 \\ 2 & 4 & 0 & 3 & 19 & 2 & 4 \\ 3 & 4 & -1 & 3 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{11}2^3 1^1 5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 8 & 3 \\ 2 & 2 & 1 & 3 & 7 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 10 & 3 \\ 5 & 7 & 0 & 6 & 14 & 26 & 8 \\ 8 & 12 & 0 & 10 & 26 & 42 & 14 \\ 3 & 4 & -1 & 3 & 8 & 14 & 4 \end{pmatrix}_{b_0}$$

4.704 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{11}3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12}3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 4 & 8 & 0 & 4 & 9 & 5 \\ 3 & 4 & -2 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 6 & 10 & 5 \\ 11 & 20 & 11 & 17 & 8 \\ 6 & 11 & 4 & 9 & 4 \\ 10 & 17 & 9 & 12 & 6 \\ 5 & 8 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 6 & 1 & 5 \\ 11 & 20 & 11 & 0 & 8 \\ 6 & 11 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{13}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & -2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 4 & -2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 6 & 9 & 5 \\ 11 & 20 & 11 & 17 & 8 \\ 6 & 11 & 4 & 9 & 4 \\ 9 & 17 & 9 & 12 & 6 \\ 5 & 8 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 8 & 12 & 6 \\ 15 & 20 & 11 & 17 & 8 \\ 8 & 11 & 4 & 9 & 4 \\ 12 & 17 & 9 & 12 & 6 \\ 6 & 8 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 21^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 121, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.11 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 9 & 15 & 7 \\ 17 & 20 & 11 & 17 & 8 \\ 9 & 11 & 4 & 9 & 4 \\ 15 & 17 & 9 & 12 & 6 \\ 7 & 8 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 21^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 16 & 8 & 12 & 6 \\ 16 & 20 & 11 & 17 & 8 \\ 8 & 11 & 4 & 9 & 4 \\ 12 & 17 & 9 & 12 & 6 \\ 6 & 8 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 7 & 13 & 0 & 7 & 27 & 9 \\ 3 & 4 & -2 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{13}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 230, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 9 & 17 & 0 & 9 & 48 & 12 \\ 3 & 4 & -2 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [13] \ 1^{15}13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -21, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [14] \ 1^141^{13}1^151^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 161, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 & 1 & 2 \\ 3 & 4 & 1 & 2 & 3 & 0 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 1 & 0 & 0 \\ 2 & 3 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

4.705 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{f}_4$

$$T = 9 \quad [0] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 6 & 9 & 1 & 5 \\ 11 & 20 & 11 & 17 & 0 & 8 \\ 6 & 11 & 4 & 9 & 0 & 4 \\ 9 & 17 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 4 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

4.706 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 12 & 6 \\ 5 & 8 & -2 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

4.707 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 12 \quad [0] \ 1^{11} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 3 & 3 & 0 & 7 & 4 \\ 2 & 3 & 0 & 3 & 0 & 6 & 2 \\ 2 & 3 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 7 & 6 & 5 & 0 & 9 & 7 \\ 3 & 4 & 2 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{13} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 3 & 3 & 0 & 15 & 4 \\ 2 & 3 & 0 & 3 & 0 & 10 & 2 \\ 2 & 3 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 15 & 10 & 9 & 0 & 41 & 13 \\ 3 & 4 & 2 & 3 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 50, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 3 \\ 3 & 4 & 3 & 3 & 0 & 11 & 4 \\ 2 & 3 & 0 & 3 & 0 & 8 & 2 \\ 2 & 3 & 3 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 11 & 8 & 7 & 0 & 22 & 10 \\ 3 & 4 & 2 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

4.708 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \ 1^{11} 3^{11} 1^{12} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.22 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 9 & 1 & 7 \\ 17 & 20 & 11 & 0 & 8 \\ 9 & 11 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1}3^{\overline{1}1}2^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 13, \ \Delta = -95, \ \det = 325, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.22 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 9 & 1 & 7 \\ 20 & 25 & 12 & 0 & 9 \\ 9 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

4.709 NHC(-2)(-3) + so₈ + su_{2n3mix}

$$T = 7 \quad [0] \ 1^{\overline{1}1^{\overline{1}1}3^{\overline{1}1}1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -8, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.50 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 5 & 12 & 17 & 5 \\ 12 & 25 & 32 & 9 \\ 17 & 32 & 37 & 11 \\ 5 & 9 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 21^{\overline{1}1^{\overline{1}1}3^{\overline{1}1}1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -8, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 10 & 16 & 20 & 6 \\ 16 & 20 & 28 & 8 \\ 20 & 28 & 37 & 11 \\ 6 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 13^{\overline{1}1^{\overline{1}2}1^{\overline{1}1}1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -15, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.40 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 24 & 5 \\ 11 & 20 & 45 & 8 \\ 24 & 45 & 100 & 18 \\ 5 & 8 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1}1^{\overline{1}1}3^{\overline{1}1}1^{\overline{1}1}1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -17, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 10 & 16 & 31 & 6 \\ 16 & 25 & 48 & 9 \\ 31 & 48 & 89 & 17 \\ 6 & 9 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 21^{\overline{1}1^{\overline{1}1}3^{\overline{1}1}1^{\overline{1}1}1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -18, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.33 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 10 & 16 & 20 & 6 \\ 16 & 25 & 32 & 9 \\ 20 & 32 & 37 & 11 \\ 6 & 9 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1}1^{\overline{1}1}3^{\overline{1}1}1^{\overline{1}1}1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -18, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.33 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 10 & 16 & 20 & 6 \\ 16 & 25 & 32 & 9 \\ 20 & 32 & 37 & 11 \\ 6 & 9 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1}1^{\overline{1}1}3^{\overline{1}1}1^{\overline{1}2}1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -20, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.80 \rightarrow 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 10 & 15 & 33 & 6 \\ 15 & 20 & 45 & 8 \\ 33 & 45 & 100 & 18 \\ 6 & 8 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1}1^{\overline{1}1}3^{\overline{1}1}1^{\overline{1}2}1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -25, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.16 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 25 & 5 \\ 11 & 20 & 45 & 8 \\ 25 & 45 & 100 & 18 \\ 5 & 8 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 21^{\overline{1}1^{\overline{1}2}3^{\overline{1}1}1^{\overline{1}1}1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -25, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 13 & 17 & 39 & 7 \\ 17 & 20 & 45 & 8 \\ 39 & 45 & 100 & 18 \\ 7 & 8 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1}1^{\overline{1}1}3^{\overline{1}1}1^{\overline{1}1}1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -25, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.16 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 7 & 5 \\ 11 & 20 & 11 & 8 \\ 7 & 11 & 4 & 4 \\ 5 & 8 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1}1^{\overline{1}1}3^{\overline{1}1}1^{\overline{1}4}1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -27, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 17, \ T_{\max} = 24, \ T \in [17, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 23 & 0 & 120 & 18 \\ 3 & 4 & -2 & 18 & -4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, \quad T_{\max} = 24, \quad \Delta = -115, \quad \det = -27, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.37 \rightarrow 12, \quad T_{\max} = 24, \quad T \in [12, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, \quad T_{\max} = 24, \quad \Delta = -115, \quad \det = -35, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.83 \rightarrow 11, \quad T_{\max} = 24, \quad T \in [11, 24]$$

$$\begin{pmatrix} 10 & 15 & 20 & 6 \\ 15 & 20 & 28 & 8 \\ 20 & 28 & 37 & 11 \\ 6 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{\overline{11}} 3^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, \quad T_{\max} = 24, \quad \Delta = -115, \quad \det = -37, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.54 \rightarrow 12, \quad T_{\max} = 24, \quad T \in [12, 24]$$

$$\begin{pmatrix} 5 & 12 & 24 & 5 \\ 12 & 25 & 48 & 9 \\ 24 & 48 & 89 & 17 \\ 5 & 9 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, \quad T_{\max} = 24, \quad \Delta = -115, \quad \det = -39, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.15 \rightarrow 12, \quad T_{\max} = 24, \quad T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 16 & 5 \\ 11 & 20 & 28 & 8 \\ 16 & 28 & 37 & 11 \\ 5 & 8 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, \quad T_{\max} = 24, \quad \Delta = -115, \quad \det = -40, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 24, \quad T \in [11, 24]$$

$$\begin{pmatrix} 10 & 15 & 9 & 6 \\ 15 & 20 & 11 & 8 \\ 9 & 11 & 4 & 4 \\ 6 & 8 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 13^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, \quad T_{\max} = 24, \quad \Delta = -115, \quad \det = -43, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.23 \rightarrow 12, \quad T_{\max} = 24, \quad T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 15 & 5 \\ 11 & 20 & 28 & 8 \\ 15 & 28 & 37 & 11 \\ 5 & 8 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, \quad T_{\max} = 24, \quad \Delta = -115, \quad \det = -44, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, \quad T_{\max} = 24, \quad T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 18 & 0 & 72 & 14 \\ 3 & 4 & -2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, \quad T_{\max} = 24, \quad \Delta = -115, \quad \det = -48, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 24, \quad T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 8 & 0 & 12 & 6 \\ 3 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, \quad T_{\max} = 24, \quad \Delta = -115, \quad \det = -51, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.67 \rightarrow 12, \quad T_{\max} = 24, \quad T \in [12, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 13 & 0 & 36 & 10 \\ 3 & 4 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, \quad T_{\max} = 24, \quad \Delta = -115, \quad \det = -55, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.82 \rightarrow 11, \quad T_{\max} = 24, \quad T \in [11, 24]$$

$$\begin{pmatrix} 10 & 15 & 21 & 6 \\ 15 & 20 & 28 & 8 \\ 21 & 28 & 37 & 11 \\ 6 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 7, \quad T_{\max} = 24, \quad \Delta = -115, \quad \det = -59, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.76 \rightarrow 11, \quad T_{\max} = 24, \quad T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 4 & 7 & 0 & 7 & 5 \\ 3 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[22] $13^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 5 & 12 & 23 & 5 \\ 12 & 25 & 48 & 9 \\ 23 & 48 & 89 & 17 \\ 5 & 9 & 17 & 2 \end{pmatrix}_{b_0}$$

[23] $21^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$

$$\begin{pmatrix} 13 & 17 & 24 & 7 \\ 17 & 20 & 28 & 8 \\ 24 & 28 & 37 & 11 \\ 7 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

[24] $1^1 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 8 & 0 & 12 & 6 \\ 3 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 5 & 12 & 16 & 5 \\ 12 & 25 & 32 & 9 \\ 16 & 32 & 37 & 11 \\ 5 & 9 & 11 & 2 \end{pmatrix}_{b_0}$$

[26] $13^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 5 & 12 & 15 & 5 \\ 12 & 25 & 32 & 9 \\ 15 & 32 & 37 & 11 \\ 5 & 9 & 11 & 2 \end{pmatrix}_{b_0}$$

[28] $1^{12} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 11 & 0 & 20 & 8 \\ 3 & 4 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 7 & 0 & 7 & 5 \\ 3 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[29] $1^1 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 13 & 0 & 36 & 10 \\ 3 & 4 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[30] $21^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -93, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$

$$\begin{pmatrix} 13 & 20 & 36 & 7 \\ 20 & 25 & 48 & 9 \\ 36 & 48 & 89 & 17 \\ 7 & 9 & 17 & 3 \end{pmatrix}_{b_0}$$

[31] $1^{13} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 15 & 0 & 39 & 11 \\ 3 & 4 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

[32] $1^1 3^1 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 18 & 0 & 72 & 14 \\ 3 & 4 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

- [33] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -112$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 12 & 0 & 25 & 9 \\ 3 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$
- [34] $1^{11}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -115$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.34 \rightarrow 13$, $T_{\max} = 24$, $T \in [13, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 23 & 0 & 120 & 18 \\ 3 & 4 & -2 & 18 & 0 \end{pmatrix}_{b_0}$$
- [35] $1^{14}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -131$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 24$, $T \in [12, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 19 & 0 & 64 & 14 \\ 3 & 4 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$
- [36] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -139$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 11 & 0 & 20 & 8 \\ 3 & 4 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$
- [37] $1^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -155$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 17 & 0 & 55 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$
- [38] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -160$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 12 & 0 & 25 & 9 \\ 3 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$
- [39] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -176$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 16 & 0 & 44 & 12 \\ 3 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$
- [40] $1^{11}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -188$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 24$, $T \in [12, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 22 & 0 & 97 & 17 \\ 3 & 4 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$
- [41] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -195$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 15 & 0 & 39 & 11 \\ 3 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$
- [42] $1^{11}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -211$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 24$, $T \in [12, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 27 & 0 & 151 & 21 \\ 3 & 4 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$
- [43] $1^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 7$, $T_{\max} = 24$, $\Delta = -115$, $\det = -227$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 24$, $T \in [11, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 17 & 0 & 55 & 13 \\ 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[44] $21^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -233, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$

$$\begin{pmatrix} 13 & 20 & 24 & 7 \\ 20 & 25 & 32 & 9 \\ 24 & 32 & 37 & 11 \\ 7 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

[45] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -240, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 20 & 0 & 69 & 15 \\ 3 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[46] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -251, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 19 & 0 & 64 & 14 \\ 3 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[47] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 16 & 0 & 44 & 12 \\ 3 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[48] $1^{11}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -259, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 21 & 0 & 80 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[49] $1^{11}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -284, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 22 & 0 & 97 & 17 \\ 3 & 4 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 24 & 0 & 100 & 18 \\ 3 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{11}3^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 27 & 0 & 151 & 21 \\ 3 & 4 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{11}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -332, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 26 & 0 & 128 & 20 \\ 3 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -352, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 20 & 0 & 69 & 15 \\ 3 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[55] $1^{11}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -363, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 17 & 29 & 0 & 148 & 22 \\ 3 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{11} 1^{13} 3^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -636, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 18 & 30 & 0 & 165 & 23 \\ 3 & 4 & -2 & 23 & 2 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{11} 1^{14} 3^{11} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -763, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 39 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 22 & 39 & 0 & 280 & 30 \\ 3 & 4 & -2 & 30 & 2 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{11} 1^{13} 3^{11} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -763, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 35 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 21 & 35 & 0 & 231 & 27 \\ 3 & 4 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{14} 3^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -812, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 34 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 20 & 34 & 0 & 208 & 26 \\ 3 & 4 & -2 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{14} 3^{11} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -979, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 39 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 23 & 39 & 0 & 280 & 30 \\ 3 & 4 & -2 & 30 & 2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [71] \quad 1^{11} 3^{11} 1^{13} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 9 & 3 \\ 4 & 10 & 1 & 20 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 20 & 0 & 37 & 11 \\ 3 & 6 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[72] \quad 21^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 12 & 4 \\ 5 & 5 & 5 & 13 & 5 \\ 3 & 5 & 0 & 9 & 2 \\ 12 & 13 & 9 & 26 & 10 \\ 4 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{13} 3^{11} 1^{13} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 11 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 11 & 12 & 0 & 39 & 11 \\ 2 & 3 & 3 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{12} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 3 & 5 & 0 & 12 & 2 \\ 10 & 18 & 12 & 56 & 14 \\ 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{13} 3^{11} 1^{13} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

- [76] $1^{12} 13^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 2 & 5 & 0 & 12 & 2 \\ 10 & 18 & 12 & 56 & 14 \\ 3 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$
- [77] $1^{11} 3^1 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 1 & 4 & 3 \\ 4 & 10 & 1 & 8 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 8 & 0 & 4 & 4 \\ 3 & 6 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$
- [78] $1^{12} 31^1 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 8 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 7 & 0 & 15 & 7 \\ 2 & 4 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$
- [79] $1^{11} 3^1 1^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 6 & 5 & 0 & 7 & 5 \\ 2 & 4 & 2 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$
- [80] $1^{111} 3^1 1^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 8 & 3 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 8 & 0 & 25 & 9 \\ 2 & 3 & 2 & -2 & 9 & 0 \end{pmatrix}_{b_0}$$
- [81] $1^{111} 313^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 1 & 8 & 3 \\ 4 & 10 & 1 & 16 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 16 & 0 & 20 & 8 \\ 3 & 6 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$
- [82] $1^{112} 313^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 21$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 7 & 3 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 7 & 0 & 15 & 7 \\ 2 & 3 & 2 & -2 & 7 & 0 \end{pmatrix}_{b_0}$$
- [83] $1^{11} 3^1 1^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 21$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 1 & 8 & 3 \\ 4 & 10 & 1 & 20 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 20 & 0 & 37 & 11 \\ 3 & 6 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$
- [84] $1^{11} 3^1 1^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 22$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 4 & 3 \\ 4 & 10 & 13 & 8 & 6 \\ 5 & 13 & 12 & 9 & 6 \\ 4 & 8 & 9 & 4 & 4 \\ 3 & 6 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$
- [85] $1^{111} 31^3 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 26$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 1 & 23 & 3 \\ 4 & 10 & 1 & 52 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 52 & 0 & 257 & 29 \\ 3 & 6 & -1 & 29 & 1 \end{pmatrix}_{b_0}$$
- [88] $1^1 3^1 1^1 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 28$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 2 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [89] $1^1 3^1 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 28$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 5 & 0 & 18 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 18 & 0 & 48 & 14 \\ 3 & 5 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 6 & 0 & 20 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{13}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 5 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 4 & 0 & 4 & 4 \\ 2 & 4 & 2 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{\overline{13}} 3^{\overline{11}} 1^{\overline{13}} 1^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 7 & 2 \\ 1 & 1 & 4 & 0 & 11 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 11 & 12 & 0 & 39 & 11 \\ 2 & 3 & 3 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{13}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 5 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 5 & 6 & 0 & 7 & 5 \\ 2 & 3 & 3 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{\overline{12}} 3^{\overline{11}} 1^{\overline{13}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 7 & 3 \\ 2 & 4 & 3 & 0 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 7 & 8 & 0 & 15 & 7 \\ 2 & 3 & 3 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[97] \quad 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 2 & 0 & 0 & 2 \\ 3 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[96] \quad 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 4 & 3 \\ 1 & 4 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & -2 & 2 & 0 \\ 2 & 4 & 4 & 2 & 4 & 4 \\ 2 & 3 & 2 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{13}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 8 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 4 & 0 & 9 & 5 \\ 2 & 4 & 2 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 4 & 3 \\ 1 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 2 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 2 & 4 & 0 & 4 & 4 \\ 2 & 2 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[98] \quad 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$\begin{aligned}
& T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 2 & 0 & 3 & 3 \\ 2 & 2 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0} \\
[99] & 21^{11}3^{12}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 5 & 5 & 18 & 5 \\ 3 & 5 & 0 & 13 & 2 \\ 15 & 18 & 13 & 48 & 14 \\ 4 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[100] & 1^{11}3^{13}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.46 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 4 & 3 \\ 2 & 4 & 3 & 0 & 5 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 2 & 3 & 3 & -2 & 4 & 1 \end{pmatrix}_{b_0} \\
[101] & 1^{14}3^{11}1^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.46 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 7 & 2 \\ 1 & 1 & 4 & 0 & 14 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 14 & 15 & 0 & 64 & 14 \\ 2 & 3 & 3 & -2 & 14 & 1 \end{pmatrix}_{b_0} \\
[102] & 1^{11}3^{11}1^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 4 & 7 & 8 & 6 & 4 \\ 7 & 10 & 13 & 8 & 6 \\ 8 & 13 & 12 & 9 & 6 \\ 6 & 8 & 9 & 4 & 4 \\ 4 & 6 & 6 & 4 & 2 \end{pmatrix}_{b_0} \\
[103] & 1^{13}3^{12}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 5 & 5 & 13 & 5 \\ 2 & 5 & 0 & 10 & 2 \\ 9 & 13 & 10 & 24 & 10 \\ 3 & 5 & 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[104] & 1^{11}3^{12}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 5 & 0 & 7 & 5 \\ 3 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0} \\
[105] & 1^{13}3^{12}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 5 & 0 & 8 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 4 & 8 & 0 & 8 & 6 \\ 3 & 5 & -2 & 6 & 3 \end{pmatrix}_{b_0} \\
[107] & 1^{11}3^{12}21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 9 & 0 & 35 & 11 \\ 3 & 3 & -1 & 11 & 3 \end{pmatrix}_{b_0} \\
[106] & 1^{13}3^{12}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 5 & 0 & 13 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 13 & 0 & 24 & 10 \\ 3 & 5 & -2 & 10 & 3 \end{pmatrix}_{b_0} \\
[109] & 21^{12}3^{12}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 3 & 21 & 4 \\ 5 & 5 & 5 & 23 & 5 \\ 3 & 5 & 0 & 16 & 2 \\ 21 & 23 & 16 & 84 & 18 \\ 4 & 5 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[108] & 1^{11}3^{12}21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 5 & 5 & 13 & 5 \\ 2 & 5 & 0 & 9 & 2 \\ 8 & 13 & 9 & 26 & 10 \\ 3 & 5 & 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[110] & 1^{11}3^{11}131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 5 \\ 1 & 4 & 0 & 0 & 5 \\ 1 & 0 & 0 & -4 & 0 \\ 2 & 5 & 5 & 0 & 7 \\ 2 & 3 & 2 & -2 & 5 \end{pmatrix}_{b_0} \\
[111] & 1^{11}3^{11}13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 5 & 7 & 3 \\ 4 & 10 & 13 & 16 & 6 \\ 5 & 13 & 12 & 17 & 6 \\ 7 & 16 & 17 & 20 & 8 \\ 3 & 6 & 6 & 8 & 1 \end{pmatrix}_{b_0} \\
[112] & 1^{11}3^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 1 & 21 & 3 \\ 4 & 10 & 1 & 52 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 52 & 0 & 257 & 29 \\ 3 & 6 & -1 & 29 & 2 \end{pmatrix}_{b_0} \\
[113] & 1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 5 & 5 & 8 & 5 \\ 2 & 5 & 0 & 6 & 2 \\ 5 & 8 & 6 & 8 & 6 \\ 3 & 5 & 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[114] & 1^{11}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 23 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 23 & 0 & 243 & 29 \\ 3 & 3 & -1 & 29 & 2 \end{pmatrix}_{b_0} \\
[115] & 1^{12}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 6 & 0 & 20 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0} \\
[118] & 1^{12}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 8 & 0 & 20 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0} \\
[117] & 1^{12}3^{11}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 16 & 0 & 116 & 20 \\ 3 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0} \\
[116] & 1^{11}3^{11}131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 5 & 0 & 7 & 5 \\ 2 & 4 & 2 & -2 & 5 & 1 \end{pmatrix}_{b_0} \\
[119] & 1^{11}3^{11}13^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.44 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 1 & 13 & 3 \\ 4 & 10 & 1 & 28 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 28 & 0 & 69 & 15 \\ 3 & 6 & -1 & 15 & 2 \end{pmatrix}_{b_0} \\
[125] & 1^{11}3^{11}23^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 1 & 18 & 3 \\ 4 & 10 & 1 & 40 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 40 & 0 & 148 & 22 \\ 3 & 6 & -1 & 22 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [124] $1^{14}3^{11}1^{11}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 17 & 0 & 100 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$
- [123] $1^{14}3^{11}1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 21 & 17 & 0 & 100 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$
- [122] $1^{112}31^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.21 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 10 & 3 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 10 & 10 & 0 & 37 & 11 \\ 2 & 3 & 2 & -2 & 11 & 0 \end{pmatrix}_{b_0}$
- [121] $1^{113}3^{121}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 2 & 5 & 0 & 13 & 2 \\ 12 & 18 & 13 & 48 & 14 \\ 3 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [120] $1^{111}313^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 1 & 6 & 3 \\ 4 & 10 & 1 & 16 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 16 & 0 & 20 & 8 \\ 3 & 6 & -1 & 8 & 2 \end{pmatrix}_{b_0}$
- [126] $13^{121}1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 5 & 5 & 8 & 5 \\ 2 & 5 & 0 & 6 & 2 \\ 4 & 8 & 6 & 8 & 6 \\ 3 & 5 & 2 & 6 & 2 \end{pmatrix}_{b_0}$
- [128] $1^{111}3^{121}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 57$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 13 & 0 & 87 & 17 \\ 3 & 3 & -1 & 17 & 2 \end{pmatrix}_{b_0}$
- [129] $1^{111}31^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 57$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 4 & 0 & 4 & 3 \\ 2 & 4 & 3 & 0 & 5 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 5 & 0 & 4 & 4 \\ 2 & 3 & 3 & -2 & 4 & 1 \end{pmatrix}_{b_0}$
- [127] $1^{113}31^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 57$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 0 & 13 & 3 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 13 & 0 & 60 & 14 \\ 2 & 3 & 2 & -2 & 14 & 0 \end{pmatrix}_{b_0}$
- [130] $1^{14}3^{11}1^{13}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 57$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 8 & 2 \\ 1 & 1 & 4 & 0 & 14 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 15 & 0 & 64 & 14 \\ 2 & 3 & 3 & -2 & 14 & 1 \end{pmatrix}_{b_0}$
- [131] $1^{111}31^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 60$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 5 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 5 & 4 & 0 & 4 & 4 \\ 2 & 4 & 2 & -2 & 4 & 1 \end{pmatrix}_{b_0}$
- [133] $1^{123}1^{13}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 61$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$

$$\begin{aligned}
& T_{\min} = 11.28 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 8 & 3 \\ 2 & 4 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 9 & 0 & 20 & 8 \\ 2 & 3 & 3 & -2 & 8 & 1 \end{pmatrix}_{b_0} \\
[132] & 1^{13} 3^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.28 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 10 & 3 \\ 2 & 4 & 3 & 0 & 11 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 10 & 11 & 0 & 32 & 10 \\ 2 & 3 & 3 & -2 & 10 & 1 \end{pmatrix}_{b_0} \\
[134] & 1^{13} 3^{12} 2^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 11 & 0 & 39 & 11 \\ 3 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0} \\
[135] & 1^{13} 3^{12} 2^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 5 & 5 & 8 & 5 \\ 3 & 5 & 0 & 6 & 2 \\ 4 & 8 & 6 & 8 & 6 \\ 3 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0} \\
[136] & 1^{11} 3^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 4 & 7 & 8 & 11 & 4 \\ 7 & 10 & 13 & 16 & 6 \\ 8 & 13 & 12 & 17 & 6 \\ 11 & 16 & 17 & 20 & 8 \\ 4 & 6 & 6 & 8 & 2 \end{pmatrix}_{b_0} \\
[137] & 1^{12} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 8 & 3 \\ 2 & 4 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 9 & 0 & 20 & 8 \\ 2 & 3 & 3 & -2 & 8 & 1 \end{pmatrix}_{b_0} \\
[138] & 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 5 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 6 & 0 & 7 & 5 \\ 2 & 3 & 3 & -2 & 5 & 1 \end{pmatrix}_{b_0} \\
[139] & 1^{12} 3^{12} 2^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 24 & 3 \\ 2 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 24 & 0 & 308 & 32 \\ 3 & 3 & -1 & 32 & 2 \end{pmatrix}_{b_0} \\
[142] & 1^{11} 3^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.48 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 1 & 7 & 3 \\ 4 & 10 & 1 & 16 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 16 & 0 & 20 & 8 \\ 3 & 6 & -1 & 8 & 2 \end{pmatrix}_{b_0} \\
[141] & 1^{11} 3^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.48 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 1 & 22 & 3 \\ 4 & 10 & 1 & 52 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 52 & 0 & 257 & 29 \\ 3 & 6 & -1 & 29 & 2 \end{pmatrix}_{b_0} \\
[140] & 1^{11} 3^{12} 2^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 5 & 5 & 13 & 5 \\ 2 & 5 & 0 & 9 & 2 \\ 7 & 13 & 9 & 26 & 10 \\ 3 & 5 & 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[143] & 1^{11} 3^{12} 2^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 5 & 0 & 13 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 13 & 0 & 26 & 10 \\ 3 & 5 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[144] $1^{11}3^{11}21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 5 & 0 & 7 & 5 \\ 3 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[145] $13^11^{11}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 6 & 3 \\ 4 & 10 & 13 & 16 & 6 \\ 5 & 13 & 12 & 17 & 6 \\ 6 & 16 & 17 & 20 & 8 \\ 3 & 6 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

[146] $1^{13}3^{11}1^{13}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 11 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 12 & 0 & 39 & 11 \\ 2 & 3 & 3 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{12}3^{11}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 7 & 3 \\ 2 & 4 & 3 & 0 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 8 & 0 & 15 & 7 \\ 2 & 3 & 3 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 5 & 5 & 13 & 5 \\ 3 & 5 & 0 & 9 & 2 \\ 7 & 13 & 9 & 26 & 10 \\ 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[149] $1^{11}3^{11}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.47 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 3 & 5 & 0 & 14 & 2 \\ 8 & 18 & 14 & 48 & 14 \\ 3 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[150] $1^{12}3^{11}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 19 & 3 \\ 2 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 20 & 0 & 200 & 26 \\ 3 & 3 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

[151] $1^{11}2^{13}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 7 & 3 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 7 & 7 & 0 & 15 & 7 \\ 2 & 3 & 2 & -2 & 7 & 0 \end{pmatrix}_{b_0}$$

[153] $1^{11}3^{11}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 4 & 7 & 8 & 10 & 4 \\ 7 & 10 & 13 & 16 & 6 \\ 8 & 13 & 12 & 17 & 6 \\ 10 & 16 & 17 & 20 & 8 \\ 4 & 6 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

[152] $1^{11}3^{11}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 5 & 5 & 13 & 5 \\ 3 & 5 & 0 & 10 & 2 \\ 6 & 13 & 10 & 24 & 10 \\ 3 & 5 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[154] $1^{11}3^{11}2^{13}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 0 & 16 & 3 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 16 & 0 & 100 & 18 \\ 2 & 3 & 2 & -2 & 18 & 0 \end{pmatrix}_{b_0}$$

[155] $1^{11}3^{12}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 1 & 16 & 3 \\ 4 & 10 & 1 & 40 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 40 & 0 & 148 & 22 \\ 3 & 6 & -1 & 22 & 2 \end{pmatrix}_{b_0}$

[156] $1^{11}3^{12}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 6 \\ 2 & 2 & 2 & -2 & 6 & 2 \end{pmatrix}_{b_0}$

[157] $1^{12}3^{12}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 5 & 0 & 18 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 18 & 0 & 56 & 14 \\ 3 & 5 & -2 & 14 & 2 \end{pmatrix}_{b_0}$

[158] $1^{13}3^{12}21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 25 & 3 \\ 2 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 27 & 0 & 359 & 35 \\ 3 & 3 & -1 & 35 & 2 \end{pmatrix}_{b_0}$

[159] $1^{11}3^{13}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 1 & 11 & 3 \\ 4 & 10 & 1 & 28 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 28 & 0 & 69 & 15 \\ 3 & 6 & -1 & 15 & 2 \end{pmatrix}_{b_0}$

[160] $1^{12}3^{12}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 12 & 0 & 56 & 14 \\ 3 & 3 & -1 & 14 & 3 \end{pmatrix}_{b_0}$

[161] $1^{11}3^{12}21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 6 & 11 & 0 & 42 & 12 \\ 2 & 2 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$

[162] $1^{13}3^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 8 & 4 \\ 1 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 4 & 0 & 9 & 5 \\ 2 & 4 & 2 & -2 & 5 & 1 \end{pmatrix}_{b_0}$

[163] $1^{11}3^{12}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 5 & 0 & 7 & 5 \\ 3 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$

[166] $1^{11}3^{12}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 1 & 17 & 3 \\ 4 & 10 & 1 & 40 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 40 & 0 & 148 & 22 \\ 3 & 6 & -1 & 22 & 2 \end{pmatrix}_{b_0}$

[164] $1^{11}3^{13}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 1 & 12 & 3 \\ 4 & 10 & 1 & 28 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 28 & 0 & 69 & 15 \\ 3 & 6 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[165] \quad 1^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\ T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 9 & 0 & 35 & 11 \\ 3 & 3 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[169] \quad 21^{123} 3^{11} 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\ T_{\min} = 11.19 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 24 & 4 \\ 5 & 5 & 5 & 28 & 5 \\ 3 & 5 & 0 & 20 & 2 \\ 24 & 28 & 20 & 120 & 22 \\ 4 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[170] \quad 1^{14} 3^{11} 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\ T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 2 & 5 & 0 & 14 & 2 \\ 12 & 18 & 14 & 48 & 14 \\ 3 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[167] \quad 1^{11} 3^{11} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\ T_{\min} = 10.96 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 8 & 0 & 23 & 9 \\ 2 & 2 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[168] \quad 1^{12} 3^{11} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\ T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 5 & 5 & 13 & 5 \\ 2 & 5 & 0 & 10 & 2 \\ 8 & 13 & 10 & 24 & 10 \\ 3 & 5 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[171] \quad 1^{12} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\ T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 8 & 0 & 20 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[172] \quad 1^{113} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\ T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 10 & 3 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 10 & 10 & 0 & 32 & 10 \\ 2 & 3 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[173] \quad 1^{12} 3^{11} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\ T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 0 & 16 & 2 \\ 14 & 23 & 16 & 84 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[174] \quad 1^{11} 3^{11} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\ T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 5 & 5 & 13 & 5 \\ 2 & 5 & 0 & 10 & 2 \\ 7 & 13 & 10 & 24 & 10 \\ 3 & 5 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[180] \quad 1^{12} 3^{11} 1^{11} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\ T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 15 & 11 & 0 & 44 & 12 \\ 2 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[179] \quad 1^{13} 3^{11} 1^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\ T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 9 & 5 & 0 & 12 & 6 \\ 2 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{13}3^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 9 & 5 & 0 & 12 & 6 \\ 2 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[175] $1^{13}31^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 17 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 17 & 13 & 0 & 60 & 14 \\ 2 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[176] $1^{13}3^{12}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 5 & 0 & 18 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 18 & 0 & 48 & 14 \\ 3 & 5 & -2 & 14 & 3 \end{pmatrix}_{b_0}$$

[177] $1^{13}3^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 15 & 11 & 0 & 44 & 12 \\ 2 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[181] $1^{13}3^{12}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 19 & 0 & 151 & 23 \\ 3 & 3 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

[182] $1^{12}3^{12}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 6 & 6 & 0 & 20 & 8 \\ 2 & 2 & 2 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[183] $1^{13}3^{12}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 1 & 10 & 3 \\ 1 & 4 & 0 & 1 & 10 & 2 \\ 1 & 1 & 1 & -2 & 4 & 0 \\ 5 & 10 & 10 & 4 & 32 & 10 \\ 2 & 3 & 2 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[184] $1^{11}3^{12}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 13 & 0 & 87 & 17 \\ 3 & 3 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[185] $1^{13}3^{12}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 15 & 0 & 83 & 17 \\ 3 & 3 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[186] $1^{11}3^{13}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 0 & 10 & 3 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 10 & 10 & 0 & 32 & 10 \\ 2 & 3 & 2 & -2 & 10 & 0 \end{pmatrix}_{b_0}$$

[187] $1^{12}1^{13}3^{12}21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 141$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 2 & 5 & 0 & 13 & 2 \\ 11 & 18 & 13 & 48 & 14 \\ 3 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0}$

[188] $1^{13}3^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.49 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 14 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 14 & 7 & 0 & 31 & 9 \\ 2 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$

[190] $1^{11}3^{12}21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 149$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 5 & 0 & 23 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 23 & 0 & 78 & 18 \\ 3 & 5 & -2 & 18 & 3 \end{pmatrix}_{b_0}$

[189] $1^{11}3^{12}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 149$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 11 & 0 & 39 & 11 \\ 3 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0}$

[191] $1^{12}1^{13}3^{12}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 151$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.03 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 9 & 0 & 35 & 11 \\ 3 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0}$

[192] $1^{11}3^{12}21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 155$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 3 & 5 & 0 & 13 & 2 \\ 9 & 18 & 13 & 48 & 14 \\ 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[194] $1^{11}3^{12}21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 156$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.05 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 22 & 3 \\ 2 & 1 & 1 & 26 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 26 & 0 & 292 & 32 \\ 3 & 3 & -1 & 32 & 2 \end{pmatrix}_{b_0}$

[195] $1^{13}3^{11}4^{14}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 156$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 27 & 14 & 0 & 120 & 18 \\ 2 & 4 & 2 & -2 & 18 & 0 \end{pmatrix}_{b_0}$

[193] $1^{13}3^{11}2^{12}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 156$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 15 & 8 & 0 & 36 & 10 \\ 2 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$

[196] $1^{12}3^{12}21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 159$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 3 & 5 & 0 & 16 & 2 \\ 12 & 23 & 16 & 84 & 18 \\ 3 & 5 & 2 & 18 & 3 \end{pmatrix}_{b_0}$

[197] $1^{11}3^{11}1^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 160$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.20 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 12 & 8 & 0 & 25 & 9 \\ 2 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$

- [198] $1^{11}3^{11}1^{11}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 160$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.20 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 8 & 0 & 25 & 9 \\ 2 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$
- [199] $1^{14}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 160$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.20 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 20 & 4 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 20 & 16 & 0 & 89 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$
- [200] $1^{12}3^{11}21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 160$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 16 & 0 & 116 & 20 \\ 3 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$
- [201] $1^{11}3^{11}21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 160$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 5 & 0 & 28 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 28 & 0 & 116 & 22 \\ 3 & 5 & -2 & 22 & 3 \end{pmatrix}_{b_0}$$
- [203] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 161$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 13 & 3 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 13 & 13 & 0 & 60 & 14 \\ 2 & 3 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$
- [202] $13^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 161$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 2 & 5 & 0 & 13 & 2 \\ 9 & 18 & 13 & 48 & 14 \\ 3 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$
- [204] $1^{11}3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 164$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 7 & 0 & 34 & 10 \\ 2 & 2 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$
- [205] $1^{14}3^{11}21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 164$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 2 & 5 & 0 & 20 & 2 \\ 18 & 28 & 20 & 120 & 22 \\ 3 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$
- [206] $1^{12}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 164$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.34 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 8 & 0 & 20 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$
- [207] $1^{11}3^{11}21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 165$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.49 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 0 & 17 & 2 \\ 15 & 23 & 17 & 78 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$
- [208] $1^1 3^{11} 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 168$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 21 & 11 & 0 & 72 & 14 \\ 2 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[209] $1^{11}3^{12}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 0 & 16 & 2 \\ 13 & 23 & 16 & 84 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[210] $1^{11}3^{12}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 12 & 0 & 56 & 14 \\ 3 & 3 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

[211] $1^{14}3^{12}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 18 & 0 & 116 & 20 \\ 3 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[212] $1^{12}3^{12}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 5 & 0 & 23 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 23 & 0 & 84 & 18 \\ 3 & 5 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[213] $1^{12}3^{12}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 0 & 16 & 2 \\ 12 & 23 & 16 & 84 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[214] $1^{11}3^{12}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 2 & 5 & 0 & 13 & 2 \\ 10 & 18 & 13 & 48 & 14 \\ 3 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[215] $1^{11}3^{12}21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 2 & 5 & 0 & 14 & 2 \\ 9 & 18 & 14 & 48 & 14 \\ 3 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[216] $1^{13}3^{12}21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 2 & 5 & 0 & 14 & 2 \\ 11 & 18 & 14 & 48 & 14 \\ 3 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[217] $1^{11}3^{12}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 14 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 14 & 7 & 0 & 31 & 9 \\ 2 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[218] $1^{11}3^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 11 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 11 & 7 & 0 & 20 & 8 \\ 2 & 4 & 2 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

[219] $1^{11}3^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 11 & 4 \\ 1 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 11 & 7 & 0 & 20 & 8 \\ 2 & 4 & 2 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

[220] $1^{14}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 14 & 0 & 64 & 14 \\ 3 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[221] $1^{14}3^{11}1131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 21 & 17 & 0 & 100 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

[222] $1^{14}3^{11}1131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 21 & 17 & 0 & 100 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

[227] $1^{13}3^{11}1131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 18 & 14 & 0 & 69 & 15 \\ 2 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[223] $1^{13}3^{11}1131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 18 & 14 & 0 & 69 & 15 \\ 2 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[224] $1^{12}3^{11}113^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 14 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 14 & 10 & 0 & 37 & 11 \\ 2 & 4 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

[225] $1^{14}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 8 & 0 & 52 & 12 \\ 2 & 2 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[226] $1^{12}3^{11}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 11 \\ 2 & 2 & 2 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[228] $1^{11}4^313^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 197, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 0 & 13 & 3 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 13 & 13 & 0 & 55 & 13 \\ 2 & 3 & 2 & -2 & 13 & 0 \end{pmatrix}_{b_0}$$

[229] $1^{12}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 20 & 0 & 200 & 26 \\ 3 & 3 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

[230] $1^{14}3^{11}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 22 & 0 & 192 & 26 \\ 3 & 3 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[231] \ 1^{12} 3^{12} 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 2 & 5 & 0 & 14 & 2 \\ 10 & 18 & 14 & 48 & 14 \\ 3 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[232] \ 1^{11} 3^{12} 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 21 & 3 \\ 2 & 1 & 1 & 23 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 23 & 0 & 243 & 29 \\ 3 & 3 & -1 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[233] \ 1^{114} 3^{11} 3^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 0 & 16 & 3 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 16 & 0 & 89 & 17 \\ 2 & 3 & 2 & -2 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[234] \ 1^{11} 3^{12} 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 3 & 5 & 0 & 17 & 2 \\ 11 & 23 & 17 & 78 & 18 \\ 3 & 5 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[235] \ 1^{14} 3^{11} 1^{12} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 27 & 20 & 0 & 148 & 22 \\ 2 & 4 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[236] \ 1^{13} 3^{112} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 15 & 8 & 0 & 36 & 10 \\ 2 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[237] \ 1^{114} 3^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 0 & 13 & 3 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 13 & 13 & 0 & 55 & 13 \\ 2 & 3 & 2 & -2 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[240] \ 1^{12} 3^{12} 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 15 & 0 & 83 & 17 \\ 2 & 2 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[238] \ 1^{11} 3^{11} 1^{11} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 8 & 0 & 25 & 9 \\ 2 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[239] \ 1^{11} 3^{12} 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 12 & 0 & 56 & 14 \\ 2 & 2 & 2 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[241] $1^{12}3^{12}21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 227$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.34 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 11 & 0 & 39 & 11 \\ 3 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[242] $1^{11}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 241$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 15 & 0 & 83 & 17 \\ 3 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[243] $1^{11}3^{11}1^{12}31^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 244$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 18 & 11 & 0 & 55 & 13 \\ 2 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[244] $1^{11}4^{11}31^{13}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 245$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.05 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 4 & 0 & 19 & 3 \\ 1 & 4 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 19 & 19 & 0 & 135 & 21 \\ 2 & 3 & 2 & -2 & 21 & 0 \end{pmatrix}_{b_0}$$

[245] $1^{11}3^{11}21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 245$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 0 & 17 & 2 \\ 11 & 23 & 17 & 78 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[246] $1^{13}3^{11}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 248$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 17 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 17 & 13 & 0 & 60 & 14 \\ 2 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[247] $1^{14}3^{11}21^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 248$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 27 & 3 \\ 2 & 1 & 1 & 30 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 30 & 0 & 416 & 38 \\ 3 & 3 & -1 & 38 & 3 \end{pmatrix}_{b_0}$$

[248] $1^{12}3^{11}1^{11}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 248$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 15 & 11 & 0 & 44 & 12 \\ 2 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[249] $1^{12}3^{11}1^{11}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 248$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 15 & 11 & 0 & 44 & 12 \\ 2 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[250] $1^{12}3^{11}21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 248$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 12 & 0 & 56 & 14 \\ 3 & 3 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

[251] $1^{12}3^{11}21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 248$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 5 & 0 & 28 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 28 & 0 & 120 & 22 \\ 3 & 5 & -2 & 22 & 3 \end{pmatrix}_{b_0}$$

- [252] $1^{12}3^{11}1^{12}3^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 252$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 14 & 0 & 80 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$
- [253] $1^{13}3^{11}1^{12}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 252$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 15 & 4 \\ 1 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 15 & 8 & 0 & 36 & 10 \\ 2 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$
- [254] $1^{13}3^{11}1^{11}3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 256$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 18 & 14 & 0 & 69 & 15 \\ 2 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$
- [255] $1^{12}3^{11}1^{13}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 256$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 14 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 14 & 10 & 0 & 37 & 11 \\ 2 & 4 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$
- [256] $1^{13}3^{11}2^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 259$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.99 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 26 & 3 \\ 2 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 27 & 0 & 359 & 35 \\ 3 & 3 & -1 & 35 & 3 \end{pmatrix}_{b_0}$$
- [257] $1^{11}3^{11}2^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 259$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.05 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 3 & 5 & 0 & 21 & 2 \\ 13 & 28 & 21 & 116 & 22 \\ 3 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$
- [258] $1^{12}3^{11}2^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 260$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.91 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 16 & 0 & 116 & 20 \\ 3 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$
- [259] $1^{11}3^{11}2^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 260$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 0 & 17 & 2 \\ 14 & 23 & 17 & 78 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$
- [260] $1^{13}3^{11}2^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 261$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 19 & 0 & 151 & 23 \\ 3 & 3 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$
- [261] $1^{13}3^{11}2^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 272$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 10 & 0 & 51 & 13 \\ 2 & 2 & 2 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$
- [262] $1^{13}3^{11}2^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -88$, $\det = 276$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 0 & 18 & 2 \\ 14 & 23 & 18 & 80 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[263] $1^{11}431^{13}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 0 & 16 & 3 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 16 & 16 & 0 & 89 & 17 \\ 2 & 3 & 2 & -2 & 17 & 0 \end{pmatrix}_{b_0}$$

$$1^{13}$$

[264] $1^{12}3^{12}21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 279, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 2 & 5 & 0 & 20 & 2 \\ 17 & 28 & 20 & 120 & 22 \\ 3 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$1^1$$

[265] $1^{12}3^{12}21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 284, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 3 & 5 & 0 & 20 & 2 \\ 14 & 28 & 20 & 120 & 22 \\ 3 & 5 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$1^{12}$$

[266] $1^{14}3^{12}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 14 & 0 & 64 & 14 \\ 3 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$1^2$$

[267] $1^{13}3^{12}21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 0 & 18 & 2 \\ 12 & 23 & 18 & 80 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$1^1$$

[268] $1^{11}3^{12}21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 300, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 0 & 17 & 2 \\ 12 & 23 & 17 & 78 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$1^1$$

[269] $1^{12}3^{12}21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 303, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 5 & 0 & 33 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 16 & 33 & 0 & 164 & 26 \\ 3 & 5 & -2 & 26 & 3 \end{pmatrix}_{b_0}$$

[272] $1^{13}3^{12}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 20 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 20 & 10 & 0 & 65 & 13 \\ 2 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[271] $1^{11}3^{11}1331^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 24 & 14 & 0 & 97 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

[270] $1^{13}3^{12}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 20 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 20 & 10 & 0 & 65 & 13 \\ 2 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$1^2$$

[273] $1^{11}3^{12}21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 305, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 0 & 17 & 2 \\ 13 & 23 & 17 & 78 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[274] \quad 1^{14}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 308, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.03 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 18 & 0 & 116 & 20 \\ 3 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[275] \quad 1^{13}3^{11}21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 311, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 0 & 18 & 2 \\ 13 & 23 & 18 & 80 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[276] \quad 1^{12}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 324, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 2 & 5 & 0 & 20 & 2 \\ 14 & 28 & 20 & 120 & 22 \\ 3 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[278] \quad 1^{11}3^{11}1^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 18 & 11 & 0 & 55 & 13 \\ 2 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[279] \quad 1^{13}3^{11}1^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 24 & 17 & 0 & 111 & 19 \\ 2 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[277] \quad 1^{14}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 23 & 3 \\ 2 & 1 & 1 & 26 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 26 & 0 & 292 & 32 \\ 3 & 3 & -1 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[280] \quad 1^{12}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 15 & 0 & 83 & 17 \\ 3 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[282] \quad 1^{12}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 344, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 2 & 5 & 0 & 20 & 2 \\ 16 & 28 & 20 & 120 & 22 \\ 3 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[281] \quad 1^{14}3^{11}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 344, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 11 & 11 & 0 & 71 & 15 \\ 2 & 2 & 2 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[284] \quad 1^{13}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 348, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 10 & 13 & 0 & 74 & 16 \\ 2 & 2 & 2 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[283] \quad 1^{11}31^{12}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 348, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 17 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 17 & 10 & 0 & 48 & 12 \\ 2 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[286] \quad 1^{12} 3^{11} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 359, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 2 & 5 & 0 & 20 & 2 \\ 15 & 28 & 20 & 120 & 22 \\ 3 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[285] \quad 1^{12} 3^{11} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 359, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 22 & 3 \\ 2 & 1 & 1 & 23 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 23 & 0 & 243 & 29 \\ 3 & 3 & -1 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[289] \quad 1^{14} 3^{11} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 19 & 3 \\ 2 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 22 & 0 & 192 & 26 \\ 3 & 3 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[288] \quad 1^{13} 3^{11} 1331^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.36 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 21 & 11 & 0 & 72 & 14 \\ 2 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[287] \quad 1^{13} 3^{11} 1331^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.36 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 21 & 11 & 0 & 72 & 14 \\ 2 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[290] \quad 1^{14} 31^{12} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.19 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 26 & 4 \\ 1 & 4 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 26 & 19 & 0 & 135 & 21 \\ 2 & 4 & 2 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[292] \quad 1^{12} 3^{11} 1231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 380, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.19 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 21 & 14 & 0 & 80 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[291] \quad 1^{11} 31^{12} 3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 380, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.19 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 17 & 4 \\ 1 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 17 & 10 & 0 & 48 & 12 \\ 2 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[293] \quad 1^{12} 3^{11} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 387, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 19 & 0 & 151 & 23 \\ 3 & 3 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[294] \quad 31^{111} 23^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 14, \Delta = -96, \det = 388, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.18 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 4 & 2 \\ 3 & 1 & 4 & 1 & 4 & 3 \\ 3 & 4 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & -2 & 2 & 0 \\ 4 & 4 & 4 & 2 & 4 & 4 \\ 2 & 3 & 2 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[295] \quad 1^{12} 3^{11} 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 391, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 5 & 5 & 33 & 5 \\ 3 & 5 & 0 & 24 & 2 \\ 16 & 33 & 24 & 164 & 26 \\ 3 & 5 & 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[296] \quad 1^{13}3^{12}21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 11 & 16 & 0 & 103 & 19 \\ 2 & 2 & 2 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[297] \quad 1^{14}3^{12}21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 401, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 2 & 5 & 0 & 21 & 2 \\ 17 & 28 & 21 & 116 & 22 \\ 3 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[298] \quad 1^{11}3^{11}1^{12}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 18 & 11 & 0 & 55 & 13 \\ 2 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[299] \quad 1^{11}3^{12}21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 2 & 5 & 0 & 21 & 2 \\ 14 & 28 & 21 & 116 & 22 \\ 3 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[300] \quad 1^{13}3^{11}1^{13}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 21 & 11 & 0 & 72 & 14 \\ 2 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[302] \quad 1^{14}3^{12}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.02 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 18 & 0 & 116 & 20 \\ 3 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[303] \quad 1^{11}3^{11}1^{12}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 18 & 4 \\ 1 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 18 & 11 & 0 & 55 & 13 \\ 2 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[301] \quad 1^{12}3^{12}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 0 & 20 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 20 & 13 & 0 & 71 & 15 \\ 2 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[304] \quad 1^{13}3^{11}1^{14}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 444, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 27 & 14 & 0 & 120 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[305] \quad 1^{13}3^{12}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 444, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 23 & 4 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 23 & 16 & 0 & 100 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[306] \quad 1^{14}3^{12}21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 28 & 3 \\ 2 & 1 & 1 & 30 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 30 & 0 & 416 & 38 \\ 3 & 3 & -1 & 38 & 3 \end{pmatrix}_{b_0}$$

 1^3

$$[307] \ 1^{11}3^{121}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 2 & 5 & 0 & 21 & 2 \\ 16 & 28 & 21 & 116 & 22 \\ 3 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

 1^2

$$[308] \ 1^{11}3^{121}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 461, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 2 & 5 & 0 & 21 & 2 \\ 15 & 28 & 21 & 116 & 22 \\ 3 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

 1^1

$$[309] \ 1^{14}3^{121}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 14 & 0 & 96 & 18 \\ 2 & 2 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

 1^4

$$[310] \ 1^{12}3^{121}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 5 & 5 & 33 & 5 \\ 2 & 5 & 0 & 24 & 2 \\ 20 & 33 & 24 & 164 & 26 \\ 3 & 5 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

 1^{12}

$$[311] \ 13^{121}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 5 & 5 & 33 & 5 \\ 2 & 5 & 0 & 24 & 2 \\ 16 & 33 & 24 & 164 & 26 \\ 3 & 5 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[312] \ 1^{13}3^{112}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 468, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 24 & 17 & 0 & 111 & 19 \\ 2 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[313] \ 1^131^{14}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 468, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 26 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 26 & 13 & 0 & 111 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[314] \ 1^{12}3^{112}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 476, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 21 & 14 & 0 & 80 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[315] \ 1^131^{14}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 500, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 26 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 26 & 13 & 0 & 111 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[316] \ 1^{14}3^{112}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 508, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 27 & 20 & 0 & 148 & 22 \\ 2 & 4 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

 1^{12}

$$[317] \ 1^{14}3^{121}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 512, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 22 & 0 & 192 & 26 \\ 3 & 3 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

 12

$$[318] \ 1^{14}3^{12}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 516, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 24 & 3 \\ 2 & 1 & 1 & 26 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 26 & 0 & 292 & 32 \\ 3 & 3 & -1 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[319] \ 1^{12}3^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 532, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 20 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 20 & 13 & 0 & 71 & 15 \\ 2 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

 13

$$[321] \ 1^{12}3^{12}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 539, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 5 & 5 & 33 & 5 \\ 2 & 5 & 0 & 24 & 2 \\ 19 & 33 & 24 & 164 & 26 \\ 3 & 5 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

 11

$$[320] \ 1^{12}3^{12}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 539, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 5 & 5 & 33 & 5 \\ 2 & 5 & 0 & 24 & 2 \\ 17 & 33 & 24 & 164 & 26 \\ 3 & 5 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[323] \ 1^{12}3^{11}1^{12}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 540, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 21 & 14 & 0 & 80 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[322] \ 1^{13}3^{11}1^{14}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 540, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 27 & 14 & 0 & 120 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[325] \ 1^{11}3^{11}1^{13}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 552, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 23 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 23 & 13 & 0 & 88 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

 11

$$[324] \ 1^{14}3^{12}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 552, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 13 & 2 \\ 1 & 2 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 13 & 17 & 0 & 127 & 21 \\ 2 & 2 & 2 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[326] \ 1^{11}3^{11}1^{13}3^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 560, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 24 & 14 & 0 & 97 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[327] \ 1^{11}3^{11}1^{13}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 560, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 24 & 14 & 0 & 97 & 17 \\ 2 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

 12

$$[328] \ 1^{12}3^{12}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$T_{\min} = 11.47 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 5 & 5 & 33 & 5 \\ 2 & 5 & 0 & 24 & 2 \\ 18 & 33 & 24 & 164 & 26 \\ 3 & 5 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

[329] $1^{13}3^{11}1^{12}31^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2}$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 564$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$

$T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 24 & 17 & 0 & 111 & 19 \\ 2 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[330] $1^{12}3^{11}1^{12}31^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2}$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 572$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$

$T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 21 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 21 & 14 & 0 & 80 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[331] $1^{14}3^{12}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2}$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 596$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$

$T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 26 & 4 \\ 1 & 4 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 26 & 19 & 0 & 135 & 21 \\ 2 & 4 & 2 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

[334] $1^{14}3^{11}1^{12}31^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2}$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 604$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$

$T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 27 & 20 & 0 & 148 & 22 \\ 2 & 4 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

[333] $1^{13}1^{14}31^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2}$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 604$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$

$T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 27 & 14 & 0 & 120 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

[332] $1^{13}3^{12}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2}$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 604$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$

$T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 23 & 4 \\ 1 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 23 & 16 & 0 & 100 & 18 \\ 2 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

[335] $1^{11}3^{13}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2}$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 616$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$

$T_{\min} = 11.21 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 0 & 23 & 4 \\ 1 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 23 & 13 & 0 & 88 & 16 \\ 2 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[336] $1^{13}3^{11}1^{12}31^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2}$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 628$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$

$T_{\min} = 11.11 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 0 & 24 & 4 \\ 1 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 24 & 17 & 0 & 111 & 19 \\ 2 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[337] $1^{13}1^{14}31^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{orange}{\mathfrak{su}_2}$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 636$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$

$T_{\min} = 11.31 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 0 & 27 & 4 \\ 1 & 4 & 0 & 0 & 14$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 3 & 2 \\ 2 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 2 & 6 & 0 & 4 & 4 \\ 2 & 2 & 3 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[409] \ 1^{\overline{11}} 4^{\overline{11} 3^{\overline{11}} 1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 1 & 7 & 3 \\ 4 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 7 & 8 & 1 & 14 & 8 \\ 3 & 4 & -1 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[406] \ 1^{\overline{11}} 4^{\overline{11} 3^{\overline{11}} 1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 1 & 6 & 3 \\ 4 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 8 & 0 & 13 & 7 \\ 3 & 4 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[408] \ 1^{\overline{11}} 4^{\overline{11} 1^{\overline{1}} 2^{\overline{1}} 3^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 & 2 \\ 1 & 0 & 3 & 0 & 2 & 2 \\ 3 & 3 & 3 & 0 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 2 & 0 & 0 & 2 \\ 2 & 2 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[407] \ 1^{\overline{11}} 4^{\overline{11} 3^{\overline{11}} 1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 3 & 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[410] \ 1^{\overline{12}} 4^{\overline{12}} 1^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 6 & 8 & 1 & 11 & 7 \\ 3 & 4 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[411] \ 1^{\overline{12}} 4^{\overline{12}} 1^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 8 & 0 & 10 & 6 \\ 3 & 4 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[413] \ 1^{\overline{11}} 4^{\overline{11} 3^{\overline{11}} 1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 2 & 1 & 2 & 4 \\ 3 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[412] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 7 & 9 & 1 & 14 & 8 \\ 3 & 4 & -1 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[414] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 3 & 2 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[417] \ 1^{\overline{13}} 4^{\overline{13} 3^{\overline{11}} 1^{\overline{1}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -73, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 4 & 1 & 18 & 8 \\ 3 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[415] $1^{11}4^{11}13^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -73$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 9 & 0 & 13 & 7 \\ 3 & 4 & -1 & 7 & 3 \end{pmatrix}_{b_0}$

[416] $1^{11}4^{11}13^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -73$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 3 & 1 & 2 & 4 \\ 3 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$

[418] $1^{12}4^{11}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -73$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 9 & 11 & 1 & 22 & 10 \\ 3 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$

[419] $1^{11}4^{11}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -74$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 10 & 12 & 1 & 27 & 11 \\ 3 & 4 & -1 & 11 & 3 \end{pmatrix}_{b_0}$

[420] $1^{12}4^{11}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -76$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 3 & 1 & 8 & 6 \\ 3 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$

[421] $1^{11}4^{11}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 4 & 3 & 0 & 2 & 4 & 4 \\ 2 & 2 & 0 & 0 & 4 & 2 \end{pmatrix}_{b_0}$

[422] $1^{11}4^{11}13^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -88$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 10 & 6 & 1 & 32 & 10 \\ 3 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$

[423] $1^{11}4^{11}13^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 4 & 1 & 8 & 6 \\ 3 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$

[424] $1^{13}4^{11}13^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -101$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 8 & 12 & 1 & 26 & 10 \\ 3 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$

[425] $1^{12}4^{11}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -101$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 11 & 0 & 21 & 9 \\ 3 & 4 & -1 & 9 & 3 \end{pmatrix}_{b_0}$

[426] $1^{11}4^{11}13^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -101$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 8 & 5 & 1 & 18 & 8 \\ 3 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$

- [427] $1^{11} 1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -110$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 12 & 0 & 26 & 10 \\ 3 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$
- [428] $1^{12} 4^{11} 13^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -120$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 13 & 0 & 28 & 10 \\ 3 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$
- [430] $1^{11} 1^{11} 3^{11} 1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -128$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 5 & 3 & 1 & 2 & 5 & 5 \\ 2 & 2 & 0 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$
- [429] $1^{12} 4^{11} 13^{11} 1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -128$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 4 & 4 & 1 & 2 & 7 & 5 \\ 2 & 2 & 0 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$
- [431] $1^{11} 1^{11} 3^{11} 1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -138$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 4 & 2 & 6 & 6 \\ 3 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$
- [434] $1^{12} 1^{11} 3^{11} 1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 5 & 0 & 3 & 12 & 6 \\ 2 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$
- [433] $1^{11} 1^{12} 3^{11} 1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 6 & 3 & 2 & 2 & 8 & 6 \\ 2 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$
- [432] $1^{12} 4^{11} 13^{11} 1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 9 & 13 & 1 & 29 & 11 \\ 3 & 4 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$
- [435] $1^{14} 4^{11} 13^{11} 1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -150$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 10 & 16 & 1 & 47 & 13 \\ 3 & 4 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$
- [436] $1^{13} 1^{11} 3^{11} 1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -160$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 15 & 0 & 40 & 12 \\ 3 & 4 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$
- [438] $1^{14} 1^{11} 23^{11} 1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -164$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 6 & 3 \\ 6 & 13 & 0 & 12 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 12 & 0 & 8 & 6 \\ 3 & 7 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[437] $1^{13} 4^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 11 & 15 & 1 & 41 & 13 \\ 3 & 4 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

[439] $1^{12} 4^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 5 & 4 & 2 & 2 & 10 & 6 \\ 2 & 2 & 0 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[440] $1^{12} 4^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 6 & 4 & 3 & 2 & 15 & 7 \\ 2 & 2 & 0 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[441] $1^{13} 4^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 5 & 6 & 1 & 3 & 17 & 7 \\ 2 & 2 & 0 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[442] $1^{14} 1^{11} 23^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 5 & 2 \\ 3 & 4 & 1 & 2 & 6 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & -2 & 2 & 0 \\ 5 & 6 & 1 & 2 & 5 & 5 \\ 2 & 4 & 0 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[443] $1^{12} 4^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -193, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 12 & 16 & 1 & 46 & 14 \\ 3 & 4 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

[444] $1^{12} 4^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -197, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 16 & 0 & 45 & 13 \\ 3 & 4 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

[445] $1^{11} 4^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -197, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 5 & 2 & 12 & 8 \\ 3 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[446] $1^{14} 1^{11} 23^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 4 & 2 \\ 3 & 4 & 1 & 2 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 2 & 0 \\ 4 & 6 & 0 & 2 & 4 & 4 \\ 2 & 4 & 0 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

[447] $1^{14} 4^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -217, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 19 & 0 & 65 & 15 \\ 3 & 4 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[448] $1^{13} 4^{13} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -217, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 11 & 17 & 1 & 50 & 14 \\ 3 & 4 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[449] \ 1^{12} 4^{113^1 1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 6 & 5 & 1 & 3 & 13 & 7 \\ 2 & 2 & 0 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[450] \ 1^{11} 4^{113^1 1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 6 & 2 & 22 & 10 \\ 3 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[451] \ 1^{41} 1^{11} 31^{13^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -250, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 4 & 2 \\ 3 & 4 & 4 & 2 & 6 & 4 \\ 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & -2 & 2 & 0 \\ 4 & 6 & 3 & 2 & 4 & 4 \\ 2 & 4 & 2 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[452] \ 1^{14} 4^{113^1 1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -253, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 13 & 19 & 1 & 66 & 16 \\ 3 & 4 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[453] \ 1^{13} 4^{11} 13^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 6 & 6 & 2 & 3 & 20 & 8 \\ 2 & 2 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[454] \ 1^{11} 4^{114^1 1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -261, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 7 & 2 & 36 & 12 \\ 3 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[455] \ 1^{14} 1^{11} 23^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -265, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 1 & 9 & 3 \\ 6 & 13 & 0 & 20 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 20 & 0 & 26 & 10 \\ 3 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[456] \ 1^{13} 4^{113^1 1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -282, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 20 & 0 & 70 & 16 \\ 3 & 4 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[457] \ 1^{12} 4^{112^1 1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 7 & 5 & 2 & 3 & 16 & 8 \\ 2 & 2 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[458] \ 1^{14} 4^{13^1} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 13 & 21 & 1 & 77 & 17 \\ 3 & 4 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[459] $1^{12} 4^{1431^1} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -300$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 9 & 5 & 4 & 3 & 28 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[461] $1^{12} 4^{1331^1} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -308$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 8 & 5 & 3 & 3 & 21 & 9 \\ 2 & 2 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[460] $1^{13} 4^{13} 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -308$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 7 & 6 & 3 & 3 & 25 & 9 \\ 2 & 2 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[462] $1^{13} 4^{1131^1} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -310$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 14 & 20 & 1 & 71 & 17 \\ 3 & 4 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[463] $1^{13} 4^{1131^1} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -320$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 7 & 7 & 1 & 4 & 25 & 9 \\ 2 & 2 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[464] $1^{13} 4^{13} 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -324$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 \\ 8 & 6 & 4 & 3 & 32 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[465] $1^{14} 4^{13} 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -348$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 7 & 8 & 2 & 4 & 34 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[466] $1^{14} 4^{1231^1} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -353$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 11 & 3 \\ 6 & 13 & 0 & 24 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 24 & 0 & 38 & 12 \\ 3 & 7 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

[467] $1^{14} 4^{131^1} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -365$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 24 & 0 & 101 & 19 \\ 3 & 4 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

[469] $1^{14} 4^{1231^1} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -404$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 7 & 2 \\ 3 & 4 & 1 & 2 & 12 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 4 & 0 \\ 7 & 12 & 0 & 4 & 22 & 8 \\ 2 & 4 & 0 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

[468] $1^{14} 4^{1131^1} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -404$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 8 & 9 & 1 & 5 & 41 & 11 \\ 2 & 2 & 0 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[470] $1^{13} 4^{1^{12} 3^1 1^1} 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -420, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 8 & 7 & 2 & 4 & 28 & 10 \\ 2 & 2 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[471] $1^{14} 4^{1^{11} 3^1 1^1} 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -425, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 16 & 24 & 1 & 102 & 20 \\ 3 & 4 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[472] $1^{14} 4^{13} 1^{11} 1^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 8 & 8 & 3 & 4 & 39 & 11 \\ 2 & 2 & 0 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[473] $1^{14} 1^{41} 1^{23} 1^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 9 & 2 \\ 3 & 4 & 1 & 2 & 12 & 4 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & -2 & 4 & 0 \\ 9 & 12 & 2 & 4 & 26 & 10 \\ 2 & 4 & 0 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[474] $1^{13} 4^{1^{13} 3^1 1^1} 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -488, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 9 & 7 & 3 & 4 & 33 & 11 \\ 2 & 2 & 0 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[475] $1^{14} 4^{13} 1^{11} 1^{14} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 9 & 8 & 4 & 4 & 46 & 12 \\ 2 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[476] $1^{13} 4^{1^{14} 3^1 1^1} 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 \\ 10 & 7 & 4 & 4 & 40 & 12 \\ 2 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[477] $1^{14} 4^{1^{12} 3^1 1^1} 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -548, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 9 & 9 & 2 & 5 & 44 & 12 \\ 2 & 2 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[478] $1^{14} 1^{41} 1^{23} 1^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -549, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 8 & 2 \\ 3 & 4 & 1 & 2 & 12 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & -2 & 4 & 0 \\ 8 & 12 & 1 & 4 & 23 & 9 \\ 2 & 4 & 0 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[479] $1^{14} 1^{41} 1^{23} 1^{11} 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -580, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 1 & 14 & 3 \\ 6 & 13 & 0 & 32 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 32 & 0 & 68 & 16 \\ 3 & 7 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

- [480] $1^{141}1^{13}23^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -640$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 1 & 16 & 3 \\ 6 & 13 & 0 & 36 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 36 & 0 & 88 & 18 \\ 3 & 7 & -1 & 18 & 3 \end{pmatrix}_{b_0}$
- [481] $1^{141}1^{13}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -660$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 3 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 10 & 9 & 3 & 5 & 49 & 13 \\ 2 & 2 & 0 & 0 & 13 & 2 \end{pmatrix}_{b_0}$
- [482] $1^{141}1^{13}23^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -684$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.99 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 1 & 1 & 10 & 2 \\ 3 & 4 & 1 & 2 & 18 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 6 & 0 \\ 10 & 18 & 0 & 6 & 52 & 12 \\ 2 & 4 & 0 & 0 & 12 & 2 \end{pmatrix}_{b_0}$
- [483] $1^{141}1^{14}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -740$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 4 & 0 \\ 1 & 1 & 0 & -2 & 5 & 0 \\ 11 & 9 & 4 & 5 & 56 & 14 \\ 2 & 2 & 0 & 0 & 14 & 2 \end{pmatrix}_{b_0}$
- [484] $1^{141}1^{13}23^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -989$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 1 & 1 & 11 & 2 \\ 3 & 4 & 1 & 2 & 18 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & -2 & 6 & 0 \\ 11 & 18 & 1 & 6 & 53 & 13 \\ 2 & 4 & 0 & 0 & 13 & 2 \end{pmatrix}_{b_0}$
- [485] $1^{11}1^{141}1^{13}23^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -993$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 1 & 19 & 3 \\ 6 & 13 & 0 & 44 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 44 & 0 & 130 & 22 \\ 3 & 7 & -1 & 22 & 3 \end{pmatrix}_{b_0}$
- [486] $1^{141}1^{14}23^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -1025$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 1 & 21 & 3 \\ 6 & 13 & 0 & 48 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 48 & 0 & 158 & 24 \\ 3 & 7 & -1 & 24 & 3 \end{pmatrix}_{b_0}$
- [487] $1^{141}1^{14}23^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -1044$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.99 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 1 & 1 & 13 & 2 \\ 3 & 4 & 1 & 2 & 24 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 8 & 0 \\ 13 & 24 & 0 & 8 & 94 & 16 \\ 2 & 4 & 0 & 0 & 16 & 2 \end{pmatrix}_{b_0}$
- [488] $1^{141}1^{13}23^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -1064$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 1 & 1 & 12 & 2 \\ 3 & 4 & 1 & 2 & 18 & 4 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & -2 & 6 & 0 \\ 12 & 18 & 2 & 6 & 56 & 14 \\ 2 & 4 & 0 & 0 & 14 & 2 \end{pmatrix}_{b_0}$
- [489] $1^{11}1^{141}1^{14}23^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -1504$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 1 & 24 & 3 \\ 6 & 13 & 0 & 56 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 56 & 0 & 212 & 28 \\ 3 & 7 & -1 & 28 & 3 \end{pmatrix}_{b_0}$
- [490] $1^{141}1^{14}23^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -1509$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 1 & 1 & 14 & 2 \\ 3 & 4 & 1 & 2 & 24 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & -2 & 8 & 0 \\ 14 & 24 & 1 & 8 & 95 & 17 \\ 2 & 4 & 0 & 0 & 17 & 2 \end{pmatrix}_{b_0}$
- [491] $1^{141}1^{14}23^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -87$, $\det = -1744$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 15 & 2 \\ 3 & 4 & 1 & 2 & 24 & 4 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & -2 & 8 & 0 \\ 15 & 24 & 2 & 8 & 98 & 18 \\ 2 & 4 & 0 & 0 & 18 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [492] \quad 21^{1^1} 5 13^{1^1} 1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -80, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 5 & 5 & 7 & 5 \\ 3 & 5 & 0 & 5 & 2 \\ 6 & 7 & 5 & 8 & 6 \\ 4 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[493] \quad 1^{1^1} 4 13^{1^1} 21^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 4 & 2 & 5 \\ 4 & 2 & 2 & 4 \\ 2 & 2 & 0 & 2 \\ 5 & 4 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[494] \quad 1^{1^3} 5 13^{1^1} 1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -80, \quad \det = 6, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 9 & 10 & 9 & 22 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[495] \quad 1^{1^1} 5 13^{1^1} 1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -80, \quad \det = 7, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.57 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[496] \quad 21^{1^1} 5 13^{1^1} 1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -80, \quad \det = 12, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.67 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 3 & 7 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 6 & 7 & 5 & 8 & 6 \\ 4 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[497] \quad 1^{1^1} 4 13^{1^1} 21^{1^3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 15, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 4 & 13 & 5 \\ 4 & 2 & 11 & 4 \\ 13 & 11 & 31 & 13 \\ 5 & 4 & 13 & 5 \end{pmatrix}_{b_0}$$

$$[499] \quad 1^{1^1} 5 13^{1^1} 1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -80, \quad \det = 18, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 7 & 10 & 9 & 22 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[498] \quad 1^{1^3} 5 13^{1^1} 1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -80, \quad \det = 18, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 7 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[500] \quad 1^{1^2} 5 13^{1^1} 1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -80, \quad \det = 21, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.67 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 8 & 10 & 9 & 22 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[503] \quad 1^{3^1} 4 \quad 141^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 33, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 & 2 & 2 \\ 3 & 13 & 1 & 9 & 5 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 9 & 0 & 3 & 3 & 3 \\ 2 & 5 & 0 & 3 & 0 & 2 \\ 2 & 7 & -1 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[501] $1 \overset{1}{5} 13 \overset{1}{1} 1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 4 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[502] $1 \overset{1}{11} 5 123 \overset{1}{1} 1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 6 & 0 & 13 & 7 \\ 3 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0}$

[506] $1 \overset{1}{11} \overset{2}{5} 13 \overset{1}{1} 1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 6 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[505] $1 \overset{1}{41} \overset{1}{13} \overset{1}{1} 141 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 8 & 1 & 12 & 3 \\ 8 & 38 & 1 & 56 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 56 & 0 & 81 & 17 \\ 3 & 12 & -1 & 17 & 3 \end{pmatrix}_{b_0}$

[504] $1 \overset{1}{41} \overset{1}{13} \overset{1}{1} 141 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 8 & 1 & 6 & 3 \\ 8 & 38 & 1 & 24 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 24 & 0 & 13 & 7 \\ 3 & 12 & -1 & 7 & 3 \end{pmatrix}_{b_0}$

[507] $1 \overset{1}{11} 4 13 \overset{1}{21} \overset{1}{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 4 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 4 & 1 & 6 & 6 \\ 3 & 3 & 0 & 6 & 4 \end{pmatrix}_{b_0}$

[508] $1 \overset{1}{11} \overset{1}{5} 13 \overset{1}{1} 1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 46, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 5 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[509] $1 \overset{1}{5} 13 \overset{1}{1} 1 \overset{1}{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 8 & 14 & 11 & 39 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$

[511] $1 \overset{1}{5} 13 \overset{1}{1} 1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 6 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[510] $1 \overset{1}{12} 5 123 \overset{1}{1} 1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 28 & 10 \\ 3 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$

[512] $1 \overset{1}{11} 4 13 \overset{1}{21} \overset{1}{11} 1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 57, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 4 & 3 \\ 3 & 3 & 1 & 5 & 3 \\ 5 & 4 & 5 & 6 & 6 \\ 3 & 3 & 3 & 6 & 4 \end{pmatrix}_{b_0}$

[513] $1^{12} \overset{1^4}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 62, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 10 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[514] $1^{11} \overset{1}{5} 1^{23} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 13 & 0 & 79 & 17 \\ 3 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$

[515] $1^{12} \overset{1}{4} 13^{121} 1^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 10 & 2 & 60 & 16 \\ 3 & 3 & 0 & 16 & 4 \end{pmatrix}_{b_0}$

[517] $1^{11} \overset{1}{5} 1^{223} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 65, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 19 & 3 \\ 2 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 20 & 0 & 201 & 27 \\ 3 & 3 & -1 & 27 & 3 \end{pmatrix}_{b_0}$

[516] $21^{11} \overset{1}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 65, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 5 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[518] $21^{12} \overset{1}{5} 13^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 67, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 2 & 10 & 3 \\ 4 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 10 & 14 & 11 & 39 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$

[519] $1^{11} \overset{1}{4} 1^{13} 1^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 76, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 3 & 1 & 1 & 6 & 2 \\ 3 & 13 & 6 & 0 & 26 & 7 \\ 1 & 6 & 0 & 0 & 10 & -2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 26 & 10 & 0 & 49 & 13 \\ 2 & 7 & 2 & -2 & 13 & 2 \end{pmatrix}_{b_0}$

[520] $1^{12} \overset{1}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 83, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 7 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[522] $1^{12} \overset{1^3}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 87, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.69 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 9 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[521] $1^{12} \overset{1}{5} 13^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 87, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.69 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 9 & 14 & 11 & 39 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$

[523] $1^{12} \overset{1^4}{5} 13^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 90, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 12 & 14 & 11 & 39 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[524] $1^{12}5^{12}13^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 8 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[525] $1^{12}4^{11}13^{12}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 101, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -2 & 2 & 0 \\ 9 & 8 & 2 & 30 & 12 \\ 3 & 3 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

[526] $1^{11}5^{13}13^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 101, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 13 & 2 \\ 10 & 18 & 13 & 62 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[527] $1^{11}4^{11}13^{12}21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 105, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 6 & 3 \\ 2 & 1 & 3 & 6 & 3 \\ 3 & 3 & 1 & 7 & 3 \\ 6 & 6 & 7 & 12 & 8 \\ 3 & 3 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[528] $1^{12}5^{13}13^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 106, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 10 & 14 & 11 & 39 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[529] $1^{12}5^{13}13^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 107, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 11 & 14 & 11 & 39 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[530] $1^{11}4^{11}13^{11}14^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 3 & 2 \\ 3 & 13 & 6 & 0 & 14 & 7 \\ 1 & 6 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 14 & 6 & 0 & 13 & 7 \\ 2 & 7 & 2 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[531] $1^{13}5^{13}13^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 118, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 9 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[532] $1^{12}5^{11}123^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.71 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 16 & 0 & 108 & 20 \\ 3 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[533] $1^{13}4^{11}13^{12}21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 141, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -2 & 3 & 0 \\ 16 & 14 & 3 & 112 & 22 \\ 3 & 3 & 0 & 22 & 4 \end{pmatrix}_{b_0}$$

[534] $21^{12}5^{13}13^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 143, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 3 \\ 4 & 4 & 3 & 1 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 8 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[535] \ 1^{13} \overset{1^2}{5} 13^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 145, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 10 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[536] \ 1^{13} \overset{1^4}{5} 13^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 145, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 12 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[537] \ 1^{11} \overset{3^1}{4} 141^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 148, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 2 & 1 & 3 & 2 \\ 3 & 13 & 10 & 0 & 14 & 7 \\ 2 & 10 & 4 & 0 & 10 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 14 & 10 & 0 & 13 & 7 \\ 2 & 7 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[538] \ 1^{13} \overset{1^3}{5} 13^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 154, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 11 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[539] \ 1^{13} \overset{1^1}{5} 13^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 154, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 13 & 2 \\ 11 & 18 & 13 & 62 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[540] \ 1^{13} \overset{1^1}{4} 13^{12} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -2 & 3 & 0 \\ 10 & 10 & 3 & 40 & 14 \\ 3 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[541] \ 1^{13} \overset{1^1}{4} 13^{12} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 12 & 3 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 12 & 3 & 70 & 18 \\ 3 & 3 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[542] \ 1^{13} \overset{1^2}{5} 13^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 189, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 13 & 2 \\ 12 & 18 & 13 & 62 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[543] \ 1^{12} \overset{1^1}{5} 1^{22} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 198, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.68 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 21 & 3 \\ 2 & 1 & 1 & 23 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 23 & 0 & 244 & 30 \\ 3 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[544] \ 1^{13} \overset{1^4}{5} 13^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 205, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 13 & 2 \\ 14 & 18 & 13 & 62 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[545] $1^{13}5^{13}1^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -80$, $\det = 206$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 13 & 2 \\ 13 & 18 & 13 & 62 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[546] $1^{13}5^{13}1^{23}1^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 215$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 19 & 0 & 143 & 23 \\ 3 & 3 & -1 & 23 & 3 \end{pmatrix}_{b_0}$

[547] $1^{12}5^{13}23^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 228$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.68 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 28 & 3 \\ 2 & 1 & 1 & 30 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 30 & 0 & 436 & 40 \\ 3 & 3 & -1 & 40 & 3 \end{pmatrix}_{b_0}$

[548] $21^{13}5^{13}1^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -80$, $\det = 229$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 2 & 13 & 3 \\ 4 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 13 & 2 \\ 13 & 18 & 13 & 62 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[549] $21^{13}5^{13}1^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -80$, $\det = 241$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 2 & 11 & 3 \\ 4 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 11 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[550] $1^{14}4^{13}1^{21}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 288$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -2 & 4 & 0 \\ 14 & 14 & 4 & 84 & 20 \\ 3 & 3 & 0 & 20 & 4 \end{pmatrix}_{b_0}$

[551] $1^{14}4^{13}1^{21}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 293$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & -2 & 4 & 0 \\ 17 & 16 & 4 & 126 & 24 \\ 3 & 3 & 0 & 24 & 4 \end{pmatrix}_{b_0}$

[552] $1^{13}5^{13}1^{23}1^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 329$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.69 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 23 & 3 \\ 2 & 1 & 1 & 26 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 26 & 0 & 293 & 33 \\ 3 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$

[553] $1^{13}5^{13}1^{323}1^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 415$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.67 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 30 & 3 \\ 2 & 1 & 1 & 33 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 33 & 0 & 499 & 43 \\ 3 & 3 & -1 & 43 & 3 \end{pmatrix}_{b_0}$

$T = 11$ [554] $1^{14}1^{13}1^{15}1^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 14$, $\Delta = -79$, $\det = -6$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 8.0$

$T_{\min} = 14$, $T_{\max} = 14$, $T \in [14, 14]$

$\begin{pmatrix} 5 & 1 & 9 & 5 \\ 1 & -2 & 3 & 0 \\ 9 & 3 & 15 & 9 \\ 5 & 0 & 9 & 4 \end{pmatrix}_{b_0}$

[556] $1^{111}1^{315}1^{13}1^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -59$, $\det = -12$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 8.0$

$T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 4 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 11 & 4 \\ 3 & 5 & 0 & 11 & 14 & 8 \\ 2 & 3 & -1 & 4 & 8 & 3 \end{pmatrix}_{b_0}$$

[555] $1^{11}3^{11}15^{11}13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 4 & 7 & 4 \\ 2 & 3 & 0 & 7 & 5 & 5 \\ 2 & 3 & -1 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[557] $1^{11}3^{11}15^{11}13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 7 & 3 \\ 5 & 0 & 7 & 14 & 8 \\ 3 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[558] $1^{11}3^{11}15^{11}13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -21, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 2 & 1 & 7 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 7 & 0 & 14 & 8 \\ 3 & 4 & -1 & 8 & 4 \end{pmatrix}_{b_0}$$

[560] $1^{11}3^{11}15^{11}13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 2 & 1 & 11 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 11 & 0 & 39 & 13 \\ 3 & 4 & -1 & 13 & 4 \end{pmatrix}_{b_0}$$

[559] $1^{12}3^{11}15^{11}13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 2 & 1 & 11 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 11 & 0 & 32 & 12 \\ 3 & 4 & -1 & 12 & 4 \end{pmatrix}_{b_0}$$

[561] $1^{14}1^{13}1^{11}15^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -39, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13.92 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 3 & 2 \\ 3 & 4 & 1 & 2 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 1 & 0 \\ 3 & 3 & 0 & 1 & 1 & 3 \\ 2 & 4 & 0 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[562] $1^{12}3^{11}15^{11}13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 2 & 1 & 15 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 65 & 17 \\ 3 & 4 & -1 & 17 & 4 \end{pmatrix}_{b_0}$$

[563] $1^{14}1^{13}1^{11}15^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -46, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13.65 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 5 & 1 & 6 & 5 \\ 1 & -2 & 1 & 0 \\ 6 & 1 & 3 & 5 \\ 5 & 0 & 5 & 4 \end{pmatrix}_{b_0}$$

[564] $1^{11}3^{11}15^{11}13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -53, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 6 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 7 & 3 \\ 6 & 0 & 7 & 14 & 8 \\ 3 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[565] $1^{13}3^{11}15^{11}13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -61, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 2 & 1 & 19 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 19 & 0 & 110 & 22 \\ 3 & 4 & -1 & 22 & 4 \end{pmatrix}_{b_0}$$

[566] $1^{11}3^{11}15^{11}13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$

8.0

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 11 & 3 \\ 9 & 0 & 11 & 39 & 13 \\ 3 & 0 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[568] \ 1^{13} 3^{11} 5^{13} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -75, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 2 & 1 & 23 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 23 & 0 & 167 & 27 \\ 3 & 4 & -1 & 27 & 4 \end{pmatrix}_{b_0}$$

$$[567] \ 1^{14} 3^{11} 5^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -75, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 2 & 1 & 23 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 23 & 0 & 152 & 26 \\ 3 & 4 & -1 & 26 & 4 \end{pmatrix}_{b_0}$$

$$[569] \ 1^{14} 13^{11} 5^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -83, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 1 & 10 & 5 \\ 1 & -2 & 1 & 0 \\ 10 & 1 & 12 & 8 \\ 5 & 0 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[570] \ 1^{12} 3^{11} 5^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -85, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 12 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 15 & 3 \\ 12 & 0 & 15 & 76 & 18 \\ 3 & 0 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[571] \ 1^{14} 3^{11} 5^{13} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -101, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 2 & 1 & 27 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 27 & 0 & 217 & 31 \\ 3 & 4 & -1 & 31 & 4 \end{pmatrix}_{b_0}$$

$$[572] \ 1^{14} 13^{11} 5^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -107, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 1 & 13 & 5 \\ 1 & -2 & 3 & 0 \\ 13 & 3 & 24 & 12 \\ 5 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[573] \ 1^{14} 13^{11} 5^{11} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -118, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 1 & 14 & 5 \\ 1 & -2 & 1 & 0 \\ 14 & 1 & 27 & 11 \\ 5 & 0 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[574] \ 1^{11} 3^{11} 5^{12} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -125, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 11 & 3 \\ 10 & 0 & 11 & 39 & 13 \\ 3 & 0 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[575] \ 1^{12} 3^{11} 5^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -160, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 15 & 3 \\ 13 & 0 & 15 & 76 & 18 \\ 3 & 0 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[576] \ 1^{12} 3^{11} 5^{12} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -173, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 12 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 15 & 3 \\ 12 & 0 & 15 & 65 & 17 \\ 3 & 0 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[577] \ 1^{12} 3^1 1^5 13^{12} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -189, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 16 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 19 & 3 \\ 16 & 0 & 19 & 125 & 23 \\ 3 & 0 & 3 & 23 & 4 \end{pmatrix}_{b_0}$$

$$[578] \ 1^1 41^{13} 1^5 1^{13} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -206, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.63 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 1 & 17 & 5 \\ 1 & -2 & 3 & 0 \\ 17 & 3 & 39 & 15 \\ 5 & 0 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[579] \ 1^{13} 3^1 5 13^1 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -212, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 15 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 19 & 3 \\ 15 & 0 & 19 & 110 & 22 \\ 3 & 0 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[580] \ 1^{13} 3^1 1^5 13^{13} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -229, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 22 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 27 & 3 \\ 22 & 0 & 27 & 259 & 33 \\ 3 & 0 & 3 & 33 & 4 \end{pmatrix}_{b_0}$$

$$[581] \ 1^{12} 3^1 5 13^1 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -229, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 14 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 15 & 3 \\ 14 & 0 & 15 & 76 & 18 \\ 3 & 0 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[582] \ 1^{13} 3^1 1^5 13^{12} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -245, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 18 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 23 & 3 \\ 18 & 0 & 23 & 167 & 27 \\ 3 & 0 & 3 & 27 & 4 \end{pmatrix}_{b_0}$$

$$[583] \ 1^{12} 3^1 1^5 13^{12} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -280, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 17 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 19 & 3 \\ 17 & 0 & 19 & 125 & 23 \\ 3 & 0 & 3 & 23 & 4 \end{pmatrix}_{b_0}$$

$$[584] \ 1^{14} 3^1 1^5 13^{13} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -293, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 24 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 31 & 3 \\ 24 & 0 & 31 & 317 & 37 \\ 3 & 0 & 3 & 37 & 4 \end{pmatrix}_{b_0}$$

$$[585] \ 1^{13} 3^1 5 13^1 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{o}_8 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -309, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 16 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 19 & 3 \\ 16 & 0 & 19 & 110 & 22 \\ 3 & 0 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[586] \ 1^{13} 3^1 1^5 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 13, \ \Delta = -59, \ \det = -364, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 19 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 23 & 3 \\ 19 & 0 & 23 & 167 & 27 \\ 3 & 0 & 3 & 27 & 4 \end{pmatrix}_{b_0}$$

$$[587] \ 1^{13} 3^1 1^5 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 13, \ \Delta = -59, \ \det = -364, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 23 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 27 & 3 \\ 23 & 0 & 27 & 259 & 33 \\ 3 & 0 & 3 & 33 & 4 \end{pmatrix}_{b_0}$$

$$[588] \ 1^{14} 3^1 1^5 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 13, \ \Delta = -59, \ \det = -365, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.99 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 18 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 23 & 3 \\ 18 & 0 & 23 & 152 & 26 \\ 3 & 0 & 3 & 26 & 4 \end{pmatrix}_{b_0}$$

$$[590] \ 1^{14} 3^1 1^5 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 13, \ \Delta = -59, \ \det = -424, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.97 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 29 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 35 & 3 \\ 29 & 0 & 35 & 441 & 43 \\ 3 & 0 & 3 & 43 & 4 \end{pmatrix}_{b_0}$$

$$[589] \ 1^{14} 3^1 1^5 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 13, \ \Delta = -59, \ \det = -424, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.97 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 21 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 27 & 3 \\ 21 & 0 & 27 & 217 & 31 \\ 3 & 0 & 3 & 31 & 4 \end{pmatrix}_{b_0}$$

$$[591] \ 1^{14} 3^1 1^5 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 13, \ \Delta = -59, \ \det = -456, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 25 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 31 & 3 \\ 25 & 0 & 31 & 317 & 37 \\ 3 & 0 & 3 & 37 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [592] \ 1^{13} 1^5 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -31, \ \det = 8, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 8 & 4 \\ 5 & 0 & 8 & 15 & 9 \\ 3 & 0 & 4 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[593] \ 1^{13} 3^1 21^1 6^1 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 12, \ T_{\max} = 14, \ \Delta = -58, \ \det = 16, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 1 & 3 & 0 & 1 & 3 \\ 2 & 2 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[594] \ 1^{13} 3^1 21^1 6^1 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 12, \ T_{\max} = 14, \ \Delta = -58, \ \det = 53, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.96 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[596] \ 1^{13} 3^1 21^1 6^1 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 \quad T = 12, \ T_{\max} = 14, \ \Delta = -58, \ \det = 64, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 4 & 8 & 0 & 24 & 10 \\ 2 & 2 & 2 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[595] $1^{11} 6^{11} 131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 12, T_{\max} = 14, \Delta = -58, \det = 64, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 8 & 1 & 37 & 13 \\ 3 & 3 & 0 & 13 & 4 \end{pmatrix}_{b_0}$$

[597] $1^{11} 6^{11} 131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 12, T_{\max} = 14, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13.94 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 8 & 7 & 1 & 22 & 10 \\ 3 & 3 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

[599] $1^{11} 6^{11} 131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 12, T_{\max} = 14, \Delta = -58, \det = 85, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13.98 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 5 & 6 & 0 & 19 & 9 \\ 2 & 2 & 2 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

[598] $1^{11} 6^{11} 131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 12, T_{\max} = 14, \Delta = -58, \det = 85, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13.98 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 12 & 11 & 0 & 55 & 15 \\ 3 & 3 & 0 & 15 & 4 \end{pmatrix}_{b_0}$$

[600] $1^{11} 3^{11} 1^5 1321^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -31, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 15 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 22 & 4 \\ 15 & 0 & 22 & 119 & 25 \\ 3 & 0 & 4 & 25 & 5 \end{pmatrix}_{b_0}$$

[601] $1^{11} 6^{11} 131^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 12, T_{\max} = 14, \Delta = -58, \det = 141, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13.93 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 15 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 17 & 15 & 1 & 116 & 22 \\ 3 & 3 & 0 & 22 & 4 \end{pmatrix}_{b_0}$$

[602] $1^{11} 6^{11} 131^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 12, T_{\max} = 14, \Delta = -58, \det = 157, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13.89 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 13 & 11 & 1 & 56 & 16 \\ 3 & 3 & 0 & 16 & 4 \end{pmatrix}_{b_0}$$

[603] $1^{11} 6^{11} 131^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 12, T_{\max} = 14, \Delta = -58, \det = 157, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13.89 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 7 & 6 & 9 & 0 & 34 & 12 \\ 2 & 2 & 2 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

[604] $1^{11} 6^{11} 131^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 12, T_{\max} = 14, \Delta = -58, \det = 173, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13.99 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 2 & 1 & 22 & 3 \\ 2 & 1 & 1 & 19 & 3 \\ 1 & 1 & -2 & 2 & 0 \\ 22 & 19 & 2 & 203 & 29 \\ 3 & 3 & 0 & 29 & 4 \end{pmatrix}_{b_0}$$

[606] $1^{11} 6^{11} 131^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 12, T_{\max} = 14, \Delta = -58, \det = 205, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 7 & 12 & 0 & 55 & 15 \\ 2 & 2 & 2 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[605] \ 1^{11} 1^{12} 3^{121} 1^{12} 6^{11} 131^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 205, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 11 & 2 & 59 & 17 \\ 3 & 3 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[607] \ 1^{11} 1^{12} 3^{121} 1^{12} 6^{11} 131^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 221, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 1 & 1 & 15 & 3 \\ 1 & 1 & -2 & 2 & 0 \\ 18 & 15 & 2 & 119 & 23 \\ 3 & 3 & 0 & 23 & 4 \end{pmatrix}_{b_0}$$

$$[608] \ 1^{11} 1^{13} 3^{121} 1^{12} 6^{11} 131^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 229, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.99 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 8 & 15 & 0 & 82 & 18 \\ 2 & 2 & 2 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

4.710 NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{f}_4$

$$T = 8 \quad [0] \ 1^{11} 3^{11} 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 24 & 1 & 5 \\ 11 & 20 & 45 & 0 & 8 \\ 24 & 45 & 100 & 0 & 18 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 18 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 15 & 1 & 5 \\ 11 & 20 & 28 & 0 & 8 \\ 15 & 28 & 37 & 0 & 11 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 11 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 23 & 1 & 5 \\ 12 & 25 & 48 & 0 & 9 \\ 23 & 48 & 89 & 0 & 17 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 17 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 15 & 1 & 5 \\ 12 & 25 & 32 & 0 & 9 \\ 15 & 32 & 37 & 0 & 11 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 11 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \ 1^{11} 3^{11} 31^{13} 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 4 & 5 & 6 & 1 & 3 \\ 4 & 10 & 13 & 16 & 0 & 6 \\ 5 & 13 & 12 & 17 & 0 & 6 \\ 6 & 16 & 17 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

4.711 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{f}_4 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -212, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.64 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 15 & 1 & 9 & 5 \\ 11 & 20 & 28 & 0 & 17 & 8 \\ 15 & 28 & 37 & 0 & 24 & 11 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 24 & 0 & 13 & 7 \\ 5 & 8 & 11 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

4.712 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 35, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 24 & 1 & 5 \\ 11 & 20 & 45 & 0 & 8 \\ 24 & 45 & 100 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 59, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 7 & 1 & 5 \\ 11 & 20 & 11 & 0 & 8 \\ 7 & 11 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 69, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 12 & 24 & 1 & 5 \\ 12 & 25 & 48 & 0 & 9 \\ 24 & 48 & 89 & 0 & 17 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.81 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 25 & 1 & 5 \\ 11 & 20 & 45 & 0 & 8 \\ 25 & 45 & 100 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 112, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.82 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 16 & 1 & 5 \\ 11 & 20 & 28 & 0 & 8 \\ 16 & 28 & 37 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 128, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.84 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 15 & 1 & 5 \\ 11 & 20 & 28 & 0 & 8 \\ 15 & 28 & 37 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 161, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.85 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 12 & 23 & 1 & 5 \\ 12 & 25 & 48 & 0 & 9 \\ 23 & 48 & 89 & 0 & 17 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 24, \ \Delta = -114, \ \det = 189, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 12 & 15 & 1 & 5 \\ 12 & 25 & 32 & 0 & 9 \\ 15 & 32 & 37 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \quad 1^1 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 1 & 4 & 0 & 3 \\ 1 & 4 & 0 & 1 & 4 & 0 & 2 \\ 1 & 1 & 1 & -2 & 2 & 0 & 0 \\ 2 & 4 & 4 & 2 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 0 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{12} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 5 & 5 & 18 & 0 & 5 \\ 2 & 5 & 0 & 12 & 0 & 2 \\ 10 & 18 & 12 & 56 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 14 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{111} 3^1 1^1 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 7 & 3 \\ 4 & 10 & 13 & 0 & 16 & 6 \\ 5 & 13 & 12 & 0 & 17 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 16 & 17 & 0 & 20 & 8 \\ 3 & 6 & 6 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 5 & 5 & 13 & 0 & 5 \\ 2 & 5 & 0 & 9 & 0 & 2 \\ 7 & 13 & 9 & 26 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111} 3^1 1^1 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 6 & 3 \\ 4 & 10 & 13 & 0 & 16 & 6 \\ 5 & 13 & 12 & 0 & 17 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 16 & 17 & 0 & 20 & 8 \\ 3 & 6 & 6 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -349, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 5 & 0 & 18 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 18 & 0 & 48 & 13 & 14 \\ 2 & 5 & 0 & 13 & 0 & 2 \\ 3 & 5 & -2 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{12} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 5 & 0 & 23 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 23 & 0 & 84 & 16 & 18 \\ 2 & 5 & 0 & 16 & 0 & 2 \\ 3 & 5 & -2 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{12} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 5 & 0 & 28 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 28 & 0 & 120 & 20 & 22 \\ 2 & 5 & 0 & 20 & 0 & 2 \\ 3 & 5 & -2 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

4.713 NHC(-2)(-3) + so₈ + su₂n3mix + so₈ + su₂

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 15 & 1 & 1 & 5 \\ 11 & 20 & 28 & 0 & 0 & 8 \\ 15 & 28 & 37 & 0 & 0 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 8 & 11 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -117$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 15 & 1 & 10 & 5 \\ 11 & 20 & 28 & 0 & 17 & 8 \\ 15 & 28 & 37 & 0 & 24 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 24 & 0 & 13 & 7 \\ 5 & 8 & 11 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -145$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 16 & 1 & 9 & 5 \\ 11 & 20 & 28 & 0 & 17 & 8 \\ 16 & 28 & 37 & 0 & 24 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 24 & 0 & 13 & 7 \\ 5 & 8 & 11 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -157$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 15 & 1 & 9 & 5 \\ 11 & 20 & 28 & 0 & 17 & 8 \\ 15 & 28 & 37 & 0 & 24 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 24 & 0 & 13 & 7 \\ 5 & 8 & 11 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

4.714 NHC(-2)(-3) + so₈ + su₂n3mix + so₈ + su₃

$T = 9$ [0] $1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 7 & 1 & 9 & 5 \\ 11 & 20 & 11 & 0 & 17 & 8 \\ 7 & 11 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 9 & 0 & 12 & 6 \\ 5 & 8 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

4.715 NHC(-2)(-3) + so₈ + su₂n3mix + su₂

$T = 8$ [0] $1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 19$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 20 & 12 & 6 \\ 15 & 20 & 28 & 17 & 8 \\ 20 & 28 & 37 & 24 & 11 \\ 12 & 17 & 24 & 13 & 7 \\ 6 & 8 & 11 & 7 & 3 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 26$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 20 & 1 & 6 \\ 15 & 20 & 28 & 0 & 8 \\ 20 & 28 & 37 & 0 & 11 \\ 1 & 0 & 0 & -2 & 0 \\ 6 & 8 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

[3] $1^{144} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 9 & 3 \\ 3 & 4 & 0 & 24 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 68 & 18 \\ 9 & 16 & 0 & 68 & 46 & 12 \\ 3 & 4 & -2 & 18 & 12 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 3 \\ 3 & 4 & 0 & 8 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

- [2] $1^{11} 1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 1 & 3 \\ 3 & 4 & 0 & 16 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 12 & 0 & 2 \end{pmatrix}_{b_0}$
- [5] $1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 7 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 17 & 5 \\ 8 & 14 & 0 & 17 & 37 & 11 \\ 3 & 4 & -2 & 5 & 11 & 2 \end{pmatrix}_{b_0}$
- [6] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 7 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 5 & 0 & 2 \end{pmatrix}_{b_0}$
- [7] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 7 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 6 & 5 \\ 3 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -2 & 5 & 4 & 2 \end{pmatrix}_{b_0}$
- [8] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 1 & 3 \\ 3 & 4 & 0 & 12 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 9 & 0 & 2 \end{pmatrix}_{b_0}$
- [10] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 10 & 15 & 20 & 13 & 6 \\ 15 & 20 & 28 & 17 & 8 \\ 20 & 28 & 37 & 24 & 11 \\ 13 & 17 & 24 & 13 & 7 \\ 6 & 8 & 11 & 7 & 3 \end{pmatrix}_{b_0}$
- [9] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 12 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 12 & 9 \\ 4 & 5 & 0 & 12 & 4 & 4 \\ 3 & 4 & -2 & 9 & 4 & 2 \end{pmatrix}_{b_0}$
- [11] $1^{11} 3^{12} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 14 & 4 & 3 \\ 3 & 4 & 0 & 25 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 25 & 0 & 111 & 34 & 19 \\ 4 & 8 & 0 & 34 & 10 & 6 \\ 3 & 4 & -2 & 19 & 6 & 2 \end{pmatrix}_{b_0}$
- [12] $1^{144} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 14 & 11 & 3 \\ 3 & 4 & 0 & 24 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 24 & 0 & 100 & 89 & 18 \\ 11 & 21 & 0 & 89 & 76 & 16 \\ 3 & 4 & -2 & 18 & 16 & 2 \end{pmatrix}_{b_0}$
- [13] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 10 & 15 & 21 & 12 & 6 \\ 15 & 20 & 28 & 17 & 8 \\ 21 & 28 & 37 & 24 & 11 \\ 12 & 17 & 24 & 13 & 7 \\ 6 & 8 & 11 & 7 & 3 \end{pmatrix}_{b_0}$

[14] $1^{1123}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 13 & 3 \\ 3 & 4 & 0 & 7 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 28 & 5 \\ 13 & 23 & 0 & 28 & 100 & 18 \\ 3 & 4 & -2 & 5 & 18 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 7 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 17 & 5 \\ 9 & 14 & 0 & 17 & 37 & 11 \\ 3 & 4 & -2 & 5 & 11 & 2 \end{pmatrix}_{b_0}$$

[16] $21^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 17 & 24 & 15 & 7 \\ 17 & 20 & 28 & 17 & 8 \\ 24 & 28 & 37 & 24 & 11 \\ 15 & 17 & 24 & 13 & 7 \\ 7 & 8 & 11 & 7 & 3 \end{pmatrix}_{b_0}$$

[18] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 8 & 9 \\ 3 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -2 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{133}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 7 & 3 \\ 3 & 4 & 0 & 20 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 42 & 15 \\ 7 & 12 & 0 & 42 & 25 & 9 \\ 3 & 4 & -2 & 15 & 9 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{122}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 11 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 0 & 20 & 16 & 8 \\ 4 & 8 & 0 & 16 & 10 & 6 \\ 3 & 4 & -2 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{112}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 7 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 10 & 5 \\ 4 & 8 & 0 & 10 & 10 & 6 \\ 3 & 4 & -2 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[22] $1^{122}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 0 & 16 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 22 & 12 \\ 4 & 8 & 0 & 22 & 10 & 6 \\ 3 & 4 & -2 & 12 & 6 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{122}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 16 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 22 & 12 \\ 5 & 8 & 0 & 22 & 10 & 6 \\ 3 & 4 & -2 & 12 & 6 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{121}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 & 3 \\ 3 & 4 & 0 & 11 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 8 & 8 \\ 3 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -2 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[24] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 109$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 5 & 5 \\ 2 & 4 & 0 & 5 & 1 & 3 \\ 3 & 4 & -2 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 113$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 5 & 5 \\ 3 & 4 & 0 & 5 & 1 & 3 \\ 3 & 4 & -2 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{142}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 114$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 4 & 3 \\ 3 & 4 & 0 & 29 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 29 & 0 & 148 & 40 & 22 \\ 4 & 8 & 0 & 40 & 10 & 6 \\ 3 & 4 & -2 & 22 & 6 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 115$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 11 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 17 & 8 \\ 5 & 9 & 0 & 17 & 13 & 7 \\ 3 & 4 & -2 & 8 & 7 & 2 \end{pmatrix}_{b_0}$$

[28] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 117$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 7 & 5 & 5 \\ 2 & 4 & 0 & 5 & 1 & 3 \\ 3 & 4 & -2 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{111}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 120$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 12 & 14 & 6 \\ 5 & 9 & 0 & 14 & 13 & 7 \\ 3 & 4 & -2 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{132}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 128$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 4 & 3 \\ 3 & 4 & 0 & 20 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 20 & 0 & 69 & 28 & 15 \\ 4 & 8 & 0 & 28 & 10 & 6 \\ 3 & 4 & -2 & 15 & 6 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 128$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 11 & 12 \\ 3 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & -2 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{113}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 129$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 12 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 36 & 9 \\ 9 & 17 & 0 & 36 & 49 & 13 \\ 3 & 4 & -2 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 142$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 11 & 5 \\ 5 & 9 & 0 & 11 & 13 & 7 \\ 3 & 4 & -2 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{11} 1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 14 & 15 \\ 3 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -2 & 15 & 3 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{11} 1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 145$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 12 & 3 \\ 3 & 4 & 0 & 12 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 44 & 9 \\ 12 & 21 & 0 & 44 & 76 & 16 \\ 3 & 4 & -2 & 9 & 16 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{11} 1^{13} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 149$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 18 & 3 \\ 3 & 4 & 0 & 7 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 39 & 5 \\ 18 & 32 & 0 & 39 & 193 & 25 \\ 3 & 4 & -2 & 5 & 25 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{11} 1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 149$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 20 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 18 & 15 \\ 3 & 5 & 0 & 18 & 4 & 4 \\ 3 & 4 & -2 & 15 & 4 & 2 \end{pmatrix}_{b_0}$$

[38] $1^{11} 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 152$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 12 & 14 & 6 \\ 6 & 9 & 0 & 14 & 13 & 7 \\ 3 & 4 & -2 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{11} 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 153$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 12 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 12 & 9 \\ 3 & 5 & 0 & 12 & 4 & 4 \\ 3 & 4 & -2 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{121} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 157$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 11 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 8 & 8 \\ 2 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -2 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 160$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 8 & 9 \\ 2 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -2 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{11} 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 162$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 11 & 5 \\ 6 & 9 & 0 & 11 & 13 & 7 \\ 3 & 4 & -2 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{112} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 174$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 7 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 10 & 5 \\ 5 & 8 & 0 & 10 & 10 & 6 \\ 3 & 4 & -2 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[44] $1^{12}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 16 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 15 & 12 \\ 3 & 5 & 0 & 15 & 4 & 4 \\ 3 & 4 & -2 & 12 & 4 & 2 \end{pmatrix}_{b_0}$$

[45] $1^{132}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 5 & 3 \\ 3 & 4 & 0 & 20 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 28 & 15 \\ 5 & 8 & 0 & 28 & 10 & 6 \\ 3 & 4 & -2 & 15 & 6 & 2 \end{pmatrix}_{b_0}$$

[46] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 0 & 25 & 8 & 9 \\ 2 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -2 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[47] $1^{14}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 & 3 \\ 3 & 4 & 0 & 24 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 24 & 0 & 100 & 21 & 18 \\ 3 & 5 & 0 & 21 & 4 & 4 \\ 3 & 4 & -2 & 18 & 4 & 2 \end{pmatrix}_{b_0}$$

[48] $1^{123}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 12 & 3 \\ 3 & 4 & 0 & 11 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 42 & 8 \\ 12 & 22 & 0 & 42 & 85 & 17 \\ 3 & 4 & -2 & 8 & 17 & 2 \end{pmatrix}_{b_0}$$

[49] $1^{122}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 11 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 25 & 8 \\ 8 & 13 & 0 & 25 & 28 & 10 \\ 3 & 4 & -2 & 8 & 10 & 2 \end{pmatrix}_{b_0}$$

[50] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 & 3 \\ 3 & 4 & 0 & 12 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 0 & 25 & 12 & 9 \\ 3 & 5 & 0 & 12 & 4 & 4 \\ 3 & 4 & -2 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{122}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 190, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 11 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 16 & 8 \\ 5 & 8 & 0 & 16 & 10 & 6 \\ 3 & 4 & -2 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[52] $1^{143}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 7 & 3 \\ 3 & 4 & 0 & 24 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 51 & 18 \\ 7 & 12 & 0 & 51 & 25 & 9 \\ 3 & 4 & -2 & 18 & 9 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{132}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 0 & 15 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 15 & 0 & 39 & 22 & 11 \\ 4 & 8 & 0 & 22 & 10 & 6 \\ 3 & 4 & -2 & 11 & 6 & 2 \end{pmatrix}_{b_0}$$

[54] $1^{112}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 207$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 7 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 28 & 5 \\ 14 & 23 & 0 & 28 & 100 & 18 \\ 3 & 4 & -2 & 5 & 18 & 2 \end{pmatrix}_{b_0}$$

[55] $1^{124}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 208$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 11 & 3 \\ 3 & 4 & 0 & 16 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 59 & 12 \\ 11 & 21 & 0 & 59 & 76 & 16 \\ 3 & 4 & -2 & 12 & 16 & 2 \end{pmatrix}_{b_0}$$

[56] $1^{133}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 209$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 9 & 3 \\ 3 & 4 & 0 & 20 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 20 & 0 & 69 & 60 & 15 \\ 9 & 17 & 0 & 60 & 49 & 13 \\ 3 & 4 & -2 & 15 & 13 & 2 \end{pmatrix}_{b_0}$$

[57] $1^{112}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 214$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 7 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 22 & 5 \\ 10 & 18 & 0 & 22 & 58 & 14 \\ 3 & 4 & -2 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$

[58] $1^{121}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 216$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 6 & 3 \\ 3 & 4 & 0 & 16 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 26 & 12 \\ 6 & 9 & 0 & 26 & 13 & 7 \\ 3 & 4 & -2 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[59] $1^{113}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 217$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 15 & 5 \\ 7 & 12 & 0 & 15 & 25 & 9 \\ 3 & 4 & -2 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[60] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 219$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 7 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 16 & 5 \\ 7 & 13 & 0 & 16 & 28 & 10 \\ 3 & 4 & -2 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

[61] $1^{122}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 222$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 11 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 25 & 8 \\ 7 & 13 & 0 & 25 & 28 & 10 \\ 3 & 4 & -2 & 8 & 10 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{142}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 224$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 5 & 3 \\ 3 & 4 & 0 & 24 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 34 & 18 \\ 5 & 8 & 0 & 34 & 10 & 6 \\ 3 & 4 & -2 & 18 & 6 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{132}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 224$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 4 & 3 \\ 3 & 4 & 0 & 20 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 28 & 15 \\ 4 & 8 & 0 & 28 & 10 & 6 \\ 3 & 4 & -2 & 15 & 6 & 2 \end{pmatrix}_{b_0}$$

[64] $1^{111}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 229$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 5 & 3 \\ 3 & 4 & 0 & 12 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 0 & 25 & 20 & 9 \\ 5 & 9 & 0 & 20 & 13 & 7 \\ 3 & 4 & -2 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{121}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 229$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 11 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 0 & 20 & 8 & 8 \\ 2 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & -2 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{113}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 233$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 & 3 \\ 3 & 4 & 0 & 20 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 20 & 0 & 69 & 18 & 15 \\ 3 & 5 & 0 & 18 & 4 & 4 \\ 3 & 4 & -2 & 15 & 4 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{123}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 236$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 & 3 \\ 3 & 4 & 0 & 16 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 16 & 0 & 44 & 15 & 12 \\ 3 & 5 & 0 & 15 & 4 & 4 \\ 3 & 4 & -2 & 12 & 4 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{114}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 242$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 20 & 5 \\ 9 & 16 & 0 & 20 & 46 & 12 \\ 3 & 4 & -2 & 5 & 12 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{123}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 257$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 11 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 24 & 8 \\ 7 & 12 & 0 & 24 & 25 & 9 \\ 3 & 4 & -2 & 8 & 9 & 2 \end{pmatrix}_{b_0}$$

[70] $1^{142}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 258$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 4 & 3 \\ 3 & 4 & 0 & 29 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 29 & 0 & 148 & 40 & 22 \\ 4 & 8 & 0 & 40 & 10 & 6 \\ 3 & 4 & -2 & 22 & 6 & 2 \end{pmatrix}_{b_0}$$

[71] $1^{111}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 265$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 0 & 12 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 20 & 9 \\ 6 & 9 & 0 & 20 & 13 & 7 \\ 3 & 4 & -2 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{112}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 265$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 12 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 28 & 9 \\ 7 & 13 & 0 & 28 & 28 & 10 \\ 3 & 4 & -2 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{122}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 268$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 7 & 3 \\ 3 & 4 & 0 & 16 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 16 & 0 & 44 & 37 & 12 \\ 7 & 13 & 0 & 37 & 28 & 10 \\ 3 & 4 & -2 & 12 & 10 & 2 \end{pmatrix}_{b_0}$$

[74] $1^{11}3^{111}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 273$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 12 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 20 & 9 \\ 5 & 9 & 0 & 20 & 13 & 7 \\ 3 & 4 & -2 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[75] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 278$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 7 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 21 & 5 \\ 9 & 17 & 0 & 21 & 49 & 13 \\ 3 & 4 & -2 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[76] $1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 288$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 11 & 12 \\ 2 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & -2 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[77] $1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 311$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 9 & 3 \\ 3 & 4 & 0 & 11 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 33 & 8 \\ 9 & 17 & 0 & 33 & 49 & 13 \\ 3 & 4 & -2 & 8 & 13 & 2 \end{pmatrix}_{b_0}$$

[78] $1^{113}3^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 313$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 0 & 12 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 36 & 9 \\ 10 & 17 & 0 & 36 & 49 & 13 \\ 3 & 4 & -2 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[79] $1^{114}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 319$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 0 & 7 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 26 & 5 \\ 11 & 21 & 0 & 26 & 76 & 16 \\ 3 & 4 & -2 & 5 & 16 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{142}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 320$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 4 & 3 \\ 3 & 4 & 0 & 24 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 24 & 0 & 100 & 34 & 18 \\ 4 & 8 & 0 & 34 & 10 & 6 \\ 3 & 4 & -2 & 18 & 6 & 2 \end{pmatrix}_{b_0}$$

[81] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 327$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 7 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 16 & 5 \\ 8 & 13 & 0 & 16 & 28 & 10 \\ 3 & 4 & -2 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

[82] $1^{113}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 334$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 15 & 3 \\ 3 & 4 & 0 & 7 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 33 & 5 \\ 15 & 27 & 0 & 33 & 133 & 21 \\ 3 & 4 & -2 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{112}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 338$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 0 & 7 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 22 & 5 \\ 11 & 18 & 0 & 22 & 58 & 14 \\ 3 & 4 & -2 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$

[84] $1^{11} 1^{131} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 341, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 15 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 15 & 0 & 39 & 11 & 11 \\ 2 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & -2 & 11 & 3 & 2 \end{pmatrix}_{b_0}$$

[85] $1^{11} 1^{121} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 349, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 80 & 14 & 16 \\ 2 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -2 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[86] $1^{142} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 4 & 3 \\ 3 & 4 & 0 & 24 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 34 & 18 \\ 4 & 8 & 0 & 34 & 10 & 6 \\ 3 & 4 & -2 & 18 & 6 & 2 \end{pmatrix}_{b_0}$$

[87] $1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 353, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 8 & 3 \\ 3 & 4 & 0 & 12 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 28 & 9 \\ 8 & 13 & 0 & 28 & 28 & 10 \\ 3 & 4 & -2 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[88] $1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 16 & 0 & 44 & 11 & 12 \\ 2 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & -2 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[89] $1^{121} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 16 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 26 & 12 \\ 5 & 9 & 0 & 26 & 13 & 7 \\ 3 & 4 & -2 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[90] $1^{113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 377, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 7 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 27 & 5 \\ 12 & 22 & 0 & 27 & 85 & 17 \\ 3 & 4 & -2 & 5 & 17 & 2 \end{pmatrix}_{b_0}$$

[91] $1^{132} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 8 & 3 \\ 3 & 4 & 0 & 20 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 46 & 15 \\ 8 & 13 & 0 & 46 & 28 & 10 \\ 3 & 4 & -2 & 15 & 10 & 2 \end{pmatrix}_{b_0}$$

[92] $1^{124} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 11 & 3 \\ 3 & 4 & 0 & 11 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 41 & 8 \\ 11 & 21 & 0 & 41 & 76 & 16 \\ 3 & 4 & -2 & 8 & 16 & 2 \end{pmatrix}_{b_0}$$

[93] $1^{123} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 387, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 10 & 3 \\ 3 & 4 & 0 & 11 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 33 & 8 \\ 10 & 17 & 0 & 33 & 49 & 13 \\ 3 & 4 & -2 & 8 & 13 & 2 \end{pmatrix}_{b_0}$$

[94] $1^{124}3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 14 & 3 \\ 3 & 4 & 0 & 11 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 50 & 8 \\ 14 & 26 & 0 & 50 & 118 & 20 \\ 3 & 4 & -2 & 8 & 20 & 2 \end{pmatrix}_{b_0}$$

[95] $1^{123}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 0 & 16 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 48 & 12 \\ 9 & 17 & 0 & 48 & 49 & 13 \\ 3 & 4 & -2 & 12 & 13 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{111}3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 393, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 19 & 3 \\ 3 & 4 & 0 & 7 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 39 & 5 \\ 19 & 32 & 0 & 39 & 193 & 25 \\ 3 & 4 & -2 & 5 & 25 & 2 \end{pmatrix}_{b_0}$$

[97] $1^{121}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 396, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 5 & 3 \\ 3 & 4 & 0 & 16 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 16 & 0 & 44 & 26 & 12 \\ 5 & 9 & 0 & 26 & 13 & 7 \\ 3 & 4 & -2 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[98] $1^{131}3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 14 & 15 \\ 2 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -2 & 15 & 3 & 2 \end{pmatrix}_{b_0}$$

[99] $1^{131}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 5 & 3 \\ 3 & 4 & 0 & 20 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 32 & 15 \\ 5 & 9 & 0 & 32 & 13 & 7 \\ 3 & 4 & -2 & 15 & 7 & 2 \end{pmatrix}_{b_0}$$

[100] $1^{141}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 5 & 3 \\ 3 & 4 & 0 & 24 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 38 & 18 \\ 5 & 9 & 0 & 38 & 13 & 7 \\ 3 & 4 & -2 & 18 & 7 & 2 \end{pmatrix}_{b_0}$$

[101] $1^{121}3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 26 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 26 & 0 & 128 & 17 & 20 \\ 2 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & -2 & 20 & 3 & 2 \end{pmatrix}_{b_0}$$

[102] $1^{121}3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 437, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 21 & 0 & 80 & 14 & 16 \\ 2 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -2 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[103] $1^{122}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 8 & 3 \\ 3 & 4 & 0 & 16 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 37 & 12 \\ 8 & 13 & 0 & 37 & 28 & 10 \\ 3 & 4 & -2 & 12 & 10 & 2 \end{pmatrix}_{b_0}$$

[104] $1^{122}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 448$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 7 & 3 \\ 3 & 4 & 0 & 16 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 37 & 12 \\ 7 & 13 & 0 & 37 & 28 & 10 \\ 3 & 4 & -2 & 12 & 10 & 2 \end{pmatrix}_{b_0}$$

[105] $1^{141}3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 453$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 19 & 0 & 64 & 14 & 14 \\ 2 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -2 & 14 & 3 & 2 \end{pmatrix}_{b_0}$$

[106] $1^{113}3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 474$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 7 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 21 & 5 \\ 10 & 17 & 0 & 21 & 49 & 13 \\ 3 & 4 & -2 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[107] $1^{124}3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 482$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 15 & 3 \\ 3 & 4 & 0 & 11 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 50 & 8 \\ 15 & 26 & 0 & 50 & 118 & 20 \\ 3 & 4 & -2 & 8 & 20 & 2 \end{pmatrix}_{b_0}$$

[108] $1^{134}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 501$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.87 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 11 & 3 \\ 3 & 4 & 0 & 20 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 74 & 15 \\ 11 & 21 & 0 & 74 & 76 & 16 \\ 3 & 4 & -2 & 15 & 16 & 2 \end{pmatrix}_{b_0}$$

[109] $1^{114}3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 502$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 20 & 3 \\ 3 & 4 & 0 & 7 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 44 & 5 \\ 20 & 36 & 0 & 44 & 238 & 28 \\ 3 & 4 & -2 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$

[110] $1^{132}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 505$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 7 & 3 \\ 3 & 4 & 0 & 20 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 20 & 0 & 69 & 46 & 15 \\ 7 & 13 & 0 & 46 & 28 & 10 \\ 3 & 4 & -2 & 15 & 10 & 2 \end{pmatrix}_{b_0}$$

[111] $1^{131}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 513$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 5 & 3 \\ 3 & 4 & 0 & 20 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 20 & 0 & 69 & 32 & 15 \\ 5 & 9 & 0 & 32 & 13 & 7 \\ 3 & 4 & -2 & 15 & 7 & 2 \end{pmatrix}_{b_0}$$

[112] $1^{143}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 516$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 9 & 3 \\ 3 & 4 & 0 & 24 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 24 & 0 & 100 & 72 & 18 \\ 9 & 17 & 0 & 72 & 49 & 13 \\ 3 & 4 & -2 & 18 & 13 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{114}3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 522$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 7 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 32 & 5 \\ 14 & 26 & 0 & 32 & 118 & 20 \\ 3 & 4 & -2 & 5 & 20 & 2 \end{pmatrix}_{b_0}$$

[114] $1^{1^{11}} 1^{124} 3^1 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 12 & 3 \\ 3 & 4 & 0 & 16 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 59 & 12 \\ 12 & 21 & 0 & 59 & 76 & 16 \\ 3 & 4 & -2 & 12 & 16 & 2 \end{pmatrix}_{b_0}$$

[115] $1^{1^{11}} 1^{141} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 17 & 18 \\ 2 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & -2 & 18 & 3 & 2 \end{pmatrix}_{b_0}$$

[116] $1^{1^{11}} 1^{143} 3^1 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 10 & 3 \\ 3 & 4 & 0 & 24 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 72 & 18 \\ 10 & 17 & 0 & 72 & 49 & 13 \\ 3 & 4 & -2 & 18 & 13 & 2 \end{pmatrix}_{b_0}$$

[117] $1^{1^{11}} 1^{123} 3^1 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 552, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 10 & 3 \\ 3 & 4 & 0 & 16 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 48 & 12 \\ 10 & 17 & 0 & 48 & 49 & 13 \\ 3 & 4 & -2 & 12 & 13 & 2 \end{pmatrix}_{b_0}$$

[118] $1^{1^{11}} 1^{131} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 557, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 25 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 25 & 0 & 111 & 17 & 19 \\ 2 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & -2 & 19 & 3 & 2 \end{pmatrix}_{b_0}$$

[119] $1^{1^{11}} 1^{131} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 560, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 20 & 0 & 69 & 14 & 15 \\ 2 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & -2 & 15 & 3 & 2 \end{pmatrix}_{b_0}$$

[120] $1^{1^{11}} 1^{113} 3^1 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 562, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 16 & 3 \\ 3 & 4 & 0 & 7 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 33 & 5 \\ 16 & 27 & 0 & 33 & 133 & 21 \\ 3 & 4 & -2 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$

[121] $1^{1^{11}} 1^{124} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 570, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 12 & 3 \\ 3 & 4 & 0 & 11 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 41 & 8 \\ 12 & 21 & 0 & 41 & 76 & 16 \\ 3 & 4 & -2 & 8 & 16 & 2 \end{pmatrix}_{b_0}$$

[122] $1^{1^{11}} 1^{141} 3^1 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 5 & 3 \\ 3 & 4 & 0 & 24 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 24 & 0 & 100 & 38 & 18 \\ 5 & 9 & 0 & 38 & 13 & 7 \\ 3 & 4 & -2 & 18 & 7 & 2 \end{pmatrix}_{b_0}$$

[123] $1^{1^{11}} 1^{132} 3^1 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 581, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 7 & 3 \\ 3 & 4 & 0 & 20 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 46 & 15 \\ 7 & 13 & 0 & 46 & 28 & 10 \\ 3 & 4 & -2 & 15 & 10 & 2 \end{pmatrix}_{b_0}$$

[124] $1^{114}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 583, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 17 & 3 \\ 3 & 4 & 0 & 7 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 38 & 5 \\ 17 & 31 & 0 & 38 & 172 & 24 \\ 3 & 4 & -2 & 5 & 24 & 2 \end{pmatrix}_{b_0}$$

[125] $1^{113}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 589, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 13 & 3 \\ 3 & 4 & 0 & 7 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 27 & 5 \\ 13 & 22 & 0 & 27 & 85 & 17 \\ 3 & 4 & -2 & 5 & 17 & 2 \end{pmatrix}_{b_0}$$

[126] $1^{114}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 7 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 26 & 5 \\ 12 & 21 & 0 & 26 & 76 & 16 \\ 3 & 4 & -2 & 5 & 16 & 2 \end{pmatrix}_{b_0}$$

[127] $1^{133}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 605, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 9 & 3 \\ 3 & 4 & 0 & 20 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 60 & 15 \\ 9 & 17 & 0 & 60 & 49 & 13 \\ 3 & 4 & -2 & 15 & 13 & 2 \end{pmatrix}_{b_0}$$

[128] $1^{131}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 628, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 30 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 30 & 0 & 165 & 20 & 23 \\ 2 & 4 & 0 & 20 & 1 & 3 \\ 3 & 4 & -2 & 23 & 3 & 2 \end{pmatrix}_{b_0}$$

[129] $1^{131}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 0 & 35 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 20 & 35 & 0 & 231 & 23 & 27 \\ 2 & 4 & 0 & 23 & 1 & 3 \\ 3 & 4 & -2 & 27 & 3 & 2 \end{pmatrix}_{b_0}$$

[130] $1^{133}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 10 & 3 \\ 3 & 4 & 0 & 20 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 60 & 15 \\ 10 & 17 & 0 & 60 & 49 & 13 \\ 3 & 4 & -2 & 15 & 13 & 2 \end{pmatrix}_{b_0}$$

[131] $1^{142}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 664, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 7 & 3 \\ 3 & 4 & 0 & 24 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 55 & 18 \\ 7 & 13 & 0 & 55 & 28 & 10 \\ 3 & 4 & -2 & 18 & 10 & 2 \end{pmatrix}_{b_0}$$

[132] $1^{142}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 692, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 7 & 3 \\ 3 & 4 & 0 & 24 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 24 & 0 & 100 & 55 & 18 \\ 7 & 13 & 0 & 55 & 28 & 10 \\ 3 & 4 & -2 & 18 & 10 & 2 \end{pmatrix}_{b_0}$$

[133] $1^{131}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 709, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 25 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 25 & 0 & 111 & 17 & 19 \\ 2 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & -2 & 19 & 3 & 2 \end{pmatrix}_{b_0}$$

[134] $1^{144}3^11^{111}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{blue}{so}_8+\textcolor{blue}{su}_2+\textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 744, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 13 & 11 & 3 \\ 3 & 4 & 0 & 24 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 89 & 18 \\ 11 & 21 & 0 & 89 & 76 & 16 \\ 3 & 4 & -2 & 18 & 16 & 2 \end{pmatrix}_{b_0}$

[135] $1^{134}3^11^{111}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{blue}{so}_8+\textcolor{blue}{su}_2+\textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 749, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 11 & 12 & 3 \\ 3 & 4 & 0 & 20 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 74 & 15 \\ 12 & 21 & 0 & 74 & 76 & 16 \\ 3 & 4 & -2 & 15 & 16 & 2 \end{pmatrix}_{b_0}$

[136] $1^{141}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{blue}{so}_8+\textcolor{blue}{su}_2+\textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 752, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 24 & 0 & 100 & 17 & 18 \\ 2 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & -2 & 18 & 3 & 2 \end{pmatrix}_{b_0}$

[137] $1^{141}3^11^{12}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{blue}{so}_8+\textcolor{blue}{su}_2+\textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 765, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 29 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 29 & 0 & 148 & 20 & 22 \\ 2 & 4 & 0 & 20 & 1 & 3 \\ 3 & 4 & -2 & 22 & 3 & 2 \end{pmatrix}_{b_0}$

[138] $1^{143}3^11^{111}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{blue}{so}_8+\textcolor{blue}{su}_2+\textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 13 & 9 & 3 \\ 3 & 4 & 0 & 24 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 72 & 18 \\ 9 & 17 & 0 & 72 & 49 & 13 \\ 3 & 4 & -2 & 18 & 13 & 2 \end{pmatrix}_{b_0}$

[139] $1^{131}3^11^{13}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{blue}{so}_8+\textcolor{blue}{su}_2+\textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 788, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 0 & 30 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 18 & 30 & 0 & 165 & 20 & 23 \\ 2 & 4 & 0 & 20 & 1 & 3 \\ 3 & 4 & -2 & 23 & 3 & 2 \end{pmatrix}_{b_0}$

[140] $1^{131}3^11^{14}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{blue}{so}_8+\textcolor{blue}{su}_2+\textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 797, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 21 & 2 & 3 \\ 3 & 4 & 0 & 35 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 21 & 35 & 0 & 231 & 23 & 27 \\ 2 & 4 & 0 & 23 & 1 & 3 \\ 3 & 4 & -2 & 27 & 3 & 2 \end{pmatrix}_{b_0}$

[141] $1^{144}3^11^{111}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{blue}{so}_8+\textcolor{blue}{su}_2+\textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 808, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 13 & 12 & 3 \\ 3 & 4 & 0 & 24 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 89 & 18 \\ 12 & 21 & 0 & 89 & 76 & 16 \\ 3 & 4 & -2 & 18 & 16 & 2 \end{pmatrix}_{b_0}$

[142] $1^{114}3^11^{12}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{blue}{so}_8+\textcolor{blue}{su}_2+\textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 822, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 15 & 3 \\ 3 & 4 & 0 & 7 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 32 & 5 \\ 15 & 26 & 0 & 32 & 118 & 20 \\ 3 & 4 & -2 & 5 & 20 & 2 \end{pmatrix}_{b_0}$

[143] $1^{114}3^11^{14}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{blue}{so}_8+\textcolor{blue}{su}_2+\textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 834, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 21 & 3 \\ 3 & 4 & 0 & 7 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 44 & 5 \\ 21 & 36 & 0 & 44 & 238 & 28 \\ 3 & 4 & -2 & 5 & 28 & 2 \end{pmatrix}_{b_0}$

[144] $1^{11} 1^{14} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 899, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 18 & 3 \\ 3 & 4 & 0 & 7 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 38 & 5 \\ 18 & 31 & 0 & 38 & 172 & 24 \\ 3 & 4 & -2 & 5 & 24 & 2 \end{pmatrix}_{b_0}$$

[145] $1^{11} 1^{14} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 916, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 2 & 3 \\ 3 & 4 & 0 & 34 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 19 & 34 & 0 & 208 & 23 & 26 \\ 2 & 4 & 0 & 23 & 1 & 3 \\ 3 & 4 & -2 & 26 & 3 & 2 \end{pmatrix}_{b_0}$$

[146] $1^{11} 1^{14} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 981, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 29 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 29 & 0 & 148 & 20 & 22 \\ 2 & 4 & 0 & 20 & 1 & 3 \\ 3 & 4 & -2 & 22 & 3 & 2 \end{pmatrix}_{b_0}$$

[147] $1^{11} 1^{14} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 997, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 2 & 3 \\ 3 & 4 & 0 & 39 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 22 & 39 & 0 & 280 & 26 & 30 \\ 2 & 4 & 0 & 26 & 1 & 3 \\ 3 & 4 & -2 & 30 & 3 & 2 \end{pmatrix}_{b_0}$$

[148] $1^{11} 1^{14} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 1140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 0 & 34 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 20 & 34 & 0 & 208 & 23 & 26 \\ 2 & 4 & 0 & 23 & 1 & 3 \\ 3 & 4 & -2 & 26 & 3 & 2 \end{pmatrix}_{b_0}$$

[149] $1^{11} 1^{14} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 1229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 2 & 3 \\ 3 & 4 & 0 & 39 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 23 & 39 & 0 & 280 & 26 & 30 \\ 2 & 4 & 0 & 26 & 1 & 3 \\ 3 & 4 & -2 & 30 & 3 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [150] $1^{11} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 5 & 3 \\ 2 & 1 & 1 & 9 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 20 & 11 \\ 5 & 4 & 0 & 20 & 10 & 6 \\ 3 & 3 & -1 & 11 & 6 & 3 \end{pmatrix}_{b_0}$$

[151] $1^{11} 1^{11} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 4 \\ 1 & 2 & 0 & 0 & 5 & 7 \\ 1 & 0 & 0 & -4 & 0 & 0 \\ 3 & 4 & 5 & 0 & 10 & 14 \\ 3 & 4 & 7 & 0 & 14 & 16 \\ 2 & 2 & 2 & -2 & 6 & 8 \end{pmatrix}_{b_0}$$

[152] $1^{11} 1^{122} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 7 & 2 \\ 1 & 2 & 0 & 0 & 9 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 41 & 11 \\ 5 & 7 & 11 & 0 & 41 & 45 & 13 \\ 2 & 2 & 2 & -2 & 11 & 13 & 2 \end{pmatrix}_{b_0}$$

[153] $1^{11} 1^{13} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 2 & 3 \\ 3 & 5 & 5 & 13 & 5 & 5 \\ 3 & 5 & 0 & 9 & 4 & 2 \\ 7 & 13 & 9 & 26 & 9 & 10 \\ 2 & 5 & 4 & 9 & 2 & 4 \\ 3 & 5 & 2 & 10 & 4 & 3 \end{pmatrix}_{b_0}$$

[154] $1^{113}3^{121}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -27$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.8$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 11 & 3 \\ 2 & 1 & 1 & 9 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 47 & 11 \\ 11 & 13 & 0 & 47 & 61 & 15 \\ 3 & 3 & -1 & 11 & 15 & 3 \end{pmatrix}_{b_0}$$

[155] $1^{11}3^{121}1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -27$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 5 & 0 & 13 & 10 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 13 & 0 & 24 & 19 & 10 \\ 5 & 10 & 0 & 19 & 14 & 8 \\ 3 & 5 & -2 & 10 & 8 & 3 \end{pmatrix}_{b_0}$$

[156] $1^{11}3^{121}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 0 & 4 & 1 & 2 \\ 1 & 2 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 5 & 6 \\ 1 & 1 & 3 & 0 & 5 & 1 & 3 \\ 2 & 2 & 2 & -2 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[157] $1^{112}3^{121}1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.5$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 0 & 5 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 23 & 6 \\ 5 & 7 & 11 & 0 & 23 & 45 & 13 \\ 2 & 2 & 2 & -2 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[158] $1^{121}3^{121}1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 2 & 0 & 7 & 4 & 2 \\ 1 & 2 & 0 & 0 & 9 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 25 & 11 \\ 3 & 4 & 7 & 0 & 25 & 16 & 8 \\ 2 & 2 & 2 & -2 & 11 & 8 & 2 \end{pmatrix}_{b_0}$$

[159] $1^{113}3^{121}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -30$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.8$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 10 & 3 \\ 2 & 1 & 1 & 9 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 47 & 11 \\ 10 & 13 & 0 & 47 & 61 & 15 \\ 3 & 3 & -1 & 11 & 15 & 3 \end{pmatrix}_{b_0}$$

[160] $1^{112}3^{121}1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -30$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 12 & 3 \\ 2 & 1 & 1 & 9 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 49 & 11 \\ 12 & 11 & 0 & 49 & 65 & 15 \\ 3 & 3 & -1 & 11 & 15 & 3 \end{pmatrix}_{b_0}$$

[161] $1^{111}3^{121}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 2 & 3 \\ 2 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 9 & 11 \\ 2 & 3 & 0 & 9 & 1 & 3 \\ 3 & 3 & -1 & 11 & 3 & 3 \end{pmatrix}_{b_0}$$

[162] $1^{132}3^{121}1^{133} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -37$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.5$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 5 & 2 \\ 1 & 0 & 2 & 0 & 10 & 7 & 2 \\ 1 & 2 & 0 & 0 & 13 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 10 & 13 & 0 & 74 & 59 & 16 \\ 5 & 7 & 11 & 0 & 59 & 45 & 13 \\ 2 & 2 & 2 & -2 & 16 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[163] \ 1^{11} 3^{11} 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -44, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 5 & 0 & 8 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 8 & 6 & 6 \\ 2 & 5 & 0 & 6 & 2 & 4 \\ 3 & 5 & -2 & 6 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[164] \ 1^{11} 3^{11} 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -50, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.96 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 5 & 0 & 13 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 13 & 0 & 24 & 10 & 10 \\ 2 & 5 & 0 & 10 & 2 & 4 \\ 3 & 5 & -2 & 10 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[165] \ 1^{11} 3^{11} 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -53, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.96 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 17 & 6 \\ 4 & 7 & 6 & 0 & 17 & 25 & 9 \\ 2 & 2 & 2 & -2 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[166] \ 1^{123} 3^{11} 21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -53, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.96 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 10 & 2 \\ 1 & 2 & 0 & 0 & 9 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 57 & 11 \\ 7 & 10 & 15 & 0 & 57 & 88 & 18 \\ 2 & 2 & 2 & -2 & 11 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[167] \ 1^{123} 3^{11} 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -62, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.97 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 12 & 3 \\ 3 & 5 & 0 & 13 & 25 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 13 & 0 & 24 & 48 & 10 \\ 12 & 25 & 0 & 48 & 94 & 20 \\ 3 & 5 & -2 & 10 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[168] \ 1^{111} 3^{11} 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -62, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 6 & 3 \\ 2 & 1 & 1 & 9 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 29 & 11 \\ 6 & 7 & 0 & 29 & 21 & 9 \\ 3 & 3 & -1 & 11 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[169] \ 1^{111} 3^{11} 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -63, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 6 & 3 \\ 2 & 1 & 1 & 9 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 9 & 9 & 0 & 35 & 29 & 11 \\ 6 & 7 & 0 & 29 & 21 & 9 \\ 3 & 3 & -1 & 11 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[170] \ 1^{121} 3^{11} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -69, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 2 & 3 \\ 2 & 1 & 1 & 16 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 15 & 16 & 0 & 116 & 16 & 20 \\ 2 & 3 & 0 & 16 & 1 & 3 \\ 3 & 3 & -1 & 20 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[171] \ 1^{111} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -75, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.95 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 2 & 3 \\ 2 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 9 & 9 & 0 & 35 & 9 & 11 \\ 2 & 3 & 0 & 9 & 1 & 3 \\ 3 & 3 & -1 & 11 & 3 & 3 \end{pmatrix}_{b_0}$$

[172] $1^{11}3^{124}21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 11 & 3 \\ 3 & 5 & 0 & 13 & 25 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 13 & 0 & 24 & 49 & 10 \\ 11 & 25 & 0 & 49 & 92 & 20 \\ 3 & 5 & -2 & 10 & 20 & 3 \end{pmatrix}_{b_0}$$

[173] $1^{131}3^{121}21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 2 & 0 & 10 & 4 & 2 \\ 1 & 2 & 0 & 0 & 13 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 10 & 13 & 0 & 74 & 36 & 16 \\ 3 & 4 & 7 & 0 & 36 & 16 & 8 \\ 2 & 2 & 2 & -2 & 16 & 8 & 2 \end{pmatrix}_{b_0}$$

[174] $1^{11}3^{121}21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 5 & 0 & 8 & 20 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 8 & 24 & 6 \\ 9 & 20 & 0 & 24 & 58 & 16 \\ 3 & 5 & -2 & 6 & 16 & 3 \end{pmatrix}_{b_0}$$

[175] $1^{11}3^{121}21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 4 & 0 & 4 & 6 & 4 \\ 2 & 3 & 4 & 0 & 6 & 5 & 5 \\ 2 & 2 & 2 & -2 & 4 & 5 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{123}3^{121}21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 0 & 7 & 1 & 2 \\ 1 & 2 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 9 & 11 \\ 1 & 1 & 3 & 0 & 9 & 1 & 3 \\ 2 & 2 & 2 & -2 & 11 & 3 & 2 \end{pmatrix}_{b_0}$$

[177] $1^{142}3^{121}21^{143} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 5 & 2 \\ 1 & 0 & 2 & 0 & 13 & 7 & 2 \\ 1 & 2 & 0 & 0 & 17 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 13 & 17 & 0 & 127 & 77 & 21 \\ 5 & 7 & 11 & 0 & 77 & 45 & 13 \\ 2 & 2 & 2 & -2 & 21 & 13 & 2 \end{pmatrix}_{b_0}$$

[178] $1^{124}3^{121}21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 10 & 2 \\ 1 & 2 & 0 & 0 & 9 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 46 & 11 \\ 6 & 10 & 10 & 0 & 46 & 58 & 14 \\ 2 & 2 & 2 & -2 & 11 & 14 & 2 \end{pmatrix}_{b_0}$$

[179] $1^{111}3^{121}21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 & 2 \\ 1 & 2 & 0 & 0 & 5 & 1 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 4 & 6 \\ 1 & 2 & 1 & 0 & 4 & 0 & 2 \\ 2 & 2 & 2 & -2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[180] $1^{121}3^{121}21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 6 & 3 \\ 2 & 1 & 1 & 12 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 36 & 14 \\ 6 & 7 & 0 & 36 & 21 & 9 \\ 3 & 3 & -1 & 14 & 9 & 3 \end{pmatrix}_{b_0}$$

[181] $1^{123}3^{121}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$\begin{aligned}
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 10 & 14 & 3 \\ 2 & 1 & 1 & 12 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 84 & 14 \\ 14 & 17 & 0 & 84 & 121 & 21 \\ 3 & 3 & -1 & 14 & 21 & 3 \end{pmatrix}_{b_0} \\
& [182] \ 1^{\overset{1^1}{113}3^1 21^{\overset{1^1}{114}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8 \\
& T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 0 & 5 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 32 & 6 \\ 7 & 10 & 15 & 0 & 32 & 88 & 18 \\ 2 & 2 & 2 & -2 & 6 & 18 & 2 \end{pmatrix}_{b_0} \\
& [183] \ 1^{\overset{1^1}{113}3^1 21^{\overset{1^1}{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 5 & 0 & 8 & 10 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 8 & 12 & 6 \\ 5 & 10 & 0 & 12 & 14 & 8 \\ 3 & 5 & -2 & 6 & 8 & 3 \end{pmatrix}_{b_0} \\
& [184] \ 1^{\overset{1^1}{112}3^1 21^{\overset{1^1}{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 8 & 8 & 3 \\ 2 & 1 & 1 & 9 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 38 & 11 \\ 8 & 10 & 0 & 38 & 38 & 12 \\ 3 & 3 & -1 & 11 & 12 & 3 \end{pmatrix}_{b_0} \\
& [185] \ 1^{\overset{1^{11}}{122}3^1 21^{\overset{1^{11}}{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 11 & 8 & 3 \\ 2 & 1 & 1 & 12 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 11 & 12 & 0 & 56 & 48 & 14 \\ 8 & 10 & 0 & 48 & 38 & 12 \\ 3 & 3 & -1 & 14 & 12 & 3 \end{pmatrix}_{b_0} \\
& [186] \ 1^{\overset{1^1}{113}3^1 21^{\overset{1^1}{112}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 8 & 14 & 3 \\ 2 & 1 & 1 & 9 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 67 & 11 \\ 14 & 17 & 0 & 67 & 121 & 21 \\ 3 & 3 & -1 & 11 & 21 & 3 \end{pmatrix}_{b_0} \\
& [187] \ 1^{\overset{1^{11}}{111}3^1 21^{\overset{1^{11}}{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 8 & 7 & 3 \\ 2 & 1 & 1 & 9 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 29 & 11 \\ 7 & 7 & 0 & 29 & 21 & 9 \\ 3 & 3 & -1 & 11 & 9 & 3 \end{pmatrix}_{b_0} \\
& [188] \ 1^{\overset{1^1}{113}3^1 21^{\overset{1^1}{122}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 5 & 0 & 13 & 15 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 13 & 0 & 24 & 29 & 10 \\ 7 & 15 & 0 & 29 & 32 & 12 \\ 3 & 5 & -2 & 10 & 12 & 3 \end{pmatrix}_{b_0} \\
& [189] \ 1^{\overset{1^{11}}{122}3^1 21^{\overset{1^{11}}{121}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 15 & 8 & 3 \\ 2 & 1 & 1 & 16 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 15 & 16 & 0 & 116 & 68 & 20 \\ 8 & 10 & 0 & 68 & 38 & 12 \\ 3 & 3 & -1 & 20 & 12 & 3 \end{pmatrix}_{b_0} \\
& [190] \ 1^{\overset{1^1}{131}3^1 21^{\overset{1^1}{12}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0
\end{aligned}$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 2 & 3 \\ 2 & 1 & 1 & 19 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 16 & 19 & 0 & 151 & 19 & 23 \\ 2 & 3 & 0 & 19 & 1 & 3 \\ 3 & 3 & -1 & 23 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[191] \ 1^{1^{12}} 1^{13} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 22 & 2 & 3 \\ 2 & 1 & 1 & 23 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 22 & 23 & 0 & 243 & 23 & 29 \\ 2 & 3 & 0 & 23 & 1 & 3 \\ 3 & 3 & -1 & 29 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[192] \ 1^{1^{11}} 1^{12} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 9 & 3 \\ 2 & 1 & 1 & 9 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 38 & 11 \\ 9 & 10 & 0 & 38 & 38 & 12 \\ 3 & 3 & -1 & 11 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[193] \ 1^{1^{21}} 1^{21} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 2 & 3 \\ 2 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 12 & 14 \\ 2 & 3 & 0 & 12 & 1 & 3 \\ 3 & 3 & -1 & 14 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[194] \ 1^{1^{13}} 1^{21} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 5 & 0 & 8 & 15 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 8 & 18 & 6 \\ 7 & 15 & 0 & 18 & 32 & 12 \\ 3 & 5 & -2 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[195] \ 1^{1^{12}} 1^{21} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 2 & 3 \\ 2 & 1 & 1 & 16 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 16 & 16 & 0 & 116 & 16 & 20 \\ 2 & 3 & 0 & 16 & 1 & 3 \\ 3 & 3 & -1 & 20 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[196] \ 1^{1^{13}} 1^{21} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 0 & 4 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 4 & 0 & 4 & 12 & 4 \\ 4 & 6 & 8 & 0 & 12 & 26 & 10 \\ 2 & 2 & 2 & -2 & 4 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[197] \ 1^{1^{13}} 1^{21} 3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 9 & 3 \\ 3 & 5 & 0 & 13 & 20 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 13 & 0 & 24 & 39 & 10 \\ 9 & 20 & 0 & 39 & 58 & 16 \\ 3 & 5 & -2 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[199] \ 1^{1^{12}} 1^{12} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 18 & 3 \\ 2 & 1 & 1 & 9 & 18 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 78 & 11 \\ 18 & 18 & 0 & 78 & 166 & 24 \\ 3 & 3 & -1 & 11 & 24 & 3 \end{pmatrix}_{b_0}$$

[198] $1^{114}3^{121}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 8 & 16 & 3 \\ 2 & 1 & 1 & 9 & 20 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 76 & 11 \\ 16 & 20 & 0 & 76 & 158 & 24 \\ 3 & 3 & -1 & 11 & 24 & 3 \end{pmatrix}_{b_0}$

[200] $1^{123}3^{121}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 5 & 0 & 8 & 20 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 8 & 24 & 6 \\ 10 & 20 & 0 & 24 & 62 & 16 \\ 3 & 5 & -2 & 6 & 16 & 3 \end{pmatrix}_{b_0}$

[201] $1^{123}3^{121}21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 0 & 8 & 1 & 2 \\ 1 & 2 & 0 & 0 & 12 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 12 & 0 & 56 & 12 & 14 \\ 1 & 1 & 3 & 0 & 12 & 1 & 3 \\ 2 & 2 & 2 & -2 & 14 & 3 & 2 \end{pmatrix}_{b_0}$

[202] $1^{141}3^{121}21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -149, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 3 & 2 \\ 1 & 0 & 2 & 0 & 13 & 4 & 2 \\ 1 & 2 & 0 & 0 & 17 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 13 & 17 & 0 & 127 & 47 & 21 \\ 3 & 4 & 7 & 0 & 47 & 16 & 8 \\ 2 & 2 & 2 & -2 & 21 & 8 & 2 \end{pmatrix}_{b_0}$

[203] $1^{121}3^{112}21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 16 & 6 & 3 \\ 2 & 1 & 1 & 16 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 16 & 16 & 0 & 116 & 52 & 20 \\ 6 & 7 & 0 & 52 & 21 & 9 \\ 3 & 3 & -1 & 20 & 9 & 3 \end{pmatrix}_{b_0}$

[204] $1^{131}3^{112}21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 0 & 10 & 1 & 2 \\ 1 & 2 & 0 & 0 & 13 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 10 & 13 & 0 & 74 & 13 & 16 \\ 1 & 1 & 3 & 0 & 13 & 1 & 3 \\ 2 & 2 & 2 & -2 & 16 & 3 & 2 \end{pmatrix}_{b_0}$

[205] $1^{113}3^{121}21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 5 & 0 & 23 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 23 & 0 & 78 & 17 & 18 \\ 2 & 5 & 0 & 17 & 2 & 4 \\ 3 & 5 & -2 & 18 & 4 & 3 \end{pmatrix}_{b_0}$

[206] $1^{131}3^{121}21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -170, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 16 & 6 & 3 \\ 2 & 1 & 1 & 19 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 16 & 19 & 0 & 151 & 59 & 23 \\ 6 & 7 & 0 & 59 & 21 & 9 \\ 3 & 3 & -1 & 23 & 9 & 3 \end{pmatrix}_{b_0}$

[207] $1^{132}3^{111}21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -170, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$\begin{aligned}
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 21 & 8 & 3 \\ 2 & 1 & 1 & 23 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 21 & 23 & 0 & 243 & 98 & 29 \\ 8 & 10 & 0 & 98 & 38 & 12 \\ 3 & 3 & -1 & 29 & 12 & 3 \end{pmatrix}_{b_0} \\
[208] & 1^{111} 3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 4 \\ 1 & 2 & 0 & 0 & 8 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 8 & 0 & 23 & 21 \\ 3 & 4 & 7 & 0 & 21 & 16 \\ 2 & 2 & 2 & -2 & 9 & 8 \end{pmatrix}_{b_0} \\
[209] & 1^{121} 3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 9 & 1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 7 \\ 1 & 2 & 1 & 0 & 7 & 0 \\ 2 & 2 & 2 & -2 & 11 & 2 \end{pmatrix}_{b_0} \\
[210] & 1^{121} 3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 15 & 6 & 3 \\ 2 & 1 & 1 & 16 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 15 & 16 & 0 & 116 & 52 & 20 \\ 6 & 7 & 0 & 52 & 21 & 9 \\ 3 & 3 & -1 & 20 & 9 & 3 \end{pmatrix}_{b_0} \\
[211] & 1^{123} 3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 10 & 10 & 3 \\ 2 & 1 & 1 & 12 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 60 & 14 \\ 10 & 13 & 0 & 60 & 61 & 15 \\ 3 & 3 & -1 & 14 & 15 & 3 \end{pmatrix}_{b_0} \\
[212] & 1^{11} 3^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 5 & 0 & 28 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 28 & 0 & 116 & 21 & 22 \\ 2 & 5 & 0 & 21 & 2 & 4 \\ 3 & 5 & -2 & 22 & 4 & 3 \end{pmatrix}_{b_0} \\
[213] & 1^{111} 3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 5 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 9 \\ 2 & 3 & 4 & 0 & 9 & 5 \\ 2 & 2 & 2 & -2 & 6 & 5 \end{pmatrix}_{b_0} \\
[214] & 1^{113} 3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8 \\
& T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 8 & 24 & 3 \\ 2 & 1 & 1 & 9 & 25 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 107 & 11 \\ 24 & 25 & 0 & 107 & 313 & 33 \\ 3 & 3 & -1 & 11 & 33 & 3 \end{pmatrix}_{b_0} \\
[215] & 1^{131} 3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 6 & 3 \\ 2 & 1 & 1 & 19 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 18 & 19 & 0 & 151 & 59 & 23 \\ 6 & 7 & 0 & 59 & 21 & 9 \\ 3 & 3 & -1 & 23 & 9 & 3 \end{pmatrix}_{b_0}$$

[216] $1^{11} 1^{124} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 13 & 3 \\ 2 & 1 & 1 & 12 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 72 & 14 \\ 13 & 16 & 0 & 72 & 90 & 18 \\ 3 & 3 & -1 & 14 & 18 & 3 \end{pmatrix}_{b_0}$$

[217] $1^{11} 1^{122} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 8 & 3 \\ 2 & 1 & 1 & 12 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 48 & 14 \\ 8 & 10 & 0 & 48 & 38 & 12 \\ 3 & 3 & -1 & 14 & 12 & 3 \end{pmatrix}_{b_0}$$

[218] $1^{12} 1^{142} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 28 & 8 & 3 \\ 2 & 1 & 1 & 30 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 28 & 30 & 0 & 416 & 128 & 38 \\ 8 & 10 & 0 & 128 & 38 & 12 \\ 3 & 3 & -1 & 38 & 12 & 3 \end{pmatrix}_{b_0}$$

[219] $1^{11} 1^{112} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -197, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 13 & 6 \\ 3 & 5 & 5 & 0 & 13 & 13 & 7 \\ 2 & 2 & 2 & -2 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[220] $1^{11} 1^{112} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 13 & 3 \\ 2 & 1 & 1 & 9 & 14 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 58 & 11 \\ 13 & 14 & 0 & 58 & 90 & 18 \\ 3 & 3 & -1 & 11 & 18 & 3 \end{pmatrix}_{b_0}$$

[222] $1^{11} 1^{123} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 14 & 3 \\ 3 & 5 & 0 & 13 & 30 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 13 & 0 & 24 & 58 & 10 \\ 14 & 30 & 0 & 58 & 134 & 24 \\ 3 & 5 & -2 & 10 & 24 & 3 \end{pmatrix}_{b_0}$$

[221] $1^{12} 1^{112} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 14 & 3 \\ 2 & 1 & 1 & 9 & 14 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 58 & 11 \\ 14 & 14 & 0 & 58 & 90 & 18 \\ 3 & 3 & -1 & 11 & 18 & 3 \end{pmatrix}_{b_0}$$

[223] $1^{11} 1^{113} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 0 & 4 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 4 & 0 & 4 & 18 & 4 \\ 6 & 9 & 12 & 0 & 18 & 61 & 15 \\ 2 & 2 & 2 & -2 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$

[225] $1^{1^{12}} 1^{121} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 20 & 6 & 3 \\ 2 & 1 & 1 & 20 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 20 & 20 & 0 & 200 & 68 & 26 \\ 6 & 7 & 0 & 68 & 21 & 9 \\ 3 & 3 & -1 & 26 & 9 & 3 \end{pmatrix}_{b_0}$

[224] $1^{1^1} 1^{141} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 18 & 6 & 3 \\ 2 & 1 & 1 & 22 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 18 & 22 & 0 & 192 & 66 & 26 \\ 6 & 7 & 0 & 66 & 21 & 9 \\ 3 & 3 & -1 & 26 & 9 & 3 \end{pmatrix}_{b_0}$

[226] $1^{1^{11}} 1^{121} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -217, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 11 & 2 & 3 \\ 2 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 11 & 12 & 0 & 56 & 12 & 14 \\ 2 & 3 & 0 & 12 & 1 & 3 \\ 3 & 3 & -1 & 14 & 3 & 3 \end{pmatrix}_{b_0}$

[227] $1^{1^1} 1^{111} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 2 & 0 & 5 & 3 & 2 \\ 1 & 2 & 0 & 0 & 8 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 8 & 0 & 23 & 13 & 9 \\ 2 & 3 & 4 & 0 & 13 & 5 & 5 \\ 2 & 2 & 2 & -2 & 9 & 5 & 2 \end{pmatrix}_{b_0}$

[228] $1^{1^{12}} 1^{114} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 8 & 18 & 3 \\ 2 & 1 & 1 & 9 & 20 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 76 & 11 \\ 18 & 20 & 0 & 76 & 158 & 24 \\ 3 & 3 & -1 & 11 & 24 & 3 \end{pmatrix}_{b_0}$

[229] $1^{1^{11}} 1^{113} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 8 & 19 & 3 \\ 2 & 1 & 1 & 9 & 21 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 87 & 11 \\ 19 & 21 & 0 & 87 & 205 & 27 \\ 3 & 3 & -1 & 11 & 27 & 3 \end{pmatrix}_{b_0}$

[230] $1^{1^1} 1^{123} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 5 & 0 & 8 & 25 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 8 & 30 & 6 \\ 12 & 25 & 0 & 30 & 94 & 20 \\ 3 & 5 & -2 & 6 & 20 & 3 \end{pmatrix}_{b_0}$

[232] $1^{1^{11}} 1^{122} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 10 & 9 & 3 \\ 2 & 1 & 1 & 12 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 48 & 14 \\ 9 & 10 & 0 & 48 & 38 & 12 \\ 3 & 3 & -1 & 14 & 12 & 3 \end{pmatrix}_{b_0}$

[231] $1^{1^1} 1^{124} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 16 & 3 \\ 2 & 1 & 1 & 12 & 20 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 96 & 14 \\ 16 & 20 & 0 & 96 & 158 & 24 \\ 3 & 3 & -1 & 14 & 24 & 3 \end{pmatrix}_{b_0}$$

[233] $1^{11} 1^{13} 3^1 21^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 6 & 3 \\ 2 & 1 & 1 & 19 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 17 & 19 & 0 & 151 & 59 & 23 \\ 6 & 7 & 0 & 59 & 21 & 9 \\ 3 & 3 & -1 & 23 & 9 & 3 \end{pmatrix}_{b_0}$$

[234] $1^{11} 1^{13} 3^1 21^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 0 & 11 & 1 & 2 \\ 1 & 2 & 0 & 0 & 16 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 11 & 16 & 0 & 103 & 16 & 19 \\ 1 & 1 & 3 & 0 & 16 & 1 & 3 \\ 2 & 2 & 2 & -2 & 19 & 3 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{11} 1^{12} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -252, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 6 & 2 \\ 1 & 2 & 0 & 0 & 5 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 18 & 6 \\ 4 & 6 & 8 & 0 & 18 & 26 & 10 \\ 2 & 2 & 2 & -2 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[237] $1^{11} 1^{12} 3^1 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 5 & 0 & 8 & 30 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 8 & 36 & 6 \\ 14 & 30 & 0 & 36 & 134 & 24 \\ 3 & 5 & -2 & 6 & 24 & 3 \end{pmatrix}_{b_0}$$

[236] $1^{11} 1^{12} 3^1 21^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 8 & 3 \\ 2 & 1 & 1 & 22 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 20 & 22 & 0 & 192 & 88 & 26 \\ 8 & 10 & 0 & 88 & 38 & 12 \\ 3 & 3 & -1 & 26 & 12 & 3 \end{pmatrix}_{b_0}$$

[238] $1^{11} 1^{14} 3^1 21^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 0 & 13 & 1 & 2 \\ 1 & 2 & 0 & 0 & 17 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 13 & 17 & 0 & 127 & 17 & 21 \\ 1 & 1 & 3 & 0 & 17 & 1 & 3 \\ 2 & 2 & 2 & -2 & 21 & 3 & 2 \end{pmatrix}_{b_0}$$

[239] $1^{11} 1^{14} 3^1 21^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -261, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 19 & 6 & 3 \\ 2 & 1 & 1 & 22 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 19 & 22 & 0 & 192 & 66 & 26 \\ 6 & 7 & 0 & 66 & 21 & 9 \\ 3 & 3 & -1 & 26 & 9 & 3 \end{pmatrix}_{b_0}$$

[241] $1^{11} 1^{14} 3^1 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 2 & 3 \\ 2 & 1 & 1 & 22 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 18 & 22 & 0 & 192 & 22 & 26 \\ 2 & 3 & 0 & 22 & 1 & 3 \\ 3 & 3 & -1 & 26 & 3 & 3 \end{pmatrix}_{b_0}$$

[240] $1^{1^{12}} 1^{121} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 2 & 3 \\ 2 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 12 & 12 & 0 & 56 & 12 & 14 \\ 2 & 3 & 0 & 12 & 1 & 3 \\ 3 & 3 & -1 & 14 & 3 & 3 \end{pmatrix}_{b_0}$$

[242] $1^{1^{11}} 1^{141} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -265, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 23 & 2 & 3 \\ 2 & 1 & 1 & 26 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 23 & 26 & 0 & 292 & 26 & 32 \\ 2 & 3 & 0 & 26 & 1 & 3 \\ 3 & 3 & -1 & 32 & 3 & 3 \end{pmatrix}_{b_0}$$

[243] $1^{1^{13}} 1^{132} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -266, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 8 & 3 \\ 2 & 1 & 1 & 19 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 16 & 19 & 0 & 151 & 78 & 23 \\ 8 & 10 & 0 & 78 & 38 & 12 \\ 3 & 3 & -1 & 23 & 12 & 3 \end{pmatrix}_{b_0}$$

[244] $1^{1^{11}} 1^{123} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 11 & 3 \\ 2 & 1 & 1 & 12 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 60 & 14 \\ 11 & 13 & 0 & 60 & 61 & 15 \\ 3 & 3 & -1 & 14 & 15 & 3 \end{pmatrix}_{b_0}$$

[245] $1^{1^{12}} 1^{113} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -270, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 16 & 3 \\ 2 & 1 & 1 & 9 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 67 & 11 \\ 16 & 17 & 0 & 67 & 121 & 21 \\ 3 & 3 & -1 & 11 & 21 & 3 \end{pmatrix}_{b_0}$$

[246] $1^{1^{11}} 1^{131} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 2 & 3 \\ 2 & 1 & 1 & 19 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 17 & 19 & 0 & 151 & 19 & 23 \\ 2 & 3 & 0 & 19 & 1 & 3 \\ 3 & 3 & -1 & 23 & 3 & 3 \end{pmatrix}_{b_0}$$

[247] $1^{1^{11}} 1^{112} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 6 \\ 1 & 2 & 0 & 0 & 8 & 8 \\ 1 & 0 & 0 & -4 & 0 & 0 \\ 4 & 5 & 8 & 0 & 23 & 26 \\ 4 & 6 & 8 & 0 & 26 & 26 \\ 2 & 2 & 2 & -2 & 9 & 10 \end{pmatrix}_{b_0}$$

[248] $1^{1^{11}} 1^{131} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -279, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 21 & 6 & 3 \\ 2 & 1 & 1 & 23 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 21 & 23 & 0 & 243 & 75 & 29 \\ 6 & 7 & 0 & 75 & 21 & 9 \\ 3 & 3 & -1 & 29 & 9 & 3 \end{pmatrix}_{b_0}$$

[249] $1^{1^{11}} 1^{131} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 & 2 \\ 1 & 2 & 0 & 0 & 13 & 1 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 10 & 13 & 0 & 74 & 10 & 16 \\ 1 & 2 & 1 & 0 & 10 & 0 & 2 \\ 2 & 2 & 2 & -2 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[250] $1^{114}3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 0 & 5 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 26 & 6 \\ 6 & 10 & 10 & 0 & 26 & 58 & 14 \\ 2 & 2 & 2 & -2 & 6 & 14 & 2 \end{pmatrix}_{b_0}$$

[251] $1^{114}3^{11} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 17 & 3 \\ 2 & 1 & 1 & 9 & 20 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 76 & 11 \\ 17 & 20 & 0 & 76 & 158 & 24 \\ 3 & 3 & -1 & 11 & 24 & 3 \end{pmatrix}_{b_0}$$

[252] $1^{121}3^1 21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 2 & 0 & 8 & 4 & 2 \\ 1 & 2 & 0 & 0 & 12 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 12 & 0 & 56 & 32 & 14 \\ 3 & 4 & 7 & 0 & 32 & 16 & 8 \\ 2 & 2 & 2 & -2 & 14 & 8 & 2 \end{pmatrix}_{b_0}$$

[253] $1^{113}3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 15 & 3 \\ 2 & 1 & 1 & 9 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 67 & 11 \\ 15 & 17 & 0 & 67 & 121 & 21 \\ 3 & 3 & -1 & 11 & 21 & 3 \end{pmatrix}_{b_0}$$

[254] $1^{122}3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 2 & 0 & 7 & 5 & 2 \\ 1 & 2 & 0 & 0 & 9 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 23 & 11 \\ 3 & 5 & 5 & 0 & 23 & 13 & 7 \\ 2 & 2 & 2 & -2 & 11 & 7 & 2 \end{pmatrix}_{b_0}$$

[255] $1^{112}3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -309, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 7 & 2 \\ 1 & 2 & 0 & 0 & 8 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 8 & 0 & 23 & 34 & 9 \\ 5 & 7 & 11 & 0 & 34 & 45 & 13 \\ 2 & 2 & 2 & -2 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[256] $1^{123}3^1 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -310, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 5 & 0 & 33 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 33 & 0 & 164 & 24 & 26 \\ 2 & 5 & 0 & 24 & 2 & 4 \\ 3 & 5 & -2 & 26 & 4 & 3 \end{pmatrix}_{b_0}$$

[258] $1^{14}3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 0 & 4 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 4 & 0 & 4 & 24 & 4 \\ 8 & 12 & 16 & 0 & 24 & 110 & 20 \\ 2 & 2 & 2 & -2 & 4 & 20 & 2 \end{pmatrix}_{b_0}$$

[259] $1^{122}3^121^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 2 & 0 & 6 & 5 & 2 \\ 1 & 2 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 6 & 0 & 20 & 18 & 8 \\ 3 & 5 & 5 & 0 & 18 & 13 & 7 \\ 2 & 2 & 2 & -2 & 8 & 7 & 2 \end{pmatrix}_{b_0}$$

[257] $1^{124}3^121^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 21 & 3 \\ 2 & 1 & 1 & 12 & 24 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 120 & 14 \\ 21 & 24 & 0 & 120 & 250 & 30 \\ 3 & 3 & -1 & 14 & 30 & 3 \end{pmatrix}_{b_0}$$

[260] $1^{132}3^121^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 8 & 3 \\ 2 & 1 & 1 & 19 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 17 & 19 & 0 & 151 & 78 & 23 \\ 8 & 10 & 0 & 78 & 38 & 12 \\ 3 & 3 & -1 & 23 & 12 & 3 \end{pmatrix}_{b_0}$$

[261] $1^{131}3^121^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 & 2 \\ 1 & 2 & 0 & 0 & 10 & 1 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 10 & 0 & 51 & 9 & 13 \\ 1 & 2 & 1 & 0 & 9 & 0 & 2 \\ 2 & 2 & 2 & -2 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[262] $1^{131}3^121^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 26 & 6 & 3 \\ 2 & 1 & 1 & 27 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 26 & 27 & 0 & 359 & 91 & 35 \\ 6 & 7 & 0 & 91 & 21 & 9 \\ 3 & 3 & -1 & 35 & 9 & 3 \end{pmatrix}_{b_0}$$

[263] $1^{113}3^121^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 5 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 22 & 6 \\ 5 & 8 & 9 & 0 & 22 & 40 & 12 \\ 2 & 2 & 2 & -2 & 6 & 12 & 2 \end{pmatrix}_{b_0}$$

[264] $1^{113}3^121^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 9 & 2 \\ 1 & 2 & 0 & 0 & 8 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 8 & 0 & 23 & 39 & 9 \\ 6 & 9 & 12 & 0 & 39 & 61 & 15 \\ 2 & 2 & 2 & -2 & 9 & 15 & 2 \end{pmatrix}_{b_0}$$

[265] $1^{131}3^121^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 22 & 6 & 3 \\ 2 & 1 & 1 & 23 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 22 & 23 & 0 & 243 & 75 & 29 \\ 6 & 7 & 0 & 75 & 21 & 9 \\ 3 & 3 & -1 & 29 & 9 & 3 \end{pmatrix}_{b_0}$$

[266] $1^{11} 1^{131} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 2 & 3 \\ 2 & 1 & 1 & 15 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 13 & 15 & 0 & 83 & 15 & 17 \\ 2 & 3 & 0 & 15 & 1 & 3 \\ 3 & 3 & -1 & 17 & 3 & 3 \end{pmatrix}_{b_0}$$

[267] $1^{11} 1^{113} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -357, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 9 & 2 \\ 1 & 2 & 0 & 0 & 5 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 27 & 6 \\ 6 & 9 & 12 & 0 & 27 & 61 & 15 \\ 2 & 2 & 2 & -2 & 6 & 15 & 2 \end{pmatrix}_{b_0}$$

[268] $1^{11} 1^{123} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 15 & 3 \\ 2 & 1 & 1 & 12 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 84 & 14 \\ 15 & 17 & 0 & 84 & 121 & 21 \\ 3 & 3 & -1 & 14 & 21 & 3 \end{pmatrix}_{b_0}$$

[269] $1^{11} 1^{132} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 0 & 9 & 6 & 2 \\ 1 & 2 & 0 & 0 & 10 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 10 & 0 & 51 & 38 & 13 \\ 4 & 6 & 8 & 0 & 38 & 26 & 10 \\ 2 & 2 & 2 & -2 & 13 & 10 & 2 \end{pmatrix}_{b_0}$$

[270] $1^{12} 1^{142} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 24 & 8 & 3 \\ 2 & 1 & 1 & 26 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 24 & 26 & 0 & 292 & 108 & 32 \\ 8 & 10 & 0 & 108 & 38 & 12 \\ 3 & 3 & -1 & 32 & 12 & 3 \end{pmatrix}_{b_0}$$

[271] $1^{12} 1^{131} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -378, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 2 & 3 \\ 2 & 1 & 1 & 19 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 18 & 19 & 0 & 151 & 19 & 23 \\ 2 & 3 & 0 & 19 & 1 & 3 \\ 3 & 3 & -1 & 23 & 3 & 3 \end{pmatrix}_{b_0}$$

[272] $1^{12} 1^{113} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 20 & 3 \\ 2 & 1 & 1 & 9 & 21 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 87 & 11 \\ 20 & 21 & 0 & 87 & 205 & 27 \\ 3 & 3 & -1 & 11 & 27 & 3 \end{pmatrix}_{b_0}$$

[273] $1^{11} 1^{141} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 23 & 6 & 3 \\ 2 & 1 & 1 & 26 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 23 & 26 & 0 & 292 & 82 & 32 \\ 6 & 7 & 0 & 82 & 21 & 9 \\ 3 & 3 & -1 & 32 & 9 & 3 \end{pmatrix}_{b_0}$$

[274] $1^{121}3^121^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -405, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 2 & 0 & 7 & 3 & 2 \\ 1 & 2 & 0 & 0 & 9 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 16 & 11 \\ 2 & 3 & 4 & 0 & 16 & 5 & 5 \\ 2 & 2 & 2 & -2 & 11 & 5 & 2 \end{pmatrix}_{b_0}$$

[275] $1^{141}3^121^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -413, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 0 & 13 & 2 & 2 \\ 1 & 2 & 0 & 0 & 17 & 1 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 13 & 17 & 0 & 127 & 13 & 21 \\ 1 & 2 & 1 & 0 & 13 & 0 & 2 \\ 2 & 2 & 2 & -2 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[276] $1^{142}3^121^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 8 & 3 \\ 2 & 1 & 1 & 22 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 18 & 22 & 0 & 192 & 88 & 26 \\ 8 & 10 & 0 & 88 & 38 & 12 \\ 3 & 3 & -1 & 26 & 12 & 3 \end{pmatrix}_{b_0}$$

[277] $1^{111}3^121^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -419, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 21 & 3 \\ 2 & 1 & 1 & 9 & 24 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 96 & 11 \\ 21 & 24 & 0 & 96 & 250 & 30 \\ 3 & 3 & -1 & 11 & 30 & 3 \end{pmatrix}_{b_0}$$

[278] $1^{131}3^121^{142} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -429, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 3 & 2 \\ 1 & 0 & 2 & 0 & 11 & 4 & 2 \\ 1 & 2 & 0 & 0 & 16 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 11 & 16 & 0 & 103 & 43 & 19 \\ 3 & 4 & 7 & 0 & 43 & 16 & 8 \\ 2 & 2 & 2 & -2 & 19 & 8 & 2 \end{pmatrix}_{b_0}$$

[279] $1^{141}3^121^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 24 & 2 & 3 \\ 2 & 1 & 1 & 26 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 24 & 26 & 0 & 292 & 26 & 32 \\ 2 & 3 & 0 & 26 & 1 & 3 \\ 3 & 3 & -1 & 32 & 3 & 3 \end{pmatrix}_{b_0}$$

[280] $1^{124}3^121^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 18 & 3 \\ 2 & 1 & 1 & 12 & 20 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 96 & 14 \\ 18 & 20 & 0 & 96 & 158 & 24 \\ 3 & 3 & -1 & 14 & 24 & 3 \end{pmatrix}_{b_0}$$

[281] $1^{132}3^121^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -445, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 2 & 0 & 10 & 5 & 2 \\ 1 & 2 & 0 & 0 & 13 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 10 & 13 & 0 & 74 & 33 & 16 \\ 3 & 5 & 5 & 0 & 33 & 13 & 7 \\ 2 & 2 & 2 & -2 & 16 & 7 & 2 \end{pmatrix}_{b_0}$$

[282] $1^{141}3^121^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -448, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 24 & 6 & 3 \\ 2 & 1 & 1 & 26 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 24 & 26 & 0 & 292 & 82 & 32 \\ 6 & 7 & 0 & 82 & 21 & 9 \\ 3 & 3 & -1 & 32 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[283] \ 1^{1^{12}} 1^{131} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -454, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.00 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 2 & 3 \\ 2 & 1 & 1 & 15 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 14 & 15 & 0 & 83 & 15 & 17 \\ 2 & 3 & 0 & 15 & 1 & 3 \\ 3 & 3 & -1 & 17 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[285] \ 1^{1^{12}} 1^{123} 3^{11} 21^{133} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 7 & 2 \\ 1 & 2 & 0 & 0 & 12 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 12 & 0 & 56 & 52 & 14 \\ 5 & 7 & 11 & 0 & 52 & 45 & 13 \\ 2 & 2 & 2 & -2 & 14 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[286] \ 1^{1^{12}} 1^{123} 3^{11} 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 2 & 0 & 8 & 3 & 2 \\ 1 & 2 & 0 & 0 & 12 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 12 & 0 & 56 & 20 & 14 \\ 2 & 3 & 4 & 0 & 20 & 5 & 5 \\ 2 & 2 & 2 & -2 & 14 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[284] \ 1^{1^{14}} 3^{11} 21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 2 & 0 & 5 & 12 & 2 \\ 1 & 2 & 0 & 0 & 8 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 8 & 0 & 23 & 52 & 9 \\ 8 & 12 & 16 & 0 & 52 & 110 & 20 \\ 2 & 2 & 2 & -2 & 9 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[287] \ 1^{1^{12}} 1^{123} 3^{11} 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -468, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 6 & 2 \\ 1 & 2 & 0 & 0 & 9 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 32 & 11 \\ 4 & 6 & 8 & 0 & 32 & 26 & 10 \\ 2 & 2 & 2 & -2 & 11 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[288] \ 1^{1^{11}} 1^{141} 3^{11} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -469, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 19 & 2 & 3 \\ 2 & 1 & 1 & 22 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 19 & 22 & 0 & 192 & 22 & 26 \\ 2 & 3 & 0 & 22 & 1 & 3 \\ 3 & 3 & -1 & 26 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[289] \ 1^{1^{11}} 1^{142} 3^{11} 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 23 & 8 & 3 \\ 2 & 1 & 1 & 26 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 23 & 26 & 0 & 292 & 108 & 32 \\ 8 & 10 & 0 & 108 & 38 & 12 \\ 3 & 3 & -1 & 32 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[290] \ 1^{1^{13}} 3^{11} 21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -477, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 0 & 5 & 10 & 2 \\ 1 & 2 & 0 & 0 & 8 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 8 & 0 & 23 & 47 & 9 \\ 7 & 10 & 15 & 0 & 47 & 88 & 18 \\ 2 & 2 & 2 & -2 & 9 & 18 & 2 \end{pmatrix}_{b_0}$$

[291] $1^{11} 1^{142} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -484, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 19 & 8 & 3 \\ 2 & 1 & 1 & 22 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 19 & 22 & 0 & 192 & 88 & 26 \\ 8 & 10 & 0 & 88 & 38 & 12 \\ 3 & 3 & -1 & 26 & 12 & 3 \end{pmatrix}_{b_0}$$

[292] $1^{11} 1^{123} 3^1 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -485, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 8 & 2 \\ 1 & 2 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 39 & 11 \\ 5 & 8 & 9 & 0 & 39 & 40 & 12 \\ 2 & 2 & 2 & -2 & 11 & 12 & 2 \end{pmatrix}_{b_0}$$

[293] $1^{11} 1^{131} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 2 & 0 & 9 & 3 & 2 \\ 1 & 2 & 0 & 0 & 10 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 10 & 0 & 51 & 19 & 13 \\ 2 & 3 & 4 & 0 & 19 & 5 & 5 \\ 2 & 2 & 2 & -2 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[294] $1^{11} 1^{114} 3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 8 & 2 \\ 1 & 0 & 2 & 0 & 4 & 12 & 2 \\ 1 & 2 & 0 & 0 & 5 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 36 & 6 \\ 8 & 12 & 16 & 0 & 36 & 110 & 20 \\ 2 & 2 & 2 & -2 & 6 & 20 & 2 \end{pmatrix}_{b_0}$$

[295] $1^{11} 1^{124} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 10 & 2 \\ 1 & 2 & 0 & 0 & 6 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 6 & 0 & 20 & 36 & 8 \\ 6 & 10 & 10 & 0 & 36 & 58 & 14 \\ 2 & 2 & 2 & -2 & 8 & 14 & 2 \end{pmatrix}_{b_0}$$

[296] $1^{11} 1^{122} 3^1 21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 0 & 8 & 6 & 2 \\ 1 & 2 & 0 & 0 & 12 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 12 & 0 & 56 & 40 & 14 \\ 4 & 6 & 8 & 0 & 40 & 26 & 10 \\ 2 & 2 & 2 & -2 & 14 & 10 & 2 \end{pmatrix}_{b_0}$$

[297] $1^{11} 1^{141} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 & 2 \\ 1 & 2 & 0 & 0 & 14 & 1 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 14 & 0 & 96 & 12 & 18 \\ 1 & 2 & 1 & 0 & 12 & 0 & 2 \\ 2 & 2 & 2 & -2 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[298] $1^{12} 1^{141} 3^1 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -532, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 28 & 6 & 3 \\ 2 & 1 & 1 & 30 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 28 & 30 & 0 & 416 & 98 & 38 \\ 6 & 7 & 0 & 98 & 21 & 9 \\ 3 & 3 & -1 & 38 & 9 & 3 \end{pmatrix}_{b_0}$$

[299] $1^{114}3^{121}1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -533, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 11 & 2 \\ 1 & 2 & 0 & 0 & 5 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 5 & 0 & 10 & 31 & 6 \\ 7 & 11 & 13 & 0 & 31 & 81 & 17 \\ 2 & 2 & 2 & -2 & 6 & 17 & 2 \end{pmatrix}_{b_0}$$

[300] $1^{112}3^{114}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -538, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 26 & 3 \\ 2 & 1 & 1 & 9 & 28 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 116 & 11 \\ 26 & 28 & 0 & 116 & 366 & 36 \\ 3 & 3 & -1 & 11 & 36 & 3 \end{pmatrix}_{b_0}$$

[301] $1^{111}3^{112}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 17 & 3 \\ 2 & 1 & 1 & 12 & 20 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 56 & 96 & 14 \\ 17 & 20 & 0 & 96 & 158 & 24 \\ 3 & 3 & -1 & 14 & 24 & 3 \end{pmatrix}_{b_0}$$

[302] $1^{112}3^{113}21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -550, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 22 & 3 \\ 2 & 1 & 1 & 9 & 24 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 35 & 96 & 11 \\ 22 & 24 & 0 & 96 & 250 & 30 \\ 3 & 3 & -1 & 11 & 30 & 3 \end{pmatrix}_{b_0}$$

[303] $1^{132}3^{121}1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 2 & 0 & 9 & 5 & 2 \\ 1 & 2 & 0 & 0 & 10 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 10 & 0 & 51 & 28 & 13 \\ 3 & 5 & 5 & 0 & 28 & 13 & 7 \\ 2 & 2 & 2 & -2 & 13 & 7 & 2 \end{pmatrix}_{b_0}$$

[304] $1^{123}3^{121}1^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 9 & 2 \\ 1 & 2 & 0 & 0 & 9 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 48 & 11 \\ 6 & 9 & 12 & 0 & 48 & 61 & 15 \\ 2 & 2 & 2 & -2 & 11 & 15 & 2 \end{pmatrix}_{b_0}$$

[305] $1^{123}3^{121}1^{133} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -592, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 9 & 2 \\ 1 & 2 & 0 & 0 & 12 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 12 & 0 & 56 & 60 & 14 \\ 6 & 9 & 12 & 0 & 60 & 61 & 15 \\ 2 & 2 & 2 & -2 & 14 & 15 & 2 \end{pmatrix}_{b_0}$$

[306] $1^{142}3^{121}1^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 3 & 2 \\ 1 & 0 & 2 & 0 & 13 & 5 & 2 \\ 1 & 2 & 0 & 0 & 17 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 13 & 17 & 0 & 127 & 43 & 21 \\ 3 & 5 & 5 & 0 & 43 & 13 & 7 \\ 2 & 2 & 2 & -2 & 21 & 7 & 2 \end{pmatrix}_{b_0}$$

[307] $1^{141}3^{12}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -628, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 2 & 3 \\ 2 & 1 & 1 & 22 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 20 & 22 & 0 & 192 & 22 & 26 \\ 2 & 3 & 0 & 22 & 1 & 3 \\ 3 & 3 & -1 & 26 & 3 & 3 \end{pmatrix}_{b_0}$$

[308] $1^{132}3^{11}21^{143} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -645, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 5 & 2 \\ 1 & 0 & 2 & 0 & 11 & 7 & 2 \\ 1 & 2 & 0 & 0 & 16 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 11 & 16 & 0 & 103 & 70 & 19 \\ 5 & 7 & 11 & 0 & 70 & 45 & 13 \\ 2 & 2 & 2 & -2 & 19 & 13 & 2 \end{pmatrix}_{b_0}$$

[309] $1^{123}3^{11}21^{134} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -672, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 7 & 2 \\ 1 & 0 & 2 & 0 & 8 & 10 & 2 \\ 1 & 2 & 0 & 0 & 12 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 12 & 0 & 56 & 72 & 14 \\ 7 & 10 & 15 & 0 & 72 & 88 & 18 \\ 2 & 2 & 2 & -2 & 14 & 18 & 2 \end{pmatrix}_{b_0}$$

[311] $1^{124}3^{11}21^{134} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 8 & 2 \\ 1 & 0 & 2 & 0 & 8 & 12 & 2 \\ 1 & 2 & 0 & 0 & 12 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 12 & 0 & 56 & 80 & 14 \\ 8 & 12 & 16 & 0 & 80 & 110 & 20 \\ 2 & 2 & 2 & -2 & 14 & 20 & 2 \end{pmatrix}_{b_0}$$

[310] $1^{124}3^{11}21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 11 & 2 \\ 1 & 2 & 0 & 0 & 9 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 55 & 11 \\ 7 & 11 & 13 & 0 & 55 & 81 & 17 \\ 2 & 2 & 2 & -2 & 11 & 17 & 2 \end{pmatrix}_{b_0}$$

[312] $1^{131}3^{11}21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -717, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 2 & 0 & 10 & 3 & 2 \\ 1 & 2 & 0 & 0 & 13 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 10 & 13 & 0 & 74 & 23 & 16 \\ 2 & 3 & 4 & 0 & 23 & 5 & 5 \\ 2 & 2 & 2 & -2 & 16 & 5 & 2 \end{pmatrix}_{b_0}$$

[313] $1^{124}3^{11}21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -720, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 0 & 7 & 12 & 2 \\ 1 & 2 & 0 & 0 & 9 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 9 & 0 & 35 & 64 & 11 \\ 8 & 12 & 16 & 0 & 64 & 110 & 20 \\ 2 & 2 & 2 & -2 & 11 & 20 & 2 \end{pmatrix}_{b_0}$$

[315] $1^{142}3^{11}21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -752, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 0 & 12 & 6 & 2 \\ 1 & 2 & 0 & 0 & 14 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 14 & 0 & 96 & 52 & 18 \\ 4 & 6 & 8 & 0 & 52 & 26 & 10 \\ 2 & 2 & 2 & -2 & 18 & 10 & 2 \end{pmatrix}_{b_0}$$

[314] $1^{142}3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -752, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 2 & 0 & 11 & 5 & 2 \\ 1 & 2 & 0 & 0 & 11 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 11 & 0 & 71 & 33 & 15 \\ 3 & 5 & 5 & 0 & 33 & 13 & 7 \\ 2 & 2 & 2 & -2 & 15 & 7 & 2 \end{pmatrix}_{b_0}$$

[316] $1^{132}3^1 21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -780, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 0 & 10 & 6 & 2 \\ 1 & 2 & 0 & 0 & 13 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 10 & 13 & 0 & 74 & 46 & 16 \\ 4 & 6 & 8 & 0 & 46 & 26 & 10 \\ 2 & 2 & 2 & -2 & 16 & 10 & 2 \end{pmatrix}_{b_0}$$

[317] $1^{131}3^1 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -800, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 2 & 0 & 11 & 3 & 2 \\ 1 & 2 & 0 & 0 & 16 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 11 & 16 & 0 & 103 & 27 & 19 \\ 2 & 3 & 4 & 0 & 27 & 5 & 5 \\ 2 & 2 & 2 & -2 & 19 & 5 & 2 \end{pmatrix}_{b_0}$$

[318] $1^{132}3^1 21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -848, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 0 & 11 & 6 & 2 \\ 1 & 2 & 0 & 0 & 16 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 11 & 16 & 0 & 103 & 54 & 19 \\ 4 & 6 & 8 & 0 & 54 & 26 & 10 \\ 2 & 2 & 2 & -2 & 19 & 10 & 2 \end{pmatrix}_{b_0}$$

[319] $1^{142}3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -852, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 3 & 2 \\ 1 & 0 & 2 & 0 & 12 & 5 & 2 \\ 1 & 2 & 0 & 0 & 14 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 14 & 0 & 96 & 38 & 18 \\ 3 & 5 & 5 & 0 & 38 & 13 & 7 \\ 2 & 2 & 2 & -2 & 18 & 7 & 2 \end{pmatrix}_{b_0}$$

[320] $1^{141}3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -884, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 2 & 0 & 12 & 3 & 2 \\ 1 & 2 & 0 & 0 & 14 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 14 & 0 & 96 & 26 & 18 \\ 2 & 3 & 4 & 0 & 26 & 5 & 5 \\ 2 & 2 & 2 & -2 & 18 & 5 & 2 \end{pmatrix}_{b_0}$$

[321] $1^{141}3^1 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -1125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 & 2 \\ 1 & 0 & 2 & 0 & 13 & 3 & 2 \\ 1 & 2 & 0 & 0 & 17 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 13 & 17 & 0 & 127 & 30 & 21 \\ 2 & 3 & 4 & 0 & 30 & 5 & 5 \\ 2 & 2 & 2 & -2 & 21 & 5 & 2 \end{pmatrix}_{b_0}$$

[322] $1^{142}3^1 21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -1188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.5$

$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 0 & 13 & 6 & 2 \\ 1 & 2 & 0 & 0 & 17 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 13 & 17 & 0 & 127 & 60 & 21 \\ 4 & 6 & 8 & 0 & 60 & 26 & 10 \\ 2 & 2 & 2 & -2 & 21 & 10 & 2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [323] \quad 1^{11} 1^1 3^1 5^1 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 6 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 7 & 4 & 3 \\ 6 & 0 & 7 & 14 & 7 & 8 \\ 2 & 0 & 4 & 7 & 2 & 4 \\ 3 & 0 & 3 & 8 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[324] \quad 1^{12} 1^1 3^1 5^1 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 15 & 4 & 3 \\ 12 & 0 & 15 & 76 & 15 & 18 \\ 2 & 0 & 4 & 15 & 2 & 4 \\ 3 & 0 & 3 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[325] \quad 1^{11} 1^1 3^1 5^1 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 9 & 6 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 11 & 8 & 3 \\ 9 & 0 & 11 & 39 & 31 & 13 \\ 6 & 0 & 8 & 31 & 22 & 10 \\ 3 & 0 & 3 & 13 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[326] \quad 1^{11} 1^1 3^1 5^1 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 9 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 11 & 4 & 3 \\ 9 & 0 & 11 & 39 & 11 & 13 \\ 2 & 0 & 4 & 11 & 2 & 4 \\ 3 & 0 & 3 & 13 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[327] \quad 1^{11} 1^1 3^1 5^1 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 7 & 4 & 3 \\ 5 & 0 & 7 & 14 & 7 & 8 \\ 3 & 0 & 4 & 7 & 2 & 4 \\ 3 & 0 & 3 & 8 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[328] \quad 1^{13} 1^1 3^1 5^1 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 10, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 16 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 19 & 4 & 3 \\ 16 & 0 & 19 & 110 & 19 & 22 \\ 2 & 0 & 4 & 19 & 2 & 4 \\ 3 & 0 & 3 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[329] \quad 1^{11} 1^1 3^1 5^1 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 3 \\ 2 & 2 & 1 & 7 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 7 & 8 \\ 2 & 4 & 0 & 7 & 2 & 4 \\ 3 & 4 & -1 & 8 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[330] \quad 1^{11} 1^1 3^1 5^1 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 10 & 3 \\ 2 & 2 & 1 & 7 & 16 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 33 & 8 \\ 10 & 16 & 0 & 33 & 70 & 18 \\ 3 & 4 & -1 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[331] \quad & 1^{\overline{111}} 31^{\overline{1}} 513^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 23, \quad (n_+, n_-, n_0) = \\
& (1, 12, 0), \quad c_{\text{ext}} = 9.0 \\
& T_{\min} = 12.96 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 5 & 6 & 3 \\ 2 & 2 & 1 & 7 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 19 & 8 \\ 6 & 8 & 0 & 19 & 22 & 10 \\ 3 & 4 & -1 & 8 & 10 & 4 \end{pmatrix}_{b_0} \\
[332] \quad & 1^{\overline{121}} 31^{\overline{1}} 5131^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 23, \quad (n_+, n_-, n_0) = \\
& (1, 12, 0), \quad c_{\text{ext}} = 9.0 \\
& T_{\min} = 12.96 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 10 & 6 & 3 \\ 2 & 2 & 1 & 15 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 15 & 0 & 65 & 39 & 17 \\ 6 & 8 & 0 & 39 & 22 & 10 \\ 3 & 4 & -1 & 17 & 10 & 4 \end{pmatrix}_{b_0} \\
[334] \quad & 1^{\overline{131}} 31^{\overline{1}} 513^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 35, \quad (n_+, n_-, n_0) = \\
& (1, 12, 0), \quad c_{\text{ext}} = 9.0 \\
& T_{\min} = 12.97 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 13 & 2 & 3 \\ 2 & 2 & 1 & 19 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 13 & 19 & 0 & 110 & 19 & 22 \\ 2 & 4 & 0 & 19 & 2 & 4 \\ 3 & 4 & -1 & 22 & 4 & 4 \end{pmatrix}_{b_0} \\
[333] \quad & 1^{\overline{121}} 31^{\overline{1}} 5131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 35, \quad (n_+, n_-, n_0) = \\
& (1, 12, 0), \quad c_{\text{ext}} = 9.0 \\
& T_{\min} = 12.97 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 10 & 2 & 3 \\ 2 & 2 & 1 & 15 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 15 & 0 & 65 & 15 & 17 \\ 2 & 4 & 0 & 15 & 2 & 4 \\ 3 & 4 & -1 & 17 & 4 & 4 \end{pmatrix}_{b_0} \\
[336] \quad & 1^{\overline{113}} 31^{\overline{1}} 513^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 40, \quad (n_+, n_-, n_0) = \\
& (1, 12, 0), \quad c_{\text{ext}} = 9.8 \\
& T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 1 & 2 & 5 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 7 & 16 & 3 \\ 5 & 0 & 7 & 14 & 33 & 8 \\ 13 & 0 & 16 & 33 & 70 & 18 \\ 3 & 0 & 3 & 8 & 18 & 4 \end{pmatrix}_{b_0} \\
[335] \quad & 1^{\overline{111}} 31^{\overline{1}} 513^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 40, \quad (n_+, n_-, n_0) = \\
& (1, 12, 0), \quad c_{\text{ext}} = 9.0 \\
& T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 1 & 2 & 5 & 7 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 7 & 8 & 3 \\ 5 & 0 & 7 & 14 & 19 & 8 \\ 7 & 0 & 8 & 19 & 22 & 10 \\ 3 & 0 & 3 & 8 & 10 & 4 \end{pmatrix}_{b_0} \\
[337] \quad & 1^{\overline{112}} 31^{\overline{1}} 513^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 50, \quad (n_+, n_-, n_0) = \\
& (1, 12, 0), \quad c_{\text{ext}} = 9.5 \\
& T_{\min} = 12.92 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 5 & 8 & 3 \\ 2 & 2 & 1 & 7 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 26 & 8 \\ 8 & 12 & 0 & 26 & 42 & 14 \\ 3 & 4 & -1 & 8 & 14 & 4 \end{pmatrix}_{b_0} \\
[339] \quad & 1^{\overline{122}} 31^{\overline{1}} 513^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 54, \quad (n_+, n_-, n_0) = \\
& (1, 12, 0), \quad c_{\text{ext}} = 9.5 \\
& T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 7 & 8 & 3 \\ 2 & 2 & 1 & 11 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 7 & 11 & 0 & 32 & 38 & 12 \\ 8 & 12 & 0 & 38 & 42 & 14 \\ 3 & 4 & -1 & 12 & 14 & 4 \end{pmatrix}_{b_0} \\
[338] \quad & 1^{\overline{112}} 31^{\overline{1}} 5131^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 54, \quad (n_+, n_-, n_0) = \\
& (1, 12, 0), \quad c_{\text{ext}} = 9.5 \\
& T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 8 & 3 \\ 2 & 2 & 1 & 11 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 11 & 0 & 39 & 42 & 13 \\ 8 & 12 & 0 & 42 & 42 & 14 \\ 3 & 4 & -1 & 13 & 14 & 4 \end{pmatrix}_{b_0}$$

[340] $1^{123}3^{11}5^{131}1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 63, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 10 & 3 \\ 2 & 2 & 1 & 15 & 16 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 15 & 0 & 65 & 69 & 17 \\ 10 & 16 & 0 & 69 & 70 & 18 \\ 3 & 4 & -1 & 17 & 18 & 4 \end{pmatrix}_{b_0}$$

[341] $1^{112}3^{11}5^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 64, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 7 & 12 & 3 \\ 5 & 0 & 7 & 14 & 26 & 8 \\ 10 & 0 & 12 & 26 & 42 & 14 \\ 3 & 0 & 3 & 8 & 14 & 4 \end{pmatrix}_{b_0}$$

[342] $1^{141}3^{11}5^{131}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 75, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 2 & 3 \\ 2 & 2 & 1 & 27 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 18 & 27 & 0 & 217 & 27 & 31 \\ 2 & 4 & 0 & 27 & 2 & 4 \\ 3 & 4 & -1 & 31 & 4 & 4 \end{pmatrix}_{b_0}$$

[343] $1^{122}3^{11}5^{131}1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 98, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 8 & 3 \\ 2 & 2 & 1 & 15 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 15 & 0 & 65 & 54 & 17 \\ 8 & 12 & 0 & 54 & 42 & 14 \\ 3 & 4 & -1 & 17 & 14 & 4 \end{pmatrix}_{b_0}$$

$T = 13$ [344] $1^{13}3^{11}5^{1321}1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 8 & 5 & 4 \\ 5 & 0 & 8 & 15 & 8 & 9 \\ 3 & 0 & 5 & 8 & 3 & 5 \\ 3 & 0 & 4 & 9 & 5 & 5 \end{pmatrix}_{b_0}$$

[345] $1^{113}3^{11}5^{1321}1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 7 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 8 & 9 & 4 \\ 5 & 0 & 8 & 15 & 20 & 9 \\ 7 & 0 & 9 & 20 & 23 & 11 \\ 3 & 0 & 4 & 9 & 11 & 5 \end{pmatrix}_{b_0}$$

[346] $1^{113}3^{11}5^{1321}1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 13 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 8 & 19 & 4 \\ 5 & 0 & 8 & 15 & 36 & 9 \\ 13 & 0 & 19 & 36 & 79 & 21 \\ 3 & 0 & 4 & 9 & 21 & 5 \end{pmatrix}_{b_0}$$

[347] $1^{113}3^{11}5^{1321}1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 8 & 14 & 4 \\ 5 & 0 & 8 & 15 & 28 & 9 \\ 10 & 0 & 14 & 28 & 46 & 16 \\ 3 & 0 & 4 & 9 & 16 & 5 \end{pmatrix}_{b_0}$$

4.716 NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{su}_2$ + $\mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 1 & 5 \\ 11 & 20 & 0 & 28 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 15 & 28 & 0 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 8 & -2 & 11 & 0 & 2 \end{pmatrix}_{b_0}$$

4.717 NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 2 & 3 \\ 3 & 4 & 0 & 7 & 14 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 17 & 4 & 5 \\ 8 & 14 & 0 & 17 & 37 & 8 & 11 \\ 2 & 3 & 0 & 4 & 8 & 0 & 2 \\ 3 & 4 & -2 & 5 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{122} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 4 & 2 & 3 \\ 3 & 4 & 0 & 16 & 8 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 22 & 9 & 12 \\ 4 & 8 & 0 & 22 & 10 & 4 & 6 \\ 2 & 3 & 0 & 9 & 4 & 0 & 2 \\ 3 & 4 & -2 & 12 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{112} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 2 & 3 \\ 3 & 4 & 0 & 7 & 8 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 10 & 4 & 5 \\ 4 & 8 & 0 & 10 & 10 & 4 & 6 \\ 2 & 3 & 0 & 4 & 4 & 0 & 2 \\ 3 & 4 & -2 & 5 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 2 & 3 \\ 3 & 4 & 0 & 7 & 5 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 6 & 4 & 5 \\ 3 & 5 & 0 & 6 & 4 & 3 & 4 \\ 2 & 3 & 0 & 4 & 3 & 0 & 2 \\ 3 & 4 & -2 & 5 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 5 & 4 & 5 \\ 2 & 4 & 0 & 5 & 1 & 2 & 3 \\ 2 & 3 & 0 & 4 & 2 & 0 & 2 \\ 3 & 4 & -2 & 5 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{121} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 2 & 3 \\ 3 & 4 & 0 & 11 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 8 & 6 & 8 \\ 2 & 4 & 0 & 8 & 1 & 2 & 3 \\ 2 & 3 & 0 & 6 & 2 & 0 & 2 \\ 3 & 4 & -2 & 8 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{132} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 4 & 2 & 3 \\ 3 & 4 & 0 & 20 & 8 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 28 & 11 & 15 \\ 4 & 8 & 0 & 28 & 10 & 4 & 6 \\ 2 & 3 & 0 & 11 & 4 & 0 & 2 \\ 3 & 4 & -2 & 15 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -251, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 2 & 3 \\ 3 & 4 & 0 & 7 & 9 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 11 & 4 & 5 \\ 5 & 9 & 0 & 11 & 13 & 5 & 7 \\ 2 & 3 & 0 & 4 & 5 & 0 & 2 \\ 3 & 4 & -2 & 5 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{142}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -264, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 4 & 2 & 3 \\ 3 & 4 & 0 & 29 & 8 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 16 & 29 & 0 & 148 & 40 & 16 & 22 \\ 4 & 8 & 0 & 40 & 10 & 4 & 6 \\ 2 & 3 & 0 & 16 & 4 & 0 & 2 \\ 3 & 4 & -2 & 22 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -287, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 8 & 7 & 9 \\ 2 & 4 & 0 & 8 & 1 & 2 & 3 \\ 2 & 3 & 0 & 7 & 2 & 0 & 2 \\ 3 & 4 & -2 & 9 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{114}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -299, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.92 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 2 & 3 \\ 3 & 4 & 0 & 7 & 21 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 26 & 4 & 5 \\ 11 & 21 & 0 & 26 & 76 & 11 & 16 \\ 2 & 3 & 0 & 4 & 11 & 0 & 2 \\ 3 & 4 & -2 & 5 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{112}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -314, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 2 & 3 \\ 3 & 4 & 0 & 7 & 18 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 22 & 4 & 5 \\ 10 & 18 & 0 & 22 & 58 & 10 & 14 \\ 2 & 3 & 0 & 4 & 10 & 0 & 2 \\ 3 & 4 & -2 & 5 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{122}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -336, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 2 & 3 \\ 3 & 4 & 0 & 11 & 13 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 25 & 6 & 8 \\ 7 & 13 & 0 & 25 & 28 & 7 & 10 \\ 2 & 3 & 0 & 6 & 7 & 0 & 2 \\ 3 & 4 & -2 & 8 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -339, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 2 & 3 \\ 3 & 4 & 0 & 7 & 13 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 16 & 4 & 5 \\ 7 & 13 & 0 & 16 & 28 & 7 & 10 \\ 2 & 3 & 0 & 4 & 7 & 0 & 2 \\ 3 & 4 & -2 & 5 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{113}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -355, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 2 & 3 \\ 3 & 4 & 0 & 7 & 17 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 21 & 4 & 5 \\ 9 & 17 & 0 & 21 & 49 & 9 & 13 \\ 2 & 3 & 0 & 4 & 9 & 0 & 2 \\ 3 & 4 & -2 & 5 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{142}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 4 & 2 & 3 \\ 3 & 4 & 0 & 24 & 8 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 34 & 13 & 18 \\ 4 & 8 & 0 & 34 & 10 & 4 & 6 \\ 2 & 3 & 0 & 13 & 4 & 0 & 2 \\ 3 & 4 & -2 & 18 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{123}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 9 & 2 & 3 \\ 3 & 4 & 0 & 11 & 17 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 33 & 6 & 8 \\ 9 & 17 & 0 & 33 & 49 & 9 & 13 \\ 2 & 3 & 0 & 6 & 9 & 0 & 2 \\ 3 & 4 & -2 & 8 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{124}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 11 & 2 & 3 \\ 3 & 4 & 0 & 11 & 21 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 41 & 6 & 8 \\ 11 & 21 & 0 & 41 & 76 & 11 & 16 \\ 2 & 3 & 0 & 6 & 11 & 0 & 2 \\ 3 & 4 & -2 & 8 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{121}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -477, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 2 & 3 \\ 3 & 4 & 0 & 16 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 11 & 9 & 12 \\ 2 & 4 & 0 & 11 & 1 & 2 & 3 \\ 2 & 3 & 0 & 9 & 2 & 0 & 2 \\ 3 & 4 & -2 & 12 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{113}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -526, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 2 & 3 \\ 3 & 4 & 0 & 7 & 22 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 27 & 4 & 5 \\ 12 & 22 & 0 & 27 & 85 & 12 & 17 \\ 2 & 3 & 0 & 4 & 12 & 0 & 2 \\ 3 & 4 & -2 & 5 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{124}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 14 & 2 & 3 \\ 3 & 4 & 0 & 11 & 26 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 50 & 6 & 8 \\ 14 & 26 & 0 & 50 & 118 & 14 & 20 \\ 2 & 3 & 0 & 6 & 14 & 0 & 2 \\ 3 & 4 & -2 & 8 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{121}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 2 & 3 \\ 3 & 4 & 0 & 21 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 12 & 21 & 0 & 80 & 14 & 12 & 16 \\ 2 & 4 & 0 & 14 & 1 & 2 & 3 \\ 2 & 3 & 0 & 12 & 2 & 0 & 2 \\ 3 & 4 & -2 & 16 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{131}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -631, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 2 & 3 \\ 3 & 4 & 0 & 20 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 14 & 11 & 15 \\ 2 & 4 & 0 & 14 & 1 & 2 & 3 \\ 2 & 3 & 0 & 11 & 2 & 0 & 2 \\ 3 & 4 & -2 & 15 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{114}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -666, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 2 & 3 \\ 3 & 4 & 0 & 7 & 26 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 32 & 4 & 5 \\ 14 & 26 & 0 & 32 & 118 & 14 & 20 \\ 2 & 3 & 0 & 4 & 14 & 0 & 2 \\ 3 & 4 & -2 & 5 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{141}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -749, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 2 & 3 \\ 3 & 4 & 0 & 24 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 17 & 13 & 18 \\ 2 & 4 & 0 & 17 & 1 & 2 & 3 \\ 2 & 3 & 0 & 13 & 2 & 0 & 2 \\ 3 & 4 & -2 & 18 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{131}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -886, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 2 & 3 \\ 3 & 4 & 0 & 25 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 14 & 25 & 0 & 111 & 17 & 14 & 19 \\ 2 & 4 & 0 & 17 & 1 & 2 & 3 \\ 2 & 3 & 0 & 14 & 2 & 0 & 2 \\ 3 & 4 & -2 & 19 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{141}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -1152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 2 & 3 \\ 3 & 4 & 0 & 29 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 16 & 29 & 0 & 148 & 20 & 16 & 22 \\ 2 & 4 & 0 & 20 & 1 & 2 & 3 \\ 2 & 3 & 0 & 16 & 2 & 0 & 2 \\ 3 & 4 & -2 & 22 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [29] \quad 1^{111}3^{11}5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 2 & 3 \\ 2 & 2 & 1 & 7 & 4 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 7 & 6 & 8 \\ 2 & 4 & 0 & 7 & 2 & 3 & 4 \\ 2 & 3 & 0 & 6 & 3 & 1 & 3 \\ 3 & 4 & -1 & 8 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{113}3^{11}5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 10 & 2 & 3 \\ 2 & 2 & 1 & 7 & 16 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 33 & 6 & 8 \\ 10 & 16 & 0 & 33 & 70 & 13 & 18 \\ 2 & 3 & 0 & 6 & 13 & 1 & 3 \\ 3 & 4 & -1 & 8 & 18 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{111}3^{11}5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 6 & 2 & 3 \\ 2 & 2 & 1 & 7 & 8 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 19 & 6 & 8 \\ 6 & 8 & 0 & 19 & 22 & 7 & 10 \\ 2 & 3 & 0 & 6 & 7 & 1 & 3 \\ 3 & 4 & -1 & 8 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{131}3^{11}5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 2 & 2 & 3 \\ 2 & 2 & 1 & 19 & 4 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 13 & 19 & 0 & 110 & 19 & 16 & 22 \\ 2 & 4 & 0 & 19 & 2 & 3 & 4 \\ 2 & 3 & 0 & 16 & 3 & 1 & 3 \\ 3 & 4 & -1 & 22 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{112}3^{11}15^{11}13^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 8 & 2 & 3 \\ 2 & 2 & 1 & 7 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 26 & 6 & 8 \\ 8 & 12 & 0 & 26 & 42 & 10 & 14 \\ 2 & 3 & 0 & 6 & 10 & 1 & 3 \\ 3 & 4 & -1 & 8 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

4.718 NHC(-2)(-3) + so₈ + su₂n3mix + su₂ + su₂n3mix

$$T = 9 \quad [0] \ 1^{121}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 7 & 3 \\ 3 & 4 & 0 & 11 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 8 & 23 & 8 \\ 2 & 4 & 0 & 8 & 1 & 8 & 3 \\ 7 & 12 & 0 & 23 & 8 & 25 & 9 \\ 3 & 4 & -2 & 8 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 4 & 3 \\ 3 & 4 & 0 & 7 & 8 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 10 & 9 & 5 \\ 4 & 8 & 0 & 10 & 10 & 10 & 6 \\ 4 & 7 & 0 & 9 & 10 & 7 & 5 \\ 3 & 4 & -2 & 5 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 4 & 3 \\ 3 & 4 & 0 & 7 & 9 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 11 & 9 & 5 \\ 5 & 9 & 0 & 11 & 13 & 11 & 7 \\ 4 & 7 & 0 & 9 & 11 & 7 & 5 \\ 3 & 4 & -2 & 5 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -303, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 7 & 3 \\ 3 & 4 & 0 & 7 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 5 & 15 & 5 \\ 2 & 4 & 0 & 5 & 1 & 8 & 3 \\ 7 & 12 & 0 & 15 & 8 & 25 & 9 \\ 3 & 4 & -2 & 5 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 4 & 3 \\ 3 & 4 & 0 & 7 & 4 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 5 & 9 & 5 \\ 2 & 4 & 0 & 5 & 1 & 5 & 3 \\ 4 & 7 & 0 & 9 & 5 & 7 & 5 \\ 3 & 4 & -2 & 5 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -438, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 12 & 3 \\ 3 & 4 & 0 & 7 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 5 & 26 & 5 \\ 2 & 4 & 0 & 5 & 1 & 14 & 3 \\ 12 & 21 & 0 & 26 & 14 & 80 & 16 \\ 3 & 4 & -2 & 5 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -448, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 4 & 3 \\ 3 & 4 & 0 & 7 & 13 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 16 & 9 & 5 \\ 7 & 13 & 0 & 16 & 28 & 16 & 10 \\ 4 & 7 & 0 & 9 & 16 & 7 & 5 \\ 3 & 4 & -2 & 5 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11123}1^1}1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -589, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 9 & 3 \\ 3 & 4 & 0 & 7 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 5 & 20 & 5 \\ 2 & 4 & 0 & 5 & 1 & 11 & 3 \\ 9 & 16 & 0 & 20 & 11 & 44 & 12 \\ 3 & 4 & -2 & 5 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [8] \ 1^{\overline{11123}1^1}1^{\overline{15}13^1}1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 10 & 3 \\ 2 & 2 & 1 & 7 & 4 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 7 & 32 & 8 \\ 2 & 4 & 0 & 7 & 2 & 15 & 4 \\ 10 & 15 & 0 & 32 & 15 & 65 & 17 \\ 3 & 4 & -1 & 8 & 4 & 17 & 4 \end{pmatrix}_{b_0}$$

4.719 NHC(-2)(-3) + so₈ + su_{2n3mix} + su₂→(-3)

$$T = 8 \quad [0] \ 1^{\overline{13}1^1}1^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 13 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 36 & 8 & 10 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{13}1^1}1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 12 & 5 & 6 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{113}1^1}1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 7 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 5 \\ 2 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{14}3^1}1^{\overline{1}}1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 19 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 19 & 0 & 64 & 10 & 14 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 3 & 4 & -2 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{12}3^1}1^{\overline{1}}1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 11 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 6 & 8 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & -2 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{13}3^1}1^{\overline{1}}1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} =$$

9.0

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 15 & 0 & 39 & 8 & 11 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 7 & 9 \\ 2 & 3 & 0 & 7 & 0 & 2 \\ 3 & 4 & -2 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.02 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 17 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 55 & 10 & 13 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 3 & 4 & -2 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 16 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 9 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 20 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 11 & 15 \\ 2 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & -2 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 80 & 12 & 16 \\ 2 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & -2 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{14} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 24 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 13 & 18 \\ 2 & 3 & 0 & 13 & 0 & 2 \\ 3 & 4 & -2 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{13} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 591, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 25 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 25 & 0 & 111 & 14 & 19 \\ 2 & 3 & 0 & 14 & 0 & 2 \\ 3 & 4 & -2 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{14} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 29 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 29 & 0 & 148 & 16 & 22 \\ 2 & 3 & 0 & 16 & 0 & 2 \\ 3 & 4 & -2 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [14] \quad 1^{11} 4 13^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11} 4 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 34, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 1 & 3 \\ 2 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 2 & 0 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{12} 4 13^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 3 \\ 3 & 4 & 0 & 8 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 5 & 8 & 0 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 4 13^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 59, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 1 & 3 \\ 3 & 4 & 0 & 9 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 0 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11} 4 13^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 65, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 & 0 \\ 2 & 2 & 0 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{12} 4 13^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 1 & 3 \\ 3 & 4 & 0 & 13 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 8 & 13 & 0 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11} 4 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 & 0 \\ 3 & 2 & 1 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11} 4 13^{11} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 1 & 3 \\ 2 & 0 & 1 & 6 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 10 & 6 & 1 & 32 & 0 & 10 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 2 & 0 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[21] $1^{11}413^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 1 & 3 \\ 2 & 0 & 1 & 3 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 4 & 3 & 1 & 2 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 2 & 0 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

[23] $1^{11}413^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 1 & 3 \\ 2 & 0 & 1 & 5 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 8 & 5 & 1 & 18 & 0 & 8 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 2 & 0 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

[24] $1^{11}413^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 1 & 3 \\ 2 & 0 & 1 & 4 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 6 & 4 & 1 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 2 & 0 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[25] $1^{12}413^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 0 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 & 0 \\ 4 & 4 & 1 & 2 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 5 & 0 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{13}413^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 6 & 0 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 & 0 \\ 5 & 6 & 1 & 3 & 17 & 0 & 7 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 7 & 0 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{12}413^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 4 & 0 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 & 0 \\ 5 & 4 & 2 & 2 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

[28] $1^{13}413^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 337, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 1 & 6 & 0 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 & 0 \\ 1 & 1 & 0 & -2 & 3 & 0 & 0 \\ 6 & 6 & 2 & 3 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 8 & 0 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{14}413^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 388, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 1 & 8 & 0 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 & 0 \\ 1 & 1 & 0 & -2 & 4 & 0 & 0 \\ 7 & 8 & 2 & 4 & 34 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 10 & 0 & 2 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [30] \quad 1^{12} 3^1 1^5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 12 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 15 & 3 & 3 \\ 12 & 0 & 15 & 65 & 17 & 17 \\ 3 & 0 & 3 & 17 & 4 & 4 \\ 3 & 0 & 3 & 17 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{11} 3^1 1^5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 3 \\ 2 & 2 & 1 & 7 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 6 & 8 \\ 2 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & -1 & 8 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{13} 3^1 1^5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 2 & 3 \\ 2 & 2 & 1 & 19 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 13 & 19 & 0 & 110 & 16 & 22 \\ 2 & 3 & 0 & 16 & 1 & 3 \\ 3 & 4 & -1 & 22 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{12} 3^1 1^5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 162, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 13 & 2 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 15 & 3 & 3 \\ 13 & 0 & 15 & 76 & 13 & 18 \\ 2 & 0 & 3 & 13 & 1 & 3 \\ 3 & 0 & 3 & 18 & 3 & 4 \end{pmatrix}_{b_0}$$

4.720 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 8 & 3 \\ 3 & 4 & 0 & 7 & 3 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 17 & 5 \\ 2 & 3 & 0 & 4 & 0 & 8 & 2 \\ 8 & 14 & 0 & 17 & 8 & 37 & 11 \\ 3 & 4 & -2 & 5 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 4 & 3 \\ 3 & 4 & 0 & 16 & 3 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 9 & 22 & 12 \\ 2 & 3 & 0 & 9 & 0 & 4 & 2 \\ 4 & 8 & 0 & 22 & 4 & 10 & 6 \\ 3 & 4 & -2 & 12 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 4 & 3 \\ 3 & 4 & 0 & 7 & 3 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 10 & 5 \\ 2 & 3 & 0 & 4 & 0 & 4 & 2 \\ 4 & 8 & 0 & 10 & 4 & 10 & 6 \\ 3 & 4 & -2 & 5 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 & 3 \\ 3 & 4 & 0 & 7 & 3 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 6 & 5 \\ 2 & 3 & 0 & 4 & 0 & 3 & 2 \\ 3 & 5 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & -2 & 5 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 2 & 3 \\ 3 & 4 & 0 & 7 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 5 & 5 \\ 2 & 3 & 0 & 4 & 0 & 2 & 2 \\ 2 & 4 & 0 & 5 & 2 & 1 & 3 \\ 3 & 4 & -2 & 5 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 2 & 3 \\ 3 & 4 & 0 & 11 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 6 & 8 & 8 \\ 2 & 3 & 0 & 6 & 0 & 2 & 2 \\ 2 & 4 & 0 & 8 & 2 & 1 & 3 \\ 3 & 4 & -2 & 8 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{132}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 4 & 3 \\ 3 & 4 & 0 & 20 & 3 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 11 & 28 & 15 \\ 2 & 3 & 0 & 11 & 0 & 4 & 2 \\ 4 & 8 & 0 & 28 & 4 & 10 & 6 \\ 3 & 4 & -2 & 15 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 5 & 3 \\ 3 & 4 & 0 & 7 & 3 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 11 & 5 \\ 2 & 3 & 0 & 4 & 0 & 5 & 2 \\ 5 & 9 & 0 & 11 & 5 & 13 & 7 \\ 3 & 4 & -2 & 5 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{142}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 4 & 3 \\ 3 & 4 & 0 & 29 & 3 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 16 & 29 & 0 & 148 & 16 & 40 & 22 \\ 2 & 3 & 0 & 16 & 0 & 4 & 2 \\ 4 & 8 & 0 & 40 & 4 & 10 & 6 \\ 3 & 4 & -2 & 22 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 0 & 12 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 7 & 8 & 9 \\ 2 & 3 & 0 & 7 & 0 & 2 & 2 \\ 2 & 4 & 0 & 8 & 2 & 1 & 3 \\ 3 & 4 & -2 & 9 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{114}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 11 & 3 \\ 3 & 4 & 0 & 7 & 3 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 26 & 5 \\ 2 & 3 & 0 & 4 & 0 & 11 & 2 \\ 11 & 21 & 0 & 26 & 11 & 76 & 16 \\ 3 & 4 & -2 & 5 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{112}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 10 & 3 \\ 3 & 4 & 0 & 7 & 3 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 22 & 5 \\ 2 & 3 & 0 & 4 & 0 & 10 & 2 \\ 10 & 18 & 0 & 22 & 10 & 58 & 14 \\ 3 & 4 & -2 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[12] \quad & 1^{123}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 7 & 3 \\ 3 & 4 & 0 & 11 & 3 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 6 & 25 & 8 \\ 2 & 3 & 0 & 6 & 0 & 7 & 2 \\ 7 & 13 & 0 & 25 & 7 & 28 & 10 \\ 3 & 4 & -2 & 8 & 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[13] \quad & 1^{1123}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 7 & 3 \\ 3 & 4 & 0 & 7 & 3 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 16 & 5 \\ 2 & 3 & 0 & 4 & 0 & 7 & 2 \\ 7 & 13 & 0 & 16 & 7 & 28 & 10 \\ 3 & 4 & -2 & 5 & 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[14] \quad & 1^{113}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 9 & 3 \\ 3 & 4 & 0 & 7 & 3 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 21 & 5 \\ 2 & 3 & 0 & 4 & 0 & 9 & 2 \\ 9 & 17 & 0 & 21 & 9 & 49 & 13 \\ 3 & 4 & -2 & 5 & 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[15] \quad & 1^{1423}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 4 & 3 \\ 3 & 4 & 0 & 24 & 3 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 13 & 34 & 18 \\ 2 & 3 & 0 & 13 & 0 & 4 & 2 \\ 4 & 8 & 0 & 34 & 4 & 10 & 6 \\ 3 & 4 & -2 & 18 & 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[16] \quad & 1^{123}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 9 & 3 \\ 3 & 4 & 0 & 11 & 3 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 6 & 33 & 8 \\ 2 & 3 & 0 & 6 & 0 & 9 & 2 \\ 9 & 17 & 0 & 33 & 9 & 49 & 13 \\ 3 & 4 & -2 & 8 & 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[17] \quad & 1^{1243}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 11 & 3 \\ 3 & 4 & 0 & 11 & 3 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 6 & 41 & 8 \\ 2 & 3 & 0 & 6 & 0 & 11 & 2 \\ 11 & 21 & 0 & 41 & 11 & 76 & 16 \\ 3 & 4 & -2 & 8 & 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[18] \quad & 1^{1213}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -477, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 2 & 3 \\ 3 & 4 & 0 & 16 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 9 & 16 & 0 & 44 & 9 & 11 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 & 2 \\ 2 & 4 & 0 & 11 & 2 & 1 & 3 \\ 3 & 4 & -2 & 12 & 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[19] \quad & 1^{113}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -526, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 12 & 3 \\ 3 & 4 & 0 & 7 & 3 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 27 & 5 \\ 2 & 3 & 0 & 4 & 0 & 12 & 2 \\ 12 & 22 & 0 & 27 & 12 & 85 & 17 \\ 3 & 4 & -2 & 5 & 2 & 17 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[20] \quad 1^{124}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 14 & 3 \\ 3 & 4 & 0 & 11 & 3 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 6 & 50 & 8 \\ 2 & 3 & 0 & 6 & 0 & 14 & 2 \\ 14 & 26 & 0 & 50 & 14 & 118 & 20 \\ 3 & 4 & -2 & 8 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{124}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 2 & 3 \\ 3 & 4 & 0 & 21 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 12 & 21 & 0 & 80 & 12 & 14 & 16 \\ 2 & 3 & 0 & 12 & 0 & 2 & 2 \\ 2 & 4 & 0 & 14 & 2 & 1 & 3 \\ 3 & 4 & -2 & 16 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{134}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -631, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 2 & 3 \\ 3 & 4 & 0 & 20 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 11 & 20 & 0 & 69 & 11 & 14 & 15 \\ 2 & 3 & 0 & 11 & 0 & 2 & 2 \\ 2 & 4 & 0 & 14 & 2 & 1 & 3 \\ 3 & 4 & -2 & 15 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{114}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -666, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 14 & 3 \\ 3 & 4 & 0 & 7 & 3 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 4 & 32 & 5 \\ 2 & 3 & 0 & 4 & 0 & 14 & 2 \\ 14 & 26 & 0 & 32 & 14 & 118 & 20 \\ 3 & 4 & -2 & 5 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{141}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -749, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 2 & 3 \\ 3 & 4 & 0 & 24 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 13 & 24 & 0 & 100 & 13 & 17 & 18 \\ 2 & 3 & 0 & 13 & 0 & 2 & 2 \\ 2 & 4 & 0 & 17 & 2 & 1 & 3 \\ 3 & 4 & -2 & 18 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{134}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -886, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 2 & 3 \\ 3 & 4 & 0 & 25 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 14 & 25 & 0 & 111 & 14 & 17 & 19 \\ 2 & 3 & 0 & 14 & 0 & 2 & 2 \\ 2 & 4 & 0 & 17 & 2 & 1 & 3 \\ 3 & 4 & -2 & 19 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{141}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -1152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 2 & 3 \\ 3 & 4 & 0 & 29 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 16 & 29 & 0 & 148 & 16 & 20 & 22 \\ 2 & 3 & 0 & 16 & 0 & 2 & 2 \\ 2 & 4 & 0 & 20 & 2 & 1 & 3 \\ 3 & 4 & -2 & 22 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [27] \quad 1^{113}3^{11}1513^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 10 & 3 \\ 2 & 2 & 1 & 7 & 3 & 16 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 6 & 33 & 8 \\ 2 & 3 & 0 & 6 & 1 & 13 & 3 \\ 10 & 16 & 0 & 33 & 13 & 70 & 18 \\ 3 & 4 & -1 & 8 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{111}3^{11}15^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 6 & 3 \\ 2 & 2 & 1 & 7 & 3 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 6 & 19 & 8 \\ 2 & 3 & 0 & 6 & 1 & 7 & 3 \\ 6 & 8 & 0 & 19 & 7 & 22 & 10 \\ 3 & 4 & -1 & 8 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{111}3^{11}15^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 2 & 3 \\ 2 & 2 & 1 & 7 & 3 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 6 & 7 & 8 \\ 2 & 3 & 0 & 6 & 1 & 3 & 3 \\ 2 & 4 & 0 & 7 & 3 & 2 & 4 \\ 3 & 4 & -1 & 8 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{131}3^{11}15^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 2 & 2 & 3 \\ 2 & 2 & 1 & 19 & 3 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 13 & 19 & 0 & 110 & 16 & 19 & 22 \\ 2 & 3 & 0 & 16 & 1 & 3 & 3 \\ 2 & 4 & 0 & 19 & 3 & 2 & 4 \\ 3 & 4 & -1 & 22 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{112}3^{11}15^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 8 & 3 \\ 2 & 2 & 1 & 7 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 6 & 26 & 8 \\ 2 & 3 & 0 & 6 & 1 & 10 & 3 \\ 8 & 12 & 0 & 26 & 10 & 42 & 14 \\ 3 & 4 & -1 & 8 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

4.721 NHC(-2)(-3) + so₈ + su₂n₃mix + su₂n₃mix

$$T = 8 \quad [0] \ 1^{113}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 12 & 10 & 6 \\ 4 & 7 & 0 & 10 & 7 & 5 \\ 3 & 4 & -2 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{132}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 9 & 3 \\ 3 & 4 & 0 & 15 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 15 & 0 & 39 & 42 & 11 \\ 9 & 16 & 0 & 42 & 44 & 12 \\ 3 & 4 & -2 & 11 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{121}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 11 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 23 & 8 \\ 7 & 12 & 0 & 23 & 25 & 9 \\ 3 & 4 & -2 & 8 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 7 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 21 & 5 \\ 10 & 17 & 0 & 21 & 55 & 13 \\ 3 & 4 & -2 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 7 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 9 & 5 \\ 4 & 7 & 0 & 9 & 7 & 5 \\ 3 & 4 & -2 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{112}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.02 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 0 & 7 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 14 & 5 \\ 6 & 11 & 0 & 14 & 20 & 8 \\ 3 & 4 & -2 & 5 & 8 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 15 & 5 \\ 7 & 12 & 0 & 15 & 25 & 9 \\ 3 & 4 & -2 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[7] $1^{113}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 7 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 19 & 5 \\ 8 & 15 & 0 & 19 & 39 & 11 \\ 3 & 4 & -2 & 5 & 11 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{114}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 235, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 7 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 24 & 5 \\ 10 & 19 & 0 & 24 & 64 & 14 \\ 3 & 4 & -2 & 5 & 14 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{123}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 263, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 14 & 3 \\ 3 & 4 & 0 & 11 & 25 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 48 & 8 \\ 14 & 25 & 0 & 48 & 111 & 19 \\ 3 & 4 & -2 & 8 & 19 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{122}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 0 & 11 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 22 & 8 \\ 6 & 11 & 0 & 22 & 20 & 8 \\ 3 & 4 & -2 & 8 & 8 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{123}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 9 & 3 \\ 3 & 4 & 0 & 11 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 31 & 8 \\ 9 & 16 & 0 & 31 & 44 & 12 \\ 3 & 4 & -2 & 8 & 12 & 2 \end{pmatrix}_{b_0}$$

[12] $1^{123}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 11 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 30 & 8 \\ 8 & 15 & 0 & 30 & 39 & 11 \\ 3 & 4 & -2 & 8 & 11 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{124}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 10 & 3 \\ 3 & 4 & 0 & 11 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 38 & 8 \\ 10 & 19 & 0 & 38 & 64 & 14 \\ 3 & 4 & -2 & 8 & 14 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 20 & 5 \\ 9 & 16 & 0 & 20 & 44 & 12 \\ 3 & 4 & -2 & 5 & 12 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{1123}3^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 367, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 7 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 26 & 5 \\ 12 & 21 & 0 & 26 & 80 & 16 \\ 3 & 4 & -2 & 5 & 16 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{123}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 461, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 11 & 3 \\ 3 & 4 & 0 & 11 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 39 & 8 \\ 11 & 20 & 0 & 39 & 69 & 15 \\ 3 & 4 & -2 & 8 & 15 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{113}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 485, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 0 & 7 & 20 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 25 & 5 \\ 11 & 20 & 0 & 25 & 69 & 15 \\ 3 & 4 & -2 & 5 & 15 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{124}3^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 552, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 16 & 3 \\ 3 & 4 & 0 & 11 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 56 & 8 \\ 16 & 29 & 0 & 56 & 148 & 22 \\ 3 & 4 & -2 & 8 & 22 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{114}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 592, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 13 & 3 \\ 3 & 4 & 0 & 7 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 30 & 5 \\ 13 & 24 & 0 & 30 & 100 & 18 \\ 3 & 4 & -2 & 5 & 18 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{113}3^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 0 & 7 & 25 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 31 & 5 \\ 14 & 25 & 0 & 31 & 111 & 19 \\ 3 & 4 & -2 & 5 & 19 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{124}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 616, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 13 & 3 \\ 3 & 4 & 0 & 11 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 47 & 8 \\ 13 & 24 & 0 & 47 & 100 & 18 \\ 3 & 4 & -2 & 8 & 18 & 2 \end{pmatrix}_{b_0}$$

[22] $1^{114}3^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 807, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 16 & 3 \\ 3 & 4 & 0 & 7 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 36 & 5 \\ 16 & 29 & 0 & 36 & 148 & 22 \\ 3 & 4 & -2 & 5 & 22 & 2 \end{pmatrix}_{b_0}$$

$T = 10$ [23] $1^{11}4^{13}11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & 4 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

- [24] $1^{11} 4^{11} 13^{11} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 43, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 4 & 3 \\ 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 1 & 0 & 0 & 2 & 2 \\ 4 & 3 & 1 & 2 & 2 & 4 \\ 3 & 2 & 0 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$
- [25] $1^{121} 4^{11} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 5 & 8 & 0 & 10 & 12 & 6 \\ 6 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -1 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$
- [26] $1^{111} 4^{11} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 88, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$
- [27] $1^{11} 4^{11} 13^{11} 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 109, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 6 & 3 \\ 2 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 1 & 0 & 0 & 4 & 2 \\ 6 & 4 & 1 & 4 & 8 & 6 \\ 3 & 2 & 0 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$
- [28] $1^{111} 4^{11} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 1 & 0 \\ 2 & 2 & 0 & 1 & 0 & 2 & 2 \\ 3 & 2 & 1 & 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 0 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$
- [29] $1^{111} 4^{11} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 120, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$
- [30] $1^{111} 4^{11} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 1 & 1 & 0 \\ 2 & 2 & 0 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [31] $1^{11} 4^{11} 13^{11} 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 8 & 3 \\ 2 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 1 & 0 & 0 & 6 & 2 \\ 8 & 5 & 1 & 6 & 18 & 8 \\ 3 & 2 & 0 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$
- [32] $1^{112} 4^{11} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 9 & 3 \\ 8 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[33] $1^{112}413^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 193, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 5 & 4 & 2 & 2 & 4 & 10 & 6 \\ 2 & 2 & 0 & 0 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{112}413^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 261, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 2 & 0 \\ 2 & 2 & 0 & 1 & 0 & 4 & 2 \\ 4 & 4 & 1 & 2 & 4 & 7 & 5 \\ 2 & 2 & 0 & 0 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{113}413^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 3 & 0 \\ 2 & 2 & 0 & 1 & 0 & 6 & 2 \\ 5 & 6 & 1 & 3 & 6 & 17 & 7 \\ 2 & 2 & 0 & 0 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{113}413^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 420, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 1 & 3 & 0 \\ 2 & 2 & 0 & 1 & 0 & 6 & 2 \\ 6 & 6 & 2 & 3 & 6 & 20 & 8 \\ 2 & 2 & 0 & 0 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{114}413^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 601, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & -2 & 0 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 1 & 4 & 0 \\ 2 & 2 & 0 & 1 & 0 & 8 & 2 \\ 7 & 8 & 2 & 4 & 8 & 34 & 10 \\ 2 & 2 & 0 & 0 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$T = 12$ [38] $1^{111}3^{11}513^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -31, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 1 & 2 & 5 & 10 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 1 & 7 & 11 & 3 \\ 5 & 0 & 7 & 14 & 25 & 8 \\ 10 & 0 & 11 & 25 & 39 & 13 \\ 3 & 0 & 3 & 8 & 13 & 4 \end{pmatrix}_{b_0}$$

[39] $1^{112}3^{11}513^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -31, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 10 & 3 \\ 2 & 2 & 1 & 7 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 32 & 8 \\ 10 & 15 & 0 & 32 & 65 & 17 \\ 3 & 4 & -1 & 8 & 17 & 4 \end{pmatrix}_{b_0}$$

4.722 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{su}_2$

$T = 9$ [0] $1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 2 & 3 \\ 3 & 4 & 0 & 11 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 23 & 8 & 8 \\ 7 & 12 & 0 & 23 & 25 & 8 & 9 \\ 2 & 4 & 0 & 8 & 8 & 1 & 3 \\ 3 & 4 & -2 & 8 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11123}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 4 & 3 \\ 3 & 4 & 0 & 7 & 7 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 9 & 10 & 5 \\ 4 & 7 & 0 & 9 & 7 & 10 & 5 \\ 4 & 8 & 0 & 10 & 10 & 10 & 6 \\ 3 & 4 & -2 & 5 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11113}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 5 & 3 \\ 3 & 4 & 0 & 7 & 7 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 9 & 11 & 5 \\ 4 & 7 & 0 & 9 & 7 & 11 & 5 \\ 5 & 9 & 0 & 11 & 11 & 13 & 7 \\ 3 & 4 & -2 & 5 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11113}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -303, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 2 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 15 & 5 & 5 \\ 7 & 12 & 0 & 15 & 25 & 8 & 9 \\ 2 & 4 & 0 & 5 & 8 & 1 & 3 \\ 3 & 4 & -2 & 5 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11113}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 2 & 3 \\ 3 & 4 & 0 & 7 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 9 & 5 & 5 \\ 4 & 7 & 0 & 9 & 7 & 5 & 5 \\ 2 & 4 & 0 & 5 & 5 & 1 & 3 \\ 3 & 4 & -2 & 5 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11213}} 3^{\overline{11}} 1^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -438, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 2 & 3 \\ 3 & 4 & 0 & 7 & 21 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 26 & 5 & 5 \\ 12 & 21 & 0 & 26 & 80 & 14 & 16 \\ 2 & 4 & 0 & 5 & 14 & 1 & 3 \\ 3 & 4 & -2 & 5 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{11123}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -448, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 7 & 3 \\ 3 & 4 & 0 & 7 & 7 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 9 & 16 & 5 \\ 4 & 7 & 0 & 9 & 7 & 16 & 5 \\ 7 & 13 & 0 & 16 & 16 & 28 & 10 \\ 3 & 4 & -2 & 5 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11213}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -589, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 2 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 20 & 5 & 5 \\ 9 & 16 & 0 & 20 & 44 & 11 & 12 \\ 2 & 4 & 0 & 5 & 11 & 1 & 3 \\ 3 & 4 & -2 & 5 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [8] \ 1^{\overline{11213}} 3^{\overline{11}} 1^{\overline{15}} 1^{\overline{13}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 10 & 2 & 3 \\ 2 & 2 & 1 & 7 & 15 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 7 & 0 & 14 & 32 & 7 & 8 \\ 10 & 15 & 0 & 32 & 65 & 15 & 17 \\ 2 & 4 & 0 & 7 & 15 & 2 & 4 \\ 3 & 4 & -1 & 8 & 17 & 4 & 4 \end{pmatrix}_{b_0}$$

4.723 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 24 & 1 & 5 \\ 11 & 20 & 45 & 0 & 8 \\ 24 & 45 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 18 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 12 & 24 & 1 & 5 \\ 12 & 25 & 48 & 0 & 9 \\ 24 & 48 & 89 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 17 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 4 & -2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{11}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 7 & 1 & 5 \\ 11 & 20 & 11 & 0 & 8 \\ 7 & 11 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{14}} 3^{\overline{11}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 4 & 0 & 19 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 19 & 0 & 64 & 15 & 14 \\ 3 & 5 & 0 & 15 & 3 & 3 \\ 3 & 4 & -2 & 14 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{14}} 3^{\overline{11}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 42, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 5 & 3 \\ 3 & 4 & 0 & 19 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 19 & 0 & 64 & 29 & 14 \\ 5 & 9 & 0 & 29 & 12 & 6 \\ 3 & 4 & -2 & 14 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{11}} 3^{\overline{11}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & -2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}} 3^{\overline{11}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 8 & 0 & 12 & 5 & 6 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{11}} 3^{\overline{11}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{aligned}
& \begin{pmatrix} 5 & 11 & 7 & 9 & 5 \\ 11 & 20 & 11 & 17 & 8 \\ 7 & 11 & 4 & 9 & 4 \\ 9 & 17 & 9 & 12 & 6 \\ 5 & 8 & 4 & 6 & 2 \end{pmatrix}_{b_0} \\
[9] & 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 5 & 11 & 7 & 10 & 5 \\ 11 & 20 & 11 & 17 & 8 \\ 7 & 11 & 4 & 9 & 4 \\ 10 & 17 & 9 & 12 & 6 \\ 5 & 8 & 4 & 6 & 2 \end{pmatrix}_{b_0} \\
[10] & 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 7 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 7 & 16 & 5 \\ 7 & 13 & 0 & 16 & 27 & 9 \\ 3 & 4 & -2 & 5 & 9 & 1 \end{pmatrix}_{b_0} \\
[11] & 1^{11} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 5 & 11 & 25 & 1 & 5 \\ 11 & 20 & 45 & 0 & 8 \\ 25 & 45 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 18 & -1 & 2 \end{pmatrix}_{b_0} \\
[12] & 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 7 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 7 & 10 & 5 \\ 4 & 8 & 0 & 10 & 9 & 5 \\ 3 & 4 & -2 & 5 & 5 & 1 \end{pmatrix}_{b_0} \\
[13] & 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 10 & 15 & 9 & 12 & 6 \\ 15 & 20 & 11 & 17 & 8 \\ 9 & 11 & 4 & 9 & 4 \\ 12 & 17 & 9 & 12 & 6 \\ 6 & 8 & 4 & 6 & 2 \end{pmatrix}_{b_0} \\
[14] & 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 5 & 11 & 16 & 1 & 5 \\ 11 & 20 & 28 & 0 & 8 \\ 16 & 28 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 11 & -1 & 2 \end{pmatrix}_{b_0} \\
[15] & 1^{12} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 6 & 3 & 3 \\ 3 & 4 & 0 & 11 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 9 & 8 \\ 3 & 5 & 0 & 9 & 3 & 3 \\ 3 & 4 & -2 & 8 & 3 & 1 \end{pmatrix}_{b_0} \\
[16] & 1^{12} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.85 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 12 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 16 & 9 \\ 4 & 8 & 0 & 16 & 9 & 5 \\ 3 & 4 & -2 & 9 & 5 & 1 \end{pmatrix}_{b_0} \\
[17] & 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 12 & 5 & 6 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 1 \end{pmatrix}_{b_0} \\
[18] & 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 5 & 11 & 15 & 1 & 5 \\ 11 & 20 & 28 & 0 & 8 \\ 15 & 28 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 11 & -1 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [19] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 5 & 12 & 16 & 1 & 5 \\ 12 & 25 & 32 & 0 & 9 \\ 16 & 32 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 11 & -1 & 2 \end{pmatrix}_{b_0}$
- [20] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 101$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 5 & 12 & 23 & 1 & 5 \\ 12 & 25 & 48 & 0 & 9 \\ 23 & 48 & 89 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 17 & -1 & 2 \end{pmatrix}_{b_0}$
- [21] $1^{11}3^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 104$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 27 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 16 & 27 & 0 & 151 & 17 & 21 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 3 & 4 & -2 & 21 & 2 & 1 \end{pmatrix}_{b_0}$
- [22] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 106$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 7 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 6 & 5 \\ 3 & 5 & 0 & 6 & 3 & 3 \\ 3 & 4 & -2 & 5 & 3 & 1 \end{pmatrix}_{b_0}$
- [23] $1^{13}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 114$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 15 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 15 & 0 & 39 & 12 & 11 \\ 3 & 5 & 0 & 12 & 3 & 3 \\ 3 & 4 & -2 & 11 & 3 & 1 \end{pmatrix}_{b_0}$
- [24] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 117$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 5 & 12 & 15 & 1 & 5 \\ 12 & 25 & 32 & 0 & 9 \\ 15 & 32 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 11 & -1 & 2 \end{pmatrix}_{b_0}$
- [25] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 118$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 7 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 7 & 6 & 5 \\ 3 & 5 & 0 & 6 & 3 & 3 \\ 3 & 4 & -2 & 5 & 3 & 1 \end{pmatrix}_{b_0}$
- [26] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 133$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 7 & 11 & 5 \\ 5 & 9 & 0 & 11 & 12 & 6 \\ 3 & 4 & -2 & 5 & 6 & 1 \end{pmatrix}_{b_0}$
- [28] $1^{12}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 141$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 11 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 0 & 20 & 9 & 8 \\ 3 & 5 & 0 & 9 & 3 & 3 \\ 3 & 4 & -2 & 8 & 3 & 1 \end{pmatrix}_{b_0}$
- [27] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 141$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 7 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 10 & 5 \\ 4 & 8 & 0 & 10 & 9 & 5 \\ 3 & 4 & -2 & 5 & 5 & 1 \end{pmatrix}_{b_0}$

- [29] $1^{11} 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 5 & 3 \\ 3 & 4 & 0 & 15 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 15 & 0 & 39 & 23 & 11 \\ 5 & 9 & 0 & 23 & 12 & 6 \\ 3 & 4 & -2 & 11 & 6 & 1 \end{pmatrix}_{b_0}$$
- [30] $1^{11} 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 5 & 5 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$
- [31] $1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 11 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 17 & 8 \\ 5 & 9 & 0 & 17 & 12 & 6 \\ 3 & 4 & -2 & 8 & 6 & 1 \end{pmatrix}_{b_0}$$
- [32] $1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 0 & 7 & 5 & 5 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$
- [33] $1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 11 & 5 \\ 5 & 9 & 0 & 11 & 12 & 6 \\ 3 & 4 & -2 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$
- [34] $1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 11 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 0 & 20 & 17 & 8 \\ 5 & 9 & 0 & 17 & 12 & 6 \\ 3 & 4 & -2 & 8 & 6 & 1 \end{pmatrix}_{b_0}$$
- [35] $1^{111} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 22 & 0 & 97 & 14 & 17 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 3 & 4 & -2 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$
- [36] $1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 11 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 0 & 20 & 8 & 8 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$
- [37] $1^{141} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 262, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 5 & 3 \\ 3 & 4 & 0 & 19 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 19 & 0 & 64 & 29 & 14 \\ 5 & 9 & 0 & 29 & 12 & 6 \\ 3 & 4 & -2 & 14 & 6 & 1 \end{pmatrix}_{b_0}$$
- [38] $1^{111} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 27 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 27 & 0 & 151 & 17 & 21 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 3 & 4 & -2 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

- [39] $1^{131}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 269$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 15 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 15 & 0 & 39 & 23 & 11 \\ 5 & 9 & 0 & 23 & 12 & 6 \\ 3 & 4 & -2 & 11 & 6 & 1 \end{pmatrix}_{b_0}$$
- [40] $1^{111}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 272$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 25 & 8 & 9 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$
- [41] $1^{112}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 278$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.17 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 7 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 0 & 7 & 16 & 5 \\ 7 & 13 & 0 & 16 & 27 & 9 \\ 3 & 4 & -2 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$
- [42] $1^{111}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 296$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 55 & 11 & 13 \\ 2 & 4 & 0 & 11 & 0 & 2 \\ 3 & 4 & -2 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$
- [43] $1^{131}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 312$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 15 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 15 & 0 & 39 & 11 & 11 \\ 2 & 4 & 0 & 11 & 0 & 2 \\ 3 & 4 & -2 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$
- [44] $1^{111}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 336$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 0 & 25 & 8 & 9 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$
- [45] $1^{111}3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 368$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 14 & 22 & 0 & 97 & 14 & 17 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 3 & 4 & -2 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$
- [46] $1^{121}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 368$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 11 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 0 & 20 & 8 & 8 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 3 & 4 & -2 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$
- [47] $1^{141}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 384$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 19 & 0 & 64 & 14 & 14 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 3 & 4 & -2 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$
- [48] $1^{111}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 392$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 17 & 0 & 55 & 11 & 13 \\ 2 & 4 & 0 & 11 & 0 & 2 \\ 3 & 4 & -2 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{141}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 1536$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 2 & 3 \\ 3 & 4 & 0 & 34 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 19 & 34 & 0 & 208 & 23 & 26 \\ 2 & 4 & 0 & 23 & 0 & 2 \\ 3 & 4 & -2 & 26 & 2 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{131}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 1560$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 2 & 3 \\ 3 & 4 & 0 & 35 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 21 & 35 & 0 & 231 & 23 & 27 \\ 2 & 4 & 0 & 23 & 0 & 2 \\ 3 & 4 & -2 & 27 & 2 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{141}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 1616$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.19 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 29 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 17 & 29 & 0 & 148 & 20 & 22 \\ 2 & 4 & 0 & 20 & 0 & 2 \\ 3 & 4 & -2 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{141}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 1760$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.17 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 2 & 3 \\ 3 & 4 & 0 & 39 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 22 & 39 & 0 & 280 & 26 & 30 \\ 2 & 4 & 0 & 26 & 0 & 2 \\ 3 & 4 & -2 & 30 & 2 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{141}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 1952$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 0 & 34 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 20 & 34 & 0 & 208 & 23 & 26 \\ 2 & 4 & 0 & 23 & 0 & 2 \\ 3 & 4 & -2 & 26 & 2 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{141}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 2208$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 2 & 3 \\ 3 & 4 & 0 & 39 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 23 & 39 & 0 & 280 & 26 & 30 \\ 2 & 4 & 0 & 26 & 0 & 2 \\ 3 & 4 & -2 & 30 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 11$ [76] $1^{12}5^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 11$, $T_{\max} = 13$, $\Delta = -59$, $\det = -61$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 14 & 0 & 4 \\ 2 & 3 & 0 & 11 & 0 & 2 \\ 8 & 14 & 11 & 39 & 0 & 13 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 13 & -1 & 3 \end{pmatrix}_{b_0}$$

[75] $1^{11}5^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 11$, $T_{\max} = 13$, $\Delta = -59$, $\det = -61$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 7 & 0 & 4 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 4 & 7 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

[77] $1^{13}5^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3$ $T = 11$, $T_{\max} = 13$, $\Delta = -59$, $\det = -133$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.98 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 3 & 18 & 0 & 4 \\ 2 & 3 & 0 & 13 & 0 & 2 \\ 10 & 18 & 13 & 62 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

4.724 NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{su}_3$ + $\mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 1 & 7 & 9 & 5 \\ 11 & 20 & 0 & 11 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 11 & 0 & 4 & 9 & 4 \\ 9 & 17 & 0 & 9 & 12 & 6 \\ 5 & 8 & -2 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

4.725 NHC(-2)(-3) + $\mathfrak{so}_8$ + su2n3mix + (-3)→(-2)

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 24 & 1 & 7 \\ 17 & 20 & 28 & 0 & 8 \\ 24 & 28 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$

4.726 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$

$$T = 7 \quad [0] \quad 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 14.14 \rightarrow 15, T_{\max} = 71, T \in [15, 71]$$

$$\begin{pmatrix} 5 & 11 & 18 & 5 \\ 11 & 20 & 34 & 8 \\ 18 & 34 & 57 & 13 \\ 5 & 8 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 21^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 14, T_{\max} = 71, T \in [14, 71]$$

$$\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 25 & 20 & 9 \\ 12 & 20 & 12 & 6 \\ 6 & 9 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 14, T_{\max} = 71, T \in [14, 71]$$

$$\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 25 & 20 & 9 \\ 12 & 20 & 12 & 6 \\ 6 & 9 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 10 & 15 & 25 & 6 \\ 15 & 20 & 34 & 8 \\ 25 & 34 & 57 & 13 \\ 6 & 8 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 31^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 5 & 11 & 19 & 5 \\ 11 & 20 & 34 & 8 \\ 19 & 34 & 57 & 13 \\ 5 & 8 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 5 & 12 & 8 & 5 \\ 12 & 25 & 16 & 9 \\ 8 & 16 & 9 & 5 \\ 5 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 31^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 10 & 16 & 23 & 6 \\ 16 & 25 & 36 & 9 \\ 23 & 36 & 48 & 12 \\ 6 & 9 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 31^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 5 & 12 & 11 & 5 \\ 12 & 25 & 20 & 9 \\ 11 & 20 & 12 & 6 \\ 5 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

- [8] $1^1 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 71$, $T \in [14, 71]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 10 & 0 & 21 & 7 \\ 3 & 4 & -2 & 7 & -1 \end{pmatrix}_{b_0}$
- [9] $1^1 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & -1 & 1 \end{pmatrix}_{b_0}$
- [10] $1^1 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -3 & -1 \\ 5 & 9 & -1 & 2 \end{pmatrix}_{b_0}$
- [11] $1^1 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -43$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 5 & 0 & 3 & 3 \\ 3 & 4 & -2 & 3 & 1 \end{pmatrix}_{b_0}$
- [12] $1^1 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -43$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 5 & 11 & 1 & 5 \\ 11 & 20 & 0 & 8 \\ 1 & 0 & -3 & -1 \\ 5 & 8 & -1 & 2 \end{pmatrix}_{b_0}$
- [13] $1^1 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -43$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 5 & 11 & 10 & 5 \\ 11 & 20 & 17 & 8 \\ 10 & 17 & 12 & 6 \\ 5 & 8 & 6 & 2 \end{pmatrix}_{b_0}$
- [18] $1^{15} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.33 \rightarrow 15$, $T_{\max} = 71$, $T \in [15, 71]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 20 & 0 & 72 & 14 \\ 3 & 4 & -2 & 14 & -2 \end{pmatrix}_{b_0}$
- [16] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.33 \rightarrow 13$, $T_{\max} = 71$, $T \in [13, 71]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 12 & 0 & 24 & 8 \\ 3 & 4 & -2 & 8 & 0 \end{pmatrix}_{b_0}$
- [15] $1^{14} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.6$
 $T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 71$, $T \in [14, 71]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 16 & 0 & 45 & 11 \\ 3 & 4 & -2 & 11 & -1 \end{pmatrix}_{b_0}$
- [14] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 71$, $T \in [11, 71]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$
- [17] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 4 & 8 & 0 & 9 & 5 \\ 3 & 4 & -2 & 5 & 1 \end{pmatrix}_{b_0}$
- [19] $1^1 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -49$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 5 & 11 & 9 & 5 \\ 11 & 20 & 17 & 8 \\ 9 & 17 & 12 & 6 \\ 5 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 10 & 15 & 12 & 6 \\ 15 & 20 & 17 & 8 \\ 12 & 17 & 12 & 6 \\ 6 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 4 & 5 & 0 & 3 & 3 \\ 3 & 4 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 10 & 15 & 1 & 6 \\ 15 & 20 & 0 & 8 \\ 1 & 0 & -3 & -1 \\ 6 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 10 & 15 & 13 & 6 \\ 15 & 20 & 17 & 8 \\ 13 & 17 & 12 & 6 \\ 6 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 5 & 12 & 18 & 5 \\ 12 & 25 & 36 & 9 \\ 18 & 36 & 48 & 12 \\ 5 & 9 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{11} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 71, T \in [13, 71]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 10 & 0 & 21 & 7 \\ 3 & 4 & -2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 21^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.06 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 13 & 17 & 15 & 7 \\ 17 & 20 & 17 & 8 \\ 15 & 17 & 12 & 6 \\ 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 13 1^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 71, T \in [13, 71]$$

$$\begin{pmatrix} 5 & 12 & 9 & 5 \\ 12 & 25 & 20 & 9 \\ 9 & 20 & 12 & 6 \\ 5 & 9 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 13 1^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 5 & 12 & 17 & 5 \\ 12 & 25 & 36 & 9 \\ 17 & 36 & 48 & 12 \\ 5 & 9 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 5 & 12 & 10 & 5 \\ 12 & 25 & 20 & 9 \\ 10 & 20 & 12 & 6 \\ 5 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[31] \ 21^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -77, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 13 & 20 & 12 & 7 \\ 20 & 25 & 16 & 9 \\ 12 & 16 & 9 & 5 \\ 7 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 9 & 0 & 12 & 6 \\ 3 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[33] \ 21^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 20 & 17 & 8 \\ 12 & 17 & 12 & 6 \\ 6 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{12} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 8 & 0 & 9 & 5 \\ 3 & 4 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{11} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 14 & 0 & 36 & 10 \\ 3 & 4 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{12} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 13 & 0 & 27 & 9 \\ 3 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 6 & 9 & 0 & 12 & 6 \\ 3 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{13} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 12 & 0 & 24 & 8 \\ 3 & 4 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{14} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.6$$

$$T_{\min} = 11.48 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 16 & 0 & 45 & 11 \\ 3 & 4 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{13} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 48 & 12 \\ 3 & 4 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{11} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 14 & 0 & 36 & 10 \\ 3 & 4 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{12} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -188, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 18 & 0 & 57 & 13 \\ 3 & 4 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -195, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 8 & 13 & 0 & 27 & 9 \\ 3 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -203, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.6$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 21 & 0 & 75 & 15 \\ 3 & 4 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[45] $21^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.03 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 13 & 20 & 27 & 7 \\ 20 & 25 & 36 & 9 \\ 27 & 36 & 48 & 12 \\ 7 & 9 & 12 & 2 \end{pmatrix}_{b_0}$$

[46] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -267, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 10 & 17 & 0 & 48 & 12 \\ 3 & 4 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 22 & 0 & 84 & 16 \\ 3 & 4 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[48] $21^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -275, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 13 & 20 & 15 & 7 \\ 20 & 25 & 20 & 9 \\ 15 & 20 & 12 & 6 \\ 7 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[49] $1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -284, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 11 & 18 & 0 & 57 & 13 \\ 3 & 4 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -339, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.6$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 12 & 21 & 0 & 75 & 15 \\ 3 & 4 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{14} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -348, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.6$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 14 & 26 & 0 & 117 & 19 \\ 3 & 4 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -396, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 71, T \in [11, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 13 & 22 & 0 & 84 & 16 \\ 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{14} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -508, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.6$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 71, T \in [11, 71]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & -2 \\ 15 & 26 & 0 & 117 & 19 \\ 3 & 4 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& T = 8 \quad [54] \quad 1^{\overline{11}} 4^{\overline{11}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 18, T_{\max} = 23, T \in [18, 23] \\
& \begin{pmatrix} 5 & 9 & 9 & 5 \\ 9 & 13 & 10 & 7 \\ 9 & 10 & 4 & 4 \\ 5 & 7 & 4 & -3 \end{pmatrix}_{b_0} \\
& [55] \quad 21^{\overline{11}} 4^{\overline{11}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12, T_{\max} = 23, T \in [12, 23] \\
& \begin{pmatrix} 10 & 16 & 8 & 6 \\ 16 & 25 & 12 & 9 \\ 8 & 12 & 4 & 4 \\ 6 & 9 & 4 & 3 \end{pmatrix}_{b_0} \\
& [56] \quad 1^{\overline{11}} 4^{\overline{11}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12, T_{\max} = 23, T \in [12, 23] \\
& \begin{pmatrix} 1 & 4 & 4 & 1 & 3 \\ 4 & 9 & 8 & 0 & 5 \\ 4 & 8 & 6 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & 2 \end{pmatrix}_{b_0} \\
& [57] \quad 131^{\overline{11}} 31^{\overline{11}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 15, T_{\max} = 20, T \in [15, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 3 & 3 \\ 1 & 4 & 0 & 3 & 3 & 2 \\ 2 & 3 & 3 & 2 & 3 & 2 \\ 1 & 3 & 3 & 3 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 & -2 \end{pmatrix}_{b_0} \\
& [58] \quad 1^{\overline{12}} 321^{\overline{11}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 5 & 5 & 10 & 5 \\ 2 & 5 & 0 & 7 & 2 \\ 7 & 10 & 7 & 13 & 7 \\ 3 & 5 & 2 & 7 & 2 \end{pmatrix}_{b_0} \\
& [59] \quad 1^{\overline{11}} 321^{\overline{11}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 1 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 1 \end{pmatrix}_{b_0} \\
& [60] \quad 1^{\overline{11}} 3131^{\overline{11}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 4 & 1 & 1 & 3 \\ 4 & 10 & 1 & 0 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & 2 \end{pmatrix}_{b_0} \\
& [61] \quad 1^{\overline{11}} 321^{\overline{11}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 2 & 5 & 0 & 3 & 2 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 5 & 2 & 3 & 2 \end{pmatrix}_{b_0} \\
& [62] \quad 1^{\overline{11}} 321^{\overline{11}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 4 & 0 & 9 & 5 \\ 3 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0} \\
& [63] \quad 1^{\overline{11}} 321^{\overline{11}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 5 & 3 & 3 \\ 4 & 5 & 9 & 4 & 5 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 3 & 5 & 2 & 2 \end{pmatrix}_{b_0} \\
& [64] \quad 1^{\overline{11}} 31^{\overline{11}} 31^{\overline{11}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 13.41 \rightarrow 14, T_{\max} = 20, T \in [14, 20] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 0 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 3 & -2 & -1 & -1 \end{pmatrix}_{b_0} \\
& [65] \quad 1^{\overline{11}} 31^{\overline{11}} 31^{\overline{11}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0
\end{aligned}$$

$$T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 4 & 3 & 3 & 3 \\ 1 & 4 & 0 & 3 & 3 & 2 \\ 2 & 3 & 3 & 2 & 3 & 2 \\ 2 & 3 & 3 & 3 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{11} 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 4 & 3 & 7 & 3 & 4 \\ 3 & 1 & 5 & 3 & 3 \\ 7 & 5 & 9 & 4 & 5 \\ 3 & 3 & 4 & 0 & 2 \\ 4 & 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{11} 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 4 & 3 \\ 1 & 1 & 0 & 4 & 0 & 2 \\ 2 & 2 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{11} 3 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 6 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 11 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & -1 & 5 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{11} 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 1 & 3 & 0 & 0 & 2 \\ 2 & 2 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{11} 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & -2 & 1 & 0 \\ 2 & 1 & 3 & 1 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{11} 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 4 & 5 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{11} 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{13} 3 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 9 & 2 \\ 4 & 9 & 0 & 9 & 24 & 8 \\ 2 & 3 & -1 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{12} 3 2 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 1 & 6 & 3 \\ 1 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -2 & 2 & 0 \\ 4 & 6 & 6 & 2 & 9 & 5 \\ 2 & 3 & 2 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{11} 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

- [76] $1^1 1^2 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 21$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 2 & 5 & 0 & 4 & 2 \\ 4 & 5 & 4 & 1 & 3 \\ 3 & 5 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [80] $2^1 2^1 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 22$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.55 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 4 & 3 & 8 & 3 & 4 \\ 3 & 1 & 5 & 3 & 3 \\ 8 & 5 & 9 & 4 & 5 \\ 3 & 3 & 4 & 0 & 2 \\ 4 & 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$
- [79] $1^1 1^1 3 2 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 22$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$
- [77] $1^1 1^1 3 2 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 22$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 3 & 5 & 0 & 3 & 3 \\ 3 & 5 & -2 & 3 & 1 \end{pmatrix}_{b_0}$
- [78] $1^1 1^1 3 2 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 22$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 1 & 0 \\ 2 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$
- [81] $1^1 1^1 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 24$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 13$, $T \in [11, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 5 & 0 & 1 & 3 \\ 3 & 5 & -2 & 3 & 3 \end{pmatrix}_{b_0}$
- [82] $1^1 1^1 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 24$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 3 & 5 & 0 & 3 & 2 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 5 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [83] $2 1^1 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 25$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.20 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 4 & 5 & 3 & 9 & 4 \\ 5 & 5 & 5 & 10 & 5 \\ 3 & 5 & 0 & 7 & 2 \\ 9 & 10 & 7 & 13 & 7 \\ 4 & 5 & 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [84] $1^1 3 1^1 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 26$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & -1 & 2 & -1 & 0 \end{pmatrix}_{b_0}$
- [85] $1^1 1^1 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 28$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 7 & 0 & 20 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$
- [90] $1^1 1^1 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 29$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & -1 & -1 & 2 \end{pmatrix}_{b_0}$

- [89] $1^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 29$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 0 & -1 & 1 \end{pmatrix}_{b_0}$
- [88] $1^{12}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 29$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.41 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 6 & 0 & 9 & 5 \\ 3 & 3 & -1 & 5 & 1 \end{pmatrix}_{b_0}$
- [87] $1^{11}31^131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 29$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 3 & 2 \\ 1 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [86] $1^{11}31^131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 29$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 12 & 0 & 12 & 6 \\ 3 & 6 & -1 & 6 & 2 \end{pmatrix}_{b_0}$
- [91] $2^231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 30$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.27 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 2 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & -2 & 1 & 0 \\ 2 & 1 & 3 & 1 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 1 \end{pmatrix}_{b_0}$
- [95] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 1 & 3 & 3 \\ 1 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 2 & 3 & 3 & 1 & 0 & 2 \\ 2 & 3 & 2 & 0 & 2 & 1 \end{pmatrix}_{b_0}$
- [94] $1^{11}41^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 23$, $\Delta = -113$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 23$, $T \in [12, 23]$
 $\begin{pmatrix} 10 & 12 & 8 & 6 \\ 12 & 13 & 10 & 7 \\ 8 & 10 & 4 & 4 \\ 6 & 7 & 4 & 3 \end{pmatrix}_{b_0}$
- [93] $1^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & -2 & 2 \end{pmatrix}_{b_0}$
- [97] $21^141^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 23$, $\Delta = -113$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 23$, $T \in [12, 23]$
 $\begin{pmatrix} 10 & 12 & 8 & 6 \\ 12 & 13 & 10 & 7 \\ 8 & 10 & 4 & 4 \\ 6 & 7 & 4 & 3 \end{pmatrix}_{b_0}$
- [92] $1^{13}1^131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 0 & 4 \\ 1 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 2 & -2 & -1 & 1 \end{pmatrix}_{b_0}$
- [96] $1^141^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 23$, $\Delta = -113$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$
 $\begin{pmatrix} 5 & 12 & 6 & 5 \\ 12 & 25 & 12 & 9 \\ 6 & 12 & 4 & 4 \\ 5 & 9 & 4 & 2 \end{pmatrix}_{b_0}$

- [98] $1^{11}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$
- [99] $1^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 23$, $\Delta = -113$, $\det = 35$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 23$, $T \in [12, 23]$
 $\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 0 & 7 \\ 1 & 0 & -3 & -1 \\ 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$
- [101] $1^{11}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 36$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 3 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & 3 & -1 & 1 \end{pmatrix}_{b_0}$
- [100] $1^{11}31^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 36$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 6 & 0 & 12 & 6 \\ 2 & 3 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$
- [102] $1^{11}31^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 38$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 10 & 13 & 0 & 6 \\ 5 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 6 & 6 & -1 & 2 \end{pmatrix}_{b_0}$
- [103] $1^{11}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 39$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 4 & 3 \\ 2 & 1 & 0 & 4 & 0 & 2 \\ 2 & 2 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [104] $1^{11}3131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 41$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 13.17 \rightarrow 14$, $T_{\max} = 20$, $T \in [14, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 2 & -2 & 2 & -1 \end{pmatrix}_{b_0}$
- [105] $1^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 23$, $\Delta = -113$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 23$, $T \in [12, 23]$
 $\begin{pmatrix} 5 & 9 & 8 & 5 \\ 9 & 13 & 10 & 7 \\ 8 & 10 & 4 & 4 \\ 5 & 7 & 4 & 3 \end{pmatrix}_{b_0}$
- [106] $1^{11}31^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 4 & 8 & 8 & 7 & 4 \\ 8 & 13 & 16 & 12 & 7 \\ 8 & 16 & 12 & 12 & 6 \\ 7 & 12 & 12 & 9 & 5 \\ 4 & 7 & 6 & 5 & 2 \end{pmatrix}_{b_0}$
- [107] $1^{11}31^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & -2 & 2 & 1 \end{pmatrix}_{b_0}$
- [109] $1^{11}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 3 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 3 & -1 & 1 \end{pmatrix}_{b_0}$
- [108] $1^{11}31^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$

$$\begin{aligned}
& T_{\min} = 11.73 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 3 & 2 \\ 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[110] & 1^{12} 3^1 1^1 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 6 & 0 & 9 & 5 \\ 2 & 3 & 3 & -2 & 5 & 1 \end{pmatrix}_{b_0} \\
[111] & 1^{11} 3^1 1^1 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 4 & 7 & 8 & 1 & 4 & 4 \\ 7 & 10 & 13 & 0 & 6 & 6 \\ 8 & 13 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 4 & 6 & 6 & -1 & 2 & 2 \end{pmatrix}_{b_0} \\
[112] & 1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12, T_{\max} = 23, T \in [12, 23] \\
& \begin{pmatrix} 5 & 9 & 6 & 5 & 5 \\ 9 & 13 & 10 & 7 & 7 \\ 6 & 10 & 4 & 4 & 4 \\ 5 & 7 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[113] & 1^{11} 12^1 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0} \\
[114] & 3^1 1^{11} 2^1 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 5 & 4 & 3 & 3 & 3 \\ 5 & 5 & 5 & 5 & 5 & 5 \\ 4 & 5 & 0 & 3 & 2 & 2 \\ 3 & 5 & 3 & 3 & 3 & 3 \\ 3 & 5 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[116] & 1^{11} 1^1 3^1 2^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 & 5 \\ 3 & 5 & 0 & 4 & 2 & 2 \\ 2 & 5 & 4 & 1 & 3 & 3 \\ 3 & 5 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[115] & 1^{11} 2^1 2^1 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 3 & 3 & 3 & 3 \\ 4 & 5 & 5 & 5 & 5 & 5 \\ 3 & 5 & 0 & 3 & 2 & 2 \\ 3 & 5 & 3 & 3 & 3 & 3 \\ 3 & 5 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[119] & 1^{11} 1^{11} 3^1 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 5 & 0 & 25 & 5 & 5 \\ 1 & 0 & -4 & 0 & -2 & -2 \\ 11 & 25 & 0 & 91 & 19 & 19 \\ 3 & 5 & -2 & 19 & 2 & 2 \end{pmatrix}_{b_0} \\
[118] & 1^{12} 3^1 1^1 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 3 & 6 & 0 & 6 & 9 & 5 \\ 2 & 3 & -1 & 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[117] & 1^{11} 1^1 3^1 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 & -1 \\ 3 & 3 & 0 & 0 & 2 & 2 \\ 3 & 3 & -1 & 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[123] & 1^{11} 1^1 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0} \\
[122] & 1^{11} 31^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 3 & 2 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[121] & 1^{11} 321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.79 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 2 & 5 & 0 & 4 & 2 \\ 3 & 5 & 4 & 1 & 3 \\ 3 & 5 & 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[120] & 1^{11} 321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 7 & 0 & 15 & 7 \\ 2 & 2 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0} \\
[125] & 1^{12} 321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 10 & 0 & 37 & 11 \\ 3 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0} \\
[124] & 1^{11} 321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 5 & 0 & 10 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 5 & 10 & 0 & 13 & 7 \\ 3 & 5 & -2 & 7 & 2 \end{pmatrix}_{b_0} \\
[127] & 1^{11} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.76 \rightarrow 12, T_{\max} = 23, T \in [12, 23] \\
& \begin{pmatrix} 5 & 9 & 7 & 5 \\ 9 & 13 & 10 & 7 \\ 7 & 10 & 4 & 4 \\ 5 & 7 & 4 & 3 \end{pmatrix}_{b_0} \\
[126] & 21^{11} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.76 \rightarrow 12, T_{\max} = 23, T \in [12, 23] \\
& \begin{pmatrix} 13 & 14 & 9 & 7 \\ 14 & 13 & 10 & 7 \\ 9 & 10 & 4 & 4 \\ 7 & 7 & 4 & 3 \end{pmatrix}_{b_0} \\
[128] & 1^{11} 321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 0 & 4 & 4 \\ 2 & 2 & 2 & -2 & 4 & 1 \end{pmatrix}_{b_0} \\
[129] & 1^{11} 321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.48 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 5 & 5 & 10 & 5 \\ 2 & 5 & 0 & 7 & 2 \\ 6 & 10 & 7 & 13 & 7 \\ 3 & 5 & 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[130] & 1^{11} 321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 5 & 0 & 20 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 20 & 0 & 57 & 15 \\ 3 & 5 & -2 & 15 & 2 \end{pmatrix}_{b_0} \\
[131] & 1^{11} 321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 5 & 0 & 15 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 7 & 15 & 0 & 31 & 11 \\ 3 & 5 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 74, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0 \\ T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 5 & 5 & 10 & 5 \\ 2 & 5 & 0 & 7 & 2 \\ 5 & 10 & 7 & 13 & 7 \\ 3 & 5 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 74, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0 \\ T_{\min} = 11.16 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 7 & 0 & 20 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{11} 3^{11} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 77, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2 \\ T_{\min} = 12.16 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 1 & 6 & 3 \\ 1 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -2 & 2 & 0 \\ 3 & 6 & 6 & 2 & 9 & 5 \\ 2 & 3 & 2 & 0 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[135] \ 131^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 77, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0 \\ T_{\min} = 12.26 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 3 \\ 5 & 13 & 16 & 12 & 7 \\ 5 & 16 & 12 & 12 & 6 \\ 4 & 12 & 12 & 9 & 5 \\ 3 & 7 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[136] \ 1^{12} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 77, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2 \\ T_{\min} = 11.74 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 6 & 0 & 9 & 5 \\ 3 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 79, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0 \\ T_{\min} = 11.16 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 5 & 5 & 10 & 5 \\ 3 & 5 & 0 & 7 & 2 \\ 5 & 10 & 7 & 13 & 7 \\ 3 & 5 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[138] \ 1^{14} 31^1 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 81, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.6 \\ T_{\min} = 11.41 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 12 & 2 \\ 6 & 12 & 0 & 12 & 45 & 11 \\ 2 & 3 & -1 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{13} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 84, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0 \\ T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 13 & 0 & 60 & 14 \\ 3 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{13} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 84, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0 \\ T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[141] \ 1^{12} 31^1 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 89, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2 \\ T_{\min} = 11.46 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 4 & 6 & 0 & 6 & 9 & 5 \\ 2 & 3 & -1 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{12} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 92, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2 \\ T_{\min} = 11.26 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 5 & 0 & 12 & 6 \\ 2 & 2 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[143] $1^{13}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 24 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[144] $1^{13}31^{13}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 9 & 2 \\ 5 & 9 & 0 & 9 & 24 & 8 \\ 2 & 3 & -1 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{13}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 6 & 0 & 24 & 8 \\ 2 & 2 & 2 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

[146] $1^{12}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 12 & 9 & 0 & 27 & 9 \\ 2 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[145] $1^131^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 6 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[149] $21^{11}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 13 & 20 & 9 & 7 \\ 20 & 25 & 12 & 9 \\ 9 & 12 & 4 & 4 \\ 7 & 9 & 4 & 3 \end{pmatrix}_{b_0}$$

[148] $21^{11}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 5 & 10 & 7 & 5 \\ 10 & 13 & 10 & 7 \\ 7 & 10 & 4 & 4 \\ 5 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[150] $1^131^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 125, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 3 \\ 5 & 13 & 16 & 0 & 7 \\ 5 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[151] $1^{12}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 10 & 0 & 37 & 11 \\ 3 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[152] $1^{12}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 6 & 8 & 0 & 25 & 9 \\ 2 & 2 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[153] $1^{12}321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 11 & 0 & 44 & 12 \\ 2 & 2 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0} \\
[154] \quad & 1^{11} 31^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.46 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 9 & 6 & 0 & 12 & 6 \\ 2 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0} \\
[155] \quad & 1^{11} 2 \cdot 2 \cdot 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.32 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 5 & 5 & 5 & 5 \\ 3 & 5 & 0 & 4 & 2 \\ 3 & 5 & 4 & 1 & 3 \\ 3 & 5 & 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[156] \quad & 1^{13} 3 \cdot 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 182, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 13 & 0 & 60 & 14 \\ 3 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0} \\
[157] \quad & 1^{11} 31^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 9 & 6 & 0 & 12 & 6 \\ 2 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0} \\
[158] \quad & 1^{13} 3 \cdot 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 9 & 0 & 39 & 11 \\ 2 & 2 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0} \\
[159] \quad & 1^{11} 2 \cdot 2 \cdot 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.08 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 5 & 5 & 10 & 5 \\ 3 & 5 & 0 & 7 & 2 \\ 6 & 10 & 7 & 13 & 7 \\ 3 & 5 & 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[160] \quad & 1^{12} 31^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.34 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 9 & 0 & 27 & 9 \\ 2 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0} \\
[161] \quad & 1^{11} 2 \cdot 2 \cdot 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 249, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 2 & 1 & 2 & 2 \\ 2 & 1 & 4 & 1 & 3 & 3 \\ 2 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 2 & 3 & 3 & 1 & 0 & 2 \\ 2 & 3 & 2 & 0 & 2 & 1 \end{pmatrix}_{b_0} \\
[162] \quad & 1^{13} 3 \cdot 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.10 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 12 & 0 & 60 & 14 \\ 2 & 2 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0} \\
[163] \quad & 1^{112} 2 \cdot 2 \cdot 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.09 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 2 & 1 & 4 & 2 \\ 2 & 1 & 4 & 1 & 6 & 3 \\ 2 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & -2 & 2 & 0 \\ 4 & 6 & 6 & 2 & 9 & 5 \\ 2 & 3 & 2 & 0 & 5 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

$$T = 9 \quad [164] \ 21^1 41^1 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 7 & 2 & 5 \\ 7 & 8 & 1 & 6 \\ 2 & 1 & -2 & 0 \\ 5 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[165] \ 1^1 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.44 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[168] \ 1^{11} 4 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 5 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[166] \ 1^{11} 4 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[167] \ 1^1 4 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 2 & 3 & 2 & 4 \\ 3 & 1 & 1 & 3 \\ 2 & 1 & -2 & 0 \\ 4 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[169] \ 1^1 41^1 231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 6 & 1 & 5 \\ 6 & 5 & 2 & 5 \\ 1 & 2 & -2 & 0 \\ 5 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[170] \ 1^1 4 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 4 & 1 & 5 \\ 4 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 5 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[171] \ 1^{11} 4 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 3 & 4 & 1 & 1 & 3 \\ 3 & 4 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{11} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 6 & 7 & 1 & 8 & 6 \\ 3 & 4 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[173] \ 1^1 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 4 & 2 & 5 \\ 4 & 1 & 1 & 3 \\ 2 & 1 & -2 & 0 \\ 5 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[174] \ 1^1 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 14 & 10 & 8 \\ 10 & 5 & 5 \\ 8 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[175] \ 1^1 41^1 231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 2 & 6 & 2 & 4 \\ 6 & 8 & 1 & 6 \\ 2 & 1 & -2 & 0 \\ 4 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned} [178] \quad & 1^{\overline{11}31^1} 4^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\ & T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 1 & 4 & 1 & 1 & 3 \\ 4 & 4 & 0 & 0 & 4 \\ 1 & 0 & -3 & 1 & -1 \\ 1 & 0 & 1 & -2 & 0 \\ 3 & 4 & -1 & 0 & 2 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [179] \quad & 1^{\overline{1}} 4 1 3 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\ & T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 2 & 0 & 3 & 2 \\ 1 & 0 & -2 & 1 & 0 \\ 3 & 3 & 1 & 1 & 3 \\ 3 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [180] \quad & 1^{\overline{1}} 4 1^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\ & T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 13 & 9 & 1 & 7 \\ 9 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 7 & 5 & 0 & 3 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [177] \quad & 1^{\overline{11}} 4 1 3 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\ & T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 5 & 7 & 0 & 3 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [176] \quad & 1^{\overline{11}} 4 1 3 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\ & T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 4 & -1 & 2 & 2 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [181] \quad & 1^{\overline{1}} 4^{\overline{11}31^1} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\ & T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 3 & 3 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 3 & 4 & 2 & 0 & 2 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [182] \quad & 1^{\overline{11}31^1} 4^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\ & T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 2 & 0 & 3 & 2 \\ 1 & 0 & -2 & 1 & 0 \\ 4 & 3 & 1 & 1 & 3 \\ 3 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [183] \quad & 1^{\overline{1}} 4^{\overline{11}31^1} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\ & T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 3 & 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [184] \quad & 1^{\overline{1}} 4^{\overline{11}31^1} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\ & T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 5 & 4 & 5 & 5 \\ 4 & 1 & 3 & 3 \\ 5 & 3 & 1 & 3 \\ 5 & 3 & 3 & 3 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [185] \quad & 1^{\overline{1}} 4 1^{\overline{1}} 2 3 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\ & T_{\min} = 12.40 \rightarrow 13, T_{\max} = 14, T \in [13, 14] \\ & \begin{pmatrix} 0 & 3 & 1 & 1 & 1 & 2 \\ 3 & 4 & 1 & 2 & 0 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & -2 & 0 & 0 \\ 1 & 0 & 1 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [186] \quad & 1^{\overline{1}} 4^{\overline{11}31^1} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\ & T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 5 & 7 & 1 & 5 \\ 7 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 5 & 5 & 0 & 3 \end{pmatrix}_{b_0} \end{aligned}$$

[188] $1^{11}4^{131^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 3 & 2 & 2 & 1 & 3 & 3 \\ 2 & 2 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[187] $1^{11}4^{131^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 0 & 1 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[191] $1^{11}4^{131^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 2 & 1 & 1 & 3 \\ 3 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[192] $1^{11}4^{131^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 13 & 0 & 7 \\ 1 & 0 & -2 & 0 \\ 5 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

[190] $1^{11}4^{131^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 7 & 0 & 7 & 5 \\ 3 & 4 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[189] $1^{11}4^{131^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 8 & 4 & 1 & 17 & 7 \\ 3 & 2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[193] $1^{11}4^{131^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[194] $1^{11}4^{131^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 4 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[195] $1^{11}4^{131^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 1 & 0 \\ 2 & 2 & 1 & 1 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[196] $1^{11}4^{131^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 3 & 1 & 7 & 5 \\ 3 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[197] $1^{11}4^{131^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 3 & 3 & 0 & 2 & 3 & 3 \\ 2 & 2 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

- [198] $31^{11}31^{141^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3$ $T = 9$, $T_{\max} = 21$, $\Delta = -100$, $\det = -72$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.44 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 3 & 2 \\ 3 & 1 & 4 & 1 & 3 & 3 \\ 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & -2 & 2 & 0 \\ 3 & 3 & 3 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$
- [199] $1^{11}4^{11}31^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 4 & 3 & 1 & 2 & 4 & 4 \\ 2 & 2 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [200] $1^{11}4^{12}31^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -132$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 1 & 0 & -2 & 2 & 0 \\ 5 & 3 & 2 & 2 & 7 & 5 \\ 2 & 2 & 0 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$
- [201] $2^{141^1}3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -186$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.11 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 3 & 2 & 1 & 2 & 2 \\ 3 & 4 & 4 & 2 & 3 & 4 \\ 2 & 4 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & -2 & 1 & 0 \\ 2 & 3 & 3 & 1 & 0 & 2 \\ 2 & 4 & 2 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$
- [202] $1^{141}1^{11}231^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -205$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.32 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 3 & 2 \\ 3 & 4 & 1 & 2 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 2 & 0 \\ 3 & 6 & 0 & 2 & 3 & 3 \\ 2 & 4 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$
- [203] $1^{141}1^{11}231^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -217$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.19 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 6 & 1 & 5 & 3 \\ 6 & 13 & 0 & 12 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 12 & 0 & 7 & 5 \\ 3 & 7 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$
- [204] $1^{141}1^{11}231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -250$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.08 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 4 & 2 \\ 3 & 4 & 1 & 2 & 6 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & -2 & 2 & 0 \\ 4 & 6 & 1 & 2 & 4 & 4 \\ 2 & 4 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- $T = 10$ [205] $21^{1^1}5^{11}131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 4$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 5 & 5 & 3 & 5 \\ 3 & 5 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 4 & 5 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$
- [206] $1^{11}3221^{151^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -88$, $\det = 6$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 10 & 21 & 0 & 6 \\ 9 & 21 & 42 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 6 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$
- [207] $1^{1^1}4^{1^1}141^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -83$, $\det = 10$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 1 & 0 & 3 \\ 1 & 3 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & 0 & -1 & 2 \end{pmatrix}_{b_0}$$
- [209] $1^{11}4^{1^1}141^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -83$, $\det = 18$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 4 & 5 & 5 \\ 3 & 4 & 0 & 3 & 2 \\ 2 & 5 & 3 & 1 & 3 \\ 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[208] $1^{11}4^{11}141^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 5 & 0 & 9 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 4 & 9 & 0 & 11 & 7 \\ 3 & 5 & -2 & 7 & 3 \end{pmatrix}_{b_0}$$

[211] $1^{11}4^{11}141^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[210] $1^{11}5^{11}1231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[212] $1^{11}4^{11}141^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 5 & -2 & 3 & 3 \end{pmatrix}_{b_0}$$

[215] $1^{11}4^{11}141^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 5 & 4 & 5 & 5 \\ 2 & 4 & 0 & 3 & 2 \\ 4 & 5 & 3 & 1 & 3 \\ 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[214] $1^{11}4^{11}141^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -4 & 0 & -2 \\ 2 & 5 & 0 & 1 & 3 \\ 3 & 5 & -2 & 3 & 3 \end{pmatrix}_{b_0}$$

[213] $1^{11}5^{11}131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[216] $21^{11}4^{11}141^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[217] $1^{11}4^{11}141^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 3 & 1 & 3 & 3 \\ 1 & 3 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 2 & 3 & 2 & 1 & 0 & 2 \\ 2 & 3 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{11}4^{11}131^{11}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 3 & 2 \\ 3 & 13 & 6 & 0 & 12 & 7 \\ 1 & 6 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 12 & 4 & 0 & 9 & 5 \\ 2 & 7 & 2 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[219] $1^{11}4^{11}141^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 5 & 4 & 5 & 5 \\ 2 & 4 & 0 & 3 & 2 \\ 3 & 5 & 3 & 1 & 3 \\ 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[220] $31^{\overline{11}}4141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 57, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.16 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 3 & 2 \\ 2 & 5 & 8 & 0 & 9 & 5 \\ 2 & 8 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 9 & 9 & 0 & 12 & 6 \\ 2 & 5 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[221] $231^{\overline{11}}51^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 79, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 5 & 5 & 3 & 5 \\ 3 & 5 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 5 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[222] $1^{\overline{1}}41^{\overline{1}}31^{\overline{1}}41^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 8 & 1 & 7 & 3 \\ 8 & 38 & 1 & 32 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 32 & 0 & 25 & 9 \\ 3 & 12 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[223] $1^{\overline{1}}51^{\overline{1}}231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 7 & 2 & 1 & 3 \\ 7 & 26 & 8 & 0 & 10 \\ 2 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 10 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$T = 11$ [224] $1^{\overline{11}}41^{\overline{1}}4141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 4 & 3 & 1 & 3 & 4 \\ 3 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 \\ 4 & 3 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[225] $1^{\overline{11}}41^{\overline{1}}4141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.79 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[226] $1^{\overline{1}}41^{\overline{1}}31^{\overline{1}}51^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -86, \det = -84, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 1 & 2 \\ 3 & 4 & 1 & 2 & 0 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[228] $1^{\overline{11}}41^{\overline{1}}4141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -97, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.85 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 1 & 4 & 1 & 0 & 3 \\ 2 & 4 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[227] $1^{\overline{11}}41^{\overline{1}}4141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -97, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.76 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 \\ 4 & 5 & 0 & 5 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 5 & 0 & 1 & 3 \\ 3 & 5 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$T = 12$ [229] $1^{\overline{1}}61^{\overline{1}}31^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 \quad T = 12, T_{\max} = 14, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

4.727 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6$

$$T = 8 \quad [1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, \ T_{\max} = 101, \ \Delta = -171, \ \det = 95, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.96 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 5 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, \ T_{\max} = 101, \ \Delta = -171, \ \det = 245, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, \ T_{\max} = 101, \ \Delta = -171, \ \det = 302, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.86 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, \ T_{\max} = 101, \ \Delta = -171, \ \det = 330, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.68 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 48 & 0 & 12 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 12 & -4 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \ 1^{\overline{11}} 3^{\overline{11}} 2^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 9, \ T_{\max} = 99, \ \Delta = -143, \ \det = -70, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.83 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 5 & 0 & 3 & -2 & 0 \\ 3 & 5 & -2 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{11}} 3^{\overline{11}} 2^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 9, \ T_{\max} = 99, \ \Delta = -143, \ \det = -196, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.88 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 0 & 1 & 0 & -3 & -1 \\ 2 & 2 & -1 & 0 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 9, \ T_{\max} = 100, \ \Delta = -150, \ \det = -354, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.07 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 1 & 3 \\ 5 & 13 & 16 & 12 & 0 & 7 \\ 5 & 16 & 12 & 12 & 0 & 6 \\ 4 & 12 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 7 & 6 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

4.728 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -115, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, \ T_{\max} = 99, \ T \in [15, 99]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -271, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} =$$

16.0

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

4.729 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_7$

$$T = 9 \quad [0] 1^{11} 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_7 \quad T = 9, T_{\max} = 103, \Delta = -198, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.63 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 0 & 1 & 0 & -5 & -3 \\ 2 & 2 & -1 & 0 & -1 & -3 & -3 \end{pmatrix}_{b_0}$$

4.730 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{blue}{su}_3 + \textcolor{red}{e}'_7$

$$T = 9 \quad [0] 1^{11} 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{e}'_7 \quad T = 9, T_{\max} = 102, \Delta = -170, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.73 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 0 & 1 & 0 & -4 & -2 \\ 2 & 2 & -1 & 0 & -1 & -2 & -2 \end{pmatrix}_{b_0}$$

4.731 $\text{NHC}(-2)(-3) + \textcolor{green}{so}_8 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4$

$$T = 8 \quad [0] 1^{11} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 98, T \in [15, 98]$$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 18 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 13 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[1] 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[2] 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 235, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 259, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 48 & 0 & 12 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [5] $1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 2 & 5 & 0 & 3 & 0 & 2 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.13 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 1 & 2 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 3 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 0 & 2 & -3 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 5 & 13 & 1 & 9 & 5 \\ 13 & 30 & 0 & 20 & 10 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 20 & 0 & 12 & 6 \\ 5 & 10 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.10 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 \\ 4 & 5 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 5 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.12 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & -1 & 0 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 5 & 5 & 10 & 0 & 5 \\ 2 & 5 & 0 & 7 & 0 & 2 \\ 5 & 10 & 7 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 82, \Delta = -124, \det = -277, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 82, T \in [14, 82]$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 1 & 3 \\ 5 & 13 & 16 & 12 & 0 & 7 \\ 5 & 16 & 12 & 12 & 0 & 6 \\ 4 & 12 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 7 & 6 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

4.732 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$

$T = 9$ [0] $1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[2] 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

4.733 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 6 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

4.734 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 8 \quad [0] 1^{11} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 67, T \in [15, 67]$$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 18 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[1] 1^{11} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 19 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 67, \Delta = -121, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 67, \ \Delta = -121, \ \det = 123, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 67, \ T \in [13, 67]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 10 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 67, \ \Delta = -121, \ \det = 144, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12, \ T_{\max} = 67, \ T \in [12, 67]$$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 18 & 36 & 48 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 67, \ \Delta = -121, \ \det = 147, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 67, \ T \in [13, 67]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 12 & 6 \\ 5 & 8 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 67, \ \Delta = -121, \ \det = 168, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.43 \rightarrow 13, \ T_{\max} = 67, \ T \in [13, 67]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 67, \ \Delta = -121, \ \det = 188, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 67, \ T \in [13, 67]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 48 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [9] \ 1^{11} 3 \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -100, \ \det = -5, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 15.40 \rightarrow 16, \ T_{\max} = 21, \ T \in [16, 21]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 1 & 3 \\ 4 & 10 & 13 & 0 & 0 & 6 \\ 5 & 13 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & 6 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3 \ 2 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 14, \ \Delta = -93, \ \det = -26, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.46 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 2 & 5 & 0 & 3 & 0 & 2 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 3 \ 2 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 14, \ \Delta = -93, \ \det = -33, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.27 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 0 \\ 1 & 2 & 0 & 1 & 3 & 0 \\ 1 & 2 & 1 & -2 & 1 & 0 \\ 1 & 1 & 3 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -4 \\ 2 & 2 & 2 & 0 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 3 \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -100, \ \det = -41, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.41 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 1 & 1 & 5 & 3 \\ 4 & 10 & 1 & 0 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 0 & 0 & 11 & 5 \\ 3 & 6 & -1 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3 \ 2 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 9, \ T_{\max} = 14, \ \Delta = -93, \ \det = -45, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 \\ 4 & 5 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 5 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{11} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 14, \Delta = -93, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 12 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 6 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{11} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 14, \Delta = -93, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 1 & 4 & 3 \\ 2 & 5 & 0 & 4 & 0 & 2 \\ 3 & 5 & -2 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 23, \Delta = -112, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{11} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 14, \Delta = -93, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 5 & 8 & 1 & 1 & 5 \\ 8 & 10 & 1 & 0 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 6 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 23, \Delta = -112, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 5 \\ 9 & 13 & 10 & 0 & 7 \\ 6 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{11} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 14, \Delta = -93, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.19 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 5 & 0 & 10 & 5 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 10 & 0 & 13 & 7 & 7 \\ 2 & 5 & 0 & 7 & 0 & 2 \\ 3 & 5 & -2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{11} 31^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 21, \Delta = -100, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 1 & 3 \\ 5 & 13 & 16 & 12 & 0 & 7 \\ 5 & 16 & 12 & 12 & 0 & 6 \\ 4 & 12 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 7 & 6 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [21] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.12 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 14 & 10 & 2 & 8 \\ 10 & 5 & 1 & 5 \\ 2 & 1 & -2 & 0 \\ 8 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

[22] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 5 & 10 & 1 & 1 & 5 \\ 10 & 13 & 0 & 0 & 7 \\ 1 & 0 & -3 & 1 & -1 \\ 1 & 0 & 1 & -2 & 0 \\ 5 & 7 & -1 & 0 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 129, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.07 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 0 & 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & -2 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [24] \quad 1^{11} 4141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -69, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 4 & 5 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 5 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 5 & -2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

4.735 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 28 & 5 \\ 12 & 25 & 16 & 0 & 60 & 9 \\ 8 & 16 & 9 & 0 & 36 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 28 & 60 & 36 & 0 & 133 & 21 \\ 5 & 9 & 5 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 1 & 5 \\ 11 & 20 & 0 & 17 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 8 & -2 & 6 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 10 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 10 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 15 & 6 \\ 10 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -2 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -133, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 13 & 7 \\ 5 & 8 & -2 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 10 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 15 & 6 \\ 9 & 17 & 0 & 15 & 13 & 7 \\ 5 & 8 & -2 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

$$3] 1^{111} 3 1^1 + [su_2 \cdot g_2] + s$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 8 & 5 \\ 11 & 20 & 0 & 0 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 16 & 0 & 0 & 10 & 6 \\ 5 & 8 & -2 & -1 & 6 & 1 \end{pmatrix} b$$

9] $1^{112} 3 1^{11} + [su_2 \cdot g_2] + g_2$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 19 & 5 \\ 12 & 25 & 16 & 0 & 40 & 9 \\ 8 & 16 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 40 & 24 & 0 & 58 & 14 \\ 5 & 9 & 5 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$10] \ 1^{112} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 20 & 5 \\ 12 & 25 & 16 & 0 & 40 & 9 \\ 8 & 16 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 40 & 24 & 0 & 58 & 14 \\ 5 & 9 & 5 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$11] \ 1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \dots$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$12] \ 1^{111} 3 \ 1^{111} + [5u_2 \cdot g$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 10 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111} 3^1 1^{11} + [5u_2 \cdot g]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 9 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{111} 3^1 1^{12} + [5u_2 \cdot g]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 26 & 5 \\ 11 & 20 & 0 & 0 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 26 & 50 & 0 & 0 & 118 & 20 \\ 5 & 8 & -2 & -1 & 20 & 1 \end{pmatrix},$$

$$15] \ 1^{111} 3 \ 1^{11} + [\text{su}_2 \cdot \text{g}$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 17 & 5 \\ 11 & 20 & 0 & 0 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 17 & 33 & 0 & 0 & 49 & 13 \\ 5 & 8 & -2 & -1 & 13 & 1 \end{pmatrix}_{b_1}$$

[16] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 30 & 5 \\ 12 & 25 & 16 & 0 & 60 & 9 \\ 8 & 16 & 9 & 0 & 36 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 30 & 60 & 36 & 0 & 133 & 21 \\ 5 & 9 & 5 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 29 & 5 \\ 12 & 25 & 16 & 0 & 60 & 9 \\ 8 & 16 & 9 & 0 & 36 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 29 & 60 & 36 & 0 & 133 & 21 \\ 5 & 9 & 5 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 38 & 5 \\ 12 & 25 & 16 & 0 & 80 & 9 \\ 8 & 16 & 9 & 0 & 48 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 38 & 80 & 48 & 0 & 238 & 28 \\ 5 & 9 & 5 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 40 & 5 \\ 12 & 25 & 16 & 0 & 80 & 9 \\ 8 & 16 & 9 & 0 & 48 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 40 & 80 & 48 & 0 & 238 & 28 \\ 5 & 9 & 5 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 39 & 5 \\ 12 & 25 & 16 & 0 & 80 & 9 \\ 8 & 16 & 9 & 0 & 48 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 39 & 80 & 48 & 0 & 238 & 28 \\ 5 & 9 & 5 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 19 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 19 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 18 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 18 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [24] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 9 & 5 \\ 9 & 13 & 10 & 0 & 14 & 7 \\ 6 & 10 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 9 & 0 & 13 & 7 \\ 5 & 7 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{\overline{111}} 4 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 261, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 6 & 5 \\ 9 & 13 & 10 & 0 & 10 & 7 \\ 6 & 10 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 7 & 0 & 5 & 5 \\ 5 & 7 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{\overline{111}} 4 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 480, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 15 & 5 \\ 9 & 13 & 10 & 0 & 24 & 7 \\ 6 & 10 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 24 & 16 & 0 & 38 & 12 \\ 5 & 7 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

4.736 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 14, \Delta = -93, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 9 & 6 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -2 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

4.737 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \mathfrak{n} 3 \text{mix}$

$T = 9$ [0] $1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9, T_{\max} = 14, \Delta = -93, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 7 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 9 & 6 \\ 7 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -2 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

4.738 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{so}_8$ + $\mathfrak{su}_3$

$T = 9$ [0] $1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9, T_{\max} = 21, \Delta = -100, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 1 & 5 \\ 12 & 25 & 16 & 0 & 0 & 9 \\ 8 & 16 & 9 & 0 & 0 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 5 & -2 & -1 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9, T_{\max} = 21, \Delta = -100, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 20 & 12 & 0 & 12 & 6 \\ 5 & 9 & 5 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9, T_{\max} = 21, \Delta = -100, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 10 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 20 & 12 & 0 & 12 & 6 \\ 5 & 9 & 5 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 21, \ \Delta = -100, \ \det = -245, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.43 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \ 1^{11} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 45, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.20 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 6 & 5 \\ 12 & 25 & 12 & 0 & 12 & 9 \\ 6 & 12 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 7 & 0 & 4 & 4 \\ 5 & 9 & 4 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -79, \ \det = 168, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.71 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 1 & 1 & 3 \\ 5 & 13 & 16 & 12 & 0 & 0 & 7 \\ 5 & 16 & 12 & 12 & 0 & 0 & 6 \\ 4 & 12 & 12 & 9 & 0 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 6 & 5 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

4.739 $\text{NHC}(-2)(-3) + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$

$$T = 8 \quad [0] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 9, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.56 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 12 & 14 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 12 & 15 & 6 \\ 14 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 14, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 19 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 8 & 13 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 18, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.44 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 24 & 6 \\ 9 & 16 & 0 & 24 & 46 & 12 \\ 3 & 4 & -2 & 6 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 0 & 13 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 27 & 9 \\ 6 & 12 & 0 & 27 & 25 & 9 \\ 3 & 4 & -2 & 9 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 24, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 25, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 12 & 15 & 6 \\ 12 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

- [6] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 10 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 10 & 17 & 12 & 15 & 6 \\ 10 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$
- [7] $1^{11} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 8 & -1 & 0 & 2 \end{pmatrix}_{b_0}$
- [8] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 1 & 20 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$
- [10] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 8 & 9 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 9 & 20 & 12 & 13 & 7 \\ 5 & 9 & 5 & 7 & 2 \end{pmatrix}_{b_0}$
- [11] $1^{11} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 4 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$
- [12] $1^{11} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 17 & 24 & 0 & 15 & 121 & 19 \\ 3 & 4 & -2 & 2 & 19 & 1 \end{pmatrix}_{b_0}$
- [9] $1^{11} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 2 & 0 & 2 \end{pmatrix}_{b_0}$
- [13] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 5 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 3 & 0 & 1 \end{pmatrix}_{b_0}$
- [14] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 3 \\ 3 & 4 & 0 & 9 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -2 & 6 & 0 & 1 \end{pmatrix}_{b_0}$
- [15] $1^{11} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 16 & 0 & 10 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$
- [16] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 10 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 8 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 37$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 17 & 0 & 13 & 7 \\ 5 & 8 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 39$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 13 & 11 & 6 \\ 15 & 20 & 17 & 16 & 8 \\ 13 & 17 & 12 & 12 & 6 \\ 11 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{11} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 11 & 3 \\ 3 & 4 & 0 & 0 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 11 & 19 & 0 & 0 & 73 & 15 \\ 3 & 4 & -2 & -1 & 15 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 0 & 0 & 4 & 4 \\ 3 & 4 & -2 & -1 & 4 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{11} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 59 & 5 \\ 12 & 25 & 0 & 124 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 59 & 124 & 0 & 613 & 45 \\ 5 & 9 & -1 & 45 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 17 & 0 & 13 & 7 \\ 6 & 8 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[24] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 10 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 10 & 17 & 12 & 12 & 6 \\ 10 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 6 & 3 \\ 5 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -2 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{11} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 60 & 5 \\ 12 & 25 & 0 & 124 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 60 & 124 & 0 & 613 & 45 \\ 5 & 9 & -1 & 45 & 2 \end{pmatrix}_{b_0}$$

- [27] $1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & -2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$
- [26] $1^{123} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 9 & 18 & 5 \\ 6 & 12 & 0 & 18 & 25 & 9 \\ 3 & 4 & -2 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$
- [29] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 12 & 15 & 6 \\ 10 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$
- [30] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 37 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 37 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$
- [32] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 1 & 6 \\ 15 & 20 & 17 & 0 & 8 \\ 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 6 & 8 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$
- [31] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 10 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 10 & 20 & 12 & 13 & 7 \\ 5 & 9 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$
- [33] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 18 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 36 & 0 & 49 & 13 \\ 5 & 9 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$
- [34] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 13 & 7 \\ 5 & 8 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$
- [35] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 10 & 17 & 12 & 15 & 6 \\ 9 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$
- [36] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 13 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 13 & 17 & 12 & 15 & 6 \\ 12 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$
- [37] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[38] $1^{111} 1^3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 12 & 15 & 6 \\ 9 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[40] $21^{111} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 17 & 15 & 15 & 7 \\ 17 & 20 & 17 & 17 & 8 \\ 15 & 17 & 12 & 15 & 6 \\ 15 & 17 & 15 & 13 & 7 \\ 7 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 6 & 2 \\ 4 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{11} 3^1 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 12 & 15 & 6 \\ 13 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[46] $1^{11} 3^1 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 9 & 6 \\ 4 & 5 & 0 & 9 & 4 & 4 \\ 3 & 4 & -2 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 12 & 6 \\ 4 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 4 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 27 & 18 & 9 \\ 4 & 8 & 0 & 18 & 10 & 6 \\ 3 & 4 & -2 & 9 & 6 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 9 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

- [48] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 9 & 3 \\ 3 & 4 & 0 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 49 & 13 \\ 3 & 4 & -2 & -1 & 13 & 1 \end{pmatrix}_{b_0}$$
- [49] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 9 & 17 & 12 & 12 & 6 \\ 8 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$
- [50] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 28 & 5 \\ 12 & 25 & 16 & 60 & 9 \\ 8 & 16 & 9 & 36 & 5 \\ 28 & 60 & 36 & 133 & 21 \\ 5 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$
- [51] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 6 & 3 \\ 3 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -2 & 3 & 4 & 1 \end{pmatrix}_{b_0}$$
- [52] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 19 & 5 \\ 12 & 25 & 16 & 40 & 9 \\ 8 & 16 & 9 & 24 & 5 \\ 19 & 40 & 24 & 58 & 14 \\ 5 & 9 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$
- [53] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 10 & 17 & 12 & 12 & 6 \\ 9 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$
- [54] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 11 & 6 \\ 15 & 20 & 17 & 16 & 8 \\ 12 & 17 & 12 & 12 & 6 \\ 11 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$
- [55] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 17 & 16 & 8 \\ 12 & 17 & 12 & 12 & 6 \\ 12 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$
- [56] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 9 & 17 & 12 & 12 & 6 \\ 9 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$
- [57] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 20 & 5 \\ 12 & 25 & 16 & 40 & 9 \\ 8 & 16 & 9 & 24 & 5 \\ 20 & 40 & 24 & 58 & 14 \\ 5 & 9 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$
- [58] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 15 & 6 \\ 7 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -2 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{121} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 6 & 5 \\ 3 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -2 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 11 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & -2 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[61] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 9 & 3 \\ 8 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -2 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 3 & 6 & 3 \\ 3 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -2 & 3 & 4 & 1 \end{pmatrix}_{b_0}$$

[64] $1^{11} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 3 \\ 3 & 4 & 0 & 0 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 14 & 0 & 0 & 37 & 11 \\ 3 & 4 & -2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 10 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 20 & 0 & 13 & 7 \\ 5 & 9 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 17 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 36 & 0 & 49 & 13 \\ 5 & 9 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 9 & 0 & 0 & 13 & 7 \\ 3 & 4 & -2 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

[68] $1^{13} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 48 & 15 & 12 \\ 3 & 5 & 0 & 15 & 4 & 4 \\ 3 & 4 & -2 & 12 & 4 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{131} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 48 & 12 & 12 \\ 3 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -2 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 6 & 3 \\ 4 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & -2 & 3 & 4 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 7 & 3 \\ 3 & 4 & 0 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 7 & 13 & 0 & 0 & 28 & 10 \\ 3 & 4 & -2 & -1 & 10 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 20 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 20 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 1 & 11 & 6 \\ 15 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 16 & 0 & 10 & 6 \\ 6 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[76] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 11 & 19 & 0 & 12 & 73 & 15 \\ 3 & 4 & -2 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{112} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 7 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 14 & 19 & 0 & 12 & 73 & 15 \\ 3 & 4 & -2 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[80] $1^{121} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 6 & 5 \\ 2 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -2 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 15 & 24 & 0 & 15 & 121 & 19 \\ 3 & 4 & -2 & 2 & 19 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 4 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & -2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 18 & 6 \\ 7 & 12 & 0 & 18 & 25 & 9 \\ 3 & 4 & -2 & 6 & 9 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 48 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 48 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 0 & 3 & 9 & 3 \\ 5 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -2 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[84] $1^{14} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 & 3 \\ 3 & 4 & 0 & 21 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 0 & 75 & 18 & 15 \\ 3 & 5 & 0 & 18 & 4 & 4 \\ 3 & 4 & -2 & 15 & 4 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 12 & 5 \\ 5 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -2 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 6 & 6 \\ 3 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -2 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 12 & 2 \\ 10 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -2 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 9 & 12 & 5 \\ 5 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -2 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 9 & 9 \\ 3 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -2 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 17 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 17 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[92] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 12 & 3 \\ 11 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -2 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 35 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 35 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[95] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 14 & 0 & 36 & 9 & 10 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -2 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 1 & 23 & 6 \\ 15 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 33 & 0 & 49 & 13 \\ 6 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[97] $1^{114} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 33 & 6 \\ 15 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -2 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

[96] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 16 & 0 & 10 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[98] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 30 & 5 \\ 12 & 25 & 16 & 60 & 9 \\ 8 & 16 & 9 & 36 & 5 \\ 30 & 60 & 36 & 133 & 21 \\ 5 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$

- [99] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$
- [100] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 9 & 6 \\ 3 & 5 & 0 & 9 & 4 & 4 \\ 3 & 4 & -2 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$
- [101] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 12 & 6 \\ 5 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$
- [102] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 29 & 5 \\ 12 & 25 & 16 & 60 & 9 \\ 8 & 16 & 9 & 36 & 5 \\ 29 & 60 & 36 & 133 & 21 \\ 5 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$
- [103] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [104] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 9 & 12 & 5 \\ 4 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -2 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$
- [105] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 16 & 24 & 0 & 15 & 121 & 19 \\ 3 & 4 & -2 & 2 & 19 & 1 \end{pmatrix}_{b_0}$$
- [106] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 26 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 26 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [107] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 12 & 9 \\ 3 & 5 & 0 & 12 & 4 & 4 \\ 3 & 4 & -2 & 9 & 4 & 1 \end{pmatrix}_{b_0}$$
- [108] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 13 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 13 & 17 & 12 & 27 & 6 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [109] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 6 & 6 \\ 2 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -2 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{112} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 144, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.39 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 20 & 5 \\ 12 & 25 & 0 & 40 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 40 & 0 & 58 & 14 \\ 5 & 9 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{112} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 144, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.50 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 28 & 5 \\ 12 & 25 & 0 & 56 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 28 & 56 & 0 & 118 & 20 \\ 5 & 9 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{112} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 144, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.39 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 16 & 3 \\ 3 & 4 & 0 & 0 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 16 & 28 & 0 & 0 & 154 & 22 \\ 3 & 4 & -2 & -1 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{122} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 146, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.16 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 18 & 9 \\ 4 & 8 & 0 & 18 & 10 & 6 \\ 3 & 4 & -2 & 9 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{111} 3^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 148, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 14 & 0 & 36 & 9 & 10 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -2 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{112} 3^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 148, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.41 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 34 & 5 \\ 12 & 25 & 0 & 72 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 34 & 72 & 0 & 202 & 26 \\ 5 & 9 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[116] \ 21^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 148, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.08 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 16 & 12 & 12 & 6 \\ 16 & 20 & 17 & 16 & 8 \\ 12 & 17 & 12 & 12 & 6 \\ 12 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[114] \ 1^{122} 3^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 148, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.41 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 10 & 3 \\ 3 & 4 & 0 & 0 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 18 & 0 & 0 & 58 & 14 \\ 3 & 4 & -2 & -1 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 150, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.13 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 12 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{14} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 151, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.25 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 18 & 3 \\ 11 & 21 & 0 & 18 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

- [120] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$
- [121] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 157, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 35 & 5 \\ 12 & 25 & 0 & 72 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 35 & 72 & 0 & 202 & 26 \\ 5 & 9 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$
- [122] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 158, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 38 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 38 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$
- [123] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 12 & 9 & 6 \\ 3 & 5 & 0 & 9 & 4 & 4 \\ 3 & 4 & -2 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$
- [124] $1^{12}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 13 & 3 \\ 3 & 4 & 0 & 0 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 13 & 23 & 0 & 0 & 100 & 18 \\ 3 & 4 & -2 & -1 & 18 & 1 \end{pmatrix}_{b_0}$$
- [125] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 12 & 6 & 6 \\ 2 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -2 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$
- [126] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 26 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$
- [127] $1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 9 & 3 \\ 5 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -2 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$
- [128] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 29 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 29 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$
- [129] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 11 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$
- [130] $1^{13}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 170, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 & 3 \\ 3 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 48 & 15 & 12 \\ 3 & 5 & 0 & 15 & 4 & 4 \\ 3 & 4 & -2 & 12 & 4 & 1 \end{pmatrix}_{b_0}$$

[131] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 9 & 2 \\ 7 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 15 & 2 \\ 11 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -2 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 13 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 18 & 9 \\ 5 & 8 & 0 & 18 & 10 & 6 \\ 3 & 4 & -2 & 9 & 6 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 7 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[138] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[136] $1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 9 & 6 & 5 \\ 2 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & -2 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[135] $1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & -2 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 12 & 5 \\ 5 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -2 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 46 & 5 \\ 12 & 25 & 0 & 92 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 46 & 92 & 0 & 325 & 33 \\ 5 & 9 & -1 & 33 & 2 \end{pmatrix}_{b_0}$$

[142] $1^{14} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 14 & 3 \\ 3 & 4 & 0 & 0 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 26 & 0 & 0 & 118 & 20 \\ 3 & 4 & -2 & -1 & 20 & 1 \end{pmatrix}_{b_0}$$

[141] $1^{13} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 12 & 3 \\ 3 & 4 & 0 & 0 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 22 & 0 & 0 & 85 & 17 \\ 3 & 4 & -2 & -1 & 17 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 26 & 5 \\ 12 & 25 & 0 & 56 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 26 & 56 & 0 & 118 & 20 \\ 5 & 9 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[143] $21^{111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 15 & 7 \\ 20 & 25 & 16 & 20 & 9 \\ 12 & 16 & 9 & 12 & 5 \\ 15 & 20 & 12 & 13 & 7 \\ 7 & 9 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

[144] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 27 & 9 \\ 7 & 12 & 0 & 27 & 25 & 9 \\ 3 & 4 & -2 & 9 & 9 & 1 \end{pmatrix}_{b_0}$$

[145] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 40 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 40 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$

[146] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 12 & 2 \\ 9 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{112} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 19 & 5 \\ 12 & 25 & 0 & 40 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 40 & 0 & 58 & 14 \\ 5 & 9 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[148] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 19 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[149] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 & 3 \\ 3 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 27 & 12 & 9 \\ 3 & 5 & 0 & 12 & 4 & 4 \\ 3 & 4 & -2 & 9 & 4 & 1 \end{pmatrix}_{b_0}$$

[151] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 19 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 9 & 17 & 12 & 27 & 6 \\ 19 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

- [150] $1^{111} 1^3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 9 & 17 & 12 & 27 & 6 \\ 17 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [152] $1^{111} 1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 19 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 19 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [153] $1^{122} 1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 9 & 18 & 5 \\ 7 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & -2 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$
- [154] $1^{114} 1^3 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 39 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 39 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$
- [155] $1^{113} 1^{12} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 205, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 53 & 5 \\ 12 & 25 & 0 & 108 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 53 & 108 & 0 & 457 & 39 \\ 5 & 9 & -1 & 39 & 2 \end{pmatrix}_{b_0}$$
- [156] $1^{13} 1^1 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 15 & 3 \\ 9 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$
- [157] $1^{111} 1^1 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 18 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 18 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [158] $1^{111} 1^1 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 45 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 45 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$
- [159] $1^{112} 1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 12 & 21 & 6 \\ 7 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -2 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$
- [160] $1^{121} 1^1 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 21 & 9 \\ 6 & 9 & 0 & 21 & 13 & 7 \\ 3 & 4 & -2 & 9 & 7 & 1 \end{pmatrix}_{b_0}$$
- [161] $1^{12} 1^1 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 10 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -2 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[172] $1^{123} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 224$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 18 & 5 \\ 8 & 12 & 0 & 18 & 25 & 9 \\ 3 & 4 & -2 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

[173] $1^{114} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 229$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 71 & 5 \\ 12 & 25 & 0 & 144 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 71 & 144 & 0 & 814 & 52 \\ 5 & 9 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[174] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 233$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 18 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[175] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 234$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 25 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 25 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 237$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 27 & 5 \\ 12 & 25 & 0 & 56 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 27 & 56 & 0 & 118 & 20 \\ 5 & 9 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[177] $1^{131} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 240$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 0 & 24 & 9 & 8 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -2 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 240$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[179] $1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 240$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 10 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & -2 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[180] $1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 241$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 38 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 38 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[181] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 241$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 24 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

- [182] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 242, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 9 & 3 \\ 6 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -2 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$
- [183] $1^{113} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 30 & 5 \\ 12 & 25 & 0 & 60 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 30 & 60 & 0 & 133 & 21 \\ 5 & 9 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$
- [184] $21^{111} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 17 & 15 & 27 & 7 \\ 17 & 20 & 17 & 33 & 8 \\ 15 & 17 & 12 & 27 & 6 \\ 27 & 33 & 27 & 49 & 13 \\ 7 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [185] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 18 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 9 & 17 & 12 & 27 & 6 \\ 18 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [186] $1^{113} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 51 & 5 \\ 12 & 25 & 0 & 108 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 51 & 108 & 0 & 457 & 39 \\ 5 & 9 & -1 & 39 & 2 \end{pmatrix}_{b_0}$$
- [187] $1^{113} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 15 & 3 \\ 3 & 4 & 0 & 0 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 27 & 0 & 0 & 133 & 21 \\ 3 & 4 & -2 & -1 & 21 & 1 \end{pmatrix}_{b_0}$$
- [188] $1^{111} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 47 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 47 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$
- [189] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 0 & 9 & 24 & 5 \\ 9 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -2 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$
- [190] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 9 & 0 & 12 & 15 & 6 \\ 5 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -2 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$
- [191] $1^{114} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 21 & 2 \\ 17 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -2 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$
- [192] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 15 & 6 \\ 5 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -2 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

[194] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 18 & 5 \\ 7 & 12 & 0 & 18 & 25 & 9 \\ 3 & 4 & -2 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

[193] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 9 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & -2 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[195] $21^{112} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 30 & 7 \\ 20 & 25 & 16 & 40 & 9 \\ 12 & 16 & 9 & 24 & 5 \\ 30 & 40 & 24 & 58 & 14 \\ 7 & 9 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$

[196] $1^{121} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 18 & 0 & 57 & 12 & 13 \\ 2 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -2 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$

[197] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 38 & 5 \\ 12 & 25 & 0 & 76 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 38 & 76 & 0 & 217 & 27 \\ 5 & 9 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[198] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 277, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 36 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 36 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[199] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 285, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 35 & 5 \\ 12 & 25 & 0 & 76 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 35 & 76 & 0 & 217 & 27 \\ 5 & 9 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[200] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 28 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 28 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[201] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 27 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 27 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[202] $21^{111} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 16 & 12 & 24 & 6 \\ 16 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{13} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 18 & 3 \\ 3 & 4 & 0 & 0 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 18 & 32 & 0 & 0 & 193 & 25 \\ 3 & 4 & -2 & -1 & 25 & 1 \end{pmatrix}_{b_0}$$

[204] $1^{111} 3^{12} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 290$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 46 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 46 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[205] $1^{123} 3^{111} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 290$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 9 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 27 & 39 & 9 \\ 9 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -2 & 9 & 13 & 1 \end{pmatrix}_{b_0}$$

[206] $1^{123} 3^{111} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 293$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 24 & 5 \\ 11 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -2 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{123} 3^{111} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 295$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 12 & 3 \\ 10 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -2 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[208] $1^{112} 3^{111} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 296$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 7 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[209] $1^{131} 3^{111} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 301$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.22 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 48 & 12 & 12 \\ 2 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -2 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

[210] $1^{113} 3^{111} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 303$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.17 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 27 & 6 \\ 12 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

[211] $1^{114} 3^{112} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 304$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 18 & 2 \\ 14 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & -2 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

[212] $1^{113} 3^{113} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 304$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 52 & 5 \\ 12 & 25 & 0 & 108 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 52 & 108 & 0 & 457 & 39 \\ 5 & 9 & -1 & 39 & 2 \end{pmatrix}_{b_0}$$

[213] $1^{113} 3^{111} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 304$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 28 & 5 \\ 12 & 25 & 0 & 60 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 28 & 60 & 0 & 133 & 21 \\ 5 & 9 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[214] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 307$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 15 & 6 \\ 6 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -2 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

[215] $1^{14} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 309$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.44 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 17 & 3 \\ 3 & 4 & 0 & 0 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 17 & 31 & 0 & 0 & 172 & 24 \\ 3 & 4 & -2 & -1 & 24 & 1 \end{pmatrix}_{b_0}$$

[216] $1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 309$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 27 & 9 & 9 \\ 2 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & -2 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[217] $1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 319$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 37 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 37 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 320$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.34 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 7 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -2 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[219] $1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 326$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 21 & 9 \\ 5 & 9 & 0 & 21 & 13 & 7 \\ 3 & 4 & -2 & 9 & 7 & 1 \end{pmatrix}_{b_0}$$

[220] $1^{131} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 331$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 48 & 27 & 12 \\ 5 & 9 & 0 & 27 & 13 & 7 \\ 3 & 4 & -2 & 12 & 7 & 1 \end{pmatrix}_{b_0}$$

[221] $1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 333$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 43 & 5 \\ 12 & 25 & 0 & 92 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 43 & 92 & 0 & 325 & 33 \\ 5 & 9 & -1 & 33 & 2 \end{pmatrix}_{b_0}$$

[222] $1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 334$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 13 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 15 & 3 \\ 13 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

[223] $1^{123} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 341$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.61 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 24 & 5 \\ 9 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -2 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

[224] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 12 & 3 \\ 8 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -2 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[225] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 349, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 29 & 5 \\ 12 & 25 & 0 & 60 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 29 & 60 & 0 & 133 & 21 \\ 5 & 9 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[226] $1^{123} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 13 & 3 \\ 3 & 4 & 0 & 8 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 30 & 5 \\ 13 & 22 & 0 & 30 & 85 & 17 \\ 3 & 4 & -2 & 5 & 17 & 1 \end{pmatrix}_{b_0}$$

[227] $1^{123} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 8 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 30 & 5 \\ 12 & 22 & 0 & 30 & 85 & 17 \\ 3 & 4 & -2 & 5 & 17 & 1 \end{pmatrix}_{b_0}$$

[228] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 21 & 6 \\ 7 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -2 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

[229] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 21 & 6 \\ 9 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -2 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

[230] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 371, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 12 & 3 \\ 9 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -2 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[231] $1^{121} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 11 & 18 & 0 & 57 & 12 & 13 \\ 2 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -2 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$

[232] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 45 & 5 \\ 12 & 25 & 0 & 92 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 45 & 92 & 0 & 325 & 33 \\ 5 & 9 & -1 & 33 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 56 & 5 \\ 12 & 25 & 0 & 112 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 56 & 112 & 0 & 478 & 40 \\ 5 & 9 & -1 & 40 & 2 \end{pmatrix}_{b_0}$$

[233] $1^{114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 40 & 5 \\ 12 & 25 & 0 & 80 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 40 & 80 & 0 & 238 & 28 \\ 5 & 9 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[234] $1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 9 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -2 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[236] $1^{121} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 386, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 5 & 3 \\ 3 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 27 & 21 & 9 \\ 5 & 9 & 0 & 21 & 13 & 7 \\ 3 & 4 & -2 & 9 & 7 & 1 \end{pmatrix}_{b_0}$$

[237] $1^{123} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 39 & 9 \\ 11 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -2 & 9 & 13 & 1 \end{pmatrix}_{b_0}$$

[238] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 10 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[239] $1^{13} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 394, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 15 & 3 \\ 10 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

[240] $1^{14} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 395, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 18 & 3 \\ 12 & 21 & 0 & 18 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

[243] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 44 & 5 \\ 12 & 25 & 0 & 96 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 44 & 96 & 0 & 346 & 34 \\ 5 & 9 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[242] $1^{114} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 68 & 5 \\ 12 & 25 & 0 & 144 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 68 & 144 & 0 & 814 & 52 \\ 5 & 9 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[241] $1^{14} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 20 & 3 \\ 3 & 4 & 0 & 0 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 20 & 36 & 0 & 0 & 238 & 28 \\ 3 & 4 & -2 & -1 & 28 & 1 \end{pmatrix}_{b_0}$$

[244] $1^{114}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 33 & 6 \\ 11 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -2 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

[245] $1^{113}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 411, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 27 & 6 \\ 9 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

[247] $1^{114}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 9 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -2 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[248] $1^{112}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 16 & 28 & 0 & 18 & 154 & 22 \\ 3 & 4 & -2 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

[249] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 14 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -2 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[250] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 9 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[251] $1^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 8 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & -2 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[246] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 10 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -2 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[252] $1^{122}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 8 & 13 & 0 & 27 & 30 & 9 \\ 7 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & -2 & 9 & 10 & 1 \end{pmatrix}_{b_0}$$

[253] $21^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 45 & 7 \\ 20 & 25 & 16 & 60 & 9 \\ 12 & 16 & 9 & 36 & 5 \\ 45 & 60 & 36 & 133 & 21 \\ 7 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$

[254] $1^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 37 & 5 \\ 12 & 25 & 0 & 76 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 37 & 76 & 0 & 217 & 27 \\ 5 & 9 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[255] 1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 24 & 5 \\ 10 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & -2 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[256] 1^{114} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 37 & 5 \\ 12 & 25 & 0 & 80 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 37 & 80 & 0 & 238 & 28 \\ 5 & 9 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[257] 1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 36 & 5 \\ 12 & 25 & 0 & 76 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 36 & 76 & 0 & 217 & 27 \\ 5 & 9 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[258] 1^{113} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 44 & 5 \\ 12 & 25 & 0 & 92 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 44 & 92 & 0 & 325 & 33 \\ 5 & 9 & -1 & 33 & 2 \end{pmatrix}_{b_0}$$

$$[259] 1^{131} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 12 & 22 & 0 & 84 & 15 & 16 \\ 2 & 4 & 0 & 15 & 1 & 3 \\ 3 & 4 & -2 & 16 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[260] 1^{114} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 70 & 5 \\ 12 & 25 & 0 & 144 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 70 & 144 & 0 & 814 & 52 \\ 5 & 9 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

$$[261] 1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 48 & 5 \\ 12 & 25 & 0 & 96 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 48 & 96 & 0 & 346 & 34 \\ 5 & 9 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[262] 1^{131} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 453, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 48 & 12 & 12 \\ 2 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & -2 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[263] 1^{131} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 455, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 5 & 3 \\ 3 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 48 & 27 & 12 \\ 5 & 9 & 0 & 27 & 13 & 7 \\ 3 & 4 & -2 & 12 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[264] 1^{112} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 13 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & -2 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[265] $1^{112} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 456$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.11 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 8 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$

[266] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 458$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 15 & 3 \\ 12 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 1 \end{pmatrix}_{b_0}$

[267] $1^{122}3^1{}^{111}+[\textcolor{red}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{so}}_8+\textcolor{brown}{\mathfrak{su}}_3+\textcolor{red}{\mathfrak{su}}_2$ $T=8$, $T_{\max}=12$, $\Delta=-92$, $\det=471$, $(n_+, n_-, n_0)=(1, 8, 0)$, $c_{\text{ext}}=11.7$
 $T_{\min}=11.15 \rightarrow 12$, $T_{\max}=12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 30 & 9 \\ 7 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & -2 & 9 & 10 & 1 \end{pmatrix}_{b_0}$

[268] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 478$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 15 & 3 \\ 11 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & -2 & 3 & 13 & 1 \end{pmatrix}_{b_0}$

[269] $1^{\overline{112}} 3^{\overline{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 488$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 9 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$

[270] $1^{122} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 491$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.19 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 8 & 3 \\ 3 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 30 & 9 \\ 8 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & -2 & 9 & 10 & 1 \end{pmatrix}_{b_0}$

$$[271] \ 1^{114} 3^{114} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{green}{\mathbf{so}}_8 + \textcolor{orange}{\mathbf{su}}_3 + \textcolor{red}{\mathbf{su}}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 493, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 69 & 5 \\ 12 & 25 & 0 & 144 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 69 & 144 & 0 & 814 & 52 \\ 5 & 9 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[272] $1^{\overline{112}} 3^{\overline{111}} + [\textcolor{red}{\mathbf{su}_2} \cdot \textcolor{blue}{\mathbf{g}_2}] + \textcolor{green}{\mathbf{so}_8} + \textcolor{brown}{\mathbf{su}_3} + \textcolor{red}{\mathbf{su}_2}$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 498$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 21 & 6 \\ 8 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -2 & 6 & 10 & 1 \end{pmatrix}_{b_0}$

[273] $1^{114} 3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 501$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.47 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 1 & 63 & 5 \\ 12 & 25 & 0 & 128 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 63 & 128 & 0 & 634 & 46 \\ 5 & 9 & -1 & 46 & 2 \end{pmatrix}_{b_0}$

[274] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 503$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 18 & 3 \\ 15 & 21 & 0 & 18 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 1 \end{pmatrix}_{b_0}$

[275] $1^{114}3^11^{13} + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}] + \textcolor{green}{5}\textcolor{blue}{o}_8 + \textcolor{brown}{5}\textcolor{blue}{u}_3 + \textcolor{red}{5}\textcolor{blue}{u}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.47 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 60 & 5 \\ 12 & 25 & 0 & 128 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 60 & 128 & 0 & 634 & 46 \\ 5 & 9 & -1 & 46 & 2 \end{pmatrix}_{b_0}$$

$$[276] \ 1^{114} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 52 & 5 \\ 12 & 25 & 0 & 112 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 52 & 112 & 0 & 478 & 40 \\ 5 & 9 & -1 & 40 & 2 \end{pmatrix}_{b_0}$$

$$[277] \ 1^{14} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 535, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 13 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 18 & 3 \\ 13 & 21 & 0 & 18 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[278] \ 1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 11 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -2 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[279] \ 1^{114} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 549, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 39 & 5 \\ 12 & 25 & 0 & 80 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 39 & 80 & 0 & 238 & 28 \\ 5 & 9 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[280] \ 1^{114} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 38 & 5 \\ 12 & 25 & 0 & 80 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 38 & 80 & 0 & 238 & 28 \\ 5 & 9 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[281] \ 1^{123} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 566, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 39 & 9 \\ 9 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -2 & 9 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[282] \ 1^{14} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 571, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 18 & 3 \\ 14 & 21 & 0 & 18 & 76 & 16 \\ 3 & 4 & -2 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[283] \ 1^{112} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 584, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 17 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & -2 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[284] \ 1^{114} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 586, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 33 & 6 \\ 14 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -2 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[285] \ 1^{131} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 13 & 22 & 0 & 84 & 15 & 16 \\ 2 & 4 & 0 & 15 & 1 & 3 \\ 3 & 4 & -2 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[286] $1^{113} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 603$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 27 & 6 \\ 11 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

[290] $1^{114} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 608$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.32 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 10 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -2 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[289] $1^{114} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 608$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.32 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 12 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -2 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[288] $1^{112} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 608$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 13 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -2 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[287] $1^{112} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 608$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 11 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -2 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[291] $21^{114} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 634$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 60 & 7 \\ 20 & 25 & 16 & 80 & 9 \\ 12 & 16 & 9 & 48 & 5 \\ 60 & 80 & 48 & 238 & 28 \\ 7 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$

[292] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 637$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 45 & 5 \\ 12 & 25 & 0 & 96 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 45 & 96 & 0 & 346 & 34 \\ 5 & 9 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[293] $1^{113} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 639$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 27 & 6 \\ 10 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

[294] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 645$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 55 & 5 \\ 12 & 25 & 0 & 112 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 55 & 112 & 0 & 478 & 40 \\ 5 & 9 & -1 & 40 & 2 \end{pmatrix}_{b_0}$$

[295] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 656$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 12 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -2 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

[296] $1^{114} 3^{113} + [\textcolor{red}{5}\textcolor{blue}{2} \cdot \textcolor{blue}{9}\textcolor{red}{2}] + \textcolor{green}{5}\textcolor{blue}{0}\textcolor{red}{8} + \textcolor{red}{5}\textcolor{blue}{0}\textcolor{red}{3} + \textcolor{red}{5}\textcolor{blue}{0}\textcolor{red}{2}$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 660$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 1 & 62 & 5 \\ 12 & 25 & 0 & 128 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 62 & 128 & 0 & 634 & 46 \\ 5 & 9 & -1 & 46 & 2 \end{pmatrix}_{b_0}$

[297] $1^{114} 3^1 1^{11} + [\text{red}_2 \cdot \text{red}_2] + \text{red}_8 + \text{red}_3 + \text{red}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 661$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 1 & 47 & 5 \\ 12 & 25 & 0 & 96 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 47 & 96 & 0 & 346 & 34 \\ 5 & 9 & -1 & 34 & 2 \end{pmatrix}_{b_0}$

[298] $1^{114} 3^{113} + [\textcolor{red}{5}\textcolor{blue}{2} \cdot \textcolor{blue}{9}\textcolor{red}{2}] + \textcolor{green}{5}\textcolor{green}{8} + \textcolor{orange}{5}\textcolor{orange}{3} + \textcolor{red}{5}\textcolor{red}{2}$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 669$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 1 & 61 & 5 \\ 12 & 25 & 0 & 128 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 61 & 128 & 0 & 634 & 46 \\ 5 & 9 & -1 & 46 & 2 \end{pmatrix}_{b_0}$

[303] $1^{112}3^{114} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{green}{\mathbf{so}}_8 + \textcolor{brown}{\mathbf{su}}_3 + \textcolor{red}{\mathbf{su}}_2 \quad T = 8, \quad T_{\max} = 12, \quad \Delta = -92, \quad \det = 672, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 20 & 28 & 0 & 18 & 154 & 22 \\ 3 & 4 & -2 & 2 & 22 & 1 \end{pmatrix}_{b_0}$

[299] $1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{3}}3^{\textcolor{blue}{1}\textcolor{red}{1}} + [\textcolor{red}{5}\textcolor{blue}{2}, \textcolor{blue}{9}\textcolor{red}{2}] + [\textcolor{green}{5}\textcolor{blue}{0}\textcolor{red}{8} + \textcolor{red}{5}\textcolor{blue}{3} + \textcolor{red}{5}\textcolor{blue}{2}] \quad T = 8, \quad T_{\max} = 12, \quad \Delta = -92, \quad \det = 672, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$
 $\left(\begin{array}{cccccc} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 13 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 2 & 13 & 1 \end{array} \right)_{b_0}$

$$[300] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{4}} 3 \ 1^{\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}\textcolor{red}{2} \cdot \textcolor{blue}{g}\textcolor{red}{2}] + \textcolor{green}{s}\textcolor{blue}{o}\textcolor{red}{8} + \textcolor{red}{s}\textcolor{blue}{u}\textcolor{red}{3} + \textcolor{red}{s}\textcolor{blue}{u}\textcolor{red}{2} \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 672, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.29 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 11 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & -2 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[301] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{2}} 3^{\textcolor{blue}{1}\textcolor{red}{2}} + [\textcolor{red}{5}\textcolor{blue}{2} \cdot \textcolor{blue}{9}\textcolor{red}{2}] + \textcolor{blue}{5}\textcolor{green}{0}\textcolor{blue}{8} + \textcolor{red}{5}\textcolor{blue}{3} + \textcolor{red}{5}\textcolor{blue}{2} \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 672, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.07 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 12 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & -2 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[302] $1^{112}3^{114} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{so}_8} + \textcolor{brown}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2}$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 672$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 17 & 28 & 0 & 18 & 154 & 22 \\ 3 & 4 & -2 & 2 & 22 & 1 \end{pmatrix}_{b_0}$

[304] $1^{113}3^{111} + [\textcolor{red}{\mathbf{su}_2} \cdot \textcolor{blue}{\mathbf{g}_2}] + \textcolor{green}{\mathbf{so}_8} + \textcolor{brown}{\mathbf{su}_3} + \textcolor{red}{\mathbf{su}_2}$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 672$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 10 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 2 & 13 & 1 \end{pmatrix}_{b_0}$

[305] $1^{\textcolor{red}{1}\textcolor{blue}{1}2}3^{\textcolor{blue}{1}\textcolor{red}{1}3} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 12, \quad \Delta = -92, \quad \det = 680, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 14 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & -2 & 2 & 18 & 1 \end{pmatrix}_{b_0}$

[306] $1^{114} 3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 717, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 53 & 5 \\ 12 & 25 & 0 & 112 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 53 & 112 & 0 & 478 & 40 \\ 5 & 9 & -1 & 40 & 2 \end{pmatrix}_{b_0}$$

[307] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 722$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.17 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 39 & 9 \\ 10 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & -2 & 9 & 13 & 1 \end{pmatrix}_{b_0}$$

[308] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 724$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 46 & 5 \\ 12 & 25 & 0 & 96 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 46 & 96 & 0 & 346 & 34 \\ 5 & 9 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[309] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 730$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 33 & 6 \\ 12 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -2 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

[310] $1^{112} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 744$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 16 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & -2 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[311] $1^{114} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 756$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 1 & 54 & 5 \\ 12 & 25 & 0 & 112 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 54 & 112 & 0 & 478 & 40 \\ 5 & 9 & -1 & 40 & 2 \end{pmatrix}_{b_0}$$

[312] $1^{112} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 776$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 15 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & -2 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[313] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 790$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 33 & 6 \\ 13 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & -2 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

[316] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 800$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 11 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[315] $1^{112} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 800$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 19 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 19 & 28 & 0 & 18 & 154 & 22 \\ 3 & 4 & -2 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

[317] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 800$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 12 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & -2 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[314] $1^{112}3^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 800$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 18 & 28 & 0 & 18 & 154 & 22 \\ 3 & 4 & -2 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

[318] $1^{113}3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 816$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 15 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & -2 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

[319] $1^{114}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 888$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.22 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 12 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & -2 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

[320] $1^{113}3^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 896$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 18 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & -2 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

[321] $1^{114}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 896$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.21 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 14 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & -2 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

[322] $21^{115}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 904$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.1$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 75 & 7 \\ 20 & 25 & 16 & 100 & 9 \\ 12 & 16 & 9 & 60 & 5 \\ 75 & 100 & 60 & 373 & 35 \\ 7 & 9 & 5 & 35 & 2 \end{pmatrix}_{b_0}$$

[323] $1^{113}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 944$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.11 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 13 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -2 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

[324] $1^{113}3^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 1040$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 16 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -2 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

[325] $1^{113}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 1104$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 14 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & -2 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

[326] $1^{114}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 1112$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.17 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 19 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 19 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -2 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

[347] $1^{114} 3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 2040$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.09 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 20 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -2 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

[348] $1^{114} 3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 2072$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.09 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 21 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 21 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & -2 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

[349] $1^{114} 3^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 2144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.09 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 22 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 22 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & -2 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

[350] $1^{114} 3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 2336$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 23 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 23 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & -2 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

[351] $1^{114} 3^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 2400$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 24 & 3 \\ 3 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 24 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & -2 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [352] $1^{111} 4^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -14$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 7 & 19 & 5 \\ 9 & 13 & 10 & 24 & 7 \\ 7 & 10 & 4 & 16 & 4 \\ 19 & 24 & 16 & 38 & 12 \\ 5 & 7 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[353] $1^{11} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 5 \\ 9 & 13 & 0 & 0 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 7 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

[354] $1^{12} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 1 & 12 & 5 \\ 9 & 13 & 0 & 20 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 20 & 0 & 26 & 10 \\ 5 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[358] $21^1 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 8 & 12 & 6 \\ 12 & 13 & 10 & 14 & 7 \\ 8 & 10 & 4 & 9 & 4 \\ 12 & 14 & 9 & 13 & 7 \\ 6 & 7 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[357] $1^{11} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 8 & 12 & 6 \\ 12 & 13 & 10 & 14 & 7 \\ 8 & 10 & 4 & 9 & 4 \\ 12 & 14 & 9 & 13 & 7 \\ 6 & 7 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[356] $21^{1^1} 41^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 8 & 8 & 6 \\ 12 & 13 & 10 & 10 & 7 \\ 8 & 10 & 4 & 7 & 4 \\ 8 & 10 & 7 & 5 & 5 \\ 6 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[355] $1^{111} 41^{1^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 8 & 8 & 6 \\ 12 & 13 & 10 & 10 & 7 \\ 8 & 10 & 4 & 7 & 4 \\ 8 & 10 & 7 & 5 & 5 \\ 6 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[359] $21^{11} 41^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 14 & 9 & 15 & 7 \\ 14 & 13 & 10 & 14 & 7 \\ 9 & 10 & 4 & 9 & 4 \\ 15 & 14 & 9 & 13 & 7 \\ 7 & 7 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[360] $1^{11} 41^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 7 & 9 & 5 \\ 9 & 13 & 10 & 14 & 7 \\ 7 & 10 & 4 & 9 & 4 \\ 9 & 14 & 9 & 13 & 7 \\ 5 & 7 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[361] $1^{11} 41^{1^2} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -49, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 1 & 8 & 5 \\ 9 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 10 & 0 & 5 & 5 \\ 5 & 7 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

[363] $21^{1^1} 41^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 14 & 9 & 9 & 7 \\ 14 & 13 & 10 & 10 & 7 \\ 9 & 10 & 4 & 7 & 4 \\ 9 & 10 & 7 & 5 & 5 \\ 7 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[362] $1^{111} 41^{1^2} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 8 & 6 & 5 \\ 9 & 13 & 10 & 10 & 7 \\ 8 & 10 & 4 & 7 & 4 \\ 6 & 10 & 7 & 5 & 5 \\ 5 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[364] $1^{1^1} 41^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 1 & 9 & 5 \\ 9 & 13 & 0 & 14 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 14 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[365] $1^1 41^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 6 & 9 & 5 \\ 9 & 13 & 10 & 14 & 7 \\ 6 & 10 & 4 & 9 & 4 \\ 9 & 14 & 9 & 13 & 7 \\ 5 & 7 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[366] $1^{11} 41^{1^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 1 & 6 & 5 \\ 9 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 10 & 0 & 5 & 5 \\ 5 & 7 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

[368] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 12 & 8 & 20 & 6 \\ 12 & 13 & 10 & 24 & 7 \\ 8 & 10 & 4 & 16 & 4 \\ 20 & 24 & 16 & 38 & 12 \\ 6 & 7 & 4 & 12 & 3 \end{pmatrix}_{b_0}$

[367] $21^{11} 4 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 12 & 8 & 20 & 6 \\ 12 & 13 & 10 & 24 & 7 \\ 8 & 10 & 4 & 16 & 4 \\ 20 & 24 & 16 & 38 & 12 \\ 6 & 7 & 4 & 12 & 3 \end{pmatrix}_{b_0}$

[370] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 1 & 10 & 5 \\ 9 & 13 & 0 & 14 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 14 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 3 \end{pmatrix}_{b_0}$

[369] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 7 & 10 & 5 \\ 9 & 13 & 10 & 14 & 7 \\ 7 & 10 & 4 & 9 & 4 \\ 10 & 14 & 9 & 13 & 7 \\ 5 & 7 & 4 & 7 & 3 \end{pmatrix}_{b_0}$

[371] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -70$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 7 & 1 & 5 \\ 9 & 13 & 10 & 0 & 7 \\ 7 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 7 & 4 & 0 & 3 \end{pmatrix}_{b_0}$

[372] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.66 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 1 & 7 & 5 \\ 9 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 10 & 0 & 5 & 5 \\ 5 & 7 & -1 & 5 & 3 \end{pmatrix}_{b_0}$

[373] $1^{111} 4 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -97$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 7 & 8 & 5 \\ 9 & 13 & 10 & 10 & 7 \\ 7 & 10 & 4 & 7 & 4 \\ 8 & 10 & 7 & 5 & 5 \\ 5 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[374] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -109$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 7 & 6 & 5 \\ 9 & 13 & 10 & 10 & 7 \\ 7 & 10 & 4 & 7 & 4 \\ 6 & 10 & 7 & 5 & 5 \\ 5 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[375] $1^{12} 4 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 1 & 15 & 5 \\ 9 & 13 & 0 & 20 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 20 & 0 & 26 & 10 \\ 5 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0}$

[376] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -119$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.71 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 9 & 1 & 15 & 5 \\ 9 & 13 & 0 & 24 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 24 & 0 & 38 & 12 \\ 5 & 7 & -1 & 12 & 3 \end{pmatrix}_{b_0}$

[377] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -120$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 1 & 21 & 5 \\ 9 & 13 & 0 & 30 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 30 & 0 & 61 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[398] $21^{111} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 7 & 7 & 5 \\ 10 & 13 & 10 & 10 & 7 \\ 7 & 10 & 4 & 7 & 4 \\ 7 & 10 & 7 & 5 & 5 \\ 5 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[399] $1^{12} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -325, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 22 & 5 \\ 9 & 13 & 0 & 34 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 22 & 34 & 0 & 77 & 17 \\ 5 & 7 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[400] $1^{12} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 25 & 5 \\ 9 & 13 & 0 & 34 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 25 & 34 & 0 & 77 & 17 \\ 5 & 7 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[401] $1^{14} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 26 & 5 \\ 9 & 13 & 0 & 40 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 26 & 40 & 0 & 110 & 20 \\ 5 & 7 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[402] $21^{111} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -382, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 7 & 16 & 5 \\ 10 & 13 & 10 & 24 & 7 \\ 7 & 10 & 4 & 16 & 4 \\ 16 & 24 & 16 & 38 & 12 \\ 5 & 7 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[403] $1^{12} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -406, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 23 & 5 \\ 9 & 13 & 0 & 34 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 34 & 0 & 77 & 17 \\ 5 & 7 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[404] $1^{12} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -409, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 24 & 5 \\ 9 & 13 & 0 & 34 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 34 & 0 & 77 & 17 \\ 5 & 7 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[405] $1^{13} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 28 & 5 \\ 9 & 13 & 0 & 44 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 28 & 44 & 0 & 130 & 22 \\ 5 & 7 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[406] $1^{14} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -439, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 27 & 5 \\ 9 & 13 & 0 & 40 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 27 & 40 & 0 & 110 & 20 \\ 5 & 7 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[407] $1^{14} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -454, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 28 & 5 \\ 9 & 13 & 0 & 40 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 28 & 40 & 0 & 110 & 20 \\ 5 & 7 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[408] \ 1^{14} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -509, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.73 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 34 & 5 \\ 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 34 & 54 & 0 & 197 & 27 \\ 5 & 7 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[409] \ 1^{13} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -579, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.72 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 29 & 5 \\ 9 & 13 & 0 & 44 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 29 & 44 & 0 & 130 & 22 \\ 5 & 7 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[410] \ 1^{13} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -651, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 31 & 5 \\ 9 & 13 & 0 & 44 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 31 & 44 & 0 & 130 & 22 \\ 5 & 7 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[411] \ 1^{13} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -654, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.74 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 30 & 5 \\ 9 & 13 & 0 & 44 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 30 & 44 & 0 & 130 & 22 \\ 5 & 7 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[412] \ 1^{14} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -734, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.72 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 35 & 5 \\ 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 35 & 54 & 0 & 197 & 27 \\ 5 & 7 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[413] \ 1^{14} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -881, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.73 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 36 & 5 \\ 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 36 & 54 & 0 & 197 & 27 \\ 5 & 7 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[414] \ 1^{14} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -950, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.75 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 37 & 5 \\ 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 37 & 54 & 0 & 197 & 27 \\ 5 & 7 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [415] \ 1^{11} 4 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -62, \ \det = 12, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 4 & 1 & 2 & 5 \\ 4 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 0 & 2 \\ 5 & 3 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

4.740 NHC(-2)(-3) + so₈ + su₃ + su₂ + e₆

$$T = 9 \quad [0] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{e}_6 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -115, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, \ T_{\max} = 99, \ T \in [15, 99]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 9 & 5 & 7 & -4 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{c}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{c}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -334$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 1 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 8 & 16 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{c}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -931$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

4.741 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{f}_4$

$T = 9$ [0] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 13.27 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 9 & 5 & 7 & -3 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -215$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -264$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 1 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 8 & 16 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -735$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

4.742 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$

$T = 9$ [0] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.33 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

- [1] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 8 & 28 & 1 & 5 \\ 12 & 25 & 16 & 60 & 0 & 9 \\ 8 & 16 & 9 & 36 & 0 & 5 \\ 28 & 60 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$
- [2] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 1 & 5 \\ 11 & 20 & 0 & 17 & 0 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 8 & -2 & 6 & 0 & 1 \end{pmatrix}_{b_0}$$
- [3] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 8 & 10 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 10 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [4] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 10 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [5] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -133, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 13 & 7 \\ 5 & 8 & -2 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$
- [6] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 10 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [8] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 8 & 5 \\ 11 & 20 & 0 & 0 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 16 & 0 & 0 & 10 & 6 \\ 5 & 8 & -2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 8 & 19 & 1 & 5 \\ 12 & 25 & 16 & 40 & 0 & 9 \\ 8 & 16 & 9 & 24 & 0 & 5 \\ 19 & 40 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 20 & 1 & 5 \\ 12 & 25 & 16 & 40 & 0 & 9 \\ 8 & 16 & 9 & 24 & 0 & 5 \\ 20 & 40 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 8 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 8 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 10 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 9 & 5 \\ 11 & 20 & 0 & 17 & 16 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 12 & 6 \\ 9 & 16 & 0 & 12 & 10 & 6 \\ 5 & 8 & -2 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 26 & 5 \\ 11 & 20 & 0 & 0 & 50 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 26 & 50 & 0 & 0 & 118 & 20 \\ 5 & 8 & -2 & -1 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 17 & 5 \\ 11 & 20 & 0 & 0 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 17 & 33 & 0 & 0 & 49 & 13 \\ 5 & 8 & -2 & -1 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 30 & 1 & 5 \\ 12 & 25 & 16 & 60 & 0 & 9 \\ 8 & 16 & 9 & 36 & 0 & 5 \\ 30 & 60 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 29 & 1 & 5 \\ 12 & 25 & 16 & 60 & 0 & 9 \\ 8 & 16 & 9 & 36 & 0 & 5 \\ 29 & 60 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -330, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 38 & 1 & 5 \\ 12 & 25 & 16 & 80 & 0 & 9 \\ 8 & 16 & 9 & 48 & 0 & 5 \\ 38 & 80 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{114} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 40 & 1 & 5 \\ 12 & 25 & 16 & 80 & 0 & 9 \\ 8 & 16 & 9 & 48 & 0 & 5 \\ 40 & 80 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{114} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 39 & 1 & 5 \\ 12 & 25 & 16 & 80 & 0 & 9 \\ 8 & 16 & 9 & 48 & 0 & 5 \\ 39 & 80 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 19 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 19 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 17 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 17 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 18 & 5 \\ 11 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 27 & 6 \\ 18 & 33 & 0 & 27 & 49 & 13 \\ 5 & 8 & -2 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \ 1^{11} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 6 & 9 & 1 & 5 \\ 9 & 13 & 10 & 14 & 0 & 7 \\ 6 & 10 & 4 & 9 & 0 & 4 \\ 9 & 14 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 7 & 4 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

4.743 NHC(-2)(-3) + so₈ + su₃ + su₂ + su₂

$$T = 9 \quad [0] \ 1^{11} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 4 & 5 & 0 & 3 & 4 & 6 & 4 \\ 3 & 5 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -2 & 2 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 6 & 4 \\ 3 & 5 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -2 & 2 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -2 & 2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

4.744 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \ 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 5 \\ 12 & 25 & 0 & 12 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 12 & 0 & 4 & 4 \\ 5 & 9 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 0 & 8 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 4 & 5 \\ 2 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & -2 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 6 & 5 \\ 11 & 20 & 17 & 11 & 8 \\ 10 & 17 & 12 & 9 & 6 \\ 6 & 11 & 9 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 5 \\ 11 & 20 & 0 & 11 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 11 & 0 & 4 & 4 \\ 5 & 8 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 6 & 5 \\ 11 & 20 & 17 & 11 & 8 \\ 9 & 17 & 12 & 9 & 6 \\ 6 & 11 & 9 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 99, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.17 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 12 & 8 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 12 & 17 & 12 & 9 & 6 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 21^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 121, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.11 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 15 & 9 & 7 \\ 17 & 20 & 17 & 11 & 8 \\ 15 & 17 & 12 & 9 & 6 \\ 9 & 11 & 9 & 4 & 4 \\ 7 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 154, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.18 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 5 & 6 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & -2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 21^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 156, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.13 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 10 & 16 & 12 & 8 & 6 \\ 16 & 20 & 17 & 11 & 8 \\ 12 & 17 & 12 & 9 & 6 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 201, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.16 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 0 & 13 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 7 & 9 \\ 2 & 3 & 0 & 7 & 0 & 2 \\ 3 & 4 & -2 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{13}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 230, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.21 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 17 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 48 & 9 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & -2 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [13] \ 1^{11}5^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 13, \ \Delta = -59, \ \det = -21, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.90 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [14] \ 1^14^{13}1^15^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 14, \ \Delta = -58, \ \det = 161, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.96 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 1 & 2 & 2 \\ 3 & 4 & 1 & 2 & 0 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 0 & 0 & 2 \\ 2 & 4 & 0 & 0 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

4.745 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{f}_4$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 9 & 6 & 1 & 5 \\ 11 & 20 & 17 & 11 & 0 & 8 \\ 9 & 17 & 12 & 9 & 0 & 6 \\ 6 & 11 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

4.746 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 6 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 9 & 6 \\ 6 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -2 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

4.747 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 12 \quad [0] \quad 1^{11} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 2 & 3 \\ 2 & 4 & 2 & 0 & 2 & 1 & 3 \\ 3 & 4 & 2 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 4 & 3 & 0 & 3 & 7 & 4 \\ 2 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 5 & 3 \\ 4 & 7 & 6 & 0 & 5 & 9 & 7 \\ 3 & 4 & 2 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{13} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 3 & 0 & 3 & 15 & 4 \\ 2 & 3 & 0 & 0 & 3 & 10 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 9 & 3 \\ 8 & 15 & 10 & 0 & 9 & 41 & 13 \\ 3 & 4 & 2 & -1 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{12} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 50, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 6 & 3 \\ 3 & 4 & 3 & 0 & 3 & 11 & 4 \\ 2 & 3 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 7 & 3 \\ 6 & 11 & 8 & 0 & 7 & 22 & 10 \\ 3 & 4 & 2 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

4.748 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \mathfrak{n} 3 \mathfrak{mix}$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 24 & 5 \\ 11 & 20 & 0 & 45 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 45 & 0 & 100 & 18 \\ 5 & 8 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 2 & 2 \\ 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 24 & 5 \\ 12 & 25 & 0 & 48 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 48 & 0 & 89 & 17 \\ 5 & 9 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 7 & 5 \\ 11 & 20 & 0 & 11 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 11 & 0 & 4 & 4 \\ 5 & 8 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{14}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 5 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 15 & 3 \\ 11 & 19 & 0 & 15 & 64 & 14 \\ 3 & 4 & -2 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{114}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 42, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 9 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 29 & 6 \\ 10 & 19 & 0 & 29 & 64 & 14 \\ 3 & 4 & -2 & 6 & 14 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 6 & 8 & 0 & 5 & 12 & 6 \\ 3 & 4 & -2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 11 & 9 & 7 & 5 \\ 11 & 20 & 17 & 11 & 8 \\ 9 & 17 & 12 & 9 & 6 \\ 7 & 11 & 9 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 11 & 10 & 7 & 5 \\ 11 & 20 & 17 & 11 & 8 \\ 10 & 17 & 12 & 9 & 6 \\ 7 & 11 & 9 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 25 & 5 \\ 11 & 20 & 0 & 45 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 25 & 45 & 0 & 100 & 18 \\ 5 & 8 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{11} 1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 50$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 0 & 13 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 16 & 9 \\ 5 & 7 & 0 & 16 & 7 & 5 \\ 3 & 4 & -2 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{11} 1^{121} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 53$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 10 & 5 \\ 5 & 7 & 0 & 10 & 7 & 5 \\ 3 & 4 & -2 & 5 & 5 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{11} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 64$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 10 & 15 & 12 & 9 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 12 & 17 & 12 & 9 & 6 \\ 9 & 11 & 9 & 4 & 4 \\ 6 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{11} 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 73$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 5 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 9 & 3 \\ 6 & 11 & 0 & 9 & 20 & 8 \\ 3 & 4 & -2 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{11} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 73$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 16 & 5 \\ 11 & 20 & 0 & 28 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 28 & 0 & 37 & 11 \\ 5 & 8 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{11} 1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 80$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 16 & 5 \\ 7 & 12 & 0 & 16 & 25 & 9 \\ 3 & 4 & -2 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{11} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 80$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 5 & 8 & 0 & 5 & 12 & 6 \\ 3 & 4 & -2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{11} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 85$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 5 \\ 11 & 20 & 0 & 28 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 28 & 0 & 37 & 11 \\ 5 & 8 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{11} 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 16 & 5 \\ 12 & 25 & 0 & 32 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 32 & 0 & 37 & 11 \\ 5 & 9 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{11} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 101$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 23 & 5 \\ 12 & 25 & 0 & 48 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 48 & 0 & 89 & 17 \\ 5 & 9 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{11} 1^{11} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 104$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 16 & 27 & 0 & 17 & 151 & 21 \\ 3 & 4 & -2 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 5 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 6 & 3 \\ 4 & 7 & 0 & 6 & 7 & 5 \\ 3 & 4 & -2 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 12 & 3 \\ 9 & 15 & 0 & 12 & 39 & 11 \\ 3 & 4 & -2 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 15 & 5 \\ 12 & 25 & 0 & 32 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 32 & 0 & 37 & 11 \\ 5 & 9 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 6 & 3 \\ 5 & 7 & 0 & 6 & 7 & 5 \\ 3 & 4 & -2 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 133, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 11 & 6 \\ 5 & 7 & 0 & 11 & 7 & 5 \\ 3 & 4 & -2 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 9 & 3 \\ 7 & 11 & 0 & 9 & 20 & 8 \\ 3 & 4 & -2 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 0 & 9 & 10 & 5 \\ 4 & 7 & 0 & 10 & 7 & 5 \\ 3 & 4 & -2 & 5 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 9 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 23 & 6 \\ 8 & 15 & 0 & 23 & 39 & 11 \\ 3 & 4 & -2 & 6 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 4 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & -2 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 9 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 17 & 6 \\ 6 & 11 & 0 & 17 & 20 & 8 \\ 3 & 4 & -2 & 6 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 5 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & -2 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 0 & 9 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 11 & 6 \\ 4 & 7 & 0 & 11 & 7 & 5 \\ 3 & 4 & -2 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 0 & 9 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 17 & 6 \\ 7 & 11 & 0 & 17 & 20 & 8 \\ 3 & 4 & -2 & 6 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 6 & 11 & 0 & 8 & 20 & 8 \\ 3 & 4 & -2 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 13 & 22 & 0 & 14 & 97 & 17 \\ 3 & 4 & -2 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 262, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 9 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 29 & 6 \\ 11 & 19 & 0 & 29 & 64 & 14 \\ 3 & 4 & -2 & 6 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{111} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 17 & 27 & 0 & 17 & 151 & 21 \\ 3 & 4 & -2 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 5 & 9 & 0 & 12 & 23 & 6 \\ 9 & 15 & 0 & 23 & 39 & 11 \\ 3 & 4 & -2 & 6 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 7 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & -2 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 278, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 0 & 13 & 7 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 7 & 13 & 0 & 27 & 16 & 9 \\ 4 & 7 & 0 & 16 & 7 & 5 \\ 3 & 4 & -2 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 10 & 17 & 0 & 11 & 55 & 13 \\ 3 & 4 & -2 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 8 & 15 & 0 & 11 & 39 & 11 \\ 3 & 4 & -2 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 8 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & -2 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 11 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 7 & 11 & 0 & 8 & 20 & 8 \\ 3 & 4 & -2 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 14 & 22 & 0 & 14 & 97 & 17 \\ 3 & 4 & -2 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 10 & 19 & 0 & 14 & 64 & 14 \\ 3 & 4 & -2 & 2 & 14 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 11 & 17 & 0 & 11 & 55 & 13 \\ 3 & 4 & -2 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 9 & 16 & 0 & 11 & 44 & 12 \\ 3 & 4 & -2 & 2 & 12 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 11.36 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 9 & 15 & 0 & 11 & 39 & 11 \\ 3 & 4 & -2 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$[52] \quad 1^{11} 2^{11} 3^{11} 1^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}o_8 + \textcolor{brown}{s}u_3 + \textcolor{red}{s}u_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 624, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$[54] \quad 1^{113} 3^{11} 1^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}o_8 + \textcolor{brown}{s}u_3 + \textcolor{red}{s}u_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$[53] \quad 1^{11}2^33^11^{14} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}o_8 + \textcolor{brown}{s}u_3 + \textcolor{red}{s}u_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

[55] $1^{112}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

[56] $1^{114}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 704, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$[57] \quad 1^{11} 1^{12} 3^1 1^{12} + [\textcolor{red}{s} \textcolor{red}{u} \textcolor{red}{2} \cdot \textcolor{blue}{g} \textcolor{blue}{2}] + [\textcolor{green}{s} \textcolor{green}{0} \textcolor{green}{8} + \textcolor{orange}{s} \textcolor{orange}{u} \textcolor{orange}{3} + \textcolor{red}{s} \textcolor{red}{u} \textcolor{red}{2}] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.13 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 4 & 23 & 4 \end{pmatrix}$$

$$\begin{pmatrix} 3 & 0 & 0 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 13 & 21 & 0 & 14 & 80 & 16 \\ 2 & 2 & 2 & 2 & 2 & 2 \end{pmatrix}.$$

$$[58] \quad 1^{11} 1^{14} 3^{11} 1^{11} + [6111_1, 610] + 6090 + 6110 + 6110 \quad T = 8, \quad T_{\text{min}} = 13, \quad \Delta = -94, \quad \det = 848, \quad (n_1, n_2, n_3) = (1, 8, 0), \quad c_1, c_2 = 10, 0$$

$T_{\min} = 11.35 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 12 & 4 & 0 & 0 & 7 & 10 \end{pmatrix}$$

$$\begin{pmatrix} 13 & 24 & 0 & 17 & 100 & 18 \\ 3 & 4 & -2 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{112} 3^1 1^{13} + [\textcolor{red}{s}u_2 \cdot \textcolor{red}{g}_2] + \textcolor{green}{s}o_8 + \textcolor{brown}{s}u_3 + \textcolor{red}{s}u_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 896, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 4 & 0 & 0 & 17 & 2 \\ 16 & 26 & 0 & 17 & 128 & 20 \\ 3 & 4 & -2 & 2 & 20 & 1 \end{pmatrix} b_0$$

[60] $1^{113} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 912, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$l_{\min} = 11.21 \rightarrow 12, \quad l_{\max} = 13, \quad t \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 4 & 20 & 4 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 19 & 3 \\ 3 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 20 & 2 \\ 19 & 31 & 0 & 20 & 188 & 24 \\ 3 & 4 & -2 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 25 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 14 & 25 & 0 & 17 & 111 & 19 \\ 3 & 4 & -2 & 2 & 19 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 30 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 20 & 2 \\ 17 & 30 & 0 & 20 & 165 & 23 \\ 3 & 4 & -2 & 2 & 23 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 14 & 24 & 0 & 17 & 100 & 18 \\ 3 & 4 & -2 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 35 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 23 & 2 \\ 20 & 35 & 0 & 23 & 231 & 27 \\ 3 & 4 & -2 & 2 & 27 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 4 & 25 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 15 & 25 & 0 & 17 & 111 & 19 \\ 3 & 4 & -2 & 2 & 19 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 4 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 20 & 2 \\ 16 & 29 & 0 & 20 & 148 & 22 \\ 3 & 4 & -2 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 4 & 30 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 20 & 2 \\ 18 & 30 & 0 & 20 & 165 & 23 \\ 3 & 4 & -2 & 2 & 23 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 19 & 3 \\ 3 & 4 & 0 & 4 & 34 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 23 & 2 \\ 19 & 34 & 0 & 23 & 208 & 26 \\ 3 & 4 & -2 & 2 & 26 & 1 \end{pmatrix}_b$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 21 & 3 \\ 3 & 4 & 0 & 4 & 35 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 23 & 2 \\ 21 & 35 & 0 & 23 & 231 & 27 \\ 3 & 4 & -2 & 2 & 27 & 1 \end{pmatrix}.$$

$$[\begin{matrix} \text{red} & \text{blue} \\ \text{2} & \text{2} \end{matrix}] \mid \begin{matrix} \text{green} & \text{orange} \\ \text{2} & \text{2} \end{matrix} \mid \begin{matrix} \text{red} & \text{blue} \\ \text{2} & \text{2} \end{matrix}]$$

$$T_{\min} = 11.19 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 4 & 29 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 20 & 2 \\ 17 & 29 & 0 & 20 & 148 & 22 \\ 3 & 4 & -2 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{\overline{114}3^{\overline{11}}1^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2} \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 1760, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 22 & 3 \\ 3 & 4 & 0 & 4 & 39 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 26 & 2 \\ 22 & 39 & 0 & 26 & 280 & 30 \\ 3 & 4 & -2 & 2 & 30 & 1 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{\overline{114}3^{\overline{11}}1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2} \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 1952, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 4 & 34 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 23 & 2 \\ 20 & 34 & 0 & 23 & 208 & 26 \\ 3 & 4 & -2 & 2 & 26 & 1 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{\overline{114}3^{\overline{11}}1^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2} \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 2208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 23 & 3 \\ 3 & 4 & 0 & 4 & 39 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 26 & 2 \\ 23 & 39 & 0 & 26 & 280 & 30 \\ 3 & 4 & -2 & 2 & 30 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [75] \ 1^{\overline{115}3^{\overline{11}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2} \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -61, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 3 & 0 & 7 & 4 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 7 & 5 & 0 & 8 & 6 \\ 3 & 4 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{\overline{125}3^{\overline{11}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2} \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -61, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 3 & 0 & 14 & 4 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 14 & 11 & 0 & 39 & 13 \\ 3 & 4 & 2 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[77] \ 1^{\overline{135}3^{\overline{11}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2} \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -133, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 3 & 0 & 18 & 4 \\ 2 & 3 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 18 & 13 & 0 & 62 & 16 \\ 3 & 4 & 2 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

4.749 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_{2n3\text{mix}} + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{\overline{111}3^{\overline{11}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8} \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 7 & 5 \\ 11 & 20 & 0 & 17 & 11 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 9 & 17 & 0 & 12 & 9 & 6 \\ 7 & 11 & 0 & 9 & 4 & 4 \\ 5 & 8 & -2 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

4.750 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{\overline{111}3^{\overline{11}}1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3} \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 17.60 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 18 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & -4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{13} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 0 & 5 & 12 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 9 & 3 \\ 7 & 12 & 0 & 9 & 24 & 8 \\ 3 & 4 & -2 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{13} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 1 & 3 \\ 3 & 4 & 0 & 10 & 0 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 6 & 10 & 0 & 21 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & -2 & 7 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{13} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 12 & 1 & 1 & 5 \\ 12 & 25 & 0 & 0 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 12 & 8 & 11 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 11 & 20 & 12 & 12 & 6 \\ 5 & 9 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 19 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{12} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 5 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 6 & 3 \\ 4 & 8 & 0 & 6 & 9 & 5 \\ 3 & 4 & -2 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 4 & 0 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{12} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 4 & 3 \\ 3 & 4 & 0 & 0 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 0 & 0 & 9 & 5 \\ 3 & 4 & -2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{14} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.6$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 11 & 3 \\ 3 & 4 & 0 & 0 & 21 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 11 & 21 & 0 & 0 & 75 & 15 \\ 3 & 4 & -2 & -1 & 15 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11} 1^3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 9 & 20 & 12 & 12 & 6 \\ 5 & 9 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{11} 1^3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

[14] $1^{11} 1^3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 12 & 8 & 10 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 10 & 20 & 12 & 12 & 6 \\ 5 & 9 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{11} 1^3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 4 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & -2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{11} 1^3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 0 & 0 & 3 & 3 \\ 3 & 4 & -2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{11} 1^3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 3 & 4 & -2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{11} 1^3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 17 & 0 & 12 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{11} 1^3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 12 & 1 & 18 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 36 & 0 & 48 & 12 \\ 5 & 9 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{12} 1^3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 8 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 6 & 3 \\ 5 & 8 & 0 & 6 & 9 & 5 \\ 3 & 4 & -2 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{11} 1^3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 12 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

- [23] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 101$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 10 & 15 & 12 & 1 & 6 \\ 15 & 20 & 17 & 0 & 8 \\ 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$
- [22] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 101$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 9 & 3 \\ 3 & 4 & 0 & 0 & 17 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 48 & 12 \\ 3 & 4 & -2 & -1 & 12 & 1 \end{pmatrix}_{b_0}$
- [21] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 101$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$
 $\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 20 & 0 & 12 & 6 \\ 5 & 9 & -1 & 6 & 1 \end{pmatrix}_{b_0}$
- [24] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 104$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [25] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 108$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [26] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 116$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 5 & 12 & 1 & 10 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 20 & 0 & 12 & 6 \\ 5 & 9 & -1 & 6 & 2 \end{pmatrix}_{b_0}$
- [27] $1^{11} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 116$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 3 \\ 3 & 4 & 0 & 0 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 14 & 0 & 0 & 36 & 10 \\ 3 & 4 & -2 & -1 & 10 & 1 \end{pmatrix}_{b_0}$
- [28] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 117$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 5 & 12 & 1 & 17 & 5 \\ 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 36 & 0 & 48 & 12 \\ 5 & 9 & -1 & 12 & 2 \end{pmatrix}_{b_0}$
- [29] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 117$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 9 & 0 & 0 & 12 & 6 \\ 3 & 4 & -2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$
- [30] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 125$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 7 & 3 \\ 3 & 4 & 0 & 0 & 13 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 7 & 13 & 0 & 0 & 27 & 9 \\ 3 & 4 & -2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$
- [31] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 9 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [42] \quad 1 \overset{11}{4} 1 \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 6 & 6 & 5 \\ 12 & 25 & 12 & 12 & 9 \\ 6 & 12 & 4 & 7 & 4 \\ 6 & 12 & 7 & 4 & 4 \\ 5 & 9 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[44] \quad 1 \overset{11}{1} 31 \overset{11}{1} 31 \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 8 & 8 & 7 & 1 & 4 \\ 8 & 13 & 16 & 12 & 0 & 7 \\ 8 & 16 & 12 & 12 & 0 & 6 \\ 7 & 12 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 7 & 6 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[43] \quad 1 \overset{11}{1} 31 \overset{11}{1} 31 \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 0 & 3 \\ 1 & 4 & 0 & 0 & 0 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & -2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[45] \quad 1 \overset{11}{1} 31 \overset{11}{1} 31 \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 1 & 1 & 5 & 3 \\ 4 & 10 & 1 & 0 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 0 & 0 & 12 & 6 \\ 3 & 6 & -1 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[46] \quad 1 \overset{11}{1} 31 \overset{11}{1} 31 \overset{12}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 3 & 2 \\ 1 & 3 & 0 & 3 & 0 & 3 \\ 2 & 3 & 0 & 3 & 3 & 2 \\ 2 & 3 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1 \overset{11}{1} 41 \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[48] \quad 21 \overset{11}{1} 41 \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 13 & 20 & 9 & 9 & 7 \\ 20 & 25 & 12 & 12 & 9 \\ 9 & 12 & 4 & 7 & 4 \\ 9 & 12 & 7 & 4 & 4 \\ 7 & 9 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[49] \quad 1 \overset{11}{1} 41 \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 5 \\ 9 & 13 & 0 & 0 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[50] \quad 1 \overset{11}{1} 31 \overset{11}{1} 31 \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 5 & 1 & 1 & 3 \\ 5 & 13 & 16 & 0 & 0 & 7 \\ 5 & 16 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 6 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[51] \quad 1 \overset{11}{1} 31 \overset{11}{1} 31 \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 9 & 6 & 0 & 0 & 12 & 6 \\ 2 & 4 & 2 & -2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{11} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 6 & 5 \\ 9 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 10 & 0 & 4 & 4 \\ 5 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{11} 3^1 1^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -105, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 3 & 3 & 2 \\ 1 & 3 & 0 & 3 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & 2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{11} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 7 & 5 \\ 9 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 10 & 0 & 4 & 4 \\ 5 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{11} 3^1 1^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 2 & -2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

4.751 NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 1 & 5 \\ 12 & 25 & 16 & 0 & 0 & 9 \\ 8 & 16 & 9 & 0 & 0 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & -1 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 10 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 10 & 20 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \quad 1^{11} 4 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.20 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 12 & 6 & 6 & 1 & 5 \\ 12 & 25 & 12 & 12 & 0 & 9 \\ 6 & 12 & 4 & 7 & 0 & 4 \\ 6 & 12 & 7 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 4 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

4.752 NHC(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{111} 3 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 1 & 10 & 5 \\ 12 & 25 & 0 & 0 & 20 & 9 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 20 & 0 & 0 & 12 & 6 \\ 5 & 9 & -1 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 1 & 1 & 17 & 5 \\ 12 & 25 & 0 & 0 & 36 & 9 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 17 & 36 & 0 & 0 & 48 & 12 \\ 5 & 9 & -1 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 10 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 20 & 12 & 0 & 12 & 6 \\ 5 & 9 & 5 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -1 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 0 & 3 & 3 & 3 \\ 2 & 4 & 0 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 3 & 2 & 2 \\ 2 & 4 & 0 & 0 & 3 & 0 & 2 \\ 3 & 4 & -2 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -237, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3 \ 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -240, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.68 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 0 & 4 & 14 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 0 & 9 & 2 \\ 8 & 14 & 0 & 0 & 9 & 36 & 10 \\ 3 & 4 & -2 & -1 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -261, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.70 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 4 & 0 & 3 & 0 & 0 & 2 \\ 3 & 4 & -2 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -288, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.74 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 0 & 4 & 9 & 4 \\ 1 & 0 & -4 & 0 & 0 & 0 & -2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 0 & 6 & 2 \\ 5 & 9 & 0 & 0 & 6 & 12 & 6 \\ 3 & 4 & -2 & -1 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

4.753 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \ 1^{11} 31^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 20, \ \Delta = -102, \ \det = 146, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.49 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 31^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 20, \ \Delta = -102, \ \det = 200, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.56 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 13 & 20 & 12 & 1 & 7 \\ 20 & 25 & 16 & 0 & 9 \\ 12 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 31^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 20, \ \Delta = -102, \ \det = 209, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 10 & 16 & 12 & 1 & 6 \\ 16 & 20 & 17 & 0 & 8 \\ 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \ 1^{11} 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 9, \ T_{\max} = 14, \ \Delta = -93, \ \det = -129, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 13 & 14 & 9 & 1 & 7 \\ 14 & 13 & 10 & 0 & 7 \\ 9 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{21} 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) \quad T = 9, \ T_{\max} = 14, \ \Delta = -93, \ \det = -321, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.18 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 5 & 10 & 7 & 1 & 5 \\ 10 & 13 & 10 & 0 & 7 \\ 7 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

4.754 $\text{NHC}(-2)(-3) + \mathfrak{so}_8 + (-3) \rightarrow (-2)$

$$T = 7 \quad [0] \ 1^{11} 31^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) \quad T = 7, \ T_{\max} = 77, \ \Delta = -123, \ \det = -67, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.0$$

$$\begin{aligned}
& T_{\min} = 11.15 \rightarrow 12, T_{\max} = 77, T \in [12, 77] \\
& \begin{pmatrix} 13 & 17 & 1 & 7 \\ 17 & 20 & 0 & 8 \\ 1 & 0 & -3 & -1 \\ 7 & 8 & -1 & 2 \end{pmatrix}_{b_0} \\
[1] & 1^1 31^{11} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.35 \rightarrow 12, T_{\max} = 77, T \in [12, 77] \\
& \begin{pmatrix} 10 & 16 & 1 & 6 \\ 16 & 20 & 0 & 8 \\ 1 & 0 & -3 & -1 \\ 6 & 8 & -1 & 2 \end{pmatrix}_{b_0} \\
[2] & 1^1 31^{11} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.20 \rightarrow 12, T_{\max} = 77, T \in [12, 77] \\
& \begin{pmatrix} 13 & 20 & 1 & 7 \\ 20 & 25 & 0 & 9 \\ 1 & 0 & -3 & -1 \\ 7 & 9 & -1 & 2 \end{pmatrix}_{b_0} \\
\\
T = 8 & [3] 1^{11} 412^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12.38 \rightarrow 13, T_{\max} = 24, T \in [13, 24] \\
& \begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -3 & -1 \\ 5 & 9 & -1 & 2 \end{pmatrix}_{b_0} \\
[4] & 1^{11} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12, T_{\max} = 24, T \in [12, 24] \\
& \begin{pmatrix} 13 & 14 & 1 & 7 \\ 14 & 13 & 0 & 7 \\ 1 & 0 & -3 & -1 \\ 7 & 7 & -1 & 3 \end{pmatrix}_{b_0} \\
[5] & 1^{11} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.78 \rightarrow 12, T_{\max} = 24, T \in [12, 24] \\
& \begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 13 & 0 & 7 \\ 1 & 0 & -3 & -1 \\ 5 & 7 & -1 & 3 \end{pmatrix}_{b_0} \\
[6] & 1^{11} 231^{11} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 125, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.78 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0} \\
[7] & 1^{11} 41^{11} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 417, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.79 \rightarrow 12, T_{\max} = 24, T \in [12, 24] \\
& \begin{pmatrix} 0 & 3 & 3 & 1 & 1 & 2 \\ 3 & 4 & 7 & 2 & 0 & 4 \\ 3 & 7 & 3 & 2 & 0 & 3 \\ 1 & 2 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & 0 & -1 & 1 \end{pmatrix}_{b_0} \\
\\
T = 9 & [8] 2^{11} 1^1 41^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12.38 \rightarrow 13, T_{\max} = 14, T \in [13, 14] \\
& \begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 1 & 6 & 0 & 3 \\ 5 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 6 & -1 & 2 \end{pmatrix}_{b_0} \\
\\
T = 10 & [9] 1^{11} 5^{11} 12^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 13.40 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0} \\
[10] & 1^{11} 4141^{11} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 1 & 4 & 0 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{11}} 5^{\overline{11}} 131^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 14, \ \Delta = -88, \ \det = 60, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 13.13 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

4.755 NHC(-2)(-3) + so₈ + (-3)→(-2) + su₂

$$T = 8 \quad [0] \ 1^{\overline{11}} 31^{\overline{1}} 2^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 12 & 7 \\ 17 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 16 & 0 & 10 & 6 \\ 7 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 31^{\overline{1}} 2^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 61, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.46 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 10 & 16 & 1 & 12 & 6 \\ 16 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 17 & 0 & 13 & 7 \\ 6 & 8 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 31^{\overline{1}} 2^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 79, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 15 & 7 \\ 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 17 & 0 & 13 & 7 \\ 7 & 8 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 31^{\overline{1}} 2^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 299, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 27 & 7 \\ 17 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 27 & 33 & 0 & 49 & 13 \\ 7 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}} 31^{\overline{1}} 2^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 325, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.30 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 1 & 27 & 7 \\ 20 & 25 & 0 & 36 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 27 & 36 & 0 & 49 & 13 \\ 7 & 9 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}} 31^{\overline{1}} 2^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 434, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.17 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 72 & 7 \\ 17 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 72 & 84 & 0 & 346 & 34 \\ 7 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{11}} 31^{\overline{1}} 2^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 454, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.15 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 42 & 7 \\ 17 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 42 & 50 & 0 & 118 & 20 \\ 7 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}} 31^{\overline{1}} 2^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 499, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.15 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 57 & 7 \\ 17 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 57 & 67 & 0 & 217 & 27 \\ 7 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{11}} 31^{\overline{1}} 2^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + (-3) \rightarrow (-2) + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 525, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 1 & 15 & 7 \\ 20 & 25 & 0 & 20 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 20 & 0 & 13 & 7 \\ 7 & 9 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [9] \quad 1^{11} 4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 9 & 5 \\ 10 & 13 & 0 & 14 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 14 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 7 & 5 \\ 10 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 10 & 0 & 5 & 5 \\ 5 & 7 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11} 4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 16 & 5 \\ 10 & 13 & 0 & 24 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 24 & 0 & 38 & 12 \\ 5 & 7 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11} 4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 14 & 5 \\ 10 & 13 & 0 & 20 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 20 & 0 & 26 & 10 \\ 5 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11} 4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -589, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 21 & 5 \\ 10 & 13 & 0 & 30 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 30 & 0 & 61 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -709, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 23 & 5 \\ 10 & 13 & 0 & 34 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 34 & 0 & 77 & 17 \\ 5 & 7 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11} 4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -904, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 28 & 5 \\ 10 & 13 & 0 & 40 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 28 & 40 & 0 & 110 & 20 \\ 5 & 7 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11} 4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -1164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 30 & 5 \\ 10 & 13 & 0 & 44 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 30 & 44 & 0 & 130 & 22 \\ 5 & 7 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 4 \quad 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{so}_8 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -1709, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 1 & 37 & 5 \\ 10 & 13 & 0 & 54 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 37 & 54 & 0 & 197 & 27 \\ 5 & 7 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

4.756 NHC(-2)(-3) + $\mathfrak{so}_8$ + (-3)→(-2) + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \ 1^{\overline{11}3}1^{\overline{11}2}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.22 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 9 & 7 \\ 17 & 20 & 0 & 11 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 11 & 0 & 4 & 4 \\ 7 & 8 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}3}1^{\overline{11}2}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 325, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.22 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 1 & 9 & 7 \\ 20 & 25 & 0 & 12 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 12 & 0 & 4 & 4 \\ 7 & 9 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

4.757 NHC(-2)(-3) + $\mathfrak{so}_8$ + (-3)→(-2) + $\mathfrak{su}_2 \mathfrak{n}3\text{mix}$

$$T = 8 \quad [0] \ 1^{\overline{11}3}1^{\overline{11}2}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 1 & 24 & 7 \\ 17 & 20 & 0 & 28 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 28 & 0 & 37 & 11 \\ 7 & 8 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

4.758 NHC(-2)(-3) + $\mathfrak{so}_8$ + (-3)→(-2) + $\mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{\overline{11}3}1^{\overline{11}2}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 17 & 1 & 15 & 7 \\ 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 17 & 0 & 12 & 6 \\ 7 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}3}1^{\overline{11}2}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 20 & 1 & 12 & 7 \\ 20 & 25 & 0 & 16 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 16 & 0 & 9 & 5 \\ 7 & 9 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}3}1^{\overline{11}2}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 16 & 1 & 12 & 6 \\ 16 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 17 & 0 & 12 & 6 \\ 6 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \ 1^{\overline{21}1}4^{\overline{11}}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 13 & 14 & 1 & 9 & 7 \\ 14 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 10 & 0 & 4 & 4 \\ 7 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{21}1}4^{\overline{11}}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 10 & 1 & 7 & 5 \\ 10 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 10 & 0 & 4 & 4 \\ 5 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

4.759 NHC(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_{3n2mix}$

$$T = 7 \quad [0] \ 1^{\overline{11}} 3 1^{\overline{11}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 77, \Delta = -123, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.08 \rightarrow 12, T_{\max} = 77, T \in [12, 77]$$

$$\begin{pmatrix} 10 & 16 & 13 & 6 \\ 16 & 20 & 17 & 8 \\ 13 & 17 & 12 & 6 \\ 6 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [1] \ 1^{\overline{11}} 4 \ 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 13 & 14 & 10 & 7 \\ 14 & 13 & 10 & 7 \\ 10 & 10 & 4 & 4 \\ 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [2] \ 1^{\overline{11}} 4 14 1^{\overline{1}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 & 4 \\ 5 & 5 & 4 & 5 & 5 \\ 3 & 4 & 0 & 3 & 2 \\ 4 & 5 & 3 & 1 & 3 \\ 4 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

4.760 NHC(-2)(-3) + $\mathfrak{su}_2$

$$T = 6 \quad [0] \ 1^{\overline{12}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 7 & 3 \\ 4 & 9 & 16 & 5 \\ 7 & 16 & 28 & 10 \\ 3 & 5 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{12}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 5 & 4 \\ 5 & 5 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 5, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & 1 & 3 \\ 3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{13}} 3 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 6, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 10 & 4 \\ 9 & 10 & 22 & 8 \\ 3 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{12}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 6, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 10 & 4 \\ 6 & 10 & 22 & 8 \\ 3 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 6, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9, T_{\max} = 22, T \in [9, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.56 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 5 & 4 \\ 3 & 5 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{14}} 3 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 9, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 15 & 4 \\ 13 & 15 & 52 & 12 \\ 3 & 4 & 12 & -1 \end{pmatrix}_{b_0} \\
[12] \quad & 21^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 11, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9.45 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 4 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[11] \quad & 21^{11} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 11, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.36 \rightarrow 13, T_{\max} = 22, T \in [13, 22] \\
& \begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 19 & 4 \\ 12 & 19 & 73 & 15 \\ 3 & 4 & 15 & 0 \end{pmatrix}_{b_0} \\
[10] \quad & 1^{12} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 11, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9.45 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 4 & 4 \\ 4 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[9] \quad & 1^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 11, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9.45 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[8] \quad & 1^1 31^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 11, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9.45 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 5 & 4 \\ 4 & 5 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0} \\
[13] \quad & 1^{12} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 12, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 13.33 \rightarrow 14, T_{\max} = 22, T \in [14, 22] \\
& \begin{pmatrix} 1 & 4 & 11 & 3 \\ 4 & 9 & 20 & 5 \\ 11 & 20 & 37 & 11 \\ 3 & 5 & 11 & -1 \end{pmatrix}_{b_0} \\
[14] \quad & 1^1 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 13, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.54 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 9 & 28 & 5 \\ 12 & 28 & 85 & 17 \\ 3 & 5 & 17 & 1 \end{pmatrix}_{b_0} \\
[17] \quad & 1^1 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 14, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.43 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 8 & 4 \\ 4 & 8 & 10 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0} \\
[16] \quad & 1^{12} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 14, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.43 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 8 & 4 \\ 8 & 8 & 10 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0} \\
[15] \quad & 1^1 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 14, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.43 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 10 & 4 \\ 7 & 10 & 22 & 8 \\ 3 & 4 & 8 & 2 \end{pmatrix}_{b_0} \\
[19] \quad & 1^1 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 14, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 14.71 \rightarrow 15, T_{\max} = 22, T \in [15, 22] \\
& \begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 20 & 4 \\ 13 & 20 & 94 & 16 \\ 3 & 4 & 16 & -2 \end{pmatrix}_{b_0} \\
[18] \quad & 1^{12} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 14, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.43 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 10 & 4 \\ 8 & 10 & 22 & 8 \\ 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [20] $1^1 3^1 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 15$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 15 & 4 \\ 10 & 15 & 52 & 12 \\ 3 & 4 & 12 & 1 \end{pmatrix}_{b_0}$
- [21] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 15$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.33 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [22] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 16$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.25 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 9 & 4 \\ 9 & 9 & 13 & 7 \\ 3 & 4 & 7 & 2 \end{pmatrix}_{b_0}$
- [23] $1^{111} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 17$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 9 & 12 & 5 \\ 6 & 12 & 13 & 7 \\ 3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$
- [24] $21^1 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 17$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 9 & 12 & 5 \\ 6 & 12 & 13 & 7 \\ 3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$
- [25] $2131^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 19$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 8 & 5 \\ 3 & 8 & 4 & 4 \\ 3 & 5 & 4 & 1 \end{pmatrix}_{b_0}$
- [26] $1^{111} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 19$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 8 & 5 \\ 3 & 8 & 4 & 4 \\ 3 & 5 & 4 & 1 \end{pmatrix}_{b_0}$
- [27] $1^1 3^1 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 19$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 15 & 4 \\ 12 & 15 & 52 & 12 \\ 3 & 4 & 12 & 1 \end{pmatrix}_{b_0}$
- [28] $1^1 3^1 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 19$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 12 & 4 \\ 6 & 12 & 25 & 9 \\ 3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$
- [29] $1^1 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 20$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 9 & 12 & 5 \\ 5 & 12 & 13 & 7 \\ 3 & 5 & 7 & 2 \end{pmatrix}_{b_0}$
- [30] $21^{11} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 20$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 4 & 5 & 6 & 4 \\ 5 & 4 & 5 & 4 \\ 6 & 5 & 4 & 4 \\ 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [31] $1^1 3^1 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 21$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 15 & 4 \\ 11 & 15 & 52 & 12 \\ 3 & 4 & 12 & 1 \end{pmatrix}_{b_0}$
- [32] $1^1 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 24$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 9 & 4 \\ 5 & 9 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[37] $1^{12}3^11^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 26, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 8 & 4 \\ 5 & 8 & 10 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

[38] $1^{12}3^11^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 26, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 20 & 4 \\ 14 & 20 & 94 & 16 \\ 3 & 4 & 16 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{14}3^11^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 26, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 20 & 4 \\ 16 & 20 & 94 & 16 \\ 3 & 4 & 16 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{12}3^11^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 26, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 8 & 4 \\ 7 & 8 & 10 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

[35] $21^{12}3^11^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 26, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 4 & 5 & 12 & 4 \\ 5 & 4 & 10 & 4 \\ 12 & 10 & 22 & 8 \\ 4 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{112}3^11^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 26, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 9 & 24 & 5 \\ 12 & 24 & 58 & 14 \\ 3 & 5 & 14 & 0 \end{pmatrix}_{b_0}$$

[34] $1^13^11^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 26, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 16 & 4 \\ 8 & 16 & 46 & 12 \\ 3 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

[33] $21^23^11^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 26, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 9 & 24 & 5 \\ 12 & 24 & 58 & 14 \\ 3 & 5 & 14 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{113}3^11^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 27, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 9 & 32 & 5 \\ 17 & 32 & 100 & 18 \\ 3 & 5 & 18 & -1 \end{pmatrix}_{b_0}$$

[43] $1^{13}3^11^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 30, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 20 & 4 \\ 15 & 20 & 94 & 16 \\ 3 & 4 & 16 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{113}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 30, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 9 & 4 \\ 8 & 9 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[42] $1^{12}3^11^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 30, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 8 & 4 \\ 6 & 8 & 10 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

- [45] $21^{11}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 32, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.50 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [48] $1^{112} 1^1 3^1 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 34, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 9 & 16 & 5 \\ 6 & 16 & 22 & 8 \\ 3 & 5 & 8 & 1 \end{pmatrix}_{b_0}$
- [47] $2131^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 34, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 9 & 16 & 5 \\ 6 & 16 & 22 & 8 \\ 3 & 5 & 8 & 1 \end{pmatrix}_{b_0}$
- [46] $1^{11} 1^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 34, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.59 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 9 & 4 \\ 6 & 9 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$
- [50] $1^1 1^1 31^{15} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 35, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.1$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 20 & 4 \\ 10 & 20 & 73 & 15 \\ 3 & 4 & 15 & 0 \end{pmatrix}_{b_0}$
- [49] $1^{112} 1^1 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 35, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 14 & 4 \\ 8 & 14 & 37 & 11 \\ 3 & 4 & 11 & 2 \end{pmatrix}_{b_0}$
- [51] $1^4 1^1 31^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 35, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 9 & 40 & 5 \\ 17 & 40 & 172 & 24 \\ 3 & 5 & 24 & 1 \end{pmatrix}_{b_0}$
- [54] $1^{11} 1^{12} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.56 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 9 & 4 \\ 7 & 9 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$
- [53] $21^{11} 31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 4 & 5 & 18 & 4 \\ 5 & 4 & 21 & 4 \\ 18 & 21 & 76 & 16 \\ 4 & 4 & 16 & 3 \end{pmatrix}_{b_0}$
- [52] $21^{13} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 36, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 4 & 5 & 18 & 4 \\ 5 & 4 & 15 & 4 \\ 18 & 15 & 52 & 12 \\ 4 & 4 & 12 & 1 \end{pmatrix}_{b_0}$
- [55] $1^2 1^1 31^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 38, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 4 & 10 & 3 \\ 4 & 9 & 24 & 5 \\ 10 & 24 & 58 & 14 \\ 3 & 5 & 14 & 2 \end{pmatrix}_{b_0}$
- [56] $1^{13} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 39, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 12 & 4 \\ 7 & 12 & 25 & 9 \\ 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$
- [57] $1^{113} 1^4 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 40, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

- [58] $21^3 31^{113} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 41, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.54 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 9 & 28 & 5 \\ 15 & 28 & 73 & 15 \\ 3 & 5 & 15 & 1 \end{pmatrix}_{b_0}$$
- [59] $1^{113} 31^3 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 41, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.54 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 9 & 36 & 5 \\ 18 & 36 & 133 & 21 \\ 3 & 5 & 21 & 0 \end{pmatrix}_{b_0}$$
- [60] $1^{112} 31^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 41, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 10 & 3 \\ 4 & 9 & 20 & 5 \\ 10 & 20 & 37 & 11 \\ 3 & 5 & 11 & 1 \end{pmatrix}_{b_0}$$
- [61] $1^{11} 1^3 1^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 41, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 13 & 4 \\ 7 & 13 & 28 & 10 \\ 3 & 4 & 10 & 2 \end{pmatrix}_{b_0}$$
- [62] $1^{13} 1^3 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 44, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 19 & 4 \\ 11 & 19 & 73 & 15 \\ 3 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$
- [63] $21^{113} 1^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 44, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 4 & 14 & 4 \\ 9 & 14 & 37 & 11 \\ 3 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$
- [64] $21^{113} 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 45, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.64 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 4 & 5 & 9 & 4 \\ 5 & 4 & 9 & 4 \\ 9 & 9 & 13 & 7 \\ 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$
- [65] $1^1 1^3 1^{112} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 45, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 8 & 3 \\ 4 & 9 & 20 & 5 \\ 8 & 20 & 37 & 11 \\ 3 & 5 & 11 & 1 \end{pmatrix}_{b_0}$$
- [66] $1^1 1^3 1^{16} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 46, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.2$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 24 & 4 \\ 12 & 24 & 106 & 18 \\ 3 & 4 & 18 & 0 \end{pmatrix}_{b_0}$$
- [67] $21^{14} 31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 50, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 4 & 5 & 24 & 4 \\ 5 & 4 & 20 & 4 \\ 24 & 20 & 94 & 16 \\ 4 & 4 & 16 & 1 \end{pmatrix}_{b_0}$$
- [72] $1^{13} 31^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 51, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 12 & 4 \\ 8 & 12 & 25 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$
- [71] $1^{14} 1^3 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 51, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 24 & 4 \\ 14 & 24 & 121 & 19 \\ 3 & 4 & 19 & 1 \end{pmatrix}_{b_0}$$

- [73] $1^{11}3^11^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 51$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 14 & 4 \\ 9 & 14 & 37 & 11 \\ 3 & 4 & 11 & 3 \end{pmatrix}_{b_0}$
- [68] $1^{13}3^11^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 51$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 12 & 4 \\ 10 & 12 & 25 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$
- [69] $21^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 51$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.57 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 4 & 9 & 4 \\ 6 & 9 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$
- [70] $1^{11}3^11^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 51$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 14 & 4 \\ 12 & 14 & 37 & 11 \\ 3 & 4 & 11 & 3 \end{pmatrix}_{b_0}$
- [75] $21^131^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 52$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 9 & 20 & 5 \\ 9 & 20 & 37 & 11 \\ 3 & 5 & 11 & 2 \end{pmatrix}_{b_0}$
- [74] $1^{112}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 52$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 9 & 20 & 5 \\ 9 & 20 & 37 & 11 \\ 3 & 5 & 11 & 2 \end{pmatrix}_{b_0}$
- [76] $21^{11}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 53$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.70 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 4 & 5 & 15 & 4 \\ 5 & 4 & 17 & 4 \\ 15 & 17 & 49 & 13 \\ 4 & 4 & 13 & 3 \end{pmatrix}_{b_0}$
- [77] $1^{14}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 54$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.48 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 16 & 4 \\ 9 & 16 & 46 & 12 \\ 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$
- [78] $1^{13}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 55$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.82 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 12 & 4 \\ 9 & 12 & 25 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$
- [80] $21^{11}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 56$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.64 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 4 & 5 & 12 & 4 \\ 5 & 4 & 13 & 4 \\ 12 & 13 & 28 & 10 \\ 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$
- [79] $21^{12}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 56$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 9.71 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 4 & 8 & 4 \\ 6 & 8 & 10 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$
- [83] $1^{12}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 59$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 9.76 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 13 & 4 \\ 8 & 13 & 28 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$
- [87] $13^11^{17} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 59$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.3$
 $T_{\min} = 13.15 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 28 & 4 \\ 14 & 28 & 145 & 21 \\ 3 & 4 & 21 & -1 \end{pmatrix}_{b_0} \\
[86] \quad & 21^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 59, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.44 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 4 & 8 & 9 & 4 \\ 8 & 9 & 12 & 5 \\ 9 & 12 & 13 & 7 \\ 4 & 5 & 7 & 2 \end{pmatrix}_{b_0} \\
[84] \quad & 1^{113} 1 \ 3 \ 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 59, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.07 \rightarrow 13, T_{\max} = 22, T \in [13, 22] \\
& \begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 9 & 24 & 5 \\ 9 & 24 & 52 & 12 \\ 3 & 5 & 12 & 0 \end{pmatrix}_{b_0} \\
[85] \quad & 1^{113} 1^{13} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 59, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9.76 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 14 & 4 \\ 11 & 14 & 37 & 11 \\ 3 & 4 & 11 & 3 \end{pmatrix}_{b_0} \\
[81] \quad & 2131^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 59, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.07 \rightarrow 13, T_{\max} = 22, T \in [13, 22] \\
& \begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 9 & 24 & 5 \\ 9 & 24 & 52 & 12 \\ 3 & 5 & 12 & 0 \end{pmatrix}_{b_0} \\
[82] \quad & 1^{112} 1^{11} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 59, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9.76 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 14 & 4 \\ 10 & 14 & 37 & 11 \\ 3 & 4 & 11 & 3 \end{pmatrix}_{b_0} \\
[88] \quad & 1^{11} 1 \ 3 \ 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 60, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.33 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 17 & 4 \\ 9 & 17 & 49 & 13 \\ 3 & 4 & 13 & 2 \end{pmatrix}_{b_0} \\
[90] \quad & 1^{114} 1^{14} 31^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 62, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.81 \rightarrow 13, T_{\max} = 22, T \in [13, 22] \\
& \begin{pmatrix} 1 & 4 & 24 & 3 \\ 4 & 9 & 48 & 5 \\ 24 & 48 & 238 & 28 \\ 3 & 5 & 28 & 0 \end{pmatrix}_{b_0} \\
[89] \quad & 21^4 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 62, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.81 \rightarrow 13, T_{\max} = 22, T \in [13, 22] \\
& \begin{pmatrix} 1 & 4 & 24 & 3 \\ 4 & 9 & 48 & 5 \\ 24 & 48 & 238 & 28 \\ 3 & 5 & 28 & 0 \end{pmatrix}_{b_0} \\
[91] \quad & 1^{12} 1^{14} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 65, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 9.69 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 13 & 4 \\ 11 & 13 & 28 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0} \\
[93] \quad & 1^{12} 1 \ 3 \ 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 66, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.21 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 18 & 4 \\ 10 & 18 & 58 & 14 \\ 3 & 4 & 14 & 2 \end{pmatrix}_{b_0} \\
[92] \quad & 1^{11} 1^{11} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 66, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.21 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 19 & 4 \\ 12 & 19 & 73 & 15 \\ 3 & 4 & 15 & 2 \end{pmatrix}_{b_0} \\
[97] \quad & 21^{15} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 68, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.1 \\
& T_{\min} = 12.06 \rightarrow 13, T_{\max} = 22, T \in [13, 22] \\
& \begin{pmatrix} 4 & 5 & 30 & 4 \\ 5 & 4 & 25 & 4 \\ 30 & 25 & 148 & 20 \\ 4 & 4 & 20 & 0 \end{pmatrix}_{b_0}
\end{aligned}$$

- [96] $1^3 1^3 1^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 68$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 9 & 36 & 5 \\ 15 & 36 & 133 & 21 \\ 3 & 5 & 21 & 2 \end{pmatrix}_{b_0}$
- [95] $21^{112} 31^2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 68$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 4 & 8 & 18 & 4 \\ 8 & 9 & 24 & 5 \\ 18 & 24 & 58 & 14 \\ 4 & 5 & 14 & 2 \end{pmatrix}_{b_0}$
- [94] $1^{113} 31^2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 68$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 16 & 3 \\ 4 & 9 & 32 & 5 \\ 16 & 32 & 100 & 18 \\ 3 & 5 & 18 & 1 \end{pmatrix}_{b_0}$
- [98] $1^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 69$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 9.65 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 13 & 4 \\ 9 & 13 & 28 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$
- [99] $1^{12} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 71$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 9.63 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 13 & 4 \\ 10 & 13 & 28 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$
- [100] $1^{14} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 74$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.08 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 16 & 4 \\ 10 & 16 & 46 & 12 \\ 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$
- [101] $131^{111} 2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 76$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.32 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 4 & 8 & 6 & 4 \\ 8 & 9 & 8 & 5 \\ 6 & 8 & 4 & 4 \\ 4 & 5 & 4 & 2 \end{pmatrix}_{b_0}$
- [102] $1^{113} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 77$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.09 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 14 & 3 \\ 4 & 9 & 28 & 5 \\ 14 & 28 & 73 & 15 \\ 3 & 5 & 15 & 1 \end{pmatrix}_{b_0}$
- [103] $1^{11} 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 79$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.58 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 24 & 4 \\ 15 & 24 & 121 & 19 \\ 3 & 4 & 19 & 2 \end{pmatrix}_{b_0}$
- [104] $1^1 1^3 1^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 80$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.35 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 11 & 3 \\ 4 & 9 & 28 & 5 \\ 11 & 28 & 73 & 15 \\ 3 & 5 & 15 & 1 \end{pmatrix}_{b_0}$
- [105] $1^{11} 31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 80$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 19 & 4 \\ 13 & 19 & 73 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$
- [106] $1^1 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 81$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.54 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 21 & 4 \\ 11 & 21 & 76 & 16 \\ 3 & 4 & 16 & 2 \end{pmatrix}_{b_0}$
- [109] $21^{113} 31^3 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 83$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.28 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{aligned}
& \begin{pmatrix} 4 & 8 & 27 & 4 \\ 8 & 9 & 36 & 5 \\ 27 & 36 & 133 & 21 \\ 4 & 5 & 21 & 1 \end{pmatrix}_{b_0} \\
[108] \quad & 21^{12}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 83, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 4 & 5 & 15 & 4 \\ 5 & 4 & 14 & 4 \\ 15 & 14 & 37 & 11 \\ 4 & 4 & 11 & 3 \end{pmatrix}_{b_0} \\
[107] \quad & 1^2 131^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 83, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.08 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 1 & 4 & 13 & 3 \\ 4 & 9 & 32 & 5 \\ 13 & 32 & 100 & 18 \\ 3 & 5 & 18 & 1 \end{pmatrix}_{b_0} \\
[110] \quad & 1^{14} 1^3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 84, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 19 & 4 \\ 15 & 19 & 73 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0} \\
[113] \quad & 1^{14} 1^3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 86, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 16 & 4 \\ 11 & 16 & 46 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0} \\
[112] \quad & 1^{13} 1^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 86, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 17 & 4 \\ 10 & 17 & 49 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0} \\
[111] \quad & 1^{13} 1^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 86, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 19 & 4 \\ 14 & 19 & 73 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0} \\
[114] \quad & 1^{13} 1^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 89, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.40 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 23 & 4 \\ 13 & 23 & 100 & 18 \\ 3 & 4 & 18 & 2 \end{pmatrix}_{b_0} \\
[115] \quad & 1^{14} 1^4 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 90, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 16 & 4 \\ 12 & 16 & 46 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0} \\
[116] \quad & 21^{16}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 90, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.2 \\
& T_{\min} = 12.29 \rightarrow 13, T_{\max} = 22, T \in [13, 22] \\
& \begin{pmatrix} 4 & 5 & 36 & 4 \\ 5 & 4 & 30 & 4 \\ 36 & 30 & 214 & 24 \\ 4 & 4 & 24 & 0 \end{pmatrix}_{b_0} \\
[117] \quad & 1^{12} 1^1 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 90, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 18 & 4 \\ 11 & 18 & 58 & 14 \\ 3 & 4 & 14 & 3 \end{pmatrix}_{b_0} \\
[118] \quad & 1^{113} 1^2 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 91, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.08 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 9 & 32 & 5 \\ 15 & 32 & 100 & 18 \\ 3 & 5 & 18 & 1 \end{pmatrix}_{b_0} \\
[119] \quad & 21^2 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 91, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.08 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 9 & 32 & 5 \\ 15 & 32 & 100 & 18 \\ 3 & 5 & 18 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[120] \ 2131^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 94, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 9 & 32 & 5 \\ 12 & 32 & 94 & 16 \\ 3 & 5 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{114} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 94, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 9 & 32 & 5 \\ 12 & 32 & 94 & 16 \\ 3 & 5 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[122] \ 21^{13} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 96, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 4 & 12 & 4 \\ 9 & 12 & 25 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{113} 31^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 96, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 4 & 13 & 3 \\ 4 & 9 & 28 & 5 \\ 13 & 28 & 73 & 15 \\ 3 & 5 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{113} 31^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 97, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 9 & 28 & 5 \\ 12 & 28 & 73 & 15 \\ 3 & 5 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[125] \ 21^1 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 97, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 9 & 28 & 5 \\ 12 & 28 & 73 & 15 \\ 3 & 5 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{12} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 99, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 22 & 4 \\ 12 & 22 & 85 & 17 \\ 3 & 4 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{11} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 99, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 24 & 4 \\ 16 & 24 & 121 & 19 \\ 3 & 4 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{114} 31^{1^4} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 101, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 4 & 22 & 3 \\ 4 & 9 & 44 & 5 \\ 22 & 44 & 193 & 25 \\ 3 & 5 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[129] \ 21^{114} 31^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 104, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 4 & 8 & 36 & 4 \\ 8 & 9 & 48 & 5 \\ 36 & 48 & 238 & 28 \\ 4 & 5 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{13} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 104, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 17 & 4 \\ 11 & 17 & 49 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[130] \ 21^{12} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 104, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.15 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 4 & 28 & 4 \\ 18 & 28 & 154 & 22 \\ 3 & 4 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{12} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 106, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.75 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 18 & 4 \\ 12 & 18 & 58 & 14 \\ 3 & 4 & 14 & 3 \end{pmatrix}_{b_0} \\
[133] & 21^{12}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 107, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 9.64 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 4 & 13 & 4 \\ 9 & 13 & 28 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0} \\
[134] & 1^{14} 1^3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 110, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.64 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 28 & 4 \\ 16 & 28 & 154 & 22 \\ 3 & 4 & 22 & 2 \end{pmatrix}_{b_0} \\
[135] & 1^4 1^3 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 110, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 4 & 20 & 3 \\ 4 & 9 & 48 & 5 \\ 20 & 48 & 238 & 28 \\ 3 & 5 & 28 & 2 \end{pmatrix}_{b_0} \\
[136] & 1^{11} 1^3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 111, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.13 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 24 & 4 \\ 17 & 24 & 121 & 19 \\ 3 & 4 & 19 & 2 \end{pmatrix}_{b_0} \\
[138] & 1^{13} 1^3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 114, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 9.70 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 17 & 4 \\ 12 & 17 & 49 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0} \\
[139] & 1^{12} 1^3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 114, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 9.70 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 18 & 4 \\ 13 & 18 & 58 & 14 \\ 3 & 4 & 14 & 3 \end{pmatrix}_{b_0} \\
[137] & 1^{12} 1^4 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 114, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 9.70 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 18 & 4 \\ 14 & 18 & 58 & 14 \\ 3 & 4 & 14 & 3 \end{pmatrix}_{b_0} \\
[141] & 1^{14} 1^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 115, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.09 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 21 & 4 \\ 12 & 21 & 76 & 16 \\ 3 & 4 & 16 & 2 \end{pmatrix}_{b_0} \\
[140] & 1^{11} 1^4 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 115, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.09 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 24 & 4 \\ 18 & 24 & 121 & 19 \\ 3 & 4 & 19 & 2 \end{pmatrix}_{b_0} \\
[142] & 1^{13} 1^4 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 116, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 9.69 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 17 & 4 \\ 13 & 17 & 49 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0} \\
[143] & 21^{17} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 116, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.3 \\
& T_{\min} = 12.45 \rightarrow 13, T_{\max} = 22, T \in [13, 22] \\
& \begin{pmatrix} 4 & 5 & 42 & 4 \\ 5 & 4 & 35 & 4 \\ 42 & 35 & 292 & 28 \\ 4 & 4 & 28 & 0 \end{pmatrix}_{b_0} \\
[144] & 1^{12} 1^1 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 119, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.05 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 23 & 4 \\ 14 & 23 & 100 & 18 \\ 3 & 4 & 18 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[145] $1^{1\bar{1}4}31^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 122$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 20 & 3 \\ 4 & 9 & 40 & 5 \\ 20 & 40 & 154 & 22 \\ 3 & 5 & 22 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{1\bar{1}4}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 125$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 9 & 36 & 5 \\ 18 & 36 & 121 & 19 \\ 3 & 5 & 19 & 1 \end{pmatrix}_{b_0}$$

[146] $21^{13}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 125$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10$, $T_{\max} = 22$, $T \in [10, 22]$

$$\begin{pmatrix} 4 & 5 & 21 & 4 \\ 5 & 4 & 19 & 4 \\ 21 & 19 & 73 & 15 \\ 4 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[147] $1^1 131^{1\bar{1}4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 125$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 14 & 3 \\ 4 & 9 & 36 & 5 \\ 14 & 36 & 121 & 19 \\ 3 & 5 & 19 & 1 \end{pmatrix}_{b_0}$$

[149] $21^{12}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 126$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 9.73 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$

$$\begin{pmatrix} 4 & 5 & 18 & 4 \\ 5 & 4 & 18 & 4 \\ 18 & 18 & 58 & 14 \\ 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[150] $21^{12}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 131$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 23 & 4 \\ 15 & 23 & 100 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[151] $1^{13}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 131$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 22 & 4 \\ 13 & 22 & 85 & 17 \\ 3 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

[152] $21^{12}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 132$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 9.70 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 18 & 4 \\ 12 & 18 & 58 & 14 \\ 3 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[153] $1^3 131^{1\bar{1}4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 133$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.05 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 9 & 44 & 5 \\ 18 & 44 & 193 & 25 \\ 3 & 5 & 25 & 1 \end{pmatrix}_{b_0}$$

[154] $1^3 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 134$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.34 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 26 & 4 \\ 14 & 26 & 118 & 20 \\ 3 & 4 & 20 & 2 \end{pmatrix}_{b_0}$$

[155] $1^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 136$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.32 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 27 & 4 \\ 15 & 27 & 133 & 21 \\ 3 & 4 & 21 & 2 \end{pmatrix}_{b_0}$$

[156] $21^3 31^{1\bar{1}4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 136$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.21 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 9 & 44 & 5 \\ 21 & 44 & 193 & 25 \\ 3 & 5 & 25 & 1 \end{pmatrix}_{b_0}$$

[157] $1^{1\bar{1}4}31^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 136$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.21 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

- [158] $131^{12}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 136, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 9 & 44 & 5 \\ 21 & 44 & 193 & 25 \\ 3 & 5 & 25 & 1 \end{pmatrix}_{b_0}$$
- [159] $131^{12}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 138, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 16 & 3 \\ 4 & 9 & 40 & 5 \\ 16 & 40 & 154 & 22 \\ 3 & 5 & 22 & 1 \end{pmatrix}_{b_0}$$
- [162] $2131^{15} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 139, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.1$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 9 & 40 & 5 \\ 15 & 40 & 148 & 20 \\ 3 & 5 & 20 & 0 \end{pmatrix}_{b_0}$$
- [161] $21^{12}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 139, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 4 & 8 & 15 & 4 \\ 8 & 9 & 20 & 5 \\ 15 & 20 & 37 & 11 \\ 4 & 5 & 11 & 2 \end{pmatrix}_{b_0}$$
- [160] $1^{15}31 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 139, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.1$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 9 & 40 & 5 \\ 15 & 40 & 148 & 20 \\ 3 & 5 & 20 & 0 \end{pmatrix}_{b_0}$$
- [163] $1^{12}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 141, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 23 & 4 \\ 15 & 23 & 100 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$
- [164] $1^{14}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 141, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 21 & 4 \\ 13 & 21 & 76 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$
- [165] $1^{12}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 146, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 28 & 4 \\ 17 & 28 & 154 & 22 \\ 3 & 4 & 22 & 2 \end{pmatrix}_{b_0}$$
- [167] $1^{14}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 152, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 9 & 36 & 5 \\ 17 & 36 & 121 & 19 \\ 3 & 5 & 19 & 2 \end{pmatrix}_{b_0}$$
- [169] $1^{14}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 152, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 9 & 36 & 5 \\ 15 & 36 & 121 & 19 \\ 3 & 5 & 19 & 1 \end{pmatrix}_{b_0}$$
- [168] $21^131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 152, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 9 & 36 & 5 \\ 15 & 36 & 121 & 19 \\ 3 & 5 & 19 & 1 \end{pmatrix}_{b_0}$$
- [166] $21^{14}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 152, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 16 & 4 \\ 12 & 16 & 46 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$
- [170] $1^{14}31^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 153, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11.05 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 4 & 19 & 3 \\ 4 & 9 & 40 & 5 \\ 19 & 40 & 154 & 22 \\ 3 & 5 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{13}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 155, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 22 & 4 \\ 14 & 22 & 85 & 17 \\ 3 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[171] \ 1^{13}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 155, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 23 & 4 \\ 16 & 23 & 100 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[173] \ 21^{12}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 155, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.70 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 4 & 5 & 21 & 4 \\ 5 & 4 & 22 & 4 \\ 21 & 22 & 85 & 17 \\ 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[174] \ 1^{14}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 159, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 21 & 4 \\ 14 & 21 & 76 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[175] \ 1^{12}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 161, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 23 & 4 \\ 17 & 23 & 100 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[176] \ 1^{14}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 161, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.04 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 4 & 16 & 3 \\ 4 & 9 & 36 & 5 \\ 16 & 36 & 121 & 19 \\ 3 & 5 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[177] \ 1^{14}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 166, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 9 & 40 & 5 \\ 18 & 40 & 154 & 22 \\ 3 & 5 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[178] \ 21^{23}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 166, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 9 & 40 & 5 \\ 18 & 40 & 154 & 22 \\ 3 & 5 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[179] \ 1^{14}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 169, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.74 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 21 & 4 \\ 15 & 21 & 76 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[180] \ 21^{12}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 170, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.71 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 4 & 5 & 24 & 4 \\ 5 & 4 & 26 & 4 \\ 24 & 26 & 118 & 20 \\ 4 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[182] \ 221^{11}3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 171, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.73 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 3 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[183] \ 1^{13}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 171, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.73 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 22 & 4 \\ 15 & 22 & 85 & 17 \\ 3 & 4 & 17 & 3 \end{pmatrix}_{b_0} \\
[181] & 1^{\overline{14}} 1^3 1^{\overline{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 171, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.43 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 32 & 4 \\ 18 & 32 & 193 & 25 \\ 3 & 4 & 25 & 2 \end{pmatrix}_{b_0} \\
[184] & 21^{\overline{14}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 171, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.15 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 4 & 5 & 27 & 4 \\ 5 & 4 & 24 & 4 \\ 27 & 24 & 121 & 19 \\ 4 & 4 & 19 & 2 \end{pmatrix}_{b_0} \\
[187] & 1^{\overline{12}} 1^{\overline{2}} 31^{\overline{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 174, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.03 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 28 & 4 \\ 18 & 28 & 154 & 22 \\ 3 & 4 & 22 & 2 \end{pmatrix}_{b_0} \\
[186] & 1^{\overline{14}} 1^{\overline{1}} 31^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 174, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.03 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 26 & 4 \\ 15 & 26 & 118 & 20 \\ 3 & 4 & 20 & 2 \end{pmatrix}_{b_0} \\
[185] & 1^{\overline{13}} 1^{\overline{1}} 31^{\overline{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 174, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.03 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 27 & 4 \\ 16 & 27 & 133 & 21 \\ 3 & 4 & 21 & 2 \end{pmatrix}_{b_0} \\
[189] & 21^{\overline{13}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 179, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 9.70 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 17 & 4 \\ 12 & 17 & 49 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0} \\
[188] & 1^{\overline{13}} 1^{\overline{4}} 31^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 179, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 9.70 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 22 & 4 \\ 16 & 22 & 85 & 17 \\ 3 & 4 & 17 & 3 \end{pmatrix}_{b_0} \\
[190] & 1^{\overline{13}} 1^3 1^{\overline{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 185, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.32 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 31 & 4 \\ 17 & 31 & 172 & 24 \\ 3 & 4 & 24 & 2 \end{pmatrix}_{b_0} \\
[191] & 1^{\overline{12}} 1^{\overline{3}} 31^{\overline{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 194, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 28 & 4 \\ 19 & 28 & 154 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0} \\
[192] & 21^{\overline{13}} 31^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 200, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 9.82 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 4 & 5 & 24 & 4 \\ 5 & 4 & 23 & 4 \\ 24 & 23 & 100 & 18 \\ 4 & 4 & 18 & 3 \end{pmatrix}_{b_0} \\
[193] & 1^{\overline{12}} 1^{\overline{2}} 31^{\overline{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 204, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 27 & 4 \\ 17 & 27 & 133 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0} \\
[195] & 1^{\overline{14}} 1^{\overline{2}} 31^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 206, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.87 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 26 & 4 \\ 16 & 26 & 118 & 20 \\ 3 & 4 & 20 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [194] $1^{12}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 206$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 9.87 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 28 & 4 \\ 20 & 28 & 154 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$
- [196] $21^{113}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 208$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.38 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 4 & 8 & 24 & 4 \\ 8 & 9 & 32 & 5 \\ 24 & 32 & 100 & 18 \\ 4 & 5 & 18 & 2 \end{pmatrix}_{b_0}$
- [197] $1^{13}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 215$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.14 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 32 & 4 \\ 19 & 32 & 193 & 25 \\ 3 & 4 & 25 & 2 \end{pmatrix}_{b_0}$
- [198] $21^{15}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 224$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.1$
 $T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 20 & 4 \\ 15 & 20 & 73 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$
- [199] $1^{13}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 226$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 27 & 4 \\ 18 & 27 & 133 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$
- [200] $1^{14}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 230$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.78 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 26 & 4 \\ 17 & 26 & 118 & 20 \\ 3 & 4 & 20 & 3 \end{pmatrix}_{b_0}$
- [201] $1^{14}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 231$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.06 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 31 & 4 \\ 18 & 31 & 172 & 24 \\ 3 & 4 & 24 & 2 \end{pmatrix}_{b_0}$
- [202] $1^3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 234$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.37 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 36 & 4 \\ 20 & 36 & 238 & 28 \\ 3 & 4 & 28 & 2 \end{pmatrix}_{b_0}$
- [203] $131^{113}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 236$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.37 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 4 & 8 & 18 & 4 \\ 8 & 9 & 24 & 5 \\ 18 & 24 & 52 & 12 \\ 4 & 5 & 12 & 2 \end{pmatrix}_{b_0}$
- [204] $21^{13}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 236$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.69 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 22 & 4 \\ 15 & 22 & 85 & 17 \\ 3 & 4 & 17 & 3 \end{pmatrix}_{b_0}$
- [205] $1^{13}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 240$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.75 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 27 & 4 \\ 19 & 27 & 133 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$
- [206] $1^{14}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 246$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.73 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 26 & 4 \\ 18 & 26 & 118 & 20 \\ 3 & 4 & 20 & 3 \end{pmatrix}_{b_0}$
- [207] $1^{13}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 6$, $T_{\max} = 22$, $\Delta = -114$, $\det = 251$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 22$, $T \in [10, 22]$

- [208] $21^{113}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 259, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 32 & 4 \\ 20 & 32 & 193 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0}$
- [209] $21^{13}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 261, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 4 & 8 & 21 & 4 \\ 8 & 9 & 28 & 5 \\ 21 & 28 & 73 & 15 \\ 4 & 5 & 15 & 2 \end{pmatrix}_{b_0}$
- [210] $21^{14}31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 267, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 21 & 4 \\ 15 & 21 & 76 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$
- [211] $21^{13}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 267, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 4 & 27 & 4 \\ 18 & 27 & 133 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$
- [212] $1^{12}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 269, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 31 & 4 \\ 19 & 31 & 172 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$
- [213] $21^{13}31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 272, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 4 & 32 & 4 \\ 21 & 32 & 193 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0}$
- [214] $21^{14}31^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 278, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 4 & 5 & 30 & 4 \\ 5 & 4 & 28 & 4 \\ 30 & 28 & 154 & 22 \\ 4 & 4 & 22 & 3 \end{pmatrix}_{b_0}$
- [215] $1^{13}31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 279, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 32 & 4 \\ 21 & 32 & 193 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0}$
- [216] $21^{114}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 283, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 4 & 8 & 33 & 4 \\ 8 & 9 & 44 & 5 \\ 33 & 44 & 193 & 25 \\ 4 & 5 & 25 & 2 \end{pmatrix}_{b_0}$
- [217] $1^{14}31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 286, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 36 & 4 \\ 21 & 36 & 238 & 28 \\ 3 & 4 & 28 & 2 \end{pmatrix}_{b_0}$
- [219] $1^{14}31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 299, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 32 & 4 \\ 22 & 32 & 193 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0}$
- [218] $1^{14}31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 299, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 31 & 4 \\ 20 & 31 & 172 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$
- [220] $221^{112}3 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 306, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 22, T \in [10, 22]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 3 & 3 & 6 & 2 \\ 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 0 & 6 & 2 \\ 6 & 6 & 6 & 10 & 6 \\ 2 & 3 & 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[221] \quad & 21^{13}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 308, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 9.73 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 4 & 5 & 30 & 4 \\ 5 & 4 & 31 & 4 \\ 30 & 31 & 172 & 24 \\ 4 & 4 & 24 & 3 \end{pmatrix}_{b_0} \\
[222] \quad & 21^{16}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 312, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.2 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 4 & 24 & 4 \\ 18 & 24 & 106 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0} \\
[223] \quad & 1^{14}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 321, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.76 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 31 & 4 \\ 21 & 31 & 172 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0} \\
[224] \quad & 1^{14}31^{12}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 330, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 36 & 4 \\ 22 & 36 & 238 & 28 \\ 3 & 4 & 28 & 3 \end{pmatrix}_{b_0} \\
[225] \quad & 21^{14}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 356, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.71 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 4 & 26 & 4 \\ 18 & 26 & 118 & 20 \\ 3 & 4 & 20 & 3 \end{pmatrix}_{b_0} \\
[226] \quad & 1^{14}31^{13}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 366, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.87 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 23 & 3 \\ 3 & 4 & 36 & 4 \\ 23 & 36 & 238 & 28 \\ 3 & 4 & 28 & 3 \end{pmatrix}_{b_0} \\
[227] \quad & 21^{14}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 371, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.81 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 4 & 5 & 33 & 4 \\ 5 & 4 & 32 & 4 \\ 33 & 32 & 193 & 25 \\ 4 & 4 & 25 & 3 \end{pmatrix}_{b_0} \\
[228] \quad & 131^{14}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 376, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.38 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 4 & 8 & 24 & 4 \\ 8 & 9 & 32 & 5 \\ 24 & 32 & 94 & 16 \\ 4 & 5 & 16 & 2 \end{pmatrix}_{b_0} \\
[229] \quad & 21^{114}31^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 388, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.31 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 4 & 8 & 30 & 4 \\ 8 & 9 & 40 & 5 \\ 30 & 40 & 154 & 22 \\ 4 & 5 & 22 & 2 \end{pmatrix}_{b_0} \\
[230] \quad & 1^{14}31^{14}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 394, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.81 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 3 & 24 & 3 \\ 3 & 4 & 36 & 4 \\ 24 & 36 & 238 & 28 \\ 3 & 4 & 28 & 3 \end{pmatrix}_{b_0} \\
[231] \quad & 21^{17}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 416, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.3 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 4 & 28 & 4 \\ 21 & 28 & 145 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0} \\
[232] \quad & 21^{114}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 419, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.29 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 4 & 8 & 27 & 4 \\ 8 & 9 & 36 & 5 \\ 27 & 36 & 121 & 19 \\ 4 & 5 & 19 & 2 \end{pmatrix}_{b_0} \\
[233] \quad & 21^{14}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 419, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.72 \rightarrow 10, T_{\max} = 22, T \in [10, 22]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 4 & 31 & 4 \\ 21 & 31 & 172 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0} \\
[234] \quad & 21^{14}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 450, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.76 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 4 & 5 & 36 & 4 \\ 5 & 4 & 36 & 4 \\ 36 & 36 & 238 & 28 \\ 4 & 4 & 28 & 3 \end{pmatrix}_{b_0} \\
[235] \quad & 21^{14}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 456, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.75 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 1 & 4 & 24 & 3 \\ 4 & 4 & 36 & 4 \\ 24 & 36 & 238 & 28 \\ 3 & 4 & 28 & 3 \end{pmatrix}_{b_0} \\
[236] \quad & 221^{113}3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 531, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 9.73 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 0 & 3 & 3 & 9 & 2 \\ 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 0 & 9 & 2 \\ 9 & 9 & 9 & 25 & 9 \\ 2 & 3 & 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[237] \quad & 131^{115}2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 556, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.1 \\
& T_{\min} = 10.39 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 4 & 8 & 30 & 4 \\ 8 & 9 & 40 & 5 \\ 30 & 40 & 148 & 20 \\ 4 & 5 & 20 & 2 \end{pmatrix}_{b_0} \\
[238] \quad & 221^{114}3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 846, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.73 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 0 & 3 & 3 & 12 & 2 \\ 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 0 & 12 & 2 \\ 12 & 12 & 12 & 46 & 12 \\ 2 & 3 & 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[239] \quad & 221^{115}3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 6, T_{\max} = 22, \Delta = -114, \det = 1251, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.1 \\
& T_{\min} = 9.73 \rightarrow 10, T_{\max} = 22, T \in [10, 22] \\
& \begin{pmatrix} 0 & 3 & 3 & 15 & 2 \\ 3 & 1 & 4 & 15 & 3 \\ 3 & 4 & 0 & 15 & 2 \\ 15 & 15 & 15 & 73 & 15 \\ 2 & 3 & 2 & 15 & 2 \end{pmatrix}_{b_0} \\
\\
T = 7 \quad [240] \quad & 1^1 3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -3, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9, T_{\max} = 10, T \in [9, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -2 & 0 \\ 3 & 5 & 0 & 5 \end{pmatrix}_{b_0} \\
[248] \quad & 1^1 3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9, T_{\max} = 10, T \in [9, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[241] \quad & 1^1 3 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 12, T \in [10, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & 0 & 3 \end{pmatrix}_{b_0} \\
[242] \quad & 1^1 3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 \\ 3 & 5 & 4 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0} \\
[243] \quad & 1^1 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 19, T \in [10, 19] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 \\ 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0} \\
[244] \quad & 1^{11} 3 21^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 6 & 0 & 56 & 14 \\ 2 & 2 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

[245] $21^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 19, T \in [10, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 \\ 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[246] $1^1 321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 3 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[247] $1^1 321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 1 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[249] $1^1 31^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 \\ 2 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

[250] $1^1 1^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 13 & 4 \\ 11 & 13 & 40 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[251] $21^1 341^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 13 & 4 \\ 11 & 13 & 40 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[252] $1^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 11 & 4 \\ 7 & 11 & 28 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[253] $1^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 19, T \in [14, 19]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & 1 & 3 \\ 3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

[254] $21^1 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 19, T \in [14, 19]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & 1 & 3 \\ 3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

[255] $2^1 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 4 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

[256] $1^1 3^1 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[257] \ 1^1 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -6, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, \ T_{\max} = 19, \ T \in [10, 19]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 \\ 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[260] \ 1^1 3 2 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -7, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.57 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[258] \ 2^1 2 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -7, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.57 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[264] \ 1^1 1^1 3 1 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -7, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.57 \rightarrow 10, \ T_{\max} = 12, \ T \in [10, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[263] \ 1^1 1^1 3 1^1 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -7, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.57 \rightarrow 10, \ T_{\max} = 12, \ T \in [10, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 4 & 4 \\ 2 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[262] \ 1^1 1^2 3 2 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -7, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.57 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 5 & 5 & 5 \\ 4 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[259] \ 1^1 1^1 3 2 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -7, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.57 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[261] \ 1^1 3 2 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -7, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.57 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[269] \ 2 1^1 1^4 4 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -8, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.50 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 24 & 4 \\ 5 & 4 & 25 & 4 \\ 24 & 25 & 132 & 22 \\ 4 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[268] \ 1^1 3 1^1 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -8, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 4 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[267] \ 2 1^1 1^2 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -8, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.50 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 12 & 4 \\ 5 & 4 & 11 & 4 \\ 12 & 11 & 28 & 10 \\ 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[266] \ 2 1^1 1^1 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -8, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.50 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 6 & 4 \\ 5 & 4 & 7 & 4 \\ 6 & 7 & 8 & 6 \\ 4 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[265] \ 1^{\overline{1}1} 3 2 1^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -8, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.50 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[276] \ 1^{\overline{1}1} 3 1^{\overline{1}1} 3 1^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -9, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.56 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 2 & 4 & 3 & 6 & 3 \\ 3 & 3 & 6 & 4 & 4 \\ 2 & 3 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[275] \ 1^{\overline{1}2} 3 1^{\overline{1}1} 3 1^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -9, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.56 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 1 & 4 & 6 & 3 \\ 2 & 4 & 3 & 6 & 3 \\ 3 & 6 & 6 & 10 & 6 \\ 2 & 3 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[274] \ 2 1^{\overline{1}1} 4 1^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -9, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.56 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 11 & 4 \\ 9 & 11 & 28 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[273] \ 1^{\overline{1}1} 4 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -9, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.56 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 5 & 4 \\ 3 & 5 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[272] \ 1^{\overline{1}2} 4 1^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -9, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.56 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 11 & 4 \\ 9 & 11 & 28 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[271] \ 1^{\overline{1}3} 4 1^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -9, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.56 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 7 & 4 \\ 7 & 7 & 8 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[270] \ 2 1^{\overline{1}1} 4 1^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -9, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.56 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 5 & 4 \\ 3 & 5 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[280] \ 1^{\overline{1}4} 3 2 1^{\overline{1}4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -10, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 12 & 0 & 110 & 20 \\ 2 & 2 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[279] \ 1^{\overline{1}1} 3 2 1^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -10, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 5 & 5 \\ 2 & 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[278] \ 1^{\overline{1}2} 3 2 1^{\overline{1}2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -10, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 6 & 0 & 26 & 10 \\ 2 & 2 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[277] \ 1^{13}321^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -10, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 61 & 15 \\ 2 & 2 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[281] \ 131^{13}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -11, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 4 \\ 1 & 4 & 0 & 3 & 2 \\ 1 & 3 & 3 & 1 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[282] \ 131^{13}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -11, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 2 & 4 & 3 & 3 & 3 \\ 1 & 3 & 3 & 1 & 3 \\ 2 & 3 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[283] \ 1^{12}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -11, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 10 & 4 \\ 6 & 10 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[284] \ 221^{14}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -11, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 3 & 2 & 3 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[285] \ 1^4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -11, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 8 & 4 \\ 5 & 8 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[286] \ 2141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -11, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[287] \ 1^4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -11, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[288] \ 1^{13}1^{13}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -11, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & 4 & 4 \\ 2 & 3 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[289] \ 1^{13}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -11, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 10 & 4 \\ 9 & 10 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[290] \ 1^{12}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -11, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 8 & 4 \\ 7 & 8 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[291] \ 1^{11}31^{13}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -11, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 4 \\ 1 & 4 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

- [292] $1 \overset{1}{4} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [293] $21 \overset{1}{1} 41 \overset{1}{2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 4 & 8 & 6 & 4 \\ 8 & 13 & 12 & 7 \\ 6 & 12 & 8 & 6 \\ 4 & 7 & 6 & 3 \end{pmatrix}_{b_0}$
- [294] $21 \overset{1}{1} 41 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 7 & 3 \\ 4 & 4 & 10 & 4 \\ 7 & 10 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$
- [295] $1 \overset{1}{3} 321 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.33 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 \\ 3 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$
- [299] $2 \overset{1}{3} 1 \overset{1}{3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 3 & 4 & 2 \\ 2 & 1 & 3 & 6 & 3 \\ 3 & 3 & 0 & 4 & 2 \\ 4 & 6 & 4 & 10 & 6 \\ 2 & 3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$
- [298] $1 \overset{1}{1} 31 \overset{1}{3} 1 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 2 & 4 & 3 & 3 & 3 \\ 2 & 3 & 3 & 1 & 3 \\ 2 & 3 & 3 & 3 & 2 \end{pmatrix}_{b_0}$
- [301] $141 \overset{1}{1} \overset{1}{2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 20 & 4 \\ 12 & 20 & 90 & 18 \\ 3 & 4 & 18 & 2 \end{pmatrix}_{b_0}$
- [297] $2141 \overset{1}{1} \overset{1}{2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 8 & 4 \\ 4 & 8 & 10 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$
- [300] $1 \overset{1}{1} 321 \overset{1}{3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 5 & 0 & 33 & 11 \\ 2 & 2 & -1 & 11 & 3 \end{pmatrix}_{b_0}$
- [296] $1 \overset{1}{4} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 8 & 4 \\ 4 & 8 & 10 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$
- [302] $221 \overset{1}{1} 41 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 6 & 3 & 3 \\ 6 & 13 & 8 & 7 \\ 3 & 8 & 4 & 4 \\ 3 & 7 & 4 & 3 \end{pmatrix}_{b_0}$
- [303] $31 \overset{1}{1} 231 \overset{1}{2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 5 & 10 & 3 \\ 5 & 5 & 15 & 5 \\ 10 & 15 & 38 & 12 \\ 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$

$$[304] \ 1 \overset{1}{\underset{1}{4}} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -15, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 8 & 4 \\ 6 & 8 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[305] \ 2 \ 2 \ 3 \overset{1}{\underset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -15, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.80 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[306] \ 1 \overset{1}{\underset{1}{4}} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -15, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 7 & 4 \\ 4 & 7 & 8 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[307] \ 2 \overset{1}{\underset{1}{1}} 4 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -15, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 8 & 4 \\ 6 & 8 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[308] \ 1 \overset{1}{\underset{1}{4}} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -16, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.25 \rightarrow 12, \ T_{\max} = 19, \ T \in [12, 19]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 21 & 4 \\ 13 & 21 & 101 & 19 \\ 3 & 4 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[309] \ 2 \overset{1}{\underset{1}{1}} 3 \overset{1}{\underset{1}{2}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -16, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[310] \ 1 \overset{1}{\underset{1}{3}} 1 \overset{2}{\underset{1}{3}} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -16, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.25 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 6 & 0 & 22 & 8 \\ 2 & 3 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[311] \ 1 \overset{1}{\underset{1}{1}} 1^{11} 3 \overset{2}{\underset{1}{2}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -16, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 4 & 0 & 16 & 8 \\ 2 & 2 & -1 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[312] \ 2 \overset{1}{\underset{1}{1}} 1^{13} 4 \overset{1}{\underset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -18, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.44 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 18 & 4 \\ 5 & 4 & 18 & 4 \\ 18 & 18 & 70 & 16 \\ 4 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[313] \ 1 \overset{1}{\underset{1}{3}} 2 \overset{1}{\underset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -19, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.84 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 10 & 5 \\ 5 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[316] \ 1 \overset{1}{\underset{1}{1}} 1^{12} 4 \overset{1}{\underset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -19, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.26 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 7 & 4 \\ 6 & 7 & 8 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[317] \ 1 \overset{1}{\underset{1}{1}} 1^{12} 4 \overset{1}{\underset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -19, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.26 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 10 & 4 \\ 8 & 10 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[318] $1^{\overline{21}} 4^{\overline{11}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 19, T \in [13, 19]$
 $\begin{pmatrix} 4 & 8 & 3 & 4 \\ 8 & 9 & 4 & 5 \\ 3 & 4 & 1 & 3 \\ 4 & 5 & 3 & 1 \end{pmatrix}_{b_0}$

[319] $21^{\overline{1}} 41^{\overline{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 19, T \in [13, 19]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 12 & 4 \\ 6 & 12 & 25 & 9 \\ 3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$

[320] $1^{\overline{1}} 321^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 10 & 5 \\ 5 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$

[321] $1^{\overline{1}} 321^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[322] $1^{\overline{1}} 321^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 5 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[323] $21^{\overline{2}} 41^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 10 & 4 \\ 8 & 10 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$

[324] $1^{\overline{1}} 321^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 10 & 5 \\ 7 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$

[315] $1^{\overline{1}} 4^{\overline{13}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 19, T \in [13, 19]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 12 & 4 \\ 6 & 12 & 25 & 9 \\ 3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$

[314] $1^{\overline{11}} 321^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 10 & 5 \\ 7 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$

[325] $1^{\overline{11}} 3131^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$

[326] $2^{\overline{1}} 3^{\overline{11}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 2 & 3 \\ 2 & 4 & 0 & 2 & 2 \\ 2 & 2 & 2 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$

[329] $1^{\overline{12}} 31^{\overline{11}} 31^{\overline{3}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$

$$\begin{aligned}
& T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 12 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 7 & 12 & 9 & 28 & 10 \\ 2 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[330] & 21^{11}321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 12 & 4 \\ 5 & 5 & 15 & 5 \\ 12 & 15 & 32 & 12 \\ 4 & 5 & 12 & 4 \end{pmatrix}_{b_0} \\
[331] & 1^{11}31^{13}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9.80 \rightarrow 10, T_{\max} = 12, T \in [10, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 6 & 0 & 13 & 7 \\ 2 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0} \\
[328] & 21^{11}1^14^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 4 & 5 & 6 & 4 \\ 5 & 4 & 5 & 4 \\ 6 & 5 & 4 & 4 \\ 4 & 4 & 4 & 3 \end{pmatrix}_{b_0} \\
[327] & 131^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 2 & 4 & 3 & 6 & 3 \\ 2 & 3 & 6 & 4 & 4 \\ 2 & 3 & 3 & 4 & 2 \end{pmatrix}_{b_0} \\
[333] & 21^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.24 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 7 & 4 \\ 5 & 7 & 8 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0} \\
[332] & 1^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.24 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 7 & 4 \\ 5 & 7 & 8 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0} \\
[334] & 21^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.57 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 4 & 5 & 9 & 4 \\ 5 & 4 & 8 & 4 \\ 9 & 8 & 13 & 7 \\ 4 & 4 & 7 & 3 \end{pmatrix}_{b_0} \\
[335] & 1^{11}3131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 2 \end{pmatrix}_{b_0} \\
[336] & 21^14^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 9 & 4 \\ 5 & 9 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0} \\
[337] & 1^131^{13}31^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 18 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 9 & 18 & 9 & 52 & 12 \\ 2 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[338] & 1^131^{12}31^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 12 & 4 \\ 1 & 4 & 0 & 6 & 2 \\ 6 & 12 & 6 & 22 & 8 \\ 2 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[339] & 1^131^{14}31^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 12 & 24 & 12 & 94 & 16 \\ 2 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[340] $1^1 4^3 31^{11} 31^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 18 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 9 & 18 & 15 & 76 & 16 \\ 2 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[341] $1^1 3^2 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 5 & 10 & 5 \\ 6 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$$

[342] $1^1 3^1 321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 5 & 10 & 5 \\ 6 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$$

[344] $1^1 4^1 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 9 & 4 \\ 5 & 9 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[345] $1^1 3^1 1^1 31^{11} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 4 \\ 1 & 4 & 0 & 3 & 2 \\ 3 & 6 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[346] $31^{11} 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 3 & 2 & 0 & 3 & 2 \\ 3 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[343] $2^1 3^2 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 4 & 3 \\ 2 & 3 & 0 & 6 & 2 \\ 4 & 4 & 6 & 10 & 6 \\ 2 & 3 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[347] $1^1 4^1 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.08 \rightarrow 14, T_{\max} = 19, T \in [14, 19]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 16 & 4 \\ 8 & 16 & 46 & 12 \\ 3 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

[348] $21^1 41^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.08 \rightarrow 14, T_{\max} = 19, T \in [14, 19]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 16 & 4 \\ 8 & 16 & 46 & 12 \\ 3 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

[349] $1^1 2^3 3^2 21^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 7 & 0 & 45 & 13 \\ 2 & 2 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

[350] $1^3 4^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 17 & 4 \\ 10 & 17 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

- [351] $1^1 1^1 4^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 11 & 4 \\ 6 & 11 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$
- [352] $1^1 1^1 3^1 2^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.59 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$
- [353] $1^1 1^1 3^1 2^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.59 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$
- [354] $2^1 2^1 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$
- [355] $2^1 1^1 1^1 4^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 4 & 5 & 27 & 4 \\ 5 & 4 & 29 & 4 \\ 27 & 29 & 169 & 25 \\ 4 & 4 & 25 & 3 \end{pmatrix}_{b_0}$
- [356] $1^1 1^1 3^1 1^1 3^1 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 1 & 4 & 6 & 3 \\ 2 & 4 & 3 & 9 & 3 \\ 5 & 6 & 9 & 13 & 7 \\ 2 & 3 & 3 & 7 & 2 \end{pmatrix}_{b_0}$
- [357] $1^1 1^1 4^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 14 & 4 \\ 8 & 14 & 38 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$
- [359] $1^1 1^1 4^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 18 & 4 \\ 11 & 18 & 70 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$
- [358] $2^1 2^1 1^1 4^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 3 & 8 & 2 \\ 3 & 3 & 3 & 12 & 3 \\ 6 & 8 & 12 & 34 & 10 \\ 2 & 2 & 3 & 10 & 2 \end{pmatrix}_{b_0}$
- [360] $2^1 1^1 3^1 2^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$
- [361] $2^1 1^1 2^1 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [362] $1^1 1^1 3^1 2^1 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 9 & 0 & 37 & 11 \\ 2 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0} \\
[363] & 1^{12}31^{13}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 9 & 0 & 28 & 10 \\ 2 & 3 & -1 & 10 & 2 \end{pmatrix}_{b_0} \\
[364] & 1^{14}31^{11}31^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 18 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 8 & 18 & 15 & 76 & 16 \\ 2 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0} \\
& \begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[365] & 1^{21}4^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[366] & 1^{12}3131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 6 & 0 & 10 & 6 \\ 2 & 3 & -1 & 6 & 2 \end{pmatrix}_{b_0} \\
[367] & 131^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 4 \\ 1 & 4 & 0 & 3 & 2 \\ 2 & 6 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[370] & 1^{11}4^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.59 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 12 & 4 \\ 7 & 12 & 26 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0} \\
[369] & 1^{12}4^{14}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.59 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 11 & 4 \\ 10 & 11 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0} \\
[368] & 1^{11}4^{14}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.59 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 12 & 4 \\ 11 & 12 & 26 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0} \\
[371] & 1^{12}3^{12}21^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 8 & 0 & 70 & 16 \\ 2 & 2 & -1 & 16 & 3 \end{pmatrix}_{b_0} \\
[372] & 1^{13}31^{11}31^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 15 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 8 & 15 & 12 & 49 & 13 \\ 2 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[373] & 21^{11}4^{12}4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 4 & 10 & 3 \\ 4 & 4 & 15 & 4 \\ 10 & 15 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0} \\
[374] & 21^{12}4^{13}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 17 & 4 \\ 12 & 17 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0} \\
[375] & 21 \overset{1}{4} 1^{\overset{1}{15}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.1 \\
& T_{\min} = 13.57 \rightarrow 14, T_{\max} = 19, T \in [14, 19] \\
& \begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 20 & 4 \\ 10 & 20 & 73 & 15 \\ 3 & 4 & 15 & 0 \end{pmatrix}_{b_0} \\
[376] & 1 \overset{1}{4} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.1 \\
& T_{\min} = 13.57 \rightarrow 14, T_{\max} = 19, T \in [14, 19] \\
& \begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 20 & 4 \\ 10 & 20 & 73 & 15 \\ 3 & 4 & 15 & 0 \end{pmatrix}_{b_0} \\
[377] & 21 \overset{1}{4} 1^{\overset{4}{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 4 & 25 & 4 \\ 17 & 25 & 132 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0} \\
[378] & 1 \overset{1}{4} 1^{\overset{1}{15}} 31^{\overset{1}{15}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 9 & 4 \\ 1 & 4 & 0 & 6 & 2 \\ 5 & 9 & 6 & 13 & 7 \\ 2 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[379] & 21 \overset{1}{4} 1^{\overset{1}{15}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 4 & 7 & 4 \\ 5 & 7 & 8 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0} \\
[385] & 2 \overset{1}{2} 2 31^{\overset{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 5 & 7 & 6 & 25 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[383] & 31 \overset{1}{4} 1^{\overset{1}{15}} 31^{\overset{1}{15}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.56 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 3 & 3 & 6 & 2 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 6 & 12 & 6 & 22 & 8 \\ 2 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[384] & 31 \overset{1}{4} 1^{\overset{1}{15}} 31^{\overset{1}{15}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.56 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 3 & 3 & 1 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[382] & 1 \overset{1}{4} 1^{\overset{1}{15}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.56 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 15 & 4 \\ 9 & 15 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0} \\
[386] & 1 \overset{2}{4} 1^{\overset{1}{15}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 4 & 5 & 18 & 4 \\ 5 & 4 & 21 & 4 \\ 18 & 21 & 76 & 16 \\ 4 & 4 & 16 & 3 \end{pmatrix}_{b_0} \\
[387] & 131 \overset{1}{4} 1^{\overset{1}{15}} 31^{\overset{1}{15}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 1 & 4 & 15 & 3 \\ 2 & 4 & 3 & 18 & 3 \\ 6 & 15 & 18 & 76 & 16 \\ 2 & 3 & 3 & 16 & 2 \end{pmatrix}_{b_0} \\
[388] & 1 \overset{1}{4} 1^{\overset{1}{15}} 3 21^{\overset{1}{15}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8
\end{aligned}$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 10 & 0 & 88 & 18 \\ 2 & 2 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[380] \quad 1^{11} 2^3 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 2 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[381] \quad 21^{11} 41^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 30 & 4 \\ 5 & 4 & 33 & 4 \\ 30 & 33 & 212 & 28 \\ 4 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[389] \quad 1^{11} 2^3 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 6 & 3 \\ 1 & 4 & 0 & 6 & 2 \\ 3 & 6 & 6 & 10 & 6 \\ 2 & 3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[397] \quad 1^{11} 321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 4 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[393] \quad 1^{12} 321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 5 & 15 & 5 \\ 8 & 15 & 34 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[394] \quad 1^{14} 321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 7 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[395] \quad 221^{11} 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.54 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 3 & 6 & 3 \\ 3 & 2 & 6 & 4 & 4 \\ 2 & 2 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[396] \quad 1^{14} 321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 15 & 5 \\ 10 & 15 & 34 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[392] \quad 1^{11} 141^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 15 & 4 \\ 8 & 15 & 40 & 12 \\ 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[390] \quad 2^3 1^{11} 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 2 & 2 \\ 3 & 1 & 3 & 2 & 3 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 2 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[391] \quad 2^3 1^{14} 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 11 & 2 \\ 3 & 1 & 3 & 11 & 3 \\ 2 & 3 & 0 & 14 & 2 \\ 11 & 11 & 14 & 65 & 15 \\ 2 & 3 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[398] $2 \overset{1^{11}}{2} 31^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[399] $1^{\overset{11}{1}} 31^{\overset{3}{1}} 31^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 12 & 0 & 73 & 15 \\ 2 & 3 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[400] $1^{\overset{11}{1}} 31^{\overset{1}{1}} 31^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 6 & 0 & 13 & 7 \\ 2 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[401] $1^{\overset{12}{1}} 41^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 11 & 4 \\ 7 & 11 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[402] $21^{\overset{14}{4}} 41^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 21 & 4 \\ 17 & 21 & 101 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

[403] $1^{\overset{13}{1}} 31^{\overset{1}{1}} 31^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 12 & 0 & 49 & 13 \\ 2 & 3 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[404] $1^{\overset{14}{1}} 41^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 21 & 4 \\ 17 & 21 & 101 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

[405] $21^{\overset{1}{1}} 41^{\overset{12}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 11 & 4 \\ 7 & 11 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[406] $1^{\overset{12}{1}} 31^{\overset{11}{1}} 31^{\overset{3}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.51 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 1 & 4 & 9 & 3 \\ 2 & 4 & 3 & 12 & 3 \\ 7 & 9 & 12 & 28 & 10 \\ 2 & 3 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[407] $21^{\overset{11}{1}} 41^{\overset{12}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 13 & 4 \\ 7 & 13 & 28 & 10 \\ 3 & 4 & 10 & 2 \end{pmatrix}_{b_0}$$

[408] $1^{\overset{11}{1}} 41^{\overset{12}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 13 & 4 \\ 7 & 13 & 28 & 10 \\ 3 & 4 & 10 & 2 \end{pmatrix}_{b_0}$$

[409] $1^{\overset{13}{1}} 3131^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 9 & 0 & 25 & 9 \\ 2 & 3 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[411] $1^{12}3^{11}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[410] $1^{11}4^{14}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 15 & 4 \\ 13 & 15 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[412] $1^{13}3^{13}21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 15 & 5 \\ 9 & 15 & 34 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

[413] $2^3 3^{11} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 3 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[414] $1^{12}3^{13}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 6 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[415] $31^{11}231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 3 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[416] $1^{12}4^{13}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 11 & 4 \\ 9 & 11 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[417] $131^{11}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 12 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 4 & 12 & 9 & 28 & 10 \\ 2 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[418] $2^3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[420] $21^{11}4^{12}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.47 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 15 & 4 \\ 5 & 4 & 15 & 4 \\ 15 & 15 & 45 & 13 \\ 4 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[419] $131^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 1 & 4 & 6 & 3 \\ 2 & 4 & 3 & 9 & 3 \\ 3 & 6 & 9 & 13 & 7 \\ 2 & 3 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

- [422] $1 \overset{21}{4} \overset{11}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 4 & 5 & 9 & 4 \\ 5 & 4 & 9 & 4 \\ 9 & 9 & 13 & 7 \\ 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$
- [421] $1 \overset{13}{3} 1 \overset{11}{3} 1 \overset{4}{4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 1 & 4 & 12 & 3 \\ 2 & 4 & 3 & 15 & 3 \\ 9 & 12 & 15 & 49 & 13 \\ 2 & 3 & 3 & 13 & 2 \end{pmatrix}_{b_0}$
- [427] $21 \overset{11}{4} 1 \overset{11}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 12 & 4 \\ 8 & 12 & 26 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$
- [428] $1 \overset{13}{1} \overset{11}{4} 1 \overset{11}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 12 & 4 \\ 10 & 12 & 26 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$
- [429] $1 \overset{1}{1} \overset{16}{4} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.2$
 $T_{\min} = 13.91 \rightarrow 14, T_{\max} = 19, T \in [14, 19]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 24 & 4 \\ 12 & 24 & 106 & 18 \\ 3 & 4 & 18 & 0 \end{pmatrix}_{b_0}$
- [430] $2 \overset{11}{2} 2 31 \overset{11}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$
- [431] $1 \overset{14}{1} \overset{12}{2} 4 1 \overset{11}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 20 & 4 \\ 16 & 20 & 90 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$
- [423] $21 \overset{1}{4} 1 \overset{16}{4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.2$
 $T_{\min} = 13.91 \rightarrow 14, T_{\max} = 19, T \in [14, 19]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 24 & 4 \\ 12 & 24 & 106 & 18 \\ 3 & 4 & 18 & 0 \end{pmatrix}_{b_0}$
- [424] $1 \overset{14}{1} \overset{11}{1} \overset{13}{4} 1 \overset{11}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 18 & 4 \\ 15 & 18 & 70 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$
- [425] $21 \overset{1}{4} 4 1 \overset{12}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 20 & 4 \\ 16 & 20 & 90 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$
- [426] $2 \overset{12}{2} 2 31 \overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$
- [432] $1 \overset{11}{1} \overset{1}{1} \overset{11}{4} 1 \overset{11}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 12 & 4 \\ 8 & 12 & 26 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[433] \ 1^{12}4^{11}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 11 & 4 \\ 8 & 11 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[434] \ 1^{12}4^{11}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 14 & 4 \\ 12 & 14 & 38 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[435] \ 2^2 3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[443] \ 221^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 4 & 7 & 2 & 4 \\ 3 & 7 & 3 & 2 & 3 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 4 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[442] \ 1^{11}3^{13}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 6 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[441] \ 21^{11}41^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 21 & 4 \\ 5 & 4 & 22 & 4 \\ 21 & 22 & 97 & 19 \\ 4 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[440] \ 1^{11}3^{12}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 5 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[439] \ 1^{11}3^{12}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 5 & 15 & 5 \\ 8 & 15 & 32 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[438] \ 1^{11}3^{12}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 15 & 5 \\ 10 & 15 & 32 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[437] \ 1^{14}31^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 15 & 0 & 76 & 16 \\ 2 & 3 & -1 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[436] \ 1^{11}31^{14}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 121 & 19 \\ 2 & 3 & -1 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[444] \ 1^{12}3^{12}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[448] \ 1^{12}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 17 & 4 \\ 14 & 17 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[445] \ 1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 1 & 4 & 6 & 3 \\ 2 & 4 & 3 & 9 & 3 \\ 4 & 6 & 9 & 13 & 7 \\ 2 & 3 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[446] \ 21^{12}321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 24 & 4 \\ 5 & 5 & 30 & 5 \\ 24 & 30 & 134 & 24 \\ 4 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[447] \ 1^{12}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 12 & 4 \\ 9 & 12 & 26 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[453] \ 1^4 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 4 & 9 & 4 \\ 6 & 9 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[452] \ 141^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 18 & 4 \\ 10 & 18 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[451] \ 141^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 24 & 4 \\ 14 & 24 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[449] \ 1^{13}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 15 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 6 & 15 & 12 & 49 & 13 \\ 2 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[450] \ 131^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 4 \\ 1 & 4 & 0 & 6 & 2 \\ 3 & 9 & 6 & 13 & 7 \\ 2 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[455] \ 131^{11}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 9 & 3 \\ 1 & 4 & 0 & 9 & 2 \\ 4 & 9 & 9 & 28 & 10 \\ 2 & 3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[454] \ 1^{14}3131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 12 & 0 & 46 & 12 \\ 2 & 3 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[456] \ 1^{11}31^{11}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 6 & 0 & 13 & 7 \\ 2 & 3 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[459] \ 1^{1^2} 1^1 3^1 2^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.69 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 15 & 5 \\ 9 & 15 & 32 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[458] \ 1^{1^1} 3^1 1^1 3^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 6 & 0 & 13 & 7 \\ 2 & 3 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[457] \ 1^{1^2} 3^1 1^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 6 & 0 & 10 & 6 \\ 2 & 3 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[461] \ 13^1 1^{1^1} 3^1 1^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -53, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 1 & 4 & 12 & 3 \\ 2 & 4 & 3 & 15 & 3 \\ 5 & 12 & 15 & 49 & 13 \\ 2 & 3 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[460] \ 1^{21} 4^1 1^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -53, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 15 & 4 \\ 5 & 4 & 17 & 4 \\ 15 & 17 & 49 & 13 \\ 4 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[464] \ 21^3 4^1 1^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 18 & 4 \\ 14 & 18 & 70 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[463] \ 1^{1^2} 4^1 1^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 14 & 4 \\ 10 & 14 & 38 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[465] \ 1^{1^2} 4^1 1^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 14 & 4 \\ 11 & 14 & 38 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[466] \ 1^{1^1} 4^1 1^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 15 & 4 \\ 12 & 15 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[467] \ 21^2 4^1 1^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 14 & 4 \\ 10 & 14 & 38 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[468] \ 1^1 4^1 1^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.48 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 19 & 4 \\ 10 & 19 & 65 & 15 \\ 3 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$

[469] $1^{11} \overset{13}{4} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 18 & 4 \\ 14 & 18 & 70 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$

[462] $21^{11} \overset{1}{4} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 4 & 5 & 12 & 4 \\ 5 & 4 & 12 & 4 \\ 12 & 12 & 26 & 10 \\ 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$

[470] $1^{13} \overset{1}{4} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 21 & 4 \\ 12 & 21 & 88 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[471] $31^{11} 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 5 & 4 & 3 \\ 5 & 5 & 5 & 5 \\ 4 & 5 & 2 & 4 \\ 3 & 5 & 4 & 3 \end{pmatrix}_{b_0}$

[472] $1^{12} \overset{1}{4} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 15 & 4 \\ 11 & 15 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$

[473] $21^{11} \overset{1}{4} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 4 & 8 & 12 & 4 \\ 8 & 13 & 20 & 7 \\ 12 & 20 & 26 & 10 \\ 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$

[474] $1^4 1^{12} \overset{1}{4} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 4 & 5 & 12 & 4 \\ 5 & 4 & 13 & 4 \\ 12 & 13 & 28 & 10 \\ 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$

[475] $21^{11} \overset{1}{4} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 8 & 3 \\ 4 & 4 & 12 & 4 \\ 8 & 12 & 26 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$

[476] $1^4 1^{12} \overset{1}{4} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 4 & 8 & 4 \\ 6 & 8 & 10 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$

[477] $1^{11} 31^1 31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 6 & 0 & 13 & 7 \\ 2 & 3 & -1 & 7 & 2 \end{pmatrix}_{b_0}$

[478] $21^3 \overset{1}{4} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 17 & 4 \\ 13 & 17 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$

[479] $1^1 31^{12} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 12 & 4 \\ 1 & 4 & 0 & 6 & 2 \\ 5 & 12 & 6 & 22 & 8 \\ 2 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$

[480] $1^{12} 31^2 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$

$$\begin{aligned}
& T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 12 & 0 & 58 & 14 \\ 2 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0} \\
[481] \quad & 131^{12}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.71 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 12 & 4 \\ 1 & 4 & 0 & 6 & 2 \\ 4 & 12 & 6 & 22 & 8 \\ 2 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0} \\
[482] \quad & 2^1 3^2 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 9.86 \rightarrow 10, T_{\max} = 11, T \in [10, 11] \\
& \begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 4 & 2 \\ 4 & 6 & 4 & 10 & 6 \\ 2 & 3 & 2 & 6 & 3 \end{pmatrix}_{b_0} \\
[483] \quad & 21^2 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.36 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 15 & 4 \\ 11 & 15 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0} \\
[484] \quad & 131^{11}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 1 & 4 & 9 & 3 \\ 2 & 4 & 3 & 12 & 3 \\ 4 & 9 & 12 & 28 & 10 \\ 2 & 3 & 3 & 10 & 2 \end{pmatrix}_{b_0} \\
[485] \quad & 1^{12}4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.36 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 17 & 4 \\ 13 & 17 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0} \\
[492] \quad & 1^{11}31^{12}31^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 15 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 8 & 15 & 9 & 37 & 11 \\ 2 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[491] \quad & 1^{13}31^{11}31^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 15 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 7 & 15 & 12 & 49 & 13 \\ 2 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[490] \quad & 1^1 4^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.76 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 16 & 4 \\ 9 & 16 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0} \\
[488] \quad & 1^1 4^1 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.3 \\
& T_{\min} = 14.15 \rightarrow 15, T_{\max} = 19, T \in [15, 19] \\
& \begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 28 & 4 \\ 14 & 28 & 145 & 21 \\ 3 & 4 & 21 & -1 \end{pmatrix}_{b_0} \\
[486] \quad & 31^{11}231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 3 & 2 & 0 & 4 & 2 \\ 4 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[489] \quad & 1^{11}31^2 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 37 & 11 \\ 2 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[487] $21\overset{1}{4}1\overset{17}{+}[\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 14.15 \rightarrow 15, T_{\max} = 19, T \in [15, 19]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 28 & 4 \\ 14 & 28 & 145 & 21 \\ 3 & 4 & 21 & -1 \end{pmatrix}_{b_0}$$

[493] $1\overset{11}{3}1\overset{11}{3}1\overset{1}{+}[\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 9 & 4 \\ 1 & 4 & 0 & 6 & 2 \\ 4 & 9 & 6 & 13 & 7 \\ 2 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[494] $1\overset{11}{4}1\overset{13}{+}[\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 17 & 4 \\ 9 & 17 & 49 & 13 \\ 3 & 4 & 13 & 2 \end{pmatrix}_{b_0}$$

[495] $1\overset{12}{3}1\overset{11}{3}1\overset{2}{+}[\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 12 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 6 & 12 & 9 & 28 & 10 \\ 2 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[496] $21\overset{11}{4}1\overset{13}{+}[\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 17 & 4 \\ 9 & 17 & 49 & 13 \\ 3 & 4 & 13 & 2 \end{pmatrix}_{b_0}$$

[498] $1\overset{13}{4}1\overset{1}{+}[\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 15 & 4 \\ 9 & 15 & 40 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[497] $21\overset{11}{4}1\overset{13}{+}[\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 15 & 4 \\ 9 & 15 & 40 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[499] $21\overset{11}{4}1\overset{13}{+}[\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -62, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 24 & 4 \\ 5 & 4 & 26 & 4 \\ 24 & 26 & 130 & 22 \\ 4 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

[502] $21\overset{11}{4}1\overset{14}{+}[\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 21 & 4 \\ 5 & 4 & 24 & 4 \\ 21 & 24 & 101 & 19 \\ 4 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

[501] $21\overset{11}{4}1\overset{14}{+}[\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 27 & 4 \\ 5 & 4 & 30 & 4 \\ 27 & 30 & 169 & 25 \\ 4 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

[500] $31\overset{11}{3}1\overset{1}{3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 6 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 6 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[503] $1\overset{12}{3}1\overset{13}{3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 6 & 0 & 10 & 6 \\ 2 & 3 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[504] \ 1^4 4^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -65, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.69 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 25 & 4 \\ 15 & 25 & 132 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[507] \ 1^{13} 3^{12} 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -67, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 7 & 6 & 25 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[505] \ 1^{12} 3^{11} 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -67, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 20 & 5 \\ 11 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[506] \ 1^{12} 3^{13} 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -67, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 20 & 5 \\ 13 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[513] \ 1^{13} 3^{12} 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -67, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 7 & 6 & 25 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[512] \ 1^{14} 3^{13} 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -67, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 5 & 10 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[511] \ 1^{11} 3^{13} 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -67, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 7 & 5 & 10 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[510] \ 221^{11} 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -67, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.31 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 6 & 5 & 3 \\ 6 & 13 & 12 & 7 \\ 5 & 12 & 8 & 6 \\ 3 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[509] \ 1^{12} 3^{13} 1^{31} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -67, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 12, \ T \in [10, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 9 & 0 & 28 & 10 \\ 2 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[508] \ 1^{13} 3^{14} 21^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -67, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 20 & 5 \\ 11 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[517] \ 1^{12} 3^{11} 1^{31} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -68, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.41 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 12 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 5 & 12 & 9 & 28 & 10 \\ 2 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[516] \ 1^{12} 3^{13} 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -68, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 6 & 0 & 10 & 6 \\ 2 & 3 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[515] $1^{11}321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 20 & 5 \\ 10 & 20 & 58 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

[519] $21^{12}41^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 10 & 3 \\ 4 & 4 & 14 & 4 \\ 10 & 14 & 38 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[514] $21^{11}41^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.47 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 18 & 4 \\ 5 & 4 & 19 & 4 \\ 18 & 19 & 68 & 16 \\ 4 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

[518] $1^{12}31^{12}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 18 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 10 & 18 & 12 & 58 & 14 \\ 2 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[520] $1^2 1^{11}41^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -69, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 19 & 4 \\ 11 & 19 & 68 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

[524] $21^{11}41^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 15 & 4 \\ 5 & 4 & 16 & 4 \\ 15 & 16 & 45 & 13 \\ 4 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[521] $2^2 2^3 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 5 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[522] $221^{11}41^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 3 & 5 & 2 \\ 1 & 0 & 3 & 6 & 2 \\ 3 & 3 & 3 & 10 & 3 \\ 5 & 6 & 10 & 20 & 8 \\ 2 & 2 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[523] $1^3 1^{11}41^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.63 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 22 & 4 \\ 13 & 22 & 97 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

[525] $21^{12}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[530] $1^4 1^3 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 20 & 5 \\ 12 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

[529] $21^{11}41^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 24 & 4 \\ 5 & 4 & 27 & 4 \\ 24 & 27 & 132 & 22 \\ 4 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[528] \ 1^{12} 3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 5 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[527] \ 1^{12} 3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 20 & 5 \\ 12 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[526] \ 1^{12} 31^{11} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 1 & 4 & 9 & 3 \\ 2 & 4 & 3 & 12 & 3 \\ 6 & 9 & 12 & 28 & 10 \\ 2 & 3 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[533] \ 141^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.08 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 22 & 4 \\ 12 & 22 & 90 & 18 \\ 3 & 4 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[532] \ 221^{11} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 3 & 4 & 2 \\ 1 & 0 & 3 & 4 & 2 \\ 3 & 3 & 3 & 8 & 3 \\ 4 & 4 & 8 & 10 & 6 \\ 2 & 2 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[531] \ 21^{11} 41^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.54 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 18 & 4 \\ 5 & 4 & 20 & 4 \\ 18 & 20 & 70 & 16 \\ 4 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[534] \ 21^{12} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.49 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 8 & 3 \\ 4 & 4 & 11 & 4 \\ 8 & 11 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[536] \ 31^{11} 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 5 & 7 & 3 \\ 5 & 5 & 10 & 5 \\ 7 & 10 & 14 & 8 \\ 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[535] \ 1^{13} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 15 & 4 \\ 10 & 15 & 40 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[537] \ 1^{11} 31^3 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 12 & 0 & 73 & 15 \\ 2 & 3 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[538] \ 2231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 5 & 15 & 5 \\ 9 & 15 & 34 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[539] \ 21^{12} 321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[540] $2 \overset{1^{11}}{3} 1^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 5 & 2 \\ 2 & 1 & 4 & 5 & 3 \\ 2 & 4 & 0 & 5 & 2 \\ 5 & 5 & 5 & 8 & 6 \\ 2 & 3 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[541] $2 \overset{1^{11}}{3} 13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[542] $2 \overset{1^1}{3} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 0 & 6 & 2 \\ 6 & 9 & 6 & 25 & 9 \\ 2 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[544] $1^{12} 31^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -77, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 1 & 4 & 9 & 3 \\ 2 & 4 & 3 & 12 & 3 \\ 5 & 9 & 12 & 28 & 10 \\ 2 & 3 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[543] $21^{11} 41^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -77, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 21 & 4 \\ 5 & 4 & 23 & 4 \\ 21 & 23 & 97 & 19 \\ 4 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

[550] $1^{13} 31^2 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 15 & 0 & 85 & 17 \\ 2 & 3 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[549] $1^{12} 31^3 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 15 & 0 & 100 & 18 \\ 2 & 3 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[551] $1^{12} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 16 & 4 \\ 10 & 16 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[552] $2 31^{11} 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 3 & 2 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[548] $1^{13} 3131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 9 & 0 & 25 & 9 \\ 2 & 3 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[547] $1^{13} 41^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 15 & 4 \\ 12 & 15 & 40 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[546] $21^{1^{11}}41^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -79$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.57 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 16 & 4 \\ 10 & 16 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$

[545] $1^{11}31^{231^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -79$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.27 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 37 & 11 \\ 2 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0}$

[554] $1^{131^{1331^2}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -80$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.40 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 18 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 8 & 18 & 9 & 52 & 12 \\ 2 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$

[553] $21^{1^4}41^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -80$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.40 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 4 & 24 & 4 \\ 17 & 24 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$

[555] $1^{11}41^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 19$, $T \in [12, 19]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 21 & 4 \\ 11 & 21 & 76 & 16 \\ 3 & 4 & 16 & 2 \end{pmatrix}_{b_0}$

[556] $21^{11}41^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 19$, $T \in [12, 19]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 21 & 4 \\ 11 & 21 & 76 & 16 \\ 3 & 4 & 16 & 2 \end{pmatrix}_{b_0}$

[557] $1^{13}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -81$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.56 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 15 & 4 \\ 11 & 15 & 40 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$

[560] $1^{11}31^{1331^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -83$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.35 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 21 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 11 & 21 & 12 & 73 & 15 \\ 2 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$

[558] $1^{11}321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -83$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.77 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 20 & 5 \\ 13 & 20 & 58 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$

[559] $1^{11}321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -83$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.77 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 20 & 5 \\ 11 & 20 & 58 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$

[561] $1^{12}321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -84$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 9.76 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$

[564] $21^{11}41^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -84$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 19 & 4 \\ 11 & 19 & 65 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$

$$[563] \ 1^{14} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -84, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.95 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 19 & 4 \\ 11 & 19 & 65 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[562] \ 1^{12} 3 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -84, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.76 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 8 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[567] \ 1^3 4 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -86, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 25 & 4 \\ 14 & 25 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[566] \ 1^{13} 3 1^{11} 3 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -86, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.47 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 1 & 4 & 12 & 3 \\ 2 & 4 & 3 & 15 & 3 \\ 8 & 12 & 15 & 49 & 13 \\ 2 & 3 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[565] \ 1^1 4 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -86, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 20 & 4 \\ 11 & 20 & 70 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[569] \ 1^{11} 3 2 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -88, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.73 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 20 & 5 \\ 12 & 20 & 58 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[568] \ 1^{12} 3 2 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -88, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.73 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 7 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[570] \ 1^{14} 3 1^{11} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -89, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.51 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 1 & 4 & 15 & 3 \\ 2 & 4 & 3 & 18 & 3 \\ 7 & 15 & 18 & 76 & 16 \\ 2 & 3 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[572] \ 1^{13} 3 1^{11} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -90, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.49 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 1 & 4 & 12 & 3 \\ 2 & 4 & 3 & 15 & 3 \\ 6 & 12 & 15 & 49 & 13 \\ 2 & 3 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[571] \ 2 2 1^{11} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -90, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.36 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 6 & 8 & 3 \\ 6 & 13 & 20 & 7 \\ 8 & 20 & 26 & 10 \\ 3 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[573] \ 1^4 4 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -90, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.89 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 28 & 4 \\ 16 & 28 & 158 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[579] \ 1^{11} 3 1^2 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -91, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.58 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 9 & 0 & 37 & 11 \\ 2 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[578] \ 1^{13}4^{12}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -91, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.49 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 18 & 4 \\ 12 & 18 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[577] \ 1^{11}31^{43}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -91, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.58 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 121 & 19 \\ 2 & 3 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[576] \ 1^{12}4^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -91, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.49 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 16 & 4 \\ 11 & 16 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[575] \ 1^{12}4^{14}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -91, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.49 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 16 & 4 \\ 13 & 16 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[574] \ 21^24^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -91, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.49 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 18 & 4 \\ 12 & 18 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[582] \ 21^{12}41^{11}2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -92, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.39 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 4 & 8 & 18 & 4 \\ 8 & 13 & 32 & 7 \\ 18 & 32 & 68 & 16 \\ 4 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[581] \ 1^{14}31^{11}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -92, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.52 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 1 & 4 & 15 & 3 \\ 2 & 4 & 3 & 18 & 3 \\ 10 & 15 & 18 & 76 & 16 \\ 2 & 3 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[584] \ 1^{13}31^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -92, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.09 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 12 & 0 & 49 & 13 \\ 2 & 3 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[583] \ 21^{12}4^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -92, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.39 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 22 & 4 \\ 15 & 22 & 97 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[580] \ 21^{14}41^{11}2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -92, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.39 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 4 & 8 & 30 & 4 \\ 8 & 13 & 56 & 7 \\ 30 & 56 & 212 & 28 \\ 4 & 7 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[587] \ 1^{12}31^{13}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -95, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.05 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 28 & 10 \\ 2 & 3 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[586] \ 1^{12}4^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 19, \ \Delta = -105, \ \det = -95, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.47 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 16 & 4 \\ 12 & 16 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[585] \ 1^{11}31^{23}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -95, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 4.5$$

$$\begin{aligned}
& T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 9 & 0 & 37 & 11 \\ 2 & 3 & -1 & 11 & 1 \end{pmatrix}_{b_0} \\
[591] & 131^{13}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 18 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 6 & 18 & 9 & 52 & 12 \\ 2 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[590] & 2^1 3^1 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 11, T \in [10, 11] \\
& \begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 0 & 5 & 2 \\ 4 & 5 & 5 & 8 & 6 \\ 2 & 3 & 2 & 6 & 3 \end{pmatrix}_{b_0} \\
[589] & 21^{13} 1^4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 4 & 12 & 4 \\ 9 & 12 & 25 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0} \\
[588] & 21^{13} 321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 10 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[592] & 2^1 2^1 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -98, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 5 & 15 & 5 \\ 9 & 15 & 32 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0} \\
[594] & 1^{13} 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 18 & 4 \\ 14 & 18 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0} \\
[593] & 1^{13} 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 18 & 4 \\ 13 & 18 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0} \\
[600] & 31^{12} 231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 5 & 8 & 3 \\ 5 & 5 & 10 & 5 \\ 8 & 10 & 14 & 8 \\ 3 & 5 & 8 & 3 \end{pmatrix}_{b_0} \\
[599] & 31^{12} 231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 5 & 17 & 3 \\ 5 & 5 & 25 & 5 \\ 17 & 25 & 104 & 20 \\ 3 & 5 & 20 & 3 \end{pmatrix}_{b_0} \\
[598] & 1^{12} 31^4 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 18 & 0 & 154 & 22 \\ 2 & 3 & -1 & 22 & 2 \end{pmatrix}_{b_0} \\
[597] & 2^1 2^1 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 7 & 5 & 10 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[596] & 1^{12} 31^2 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5
\end{aligned}$$

$$T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 12 & 0 & 58 & 14 \\ 2 & 3 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[595] \ 1^{14}31^{23}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 18 & 0 & 118 & 20 \\ 2 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[602] \ 31^{11}231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 3 & 4 & 2 \\ 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 0 & 4 & 2 \\ 4 & 4 & 4 & 5 & 5 \\ 2 & 3 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[601] \ 1^{13}31^{11}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 1 & 4 & 12 & 3 \\ 2 & 4 & 3 & 15 & 3 \\ 7 & 12 & 15 & 49 & 13 \\ 2 & 3 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[603] \ 1^{13}321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 6 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[605] \ 131^{12}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 8 & 24 & 18 & 118 & 20 \\ 2 & 4 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[608] \ 1^131^{14}31^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 11 & 24 & 12 & 94 & 16 \\ 2 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[611] \ 1^24^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 23 & 4 \\ 13 & 23 & 97 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[612] \ 1^{14}3131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 12 & 0 & 46 & 12 \\ 2 & 3 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[613] \ 1^{12}321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 25 & 5 \\ 13 & 25 & 94 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[606] \ 31^{12}231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 3 & 2 & 0 & 10 & 2 \\ 10 & 10 & 10 & 58 & 14 \\ 2 & 2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[610] \ 1^{11}31^{33}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 12 & 0 & 73 & 15 \\ 2 & 3 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[609] \ 1^131^{13}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 18 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 7 & 18 & 9 & 52 & 12 \\ 2 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[607] & 21^{13}41^{11}2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 4 & 8 & 24 & 4 \\ 8 & 13 & 44 & 7 \\ 24 & 44 & 130 & 22 \\ 4 & 7 & 22 & 3 \end{pmatrix}_{b_0} \\
[604] & 2^{11}3^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 11, T \in [10, 11] \\
& \begin{pmatrix} 0 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 0 & 8 & 2 \\ 8 & 12 & 8 & 46 & 12 \\ 2 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[615] & 1^{12}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 19 & 4 \\ 13 & 19 & 68 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0} \\
[614] & 21^{11}4^21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 19 & 4 \\ 13 & 19 & 68 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0} \\
[616] & 1^{14}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.75 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 19 & 4 \\ 12 & 19 & 65 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0} \\
[620] & 1^{13}3131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 25 & 9 \\ 2 & 3 & -1 & 9 & 2 \end{pmatrix}_{b_0} \\
[619] & 21^{12}1^41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.64 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 4 & 13 & 4 \\ 9 & 13 & 28 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0} \\
[618] & 1^{11}31^{12}31^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 15 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 7 & 15 & 9 & 37 & 11 \\ 2 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[617] & 131^{12}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 15 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 5 & 15 & 9 & 37 & 11 \\ 2 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[623] & 1^{13}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 21 & 4 \\ 15 & 21 & 88 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0} \\
[622] & 21^{13}41^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 21 & 4 \\ 15 & 21 & 88 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0} \\
[621] & 1^{12}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5
\end{aligned}$$

$$T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 19 & 4 \\ 15 & 19 & 68 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[624] \ 1^{14} 3^{13} 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -110, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 12 & 3 \\ 1 & 4 & 0 & 12 & 2 \\ 6 & 12 & 12 & 46 & 12 \\ 2 & 3 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[625] \ 1^{13} 4^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 19 & 4 \\ 14 & 19 & 68 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[626] \ 1^{13} 4^{11} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 21 & 4 \\ 16 & 21 & 88 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[627] \ 1^{11} 3^{12} 2^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 25 & 5 \\ 13 & 25 & 92 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[628] \ 2^2 2^{12} 3^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[629] \ 1^{12} 3^{12} 2^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[630] \ 1^{13} 4^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 20 & 4 \\ 12 & 20 & 70 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[631] \ 1^3 4^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 26 & 4 \\ 15 & 26 & 130 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[632] \ 21^1 4^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 20 & 4 \\ 12 & 20 & 70 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[637] \ 1^{12} 3^{11} 3^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 28 & 10 \\ 2 & 3 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[633] \ 1^{13} 4^{11} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 24 & 4 \\ 18 & 24 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

- [634] $1^{14}31^{13}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 15 & 0 & 76 & 16 \\ 2 & 3 & -1 & 16 & 2 \end{pmatrix}_{b_0}$
- [635] $1^{11}4^{14}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 24 & 4 \\ 13 & 24 & 101 & 19 \\ 3 & 4 & 19 & 2 \end{pmatrix}_{b_0}$
- [636] $21^{14}4^{13}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 24 & 4 \\ 18 & 24 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$
- [640] $1^{13}31^{13}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 18 & 0 & 133 & 21 \\ 2 & 3 & -1 & 21 & 2 \end{pmatrix}_{b_0}$
- [639] $1^{14}4^{13}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 29 & 4 \\ 17 & 29 & 169 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0}$
- [638] $1^{14}31^{13}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 1 & 4 & 15 & 3 \\ 2 & 4 & 3 & 18 & 3 \\ 8 & 15 & 18 & 76 & 16 \\ 2 & 3 & 3 & 16 & 2 \end{pmatrix}_{b_0}$
- [641] $1^{14}31^{13}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 1 & 4 & 15 & 3 \\ 2 & 4 & 3 & 18 & 3 \\ 9 & 15 & 18 & 76 & 16 \\ 2 & 3 & 3 & 16 & 2 \end{pmatrix}_{b_0}$
- [643] $2^{14}2^{13}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 5 & 20 & 5 \\ 12 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$
- [642] $2^{12}2^{13}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 5 & 20 & 5 \\ 12 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$
- [650] $21^{13}3^{11}4^{12}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -119, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 22 & 4 \\ 16 & 22 & 97 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$
- [644] $1^{12}3^{13}4^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -119, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 22 & 4 \\ 16 & 22 & 97 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$
- [645] $1^{13}3^{12}21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -119, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

- [646] $1^{12}3^{14}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -119$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 9.84 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 5 & 25 & 5 \\ 16 & 25 & 94 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$
- [647] $1^{12}4^{14}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -119$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.38 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 22 & 4 \\ 17 & 22 & 97 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$
- [648] $1^{12}3^{12}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -119$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 9.84 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 25 & 5 \\ 14 & 25 & 94 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$
- [649] $31^{12}231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -119$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.51 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 3 & 2 & 0 & 7 & 2 \\ 7 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$
- [651] $1^{14}4^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -120$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.67 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 19 & 4 \\ 13 & 19 & 65 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$
- [652] $1^{12}4^{14}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -121$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.37 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 25 & 4 \\ 19 & 25 & 132 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$
- [653] $21^{14}4^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -121$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.37 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 25 & 4 \\ 19 & 25 & 132 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$
- [654] $21^{13}4^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -123$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 21 & 4 \\ 15 & 21 & 88 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$
- [655] $1^{11}31^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -123$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 15 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 6 & 15 & 9 & 37 & 11 \\ 2 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$
- [656] $21^{12}4^{12}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -123$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 13 & 3 \\ 4 & 4 & 19 & 4 \\ 13 & 19 & 68 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$
- [661] $1^{12}3^{13}21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -124$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 9.81 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 5 & 25 & 5 \\ 15 & 25 & 94 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$
- [659] $2^3 3^1 1^{13} 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -124$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.13 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 3 & 2 & 9 & 2 \\ 3 & 1 & 3 & 9 & 3 \\ 2 & 3 & 0 & 11 & 2 \\ 9 & 9 & 11 & 40 & 12 \\ 2 & 3 & 2 & 12 & 2 \end{pmatrix}_{b_0}$

$$[658] \quad 1^{1^1} 231^{12} 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 6 & 2 \\ 3 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 4 & 2 \\ 6 & 6 & 4 & 10 & 6 \\ 2 & 3 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[657] \quad 1^{1^3} 1^{12} 321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 8 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[660] \quad 1^{1^1} 1^{11} 31^{13} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.58 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 12 & 0 & 73 & 15 \\ 2 & 3 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[662] \quad 1^{1^4} 1^{14} 41^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.63 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 19 & 4 \\ 14 & 19 & 65 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[668] \quad 1^{1^3} 1^{13} 321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[663] \quad 1^{1^2} 31^{13} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 9 & 0 & 28 & 10 \\ 2 & 3 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[664] \quad 1^{1^3} 3131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 25 & 9 \\ 2 & 3 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[665] \quad 1^{1^3} 1^{13} 321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 9 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[666] \quad 1^{1^1} 1^{11} 321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 25 & 5 \\ 14 & 25 & 92 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[667] \quad 1^{1^1} 1^{11} 31^{14} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 15 & 0 & 121 & 19 \\ 2 & 3 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[669] \quad 21^{1^2} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 11 & 3 \\ 4 & 4 & 16 & 4 \\ 11 & 16 & 45 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[670] \quad 1^{1^2} 1^{12} 321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

- [671] $221 \begin{smallmatrix} 1^2 & 4^1 & 1^1 \\ 1^2 & 1^2 & 1^1 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 6 & 10 & 3 \\ 6 & 13 & 24 & 7 \\ 10 & 24 & 38 & 12 \\ 3 & 7 & 12 & 3 \end{pmatrix}_{b_0}$
- [672] $1 \begin{smallmatrix} 1^2 & 4^1 & 1^1 \\ 1^2 & 1^2 & 1^1 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 22 & 4 \\ 14 & 22 & 90 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$
- [673] $21 \begin{smallmatrix} 1^1 & 4^1 & 1^4 \\ 1^1 & 1^2 & 1^4 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 22 & 4 \\ 14 & 22 & 90 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$
- [674] $131 \begin{smallmatrix} 1^2 & 3^1 & 1^3 \\ 1^2 & 1^2 & 1^3 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 21 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 7 & 21 & 15 & 85 & 17 \\ 2 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [675] $21 \begin{smallmatrix} 1^3 & 4^1 & 1^1 \\ 1^3 & 1^2 & 1^1 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 11 & 3 \\ 4 & 4 & 15 & 4 \\ 11 & 15 & 40 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$
- [676] $1 \begin{smallmatrix} 1^3 & 3^1 & 1^2 & 3^1 & 1^4 \\ 1^3 & 1^2 & 1^2 & 1^2 & 1^4 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 21 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 11 & 21 & 15 & 85 & 17 \\ 2 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$
- [677] $1 \begin{smallmatrix} 1^2 & 3^1 & 3^1 & 1^1 & 1^1 \\ 1^2 & 1^2 & 1^2 & 1^1 & 1^1 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 100 & 18 \\ 2 & 3 & -1 & 18 & 2 \end{pmatrix}_{b_0}$
- [678] $1 \begin{smallmatrix} 1^2 & 3^1 & 1^2 & 3^1 & 1^3 \\ 1^2 & 1^2 & 1^2 & 1^2 & 1^3 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 18 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 9 & 18 & 12 & 58 & 14 \\ 2 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [679] $131 \begin{smallmatrix} 1^2 & 3^1 & 1^2 & 1^2 \\ 1^2 & 1^2 & 1^2 & 1^2 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 18 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 6 & 18 & 12 & 58 & 14 \\ 2 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [680] $1 \begin{smallmatrix} 1^4 & 3^1 & 2^1 & 1^4 \\ 1^4 & 1^2 & 1^2 & 1^4 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 5 & 25 & 5 \\ 15 & 25 & 92 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$
- [681] $1 \begin{smallmatrix} 1^2 & 3^1 & 4^1 & 1^1 & 1^1 \\ 1^2 & 1^2 & 1^2 & 1^1 & 1^1 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -134, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 20 & 4 \\ 13 & 20 & 70 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$
- [682] $2 \begin{smallmatrix} 1^2 & 2^1 & 3^1 & 1^3 & 1^3 \\ 1^2 & 1^2 & 1^2 & 1^3 & 1^3 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 7 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[685] & 1^{12}31^{23}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 12 & 0 & 58 & 14 \\ 2 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0} \\
[683] & 1^{11}31^{33}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 12 & 0 & 73 & 15 \\ 2 & 3 & -1 & 15 & 2 \end{pmatrix}_{b_0} \\
[684] & 1^{13}31^{13}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 12 & 0 & 49 & 13 \\ 2 & 3 & -1 & 13 & 2 \end{pmatrix}_{b_0} \\
[686] & 1^{13}31^{23}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 15 & 0 & 85 & 17 \\ 2 & 3 & -1 & 17 & 2 \end{pmatrix}_{b_0} \\
[687] & 2^{11}3^{12}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.1 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 11, T \in [10, 11] \\
& \begin{pmatrix} 0 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 0 & 10 & 2 \\ 10 & 15 & 10 & 73 & 15 \\ 2 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[688] & 21^{13}41^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.49 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 4 & 13 & 3 \\ 4 & 4 & 18 & 4 \\ 13 & 18 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0} \\
[689] & 2^{112}3^{11}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 11, T \in [10, 11] \\
& \begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 0 & 8 & 2 \\ 6 & 7 & 8 & 21 & 9 \\ 2 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[690] & 2^{112}2^{12}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 10 & 10 & 58 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[691] & 1^{12}4^{11}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 27 & 4 \\ 15 & 27 & 132 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0} \\
[692] & 2^{111}3^{11}3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 3 & 2 & 5 & 2 \\ 3 & 1 & 3 & 5 & 3 \\ 2 & 3 & 0 & 5 & 2 \\ 5 & 5 & 5 & 8 & 6 \\ 2 & 3 & 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[693] & 1^{14}4^{12}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -146, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.55 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 22 & 4 \\ 15 & 22 & 90 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[694] $1^{13}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -146, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 20 & 4 \\ 14 & 20 & 70 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$

[695] $1^{13}321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -147, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 5 & 30 & 5 \\ 17 & 30 & 142 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$

[696] $1^{13}321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -147, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 5 & 30 & 5 \\ 19 & 30 & 142 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$

[697] $1^{14}3131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 12 & 0 & 46 & 12 \\ 2 & 3 & -1 & 12 & 2 \end{pmatrix}_{b_0}$

[698] $1^{13}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.53 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 20 & 4 \\ 15 & 20 & 70 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$

[699] $1^{13}31^431^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 21 & 0 & 193 & 25 \\ 2 & 3 & -1 & 25 & 2 \end{pmatrix}_{b_0}$

[700] $1^{14}31^331^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 21 & 0 & 172 & 24 \\ 2 & 3 & -1 & 24 & 2 \end{pmatrix}_{b_0}$

[701] $1^{14}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 24 & 4 \\ 14 & 24 & 101 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$

[702] $21^141^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 24 & 4 \\ 14 & 24 & 101 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$

[703] $21^{13}321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$

[708] $1^{12}321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 9 & 8 & 14 & 70 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[704] $1^{14}321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 10 & 10 & 58 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[705] $1^{\overline{21}1^4} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 16 & 4 \\ 12 & 16 & 46 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[706] $1^{\overline{13}1^4} 3^1 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.47 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 10 & 24 & 12 & 94 & 16 \\ 2 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[707] $131^{\overline{14}3} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 8 & 24 & 12 & 94 & 16 \\ 2 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[709] $1^{\overline{13}} 3^1 2^1 1^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 5 & 30 & 5 \\ 18 & 30 & 142 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[711] $1^{\overline{14}} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 22 & 4 \\ 16 & 22 & 90 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[710] $1^{\overline{14}3} 1^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 15 & 3 \\ 1 & 4 & 0 & 15 & 2 \\ 7 & 15 & 15 & 76 & 16 \\ 2 & 3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

[713] $1^{\overline{11}3} 1^4 3^1 1^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -155, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 15 & 0 & 121 & 19 \\ 2 & 3 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[712] $1^{\overline{11}3} 1^1 3^1 1^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -155, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 21 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 10 & 21 & 12 & 73 & 15 \\ 2 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[715] $21^{\overline{13}3} 2^1 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[716] $21^{\overline{11}} 4^1 1^{\overline{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.51 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 23 & 4 \\ 15 & 23 & 97 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

[714] $1^{\overline{13}} 4^1 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.51 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 23 & 4 \\ 15 & 23 & 97 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

[717] $1^{\overline{13}} 4^1 1^{\overline{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 30 & 4 \\ 17 & 30 & 169 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0} \\
[718] & 31^{12}231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 3 & 2 & 0 & 9 & 2 \\ 9 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[719] & 2^1 3^{12} 1^{11} 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 3 & 2 & 7 & 2 \\ 3 & 1 & 3 & 7 & 3 \\ 2 & 3 & 0 & 8 & 2 \\ 7 & 7 & 8 & 21 & 9 \\ 2 & 3 & 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[720] & 1^{12} 31^{13} 431^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 18 & 0 & 154 & 22 \\ 2 & 3 & -1 & 22 & 2 \end{pmatrix}_{b_0} \\
[721] & 2^1 2^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -162, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 5 & 20 & 5 \\ 12 & 20 & 58 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0} \\
[722] & 1^{13} 3^1 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[723] & 1^{12} 31^{12} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 18 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 7 & 18 & 12 & 58 & 14 \\ 2 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[724] & 2^1 3^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 11, T \in [10, 11] \\
& \begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 0 & 7 & 2 \\ 6 & 8 & 7 & 21 & 9 \\ 2 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[725] & 1^{12} 3^1 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 5 & 30 & 5 \\ 16 & 30 & 134 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0} \\
[726] & 21^1 3^1 41^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.49 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 25 & 4 \\ 17 & 25 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0} \\
[727] & 31^{12} 231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 3 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 3 & 2 & 0 & 8 & 2 \\ 8 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[728] & 1^{12} 31^{12} 31^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 18 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 8 & 18 & 12 & 58 & 14 \\ 2 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[729] & 1^{12} 31^2 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 12 & 0 & 58 & 14 \\ 2 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[730] $1^{12}3^{14}21^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 10 & 8 & 14 & 70 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[731] $1^{14}3^{12}21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 10 & 10 & 58 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[732] $1^{14}4^{13}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 25 & 4 \\ 17 & 25 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[733] $131^{23}1^{14}4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -166, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 18 & 3 \\ 1 & 4 & 0 & 18 & 2 \\ 8 & 18 & 18 & 118 & 20 \\ 2 & 3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

[734] $1^{14}3^{12}21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 10 & 10 & 58 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[735] $1^{14}31^{12}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 9 & 24 & 18 & 118 & 20 \\ 2 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[736] $1^{14}31^{12}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 12 & 24 & 18 & 118 & 20 \\ 2 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[737] $1^{13}1^{14}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 9 & 24 & 12 & 94 & 16 \\ 2 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[738] $1^4 4^{14} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -169, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 33 & 4 \\ 19 & 33 & 212 & 28 \\ 3 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

[739] $1^{13}4^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -170, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.47 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 23 & 4 \\ 16 & 23 & 97 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

[742] $31^{11}31^{11}3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 3 & 3 & 6 & 2 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 6 & 9 & 6 & 13 & 7 \\ 2 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[741] $1321^{11}1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 3 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[740] \ 21^{13}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 20 & 3 \\ 4 & 4 & 29 & 4 \\ 20 & 29 & 169 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[744] \ 1^{13}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 12 & 0 & 49 & 13 \\ 2 & 3 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[743] \ 2^2 31^{15} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.1$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 5 & 25 & 5 \\ 15 & 25 & 98 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[745] \ 1^{14}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -174, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.46 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 25 & 4 \\ 18 & 25 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[746] \ 1^{12}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 100 & 18 \\ 2 & 3 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[747] \ 31^{13}231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 5 & 12 & 3 \\ 5 & 5 & 15 & 5 \\ 12 & 15 & 34 & 12 \\ 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[748] \ 1^{11}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 15 & 0 & 121 & 19 \\ 2 & 3 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[749] \ 2^2 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 8 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[750] \ 1^{14}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 15 & 0 & 76 & 16 \\ 2 & 3 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[751] \ 1^{13}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 23 & 4 \\ 17 & 23 & 97 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[752] \ 1^{14}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 18 & 0 & 118 & 20 \\ 2 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[754] \ 1^{13}231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$\begin{aligned}
& T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 5 & 30 & 5 \\ 17 & 30 & 134 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0} \\
[753] & 131^{13}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.70 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 21 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 7 & 21 & 12 & 73 & 15 \\ 2 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[755] & 1^{14}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.70 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 24 & 4 \\ 15 & 24 & 101 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0} \\
[756] & 1^{13}321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[757] & 1^{13}31^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 21 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 8 & 21 & 15 & 85 & 17 \\ 2 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[758] & 31^{13}231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.70 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 3 & 10 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 3 & 2 & 0 & 10 & 2 \\ 10 & 5 & 10 & 33 & 11 \\ 2 & 2 & 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[759] & 1^{21}41^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.70 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 17 & 4 \\ 12 & 17 & 49 & 13 \\ 3 & 4 & 13 & 3 \end{pmatrix}_{b_0} \\
[760] & 1^{13}31^{12}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 21 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 10 & 21 & 15 & 85 & 17 \\ 2 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[761] & 31^{12}231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 5 & 14 & 3 \\ 5 & 5 & 20 & 5 \\ 14 & 20 & 62 & 16 \\ 3 & 5 & 16 & 3 \end{pmatrix}_{b_0} \\
[762] & 31^{12}231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 5 & 11 & 3 \\ 5 & 5 & 15 & 5 \\ 11 & 15 & 32 & 12 \\ 3 & 5 & 12 & 3 \end{pmatrix}_{b_0} \\
[767] & 1^{14}3131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 12 & 0 & 46 & 12 \\ 2 & 3 & -1 & 12 & 2 \end{pmatrix}_{b_0} \\
[766] & 2^132 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.2 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 11, T \in [10, 11] \\
& \begin{pmatrix} 0 & 2 & 2 & 12 & 2 \\ 2 & 1 & 3 & 18 & 3 \\ 2 & 3 & 0 & 12 & 2 \\ 12 & 18 & 12 & 106 & 18 \\ 2 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[763] & 1^{12}31^231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 12 & 0 & 58 & 14 \\ 2 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[764] $1^{12}3^{14}21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 5 & 30 & 5 \\ 18 & 30 & 134 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[765] $1^{13}31^{33}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 18 & 0 & 133 & 21 \\ 2 & 3 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[768] $21^{41}4^{14}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -186, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 28 & 4 \\ 20 & 28 & 158 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[769] $1^{13}4^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -186, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 26 & 4 \\ 18 & 26 & 130 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

[771] $1^{14}4^{14}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -186, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 28 & 4 \\ 20 & 28 & 158 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[770] $21^{34}4^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -186, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 26 & 4 \\ 18 & 26 & 130 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

[772] $1^{13}3^{14}21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 10 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[773] $1^{13}31^{23}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -191, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.03 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 85 & 17 \\ 2 & 3 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[774] $1^{13}4^{14}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -194, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 26 & 4 \\ 19 & 26 & 130 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

[775] $1^{11}31^{13}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -195, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 21 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 9 & 21 & 12 & 73 & 15 \\ 2 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[776] $1^{13}31^{12}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -195, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 21 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 9 & 21 & 15 & 85 & 17 \\ 2 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[777] $1^{12}3^{14}21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -199, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 9 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[778] \ 1^{14}41^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -199, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.63 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 24 & 4 \\ 16 & 24 & 101 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[779] \ 231^{13}3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -199, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 9 & 2 \\ 3 & 1 & 3 & 9 & 3 \\ 2 & 3 & 0 & 6 & 2 \\ 9 & 9 & 6 & 25 & 9 \\ 2 & 3 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[783] \ 1^{14}31^{12}31^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 10 & 24 & 18 & 118 & 20 \\ 2 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[784] \ 1^{14}31^{43}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 24 & 0 & 238 & 28 \\ 2 & 3 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[780] \ 1^{14}31^{12}31^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 11 & 24 & 18 & 118 & 20 \\ 2 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[781] \ 1^{13}31^{13}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 12 & 0 & 49 & 13 \\ 2 & 3 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[782] \ 21^{14}41^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 20 & 3 \\ 4 & 4 & 28 & 4 \\ 20 & 28 & 158 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[787] \ 21^{14}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -203, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.66 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 14 & 3 \\ 4 & 4 & 19 & 4 \\ 14 & 19 & 65 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[785] \ 1^{11}31^{13}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -203, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 21 & 4 \\ 1 & 4 & 0 & 12 & 2 \\ 8 & 21 & 12 & 73 & 15 \\ 2 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[786] \ 1^{11}31^{14}31^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -203, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 4 & 4 & 27 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 13 & 27 & 15 & 121 & 19 \\ 2 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[790] \ 21^{14}41^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 29 & 4 \\ 21 & 29 & 169 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[789] \ 1^{12}31^{13}31^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 12 & 24 & 15 & 100 & 18 \\ 2 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[788] \ 1^{13}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 29 & 4 \\ 21 & 29 & 169 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[791] \ 221^{12}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 6 & 13 & 3 \\ 6 & 13 & 32 & 7 \\ 13 & 32 & 68 & 16 \\ 3 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[792] \ 1^{13}321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 5 & 35 & 5 \\ 19 & 35 & 188 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[794] \ 1^{14}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -209, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 27 & 4 \\ 17 & 27 & 132 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[793] \ 21^241^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -209, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 27 & 4 \\ 17 & 27 & 132 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[795] \ 1^{12}31^331^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 15 & 0 & 100 & 18 \\ 2 & 3 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[796] \ 1^{14}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 24 & 4 \\ 17 & 24 & 101 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[797] \ 1^{14}321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -212, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 9 & 11 & 13 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[798] \ 21^{13}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -212, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 4 & 26 & 4 \\ 18 & 26 & 130 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[799] \ 1^{12}31^431^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -212, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 18 & 0 & 154 & 22 \\ 2 & 3 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[800] \ 2231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 5 & 25 & 5 \\ 15 & 25 & 94 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

- [801] $21^{13}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -216$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.56 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 14 & 3 \\ 4 & 4 & 20 & 4 \\ 14 & 20 & 70 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$
- [802] $1^{13}321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -223$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 10 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [803] $1^{13}321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -223$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 5 & 35 & 5 \\ 20 & 35 & 188 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$
- [804] $1^{21}41^{15} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -224$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.1$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 20 & 4 \\ 15 & 20 & 73 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$
- [805] $131^{15}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -224$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.1$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 30 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 10 & 30 & 15 & 148 & 20 \\ 2 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$
- [806] $21^{14}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -227$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.48 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 4 & 25 & 4 \\ 18 & 25 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$
- [807] $21^{13}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -227$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.48 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 16 & 3 \\ 4 & 4 & 23 & 4 \\ 16 & 23 & 97 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$
- [808] $1^{14}31^{13}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -227$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.13 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 76 & 16 \\ 2 & 3 & -1 & 16 & 2 \end{pmatrix}_{b_0}$
- [809] $1^{13}31^{43}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -227$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.13 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 21 & 0 & 193 & 25 \\ 2 & 3 & -1 & 25 & 2 \end{pmatrix}_{b_0}$
- [810] $21^{14}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -228$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.54 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 16 & 3 \\ 4 & 4 & 22 & 4 \\ 16 & 22 & 90 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$
- [811] $1^{13}321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -228$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 9.86 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 5 & 35 & 5 \\ 21 & 35 & 188 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$
- [812] $1^{14}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -105$, $\det = -231$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.54 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 27 & 4 \\ 18 & 27 & 132 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$

$$[814] \quad 2 \overset{1^{13}}{2} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[813] \quad 1 \overset{1^3}{4} 321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 10 & 11 & 13 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[817] \quad 1^{13} 31^2 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 85 & 17 \\ 2 & 3 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[815] \quad 221^{13} 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 6 & 15 & 3 \\ 6 & 13 & 36 & 7 \\ 15 & 36 & 88 & 18 \\ 3 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[816] \quad 1^{14} 31^3 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 21 & 0 & 172 & 24 \\ 2 & 3 & -1 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[818] \quad 131^{13} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.69 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 8 & 24 & 15 & 100 & 18 \\ 2 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[819] \quad 2 \overset{1^{14}}{2} 2 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 10 & 8 & 14 & 70 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[820] \quad 2 \overset{1^1}{3} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.3$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 14 & 2 \\ 2 & 1 & 3 & 21 & 3 \\ 2 & 3 & 0 & 14 & 2 \\ 14 & 21 & 14 & 145 & 21 \\ 2 & 3 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[821] \quad 21^{14} 321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 10 & 10 & 58 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[824] \quad 2 \overset{1^{14}}{2} 2 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 5 & 25 & 5 \\ 15 & 25 & 92 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[822] \quad 2 \overset{1^{16}}{2} 2 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.2$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 5 & 30 & 5 \\ 18 & 30 & 142 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[823] \quad 2 \overset{1^{13}}{2} 2 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 5 & 30 & 5 \\ 18 & 30 & 142 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

- [827] $1^{13}3^{14}21^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -239, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 11 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [826] $1^{12}3^{13}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -239, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 15 & 0 & 100 & 18 \\ 2 & 3 & -1 & 18 & 2 \end{pmatrix}_{b_0}$
- [825] $31^{14}231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -239, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 13 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 3 & 2 & 0 & 13 & 2 \\ 13 & 6 & 13 & 56 & 14 \\ 2 & 2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$
- [828] $2^{11}3^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -240, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 11 & 3 \\ 2 & 3 & 0 & 9 & 2 \\ 8 & 11 & 9 & 40 & 12 \\ 2 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$
- [831] $1^{14}3^{13}21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 11 & 13 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [829] $1^{14}3^{12}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 18 & 0 & 118 & 20 \\ 2 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0}$
- [830] $1^{13}3^{13}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 18 & 0 & 133 & 21 \\ 2 & 3 & -1 & 21 & 2 \end{pmatrix}_{b_0}$
- [832] $1^{14}4^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -245, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.51 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 27 & 4 \\ 19 & 27 & 132 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0}$
- [834] $31^{13}231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -255, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 5 & 24 & 3 \\ 5 & 5 & 35 & 5 \\ 24 & 35 & 202 & 28 \\ 3 & 5 & 28 & 3 \end{pmatrix}_{b_0}$
- [835] $1^{14}4^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -255, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 30 & 4 \\ 20 & 30 & 169 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0}$
- [833] $21^{11}3^4 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -255, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 30 & 4 \\ 20 & 30 & 169 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0}$
- [836] $21^{14}3^{14}21^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 12 & 16 & 110 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[837] & 1^{12}31^{43}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 18 & 0 & 154 & 22 \\ 2 & 3 & -1 & 22 & 2 \end{pmatrix}_{b_0} \\
[838] & 1^{13}4^321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -259, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 23 & 3 \\ 3 & 5 & 40 & 5 \\ 23 & 40 & 254 & 32 \\ 3 & 5 & 32 & 4 \end{pmatrix}_{b_0} \\
[839] & 1^{12}31^{13}31^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -260, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 11 & 24 & 15 & 100 & 18 \\ 2 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0} \\
[841] & 1^{14}3^221^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 10 & 12 & 16 & 110 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[840] & 1^{14}3^221^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 24 & 3 \\ 3 & 5 & 40 & 5 \\ 24 & 40 & 254 & 32 \\ 3 & 5 & 32 & 4 \end{pmatrix}_{b_0} \\
[842] & 2^33^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 11, T \in [10, 11] \\
& \begin{pmatrix} 0 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 0 & 10 & 2 \\ 8 & 10 & 10 & 38 & 12 \\ 2 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[843] & 131^{13}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -267, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 27 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 9 & 27 & 18 & 133 & 21 \\ 2 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0} \\
[845] & 1^{11}31^{14}31^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -267, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 27 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 12 & 27 & 15 & 121 & 19 \\ 2 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0} \\
[846] & 1^44^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -267, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 21 & 4 \\ 15 & 21 & 76 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0} \\
[844] & 131^{14}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -267, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 27 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 9 & 27 & 15 & 121 & 19 \\ 2 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0} \\
[847] & 1^{14}4^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.46 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 30 & 4 \\ 21 & 30 & 169 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0} \\
[848] & 2^22^31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0
\end{aligned}$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 10 & 11 & 13 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[849] \ 21^{14}321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 11 & 13 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[850] \ 1^{13}31^{23}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 15 & 0 & 85 & 17 \\ 2 & 3 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[851] \ 1^{14}31^{13}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 76 & 16 \\ 2 & 3 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[852] \ 131^{13}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -272, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 30 & 4 \\ 1 & 4 & 0 & 21 & 2 \\ 10 & 30 & 21 & 172 & 24 \\ 2 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[853] \ 1^{12}31^{13}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.51 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 9 & 24 & 15 & 100 & 18 \\ 2 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[854] \ 2^3 3^{11} 1^{12} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -279, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 8 & 2 \\ 3 & 1 & 3 & 8 & 3 \\ 2 & 3 & 0 & 7 & 2 \\ 8 & 8 & 7 & 21 & 9 \\ 2 & 3 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[855] \ 31^{14}231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 5 & 16 & 3 \\ 5 & 5 & 20 & 5 \\ 16 & 20 & 62 & 16 \\ 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[856] \ 31^{13}231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -284, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 11 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 3 & 2 & 0 & 11 & 2 \\ 11 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[857] \ 1^{12}31^{13}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -284, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 24 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 10 & 24 & 15 & 100 & 18 \\ 2 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[858] \ 31^{12}31^{11}3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -288, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 3 & 3 & 9 & 2 \\ 3 & 4 & 4 & 15 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 9 & 15 & 9 & 37 & 11 \\ 2 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[860] \ 1^{14}4^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -289, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 23 & 3 \\ 3 & 4 & 33 & 4 \\ 23 & 33 & 212 & 28 \\ 3 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[859] \ 21^4 4^4 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -289, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 23 & 3 \\ 3 & 4 & 33 & 4 \\ 23 & 33 & 212 & 28 \\ 3 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

[861] $1^{14}3^{13}21^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 11 & 12 & 16 & 110 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[862] $1^{12}31^{43}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 18 & 0 & 154 & 22 \\ 2 & 3 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[863] $1^{14}31^{43}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 24 & 0 & 238 & 28 \\ 2 & 3 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[864] $1^{13}31^{43}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -295, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 21 & 0 & 193 & 25 \\ 2 & 3 & -1 & 25 & 2 \end{pmatrix}_{b_0}$$

[865] $1^{13}31^{33}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -296, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 18 & 0 & 133 & 21 \\ 2 & 3 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[866] $1^{13}31^{13}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -299, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 4 & 4 & 27 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 13 & 27 & 18 & 133 & 21 \\ 2 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[867] $1^{11}31^{14}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -299, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 27 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 10 & 27 & 15 & 121 & 19 \\ 2 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[868] $1^{11}31^{14}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -299, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 27 & 4 \\ 1 & 4 & 0 & 15 & 2 \\ 11 & 27 & 15 & 121 & 19 \\ 2 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[869] $1^{14}31^{23}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 18 & 0 & 118 & 20 \\ 2 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[870] $231^{14}3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 3 & 2 & 12 & 2 \\ 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 0 & 8 & 2 \\ 12 & 12 & 8 & 46 & 12 \\ 2 & 3 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[871] $1^{14}31^{33}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -311, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 21 & 0 & 172 & 24 \\ 2 & 3 & -1 & 24 & 2 \end{pmatrix}_{b_0}$$

[874] $131^{16}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -312, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.2$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 36 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 12 & 36 & 18 & 214 & 24 \\ 2 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[873] $1^{14}3^{14}21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -312, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 12 & 16 & 110 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[872] $1^4 4^1 1^{16} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -312, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.2$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 4 & 24 & 4 \\ 18 & 24 & 106 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[875] $31^{13}231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -315, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 5 & 15 & 3 \\ 5 & 5 & 20 & 5 \\ 15 & 20 & 58 & 16 \\ 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[876] $2^2 2^1 31^{17} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -316, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 5 & 35 & 5 \\ 21 & 35 & 194 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$$

[877] $21^{14}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -320, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 4 & 24 & 4 \\ 17 & 24 & 101 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

[878] $1^{13}31^{13}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -323, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 27 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 10 & 27 & 18 & 133 & 21 \\ 2 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[879] $21^{14}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -323, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 23 & 3 \\ 4 & 4 & 33 & 4 \\ 23 & 33 & 212 & 28 \\ 3 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

[880] $2^3 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -324, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 0 & 11 & 2 \\ 10 & 14 & 11 & 65 & 15 \\ 2 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[881] $2^2 2^1 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -326, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 5 & 30 & 5 \\ 18 & 30 & 134 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[882] $2^2 2^1 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -327, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 11 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[883] $31^{13}231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -339, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 3 & 2 & 0 & 12 & 2 \\ 12 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[884] $1^{13}31^{13}31^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -339, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 27 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 12 & 27 & 18 & 133 & 21 \\ 2 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0} \\
[885] & 1^{13}31^{331^{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -340, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 18 & 0 & 133 & 21 \\ 2 & 3 & -1 & 21 & 2 \end{pmatrix}_{b_0} \\
[886] & 31^{13}231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -344, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 3 & 13 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 3 & 2 & 0 & 13 & 2 \\ 13 & 11 & 13 & 81 & 17 \\ 2 & 2 & 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[887] & 1^{14}31^{13}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -344, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.58 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 30 & 4 \\ 1 & 4 & 0 & 21 & 2 \\ 11 & 30 & 21 & 172 & 24 \\ 2 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0} \\
[888] & 1^{13}31^{13}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -347, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 27 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 11 & 27 & 18 & 133 & 21 \\ 2 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0} \\
[889] & 21^{14}41^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -347, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.52 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 4 & 19 & 3 \\ 4 & 4 & 27 & 4 \\ 19 & 27 & 132 & 22 \\ 3 & 4 & 22 & 3 \end{pmatrix}_{b_0} \\
[890] & 21^{14}41^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -348, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.47 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 4 & 30 & 4 \\ 21 & 30 & 169 & 25 \\ 3 & 4 & 25 & 3 \end{pmatrix}_{b_0} \\
[891] & 31^{13}231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -355, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 5 & 21 & 3 \\ 5 & 5 & 30 & 5 \\ 21 & 30 & 142 & 24 \\ 3 & 5 & 24 & 3 \end{pmatrix}_{b_0} \\
[892] & 1^{13}31^{431^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -355, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 21 & 0 & 193 & 25 \\ 2 & 3 & -1 & 25 & 2 \end{pmatrix}_{b_0} \\
[893] & 131^{14}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -356, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.71 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 30 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 10 & 30 & 18 & 154 & 22 \\ 2 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0} \\
[894] & 1^{14}31^{231^{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -356, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 18 & 0 & 118 & 20 \\ 2 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0} \\
[895] & 1^{12}31^{14}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -356, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 4 & 4 & 30 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 14 & 30 & 18 & 154 & 22 \\ 2 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0} \\
[896] & 221^{13}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -366, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 6 & 18 & 3 \\ 6 & 13 & 44 & 7 \\ 18 & 44 & 130 & 22 \\ 3 & 7 & 22 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [897] $1^{14}31^{13}31^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -368, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 4 & 4 & 30 & 4 \\ 1 & 4 & 0 & 21 & 2 \\ 14 & 30 & 21 & 172 & 24 \\ 2 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$
- [898] $31^{13}231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -375, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 5 & 18 & 3 \\ 5 & 5 & 25 & 5 \\ 18 & 25 & 94 & 20 \\ 3 & 5 & 20 & 3 \end{pmatrix}_{b_0}$$
- [899] $1^{14}31^{13}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 24 & 0 & 238 & 28 \\ 2 & 3 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$
- [900] $1^{14}31^{13}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -379, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 21 & 0 & 172 & 24 \\ 2 & 3 & -1 & 24 & 2 \end{pmatrix}_{b_0}$$
- [901] $221^{14}41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -382, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 6 & 20 & 3 \\ 6 & 13 & 48 & 7 \\ 20 & 48 & 158 & 24 \\ 3 & 7 & 24 & 3 \end{pmatrix}_{b_0}$$
- [902] $2^1 2^1 31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -382, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 5 & 35 & 5 \\ 21 & 35 & 188 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$$
- [903] $2^1 3^1 1^{12} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 3 & 2 & 10 & 2 \\ 3 & 1 & 3 & 10 & 3 \\ 2 & 3 & 0 & 10 & 2 \\ 10 & 10 & 10 & 38 & 12 \\ 2 & 3 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$
- [904] $1^{14}31^{13}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 30 & 4 \\ 1 & 4 & 0 & 21 & 2 \\ 12 & 30 & 21 & 172 & 24 \\ 2 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$
- [905] $31^{13}31^{11} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -101, \det = -387, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 3 & 3 & 12 & 2 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 12 & 21 & 12 & 73 & 15 \\ 2 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$
- [906] $1^{14}31^{13}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -392, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 4 & 4 & 30 & 4 \\ 1 & 4 & 0 & 21 & 2 \\ 13 & 30 & 21 & 172 & 24 \\ 2 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$
- [907] $2^1 3^1 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -396, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 0 & 12 & 2 \\ 10 & 13 & 12 & 61 & 15 \\ 2 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$
- [908] $2^1 2^1 31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -400, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 0 & 16 & 2 \\ 12 & 12 & 16 & 110 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$
- [910] $1^{12}31^{14}31^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -404, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$

$$\begin{aligned}
& T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 4 & 4 & 30 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 13 & 30 & 18 & 154 & 22 \\ 2 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0} \\
[909] & 1^{12}31^{14}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -404, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 30 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 11 & 30 & 18 & 154 & 22 \\ 2 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0} \\
[911] & 31^{14}231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -404, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 3 & 14 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 3 & 2 & 0 & 14 & 2 \\ 14 & 8 & 14 & 70 & 16 \\ 2 & 2 & 2 & 16 & 2 \end{pmatrix}_{b_0} \\
[912] & 2^1 2^1 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -406, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 24 & 3 \\ 4 & 5 & 40 & 5 \\ 24 & 40 & 254 & 32 \\ 3 & 5 & 32 & 4 \end{pmatrix}_{b_0} \\
[913] & 1^{13}31^1 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -407, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 21 & 0 & 193 & 25 \\ 2 & 3 & -1 & 25 & 2 \end{pmatrix}_{b_0} \\
[914] & 31^{15}231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -415, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.1 \\
& T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 5 & 20 & 3 \\ 5 & 5 & 25 & 5 \\ 20 & 25 & 98 & 20 \\ 3 & 5 & 20 & 3 \end{pmatrix}_{b_0} \\
[915] & 1^1 4^1 1^{17} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -416, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.3 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 4 & 28 & 4 \\ 21 & 28 & 145 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0} \\
[916] & 131^{17}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -416, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.3 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 4 & 4 & 42 & 4 \\ 1 & 4 & 0 & 21 & 2 \\ 14 & 42 & 21 & 292 & 28 \\ 2 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0} \\
[917] & 131^{14}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -419, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.72 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 33 & 4 \\ 1 & 4 & 0 & 21 & 2 \\ 11 & 33 & 21 & 193 & 25 \\ 2 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0} \\
[918] & 1^{12}31^{14}31^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -420, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 30 & 4 \\ 1 & 4 & 0 & 18 & 2 \\ 12 & 30 & 18 & 154 & 22 \\ 2 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0} \\
[920] & 2^1 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 3 & 2 & 12 & 2 \\ 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 0 & 13 & 2 \\ 12 & 12 & 13 & 61 & 15 \\ 2 & 3 & 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[921] & 1^{14}31^3 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 21 & 0 & 172 & 24 \\ 2 & 3 & -1 & 24 & 2 \end{pmatrix}_{b_0} \\
[919] & 2^1 31^{15} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.1 \\
& T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 3 & 2 & 15 & 2 \\ 3 & 1 & 3 & 15 & 3 \\ 2 & 3 & 0 & 10 & 2 \\ 15 & 15 & 10 & 73 & 15 \\ 2 & 3 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[923] $2^1 3^1 1^{14} 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -444$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.14 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 0 & 3 & 2 & 14 & 2 \\ 3 & 1 & 3 & 14 & 3 \\ 2 & 3 & 0 & 16 & 2 \\ 14 & 14 & 16 & 90 & 18 \\ 2 & 3 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[922] $2^1 3^1 1^{11} 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -444$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.18 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 0 & 3 & 2 & 11 & 2 \\ 3 & 1 & 3 & 11 & 3 \\ 2 & 3 & 0 & 9 & 2 \\ 11 & 11 & 9 & 40 & 12 \\ 2 & 3 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[924] $1^{14} 31^4 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -452$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.07 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 24 & 0 & 238 & 28 \\ 2 & 3 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[925] $131^{14} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -456$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 36 & 4 \\ 1 & 4 & 0 & 24 & 2 \\ 12 & 36 & 24 & 238 & 28 \\ 2 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

[926] $31^{14} 31^{11} 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 19$, $\Delta = -101$, $\det = -468$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{pmatrix} 0 & 3 & 3 & 15 & 2 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 15 & 27 & 15 & 121 & 19 \\ 2 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[927] $31^{14} 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -480$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.17 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 5 & 19 & 3 \\ 5 & 5 & 25 & 5 \\ 19 & 25 & 92 & 20 \\ 3 & 5 & 20 & 3 \end{pmatrix}_{b_0}$$

[928] $1^{13} 31^{14} 31^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -483$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.36 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 4 & 4 & 33 & 4 \\ 1 & 4 & 0 & 21 & 2 \\ 15 & 33 & 21 & 193 & 25 \\ 2 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[929] $1^{13} 31^{14} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -483$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.56 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 33 & 4 \\ 1 & 4 & 0 & 21 & 2 \\ 12 & 33 & 21 & 193 & 25 \\ 2 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[930] $1^{13} 31^{14} 31^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -515$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.46 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 4 & 4 & 33 & 4 \\ 1 & 4 & 0 & 21 & 2 \\ 13 & 33 & 21 & 193 & 25 \\ 2 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[931] $1^{13} 31^{14} 31^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -515$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.40 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 4 & 4 & 33 & 4 \\ 1 & 4 & 0 & 21 & 2 \\ 14 & 33 & 21 & 193 & 25 \\ 2 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[932] $31^{14} 231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -519$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 15 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 3 & 2 & 0 & 15 & 2 \\ 15 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[933] $1^{14} 31^4 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -520$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.11 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 24 & 0 & 238 & 28 \\ 2 & 3 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[934] $2 \overset{1^{12}}{3} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -536, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 12 & 2 \\ 2 & 1 & 3 & 16 & 3 \\ 2 & 3 & 0 & 14 & 2 \\ 12 & 16 & 14 & 90 & 18 \\ 2 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[935] $1 \overset{1^{14}}{3} 1 \overset{1^{14}}{3} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -536, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10.58 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 4 & 4 & 36 & 4 \\ 1 & 4 & 0 & 24 & 2 \\ 13 & 36 & 24 & 238 & 28 \\ 2 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

[936] $221 \overset{1^{14}}{4} 1 \overset{1^{11}}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -567, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 6 & 23 & 3 \\ 6 & 13 & 56 & 7 \\ 23 & 56 & 212 & 28 \\ 3 & 7 & 28 & 3 \end{pmatrix}_{b_0}$$

[937] $221 \overset{1^{15}}{5} 1 \overset{1^{14}}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -571, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.1$

$$T_{\min} = 10.42 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 6 & 25 & 3 \\ 6 & 13 & 60 & 7 \\ 25 & 60 & 248 & 30 \\ 3 & 7 & 30 & 3 \end{pmatrix}_{b_0}$$

[938] $31 \overset{1^{16}}{6} 231 \overset{1^{14}}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -580, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.2$

$$T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 5 & 24 & 3 \\ 5 & 5 & 30 & 5 \\ 24 & 30 & 142 & 24 \\ 3 & 5 & 24 & 3 \end{pmatrix}_{b_0}$$

[941] $1 \overset{1^{14}}{3} 1 \overset{1^{14}}{3} 1 \overset{1^{14}}{4} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -584, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 16 & 2 \\ 2 & 4 & 4 & 36 & 4 \\ 1 & 4 & 0 & 24 & 2 \\ 16 & 36 & 24 & 238 & 28 \\ 2 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

[940] $1 \overset{1^{14}}{3} 1 \overset{1^{14}}{3} 1 \overset{1^{12}}{2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -584, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 4 & 4 & 36 & 4 \\ 1 & 4 & 0 & 24 & 2 \\ 14 & 36 & 24 & 238 & 28 \\ 2 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

[939] $31 \overset{1^{14}}{4} 231 \overset{1^{14}}{4} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -584, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 16 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 3 & 2 & 0 & 16 & 2 \\ 16 & 12 & 16 & 110 & 20 \\ 2 & 2 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[942] $31 \overset{1^{14}}{4} 231 \overset{1^{14}}{4} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 5 & 28 & 3 \\ 5 & 5 & 40 & 5 \\ 28 & 40 & 254 & 32 \\ 3 & 5 & 32 & 3 \end{pmatrix}_{b_0}$$

[943] $31 \overset{1^{14}}{4} 231 \overset{1^{12}}{2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 5 & 22 & 3 \\ 5 & 5 & 30 & 5 \\ 22 & 30 & 134 & 24 \\ 3 & 5 & 24 & 3 \end{pmatrix}_{b_0}$$

[944] $1 \overset{1^{14}}{3} 1 \overset{1^{14}}{3} 1 \overset{1^{13}}{3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 4 & 4 & 36 & 4 \\ 1 & 4 & 0 & 24 & 2 \\ 15 & 36 & 24 & 238 & 28 \\ 2 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

[945] $2 \overset{1^{11}}{3} 1 \overset{1^{14}}{3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -639, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 14 & 2 \\ 3 & 1 & 3 & 14 & 3 \\ 2 & 3 & 0 & 11 & 2 \\ 14 & 14 & 11 & 65 & 15 \\ 2 & 3 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[946] \quad 2 \ 3 \ 1^{12} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -639, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 13 & 2 \\ 3 & 1 & 3 & 13 & 3 \\ 2 & 3 & 0 & 12 & 2 \\ 13 & 13 & 12 & 61 & 15 \\ 2 & 3 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[947] \quad 31^{14} 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -640, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 5 & 25 & 3 \\ 5 & 5 & 35 & 5 \\ 25 & 35 & 188 & 28 \\ 3 & 5 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[948] \quad 31^{17} 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -775, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.3$$

$$T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 5 & 28 & 3 \\ 5 & 5 & 35 & 5 \\ 28 & 35 & 194 & 28 \\ 3 & 5 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[949] \quad 2 \ 3 \ 1^{13} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -784, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 15 & 2 \\ 3 & 1 & 3 & 15 & 3 \\ 2 & 3 & 0 & 15 & 2 \\ 15 & 15 & 15 & 88 & 18 \\ 2 & 3 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[950] \quad 221^{16} 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -802, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.2$$

$$T_{\min} = 10.42 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 6 & 30 & 3 \\ 6 & 13 & 72 & 7 \\ 30 & 72 & 358 & 36 \\ 3 & 7 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[951] \quad 2 \ 3 \ 1^{14} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -879, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 17 & 2 \\ 3 & 1 & 3 & 17 & 3 \\ 2 & 3 & 0 & 18 & 2 \\ 17 & 17 & 18 & 121 & 21 \\ 2 & 3 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[952] \quad 2 \ 3 \ 1^{12} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -924, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 16 & 2 \\ 3 & 1 & 3 & 16 & 3 \\ 2 & 3 & 0 & 14 & 2 \\ 16 & 16 & 14 & 90 & 18 \\ 2 & 3 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[953] \quad 221^{17} 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -1075, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.3$$

$$T_{\min} = 10.42 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 6 & 35 & 3 \\ 6 & 13 & 84 & 7 \\ 35 & 84 & 488 & 42 \\ 3 & 7 & 42 & 3 \end{pmatrix}_{b_0}$$

$$[954] \quad 2 \ 3 \ 1^{13} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -1159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 18 & 2 \\ 3 & 1 & 3 & 18 & 3 \\ 2 & 3 & 0 & 17 & 2 \\ 18 & 18 & 17 & 121 & 21 \\ 2 & 3 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[955] \quad 2 \ 3 \ 1^{14} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -1344, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 20 & 2 \\ 3 & 1 & 3 & 20 & 3 \\ 2 & 3 & 0 & 20 & 2 \\ 20 & 20 & 20 & 158 & 24 \\ 2 & 3 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [956] \quad 1^1 2^1 3 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9, T_{\max} = 10, T \in [9, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[957] \quad 1^1 3 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 9, \Delta = -57, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9, T_{\max} = 9, T \in [9, 9]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[958] & 1^{11}23131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 4 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 4 \end{pmatrix}_{b_0} \\
[961] & 1^123131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & 0 & 4 \end{pmatrix}_{b_0} \\
[962] & 1^{11}2321^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 5 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 5 & 0 & 17 & 9 \\ 2 & 3 & -1 & 9 & 4 \end{pmatrix}_{b_0} \\
[963] & 1^{13}2321^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 13 & 2 \\ 8 & 0 & 13 & 97 & 21 \\ 2 & 0 & 2 & 21 & 4 \end{pmatrix}_{b_0} \\
[964] & 1^12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 2 & 0 & 1 & 1 & 3 \\ 2 & 0 & 2 & 3 & 4 \end{pmatrix}_{b_0} \\
[965] & 1^14131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 11, T \in [10, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 0 & 1 & 2 \\ 1 & 1 & 0 & 2 \\ 3 & 2 & 2 & 5 \end{pmatrix}_{b_0} \\
[960] & 1^1231^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 3 & 3 \\ 2 & 0 & 3 & 4 & 4 \\ 2 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0} \\
[959] & 1^{12}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0} \\
[974] & 1^{11}51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 20, T \in [11, 20] \\
& \begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 6 & 4 \\ 4 & 6 & 8 & 6 \\ 3 & 4 & 6 & 4 \end{pmatrix}_{b_0} \\
[979] & 21^141^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 2 & 4 \\ 2 & 2 & 0 & 2 \\ 3 & 4 & 2 & 4 \end{pmatrix}_{b_0} \\
[980] & 221^151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 20, T \in [11, 20] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 4 \\ 5 & 6 & 8 & 6 \\ 3 & 4 & 6 & 4 \end{pmatrix}_{b_0} \\
[981] & 131^141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 \\ 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0} \\
[982] & 1^{\overline{1}} \overline{5}^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 20, T \in [11, 20] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 4 \\ 5 & 6 & 8 & 6 \\ 3 & 4 & 6 & 4 \end{pmatrix}_{b_0} \\
[978] & 1^{\overline{11}} 4 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 3 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[977] & 1^{\overline{1}} 4 131^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 5 & 3 & 5 \\ 3 & 1 & 3 \\ 5 & 3 & 5 \end{pmatrix}_{b_0} \\
[970] & 1^{\overline{1}} 4 1231^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 5 & 3 & 5 \\ 3 & 1 & 3 \\ 5 & 3 & 5 \end{pmatrix}_{b_0} \\
[969] & 21^{\overline{1}} \overline{5}^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 20, T \in [11, 20] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 4 \\ 5 & 6 & 8 & 6 \\ 3 & 4 & 6 & 4 \end{pmatrix}_{b_0} \\
[968] & 1^{\overline{1}} \overline{5}^{\overline{1}} 1^{\overline{1}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 20, T \in [11, 20] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 \\ 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0} \\
[967] & 41^{\overline{1}} \overline{2}^{\overline{11}} 3 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 4 & 2 & 2 & 2 \\ 4 & 10 & 5 & 4 & 6 \\ 2 & 5 & 0 & 3 & 2 \\ 2 & 4 & 3 & 1 & 3 \\ 2 & 6 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
[966] & 1^{\overline{1}} 4 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 11, T \in [10, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[976] & 1^{\overline{12}} 4 131^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 2 & 1 & 7 & 5 \\ 2 & 2 & 0 & 5 & 3 \end{pmatrix}_{b_0} \\
[975] & 1^{\overline{11}} 2 3 21^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 4 & 2 \\ 2 & 0 & 4 & 6 & 6 \\ 2 & 0 & 2 & 6 & 4 \end{pmatrix}_{b_0} \\
[971] & 1^{\overline{1}} 4 1^{\overline{12}} 231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 0 & 2 & 2 \\ 3 & 2 & 4 & 4 \\ 3 & 2 & 4 & 4 \end{pmatrix}_{b_0} \\
[972] & 21^{\overline{11}} 4 1^{\overline{4}} 31^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 16 & 4 \\ 14 & 16 & 60 & 16 \\ 3 & 4 & 16 & 4 \end{pmatrix}_{b_0} \\
[973] & 1^{11} 4 12 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 3 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[983] & 1^{11} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 0 & 2 & 2 \\ 3 & 2 & 4 & 4 \\ 3 & 2 & 4 & 4 \end{pmatrix}_{b_0} \\
[984] & 1^{11} 5 1^3 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 13 & 4 \\ 11 & 13 & 40 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0} \\
[985] & 1^{11} 5 1^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 10 & 4 \\ 10 & 10 & 19 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0} \\
[987] & 2 1^{11} 5 1^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 19 & 4 \\ 13 & 19 & 88 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0} \\
[986] & 1^{13} 5 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 22 & 4 \\ 14 & 22 & 115 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0} \\
[988] & 2 1^{11} 5 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 4 & 5 & 9 & 4 \\ 5 & 4 & 10 & 4 \\ 9 & 10 & 19 & 9 \\ 4 & 4 & 9 & 3 \end{pmatrix}_{b_0} \\
[989] & 1^{11} 2 3 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0} \\
[990] & 2 1^{11} 5 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 4 & 5 & 9 & 4 \\ 5 & 4 & 10 & 4 \\ 9 & 10 & 19 & 9 \\ 4 & 4 & 9 & 3 \end{pmatrix}_{b_0} \\
[991] & 2 3 1 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 4 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0} \\
[992] & 2 1^{11} 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 20, T \in [11, 20] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 \\ 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

- [993] $1^{\overline{11}} 4 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 5 & 4 \\ 4 & 5 & 5 & 5 \\ 3 & 4 & 5 & 4 \end{pmatrix}_{b_0}$
- [994] $1^{\overline{11}} 4 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 10 & 4 \\ 9 & 10 & 22 & 10 \\ 3 & 4 & 10 & 4 \end{pmatrix}_{b_0}$
- [995] $1^{\overline{1}} 4 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 8 & 4 \\ 7 & 8 & 14 & 8 \\ 3 & 4 & 8 & 4 \end{pmatrix}_{b_0}$
- [996] $1^{\overline{1}} 5 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 20, T \in [11, 20]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 \\ 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$
- [1002] $1^{\overline{11}} 4 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 0 & 1 & 2 \\ 2 & 1 & 0 & 2 \\ 3 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- [1003] $1^{\overline{11}} 4 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$
- [1001] $1^{\overline{11}} 4 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$
- [1006] $23141^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [1005] $14131^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 0 & 1 & 2 \\ 2 & 1 & 0 & 2 \\ 3 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- [1004] $1^{\overline{1}} 41231^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 0 & 1 & 2 \\ 2 & 1 & 0 & 2 \\ 3 & 2 & 2 & 4 \end{pmatrix}_{b_0}$
- [997] $1^{\overline{11}} 4 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 4 & 4 \\ 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$
- [998] $1^{\overline{11}} 41231^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[999] $1^{1^2} 4 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 0 & 3 & 2 \\ 4 & 3 & 8 & 6 \\ 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

[1000] $1^{1^1} 4 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 4 & 4 \\ 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

[1009] $1^{1^1} 1^3 2 3 21^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 12 & 0 & 70 & 18 \\ 2 & 3 & -1 & 18 & 4 \end{pmatrix}_{b_0}$$

[1007] $1^{1^4} 2 3 21^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 16 & 0 & 126 & 24 \\ 2 & 3 & -1 & 24 & 4 \end{pmatrix}_{b_0}$$

[1008] $1^{1^1} 4 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

[1014] $1^{1^1} 1^1 2 3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 6 & 6 \\ 2 & 3 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

[1013] $1^{1^1} 1^5 1^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 4 & 5 & 6 & 4 \\ 5 & 4 & 7 & 4 \\ 6 & 7 & 8 & 6 \\ 4 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

[1012] $1^{1^2} 2 3 21^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 8 & 0 & 30 & 12 \\ 2 & 3 & -1 & 12 & 4 \end{pmatrix}_{b_0}$$

[1011] $31^1 1^1 31^1 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 3 & 2 & 0 & 4 & 2 \\ 4 & 4 & 4 & 10 & 6 \\ 2 & 2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[1010] $1^{1^2} 1^2 3 21^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 6 & 2 \\ 6 & 0 & 6 & 34 & 12 \\ 2 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[1017] $1^{1^3} 1^5 1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 16 & 4 \\ 10 & 16 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0} \\
[1016] & 1^{\overline{11}} \overset{1^4}{5} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 16 & 4 \\ 14 & 16 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0} \\
[1015] & 1^{\overline{1}} \overset{1^1}{5} 1^{\overline{1}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 3 & 4 \\ 3 & 3 & 1 & 3 \\ 4 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[1023] & 41^{\overline{1}} \overset{1^1}{3} 22 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 3 & 4 & 2 \\ 3 & 5 & 8 & 8 & 5 \\ 3 & 8 & 4 & 8 & 4 \\ 4 & 8 & 8 & 10 & 6 \\ 2 & 5 & 4 & 6 & 3 \end{pmatrix}_{b_0} \\
[1022] & 1^{\overline{1}} \overset{1^4}{4} 131^{\overline{4}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 16 & 4 \\ 12 & 16 & 62 & 16 \\ 3 & 4 & 16 & 4 \end{pmatrix}_{b_0} \\
[1021] & 1^{\overline{13}} \overset{1^1}{4} 131^{\overline{3}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 12 & 4 \\ 9 & 12 & 34 & 12 \\ 3 & 4 & 12 & 4 \end{pmatrix}_{b_0} \\
[1020] & 1^{\overline{11}} \overset{1^2 31^3}{4} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 18 & 4 \\ 16 & 18 & 74 & 18 \\ 3 & 4 & 18 & 4 \end{pmatrix}_{b_0} \\
[1019] & 1^{\overline{12}} \overset{1^1}{4} 131^{\overline{2}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 8 & 4 \\ 6 & 8 & 14 & 8 \\ 3 & 4 & 8 & 4 \end{pmatrix}_{b_0} \\
[1018] & 1^{\overline{11}} \overset{1^1}{4} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 4 & 2 & 4 \\ 3 & 4 & 4 & 4 \end{pmatrix}_{b_0} \\
[1025] & 1^{\overline{12}} \overset{1^1 1^1}{2} 3 21^{\overline{3}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 9 & 2 \\ 6 & 0 & 9 & 49 & 15 \\ 2 & 0 & 2 & 15 & 4 \end{pmatrix}_{b_0} \\
[1037] & 21^{\overline{4}} \overset{1^4}{5} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 28 & 4 \\ 22 & 28 & 193 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0} \\
[1043] & 21^{\overline{1}} \overset{1^1}{5} 1^{\overline{11}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 2 & 2 & 3 & 2 \\ 2 & 0 & 3 & 3 & 2 \\ 2 & 3 & 3 & 4 & 3 \\ 3 & 3 & 4 & 4 & 4 \\ 2 & 2 & 3 & 4 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[1044] \quad 1 \overset{1^2}{\underset{1^1, 1^{11}}{5}} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 10 & 4 \\ 6 & 10 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1045] \quad 1 \overset{1}{\underset{1^1, 1^{11}}{5}} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1040] \quad 1 \overset{1^{11}}{1^1} 2^3 21^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 5 & 2 \\ 4 & 0 & 5 & 17 & 9 \\ 2 & 0 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[1039] \quad 21 \overset{1^2}{\underset{1^1}{5}} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 10 & 4 \\ 6 & 10 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1038] \quad 221 \overset{1^2}{\underset{1^1}{5}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 6 & 9 & 3 \\ 6 & 13 & 22 & 7 \\ 9 & 22 & 36 & 12 \\ 3 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1026] \quad 1 \overset{1^3}{\underset{1^2, 1^1}{5}} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 19 & 4 \\ 12 & 19 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1027] \quad 21 \overset{1^2}{\underset{1^1}{4}} 5 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 22 & 4 \\ 18 & 22 & 117 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1028] \quad 2221 \overset{1^1}{\underset{1^1}{5}} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 17 & 0 & 7 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 0 & 7 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[1029] \quad 221 \overset{1^1}{\underset{1^1}{4}} 5 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 28 & 4 \\ 22 & 28 & 193 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1030] \quad 21 \overset{1^2}{\underset{1^1}{4}} 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 4 & 13 & 4 \\ 9 & 13 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1031] \quad 1 \overset{1^4}{\underset{1^4, 1^{14}}{5}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 28 & 4 \\ 22 & 28 & 193 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1024] \quad 1 \overset{1^4}{\underset{1^2, 1^1}{4}} 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$\begin{aligned}
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 13 & 4 \\ 12 & 13 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0} \\
[1032] & 1^{11} 21^5 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 4 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[1033] & 1^{11} 1^{14} 5^{12} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 22 & 4 \\ 18 & 22 & 117 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0} \\
[1034] & 1^{221} 5^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 3 & 2 & 3 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[1041] & 1^{12} 2^3 3^{21} 1^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 49 & 15 \\ 2 & 3 & -1 & 15 & 4 \end{pmatrix}_{b_0} \\
[1035] & 1^{11} 1^5 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 4 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[1036] & 1^{221} 5^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[1042] & 1^1 1^5 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[1048] & 1^1 1^4 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.33 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 0 & 2 & 2 \\ 3 & 2 & 2 & 4 \\ 3 & 2 & 4 & 4 \end{pmatrix}_{b_0} \\
[1047] & 1^{21} 1^5 1^{11} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 0 & 3 & 2 & 2 \\ 2 & 3 & 3 & 2 & 3 \\ 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[1046] & 1^{12} 1^5 1^{11} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 4 & 7 & 3 \\ 4 & 4 & 10 & 4 \\ 7 & 10 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0} \\
[1059] & 1^{14} 21^5 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.43 \rightarrow 13, T_{\max} = 20, T \in [13, 20]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 20 & 4 \\ 12 & 20 & 90 & 18 \\ 3 & 4 & 18 & 2 \end{pmatrix}_{b_0}$$

[1058] $221\overset{1}{5}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 8 & 4 \\ 4 & 8 & 10 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$

[1049] $1^{12}2321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$

[1050] $1\overset{1}{5}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 8 & 4 \\ 4 & 8 & 10 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$

[1051] $1^{13}1^141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 7 & 4 \\ 6 & 7 & 9 & 7 \\ 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$

[1052] $1\overset{1}{5}12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 8 & 4 \\ 4 & 8 & 10 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$

[1053] $221\overset{1}{5}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 1 & 2 \\ 3 & 3 & 3 & 2 & 3 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$

[1054] $1^{11}231^{13}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 1 & 7 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 7 & 0 & 14 & 8 \\ 2 & 4 & -1 & 8 & 4 \end{pmatrix}_{b_0}$

[1055] $1\overset{1}{3}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$

[1056] $1\overset{1}{5}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 20 & 4 \\ 12 & 20 & 90 & 18 \\ 3 & 4 & 18 & 2 \end{pmatrix}_{b_0}$

[1057] $1^{11}23131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 0 & 4 & 2 & 4 \\ 2 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$

[1067] $2\overset{1}{3}141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[1064] $1 \overset{1}{\underset{1^1, 1^1}{5}} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 7 & 4 \\ 4 & 7 & 8 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$

[1063] $31 \overset{31^1}{\underset{1^1}{1}} 4 1^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 2 & 5 & 2 \\ 3 & 4 & 4 & 6 & 4 \\ 2 & 4 & 0 & 5 & 2 \\ 5 & 6 & 5 & 8 & 6 \\ 2 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$

[1062] $21 \overset{1^3}{\underset{1^1}{4}} 5 1^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 25 & 4 \\ 20 & 25 & 152 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$

[1061] $2 \overset{1^1}{\underset{1^1}{3}} 1^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 3 & 2 \\ 3 & 2 & 3 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$

[1060] $21 \overset{1^1}{\underset{1^1}{5}} 1^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 7 & 4 \\ 4 & 7 & 8 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$

[1065] $31 \overset{1^1}{\underset{1^1}{3}} 1^{\textcolor{blue}{1}} 41^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 3 & 2 & 0 & 3 & 2 \\ 3 & 2 & 3 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$

[1066] $1 \overset{1^1}{\underset{1^4, 1^3}{5}} 1^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 25 & 4 \\ 20 & 25 & 152 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$

[1068] $3131 \overset{1^1}{\underset{1^1}{4}} 1^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 3 & 3 \\ 6 & 13 & 8 & 7 \\ 3 & 8 & 4 & 4 \\ 3 & 7 & 4 & 3 \end{pmatrix}_{b_0}$

[1074] $1 \overset{1^1}{\underset{1^1}{1}} 4 131^{\textcolor{blue}{2}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 9 & 4 \\ 7 & 9 & 17 & 9 \\ 3 & 4 & 9 & 4 \end{pmatrix}_{b_0}$

[1073] $21 \overset{1^1}{\underset{1^1}{4}} 1^{\textcolor{blue}{2}} 31^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 4 & 1 & 10 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 6 & 10 & 0 & 20 & 8 \\ 2 & 4 & 0 & 8 & 3 \end{pmatrix}_{b_0}$

[1072] $31 \overset{31^1}{\underset{1^1}{1}} 4 1^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 2 & 2 \\ 3 & 3 & 2 & 1 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$

[1071] $1 \overset{1^2}{\underset{1^2}{4}} 1^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 17 & 4 \\ 15 & 17 & 65 & 17 \\ 3 & 4 & 17 & 4 \end{pmatrix}_{b_0} \\
[1070] & 31^{1^1}31^{1^1}41^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 5 & 5 & 3 \\ 5 & 5 & 7 & 5 \\ 5 & 7 & 8 & 6 \\ 3 & 5 & 6 & 4 \end{pmatrix}_{b_0} \\
[1069] & 21^{1^1}4131^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 4 & 1 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 6 & 0 & 4 & 4 \\ 2 & 4 & 0 & 4 & 3 \end{pmatrix}_{b_0} \\
[1076] & 1^{1^1}1^{1^1}2321^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0} \\
[1075] & 2^{1^1}3^{1^1}1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 5 & 7 & 5 \\ 5 & 7 & 8 & 6 \\ 3 & 5 & 6 & 4 \end{pmatrix}_{b_0} \\
[1098] & 21^{2^4}51^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 22 & 4 \\ 16 & 22 & 117 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0} \\
[1095] & 1^{1^1}1^{1^1}32321^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 13 & 0 & 97 & 21 \\ 2 & 3 & -1 & 21 & 4 \end{pmatrix}_{b_0} \\
[1096] & 21^{1^1}51^{1^1}22 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 4 & 5 & 6 & 4 \\ 2 & 5 & 3 & 4 & 3 \\ 3 & 6 & 4 & 4 & 4 \\ 2 & 4 & 3 & 4 & 2 \end{pmatrix}_{b_0} \\
[1097] & 321^{1^1}41^{1^1}1^{1^1}3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 3 & 5 & 3 & 2 \\ 3 & 1 & 6 & 3 & 3 \\ 5 & 6 & 12 & 5 & 6 \\ 3 & 3 & 5 & 1 & 3 \\ 2 & 3 & 6 & 3 & 2 \end{pmatrix}_{b_0} \\
[1094] & 1^{1^1}1^{1^1}4131^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 5 & 1 & 25 & 9 \\ 2 & 2 & 0 & 9 & 3 \end{pmatrix}_{b_0} \\
[1099] & 31^{1^1}31^{1^1}341^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 5 & 19 & 3 \\ 5 & 5 & 26 & 5 \\ 19 & 26 & 122 & 22 \\ 3 & 5 & 22 & 3 \end{pmatrix}_{b_0} \\
[1085] & 1^{1^1}1^{1^1}4131^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 0 & 5 & 2 \\ 5 & 5 & 14 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0} \\
[1087] & 1^{1^1}51^{1^1}12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 13.37 \rightarrow 14, T_{\max} = 20, T \in [14, 20]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 12 & 4 \\ 6 & 12 & 25 & 9 \\ 3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[1077] \ 1^{11} \begin{smallmatrix} 1^4 \\ 1^3, 1^2 \end{smallmatrix} 5 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 19, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.26 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 19 & 4 \\ 16 & 19 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1078] \ 1^{11} 41231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 19, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 0 & 3 & 2 \\ 3 & 3 & 2 & 4 \\ 3 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1079] \ 22151^{11} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 19, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 13.37 \rightarrow 14, \ T_{\max} = 20, \ T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 12 & 4 \\ 6 & 12 & 25 & 9 \\ 3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[1080] \ 1^{11} 4 \ 1^{11} 31^{11} 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 19, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 3 & 1 & 13 & 7 \\ 2 & 2 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1081] \ 1^{11} 4 \ 131^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 19, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 0 & 3 & 2 \\ 3 & 3 & 2 & 4 \\ 3 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1082] \ 1^{11} 5 \ 1^{11} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 19, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.26 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 10 & 4 \\ 8 & 10 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1083] \ 1^{12} 5 \ 1^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 19, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.26 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 16 & 4 \\ 10 & 16 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1084] \ 1^{12} 5 \ 1^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 19, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.26 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 13 & 4 \\ 8 & 13 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1088] \ 1^{11} 4 \ 1^{11} 31^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 19, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 2 & 1 & 5 & 5 \\ 2 & 2 & 0 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1089] \ 1^{11} 4 \ 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 19, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 0 & 3 & 2 \\ 5 & 3 & 8 & 6 \\ 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1090] \ 1^{11} 5 \ 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 19, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 13.37 \rightarrow 14, \ T_{\max} = 20, \ T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 12 & 4 \\ 6 & 12 & 25 & 9 \\ 3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

[1091] $1^{\overline{1}^3} 4 131^{\overline{1}^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 19$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 5 & 2 \\ 7 & 5 & 24 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$

[1092] $1^{\overline{1}^1} 4 131^{\overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 19$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 2 & 1 & 1 & 3 \\ 2 & 2 & 0 & 3 & 3 \end{pmatrix}_{b_0}$

[1093] $1^{\overline{1}^1} 5^{\overline{1}^4} 1^{\overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -100$, $\det = 19$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 16 & 4 \\ 14 & 16 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$

[1086] $1^{\overline{1}^1} 41231^{\overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 19$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 2 & 1 & 1 & 3 \\ 2 & 2 & 0 & 3 & 3 \end{pmatrix}_{b_0}$

[1100] $21^{\overline{1}^1} 4131^{\overline{1}^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 8 & 4 \\ 6 & 8 & 12 & 8 \\ 3 & 4 & 8 & 4 \end{pmatrix}_{b_0}$

[1106] $221^{\overline{1}^1} 51^{\overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -100$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 6 & 2 & 3 \\ 6 & 13 & 5 & 7 \\ 2 & 5 & 1 & 3 \\ 3 & 7 & 3 & 3 \end{pmatrix}_{b_0}$

[1105] $1^{\overline{1}^2} 2^{\overline{1}^2} 3 21^{\overline{1}^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 10 & 2 \\ 8 & 0 & 10 & 74 & 18 \\ 2 & 0 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

[1104] $1^{\overline{1}^1} 31^{\overline{1}^1} 4 1^{\overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 3 & 0 & 4 & 4 \\ 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$

[1103] $41^{\overline{1}^1} 231^{\overline{1}^1} 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 13$, $\Delta = -93$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 4 & 3 & 4 & 2 \\ 4 & 10 & 5 & 12 & 6 \\ 3 & 5 & 0 & 4 & 2 \\ 4 & 12 & 4 & 10 & 6 \\ 2 & 6 & 2 & 6 & 2 \end{pmatrix}_{b_0}$

[1102] $1^{\overline{1}^1} 31^{\overline{1}^1} 4 1^{\overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 0 & 2 & 2 \\ 4 & 2 & 2 & 4 \\ 3 & 2 & 4 & 4 \end{pmatrix}_{b_0}$

[1101] $1^{\overline{1}^1} 31^{\overline{1}^1} 4 1^{\overline{1}^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 5 & 0 & 20 & 8 \\ 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$

[1115] $1^{\overline{1}^1} 5^{\overline{1}^3} 1^{\overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -100$, $\det = 21$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 10 & 4 \\ 9 & 10 & 19 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1116] \ 1 \overset{1}{\underset{1^1, 1^1}{5}} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 10 & 4 \\ 6 & 10 & 19 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1107] \ 31 \overset{1}{\underset{1^1, 1^1}{3}} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 7 & 3 \\ 6 & 13 & 14 & 7 \\ 7 & 14 & 14 & 8 \\ 3 & 7 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[1108] \ 1 \overset{1}{\underset{1^{11}}{5}} 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 7 & 4 \\ 5 & 7 & 8 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1109] \ 41 \overset{12}{\underset{1^1}{2}} 2231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 1 & 11 & 2 \\ 4 & 26 & 1 & 56 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 56 & 0 & 110 & 20 \\ 2 & 10 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[1110] \ 21 \overset{14}{\underset{1^1}{3}} 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 25 & 4 \\ 19 & 25 & 152 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[1111] \ 21 \overset{13}{\underset{1^1}{1}} 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 16 & 4 \\ 11 & 16 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1112] \ 1 \overset{14}{\underset{1^3, 1^3}{11}} 5 \overset{1}{\underset{1^3, 1^3}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 22 & 4 \\ 18 & 22 & 115 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1113] \ 21 \overset{14}{\underset{1^1}{3}} 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 16 & 4 \\ 13 & 16 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1114] \ 1 \overset{13}{\underset{1^3, 1^1}{11}} 5 \overset{1}{\underset{1^3, 1^1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 16 & 4 \\ 13 & 16 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1125] \ 1 \overset{11}{\underset{1^1}{1}} \overset{11}{\underset{1^1}{3}} 4 \overset{1}{\underset{1^1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 4 & 0 & 10 & 6 \\ 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1124] \ 1 \overset{11}{\underset{1^1}{1}} \overset{11}{\underset{1^1}{3}} 4 \overset{1}{\underset{1^2}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 0 & 4 & 2 \\ 6 & 4 & 14 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1123] \ 31 \overset{11}{\underset{1^1}{1}} 3141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 3 & 2 & 0 & 3 & 2 \\ 3 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1122] \ 1^{11} 4^{12} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 15 & 4 \\ 13 & 15 & 49 & 15 \\ 3 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1121] \ 1^{12} 4^{13} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 0 & 4 & 2 \\ 6 & 4 & 14 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1120] \ 1^{12} 4^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 11 & 4 \\ 9 & 11 & 25 & 11 \\ 3 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1119] \ 1^{11} 4^{12} 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 4 & 0 & 10 & 6 \\ 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1118] \ 31^{11} 31^{14} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 15 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 3 & 2 & 0 & 15 & 2 \\ 15 & 9 & 15 & 121 & 19 \\ 2 & 2 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[1117] \ 1^{11} 4^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 0 & 4 & 2 \\ 4 & 4 & 6 & 6 \\ 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1126] \ 1^{11} 5^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 13 & 4 \\ 11 & 13 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1133] \ 21^{11} 4^{13} 31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 14 & 4 \\ 12 & 14 & 42 & 14 \\ 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1127] \ 21^{13} 5^{12} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 17 & 4 \\ 10 & 17 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1128] \ 1^{11} 4^{12} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 12 & 4 \\ 10 & 12 & 30 & 12 \\ 3 & 4 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1129] \ 1^{13} 5^{12} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 17 & 4 \\ 10 & 17 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1130] \ 1^{12} 4^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 14 & 4 \\ 12 & 14 & 42 & 14 \\ 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1131] \quad 1 \overset{1^1}{\underset{1^1, 1^{12}}{5}} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 11 & 4 \\ 6 & 11 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1132] \quad 21 \overset{1^1}{\underset{1^1}{5}} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 11 & 4 \\ 6 & 11 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1134] \quad 1 \overset{1^{11}}{\underset{1^1}{2}} 4 131^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 13 & 4 \\ 10 & 13 & 37 & 13 \\ 3 & 4 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[1135] \quad 1^4 1^4 1^{23} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 0 & 3 & 2 \\ 5 & 3 & 6 & 6 \\ 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1136] \quad 1 \overset{1^1}{\underset{1^1}{1}} 4 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 3 & 1 & 5 & 5 \\ 2 & 2 & 0 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1137] \quad 1 \overset{1^2}{\underset{1^1}{1}} 4 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 4 & 1 & 13 & 7 \\ 2 & 2 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1138] \quad 1 \overset{1^{11}}{\underset{1^1}{1}} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 0 & 3 & 2 \\ 5 & 3 & 6 & 6 \\ 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1139] \quad 2 \overset{1^{11}}{\underset{1^1}{3}} 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1145] \quad 21 \overset{1^2}{\underset{1^1}{5}} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 13 & 4 \\ 9 & 13 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1140] \quad 221 \overset{1^1}{\underset{1^1}{5}} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 10 & 4 \\ 7 & 10 & 19 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1141] \quad 1 \overset{1^3}{\underset{1^3, 1^3}{5}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 22 & 4 \\ 17 & 22 & 115 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[1142] $1^{11} \overset{1^3}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 19 & 4 \\ 15 & 19 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[1143] $1^{11} 231^1 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 7 & 3 \\ 4 & 0 & 7 & 14 & 8 \\ 2 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0}$

[1151] $21^3 \overset{1^2}{5} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 19 & 4 \\ 15 & 19 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[1150] $21^3 \overset{1^3}{5} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 22 & 4 \\ 17 & 22 & 115 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$

[1149] $1^{12} 2321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 6 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$

[1148] $1^{11} \overset{1^1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 10 & 4 \\ 7 & 10 & 19 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$

[1147] $1^{11} \overset{1^2}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 10 & 4 \\ 8 & 10 & 19 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$

[1146] $221^3 \overset{1^1}{5} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 22 & 4 \\ 17 & 22 & 115 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$

[1144] $21^1 \overset{1^1}{5} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 10 & 4 \\ 7 & 10 & 19 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$

[1153] $1 \overset{1^2}{\underset{1^1, 1^{12}}{5}} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 14 & 4 \\ 8 & 14 & 38 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$

[1152] $21^1 \overset{1^2}{\underset{1^1}{5}} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 14 & 4 \\ 8 & 14 & 38 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$

[1163] $2 \overset{1^{12} 41^1}{3} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$

$$\begin{aligned}
& T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 5 & 6 & 5 & 20 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[1162] & 1^{\overline{11}} \overset{1^3}{5} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -100, \quad \det = 31, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.16 \rightarrow 12, \quad T_{\max} = 20, \quad T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 16 & 4 \\ 13 & 16 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0} \\
[1161] & 21^{\overline{12}} 4131^{\overline{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 31, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 4 & 1 & 12 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 7 & 12 & 3 & 31 & 11 \\ 2 & 4 & 0 & 11 & 3 \end{pmatrix}_{b_0} \\
[1160] & 21^{\overline{12}} 41^{\overline{13}} 131^{\overline{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 31, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 1 & 14 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 9 & 14 & 3 & 41 & 13 \\ 2 & 4 & 0 & 13 & 3 \end{pmatrix}_{b_0} \\
[1159] & 31^{\overline{11}} 4 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, \quad T_{\max} = 12, \quad \Delta = -92, \quad \det = 31, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.16 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 2 & 9 & 2 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 0 & 7 & 2 \\ 9 & 12 & 7 & 28 & 10 \\ 2 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0} \\
[1158] & 1^{\overline{11}} \overset{1^2}{5} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -100, \quad \det = 31, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.16 \rightarrow 12, \quad T_{\max} = 20, \quad T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 13 & 4 \\ 10 & 13 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0} \\
[1157] & 21^{\overline{11}} 41^{\overline{13}} 131^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 31, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.65 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 4 & 1 & 8 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 8 & 1 & 11 & 7 \\ 2 & 4 & 0 & 7 & 3 \end{pmatrix}_{b_0} \\
[1156] & 21^{\overline{2}} \overset{1^1}{5} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -100, \quad \det = 31, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.16 \rightarrow 12, \quad T_{\max} = 20, \quad T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 13 & 4 \\ 10 & 13 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0} \\
[1155] & 21^{\overline{2}} \overset{1^3}{5} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -100, \quad \det = 31, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.16 \rightarrow 12, \quad T_{\max} = 20, \quad T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 19 & 4 \\ 14 & 19 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0} \\
[1154] & 21^{\overline{11}} 4131^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 31, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.65 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 4 & 1 & 6 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 6 & 1 & 5 & 5 \\ 2 & 4 & 0 & 5 & 3 \end{pmatrix}_{b_0} \\
[1165] & 21^{\overline{4}} \overset{1^1}{5} 1^{\overline{13}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -100, \quad \det = 32, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.50 \rightarrow 12, \quad T_{\max} = 20, \quad T \in [12, 20] \\
& \begin{pmatrix} 1 & 4 & 25 & 3 \\ 4 & 4 & 36 & 4 \\ 25 & 36 & 281 & 33 \\ 3 & 4 & 33 & 3 \end{pmatrix}_{b_0} \\
[1166] & 1^{\overline{1}} \overset{1^1}{21} 5 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, \quad T_{\max} = 20, \quad \Delta = -100, \quad \det = 32, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.50 \rightarrow 12, \quad T_{\max} = 20, \quad T \in [12, 20] \\
& \begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[1167] \ 21^{13}_{14} 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 25 & 3 \\ 4 & 4 & 36 & 4 \\ 25 & 36 & 281 & 33 \\ 3 & 4 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[1168] \ 21^{11} 41^{13} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 4 & 1 & 8 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 8 & 0 & 10 & 6 \\ 2 & 4 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1169] \ 2^{11} 3^{14} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 4 & 4 & 4 & 10 & 6 \\ 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1164] \ 21^1 51^{11} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1170] \ 1^{12} 2^1 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[1172] \ 1^{13} 4^{13} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 24 & 4 \\ 21 & 24 & 130 & 24 \\ 3 & 4 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[1173] \ 1^{13} 4^{13} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 15 & 4 \\ 12 & 15 & 49 & 15 \\ 3 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1174] \ 1^{12} 4^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 25 & 4 \\ 22 & 25 & 141 & 25 \\ 3 & 4 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[1175] \ 21^{11} 41^{13} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 10 & 4 \\ 8 & 10 & 18 & 10 \\ 3 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1171] \ 1^{14}_{1^1, 1^1} 5 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 23 & 4 \\ 14 & 23 & 117 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1176] \ 21^{12}_{1^2} 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 16 & 3 \\ 4 & 4 & 23 & 4 \\ 16 & 23 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1177] \ 1^{11}_{1^2, 1^2} 5^{12} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 16 & 4 \\ 12 & 16 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0} \\
[1178] \quad & 31^{1^1 1^1 1^1 1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 3 & 4 & 2 \\ 3 & 4 & 4 & 5 & 4 \\ 3 & 4 & 0 & 4 & 2 \\ 4 & 5 & 4 & 5 & 5 \\ 2 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0} \\
[1179] \quad & 1^{1^2} 2^3 21^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0} \\
[1180] \quad & 1^3 1^2 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 17 & 4 \\ 12 & 17 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0} \\
[1181] \quad & 21^{2^1} 51^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.14 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 16 & 4 \\ 12 & 16 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0} \\
[1182] \quad & 221^{1^1} 51^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.14 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 16 & 4 \\ 12 & 16 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0} \\
[1183] \quad & 1^1 1^1 1^1 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 4 & 7 & 4 \\ 5 & 7 & 8 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0} \\
[1190] \quad & 21^{1^1} 51^{1^1} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 4 & 7 & 3 \\ 4 & 4 & 10 & 4 \\ 7 & 10 & 19 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0} \\
[1189] \quad & 21^{1^1} 41^2 31^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 12 & 4 \\ 10 & 12 & 28 & 12 \\ 3 & 4 & 12 & 4 \end{pmatrix}_{b_0} \\
[1184] \quad & 141^{1^1 1^1 1^1} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 3 & 3 & 1 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[1185] \quad & 2^3 1^{1^1} 41^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 5 & 3 & 6 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0} \\
[1186] \quad & 1^{1^3} 4^4 131^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 17 & 4 \\ 13 & 17 & 65 & 17 \\ 3 & 4 & 17 & 4 \end{pmatrix}_{b_0} \\
[1187] \quad & 21^{1^1} 51^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0
\end{aligned}$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 7 & 3 \\ 4 & 4 & 10 & 4 \\ 7 & 10 & 19 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1188] \ 2 \overset{11}{3} 141^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 5 & 7 & 6 & 25 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1200] \ 1^{11} 41^2 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 7 & 6 & 0 & 32 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1197] \ 1 \overset{1}{5} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 15 & 4 \\ 8 & 15 & 40 & 12 \\ 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[1191] \ 1^4 1^{23} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 5 & 2 \\ 8 & 5 & 24 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1192] \ 1^{11} 4 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 4 & 3 & 2 & 8 & 6 \\ 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1193] \ 21 \overset{1}{5} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 15 & 4 \\ 8 & 15 & 40 & 12 \\ 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[1194] \ 1^{11} 4 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 7 & 6 & 0 & 32 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1195] \ 1^{11} 4 \overset{12}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 5 & 2 \\ 8 & 5 & 24 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1198] \ 1^4 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 5 & 2 \\ 8 & 5 & 24 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1202] \ 31^{11} 31^2 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 22 & 3 \\ 5 & 5 & 31 & 5 \\ 22 & 31 & 170 & 26 \\ 3 & 5 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[1203] \ 1^{11} 4 \overset{12}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 7 & 6 & 0 & 32 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[1204] $1^{1^1 31^1} 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 6 & 0 & 20 & 8 \\ 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$

[1205] $1^{1^2 31^1} 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 6 & 7 & 0 & 32 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$

[1206] $1^{1^4} 4^{131^{13}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 7 & 2 \\ 10 & 7 & 48 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$

[1199] $1^{1^1 31^2} 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 6 & 5 & 0 & 20 & 8 \\ 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$

[1201] $31^{1^1} 4^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 2 & 2 & 2 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$

[1196] $1^{1^{12} 31^1} 4^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 7 & 2 \\ 10 & 7 & 48 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$

[1213] $1^4 41^{14} 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 3 & 1 & 16 & 2 \\ 3 & 4 & 1 & 32 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 16 & 32 & 0 & 190 & 24 \\ 2 & 4 & 0 & 24 & 3 \end{pmatrix}_{b_0}$

[1211] $1^3 41^{13} 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 3 & 1 & 12 & 2 \\ 3 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 12 & 24 & 0 & 106 & 18 \\ 2 & 4 & 0 & 18 & 3 \end{pmatrix}_{b_0}$

[1212] $2^3 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[1210] $1^2 41^{12} 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 3 & 1 & 8 & 2 \\ 3 & 4 & 1 & 16 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 8 & 16 & 0 & 46 & 12 \\ 2 & 4 & 0 & 12 & 3 \end{pmatrix}_{b_0}$

[1214] $1^4 41^{11} 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 3 & 1 & 6 & 2 \\ 3 & 4 & 1 & 8 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 8 & 2 & 14 & 8 \\ 2 & 4 & 0 & 8 & 3 \end{pmatrix}_{b_0}$

[1215] $1^2 41^{12} 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 1 & 12 & 2 \\ 3 & 4 & 1 & 16 & 4 \\ 1 & 1 & -2 & 4 & 0 \\ 12 & 16 & 4 & 62 & 16 \\ 2 & 4 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1216] $1^{\overline{1}1131^2} 4^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 4 & 2 \\ 7 & 4 & 14 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

[1217] $1^{\overline{1}} \underset{1^1, 1^1}{5} 1^{\overline{1}2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 14 & 4 \\ 8 & 14 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[1218] $221^{\overline{1}2} \underset{1^3}{5} 1^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 6 & 16 & 3 \\ 6 & 13 & 39 & 7 \\ 16 & 39 & 113 & 21 \\ 3 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[1219] $1^{\overline{1}} \underset{1^{\overline{1}2}}{5} 1^{\overline{1}2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 11 & 4 \\ 7 & 11 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[1220] $1^{\overline{1}1131^1} 4^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 3 & 1 & 5 & 5 \\ 2 & 2 & 0 & 5 & 3 \end{pmatrix}_{b_0}$$

[1208] $1^{\overline{1}} 41^{\overline{1}1} \underset{1^{\overline{1}2}}{231} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 1 & 4 & 2 \\ 3 & 4 & 1 & 8 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 8 & 0 & 10 & 6 \\ 2 & 4 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[1207] $1^{\overline{1}1131^1} 4^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 0 & 4 & 2 \\ 5 & 4 & 6 & 6 \\ 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

[1209] $1^{\overline{1}1131^1} 4^{\overline{1}1} 1^{\overline{1}4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 8 & 7 & 1 & 53 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[1221] $1^{\overline{1}1131^1} 4^{\overline{1}1} 1^{\overline{1}2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 6 & 2 \\ 7 & 6 & 22 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

[1222] $1^{\overline{1}3} \underset{1^1, 1^1}{5} 1^{\overline{1}2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 20 & 4 \\ 12 & 20 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[1223] $1^{\overline{1}3} 4131^{\overline{1}3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 6 & 2 \\ 9 & 6 & 34 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[1224] $1^{1^4 3 1^1} 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 19 & 4 \\ 15 & 19 & 81 & 19 \\ 3 & 4 & 19 & 4 \end{pmatrix}_{b_0}$

[1225] $1^{1^2 1^3} 4 1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 7 & 1 & 37 & 11 \\ 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$

[1226] $1^{1^2 1^3} 4 1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 5 & 1 & 13 & 7 \\ 2 & 2 & 0 & 7 & 3 \end{pmatrix}_{b_0}$

[1227] $3 1^{1^3} 1 4 1^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 3 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$

[1235] $1^{1^1} 5 1^{1^4} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 20 & 4 \\ 16 & 20 & 90 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[1228] $1^{1^2 1^3 3 1^1} 4 \ 1^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 16 & 4 \\ 13 & 16 & 54 & 16 \\ 3 & 4 & 16 & 4 \end{pmatrix}_{b_0}$

[1236] $1^{1^4} 5 1^{1^2} 1^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 26 & 4 \\ 16 & 26 & 150 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$

[1229] $1^{1^2} 5 1^{1^2} 1^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 17 & 4 \\ 10 & 17 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$

[1230] $2 \ 3 \ 1 4 1^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$

[1231] $1^{1^3 3 1^3} 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 21 & 4 \\ 18 & 21 & 97 & 21 \\ 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$

[1232] $2 \ 3 \ 1 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$

[1233] $1^{1^3 3 1^2} 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 18 & 4 \\ 15 & 18 & 70 & 18 \\ 3 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1234] \ 1 \overset{1^4}{\underset{1^4,1^1}{5}} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 46, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 32 & 4 \\ 20 & 32 & 234 & 30 \\ 3 & 4 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[1237] \ 1 \overset{1^3}{\underset{1^4}{11}} 4 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 47, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 5 & 2 & 28 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1238] \ 2 \overset{1^{11}}{\underset{1^4}{3}} 141^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 47, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1239] \ 1 \overset{1^2}{\underset{1^3}{11}} 4 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 47, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 5 & 4 & 2 & 16 & 8 \\ 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[1240] \ 1 \overset{1^3}{\underset{1^4}{11}} 4 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 47, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 7 & 2 \\ 8 & 7 & 32 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1241] \ 1 \overset{1^4}{\underset{1^4}{11}} 41^{11} 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 47, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 0 & 5 & 2 \\ 6 & 5 & 12 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1242] \ 1 \overset{1^1}{\underset{1^4}{11}} 4 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 47, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 0 & 5 & 2 \\ 6 & 5 & 12 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1243] \ 1 \overset{1^1}{\underset{1^4}{11}} \overset{1^1}{\underset{1^4}{11}} 4 \overset{1^1}{\underset{1^4}{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 47, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 0 & 5 & 2 \\ 6 & 5 & 12 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1245] \ 1 \overset{1^2}{\underset{1^4}{11}} 231^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -65, \ \det = 48, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 2 & 1 & 11 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 11 & 0 & 32 & 12 \\ 2 & 4 & -1 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1244] \ 1 \overset{1^2}{\underset{1^4}{11}} 4 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 48, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.75 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 6 & 1 & 23 & 9 \\ 2 & 2 & 0 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1246] \ 21 \overset{1^3}{\underset{1^4}{11}} 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 48, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.33 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 14 & 3 \\ 4 & 4 & 20 & 4 \\ 14 & 20 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1247] \ 1^1 41^{11} 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 48, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 4 & 2 \\ 7 & 4 & 12 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1248] \ 1^{1^{12}31^1} 4 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 4 & 2 \\ 7 & 4 & 12 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1249] \ 1^{1^{12}31^3} 4 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 22 & 4 \\ 19 & 22 & 106 & 22 \\ 3 & 4 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[1250] \ 1^{1^{12}2321^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 5 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1251] \ 1^{1^{12}2321^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 7 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1252] \ 1^{1^{13}2321^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 10 & 3 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 10 & 6 & 34 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1253] \ 1^{1^{13}2321^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 10 & 3 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 10 & 6 & 34 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1254] \ 1^{1^4} 5 \ 1^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 29 & 4 \\ 18 & 29 & 189 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1255] \ 31^{1^1 31^1 41^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 3 & 2 & 0 & 6 & 2 \\ 6 & 3 & 6 & 13 & 7 \\ 2 & 2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1256] \ 31^{1^1 31^3 41^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 3 & 2 & 0 & 12 & 2 \\ 12 & 7 & 12 & 73 & 15 \\ 2 & 2 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[1257] \ 21^1 5 \ 1^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 18 & 4 \\ 10 & 18 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1258] \ 21^1 5 \ 1^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 24 & 4 \\ 14 & 24 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1259] \ 1 \overset{1^2}{\underset{1^1, 1^3}{5}} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 51, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 18 & 4 \\ 10 & 18 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1260] \ 1 \overset{1^4}{\underset{1^1, 1^3}{5}} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 51, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 24 & 4 \\ 14 & 24 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1262] \ 1 \overset{1^{11} 31^1}{1^1} 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 52, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.69 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 4 & 1 & 11 & 7 \\ 2 & 2 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1261] \ 1 \overset{1^2}{1^1} 4 1 31 \overset{1^2}{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 52, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.69 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 6 & 2 \\ 7 & 6 & 20 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1265] \ 1 \overset{1^{12} 31^1}{1^1} 4 \ 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 52, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.69 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 6 & 2 \\ 9 & 6 & 32 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1264] \ 1 \overset{1^1}{1^1} 41 \overset{1^1}{1^1} 231 \overset{1^1}{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 52, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.69 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 4 & 1 & 11 & 7 \\ 2 & 2 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1263] \ 1 \overset{1^{11} 31^1}{1^1} 4 \ 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 52, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.69 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 6 & 1 & 35 & 11 \\ 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1270] \ 1 \overset{1^3 31^2}{1^2} 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 54, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 19 & 4 \\ 16 & 19 & 77 & 19 \\ 3 & 4 & 19 & 4 \end{pmatrix}_{b_0}$$

$$[1269] \ 1 \overset{1^1}{1^1} 5 \overset{1^2}{1^2} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 54, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.37 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 14 & 4 \\ 10 & 14 & 38 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1268] \ 1 \overset{1^3}{1^2, 1^1} 5 \overset{1^2}{1^2} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 54, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.37 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 23 & 4 \\ 14 & 23 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1267] \ 1 \overset{1^2}{1^2} 2 3 21 \overset{1^2}{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 54, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1266] \ 1 \overset{1^2}{1^2, 1^1} 5 \overset{1^2}{1^2} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 54, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 20 & 4 \\ 12 & 20 & 82 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1274] \quad 1 \overset{1^3}{\underset{1^1, 1^{13}}{5}} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 21 & 4 \\ 12 & 21 & 88 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1273] \quad 21 \overset{1^3}{\underset{1^1}{5}} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 21 & 4 \\ 12 & 21 & 88 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1272] \quad 1 \overset{1^{12} 3^1}{\underset{1^1}{4}} 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 5 & 2 \\ 8 & 5 & 20 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1271] \quad 31 \overset{1^1}{\underset{1^1}{3}} 141 \overset{1^1}{\underset{1^1}{4}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 4 & 3 \\ 5 & 5 & 5 & 5 \\ 4 & 5 & 2 & 4 \\ 3 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1283] \quad 1 \overset{1^1}{\underset{1^{12}}{5}} 1^3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.36 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 17 & 4 \\ 13 & 17 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1282] \quad 1 \overset{1^3}{\underset{1^3, 1^1}{5}} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.36 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 26 & 4 \\ 16 & 26 & 148 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[1284] \quad 221 \overset{1^1}{\underset{1^1}{5}} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.36 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 14 & 4 \\ 9 & 14 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1275] \quad 1 \overset{1^4}{\underset{1^1, 1^1}{12}} 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.36 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 14 & 4 \\ 12 & 14 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1281] \quad 21 \overset{1^4}{\underset{1^1}{5}} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.36 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 23 & 4 \\ 15 & 23 & 117 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1276] \quad 21 \overset{1^1}{\underset{1^1}{5}} 1^{12} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 4 & 8 & 4 \\ 6 & 8 & 10 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1280] \quad 21 \overset{1^1}{\underset{1^1}{5}} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.36 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 14 & 4 \\ 9 & 14 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[1279] $1^{12} \begin{smallmatrix} 1^1 \\ 5 \\ 1^1, 1^1 \end{smallmatrix} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 14 & 4 \\ 9 & 14 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$

[1278] $1 \begin{smallmatrix} 1^1 \\ 5 \\ 21^{12} \end{smallmatrix} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 4 & 8 & 4 \\ 6 & 8 & 10 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$

[1277] $1^{11} \begin{smallmatrix} 1^{11} 31^1 \\ 4 \end{smallmatrix} 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 5 & 1 & 21 & 9 \\ 2 & 2 & 0 & 9 & 3 \end{pmatrix}_{b_0}$

[1285] $31^{11} 3141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 3 & 2 & 0 & 4 & 2 \\ 4 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$

[1286] $31^{11} 31^2 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 3 & 2 & 0 & 9 & 2 \\ 9 & 5 & 9 & 37 & 11 \\ 2 & 2 & 2 & 11 & 2 \end{pmatrix}_{b_0}$

[1287] $31^{11} 31^1 41^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 3 & 2 & 0 & 9 & 2 \\ 9 & 9 & 9 & 49 & 13 \\ 2 & 2 & 2 & 13 & 2 \end{pmatrix}_{b_0}$

[1288] $1 \begin{smallmatrix} 1^1 \\ 5 \\ 1^1, 1^1 \end{smallmatrix} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 18 & 4 \\ 10 & 18 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$

[1289] $31^{11} 31^{11} 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 9 & 3 \\ 6 & 13 & 18 & 7 \\ 9 & 18 & 22 & 10 \\ 3 & 7 & 10 & 3 \end{pmatrix}_{b_0}$

[1290] $1 \begin{smallmatrix} 1^1 \\ 5 \\ 1^{13} \end{smallmatrix} 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 15 & 4 \\ 9 & 15 & 40 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$

[1291] $2 \begin{smallmatrix} 1^{14} 1^1 \\ 3 \end{smallmatrix} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 8 & 3 \\ 4 & 5 & 12 & 5 \\ 8 & 12 & 22 & 10 \\ 3 & 5 & 10 & 4 \end{pmatrix}_{b_0}$

[1292] $141^{11} 31^1 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 6 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 6 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$

[1295] $1^{12} \begin{smallmatrix} 1^3 \\ 5 \\ 1^1, 1^1 \end{smallmatrix} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 14 & 4 \\ 11 & 14 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0} \\
[1298] \quad & 21^{1^2 4 1^1 3 1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 4 & 1 & 14 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 6 & 14 & 0 & 32 & 10 \\ 2 & 4 & 0 & 10 & 3 \end{pmatrix}_{b_0} \\
[1297] \quad & 21^{1^2 4 1^1 3 1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 1 & 18 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 10 & 18 & 0 & 64 & 14 \\ 2 & 4 & 0 & 14 & 3 \end{pmatrix}_{b_0} \\
[1294] \quad & 21^{1^1 5 1^{1^2}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.31 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 17 & 4 \\ 11 & 17 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0} \\
[1293] \quad & 21^{1^1 5 1^{1^2}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.31 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 20 & 4 \\ 13 & 20 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0} \\
[1296] \quad & 1^{1^2} 5^{1^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.31 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 14 & 4 \\ 10 & 14 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0} \\
[1299] \quad & 221^{1^1 5 1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 6 & 7 & 3 \\ 6 & 13 & 17 & 7 \\ 7 & 17 & 19 & 9 \\ 3 & 7 & 9 & 3 \end{pmatrix}_{b_0} \\
[1300] \quad & 1^{1^3} 4^{1^1} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 20 & 4 \\ 16 & 20 & 86 & 20 \\ 3 & 4 & 20 & 4 \end{pmatrix}_{b_0} \\
[1301] \quad & 1^{1^4} 4^{1^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 22 & 4 \\ 18 & 22 & 106 & 22 \\ 3 & 4 & 22 & 4 \end{pmatrix}_{b_0} \\
[1309] \quad & 1^{1^1} 4^{1^1} 1^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 7 & 2 \\ 9 & 7 & 34 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0} \\
[1314] \quad & 1^{1^1} 4^{1^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 5 & 2 \\ 9 & 5 & 22 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0} \\
[1302] \quad & 3131^{1^1} 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 6 & 5 & 3 \\ 6 & 13 & 12 & 7 \\ 5 & 12 & 8 & 6 \\ 3 & 7 & 6 & 3 \end{pmatrix}_{b_0} \\
[1303] \quad & 1^{1^1} 4^{1^3} 131^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 7 & 5 & 3 & 33 & 11 \\ 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

[1313] $1^{11}4^{14}13^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 8 & 6 & 3 & 49 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[1312] $1^{13}4^{13}13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 9 & 1 & 53 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[1311] $1^{11}4^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 5 & 1 & 13 & 7 \\ 2 & 2 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

[1310] $1^{22}1^{11}5^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 6 & 5 & 3 \\ 6 & 13 & 12 & 7 \\ 5 & 12 & 8 & 6 \\ 3 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

[1304] $1^{13}4^{14}13^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 7 & 2 \\ 11 & 7 & 48 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

[1305] $1^{11}4^{12}23^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 7 & 2 \\ 9 & 7 & 34 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[1306] $1^{12}3^{11}4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 5 & 2 \\ 7 & 5 & 12 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

[1307] $1^{14}1^{12}23^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 5 & 2 \\ 9 & 5 & 22 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

[1308] $1^{13}4^{12}13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 8 & 1 & 37 & 11 \\ 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

[1317] $1^{11}4^{13}13^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 6 & 2 \\ 8 & 6 & 22 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

[1319] $1^{12}4^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 7 & 7 & 0 & 32 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[1318] $1^{12}1^{12}5^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 10 & 3 \\ 4 & 4 & 14 & 4 \\ 10 & 14 & 38 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1315] \ 1^{12} 4 \ 1^{31^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 9 & 9 & 0 & 64 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1316] \ 1^{11} 4 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 6 & 2 \\ 8 & 6 & 22 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1320] \ 1^{12} 5 \ 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 17 & 4 \\ 14 & 17 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1322] \ 21^{12} 4131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 4 & 1 & 12 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 12 & 1 & 23 & 9 \\ 2 & 4 & 0 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1321] \ 2 \ 3 \ 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 5 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[1323] \ 21^{12} 41^3 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 1 & 18 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 18 & 1 & 65 & 15 \\ 2 & 4 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1331] \ 1^{12} 4 \ 1^{2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 8 & 8 & 0 & 46 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1326] \ 1^{12} 4 \ 1^{4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 8 & 2 \\ 12 & 8 & 62 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[1325] \ 21^2 51^{12} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 14 & 3 \\ 4 & 4 & 20 & 4 \\ 14 & 20 & 82 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1324] \ 21^{12} 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 14 & 3 \\ 4 & 4 & 20 & 4 \\ 14 & 20 & 82 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1332] \ 1^{11} 41^1 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 4 & 2 & 14 & 8 \\ 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[1327] $1^{12}41^2231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 8 & 8 & 0 & 46 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$

[1328] $1^{12}4131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 5 & 5 & 2 & 16 & 8 \\ 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$

[1329] $1^{14}4131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 8 & 2 \\ 12 & 8 & 62 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$

[1330] $1^{11}1^{12}31^14^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 4 & 2 & 14 & 8 \\ 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$

[1335] $1^{14}51^44^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 24 & 3 \\ 3 & 4 & 28 & 4 \\ 24 & 28 & 174 & 28 \\ 3 & 4 & 28 & 4 \end{pmatrix}_{b_0}$

[1336] $1^{12}5^{1^2}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 17 & 4 \\ 12 & 17 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$

[1337] $21^251^{1^2}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 17 & 4 \\ 12 & 17 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$

[1333] $1^{12}5^{1^4}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 23 & 4 \\ 18 & 23 & 117 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$

[1334] $21^451^{1^2}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 23 & 4 \\ 18 & 23 & 117 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$

[1338] $1^{11}1^{11}31^24^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 10 & 8 & 1 & 69 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$

[1339] $1^{11}1^{11}31^24^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 5 & 1 & 21 & 9 \\ 2 & 2 & 0 & 9 & 3 \end{pmatrix}_{b_0}$

[1340] $31^{11}3141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 7 & 3 \\ 5 & 5 & 10 & 5 \\ 7 & 10 & 14 & 8 \\ 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[1341] $1^{1^{12}31^2} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 5 & 2 \\ 9 & 5 & 20 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

[1342] $1^{1^{12}31^1} 4^1 1^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 9 & 2 \\ 11 & 9 & 60 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1344] $1^{1^2} 5^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 21 & 4 \\ 12 & 21 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[1343] $15^1 1^{12} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 8 & 3 \\ 4 & 4 & 11 & 4 \\ 8 & 11 & 21 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[1345] $21^{12} 4131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 4 & 1 & 12 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 12 & 2 & 26 & 10 \\ 2 & 4 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[1346] $1^{1^4} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 25 & 4 \\ 21 & 25 & 137 & 25 \\ 3 & 4 & 25 & 4 \end{pmatrix}_{b_0}$$

[1347] $2^1 3^1 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 5 & 15 & 5 \\ 9 & 15 & 34 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

[1348] $1^{1^2} 5^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 20 & 4 \\ 16 & 20 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[1349] $21^{12} 41^2 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 1 & 16 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 16 & 2 & 50 & 14 \\ 2 & 4 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[1350] $1^{1^2} 5^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 17 & 4 \\ 13 & 17 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[1351] $1^{1^3} 2^1 3^1 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 6 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

[1352] $31^{1^1} 31^2 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$

$$\begin{aligned}
& T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 5 & 14 & 3 \\ 5 & 5 & 19 & 5 \\ 14 & 19 & 62 & 16 \\ 3 & 5 & 16 & 3 \end{pmatrix}_{b_0} \\
[1353] & 31^{1^1} 413 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 4 & 4 & 6 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 3 & 6 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[1354] & 1^{1^4} 5^{1^3} 1^{1^3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 27 & 4 \\ 16 & 27 & 152 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0} \\
[1357] & 41^{1^{12}} 2^3 3^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 4 & 2 & 8 & 2 \\ 4 & 10 & 5 & 20 & 6 \\ 2 & 5 & 0 & 10 & 2 \\ 8 & 20 & 10 & 38 & 12 \\ 2 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1358] & 1^{1^2} 5^{1^3} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.25 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 20 & 4 \\ 15 & 20 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0} \\
[1356] & 21^{1^2} 41^{1^2} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 1 & 16 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 8 & 16 & 0 & 46 & 12 \\ 2 & 4 & 0 & 12 & 3 \end{pmatrix}_{b_0} \\
[1355] & 21^{1^3} 5^{1^2} 1^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.25 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 20 & 4 \\ 15 & 20 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0} \\
[1359] & 1^{1^4} 5^{1^3} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 4 & 24 & 4 \\ 17 & 24 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0} \\
[1360] & 1^{1^4} 41^{1^2} 231^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 5 & 2 \\ 3 & 4 & 1 & 8 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 8 & 1 & 11 & 7 \\ 2 & 4 & 0 & 7 & 3 \end{pmatrix}_{b_0} \\
[1361] & 1^{1^3} 5^{1^2} 1^{1^3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 24 & 4 \\ 14 & 24 & 115 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0} \\
[1368] & 1^{1^2} 1^{1^2} 4^{1^4} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 9 & 1 & 69 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0} \\
[1372] & 21^{1^2} 5^{1^2} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.35 \rightarrow 12, T_{\max} = 20, T \in [12, 20]
\end{aligned}$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 17 & 4 \\ 12 & 17 & 57 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[1362] $1^{11} 4^{12} 13^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 7 & 2 \\ 9 & 7 & 30 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[1363] $21^{13} 5^{14} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 4 & 23 & 3 \\ 4 & 4 & 33 & 4 \\ 23 & 33 & 232 & 30 \\ 3 & 4 & 30 & 3 \end{pmatrix}_{b_0}$

[1364] $1^{11} 4^{14} 13^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 9 & 2 \\ 11 & 9 & 58 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$

[1365] $1^{11} 4^{11} 31^{12} 4^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 8 & 6 & 1 & 33 & 11 \\ 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$

[1366] $1^{13} 2^3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 9 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$

[1367] $1^{12} 4^{11} 31^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 6 & 1 & 21 & 9 \\ 2 & 2 & 0 & 9 & 3 \end{pmatrix}_{b_0}$

[1369] $1^{12} 31^{12} 4^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 7 & 2 \\ 11 & 7 & 44 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$

[1370] $1^{11} 4^{11} 31^{12} 4^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 7 & 1 & 49 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$

[1371] $1^{11} 41^2 231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 8 & 6 & 1 & 33 & 11 \\ 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$

[1384] $1^{12} 4^{14} 13^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 8 & 2 & 58 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$

[1375] $1^{12} 31^{12} 4^{12} 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 6 & 2 \\ 10 & 6 & 30 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[1378] $1^{1^4 3^1 4^4} 4^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 25 & 3 \\ 3 & 4 & 29 & 4 \\ 25 & 29 & 185 & 29 \\ 3 & 4 & 29 & 4 \end{pmatrix}_{b_0}$

[1382] $1^{1^1 1^2 3^1 1^1} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 6 & 2 \\ 8 & 6 & 18 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$

[1377] $1^{1^1 4^1 1^2 3^1 1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 6 & 2 \\ 8 & 6 & 18 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$

[1379] $1^{1^1 1^3 3^1 1^1} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 6 & 2 \\ 10 & 6 & 30 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[1376] $1^{1^1 1^2 3^1 1^1} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 7 & 5 & 2 & 24 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$

[1381] $1^{1^2 4^1 1^3 1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 6 & 2 & 26 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$

[1380] $1^{1^1 1^2 3^1 1^1} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 9 & 7 & 2 & 56 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$

[1383] $4^1 1^{1^2 2^3 1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 28 & 0 & 26 & 10 \\ 2 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$

[1373] $1^1 4^1 2^3 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 6 & 2 \\ 10 & 6 & 30 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[1374] $1^{1^3 4^1 3^1 2^2} 4^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 23 & 4 \\ 19 & 23 & 113 & 23 \\ 3 & 4 & 23 & 4 \end{pmatrix}_{b_0}$

[1385] $1^{1^3 5^1 1^1} 1^{1^1, 1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 18 & 4 \\ 11 & 18 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$

[1386] $4^1 1^1 3^1 2^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 4 & 3 & 3 & 2 \\ 4 & 10 & 9 & 8 & 6 \\ 3 & 9 & 4 & 7 & 4 \\ 3 & 8 & 7 & 5 & 5 \\ 2 & 6 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[1387] $221^{1^1}51^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 18 & 4 \\ 11 & 18 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$

[1388] $21^{1^1}51^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 18 & 4 \\ 11 & 18 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$

[1389] $1^{1^2}5^{1^2}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 20 & 4 \\ 14 & 20 & 82 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[1390] $221^{2^1}51^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 20 & 4 \\ 14 & 20 & 82 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[1391] $21^{2^1}51^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 20 & 4 \\ 14 & 20 & 82 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[1392] $21^{2^1}51^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 26 & 4 \\ 18 & 26 & 150 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$

[1393] $1^{1^2}5^{1^4}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 20 & 4 \\ 16 & 20 & 82 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[1395] $21^{1^3}41^{4^3}31^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 17 & 2 \\ 2 & 4 & 1 & 26 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 17 & 26 & 3 & 141 & 23 \\ 2 & 4 & 0 & 23 & 3 \end{pmatrix}_{b_0}$

[1396] $2^{1^1}3^{1^1}1^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 6 & 5 & 7 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[1394] $1^{1^2}2^321^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$

[1397] $1^{1^2}4^{1^3}131^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 7 & 7 & 2 & 40 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$

[1398] $1^{1^1}4^{1^3}1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 6 & 2 & 38 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[1399] $1^{11} 1^{12} 31^1 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 8 & 2 \\ 10 & 8 & 42 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

[1400] $1^1 1^{13} 31^1 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 8 & 2 \\ 12 & 8 & 58 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1401] $1^{11} 4^{13} 131^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 8 & 2 \\ 10 & 8 & 42 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

[1402] $1^{13} 2^3 21^{12} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

[1403] $1^{13} 1^{14} 31^3 4^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 26 & 4 \\ 22 & 26 & 146 & 26 \\ 3 & 4 & 26 & 4 \end{pmatrix}_{b_0}$$

[1404] $1^{12} 1^3 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 20 & 4 \\ 15 & 20 & 82 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[1405] $31^2 31^{11} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 16 & 3 \\ 6 & 13 & 32 & 7 \\ 16 & 32 & 74 & 18 \\ 3 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[1406] $3131^{11} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 8 & 3 \\ 6 & 13 & 20 & 7 \\ 8 & 20 & 26 & 10 \\ 3 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[1407] $21^2 5^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 23 & 4 \\ 16 & 23 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[1408] $1^{12} 1^5 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 24 & 4 \\ 14 & 24 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[1409] $31^{11} 31^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 9 & 3 \\ 5 & 5 & 12 & 5 \\ 9 & 12 & 22 & 10 \\ 3 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[1410] $1^1 5^1 1^{22} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 18 & 4 \\ 12 & 18 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[1411] $1^{12} 1^{11} 31^1 4^1 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 7 & 1 & 33 & 11 \\ 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

[1412] $1^{11} 1^{12} 31^1 4^1 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 7 & 2 \\ 9 & 7 & 28 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[1413] $1^{12} 1^{11} 31^1 4^1 1^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 8 & 8 & 1 & 49 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[1414] $1^{13} 31^1 4^1 1^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 7 & 2 \\ 11 & 7 & 42 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

[1417] $21^{12} 41^{13} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 1 & 14 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 14 & 2 & 36 & 12 \\ 2 & 4 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[1415] $21^{12} 51^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 10 & 3 \\ 4 & 4 & 14 & 4 \\ 10 & 14 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[1416] $21^{11} 51^{12} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 10 & 3 \\ 4 & 4 & 14 & 4 \\ 10 & 14 & 36 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[1420] $1^{12} 5^{14} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 23 & 4 \\ 18 & 23 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[1419] $1^{12} 5^{14} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 26 & 4 \\ 20 & 26 & 150 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[1418] $21^{12} 5^{14} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 26 & 4 \\ 20 & 26 & 150 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[1422] $2^{11} 3^{14} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 11 & 3 \\ 4 & 5 & 17 & 5 \\ 11 & 17 & 46 & 14 \\ 3 & 5 & 14 & 4 \end{pmatrix}_{b_0}$$

[1425] $21^{13} 4131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 1 & 18 & 4 \\ 1 & 1 & -2 & 4 & 0 \\ 10 & 18 & 4 & 68 & 16 \\ 2 & 4 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1426] \quad 41 \overset{1}{1^1} 2^3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 2 & 4 & 2 \\ 4 & 10 & 5 & 12 & 6 \\ 2 & 5 & 0 & 4 & 2 \\ 4 & 12 & 4 & 10 & 6 \\ 2 & 6 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1427] \quad 21 \overset{1}{5} 1^{13} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 4 & 12 & 4 \\ 9 & 12 & 25 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1428] \quad 21 \overset{1}{3} \overset{1}{5} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 23 & 4 \\ 17 & 23 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1424] \quad 1^{12} \overset{1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 23 & 4 \\ 17 & 23 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1421] \quad 21 \overset{1}{1^3} 41^2 31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 4 & 1 & 22 & 4 \\ 1 & 1 & -2 & 4 & 0 \\ 14 & 22 & 4 & 100 & 20 \\ 2 & 4 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[1423] \quad 1 \overset{1}{5} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 4 & 12 & 4 \\ 9 & 12 & 25 & 9 \\ 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1429] \quad 2 \overset{1}{3} 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 5 & 15 & 5 \\ 9 & 15 & 32 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1430] \quad 1 \overset{1}{5} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 30 & 4 \\ 18 & 30 & 189 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1431] \quad 31 \overset{1}{1^3} 31^2 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 17 & 3 \\ 5 & 5 & 24 & 5 \\ 17 & 24 & 98 & 20 \\ 3 & 5 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[1432] \quad 1 \overset{1}{5} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 27 & 4 \\ 16 & 27 & 148 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[1434] \quad 2 \overset{1}{3} 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 7 & 5 & 10 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

- [1435] $31^{12}3141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 8 & 3 \\ 5 & 5 & 10 & 5 \\ 8 & 10 & 14 & 8 \\ 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$
- [1436] $1^{12}31^2 4^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 6 & 2 \\ 9 & 6 & 20 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$
- [1437] $1^{12}31^3 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 6 & 2 \\ 11 & 6 & 32 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$
- [1433] $1^{11}31^2 4^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 6 & 1 & 23 & 9 \\ 2 & 2 & 0 & 9 & 3 \end{pmatrix}_{b_0}$
- [1438] $1^{13} 5^{12} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 18 & 4 \\ 12 & 18 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$
- [1439] $21^{12} 5^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 21 & 4 \\ 13 & 21 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$
- [1440] $41^{11} 2231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 4 & 1 & 5 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 28 & 0 & 26 & 10 \\ 2 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$
- [1448] $1^{13} 4^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 9 & 10 & 0 & 64 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$
- [1445] $1^{12} 4^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 10 & 9 & 0 & 64 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$
- [1446] $1^{12}31^3 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 9 & 2 \\ 14 & 9 & 80 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$
- [1447] $1^{12} 4^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 11 & 10 & 0 & 84 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$
- [1444] $1^{11} 4^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 5 & 2 & 24 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0} \\
[1449] & 21^{1^2 4^1 2^3 1^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 1 & 16 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 16 & 1 & 47 & 13 \\ 2 & 4 & 0 & 13 & 3 \end{pmatrix}_{b_0} \\
[1450] & 1^{1^3} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 10 & 11 & 0 & 84 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0} \\
[1451] & 21^{1^2 4^1 1^3 1^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 4 & 1 & 14 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 14 & 1 & 33 & 11 \\ 2 & 4 & 0 & 11 & 3 \end{pmatrix}_{b_0} \\
[1442] & 1^{1^3} 4^1 1^3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 7 & 2 & 28 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0} \\
[1441] & 31^{1^1} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 2 & 6 & 2 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 5 & 2 \\ 6 & 9 & 5 & 13 & 7 \\ 2 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[1443] & 1^{1^2 4^1 3^2 31^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 11 & 10 & 0 & 84 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0} \\
[1452] & 1^{1^1} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 8 & 5 & 3 & 29 & 11 \\ 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0} \\
[1453] & 31^{1^1} 31^{1^1} 41^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.35 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 3 & 2 & 0 & 7 & 2 \\ 7 & 5 & 7 & 21 & 9 \\ 2 & 2 & 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[1454] & 1^{1^2} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.19 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 26 & 4 \\ 19 & 26 & 148 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0} \\
[1455] & 1^{1^2} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.19 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 26 & 4 \\ 20 & 26 & 148 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0} \\
[1456] & 21^{1^3} 5^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.19 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 26 & 4 \\ 19 & 26 & 148 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[1457] \ 21^3 \overset{1^4}{5} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 104, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.19 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 29 & 4 \\ 21 & 29 & 189 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1458] \ 221^3 \overset{1^1}{5} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 104, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.19 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 26 & 4 \\ 19 & 26 & 148 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[1460] \ 1^{12} \overset{1^4}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 106, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.19 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 29 & 4 \\ 22 & 29 & 189 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1459] \ 21^4 \overset{1^3}{5} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 106, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.19 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 29 & 4 \\ 22 & 29 & 189 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1461] \ 31^{11} 31^3 41^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 107, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 16 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 3 & 2 & 0 & 16 & 2 \\ 16 & 15 & 16 & 161 & 23 \\ 2 & 2 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

$$[1464] \ 31^{11} 31^3 41^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 107, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.35 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 3 & 2 & 0 & 8 & 2 \\ 8 & 7 & 8 & 33 & 11 \\ 2 & 2 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1462] \ 221^{13} \overset{1^4}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 107, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.27 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 6 & 23 & 3 \\ 6 & 13 & 56 & 7 \\ 23 & 56 & 232 & 30 \\ 3 & 7 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[1463] \ 21^{13} \overset{1^4}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 107, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.35 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 19 & 3 \\ 4 & 4 & 27 & 4 \\ 19 & 27 & 152 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[1465] \ 1^{11} \overset{1^{11} 31^3}{4} 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 108, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 10 & 7 & 1 & 49 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1466] \ 21^{13} \overset{1^4}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -100, \ \det = 108, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.33 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 4 & 30 & 4 \\ 21 & 30 & 189 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1467] \ 1^{11} \overset{1^{11} 31^3}{4} 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 108, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 12 & 9 & 1 & 89 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1468] \ 1^{11} \overset{1^{13} 31^2}{4} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 108, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 6 & 2 \\ 11 & 6 & 30 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[1469] $1 \overset{1}{\underset{1^3}{5}} 1^{13} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 21 & 4 \\ 15 & 21 & 88 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[1473] $21 \overset{1^4}{\underset{1^1}{5}} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 27 & 4 \\ 17 & 27 & 152 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[1474] $1 \overset{1^3}{\underset{1^1, 1^1}{5}} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 18 & 4 \\ 14 & 18 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[1472] $1 \overset{1^3}{\underset{1^1, 1^1}{5}} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 18 & 4 \\ 13 & 18 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[1470] $21 \overset{1^3}{\underset{1^1}{5}} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 24 & 4 \\ 15 & 24 & 115 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[1471] $1 \overset{1^3}{\underset{1^3, 1^1}{5}} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 30 & 4 \\ 18 & 30 & 187 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

[1478] $21 \overset{1^4}{\underset{1^1}{5}} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 24 & 3 \\ 3 & 4 & 32 & 4 \\ 24 & 32 & 234 & 30 \\ 3 & 4 & 30 & 3 \end{pmatrix}_{b_0}$$

[1479] $1 \overset{1^2}{\underset{1^4, 1^4}{5}} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 24 & 3 \\ 3 & 4 & 32 & 4 \\ 24 & 32 & 234 & 30 \\ 3 & 4 & 30 & 3 \end{pmatrix}_{b_0}$$

[1477] $1 \overset{1^2}{\underset{1^3}{2}} 3 2 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[1475] $1 \overset{1^4}{\underset{1^3}{2}} 3 2 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 10 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[1476] $221 \overset{1^1}{\underset{1^4}{5}} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 24 & 3 \\ 3 & 4 & 32 & 4 \\ 24 & 32 & 234 & 30 \\ 3 & 4 & 30 & 3 \end{pmatrix}_{b_0}$$

[1480] $1^{\overline{1}^{12}31^3} 4^{\overline{1}^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 7 & 2 \\ 12 & 7 & 44 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

[1481] $1^{\overline{1}^4} 5^{\overline{1}^{13}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 33 & 4 \\ 20 & 33 & 232 & 30 \\ 3 & 4 & 30 & 3 \end{pmatrix}_{b_0}$$

[1482] $31^{\overline{1}^{13}1^4} 41^{\overline{1}^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 12 & 3 \\ 5 & 5 & 17 & 5 \\ 12 & 17 & 46 & 14 \\ 3 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

[1483] $1^{\overline{1}^{14}31^1} 4^{\overline{1}^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 7 & 2 \\ 12 & 7 & 44 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

[1484] $1^{\overline{1}^{11}41^3} 231^{\overline{1}^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 8 & 1 & 67 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[1485] $1^{\overline{1}^{12}} 4131^{\overline{1}^{14}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 8 & 2 \\ 11 & 8 & 44 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

[1486] $1^{\overline{1}^{12}31^3} 4^{\overline{1}^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 8 & 2 \\ 13 & 8 & 60 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1487] $1^{\overline{1}^{11}31^3} 4^{\overline{1}^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 8 & 1 & 67 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[1488] $21^{\overline{1}^{13}41^1} 31^{\overline{1}^{14}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 1 & 20 & 4 \\ 1 & 1 & -2 & 4 & 0 \\ 12 & 20 & 4 & 82 & 18 \\ 2 & 4 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1489] $1^{\overline{1}^4} 41^3 231^{\overline{1}^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 8 & 2 \\ 13 & 8 & 60 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1490] $1^{\overline{1}^{13}} 4^{\overline{1}^{11}31^1} 1^{\overline{1}^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 8 & 1 & 35 & 11 \\ 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

[1491] $2^{\overline{1}^{12}} 3^{\overline{1}^{14}} 141^{\overline{1}^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1492] $1^{1^2} 4^{1^2} 13^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 7 & 6 & 3 & 31 & 11 \\ 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

[1493] $1^{1^3} 2^3 21^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[1494] $1^{1^2} 2^3 21^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[1495] $1^{1^4} 5^{1^3} 1^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 36 & 4 \\ 22 & 36 & 281 & 33 \\ 3 & 4 & 33 & 3 \end{pmatrix}_{b_0}$$

[1496] $1^{1^1} 5^{1^4} 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 24 & 4 \\ 18 & 24 & 121 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[1499] $2221^{1^1} 51^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 18 & 1 & 20 & 8 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 20 & 0 & 18 & 8 \\ 2 & 8 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[1500] $1^{1^1} 1^{1^2} 31^{1^2} 4^{1^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 10 & 2 \\ 13 & 10 & 76 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[1497] $1^{1^1} 1^{1^3} 31^{1^1} 4^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 9 & 6 & 3 & 43 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[1498] $41^{1^1} 31^{1^3} 32 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 4 & 5 & 4 & 2 \\ 4 & 10 & 13 & 12 & 6 \\ 5 & 13 & 12 & 12 & 6 \\ 4 & 12 & 12 & 10 & 6 \\ 2 & 6 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[1501] $2^{1^1} 3^{1^1} 1^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 11 & 3 \\ 4 & 5 & 17 & 5 \\ 11 & 17 & 44 & 14 \\ 3 & 5 & 14 & 4 \end{pmatrix}_{b_0}$$

[1502] $2^{1^1} 3^{1^4} 141^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 5 & 20 & 5 \\ 12 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

- [1503] $1^{11}41^2231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 119$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 9 & 6 & 2 & 36 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$
- [1504] $1^{11}4^{12}31^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 119$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 9 & 6 & 2 & 36 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$
- [1505] $1^{11}4^{13}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 119$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 7 & 2 \\ 10 & 7 & 28 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$
- [1506] $1^{11}41^1231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 119$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 7 & 2 \\ 10 & 7 & 28 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$
- [1507] $1^{13}5^{12}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -100$, $\det = 119$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 21 & 4 \\ 14 & 21 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$
- [1508] $1^{11}4^{12}31^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 119$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 7 & 2 \\ 10 & 7 & 28 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$
- [1509] $21^25^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -100$, $\det = 119$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 21 & 4 \\ 14 & 21 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$
- [1510] $1^{12}4^{13}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 119$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 7 & 6 & 2 & 24 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$
- [1511] $1^{13}4^{12}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 119$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 7 & 8 & 2 & 40 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$
- [1512] $1^{12}4^{11}31^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 120$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 8 & 7 & 1 & 33 & 11 \\ 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$
- [1513] $1^{11}4^{13}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 120$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 10 & 7 & 3 & 61 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$
- [1514] $2^{12}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 120$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 7 & 7 & 8 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1516] $1^{14}2^321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 122$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[1517] $1^{14}2^321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 122$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[1515] $221^{12}51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -100$, $\det = 122$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 6 & 14 & 3 \\ 6 & 13 & 34 & 7 \\ 14 & 34 & 82 & 18 \\ 3 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[1518] $151^{13}2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -100$, $\det = 123$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 21 & 4 \\ 15 & 21 & 88 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[1519] $31^{13}3^{24}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 123$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 3 & 2 & 0 & 12 & 2 \\ 12 & 11 & 12 & 85 & 17 \\ 2 & 2 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[1520] $1^{13}4^14 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 124$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 10 & 11 & 1 & 89 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[1521] $1^{12}4131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 124$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 8 & 7 & 3 & 45 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[1522] $1^{13}4^12 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 124$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 8 & 9 & 1 & 49 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[1523] $1^{11}4^13^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 127$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 7 & 2 & 52 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[1524] $1^{13}4131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 127$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 9 & 2 & 56 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[1525] $1^{14}1^{22}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 127$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 7 & 2 \\ 12 & 7 & 40 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

[1526] $1^{11} 4^{13} 13^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 9 & 2 \\ 12 & 9 & 56 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$

[1527] $1^{11} 4^{13} 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 7 & 2 \\ 12 & 7 & 40 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$

[1528] $1^{11} 4^{12} 3^{12} 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 11 & 8 & 2 & 72 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$

[1529] $1^{11} 4^{12} 3^{12} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 9 & 2 \\ 12 & 9 & 56 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$

[1530] $1^{13} 4^{14} 13^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 9 & 10 & 2 & 76 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$

[1531] $1^{11} 4^{14} 3^{11} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 8 & 2 \\ 13 & 8 & 56 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$

[1532] $1^{11} 4^{12} 23^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 8 & 2 \\ 11 & 8 & 40 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$

[1533] $1^{13} 4^{11} 3^{11} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 10 & 1 & 67 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$

[1534] $1^{12} 4^{14} 13^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 9 & 8 & 3 & 63 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$

[1535] $1^{11} 4^{12} 3^{12} 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 8 & 2 \\ 11 & 8 & 40 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$

[1536] $1^{13} 5^{12,1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 21 & 4 \\ 15 & 21 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[1537] $1^{22} 5^{12} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 6 & 10 & 3 \\ 6 & 13 & 24 & 7 \\ 10 & 24 & 38 & 12 \\ 3 & 7 & 12 & 3 \end{pmatrix}_{b_0} \\
[1541] \quad & 31^{13}1^{24}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 3 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 3 & 2 & 0 & 10 & 2 \\ 10 & 7 & 10 & 49 & 13 \\ 2 & 2 & 2 & 13 & 2 \end{pmatrix}_{b_0} \\
[1540] \quad & 1^{13}5^{14}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.34 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 21 & 4 \\ 16 & 21 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0} \\
[1539] \quad & 1^{13}1^{13}2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 4 & 11 & 3 \\ 4 & 4 & 15 & 4 \\ 11 & 15 & 40 & 12 \\ 3 & 4 & 12 & 3 \end{pmatrix}_{b_0} \\
[1538] \quad & 31^{13}1^{34}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 3 & 16 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 3 & 2 & 0 & 16 & 2 \\ 16 & 11 & 16 & 141 & 21 \\ 2 & 2 & 2 & 21 & 2 \end{pmatrix}_{b_0} \\
[1542] \quad & 1^{13}1^{14}1^{13}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 10 & 2 \\ 13 & 10 & 72 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0} \\
[1544] \quad & 21^{13}5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 4 & 30 & 4 \\ 21 & 30 & 187 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0} \\
[1545] \quad & 2^{14}3^{14}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 9 & 11 & 10 & 69 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1546] \quad & 1^{12}1^{11}31^{12}4^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 8 & 1 & 47 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0} \\
[1547] \quad & 1^{12}41^{22}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 8 & 1 & 47 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0} \\
[1548] \quad & 21^{35}1^{13}2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 4 & 30 & 4 \\ 21 & 30 & 187 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0} \\
[1543] \quad & 1^{12}1^{11}31^{12}4^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 10 & 1 & 87 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0} \\
[1549] \quad & 1^{12}1^{12}31^{11}4^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 7 & 2 & 36 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0} \\
[1550] & 2^{\overset{1}{1}\overset{13}{3}\overset{41}{1}} 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 8 & 9 & 9 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1551] & 1^{\overset{1}{1}\overset{14}{4}\overset{31}{1}} 1^{\overset{4}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 9 & 2 \\ 14 & 9 & 72 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0} \\
[1552] & 1^{\overset{1}{1}\overset{13}{4}\overset{31}{2}} 1^{\overset{4}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 9 & 2 \\ 14 & 9 & 72 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0} \\
[1553] & 2^{\overset{1}{2}\overset{12}{3}} 141^{\overset{13}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 7 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1554] & 1^{\overset{1}{1}\overset{12}{4}\overset{11}{3}\overset{31}{2}} 1^{\overset{3}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 10 & 9 & 1 & 65 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0} \\
[1555] & 1^{\overset{1}{1}\overset{13}{4}\overset{31}{2}} 1^{\overset{3}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 8 & 2 \\ 13 & 8 & 54 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0} \\
[1556] & 1^{\overset{1}{1}\overset{11}{4}\overset{13}{3}\overset{31}{1}} 1^{\overset{2}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 8 & 2 \\ 11 & 8 & 38 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0} \\
[1557] & 31^{\overset{1}{3}\overset{13}{31}\overset{11}{41}\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 138, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 6 & 12 & 3 \\ 6 & 13 & 26 & 7 \\ 12 & 26 & 46 & 14 \\ 3 & 7 & 14 & 3 \end{pmatrix}_{b_0} \\
[1558] & 21^{\overset{2}{2}\overset{5}{5}\overset{1}{1}\overset{13}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.32 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 24 & 4 \\ 16 & 24 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0} \\
[1559] & 221^{\overset{1}{2}\overset{2}{5}\overset{1}{1}\overset{13}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.32 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 24 & 4 \\ 16 & 24 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0} \\
[1560] & 1^{\overset{1}{1}\overset{13}{5}\overset{1}{1}\overset{2}{12}, \overset{2}{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.32 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 24 & 4 \\ 16 & 24 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[1565] \begin{smallmatrix} 1^2 \\ 1^1 \end{smallmatrix} 1^2 1^3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 13 & 3 \\ 4 & 4 & 18 & 4 \\ 13 & 18 & 61 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1562] \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 1^3 31^1 4 \begin{smallmatrix} 1^4 \\ 1^4 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 10 & 2 \\ 13 & 10 & 70 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1563] 31^1 1^3 31^3 41^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 13 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 3 & 2 & 0 & 13 & 2 \\ 13 & 9 & 13 & 89 & 17 \\ 2 & 2 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[1564] 2 \begin{smallmatrix} 1^2 \\ 1^2 \end{smallmatrix} 3 141^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 10 & 10 & 58 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1561] 2 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 3 1^4 1^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 6 & 10 & 38 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1566] \begin{smallmatrix} 1^3 \\ 1^3, 1^1 \end{smallmatrix} 1^3 5 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 24 & 4 \\ 17 & 24 & 115 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1567] 21^3 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 5 1^3 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 24 & 4 \\ 17 & 24 & 115 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1568] 1^1 1^2 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 31^1 4 \begin{smallmatrix} 1^4 \\ 1^4 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 143, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 9 & 2 & 72 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1569] 1^1 1^2 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 31^1 4 \begin{smallmatrix} 1^3 \\ 1^3 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 143, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 9 & 8 & 2 & 52 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1570] 1^1 1^1 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} 31^1 4 \begin{smallmatrix} 1^3 \\ 1^3 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 143, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 9 & 2 \\ 12 & 9 & 52 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[1571] 41^1 1^2 232 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 2 & 8 & 2 \\ 4 & 10 & 5 & 24 & 6 \\ 2 & 5 & 0 & 8 & 2 \\ 8 & 24 & 8 & 46 & 12 \\ 2 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1572] 21^1 1^3 41^2 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 1 & 22 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 10 & 22 & 0 & 84 & 16 \\ 2 & 4 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1573] $31^{11} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 2 & 11 & 2 \\ 3 & 4 & 4 & 15 & 4 \\ 2 & 4 & 0 & 10 & 2 \\ 11 & 15 & 10 & 45 & 13 \\ 2 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[1574] $31^{11} 4 13 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 2 & 5 & 2 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 5 & 9 & 6 & 13 & 7 \\ 2 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1575] $21^{13} 41^4 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 4 & 1 & 26 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 14 & 26 & 0 & 132 & 20 \\ 2 & 4 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1576] $1^{13} 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 24 & 4 \\ 18 & 24 & 115 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[1577] $1^{11} 4 1^{13} 31^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 10 & 6 & 3 & 41 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[1578] $1^{11} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 7 & 2 \\ 13 & 7 & 42 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

[1579] $1^{11} 41^2 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 10 & 6 & 3 & 41 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[1580] $1^{12} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 8 & 6 & 3 & 29 & 11 \\ 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

[1581] $1^{13} 4 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 13 & 13 & 0 & 132 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1582] $1^{11} 4 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 11 & 7 & 4 & 68 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1583] $1^{13} 4 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 11 & 11 & 0 & 84 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

- [1584] $21^{13}_{1^1} 5^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 27 & 4 \\ 18 & 27 & 148 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$
- [1585] $1^{13}_{1^2, 1^2} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 24 & 4 \\ 17 & 24 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$
- [1586] $1^{13}_{1^4, 1^1} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 27 & 4 \\ 20 & 27 & 152 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$
- [1587] $21^{13}_{1^1} 4^1 5^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 27 & 4 \\ 20 & 27 & 152 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$
- [1588] $21^{13} 4131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 1 & 18 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 9 & 18 & 3 & 61 & 15 \\ 2 & 4 & 0 & 15 & 3 \end{pmatrix}_{b_0}$
- [1589] $21^{13} 41^3 31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 15 & 24 & 3 & 115 & 21 \\ 2 & 4 & 0 & 21 & 3 \end{pmatrix}_{b_0}$
- [1592] $1^{11} 41^1 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 8 & 2 \\ 12 & 8 & 42 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$
- [1594] $1^{12} 4131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 9 & 7 & 4 & 52 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$
- [1595] $31^{11} 31^4 41^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 19 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 3 & 2 & 0 & 19 & 2 \\ 19 & 17 & 19 & 225 & 27 \\ 2 & 2 & 2 & 27 & 2 \end{pmatrix}_{b_0}$
- [1596] $1^{13}_{1^1} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 12 & 12 & 0 & 106 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$
- [1597] $31^{11} 31^2 41^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 3 & 2 & 0 & 11 & 2 \\ 11 & 9 & 11 & 65 & 15 \\ 2 & 2 & 2 & 15 & 2 \end{pmatrix}_{b_0}$
- [1598] $1^{14} 4131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 10 & 2 & 58 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[1599] $1^{11} 1^{14} 31^1 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 8 & 2 \\ 12 & 8 & 42 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

[1600] $1^{11} 1^{12} 31^3 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 8 & 2 \\ 12 & 8 & 42 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

[1601] $1^{13} 41^3 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 12 & 12 & 0 & 106 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1593] $221^{12} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 6 & 12 & 3 \\ 6 & 13 & 29 & 7 \\ 12 & 29 & 57 & 15 \\ 3 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[1591] $21^{13} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 4 & 24 & 4 \\ 17 & 24 & 115 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[1590] $1^{11} 1^{12} 31^3 4^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 11 & 7 & 2 & 52 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[1602] $1^{11} 1^{12} 31^3 4^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 11 & 2 \\ 15 & 11 & 96 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[1603] $21^2 51^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 30 & 4 \\ 20 & 30 & 189 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

[1604] $1^{12} 1^{11} 31^3 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 10 & 8 & 1 & 49 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[1605] $1^{14} 2321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 9 & 15 & 13 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

[1606] $1^{13} 5^{14} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 24 & 4 \\ 18 & 24 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

- [1607] $21^{13}51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 4 & 19 & 3 \\ 4 & 4 & 27 & 4 \\ 19 & 27 & 148 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$
- [1608] $1^{14}4^{11}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 11 & 1 & 69 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$
- [1609] $41^{12}2231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 4 & 1 & 8 & 2 \\ 4 & 26 & 1 & 56 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 56 & 0 & 110 & 20 \\ 2 & 10 & -1 & 20 & 3 \end{pmatrix}_{b_0}$
- [1610] $21^{13}4131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 4 & 1 & 18 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 18 & 2 & 56 & 14 \\ 2 & 4 & 0 & 14 & 3 \end{pmatrix}_{b_0}$
- [1611] $21^{13}41^431^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 16 & 2 \\ 2 & 4 & 1 & 26 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 16 & 26 & 2 & 136 & 22 \\ 2 & 4 & 0 & 22 & 3 \end{pmatrix}_{b_0}$
- [1613] $1^{24}1^{12}231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 3 & 1 & 9 & 2 \\ 3 & 4 & 1 & 16 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 16 & 1 & 47 & 13 \\ 2 & 4 & 0 & 13 & 3 \end{pmatrix}_{b_0}$
- [1614] $1^{24}1^{12}231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 3 & 1 & 11 & 2 \\ 3 & 4 & 1 & 16 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 16 & 3 & 55 & 15 \\ 2 & 4 & 0 & 15 & 3 \end{pmatrix}_{b_0}$
- [1612] $21^{13}41^331^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 12 & 24 & 0 & 106 & 18 \\ 2 & 4 & 0 & 18 & 3 \end{pmatrix}_{b_0}$
- [1615] $2^13^{13}141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 5 & 20 & 5 \\ 12 & 20 & 58 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$
- [1619] $1^{14}4^{11}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 13 & 1 & 113 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$
- [1620] $1^{13}4^{11}31^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 9 & 1 & 49 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$
- [1621] $1^{11}4^{13}31^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 7 & 3 & 57 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[1618] $1^{13}4^{12}13^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 8 & 8 & 3 & 45 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[1616] $1^{14}4^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 10 & 12 & 1 & 89 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[1617] $1^{11}4^{12}3^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 9 & 2 \\ 13 & 9 & 56 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1622] $1^{11}4^{13}23^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 8 & 2 & 70 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1623] $41^{11}2^3 3^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 2 & 6 & 2 \\ 4 & 10 & 5 & 16 & 6 \\ 2 & 5 & 0 & 7 & 2 \\ 6 & 16 & 7 & 21 & 9 \\ 2 & 6 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1624] $1^{13}4^{12}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 8 & 2 & 38 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[1625] $21^{11}5^{13}2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 13 & 3 \\ 4 & 4 & 18 & 4 \\ 13 & 18 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[1626] $1^{12}4^{12}3^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 9 & 7 & 2 & 36 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[1627] $1^{11}4^{12}3^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 8 & 2 & 70 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1628] $1^{14}4^{13}13^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 9 & 11 & 2 & 76 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1630] $41^{13}223^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 1 & 16 & 2 \\ 4 & 26 & 1 & 84 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 84 & 0 & 250 & 30 \\ 2 & 10 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[1631] $1^{11}41^3231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 10 & 2 \\ 14 & 10 & 74 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

[1632] $1^{11}41^3231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 10 & 2 \\ 14 & 10 & 74 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

[1633] $21^{13}51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 4 & 13 & 3 \\ 4 & 4 & 18 & 4 \\ 13 & 18 & 59 & 15 \\ 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$

[1634] $31^{11}31^{11}41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 3 & 7 & 2 \\ 3 & 4 & 4 & 11 & 4 \\ 3 & 4 & 0 & 7 & 2 \\ 7 & 11 & 7 & 21 & 9 \\ 2 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$

[1629] $1^{12}4131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 10 & 8 & 4 & 70 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$

[1635] $21^{12}51^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 27 & 4 \\ 19 & 27 & 148 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$

[1636] $1^{14}2321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 10 & 15 & 13 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$

[1637] $1^{13}51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 4 & 27 & 4 \\ 19 & 27 & 148 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$

[1638] $1^{14}2321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 167, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 15 & 13 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$

[1639] $1^{11}41^3231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 8 & 2 \\ 12 & 8 & 38 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$

[1643] $1^{14}4131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 12 & 2 & 98 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$

[1644] $31^{11}31^{11}41^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 15 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 3 & 2 & 0 & 15 & 2 \\ 15 & 13 & 15 & 133 & 21 \\ 2 & 2 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[1645] $1^4 1^{2231} 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 8 & 2 \\ 14 & 8 & 54 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1642] $1^{11} 1^{1231^3} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 13 & 9 & 2 & 92 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1641] $1^{11} 1^{1431^2} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 8 & 2 \\ 14 & 8 & 54 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1640] $1^{11} 1^{1331^3} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 8 & 2 \\ 14 & 8 & 54 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1646] $21^{13} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 21 & 4 \\ 15 & 21 & 84 & 18 \\ 3 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[1647] $1^{13} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 4 & 27 & 4 \\ 20 & 27 & 148 & 24 \\ 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[1648] $1^{12} 1^{1331^1} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 9 & 7 & 3 & 41 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[1649] $1^{12} 1^{1131^3} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 9 & 1 & 65 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[1650] $41^1 3^1 22 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 5 & 8 & 7 & 5 \\ 3 & 8 & 4 & 7 & 4 \\ 3 & 7 & 7 & 5 & 5 \\ 2 & 5 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[1651] $1^{11} 1^{1331^2} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 12 & 8 & 3 & 77 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[1652] $141^{11} 31^{11} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 3 & 6 & 2 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 6 & 9 & 6 & 13 & 7 \\ 2 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1653] $2^1 3^{15} 141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 172$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.1$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 5 & 25 & 5 \\ 15 & 25 & 98 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$

[1654] $2^1 3^{13} 141^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 175$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 8 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$

[1655] $21^{13} 41^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 175$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 4 & 1 & 20 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 20 & 1 & 67 & 15 \\ 2 & 4 & 0 & 15 & 3 \end{pmatrix}_{b_0}$

[1656] $41^{11} 3^{11} 22 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 175$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 3 & 3 & 7 & 2 \\ 3 & 5 & 8 & 15 & 5 \\ 3 & 8 & 4 & 15 & 4 \\ 7 & 15 & 15 & 33 & 11 \\ 2 & 5 & 4 & 11 & 3 \end{pmatrix}_{b_0}$

[1657] $21^{13} 41^{13} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 175$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 4 & 1 & 26 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 15 & 26 & 1 & 133 & 21 \\ 2 & 4 & 0 & 21 & 3 \end{pmatrix}_{b_0}$

[1658] $31^{13} 3141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 175$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 12 & 3 \\ 5 & 5 & 15 & 5 \\ 12 & 15 & 34 & 12 \\ 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$

[1659] $21^2 5^1 1^{13} 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -100$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.36 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 4 & 24 & 4 \\ 17 & 24 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$

[1660] $21^{12} 5^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -100$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.36 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 4 & 24 & 4 \\ 17 & 24 & 113 & 21 \\ 3 & 4 & 21 & 3 \end{pmatrix}_{b_0}$

[1662] $1^{13} 4^{11} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 179$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 10 & 10 & 1 & 65 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$

[1661] $21^4 5^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -100$, $\det = 179$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 30 & 4 \\ 22 & 30 & 189 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$

[1666] $1^1 41^{32} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 179$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 9 & 2 \\ 15 & 9 & 70 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

[1667] $1^1 4^{13} 31^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 179$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 9 & 2 \\ 15 & 9 & 70 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

- [1668] $1^{12}41^3231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 12 & 10 & 1 & 85 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$
- [1669] $1^{12}4^{11}31^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 12 & 10 & 1 & 85 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$
- [1670] $1^{12}4^{11}31^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 13 & 11 & 1 & 109 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$
- [1671] $31^{11}31^341^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 14 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 3 & 2 & 0 & 14 & 2 \\ 14 & 11 & 14 & 109 & 19 \\ 2 & 2 & 2 & 19 & 2 \end{pmatrix}_{b_0}$
- [1663] $1^{11}4^{14}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 9 & 2 \\ 13 & 9 & 52 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$
- [1664] $1^{13}4131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 9 & 9 & 3 & 61 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$
- [1665] $1^{13}5^{14}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 30 & 4 \\ 22 & 30 & 189 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$
- [1672] $31^{12}3141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 11 & 3 \\ 5 & 5 & 15 & 5 \\ 11 & 15 & 32 & 12 \\ 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$
- [1673] $1^{13}4^{13}31^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 10 & 2 \\ 16 & 10 & 90 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$
- [1674] $1^{13}5^{13}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 30 & 4 \\ 21 & 30 & 187 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$
- [1676] $221^351^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 30 & 4 \\ 21 & 30 & 187 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$
- [1675] $21^{13}51^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 4 & 30 & 4 \\ 21 & 30 & 187 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0} \\
[1677] \quad & 21^{13} 41^2 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 4 & 1 & 22 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 22 & 3 & 93 & 19 \\ 2 & 4 & 0 & 19 & 3 \end{pmatrix}_{b_0} \\
[1678] \quad & 21^{13} 41^1 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 1 & 20 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 20 & 3 & 75 & 17 \\ 2 & 4 & 0 & 17 & 3 \end{pmatrix}_{b_0} \\
[1679] \quad & 1^{12} 41^2 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 8 & 2 & 50 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0} \\
[1680] \quad & 1^{12} 4^1 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 8 & 2 & 50 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0} \\
[1681] \quad & 1^{13} 4^1 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 9 & 9 & 2 & 52 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0} \\
[1682] \quad & 2^{11} 41^1 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 14 & 3 \\ 4 & 5 & 22 & 5 \\ 14 & 22 & 74 & 18 \\ 3 & 5 & 18 & 4 \end{pmatrix}_{b_0} \\
[1683] \quad & 2^{11} 41^1 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 14 & 3 \\ 4 & 5 & 22 & 5 \\ 14 & 22 & 74 & 18 \\ 3 & 5 & 18 & 4 \end{pmatrix}_{b_0} \\
[1688] \quad & 1^{13} 4^1 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 10 & 10 & 3 & 81 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0} \\
[1687] \quad & 1^{11} 4^1 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 9 & 2 \\ 13 & 9 & 50 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0} \\
[1689] \quad & 1^{14} 31^2 4^1 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 9 & 2 \\ 15 & 9 & 68 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0} \\
[1690] \quad & 1^{13} 4^1 1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 12 & 12 & 1 & 109 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0} \\
[1684] \quad & 1^{11} 41^2 231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 9 & 2 \\ 13 & 9 & 50 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1686] $1^{13} 4^1 1^{31} 2^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 11 & 1 & 85 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[1685] $1^{12} 4^1 1^{31} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 10 & 8 & 3 & 57 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[1691] $31^{13} 31^4 41^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 17 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 3 & 2 & 0 & 17 & 2 \\ 17 & 13 & 17 & 165 & 23 \\ 2 & 2 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

[1693] $1^{13} 5^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 4 & 30 & 4 \\ 22 & 30 & 187 & 27 \\ 3 & 4 & 27 & 3 \end{pmatrix}_{b_0}$$

[1692] $21^3 5^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 23 & 3 \\ 3 & 4 & 33 & 4 \\ 23 & 33 & 232 & 30 \\ 3 & 4 & 30 & 3 \end{pmatrix}_{b_0}$$

[1694] $2^1 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 10 & 7 & 13 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1695] $31^{11} 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 2 & 8 & 2 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 0 & 8 & 2 \\ 8 & 12 & 8 & 26 & 10 \\ 2 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[1696] $1^{14} 2^3 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 1 & 2 & 0 & 16 & 2 \\ 11 & 16 & 16 & 126 & 24 \\ 2 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[1697] $31^{13} 31^4 41^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 18 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 3 & 2 & 0 & 18 & 2 \\ 18 & 15 & 18 & 193 & 25 \\ 2 & 2 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[1698] $1^{12} 4^1 1^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 9 & 3 & 77 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[1699] $1^{11} 4^1 1^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 11 & 2 \\ 15 & 11 & 86 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[1701] $1^{11} 4^{1^{13}31^2} 1^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 10 & 2 \\ 14 & 10 & 66 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

[1702] $31^{12}31^141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 13 & 3 \\ 5 & 5 & 17 & 5 \\ 13 & 17 & 44 & 14 \\ 3 & 5 & 14 & 3 \end{pmatrix}_{b_0}$

[1703] $1^{14}31^2 4^{1^4} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 10 & 2 \\ 16 & 10 & 86 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$

[1704] $1^{12}31^141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 11 & 9 & 2 & 68 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$

[1705] $1^{11} 4^{1^{14}31^1} 1^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 10 & 2 \\ 14 & 10 & 66 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

[1706] $1^{13} 4^{1^{12}31^1} 1^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 10 & 2 & 70 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$

[1707] $1^{11} 4^{1^{12}31^4} 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 10 & 2 \\ 15 & 10 & 76 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

[1700] $31^{12}31^141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 28 & 3 \\ 5 & 5 & 38 & 5 \\ 28 & 38 & 254 & 32 \\ 3 & 5 & 32 & 3 \end{pmatrix}_{b_0}$

[1708] $1^{14}2321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 16 & 16 & 126 & 24 \\ 2 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$

[1719] $1^{11} 4^{1^{12}31^4} 1^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 11 & 2 \\ 16 & 11 & 96 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$

[1713] $1^{13} 4^{1^131^4} 1^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 14 & 13 & 0 & 132 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$

[1714] $1^{11} 4^{1^{12}31^4} 1^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 9 & 2 & 92 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$

[1715] $21^4 \overset{13}{5} 1^{\overset{13}{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 24 & 3 \\ 3 & 4 & 33 & 4 \\ 24 & 33 & 232 & 30 \\ 3 & 4 & 30 & 3 \end{pmatrix}_{b_0}$

[1716] $1^{\overset{13}{1}} 41^4 231^{\overset{13}{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 15 & 14 & 0 & 160 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$

[1717] $1^{\overset{13}{1}} 4131^{\overset{14}{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 9 & 8 & 4 & 52 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$

[1718] $1^{\overset{12}{1}} 4^{\overset{14}{1}} 1^{\overset{12}{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 10 & 7 & 4 & 48 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$

[1712] $1^{\overset{13}{1}} 4^{\overset{14}{1}} 1^{\overset{14}{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 15 & 14 & 0 & 160 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$

[1711] $1^{\overset{14}{1}} 4^{\overset{13}{1}} 1^{\overset{13}{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 13 & 14 & 0 & 132 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$

[1709] $1^{\overset{13}{1}} 5^{\overset{14}{1}} 1^{\overset{13}{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 24 & 3 \\ 3 & 4 & 33 & 4 \\ 24 & 33 & 232 & 30 \\ 3 & 4 & 30 & 3 \end{pmatrix}_{b_0}$

[1710] $1^{\overset{14}{1}} 4^{\overset{13}{1}} 1^{\overset{13}{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 14 & 15 & 0 & 160 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$

[1720] $1^{\overset{12}{1}} 4^{\overset{13}{1}} 1^{\overset{12}{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 10 & 7 & 3 & 41 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$

[1723] $1^{\overset{12}{1}} 4^{\overset{12}{1}} 1^{\overset{12}{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 10 & 2 & 90 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$

[1721] $1^{\overset{2}{1}} 41^{\overset{12}{1}} 231^{\overset{12}{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 3 & 1 & 10 & 2 \\ 3 & 4 & 1 & 16 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 16 & 2 & 50 & 14 \\ 2 & 4 & 0 & 14 & 3 \end{pmatrix}_{b_0}$

[1722] $1^{\overset{13}{1}} 4^{\overset{12}{1}} 1^{\overset{12}{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 11 & 11 & 2 & 92 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1724] \ 1^{11} 1^{14} 31^1 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 11 & 2 \\ 15 & 11 & 84 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[1725] \ 1^{14} 1^{11} 31^2 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 10 & 11 & 1 & 69 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1726] \ 21^{13} 41^{13} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 1 & 20 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 20 & 2 & 70 & 16 \\ 2 & 4 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1727] \ 21^{13} 41^{33} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 24 & 2 & 110 & 20 \\ 2 & 4 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[1728] \ 1^{11} 1^{12} 31^4 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 15 & 10 & 2 & 116 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[1729] \ 21^{13} 41^{33} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 13 & 24 & 1 & 107 & 19 \\ 2 & 4 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1730] \ 1^{11} 1^{14} 31^2 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 13 & 8 & 4 & 84 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1731] \ 1^{14} 1^{12} 31^1 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 9 & 10 & 2 & 56 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1732] \ 2^1 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 9 & 8 & 11 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1733] \ 21^{13} 41^{23} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 1 & 22 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 22 & 1 & 85 & 17 \\ 2 & 4 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1734] \ 1^{12} 1^{12} 31^3 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 11 & 8 & 2 & 52 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

- [1735] $1^{11}41^4231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 15 & 10 & 2 & 116 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$
- [1736] $1^{11331^4} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 9 & 2 \\ 16 & 9 & 72 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$
- [1737] $1^{12} 4^{11331^4} 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 13 & 10 & 1 & 87 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$
- [1738] $1^{11} 4^{11331^3} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 8 & 3 & 75 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$
- [1739] $1^{11}41^3231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 8 & 3 & 75 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$
- [1740] $1^{13} 4^{11331^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 9 & 8 & 3 & 43 & 13 \\ 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$
- [1741] $21^4 51^{13} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 26 & 3 \\ 3 & 4 & 36 & 4 \\ 26 & 36 & 281 & 33 \\ 3 & 4 & 33 & 3 \end{pmatrix}_{b_0}$
- [1742] $221^4 51^{13} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 26 & 3 \\ 3 & 4 & 36 & 4 \\ 26 & 36 & 281 & 33 \\ 3 & 4 & 33 & 3 \end{pmatrix}_{b_0}$
- [1743] $1^{13} 5^{14} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 26 & 3 \\ 3 & 4 & 36 & 4 \\ 26 & 36 & 281 & 33 \\ 3 & 4 & 33 & 3 \end{pmatrix}_{b_0}$
- [1744] $1^{13} 4^{11331^3} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 10 & 1 & 67 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$
- [1745] $1^{14} 4131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 9 & 10 & 3 & 63 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$
- [1747] $1^{11} 4^{11331^3} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 215, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 9 & 2 \\ 14 & 9 & 52 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1746] $1^{11} 1^{14} 31^2 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 215, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 9 & 2 \\ 14 & 9 & 52 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1748] $221^{13} 51^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 6 & 21 & 3 \\ 6 & 13 & 51 & 7 \\ 21 & 51 & 187 & 27 \\ 3 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

[1749] $1^{11} 1^{13} 31^3 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 14 & 9 & 3 & 97 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

[1752] $1^{12} 1^{13} 31^4 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 14 & 11 & 1 & 109 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

[1751] $1^{13} 1^{13} 31^4 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 0 & 10 & 2 \\ 17 & 10 & 90 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[1750] $21^{13} 41^2 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 1 & 22 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 22 & 2 & 88 & 18 \\ 2 & 4 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1753] $1^4 41^2 231^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 223, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 0 & 11 & 2 \\ 18 & 11 & 112 & 22 \\ 3 & 2 & 22 & 4 \end{pmatrix}_{b_0}$$

[1754] $1^{13} 1^{13} 31^4 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 223, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 0 & 11 & 2 \\ 18 & 11 & 112 & 22 \\ 3 & 2 & 22 & 4 \end{pmatrix}_{b_0}$$

[1755] $1^{13} 4131^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 223, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 10 & 9 & 4 & 68 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1756] $1^{12} 1^{14} 31^1 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 223, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 11 & 8 & 4 & 64 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1757] $1^{13} 1^{12} 31^2 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 223, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 9 & 2 & 52 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[1758] $1^{14} 1^{14} 31^3 4^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 223, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 9 & 2 \\ 16 & 9 & 68 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1759] \ 1^{12} 4^{11} 31^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 15 & 12 & 1 & 135 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1760] \ 1^{12} 41^4 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 15 & 12 & 1 & 135 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1762] \ 41^{12} 2231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 1 & 10 & 2 \\ 4 & 26 & 1 & 56 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 56 & 0 & 110 & 20 \\ 2 & 10 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[1763] \ 41^{13} 232 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 2 & 12 & 2 \\ 4 & 10 & 5 & 36 & 6 \\ 2 & 5 & 0 & 12 & 2 \\ 12 & 36 & 12 & 106 & 18 \\ 2 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1761] \ 1^{14} 4^{11} 31^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 12 & 1 & 87 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1764] \ 1^{12} 4^{13} 31^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 8 & 3 & 55 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1765] \ 1^{12} 41^2 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 8 & 3 & 55 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1766] \ 1^{14} 4^{12} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 231, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 11 & 2 & 72 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1767] \ 1^{12} 4^{12} 31^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 231, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 9 & 2 & 68 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1768] \ 1^{14} 4131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 10 & 11 & 3 & 81 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1769] \ 2^3 141^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1770] $1^{13} 4^{1^{11} 31^3} 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 232$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 12 & 11 & 1 & 85 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[1771] $221^{13} 5^{1^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -100$, $\det = 235$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 6 & 15 & 3 \\ 6 & 13 & 36 & 7 \\ 15 & 36 & 88 & 18 \\ 3 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[1775] $21^{14} 4131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 236$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 10 & 24 & 2 & 98 & 18 \\ 2 & 4 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1776] $1^{14} 4^{1^{11} 31^2} 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 236$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 12 & 13 & 1 & 109 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

[1777] $2221^{12} 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -100$, $\det = 236$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.19 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 4 & 1 & 8 & 2 \\ 4 & 18 & 1 & 40 & 8 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 40 & 0 & 78 & 16 \\ 2 & 8 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[1774] $1^{11} 4^{1^{13} 31^3} 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 236$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 10 & 2 \\ 15 & 10 & 66 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[1773] $1^{13} 4^{1^{13} 31^1} 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 236$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 10 & 9 & 3 & 57 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[1772] $2^3 141^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 236$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 10 & 8 & 14 & 70 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1778] $2^3 141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 238$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 5 & 25 & 5 \\ 15 & 25 & 92 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

[1779] $2^3 141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 238$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.2$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 5 & 30 & 5 \\ 18 & 30 & 142 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[1780] $1^{13} 4^{1^4} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 239$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 11 & 10 & 4 & 88 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1781] $1^{12} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 239, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 12 & 9 & 4 & 84 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$

[1783] $1^{34} 1^{13} 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 3 & 1 & 13 & 2 \\ 3 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 13 & 24 & 1 & 107 & 19 \\ 2 & 4 & 0 & 19 & 3 \end{pmatrix}_{b_0}$

[1782] $1^{14} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 13 & 14 & 1 & 135 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$

[1785] $1^{14} 4^{13} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 12 & 3 & 103 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$

[1786] $31^{12} 31^{34} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 23 & 3 \\ 5 & 5 & 31 & 5 \\ 23 & 31 & 164 & 26 \\ 3 & 5 & 26 & 3 \end{pmatrix}_{b_0}$

[1787] $31^{12} 31^{24} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 18 & 3 \\ 5 & 5 & 24 & 5 \\ 18 & 24 & 94 & 20 \\ 3 & 5 & 20 & 3 \end{pmatrix}_{b_0}$

[1788] $1^{14} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 0 & 10 & 2 \\ 17 & 10 & 84 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$

[1789] $1^{11} 41^{23} 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 10 & 2 \\ 15 & 10 & 64 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

[1790] $1^{14} 41^{32} 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 0 & 10 & 2 \\ 17 & 10 & 84 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$

[1791] $1^{11} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 10 & 2 \\ 15 & 10 & 64 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

[1792] $1^{13} 41^{32} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 13 & 12 & 1 & 107 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$

[1784] $1^{13} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 13 & 12 & 1 & 107 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$

[1793] $1^{11} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$

$$\begin{aligned}
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 11 & 2 \\ 16 & 11 & 84 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0} \\
[1794] & 1^{11} 4^{13} 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 11 & 2 \\ 16 & 11 & 84 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0} \\
[1795] & 1^{12} 4^{13} 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 13 & 10 & 2 & 88 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0} \\
[1796] & 21^{14} 4^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 1 & 26 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 26 & 1 & 113 & 19 \\ 2 & 4 & 0 & 19 & 3 \end{pmatrix}_{b_0} \\
[1797] & 1^{14} 4^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 11 & 12 & 2 & 92 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0} \\
[1798] & 1^{13} 4^{12} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 11 & 10 & 2 & 68 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0} \\
[1799] & 1^{12} 4^{13} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 13 & 10 & 2 & 88 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0} \\
[1800] & 1^{11} 4^{14} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 0 & 12 & 2 \\ 17 & 12 & 106 & 22 \\ 3 & 2 & 22 & 4 \end{pmatrix}_{b_0} \\
[1801] & 1^{13} 4^{14} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 14 & 13 & 1 & 133 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0} \\
[1802] & 1^{12} 4^{13} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 12 & 9 & 3 & 73 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0} \\
[1805] & 1^{14} 4^{13} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 255, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 0 & 11 & 2 \\ 18 & 11 & 104 & 22 \\ 3 & 2 & 22 & 4 \end{pmatrix}_{b_0} \\
[1803] & 1^{12} 4^{14} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 255, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 11 & 2 & 112 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1804] $1^{14} 1^{12} 31^1 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 255$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 13 & 2 & 116 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$
- [1807] $21^{14} 41^3 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 256$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 4 & 1 & 30 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 14 & 30 & 0 & 160 & 22 \\ 2 & 4 & 0 & 22 & 3 \end{pmatrix}_{b_0}$
- [1808] $1^{13} 1^{13} 31^1 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 256$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 10 & 3 & 75 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$
- [1809] $21^{14} 4131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 256$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 4 & 0 \\ 12 & 24 & 4 & 110 & 20 \\ 2 & 4 & 0 & 20 & 3 \end{pmatrix}_{b_0}$
- [1810] $2^{13} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 256$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 10 & 10 & 12 & 70 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$
- [1806] $21^{14} 41^4 31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 256$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 20 & 2 \\ 2 & 4 & 1 & 32 & 4 \\ 1 & 1 & -2 & 4 & 0 \\ 20 & 32 & 4 & 206 & 28 \\ 2 & 4 & 0 & 28 & 3 \end{pmatrix}_{b_0}$
- [1811] $1^{13} 1^{11} 31^4 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 259$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.99 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 13 & 11 & 1 & 89 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$
- [1812] $1^{12} 1^{13} 31^2 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 260$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 10 & 3 & 95 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$
- [1813] $1^{14} 1^{13} 31^4 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 260$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 15 & 15 & 0 & 160 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$
- [1814] $1^{11} 1^{13} 31^4 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 260$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 10 & 2 \\ 16 & 10 & 70 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$
- [1815] $1^{11} 1^{14} 31^2 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 263$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 11 & 2 \\ 16 & 11 & 80 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$
- [1816] $1^{13} 1^{12} 31^2 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 263$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 11 & 2 & 88 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1817] $1^{11} 4^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 15 & 9 & 4 & 104 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1818] $1^{14} 4^{13} 1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 10 & 10 & 4 & 70 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1819] $1^{12} 4^1 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 12 & 8 & 4 & 62 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1820] $1^{14} 4^{12} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 16 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 16 & 16 & 0 & 190 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}$$

[1821] $1^{12} 4^{12} 31^1 1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 12 & 8 & 4 & 62 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1822] $1^{14} 4^1 1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 16 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 16 & 16 & 0 & 190 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}$$

[1823] $2^3 1^{14} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 5 & 27 & 5 \\ 17 & 27 & 112 & 22 \\ 3 & 5 & 22 & 4 \end{pmatrix}_{b_0}$$

[1826] $1^{14} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 12 & 12 & 1 & 89 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[1827] $1^{11} 4^1 1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 16 & 10 & 3 & 121 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

[1824] $1^{11} 4^{12} 31^1 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 16 & 10 & 3 & 121 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

[1825] $1^{14} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 10 & 10 & 3 & 61 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[1828] $1^{13} 4^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 12 & 11 & 3 & 97 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

[1829] $41^{12} 2231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 1 & 9 & 2 \\ 4 & 26 & 1 & 56 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 56 & 0 & 110 & 20 \\ 2 & 10 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[1830] $21^{14} 4131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 24 & 3 & 103 & 19 \\ 2 & 4 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

[1831] $1^{13} 4^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 13 & 12 & 2 & 112 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1832] $2^3 141^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 10 & 11 & 13 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1833] $1^{11} 4^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 0 & 12 & 2 \\ 17 & 12 & 100 & 22 \\ 3 & 2 & 22 & 4 \end{pmatrix}_{b_0}$$

[1834] $21^{14} 41^4 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 16 & 2 \\ 2 & 4 & 1 & 32 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 16 & 32 & 0 & 190 & 24 \\ 2 & 4 & 0 & 24 & 3 \end{pmatrix}_{b_0}$$

[1835] $221^{13} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 275, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 6 & 19 & 3 \\ 6 & 13 & 46 & 7 \\ 19 & 46 & 148 & 24 \\ 3 & 7 & 24 & 3 \end{pmatrix}_{b_0}$$

[1836] $1^{13} 4^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 275, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 9 & 3 & 57 & 15 \\ 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[1837] $1^{14} 4^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 0 & 10 & 2 \\ 18 & 10 & 86 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[1838] $41^{13} 2231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 1 & 12 & 2 \\ 4 & 26 & 1 & 84 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 84 & 0 & 250 & 30 \\ 2 & 10 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[1839] $31^{12} 31^4 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 31 & 3 \\ 5 & 5 & 43 & 5 \\ 31 & 43 & 320 & 36 \\ 3 & 5 & 36 & 3 \end{pmatrix}_{b_0}$$

- [1840] $31^{12}31^{14}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 16 & 3 \\ 5 & 5 & 22 & 5 \\ 16 & 22 & 74 & 18 \\ 3 & 5 & 18 & 3 \end{pmatrix}_{b_0}$
- [1841] $1^{12}4^{12}31^{14}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 10 & 2 & 90 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$
- [1842] $1^{11}4^{14}31^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 10 & 2 \\ 16 & 10 & 66 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$
- [1843] $1^{13}4^{14}31^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 11 & 9 & 4 & 64 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$
- [1844] $1^{14}4^{12}31^{12}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 11 & 11 & 2 & 72 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$
- [1845] $1^{13}4^{12}31^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 10 & 2 & 70 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$
- [1846] $31^{14}3141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 16 & 3 \\ 5 & 5 & 20 & 5 \\ 16 & 20 & 62 & 16 \\ 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$
- [1847] $221^{13}51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 281, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 6 & 17 & 3 \\ 6 & 13 & 41 & 7 \\ 17 & 41 & 115 & 21 \\ 3 & 7 & 21 & 3 \end{pmatrix}_{b_0}$
- [1848] $1^{11}4^{13}31^{14}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 283, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 0 & 11 & 2 \\ 17 & 11 & 86 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$
- [1849] $1^{13}4^{11}31^{14}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 283, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 14 & 12 & 1 & 109 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$
- [1850] $1^{12}4^{13}31^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 283, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 9 & 3 & 73 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$
- [1851] $2^{14}3^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 287, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 11 & 12 & 13 & 92 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1852] $2^{\overset{11}{1}\overset{41}{3}} 1^{\overset{13}{+}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 5 & 27 & 5 \\ 17 & 27 & 110 & 22 \\ 3 & 5 & 22 & 4 \end{pmatrix}_{b_0}$$

[1853] $1^{\overset{11}{1}\overset{31}{4}} 4^{\overset{12}{+}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 291, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 13 & 13 & 1 & 109 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

[1856] $2^{\overset{12}{1}\overset{41}{3}} 1^{\overset{13}{+}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 11 & 9 & 14 & 77 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[1855] $1^{\overset{12}{1}\overset{14}{3}\overset{31}{2}} 4^{\overset{13}{+}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 13 & 9 & 4 & 80 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1854] $1^{\overset{13}{1}\overset{14}{4}} 131^{\overset{14}{+}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 11 & 11 & 4 & 88 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1857] $1^{\overset{12}{1}\overset{13}{4}} 4^{\overset{13}{+}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 15 & 11 & 2 & 112 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1858] $1^{\overset{11}{1}\overset{13}{3}\overset{31}{4}} 4^{\overset{13}{+}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 0 & 12 & 2 \\ 18 & 12 & 106 & 22 \\ 3 & 2 & 22 & 4 \end{pmatrix}_{b_0}$$

[1859] $1^{\overset{11}{1}\overset{13}{3}\overset{31}{4}} 4^{\overset{14}{+}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 0 & 13 & 2 \\ 19 & 13 & 130 & 24 \\ 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[1860] $1^{\overset{11}{1}\overset{14}{4}\overset{231}{13}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 0 & 13 & 2 \\ 19 & 13 & 130 & 24 \\ 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[1861] $1^{\overset{14}{1}\overset{31}{4}} 4^{\overset{13}{+}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 0 & 11 & 2 \\ 19 & 11 & 104 & 22 \\ 3 & 2 & 22 & 4 \end{pmatrix}_{b_0}$$

[1862] $1^{\overset{14}{1}\overset{13}{3}\overset{31}{1}} 4^{\overset{12}{+}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 11 & 11 & 3 & 77 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[1863] $1^{13} 4^1 1^3 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 15 & 13 & 1 & 133 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$

[1864] $21^{14} 41^3 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 4 & 1 & 26 & 4 \\ 1 & 1 & -2 & 4 & 0 \\ 14 & 26 & 4 & 128 & 22 \\ 2 & 4 & 0 & 22 & 3 \end{pmatrix}_{b_0}$

[1865] $21^{14} 41^3 31^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 18 & 2 \\ 2 & 4 & 1 & 30 & 4 \\ 1 & 1 & -2 & 4 & 0 \\ 18 & 30 & 4 & 176 & 26 \\ 2 & 4 & 0 & 26 & 3 \end{pmatrix}_{b_0}$

[1866] $1^{13} 4^1 1^4 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 16 & 14 & 1 & 161 & 23 \\ 2 & 2 & 0 & 23 & 3 \end{pmatrix}_{b_0}$

[1867] $1^{14} 4^1 1^3 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 14 & 14 & 1 & 133 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$

[1868] $1^{12} 41^3 231^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 14 & 10 & 3 & 93 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$

[1869] $1^{13} 41^4 231^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 16 & 14 & 1 & 161 & 23 \\ 2 & 2 & 0 & 23 & 3 \end{pmatrix}_{b_0}$

[1870] $1^{11} 4^1 1^2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 0 & 11 & 2 \\ 17 & 11 & 80 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$

[1871] $1^{13} 4^1 1^2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 12 & 10 & 3 & 73 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$

[1872] $1^{12} 4^1 1^3 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 14 & 10 & 3 & 93 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$

[1875] $1^{12} 4^1 1^4 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 308, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 16 & 12 & 2 & 138 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$

[1874] $1^{13} 4^1 1^2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 308, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 13 & 11 & 2 & 88 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1877] $1^{14} 4^1 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 308, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 12 & 2 & 90 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1876] $1^{13} 4^1 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 308, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 12 & 10 & 4 & 82 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1873] $1^{12} 41^4 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 308, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 16 & 12 & 2 & 138 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

[1878] $21^{14} 41^2 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 311, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 4 & 1 & 28 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 13 & 28 & 1 & 135 & 21 \\ 2 & 4 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

[1881] $1^4 41^2 231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 20 & 3 \\ 2 & 0 & 12 & 2 \\ 20 & 12 & 126 & 24 \\ 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[1882] $1^{14} 4131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 12 & 12 & 4 & 110 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1880] $1^{12} 4^1 1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 14 & 10 & 4 & 102 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1879] $1^{14} 31^4 4^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 20 & 3 \\ 2 & 0 & 12 & 2 \\ 20 & 12 & 126 & 24 \\ 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[1883] $1^{14} 4^1 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 315, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 15 & 15 & 1 & 161 & 23 \\ 2 & 2 & 0 & 23 & 3 \end{pmatrix}_{b_0}$$

[1884] $31^{13} 3141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 315, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 15 & 3 \\ 5 & 5 & 20 & 5 \\ 15 & 20 & 58 & 16 \\ 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1885] $2^3 141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 5 & 35 & 5 \\ 21 & 35 & 194 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$$

[1886] $21^{14} 41^1 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 4 & 1 & 26 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 12 & 26 & 2 & 116 & 20 \\ 2 & 4 & 0 & 20 & 3 \end{pmatrix}_{b_0} \\
[1887] & 1^{441^{14}231^{11}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 17 & 2 \\ 3 & 4 & 1 & 32 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 17 & 32 & 1 & 191 & 25 \\ 2 & 4 & 0 & 25 & 3 \end{pmatrix}_{b_0} \\
[1888] & 21^{14}41^{231^{14}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 16 & 2 \\ 2 & 4 & 1 & 28 & 4 \\ 1 & 1 & -2 & 4 & 0 \\ 16 & 28 & 4 & 150 & 24 \\ 2 & 4 & 0 & 24 & 3 \end{pmatrix}_{b_0} \\
[1889] & 1^{12}4^{1^{13}31^3}4^{1^4} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 323, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 15 & 11 & 3 & 117 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0} \\
[1890] & 1^{14}4^{1^{13}31^1}4^{1^3} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 323, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 12 & 12 & 3 & 97 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0} \\
[1892] & 31^{12}31^341^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 324, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 5 & 26 & 3 \\ 5 & 5 & 36 & 5 \\ 26 & 36 & 218 & 30 \\ 3 & 5 & 30 & 3 \end{pmatrix}_{b_0} \\
[1891] & 31^{1231^2}41^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 324, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 5 & 21 & 3 \\ 5 & 5 & 29 & 5 \\ 21 & 29 & 136 & 24 \\ 3 & 5 & 24 & 3 \end{pmatrix}_{b_0} \\
[1893] & 1^{12}4^{1^{14}31^3}4^{1^2} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 327, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 14 & 9 & 4 & 80 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0} \\
[1894] & 2^3 3^{141^{13}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 327, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 11 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[1895] & 1^{14}4^{1^{11}31^4}4^{1^1} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 14 & 13 & 1 & 113 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0} \\
[1896] & 1^{13}41^3231^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 12 & 2 & 110 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0} \\
[1897] & 1^{13}4^{1^{14}31^1}4^{1^4} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 13 & 11 & 4 & 104 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0} \\
[1898] & 1^{13}4^{1^{12}31^3}4^{1^3} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8
\end{aligned}$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 12 & 2 & 110 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[1899] \ 1^{14} 4 \ 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 13 & 13 & 2 & 112 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[1900] \ 1^{11} 4 \ 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 0 & 12 & 2 \\ 18 & 12 & 98 & 22 \\ 3 & 2 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[1901] \ 1^{11} 41^3 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 0 & 12 & 2 \\ 18 & 12 & 98 & 22 \\ 3 & 2 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[1902] \ 31^{13} 31^{14} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 17 & 3 \\ 5 & 5 & 22 & 5 \\ 17 & 22 & 74 & 18 \\ 3 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1903] \ 1^{13} 4 \ 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 11 & 3 & 93 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1905] \ 21^{14} 41^3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 335, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 4 & 1 & 26 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 26 & 3 & 121 & 21 \\ 2 & 4 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1907] \ 1^{11} 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 335, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 0 & 11 & 2 \\ 18 & 11 & 84 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[1904] \ 21^{14} 41^4 31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 335, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 19 & 2 \\ 2 & 4 & 1 & 32 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 19 & 32 & 3 & 199 & 27 \\ 2 & 4 & 0 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[1906] \ 1^{13} 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 335, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 11 & 2 & 92 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1909] \ 41^{14} 232 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 2 & 16 & 2 \\ 4 & 10 & 5 & 48 & 6 \\ 2 & 5 & 0 & 16 & 2 \\ 16 & 48 & 16 & 190 & 24 \\ 2 & 6 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[1908] \ 1^{12} 4 \ 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 15 & 10 & 3 & 95 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

[1910] $1^{11} 1^{14} 31^3 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 339, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 0 & 13 & 2 \\ 19 & 13 & 120 & 24 \\ 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[1911] $1^{14} 1^{13} 31^1 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 339, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 13 & 3 & 121 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

[1912] $1^{13} 1^{13} 31^3 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 10 & 3 & 75 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[1913] $1^{13} 1^{12} 31^3 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 15 & 13 & 2 & 136 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

[1914] $1^{14} 1^{12} 31^2 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 14 & 2 & 138 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

[1915] $21^{14} 41^3 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 4 & 1 & 30 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 15 & 30 & 1 & 161 & 23 \\ 2 & 4 & 0 & 23 & 3 \end{pmatrix}_{b_0}$$

[1916] $21^{14} 41^4 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 17 & 2 \\ 2 & 4 & 1 & 32 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 17 & 32 & 1 & 191 & 25 \\ 2 & 4 & 0 & 25 & 3 \end{pmatrix}_{b_0}$$

[1917] $1^{13} 1^{13} 31^2 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 347, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 14 & 12 & 3 & 117 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

[1918] $1^{14} 1^{13} 31^2 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 348, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 12 & 11 & 3 & 77 & 17 \\ 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[1919] $1^{14} 1^{12} 31^3 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 351, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 13 & 12 & 2 & 92 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1921] $41^{12} 2^3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 2 & 10 & 2 \\ 4 & 10 & 5 & 28 & 6 \\ 2 & 5 & 0 & 11 & 2 \\ 10 & 28 & 11 & 65 & 15 \\ 2 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1920] $1^{14} 1^{11} 31^4 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 15 & 14 & 1 & 135 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1924] \ 1^{12} 1^{14} 3^1 4^1 1^3 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 359, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 15 & 10 & 4 & 100 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[1925] \ 1^{14} 1^{14} 3^1 4^1 1^2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 359, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 12 & 11 & 4 & 84 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1923] \ 1^{12} 4^1 3^{231} 1^4 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 359, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 15 & 10 & 4 & 100 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[1922] \ 1^{13} 1^{14} 3^1 4^1 1^2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 359, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 13 & 10 & 4 & 80 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1926] \ 1^{34} 1^{13} 231^1 4^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 3 & 1 & 16 & 2 \\ 3 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 4 & 0 \\ 16 & 24 & 4 & 122 & 22 \\ 2 & 4 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[1927] \ 1^{34} 1^{13} 231^1 4^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 3 & 1 & 14 & 2 \\ 3 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 24 & 2 & 110 & 20 \\ 2 & 4 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[1928] \ 1^{12} 1^{13} 3^1 4^1 1^3 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 16 & 11 & 3 & 117 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1929] \ 21^{14} 4^1 3^1 1^2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 4 & 1 & 28 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 28 & 2 & 138 & 22 \\ 2 & 4 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[1930] \ 21^{14} 4^1 3^1 1^2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 18 & 2 \\ 2 & 4 & 1 & 32 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 18 & 32 & 2 & 194 & 26 \\ 2 & 4 & 0 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[1931] \ 1^{13} 1^{12} 3^1 4^1 1^2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 367, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 15 & 12 & 2 & 112 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[1932] \ 21^{14} 4^1 3^1 1^3 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 367, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 17 & 2 \\ 2 & 4 & 1 & 30 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 17 & 30 & 3 & 169 & 25 \\ 2 & 4 & 0 & 25 & 3 \end{pmatrix}_{b_0}$$

[1933] $21^{14}41^231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 367, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 4 & 1 & 28 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 15 & 28 & 3 & 143 & 23 \\ 2 & 4 & 0 & 23 & 3 \end{pmatrix}_{b_0}$$

[1934] $1^{11}4^{14}31^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 0 & 12 & 2 \\ 19 & 12 & 100 & 22 \\ 3 & 2 & 22 & 4 \end{pmatrix}_{b_0}$$

[1935] $2^{12}3^{14}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 375, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 17 & 2 \\ 13 & 10 & 17 & 108 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[1936] $2^{13}3^{14}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 375, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 12 & 11 & 15 & 97 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1937] $1^{14}4^{11}31^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 16 & 15 & 1 & 161 & 23 \\ 2 & 2 & 0 & 23 & 3 \end{pmatrix}_{b_0}$$

[1938] $1^{13}4^{13}1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 14 & 11 & 3 & 93 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

[1939] $21^{14}41^331^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 380, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 16 & 2 \\ 2 & 4 & 1 & 30 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 16 & 30 & 2 & 164 & 24 \\ 2 & 4 & 0 & 24 & 3 \end{pmatrix}_{b_0}$$

[1940] $1^{12}4^{14}31^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 383, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 16 & 11 & 4 & 124 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

[1941] $1^{14}4^{12}31^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 383, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 14 & 13 & 2 & 112 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1942] $1^{12}41^4231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 17 & 12 & 3 & 143 & 23 \\ 2 & 2 & 0 & 23 & 3 \end{pmatrix}_{b_0}$$

[1943] $1^{12}4^{13}31^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 17 & 12 & 3 & 143 & 23 \\ 2 & 2 & 0 & 23 & 3 \end{pmatrix}_{b_0}$$

[1944] $1^{14}4^{13}31^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 13 & 12 & 3 & 95 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0} \\
[1945] & 1^{14} 4^{14} 231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 388, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 16 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 17 & 16 & 1 & 191 & 25 \\ 2 & 2 & 0 & 25 & 3 \end{pmatrix}_{b_0} \\
[1946] & 1^{14} 4^{14} 1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 388, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 16 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 17 & 16 & 1 & 191 & 25 \\ 2 & 2 & 0 & 25 & 3 \end{pmatrix}_{b_0} \\
[1947] & 1^{14} 4^{14} 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 391, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 13 & 12 & 4 & 104 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0} \\
[1948] & 1^{11} 4^{14} 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 391, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 20 & 3 \\ 2 & 0 & 13 & 2 \\ 20 & 13 & 120 & 24 \\ 3 & 2 & 24 & 4 \end{pmatrix}_{b_0} \\
[1949] & 1^{13} 4^{12} 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 391, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 16 & 13 & 2 & 136 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0} \\
[1950] & 1^{13} 4^{14} 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 391, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 14 & 11 & 4 & 100 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0} \\
[1951] & 1^3 4^{13} 231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 15 & 2 \\ 3 & 4 & 1 & 24 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 15 & 24 & 3 & 115 & 21 \\ 2 & 4 & 0 & 21 & 3 \end{pmatrix}_{b_0} \\
[1952] & 2^3 141^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 0 & 16 & 2 \\ 12 & 12 & 16 & 110 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1953] & 2^{11} 3^{14} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 402, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 20 & 3 \\ 4 & 5 & 32 & 5 \\ 20 & 32 & 154 & 26 \\ 3 & 5 & 26 & 4 \end{pmatrix}_{b_0} \\
[1954] & 1^{13} 4^{13} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 403, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 15 & 11 & 3 & 97 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0} \\
[1955] & 1^{13} 4^{13} 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 15 & 12 & 3 & 115 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1956] $1^{11} 1^{14} 31^4 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 404$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 21 & 3 \\ 2 & 0 & 14 & 2 \\ 21 & 14 & 144 & 26 \\ 3 & 2 & 26 & 4 \end{pmatrix}_{b_0}$
- [1957] $1^{13} 41^3 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 404$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 15 & 12 & 3 & 115 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$
- [1958] $1^{11} 41^4 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 404$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 21 & 3 \\ 2 & 0 & 14 & 2 \\ 21 & 14 & 144 & 26 \\ 3 & 2 & 26 & 4 \end{pmatrix}_{b_0}$
- [1961] $1^{13} 1^{12} 31^4 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 407$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 17 & 14 & 2 & 164 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}$
- [1960] $1^{13} 41^4 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 407$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 17 & 14 & 2 & 164 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}$
- [1959] $1^{14} 1^{12} 31^3 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 407$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 15 & 14 & 2 & 136 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$
- [1962] $1^{14} 1^{14} 31^2 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 408$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 13 & 11 & 4 & 84 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$
- [1963] $1^{14} 1^{13} 31^2 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 412$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 14 & 13 & 3 & 117 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$
- [1964] $1^{13} 1^{14} 31^2 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 415$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 15 & 12 & 4 & 124 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$
- [1965] $31^{15} 3141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 415$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.1$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 20 & 3 \\ 5 & 5 & 25 & 5 \\ 20 & 25 & 98 & 20 \\ 3 & 5 & 20 & 3 \end{pmatrix}_{b_0}$
- [1966] $1^{14} 1^{14} 31^1 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 415$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 14 & 13 & 4 & 128 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$
- [1967] $41^{12} 2^3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -87$, $\det = 416$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 2 & 12 & 2 \\ 4 & 10 & 5 & 32 & 6 \\ 2 & 5 & 0 & 14 & 2 \\ 12 & 32 & 14 & 90 & 18 \\ 2 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1968] $1^{14} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 420, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 15 & 13 & 2 & 116 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1969] $1^{14} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 423, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 16 & 15 & 2 & 164 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}$$

[1970] $1^{13} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 16 & 13 & 3 & 141 & 23 \\ 2 & 2 & 0 & 23 & 3 \end{pmatrix}_{b_0}$$

[1971] $1^{12} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 17 & 11 & 4 & 124 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

[1972] $1^{14} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 427, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 14 & 12 & 3 & 97 & 19 \\ 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

[1973] $41^{13} 2231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 1 & 15 & 2 \\ 4 & 26 & 1 & 84 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 84 & 0 & 250 & 30 \\ 2 & 10 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[1974] $1^{14} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 15 & 14 & 3 & 143 & 23 \\ 2 & 2 & 0 & 23 & 3 \end{pmatrix}_{b_0}$$

[1975] $1^{13} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 15 & 11 & 4 & 100 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1976] $2221^{13} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -100, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.19 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 4 & 1 & 12 & 2 \\ 4 & 18 & 1 & 60 & 8 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 60 & 0 & 178 & 24 \\ 2 & 8 & -1 & 24 & 2 \end{pmatrix}_{b_0}$$

[1977] $31^{13} 31^2 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 439, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 22 & 3 \\ 5 & 5 & 29 & 5 \\ 22 & 29 & 134 & 24 \\ 3 & 5 & 24 & 3 \end{pmatrix}_{b_0}$$

[1978] $2^1 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 2 & 2 & 0 & 16 & 2 \\ 13 & 13 & 16 & 121 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

- [1979] $1^{13} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 443$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 16 & 12 & 3 & 117 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$
- [1980] $41^{14} 2231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 444$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 4 & 1 & 16 & 2 \\ 4 & 26 & 1 & 112 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 112 & 0 & 446 & 40 \\ 2 & 10 & -1 & 40 & 3 \end{pmatrix}_{b_0}$
- [1982] $1^{12} 4^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 452$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 18 & 12 & 4 & 150 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}$
- [1981] $1^{12} 41^4 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 452$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 18 & 12 & 4 & 150 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}$
- [1983] $1^{14} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 452$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 14 & 12 & 4 & 102 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$
- [1984] $1^{14} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 456$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 16 & 14 & 2 & 138 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$
- [1985] $1^{14} 4^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 467$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.86 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 15 & 13 & 3 & 117 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$
- [1986] $31^{13} 31^{14} 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 471$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 20 & 3 \\ 5 & 5 & 27 & 5 \\ 20 & 27 & 110 & 22 \\ 3 & 5 & 22 & 3 \end{pmatrix}_{b_0}$
- [1987] $1^{13} 41^3 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 472$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 16 & 12 & 4 & 122 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$
- [1988] $1^{13} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 472$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 16 & 12 & 4 & 122 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$
- [1989] $1^{13} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -84$, $\det = 475$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 17 & 13 & 3 & 141 & 23 \\ 2 & 2 & 0 & 23 & 3 \end{pmatrix}_{b_0}$
- [1990] $31^{14} 3141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 480$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.17 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 19 & 3 \\ 5 & 5 & 25 & 5 \\ 19 & 25 & 92 & 20 \\ 3 & 5 & 20 & 3 \end{pmatrix}_{b_0}$$

[1991] $1^{14} 4^1 1^{1231^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 484, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 17 & 15 & 2 & 164 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}$$

[1992] $41^{13} 2231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 485, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 1 & 13 & 2 \\ 4 & 26 & 1 & 84 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 84 & 0 & 250 & 30 \\ 2 & 10 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[1993] $1^{14} 4^1 1^{1431^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 488, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 15 & 13 & 4 & 124 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

[1994] $2^1 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 492, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 14 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 0 & 18 & 2 \\ 14 & 12 & 18 & 130 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1995] $1^{14} 4^1 1^{1431^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 495, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 15 & 12 & 4 & 104 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[1996] $31^{14} 31^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 496, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 21 & 3 \\ 5 & 5 & 27 & 5 \\ 21 & 27 & 112 & 22 \\ 3 & 5 & 22 & 3 \end{pmatrix}_{b_0}$$

[1999] $31^{13} 31^3 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 499, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 27 & 3 \\ 5 & 5 & 36 & 5 \\ 27 & 36 & 214 & 30 \\ 3 & 5 & 30 & 3 \end{pmatrix}_{b_0}$$

[2000] $1^{14} 4^1 1^{1331^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 499, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 16 & 14 & 3 & 141 & 23 \\ 2 & 2 & 0 & 23 & 3 \end{pmatrix}_{b_0}$$

[1998] $1^{13} 4^1 1^{1331^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 499, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 18 & 14 & 3 & 169 & 25 \\ 2 & 2 & 0 & 25 & 3 \end{pmatrix}_{b_0}$$

[1997] $1^{13} 41^4 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 499, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 18 & 14 & 3 & 169 & 25 \\ 2 & 2 & 0 & 25 & 3 \end{pmatrix}_{b_0}$$

[2001] $1^{13} 4^1 1^{1431^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 500, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 17 & 13 & 4 & 148 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}$$

[2002] $1^{14} 4^1 1^{1231^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 504, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$

$$\begin{aligned}
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 16 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 18 & 16 & 2 & 194 & 26 \\ 2 & 2 & 0 & 26 & 3 \end{pmatrix}_{b_0} \\
[2003] & 1^{14} 4^4 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 504, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 16 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 18 & 16 & 2 & 194 & 26 \\ 2 & 2 & 0 & 26 & 3 \end{pmatrix}_{b_0} \\
[2004] & 1^{14} 4^4 1^{13} 31^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 504, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 16 & 13 & 3 & 121 & 21 \\ 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0} \\
[2005] & 31^{13} 31^4 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 511, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 5 & 32 & 3 \\ 5 & 5 & 43 & 5 \\ 32 & 43 & 314 & 36 \\ 3 & 5 & 36 & 3 \end{pmatrix}_{b_0} \\
[2006] & 1^{13} 4^4 1^{14} 31^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 511, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 17 & 12 & 4 & 124 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0} \\
[2007] & 1^{14} 4^4 1^{14} 31^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 516, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 16 & 14 & 4 & 150 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0} \\
[2008] & 1^{14} 4^4 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 520, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 18 & 2 \\ 3 & 4 & 1 & 32 & 4 \\ 1 & 1 & -2 & 2 & 0 \\ 18 & 32 & 2 & 194 & 26 \\ 2 & 4 & 0 & 26 & 3 \end{pmatrix}_{b_0} \\
[2009] & 1^{14} 4^4 1^{13} 31^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 523, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 17 & 15 & 3 & 169 & 25 \\ 2 & 2 & 0 & 25 & 3 \end{pmatrix}_{b_0} \\
[2010] & 41^{13} 2231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 4 & 1 & 14 & 2 \\ 4 & 26 & 1 & 84 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 84 & 0 & 250 & 30 \\ 2 & 10 & -1 & 30 & 3 \end{pmatrix}_{b_0} \\
[2011] & 1^{14} 4^4 1^{14} 31^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 543, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 16 & 13 & 4 & 124 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0} \\
[2012] & 1^{14} 4^4 1^{13} 31^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 17 & 14 & 3 & 143 & 23 \\ 2 & 2 & 0 & 23 & 3 \end{pmatrix}_{b_0} \\
[2013] & 1^{13} 4^4 1^{14} 31^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 551, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 18 & 13 & 4 & 148 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2014] $31^{13}31^241^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 579, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 25 & 3 \\ 5 & 5 & 34 & 5 \\ 25 & 34 & 182 & 28 \\ 3 & 5 & 28 & 3 \end{pmatrix}_{b_0}$$
- [2015] $31^{16}3141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.2$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 24 & 3 \\ 5 & 5 & 30 & 5 \\ 24 & 30 & 142 & 24 \\ 3 & 5 & 24 & 3 \end{pmatrix}_{b_0}$$
- [2016] $41^{13}2^32 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 2 & 14 & 2 \\ 4 & 10 & 5 & 40 & 6 \\ 2 & 5 & 0 & 15 & 2 \\ 14 & 40 & 15 & 133 & 21 \\ 2 & 6 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$
- [2017] $1^{14}4^11 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 17 & 13 & 4 & 128 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$
- [2018] $1^{13}4^14 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 583, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 19 & 14 & 4 & 176 & 26 \\ 2 & 2 & 0 & 26 & 3 \end{pmatrix}_{b_0}$$
- [2019] $1^{14}4^13 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 583, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 17 & 14 & 4 & 148 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}$$
- [2020] $1^{13}41^4231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 583, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 19 & 14 & 4 & 176 & 26 \\ 2 & 2 & 0 & 26 & 3 \end{pmatrix}_{b_0}$$
- [2021] $1^{14}4^13 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 584, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 18 & 15 & 3 & 169 & 25 \\ 2 & 2 & 0 & 25 & 3 \end{pmatrix}_{b_0}$$
- [2022] $2^1441^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 591, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 15 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 2 & 2 & 0 & 19 & 2 \\ 15 & 14 & 19 & 156 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$
- [2023] $1^{14}4^14 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 612, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 0 & 1 & 16 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 19 & 16 & 3 & 199 & 27 \\ 2 & 2 & 0 & 27 & 3 \end{pmatrix}_{b_0}$$
- [2024] $1^{14}41^4231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 612, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 0 & 1 & 16 & 2 \\ 1 & 1 & -2 & 3 & 0 \\ 19 & 16 & 3 & 199 & 27 \\ 2 & 2 & 0 & 27 & 3 \end{pmatrix}_{b_0}$$
- [2025] $1^{14}4^14 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 615, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 18 & 15 & 4 & 176 & 26 \\ 2 & 2 & 0 & 26 & 3 \end{pmatrix}_{b_0}$$

[2026] $1^{14} 4^1 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 632, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 14 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 18 & 14 & 4 & 150 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}$$

[2027] $31^{13} 31^3 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 639, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 30 & 3 \\ 5 & 5 & 41 & 5 \\ 30 & 41 & 274 & 34 \\ 3 & 5 & 34 & 3 \end{pmatrix}_{b_0}$$

[2028] $1^4 41^{14} 231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 640, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 1 & 19 & 2 \\ 3 & 4 & 1 & 32 & 4 \\ 1 & 1 & -2 & 3 & 0 \\ 19 & 32 & 3 & 199 & 27 \\ 2 & 4 & 0 & 27 & 3 \end{pmatrix}_{b_0}$$

[2029] $31^{13} 31^4 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 651, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 35 & 3 \\ 5 & 5 & 48 & 5 \\ 35 & 48 & 386 & 40 \\ 3 & 5 & 40 & 3 \end{pmatrix}_{b_0}$$

[2030] $41^{15} 2231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 660, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.1$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 1 & 20 & 2 \\ 4 & 26 & 1 & 140 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 140 & 0 & 698 & 50 \\ 2 & 10 & -1 & 50 & 3 \end{pmatrix}_{b_0}$$

[2031] $31^{14} 31^2 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 664, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 26 & 3 \\ 5 & 5 & 34 & 5 \\ 26 & 34 & 182 & 28 \\ 3 & 5 & 28 & 3 \end{pmatrix}_{b_0}$$

[2032] $1^{14} 4^1 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 676, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 19 & 15 & 4 & 176 & 26 \\ 2 & 2 & 0 & 26 & 3 \end{pmatrix}_{b_0}$$

[2033] $1^4 41^{14} 231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 680, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 1 & 20 & 2 \\ 3 & 4 & 1 & 32 & 4 \\ 1 & 1 & -2 & 4 & 0 \\ 20 & 32 & 4 & 206 & 28 \\ 2 & 4 & 0 & 28 & 3 \end{pmatrix}_{b_0}$$

[2034] $31^{14} 31^1 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 696, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 24 & 3 \\ 5 & 5 & 32 & 5 \\ 24 & 32 & 154 & 26 \\ 3 & 5 & 26 & 3 \end{pmatrix}_{b_0}$$

[2036] $1^{14} 4^1 1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 712, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & 0 & 1 & 16 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 20 & 16 & 4 & 206 & 28 \\ 2 & 2 & 0 & 28 & 3 \end{pmatrix}_{b_0}$$

[2035] $1^4 41^4 231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 712, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & 0 & 1 & 16 & 2 \\ 1 & 1 & -2 & 4 & 0 \\ 20 & 16 & 4 & 206 & 28 \\ 2 & 2 & 0 & 28 & 3 \end{pmatrix}_{b_0}$$

[2037] $41^{14} 2231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 716, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 1 & 20 & 2 \\ 4 & 26 & 1 & 112 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 112 & 0 & 446 & 40 \\ 2 & 10 & -1 & 40 & 3 \end{pmatrix}_{b_0}$$

- [2038] $41^{14}2231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 749, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 1 & 17 & 2 \\ 4 & 26 & 1 & 112 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 112 & 0 & 446 & 40 \\ 2 & 10 & -1 & 40 & 3 \end{pmatrix}_{b_0}$$
- [2039] $31^{17}3141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 775, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 28 & 3 \\ 5 & 5 & 35 & 5 \\ 28 & 35 & 194 & 28 \\ 3 & 5 & 28 & 3 \end{pmatrix}_{b_0}$$
- [2040] $31^{14}31^341^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 784, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 31 & 3 \\ 5 & 5 & 41 & 5 \\ 31 & 41 & 272 & 34 \\ 3 & 5 & 34 & 3 \end{pmatrix}_{b_0}$$
- [2041] $41^{13}2^332 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 784, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 2 & 16 & 2 \\ 4 & 10 & 5 & 44 & 6 \\ 2 & 5 & 0 & 18 & 2 \\ 16 & 44 & 18 & 166 & 24 \\ 2 & 6 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$
- [2042] $41^{14}2^332 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 836, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 2 & 18 & 2 \\ 4 & 10 & 5 & 52 & 6 \\ 2 & 5 & 0 & 19 & 2 \\ 18 & 52 & 19 & 225 & 27 \\ 2 & 6 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$
- [2043] $31^{14}31^441^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 856, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 36 & 3 \\ 5 & 5 & 48 & 5 \\ 36 & 48 & 382 & 40 \\ 3 & 5 & 40 & 3 \end{pmatrix}_{b_0}$$
- [2044] $31^{14}31^241^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 864, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 29 & 3 \\ 5 & 5 & 39 & 5 \\ 29 & 39 & 236 & 32 \\ 3 & 5 & 32 & 3 \end{pmatrix}_{b_0}$$
- [2045] $41^{14}2231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 885, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 1 & 19 & 2 \\ 4 & 26 & 1 & 112 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 112 & 0 & 446 & 40 \\ 2 & 10 & -1 & 40 & 3 \end{pmatrix}_{b_0}$$
- [2046] $41^{14}2231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 896, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 1 & 18 & 2 \\ 4 & 26 & 1 & 112 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 112 & 0 & 446 & 40 \\ 2 & 10 & -1 & 40 & 3 \end{pmatrix}_{b_0}$$
- [2047] $41^{16}2231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 924, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.2$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 1 & 24 & 2 \\ 4 & 26 & 1 & 168 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 168 & 0 & 1006 & 60 \\ 2 & 10 & -1 & 60 & 3 \end{pmatrix}_{b_0}$$
- [2048] $31^{14}31^341^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 984, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 34 & 3 \\ 5 & 5 & 46 & 5 \\ 34 & 46 & 338 & 38 \\ 3 & 5 & 38 & 3 \end{pmatrix}_{b_0}$$
- [2049] $31^{14}31^441^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 1056, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 39 & 3 \\ 5 & 5 & 53 & 5 \\ 39 & 53 & 460 & 44 \\ 3 & 5 & 44 & 3 \end{pmatrix}_{b_0}$$
- [2050] $41^{14}2^332 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 1184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 2 & 20 & 2 \\ 4 & 10 & 5 & 56 & 6 \\ 2 & 5 & 0 & 22 & 2 \\ 20 & 56 & 22 & 266 & 30 \\ 2 & 6 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[2051] \ 41^{1\bar{7}2231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2} \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 1236, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.3$$

$$T_{\min} = 10.99 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 1 & 28 & 2 \\ 4 & 26 & 1 & 196 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 196 & 0 & 1370 & 70 \\ 2 & 10 & -1 & 70 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2052] \ 1^1 \bar{4} 1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -56, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[2055] \ 1^1 \bar{4} 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -75, \ \det = -3, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 4 & 5 \\ 3 & 4 & 2 & 4 \\ 3 & 5 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[2056] \ 1^1 \bar{5} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -3, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 3 & 5 \\ 2 & 3 & 1 & 3 \\ 3 & 5 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[2057] \ 2^1 \bar{3} 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -3, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 1 & 1 & 3 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[2058] \ 1^1 \bar{4} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -3, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 3 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[2059] \ 1^1 \bar{5} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -3, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -2 & 0 \\ 3 & 5 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[2060] \ 1^1 \bar{3} 14131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -63, \ \det = -3, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 3 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[2061] \ 1^1 \bar{3} 1^1 4131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -63, \ \det = -3, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2054] \ 1^1 \bar{4} 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -75, \ \det = -3, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -2 & 0 \\ 3 & 5 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[2053] \ 1^1 \bar{4} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -3, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 3 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[2071] \ 21 \bar{5} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -4, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 1 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
[2066] \quad 1^1 4 13 1^1 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 4 & 0 & 10 & 6 \\ 2 & 3 & -1 & 6 & 3 \end{pmatrix}_{b_0} \\
[2067] \quad 1^1 5 13 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 3 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0} \\
[2068] \quad 1^1 5 1^1 3 2 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 5 & 10 & 5 \\ 6 & 10 & 19 & 9 \\ 3 & 5 & 9 & 4 \end{pmatrix}_{b_0} \\
[2069] \quad 1^1 4 13 14 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & -1 & 0 & 3 \end{pmatrix}_{b_0} \\
[2070] \quad 1^1 5 12 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 3 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
[2065] \quad 2 1^1 5 13 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 5 & 3 & 5 \\ 3 & 3 & 1 & 3 \\ 4 & 5 & 3 & 4 \end{pmatrix}_{b_0} \\
[2072] \quad 15 1^1 2 3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 2 & 2 & 2 \\ 4 & 10 & 5 & 4 & 6 \\ 2 & 5 & 0 & 3 & 2 \\ 2 & 4 & 3 & 1 & 3 \\ 2 & 6 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
[2073] \quad 1 4 1 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -75, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 4 & 5 \\ 2 & 4 & 2 & 4 \\ 3 & 5 & 4 & 5 \end{pmatrix}_{b_0} \\
[2074] \quad 13 2 1^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & 10 & 6 \\ 2 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0} \\
[2075] \quad 2 1^1 5 13 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 3 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0} \\
[2076] \quad 1^1 5 12 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[2077] $1^{11}322151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 3 & 2 \\ 1 & 0 & 3 & 1 & 3 \\ 2 & 0 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[2078] $1^{11}5131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[2079] $1^{11}5131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 1 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[2080] $1^{12}5131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 3 & 3 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$

[2064] $31^{11}4141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 9 & 4 \\ 3 & 0 & 9 & 13 & 7 \\ 2 & 0 & 4 & 7 & 3 \end{pmatrix}_{b_0}$

[2062] $1^{11}51231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 1 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[2063] $1^{13}5131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 7 & 4 \\ 7 & 7 & 9 & 7 \\ 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$

[2081] $1^{11}2321^151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 7 & 3 \\ 4 & 5 & 10 & 5 \\ 7 & 10 & 19 & 9 \\ 3 & 5 & 9 & 4 \end{pmatrix}_{b_0}$

[2082] $31^{11}4141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 5 & 1 & 0 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 5 & -1 & 0 & 3 \end{pmatrix}_{b_0}$

[2083] $131^{11}51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 4 \end{pmatrix}_{b_0}$

[2093] $2^13^151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 3 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$

$$[2085] \quad 1 \overset{11}{5} 1 \overset{1}{3} 2 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2108] \quad 1 \overset{12}{5} 1231 \overset{11}{1} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 5 & 5 & 5 \\ 4 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2107] \quad 21 \overset{11}{5} 131 \overset{1}{1} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 3 & 3 & 2 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2106] \quad 1 \overset{1}{5} 131 \overset{11}{1} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2105] \quad 2 \overset{11}{3} 1 \overset{13}{5} 1 \overset{11}{1} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 6 & 6 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2104] \quad 1 \overset{11}{4} 13141 \overset{1}{1} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2103] \quad 231 \overset{1}{5} 1 \overset{1}{1} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2102] \quad 1 \overset{11}{5} 131 \overset{1}{1} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 3 & 3 & 2 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2101] \quad 1 \overset{11}{3} 22151 \overset{1}{1} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2094] \quad 21 \overset{11}{5} 131 \overset{12}{2} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 5 & 5 & 8 & 24 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2095] \quad 1 \overset{1}{3} 221 \overset{1}{5} 1 \overset{1}{1} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 5 & 2 \\ 4 & 0 & 5 & 18 & 8 \\ 2 & 0 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[2096] \quad 1 \overset{1}{5} 1231 \overset{11}{1} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2097] \quad 223151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 4 & 2 & 4 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2090] \quad 215131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2091] \quad 1^1 41^1 3141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 5 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 5 & 0 & 13 & 7 \\ 2 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2092] \quad 215131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2089] \quad 15131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2088] \quad 1^2 5131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 5 & 11 & 5 \\ 6 & 11 & 22 & 10 \\ 3 & 5 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[2100] \quad 232151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2099] \quad 1^{11} 51231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2098] \quad 314141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 0 & 4 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 0 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[2087] \quad 1^1 322151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 0 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[2086] \quad 1^1 5131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[2084] $1^{\overline{11}} 5^{11} 1231^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[2113] $1^{\overline{11}} 5^{11} 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 2 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[2115] $2^{\overline{11}} 3^{11} 151^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 7 & 3 \\ 4 & 5 & 11 & 5 \\ 7 & 11 & 22 & 10 \\ 3 & 5 & 10 & 4 \end{pmatrix}_{b_0}$$

[2109] $1^{\overline{11}} 2231^{\overline{11}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -36, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 3 & 1 & 8 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 8 & 0 & 15 & 9 \\ 2 & 5 & -1 & 9 & 5 \end{pmatrix}_{b_0}$$

[2110] $2^{\overline{11}} 3^{11} 151^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 5 & 3 & 5 \\ 2 & 3 & 1 & 3 \\ 3 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

[2111] $1^{\overline{12}} 5^{12} 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 2 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[2114] $1^{\overline{12}} 5^{12} 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 5 & 8 & 5 \\ 6 & 8 & 9 & 7 \\ 3 & 5 & 7 & 4 \end{pmatrix}_{b_0}$$

[2112] $1^{\overline{11}} 5^{11} 1231^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[2118] $1^{\overline{11}} 223131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -36, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & -1 & 5 & 5 \end{pmatrix}_{b_0}$$

[2117] $1^{\overline{11}} 231321^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -37, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & -1 & 5 & 5 \end{pmatrix}_{b_0}$$

[2116] $1^{\overline{11}} 223131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -36, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 5 & 4 \\ 2 & 0 & 5 & 3 & 5 \\ 2 & 0 & 4 & 5 & 5 \end{pmatrix}_{b_0}$$

[2126] $31^{\overset{1}{1}}4141^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 5 & 1 & 9 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 9 & 0 & 13 & 7 \\ 2 & 5 & -1 & 7 & 3 \end{pmatrix}_{b_0}$

[2125] $2^{\overset{1}{1}}3^{\overset{1}{1}}21^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 1 & 4 & 3 \\ 2 & 1 & 4 & 2 & 4 \\ 2 & 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}$

[2124] $1^{\overset{1}{1}}5^{\overset{1}{1}}131^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 7 & 4 \\ 4 & 7 & 9 & 7 \\ 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$

[2123] $1^{\overset{1}{1}}1^{\overset{1}{1}}5^{\overset{1}{1}}1231^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 5 & 5 \\ 2 & 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$

[2122] $1^{\overset{1}{1}}1^{\overset{1}{1}}5^{\overset{1}{1}}1231^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 6 & 0 & 26 & 10 \\ 2 & 2 & -1 & 10 & 3 \end{pmatrix}_{b_0}$

[2121] $1^{\overset{1}{1}}5^{\overset{1}{1}}131^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 14 & 4 \\ 8 & 14 & 42 & 14 \\ 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$

[2120] $1^{\overset{1}{1}}5^{\overset{1}{1}}131^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 21 & 4 \\ 12 & 21 & 97 & 21 \\ 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$

[2119] $1^{\overset{1}{1}}1^{\overset{1}{1}}5^{\overset{1}{1}}1231^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 61 & 15 \\ 2 & 2 & -1 & 15 & 3 \end{pmatrix}_{b_0}$

[2127] $1^{\overset{1}{1}}1^{\overset{1}{1}}5^{\overset{1}{1}}131^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 2 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[2129] $1^{\overset{1}{1}}1^{\overset{1}{1}}5^{\overset{1}{1}}1231^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 \\ 3 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$

[2128] $2^{\overset{1}{1}}3^{\overset{1}{1}}21^{\overset{1}{1}}51^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 4 & 3 & 4 & 10 & 6 \\ 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

$$[2130] \begin{smallmatrix} 1^1 \\ 1 \end{smallmatrix} 1^1 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 5 & 8 & 5 \\ 4 & 8 & 9 & 7 \\ 3 & 5 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[2131] 21^1 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 2 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[2132] 22^1 3 151^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 3 & 8 & 2 \\ 3 & 3 & 4 & 13 & 4 \\ 6 & 8 & 13 & 35 & 11 \\ 2 & 2 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2140] 31^1 31^1 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 7 & 4 \\ 6 & 7 & 9 & 7 \\ 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[2139] 1^1 5 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 7 & 4 \\ 6 & 7 & 9 & 7 \\ 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[2138] 1^1 412321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 4 & 3 \\ 4 & 4 & 6 & 6 \\ 3 & 3 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[2137] 1^1 314131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 3 & 1 & 2 & 4 \\ 2 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2136] 1^1 41^1 2321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 1 & 4 & 3 \\ 6 & 4 & 10 & 8 \\ 3 & 3 & 8 & 5 \end{pmatrix}_{b_0}$$

$$[2135] 1^1 412321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 4 & 3 \\ 4 & 4 & 6 & 6 \\ 3 & 3 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[2134] 1^1 3141^1 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 4 & 0 & 5 & 5 \\ 2 & 3 & 0 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[2133] 1^1 41^1 2321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 1 & 4 & 3 \\ 6 & 4 & 10 & 8 \\ 3 & 3 & 8 & 5 \end{pmatrix}_{b_0}$$

$$[2141] 31^1 4 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 7 & 4 \\ 2 & 0 & 7 & 5 & 5 \\ 2 & 0 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[2149] $21^{11}5131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 8 & 5 \\ 5 & 8 & 9 & 7 \\ 3 & 5 & 7 & 4 \end{pmatrix}_{b_0}$

[2148] $23^{11}151^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 8 & 10 & 9 & 67 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2147] $151^132 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[2146] $23^{12}151^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 7 & 5 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[2144] $21^{11}5131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 3 & 3 & 2 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$

[2142] $21^{12}5131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 9 & 9 & 9 & 69 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2143] $1^15131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 8 & 5 \\ 5 & 8 & 9 & 7 \\ 3 & 5 & 7 & 4 \end{pmatrix}_{b_0}$

[2145] $23^{11}2151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[2150] $1^15131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 6 & 3 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[2159] $1^15131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 19 & 5 \\ 13 & 19 & 65 & 17 \\ 3 & 5 & 17 & 4 \end{pmatrix}_{b_0}$

[2158] $21^{11}51231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[2157] $21^35131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 5 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2156] \ 1^{11} 5^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 7 & 4 \\ 5 & 7 & 9 & 7 \\ 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[2155] \ 1^{11} 5^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 7 & 4 \\ 5 & 7 & 9 & 7 \\ 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[2160] \ 1^{11} 5^{11} 1^{32} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 12 & 0 & 155 & 23 \\ 2 & 2 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[2163] \ 1^{11} 5^{11} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 4 & 3 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[2162] \ 21^{11} 5^{11} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 4 & 3 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[2154] \ 1^{11} 41^{11} 31^{44} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 17 & 2 \\ 3 & 13 & 1 & 68 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 68 & 0 & 346 & 34 \\ 2 & 7 & -1 & 34 & 3 \end{pmatrix}_{b_0}$$

$$[2153] \ 141^{11} 31^{14} 1^{41} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 13 & 1 & 12 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 12 & 0 & 10 & 6 \\ 2 & 7 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[2152] \ 21^{11} 5^{11} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 19 & 5 \\ 13 & 19 & 65 & 17 \\ 3 & 5 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[2151] \ 1^{21} 5^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2161] \ 1^{13} 5^{13} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 5 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2164] \ 1^{11} 2231^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -36, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 8 & 4 \\ 4 & 0 & 8 & 15 & 9 \\ 2 & 0 & 4 & 9 & 5 \end{pmatrix}_{b_0} \\
[2175] & 1^{11}322151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 0 & 3 & 1 & 3 \\ 2 & 0 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
[2165] & 1^{14}1^{13}141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 5 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 5 & 0 & 13 & 7 \\ 2 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0} \\
[2166] & 1^{13}51231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 5 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[2167] & 1^{13}51231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 10 & 5 \\ 7 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0} \\
[2168] & 1^{11}51231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[2169] & 1^{11}413141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0} \\
[2170] & 21^35131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 8 & 6 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2171] & 1^{11}5131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 8 & 6 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2172] & 1^{11}51231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 10 & 5 \\ 5 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0} \\
[2173] & 1^35131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 19 & 5 \\ 10 & 19 & 65 & 17 \\ 3 & 5 & 17 & 4 \end{pmatrix}_{b_0} \\
[2174] & 21^35131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[2176] \quad 1 \overset{1}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 5 & 13 & 5 \\ 6 & 13 & 25 & 11 \\ 3 & 5 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[2177] \quad 1 \overset{1}{5} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 5 & 6 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2178] \quad 1 \overset{1}{5} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[2179] \quad 31^1 321^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 4 & 3 & 1 & 2 \\ 4 & 10 & 5 & 3 & 6 \\ 3 & 5 & 0 & 1 & 2 \\ 1 & 3 & 1 & 0 & 2 \\ 2 & 6 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2183] \quad 2 \overset{11}{3} 1 \overset{13}{5} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 7 & 8 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2182] \quad 1 \overset{1}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 7 & 4 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2181] \quad 2 \overset{11}{3} 1 \overset{1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 2 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[2180] \quad 1 \overset{1}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 5 & 3 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[2184] \quad 151^{12} 2231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 11 & 2 \\ 4 & 18 & 6 & 55 & 8 \\ 1 & 6 & 0 & 19 & 2 \\ 11 & 55 & 19 & 165 & 25 \\ 2 & 8 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[2186] \quad 21^{12} 5131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 11 & 11 & 15 & 117 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[2185] \quad 21^{12} 5131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 5 & 5 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[2188] $1^{11}5^{12}1^{12}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 6 & 5 & 5 & 18 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[2187] $2^13^{141}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 6 & 6 & 6 \\ 3 & 6 & 3 & 5 \\ 3 & 6 & 5 & 5 \end{pmatrix}_{b_0}$

[2189] $31^{11}4^{141}141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 5 & 1 & 9 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 9 & 0 & 13 & 7 \\ 2 & 5 & -1 & 7 & 3 \end{pmatrix}_{b_0}$

[2190] $1^{12}5^{12}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 5 & 16 & 5 \\ 8 & 16 & 42 & 14 \\ 3 & 5 & 14 & 4 \end{pmatrix}_{b_0}$

[2191] $1^{12}5^{12}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 4 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$

[2192] $21^{12}5^{12}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 4 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$

[2193] $1^{12}5^{12}1231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 5 & 10 & 5 \\ 6 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$

[2194] $1^{13}2221^{15}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 18 & 6 & 0 & 8 \\ 1 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 8 & 2 & 0 & 3 \end{pmatrix}_{b_0}$

[2195] $1^{13}21^{1321}5^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 5 & 5 & 18 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[2196] $1^{11}5^{12}1^{12}231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 6 & 0 & 31 & 11 \\ 2 & 2 & -1 & 11 & 3 \end{pmatrix}_{b_0}$

[2197] $1^{11}41^{13}141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 8 & 0 & 26 & 10 \\ 2 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$

[2201] $1^{12}5^{14}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 14 & 4 \\ 12 & 14 & 42 & 14 \\ 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[2200] \ 1^{11} 5^{12} 1^{23} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 9 & 0 & 81 & 17 \\ 2 & 2 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2199] \ 1^{12} 5^{13} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 14 & 4 \\ 9 & 14 & 42 & 14 \\ 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[2198] \ 31^{23} 1^{12} 5^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 14 & 4 \\ 12 & 14 & 42 & 14 \\ 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[2208] \ 1^{11} 5^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 6 & 4 & 5 & 11 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2209] \ 1^{11} 5^{12} 1^{23} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[2207] \ 1^{11} 41^{13} 1^{41} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 7 & 0 & 21 & 9 \\ 2 & 3 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2212] \ 1^{11} 5^{12} 1^{23} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[2211] \ 1^{11} 5^{12} 1^{23} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 15 & 5 \\ 10 & 15 & 37 & 13 \\ 3 & 5 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[2210] \ 1^{11} 5^{32} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 5 & 15 & 5 \\ 8 & 15 & 37 & 13 \\ 3 & 5 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[2202] \ 1^{11} 5^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 5 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2203] \ 1^{11} 32^{21} 1^{51} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 8 & 2 \\ 5 & 0 & 8 & 31 & 11 \\ 2 & 0 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2204] & 1 \overset{1^3}{\underset{1^2}{5}} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 16 & 5 \\ 11 & 16 & 42 & 14 \\ 3 & 5 & 14 & 4 \end{pmatrix}_{b_0} \\
[2205] & 1 \overset{1^2}{\underset{1^{11}}{5}} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 4 & 5 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[2206] & 1 \overset{1^1}{\underset{1^{11}}{5}} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 3 & 4 & 5 & 11 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0} \\
[2219] & 2 \overset{1^{12}}{\underset{1^{13}}{3}} 151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 5 & 19 & 5 \\ 12 & 19 & 65 & 17 \\ 3 & 5 & 17 & 4 \end{pmatrix}_{b_0} \\
[2218] & 1^1 3221^1 \overset{1^{11}}{51} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 5 & 2 \\ 5 & 0 & 5 & 18 & 8 \\ 2 & 0 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[2217] & 1 \overset{1^1}{\underset{1^{11}}{5}} 1^1 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 18 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[2216] & 2 \overset{1^{11}}{\underset{1^{11}}{3}} 2151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0} \\
[2215] & 31 \overset{1^{11}}{\underset{1^{11}}{4}} 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 7 & 0 & 5 & 5 \\ 2 & 5 & -1 & 5 & 3 \end{pmatrix}_{b_0} \\
[2214] & 1 \overset{1^3}{\underset{1^1}{5}} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 13 & 5 \\ 9 & 13 & 25 & 11 \\ 3 & 5 & 11 & 4 \end{pmatrix}_{b_0} \\
[2213] & 1^1 51 \overset{1^{11}}{\underset{1^{11}}{3}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0} \\
[2220] & 1 \overset{1^1}{\underset{1^{11}}{3}} 1^1 4131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 6 & 1 & 14 & 8 \\ 2 & 3 & 0 & 8 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[2221] $31^2 4141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 18 & 4 \\ 7 & 0 & 18 & 58 & 14 \\ 2 & 0 & 4 & 14 & 3 \end{pmatrix}_{b_0}$

[2235] $21^{11} 51^{12} 231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 18 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[2229] $1^2 5^{12} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$

[2227] $2^3 3^{12} 151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 5 & 3 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[2230] $2^3 3^{13} 151^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 10 & 7 & 13 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2231] $21^{11} 51231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[2232] $21^{12} 51^2 231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 13 & 13 & 19 & 165 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0}$

[2223] $21^{12} 51231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$

[2225] $221^{11} 5131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 4 & 3 & 6 & 4 \\ 1 & 3 & 0 & 2 & 2 \\ 3 & 6 & 2 & 4 & 4 \\ 2 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$

[2226] $1^2 5^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[2224] $21^{12} 51^3 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 20 & 2 \\ 15 & 15 & 20 & 218 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0}$

[2222] $31^1 4141^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 9 & 4 \\ 6 & 0 & 9 & 13 & 7 \\ 2 & 0 & 4 & 7 & 3 \end{pmatrix}_{b_0} \\
[2233] \quad & 21^{1251^3}231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 13 & 13 & 14 & 150 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[2228] \quad & 31^{11}4141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 9 & 4 \\ 4 & 0 & 9 & 13 & 7 \\ 2 & 0 & 4 & 7 & 3 \end{pmatrix}_{b_0} \\
[2234] \quad & 2321^{11}51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 18 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[2237] \quad & 21^{11}5131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 13 & 5 \\ 7 & 13 & 25 & 11 \\ 3 & 5 & 11 & 4 \end{pmatrix}_{b_0} \\
[2236] \quad & 1^{11}51^{11}231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 15 & 5 \\ 9 & 15 & 37 & 13 \\ 3 & 5 & 13 & 4 \end{pmatrix}_{b_0} \\
[2239] \quad & 2^13^1151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 5 & 8 & 5 \\ 5 & 8 & 9 & 7 \\ 3 & 5 & 7 & 4 \end{pmatrix}_{b_0} \\
[2242] \quad & 2^13^1151^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 6 & 4 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0} \\
[2238] \quad & 1^{11}5131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 13 & 5 \\ 7 & 13 & 25 & 11 \\ 3 & 5 & 11 & 4 \end{pmatrix}_{b_0} \\
[2241] \quad & 1^{11}3221^{11}51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 8 & 0 & 31 & 11 \\ 2 & 3 & -1 & 11 & 3 \end{pmatrix}_{b_0} \\
[2240] \quad & 2^13^1151^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 6 & 7 & 6 & 30 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0} \\
[2243] \quad & 1^{12}5131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 14 & 4 \\ 10 & 14 & 42 & 14 \\ 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[2244] \quad 2 \overset{1^2}{3} 1 \overset{1^1}{5} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 14 & 4 \\ 10 & 14 & 42 & 14 \\ 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[2245] \quad 22 \overset{1^{11}}{3} 151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 4 & 7 & 4 \\ 3 & 2 & 7 & 5 & 5 \\ 2 & 2 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[2246] \quad 1 \overset{1^3}{2} 5 131^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 14 & 4 \\ 11 & 14 & 42 & 14 \\ 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[2249] \quad 1 \overset{1^2}{2} 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 4 & 5 & 11 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2250] \quad 2 \overset{1^{11}41}{3} 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 1 & 5 & 3 \\ 3 & 3 & 5 & 6 & 6 \\ 2 & 2 & 3 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[2251] \quad 21 \overset{1^{11}}{1} 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 4 & 5 & 11 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2252] \quad 1 \overset{1^2}{2} 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 5 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2253] \quad 1 \overset{1^2}{2} 5 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 5 & 13 & 5 \\ 8 & 13 & 25 & 11 \\ 3 & 5 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[2254] \quad 21 \overset{1^{11}}{2} 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 5 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2255] \quad 1 \overset{1^{11}41}{2} 12321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 1 & 6 & 3 \\ 7 & 6 & 16 & 10 \\ 3 & 3 & 10 & 5 \end{pmatrix}_{b_0}$$

$$[2248] \quad 1 \overset{1^1}{2} 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 4 & 5 & 11 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2247] \quad 1 \overset{1^1}{4} 1^{12} 2321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$\begin{aligned}
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 1 & 6 & 3 \\ 7 & 6 & 16 & 10 \\ 3 & 3 & 10 & 5 \end{pmatrix}_{b_0} \\
[2262] & \begin{matrix} 1^{13} & 1^{11} \\ 1 & 5 & 1 & 3 & 2 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 5 & 7 & 6 & 25 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[2259] & \begin{matrix} 1^{11} \\ 31 & 4 & 141 \end{matrix} \begin{matrix} 1^{13} \\ 1^{11} \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 7 & 4 \\ 5 & 0 & 7 & 5 & 5 \\ 2 & 0 & 4 & 5 & 3 \end{pmatrix}_{b_0} \\
[2258] & 21 \begin{matrix} 1^{11} \\ 1^{13} \end{matrix} 51 \begin{matrix} 1^{11} \\ 1^{13} \end{matrix} 231 \begin{matrix} 1^{11} \\ 1^{13} \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 31 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2257] & 21 \begin{matrix} 1^{11} \\ 2^{12} \end{matrix} 5131 \begin{matrix} 1^{13} \\ 1^{13} \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 5 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2256] & 1 \begin{matrix} 1^4 \\ 1^2 \end{matrix} 5131 \begin{matrix} 1^{13} \\ 1^{13} \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 9 & 5 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2266] & 1 \begin{matrix} 1^1 \\ 1^3 \end{matrix} 5131 \begin{matrix} 1^1 \\ 1^3 \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 21 & 4 \\ 13 & 21 & 97 & 21 \\ 3 & 4 & 21 & 4 \end{pmatrix}_{b_0} \\
[2273] & 1 \begin{matrix} 1^2 \\ 1^2 \end{matrix} 5131 \begin{matrix} 1^{12} \\ 1^{12} \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 16 & 5 \\ 10 & 16 & 42 & 14 \\ 3 & 5 & 14 & 4 \end{pmatrix}_{b_0} \\
[2260] & 21 \begin{matrix} 1^{11} \\ 1^{11} \end{matrix} 51 \begin{matrix} 1^{11} \\ 1^{11} \end{matrix} 3 \begin{matrix} 1^{11} \\ 1^{11} \end{matrix} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 4 & 3 & 8 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 4 & 8 & 5 & 11 & 7 \\ 2 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0} \\
[2261] & 1 \begin{matrix} 1^2 \\ 1^2 \end{matrix} 5131 \begin{matrix} 1^{13} \\ 1^{13} \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 5 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2272] & 2 \begin{matrix} 1^{11} \\ 1^{11} \end{matrix} 3 \begin{matrix} 1^{11} \\ 1^{11} \end{matrix} 15 \begin{matrix} 1^{11} \\ 1^{11} \end{matrix} 1 \begin{matrix} 1^{11} \\ 1^{11} \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 5 & 5 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[2271] & 21 \begin{matrix} 1^1 \\ 4^1 \end{matrix} 5131 \begin{matrix} 1^{13} \\ 1^{13} \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 7 & 13 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2270] \ 2 \overset{1}{\overset{41}{3}} 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -36, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.78 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 6 & 12 & 6 \\ 6 & 12 & 18 & 10 \\ 3 & 6 & 10 & 5 \end{pmatrix}_{b_0}$$

$$[2269] \ 2 \overset{1}{\overset{11}{3}} \overset{1}{\overset{41}{2}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -36, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.89 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 3 & 4 \\ 2 & 4 & 1 & 4 & 3 \\ 2 & 3 & 4 & 2 & 4 \\ 2 & 4 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2268] \ 21 \overset{1}{\overset{1}{2}} 5131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -36, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.44 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 16 & 5 \\ 10 & 16 & 42 & 14 \\ 3 & 5 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[2265] \ 21 \overset{1}{\overset{11}{3}} 51 \overset{1}{\overset{11}{3}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -36, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 8 & 2 \\ 2 & 4 & 3 & 20 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 8 & 20 & 9 & 67 & 15 \\ 2 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2264] \ 1 \overset{1}{\overset{4}{5}} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -36, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 7 & 13 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2267] \ 21 \overset{1}{\overset{11}{5}} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -36, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.56 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 4 & 5 & 11 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2263] \ 2 \overset{1}{\overset{11}{3}} 2151^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -36, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 5 & 7 & 6 & 25 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2274] \ 2 \overset{1}{\overset{12}{3}} \overset{41}{21}^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -38, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.95 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 1 & 6 & 3 \\ 4 & 5 & 6 & 14 & 8 \\ 2 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[2291] \ 1 \overset{1}{\overset{4}{5}} 1231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -39, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 15 & 5 \\ 10 & 15 & 34 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[2275] \ 1 \overset{1}{\overset{4}{5}} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -39, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 8 & 11 & 72 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2290] \ 1 \overset{1}{\overset{12}{4}} \overset{1}{\overset{11}{3}} 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -39, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 11 & 0 & 45 & 13 \\ 2 & 3 & -1 & 13 & 3 \end{pmatrix}_{b_0} \\
[2281] & 1^{11} 4131^{24} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 11 & 0 & 65 & 15 \\ 2 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0} \\
[2280] & 1^{14} 15131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 27 & 5 \\ 14 & 27 & 130 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0} \\
[2279] & 1^{13} 151^{23} 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 10 & 8 & 9 & 60 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2278] & 1^{12} 151231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 5 & 15 & 5 \\ 8 & 15 & 34 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0} \\
[2277] & 1^{12} 3221^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 11 & 2 \\ 6 & 0 & 11 & 50 & 14 \\ 2 & 0 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2276] & 1^{11} 3221^{25} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 13 & 0 & 101 & 19 \\ 2 & 3 & -1 & 19 & 3 \end{pmatrix}_{b_0} \\
[2284] & 1^{13} 21^2 51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 8 & 9 & 60 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2283] & 1^{12} 151^{23} 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 5 & 25 & 5 \\ 16 & 25 & 114 & 22 \\ 3 & 5 & 22 & 4 \end{pmatrix}_{b_0} \\
[2282] & 1^{14} 151231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 7 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[2287] & 1^{12} 15131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 21 & 5 \\ 10 & 21 & 70 & 18 \\ 3 & 5 & 18 & 4 \end{pmatrix}_{b_0} \\
[2288] & 1^{11} 15131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 4 & 5 & 8 & 24 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2286] $1^{11}41^23141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 13 & 0 & 81 & 17 \\ 2 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$

[2289] $1^{11}51231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 4 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[2285] $1^{11}321^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 25 & 5 \\ 14 & 25 & 114 & 22 \\ 3 & 5 & 22 & 4 \end{pmatrix}_{b_0}$

[2300] $314141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 7 & 4 \\ 3 & 0 & 7 & 5 & 5 \\ 2 & 0 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[2299] $1^{13}5131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 8 & 5 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$

[2298] $232151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[2296] $31^14141^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 9 & 4 \\ 5 & 0 & 9 & 13 & 7 \\ 2 & 0 & 4 & 7 & 3 \end{pmatrix}_{b_0}$

[2295] $23151^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 7 & 9 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$

[2294] $15132 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[2293] $1^{14}5131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 10 & 6 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2292] $31^24141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 5 & 1 & 25 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 25 & 0 & 101 & 19 \\ 2 & 5 & -1 & 19 & 3 \end{pmatrix}_{b_0}$

[2297] $1^141^{11}3141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 4 & 2 \\ 3 & 13 & 1 & 20 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 20 & 0 & 26 & 10 \\ 2 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2301] \quad 2 \overset{1^{11}41}{3} 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 6 & 11 & 6 \\ 6 & 11 & 15 & 9 \\ 3 & 6 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[2315] \quad 1 \overset{1^3}{5} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 6 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2307] \quad 21 \overset{1^1}{3} 5 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 6 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2308] \quad 1 \overset{1^3}{12} 5 1231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 6 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2309] \quad 1 \overset{1^1}{1^1} 5 1^2 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 5 & 25 & 5 \\ 15 & 25 & 114 & 22 \\ 3 & 5 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[2310] \quad 1 \overset{1^4}{12} 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 8 & 6 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2311] \quad 1 \overset{1^1}{12} 5 1231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2312] \quad 31 \overset{1^1}{1^{11}} 13 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2313] \quad 1 \overset{1^1}{12} 5 1^1 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 66 & 16 \\ 2 & 2 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2306] \quad 1 \overset{1^3}{5} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 24 & 5 \\ 12 & 24 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[2305] \quad 21 \overset{1^4}{12} 5 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 10 & 10 & 12 & 90 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[2304] \quad 2 \overset{1^1 1^1}{3} 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 2 & 4 & 3 & 2 & 4 \\ 2 & 4 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2303] \quad 1 \overset{1^1}{5} 131 \overset{1^1}{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 4 & 6 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2314] \quad 1 \overset{1^4}{5} 131 \overset{1^1}{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 10 & 8 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2302] \quad 1 \overset{1^3}{5} 131 \overset{1^1}{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 8 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2316] \quad 1 \overset{1^3}{5} 1231 \overset{1^3}{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 15 & 5 \\ 9 & 15 & 34 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[2317] \quad 221 \overset{1^1 1^2}{5} 131 \overset{1^1}{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 4 & 2 \\ 3 & 4 & 3 & 8 & 4 \\ 1 & 3 & 0 & 5 & 2 \\ 4 & 8 & 5 & 11 & 7 \\ 2 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2323] \quad 31 \overset{1^1 1^1}{4} 141 \overset{1^1}{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 7 & 0 & 5 & 5 \\ 2 & 5 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[2322] \quad 2 \overset{1^1 1^2}{3} 2151 \overset{1^1}{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[2321] \quad 1 \overset{1^1}{5} 1 \overset{1^1 1^2}{3} 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[2320] \quad 31 \overset{1^1 1^1}{4} 141 \overset{1^1}{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 5 & 1 & 16 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 16 & 0 & 38 & 12 \\ 2 & 5 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2319] \quad 31 \overset{1^1 1^2}{4} 141 \overset{1^1}{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 5 & 1 & 14 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 14 & 0 & 26 & 10 \\ 2 & 5 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[2318] $31^3 31^1 51^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 21 & 4 \\ 18 & 21 & 97 & 21 \\ 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

[2344] $2^3 1^1 51^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 6 & 8 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2336] $1^1 3221^2 51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 10 & 2 \\ 9 & 0 & 10 & 78 & 16 \\ 2 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2334] $2^3 1^1 51^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 6 & 7 & 7 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2343] $21^4 5131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 5 & 27 & 5 \\ 18 & 27 & 130 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[2342] $1^1 5^4 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 5 & 27 & 5 \\ 18 & 27 & 130 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[2341] $2^3 21^1 51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 6 & 4 & 7 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[2340] $2^3 1^1 51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 4 & 5 & 11 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[2339] $1^1 3221^1 51^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 5 & 2 \\ 6 & 0 & 5 & 18 & 8 \\ 2 & 0 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[2333] $21^1 5131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 18 & 5 \\ 9 & 18 & 49 & 15 \\ 3 & 5 & 15 & 4 \end{pmatrix}_{b_0}$$

[2332] $1^3 5131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 6 & 7 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2331] $1^{11}5^{13}1231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -47$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 6 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[2330] $1^{12}5^{11}132 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -47$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$

[2329] $2321^1 251^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -47$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 9 & 8 & 9 & 60 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[2328] $1^{11}5^{12}231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -47$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 8 & 9 & 60 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[2327] $23151^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -47$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 9 & 8 & 11 & 72 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2326] $232151^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -47$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$

[2325] $314141^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -47$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 7 & 4 \\ 4 & 0 & 7 & 5 & 5 \\ 2 & 0 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[2338] $1^{11}4131^{14}141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -47$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 7 & 0 & 21 & 9 \\ 2 & 3 & -1 & 9 & 3 \end{pmatrix}_{b_0}$

[2337] $1^{11}5^{13}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -47$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 18 & 5 \\ 9 & 18 & 49 & 15 \\ 3 & 5 & 15 & 4 \end{pmatrix}_{b_0}$

[2324] $1^{11}5^{12}231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -47$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 9 & 8 & 9 & 60 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[2345] $1^{11}5^{12}1231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -47$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 5 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

$$[2335] \ 1 \overset{1^2}{\underset{1^{11}}{5}} 131^{12} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 5 & 6 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2348] \ 1 \overset{1^2}{\underset{1^{12}}{5}} 131^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 5 & 7 & 7 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2351] \ 1^{12} 3141^{13} 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 7 & 1 & 17 & 9 \\ 2 & 3 & 0 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[2346] \ 1^{12} 4131^{14} 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 10 & 0 & 38 & 12 \\ 2 & 3 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2352] \ 1^{11} 3221^{12} 51^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 13 & 2 \\ 9 & 0 & 13 & 101 & 19 \\ 2 & 0 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[2353] \ 1^{14} 51^{14} 231^{12} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 20 & 5 \\ 14 & 20 & 63 & 17 \\ 3 & 5 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[2350] \ 1^{12} \overset{1^4}{\underset{1^2}{5}} 131^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 9 & 7 & 7 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2349] \ 1^{12} \overset{1^2}{\underset{1^2}{5}} 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 20 & 5 \\ 10 & 20 & 63 & 17 \\ 3 & 5 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[2347] \ 1^{12} \overset{1^2}{\underset{1^2}{5}} 1231^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2354] \ 1^{12} 41^{13} 2321^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 1 & 8 & 3 \\ 8 & 8 & 26 & 12 \\ 3 & 3 & 12 & 5 \end{pmatrix}_{b_0}$$

$$[2355] \ 1^{14} 41^{13} 2321^{13} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 1 & 8 & 3 \\ 8 & 8 & 26 & 12 \\ 3 & 3 & 12 & 5 \end{pmatrix}_{b_0}$$

$$[2356] \ 21^{12} \overset{1^3}{\underset{1^3}{5}} 131^{13} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 10 & 10 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[2357] \quad 2 \overset{1^{13}}{3} 1 \overset{1^3}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 9 & 6 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2358] \quad 22 \overset{1^{12}}{3} 151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 6 & 2 \\ 1 & 0 & 3 & 4 & 2 \\ 3 & 3 & 4 & 14 & 4 \\ 6 & 4 & 14 & 26 & 10 \\ 2 & 2 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2359] \quad 1^{\overset{1^{11}}{4}} 41^1 3141^{\overset{1^{11}}{4}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 8 & 0 & 26 & 10 \\ 2 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2360] \quad 1^{\overset{1^4}{11}} 51^1 231^{\overset{1^{11}}{4}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 9 & 6 & 8 & 31 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2361] \quad 1^{\overset{1^{11}}{321^1}} 51^{\overset{1^{11}}{4}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 5 & 6 & 8 & 31 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2362] \quad 1^{\overset{1^{11}}{4}} 41^1 31^{\overset{1^{11}}{4}} 41^{\overset{1^{11}}{4}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 12 & 0 & 70 & 16 \\ 2 & 3 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2365] \quad 1^{\overset{1^{11}}{4}} 41^1 2321^{\overset{1^{12}}{5}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 1 & 8 & 3 \\ 10 & 8 & 34 & 14 \\ 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$$

$$[2363] \quad 1^{\overset{1^2}{13}} 51^{\overset{1^2}{13}} 131^{\overset{1^2}{13}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 21 & 4 \\ 14 & 21 & 97 & 21 \\ 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[2364] \quad 1^{\overset{1^{11}}{4}} 41^1 2321^{\overset{1^{12}}{5}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 1 & 8 & 3 \\ 10 & 8 & 34 & 14 \\ 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$$

$$[2371] \quad 21^{\overset{1^{11}}{4}} 51^{\overset{1^{11}}{4}} 131^{\overset{1^{12}}{5}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 11 & 8 & 11 & 72 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2368] \quad 1^{\overset{1^1}{11}} 51^{\overset{1^1}{11}} 131^{\overset{1^2}{5}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 5 & 5 & 8 & 24 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[2367] $1^{11} \overset{14}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 8 & 5 & 8 & 24 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$

[2366] $1^{11} \overset{14}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 11 & 8 & 11 & 72 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2369] $31^{12} 321^{35} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 29 & 3 \\ 5 & 5 & 40 & 5 \\ 29 & 40 & 287 & 35 \\ 3 & 5 & 35 & 3 \end{pmatrix}_{b_0}$

[2370] $2321^3 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 13 & 11 & 13 & 126 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2372] $21^1 \overset{11}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 5 & 5 & 8 & 24 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$

[2373] $231^{13} 151^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 8 & 5 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$

[2374] $31^{12} 32151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 4 & 3 \\ 5 & 5 & 5 & 5 \\ 4 & 5 & 2 & 4 \\ 3 & 5 & 4 & 3 \end{pmatrix}_{b_0}$

[2385] $21^{12} \overset{13}{5} 132 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 2 & 10 & 2 \\ 2 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 13 & 2 \\ 10 & 18 & 13 & 73 & 17 \\ 2 & 4 & 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2391] $31^1 \overset{12}{5} 13 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 4 & 2 \\ 4 & 6 & 4 & 10 & 6 \\ 2 & 3 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[2377] $231^{11} \overset{11}{5} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 5 & 6 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[2378] $2^1 \overset{12}{3} \overset{11}{2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 2 & 4 & 8 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 4 & 8 & 6 & 14 & 8 \\ 2 & 4 & 3 & 8 & 4 \end{pmatrix}_{b_0}$

[2386] $21^{\overline{3}1^{12}}5131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 9 & 8 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2379] $21^{\overline{1}2^1}5131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 8 & 8 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2380] $31^{\overline{1}1}31^{\overline{1}1}51^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 6 & 3 \\ 5 & 5 & 8 & 5 \\ 6 & 8 & 9 & 7 \\ 3 & 5 & 7 & 4 \end{pmatrix}_{b_0}$$

[2381] $21^{\overline{1}2^1\overline{5}1^3}231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 14 & 14 & 17 & 181 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2384] $1^{\overline{1}2^1\overline{5}1^3}5131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 9 & 8 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2382] $1^{\overline{1}2^1\overline{5}1^3}5131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 5 & 6 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[2375] $21^{\overline{1}2^1\overline{5}1^3}1^{\overline{1}2^1}231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 10 & 10 & 15 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2383] $2^{\overline{1}41}3^{\overline{1}2}21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 6 & 18 & 6 \\ 9 & 18 & 43 & 15 \\ 3 & 6 & 15 & 5 \end{pmatrix}_{b_0}$$

[2387] $21^{\overline{1}1}5^{\overline{1}2}5131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 5 & 6 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[2388] $1^{\overline{1}2^1\overline{5}1^3}5131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 6 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[2376] $21^{\overline{1}2^1\overline{5}1^3}5131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 8 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2389] $21^{\overline{1}2^1\overline{5}1^3}3^{\overline{1}1}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 6 & 14 & 7 & 33 & 11 \\ 2 & 4 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2390] \quad 1^1 \overset{1^4}{5} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 18 & 5 \\ 12 & 18 & 49 & 15 \\ 3 & 5 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[2392] \quad 2^1 \overset{1^1 41}{3} 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 1 & 9 & 3 \\ 5 & 4 & 9 & 20 & 10 \\ 2 & 2 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[2395] \quad 1^1 3221^{13} 51^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 15 & 2 \\ 4 & 18 & 6 & 60 & 8 \\ 1 & 6 & 0 & 15 & 2 \\ 15 & 60 & 15 & 178 & 24 \\ 2 & 8 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[2393] \quad 1^1 \overset{1^2}{5} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 18 & 5 \\ 10 & 18 & 49 & 15 \\ 3 & 5 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[2394] \quad 1^1 \overset{1^2}{5} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 6 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2396] \quad 41^1 \overset{1^1}{2} 31^{14} 11 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 8 & 2 \\ 4 & 26 & 1 & 44 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 44 & 0 & 70 & 16 \\ 2 & 10 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2397] \quad 1^1 \overset{1^3}{5} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 8 & 7 & 7 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2398] \quad 1^1 \overset{1^4}{5} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 5 & 24 & 5 \\ 16 & 24 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[2400] \quad 1^1 3221^{12} 51^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 10 & 2 \\ 4 & 18 & 6 & 40 & 8 \\ 1 & 6 & 0 & 10 & 2 \\ 10 & 40 & 10 & 78 & 16 \\ 2 & 8 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2399] \quad 1^1 3221^{11} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 5 & 2 \\ 4 & 18 & 6 & 20 & 8 \\ 1 & 6 & 0 & 5 & 2 \\ 5 & 20 & 5 & 18 & 8 \\ 2 & 8 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[2401] \quad 2^1 \overset{1^2}{3} 15^1 \overset{1^2}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 10 & 3 \\ 4 & 5 & 16 & 5 \\ 10 & 16 & 42 & 14 \\ 3 & 5 & 14 & 4 \end{pmatrix}_{b_0}$$

[2402] $22 \overset{11}{3} 151^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 5 & 2 \\ 1 & 0 & 3 & 6 & 2 \\ 3 & 3 & 4 & 11 & 4 \\ 5 & 6 & 11 & 21 & 9 \\ 2 & 2 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[2411] $1 \overset{14}{5} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 21 & 5 \\ 14 & 21 & 70 & 18 \\ 3 & 5 & 18 & 4 \end{pmatrix}_{b_0}$$

[2403] $1 \overset{11}{5} 1^{12} 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 20 & 5 \\ 11 & 20 & 63 & 17 \\ 3 & 5 & 17 & 4 \end{pmatrix}_{b_0}$$

[2412] $1 \overset{12}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 5 & 8 & 24 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[2413] $22 \overset{11}{3} 151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 5 & 3 \\ 6 & 14 & 13 & 8 \\ 5 & 13 & 9 & 7 \\ 3 & 8 & 7 & 4 \end{pmatrix}_{b_0}$$

[2409] $1^{12} 3221^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 11 & 0 & 50 & 14 \\ 2 & 3 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

[2408] $1 \overset{13}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 7 & 5 & 8 & 24 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[2407] $1 \overset{14}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 7 & 0 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2406] $1 \overset{13}{5} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 18 & 5 \\ 11 & 18 & 49 & 15 \\ 3 & 5 & 15 & 4 \end{pmatrix}_{b_0}$$

[2405] $1 \overset{14}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 9 & 6 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2404] $1 \overset{13}{5} 1^{12} 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 20 & 5 \\ 13 & 20 & 63 & 17 \\ 3 & 5 & 17 & 4 \end{pmatrix}_{b_0}$$

[2410] $21^{1^2}5^{12}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 9 & 9 & 13 & 73 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2418] $1^{1^3}3221^351^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 15 & 2 \\ 13 & 0 & 15 & 178 & 24 \\ 2 & 0 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[2417] $1^{1^3}5^{1^3}1^{1^3}231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 8 & 6 & 8 & 31 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2419] $1^{1^2}5^{1^2}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 7 & 7 & 7 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2420] $1^{1^3}3221^151^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 5 & 2 \\ 7 & 0 & 5 & 18 & 8 \\ 2 & 0 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[2421] $1^{1^3}5^{1^3}131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 10 & 7 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2422] $21^{1^2}5^{1^2}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 7 & 7 & 7 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2423] $31^{1^1}4^{1^1}141^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 5 & 2 \\ 3 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 24 & 0 & 38 & 12 \\ 2 & 7 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

[2424] $1^{1^3}5^{1^3}131^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 21 & 4 \\ 15 & 21 & 97 & 21 \\ 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

[2425] $1^{1^4}4^{1^1}31^341^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 13 & 2 \\ 3 & 13 & 1 & 56 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 56 & 0 & 230 & 28 \\ 2 & 7 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[2426] $1^{1^1}5^{1^1}1^{1^1}231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 31 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2427] $1^{11}51^3231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 11 & 13 & 126 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2428] $21^35131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 10 & 7 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2429] $1^{13}51^1231^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 12 & 0 & 115 & 21 \\ 2 & 2 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[2416] $1^{11}3221^151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 8 & 2 \\ 6 & 0 & 8 & 31 & 11 \\ 2 & 0 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2415] $1^{11}51^3231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 13 & 11 & 13 & 126 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2414] $231^351^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 21 & 4 \\ 15 & 21 & 97 & 21 \\ 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

[2430] $1^{12}51^2231^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 12 & 0 & 130 & 22 \\ 2 & 2 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[2431] $1^{13}5131^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 21 & 4 \\ 16 & 21 & 97 & 21 \\ 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

[2444] $1^35131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 29 & 5 \\ 14 & 29 & 137 & 25 \\ 3 & 5 & 25 & 4 \end{pmatrix}_{b_0}$$

[2441] $1^{11}4131^241^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 11 & 0 & 65 & 15 \\ 2 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[2437] $1^45131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 8 & 9 & 14 & 97 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2438] $1^{13}51231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 7 & 6 & 25 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2439] \ 21^{\overset{11}{1}} 5 \ 131^{\overset{13}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 6 & 6 & 11 & 43 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2440] \ 1^{\overset{13}{1}} 5 \ 1231^{\overset{11}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 7 & 6 & 25 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2446] \ 1^{\overset{13}{1}} 5 \ 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 25 & 5 \\ 12 & 25 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[2447] \ 1^{\overset{13}{1}} 5 \ 1231^{\overset{13}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 7 & 5 & 10 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2443] \ 1^{\overset{13}{1}} 5 \ 131^{\overset{13}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 6 & 6 & 11 & 43 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2432] \ 1^{\overset{13}{1}} 4131^{\overset{1}{1}} 41^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 13 & 0 & 61 & 15 \\ 2 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2433] \ 1^{\overset{13}{1}} 5 \ 1231^{\overset{14}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 20 & 5 \\ 11 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[2434] \ 1^{\overset{13}{1}} 5 \ 131^{\overset{13}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 7 & 8 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2435] \ 1^{\overset{11}{1}} 3221^{\overset{3}{1}} 51^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 18 & 2 \\ 13 & 0 & 18 & 211 & 27 \\ 2 & 0 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[2436] \ 1^{\overset{14}{1}} 5 \ 1231^{\overset{13}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 5 & 10 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2442] \ 1^{\overset{13}{1}} 5 \ 131^{\overset{11}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 10 & 9 & 67 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2445] $1 \overset{1^2}{\underset{1^{13}}{5}} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 9 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2448] $1 \overset{1^2}{\underset{1^{13}}{5}} 1^3 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 12 & 11 & 13 & 126 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2450] $1 \overset{1^4}{\underset{1^3}{5}} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 11 & 7 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2451] $2 \overset{1^{12}}{\underset{1^{13}}{3}} 1 \overset{1^1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 8 & 3 \\ 4 & 5 & 13 & 5 \\ 8 & 13 & 25 & 11 \\ 3 & 5 & 11 & 4 \end{pmatrix}_{b_0}$$

[2452] $2 \overset{1^{13}}{\underset{1^{14}}{3}} 1 \overset{1^1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 5 & 27 & 5 \\ 17 & 27 & 130 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[2453] $1 \overset{1^2}{\underset{1^{11}}{5}} 1^1 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 11.53 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 7 & 6 & 8 & 31 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2454] $2 \overset{1^{11}}{\underset{1^{11}}{3}} 21 \overset{1^1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 10 & 3 \\ 4 & 5 & 15 & 5 \\ 10 & 15 & 37 & 13 \\ 3 & 5 & 13 & 4 \end{pmatrix}_{b_0}$$

[2449] $1 \overset{1^2}{\underset{1^{12}}{5}} 1^1 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 11.53 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 20 & 5 \\ 12 & 20 & 63 & 17 \\ 3 & 5 & 17 & 4 \end{pmatrix}_{b_0}$$

[2455] $22 \overset{1^{11}}{\underset{1^{11}}{3}} 151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 4 & 2 \\ 1 & 0 & 3 & 4 & 2 \\ 3 & 3 & 4 & 9 & 4 \\ 4 & 4 & 9 & 11 & 7 \\ 2 & 2 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[2456] $1 \overset{1^3}{\underset{1^{13}}{3}} 221^{12} 51^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 13 & 2 \\ 4 & 18 & 6 & 55 & 8 \\ 1 & 6 & 0 & 19 & 2 \\ 13 & 55 & 19 & 165 & 25 \\ 2 & 8 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2471] $21 \overset{1^2}{\underset{1^{12}}{5}} 1^1 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 7 & 6 & 30 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[2464] $21^2 \overset{11}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 6 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2470] $2^1 3 2151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 5 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[2469] $2^1 3 21^2 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 10 & 7 & 11 & 65 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2457] $1^1 \overset{12}{5} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 21 & 5 \\ 12 & 21 & 70 & 18 \\ 3 & 5 & 18 & 4 \end{pmatrix}_{b_0}$$

[2458] $1^1 \overset{13}{5} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 5 & 24 & 5 \\ 15 & 24 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

[2459] $21^3 \overset{11}{5} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 5 & 24 & 5 \\ 15 & 24 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

[2460] $21^2 \overset{11}{5} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 21 & 5 \\ 12 & 21 & 70 & 18 \\ 3 & 5 & 18 & 4 \end{pmatrix}_{b_0}$$

[2461] $21^{12} 51231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[2462] $1^1 \overset{11}{5} 1^1 \overset{12}{3} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 5 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[2463] $21^3 \overset{11}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 7 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2468] $1^1 \overset{13}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 7 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2467] $1^1 \overset{12}{5} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2466] \ 1^{1^3} 5^{1^2} 131^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 6 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2465] \ 1^{1^2} 5^{1^2} 131^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 6 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2480] \ 1^{1^2} 5^{321^1} 1^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 6 & 7 & 11 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2477] \ 1^{1^3} 5^{1^2} 131^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 21 & 5 \\ 13 & 21 & 70 & 18 \\ 3 & 5 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[2478] \ 21^{1^4} 5^{1^4} 131^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 8 & 16 & 110 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2479] \ 1^{1^1} 5^{1^2} 131^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 5 & 7 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2476] \ 1^{1^4} 5^{1^4} 1231^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 20 & 5 \\ 12 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[2475] \ 1^{1^1} 41^{1^1} 31^{2^1} 41^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 16 & 0 & 138 & 22 \\ 2 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[2474] \ 1^{1^4} 5^{1^4} 131^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 8 & 16 & 110 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2473] \ 1^{1^2} 5^{1^1} 1231^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 5 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2472] \ 1^{1^4} 5^{1^1} 131^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 5 & 32 & 5 \\ 16 & 32 & 174 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[2481] \ 1^{1^1 321^2} 1^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 10 & 10 & 78 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2482] \ 1^{1^2 41^2 3141^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 16 & 0 & 110 & 20 \\ 2 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2483] \ 1^{1^2 51^2 231^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 12 & 10 & 10 & 78 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2487] \ 1^{1^3 51^4} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 9 & 8 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2486] \ 1^{1^2 31^1 41^1 31^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 10 & 1 & 39 & 13 \\ 2 & 3 & 0 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[2485] \ 1^{1^3 51^4} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 6 & 8 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2484] \ 1^{1^3 51^4} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 12 & 11 & 10 & 89 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2488] \ 21^4 51^4 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 12 & 11 & 10 & 89 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2489] \ 21^4 51^4 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 6 & 8 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2490] \ 1^{1^1 41^2 31^1 41^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 17 & 0 & 149 & 23 \\ 2 & 3 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[2491] \ 2^3 21^{1^1 41^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 1 & 13 & 3 \\ 7 & 5 & 13 & 42 & 14 \\ 2 & 2 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[2493] $2 \overset{1}{3} 2151^{\overset{13}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 5 & 15 & 5 \\ 9 & 15 & 34 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

[2492] $2 \overset{1}{3} 21^{\overset{13}{2}} 51^{\overset{12}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 10 & 10 & 10 & 78 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2494] $1 \overset{21}{5} 131^{\overset{12}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[2495] $2 \overset{1}{3} \overset{13}{4} 1^{\overset{1}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 2 & 4 & 12 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 6 & 12 & 9 & 34 & 12 \\ 2 & 4 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[2496] $31 \overset{1}{4} 141^{\overset{13}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 5 & 1 & 21 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 21 & 0 & 61 & 15 \\ 2 & 5 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[2497] $21^{\overset{12}{5}} 1231^{\overset{12}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[2498] $31 \overset{1}{5} 13 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 0 & 6 & 2 \\ 6 & 9 & 6 & 25 & 9 \\ 2 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[2499] $2 \overset{1}{3} 21^{\overset{13}{2}} 51^{\overset{1}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 17 & 3 \\ 4 & 5 & 25 & 5 \\ 17 & 25 & 114 & 22 \\ 3 & 5 & 22 & 4 \end{pmatrix}_{b_0}$$

[2500] $21^{\overset{12}{5}} 51^{\overset{2}{2}} 231^{\overset{1}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 10 & 10 & 78 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2501] $1 \overset{1}{5} 1^{\overset{13}{1}} 3 \overset{1}{2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 5 & 15 & 5 \\ 9 & 15 & 34 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

[2502] $31^{\overset{1}{1}} 4 \overset{11}{141}^{\overset{13}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 16 & 4 \\ 6 & 0 & 16 & 38 & 12 \\ 2 & 0 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[2503] $31^{12}4141^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -76$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 18 & 4 \\ 8 & 0 & 18 & 58 & 14 \\ 2 & 0 & 4 & 14 & 3 \end{pmatrix}_{b_0}$

[2504] $1^{12}51^{13}231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -76$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 15 & 0 & 218 & 28 \\ 2 & 2 & -1 & 28 & 3 \end{pmatrix}_{b_0}$

[2505] $21^{12}5131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -79$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.66 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 7 & 7 & 7 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$

[2506] $21^{12}5131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -79$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 9 & 9 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2507] $1^{14}1^{11}3141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 1 & 5 & 2 \\ 3 & 13 & 1 & 20 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 20 & 0 & 26 & 10 \\ 2 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0}$

[2508] $2^13^151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 7 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[2509] $31^{11}321^{15}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 5 & 11 & 3 \\ 5 & 5 & 15 & 5 \\ 11 & 15 & 37 & 13 \\ 3 & 5 & 13 & 4 \end{pmatrix}_{b_0}$

[2510] $2^13^121^{15}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 5 & 10 & 38 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[2511] $2^13^151^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -80$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 16 & 2 \\ 12 & 8 & 16 & 110 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2512] $1^{13}3221^{12}51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -81$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 1 & 12 & 2 \\ 4 & 18 & 6 & 55 & 8 \\ 1 & 6 & 0 & 19 & 2 \\ 12 & 55 & 19 & 165 & 25 \\ 2 & 8 & 2 & 25 & 3 \end{pmatrix}_{b_0}$

[2513] $22^13^151^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -82$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 3 & 6 & 2 \\ 3 & 3 & 4 & 21 & 4 \\ 9 & 6 & 21 & 61 & 15 \\ 2 & 2 & 4 & 15 & 3 \end{pmatrix}_{b_0}$

[2523] $1^{11}51^{12}131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 6 & 11 & 43 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2524] & 1^{11} 41^2 3141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 13 & 0 & 81 & 17 \\ 2 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0} \\
[2514] & 1^{13} 5^{14} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 10 & 9 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[2515] & 1^{11} 41^2 2321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 1 & 10 & 3 \\ 13 & 10 & 56 & 18 \\ 3 & 3 & 18 & 5 \end{pmatrix}_{b_0} \\
[2516] & 1^{11} 5^{12} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 23 & 5 \\ 12 & 23 & 81 & 19 \\ 3 & 5 & 19 & 4 \end{pmatrix}_{b_0} \\
[2517] & 1^{13} 5^{12} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 7 & 8 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2518] & 1^{11} 5^{12} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 9 & 12 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2519] & 1^{13} 5^{14} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 11 & 10 & 9 & 67 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2520] & 1^{13} 5^{13} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 8 & 8 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[2521] & 1^{11} 41^2 2321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 1 & 10 & 3 \\ 13 & 10 & 56 & 18 \\ 3 & 3 & 18 & 5 \end{pmatrix}_{b_0} \\
[2522] & 1^{12} 41^1 3141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 11 & 0 & 45 & 13 \\ 2 & 3 & -1 & 13 & 3 \end{pmatrix}_{b_0} \\
[2535] & 151^{11} 2231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 18 & 6 & 20 & 8 \\ 1 & 6 & 0 & 5 & 2 \\ 4 & 20 & 5 & 18 & 8 \\ 2 & 8 & 2 & 8 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2526] $41^{1\bar{1}23141^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 28 & 0 & 26 & 10 \\ 2 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$

[2527] $1^{1\bar{2}4131^141^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 10 & 0 & 38 & 12 \\ 2 & 3 & -1 & 12 & 3 \end{pmatrix}_{b_0}$

[2528] $1^{1\bar{2}5^{14}1231^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 8 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$

[2529] $1^{1\bar{2}5^{13}1^{23}231^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 11 & 10 & 10 & 78 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2530] $1^{1\bar{2}3221^{25}1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 16 & 2 \\ 10 & 0 & 16 & 130 & 22 \\ 2 & 0 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[2531] $1^{1\bar{2}321^2} 5^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 5 & 30 & 5 \\ 16 & 30 & 154 & 26 \\ 3 & 5 & 26 & 4 \end{pmatrix}_{b_0}$

[2532] $1^{1\bar{2}} 5^{1\bar{2}} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 6 & 8 & 10 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[2533] $1^{1\bar{2}} 5^{1\bar{2}} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 10 & 12 & 90 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$

[2534] $1^1 5^{1\bar{2}} 1^{23} 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 5 & 30 & 5 \\ 20 & 30 & 154 & 26 \\ 3 & 5 & 26 & 4 \end{pmatrix}_{b_0}$

[2539] $1^{1\bar{2}5^1 1^{23} 231^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 10 & 10 & 78 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2538] $21^{1\bar{2}5^1 131^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 6 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[2525] $1^{1\bar{2}4131^{24}1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 14 & 0 & 90 & 18 \\ 2 & 3 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[2537] $2 \overset{1}{3} \overset{41}{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 6 & 24 & 6 \\ 12 & 24 & 78 & 20 \\ 3 & 6 & 20 & 5 \end{pmatrix}_{b_0}$$

[2536] $1 \overset{1}{2} \overset{2}{5} 1231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[2545] $2 \overset{1}{3} \overset{11}{21} \overset{3}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 14 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 14 & 10 & 15 & 133 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2540] $1 \overset{1}{2} \overset{41}{41} \overset{1}{2} 321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 1 & 10 & 3 \\ 11 & 10 & 44 & 16 \\ 3 & 3 & 16 & 5 \end{pmatrix}_{b_0}$$

[2541] $1 \overset{1}{1} \overset{41}{41} \overset{1}{2} 321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 1 & 10 & 3 \\ 11 & 10 & 44 & 16 \\ 3 & 3 & 16 & 5 \end{pmatrix}_{b_0}$$

[2542] $2 \overset{1}{3} \overset{14}{15} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 11 & 7 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2543] $2 \overset{1}{3} \overset{21}{21} \overset{2}{5} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 11 & 12 & 11 & 100 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2544] $21 \overset{1}{13} \overset{4}{5} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 11 & 11 & 10 & 89 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2552] $1 \overset{1}{5} \overset{3}{13} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 9 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2551] $1^1 3221^2 \overset{1}{5} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 10 & 2 \\ 10 & 0 & 10 & 78 & 16 \\ 2 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2550] $1 \overset{1}{2} \overset{3}{5} 1231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 7 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[2549] $31^3 4^{11} 141^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -88$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 27 & 4 \\ 11 & 0 & 27 & 133 & 21 \\ 2 & 0 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[2548] $31^2 4^{11} 141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -88$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 25 & 4 \\ 9 & 0 & 25 & 101 & 19 \\ 2 & 0 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

[2547] $1^{12} 5^{12} 231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -88$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 10 & 10 & 78 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2546] $1^{11} 41^{13} 31^{14} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -88$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 12 & 0 & 70 & 16 \\ 2 & 3 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[2554] $2^3 3^{11} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -89$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 6 & 17 & 6 \\ 9 & 17 & 36 & 14 \\ 3 & 6 & 14 & 5 \end{pmatrix}_{b_0}$$

[2553] $1^{11} 3221^{12} 51^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -89$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 12 & 2 \\ 4 & 18 & 6 & 45 & 8 \\ 1 & 6 & 0 & 13 & 2 \\ 12 & 45 & 13 & 101 & 19 \\ 2 & 8 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2558] $1^{11} 5^{14} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -91$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 9 & 6 & 11 & 43 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2557] $1^{12} 41^{13} 31^{14} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -91$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 15 & 0 & 97 & 19 \\ 2 & 3 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

[2556] $21^3 5^{13} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -91$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 10 & 10 & 9 & 67 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2555] $1^{13} 5^{12} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -91$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 8 & 9 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2559] $1^{11} 5^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -91$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 8 & 6 & 11 & 43 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2560] $1^{13} 5^{13} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -91$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 9 & 9 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2561] \ 1^{13} 5^{13} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 10 & 10 & 9 & 67 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2562] \ 41^{231} 4^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 5 & 3 & 3 & 2 \\ 5 & 13 & 10 & 9 & 7 \\ 3 & 10 & 4 & 7 & 4 \\ 3 & 9 & 7 & 5 & 5 \\ 2 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[2563] \ 21^{13} 5^{13} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 8 & 9 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2566] \ 1^{13} 3221^{15} 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 14 & 0 & 75 & 17 \\ 2 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2564] \ 1^{11} 5^{11} 1^{231} 13^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 25 & 5 \\ 13 & 25 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[2567] \ 21^{11} 5^{12} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 6 & 7 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2565] \ 1^{12} 5^{11} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 6 & 7 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2568] \ 2^{112} 3^{11} 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 6 & 5 & 8 & 24 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2569] \ 1^{12} 3221^{15} 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 11 & 2 \\ 7 & 0 & 11 & 50 & 14 \\ 2 & 0 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2570] \ 1^{11} 5^{11} 1^{231} 12^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 7 & 11 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2571] \ 31^{112} 4^{11} 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 5 & 1 & 23 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 23 & 0 & 77 & 17 \\ 2 & 5 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

[2572] $31^{1^{11}}4^{11}141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 5 & 1 & 16 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 16 & 0 & 38 & 12 \\ 2 & 5 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

[2573] $23^{1^{12}}151^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 7 & 6 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2574] $21^{1^{12}}5^{12}131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 7 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2583] $1^15^{1^3}131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 23 & 5 \\ 13 & 23 & 81 & 19 \\ 3 & 5 & 19 & 4 \end{pmatrix}_{b_0}$$

[2584] $21^{1^{12}}5^{12}132 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 8 & 2 \\ 2 & 4 & 3 & 16 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 8 & 16 & 10 & 50 & 14 \\ 2 & 4 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2581] $21^{1^{13}}5^{13}1231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 10 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2580] $21^{1^{12}}5^{12}131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 8 & 10 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2579] $21^{1^{13}}5^{13}131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 10 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2575] $21^{1^{12}}5^{12}231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 12 & 16 & 130 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[2582] $23^{1^{12}}151^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 10 & 10 & 12 & 90 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2576] $2^{1^{11}}3^{1^4}2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 2 & 4 & 7 & 4 \\ 2 & 4 & 1 & 7 & 3 \\ 4 & 7 & 7 & 12 & 8 \\ 2 & 4 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[2577] \ 221^{12}5131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -96, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.58 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 8 & 2 \\ 3 & 4 & 3 & 16 & 4 \\ 1 & 3 & 0 & 10 & 2 \\ 8 & 16 & 10 & 50 & 14 \\ 2 & 4 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2578] \ 1^{14}1^{11}31^{14}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -96, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 7 & 2 \\ 3 & 13 & 1 & 32 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 32 & 0 & 70 & 16 \\ 2 & 7 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2585] \ 2 \ 3 \ 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -98, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.86 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 1 & 10 & 3 \\ 6 & 6 & 10 & 30 & 12 \\ 2 & 2 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[2586] \ 31 \ 4 \ 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -98, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 5 & 1 & 14 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 14 & 0 & 26 & 10 \\ 2 & 5 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2588] \ 1^1 \ 5^{14} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -99, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.65 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 23 & 5 \\ 14 & 23 & 81 & 19 \\ 3 & 5 & 19 & 4 \end{pmatrix}_{b_0}$$

$$[2594] \ 1^{11} 3221^{15} 51^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -99, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.65 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 8 & 2 \\ 7 & 0 & 8 & 31 & 11 \\ 2 & 0 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2593] \ 1^{11} 5^{14} 1^{23} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -99, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.65 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 12 & 9 & 12 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2592] \ 21^2 \ 5^{11} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -99, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.65 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 7 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2591] \ 1^1 \ 5^{12} 1^{23} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -99, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.65 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 5 & 30 & 5 \\ 17 & 30 & 154 & 26 \\ 3 & 5 & 26 & 4 \end{pmatrix}_{b_0}$$

$$[2590] \ 1^{11} \ 5^{12} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -99, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.65 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 7 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2589] \ 1^1 \ 5^{13} 1^{23} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -99, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.65 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 5 & 30 & 5 \\ 19 & 30 & 154 & 26 \\ 3 & 5 & 26 & 4 \end{pmatrix}_{b_0}$$

[2587] $1^{11}51^2231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2596] $1^{11}51^32 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 7 & 5 & 10 & 33 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$

[2595] $31^{12}32151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.20 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 8 & 3 \\ 5 & 5 & 10 & 5 \\ 8 & 10 & 14 & 8 \\ 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$

[2597] $2^1321^{11}51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 7 & 6 & 8 & 31 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$

[2600] $1^{11}3221^{11}51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 1 & 6 & 2 \\ 4 & 18 & 6 & 25 & 8 \\ 1 & 6 & 0 & 8 & 2 \\ 6 & 25 & 8 & 31 & 11 \\ 2 & 8 & 2 & 11 & 3 \end{pmatrix}_{b_0}$

[2599] $151^{11}2231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 1 & 5 & 2 \\ 4 & 18 & 6 & 25 & 8 \\ 1 & 6 & 0 & 8 & 2 \\ 5 & 25 & 8 & 31 & 11 \\ 2 & 8 & 2 & 11 & 3 \end{pmatrix}_{b_0}$

[2598] $41^{11}321^141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 1 & 7 & 2 \\ 4 & 26 & 1 & 44 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 44 & 0 & 70 & 16 \\ 2 & 10 & -1 & 16 & 3 \end{pmatrix}_{b_0}$

[2603] $1^{11}321^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 11 & 13 & 14 & 150 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[2607] $1^{12}41^33141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 21 & 0 & 205 & 27 \\ 2 & 3 & -1 & 27 & 3 \end{pmatrix}_{b_0}$

[2602] $31^24141^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 18 & 4 \\ 9 & 0 & 18 & 58 & 14 \\ 2 & 0 & 4 & 14 & 3 \end{pmatrix}_{b_0}$

[2604] $1^{13}41^23141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 19 & 0 & 145 & 23 \\ 2 & 3 & -1 & 23 & 3 \end{pmatrix}_{b_0}$

[2601] $1^{11}41^231^241^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 21 & 0 & 241 & 29 \\ 2 & 3 & -1 & 29 & 3 \end{pmatrix}_{b_0}$

[2605] $1^1 \bar{5} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 6 & 9 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2606] $1^1 \bar{3} \bar{5} 1231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 6 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2610] $21^2 \bar{5} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 26 & 5 \\ 14 & 26 & 106 & 22 \\ 3 & 5 & 22 & 4 \end{pmatrix}_{b_0}$$

[2608] $1^1 \bar{5} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 26 & 5 \\ 14 & 26 & 106 & 22 \\ 3 & 5 & 22 & 4 \end{pmatrix}_{b_0}$$

[2611] $1^1 \bar{4} 1^{11} 31^{24} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 10 & 2 \\ 3 & 13 & 1 & 44 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 44 & 0 & 138 & 22 \\ 2 & 7 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[2612] $1^{12} 3221^{25} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 16 & 0 & 130 & 22 \\ 2 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[2613] $1^{11} \bar{5} 1^{14} 1231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 10 & 7 & 11 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2614] $1^{11} \bar{5} 1^{12} 1231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 8 & 7 & 11 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2615] $31^{12} 31^5 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 12 & 3 \\ 5 & 5 & 16 & 5 \\ 12 & 16 & 42 & 14 \\ 3 & 5 & 14 & 4 \end{pmatrix}_{b_0}$$

[2616] $1^1 \bar{4} 1^{11} 31^3 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 14 & 2 \\ 3 & 13 & 1 & 56 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 56 & 0 & 230 & 28 \\ 2 & 7 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[2617] $1^{12} \bar{5} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 9 & 7 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2618] $31^{11} \bar{4} 141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 6 & 2 \\ 3 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 24 & 0 & 38 & 12 \\ 2 & 7 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

[2619] $1^{12}5^{12}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 7 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2620] $1^{12}5^{12}231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 5 & 30 & 5 \\ 18 & 30 & 154 & 26 \\ 3 & 5 & 26 & 4 \end{pmatrix}_{b_0}$$

[2609] $2^{14}3^{14}2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 8 & 2 \\ 2 & 2 & 4 & 16 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 8 & 16 & 12 & 62 & 16 \\ 2 & 4 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

[2621] $1^{13}5^{12}231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 15 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 15 & 0 & 193 & 27 \\ 2 & 2 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

[2623] $1^{11}5^{13}231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 11 & 9 & 12 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2624] $1^{14}5^{12}231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 5 & 25 & 5 \\ 16 & 25 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

[2625] $1^{11}5^{13}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 7 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2626] $1^{11}5^{14}131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 10 & 7 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2627] $1^{12}5^{12}231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 10 & 9 & 12 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2628] $1^{12}5^{12}1231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 25 & 5 \\ 14 & 25 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

[2622] $1^{11}3221^{25}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 13 & 2 \\ 10 & 0 & 13 & 101 & 19 \\ 2 & 0 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2629] $1^{12}5^{13}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 7 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2630] $1^{11}5^{13}1^{12}231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 9 & 7 & 11 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2631] $1^{11}41^{23}2321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 1 & 12 & 3 \\ 16 & 12 & 86 & 22 \\ 3 & 3 & 22 & 5 \end{pmatrix}_{b_0}$$

[2632] $1^{11}41^{23}2321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 1 & 12 & 3 \\ 16 & 12 & 86 & 22 \\ 3 & 3 & 22 & 5 \end{pmatrix}_{b_0}$$

[2633] $1^{12}41^{13}2321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 1 & 12 & 3 \\ 14 & 12 & 74 & 20 \\ 3 & 3 & 20 & 5 \end{pmatrix}_{b_0}$$

[2637] $21^{12}51^{13}1^{12}231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 66 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2636] $1^{11}5^{13}131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 9 & 7 & 14 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2634] $21^{12}51^{23}231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 11 & 13 & 101 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2635] $21^{13}51^{33}231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 19 & 2 \\ 2 & 0 & 2 & 19 & 2 \\ 1 & 2 & 0 & 27 & 2 \\ 19 & 19 & 27 & 346 & 36 \\ 2 & 2 & 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[2641] $2^3 3^1 2151^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[2638] $31^{12}4141^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 18 & 4 \\ 10 & 0 & 18 & 58 & 14 \\ 2 & 0 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[2642] $1^{11}41^{33}3141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 18 & 0 & 166 & 24 \\ 2 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[2643] $1^{1^2 321^3} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 5 & 40 & 5 \\ 22 & 40 & 287 & 35 \\ 3 & 5 & 35 & 4 \end{pmatrix}_{b_0}$$

[2644] $1^{1^3} 1^5 1^{231^{13}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 5 & 25 & 5 \\ 15 & 25 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

[2645] $1^{1^4} 1^5 1^{3231^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 26 & 3 \\ 3 & 5 & 40 & 5 \\ 26 & 40 & 287 & 35 \\ 3 & 5 & 35 & 4 \end{pmatrix}_{b_0}$$

[2646] $1^{1^2 4131^3} 41^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 18 & 0 & 166 & 24 \\ 2 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[2647] $1^{1^2} 1^5 1^{31^3} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 7 & 9 & 13 & 73 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2648] $1^{1^3} 3221^{151^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 14 & 2 \\ 8 & 0 & 14 & 75 & 17 \\ 2 & 0 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2654] $2^3 3^1 5^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 5 & 24 & 5 \\ 15 & 24 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

[2653] $1^{1^3} 3221^{251^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 19 & 2 \\ 11 & 0 & 19 & 165 & 25 \\ 2 & 0 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[2639] $1^{1^1} 1^5 1^{1231^{13}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 8 & 8 & 14 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2652] $1^{1^3} 41^1 3141^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 14 & 0 & 70 & 16 \\ 2 & 3 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[2640] $1^{1^2} 1^5 1^{231^{13}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2651] $2^3 21^3 51^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 14 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 14 & 13 & 14 & 150 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[2650] \quad 1 \overset{12}{5} 1 \overset{12}{3} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2649] \quad 1 \overset{11}{5} 1 \overset{11}{3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 11 & 12 & 16 & 155 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[2656] \quad 1 \overset{13}{4} 1 \overset{12}{2} 3 2 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 1 & 12 & 3 \\ 12 & 12 & 58 & 18 \\ 3 & 3 & 18 & 5 \end{pmatrix}_{b_0}$$

$$[2655] \quad 1 \overset{11}{4} 1 \overset{13}{2} 3 2 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 1 & 12 & 3 \\ 12 & 12 & 58 & 18 \\ 3 & 3 & 18 & 5 \end{pmatrix}_{b_0}$$

$$[2661] \quad 2 1 \overset{11}{4} 5 1 3 1 \overset{13}{3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 12 & 9 & 14 & 97 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[2664] \quad 1 \overset{11}{5} 1 \overset{13}{3} 1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 8 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2663] \quad 2 \overset{13}{3} 2 1 \overset{12}{2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 1 & 11 & 3 \\ 7 & 8 & 11 & 44 & 14 \\ 2 & 2 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[2662] \quad 1 \overset{13}{5} 1 \overset{13}{3} 1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 10 & 8 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2660] \quad 2 1 \overset{11}{3} 5 1 3 1 \overset{13}{3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 10 & 8 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2658] \quad 2 2 \overset{11}{3} 3 1 5 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 3 & 4 & 2 \\ 3 & 5 & 8 & 9 & 5 \\ 3 & 8 & 4 & 9 & 4 \\ 4 & 9 & 9 & 11 & 7 \\ 2 & 5 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[2657] \quad 1 \overset{11}{5} 1 \overset{13}{3} 1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 12 & 9 & 14 & 97 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2659] $1^{1^3} 5^{1^2} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 5 & 26 & 5 \\ 15 & 26 & 106 & 22 \\ 3 & 5 & 22 & 4 \end{pmatrix}_{b_0}$$

[2674] $1^{11} 41^{1^2} 31^2 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 16 & 0 & 138 & 22 \\ 2 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[2665] $1^{1^4} 5^{1^2} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 5 & 26 & 5 \\ 16 & 26 & 106 & 22 \\ 3 & 5 & 22 & 4 \end{pmatrix}_{b_0}$$

[2667] $1^{1^2} 5^{1^2} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 8 & 10 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2668] $2^3 1^3 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 11 & 3 \\ 4 & 5 & 18 & 5 \\ 11 & 18 & 49 & 15 \\ 3 & 5 & 15 & 4 \end{pmatrix}_{b_0}$$

[2669] $21^{1^2} 5^{1^2} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 8 & 10 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2670] $21^{1^2} 5^1 1^2 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 8 & 9 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2673] $21^{1^2} 4^1 5^{1^2} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 12 & 10 & 12 & 90 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2672] $1^{1^2} 3221^{1^2} 5^1 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 13 & 2 \\ 4 & 18 & 6 & 50 & 8 \\ 1 & 6 & 0 & 16 & 2 \\ 13 & 50 & 16 & 130 & 22 \\ 2 & 8 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[2666] $1^{1^4} 5^{1^2} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 12 & 10 & 12 & 90 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2671] $1^{1^2} 5^{1^4} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 8 & 10 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2675] $2 \overset{1}{3} 2151^{\overset{14}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 5 & 20 & 5 \\ 12 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

[2680] $31 \overset{14}{4} 141^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 5 & 1 & 28 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 28 & 0 & 110 & 20 \\ 2 & 5 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[2678] $31 \overset{11}{3} 4 141^{\overset{11}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 5 & 1 & 34 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 34 & 0 & 194 & 26 \\ 2 & 5 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

[2677] $31 \overset{12}{2} 4 141^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 5 & 1 & 32 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 32 & 0 & 158 & 24 \\ 2 & 5 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[2676] $1 \overset{1}{5} 1 \overset{14}{3} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 5 & 20 & 5 \\ 12 & 20 & 62 & 16 \\ 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

[2681] $31 \overset{12}{3} 4 141^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 5 & 1 & 41 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 41 & 0 & 269 & 31 \\ 2 & 5 & -1 & 31 & 3 \end{pmatrix}_{b_0}$$

[2679] $31 \overset{11}{2} 4 141^{\overset{11}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 5 & 1 & 25 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 25 & 0 & 101 & 19 \\ 2 & 5 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

[2691] $1 \overset{14}{11} 5 131^{\overset{14}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 10 & 7 & 14 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2693] $1 \overset{12}{5} 1 \overset{11}{3} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 9 & 10 & 15 & 108 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2692] $1 \overset{2}{5} 131^{\overset{12}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 10 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2690] $1 \overset{12}{11} 5 131^{\overset{14}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 9 & 8 & 15 & 84 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2689] $1^{11}5^{11}231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 5 & 30 & 5 \\ 15 & 30 & 139 & 25 \\ 3 & 5 & 25 & 4 \end{pmatrix}_{b_0}$$

[2688] $21^{11}5^{11}131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 9 & 8 & 15 & 84 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2687] $1^{13}5^{12}1231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2686] $1^{14}3221^{15}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 17 & 0 & 106 & 20 \\ 2 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[2685] $1^{12}5^{14}3231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 15 & 13 & 14 & 150 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[2684] $1^{13}4131^{14}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 13 & 0 & 61 & 15 \\ 2 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[2683] $31^{11}4^{11}141^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 16 & 4 \\ 7 & 0 & 16 & 38 & 12 \\ 2 & 0 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[2682] $1^{12}5^{13}231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 12 & 13 & 14 & 150 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[2699] $22^13^{12}151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 10 & 3 \\ 6 & 14 & 26 & 8 \\ 10 & 26 & 42 & 14 \\ 3 & 8 & 14 & 4 \end{pmatrix}_{b_0}$$

[2698] $1^{14}5^{13}131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 5 & 32 & 5 \\ 20 & 32 & 174 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$$

[2697] $1^{12}5^{11}131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 7 & 8 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2696] $21^4 \overset{1}{5} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 5 & 32 & 5 \\ 20 & 32 & 174 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$

[2695] $1^{12} \overset{13}{5} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 8 & 10 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[2694] $21^{13} \overset{13}{5} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 10 & 10 & 9 & 67 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2701] $2 \overset{1}{3} 21^{15} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.1$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 6 & 30 & 6 \\ 15 & 30 & 123 & 25 \\ 3 & 6 & 25 & 5 \end{pmatrix}_{b_0}$

[2702] $21^1 \overset{12}{5} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 7 & 8 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$

[2700] $1^{12} \overset{14}{5} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 11 & 9 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2703] $1^{12} \overset{14}{41} 2321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -122, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 1 & 12 & 3 \\ 14 & 12 & 70 & 20 \\ 3 & 3 & 20 & 5 \end{pmatrix}_{b_0}$

[2704] $1^{14} \overset{12}{41} 2321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -122, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 1 & 12 & 3 \\ 14 & 12 & 70 & 20 \\ 3 & 3 & 20 & 5 \end{pmatrix}_{b_0}$

[2705] $1^1 \overset{14}{5} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 5 & 29 & 5 \\ 18 & 29 & 137 & 25 \\ 3 & 5 & 25 & 4 \end{pmatrix}_{b_0}$

[2707] $21^3 \overset{1}{5} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 5 & 29 & 5 \\ 17 & 29 & 137 & 25 \\ 3 & 5 & 25 & 4 \end{pmatrix}_{b_0}$

[2708] $1^{12} \overset{13}{5} 1231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 8 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$

[2709] $22 \overset{1}{3} 151^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 3 & 8 & 2 \\ 3 & 3 & 4 & 28 & 4 \\ 12 & 8 & 28 & 110 & 20 \\ 2 & 2 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2710] \quad 1 \overset{1^3}{\underset{1^{12}}{5}} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 8 & 10 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[2711] \quad 1 \overset{1^3}{\underset{1^3}{12}} 5 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 10 & 9 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2712] \quad 1 \overset{1^3}{\underset{1^3}{12}} 5 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 5 & 29 & 5 \\ 17 & 29 & 137 & 25 \\ 3 & 5 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[2706] \quad 21 \overset{1^{12}}{\underset{3}{5}} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 10 & 9 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2713] \quad 221 \overset{1^{12}}{\underset{1^3}{5}} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 7 & 2 \\ 3 & 4 & 3 & 14 & 4 \\ 1 & 3 & 0 & 7 & 2 \\ 7 & 14 & 7 & 33 & 11 \\ 2 & 4 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[2716] \quad 21 \overset{1^3}{\underset{1^3}{12}} 5 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 7 & 9 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2717] \quad 1 \overset{1^1}{\underset{1^3}{5}} 1^3 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 23 & 3 \\ 3 & 5 & 40 & 5 \\ 23 & 40 & 287 & 35 \\ 3 & 5 & 35 & 4 \end{pmatrix}_{b_0}$$

$$[2718] \quad 1 \overset{1^2}{\underset{1^2}{12}} 5 1^3 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 13 & 13 & 14 & 150 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[2719] \quad 1 \overset{1^2}{\underset{1^2}{12}} 3221^3 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 21 & 0 & 250 & 30 \\ 2 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[2720] \quad 1 \overset{1^3}{\underset{1^3}{12}} 5 1231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2721] \quad 1 \overset{1^3}{\underset{1^1}{12}} 5 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 7 & 9 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2722] $1^{1^3 321^2} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 5 & 35 & 5 \\ 18 & 35 & 202 & 30 \\ 3 & 5 & 30 & 4 \end{pmatrix}_{b_0}$$

[2714] $1^{1^3 221^3} 51^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 15 & 2 \\ 14 & 0 & 15 & 178 & 24 \\ 2 & 0 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[2724] $1^{1^4 41^2} 31^{14} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 17 & 0 & 149 & 23 \\ 2 & 3 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

[2725] $2^3 3^{1^2} 15^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 9 & 9 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2726] $1^{1^2 5^1} 3^{231^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 14 & 13 & 14 & 150 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[2727] $1^{1^2 3221^3} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 21 & 2 \\ 14 & 0 & 21 & 250 & 30 \\ 2 & 0 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

[2728] $1^{1^3 5^1} 1231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 9 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2729] $2^3 3^{1^2} 15^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 11 & 12 & 13 & 112 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2730] $1^{1^3 3221^2} 51^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 10 & 2 \\ 11 & 0 & 10 & 78 & 16 \\ 2 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2731] $1^{1^3 4131^2} 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 17 & 0 & 121 & 21 \\ 2 & 3 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[2732] $1^{1^3} 5^{13} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 11 & 12 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2715] $1^4 5 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 9 & 12 & 13 & 112 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2723] $1^1 5^3 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 25 & 3 \\ 3 & 5 & 40 & 5 \\ 25 & 40 & 287 & 35 \\ 3 & 5 & 35 & 4 \end{pmatrix}_{b_0}$

[2737] $1^{12} 4131^2 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 14 & 0 & 90 & 18 \\ 2 & 3 & -1 & 18 & 3 \end{pmatrix}_{b_0}$

[2733] $1^{13} 41^1 31^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 18 & 0 & 130 & 22 \\ 2 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0}$

[2736] $2^1 3^1 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 13 & 3 \\ 4 & 5 & 21 & 5 \\ 13 & 21 & 70 & 18 \\ 3 & 5 & 18 & 4 \end{pmatrix}_{b_0}$

[2735] $1^4 5 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 9 & 11 & 15 & 117 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$

[2734] $1^{12} 5^4 1231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 9 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$

[2738] $1^{11} 3221^1 51^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 8 & 2 \\ 8 & 0 & 8 & 31 & 11 \\ 2 & 0 & 2 & 11 & 3 \end{pmatrix}_{b_0}$

[2739] $1^{12} 41^2 3141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 16 & 0 & 110 & 20 \\ 2 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0}$

[2740] $1^{12} 5^1 3^2 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 24 & 3 \\ 3 & 5 & 40 & 5 \\ 24 & 40 & 287 & 35 \\ 3 & 5 & 35 & 4 \end{pmatrix}_{b_0}$

[2745] $1^{12} 41^2 31^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 20 & 0 & 186 & 26 \\ 2 & 3 & -1 & 26 & 3 \end{pmatrix}_{b_0}$

[2746] $1^{11} 5^1 1^2 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 9 & 8 & 14 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2741] $1^{11}5^{13}1^{11}3^{23}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 12 & 16 & 155 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[2742] $21^{13}5^{13}1^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 8 & 8 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[2743] $2^3 3^{13} 1^{14} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 11 & 9 & 14 & 97 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2744] $1^{11}3^{21}2^{12} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 9 & 11 & 13 & 101 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2755] $2^3 3^{11} 21^{15} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 8 & 8 & 9 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2751] $1^{12} 41^{13} 31^{24} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 19 & 0 & 173 & 25 \\ 2 & 3 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

[2753] $2^3 3^{12} 1^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 7 & 7 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2754] $1^{11}5^{13}1^{14} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 10 & 8 & 15 & 84 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2750] $1^{13} 5^{13} 3^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 7 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2749] $2^3 3^{13} 1^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 8 & 6 & 11 & 43 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2748] $21^{13} 5^{12} 1^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 9 & 9 & 8 & 49 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[2747] $21^{13}5^{11}131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 11 & 11 & 16 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2752] $23^{12}2151^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 7 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[2756] $1^{14}41^{11}31^{14}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 8 & 2 \\ 3 & 13 & 1 & 32 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 32 & 0 & 70 & 16 \\ 2 & 7 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[2757] $1^{13}5^{13}231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 18 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 18 & 0 & 295 & 33 \\ 2 & 2 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

[2758] $321^{11}4141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 8 & 7 & 3 \\ 8 & 17 & 16 & 9 \\ 7 & 16 & 12 & 8 \\ 3 & 9 & 8 & 4 \end{pmatrix}_{b_0}$$

[2759] $1^{12}5^{12}131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 8 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2760] $23^{12}151^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 8 & 10 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2762] $23^{12}21^{25}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 12 & 8 & 14 & 90 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[2761] $2^{15}3^{14}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.1$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 10 & 2 \\ 2 & 2 & 4 & 20 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 10 & 20 & 15 & 98 & 20 \\ 2 & 4 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

[2763] $23^{12}21^{15}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -142, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 13 & 3 \\ 4 & 5 & 20 & 5 \\ 13 & 20 & 63 & 17 \\ 3 & 5 & 17 & 4 \end{pmatrix}_{b_0}$$

[2764] $1^{12}41^{13}31^{14}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -143, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 97 & 19 \\ 2 & 3 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

[2765] $22 \overset{1^{12}}{3} 151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -143, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 7 & 2 \\ 1 & 0 & 3 & 6 & 2 \\ 3 & 3 & 4 & 16 & 4 \\ 7 & 6 & 16 & 36 & 12 \\ 2 & 2 & 4 & 12 & 3 \end{pmatrix}_{b_0}$

[2766] $1 \overset{1^2}{\overset{1^3}{5}} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -143, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 9 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[2767] $1 \overset{1^2}{\overset{1^3}{3221}} 51^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -143, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 11 & 2 \\ 8 & 0 & 11 & 50 & 14 \\ 2 & 0 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[2768] $1 \overset{1^4}{\overset{1^2}{5}} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -143, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 11 & 8 & 15 & 84 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$

[2769] $1 \overset{1^1}{\overset{1^2}{5}} 1^{2231} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -143, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 10 & 10 & 15 & 108 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2770] $1 \overset{1^4}{\overset{1^1}{5}} 1^{3231} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 15 & 12 & 16 & 155 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$

[2771] $1 \overset{1^4}{\overset{1^1}{41}} 1^{131} 241^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 1 & 11 & 2 \\ 3 & 13 & 1 & 44 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 44 & 0 & 138 & 22 \\ 2 & 7 & -1 & 22 & 3 \end{pmatrix}_{b_0}$

[2772] $1 \overset{1^2}{\overset{1^1}{5}} 1^{3231} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 13 & 12 & 16 & 155 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$

[2773] $1 \overset{1^3}{\overset{1^1}{5}} 1^{1231} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 10 & 8 & 14 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2774] $21 \overset{1^4}{\overset{1^3}{5}} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 14 & 14 & 19 & 176 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0}$

[2775] $31 \overset{1^{11}}{1} 4 141^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 16 & 4 \\ 8 & 0 & 16 & 38 & 12 \\ 2 & 0 & 4 & 12 & 3 \end{pmatrix}_{b_0}$

[2776] $1 \overset{1^1}{3221} 51^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 9 & 2 \\ 4 & 18 & 6 & 40 & 8 \\ 1 & 6 & 0 & 10 & 2 \\ 9 & 40 & 10 & 78 & 16 \\ 2 & 8 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2777] \ 1^1 5^1 1^2 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 5 & 30 & 5 \\ 16 & 30 & 139 & 25 \\ 3 & 5 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[2778] \ 2^1 3^1 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 6 & 23 & 6 \\ 12 & 23 & 67 & 19 \\ 3 & 6 & 19 & 5 \end{pmatrix}_{b_0}$$

$$[2779] \ 1^1 5^1 1^{13} 321^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 10 & 11 & 18 & 141 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[2780] \ 1^1 41^3 2321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 1 & 14 & 3 \\ 19 & 14 & 120 & 26 \\ 3 & 3 & 26 & 5 \end{pmatrix}_{b_0}$$

$$[2781] \ 1^1 41^3 2321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 1 & 14 & 3 \\ 19 & 14 & 120 & 26 \\ 3 & 3 & 26 & 5 \end{pmatrix}_{b_0}$$

$$[2782] \ 1^1 5^1 1^4 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 11 & 8 & 14 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2783] \ 1^1 3221^3 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 18 & 2 \\ 14 & 0 & 18 & 211 & 27 \\ 2 & 0 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[2784] \ 1^1 5^1 1^3 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 14 & 12 & 16 & 155 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[2785] \ 1^1 3221^3 51^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -149, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 17 & 2 \\ 4 & 18 & 6 & 65 & 8 \\ 1 & 6 & 0 & 18 & 2 \\ 17 & 65 & 18 & 211 & 27 \\ 2 & 8 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[2786] \ 2^1 3^1 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 1 & 14 & 3 \\ 8 & 7 & 14 & 54 & 16 \\ 2 & 2 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[2796] \ 21^1 51^3 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 15 & 15 & 15 & 178 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[2797] $1^{11} \overset{13}{5} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 11 & 9 & 16 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2787] $1^{13} \overset{13}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 9 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2788] $21^3 \overset{11}{5} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 11 & 9 & 16 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2789] $21^{13} \overset{11}{5} 1^{12} 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 13 & 13 & 19 & 150 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[2790] $21^{13} \overset{13}{5} 1231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2791] $1^{13} \overset{14}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 9 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2792] $2321^1 \overset{13}{3} 51^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 15 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 15 & 15 & 15 & 178 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[2793] $31^3 \overset{11}{4} 141^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 27 & 4 \\ 12 & 0 & 27 & 133 & 21 \\ 2 & 0 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[2798] $2^3 \overset{13}{3} 1^3 \overset{13}{5} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 10 & 8 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2794] $31^1 \overset{11}{4} 141^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 16 & 4 \\ 9 & 0 & 16 & 38 & 12 \\ 2 & 0 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[2795] $21^{13} 1^5 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2808] $1^{1321^3} 5^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 12 & 15 & 15 & 178 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[2809] $1^{11^2} 5^1 1^{1231} 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 10 & 9 & 17 & 106 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[2810] $2^3 3^1 5^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 9 & 7 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2799] $1^{12^1} 5^1 1^{2231} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 10 & 11 & 13 & 101 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2800] $2^3 3^1 21^2 5^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 11 & 9 & 12 & 81 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2801] $2^3 3^1 21^2 5^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 9 & 10 & 10 & 63 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2802] $1^1 5^1 1^{2231} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 3 \\ 3 & 5 & 35 & 5 \\ 19 & 35 & 202 & 30 \\ 3 & 5 & 30 & 4 \end{pmatrix}_{b_0}$$

[2803] $1^{12^3} 5^1 1^{231} 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 9 & 8 & 14 & 70 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2804] $1^5 13^1 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 9 & 11 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[2805] $1^{1341^3} 3^1 14^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 24 & 0 & 250 & 30 \\ 2 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[2806] $1^{14} 413^1 14^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 16 & 0 & 90 & 18 \\ 2 & 3 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[2807] $1^{12^3} 5^1 13^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 8 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2811] \quad 31^{1^3} 4^{1^3} 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 5 & 1 & 30 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 30 & 0 & 130 & 22 \\ 2 & 5 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[2812] \quad 321^{1^3} 4141^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 8 & 10 & 3 \\ 8 & 17 & 24 & 9 \\ 10 & 24 & 30 & 12 \\ 3 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[2814] \quad 1^{1^3} 41^2 2321^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 1 & 14 & 3 \\ 15 & 14 & 88 & 22 \\ 3 & 3 & 22 & 5 \end{pmatrix}_{b_0}$$

$$[2813] \quad 1^{1^3} 41^1 2321^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 1 & 14 & 3 \\ 15 & 14 & 88 & 22 \\ 3 & 3 & 22 & 5 \end{pmatrix}_{b_0}$$

$$[2825] \quad 151^{1^2} 2231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 8 & 2 \\ 4 & 18 & 6 & 40 & 8 \\ 1 & 6 & 0 & 10 & 2 \\ 8 & 40 & 10 & 78 & 16 \\ 2 & 8 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2823] \quad 2^3 3^{1^2} 151^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 10 & 11 & 12 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[2822] \quad 1^{21} 5^{1^3} 131^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2821] \quad 21^{1^3} 51^2 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 12 & 12 & 11 & 100 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[2820] \quad 41^{1^2} 23141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 8 & 2 \\ 4 & 26 & 1 & 56 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 56 & 0 & 110 & 20 \\ 2 & 10 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2819] \quad 21^{1^3} 51231^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 8 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2826] \quad 1^{1^4} 12^5 131^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 10 & 8 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[2818] \quad 21^{1^3} 51^1 3^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 14 & 2 \\ 2 & 4 & 3 & 26 & 4 \\ 2 & 3 & 0 & 18 & 2 \\ 14 & 26 & 18 & 148 & 24 \\ 2 & 4 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[2817] \ 21^{13}5^{14}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 12 & 12 & 13 & 112 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2816] \ 21^{13}5^{12}131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 12 & 12 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[2815] \ 21^{13}5^{12}231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 16 & 2 \\ 2 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 23 & 2 \\ 16 & 16 & 23 & 236 & 30 \\ 2 & 2 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[2824] \ 21^{13}5^{12}13^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 10 & 2 \\ 2 & 4 & 3 & 22 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 10 & 22 & 12 & 88 & 18 \\ 2 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[2827] \ 21^{13}5^{13}131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 13 & 13 & 18 & 148 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[2828] \ 1^{12}3221^{25}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 16 & 2 \\ 11 & 0 & 16 & 130 & 22 \\ 2 & 0 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[2829] \ 1^{13}5^{13}1^{12}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 5 & 30 & 5 \\ 17 & 30 & 139 & 25 \\ 3 & 5 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[2830] \ 1^{11}5^{14}131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 9 & 16 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2831] \ 1^{13}5^{12}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 9 & 10 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2832] \ 1^{11}5^{12}1^{22}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 11 & 10 & 15 & 108 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2833] \ 21^{213}5^{13}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 9 & 10 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[2834] \quad & 231^{1^1 3^1 15^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 5 & 1 & 5 & 2 \\ 5 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 35 & 0 & 33 & 11 \\ 2 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0} \\
[2835] \quad & 2^1 3^1 15^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 9 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[2836] \quad & 1^{1^2} 5^{1^2} 13^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 9 & 9 & 13 & 73 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2837] \quad & 21^{1^2} 5^{1^2} 13^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 9 & 9 & 13 & 73 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2838] \quad & 31^1 4^1 141^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 5 & 1 & 21 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 21 & 0 & 61 & 15 \\ 2 & 5 & -1 & 15 & 3 \end{pmatrix}_{b_0} \\
[2840] \quad & 1^{1^1} 41^{1^2} 31^{1^2} 41^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 21 & 0 & 241 & 29 \\ 2 & 3 & -1 & 29 & 3 \end{pmatrix}_{b_0} \\
[2839] \quad & 1^2 5^{1^2} 13^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 8 & 11 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2842] \quad & 1^{1^3} 5^{1^2} 1231^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2841] \quad & 1^{1^4} 41^{1^3} 31^{1^1} 41^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 21 & 0 & 169 & 25 \\ 2 & 3 & -1 & 25 & 3 \end{pmatrix}_{b_0} \\
[2846] \quad & 1^4 5^{1^2} 13^{1^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 10 & 12 & 18 & 150 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2847] \quad & 1^{1^1} 3221^{1^2} 51^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 13 & 2 \\ 11 & 0 & 13 & 101 & 19 \\ 2 & 0 & 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2845] & 151^{13}2231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 16 & 2 \\ 4 & 18 & 6 & 80 & 8 \\ 1 & 6 & 0 & 27 & 2 \\ 16 & 80 & 27 & 346 & 36 \\ 2 & 8 & 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[2844] & 1^{11}41^331^141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 22 & 0 & 258 & 30 \\ 2 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0} \\
[2848] & 1^{12}51^{2231}^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 13 & 11 & 13 & 101 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2843] & 1^{14}51^{1231}^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 5 & 30 & 5 \\ 18 & 30 & 139 & 25 \\ 3 & 5 & 25 & 4 \end{pmatrix}_{b_0} \\
[2849] & 1^{12}4131^341^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 18 & 0 & 166 & 24 \\ 2 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0} \\
[2850] & 2^{12}3^{11}4^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 2 & 4 & 11 & 4 \\ 2 & 4 & 1 & 10 & 3 \\ 6 & 11 & 10 & 30 & 12 \\ 2 & 4 & 3 & 12 & 4 \end{pmatrix}_{b_0} \\
[2851] & 1^{12}321^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 12 & 13 & 19 & 190 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[2852] & 1^{12}51^{2231}^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 11 & 13 & 101 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2853] & 1^13221^451^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 20 & 2 \\ 18 & 0 & 20 & 318 & 32 \\ 2 & 0 & 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[2854] & 2^{14}3^{21}^{16} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.2 \\
& T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 6 & 36 & 6 \\ 18 & 36 & 178 & 30 \\ 3 & 6 & 30 & 5 \end{pmatrix}_{b_0} \\
[2855] & 1^{12}51^{1231}^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 10 & 8 & 14 & 70 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[2857] $1^{12} 5^1 1^2 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 5 & 35 & 5 \\ 20 & 35 & 202 & 30 \\ 3 & 5 & 30 & 4 \end{pmatrix}_{b_0}$

[2858] $2^1 3^1 1^2 5^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 9 & 10 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2856] $1^{14} 5^1 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 13 & 10 & 17 & 128 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[2859] $1^{14} 5^1 1^2 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 5 & 35 & 5 \\ 22 & 35 & 202 & 30 \\ 3 & 5 & 30 & 4 \end{pmatrix}_{b_0}$

[2860] $1^{13} 5^1 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 10 & 10 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2861] $1^{14} 5^1 1^2 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 13 & 10 & 15 & 108 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2862] $1^{13} 5^1 1^2 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 12 & 10 & 15 & 108 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2863] $1^{14} 5^1 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 11 & 10 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[2864] $2^1 3^1 21^1 5^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 9 & 7 & 11 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[2865] $21^4 5^1 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 13 & 10 & 17 & 128 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[2869] $31^2 4^1 141^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 25 & 4 \\ 10 & 0 & 25 & 101 & 19 \\ 2 & 0 & 4 & 19 & 3 \end{pmatrix}_{b_0}$

[2868] $1^{11}41^22321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -167$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 1 & 14 & 3 \\ 17 & 14 & 100 & 24 \\ 3 & 3 & 24 & 5 \end{pmatrix}_{b_0}$

[2866] $1^{13}3221^251^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -167$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 19 & 0 & 165 & 25 \\ 2 & 3 & -1 & 25 & 3 \end{pmatrix}_{b_0}$

[2867] $1^{12}41^22321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -167$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 1 & 14 & 3 \\ 17 & 14 & 100 & 24 \\ 3 & 3 & 24 & 5 \end{pmatrix}_{b_0}$

[2871] $1^{14}4131^241^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -168$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 20 & 0 & 158 & 24 \\ 2 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0}$

[2870] $1^{12}3221^451^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -168$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 26 & 2 \\ 18 & 0 & 26 & 410 & 38 \\ 2 & 0 & 2 & 38 & 3 \end{pmatrix}_{b_0}$

[2872] $1^{14}321^251^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -168$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 5 & 40 & 5 \\ 20 & 40 & 258 & 34 \\ 3 & 5 & 34 & 4 \end{pmatrix}_{b_0}$

[2874] $1^{12}51^2131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -168$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 9 & 9 & 15 & 84 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$

[2873] $1^{12}51^32231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -168$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 12 & 11 & 13 & 101 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$

[2879] $21^{11}5131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -171$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 8 & 10 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2878] $1^{12}41^131^341^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -171$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 23 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 23 & 0 & 273 & 31 \\ 2 & 3 & -1 & 31 & 3 \end{pmatrix}_{b_0}$

[2877] $1^{13}51^2131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -171$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 8 & 10 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2876] $1^{12}41^331^141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -171$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 25 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 25 & 0 & 305 & 33 \\ 2 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$

[2875] $151^{\overline{111}} 3 22 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 5 & 8 & 7 & 5 \\ 3 & 8 & 4 & 7 & 4 \\ 3 & 7 & 7 & 5 & 5 \\ 2 & 5 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[2881] $2 3 2151^{\overline{15}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.1$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 5 & 25 & 5 \\ 15 & 25 & 98 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$

[2880] $31^{\overline{15}} 4 141^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.1$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 5 & 1 & 35 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 35 & 0 & 173 & 25 \\ 2 & 5 & -1 & 25 & 3 \end{pmatrix}_{b_0}$

[2886] $1^{\overline{12}} 5^{\overline{13}} 131^{\overline{13}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 10 & 9 & 13 & 73 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$

[2885] $151^{\overline{1}} 3 2 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.1$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 5 & 25 & 5 \\ 15 & 25 & 98 & 20 \\ 3 & 5 & 20 & 4 \end{pmatrix}_{b_0}$

[2884] $1^{\overline{11}} 5^{\overline{13}} 1^{\overline{23}} 1^{\overline{14}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 11 & 9 & 17 & 106 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2883] $1^{\overline{15}} 5^{\overline{12}} 231^{\overline{13}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 5 & 35 & 5 \\ 21 & 35 & 202 & 30 \\ 3 & 5 & 30 & 4 \end{pmatrix}_{b_0}$

[2882] $1^{\overline{13}} 5^{\overline{1}} 1^{\overline{3}} 231^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 13 & 15 & 15 & 178 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$

[2891] $21^{\overline{13}} 5^{\overline{1}} 131^{\overline{12}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 13 & 12 & 13 & 112 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$

[2890] $21^{\overline{13}} 5^{\overline{14}} 131^{\overline{13}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 13 & 13 & 16 & 141 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$

[2889] $21^{\overline{13}} 5^{\overline{13}} 131^{\overline{12}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 11 & 11 & 12 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$

[2892] $1^{\overline{13}} 5^{\overline{13}} 131^{\overline{12}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 11 & 11 & 12 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[2893] \quad 1 \overset{12}{5} 1 \overset{13}{3} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 8 & 7 & 11 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[2894] \quad 31 \overset{1}{4} 141 \overset{13}{3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 15 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 36 & 4 \\ 15 & 0 & 36 & 238 & 28 \\ 2 & 0 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[2895] \quad 1 \overset{13}{5} 131 \overset{12}{2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 13 & 12 & 13 & 112 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2888] \quad 31 \overset{13}{3} 32151 \overset{1}{4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 12 & 3 \\ 5 & 5 & 15 & 5 \\ 12 & 15 & 34 & 12 \\ 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2887] \quad 1 \overset{13}{5} 131 \overset{12}{2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 12 & 11 & 12 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[2896] \quad 2 \overset{14}{3} 1 \overset{11}{5} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 10 & 7 & 14 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[2899] \quad 2 \overset{14}{3} 1 \overset{11}{5} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 14 & 3 \\ 4 & 5 & 23 & 5 \\ 14 & 23 & 81 & 19 \\ 3 & 5 & 19 & 4 \end{pmatrix}_{b_0}$$

$$[2898] \quad 31 \overset{1}{4} 141 \overset{11}{3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 5 & 1 & 23 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 23 & 0 & 77 & 17 \\ 2 & 5 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2897] \quad 1 \overset{12}{5} 131 \overset{13}{3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 9 & 13 & 73 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2900] \quad 22 \overset{15}{3} 151 \overset{1}{4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -178, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.1$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 15 & 2 \\ 1 & 0 & 3 & 10 & 2 \\ 3 & 3 & 4 & 35 & 4 \\ 15 & 10 & 35 & 173 & 25 \\ 2 & 2 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[2910] \quad 1 \overset{11}{5} 1 \overset{12}{3} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 12 & 14 & 17 & 181 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[2909] & 1^{12}41^33141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 21 & 0 & 205 & 27 \\ 2 & 3 & -1 & 27 & 3 \end{pmatrix}_{b_0} \\
[2908] & 1^{13}4131^341^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 21 & 0 & 205 & 27 \\ 2 & 3 & -1 & 27 & 3 \end{pmatrix}_{b_0} \\
[2907] & 1^{13}41^23141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 19 & 0 & 145 & 23 \\ 2 & 3 & -1 & 23 & 3 \end{pmatrix}_{b_0} \\
[2906] & 1^{13}3221^351^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 24 & 2 \\ 15 & 0 & 24 & 295 & 33 \\ 2 & 0 & 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[2905] & 1^{13}321^3 \quad 5 \quad 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 24 & 3 \\ 3 & 5 & 45 & 5 \\ 24 & 45 & 349 & 39 \\ 3 & 5 & 39 & 4 \end{pmatrix}_{b_0} \\
[2904] & 1^{13}51231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2903] & 1^3 \quad 5 \quad 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 9 & 12 & 15 & 115 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2902] & 1^{13}3221^151^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 14 & 2 \\ 9 & 0 & 14 & 75 & 17 \\ 2 & 0 & 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[2901] & 1^{11}51^2231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 11 & 11 & 18 & 141 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2911] & 221^{13}5131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 1 & 12 & 2 \\ 3 & 4 & 3 & 24 & 4 \\ 1 & 3 & 0 & 15 & 2 \\ 12 & 24 & 15 & 115 & 21 \\ 2 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2912] & 2^3 \quad 3 \quad 15 \quad 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 20 & 3 \\ 4 & 5 & 32 & 5 \\ 20 & 32 & 174 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0} \\
[2914] & 21^{13}5131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -183, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8
\end{aligned}$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 11 & 11 & 12 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[2913] \ 21^{13}5^{11}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -183, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 9 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2915] \ 21^{312}5^{11}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 11 & 10 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[2916] \ 2^{16}3^{14}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.2$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 12 & 2 \\ 2 & 2 & 4 & 24 & 4 \\ 2 & 4 & 1 & 18 & 3 \\ 12 & 24 & 18 & 142 & 24 \\ 2 & 4 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[2920] \ 31^{13}31^{11}5^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 18 & 3 \\ 5 & 5 & 24 & 5 \\ 18 & 24 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[2921] \ 1^{12}41^{23}1^{24}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 24 & 0 & 286 & 32 \\ 2 & 3 & -1 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[2922] \ 1^{12}5^{13}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 11 & 10 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[2923] \ 1^{13}5^{14}1^{32}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 16 & 15 & 15 & 178 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[2924] \ 21^{13}5^{11}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 10 & 10 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[2917] \ 1^{11}5^{14}1^{12}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 12 & 9 & 17 & 106 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[2918] \ 1^{12}321^{21}5^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 10 & 12 & 16 & 130 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[2919] \ 1^{13}5^{12}1^{32}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$\begin{aligned}
& T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 14 & 15 & 15 & 178 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2929] & 1^{13} 41^2 31^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 23 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 23 & 0 & 229 & 29 \\ 2 & 3 & -1 & 29 & 3 \end{pmatrix}_{b_0} \\
[2928] & 1^{11} 3221^4 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 23 & 2 \\ 18 & 0 & 23 & 361 & 35 \\ 2 & 0 & 2 & 35 & 3 \end{pmatrix}_{b_0} \\
[2927] & 1^{13} 4131^2 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 17 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 17 & 0 & 121 & 21 \\ 2 & 3 & -1 & 21 & 3 \end{pmatrix}_{b_0} \\
[2926] & 1^{13} 5^{14} 1231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 10 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2925] & 1^{14} 5^{13} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 10 & 13 & 16 & 141 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2932] & 1^{12} 5^{14} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.53 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 12 & 10 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[2931] & 1^{12} 5^{13} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 10 & 9 & 15 & 84 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[2930] & 1^1 3221^3 51^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 15 & 2 \\ 15 & 0 & 15 & 178 & 24 \\ 2 & 0 & 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2933] & 1^{13} 5^{13} 1^3 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 15 & 15 & 15 & 178 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2935] & 31^3 4^{11} 141^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 34 & 4 \\ 13 & 0 & 34 & 194 & 26 \\ 2 & 0 & 4 & 26 & 3 \end{pmatrix}_{b_0} \\
[2934] & 31^4 4^{11} 141^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 2 & 16 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 43 & 4 \\ 16 & 0 & 43 & 317 & 33 \\ 2 & 0 & 4 & 33 & 3 \end{pmatrix}_{b_0}$$

[2938] $2^1 3^{12} 21^3 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 16 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 18 & 2 \\ 16 & 11 & 18 & 166 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[2936] $21^4 5^{12} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 13 & 11 & 15 & 117 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2937] $22^1 3^{12} 151^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 8 & 2 \\ 1 & 0 & 3 & 8 & 2 \\ 3 & 3 & 4 & 18 & 4 \\ 8 & 8 & 18 & 50 & 14 \\ 2 & 2 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[2939] $1^{12} 3221^1 51^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 11 & 2 \\ 9 & 0 & 11 & 50 & 14 \\ 2 & 0 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[2940] $1^{12} 5^{14} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 13 & 11 & 15 & 117 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2941] $1^{11} 5^1 3231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 13 & 13 & 19 & 190 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[2942] $21^{13} 5^{12} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 10 & 10 & 11 & 68 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[2944] $1^{14} 41^1 2321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -194$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.99 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 1 & 16 & 3 \\ 16 & 16 & 106 & 24 \\ 3 & 3 & 24 & 5 \end{pmatrix}_{b_0}$$

[2943] $2^1 3^{12} 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -194$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 1 & 18 & 3 \\ 10 & 8 & 18 & 86 & 20 \\ 2 & 2 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

[2946] $1^{13} 5^{12} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -195$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 9 & 10 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[2945] $1^{12} 41^1 31^2 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -195$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 19 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 19 & 0 & 173 & 25 \\ 2 & 3 & -1 & 25 & 3 \end{pmatrix}_{b_0} \\
[2956] & 21^{13}51^3231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 18 & 2 \\ 2 & 0 & 2 & 18 & 2 \\ 1 & 2 & 0 & 24 & 2 \\ 18 & 18 & 24 & 295 & 33 \\ 2 & 2 & 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[2955] & 31^{11}34141^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 27 & 4 \\ 13 & 0 & 27 & 133 & 21 \\ 2 & 0 & 4 & 21 & 3 \end{pmatrix}_{b_0} \\
[2954] & 2^3 21^{11}351^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 15 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 0 & 16 & 2 \\ 15 & 12 & 16 & 155 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2953] & 1^{13}41^{13}31^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 18 & 0 & 130 & 22 \\ 2 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0} \\
[2952] & 1^{13}41^{13}31^{24}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 22 & 0 & 214 & 28 \\ 2 & 3 & -1 & 28 & 3 \end{pmatrix}_{b_0} \\
[2951] & 21^{13}51^3 3 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 2 & 12 & 2 \\ 2 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 15 & 2 \\ 12 & 24 & 15 & 115 & 21 \\ 2 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2950] & 1^{12}5^{12}131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 10 & 10 & 16 & 102 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2949] & 21^{13}51^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 10 & 10 & 63 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[2948] & 21^{12}5^{12}131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 10 & 10 & 16 & 102 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2947] & 21^{13}5^{13}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 12 & 12 & 15 & 115 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[2957] & 1^{12}41^{22}321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 20 & 3 \\ 2 & 1 & 16 & 3 \\ 20 & 16 & 142 & 28 \\ 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

[2963] $1^{13}5^{12}1231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 9 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$

[2965] $1^{12}41^{13}31^{44}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 27 & 0 & 397 & 37 \\ 2 & 3 & -1 & 37 & 3 \end{pmatrix}_{b_0}$

[2964] $1^{12}5^{13}131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 9 & 12 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[2958] $1^{14}41^{33}141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 27 & 0 & 301 & 33 \\ 2 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$

[2959] $1^{11}41^{43}31^{41}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 27 & 0 & 397 & 37 \\ 2 & 3 & -1 & 37 & 3 \end{pmatrix}_{b_0}$

[2960] $21^{13}5^{12}131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 11 & 11 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$

[2961] $1^{13}41^{43}141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 29 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 29 & 0 & 385 & 37 \\ 2 & 3 & -1 & 37 & 3 \end{pmatrix}_{b_0}$

[2962] $22^13^{12}151^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 3 & 10 & 2 \\ 3 & 3 & 4 & 20 & 4 \\ 9 & 10 & 20 & 68 & 16 \\ 2 & 2 & 4 & 16 & 3 \end{pmatrix}_{b_0}$

[2966] $31^{12}321^{25}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 5 & 22 & 3 \\ 5 & 5 & 30 & 5 \\ 22 & 30 & 154 & 26 \\ 3 & 5 & 26 & 4 \end{pmatrix}_{b_0}$

[2967] $1^{12}5^{14}131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 11 & 9 & 15 & 84 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$

[2968] $1^{12}41^{23}1^{41}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 20 & 0 & 186 & 26 \\ 2 & 3 & -1 & 26 & 3 \end{pmatrix}_{b_0}$

[2971] $1^{12}5^{11}1^{32}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 13 & 14 & 17 & 181 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[2970] & 1^{13}3221^{25}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 19 & 2 \\ 12 & 0 & 19 & 165 & 25 \\ 2 & 0 & 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[2969] & 1^{11}5^{12}231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 12 & 11 & 18 & 141 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0} \\
[2972] & 2^{13}3^{41}21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 1 & 15 & 3 \\ 9 & 9 & 15 & 70 & 18 \\ 2 & 2 & 3 & 18 & 4 \end{pmatrix}_{b_0} \\
[2973] & 1^{11}5^{12}231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 21 & 3 \\ 3 & 5 & 40 & 5 \\ 21 & 40 & 258 & 34 \\ 3 & 5 & 34 & 4 \end{pmatrix}_{b_0} \\
[2974] & 1^{11}5^{13}231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 25 & 3 \\ 3 & 5 & 45 & 5 \\ 25 & 45 & 349 & 39 \\ 3 & 5 & 39 & 4 \end{pmatrix}_{b_0} \\
[2975] & 1^{13}221^{13}51^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 14 & 2 \\ 4 & 18 & 6 & 60 & 8 \\ 1 & 6 & 0 & 15 & 2 \\ 14 & 60 & 15 & 178 & 24 \\ 2 & 8 & 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[2976] & 2^{14}3^{12}151^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 16 & 3 \\ 4 & 5 & 26 & 5 \\ 16 & 26 & 106 & 22 \\ 3 & 5 & 22 & 4 \end{pmatrix}_{b_0} \\
[2977] & 2^{12}3^{21}251^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 20 & 3 \\ 4 & 5 & 30 & 5 \\ 20 & 30 & 154 & 26 \\ 3 & 5 & 26 & 4 \end{pmatrix}_{b_0} \\
[2978] & 1^{14}3221^{15}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -207, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 17 & 2 \\ 10 & 0 & 17 & 106 & 20 \\ 2 & 0 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2982] & 21^{13}5^{13}131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -207, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 9 & 11 & 16 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[2981] & 1^{13}5^{11}131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -207, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 9 & 11 & 16 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [2980] $2^{\overline{14}} 3^{\overline{12}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -207$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 11 & 8 & 15 & 84 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$
- [2979] $1^{\overline{11}} 5^{\overline{12}} 231^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -207$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 21 & 2 \\ 12 & 12 & 21 & 180 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [2987] $1^{\overline{11}} 41^{\overline{3}} 31^{\overline{2}} 41^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -208$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 1 & 1 & 26 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 26 & 0 & 374 & 36 \\ 2 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0}$
- [2986] $1^{\overline{12}} 41^{\overline{4}} 31^{\overline{1}} 41^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -208$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 1 & 1 & 30 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 30 & 0 & 454 & 40 \\ 2 & 3 & -1 & 40 & 3 \end{pmatrix}_{b_0}$
- [2983] $1^{\overline{13}} 5^{\overline{321}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -208$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 22 & 2 \\ 13 & 14 & 22 & 231 & 29 \\ 2 & 2 & 2 & 29 & 3 \end{pmatrix}_{b_0}$
- [2984] $31^{\overline{2}} 4^{\overline{13}} 141^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -208$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 5 & 1 & 39 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 39 & 0 & 229 & 29 \\ 2 & 5 & -1 & 29 & 3 \end{pmatrix}_{b_0}$
- [2985] $2^{\overline{14}} 3^{\overline{13}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -208$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 5 & 29 & 5 \\ 18 & 29 & 137 & 25 \\ 3 & 5 & 25 & 4 \end{pmatrix}_{b_0}$
- [2988] $2^{\overline{11}} 3^{\overline{14}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -209$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 6 & 29 & 6 \\ 15 & 29 & 108 & 24 \\ 3 & 6 & 24 & 5 \end{pmatrix}_{b_0}$
- [2989] $1^{\overline{12}} 41^{\overline{3}} 2321^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -210$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 10.91 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 20 & 3 \\ 2 & 1 & 16 & 3 \\ 20 & 16 & 138 & 28 \\ 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}$
- [2990] $1^{\overline{13}} 5^{\overline{131}} 131^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -211$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.68 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 10 & 10 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [2993] $2^{\overline{13}} 3^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -212$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 9 & 8 & 12 & 57 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$
- [2992] $1^{\overline{12}} 5^{\overline{1}} 2^{\overline{231}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -212$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.68 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 11 & 12 & 16 & 130 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[2991] $1^{11} 5^{12} 1^3 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 14 & 13 & 19 & 190 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[2994] $221^{13} 5131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -213, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 10 & 2 \\ 3 & 4 & 3 & 20 & 4 \\ 1 & 3 & 0 & 9 & 2 \\ 10 & 20 & 9 & 67 & 15 \\ 2 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[2999] $2^3 3^{11} 21^{25} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 12 & 11 & 13 & 101 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[2998] $31^{12} 321^{15} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 15 & 3 \\ 5 & 5 & 20 & 5 \\ 15 & 20 & 63 & 17 \\ 3 & 5 & 17 & 3 \end{pmatrix}_{b_0}$$

[2997] $22^3 3^{13} 151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 15 & 3 \\ 6 & 14 & 39 & 8 \\ 15 & 39 & 97 & 21 \\ 3 & 8 & 21 & 4 \end{pmatrix}_{b_0}$$

[2996] $2^3 3^{13} 1^4 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 12 & 11 & 15 & 117 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[2995] $2^3 3^{14} 1^4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 17 & 2 \\ 13 & 10 & 17 & 128 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[3003] $2^3 3^{14} 21^{17} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 6 & 42 & 6 \\ 21 & 42 & 243 & 35 \\ 3 & 6 & 35 & 5 \end{pmatrix}_{b_0}$$

[3002] $1^{12} 5^{13} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 11 & 10 & 16 & 102 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[3001] $21^{13} 51^1 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 12 & 16 & 115 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[3000] $21^{13} 51^3 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 16 & 2 \\ 2 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 16 & 16 & 18 & 211 & 27 \\ 2 & 2 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[3004] $1^{13} 41^2 2321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 1 & 16 & 3 \\ 18 & 16 & 118 & 26 \\ 3 & 3 & 26 & 5 \end{pmatrix}_{b_0}$$

$$[3009] \ 21^2 5^{13} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 10 & 11 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[3010] \ 1^{11} 3221^2 51^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 13 & 2 \\ 12 & 0 & 13 & 101 & 19 \\ 2 & 0 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[3011] \ 1^{11} 51^2 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 13 & 11 & 18 & 141 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[3012] \ 1^{13} 5^{12} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 10 & 11 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[3005] \ 1^{12} 51^2 3231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 14 & 14 & 17 & 181 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[3006] \ 1^1 5^2 3231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 26 & 3 \\ 3 & 5 & 45 & 5 \\ 26 & 45 & 349 & 39 \\ 3 & 5 & 39 & 4 \end{pmatrix}_{b_0}$$

$$[3007] \ 1^{13} 5^{14} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 10 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[3008] \ 1^1 5^3 3231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 28 & 3 \\ 3 & 5 & 45 & 5 \\ 28 & 45 & 349 & 39 \\ 3 & 5 & 39 & 4 \end{pmatrix}_{b_0}$$

$$[3013] \ 2^3 3^1 5^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 16 & 2 \\ 12 & 9 & 16 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[3014] \ 1^{11} 3221^2 51^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 11 & 2 \\ 4 & 18 & 6 & 45 & 8 \\ 1 & 6 & 0 & 13 & 2 \\ 11 & 45 & 13 & 101 & 19 \\ 2 & 8 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[3015] \ 1^{13} 4131^4 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -223, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 25 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 25 & 0 & 313 & 33 \\ 2 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

[3016] $1^{13}5^{13}1231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -223, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 10 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[3017] $1^{14}3221^3 51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -223, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 27 & 2 \\ 16 & 0 & 27 & 346 & 36 \\ 2 & 0 & 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[3018] $1^{13}5^{13}131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -223, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 10 & 13 & 18 & 148 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[3019] $1^{12}3221^3 51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 21 & 2 \\ 15 & 0 & 21 & 250 & 30 \\ 2 & 0 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

[3020] $1^{13}5^{13}3231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 27 & 3 \\ 3 & 5 & 45 & 5 \\ 27 & 45 & 349 & 39 \\ 3 & 5 & 39 & 4 \end{pmatrix}_{b_0}$$

[3021] $151^{12}2231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 10 & 2 \\ 4 & 18 & 6 & 50 & 8 \\ 1 & 6 & 0 & 16 & 2 \\ 10 & 50 & 16 & 130 & 22 \\ 2 & 8 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[3022] $1^{12}41^4 3141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 26 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 26 & 0 & 330 & 34 \\ 2 & 3 & -1 & 34 & 3 \end{pmatrix}_{b_0}$$

[3023] $1^{12}3221^{12} 51^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 12 & 2 \\ 4 & 18 & 6 & 50 & 8 \\ 1 & 6 & 0 & 16 & 2 \\ 12 & 50 & 16 & 130 & 22 \\ 2 & 8 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[3024] $1^{13}3221^3 51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 24 & 0 & 295 & 33 \\ 2 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

[3025] $1^{11}5^{13}3231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 15 & 13 & 19 & 190 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[3026] $1^{14}41^2 3141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 22 & 0 & 186 & 26 \\ 2 & 3 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

[3027] $31^{14}4^{14} 141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 5 & 1 & 37 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 37 & 0 & 197 & 27 \\ 2 & 5 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[3029] \ 1^{12} 5^1 3^{231} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 15 & 14 & 17 & 181 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[3030] \ 1^{12} 5^1 3^{231} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 16 & 14 & 17 & 181 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[3031] \ 1^{11} 5^1 2^{231} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 14 & 11 & 18 & 141 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[3028] \ 1^{11} 3^{221} 5^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 18 & 2 \\ 15 & 0 & 18 & 211 & 27 \\ 2 & 0 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[3032] \ 2^3 3^{151} 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 16 & 3 \\ 4 & 5 & 25 & 5 \\ 16 & 25 & 97 & 21 \\ 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[3033] \ 1^{13} 5^1 3^{21} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 11 & 13 & 19 & 165 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[3034] \ 1^{12} 5^1 1^{14} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 10 & 16 & 102 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[3035] \ 1^{14} 3^{221} 5^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 22 & 0 & 206 & 28 \\ 2 & 3 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[3036] \ 1^{12} 5^1 1^{231} 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 22 & 3 \\ 3 & 5 & 40 & 5 \\ 22 & 40 & 258 & 34 \\ 3 & 5 & 34 & 4 \end{pmatrix}_{b_0}$$

$$[3037] \ 1^{11} 5^1 3^{231} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 16 & 13 & 19 & 190 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[3038] \ 1^{12} 4^1 2^3 3^1 4^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 1 & 1 & 28 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 28 & 0 & 410 & 38 \\ 2 & 3 & -1 & 38 & 3 \end{pmatrix}_{b_0}$$

[3039] $31^{2^{11}}4^{11}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 25 & 4 \\ 11 & 0 & 25 & 101 & 19 \\ 2 & 0 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

[3040] $22^{1^{13}}3^{13}151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 10 & 2 \\ 1 & 0 & 3 & 8 & 2 \\ 3 & 3 & 4 & 23 & 4 \\ 10 & 8 & 23 & 75 & 17 \\ 2 & 2 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

[3041] $31^{2^{12}}4^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -230, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 5 & 1 & 32 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 32 & 0 & 158 & 24 \\ 2 & 5 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

[3042] $2^{1^{13}}3^{1^2}151^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 10 & 9 & 13 & 73 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[3043] $21^{1^{13}}51^{1^2}231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 11 & 13 & 86 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[3044] $21^{1^{13}}51^{2^2}231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 13 & 13 & 14 & 125 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[3046] $21^{1^{13}}51^{3^2}231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 17 & 2 \\ 2 & 0 & 2 & 17 & 2 \\ 1 & 2 & 0 & 21 & 2 \\ 17 & 17 & 21 & 250 & 30 \\ 2 & 2 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

[3047] $1^{12}41^{3^2}31^{2^2}41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 1 & 1 & 29 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 29 & 0 & 429 & 39 \\ 2 & 3 & -1 & 39 & 3 \end{pmatrix}_{b_0}$$

[3045] $21^{1^{13}}51^{2^2}231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 20 & 2 \\ 15 & 15 & 20 & 193 & 27 \\ 2 & 2 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[3050] $1^{12}51^{1^2}231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 12 & 16 & 130 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[3048] $1^{13}51^{1^3}3^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 61 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[3051] $2 \overset{13}{3} 1 \overset{13}{5} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 11 & 10 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$

[3049] $1^{12} 3 2 2 1^{25} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 16 & 2 \\ 12 & 0 & 16 & 130 & 22 \\ 2 & 0 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[3052] $2 \overset{14}{3} 2 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 1 & 16 & 3 \\ 10 & 11 & 16 & 90 & 20 \\ 2 & 2 & 3 & 20 & 4 \end{pmatrix}_{b_0}$

[3053] $1^{13} \overset{13}{5} 1 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 11 & 11 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$

[3058] $2 \overset{17}{3} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 14 & 2 \\ 2 & 2 & 4 & 28 & 4 \\ 2 & 4 & 1 & 21 & 3 \\ 14 & 28 & 21 & 194 & 28 \\ 2 & 4 & 3 & 28 & 4 \end{pmatrix}_{b_0}$

[3057] $1^{12} \overset{13}{3} 5 1 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 12 & 11 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[3056] $2 \overset{12}{3} 2 1^{15} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 10 & 9 & 12 & 66 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[3055] $2 1^{12} \overset{3}{5} 1 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 12 & 11 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[3054] $1^{12} \overset{1}{5} 1 3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 10 & 8 & 14 & 70 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[3059] $1^{14} 3 2 2 1^{13} 5 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 1 & 19 & 2 \\ 4 & 18 & 6 & 80 & 8 \\ 1 & 6 & 0 & 27 & 2 \\ 19 & 80 & 27 & 346 & 36 \\ 2 & 8 & 2 & 36 & 3 \end{pmatrix}_{b_0}$

[3062] $2 \overset{16}{3} 2 1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 5 & 30 & 5 \\ 18 & 30 & 142 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$

[3063] $2^1 3^{12} 21^3 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 27 & 3 \\ 4 & 5 & 40 & 5 \\ 27 & 40 & 287 & 35 \\ 3 & 5 & 35 & 4 \end{pmatrix}_{b_0}$

[3061] $31^3 4^{13} 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 5 & 1 & 48 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 48 & 0 & 358 & 36 \\ 2 & 5 & -1 & 36 & 3 \end{pmatrix}_{b_0}$

[3060] $31^1 4^{14} 141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 5 & 1 & 28 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 28 & 0 & 110 & 20 \\ 2 & 5 & -1 & 20 & 3 \end{pmatrix}_{b_0}$

[3064] $1^1 5^1 1^3 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 5 & 30 & 5 \\ 18 & 30 & 142 & 24 \\ 3 & 5 & 24 & 4 \end{pmatrix}_{b_0}$

[3065] $31^1 4^{16} 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 5 & 1 & 42 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 42 & 0 & 250 & 30 \\ 2 & 5 & -1 & 30 & 3 \end{pmatrix}_{b_0}$

[3068] $1^{14} 3221^2 51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -239, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 22 & 2 \\ 13 & 0 & 22 & 206 & 28 \\ 2 & 0 & 2 & 28 & 3 \end{pmatrix}_{b_0}$

[3067] $1^{12} 41^1 31^3 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -239, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 23 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 23 & 0 & 273 & 31 \\ 2 & 3 & -1 & 31 & 3 \end{pmatrix}_{b_0}$

[3066] $1^{11} 5^1 231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -239, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 21 & 2 \\ 13 & 12 & 21 & 180 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0}$

[3071] $1^{13} 5^{14} 1231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -239, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 11 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$

[3070] $1^{14} 5^{13} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -239, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 11 & 14 & 19 & 176 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0}$

[3069] $1^{13} 5^{13} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -239, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 10 & 11 & 16 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$

[3073] $1^{14} 41^2 31^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 26 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 26 & 0 & 278 & 32 \\ 2 & 3 & -1 & 32 & 3 \end{pmatrix}_{b_0}$$

[3072] $2^{1^{13}4^1} 3^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 8 & 2 \\ 2 & 2 & 4 & 15 & 4 \\ 2 & 4 & 1 & 13 & 3 \\ 8 & 15 & 13 & 56 & 16 \\ 2 & 4 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

[3074] $1^{1^3} 5^1 2^{231} 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 23 & 3 \\ 3 & 5 & 40 & 5 \\ 23 & 40 & 258 & 34 \\ 3 & 5 & 34 & 4 \end{pmatrix}_{b_0}$$

[3075] $1^{1^3} 5^{1^4} 13^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 14 & 2 \\ 12 & 11 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[3080] $31^{1^3} 4^{1^4} 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 18 & 2 \\ 2 & 5 & 1 & 57 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 57 & 0 & 517 & 43 \\ 2 & 5 & -1 & 43 & 3 \end{pmatrix}_{b_0}$$

[3084] $1^{1^4} 4^{1^3} 13^{1^4} 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 24 & 0 & 250 & 30 \\ 2 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[3085] $1^{1^3} 5^{1^2} 2^{231} 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 13 & 12 & 16 & 130 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[3086] $1^{1^3} 3^{221} 4^{1^5} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 29 & 2 \\ 19 & 0 & 29 & 465 & 41 \\ 2 & 0 & 2 & 41 & 3 \end{pmatrix}_{b_0}$$

[3087] $1^{1^3} 4^{1^3} 3^{1^4} 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 28 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 28 & 0 & 358 & 36 \\ 2 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

[3083] $1^{1^1} 5^{1^3} 2^{231} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 22 & 2 \\ 14 & 14 & 22 & 231 & 29 \\ 2 & 2 & 2 & 29 & 3 \end{pmatrix}_{b_0}$$

[3082] $22^1 3^{1^6} 151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 18 & 2 \\ 1 & 0 & 3 & 12 & 2 \\ 3 & 3 & 4 & 42 & 4 \\ 18 & 12 & 42 & 250 & 30 \\ 2 & 2 & 4 & 30 & 3 \end{pmatrix}_{b_0}$$

[3081] $1^{1^4} 4^{1^3} 2^{241} 41^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 20 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 20 & 0 & 158 & 24 \\ 2 & 3 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

- [3079] $1^{1^4 3 2 1^3} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 26 & 3 \\ 3 & 5 & 50 & 5 \\ 26 & 50 & 419 & 43 \\ 3 & 5 & 43 & 4 \end{pmatrix}_{b_0}$
- [3078] $1^{1^1 3 2 1^3} 5^{1^3} 1^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 13 & 16 & 18 & 211 & 27 \\ 2 & 2 & 2 & 27 & 3 \end{pmatrix}_{b_0}$
- [3077] $1^{1^3 3 2 2 1^5} 5 1^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 14 & 2 \\ 10 & 0 & 14 & 75 & 17 \\ 2 & 0 & 2 & 17 & 3 \end{pmatrix}_{b_0}$
- [3076] $1^{1^3 4 1^3} 3 1 4 1^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 24 & 0 & 250 & 30 \\ 2 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0}$
- [3088] $1^{1^4 4 1^1} 3 1^1 4 1^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -247, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 21 & 0 & 169 & 25 \\ 2 & 3 & -1 & 25 & 3 \end{pmatrix}_{b_0}$
- [3089] $1^{1^1 3 2 2 1^4} 5 1^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -247, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 20 & 2 \\ 19 & 0 & 20 & 318 & 32 \\ 2 & 0 & 2 & 32 & 3 \end{pmatrix}_{b_0}$
- [3090] $1^{1^1 3 2 1^3} 5^1 1^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -247, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 20 & 2 \\ 13 & 15 & 20 & 218 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [3091] $1^{1^3 4 1 3 1^3} 4 1^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -247, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 21 & 0 & 205 & 27 \\ 2 & 3 & -1 & 27 & 3 \end{pmatrix}_{b_0}$
- [3092] $1^{1^1 5^1 2} 2 3 1^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 24 & 3 \\ 3 & 5 & 40 & 5 \\ 24 & 40 & 258 & 34 \\ 3 & 5 & 34 & 4 \end{pmatrix}_{b_0}$
- [3093] $1^{1^2 5^1 2} 2 3 1^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 14 & 12 & 16 & 130 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$
- [3094] $1^{1^2 5^1 3 1^4} 1^{1^3} 1^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 13 & 11 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$
- [3095] $1^{1^3 5^1 3 1^3} 1^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 12 & 12 & 15 & 115 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[3096] $21^3 5^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -251$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 12 & 12 & 15 & 115 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[3100] $1^{12} 41^3 31^{14} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -255$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 25 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 25 & 0 & 305 & 33 \\ 2 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

[3099] $2^3 21^{12} 51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -255$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.74 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 13 & 10 & 15 & 108 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[3097] $221^{13} 5131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -255$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.64 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 11 & 2 \\ 3 & 4 & 3 & 22 & 4 \\ 1 & 3 & 0 & 12 & 2 \\ 11 & 22 & 12 & 88 & 18 \\ 2 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[3098] $1^{14} 41^1 31^{24} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -255$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 25 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 25 & 0 & 261 & 31 \\ 2 & 3 & -1 & 31 & 3 \end{pmatrix}_{b_0}$$

[3101] $21^{13} 51^2 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 14 & 14 & 17 & 156 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[3102] $1^{13} 41^1 31^{34} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 26 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 26 & 0 & 322 & 34 \\ 2 & 3 & -1 & 34 & 3 \end{pmatrix}_{b_0}$$

[3103] $31^4 4141^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 16 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 36 & 4 \\ 16 & 0 & 36 & 238 & 28 \\ 2 & 0 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

[3104] $1^{13} 5^{14} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -259$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 15 & 2 \\ 13 & 12 & 15 & 115 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[3111] $1^{12} 41^2 31^{24} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -260$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 24 & 0 & 286 & 32 \\ 2 & 3 & -1 & 32 & 3 \end{pmatrix}_{b_0}$$

[3105] $31^3 4^{12} 141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -260$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 5 & 1 & 41 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 41 & 0 & 269 & 31 \\ 2 & 5 & -1 & 31 & 3 \end{pmatrix}_{b_0}$$

- [3106] $1^{12} \overset{14}{5} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 14 & 12 & 18 & 150 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [3107] $1^{12} 3221^{12} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 1 & 11 & 2 \\ 4 & 18 & 6 & 50 & 8 \\ 1 & 6 & 0 & 16 & 2 \\ 11 & 50 & 16 & 130 & 22 \\ 2 & 8 & 2 & 22 & 3 \end{pmatrix}_{b_0}$
- [3108] $2^1 3 21^{25} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 13 & 13 & 14 & 125 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$
- [3109] $21^4 \overset{12}{5} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 14 & 12 & 18 & 150 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [3110] $1^{14} 3221^{13} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 1 & 17 & 2 \\ 4 & 18 & 6 & 80 & 8 \\ 1 & 6 & 0 & 27 & 2 \\ 17 & 80 & 27 & 346 & 36 \\ 2 & 8 & 2 & 36 & 3 \end{pmatrix}_{b_0}$
- [3112] $1^{13} 41^2 31^2 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 27 & 0 & 337 & 35 \\ 2 & 3 & -1 & 35 & 3 \end{pmatrix}_{b_0}$
- [3113] $1^{13} 3221^4 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 1 & 1 & 29 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 29 & 0 & 465 & 41 \\ 2 & 3 & -1 & 41 & 3 \end{pmatrix}_{b_0}$
- [3114] $1^{14} 41^2 2321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 1 & 18 & 3 \\ 19 & 18 & 140 & 28 \\ 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}$
- [3115] $1^{11} \overset{13}{5} 1^2 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 21 & 2 \\ 14 & 12 & 21 & 180 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0}$
- [3116] $1^{12} 41^3 2321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 23 & 3 \\ 2 & 1 & 18 & 3 \\ 23 & 18 & 180 & 32 \\ 3 & 3 & 32 & 5 \end{pmatrix}_{b_0}$
- [3117] $1^{13} \overset{13}{5} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 11 & 11 & 16 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [3118] $1^{14} \overset{12}{5} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 22 & 2 \\ 12 & 14 & 22 & 206 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[3119] $1^{12}5^1 1^2 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 12 & 13 & 19 & 165 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[3120] $1^{12}41^2 31^4 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & 1 & 1 & 32 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 32 & 0 & 558 & 44 \\ 2 & 3 & -1 & 44 & 3 \end{pmatrix}_{b_0}$$

[3121] $1^{14}41^4 3141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 32 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 32 & 0 & 446 & 40 \\ 2 & 3 & -1 & 40 & 3 \end{pmatrix}_{b_0}$$

[3122] $31^{12}4^1 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -266, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 17 & 2 \\ 2 & 5 & 1 & 50 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 50 & 0 & 410 & 38 \\ 2 & 5 & -1 & 38 & 3 \end{pmatrix}_{b_0}$$

[3124] $1^{13}41^2 2321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -267, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 21 & 3 \\ 2 & 1 & 18 & 3 \\ 21 & 18 & 160 & 30 \\ 3 & 3 & 30 & 5 \end{pmatrix}_{b_0}$$

[3125] $21^{13}4^1 5^1 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -267, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 14 & 13 & 16 & 141 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[3126] $1^{13}5^1 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -267, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 14 & 13 & 16 & 141 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[3123] $1^{13}41^3 2321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -267, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 21 & 3 \\ 2 & 1 & 18 & 3 \\ 21 & 18 & 160 & 30 \\ 3 & 3 & 30 & 5 \end{pmatrix}_{b_0}$$

[3127] $151^{12} 2231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 9 & 2 \\ 4 & 18 & 6 & 45 & 8 \\ 1 & 6 & 0 & 13 & 2 \\ 9 & 45 & 13 & 101 & 19 \\ 2 & 8 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[3128] $41^{12} 231^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 11 & 2 \\ 4 & 26 & 1 & 72 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 72 & 0 & 186 & 26 \\ 2 & 10 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

[3129] $31^{13} 4^1 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -270, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 5 & 1 & 30 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 30 & 0 & 130 & 22 \\ 2 & 5 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[3130] $31^{11} 2^1 4^1 141^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 25 & 4 \\ 12 & 0 & 25 & 101 & 19 \\ 2 & 0 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[3131] \ 2^{13} 5^{13} 13^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 11 & 12 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[3132] \ 2^{13} 3^{13} 15^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 10 & 10 & 13 & 75 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[3133] \ 1^{13} 5^{12} 13^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 11 & 12 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[3134] \ 1^{13} 4^{12} 31^{14} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 23 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 23 & 0 & 229 & 29 \\ 2 & 3 & -1 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[3135] \ 1^{11} 5^{12} 3^{23} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 22 & 2 \\ 15 & 14 & 22 & 231 & 29 \\ 2 & 2 & 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[3136] \ 1^{13} 5^{11} 3^{23} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 14 & 16 & 18 & 211 & 27 \\ 2 & 2 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[3137] \ 1^{13} 4^{11} 31^{24} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 1 & 1 & 22 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 22 & 0 & 214 & 28 \\ 2 & 3 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[3138] \ 32^{12} 4^{14} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 8 & 14 & 3 \\ 8 & 17 & 32 & 9 \\ 14 & 32 & 54 & 16 \\ 3 & 9 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[3139] \ 1^{12} 4^{11} 31^{44} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 27 & 0 & 397 & 37 \\ 2 & 3 & -1 & 37 & 3 \end{pmatrix}_{b_0}$$

$$[3142] \ 4^{13} 2^{31} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 12 & 2 \\ 4 & 26 & 1 & 84 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 84 & 0 & 250 & 30 \\ 2 & 10 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[3143] \ 1^{12} 4^{14} 31^{24} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & 1 & 1 & 34 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 34 & 0 & 602 & 46 \\ 2 & 3 & -1 & 46 & 3 \end{pmatrix}_{b_0}$$

[3141] $1^1 3221^{13} 51^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 13 & 2 \\ 4 & 18 & 6 & 60 & 8 \\ 1 & 6 & 0 & 15 & 2 \\ 13 & 60 & 15 & 178 & 24 \\ 2 & 8 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[3140] $151^{13} 2231^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 12 & 2 \\ 4 & 18 & 6 & 60 & 8 \\ 1 & 6 & 0 & 15 & 2 \\ 12 & 60 & 15 & 178 & 24 \\ 2 & 8 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[3146] $1^{12} 51^1 3231^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -279, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 20 & 2 \\ 14 & 15 & 20 & 218 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[3145] $1^{11} 51^{14} 2231^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -279, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 21 & 2 \\ 15 & 12 & 21 & 180 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[3144] $1^1 51^1 3231^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -279, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 27 & 3 \\ 3 & 5 & 50 & 5 \\ 27 & 50 & 419 & 43 \\ 3 & 5 & 43 & 4 \end{pmatrix}_{b_0}$$

[3147] $1^{13} 5^{14} 131^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -279, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 11 & 16 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[3149] $2^1 3^1 151^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 2 & 2 & 0 & 16 & 2 \\ 13 & 13 & 16 & 141 & 23 \\ 2 & 2 & 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[3148] $31^{14} 32151^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 16 & 3 \\ 5 & 5 & 20 & 5 \\ 16 & 20 & 62 & 16 \\ 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[3150] $1^{11} 3221^{12} 51^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -281, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 10 & 2 \\ 4 & 18 & 6 & 45 & 8 \\ 1 & 6 & 0 & 13 & 2 \\ 10 & 45 & 13 & 101 & 19 \\ 2 & 8 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[3152] $2^1 3^1 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 1 & 19 & 3 \\ 11 & 10 & 19 & 104 & 22 \\ 2 & 2 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[3151] $1^{12} 41^3 31^3 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & 1 & 1 & 33 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 33 & 0 & 577 & 45 \\ 2 & 3 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

[3153] $1^{14} 3221^{13} 51^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 4 & 1 & 18 & 2 \\ 4 & 18 & 6 & 80 & 8 \\ 1 & 6 & 0 & 27 & 2 \\ 18 & 80 & 27 & 346 & 36 \\ 2 & 8 & 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[3154] & 1^{13}3221^{13}51^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -285, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 19 & 2 \\ 4 & 18 & 6 & 75 & 8 \\ 1 & 6 & 0 & 24 & 2 \\ 19 & 75 & 24 & 295 & 33 \\ 2 & 8 & 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[3158] & 1^{12}3221^451^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 26 & 2 \\ 19 & 0 & 26 & 410 & 38 \\ 2 & 0 & 2 & 38 & 3 \end{pmatrix}_{b_0} \\
& 1^{14} \quad 1^1 \\
[3156] & 2^3 1^5 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 11 & 9 & 15 & 84 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
& 1^{11} \\
[3155] & 2^3 21^351^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 16 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 2 & 2 & 0 & 17 & 2 \\ 16 & 14 & 17 & 181 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0} \\
& 1^{12} \\
[3157] & 2^3 21^{15}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 11 & 11 & 13 & 86 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
& 1^{11} \\
[3162] & 1^{11}51^3231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 25 & 2 \\ 15 & 15 & 25 & 278 & 32 \\ 2 & 2 & 2 & 32 & 3 \end{pmatrix}_{b_0} \\
[3161] & 1^{14}3221^{15}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 17 & 2 \\ 11 & 0 & 17 & 106 & 20 \\ 2 & 0 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
& 1^{11} \\
[3159] & 31^3 4 141^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 34 & 4 \\ 14 & 0 & 34 & 194 & 26 \\ 2 & 0 & 4 & 26 & 3 \end{pmatrix}_{b_0} \\
[3160] & 1^{11}3221^451^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 23 & 2 \\ 19 & 0 & 23 & 361 & 35 \\ 2 & 0 & 2 & 35 & 3 \end{pmatrix}_{b_0} \\
& 1^{13} \quad 1^2 \\
[3163] & 2^3 1^5 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 11 & 11 & 14 & 93 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[3164] & 1^{13}3221^{25}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 19 & 2 \\ 13 & 0 & 19 & 165 & 25 \\ 2 & 0 & 2 & 25 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [3165] $1^{11}51^3231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 22 & 2 \\ 16 & 14 & 22 & 231 & 29 \\ 2 & 2 & 2 & 29 & 3 \end{pmatrix}_{b_0}$$
- [3166] $1^{12}51^2231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 13 & 13 & 19 & 165 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$
- [3167] $1^{13}3221^351^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 24 & 2 \\ 16 & 0 & 24 & 295 & 33 \\ 2 & 0 & 2 & 33 & 3 \end{pmatrix}_{b_0}$$
- [3168] $1^{13}51^3231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 15 & 16 & 18 & 211 & 27 \\ 2 & 2 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$
- [3169] $2^33151^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 12 & 12 & 15 & 115 & 21 \\ 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$
- [3170] $1^{13}51^3131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 12 & 12 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$
- [3171] $1^{12}3221^{13}51^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 18 & 2 \\ 4 & 18 & 6 & 70 & 8 \\ 1 & 6 & 0 & 21 & 2 \\ 18 & 70 & 21 & 250 & 30 \\ 2 & 8 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$
- [3172] $31^{14}3151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 24 & 3 \\ 5 & 5 & 32 & 5 \\ 24 & 32 & 174 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$$
- [3173] $1^{13}41^431^{14}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 1 & 1 & 33 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 33 & 0 & 517 & 43 \\ 2 & 3 & -1 & 43 & 3 \end{pmatrix}_{b_0}$$
- [3174] $1^{13}4131^441^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 25 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 25 & 0 & 313 & 33 \\ 2 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$
- [3175] $1^{13}41^42321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 24 & 3 \\ 2 & 1 & 20 & 3 \\ 24 & 20 & 214 & 34 \\ 3 & 3 & 34 & 5 \end{pmatrix}_{b_0}$$
- [3177] $1^{12}51^3231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -303, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 20 & 2 \\ 15 & 15 & 20 & 218 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[3176] & 1^{12}3221^251^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -303, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 16 & 2 \\ 13 & 0 & 16 & 130 & 22 \\ 2 & 0 & 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[3181] & 1^{11}3221^351^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 18 & 2 \\ 16 & 0 & 18 & 211 & 27 \\ 2 & 0 & 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[3179] & 1^{11}51^43231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 22 & 2 \\ 17 & 14 & 22 & 231 & 29 \\ 2 & 2 & 2 & 29 & 3 \end{pmatrix}_{b_0} \\
[3180] & 1^{12}51^3231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 28 & 3 \\ 3 & 5 & 50 & 5 \\ 28 & 50 & 419 & 43 \\ 3 & 5 & 43 & 4 \end{pmatrix}_{b_0} \\
[3178] & 1^{13}51^3231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 16 & 16 & 18 & 211 & 27 \\ 2 & 2 & 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[3182] & 1^{13}41^43141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 29 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 29 & 0 & 385 & 37 \\ 2 & 3 & -1 & 37 & 3 \end{pmatrix}_{b_0} \\
[3183] & 1^{13}321^351^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 23 & 2 \\ 14 & 16 & 23 & 261 & 31 \\ 2 & 2 & 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[3184] & 1^{14}41^33141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 27 & 0 & 301 & 33 \\ 2 & 3 & -1 & 33 & 3 \end{pmatrix}_{b_0} \\
[3188] & 1^{12}51^2231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 22 & 2 \\ 13 & 14 & 22 & 206 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[3185] & 1^{13}41^13141^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 1 & 1 & 30 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 30 & 0 & 454 & 40 \\ 2 & 3 & -1 & 40 & 3 \end{pmatrix}_{b_0} \\
[3186] & 1^{13}51^43231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 17 & 16 & 18 & 211 & 27 \\ 2 & 2 & 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[3187] & 1^{12}41^43141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0
\end{aligned}$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 1 & 1 & 30 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 30 & 0 & 454 & 40 \\ 2 & 3 & -1 & 40 & 3 \end{pmatrix}_{b_0}$$

$$[3189] \ 31^{14} 4^{14} 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -310, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 5 & 1 & 46 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 46 & 0 & 314 & 34 \\ 2 & 5 & -1 & 34 & 3 \end{pmatrix}_{b_0}$$

$$[3190] \ 1^{13} 5^{14} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -311, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 17 & 2 \\ 13 & 12 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[3191] \ 1^{12} 41^{23} 31^4 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -312, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 1 & 1 & 28 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 28 & 0 & 410 & 38 \\ 2 & 3 & -1 & 38 & 3 \end{pmatrix}_{b_0}$$

$$[3192] \ 1^{14} 4131^4 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -312, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 28 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 28 & 0 & 366 & 36 \\ 2 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[3193] \ 1^{14} 3221^4 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -312, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 32 & 2 \\ 20 & 0 & 32 & 526 & 44 \\ 2 & 0 & 2 & 44 & 3 \end{pmatrix}_{b_0}$$

$$[3194] \ 1^{12} 5^{13} 1^{23} 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -312, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 14 & 13 & 19 & 165 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[3195] \ 1^{12} 41^4 2321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 26 & 3 \\ 2 & 1 & 20 & 3 \\ 26 & 20 & 230 & 36 \\ 3 & 3 & 36 & 5 \end{pmatrix}_{b_0}$$

$$[3196] \ 1^{14} 41^3 31^4 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 31 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 31 & 0 & 417 & 39 \\ 2 & 3 & -1 & 39 & 3 \end{pmatrix}_{b_0}$$

$$[3198] \ 31^1 4^{17} 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.3$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 5 & 1 & 49 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 49 & 0 & 341 & 35 \\ 2 & 5 & -1 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[3199] \ 1^1 5^1 3^1 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.3$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 5 & 35 & 5 \\ 21 & 35 & 194 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[3197] \ 2^1 3^1 2151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.3$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 5 & 35 & 5 \\ 21 & 35 & 194 & 28 \\ 3 & 5 & 28 & 4 \end{pmatrix}_{b_0}$$

- [3200] $1^{14}41^32321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -318, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 22 & 3 \\ 2 & 1 & 20 & 3 \\ 22 & 20 & 186 & 32 \\ 3 & 3 & 32 & 5 \end{pmatrix}_{b_0}$
- [3201] $1^{13}51^3231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 29 & 3 \\ 3 & 5 & 50 & 5 \\ 29 & 50 & 419 & 43 \\ 3 & 5 & 43 & 4 \end{pmatrix}_{b_0}$
- [3202] $21^35131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 13 & 13 & 18 & 148 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [3203] $1^{12}3221^351^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 21 & 2 \\ 16 & 0 & 21 & 250 & 30 \\ 2 & 0 & 2 & 30 & 3 \end{pmatrix}_{b_0}$
- [3204] $1^{13}5131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 13 & 13 & 18 & 148 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$
- [3205] $1^{12}51^3231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 20 & 2 \\ 16 & 15 & 20 & 218 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0}$
- [3206] $1^{14}3221^351^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 27 & 0 & 346 & 36 \\ 2 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0}$
- [3207] $2^33151^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 16 & 2 \\ 12 & 10 & 16 & 102 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$
- [3208] $22^33151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.3$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 21 & 2 \\ 1 & 0 & 3 & 14 & 2 \\ 3 & 3 & 4 & 49 & 4 \\ 21 & 14 & 49 & 341 & 35 \\ 2 & 2 & 4 & 35 & 3 \end{pmatrix}_{b_0}$
- [3209] $1^{12}41^331^241^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -323, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 1 & 1 & 29 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 29 & 0 & 429 & 39 \\ 2 & 3 & -1 & 39 & 3 \end{pmatrix}_{b_0}$
- [3210] $31^44141^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 17 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 43 & 4 \\ 17 & 0 & 43 & 317 & 33 \\ 2 & 0 & 4 & 33 & 3 \end{pmatrix}_{b_0}$
- [3214] $2^{14}3^{11}2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 10 & 2 \\ 2 & 2 & 4 & 19 & 4 \\ 2 & 4 & 1 & 16 & 3 \\ 10 & 19 & 16 & 90 & 20 \\ 2 & 4 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

[3211] $1^{12}5^{14}1^{23}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 15 & 13 & 19 & 165 & 25 \\ 2 & 2 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[3212] $1^{14}5^{13}231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 30 & 3 \\ 3 & 5 & 50 & 5 \\ 30 & 50 & 419 & 43 \\ 3 & 5 & 43 & 4 \end{pmatrix}_{b_0}$$

[3213] $41^{12}231^{14}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 12 & 2 \\ 4 & 26 & 1 & 72 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 72 & 0 & 186 & 26 \\ 2 & 10 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

[3216] $1^{11}5^{12}1^{32}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 25 & 2 \\ 16 & 15 & 25 & 278 & 32 \\ 2 & 2 & 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[3215] $231^{12}3151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 5 & 1 & 10 & 2 \\ 5 & 33 & 1 & 70 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 70 & 0 & 138 & 22 \\ 2 & 11 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[3217] $2^3 3 21^{15}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -326, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 19 & 3 \\ 4 & 5 & 30 & 5 \\ 19 & 30 & 139 & 25 \\ 3 & 5 & 25 & 4 \end{pmatrix}_{b_0}$$

[3221] $22^3 3 151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -327, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 13 & 2 \\ 1 & 0 & 3 & 10 & 2 \\ 3 & 3 & 4 & 30 & 4 \\ 13 & 10 & 30 & 128 & 22 \\ 2 & 2 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

[3220] $1^{12}5^{14}1^{32}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -327, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 20 & 2 \\ 17 & 15 & 20 & 218 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[3219] $1^{13}5^{13}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -327, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 11 & 10 & 15 & 88 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[3218] $1^{12}41^{33}1^{44}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -327, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 23 & 2 \\ 1 & 1 & 1 & 37 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 37 & 0 & 749 & 51 \\ 2 & 3 & -1 & 51 & 3 \end{pmatrix}_{b_0}$$

[3224] $1^{14}4131^{34}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -328, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 24 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 24 & 0 & 250 & 30 \\ 2 & 3 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[3223] $1^{13}41^{33}1^{24}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -328, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 1 & 1 & 32 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 32 & 0 & 490 & 42 \\ 2 & 3 & -1 & 42 & 3 \end{pmatrix}_{b_0}$$

$$[3222] \ 1^{1^2 3 2 1^3} 5^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -328, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 1 & 2 & 0 & 21 & 2 \\ 14 & 17 & 21 & 250 & 30 \\ 2 & 2 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[3225] \ 1^{1^4 4 1^2 2 3 2 1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 22 & 3 \\ 2 & 1 & 20 & 3 \\ 22 & 20 & 182 & 32 \\ 3 & 3 & 32 & 5 \end{pmatrix}_{b_0}$$

$$[3226] \ 1^{1^3 4 1^2 3 1^3 4 1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 1 & 1 & 31 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 31 & 0 & 469 & 41 \\ 2 & 3 & -1 & 41 & 3 \end{pmatrix}_{b_0}$$

$$[3227] \ 2^{1^4} 3^{1^4} 15^{1^2} 1^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -332, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 14 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 0 & 18 & 2 \\ 14 & 12 & 18 & 150 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[3228] \ 22^1 3^{1^3} 151^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 11 & 2 \\ 1 & 0 & 3 & 10 & 2 \\ 3 & 3 & 4 & 25 & 4 \\ 11 & 10 & 25 & 93 & 19 \\ 2 & 2 & 4 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[3231] \ 1^{1^3} 5^{1^4} 131^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -335, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 18 & 2 \\ 14 & 13 & 18 & 148 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[3230] \ 2^1 3^{1^2} 21^{3^5} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -335, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 17 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 2 & 2 & 0 & 19 & 2 \\ 17 & 13 & 19 & 190 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[3229] \ 2^{1^4} 3^{1^3} 15^{1^2} 1^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -335, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 17 & 2 \\ 13 & 11 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[3232] \ 1^{1^2} 41^{4^3} 31^{3^4} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 23 & 2 \\ 1 & 1 & 1 & 38 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 38 & 0 & 774 & 52 \\ 2 & 3 & -1 & 52 & 3 \end{pmatrix}_{b_0}$$

$$[3233] \ 1^{1^3} 41^{3^2} 2321^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 24 & 3 \\ 2 & 1 & 20 & 3 \\ 24 & 20 & 202 & 34 \\ 3 & 3 & 34 & 5 \end{pmatrix}_{b_0}$$

$$[3234] \ 1^{1^4} 41^{1^3} 31^{3^4} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 29 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 29 & 0 & 377 & 37 \\ 2 & 3 & -1 & 37 & 3 \end{pmatrix}_{b_0}$$

[3236] $1^{13}41^131^341^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -340, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 26 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 26 & 0 & 322 & 34 \\ 2 & 3 & -1 & 34 & 3 \end{pmatrix}_{b_0}$$

[3235] $1^{14}41^231^241^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -340, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 30 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 30 & 0 & 394 & 38 \\ 2 & 3 & -1 & 38 & 3 \end{pmatrix}_{b_0}$$

[3237] $1^{14}41^231^141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -340, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 26 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 26 & 0 & 278 & 32 \\ 2 & 3 & -1 & 32 & 3 \end{pmatrix}_{b_0}$$

[3238] $1^{11}3221^{13}51^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -341, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 16 & 2 \\ 4 & 18 & 6 & 65 & 8 \\ 1 & 6 & 0 & 18 & 2 \\ 16 & 65 & 18 & 211 & 27 \\ 2 & 8 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[3239] $1^{13}41^331^141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 28 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 28 & 0 & 358 & 36 \\ 2 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

[3240] $1^{14}3221^251^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 22 & 2 \\ 14 & 0 & 22 & 206 & 28 \\ 2 & 0 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[3241] $1^{12}51^2231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 22 & 2 \\ 14 & 14 & 22 & 206 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[3243] $1^{12}51^3231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -347, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 23 & 2 \\ 15 & 16 & 23 & 261 & 31 \\ 2 & 2 & 2 & 31 & 3 \end{pmatrix}_{b_0}$$

[3242] $1^{14}41^131^241^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -347, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 1 & 1 & 25 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 25 & 0 & 261 & 31 \\ 2 & 3 & -1 & 31 & 3 \end{pmatrix}_{b_0}$$

[3245] $2^1321^251^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -348, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 23 & 3 \\ 4 & 5 & 35 & 5 \\ 23 & 35 & 202 & 30 \\ 3 & 5 & 30 & 4 \end{pmatrix}_{b_0}$$

[3244] $22^13151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -348, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 20 & 3 \\ 6 & 14 & 52 & 8 \\ 20 & 52 & 174 & 28 \\ 3 & 8 & 28 & 4 \end{pmatrix}_{b_0}$$

[3246] $2^{14}321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -350, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 1 & 20 & 3 \\ 12 & 12 & 20 & 126 & 24 \\ 2 & 2 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[3248] $21^4 5^{13} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -351, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 15 & 14 & 19 & 176 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[3247] $1^{13} 5^{14} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -351, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 19 & 2 \\ 15 & 14 & 19 & 176 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[3249] $1^{11} 5^{13} 1^3 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 25 & 2 \\ 17 & 15 & 25 & 278 & 32 \\ 2 & 2 & 2 & 32 & 3 \end{pmatrix}_{b_0}$$

[3250] $1^{14} 3221^3 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 27 & 2 \\ 17 & 0 & 27 & 346 & 36 \\ 2 & 0 & 2 & 36 & 3 \end{pmatrix}_{b_0}$$

[3251] $2^3 3^{12} 21^{25} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 14 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 0 & 16 & 2 \\ 14 & 12 & 16 & 130 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[3252] $31^2 4^{13} 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 5 & 1 & 39 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 39 & 0 & 229 & 29 \\ 2 & 5 & -1 & 29 & 3 \end{pmatrix}_{b_0}$$

[3253] $1^{13} 41^2 31^2 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & 1 & 27 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 27 & 0 & 337 & 35 \\ 2 & 3 & -1 & 35 & 3 \end{pmatrix}_{b_0}$$

[3254] $1^{12} 41^2 31^4 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 21 & 2 \\ 1 & 1 & 1 & 32 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 32 & 0 & 558 & 44 \\ 2 & 3 & -1 & 44 & 3 \end{pmatrix}_{b_0}$$

[3255] $1^{14} 321^3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -359, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 1 & 2 & 0 & 26 & 2 \\ 15 & 17 & 26 & 310 & 34 \\ 2 & 2 & 2 & 34 & 3 \end{pmatrix}_{b_0}$$

[3256] $2^3 3^{11} 21^3 51^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -360, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 17 & 2 \\ 1 & 0 & 2 & 16 & 2 \\ 2 & 2 & 0 & 18 & 2 \\ 17 & 16 & 18 & 211 & 27 \\ 2 & 2 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[3257] $1^{13} 5^{11} 1^3 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 1 & 2 & 0 & 21 & 2 \\ 15 & 17 & 21 & 250 & 30 \\ 2 & 2 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

[3258] $31^3 4^{11} 141^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -367, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 15 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 34 & 4 \\ 15 & 0 & 34 & 194 & 26 \\ 2 & 0 & 4 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[3259] \ 31^3 4^{14} 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -370, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 17 & 2 \\ 2 & 5 & 1 & 55 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 55 & 0 & 461 & 41 \\ 2 & 5 & -1 & 41 & 3 \end{pmatrix}_{b_0}$$

$$[3260] \ 1^{12} 5^3 1^{23} 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 22 & 2 \\ 15 & 14 & 22 & 206 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[3261] \ 1^{11} 5^{14} 3^{23} 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 1 & 2 & 0 & 25 & 2 \\ 18 & 15 & 25 & 278 & 32 \\ 2 & 2 & 2 & 32 & 3 \end{pmatrix}_{b_0}$$

$$[3262] \ 31^1 4^{14} 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 5 & 1 & 37 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 37 & 0 & 197 & 27 \\ 2 & 5 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[3263] \ 1^{13} 3221^4 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -379, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 29 & 2 \\ 20 & 0 & 29 & 465 & 41 \\ 2 & 0 & 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$$[3264] \ 1^{13} 3221^2 51^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -379, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 19 & 2 \\ 14 & 0 & 19 & 165 & 25 \\ 2 & 0 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[3265] \ 1^{12} 5^3 1^{32} 31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -379, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 23 & 2 \\ 16 & 16 & 23 & 261 & 31 \\ 2 & 2 & 2 & 31 & 3 \end{pmatrix}_{b_0}$$

$$[3266] \ 2^3 3^{14} 15^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 16 & 2 \\ 12 & 11 & 16 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[3267] \ 31^{13} 321^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 19 & 3 \\ 5 & 5 & 25 & 5 \\ 19 & 25 & 97 & 21 \\ 3 & 5 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[3268] \ 1^{12} 41^3 31^3 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -383, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 21 & 2 \\ 1 & 1 & 1 & 33 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 33 & 0 & 577 & 45 \\ 2 & 3 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[3269] \ 1^{12} 41^4 31^2 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 21 & 2 \\ 1 & 1 & 1 & 34 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 34 & 0 & 602 & 46 \\ 2 & 3 & -1 & 46 & 3 \end{pmatrix}_{b_0}$$

[3270] $1^{11}3221^451^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -387$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.66 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 23 & 2 \\ 20 & 0 & 23 & 361 & 35 \\ 2 & 0 & 2 & 35 & 3 \end{pmatrix}_{b_0}$

[3271] $1^{14}41^43141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -388$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 1 & 1 & 32 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 32 & 0 & 446 & 40 \\ 2 & 3 & -1 & 40 & 3 \end{pmatrix}_{b_0}$

[3272] $1^{14}41^431^141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -388$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 1 & 1 & 36 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 36 & 0 & 586 & 46 \\ 2 & 3 & -1 & 46 & 3 \end{pmatrix}_{b_0}$

[3273] $1^{13}41^231^441^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -391$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 21 & 2 \\ 1 & 1 & 1 & 35 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 35 & 0 & 625 & 47 \\ 2 & 3 & -1 & 47 & 3 \end{pmatrix}_{b_0}$

[3274] $1^{13}41^431^241^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -391$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 21 & 2 \\ 1 & 1 & 1 & 37 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 37 & 0 & 673 & 49 \\ 2 & 3 & -1 & 49 & 3 \end{pmatrix}_{b_0}$

[3276] $1^{13}5^13^231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -392$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 1 & 2 & 0 & 21 & 2 \\ 16 & 17 & 21 & 250 & 30 \\ 2 & 2 & 2 & 30 & 3 \end{pmatrix}_{b_0}$

[3277] $1^{14}3221^451^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -392$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & 1 & 1 & 32 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 32 & 0 & 526 & 44 \\ 2 & 3 & -1 & 44 & 3 \end{pmatrix}_{b_0}$

[3275] $1^{12}5^12^231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -392$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 1 & 2 & 0 & 22 & 2 \\ 16 & 14 & 22 & 206 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0}$

[3278] $1^{13}41^42321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -395$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 27 & 3 \\ 2 & 1 & 22 & 3 \\ 27 & 22 & 256 & 38 \\ 3 & 3 & 38 & 5 \end{pmatrix}_{b_0}$

[3279] $22^13^151^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -397$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 3 & 12 & 2 \\ 1 & 0 & 3 & 12 & 2 \\ 3 & 3 & 4 & 27 & 4 \\ 12 & 12 & 27 & 115 & 21 \\ 2 & 2 & 4 & 21 & 3 \end{pmatrix}_{b_0}$

[3281] $31^{13}321^351^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -400$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 5 & 33 & 3 \\ 5 & 5 & 45 & 5 \\ 33 & 45 & 349 & 39 \\ 3 & 5 & 39 & 4 \end{pmatrix}_{b_0}$

[3280] $1^{13}41^131^441^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -400$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 1 & 1 & 30 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 30 & 0 & 454 & 40 \\ 2 & 3 & -1 & 40 & 3 \end{pmatrix}_{b_0}$

[3283] $1^{12}5^13^{231}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -403, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 23 & 2 \\ 17 & 16 & 23 & 261 & 31 \\ 2 & 2 & 2 & 31 & 3 \end{pmatrix}_{b_0}$

[3282] $1^{13}3221^351^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -403, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 24 & 2 \\ 17 & 0 & 24 & 295 & 33 \\ 2 & 0 & 2 & 33 & 3 \end{pmatrix}_{b_0}$

[3284] $1^{13}321^351^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 18 & 2 \\ 1 & 2 & 0 & 24 & 2 \\ 15 & 18 & 24 & 295 & 33 \\ 2 & 2 & 2 & 33 & 3 \end{pmatrix}_{b_0}$

[3285] $1^{13}41^331^341^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 21 & 2 \\ 1 & 1 & 1 & 36 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 36 & 0 & 646 & 48 \\ 2 & 3 & -1 & 48 & 3 \end{pmatrix}_{b_0}$

[3286] $1^{12}3221^451^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 26 & 2 \\ 20 & 0 & 26 & 410 & 38 \\ 2 & 0 & 2 & 38 & 3 \end{pmatrix}_{b_0}$

[3287] $1^{14}4131^441^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 1 & 1 & 28 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 28 & 0 & 366 & 36 \\ 2 & 3 & -1 & 36 & 3 \end{pmatrix}_{b_0}$

[3288] $31^44^{14}141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -406, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 20 & 2 \\ 2 & 5 & 1 & 64 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 64 & 0 & 638 & 48 \\ 2 & 5 & -1 & 48 & 3 \end{pmatrix}_{b_0}$

[3289] $1^{14}41^32321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -407, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 25 & 3 \\ 2 & 1 & 22 & 3 \\ 25 & 22 & 228 & 36 \\ 3 & 3 & 36 & 5 \end{pmatrix}_{b_0}$

[3290] $1^{12}5^13^{231}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -407, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 1 & 2 & 0 & 26 & 2 \\ 16 & 17 & 26 & 310 & 34 \\ 2 & 2 & 2 & 34 & 3 \end{pmatrix}_{b_0}$

[3291] $1^{12}3221^351^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 21 & 2 \\ 17 & 0 & 21 & 250 & 30 \\ 2 & 0 & 2 & 30 & 3 \end{pmatrix}_{b_0}$

[3292] $1^{13}5^13^{231}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 1 & 2 & 0 & 21 & 2 \\ 17 & 17 & 21 & 250 & 30 \\ 2 & 2 & 2 & 30 & 3 \end{pmatrix}_{b_0}$

[3293] $31^34^{13}141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -414, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 16 & 2 \\ 2 & 5 & 1 & 48 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 48 & 0 & 358 & 36 \\ 2 & 5 & -1 & 36 & 3 \end{pmatrix}_{b_0} \\
[3294] & 1^{14} 41^1 31^4 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -415, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 1 & 1 & 33 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 33 & 0 & 517 & 43 \\ 2 & 3 & -1 & 43 & 3 \end{pmatrix}_{b_0} \\
[3295] & 31^{15} 32151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -415, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.1 \\
& T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 20 & 3 \\ 5 & 5 & 25 & 5 \\ 20 & 25 & 98 & 20 \\ 3 & 5 & 20 & 3 \end{pmatrix}_{b_0} \\
[3296] & 2^{14} 3^{12} 151^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -415, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 12 & 2 \\ 2 & 2 & 0 & 17 & 2 \\ 13 & 12 & 17 & 124 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[3297] & 1^{13} 41^4 31^1 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -415, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 1 & 1 & 33 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 33 & 0 & 517 & 43 \\ 2 & 3 & -1 & 43 & 3 \end{pmatrix}_{b_0} \\
[3298] & 321^{12} 4141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -419, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 8 & 17 & 3 \\ 8 & 17 & 40 & 9 \\ 17 & 40 & 84 & 20 \\ 3 & 9 & 20 & 4 \end{pmatrix}_{b_0} \\
[3299] & 1^{12} 5^{14} 1^3 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -419, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 2 & 16 & 2 \\ 1 & 2 & 0 & 23 & 2 \\ 18 & 16 & 23 & 261 & 31 \\ 2 & 2 & 2 & 31 & 3 \end{pmatrix}_{b_0} \\
[3300] & 1^{14} 41^3 31^2 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -423, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 1 & 1 & 35 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 35 & 0 & 557 & 45 \\ 2 & 3 & -1 & 45 & 3 \end{pmatrix}_{b_0} \\
[3301] & 1^{13} 5^{14} 1^3 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 1 & 2 & 0 & 21 & 2 \\ 18 & 17 & 21 & 250 & 30 \\ 2 & 2 & 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[3303] & 1^{13} 3221^{13} 51^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -429, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 18 & 2 \\ 4 & 18 & 6 & 75 & 8 \\ 1 & 6 & 0 & 24 & 2 \\ 18 & 75 & 24 & 295 & 33 \\ 2 & 8 & 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[3302] & 151^{13} 2231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -429, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 15 & 2 \\ 4 & 18 & 6 & 75 & 8 \\ 1 & 6 & 0 & 24 & 2 \\ 15 & 75 & 24 & 295 & 33 \\ 2 & 8 & 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[3307] & 2^{12} 3^{21} 51^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -431, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 15 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 2 & 2 & 0 & 17 & 2 \\ 15 & 14 & 17 & 156 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[3306] & 1^{13} 41^2 31^3 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -431, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 1 & 1 & 31 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 31 & 0 & 469 & 41 \\ 2 & 3 & -1 & 41 & 3 \end{pmatrix}_{b_0}$$

[3304] $2^1 3^{14} 15^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -431, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 15 & 2 \\ 1 & 0 & 2 & 14 & 2 \\ 2 & 2 & 0 & 19 & 2 \\ 15 & 14 & 19 & 176 & 26 \\ 2 & 2 & 2 & 26 & 3 \end{pmatrix}_{b_0}$$

[3305] $1^{14} 41^3 31^1 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -431, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 1 & 1 & 31 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 31 & 0 & 417 & 39 \\ 2 & 3 & -1 & 39 & 3 \end{pmatrix}_{b_0}$$

[3309] $2^1 3^{14} 15^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 14 & 2 \\ 1 & 0 & 2 & 13 & 2 \\ 2 & 2 & 0 & 18 & 2 \\ 14 & 13 & 18 & 148 & 24 \\ 2 & 2 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[3308] $1^{14} 41^2 31^3 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 1 & 1 & 34 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 34 & 0 & 534 & 44 \\ 2 & 3 & -1 & 44 & 3 \end{pmatrix}_{b_0}$$

[3310] $1^{12} 41^3 31^4 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 24 & 2 \\ 1 & 1 & 1 & 37 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 37 & 0 & 749 & 51 \\ 2 & 3 & -1 & 51 & 3 \end{pmatrix}_{b_0}$$

[3311] $2^1 3^{13} 21^3 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 30 & 3 \\ 4 & 5 & 45 & 5 \\ 30 & 45 & 349 & 39 \\ 3 & 5 & 39 & 4 \end{pmatrix}_{b_0}$$

[3312] $1^{13} 41^3 31^2 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 1 & 1 & 32 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 32 & 0 & 490 & 42 \\ 2 & 3 & -1 & 42 & 3 \end{pmatrix}_{b_0}$$

[3314] $1^{14} 41^1 31^3 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -439, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 1 & 1 & 29 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 29 & 0 & 377 & 37 \\ 2 & 3 & -1 & 37 & 3 \end{pmatrix}_{b_0}$$

[3313] $31^4 4^{11} 141^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -439, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 18 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 43 & 4 \\ 18 & 0 & 43 & 317 & 33 \\ 2 & 0 & 4 & 33 & 3 \end{pmatrix}_{b_0}$$

[3315] $41^{14} 23141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 16 & 2 \\ 4 & 26 & 1 & 112 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 112 & 0 & 446 & 40 \\ 2 & 10 & -1 & 40 & 3 \end{pmatrix}_{b_0}$$

[3316] $1^{12} 5^1 3^{23} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -447, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 1 & 2 & 0 & 26 & 2 \\ 17 & 17 & 26 & 310 & 34 \\ 2 & 2 & 2 & 34 & 3 \end{pmatrix}_{b_0}$$

[3317] $1^{14} 3221^2 51^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -447, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$

$$\begin{aligned}
& T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 22 & 2 \\ 15 & 0 & 22 & 206 & 28 \\ 2 & 0 & 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[3319] & 1^{14}41^231^241^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -448, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 1 & 1 & 30 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 30 & 0 & 394 & 38 \\ 2 & 3 & -1 & 38 & 3 \end{pmatrix}_{b_0} \\
[3318] & 1^{13}51^3231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -448, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 18 & 2 \\ 1 & 2 & 0 & 24 & 2 \\ 16 & 18 & 24 & 295 & 33 \\ 2 & 2 & 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[3320] & 31^44141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -450, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 19 & 2 \\ 2 & 5 & 1 & 57 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 57 & 0 & 517 & 43 \\ 2 & 5 & -1 & 43 & 3 \end{pmatrix}_{b_0} \\
[3321] & 1^{12}41^431^341^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -452, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 24 & 2 \\ 1 & 1 & 1 & 38 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 38 & 0 & 774 & 52 \\ 2 & 3 & -1 & 52 & 3 \end{pmatrix}_{b_0} \\
[3322] & 31^{13}321^251^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -455, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 26 & 3 \\ 5 & 5 & 35 & 5 \\ 26 & 35 & 202 & 30 \\ 3 & 5 & 30 & 3 \end{pmatrix}_{b_0} \\
[3323] & 2^1321^351^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -460, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 18 & 2 \\ 1 & 0 & 2 & 15 & 2 \\ 2 & 2 & 0 & 20 & 2 \\ 18 & 15 & 20 & 218 & 28 \\ 2 & 2 & 2 & 28 & 3 \end{pmatrix}_{b_0} \\
[3324] & 1^{11}3221^{13}51^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -461, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 15 & 2 \\ 4 & 18 & 6 & 65 & 8 \\ 1 & 6 & 0 & 18 & 2 \\ 15 & 65 & 18 & 211 & 27 \\ 2 & 8 & 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[3325] & 1^{14}3221^451^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -463, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 21 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 32 & 2 \\ 21 & 0 & 32 & 526 & 44 \\ 2 & 0 & 2 & 44 & 3 \end{pmatrix}_{b_0} \\
[3326] & 1^{12}3221^{13}51^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 17 & 2 \\ 4 & 18 & 6 & 70 & 8 \\ 1 & 6 & 0 & 21 & 2 \\ 17 & 70 & 21 & 250 & 30 \\ 2 & 8 & 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[3327] & 1^{14}41^42321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -470, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 28 & 3 \\ 2 & 1 & 24 & 3 \\ 28 & 24 & 286 & 40 \\ 3 & 3 & 40 & 5 \end{pmatrix}_{b_0} \\
[3329] & 1^{14}5^{13}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -472, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 19 & 2 \\ 1 & 2 & 0 & 27 & 2 \\ 16 & 19 & 27 & 346 & 36 \\ 2 & 2 & 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[3328] & 1^{13}41^331^441^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -472, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 24 & 2 \\ 1 & 1 & 1 & 40 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 40 & 0 & 826 & 54 \\ 2 & 3 & -1 & 54 & 3 \end{pmatrix}_{b_0} \\
[3330] & 1^{13}41^431^341^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -475, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 24 & 2 \\ 1 & 1 & 1 & 41 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 41 & 0 & 853 & 55 \\ 2 & 3 & -1 & 55 & 3 \end{pmatrix}_{b_0} \\
[3332] & 1^{14}3221^351^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -479, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 27 & 2 \\ 18 & 0 & 27 & 346 & 36 \\ 2 & 0 & 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[3331] & 1^{12}51^3231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -479, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 1 & 2 & 0 & 26 & 2 \\ 18 & 17 & 26 & 310 & 34 \\ 2 & 2 & 2 & 34 & 3 \end{pmatrix}_{b_0} \\
[3333] & 1^{13}51^23231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -484, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 2 & 18 & 2 \\ 1 & 2 & 0 & 24 & 2 \\ 17 & 18 & 24 & 295 & 33 \\ 2 & 2 & 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[3334] & 41^{13}231^{14}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -485, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 15 & 2 \\ 4 & 26 & 1 & 100 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 100 & 0 & 358 & 36 \\ 2 & 10 & -1 & 36 & 3 \end{pmatrix}_{b_0} \\
[3335] & 151^{13}2231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -485, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 13 & 2 \\ 4 & 18 & 6 & 65 & 8 \\ 1 & 6 & 0 & 18 & 2 \\ 13 & 65 & 18 & 211 & 27 \\ 2 & 8 & 2 & 27 & 3 \end{pmatrix}_{b_0} \\
[3336] & 22^13151^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -488, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 3 & 14 & 2 \\ 1 & 0 & 3 & 12 & 2 \\ 3 & 3 & 4 & 32 & 4 \\ 14 & 12 & 32 & 150 & 24 \\ 2 & 2 & 4 & 24 & 3 \end{pmatrix}_{b_0} \\
[3337] & 31^24141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -490, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 5 & 1 & 46 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 46 & 0 & 314 & 34 \\ 2 & 5 & -1 & 34 & 3 \end{pmatrix}_{b_0} \\
[3339] & 1^{13}41^231^441^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -499, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 22 & 2 \\ 1 & 1 & 1 & 35 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 35 & 0 & 625 & 47 \\ 2 & 3 & -1 & 47 & 3 \end{pmatrix}_{b_0} \\
[3338] & 321^{13}4141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -499, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 8 & 21 & 3 \\ 8 & 17 & 48 & 9 \\ 21 & 48 & 124 & 24 \\ 3 & 9 & 24 & 4 \end{pmatrix}_{b_0} \\
[3340] & 1^{13}3221^{13}51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -501, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 16 & 2 \\ 4 & 18 & 6 & 75 & 8 \\ 1 & 6 & 0 & 24 & 2 \\ 16 & 75 & 24 & 295 & 33 \\ 2 & 8 & 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[3341] & 1^{13}3221^{13}51^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -501, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8
\end{aligned}$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 17 & 2 \\ 4 & 18 & 6 & 75 & 8 \\ 1 & 6 & 0 & 24 & 2 \\ 17 & 75 & 24 & 295 & 33 \\ 2 & 8 & 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[3342] \quad 1^{14} 2^3 21^{25} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -502, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 26 & 3 \\ 4 & 5 & 40 & 5 \\ 26 & 40 & 258 & 34 \\ 3 & 5 & 34 & 4 \end{pmatrix}_{b_0}$$

$$[3343] \quad 1^{12} 5^1 1^3 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -503, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 1 & 2 & 0 & 26 & 2 \\ 19 & 17 & 26 & 310 & 34 \\ 2 & 2 & 2 & 34 & 3 \end{pmatrix}_{b_0}$$

$$[3344] \quad 1^{14} 41^4 31^2 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 22 & 2 \\ 1 & 1 & 1 & 40 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 40 & 0 & 750 & 52 \\ 2 & 3 & -1 & 52 & 3 \end{pmatrix}_{b_0}$$

$$[3345] \quad 1^{11} 3221^{13} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 14 & 2 \\ 4 & 18 & 6 & 65 & 8 \\ 1 & 6 & 0 & 18 & 2 \\ 14 & 65 & 18 & 211 & 27 \\ 2 & 8 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[3346] \quad 1^{13} 3221^4 51^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 21 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 29 & 2 \\ 21 & 0 & 29 & 465 & 41 \\ 2 & 0 & 2 & 41 & 3 \end{pmatrix}_{b_0}$$

$$[3347] \quad 1^{13} 3221^3 51^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 24 & 2 \\ 18 & 0 & 24 & 295 & 33 \\ 2 & 0 & 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[3348] \quad 1^{12} 41^4 31^4 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 27 & 2 \\ 1 & 1 & 1 & 42 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 42 & 0 & 970 & 58 \\ 2 & 3 & -1 & 58 & 3 \end{pmatrix}_{b_0}$$

$$[3349] \quad 1^{13} 5^1 1^3 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 2 & 18 & 2 \\ 1 & 2 & 0 & 24 & 2 \\ 18 & 18 & 24 & 295 & 33 \\ 2 & 2 & 2 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[3350] \quad 1^{13} 41^4 31^2 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -515, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 22 & 2 \\ 1 & 1 & 1 & 37 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 37 & 0 & 673 & 49 \\ 2 & 3 & -1 & 49 & 3 \end{pmatrix}_{b_0}$$

$$[3351] \quad 1^{14} 41^2 31^4 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 22 & 2 \\ 1 & 1 & 1 & 38 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 38 & 0 & 698 & 50 \\ 2 & 3 & -1 & 50 & 3 \end{pmatrix}_{b_0}$$

$$[3353] \quad 1^{12} 3221^4 51^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -519, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 21 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 26 & 2 \\ 21 & 0 & 26 & 410 & 38 \\ 2 & 0 & 2 & 38 & 3 \end{pmatrix}_{b_0}$$

$$[3352] \quad 22^1 3^1 151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -519, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.1$$

$$\begin{aligned}
& T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 6 & 25 & 3 \\ 6 & 14 & 65 & 8 \\ 25 & 65 & 273 & 35 \\ 3 & 8 & 35 & 4 \end{pmatrix}_{b_0} \\
[3355] & 1^{14}41^431^41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -520, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & 1 & 1 & 36 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 36 & 0 & 586 & 46 \\ 2 & 3 & -1 & 46 & 3 \end{pmatrix}_{b_0} \\
[3354] & 1^{13}41^331^341^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -520, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 22 & 2 \\ 1 & 1 & 1 & 36 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 36 & 0 & 646 & 48 \\ 2 & 3 & -1 & 48 & 3 \end{pmatrix}_{b_0} \\
[3357] & 1^{14}41^331^341^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -523, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 22 & 2 \\ 1 & 1 & 1 & 39 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 39 & 0 & 721 & 51 \\ 2 & 3 & -1 & 51 & 3 \end{pmatrix}_{b_0} \\
[3356] & 1^{14}41^131^441^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -523, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & 1 & 1 & 33 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 33 & 0 & 517 & 43 \\ 2 & 3 & -1 & 43 & 3 \end{pmatrix}_{b_0} \\
[3358] & 1^{13}5^11^3231^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 2 & 19 & 2 \\ 1 & 2 & 0 & 27 & 2 \\ 17 & 19 & 27 & 346 & 36 \\ 2 & 2 & 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[3359] & 1^{13}5^11^3231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -532, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 0 & 2 & 18 & 2 \\ 1 & 2 & 0 & 24 & 2 \\ 19 & 18 & 24 & 295 & 33 \\ 2 & 2 & 2 & 33 & 3 \end{pmatrix}_{b_0} \\
[3360] & 151^{13}2231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 14 & 2 \\ 4 & 18 & 6 & 70 & 8 \\ 1 & 6 & 0 & 21 & 2 \\ 14 & 70 & 21 & 250 & 30 \\ 2 & 8 & 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[3361] & 1^{14}41^331^241^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -547, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & 1 & 1 & 35 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 35 & 0 & 557 & 45 \\ 2 & 3 & -1 & 45 & 3 \end{pmatrix}_{b_0} \\
[3362] & 1^{14}41^231^341^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -548, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & 1 & 1 & 34 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 34 & 0 & 534 & 44 \\ 2 & 3 & -1 & 44 & 3 \end{pmatrix}_{b_0} \\
[3363] & 1^{13}41^431^441^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -551, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 27 & 2 \\ 1 & 1 & 1 & 45 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 45 & 0 & 1057 & 61 \\ 2 & 3 & -1 & 61 & 3 \end{pmatrix}_{b_0} \\
[3364] & 1^{12}3221^{13}51^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 16 & 2 \\ 4 & 18 & 6 & 70 & 8 \\ 1 & 6 & 0 & 21 & 2 \\ 16 & 70 & 21 & 250 & 30 \\ 2 & 8 & 2 & 30 & 3 \end{pmatrix}_{b_0} \\
[3365] & 2^1321^{35}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -567, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8
\end{aligned}$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 19 & 2 \\ 1 & 0 & 2 & 17 & 2 \\ 2 & 2 & 0 & 21 & 2 \\ 19 & 17 & 21 & 250 & 30 \\ 2 & 2 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[3366] \ 1^{13} 5^1 1^3 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -568, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 2 & 19 & 2 \\ 1 & 2 & 0 & 27 & 2 \\ 18 & 19 & 27 & 346 & 36 \\ 2 & 2 & 2 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[3367] \ 31^{14} 321^{15} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -575, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 23 & 3 \\ 5 & 5 & 30 & 5 \\ 23 & 30 & 139 & 25 \\ 3 & 5 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[3368] \ 31^{14} 4^{14} 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -580, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 18 & 2 \\ 2 & 5 & 1 & 55 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 55 & 0 & 461 & 41 \\ 2 & 5 & -1 & 41 & 3 \end{pmatrix}_{b_0}$$

$$[3369] \ 1^{12} 3221^{13} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 15 & 2 \\ 4 & 18 & 6 & 70 & 8 \\ 1 & 6 & 0 & 21 & 2 \\ 15 & 70 & 21 & 250 & 30 \\ 2 & 8 & 2 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[3370] \ 1^{13} 41^3 31^4 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -596, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 25 & 2 \\ 1 & 1 & 1 & 40 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 40 & 0 & 826 & 54 \\ 2 & 3 & -1 & 54 & 3 \end{pmatrix}_{b_0}$$

$$[3371] \ 231^{13} 3151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -599, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 5 & 1 & 15 & 2 \\ 5 & 33 & 1 & 105 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 105 & 0 & 313 & 33 \\ 2 & 11 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[3372] \ 1^{13} 5^1 1^3 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -604, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 0 & 2 & 19 & 2 \\ 1 & 2 & 0 & 27 & 2 \\ 19 & 19 & 27 & 346 & 36 \\ 2 & 2 & 2 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[3373] \ 1^{14} 3221^3 51^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -604, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 27 & 2 \\ 19 & 0 & 27 & 346 & 36 \\ 2 & 0 & 2 & 36 & 3 \end{pmatrix}_{b_0}$$

$$[3374] \ 22 \ 3 \ 151^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -607, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 3 & 15 & 2 \\ 1 & 0 & 3 & 14 & 2 \\ 3 & 3 & 4 & 34 & 4 \\ 15 & 14 & 34 & 176 & 26 \\ 2 & 2 & 4 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[3375] \ 1^{13} 41^4 31^3 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -607, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 25 & 2 \\ 1 & 1 & 1 & 41 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 41 & 0 & 853 & 55 \\ 2 & 3 & -1 & 55 & 3 \end{pmatrix}_{b_0}$$

$$[3376] \ 1^{14} 41^4 31^3 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -612, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 25 & 2 \\ 1 & 1 & 1 & 44 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 44 & 0 & 938 & 58 \\ 2 & 3 & -1 & 58 & 3 \end{pmatrix}_{b_0}$$

$$[3377] \ 1^{14} 3221^4 51^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -612, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$\begin{aligned}
& T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 22 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 32 & 2 \\ 22 & 0 & 32 & 526 & 44 \\ 2 & 0 & 2 & 44 & 3 \end{pmatrix}_{b_0} \\
[3378] & 1^{14} 41^3 31^4 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -615, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 25 & 2 \\ 1 & 1 & 1 & 43 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 43 & 0 & 909 & 57 \\ 2 & 3 & -1 & 57 & 3 \end{pmatrix}_{b_0} \\
[3379] & 1^{13} 51^3 231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -632, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 20 & 2 \\ 1 & 0 & 2 & 19 & 2 \\ 1 & 2 & 0 & 27 & 2 \\ 20 & 19 & 27 & 346 & 36 \\ 2 & 2 & 2 & 36 & 3 \end{pmatrix}_{b_0} \\
[3380] & 41^{13} 231^{14} 11 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -636, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 16 & 2 \\ 4 & 26 & 1 & 100 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 100 & 0 & 358 & 36 \\ 2 & 10 & -1 & 36 & 3 \end{pmatrix}_{b_0} \\
[3381] & 1^{14} 41^2 31^4 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -640, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 23 & 2 \\ 1 & 1 & 1 & 38 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 38 & 0 & 698 & 50 \\ 2 & 3 & -1 & 50 & 3 \end{pmatrix}_{b_0} \\
[3382] & 1^{13} 3221^4 51^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -643, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 22 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 29 & 2 \\ 22 & 0 & 29 & 465 & 41 \\ 2 & 0 & 2 & 41 & 3 \end{pmatrix}_{b_0} \\
[3383] & 1^{14} 41^4 31^2 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -644, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 23 & 2 \\ 1 & 1 & 1 & 40 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 40 & 0 & 750 & 52 \\ 2 & 3 & -1 & 52 & 3 \end{pmatrix}_{b_0} \\
[3384] & 31^4 4141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -646, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 21 & 2 \\ 2 & 5 & 1 & 64 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 64 & 0 & 638 & 48 \\ 2 & 5 & -1 & 48 & 3 \end{pmatrix}_{b_0} \\
[3385] & 2^3 21^3 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -646, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 33 & 3 \\ 4 & 5 & 50 & 5 \\ 33 & 50 & 419 & 43 \\ 3 & 5 & 43 & 4 \end{pmatrix}_{b_0} \\
[3386] & 1^{14} 41^3 31^4 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -655, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 23 & 2 \\ 1 & 1 & 1 & 39 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 39 & 0 & 721 & 51 \\ 2 & 3 & -1 & 51 & 3 \end{pmatrix}_{b_0} \\
[3387] & 1^{13} 41^4 31^4 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -691, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 28 & 2 \\ 1 & 1 & 1 & 45 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 45 & 0 & 1057 & 61 \\ 2 & 3 & -1 & 61 & 3 \end{pmatrix}_{b_0} \\
[3388] & 1^{14} 41^4 31^4 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -712, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 28 & 2 \\ 1 & 1 & 1 & 48 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 48 & 0 & 1150 & 64 \\ 2 & 3 & -1 & 64 & 3 \end{pmatrix}_{b_0} \\
[3389] & 22^3 151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -728, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.2
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 6 & 30 & 3 \\ 6 & 14 & 78 & 8 \\ 30 & 78 & 394 & 42 \\ 3 & 8 & 42 & 4 \end{pmatrix}_{b_0} \\
[3390] & 31^{14}321^{25}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -740, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 30 & 3 \\ 5 & 5 & 40 & 5 \\ 30 & 40 & 258 & 34 \\ 3 & 5 & 34 & 3 \end{pmatrix}_{b_0} \\
[3391] & 41^{14}231^{14}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -749, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 19 & 2 \\ 4 & 26 & 1 & 128 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 128 & 0 & 586 & 46 \\ 2 & 10 & -1 & 46 & 3 \end{pmatrix}_{b_0} \\
[3392] & 1^{14}41^331^441^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -755, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 26 & 2 \\ 1 & 1 & 1 & 43 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 43 & 0 & 909 & 57 \\ 2 & 3 & -1 & 57 & 3 \end{pmatrix}_{b_0} \\
[3393] & 1^{14}3221^{45}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -759, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 23 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 32 & 2 \\ 23 & 0 & 32 & 526 & 44 \\ 2 & 0 & 2 & 44 & 3 \end{pmatrix}_{b_0} \\
[3394] & 1^{14}41^431^341^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -760, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 26 & 2 \\ 1 & 1 & 1 & 44 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 44 & 0 & 938 & 58 \\ 2 & 3 & -1 & 58 & 3 \end{pmatrix}_{b_0} \\
[3395] & 321^{13}4141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -774, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 8 & 24 & 3 \\ 8 & 17 & 56 & 9 \\ 24 & 56 & 166 & 28 \\ 3 & 9 & 28 & 4 \end{pmatrix}_{b_0} \\
[3396] & 31^{14}321^{35}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -775, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 37 & 3 \\ 5 & 5 & 50 & 5 \\ 37 & 50 & 419 & 43 \\ 3 & 5 & 43 & 3 \end{pmatrix}_{b_0} \\
[3397] & 321^{14}4141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -814, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 8 & 28 & 3 \\ 8 & 17 & 64 & 9 \\ 28 & 64 & 222 & 32 \\ 3 & 9 & 32 & 4 \end{pmatrix}_{b_0} \\
[3398] & 1^{14}41^431^441^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -868, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 29 & 2 \\ 1 & 1 & 1 & 48 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 48 & 0 & 1150 & 64 \\ 2 & 3 & -1 & 64 & 3 \end{pmatrix}_{b_0} \\
[3399] & 1^{17}22\ 3\ 151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -975, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.3 \\
& T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 6 & 35 & 3 \\ 6 & 14 & 91 & 8 \\ 35 & 91 & 537 & 49 \\ 3 & 8 & 49 & 4 \end{pmatrix}_{b_0} \\
[3400] & 231^{14}3151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -984, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 5 & 1 & 20 & 2 \\ 5 & 33 & 1 & 140 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 140 & 0 & 558 & 44 \\ 2 & 11 & -1 & 44 & 3 \end{pmatrix}_{b_0} \\
[3401] & 41^{14}231^{14}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -996, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 20 & 2 \\ 4 & 26 & 1 & 128 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 128 & 0 & 586 & 46 \\ 2 & 10 & -1 & 46 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[3402] $321^{14}4141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -1219$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 8 & 31 & 3 \\ 8 & 17 & 72 & 9 \\ 31 & 72 & 276 & 36 \\ 3 & 9 & 36 & 4 \end{pmatrix}_{b_0}$

[3403] $231^{15}3151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1479$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.1$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 5 & 1 & 25 & 2 \\ 5 & 33 & 1 & 175 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 175 & 0 & 873 & 55 \\ 2 & 11 & -1 & 55 & 3 \end{pmatrix}_{b_0}$

$T = 10$ [3406] $1^{13}15131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 2$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$

[3407] $1^{14}14131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -54$, $\det = 2$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[3404] $15^{11}1321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -51$, $\det = 2$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[3405] $1^{15}12321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -53$, $\det = 2$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[3412] $1^{14}13151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -78$, $\det = 3$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$

[3413] $1^{13}15131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 3$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0}$

[3409] $1^{13}151231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 3$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & 0 & 4 \end{pmatrix}_{b_0}$

[3408] $1^{13}15131^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 3$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 13 & 2 \\ 8 & 0 & 13 & 97 & 21 \\ 2 & 0 & 2 & 21 & 4 \end{pmatrix}_{b_0}$

[3414] $1^{14}14141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -74$, $\det = 3$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$

[3410] $1^{11}13151231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 3$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 4 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[3411] \quad 1^1 31 \overset{11}{5} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 2 & 0 & 1 & 1 & 3 \\ 2 & 0 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[3415] \quad 2161^1 2 \overset{11}{3} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 4 & 2 & 2 & 2 \\ 4 & 10 & 5 & 4 & 6 \\ 2 & 5 & 0 & 3 & 2 \\ 2 & 4 & 3 & 1 & 3 \\ 2 & 6 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3416] \quad 1^1 11 2231321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -8, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 4 & 1 & 6 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 4 & 6 \\ 2 & 6 & -1 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[3417] \quad 1^1 4131 \overset{11}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -78, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 2 & 2 & 4 \\ 2 & 0 & 2 \\ 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[3418] \quad 1^1 11 4132151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 \\ 3 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[3419] \quad 21 \overset{11}{6} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -76, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[3420] \quad 1^1 11 612231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -77, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[3421] \quad 1^1 11 61231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -77, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[3422] \quad 1^1 \overset{11}{2} 3 \overset{11}{21} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 4 & 2 & 2 & 2 \\ 4 & 10 & 5 & 4 & 6 \\ 2 & 5 & 0 & 3 & 2 \\ 2 & 4 & 3 & 1 & 3 \\ 2 & 6 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3423] \quad 1^1 11 31 \overset{11}{5} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 4 & 2 \\ 2 & 0 & 4 & 6 & 6 \\ 2 & 0 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[3424] \quad 1^1 11 2231321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -8, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 3 & 6 & 5 \\ 2 & 0 & 6 & 4 & 6 \\ 2 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0}$$

[3425] $1^{11}315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 5$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

[3426] $2321^{12}61^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 7$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 7 & 7 & 7 & 42 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[3428] $1^{11}4132151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 7$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[3429] $2321^{11}61^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 7$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 6 & 6 & 7 & 35 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3430] $2321^{11}61^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 7$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 5 & 4 & 5 & 18 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[3431] $161232 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 7$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[3432] $2321^{11}61^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 7$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 3 & 5 & 13 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[3433] $1^14132151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 7$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 0 & 1 & 2 \\ 2 & 1 & 0 & 2 \\ 3 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

[3434] $2321^{11}61^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 7$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[3427] $1^141^{13}32151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 7$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 4 & 4 \\ 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

[3435] $1^{11}315131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 8$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 6 & 2 \\ 6 & 0 & 6 & 34 & 12 \\ 2 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[3436] $1^{12}31^15131^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 8 & 0 & 30 & 12 \\ 2 & 3 & -1 & 12 & 4 \end{pmatrix}_{b_0}$

[3437] $1^{13}31^15131^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 12 & 0 & 70 & 18 \\ 2 & 3 & -1 & 18 & 4 \end{pmatrix}_{b_0}$

[3438] $1^131^141321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 11, \Delta = -35, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 2 & 3 & 6 & 6 \\ 2 & 2 & 2 & 6 & 5 \end{pmatrix}_{b_0}$

[3439] $1^{14}31^15131^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 16 & 0 & 126 & 24 \\ 2 & 3 & -1 & 24 & 4 \end{pmatrix}_{b_0}$

[3440] $1^{11}31^15131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 6 & 6 \\ 2 & 3 & -1 & 6 & 4 \end{pmatrix}_{b_0}$

[3441] $1^{11}3141321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 11, \Delta = -35, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 2 & 5 & 5 \end{pmatrix}_{b_0}$

[3442] $1^{11}3141321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 11, \Delta = -35, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 2 & 5 & 4 \\ 4 & 5 & 7 & 7 \\ 3 & 4 & 7 & 6 \end{pmatrix}_{b_0}$

[3443] $41^131^14141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 10, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 5 & 1 & 4 & 2 \\ 5 & 38 & 1 & 32 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 32 & 0 & 26 & 10 \\ 2 & 12 & -1 & 10 & 3 \end{pmatrix}_{b_0}$

[3444] $2^3315131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$

[3445] $1^{12}31^15131^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 9 & 2 \\ 6 & 0 & 9 & 49 & 15 \\ 2 & 0 & 2 & 15 & 4 \end{pmatrix}_{b_0}$

[3446] $1^{11}31^15131^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 5 & 2 \\ 4 & 0 & 5 & 17 & 9 \\ 2 & 0 & 2 & 9 & 4 \end{pmatrix}_{b_0}$

[3447] $2^3315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

[3448] $1^{11} 6^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 4 & 3 & 4 & 10 & 6 \\ 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[3449] $1^{11} 13151^{11} 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 2 & 1 & 9 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 9 & 0 & 27 & 11 \\ 2 & 4 & -1 & 11 & 4 \end{pmatrix}_{b_0}$$

[3450] $1^{11} 13151231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 0 & 4 & 2 & 4 \\ 2 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[3451] $1^{11} 315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

[3452] $1^{11} 5^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

[3453] $1^{11} 13151231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 14, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

[3454] $161232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 1 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[3455] $232161^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 7 & 5 & 9 & 40 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[3456] $1^{11} 31^{11} 41321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 2 & 8 & 4 \\ 6 & 8 & 19 & 11 \\ 3 & 4 & 11 & 6 \end{pmatrix}_{b_0}$$

[3457] $1^{11} 315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

[3458] $1^{11} 4132151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 0 & 3 & 2 \\ 3 & 3 & 2 & 4 \\ 3 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

[3459] $2^{\overline{1}12} 3 15 13 1^{\overline{1}4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 14 & 10 & 74 & 18 \\ 2 & 2 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[3460] $2^{\overline{1}11} 3 21 6 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 2 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[3461] $2^{\overline{1}11} 3 21^{\overline{1}3} 6 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 9 & 7 & 10 & 63 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[3462] $1^{\overline{1}2} 3 1^{\overline{1}2} 5 13 1^{\overline{1}4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 10 & 2 \\ 8 & 0 & 10 & 74 & 18 \\ 2 & 0 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[3463] $4 1^{\overline{1}3} 1^{\overline{1}4} 14 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 5 & 2 \\ 4 & 26 & 1 & 32 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 32 & 0 & 38 & 12 \\ 2 & 10 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

[3464] $4 1^{\overline{1}3} 1^{\overline{1}3} 4 14 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 19 & 2 \\ 4 & 26 & 1 & 124 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 124 & 0 & 578 & 46 \\ 2 & 10 & -1 & 46 & 3 \end{pmatrix}_{b_0}$$

[3466] $1^{\overline{1}1} 4 1^{\overline{1}3} 2 15 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 4 & 0 & 10 & 6 \\ 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[3465] $1^{\overline{1}1} 4 13 2 1^{\overline{1}5} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 2 & 6 & 4 \\ 6 & 6 & 6 \\ 4 & 6 & 5 \end{pmatrix}_{b_0}$$

[3467] $2^{\overline{1}11} 3 15 13 1^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

[3468] $4 1^{\overline{1}1} 4 14 1^{\overline{1}1} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 3 & 3 & 1 & 3 \\ 2 & 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

[3469] $1^{\overline{1}1} 4 13 2 1^{\overline{1}5} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 0 & 3 & 2 \\ 5 & 3 & 6 & 6 \\ 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

[3470] $2 1^{\overline{1}1} 1^{\overline{1}1} 2 3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 6 & 12 & 6 & 28 & 10 \\ 2 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[3471] $1^{1^1} 2^1 3^1 2^1 1^1 6^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 6 & 6 & 6 & 28 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[3472] $1^{1^1} 4^1 1^1 3^1 2^1 1^1 5^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 0 & 2 & 2 \\ 6 & 2 & 4 & 6 \\ 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

[3474] $1^{1^3} 2^1 3^1 1^1 5^1 1^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 6 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

[3473] $1^{1^2} 3^1 1^1 5^1 1^1 2^1 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 2 & 1 & 13 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 13 & 0 & 49 & 15 \\ 2 & 4 & -1 & 15 & 4 \end{pmatrix}_{b_0}$$

[3476] $1^{1^2} 2^1 3^1 2^1 1^1 6^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 5 & 3 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[3475] $1^{2^1} 3^1 2^1 1^1 6^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 18 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[3478] $1^{1^1} 4^1 1^1 3^1 2^1 1^1 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 3 & 0 & 6 & 6 \\ 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[3477] $2^{1^2} 3^1 2^1 1^1 6^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 6 & 4 & 8 & 26 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[3480] $1^{1^2} 3^1 1^1 5^1 1^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

[3479] $1^{1^1} 3^1 1^1 5^1 1^1 2^1 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 9 & 3 \\ 5 & 0 & 9 & 27 & 11 \\ 2 & 0 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[3481] $4^1 1^1 2^1 3^1 5^1 1^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 34, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 8 & 0 & 50 & 14 \\ 2 & 2 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

[3482] $1^{12}315131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

[3485] $21^{11}61232 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 4 & 3 & 8 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 4 & 8 & 5 & 11 & 7 \\ 2 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[3486] $2321^{11}61^{14}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 11 & 10 & 12 & 103 & 19 \\ 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[3483] $232161^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 5 & 5 & 6 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[3484] $21^{11}61^{11}232 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 8 & 2 \\ 2 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 8 & 14 & 9 & 43 & 13 \\ 2 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[3488] $1^{12}41^{11}32151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 6 & 0 & 20 & 8 \\ 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[3487] $1^{11}41321^{25}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 9 & 4 & 0 & 20 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[3489] $161232 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 5 & 5 \\ 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[3490] $41^{11}2321^{15}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 8 & 2 \\ 4 & 26 & 1 & 48 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 48 & 0 & 86 & 18 \\ 2 & 10 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[3491] $1^{11}2321^{11}61^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 47, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 6 & 4 & 7 & 21 & 9 \\ 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[3492] $1^141321^{25}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 47, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 3 & 2 \\ 10 & 3 & 16 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

[3493] $232161^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 47, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 4 & 5 & 11 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[3494] $2^1 3^1 21^1 6^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 47, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 8 & 6 & 9 & 45 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[3495] $1^1 41^1 32151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 47, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 0 & 5 & 2 \\ 6 & 5 & 12 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

[3496] $2^1 3^1 2161^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 47, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 9 & 8 & 11 & 72 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[3497] $1^1 41321^2 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 4 & 2 \\ 9 & 4 & 20 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

[3498] $1^1 331^1 5131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 10 & 3 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 10 & 6 & 34 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[3499] $1^1 331^1 5131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 10 & 3 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 10 & 6 & 34 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[3500] $1^1 331^1 5131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 5 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[3501] $1^1 331^1 5131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 7 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[3502] $1^1 331^1 5131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[3503] $1^1 3151^1 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 9 & 3 \\ 6 & 0 & 9 & 27 & 11 \\ 2 & 0 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[3505] $2^1 3^1 21^1 6^1 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 7 & 5 & 8 & 31 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[3504] $1^{12}3151^{12}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 2 & 1 & 13 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 13 & 0 & 49 & 15 \\ 2 & 4 & -1 & 15 & 4 \end{pmatrix}_{b_0}$

[3507] $1^{12}2315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 62, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$

[3506] $1^{12}2315131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 62, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[3509] $1^{11}41^{13}321^{15}51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 8 & 5 & 0 & 22 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$

[3508] $1^{12}41321^{15}51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 6 & 5 & 0 & 14 & 8 \\ 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$

[3512] $1^{12}2321^{16}61^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 12 & 9 & 14 & 110 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$

[3510] $1^{14}41321^{35}51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 4 & 2 \\ 14 & 4 & 36 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$

[3511] $1^{11}41321^{15}51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 4 & 2 \\ 7 & 4 & 8 & 8 \\ 3 & 2 & 8 & 4 \end{pmatrix}_{b_0}$

[3513] $1^{11}41321^{35}51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 67, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 5 & 2 \\ 13 & 5 & 42 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$

[3514] $1^{11}61232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 71, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 5 & 4 & 7 & 16 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[3515] $1^{13}315131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 77, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 6 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$

[3516] $1^{11}2321^{16}61^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 79, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 10 & 9 & 11 & 79 & 17 \\ 2 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$

[3518] $1^{1^{12}} 2^3 3^{21^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 5 & 10 & 38 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[3517] $1^{1^{12}} 2^3 2161^{1^3} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 7 & 10 & 52 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[3519] $41^{1^1} 23151^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 81, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 9 & 9 & 3 \\ 9 & 51 & 48 & 15 \\ 9 & 48 & 42 & 14 \\ 3 & 15 & 14 & 4 \end{pmatrix}_{b_0}$

[3520] $1^{1^4} 1^3 315131^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 83, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 9 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$

[3521] $1^{1^3} 3151^{1^2} 231^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 83, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 2 & 1 & 17 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 17 & 0 & 79 & 19 \\ 2 & 4 & -1 & 19 & 4 \end{pmatrix}_{b_0}$

[3522] $1^{1^2} 3151^{1^2} 231^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 83, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 13 & 3 \\ 7 & 0 & 13 & 49 & 15 \\ 2 & 0 & 3 & 15 & 4 \end{pmatrix}_{b_0}$

[3526] $41^{1^1} 232151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 28 & 0 & 26 & 10 \\ 2 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$

[3527] $1^{1^1} 41^{1^4} 32151^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 6 & 2 \\ 8 & 6 & 18 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$

[3523] $41^{1^1} 314141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 28 & 0 & 26 & 10 \\ 2 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$

[3524] $41^{1^1} 23151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 9 & 8 & 3 \\ 9 & 51 & 48 & 15 \\ 8 & 48 & 42 & 14 \\ 3 & 15 & 14 & 4 \end{pmatrix}_{b_0}$

[3525] $1^{1^1} 41^{1^4} 321^{2^1} 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 12 & 6 & 0 & 42 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$

[3528] $1^{1^2} 1^3 315131^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 87, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$

[3529] $1^1 4 1 3 2 1^2 5 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 88$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 4 & 2 \\ 12 & 4 & 22 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[3530] $1^{13} 3 1^1 5 1 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 89$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$

[3531] $2^1 3^1 2 1 6 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 92$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.70 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 6 & 5 & 8 & 24 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$

[3532] $1^{13} 4 1 3 2 1^1 5 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 92$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 12.70 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 7 & 7 & 0 & 26 & 10 \\ 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$

[3534] $2^1 3^1 2 1^1 6 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 95$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 11 & 8 & 13 & 86 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$

[3533] $2^1 3^1 2 1 6 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 95$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 7 & 6 & 9 & 36 & 12 \\ 2 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[3535] $4 1^{11} 3 1^1 5 1 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -86$, $\det = 96$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 4 & 2 & 4 & 2 \\ 4 & 10 & 5 & 12 & 6 \\ 2 & 5 & 0 & 4 & 2 \\ 4 & 12 & 4 & 10 & 6 \\ 2 & 6 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[3536] $1^{11} 4 1 3 2 1^1 5 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 99$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.65 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 5 & 2 \\ 9 & 5 & 14 & 10 \\ 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$

[3537] $1^{11} 6^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 100$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.68 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 7 & 6 & 8 & 31 & 11 \\ 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$

[3538] $4 1^{11} 3 1 4 1 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 101$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 4 & 1 & 5 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 28 & 0 & 26 & 10 \\ 2 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$

[3539] $2^1 3^1 2 1^1 6 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 103$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 9 & 6 & 11 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[3542] $2^1 3^1 2 1^1 6 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 104$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 9 & 8 & 10 & 59 & 15 \\ 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[3540] & 31^{11}41^13151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 3 & 1 & 8 & 2 \\ 3 & 4 & 1 & 12 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 8 & 12 & 0 & 26 & 10 \\ 2 & 4 & 0 & 10 & 3 \end{pmatrix}_{b_0} \\
[3541] & 1^{12}41321^{251^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 10 & 6 & 0 & 30 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0} \\
[3543] & 51^{11}2231^{141^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 105, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 7 & 11 & 3 \\ 7 & 7 & 14 & 7 \\ 11 & 14 & 24 & 12 \\ 3 & 7 & 12 & 4 \end{pmatrix}_{b_0} \\
[3544] & 1^{11}41321^{251^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 107, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 5 & 2 \\ 11 & 5 & 22 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0} \\
[3546] & 2^1321^{12}61^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 10 & 7 & 12 & 66 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[3545] & 1^{11}41^1321^{151^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 6 & 2 \\ 10 & 6 & 24 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0} \\
[3548] & 1^{13}31^15131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 110, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 10 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0} \\
[3547] & 1^{13}31^15131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 110, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0} \\
[3549] & 2^1321^{11}61^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 111, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.61 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 8 & 7 & 9 & 43 & 13 \\ 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[3550] & 1^{12}41^1321^{151^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 9 & 7 & 0 & 34 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0} \\
[3552] & 1^{13}31^15131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0} \\
[3551] & 1^{12}3151^{2231^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 18 & 3 \\ 10 & 0 & 18 & 114 & 22 \\ 2 & 0 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[3553] $1^{14}315131^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[3554] $1^{13}3151^231^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 2 & 1 & 22 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 22 & 0 & 154 & 26 \\ 2 & 4 & -1 & 26 & 4 \end{pmatrix}_{b_0}$$

[3555] $1^{13}3151^1231^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 2 & 1 & 17 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 17 & 0 & 79 & 19 \\ 2 & 4 & -1 & 19 & 4 \end{pmatrix}_{b_0}$$

[3556] $1^{12}3151^1231^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 13 & 3 \\ 8 & 0 & 13 & 49 & 15 \\ 2 & 0 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[3557] $51^{11}223141^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 119, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.71 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 7 & 6 & 3 \\ 7 & 7 & 7 & 7 \\ 6 & 7 & 4 & 6 \\ 3 & 7 & 6 & 4 \end{pmatrix}_{b_0}$$

[3558] $1^{14}41321^{15}1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 119, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 8 & 9 & 0 & 42 & 12 \\ 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[3559] $1^{11}41^132151^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 119, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 7 & 2 \\ 10 & 7 & 28 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[3561] $31^141^{11}3151^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 120, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 5 & 1 & 8 & 2 \\ 5 & 8 & 1 & 16 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 8 & 16 & 0 & 26 & 10 \\ 2 & 6 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[3560] $41^{11}31^24141^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 120, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 14 & 2 \\ 4 & 26 & 1 & 92 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 92 & 0 & 314 & 34 \\ 2 & 10 & -1 & 34 & 3 \end{pmatrix}_{b_0}$$

[3562] $1^{14}315131^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 122, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[3563] $1^{14}315131^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 122, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[3564] $2^315131^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 122, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[3565] $41^1 4^{11} 141^{11} 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 6 & 2 \\ 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 4 & 10 & 4 \\ 6 & 6 & 10 & 14 & 8 \\ 2 & 3 & 4 & 8 & 3 \end{pmatrix}_{b_0}$$

[3568] $41^1 31^{11} 4141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 127, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 5 & 1 & 9 & 2 \\ 5 & 38 & 1 & 72 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 72 & 0 & 130 & 22 \\ 2 & 12 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[3567] $1^1 41321^3 51^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 127, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 5 & 2 \\ 16 & 5 & 42 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[3566] $1^{12} 41321^3 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 127, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 14 & 7 & 0 & 54 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[3569] $51^{11} 223141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 129, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 7 & 9 & 3 \\ 7 & 7 & 12 & 7 \\ 9 & 12 & 16 & 10 \\ 3 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

[3570] $1^{11} 41321^{15} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 132, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 6 & 2 \\ 11 & 6 & 24 & 12 \\ 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[3571] $2^3 3^{13} 15131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

[3572] $4131^{11} 4141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 5 & 1 & 5 & 2 \\ 5 & 38 & 1 & 40 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 40 & 0 & 38 & 12 \\ 2 & 12 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

[3573] $41^{11} 31^1 4141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 141, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 9 & 2 \\ 4 & 26 & 1 & 60 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 60 & 0 & 130 & 22 \\ 2 & 10 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[3574] $41^{12} 231^1 5^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 18 & 3 \\ 9 & 51 & 96 & 15 \\ 18 & 96 & 174 & 28 \\ 3 & 15 & 28 & 4 \end{pmatrix}_{b_0}$$

[3575] $1^{11} 41321^3 51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 148, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.68 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 6 & 2 \\ 15 & 6 & 44 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[3576] $1^{13} 3151^1 231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 149, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 17 & 3 \\ 9 & 0 & 17 & 79 & 19 \\ 2 & 0 & 3 & 19 & 4 \end{pmatrix}_{b_0}$$

[3577] $1^{14}3151^1231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 153, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 2 & 1 & 21 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 21 & 0 & 117 & 23 \\ 2 & 4 & -1 & 23 & 4 \end{pmatrix}_{b_0}$$

[3578] $4131^{11}4141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 154, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 5 & 1 & 6 & 2 \\ 5 & 38 & 1 & 40 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 40 & 0 & 38 & 12 \\ 2 & 12 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

[3579] $1^{14}31^25131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 155, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 9 & 15 & 13 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

[3581] $41^{12}231^151 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 156, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 16 & 3 \\ 9 & 51 & 96 & 15 \\ 16 & 96 & 174 & 28 \\ 3 & 15 & 28 & 4 \end{pmatrix}_{b_0}$$

[3580] $41^{12}232151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 156, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 8 & 2 \\ 4 & 26 & 1 & 56 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 56 & 0 & 110 & 20 \\ 2 & 10 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[3583] $1^{14}3151^2231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 159, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 2 & 1 & 26 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 26 & 0 & 202 & 30 \\ 2 & 4 & -1 & 30 & 4 \end{pmatrix}_{b_0}$$

[3586] $1^{13}41^1321^{15}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 159, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 10 & 9 & 0 & 50 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[3585] $1^{12}3151^2231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 159, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 18 & 3 \\ 11 & 0 & 18 & 114 & 22 \\ 2 & 0 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[3582] $3141^{11}3151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 159, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 5 & 1 & 5 & 2 \\ 5 & 8 & 1 & 10 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 10 & 0 & 8 & 6 \\ 2 & 6 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[3584] $2^13^121^161^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 159, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 11 & 2 \\ 2 & 2 & 0 & 15 & 2 \\ 13 & 11 & 15 & 130 & 22 \\ 2 & 2 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[3587] $1^{11}41321^251^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 164, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 6 & 2 \\ 13 & 6 & 28 & 14 \\ 3 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

[3588] $41^{11}31^34141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 164, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 20 & 2 \\ 4 & 26 & 1 & 124 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 124 & 0 & 578 & 46 \\ 2 & 10 & -1 & 46 & 3 \end{pmatrix}_{b_0}$$

[3589] $1^{14}31^15131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 10 & 15 & 13 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$

[3590] $1^{11}41^1321^{25}51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 167, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 0 & 7 & 2 \\ 14 & 7 & 44 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0}$

[3591] $1^{13}41321^{25}51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 167, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 11 & 8 & 0 & 44 & 14 \\ 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$

[3592] $1^{12}2^33^{21}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 167, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 9 & 7 & 11 & 50 & 14 \\ 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[3593] $1^{14}31^15131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 167, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 15 & 13 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$

[3594] $1^{12}41^1321^{25}51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 167, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 13 & 8 & 0 & 56 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$

[3595] $41^{23}31^{11}4141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 170, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 5 & 1 & 14 & 2 \\ 5 & 38 & 1 & 104 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 104 & 0 & 278 & 32 \\ 2 & 12 & -1 & 32 & 3 \end{pmatrix}_{b_0}$

[3596] $2^33^{12}21^{13}61^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 188, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 10 & 2 \\ 2 & 2 & 0 & 14 & 2 \\ 12 & 10 & 14 & 104 & 20 \\ 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$

[3597] $1^141321^351^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 188, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 0 & 6 & 2 \\ 18 & 6 & 52 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

[3598] $2^33^{12}21^{13}61^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 10 & 8 & 12 & 64 & 16 \\ 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[3599] $1^{14}31^15131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 194, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 1 & 2 & 0 & 16 & 2 \\ 11 & 16 & 16 & 126 & 24 \\ 2 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$

[3600] $1^{13}3151^1231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 194, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 17 & 3 \\ 10 & 0 & 17 & 79 & 19 \\ 2 & 0 & 3 & 19 & 4 \end{pmatrix}_{b_0}$$

[3601] $1^{14}315131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 198, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 16 & 16 & 126 & 24 \\ 2 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[3602] $1^{12}3151^{231}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 198, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 18 & 3 \\ 12 & 0 & 18 & 114 & 22 \\ 2 & 0 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[3603] $2^1321^{12}61^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 199, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 13 & 2 \\ 11 & 9 & 13 & 82 & 18 \\ 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[3604] $41^{11}31^{14}14141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 10 & 2 \\ 4 & 26 & 1 & 60 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 60 & 0 & 130 & 22 \\ 2 & 10 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[3605] $41^{12}23151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 201, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 17 & 3 \\ 9 & 51 & 96 & 15 \\ 17 & 96 & 174 & 28 \\ 3 & 15 & 28 & 4 \end{pmatrix}_{b_0}$$

[3606] $1^{14}41^{13}21^{15}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 204, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.71 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 11 & 11 & 0 & 70 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[3607] $1^{12}41^{13}21^{35}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 204, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.71 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 17 & 9 & 0 & 86 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[3610] $1^{13}3151^{231}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 206, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 22 & 3 \\ 12 & 0 & 22 & 154 & 26 \\ 2 & 0 & 3 & 26 & 4 \end{pmatrix}_{b_0}$$

[3608] $2^1315131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 206, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 1 & 2 & 16 & 3 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 16 & 16 & 126 & 24 \\ 2 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[3611] $2^1315131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 206, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 1 & 2 & 14 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[3609] $1^{14}3151^{231}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 206, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 2 & 1 & 26 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 26 & 0 & 202 & 30 \\ 2 & 4 & -1 & 30 & 4 \end{pmatrix}_{b_0}$$

[3612] $1^{11}41321^{251}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 219, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.66 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 0 & 7 & 2 \\ 15 & 7 & 38 & 16 \\ 3 & 2 & 16 & 4 \end{pmatrix}_{b_0} \\
[3613] & 41^{13}1^{24}141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 221, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 1 & 15 & 2 \\ 4 & 26 & 1 & 92 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 92 & 0 & 314 & 34 \\ 2 & 10 & -1 & 34 & 3 \end{pmatrix}_{b_0} \\
[3615] & 1^{13}41321^{35}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 15 & 9 & 0 & 70 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0} \\
[3614] & 1^{11}41^{13}21^{35}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 0 & 8 & 2 \\ 18 & 8 & 72 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0} \\
[3616] & 51^{12}223141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 7 & 12 & 3 \\ 7 & 7 & 14 & 7 \\ 12 & 14 & 22 & 12 \\ 3 & 7 & 12 & 4 \end{pmatrix}_{b_0} \\
[3617] & 1^{11}41321^{35}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 227, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 0 & 7 & 2 \\ 17 & 7 & 50 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0} \\
[3618] & 1^{14}41321^{25}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 12 & 10 & 0 & 62 & 16 \\ 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0} \\
[3619] & 2^3 3^{14}15131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 234, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 15 & 13 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0} \\
[3620] & 41^{13}1^{11}4141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 246, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 5 & 1 & 10 & 2 \\ 5 & 38 & 1 & 72 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 72 & 0 & 130 & 22 \\ 2 & 12 & -1 & 22 & 3 \end{pmatrix}_{b_0} \\
[3622] & 1^{13}41^{13}21^{25}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 248, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.58 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 14 & 10 & 0 & 74 & 18 \\ 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0} \\
[3621] & 1^{11}41^{13}21^{25}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 248, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.58 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 0 & 8 & 2 \\ 16 & 8 & 50 & 18 \\ 3 & 2 & 18 & 4 \end{pmatrix}_{b_0} \\
[3624] & 41^{13}231^{13}51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 9 & 27 & 3 \\ 9 & 51 & 144 & 15 \\ 27 & 144 & 394 & 42 \\ 3 & 15 & 42 & 4 \end{pmatrix}_{b_0} \\
[3623] & 1^{13}3151^{32}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 27 & 3 \\ 15 & 0 & 27 & 259 & 33 \\ 2 & 0 & 3 & 33 & 4 \end{pmatrix}_{b_0} \\
[3625] & 41^{12}2321^{15}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 260, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 1 & 12 & 2 \\ 4 & 26 & 1 & 76 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 76 & 0 & 210 & 28 \\ 2 & 10 & -1 & 28 & 3 \end{pmatrix}_{b_0} \\
[3626] & 1^{13}3151^2231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 263, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 22 & 3 \\ 13 & 0 & 22 & 154 & 26 \\ 2 & 0 & 3 & 26 & 4 \end{pmatrix}_{b_0} \\
[3628] & 41^{13}232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 1 & 12 & 2 \\ 4 & 26 & 1 & 84 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 84 & 0 & 250 & 30 \\ 2 & 10 & -1 & 30 & 3 \end{pmatrix}_{b_0} \\
[3627] & 41^{13}231\bar{5}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 9 & 24 & 3 \\ 9 & 51 & 144 & 15 \\ 24 & 144 & 394 & 42 \\ 3 & 15 & 42 & 4 \end{pmatrix}_{b_0} \\
[3629] & 1^{11}41321^351^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 304, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.64 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 0 & 8 & 2 \\ 19 & 8 & 60 & 20 \\ 3 & 2 & 20 & 4 \end{pmatrix}_{b_0} \\
[3630] & 1^{14}3151^2231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 314, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 26 & 3 \\ 14 & 0 & 26 & 202 & 30 \\ 2 & 0 & 3 & 30 & 4 \end{pmatrix}_{b_0} \\
[3631] & 1^{13}3151^3231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 318, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 27 & 3 \\ 16 & 0 & 27 & 259 & 33 \\ 2 & 0 & 3 & 33 & 4 \end{pmatrix}_{b_0} \\
[3632] & 1^{13}41^1321^351^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 319, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.61 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 18 & 11 & 0 & 106 & 22 \\ 2 & 2 & 0 & 22 & 3 \end{pmatrix}_{b_0} \\
[3633] & 1^{14}41321^351^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 319, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.61 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 11 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 16 & 11 & 0 & 90 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0} \\
[3635] & 1^{11}41^1321^351^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 327, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 20 & 3 \\ 2 & 0 & 9 & 2 \\ 20 & 9 & 78 & 22 \\ 3 & 2 & 22 & 4 \end{pmatrix}_{b_0} \\
[3634] & 1^{14}41^1321^251^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 327, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 15 & 12 & 0 & 96 & 20 \\ 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0} \\
[3636] & 51^{12}2231^141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 350, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.68 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 7 & 17 & 3 \\ 7 & 7 & 21 & 7 \\ 17 & 21 & 52 & 18 \\ 3 & 7 & 18 & 4 \end{pmatrix}_{b_0} \\
[3637] & 51^{12}2231^141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 7 & 20 & 3 \\ 7 & 7 & 26 & 7 \\ 20 & 26 & 82 & 22 \\ 3 & 7 & 22 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

- [3638] $41^{1^2}1^323151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 26 & 3 \\ 9 & 51 & 144 & 15 \\ 26 & 144 & 394 & 42 \\ 3 & 15 & 42 & 4 \end{pmatrix}_{b_0}$$
- [3639] $41^{1^1}1^323151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 369, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 25 & 3 \\ 9 & 51 & 144 & 15 \\ 25 & 144 & 394 & 42 \\ 3 & 15 & 42 & 4 \end{pmatrix}_{b_0}$$
- [3640] $51^{1^2}223141^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 374, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.68 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 7 & 15 & 3 \\ 7 & 7 & 19 & 7 \\ 15 & 19 & 40 & 16 \\ 3 & 7 & 16 & 4 \end{pmatrix}_{b_0}$$
- [3641] $1^{1^4}3151^3231^{1^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 383, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 31 & 3 \\ 17 & 0 & 31 & 317 & 37 \\ 2 & 0 & 3 & 37 & 4 \end{pmatrix}_{b_0}$$
- [3642] $41^{1^4}23151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 396, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 36 & 3 \\ 9 & 51 & 192 & 15 \\ 36 & 192 & 702 & 56 \\ 3 & 15 & 56 & 4 \end{pmatrix}_{b_0}$$
- [3643] $51^{1^3}223141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 399, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 7 & 18 & 3 \\ 7 & 7 & 21 & 7 \\ 18 & 21 & 52 & 18 \\ 3 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$
- [3644] $41^{1^3}2321^{2^5}51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 404, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 20 & 2 \\ 4 & 26 & 1 & 124 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 124 & 0 & 570 & 46 \\ 2 & 10 & -1 & 46 & 3 \end{pmatrix}_{b_0}$$
- [3645] $1^{1^1}41^1321^351^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 428, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 22 & 3 \\ 2 & 0 & 10 & 2 \\ 22 & 10 & 88 & 24 \\ 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$
- [3646] $1^{1^4}41^1321^351^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 432, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 19 & 2 \\ 1 & 0 & 1 & 13 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 19 & 13 & 0 & 130 & 24 \\ 2 & 2 & 0 & 24 & 3 \end{pmatrix}_{b_0}$$
- [3648] $41^{1^4}232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 444, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 16 & 2 \\ 4 & 26 & 1 & 112 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 112 & 0 & 446 & 40 \\ 2 & 10 & -1 & 40 & 3 \end{pmatrix}_{b_0}$$
- [3647] $41^{1^4}23151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 444, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 32 & 3 \\ 9 & 51 & 192 & 15 \\ 32 & 192 & 702 & 56 \\ 3 & 15 & 56 & 4 \end{pmatrix}_{b_0}$$
- [3649] $41^{1^4}2321^{3^5}51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 516, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 28 & 2 \\ 4 & 26 & 1 & 172 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 172 & 0 & 1106 & 64 \\ 2 & 10 & -1 & 64 & 3 \end{pmatrix}_{b_0}$$
- [3650] $41^{1^3}2321^{1^5}51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 524, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 16 & 2 \\ 4 & 26 & 1 & 104 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 104 & 0 & 390 & 38 \\ 2 & 10 & -1 & 38 & 3 \end{pmatrix}_{b_0}$$

[3651] $41^{14}23151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 561, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 35 & 3 \\ 9 & 51 & 192 & 15 \\ 35 & 192 & 702 & 56 \\ 3 & 15 & 56 & 4 \end{pmatrix}_{b_0}$$

[3652] $41^{14}23151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 585, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 33 & 3 \\ 9 & 51 & 192 & 15 \\ 33 & 192 & 702 & 56 \\ 3 & 15 & 56 & 4 \end{pmatrix}_{b_0}$$

[3653] $41^{14}23151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 624, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 34 & 3 \\ 9 & 51 & 192 & 15 \\ 34 & 192 & 702 & 56 \\ 3 & 15 & 56 & 4 \end{pmatrix}_{b_0}$$

[3654] $51^{13}2231^{14}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 665, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.69 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 7 & 23 & 3 \\ 7 & 7 & 28 & 7 \\ 23 & 28 & 92 & 24 \\ 3 & 7 & 24 & 4 \end{pmatrix}_{b_0}$$

[3655] $51^{13}223141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 689, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.69 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 7 & 21 & 3 \\ 7 & 7 & 26 & 7 \\ 21 & 26 & 76 & 22 \\ 3 & 7 & 22 & 4 \end{pmatrix}_{b_0}$$

[3656] $51^{13}2231^{14}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 815, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 7 & 26 & 3 \\ 7 & 7 & 33 & 7 \\ 26 & 33 & 128 & 28 \\ 3 & 7 & 28 & 4 \end{pmatrix}_{b_0}$$

[3657] $41^{14}2321^{15}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 836, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 20 & 2 \\ 4 & 26 & 1 & 132 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 132 & 0 & 626 & 48 \\ 2 & 10 & -1 & 48 & 3 \end{pmatrix}_{b_0}$$

[3658] $41^{14}2321^{25}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 860, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 24 & 2 \\ 4 & 26 & 1 & 152 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 152 & 0 & 846 & 56 \\ 2 & 10 & -1 & 56 & 3 \end{pmatrix}_{b_0}$$

$T = 11$ [3660] $1^123151321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 4.0$

[3661] $1^131513221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 4.0$

[3659] $1^1231513221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = 0, (n_+, n_-, n_0) = (1, 11, 2), c_{\text{ext}} = 4.0$

[3668] $1^1414141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 6 & 5 & 6 \\ 2 & 5 & 3 & 5 \\ 3 & 6 & 5 & 6 \end{pmatrix}_{b_0}$$

[3669] $1^13151321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 5 & 3 \\ 2 & 0 & 5 & 7 & 7 \\ 2 & 0 & 3 & 7 & 5 \end{pmatrix}_{b_0}$$

[3670] $1^43151321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 16 & 3 \\ 8 & 0 & 16 & 106 & 24 \\ 2 & 0 & 3 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[3663] \ 1^1 \bar{6} 12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[3664] \ 1^1 \bar{6}^{1321} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[3665] \ 1^1 \bar{6}^{1321} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -56, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[3666] \ 1^1 \bar{4} 141 \bar{4} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 6 & 5 & 6 \\ 3 & 5 & 3 & 5 \\ 3 & 6 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[3667] \ 1^1 \bar{6}^{131} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -55, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[3672] \ 1^1 \bar{5}^{131} 131^1 \bar{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 4 & 5 \\ 3 & 4 & 2 & 4 \\ 3 & 5 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[3673] \ 1^1 \bar{3} 15^{1321} \bar{1}^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 4 & 1 & 2 & 4 \\ 2 & 4 & 2 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[3674] \ 1^1 \bar{5}^{1321} 151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -74, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -2 & 0 \\ 3 & 5 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[3671] \ 1^1 \bar{5}^{131} 151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -73, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -2 & 0 \\ 3 & 5 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[3681] \ 1^1 \bar{3} 1413221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 11, \Delta = -6, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[3682] \ 1^1 \bar{7} 12231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -75, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

[3683] $1^1 5 131 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 13, \Delta = -73, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 4 & 5 \\ 2 & 4 & 2 & 4 \\ 3 & 5 & 4 & 5 \end{pmatrix}_{b_0}$

[3675] $1^1 6 123141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 13, \Delta = -76, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0}$

[3676] $1^{11} 23141321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 11, \Delta = -7, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0}$

[3677] $1^1 5 132151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 13, \Delta = -74, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[3678] $1^{11} 31413221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 11, \Delta = -6, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0}$

[3679] $31^1 316 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 13, \Delta = -63, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 2 & 4 & 6 & 6 \\ 2 & 2 & 2 & 6 & 4 \end{pmatrix}_{b_0}$

[3680] $21^1 7 1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 13, \Delta = -75, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[3692] $2^1 3 151321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 7 & 7 \\ 2 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$

[3693] $2^1 3 151321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 1 & 2 & 8 & 3 \\ 1 & 2 & 1 & 10 & 3 \\ 6 & 8 & 10 & 34 & 14 \\ 2 & 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$

[3691] $1^4 31 5 1321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 12 & 3 \\ 8 & 0 & 12 & 78 & 20 \\ 2 & 0 & 3 & 20 & 5 \end{pmatrix}_{b_0}$

[3684] $2^1 3 151321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 1 & 2 & 16 & 3 \\ 1 & 2 & 1 & 20 & 3 \\ 12 & 16 & 20 & 142 & 28 \\ 2 & 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}$

[3685] $1^1 31 5 1321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$

$$\begin{aligned}
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 5 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 7 & 7 \\ 2 & 4 & -1 & 7 & 5 \end{pmatrix}_{b_0} \\
[3686] & \begin{matrix} 1^{13} \\ 2^3 31 151321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \end{matrix} \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 1 & 2 & 12 & 3 \\ 1 & 2 & 1 & 15 & 3 \\ 9 & 12 & 15 & 79 & 21 \\ 2 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0} \\
[3687] & \begin{matrix} 1^1 \\ 1^2 31 5 1321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \end{matrix} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 2 & 1 & 10 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 10 & 0 & 34 & 14 \\ 2 & 4 & -1 & 14 & 5 \end{pmatrix}_{b_0} \\
[3688] & \begin{matrix} 1^1 \\ 1^3 31 5 1321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \end{matrix} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 2 & 1 & 15 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 15 & 0 & 79 & 21 \\ 2 & 4 & -1 & 21 & 5 \end{pmatrix}_{b_0} \\
[3689] & \begin{matrix} 1^1 \\ 1^4 31 5 1321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \end{matrix} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 2 & 1 & 20 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 20 & 0 & 142 & 28 \\ 2 & 4 & -1 & 28 & 5 \end{pmatrix}_{b_0} \\
[3690] & \begin{matrix} 1^{11} \\ 1^2 31 5 1321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \end{matrix} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 6 & 3 \\ 4 & 0 & 6 & 18 & 10 \\ 2 & 0 & 3 & 10 & 5 \end{pmatrix}_{b_0} \\
[3694] & \begin{matrix} 1^1 \\ 31 5 13151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \end{matrix} \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[3697] & \begin{matrix} 1^{11} \\ 1^3 31 5 1321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \end{matrix} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 11 & 3 \\ 6 & 0 & 11 & 53 & 17 \\ 2 & 0 & 3 & 17 & 5 \end{pmatrix}_{b_0} \\
[3696] & \begin{matrix} 1^1 \\ 2^3 21^1 6131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \end{matrix} \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 1 & 2 & 9 & 3 \\ 2 & 2 & 0 & 6 & 2 \\ 6 & 9 & 6 & 34 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0} \\
[3695] & \begin{matrix} 1^{11} \\ 1^1 314141^{13} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \end{matrix} \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 1 & 5 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 5 & 0 & 6 & 6 \\ 2 & 4 & 0 & 6 & 5 \end{pmatrix}_{b_0} \\
[3705] & \begin{matrix} 1^2 \\ 1^{11} 31 5 1321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \end{matrix} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 4 & 4 & 5 & 7 & 7 \\ 2 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0} \\
[3706] & \begin{matrix} 1^{11} 231 \\ 1^1 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \end{matrix} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 2 & 4 & 5 & 7 & 7 \\ 2 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$$

[3698] $1^{11} 5^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 5 & 4 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 5 & 4 & 7 & 7 \\ 2 & 4 & 2 & 7 & 5 \end{pmatrix}_{b_0}$$

[3699] $1^{11} 23151231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & -1 & 5 & 5 \end{pmatrix}_{b_0}$$

[3700] $1^{11} 23215131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & -1 & 5 & 5 \end{pmatrix}_{b_0}$$

[3701] $1^{11} 23151231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 5 & 4 \\ 2 & 0 & 5 & 3 & 5 \\ 2 & 0 & 4 & 5 & 5 \end{pmatrix}_{b_0}$$

[3702] $1^{11} 23215131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 5 & 4 \\ 2 & 0 & 5 & 3 & 5 \\ 2 & 0 & 4 & 5 & 5 \end{pmatrix}_{b_0}$$

[3703] $1^{11} 13151321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 2 & 5 & 4 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 5 & 4 & 7 & 7 \\ 2 & 4 & 2 & 7 & 5 \end{pmatrix}_{b_0}$$

[3704] $1^{11} 31414131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 4 & 1 & 3 & 5 \\ 2 & 4 & 0 & 5 & 5 \end{pmatrix}_{b_0}$$

[3707] $2^1 3^{11} 216131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

[3709] $2^1 3^{11} 216131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

[3708] $1^{11} 13151321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 7 & 7 \\ 2 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$$

[3710] $31^1 6^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -63, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 \\ 2 & 2 & 4 & 2 & 4 \\ 2 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[3711] $1^{11}31^1414131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 2 & 1 & 9 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 9 & 1 & 23 & 11 \\ 2 & 4 & 0 & 11 & 5 \end{pmatrix}_{b_0}$

[3712] $1^{12}31^151321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 1 & 6 & 3 \\ 5 & 7 & 6 & 18 & 10 \\ 2 & 3 & 3 & 10 & 5 \end{pmatrix}_{b_0}$

[3714] $1^{12}23151^1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 3 & 1 & 15 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 15 & 0 & 53 & 17 \\ 2 & 5 & -1 & 17 & 5 \end{pmatrix}_{b_0}$

[3715] $1^{12}31^151321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 1 & 6 & 3 \\ 4 & 7 & 6 & 18 & 10 \\ 2 & 3 & 3 & 10 & 5 \end{pmatrix}_{b_0}$

[3713] $1^{11}23151^1231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 10 & 4 \\ 5 & 0 & 10 & 28 & 12 \\ 2 & 0 & 4 & 12 & 5 \end{pmatrix}_{b_0}$

[3716] $2^13^1216131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 14 & 3 \\ 2 & 2 & 0 & 10 & 2 \\ 8 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$

[3718] $1^{11}31^151321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 2 & 2 & 9 & 4 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 9 & 5 & 21 & 11 \\ 2 & 4 & 2 & 11 & 5 \end{pmatrix}_{b_0}$

[3717] $1^{11}31^151321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 2 & 2 & 9 & 4 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 9 & 5 & 21 & 11 \\ 2 & 4 & 2 & 11 & 5 \end{pmatrix}_{b_0}$

[3719] $2^13^1216131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$

[3720] $31^15^113151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$

[3721] $1^{11}2321^15131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 12 & 4 \\ 6 & 0 & 12 & 38 & 14 \\ 2 & 0 & 4 & 14 & 5 \end{pmatrix}_{b_0}$

[3726] $1^{12}31^151321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 1 & 10 & 3 \\ 5 & 8 & 10 & 34 & 14 \\ 2 & 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$$

$$[3722] \quad 1^{1^3} 31^1 51321^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5 \\ T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 1 & 10 & 3 \\ 7 & 8 & 10 & 34 & 14 \\ 2 & 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$$

$$[3723] \quad 1^{1^2} 2321^1 5131^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 3 & 1 & 17 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 17 & 0 & 67 & 19 \\ 2 & 5 & -1 & 19 & 5 \end{pmatrix}_{b_0}$$

$$[3724] \quad 1^{1^3} 31^1 51321^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 2 & 2 & 13 & 4 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 13 & 6 & 43 & 15 \\ 2 & 4 & 2 & 15 & 5 \end{pmatrix}_{b_0}$$

$$[3725] \quad 1^{1^2} 31^1 51321^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\ T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 2 & 2 & 13 & 4 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 13 & 6 & 43 & 15 \\ 2 & 4 & 2 & 15 & 5 \end{pmatrix}_{b_0}$$

$$[3727] \quad 2^1 3^1 21^1 6131^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -35, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0 \\ T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 2 & 2 & 0 & 11 & 2 \\ 10 & 16 & 11 & 101 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[3728] \quad 1^{1^2} 31^1 51321^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5 \\ T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 1 & 10 & 3 \\ 6 & 8 & 10 & 34 & 14 \\ 2 & 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$$

$$[3729] \quad 31^1 513151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -46, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5 \\ T_{\min} = 13.74 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 14 & 8 \\ 3 & 5 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[3730] \quad 2^1 3^1 21^1 6131^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -48, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5 \\ T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 1 & 2 & 9 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 9 & 10 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[3732] \quad 1^{1^2} 31^1 4141^1 31^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5 \\ T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 2 & 1 & 14 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 14 & 1 & 55 & 17 \\ 2 & 4 & 0 & 17 & 5 \end{pmatrix}_{b_0}$$

$$[3731] \quad 1^{1^3} 31^1 51321^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8 \\ T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 1 & 11 & 3 \\ 6 & 11 & 11 & 53 & 17 \\ 2 & 3 & 3 & 17 & 5 \end{pmatrix}_{b_0}$$

$$[3733] \quad 1^{1^2} 31^1 51321^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -51, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5 \\ T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 2 & 2 & 14 & 4 \\ 1 & 2 & 0 & 9 & 2 \\ 6 & 14 & 9 & 58 & 18 \\ 2 & 4 & 2 & 18 & 5 \end{pmatrix}_{b_0}$$

[3734] $1^{1^3}3151321^{1^3} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -51, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 2 & 2 & 14 & 4 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 14 & 9 & 58 & 18 \\ 2 & 4 & 2 & 18 & 5 \end{pmatrix}_{b_0}$$

[3735] $1^{1^3}23151^{1^3}231^{1^1} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 3 & 1 & 20 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 20 & 0 & 88 & 22 \\ 2 & 5 & -1 & 22 & 5 \end{pmatrix}_{b_0}$$

[3736] $1^{1^3}3151321^{1^2} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 1 & 11 & 3 \\ 8 & 11 & 11 & 53 & 17 \\ 2 & 3 & 3 & 17 & 5 \end{pmatrix}_{b_0}$$

[3737] $1^{1^2}23151^{1^2}231^{1^2} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 15 & 4 \\ 7 & 0 & 15 & 53 & 17 \\ 2 & 0 & 4 & 17 & 5 \end{pmatrix}_{b_0}$$

[3738] $1^{1^2}3151321^{1^2} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -55, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 1 & 11 & 3 \\ 7 & 11 & 11 & 53 & 17 \\ 2 & 3 & 3 & 17 & 5 \end{pmatrix}_{b_0}$$

[3739] $1^{1^2}3151321^{1^3} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -55, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 2 & 2 & 14 & 4 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 14 & 9 & 58 & 18 \\ 2 & 4 & 2 & 18 & 5 \end{pmatrix}_{b_0}$$

[3740] $2^13216131^{1^1} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -62, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 7 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

[3741] $2^13216131^{1^2} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -62, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 8 & 8 & 30 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[3742] $2^1321^16131^{1^1} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -62, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 1 & 2 & 8 & 3 \\ 1 & 2 & 0 & 7 & 2 \\ 7 & 8 & 7 & 32 & 12 \\ 2 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[3743] $1^{1^3}2321^15131^{1^2} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 3 & 1 & 22 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 22 & 0 & 106 & 24 \\ 2 & 5 & -1 & 24 & 5 \end{pmatrix}_{b_0}$$

[3744] $1^{1^2}3151321^{1^3} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -73, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 1 & 15 & 3 \\ 8 & 12 & 15 & 79 & 21 \\ 2 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0}$$

[3745] $1^{1^4}3151321^{1^3} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -73, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$

$$\begin{aligned}
& T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 1 & 15 & 3 \\ 10 & 12 & 15 & 79 & 21 \\ 2 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0} \\
[3746] & 1^{12} 2321^1 5131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -73, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 17 & 4 \\ 8 & 0 & 17 & 67 & 19 \\ 2 & 0 & 4 & 19 & 5 \end{pmatrix}_{b_0} \\
[3747] & 1^{14} 31^1 51321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 2 & 2 & 18 & 4 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 18 & 10 & 90 & 22 \\ 2 & 4 & 2 & 22 & 5 \end{pmatrix}_{b_0} \\
[3748] & 1^{13} 31^1 513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 13.84 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 5 & 15 & 5 \\ 9 & 15 & 34 & 12 \\ 3 & 5 & 12 & 4 \end{pmatrix}_{b_0} \\
[3749] & 1^{13} 31^1 51321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 1 & 15 & 3 \\ 9 & 12 & 15 & 79 & 21 \\ 2 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0} \\
[3750] & 1^{12} 31^1 51321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 2 & 2 & 18 & 4 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 18 & 10 & 90 & 22 \\ 2 & 4 & 2 & 22 & 5 \end{pmatrix}_{b_0} \\
[3751] & 1^{12} 31^1 51321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -80, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 2 & 2 & 18 & 4 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 18 & 10 & 90 & 22 \\ 2 & 4 & 2 & 22 & 5 \end{pmatrix}_{b_0} \\
[3753] & 1^{12} 31^1 51321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 2 & 2 & 19 & 4 \\ 1 & 2 & 0 & 13 & 2 \\ 9 & 19 & 13 & 113 & 25 \\ 2 & 4 & 2 & 25 & 5 \end{pmatrix}_{b_0} \\
[3752] & 1^{13} 31^1 51321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 2 & 2 & 19 & 4 \\ 1 & 2 & 0 & 13 & 2 \\ 10 & 19 & 13 & 113 & 25 \\ 2 & 4 & 2 & 25 & 5 \end{pmatrix}_{b_0} \\
[3755] & 1^{14} 23151^1 231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -99, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 3 & 1 & 25 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 25 & 0 & 133 & 27 \\ 2 & 5 & -1 & 27 & 5 \end{pmatrix}_{b_0} \\
[3754] & 1^{13} 23151^1 231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -99, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 20 & 4 \\ 9 & 0 & 20 & 88 & 22 \\ 2 & 0 & 4 & 22 & 5 \end{pmatrix}_{b_0} \\
[3756] & 1^{12} 31^1 51321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -102, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 1 & 16 & 3 \\ 9 & 15 & 16 & 106 & 24 \\ 2 & 3 & 3 & 24 & 5 \end{pmatrix}_{b_0}$$

[3757] $1^{14}3151321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -106$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 1 & 16 & 3 \\ 11 & 15 & 16 & 106 & 24 \\ 2 & 3 & 3 & 24 & 5 \end{pmatrix}_{b_0}$$

[3758] $1^{14}3151321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -107$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 1 & 16 & 3 \\ 10 & 15 & 16 & 106 & 24 \\ 2 & 3 & 3 & 24 & 5 \end{pmatrix}_{b_0}$$

[3760] $2^3 3 21^{16}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -58$, $\det = -122$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 1 & 2 & 13 & 3 \\ 1 & 2 & 0 & 14 & 2 \\ 11 & 13 & 14 & 95 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

[3761] $2^3 3 216131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -58$, $\det = -122$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 12 & 12 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[3759] $2^3 3 21^{16}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -58$, $\det = -122$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 8 & 2 \\ 9 & 11 & 8 & 51 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

[3763] $1^{13}23151^{23}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -31$, $\det = -129$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.99 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 25 & 4 \\ 12 & 0 & 25 & 163 & 29 \\ 2 & 0 & 4 & 29 & 5 \end{pmatrix}_{b_0}$$

[3762] $1^{14}3151321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -129$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.99 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 1 & 2 & 1 & 20 & 3 \\ 11 & 16 & 20 & 142 & 28 \\ 2 & 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}$$

[3764] $1^{14}3151321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -132$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 1 & 2 & 1 & 20 & 3 \\ 12 & 16 & 20 & 142 & 28 \\ 2 & 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}$$

[3765] $2^3 3 216131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -58$, $\det = -136$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 12.91 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 1 & 2 & 11 & 3 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 11 & 9 & 49 & 15 \\ 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

[3766] $2^3 3 21^{16}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -58$, $\det = -150$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 1 & 2 & 12 & 3 \\ 1 & 2 & 0 & 11 & 2 \\ 10 & 12 & 11 & 70 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[3767] $1^{14}23151^{23}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -31$, $\det = -204$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 30 & 4 \\ 14 & 0 & 30 & 218 & 34 \\ 2 & 0 & 4 & 34 & 5 \end{pmatrix}_{b_0}$$

[3769] $2 \overset{14}{3} 216131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -206, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 1 & 2 & 16 & 3 \\ 1 & 2 & 0 & 16 & 2 \\ 12 & 16 & 16 & 126 & 24 \\ 2 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[3768] $2 \overset{14}{3} 216131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -206, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 1 & 2 & 14 & 3 \\ 1 & 2 & 0 & 10 & 2 \\ 10 & 14 & 10 & 74 & 18 \\ 2 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[3770] $2 \overset{13}{3} 21^{16}131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -220, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 1 & 2 & 17 & 3 \\ 1 & 2 & 0 & 18 & 2 \\ 14 & 17 & 18 & 157 & 27 \\ 2 & 3 & 2 & 27 & 4 \end{pmatrix}_{b_0}$$

[3771] $2 \overset{14}{3} 216131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -234, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 2 \\ 2 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 11 & 15 & 13 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

[3772] $2 \overset{13}{3} 21^{16}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -248, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 1 & 2 & 15 & 3 \\ 1 & 2 & 0 & 12 & 2 \\ 12 & 15 & 12 & 97 & 21 \\ 2 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

[3773] $2 \overset{13}{3} 21^{16}131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -262, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 1 & 2 & 16 & 3 \\ 1 & 2 & 0 & 15 & 2 \\ 13 & 16 & 15 & 124 & 24 \\ 2 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[3774] $2 \overset{14}{3} 21^{16}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -342, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 2 \\ 2 & 1 & 2 & 18 & 3 \\ 1 & 2 & 0 & 13 & 2 \\ 14 & 18 & 13 & 130 & 24 \\ 2 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[3775] $2 \overset{14}{3} 21^{16}131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -398, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 16 & 2 \\ 2 & 1 & 2 & 20 & 3 \\ 1 & 2 & 0 & 19 & 2 \\ 16 & 20 & 19 & 194 & 30 \\ 2 & 3 & 2 & 30 & 4 \end{pmatrix}_{b_0}$$

[3776] $2 \overset{14}{3} 21^{16}131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -398, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 15 & 2 \\ 2 & 1 & 2 & 19 & 3 \\ 1 & 2 & 0 & 16 & 2 \\ 15 & 19 & 16 & 159 & 27 \\ 2 & 3 & 2 & 27 & 4 \end{pmatrix}_{b_0}$$

$T = 12$ [3780] $1^{14}231 \overset{13}{5} 13221^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$

[3779] $1^{11}231 \overset{12}{5} 13221^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 4.0$

[3778] $1^{12}231 \overset{11}{5} 13221^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 4.5$

[3777] $1^{14}231 \overset{12}{5} 13221^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$

[3785] $1^{\overline{1}131} 6^{11} 13141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 4 & -1 & 3 & 5 \end{pmatrix}_{b_0}$

[3794] $1^{\overline{1}112321^1} 6^{11} 131^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 1 & 6 & 3 \\ 8 & 6 & 30 & 14 \\ 3 & 3 & 14 & 6 \end{pmatrix}_{b_0}$

[3796] $1^{\overline{1}1321} 7^{11} 1231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -55, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0}$

[3795] $1^{\overline{1}132321^1} 6^{11} 131^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 1 & 12 & 3 \\ 16 & 12 & 122 & 28 \\ 3 & 3 & 28 & 6 \end{pmatrix}_{b_0}$

[3786] $1^{\overline{1}131} 6^{11} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -34, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 3 & 1 & 2 & 4 \\ 2 & 3 & 2 & 4 & 5 \end{pmatrix}_{b_0}$

[3787] $1^{\overline{1}131} 6^{11} 13141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -54, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 1 & 3 \\ 2 & 0 & 3 & 3 & 5 \end{pmatrix}_{b_0}$

[3788] $1^{\overline{1}131} 6^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -34, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 4 & 4 & 10 & 8 \\ 2 & 3 & 2 & 8 & 5 \end{pmatrix}_{b_0}$

[3789] $21^{\overline{1}1} 7^{11} 12321^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[3790] $1^{\overline{1}1231} 6^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -34, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 \\ 6 & 7 & 5 & 26 & 12 \\ 2 & 3 & 2 & 12 & 5 \end{pmatrix}_{b_0}$

[3791] $1^{\overline{1}112321} 6^{11} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 3 \\ 4 & 3 & 7 & 7 \\ 3 & 3 & 7 & 6 \end{pmatrix}_{b_0}$

[3792] $1^{\overline{1}1} 5^{11} 1315131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -52, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[3793] $1^{\overline{1}11} 6^{11} 131^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 1 & 12 & 3 \\ 14 & 12 & 110 & 26 \\ 3 & 3 & 26 & 6 \end{pmatrix}_{b_0}$$

$$[3805] \quad 1^1 6 131 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 14, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[3804] \quad 1^1 6 12231 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 14, \Delta = -71, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[3803] \quad 1^1 6 1231 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 14, \Delta = -71, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[3802] \quad 1^1 6 1231 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 14, \Delta = -71, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[3801] \quad 1^1 6 131 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 14, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 2 & 3 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[3800] \quad 1^1 6 131 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 14, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 3 & 2 & 3 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[3799] \quad 21^1 6 131 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 14, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[3798] \quad 1^{11} 6^{2321} 131 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 2 & 2 & 1 & 2 & 4 \\ 2 & 2 & 2 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[3797] \quad 21^1 6 131 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 14, \Delta = -70, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[3825] \quad 1^{11} 314141321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -5, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 2 & 4 & 4 & 6 \\ 2 & 2 & 3 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[3827] \quad 1^{11} 314141321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -5, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0} \\
[3823] \quad & 1^{11}223151231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 3 & 6 & 5 \\ 2 & 0 & 6 & 4 & 6 \\ 2 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0} \\
[3822] \quad & 1^{11}231512321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 3 & 6 & 5 \\ 2 & 0 & 6 & 4 & 6 \\ 2 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0} \\
[3821] \quad & 1^{11}2231 \quad 5^{11}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 2 & 6 & 4 \\ 2 & 4 & 6 & 8 & 8 \\ 2 & 3 & 4 & 8 & 6 \end{pmatrix}_{b_0} \\
[3820] \quad & 1^{12}315^{12}13221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 2 & 6 & 4 \\ 4 & 4 & 6 & 8 & 8 \\ 2 & 3 & 4 & 8 & 6 \end{pmatrix}_{b_0} \\
[3819] \quad & 1^1315^{11}13221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 6 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 8 & 8 \\ 2 & 5 & -1 & 8 & 6 \end{pmatrix}_{b_0} \\
[3818] \quad & 1^2315^{11}13221^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 3 & 1 & 12 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 12 & 0 & 38 & 16 \\ 2 & 5 & -1 & 16 & 6 \end{pmatrix}_{b_0} \\
[3817] \quad & 1^3315^{11}13221^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 3 & 1 & 18 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 18 & 0 & 88 & 24 \\ 2 & 5 & -1 & 24 & 6 \end{pmatrix}_{b_0} \\
[3816] \quad & 1^4315^{11}13221^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 3 & 1 & 24 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 24 & 0 & 158 & 32 \\ 2 & 5 & -1 & 32 & 6 \end{pmatrix}_{b_0} \\
[3815] \quad & 1^{11}223151231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 4 & 1 & 6 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 4 & 6 \\ 2 & 6 & -1 & 6 & 6 \end{pmatrix}_{b_0} \\
[3814] \quad & 1^{11}2231 \quad 5^{11}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 2 & 6 & 5 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 6 & 4 & 8 & 8 \\ 2 & 5 & 2 & 8 & 6 \end{pmatrix}_{b_0} \\
[3806] \quad & 1^{11}2315^{11}1321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 2 & 5 & 4 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 5 & 5 & 8 & 8 \\ 2 & 4 & 3 & 8 & 6 \end{pmatrix}_{b_0}
\end{aligned}$$

[3807] $1\overset{1}{5}131\overset{1}{6}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -70, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[3808] $221\overset{1}{8}1231\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -74, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[3809] $1\overset{1}{5}1321\overset{1}{6}1\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -71, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

[3810] $1\overset{1}{6}1314141\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -74, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0}$

[3811] $21\overset{1}{8}12231\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -74, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[3826] $1\overset{1}{5}1321\overset{1}{1}\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 5 & 4 \\ 1 & 2 & 1 & 5 & 3 \\ 2 & 5 & 5 & 8 & 8 \\ 2 & 4 & 3 & 8 & 6 \end{pmatrix}_{b_0}$

[3813] $1\overset{1}{6}1232151\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -73, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

[3812] $1\overset{1}{7}1223141\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -74, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0}$

[3824] $1\overset{1}{1}231512321\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 4 & 1 & 6 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 4 & 6 \\ 2 & 6 & -1 & 6 & 6 \end{pmatrix}_{b_0}$

[3828] $1\overset{1}{3}1\overset{1}{5}13221\overset{1}{1}\overset{1}{2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 13 & 4 \\ 6 & 0 & 13 & 57 & 19 \\ 2 & 0 & 4 & 19 & 6 \end{pmatrix}_{b_0}$

[3829] $1\overset{1}{1}\overset{1}{1}2321\overset{1}{6}131\overset{1}{2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 1 & 6 & 3 \\ 6 & 6 & 22 & 12 \\ 3 & 3 & 12 & 6 \end{pmatrix}_{b_0}$

[3830] $1^{\overline{1}^{12}2321^{\overline{1}}}$ $6^{131^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 1 & 9 & 3 \\ 12 & 9 & 67 & 21 \\ 3 & 3 & 21 & 6 \end{pmatrix}_{b_0}$

[3831] $1^{\overline{1}^{11}31513221^{\overline{1}1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 3 \\ 1 & 2 & 2 & 6 & 4 \\ 3 & 4 & 6 & 8 & 8 \\ 2 & 3 & 4 & 8 & 6 \end{pmatrix}_{b_0}$

[3832] $31^{\overline{1}}5132161^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 15, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[3833] $231^{\overline{1}}513161^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 15, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 4 & 2 & 4 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$

[3834] $321^{\overline{1}}613151^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 15, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 5 & 0 & 13 & 7 \\ 2 & 3 & 0 & 7 & 3 \end{pmatrix}_{b_0}$

[3835] $1^{\overline{1}^{11}2321^{\overline{1}}}$ $6^{131^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 1 & 9 & 3 \\ 10 & 9 & 57 & 19 \\ 3 & 3 & 19 & 6 \end{pmatrix}_{b_0}$

[3848] $1^{\overline{1}^{11}2321^{\overline{1}}}$ $6^{131^{\overline{1}1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 2 & 5 & 5 \end{pmatrix}_{b_0}$

[3840] $1^{\overline{1}^{11}413151321^{\overline{1}1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 2 & 5 & 4 \\ 4 & 5 & 7 & 7 \\ 3 & 4 & 7 & 6 \end{pmatrix}_{b_0}$

[3839] $1^{\overline{1}^{12}23151321^{\overline{1}1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 2 & 2 & 9 & 4 \\ 1 & 2 & 1 & 6 & 3 \\ 4 & 9 & 6 & 22 & 12 \\ 2 & 4 & 3 & 12 & 6 \end{pmatrix}_{b_0}$

[3837] $1^{\overline{1}^{11}321612321^{\overline{1}1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -30, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 2 & 5 & 4 \\ 4 & 5 & 7 & 7 \\ 3 & 4 & 7 & 6 \end{pmatrix}_{b_0}$

[3836] $1^{\overline{1}^{11}413151321^{\overline{1}1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 4 & 2 & 3 & 7 & 5 \\ 2 & 2 & 2 & 7 & 5 \end{pmatrix}_{b_0}$

[3841] $1^{\overline{1}^{11}31513221^{\overline{1}2}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 3 & 2 & 11 & 5 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 11 & 5 & 25 & 13 \\ 2 & 5 & 2 & 13 & 6 \end{pmatrix}_{b_0}$$

$$[3842] \ 1^{11}413151321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[3843] \ 1^{11}2321^{11}6^{11}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 2 & 5 & 4 \\ 4 & 5 & 7 & 7 \\ 3 & 4 & 7 & 6 \end{pmatrix}_{b_0}$$

$$[3844] \ 1^{11}2321^{11}6^{11}2^{11}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 2 & 7 & 4 \\ 6 & 7 & 19 & 11 \\ 3 & 4 & 11 & 6 \end{pmatrix}_{b_0}$$

$$[3845] \ 1^{12}231^{12}5^{12}1321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 2 & 2 & 9 & 4 \\ 1 & 2 & 1 & 6 & 3 \\ 5 & 9 & 6 & 22 & 12 \\ 2 & 4 & 3 & 12 & 6 \end{pmatrix}_{b_0}$$

$$[3846] \ 1^{11}2321^{11}6^{11}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 4 & 2 \\ 3 & 2 & 4 & 7 & 7 \\ 2 & 2 & 2 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[3847] \ 1^{12}223151^{12}231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 4 & 1 & 17 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 17 & 0 & 57 & 19 \\ 2 & 6 & -1 & 19 & 6 \end{pmatrix}_{b_0}$$

$$[3838] \ 1^{11}321612321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 2 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[3849] \ 1^{11}31^{11}5^{11}13221^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 3 & 2 & 11 & 5 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 11 & 5 & 25 & 13 \\ 2 & 5 & 2 & 13 & 6 \end{pmatrix}_{b_0}$$

$$[3850] \ 1^{11}2321^{11}6^{11}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 2 & 6 & 4 \\ 5 & 6 & 11 & 9 \\ 3 & 4 & 9 & 6 \end{pmatrix}_{b_0}$$

$$[3851] \ 1^{11}2321^{11}6^{11}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 14, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 1 & 12 & 3 \\ 14 & 12 & 106 & 26 \\ 3 & 3 & 26 & 6 \end{pmatrix}_{b_0}$$

$$[3852] \ 1^{12}31^{12}5^{12}13221^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 2 & 12 & 4 \\ 7 & 8 & 12 & 38 & 16 \\ 2 & 3 & 4 & 16 & 6 \end{pmatrix}_{b_0}$$

$$[3853] \ 1^{12}31^{12}5^{12}13221^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 2 & 12 & 4 \\ 5 & 8 & 12 & 38 & 16 \\ 2 & 3 & 4 & 16 & 6 \end{pmatrix}_{b_0}$$

$$[3854] \ 1^{11}31513221^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 3 & 2 & 16 & 5 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 16 & 6 & 52 & 18 \\ 2 & 5 & 2 & 18 & 6 \end{pmatrix}_{b_0}$$

$$[3855] \ 1^{11}31513221^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 3 & 2 & 16 & 5 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 16 & 6 & 52 & 18 \\ 2 & 5 & 2 & 18 & 6 \end{pmatrix}_{b_0}$$

$$[3858] \ 1^{11}41^{13}151321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 5 & 4 & 3 & 13 & 9 \\ 2 & 2 & 2 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[3857] \ 1^{12}31513221^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 8 & 3 \\ 1 & 2 & 2 & 12 & 4 \\ 6 & 8 & 12 & 38 & 16 \\ 2 & 3 & 4 & 16 & 6 \end{pmatrix}_{b_0}$$

$$[3856] \ 1^21^{11}2321^{12}6^{13}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 2 & 10 & 4 \\ 7 & 10 & 32 & 14 \\ 3 & 4 & 14 & 6 \end{pmatrix}_{b_0}$$

$$[3861] \ 1^1321^{16}1612321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 19, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 4 & 2 \\ 5 & 4 & 4 & 18 & 10 \\ 2 & 2 & 2 & 10 & 5 \end{pmatrix}_{b_0}$$

$$[3860] \ 1^{11}2321^{12}6^{13}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 19, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 2 & 9 & 4 \\ 6 & 9 & 22 & 12 \\ 3 & 4 & 12 & 6 \end{pmatrix}_{b_0}$$

$$[3859] \ 1^{11}321^{16}1612321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 19, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 2 & 9 & 4 \\ 7 & 9 & 27 & 13 \\ 3 & 4 & 13 & 6 \end{pmatrix}_{b_0}$$

$$[3864] \ 1^21^{12}2321^{12}6^{13}131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 5 & 2 \\ 5 & 4 & 5 & 21 & 11 \\ 2 & 2 & 2 & 11 & 5 \end{pmatrix}_{b_0}$$

$$[3863] \ 1^141^{13}151321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 2 & 8 & 4 \\ 7 & 8 & 20 & 12 \\ 3 & 4 & 12 & 6 \end{pmatrix}_{b_0}$$

$$[3862] \ 1^{11}2321^{12}6^{13}131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 4 & 2 \\ 4 & 4 & 4 & 13 & 9 \\ 2 & 2 & 2 & 9 & 5 \end{pmatrix}_{b_0}$$

[3867] $1^{13}231\overset{13}{5}1321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -2$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 2 & 2 & 14 & 4 \\ 1 & 2 & 1 & 11 & 3 \\ 8 & 14 & 11 & 62 & 20 \\ 2 & 4 & 3 & 20 & 6 \end{pmatrix}_{b_0}$

[3868] $1^{13}231\overset{1}{5}1321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -2$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 2 & 2 & 14 & 4 \\ 1 & 2 & 1 & 11 & 3 \\ 6 & 14 & 11 & 62 & 20 \\ 2 & 4 & 3 & 20 & 6 \end{pmatrix}_{b_0}$

[3869] $1^{12}31\overset{1}{5}13221^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -1$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 3 & 2 & 17 & 5 \\ 1 & 2 & 0 & 9 & 2 \\ 6 & 17 & 9 & 67 & 21 \\ 2 & 5 & 2 & 21 & 6 \end{pmatrix}_{b_0}$

[3865] $1^{13}223151^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -2$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 4 & 1 & 23 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 23 & 0 & 97 & 25 \\ 2 & 6 & -1 & 25 & 6 \end{pmatrix}_{b_0}$

[3866] $1^{12}223151^{12}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -2$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 3 & 17 & 5 \\ 7 & 0 & 17 & 57 & 19 \\ 2 & 0 & 5 & 19 & 6 \end{pmatrix}_{b_0}$

[3870] $1^{13}231\overset{12}{5}1321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -2$, $\det = 27$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 2 & 2 & 14 & 4 \\ 1 & 2 & 1 & 11 & 3 \\ 7 & 14 & 11 & 62 & 20 \\ 2 & 4 & 3 & 20 & 6 \end{pmatrix}_{b_0}$

[3872] $321\overset{11}{6}13151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 15$, $\Delta = -79$, $\det = 28$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.57 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 3 \end{pmatrix}_{b_0}$

[3871] $31\overset{1}{5}1321^{16}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 15$, $\Delta = -79$, $\det = 28$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.57 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 6 & 6 & 6 & 28 & 10 \\ 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$

[3873] $31\overset{11}{6}123151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 15$, $\Delta = -79$, $\det = 28$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.57 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 3 \end{pmatrix}_{b_0}$

[3875] $1^{12}32161^{12}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -30$, $\det = 32$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 2 & 10 & 4 \\ 8 & 10 & 30 & 14 \\ 3 & 4 & 14 & 6 \end{pmatrix}_{b_0}$

[3874] $1^{11}2321^{11}6^{12}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 32$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 2 & 10 & 4 \\ 8 & 10 & 30 & 14 \\ 3 & 4 & 14 & 6 \end{pmatrix}_{b_0}$

[3884] $1^{11}2321^{12}6^{12}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 13$, $\Delta = -29$, $\det = 33$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 2 & 11 & 4 \\ 9 & 11 & 40 & 16 \\ 3 & 4 & 16 & 6 \end{pmatrix}_{b_0} \\
[3883] & 1^{\overline{1}^{11}2321^1} 6^{13} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 2 & 11 & 4 \\ 9 & 11 & 40 & 16 \\ 3 & 4 & 16 & 6 \end{pmatrix}_{b_0} \\
[3882] & 1^{\overline{1}^{11}2321^1} 41^{13} 13151321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 2 & 11 & 4 \\ 8 & 11 & 35 & 15 \\ 3 & 4 & 15 & 6 \end{pmatrix}_{b_0} \\
[3881] & 1^{\overline{1}^{12}2321^1} 3^{\overline{1}^{13}2321^1} 6^{13} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 2 & 13 & 4 \\ 11 & 13 & 64 & 20 \\ 3 & 4 & 20 & 6 \end{pmatrix}_{b_0} \\
[3879] & 1^{\overline{1}^{13}2321^1} 3^{\overline{1}^{13}2321^1} 6^{13} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 7 & 6 & 6 & 43 & 15 \\ 2 & 2 & 2 & 15 & 5 \end{pmatrix}_{b_0} \\
[3880] & 1^{\overline{1}^{13}2321^1} 2^{\overline{1}^{13}2321^1} 6^{13} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 5 & 2 \\ 6 & 6 & 5 & 31 & 13 \\ 2 & 2 & 2 & 13 & 5 \end{pmatrix}_{b_0} \\
[3878] & 1^{\overline{1}^{11}32161^1} 32161^{12} 2321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 5 & 2 \\ 6 & 4 & 5 & 19 & 11 \\ 2 & 2 & 2 & 11 & 5 \end{pmatrix}_{b_0} \\
[3877] & 1^{\overline{1}^{12}2321^1} 1^{\overline{1}^{12}2321^1} 6^{13} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 7 & 4 & 6 & 27 & 13 \\ 2 & 2 & 2 & 13 & 5 \end{pmatrix}_{b_0} \\
[3876] & 1^{\overline{1}^{12}2321^1} 1^{\overline{1}^{12}2321^1} 6^{13} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 5 & 2 \\ 6 & 4 & 5 & 19 & 11 \\ 2 & 2 & 2 & 11 & 5 \end{pmatrix}_{b_0} \\
[3888] & 1^{\overline{1}^{13}31513221^1} 1^{\overline{1}^{13}31513221^1} 5^{13} 13221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 2 & 18 & 4 \\ 10 & 12 & 18 & 88 & 24 \\ 2 & 3 & 4 & 24 & 6 \end{pmatrix}_{b_0} \\
[3887] & 1^{\overline{1}^{13}31513221^1} 1^{\overline{1}^{13}31513221^1} 5^{13} 13221^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 2 & 18 & 4 \\ 8 & 12 & 18 & 88 & 24 \\ 2 & 3 & 4 & 24 & 6 \end{pmatrix}_{b_0} \\
[3886] & 1^{\overline{1}^{12}2321^1} 2^{\overline{1}^{12}2321^1} 6^{13} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 2 & 12 & 4 \\ 10 & 12 & 50 & 18 \\ 3 & 4 & 18 & 6 \end{pmatrix}_{b_0} \\
[3885] & 1^{\overline{1}^{11}41^1} 3151321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0
\end{aligned}$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 6 & 4 & 6 & 22 & 12 \\ 2 & 2 & 2 & 12 & 5 \end{pmatrix}_{b_0}$$

$$[3890] \quad 1^{1^3} 31513221^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 3 & 2 & 22 & 5 \\ 1 & 2 & 0 & 10 & 2 \\ 9 & 22 & 10 & 106 & 26 \\ 2 & 5 & 2 & 26 & 6 \end{pmatrix}_{b_0}$$

$$[3891] \quad 1^{1^3} 31513221^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 12 & 3 \\ 1 & 2 & 2 & 18 & 4 \\ 9 & 12 & 18 & 88 & 24 \\ 2 & 3 & 4 & 24 & 6 \end{pmatrix}_{b_0}$$

$$[3889] \quad 1^{1^2} 31513221^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 3 & 2 & 22 & 5 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 22 & 10 & 106 & 26 \\ 2 & 5 & 2 & 26 & 6 \end{pmatrix}_{b_0}$$

$$[3892] \quad 3151321^{1^1} 61^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 39, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.92 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 5 & 2 & 1 & 2 \\ 5 & 18 & 6 & 4 & 8 \\ 2 & 6 & 0 & 1 & 2 \\ 1 & 4 & 1 & 0 & 2 \\ 2 & 8 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3893] \quad 321^{1^1} 613151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 40, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 5 & 1 & 2 & 2 \\ 5 & 13 & 1 & 5 & 7 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 1 & 3 \\ 2 & 7 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3895] \quad 1^{1^3} 31513221^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 2 & 19 & 4 \\ 10 & 15 & 19 & 115 & 27 \\ 2 & 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$$

$$[3894] \quad 1^{1^2} 31513221^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 15 & 3 \\ 1 & 2 & 2 & 19 & 4 \\ 9 & 15 & 19 & 115 & 27 \\ 2 & 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$$

$$[3896] \quad 1^{1^3} 223151^{1^1} 231^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 3 & 23 & 5 \\ 9 & 0 & 23 & 97 & 25 \\ 2 & 0 & 5 & 25 & 6 \end{pmatrix}_{b_0}$$

$$[3897] \quad 1^{1^2} 423151321^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 2 & 2 & 19 & 4 \\ 1 & 2 & 1 & 16 & 3 \\ 9 & 19 & 16 & 122 & 28 \\ 2 & 4 & 3 & 28 & 6 \end{pmatrix}_{b_0}$$

$$[3898] \quad 1^{1^3} 423151321^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 2 & 2 & 19 & 4 \\ 1 & 2 & 1 & 16 & 3 \\ 10 & 19 & 16 & 122 & 28 \\ 2 & 4 & 3 & 28 & 6 \end{pmatrix}_{b_0}$$

$$[3899] \quad 1^{1^2} 6^{1^2} 131^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 50, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 2 & 14 & 4 \\ 13 & 14 & 72 & 22 \\ 3 & 4 & 22 & 6 \end{pmatrix}_{b_0}$$

[3900] $1^{1^{12}2321} 6^{131^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 50, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 2 & 14 & 4 \\ 10 & 14 & 57 & 19 \\ 3 & 4 & 19 & 6 \end{pmatrix}_{b_0}$$

[3901] $1^{1^{13}2321} 6^{131^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 51, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 7 & 2 \\ 6 & 6 & 7 & 34 & 14 \\ 2 & 2 & 2 & 14 & 5 \end{pmatrix}_{b_0}$$

[3902] $1^{3^{13}2321} 6^{131^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 51, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 8 & 6 & 9 & 58 & 18 \\ 2 & 2 & 2 & 18 & 5 \end{pmatrix}_{b_0}$$

[3903] $1^{1^{12}2321^1} 6^{131^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 51, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 2 & 13 & 4 \\ 12 & 13 & 58 & 20 \\ 3 & 4 & 20 & 6 \end{pmatrix}_{b_0}$$

[3907] $1^{2^{13}2321^1} 6^{131^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 7 & 2 \\ 9 & 6 & 7 & 49 & 17 \\ 2 & 2 & 2 & 17 & 5 \end{pmatrix}_{b_0}$$

[3906] $1^{1^{13}2321^1} 6^{131^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 8 & 6 & 6 & 37 & 15 \\ 2 & 2 & 2 & 15 & 5 \end{pmatrix}_{b_0}$$

[3908] $1^{12}321^{16}12321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 2 & 14 & 4 \\ 11 & 14 & 62 & 20 \\ 3 & 4 & 20 & 6 \end{pmatrix}_{b_0}$$

[3904] $1^{3^{12}2321} 6^{131^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 2 & 16 & 4 \\ 12 & 16 & 85 & 23 \\ 3 & 4 & 23 & 6 \end{pmatrix}_{b_0}$$

[3905] $1^{11}321^{16}12321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 7 & 2 \\ 7 & 6 & 7 & 39 & 15 \\ 2 & 2 & 2 & 15 & 5 \end{pmatrix}_{b_0}$$

[3909] $1^{12}41^{13}151321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 55, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 7 & 6 & 6 & 32 & 14 \\ 2 & 2 & 2 & 14 & 5 \end{pmatrix}_{b_0}$$

[3910] $1^{2^{12}2321} 6^{131^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 55, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 2 & 15 & 4 \\ 11 & 15 & 69 & 21 \\ 3 & 4 & 21 & 6 \end{pmatrix}_{b_0}$$

[3911] $1^{2^{13}2321} 6^{131^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 55, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 8 & 2 \\ 7 & 6 & 8 & 44 & 16 \\ 2 & 2 & 2 & 16 & 5 \end{pmatrix}_{b_0} \\
[3912] \quad & 1^{11} 41^{13} 13151321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 55, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 2 & 13 & 4 \\ 11 & 13 & 53 & 19 \\ 3 & 4 & 19 & 6 \end{pmatrix}_{b_0} \\
[3913] \quad & 321^{11} 6^{11} 13151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 63, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 7 & 8 & 0 & 26 & 10 \\ 2 & 3 & 0 & 10 & 3 \end{pmatrix}_{b_0} \\
[3914] \quad & 1^{13} 31^{14} 513221^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 64, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 1 & 2 & 2 & 24 & 4 \\ 11 & 16 & 24 & 158 & 32 \\ 2 & 3 & 4 & 32 & 6 \end{pmatrix}_{b_0} \\
[3915] \quad & 1^{14} 31^{14} 513221^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 66, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 2 & 16 & 3 \\ 1 & 2 & 2 & 24 & 4 \\ 12 & 16 & 24 & 158 & 32 \\ 2 & 3 & 4 & 32 & 6 \end{pmatrix}_{b_0} \\
[3916] \quad & 1^{213} 2321^{13} 6^{13} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 10 & 2 \\ 10 & 6 & 10 & 66 & 20 \\ 2 & 2 & 2 & 20 & 5 \end{pmatrix}_{b_0} \\
[3919] \quad & 1^{12} 32161^{12} 2321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 8 & 2 \\ 8 & 6 & 8 & 42 & 16 \\ 2 & 2 & 2 & 16 & 5 \end{pmatrix}_{b_0} \\
[3918] \quad & 1^{12} 321^{12} 61^{12} 2321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 2 & 16 & 4 \\ 14 & 16 & 88 & 24 \\ 3 & 4 & 24 & 6 \end{pmatrix}_{b_0} \\
[3917] \quad & 1^{13} 2321^{13} 6^{13} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 8 & 2 \\ 8 & 6 & 8 & 42 & 16 \\ 2 & 2 & 2 & 16 & 5 \end{pmatrix}_{b_0} \\
[3925] \quad & 1^{214} 2321^{14} 6^{14} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 7 & 2 \\ 10 & 8 & 7 & 63 & 19 \\ 2 & 2 & 2 & 19 & 5 \end{pmatrix}_{b_0} \\
[3924] \quad & 1^{314} 2321^{14} 6^{14} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 8 & 2 \\ 11 & 8 & 8 & 79 & 21 \\ 2 & 2 & 2 & 21 & 5 \end{pmatrix}_{b_0} \\
[3923] \quad & 1^{12} 2321^{12} 6^{12} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 2 & 15 & 4 \\ 12 & 15 & 67 & 21 \\ 3 & 4 & 21 & 6 \end{pmatrix}_{b_0} \\
[3922] \quad & 1^{11} 41^{23} 13151321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0
\end{aligned}$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 10 & 6 & 9 & 59 & 19 \\ 2 & 2 & 2 & 19 & 5 \end{pmatrix}_{b_0}$$

$$[3921] \ 1^2 \overset{13}{6} \ 131^{\overset{13}{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 2 & 17 & 4 \\ 14 & 17 & 99 & 25 \\ 3 & 4 & 25 & 6 \end{pmatrix}_{b_0}$$

$$[3920] \ 1^4 \overset{13}{6} \ 131^{\overset{13}{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 2 & 19 & 4 \\ 16 & 19 & 135 & 29 \\ 3 & 4 & 29 & 6 \end{pmatrix}_{b_0}$$

$$[3926] \ 1^{\overset{13}{13}} 32161^{\overset{12}{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 2 & 15 & 4 \\ 12 & 15 & 67 & 21 \\ 3 & 4 & 21 & 6 \end{pmatrix}_{b_0}$$

$$[3927] \ 1^{\overset{12}{12}} 41^{\overset{1}{1}} 3151321^{\overset{13}{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 8 & 6 & 9 & 47 & 17 \\ 2 & 2 & 2 & 17 & 5 \end{pmatrix}_{b_0}$$

$$[3928] \ 31^{\overset{15}{15}} 6^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 75, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.96 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 & 2 \\ 7 & 19 & 1 & 6 & 9 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 6 & 0 & 1 & 3 \\ 2 & 9 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3934] \ 1^3 \overset{13}{6} \ 131^{\overset{13}{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 76, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 2 & 18 & 4 \\ 15 & 18 & 115 & 27 \\ 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$$

$$[3935] \ 1^{\overset{11}{11}} 41^{\overset{1}{1}} 3151321^{\overset{14}{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 76, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 2 & 16 & 4 \\ 12 & 16 & 74 & 22 \\ 3 & 4 & 22 & 6 \end{pmatrix}_{b_0}$$

$$[3932] \ 1^4 \overset{14}{6} \ 131^{\overset{12}{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 76, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 10 & 2 \\ 10 & 8 & 10 & 90 & 22 \\ 2 & 2 & 2 & 22 & 5 \end{pmatrix}_{b_0}$$

$$[3933] \ 1^{\overset{13}{13}} \overset{1}{6} \ 131^{\overset{12}{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 76, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 9 & 6 & 9 & 52 & 18 \\ 2 & 2 & 2 & 18 & 5 \end{pmatrix}_{b_0}$$

$$[3929] \ 1^3 \overset{12}{6} \ 131^{\overset{13}{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 76, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 2 & 18 & 4 \\ 15 & 18 & 115 & 27 \\ 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$$

$$[3930] \ 1^2 \overset{14}{6} \ 131^{\overset{12}{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 76, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 8 & 2 \\ 8 & 8 & 8 & 58 & 18 \\ 2 & 2 & 2 & 18 & 5 \end{pmatrix}_{b_0}$$

$$[3931] \ 1^{\overset{1}{1}} 41^{\overset{2}{2}} 3151321^{\overset{14}{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 76, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 2 & 16 & 4 \\ 14 & 16 & 86 & 24 \\ 3 & 4 & 24 & 6 \end{pmatrix}_{b_0}$$

$$[3937] \ 1^{1^{12}2321^1} 6^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 80, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 2 & 16 & 4 \\ 13 & 16 & 79 & 23 \\ 3 & 4 & 23 & 6 \end{pmatrix}_{b_0}$$

$$[3936] \ 1^{3^{14}2321} 6^{12} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 80, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 9 & 8 & 9 & 72 & 20 \\ 2 & 2 & 2 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[3938] \ 1^{2^{12}2321^1} 6^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 81, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 2 & 17 & 4 \\ 14 & 17 & 95 & 25 \\ 3 & 4 & 25 & 6 \end{pmatrix}_{b_0}$$

$$[3939] \ 41^{1^{13}1513151^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 84, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 28 & 0 & 26 & 10 \\ 2 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[3942] \ 1^{3^{14}2321} 6^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 98, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 10 & 8 & 12 & 93 & 23 \\ 2 & 2 & 2 & 23 & 5 \end{pmatrix}_{b_0}$$

$$[3941] \ 1^{2^{14}2321} 6^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 98, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 11 & 2 \\ 9 & 8 & 11 & 77 & 21 \\ 2 & 2 & 2 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[3940] \ 1^{1^{13}2321^1} 6^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 98, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 2 & 18 & 4 \\ 16 & 18 & 109 & 27 \\ 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$$

$$[3947] \ 1^{12}321^{16}12321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 99, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 10 & 2 \\ 9 & 8 & 10 & 70 & 20 \\ 2 & 2 & 2 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[3946] \ 1^{13}321^{16}1^{12}321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 99, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 2 & 19 & 4 \\ 15 & 19 & 115 & 27 \\ 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$$

$$[3945] \ 1^{13}321^{16}12321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 99, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 2 & 19 & 4 \\ 15 & 19 & 115 & 27 \\ 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$$

$$[3944] \ 1^{1^{12}2321^1} 6^{13} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 99, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 2 & 19 & 4 \\ 14 & 19 & 106 & 26 \\ 3 & 4 & 26 & 6 \end{pmatrix}_{b_0}$$

$$[3943] \ 1^{3^{12}2321^1} 6^{13} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 99, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 2 & 21 & 4 \\ 16 & 21 & 146 & 30 \\ 3 & 4 & 30 & 6 \end{pmatrix}_{b_0}$$

$$[3949] \ 1^2 \overset{13}{6} \overset{2321}{131} \overset{14}{+} [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 2 & 20 & 4 \\ 15 & 20 & 124 & 28 \\ 3 & 4 & 28 & 6 \end{pmatrix}_{b_0}$$

$$[3948] \ 1^{\overset{13}{1}} \overset{13}{41} \overset{1}{3151321} \overset{13}{+} [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 9 & 8 & 9 & 61 & 19 \\ 2 & 2 & 2 & 19 & 5 \end{pmatrix}_{b_0}$$

$$[3950] \ 1^2 \overset{12}{6} \overset{2321}{131} \overset{14}{+} [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 2 & 20 & 4 \\ 15 & 20 & 124 & 28 \\ 3 & 4 & 28 & 6 \end{pmatrix}_{b_0}$$

$$[3951] \ 1^{\overset{13}{3}} \overset{13}{6} \overset{2321}{131} \overset{13}{+} [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 2 & 20 & 4 \\ 18 & 20 & 145 & 31 \\ 3 & 4 & 31 & 6 \end{pmatrix}_{b_0}$$

$$[3952] \ 1^{\overset{11}{1}} \overset{11}{321} \overset{161}{161} \overset{2321}{14} \overset{14}{+} [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 11 & 8 & 9 & 75 & 21 \\ 2 & 2 & 2 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[3953] \ 1^2 \overset{13}{6} \overset{2321}{131} \overset{13}{+} [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 103, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 2 & 19 & 4 \\ 17 & 19 & 125 & 29 \\ 3 & 4 & 29 & 6 \end{pmatrix}_{b_0}$$

$$[3954] \ 1^{\overset{12}{1}} \overset{12}{41} \overset{2}{3151321} \overset{13}{+} [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 106, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 11 & 8 & 9 & 73 & 21 \\ 2 & 2 & 2 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[3955] \ 1^4 \overset{13}{6} \overset{2321}{131} \overset{14}{+} [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 106, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 2 & 22 & 4 \\ 17 & 22 & 164 & 32 \\ 3 & 4 & 32 & 6 \end{pmatrix}_{b_0}$$

$$[3956] \ 1^{\overset{13}{3}} \overset{13}{6} \overset{2321}{131} \overset{14}{+} [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 107, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 2 & 21 & 4 \\ 16 & 21 & 142 & 30 \\ 3 & 4 & 30 & 6 \end{pmatrix}_{b_0}$$

$$[3957] \ 1^{\overset{14}{3}} \overset{14}{6} \overset{2321}{131} \overset{12}{+} [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 107, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 11 & 2 \\ 12 & 8 & 11 & 98 & 24 \\ 2 & 2 & 2 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[3958] \ 1^{\overset{14}{1}} \overset{14}{6} \overset{2321}{131} \overset{12}{+} [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 107, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 10 & 8 & 9 & 66 & 20 \\ 2 & 2 & 2 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[3959] \ 1^2 \overset{14}{6} \overset{2321}{131} \overset{12}{+} [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 111, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 10 & 2 \\ 11 & 8 & 10 & 80 & 22 \\ 2 & 2 & 2 & 22 & 5 \end{pmatrix}_{b_0}$$

$$[3960] \quad 31 \overset{1}{\overset{151321}{6}} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 112, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 9 & 9 & 0 & 34 & 12 \\ 2 & 3 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[3961] \quad 61 \overset{1}{\overset{11}{22315}} 13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 124, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 6 & 2 & 6 & 2 \\ 6 & 28 & 7 & 30 & 10 \\ 2 & 7 & 0 & 6 & 2 \\ 6 & 30 & 6 & 28 & 10 \\ 2 & 10 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[3962] \quad 1 \overset{12}{\overset{321}{61}} 1^{14} 2321 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 12 & 8 & 12 & 96 & 24 \\ 2 & 2 & 2 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[3964] \quad 1 \overset{3}{\overset{14}{2321}} 6^{13} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 129, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 2 & 23 & 4 \\ 19 & 23 & 184 & 34 \\ 3 & 4 & 34 & 6 \end{pmatrix}_{b_0}$$

$$[3965] \quad 1 \overset{13}{\overset{41^2}{3151321}} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 129, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 12 & 10 & 9 & 91 & 23 \\ 2 & 2 & 2 & 23 & 5 \end{pmatrix}_{b_0}$$

$$[3966] \quad 1 \overset{1}{\overset{14}{2321}} 6^{13} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 129, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 11 & 8 & 12 & 87 & 23 \\ 2 & 2 & 2 & 23 & 5 \end{pmatrix}_{b_0}$$

$$[3963] \quad 1 \overset{2}{\overset{14}{2321}} 6^{13} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 129, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 13 & 2 \\ 12 & 8 & 13 & 103 & 25 \\ 2 & 2 & 2 & 25 & 5 \end{pmatrix}_{b_0}$$

$$[3968] \quad 1 \overset{12}{\overset{41^2}{3151321}} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 132, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 12 & 8 & 12 & 94 & 24 \\ 2 & 2 & 2 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[3967] \quad 1 \overset{4}{\overset{14}{2321}} 6^{13} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 132, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 20 & 3 \\ 2 & 2 & 24 & 4 \\ 20 & 24 & 206 & 36 \\ 3 & 4 & 36 & 6 \end{pmatrix}_{b_0}$$

$$[3969] \quad 1 \overset{13}{\overset{321}{61}} 1^{13} 2321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -30, \det = 133, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 2 & 21 & 4 \\ 18 & 21 & 145 & 31 \\ 3 & 4 & 31 & 6 \end{pmatrix}_{b_0}$$

$$[3970] \quad 1 \overset{1}{\overset{13}{2321}} 6^{13} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 137, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 2 & 21 & 4 \\ 17 & 21 & 136 & 30 \\ 3 & 4 & 30 & 6 \end{pmatrix}_{b_0}$$

[3971] $1^4 \overset{13}{6} \overset{2321}{131} \overset{14}{+ [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2}$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 140, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 20 & 3 \\ 2 & 2 & 24 & 4 \\ 20 & 24 & 202 & 36 \\ 3 & 4 & 36 & 6 \end{pmatrix}_{b_0}$

[3972] $1^2 \overset{13}{6} \overset{2321}{131} \overset{14}{+ [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2}$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 144, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 18 & 3 \\ 2 & 2 & 22 & 4 \\ 18 & 22 & 154 & 32 \\ 3 & 4 & 32 & 6 \end{pmatrix}_{b_0}$

[3973] $1^3 \overset{13}{6} \overset{2321}{131} \overset{14}{+ [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2}$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 145, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 2 & 23 & 4 \\ 19 & 23 & 176 & 34 \\ 3 & 4 & 34 & 6 \end{pmatrix}_{b_0}$

[3974] $1^2 \overset{14}{6} \overset{2321}{131} \overset{14}{+ [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2}$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 166, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 21 & 3 \\ 2 & 2 & 24 & 4 \\ 21 & 24 & 196 & 36 \\ 3 & 4 & 36 & 6 \end{pmatrix}_{b_0}$

[3975] $1^{\textcolor{red}{12}} 41^3 \overset{3151321}{14} \overset{14}{+ [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2}$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 167, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 15 & 10 & 12 & 132 & 28 \\ 2 & 2 & 2 & 28 & 5 \end{pmatrix}_{b_0}$

[3977] $1^4 \overset{14}{6} \overset{2321}{131} \overset{14}{+ [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2}$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 170, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 23 & 3 \\ 2 & 2 & 26 & 4 \\ 23 & 26 & 244 & 40 \\ 3 & 4 & 40 & 6 \end{pmatrix}_{b_0}$

[3976] $1^{\textcolor{red}{14}} 321^{\textcolor{red}{16}} 1^{\textcolor{red}{12}} 2321^{\textcolor{red}{13}} \overset{14}{+ [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2}$ $T = 12, T_{\max} = 13, \Delta = -30, \det = 170, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 2 & 24 & 4 \\ 19 & 24 & 178 & 34 \\ 3 & 4 & 34 & 6 \end{pmatrix}_{b_0}$

[3978] $1^3 \overset{14}{6} \overset{2321}{131} \overset{14}{+ [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2}$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 171, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 22 & 3 \\ 2 & 2 & 25 & 4 \\ 22 & 25 & 218 & 38 \\ 3 & 4 & 38 & 6 \end{pmatrix}_{b_0}$

[3979] $1^{\textcolor{red}{13}} 41^2 \overset{3151321}{14} \overset{14}{+ [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2}$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 175, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 13 & 10 & 12 & 112 & 26 \\ 2 & 2 & 2 & 26 & 5 \end{pmatrix}_{b_0}$

[3981] $1^{\textcolor{red}{13}} 41^3 \overset{3151321}{14} \overset{14}{+ [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2}$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 204, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 16 & 12 & 12 & 154 & 30 \\ 2 & 2 & 2 & 30 & 5 \end{pmatrix}_{b_0}$

[3980] $1^{\textcolor{red}{14}} 321^{\textcolor{red}{16}} 1^{\textcolor{red}{12}} 2321^{\textcolor{red}{14}} \overset{14}{+ [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2}$ $T = 12, T_{\max} = 13, \Delta = -30, \det = 204, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 22 & 3 \\ 2 & 2 & 26 & 4 \\ 22 & 26 & 220 & 38 \\ 3 & 4 & 38 & 6 \end{pmatrix}_{b_0}$

[3982] $1^{\textcolor{red}{14}} 41^2 \overset{3151321}{14} \overset{14}{+ [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2}$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 208, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 12 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 14 & 12 & 12 & 134 & 28 \\ 2 & 2 & 2 & 28 & 5 \end{pmatrix}_{b_0}$

[3983] $3221^{\textcolor{red}{11}} 613151^{\textcolor{red}{1}} \overset{14}{+ [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2}$ $T = 12, T_{\max} = 15, \Delta = -79, \det = 263, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.65 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 7 & 1 & 7 & 2 \\ 7 & 26 & 1 & 28 & 10 \\ 1 & 1 & -2 & 0 & 0 \\ 7 & 28 & 0 & 26 & 10 \\ 2 & 10 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[3984] $3161^{11}223151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 463, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.63 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 9 & 1 & 9 & 2 \\ 9 & 34 & 1 & 36 & 12 \\ 1 & 1 & -2 & 0 & 0 \\ 9 & 36 & 0 & 34 & 12 \\ 2 & 12 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

$T = 13$ [3985] $1^{11}7^{12321}12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 14, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 4.0$

[3993] $15^{11}1316^{11}131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 14, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

[3992] $1^{11}8^{13221}1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 14, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0}$$

[3997] $1^{11}6^{12315131}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 14, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

[3998] $221^{11}8^{12321}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 14, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

[3995] $1^{11}32161315^{11}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 14, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$$

[3988] $1^{11}5^{11}131^{11}131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 14, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$$

[3989] $1^{11}6^{1315131}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 14, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

[3990] $1^{11}8^{13221}1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 14, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

[3996] $1^{11}8^{1321}1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 14, \Delta = -54, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 3 & 2 & 2 & 5 \end{pmatrix}_{b_0}$$

[3991] $15^{11}1316^{11}131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 14, \Delta = -49, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$

$$\begin{aligned}
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0} \\
[3994] & 1^1 1^1 3161231 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0} \\
[4005] & 1^1 1^1 712231 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0} \\
[3999] & 1^1 1^1 71231 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 3 & 2 & 3 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 \end{pmatrix}_{b_0} \\
[4000] & 21^1 1^1 71231 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0} \\
[4001] & 21^1 1^1 71231 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \\
[4002] & 1^1 1^1 71231 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 2 & 3 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 \end{pmatrix}_{b_0} \\
[4003] & 1^1 1^1 71231 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 3 & 4 \end{pmatrix}_{b_0} \\
[4004] & 1^1 1^1 712231 5 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \\
[4014] & 1^1 1^1 2321612321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -3, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0} \\
[4013] & 31^1 1^1 32181231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -62, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 2 & 4 & 6 & 6 \\ 2 & 2 & 2 & 6 & 4 \end{pmatrix}_{b_0} \\
[4015] & 1^1 1^1 4131513221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0}
\end{aligned}$$

[4006] $1^{11}3141414131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -3, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 5 & 1 & 4 & 6 \\ 2 & 5 & 0 & 6 & 6 \end{pmatrix}_{b_0}$$

[4012] $1^{11}71232151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -71, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

[4011] $1^{14}41^{13}1513221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 4 & 2 & 4 & 8 & 8 \\ 2 & 2 & 3 & 8 & 6 \end{pmatrix}_{b_0}$$

[4010] $1^{11}4131513221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 2 & 4 & 4 & 6 \\ 2 & 2 & 3 & 6 & 6 \end{pmatrix}_{b_0}$$

[4009] $1^{11}51321^{11}71^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -69, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

[4008] $1^{11}712314141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -73, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

[4007] $1^{11}513221^{11}71^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -69, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

[4016] $2^{14}321712321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 1 & 2 & 16 & 3 \\ 1 & 2 & 1 & 20 & 3 \\ 12 & 16 & 20 & 142 & 28 \\ 2 & 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}$$

[4017] $2^{13}321712321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 1 & 2 & 12 & 3 \\ 1 & 2 & 1 & 15 & 3 \\ 9 & 12 & 15 & 79 & 21 \\ 2 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0}$$

[4018] $2^{12}321712321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 1 & 2 & 8 & 3 \\ 1 & 2 & 1 & 10 & 3 \\ 6 & 8 & 10 & 34 & 14 \\ 2 & 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$$

[4019] $2^{11}321712321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 7 & 7 \\ 2 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$$

[4020] $1^{11}3151315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -31, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$

$$\begin{aligned}
& T_{\min} = 13.89 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & -1 & 5 & 5 \end{pmatrix}_{b_0} \\
[4021] & 1^{14} 41^{13} 1513221^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 3 & 10 & 5 \\ 7 & 10 & 24 & 14 \\ 3 & 5 & 14 & 7 \end{pmatrix}_{b_0} \\
[4022] & 1^{14} 41^{13} 1513221^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 3 & 14 & 5 \\ 8 & 14 & 44 & 18 \\ 3 & 5 & 18 & 7 \end{pmatrix}_{b_0} \\
[4023] & 1^{11} 41^{13} 1513221^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -18, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 6 & 4 & 8 & 26 & 14 \\ 2 & 2 & 3 & 14 & 6 \end{pmatrix}_{b_0} \\
[4024] & 2^3 3^{12} 21812^3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -20, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 5 & 3 & 16 & 5 \\ 2 & 3 & 0 & 7 & 2 \\ 6 & 16 & 7 & 37 & 13 \\ 2 & 5 & 2 & 13 & 4 \end{pmatrix}_{b_0} \\
[4025] & 1^{12} 31^{14} 1414141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = -3, \det = -25, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 3 & 1 & 18 & 5 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 18 & 1 & 71 & 21 \\ 2 & 5 & 0 & 21 & 6 \end{pmatrix}_{b_0} \\
[4026] & 1^{11} 41^{13} 1513221^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 10 & 6 & 12 & 68 & 22 \\ 2 & 2 & 3 & 22 & 6 \end{pmatrix}_{b_0} \\
[4027] & 1^{12} 41^{13} 1513221^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 8 & 6 & 12 & 56 & 20 \\ 2 & 2 & 3 & 20 & 6 \end{pmatrix}_{b_0} \\
[4028] & 1^{11} 41^{13} 1513221^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -38, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 3 & 20 & 5 \\ 12 & 20 & 90 & 26 \\ 3 & 5 & 26 & 7 \end{pmatrix}_{b_0} \\
[4029] & 1^{13} 41^{13} 1513221^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 9 & 8 & 12 & 70 & 22 \\ 2 & 2 & 3 & 22 & 6 \end{pmatrix}_{b_0} \\
[4030] & 2^3 3^{14} 21812^3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -56, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 2 & 12 & 2 \\ 2 & 5 & 3 & 32 & 5 \\ 2 & 3 & 0 & 14 & 2 \\ 12 & 32 & 14 & 154 & 26 \\ 2 & 5 & 2 & 26 & 4 \end{pmatrix}_{b_0} \\
[4031] & 1^{12} 41^{13} 1513221^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -66, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 12 & 8 & 16 & 110 & 28 \\ 2 & 2 & 3 & 28 & 6 \end{pmatrix}_{b_0} \\
[4032] & 1^{13} 41^{13} 1513221^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -83, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]
\end{aligned}$$

[illegible]

$$\begin{aligned}
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 3 & 2 & 3 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 \end{pmatrix}_{b_0} \\
[4055] & 21^1 \overset{1^1}{8} 1231 \overset{1^1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 16, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 3 & 4 & 11 & 7 \\ 2 & 2 & 2 & 7 & 4 \end{pmatrix}_{b_0} \\
[4062] & 1^1 \overset{1^1}{6} 131513151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 16, \Delta = -72, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[4063] & 1^1 \overset{1^1}{7}^{2321} 12321^{\textcolor{red}{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 14, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0} \\
[4064] & 1^1 \overset{1^1}{8} 122314141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 16, \Delta = -72, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0} \\
[4065] & 1^1 \overset{1^1}{5} 13221 \overset{1^1}{8} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 16, \Delta = -68, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[4066] & 1^1 \overset{1^1}{5} 1321 \overset{1^1}{8} 12 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 16, \Delta = -68, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[4067] & 21^1 \overset{1^1}{8} 1232151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 16, \Delta = -70, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[4068] & 1^1 \overset{1^1}{2} 3151315131^{\textcolor{red}{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 14, \Delta = -3, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 4 & 1 & 6 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 4 & 6 \\ 2 & 6 & -1 & 6 & 6 \end{pmatrix}_{b_0} \\
[4069] & 1^1 \overset{1^1}{2} 3151315131^{\textcolor{red}{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 14, \Delta = -3, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 3 & 6 & 5 \\ 2 & 0 & 6 & 4 & 6 \\ 2 & 0 & 5 & 6 & 6 \end{pmatrix}_{b_0} \\
[4070] & 1^1 \overset{1^1}{7}^{2321} 12321^{\textcolor{red}{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 14, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 3 & 7 & 5 \\ 5 & 7 & 12 & 10 \\ 3 & 5 & 10 & 7 \end{pmatrix}_{b_0} \\
[4071] & 1^1 \overset{1^1}{2} \overset{1^1}{3} \overset{1^1}{21} 12321^{\textcolor{red}{1}3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.8
\end{aligned}$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 1 & 2 & 12 & 3 \\ 1 & 2 & 1 & 15 & 3 \\ 9 & 12 & 15 & 79 & 21 \\ 2 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[4072] \quad 2 \overset{1^4}{3} 2 \overset{1^1}{1} 8 12321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 1 & 20 & 3 \\ 12 & 8 & 20 & 142 & 28 \\ 2 & 2 & 3 & 28 & 5 \end{pmatrix}_{b_0}$$

$$[4073] \quad 2 \overset{1^3}{3} 2 \overset{1^1}{1} 8 12321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.8$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 1 & 15 & 3 \\ 9 & 6 & 15 & 79 & 21 \\ 2 & 2 & 3 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[4074] \quad 2 \overset{1^2}{3} 2 \overset{1^1}{1} 8 12321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 1 & 10 & 3 \\ 6 & 4 & 10 & 34 & 14 \\ 2 & 2 & 3 & 14 & 5 \end{pmatrix}_{b_0}$$

$$[4075] \quad 1 \overset{1^{12}}{2} \overset{2^3}{3} 21 12321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.5$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 1 & 2 & 8 & 3 \\ 1 & 2 & 1 & 10 & 3 \\ 6 & 8 & 10 & 34 & 14 \\ 2 & 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$$

$$[4076] \quad 2 \overset{1^1}{3} 2 \overset{1^1}{1} 8 12321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 2 & 5 & 7 & 7 \\ 2 & 2 & 3 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[4077] \quad 1 \overset{1^{14}}{2} \overset{2^3}{3} 21 12321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 2 \\ 2 & 1 & 2 & 16 & 3 \\ 1 & 2 & 1 & 20 & 3 \\ 12 & 16 & 20 & 142 & 28 \\ 2 & 3 & 3 & 28 & 5 \end{pmatrix}_{b_0}$$

$$[4078] \quad 1 \overset{1^{11}}{2} \overset{2^3}{3} 21 12321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 7 & 7 \\ 2 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[4079] \quad 2 \overset{1^{11}}{3} 2 \overset{1^1}{1} 8 12321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 1 & 2 & 5 & 3 \\ 1 & 2 & 1 & 6 & 3 \\ 4 & 5 & 6 & 13 & 9 \\ 2 & 3 & 3 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[4083] \quad 1 \overset{1^{11}}{3} 1612315131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.89 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 2 & 5 & 4 \\ 4 & 5 & 7 & 7 \\ 3 & 4 & 7 & 6 \end{pmatrix}_{b_0}$$

$$[4080] \quad 1 \overset{1^{11}}{3} 7 12321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 14, \Delta = -1, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 3 & 11 & 5 \\ 6 & 11 & 26 & 14 \\ 3 & 5 & 14 & 7 \end{pmatrix}_{b_0}$$

$$[4081] \quad 1 \overset{1^{11}}{3} 2161315131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.89 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 2 & 5 & 5 \end{pmatrix}_{b_0}$$

[4082] $1^{11}31612315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.89 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 2 & 5 & 5 \end{pmatrix}_{b_0}$$

[4084] $1^{11}32161315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 9$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.89 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 2 & 5 & 4 \\ 4 & 5 & 7 & 7 \\ 3 & 4 & 7 & 6 \end{pmatrix}_{b_0}$$

[4085] $2^{\textcolor{red}{12}}3^{\textcolor{blue}{1}}21812321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 14$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 1 & 2 & 9 & 3 \\ 1 & 2 & 1 & 11 & 3 \\ 7 & 9 & 11 & 44 & 16 \\ 2 & 3 & 3 & 16 & 5 \end{pmatrix}_{b_0}$$

[4086] $1^{11}321^{\textcolor{blue}{1}}61315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 19$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.95 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 2 & 9 & 4 \\ 7 & 9 & 27 & 13 \\ 3 & 4 & 13 & 6 \end{pmatrix}_{b_0}$$

[4087] $1^{\textcolor{blue}{1}}321^{\textcolor{blue}{1}}61315131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 19$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.95 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 4 & 2 \\ 5 & 4 & 4 & 18 & 10 \\ 2 & 2 & 2 & 10 & 5 \end{pmatrix}_{b_0}$$

[4088] $2^{\textcolor{red}{13}}3^{\textcolor{blue}{1}}21812321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 20$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 2 \\ 2 & 1 & 2 & 13 & 3 \\ 1 & 2 & 1 & 16 & 3 \\ 10 & 13 & 16 & 93 & 23 \\ 2 & 3 & 3 & 23 & 5 \end{pmatrix}_{b_0}$$

[4089] $1^{12}7^{2321}12321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 14$, $\Delta = -1$, $\det = 25$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 3 & 17 & 5 \\ 10 & 17 & 66 & 22 \\ 3 & 5 & 22 & 7 \end{pmatrix}_{b_0}$$

[4090] $2^{\textcolor{red}{14}}3^{\textcolor{blue}{1}}21812321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 26$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 2 \\ 2 & 1 & 2 & 17 & 3 \\ 1 & 2 & 1 & 21 & 3 \\ 13 & 17 & 21 & 160 & 30 \\ 2 & 3 & 3 & 30 & 5 \end{pmatrix}_{b_0}$$

[4091] $1^{12}3161^{\textcolor{blue}{1}}2315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 33$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 14.97 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 2 & 11 & 4 \\ 9 & 11 & 40 & 16 \\ 3 & 4 & 16 & 6 \end{pmatrix}_{b_0}$$

[4092] $1^{11}3161^{\textcolor{blue}{1}}2315131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 33$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.97 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 7 & 4 & 6 & 27 & 13 \\ 2 & 2 & 2 & 13 & 5 \end{pmatrix}_{b_0}$$

[4093] $1^{12}3161^{\textcolor{blue}{1}}2315131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 51$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 14.98 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 2 & 13 & 4 \\ 12 & 13 & 58 & 20 \\ 3 & 4 & 20 & 6 \end{pmatrix}_{b_0}$$

[4094] $1^{11}321^{\textcolor{blue}{1}}61315131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 54$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.93 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 7 & 2 \\ 7 & 6 & 7 & 39 & 15 \\ 2 & 2 & 2 & 15 & 5 \end{pmatrix}_{b_0}$$

[4095] $1^{12}321^161315131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 54$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 14.93 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 2 & 14 & 4 \\ 11 & 14 & 62 & 20 \\ 3 & 4 & 20 & 6 \end{pmatrix}_{b_0}$

[4096] $1^{11}3161^12315131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 54$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.93 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 8 & 6 & 6 & 37 & 15 \\ 2 & 2 & 2 & 15 & 5 \end{pmatrix}_{b_0}$

[4097] $1^{12}3161^12315131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 76$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 14.95 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 9 & 6 & 9 & 52 & 18 \\ 2 & 2 & 2 & 18 & 5 \end{pmatrix}_{b_0}$

[4098] $1^{13}3161^12315131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 80$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 14.90 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 2 & 16 & 4 \\ 13 & 16 & 79 & 23 \\ 3 & 4 & 23 & 6 \end{pmatrix}_{b_0}$

[4099] $1^{13}3161^12315131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 98$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 2 & 18 & 4 \\ 16 & 18 & 109 & 27 \\ 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$

[4100] $1^{12}321^161315131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 99$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 14.99 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 10 & 2 \\ 9 & 8 & 10 & 70 & 20 \\ 2 & 2 & 2 & 20 & 5 \end{pmatrix}_{b_0}$

[4101] $1^{13}321^161315131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 99$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 14.99 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 2 & 19 & 4 \\ 15 & 19 & 115 & 27 \\ 3 & 4 & 27 & 6 \end{pmatrix}_{b_0}$

[4102] $1^{14}3161^12315131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 99$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 14.99 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 14 & 3 \\ 2 & 2 & 19 & 4 \\ 14 & 19 & 106 & 26 \\ 3 & 4 & 26 & 6 \end{pmatrix}_{b_0}$

[4103] $1^{12}3161^12315131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 107$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.5$
 $T_{\min} = 14.92 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 9 & 2 \\ 10 & 8 & 9 & 66 & 20 \\ 2 & 2 & 2 & 20 & 5 \end{pmatrix}_{b_0}$

[4104] $1^{13}3161^12315131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 129$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.8$
 $T_{\min} = 14.99 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 0 & 12 & 2 \\ 11 & 8 & 12 & 87 & 23 \\ 2 & 2 & 2 & 23 & 5 \end{pmatrix}_{b_0}$

[4105] $1^{14}3161^12315131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 137$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 14.93 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 2 & 21 & 4 \\ 17 & 21 & 136 & 30 \\ 3 & 4 & 30 & 6 \end{pmatrix}_{b_0}$

$T = 15$ [4107] $1^{12}6^{14}22315131^{13}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 15$, $T_{\max} = 15$, $\Delta = 0$, $\det = 0$, $(n_+, n_-, n_0) = (1, 15, 1)$, $c_{\text{ext}} = 4.5$

[4108] $1^{12}6^{12}22315131^{13}131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 15$, $T_{\max} = 15$, $\Delta = 0$, $\det = 0$, $(n_+, n_-, n_0) = (1, 15, 1)$, $c_{\text{ext}} = 4.5$

[4109] $1^{13}6^{13}22315131^{13}131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2$ $T = 15$, $T_{\max} = 15$, $\Delta = 0$, $\det = 0$, $(n_+, n_-, n_0) = (1, 15, 1)$, $c_{\text{ext}} = 4.8$

[4110] $1^4 \overset{14}{6} \overset{22315131}{131^{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 5.0$

[4111] $1^1 \overset{131}{6} \overset{131612321^1}{131^{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 15, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 4.0$

[4112] $1^1 \overset{13}{6} \overset{1322315131}{131^4} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 4.0$

[4113] $1^1 \overset{12}{6} \overset{122315131}{131^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 4.0$

[4114] $1^1 \overset{13}{6} \overset{13161^1 231513221^{13}}{131^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 15, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 4.8$

[4106] $1^1 \overset{11}{6} \overset{122315131}{131^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 4.0$

[4121] $1^1 \overset{1}{5} \overset{1}{1321} \overset{1}{8} \overset{1}{1231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 16, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0}$

[4122] $1^1 \overset{1}{321812231} \overset{1}{5} \overset{1}{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 16, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$

[4127] $1^1 \overset{1}{5} \overset{1}{1316131} \overset{1}{5} \overset{1}{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 16, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[4129] $1^1 \overset{1}{322181231} \overset{1}{5} \overset{1}{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 16, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$

[4130] $1^1 \overset{11}{31} \overset{11}{6} \overset{1316131^{12}}{131^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 16, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 6 & 10 & 6 \\ 6 & 10 & 14 & 10 \\ 3 & 6 & 10 & 6 \end{pmatrix}_{b_0}$

[4128] $1^1 \overset{1}{21} \overset{1}{8} \overset{1}{12315131^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 16, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[4123] $1^1 \overset{11}{512321} \overset{11}{8} \overset{11}{1231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 16, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$

[4124] $1^1 \overset{11}{512321} \overset{11}{8} \overset{11}{1231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 16, \Delta = -50, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0}$

[4125] $1^1 \overset{1}{18} \overset{1}{122315131^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 16, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[4131] $1^1 \overset{11}{51321} \overset{11}{8} \overset{11}{1231^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 16, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{aligned}
& \begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0} \\
[4132] & 1^{\overline{1}} 5^{\overline{1}} 1321^{\overline{1}} 81231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0} \\
[4133] & 1^{\overline{1}} 6^{\overline{1}} 1232171231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -48, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[4126] & 1^{\overline{1}} 5^{\overline{1}} 1316131^{\overline{1}} 51^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0} \\
[4134] & 1^{\overline{1}} 6^{\overline{1}} 131613151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 17, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0} \\
[4135] & 1^{\overline{1}} 6^{\overline{1}} 131613151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 17, \Delta = -69, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0} \\
[4149] & 21^{\overline{11}} 8^{\overline{2321}} 12321^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0} \\
[4138] & 1^{\overline{2}} 6^{\overline{112315131}} 131^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 5 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 4 & 5 & 3 & 16 & 10 \\ 2 & 3 & 2 & 10 & 6 \end{pmatrix}_{b_0} \\
[4139] & 1^{\overline{11}} 316123151321^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \\
[4140] & 1^{\overline{11}} 321613151321^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \\
[4141] & 1^{\overline{11}} 316123151321^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \\
[4142] & 1^{\overline{11}} 232161315131^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -2, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 2 & 4 & 4 & 6 \\ 2 & 2 & 3 & 6 & 6 \end{pmatrix}_{b_0} \\
[4143] & 1^{\overline{1}} 7^{\overline{1}} 1231513151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 17, \Delta = -71, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[4144] & 1 \overset{1}{1^{11}2321} 8^{12321^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 5 & 4 \\ 1 & 2 & 1 & 5 & 3 \\ 2 & 5 & 5 & 8 & 8 \\ 2 & 4 & 3 & 8 & 6 \end{pmatrix}_{b_0} \\
[4145] & 1 \overset{1}{1^{11}8^{1231414141^1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 17, \Delta = -71, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0} \\
[4146] & 1 \overset{1}{1^{11}2321} 8^{12321^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 2 & 2 & 6 & 4 \\ 1 & 2 & 1 & 6 & 3 \\ 3 & 6 & 6 & 14 & 10 \\ 2 & 4 & 3 & 10 & 6 \end{pmatrix}_{b_0} \\
[4147] & 1 \overset{1}{1^{11}321613151321^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \\
[4137] & 1 \overset{1}{1^{15131}} 6^{13161^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 17, \Delta = -69, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[4136] & 1 \overset{1}{1^{11}232161315131^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -2, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0} \\
[4148] & 1 \overset{1}{1^{11}2315131} 6^{131^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \\
[4151] & 1 \overset{1}{1^{11}6^{1316123151^1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 17, \Delta = -70, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[4150] & 1 \overset{1}{1^{11}2315131} 6^{131^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 2 & 3 & 3 & 4 & 6 \\ 2 & 3 & 2 & 6 & 6 \end{pmatrix}_{b_0} \\
[4152] & 1 \overset{1}{1^{11}6^{1122315131}} 131^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -5, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & 0 & 6 & 2 \\ 6 & 5 & 6 & 45 & 17 \\ 2 & 2 & 2 & 17 & 6 \end{pmatrix}_{b_0} \\
[4153] & 1 \overset{1}{1^{11}2315131} 6^{131^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -6, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 4 & 3 & 8 & 8 \\ 2 & 3 & 2 & 8 & 6 \end{pmatrix}_{b_0}$$

[4154] $1^{11} 31^1 6^{11} 1316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 16, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 6 & 1 & 8 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 8 & 0 & 6 & 6 \\ 2 & 6 & 0 & 6 & 5 \end{pmatrix}_{b_0}$$

[4156] $1^{11} 8^{11} 12321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 2 & 2 & 10 & 4 \\ 1 & 2 & 1 & 7 & 3 \\ 4 & 10 & 7 & 30 & 14 \\ 2 & 4 & 3 & 14 & 6 \end{pmatrix}_{b_0}$$

[4155] $1^1 321^1 613151321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 15, \Delta = -1, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & 0 & 4 & 2 \\ 5 & 6 & 4 & 22 & 12 \\ 2 & 3 & 2 & 12 & 6 \end{pmatrix}_{b_0}$$

[4157] $1^2 6^{12} 2315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & 0 & 4 & 2 \\ 5 & 6 & 5 & 25 & 13 \\ 2 & 3 & 2 & 13 & 6 \end{pmatrix}_{b_0}$$

[4158] $1^1 8^{11} 12321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -11, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.82 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 2 & 2 & 10 & 4 \\ 1 & 2 & 1 & 7 & 3 \\ 5 & 10 & 7 & 30 & 14 \\ 2 & 4 & 3 & 14 & 6 \end{pmatrix}_{b_0}$$

[4159] $21^1 8^{11} 12321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -11, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.82 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 3 & 12 & 5 \\ 7 & 12 & 34 & 16 \\ 3 & 5 & 16 & 7 \end{pmatrix}_{b_0}$$

[4160] $1^1 6^{12} 2315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -11, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.82 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & 0 & 4 & 2 \\ 4 & 6 & 4 & 17 & 11 \\ 2 & 3 & 2 & 11 & 6 \end{pmatrix}_{b_0}$$

[4161] $1^2 6^{13} 2315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 5 & 2 \\ 6 & 9 & 5 & 40 & 16 \\ 2 & 3 & 2 & 16 & 6 \end{pmatrix}_{b_0}$$

[4162] $1^1 6^{12} 2315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 5 & 7 & 6 & 28 & 14 \\ 2 & 3 & 2 & 14 & 6 \end{pmatrix}_{b_0}$$

[4163] $1^3 6^{12} 2315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 7 & 9 & 6 & 52 & 18 \\ 2 & 3 & 2 & 18 & 6 \end{pmatrix}_{b_0}$$

[4164] $1^3 6^{13} 2315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 7 & 9 & 6 & 52 & 18 \\ 2 & 3 & 2 & 18 & 6 \end{pmatrix}_{b_0} \\
[4165] & 31 \overset{31}{6} 131 \overset{1^1}{6} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 3 & 4 \\ 2 & 4 & 2 & 5 & 4 \\ 2 & 3 & 5 & 3 & 5 \\ 2 & 4 & 4 & 5 & 5 \end{pmatrix}_{b_0} \\
[4166] & 1 \overset{1^{12}}{2} \overset{2315131}{6} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -18, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 14.89 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 6 & 8 & 6 & 38 & 16 \\ 2 & 3 & 2 & 16 & 6 \end{pmatrix}_{b_0} \\
[4167] & 31 \overset{1^{11}31}{6} 1316131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 16, \Delta = -33, \det = -21, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15.95 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 6 & 1 & 8 & 6 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 8 & 1 & 7 & 7 \\ 2 & 6 & 0 & 7 & 5 \end{pmatrix}_{b_0} \\
[4168] & 1^{12} 321^1 613151321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 7 & 10 & 6 & 50 & 18 \\ 2 & 3 & 2 & 18 & 6 \end{pmatrix}_{b_0} \\
[4169] & 1^{11} 321^1 613151321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 7 & 2 \\ 7 & 9 & 7 & 48 & 18 \\ 2 & 3 & 2 & 18 & 6 \end{pmatrix}_{b_0} \\
[4170] & 1 \overset{1^{12}2321}{8} \overset{1^1}{12} 321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 2 & 2 & 15 & 4 \\ 1 & 2 & 1 & 12 & 3 \\ 6 & 15 & 12 & 74 & 22 \\ 2 & 4 & 3 & 22 & 6 \end{pmatrix}_{b_0} \\
[4171] & 1 \overset{1^{13}2315131}{6} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 7 & 2 \\ 6 & 9 & 7 & 43 & 17 \\ 2 & 3 & 2 & 17 & 6 \end{pmatrix}_{b_0} \\
[4172] & 1^{11} 3161^1 23151321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 8 & 9 & 6 & 46 & 18 \\ 2 & 3 & 2 & 18 & 6 \end{pmatrix}_{b_0} \\
[4173] & 1^{12} 3161^1 23151321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -25, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 8 & 9 & 6 & 46 & 18 \\ 2 & 3 & 2 & 18 & 6 \end{pmatrix}_{b_0} \\
[4174] & 1 \overset{1^1}{8} \overset{1^{12}2321}{12} 321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -29, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.86 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 2 & 2 & 15 & 4 \\ 1 & 2 & 1 & 12 & 3 \\ 7 & 15 & 12 & 74 & 22 \\ 2 & 4 & 3 & 22 & 6 \end{pmatrix}_{b_0}
\end{aligned}$$

[4175] $21^1 8^{1^{12}2321} 12321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -29, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.86 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 3 & 18 & 5 \\ 11 & 18 & 76 & 24 \\ 3 & 5 & 24 & 7 \end{pmatrix}_{b_0}$$

[4176] $1^4 6^{1^{13}2315131} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -36, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 10 & 13 & 9 & 108 & 26 \\ 2 & 3 & 2 & 26 & 6 \end{pmatrix}_{b_0}$$

[4177] $1^2 6^{1^{13}2315131} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -36, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 8 & 11 & 9 & 72 & 22 \\ 2 & 3 & 2 & 22 & 6 \end{pmatrix}_{b_0}$$

[4178] $1^3 6^{1^{13}2315131} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -38, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 14.95 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 9 & 12 & 9 & 88 & 24 \\ 2 & 3 & 2 & 24 & 6 \end{pmatrix}_{b_0}$$

[4179] $1^2 6^{1^{14}2315131} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -38, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 14.95 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & 0 & 8 & 2 \\ 8 & 12 & 8 & 74 & 22 \\ 2 & 3 & 2 & 22 & 6 \end{pmatrix}_{b_0}$$

[4180] $1^3 6^{1^{14}2315131} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -38, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 14.95 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 9 & 12 & 9 & 88 & 24 \\ 2 & 3 & 2 & 24 & 6 \end{pmatrix}_{b_0}$$

[4182] $1^2 6^{1^{13}2315131} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -49, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 14 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 9 & 14 & 9 & 97 & 25 \\ 2 & 3 & 2 & 25 & 6 \end{pmatrix}_{b_0}$$

[4184] $1^{13} 321^1 613151321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -49, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 13 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 9 & 13 & 9 & 88 & 24 \\ 2 & 3 & 2 & 24 & 6 \end{pmatrix}_{b_0}$$

[4181] $1^{13} 3161^1 23151321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -49, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 10 & 12 & 9 & 82 & 24 \\ 2 & 3 & 2 & 24 & 6 \end{pmatrix}_{b_0}$$

[4185] $1^{12} 3161^1 23151321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -49, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 10 & 12 & 9 & 82 & 24 \\ 2 & 3 & 2 & 24 & 6 \end{pmatrix}_{b_0}$$

[4183] $1^3 6^{1^{13}2315131} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -49, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 10 & 15 & 9 & 115 & 27 \\ 2 & 3 & 2 & 27 & 6 \end{pmatrix}_{b_0}$$

[4186] $1^{\overline{1}^1} \begin{smallmatrix} 8 \\ 1^3 2321 \end{smallmatrix} 12321^{\overline{1}^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -51, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.96 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 2 & 2 & 20 & 4 \\ 1 & 2 & 1 & 17 & 3 \\ 9 & 20 & 17 & 138 & 30 \\ 2 & 4 & 3 & 30 & 6 \end{pmatrix}_{b_0}$

[4187] $21^{\overline{1}^1} \begin{smallmatrix} 8 \\ 1^3 2321 \end{smallmatrix} 12321^{\overline{1}^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -51, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.96 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 3 & 24 & 5 \\ 15 & 24 & 138 & 32 \\ 3 & 5 & 32 & 7 \end{pmatrix}_{b_0}$

[4188] $31^{\overline{1}^2 31} \begin{smallmatrix} 6 \\ 1316131 \end{smallmatrix} 1^{\overline{1}^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 16, \Delta = -33, \det = -53, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.5$
 $T_{\min} = 15.98 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 6 & 1 & 16 & 6 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 16 & 1 & 31 & 13 \\ 2 & 6 & 0 & 13 & 5 \end{pmatrix}_{b_0}$

[4189] $1^{\overline{1}^4} \begin{smallmatrix} 3161^1 23151321^{\overline{1}^3} \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 11 & 15 & 9 & 109 & 27 \\ 2 & 3 & 2 & 27 & 6 \end{pmatrix}_{b_0}$

[4190] $1^{\overline{3}^1} \begin{smallmatrix} 1^4 2315131 \end{smallmatrix} \begin{smallmatrix} 6 \\ 131^{\overline{1}^4} \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -64, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.8$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & 0 & 12 & 2 \\ 11 & 15 & 12 & 136 & 30 \\ 2 & 3 & 2 & 30 & 6 \end{pmatrix}_{b_0}$

[4191] $1^{\overline{4}^1} \begin{smallmatrix} 1^4 2315131 \end{smallmatrix} \begin{smallmatrix} 6 \\ 131^{\overline{1}^4} \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -66, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.97 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 1 & 1 & 0 & 12 & 2 \\ 12 & 16 & 12 & 158 & 32 \\ 2 & 3 & 2 & 32 & 6 \end{pmatrix}_{b_0}$

$T = 16$ [4192] $1^{\overline{1}^1} \begin{smallmatrix} 1^1 23151321 \end{smallmatrix} \begin{smallmatrix} 8 \\ 1231^{\overline{1}^1} \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 4.0$

[4193] $1^{\overline{1}^1} \begin{smallmatrix} 1^1 3151321 \end{smallmatrix} \begin{smallmatrix} 8 \\ 12321^{\overline{1}^1} \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 4.0$

[4194] $1^{\overline{1}^1} \begin{smallmatrix} 1^1 23151321 \end{smallmatrix} \begin{smallmatrix} 8 \\ 12321^{\overline{1}^1} \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 17, \Delta = -30, \det = 0, (n_+, n_-, n_0) = (1, 16, 2), c_{\text{ext}} = 4.0$

[4195] $1^{\overline{1}^1} \begin{smallmatrix} 1^1 3151321 \end{smallmatrix} \begin{smallmatrix} 8 \\ 12321^{\overline{1}^1} \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 4.0$

[4205] $1^{\overline{1}^1} \begin{smallmatrix} 1^1 321 \end{smallmatrix} \begin{smallmatrix} 1^1 \end{smallmatrix} \begin{smallmatrix} 8 \\ 1232161^{\overline{1}^1} \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 17, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$

[4200] $1^{\overline{1}^1} 413218123151^{\overline{1}^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 17, \Delta = -47, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$

[4202] $21^{\overline{1}^1} 6131613151^{\overline{1}^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 17, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 4 & 0 & 9 & 7 \\ 2 & 3 & 0 & 7 & 5 \end{pmatrix}_{b_0}$

[4203] $1^{\overline{1}^1} 613151316131^{\overline{1}^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 17, \Delta = -48, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{aligned}
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 2 & 2 & 4 & 4 & 6 \\ 2 & 2 & 3 & 6 & 6 \end{pmatrix}_{b_0} \\
[4216] & 2 \ 3 \ 21812315131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 6 & 27 & 6 \\ 15 & 27 & 115 & 27 \\ 3 & 6 & 27 & 6 \end{pmatrix}_{b_0} \\
[4217] & 31 \ 5 \ 1321812321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 1 & 2 & 12 & 3 \\ 1 & 2 & 1 & 15 & 3 \\ 9 & 12 & 15 & 79 & 21 \\ 2 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0} \\
[4218] & 2 \ 3 \ 21812315131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 3 & 20 & 3 \\ 3 & 6 & 36 & 6 \\ 20 & 36 & 206 & 36 \\ 3 & 6 & 36 & 6 \end{pmatrix}_{b_0} \\
[4219] & 31 \ 5 \ 1321812321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 1 & 2 & 8 & 3 \\ 1 & 2 & 1 & 10 & 3 \\ 6 & 8 & 10 & 34 & 14 \\ 2 & 3 & 3 & 14 & 5 \end{pmatrix}_{b_0} \\
[4220] & 31 \ 5 \ 1321812321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 1 & 2 & 4 & 3 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 7 & 7 \\ 2 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0} \\
[4221] & 2 \ 3 \ 21812315131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 6 & 9 & 6 \\ 5 & 9 & 11 & 9 \\ 3 & 6 & 9 & 6 \end{pmatrix}_{b_0} \\
[4222] & 2 \ 3 \ 21812315131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 6 & 18 & 6 \\ 10 & 18 & 50 & 18 \\ 3 & 6 & 18 & 6 \end{pmatrix}_{b_0} \\
[4223] & 1^1 321^1 6131612321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & 7 & 3 \\ 5 & 4 & 7 & 27 & 13 \\ 2 & 2 & 3 & 13 & 6 \end{pmatrix}_{b_0} \\
[4225] & 1^{12} 321^1 6131612321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 3 & 19 & 5 \\ 11 & 19 & 87 & 25 \\ 3 & 5 & 25 & 7 \end{pmatrix}_{b_0} \\
[4224] & 1^{11} 321^1 6131612321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & 11 & 3 \\ 7 & 6 & 11 & 55 & 19 \\ 2 & 2 & 3 & 19 & 6 \end{pmatrix}_{b_0} \\
[4228] & 1^{13} 321^1 613161^{12} 2321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 3 & 25 & 5 \\ 15 & 25 & 151 & 33 \\ 3 & 5 & 33 & 7 \end{pmatrix}_{b_0} \\
[4226] & 1^{13} 321^1 6131612321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 3 & 25 & 5 \\ 15 & 25 & 151 & 33 \\ 3 & 5 & 33 & 7 \end{pmatrix}_{b_0} \\
[4227] & 1^{11} 321^1 613161^1 2321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 1 & 13 & 3 \\ 11 & 8 & 13 & 91 & 25 \\ 2 & 2 & 3 & 25 & 6 \end{pmatrix}_{b_0} \\
[4229] & 1^{14} 321^1 613161^1 2321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 85, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 15.95 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 1 & 2 & 19 & 3 \\ 2 & 3 & 31 & 5 \\ 19 & 31 & 227 & 41 \\ 3 & 5 & 41 & 7 \end{pmatrix}_{b_0} \\
\\
T = 17 & [4233] 1^{11} 6^{131} 13161315131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -27, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 4.0 \\
[4230] & 1^{11} 8^{2321} 123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 4.0 \\
[4231] & 1^{11} 8^{2321} 123216131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 4.0 \\
[4232] & 1^{113} 8^{23151321} 12321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 4.0 \\
[4242] & 1^{11} 6^{316131} 13161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 18, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0} \\
[4243] & 1^{11} 7^{12} 123151316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 18, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[4241] & 1^{11} 6^{12} 123218123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 18, \Delta = -46, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0} \\
[4237] & 1^{11} 5^{13} 12181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 18, \Delta = -44, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[4238] & 1^{11} 5^{13} 1218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 18, \Delta = -42, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0} \\
[4239] & 1^{11} 5^{13} 1218123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 18, \Delta = -42, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
[4240] & 21^1 7^{12} 123218^{11} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 18, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \\
[4244] & 1^{11} 8^{151321} 1232161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 19, \Delta = -70, \det = -3, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19]
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 3 & 11 & 5 \\ 6 & 11 & 26 & 14 \\ 3 & 5 & 14 & 7 \end{pmatrix}_{b_0} \\
[4258] & 1^{1^{13}3151321} 8^{12321^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 6 & 6 & 9 & 38 & 16 \\ 2 & 2 & 3 & 16 & 6 \end{pmatrix}_{b_0} \\
[4260] & 1^{1^{12}3151321} 8^{12321^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 3 & 17 & 5 \\ 10 & 17 & 66 & 22 \\ 3 & 5 & 22 & 7 \end{pmatrix}_{b_0} \\
[4259] & 1^{123161^12321712321^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 3 & 19 & 5 \\ 12 & 19 & 94 & 26 \\ 3 & 5 & 26 & 7 \end{pmatrix}_{b_0} \\
[4257] & 1^{113161^12321712321^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & 11 & 3 \\ 8 & 6 & 11 & 62 & 20 \\ 2 & 2 & 3 & 20 & 6 \end{pmatrix}_{b_0} \\
[4261] & 1^{1232181^122315131^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -36, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.5 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 10 & 6 & 12 & 68 & 22 \\ 2 & 2 & 3 & 22 & 6 \end{pmatrix}_{b_0} \\
[4263] & 1^{133161^12321712321^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 16 & 3 \\ 2 & 3 & 25 & 5 \\ 16 & 25 & 158 & 34 \\ 3 & 5 & 34 & 7 \end{pmatrix}_{b_0} \\
[4262] & 1^{123161^12321712321^{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -49, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 10 & 8 & 15 & 102 & 26 \\ 2 & 2 & 3 & 26 & 6 \end{pmatrix}_{b_0} \\
[4264] & 1^{14232181^122315131^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -51, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 16.96 \rightarrow 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 15 & 3 \\ 2 & 3 & 24 & 5 \\ 15 & 24 & 138 & 32 \\ 3 & 5 & 32 & 7 \end{pmatrix}_{b_0} \\
[4265] & 1^{143161^12321712321^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 2 & 17 & 3 \\ 2 & 3 & 29 & 5 \\ 17 & 29 & 200 & 38 \\ 3 & 5 & 38 & 7 \end{pmatrix}_{b_0} \\
[4266] & 1^{13232181^122315131^{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -66, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.8 \\
& T_{\min} = 16.97 \rightarrow 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & 1 & 16 & 3 \\ 12 & 8 & 16 & 110 & 28 \\ 2 & 2 & 3 & 28 & 6 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

[4280] $1^{11}3221^{11}(12)12231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 18, T_{\max} = 19, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 2 & 3 & 1 & 2 & 4 \\ 2 & 3 & 2 & 4 & 5 \end{pmatrix}_{b_0}$$

[4285] $1^{11}712321^{11}8123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 18, T_{\max} = 20, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

[4287] $1^{11}61316^{11}131613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 18, T_{\max} = 20, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0}$$

[4286] $1^{11}712321^{11}8123151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 18, T_{\max} = 20, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0}$$

[4289] $1^{11}71232181223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 18, T_{\max} = 20, \Delta = -68, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

[4288] $1^{11}51321^{11}8^{11}1232171^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 18, T_{\max} = 20, \Delta = -68, \det = 4, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$T = 19$ [4292] $1^{11}2321^{11}8^{11}12321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 19, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0$

[4290] $1^{11}321^{11}8^{11}12321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 19, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0$

[4291] $1^{11}321^{11}8^{11}12321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 19, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0$

[4294] $1^{11}2321^{11}8^{11}12321712321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 19, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0$

[4295] $1^{11}31^{11}6^{11}1316131613141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 19, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0$

[4296] $1^{11}2321^{11}8^{11}12321812321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 19, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 19, 2), c_{\text{ext}} = 4.0$

[4297] $1^{11}2321^{11}8^{11}1232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 19, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0$

[4293] $1^{11}2321^{11}8^{11}12321812231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 19, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0$

[4302] $1^{11}712315132181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2$ $T = 19, T_{\max} = 20, \Delta = -46, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[4303] \ 21^{1\bar{1}} 7^{1\bar{1}} 123218123151^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 19, T_{\max} = 20, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 4 & 0 & 9 & 7 \\ 2 & 3 & 0 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[4304] \ 1^{21} (12) \ 12231^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 19, T_{\max} = 20, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[4305] \ 1^{1\bar{1}} 6^{1\bar{1}} 13161232181231^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 19, T_{\max} = 20, \Delta = -46, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[4306] \ 221^{1\bar{1}} (12) \ 12231^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 19, T_{\max} = 20, \Delta = -47, \det = -2, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$$

$$[4307] \ 21^{1\bar{1}} 8^{1\bar{1}} 12321^{1\bar{1}} 8^{1\bar{1}} 123151^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 19, T_{\max} = 21, \Delta = -67, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[4308] \ 21^{1\bar{1}} 8^{1\bar{1}} 12321^{1\bar{1}} 8^{1\bar{1}} 123151^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 19, T_{\max} = 21, \Delta = -67, \det = -3, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 3 & 0 & 11 & 7 \\ 2 & 2 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[4309] \ 1^{1\bar{1}} 31^{1\bar{1}} 51^{1\bar{1}} 321^{1\bar{1}} 81^{1\bar{1}} 231^{1\bar{1}} 51^{1\bar{1}} 31^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 19, T_{\max} = 19, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 3 & 6 & 5 \\ 4 & 6 & 8 & 8 \\ 3 & 5 & 8 & 7 \end{pmatrix}_{b_0}$$

$$[4310] \ 1^{1\bar{1}} 8^{1\bar{1}} 123151^{1\bar{1}} 31^{1\bar{1}} 61^{1\bar{1}} 31^{1\bar{1}} 51^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 19, T_{\max} = 21, \Delta = -69, \det = -4, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[4311] \ 31^{1\bar{1}} 6^{1\bar{1}} 131^{1\bar{1}} 61^{1\bar{1}} 31^{1\bar{1}} 6^{1\bar{1}} 131^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 19, T_{\max} = 19, \Delta = -3, \det = -16, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 3 & 5 & 4 & 5 \\ 2 & 5 & 3 & 6 & 5 \\ 2 & 4 & 6 & 4 & 6 \\ 2 & 5 & 5 & 6 & 6 \end{pmatrix}_{b_0}$$

$$T = 20 \quad [4312] \ 1^{1\bar{3}} 31^{1\bar{1}} 61^{1\bar{1}} 2321^{1\bar{1}} 81^{1\bar{1}} 231^{1\bar{1}} 51^{1\bar{1}} 321^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 20, T_{\max} = 20, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 4.0$$

$$[4313] \ 1^{1\bar{1}} 6^{1\bar{1}} 131^{1\bar{1}} 61^{1\bar{1}} 31^{1\bar{1}} 6^{1\bar{1}} 131^{1\bar{1}} 51^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 20, T_{\max} = 20, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 4.0$$

[4314] $1^{\overline{1}^{12321}} 8^{\overline{1}} 123218123141^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 20, T_{\max} = 20, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 4.0$

[4315] $1^{\overline{1}^{12321}} 8^{\overline{1}} 123218123141^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 20, T_{\max} = 20, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 4.0$

[4316] $1^{\overline{1}} 5^{\overline{1}} 13161316131^{\overline{1}} 6^{\overline{1}} 131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 20, T_{\max} = 20, \Delta = -18, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 4.0$

[4319] $1^{\overline{1}^{\overline{1}}} 8^{\overline{1}} 122315132181231^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 20, T_{\max} = 21, \Delta = -45, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[4320] $21^{\overline{1}^{\overline{1}}} (12)^{\overline{1}^{\overline{1}}} 1223141^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 20, T_{\max} = 21, \Delta = -46, \det = 2, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0}$

[4321] $1^{\overline{1}^{\overline{1}}} 8^{\overline{1}} 123216131613151^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 20, T_{\max} = 22, \Delta = -68, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

$T = 21$ [4327] $1^{\overline{1}^{12321}} 8^{\overline{1}} 1232181232151^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 21, T_{\max} = 21, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 4.0$

[4322] $1^{\overline{1}^{\overline{1}}} 8^{\overline{1}^{\overline{1}}} 12321812315131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 21, T_{\max} = 21, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 4.0$

[4323] $1^{\overline{1}^{12321}} 8^{\overline{1}} 12321812315131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 21, T_{\max} = 21, \Delta = -28, \det = 0, (n_+, n_-, n_0) = (1, 21, 2), c_{\text{ext}} = 4.0$

[4324] $1^{\overline{1}^{12321}} 8^{\overline{1}} 1232181232151^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 21, T_{\max} = 21, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 4.0$

[4325] $1^{\overline{1}} 5^{\overline{1}} 1321812321^{\overline{1}} 8^{\overline{1}} 12321^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 21, T_{\max} = 21, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 4.0$

[4326] $1^{\overline{1}} 5^{\overline{1}} 1321812321^{\overline{1}} 8^{\overline{1}} 12321^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 21, T_{\max} = 21, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 4.0$

[4332] $1^{\overline{1}^{\overline{1}}} 8^{\overline{1}} 1231513161316131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 21, T_{\max} = 22, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[4330] $21^{\overline{1}^{\overline{1}}} (12)^{\overline{1}^{\overline{1}}} 12231^{\overline{1}} 5^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 21, T_{\max} = 22, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[4331] $1^{\overline{1}} (12)^{\overline{1}} 12231^{\overline{1}} 5^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 21, T_{\max} = 22, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[4333] \quad 1^{1^1 5^{13221}} (12) \quad 122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 21, \quad T_{\max} = 22, \quad \Delta = -45, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 21, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 22, \quad T_{\max} = 22, \quad T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[4334] \quad 1^{1^1} (12) \quad 1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 21, \quad T_{\max} = 22, \quad \Delta = -41, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 21, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 22, \quad T_{\max} = 22, \quad T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 3 & 3 & 6 & 6 \\ 2 & 3 & 2 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[4335] \quad 21^{1^1 5^{13221}} (12) \quad 1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 21, \quad T_{\max} = 22, \quad \Delta = -41, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 21, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 22, \quad T_{\max} = 22, \quad T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[4336] \quad 1^{1^1} 7^{12321} 17^{12321} 18^{1231} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 21, \quad T_{\max} = 22, \quad \Delta = -45, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 21, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 22, \quad T_{\max} = 22, \quad T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[4337] \quad 1^{1^1} 8^{12315} 13^{1218} 12315^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 21, \quad T_{\max} = 23, \quad \Delta = -68, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 21, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 23, \quad T_{\max} = 23, \quad T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 22 \quad [4338] \quad 16^{1^1} 123218^{1^1} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 22, \quad T_{\max} = 22, \quad \Delta = -19, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 22, 1), \quad c_{\text{ext}} = 4.0$$

$$[4339] \quad 1^{1^1 2321} 8^{12321} 123218^{1^1} 123216^{1^1} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 22, \quad T_{\max} = 22, \quad \Delta = -27, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 22, 2), \quad c_{\text{ext}} = 4.0$$

$$[4340] \quad 1^{1^1 321} 8^{12321} 123218^{1^1} 123216^{1^1} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 22, \quad T_{\max} = 22, \quad \Delta = -25, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 22, 1), \quad c_{\text{ext}} = 4.0$$

$$[4341] \quad 1^{1^1} 5^{1316} 13^{1613} 16^{1316} 131^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 22, \quad T_{\max} = 22, \quad \Delta = -12, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 22, 1), \quad c_{\text{ext}} = 4.0$$

$$[4345] \quad 1^1 4141413221(12)1223151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 22, \quad T_{\max} = 23, \quad \Delta = -44, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 22, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 23, \quad T_{\max} = 23, \quad T \in [23, 23]$$

$$\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$$

$$[4342] \quad 21^{1^1 3221} (12) \quad 1223151316^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 22, \quad T_{\max} = 23, \quad \Delta = -46, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 22, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 23, \quad T_{\max} = 23, \quad T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[4343] \quad 1^{1^1 321812321} 8^{12321} 7^{1^1} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 22, \quad T_{\max} = 23, \quad \Delta = -44, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 22, 0), \quad c_{\text{ext}} =$$

4.0

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[4344] \ 1^{\overline{1}} 8^{1^1} 12321613161316131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 22, T_{\max} = 23, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$T = 23 \quad [4349] \ 21^{\overline{1}} 6^{1^1} 131613161316131^{\overline{1}} 51^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\max} = 23, \Delta = -9, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 4.0$$

$$[4346] \ 1^{\overline{1}} 8^{1^1} 1232181232171231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\max} = 23, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 23, 2), c_{\text{ext}} = 4.0$$

$$[4347] \ 21^{\overline{1}} 7^{1^1} 12321812321^{\overline{1}} 8^{1^1} 12321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\max} = 23, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 4.0$$

$$[4348] \ 1^{\overline{1}} 8^{1^1} 1232181232171231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\max} = 23, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 4.0$$

$$[4351] \ 1^{\overline{1}} 5^{1^1} 131513221^{\overline{1}} (12) \ 1223151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\max} = 24, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[4352] \ 1^{\overline{1}} 4^{1^1} 13221^{\overline{1}} (12) \ 12231513161^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\max} = 24, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[4353] \ 21^{\overline{1}} 3^{1^1} 221^{\overline{1}} (12) \ 122315132171^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\max} = 24, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[4354] \ 1^{\overline{1}} 7^{1^1} 123218^{1^1} 123218123151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\max} = 25, \Delta = -67, \det = -3, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[4355] \ 1^{\overline{1}} 8^{1^1} 123217123218123151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 23, T_{\max} = 25, \Delta = -67, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 24 \quad [4356] \ 221^{\overline{1}} 8^{1^1} 12321812321^{\overline{1}} 8^{1^1} 12321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 24, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 4.0$$

$$[4357] \ 1^{\overline{1}} 8^{1^1} 12321812321^{\overline{1}} 8^{1^1} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 24, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 4.0$$

[4358] $1^{\overline{1}1321} 8^{\overline{1}} 12321812321812231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 24, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 4.0$

[4359] $1^{\overline{1}1321} 8^{\overline{1}} 123218123218^{\overline{1}} 12321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 24, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 24, 2), c_{\text{ext}} = 4.0$

[4360] $1^{\overline{1}12321} 8^{\overline{1}} 12321812321812231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 24, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 24, 2), c_{\text{ext}} = 4.0$

[4365] $21^{\overline{1}13221} (12)^{\overline{1}} 122315132218^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 25, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$

[4362] $1^{\overline{1}1} 6^{\overline{1}} 131513221(12)^{\overline{1}} 122314141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 25, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0}$

[4363] $1^{\overline{1}} 15^{\overline{1}} 13221(12)^{\overline{1}} 1223151316^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 25, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$

[4366] $1^{\overline{1}413221} 1^{\overline{1}} (12)^{\overline{1}} 12231513217^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 25, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$

[4364] $1^{\overline{1}1513221} 1^{\overline{1}} (12)^{\overline{1}} 1223151316^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 25, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 3 & 3 & 6 & 6 \\ 2 & 3 & 2 & 6 & 5 \end{pmatrix}_{b_0}$

[4367] $1^{\overline{1}} 18^{\overline{1}} 123218^{\overline{1}} 123218123151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 26, \Delta = -66, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 5 & 5 \\ 2 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0}$

[4368] $1^{\overline{1}} 8^{\overline{1}} 123218^{\overline{1}} 123218123151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 26, \Delta = -66, \det = 3, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 3 & 0 & 11 & 7 \\ 2 & 2 & 0 & 7 & 4 \end{pmatrix}_{b_0}$

[4369] $1^{\overline{1}51321812321} 8^{\overline{1}} 123218^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 24, T_{\max} = 26, \Delta = -66, \det = 4, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 26, T_{\max} = 26, T \in [26, 26]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

$$\begin{aligned}
& T = 25 \quad [4371] \ 1^{\overline{1}1} 8^{\overline{1}2321} 123218123218123141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 25, T_{\max} = 25, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 25, 2), c_{\text{ext}} = 4.0 \\
& [4370] \ 1^{\overline{1}1} 8^{\overline{1}321} 123218123218123141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 25, T_{\max} = 25, \Delta = -22, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 4.0 \\
& [4373] \ 1^{\overline{1}1} (12) 122315131613161^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 25, T_{\max} = 26, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 4.0 \\
& \quad T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0} \\
& [4374] \ 1^{\overline{1}1} 5^{\overline{1}1} 13221^{\overline{1}} (12) 122315132171^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 25, T_{\max} = 26, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 4.0 \\
& \quad T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 3 & 3 & 6 & 6 \\ 2 & 3 & 2 & 6 & 5 \end{pmatrix}_{b_0} \\
& [4375] \ 1^{\overline{1}1} 15^{\overline{1}} 13221^{\overline{1}} (12) 122315132171^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 25, T_{\max} = 26, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 4.0 \\
& \quad T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0} \\
& [4376] \ 1^{\overline{1}1} 7^{\overline{1}} 1231513221(12)122314141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 25, T_{\max} = 26, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 4.0 \\
& \quad T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0} \\
& [4377] \ 1^{\overline{1}1} 8^{\overline{1}} 12315132181232181231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 25, T_{\max} = 26, \Delta = -44, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 4.0 \\
& \quad T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
& [4378] \ 1^{\overline{1}1} 413221^{\overline{1}} (12) 1223151322181^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 25, T_{\max} = 26, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 4.0 \\
& \quad T_{\min} = 26, T_{\max} = 26, T \in [26, 26] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0} \\
& [4379] \ 21^{\overline{1}1} 513221^{\overline{1}} (12) 122315132181^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 25, T_{\max} = 27, \Delta = -67, \det = -4, (n_+, n_-, n_0) = (1, 25, 0), c_{\text{ext}} = 4.0 \\
& \quad T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
& T = 26 \quad [4380] \ 15^{\overline{1}1} 132181232181232181231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 26, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 4.0 \\
& [4381] \ 1^{\overline{1}1} 8^{\overline{1}321} 1232181232181232151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 26, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 4.0 \\
& [4382] \ 1^{\overline{1}1} 8^{\overline{1}321} 123218123218123151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 26, \Delta = -17, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 4.0
\end{aligned}$$

$$\begin{aligned}
[4383] & 1 \overset{1}{5} 1321812321812321 \overset{1}{8} 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 26, \Delta = -19, \det = 0, (n_+, n_-, n_0) = \\
& (1, 26, 2), c_{\text{ext}} = 4.0 \\
[4384] & 1 \overset{1}{8} 1232181232181232151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 26, \Delta = -21, \det = 0, (n_+, n_-, n_0) = \\
& (1, 26, 2), c_{\text{ext}} = 4.0 \\
[4390] & 1 \overset{1}{5} 13221(12)12231513221 \overset{1}{8} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 27, \Delta = -38, \det = 2, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0} \\
[4391] & 21 \overset{1}{8} 1231513221(12)12231 \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 27, \Delta = -38, \det = 2, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 5 & 3 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 5 & 4 & 15 & 9 \\ 2 & 3 & 2 & 9 & 5 \end{pmatrix}_{b_0} \\
[4389] & 1 \overset{1}{5} 1316131513221(12)12231 \overset{1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 27, \Delta = -42, \det = 2, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0} \\
[4385] & 1 \overset{1}{5} 13221(12)12231513221 \overset{1}{8} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 27, \Delta = -38, \det = 2, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 3 & 3 & 6 & 6 \\ 2 & 3 & 2 & 6 & 5 \end{pmatrix}_{b_0} \\
[4386] & 1 \overset{1}{8} 12231513221(12)122314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 27, \Delta = -42, \det = 2, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0} \\
[4387] & 1 \overset{1}{6} 1316131513221(12)1223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 27, \Delta = -44, \det = 2, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0} \\
[4388] & 1 \overset{1}{6} 131513221(12)12231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 26, T_{\max} = 27, \Delta = -42, \det = 2, (n_+, n_-, n_0) = \\
& (1, 26, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 27, T_{\max} = 27, T \in [27, 27] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0} \\
T = 27 & [4392] 1 \overset{1}{6} 12321812321812321 \overset{1}{8} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 27, T_{\max} = 27, \Delta = -16, \det = 0, (n_+, n_-, n_0) = \\
& (1, 27, 1), c_{\text{ext}} = 4.0 \\
[4393] & 1 \overset{1}{8} 1232171232181232181231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 27, T_{\max} = 28, \Delta = -43, \det = -2, (n_+, n_-, n_0) = \\
& (1, 27, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 28, T_{\max} = 28, T \in [28, 28] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

[4395] $1^1 5 13221(12)122315131613161^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 27, T_{\max} = 28, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[4394] $1^1 6 1316131513221(12)1223151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 27, T_{\max} = 28, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$

[4396] $1^1 7 1231513221(12)12231513151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 27, T_{\max} = 28, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 27, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

$T = 28$ [4397] $1^1 5 13218123218123218123151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 28, T_{\max} = 28, \Delta = -11, \det = 0, (n_+, n_-, n_0) = (1, 28, 1), c_{\text{ext}} = 4.0$

[4402] $1^1 3221 (12) 122315132181232171^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 28, T_{\max} = 29, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$

[4401] $1^1 8 12231513221(12)12231513151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 28, T_{\max} = 29, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[4400] $1^1 6 13161316131513221(12)12231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 28, T_{\max} = 29, \Delta = -44, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[4399] $1^1 321812321812321 8 1232181^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 28, T_{\max} = 29, \Delta = -42, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$

[4398] $1^1 5 132181231513221(12)1223151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 28, T_{\max} = 29, \Delta = -41, \det = 2, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 29, T_{\max} = 29, T \in [29, 29]$
 $\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$

$T = 29$ [4403] $1^1 6 123218123218123218123151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 29, T_{\max} = 29, \Delta = -10, \det = 0, (n_+, n_-, n_0) = (1, 29, 1), c_{\text{ext}} = 4.0$

[4404] $1^1 7 1232181231513221(12)1223141^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2$ $T = 29, T_{\max} = 30, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 30, T_{\max} = 30, T \in [30, 30]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[4405] \quad 1^1 8 12321 8^{1^1} 123218123218123151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 29, T_{\max} = 31, \Delta = -65, \det = -3, (n_+, n_-, n_0) = (1, 29, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 3 & 0 & 11 & 7 \\ 2 & 2 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

$$T = 30 \quad [4406] \quad 21^{1^1} 7 123218123218123218123151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 30, T_{\max} = 30, \Delta = -8, \det = 0, (n_+, n_-, n_0) = (1, 30, 1), c_{\text{ext}} = 4.0$$

$$[4408] \quad 1^1 7 1232181231513221(12)1223151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 30, T_{\max} = 31, \Delta = -38, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[4407] \quad 1^1 5 13221(12)122315132181232171^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 30, T_{\max} = 31, \Delta = -38, \det = 2, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 31, T_{\max} = 31, T \in [31, 31]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[4409] \quad 1^{1^1} 5 13221(12)122315132181232181^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 30, T_{\max} = 32, \Delta = -66, \det = 4, (n_+, n_-, n_0) = (1, 30, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 31 \quad [4410] \quad 1^{1^1} 8^{1^1} 123218123218123218123151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 31, T_{\max} = 31, \Delta = -16, \det = 0, (n_+, n_-, n_0) = (1, 31, 2), c_{\text{ext}} = 4.0$$

$$[4411] \quad 221^{1^1} 8^{1^2} 123218123218123218123151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 31, T_{\max} = 31, \Delta = -7, \det = 0, (n_+, n_-, n_0) = (1, 31, 1), c_{\text{ext}} = 4.0$$

$$[4412] \quad 21^{1^1} 8^{1^1} 1232181231513221(12)1223151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 31, T_{\max} = 32, \Delta = -37, \det = -2, (n_+, n_-, n_0) = (1, 31, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 32, T_{\max} = 32, T \in [32, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 4 & 0 & 9 & 7 \\ 2 & 3 & 0 & 7 & 5 \end{pmatrix}_{b_0}$$

$$T = 32 \quad [4413] \quad 1^{1^1} 6^{1^1} 1232181232181232181232181231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 32, T_{\max} = 32, \Delta = -15, \det = 0, (n_+, n_-, n_0) = (1, 32, 2), c_{\text{ext}} = 4.0$$

$$[4414] \quad 21^{1^1} 5 13221(12)1223151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 32, T_{\max} = 33, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 32, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 33, T_{\max} = 33, T \in [33, 33]$$

$$\begin{pmatrix} 2 & 2 & 4 \\ 2 & 1 & 3 \\ 4 & 3 & 7 \end{pmatrix}_{b_0}$$

$$T = 33 \quad [4416] \quad 1^{1^1} 7^{1^1} 1232181232181231513221(12)12231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \quad T = 33, T_{\max} = 34, \Delta = -43, \det = -2, (n_+, n_-, n_0) = (1, 33, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 34, T_{\max} = 34, T \in [34, 34]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$T = 35 \quad [4417] \quad 1 \quad \begin{matrix} 1^1 3221(12)12231513221 \\ (12) \end{matrix} \quad 122315132181^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 35, \quad T_{\max} = 36, \quad \Delta = -43, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 35, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 36, \quad T_{\max} = 36, \quad T \in [36, 36]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[4418] \quad 1^1 81232181232181231513221(12)1223151^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 35, \quad T_{\max} = 37, \quad \Delta = -65, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 35, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 37, \quad T_{\max} = 37, \quad T \in [37, 37]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 36 \quad [4419] \quad 221^1 8123218123218123218123218123151^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 36, \quad T_{\max} = 36, \quad \Delta = -6, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 36, 2), \quad c_{\text{ext}} = 4.0$$

$$[4421] \quad 1 \quad \begin{matrix} 1^1 413221(12)12231513221 \\ (12) \end{matrix} \quad 122315132181^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 36, \quad T_{\max} = 37, \quad \Delta = -42, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 36, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 37, \quad T_{\max} = 37, \quad T \in [37, 37]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$$T = 37 \quad [4424] \quad 1^1 513221(12)12231513221(12)122315132181^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 37, \quad T_{\max} = 38, \quad \Delta = -37, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 37, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 38, \quad T_{\max} = 38, \quad T \in [38, 38]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[4423] \quad 1 \quad \begin{matrix} 1^1 5^1 13221(12)12231513221 \\ (12) \end{matrix} \quad 122315132181^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 37, \quad T_{\max} = 38, \quad \Delta = -37, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 37, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 38, \quad T_{\max} = 38, \quad T \in [38, 38]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 2 & 3 \\ 3 & 2 & 3 & 5 \\ 3 & 3 & 5 & 6 \end{pmatrix}_{b_0}$$

$$T = 40 \quad [4427] \quad 1 \quad \begin{matrix} 1^1 3221 \\ (12) \end{matrix} \quad 12231513221(12)122315132181232181^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 40, \quad T_{\max} = 41, \quad \Delta = -42, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 40, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 41, \quad T_{\max} = 41, \quad T \in [41, 41]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$T = 41 \quad [4428] \quad 1^1 81232181231513221(12)12231513221(12)1223141^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 41, \quad T_{\max} = 42, \quad \Delta = -41, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 41, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 42, \quad T_{\max} = 42, \quad T \in [42, 42]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 2 & 4 \\ 2 & 4 & -1 & 4 & 5 \end{pmatrix}_{b_0}$$

$$T = 42 \quad [4429] \quad 1^1 513221(12)12231513221(12)122315132181232181^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 \quad T = 42, \quad T_{\max} = 43, \quad \Delta = -36, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 42, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 43, \quad T_{\max} = 43, \quad T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[4430] \quad 1^{\overline{1}} 8^{1^1} 1232181231513221(12)12231513221(12)1223151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 42, T_{\max} = 43, \Delta = -36, \det = 2, (n_+, n_-, n_0) = (1, 42, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 43, T_{\max} = 43, T \in [43, 43]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$T = 44 \quad [4431] \quad 1^{\overline{1}} 5^{1^1} 13221(12)12231513221(12)12231513221(12)1223151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 44, T_{\max} = 44, \Delta = -10, \det = 0, (n_+, n_-, n_0) = (1, 44, 1), c_{\text{ext}} = 4.0$$

$$T = 46 \quad [4433] \quad 1^{\overline{1}} 3221(12)12231513221(12)122315132181^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 46, T_{\max} = 47, \Delta = -42, \det = 2, (n_+, n_-, n_0) = (1, 46, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 47, T_{\max} = 47, T \in [47, 47]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$$T = 47 \quad [4434] \quad 1^{\overline{1}} 5^{1^1} 13221(12)12231513221(12)12231513161^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 47, T_{\max} = 47, \Delta = -9, \det = 0, (n_+, n_-, n_0) = (1, 47, 1), c_{\text{ext}} = 4.0$$

$$T = 48 \quad [4435] \quad 1^{\overline{1}} 5^{1^1} 13221(12)12231513221(12)122315132171^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 48, T_{\max} = 48, \Delta = -8, \det = 0, (n_+, n_-, n_0) = (1, 48, 1), c_{\text{ext}} = 4.0$$

$$T = 49 \quad [4436] \quad 21^{\overline{1}} 8^{1^1} 1231513221(12)12231513221(12)12231513221(12)1223151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 49, T_{\max} = 49, \Delta = -7, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), c_{\text{ext}} = 4.0$$

$$[4437] \quad 1^{\overline{1}} 5^{1^1} 13221(12)12231513221(12)12231513221(12)1223151322181^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 \quad T = 49, T_{\max} = 49, \Delta = -7, \det = 0, (n_+, n_-, n_0) = (1, 49, 1), c_{\text{ext}} = 4.0$$

4.761 NHC(-2)(-3) + $\textcolor{red}{su}_2 + \textcolor{brown}{e}_6$

$$T = 7 \quad [0] \quad 1^{\overline{1}} 1^{1^1} 31^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.71 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 1 & 4 & 7 & 1 & 3 \\ 4 & 9 & 16 & 0 & 5 \\ 7 & 16 & 28 & 0 & 10 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 10 & -4 & -3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1}} 31^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1}} 31^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 6 & 10 & 22 & 0 & 8 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 8 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1}} 1^{1^1} 31^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 9 & 16 & 4 & 5 \\ 6 & 16 & 22 & 6 & 8 \\ 3 & 4 & 6 & -3 & -1 \\ 3 & 5 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 12 & 28 & 85 & 0 & 17 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 17 & -4 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{12}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 4 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{13}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 6 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{14}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 8 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 12 & -4 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{13}1^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 4 & 3 & 4 \\ 2 & 4 & 3 & -3 & -1 \\ 3 & 5 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{111}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -93, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 9 & 12 & 0 & 5 \\ 5 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{15}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.1$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 10 & 20 & 73 & 0 & 15 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 15 & -4 & -1 \end{pmatrix}_{b_0}$$

[22] $1^{11}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 5 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{16}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 13.19 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 12 & 24 & 106 & 0 & 18 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 18 & -4 & -1 \end{pmatrix}_{b_0}$$

[24] $1^{13}1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -146, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 3 \\ 4 & 9 & 16 & 4 & 5 \\ 6 & 16 & 22 & 6 & 8 \\ 2 & 4 & 6 & -3 & -1 \\ 3 & 5 & 8 & -1 & -1 \end{pmatrix}_{b_0}$$

[25] $1^{17}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 14 & 28 & 145 & 0 & 21 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 21 & -4 & -1 \end{pmatrix}_{b_0}$$

[26] $1^{113}3^11^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -158, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 17 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 17 & 40 & 172 & 0 & 24 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 24 & -4 & -1 \end{pmatrix}_{b_0}$$

[27] $1^{11}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -162, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 11 & -4 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{112}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -174, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 3 \\ 4 & 9 & 24 & 0 & 5 \\ 10 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 14 & -4 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{12}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -189, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.31 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 7 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 10 & -4 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{11}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -195, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 11 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 15 & -4 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{112}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -203, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 3 \\ 4 & 9 & 20 & 0 & 5 \\ 8 & 20 & 37 & 0 & 11 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 11 & -4 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{11}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 19 & -4 & 0 \end{pmatrix}_{b_0}$$

[33] $1^113^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -246, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 13.27 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 9 & 24 & 4 & 5 \\ 9 & 24 & 52 & 9 & 12 \\ 2 & 4 & 9 & -3 & -1 \\ 3 & 5 & 12 & -1 & -1 \end{pmatrix}_{b_0}$$

[34] $1^{13}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -267, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 9 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{12}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 10 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 14 & -4 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{113}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -309, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 15 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 21 & -4 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{14}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -349, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 11 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 16 & -4 & 0 \end{pmatrix}_{b_0}$$

[38] $21^{1113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -352, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 4 \\ 8 & 9 & 8 & 4 & 5 \\ 6 & 8 & 4 & 3 & 4 \\ 3 & 4 & 3 & -3 & -1 \\ 4 & 5 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -353, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 11 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 11 & 28 & 73 & 0 & 15 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 15 & -4 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{113}31^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -374, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 3 \\ 4 & 9 & 32 & 0 & 5 \\ 13 & 32 & 100 & 0 & 18 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 18 & -4 & 0 \end{pmatrix}_{b_0}$$

[41] $131^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -386, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.32 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 3 \\ 4 & 9 & 32 & 4 & 5 \\ 12 & 32 & 94 & 12 & 16 \\ 2 & 4 & 12 & -3 & -1 \\ 3 & 5 & 16 & -1 & -1 \end{pmatrix}_{b_0}$$

[42] $1^{12}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -405, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.44 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 13 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 18 & -4 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{13}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -450, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 12 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 17 & -4 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{12}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -492, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.53 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 16 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 22 & -4 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{114}31^4 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -498, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 3 \\ 4 & 9 & 48 & 0 & 5 \\ 20 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 28 & -4 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -543, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 14 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 19 & -4 & -1 \end{pmatrix}_{b_0}$$

[47] $131^{115} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -566, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.1$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 9 & 40 & 4 & 5 \\ 15 & 40 & 148 & 15 & 20 \\ 2 & 4 & 15 & -3 & -1 \\ 3 & 5 & 20 & -1 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{114}31^3 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -599, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 9 & 44 & 0 & 5 \\ 18 & 44 & 193 & 0 & 25 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 25 & -4 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{14}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 14 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 20 & -4 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{114}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -614, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 16 & 40 & 154 & 0 & 22 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 22 & -4 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{13}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -619, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 15 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 21 & -4 & 0 \end{pmatrix}_{b_0}$$

[52] $21^{112}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -622, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 4 & 8 & 12 & 3 & 4 \\ 8 & 9 & 16 & 4 & 5 \\ 12 & 16 & 22 & 6 & 8 \\ 3 & 4 & 6 & -3 & -1 \\ 4 & 5 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{13}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -774, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.44 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 18 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 25 & -4 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{14}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -837, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 31 & 0 & 4 \\ 17 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 24 & -4 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{14}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -1060, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 20 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 28 & -4 & 0 \end{pmatrix}_{b_0}$$

[56] $21^{113}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -1072, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 4 & 8 & 18 & 3 & 4 \\ 8 & 9 & 24 & 4 & 5 \\ 18 & 24 & 52 & 9 & 12 \\ 3 & 4 & 9 & -3 & -1 \\ 4 & 5 & 12 & -1 & 0 \end{pmatrix}_{b_0}$$

[57] $21^{114}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -1702, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 4 & 8 & 24 & 3 & 4 \\ 8 & 9 & 32 & 4 & 5 \\ 24 & 32 & 94 & 12 & 16 \\ 3 & 4 & 12 & -3 & -1 \\ 4 & 5 & 16 & -1 & 0 \end{pmatrix}_{b_0}$$

[58] $21^{115}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -2512, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.1$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$

$$\begin{pmatrix} 4 & 8 & 30 & 3 & 4 \\ 8 & 9 & 40 & 4 & 5 \\ 30 & 40 & 148 & 15 & 20 \\ 3 & 4 & 15 & -3 & -1 \\ 4 & 5 & 20 & -1 & 0 \end{pmatrix}_{b_0}$$

$T = 8$ [59] $1^33^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.14 \rightarrow 16, T_{\max} = 94, T \in [16, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 3 & 5 & 4 & 3 & 4 \\ 2 & 5 & 3 & -2 & 0 \\ 3 & 5 & 4 & 0 & -2 \end{pmatrix}_{b_0}$$

[60] $1^{\overline{1}1} 3 2 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.82 \rightarrow 16, T_{\max} = 94, T \in [16, 94]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 5 & 0 & -2 & 0 \\ 3 & 5 & 0 & 0 & -2 \end{pmatrix}_{b_0}$$

[61] $1^{\overline{1}1} 2 2 3 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 14.15 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 5 & 5 & 5 & 5 \\ 3 & 5 & 4 & 3 & 4 \\ 3 & 5 & 3 & -2 & 0 \\ 3 & 5 & 4 & 0 & -1 \end{pmatrix}_{b_0}$$

[62] $1^{\overline{1}1} 3 2 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 1 & 0 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & -3 & -1 \\ 2 & 2 & -1 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[63] $1^{\overline{1}1} 3 2 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 4 & 5 \\ 1 & 1 & 0 & 4 & -3 & -1 \\ 2 & 2 & -1 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{\overline{1}4} 3 1^{\overline{1}1} 3 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 15.02 \rightarrow 16, T_{\max} = 98, T \in [16, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 1 & 2 \\ 1 & 1 & 4 & 15 & 0 & 3 \\ 2 & 4 & 3 & 18 & 0 & 3 \\ 6 & 15 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 3 & 16 & -4 & -3 \end{pmatrix}_{b_0}$$

[64] $1^{\overline{1}1} 3 1^{\overline{1}1} 3 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.35 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 3 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

[66] $1^{\overline{1}1} 3 1^{\overline{1}1} 3 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 1 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{\overline{1}1} 3 1^{\overline{1}1} 3 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 6 & 0 & 3 \\ 2 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 3 & 4 & -4 & -1 \end{pmatrix}_{b_0}$$

[68] $1^{\overline{1}1} 3 1^{\overline{1}1} 3 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 2 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{\overline{1}2} 3 1^{\overline{1}1} 3 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 4 & 12 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 10 & -4 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{\overline{1}3} 3 1^{\overline{1}1} 3 1^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.23 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

- [71] $1^{112}31^{131}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.08 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 1 & 2 \\ 1 & 1 & 4 & 12 & 0 & 3 \\ 2 & 4 & 3 & 15 & 0 & 3 \\ 5 & 12 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 3 & 13 & -4 & -1 \end{pmatrix}_{b_0}$$
- [72] $1^{113}31^{113}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 4 & 9 & 0 & 3 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 4 & 9 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 2 & 10 & -4 & -1 \end{pmatrix}_{b_0}$$
- [73] $1^{123}31^{113}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 223, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 13.07 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 1 & 2 \\ 1 & 1 & 4 & 9 & 0 & 3 \\ 2 & 4 & 3 & 12 & 0 & 3 \\ 4 & 9 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 3 & 10 & -4 & -1 \end{pmatrix}_{b_0}$$
- [74] $1^{113}31^{113}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 3 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$
- [75] $1^{131}^{123}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 4 & 12 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 8 & -4 & 0 \end{pmatrix}_{b_0}$$
- [76] $1^{143}31^{123}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 8 & 24 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 20 & -4 & 0 \end{pmatrix}_{b_0}$$
- [77] $1^{131}^{133}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 6 & 18 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 12 & -4 & 0 \end{pmatrix}_{b_0}$$
- [78] $1^{113}31^{123}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 5 & 15 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 11 & -4 & 0 \end{pmatrix}_{b_0}$$
- [79] $1^{133}31^{123}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 526, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 7 & 21 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 17 & -4 & 0 \end{pmatrix}_{b_0}$$
- [80] $1^{123}31^{123}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 568, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 6 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 14 & -4 & 0 \end{pmatrix}_{b_0}$$
- [81] $1^{114}31^{123}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 616, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.08 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 1 & 2 \\ 2 & 4 & 4 & 42 & 0 & 4 \\ 1 & 4 & 0 & 21 & 0 & 2 \\ 14 & 42 & 21 & 292 & 0 & 28 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 28 & -4 & 0 \end{pmatrix}_{b_0}$$

[93] $1^{14}31^{14}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1936$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 1 & 2 \\ 2 & 4 & 4 & 36 & 0 & 4 \\ 1 & 4 & 0 & 24 & 0 & 2 \\ 12 & 36 & 24 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 28 & -4 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [94] $1^{12}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -3$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$

$T_{\min} = 19$, $T_{\max} = 21$, $T \in [19, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 2 \\ 2 & 0 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & 3 & -3 & -1 \\ 2 & 0 & 2 & 3 & -1 & -5 \end{pmatrix}_{b_0}$$

[95] $1^{11}2321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 16.14 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 3 & 1 & 0 & -3 & -1 \\ 2 & 3 & 2 & 0 & -1 & -3 \end{pmatrix}_{b_0}$$

[96] $1^{11}2321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -12$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 1 & 2 \\ 2 & 0 & 4 & 6 & 4 & 6 \\ 1 & 0 & 1 & 4 & -3 & -1 \\ 2 & 0 & 2 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{11}4^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6$ $T = 9$, $T_{\max} = 93$, $\Delta = -133$, $\det = -15$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[98] $1^{11}2321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -26$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 4 & 6 \\ 1 & 1 & 0 & 4 & -3 & -1 \\ 2 & 3 & -1 & 6 & -1 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [99] $1^{11}5^{11}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6$ $T = 10$, $T_{\max} = 91$, $\Delta = -128$, $\det = 25$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.0$

$T_{\min} = 14.16 \rightarrow 15$, $T_{\max} = 91$, $T \in [15, 91]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & -2 & 0 \\ 3 & 4 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

4.762 NHC(-2)(-3) + $\textcolor{red}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{orange}{e}_6$

$T = 8$ [0] $1^{111}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{orange}{e}_6$ $T = 8$, $T_{\max} = 103$, $\Delta = -212$, $\det = 385$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 15.06 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 4 & -1 & -4 & -3 \end{pmatrix}_{b_0}$$

4.763 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{f}_4$

$$T = 8 \quad [0] \ 1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 4 & -1 & -3 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{112}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.7$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 1 & 3 \\ 4 & 9 & 16 & 4 & 0 & 5 \\ 6 & 16 & 22 & 6 & 0 & 8 \\ 2 & 4 & 6 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 8 & -1 & -3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{113}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 739, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 1 & 3 \\ 4 & 9 & 24 & 4 & 0 & 5 \\ 9 & 24 & 52 & 9 & 0 & 12 \\ 2 & 4 & 9 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 12 & -1 & -3 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \ 1^{\overline{11}} 3^{\overline{21}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 5 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 4 & 0 & -3 & -2 \end{pmatrix}_{b_0}$$

4.764 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -210, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 4 & 3 \\ 3 & 5 & 3 & 0 & 4 & 3 & 3 \\ 3 & 4 & 3 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 & 2 & 3 \\ 4 & 9 & 12 & 0 & 8 & 4 & 5 \\ 5 & 12 & 13 & 0 & 9 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 8 & 9 & 0 & 4 & 4 & 4 \\ 2 & 4 & 6 & 0 & 4 & 0 & 2 \\ 3 & 5 & 7 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -298, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 4 & 3 \\ 2 & 4 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$(1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 & 2 & 3 \\ 3 & 4 & 36 & 0 & 5 & 4 & 4 \\ 20 & 36 & 238 & 0 & 36 & 24 & 28 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 36 & 0 & 4 & 4 & 4 \\ 2 & 4 & 24 & 0 & 4 & 0 & 2 \\ 3 & 4 & 28 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [13] \quad 1^{11}31^{121}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 170, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 3 & 2 \\ 4 & 12 & 6 & 22 & 0 & 12 & 6 & 8 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 12 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 6 & 0 & 4 & 0 & 2 \\ 2 & 4 & 2 & 8 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11}31^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 440, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 3 & 0 & 4 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

4.765 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \text{NHC}(-2)(-3) + \textcolor{blue}{su}_2$

$$T = 10 \quad [0] \quad 1^{11}31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 5 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 6 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 4 & 6 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & 4 & -4 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 429, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 4 & 9 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111}31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 1264, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 & 9 & 4 \\ 5 & 9 & 13 & 0 & 9 & 6 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 9 & 0 & 4 & 4 & 9 & 4 \\ 2 & 4 & 6 & 0 & 4 & 0 & 6 & 2 \\ 5 & 9 & 15 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & 7 & -4 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

4.766 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \text{NHC}(-2)(-3) + \textcolor{blue}{su}_3$

$$T = 10 \quad [0] \quad 1^{1111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 3 & 0 & 4 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.767 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 6 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.71 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 44 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 44 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 34 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 59 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 59 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 45 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.53 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 318, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 1 & 3 \\ 4 & 9 & 16 & 4 & 0 & 5 \\ 6 & 16 & 22 & 6 & 0 & 8 \\ 2 & 4 & 6 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 8 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 379, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 35 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 35 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 27 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 17 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3^1 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 430, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 34 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 26 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 26 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 20 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{113} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 493, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 1 & 3 \\ 4 & 9 & 24 & 4 & 0 & 5 \\ 9 & 24 & 52 & 9 & 0 & 12 \\ 2 & 4 & 9 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 12 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 546, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 26 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 20 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{114} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 738, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 1 & 3 \\ 4 & 9 & 32 & 4 & 0 & 5 \\ 12 & 32 & 94 & 12 & 0 & 16 \\ 2 & 4 & 12 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 16 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{113} 3^1 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 745, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 51 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 39 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{113} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 921, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 35 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 27 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{113} 3^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1005, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 43 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 43 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 33 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{115} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1053, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.1$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 1 & 3 \\ 4 & 9 & 40 & 4 & 0 & 5 \\ 15 & 40 & 148 & 15 & 0 & 20 \\ 2 & 4 & 15 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 20 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{114} 3^1 4^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 68 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 52 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{114} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 44 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 44 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 34 & -4 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{114}31^{31} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{so}}_8$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1590$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 60 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 46 & -4 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{114}31^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{so}}_8$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1650$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 52 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 40 & -4 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [23] $1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -13$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 15.15 \rightarrow 16$, $T_{\max} = 94$, $T \in [16, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 0 & 5 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 5 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & 0 & -2 & -2 \end{pmatrix}_{b_0}$$

[24] $1^{111}31^{131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 5 & 4 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 12 & 0 & 6 \\ 4 & 12 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{11}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -151$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 15 & 0 & 6 \\ 5 & 12 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{111}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -571$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 5 & 9 & 1 & 3 \\ 4 & 10 & 13 & 24 & 0 & 6 \\ 5 & 13 & 12 & 27 & 0 & 6 \\ 9 & 24 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [27] $1^{111}2321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{so}}_8$ $T = 10$, $T_{\max} = 22$, $\Delta = -107$, $\det = 60$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.60 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 4 & 1 & 6 & 2 & 3 \\ 4 & 10 & 1 & 16 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 16 & 0 & 22 & 8 & 10 \\ 2 & 4 & 0 & 8 & -2 & 0 \\ 3 & 6 & -1 & 10 & 0 & 1 \end{pmatrix}_{b_0}$$

4.768 $\text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$

$T = 9$ [0] $1^{112}31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -28$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.5$
 $T_{\min} = 14.71 \rightarrow 15$, $T_{\max} = 93$, $T \in [15, 93]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 26 & 5 \\ 12 & 25 & 16 & 0 & 56 & 9 \\ 8 & 16 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 56 & 36 & 0 & 118 & 20 \\ 5 & 9 & 6 & -4 & 20 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -114$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.89 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 7 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -132$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -372$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 4 & 3 \\ 4 & 9 & 8 & 4 & 0 & 8 & 5 \\ 3 & 8 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 6 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^11^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -432$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 35 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 35 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -496$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 & 3 \\ 4 & 9 & 8 & 4 & 0 & 8 & 5 \\ 3 & 8 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 6 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{111}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -512$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -608$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 17 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111}3^11^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -618$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 26 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

4.769 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{brown}{e}_6 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$

$T = 10$ [0] $1^{111}3^11^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{e}_6 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -102$, $\det = 372$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 17 & 5 \\ 11 & 20 & 17 & 0 & 17 & 33 & 8 \\ 9 & 17 & 13 & 0 & 15 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 27 & 7 \\ 17 & 33 & 27 & 0 & 27 & 49 & 13 \\ 5 & 8 & 7 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

4.770 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{e}_6$ + $\mathfrak{so}_8$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 0 & 4 \\ 2 & 4 & 3 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & -2 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 8 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

4.771 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{e}_6$ + $\mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^{\overline{132}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$$

$$T_{\min} = 15.11 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 7 & 3 \\ 3 & 4 & 12 & 0 & 13 & 4 \\ 6 & 12 & 25 & 0 & 27 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 27 & 0 & 28 & 10 \\ 3 & 4 & 9 & -4 & 10 & -3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 21.15 \rightarrow 22, T_{\max} = 92, T \in [22, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 3 & -4 & 8 & -9 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 3^{\overline{121}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 & 3 \\ 3 & 4 & 10 & 0 & 5 & 4 \\ 6 & 10 & 22 & 0 & 12 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 12 & 0 & 4 & 4 \\ 3 & 4 & 8 & -4 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3^{\overline{121}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 15.78 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$$

$$\begin{pmatrix} 1 & 4 & 7 & 1 & 5 & 3 \\ 4 & 9 & 16 & 0 & 12 & 5 \\ 7 & 16 & 28 & 0 & 21 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 21 & 0 & 13 & 7 \\ 3 & 5 & 10 & -4 & 7 & -3 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 15.02 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 3 & -4 & 4 & -3 \end{pmatrix}_{b_0}$$

[5] $1^{144}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 42, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 14.81 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 11 & 3 \\ 3 & 4 & 16 & 0 & 21 & 4 \\ 8 & 16 & 46 & 0 & 60 & 12 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 60 & 0 & 76 & 16 \\ 3 & 4 & 12 & -4 & 16 & -2 \end{pmatrix}_{b_0}$$

[6] $1^{133}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 15 & 3 \\ 3 & 4 & 17 & 0 & 27 & 4 \\ 9 & 17 & 49 & 0 & 81 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 81 & 0 & 133 & 21 \\ 3 & 4 & 13 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.22 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 6 & 0 & 13 & 7 \\ 3 & 5 & 3 & -4 & 7 & -2 \end{pmatrix}_{b_0}$$

[9] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 7 \\ 3 & 4 & 6 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{124}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 17 & 3 \\ 3 & 4 & 8 & 0 & 31 & 4 \\ 4 & 8 & 10 & 0 & 42 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 42 & 0 & 172 & 24 \\ 3 & 4 & 6 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.47 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 4 \\ 3 & 4 & 4 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{1111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 & 3 \\ 4 & 9 & 8 & 4 & 8 & 5 \\ 3 & 8 & 4 & 3 & 6 & 4 \\ 3 & 4 & 3 & -3 & 3 & -1 \\ 3 & 8 & 6 & 3 & 4 & 4 \\ 3 & 5 & 4 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[13] $21^131^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 & 3 \\ 4 & 9 & 8 & 4 & 8 & 5 \\ 3 & 8 & 4 & 3 & 6 & 4 \\ 3 & 4 & 3 & -3 & 3 & -1 \\ 3 & 8 & 6 & 3 & 4 & 4 \\ 3 & 5 & 4 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{112}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 24 & 12 & 0 & 58 & 14 \\ 3 & 5 & 3 & -4 & 14 & -2 \end{pmatrix}_{b_0}$$

[15] $1^{111}3^11^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 8 & 3 \\ 3 & 4 & 10 & 0 & 14 & 4 \\ 6 & 10 & 22 & 0 & 30 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 30 & 0 & 37 & 11 \\ 3 & 4 & 8 & -4 & 11 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{111}3^11 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{13}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 49 & 13 \\ 3 & 4 & 4 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{112}3^11 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 3 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{113}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 15.23 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 36 & 18 & 0 & 133 & 21 \\ 3 & 5 & 3 & -4 & 21 & -3 \end{pmatrix}_{b_0}$$

[20] $1^{122}3^11 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.53 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.91 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 20 & 5 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 20 & 9 & 0 & 37 & 11 \\ 3 & 5 & 3 & -4 & 11 & -1 \end{pmatrix}_{b_0}$$

[22] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 3 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{111}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 3 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{111}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.74 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 14 & 3 \\ 3 & 4 & 9 & 0 & 24 & 4 \\ 5 & 9 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 24 & 42 & 0 & 121 & 19 \\ 3 & 4 & 7 & -4 & 19 & -1 \end{pmatrix}_{b_0}$$

[25] $1^{123}3^11 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 6 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 18 & 0 & 25 & 9 \\ 3 & 4 & 6 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{133}3^11 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 9 & 3 \\ 3 & 4 & 12 & 0 & 17 & 4 \\ 6 & 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 36 & 0 & 49 & 13 \\ 3 & 4 & 9 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.34 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{114}3^11 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 3 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{133}3^11 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 6 & 3 \\ 3 & 4 & 12 & 0 & 12 & 4 \\ 6 & 12 & 25 & 0 & 27 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 27 & 0 & 25 & 9 \\ 3 & 4 & 9 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{111}3^11 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.17 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 4 & 3 \\ 4 & 9 & 8 & 4 & 8 & 5 \\ 3 & 8 & 4 & 3 & 6 & 4 \\ 2 & 4 & 3 & -3 & 3 & -1 \\ 4 & 8 & 6 & 3 & 4 & 4 \\ 3 & 5 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[32] $1^{124}3^11 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 4 & 8 & 0 & 16 & 4 \\ 4 & 8 & 10 & 0 & 24 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 24 & 0 & 46 & 12 \\ 3 & 4 & 6 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{12}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 4 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{111}3^11 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.28 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{114}3^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 190, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 15.48 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 9 & 4 & 0 & 48 & 5 \\ 2 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 20 & 48 & 24 & 0 & 238 & 28 \\ 3 & 5 & 3 & -4 & 28 & -3 \end{pmatrix}_{b_0}$$

[35] $1^{112}3^{1121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.44 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 7 & 3 \\ 3 & 4 & 14 & 0 & 13 & 4 \\ 8 & 14 & 37 & 0 & 33 & 11 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 33 & 0 & 28 & 10 \\ 3 & 4 & 11 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{123}3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 3 \\ 3 & 4 & 8 & 0 & 22 & 4 \\ 4 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 30 & 0 & 85 & 17 \\ 3 & 4 & 6 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{134}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 8 & 3 \\ 3 & 4 & 12 & 0 & 16 & 4 \\ 6 & 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 36 & 0 & 46 & 12 \\ 3 & 4 & 9 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$

[38] $1^{115}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.1$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 20 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 20 & 15 & 0 & 73 & 15 \\ 3 & 4 & 3 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

[39] $1^{11}13^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 3 \\ 4 & 9 & 8 & 4 & 8 & 5 \\ 3 & 8 & 4 & 3 & 6 & 4 \\ 2 & 4 & 3 & -3 & 3 & -1 \\ 3 & 8 & 6 & 3 & 4 & 4 \\ 3 & 5 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[40] $1^{122}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 18 & 0 & 28 & 10 \\ 3 & 4 & 6 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{113}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 215, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 14.28 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 28 & 5 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 28 & 12 & 0 & 73 & 15 \\ 3 & 5 & 3 & -4 & 15 & -2 \end{pmatrix}_{b_0}$$

[42] $1^{125}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 10 & 3 \\ 3 & 4 & 8 & 0 & 20 & 4 \\ 4 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 20 & 30 & 0 & 73 & 15 \\ 3 & 4 & 6 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

[43] $1^{144}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 8 & 3 \\ 3 & 4 & 16 & 0 & 16 & 4 \\ 8 & 16 & 46 & 0 & 48 & 12 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 48 & 0 & 46 & 12 \\ 3 & 4 & 12 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{135}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 13.17 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 10 & 3 \\ 3 & 4 & 12 & 0 & 20 & 4 \\ 6 & 12 & 25 & 0 & 45 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 20 & 45 & 0 & 73 & 15 \\ 3 & 4 & 9 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

[44] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 3 \\ 4 & 9 & 12 & 0 & 12 & 5 \\ 5 & 12 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 15 & 0 & 13 & 7 \\ 3 & 5 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{116}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.21 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 24 & 18 & 0 & 106 & 18 \\ 3 & 4 & 3 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

[47] $1^{113}3^11^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 13.88 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 13 & 32 & 15 & 0 & 100 & 18 \\ 3 & 5 & 3 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{126}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.24 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 3 \\ 3 & 4 & 8 & 0 & 24 & 4 \\ 4 & 8 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 24 & 36 & 0 & 106 & 18 \\ 3 & 4 & 6 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

[49] $1^{111}3^11^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 5 & 0 & 14 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 15 & 0 & 37 & 11 \\ 3 & 4 & 4 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{145}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 10 & 3 \\ 3 & 4 & 16 & 0 & 20 & 4 \\ 8 & 16 & 46 & 0 & 60 & 12 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 20 & 60 & 0 & 73 & 15 \\ 3 & 4 & 12 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

[50] $1^{112}3^11^4 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 28 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 16 & 28 & 18 & 0 & 154 & 22 \\ 3 & 4 & 3 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{114}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 14.81 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 36 & 15 & 0 & 121 & 19 \\ 3 & 5 & 3 & -4 & 19 & -2 \end{pmatrix}_{b_0}$$

[53] $1^{136}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 284, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.30 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 12 & 3 \\ 3 & 4 & 12 & 0 & 24 & 4 \\ 6 & 12 & 25 & 0 & 54 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 24 & 54 & 0 & 106 & 18 \\ 3 & 4 & 9 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

[55] $1^{117}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 28 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 28 & 21 & 0 & 145 & 21 \\ 3 & 4 & 3 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

[54] $1^{155}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 10 & 3 \\ 3 & 4 & 20 & 0 & 20 & 4 \\ 10 & 20 & 73 & 0 & 75 & 15 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 20 & 75 & 0 & 73 & 15 \\ 3 & 4 & 15 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

[56] $1^{146}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 12 & 3 \\ 3 & 4 & 16 & 0 & 24 & 4 \\ 8 & 16 & 46 & 0 & 72 & 12 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 24 & 72 & 0 & 106 & 18 \\ 3 & 4 & 12 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

[57] $1^{127}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 14 & 3 \\ 3 & 4 & 8 & 0 & 28 & 4 \\ 4 & 8 & 10 & 0 & 42 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 28 & 42 & 0 & 145 & 21 \\ 3 & 4 & 6 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

[58] $1^{132}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 318, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 10 & 3 \\ 3 & 4 & 17 & 0 & 18 & 4 \\ 9 & 17 & 49 & 0 & 54 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 54 & 0 & 58 & 14 \\ 3 & 4 & 13 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{134}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 329, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 11 & 3 \\ 3 & 4 & 12 & 0 & 21 & 4 \\ 6 & 12 & 25 & 0 & 45 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 45 & 0 & 76 & 16 \\ 3 & 4 & 9 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{11}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.71 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 19 & 4 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 19 & 21 & 0 & 73 & 15 \\ 3 & 4 & 4 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

[62] $1^{137}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 14 & 3 \\ 3 & 4 & 12 & 0 & 28 & 4 \\ 6 & 12 & 25 & 0 & 63 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 28 & 63 & 0 & 145 & 21 \\ 3 & 4 & 9 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

[61] $1^{111}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 19 & 4 \\ 5 & 9 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 19 & 33 & 0 & 73 & 15 \\ 3 & 4 & 7 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

[63] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 337, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.32 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{156}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 348, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.4$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 12 & 3 \\ 3 & 4 & 20 & 0 & 24 & 4 \\ 10 & 20 & 73 & 0 & 90 & 15 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 24 & 90 & 0 & 106 & 18 \\ 3 & 4 & 15 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

[65] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[66] $1^{122}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 13 & 3 \\ 3 & 4 & 13 & 0 & 23 & 4 \\ 7 & 13 & 28 & 0 & 54 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 54 & 0 & 100 & 18 \\ 3 & 4 & 10 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{147}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 14 & 3 \\ 3 & 4 & 16 & 0 & 28 & 4 \\ 8 & 16 & 46 & 0 & 84 & 12 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 28 & 84 & 0 & 145 & 21 \\ 3 & 4 & 12 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

[68] $1^{143}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 12 & 3 \\ 3 & 4 & 21 & 0 & 22 & 4 \\ 11 & 21 & 76 & 0 & 81 & 16 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 81 & 0 & 85 & 17 \\ 3 & 4 & 16 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{111}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 5 & 0 & 24 & 4 \\ 3 & 5 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 24 & 27 & 0 & 121 & 19 \\ 3 & 4 & 4 & -4 & 19 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{123}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 17 & 4 \\ 4 & 8 & 10 & 0 & 24 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 24 & 0 & 49 & 13 \\ 3 & 4 & 6 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{114}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.97 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 3 \\ 4 & 9 & 4 & 0 & 44 & 5 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 18 & 44 & 21 & 0 & 193 & 25 \\ 3 & 5 & 3 & -4 & 25 & -1 \end{pmatrix}_{b_0}$$

[72] $1^{166}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 12 & 3 \\ 3 & 4 & 24 & 0 & 24 & 4 \\ 12 & 24 & 106 & 0 & 108 & 18 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 24 & 108 & 0 & 106 & 18 \\ 3 & 4 & 18 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

[73] $1^{113}3^11^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 12 & 3 \\ 3 & 4 & 14 & 0 & 22 & 4 \\ 8 & 14 & 37 & 0 & 57 & 11 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 57 & 0 & 85 & 17 \\ 3 & 4 & 11 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{157}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.5$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 14 & 3 \\ 3 & 4 & 20 & 0 & 28 & 4 \\ 10 & 20 & 73 & 0 & 105 & 15 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 28 & 105 & 0 & 145 & 21 \\ 3 & 4 & 15 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

[75] $1^{111}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 405, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 3 \\ 3 & 4 & 9 & 0 & 14 & 4 \\ 5 & 9 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 24 & 0 & 37 & 11 \\ 3 & 4 & 7 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{112}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 409, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 3 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{12}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 414, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 5 & 0 & 18 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[78] $1^{114}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 21 & 4 \\ 5 & 9 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 33 & 0 & 76 & 16 \\ 3 & 4 & 7 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{114}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 422, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.93 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 16 & 40 & 18 & 0 & 154 & 22 \\ 3 & 5 & 3 & -4 & 22 & -1 \end{pmatrix}_{b_0}$$

[80] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 3 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[81] $1^{167}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 444, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 14 & 3 \\ 3 & 4 & 24 & 0 & 28 & 4 \\ 12 & 24 & 106 & 0 & 126 & 18 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 28 & 126 & 0 & 145 & 21 \\ 3 & 4 & 18 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

[82] $1^{112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 459, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.34 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 7 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[83] $1^{14}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 466, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 5 & 0 & 26 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 24 & 0 & 118 & 20 \\ 3 & 4 & 4 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[84] $1^{113}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 27 & 0 & 49 & 13 \\ 3 & 4 & 7 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[85] $1^{13}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 481, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 5 & 0 & 22 & 4 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 21 & 0 & 85 & 17 \\ 3 & 4 & 4 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[86] $1^{113}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 493, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[87] $1^{124}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 496, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 14 & 3 \\ 3 & 4 & 8 & 0 & 26 & 4 \\ 4 & 8 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 36 & 0 & 118 & 20 \\ 3 & 4 & 6 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[88] $1^{177}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 496, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.60 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 14 & 3 \\ 3 & 4 & 28 & 0 & 28 & 4 \\ 14 & 28 & 145 & 0 & 147 & 21 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 28 & 147 & 0 & 145 & 21 \\ 3 & 4 & 21 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

[89] $1^{124}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 534, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 11 & 3 \\ 3 & 4 & 8 & 0 & 21 & 4 \\ 4 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 30 & 0 & 76 & 16 \\ 3 & 4 & 6 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[90] $1^{111}3^11^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 8 & 3 \\ 3 & 4 & 14 & 0 & 14 & 4 \\ 8 & 14 & 37 & 0 & 39 & 11 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 39 & 0 & 37 & 11 \\ 3 & 4 & 11 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[91] $1^{12}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 616, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 5 & 0 & 23 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 24 & 0 & 100 & 18 \\ 3 & 4 & 4 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

[92] $1^{112}3^11^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 648, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 10 & 3 \\ 3 & 4 & 14 & 0 & 18 & 4 \\ 8 & 14 & 37 & 0 & 48 & 11 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 48 & 0 & 58 & 14 \\ 3 & 4 & 11 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[93] $1^{122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 652, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 30 & 0 & 28 & 10 \\ 3 & 4 & 10 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[94] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 657, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 3 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[95] $1^{123} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 697, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 18 & 3 \\ 3 & 4 & 13 & 0 & 32 & 4 \\ 7 & 13 & 28 & 0 & 75 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 75 & 0 & 193 & 25 \\ 3 & 4 & 10 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[96] $1^{122} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 702, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 10 & 3 \\ 3 & 4 & 13 & 0 & 18 & 4 \\ 7 & 13 & 28 & 0 & 42 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 42 & 0 & 58 & 14 \\ 3 & 4 & 10 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[97] $1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 3 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[98] $1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 734, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 10 & 3 \\ 3 & 4 & 9 & 0 & 18 & 4 \\ 5 & 9 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 30 & 0 & 58 & 14 \\ 3 & 4 & 7 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[99] $1^{123} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 763, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 9 & 3 \\ 3 & 4 & 13 & 0 & 17 & 4 \\ 7 & 13 & 28 & 0 & 39 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 39 & 0 & 49 & 13 \\ 3 & 4 & 10 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[100] $1^{12} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 790, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 5 & 0 & 28 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 16 & 28 & 30 & 0 & 154 & 22 \\ 3 & 4 & 4 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

[101] $1^{124} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 792, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 11 & 3 \\ 3 & 4 & 13 & 0 & 21 & 4 \\ 7 & 13 & 28 & 0 & 48 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 48 & 0 & 76 & 16 \\ 3 & 4 & 10 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[102] $1^{13} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 819, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 5 & 0 & 27 & 4 \\ 3 & 5 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[103] $1^{113} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 832, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 21 & 0 & 193 & 25 \\ 3 & 4 & 3 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[104] $1^{113}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 837, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 3 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[105] $21^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 884, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 6 & 4 \\ 8 & 9 & 8 & 4 & 8 & 5 \\ 6 & 8 & 4 & 3 & 6 & 4 \\ 3 & 4 & 3 & -3 & 3 & -1 \\ 6 & 8 & 6 & 3 & 4 & 4 \\ 4 & 5 & 4 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[106] $1^{112}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 891, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.44 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 13 & 3 \\ 3 & 4 & 9 & 0 & 23 & 4 \\ 5 & 9 & 13 & 0 & 39 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 39 & 0 & 100 & 18 \\ 3 & 4 & 7 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

[107] $1^{133}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 925, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 12 & 3 \\ 3 & 4 & 17 & 0 & 22 & 4 \\ 9 & 17 & 49 & 0 & 66 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 66 & 0 & 85 & 17 \\ 3 & 4 & 13 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[108] $1^{112}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 930, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.54 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 16 & 3 \\ 3 & 4 & 9 & 0 & 28 & 4 \\ 5 & 9 & 13 & 0 & 48 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 16 & 28 & 48 & 0 & 154 & 22 \\ 3 & 4 & 7 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

[109] $1^{14}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 940, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 5 & 0 & 31 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 30 & 0 & 172 & 24 \\ 3 & 4 & 4 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[110] $1^{133}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 964, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 9 & 3 \\ 3 & 4 & 17 & 0 & 17 & 4 \\ 9 & 17 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 51 & 0 & 49 & 13 \\ 3 & 4 & 13 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[111] $1^{134}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 979, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 17 & 3 \\ 3 & 4 & 17 & 0 & 31 & 4 \\ 9 & 17 & 49 & 0 & 93 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 93 & 0 & 172 & 24 \\ 3 & 4 & 13 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[112] $1^{113}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 981, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 12 & 3 \\ 3 & 4 & 9 & 0 & 22 & 4 \\ 5 & 9 & 13 & 0 & 36 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 36 & 0 & 85 & 17 \\ 3 & 4 & 7 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[113] $1^{124}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 984, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 14 & 3 \\ 3 & 4 & 18 & 0 & 26 & 4 \\ 10 & 18 & 58 & 0 & 84 & 14 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 84 & 0 & 118 & 20 \\ 3 & 4 & 14 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[114] $1^{114}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1024, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 3 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[115] $1^{144}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1074, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 14 & 3 \\ 3 & 4 & 21 & 0 & 26 & 4 \\ 11 & 21 & 76 & 0 & 96 & 16 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 96 & 0 & 118 & 20 \\ 3 & 4 & 16 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[116] $1^{134}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1083, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 11 & 3 \\ 3 & 4 & 17 & 0 & 21 & 4 \\ 9 & 17 & 49 & 0 & 63 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 63 & 0 & 76 & 16 \\ 3 & 4 & 13 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[117] $1^{123}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 15 & 3 \\ 3 & 4 & 13 & 0 & 27 & 4 \\ 7 & 13 & 28 & 0 & 63 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 63 & 0 & 133 & 21 \\ 3 & 4 & 10 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[118] $1^{122}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 10 & 3 \\ 3 & 4 & 18 & 0 & 18 & 4 \\ 10 & 18 & 58 & 0 & 60 & 14 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 60 & 0 & 58 & 14 \\ 3 & 4 & 14 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[119] $1^{13}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.48 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 5 & 0 & 32 & 4 \\ 3 & 5 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 33 & 0 & 193 & 25 \\ 3 & 4 & 4 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[120] $1^{123}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 12 & 3 \\ 3 & 4 & 13 & 0 & 22 & 4 \\ 7 & 13 & 28 & 0 & 51 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 51 & 0 & 85 & 17 \\ 3 & 4 & 10 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[121] $1^{114}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 14 & 3 \\ 3 & 4 & 9 & 0 & 26 & 4 \\ 5 & 9 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 42 & 0 & 118 & 20 \\ 3 & 4 & 7 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[122] $1^{123}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 12 & 3 \\ 3 & 4 & 18 & 0 & 22 & 4 \\ 10 & 18 & 58 & 0 & 72 & 14 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 72 & 0 & 85 & 17 \\ 3 & 4 & 14 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[123] $1^{114}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1273, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 3 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[124] $1^{144}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 11 & 3 \\ 3 & 4 & 21 & 0 & 21 & 4 \\ 11 & 21 & 76 & 0 & 78 & 16 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 78 & 0 & 76 & 16 \\ 3 & 4 & 16 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[125] $1^{113}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 15 & 3 \\ 3 & 4 & 9 & 0 & 27 & 4 \\ 5 & 9 & 13 & 0 & 45 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 45 & 0 & 133 & 21 \\ 3 & 4 & 7 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[126] $1^{14}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1386, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 5 & 0 & 36 & 4 \\ 3 & 5 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

[127] $1^{114}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 3 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

[128] $1^{134}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 14 & 3 \\ 3 & 4 & 17 & 0 & 26 & 4 \\ 9 & 17 & 49 & 0 & 78 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 78 & 0 & 118 & 20 \\ 3 & 4 & 13 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[129] $1^{124}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1474, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 14 & 3 \\ 3 & 4 & 13 & 0 & 26 & 4 \\ 7 & 13 & 28 & 0 & 60 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 60 & 0 & 118 & 20 \\ 3 & 4 & 10 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[130] $1^{113}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.44 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 18 & 3 \\ 3 & 4 & 9 & 0 & 32 & 4 \\ 5 & 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 54 & 0 & 193 & 25 \\ 3 & 4 & 7 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[131] $1^{124}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1674, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 20 & 3 \\ 3 & 4 & 13 & 0 & 36 & 4 \\ 7 & 13 & 28 & 0 & 84 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 84 & 0 & 238 & 28 \\ 3 & 4 & 10 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

[132] $1^{133} 31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 1696, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 12 & 3 \\ 3 & 4 & 22 & 0 & 22 & 4 \\ 12 & 22 & 85 & 0 & 87 & 17 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 87 & 0 & 85 & 17 \\ 3 & 4 & 17 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[133] $1^{114} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 1755, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 17 & 3 \\ 3 & 4 & 9 & 0 & 31 & 4 \\ 5 & 9 & 13 & 0 & 51 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 51 & 0 & 172 & 24 \\ 3 & 4 & 7 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[134] $1^{124} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 1768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 17 & 3 \\ 3 & 4 & 13 & 0 & 31 & 4 \\ 7 & 13 & 28 & 0 & 72 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 72 & 0 & 172 & 24 \\ 3 & 4 & 10 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[135] $1^{134} 31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 1816, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 14 & 3 \\ 3 & 4 & 22 & 0 & 26 & 4 \\ 12 & 22 & 85 & 0 & 102 & 17 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 102 & 0 & 118 & 20 \\ 3 & 4 & 17 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[136] $1^{114} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 2246, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.41 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 20 & 3 \\ 3 & 4 & 9 & 0 & 36 & 4 \\ 5 & 9 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 60 & 0 & 238 & 28 \\ 3 & 4 & 7 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

[137] $1^{144} 31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 92, \Delta = -133, \det = 2296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 14 & 3 \\ 3 & 4 & 26 & 0 & 26 & 4 \\ 14 & 26 & 118 & 0 & 120 & 20 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 120 & 0 & 118 & 20 \\ 3 & 4 & 20 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [138] $1^{111} 31^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 23, \Delta = -112, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 19.50 \rightarrow 20, T_{\max} = 23, T \in [20, 23]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 \\ 1 & 3 & 3 & 3 & 0 & 1 \\ 2 & 3 & 3 & 3 & -4 & 3 \end{pmatrix}_{b_0}$$

[139] $1^{111} 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 21, \Delta = -105, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 1 & 0 & 1 & 3 & 0 & 3 \\ 1 & 1 & 0 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 3 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

[140] $1^{111} 31^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 23, \Delta = -112, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 1 & 3 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 4 & 4 \\ 2 & 4 & 2 & 3 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{12}31^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 8 & 2 \\ 2 & 4 & 4 & 12 & 0 & 24 & 4 \\ 1 & 4 & 0 & 9 & 0 & 15 & 2 \\ 4 & 12 & 9 & 28 & 0 & 54 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 15 & 54 & 0 & 100 & 18 \\ 2 & 4 & 2 & 10 & -4 & 18 & -2 \end{pmatrix}_{b_0}$$

$$[141] \ 1^{14}31^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 3 & 0 & 24 & 4 \\ 1 & 4 & 0 & 3 & 0 & 18 & 2 \\ 1 & 3 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 18 & 18 & 0 & 118 & 20 \\ 2 & 4 & 2 & 3 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[143] \ 1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 4 & 3 & 5 \\ 1 & 1 & 0 & 4 & -3 & 1 & -1 \\ 1 & 2 & 0 & 3 & 1 & 0 & 2 \\ 2 & 2 & -1 & 5 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[144] \ 1^{12}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 3 & 0 & 12 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 \\ 1 & 3 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 9 & 9 & 0 & 28 & 10 \\ 2 & 4 & 2 & 3 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[145] \ 1^{111}31^{14}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 3 & 0 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 15 & 2 \\ 1 & 3 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 15 & 15 & 0 & 121 & 19 \\ 2 & 4 & 2 & 3 & -4 & 19 & -2 \end{pmatrix}_{b_0}$$

$$[146] \ 1^{14}31^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 6 & 0 & 30 & 4 \\ 1 & 4 & 0 & 3 & 0 & 21 & 2 \\ 2 & 6 & 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 21 & 30 & 0 & 172 & 24 \\ 2 & 4 & 2 & 4 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[147] \ 1^{13}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 3 & 0 & 15 & 3 \\ 1 & 3 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 15 & 12 & 0 & 49 & 13 \\ 2 & 3 & 3 & 3 & -4 & 13 & -2 \end{pmatrix}_{b_0}$$

$$[148] \ 1^{111}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 3 & 0 & 9 & 3 \\ 1 & 3 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 6 & 9 & 6 & 0 & 13 & 7 \\ 2 & 3 & 3 & 3 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[149] \ 1^{111}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -185, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 1 & 3 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & 3 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[150] \ 1^{11}31^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 4 & 2 \\ 2 & 4 & 4 & 9 & 0 & 12 & 4 \\ 1 & 4 & 0 & 6 & 0 & 6 & 2 \\ 3 & 9 & 6 & 13 & 0 & 18 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 6 & 18 & 0 & 22 & 8 \\ 2 & 4 & 2 & 7 & -4 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[151] \ 1^{112}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -233, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 3 & 0 & 12 & 3 \\ 1 & 3 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 9 & 12 & 9 & 0 & 28 & 10 \\ 2 & 3 & 3 & 3 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[152] \ 1^{111}31^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -257, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 13.20 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 3 & 0 & 21 & 4 \\ 1 & 4 & 0 & 3 & 0 & 12 & 2 \\ 1 & 3 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 12 & 12 & 0 & 73 & 15 \\ 2 & 4 & 2 & 3 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[153] \ 1^{111}31^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 3 & 0 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 \\ 1 & 3 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 9 & 9 & 0 & 37 & 11 \\ 2 & 4 & 2 & 3 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[154] \ 1^131^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 4 \\ 2 & 4 & 2 & 4 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[155] \ 1^{113}31^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 3 & 0 & 21 & 4 \\ 1 & 4 & 0 & 3 & 0 & 15 & 2 \\ 1 & 3 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 15 & 15 & 0 & 85 & 17 \\ 2 & 4 & 2 & 3 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[156] \ 1^{11}31^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -389, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 6 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 9 & 0 & 13 & 7 \\ 2 & 4 & 2 & 4 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[157] \ 1^{114}31^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 8 & 2 \\ 2 & 4 & 4 & 9 & 0 & 24 & 4 \\ 1 & 4 & 0 & 6 & 0 & 18 & 2 \\ 3 & 9 & 6 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 18 & 42 & 0 & 118 & 20 \\ 2 & 4 & 2 & 7 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[158] \ 1^{121}31^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -437, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

11.0

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 3 & 2 \\ 2 & 4 & 4 & 12 & 0 & 9 & 4 \\ 1 & 4 & 0 & 9 & 0 & 6 & 2 \\ 4 & 12 & 9 & 28 & 0 & 21 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 21 & 0 & 13 & 7 \\ 2 & 4 & 2 & 10 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[160] \ 1^{122}31^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 4 & 2 \\ 2 & 4 & 4 & 12 & 0 & 12 & 4 \\ 1 & 4 & 0 & 9 & 0 & 9 & 2 \\ 4 & 12 & 9 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 9 & 30 & 0 & 28 & 10 \\ 2 & 4 & 2 & 10 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[159] \ 1^{122}31^{123}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 3 & 0 & 18 & 4 \\ 1 & 4 & 0 & 3 & 0 & 12 & 2 \\ 1 & 3 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 12 & 12 & 0 & 58 & 14 \\ 2 & 4 & 2 & 3 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[161] \ 1^131^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 12 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 6 & 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 6 & 12 & 0 & 22 & 8 \\ 2 & 4 & 2 & 4 & -4 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[162] \ 1^{123}1^{123}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -668, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 6 & 0 & 18 & 4 \\ 1 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 6 & 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 12 & 18 & 0 & 58 & 14 \\ 2 & 4 & 2 & 4 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[163] \ 1^{111}31^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -680, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 9 & 4 \\ 1 & 4 & 0 & 6 & 0 & 6 & 2 \\ 3 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 15 & 0 & 13 & 7 \\ 2 & 4 & 2 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[164] \ 1^{11}31^{122}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 5 & 2 \\ 2 & 4 & 4 & 12 & 0 & 15 & 4 \\ 1 & 4 & 0 & 6 & 0 & 9 & 2 \\ 4 & 12 & 6 & 22 & 0 & 30 & 8 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 9 & 30 & 0 & 37 & 11 \\ 2 & 4 & 2 & 8 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[165] \ 1^{122}31^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -716, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 6 & 2 \\ 2 & 4 & 4 & 12 & 0 & 18 & 4 \\ 1 & 4 & 0 & 9 & 0 & 12 & 2 \\ 4 & 12 & 9 & 28 & 0 & 42 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 12 & 42 & 0 & 58 & 14 \\ 2 & 4 & 2 & 10 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[166] \ 1^{122}31^{133}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -725, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 3 & 0 & 24 & 4 \\ 1 & 4 & 0 & 3 & 0 & 15 & 2 \\ 1 & 3 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 15 & 15 & 0 & 100 & 18 \\ 2 & 4 & 2 & 3 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

[167] $1^1 31^{122} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -800, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 4 & 2 \\ 2 & 4 & 4 & 12 & 0 & 12 & 4 \\ 1 & 4 & 0 & 6 & 0 & 6 & 2 \\ 4 & 12 & 6 & 22 & 0 & 24 & 8 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 6 & 24 & 0 & 22 & 8 \\ 2 & 4 & 2 & 8 & -4 & 8 & 0 \end{pmatrix}_{b_0}$$

[168] $1^{123} 31^{143} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -860, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 3 & 0 & 30 & 4 \\ 1 & 4 & 0 & 3 & 0 & 18 & 2 \\ 1 & 3 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 18 & 18 & 0 & 154 & 22 \\ 2 & 4 & 2 & 3 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

[169] $1^{144} 31^{133} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -869, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 3 & 0 & 30 & 4 \\ 1 & 4 & 0 & 3 & 0 & 21 & 2 \\ 1 & 3 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 21 & 21 & 0 & 172 & 24 \\ 2 & 4 & 2 & 3 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[170] $1^{113} 31^{123} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -893, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 6 & 0 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 6 & 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 9 & 15 & 0 & 37 & 11 \\ 2 & 4 & 2 & 4 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[171] $1^{133} 31^{133} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -929, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 3 & 0 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 18 & 2 \\ 1 & 3 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 18 & 18 & 0 & 133 & 21 \\ 2 & 4 & 2 & 3 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[172] $1^1 31^{113} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 6 & 0 & 18 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 6 & 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 9 & 18 & 0 & 52 & 12 \\ 2 & 4 & 2 & 4 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

[173] $1^{111} 31^{123} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1049, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 5 & 2 \\ 2 & 4 & 4 & 9 & 0 & 15 & 4 \\ 1 & 4 & 0 & 6 & 0 & 9 & 2 \\ 3 & 9 & 6 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 9 & 24 & 0 & 37 & 11 \\ 2 & 4 & 2 & 7 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[174] $1^{133} 31^{133} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1133, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 6 & 0 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 18 & 2 \\ 2 & 6 & 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 18 & 27 & 0 & 133 & 21 \\ 2 & 4 & 2 & 4 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[175] $1^{133} 31^{123} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 7 & 2 \\ 2 & 4 & 4 & 9 & 0 & 21 & 4 \\ 1 & 4 & 0 & 6 & 0 & 15 & 2 \\ 3 & 9 & 6 & 13 & 0 & 36 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 15 & 36 & 0 & 85 & 17 \\ 2 & 4 & 2 & 7 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[176] \ 1^{111}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{c}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1292, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 7 & 2 \\ 2 & 4 & 4 & 9 & 0 & 21 & 4 \\ 1 & 4 & 0 & 6 & 0 & 12 & 2 \\ 3 & 9 & 6 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 12 & 33 & 0 & 73 & 15 \\ 2 & 4 & 2 & 7 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[177] \ 1^{112}31^{122}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{c}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1328, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 6 & 2 \\ 2 & 4 & 4 & 15 & 0 & 18 & 4 \\ 1 & 4 & 0 & 9 & 0 & 12 & 2 \\ 5 & 15 & 9 & 37 & 0 & 48 & 11 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 12 & 48 & 0 & 58 & 14 \\ 2 & 4 & 2 & 11 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[178] \ 1^{113}31^{143}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{c}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1352, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.82 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 3 & 0 & 33 & 4 \\ 1 & 4 & 0 & 3 & 0 & 21 & 2 \\ 1 & 3 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 33 & 21 & 21 & 0 & 193 & 25 \\ 2 & 4 & 2 & 3 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[179] \ 1^{112}31^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{c}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1364, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.75 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 6 & 2 \\ 2 & 4 & 4 & 9 & 0 & 18 & 4 \\ 1 & 4 & 0 & 6 & 0 & 12 & 2 \\ 3 & 9 & 6 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 12 & 30 & 0 & 58 & 14 \\ 2 & 4 & 2 & 7 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[180] \ 1^{111}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{c}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1409, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.90 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 9 & 2 \\ 2 & 4 & 4 & 9 & 0 & 27 & 4 \\ 1 & 4 & 0 & 6 & 0 & 15 & 2 \\ 3 & 9 & 6 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 15 & 42 & 0 & 121 & 19 \\ 2 & 4 & 2 & 7 & -4 & 19 & 0 \end{pmatrix}_{b_0}$$

$$[181] \ 1^{113}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{c}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1448, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 6 & 0 & 24 & 4 \\ 1 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 6 & 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 12 & 24 & 0 & 94 & 16 \\ 2 & 4 & 2 & 4 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[182] \ 1^{113}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{c}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1541, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.82 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 6 & 0 & 21 & 4 \\ 1 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 6 & 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 12 & 21 & 0 & 73 & 15 \\ 2 & 4 & 2 & 4 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[183] \ 1^{114}31^{143}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{c}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1580, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 3 & 0 & 36 & 4 \\ 1 & 4 & 0 & 3 & 0 & 24 & 2 \\ 1 & 3 & 3 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 36 & 24 & 24 & 0 & 238 & 28 \\ 2 & 4 & 2 & 3 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[184] \ 1^{123}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{c}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1604, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.79 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 6 & 0 & 24 & 4 \\ 1 & 4 & 0 & 3 & 0 & 15 & 2 \\ 2 & 6 & 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 15 & 24 & 0 & 100 & 18 \\ 2 & 4 & 2 & 4 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[185] \ 1^{111}31^{122}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1688, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 5 & 2 \\ 2 & 4 & 4 & 15 & 0 & 15 & 4 \\ 1 & 4 & 0 & 9 & 0 & 9 & 2 \\ 5 & 15 & 9 & 37 & 0 & 39 & 11 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 9 & 39 & 0 & 37 & 11 \\ 2 & 4 & 2 & 11 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[186] \ 1^{143}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -1784, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 6 & 0 & 36 & 4 \\ 1 & 4 & 0 & 3 & 0 & 24 & 2 \\ 2 & 6 & 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 36 & 24 & 36 & 0 & 238 & 28 \\ 2 & 4 & 2 & 4 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[187] \ 1^{122}31^{122}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -2048, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.75 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 6 & 2 \\ 2 & 4 & 4 & 18 & 0 & 18 & 4 \\ 1 & 4 & 0 & 12 & 0 & 12 & 2 \\ 6 & 18 & 12 & 58 & 0 & 60 & 14 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 12 & 60 & 0 & 58 & 14 \\ 2 & 4 & 2 & 14 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[188] \ 1^{112}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -2165, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 8 & 2 \\ 2 & 4 & 4 & 9 & 0 & 24 & 4 \\ 1 & 4 & 0 & 6 & 0 & 15 & 2 \\ 3 & 9 & 6 & 13 & 0 & 39 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 15 & 39 & 0 & 100 & 18 \\ 2 & 4 & 2 & 7 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[189] \ 1^{114}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -2309, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 10 & 2 \\ 2 & 4 & 4 & 9 & 0 & 30 & 4 \\ 1 & 4 & 0 & 6 & 0 & 21 & 2 \\ 3 & 9 & 6 & 13 & 0 & 51 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 21 & 51 & 0 & 172 & 24 \\ 2 & 4 & 2 & 7 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[190] \ 1^{113}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -2333, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.84 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 6 & 0 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 15 & 2 \\ 2 & 6 & 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 15 & 27 & 0 & 121 & 19 \\ 2 & 4 & 2 & 4 & -4 & 19 & 0 \end{pmatrix}_{b_0}$$

$$[191] \ 1^{133}1^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -2501, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 6 & 0 & 33 & 4 \\ 1 & 4 & 0 & 3 & 0 & 21 & 2 \\ 2 & 6 & 3 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 33 & 21 & 33 & 0 & 193 & 25 \\ 2 & 4 & 2 & 4 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[192] \ 1^{113}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -2504, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.75 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 9 & 2 \\ 2 & 4 & 4 & 9 & 0 & 27 & 4 \\ 1 & 4 & 0 & 6 & 0 & 18 & 2 \\ 3 & 9 & 6 & 13 & 0 & 45 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 18 & 45 & 0 & 133 & 21 \\ 2 & 4 & 2 & 7 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[193] \ 1^{123}1^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -2684, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 6 & 0 & 30 & 4 \\ 1 & 4 & 0 & 3 & 0 & 18 & 2 \\ 2 & 6 & 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 18 & 30 & 0 & 154 & 22 \\ 2 & 4 & 2 & 4 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[194] \quad 1^{112}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2840, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 10 & 2 \\ 2 & 4 & 4 & 9 & 0 & 30 & 4 \\ 1 & 4 & 0 & 6 & 0 & 18 & 2 \\ 3 & 9 & 6 & 13 & 0 & 48 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 18 & 48 & 0 & 154 & 22 \\ 2 & 4 & 2 & 7 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[195] \quad 1^{113}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3737, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 11 & 2 \\ 2 & 4 & 4 & 9 & 0 & 33 & 4 \\ 1 & 4 & 0 & 6 & 0 & 21 & 2 \\ 3 & 9 & 6 & 13 & 0 & 54 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 33 & 21 & 54 & 0 & 193 & 25 \\ 2 & 4 & 2 & 7 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[196] \quad 1^{114}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -4100, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 12 & 2 \\ 2 & 4 & 4 & 9 & 0 & 36 & 4 \\ 1 & 4 & 0 & 6 & 0 & 24 & 2 \\ 3 & 9 & 6 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 36 & 24 & 60 & 0 & 238 & 28 \\ 2 & 4 & 2 & 7 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [197] \quad 1^{12}4^{1131^{11}}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -103, \quad \det = 20, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.5$$

$$T_{\min} = 13.40 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 3 & 0 & 11 & 4 \\ 3 & 3 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 7 & 0 & 21 & 9 \\ 3 & 4 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[198] \quad 1^{11}4^{1131^{11}}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -103, \quad \det = 45, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.73 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 3 & 0 & 8 & 4 \\ 3 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 8 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[199] \quad 1^{11}4^{1131^{11}}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -103, \quad \det = 59, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.49 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 3 & 0 & 7 & 4 \\ 3 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 7 & 5 & 0 & 8 & 6 \\ 3 & 4 & 3 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[200] \quad 1^{11}4^{1131^{11}}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -103, \quad \det = 74, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.43 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 3 & 0 & 12 & 4 \\ 3 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 8 & 0 & 26 & 10 \\ 3 & 4 & 3 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [201] \quad 1^{11}5^{11}131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 18, \quad \Delta = -98, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.25 \rightarrow 15, \quad T_{\max} = 18, \quad T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[202] \quad 1^{11}5^{11}131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 18, \quad \Delta = -98, \quad \det = -31, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 11.0$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & -2 & 2 & 0 \\ 4 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 18, T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & -2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 14.19 \rightarrow 15, T_{\max} = 18, T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & -2 & 2 & 0 \\ 3 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & 0 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 14.19 \rightarrow 15, T_{\max} = 18, T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & -2 & 4 & 0 \\ 7 & 8 & 0 & 4 & 10 & 6 \\ 3 & 4 & 0 & 0 & 6 & 1 \end{pmatrix}_b$$

$$T_{\min} = 14.32 \rightarrow 15, T_{\max} = 18, T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & -2 & 4 & 0 \\ 5 & 8 & 0 & 4 & 10 & 6 \\ 3 & 4 & 0 & 0 & 6 & 1 \end{pmatrix}_b$$

$$T_{\min} = 14.56 \rightarrow 15, T_{\max} = 18, T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 7 & 12 & 0 & 6 & 25 & 9 \\ 3 & 4 & 0 & 0 & 9 & 1 \end{pmatrix}$$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 18, T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & -2 & 4 & 0 \\ 6 & 8 & 0 & 4 & 10 & 6 \\ 3 & 4 & 0 & 0 & 6 & 1 \end{pmatrix}_b$$

$$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 18, T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 8 & 12 & 0 & 6 & 25 & 9 \\ 3 & 4 & 0 & 0 & 9 & 1 \end{pmatrix}$$

$$T_{\min} = 14.24 \rightarrow 15, T_{\max} = 18, T \in [15, 18]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 9 & 12 & 0 & 6 & 25 & 9 \\ 3 & 4 & 0 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

4.772 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \quad 1^{11} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 5 & 4 & 4 \\ 3 & 5 & 4 & 0 & 6 & 6 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 6 & 3 & 4 \\ 3 & 5 & 6 & 0 & 6 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 429, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 5 & 4 & 4 \\ 3 & 5 & 4 & 0 & 9 & 6 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 9 & 6 & 7 \\ 3 & 5 & 6 & 0 & 9 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 6 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 7 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 1264, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 9 & 0 & 9 & 5 & 4 & 4 \\ 5 & 9 & 13 & 0 & 15 & 9 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 15 & 0 & 13 & 9 & 6 & 7 \\ 3 & 5 & 9 & 0 & 9 & 4 & 4 & 4 \\ 2 & 4 & 6 & 0 & 6 & 4 & 0 & 2 \\ 3 & 4 & 7 & -4 & 7 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

4.773 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 14.71 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 26 & 5 \\ 12 & 25 & 16 & 0 & 56 & 9 \\ 8 & 16 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 56 & 36 & 0 & 118 & 20 \\ 5 & 9 & 6 & -4 & 20 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 7 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 4 & 1 & 3 \\ 4 & 9 & 8 & 4 & 8 & 0 & 5 \\ 3 & 8 & 4 & 3 & 6 & 0 & 4 \\ 2 & 4 & 3 & -3 & 3 & 0 & -1 \\ 4 & 8 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & -1 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{111}} 3 1^{\overline{113}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -432$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.58 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 35 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 35 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{111}} 3 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -496$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 1 & 3 \\ 4 & 9 & 8 & 4 & 8 & 0 & 5 \\ 3 & 8 & 4 & 3 & 6 & 0 & 4 \\ 2 & 4 & 3 & -3 & 3 & 0 & -1 \\ 3 & 8 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & -1 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{111}} 3 1^{\overline{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -512$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.52 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{\overline{111}} 3 1^{\overline{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -608$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 17 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{\overline{111}} 3 1^{\overline{112}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -618$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 26 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

4.774 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$

$T = 10$ [0] $1^{\overline{111}} 3 1^{\overline{1111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -102$, $\det = 372$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 17 & 5 \\ 11 & 20 & 17 & 0 & 17 & 33 & 8 \\ 9 & 17 & 13 & 0 & 15 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 27 & 7 \\ 17 & 33 & 27 & 0 & 27 & 49 & 13 \\ 5 & 8 & 7 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

4.775 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$

$T = 9$ [0] $1^{\overline{1112}} 3 1^{\overline{13}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 18.25 \rightarrow 19$, $T_{\max} = 21$, $T \in [19, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 13 & 3 \\ 3 & 4 & 4 & 0 & 4 & 23 & 4 \\ 2 & 4 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 13 & 23 & 15 & 0 & 15 & 100 & 18 \\ 3 & 4 & 3 & -4 & 3 & 18 & -6 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{111}} 3 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.25 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{1211}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 9 & 4 \\ 4 & 8 & 10 & 0 & 12 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 15 & 7 \\ 5 & 9 & 12 & 0 & 15 & 13 & 7 \\ 3 & 4 & 6 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{131}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 5 & 3 \\ 3 & 4 & 5 & 0 & 17 & 9 & 4 \\ 3 & 5 & 4 & 0 & 15 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 49 & 27 & 13 \\ 5 & 9 & 9 & 0 & 27 & 13 & 7 \\ 3 & 4 & 4 & -4 & 13 & 7 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{123}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.3$

$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 12 & 3 \\ 3 & 4 & 4 & 0 & 8 & 22 & 4 \\ 2 & 4 & 1 & 0 & 6 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 30 & 6 \\ 12 & 22 & 15 & 0 & 30 & 85 & 17 \\ 3 & 4 & 3 & -4 & 6 & 17 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{1211}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 2 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 6 & 1 & 3 \\ 3 & 4 & 6 & -4 & 7 & 3 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{112}3^11^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 17.66 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 5 & 0 & 4 & 28 & 4 \\ 3 & 5 & 4 & 0 & 3 & 30 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 18 & 3 \\ 16 & 28 & 30 & 0 & 18 & 154 & 22 \\ 3 & 4 & 4 & -4 & 3 & 22 & -5 \end{pmatrix}_{b_0}$$

[7] $1^{113}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 5 & 0 & 4 & 17 & 4 \\ 3 & 5 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 9 & 17 & 15 & 0 & 12 & 49 & 13 \\ 3 & 4 & 4 & -4 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{1111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 16.50 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 8 & 3 & 0 & 4 & 6 & 4 \\ 3 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & 3 & -4 & 4 & 4 & -4 \end{pmatrix}_{b_0}$$

[10] $1^{111}3^11^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 5 & 0 & 4 & 14 & 4 \\ 3 & 5 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 8 & 14 & 15 & 0 & 9 & 37 & 11 \\ 3 & 4 & 4 & -4 & 3 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1113} 31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -88, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 12 & 3 \\ 3 & 4 & 4 & 0 & 14 & 22 & 4 \\ 2 & 4 & 1 & 0 & 9 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 57 & 11 \\ 12 & 22 & 15 & 0 & 57 & 85 & 17 \\ 3 & 4 & 3 & -4 & 11 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1222} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -100, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 7 & 3 \\ 3 & 4 & 8 & 0 & 8 & 13 & 4 \\ 4 & 8 & 10 & 0 & 12 & 18 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 18 & 6 \\ 7 & 13 & 18 & 0 & 18 & 28 & 10 \\ 3 & 4 & 6 & -4 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.81 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 5 & 4 \\ 3 & 5 & 4 & 0 & 6 & 6 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 6 & 4 \\ 3 & 5 & 6 & 0 & 6 & 4 & 4 \\ 3 & 4 & 4 & -4 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1111} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.47 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 12 & 5 \\ 2 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 12 & 6 & 0 & 13 & 15 & 7 \\ 5 & 12 & 6 & 0 & 15 & 13 & 7 \\ 3 & 5 & 3 & -4 & 7 & 7 & -2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1214} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -150, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 11 & 3 \\ 3 & 4 & 8 & 0 & 9 & 21 & 4 \\ 4 & 8 & 10 & 0 & 12 & 30 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 33 & 7 \\ 11 & 21 & 30 & 0 & 33 & 76 & 16 \\ 3 & 4 & 6 & -4 & 7 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -170, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.35 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1213} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -194, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.37 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 12 & 3 \\ 3 & 4 & 8 & 0 & 9 & 22 & 4 \\ 4 & 8 & 10 & 0 & 12 & 30 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 36 & 7 \\ 12 & 22 & 30 & 0 & 36 & 85 & 17 \\ 3 & 4 & 6 & -4 & 7 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{112} 31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -196, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.33 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 13 & 4 \\ 3 & 5 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 7 & 13 & 12 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & -4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{111} 31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.48 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 5 & 0 & 5 & 9 & 4 \\ 3 & 5 & 4 & 0 & 6 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 9 & 4 \\ 5 & 9 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & 4 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1212} 31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -230, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 7 & 3 \\ 3 & 4 & 8 & 0 & 9 & 13 & 4 \\ 4 & 8 & 10 & 0 & 12 & 18 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 21 & 7 \\ 7 & 13 & 18 & 0 & 21 & 28 & 10 \\ 3 & 4 & 6 & -4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1111} 31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.28 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1113} 31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.10 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 18 & 3 \\ 3 & 4 & 4 & 0 & 4 & 32 & 4 \\ 2 & 4 & 1 & 0 & 3 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 21 & 3 \\ 18 & 32 & 21 & 0 & 21 & 193 & 25 \\ 3 & 4 & 3 & -4 & 3 & 25 & -1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1133} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -248, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 9 & 3 \\ 3 & 4 & 4 & 0 & 12 & 17 & 4 \\ 2 & 4 & 1 & 0 & 9 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 9 & 0 & 25 & 36 & 9 \\ 9 & 17 & 12 & 0 & 36 & 49 & 13 \\ 3 & 4 & 3 & -4 & 9 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1112} 31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.44 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 4 & 4 & 0 & 4 & 8 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 4 & 8 & 6 & 0 & 6 & 10 & 6 \\ 3 & 4 & 3 & -4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1213} 31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -272, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.44 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 9 & 3 \\ 3 & 4 & 8 & 0 & 9 & 17 & 4 \\ 4 & 8 & 10 & 0 & 12 & 24 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 27 & 7 \\ 9 & 17 & 24 & 0 & 27 & 49 & 13 \\ 3 & 4 & 6 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1112} 31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -280, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.57 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 8 & 4 \\ 2 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 12 & 6 \\ 4 & 8 & 6 & 0 & 12 & 10 & 6 \\ 3 & 4 & 3 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{133}31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -280$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.6$

$T_{\min} = 12.44 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 5 & 0 & 17 & 17 & 4 \\ 3 & 5 & 4 & 0 & 15 & 15 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 49 & 51 & 13 \\ 9 & 17 & 15 & 0 & 51 & 49 & 13 \\ 3 & 4 & 4 & -4 & 13 & 13 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{1114}31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -280$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.44 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 11 & 3 \\ 3 & 4 & 9 & 0 & 9 & 21 & 4 \\ 5 & 9 & 13 & 0 & 15 & 33 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 15 & 0 & 13 & 33 & 7 \\ 11 & 21 & 33 & 0 & 33 & 76 & 16 \\ 3 & 4 & 7 & -4 & 7 & 16 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{1113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -296$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 12.65 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 6 & 3 \\ 3 & 4 & 4 & 0 & 4 & 12 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 6 & 12 & 9 & 0 & 9 & 25 & 9 \\ 3 & 4 & 3 & -4 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{1111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -296$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{1431}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -296$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 12 & 2 & 3 \\ 3 & 4 & 21 & 0 & 22 & 4 & 4 \\ 11 & 21 & 76 & 0 & 81 & 15 & 16 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 81 & 0 & 85 & 15 & 17 \\ 2 & 4 & 15 & 0 & 15 & 1 & 3 \\ 3 & 4 & 16 & -4 & 17 & 3 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{1222}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -304$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 12.68 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 4 & 3 \\ 3 & 4 & 8 & 0 & 8 & 8 & 4 \\ 4 & 8 & 10 & 0 & 12 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 12 & 6 \\ 4 & 8 & 12 & 0 & 12 & 10 & 6 \\ 3 & 4 & 6 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{1123}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -320$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 6 & 3 \\ 3 & 4 & 4 & 0 & 8 & 12 & 4 \\ 2 & 4 & 1 & 0 & 6 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 18 & 6 \\ 6 & 12 & 9 & 0 & 18 & 25 & 9 \\ 3 & 4 & 3 & -4 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{132}31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -328$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 7 & 3 \\ 3 & 4 & 5 & 0 & 17 & 13 & 4 \\ 3 & 5 & 4 & 0 & 15 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 49 & 39 & 13 \\ 7 & 13 & 12 & 0 & 39 & 28 & 10 \\ 3 & 4 & 4 & -4 & 13 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1112} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -338, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.21 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 16 & 3 \\ 3 & 4 & 4 & 0 & 9 & 28 & 4 \\ 2 & 4 & 1 & 0 & 6 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 48 & 7 \\ 16 & 28 & 18 & 0 & 48 & 154 & 22 \\ 3 & 4 & 3 & -4 & 7 & 22 & -1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{112} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -340, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 5 & 0 & 4 & 23 & 4 \\ 3 & 5 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 13 & 23 & 24 & 0 & 15 & 100 & 18 \\ 3 & 4 & 4 & -4 & 3 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1223} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -344, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.84 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 6 & 3 \\ 3 & 4 & 8 & 0 & 8 & 12 & 4 \\ 4 & 8 & 10 & 0 & 12 & 18 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 18 & 6 \\ 6 & 12 & 18 & 0 & 18 & 25 & 9 \\ 3 & 4 & 6 & -4 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -346, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 7 & 3 \\ 3 & 4 & 4 & 0 & 8 & 13 & 4 \\ 2 & 4 & 1 & 0 & 6 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 18 & 6 \\ 7 & 13 & 9 & 0 & 18 & 28 & 10 \\ 3 & 4 & 3 & -4 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -352, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.86 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 16 & 4 \\ 2 & 4 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 8 & 16 & 12 & 0 & 12 & 46 & 12 \\ 3 & 4 & 3 & -4 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1133} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -360, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.89 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 6 & 3 \\ 3 & 4 & 4 & 0 & 12 & 12 & 4 \\ 2 & 4 & 1 & 0 & 9 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 9 & 0 & 25 & 27 & 9 \\ 6 & 12 & 9 & 0 & 27 & 25 & 9 \\ 3 & 4 & 3 & -4 & 9 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1111} 3^1 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -368, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.49 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 5 & 3 \\ 3 & 4 & 4 & 0 & 14 & 9 & 4 \\ 2 & 4 & 1 & 0 & 9 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 24 & 11 \\ 5 & 9 & 6 & 0 & 24 & 13 & 7 \\ 3 & 4 & 3 & -4 & 11 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{1124} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 8 & 3 \\ 3 & 4 & 4 & 0 & 8 & 16 & 4 \\ 2 & 4 & 1 & 0 & 6 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 24 & 6 \\ 8 & 16 & 12 & 0 & 24 & 46 & 12 \\ 3 & 4 & 3 & -4 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1111} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.68 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 14 & 4 \\ 2 & 4 & 1 & 0 & 9 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 39 & 11 \\ 8 & 14 & 9 & 0 & 39 & 37 & 11 \\ 3 & 4 & 3 & -4 & 11 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{112} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -386, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.44 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 5 & 0 & 4 & 18 & 4 \\ 3 & 5 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 10 & 18 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & -4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1224} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -400, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 8 & 3 \\ 3 & 4 & 8 & 0 & 8 & 16 & 4 \\ 4 & 8 & 10 & 0 & 12 & 24 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 24 & 6 \\ 8 & 16 & 24 & 0 & 24 & 46 & 12 \\ 3 & 4 & 6 & -4 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1223} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -412, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.53 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 9 & 3 \\ 3 & 4 & 8 & 0 & 8 & 17 & 4 \\ 4 & 8 & 10 & 0 & 12 & 24 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 24 & 6 \\ 9 & 17 & 24 & 0 & 24 & 49 & 13 \\ 3 & 4 & 6 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1134} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.04 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 8 & 3 \\ 3 & 4 & 4 & 0 & 12 & 16 & 4 \\ 2 & 4 & 1 & 0 & 9 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 9 & 0 & 25 & 36 & 9 \\ 8 & 16 & 12 & 0 & 36 & 46 & 12 \\ 3 & 4 & 3 & -4 & 9 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1115} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -424, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.1$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 4 & 20 & 4 \\ 2 & 4 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 10 & 20 & 15 & 0 & 15 & 73 & 15 \\ 3 & 4 & 3 & -4 & 3 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{112} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -440, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.35 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 5 & 0 & 9 & 13 & 4 \\ 3 & 5 & 4 & 0 & 9 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 21 & 7 \\ 7 & 13 & 12 & 0 & 21 & 28 & 10 \\ 3 & 4 & 4 & -4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1125} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -448, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 13.11 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 10 & 3 \\ 3 & 4 & 4 & 0 & 8 & 20 & 4 \\ 2 & 4 & 1 & 0 & 6 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 30 & 6 \\ 10 & 20 & 15 & 0 & 30 & 73 & 15 \\ 3 & 4 & 3 & -4 & 6 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{1144}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -472, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 8 & 3 \\ 3 & 4 & 4 & 0 & 16 & 16 & 4 \\ 2 & 4 & 1 & 0 & 12 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 12 & 0 & 46 & 48 & 12 \\ 8 & 16 & 12 & 0 & 48 & 46 & 12 \\ 3 & 4 & 3 & -4 & 12 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{1135}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -488, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.9$$

$$T_{\min} = 13.18 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 10 & 3 \\ 3 & 4 & 4 & 0 & 12 & 20 & 4 \\ 2 & 4 & 1 & 0 & 9 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 9 & 0 & 25 & 45 & 9 \\ 10 & 20 & 15 & 0 & 45 & 73 & 15 \\ 3 & 4 & 3 & -4 & 9 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{114}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -490, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 5 & 0 & 4 & 26 & 4 \\ 3 & 5 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 18 & 3 \\ 14 & 26 & 24 & 0 & 18 & 118 & 20 \\ 3 & 4 & 4 & -4 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{1232}31^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -490, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 7 & 3 \\ 3 & 4 & 8 & 0 & 22 & 13 & 4 \\ 4 & 8 & 10 & 0 & 30 & 18 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 30 & 0 & 85 & 51 & 17 \\ 7 & 13 & 18 & 0 & 51 & 28 & 10 \\ 3 & 4 & 6 & -4 & 17 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{111}31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -504, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.37 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & 4 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{1116}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -512, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.22 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 4 & 0 & 4 & 24 & 4 \\ 2 & 4 & 1 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 18 & 3 \\ 12 & 24 & 18 & 0 & 18 & 106 & 18 \\ 3 & 4 & 3 & -4 & 3 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[55] \ 1^{1112}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -512, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 4 & 18 & 4 \\ 2 & 4 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 10 & 18 & 12 & 0 & 12 & 58 & 14 \\ 3 & 4 & 3 & -4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{113}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -520, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 5 & 0 & 4 & 22 & 4 \\ 3 & 5 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 12 & 22 & 21 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & -4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1214} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 14 & 3 \\ 3 & 4 & 8 & 0 & 9 & 26 & 4 \\ 4 & 8 & 10 & 0 & 12 & 36 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 42 & 7 \\ 14 & 26 & 36 & 0 & 42 & 118 & 20 \\ 3 & 4 & 6 & -4 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{1126} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -536, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.25 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 12 & 3 \\ 3 & 4 & 4 & 0 & 8 & 24 & 4 \\ 2 & 4 & 1 & 0 & 6 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 36 & 6 \\ 12 & 24 & 18 & 0 & 36 & 106 & 18 \\ 3 & 4 & 3 & -4 & 6 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{1111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -540, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 3 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{122} 31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -556, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.35 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 5 & 0 & 13 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 12 & 0 & 28 & 30 & 10 \\ 7 & 13 & 12 & 0 & 30 & 28 & 10 \\ 3 & 4 & 4 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{1134} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -576, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.62 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 11 & 3 \\ 3 & 4 & 4 & 0 & 12 & 21 & 4 \\ 2 & 4 & 1 & 0 & 9 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 9 & 0 & 25 & 45 & 9 \\ 11 & 21 & 15 & 0 & 45 & 76 & 16 \\ 3 & 4 & 3 & -4 & 9 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{1112} 31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -584, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.49 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 10 & 3 \\ 3 & 4 & 4 & 0 & 14 & 18 & 4 \\ 2 & 4 & 1 & 0 & 9 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 48 & 11 \\ 10 & 18 & 12 & 0 & 48 & 58 & 14 \\ 3 & 4 & 3 & -4 & 11 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{1112} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -592, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 4 & 13 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 7 & 13 & 9 & 0 & 9 & 28 & 10 \\ 3 & 4 & 3 & -4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{1222} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -616, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 13 & 4 \\ 4 & 8 & 10 & 0 & 18 & 18 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 18 & 0 & 28 & 30 & 10 \\ 7 & 13 & 18 & 0 & 30 & 28 & 10 \\ 3 & 4 & 6 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{1117}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -616$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 13.35 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 4 & 4 & 0 & 4 & 28 & 4 \\ 2 & 4 & 1 & 0 & 3 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 21 & 3 \\ 14 & 28 & 21 & 0 & 21 & 145 & 21 \\ 3 & 4 & 3 & -4 & 3 & 21 & -1 \end{pmatrix}_{b_0}$$

[68] $1^{1127}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -640$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.37 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 14 & 3 \\ 3 & 4 & 4 & 0 & 8 & 28 & 4 \\ 2 & 4 & 1 & 0 & 6 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 42 & 6 \\ 14 & 28 & 21 & 0 & 42 & 145 & 21 \\ 3 & 4 & 3 & -4 & 6 & 21 & -1 \end{pmatrix}_{b_0}$$

[67] $1^{1124}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -640$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 14 & 3 \\ 3 & 4 & 4 & 0 & 8 & 26 & 4 \\ 2 & 4 & 1 & 0 & 6 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 36 & 6 \\ 14 & 26 & 18 & 0 & 36 & 118 & 20 \\ 3 & 4 & 3 & -4 & 6 & 20 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{1123}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -658$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 9 & 3 \\ 3 & 4 & 4 & 0 & 8 & 17 & 4 \\ 2 & 4 & 1 & 0 & 6 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 24 & 6 \\ 9 & 17 & 12 & 0 & 24 & 49 & 13 \\ 3 & 4 & 3 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{1114}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -666$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 11 & 3 \\ 3 & 4 & 4 & 0 & 9 & 21 & 4 \\ 2 & 4 & 1 & 0 & 6 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 33 & 7 \\ 11 & 21 & 15 & 0 & 33 & 76 & 16 \\ 3 & 4 & 3 & -4 & 7 & 16 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{113}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -736$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 12.66 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 5 & 0 & 4 & 32 & 4 \\ 3 & 5 & 4 & 0 & 3 & 33 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 21 & 3 \\ 18 & 32 & 33 & 0 & 21 & 193 & 25 \\ 3 & 4 & 4 & -4 & 3 & 25 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{1224}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -740$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.54 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 11 & 3 \\ 3 & 4 & 8 & 0 & 8 & 21 & 4 \\ 4 & 8 & 10 & 0 & 12 & 30 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 30 & 6 \\ 11 & 21 & 30 & 0 & 30 & 76 & 16 \\ 3 & 4 & 6 & -4 & 6 & 16 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{113}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -746$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 12.47 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 5 & 0 & 4 & 27 & 4 \\ 3 & 5 & 4 & 0 & 3 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 18 & 3 \\ 15 & 27 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & -4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{11123} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -746, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.34 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 7 & 3 \\ 3 & 4 & 4 & 0 & 9 & 13 & 4 \\ 2 & 4 & 1 & 0 & 6 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 21 & 7 \\ 7 & 13 & 9 & 0 & 21 & 28 & 10 \\ 3 & 4 & 3 & -4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{1113} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -788, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 9 & 3 \\ 3 & 4 & 4 & 0 & 9 & 17 & 4 \\ 2 & 4 & 1 & 0 & 6 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 27 & 7 \\ 9 & 17 & 12 & 0 & 27 & 49 & 13 \\ 3 & 4 & 3 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{1223} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -838, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.41 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 9 & 3 \\ 3 & 4 & 8 & 0 & 13 & 17 & 4 \\ 4 & 8 & 10 & 0 & 18 & 24 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 18 & 0 & 28 & 39 & 10 \\ 9 & 17 & 24 & 0 & 39 & 49 & 13 \\ 3 & 4 & 6 & -4 & 10 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{1113} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -852, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.36 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 9 & 3 \\ 3 & 4 & 9 & 0 & 9 & 17 & 4 \\ 5 & 9 & 13 & 0 & 15 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 15 & 0 & 13 & 27 & 7 \\ 9 & 17 & 27 & 0 & 27 & 49 & 13 \\ 3 & 4 & 7 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{1113} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -872, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 15 & 3 \\ 3 & 4 & 4 & 0 & 4 & 27 & 4 \\ 2 & 4 & 1 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 18 & 3 \\ 15 & 27 & 18 & 0 & 18 & 133 & 21 \\ 3 & 4 & 3 & -4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{1112} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -890, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 13 & 3 \\ 3 & 4 & 4 & 0 & 9 & 23 & 4 \\ 2 & 4 & 1 & 0 & 6 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 39 & 7 \\ 13 & 23 & 15 & 0 & 39 & 100 & 18 \\ 3 & 4 & 3 & -4 & 7 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{1224} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 11 & 3 \\ 3 & 4 & 8 & 0 & 13 & 21 & 4 \\ 4 & 8 & 10 & 0 & 18 & 30 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 18 & 0 & 28 & 48 & 10 \\ 11 & 21 & 30 & 0 & 48 & 76 & 16 \\ 3 & 4 & 6 & -4 & 10 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{1113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -904, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 4 & 4 & 0 & 4 & 17 & 4 \\ 2 & 4 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 9 & 17 & 12 & 0 & 12 & 49 & 13 \\ 3 & 4 & 3 & -4 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{1142}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -936, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.41 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 7 & 3 \\ 3 & 4 & 9 & 0 & 21 & 13 & 4 \\ 5 & 9 & 13 & 0 & 33 & 21 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 33 & 0 & 76 & 48 & 16 \\ 7 & 13 & 21 & 0 & 48 & 28 & 10 \\ 3 & 4 & 7 & -4 & 16 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{1122}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -962, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 10 & 3 \\ 3 & 4 & 4 & 0 & 13 & 18 & 4 \\ 2 & 4 & 1 & 0 & 9 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 42 & 10 \\ 10 & 18 & 12 & 0 & 42 & 58 & 14 \\ 3 & 4 & 3 & -4 & 10 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{1111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -964, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.35 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 9 & 4 \\ 5 & 9 & 13 & 0 & 15 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 15 & 0 & 13 & 15 & 7 \\ 5 & 9 & 15 & 0 & 15 & 13 & 7 \\ 3 & 4 & 7 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{1124}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -986, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 11 & 3 \\ 3 & 4 & 4 & 0 & 8 & 21 & 4 \\ 2 & 4 & 1 & 0 & 6 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 30 & 6 \\ 11 & 21 & 15 & 0 & 30 & 76 & 16 \\ 3 & 4 & 3 & -4 & 6 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{114}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -988, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.40 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 5 & 0 & 4 & 31 & 4 \\ 3 & 5 & 4 & 0 & 3 & 30 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 21 & 3 \\ 17 & 31 & 30 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & -4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{1112}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1026, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 10 & 3 \\ 3 & 4 & 4 & 0 & 9 & 18 & 4 \\ 2 & 4 & 1 & 0 & 6 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 30 & 7 \\ 10 & 18 & 12 & 0 & 30 & 58 & 14 \\ 3 & 4 & 3 & -4 & 7 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{1134}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1066, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 17 & 3 \\ 3 & 4 & 4 & 0 & 17 & 31 & 4 \\ 2 & 4 & 1 & 0 & 12 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 12 & 0 & 49 & 93 & 13 \\ 17 & 31 & 21 & 0 & 93 & 172 & 24 \\ 3 & 4 & 3 & -4 & 13 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{1112}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1080, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.35 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 7 & 3 \\ 3 & 4 & 9 & 0 & 9 & 13 & 4 \\ 5 & 9 & 13 & 0 & 15 & 21 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 15 & 0 & 13 & 21 & 7 \\ 7 & 13 & 21 & 0 & 21 & 28 & 10 \\ 3 & 4 & 7 & -4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{1113} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1096, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.36 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 4 & 0 & 4 & 22 & 4 \\ 2 & 4 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 12 & 22 & 15 & 0 & 15 & 85 & 17 \\ 3 & 4 & 3 & -4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{1122} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1132, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.35 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 13 & 4 \\ 2 & 4 & 1 & 0 & 9 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 30 & 10 \\ 7 & 13 & 9 & 0 & 30 & 28 & 10 \\ 3 & 4 & 3 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{1224} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1180, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 14 & 3 \\ 3 & 4 & 8 & 0 & 13 & 26 & 4 \\ 4 & 8 & 10 & 0 & 18 & 36 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 18 & 0 & 28 & 60 & 10 \\ 14 & 26 & 36 & 0 & 60 & 118 & 20 \\ 3 & 4 & 6 & -4 & 10 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{1144} 31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1216, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 11 & 3 \\ 3 & 4 & 9 & 0 & 21 & 21 & 4 \\ 5 & 9 & 13 & 0 & 33 & 33 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 33 & 0 & 76 & 78 & 16 \\ 11 & 21 & 33 & 0 & 78 & 76 & 16 \\ 3 & 4 & 7 & -4 & 16 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{1114} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.42 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 11 & 3 \\ 3 & 4 & 4 & 0 & 4 & 21 & 4 \\ 2 & 4 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 11 & 21 & 15 & 0 & 15 & 76 & 16 \\ 3 & 4 & 3 & -4 & 3 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{1143} 31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1248, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.42 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 9 & 3 \\ 3 & 4 & 9 & 0 & 21 & 17 & 4 \\ 5 & 9 & 13 & 0 & 33 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 33 & 0 & 76 & 63 & 16 \\ 9 & 17 & 27 & 0 & 63 & 49 & 13 \\ 3 & 4 & 7 & -4 & 16 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[96] \quad 1^{114} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1250, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.48 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 5 & 0 & 4 & 36 & 4 \\ 3 & 5 & 4 & 0 & 3 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 24 & 3 \\ 20 & 36 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & -4 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[97] \quad 1^{1124} 31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 14 & 3 \\ 3 & 4 & 4 & 0 & 18 & 26 & 4 \\ 2 & 4 & 1 & 0 & 12 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 12 & 0 & 58 & 84 & 14 \\ 14 & 26 & 18 & 0 & 84 & 118 & 20 \\ 3 & 4 & 3 & -4 & 14 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[98] \quad 1^{1123} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1322, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.41 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 15 & 3 \\ 3 & 4 & 4 & 0 & 13 & 27 & 4 \\ 2 & 4 & 1 & 0 & 9 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 63 & 10 \\ 15 & 27 & 18 & 0 & 63 & 133 & 21 \\ 3 & 4 & 3 & -4 & 10 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[99] \quad 1^{1123} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1328, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 9 & 3 \\ 3 & 4 & 9 & 0 & 13 & 17 & 4 \\ 5 & 9 & 13 & 0 & 21 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 21 & 0 & 28 & 39 & 10 \\ 9 & 17 & 27 & 0 & 39 & 49 & 13 \\ 3 & 4 & 7 & -4 & 10 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[100] \quad 1^{1123} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1354, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 9 & 3 \\ 3 & 4 & 4 & 0 & 13 & 17 & 4 \\ 2 & 4 & 1 & 0 & 9 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 39 & 10 \\ 9 & 17 & 12 & 0 & 39 & 49 & 13 \\ 3 & 4 & 3 & -4 & 10 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[102] \quad 1^{1122} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.36 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 7 & 3 \\ 3 & 4 & 9 & 0 & 13 & 13 & 4 \\ 5 & 9 & 13 & 0 & 21 & 21 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 21 & 0 & 28 & 30 & 10 \\ 7 & 13 & 21 & 0 & 30 & 28 & 10 \\ 3 & 4 & 7 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[101] \quad 1^{1114} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.45 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 20 & 3 \\ 3 & 4 & 4 & 0 & 4 & 36 & 4 \\ 2 & 4 & 1 & 0 & 3 & 24 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 24 & 3 \\ 20 & 36 & 24 & 0 & 24 & 238 & 28 \\ 3 & 4 & 3 & -4 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[103] \quad 1^{1133} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1408, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 12 & 3 \\ 3 & 4 & 4 & 0 & 17 & 22 & 4 \\ 2 & 4 & 1 & 0 & 12 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 12 & 0 & 49 & 66 & 13 \\ 12 & 22 & 15 & 0 & 66 & 85 & 17 \\ 3 & 4 & 3 & -4 & 13 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[104] \quad 1^{1124} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1412, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.41 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 11 & 3 \\ 3 & 4 & 4 & 0 & 13 & 21 & 4 \\ 2 & 4 & 1 & 0 & 9 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 48 & 10 \\ 11 & 21 & 15 & 0 & 48 & 76 & 16 \\ 3 & 4 & 3 & -4 & 10 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[105] \quad 1^{1133} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1460, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 9 & 3 \\ 3 & 4 & 9 & 0 & 17 & 17 & 4 \\ 5 & 9 & 13 & 0 & 27 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 27 & 0 & 49 & 51 & 13 \\ 9 & 17 & 27 & 0 & 51 & 49 & 13 \\ 3 & 4 & 7 & -4 & 13 & 13 & 0 \end{pmatrix}_{b_0}$$

[106] $1^{1122}31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1512$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 18 & 4 \\ 2 & 4 & 1 & 0 & 12 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 12 & 0 & 58 & 60 & 14 \\ 10 & 18 & 12 & 0 & 60 & 58 & 14 \\ 3 & 4 & 3 & -4 & 14 & 14 & 0 \end{pmatrix}_{b_0}$$

[107] $1^{1113}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1520$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 12 & 3 \\ 3 & 4 & 4 & 0 & 9 & 22 & 4 \\ 2 & 4 & 1 & 0 & 6 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 36 & 7 \\ 12 & 22 & 15 & 0 & 36 & 85 & 17 \\ 3 & 4 & 3 & -4 & 7 & 17 & 0 \end{pmatrix}_{b_0}$$

[108] $1^{1114}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1696$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 4 & 4 & 0 & 4 & 26 & 4 \\ 2 & 4 & 1 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 18 & 3 \\ 14 & 26 & 18 & 0 & 18 & 118 & 20 \\ 3 & 4 & 3 & -4 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

[109] $1^{1144}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1706$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.41 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 14 & 3 \\ 3 & 4 & 4 & 0 & 21 & 26 & 4 \\ 2 & 4 & 1 & 0 & 15 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 15 & 0 & 76 & 96 & 16 \\ 14 & 26 & 18 & 0 & 96 & 118 & 20 \\ 3 & 4 & 3 & -4 & 16 & 20 & 0 \end{pmatrix}_{b_0}$$

[110] $1^{1123}31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1736$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.3$

$T_{\min} = 12.39 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 12 & 3 \\ 3 & 4 & 4 & 0 & 18 & 22 & 4 \\ 2 & 4 & 1 & 0 & 12 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 12 & 0 & 58 & 72 & 14 \\ 12 & 22 & 15 & 0 & 72 & 85 & 17 \\ 3 & 4 & 3 & -4 & 14 & 17 & 0 \end{pmatrix}_{b_0}$$

[111] $1^{1113}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1736$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$T_{\min} = 12.49 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 18 & 3 \\ 3 & 4 & 4 & 0 & 9 & 32 & 4 \\ 2 & 4 & 1 & 0 & 6 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 54 & 7 \\ 18 & 32 & 21 & 0 & 54 & 193 & 25 \\ 3 & 4 & 3 & -4 & 7 & 25 & 0 \end{pmatrix}_{b_0}$$

[112] $1^{1114}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1744$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.39 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 17 & 3 \\ 3 & 4 & 4 & 0 & 4 & 31 & 4 \\ 2 & 4 & 1 & 0 & 3 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 21 & 3 \\ 17 & 31 & 21 & 0 & 21 & 172 & 24 \\ 3 & 4 & 3 & -4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

[113] $1^{1133}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1756$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.6$

$T_{\min} = 12.39 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 17 & 4 \\ 2 & 4 & 1 & 0 & 12 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 12 & 0 & 49 & 51 & 13 \\ 9 & 17 & 12 & 0 & 51 & 49 & 13 \\ 3 & 4 & 3 & -4 & 13 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[114] \quad 1^{1224}31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1772, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.40 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 11 & 3 \\ 3 & 4 & 13 & 0 & 13 & 21 & 4 \\ 7 & 13 & 28 & 0 & 30 & 48 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 30 & 0 & 28 & 48 & 10 \\ 11 & 21 & 48 & 0 & 48 & 76 & 16 \\ 3 & 4 & 10 & -4 & 10 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[115] \quad 1^{1123}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1816, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 12 & 3 \\ 3 & 4 & 4 & 0 & 13 & 22 & 4 \\ 2 & 4 & 1 & 0 & 9 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 51 & 10 \\ 12 & 22 & 15 & 0 & 51 & 85 & 17 \\ 3 & 4 & 3 & -4 & 10 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[116] \quad 1^{1124}31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1826, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 20 & 3 \\ 3 & 4 & 4 & 0 & 13 & 36 & 4 \\ 2 & 4 & 1 & 0 & 9 & 24 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 84 & 10 \\ 20 & 36 & 24 & 0 & 84 & 238 & 28 \\ 3 & 4 & 3 & -4 & 10 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[117] \quad 1^{1113}31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1836, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.41 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 15 & 3 \\ 3 & 4 & 4 & 0 & 9 & 27 & 4 \\ 2 & 4 & 1 & 0 & 6 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 45 & 7 \\ 15 & 27 & 18 & 0 & 45 & 133 & 21 \\ 3 & 4 & 3 & -4 & 7 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[118] \quad 1^{1114}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1850, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 14 & 3 \\ 3 & 4 & 4 & 0 & 9 & 26 & 4 \\ 2 & 4 & 1 & 0 & 6 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 42 & 7 \\ 14 & 26 & 18 & 0 & 42 & 118 & 20 \\ 3 & 4 & 3 & -4 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[119] \quad 1^{1222}31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1852, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 13 & 4 \\ 7 & 13 & 28 & 0 & 30 & 30 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 30 & 0 & 28 & 30 & 10 \\ 7 & 13 & 30 & 0 & 30 & 28 & 10 \\ 3 & 4 & 10 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[120] \quad 1^{1223}31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1984, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 9 & 3 \\ 3 & 4 & 13 & 0 & 13 & 17 & 4 \\ 7 & 13 & 28 & 0 & 30 & 39 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 30 & 0 & 28 & 39 & 10 \\ 9 & 17 & 39 & 0 & 39 & 49 & 13 \\ 3 & 4 & 10 & -4 & 10 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[121] \quad 1^{1134}31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1994, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.42 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 11 & 3 \\ 3 & 4 & 4 & 0 & 17 & 21 & 4 \\ 2 & 4 & 1 & 0 & 12 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 12 & 0 & 49 & 63 & 13 \\ 11 & 21 & 15 & 0 & 63 & 76 & 16 \\ 3 & 4 & 3 & -4 & 13 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[122] \quad 1^{1144} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2412, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 11 & 3 \\ 3 & 4 & 4 & 0 & 21 & 21 & 4 \\ 2 & 4 & 1 & 0 & 15 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 15 & 0 & 76 & 78 & 16 \\ 11 & 21 & 15 & 0 & 78 & 76 & 16 \\ 3 & 4 & 3 & -4 & 16 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[123] \quad 1^{1134} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2458, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 14 & 3 \\ 3 & 4 & 4 & 0 & 17 & 26 & 4 \\ 2 & 4 & 1 & 0 & 12 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 12 & 0 & 49 & 78 & 13 \\ 14 & 26 & 18 & 0 & 78 & 118 & 20 \\ 3 & 4 & 3 & -4 & 13 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[124] \quad 1^{1124} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2506, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 14 & 3 \\ 3 & 4 & 4 & 0 & 13 & 26 & 4 \\ 2 & 4 & 1 & 0 & 9 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 60 & 10 \\ 14 & 26 & 18 & 0 & 60 & 118 & 20 \\ 3 & 4 & 3 & -4 & 10 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[125] \quad 1^{1114} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2618, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 17 & 3 \\ 3 & 4 & 4 & 0 & 9 & 31 & 4 \\ 2 & 4 & 1 & 0 & 6 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 51 & 7 \\ 17 & 31 & 21 & 0 & 51 & 172 & 24 \\ 3 & 4 & 3 & -4 & 7 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[126] \quad 1^{1124} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2644, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 17 & 3 \\ 3 & 4 & 4 & 0 & 13 & 31 & 4 \\ 2 & 4 & 1 & 0 & 9 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 72 & 10 \\ 17 & 31 & 21 & 0 & 72 & 172 & 24 \\ 3 & 4 & 3 & -4 & 10 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[127] \quad 1^{1133} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2680, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 12 & 3 \\ 3 & 4 & 4 & 0 & 22 & 22 & 4 \\ 2 & 4 & 1 & 0 & 15 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 15 & 0 & 85 & 87 & 17 \\ 12 & 22 & 15 & 0 & 87 & 85 & 17 \\ 3 & 4 & 3 & -4 & 17 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[128] \quad 1^{1134} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2920, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 14 & 3 \\ 3 & 4 & 4 & 0 & 22 & 26 & 4 \\ 2 & 4 & 1 & 0 & 15 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 15 & 0 & 85 & 102 & 17 \\ 14 & 26 & 18 & 0 & 102 & 118 & 20 \\ 3 & 4 & 3 & -4 & 17 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[129] \quad 1^{1114} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2970, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.42 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 20 & 3 \\ 3 & 4 & 4 & 0 & 9 & 36 & 4 \\ 2 & 4 & 1 & 0 & 6 & 24 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 60 & 7 \\ 20 & 36 & 24 & 0 & 60 & 238 & 28 \\ 3 & 4 & 3 & -4 & 7 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[130] \quad 1^{1144}31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -3880, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 14 & 3 \\ 3 & 4 & 4 & 0 & 26 & 26 & 4 \\ 2 & 4 & 1 & 0 & 18 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 14 & 26 & 18 & 0 & 118 & 120 & 20 \\ 14 & 26 & 18 & 0 & 120 & 118 & 20 \\ 3 & 4 & 3 & -4 & 20 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [131] \quad 1^{1111}31^{112}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 484, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 5 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 9 & 2 \\ 1 & 3 & 3 & 1 & 0 & 6 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 6 & 0 & 13 & 24 & 7 \\ 5 & 15 & 9 & 9 & 0 & 24 & 37 & 11 \\ 2 & 4 & 2 & 3 & -4 & 7 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[132] \quad 1^{1111}31^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 556, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 6 & 2 \\ 1 & 3 & 3 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 6 & 0 & 13 & 15 & 7 \\ 3 & 9 & 6 & 6 & 0 & 15 & 13 & 7 \\ 2 & 4 & 2 & 3 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[133] \quad 1^{11}31^{1111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 652, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.85 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 2 & 6 & 3 & 4 & 0 & 6 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 9 & 4 \\ 3 & 9 & 6 & 9 & 0 & 9 & 13 & 7 \\ 2 & 4 & 2 & 4 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[134] \quad 1^131^{1111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 880, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.85 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 6 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 6 & 4 \\ 2 & 6 & 3 & 6 & 0 & 6 & 4 & 4 \\ 2 & 4 & 2 & 4 & -4 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[135] \quad 1^{1111}31^{122}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 952, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.92 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 5 & 2 \\ 2 & 4 & 4 & 3 & 0 & 15 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 9 & 2 \\ 1 & 3 & 3 & 1 & 0 & 9 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 15 & 9 & 9 & 0 & 37 & 39 & 11 \\ 5 & 15 & 9 & 9 & 0 & 39 & 37 & 11 \\ 2 & 4 & 2 & 3 & -4 & 11 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[136] \quad 1^{111}31^{1111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 964, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 6 & 2 \\ 2 & 6 & 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 9 & 0 & 13 & 15 & 7 \\ 3 & 9 & 6 & 9 & 0 & 15 & 13 & 7 \\ 2 & 4 & 2 & 4 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[137] \quad 1^{1111}31^{1111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 1816, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 6 & 0 & 6 & 6 & 2 \\ 3 & 9 & 6 & 13 & 0 & 15 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 15 & 0 & 13 & 15 & 7 \\ 3 & 9 & 6 & 15 & 0 & 15 & 13 & 7 \\ 2 & 4 & 2 & 7 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

4.776 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 372, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 17 & 5 \\ 11 & 20 & 17 & 0 & 17 & 33 & 8 \\ 9 & 17 & 13 & 0 & 15 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 27 & 7 \\ 17 & 33 & 27 & 0 & 27 & 49 & 13 \\ 5 & 8 & 7 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

4.777 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 148, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 9 & 4 & 4 \\ 3 & 5 & 4 & 0 & 6 & 9 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 9 & 0 & 9 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 14 & 9 & 4 & 4 \\ 2 & 4 & 1 & 0 & 9 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 24 & 9 & 11 \\ 5 & 9 & 6 & 0 & 24 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 9 & 6 & 0 & 2 \\ 3 & 4 & 3 & -4 & 11 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1121}} 3^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 274, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 18 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 18 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 & 3 \\ 10 & 18 & 18 & 0 & 12 & 58 & 12 & 14 \\ 2 & 4 & 3 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1131}} 3^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 454, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 27 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 27 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 18 & 3 & 3 \\ 15 & 27 & 27 & 0 & 18 & 133 & 18 & 21 \\ 2 & 4 & 3 & 0 & 3 & 18 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & 3 & -4 & 3 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1131} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 676, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.8$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 22 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 21 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 & 3 \\ 12 & 22 & 21 & 0 & 15 & 85 & 15 & 17 \\ 2 & 4 & 3 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1141} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 706, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.99 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 36 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 36 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 24 & 3 & 3 \\ 20 & 36 & 36 & 0 & 24 & 238 & 24 & 28 \\ 2 & 4 & 3 & 0 & 3 & 24 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 712, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.87 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 2 \\ 3 & 4 & 3 & -4 & 3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 724, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 6 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1121} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 796, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.84 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 13 & 4 & 4 \\ 3 & 5 & 4 & 0 & 9 & 12 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 21 & 6 & 7 \\ 7 & 13 & 12 & 0 & 21 & 28 & 9 & 10 \\ 2 & 4 & 3 & 0 & 6 & 9 & 0 & 2 \\ 3 & 4 & 4 & -4 & 7 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 988, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 9 & 4 & 4 \\ 3 & 5 & 4 & 0 & 9 & 9 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 15 & 6 & 7 \\ 5 & 9 & 9 & 0 & 15 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 7 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1032, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 6 & 3 \\ 5 & 9 & 6 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 6 & 0 & 6 & 15 & 12 & 6 \\ 3 & 4 & 3 & -4 & 3 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1221} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1144, \ (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 13 & 13 & 4 & 4 \\ 3 & 5 & 4 & 0 & 12 & 12 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 12 & 0 & 28 & 30 & 9 & 10 \\ 7 & 13 & 12 & 0 & 30 & 28 & 9 & 10 \\ 2 & 4 & 3 & 0 & 9 & 9 & 0 & 2 \\ 3 & 4 & 4 & -4 & 10 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1141} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1180, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 17 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 31 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 30 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 21 & 3 & 3 \\ 17 & 31 & 30 & 0 & 21 & 172 & 21 & 24 \\ 2 & 4 & 3 & 0 & 3 & 21 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11111} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 & 4 \\ 2 & 4 & 1 & 0 & 6 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 15 & 6 & 7 \\ 5 & 9 & 6 & 0 & 15 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 6 & 0 & 2 \\ 3 & 4 & 3 & -4 & 7 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{11111} 31^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 2368, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 5 & 2 & 3 \\ 3 & 4 & 9 & 0 & 9 & 9 & 4 & 4 \\ 5 & 9 & 13 & 0 & 15 & 15 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 15 & 0 & 13 & 15 & 6 & 7 \\ 5 & 9 & 15 & 0 & 15 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 6 & 6 & 0 & 2 \\ 3 & 4 & 7 & -4 & 7 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

4.778 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{111} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 329, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 & 2 \\ 1 & 3 & 3 & 1 & 0 & 6 & 2 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 6 & 0 & 13 & 5 & 7 \\ 1 & 3 & 2 & 2 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & 3 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 31^{12} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 464, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 & 2 \\ 1 & 3 & 3 & 1 & 0 & 9 & 2 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 15 & 9 & 9 & 0 & 37 & 8 & 11 \\ 1 & 3 & 2 & 2 & 0 & 8 & 0 & 2 \\ 2 & 4 & 2 & 3 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{131} 1^{111} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 604, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 & 2 \\ 2 & 6 & 3 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 3 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{113} 1^{111} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 669, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 & 2 \\ 2 & 6 & 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 9 & 0 & 13 & 5 & 7 \\ 1 & 3 & 2 & 3 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^11^{11}3^11 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 1184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 6 & 2 & 2 \\ 3 & 9 & 6 & 13 & 0 & 15 & 5 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 15 & 0 & 13 & 5 & 7 \\ 1 & 3 & 2 & 5 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & 7 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

4.779 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 9 \quad [0] \quad 1^{11}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 16.86 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 3 & 5 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & 4 & -4 & 3 & 3 & -4 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 16.71 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 8 & 14 & 9 & 0 & 9 & 36 & 10 \\ 3 & 4 & 3 & -4 & 3 & 10 & -4 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{1211}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 9 & 4 \\ 4 & 8 & 10 & 0 & 12 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 15 & 7 \\ 5 & 9 & 12 & 0 & 15 & 12 & 6 \\ 3 & 4 & 6 & -4 & 7 & 6 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{1111}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 9 & 11 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 3 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 5 & 0 & 19 & 4 & 4 \\ 3 & 5 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 19 & 21 & 0 & 73 & 12 & 15 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -4 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{113} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 5 & 0 & 4 & 17 & 4 \\ 3 & 5 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 9 & 17 & 15 & 0 & 12 & 48 & 12 \\ 3 & 4 & 4 & -4 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3^1 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 2 & 3 \\ 3 & 4 & 9 & 0 & 19 & 4 & 4 \\ 5 & 9 & 13 & 0 & 33 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 19 & 33 & 0 & 73 & 12 & 15 \\ 2 & 4 & 6 & 0 & 12 & 0 & 2 \\ 3 & 4 & 7 & -4 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1211} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 2 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 6 & 0 & 2 \\ 3 & 4 & 6 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 3 & 4 & 3 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1121} 3^1 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -137, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 7 & 2 & 3 \\ 3 & 4 & 14 & 0 & 13 & 4 & 4 \\ 8 & 14 & 37 & 0 & 33 & 9 & 11 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 33 & 0 & 28 & 9 & 10 \\ 2 & 4 & 9 & 0 & 9 & 0 & 2 \\ 3 & 4 & 11 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{131} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 & 4 \\ 3 & 5 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 49 & 12 & 13 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -4 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1121} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 5 & 3 \\ 3 & 4 & 4 & 0 & 8 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 12 & 6 \\ 5 & 9 & 6 & 0 & 12 & 12 & 6 \\ 3 & 4 & 3 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 4 & 1 & 3 \\ 4 & 9 & 8 & 4 & 8 & 0 & 5 \\ 3 & 8 & 4 & 3 & 6 & 0 & 4 \\ 2 & 4 & 3 & -3 & 3 & 0 & -1 \\ 4 & 8 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1221}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 13 & 2 & 3 \\ 3 & 4 & 13 & 0 & 23 & 4 & 4 \\ 7 & 13 & 28 & 0 & 54 & 9 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 13 & 23 & 54 & 0 & 100 & 15 & 18 \\ 2 & 4 & 9 & 0 & 15 & 0 & 2 \\ 3 & 4 & 10 & -4 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -219, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & -4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{112}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -245, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 13 & 4 \\ 3 & 5 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 7 & 13 & 12 & 0 & 9 & 27 & 9 \\ 3 & 4 & 4 & -4 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -284, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 1 & 3 \\ 4 & 9 & 8 & 4 & 8 & 0 & 5 \\ 3 & 8 & 4 & 3 & 6 & 0 & 4 \\ 2 & 4 & 3 & -3 & 3 & 0 & -1 \\ 3 & 8 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -341, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 & 4 \\ 3 & 5 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -344, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.81 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -353, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 5 & 0 & 14 & 4 & 4 \\ 3 & 5 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 15 & 0 & 37 & 9 & 11 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -359, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 & 3 \\ 3 & 4 & 4 & 0 & 9 & 5 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 9 & 7 \\ 3 & 5 & 3 & 0 & 9 & 3 & 3 \\ 3 & 4 & 3 & -4 & 7 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -368, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 2 & 3 \\ 4 & 9 & 12 & 0 & 12 & 4 & 5 \\ 5 & 12 & 13 & 0 & 15 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 12 & 15 & 0 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 6 & 0 & 2 \\ 3 & 5 & 7 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1121} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -380, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.96 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 6 & 6 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & -4 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1211} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -380, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 & 4 \\ 3 & 5 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 12 & 0 & 28 & 9 & 10 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -408, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 6 & 12 & 6 \\ 3 & 4 & 3 & -4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1141} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -413, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 & 3 \\ 3 & 4 & 4 & 0 & 21 & 5 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 15 & 0 & 76 & 18 & 16 \\ 3 & 5 & 3 & 0 & 18 & 3 & 3 \\ 3 & 4 & 3 & -4 & 16 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1221} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 2 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 6 & 6 \\ 2 & 4 & 6 & 0 & 6 & 0 & 2 \\ 3 & 4 & 6 & -4 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1121} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -425, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 2 & 3 \\ 3 & 4 & 4 & 0 & 23 & 4 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 13 & 23 & 15 & 0 & 100 & 15 & 18 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 3 & -4 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1131} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -440, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.15 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 2 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 9 & 0 & 25 & 9 & 9 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 3 & -4 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1221} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -444, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 5 & 3 \\ 3 & 4 & 8 & 0 & 13 & 9 & 4 \\ 4 & 8 & 10 & 0 & 18 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 18 & 0 & 28 & 21 & 10 \\ 5 & 9 & 12 & 0 & 21 & 12 & 6 \\ 3 & 4 & 6 & -4 & 10 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1111} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -485, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1131} 31^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -488, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.87 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 12 & 2 & 3 \\ 3 & 4 & 14 & 0 & 22 & 4 & 4 \\ 8 & 14 & 37 & 0 & 57 & 9 & 11 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 57 & 0 & 85 & 15 & 17 \\ 2 & 4 & 9 & 0 & 15 & 0 & 2 \\ 3 & 4 & 11 & -4 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1141} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 12 & 0 & 46 & 12 & 12 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 3 & -4 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1121} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -533, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.03 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 & 3 \\ 3 & 4 & 4 & 0 & 13 & 5 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 12 & 10 \\ 3 & 5 & 3 & 0 & 12 & 3 & 3 \\ 3 & 4 & 3 & -4 & 10 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1131} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -551, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 & 3 \\ 3 & 4 & 4 & 0 & 17 & 5 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 12 & 0 & 49 & 15 & 13 \\ 3 & 5 & 3 & 0 & 15 & 3 & 3 \\ 3 & 4 & 3 & -4 & 13 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1221} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -560, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 2 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 18 & 0 & 28 & 9 & 10 \\ 2 & 4 & 6 & 0 & 9 & 0 & 2 \\ 3 & 4 & 6 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1111} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -729, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 0 & 15 & 12 & 6 \\ 3 & 4 & 3 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1111} 31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -773, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 2 & 3 \\ 3 & 4 & 9 & 0 & 14 & 4 & 4 \\ 5 & 9 & 13 & 0 & 24 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 24 & 0 & 37 & 9 & 11 \\ 2 & 4 & 6 & 0 & 9 & 0 & 2 \\ 3 & 4 & 7 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{121} 31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -800, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 5 & 0 & 18 & 4 & 4 \\ 3 & 5 & 4 & 0 & 18 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{121} 31^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -848, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 5 & 0 & 28 & 4 & 4 \\ 3 & 5 & 4 & 0 & 30 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 16 & 28 & 30 & 0 & 154 & 18 & 22 \\ 2 & 4 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -4 & 22 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1111} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 2 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 & 4 \\ 5 & 9 & 13 & 0 & 15 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 15 & 0 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 6 & 0 & 2 \\ 3 & 4 & 7 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1111} 31^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -920, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 8 & 2 & 3 \\ 3 & 4 & 14 & 0 & 14 & 4 & 4 \\ 8 & 14 & 37 & 0 & 39 & 9 & 11 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 39 & 0 & 37 & 9 & 11 \\ 2 & 4 & 9 & 0 & 9 & 0 & 2 \\ 3 & 4 & 11 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1121} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -929, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 9 & 10 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 3 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1121} 31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 2 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 12 & 0 & 58 & 12 & 14 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 3 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{141} 31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.91 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 5 & 0 & 26 & 4 & 4 \\ 3 & 5 & 4 & 0 & 24 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 14 & 26 & 24 & 0 & 118 & 18 & 20 \\ 2 & 4 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -4 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{12\bar{1}}31^{1\bar{1}3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 5 & 0 & 23 & 4 & 4 \\ 3 & 5 & 4 & 0 & 24 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 13 & 23 & 24 & 0 & 100 & 15 & 18 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -4 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{13\bar{1}}31^{1\bar{1}2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1001, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 5 & 0 & 22 & 4 & 4 \\ 3 & 5 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 21 & 0 & 85 & 15 & 17 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -4 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{11\bar{3}}31^{1\bar{1}4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 2 & 3 \\ 3 & 4 & 4 & 0 & 32 & 4 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 18 & 32 & 21 & 0 & 193 & 21 & 25 \\ 2 & 4 & 3 & 0 & 21 & 0 & 2 \\ 3 & 4 & 3 & -4 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{11\bar{2}}31^{1\bar{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1083, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 5 & 3 \\ 3 & 4 & 4 & 0 & 13 & 9 & 4 \\ 2 & 4 & 1 & 0 & 9 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 21 & 10 \\ 5 & 9 & 6 & 0 & 21 & 12 & 6 \\ 3 & 4 & 3 & -4 & 10 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{11\bar{4}}31^{1\bar{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1085, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 2 & 3 \\ 3 & 4 & 9 & 0 & 21 & 4 & 4 \\ 5 & 9 & 13 & 0 & 33 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 33 & 0 & 76 & 15 & 16 \\ 2 & 4 & 6 & 0 & 15 & 0 & 2 \\ 3 & 4 & 7 & -4 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{11\bar{2}}31^{1\bar{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1205, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 2 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 & 4 \\ 5 & 9 & 13 & 0 & 21 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 21 & 0 & 28 & 9 & 10 \\ 2 & 4 & 6 & 0 & 9 & 0 & 2 \\ 3 & 4 & 7 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{11\bar{2}}31^{1\bar{2}2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 10 & 2 & 3 \\ 3 & 4 & 14 & 0 & 18 & 4 & 4 \\ 8 & 14 & 37 & 0 & 48 & 9 & 11 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 48 & 0 & 58 & 12 & 14 \\ 2 & 4 & 9 & 0 & 12 & 0 & 2 \\ 3 & 4 & 11 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{11\bar{3}}31^{1\bar{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 2 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 & 4 \\ 5 & 9 & 13 & 0 & 27 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 27 & 0 & 49 & 12 & 13 \\ 2 & 4 & 6 & 0 & 12 & 0 & 2 \\ 3 & 4 & 7 & -4 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1121} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 16 & 2 & 3 \\ 3 & 4 & 9 & 0 & 28 & 4 & 4 \\ 5 & 9 & 13 & 0 & 48 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 16 & 28 & 48 & 0 & 154 & 18 & 22 \\ 2 & 4 & 6 & 0 & 18 & 0 & 2 \\ 3 & 4 & 7 & -4 & 22 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1131} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1281, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 5 & 3 \\ 3 & 4 & 4 & 0 & 17 & 9 & 4 \\ 2 & 4 & 1 & 0 & 12 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 12 & 0 & 49 & 27 & 13 \\ 5 & 9 & 6 & 0 & 27 & 12 & 6 \\ 3 & 4 & 3 & -4 & 13 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1141} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1323, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 5 & 3 \\ 3 & 4 & 4 & 0 & 21 & 9 & 4 \\ 2 & 4 & 1 & 0 & 15 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 15 & 0 & 76 & 33 & 16 \\ 5 & 9 & 6 & 0 & 33 & 12 & 6 \\ 3 & 4 & 3 & -4 & 16 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1131} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1397, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 2 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 12 & 0 & 49 & 12 & 13 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 3 & -4 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{1311} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1565, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 5 & 0 & 27 & 4 & 4 \\ 3 & 5 & 4 & 0 & 27 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 15 & 27 & 27 & 0 & 133 & 18 & 21 \\ 2 & 4 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -4 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{1221} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 10 & 2 & 3 \\ 3 & 4 & 13 & 0 & 18 & 4 & 4 \\ 7 & 13 & 28 & 0 & 42 & 9 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 42 & 0 & 58 & 12 & 14 \\ 2 & 4 & 9 & 0 & 12 & 0 & 2 \\ 3 & 4 & 10 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{1131} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 2 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 15 & 27 & 18 & 0 & 133 & 18 & 21 \\ 2 & 4 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & 3 & -4 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{1121} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1760, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 10 & 2 & 3 \\ 3 & 4 & 9 & 0 & 18 & 4 & 4 \\ 5 & 9 & 13 & 0 & 30 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 30 & 0 & 58 & 12 & 14 \\ 2 & 4 & 6 & 0 & 12 & 0 & 2 \\ 3 & 4 & 7 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{112\bar{1}} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1781, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 13 & 2 & 3 \\ 3 & 4 & 9 & 0 & 23 & 4 & 4 \\ 5 & 9 & 13 & 0 & 39 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 13 & 23 & 39 & 0 & 100 & 15 & 18 \\ 2 & 4 & 6 & 0 & 15 & 0 & 2 \\ 3 & 4 & 7 & -4 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{122\bar{1}} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1784, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 2 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 & 4 \\ 7 & 13 & 28 & 0 & 30 & 9 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 30 & 0 & 28 & 9 & 10 \\ 2 & 4 & 9 & 0 & 9 & 0 & 2 \\ 3 & 4 & 10 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{113\bar{1}} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1820, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 2 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 15 & 0 & 85 & 15 & 17 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 3 & -4 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{13\bar{1}} 3 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1865, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 5 & 0 & 32 & 4 & 4 \\ 3 & 5 & 4 & 0 & 33 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 18 & 32 & 33 & 0 & 193 & 21 & 25 \\ 2 & 4 & 3 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -4 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{114\bar{1}} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1889, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.86 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 2 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 15 & 0 & 76 & 15 & 16 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 3 & -4 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{14\bar{1}} 3 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1928, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 5 & 0 & 31 & 4 & 4 \\ 3 & 5 & 4 & 0 & 30 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 17 & 31 & 30 & 0 & 172 & 21 & 24 \\ 2 & 4 & 3 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -4 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{113\bar{1}} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2501, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 12 & 2 & 3 \\ 3 & 4 & 9 & 0 & 22 & 4 & 4 \\ 5 & 9 & 13 & 0 & 36 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 36 & 0 & 85 & 15 & 17 \\ 2 & 4 & 6 & 0 & 15 & 0 & 2 \\ 3 & 4 & 7 & -4 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{122\bar{1}} 3 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2624, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 10 & 2 & 3 \\ 3 & 4 & 18 & 0 & 18 & 4 & 4 \\ 10 & 18 & 58 & 0 & 60 & 12 & 14 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 60 & 0 & 58 & 12 & 14 \\ 2 & 4 & 12 & 0 & 12 & 0 & 2 \\ 3 & 4 & 14 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{141} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2636, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 5 & 0 & 36 & 4 & 4 \\ 3 & 5 & 4 & 0 & 36 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 20 & 36 & 36 & 0 & 238 & 24 & 28 \\ 2 & 4 & 3 & 0 & 24 & 0 & 2 \\ 3 & 4 & 4 & -4 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{1141} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2720, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 2 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 14 & 26 & 18 & 0 & 118 & 18 & 20 \\ 2 & 4 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & 3 & -4 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{1141} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2780, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 2 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 & 4 \\ 2 & 4 & 1 & 0 & 24 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 20 & 36 & 24 & 0 & 238 & 24 & 28 \\ 2 & 4 & 3 & 0 & 24 & 0 & 2 \\ 3 & 4 & 3 & -4 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{1141} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2996, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 14 & 2 & 3 \\ 3 & 4 & 9 & 0 & 26 & 4 & 4 \\ 5 & 9 & 13 & 0 & 42 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 14 & 26 & 42 & 0 & 118 & 18 & 20 \\ 2 & 4 & 6 & 0 & 18 & 0 & 2 \\ 3 & 4 & 7 & -4 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{1141} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3017, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 2 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 17 & 31 & 21 & 0 & 172 & 21 & 24 \\ 2 & 4 & 3 & 0 & 21 & 0 & 2 \\ 3 & 4 & 3 & -4 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{1131} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 15 & 2 & 3 \\ 3 & 4 & 9 & 0 & 27 & 4 & 4 \\ 5 & 9 & 13 & 0 & 45 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 15 & 27 & 45 & 0 & 133 & 18 & 21 \\ 2 & 4 & 6 & 0 & 18 & 0 & 2 \\ 3 & 4 & 7 & -4 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{1131} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3365, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 18 & 2 & 3 \\ 3 & 4 & 9 & 0 & 32 & 4 & 4 \\ 5 & 9 & 13 & 0 & 54 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 18 & 32 & 54 & 0 & 193 & 21 & 25 \\ 2 & 4 & 6 & 0 & 21 & 0 & 2 \\ 3 & 4 & 7 & -4 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{1141} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -4373, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 17 & 2 & 3 \\ 3 & 4 & 9 & 0 & 31 & 4 & 4 \\ 5 & 9 & 13 & 0 & 51 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 17 & 31 & 51 & 0 & 172 & 21 & 24 \\ 2 & 4 & 6 & 0 & 21 & 0 & 2 \\ 3 & 4 & 7 & -4 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{1141} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -5216, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 20 & 2 & 3 \\ 3 & 4 & 9 & 0 & 36 & 4 & 4 \\ 5 & 9 & 13 & 0 & 60 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 20 & 36 & 60 & 0 & 238 & 24 & 28 \\ 2 & 4 & 6 & 0 & 24 & 0 & 2 \\ 3 & 4 & 7 & -4 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [80] \quad 1^{1111} 321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 82, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.90 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 & 3 \\ 3 & 4 & 1 & 9 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 9 & 0 & 14 & 7 & 9 & 8 \\ 2 & 3 & 0 & 7 & -2 & 3 & 0 \\ 3 & 6 & 0 & 9 & 3 & 4 & 4 \\ 3 & 4 & -1 & 8 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{1111} 31^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 111, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.62 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 9 & 3 & 3 \\ 1 & 3 & 3 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 9 & 6 & 0 & 13 & 6 & 7 \\ 1 & 3 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 3 & 3 & 3 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{11} 31^{112} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 138, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.35 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 6 & 12 & 0 & 22 & 6 & 8 \\ 1 & 3 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{1121} 31^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 273, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.38 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 12 & 3 & 3 \\ 1 & 3 & 3 & 1 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 9 & 12 & 9 & 0 & 28 & 9 & 10 \\ 1 & 3 & 3 & 3 & 0 & 9 & 0 & 2 \\ 2 & 3 & 3 & 3 & -4 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{11} 31^{111} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 300, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 3 & 4 \\ 1 & 3 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{111} 31^{111} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 363, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 9 & 0 & 13 & 6 & 7 \\ 1 & 3 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{111} 31^{112} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 1065, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 15 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 15 & 9 & 15 & 0 & 37 & 9 & 11 \\ 1 & 3 & 3 & 3 & 0 & 9 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 11 & 2 & -1 \end{pmatrix}_{b_0}$$

[87] $1^{111}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 1659, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 21 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 12 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 21 & 12 & 21 & 0 & 73 & 12 & 15 \\ 1 & 3 & 3 & 3 & 0 & 12 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 15 & 2 & -1 \end{pmatrix}_{b_0}$$

[88] $1^{111}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 2145, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 27 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 15 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 27 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 27 & 15 & 27 & 0 & 121 & 15 & 19 \\ 1 & 3 & 3 & 3 & 0 & 15 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 19 & 2 & -1 \end{pmatrix}_{b_0}$$

4.780 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$

$T = 10$ [0] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 148, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 3 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 9 & 24 & 11 \\ 2 & 4 & 3 & 0 & 9 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 24 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 11 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 274, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 18 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 12 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 12 & 2 \\ 10 & 18 & 18 & 0 & 12 & 12 & 58 & 14 \\ 3 & 4 & 4 & -4 & 3 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 454, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 27 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 18 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 18 & 2 \\ 15 & 27 & 27 & 0 & 18 & 18 & 133 & 21 \\ 3 & 4 & 4 & -4 & 3 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

- [5] $1^{\overline{11111}}31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 6 & 0 & 6 & 12 & 6 & 6 \\ 2 & 4 & 3 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & 3 & -4 & 3 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$
- [6] $1^{\overline{1113}}31^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 22 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 15 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 15 & 2 \\ 12 & 22 & 21 & 0 & 15 & 15 & 85 & 17 \\ 3 & 4 & 4 & -4 & 3 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$
- [7] $1^{\overline{1114}}31^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 706, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 36 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 24 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 24 & 2 \\ 20 & 36 & 36 & 0 & 24 & 24 & 238 & 28 \\ 3 & 4 & 4 & -4 & 3 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{\overline{11111}}31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 712, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 & 3 \\ 3 & 4 & 3 & -4 & 3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{\overline{11111}}31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 724, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{\overline{1112}}31^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 796, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 & 13 & 4 \\ 3 & 5 & 4 & 0 & 9 & 3 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 6 & 21 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 9 & 2 \\ 7 & 13 & 12 & 0 & 21 & 9 & 28 & 10 \\ 3 & 4 & 4 & -4 & 7 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{\overline{11111}}31^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 6 & 15 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 15 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 7 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{\overline{11111}}31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1032, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 6 & 3 \\ 5 & 9 & 6 & 0 & 6 & 12 & 15 & 6 \\ 5 & 9 & 6 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & 3 & -4 & 3 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1212}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1144, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.87 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 3 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 12 & 0 & 28 & 9 & 30 & 10 \\ 2 & 4 & 3 & 0 & 9 & 0 & 9 & 2 \\ 7 & 13 & 12 & 0 & 30 & 9 & 28 & 10 \\ 3 & 4 & 4 & -4 & 10 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1114}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1180, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 31 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 30 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 21 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 21 & 2 \\ 17 & 31 & 30 & 0 & 21 & 21 & 172 & 24 \\ 3 & 4 & 4 & -4 & 3 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1276, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 6 & 15 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 15 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 7 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 2368, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 & 9 & 4 \\ 5 & 9 & 13 & 0 & 15 & 6 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 15 & 0 & 13 & 6 & 15 & 7 \\ 2 & 4 & 6 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 15 & 0 & 15 & 6 & 13 & 7 \\ 3 & 4 & 7 & -4 & 7 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

4.781 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{1111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 48, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 2 & 2 & 3 \\ 4 & 9 & 12 & 0 & 12 & 4 & 4 & 5 \\ 5 & 12 & 13 & 0 & 15 & 6 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 12 & 15 & 0 & 13 & 6 & 6 & 7 \\ 2 & 4 & 6 & 0 & 6 & 0 & 3 & 2 \\ 2 & 4 & 6 & 0 & 6 & 3 & 0 & 2 \\ 3 & 5 & 7 & -4 & 7 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 228, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 & 3 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 231, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & 4 & -4 & 3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 507, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 & 4 & 4 \\ 3 & 5 & 4 & 0 & 9 & 3 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 6 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 6 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 7 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 651, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 5 & 4 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 9 & 6 & 7 \\ 3 & 5 & 3 & 0 & 9 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 7 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{1141}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 813, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 21 & 5 & 4 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 11 & 21 & 15 & 0 & 76 & 18 & 15 & 16 \\ 3 & 5 & 3 & 0 & 18 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 15 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 16 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{1121}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 1173, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 13 & 5 & 4 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 12 & 9 & 10 \\ 3 & 5 & 3 & 0 & 12 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 9 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 10 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\overline{1131}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 1227, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 17 & 5 & 4 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 9 & 17 & 12 & 0 & 49 & 15 & 12 & 13 \\ 3 & 5 & 3 & 0 & 15 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 12 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 13 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.782 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 13^{\overline{1}} 21^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 16.67 \rightarrow 17, \quad T_{\max} = 21, \quad T \in [17, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 5 & 5 & 5 & 3 & 5 \\ 3 & 5 & 4 & 3 & 3 & 4 \\ 2 & 5 & 3 & -2 & 2 & 0 \\ 2 & 3 & 3 & 2 & 0 & 2 \\ 3 & 5 & 4 & 0 & 2 & -3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1}} 3^{\overline{1}} 13^{\overline{1}} 21^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -10, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 17.60 \rightarrow 18, \quad T_{\max} = 21, \quad T \in [18, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 3 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 5 & 0 & -2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 5 & 0 & 0 & 2 & -4 \end{pmatrix}_{b_0}$$

- [2] $1^{11}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 & 2 & 3 \\ 2 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 2 & 0 & 2 & 4 & 4 \\ 2 & 2 & 0 & 4 & -2 & 0 \\ 3 & 3 & -1 & 4 & 0 & 1 \end{pmatrix}_{b_0}$
- [3] $1^{11}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -58$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.55 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 2 & 1 & 4 & 2 & 3 \\ 2 & 1 & 1 & 5 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & 9 & 6 & 7 \\ 2 & 2 & 0 & 6 & -2 & 0 \\ 3 & 3 & -1 & 7 & 0 & 1 \end{pmatrix}_{b_0}$
- [4] $1^{11}3^11^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -82$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 2 & 3 & 2 & 0 & 0 & 2 \\ 2 & 3 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$
- [5] $1^{11}3^11^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -87$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 2 \\ 1 & 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 2 & 0 & 0 & 2 \\ 2 & 4 & 2 & 3 & -4 & 2 & 0 \end{pmatrix}_{b_0}$
- [6] $1^{13}3^11^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -257$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 2 & 0 \end{pmatrix}_{b_0}$
- [7] $1^{12}3^11^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -297$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 3 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 & 2 \\ 4 & 12 & 9 & 28 & 0 & 7 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 7 & 0 & 0 & 2 \\ 2 & 4 & 2 & 10 & -4 & 2 & 0 \end{pmatrix}_{b_0}$
- [8] $1^{13}3^11^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -392$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 & 2 \\ 4 & 12 & 6 & 22 & 0 & 6 & 8 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 6 & 0 & 0 & 2 \\ 2 & 4 & 2 & 8 & -4 & 2 & 0 \end{pmatrix}_{b_0}$
- [9] $1^{11}3^11^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -402$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 & 2 \\ 3 & 9 & 6 & 13 & 0 & 5 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 5 & 0 & 0 & 2 \\ 2 & 4 & 2 & 7 & -4 & 2 & 0 \end{pmatrix}_{b_0}$
- [10] $1^{14}3^11^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -552$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 1 & 2 \\ 2 & 4 & 4 & 30 & 0 & 3 & 4 \\ 1 & 4 & 0 & 21 & 0 & 2 & 2 \\ 10 & 30 & 21 & 172 & 0 & 17 & 24 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 17 & 0 & 0 & 2 \\ 2 & 4 & 2 & 24 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{11}3^{11}1^{14}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1887, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 3 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 & 2 \\ 9 & 27 & 15 & 121 & 0 & 14 & 19 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 14 & 0 & 0 & 2 \\ 2 & 4 & 2 & 19 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{13}3^{11}1^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1922, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 3 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 & 2 \\ 9 & 27 & 18 & 133 & 0 & 15 & 21 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 15 & 0 & 0 & 2 \\ 2 & 4 & 2 & 21 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{12}3^{11}1^{14}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -2592, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 1 & 2 \\ 2 & 4 & 4 & 30 & 0 & 3 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 & 2 \\ 10 & 30 & 18 & 154 & 0 & 16 & 22 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 16 & 0 & 0 & 2 \\ 2 & 4 & 2 & 22 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{13}3^{11}1^{14}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -3047, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 1 & 1 & 2 \\ 2 & 4 & 4 & 33 & 0 & 3 & 4 \\ 1 & 4 & 0 & 21 & 0 & 2 & 2 \\ 11 & 33 & 21 & 193 & 0 & 18 & 25 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 18 & 0 & 0 & 2 \\ 2 & 4 & 2 & 25 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{14}3^{11}1^{14}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -3252, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 1 & 1 & 2 \\ 2 & 4 & 4 & 36 & 0 & 3 & 4 \\ 1 & 4 & 0 & 24 & 0 & 2 & 2 \\ 12 & 36 & 24 & 238 & 0 & 20 & 28 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 20 & 0 & 0 & 2 \\ 2 & 4 & 2 & 28 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

4.783 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$

$T = 10 \quad [0] \quad 1^{111}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 329, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 6 & 2 \\ 1 & 3 & 3 & 1 & 0 & 2 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 2 & 0 & 0 & 5 & 2 \\ 3 & 9 & 6 & 6 & 0 & 5 & 13 & 7 \\ 2 & 4 & 2 & 3 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^{11}1^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 464, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 9 & 2 \\ 1 & 3 & 3 & 1 & 0 & 2 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 2 & 0 & 0 & 8 & 2 \\ 5 & 15 & 9 & 9 & 0 & 8 & 37 & 11 \\ 2 & 4 & 2 & 3 & -4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{13}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 604, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & -4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^11^{111}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{c}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 669, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.85 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 6 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 3 & 0 & 0 & 5 & 2 \\ 3 & 9 & 6 & 9 & 0 & 5 & 13 & 7 \\ 2 & 4 & 2 & 4 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^{111}31^{111}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{c}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 1184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 & 6 & 2 \\ 3 & 9 & 6 & 13 & 0 & 5 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 5 & 0 & 0 & 5 & 2 \\ 3 & 9 & 6 & 15 & 0 & 5 & 13 & 7 \\ 2 & 4 & 2 & 7 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

4.784 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_{2 \rightarrow (-3)} + \mathfrak{su}_{2 \rightarrow (-3)}$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 2 & 2 \\ 1 & 3 & 3 & 1 & 0 & 2 & 2 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 2 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 2 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 3 & -4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{13}1^{11}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 436$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 2 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{12}3^{11}1^{11}31^1 + [\textcolor{red}{\text{su}}_2 \cdot \textcolor{blue}{\text{g}}_2] + \textcolor{green}{\text{su}}_2 + \textcolor{brown}{\text{c}}_6 + \textcolor{red}{\text{su}}_2 \rightarrow (-3) + \textcolor{blue}{\text{su}}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 500$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 & 2 & 2 \\ 4 & 12 & 9 & 28 & 0 & 7 & 7 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 7 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 7 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 10 & -4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{13}1^{11}1^{12}31^1 + [\textcolor{red}{\mathbf{s}}\mathbf{u}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{green}{\mathbf{s}}\mathbf{u}_2 + \textcolor{brown}{\mathbf{c}}_6 + \textcolor{red}{\mathbf{s}}\mathbf{u}_2 \rightarrow (-3) + \textcolor{blue}{\mathbf{s}}\mathbf{u}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 544$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 & 2 & 2 \\ 4 & 12 & 6 & 22 & 0 & 6 & 6 & 8 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 6 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 6 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 8 & -4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{11}3^{11}1^{11}31^1 + [\textcolor{red}{\text{su}}_2 \cdot \textcolor{blue}{\text{g}}_2] + \textcolor{teal}{\text{su}}_2 + \textcolor{brown}{\text{c}}_6 + \textcolor{red}{\text{su}}_2 \rightarrow (-3) + \textcolor{teal}{\text{su}}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 704$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 & 3 & 2 \\ 3 & 9 & 6 & 13 & 0 & 5 & 5 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 5 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 5 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 7 & -4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{11}3^{11}1^{12}31^1 + [\textcolor{red}{\mathbf{s}}\textcolor{red}{\mathbf{u}}_2 - \textcolor{red}{\mathbf{g}}_2] + \textcolor{teal}{\mathbf{s}}\textcolor{teal}{\mathbf{u}}_2 + \textcolor{brown}{\mathbf{c}}_6 + \textcolor{red}{\mathbf{s}}\textcolor{red}{\mathbf{u}}_2 \rightarrow (-3) + \textcolor{teal}{\mathbf{s}}\textcolor{teal}{\mathbf{u}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 1316, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 & 2 & 2 \\ 5 & 15 & 9 & 37 & 0 & 8 & 8 & 11 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 8 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 8 & 0 & 0 & 2 & 2 \\ 2 & 4 & 2 & 11 & -4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{123}3^{11}1^{123}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 1616$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 & 2 & 2 \\ 6 & 18 & 12 & 58 & 0 & 10 & 10 & 14 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 10 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 10 & 0 & 0 & 2 & 2 \\ 2 & 4 & 2 & 14 & -4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

4.785 NHC(-2)(-3) + su₂ + e₆ + su₂ → (-3) + su₃

$T = 10$ [0] $1^{111}3^11^{131}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_3$ $T = 10$, $T_{\max} = 15$, $\Delta = -93$, $\det = 86$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.56 \rightarrow 14$, $T_{\max} = 15$, $T \in [14, 15]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 & 3 \\ 1 & 3 & 3 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 3 & 3 & 3 & 0 & 2 & 0 & 2 \\ 2 & 3 & 3 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{113}3^11^{113}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_3$ $T = 10$, $T_{\max} = 15$, $\Delta = -93$, $\det = 126$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.43 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 3 & 2 \\ 1 & 3 & 3 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 2 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 2 & 3 & -4 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^12^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -87$, $\det = 130$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 2 & 2 & 3 & 3 \\ 2 & 1 & 1 & 5 & 2 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 & -1 \\ 4 & 5 & 0 & 9 & 6 & 5 & 7 & 7 \\ 2 & 2 & 0 & 6 & -2 & 2 & 0 & 0 \\ 2 & 3 & 0 & 5 & 2 & 0 & 2 & 1 \\ 3 & 3 & -1 & 7 & 0 & 2 & 1 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{113}3^11^{113}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2 \rightarrow (-3) + \text{su}_3$ $T = 10$, $T_{\max} = 15$, $\Delta = -93$, $\det = 296$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.27 \rightarrow 14$, $T_{\max} = 15$, $T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 3 & 3 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.786 NHC(-2)(-3) + su₂ + e₆ + su₂n3mix

$T = 9$ [0] $1^{111}3^12^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 3 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 & -1 \\ 2 & 3 & 0 & 5 & 4 & 5 & 5 & 5 \\ 1 & 1 & 0 & 4 & -3 & 2 & -1 & -1 \\ 2 & 3 & 0 & 5 & 2 & 3 & 3 & 3 \\ 2 & 2 & -1 & 5 & -1 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^11^{131}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 13.58 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 5 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 6 & 3 & 3 \\ 1 & 3 & 3 & 1 & 0 & 5 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 & -4 \\ 2 & 5 & 6 & 5 & 0 & 7 & 5 & 5 \\ 2 & 3 & 3 & 3 & -4 & 5 & -1 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{113}3^11^{113}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -78$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 5 & 2 \\ 1 & 3 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 5 & 5 & 0 & 12 & 6 \\ 2 & 4 & 2 & 3 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 & 2 \\ 2 & 4 & 4 & 9 & 0 & 6 & 4 \\ 1 & 4 & 0 & 6 & 0 & 5 & 2 \\ 3 & 9 & 6 & 13 & 0 & 11 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 5 & 11 & 0 & 7 & 5 \\ 2 & 4 & 2 & 7 & -4 & 5 & 0 \end{pmatrix}_b$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 3 & 0 & 12 & 4 \\ 1 & 4 & 0 & 3 & 0 & 8 & 2 \\ 1 & 3 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 8 & 8 & 0 & 25 & 9 \\ 2 & 4 & 2 & 3 & -4 & 9 & 0 \end{pmatrix}_b$$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 3 & 0 & 30 & 4 \\ 1 & 4 & 0 & 3 & 0 & 17 & 2 \\ 1 & 3 & 3 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 17 & 17 & 0 & 151 & 21 \\ 2 & 4 & 2 & 3 & -4 & 21 & -1 \end{pmatrix} b_0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 3 & 0 & 18 & 4 \\ 1 & 4 & 0 & 3 & 0 & 11 & 2 \\ 1 & 3 & 3 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 11 & 11 & 0 & 55 & 13 \\ 2 & 4 & 2 & 3 & -4 & 13 & 0 \end{pmatrix} b_0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 12 & 4 \\ 1 & 4 & 0 & 3 & 0 & 8 & 2 \\ 2 & 6 & 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 8 & 12 & 0 & 25 & 9 \\ 2 & 4 & 2 & 4 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

$$T_{\min} = 13.19 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 3 & 0 & 24 & 4 \\ 1 & 4 & 0 & 3 & 0 & 14 & 2 \\ 1 & 3 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 14 & 14 & 0 & 97 & 17 \\ 2 & 4 & 2 & 3 & -4 & 17 & -1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 5 & 2 \\ 2 & 6 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 5 & 9 & 0 & 12 & 6 \\ 2 & 4 & 2 & 4 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 9 & 4 \\ 1 & 4 & 0 & 6 & 0 & 5 & 2 \\ 3 & 9 & 6 & 13 & 0 & 14 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 5 & 14 & 0 & 12 & 6 \\ 2 & 4 & 2 & 7 & -4 & 6 & 0 \end{pmatrix}_b$$

4225

11.0

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 4 & 2 \\ 2 & 4 & 4 & 9 & 0 & 12 & 4 \\ 1 & 4 & 0 & 6 & 0 & 8 & 2 \\ 3 & 9 & 6 & 13 & 0 & 20 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 8 & 20 & 0 & 25 & 9 \\ 2 & 4 & 2 & 7 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

4.787 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_{2n3\text{mix}} + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{11}3^{11}1^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 494, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 5 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 5 & 9 & 0 & 12 & 5 & 6 \\ 1 & 3 & 3 & 3 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

4.788 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{14}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.6$$

$$T_{\min} = 24, T_{\max} = 98, T \in [24, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 18 & 0 & 75 & 15 \\ 3 & 4 & 4 & -4 & 15 & -11 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16, T_{\max} = 98, T \in [16, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 3 & -4 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 6 & 0 & 12 & 6 \\ 3 & 5 & 3 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{13}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 7 & 3 \\ 3 & 4 & 12 & 0 & 13 & 4 \\ 6 & 12 & 25 & 0 & 27 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 27 & 0 & 27 & 9 \\ 3 & 4 & 9 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.03 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 3 & 4 & 3 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 121, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 6 & 0 & 3 & 3 \\ 3 & 4 & 4 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 126, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.81 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 12 & 0 & 12 & 6 \\ 3 & 4 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 127, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.02 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 148, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 3 \\ 4 & 9 & 12 & 0 & 4 & 5 \\ 5 & 12 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 5 & 7 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 160, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 48 & 12 \\ 3 & 4 & 4 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 166, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 172, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.21 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 1 & 3 \\ 4 & 9 & 16 & 4 & 0 & 5 \\ 6 & 16 & 22 & 6 & 0 & 8 \\ 2 & 4 & 6 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 8 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 178, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.07 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 181, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 12 & 0 & 9 & 5 \\ 3 & 4 & 7 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 184, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 19 & 0 & 4 & 4 \\ 11 & 19 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 15 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 184, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.91 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 4 & 8 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 6 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{122}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 12 & 0 & 9 & 5 \\ 3 & 4 & 6 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

[18] $1^{113}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 9 & 0 & 24 & 8 \\ 3 & 4 & 3 & -4 & 8 & -1 \end{pmatrix}_{b_0}$$

[19] $1^{111}3^11^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 0 & 36 & 10 \\ 3 & 4 & 3 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{122}3^11^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.54 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 4 & 3 \\ 3 & 4 & 18 & 0 & 8 & 4 \\ 10 & 18 & 58 & 0 & 24 & 14 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 24 & 0 & 9 & 5 \\ 3 & 4 & 14 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

[21] $1^{131}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 4 & 4 \\ 6 & 12 & 25 & 0 & 9 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 9 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[22] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{132}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 4 & 3 \\ 3 & 4 & 12 & 0 & 8 & 4 \\ 6 & 12 & 25 & 0 & 18 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 18 & 0 & 9 & 5 \\ 3 & 4 & 9 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

[24] $1^{123}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 12 & 0 & 27 & 9 \\ 3 & 4 & 4 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 3 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{113}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 1 & 3 \\ 4 & 9 & 24 & 4 & 0 & 5 \\ 9 & 24 & 52 & 9 & 0 & 12 \\ 2 & 4 & 9 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 12 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[27] $1^{132}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.98 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 4 & 3 \\ 3 & 4 & 27 & 0 & 8 & 4 \\ 15 & 27 & 133 & 0 & 36 & 21 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 36 & 0 & 9 & 5 \\ 3 & 4 & 21 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

[29] $1^{112}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 2 & 3 \\ 4 & 9 & 24 & 0 & 4 & 5 \\ 10 & 24 & 58 & 0 & 12 & 14 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 5 & 14 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{141}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.31 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 16 & 0 & 4 & 4 \\ 8 & 16 & 46 & 0 & 12 & 12 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 12 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{122}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 18 & 0 & 27 & 9 \\ 3 & 4 & 6 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 5 & 9 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 9 & 0 & 3 & 3 \\ 3 & 4 & 7 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 \\ 5 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 7 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{151}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 20 & 0 & 4 & 4 \\ 10 & 20 & 73 & 0 & 15 & 15 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 15 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[33] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 \\ 8 & 14 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 11 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{141}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 313, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 & 3 \\ 3 & 4 & 21 & 0 & 5 & 4 \\ 11 & 21 & 76 & 0 & 18 & 16 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 18 & 0 & 3 & 3 \\ 3 & 4 & 16 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[36] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 2 & 3 \\ 4 & 9 & 20 & 0 & 4 & 5 \\ 8 & 20 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 5 & 11 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 1 & 3 \\ 4 & 9 & 32 & 4 & 0 & 5 \\ 12 & 32 & 94 & 12 & 0 & 16 \\ 2 & 4 & 12 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 16 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[38] $1^{12}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 373, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 \\ 7 & 13 & 28 & 0 & 12 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 12 & 0 & 3 & 3 \\ 3 & 4 & 10 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{161}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.64 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 24 & 0 & 4 & 4 \\ 12 & 24 & 106 & 0 & 18 & 18 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 18 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[40] $1^{13}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 \\ 9 & 17 & 49 & 0 & 15 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 15 & 0 & 3 & 3 \\ 3 & 4 & 13 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 391, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 0 & 27 & 9 \\ 3 & 4 & 3 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{122}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 403, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 8 & 4 \\ 7 & 13 & 28 & 0 & 18 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 18 & 0 & 9 & 5 \\ 3 & 4 & 10 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

[43] $1^{113}31^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 2 & 3 \\ 4 & 9 & 36 & 0 & 4 & 5 \\ 15 & 36 & 133 & 0 & 18 & 21 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 5 & 21 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{171}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 13.76 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 4 & 4 \\ 14 & 28 & 145 & 0 & 21 & 21 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 3 & 4 & 21 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 15 & 0 & 12 & 6 \\ 3 & 4 & 7 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 486, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 12 & 0 & 57 & 13 \\ 3 & 4 & 3 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[47] $1^{115}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 487, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 1 & 3 \\ 4 & 9 & 40 & 4 & 0 & 5 \\ 15 & 40 & 148 & 15 & 0 & 20 \\ 2 & 4 & 15 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 20 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 526, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 7 & 13 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 10 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 547, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 0 & 48 & 12 \\ 3 & 4 & 3 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 568, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 11 & 1 & 2 & 3 \\ 4 & 9 & 28 & 0 & 4 & 5 \\ 11 & 28 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 5 & 15 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 574, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 21 & 0 & 27 & 9 \\ 3 & 4 & 7 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{113}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 595, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 27 & 0 & 48 & 12 \\ 3 & 4 & 7 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{113}31^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 631, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 2 & 3 \\ 4 & 9 & 32 & 0 & 4 & 5 \\ 13 & 32 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 5 & 18 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{132}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 4 & 3 \\ 3 & 4 & 17 & 0 & 8 & 4 \\ 9 & 17 & 49 & 0 & 24 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 24 & 0 & 9 & 5 \\ 3 & 4 & 13 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

[55] $1^{132}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 4 & 3 \\ 3 & 4 & 22 & 0 & 8 & 4 \\ 12 & 22 & 85 & 0 & 30 & 17 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 30 & 0 & 9 & 5 \\ 3 & 4 & 17 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

[56] $1^{121}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 648, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 7 & 13 & 28 & 0 & 21 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 21 & 0 & 12 & 6 \\ 3 & 4 & 10 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{114}31^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 688, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 2 & 3 \\ 4 & 9 & 48 & 0 & 4 & 5 \\ 20 & 48 & 238 & 0 & 24 & 28 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 24 & 0 & 0 & 2 \\ 3 & 5 & 28 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[58] $1^{121}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 736, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 4 & 4 \\ 10 & 18 & 58 & 0 & 12 & 14 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 14 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{131}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 747, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 5 & 3 \\ 3 & 4 & 17 & 0 & 9 & 4 \\ 9 & 17 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 27 & 0 & 12 & 6 \\ 3 & 4 & 13 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[61] $1^{131}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 760, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 9 & 17 & 49 & 0 & 12 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 13 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{121}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 760, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 4 & 4 \\ 16 & 28 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 22 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[62] $1^{141}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 5 & 3 \\ 3 & 4 & 21 & 0 & 9 & 4 \\ 11 & 21 & 76 & 0 & 33 & 16 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 33 & 0 & 12 & 6 \\ 3 & 4 & 16 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[63] $1^{113}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 778, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.66 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 15 & 0 & 84 & 16 \\ 3 & 4 & 3 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{121}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 814, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 23 & 0 & 4 & 4 \\ 13 & 23 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 18 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{114}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 823, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 2 & 3 \\ 4 & 9 & 36 & 0 & 4 & 5 \\ 14 & 36 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 5 & 19 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[66] $1^{122}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 841, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 30 & 0 & 27 & 9 \\ 3 & 4 & 10 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{114} 3^1 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 991, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 2 & 3 \\ 4 & 9 & 44 & 0 & 4 & 5 \\ 18 & 44 & 193 & 0 & 21 & 25 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 3 & 5 & 25 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{141} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1006, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 4 & 4 \\ 11 & 21 & 76 & 0 & 15 & 16 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 16 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{132} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1030, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 7 & 3 \\ 3 & 4 & 17 & 0 & 13 & 4 \\ 9 & 17 & 49 & 0 & 39 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 39 & 0 & 27 & 9 \\ 3 & 4 & 13 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{114} 3^1 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1036, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 2 & 3 \\ 4 & 9 & 40 & 0 & 4 & 5 \\ 16 & 40 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 5 & 22 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{131} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 4 & 4 \\ 12 & 22 & 85 & 0 & 15 & 17 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 17 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{131} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 27 & 0 & 4 & 4 \\ 15 & 27 & 133 & 0 & 18 & 21 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 21 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{131} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1606, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 2 & 3 \\ 3 & 4 & 32 & 0 & 4 & 4 \\ 18 & 32 & 193 & 0 & 21 & 25 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 3 & 4 & 25 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{141} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1624, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 26 & 0 & 4 & 4 \\ 14 & 26 & 118 & 0 & 18 & 20 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 20 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{141} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 2110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 3 \\ 3 & 4 & 31 & 0 & 4 & 4 \\ 17 & 31 & 172 & 0 & 21 & 24 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 3 & 4 & 24 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{141} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 2464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 2 & 3 \\ 3 & 4 & 36 & 0 & 4 & 4 \\ 20 & 36 & 238 & 0 & 24 & 28 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 24 & 0 & 0 & 2 \\ 3 & 4 & 28 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [77] \quad 1^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 4 & 5 \\ 1 & 1 & 0 & 4 & -3 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & 5 & -1 & -4 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{11} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16, T_{\max} = 68, T \in [16, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 \\ 1 & 4 & 0 & 3 & 0 & 3 \\ 1 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 1 & 3 & 3 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 & -4 & -3 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{11} 321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16.67 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 5 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & 0 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{11} 321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16.40 \rightarrow 17, T_{\max} = 25, T \in [17, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 5 & 0 & -2 & 3 & 0 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 0 & 0 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{12} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.17 \rightarrow 16, T_{\max} = 68, T \in [16, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 12 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 6 & 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 9 & 12 & 0 & 27 & 9 \\ 2 & 4 & 2 & 4 & -4 & 9 & -3 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{12} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 18.58 \rightarrow 19, T_{\max} = 68, T \in [19, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 & 2 \\ 2 & 4 & 4 & 12 & 0 & 6 & 4 \\ 1 & 4 & 0 & 9 & 0 & 3 & 2 \\ 4 & 12 & 9 & 28 & 0 & 12 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 12 & 0 & 3 & 3 \\ 2 & 4 & 2 & 10 & -4 & 3 & -6 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{11} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{11} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 6 & 0 & 3 & 3 \\ 2 & 3 & 6 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & 4 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{11} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 68, T \in [15, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 1 & 3 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & 3 & -4 & 3 & -2 \end{pmatrix}_{b_0}$$

12.0

$$T_{\min} = 13.04 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 6 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 9 & 0 & 12 & 6 \\ 2 & 4 & 2 & 4 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

[96] $1^{11}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -615, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 & 2 \\ 2 & 4 & 4 & 9 & 0 & 6 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 2 \\ 3 & 9 & 6 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 9 & 0 & 3 & 3 \\ 2 & 4 & 2 & 7 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[97] $1^{11}31^{12}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -762, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 3 & 4 \\ 1 & 4 & 0 & 9 & 0 & 3 & 2 \\ 5 & 15 & 9 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 9 & 0 & 0 & 2 \\ 2 & 4 & 2 & 11 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[98] $1^{11}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -906, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 9 & 4 \\ 1 & 4 & 0 & 6 & 0 & 6 & 2 \\ 3 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 15 & 0 & 12 & 6 \\ 2 & 4 & 2 & 7 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

[99] $1^{11}31^{13}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -1059, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 3 & 4 \\ 1 & 4 & 0 & 12 & 0 & 3 & 2 \\ 7 & 21 & 12 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 12 & 0 & 0 & 2 \\ 2 & 4 & 2 & 15 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[100] $1^{11}31^{14}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -1302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 3 & 4 \\ 1 & 4 & 0 & 15 & 0 & 3 & 2 \\ 9 & 27 & 15 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 15 & 0 & 0 & 2 \\ 2 & 4 & 2 & 19 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$T = 10 \quad [101] \quad 1^{11}2^1 2321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 34, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 1 & 4 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 4 & 4 & 6 \\ 1 & 1 & 0 & 4 & -3 & 1 & -1 \\ 1 & 3 & 0 & 4 & 1 & 0 & 2 \\ 2 & 3 & -1 & 6 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

4.789 $\text{NHC}(-2)(-3) + \textcolor{red}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \text{NHC}(-2)(-3)$

$T = 10 \quad [0] \quad 1^{11}1^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + [\textcolor{blue}{su}_2 \cdot \textcolor{magenta}{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 5 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 4 & 3 \\ 3 & 5 & 3 & 0 & 3 & 4 & 3 & 3 \\ 3 & 4 & 3 & -4 & 2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

4.790 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 0 & 4 \\ 2 & 4 & 3 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & -1 & -1 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 8 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

4.791 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.86 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 5 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 6 & 3 \\ 3 & 5 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & 3 & -4 & 3 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.71 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 36 & 9 & 10 \\ 2 & 4 & 3 & 0 & 9 & 1 & 3 \\ 3 & 4 & 3 & -4 & 10 & 3 & -4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{1211}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 9 & 4 \\ 4 & 8 & 10 & 0 & 12 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 12 & 15 & 6 \\ 5 & 9 & 12 & 0 & 15 & 13 & 7 \\ 3 & 4 & 6 & -4 & 6 & 7 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{1111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 8 & 14 & 9 & 0 & 9 & 37 & 11 \\ 3 & 4 & 3 & -4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 5 & 0 & 4 & 19 & 4 \\ 3 & 5 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 11 & 19 & 21 & 0 & 12 & 73 & 15 \\ 3 & 4 & 4 & -4 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{131}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 & 4 \\ 3 & 5 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 48 & 12 & 12 \\ 2 & 4 & 3 & 0 & 12 & 1 & 3 \\ 3 & 4 & 4 & -4 & 12 & 3 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{1111}31^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 5 & 3 \\ 3 & 4 & 19 & 0 & 4 & 9 & 4 \\ 11 & 19 & 73 & 0 & 12 & 33 & 15 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 12 & 0 & 0 & 6 & 2 \\ 5 & 9 & 33 & 0 & 6 & 13 & 7 \\ 3 & 4 & 15 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{1211}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 5 & 3 \\ 3 & 4 & 8 & 0 & 4 & 9 & 4 \\ 4 & 8 & 10 & 0 & 6 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 5 & 9 & 12 & 0 & 6 & 13 & 7 \\ 3 & 4 & 6 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{1112}31^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -137, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 7 & 3 \\ 3 & 4 & 14 & 0 & 4 & 13 & 4 \\ 8 & 14 & 37 & 0 & 9 & 33 & 11 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 9 & 2 \\ 7 & 13 & 33 & 0 & 9 & 28 & 10 \\ 3 & 4 & 11 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -173, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.39 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 5 & 0 & 4 & 17 & 4 \\ 3 & 5 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 9 & 17 & 15 & 0 & 12 & 49 & 13 \\ 3 & 4 & 4 & -4 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{121} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -174, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.85 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 2 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 12 & 6 & 6 \\ 2 & 4 & 6 & 0 & 6 & 1 & 3 \\ 3 & 4 & 6 & -4 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -191, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 4 & 3 \\ 4 & 9 & 8 & 4 & 0 & 8 & 5 \\ 3 & 8 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 6 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{121} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 13 & 3 \\ 3 & 4 & 13 & 0 & 4 & 23 & 4 \\ 7 & 13 & 28 & 0 & 9 & 54 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 15 & 2 \\ 13 & 23 & 54 & 0 & 15 & 100 & 18 \\ 3 & 4 & 10 & -4 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -219, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 & 4 \\ 3 & 5 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 12 & 6 & 6 \\ 2 & 4 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 4 & -4 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{121} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -245, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 & 4 \\ 3 & 5 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 12 & 0 & 27 & 9 & 9 \\ 2 & 4 & 3 & 0 & 9 & 1 & 3 \\ 3 & 4 & 4 & -4 & 9 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -284, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 & 3 \\ 4 & 9 & 8 & 4 & 0 & 8 & 5 \\ 3 & 8 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 8 & 6 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -341, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1111}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -344, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.81 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -353, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 5 & 0 & 4 & 14 & 4 \\ 3 & 5 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 8 & 14 & 15 & 0 & 9 & 37 & 11 \\ 3 & 4 & 4 & -4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -359, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 9 & 3 \\ 5 & 9 & 6 & 0 & 9 & 13 & 7 \\ 3 & 4 & 3 & -4 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -368, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 5 & 3 \\ 4 & 9 & 12 & 0 & 4 & 12 & 5 \\ 5 & 12 & 13 & 0 & 6 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 5 & 12 & 15 & 0 & 6 & 13 & 7 \\ 3 & 5 & 7 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1112}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -380, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.96 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 4 & 4 & 0 & 4 & 8 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 4 & 8 & 6 & 0 & 6 & 10 & 6 \\ 3 & 4 & 3 & -4 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{112}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -380, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 13 & 4 \\ 3 & 5 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 7 & 13 & 12 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -408, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 6 & 6 \\ 2 & 4 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 3 & -4 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{114}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -413, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 11 & 3 \\ 3 & 4 & 4 & 0 & 5 & 21 & 4 \\ 2 & 4 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 18 & 3 \\ 11 & 21 & 15 & 0 & 18 & 76 & 16 \\ 3 & 4 & 3 & -4 & 3 & 16 & -1 \end{pmatrix}_{b_0}$$

[28] $1^{1112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 4 & 3 \\ 3 & 4 & 8 & 0 & 4 & 8 & 4 \\ 4 & 8 & 10 & 0 & 6 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 4 & 8 & 12 & 0 & 6 & 10 & 6 \\ 3 & 4 & 6 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[29] $1^{1112}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -425, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 13 & 3 \\ 3 & 4 & 4 & 0 & 4 & 23 & 4 \\ 2 & 4 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 13 & 23 & 15 & 0 & 15 & 100 & 18 \\ 3 & 4 & 3 & -4 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{1113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 6 & 3 \\ 3 & 4 & 4 & 0 & 4 & 12 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 6 & 12 & 9 & 0 & 9 & 25 & 9 \\ 3 & 4 & 3 & -4 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

[31] $1^{1212}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 7 & 3 \\ 3 & 4 & 8 & 0 & 9 & 13 & 4 \\ 4 & 8 & 10 & 0 & 12 & 18 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 12 & 21 & 6 \\ 7 & 13 & 18 & 0 & 21 & 28 & 10 \\ 3 & 4 & 6 & -4 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{1111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -485, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{1113}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -488, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 12 & 3 \\ 3 & 4 & 14 & 0 & 4 & 22 & 4 \\ 8 & 14 & 37 & 0 & 9 & 57 & 11 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 15 & 2 \\ 12 & 22 & 57 & 0 & 15 & 85 & 17 \\ 3 & 4 & 11 & -4 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{1114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 16 & 4 \\ 2 & 4 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 8 & 16 & 12 & 0 & 12 & 46 & 12 \\ 3 & 4 & 3 & -4 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

[35] $1^{1112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -533, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 13.03 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 7 & 3 \\ 3 & 4 & 4 & 0 & 5 & 13 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 12 & 3 \\ 7 & 13 & 9 & 0 & 12 & 28 & 10 \\ 3 & 4 & 3 & -4 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -551, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 9 & 3 \\ 3 & 4 & 4 & 0 & 5 & 17 & 4 \\ 2 & 4 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 15 & 3 \\ 9 & 17 & 12 & 0 & 15 & 49 & 13 \\ 3 & 4 & 3 & -4 & 3 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1212} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -560, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 7 & 3 \\ 3 & 4 & 8 & 0 & 4 & 13 & 4 \\ 4 & 8 & 10 & 0 & 6 & 18 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 9 & 2 \\ 7 & 13 & 18 & 0 & 9 & 28 & 10 \\ 3 & 4 & 6 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -729, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 15 & 6 \\ 5 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 3 & -4 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1111} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -773, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 8 & 3 \\ 3 & 4 & 9 & 0 & 4 & 14 & 4 \\ 5 & 9 & 13 & 0 & 6 & 24 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 9 & 2 \\ 8 & 14 & 24 & 0 & 9 & 37 & 11 \\ 3 & 4 & 7 & -4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{112} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -800, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 5 & 0 & 4 & 18 & 4 \\ 3 & 5 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 10 & 18 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & -4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{112} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -848, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 5 & 0 & 4 & 28 & 4 \\ 3 & 5 & 4 & 0 & 3 & 30 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 16 & 28 & 30 & 0 & 18 & 154 & 22 \\ 3 & 4 & 4 & -4 & 2 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 4 & 9 & 4 \\ 5 & 9 & 13 & 0 & 6 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 5 & 9 & 15 & 0 & 6 & 13 & 7 \\ 3 & 4 & 7 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1111} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -920, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 8 & 3 \\ 3 & 4 & 14 & 0 & 4 & 14 & 4 \\ 8 & 14 & 37 & 0 & 9 & 39 & 11 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 9 & 2 \\ 8 & 14 & 39 & 0 & 9 & 37 & 11 \\ 3 & 4 & 11 & -4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -929, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 4 & 13 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 7 & 13 & 9 & 0 & 9 & 28 & 10 \\ 3 & 4 & 3 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1112} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 4 & 18 & 4 \\ 2 & 4 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 10 & 18 & 12 & 0 & 12 & 58 & 14 \\ 3 & 4 & 3 & -4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1112} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.91 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 5 & 0 & 4 & 26 & 4 \\ 3 & 5 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 14 & 26 & 24 & 0 & 18 & 118 & 20 \\ 3 & 4 & 4 & -4 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1112} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 5 & 0 & 4 & 23 & 4 \\ 3 & 5 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 13 & 23 & 24 & 0 & 15 & 100 & 18 \\ 3 & 4 & 4 & -4 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1001, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 5 & 0 & 4 & 22 & 4 \\ 3 & 5 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 12 & 22 & 21 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & -4 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1113} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 18 & 3 \\ 3 & 4 & 4 & 0 & 4 & 32 & 4 \\ 2 & 4 & 1 & 0 & 3 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 18 & 32 & 21 & 0 & 21 & 193 & 25 \\ 3 & 4 & 3 & -4 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1083, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 7 & 3 \\ 3 & 4 & 4 & 0 & 9 & 13 & 4 \\ 2 & 4 & 1 & 0 & 6 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 21 & 6 \\ 7 & 13 & 9 & 0 & 21 & 28 & 10 \\ 3 & 4 & 3 & -4 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1114} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1085, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 11 & 3 \\ 3 & 4 & 9 & 0 & 4 & 21 & 4 \\ 5 & 9 & 13 & 0 & 6 & 33 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 15 & 2 \\ 11 & 21 & 33 & 0 & 15 & 76 & 16 \\ 3 & 4 & 7 & -4 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1205, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 7 & 3 \\ 3 & 4 & 9 & 0 & 4 & 13 & 4 \\ 5 & 9 & 13 & 0 & 6 & 21 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 9 & 2 \\ 7 & 13 & 21 & 0 & 9 & 28 & 10 \\ 3 & 4 & 7 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1112} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 10 & 3 \\ 3 & 4 & 14 & 0 & 4 & 18 & 4 \\ 8 & 14 & 37 & 0 & 9 & 48 & 11 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 12 & 2 \\ 10 & 18 & 48 & 0 & 12 & 58 & 14 \\ 3 & 4 & 11 & -4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 9 & 3 \\ 3 & 4 & 9 & 0 & 4 & 17 & 4 \\ 5 & 9 & 13 & 0 & 6 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 12 & 2 \\ 9 & 17 & 27 & 0 & 12 & 49 & 13 \\ 3 & 4 & 7 & -4 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1112} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 16 & 3 \\ 3 & 4 & 9 & 0 & 4 & 28 & 4 \\ 5 & 9 & 13 & 0 & 6 & 48 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 18 & 2 \\ 16 & 28 & 48 & 0 & 18 & 154 & 22 \\ 3 & 4 & 7 & -4 & 2 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1281, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 9 & 3 \\ 3 & 4 & 4 & 0 & 9 & 17 & 4 \\ 2 & 4 & 1 & 0 & 6 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 27 & 6 \\ 9 & 17 & 12 & 0 & 27 & 49 & 13 \\ 3 & 4 & 3 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1323, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 11 & 3 \\ 3 & 4 & 4 & 0 & 9 & 21 & 4 \\ 2 & 4 & 1 & 0 & 6 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 33 & 6 \\ 11 & 21 & 15 & 0 & 33 & 76 & 16 \\ 3 & 4 & 3 & -4 & 6 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1113} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1397, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 4 & 4 & 0 & 4 & 17 & 4 \\ 2 & 4 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 9 & 17 & 12 & 0 & 12 & 49 & 13 \\ 3 & 4 & 3 & -4 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{113} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1565, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 5 & 0 & 4 & 27 & 4 \\ 3 & 5 & 4 & 0 & 3 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 15 & 27 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & -4 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{1212} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 10 & 3 \\ 3 & 4 & 13 & 0 & 4 & 18 & 4 \\ 7 & 13 & 28 & 0 & 9 & 42 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 12 & 2 \\ 10 & 18 & 42 & 0 & 12 & 58 & 14 \\ 3 & 4 & 10 & -4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{1113} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 15 & 3 \\ 3 & 4 & 4 & 0 & 4 & 27 & 4 \\ 2 & 4 & 1 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 15 & 27 & 18 & 0 & 18 & 133 & 21 \\ 3 & 4 & 3 & -4 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{1112} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1760, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 10 & 3 \\ 3 & 4 & 9 & 0 & 4 & 18 & 4 \\ 5 & 9 & 13 & 0 & 6 & 30 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 12 & 2 \\ 10 & 18 & 30 & 0 & 12 & 58 & 14 \\ 3 & 4 & 7 & -4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{1112} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1781, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 13 & 3 \\ 3 & 4 & 9 & 0 & 4 & 23 & 4 \\ 5 & 9 & 13 & 0 & 6 & 39 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 15 & 2 \\ 13 & 23 & 39 & 0 & 15 & 100 & 18 \\ 3 & 4 & 7 & -4 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{1212} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1784, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 7 & 3 \\ 3 & 4 & 13 & 0 & 4 & 13 & 4 \\ 7 & 13 & 28 & 0 & 9 & 30 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 9 & 2 \\ 7 & 13 & 30 & 0 & 9 & 28 & 10 \\ 3 & 4 & 10 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{1113} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1820, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 4 & 0 & 4 & 22 & 4 \\ 2 & 4 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 12 & 22 & 15 & 0 & 15 & 85 & 17 \\ 3 & 4 & 3 & -4 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{113} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1865, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 5 & 0 & 4 & 32 & 4 \\ 3 & 5 & 4 & 0 & 3 & 33 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 18 & 32 & 33 & 0 & 21 & 193 & 25 \\ 3 & 4 & 4 & -4 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{1114} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1889, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.86 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 11 & 3 \\ 3 & 4 & 4 & 0 & 4 & 21 & 4 \\ 2 & 4 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 11 & 21 & 15 & 0 & 15 & 76 & 16 \\ 3 & 4 & 3 & -4 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{114} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1928, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 5 & 0 & 4 & 31 & 4 \\ 3 & 5 & 4 & 0 & 3 & 30 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 17 & 31 & 30 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & -4 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{1113} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2501, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 12 & 3 \\ 3 & 4 & 9 & 0 & 4 & 22 & 4 \\ 5 & 9 & 13 & 0 & 6 & 36 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 15 & 2 \\ 12 & 22 & 36 & 0 & 15 & 85 & 17 \\ 3 & 4 & 7 & -4 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{1212} 31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2624, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 10 & 3 \\ 3 & 4 & 18 & 0 & 4 & 18 & 4 \\ 10 & 18 & 58 & 0 & 12 & 60 & 14 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 12 & 0 & 0 & 12 & 2 \\ 10 & 18 & 60 & 0 & 12 & 58 & 14 \\ 3 & 4 & 14 & -4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{114} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2636, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 5 & 0 & 4 & 36 & 4 \\ 3 & 5 & 4 & 0 & 3 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 24 & 2 \\ 20 & 36 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & -4 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{1114} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2720, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 4 & 4 & 0 & 4 & 26 & 4 \\ 2 & 4 & 1 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 14 & 26 & 18 & 0 & 18 & 118 & 20 \\ 3 & 4 & 3 & -4 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{1114} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2780, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 20 & 3 \\ 3 & 4 & 4 & 0 & 4 & 36 & 4 \\ 2 & 4 & 1 & 0 & 3 & 24 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 24 & 2 \\ 20 & 36 & 24 & 0 & 24 & 238 & 28 \\ 3 & 4 & 3 & -4 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{1114} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2996, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 14 & 3 \\ 3 & 4 & 9 & 0 & 4 & 26 & 4 \\ 5 & 9 & 13 & 0 & 6 & 42 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 18 & 2 \\ 14 & 26 & 42 & 0 & 18 & 118 & 20 \\ 3 & 4 & 7 & -4 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{1114} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3017, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 17 & 3 \\ 3 & 4 & 4 & 0 & 4 & 31 & 4 \\ 2 & 4 & 1 & 0 & 3 & 21 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 17 & 31 & 21 & 0 & 21 & 172 & 24 \\ 3 & 4 & 3 & -4 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{1113}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -3200$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 15 & 3 \\ 3 & 4 & 9 & 0 & 4 & 27 & 4 \\ 5 & 9 & 13 & 0 & 6 & 45 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 18 & 2 \\ 15 & 27 & 45 & 0 & 18 & 133 & 21 \\ 3 & 4 & 7 & -4 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{1113}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -3365$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 18 & 3 \\ 3 & 4 & 9 & 0 & 4 & 32 & 4 \\ 5 & 9 & 13 & 0 & 6 & 54 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 21 & 2 \\ 18 & 32 & 54 & 0 & 21 & 193 & 25 \\ 3 & 4 & 7 & -4 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

[78] $1^{1114}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -4373$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 17 & 3 \\ 3 & 4 & 9 & 0 & 4 & 31 & 4 \\ 5 & 9 & 13 & 0 & 6 & 51 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 21 & 2 \\ 17 & 31 & 51 & 0 & 21 & 172 & 24 \\ 3 & 4 & 7 & -4 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{1114}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -5216$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 20 & 3 \\ 3 & 4 & 9 & 0 & 4 & 36 & 4 \\ 5 & 9 & 13 & 0 & 6 & 60 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 24 & 2 \\ 20 & 36 & 60 & 0 & 24 & 238 & 28 \\ 3 & 4 & 7 & -4 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [80] $1^{1111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -84$, $\det = 82$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 & 3 \\ 3 & 4 & 1 & 9 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 9 & 0 & 14 & 7 & 9 & 8 \\ 2 & 3 & 0 & 7 & -2 & 3 & 0 \\ 3 & 6 & 0 & 9 & 3 & 4 & 4 \\ 3 & 4 & -1 & 8 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{1111}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 111$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.62 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 6 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 9 & 3 \\ 1 & 3 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 6 & 2 \\ 3 & 6 & 9 & 6 & 0 & 6 & 13 & 7 \\ 2 & 3 & 3 & 3 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

[82] $1^{11}31^{121}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 138$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.35 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 12 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 3 & 2 \\ 4 & 12 & 6 & 22 & 0 & 6 & 12 & 8 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 6 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 12 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & 8 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[83] $1^{1112}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 273$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.38 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 9 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 12 & 3 \\ 1 & 3 & 3 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 9 & 2 \\ 4 & 9 & 12 & 9 & 0 & 9 & 28 & 10 \\ 2 & 3 & 3 & 3 & -4 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{111}31^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 300, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{111}31^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 363, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 6 & 2 \\ 3 & 9 & 6 & 9 & 0 & 6 & 13 & 7 \\ 2 & 4 & 2 & 4 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{111}31^{112}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 1065, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 9 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 9 & 2 \\ 5 & 15 & 9 & 15 & 0 & 9 & 37 & 11 \\ 2 & 4 & 2 & 4 & -4 & 2 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{111}31^{113}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 1659, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 21 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 12 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 12 & 2 \\ 7 & 21 & 12 & 21 & 0 & 12 & 73 & 15 \\ 2 & 4 & 2 & 4 & -4 & 2 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{111}31^{114}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 2145, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 15 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 27 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 15 & 2 \\ 9 & 27 & 15 & 27 & 0 & 15 & 121 & 19 \\ 2 & 4 & 2 & 4 & -4 & 2 & 19 & -1 \end{pmatrix}_{b_0}$$

4.792 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{1111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 148, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 9 & 4 \\ 5 & 9 & 9 & 0 & 6 & 9 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111}31^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 220, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 6 & 2 \\ 8 & 14 & 9 & 0 & 9 & 37 & 24 & 11 \\ 5 & 9 & 6 & 0 & 6 & 24 & 13 & 7 \\ 3 & 4 & 3 & -4 & 2 & 11 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1112} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 274, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 18 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 12 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 12 & 3 \\ 10 & 18 & 18 & 0 & 12 & 12 & 58 & 14 \\ 3 & 4 & 4 & -4 & 2 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1113} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 454, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 27 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 18 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 18 & 3 \\ 15 & 27 & 27 & 0 & 18 & 18 & 133 & 21 \\ 3 & 4 & 4 & -4 & 2 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 6 & 6 & 6 \\ 2 & 4 & 3 & 0 & 6 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 & 1 & 3 \\ 3 & 4 & 3 & -4 & 6 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1113} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 22 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 15 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 15 & 3 \\ 12 & 22 & 21 & 0 & 15 & 15 & 85 & 17 \\ 3 & 4 & 4 & -4 & 2 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1114} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 706, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 36 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 24 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 24 & 3 \\ 20 & 36 & 36 & 0 & 24 & 24 & 238 & 28 \\ 3 & 4 & 4 & -4 & 2 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 712, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & -4 & 2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{c}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 724, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 2 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1112} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 796, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.84 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 13 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 9 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 21 & 7 \\ 7 & 13 & 12 & 0 & 9 & 21 & 28 & 10 \\ 3 & 4 & 4 & -4 & 2 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 988, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1032, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 6 & 15 & 6 \\ 2 & 4 & 3 & 0 & 6 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 15 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 6 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1122} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1144, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.87 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 13 & 13 & 4 \\ 3 & 5 & 4 & 0 & 3 & 12 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 9 & 2 \\ 7 & 13 & 12 & 0 & 9 & 28 & 30 & 10 \\ 7 & 13 & 12 & 0 & 9 & 30 & 28 & 10 \\ 3 & 4 & 4 & -4 & 2 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1114} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1180, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 31 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 30 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 21 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 21 & 3 \\ 17 & 31 & 30 & 0 & 21 & 21 & 172 & 24 \\ 3 & 4 & 4 & -4 & 2 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1276, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 6 & 2 \\ 5 & 9 & 6 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 6 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & 3 & -4 & 2 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 2368, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 9 & 0 & 4 & 9 & 9 & 4 \\ 5 & 9 & 13 & 0 & 6 & 15 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 6 & 6 & 2 \\ 5 & 9 & 15 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 15 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & 7 & -4 & 2 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

4.793 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 31^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 5 & 2 & 3 \\ 4 & 9 & 12 & 0 & 4 & 12 & 4 & 5 \\ 5 & 12 & 13 & 0 & 6 & 15 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 6 & 3 & 2 \\ 5 & 12 & 15 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 3 & 6 & 0 & 2 \\ 3 & 5 & 7 & -4 & 2 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 231, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & 4 & -4 & 2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 507, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 3 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 2 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 651, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 9 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 9 & 3 & 3 \\ 5 & 9 & 6 & 0 & 9 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1141}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 813, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 21 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 15 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 18 & 3 & 3 \\ 11 & 21 & 15 & 0 & 18 & 76 & 15 & 16 \\ 2 & 4 & 3 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 16 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1121}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 1173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 13 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 12 & 3 & 3 \\ 7 & 13 & 9 & 0 & 12 & 28 & 9 & 10 \\ 2 & 4 & 3 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1131}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 1227, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 17 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 12 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 15 & 3 & 3 \\ 9 & 17 & 12 & 0 & 15 & 49 & 12 & 13 \\ 2 & 4 & 3 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

4.794 NHC(-2)(-3) + su₂ + e₆ + su₃ + su₂→(-3)

$$T = 10 \quad [0] \quad 1^{111}3^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 2 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 & 3 \\ 1 & 3 & 3 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 3 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 & 2 \\ 1 & 3 & 3 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 & 3 \\ 1 & 3 & 2 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 & -4 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 130, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 1 & 9 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 9 & 0 & 14 & 7 & 5 & 8 \\ 2 & 3 & 0 & 7 & -2 & 2 & 0 \\ 2 & 3 & 0 & 5 & 2 & 0 & 2 \\ 3 & 4 & -1 & 8 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}3^{11}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.795 NHC(-2)(-3) + su₂ + e₆ + su₃ + su₂n3mix

$$T = 10 \quad [0] \quad 1^{111}3^{11}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 494, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 5 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 5 & 2 \\ 3 & 9 & 5 & 9 & 0 & 5 & 12 & 6 \\ 2 & 4 & 2 & 4 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

4.796 NHC(-2)(-3) + su₂ + e₆ + su₃ + su₃

$$T = 9 \quad [0] \quad 1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 & 3 \\ 3 & 4 & 9 & 0 & 8 & 5 & 4 \\ 5 & 9 & 13 & 0 & 12 & 9 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 0 & 9 & 6 & 5 \\ 3 & 5 & 9 & 0 & 6 & 3 & 3 \\ 3 & 4 & 7 & -4 & 5 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{e}_6 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 2 & 3 \\ 4 & 9 & 12 & 0 & 4 & 4 & 5 \\ 5 & 12 & 13 & 0 & 6 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 3 & 2 \\ 2 & 4 & 6 & 0 & 3 & 0 & 2 \\ 3 & 5 & 7 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{114} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -51, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 2 & 2 & 3 \\ 4 & 9 & 36 & 0 & 4 & 4 & 5 \\ 14 & 36 & 121 & 0 & 15 & 15 & 19 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 15 & 0 & 0 & 3 & 2 \\ 2 & 4 & 15 & 0 & 3 & 0 & 2 \\ 3 & 5 & 19 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -150, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & 4 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -159, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.87 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{113} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -231, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.08 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 11 & 1 & 2 & 2 & 3 \\ 4 & 9 & 28 & 0 & 4 & 4 & 5 \\ 11 & 28 & 73 & 0 & 12 & 12 & 15 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 12 & 0 & 0 & 3 & 2 \\ 2 & 4 & 12 & 0 & 3 & 0 & 2 \\ 3 & 5 & 15 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -231, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.08 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 2 & 2 & 3 \\ 4 & 9 & 20 & 0 & 4 & 4 & 5 \\ 8 & 20 & 37 & 0 & 9 & 9 & 11 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 3 & 2 \\ 2 & 4 & 9 & 0 & 3 & 0 & 2 \\ 3 & 5 & 11 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -231, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -258, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.18 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 8 & 14 & 9 & 0 & 9 & 36 & 10 \\ 3 & 4 & 3 & -4 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{122} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -267, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.7$$

$$T_{\min} = 13.07 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 4 & 3 \\ 3 & 4 & 13 & 0 & 5 & 8 & 4 \\ 7 & 13 & 28 & 0 & 12 & 18 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 12 & 0 & 3 & 6 & 3 \\ 4 & 8 & 18 & 0 & 6 & 9 & 5 \\ 3 & 4 & 10 & -4 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{132} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -294, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 4 & 3 \\ 3 & 4 & 17 & 0 & 5 & 8 & 4 \\ 9 & 17 & 49 & 0 & 15 & 24 & 13 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 15 & 0 & 3 & 6 & 3 \\ 4 & 8 & 24 & 0 & 6 & 9 & 5 \\ 3 & 4 & 13 & -4 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{111} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -402, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.79 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 & 4 \\ 8 & 14 & 37 & 0 & 9 & 9 & 11 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 9 & 0 & 0 & 3 & 2 \\ 2 & 4 & 9 & 0 & 3 & 0 & 2 \\ 3 & 4 & 11 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{113} 31^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -420, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.09 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 2 & 2 & 3 \\ 4 & 9 & 32 & 0 & 4 & 4 & 5 \\ 13 & 32 & 100 & 0 & 15 & 15 & 18 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 15 & 0 & 0 & 3 & 2 \\ 2 & 4 & 15 & 0 & 3 & 0 & 2 \\ 3 & 5 & 18 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -435, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -510, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{111} 31^{311} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -555, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.10 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 2 & 2 & 3 \\ 4 & 9 & 44 & 0 & 4 & 4 & 5 \\ 18 & 44 & 193 & 0 & 21 & 21 & 25 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 21 & 0 & 0 & 3 & 2 \\ 2 & 4 & 21 & 0 & 3 & 0 & 2 \\ 3 & 5 & 25 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -645, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 2 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 & 4 \\ 5 & 9 & 13 & 0 & 9 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 9 & 0 & 3 & 3 & 3 \\ 2 & 4 & 6 & 0 & 3 & 0 & 2 \\ 3 & 4 & 7 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -651, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.05 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 6 & 12 & 6 \\ 3 & 4 & 3 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{114} 31^{211} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -690, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.08 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 2 & 2 & 3 \\ 4 & 9 & 40 & 0 & 4 & 4 & 5 \\ 16 & 40 & 154 & 0 & 18 & 18 & 22 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 18 & 0 & 0 & 3 & 2 \\ 2 & 4 & 18 & 0 & 3 & 0 & 2 \\ 3 & 5 & 22 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{141} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -726, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 & 2 & 3 \\ 3 & 4 & 21 & 0 & 5 & 4 & 4 \\ 11 & 21 & 76 & 0 & 18 & 15 & 16 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 18 & 0 & 3 & 3 & 3 \\ 2 & 4 & 15 & 0 & 3 & 0 & 2 \\ 3 & 4 & 16 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -789, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 & 4 \\ 5 & 9 & 13 & 0 & 6 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 3 & 2 \\ 2 & 4 & 6 & 0 & 3 & 0 & 2 \\ 3 & 4 & 7 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{121} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -906, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 2 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 & 4 \\ 7 & 13 & 28 & 0 & 12 & 9 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 12 & 0 & 3 & 3 & 3 \\ 2 & 4 & 9 & 0 & 3 & 0 & 2 \\ 3 & 4 & 10 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{131} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -933, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 2 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 & 4 \\ 9 & 17 & 49 & 0 & 15 & 12 & 13 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 15 & 0 & 3 & 3 & 3 \\ 2 & 4 & 12 & 0 & 3 & 0 & 2 \\ 3 & 4 & 13 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -1200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 4 & 9 & 4 \\ 5 & 9 & 13 & 0 & 6 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 5 & 9 & 15 & 0 & 6 & 12 & 6 \\ 3 & 4 & 7 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \quad 1^{11} 31^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -100, \quad \det = 477, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.37 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & 2 & 4 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{11} 31^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -100, \quad \det = 486, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.44 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 9 & 0 & 3 & 3 & 3 \\ 3 & 6 & 9 & 13 & 0 & 6 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 6 & 0 & 0 & 3 & 2 \\ 1 & 3 & 3 & 6 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 7 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{\overline{111}} 31^{\overline{111}} 31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 774, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.40 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 3 & 2 \\ 3 & 9 & 6 & 13 & 0 & 6 & 9 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 6 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 9 & 0 & 3 & 3 & 3 \\ 2 & 4 & 2 & 7 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

4.797 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{111}} 31^{\overline{111}} 31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 48, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 2 & 5 & 3 \\ 4 & 9 & 12 & 0 & 4 & 4 & 12 & 5 \\ 5 & 12 & 13 & 0 & 6 & 6 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 6 & 0 & 3 & 0 & 6 & 2 \\ 5 & 12 & 15 & 0 & 6 & 6 & 13 & 7 \\ 3 & 5 & 7 & -4 & 2 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 31^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 228, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & -4 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 31^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 231, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 3 & 2 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 & 6 \\ 2 & 4 & 3 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & 4 & -4 & 2 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 31^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 507, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 3 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 31^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 651, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 9 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 3 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 31^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 813, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 21 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 18 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 15 & 2 \\ 11 & 21 & 15 & 0 & 18 & 15 & 76 & 16 \\ 3 & 4 & 3 & -4 & 3 & 2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}\textcolor{red}{2}} 3 1^{\textcolor{blue}{1}\textcolor{blue}{1}\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{red}{e}_6 + \textcolor{red}{s}\textcolor{blue}{u}_3 + \textcolor{blue}{s}\textcolor{blue}{u}_3 + \textcolor{red}{s}\textcolor{blue}{u}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 1173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 13 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 12 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 9 & 2 \\ 7 & 13 & 9 & 0 & 12 & 9 & 28 & 10 \\ 3 & 4 & 3 & -4 & 3 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}\textcolor{red}{3}} 3 1^{\textcolor{blue}{1}\textcolor{blue}{1}\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{red}{e}_6 + \textcolor{red}{s}\textcolor{blue}{u}_3 + \textcolor{blue}{s}\textcolor{blue}{u}_3 + \textcolor{red}{s}\textcolor{blue}{u}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 1227, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 17 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 15 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 12 & 2 \\ 9 & 17 & 12 & 0 & 15 & 12 & 49 & 13 \\ 3 & 4 & 3 & -4 & 3 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

4.798 NHC(-2)(-3) + $\textcolor{red}{s}\textcolor{blue}{u}_2 + \textcolor{red}{e}_6 + \textcolor{red}{s}\textcolor{blue}{u}_3 + \textcolor{blue}{s}\textcolor{blue}{u}_3 + \textcolor{red}{s}\textcolor{blue}{u}_3$

$$T = 10 \quad [0] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}\textcolor{red}{1}} 3 1^{\textcolor{blue}{1}\textcolor{blue}{1}\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{red}{e}_6 + \textcolor{red}{s}\textcolor{blue}{u}_3 + \textcolor{blue}{s}\textcolor{blue}{u}_3 + \textcolor{red}{s}\textcolor{blue}{u}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 459, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}\textcolor{red}{1}} 3 1^{\textcolor{blue}{1}\textcolor{blue}{1}\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{red}{e}_6 + \textcolor{red}{s}\textcolor{blue}{u}_3 + \textcolor{blue}{s}\textcolor{blue}{u}_3 + \textcolor{red}{s}\textcolor{blue}{u}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 1296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 & 4 & 4 \\ 5 & 9 & 13 & 0 & 9 & 6 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 9 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 6 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 6 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 7 & -4 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.799 NHC(-2)(-3) + $\textcolor{red}{s}\textcolor{blue}{u}_2 + \textcolor{red}{e}_6 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \ 1^{\textcolor{red}{1}\textcolor{blue}{3}} 3 1^{\textcolor{blue}{1}\textcolor{blue}{1}\textcolor{green}{2}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{red}{e}_6 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 98, \Delta = -143, \det = 925, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 1 & 4 \\ 8 & 9 & 8 & 4 & 0 & 5 \\ 6 & 8 & 4 & 3 & 0 & 4 \\ 3 & 4 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

4.800 NHC(-2)(-3) + $\textcolor{red}{s}\textcolor{blue}{u}_2 + \textcolor{red}{f}_4$

$$T = 7 \quad [0] \ 2 1^{\textcolor{red}{1}} 3 1^{\textcolor{blue}{1}\textcolor{blue}{2}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 9 & 16 & 4 & 5 \\ 6 & 16 & 22 & 6 & 8 \\ 3 & 4 & 6 & -2 & 0 \\ 3 & 5 & 8 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\textcolor{red}{1}\textcolor{blue}{2}} 3 1^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$$

$$T_{\min} = 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 9 & 16 & 4 & 5 \\ 6 & 16 & 22 & 6 & 8 \\ 3 & 4 & 6 & -2 & 0 \\ 3 & 5 & 8 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}} 3 1^{\textcolor{red}{2}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 14.82 \rightarrow 15, T_{\max} = 97, T \in [15, 97]$$

$$\begin{pmatrix} 1 & 4 & 7 & 1 & 3 \\ 4 & 9 & 16 & 0 & 5 \\ 7 & 16 & 28 & 0 & 10 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 10 & -3 & -2 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 6 & 10 & 22 & 0 & 8 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 8 & -3 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 97, T \in [14, 97]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 3 & -3 & -1 \end{pmatrix}_{b_0}$$

[5] $21^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 4 & 3 & 4 \\ 3 & 4 & 3 & -2 & 0 \\ 3 & 5 & 4 & 0 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 4 & 3 & 4 \\ 3 & 4 & 3 & -2 & 0 \\ 3 & 5 & 4 & 0 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 4 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{112}31^3 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 97, T \in [14, 97]$

$$\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 12 & 28 & 85 & 0 & 17 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 17 & -3 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 6 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{14}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 8 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 12 & -3 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{15}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 10 & 20 & 73 & 0 & 15 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 15 & -3 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{11}3^{11}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 4 & 3 & 4 \\ 2 & 4 & 3 & -2 & 0 \\ 3 & 5 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}3^{11}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 9 & 12 & 0 & 5 \\ 5 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 7 & -3 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{16}3^{11}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.4$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 12 & 24 & 106 & 0 & 18 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 18 & -3 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{17}3^{11}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 13.22 \rightarrow 14, T_{\max} = 97, T \in [14, 97]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 14 & 28 & 145 & 0 & 21 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 21 & -3 & -1 \end{pmatrix}_{b_0}$$

[17] $1^{11}3^{11}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 5 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{11}3^{112}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 3 \\ 4 & 9 & 16 & 4 & 5 \\ 6 & 16 & 22 & 6 & 8 \\ 2 & 4 & 6 & -2 & 0 \\ 3 & 5 & 8 & 0 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{113}3^{14}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 17 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 17 & 40 & 172 & 0 & 24 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 24 & -3 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{11}3^{112}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 11 & -3 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{112}3^{12}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 3 \\ 4 & 9 & 24 & 0 & 5 \\ 10 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 14 & -3 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{12}3^{11}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 7 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 10 & -3 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{11}3^{113}1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 11 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 15 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{112} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -137, \quad \det = -158, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 9.7$$

$$T_{\min} = 12.10 \rightarrow 13, \quad T_{\max} = 97, \quad T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 3 \\ 4 & 9 & 20 & 0 & 5 \\ 8 & 20 & 37 & 0 & 11 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 11 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{11} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -137, \quad \det = -163, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.09 \rightarrow 13, \quad T_{\max} = 97, \quad T \in [13, 97]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 19 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^1 13^1 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -137, \quad \det = -187, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.55 \rightarrow 13, \quad T_{\max} = 97, \quad T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 9 & 24 & 4 & 5 \\ 9 & 24 & 52 & 9 & 12 \\ 2 & 4 & 9 & -2 & 0 \\ 3 & 5 & 12 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{13} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -137, \quad \det = -207, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.49 \rightarrow 12, \quad T_{\max} = 97, \quad T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 9 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{12} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -137, \quad \det = -238, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 9.7$$

$$T_{\min} = 11.50 \rightarrow 12, \quad T_{\max} = 97, \quad T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 10 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 14 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{113} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -137, \quad \det = -241, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.13 \rightarrow 13, \quad T_{\max} = 97, \quad T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 15 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 21 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{14} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -137, \quad \det = -268, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.55 \rightarrow 12, \quad T_{\max} = 97, \quad T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 11 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 16 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{113} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -137, \quad \det = -273, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.17 \rightarrow 13, \quad T_{\max} = 97, \quad T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 11 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 11 & 28 & 73 & 0 & 15 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 15 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 21^{111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -137, \quad \det = -276, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.54 \rightarrow 12, \quad T_{\max} = 97, \quad T \in [12, 97]$$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 4 \\ 8 & 9 & 8 & 4 & 5 \\ 6 & 8 & 4 & 3 & 4 \\ 3 & 4 & 3 & -2 & 0 \\ 4 & 5 & 4 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{113} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -137, \quad \det = -291, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.08 \rightarrow 13, \quad T_{\max} = 97, \quad T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 3 \\ 4 & 9 & 32 & 0 & 5 \\ 13 & 32 & 100 & 0 & 18 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 18 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[35] \quad 13^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, \quad T_{\max} = 97, \quad \Delta = -137, \quad \det = -292, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.62 \rightarrow 13, \quad T_{\max} = 97, \quad T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 3 \\ 4 & 9 & 32 & 4 & 5 \\ 12 & 32 & 94 & 12 & 16 \\ 2 & 4 & 12 & -2 & 0 \\ 3 & 5 & 16 & 0 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{12}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -316, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 13 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 18 & -3 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{13}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -351, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 12 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 17 & -3 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{12}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -382, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 16 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 22 & -3 & 1 \end{pmatrix}_{b_0}$$

[39] $1^{114}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -388, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 3 \\ 4 & 9 & 48 & 0 & 5 \\ 20 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 28 & -3 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{114}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -418, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 14 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 19 & -3 & 0 \end{pmatrix}_{b_0}$$

[41] $1^113^11^{15} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -427, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 12.65 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 9 & 40 & 4 & 5 \\ 15 & 40 & 148 & 15 & 20 \\ 2 & 4 & 15 & -2 & 0 \\ 3 & 5 & 20 & 0 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{14}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -466, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 14 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 20 & -3 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{114}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -466, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 9 & 44 & 0 & 5 \\ 18 & 44 & 193 & 0 & 25 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 25 & -3 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{114}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -476, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 16 & 40 & 154 & 0 & 22 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 22 & -3 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{13}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -483, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 15 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 21 & -3 & 1 \end{pmatrix}_{b_0}$$

[46] $21^{112}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -486, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 4 & 8 & 12 & 3 & 4 \\ 8 & 9 & 16 & 4 & 5 \\ 12 & 16 & 22 & 6 & 8 \\ 3 & 4 & 6 & -2 & 0 \\ 4 & 5 & 8 & 0 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{13}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -603, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 18 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 25 & -3 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{14}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -652, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 31 & 0 & 4 \\ 17 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 24 & -3 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{14}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -826, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 20 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 28 & -3 & 1 \end{pmatrix}_{b_0}$$

[50] $21^{113}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -836, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 4 & 8 & 18 & 3 & 4 \\ 8 & 9 & 24 & 4 & 5 \\ 18 & 24 & 52 & 9 & 12 \\ 3 & 4 & 9 & -2 & 0 \\ 4 & 5 & 12 & 0 & 1 \end{pmatrix}_{b_0}$$

[51] $21^{114}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -1326, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 4 & 8 & 24 & 3 & 4 \\ 8 & 9 & 32 & 4 & 5 \\ 24 & 32 & 94 & 12 & 16 \\ 3 & 4 & 12 & -2 & 0 \\ 4 & 5 & 16 & 0 & 1 \end{pmatrix}_{b_0}$$

[52] $21^{115}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -1956, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$

$$\begin{pmatrix} 4 & 8 & 30 & 3 & 4 \\ 8 & 9 & 40 & 4 & 5 \\ 30 & 40 & 148 & 15 & 20 \\ 3 & 4 & 15 & -2 & 0 \\ 4 & 5 & 20 & 0 & 1 \end{pmatrix}_{b_0}$$

$T = 8$ [53] $1^13^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 15.33 \rightarrow 16, T_{\max} = 20, T \in [16, 20]$

$$\begin{pmatrix} 5 & 13 & 9 & 5 \\ 13 & 30 & 20 & 10 \\ 9 & 20 & 13 & 7 \\ 5 & 10 & 7 & -1 \end{pmatrix}_{b_0}$$

[54] $1^{14}3^11^{13}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 18.57 \rightarrow 19, T_{\max} = 28, T \in [19, 28]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 1 & 2 \\ 1 & 1 & 4 & 15 & 0 & 3 \\ 2 & 4 & 3 & 18 & 0 & 3 \\ 6 & 15 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 3 & 16 & -3 & -6 \end{pmatrix}_{b_0}$$

[55] $1^{11}2^23^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 10 & 19 & 12 & 6 \\ 19 & 30 & 20 & 10 \\ 12 & 20 & 13 & 7 \\ 6 & 10 & 7 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{11}3^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 18 & 12 & 6 \\ 18 & 30 & 20 & 10 \\ 12 & 20 & 13 & 7 \\ 6 & 10 & 7 & 3 \end{pmatrix}_{b_0}$$

[57] $1^13^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 5 & 13 & 1 & 5 \\ 13 & 30 & 0 & 10 \\ 1 & 0 & -2 & 0 \\ 5 & 10 & 0 & 0 \end{pmatrix}_{b_0}$$

- [58] $1^1 1^1 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 1 & 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 3 & -3 & 1 \end{pmatrix}_{b_0}$
- [59] $1^1 1^1 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 1 & 0 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & -2 & 0 \\ 2 & 2 & -1 & 3 & 0 & 1 \end{pmatrix}_{b_0}$
- [60] $1^{12} 3 2 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 13.33 \rightarrow 14$, $T_{\max} = 20$, $T \in [14, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 6 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 8 & 10 \\ 1 & 1 & 0 & 8 & -2 & 0 \\ 2 & 2 & -1 & 10 & 0 & -1 \end{pmatrix}_{b_0}$
- [61] $1^{11} 3 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 15$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 3 & 5 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & -3 & -3 \\ 3 & 5 & 4 & -3 & 1 & 1 \end{pmatrix}_{b_0}$
- [62] $1^1 1^1 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 20$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & 2 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & -3 & -3 \\ 3 & 5 & 4 & -3 & 2 & 2 \end{pmatrix}_{b_0}$
- [63] $1^{11} 1^{11} 3 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 2 & -3 & 1 \end{pmatrix}_{b_0}$
- [64] $1^{11} 1^{11} 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 4 & 5 \\ 1 & 1 & 0 & 4 & -2 & 0 \\ 2 & 2 & -1 & 5 & 0 & 1 \end{pmatrix}_{b_0}$
- [65] $1^{11} 1^{11} 3 2 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 4 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 4 & 0 & 16 & 7 & 8 \\ 1 & 1 & 0 & 7 & -2 & 0 \\ 2 & 2 & -1 & 8 & 0 & 0 \end{pmatrix}_{b_0}$
- [66] $1^{12} 3 2 1^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.7$
 $T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 20$, $T \in [14, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 1 & 7 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 45 & 11 & 13 \\ 1 & 1 & 0 & 11 & -2 & 0 \\ 2 & 2 & -1 & 13 & 0 & -1 \end{pmatrix}_{b_0}$
- [67] $1^{11} 3 1^1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 3 & 3 & -3 & 0 \end{pmatrix}_{b_0}$
- [68] $1^{11} 3 1^1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 35$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.2$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 1 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{11}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 5 & 10 & 0 & 5 \\ 5 & 10 & 14 & 0 & 8 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 8 & -3 & 2 \end{pmatrix}_{b_0}$$

[70] $1^{11}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 3 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 5 & -3 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{13}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 6 & 0 & 3 \\ 2 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 3 & 4 & -3 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{13}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 2 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{13}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 1 & 2 \\ 1 & 1 & 4 & 12 & 0 & 3 \\ 2 & 4 & 3 & 15 & 0 & 3 \\ 5 & 12 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 3 & 13 & -3 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{12}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 4 & 12 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 10 & -3 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{112}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 4 & 9 & 0 & 3 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 4 & 9 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & 10 & -3 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.15 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 \\ 2 & 4 & 3 & 9 & 0 & 3 \\ 3 & 6 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 3 & 7 & -3 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{12}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 167, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 1 & 2 \\ 1 & 1 & 4 & 9 & 0 & 3 \\ 2 & 4 & 3 & 12 & 0 & 3 \\ 4 & 9 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 3 & 10 & -3 & 0 \end{pmatrix}_{b_0}$$

[78] $1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 3 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{13}1^{12}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12, T_{\max} = 28, T \in [12, 28]$

- [80] $1^{14}31^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 4 & 12 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$
- [81] $1^{13}31^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 8 & 24 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 20 & -3 & 1 \end{pmatrix}_{b_0}$$
- [82] $1^{11}31^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 371, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 6 & 18 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$
- [83] $1^{13}31^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 395, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 7 & 21 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 17 & -3 & 1 \end{pmatrix}_{b_0}$$
- [84] $1^{12}31^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 6 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 14 & -3 & 1 \end{pmatrix}_{b_0}$$
- [85] $1^{114}31^{23}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.7$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 1 & 4 & 18 & 0 & 3 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 8 & 18 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & 20 & -3 & 0 \end{pmatrix}_{b_0}$$
- [86] $1^{13}31^{14}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 8 & 24 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 16 & -3 & 0 \end{pmatrix}_{b_0}$$
- [87] $31^{111}231^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 31, T \in [12, 31]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 3 & 1 & 3 \\ 3 & 4 & 0 & 3 & 1 & 2 \\ 3 & 3 & 3 & 1 & 1 & 3 \\ 1 & 1 & 1 & 1 & -3 & -1 \\ 2 & 3 & 2 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$
- [88] $1^{11}31^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 623, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 7 & 21 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 15 & -3 & 1 \end{pmatrix}_{b_0}$$
- [89] $1^{13}31^{15}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 788, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 30 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 10 & 30 & 15 & 148 & 0 & 20 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 20 & -3 & 0 \end{pmatrix}_{b_0}$$
- [90] $1^{12}31^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 8 & 24 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 18 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[91] \ 1^{14}31^{13}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 836, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.87 \rightarrow 12, \ T_{\max} = 28, \ T \in [12, 28]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 30 & 0 & 4 \\ 1 & 4 & 0 & 21 & 0 & 2 \\ 10 & 30 & 21 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 24 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[92] \ 1^{13}31^{13}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 871, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 28, \ T \in [12, 28]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 9 & 27 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 21 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[93] \ 1^{11}31^{14}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 931, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 28, \ T \in [12, 28]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 9 & 27 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 19 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[94] \ 1^{13}31^{16}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 1096, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.4$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 28, \ T \in [13, 28]$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 1 & 2 \\ 2 & 4 & 4 & 36 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 12 & 36 & 18 & 214 & 0 & 24 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 24 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[95] \ 1^{12}31^{14}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 1220, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 28, \ T \in [12, 28]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 30 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 10 & 30 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 22 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[96] \ 1^{13}31^{14}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 1403, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.89 \rightarrow 12, \ T_{\max} = 28, \ T \in [12, 28]$$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 1 & 2 \\ 2 & 4 & 4 & 33 & 0 & 4 \\ 1 & 4 & 0 & 21 & 0 & 2 \\ 11 & 33 & 21 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 25 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[97] \ 1^{13}31^{17}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 1460, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 28, \ T \in [13, 28]$$

$$\begin{pmatrix} 0 & 2 & 1 & 14 & 1 & 2 \\ 2 & 4 & 4 & 42 & 0 & 4 \\ 1 & 4 & 0 & 21 & 0 & 2 \\ 14 & 42 & 21 & 292 & 0 & 28 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 28 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[98] \ 1^{14}31^{14}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 1480, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 28, \ T \in [12, 28]$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 1 & 2 \\ 2 & 4 & 4 & 36 & 0 & 4 \\ 1 & 4 & 0 & 24 & 0 & 2 \\ 12 & 36 & 24 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 28 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [99] \ 1^{11}2321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -8, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 1 & 2 \\ 2 & 0 & 4 & 6 & 4 & 6 \\ 1 & 0 & 1 & 4 & -2 & 0 \\ 2 & 0 & 2 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[100] \ 1^{11}4^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 9, \ T_{\max} = 21, \ \Delta = -107, \ \det = -11, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.45 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[101] \ 1^{11}2321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -18, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 4 & 6 \\ 1 & 1 & 0 & 4 & -2 & 0 \\ 2 & 3 & -1 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [102] \quad 1^1 \overset{1}{5} 1^1 231^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 10, T_{\max} = 22, \Delta = -103, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & 10 & 4 & 6 \\ 1 & 1 & 0 & 4 & -2 & 0 \\ 2 & 2 & -1 & 6 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[103] \quad 1^1 \overset{1}{5} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.26 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 0 & 7 \\ 1 & 0 & -2 & 0 \\ 5 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

4.801 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{e}_6$

$$T = 8 \quad [0] \quad 1^1 \overset{1}{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 4 & 0 & -4 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 \overset{1}{12} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.7$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 1 & 3 \\ 4 & 9 & 16 & 4 & 0 & 5 \\ 6 & 16 & 22 & 6 & 0 & 8 \\ 2 & 4 & 6 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 8 & 0 & -4 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 \overset{1}{13} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 739, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 1 & 3 \\ 4 & 9 & 24 & 4 & 0 & 5 \\ 9 & 24 & 52 & 9 & 0 & 12 \\ 2 & 4 & 9 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 12 & 0 & -4 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \quad 1^1 \overset{1}{11} 321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{e}_6 \quad T = 9, T_{\max} = 101, \Delta = -158, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 3 & 0 & -2 & 0 \\ 3 & 5 & 4 & -3 & 0 & -2 \end{pmatrix}_{b_0}$$

4.802 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{f}_4$

$$T = 8 \quad [0] \quad 1^1 \overset{1}{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 100, \Delta = -160, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 4 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 \overset{1}{12} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 100, \Delta = -160, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.9$$

$$T_{\min} = 13.36 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 1 & 3 \\ 4 & 9 & 16 & 4 & 0 & 5 \\ 6 & 16 & 22 & 6 & 0 & 8 \\ 2 & 4 & 6 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 8 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{113}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 100, \Delta = -160, \det = 552, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 1 & 3 \\ 4 & 9 & 24 & 4 & 0 & 5 \\ 9 & 24 & 52 & 9 & 0 & 12 \\ 2 & 4 & 9 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 12 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{114}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 100, \Delta = -160, \det = 832, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.4$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 1 & 3 \\ 4 & 9 & 32 & 4 & 0 & 5 \\ 12 & 32 & 94 & 12 & 0 & 16 \\ 2 & 4 & 12 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 16 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{115}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 100, \Delta = -160, \det = 1192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.5$
 $T_{\min} = 13.39 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 1 & 3 \\ 4 & 9 & 40 & 4 & 0 & 5 \\ 15 & 40 & 148 & 15 & 0 & 20 \\ 2 & 4 & 15 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 20 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$T = 9$ [5] $1^{11}321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$

$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 93, T \in [16, 93]$

$$\begin{pmatrix} 5 & 13 & 9 & 1 & 5 \\ 13 & 30 & 20 & 0 & 10 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 10 & 7 & -3 & -1 \end{pmatrix}_{b_0}$$

4.803 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_2$

$T = 9$ [0] $1^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$

$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 & 3 \\ 4 & 9 & 8 & 4 & 0 & 8 & 5 \\ 3 & 8 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 6 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & -3 & 4 & -1 \end{pmatrix}_{b_0}$$

4.804 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$

$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 2 & 3 \\ 4 & 9 & 8 & 4 & 0 & 4 & 5 \\ 3 & 8 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 3 & 5 & 4 & 0 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.805 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \text{NHC}(-2)(-3)$

$T = 9$ [0] $1^{11}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 6 & 0 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

- [1] $1^{111}31^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + [\text{su}_2 \cdot \text{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -87, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 & 2 & 3 \\ 3 & 4 & 19 & 0 & 5 & 4 & 4 \\ 11 & 19 & 73 & 0 & 21 & 12 & 15 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 21 & 0 & 4 & 4 & 4 \\ 2 & 4 & 12 & 0 & 4 & 0 & 2 \\ 3 & 4 & 15 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$
- [2] $1^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + [\text{su}_2 \cdot \text{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -87, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 2 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 & 4 \\ 5 & 9 & 13 & 0 & 9 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 9 & 0 & 4 & 4 & 4 \\ 2 & 4 & 6 & 0 & 4 & 0 & 2 \\ 3 & 4 & 7 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$
- [3] $1^{111}31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + [\text{su}_2 \cdot \text{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -87, \det = -396, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 & 2 & 3 \\ 3 & 4 & 14 & 0 & 5 & 4 & 4 \\ 8 & 14 & 37 & 0 & 15 & 9 & 11 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 15 & 0 & 4 & 4 & 4 \\ 2 & 4 & 9 & 0 & 4 & 0 & 2 \\ 3 & 4 & 11 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$
- [4] $1^{121}31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + [\text{su}_2 \cdot \text{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -87, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 & 2 & 3 \\ 3 & 4 & 18 & 0 & 5 & 4 & 4 \\ 10 & 18 & 58 & 0 & 18 & 12 & 14 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 18 & 0 & 4 & 4 & 4 \\ 2 & 4 & 12 & 0 & 4 & 0 & 2 \\ 3 & 4 & 14 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$
- [5] $1^{121}31^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + [\text{su}_2 \cdot \text{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -87, \det = -936, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 & 2 & 3 \\ 3 & 4 & 28 & 0 & 5 & 4 & 4 \\ 16 & 28 & 154 & 0 & 30 & 18 & 22 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 30 & 0 & 4 & 4 & 4 \\ 2 & 4 & 18 & 0 & 4 & 0 & 2 \\ 3 & 4 & 22 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$
- [6] $1^{121}31^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + [\text{su}_2 \cdot \text{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -87, \det = -956, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 & 2 & 3 \\ 3 & 4 & 23 & 0 & 5 & 4 & 4 \\ 13 & 23 & 100 & 0 & 24 & 15 & 18 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 24 & 0 & 4 & 4 & 4 \\ 2 & 4 & 15 & 0 & 4 & 0 & 2 \\ 3 & 4 & 18 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{131}31^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + [\text{su}_2 \cdot \text{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -87, \det = -1476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 & 2 & 3 \\ 3 & 4 & 27 & 0 & 5 & 4 & 4 \\ 15 & 27 & 133 & 0 & 27 & 18 & 21 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 27 & 0 & 4 & 4 & 4 \\ 2 & 4 & 18 & 0 & 4 & 0 & 2 \\ 3 & 4 & 21 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$
- [8] $1^{131}31^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + [\text{su}_2 \cdot \text{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -87, \det = -1796, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 & 2 & 3 \\ 3 & 4 & 32 & 0 & 5 & 4 & 4 \\ 18 & 32 & 193 & 0 & 33 & 21 & 25 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 33 & 0 & 4 & 4 & 4 \\ 2 & 4 & 21 & 0 & 4 & 0 & 2 \\ 3 & 4 & 25 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{141}31^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + [\text{su}_2 \cdot \text{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -87, \det = -2456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

13.2

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 & 2 & 3 \\ 3 & 4 & 36 & 0 & 5 & 4 & 4 \\ 20 & 36 & 238 & 0 & 36 & 24 & 28 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 36 & 0 & 4 & 4 & 4 \\ 2 & 4 & 24 & 0 & 4 & 0 & 2 \\ 3 & 4 & 28 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

4.806 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \quad 1^{11} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 3 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 6 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 600, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 & 3 & 4 \\ 5 & 9 & 13 & 0 & 9 & 6 & 5 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 9 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 6 & 0 & 4 & 0 & 2 & 2 \\ 2 & 3 & 5 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & 7 & -3 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.807 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8$

$$T = 8 \quad [0] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 44 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 44 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 34 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{113} 3^1 1^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 59 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 59 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 45 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.68 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 1 & 3 \\ 4 & 9 & 16 & 4 & 0 & 5 \\ 6 & 16 & 22 & 6 & 0 & 8 \\ 2 & 4 & 6 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 8 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 291, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 35 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 35 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 27 & -3 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.61 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 17 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{112}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 34 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 26 & -3 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 365, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.71 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 1 & 3 \\ 4 & 9 & 24 & 4 & 0 & 5 \\ 9 & 24 & 52 & 9 & 0 & 12 \\ 2 & 4 & 9 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 12 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 26 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 20 & -3 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 26 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 20 & -3 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 540, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 1 & 3 \\ 4 & 9 & 32 & 4 & 0 & 5 \\ 12 & 32 & 94 & 12 & 0 & 16 \\ 2 & 4 & 12 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 16 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{113}31^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 581, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 51 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 39 & -3 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 709, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 35 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 27 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{115}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 765, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.3$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 1 & 3 \\ 4 & 9 & 40 & 4 & 0 & 5 \\ 15 & 40 & 148 & 15 & 0 & 20 \\ 2 & 4 & 15 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 20 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{113}31^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 781, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 43 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 43 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 33 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{114}31^{41} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 924, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 68 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 52 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{114}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 1044, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 44 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 44 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 34 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{114}31^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 1236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 60 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 46 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{114}31^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 1276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 52 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 40 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [23] \ 1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 5 & 13 & 9 & 1 & 5 \\ 13 & 30 & 20 & 0 & 10 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 10 & 7 & -2 & -3 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 15.11 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 \\ 4 & 5 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 5 & 3 & -3 & -2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & 2 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 0 & 2 \\ 3 & 5 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{111}31^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 1 & 4 & 5 & 4 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 12 & 0 & 6 \\ 4 & 12 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{11}31^{11}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 26, \ \Delta = -115, \ \det = -115, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 26, \ T \in [13, 26]$$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 15 & 0 & 6 \\ 5 & 12 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{11}321^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -190, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.61 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 5 & 10 & 0 & 5 & 5 \\ 5 & 10 & 14 & 0 & 7 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 7 & 0 & 0 & 2 \\ 3 & 5 & 8 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{11}31^{11}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 26, \ \Delta = -115, \ \det = -435, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 26, \ T \in [13, 26]$$

$$\begin{pmatrix} 1 & 4 & 5 & 9 & 1 & 3 \\ 4 & 10 & 13 & 24 & 0 & 6 \\ 5 & 13 & 12 & 27 & 0 & 6 \\ 9 & 24 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [30] \ 1^{111}2321^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 10, \ T_{\max} = 13, \ \Delta = -81, \ \det = 34, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 22 & 5 \\ 12 & 26 & 1 & 48 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 48 & 0 & 86 & 18 \\ 5 & 10 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

4.808 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{112}31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -106, \ \det = -16, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.7$$

$$T_{\min} = 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 26 & 5 \\ 12 & 25 & 16 & 0 & 56 & 9 \\ 8 & 16 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 56 & 36 & 0 & 118 & 20 \\ 5 & 9 & 6 & -3 & 20 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -106, \ \det = -90, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.07 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 7 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -106, \ \det = -104, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1111}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -106, \ \det = -288, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 4 & 3 \\ 4 & 9 & 8 & 4 & 0 & 8 & 5 \\ 3 & 8 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 6 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -324$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 12.67 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 35 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 35 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -380$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 & 3 \\ 4 & 9 & 8 & 4 & 0 & 8 & 5 \\ 3 & 8 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 6 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -404$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 12.65 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{111}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -476$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 17 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -482$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 26 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

4.809 NHC(-2)(-3) + $\mathbf{su}_2 + \mathbf{f}_4 + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{111}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -218$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 7 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -222$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 12.63 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 2 & 3 \\ 4 & 9 & 8 & 4 & 0 & 4 & 5 \\ 3 & 8 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 3 & 5 & 4 & 0 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -302$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 6 & 5 \\ 12 & 25 & 36 & 0 & 12 & 9 \\ 17 & 36 & 49 & 0 & 17 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 17 & 0 & 4 & 4 \\ 5 & 9 & 13 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

4.810 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 + \text{su2n3mix}$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 15 & 5 \\ 11 & 20 & 17 & 0 & 28 & 8 \\ 9 & 17 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 28 & 24 & 0 & 37 & 11 \\ 5 & 8 & 7 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

4.811 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 0 & 4 \\ 2 & 4 & 3 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & 0 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 8 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

4.812 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^{\overline{111}} 3^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 & 3 \\ 3 & 4 & 10 & 0 & 5 & 4 \\ 6 & 10 & 22 & 0 & 12 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 12 & 0 & 4 & 4 \\ 3 & 4 & 8 & -3 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 4 & 7 & 1 & 5 & 3 \\ 4 & 9 & 16 & 0 & 12 & 5 \\ 7 & 16 & 28 & 0 & 21 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 21 & 0 & 13 & 7 \\ 3 & 5 & 10 & -3 & 7 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 14.52 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 3 & -3 & 4 & -2 \end{pmatrix}_{b_0}$$

- [4] $1^{1111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 & 3 \\ 4 & 9 & 8 & 4 & 8 & 5 \\ 3 & 8 & 4 & 3 & 6 & 4 \\ 3 & 4 & 3 & -2 & 3 & 0 \\ 3 & 8 & 6 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [3] $21^{133}31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 & 3 \\ 4 & 9 & 8 & 4 & 8 & 5 \\ 3 & 8 & 4 & 3 & 6 & 4 \\ 3 & 4 & 3 & -2 & 3 & 0 \\ 3 & 8 & 6 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [5] $1^{133}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 36$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 15 & 3 \\ 3 & 4 & 17 & 0 & 27 & 4 \\ 9 & 17 & 49 & 0 & 81 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 81 & 0 & 133 & 21 \\ 3 & 4 & 13 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$
- [6] $1^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 39$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 43$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 20$, $T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 6 & 0 & 13 & 7 \\ 3 & 5 & 3 & -3 & 7 & -1 \end{pmatrix}_{b_0}$$
- [8] $1^{121}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 50$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 12 & 0 & 13 & 7 \\ 3 & 4 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{124}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 52$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 17 & 3 \\ 3 & 4 & 8 & 0 & 31 & 4 \\ 4 & 8 & 10 & 0 & 42 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 42 & 0 & 172 & 24 \\ 3 & 4 & 6 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{11}31^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 60$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 6 & 0 & 4 & 4 \\ 3 & 4 & 4 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{112}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 64$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 14$, $T_{\max} = 20$, $T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 24 & 12 & 0 & 58 & 14 \\ 3 & 5 & 3 & -3 & 14 & -1 \end{pmatrix}_{b_0}$$
- [12] $1^{11}31^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 66$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 12.70 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 8 & 3 \\ 3 & 4 & 10 & 0 & 14 & 4 \\ 6 & 10 & 22 & 0 & 30 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 30 & 0 & 37 & 11 \\ 3 & 4 & 8 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$
- [13] $1^{13}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 87$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 11.52 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 49 & 13 \\ 3 & 4 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{112}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 3 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{113}3^11^3 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.40 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 36 & 18 & 0 & 133 & 21 \\ 3 & 5 & 3 & -3 & 21 & -2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{122}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{113}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{112}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 13.19 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 20 & 5 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 9 & 0 & 37 & 11 \\ 3 & 5 & 3 & -3 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.21 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 14 & 3 \\ 3 & 4 & 9 & 0 & 24 & 4 \\ 5 & 9 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 24 & 42 & 0 & 121 & 19 \\ 3 & 4 & 7 & -3 & 19 & -1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{123}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 6 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 18 & 0 & 25 & 9 \\ 3 & 4 & 6 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 3 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{133}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 9 & 3 \\ 3 & 4 & 12 & 0 & 17 & 4 \\ 6 & 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 36 & 0 & 49 & 13 \\ 3 & 4 & 9 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{114}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 3 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{134}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 6 & 3 \\ 3 & 4 & 12 & 0 & 12 & 4 \\ 6 & 12 & 25 & 0 & 27 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 27 & 0 & 25 & 9 \\ 3 & 4 & 9 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{124}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 4 & 8 & 0 & 16 & 4 \\ 4 & 8 & 10 & 0 & 24 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 24 & 0 & 46 & 12 \\ 3 & 4 & 6 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{134}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 134, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 8 & 3 \\ 3 & 4 & 12 & 0 & 16 & 4 \\ 6 & 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 36 & 0 & 46 & 12 \\ 3 & 4 & 9 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{113}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{115}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 20 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 20 & 15 & 0 & 73 & 15 \\ 3 & 4 & 3 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{1111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 4 & 3 \\ 4 & 9 & 8 & 4 & 8 & 5 \\ 3 & 8 & 4 & 3 & 6 & 4 \\ 2 & 4 & 3 & -2 & 3 & 0 \\ 4 & 8 & 6 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{125}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 10 & 3 \\ 3 & 4 & 8 & 0 & 20 & 4 \\ 4 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 20 & 30 & 0 & 73 & 15 \\ 3 & 4 & 6 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{12}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 4 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{114}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 14.65 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 9 & 4 & 0 & 48 & 5 \\ 2 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 48 & 24 & 0 & 238 & 28 \\ 3 & 5 & 3 & -3 & 28 & -2 \end{pmatrix}_{b_0}$$

[34] $1^{144}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 8 & 3 \\ 3 & 4 & 16 & 0 & 16 & 4 \\ 8 & 16 & 46 & 0 & 48 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 48 & 0 & 46 & 12 \\ 3 & 4 & 12 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{123}3^11^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 3 \\ 3 & 4 & 8 & 0 & 22 & 4 \\ 4 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 30 & 0 & 85 & 17 \\ 3 & 4 & 6 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{112}3^11^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 7 & 3 \\ 3 & 4 & 14 & 0 & 13 & 4 \\ 8 & 14 & 37 & 0 & 33 & 11 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 33 & 0 & 28 & 10 \\ 3 & 4 & 11 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{135}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.1$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 10 & 3 \\ 3 & 4 & 12 & 0 & 20 & 4 \\ 6 & 12 & 25 & 0 & 45 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 20 & 45 & 0 & 73 & 15 \\ 3 & 4 & 9 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{116}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 158, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 24 & 18 & 0 & 106 & 18 \\ 3 & 4 & 3 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{113}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.66 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 28 & 5 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 28 & 12 & 0 & 73 & 15 \\ 3 & 5 & 3 & -3 & 15 & -1 \end{pmatrix}_{b_0}$$

[43] $1^113^11^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.32 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 3 \\ 4 & 9 & 8 & 4 & 8 & 5 \\ 3 & 8 & 4 & 3 & 6 & 4 \\ 2 & 4 & 3 & -2 & 3 & 0 \\ 3 & 8 & 6 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{122}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 18 & 0 & 28 & 10 \\ 3 & 4 & 6 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{126}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.9$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 3 \\ 3 & 4 & 8 & 0 & 24 & 4 \\ 4 & 8 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 24 & 36 & 0 & 106 & 18 \\ 3 & 4 & 6 & -3 & 18 & -1 \end{pmatrix}_{b_0}$$

[44] $1^{145}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 13.05 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 10 & 3 \\ 3 & 4 & 16 & 0 & 20 & 4 \\ 8 & 16 & 46 & 0 & 60 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 20 & 60 & 0 & 73 & 15 \\ 3 & 4 & 12 & -3 & 15 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{136}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 12 & 3 \\ 3 & 4 & 12 & 0 & 24 & 4 \\ 6 & 12 & 25 & 0 & 54 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 24 & 54 & 0 & 106 & 18 \\ 3 & 4 & 9 & -3 & 18 & -1 \end{pmatrix}_{b_0}$$

[47] $1^{155}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.24 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 10 & 3 \\ 3 & 4 & 20 & 0 & 20 & 4 \\ 10 & 20 & 73 & 0 & 75 & 15 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 20 & 75 & 0 & 73 & 15 \\ 3 & 4 & 15 & -3 & 15 & -1 \end{pmatrix}_{b_0}$$

[46] $1^{117}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.24 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 28 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 28 & 21 & 0 & 145 & 21 \\ 3 & 4 & 3 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[49] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 5 & 3 \\ 4 & 9 & 12 & 0 & 12 & 5 \\ 5 & 12 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 15 & 0 & 13 & 7 \\ 3 & 5 & 7 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[48] $1^{146}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 13.28 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 12 & 3 \\ 3 & 4 & 16 & 0 & 24 & 4 \\ 8 & 16 & 46 & 0 & 72 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 24 & 72 & 0 & 106 & 18 \\ 3 & 4 & 12 & -3 & 18 & -1 \end{pmatrix}_{b_0}$$

[50] $1^{127}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 190, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.29 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 14 & 3 \\ 3 & 4 & 8 & 0 & 28 & 4 \\ 4 & 8 & 10 & 0 & 42 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 28 & 42 & 0 & 145 & 21 \\ 3 & 4 & 6 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[51] $1^{113}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.12 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 32 & 15 & 0 & 100 & 18 \\ 3 & 5 & 3 & -3 & 18 & -1 \end{pmatrix}_{b_0}$$

[52] $1^{137}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 14 & 3 \\ 3 & 4 & 12 & 0 & 28 & 4 \\ 6 & 12 & 25 & 0 & 63 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 28 & 63 & 0 & 145 & 21 \\ 3 & 4 & 9 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[54] $1^{112}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 28 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 28 & 18 & 0 & 154 & 22 \\ 3 & 4 & 3 & -3 & 22 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{156}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 12 & 3 \\ 3 & 4 & 20 & 0 & 24 & 4 \\ 10 & 20 & 73 & 0 & 90 & 15 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 24 & 90 & 0 & 106 & 18 \\ 3 & 4 & 15 & -3 & 18 & -1 \end{pmatrix}_{b_0}$$

[55] $1^{11}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 5 & 0 & 14 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 15 & 0 & 37 & 11 \\ 3 & 4 & 4 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{114}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 14.35 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 36 & 15 & 0 & 121 & 19 \\ 3 & 5 & 3 & -3 & 19 & -2 \end{pmatrix}_{b_0}$$

[57] $1^{147}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 14 & 3 \\ 3 & 4 & 16 & 0 & 28 & 4 \\ 8 & 16 & 46 & 0 & 84 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 28 & 84 & 0 & 145 & 21 \\ 3 & 4 & 12 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[58] $1^{166}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 13.58 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 12 & 3 \\ 3 & 4 & 24 & 0 & 24 & 4 \\ 12 & 24 & 106 & 0 & 108 & 18 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 24 & 108 & 0 & 106 & 18 \\ 3 & 4 & 18 & -3 & 18 & -1 \end{pmatrix}_{b_0}$$

[59] $1^{157}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 13.60 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 14 & 3 \\ 3 & 4 & 20 & 0 & 28 & 4 \\ 10 & 20 & 73 & 0 & 105 & 15 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 28 & 105 & 0 & 145 & 21 \\ 3 & 4 & 15 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[60] $1^{134}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 239, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 11 & 3 \\ 3 & 4 & 12 & 0 & 21 & 4 \\ 6 & 12 & 25 & 0 & 45 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 45 & 0 & 76 & 16 \\ 3 & 4 & 9 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[61] $1^{132}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 246, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 10 & 3 \\ 3 & 4 & 17 & 0 & 18 & 4 \\ 9 & 17 & 49 & 0 & 54 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 54 & 0 & 58 & 14 \\ 3 & 4 & 13 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{167}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.72 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 14 & 3 \\ 3 & 4 & 24 & 0 & 28 & 4 \\ 12 & 24 & 106 & 0 & 126 & 18 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 28 & 126 & 0 & 145 & 21 \\ 3 & 4 & 18 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[63] $1^{11}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 255, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 19 & 4 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 19 & 21 & 0 & 73 & 15 \\ 3 & 4 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[64] $1^{111} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 19 & 4 \\ 5 & 9 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 19 & 33 & 0 & 73 & 15 \\ 3 & 4 & 7 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 263, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{111} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 279, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 5 & 0 & 24 & 4 \\ 3 & 5 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 24 & 27 & 0 & 121 & 19 \\ 3 & 4 & 4 & -3 & 19 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{143} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 279, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 12 & 3 \\ 3 & 4 & 21 & 0 & 22 & 4 \\ 11 & 21 & 76 & 0 & 81 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 81 & 0 & 85 & 17 \\ 3 & 4 & 16 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[68] $1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{177} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 13.84 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 14 & 3 \\ 3 & 4 & 28 & 0 & 28 & 4 \\ 14 & 28 & 145 & 0 & 147 & 21 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 28 & 147 & 0 & 145 & 21 \\ 3 & 4 & 21 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[70] $1^{123} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 282, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 17 & 4 \\ 4 & 8 & 10 & 0 & 24 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 24 & 0 & 49 & 13 \\ 3 & 4 & 6 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{122} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 284, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 13 & 3 \\ 3 & 4 & 13 & 0 & 23 & 4 \\ 7 & 13 & 28 & 0 & 54 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 54 & 0 & 100 & 18 \\ 3 & 4 & 10 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{114} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 3 \\ 4 & 9 & 4 & 0 & 44 & 5 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 44 & 21 & 0 & 193 & 25 \\ 3 & 5 & 3 & -3 & 25 & -1 \end{pmatrix}_{b_0}$$

[73] $1^{113} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 12 & 3 \\ 3 & 4 & 14 & 0 & 22 & 4 \\ 8 & 14 & 37 & 0 & 57 & 11 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 57 & 0 & 85 & 17 \\ 3 & 4 & 11 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{114}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 317, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 3 \\ 3 & 4 & 9 & 0 & 14 & 4 \\ 5 & 9 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 24 & 0 & 37 & 11 \\ 3 & 4 & 7 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[76] $1^{114}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 21 & 4 \\ 5 & 9 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 33 & 0 & 76 & 16 \\ 3 & 4 & 7 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{112}3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 3 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{114}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 324, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 13.21 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 40 & 18 & 0 & 154 & 22 \\ 3 & 5 & 3 & -3 & 22 & -1 \end{pmatrix}_{b_0}$$

[78] $1^{12}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 324, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 5 & 0 & 18 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{112}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 338, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 3 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[80] $1^{112}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 359, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 7 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{14}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 5 & 0 & 26 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 24 & 0 & 118 & 20 \\ 3 & 4 & 4 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{113}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 27 & 0 & 49 & 13 \\ 3 & 4 & 7 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[83] $1^{13}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 375, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 5 & 0 & 22 & 4 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 21 & 0 & 85 & 17 \\ 3 & 4 & 4 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[84] $1^{124}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 380$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 14 & 3 \\ 3 & 4 & 8 & 0 & 26 & 4 \\ 4 & 8 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 36 & 0 & 118 & 20 \\ 3 & 4 & 6 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 381$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.51 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{124}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 404$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 11 & 3 \\ 3 & 4 & 8 & 0 & 21 & 4 \\ 4 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 30 & 0 & 76 & 16 \\ 3 & 4 & 6 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{111}31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 424$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 8 & 3 \\ 3 & 4 & 14 & 0 & 14 & 4 \\ 8 & 14 & 37 & 0 & 39 & 11 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 39 & 0 & 37 & 11 \\ 3 & 4 & 11 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{123}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 480$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 5 & 0 & 23 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 24 & 0 & 100 & 18 \\ 3 & 4 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{114}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 503$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{112}31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 506$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 10 & 3 \\ 3 & 4 & 14 & 0 & 18 & 4 \\ 8 & 14 & 37 & 0 & 48 & 11 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 48 & 0 & 58 & 14 \\ 3 & 4 & 11 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{122}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 508$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 30 & 0 & 28 & 10 \\ 3 & 4 & 10 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[92] $1^{123}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 543$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 11.64 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 18 & 3 \\ 3 & 4 & 13 & 0 & 32 & 4 \\ 7 & 13 & 28 & 0 & 75 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 75 & 0 & 193 & 25 \\ 3 & 4 & 10 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{122}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 548$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 10 & 3 \\ 3 & 4 & 13 & 0 & 18 & 4 \\ 7 & 13 & 28 & 0 & 42 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 42 & 0 & 58 & 14 \\ 3 & 4 & 10 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{113}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 564$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 3 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{112}3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 574$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.51 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 10 & 3 \\ 3 & 4 & 9 & 0 & 18 & 4 \\ 5 & 9 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 30 & 0 & 58 & 14 \\ 3 & 4 & 7 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[96] $1^{123}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 591$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 11.51 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 9 & 3 \\ 3 & 4 & 13 & 0 & 17 & 4 \\ 7 & 13 & 28 & 0 & 39 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 39 & 0 & 49 & 13 \\ 3 & 4 & 10 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{124}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 608$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.55 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 11 & 3 \\ 3 & 4 & 13 & 0 & 21 & 4 \\ 7 & 13 & 28 & 0 & 48 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 48 & 0 & 76 & 16 \\ 3 & 4 & 10 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[98] $1^{123}3^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 612$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 5 & 0 & 28 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 28 & 30 & 0 & 154 & 22 \\ 3 & 4 & 4 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[99] $1^{13}3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 639$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 5 & 0 & 27 & 4 \\ 3 & 5 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[100] $1^{113}3^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 648$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.64 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 21 & 0 & 193 & 25 \\ 3 & 4 & 3 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[101] $1^{113}3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 653$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 3 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[102] $21^{111}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 692$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.54 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 6 & 4 \\ 8 & 9 & 8 & 4 & 8 & 5 \\ 6 & 8 & 4 & 3 & 6 & 4 \\ 3 & 4 & 3 & -2 & 3 & 0 \\ 6 & 8 & 6 & 3 & 4 & 4 \\ 4 & 5 & 4 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[103] $1^{112}3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 695$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 13 & 3 \\ 3 & 4 & 9 & 0 & 23 & 4 \\ 5 & 9 & 13 & 0 & 39 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 39 & 0 & 100 & 18 \\ 3 & 4 & 7 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[104] $1^{133}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 717$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 11.51 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 12 & 3 \\ 3 & 4 & 17 & 0 & 22 & 4 \\ 9 & 17 & 49 & 0 & 66 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 66 & 0 & 85 & 17 \\ 3 & 4 & 13 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[105] $1^{112}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 722$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 16 & 3 \\ 3 & 4 & 9 & 0 & 28 & 4 \\ 5 & 9 & 13 & 0 & 48 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 28 & 48 & 0 & 154 & 22 \\ 3 & 4 & 7 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[106] $1^{143}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 732$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.53 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 5 & 0 & 31 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 30 & 0 & 172 & 24 \\ 3 & 4 & 4 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[107] $1^{133}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 744$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.8$
 $T_{\min} = 11.53 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 9 & 3 \\ 3 & 4 & 17 & 0 & 17 & 4 \\ 9 & 17 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 51 & 0 & 49 & 13 \\ 3 & 4 & 13 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[108] $1^{134}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 759$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 17 & 3 \\ 3 & 4 & 17 & 0 & 31 & 4 \\ 9 & 17 & 49 & 0 & 93 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 93 & 0 & 172 & 24 \\ 3 & 4 & 13 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[109] $1^{124}31^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 764$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 14 & 3 \\ 3 & 4 & 18 & 0 & 26 & 4 \\ 10 & 18 & 58 & 0 & 84 & 14 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 84 & 0 & 118 & 20 \\ 3 & 4 & 14 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[110] $1^{113}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 765$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 12 & 3 \\ 3 & 4 & 9 & 0 & 22 & 4 \\ 5 & 9 & 13 & 0 & 36 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 36 & 0 & 85 & 17 \\ 3 & 4 & 7 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[111] $1^{114}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 794$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.51 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{144}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 824$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.55 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 14 & 3 \\ 3 & 4 & 21 & 0 & 26 & 4 \\ 11 & 21 & 76 & 0 & 96 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 96 & 0 & 118 & 20 \\ 3 & 4 & 16 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[113] $1^{134}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 831$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 11 & 3 \\ 3 & 4 & 17 & 0 & 21 & 4 \\ 9 & 17 & 49 & 0 & 63 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 63 & 0 & 76 & 16 \\ 3 & 4 & 13 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[114] $1^{123}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 863, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 15 & 3 \\ 3 & 4 & 13 & 0 & 27 & 4 \\ 7 & 13 & 28 & 0 & 63 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 63 & 0 & 133 & 21 \\ 3 & 4 & 10 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[115] $1^{122}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 868, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 10 & 3 \\ 3 & 4 & 18 & 0 & 18 & 4 \\ 10 & 18 & 58 & 0 & 60 & 14 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 60 & 0 & 58 & 14 \\ 3 & 4 & 14 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[116] $1^{123}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 879, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 12 & 3 \\ 3 & 4 & 13 & 0 & 22 & 4 \\ 7 & 13 & 28 & 0 & 51 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 51 & 0 & 85 & 17 \\ 3 & 4 & 10 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[117] $1^{13}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 879, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 5 & 0 & 32 & 4 \\ 3 & 5 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 33 & 0 & 193 & 25 \\ 3 & 4 & 4 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[118] $1^{114}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 890, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 14 & 3 \\ 3 & 4 & 9 & 0 & 26 & 4 \\ 5 & 9 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 42 & 0 & 118 & 20 \\ 3 & 4 & 7 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[119] $1^{123}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 954, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 12 & 3 \\ 3 & 4 & 18 & 0 & 22 & 4 \\ 10 & 18 & 58 & 0 & 72 & 14 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 72 & 0 & 85 & 17 \\ 3 & 4 & 14 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[120] $1^{144}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 988, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 11 & 3 \\ 3 & 4 & 21 & 0 & 21 & 4 \\ 11 & 21 & 76 & 0 & 78 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 78 & 0 & 76 & 16 \\ 3 & 4 & 16 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[121] $1^{114}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 991, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[122] $1^{113}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 1064, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 15 & 3 \\ 3 & 4 & 9 & 0 & 27 & 4 \\ 5 & 9 & 13 & 0 & 45 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 45 & 0 & 133 & 21 \\ 3 & 4 & 7 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[123] $1^{14}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 1080, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 20 & 3 \\ 3 & 4 & 9 & 0 & 36 & 4 \\ 5 & 9 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 60 & 0 & 238 & 28 \\ 3 & 4 & 7 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{14} 3^1 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 1780$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.53 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 14 & 3 \\ 3 & 4 & 26 & 0 & 26 & 4 \\ 14 & 26 & 118 & 0 & 120 & 20 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 120 & 0 & 118 & 20 \\ 3 & 4 & 20 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [135] $1^{111} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -14$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 1 & 0 & 1 & 3 & 0 & 3 \\ 1 & 1 & 0 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & 3 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[136] $1^{13} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -20$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 2 & 5 & 2 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 2 & 4 \\ 3 & 5 & 4 & -3 & 4 & 2 \end{pmatrix}_{b_0}$$

[137] $1^{111} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 4 & 3 & 5 \\ 1 & 1 & 0 & 4 & -2 & 1 & 0 \\ 1 & 2 & 0 & 3 & 1 & 0 & 2 \\ 2 & 2 & -1 & 5 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[138] $1^{13} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 3 & 0 & 9 & 2 \\ 1 & 1 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 9 & 9 & 0 & 40 & 12 \\ 2 & 2 & 2 & 3 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{111} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 2 \\ 1 & 2 & 0 & 3 & 0 & 4 & 2 \\ 1 & 1 & 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & 3 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{111} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 1 & 2 & 0 & 3 & 0 & 7 & 2 \\ 1 & 1 & 3 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 4 & 7 & 7 & 0 & 16 & 8 \\ 2 & 2 & 2 & 3 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[141] $1^{111} 3^1 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 2 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[142] $1^{112} 3^1 11^1 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -45$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 3 & 0 & 12 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 \\ 1 & 3 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 9 & 9 & 0 & 28 & 10 \\ 2 & 4 & 2 & 3 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[143] \ 1^{14}31^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -48, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 6 & 0 & 30 & 4 \\ 1 & 4 & 0 & 3 & 0 & 21 & 2 \\ 2 & 6 & 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 21 & 30 & 0 & 172 & 24 \\ 2 & 4 & 2 & 4 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[144] \ 1^{11}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.73 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 5 & 5 & 0 & 10 & 5 \\ 2 & 5 & 2 & 0 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 10 & 7 & 0 & 14 & 8 \\ 3 & 5 & 4 & -3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[145] \ 1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -74, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.51 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 1 & 2 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 3 & 0 & 5 & 5 \\ 2 & 2 & 2 & 2 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[146] \ 1^{111}321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -80, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.65 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 5 & 10 & 0 & 20 & 5 \\ 5 & 10 & 14 & 0 & 30 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 30 & 0 & 58 & 16 \\ 3 & 5 & 8 & -3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[147] \ 1^{12}321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -104, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.54 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 2 \\ 1 & 2 & 0 & 3 & 0 & 8 & 2 \\ 1 & 1 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 8 & 0 & 26 & 10 \\ 2 & 2 & 2 & 3 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[149] \ 1^{111}321^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -120, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 5 & 5 & 0 & 25 & 5 \\ 2 & 5 & 2 & 0 & 19 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 25 & 19 & 0 & 92 & 20 \\ 3 & 5 & 4 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[148] \ 1^{11}321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -120, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 5 & 5 & 0 & 15 & 5 \\ 2 & 5 & 2 & 0 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 15 & 11 & 0 & 32 & 12 \\ 3 & 5 & 4 & -3 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[150] \ 1^{112}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -134, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.55 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 1 & 2 & 1 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 6 & 0 & 26 & 10 \\ 2 & 2 & 2 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[151] \ 1^{111}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -139, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 1 & 3 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & 3 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[152] \ 1^{11}321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -140, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 5 & 5 & 0 & 20 & 5 \\ 2 & 5 & 2 & 0 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 15 & 0 & 58 & 16 \\ 3 & 5 & 4 & -3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[154] \ 1^{14} 3 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 11 & 2 \\ 1 & 2 & 0 & 3 & 0 & 13 & 2 \\ 1 & 1 & 3 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 13 & 13 & 0 & 81 & 17 \\ 2 & 2 & 2 & 3 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[153] \ 1^{12} 3 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 7 & 2 \\ 1 & 2 & 0 & 3 & 0 & 11 & 2 \\ 1 & 1 & 3 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 7 & 11 & 11 & 0 & 45 & 13 \\ 2 & 2 & 2 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[155] \ 1^{111} 3 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 0 & 4 & 2 \\ 1 & 2 & 0 & 4 & 0 & 7 & 2 \\ 2 & 3 & 4 & 5 & 0 & 11 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 4 & 7 & 11 & 0 & 16 & 8 \\ 2 & 2 & 2 & 5 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[156] \ 1^{113} 3 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 0 & 7 & 2 \\ 1 & 2 & 0 & 4 & 0 & 6 & 2 \\ 2 & 3 & 4 & 5 & 0 & 13 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 13 & 0 & 25 & 9 \\ 2 & 2 & 2 & 5 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[157] \ 1^{112} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 1 & 2 & 1 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 5 & 0 & 13 & 7 \\ 2 & 2 & 2 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[158] \ 1^{111} 3 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 5 & 10 & 0 & 10 & 5 \\ 5 & 10 & 14 & 0 & 16 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 10 & 16 & 0 & 14 & 8 \\ 3 & 5 & 8 & -3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[159] \ 1^{13} 3 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 9 & 2 \\ 1 & 2 & 0 & 3 & 0 & 12 & 2 \\ 1 & 1 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 12 & 12 & 0 & 61 & 15 \\ 2 & 2 & 2 & 3 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[160] \ 1^{111} 3 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 0 & 3 & 2 \\ 1 & 2 & 0 & 4 & 0 & 4 & 2 \\ 2 & 3 & 4 & 5 & 0 & 7 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 7 & 0 & 5 & 5 \\ 2 & 2 & 2 & 5 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[161] \ 1^{111} 3 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -210, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

10.2

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 5 & 10 & 0 & 15 & 5 \\ 5 & 10 & 14 & 0 & 23 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 15 & 23 & 0 & 32 & 12 \\ 3 & 5 & 8 & -3 & 12 & 2 \end{pmatrix}_{b_0}$$

[162] $1^{112}3^{11}21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -214, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 0 & 2 & 3 & 0 & 8 & 2 \\ 1 & 2 & 0 & 4 & 0 & 14 & 2 \\ 2 & 3 & 4 & 5 & 0 & 22 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 8 & 14 & 22 & 0 & 70 & 16 \\ 2 & 2 & 2 & 5 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[163] $1^{111}3^{11}31^{12}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 3 & 0 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 \\ 1 & 3 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 9 & 9 & 0 & 37 & 11 \\ 2 & 4 & 2 & 3 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[164] $1^{113}3^{11}321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 0 & 12 & 2 \\ 1 & 2 & 1 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 12 & 9 & 0 & 61 & 15 \\ 2 & 2 & 2 & 2 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[165] $1^{113}3^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 1 & 2 & 1 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 7 & 0 & 25 & 9 \\ 2 & 2 & 2 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[166] $1^{112}3^{11}321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 0 & 5 & 2 \\ 1 & 2 & 0 & 4 & 0 & 5 & 2 \\ 2 & 3 & 4 & 5 & 0 & 10 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 10 & 0 & 13 & 7 \\ 2 & 2 & 2 & 5 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[167] $1^{114}3^{11}31^{12}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -250, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 8 & 2 \\ 2 & 4 & 4 & 9 & 0 & 24 & 4 \\ 1 & 4 & 0 & 6 & 0 & 18 & 2 \\ 3 & 9 & 6 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 18 & 42 & 0 & 118 & 20 \\ 2 & 4 & 2 & 7 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[168] $1^{12}3^{11}321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 5 & 5 & 0 & 25 & 5 \\ 2 & 5 & 2 & 0 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 25 & 18 & 0 & 94 & 20 \\ 3 & 5 & 4 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$

[169] $1^{113}3^{11}31^{12}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 3 & 0 & 21 & 4 \\ 1 & 4 & 0 & 3 & 0 & 15 & 2 \\ 1 & 3 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 15 & 15 & 0 & 85 & 17 \\ 2 & 4 & 2 & 3 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[170] $1^{13}3^{11}321^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 2 \\ 1 & 2 & 0 & 3 & 0 & 15 & 2 \\ 1 & 1 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 10 & 15 & 15 & 0 & 88 & 18 \\ 2 & 2 & 2 & 3 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[171] \ 1^{11}31^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 6 & 0 & 4 & 4 \\ 2 & 4 & 2 & 4 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{11}31^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -301, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 6 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 9 & 0 & 13 & 7 \\ 2 & 4 & 2 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[173] \ 1^{113}3^{11}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 1 & 2 & 1 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 9 & 8 & 0 & 40 & 12 \\ 2 & 2 & 2 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[174] \ 1^{121}31^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -325, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 3 & 2 \\ 2 & 4 & 4 & 12 & 0 & 9 & 4 \\ 1 & 4 & 0 & 9 & 0 & 6 & 2 \\ 4 & 12 & 9 & 28 & 0 & 21 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 21 & 0 & 13 & 7 \\ 2 & 4 & 2 & 10 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[175] \ 1^{122}31^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 4 & 2 \\ 2 & 4 & 4 & 12 & 0 & 12 & 4 \\ 1 & 4 & 0 & 9 & 0 & 9 & 2 \\ 4 & 12 & 9 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 9 & 30 & 0 & 28 & 10 \\ 2 & 4 & 2 & 10 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[176] \ 1^{14}3^{11}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 12 & 2 \\ 1 & 2 & 0 & 3 & 0 & 16 & 2 \\ 1 & 1 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 16 & 16 & 0 & 110 & 20 \\ 2 & 2 & 2 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[177] \ 1^{123}31^{123}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 3 & 0 & 18 & 4 \\ 1 & 4 & 0 & 3 & 0 & 12 & 2 \\ 1 & 3 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 12 & 12 & 0 & 58 & 14 \\ 2 & 4 & 2 & 3 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[178] \ 1^{12}3^{11}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -360, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 5 & 5 & 0 & 30 & 5 \\ 2 & 5 & 2 & 0 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 30 & 22 & 0 & 134 & 24 \\ 3 & 5 & 4 & -3 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[179] \ 1^{114}3^{11}21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -374, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 0 & 16 & 2 \\ 1 & 2 & 1 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 16 & 12 & 0 & 110 & 20 \\ 2 & 2 & 2 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[180] $1^{112}3^{11}21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -404$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 & 0 & 7 & 2 \\ 1 & 2 & 0 & 4 & 0 & 11 & 2 \\ 2 & 3 & 4 & 5 & 0 & 18 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 7 & 11 & 18 & 0 & 45 & 13 \\ 2 & 2 & 2 & 5 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[181] $1^{112}3^{11}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -414$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.55 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 0 & 6 & 2 \\ 1 & 2 & 0 & 4 & 0 & 8 & 2 \\ 2 & 3 & 4 & 5 & 0 & 14 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 14 & 0 & 26 & 10 \\ 2 & 2 & 2 & 5 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[182] $1^{13}3^{11}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -440$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 5 & 5 & 0 & 35 & 5 \\ 2 & 5 & 2 & 0 & 25 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 35 & 25 & 0 & 188 & 28 \\ 3 & 5 & 4 & -3 & 28 & 2 \end{pmatrix}_{b_0}$$

[183] $1^{114}3^{11}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -494$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 1 & 2 & 1 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 10 & 10 & 10 & 0 & 58 & 14 \\ 2 & 2 & 2 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[184] $1^{12}31^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -508$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 6 & 0 & 18 & 4 \\ 1 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 6 & 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 12 & 18 & 0 & 58 & 14 \\ 2 & 4 & 2 & 4 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[186] $1^{111}31^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -524$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 9 & 4 \\ 1 & 4 & 0 & 6 & 0 & 6 & 2 \\ 3 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 15 & 0 & 13 & 7 \\ 2 & 4 & 2 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[185] $1^{114}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -524$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 0 & 13 & 2 \\ 1 & 2 & 1 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 13 & 11 & 0 & 81 & 17 \\ 2 & 2 & 2 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[187] $1^{122}31^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -532$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 6 & 2 \\ 2 & 4 & 4 & 12 & 0 & 18 & 4 \\ 1 & 4 & 0 & 9 & 0 & 12 & 2 \\ 4 & 12 & 9 & 28 & 0 & 42 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 12 & 42 & 0 & 58 & 14 \\ 2 & 4 & 2 & 10 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[188] $1^{112}31^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -549$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 3 & 0 & 24 & 4 \\ 1 & 4 & 0 & 3 & 0 & 15 & 2 \\ 1 & 3 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 15 & 15 & 0 & 100 & 18 \\ 2 & 4 & 2 & 3 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[189] $1^{113}3^{11}21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -554$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 & 0 & 8 & 2 \\ 1 & 2 & 0 & 4 & 0 & 9 & 2 \\ 2 & 3 & 4 & 5 & 0 & 17 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 9 & 17 & 0 & 40 & 12 \\ 2 & 2 & 2 & 5 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[190] $1^{114}3^{11}21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -574$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 0 & 2 & 3 & 0 & 10 & 2 \\ 1 & 2 & 0 & 4 & 0 & 10 & 2 \\ 2 & 3 & 4 & 5 & 0 & 20 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 10 & 10 & 20 & 0 & 58 & 14 \\ 2 & 2 & 2 & 5 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[191] $1^{114}31^{13}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -621$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 3 & 0 & 30 & 4 \\ 1 & 4 & 0 & 3 & 0 & 21 & 2 \\ 1 & 3 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 21 & 21 & 0 & 172 & 24 \\ 2 & 4 & 2 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[192] $1^{113}31^{13}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -691$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 3 & 0 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 18 & 2 \\ 1 & 3 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 18 & 18 & 0 & 133 & 21 \\ 2 & 4 & 2 & 3 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[193] $1^{113}31^{12}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -693$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 6 & 0 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 6 & 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 9 & 15 & 0 & 37 & 11 \\ 2 & 4 & 2 & 4 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[194] $1^{113}3^{11}21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -764$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 0 & 2 & 3 & 0 & 9 & 2 \\ 1 & 2 & 0 & 4 & 0 & 12 & 2 \\ 2 & 3 & 4 & 5 & 0 & 21 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 12 & 21 & 0 & 61 & 15 \\ 2 & 2 & 2 & 5 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[195] $1^{113}3^{11}21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -794$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 7 & 2 \\ 1 & 0 & 2 & 3 & 0 & 10 & 2 \\ 1 & 2 & 0 & 4 & 0 & 15 & 2 \\ 2 & 3 & 4 & 5 & 0 & 25 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 10 & 15 & 25 & 0 & 88 & 18 \\ 2 & 2 & 2 & 5 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[196] $1^{111}31^{112}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -811$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 5 & 2 \\ 2 & 4 & 4 & 9 & 0 & 15 & 4 \\ 1 & 4 & 0 & 6 & 0 & 9 & 2 \\ 3 & 9 & 6 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 9 & 24 & 0 & 37 & 11 \\ 2 & 4 & 2 & 7 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[197] $1^{13}31^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -853, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 6 & 0 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 18 & 2 \\ 2 & 6 & 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 18 & 27 & 0 & 133 & 21 \\ 2 & 4 & 2 & 4 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[198] $1^{113}31^{112}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -859, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 7 & 2 \\ 2 & 4 & 4 & 9 & 0 & 21 & 4 \\ 1 & 4 & 0 & 6 & 0 & 15 & 2 \\ 3 & 9 & 6 & 13 & 0 & 36 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 15 & 36 & 0 & 85 & 17 \\ 2 & 4 & 2 & 7 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[199] $1^{111}31^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -1000, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 7 & 2 \\ 2 & 4 & 4 & 9 & 0 & 21 & 4 \\ 1 & 4 & 0 & 6 & 0 & 12 & 2 \\ 3 & 9 & 6 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 12 & 33 & 0 & 73 & 15 \\ 2 & 4 & 2 & 7 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[200] $1^{114}321^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -1004, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 7 & 2 \\ 1 & 0 & 2 & 3 & 0 & 11 & 2 \\ 1 & 2 & 0 & 4 & 0 & 13 & 2 \\ 2 & 3 & 4 & 5 & 0 & 24 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 13 & 24 & 0 & 81 & 17 \\ 2 & 2 & 2 & 5 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[201] $1^{112}31^{122}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -1018, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 6 & 2 \\ 2 & 4 & 4 & 15 & 0 & 18 & 4 \\ 1 & 4 & 0 & 9 & 0 & 12 & 2 \\ 5 & 15 & 9 & 37 & 0 & 48 & 11 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 12 & 48 & 0 & 58 & 14 \\ 2 & 4 & 2 & 11 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[202] $1^{113}31^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -1020, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 3 & 0 & 33 & 4 \\ 1 & 4 & 0 & 3 & 0 & 21 & 2 \\ 1 & 3 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 33 & 21 & 21 & 0 & 193 & 25 \\ 2 & 4 & 2 & 3 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[203] $1^{112}31^{112}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -1046, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 6 & 2 \\ 2 & 4 & 4 & 9 & 0 & 18 & 4 \\ 1 & 4 & 0 & 6 & 0 & 12 & 2 \\ 3 & 9 & 6 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 12 & 30 & 0 & 58 & 14 \\ 2 & 4 & 2 & 7 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[204] $1^{114}31^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -1174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 3 & 0 & 36 & 4 \\ 1 & 4 & 0 & 3 & 0 & 24 & 2 \\ 1 & 3 & 3 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 36 & 24 & 24 & 0 & 238 & 28 \\ 2 & 4 & 2 & 3 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[205] $1^{11}31^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -1197, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 6 & 0 & 21 & 4 \\ 1 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 6 & 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 12 & 21 & 0 & 73 & 15 \\ 2 & 4 & 2 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[206] $1^{12}31^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1236$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 6 & 0 & 24 & 4 \\ 1 & 4 & 0 & 3 & 0 & 15 & 2 \\ 2 & 6 & 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 15 & 24 & 0 & 100 & 18 \\ 2 & 4 & 2 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{114}31^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1254$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 8 & 2 \\ 1 & 0 & 2 & 3 & 0 & 12 & 2 \\ 1 & 2 & 0 & 4 & 0 & 16 & 2 \\ 2 & 3 & 4 & 5 & 0 & 28 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 16 & 28 & 0 & 110 & 20 \\ 2 & 2 & 2 & 5 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[208] $1^{111}31^{122}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1308$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 5 & 2 \\ 2 & 4 & 4 & 15 & 0 & 15 & 4 \\ 1 & 4 & 0 & 9 & 0 & 9 & 2 \\ 5 & 15 & 9 & 37 & 0 & 39 & 11 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 9 & 39 & 0 & 37 & 11 \\ 2 & 4 & 2 & 11 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[209] $1^{143}31^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1336$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 6 & 0 & 36 & 4 \\ 1 & 4 & 0 & 3 & 0 & 24 & 2 \\ 2 & 6 & 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 36 & 24 & 36 & 0 & 238 & 28 \\ 2 & 4 & 2 & 4 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[210] $1^{122}31^{122}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1568$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 6 & 2 \\ 2 & 4 & 4 & 18 & 0 & 18 & 4 \\ 1 & 4 & 0 & 12 & 0 & 12 & 2 \\ 6 & 18 & 12 & 58 & 0 & 60 & 14 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 12 & 60 & 0 & 58 & 14 \\ 2 & 4 & 2 & 14 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[211] $1^{123}31^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1669$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 8 & 2 \\ 2 & 4 & 4 & 9 & 0 & 24 & 4 \\ 1 & 4 & 0 & 6 & 0 & 15 & 2 \\ 3 & 9 & 6 & 13 & 0 & 39 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 15 & 39 & 0 & 100 & 18 \\ 2 & 4 & 2 & 7 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[212] $1^{114}31^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1741$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 10 & 2 \\ 2 & 4 & 4 & 9 & 0 & 30 & 4 \\ 1 & 4 & 0 & 6 & 0 & 21 & 2 \\ 3 & 9 & 6 & 13 & 0 & 51 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 21 & 51 & 0 & 172 & 24 \\ 2 & 4 & 2 & 7 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[213] $1^{113}31^{114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1813$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 6 & 0 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 15 & 2 \\ 2 & 6 & 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 15 & 27 & 0 & 121 & 19 \\ 2 & 4 & 2 & 4 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

[214] $1^{113}31^{113}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -1916$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 9 & 2 \\ 2 & 4 & 4 & 9 & 0 & 27 & 4 \\ 1 & 4 & 0 & 6 & 0 & 18 & 2 \\ 3 & 9 & 6 & 13 & 0 & 45 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 18 & 45 & 0 & 133 & 21 \\ 2 & 4 & 2 & 7 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[215] \ 1^{13}31^{114}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1917, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 6 & 0 & 33 & 4 \\ 1 & 4 & 0 & 3 & 0 & 21 & 2 \\ 2 & 6 & 3 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 33 & 21 & 33 & 0 & 193 & 25 \\ 2 & 4 & 2 & 4 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[216] \ 1^{12}31^{114}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -2076, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 6 & 0 & 30 & 4 \\ 1 & 4 & 0 & 3 & 0 & 18 & 2 \\ 2 & 6 & 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 18 & 30 & 0 & 154 & 22 \\ 2 & 4 & 2 & 4 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[217] \ 1^{12}31^{114}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -2194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 10 & 2 \\ 2 & 4 & 4 & 9 & 0 & 30 & 4 \\ 1 & 4 & 0 & 6 & 0 & 18 & 2 \\ 3 & 9 & 6 & 13 & 0 & 48 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 18 & 48 & 0 & 154 & 22 \\ 2 & 4 & 2 & 7 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[218] \ 1^{113}31^{114}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -2875, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 11 & 2 \\ 2 & 4 & 4 & 9 & 0 & 33 & 4 \\ 1 & 4 & 0 & 6 & 0 & 21 & 2 \\ 3 & 9 & 6 & 13 & 0 & 54 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 33 & 21 & 54 & 0 & 193 & 25 \\ 2 & 4 & 2 & 7 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[219] \ 1^{114}31^{114}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -3134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 12 & 2 \\ 2 & 4 & 4 & 9 & 0 & 36 & 4 \\ 1 & 4 & 0 & 6 & 0 & 24 & 2 \\ 3 & 9 & 6 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 36 & 24 & 60 & 0 & 238 & 28 \\ 2 & 4 & 2 & 7 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

4.813 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4$

$$T = 9 \quad [0] \ 1^{1111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 1 & 3 \\ 4 & 9 & 8 & 4 & 8 & 0 & 5 \\ 3 & 8 & 4 & 3 & 6 & 0 & 4 \\ 2 & 4 & 3 & -2 & 3 & 0 & 0 \\ 3 & 8 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 4 & 0 & 4 & -3 & -1 \end{pmatrix}_{b_0}$$

4.814 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{112}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.7$$

$$T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 26 & 5 \\ 12 & 25 & 16 & 0 & 56 & 9 \\ 8 & 16 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 56 & 36 & 0 & 118 & 20 \\ 5 & 9 & 6 & -3 & 20 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

14.2

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 7 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 4 & 1 & 3 \\ 4 & 9 & 8 & 4 & 8 & 0 & 5 \\ 3 & 8 & 4 & 3 & 6 & 0 & 4 \\ 2 & 4 & 3 & -2 & 3 & 0 & 0 \\ 4 & 8 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & 0 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 35 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 35 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 1 & 3 \\ 4 & 9 & 8 & 4 & 8 & 0 & 5 \\ 3 & 8 & 4 & 3 & 6 & 0 & 4 \\ 2 & 4 & 3 & -2 & 3 & 0 & 0 \\ 3 & 8 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & 0 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 17 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -482, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 26 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$4.815 \quad \text{NHC}(-2)(-3) + \textcolor{red}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3)$$

$$T = 9 \quad [0] \ 1^{122} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 2 & 3 \\ 3 & 4 & 8 & 0 & 8 & 3 & 4 \\ 4 & 8 & 10 & 0 & 12 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 12 & 0 & 10 & 4 & 6 \\ 2 & 3 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & 6 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{124}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 17 & 2 & 3 \\ 3 & 4 & 8 & 0 & 31 & 3 & 4 \\ 4 & 8 & 10 & 0 & 42 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 31 & 42 & 0 & 172 & 17 & 24 \\ 2 & 3 & 4 & 0 & 17 & 0 & 2 \\ 3 & 4 & 6 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{121}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 2 & 3 \\ 3 & 4 & 8 & 0 & 9 & 3 & 4 \\ 4 & 8 & 10 & 0 & 12 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 12 & 0 & 13 & 5 & 7 \\ 2 & 3 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & 6 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{132}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 5 & 0 & 17 & 3 & 4 \\ 3 & 5 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 15 & 0 & 49 & 9 & 13 \\ 2 & 3 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{131}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 3 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{112}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 0 & 8 & 3 & 4 \\ 2 & 4 & 1 & 0 & 6 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 10 & 4 & 6 \\ 2 & 3 & 2 & 0 & 4 & 0 & 2 \\ 3 & 4 & 3 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{131}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 3 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 6 & 0 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 3 & 4 \\ 2 & 4 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{123}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 2 & 3 \\ 3 & 4 & 8 & 0 & 22 & 3 & 4 \\ 4 & 8 & 10 & 0 & 30 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 30 & 0 & 85 & 12 & 17 \\ 2 & 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & 6 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{111}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 4 & 4 & 0 & 14 & 3 & 4 \\ 2 & 4 & 1 & 0 & 9 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 9 & 0 & 37 & 8 & 11 \\ 2 & 3 & 2 & 0 & 8 & 0 & 2 \\ 3 & 4 & 3 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{122}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 2 & 3 \\ 3 & 4 & 8 & 0 & 13 & 3 & 4 \\ 4 & 8 & 10 & 0 & 18 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 18 & 0 & 28 & 7 & 10 \\ 2 & 3 & 4 & 0 & 7 & 0 & 2 \\ 3 & 4 & 6 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 13 & 3 & 4 \\ 3 & 5 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 12 & 0 & 28 & 7 & 10 \\ 2 & 3 & 3 & 0 & 7 & 0 & 2 \\ 3 & 4 & 4 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{113}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 3 & 4 \\ 3 & 5 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 5 & 7 \\ 2 & 3 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 3 & 4 \\ 2 & 4 & 1 & 0 & 6 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 5 & 7 \\ 2 & 3 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 3 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 2 & 3 \\ 3 & 4 & 8 & 0 & 17 & 3 & 4 \\ 4 & 8 & 10 & 0 & 24 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 24 & 0 & 49 & 9 & 13 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 6 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{114}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 2 & 3 \\ 3 & 4 & 9 & 0 & 21 & 3 & 4 \\ 5 & 9 & 13 & 0 & 33 & 5 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 33 & 0 & 76 & 11 & 16 \\ 2 & 3 & 5 & 0 & 11 & 0 & 2 \\ 3 & 4 & 7 & -3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{124}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 11 & 2 & 3 \\ 3 & 4 & 8 & 0 & 21 & 3 & 4 \\ 4 & 8 & 10 & 0 & 30 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 30 & 0 & 76 & 11 & 16 \\ 2 & 3 & 4 & 0 & 11 & 0 & 2 \\ 3 & 4 & 6 & -3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 4 & 0 & 13 & 3 & 4 \\ 2 & 4 & 1 & 0 & 9 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 7 & 10 \\ 2 & 3 & 2 & 0 & 7 & 0 & 2 \\ 3 & 4 & 3 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{12} 4^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 14 & 2 & 3 \\ 3 & 4 & 8 & 0 & 26 & 3 & 4 \\ 4 & 8 & 10 & 0 & 36 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 36 & 0 & 118 & 14 & 20 \\ 2 & 3 & 4 & 0 & 14 & 0 & 2 \\ 3 & 4 & 6 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11} 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 2 & 3 \\ 3 & 4 & 9 & 0 & 9 & 3 & 4 \\ 5 & 9 & 13 & 0 & 15 & 5 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 15 & 0 & 13 & 5 & 7 \\ 2 & 3 & 5 & 0 & 5 & 0 & 2 \\ 3 & 4 & 7 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11} 1^{12} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -474, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 2 & 3 \\ 3 & 4 & 4 & 0 & 23 & 3 & 4 \\ 2 & 4 & 1 & 0 & 15 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 23 & 15 & 0 & 100 & 13 & 18 \\ 2 & 3 & 2 & 0 & 13 & 0 & 2 \\ 3 & 4 & 3 & -3 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 2 & 3 \\ 3 & 4 & 4 & 0 & 17 & 3 & 4 \\ 2 & 4 & 1 & 0 & 12 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 12 & 0 & 49 & 9 & 13 \\ 2 & 3 & 2 & 0 & 9 & 0 & 2 \\ 3 & 4 & 3 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 2 & 3 \\ 3 & 4 & 9 & 0 & 17 & 3 & 4 \\ 5 & 9 & 13 & 0 & 27 & 5 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 27 & 0 & 49 & 9 & 13 \\ 2 & 3 & 5 & 0 & 9 & 0 & 2 \\ 3 & 4 & 7 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11} 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 2 & 3 \\ 3 & 4 & 4 & 0 & 18 & 3 & 4 \\ 2 & 4 & 1 & 0 & 12 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 12 & 0 & 58 & 10 & 14 \\ 2 & 3 & 2 & 0 & 10 & 0 & 2 \\ 3 & 4 & 3 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 2 & 3 \\ 3 & 4 & 9 & 0 & 13 & 3 & 4 \\ 5 & 9 & 13 & 0 & 21 & 5 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 21 & 0 & 28 & 7 & 10 \\ 2 & 3 & 5 & 0 & 7 & 0 & 2 \\ 3 & 4 & 7 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{11} 1^4 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 2 & 3 \\ 3 & 4 & 4 & 0 & 21 & 3 & 4 \\ 2 & 4 & 1 & 0 & 15 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 15 & 0 & 76 & 11 & 16 \\ 2 & 3 & 2 & 0 & 11 & 0 & 2 \\ 3 & 4 & 3 & -3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{12} 4^1 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 11 & 2 & 3 \\ 3 & 4 & 13 & 0 & 21 & 3 & 4 \\ 7 & 13 & 28 & 0 & 48 & 7 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 48 & 0 & 76 & 11 & 16 \\ 2 & 3 & 7 & 0 & 11 & 0 & 2 \\ 3 & 4 & 10 & -3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{12} 2^1 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 2 & 3 \\ 3 & 4 & 13 & 0 & 13 & 3 & 4 \\ 7 & 13 & 28 & 0 & 30 & 7 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 30 & 0 & 28 & 7 & 10 \\ 2 & 3 & 7 & 0 & 7 & 0 & 2 \\ 3 & 4 & 10 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -794, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 9 & 2 & 3 \\ 3 & 4 & 13 & 0 & 17 & 3 & 4 \\ 7 & 13 & 28 & 0 & 39 & 7 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 39 & 0 & 49 & 9 & 13 \\ 2 & 3 & 7 & 0 & 9 & 0 & 2 \\ 3 & 4 & 10 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -824, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 2 & 3 \\ 3 & 4 & 4 & 0 & 22 & 3 & 4 \\ 2 & 4 & 1 & 0 & 15 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 15 & 0 & 85 & 12 & 17 \\ 2 & 3 & 2 & 0 & 12 & 0 & 2 \\ 3 & 4 & 3 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 9 & 2 & 3 \\ 3 & 4 & 17 & 0 & 17 & 3 & 4 \\ 9 & 17 & 49 & 0 & 51 & 9 & 13 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 51 & 0 & 49 & 9 & 13 \\ 2 & 3 & 9 & 0 & 9 & 0 & 2 \\ 3 & 4 & 13 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{11} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 2 & 3 \\ 3 & 4 & 4 & 0 & 32 & 3 & 4 \\ 2 & 4 & 1 & 0 & 21 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 18 & 32 & 21 & 0 & 193 & 18 & 25 \\ 2 & 3 & 2 & 0 & 18 & 0 & 2 \\ 3 & 4 & 3 & -3 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{13} 4^1 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -954, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 11 & 2 & 3 \\ 3 & 4 & 17 & 0 & 21 & 3 & 4 \\ 9 & 17 & 49 & 0 & 63 & 9 & 13 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 63 & 0 & 76 & 11 & 16 \\ 2 & 3 & 9 & 0 & 11 & 0 & 2 \\ 3 & 4 & 13 & -3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{11} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -974, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 2 & 3 \\ 3 & 4 & 4 & 0 & 27 & 3 & 4 \\ 2 & 4 & 1 & 0 & 18 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 18 & 0 & 133 & 15 & 21 \\ 2 & 3 & 2 & 0 & 15 & 0 & 2 \\ 3 & 4 & 3 & -3 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{14} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1064, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 11 & 2 & 3 \\ 3 & 4 & 21 & 0 & 21 & 3 & 4 \\ 11 & 21 & 76 & 0 & 78 & 11 & 16 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 78 & 0 & 76 & 11 & 16 \\ 2 & 3 & 11 & 0 & 11 & 0 & 2 \\ 3 & 4 & 16 & -3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{11} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1084, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 2 & 3 \\ 3 & 4 & 4 & 0 & 26 & 3 & 4 \\ 2 & 4 & 1 & 0 & 18 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 18 & 0 & 118 & 14 & 20 \\ 2 & 3 & 2 & 0 & 14 & 0 & 2 \\ 3 & 4 & 3 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{11} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1434, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 2 & 3 \\ 3 & 4 & 4 & 0 & 31 & 3 & 4 \\ 2 & 4 & 1 & 0 & 21 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 31 & 21 & 0 & 172 & 17 & 24 \\ 2 & 3 & 2 & 0 & 17 & 0 & 2 \\ 3 & 4 & 3 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{11} 3^1 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1604, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 2 & 3 \\ 3 & 4 & 4 & 0 & 36 & 3 & 4 \\ 2 & 4 & 1 & 0 & 24 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 20 & 36 & 24 & 0 & 238 & 20 & 28 \\ 2 & 3 & 2 & 0 & 20 & 0 & 2 \\ 3 & 4 & 3 & -3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [38] \quad 1^{11} 3^1 1^{11} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 247, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 & 2 \\ 1 & 3 & 3 & 1 & 0 & 6 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 6 & 0 & 13 & 5 & 7 \\ 1 & 3 & 2 & 2 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & 3 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^3 3^1 1^{11} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 472, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 & 2 \\ 2 & 6 & 3 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 6 & 3 & 6 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 3 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{11} 3^1 1^{11} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 517, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 & 2 \\ 2 & 6 & 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 9 & 0 & 13 & 5 & 7 \\ 1 & 3 & 2 & 3 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & 4 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{\overline{111}} 3^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 648, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 6 & 3 \\ 4 & 8 & 6 & 15 & 0 & 15 & 6 \\ 2 & 6 & 1 & 9 & 0 & 9 & 3 \\ 6 & 15 & 9 & 22 & 0 & 24 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 15 & 9 & 24 & 0 & 22 & 10 \\ 3 & 6 & 3 & 10 & -3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 912, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 6 & 2 & 2 \\ 3 & 9 & 6 & 13 & 0 & 15 & 5 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 9 & 6 & 15 & 0 & 13 & 5 & 7 \\ 1 & 3 & 2 & 5 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & 7 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.816 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 72, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 6 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 392, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 3 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 2 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & -3 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 462, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 9 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 5 & 6 & 7 \\ 2 & 3 & 3 & 0 & 5 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 6 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 512, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 2 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 0 & 5 & 2 \\ 5 & 9 & 6 & 0 & 6 & 5 & 12 & 6 \\ 3 & 4 & 3 & -3 & 3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 622, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 3 & 4 & 4 \\ 2 & 4 & 1 & 0 & 6 & 2 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 5 & 6 & 7 \\ 2 & 3 & 2 & 0 & 5 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 6 & 2 & 0 & 2 \\ 3 & 4 & 3 & -3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3^{\overline{1111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 9 & 3 & 4 & 4 \\ 5 & 9 & 13 & 0 & 15 & 5 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 15 & 0 & 13 & 5 & 6 & 7 \\ 2 & 3 & 5 & 0 & 5 & 0 & 2 & 2 \\ 2 & 4 & 6 & 0 & 6 & 2 & 0 & 2 \\ 3 & 4 & 7 & -3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.817 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_{2n3\text{mix}}$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 5 & 0 & 4 & 7 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 5 & 3 \\ 4 & 7 & 6 & 0 & 5 & 7 & 5 \\ 3 & 4 & 4 & -3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1211}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 4 & 3 \\ 3 & 4 & 8 & 0 & 9 & 7 & 4 \\ 4 & 8 & 10 & 0 & 12 & 10 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 12 & 0 & 13 & 11 & 7 \\ 4 & 7 & 10 & 0 & 11 & 7 & 5 \\ 3 & 4 & 6 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1122}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 9 & 3 \\ 3 & 4 & 4 & 0 & 8 & 16 & 4 \\ 2 & 4 & 1 & 0 & 6 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 10 & 22 & 6 \\ 9 & 16 & 11 & 0 & 22 & 44 & 12 \\ 3 & 4 & 3 & -3 & 6 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1131}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 4 & 3 \\ 3 & 4 & 4 & 0 & 12 & 7 & 4 \\ 2 & 4 & 1 & 0 & 9 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 9 & 0 & 25 & 15 & 9 \\ 4 & 7 & 5 & 0 & 15 & 7 & 5 \\ 3 & 4 & 3 & -3 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1113}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 5 & 0 & 9 & 7 & 4 \\ 3 & 5 & 4 & 0 & 9 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 11 & 7 \\ 4 & 7 & 6 & 0 & 11 & 7 & 5 \\ 3 & 4 & 4 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1313}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 5 & 0 & 17 & 7 & 4 \\ 3 & 5 & 4 & 0 & 15 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 15 & 0 & 49 & 21 & 13 \\ 4 & 7 & 6 & 0 & 21 & 7 & 5 \\ 3 & 4 & 4 & -3 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1212}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 9 & 3 \\ 3 & 4 & 8 & 0 & 9 & 16 & 4 \\ 4 & 8 & 10 & 0 & 12 & 22 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 12 & 0 & 13 & 26 & 7 \\ 9 & 16 & 22 & 0 & 26 & 44 & 12 \\ 3 & 4 & 6 & -3 & 7 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1221} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -156, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 4 & 3 \\ 3 & 4 & 8 & 0 & 8 & 7 & 4 \\ 4 & 8 & 10 & 0 & 12 & 10 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 12 & 0 & 10 & 10 & 6 \\ 4 & 7 & 10 & 0 & 10 & 7 & 5 \\ 3 & 4 & 6 & -3 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1131} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -164, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 7 & 3 \\ 3 & 4 & 4 & 0 & 17 & 12 & 4 \\ 2 & 4 & 1 & 0 & 12 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 12 & 0 & 49 & 36 & 13 \\ 7 & 12 & 8 & 0 & 36 & 25 & 9 \\ 3 & 4 & 3 & -3 & 13 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1113} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 6 & 3 \\ 3 & 4 & 9 & 0 & 9 & 11 & 4 \\ 5 & 9 & 13 & 0 & 15 & 17 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 15 & 0 & 13 & 17 & 7 \\ 6 & 11 & 17 & 0 & 17 & 20 & 8 \\ 3 & 4 & 7 & -3 & 7 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1113} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -194, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 12 & 4 \\ 3 & 5 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 8 & 3 \\ 7 & 12 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & 4 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1113} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -194, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 5 & 0 & 4 & 20 & 4 \\ 3 & 5 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 14 & 3 \\ 11 & 20 & 18 & 0 & 14 & 69 & 15 \\ 3 & 4 & 4 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1121} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -216, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.67 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 7 & 4 \\ 2 & 4 & 1 & 0 & 6 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 10 & 10 & 6 \\ 4 & 7 & 5 & 0 & 10 & 7 & 5 \\ 3 & 4 & 3 & -3 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1213} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -216, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.61 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 5 & 0 & 13 & 7 & 4 \\ 3 & 5 & 4 & 0 & 12 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 12 & 0 & 28 & 16 & 10 \\ 4 & 7 & 6 & 0 & 16 & 7 & 5 \\ 3 & 4 & 4 & -3 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -246, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.79 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 11 & 4 \\ 2 & 4 & 1 & 0 & 6 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 17 & 7 \\ 6 & 11 & 8 & 0 & 17 & 20 & 8 \\ 3 & 4 & 3 & -3 & 7 & 8 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{112}3^11^{111}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{brown}{f}_4+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 5 & 0 & 4 & 16 & 4 \\ 3 & 5 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 11 & 3 \\ 9 & 16 & 15 & 0 & 11 & 44 & 12 \\ 3 & 4 & 4 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{1111}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{brown}{f}_4+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 4 & 12 & 4 \\ 2 & 4 & 1 & 0 & 3 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 8 & 3 \\ 7 & 12 & 8 & 0 & 8 & 25 & 9 \\ 3 & 4 & 3 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{1111}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{brown}{f}_4+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 4 & 4 & 0 & 4 & 7 & 4 \\ 2 & 4 & 1 & 0 & 3 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 5 & 3 \\ 4 & 7 & 5 & 0 & 5 & 7 & 5 \\ 3 & 4 & 3 & -3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{1111}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{brown}{f}_4+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 4 & 3 \\ 3 & 4 & 4 & 0 & 9 & 7 & 4 \\ 2 & 4 & 1 & 0 & 6 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 11 & 7 \\ 4 & 7 & 5 & 0 & 11 & 7 & 5 \\ 3 & 4 & 3 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{1221}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{brown}{f}_4+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -326, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 4 & 3 \\ 3 & 4 & 8 & 0 & 13 & 7 & 4 \\ 4 & 8 & 10 & 0 & 18 & 10 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 18 & 0 & 28 & 16 & 10 \\ 4 & 7 & 10 & 0 & 16 & 7 & 5 \\ 3 & 4 & 6 & -3 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{1122}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{brown}{f}_4+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 6 & 3 \\ 3 & 4 & 4 & 0 & 8 & 11 & 4 \\ 2 & 4 & 1 & 0 & 6 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 10 & 16 & 6 \\ 6 & 11 & 8 & 0 & 16 & 20 & 8 \\ 3 & 4 & 3 & -3 & 6 & 8 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{1121}3^11^{12}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{brown}{f}_4+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 4 & 3 \\ 3 & 4 & 4 & 0 & 18 & 7 & 4 \\ 2 & 4 & 1 & 0 & 12 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 12 & 0 & 58 & 22 & 14 \\ 4 & 7 & 5 & 0 & 22 & 7 & 5 \\ 3 & 4 & 3 & -3 & 14 & 5 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{1112}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{brown}{f}_4+\textcolor{red}{su}_2+\textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 6 & 3 \\ 3 & 4 & 4 & 0 & 4 & 11 & 4 \\ 2 & 4 & 1 & 0 & 3 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 8 & 3 \\ 6 & 11 & 8 & 0 & 8 & 20 & 8 \\ 3 & 4 & 3 & -3 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{\overline{11123}} 3^{\overline{1}} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -426, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.74 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 6 & 3 \\ 3 & 4 & 9 & 0 & 13 & 11 & 4 \\ 5 & 9 & 13 & 0 & 21 & 17 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 21 & 0 & 28 & 25 & 10 \\ 6 & 11 & 17 & 0 & 25 & 20 & 8 \\ 3 & 4 & 7 & -3 & 10 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{\overline{11113}} 3^{\overline{1}} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -456, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 4 & 3 \\ 3 & 4 & 9 & 0 & 9 & 7 & 4 \\ 5 & 9 & 13 & 0 & 15 & 11 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 15 & 0 & 13 & 11 & 7 \\ 4 & 7 & 11 & 0 & 11 & 7 & 5 \\ 3 & 4 & 7 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{\overline{1131}} 3^{\overline{1}} 1^{\overline{13}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -476, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 4 & 3 \\ 3 & 4 & 4 & 0 & 27 & 7 & 4 \\ 2 & 4 & 1 & 0 & 18 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 18 & 0 & 133 & 33 & 21 \\ 4 & 7 & 5 & 0 & 33 & 7 & 5 \\ 3 & 4 & 3 & -3 & 21 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{\overline{11123}} 3^{\overline{1}} 1^{\overline{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -496, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 4 & 0 & 4 & 21 & 4 \\ 2 & 4 & 1 & 0 & 3 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 14 & 3 \\ 12 & 21 & 14 & 0 & 14 & 80 & 16 \\ 3 & 4 & 3 & -3 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{\overline{11123}} 3^{\overline{1}} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -514, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.63 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 12 & 4 \\ 2 & 4 & 1 & 0 & 9 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 28 & 10 \\ 7 & 12 & 8 & 0 & 28 & 25 & 9 \\ 3 & 4 & 3 & -3 & 10 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{\overline{11113}} 3^{\overline{1}} 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 7 & 3 \\ 3 & 4 & 4 & 0 & 9 & 12 & 4 \\ 2 & 4 & 1 & 0 & 6 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 20 & 7 \\ 7 & 12 & 8 & 0 & 20 & 25 & 9 \\ 3 & 4 & 3 & -3 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{\overline{1121}} 3^{\overline{1}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -536, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 4 & 3 \\ 3 & 4 & 4 & 0 & 13 & 7 & 4 \\ 2 & 4 & 1 & 0 & 9 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 16 & 10 \\ 4 & 7 & 5 & 0 & 16 & 7 & 5 \\ 3 & 4 & 3 & -3 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{\overline{1122}} 3^{\overline{1}} 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -546, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.71 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 6 & 3 \\ 3 & 4 & 4 & 0 & 13 & 11 & 4 \\ 2 & 4 & 1 & 0 & 9 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 25 & 10 \\ 6 & 11 & 8 & 0 & 25 & 20 & 8 \\ 3 & 4 & 3 & -3 & 10 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1141}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -546, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.66 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 4 & 3 \\ 3 & 4 & 9 & 0 & 21 & 7 & 4 \\ 5 & 9 & 13 & 0 & 33 & 11 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 33 & 0 & 76 & 26 & 16 \\ 4 & 7 & 11 & 0 & 26 & 7 & 5 \\ 3 & 4 & 7 & -3 & 16 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1113}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -564, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.81 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 15 & 4 \\ 2 & 4 & 1 & 0 & 3 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 11 & 3 \\ 8 & 15 & 11 & 0 & 11 & 39 & 11 \\ 3 & 4 & 3 & -3 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1141}3^11^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -616, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 4 & 3 \\ 3 & 4 & 4 & 0 & 36 & 7 & 4 \\ 2 & 4 & 1 & 0 & 24 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 20 & 36 & 24 & 0 & 238 & 44 & 28 \\ 4 & 7 & 5 & 0 & 44 & 7 & 5 \\ 3 & 4 & 3 & -3 & 28 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1112}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -620, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 4 & 4 & 0 & 4 & 16 & 4 \\ 2 & 4 & 1 & 0 & 3 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 11 & 3 \\ 9 & 16 & 11 & 0 & 11 & 44 & 12 \\ 3 & 4 & 3 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1112}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -626, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 4 & 3 \\ 3 & 4 & 9 & 0 & 13 & 7 & 4 \\ 5 & 9 & 13 & 0 & 21 & 11 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 21 & 0 & 28 & 16 & 10 \\ 4 & 7 & 11 & 0 & 16 & 7 & 5 \\ 3 & 4 & 7 & -3 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1113}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -656, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 4 & 3 \\ 3 & 4 & 9 & 0 & 17 & 7 & 4 \\ 5 & 9 & 13 & 0 & 27 & 11 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 27 & 0 & 49 & 21 & 13 \\ 4 & 7 & 11 & 0 & 21 & 7 & 5 \\ 3 & 4 & 7 & -3 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1113}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -716, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 4 & 3 \\ 3 & 4 & 4 & 0 & 17 & 7 & 4 \\ 2 & 4 & 1 & 0 & 12 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 12 & 0 & 49 & 21 & 13 \\ 4 & 7 & 5 & 0 & 21 & 7 & 5 \\ 3 & 4 & 3 & -3 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1114}3^11^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -720, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 4 & 19 & 4 \\ 2 & 4 & 1 & 0 & 3 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 14 & 3 \\ 10 & 19 & 14 & 0 & 14 & 64 & 14 \\ 3 & 4 & 3 & -3 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{\overline{1113}1} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -810, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 9 & 3 \\ 3 & 4 & 4 & 0 & 9 & 16 & 4 \\ 2 & 4 & 1 & 0 & 6 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 26 & 7 \\ 9 & 16 & 11 & 0 & 26 & 44 & 12 \\ 3 & 4 & 3 & -3 & 7 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{\overline{1131}3} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -836, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 4 & 3 \\ 3 & 4 & 4 & 0 & 22 & 7 & 4 \\ 2 & 4 & 1 & 0 & 15 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 15 & 0 & 85 & 27 & 17 \\ 4 & 7 & 5 & 0 & 27 & 7 & 5 \\ 3 & 4 & 3 & -3 & 17 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{\overline{1131}3} 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -840, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 17 & 3 \\ 3 & 4 & 4 & 0 & 4 & 30 & 4 \\ 2 & 4 & 1 & 0 & 3 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 20 & 3 \\ 17 & 30 & 20 & 0 & 20 & 165 & 23 \\ 3 & 4 & 3 & -3 & 3 & 23 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{\overline{1141}3} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -856, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 4 & 3 \\ 3 & 4 & 4 & 0 & 21 & 7 & 4 \\ 2 & 4 & 1 & 0 & 15 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 15 & 0 & 76 & 26 & 16 \\ 4 & 7 & 5 & 0 & 26 & 7 & 5 \\ 3 & 4 & 3 & -3 & 16 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{\overline{1221}3} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 4 & 3 \\ 3 & 4 & 13 & 0 & 13 & 7 & 4 \\ 7 & 13 & 28 & 0 & 30 & 16 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 30 & 0 & 28 & 16 & 10 \\ 4 & 7 & 16 & 0 & 16 & 7 & 5 \\ 3 & 4 & 10 & -3 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{\overline{1113}3} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -964, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.62 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 11 & 3 \\ 3 & 4 & 4 & 0 & 9 & 20 & 4 \\ 2 & 4 & 1 & 0 & 6 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 32 & 7 \\ 11 & 20 & 14 & 0 & 32 & 69 & 15 \\ 3 & 4 & 3 & -3 & 7 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{\overline{1113}3} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -984, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 11 & 3 \\ 3 & 4 & 4 & 0 & 4 & 20 & 4 \\ 2 & 4 & 1 & 0 & 3 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 14 & 3 \\ 11 & 20 & 14 & 0 & 14 & 69 & 15 \\ 3 & 4 & 3 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{\overline{1114}3} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -986, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.67 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 13 & 3 \\ 3 & 4 & 4 & 0 & 9 & 24 & 4 \\ 2 & 4 & 1 & 0 & 6 & 17 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 38 & 7 \\ 13 & 24 & 17 & 0 & 38 & 100 & 18 \\ 3 & 4 & 3 & -3 & 7 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{11223}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1020, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 9 & 3 \\ 3 & 4 & 4 & 0 & 13 & 16 & 4 \\ 2 & 4 & 1 & 0 & 9 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 37 & 10 \\ 9 & 16 & 11 & 0 & 37 & 44 & 12 \\ 3 & 4 & 3 & -3 & 10 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1113}3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1076, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 4 & 4 & 0 & 4 & 25 & 4 \\ 2 & 4 & 1 & 0 & 3 & 17 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 17 & 3 \\ 14 & 25 & 17 & 0 & 17 & 111 & 19 \\ 3 & 4 & 3 & -3 & 3 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1141}3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1176, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 4 & 3 \\ 3 & 4 & 4 & 0 & 31 & 7 & 4 \\ 2 & 4 & 1 & 0 & 21 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 31 & 21 & 0 & 172 & 38 & 24 \\ 4 & 7 & 5 & 0 & 38 & 7 & 5 \\ 3 & 4 & 3 & -3 & 24 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1141}3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.61 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 4 & 3 \\ 3 & 4 & 4 & 0 & 26 & 7 & 4 \\ 2 & 4 & 1 & 0 & 18 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 18 & 0 & 118 & 32 & 20 \\ 4 & 7 & 5 & 0 & 32 & 7 & 5 \\ 3 & 4 & 3 & -3 & 20 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1114}3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1296, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 22 & 3 \\ 3 & 4 & 4 & 0 & 4 & 39 & 4 \\ 2 & 4 & 1 & 0 & 3 & 26 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 26 & 3 \\ 22 & 39 & 26 & 0 & 26 & 280 & 30 \\ 3 & 4 & 3 & -3 & 3 & 30 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1114}3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1356, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 13 & 3 \\ 3 & 4 & 4 & 0 & 4 & 24 & 4 \\ 2 & 4 & 1 & 0 & 3 & 17 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 17 & 3 \\ 13 & 24 & 17 & 0 & 17 & 100 & 18 \\ 3 & 4 & 3 & -3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1114}3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1644, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 19 & 3 \\ 3 & 4 & 4 & 0 & 4 & 34 & 4 \\ 2 & 4 & 1 & 0 & 3 & 23 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 23 & 3 \\ 19 & 34 & 23 & 0 & 23 & 208 & 26 \\ 3 & 4 & 3 & -3 & 3 & 26 & 1 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1114}3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 16 & 3 \\ 3 & 4 & 4 & 0 & 4 & 29 & 4 \\ 2 & 4 & 1 & 0 & 3 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 20 & 3 \\ 16 & 29 & 20 & 0 & 20 & 148 & 22 \\ 3 & 4 & 3 & -3 & 3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [55] \quad 1^{11}31^{11}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 5 \\ 1 & 4 & 0 & 3 & 0 & 6 & 4 \\ 2 & 6 & 3 & 4 & 0 & 6 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 \\ 2 & 6 & 3 & 6 & 0 & 4 & 5 \\ 2 & 5 & 4 & 5 & 0 & 5 & 4 \\ 2 & 4 & 2 & 4 & -3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{11}31^{11}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 5 \\ 1 & 4 & 0 & 3 & 0 & 6 & 4 \\ 1 & 3 & 3 & 1 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 \\ 3 & 9 & 6 & 6 & 0 & 13 & 9 \\ 2 & 5 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 2 & 3 & -3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{11}31^{11}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 262, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 5 \\ 1 & 4 & 0 & 3 & 0 & 6 & 4 \\ 2 & 6 & 3 & 4 & 0 & 9 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 \\ 3 & 9 & 6 & 9 & 0 & 13 & 9 \\ 2 & 5 & 4 & 5 & 0 & 9 & 4 \\ 2 & 4 & 2 & 4 & -3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{11}31^{11}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 752, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 9 & 5 \\ 1 & 4 & 0 & 6 & 0 & 6 & 4 \\ 3 & 9 & 6 & 13 & 0 & 15 & 9 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 \\ 3 & 9 & 6 & 15 & 0 & 13 & 9 \\ 2 & 5 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 2 & 7 & -3 & 7 & 4 \end{pmatrix}_{b_0}$$

4.818 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11}31^{11}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 6 & 0 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}31^{11}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 5 & 0 & 19 & 4 & 4 \\ 3 & 5 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 19 & 21 & 0 & 73 & 12 & 15 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -3 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}31^{11}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 9 & 0 & 37 & 9 & 11 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 3 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}31^{11}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

12.2

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 2 & 3 \\ 3 & 4 & 9 & 0 & 19 & 4 & 4 \\ 5 & 9 & 13 & 0 & 33 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 19 & 33 & 0 & 73 & 12 & 15 \\ 2 & 4 & 6 & 0 & 12 & 0 & 2 \\ 3 & 4 & 7 & -3 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{1121}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 5 & 3 \\ 3 & 4 & 4 & 0 & 8 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 10 & 12 & 6 \\ 5 & 9 & 6 & 0 & 12 & 12 & 6 \\ 3 & 4 & 3 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & -3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{1121}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 13 & 4 \\ 3 & 5 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 7 & 13 & 12 & 0 & 9 & 27 & 9 \\ 3 & 4 & 4 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{1111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 & 4 \\ 3 & 5 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{111}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 5 & 0 & 14 & 4 & 4 \\ 3 & 5 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 15 & 0 & 37 & 9 & 11 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{121}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 & 4 \\ 3 & 5 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 12 & 0 & 28 & 9 & 10 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{1111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 6 & 12 & 6 \\ 3 & 4 & 3 & -3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1121} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 2 & 3 \\ 3 & 4 & 4 & 0 & 23 & 4 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 23 & 15 & 0 & 100 & 15 & 18 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 3 & -3 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1221} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 5 & 3 \\ 3 & 4 & 8 & 0 & 13 & 9 & 4 \\ 4 & 8 & 10 & 0 & 13 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 18 & 0 & 28 & 21 & 10 \\ 5 & 9 & 12 & 0 & 21 & 12 & 6 \\ 3 & 4 & 6 & -3 & 10 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 0 & 15 & 12 & 6 \\ 3 & 4 & 3 & -3 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -605, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 2 & 3 \\ 3 & 4 & 9 & 0 & 14 & 4 & 4 \\ 5 & 9 & 13 & 0 & 24 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 24 & 0 & 37 & 9 & 11 \\ 2 & 4 & 6 & 0 & 9 & 0 & 2 \\ 3 & 4 & 7 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{121} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 5 & 0 & 18 & 4 & 4 \\ 3 & 5 & 4 & 0 & 18 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{121} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -650, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 5 & 0 & 28 & 4 & 4 \\ 3 & 5 & 4 & 0 & 30 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 16 & 28 & 30 & 0 & 154 & 18 & 22 \\ 2 & 4 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -3 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{1111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 2 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 & 4 \\ 5 & 9 & 13 & 0 & 15 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 15 & 0 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 6 & 0 & 2 \\ 3 & 4 & 7 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{1111}31^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -716$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 8 & 2 & 3 \\ 3 & 4 & 14 & 0 & 14 & 4 & 4 \\ 8 & 14 & 37 & 0 & 39 & 9 & 11 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 39 & 0 & 37 & 9 & 11 \\ 2 & 4 & 9 & 0 & 9 & 0 & 2 \\ 3 & 4 & 11 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{1121}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -723$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.7$

$T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 9 & 10 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 3 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{1121}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -738$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.7$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 2 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 12 & 0 & 58 & 12 & 14 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 3 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{1211}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -744$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.7$

$T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 5 & 0 & 23 & 4 & 4 \\ 3 & 5 & 4 & 0 & 24 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 23 & 24 & 0 & 100 & 15 & 18 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -3 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{1311}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -779$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 5 & 0 & 22 & 4 & 4 \\ 3 & 5 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 21 & 0 & 85 & 15 & 17 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{1131}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -828$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 2 & 3 \\ 3 & 4 & 4 & 0 & 32 & 4 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 18 & 32 & 21 & 0 & 193 & 21 & 25 \\ 2 & 4 & 3 & 0 & 21 & 0 & 2 \\ 3 & 4 & 3 & -3 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{1121}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -845$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.7$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 5 & 3 \\ 3 & 4 & 4 & 0 & 13 & 9 & 4 \\ 2 & 4 & 1 & 0 & 9 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 21 & 10 \\ 5 & 9 & 6 & 0 & 21 & 12 & 6 \\ 3 & 4 & 3 & -3 & 10 & 6 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{1121}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -941$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.7$

$T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 2 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 & 4 \\ 5 & 9 & 13 & 0 & 21 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 21 & 0 & 28 & 9 & 10 \\ 2 & 4 & 6 & 0 & 9 & 0 & 2 \\ 3 & 4 & 7 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1121} 31^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -962, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 10 & 2 & 3 \\ 3 & 4 & 14 & 0 & 18 & 4 & 4 \\ 8 & 14 & 37 & 0 & 48 & 9 & 11 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 48 & 0 & 58 & 12 & 14 \\ 2 & 4 & 9 & 0 & 12 & 0 & 2 \\ 3 & 4 & 11 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1121} 31^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -980, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 16 & 2 & 3 \\ 3 & 4 & 9 & 0 & 28 & 4 & 4 \\ 5 & 9 & 13 & 0 & 48 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 16 & 28 & 48 & 0 & 154 & 18 & 22 \\ 2 & 4 & 6 & 0 & 18 & 0 & 2 \\ 3 & 4 & 7 & -3 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1131} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -980, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 2 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 & 4 \\ 5 & 9 & 13 & 0 & 27 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 27 & 0 & 49 & 12 & 13 \\ 2 & 4 & 6 & 0 & 12 & 0 & 2 \\ 3 & 4 & 7 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1131} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -989, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 5 & 3 \\ 3 & 4 & 4 & 0 & 17 & 9 & 4 \\ 2 & 4 & 1 & 0 & 12 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 12 & 0 & 49 & 27 & 13 \\ 5 & 9 & 6 & 0 & 27 & 12 & 6 \\ 3 & 4 & 3 & -3 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1141} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1005, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 5 & 3 \\ 3 & 4 & 4 & 0 & 21 & 9 & 4 \\ 2 & 4 & 1 & 0 & 15 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 15 & 0 & 76 & 33 & 16 \\ 5 & 9 & 6 & 0 & 33 & 12 & 6 \\ 3 & 4 & 3 & -3 & 16 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1131} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1077, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 2 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 12 & 0 & 49 & 12 & 13 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 3 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1311} 31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1221, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 5 & 0 & 27 & 4 & 4 \\ 3 & 5 & 4 & 0 & 27 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 27 & 0 & 133 & 18 & 21 \\ 2 & 4 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -3 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1221} 31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1298, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 10 & 2 & 3 \\ 3 & 4 & 13 & 0 & 18 & 4 & 4 \\ 7 & 13 & 28 & 0 & 42 & 9 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 42 & 0 & 58 & 12 & 14 \\ 2 & 4 & 9 & 0 & 12 & 0 & 2 \\ 3 & 4 & 10 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1131^1} 31^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1333, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 2 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 18 & 0 & 133 & 18 & 21 \\ 2 & 4 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & 3 & -3 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1121^1} 31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 10 & 2 & 3 \\ 3 & 4 & 9 & 0 & 18 & 4 & 4 \\ 5 & 9 & 13 & 0 & 30 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 30 & 0 & 58 & 12 & 14 \\ 2 & 4 & 6 & 0 & 12 & 0 & 2 \\ 3 & 4 & 7 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1221^1} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1388, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 2 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 & 4 \\ 7 & 13 & 28 & 0 & 30 & 9 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 30 & 0 & 28 & 9 & 10 \\ 2 & 4 & 9 & 0 & 9 & 0 & 2 \\ 3 & 4 & 10 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1121^1} 31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1389, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 13 & 2 & 3 \\ 3 & 4 & 9 & 0 & 23 & 4 & 4 \\ 5 & 9 & 13 & 0 & 39 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 23 & 39 & 0 & 100 & 15 & 18 \\ 2 & 4 & 6 & 0 & 15 & 0 & 2 \\ 3 & 4 & 7 & -3 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1131^1} 31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 2 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 15 & 0 & 85 & 15 & 17 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 3 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{131^1} 31^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1451, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 5 & 0 & 32 & 4 & 4 \\ 3 & 5 & 4 & 0 & 33 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 18 & 32 & 33 & 0 & 193 & 21 & 25 \\ 2 & 4 & 3 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -3 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{141^1} 31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1500, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 5 & 0 & 31 & 4 & 4 \\ 3 & 5 & 4 & 0 & 30 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 31 & 30 & 0 & 172 & 21 & 24 \\ 2 & 4 & 3 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1131^1} 31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1949, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 12 & 2 & 3 \\ 3 & 4 & 9 & 0 & 22 & 4 & 4 \\ 5 & 9 & 13 & 0 & 36 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 36 & 0 & 85 & 15 & 17 \\ 2 & 4 & 6 & 0 & 15 & 0 & 2 \\ 3 & 4 & 7 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{122\bar{1}} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2048, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 10 & 2 & 3 \\ 3 & 4 & 18 & 0 & 18 & 4 & 4 \\ 10 & 18 & 58 & 0 & 60 & 12 & 14 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 60 & 0 & 58 & 12 & 14 \\ 2 & 4 & 12 & 0 & 12 & 0 & 2 \\ 3 & 4 & 14 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{14\bar{1}} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2054, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 5 & 0 & 36 & 4 & 4 \\ 3 & 5 & 4 & 0 & 36 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 20 & 36 & 36 & 0 & 238 & 24 & 28 \\ 2 & 4 & 3 & 0 & 24 & 0 & 2 \\ 3 & 4 & 4 & -3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{114\bar{1}} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2106, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 2 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 18 & 0 & 118 & 18 & 20 \\ 2 & 4 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & 3 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{114\bar{1}} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 2 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 & 4 \\ 2 & 4 & 1 & 0 & 24 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 20 & 36 & 24 & 0 & 238 & 24 & 28 \\ 2 & 4 & 3 & 0 & 24 & 0 & 2 \\ 3 & 4 & 3 & -3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{114\bar{1}} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2324, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 14 & 2 & 3 \\ 3 & 4 & 9 & 0 & 26 & 4 & 4 \\ 5 & 9 & 13 & 0 & 42 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 42 & 0 & 118 & 18 & 20 \\ 2 & 4 & 6 & 0 & 18 & 0 & 2 \\ 3 & 4 & 7 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{114\bar{1}} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2347, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 2 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 31 & 21 & 0 & 172 & 21 & 24 \\ 2 & 4 & 3 & 0 & 21 & 0 & 2 \\ 3 & 4 & 3 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{113\bar{1}} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2496, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 15 & 2 & 3 \\ 3 & 4 & 9 & 0 & 27 & 4 & 4 \\ 5 & 9 & 13 & 0 & 45 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 45 & 0 & 133 & 18 & 21 \\ 2 & 4 & 6 & 0 & 18 & 0 & 2 \\ 3 & 4 & 7 & -3 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{113\bar{1}} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2621, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 18 & 2 & 3 \\ 3 & 4 & 9 & 0 & 32 & 4 & 4 \\ 5 & 9 & 13 & 0 & 54 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 18 & 32 & 54 & 0 & 193 & 21 & 25 \\ 2 & 4 & 6 & 0 & 21 & 0 & 2 \\ 3 & 4 & 7 & -3 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1141} 31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -3405, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 17 & 2 & 3 \\ 3 & 4 & 9 & 0 & 31 & 4 & 4 \\ 5 & 9 & 13 & 0 & 51 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 31 & 51 & 0 & 172 & 21 & 24 \\ 2 & 4 & 6 & 0 & 21 & 0 & 2 \\ 3 & 4 & 7 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1141} 31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -4064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 20 & 2 & 3 \\ 3 & 4 & 9 & 0 & 36 & 4 & 4 \\ 5 & 9 & 13 & 0 & 60 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 20 & 36 & 60 & 0 & 238 & 24 & 28 \\ 2 & 4 & 6 & 0 & 24 & 0 & 2 \\ 3 & 4 & 7 & -3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [54] \quad 1^{11} 321^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 5 & 5 & 0 & 10 & 5 & 5 \\ 2 & 5 & 2 & 0 & 7 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 10 & 7 & 0 & 14 & 7 & 8 \\ 2 & 5 & 4 & 0 & 7 & 1 & 3 \\ 3 & 5 & 4 & -3 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1111} 321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 1 & 0 & 1 & 4 & 2 \\ 1 & 2 & 1 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 2 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 3 & 0 & 3 & 4 & 4 \\ 2 & 2 & 2 & 2 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{11} 321^{1113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 160, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 5 & 5 & 0 & 5 & 20 & 5 \\ 2 & 5 & 2 & 0 & 4 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 2 & 15 & 4 \\ 9 & 20 & 15 & 0 & 15 & 57 & 15 \\ 3 & 5 & 4 & -3 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{111} 321^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 172, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 3 & 0 & 4 & 4 & 2 \\ 1 & 1 & 3 & 1 & 0 & 4 & 4 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 4 & 0 & 5 & 7 & 5 \\ 2 & 3 & 4 & 4 & 0 & 7 & 4 & 4 \\ 2 & 2 & 2 & 3 & -3 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{11} 321^{1121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 220, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 5 & 5 & 0 & 15 & 5 & 5 \\ 2 & 5 & 2 & 0 & 11 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 15 & 11 & 0 & 32 & 11 & 12 \\ 2 & 5 & 4 & 0 & 11 & 1 & 3 \\ 3 & 5 & 4 & -3 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{1111} 321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 262, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 4 & 2 \\ 1 & 2 & 1 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 3 & 0 & 5 & 7 & 5 \\ 2 & 3 & 4 & 3 & 0 & 7 & 4 & 4 \\ 2 & 2 & 2 & 2 & -3 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{12}3^{11}3^{11}2^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 340, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 2 & 3 \\ 3 & 5 & 5 & 0 & 25 & 5 & 5 \\ 2 & 5 & 2 & 0 & 18 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 25 & 18 & 0 & 94 & 18 & 20 \\ 2 & 5 & 4 & 0 & 18 & 1 & 3 \\ 3 & 5 & 4 & -3 & 20 & 3 & 2 \end{pmatrix}_{b_0}$$

[61] $1^{111}3^{11}3^{11}2^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 2 & 3 \\ 3 & 5 & 10 & 0 & 10 & 5 & 5 \\ 5 & 10 & 14 & 0 & 16 & 7 & 8 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 10 & 16 & 0 & 14 & 7 & 8 \\ 2 & 5 & 7 & 0 & 7 & 1 & 3 \\ 3 & 5 & 8 & -3 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{13}3^{11}3^{11}2^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 380, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 2 & 3 \\ 3 & 5 & 5 & 0 & 35 & 5 & 5 \\ 2 & 5 & 2 & 0 & 25 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 35 & 25 & 0 & 188 & 25 & 28 \\ 2 & 5 & 4 & 0 & 25 & 1 & 3 \\ 3 & 5 & 4 & -3 & 28 & 3 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{111}3^{11}3^{11}2^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 390, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 2 & 3 \\ 3 & 5 & 10 & 0 & 15 & 5 & 5 \\ 5 & 10 & 14 & 0 & 23 & 7 & 8 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 15 & 23 & 0 & 32 & 11 & 12 \\ 2 & 5 & 7 & 0 & 11 & 1 & 3 \\ 3 & 5 & 8 & -3 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[64] $1^{12}3^{11}3^{11}2^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 2 & 3 \\ 3 & 5 & 5 & 0 & 30 & 5 & 5 \\ 2 & 5 & 2 & 0 & 22 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 30 & 22 & 0 & 134 & 22 & 24 \\ 2 & 5 & 4 & 0 & 22 & 1 & 3 \\ 3 & 5 & 4 & -3 & 24 & 3 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{1111}3^{11}3^{11}2^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 752, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 3 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 4 & 0 & 4 & 4 & 2 \\ 2 & 3 & 4 & 5 & 0 & 7 & 7 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 7 & 0 & 5 & 7 & 5 \\ 2 & 3 & 4 & 7 & 0 & 7 & 4 & 4 \\ 2 & 2 & 2 & 5 & -3 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

4.819 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 10$ $[0] 1^{11}3^{11}1^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 3 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 6 & 0 & 4 & 3 & 3 & 4 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 392, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 3 \\ 2 & 4 & 1 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & 2 & 0 & 2 & 2 & 0 \\ 3 & 4 & 3 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 462, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 & 3 & 4 \\ 3 & 5 & 4 & 0 & 9 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 6 & 5 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 5 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 512, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 3 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 6 & 2 & 3 \\ 5 & 9 & 6 & 0 & 6 & 12 & 5 & 6 \\ 2 & 3 & 2 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & 3 & -3 & 3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 622, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 3 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 6 & 5 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 5 & 2 & 0 & 2 \\ 3 & 4 & 3 & -3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 1152, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 & 3 & 4 \\ 5 & 9 & 13 & 0 & 15 & 6 & 5 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 15 & 0 & 13 & 6 & 5 & 7 \\ 2 & 4 & 6 & 0 & 6 & 0 & 2 & 2 \\ 2 & 3 & 5 & 0 & 5 & 2 & 0 & 2 \\ 3 & 4 & 7 & -3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.820 NHC(-2)(-3) + $\mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \ 1^{\overline{13}} 3^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 2, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 15, \ T_{\max} = 20, \ T \in [15, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 3 & 4 \\ 6 & 12 & 25 & 0 & 6 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 9 & -3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{13}} 3^{\overline{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 12, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 13.33 \rightarrow 14, \ T_{\max} = 20, \ T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 10 & 0 & 3 & 4 \\ 6 & 10 & 22 & 0 & 6 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 8 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{13}} 3^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 15, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 3 & 3 & 2 & 3 \\ 4 & 9 & 8 & 4 & 4 & 5 \\ 3 & 8 & 4 & 3 & 3 & 4 \\ 3 & 4 & 3 & -2 & 2 & 0 \\ 2 & 4 & 3 & 2 & 0 & 2 \\ 3 & 5 & 4 & 0 & 2 & 1 \end{pmatrix}_{b_0} \\
[3] & 21^1 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 4 & 3 & 3 & 2 & 3 \\ 4 & 9 & 8 & 4 & 4 & 5 \\ 3 & 8 & 4 & 3 & 3 & 4 \\ 3 & 4 & 3 & -2 & 2 & 0 \\ 2 & 4 & 3 & 2 & 0 & 2 \\ 3 & 5 & 4 & 0 & 2 & 1 \end{pmatrix}_{b_0} \\
[4] & 1^{11} 3^1 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 15, T_{\max} = 20, T \in [15, 20] \\
& \begin{pmatrix} 1 & 4 & 7 & 1 & 2 & 3 \\ 4 & 9 & 16 & 0 & 4 & 5 \\ 7 & 16 & 28 & 0 & 7 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 7 & 0 & 0 & 2 \\ 3 & 5 & 10 & -3 & 2 & -2 \end{pmatrix}_{b_0} \\
[5] & 1^{11} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 13.14 \rightarrow 14, T_{\max} = 20, T \in [14, 20] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & 3 & -3 & 2 & -1 \end{pmatrix}_{b_0} \\
[6] & 1^{11} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7 \\
& T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 3 & 4 \\ 4 & 8 & 10 & 0 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 6 & -3 & 2 & 1 \end{pmatrix}_{b_0} \\
[7] & 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0} \\
[8] & 1^{11} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 1 \end{pmatrix}_{b_0} \\
[9] & 13^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 12.22 \rightarrow 13, T_{\max} = 20, T \in [13, 20] \\
& \begin{pmatrix} 1 & 4 & 3 & 2 & 2 & 3 \\ 4 & 9 & 8 & 4 & 4 & 5 \\ 3 & 8 & 4 & 3 & 3 & 4 \\ 2 & 4 & 3 & -2 & 2 & 0 \\ 2 & 4 & 3 & 2 & 0 & 2 \\ 3 & 5 & 4 & 0 & 2 & 0 \end{pmatrix}_{b_0} \\
[10] & 1^{11} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 12.12 \rightarrow 13, T_{\max} = 20, T \in [13, 20] \\
& \begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 3 \\ 4 & 9 & 12 & 0 & 4 & 5 \\ 5 & 12 & 13 & 0 & 5 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 0 & 0 & 2 \\ 3 & 5 & 7 & -3 & 2 & 0 \end{pmatrix}_{b_0} \\
[12] & 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2 \\
& T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 3 & 4 \\ 5 & 9 & 13 & 0 & 5 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 7 & -3 & 2 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

[11] $1^{11}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.14 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 24 & 0 & 3 & 4 \\ 14 & 24 & 121 & 0 & 14 & 19 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 14 & 0 & 0 & 2 \\ 3 & 4 & 19 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 4 \\ 8 & 14 & 37 & 0 & 8 & 11 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 3 & 4 & 11 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{11}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 19 & 0 & 3 & 4 \\ 11 & 19 & 73 & 0 & 11 & 15 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 11 & 0 & 0 & 2 \\ 3 & 4 & 15 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 3 & 4 \\ 7 & 13 & 28 & 0 & 7 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 7 & 0 & 0 & 2 \\ 3 & 4 & 10 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 282, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 3 & 4 \\ 9 & 17 & 49 & 0 & 9 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 9 & 0 & 0 & 2 \\ 3 & 4 & 13 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{14}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 3 & 4 \\ 11 & 21 & 76 & 0 & 11 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 11 & 0 & 0 & 2 \\ 3 & 4 & 16 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{12}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 3 & 4 \\ 10 & 18 & 58 & 0 & 10 & 14 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 3 & 4 & 14 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[19] $21^{111}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 3 & 4 \\ 8 & 9 & 8 & 4 & 4 & 5 \\ 6 & 8 & 4 & 3 & 3 & 4 \\ 3 & 4 & 3 & -2 & 2 & 0 \\ 3 & 4 & 3 & 2 & 0 & 2 \\ 4 & 5 & 4 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{12}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 23 & 0 & 3 & 4 \\ 13 & 23 & 100 & 0 & 13 & 18 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 13 & 0 & 0 & 2 \\ 3 & 4 & 18 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{12}3^11^{12} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 522, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 3 & 4 \\ 12 & 22 & 85 & 0 & 12 & 17 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 12 & 0 & 0 & 2 \\ 3 & 4 & 17 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{12}3^11^{14} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 532, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 3 & 4 \\ 16 & 28 & 154 & 0 & 16 & 22 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 16 & 0 & 0 & 2 \\ 3 & 4 & 22 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{14}3^11^{12} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 652, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 26 & 0 & 3 & 4 \\ 14 & 26 & 118 & 0 & 14 & 20 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 14 & 0 & 0 & 2 \\ 3 & 4 & 20 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{13}3^11^{13} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 722, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 27 & 0 & 3 & 4 \\ 15 & 27 & 133 & 0 & 15 & 21 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 15 & 0 & 0 & 2 \\ 3 & 4 & 21 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{13}3^11^{14} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 882, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 2 & 3 \\ 3 & 4 & 32 & 0 & 3 & 4 \\ 18 & 32 & 193 & 0 & 18 & 25 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 18 & 0 & 0 & 2 \\ 3 & 4 & 25 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{14}3^11^{13} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 952, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 3 \\ 3 & 4 & 31 & 0 & 3 & 4 \\ 17 & 31 & 172 & 0 & 17 & 24 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 17 & 0 & 0 & 2 \\ 3 & 4 & 24 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{14}3^11^{14} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 2 & 3 \\ 3 & 4 & 36 & 0 & 3 & 4 \\ 20 & 36 & 238 & 0 & 20 & 28 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 20 & 0 & 0 & 2 \\ 3 & 4 & 28 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [28] $1^{11}3^121^{11} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{u}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 6 & 1 & 11 & 5 \\ 6 & 5 & 1 & 10 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 10 & 0 & 18 & 8 \\ 5 & 5 & -1 & 8 & 3 \end{pmatrix}_{b_0}$$

- [29] $1^{11}31^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 8 & 6 & 5 & 0 & 6 \\ 2 & 6 & 1 & 4 & 0 & 3 \\ 2 & 5 & 4 & 2 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$
- [30] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 1 & 16 & 5 \\ 6 & 5 & 1 & 17 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 17 & 0 & 45 & 13 \\ 5 & 5 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$
- [31] $1^{11}3^{11}1^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.54 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 3 & 2 & 0 & 0 & 2 \\ 2 & 3 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$
- [32] $1^{12}31^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 3 \\ 4 & 8 & 6 & 20 & 0 & 6 \\ 2 & 6 & 1 & 13 & 0 & 3 \\ 8 & 20 & 13 & 44 & 0 & 14 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 14 & -3 & 1 \end{pmatrix}_{b_0}$$
- [33] $1^{11}3^{11}1^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 2 \\ 1 & 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 2 & 0 & 0 & 2 \\ 2 & 4 & 2 & 3 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$
- [34] $1^{11}31^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 8 & 6 & 10 & 0 & 6 \\ 2 & 6 & 1 & 5 & 0 & 3 \\ 4 & 10 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$
- [35] $1^{13}1^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -201, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$
- [36] $1^{12}3^{11}1^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 3 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 & 2 \\ 4 & 12 & 9 & 28 & 0 & 7 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 7 & 0 & 0 & 2 \\ 2 & 4 & 2 & 10 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$
- [37] $1^{11}31^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 8 & 6 & 15 & 0 & 6 \\ 2 & 6 & 1 & 9 & 0 & 3 \\ 6 & 15 & 9 & 22 & 0 & 10 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 10 & -3 & 2 \end{pmatrix}_{b_0}$$
- [38] $1^{13}31^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 3 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 & 2 \\ 5 & 15 & 9 & 37 & 0 & 8 & 11 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 8 & 0 & 0 & 2 \\ 2 & 4 & 2 & 11 & -3 & 2 & 1 \end{pmatrix}_{b_0} \\
[48] \quad & 1^{14}31^{13}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0 \\
& T_{\min} = 12.01 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 20 & 1 & 3 \\ 4 & 8 & 6 & 50 & 0 & 6 \\ 2 & 6 & 1 & 31 & 0 & 3 \\ 20 & 50 & 31 & 272 & 0 & 34 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 34 & -3 & 1 \end{pmatrix}_{b_0} \\
[49] \quad & 1^{13}3^{11}1^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -666, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7 \\
& T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 3 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 & 2 \\ 7 & 21 & 15 & 85 & 0 & 12 & 17 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 12 & 0 & 0 & 2 \\ 2 & 4 & 2 & 17 & -3 & 2 & 1 \end{pmatrix}_{b_0} \\
[50] \quad & 1^{13}1^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -726, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2 \\
& T_{\min} = 12.09 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 3 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 & 2 \\ 8 & 24 & 12 & 94 & 0 & 12 & 16 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 12 & 0 & 0 & 2 \\ 2 & 4 & 2 & 16 & -3 & 2 & 0 \end{pmatrix}_{b_0} \\
[51] \quad & 1^{12}3^{11}1^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -746, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.7 \\
& T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 3 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 & 2 \\ 6 & 18 & 12 & 58 & 0 & 10 & 14 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 10 & 0 & 0 & 2 \\ 2 & 4 & 2 & 14 & -3 & 2 & 1 \end{pmatrix}_{b_0} \\
[52] \quad & 1^{13}1^{14}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -784, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2 \\
& T_{\min} = 12.16 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 8 & 6 & 40 & 0 & 6 \\ 2 & 6 & 1 & 20 & 0 & 3 \\ 16 & 40 & 20 & 158 & 0 & 24 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 24 & -3 & 1 \end{pmatrix}_{b_0} \\
[53] \quad & 1^{11}31^{13}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -874, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0 \\
& T_{\min} = 12.04 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 3 \\ 4 & 8 & 6 & 35 & 0 & 6 \\ 2 & 6 & 1 & 19 & 0 & 3 \\ 14 & 35 & 19 & 122 & 0 & 22 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 22 & -3 & 1 \end{pmatrix}_{b_0} \\
[54] \quad & 1^{13}31^{13}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -954, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0 \\
& T_{\min} = 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 18 & 1 & 3 \\ 4 & 8 & 6 & 45 & 0 & 6 \\ 2 & 6 & 1 & 27 & 0 & 3 \\ 18 & 45 & 27 & 214 & 0 & 30 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 30 & -3 & 2 \end{pmatrix}_{b_0} \\
[55] \quad & 1^{11}3^{11}1^{13}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -1011, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0 \\
& T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 3 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 & 2 \\ 7 & 21 & 12 & 73 & 0 & 11 & 15 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 11 & 0 & 0 & 2 \\ 2 & 4 & 2 & 15 & -3 & 2 & 1 \end{pmatrix}_{b_0} \\
[56] \quad & 1^{12}31^{13}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -1024, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0 \\
& T_{\min} = 12.01 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]
\end{aligned}$$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 8 & 6 & 40 & 0 & 6 \\ 2 & 6 & 1 & 23 & 0 & 3 \\ 16 & 40 & 23 & 164 & 0 & 26 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 26 & -3 & 1 \end{pmatrix}_{b_0}$$

[57] $1^{13}3^{11}1^{15}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1041, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 1 & 2 \\ 2 & 4 & 4 & 30 & 0 & 3 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 & 2 \\ 10 & 30 & 15 & 148 & 0 & 15 & 20 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 15 & 0 & 0 & 2 \\ 2 & 4 & 2 & 20 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[58] $1^{13}3^{11}1^{15}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 20 & 1 & 3 \\ 4 & 8 & 6 & 50 & 0 & 6 \\ 2 & 6 & 1 & 25 & 0 & 3 \\ 20 & 50 & 25 & 248 & 0 & 30 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 30 & -3 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{11}3^{11}1^{14}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 1 & 3 \\ 4 & 8 & 6 & 45 & 0 & 6 \\ 2 & 6 & 1 & 24 & 0 & 3 \\ 18 & 45 & 24 & 202 & 0 & 28 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 28 & -3 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{12}3^{11}1^{13}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1341, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 3 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 & 2 \\ 8 & 24 & 15 & 100 & 0 & 13 & 18 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 13 & 0 & 0 & 2 \\ 2 & 4 & 2 & 18 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[61] $1^{14}3^{11}1^{13}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 1 & 2 \\ 2 & 4 & 4 & 30 & 0 & 3 & 4 \\ 1 & 4 & 0 & 21 & 0 & 2 & 2 \\ 10 & 30 & 21 & 172 & 0 & 17 & 24 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 17 & 0 & 0 & 2 \\ 2 & 4 & 2 & 24 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{11}3^{11}1^{14}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1466, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.00 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 3 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 & 2 \\ 9 & 27 & 15 & 121 & 0 & 14 & 19 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 14 & 0 & 0 & 2 \\ 2 & 4 & 2 & 19 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[63] $1^{13}3^{11}1^{13}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1471, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 3 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 & 2 \\ 9 & 27 & 18 & 133 & 0 & 15 & 21 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 15 & 0 & 0 & 2 \\ 2 & 4 & 2 & 21 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[64] $1^{14}3^{11}1^{14}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 24 & 1 & 3 \\ 4 & 8 & 6 & 60 & 0 & 6 \\ 2 & 6 & 1 & 36 & 0 & 3 \\ 24 & 60 & 36 & 382 & 0 & 40 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 40 & -3 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{12}3^{11}1^{14}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 20 & 1 & 3 \\ 4 & 8 & 6 & 50 & 0 & 6 \\ 2 & 6 & 1 & 28 & 0 & 3 \\ 20 & 50 & 28 & 254 & 0 & 32 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 32 & -3 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{13}3^1 1^{14}3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1714, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.00 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 22 & 1 & 3 \\ 4 & 8 & 6 & 55 & 0 & 6 \\ 2 & 6 & 1 & 32 & 0 & 3 \\ 22 & 55 & 32 & 314 & 0 & 36 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 36 & -3 & 1 \end{pmatrix}_{b_0}$$

[67] $1^{12}3^1 1^{14}3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2006, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 1 & 2 \\ 2 & 4 & 4 & 30 & 0 & 3 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 & 2 \\ 10 & 30 & 18 & 154 & 0 & 16 & 22 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 16 & 0 & 0 & 2 \\ 2 & 4 & 2 & 22 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[68] $1^{13}3^1 1^{14}3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 1 & 1 & 2 \\ 2 & 4 & 4 & 33 & 0 & 3 & 4 \\ 1 & 4 & 0 & 21 & 0 & 2 & 2 \\ 11 & 33 & 21 & 193 & 0 & 18 & 25 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 18 & 0 & 0 & 2 \\ 2 & 4 & 2 & 25 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{14}3^1 1^{14}3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2486, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 1 & 1 & 2 \\ 2 & 4 & 4 & 36 & 0 & 3 & 4 \\ 1 & 4 & 0 & 24 & 0 & 2 & 2 \\ 12 & 36 & 24 & 238 & 0 & 20 & 28 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 20 & 0 & 0 & 2 \\ 2 & 4 & 2 & 28 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

4.821 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4$

$T = 9 \quad [0] \quad 1^{11}3^1 1^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 3 & 0 & 4 \\ 2 & 4 & 3 & -2 & 2 & 0 & 0 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 4 & 0 & 2 & -3 & -1 \end{pmatrix}_{b_0}$$

4.822 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \text{NHC}(-2)(-3)$

$T = 10 \quad [0] \quad 1^{11}3^1 1^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 5 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 2 & 4 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^1 1^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 600, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 9 & 0 & 3 & 5 & 4 & 4 \\ 5 & 9 & 13 & 0 & 5 & 9 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 9 & 0 & 3 & 4 & 4 & 4 \\ 2 & 4 & 6 & 0 & 2 & 4 & 0 & 2 \\ 3 & 4 & 7 & -3 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

4.823 NHC(-2)(-3) + su₂ + f₄ + su₂→(-3) + so₈

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 7 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 3 & 0 & 4 \\ 2 & 4 & 3 & -2 & 2 & 0 & 0 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & 0 & 2 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 6 & 5 \\ 12 & 25 & 36 & 0 & 12 & 9 \\ 17 & 36 & 49 & 0 & 17 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 17 & 0 & 4 & 4 \\ 5 & 9 & 13 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

4.824 NHC(-2)(-3) + su₂ + f₄ + su₂→(-3) + su₂

$$T = 9 \quad [1] \ 1^{122}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 4 & 3 \\ 3 & 4 & 8 & 0 & 3 & 8 & 4 \\ 4 & 8 & 10 & 0 & 4 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 4 & 2 \\ 4 & 8 & 12 & 0 & 4 & 10 & 6 \\ 3 & 4 & 6 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{124}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 17 & 3 \\ 3 & 4 & 8 & 0 & 3 & 31 & 4 \\ 4 & 8 & 10 & 0 & 4 & 42 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 17 & 2 \\ 17 & 31 & 42 & 0 & 17 & 172 & 24 \\ 3 & 4 & 6 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 5 & 3 \\ 3 & 4 & 8 & 0 & 3 & 9 & 4 \\ 4 & 8 & 10 & 0 & 4 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 5 & 2 \\ 5 & 9 & 12 & 0 & 5 & 13 & 7 \\ 3 & 4 & 6 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 4 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 5 & 0 & 3 & 17 & 4 \\ 3 & 5 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 9 & 2 \\ 9 & 17 & 15 & 0 & 9 & 49 & 13 \\ 3 & 4 & 4 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{12}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -84, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 2 & 3 \\ 3 & 4 & 8 & 0 & 3 & 4 & 4 \\ 4 & 8 & 10 & 0 & 4 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 & 2 \\ 2 & 4 & 6 & 0 & 2 & 1 & 3 \\ 3 & 4 & 6 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{13}3^{111}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -104, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 3 & 5 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -144, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.50 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 3 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{123}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -164, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 12 & 3 \\ 3 & 4 & 8 & 0 & 3 & 22 & 4 \\ 4 & 8 & 10 & 0 & 4 & 30 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 12 & 22 & 30 & 0 & 12 & 85 & 17 \\ 3 & 4 & 6 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{111}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 3 & 14 & 4 \\ 2 & 4 & 1 & 0 & 2 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 8 & 2 \\ 8 & 14 & 9 & 0 & 8 & 37 & 11 \\ 3 & 4 & 3 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{122}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.57 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 7 & 3 \\ 3 & 4 & 8 & 0 & 3 & 13 & 4 \\ 4 & 8 & 10 & 0 & 4 & 18 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 7 & 2 \\ 7 & 13 & 18 & 0 & 7 & 28 & 10 \\ 3 & 4 & 6 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12}3^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 3 & 13 & 4 \\ 3 & 5 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 7 & 2 \\ 7 & 13 & 12 & 0 & 7 & 28 & 10 \\ 3 & 4 & 4 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11}3^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -234, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 3 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 5 & 2 \\ 5 & 9 & 9 & 0 & 5 & 13 & 7 \\ 3 & 4 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 3 & 9 & 4 \\ 2 & 4 & 1 & 0 & 2 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 \\ 5 & 9 & 6 & 0 & 5 & 13 & 7 \\ 3 & 4 & 3 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{123} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 9 & 3 \\ 3 & 4 & 8 & 0 & 3 & 17 & 4 \\ 4 & 8 & 10 & 0 & 4 & 24 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 9 & 17 & 24 & 0 & 9 & 49 & 13 \\ 3 & 4 & 6 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{111} 3^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 11 & 3 \\ 3 & 4 & 9 & 0 & 3 & 21 & 4 \\ 5 & 9 & 13 & 0 & 5 & 33 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 11 & 2 \\ 11 & 21 & 33 & 0 & 11 & 76 & 16 \\ 3 & 4 & 7 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{124} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 11 & 3 \\ 3 & 4 & 8 & 0 & 3 & 21 & 4 \\ 4 & 8 & 10 & 0 & 4 & 30 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 11 & 2 \\ 11 & 21 & 30 & 0 & 11 & 76 & 16 \\ 3 & 4 & 6 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{112} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 3 & 13 & 4 \\ 2 & 4 & 1 & 0 & 2 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 7 & 2 \\ 7 & 13 & 9 & 0 & 7 & 28 & 10 \\ 3 & 4 & 3 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{124} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 14 & 3 \\ 3 & 4 & 8 & 0 & 3 & 26 & 4 \\ 4 & 8 & 10 & 0 & 4 & 36 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 14 & 2 \\ 14 & 26 & 36 & 0 & 14 & 118 & 20 \\ 3 & 4 & 6 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{111} 3^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 3 & 9 & 4 \\ 5 & 9 & 13 & 0 & 5 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 5 & 2 \\ 5 & 9 & 15 & 0 & 5 & 13 & 7 \\ 3 & 4 & 7 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{112} 3^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -474, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 13 & 3 \\ 3 & 4 & 4 & 0 & 3 & 23 & 4 \\ 2 & 4 & 1 & 0 & 2 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 13 & 2 \\ 13 & 23 & 15 & 0 & 13 & 100 & 18 \\ 3 & 4 & 3 & -3 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{113}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 4 & 4 & 0 & 3 & 17 & 4 \\ 2 & 4 & 1 & 0 & 2 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 9 & 2 \\ 9 & 17 & 12 & 0 & 9 & 49 & 13 \\ 3 & 4 & 3 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{113}3^{11}1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 9 & 3 \\ 3 & 4 & 9 & 0 & 3 & 17 & 4 \\ 5 & 9 & 13 & 0 & 5 & 27 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 9 & 2 \\ 9 & 17 & 27 & 0 & 9 & 49 & 13 \\ 3 & 4 & 7 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{112}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 3 & 18 & 4 \\ 2 & 4 & 1 & 0 & 2 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 10 & 2 \\ 10 & 18 & 12 & 0 & 10 & 58 & 14 \\ 3 & 4 & 3 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{114}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 11 & 3 \\ 3 & 4 & 4 & 0 & 3 & 21 & 4 \\ 2 & 4 & 1 & 0 & 2 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 11 & 2 \\ 11 & 21 & 15 & 0 & 11 & 76 & 16 \\ 3 & 4 & 3 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{112}3^{11}1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 7 & 3 \\ 3 & 4 & 9 & 0 & 3 & 13 & 4 \\ 5 & 9 & 13 & 0 & 5 & 21 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 7 & 2 \\ 7 & 13 & 21 & 0 & 7 & 28 & 10 \\ 3 & 4 & 7 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{124}3^{11}1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 11 & 3 \\ 3 & 4 & 13 & 0 & 3 & 21 & 4 \\ 7 & 13 & 28 & 0 & 7 & 48 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 7 & 0 & 0 & 11 & 2 \\ 11 & 21 & 48 & 0 & 11 & 76 & 16 \\ 3 & 4 & 10 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{122}3^{11}1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 7 & 3 \\ 3 & 4 & 13 & 0 & 3 & 13 & 4 \\ 7 & 13 & 28 & 0 & 7 & 30 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 7 & 0 & 0 & 7 & 2 \\ 7 & 13 & 30 & 0 & 7 & 28 & 10 \\ 3 & 4 & 10 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{123}3^{11}1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -794, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 9 & 3 \\ 3 & 4 & 13 & 0 & 3 & 17 & 4 \\ 7 & 13 & 28 & 0 & 7 & 39 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 7 & 0 & 0 & 9 & 2 \\ 9 & 17 & 39 & 0 & 9 & 49 & 13 \\ 3 & 4 & 10 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{113}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -824, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 4 & 0 & 3 & 22 & 4 \\ 2 & 4 & 1 & 0 & 2 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 12 & 2 \\ 12 & 22 & 15 & 0 & 12 & 85 & 17 \\ 3 & 4 & 3 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{113}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 18 & 3 \\ 3 & 4 & 4 & 0 & 3 & 32 & 4 \\ 2 & 4 & 1 & 0 & 2 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 18 & 2 \\ 18 & 32 & 21 & 0 & 18 & 193 & 25 \\ 3 & 4 & 3 & -3 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{133}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 9 & 3 \\ 3 & 4 & 17 & 0 & 3 & 17 & 4 \\ 9 & 17 & 49 & 0 & 9 & 51 & 13 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 9 & 0 & 0 & 9 & 2 \\ 9 & 17 & 51 & 0 & 9 & 49 & 13 \\ 3 & 4 & 13 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{134}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -954, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 11 & 3 \\ 3 & 4 & 17 & 0 & 3 & 21 & 4 \\ 9 & 17 & 49 & 0 & 9 & 63 & 13 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 9 & 0 & 0 & 11 & 2 \\ 11 & 21 & 63 & 0 & 11 & 76 & 16 \\ 3 & 4 & 13 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{113}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -974, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 15 & 3 \\ 3 & 4 & 4 & 0 & 3 & 27 & 4 \\ 2 & 4 & 1 & 0 & 2 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 15 & 2 \\ 15 & 27 & 18 & 0 & 15 & 133 & 21 \\ 3 & 4 & 3 & -3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{144}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1064, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 11 & 3 \\ 3 & 4 & 21 & 0 & 3 & 21 & 4 \\ 11 & 21 & 76 & 0 & 11 & 78 & 16 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 11 & 0 & 0 & 11 & 2 \\ 11 & 21 & 78 & 0 & 11 & 76 & 16 \\ 3 & 4 & 16 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{114}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1084, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 4 & 4 & 0 & 3 & 26 & 4 \\ 2 & 4 & 1 & 0 & 2 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 14 & 2 \\ 14 & 26 & 18 & 0 & 14 & 118 & 20 \\ 3 & 4 & 3 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{114}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1434, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 17 & 3 \\ 3 & 4 & 4 & 0 & 3 & 31 & 4 \\ 2 & 4 & 1 & 0 & 2 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 17 & 2 \\ 17 & 31 & 21 & 0 & 17 & 172 & 24 \\ 3 & 4 & 3 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{114}3^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1604, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.63 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 20 & 3 \\ 3 & 4 & 4 & 0 & 3 & 36 & 4 \\ 2 & 4 & 1 & 0 & 2 & 24 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 20 & 2 \\ 20 & 36 & 24 & 0 & 20 & 238 & 28 \\ 3 & 4 & 3 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [38] \quad 1^{111}3^{11}1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 247, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 9 \\ 1 & 4 & 0 & 3 & 0 & 2 & 6 \\ 1 & 3 & 3 & 1 & 0 & 2 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 \\ 1 & 3 & 2 & 2 & 0 & 0 & 5 \\ 3 & 9 & 6 & 6 & 0 & 5 & 13 \\ 2 & 4 & 2 & 3 & -3 & 2 & 7 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{13}1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 472, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 6 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{113}1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 517, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 6 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 3 & 0 & 0 & 5 & 2 \\ 3 & 9 & 6 & 9 & 0 & 5 & 13 & 7 \\ 2 & 4 & 2 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{111}31^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 648, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 6 & 3 \\ 4 & 8 & 6 & 15 & 0 & 15 & 6 \\ 2 & 6 & 1 & 9 & 0 & 9 & 3 \\ 6 & 15 & 9 & 22 & 0 & 24 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 15 & 9 & 24 & 0 & 22 & 10 \\ 3 & 6 & 3 & 10 & -3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{111}31^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 912, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 & 6 & 2 \\ 3 & 9 & 6 & 13 & 0 & 5 & 15 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 5 & 0 & 0 & 5 & 2 \\ 3 & 9 & 6 & 15 & 0 & 5 & 13 & 7 \\ 2 & 4 & 2 & 7 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.825 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{111}3^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 5 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 392, \quad (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{11}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 9 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 5 & 2 & 2 \\ 5 & 9 & 9 & 0 & 5 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 5 & 2 \\ 2 & 4 & 3 & 0 & 2 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 5 & 6 & 12 & 6 \\ 3 & 4 & 3 & -3 & 2 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 9 & 4 & 4 \\ 2 & 4 & 1 & 0 & 2 & 6 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 & 2 \\ 5 & 9 & 6 & 0 & 5 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 9 & 0 & 3 & 9 & 4 & 4 \\ 5 & 9 & 13 & 0 & 5 & 15 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 5 & 2 & 2 \\ 5 & 9 & 15 & 0 & 5 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & 7 & -3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.826 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ + su2n3mix

$$T = 10 \quad [0] \ 1^{111}3^{11}1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 5 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 4 & 2 \\ 1 & 3 & 3 & 1 & 0 & 2 & 4 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 2 & 0 & 0 & 3 & 2 \\ 2 & 5 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & 3 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 175, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 5 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 4 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 3 & 0 & 0 & 3 & 2 \\ 2 & 5 & 4 & 5 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{11}1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 475, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 5 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 & 4 & 2 \\ 3 & 9 & 6 & 13 & 0 & 5 & 9 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 2 & 5 & 0 & 0 & 3 & 2 \\ 2 & 5 & 4 & 9 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & 7 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

4.827 NHC(-2)(-3) + su₂ + f₄ + su₂→(-3) + su₃

$$T = 9 \quad [0] \quad 1^{121}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 5 & 3 \\ 3 & 4 & 8 & 0 & 3 & 9 & 4 \\ 4 & 8 & 10 & 0 & 4 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 5 & 2 \\ 5 & 9 & 12 & 0 & 5 & 12 & 6 \\ 3 & 4 & 6 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 3 & 0 & 4 \\ 2 & 4 & 3 & -2 & 2 & 0 & 0 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & 0 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 2 & 3 \\ 3 & 4 & 19 & 0 & 3 & 4 & 4 \\ 11 & 19 & 73 & 0 & 11 & 12 & 15 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 11 & 0 & 0 & 2 & 2 \\ 2 & 4 & 12 & 0 & 2 & 0 & 2 \\ 3 & 4 & 15 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{121}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 2 & 3 \\ 3 & 4 & 8 & 0 & 3 & 4 & 4 \\ 4 & 8 & 10 & 0 & 4 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 & 2 \\ 2 & 4 & 6 & 0 & 2 & 0 & 2 \\ 3 & 4 & 6 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{112}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 3 & 5 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 3 & 2 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 3 & 4 & 3 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{112}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.31 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 4 & 4 & 0 & 3 & 8 & 4 \\ 2 & 4 & 1 & 0 & 2 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 4 & 2 \\ 4 & 8 & 6 & 0 & 4 & 9 & 5 \\ 3 & 4 & 3 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -184, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 2 & 3 \\ 4 & 9 & 12 & 0 & 4 & 4 & 5 \\ 5 & 12 & 13 & 0 & 5 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 5 & 0 & 0 & 2 & 2 \\ 2 & 4 & 6 & 0 & 2 & 0 & 2 \\ 3 & 5 & 7 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -206, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.37 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 4 & 3 \\ 3 & 4 & 9 & 0 & 3 & 8 & 4 \\ 5 & 9 & 13 & 0 & 5 & 12 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 4 & 2 \\ 4 & 8 & 12 & 0 & 4 & 9 & 5 \\ 3 & 4 & 7 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -226, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -336, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 3 & 9 & 4 \\ 2 & 4 & 1 & 0 & 2 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 \\ 5 & 9 & 6 & 0 & 5 & 12 & 6 \\ 3 & 4 & 3 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -386, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 4 & 4 \\ 8 & 14 & 37 & 0 & 8 & 9 & 11 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 8 & 0 & 0 & 2 & 2 \\ 2 & 4 & 9 & 0 & 2 & 0 & 2 \\ 3 & 4 & 11 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -416, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 3 & 4 & 4 \\ 5 & 9 & 13 & 0 & 5 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 2 & 2 \\ 2 & 4 & 6 & 0 & 2 & 0 & 2 \\ 3 & 4 & 7 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{121}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -476, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 3 & 13 & 4 \\ 2 & 4 & 1 & 0 & 2 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 7 & 2 \\ 7 & 13 & 9 & 0 & 7 & 27 & 9 \\ 3 & 4 & 3 & -3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{\overline{121}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -626, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 2 & 3 \\ 3 & 4 & 13 & 0 & 3 & 4 & 4 \\ 7 & 13 & 28 & 0 & 7 & 9 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 7 & 0 & 0 & 2 & 2 \\ 2 & 4 & 9 & 0 & 2 & 0 & 2 \\ 3 & 4 & 10 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{121}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 2 & 3 \\ 3 & 4 & 28 & 0 & 3 & 4 & 4 \\ 16 & 28 & 154 & 0 & 16 & 18 & 22 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 16 & 0 & 0 & 2 & 2 \\ 2 & 4 & 18 & 0 & 2 & 0 & 2 \\ 3 & 4 & 22 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{131}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 2 & 3 \\ 3 & 4 & 17 & 0 & 3 & 4 & 4 \\ 9 & 17 & 49 & 0 & 9 & 12 & 13 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 9 & 0 & 0 & 2 & 2 \\ 2 & 4 & 12 & 0 & 2 & 0 & 2 \\ 3 & 4 & 13 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{141}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 2 & 3 \\ 3 & 4 & 21 & 0 & 3 & 4 & 4 \\ 11 & 21 & 76 & 0 & 11 & 15 & 16 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 11 & 0 & 0 & 2 & 2 \\ 2 & 4 & 15 & 0 & 2 & 0 & 2 \\ 3 & 4 & 16 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{121}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 2 & 3 \\ 3 & 4 & 18 & 0 & 3 & 4 & 4 \\ 10 & 18 & 58 & 0 & 10 & 12 & 14 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 10 & 0 & 0 & 2 & 2 \\ 2 & 4 & 12 & 0 & 2 & 0 & 2 \\ 3 & 4 & 14 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{121}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 2 & 3 \\ 3 & 4 & 23 & 0 & 3 & 4 & 4 \\ 13 & 23 & 100 & 0 & 13 & 15 & 18 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 13 & 0 & 0 & 2 & 2 \\ 2 & 4 & 15 & 0 & 2 & 0 & 2 \\ 3 & 4 & 18 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{131}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 2 & 3 \\ 3 & 4 & 22 & 0 & 3 & 4 & 4 \\ 12 & 22 & 85 & 0 & 12 & 15 & 17 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 12 & 0 & 0 & 2 & 2 \\ 2 & 4 & 15 & 0 & 2 & 0 & 2 \\ 3 & 4 & 17 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{131}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 2 & 3 \\ 3 & 4 & 27 & 0 & 3 & 4 & 4 \\ 15 & 27 & 133 & 0 & 15 & 18 & 21 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 15 & 0 & 0 & 2 & 2 \\ 2 & 4 & 18 & 0 & 2 & 0 & 2 \\ 3 & 4 & 21 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{141}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 2 & 3 \\ 3 & 4 & 26 & 0 & 3 & 4 & 4 \\ 14 & 26 & 118 & 0 & 14 & 18 & 20 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 14 & 0 & 0 & 2 & 2 \\ 2 & 4 & 18 & 0 & 2 & 0 & 2 \\ 3 & 4 & 20 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{131}3^{11}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1826, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 2 & 2 & 3 \\ 3 & 4 & 32 & 0 & 3 & 4 & 4 \\ 18 & 32 & 193 & 0 & 18 & 21 & 25 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 18 & 0 & 0 & 2 & 2 \\ 2 & 4 & 21 & 0 & 2 & 0 & 2 \\ 3 & 4 & 25 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{141}3^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 2 & 3 \\ 3 & 4 & 31 & 0 & 3 & 4 & 4 \\ 17 & 31 & 172 & 0 & 17 & 21 & 24 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 17 & 0 & 0 & 2 & 2 \\ 2 & 4 & 21 & 0 & 2 & 0 & 2 \\ 3 & 4 & 24 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{141}3^{11}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 2 & 2 & 3 \\ 3 & 4 & 36 & 0 & 3 & 4 & 4 \\ 20 & 36 & 238 & 0 & 20 & 24 & 28 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 20 & 0 & 0 & 2 & 2 \\ 2 & 4 & 24 & 0 & 2 & 0 & 2 \\ 3 & 4 & 28 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.828 NHC(-2)(-3) + $\textcolor{red}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$

$$T = 10 \quad [0] \quad 1^{111}3^{11}1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 3 \\ 2 & 4 & 3 & 0 & 2 & 0 & 3 \\ 3 & 5 & 6 & 0 & 3 & 3 & 4 \\ 3 & 4 & 4 & -3 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & -3 & 2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111}3^{11}1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 5 & 2 \\ 2 & 4 & 3 & 0 & 2 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 5 & 6 & 13 & 7 \\ 3 & 4 & 4 & -3 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 9 & 4 & 4 \\ 2 & 4 & 1 & 0 & 2 & 6 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 & 2 \\ 5 & 9 & 6 & 0 & 5 & 12 & 6 & 6 \\ 2 & 4 & 3 & 0 & 2 & 6 & 1 & 3 \\ 3 & 4 & 3 & -3 & 2 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 5 & 2 \\ 2 & 4 & 3 & 0 & 2 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 5 & 6 & 13 & 7 \\ 3 & 4 & 3 & -3 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{111}1}3^{\overline{111}1}1^{\overline{111}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 3 & 4 & 9 & 4 \\ 5 & 9 & 13 & 0 & 5 & 6 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 2 & 5 & 2 \\ 2 & 4 & 6 & 0 & 2 & 0 & 6 & 2 \\ 5 & 9 & 15 & 0 & 5 & 6 & 13 & 7 \\ 3 & 4 & 7 & -3 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.829 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_{2n3\text{mix}}$

$T = 8$ [1] $1^{\overline{111}1}3^{\overline{111}1}1^{\overline{111}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 13.63 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 16 & 8 & 0 & 25 & 9 \\ 3 & 5 & 3 & -3 & 9 & -1 \end{pmatrix}_{b_0}$$

[0] $1^{\overline{11}1}3^{\overline{11}1}1^{\overline{11}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 5 & 0 & 12 & 6 \\ 3 & 4 & 3 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{111}1}3^{\overline{111}1}1^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 4 & 3 \\ 3 & 4 & 14 & 0 & 7 & 4 \\ 8 & 14 & 37 & 0 & 17 & 11 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 17 & 0 & 7 & 5 \\ 3 & 4 & 11 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{12}3}1^{\overline{111}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 8 & 4 \\ 7 & 13 & 28 & 0 & 19 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 19 & 0 & 12 & 6 \\ 3 & 4 & 10 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{13}3}1^{\overline{11}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.3$

$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 8 & 3 \\ 3 & 4 & 20 & 0 & 15 & 4 \\ 10 & 20 & 73 & 0 & 55 & 15 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 55 & 0 & 39 & 11 \\ 3 & 4 & 15 & -3 & 11 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{113}3}1^{\overline{11}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 3 \\ 3 & 4 & 9 & 0 & 15 & 4 \\ 5 & 9 & 13 & 0 & 23 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 23 & 0 & 39 & 11 \\ 3 & 4 & 7 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{\overline{11}3}1^{\overline{11}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 \\ 3 & 5 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 0 & 7 & 5 \\ 3 & 4 & 4 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{111}3}1^{\overline{11}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 5 & 0 & 12 & 6 \\ 3 & 5 & 3 & -3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{14231^1 1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 6 & 3 \\ 3 & 4 & 16 & 0 & 11 & 4 \\ 8 & 16 & 46 & 0 & 32 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 32 & 0 & 20 & 8 \\ 3 & 4 & 12 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{13131^1 1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 4 & 3 \\ 3 & 4 & 12 & 0 & 7 & 4 \\ 6 & 12 & 25 & 0 & 15 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 15 & 0 & 7 & 5 \\ 3 & 4 & 9 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{14231^1 1^{13}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 6 & 3 \\ 3 & 4 & 31 & 0 & 11 & 4 \\ 17 & 31 & 172 & 0 & 59 & 24 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 59 & 0 & 20 & 8 \\ 3 & 4 & 24 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11131^1 1^{13}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 22 & 14 & 0 & 97 & 17 \\ 3 & 4 & 3 & -3 & 17 & -1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{12331^1 1^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 14 & 3 \\ 3 & 4 & 8 & 0 & 25 & 4 \\ 4 & 8 & 10 & 0 & 34 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 25 & 34 & 0 & 111 & 19 \\ 3 & 4 & 6 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{14331^1 1^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 8 & 3 \\ 3 & 4 & 26 & 0 & 15 & 4 \\ 14 & 26 & 118 & 0 & 68 & 20 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 68 & 0 & 39 & 11 \\ 3 & 4 & 20 & -3 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{13431^1 1^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 13 & 3 \\ 3 & 4 & 12 & 0 & 24 & 4 \\ 6 & 12 & 25 & 0 & 51 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 51 & 0 & 100 & 18 \\ 3 & 4 & 9 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{12131^1 1^{13}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 4 & 3 \\ 3 & 4 & 23 & 0 & 7 & 4 \\ 13 & 23 & 100 & 0 & 28 & 18 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 28 & 0 & 7 & 5 \\ 3 & 4 & 18 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{1431^1 1^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 5 & 0 & 24 & 4 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 21 & 0 & 100 & 18 \\ 3 & 4 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{131^1 1^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 8 & 4 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{121}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 7 & 4 \\ 4 & 8 & 10 & 0 & 10 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 10 & 0 & 7 & 5 \\ 3 & 4 & 6 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{124}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 10 & 3 \\ 3 & 4 & 13 & 0 & 19 & 4 \\ 7 & 13 & 28 & 0 & 43 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 43 & 0 & 64 & 14 \\ 3 & 4 & 10 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{154}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.3$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 10 & 3 \\ 3 & 4 & 20 & 0 & 19 & 4 \\ 10 & 20 & 73 & 0 & 70 & 15 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 70 & 0 & 64 & 14 \\ 3 & 4 & 15 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{132}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 138, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 6 & 3 \\ 3 & 4 & 12 & 0 & 11 & 4 \\ 6 & 12 & 25 & 0 & 24 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 24 & 0 & 20 & 8 \\ 3 & 4 & 9 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{122}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 16 & 4 \\ 4 & 8 & 10 & 0 & 22 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 22 & 0 & 44 & 12 \\ 3 & 4 & 6 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{143}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 8 & 3 \\ 3 & 4 & 16 & 0 & 15 & 4 \\ 8 & 16 & 46 & 0 & 44 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 44 & 0 & 39 & 11 \\ 3 & 4 & 12 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 \\ 2 & 4 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 5 \\ 3 & 4 & 3 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 158, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 3 \\ 3 & 4 & 9 & 0 & 11 & 4 \\ 5 & 9 & 13 & 0 & 17 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 17 & 0 & 20 & 8 \\ 3 & 4 & 7 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 13 & 0 & 14 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 14 & 0 & 12 & 6 \\ 3 & 4 & 7 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{122}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 6 & 3 \\ 3 & 4 & 8 & 0 & 11 & 4 \\ 4 & 8 & 10 & 0 & 16 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 16 & 0 & 20 & 8 \\ 3 & 4 & 6 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{\overline{111}3^1}1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 17 & 11 & 0 & 55 & 13 \\ 3 & 4 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{\overline{131}3^1}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 7 & 3 \\ 3 & 4 & 17 & 0 & 12 & 4 \\ 9 & 17 & 49 & 0 & 36 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 36 & 0 & 25 & 9 \\ 3 & 4 & 13 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{\overline{111}3^1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 7 & 4 \\ 5 & 9 & 13 & 0 & 11 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 11 & 0 & 7 & 5 \\ 3 & 4 & 7 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{\overline{113}3^1}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 12 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{\overline{133}3^1}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 20 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 18 & 0 & 69 & 15 \\ 3 & 4 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{\overline{133}3^1}1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 8 & 3 \\ 3 & 4 & 12 & 0 & 15 & 4 \\ 6 & 12 & 25 & 0 & 33 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 33 & 0 & 39 & 11 \\ 3 & 4 & 9 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{\overline{111}3^1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 8 & 0 & 25 & 9 \\ 3 & 4 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{\overline{112}3^1}1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 218, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 11 & 4 \\ 2 & 4 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 8 & 0 & 20 & 8 \\ 3 & 4 & 3 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{\overline{144}3^1}1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 10 & 3 \\ 3 & 4 & 16 & 0 & 19 & 4 \\ 8 & 16 & 46 & 0 & 56 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 56 & 0 & 64 & 14 \\ 3 & 4 & 12 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{\overline{131}3^1}1^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 4 & 3 \\ 3 & 4 & 32 & 0 & 7 & 4 \\ 18 & 32 & 193 & 0 & 39 & 25 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 39 & 0 & 7 & 5 \\ 3 & 4 & 25 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{123}3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 16 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 15 & 0 & 44 & 12 \\ 3 & 4 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{123}3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 8 & 3 \\ 3 & 4 & 13 & 0 & 15 & 4 \\ 7 & 13 & 28 & 0 & 34 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 34 & 0 & 39 & 11 \\ 3 & 4 & 10 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{123}3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 258, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 6 & 3 \\ 3 & 4 & 22 & 0 & 11 & 4 \\ 12 & 22 & 85 & 0 & 42 & 17 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 42 & 0 & 20 & 8 \\ 3 & 4 & 17 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{123}3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 262, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 4 & 8 & 0 & 15 & 4 \\ 4 & 8 & 10 & 0 & 22 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 22 & 0 & 39 & 11 \\ 3 & 4 & 6 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{134}3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 10 & 3 \\ 3 & 4 & 12 & 0 & 19 & 4 \\ 6 & 12 & 25 & 0 & 42 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 42 & 0 & 64 & 14 \\ 3 & 4 & 9 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{113}3^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 15 & 4 \\ 2 & 4 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 11 & 0 & 39 & 11 \\ 3 & 4 & 3 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{121}3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 298, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 4 & 3 \\ 3 & 4 & 18 & 0 & 7 & 4 \\ 10 & 18 & 58 & 0 & 22 & 14 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 22 & 0 & 7 & 5 \\ 3 & 4 & 14 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{121}3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 303, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 7 & 4 \\ 7 & 13 & 28 & 0 & 16 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 16 & 0 & 7 & 5 \\ 3 & 4 & 10 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{122}3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 308, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 6 & 3 \\ 3 & 4 & 13 & 0 & 11 & 4 \\ 7 & 13 & 28 & 0 & 25 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 25 & 0 & 20 & 8 \\ 3 & 4 & 10 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{142}3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 9 & 3 \\ 3 & 4 & 21 & 0 & 16 & 4 \\ 11 & 21 & 76 & 0 & 59 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 59 & 0 & 44 & 12 \\ 3 & 4 & 16 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{123}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 322, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 11 & 3 \\ 3 & 4 & 8 & 0 & 20 & 4 \\ 4 & 8 & 10 & 0 & 28 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 28 & 0 & 69 & 15 \\ 3 & 4 & 6 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{134}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 330, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 10 & 3 \\ 3 & 4 & 17 & 0 & 19 & 4 \\ 9 & 17 & 49 & 0 & 57 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 57 & 0 & 64 & 14 \\ 3 & 4 & 13 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{124}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 10 & 3 \\ 3 & 4 & 8 & 0 & 19 & 4 \\ 4 & 8 & 10 & 0 & 28 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 28 & 0 & 64 & 14 \\ 3 & 4 & 6 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{112}3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 362, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 31 & 20 & 0 & 188 & 24 \\ 3 & 4 & 3 & -3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{121}3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 367, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 12 & 4 \\ 7 & 13 & 28 & 0 & 28 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 28 & 0 & 25 & 9 \\ 3 & 4 & 10 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{114}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 19 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 14 & 0 & 64 & 14 \\ 3 & 4 & 3 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{111}3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 12 & 4 \\ 5 & 9 & 13 & 0 & 20 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 20 & 0 & 25 & 9 \\ 3 & 4 & 7 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[55] \ 1^{124}3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 16 & 3 \\ 3 & 4 & 8 & 0 & 29 & 4 \\ 4 & 8 & 10 & 0 & 40 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 29 & 40 & 0 & 148 & 22 \\ 3 & 4 & 6 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{131}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 393, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 4 & 3 \\ 3 & 4 & 17 & 0 & 7 & 4 \\ 9 & 17 & 49 & 0 & 21 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 21 & 0 & 7 & 5 \\ 3 & 4 & 13 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{112}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 395, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 11 & 0 & 44 & 12 \\ 3 & 4 & 3 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{133}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 417, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 8 & 3 \\ 3 & 4 & 17 & 0 & 15 & 4 \\ 9 & 17 & 49 & 0 & 45 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 45 & 0 & 39 & 11 \\ 3 & 4 & 13 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{132}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 438, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 6 & 3 \\ 3 & 4 & 17 & 0 & 11 & 4 \\ 9 & 17 & 49 & 0 & 33 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 33 & 0 & 20 & 8 \\ 3 & 4 & 13 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{141}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 463, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 4 & 3 \\ 3 & 4 & 21 & 0 & 7 & 4 \\ 11 & 21 & 76 & 0 & 26 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 26 & 0 & 7 & 5 \\ 3 & 4 & 16 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{112}3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 467, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 26 & 17 & 0 & 128 & 20 \\ 3 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{131}3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 473, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 4 & 3 \\ 3 & 4 & 27 & 0 & 7 & 4 \\ 15 & 27 & 133 & 0 & 33 & 21 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 33 & 0 & 7 & 5 \\ 3 & 4 & 21 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{112}3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 21 & 14 & 0 & 80 & 16 \\ 3 & 4 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{124}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 508, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 13 & 3 \\ 3 & 4 & 8 & 0 & 24 & 4 \\ 4 & 8 & 10 & 0 & 34 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 34 & 0 & 100 & 18 \\ 3 & 4 & 6 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{112}3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 515, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 16 & 4 \\ 5 & 9 & 13 & 0 & 26 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 26 & 0 & 44 & 12 \\ 3 & 4 & 7 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{144}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 520, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 10 & 3 \\ 3 & 4 & 21 & 0 & 19 & 4 \\ 11 & 21 & 76 & 0 & 71 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 71 & 0 & 64 & 14 \\ 3 & 4 & 16 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[67] \ 1^{131}3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 4 & 3 \\ 3 & 4 & 22 & 0 & 7 & 4 \\ 12 & 22 & 85 & 0 & 27 & 17 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 27 & 0 & 7 & 5 \\ 3 & 4 & 17 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{142}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 6 & 3 \\ 3 & 4 & 21 & 0 & 11 & 4 \\ 11 & 21 & 76 & 0 & 41 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 41 & 0 & 20 & 8 \\ 3 & 4 & 16 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{142}3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 6 & 3 \\ 3 & 4 & 26 & 0 & 11 & 4 \\ 14 & 26 & 118 & 0 & 50 & 20 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 50 & 0 & 20 & 8 \\ 3 & 4 & 20 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{132}3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 555, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 9 & 3 \\ 3 & 4 & 17 & 0 & 16 & 4 \\ 9 & 17 & 49 & 0 & 48 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 48 & 0 & 44 & 12 \\ 3 & 4 & 13 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{143}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 567, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 8 & 3 \\ 3 & 4 & 21 & 0 & 15 & 4 \\ 11 & 21 & 76 & 0 & 56 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 56 & 0 & 39 & 11 \\ 3 & 4 & 16 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{113}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 577, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 20 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 14 & 0 & 69 & 15 \\ 3 & 4 & 3 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{113}3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 592, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 20 & 4 \\ 5 & 9 & 13 & 0 & 32 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 32 & 0 & 69 & 15 \\ 3 & 4 & 7 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{114}3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 13 & 3 \\ 3 & 4 & 9 & 0 & 24 & 4 \\ 5 & 9 & 13 & 0 & 38 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 38 & 0 & 100 & 18 \\ 3 & 4 & 7 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{122}3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 620, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 9 & 3 \\ 3 & 4 & 13 & 0 & 16 & 4 \\ 7 & 13 & 28 & 0 & 37 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 37 & 0 & 44 & 12 \\ 3 & 4 & 10 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{141}3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 718, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 4 & 3 \\ 3 & 4 & 36 & 0 & 7 & 4 \\ 20 & 36 & 238 & 0 & 44 & 28 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 44 & 0 & 7 & 5 \\ 3 & 4 & 28 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[77] \ 1^{143}3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 727, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 11 & 3 \\ 3 & 4 & 21 & 0 & 20 & 4 \\ 11 & 21 & 76 & 0 & 74 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 74 & 0 & 69 & 15 \\ 3 & 4 & 16 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[78] \ 1^{141}3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 738, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 4 & 3 \\ 3 & 4 & 26 & 0 & 7 & 4 \\ 14 & 26 & 118 & 0 & 32 & 20 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 32 & 0 & 7 & 5 \\ 3 & 4 & 20 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[79] \ 1^{114}3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 763, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 17 & 0 & 100 & 18 \\ 3 & 4 & 3 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[80] \ 1^{113}3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 25 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 25 & 17 & 0 & 111 & 19 \\ 3 & 4 & 3 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{123}3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 807, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 11 & 3 \\ 3 & 4 & 13 & 0 & 20 & 4 \\ 7 & 13 & 28 & 0 & 46 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 46 & 0 & 69 & 15 \\ 3 & 4 & 10 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{141}3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 823, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 4 & 3 \\ 3 & 4 & 31 & 0 & 7 & 4 \\ 17 & 31 & 172 & 0 & 38 & 24 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 38 & 0 & 7 & 5 \\ 3 & 4 & 24 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[83] \ 1^{133}3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 852, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 11 & 3 \\ 3 & 4 & 17 & 0 & 20 & 4 \\ 9 & 17 & 49 & 0 & 60 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 60 & 0 & 69 & 15 \\ 3 & 4 & 13 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{113}3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 865, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 30 & 4 \\ 2 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 30 & 20 & 0 & 165 & 23 \\ 3 & 4 & 3 & -3 & 23 & 1 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{113}3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 868, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 35 & 4 \\ 2 & 4 & 1 & 0 & 23 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 35 & 23 & 0 & 231 & 27 \\ 3 & 4 & 3 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

$$[86] \ 1^{124}3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 928, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 13 & 3 \\ 3 & 4 & 13 & 0 & 24 & 4 \\ 7 & 13 & 28 & 0 & 55 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 55 & 0 & 100 & 18 \\ 3 & 4 & 10 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[87] \ 1^{114}3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1062, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 29 & 4 \\ 2 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 29 & 20 & 0 & 148 & 22 \\ 3 & 4 & 3 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[88] \ 1^{144}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1068, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 13 & 3 \\ 3 & 4 & 21 & 0 & 24 & 4 \\ 11 & 21 & 76 & 0 & 89 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 89 & 0 & 100 & 18 \\ 3 & 4 & 16 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[89] \ 1^{134}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1083, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 13 & 3 \\ 3 & 4 & 17 & 0 & 24 & 4 \\ 9 & 17 & 49 & 0 & 72 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 72 & 0 & 100 & 18 \\ 3 & 4 & 13 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[90] \ 1^{143}3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 19 & 3 \\ 3 & 4 & 4 & 0 & 34 & 4 \\ 2 & 4 & 1 & 0 & 23 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 19 & 34 & 23 & 0 & 208 & 26 \\ 3 & 4 & 3 & -3 & 26 & 1 \end{pmatrix}_{b_0}$$

$$[91] \ 1^{143}3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 22 & 3 \\ 3 & 4 & 4 & 0 & 39 & 4 \\ 2 & 4 & 1 & 0 & 26 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 22 & 39 & 26 & 0 & 280 & 30 \\ 3 & 4 & 3 & -3 & 30 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [92] \ 1^{111}3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 1 & 3 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 4 & 5 & 5 \\ 1 & 1 & 0 & 4 & -2 & 2 & 0 \\ 2 & 3 & 0 & 5 & 2 & 3 & 3 \\ 2 & 2 & -1 & 5 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[93] \ 1^{111}3^1 1^{13} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 5 & 3 \\ 2 & 4 & 3 & 3 & 0 & 6 & 3 \\ 1 & 3 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 6 & 5 & 0 & 7 & 5 \\ 2 & 3 & 3 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[94] \ 1^{113}3^1 1^{13} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 3 & 0 & 17 & 4 \\ 1 & 4 & 0 & 3 & 0 & 13 & 2 \\ 1 & 3 & 3 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 17 & 13 & 13 & 0 & 60 & 14 \\ 2 & 4 & 2 & 3 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[95] \ 1^{111}3^1 1^{13} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 5 & 4 \\ 1 & 4 & 0 & 3 & 0 & 4 & 2 \\ 1 & 3 & 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 2 & 3 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[96] \ 1^{111}3^1 1^{13} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.49 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 5 & 2 \\ 1 & 3 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 5 & 5 & 0 & 12 & 6 \\ 2 & 4 & 2 & 3 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

- [97] $1^{111}31^{131}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 4 & 3 \\ 2 & 4 & 3 & 3 & 0 & 5 & 3 \\ 1 & 3 & 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 4 & 0 & 4 & 4 \\ 2 & 3 & 3 & 3 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$
- [98] $1^{121}31^{111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 4 & 2 \\ 2 & 4 & 4 & 12 & 0 & 11 & 4 \\ 1 & 4 & 0 & 9 & 0 & 7 & 2 \\ 4 & 12 & 9 & 28 & 0 & 25 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 11 & 7 & 25 & 0 & 20 & 8 \\ 2 & 4 & 2 & 10 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$
- [99] $1^{111}31^{111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 5 & 4 \\ 1 & 4 & 0 & 3 & 0 & 4 & 2 \\ 2 & 6 & 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 5 & 0 & 4 & 4 \\ 2 & 4 & 2 & 4 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$
- [100] $1^{111}31^{111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 & 2 \\ 2 & 4 & 4 & 9 & 0 & 6 & 4 \\ 1 & 4 & 0 & 6 & 0 & 5 & 2 \\ 3 & 9 & 6 & 13 & 0 & 11 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 5 & 11 & 0 & 7 & 5 \\ 2 & 4 & 2 & 7 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$
- [101] $1^{111}31^{111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 8 & 4 \\ 1 & 4 & 0 & 6 & 0 & 4 & 2 \\ 3 & 9 & 6 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 4 & 12 & 0 & 9 & 5 \\ 2 & 4 & 2 & 7 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$
- [102] $1^{111}31^{141}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 & 2 \\ 2 & 4 & 4 & 27 & 0 & 5 & 4 \\ 1 & 4 & 0 & 15 & 0 & 4 & 2 \\ 9 & 27 & 15 & 121 & 0 & 24 & 19 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 24 & 0 & 4 & 4 \\ 2 & 4 & 2 & 19 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$
- [103] $1^{121}31^{111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 6 & 0 & 14 & 4 \\ 1 & 4 & 0 & 3 & 0 & 10 & 2 \\ 2 & 6 & 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 14 & 10 & 14 & 0 & 37 & 11 \\ 2 & 4 & 2 & 4 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$
- [104] $1^{111}31^{111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 3 & 0 & 11 & 4 \\ 1 & 4 & 0 & 3 & 0 & 7 & 2 \\ 1 & 3 & 3 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 11 & 7 & 7 & 0 & 20 & 8 \\ 2 & 4 & 2 & 3 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$
- [105] $1^{111}31^{111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 3 & 0 & 12 & 4 \\ 1 & 4 & 0 & 3 & 0 & 8 & 2 \\ 1 & 3 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 8 & 8 & 0 & 25 & 9 \\ 2 & 4 & 2 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[106] $1^{111}31^{12}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 3 & 0 & 17 & 4 \\ 1 & 4 & 0 & 3 & 0 & 10 & 2 \\ 1 & 3 & 3 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 17 & 10 & 10 & 0 & 48 & 12 \\ 2 & 4 & 2 & 3 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

[107] $1^{112}31^{11}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -209, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 3 & 0 & 14 & 4 \\ 1 & 4 & 0 & 3 & 0 & 10 & 2 \\ 1 & 3 & 3 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 14 & 10 & 10 & 0 & 37 & 11 \\ 2 & 4 & 2 & 3 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[108] $1^{121}31^{11}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 & 2 \\ 2 & 4 & 4 & 12 & 0 & 5 & 4 \\ 1 & 4 & 0 & 9 & 0 & 4 & 2 \\ 4 & 12 & 9 & 28 & 0 & 13 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 13 & 0 & 4 & 4 \\ 2 & 4 & 2 & 10 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[109] $1^{111}3^11^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 3 & 0 & 30 & 4 \\ 1 & 4 & 0 & 3 & 0 & 17 & 2 \\ 1 & 3 & 3 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 17 & 17 & 0 & 151 & 21 \\ 2 & 4 & 2 & 3 & -3 & 21 & 0 \end{pmatrix}_{b_0}$$

[110] $1^{111}31^{11}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 & 2 \\ 2 & 4 & 4 & 9 & 0 & 5 & 4 \\ 1 & 4 & 0 & 6 & 0 & 4 & 2 \\ 3 & 9 & 6 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 9 & 0 & 4 & 4 \\ 2 & 4 & 2 & 7 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[111] $1^{113}31^{11}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 6 & 2 \\ 2 & 4 & 4 & 9 & 0 & 17 & 4 \\ 1 & 4 & 0 & 6 & 0 & 13 & 2 \\ 3 & 9 & 6 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 17 & 13 & 30 & 0 & 60 & 14 \\ 2 & 4 & 2 & 7 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{111}3^11^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 3 & 0 & 18 & 4 \\ 1 & 4 & 0 & 3 & 0 & 11 & 2 \\ 1 & 3 & 3 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 11 & 11 & 0 & 55 & 13 \\ 2 & 4 & 2 & 3 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$

[113] $1^131^{111}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 8 & 4 \\ 1 & 4 & 0 & 3 & 0 & 4 & 2 \\ 2 & 6 & 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 4 & 8 & 0 & 9 & 5 \\ 2 & 4 & 2 & 4 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[114] $1^{111}31^{13}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -311, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 2 & 2 \\ 2 & 4 & 4 & 21 & 0 & 5 & 4 \\ 1 & 4 & 0 & 12 & 0 & 4 & 2 \\ 7 & 21 & 12 & 73 & 0 & 19 & 15 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 19 & 0 & 4 & 4 \\ 2 & 4 & 2 & 15 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{\overline{11}}3^{\overline{11}}1^{\overline{11}}31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -316, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 12 & 4 \\ 1 & 4 & 0 & 3 & 0 & 8 & 2 \\ 2 & 6 & 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 8 & 12 & 0 & 25 & 9 \\ 2 & 4 & 2 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{\overline{11}}3^{\overline{11}}1^{\overline{13}}31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -321, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.28 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 3 & 0 & 24 & 4 \\ 1 & 4 & 0 & 3 & 0 & 14 & 2 \\ 1 & 3 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 14 & 14 & 0 & 97 & 17 \\ 2 & 4 & 2 & 3 & -3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{\overline{14}}31^{\overline{12}}31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -334, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 3 & 0 & 26 & 4 \\ 1 & 4 & 0 & 3 & 0 & 19 & 2 \\ 1 & 3 & 3 & 1 & 0 & 19 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 26 & 19 & 19 & 0 & 135 & 21 \\ 2 & 4 & 2 & 3 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{\overline{11}}31^{\overline{12}}31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -356, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 2 & 2 \\ 2 & 4 & 4 & 15 & 0 & 5 & 4 \\ 1 & 4 & 0 & 9 & 0 & 4 & 2 \\ 5 & 15 & 9 & 37 & 0 & 14 & 11 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 14 & 0 & 4 & 4 \\ 2 & 4 & 2 & 11 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{\overline{1}}3^{\overline{11}}1^{\overline{11}}31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -369, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.05 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 5 & 2 \\ 2 & 6 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 5 & 9 & 0 & 12 & 6 \\ 2 & 4 & 2 & 4 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{\overline{11}}3^{\overline{11}}1^{\overline{11}}31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -374, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.06 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 9 & 4 \\ 1 & 4 & 0 & 6 & 0 & 5 & 2 \\ 3 & 9 & 6 & 13 & 0 & 14 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 5 & 14 & 0 & 12 & 6 \\ 2 & 4 & 2 & 7 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{\overline{14}}31^{\overline{12}}31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -386, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 & 2 \\ 2 & 4 & 4 & 24 & 0 & 5 & 4 \\ 1 & 4 & 0 & 18 & 0 & 4 & 2 \\ 8 & 24 & 18 & 118 & 0 & 26 & 20 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 26 & 0 & 4 & 4 \\ 2 & 4 & 2 & 20 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{\overline{11}}31^{\overline{11}}31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -445, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 11 & 4 \\ 1 & 4 & 0 & 3 & 0 & 7 & 2 \\ 2 & 6 & 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 11 & 7 & 11 & 0 & 20 & 8 \\ 2 & 4 & 2 & 4 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{\overline{12}}31^{\overline{12}}31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -454, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 3 & 0 & 20 & 4 \\ 1 & 4 & 0 & 3 & 0 & 13 & 2 \\ 1 & 3 & 3 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 20 & 13 & 13 & 0 & 71 & 15 \\ 2 & 4 & 2 & 3 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{122}31^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 5 & 2 \\ 2 & 4 & 4 & 12 & 0 & 14 & 4 \\ 1 & 4 & 0 & 9 & 0 & 10 & 2 \\ 4 & 12 & 9 & 28 & 0 & 34 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 14 & 10 & 34 & 0 & 37 & 11 \\ 2 & 4 & 2 & 10 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{113}31^{123}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 3 & 0 & 23 & 4 \\ 1 & 4 & 0 & 3 & 0 & 16 & 2 \\ 1 & 3 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 23 & 16 & 16 & 0 & 100 & 18 \\ 2 & 4 & 2 & 3 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{121}31^{123}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -586, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 2 & 2 \\ 2 & 4 & 4 & 18 & 0 & 5 & 4 \\ 1 & 4 & 0 & 12 & 0 & 4 & 2 \\ 6 & 18 & 12 & 58 & 0 & 18 & 14 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 18 & 0 & 4 & 4 \\ 2 & 4 & 2 & 14 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{131}31^{123}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -596, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 2 & 2 \\ 2 & 4 & 4 & 21 & 0 & 5 & 4 \\ 1 & 4 & 0 & 15 & 0 & 4 & 2 \\ 7 & 21 & 15 & 85 & 0 & 22 & 17 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 22 & 0 & 4 & 4 \\ 2 & 4 & 2 & 17 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{111}31^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -600, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 4 & 2 \\ 2 & 4 & 4 & 9 & 0 & 11 & 4 \\ 1 & 4 & 0 & 6 & 0 & 7 & 2 \\ 3 & 9 & 6 & 13 & 0 & 18 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 11 & 7 & 18 & 0 & 20 & 8 \\ 2 & 4 & 2 & 7 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{111}3^11^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -601, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 4 & 2 \\ 2 & 4 & 4 & 9 & 0 & 12 & 4 \\ 1 & 4 & 0 & 6 & 0 & 8 & 2 \\ 3 & 9 & 6 & 13 & 0 & 20 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 8 & 20 & 0 & 25 & 9 \\ 2 & 4 & 2 & 7 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[130] \ 1^{122}31^{133}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -601, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 3 & 0 & 26 & 4 \\ 1 & 4 & 0 & 3 & 0 & 16 & 2 \\ 1 & 3 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 26 & 16 & 16 & 0 & 117 & 19 \\ 2 & 4 & 2 & 3 & -3 & 19 & 0 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{122}31^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -649, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 5 & 2 \\ 2 & 4 & 4 & 9 & 0 & 14 & 4 \\ 1 & 4 & 0 & 6 & 0 & 10 & 2 \\ 3 & 9 & 6 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 14 & 10 & 24 & 0 & 37 & 11 \\ 2 & 4 & 2 & 7 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{122}31^{143}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -650, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 3 & 0 & 32 & 4 \\ 1 & 4 & 0 & 3 & 0 & 19 & 2 \\ 1 & 3 & 3 & 1 & 0 & 19 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 32 & 19 & 19 & 0 & 175 & 23 \\ 2 & 4 & 2 & 3 & -3 & 23 & 0 \end{pmatrix}_{b_0}$$

[133] $1^{121}31^{133}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -821, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 & 2 \\ 2 & 4 & 4 & 24 & 0 & 5 & 4 \\ 1 & 4 & 0 & 15 & 0 & 4 & 2 \\ 8 & 24 & 15 & 100 & 0 & 23 & 18 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 23 & 0 & 4 & 4 \\ 2 & 4 & 2 & 18 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{133}31^{133}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -871, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 3 & 0 & 29 & 4 \\ 1 & 4 & 0 & 3 & 0 & 19 & 2 \\ 1 & 3 & 3 & 1 & 0 & 19 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 29 & 19 & 19 & 0 & 152 & 22 \\ 2 & 4 & 2 & 3 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[135] $1^{121}31^{143}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -926, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 2 & 2 \\ 2 & 4 & 4 & 30 & 0 & 5 & 4 \\ 1 & 4 & 0 & 18 & 0 & 4 & 2 \\ 10 & 30 & 18 & 154 & 0 & 28 & 22 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 28 & 0 & 4 & 4 \\ 2 & 4 & 2 & 22 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[136] $1^{114}31^{133}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -929, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 11 & 2 \\ 2 & 4 & 4 & 3 & 0 & 32 & 4 \\ 1 & 4 & 0 & 3 & 0 & 22 & 2 \\ 1 & 3 & 3 & 1 & 0 & 22 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 32 & 22 & 22 & 0 & 193 & 25 \\ 2 & 4 & 2 & 3 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{131}31^{133}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 & 2 \\ 2 & 4 & 4 & 27 & 0 & 5 & 4 \\ 1 & 4 & 0 & 18 & 0 & 4 & 2 \\ 9 & 27 & 18 & 133 & 0 & 27 & 21 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 27 & 0 & 4 & 4 \\ 2 & 4 & 2 & 21 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[138] $1^{133}31^{143}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 12 & 2 \\ 2 & 4 & 4 & 3 & 0 & 35 & 4 \\ 1 & 4 & 0 & 3 & 0 & 22 & 2 \\ 1 & 3 & 3 & 1 & 0 & 22 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 35 & 22 & 22 & 0 & 216 & 26 \\ 2 & 4 & 2 & 3 & -3 & 26 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{141}31^{133}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 2 & 2 \\ 2 & 4 & 4 & 30 & 0 & 5 & 4 \\ 1 & 4 & 0 & 21 & 0 & 4 & 2 \\ 10 & 30 & 21 & 172 & 0 & 31 & 24 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 31 & 0 & 4 & 4 \\ 2 & 4 & 2 & 24 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{114}31^{143}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 13 & 2 \\ 2 & 4 & 4 & 3 & 0 & 38 & 4 \\ 1 & 4 & 0 & 3 & 0 & 25 & 2 \\ 1 & 3 & 3 & 1 & 0 & 25 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 38 & 25 & 25 & 0 & 263 & 29 \\ 2 & 4 & 2 & 3 & -3 & 29 & 1 \end{pmatrix}_{b_0}$$

[141] $1^{131}31^{143}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 1 & 2 & 2 \\ 2 & 4 & 4 & 33 & 0 & 5 & 4 \\ 1 & 4 & 0 & 21 & 0 & 4 & 2 \\ 11 & 33 & 21 & 193 & 0 & 32 & 25 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 32 & 0 & 4 & 4 \\ 2 & 4 & 2 & 25 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[142] $1^{141}31^{14}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -1846$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 1 & 2 & 2 \\ 2 & 4 & 4 & 36 & 0 & 5 & 4 \\ 1 & 4 & 0 & 24 & 0 & 4 & 2 \\ 12 & 36 & 24 & 238 & 0 & 36 & 28 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 36 & 0 & 4 & 4 \\ 2 & 4 & 2 & 28 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [143] $1^{11}4^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 17$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 2 & 0 & 1 & 3 \\ 3 & 4 & 3 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[144] $1^{11}3^11^14^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -79$, $\det = 19$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 3 & 0 & 5 & 4 \\ 3 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 3 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

4.830 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{so}_8$

$T = 9$ [0] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -212$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 12.64 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 15 & 5 \\ 11 & 20 & 17 & 0 & 28 & 8 \\ 9 & 17 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 28 & 24 & 0 & 37 & 11 \\ 5 & 8 & 7 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

4.831 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_2$

$T = 9$ [0] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.64 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 7 & 5 & 5 \\ 2 & 4 & 3 & 0 & 5 & 1 & 3 \\ 3 & 4 & 4 & -3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{1221}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -106$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 5 & 3 \\ 3 & 4 & 8 & 0 & 7 & 9 & 4 \\ 4 & 8 & 10 & 0 & 10 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 10 & 0 & 7 & 11 & 5 \\ 5 & 9 & 12 & 0 & 11 & 13 & 7 \\ 3 & 4 & 6 & -3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{1221}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -110$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 2 & 3 \\ 3 & 4 & 8 & 0 & 16 & 4 & 4 \\ 4 & 8 & 10 & 0 & 22 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 22 & 0 & 44 & 11 & 12 \\ 2 & 4 & 6 & 0 & 11 & 1 & 3 \\ 3 & 4 & 6 & -3 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{1311}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 4 & 2 & 3 \\ 3 & 4 & 12 & 0 & 7 & 4 & 4 \\ 6 & 12 & 25 & 0 & 15 & 9 & 9 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 15 & 0 & 7 & 5 & 5 \\ 2 & 4 & 9 & 0 & 5 & 1 & 3 \\ 3 & 4 & 9 & -3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{113}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -146$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 5 & 0 & 7 & 17 & 4 \\ 3 & 5 & 4 & 0 & 6 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 7 & 21 & 5 \\ 9 & 17 & 15 & 0 & 21 & 49 & 13 \\ 3 & 4 & 4 & -3 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -146$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 5 & 0 & 7 & 9 & 4 \\ 3 & 5 & 4 & 0 & 6 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 7 & 11 & 5 \\ 5 & 9 & 9 & 0 & 11 & 13 & 7 \\ 3 & 4 & 4 & -3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{1221}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -150$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 5 & 3 \\ 3 & 4 & 8 & 0 & 16 & 9 & 4 \\ 4 & 8 & 10 & 0 & 22 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 22 & 0 & 44 & 26 & 12 \\ 5 & 9 & 12 & 0 & 26 & 13 & 7 \\ 3 & 4 & 6 & -3 & 12 & 7 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{1212}3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -156$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 4 & 3 \\ 3 & 4 & 8 & 0 & 7 & 8 & 4 \\ 4 & 8 & 10 & 0 & 10 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 10 & 0 & 7 & 10 & 5 \\ 4 & 8 & 12 & 0 & 10 & 10 & 6 \\ 3 & 4 & 6 & -3 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{1311}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -164$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.71 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 7 & 2 & 3 \\ 3 & 4 & 17 & 0 & 12 & 4 & 4 \\ 9 & 17 & 49 & 0 & 36 & 12 & 13 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 36 & 0 & 25 & 8 & 9 \\ 2 & 4 & 12 & 0 & 8 & 1 & 3 \\ 3 & 4 & 13 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{1121}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -176$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 5 & 3 \\ 3 & 4 & 9 & 0 & 11 & 9 & 4 \\ 5 & 9 & 13 & 0 & 17 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 17 & 0 & 20 & 17 & 8 \\ 5 & 9 & 15 & 0 & 17 & 13 & 7 \\ 3 & 4 & 7 & -3 & 8 & 7 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{1111}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -194$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 12 & 4 & 4 \\ 3 & 5 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 12 & 0 & 25 & 8 & 9 \\ 2 & 4 & 3 & 0 & 8 & 1 & 3 \\ 3 & 4 & 4 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{1311}3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -194$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 5 & 0 & 20 & 4 & 4 \\ 3 & 5 & 4 & 0 & 18 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 18 & 0 & 69 & 14 & 15 \\ 2 & 4 & 3 & 0 & 14 & 1 & 3 \\ 3 & 4 & 4 & -3 & 15 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{1\overline{12}1}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{brown}{f}_4+\textcolor{red}{su}_2+\textcolor{blue}{su}_2 \quad T=9, \quad T_{\max}=12, \quad \Delta=-79, \quad \det=-216, \quad (n_+, n_-, n_0)=(1, 9, 0), \quad c_{\text{ext}}=11.7$$

$$T_{\min}=11.67 \rightarrow 12, \quad T_{\max}=12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 2 & 3 \\ 3 & 4 & 8 & 0 & 7 & 4 & 4 \\ 4 & 8 & 10 & 0 & 10 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 10 & 0 & 7 & 5 & 5 \\ 2 & 4 & 6 & 0 & 5 & 1 & 3 \\ 3 & 4 & 6 & -3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1\overline{12}1}3^11^{111}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{brown}{f}_4+\textcolor{red}{su}_2+\textcolor{blue}{su}_2 \quad T=9, \quad T_{\max}=12, \quad \Delta=-79, \quad \det=-216, \quad (n_+, n_-, n_0)=(1, 9, 0), \quad c_{\text{ext}}=11.7$$

$$T_{\min}=11.61 \rightarrow 12, \quad T_{\max}=12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 5 & 0 & 7 & 13 & 4 \\ 3 & 5 & 4 & 0 & 6 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 7 & 16 & 5 \\ 7 & 13 & 12 & 0 & 16 & 28 & 10 \\ 3 & 4 & 4 & -3 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1\overline{12}1}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{brown}{f}_4+\textcolor{red}{su}_2+\textcolor{blue}{su}_2 \quad T=9, \quad T_{\max}=12, \quad \Delta=-79, \quad \det=-246, \quad (n_+, n_-, n_0)=(1, 9, 0), \quad c_{\text{ext}}=11.2$$

$$T_{\min}=11.79 \rightarrow 12, \quad T_{\max}=12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 2 & 3 \\ 3 & 4 & 9 & 0 & 11 & 4 & 4 \\ 5 & 9 & 13 & 0 & 17 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 17 & 0 & 20 & 8 & 8 \\ 2 & 4 & 6 & 0 & 8 & 1 & 3 \\ 3 & 4 & 7 & -3 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1\overline{12}1}3^11^{111}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{brown}{f}_4+\textcolor{red}{su}_2+\textcolor{blue}{su}_2 \quad T=9, \quad T_{\max}=12, \quad \Delta=-79, \quad \det=-260, \quad (n_+, n_-, n_0)=(1, 9, 0), \quad c_{\text{ext}}=11.2$$

$$T_{\min}=11.60 \rightarrow 12, \quad T_{\max}=12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 5 & 0 & 16 & 4 & 4 \\ 3 & 5 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 15 & 0 & 44 & 11 & 12 \\ 2 & 4 & 3 & 0 & 11 & 1 & 3 \\ 3 & 4 & 4 & -3 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1\overline{11}1}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{brown}{f}_4+\textcolor{red}{su}_2+\textcolor{blue}{su}_2 \quad T=9, \quad T_{\max}=12, \quad \Delta=-79, \quad \det=-264, \quad (n_+, n_-, n_0)=(1, 9, 0), \quad c_{\text{ext}}=11.2$$

$$T_{\min}=11.65 \rightarrow 12, \quad T_{\max}=12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 & 4 \\ 2 & 4 & 1 & 0 & 8 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 8 & 0 & 25 & 8 & 9 \\ 2 & 4 & 3 & 0 & 8 & 1 & 3 \\ 3 & 4 & 3 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{1\overline{11}1}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{brown}{f}_4+\textcolor{red}{su}_2+\textcolor{blue}{su}_2 \quad T=9, \quad T_{\max}=12, \quad \Delta=-79, \quad \det=-276, \quad (n_+, n_-, n_0)=(1, 9, 0), \quad c_{\text{ext}}=11.2$$

$$T_{\min}=11.61 \rightarrow 12, \quad T_{\max}=12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 0 & 7 & 9 & 4 \\ 2 & 4 & 1 & 0 & 5 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 5 & 5 \\ 2 & 4 & 3 & 0 & 5 & 1 & 3 \\ 3 & 4 & 3 & -3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1\overline{11}1}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{brown}{f}_4+\textcolor{red}{su}_2+\textcolor{blue}{su}_2 \quad T=9, \quad T_{\max}=12, \quad \Delta=-79, \quad \det=-316, \quad (n_+, n_-, n_0)=(1, 9, 0), \quad c_{\text{ext}}=11.2$$

$$T_{\min}=11.59 \rightarrow 12, \quad T_{\max}=12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 5 & 3 \\ 3 & 4 & 4 & 0 & 7 & 9 & 4 \\ 2 & 4 & 1 & 0 & 5 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 11 & 5 \\ 5 & 9 & 6 & 0 & 11 & 13 & 7 \\ 3 & 4 & 3 & -3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1\overline{12}2}3^11^{11}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{brown}{f}_4+\textcolor{red}{su}_2+\textcolor{blue}{su}_2 \quad T=9, \quad T_{\max}=12, \quad \Delta=-79, \quad \det=-326, \quad (n_+, n_-, n_0)=(1, 9, 0), \quad c_{\text{ext}}=12.2$$

$$T_{\min}=11.64 \rightarrow 12, \quad T_{\max}=12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 7 & 3 \\ 3 & 4 & 8 & 0 & 7 & 13 & 4 \\ 4 & 8 & 10 & 0 & 10 & 18 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 10 & 0 & 7 & 16 & 5 \\ 7 & 13 & 18 & 0 & 16 & 28 & 10 \\ 3 & 4 & 6 & -3 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1121}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 6 & 2 & 3 \\ 3 & 4 & 8 & 0 & 11 & 4 & 4 \\ 4 & 8 & 10 & 0 & 16 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 16 & 0 & 20 & 8 & 8 \\ 2 & 4 & 6 & 0 & 8 & 1 & 3 \\ 3 & 4 & 6 & -3 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1112}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 10 & 3 \\ 3 & 4 & 4 & 0 & 7 & 18 & 4 \\ 2 & 4 & 1 & 0 & 5 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 22 & 5 \\ 10 & 18 & 12 & 0 & 22 & 58 & 14 \\ 3 & 4 & 3 & -3 & 5 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1112}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 2 & 3 \\ 3 & 4 & 4 & 0 & 11 & 4 & 4 \\ 2 & 4 & 1 & 0 & 8 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 8 & 0 & 20 & 8 & 8 \\ 2 & 4 & 3 & 0 & 8 & 1 & 3 \\ 3 & 4 & 3 & -3 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1112}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 7 & 3 \\ 3 & 4 & 9 & 0 & 11 & 13 & 4 \\ 5 & 9 & 13 & 0 & 17 & 21 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 17 & 0 & 20 & 25 & 8 \\ 7 & 13 & 21 & 0 & 25 & 28 & 10 \\ 3 & 4 & 7 & -3 & 8 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1111}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 5 & 3 \\ 3 & 4 & 9 & 0 & 7 & 9 & 4 \\ 5 & 9 & 13 & 0 & 11 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 11 & 0 & 7 & 11 & 5 \\ 5 & 9 & 15 & 0 & 11 & 13 & 7 \\ 3 & 4 & 7 & -3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1113}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 15 & 3 \\ 3 & 4 & 4 & 0 & 7 & 27 & 4 \\ 2 & 4 & 1 & 0 & 5 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 33 & 5 \\ 15 & 27 & 18 & 0 & 33 & 133 & 21 \\ 3 & 4 & 3 & -3 & 5 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1112}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 2 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 21 & 14 & 0 & 80 & 14 & 16 \\ 2 & 4 & 3 & 0 & 14 & 1 & 3 \\ 3 & 4 & 3 & -3 & 16 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1112}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -514, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 13 & 4 \\ 2 & 4 & 1 & 0 & 8 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 8 & 0 & 25 & 28 & 9 \\ 7 & 13 & 9 & 0 & 28 & 28 & 10 \\ 3 & 4 & 3 & -3 & 9 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{\overline{1112}3^1}1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 5 & 3 \\ 3 & 4 & 4 & 0 & 12 & 9 & 4 \\ 2 & 4 & 1 & 0 & 8 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 8 & 0 & 25 & 20 & 9 \\ 5 & 9 & 6 & 0 & 20 & 13 & 7 \\ 3 & 4 & 3 & -3 & 9 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{\overline{1112}3^1}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -536, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 7 & 3 \\ 3 & 4 & 4 & 0 & 7 & 13 & 4 \\ 2 & 4 & 1 & 0 & 5 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 16 & 5 \\ 7 & 13 & 9 & 0 & 16 & 28 & 10 \\ 3 & 4 & 3 & -3 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{\overline{1114}3^1}1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -546, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.66 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 11 & 3 \\ 3 & 4 & 9 & 0 & 7 & 21 & 4 \\ 5 & 9 & 13 & 0 & 11 & 33 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 11 & 0 & 7 & 26 & 5 \\ 11 & 21 & 33 & 0 & 26 & 76 & 16 \\ 3 & 4 & 7 & -3 & 5 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{\overline{1122}3^1}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -546, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.71 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 7 & 3 \\ 3 & 4 & 4 & 0 & 11 & 13 & 4 \\ 2 & 4 & 1 & 0 & 8 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 8 & 0 & 20 & 25 & 8 \\ 7 & 13 & 9 & 0 & 25 & 28 & 10 \\ 3 & 4 & 3 & -3 & 8 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{\overline{1131}3^1}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -564, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.81 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 4 & 4 & 0 & 15 & 4 & 4 \\ 2 & 4 & 1 & 0 & 11 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 11 & 0 & 39 & 11 & 11 \\ 2 & 4 & 3 & 0 & 11 & 1 & 3 \\ 3 & 4 & 3 & -3 & 11 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{\overline{1114}3^1}1^{\overline{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -616, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 20 & 3 \\ 3 & 4 & 4 & 0 & 7 & 36 & 4 \\ 2 & 4 & 1 & 0 & 5 & 24 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 44 & 5 \\ 20 & 36 & 24 & 0 & 44 & 238 & 28 \\ 3 & 4 & 3 & -3 & 5 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{\overline{1112}3^1}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -620, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 2 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 & 4 \\ 2 & 4 & 1 & 0 & 11 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 11 & 0 & 44 & 11 & 12 \\ 2 & 4 & 3 & 0 & 11 & 1 & 3 \\ 3 & 4 & 3 & -3 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{\overline{1112}3^1}1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -626, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 7 & 3 \\ 3 & 4 & 9 & 0 & 7 & 13 & 4 \\ 5 & 9 & 13 & 0 & 11 & 21 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 11 & 0 & 7 & 16 & 5 \\ 7 & 13 & 21 & 0 & 16 & 28 & 10 \\ 3 & 4 & 7 & -3 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{\overline{1113}1^{\overline{111}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -656, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 9 & 3 \\ 3 & 4 & 9 & 0 & 7 & 17 & 4 \\ 5 & 9 & 13 & 0 & 11 & 27 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 11 & 0 & 7 & 21 & 5 \\ 9 & 17 & 27 & 0 & 21 & 49 & 13 \\ 3 & 4 & 7 & -3 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{\overline{1113}3^{\overline{111}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -716, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 9 & 3 \\ 3 & 4 & 4 & 0 & 7 & 17 & 4 \\ 2 & 4 & 1 & 0 & 5 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 21 & 5 \\ 9 & 17 & 12 & 0 & 21 & 49 & 13 \\ 3 & 4 & 3 & -3 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{\overline{1141}3^{\overline{111}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -720, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 2 & 3 \\ 3 & 4 & 4 & 0 & 19 & 4 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 14 & 0 & 64 & 14 & 14 \\ 2 & 4 & 3 & 0 & 14 & 1 & 3 \\ 3 & 4 & 3 & -3 & 14 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{\overline{1121}3^{\overline{111}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -810, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 5 & 3 \\ 3 & 4 & 4 & 0 & 16 & 9 & 4 \\ 2 & 4 & 1 & 0 & 11 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 11 & 0 & 44 & 26 & 12 \\ 5 & 9 & 6 & 0 & 26 & 13 & 7 \\ 3 & 4 & 3 & -3 & 12 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{\overline{1113}3^{\overline{112}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -836, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 12 & 3 \\ 3 & 4 & 4 & 0 & 7 & 22 & 4 \\ 2 & 4 & 1 & 0 & 5 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 27 & 5 \\ 12 & 22 & 15 & 0 & 27 & 85 & 17 \\ 3 & 4 & 3 & -3 & 5 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{\overline{1131}3^{\overline{113}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -840, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 2 & 3 \\ 3 & 4 & 4 & 0 & 30 & 4 & 4 \\ 2 & 4 & 1 & 0 & 20 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 30 & 20 & 0 & 165 & 20 & 23 \\ 2 & 4 & 3 & 0 & 20 & 1 & 3 \\ 3 & 4 & 3 & -3 & 23 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{\overline{1114}3^{\overline{111}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -856, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 11 & 3 \\ 3 & 4 & 4 & 0 & 7 & 21 & 4 \\ 2 & 4 & 1 & 0 & 5 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 26 & 5 \\ 11 & 21 & 15 & 0 & 26 & 76 & 16 \\ 3 & 4 & 3 & -3 & 5 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{\overline{1212}3^{\overline{111}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 7 & 3 \\ 3 & 4 & 13 & 0 & 7 & 13 & 4 \\ 7 & 13 & 28 & 0 & 16 & 30 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 16 & 0 & 7 & 16 & 5 \\ 7 & 13 & 30 & 0 & 16 & 28 & 10 \\ 3 & 4 & 10 & -3 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1131}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -964, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.62 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 5 & 3 \\ 3 & 4 & 4 & 0 & 20 & 9 & 4 \\ 2 & 4 & 1 & 0 & 14 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 14 & 0 & 69 & 32 & 15 \\ 5 & 9 & 6 & 0 & 32 & 13 & 7 \\ 3 & 4 & 3 & -3 & 15 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1131}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -984, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 2 & 3 \\ 3 & 4 & 4 & 0 & 20 & 4 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 14 & 0 & 69 & 14 & 15 \\ 2 & 4 & 3 & 0 & 14 & 1 & 3 \\ 3 & 4 & 3 & -3 & 15 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1141}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -986, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.67 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 13 & 2 & 3 \\ 3 & 4 & 9 & 0 & 24 & 4 & 4 \\ 5 & 9 & 13 & 0 & 38 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 38 & 0 & 100 & 17 & 18 \\ 2 & 4 & 6 & 0 & 17 & 1 & 3 \\ 3 & 4 & 7 & -3 & 18 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1122}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1020, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 7 & 3 \\ 3 & 4 & 4 & 0 & 16 & 13 & 4 \\ 2 & 4 & 1 & 0 & 11 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 11 & 0 & 44 & 37 & 12 \\ 7 & 13 & 9 & 0 & 37 & 28 & 10 \\ 3 & 4 & 3 & -3 & 12 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1131}3^11^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1076, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 2 & 3 \\ 3 & 4 & 4 & 0 & 25 & 4 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 25 & 17 & 0 & 111 & 17 & 19 \\ 2 & 4 & 3 & 0 & 17 & 1 & 3 \\ 3 & 4 & 3 & -3 & 19 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1114}3^11^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1176, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 17 & 3 \\ 3 & 4 & 4 & 0 & 7 & 31 & 4 \\ 2 & 4 & 1 & 0 & 5 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 38 & 5 \\ 17 & 31 & 21 & 0 & 38 & 172 & 24 \\ 3 & 4 & 3 & -3 & 5 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1114}3^11^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.61 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 14 & 3 \\ 3 & 4 & 4 & 0 & 7 & 26 & 4 \\ 2 & 4 & 1 & 0 & 5 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 32 & 5 \\ 14 & 26 & 18 & 0 & 32 & 118 & 20 \\ 3 & 4 & 3 & -3 & 5 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1141}3^11^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1296, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 5 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 4 & 2 & 2 \\ 1 & 3 & 3 & 1 & 0 & 4 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 4 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 175, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 5 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 4 & 2 & 2 \\ 2 & 6 & 3 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 5 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 3 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 475, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 5 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 4 & 2 & 2 \\ 3 & 9 & 6 & 13 & 0 & 9 & 5 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 9 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 5 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & 7 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

4.833 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2\mathfrak{n}3\mathfrak{mix}$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 8 & 4 & 4 \\ 3 & 5 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 9 & 0 & 12 & 5 & 6 \\ 2 & 4 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 2 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 & 4 \\ 2 & 4 & 1 & 0 & 11 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 17 & 11 & 0 & 55 & 11 & 13 \\ 2 & 4 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & 3 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 7 & 5 & 5 \\ 2 & 4 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 & 3 \\ 3 & 4 & 4 & 0 & 11 & 5 & 4 \\ 2 & 4 & 1 & 0 & 8 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 8 & 0 & 20 & 9 & 8 \\ 3 & 5 & 3 & 0 & 9 & 3 & 3 \\ 3 & 4 & 3 & -3 & 8 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 4 & 0 & 7 & 5 & 4 \\ 2 & 4 & 1 & 0 & 5 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 6 & 5 \\ 3 & 5 & 3 & 0 & 6 & 3 & 3 \\ 3 & 4 & 3 & -3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -269, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 2 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 & 4 \\ 5 & 9 & 13 & 0 & 14 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 14 & 0 & 12 & 5 & 6 \\ 2 & 4 & 6 & 0 & 5 & 0 & 2 \\ 3 & 4 & 7 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{131}} 3^{\overline{11}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 5 & 0 & 20 & 4 & 4 \\ 3 & 5 & 4 & 0 & 18 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 18 & 0 & 69 & 14 & 15 \\ 2 & 4 & 3 & 0 & 14 & 0 & 2 \\ 3 & 4 & 4 & -3 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 12 & 4 & 4 \\ 3 & 5 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 12 & 0 & 25 & 8 & 9 \\ 2 & 4 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & 4 & -3 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -424, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.07 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 & 4 \\ 2 & 4 & 1 & 0 & 5 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 5 & 5 \\ 2 & 4 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & 3 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 5 & 3 \\ 3 & 4 & 4 & 0 & 7 & 9 & 4 \\ 2 & 4 & 1 & 0 & 5 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 11 & 5 \\ 5 & 9 & 6 & 0 & 11 & 12 & 6 \\ 3 & 4 & 3 & -3 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -481, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 & 4 \\ 2 & 4 & 1 & 0 & 8 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 8 & 0 & 25 & 8 & 9 \\ 2 & 4 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & 3 & -3 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{121}} 3^{\overline{11}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -500, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 5 & 0 & 16 & 4 & 4 \\ 3 & 5 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 15 & 0 & 44 & 11 & 12 \\ 2 & 4 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & 4 & -3 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -509, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.06 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 2 & 3 \\ 3 & 4 & 9 & 0 & 7 & 4 & 4 \\ 5 & 9 & 13 & 0 & 11 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 11 & 0 & 7 & 5 & 5 \\ 2 & 4 & 6 & 0 & 5 & 0 & 2 \\ 3 & 4 & 7 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1111}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -896, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 2 & 3 \\ 3 & 4 & 9 & 0 & 12 & 4 & 4 \\ 5 & 9 & 13 & 0 & 20 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 20 & 0 & 25 & 8 & 9 \\ 2 & 4 & 6 & 0 & 8 & 0 & 2 \\ 3 & 4 & 7 & -3 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

4.834 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 49, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 15.51 \rightarrow 16, \ T_{\max} = 28, \ T \in [16, 28]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 3 & -3 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{132}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 50, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 28, \ T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 7 & 3 \\ 3 & 4 & 12 & 0 & 13 & 4 \\ 6 & 12 & 25 & 0 & 27 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 27 & 0 & 27 & 9 \\ 3 & 4 & 9 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 61, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.11 \rightarrow 14, \ T_{\max} = 28, \ T \in [14, 28]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 6 & 0 & 12 & 6 \\ 3 & 5 & 3 & -3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.81 \rightarrow 12, \ T_{\max} = 28, \ T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 82, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 28, \ T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 3 & 4 & 3 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 96, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.37 \rightarrow 13, \ T_{\max} = 28, \ T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 6 & 0 & 3 & 3 \\ 3 & 4 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 96, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 28, \ T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 12 & 0 & 12 & 6 \\ 3 & 4 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 98, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.22 \rightarrow 13, \ T_{\max} = 28, \ T \in [13, 28]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 116, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 3 \\ 4 & 9 & 12 & 0 & 4 & 5 \\ 5 & 12 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 5 & 7 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 48 & 12 \\ 3 & 4 & 4 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{112} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 1 & 3 \\ 4 & 9 & 16 & 4 & 0 & 5 \\ 6 & 16 & 22 & 6 & 0 & 8 \\ 2 & 4 & 6 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 8 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{112} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 137, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{121} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.11 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 4 & 8 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 6 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 19 & 0 & 4 & 4 \\ 11 & 19 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 15 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 12 & 0 & 9 & 5 \\ 3 & 4 & 7 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{122} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 12 & 0 & 9 & 5 \\ 3 & 4 & 6 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{113} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 9 & 0 & 24 & 8 \\ 3 & 4 & 3 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{131} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 4 & 4 \\ 6 & 12 & 25 & 0 & 9 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 9 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 159, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.34 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 0 & 36 & 10 \\ 3 & 4 & 3 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{132}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 4 & 3 \\ 3 & 4 & 12 & 0 & 8 & 4 \\ 6 & 12 & 25 & 0 & 18 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 18 & 0 & 9 & 5 \\ 3 & 4 & 9 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{122}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 4 & 3 \\ 3 & 4 & 18 & 0 & 8 & 4 \\ 10 & 18 & 58 & 0 & 24 & 14 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 24 & 0 & 9 & 5 \\ 3 & 4 & 14 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{11}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{141}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 16 & 0 & 4 & 4 \\ 8 & 16 & 46 & 0 & 12 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 12 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 1 & 3 \\ 4 & 9 & 24 & 4 & 0 & 5 \\ 9 & 24 & 52 & 9 & 0 & 12 \\ 2 & 4 & 9 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 12 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{12}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 12 & 0 & 27 & 9 \\ 3 & 4 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 3 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{132}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 197, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.16 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 4 & 3 \\ 3 & 4 & 27 & 0 & 8 & 4 \\ 15 & 27 & 133 & 0 & 36 & 21 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 36 & 0 & 9 & 5 \\ 3 & 4 & 21 & -3 & 5 & -1 \end{pmatrix}_{b_0}$$

[28] $1^{112}31^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 2 & 3 \\ 4 & 9 & 24 & 0 & 4 & 5 \\ 10 & 24 & 58 & 0 & 12 & 14 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 5 & 14 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{151}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 20 & 0 & 4 & 4 \\ 10 & 20 & 73 & 0 & 15 & 15 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 15 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{122}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 18 & 0 & 27 & 9 \\ 3 & 4 & 6 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 5 & 9 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 9 & 0 & 3 & 3 \\ 3 & 4 & 7 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{141}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 & 3 \\ 3 & 4 & 21 & 0 & 5 & 4 \\ 11 & 21 & 76 & 0 & 18 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 18 & 0 & 3 & 3 \\ 3 & 4 & 16 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{161}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 13.42 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 24 & 0 & 4 & 4 \\ 12 & 24 & 106 & 0 & 18 & 18 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 18 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

[34] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 \\ 5 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 7 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{111}3^11^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 \\ 8 & 14 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 11 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{114}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 1 & 3 \\ 4 & 9 & 32 & 4 & 0 & 5 \\ 12 & 32 & 94 & 12 & 0 & 16 \\ 2 & 4 & 12 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 16 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{112}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 2 & 3 \\ 4 & 9 & 20 & 0 & 4 & 5 \\ 8 & 20 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 5 & 11 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{171}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 275, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.69 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 4 & 4 \\ 14 & 28 & 145 & 0 & 21 & 21 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 3 & 4 & 21 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

[39] $1^{12}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 \\ 7 & 13 & 28 & 0 & 12 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 12 & 0 & 3 & 3 \\ 3 & 4 & 10 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{13}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 294, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 \\ 9 & 17 & 49 & 0 & 15 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 15 & 0 & 3 & 3 \\ 3 & 4 & 13 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 306, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 0 & 27 & 9 \\ 3 & 4 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 8 & 4 \\ 7 & 13 & 28 & 0 & 18 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 18 & 0 & 9 & 5 \\ 3 & 4 & 10 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{115}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 338, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 1 & 3 \\ 4 & 9 & 40 & 4 & 0 & 5 \\ 15 & 40 & 148 & 15 & 0 & 20 \\ 2 & 4 & 15 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 20 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[44] $1^{113}3^11^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 2 & 3 \\ 4 & 9 & 36 & 0 & 4 & 5 \\ 15 & 36 & 133 & 0 & 18 & 21 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 5 & 21 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 371, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 15 & 0 & 12 & 6 \\ 3 & 4 & 7 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{112}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 0 & 57 & 13 \\ 3 & 4 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 411, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 7 & 13 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 10 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 0 & 48 & 12 \\ 3 & 4 & 3 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 11 & 1 & 2 & 3 \\ 4 & 9 & 28 & 0 & 4 & 5 \\ 11 & 28 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 5 & 15 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 21 & 0 & 27 & 9 \\ 3 & 4 & 7 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 463, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 27 & 0 & 48 & 12 \\ 3 & 4 & 7 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{132}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 489, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 4 & 3 \\ 3 & 4 & 17 & 0 & 8 & 4 \\ 9 & 17 & 49 & 0 & 24 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 24 & 0 & 9 & 5 \\ 3 & 4 & 13 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{113}3^11^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 490, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 2 & 3 \\ 4 & 9 & 32 & 0 & 4 & 5 \\ 13 & 32 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 5 & 18 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{132}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 501, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 4 & 3 \\ 3 & 4 & 22 & 0 & 8 & 4 \\ 12 & 22 & 85 & 0 & 30 & 17 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 30 & 0 & 9 & 5 \\ 3 & 4 & 17 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 507, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 7 & 13 & 28 & 0 & 21 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 21 & 0 & 12 & 6 \\ 3 & 4 & 10 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{114}3^11^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 2 & 3 \\ 4 & 9 & 48 & 0 & 4 & 5 \\ 20 & 48 & 238 & 0 & 24 & 28 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 24 & 0 & 0 & 2 \\ 3 & 5 & 28 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{121}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 576, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 4 & 4 \\ 10 & 18 & 58 & 0 & 12 & 14 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 14 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{131}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 579, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 5 & 3 \\ 3 & 4 & 17 & 0 & 9 & 4 \\ 9 & 17 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 27 & 0 & 12 & 6 \\ 3 & 4 & 13 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{141} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 587, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 5 & 3 \\ 3 & 4 & 21 & 0 & 9 & 4 \\ 11 & 21 & 76 & 0 & 33 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 33 & 0 & 12 & 6 \\ 3 & 4 & 16 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{131} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 9 & 17 & 49 & 0 & 12 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 13 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[61] $1^{121} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 4 & 4 \\ 16 & 28 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 22 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 607, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 15 & 0 & 84 & 16 \\ 3 & 4 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[63] $1^{114} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 626, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 2 & 3 \\ 4 & 9 & 36 & 0 & 4 & 5 \\ 14 & 36 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 5 & 19 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{121} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 635, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 23 & 0 & 4 & 4 \\ 13 & 23 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 18 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{122} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 30 & 0 & 27 & 9 \\ 3 & 4 & 10 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{114} 3^1 1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 770, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 2 & 3 \\ 4 & 9 & 44 & 0 & 4 & 5 \\ 18 & 44 & 193 & 0 & 21 & 25 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 3 & 5 & 25 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{141} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 771, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 4 & 4 \\ 11 & 21 & 76 & 0 & 15 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 16 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{132} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 798, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 7 & 3 \\ 3 & 4 & 17 & 0 & 13 & 4 \\ 9 & 17 & 49 & 0 & 39 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 39 & 0 & 27 & 9 \\ 3 & 4 & 13 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{114}31^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 2 & 3 \\ 4 & 9 & 40 & 0 & 4 & 5 \\ 16 & 40 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 5 & 22 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{131}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 915, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 4 & 4 \\ 12 & 22 & 85 & 0 & 15 & 17 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 17 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{131}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 27 & 0 & 4 & 4 \\ 15 & 27 & 133 & 0 & 18 & 21 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 21 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{131}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1251, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 2 & 3 \\ 3 & 4 & 32 & 0 & 4 & 4 \\ 18 & 32 & 193 & 0 & 21 & 25 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 3 & 4 & 25 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{141}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 26 & 0 & 4 & 4 \\ 14 & 26 & 118 & 0 & 18 & 20 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 20 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{141}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1643, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 3 \\ 3 & 4 & 31 & 0 & 4 & 4 \\ 17 & 31 & 172 & 0 & 21 & 24 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 3 & 4 & 24 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{141}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1920, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 2 & 3 \\ 3 & 4 & 36 & 0 & 4 & 4 \\ 20 & 36 & 238 & 0 & 24 & 28 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 24 & 0 & 0 & 2 \\ 3 & 4 & 28 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [76] $1^{111}31^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 2 \\ 2 & 3 & 3 & 3 & -3 & 2 \end{pmatrix}_{b_0}$$

[77] $1^{11}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & 2 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 1 & 3 \\ 3 & 5 & 4 & -3 & 3 & 2 \end{pmatrix}_{b_0}$$

[78] $1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.45 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 6 & 0 & 3 & 3 \\ 2 & 3 & 6 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & 4 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 19 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[80] $1^{111}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 2 \\ 1 & 2 & 0 & 3 & 0 & 4 & 2 \\ 1 & 1 & 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 2 & 2 & 3 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{111}321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 1 & 2 & 0 & 3 & 0 & 7 & 2 \\ 1 & 1 & 3 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 4 & 7 & 7 & 0 & 15 & 7 \\ 2 & 2 & 2 & 3 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[83] $1^{111}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 1 & 3 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & 3 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

[82] $1^{13}321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 3 & 0 & 9 & 2 \\ 1 & 1 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 9 & 9 & 0 & 39 & 11 \\ 2 & 2 & 2 & 3 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

[84] $1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 6 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 3 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{11}31^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 2 \\ 4 & 12 & 6 & 22 & 0 & 6 & 8 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 6 & 0 & 0 & 2 \\ 2 & 4 & 2 & 8 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[86] $1^{111}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 5 & 5 & 0 & 10 & 5 \\ 2 & 5 & 2 & 0 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 10 & 7 & 0 & 13 & 7 \\ 3 & 5 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 1 & 2 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 3 & 0 & 4 & 4 \\ 2 & 2 & 2 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{111}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 1 & 2 \\ 1 & 2 & 0 & 4 & 0 & 3 & 2 \\ 2 & 3 & 4 & 5 & 0 & 4 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 1 & 3 & 4 & 0 & 0 & 2 \\ 2 & 2 & 2 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[89] \ 1^{12} 3^{21} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 2 \\ 1 & 2 & 0 & 3 & 0 & 8 & 2 \\ 1 & 1 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 8 & 0 & 25 & 9 \\ 2 & 2 & 2 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[90] \ 1^{11} 3^{21} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 5 & 10 & 0 & 5 & 5 \\ 5 & 10 & 14 & 0 & 7 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 7 & 0 & 1 & 3 \\ 3 & 5 & 8 & -3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[91] \ 1^{11} 3^{11} 1^{13} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -137, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & 4 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[93] \ 1^{11} 3^{21} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 5 & 5 & 0 & 15 & 5 \\ 2 & 5 & 2 & 0 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 15 & 11 & 0 & 31 & 11 \\ 3 & 5 & 4 & -3 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[92] \ 1^{11} 3^{21} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 5 & 5 & 0 & 25 & 5 \\ 2 & 5 & 2 & 0 & 19 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 25 & 19 & 0 & 91 & 19 \\ 3 & 5 & 4 & -3 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[94] \ 1^{11} 3^{21} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -141, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 1 & 7 & 0 & 3 \\ 5 & 9 & 7 & 14 & 0 & 8 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[95] \ 1^{11} 3^{21} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 5 & 5 & 0 & 20 & 5 \\ 2 & 5 & 2 & 0 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 15 & 0 & 57 & 15 \\ 3 & 5 & 4 & -3 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[96] \ 1^{12} 3^{21} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 1 & 2 & 1 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 6 & 0 & 25 & 9 \\ 2 & 2 & 2 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[97] \ 1^{11} 3^{11} 1^{13} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 1 & 3 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 6 & 0 & 12 & 6 \\ 2 & 4 & 2 & 3 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[98] \ 1^{13} 3^{11} 1^{13} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 1 & 1 & 2 \\ 1 & 1 & 4 & 12 & 0 & 3 & 3 \\ 2 & 4 & 3 & 15 & 0 & 3 & 3 \\ 5 & 12 & 15 & 49 & 0 & 12 & 13 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 12 & 0 & 0 & 2 \\ 2 & 3 & 3 & 13 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[99] \ 1^{112}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 1 & 2 & 1 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 5 & 0 & 12 & 6 \\ 2 & 2 & 2 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[100] \ 1^{111}3^{11}21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 0 & 4 & 2 \\ 1 & 2 & 0 & 4 & 0 & 7 & 2 \\ 2 & 3 & 4 & 5 & 0 & 11 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 4 & 7 & 11 & 0 & 15 & 7 \\ 2 & 2 & 2 & 5 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[101] \ 1^{111}3^{11}21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 0 & 3 & 2 \\ 1 & 2 & 0 & 4 & 0 & 4 & 2 \\ 2 & 3 & 4 & 5 & 0 & 7 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 7 & 0 & 4 & 4 \\ 2 & 2 & 2 & 5 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[102] \ 1^{111}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 1 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 & 3 \\ 2 & 4 & 3 & 9 & 0 & 3 & 3 \\ 3 & 6 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 6 & 0 & 0 & 2 \\ 2 & 3 & 3 & 7 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{112}3^{11}21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -311, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 0 & 5 & 2 \\ 1 & 2 & 0 & 4 & 0 & 5 & 2 \\ 2 & 3 & 4 & 5 & 0 & 10 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 10 & 0 & 12 & 6 \\ 2 & 2 & 2 & 5 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{111}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.31 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 2 \\ 3 & 9 & 6 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 6 & 0 & 0 & 2 \\ 2 & 4 & 2 & 7 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{121}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -332, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 1 & 1 & 2 \\ 1 & 1 & 4 & 9 & 0 & 3 & 3 \\ 2 & 4 & 3 & 12 & 0 & 3 & 3 \\ 4 & 9 & 12 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 9 & 0 & 0 & 2 \\ 2 & 3 & 3 & 10 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[106] \ 1^131^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 6 & 0 & 3 & 3 \\ 2 & 4 & 2 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[107] $1^{111}31^{111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -417, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 6 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 9 & 0 & 12 & 6 \\ 2 & 4 & 2 & 4 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[108] $1^{111}31^{111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 & 2 \\ 2 & 4 & 4 & 9 & 0 & 6 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 2 \\ 3 & 9 & 6 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 9 & 0 & 3 & 3 \\ 2 & 4 & 2 & 7 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[109] $1^{112}321^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -481, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 0 & 6 & 2 \\ 1 & 2 & 0 & 4 & 0 & 8 & 2 \\ 2 & 3 & 4 & 5 & 0 & 14 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 14 & 0 & 25 & 9 \\ 2 & 2 & 2 & 5 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[110] $1^{111}31^{123}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -587, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 3 & 4 \\ 1 & 4 & 0 & 9 & 0 & 3 & 2 \\ 5 & 15 & 9 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 9 & 0 & 0 & 2 \\ 2 & 4 & 2 & 11 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[111] $1^{111}31^{111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -699, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 9 & 4 \\ 1 & 4 & 0 & 6 & 0 & 6 & 2 \\ 3 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 15 & 0 & 12 & 6 \\ 2 & 4 & 2 & 7 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[112] $1^{111}31^{13}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -818, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 3 & 4 \\ 1 & 4 & 0 & 12 & 0 & 3 & 2 \\ 7 & 21 & 12 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 12 & 0 & 0 & 2 \\ 2 & 4 & 2 & 15 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[113] $1^{111}31^{14}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -1007, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 3 & 4 \\ 1 & 4 & 0 & 15 & 0 & 3 & 2 \\ 9 & 27 & 15 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 15 & 0 & 0 & 2 \\ 2 & 4 & 2 & 19 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [114] $1^{111}2321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3$ $T = 10, T_{\max} = 12, \Delta = -61, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 1 & 4 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 4 & 4 & 6 \\ 1 & 1 & 0 & 4 & -2 & 1 & 0 \\ 1 & 3 & 0 & 4 & 1 & 0 & 2 \\ 2 & 3 & -1 & 6 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

4.835 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 0 & 4 \\ 2 & 4 & 3 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & 0 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 8 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

4.836 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 5 & 0 & 4 & 19 & 4 \\ 3 & 5 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 11 & 19 & 21 & 0 & 12 & 73 & 15 \\ 3 & 4 & 4 & -3 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 8 & 14 & 9 & 0 & 9 & 37 & 11 \\ 3 & 4 & 3 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 5 & 3 \\ 3 & 4 & 19 & 0 & 4 & 9 & 4 \\ 11 & 19 & 73 & 0 & 12 & 33 & 15 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 6 & 2 \\ 5 & 9 & 33 & 0 & 6 & 13 & 7 \\ 3 & 4 & 15 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1211} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 2 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 12 & 0 & 12 & 6 & 6 \\ 2 & 4 & 6 & 0 & 6 & 1 & 3 \\ 3 & 4 & 6 & -3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -173, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 & 4 \\ 3 & 5 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 12 & 6 & 6 \\ 2 & 4 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 4 & -3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1211} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -191, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 & 4 \\ 3 & 5 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 12 & 0 & 27 & 9 & 9 \\ 2 & 4 & 3 & 0 & 9 & 1 & 3 \\ 3 & 4 & 4 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -268, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -269, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1111} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -275, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.84 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 5 & 0 & 4 & 14 & 4 \\ 3 & 5 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 8 & 14 & 15 & 0 & 9 & 37 & 11 \\ 3 & 4 & 4 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1121} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -296, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.99 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 13 & 4 \\ 3 & 5 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 7 & 13 & 12 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -320, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 6 & 6 \\ 2 & 4 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 3 & -3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{1112}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 13 & 3 \\ 3 & 4 & 4 & 0 & 4 & 23 & 4 \\ 2 & 4 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 13 & 23 & 15 & 0 & 15 & 100 & 18 \\ 3 & 4 & 3 & -3 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1212}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 7 & 3 \\ 3 & 4 & 8 & 0 & 9 & 13 & 4 \\ 4 & 8 & 10 & 0 & 12 & 18 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 12 & 0 & 12 & 21 & 6 \\ 7 & 13 & 18 & 0 & 21 & 28 & 10 \\ 3 & 4 & 6 & -3 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 15 & 6 \\ 5 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 3 & -3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1111}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -605, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 8 & 3 \\ 3 & 4 & 9 & 0 & 4 & 14 & 4 \\ 5 & 9 & 13 & 0 & 6 & 24 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 9 & 2 \\ 8 & 14 & 24 & 0 & 9 & 37 & 11 \\ 3 & 4 & 7 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{112}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 5 & 0 & 4 & 18 & 4 \\ 3 & 5 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 10 & 18 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{112}31^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -650, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 5 & 0 & 4 & 28 & 4 \\ 3 & 5 & 4 & 0 & 3 & 30 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 16 & 28 & 30 & 0 & 18 & 154 & 22 \\ 3 & 4 & 4 & -3 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 4 & 9 & 4 \\ 5 & 9 & 13 & 0 & 6 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 5 & 9 & 15 & 0 & 6 & 13 & 7 \\ 3 & 4 & 7 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{111111}31^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -716, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 8 & 3 \\ 3 & 4 & 14 & 0 & 4 & 14 & 4 \\ 8 & 14 & 37 & 0 & 9 & 39 & 11 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 9 & 2 \\ 8 & 14 & 39 & 0 & 9 & 37 & 11 \\ 3 & 4 & 11 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{1112}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -723, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 4 & 13 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 7 & 13 & 9 & 0 & 9 & 28 & 10 \\ 3 & 4 & 3 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{1112}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -738, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 4 & 18 & 4 \\ 2 & 4 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 10 & 18 & 12 & 0 & 12 & 58 & 14 \\ 3 & 4 & 3 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{112}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 5 & 0 & 4 & 23 & 4 \\ 3 & 5 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 13 & 23 & 24 & 0 & 15 & 100 & 18 \\ 3 & 4 & 4 & -3 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{113}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -779, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 5 & 0 & 4 & 22 & 4 \\ 3 & 5 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 12 & 22 & 21 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{1113}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -828, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 18 & 3 \\ 3 & 4 & 4 & 0 & 4 & 32 & 4 \\ 2 & 4 & 1 & 0 & 3 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 18 & 32 & 21 & 0 & 21 & 193 & 25 \\ 3 & 4 & 3 & -3 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{1112}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -845, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 7 & 3 \\ 3 & 4 & 4 & 0 & 9 & 13 & 4 \\ 2 & 4 & 1 & 0 & 6 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 21 & 6 \\ 7 & 13 & 9 & 0 & 21 & 28 & 10 \\ 3 & 4 & 3 & -3 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{1112}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -941, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 7 & 3 \\ 3 & 4 & 9 & 0 & 4 & 13 & 4 \\ 5 & 9 & 13 & 0 & 6 & 21 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 9 & 2 \\ 7 & 13 & 21 & 0 & 9 & 28 & 10 \\ 3 & 4 & 7 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1112} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -962, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 10 & 3 \\ 3 & 4 & 14 & 0 & 4 & 18 & 4 \\ 8 & 14 & 37 & 0 & 9 & 48 & 11 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 12 & 2 \\ 10 & 18 & 48 & 0 & 12 & 58 & 14 \\ 3 & 4 & 11 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -980, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 9 & 3 \\ 3 & 4 & 9 & 0 & 4 & 17 & 4 \\ 5 & 9 & 13 & 0 & 6 & 27 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 12 & 2 \\ 9 & 17 & 27 & 0 & 12 & 49 & 13 \\ 3 & 4 & 7 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1112} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -980, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 16 & 3 \\ 3 & 4 & 9 & 0 & 4 & 28 & 4 \\ 5 & 9 & 13 & 0 & 6 & 48 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 18 & 2 \\ 16 & 28 & 48 & 0 & 18 & 154 & 22 \\ 3 & 4 & 7 & -3 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -989, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 9 & 3 \\ 3 & 4 & 4 & 0 & 9 & 17 & 4 \\ 2 & 4 & 1 & 0 & 6 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 27 & 6 \\ 9 & 17 & 12 & 0 & 27 & 49 & 13 \\ 3 & 4 & 3 & -3 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1005, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 11 & 3 \\ 3 & 4 & 4 & 0 & 9 & 21 & 4 \\ 2 & 4 & 1 & 0 & 6 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 33 & 6 \\ 11 & 21 & 15 & 0 & 33 & 76 & 16 \\ 3 & 4 & 3 & -3 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1077, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 4 & 4 & 0 & 4 & 17 & 4 \\ 2 & 4 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 9 & 17 & 12 & 0 & 12 & 49 & 13 \\ 3 & 4 & 3 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{113} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1221, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 5 & 0 & 4 & 27 & 4 \\ 3 & 5 & 4 & 0 & 3 & 27 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 15 & 27 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & -3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1212} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1298, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 10 & 3 \\ 3 & 4 & 13 & 0 & 4 & 18 & 4 \\ 7 & 13 & 28 & 0 & 9 & 42 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 12 & 2 \\ 10 & 18 & 42 & 0 & 12 & 58 & 14 \\ 3 & 4 & 10 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1113} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1333, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 15 & 3 \\ 3 & 4 & 4 & 0 & 4 & 27 & 4 \\ 2 & 4 & 1 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 15 & 27 & 18 & 0 & 18 & 133 & 21 \\ 3 & 4 & 3 & -3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1112} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 10 & 3 \\ 3 & 4 & 9 & 0 & 4 & 18 & 4 \\ 5 & 9 & 13 & 0 & 6 & 30 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 12 & 2 \\ 10 & 18 & 30 & 0 & 12 & 58 & 14 \\ 3 & 4 & 7 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1212} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1388, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 7 & 3 \\ 3 & 4 & 13 & 0 & 4 & 13 & 4 \\ 7 & 13 & 28 & 0 & 9 & 30 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 9 & 2 \\ 7 & 13 & 30 & 0 & 9 & 28 & 10 \\ 3 & 4 & 10 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1112} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1389, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 13 & 3 \\ 3 & 4 & 9 & 0 & 4 & 23 & 4 \\ 5 & 9 & 13 & 0 & 6 & 39 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 15 & 2 \\ 13 & 23 & 39 & 0 & 15 & 100 & 18 \\ 3 & 4 & 7 & -3 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1113} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 4 & 0 & 4 & 22 & 4 \\ 2 & 4 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 12 & 22 & 15 & 0 & 15 & 85 & 17 \\ 3 & 4 & 3 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{1113} 3^1 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1451, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 5 & 0 & 4 & 32 & 4 \\ 3 & 5 & 4 & 0 & 3 & 33 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 18 & 32 & 33 & 0 & 21 & 193 & 25 \\ 3 & 4 & 4 & -3 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{114} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1500, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 5 & 0 & 4 & 31 & 4 \\ 3 & 5 & 4 & 0 & 3 & 30 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 17 & 31 & 30 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1113} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1949, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 12 & 3 \\ 3 & 4 & 9 & 0 & 4 & 22 & 4 \\ 5 & 9 & 13 & 0 & 6 & 36 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 15 & 2 \\ 12 & 22 & 36 & 0 & 15 & 85 & 17 \\ 3 & 4 & 7 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1212} 3^1 1^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2048, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 10 & 3 \\ 3 & 4 & 18 & 0 & 4 & 18 & 4 \\ 10 & 18 & 58 & 0 & 12 & 60 & 14 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 12 & 2 \\ 10 & 18 & 60 & 0 & 12 & 58 & 14 \\ 3 & 4 & 14 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{114} 3^1 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2054, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 5 & 0 & 4 & 36 & 4 \\ 3 & 5 & 4 & 0 & 3 & 36 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 24 & 2 \\ 20 & 36 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1114} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2106, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 4 & 4 & 0 & 4 & 26 & 4 \\ 2 & 4 & 1 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 14 & 26 & 18 & 0 & 18 & 118 & 20 \\ 3 & 4 & 3 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1114} 3^1 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 20 & 3 \\ 3 & 4 & 4 & 0 & 4 & 36 & 4 \\ 2 & 4 & 1 & 0 & 3 & 24 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 24 & 2 \\ 20 & 36 & 24 & 0 & 24 & 238 & 28 \\ 3 & 4 & 3 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1114} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2324, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 14 & 3 \\ 3 & 4 & 9 & 0 & 4 & 26 & 4 \\ 5 & 9 & 13 & 0 & 6 & 42 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 18 & 2 \\ 14 & 26 & 42 & 0 & 18 & 118 & 20 \\ 3 & 4 & 7 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1114} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2347, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 17 & 3 \\ 3 & 4 & 4 & 0 & 4 & 31 & 4 \\ 2 & 4 & 1 & 0 & 3 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 17 & 31 & 21 & 0 & 21 & 172 & 24 \\ 3 & 4 & 3 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1113} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2496, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 15 & 3 \\ 3 & 4 & 9 & 0 & 4 & 27 & 4 \\ 5 & 9 & 13 & 0 & 6 & 45 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 18 & 2 \\ 15 & 27 & 45 & 0 & 18 & 133 & 21 \\ 3 & 4 & 7 & -3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1113} 3^1 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2621, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 18 & 3 \\ 3 & 4 & 9 & 0 & 4 & 32 & 4 \\ 5 & 9 & 13 & 0 & 6 & 54 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 21 & 2 \\ 18 & 32 & 54 & 0 & 21 & 193 & 25 \\ 3 & 4 & 7 & -3 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1114} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -3405, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 17 & 3 \\ 3 & 4 & 9 & 0 & 4 & 31 & 4 \\ 5 & 9 & 13 & 0 & 6 & 51 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 21 & 2 \\ 17 & 31 & 51 & 0 & 21 & 172 & 24 \\ 3 & 4 & 7 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1114} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -4064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 20 & 3 \\ 3 & 4 & 9 & 0 & 4 & 36 & 4 \\ 5 & 9 & 13 & 0 & 6 & 60 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 24 & 2 \\ 20 & 36 & 60 & 0 & 24 & 238 & 28 \\ 3 & 4 & 7 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [54] \quad 1^{11} 321^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 5 & 5 & 0 & 5 & 10 & 5 \\ 2 & 5 & 2 & 0 & 4 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 1 & 7 & 3 \\ 5 & 10 & 7 & 0 & 7 & 14 & 8 \\ 3 & 5 & 4 & -3 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1111} 321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 3 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 1 & 2 \\ 1 & 2 & 1 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 3 & 0 & 4 & 3 & 4 \\ 1 & 2 & 1 & 2 & 0 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{11} 321^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 160, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 2 & 3 \\ 3 & 5 & 5 & 0 & 20 & 5 & 5 \\ 2 & 5 & 2 & 0 & 15 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 20 & 15 & 0 & 57 & 15 & 15 \\ 2 & 5 & 4 & 0 & 15 & 2 & 4 \\ 3 & 5 & 4 & -3 & 15 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{111} 321^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 172, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 3 & 0 & 4 & 4 & 2 \\ 1 & 1 & 3 & 1 & 0 & 4 & 4 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 4 & 0 & 4 & 7 & 4 \\ 2 & 3 & 4 & 4 & 0 & 7 & 5 & 5 \\ 2 & 2 & 2 & 3 & -3 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{11} 321^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 220, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 5 & 5 & 0 & 5 & 15 & 5 \\ 2 & 5 & 2 & 0 & 4 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 1 & 11 & 3 \\ 7 & 15 & 11 & 0 & 11 & 32 & 12 \\ 3 & 5 & 4 & -3 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{1111} 321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 262, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 4 & 2 \\ 1 & 2 & 1 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 3 & 0 & 4 & 7 & 4 \\ 2 & 3 & 4 & 3 & 0 & 7 & 5 & 5 \\ 2 & 2 & 2 & 2 & -3 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{12}321^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 340, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 5 & 5 & 0 & 5 & 25 & 5 \\ 2 & 5 & 2 & 0 & 4 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 1 & 18 & 3 \\ 12 & 25 & 18 & 0 & 18 & 94 & 20 \\ 3 & 5 & 4 & -3 & 3 & 20 & 2 \end{pmatrix}_{b_0}$$

[61] $1^{111}321^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 3 \\ 3 & 5 & 10 & 0 & 5 & 10 & 5 \\ 5 & 10 & 14 & 0 & 7 & 16 & 8 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 7 & 0 & 1 & 7 & 3 \\ 5 & 10 & 16 & 0 & 7 & 14 & 8 \\ 3 & 5 & 8 & -3 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{13}321^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 380, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 17 & 3 \\ 3 & 5 & 5 & 0 & 5 & 35 & 5 \\ 2 & 5 & 2 & 0 & 4 & 25 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 1 & 25 & 3 \\ 17 & 35 & 25 & 0 & 25 & 188 & 28 \\ 3 & 5 & 4 & -3 & 3 & 28 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{111}321^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 390, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 7 & 3 \\ 3 & 5 & 10 & 0 & 5 & 15 & 5 \\ 5 & 10 & 14 & 0 & 7 & 23 & 8 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 7 & 0 & 1 & 11 & 3 \\ 7 & 15 & 23 & 0 & 11 & 32 & 12 \\ 3 & 5 & 8 & -3 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[64] $1^{12}321^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 5 & 5 & 0 & 5 & 30 & 5 \\ 2 & 5 & 2 & 0 & 4 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 1 & 22 & 3 \\ 14 & 30 & 22 & 0 & 22 & 134 & 24 \\ 3 & 5 & 4 & -3 & 3 & 24 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{1111}321^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 752, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 3 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 4 & 0 & 4 & 4 & 2 \\ 2 & 3 & 4 & 5 & 0 & 7 & 7 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 7 & 0 & 4 & 7 & 4 \\ 2 & 3 & 4 & 7 & 0 & 7 & 5 & 5 \\ 2 & 2 & 2 & 5 & -3 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

4.837 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 10$ $[0] 1^{113}3^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 3 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 392, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 462, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 3 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 5 & 7 \\ 2 & 3 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 512, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 3 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 6 & 5 & 6 \\ 2 & 4 & 3 & 0 & 6 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 5 & 2 & 0 & 2 \\ 3 & 4 & 3 & -3 & 6 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 622, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 3 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 & 2 \\ 5 & 9 & 6 & 0 & 6 & 13 & 5 & 7 \\ 2 & 3 & 2 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 1152, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 13 & 0 & 6 & 15 & 5 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 & 2 \\ 5 & 9 & 15 & 0 & 6 & 13 & 5 & 7 \\ 2 & 3 & 5 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & 7 & -3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.838 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{\overline{121}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -96, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 2 & 3 \\ 3 & 4 & 8 & 0 & 9 & 3 & 4 \\ 4 & 8 & 10 & 0 & 12 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 12 & 0 & 12 & 5 & 6 \\ 2 & 3 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & 6 & -3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 2 & 3 \\ 4 & 9 & 8 & 4 & 0 & 4 & 5 \\ 3 & 8 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 3 & 5 & 4 & 0 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{121}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -136, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 2 & 3 \\ 3 & 4 & 8 & 0 & 4 & 3 & 4 \\ 4 & 8 & 10 & 0 & 6 & 4 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 2 & 0 & 2 \\ 3 & 4 & 6 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 2 & 3 \\ 3 & 4 & 19 & 0 & 4 & 3 & 4 \\ 11 & 19 & 73 & 0 & 12 & 11 & 15 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 2 & 2 \\ 2 & 3 & 11 & 0 & 2 & 0 & 2 \\ 3 & 4 & 15 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 3 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 3 & 4 \\ 2 & 4 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 2 & 3 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.31 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 0 & 8 & 3 & 4 \\ 2 & 4 & 1 & 0 & 6 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 9 & 4 & 5 \\ 2 & 3 & 2 & 0 & 4 & 0 & 2 \\ 3 & 4 & 3 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 2 & 3 \\ 4 & 9 & 12 & 0 & 4 & 4 & 5 \\ 5 & 12 & 13 & 0 & 6 & 5 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 2 & 2 \\ 2 & 4 & 5 & 0 & 2 & 0 & 2 \\ 3 & 5 & 7 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 2 & 3 \\ 3 & 4 & 9 & 0 & 8 & 3 & 4 \\ 5 & 9 & 13 & 0 & 12 & 5 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 12 & 0 & 9 & 4 & 5 \\ 2 & 3 & 5 & 0 & 4 & 0 & 2 \\ 3 & 4 & 7 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 3 & 4 \\ 2 & 4 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 3 & 4 \\ 2 & 4 & 1 & 0 & 6 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 5 & 6 \\ 2 & 3 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 3 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{111}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 3 & 4 \\ 8 & 14 & 37 & 0 & 9 & 8 & 11 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 2 & 2 \\ 2 & 3 & 8 & 0 & 2 & 0 & 2 \\ 3 & 4 & 11 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 3 & 4 \\ 5 & 9 & 13 & 0 & 6 & 5 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 2 & 2 \\ 2 & 3 & 5 & 0 & 2 & 0 & 2 \\ 3 & 4 & 7 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 4 & 0 & 13 & 3 & 4 \\ 2 & 4 & 1 & 0 & 9 & 2 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 9 & 0 & 27 & 7 & 9 \\ 2 & 3 & 2 & 0 & 7 & 0 & 2 \\ 3 & 4 & 3 & -3 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 3 & 4 \\ 7 & 13 & 28 & 0 & 9 & 7 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 2 & 2 \\ 2 & 3 & 7 & 0 & 2 & 0 & 2 \\ 3 & 4 & 10 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{121}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 2 & 3 \\ 3 & 4 & 28 & 0 & 4 & 3 & 4 \\ 16 & 28 & 154 & 0 & 18 & 16 & 22 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 18 & 0 & 0 & 2 & 2 \\ 2 & 3 & 16 & 0 & 2 & 0 & 2 \\ 3 & 4 & 22 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{131}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 3 & 4 \\ 9 & 17 & 49 & 0 & 12 & 9 & 13 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 2 & 2 \\ 2 & 3 & 9 & 0 & 2 & 0 & 2 \\ 3 & 4 & 13 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{141}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 2 & 3 \\ 3 & 4 & 21 & 0 & 4 & 3 & 4 \\ 11 & 21 & 76 & 0 & 15 & 11 & 16 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 15 & 0 & 0 & 2 & 2 \\ 2 & 3 & 11 & 0 & 2 & 0 & 2 \\ 3 & 4 & 16 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{121}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 2 & 3 \\ 3 & 4 & 18 & 0 & 4 & 3 & 4 \\ 10 & 18 & 58 & 0 & 12 & 10 & 14 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 2 & 2 \\ 2 & 3 & 10 & 0 & 2 & 0 & 2 \\ 3 & 4 & 14 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{121}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 2 & 3 \\ 3 & 4 & 23 & 0 & 4 & 3 & 4 \\ 13 & 23 & 100 & 0 & 15 & 13 & 18 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 15 & 0 & 0 & 2 & 2 \\ 2 & 3 & 13 & 0 & 2 & 0 & 2 \\ 3 & 4 & 18 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{131}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 2 & 3 \\ 3 & 4 & 22 & 0 & 4 & 3 & 4 \\ 12 & 22 & 85 & 0 & 15 & 12 & 17 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 15 & 0 & 0 & 2 & 2 \\ 2 & 3 & 12 & 0 & 2 & 0 & 2 \\ 3 & 4 & 17 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{131}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 2 & 3 \\ 3 & 4 & 27 & 0 & 4 & 3 & 4 \\ 15 & 27 & 133 & 0 & 18 & 15 & 21 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 18 & 0 & 0 & 2 & 2 \\ 2 & 3 & 15 & 0 & 2 & 0 & 2 \\ 3 & 4 & 21 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{141}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 2 & 3 \\ 3 & 4 & 26 & 0 & 4 & 3 & 4 \\ 14 & 26 & 118 & 0 & 18 & 14 & 20 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 18 & 0 & 0 & 2 & 2 \\ 2 & 3 & 14 & 0 & 2 & 0 & 2 \\ 3 & 4 & 20 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{131}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1826, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 2 & 2 & 3 \\ 3 & 4 & 32 & 0 & 4 & 3 & 4 \\ 18 & 32 & 193 & 0 & 21 & 18 & 25 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 21 & 0 & 0 & 2 & 2 \\ 2 & 3 & 18 & 0 & 2 & 0 & 2 \\ 3 & 4 & 25 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{141}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 2 & 3 \\ 3 & 4 & 31 & 0 & 4 & 3 & 4 \\ 17 & 31 & 172 & 0 & 21 & 17 & 24 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 21 & 0 & 0 & 2 & 2 \\ 2 & 3 & 17 & 0 & 2 & 0 & 2 \\ 3 & 4 & 24 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{141}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 2 & 2 & 3 \\ 3 & 4 & 36 & 0 & 4 & 3 & 4 \\ 20 & 36 & 238 & 0 & 24 & 20 & 28 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 24 & 0 & 0 & 2 & 2 \\ 2 & 3 & 20 & 0 & 2 & 0 & 2 \\ 3 & 4 & 28 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.839 NHC(-2)(-3) + su₂ + f₄ + su₃ + su₂→(-3) + su₂

$$T = 10 \quad [0] \quad 1^{11}3^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 3 & 5 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 3 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & 4 & -3 & 2 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 392, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 3 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 2 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 3 & 2 \\ 2 & 3 & 2 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & -3 & 2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 462, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 3 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 6 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 5 & 2 \\ 5 & 9 & 9 & 0 & 6 & 5 & 13 & 7 \\ 3 & 4 & 4 & -3 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 512, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 3 & 4 & 4 \\ 2 & 4 & 1 & 0 & 6 & 2 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 5 & 6 & 6 \\ 2 & 3 & 2 & 0 & 5 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 6 & 2 & 1 & 3 \\ 3 & 4 & 3 & -3 & 6 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 622, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 2 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 0 & 2 & 0 & 5 & 2 \\ 5 & 9 & 6 & 0 & 6 & 5 & 13 & 7 \\ 3 & 4 & 3 & -3 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 1152, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 4 & 3 & 9 & 4 \\ 5 & 9 & 13 & 0 & 6 & 5 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 2 & 6 & 2 \\ 2 & 3 & 5 & 0 & 2 & 0 & 5 & 2 \\ 5 & 9 & 15 & 0 & 6 & 5 & 13 & 7 \\ 3 & 4 & 7 & -3 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.840 NHC(-2)(-3) + su₂ + f₄ + su₃ + su₂n3mix

$$T = 9 \quad [0] \quad 1^{11}3^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -104, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 8 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 5 & 8 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & 4 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}1}3^11^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 4 & 17 & 4 \\ 2 & 4 & 1 & 0 & 3 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 11 & 2 \\ 10 & 17 & 11 & 0 & 11 & 55 & 13 \\ 3 & 4 & 3 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}3}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -119, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 5 & 0 & 4 & 7 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 4 & 7 & 6 & 0 & 5 & 7 & 5 \\ 3 & 4 & 4 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{112}3}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 6 & 3 \\ 3 & 4 & 4 & 0 & 5 & 11 & 4 \\ 2 & 4 & 1 & 0 & 3 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 3 & 9 & 3 \\ 6 & 11 & 8 & 0 & 9 & 20 & 8 \\ 3 & 4 & 3 & -3 & 3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}3}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -204, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.14 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 4 & 0 & 5 & 7 & 4 \\ 2 & 4 & 1 & 0 & 3 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 3 & 6 & 3 \\ 4 & 7 & 5 & 0 & 6 & 7 & 5 \\ 3 & 4 & 3 & -3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}3}1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -269, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 4 & 8 & 4 \\ 5 & 9 & 13 & 0 & 6 & 14 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 5 & 2 \\ 5 & 8 & 14 & 0 & 5 & 12 & 6 \\ 3 & 4 & 7 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{113}3}1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 5 & 0 & 4 & 20 & 4 \\ 3 & 5 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 14 & 2 \\ 11 & 20 & 18 & 0 & 14 & 69 & 15 \\ 3 & 4 & 4 & -3 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}3}1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 12 & 4 \\ 3 & 5 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 8 & 2 \\ 7 & 12 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & 4 & -3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}3}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -424, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.07 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 4 & 4 & 0 & 4 & 7 & 4 \\ 2 & 4 & 1 & 0 & 3 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 4 & 7 & 5 & 0 & 5 & 7 & 5 \\ 3 & 4 & 3 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1111}} 3^{\overline{111}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 4 & 3 \\ 3 & 4 & 4 & 0 & 9 & 7 & 4 \\ 2 & 4 & 1 & 0 & 6 & 5 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 11 & 6 \\ 4 & 7 & 5 & 0 & 11 & 7 & 5 \\ 3 & 4 & 3 & -3 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1111}} 3^{\overline{111}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -481, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 4 & 12 & 4 \\ 2 & 4 & 1 & 0 & 3 & 8 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 8 & 2 \\ 7 & 12 & 8 & 0 & 8 & 25 & 9 \\ 3 & 4 & 3 & -3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1111}} 3^{\overline{111}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -500, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 5 & 0 & 4 & 16 & 4 \\ 3 & 5 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 11 & 2 \\ 9 & 16 & 15 & 0 & 11 & 44 & 12 \\ 3 & 4 & 4 & -3 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1111}} 3^{\overline{111}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -509, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.06 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 4 & 3 \\ 3 & 4 & 9 & 0 & 4 & 7 & 4 \\ 5 & 9 & 13 & 0 & 6 & 11 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 5 & 2 \\ 4 & 7 & 11 & 0 & 5 & 7 & 5 \\ 3 & 4 & 7 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{1111}} 3^{\overline{111}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -896, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 7 & 3 \\ 3 & 4 & 9 & 0 & 4 & 12 & 4 \\ 5 & 9 & 13 & 0 & 6 & 20 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 8 & 2 \\ 7 & 12 & 20 & 0 & 8 & 25 & 9 \\ 3 & 4 & 7 & -3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

4.841 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{1111}} 3^{\overline{111}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -39, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 2 & 3 \\ 4 & 9 & 12 & 0 & 4 & 4 & 5 \\ 5 & 12 & 13 & 0 & 6 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 3 & 2 \\ 2 & 4 & 6 & 0 & 3 & 0 & 2 \\ 3 & 5 & 7 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{111}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -119, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & 4 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^{\overline{111}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -126, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.05 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{113} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -159, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 11 & 1 & 2 & 2 & 3 \\ 4 & 9 & 28 & 0 & 4 & 4 & 5 \\ 11 & 28 & 73 & 0 & 12 & 12 & 15 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 12 & 0 & 0 & 3 & 2 \\ 2 & 4 & 12 & 0 & 3 & 0 & 2 \\ 3 & 5 & 15 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -174, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 2 & 2 & 3 \\ 4 & 9 & 20 & 0 & 4 & 4 & 5 \\ 8 & 20 & 37 & 0 & 9 & 9 & 11 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 3 & 2 \\ 2 & 4 & 9 & 0 & 3 & 0 & 2 \\ 3 & 5 & 11 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -182, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.26 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{122} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -200, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.9$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 4 & 3 \\ 3 & 4 & 13 & 0 & 5 & 8 & 4 \\ 7 & 13 & 28 & 0 & 12 & 18 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 12 & 0 & 3 & 6 & 3 \\ 4 & 8 & 18 & 0 & 6 & 9 & 5 \\ 3 & 4 & 10 & -3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -203, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.34 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 8 & 14 & 9 & 0 & 9 & 36 & 10 \\ 3 & 4 & 3 & -3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{132} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -206, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 4 & 3 \\ 3 & 4 & 17 & 0 & 5 & 8 & 4 \\ 9 & 17 & 49 & 0 & 15 & 24 & 13 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 15 & 0 & 3 & 6 & 3 \\ 4 & 8 & 24 & 0 & 6 & 9 & 5 \\ 3 & 4 & 13 & -3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -315, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.90 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 & 4 \\ 8 & 14 & 37 & 0 & 9 & 9 & 11 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 9 & 0 & 0 & 3 & 2 \\ 2 & 4 & 9 & 0 & 3 & 0 & 2 \\ 3 & 4 & 11 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{113} 3^1 1^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -321, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 2 & 2 & 3 \\ 4 & 9 & 32 & 0 & 4 & 4 & 5 \\ 13 & 32 & 100 & 0 & 15 & 15 & 18 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 15 & 0 & 0 & 3 & 2 \\ 2 & 4 & 15 & 0 & 3 & 0 & 2 \\ 3 & 5 & 18 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -344, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -399, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.53 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{114} 3^1 1^{311} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -426, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 2 & 2 & 3 \\ 4 & 9 & 44 & 0 & 4 & 4 & 5 \\ 18 & 44 & 193 & 0 & 21 & 21 & 25 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 21 & 0 & 0 & 3 & 2 \\ 2 & 4 & 21 & 0 & 3 & 0 & 2 \\ 3 & 5 & 25 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -509, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.25 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 2 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 & 4 \\ 5 & 9 & 13 & 0 & 9 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 9 & 0 & 3 & 3 & 3 \\ 2 & 4 & 6 & 0 & 3 & 0 & 2 \\ 3 & 4 & 7 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -512, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 6 & 12 & 6 \\ 3 & 4 & 3 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{114} 3^1 1^{211} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -516, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 2 & 2 & 3 \\ 4 & 9 & 40 & 0 & 4 & 4 & 5 \\ 16 & 40 & 154 & 0 & 18 & 18 & 22 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 18 & 0 & 0 & 3 & 2 \\ 2 & 4 & 18 & 0 & 3 & 0 & 2 \\ 3 & 5 & 22 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{141} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -527, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.27 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 & 2 & 3 \\ 3 & 4 & 21 & 0 & 5 & 4 & 4 \\ 11 & 21 & 76 & 0 & 18 & 15 & 16 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 18 & 0 & 3 & 3 & 3 \\ 2 & 4 & 15 & 0 & 3 & 0 & 2 \\ 3 & 4 & 16 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -621, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.48 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 & 4 \\ 5 & 9 & 13 & 0 & 6 & 6 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 3 & 2 \\ 2 & 4 & 6 & 0 & 3 & 0 & 2 \\ 3 & 4 & 7 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -707, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 2 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 & 4 \\ 7 & 13 & 28 & 0 & 12 & 9 & 10 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 12 & 0 & 3 & 3 & 3 \\ 2 & 4 & 9 & 0 & 3 & 0 & 2 \\ 3 & 4 & 10 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{131} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -713, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.25 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 2 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 & 4 \\ 9 & 17 & 49 & 0 & 15 & 12 & 13 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 15 & 0 & 3 & 3 & 3 \\ 2 & 4 & 12 & 0 & 3 & 0 & 2 \\ 3 & 4 & 13 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -944, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.22 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 4 & 9 & 4 \\ 5 & 9 & 13 & 0 & 6 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 5 & 9 & 15 & 0 & 6 & 12 & 6 \\ 3 & 4 & 7 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

4.842 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{f}_4 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \ 1^{131} 1^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{f}_4 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 31, \ \Delta = -117, \ \det = 725, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 31, \ T \in [12, 31]$$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 1 & 4 \\ 8 & 9 & 8 & 4 & 0 & 5 \\ 6 & 8 & 4 & 3 & 0 & 4 \\ 3 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 4 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

4.843 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$

$$T = 8 \quad [0] \ 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 8, \ T_{\max} = 10, \ \Delta = -64, \ \det = 16, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 1 & 3 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 0 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \ 1^{11} 5^1 13^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -2, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 1 & 0 \\ 2 & 3 & 0 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{13} 5^1 13^1 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -2, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 3 & 4 \\ 13 & 21 & 97 & 0 & 18 & 21 \\ 1 & 0 & 0 & -2 & 1 & 0 \\ 2 & 3 & 18 & 1 & 1 & 3 \\ 3 & 4 & 21 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 5^1 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -4, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$\begin{aligned}
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 7 & 0 & 3 & 4 \\ 6 & 7 & 9 & 0 & 6 & 7 \\ 1 & 0 & 0 & -2 & 1 & 0 \\ 2 & 3 & 6 & 1 & 1 & 3 \\ 3 & 4 & 7 & 0 & 3 & 4 \end{pmatrix}_{b_0} \\
[3] & 1^{11} \bar{5}^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 4 & 1 & 0 & 2 \\ 2 & 2 & 4 & 5 & 3 & 0 & 5 \\ 1 & 1 & 1 & 3 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 2 & 2 & 2 & 5 & 2 & -1 & 3 \end{pmatrix}_{b_0} \\
[7] & 1^{11} \bar{5}^{11} 131^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 4 \\ 9 & 14 & 42 & 0 & 12 & 14 \\ 1 & 0 & 0 & -2 & 1 & 0 \\ 2 & 3 & 12 & 1 & 1 & 3 \\ 3 & 4 & 14 & 0 & 3 & 4 \end{pmatrix}_{b_0} \\
[6] & 1^{11} \bar{5}^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 7 & 0 & 3 & 4 \\ 5 & 7 & 9 & 0 & 6 & 7 \\ 1 & 0 & 0 & -2 & 1 & 0 \\ 2 & 3 & 6 & 1 & 1 & 3 \\ 3 & 4 & 7 & 0 & 3 & 4 \end{pmatrix}_{b_0} \\
[5] & 1^{12} \bar{5}^{31} 131^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 4 \\ 11 & 14 & 42 & 0 & 12 & 14 \\ 1 & 0 & 0 & -2 & 1 & 0 \\ 2 & 3 & 12 & 1 & 1 & 3 \\ 3 & 4 & 14 & 0 & 3 & 4 \end{pmatrix}_{b_0} \\
[8] & 1^{12} \bar{5}^{21} 131^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 4 \\ 10 & 14 & 42 & 0 & 12 & 14 \\ 1 & 0 & 0 & -2 & 1 & 0 \\ 2 & 3 & 12 & 1 & 1 & 3 \\ 3 & 4 & 14 & 0 & 3 & 4 \end{pmatrix}_{b_0} \\
[9] & 1^{13} \bar{5}^{21} 131^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 3 & 4 \\ 14 & 21 & 97 & 0 & 18 & 21 \\ 1 & 0 & 0 & -2 & 1 & 0 \\ 2 & 3 & 18 & 1 & 1 & 3 \\ 3 & 4 & 21 & 0 & 3 & 4 \end{pmatrix}_{b_0} \\
[14] & 1^{11} \bar{5}^{14} 131^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 16 & 2 & 1 & 3 \\ 3 & 5 & 32 & 5 & 0 & 5 \\ 16 & 32 & 174 & 24 & 0 & 28 \\ 2 & 5 & 24 & 2 & 1 & 4 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & 28 & 4 & -1 & 4 \end{pmatrix}_{b_0} \\
[15] & 1^{11} \bar{5}^{12} 231^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 16 & 2 & 1 & 3 \\ 3 & 5 & 30 & 5 & 0 & 5 \\ 16 & 30 & 154 & 22 & 0 & 26 \\ 2 & 5 & 22 & 2 & 1 & 4 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & 26 & 4 & -1 & 4 \end{pmatrix}_{b_0} \\
[16] & 1^{11} \bar{5}^{13} 131^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$[22] \quad 2 \quad 3 \quad 15 \quad 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.1$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 & 2 & 3 \\ 4 & 5 & 25 & 5 & 3 & 5 \\ 15 & 25 & 98 & 20 & 10 & 20 \\ 3 & 5 & 20 & 2 & 3 & 4 \\ 2 & 3 & 10 & 3 & 0 & 2 \\ 3 & 5 & 20 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[24] \quad 2 \quad 3 \quad 15 \quad 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 18 & 3 & 2 & 3 \\ 4 & 5 & 30 & 5 & 3 & 5 \\ 18 & 30 & 142 & 24 & 12 & 24 \\ 3 & 5 & 24 & 2 & 3 & 4 \\ 2 & 3 & 12 & 3 & 0 & 2 \\ 3 & 5 & 24 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 2 \quad 3 \quad 15 \quad 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 & 2 & 3 \\ 4 & 5 & 15 & 5 & 3 & 5 \\ 9 & 15 & 34 & 12 & 6 & 12 \\ 3 & 5 & 12 & 2 & 3 & 4 \\ 2 & 3 & 6 & 3 & 0 & 2 \\ 3 & 5 & 12 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [25] \quad 41 \quad 1^{11} 231 \quad 5 \quad 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 216, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 9 & 8 & 8 & 1 & 3 \\ 9 & 51 & 48 & 48 & 0 & 15 \\ 8 & 48 & 42 & 44 & 0 & 14 \\ 8 & 48 & 44 & 42 & 1 & 14 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 15 & 14 & 14 & -1 & 4 \end{pmatrix}_{b_0}$$

4.844 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3 \quad 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -34, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 1 & 3 & 3 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 3 & 5 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & 0 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [4] \quad 2 \quad 3 \quad 15 \quad 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 2 & 9 & 3 \\ 4 & 5 & 5 & 5 & 3 & 15 & 5 \\ 3 & 5 & 2 & 4 & 2 & 12 & 4 \\ 3 & 5 & 4 & 2 & 3 & 12 & 4 \\ 2 & 3 & 2 & 3 & 0 & 6 & 2 \\ 9 & 15 & 12 & 12 & 6 & 34 & 12 \\ 3 & 5 & 4 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 2 \quad 3 \quad 15 \quad 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 2 & 12 & 3 \\ 4 & 5 & 5 & 5 & 3 & 20 & 5 \\ 3 & 5 & 2 & 4 & 2 & 16 & 4 \\ 3 & 5 & 4 & 2 & 3 & 16 & 4 \\ 2 & 3 & 2 & 3 & 0 & 8 & 2 \\ 12 & 20 & 16 & 16 & 8 & 62 & 16 \\ 3 & 5 & 4 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[3] \quad 2 \quad 3 \quad 15 \quad 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.1$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 2 & 15 & 3 \\ 4 & 5 & 5 & 5 & 3 & 25 & 5 \\ 3 & 5 & 2 & 4 & 2 & 20 & 4 \\ 3 & 5 & 4 & 2 & 3 & 20 & 4 \\ 2 & 3 & 2 & 3 & 0 & 10 & 2 \\ 15 & 25 & 20 & 20 & 10 & 98 & 20 \\ 3 & 5 & 4 & 4 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 2 \quad 3 \quad 15 \quad 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 2 & 6 & 3 \\ 4 & 5 & 10 & 5 & 3 & 10 & 5 \\ 6 & 10 & 14 & 8 & 4 & 16 & 8 \\ 3 & 5 & 8 & 2 & 3 & 8 & 4 \\ 2 & 3 & 4 & 3 & 0 & 4 & 2 \\ 6 & 10 & 16 & 8 & 4 & 14 & 8 \\ 3 & 5 & 8 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[6] \ 2^{\overset{1}{\overset{1}{\overset{1}{1}}}} 3^{\overset{1}{\overset{1}{\overset{1}{1}}}} 15 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 2 & 3 & 3 \\ 4 & 5 & 5 & 5 & 3 & 5 & 5 \\ 3 & 5 & 2 & 4 & 2 & 4 & 4 \\ 3 & 5 & 4 & 2 & 3 & 4 & 4 \\ 2 & 3 & 2 & 3 & 0 & 2 & 2 \\ 3 & 5 & 4 & 4 & 2 & 2 & 4 \\ 3 & 5 & 4 & 4 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 2^{\overset{1}{\overset{1}{\overset{1}{1}}}} 3^{\overset{1}{\overset{1}{\overset{1}{1}}}} 15 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 2 & 6 & 3 \\ 4 & 5 & 5 & 5 & 3 & 10 & 5 \\ 3 & 5 & 2 & 4 & 2 & 8 & 4 \\ 3 & 5 & 4 & 2 & 3 & 8 & 4 \\ 2 & 3 & 2 & 3 & 0 & 4 & 2 \\ 6 & 10 & 8 & 8 & 4 & 14 & 8 \\ 3 & 5 & 4 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

4.845 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \text{NHC}(-2)(-3) + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 12 \quad [6] \ 1^{\overset{1}{\overset{1}{\overset{1}{1}}}} 2^{\overset{1}{\overset{1}{\overset{1}{1}}}} 5^{\overset{1}{\overset{1}{\overset{1}{1}}}} 13^{\overset{1}{1}} 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 3 & 4 \\ 10 & 14 & 42 & 0 & 12 & 10 & 14 \\ 1 & 0 & 0 & -2 & 1 & 0 & 0 \\ 2 & 3 & 12 & 1 & 1 & 3 & 3 \\ 2 & 3 & 10 & 0 & 3 & 1 & 3 \\ 3 & 4 & 14 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overset{1}{\overset{1}{\overset{1}{1}}}} 3^{\overset{1}{\overset{1}{\overset{1}{1}}}} 5^{\overset{1}{\overset{1}{\overset{1}{1}}}} 13^{\overset{1}{1}} 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 2 & 3 \\ 3 & 4 & 21 & 0 & 3 & 3 & 4 \\ 15 & 21 & 97 & 0 & 18 & 15 & 21 \\ 1 & 0 & 0 & -2 & 1 & 0 & 0 \\ 2 & 3 & 18 & 1 & 1 & 3 & 3 \\ 2 & 3 & 15 & 0 & 3 & 1 & 3 \\ 3 & 4 & 21 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overset{1}{\overset{1}{\overset{1}{1}}}} 3^{\overset{1}{\overset{1}{\overset{1}{1}}}} 5^{\overset{1}{\overset{1}{\overset{1}{1}}}} 13^{\overset{1}{1}} 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 7 & 0 & 3 & 3 & 4 \\ 5 & 7 & 9 & 0 & 6 & 5 & 7 \\ 1 & 0 & 0 & -2 & 1 & 0 & 0 \\ 2 & 3 & 6 & 1 & 1 & 3 & 3 \\ 2 & 3 & 5 & 0 & 3 & 1 & 3 \\ 3 & 4 & 7 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

4.846 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 7 \quad [0] \ 1^{\overset{1}{\overset{1}{\overset{1}{1}}}} 1^{\overset{1}{\overset{1}{\overset{1}{1}}}} 3^{\overset{1}{\overset{1}{\overset{1}{1}}}} 1^{\overset{1}{1}} 4^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 10 & 16 & 80 & 6 \\ 16 & 25 & 124 & 9 \\ 80 & 124 & 613 & 45 \\ 6 & 9 & 45 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overset{1}{\overset{1}{\overset{1}{1}}}} 3^{\overset{1}{\overset{1}{\overset{1}{1}}}} 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 12 & 8 & 5 \\ 12 & 25 & 16 & 9 \\ 8 & 16 & 10 & 6 \\ 5 & 9 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 21^{\overset{1}{\overset{1}{\overset{1}{1}}}} 31^{\overset{1}{\overset{1}{\overset{1}{1}}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 25 & 20 & 9 \\ 12 & 20 & 13 & 7 \\ 6 & 9 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overset{1}{\overset{1}{\overset{1}{1}}}} 1^{\overset{1}{\overset{1}{\overset{1}{1}}}} 3^{\overset{1}{\overset{1}{\overset{1}{1}}}} 1^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -2, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 25 & 20 & 9 \\ 12 & 20 & 13 & 7 \\ 6 & 9 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 14 & 6 \\ 15 & 20 & 17 & 8 \\ 14 & 17 & 13 & 7 \\ 6 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{114} 3 \ 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 10 & 16 & 74 & 6 \\ 16 & 25 & 112 & 9 \\ 74 & 112 & 478 & 40 \\ 6 & 9 & 40 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3 \ 1^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 4 & 7 & 1 & 3 \\ 4 & 9 & 16 & 0 & 5 \\ 7 & 16 & 28 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 10 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{112} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 10 & 16 & 37 & 6 \\ 16 & 25 & 56 & 9 \\ 37 & 56 & 118 & 20 \\ 6 & 9 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{121} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14.09 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 5 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{131} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 6 & 10 & 22 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 8 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{131} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 20.33 \rightarrow 21, T_{\max} = 22, T \in [21, 22]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 9 & 10 & 22 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 8 & -2 & -8 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -13, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 3 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 16 & 23 & 6 \\ 16 & 25 & 36 & 9 \\ 23 & 36 & 49 & 13 \\ 6 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{113} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12.41 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

- [15] $\begin{pmatrix} 10 & 16 & 47 & 6 \\ 16 & 25 & 76 & 9 \\ 47 & 76 & 217 & 27 \\ 6 & 9 & 27 & 1 \end{pmatrix}_{b_0}$
 $21^{31}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 12.20 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 10 & 16 & 84 & 6 \\ 16 & 25 & 128 & 9 \\ 84 & 128 & 634 & 46 \\ 6 & 9 & 46 & 1 \end{pmatrix}_{b_0}$
- [16] $1^{14}31^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.20 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 10 & 16 & 84 & 6 \\ 16 & 25 & 128 & 9 \\ 84 & 128 & 634 & 46 \\ 6 & 9 & 46 & 1 \end{pmatrix}_{b_0}$
- [19] $1^{111}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 5 & 11 & 30 & 5 \\ 11 & 20 & 50 & 8 \\ 30 & 50 & 118 & 20 \\ 5 & 8 & 20 & 0 \end{pmatrix}_{b_0}$
- [17] $1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 1 & 5 \\ 11 & 20 & 0 & 8 \\ 1 & 0 & -2 & 0 \\ 5 & 8 & 0 & 2 \end{pmatrix}_{b_0}$
- [18] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 10 & 5 \\ 11 & 20 & 17 & 8 \\ 10 & 17 & 13 & 7 \\ 5 & 8 & 7 & 2 \end{pmatrix}_{b_0}$
- [20] $1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -25, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 12 & 6 \\ 15 & 20 & 17 & 8 \\ 12 & 17 & 13 & 7 \\ 6 & 8 & 7 & 3 \end{pmatrix}_{b_0}$
- [21] $1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & -2 & 2 \end{pmatrix}_{b_0}$
- [22] $1^{112}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 10 & 16 & 35 & 6 \\ 16 & 25 & 56 & 9 \\ 35 & 56 & 118 & 20 \\ 6 & 9 & 20 & 2 \end{pmatrix}_{b_0}$
- [23] $1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 9 & 5 \\ 11 & 20 & 17 & 8 \\ 9 & 17 & 13 & 7 \\ 5 & 8 & 7 & 2 \end{pmatrix}_{b_0}$
- [24] $1^{112}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 5 & 12 & 29 & 5 \\ 12 & 25 & 56 & 9 \\ 29 & 56 & 118 & 20 \\ 5 & 9 & 20 & 0 \end{pmatrix}_{b_0}$
- [25] $21^{131}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 13 & 17 & 12 & 7 \\ 17 & 20 & 16 & 8 \\ 12 & 16 & 10 & 6 \\ 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$
- [26] $1^{111}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 10 & 5 \\ 11 & 20 & 16 & 8 \\ 10 & 16 & 10 & 6 \\ 5 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{11} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 1 & 6 \\ 15 & 20 & 0 & 8 \\ 1 & 0 & -2 & 0 \\ 6 & 8 & 0 & 3 \end{pmatrix}_{b_0}$

[38] $1^{15} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.1$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 10 & 20 & 73 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & -2 & -1 \end{pmatrix}_{b_0}$

[39] $1^{41} 1 3 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 59 & 5 \\ 12 & 25 & 124 & 9 \\ 59 & 124 & 613 & 45 \\ 5 & 9 & 45 & 2 \end{pmatrix}_{b_0}$

[28] $1^{112} 3 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 16 & 46 & 6 \\ 16 & 25 & 72 & 9 \\ 46 & 72 & 202 & 26 \\ 6 & 9 & 26 & 3 \end{pmatrix}_{b_0}$

[29] $1^{11} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14.75 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$
 $\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 13 & 20 & 94 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & -2 \end{pmatrix}_{b_0}$

[30] $1^{11} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & -2 & 2 \end{pmatrix}_{b_0}$

[31] $1^{14} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 8 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & -2 & 0 \end{pmatrix}_{b_0}$

[32] $1^{11} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 4 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & -2 & 1 \end{pmatrix}_{b_0}$

[33] $1^{13} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 6 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$

[34] $1^{12} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 14.75 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 8 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & -2 & -2 \end{pmatrix}_{b_0}$

[35] $1^{17} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.3$
 $T_{\min} = 16.37 \rightarrow 17, T_{\max} = 22, T \in [17, 22]$
 $\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 14 & 28 & 145 & 0 & 21 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 21 & -2 & -4 \end{pmatrix}_{b_0}$

- [36] $1^{16}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$
 $T_{\min} = 14.75 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$
 $\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 12 & 24 & 106 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & -2 & -2 \end{pmatrix}_{b_0}$
- [37] $1^{12}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 4 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & -2 & 2 \end{pmatrix}_{b_0}$
- [40] $1^{113}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 15 & 28 & 73 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 15 & -2 & 1 \end{pmatrix}_{b_0}$
- [41] $1^{112}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 12 & 28 & 85 & 0 & 17 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 17 & -2 & 0 \end{pmatrix}_{b_0}$
- [45] $1^{11}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 13 & 6 \\ 15 & 20 & 17 & 8 \\ 13 & 17 & 13 & 7 \\ 6 & 8 & 7 & 3 \end{pmatrix}_{b_0}$
- [42] $1^{111}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 27 & 6 \\ 15 & 20 & 33 & 8 \\ 27 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0}$
- [43] $1^{11}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 14.71 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$
 $\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & -2 & -2 \end{pmatrix}_{b_0}$
- [44] $1^{11}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 4 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & -2 & 2 \end{pmatrix}_{b_0}$
- [46] $21^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 13 & 17 & 15 & 7 \\ 17 & 20 & 17 & 8 \\ 15 & 17 & 13 & 7 \\ 7 & 8 & 7 & 3 \end{pmatrix}_{b_0}$
- [47] $1^{113}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 10 & 16 & 50 & 6 \\ 16 & 25 & 76 & 9 \\ 50 & 76 & 217 & 27 \\ 6 & 9 & 27 & 2 \end{pmatrix}_{b_0}$
- [48] $1^{111}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 18 & 5 \\ 12 & 25 & 36 & 9 \\ 18 & 36 & 49 & 13 \\ 5 & 9 & 13 & 2 \end{pmatrix}_{b_0}$
- [49] $1^13^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 10 & 15 & 52 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & -2 & 0 \end{pmatrix}_{b_0}$$

[51] $21^{11}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 16 & 36 & 6 \\ 16 & 25 & 56 & 9 \\ 36 & 56 & 118 & 20 \\ 6 & 9 & 20 & 3 \end{pmatrix}_{b_0}$$

[52] $1^1 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 7 & 10 & 22 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

[53] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 16 & 36 & 6 \\ 16 & 25 & 56 & 9 \\ 36 & 56 & 118 & 20 \\ 6 & 9 & 20 & 3 \end{pmatrix}_{b_0}$$

[50] $1^1 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 8 & 10 & 22 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 8 & -2 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 20 & 5 \\ 11 & 20 & 33 & 8 \\ 20 & 33 & 49 & 13 \\ 5 & 8 & 13 & 2 \end{pmatrix}_{b_0}$$

[56] $13^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 8 & 5 \\ 11 & 20 & 16 & 8 \\ 8 & 16 & 10 & 6 \\ 5 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[54] $1^1 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 44 & 5 \\ 11 & 20 & 84 & 8 \\ 44 & 84 & 346 & 34 \\ 5 & 8 & 34 & 2 \end{pmatrix}_{b_0}$$

[58] $1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.58 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

[57] $13^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 9 & 5 \\ 12 & 25 & 20 & 9 \\ 9 & 20 & 13 & 7 \\ 5 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{112} 3^1 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12.08 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 5 & 12 & 36 & 5 \\ 12 & 25 & 72 & 9 \\ 36 & 72 & 202 & 26 \\ 5 & 9 & 26 & 1 \end{pmatrix}_{b_0}$$

[64] $21^{21} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 16 & 60 & 6 \\ 16 & 25 & 92 & 9 \\ 60 & 92 & 325 & 33 \\ 6 & 9 & 33 & 2 \end{pmatrix}_{b_0}$$

[59] $1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 11 & 6 \\ 15 & 20 & 16 & 8 \\ 11 & 16 & 10 & 6 \\ 6 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

[63] $1^{111} 3^1 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 13 & 6 \\ 15 & 20 & 16 & 8 \\ 13 & 16 & 10 & 6 \\ 6 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

[60] $1^{113} 3^1 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 16 & 60 & 6 \\ 16 & 25 & 92 & 9 \\ 60 & 92 & 325 & 33 \\ 6 & 9 & 33 & 2 \end{pmatrix}_{b_0}$$

[61] $1^{111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 35 & 6 \\ 15 & 20 & 50 & 8 \\ 35 & 50 & 118 & 20 \\ 6 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

[65] $1^{11} 1^1 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 17 & 5 \\ 12 & 25 & 36 & 9 \\ 17 & 36 & 49 & 13 \\ 5 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -53, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 9 & 12 & 0 & 5 \\ 5 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[67] $1^{113} 3^1 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 12 & 47 & 5 \\ 12 & 25 & 92 & 9 \\ 47 & 92 & 325 & 33 \\ 5 & 9 & 33 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{113} 3^1 1^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 16 & 69 & 6 \\ 16 & 25 & 108 & 9 \\ 69 & 108 & 457 & 39 \\ 6 & 9 & 39 & 3 \end{pmatrix}_{b_0}$$

[69] $1^{111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 9 & 5 \\ 11 & 20 & 16 & 8 \\ 9 & 16 & 10 & 6 \\ 5 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[70] $1^{131} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 12 & 15 & 52 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & -2 & -1 \end{pmatrix}_{b_0}$$

[71] $1^{113} 3^1 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -62, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 16 & 58 & 6 \\ 16 & 25 & 92 & 9 \\ 58 & 92 & 325 & 33 \\ 6 & 9 & 33 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{111} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 39 & 5 \\ 11 & 20 & 67 & 8 \\ 39 & 67 & 217 & 27 \\ 5 & 8 & 27 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{113} 3^1 1^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 5 & 12 & 54 & 5 \\ 12 & 25 & 108 & 9 \\ 54 & 108 & 457 & 39 \\ 5 & 9 & 39 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{1^{21}} 3^1 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 11 & 15 & 52 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & -2 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 5 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[76] $1^{111} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 12 & 6 \\ 15 & 20 & 16 & 8 \\ 12 & 16 & 10 & 6 \\ 6 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

[78] $1^{114} 3^1 1^{21} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 16 & 70 & 6 \\ 16 & 25 & 112 & 9 \\ 70 & 112 & 478 & 40 \\ 6 & 9 & 40 & 2 \end{pmatrix}_{b_0}$$

[77] $21^{111} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 16 & 36 & 6 \\ 16 & 20 & 50 & 8 \\ 36 & 50 & 118 & 20 \\ 6 & 8 & 20 & 2 \end{pmatrix}_{b_0}$$

[79] $21^{11} 3^1 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 16 & 48 & 6 \\ 16 & 25 & 76 & 9 \\ 48 & 76 & 217 & 27 \\ 6 & 9 & 27 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 16 & 48 & 6 \\ 16 & 25 & 76 & 9 \\ 48 & 76 & 217 & 27 \\ 6 & 9 & 27 & 2 \end{pmatrix}_{b_0}$$

[82] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 16 & 63 & 6 \\ 16 & 25 & 96 & 9 \\ 63 & 96 & 346 & 34 \\ 6 & 9 & 34 & 2 \end{pmatrix}_{b_0}$$

[81] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 23 & 6 \\ 15 & 20 & 33 & 8 \\ 23 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

[83] $1^{11} 3^1 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 48 & 5 \\ 11 & 20 & 84 & 8 \\ 48 & 84 & 346 & 34 \\ 5 & 8 & 34 & 2 \end{pmatrix}_{b_0}$$

[84] $13^1 1^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 18 & 5 \\ 12 & 25 & 40 & 9 \\ 18 & 40 & 58 & 14 \\ 5 & 9 & 14 & 2 \end{pmatrix}_{b_0}$$

[85] $21^{112} 3^1 1^{21} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 20 & 54 & 7 \\ 20 & 25 & 72 & 9 \\ 54 & 72 & 202 & 26 \\ 7 & 9 & 26 & 3 \end{pmatrix}_{b_0}$$

[89] $1^{12} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 7 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{14} 3^1 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 14.30 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 16 & 20 & 94 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & -2 \end{pmatrix}_{b_0}$$

[87] $1^{12} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 5 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

[88] $1^{14} 3^1 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.70 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 14 & 20 & 94 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 0 \end{pmatrix}_{b_0}$$

[90] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 16 & 49 & 6 \\ 16 & 25 & 76 & 9 \\ 49 & 76 & 217 & 27 \\ 6 & 9 & 27 & 3 \end{pmatrix}_{b_0}$$

[91] $21^{113} 31^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 16 & 60 & 6 \\ 16 & 25 & 96 & 9 \\ 60 & 96 & 346 & 34 \\ 6 & 9 & 34 & 2 \end{pmatrix}_{b_0}$$

[92] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 16 & 60 & 6 \\ 16 & 25 & 96 & 9 \\ 60 & 96 & 346 & 34 \\ 6 & 9 & 34 & 2 \end{pmatrix}_{b_0}$$

[93] $1^{111} 3^1 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 48 & 6 \\ 15 & 20 & 67 & 8 \\ 48 & 67 & 217 & 27 \\ 6 & 8 & 27 & 3 \end{pmatrix}_{b_0}$$

[94] $1^{114} 3^1 1^{41} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 5 & 12 & 72 & 5 \\ 12 & 25 & 144 & 9 \\ 72 & 144 & 814 & 52 \\ 5 & 9 & 52 & 0 \end{pmatrix}_{b_0}$$

[96] $1^{113} 3^1 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 17 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 17 & 40 & 172 & 0 & 24 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 24 & -2 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 35 & 5 \\ 11 & 20 & 67 & 8 \\ 35 & 67 & 217 & 27 \\ 5 & 8 & 27 & 2 \end{pmatrix}_{b_0}$$

[95] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 17 & 5 \\ 11 & 20 & 33 & 8 \\ 17 & 33 & 49 & 13 \\ 5 & 8 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[98] \ 1^{111} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 15 & 61 & 6 \\ 15 & 20 & 84 & 8 \\ 61 & 84 & 346 & 34 \\ 6 & 8 & 34 & 3 \end{pmatrix}_{b_0}$$

$$[99] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 8 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[101] \ 1^{111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[100] \ 1^{114} 3^1 1^{41} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 16 & 92 & 6 \\ 16 & 25 & 144 & 9 \\ 92 & 144 & 814 & 52 \\ 6 & 9 & 52 & 3 \end{pmatrix}_{b_0}$$

$$[102] \ 1^{121} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 34 & 5 \\ 12 & 25 & 72 & 9 \\ 34 & 72 & 202 & 26 \\ 5 & 9 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{121} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 6 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{111} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 15 & 20 & 94 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 3 \\ 4 & 9 & 20 & 0 & 5 \\ 10 & 20 & 37 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{112} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -98, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 3 \\ 4 & 9 & 24 & 0 & 5 \\ 10 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[107] \ 1^{114} 3^1 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.06 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 10 & 16 & 73 & 6 \\ 16 & 25 & 112 & 9 \\ 73 & 112 & 478 & 40 \\ 6 & 9 & 40 & 2 \end{pmatrix}_{b_0}$$

$$[108] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{aligned}
& \begin{pmatrix} 5 & 11 & 19 & 5 \\ 11 & 20 & 33 & 8 \\ 19 & 33 & 49 & 13 \\ 5 & 8 & 13 & 2 \end{pmatrix}_{b_0} \\
[109] & 1^{\overline{12}} 3^{\overline{111}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.29 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 5 & 11 & 26 & 5 \\ 11 & 20 & 50 & 8 \\ 26 & 50 & 118 & 20 \\ 5 & 8 & 20 & 2 \end{pmatrix}_{b_0} \\
[110] & 1^{\overline{111}} 3^{\overline{12}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.29 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 5 & 11 & 29 & 5 \\ 11 & 20 & 50 & 8 \\ 29 & 50 & 118 & 20 \\ 5 & 8 & 20 & 2 \end{pmatrix}_{b_0} \\
[111] & 1^{\overline{111}} 3^{\overline{111}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.81 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 10 & 15 & 26 & 6 \\ 15 & 20 & 33 & 8 \\ 26 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0} \\
[112] & 1^{\overline{114}} 3^{\overline{131}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.06 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 10 & 16 & 81 & 6 \\ 16 & 25 & 128 & 9 \\ 81 & 128 & 634 & 46 \\ 6 & 9 & 46 & 2 \end{pmatrix}_{b_0} \\
[113] & 1^{\overline{11}} 3^{\overline{13}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.09 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 11 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & -2 & 1 \end{pmatrix}_{b_0} \\
[114] & 1^{\overline{12}} 3^{\overline{11}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 10.64 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 7 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & -2 & 2 \end{pmatrix}_{b_0} \\
[115] & 1^{\overline{11}} 3^{\overline{11}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.64 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 6 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & -2 & 2 \end{pmatrix}_{b_0} \\
[116] & 1^{\overline{113}} 3^{\overline{11}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 5 & 12 & 39 & 5 \\ 12 & 25 & 76 & 9 \\ 39 & 76 & 217 & 27 \\ 5 & 9 & 27 & 2 \end{pmatrix}_{b_0} \\
[117] & 1^{\overline{12}} 3^{\overline{11}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -110, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.49 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 5 & 12 & 28 & 5 \\ 12 & 25 & 56 & 9 \\ 28 & 56 & 118 & 20 \\ 5 & 9 & 20 & 2 \end{pmatrix}_{b_0} \\
[118] & 1^{\overline{11}} 3^{\overline{14}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.54 \rightarrow 12, T_{\max} = 22, T \in [12, 22] \\
& \begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & -2 & 1 \end{pmatrix}_{b_0} \\
[119] & 1^{\overline{13}} 3^{\overline{11}} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 10.82 \rightarrow 11, T_{\max} = 22, T \in [11, 22] \\
& \begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 7 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & -2 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

[120] $1^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -113, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 8 & 1 & 3 \\ 4 & 9 & 20 & 0 & 5 \\ 8 & 20 & 37 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 11 & -2 & 1 \end{pmatrix}_{b_0}$

[121] $1^{111}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 45 & 5 \\ 11 & 20 & 84 & 8 \\ 45 & 84 & 346 & 34 \\ 5 & 8 & 34 & 2 \end{pmatrix}_{b_0}$

[122] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 18 & 5 \\ 11 & 20 & 33 & 8 \\ 18 & 33 & 49 & 13 \\ 5 & 8 & 13 & 2 \end{pmatrix}_{b_0}$

[123] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 7 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & -2 & 2 \end{pmatrix}_{b_0}$

[124] $1^131^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -122, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 26 & 5 \\ 12 & 25 & 56 & 9 \\ 26 & 56 & 118 & 20 \\ 5 & 9 & 20 & 2 \end{pmatrix}_{b_0}$

[125] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -125, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 24 & 6 \\ 15 & 20 & 33 & 8 \\ 24 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0}$

[127] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 16 & 62 & 6 \\ 16 & 25 & 96 & 9 \\ 62 & 96 & 346 & 34 \\ 6 & 9 & 34 & 3 \end{pmatrix}_{b_0}$

[126] $1^131^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$
 $\begin{pmatrix} 5 & 12 & 27 & 5 \\ 12 & 25 & 60 & 9 \\ 27 & 60 & 133 & 21 \\ 5 & 9 & 21 & 1 \end{pmatrix}_{b_0}$

[130] $1^{111}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 47 & 5 \\ 11 & 20 & 84 & 8 \\ 47 & 84 & 346 & 34 \\ 5 & 8 & 34 & 2 \end{pmatrix}_{b_0}$

[129] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 36 & 6 \\ 15 & 20 & 50 & 8 \\ 36 & 50 & 118 & 20 \\ 6 & 8 & 20 & 3 \end{pmatrix}_{b_0}$

[131] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 39 & 6 \\ 15 & 20 & 50 & 8 \\ 39 & 50 & 118 & 20 \\ 6 & 8 & 20 & 3 \end{pmatrix}_{b_0}$

[128] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 38 & 5 \\ 11 & 20 & 67 & 8 \\ 38 & 67 & 217 & 27 \\ 5 & 8 & 27 & 2 \end{pmatrix}_{b_0}$

- [132] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -131$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.05 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 10 & 16 & 61 & 6 \\ 16 & 25 & 96 & 9 \\ 61 & 96 & 346 & 34 \\ 6 & 9 & 34 & 2 \end{pmatrix}_{b_0}$
- [133] $21^{111} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -132$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 13 & 20 & 27 & 7 \\ 20 & 25 & 36 & 9 \\ 27 & 36 & 49 & 13 \\ 7 & 9 & 13 & 3 \end{pmatrix}_{b_0}$
- [134] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -135$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 25 & 6 \\ 15 & 20 & 33 & 8 \\ 25 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0}$
- [135] $21^{111} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -136$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 13 & 17 & 27 & 7 \\ 17 & 20 & 33 & 8 \\ 27 & 33 & 49 & 13 \\ 7 & 8 & 13 & 3 \end{pmatrix}_{b_0}$
- [136] $21^{21} 31^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -140$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 16 & 72 & 6 \\ 16 & 25 & 112 & 9 \\ 72 & 112 & 478 & 40 \\ 6 & 9 & 40 & 3 \end{pmatrix}_{b_0}$
- [137] $1^{114} 3 1^{21} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -140$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 16 & 72 & 6 \\ 16 & 25 & 112 & 9 \\ 72 & 112 & 478 & 40 \\ 6 & 9 & 40 & 3 \end{pmatrix}_{b_0}$
- [138] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -141$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 27 & 5 \\ 12 & 25 & 56 & 9 \\ 27 & 56 & 118 & 20 \\ 5 & 9 & 20 & 2 \end{pmatrix}_{b_0}$
- [139] $1^{111} 3 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -142$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 46 & 5 \\ 11 & 20 & 84 & 8 \\ 46 & 84 & 346 & 34 \\ 5 & 8 & 34 & 2 \end{pmatrix}_{b_0}$
- [140] $1^{111} 3 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -142$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 36 & 5 \\ 11 & 20 & 67 & 8 \\ 36 & 67 & 217 & 27 \\ 5 & 8 & 27 & 2 \end{pmatrix}_{b_0}$
- [141] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -142$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 27 & 5 \\ 11 & 20 & 50 & 8 \\ 27 & 50 & 118 & 20 \\ 5 & 8 & 20 & 2 \end{pmatrix}_{b_0}$
- [142] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -142$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 11 & 28 & 5 \\ 11 & 20 & 50 & 8 \\ 28 & 50 & 118 & 20 \\ 5 & 8 & 20 & 2 \end{pmatrix}_{b_0}$
- [143] $21^{111} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -144$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 20 & 16 & 8 \\ 12 & 16 & 10 & 6 \\ 6 & 8 & 6 & 3 \end{pmatrix}_{b_0}$
- [144] $1^{14} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -144$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 9 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

[145] $1^{13} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -147, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 9 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

[146] $1^{13} 3^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 3 \\ 4 & 9 & 32 & 0 & 5 \\ 16 & 32 & 100 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{11} 3^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 62 & 6 \\ 15 & 20 & 84 & 8 \\ 62 & 84 & 346 & 34 \\ 6 & 8 & 34 & 3 \end{pmatrix}_{b_0}$$

[148] $1^{11} 3^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -155, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 49 & 6 \\ 15 & 20 & 67 & 8 \\ 49 & 67 & 217 & 27 \\ 6 & 8 & 27 & 3 \end{pmatrix}_{b_0}$$

[150] $1^{11} 3^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 11 & 37 & 5 \\ 11 & 20 & 67 & 8 \\ 37 & 67 & 217 & 27 \\ 5 & 8 & 27 & 2 \end{pmatrix}_{b_0}$$

[151] $1^{11} 3^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 9 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$

[149] $1^{11} 3^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 12 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$

[153] $1^{13} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 8 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

[152] $1^{13} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 10 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[154] $1^3 3^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 51 & 5 \\ 12 & 25 & 108 & 9 \\ 51 & 108 & 457 & 39 \\ 5 & 9 & 39 & 2 \end{pmatrix}_{b_0}$$

[156] $1^{11} 3^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -170, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 15 & 37 & 6 \\ 15 & 20 & 50 & 8 \\ 37 & 50 & 118 & 20 \\ 6 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

- [157] $1^{111} 3 1^{14} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -170$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 63 & 6 \\ 15 & 20 & 84 & 8 \\ 63 & 84 & 346 & 34 \\ 6 & 8 & 34 & 3 \end{pmatrix}_{b_0}$
- [155] $1^{111} 3 1^{12} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -170$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 38 & 6 \\ 15 & 20 & 50 & 8 \\ 38 & 50 & 118 & 20 \\ 6 & 8 & 20 & 3 \end{pmatrix}_{b_0}$
- [158] $1^{123} 3 1^{12} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -172$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.72 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 10 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & -2 & 2 \end{pmatrix}_{b_0}$
- [159] $1^{113} 3 1^{13} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -173$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 15 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 21 & -2 & 1 \end{pmatrix}_{b_0}$
- [160] $1^{113} 3 1^{21} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -173$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 46 & 5 \\ 12 & 25 & 92 & 9 \\ 46 & 92 & 325 & 33 \\ 5 & 9 & 33 & 2 \end{pmatrix}_{b_0}$
- [161] $1^{111} 3 1^{13} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -175$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 51 & 6 \\ 15 & 20 & 67 & 8 \\ 51 & 67 & 217 & 27 \\ 6 & 8 & 27 & 3 \end{pmatrix}_{b_0}$
- [162] $1^{133} 3 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -176$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 9 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$
- [163] $1^{123} 3 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -179$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.65 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 8 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & -2 & 2 \end{pmatrix}_{b_0}$
- [164] $1^{113} 3 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -181$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 14 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 14 & 28 & 73 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 15 & -2 & 1 \end{pmatrix}_{b_0}$
- [165] $1^{111} 3 1^{13} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -185$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 10 & 15 & 50 & 6 \\ 15 & 20 & 67 & 8 \\ 50 & 67 & 217 & 27 \\ 6 & 8 & 27 & 3 \end{pmatrix}_{b_0}$
- [166] $1^{143} 3 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -187$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 11 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 2 \end{pmatrix}_{b_0}$
- [168] $1^{111} 3 1^{12} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -188$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 11 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$

[167] $1^{11}3^{12}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -188, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 10 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$

[169] $21^{14}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -190, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 17 & 72 & 7 \\ 17 & 20 & 84 & 8 \\ 72 & 84 & 346 & 34 \\ 7 & 8 & 34 & 3 \end{pmatrix}_{b_0}$$

[170] $21^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -190, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 16 & 24 & 6 \\ 16 & 20 & 33 & 8 \\ 24 & 33 & 49 & 13 \\ 6 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

[171] $1^{11}3^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -193, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 11 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 11 & 28 & 73 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[172] $1^{11}3^{11}3^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -195, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 12 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & -2 & 2 \end{pmatrix}_{b_0}$$

[174] $21^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 17 & 42 & 7 \\ 17 & 20 & 50 & 8 \\ 42 & 50 & 118 & 20 \\ 7 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

[173] $131^{11}3^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.20 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 5 & 12 & 36 & 5 \\ 12 & 25 & 80 & 9 \\ 36 & 80 & 238 & 28 \\ 5 & 9 & 28 & 1 \end{pmatrix}_{b_0}$$

[175] $21^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 20 & 15 & 7 \\ 20 & 25 & 20 & 9 \\ 15 & 20 & 13 & 7 \\ 7 & 9 & 7 & 3 \end{pmatrix}_{b_0}$$

[176] $1^{12}3^{11}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -203, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 11 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[177] $1^{11}3^{14}3^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -205, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 22 & 1 & 3 \\ 4 & 9 & 44 & 0 & 5 \\ 22 & 44 & 193 & 0 & 25 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 25 & -2 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{11}3^{11}3^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 3 \\ 4 & 9 & 32 & 0 & 5 \\ 13 & 32 & 100 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

[179] $1^{11}3^{11}3^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -209, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 38 & 5 \\ 12 & 25 & 76 & 9 \\ 38 & 76 & 217 & 27 \\ 5 & 9 & 27 & 2 \end{pmatrix}_{b_0}$$

[180] $1^{\overline{11}} 1^{\overline{13}} 3^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -212, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 35 & 5 \\ 12 & 25 & 76 & 9 \\ 35 & 76 & 217 & 27 \\ 5 & 9 & 27 & 2 \end{pmatrix}_{b_0}$$

[181] $21^{\overline{13}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 17 & 57 & 7 \\ 17 & 20 & 67 & 8 \\ 57 & 67 & 217 & 27 \\ 7 & 8 & 27 & 3 \end{pmatrix}_{b_0}$$

[182] $1^{\overline{12}} 1^{\overline{21}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 9 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[183] $1^{\overline{11}} 1^{\overline{14}} 3^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 15 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

[184] $1^{\overline{14}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 10 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

[185] $1^{\overline{21}} 1^{\overline{13}} 3^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 43 & 5 \\ 12 & 25 & 92 & 9 \\ 43 & 92 & 325 & 33 \\ 5 & 9 & 33 & 2 \end{pmatrix}_{b_0}$$

[187] $1^{\overline{12}} 3^{\overline{13}} 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -227, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 13 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

[186] $1^{\overline{12}} 3^{\overline{131}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -227, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 10 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[188] $1^{\overline{113}} 3^{\overline{121}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -240, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 45 & 5 \\ 12 & 25 & 92 & 9 \\ 45 & 92 & 325 & 33 \\ 5 & 9 & 33 & 2 \end{pmatrix}_{b_0}$$

[189] $1^{\overline{114}} 3^{\overline{131}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -242, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 64 & 5 \\ 12 & 25 & 128 & 9 \\ 64 & 128 & 634 & 46 \\ 5 & 9 & 46 & 2 \end{pmatrix}_{b_0}$$

[190] $1^{\overline{13}} 3^{\overline{111}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -251, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 10 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

- [191] $1^{11} 3^{121} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -251$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 13 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & -2 & 2 \end{pmatrix}_{b_0}$
- [192] $1^{13} 3^{112} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -252$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.73 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 12 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 17 & -2 & 2 \end{pmatrix}_{b_0}$
- [193] $1^{113} 3^{11} 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -257$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 13 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 13 & 28 & 73 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 15 & -2 & 1 \end{pmatrix}_{b_0}$
- [194] $1^{113} 3^{112} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -260$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 37 & 5 \\ 12 & 25 & 76 & 9 \\ 37 & 76 & 217 & 27 \\ 5 & 9 & 27 & 2 \end{pmatrix}_{b_0}$
- [195] $1^{113} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -261$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 36 & 5 \\ 12 & 25 & 76 & 9 \\ 36 & 76 & 217 & 27 \\ 5 & 9 & 27 & 2 \end{pmatrix}_{b_0}$
- [196] $1^{41} 3^{114} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -262$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 68 & 5 \\ 12 & 25 & 144 & 9 \\ 68 & 144 & 814 & 52 \\ 5 & 9 & 52 & 2 \end{pmatrix}_{b_0}$
- [197] $1^{11} 3^{113} 1^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -267$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 15 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & -2 & 1 \end{pmatrix}_{b_0}$
- [198] $1^{12} 3^{112} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -268$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.70 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 11 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & -2 & 2 \end{pmatrix}_{b_0}$
- [200] $1^{14} 3^{131} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -272$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 11 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & -2 & 2 \end{pmatrix}_{b_0}$
- [199] $1^{12} 3^{114} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -272$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.97 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 16 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & -2 & 2 \end{pmatrix}_{b_0}$
- [201] $1^{114} 3^{12} 1^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -274$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 20 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 20 & 40 & 154 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 22 & -2 & 1 \end{pmatrix}_{b_0}$
- [202] $1^{11} 3^{113} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -275$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 14 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[203] \ 1^{14} 3^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -278, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 3 \\ 4 & 9 & 48 & 0 & 5 \\ 20 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[204] \ 1^{14} 3^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -288, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 12 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[205] \ 1^1 13^1 1^{15} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -288, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.1$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 5 & 12 & 45 & 5 \\ 12 & 25 & 100 & 9 \\ 45 & 100 & 373 & 35 \\ 5 & 9 & 35 & 1 \end{pmatrix}_{b_0}$$

$$[207] \ 1^{14} 3^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -293, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 14 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[206] \ 1^{14} 3^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -293, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 18 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[208] \ 1^{11} 3^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 16 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[209] \ 1^1 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -322, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 44 & 5 \\ 12 & 25 & 96 & 9 \\ 44 & 96 & 346 & 34 \\ 5 & 9 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[210] \ 1^{13} 3^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -323, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 11 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[211] \ 1^{14} 3^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -323, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 12 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[212] \ 1^{14} 3^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -326, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.53 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 56 & 5 \\ 12 & 25 & 112 & 9 \\ 56 & 112 & 478 & 40 \\ 5 & 9 & 40 & 2 \end{pmatrix}_{b_0}$$

$$[214] \ 1^{14} 3^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -332, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 14 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

[213] $1^{12}3^{12}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -332, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 12 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & -2 & 2 \end{pmatrix}_{b_0}$$

[215] $1^{114}3^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -333, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 9 & 44 & 0 & 5 \\ 18 & 44 & 193 & 0 & 25 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 25 & -2 & 1 \end{pmatrix}_{b_0}$$

[216] $1^{114}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -338, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 48 & 5 \\ 12 & 25 & 96 & 9 \\ 48 & 96 & 346 & 34 \\ 5 & 9 & 34 & 2 \end{pmatrix}_{b_0}$$

[217] $1^{114}3^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -338, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 16 & 40 & 154 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$

[218] $1^{114}3^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -345, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 63 & 5 \\ 12 & 25 & 128 & 9 \\ 63 & 128 & 634 & 46 \\ 5 & 9 & 46 & 2 \end{pmatrix}_{b_0}$$

[220] $1^{11}3^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -347, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 14 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

[219] $1^{13}3^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -347, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 15 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 21 & -2 & 2 \end{pmatrix}_{b_0}$$

[221] $21^{112}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -350, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 20 & 30 & 7 \\ 20 & 25 & 40 & 9 \\ 30 & 40 & 58 & 14 \\ 7 & 9 & 14 & 3 \end{pmatrix}_{b_0}$$

[222] $1^{11}3^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -352, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 17 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

[223] $1^{31}3^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -354, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.47 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 12 & 60 & 5 \\ 12 & 25 & 128 & 9 \\ 60 & 128 & 634 & 46 \\ 5 & 9 & 46 & 2 \end{pmatrix}_{b_0}$$

[224] $21^{112}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -362, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 20 & 42 & 7 \\ 20 & 25 & 56 & 9 \\ 42 & 56 & 118 & 20 \\ 7 & 9 & 20 & 3 \end{pmatrix}_{b_0}$$

- [225] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -363$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 12 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & -2 & 2 \end{pmatrix}_{b_0}$
- [226] $1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -364$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.04 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 14 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & -2 & 1 \end{pmatrix}_{b_0}$
- [227] $1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -364$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 10.87 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 13 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & -2 & 2 \end{pmatrix}_{b_0}$
- [228] $1^{11} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -368$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.34 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 18 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & -2 & 1 \end{pmatrix}_{b_0}$
- [229] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -371$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 13 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & -2 & 2 \end{pmatrix}_{b_0}$
- [230] $1^1 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -374$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.55 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 52 & 5 \\ 12 & 25 & 112 & 9 \\ 52 & 112 & 478 & 40 \\ 5 & 9 & 40 & 2 \end{pmatrix}_{b_0}$
- [231] $1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -380$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 10.69 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 13 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 17 & -2 & 2 \end{pmatrix}_{b_0}$
- [232] $1^{114} 3^1 1^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -398$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 19 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 19 & 40 & 154 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 22 & -2 & 1 \end{pmatrix}_{b_0}$
- [233] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -401$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 45 & 5 \\ 12 & 25 & 96 & 9 \\ 45 & 96 & 346 & 34 \\ 5 & 9 & 34 & 2 \end{pmatrix}_{b_0}$
- [234] $1^{114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -401$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 1 & 4 & 17 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 17 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 19 & -2 & 1 \end{pmatrix}_{b_0}$
- [235] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -409$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$
 $\begin{pmatrix} 5 & 12 & 47 & 5 \\ 12 & 25 & 96 & 9 \\ 47 & 96 & 346 & 34 \\ 5 & 9 & 34 & 2 \end{pmatrix}_{b_0}$
- [236] $1^{114} 3^1 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -413$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 55 & 5 \\ 12 & 25 & 112 & 9 \\ 55 & 112 & 478 & 40 \\ 5 & 9 & 40 & 2 \end{pmatrix}_{b_0}$$

$$[237] \ 1^{12} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -416, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 17 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[238] \ 1^{14} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -427, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 13 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[239] \ 1^{14} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -430, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 46 & 5 \\ 12 & 25 & 96 & 9 \\ 46 & 96 & 346 & 34 \\ 5 & 9 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[240] \ 1^{13} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -432, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 18 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 25 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[241] \ 1^{12} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -435, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 15 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[242] \ 1^{14} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -437, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 16 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[243] \ 1^{14} 3^1 1^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -450, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 54 & 5 \\ 12 & 25 & 112 & 9 \\ 54 & 112 & 478 & 40 \\ 5 & 9 & 40 & 2 \end{pmatrix}_{b_0}$$

$$[244] \ 1^{14} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -467, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 31 & 0 & 4 \\ 17 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 24 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[245] \ 1^{13} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -476, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 14 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[246] \ 21^{13} 31^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -488, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 13 & 20 & 69 & 7 \\ 20 & 25 & 92 & 9 \\ 69 & 92 & 325 & 33 \\ 7 & 9 & 33 & 3 \end{pmatrix}_{b_0}$$

$$[247] \ 1^{12} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -491, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 16 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

[248] $1^{14}3^{11}3^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -492, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 15 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

[249] $1^{13}3^{11}3^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -499, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 16 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 21 & -2 & 2 \end{pmatrix}_{b_0}$$

[250] $1^{14}3^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -499, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 14 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

[251] $1^{12}3^{11}3^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -515, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 17 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

[252] $1^{12}3^{11}3^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -528, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 18 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & -2 & 2 \end{pmatrix}_{b_0}$$

[253] $1^{14}3^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -539, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 15 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

[254] $1^{13}3^{11}3^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -540, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 15 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[255] $1^{13}3^{11}3^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -572, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 16 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[256] $21^{114}31^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -578, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 20 & 96 & 7 \\ 20 & 25 & 128 & 9 \\ 96 & 128 & 634 & 46 \\ 7 & 9 & 46 & 3 \end{pmatrix}_{b_0}$$

[257] $1^{14}3^{11}3^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -592, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 20 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 28 & -2 & 2 \end{pmatrix}_{b_0}$$

[258] $21^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 20 & 45 & 7 \\ 20 & 25 & 60 & 9 \\ 45 & 60 & 133 & 21 \\ 7 & 9 & 21 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 18 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[271] \ 1^{14} 3 \ 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -800, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 21 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 28 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[272] \ 1^{14} 3 \ 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -803, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 31 & 0 & 4 \\ 19 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 24 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[273] \ 1^{13} 3 \ 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -864, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 21 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 25 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[274] \ 1^{14} 3 \ 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -923, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 31 & 0 & 4 \\ 20 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 24 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[275] \ 1^{13} 3 \ 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -944, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 22 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 22 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 25 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[276] \ 21^{114} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -950, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 13 & 20 & 60 & 7 \\ 20 & 25 & 80 & 9 \\ 60 & 80 & 238 & 28 \\ 7 & 9 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[277] \ 1^{14} 3 \ 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -976, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 22 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 22 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 28 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[278] \ 21^{114} 31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -998, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 13 & 20 & 84 & 7 \\ 20 & 25 & 112 & 9 \\ 84 & 112 & 478 & 40 \\ 7 & 9 & 40 & 3 \end{pmatrix}_{b_0}$$

$$[279] \ 1^{14} 3 \ 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -1011, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 31 & 0 & 4 \\ 21 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 24 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[280] \ 1^{14} 3 \ 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -1120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 3 & 23 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 23 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 28 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[281] \ 21^{114} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -1122, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 13 & 20 & 72 & 7 \\ 20 & 25 & 96 & 9 \\ 72 & 96 & 346 & 34 \\ 7 & 9 & 34 & 3 \end{pmatrix}_{b_0}$$

[282] $1^{14}3^{14}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -1232$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 1 & 3 & 24 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 24 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 28 & -2 & 2 \end{pmatrix}_{b_0}$

[283] $21^{115}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 7$, $T_{\max} = 22$, $\Delta = -113$, $\det = -1400$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.1$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 22$, $T \in [11, 22]$
 $\begin{pmatrix} 13 & 20 & 75 & 7 \\ 20 & 25 & 100 & 9 \\ 75 & 100 & 373 & 35 \\ 7 & 9 & 35 & 3 \end{pmatrix}_{b_0}$

$T = 8$ [285] $1^{11}4^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 2$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 15$, $T_{\max} = 20$, $T \in [15, 20]$
 $\begin{pmatrix} 10 & 16 & 8 & 6 \\ 16 & 25 & 12 & 9 \\ 8 & 12 & 5 & 5 \\ 6 & 9 & 5 & 0 \end{pmatrix}_{b_0}$

[284] $21^{14}4^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 2$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 15$, $T_{\max} = 20$, $T \in [15, 20]$
 $\begin{pmatrix} 10 & 16 & 8 & 6 \\ 16 & 25 & 12 & 9 \\ 8 & 12 & 5 & 5 \\ 6 & 9 & 5 & 0 \end{pmatrix}_{b_0}$

[286] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 4$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 3 & 5 & 4 & 3 & 4 \\ 3 & 5 & 3 & 0 & 2 \\ 3 & 5 & 4 & 2 & 3 \end{pmatrix}_{b_0}$

[287] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 6$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 4 & 0 & 10 & 6 \\ 3 & 3 & -1 & 6 & 3 \end{pmatrix}_{b_0}$

[289] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 8$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.50 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 1 & 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 3 & -2 & 2 \end{pmatrix}_{b_0}$

[288] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 8$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 0 & -2 & 2 \end{pmatrix}_{b_0}$

[291] $1^{11}4^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 11$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 11 & 0 & 4 \\ 7 & 11 & 28 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & -2 & 2 \end{pmatrix}_{b_0}$

[290] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 11$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$

[293] $1^{11}4^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 13$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 13.54 \rightarrow 14$, $T_{\max} = 20$, $T \in [14, 20]$
 $\begin{pmatrix} 5 & 12 & 6 & 5 \\ 12 & 25 & 12 & 9 \\ 6 & 12 & 5 & 5 \\ 5 & 9 & 5 & 1 \end{pmatrix}_{b_0}$

- [292] $1^{\overline{1}1}321^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 13$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.46 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 2 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & -2 & 3 \end{pmatrix}_{b_0}$
- [294] $1^{\overline{1}1}31^{\overline{1}1}31^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 14$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 12 & 0 & 13 & 7 \\ 3 & 6 & -1 & 7 & 2 \end{pmatrix}_{b_0}$
- [295] $1^{\overline{1}1}31^{\overline{1}1}31^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 14$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 & 2 \\ 2 & 3 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$
- [296] $1^{\overline{1}1}321^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 15$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 0 & 2 \\ 3 & 5 & 0 & 2 & 3 \end{pmatrix}_{b_0}$
- [297] $1^{\overline{1}1}31^{\overline{1}1}31^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 15$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 4 & 7 & 8 & 1 & 4 \\ 7 & 10 & 13 & 0 & 6 \\ 8 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 4 & 6 & 6 & 0 & 2 \end{pmatrix}_{b_0}$
- [298] $1^{\overline{1}1}41^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 16$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 12 & 20 & 90 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & -2 & 2 \end{pmatrix}_{b_0}$
- [299] $1^{\overline{1}1}31^{\overline{1}1}31^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 16$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 4 & 7 & 8 & 9 & 4 \\ 7 & 10 & 13 & 12 & 6 \\ 8 & 13 & 12 & 15 & 6 \\ 9 & 12 & 15 & 13 & 7 \\ 4 & 6 & 6 & 7 & 2 \end{pmatrix}_{b_0}$
- [300] $1^{\overline{1}1}321^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 17$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.59 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 7 & 0 & 21 & 9 \\ 3 & 3 & -1 & 9 & 3 \end{pmatrix}_{b_0}$
- [301] $1^{\overline{1}1}41^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 18$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.44 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 10 & 12 & 8 & 6 \\ 12 & 13 & 10 & 7 \\ 8 & 10 & 5 & 5 \\ 6 & 7 & 5 & 3 \end{pmatrix}_{b_0}$
- [302] $21^{\overline{1}1}41^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 18$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.44 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 10 & 12 & 28 & 6 \\ 12 & 13 & 34 & 7 \\ 28 & 34 & 77 & 17 \\ 6 & 7 & 17 & 3 \end{pmatrix}_{b_0}$
- [303] $21^{\overline{1}1}41^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 18$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.44 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 10 & 12 & 8 & 6 \\ 12 & 13 & 10 & 7 \\ 8 & 10 & 5 & 5 \\ 6 & 7 & 5 & 3 \end{pmatrix}_{b_0}$
- [304] $21^{\overline{1}1}41^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 18$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.44 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 10 & 12 & 12 & 6 \\ 12 & 13 & 14 & 7 \\ 12 & 14 & 13 & 7 \\ 6 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[305] $1^{12}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 12 & 28 & 6 \\ 12 & 13 & 34 & 7 \\ 28 & 34 & 77 & 17 \\ 6 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

[306] $1^{14}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 12 & 12 & 6 \\ 12 & 13 & 14 & 7 \\ 12 & 14 & 13 & 7 \\ 6 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[307] $1^{14}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

[308] $1^{11}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

[309] $1^{12}31^{13}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 4 & 6 & 3 & 3 \\ 1 & 4 & 0 & 6 & 3 & 2 \\ 3 & 6 & 6 & 10 & 6 & 6 \\ 2 & 3 & 3 & 6 & 2 & 2 \\ 2 & 3 & 2 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[310] $1^{11}31^{13}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[311] $1^{12}31^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 \\ 2 & 4 & 3 & 6 & 0 & 3 \\ 3 & 6 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[312] $1^{11}31^{13}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 1 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

[313] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 2 & 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

[314] $1^{11}31^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

[315] $1^141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 9 & 5 \\ 9 & 13 & 14 & 7 \\ 9 & 14 & 13 & 7 \\ 5 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[316] \ 1^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -104, \ \det = 27, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.48 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 13 & 21 & 101 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[317] \ 1^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -104, \ \det = 28, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.29 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 6 & 10 & 21 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[318] \ 1^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -104, \ \det = 28, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 10 & 12 & 20 & 6 \\ 12 & 13 & 24 & 7 \\ 20 & 24 & 38 & 12 \\ 6 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[319] \ 21^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -104, \ \det = 28, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 10 & 12 & 20 & 6 \\ 12 & 13 & 24 & 7 \\ 20 & 24 & 38 & 12 \\ 6 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[322] \ 1^{11}31^{11}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 30, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.40 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 & 2 \\ 2 & 3 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[320] \ 1^{11}3131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 30, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.40 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 1 & 6 & 3 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 12 & 0 & 10 & 6 \\ 3 & 6 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[321] \ 1^{12}31^{11}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 30, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.40 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 6 & 0 & 10 & 6 & 6 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 2 & 3 & -1 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[323] \ 1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 31, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 2 & 5 & 2 & 4 & 4 \\ 3 & 5 & 4 & 0 & 2 \\ 3 & 5 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[326] \ 1^{11}41^{16} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -104, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 15.75 \rightarrow 16, \ T_{\max} = 20, \ T \in [16, 20]$$

$$\begin{pmatrix} 5 & 9 & 36 & 5 \\ 9 & 13 & 60 & 7 \\ 36 & 60 & 250 & 30 \\ 5 & 7 & 30 & -1 \end{pmatrix}_{b_0}$$

$$[327] \ 1^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -104, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 5 & 9 & 6 & 5 \\ 9 & 13 & 10 & 7 \\ 6 & 10 & 5 & 5 \\ 5 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[328] \ 1^{11}41^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -104, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 5 & 9 & 18 & 5 \\ 9 & 13 & 30 & 7 \\ 18 & 30 & 61 & 15 \\ 5 & 7 & 15 & 2 \end{pmatrix}_{b_0}$$

[329] $21^1 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 13 & 14 & 9 & 7 \\ 14 & 13 & 10 & 7 \\ 9 & 10 & 5 & 5 \\ 7 & 7 & 5 & 3 \end{pmatrix}_{b_0}$

[330] $141^{15} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.1$
 $T_{\min} = 14.38 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$
 $\begin{pmatrix} 5 & 9 & 30 & 5 \\ 9 & 13 & 50 & 7 \\ 30 & 50 & 173 & 25 \\ 5 & 7 & 25 & 0 \end{pmatrix}_{b_0}$

[331] $141^{17} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.3$
 $T_{\min} = 17.38 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$
 $\begin{pmatrix} 5 & 9 & 42 & 5 \\ 9 & 13 & 70 & 7 \\ 42 & 70 & 341 & 35 \\ 5 & 7 & 35 & -3 \end{pmatrix}_{b_0}$

[332] $141^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 12 & 5 \\ 9 & 13 & 20 & 7 \\ 12 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0}$

[325] $141^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$
 $\begin{pmatrix} 5 & 9 & 24 & 5 \\ 9 & 13 & 40 & 7 \\ 24 & 40 & 110 & 20 \\ 5 & 7 & 20 & 1 \end{pmatrix}_{b_0}$

[324] $1^1 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 5 & 8 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & -2 & 2 \end{pmatrix}_{b_0}$

[333] $1^1 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 3 & 5 & 2 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & -2 & 3 \end{pmatrix}_{b_0}$

[334] $1^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 2 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 3 & -2 & 1 \end{pmatrix}_{b_0}$

[336] $1^{11} 31^{11} 31^{41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 9 & 1 & 2 \\ 1 & 1 & 4 & 12 & 0 & 3 \\ 2 & 4 & 3 & 15 & 0 & 3 \\ 9 & 12 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 13 & -2 & 1 \end{pmatrix}_{b_0}$

[335] $1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 3 & 4 & 0 & 3 & 3 \\ 2 & 2 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$

[337] $1^{11} 31^{41} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 1 & 25 & 3 \\ 4 & 10 & 1 & 60 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 60 & 0 & 346 & 34 \\ 3 & 6 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[338] $1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 3 \\ 4 & 10 & 13 & 12 & 6 \\ 5 & 13 & 12 & 15 & 6 \\ 5 & 12 & 15 & 13 & 7 \\ 3 & 6 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[341] $1^{12}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 10 & 6 & 6 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 2 & 3 & -1 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[340] $1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 7 & 8 & 7 & 4 \\ 7 & 10 & 13 & 12 & 6 \\ 8 & 13 & 12 & 12 & 6 \\ 7 & 12 & 12 & 10 & 6 \\ 4 & 6 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[339] $1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 7 & 8 & 8 & 4 \\ 7 & 10 & 13 & 12 & 6 \\ 8 & 13 & 12 & 15 & 6 \\ 8 & 12 & 15 & 13 & 7 \\ 4 & 6 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[342] $1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 6 & 3 \\ 3 & 6 & 0 & 6 & 11 & 5 \\ 2 & 3 & -1 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[343] $1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 & 2 \\ 2 & 3 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[344] $1^{11}31^{11}31^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 42, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 \\ 2 & 4 & 3 & 9 & 0 & 3 \\ 5 & 6 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[345] $1^{11}4^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 4 & 7 & 8 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

[346] $1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 3 \\ 4 & 10 & 13 & 12 & 6 \\ 5 & 13 & 12 & 12 & 6 \\ 5 & 12 & 12 & 10 & 6 \\ 3 & 6 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[347] $1^{11}3^{11}21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 5 & 10 & 0 & 5 \\ 5 & 10 & 14 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 8 & -2 & 3 \end{pmatrix}_{b_0}$$

[348] $1^{11}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 5 & 10 & 0 & 5 \\ 5 & 10 & 14 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 8 & -2 & 3 \end{pmatrix}_{b_0}$$

- [349] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.58 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 3 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 5 & -2 & 2 \end{pmatrix}_{b_0}$
- [350] $1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.17 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 4 & 7 & 8 & 8 & 4 \\ 7 & 10 & 13 & 12 & 6 \\ 8 & 13 & 12 & 12 & 6 \\ 8 & 12 & 12 & 10 & 6 \\ 4 & 6 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [351] $1^{14}31^{11}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 4 & 12 & 3 & 3 \\ 1 & 4 & 0 & 12 & 3 & 2 \\ 6 & 12 & 12 & 46 & 12 & 12 \\ 2 & 3 & 3 & 12 & 2 & 2 \\ 2 & 3 & 2 & 12 & 2 & 1 \end{pmatrix}_{b_0}$
- [352] $21^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$
 $\begin{pmatrix} 13 & 20 & 9 & 7 \\ 20 & 25 & 12 & 9 \\ 9 & 12 & 5 & 5 \\ 7 & 9 & 5 & 2 \end{pmatrix}_{b_0}$
- [353] $1^{12}31^{12}31^{41} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.5$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 10 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$
- [354] $1^{11}31^{31}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 50$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.8$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 19 & 3 \\ 4 & 10 & 1 & 48 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 48 & 0 & 217 & 27 \\ 3 & 6 & -1 & 27 & 2 \end{pmatrix}_{b_0}$
- [355] $1^{12}31^{11}31^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 51$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 7 & 1 & 2 \\ 1 & 1 & 4 & 9 & 0 & 3 \\ 2 & 4 & 3 & 12 & 0 & 3 \\ 7 & 9 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 10 & -2 & 1 \end{pmatrix}_{b_0}$
- [356] $1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 51$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 6 & 0 & 3 \\ 2 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 4 & -2 & 1 \end{pmatrix}_{b_0}$
- [359] $1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 52$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 3 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 4 & -2 & 1 \end{pmatrix}_{b_0}$
- [357] $1^{11}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 52$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 3 \\ 3 & 6 & 0 & 6 & 11 & 5 \\ 2 & 3 & -1 & 3 & 5 & 1 \end{pmatrix}_{b_0}$
- [358] $1^{14}31^{11}31^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 52$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 9 & 18 & 15 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$
- [360] $21^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{aligned}
& \begin{pmatrix} 13 & 14 & 15 & 7 \\ 14 & 13 & 14 & 7 \\ 15 & 14 & 13 & 7 \\ 7 & 7 & 7 & 3 \end{pmatrix}_{b_0} \\
[361] & 1^{\overline{1}} \overline{3} 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 3 & 5 & 2 & 4 & 4 \\ 3 & 5 & 4 & 0 & 2 \\ 3 & 5 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[362] & 1^{\overline{1}} \overline{3} 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 5 & 10 & 5 & 5 \\ 5 & 10 & 14 & 7 & 8 \\ 3 & 5 & 7 & 0 & 2 \\ 3 & 5 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[363] & 1^{\overline{11}} \overline{3} 131^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 12 & 0 & 10 & 6 \\ 3 & 6 & -1 & 6 & 2 \end{pmatrix}_{b_0} \\
[364] & 1^{\overline{12}} \overline{4} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.36 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 10 & 17 & 61 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & -2 & 2 \end{pmatrix}_{b_0} \\
[365] & 1^{\overline{12}} \overline{3} 131^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 3 & 6 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0} \\
[366] & 1^{\overline{11}} \overline{3} 1^{\overline{11}} \overline{3} 1^{\overline{21}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 5 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 7 & -2 & 1 \end{pmatrix}_{b_0} \\
[367] & 1^{\overline{13}} \overline{3} 1^{\overline{11}} \overline{3} 1^{\overline{31}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 8 & 15 & 12 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 13 & -2 & 1 \end{pmatrix}_{b_0} \\
[368] & 1^{\overline{11}} \overline{3} 1^{\overline{13}} \overline{3} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 1 & 9 & 3 \\ 4 & 10 & 1 & 24 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 24 & 0 & 49 & 13 \\ 3 & 6 & -1 & 13 & 2 \end{pmatrix}_{b_0} \\
[369] & 1^{\overline{14}} \overline{3} 1^{\overline{13}} \overline{3} 1^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 46 & 12 & 12 \\ 2 & 3 & 0 & 12 & 2 & 2 \\ 2 & 3 & -1 & 12 & 2 & 1 \end{pmatrix}_{b_0} \\
[370] & 1^{\overline{13}} \overline{3} 1^{\overline{11}} \overline{3} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 5 & 1 & 2 \\ 1 & 1 & 4 & 12 & 0 & 3 \\ 2 & 4 & 3 & 15 & 0 & 3 \\ 5 & 12 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 13 & -2 & 1 \end{pmatrix}_{b_0} \\
[371] & 1^{\overline{11}} \overline{3} 1^{\overline{13}} \overline{3} 1^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 1 & 21 & 3 \\ 4 & 10 & 1 & 48 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 48 & 0 & 217 & 27 \\ 3 & 6 & -1 & 27 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [372] $1^{12}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 11 & 0 & 4 \\ 6 & 11 & 21 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & -2 & 2 \end{pmatrix}_{b_0}$
- [373] $1^141^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 5 & 9 & 15 & 5 \\ 9 & 13 & 24 & 7 \\ 15 & 24 & 38 & 12 \\ 5 & 7 & 12 & 3 \end{pmatrix}_{b_0}$
- [374] $1^{123}1^131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 6 & 0 & 10 & 6 & 6 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 2 & 3 & -1 & 6 & 2 & 1 \end{pmatrix}_{b_0}$
- [375] $21^{11}41^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 13 & 14 & 51 & 7 \\ 14 & 13 & 54 & 7 \\ 51 & 54 & 197 & 27 \\ 7 & 7 & 27 & 3 \end{pmatrix}_{b_0}$
- [376] $1^{11}31^{123}1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 8 & 15 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0}$
- [378] $1^{111}31^{131}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 11 & 3 \\ 4 & 10 & 1 & 24 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 24 & 0 & 49 & 13 \\ 3 & 6 & -1 & 13 & 2 \end{pmatrix}_{b_0}$
- [379] $1^{111}31^{231}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 14 & 3 \\ 4 & 10 & 1 & 36 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 36 & 0 & 118 & 20 \\ 3 & 6 & -1 & 20 & 2 \end{pmatrix}_{b_0}$
- [377] $1^{123}1^131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 10 & 6 & 6 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 2 & 3 & -1 & 6 & 2 & 1 \end{pmatrix}_{b_0}$
- [380] $1^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 11 & 18 & 70 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 2 \end{pmatrix}_{b_0}$
- [381] $1^{12}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 38 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & -2 & 2 \end{pmatrix}_{b_0}$
- [382] $1^{11}321^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 5 & 10 & 5 & 5 \\ 5 & 10 & 14 & 8 & 8 \\ 3 & 5 & 8 & 0 & 2 \\ 3 & 5 & 8 & 2 & 3 \end{pmatrix}_{b_0}$
- [383] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 3 & 3 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[384] \ 1^{11} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 5 & 10 & 5 & 5 \\ 6 & 10 & 14 & 7 & 8 \\ 3 & 5 & 7 & 0 & 2 \\ 3 & 5 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[385] \ 1^{12} 31^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 4 & 12 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[387] \ 1^{13} 1^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 2 & 6 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[386] \ 1^{14} 31^{11} 31^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 8 & 18 & 15 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[389] \ 1^{13} 31^1 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 9 & 0 & 25 & 9 & 9 \\ 2 & 3 & 0 & 9 & 2 & 2 \\ 2 & 3 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[388] \ 1^{11} 31^{23} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 1 & 16 & 3 \\ 4 & 10 & 1 & 36 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 36 & 0 & 118 & 20 \\ 3 & 6 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[390] \ 21^{11} 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 10 & 7 & 5 \\ 10 & 13 & 10 & 7 \\ 7 & 10 & 5 & 5 \\ 5 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[391] \ 1^{11} 31^3 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 1 & 20 & 3 \\ 4 & 10 & 1 & 48 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 48 & 0 & 217 & 27 \\ 3 & 6 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[392] \ 1^{11} 31^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 4 & 9 & 0 & 3 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 4 & 9 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[393] \ 1^{13} 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 8 & 15 & 40 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[394] \ 1^{11} 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.42 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 7 & 12 & 26 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & -2 & 2 \end{pmatrix}_{b_0} \\
[395] & 1^{\overline{11}1}31^{\overline{11}31}2^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 7 & 8 & 17 & 4 \\ 7 & 10 & 13 & 24 & 6 \\ 8 & 13 & 12 & 27 & 6 \\ 17 & 24 & 27 & 49 & 13 \\ 4 & 6 & 6 & 13 & 2 \end{pmatrix}_{b_0} \\
[396] & 1^{\overline{11}1}41^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 9 & 15 & 45 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & -2 & 2 \end{pmatrix}_{b_0} \\
[397] & 1^{\overline{11}1}31^{\overline{11}31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 1 & 10 & 3 \\ 4 & 10 & 1 & 24 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 24 & 0 & 49 & 13 \\ 3 & 6 & -1 & 13 & 2 \end{pmatrix}_{b_0} \\
[398] & 1^{\overline{11}1}31^{\overline{11}31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 3 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 \\ 2 & 4 & 3 & 9 & 0 & 3 \\ 3 & 6 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 7 & -2 & 1 \end{pmatrix}_{b_0} \\
[399] & 1^{\overline{11}1}41^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 9 & 21 & 5 \\ 9 & 13 & 34 & 7 \\ 21 & 34 & 77 & 17 \\ 5 & 7 & 17 & 3 \end{pmatrix}_{b_0} \\
[400] & 1^{\overline{11}1}321^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 4 & 0 & 2 \\ 1 & 2 & 0 & 7 & 0 & 2 \\ 4 & 4 & 7 & 16 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 8 & -2 & 2 \end{pmatrix}_{b_0} \\
[401] & 1^{\overline{11}1}31^{\overline{11}2}31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 1 & 15 & 3 \\ 4 & 10 & 1 & 36 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 36 & 0 & 118 & 20 \\ 3 & 6 & -1 & 20 & 2 \end{pmatrix}_{b_0} \\
[402] & 1^{\overline{12}1}31^{\overline{11}31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 4 & 1 & 2 \\ 1 & 1 & 4 & 9 & 0 & 3 \\ 2 & 4 & 3 & 12 & 0 & 3 \\ 4 & 9 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 10 & -2 & 1 \end{pmatrix}_{b_0} \\
[403] & 1^{\overline{13}1}41^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 121 & 0 & 21 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 21 & -2 & 2 \end{pmatrix}_{b_0} \\
[404] & 1^{\overline{14}1}41^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 10 & 19 & 65 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & -2 & 2 \end{pmatrix}_{b_0} \\
[405] & 1^{\overline{11}1}321^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 5 & 15 & 0 & 5 \\ 8 & 15 & 32 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 12 & -2 & 3 \end{pmatrix}_{b_0}$$

[406] $2 \overset{1}{1} \overset{1}{1} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 5 & 5 & 5 & 5 \\ 3 & 5 & 2 & 4 & 4 \\ 3 & 5 & 4 & 0 & 2 \\ 3 & 5 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[407] $1 \overset{1}{1} 131^1 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 3 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[408] $1 \overset{1}{1} 131^1 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 7 & 8 & 15 & 4 \\ 7 & 10 & 13 & 24 & 6 \\ 8 & 13 & 12 & 27 & 6 \\ 15 & 24 & 27 & 49 & 13 \\ 4 & 6 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[409] $1 \overset{1}{1} \overset{1}{3} 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 10 & 18 & 61 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & -2 & 2 \end{pmatrix}_{b_0}$$

[410] $1 \overset{1}{1} \overset{1}{3} 31^1 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 6 & 15 & 12 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[411] $21 \overset{1}{1} \overset{1}{1} 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 13 & 14 & 24 & 7 \\ 14 & 13 & 24 & 7 \\ 24 & 24 & 38 & 12 \\ 7 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

[412] $1 \overset{1}{1} 131^1 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 \\ 2 & 4 & 3 & 9 & 0 & 3 \\ 4 & 6 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[413] $1 \overset{1}{1} \overset{1}{2} 3 \overset{1}{2} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 5 & 0 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 \\ 4 & 5 & 5 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[414] $31 \overset{1}{1} \overset{1}{1} 231^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 4 & 1 & 2 \\ 3 & 1 & 4 & 4 & 1 & 2 \\ 3 & 4 & 0 & 4 & 1 & 2 \\ 4 & 4 & 4 & 5 & 3 & 5 \\ 1 & 1 & 1 & 3 & -2 & 0 \\ 2 & 3 & 2 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

[415] $1 \overset{1}{1} \overset{1}{1} 31^1 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 5 & 10 & 3 \\ 4 & 10 & 13 & 24 & 6 \\ 5 & 13 & 12 & 27 & 6 \\ 10 & 24 & 27 & 49 & 13 \\ 3 & 6 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[416] $1 \overset{1}{1} \overset{1}{3} 31^1 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 9 & 0 & 25 & 9 & 9 \\ 2 & 3 & 0 & 9 & 2 & 2 \\ 2 & 3 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[417] \ 1^{13}4^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 12 & 21 & 88 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[418] \ 1^{13}31^{131}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 134, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 7 & 9 & 0 & 25 & 9 & 9 \\ 2 & 3 & 0 & 9 & 2 & 2 \\ 2 & 3 & -1 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[419] \ 21^{11}41^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 134, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 14 & 42 & 7 \\ 14 & 13 & 44 & 7 \\ 42 & 44 & 130 & 22 \\ 7 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[420] \ 131^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 5 & 9 & 3 \\ 4 & 10 & 13 & 24 & 6 \\ 5 & 13 & 12 & 27 & 6 \\ 9 & 24 & 27 & 49 & 13 \\ 3 & 6 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[421] \ 21^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 10 & 16 & 5 \\ 10 & 13 & 24 & 7 \\ 16 & 24 & 38 & 12 \\ 5 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[422] \ 21^{12}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 10 & 14 & 5 \\ 10 & 13 & 20 & 7 \\ 14 & 20 & 26 & 10 \\ 5 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[425] \ 1^{13}1^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 5 & 12 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[423] \ 1^{13}1^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 4 & 12 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[424] \ 1^{14}31^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 8 & 24 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 20 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[426] \ 1^41^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 9 & 27 & 5 \\ 9 & 13 & 44 & 7 \\ 27 & 44 & 130 & 22 \\ 5 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[427] \ 1^{12}31^{11}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 6 & 12 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[428] $21^{11}41^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8, T_{\max} = 20, \Delta = -104, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 13 & 14 & 33 & 7 \\ 14 & 13 & 34 & 7 \\ 33 & 34 & 77 & 17 \\ 7 & 7 & 17 & 3 \end{pmatrix}_{b_0}$

[429] $1^{14}31^{13}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 7 & 12 & 0 & 46 & 12 & 12 \\ 2 & 3 & 0 & 12 & 2 & 2 \\ 2 & 3 & -1 & 12 & 2 & 1 \end{pmatrix}_{b_0}$

[430] $1^{13}31^{11}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 7 & 15 & 12 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 13 & -2 & 1 \end{pmatrix}_{b_0}$

[431] $1^{11}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 4 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 7 & -2 & 1 \end{pmatrix}_{b_0}$

[432] $1^{11}131^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 159, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 4 & 7 & 8 & 16 & 4 \\ 7 & 10 & 13 & 24 & 6 \\ 8 & 13 & 12 & 27 & 6 \\ 16 & 24 & 27 & 49 & 13 \\ 4 & 6 & 6 & 13 & 2 \end{pmatrix}_{b_0}$

[433] $1^{12}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8, T_{\max} = 20, \Delta = -104, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.47 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 9 & 16 & 45 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & -2 & 2 \end{pmatrix}_{b_0}$

[434] $2 \ 2 \ 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 5 & 10 & 5 & 5 \\ 6 & 10 & 14 & 7 & 8 \\ 3 & 5 & 7 & 0 & 2 \\ 3 & 5 & 8 & 2 & 3 \end{pmatrix}_{b_0}$

[435] $1^{11}3 \ 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 5 & 20 & 0 & 5 \\ 10 & 20 & 58 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 16 & -2 & 3 \end{pmatrix}_{b_0}$

[436] $1^{12}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8, T_{\max} = 20, \Delta = -104, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 25 & 0 & 4 \\ 15 & 25 & 132 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & -2 & 2 \end{pmatrix}_{b_0}$

[437] $1^{11}43131^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 4 & 12 & 0 & 3 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 6 & 12 & 12 & 46 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$

[438] $1^{11}3 \ 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 5 & 15 & 5 & 5 \\ 8 & 15 & 32 & 11 & 12 \\ 3 & 5 & 11 & 0 & 2 \\ 3 & 5 & 12 & 2 & 3 \end{pmatrix}_{b_0}$

- [439] $1^{14}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 12 & 22 & 90 & 0 & 18 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$
- [440] $1^{12}31^{11}31^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 1 & 2 \\ 1 & 1 & 4 & 9 & 0 & 3 \\ 2 & 4 & 3 & 12 & 0 & 3 \\ 6 & 9 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$
- [441] $1^{13}31^{13}31^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 8 & 18 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$
- [442] $1^{12}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 5 & 12 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$
- [443] $1^{14}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 1 & 2 \\ 1 & 1 & 4 & 15 & 0 & 3 \\ 2 & 4 & 3 & 18 & 0 & 3 \\ 7 & 15 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$
- [444] $1^41^{14}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 33 & 5 \\ 9 & 13 & 54 & 7 \\ 33 & 54 & 197 & 27 \\ 5 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$
- [445] $1^{12}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 11 & 19 & 68 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$
- [446] $1^{12}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 13 & 22 & 97 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & -2 & 2 \end{pmatrix}_{b_0}$$
- [447] $1^{12}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 5 & 20 & 5 & 5 \\ 11 & 20 & 62 & 14 & 16 \\ 3 & 5 & 14 & 0 & 2 \\ 3 & 5 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$
- [448] $1^{13}31^{14}31^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 11 & 24 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$
- [449] $1^{12}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 1 & 2 \\ 1 & 1 & 4 & 9 & 0 & 3 \\ 2 & 4 & 3 & 12 & 0 & 3 \\ 5 & 9 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[450] $1^{14}31^{11}31^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 10 & 1 & 2 \\ 1 & 1 & 4 & 15 & 0 & 3 \\ 2 & 4 & 3 & 18 & 0 & 3 \\ 10 & 15 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[451] $1^{121}2^231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 5 & 10 & 5 & 5 \\ 6 & 10 & 14 & 8 & 8 \\ 3 & 5 & 8 & 0 & 2 \\ 3 & 5 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[452] $1^{13}31^{11}31^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 1 & 2 \\ 1 & 1 & 4 & 12 & 0 & 3 \\ 2 & 4 & 3 & 15 & 0 & 3 \\ 8 & 12 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[453] $1^{12}3^221^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 \\ 1 & 2 & 0 & 8 & 0 & 2 \\ 5 & 6 & 8 & 26 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[454] $1^{14}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 25 & 0 & 4 \\ 14 & 25 & 121 & 0 & 21 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 21 & -2 & 2 \end{pmatrix}_{b_0}$$

[455] $1^{14}31^{11}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 12 & 0 & 46 & 12 & 12 \\ 2 & 3 & 0 & 12 & 2 & 2 \\ 2 & 3 & -1 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[456] $1^{14}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 16 & 28 & 158 & 0 & 24 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 24 & -2 & 2 \end{pmatrix}_{b_0}$$

[457] $1^{13}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 1 & 2 \\ 1 & 1 & 4 & 12 & 0 & 3 \\ 2 & 4 & 3 & 15 & 0 & 3 \\ 6 & 12 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[458] $1^{13}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 11 & 20 & 70 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

[459] $21^{13}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 10 & 21 & 5 \\ 10 & 13 & 30 & 7 \\ 21 & 30 & 61 & 15 \\ 5 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[460] $1^{11}3^221^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 5 & 20 & 5 & 5 \\ 10 & 20 & 58 & 15 & 16 \\ 3 & 5 & 15 & 0 & 2 \\ 3 & 5 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

- [461] $1^{13}31^{13}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 6 & 18 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$
- [462] $1^{114}31^{13}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 1 & 4 & 15 & 0 & 3 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 7 & 15 & 15 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$
- [463] $1^{13}31^{11}31^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 258, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 1 & 2 \\ 1 & 1 & 4 & 12 & 0 & 3 \\ 2 & 4 & 3 & 15 & 0 & 3 \\ 7 & 12 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$
- [464] $1^{12}31^{11}21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 5 & 25 & 0 & 5 \\ 13 & 25 & 94 & 0 & 20 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 20 & -2 & 3 \end{pmatrix}_{b_0}$$
- [465] $1^{11}31^{12}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 5 & 15 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$
- [466] $1^{13}31^{12}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 7 & 21 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 17 & -2 & 1 \end{pmatrix}_{b_0}$$
- [467] $1^{13}31^{12}31^{41} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 11 & 21 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 17 & -2 & 1 \end{pmatrix}_{b_0}$$
- [468] $1^{11}31^{12}31^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 7 & 15 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$
- [469] $1^{13}41^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 275, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 13 & 23 & 97 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & -2 & 2 \end{pmatrix}_{b_0}$$
- [470] $1^{13}31^{13}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 7 & 18 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$
- [471] $1^{114}31^{23}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 284, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 1 & 4 & 18 & 0 & 3 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 8 & 18 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 20 & -2 & 1 \end{pmatrix}_{b_0}$$

- [472] $1^{13}3^{11}21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 284, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 \\ 6 & 8 & 9 & 40 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$
- [473] $21^{12}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 20, \Delta = -104, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 10 & 23 & 5 \\ 10 & 13 & 34 & 7 \\ 23 & 34 & 77 & 17 \\ 5 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$
- [474] $1^{14}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 20, \Delta = -104, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 13 & 24 & 101 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & -2 & 2 \end{pmatrix}_{b_0}$$
- [475] $1^{14}31^{11}31^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 291, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 1 & 2 \\ 1 & 1 & 4 & 15 & 0 & 3 \\ 2 & 4 & 3 & 18 & 0 & 3 \\ 8 & 15 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$
- [476] $1^{14}31^{11}31^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 295, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 9 & 1 & 2 \\ 1 & 1 & 4 & 15 & 0 & 3 \\ 2 & 4 & 3 & 18 & 0 & 3 \\ 9 & 15 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$
- [477] $1^{13}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 20, \Delta = -104, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 29 & 0 & 4 \\ 17 & 29 & 169 & 0 & 25 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 25 & -2 & 2 \end{pmatrix}_{b_0}$$
- [478] $1^{13}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 20, \Delta = -104, \det = 300, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 15 & 26 & 130 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & -2 & 2 \end{pmatrix}_{b_0}$$
- [479] $1^{12}31^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 6 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$
- [480] $1^{12}31^{12}31^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 9 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$
- [481] $2^{13}2^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 5 & 15 & 5 & 5 \\ 9 & 15 & 34 & 12 & 12 \\ 3 & 5 & 12 & 0 & 2 \\ 3 & 5 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$
- [482] $1^{12}3^{11}21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 324, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 5 & 25 & 5 & 5 \\ 13 & 25 & 94 & 18 & 20 \\ 3 & 5 & 18 & 0 & 2 \\ 3 & 5 & 20 & 2 & 3 \end{pmatrix}_{b_0}$$

- [483] $1^{11}31^{12}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 1 & 2 \\ 2 & 4 & 4 & 15 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 6 & 15 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$
- [484] $2^{1121}2231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 344, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 5 & 20 & 5 & 5 \\ 12 & 20 & 62 & 14 & 16 \\ 3 & 5 & 14 & 0 & 2 \\ 3 & 5 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$
- [485] $1^{11}31^{13}31^{31} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 348, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 10 & 21 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$
- [486] $21^{14}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 358, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 10 & 28 & 5 \\ 10 & 13 & 40 & 7 \\ 28 & 40 & 110 & 20 \\ 5 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$
- [487] $1^{14}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 363, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 15 & 27 & 132 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & -2 & 2 \end{pmatrix}_{b_0}$$
- [488] $2^{1121}2231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 5 & 15 & 5 & 5 \\ 9 & 15 & 32 & 11 & 12 \\ 3 & 5 & 11 & 0 & 2 \\ 3 & 5 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$
- [489] $1^{13}31^{14}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 8 & 24 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$
- [490] $1^{13}31^{14}31^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 10 & 24 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$
- [491] $1^{14}31^{12}31^{41} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 12 & 24 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 20 & -2 & 1 \end{pmatrix}_{b_0}$$
- [492] $1^{14}31^{12}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 9 & 24 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 20 & -2 & 1 \end{pmatrix}_{b_0}$$
- [493] $1^{11}31^{14}31^{41} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 408, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 13 & 27 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$
- [494] $31^{111}231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 409, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 3 & 1 & 3 \\ 3 & 4 & 0 & 3 & 1 & 2 \\ 3 & 3 & 3 & 1 & 1 & 3 \\ 1 & 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[495] \ 1^{14}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 412, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 17 & 30 & 169 & 0 & 25 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 25 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[496] \ 1^{12}31^{12}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 7 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[497] \ 1^{12}31^{12}31^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 18 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 8 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[498] \ 1^{14}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 435, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.42 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 33 & 0 & 4 \\ 19 & 33 & 212 & 0 & 28 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 28 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[499] \ 1^{12}31^{13}31^{41} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 12 & 24 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[500] \ 1^{11}31^{13}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 444, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 7 & 21 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[501] \ 1^{11}31^{14}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 9 & 24 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[502] \ 1^{13}31^{12}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 8 & 21 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 17 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[503] \ 1^{13}31^{12}31^{31} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 10 & 21 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 17 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[504] \ 21^{13}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 470, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 10 & 30 & 5 \\ 10 & 13 & 44 & 7 \\ 30 & 44 & 130 & 22 \\ 5 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[505] \ 2 \ 2 \ 31^{141} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 484, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 5 & 20 & 5 & 5 \\ 12 & 20 & 62 & 16 & 16 \\ 3 & 5 & 16 & 2 & 3 \\ 3 & 5 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[506] \ 1^{11}31^{13}31^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 508, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 9 & 21 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[508] \ 21^{15}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 520, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.1$$

$$T_{\min} = 11.42 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 10 & 35 & 5 \\ 10 & 13 & 50 & 7 \\ 35 & 50 & 173 & 25 \\ 5 & 7 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[507] \ 1^{13}31^{12}31^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 520, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 9 & 21 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 17 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[509] \ 1^{14}31^{12}31^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 11 & 24 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 20 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[510] \ 1^{14}31^{12}31^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 10 & 24 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 20 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[511] \ 1^{11}31^{13}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 540, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 21 & 0 & 4 \\ 1 & 4 & 0 & 12 & 0 & 2 \\ 8 & 21 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[512] \ 1^{13}31^{15}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.1$$

$$T_{\min} = 11.35 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 30 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 10 & 30 & 15 & 148 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 20 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[513] \ 1^{12}31^{13}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 8 & 24 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[514] \ 1^{14}31^{13}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 30 & 0 & 4 \\ 1 & 4 & 0 & 21 & 0 & 2 \\ 10 & 30 & 21 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 24 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[515] \ 1^{13}31^{13}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 604, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 9 & 27 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[516] \ 2 \ 2 \ 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 625, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 5 & 20 & 5 & 5 \\ 12 & 20 & 58 & 15 & 16 \\ 3 & 5 & 15 & 0 & 2 \\ 3 & 5 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[517] \quad 2 \quad 2 \quad 31^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 649, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 18 & 3 & 3 \\ 4 & 5 & 30 & 5 & 5 \\ 18 & 30 & 142 & 21 & 24 \\ 3 & 5 & 21 & 0 & 2 \\ 3 & 5 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[518] \quad 1^{12}31^{13}31^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 660, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 11 & 24 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[519] \quad 1^{11}31^{14}31^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 664, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.19 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 12 & 27 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[520] \quad 1^{11}31^{14}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 664, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.19 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 9 & 27 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[521] \quad 21^{14}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 688, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 10 & 37 & 5 \\ 10 & 13 & 54 & 7 \\ 37 & 54 & 197 & 27 \\ 5 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[522] \quad 21^{16}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 718, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 10 & 42 & 5 \\ 10 & 13 & 60 & 7 \\ 42 & 60 & 250 & 30 \\ 5 & 7 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[523] \quad 1^{12}31^{13}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 9 & 24 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[524] \quad 1^{13}31^{13}31^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 732, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 13 & 1 & 2 \\ 2 & 4 & 4 & 27 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 13 & 27 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[525] \quad 2 \quad 2 \quad 31^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 736, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 & 3 \\ 4 & 5 & 25 & 5 & 5 \\ 15 & 25 & 94 & 18 & 20 \\ 3 & 5 & 18 & 0 & 2 \\ 3 & 5 & 20 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[526] \quad 1^{12}31^{13}31^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 756, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 1 & 2 \\ 2 & 4 & 4 & 24 & 0 & 4 \\ 1 & 4 & 0 & 15 & 0 & 2 \\ 10 & 24 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[527] \quad 1^{13}31^{16}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 784, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 1 & 2 \\ 2 & 4 & 4 & 36 & 0 & 4 \\ 1 & 4 & 0 & 18 & 0 & 2 \\ 12 & 36 & 18 & 214 & 0 & 24 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 24 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 3 & -2 & 3 \end{pmatrix}_{b_0}$$

$$[562] \ 1^{\overline{11}} 4 \ 131^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 4 & 5 & 5 & 1 & 5 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 4 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[563] \ 1^{\overline{11}} 4 \ 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[564] \ 1^{\overline{11}} 4 \ 131^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 3 & 1 & 5 \\ 3 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 5 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[565] \ 1^{\overline{11}} 2321^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 2 & 1 & 8 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 8 & 0 & 22 & 10 \\ 3 & 4 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[566] \ 1^{\overline{11}} 4 \ 131^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 0 & 1 & 1 & 2 \\ 2 & 1 & 0 & 0 & 2 \\ 1 & 1 & 0 & -2 & 0 \\ 3 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[567] \ 1^{\overline{11}} 4 \ 131^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -17, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 3 & 4 & 2 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 4 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[568] \ 1^{\overline{11}} 4 \ 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 6 & 7 & 9 & 1 & 7 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 4 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[569] \ 1^{\overline{11}} 4 \ 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 0 & 1 & 2 & 2 \\ 2 & 1 & 0 & 1 & 2 \\ 3 & 2 & 1 & 0 & 2 \\ 3 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[570] \ 1^{\overline{11}} 4 \ 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 0 & 1 & 1 & 2 \\ 2 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 3 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[572] \ 1^{\overline{11}} 4 \ 131^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[571] \ 1^{\overline{11}} 4 \ 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 2 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[573] \ 1^{\overline{1}} 4 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 0 & 1 & 2 & 2 \\ 2 & 1 & 0 & 2 & 2 \\ 3 & 2 & 2 & 0 & 2 \\ 3 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[574] \ 1^{\overline{121}} 2321^{\overline{21}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 1 & 8 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 16 & 12 \\ 2 & 4 & 0 & 16 & 4 & 4 \\ 2 & 3 & -1 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[575] \ 1^{\overline{131}} 2321^{\overline{31}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 1 & 12 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 24 & 18 \\ 2 & 4 & 0 & 24 & 4 & 4 \\ 2 & 3 & -1 & 18 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[576] \ 1^{\overline{111}} 2321^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 4 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 8 & 6 \\ 2 & 4 & 0 & 8 & 4 & 4 \\ 2 & 3 & -1 & 6 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[577] \ 1^{\overline{141}} 2321^{\overline{41}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & 1 & 16 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 32 & 24 \\ 2 & 4 & 0 & 32 & 4 & 4 \\ 2 & 3 & -1 & 24 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[578] \ 1^{\overline{112}} 2321^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 4 & 0 & 3 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 6 & -2 & 3 \end{pmatrix}_{b_0}$$

$$[579] \ 1^{\overline{1131}} 4^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 0 & 2 & 1 & 2 \\ 4 & 2 & 2 & 1 & 4 \\ 1 & 1 & 1 & -2 & 0 \\ 3 & 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[580] \ 1^{\overline{114}} 4 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 0 & 3 & 1 & 2 \\ 3 & 3 & 2 & 1 & 4 \\ 1 & 1 & 1 & -2 & 0 \\ 3 & 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[581] \ 1^{\overline{114}} 4 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 2 & 2 & 1 & 1 & 1 & 3 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[582] \ 2^{\overline{141}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -49, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 5 & 3 & 3 \\ 4 & 5 & 7 & 5 & 5 \\ 5 & 7 & 8 & 5 & 6 \\ 3 & 5 & 5 & 0 & 2 \\ 3 & 5 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[583] $1^{11} 4^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 2 & 4 \\ 1 & 1 & 0 & 2 & -2 & 0 \\ 2 & 2 & 0 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

[584] $1^{11} 4^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 0 & 3 & 2 & 2 \\ 3 & 3 & 2 & 3 & 4 \\ 3 & 2 & 3 & 0 & 2 \\ 3 & 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[585] $1^{11} 4^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 4 & 3 & 3 \\ 2 & 0 & 2 & 2 & 2 \\ 4 & 2 & 2 & 3 & 4 \\ 3 & 2 & 3 & 0 & 2 \\ 3 & 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[586] $31^{11} 31^{11} 41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 6 & 8 & 4 & 3 \\ 6 & 13 & 20 & 6 & 7 \\ 8 & 20 & 26 & 8 & 10 \\ 4 & 6 & 8 & 0 & 2 \\ 3 & 7 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[587] $31^{11} 31^{11} 41^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -79, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 4 & 4 & 3 & 2 & 4 \\ 3 & 4 & 0 & 3 & 1 & 2 \\ 3 & 3 & 3 & 1 & 1 & 3 \\ 1 & 2 & 1 & 1 & -2 & 0 \\ 2 & 4 & 2 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

[588] $1^{11} 4^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 5 & 3 & 3 \\ 2 & 0 & 4 & 2 & 2 \\ 5 & 4 & 6 & 4 & 6 \\ 3 & 2 & 4 & 0 & 2 \\ 3 & 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[589] $1^{11} 4^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 5 & 1 & 3 \\ 2 & 0 & 4 & 1 & 2 \\ 5 & 4 & 6 & 2 & 6 \\ 1 & 1 & 2 & -2 & 0 \\ 3 & 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[590] $1^{11} 4^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 4 & 3 & 1 & 5 & 2 & 5 \\ 1 & 1 & 0 & 2 & -2 & 0 \\ 2 & 2 & 0 & 5 & 0 & 2 \end{pmatrix}_{b_0}$$

[591] $31^{11} 31^{11} 41^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 6 & 5 & 4 & 3 \\ 6 & 13 & 12 & 6 & 7 \\ 5 & 12 & 8 & 5 & 6 \\ 4 & 6 & 5 & 0 & 2 \\ 3 & 7 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[592] $1^{11} 4^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 7 & 1 & 3 \\ 2 & 0 & 5 & 1 & 2 \\ 7 & 5 & 12 & 2 & 8 \\ 1 & 1 & 2 & -2 & 0 \\ 3 & 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

[593] $31^{11} 31^{11} 41^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 4 & 4 & 6 & 2 & 4 \\ 3 & 4 & 0 & 3 & 1 & 2 \\ 3 & 6 & 3 & 4 & 2 & 4 \\ 1 & 2 & 1 & 2 & -2 & 0 \\ 2 & 4 & 2 & 4 & 0 & 1 \end{pmatrix}_{b_0}$$

[594] $1^{12} 1^{11} 31^1 4^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 5 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 5 & 5 & 1 & 13 & 3 & 7 \\ 1 & 1 & 0 & 3 & -2 & 0 \\ 2 & 2 & 0 & 7 & 0 & 2 \end{pmatrix}_{b_0}$$

[595] $1^{11} 1^{12} 31^2 4^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -239, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 9 & 1 & 3 \\ 2 & 0 & 6 & 1 & 2 \\ 9 & 6 & 20 & 3 & 10 \\ 1 & 1 & 3 & -2 & 0 \\ 3 & 2 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[596] $1^{12} 1^{11} 31^2 4^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 6 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 7 & 6 & 1 & 23 & 4 & 9 \\ 1 & 1 & 0 & 4 & -2 & 0 \\ 2 & 2 & 0 & 9 & 0 & 2 \end{pmatrix}_{b_0}$$

[597] $31^{11} 31^{11} 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -439, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 6 & 1 & 2 \\ 3 & 4 & 4 & 9 & 2 & 4 \\ 3 & 4 & 0 & 6 & 1 & 2 \\ 6 & 9 & 6 & 13 & 3 & 7 \\ 1 & 2 & 1 & 3 & -2 & 0 \\ 2 & 4 & 2 & 7 & 0 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [598] $1^1 5^1 1^{12} 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 7 & 0 & 26 & 10 \\ 3 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[599] $1^{11} 5^1 1231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 17, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 7 & 0 & 21 & 9 \\ 3 & 3 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

[600] $2^3 3^1 151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 9 & 6 & 7 \\ 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

[601] $1^1 41^{11} 31^{11} 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 8 & 1 & 7 & 3 \\ 8 & 38 & 1 & 32 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 32 & 0 & 26 & 10 \\ 3 & 12 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[602] $1^{11} 5^1 2^{12} 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 21 & 0 & 225 & 29 \\ 3 & 3 & -1 & 29 & 3 \end{pmatrix}_{b_0}$$

[603] $1^{11} 5^1 1^{12} 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 43, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 13 & 14 & 0 & 95 & 19 \\ 3 & 3 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
T = 12 \quad [604] \quad 1^{\overline{11}} 31 \overline{5} 1321^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0 \\
T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
\begin{pmatrix} 1 & 1 & 2 & 6 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 9 & 4 \\ 6 & 0 & 9 & 23 & 11 \\ 3 & 0 & 4 & 11 & 5 \end{pmatrix}_{b_0} \\
[605] \quad 1^{\overline{11}} 31 \overline{5} 1321^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0 \\
T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 7 & 9 & 7 \\ 2 & 5 & 0 & 9 & 5 & 5 \\ 2 & 4 & -1 & 7 & 5 & 4 \end{pmatrix}_{b_0} \\
[606] \quad 1^{\overline{41}} 31 \overline{5} 1321^{\overline{141}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0 \\
T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 2 & 1 & 20 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 20 & 0 & 142 & 36 & 28 \\ 2 & 5 & 0 & 36 & 5 & 5 \\ 2 & 4 & -1 & 28 & 5 & 4 \end{pmatrix}_{b_0} \\
[607] \quad 1^{\overline{31}} 31 \overline{5} 1321^{\overline{131}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8 \\
T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 2 & 1 & 15 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 15 & 0 & 79 & 27 & 21 \\ 2 & 5 & 0 & 27 & 5 & 5 \\ 2 & 4 & -1 & 21 & 5 & 4 \end{pmatrix}_{b_0} \\
[608] \quad 1^{\overline{21}} 31 \overline{5} 1321^{\overline{121}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5 \\
T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 2 & 1 & 10 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 10 & 0 & 34 & 18 & 14 \\ 2 & 5 & 0 & 18 & 5 & 5 \\ 2 & 4 & -1 & 14 & 5 & 4 \end{pmatrix}_{b_0} \\
T = 13 \quad [609] \quad 1^{\overline{41}} 31 \overline{5} 13221^{\overline{141}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 9.0 \\
T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 3 & 1 & 24 & 6 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 24 & 0 & 158 & 40 & 32 \\ 2 & 6 & 0 & 40 & 6 & 6 \\ 2 & 5 & -1 & 32 & 6 & 5 \end{pmatrix}_{b_0} \\
[610] \quad 1^{\overline{31}} 31 \overline{5} 13221^{\overline{131}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.8 \\
T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 3 & 1 & 18 & 6 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 18 & 0 & 88 & 30 & 24 \\ 2 & 6 & 0 & 30 & 6 & 6 \\ 2 & 5 & -1 & 24 & 6 & 5 \end{pmatrix}_{b_0} \\
[611] \quad 1^{\overline{21}} 31 \overline{5} 13221^{\overline{121}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.5 \\
T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 3 & 1 & 12 & 6 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 12 & 0 & 38 & 20 & 16 \\ 2 & 6 & 0 & 20 & 6 & 6 \\ 2 & 5 & -1 & 16 & 6 & 5 \end{pmatrix}_{b_0} \\
[612] \quad 1^{\overline{11}} 31 \overline{5} 13221^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0 \\
T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 3 & 1 & 6 & 6 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & 8 & 10 & 8 \\ 2 & 6 & 0 & 10 & 6 & 6 \\ 2 & 5 & -1 & 8 & 6 & 5 \end{pmatrix}_{b_0}
\end{aligned}$$

4.847 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6$

$$T = 8 \quad [0] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 101, T \in [15, 101]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 6 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.71 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 44 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 44 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 34 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{113}} 3^{\overline{141}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 59 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 59 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 45 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.53 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 318, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 25 & 40 & 0 & 9 \\ 18 & 40 & 58 & 0 & 14 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 14 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 379, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 35 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 35 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 27 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 17 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{21}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 430, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 34 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 26 & -4 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{e}_6$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 472$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.47 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 26 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 20 & -4 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{113}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{e}_6$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 493$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 13.50 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 27 & 1 & 5 \\ 12 & 25 & 60 & 0 & 9 \\ 27 & 60 & 133 & 0 & 21 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 21 & -4 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{112}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{e}_6$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 546$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 26 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 20 & -4 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{114}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{e}_6$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 738$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 36 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 36 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 28 & -4 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{113}31^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{e}_6$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 745$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 51 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 39 & -4 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{113}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{e}_6$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 921$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 13.44 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 35 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 27 & -4 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{113}31^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{e}_6$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1005$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 43 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 43 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 33 & -4 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{115}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{e}_6$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1053$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.1$
 $T_{\min} = 13.52 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 100 & 0 & 9 \\ 45 & 100 & 373 & 0 & 35 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 35 & -4 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{114}31^{41} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{e}_6$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1186$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 68 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 52 & -4 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{114}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{e}_6$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1366$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.45 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 44 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 44 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 34 & -4 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{114}31^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{e}_6$ $T = 8$, $T_{\max} = 101$, $\Delta = -162$, $\det = 1590$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 101$, $T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 60 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 46 & -4 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{114}3^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 1650, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.43 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$
 $\begin{pmatrix} 5 & 12 & 52 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 52 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 40 & -4 & 0 \end{pmatrix}_{b_0}$

$T = 9$ [23] $1^{11}3^{11}3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -151, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$
 $\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 15 & 0 & 6 \\ 5 & 12 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & 7 & -4 & 0 \end{pmatrix}_{b_0}$

[24] $1^{111}3^{11}3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -571, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$
 $\begin{pmatrix} 1 & 4 & 5 & 9 & 1 & 3 \\ 4 & 10 & 13 & 24 & 0 & 6 \\ 5 & 13 & 12 & 27 & 0 & 6 \\ 9 & 24 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & 13 & -4 & 0 \end{pmatrix}_{b_0}$

4.848 NHC(-2)(-3) + $\textcolor{red}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2$

$T = 9$ [0] $1^{112}3^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.5$
 $T_{\min} = 14.71 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$
 $\begin{pmatrix} 5 & 12 & 8 & 1 & 26 & 5 \\ 12 & 25 & 16 & 0 & 56 & 9 \\ 8 & 16 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 56 & 36 & 0 & 118 & 20 \\ 5 & 9 & 6 & -4 & 20 & -1 \end{pmatrix}_{b_0}$

[1] $1^{111}3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.89 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$
 $\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 7 & -4 & 6 & 0 \end{pmatrix}_{b_0}$

[2] $1^{11}3^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.64 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$
 $\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$

[3] $1^{1111}3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$
 $\begin{pmatrix} 5 & 12 & 9 & 1 & 17 & 5 \\ 12 & 25 & 20 & 0 & 36 & 9 \\ 9 & 20 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 36 & 27 & 0 & 49 & 13 \\ 5 & 9 & 7 & -4 & 13 & 0 \end{pmatrix}_{b_0}$

[4] $1^{111}3^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.58 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$
 $\begin{pmatrix} 5 & 11 & 9 & 1 & 35 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 35 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -4 & 27 & 0 \end{pmatrix}_{b_0}$

[5] $1^{1111}3^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

15.0

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 9 & 5 \\ 12 & 25 & 20 & 0 & 20 & 9 \\ 9 & 20 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 20 & 15 & 0 & 13 & 7 \\ 5 & 9 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{e}_6 + \text{su}_2$ $T = 9, T_{\max} = 93, \Delta = -132, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{e}_6 + \text{su}_2$ $T = 9, T_{\max} = 93, \Delta = -132, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 17 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{e}_6 + \text{su}_2$ $T = 9, T_{\max} = 93, \Delta = -132, \det = -618, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 26 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

4.849 $\text{NHC}(-2)(-3) + \text{su}_2 + \text{so}_8 + \text{e}_6 + \text{su}_2 + \text{su}_2$

$T = 10$ [0] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{e}_6 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 22, \Delta = -102, \det = 372, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 17 & 5 \\ 11 & 20 & 17 & 0 & 17 & 33 & 8 \\ 9 & 17 & 13 & 0 & 15 & 27 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 27 & 7 \\ 17 & 33 & 27 & 0 & 27 & 49 & 13 \\ 5 & 8 & 7 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

4.850 $\text{NHC}(-2)(-3) + \text{su}_2 + \text{so}_8 + \text{e}_6 + \text{su}_3$

$T = 9$ [0] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{e}_6 + \text{su}_3$ $T = 9, T_{\max} = 99, \Delta = -141, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 8 & 5 \\ 12 & 25 & 20 & 0 & 16 & 9 \\ 9 & 20 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 9 & 5 \\ 5 & 9 & 7 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{e}_6 + \text{su}_3$ $T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{e}_6 + \text{su}_3$ $T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 8 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_6 + \textcolor{blue}{fu}_3 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -931, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

4.851 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{f}_4$

$$T = 8 \quad [0] \ 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 4, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 94, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 44 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 44 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 34 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{113} 3 1^{41} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 109, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 59 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 59 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 45 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 111, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.63 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 165, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.61 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 174, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.68 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 189, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.57 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 240, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.67 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 25 & 40 & 0 & 9 \\ 18 & 40 & 58 & 0 & 14 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 14 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4 \quad T = 8, \ T_{\max} = 98, \ \Delta = -136, \ \det = 291, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.61 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 35 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 35 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 27 & -3 & 1 \end{pmatrix}_{b_0}$$

- [9] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 331$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 11 & 17 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 17 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 13 & -3 & 1 \end{pmatrix}_{b_0}$
- [10] $1^{112}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 336$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 34 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 34 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 26 & -3 & 1 \end{pmatrix}_{b_0}$
- [11] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 365$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.71 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 27 & 1 & 5 \\ 12 & 25 & 60 & 0 & 9 \\ 27 & 60 & 133 & 0 & 21 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 21 & -3 & 1 \end{pmatrix}_{b_0}$
- [12] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 370$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 11 & 26 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 26 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 20 & -3 & 1 \end{pmatrix}_{b_0}$
- [13] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 424$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 26 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 26 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 20 & -3 & 1 \end{pmatrix}_{b_0}$
- [14] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 540$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 36 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 36 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 28 & -3 & 1 \end{pmatrix}_{b_0}$
- [15] $1^{113}31^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 581$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 51 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 51 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 39 & -3 & 1 \end{pmatrix}_{b_0}$
- [16] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 709$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.60 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 35 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 35 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 27 & -3 & 1 \end{pmatrix}_{b_0}$
- [17] $1^{115}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 765$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.3$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 100 & 0 & 9 \\ 45 & 100 & 373 & 0 & 35 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 35 & -3 & 1 \end{pmatrix}_{b_0}$
- [18] $1^{113}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 781$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 43 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 43 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 33 & -3 & 1 \end{pmatrix}_{b_0}$
- [19] $1^{114}31^{41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 924$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 5 & 12 & 68 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 68 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 52 & -3 & 1 \end{pmatrix}_{b_0}$

[20] $1^{114}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 1044$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 44 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 44 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 34 & -3 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{114}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 1236$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.58 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 60 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 46 & -3 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{114}31^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -136$, $\det = 1276$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 52 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 40 & -3 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [23] $1^{11}31^{11}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 \\ 5 & 13 & 12 & 15 & 0 \\ 5 & 12 & 15 & 13 & 0 \\ 1 & 0 & 0 & 0 & -5 \end{pmatrix}_{b_0}$$

[24] $1^{11}31^{11}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -435$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 1 & 4 & 5 & 9 & 1 & 3 \\ 4 & 10 & 13 & 24 & 0 & 6 \\ 5 & 13 & 12 & 27 & 0 & 6 \\ 9 & 24 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

4.852 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2$

$T = 9$ [0] $1^{112}31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.7$
 $T_{\min} = 15$, $T_{\max} = 21$, $T \in [15, 21]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 26 & 5 \\ 12 & 25 & 16 & 0 & 56 & 9 \\ 8 & 16 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 56 & 36 & 0 & 118 & 20 \\ 5 & 9 & 6 & -3 & 20 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -90$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.07 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 7 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -104$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -288$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 17 & 5 \\ 12 & 25 & 20 & 0 & 36 & 9 \\ 9 & 20 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 36 & 27 & 0 & 49 & 13 \\ 5 & 9 & 7 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -106, \ \det = -324, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.67 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 35 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 35 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -106, \ \det = -380, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 9 & 5 \\ 12 & 25 & 20 & 0 & 20 & 9 \\ 9 & 20 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 15 & 0 & 13 & 7 \\ 5 & 9 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -106, \ \det = -404, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.65 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -106, \ \det = -476, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.57 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 17 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3^{\overline{112}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -106, \ \det = -482, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 26 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

4.853 $\text{NHC}(-2)(-3) + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -218, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 7 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -222, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 6 & 5 \\ 12 & 25 & 20 & 0 & 12 & 9 \\ 9 & 20 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 9 & 0 & 4 & 4 \\ 5 & 9 & 7 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -302, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 6 & 5 \\ 12 & 25 & 36 & 0 & 12 & 9 \\ 17 & 36 & 49 & 0 & 17 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 17 & 0 & 4 & 4 \\ 5 & 9 & 13 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

4.854 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \text{su2n3mix}$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 15 & 5 \\ 11 & 20 & 17 & 0 & 28 & 8 \\ 9 & 17 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 28 & 24 & 0 & 37 & 11 \\ 5 & 8 & 7 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

4.855 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 8 & 5 \\ 12 & 25 & 20 & 0 & 16 & 9 \\ 9 & 20 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 9 & 5 \\ 5 & 9 & 7 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 8 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

4.856 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 2, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 6 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{113} 3^{131} 1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.71 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 12 & 54 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 54 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 39 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 0 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^{114} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 44 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 44 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 34 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 10 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 10 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112} 3 \ 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 5 & 12 & 36 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 36 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 26 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 20 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 20 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{113} 3 \ 1^{41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 59 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 59 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 45 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{113} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 39 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 39 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 27 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 18 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111} 3 \ 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 39 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 39 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 27 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 137, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 25 & 40 & 0 & 9 \\ 18 & 40 & 58 & 0 & 14 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 9 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 48 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 48 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 34 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 35 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 35 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 27 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 237, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 5 & 12 & 27 & 1 & 5 \\ 12 & 25 & 60 & 0 & 9 \\ 27 & 60 & 133 & 0 & 21 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{112} 3^1 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 242, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 34 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 26 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 17 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 28 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 28 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 29 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 29 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 26 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 19 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 26 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 45 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 45 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 34 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$$

$$\begin{pmatrix} 5 & 12 & 36 & 1 & 5 \\ 12 & 25 & 80 & 0 & 9 \\ 36 & 80 & 238 & 0 & 28 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 18 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 46 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 46 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 33 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 38 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 38 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 27 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 27 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 27 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} \equiv 11.85 \rightarrow 12, \quad T_{\max} \equiv 22, \quad T \in [12, 22]$$

$$\begin{pmatrix} 5 & 11 & 47 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 47 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 34 & -2 & 2 \end{pmatrix}_{b_1}$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 51 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 39 & -2 & 2 \end{pmatrix}_b$$

$$\begin{pmatrix} 5 & 11 & 36 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 36 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 27 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12.19 \rightarrow 12$$

$$T_{\min} = 11.81 \rightarrow 12, \quad T_{\max} = 12.19 \rightarrow 12$$

$$T_{\min} = 11.83 \rightarrow 12, \quad T_{\max} = 12.5 \rightarrow 13$$

$$T_{\min} = 11.90 \rightarrow 12, T$$
$$\begin{pmatrix} 5 & 12 & 64 & 1 & 5 \\ 12 & 25 & 128 & 0 & 9 \\ 64 & 128 & 634 & 0 & 46 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 46 & -2 & 2 \end{pmatrix}_b$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12.82 \rightarrow 13$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 12.72 \rightarrow 13$$

$$\begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 100 & 0 & 9 \\ 45 & 100 & 373 & 0 & 35 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 35 & -2 & 1 \end{pmatrix}_b$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12.83 \rightarrow 13$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12.05 \rightarrow 12.10$$

$$T_{\min} = 11.85 \rightarrow 12, \quad T_{\left(\begin{pmatrix} 5 & 12 & 43 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 43 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 33 & -2 & 2 \end{pmatrix} \right)_{b_0}}$$

$$T_{\min} = 11.83 \rightarrow 12, \begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 45 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 33 & -2 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 68 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 52 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12.57 \rightarrow 13$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12.5 \rightarrow 13$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12.16 \rightarrow 12$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12.06 \rightarrow 12, \begin{pmatrix} 5 & 12 & 44 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 44 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 34 & -2 & 2 \end{pmatrix} b_0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12.82 \rightarrow 13$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12.16 \rightarrow 12$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12.16 \rightarrow 12$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12.13 \rightarrow 12$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12.13 \rightarrow 12$$

$$T_{\min} = 11.82 \rightarrow 12, \quad T_{\max} = 12.18 \rightarrow 12, \quad \begin{pmatrix} 5 & 12 & 55 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 55 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 40 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 47 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 47 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 34 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[59] 1^{114} 3^{121} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 1154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 46 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 46 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 34 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[60] 1^{114} 3^{121} 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 22, \Delta = -112, \det = 1206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 12 & 54 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 54 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 40 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [61] 1^{13} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 18 & 1 & 5 \\ 9 & 13 & 30 & 0 & 7 \\ 18 & 30 & 61 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 15 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[62] 1^{111} 3^{11} 1^{31} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 4 & 5 & 3 \\ 4 & 10 & 1 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 12 & 0 & 10 & 12 & 6 \\ 5 & 12 & 0 & 12 & 11 & 5 \\ 3 & 6 & -1 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[63] 1^{11} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 5 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[64] 1^{111} 3^{11} 1^{31} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 5 & 4 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 12 & 0 & 6 \\ 4 & 12 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 6 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[65] 1^{12} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 12 & 1 & 5 \\ 9 & 13 & 20 & 0 & 7 \\ 12 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[66] 1^{11} 3^{11} 1^{31} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -79, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 15 & 0 & 6 \\ 5 & 12 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 6 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[67] 1^{11} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -81, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.41 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 5 \\ 9 & 13 & 14 & 0 & 7 \\ 9 & 14 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[68] 1^{11} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -93, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 5 \\ 9 & 13 & 10 & 0 & 7 \\ 6 & 10 & 5 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$

[70] $1^{\overline{11}} 3^{\overline{111}} 21^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 3 & 5 & 2 & 0 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 5 & 4 & 0 & 0 & 2 \\ 3 & 5 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{\overline{11}} 31^{\overline{1}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 12 & 0 & 6 \\ 5 & 12 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 6 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{\overline{11}} 31^{\overline{1}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 5 & 3 \\ 4 & 10 & 1 & 12 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 12 & 0 & 10 & 12 & 6 \\ 5 & 12 & 0 & 12 & 11 & 5 \\ 3 & 6 & -1 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{\overline{11}} 321^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 5 & 10 & 0 & 5 & 5 \\ 5 & 10 & 14 & 0 & 7 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 5 & 7 & 0 & 0 & 2 \\ 3 & 5 & 8 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{\overline{11}} 41^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 9 & 15 & 1 & 5 \\ 9 & 13 & 24 & 0 & 7 \\ 15 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

[74] $1^{\overline{11}} 31^{\overline{1}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 10 & 1 & 3 \\ 4 & 10 & 13 & 24 & 0 & 6 \\ 5 & 13 & 12 & 27 & 0 & 6 \\ 10 & 24 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 6 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{\overline{12}} 41^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -273, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 9 & 21 & 1 & 5 \\ 9 & 13 & 34 & 0 & 7 \\ 21 & 34 & 77 & 0 & 17 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[76] $1^{\overline{11}} 31^{\overline{1}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 9 & 1 & 3 \\ 4 & 10 & 13 & 24 & 0 & 6 \\ 5 & 13 & 12 & 27 & 0 & 6 \\ 9 & 24 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 6 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{\overline{11}} 3^{\overline{11}} 21^{\overline{121}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 5 & 15 & 0 & 5 & 5 \\ 8 & 15 & 32 & 0 & 11 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 5 & 11 & 0 & 0 & 2 \\ 3 & 5 & 12 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[78] $1^{\overline{13}} 41^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -342, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 27 & 1 & 5 \\ 9 & 13 & 44 & 0 & 7 \\ 27 & 44 & 130 & 0 & 22 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 22 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[79] 1^{14} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -393, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 33 & 1 & 5 \\ 9 & 13 & 54 & 0 & 7 \\ 33 & 54 & 197 & 0 & 27 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 27 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[80] 1^{11} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -495, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 5 & 20 & 0 & 5 & 5 \\ 10 & 20 & 58 & 0 & 15 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 5 & 15 & 0 & 0 & 2 \\ 3 & 5 & 16 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[81] 1^{12} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -706, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 5 & 25 & 0 & 5 & 5 \\ 13 & 25 & 94 & 0 & 18 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 5 & 18 & 0 & 0 & 2 \\ 3 & 5 & 20 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [82] 1^{11} 4^1 1^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 0 & 1 & 1 & 1 & 2 \\ 2 & 1 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[83] 1^{11} 4^1 131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 0 & 3 & 1 & 1 & 2 \\ 3 & 3 & 2 & 1 & 2 & 4 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 2 & 0 & -2 & 0 \\ 3 & 2 & 4 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[84] 1^{11} 4^1 1^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 139, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 0 & 2 & 1 & 1 & 2 \\ 4 & 2 & 2 & 1 & 2 & 4 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 2 & 0 & -2 & 0 \\ 3 & 2 & 4 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[85] 1^{11} 4^1 1^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 234, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 5 & 1 & 2 & 3 \\ 2 & 0 & 4 & 1 & 1 & 2 \\ 5 & 4 & 6 & 2 & 2 & 6 \\ 1 & 1 & 2 & -2 & 0 & 0 \\ 2 & 1 & 2 & 0 & -2 & 0 \\ 3 & 2 & 6 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

4.857 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$

$$T = 9 \quad [0] 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 0 & 13 & 7 \\ 5 & 8 & 0 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 10 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 39 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 39 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 45 & 5 \\ 11 & 20 & 0 & 0 & 84 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 45 & 84 & 0 & 0 & 346 & 34 \\ 5 & 8 & 0 & -2 & 34 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 16 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 0 & 0 & 10 & 6 \\ 5 & 8 & 0 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{1111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 10 & 5 \\ 12 & 25 & 36 & 0 & 20 & 9 \\ 17 & 36 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 20 & 27 & 0 & 13 & 7 \\ 5 & 9 & 13 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[7] $1^{1111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 18 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 18 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{1111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 18 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{1111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

13.0

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 17 & 5 \\ 12 & 25 & 20 & 0 & 36 & 9 \\ 9 & 20 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 36 & 27 & 0 & 49 & 13 \\ 5 & 9 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 35 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 35 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 48 & 5 \\ 11 & 20 & 17 & 0 & 84 & 8 \\ 9 & 17 & 13 & 0 & 72 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 48 & 84 & 72 & 0 & 346 & 34 \\ 5 & 8 & 7 & -2 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 18 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 10 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 36 & 5 \\ 11 & 20 & 0 & 0 & 67 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 36 & 67 & 0 & 0 & 217 & 27 \\ 5 & 8 & 0 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 19 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 18 & 5 \\ 11 & 20 & 0 & 0 & 33 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 33 & 0 & 0 & 49 & 13 \\ 5 & 8 & 0 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -262, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 29 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 29 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{111} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 9 & 5 \\ 12 & 25 & 20 & 0 & 20 & 9 \\ 9 & 20 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 20 & 15 & 0 & 13 & 7 \\ 5 & 9 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{111} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -286, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 45 & 5 \\ 11 & 20 & 17 & 0 & 84 & 8 \\ 10 & 17 & 13 & 0 & 72 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 45 & 84 & 72 & 0 & 346 & 34 \\ 5 & 8 & 7 & -2 & 34 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 27 & 5 \\ 11 & 20 & 0 & 0 & 50 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 27 & 50 & 0 & 0 & 118 & 20 \\ 5 & 8 & 0 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{111} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 45 & 5 \\ 11 & 20 & 17 & 0 & 84 & 8 \\ 9 & 17 & 13 & 0 & 72 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 45 & 84 & 72 & 0 & 346 & 34 \\ 5 & 8 & 7 & -2 & 34 & 2 \end{pmatrix}_{b_0}$$

[22] $1^{111} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -342, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 27 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 10 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 27 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 17 & 5 \\ 12 & 25 & 36 & 0 & 36 & 9 \\ 17 & 36 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 36 & 51 & 0 & 49 & 13 \\ 5 & 9 & 13 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[24] $1^{111} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 26 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 26 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{111} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -360, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 36 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 10 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 36 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 18 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{111} 3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -468, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 38 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 38 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{111} 3^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 47 & 5 \\ 11 & 20 & 17 & 0 & 84 & 8 \\ 9 & 17 & 13 & 0 & 72 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 47 & 84 & 72 & 0 & 346 & 34 \\ 5 & 8 & 7 & -2 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{111} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -522, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 28 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 28 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{111} 3^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -530, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 46 & 5 \\ 11 & 20 & 17 & 0 & 84 & 8 \\ 9 & 17 & 13 & 0 & 72 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 46 & 84 & 72 & 0 & 346 & 34 \\ 5 & 8 & 7 & -2 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{111} 3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -532, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 36 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 36 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{111} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -550, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 27 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 27 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{111} 3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -616, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 37 & 5 \\ 11 & 20 & 17 & 0 & 67 & 8 \\ 9 & 17 & 13 & 0 & 57 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 37 & 67 & 57 & 0 & 217 & 27 \\ 5 & 8 & 7 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

4.858 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 6 & 5 \\ 11 & 20 & 0 & 0 & 11 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 0 & 0 & 4 & 4 \\ 5 & 8 & 0 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -153, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 6 & 5 \\ 12 & 25 & 36 & 0 & 12 & 9 \\ 18 & 36 & 49 & 0 & 17 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 17 & 0 & 4 & 4 \\ 5 & 9 & 13 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -153, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 6 & 5 \\ 12 & 25 & 20 & 0 & 12 & 9 \\ 9 & 20 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 9 & 0 & 4 & 4 \\ 5 & 9 & 7 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -163, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 7 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -217, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 6 & 5 \\ 12 & 25 & 36 & 0 & 12 & 9 \\ 17 & 36 & 49 & 0 & 17 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 17 & 0 & 4 & 4 \\ 5 & 9 & 13 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

4.859 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_{2n3\text{mix}}$

$$T = 9 \quad [0] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -80, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 15 & 5 \\ 11 & 20 & 0 & 0 & 28 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 28 & 0 & 0 & 37 & 11 \\ 5 & 8 & 0 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -117, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 15 & 5 \\ 11 & 20 & 17 & 0 & 28 & 8 \\ 10 & 17 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 28 & 24 & 0 & 37 & 11 \\ 5 & 8 & 7 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -145, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 16 & 5 \\ 11 & 20 & 17 & 0 & 28 & 8 \\ 9 & 17 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 28 & 24 & 0 & 37 & 11 \\ 5 & 8 & 7 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -157, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.85 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 15 & 5 \\ 11 & 20 & 17 & 0 & 28 & 8 \\ 9 & 17 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 28 & 24 & 0 & 37 & 11 \\ 5 & 8 & 7 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

4.860 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 8 & 5 \\ 12 & 25 & 20 & 0 & 16 & 9 \\ 9 & 20 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 12 & 0 & 9 & 5 \\ 5 & 9 & 7 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 1 & 5 \\ 12 & 25 & 76 & 0 & 0 & 9 \\ 35 & 76 & 217 & 0 & 0 & 27 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & 0 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 1 & 5 \\ 12 & 25 & 36 & 0 & 0 & 9 \\ 17 & 36 & 49 & 0 & 0 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 13 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 10 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -133, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 1 & 5 \\ 11 & 20 & 17 & 0 & 0 & 8 \\ 9 & 17 & 13 & 0 & 0 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & 7 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 10 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 12 & 6 \\ 5 & 8 & 7 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 1 & 5 \\ 11 & 20 & 16 & 0 & 0 & 8 \\ 8 & 16 & 10 & 0 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & 6 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 1 & 5 \\ 12 & 25 & 56 & 0 & 0 & 9 \\ 26 & 56 & 118 & 0 & 0 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 20 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{112} 3 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 1 & 5 \\ 12 & 25 & 72 & 0 & 0 & 9 \\ 34 & 72 & 202 & 0 & 0 & 26 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 26 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 8 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 10 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 9 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 1 & 5 \\ 11 & 20 & 50 & 0 & 0 & 8 \\ 26 & 50 & 118 & 0 & 0 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 1 & 5 \\ 11 & 20 & 33 & 0 & 0 & 8 \\ 17 & 33 & 49 & 0 & 0 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{113} 3 1^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -295, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 1 & 5 \\ 12 & 25 & 108 & 0 & 0 & 9 \\ 51 & 108 & 457 & 0 & 0 & 39 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 39 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{113} 3^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 43 & 1 & 1 & 5 \\ 12 & 25 & 92 & 0 & 0 & 9 \\ 43 & 92 & 325 & 0 & 0 & 33 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 33 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{114} 3^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 1 & 5 \\ 12 & 25 & 112 & 0 & 0 & 9 \\ 52 & 112 & 478 & 0 & 0 & 40 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{114} 3^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 1 & 5 \\ 12 & 25 & 144 & 0 & 0 & 9 \\ 68 & 144 & 814 & 0 & 0 & 52 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 52 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{114} 3^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 1 & 5 \\ 12 & 25 & 128 & 0 & 0 & 9 \\ 60 & 128 & 634 & 0 & 0 & 46 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 46 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 19 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 18 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \quad 1^{11} 4^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 6 & 5 \\ 9 & 13 & 14 & 0 & 10 & 7 \\ 9 & 14 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 9 & 0 & 4 & 4 \\ 5 & 7 & 7 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 261, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 6 & 5 \\ 9 & 13 & 10 & 0 & 10 & 7 \\ 6 & 10 & 5 & 0 & 7 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 7 & 0 & 4 & 4 \\ 5 & 7 & 5 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{so}_8 + \text{su}_3$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 480, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 15 & 1 & 6 & 5 \\ 9 & 13 & 24 & 0 & 10 & 7 \\ 15 & 24 & 38 & 0 & 16 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 16 & 0 & 4 & 4 \\ 5 & 7 & 12 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

4.861 NHC(-2)(-3) + su₂ + so₈ + su₂

$T = 8$ [0] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 14 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 14 & 17 & 13 & 15 & 7 \\ 12 & 17 & 15 & 13 & 7 \\ 6 & 8 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & 0 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 0 & 0 & 13 & 7 \\ 3 & 4 & 0 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 0 & 0 & 4 & 4 \\ 3 & 4 & 0 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 1 & 1 & 6 \\ 15 & 20 & 0 & 0 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 6 & 8 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

[7] $1^{111} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 1 & 48 & 6 \\ 15 & 20 & 0 & 67 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 48 & 67 & 0 & 217 & 27 \\ 6 & 8 & 0 & 27 & 3 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 21$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 1 & 6 \\ 15 & 20 & 17 & 0 & 8 \\ 12 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -2 & 0 \\ 6 & 8 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{133} 3^1 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 24$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.6$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 15 & 3 \\ 3 & 4 & 17 & 0 & 27 & 4 \\ 9 & 17 & 49 & 0 & 81 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 27 & 81 & 0 & 133 & 21 \\ 3 & 4 & 13 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{111} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 25$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 35 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 35 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[10] $1^{1111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 28$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 10 & 16 & 23 & 13 & 6 \\ 16 & 25 & 36 & 20 & 9 \\ 23 & 36 & 49 & 27 & 13 \\ 13 & 20 & 27 & 13 & 7 \\ 6 & 9 & 13 & 7 & 3 \end{pmatrix}_{b_0}$$

[11] $1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 29$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.48 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{11} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 31$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 4 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[12] $1^{11} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 31$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 13 & 15 & 7 \\ 13 & 17 & 15 & 13 & 7 \\ 6 & 8 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[14] $1^{121} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 4 & 8 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 6 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{121} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 34$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.76 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 12 & 0 & 13 & 7 \\ 3 & 4 & 6 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{122} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 34$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 18 & 0 & 28 & 10 \\ 3 & 4 & 6 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

- [16] $1^{124} 3 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 17 & 3 \\ 3 & 4 & 8 & 0 & 31 & 4 \\ 4 & 8 & 10 & 0 & 42 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 31 & 42 & 0 & 172 & 24 \\ 3 & 4 & 6 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$
- [18] $1^{111} 3 1^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 48 & 6 \\ 15 & 20 & 17 & 67 & 8 \\ 12 & 17 & 13 & 57 & 7 \\ 48 & 67 & 57 & 217 & 27 \\ 6 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$
- [19] $1^{133} 3 1^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 16 & 3 \\ 3 & 4 & 17 & 0 & 27 & 4 \\ 10 & 17 & 49 & 0 & 81 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 81 & 0 & 133 & 21 \\ 3 & 4 & 13 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$
- [20] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 13 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 13 & 17 & 13 & 15 & 7 \\ 13 & 17 & 15 & 13 & 7 \\ 6 & 8 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$
- [21] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 4 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$
- [25] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 23 & 6 \\ 15 & 20 & 0 & 33 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 23 & 33 & 0 & 49 & 13 \\ 6 & 8 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$
- [24] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 13 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 17 & 0 & 13 & 7 \\ 6 & 8 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$
- [23] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 4 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$
- [22] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 3 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$
- [28] $1^{13} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 6 & 0 & 4 & 4 \\ 3 & 4 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$
- [27] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 16 & 23 & 23 & 6 \\ 16 & 25 & 36 & 36 & 9 \\ 23 & 36 & 49 & 51 & 13 \\ 23 & 36 & 51 & 49 & 13 \\ 6 & 9 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$

[26] $1^{111} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 11 & 6 \\ 15 & 20 & 0 & 16 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 16 & 0 & 10 & 6 \\ 6 & 8 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[31] $1^{11} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 4 & 5 & 0 & 14 & 4 \\ 4 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 15 & 0 & 37 & 11 \\ 3 & 4 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{111} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 13 & 35 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 13 & 17 & 13 & 42 & 7 \\ 35 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[29] $1^{113} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 12 & 3 \\ 3 & 4 & 9 & 0 & 22 & 4 \\ 6 & 9 & 13 & 0 & 36 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 36 & 0 & 85 & 17 \\ 3 & 4 & 7 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{123} 3 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 10 & 3 \\ 3 & 4 & 18 & 0 & 17 & 4 \\ 10 & 18 & 58 & 0 & 54 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 54 & 0 & 49 & 13 \\ 3 & 4 & 14 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{112} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 7 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{13} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 7 & 3 \\ 3 & 4 & 0 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 0 & 0 & 25 & 9 \\ 3 & 4 & 0 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{11} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & 0 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{11} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 6 & 0 & 4 & 4 \\ 3 & 4 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 13 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 13 & 17 & 13 & 27 & 7 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[38] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 3 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 49 & 13 \\ 3 & 4 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{111} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{12} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 0 & 0 & 10 & 6 \\ 3 & 4 & 0 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{112} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 3 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{123} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 18 & 0 & 25 & 9 \\ 3 & 4 & 6 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[44] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 3 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[46] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 12 & 3 \\ 3 & 4 & 9 & 0 & 21 & 4 \\ 6 & 9 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 33 & 0 & 76 & 16 \\ 3 & 4 & 7 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[45] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 11 & 3 \\ 3 & 4 & 13 & 0 & 17 & 4 \\ 7 & 13 & 28 & 0 & 39 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 17 & 39 & 0 & 49 & 13 \\ 3 & 4 & 10 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[47] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 3 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 3 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

- [48] $1^{14} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 18 & 0 & 76 & 16 \\ 3 & 4 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$
- [49] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 12 & 4 \\ 7 & 13 & 28 & 0 & 27 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 27 & 0 & 25 & 9 \\ 3 & 4 & 10 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$
- [50] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 13 & 27 & 7 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$
- [51] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$
- [52] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 39 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 39 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$
- [53] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 3 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$
- [54] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 16 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 16 & 0 & 10 & 6 \\ 6 & 8 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$
- [55] $1^{122} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 14 & 3 \\ 3 & 4 & 13 & 0 & 23 & 4 \\ 7 & 13 & 28 & 0 & 54 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 23 & 54 & 0 & 100 & 18 \\ 3 & 4 & 10 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$
- [56] $1^{113} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 15 & 3 \\ 3 & 4 & 9 & 0 & 27 & 4 \\ 6 & 9 & 13 & 0 & 45 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 27 & 45 & 0 & 133 & 21 \\ 3 & 4 & 7 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$
- [58] $1^{124} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 8 & 0 & 21 & 4 \\ 5 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 30 & 0 & 76 & 16 \\ 3 & 4 & 6 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$
- [59] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 5 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 12 & 0 & 13 & 7 \\ 3 & 4 & 6 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[57] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 5 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 18 & 0 & 28 & 10 \\ 3 & 4 & 6 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[60] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 6 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[61] $1^{132} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 8 & 3 \\ 3 & 4 & 12 & 0 & 13 & 4 \\ 7 & 12 & 25 & 0 & 27 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 27 & 0 & 28 & 10 \\ 3 & 4 & 9 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{14} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 12 & 3 \\ 3 & 4 & 0 & 0 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 0 & 0 & 76 & 16 \\ 3 & 4 & 0 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 6 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 0 & 0 & 13 & 7 \\ 3 & 4 & 0 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[64] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 25 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 13 & 27 & 7 \\ 25 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[66] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{121} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.66 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 6 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 7 & 13 & 28 & 0 & 21 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 21 & 0 & 13 & 7 \\ 3 & 4 & 10 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{123} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 3 \\ 3 & 4 & 8 & 0 & 22 & 4 \\ 4 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 30 & 0 & 85 & 17 \\ 3 & 4 & 6 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[70] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 21 & 18 & 0 & 76 & 16 \\ 3 & 4 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[71] $1^{112} 3^1 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 7 & 3 \\ 3 & 4 & 14 & 0 & 13 & 4 \\ 8 & 14 & 37 & 0 & 33 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 33 & 0 & 28 & 10 \\ 3 & 4 & 11 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{111} 3^1 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 6 & 3 \\ 3 & 4 & 14 & 0 & 9 & 4 \\ 8 & 14 & 37 & 0 & 24 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 24 & 0 & 13 & 7 \\ 3 & 4 & 11 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[74] $1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 17 & 4 \\ 5 & 8 & 10 & 0 & 24 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 24 & 0 & 49 & 13 \\ 3 & 4 & 6 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[75] $1^{111} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 12 & 3 \\ 3 & 4 & 0 & 0 & 19 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 19 & 0 & 0 & 73 & 15 \\ 3 & 4 & 0 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[77] $1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 18 & 0 & 28 & 10 \\ 3 & 4 & 6 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[76] $1^{111} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 62 & 6 \\ 15 & 20 & 17 & 84 & 8 \\ 12 & 17 & 13 & 72 & 7 \\ 62 & 84 & 72 & 346 & 34 \\ 6 & 8 & 7 & 34 & 3 \end{pmatrix}_{b_0}$$

[79] $1^{111} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 13 & 48 & 6 \\ 15 & 20 & 17 & 67 & 8 \\ 13 & 17 & 13 & 57 & 7 \\ 48 & 67 & 57 & 217 & 27 \\ 6 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

[78] $1^{13} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 115$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 16 & 3 \\ 3 & 4 & 5 & 0 & 27 & 4 \\ 4 & 5 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[81] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 116$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 3 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 116$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 24 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 13 & 27 & 7 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[82] $1^{134} 3 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 117$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 10.74 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 12 & 3 \\ 3 & 4 & 22 & 0 & 21 & 4 \\ 12 & 22 & 85 & 0 & 81 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 81 & 0 & 76 & 16 \\ 3 & 4 & 17 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{11} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 120$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 9 & 3 \\ 3 & 4 & 0 & 0 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 0 & 0 & 37 & 11 \\ 3 & 4 & 0 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[84] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 120$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 36 & 6 \\ 15 & 20 & 0 & 50 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 36 & 50 & 0 & 118 & 20 \\ 6 & 8 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[85] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 126$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 36 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 36 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[86] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 128$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 3 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[87] $1^{112} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 128$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 3 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[88] $1^{133} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 129$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.6$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 9 & 3 \\ 3 & 4 & 12 & 0 & 17 & 4 \\ 7 & 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 36 & 0 & 49 & 13 \\ 3 & 4 & 9 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

- [89] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 10 & 3 \\ 3 & 4 & 0 & 0 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 0 & 0 & 49 & 13 \\ 3 & 4 & 0 & -2 & 13 & 2 \end{pmatrix}_{b_0}$
- [90] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 3 \\ 3 & 4 & 0 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 0 & 0 & 28 & 10 \\ 3 & 4 & 0 & -2 & 10 & 2 \end{pmatrix}_{b_0}$
- [91] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 133, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & -2 & 7 & 2 \end{pmatrix}_{b_0}$
- [92] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.68 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$
- [93] $1^{123} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 13 & 3 \\ 3 & 4 & 8 & 0 & 22 & 4 \\ 5 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 30 & 0 & 85 & 17 \\ 3 & 4 & 6 & -2 & 17 & 2 \end{pmatrix}_{b_0}$
- [94] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 10 & 15 & 13 & 24 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 13 & 17 & 13 & 27 & 7 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$
- [95] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 4 & 1 & 6 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 4 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$
- [96] $1^{111} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 10 & 15 & 12 & 49 & 6 \\ 15 & 20 & 17 & 67 & 8 \\ 12 & 17 & 13 & 57 & 7 \\ 49 & 67 & 57 & 217 & 27 \\ 6 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$
- [97] $1^{123} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 7 & 1 & 17 & 3 \\ 3 & 4 & 13 & 0 & 27 & 4 \\ 7 & 13 & 28 & 0 & 63 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 27 & 63 & 0 & 133 & 21 \\ 3 & 4 & 10 & -2 & 21 & 2 \end{pmatrix}_{b_0}$
- [98] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 10 & 15 & 1 & 24 & 6 \\ 15 & 20 & 0 & 33 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 24 & 33 & 0 & 49 & 13 \\ 6 & 8 & 0 & 13 & 3 \end{pmatrix}_{b_0}$
- [99] $1^{132} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 143, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 8 & 3 \\ 3 & 4 & 17 & 0 & 13 & 4 \\ 9 & 17 & 49 & 0 & 39 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 39 & 0 & 28 & 10 \\ 3 & 4 & 13 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[100] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 143$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 15 & 0 & 49 & 13 \\ 3 & 4 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[101] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.72 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 3 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[102] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 147$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 7 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[103] $1^{12} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 17 & 3 \\ 3 & 4 & 13 & 0 & 31 & 4 \\ 8 & 13 & 28 & 0 & 72 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 31 & 72 & 0 & 172 & 24 \\ 3 & 4 & 10 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[104] $1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 149$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 5 & 0 & 14 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 15 & 0 & 37 & 11 \\ 3 & 4 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[105] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 151$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 10 & 3 \\ 3 & 4 & 21 & 0 & 17 & 4 \\ 11 & 21 & 76 & 0 & 63 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 63 & 0 & 49 & 13 \\ 3 & 4 & 16 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[106] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 152$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.76 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[107] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 153$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 11 & 3 \\ 3 & 4 & 12 & 0 & 21 & 4 \\ 7 & 12 & 25 & 0 & 45 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 45 & 0 & 76 & 16 \\ 3 & 4 & 9 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[108] $1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 154$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 38 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 38 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[109] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 155$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 17 & 15 & 0 & 49 & 13 \\ 3 & 4 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[110] $1^{112} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 156$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 10 & 3 \\ 3 & 4 & 14 & 0 & 18 & 4 \\ 9 & 14 & 37 & 0 & 48 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 48 & 0 & 58 & 14 \\ 3 & 4 & 11 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[111] $1^{112} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 158$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.72 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 10 & 3 \\ 3 & 4 & 9 & 0 & 18 & 4 \\ 6 & 9 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 30 & 0 & 58 & 14 \\ 3 & 4 & 7 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{113} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 160$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 13 & 3 \\ 3 & 4 & 14 & 0 & 22 & 4 \\ 8 & 14 & 37 & 0 & 57 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 57 & 0 & 85 & 17 \\ 3 & 4 & 11 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[114] $1^{123} 3^1 1^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 160$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.3$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 18 & 0 & 25 & 9 \\ 3 & 4 & 6 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[112] $1^{12} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 160$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 17 & 3 \\ 3 & 4 & 0 & 0 & 28 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 28 & 0 & 0 & 154 & 22 \\ 3 & 4 & 0 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[115] $1^{111} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 161$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 63 & 6 \\ 15 & 20 & 17 & 84 & 8 \\ 12 & 17 & 13 & 72 & 7 \\ 63 & 84 & 72 & 346 & 34 \\ 6 & 8 & 7 & 34 & 3 \end{pmatrix}_{b_0}$$

[116] $1^{144} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 162$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 11 & 3 \\ 3 & 4 & 16 & 0 & 21 & 4 \\ 9 & 16 & 46 & 0 & 60 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 60 & 0 & 76 & 16 \\ 3 & 4 & 12 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[117] $1^{12} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 164$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.66 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[119] $1^{111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 166$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 13 & 36 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 13 & 17 & 13 & 42 & 7 \\ 36 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[118] $1^{12} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 166$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 18 & 4 \\ 4 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[120] $1^{134} 3^1 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 167, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 18 & 3 \\ 3 & 4 & 17 & 0 & 31 & 4 \\ 9 & 17 & 49 & 0 & 93 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 31 & 93 & 0 & 172 & 24 \\ 3 & 4 & 13 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[122] $1^{14} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 15 & 3 \\ 3 & 4 & 0 & 0 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 0 & 0 & 118 & 20 \\ 3 & 4 & 0 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[121] $1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 11 & 3 \\ 3 & 4 & 0 & 0 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 0 & 0 & 58 & 14 \\ 3 & 4 & 0 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[123] $1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 37 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 37 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[124] $1^{111} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 51 & 6 \\ 15 & 20 & 17 & 67 & 8 \\ 12 & 17 & 13 & 57 & 7 \\ 51 & 67 & 57 & 217 & 27 \\ 6 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

[126] $1^{132} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 10 & 3 \\ 3 & 4 & 17 & 0 & 18 & 4 \\ 9 & 17 & 49 & 0 & 54 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 54 & 0 & 58 & 14 \\ 3 & 4 & 13 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[125] $1^{124} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 14 & 3 \\ 3 & 4 & 13 & 0 & 26 & 4 \\ 8 & 13 & 28 & 0 & 60 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 60 & 0 & 118 & 20 \\ 3 & 4 & 10 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[127] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 3 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 3 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[128] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 3 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 3 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[129] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 177, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 3 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[132] $1^{12} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 14 & 3 \\ 3 & 4 & 0 & 0 & 23 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 23 & 0 & 0 & 100 & 18 \\ 3 & 4 & 0 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[131] $1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 10 & 3 \\ 3 & 4 & 8 & 0 & 17 & 4 \\ 4 & 8 & 10 & 0 & 24 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 24 & 0 & 49 & 13 \\ 3 & 4 & 6 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[130] $1^{124} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 13 & 3 \\ 3 & 4 & 8 & 0 & 21 & 4 \\ 4 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 21 & 30 & 0 & 76 & 16 \\ 3 & 4 & 6 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[133] $1^{111} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 50 & 6 \\ 15 & 20 & 17 & 67 & 8 \\ 12 & 17 & 13 & 57 & 7 \\ 50 & 67 & 57 & 217 & 27 \\ 6 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$

[134] $1^{13} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 13 & 3 \\ 3 & 4 & 0 & 0 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 0 & 0 & 85 & 17 \\ 3 & 4 & 0 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[135] $1^{133} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 13 & 3 \\ 3 & 4 & 17 & 0 & 22 & 4 \\ 9 & 17 & 49 & 0 & 66 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 66 & 0 & 85 & 17 \\ 3 & 4 & 13 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[136] $1^{112} 3^1 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 8 & 3 \\ 3 & 4 & 14 & 0 & 13 & 4 \\ 9 & 14 & 37 & 0 & 33 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 33 & 0 & 28 & 10 \\ 3 & 4 & 11 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[138] $1^{143} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 12 & 3 \\ 3 & 4 & 21 & 0 & 22 & 4 \\ 11 & 21 & 76 & 0 & 81 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 81 & 0 & 85 & 17 \\ 3 & 4 & 16 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[137] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[140] $1^{124} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 10 & 3 \\ 3 & 4 & 8 & 0 & 16 & 4 \\ 4 & 8 & 10 & 0 & 24 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 16 & 24 & 0 & 46 & 12 \\ 3 & 4 & 6 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[139] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 3 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 3 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[141] $1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 17 & 4 \\ 4 & 8 & 10 & 0 & 24 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 24 & 0 & 49 & 13 \\ 3 & 4 & 6 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[142] $1^{124} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 15 & 3 \\ 3 & 4 & 8 & 0 & 26 & 4 \\ 4 & 8 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 36 & 0 & 118 & 20 \\ 3 & 4 & 6 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[143] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.66 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[144] $1^{113} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 3 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 3 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[145] $1^{122} 3^1 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 8 & 3 \\ 3 & 4 & 18 & 0 & 13 & 4 \\ 10 & 18 & 58 & 0 & 42 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 42 & 0 & 28 & 10 \\ 3 & 4 & 14 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[147] $1^{122} 3^1 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 13 & 3 \\ 3 & 4 & 13 & 0 & 23 & 4 \\ 7 & 13 & 28 & 0 & 54 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 23 & 54 & 0 & 100 & 18 \\ 3 & 4 & 10 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[146] $1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[148] $1^{113} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 19 & 3 \\ 3 & 4 & 0 & 0 & 32 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 32 & 0 & 0 & 193 & 25 \\ 3 & 4 & 0 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[149] $1^{113} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 3 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[150] $1^{11} 1^{111} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 14 & 4 \\ 4 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 15 & 0 & 37 & 11 \\ 3 & 4 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[151] $1^{13} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 16 & 3 \\ 3 & 4 & 0 & 0 & 27 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 0 & 0 & 133 & 21 \\ 3 & 4 & 0 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[152] $1^{14} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 18 & 3 \\ 3 & 4 & 0 & 0 & 31 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 31 & 0 & 0 & 172 & 24 \\ 3 & 4 & 0 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[153] $1^{123} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 14 & 3 \\ 3 & 4 & 13 & 0 & 22 & 4 \\ 7 & 13 & 28 & 0 & 51 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 22 & 51 & 0 & 85 & 17 \\ 3 & 4 & 10 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[154] $21^{114} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 17 & 15 & 72 & 7 \\ 17 & 20 & 17 & 84 & 8 \\ 15 & 17 & 13 & 72 & 7 \\ 72 & 84 & 72 & 346 & 34 \\ 7 & 8 & 7 & 34 & 3 \end{pmatrix}_{b_0}$$

[155] $1^{114} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 219, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 21 & 4 \\ 5 & 9 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 33 & 0 & 76 & 16 \\ 3 & 4 & 7 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[158] $1^{113} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 13 & 3 \\ 3 & 4 & 14 & 0 & 22 & 4 \\ 9 & 14 & 37 & 0 & 57 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 57 & 0 & 85 & 17 \\ 3 & 4 & 11 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[157] $1^{14} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 21 & 3 \\ 3 & 4 & 0 & 0 & 36 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 36 & 0 & 0 & 238 & 28 \\ 3 & 4 & 0 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[156] $1^{113} 31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 12 & 3 \\ 3 & 4 & 14 & 0 & 22 & 4 \\ 8 & 14 & 37 & 0 & 57 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 57 & 0 & 85 & 17 \\ 3 & 4 & 11 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[160] $21^{1111} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 20 & 27 & 27 & 7 \\ 20 & 25 & 36 & 36 & 9 \\ 27 & 36 & 49 & 51 & 13 \\ 27 & 36 & 51 & 49 & 13 \\ 7 & 9 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$

[159] $1^{112} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 3 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 3 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[162] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 3 \\ 3 & 4 & 9 & 0 & 14 & 4 \\ 5 & 9 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 24 & 0 & 37 & 11 \\ 3 & 4 & 7 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[161] $1^{112} 3 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 3 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[163] $21^{112} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 17 & 15 & 42 & 7 \\ 17 & 20 & 17 & 50 & 8 \\ 15 & 17 & 13 & 42 & 7 \\ 42 & 50 & 42 & 118 & 20 \\ 7 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[164] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 233, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 14 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 15 & 0 & 37 & 11 \\ 3 & 4 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[165] $1^{123} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 5 & 0 & 18 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[166] $1^{13} 3 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 19 & 3 \\ 3 & 4 & 5 & 0 & 32 & 4 \\ 4 & 5 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 32 & 33 & 0 & 193 & 25 \\ 3 & 4 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[167] $1^{123} 3 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 19 & 3 \\ 3 & 4 & 13 & 0 & 32 & 4 \\ 8 & 13 & 28 & 0 & 75 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 32 & 75 & 0 & 193 & 25 \\ 3 & 4 & 10 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[168] $1^{112} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 3 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[169] $21^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 20 & 27 & 15 & 7 \\ 20 & 25 & 36 & 20 & 9 \\ 27 & 36 & 49 & 27 & 13 \\ 15 & 20 & 27 & 13 & 7 \\ 7 & 9 & 13 & 7 & 3 \end{pmatrix}_{b_0}$$

[170] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 5 & 0 & 14 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 14 & 15 & 0 & 37 & 11 \\ 3 & 4 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[171] \ 1^{132} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 11 & 3 \\ 3 & 4 & 22 & 0 & 18 & 4 \\ 12 & 22 & 85 & 0 & 72 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 72 & 0 & 58 & 14 \\ 3 & 4 & 17 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{122} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 14 & 3 \\ 3 & 4 & 13 & 0 & 23 & 4 \\ 8 & 13 & 28 & 0 & 54 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 23 & 54 & 0 & 100 & 18 \\ 3 & 4 & 10 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[173] \ 1^{14} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 5 & 0 & 26 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 24 & 0 & 118 & 20 \\ 3 & 4 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[174] \ 1^{122} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 5 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 6 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[175] \ 1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 3 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[176] \ 1^{112} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 259, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.68 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 7 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[177] \ 1^{113} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 14 & 3 \\ 3 & 4 & 5 & 0 & 23 & 4 \\ 4 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 23 & 24 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[178] \ 1^{124} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 14 & 3 \\ 3 & 4 & 8 & 0 & 26 & 4 \\ 4 & 8 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 36 & 0 & 118 & 20 \\ 3 & 4 & 6 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[179] \ 1^{111} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 12 & 3 \\ 3 & 4 & 9 & 0 & 19 & 4 \\ 6 & 9 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 19 & 33 & 0 & 73 & 15 \\ 3 & 4 & 7 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[180] \ 1^{113} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$\begin{aligned}
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 27 & 0 & 49 & 13 \\ 3 & 4 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0} \\
[181] & 21^{113} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 13 & 17 & 15 & 57 & 7 \\ 17 & 20 & 17 & 67 & 8 \\ 15 & 17 & 13 & 57 & 7 \\ 57 & 67 & 57 & 217 & 27 \\ 7 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0} \\
[184] & 1^{123} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3 \\
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 8 & 1 & 12 & 3 \\ 3 & 4 & 13 & 0 & 22 & 4 \\ 8 & 13 & 28 & 0 & 51 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 51 & 0 & 85 & 17 \\ 3 & 4 & 10 & -2 & 17 & 2 \end{pmatrix}_{b_0} \\
[185] & 1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & -2 & 13 & 2 \end{pmatrix}_{b_0} \\
[183] & 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & -2 & 10 & 2 \end{pmatrix}_{b_0} \\
[182] & 1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8 \\
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 5 & 0 & 22 & 4 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 21 & 0 & 85 & 17 \\ 3 & 4 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0} \\
[186] & 1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 270, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 5 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 18 & 0 & 28 & 10 \\ 3 & 4 & 6 & -2 & 10 & 2 \end{pmatrix}_{b_0} \\
[188] & 1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 3 & -2 & 12 & 2 \end{pmatrix}_{b_0} \\
[187] & 1^{111} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 8 & 1 & 9 & 3 \\ 3 & 4 & 14 & 0 & 14 & 4 \\ 8 & 14 & 37 & 0 & 39 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 39 & 0 & 37 & 11 \\ 3 & 4 & 11 & -2 & 11 & 2 \end{pmatrix}_{b_0} \\
[189] & 1^{111} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 273, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 14 & 4 \\ 5 & 9 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 24 & 0 & 37 & 11 \\ 3 & 4 & 7 & -2 & 11 & 2 \end{pmatrix}_{b_0} \\
[190] & 1^{124} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 11 & 3 \\ 3 & 4 & 8 & 0 & 21 & 4 \\ 4 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 30 & 0 & 76 & 16 \\ 3 & 4 & 6 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[191] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 12 & 3 \\ 3 & 4 & 9 & 0 & 19 & 4 \\ 5 & 9 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 19 & 33 & 0 & 73 & 15 \\ 3 & 4 & 7 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[192] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 277, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.66 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[194] $1^{124} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 13 & 3 \\ 3 & 4 & 13 & 0 & 21 & 4 \\ 7 & 13 & 28 & 0 & 48 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 21 & 48 & 0 & 76 & 16 \\ 3 & 4 & 10 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[193] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 17 & 27 & 0 & 49 & 13 \\ 3 & 4 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[195] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 281, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 3 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[196] $1^{133} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 293, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 12 & 3 \\ 3 & 4 & 17 & 0 & 22 & 4 \\ 10 & 17 & 49 & 0 & 66 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 66 & 0 & 85 & 17 \\ 3 & 4 & 13 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[197] $1^{132} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 298, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 11 & 3 \\ 3 & 4 & 17 & 0 & 18 & 4 \\ 10 & 17 & 49 & 0 & 54 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 54 & 0 & 58 & 14 \\ 3 & 4 & 13 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[198] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 5 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 18 & 0 & 25 & 9 \\ 3 & 4 & 6 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[199] $1^{111} 3 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 8 & 3 \\ 3 & 4 & 14 & 0 & 14 & 4 \\ 8 & 14 & 37 & 0 & 39 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 39 & 0 & 37 & 11 \\ 3 & 4 & 11 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[200] $1^{122} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 11 & 3 \\ 3 & 4 & 13 & 0 & 18 & 4 \\ 7 & 13 & 28 & 0 & 42 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 42 & 0 & 58 & 14 \\ 3 & 4 & 10 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[201] $1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 312$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.68 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 8 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 30 & 0 & 28 & 10 \\ 3 & 4 & 10 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[202] $1^{133} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 313$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.6$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 10 & 3 \\ 3 & 4 & 12 & 0 & 17 & 4 \\ 7 & 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 36 & 0 & 49 & 13 \\ 3 & 4 & 9 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{143} 3^1 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 316$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 13 & 3 \\ 3 & 4 & 26 & 0 & 22 & 4 \\ 14 & 26 & 118 & 0 & 102 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 102 & 0 & 85 & 17 \\ 3 & 4 & 20 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[204] $1^{112} 3^1 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 316$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 11 & 3 \\ 3 & 4 & 14 & 0 & 18 & 4 \\ 8 & 14 & 37 & 0 & 48 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 48 & 0 & 58 & 14 \\ 3 & 4 & 11 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[205] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 319$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.66 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 7 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[206] $1^{124} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 320$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 16 & 4 \\ 5 & 8 & 10 & 0 & 24 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 24 & 0 & 46 & 12 \\ 3 & 4 & 6 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[207] $1^{124} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 322$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 3 \\ 3 & 4 & 8 & 0 & 21 & 4 \\ 4 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 30 & 0 & 76 & 16 \\ 3 & 4 & 6 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[208] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 328$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 10 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 6 & 9 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 27 & 0 & 49 & 13 \\ 3 & 4 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[209] $1^{112} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 333$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 3 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[210] $1^{12} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 334$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 17 & 3 \\ 3 & 4 & 5 & 0 & 28 & 4 \\ 4 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 28 & 30 & 0 & 154 & 22 \\ 3 & 4 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[211] $1^{112} 3^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 340$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.80 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 3 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[212] $1^{112} 3^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 340$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.71 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 3 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[213] $1^{112} 3^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 341$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 3 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[214] $1^{12} 3^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 342$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.73 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 18 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[215] $1^{12} 3^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 342$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.87 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 5 & 0 & 18 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[217] $1^{124} 3^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 344$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 10.72 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 15 & 3 \\ 3 & 4 & 8 & 0 & 26 & 4 \\ 5 & 8 & 10 & 0 & 36 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 36 & 0 & 118 & 20 \\ 3 & 4 & 6 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[216] $1^{12} 3^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 344$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 5 & 0 & 23 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 23 & 24 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{123} 3^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 345$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.3$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 19 & 3 \\ 3 & 4 & 13 & 0 & 32 & 4 \\ 7 & 13 & 28 & 0 & 75 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 32 & 75 & 0 & 193 & 25 \\ 3 & 4 & 10 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[219] $1^{114} 3^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 349$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.87 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 3 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[220] $1^{113} 3^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 353$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.71 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 3 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[221] $1^{114} 3^{111} 3^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 353, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 \\ 3 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 3 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[222] $1^{113} 3^{111} 3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 3 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 3 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[223] $1^{143} 3^{111} 3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 361, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 13 & 3 \\ 3 & 4 & 21 & 0 & 22 & 4 \\ 12 & 21 & 76 & 0 & 81 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 81 & 0 & 85 & 17 \\ 3 & 4 & 16 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[224] $1^{114} 3^{112} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 363, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 13 & 3 \\ 3 & 4 & 9 & 0 & 21 & 4 \\ 5 & 9 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 21 & 33 & 0 & 76 & 16 \\ 3 & 4 & 7 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[225] $1^{122} 3^{111} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 30 & 0 & 28 & 10 \\ 3 & 4 & 10 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[226] $1^{112} 3^{112} 3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 10 & 3 \\ 3 & 4 & 14 & 0 & 18 & 4 \\ 8 & 14 & 37 & 0 & 48 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 48 & 0 & 58 & 14 \\ 3 & 4 & 11 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[227] $1^{111} 3^{111} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 6 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 6 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[228] $1^{114} 3^{112} 3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 365, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 5 & 0 & 22 & 4 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 22 & 21 & 0 & 85 & 17 \\ 3 & 4 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[229] $1^{113} 3^{112} 3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 15 & 3 \\ 3 & 4 & 9 & 0 & 22 & 4 \\ 5 & 9 & 13 & 0 & 36 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 22 & 36 & 0 & 85 & 17 \\ 3 & 4 & 7 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[230] $1^{112} 3^{112} 3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 5 & 0 & 18 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[231] $1^{134} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 14 & 3 \\ 3 & 4 & 17 & 0 & 26 & 4 \\ 10 & 17 & 49 & 0 & 78 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 78 & 0 & 118 & 20 \\ 3 & 4 & 13 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[232] $1^{144} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 14 & 3 \\ 3 & 4 & 21 & 0 & 26 & 4 \\ 12 & 21 & 76 & 0 & 96 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 96 & 0 & 118 & 20 \\ 3 & 4 & 16 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[234] $1^{133} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 12 & 0 & 12 & 4 \\ 7 & 12 & 25 & 0 & 27 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 27 & 0 & 25 & 9 \\ 3 & 4 & 9 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[233] $1^{114} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 3 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{113} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 388, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 10 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 27 & 0 & 49 & 13 \\ 3 & 4 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[236] $1^{123} 3^1 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 389, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 18 & 3 \\ 3 & 4 & 13 & 0 & 32 & 4 \\ 7 & 13 & 28 & 0 & 75 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 32 & 75 & 0 & 193 & 25 \\ 3 & 4 & 10 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[237] $1^{133} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 10 & 3 \\ 3 & 4 & 17 & 0 & 17 & 4 \\ 9 & 17 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 51 & 0 & 49 & 13 \\ 3 & 4 & 13 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[238] $1^{114} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 393, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 3 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 3 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[239] $1^{111} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 393, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 14 & 4 \\ 6 & 9 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 24 & 0 & 37 & 11 \\ 3 & 4 & 7 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[240] $1^{122} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 394, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 10 & 3 \\ 3 & 4 & 13 & 0 & 18 & 4 \\ 7 & 13 & 28 & 0 & 42 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 42 & 0 & 58 & 14 \\ 3 & 4 & 10 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[241] $1^{112} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 398, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 12 & 3 \\ 3 & 4 & 9 & 0 & 18 & 4 \\ 5 & 9 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 18 & 30 & 0 & 58 & 14 \\ 3 & 4 & 7 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[242] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 3 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[243] $1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 401, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 5 & 0 & 22 & 4 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 21 & 0 & 85 & 17 \\ 3 & 4 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[244] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 3 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[245] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 407, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 12 & 3 \\ 3 & 4 & 9 & 0 & 21 & 4 \\ 5 & 9 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 33 & 0 & 76 & 16 \\ 3 & 4 & 7 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[246] $1^{14} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 410, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 5 & 0 & 26 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 24 & 0 & 118 & 20 \\ 3 & 4 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[248] $1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 414, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 10 & 3 \\ 3 & 4 & 8 & 0 & 17 & 4 \\ 5 & 8 & 10 & 0 & 24 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 24 & 0 & 49 & 13 \\ 3 & 4 & 6 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[247] $1^{112} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 414, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 10 & 3 \\ 3 & 4 & 9 & 0 & 18 & 4 \\ 5 & 9 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 30 & 0 & 58 & 14 \\ 3 & 4 & 7 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[249] $1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 9 & 3 \\ 3 & 4 & 13 & 0 & 17 & 4 \\ 7 & 13 & 28 & 0 & 39 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 39 & 0 & 49 & 13 \\ 3 & 4 & 10 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[250] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 421, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[251] $1^{124} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 11 & 3 \\ 3 & 4 & 13 & 0 & 21 & 4 \\ 7 & 13 & 28 & 0 & 48 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 48 & 0 & 76 & 16 \\ 3 & 4 & 10 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[252] $1^{12} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 427, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 8 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 6 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 7 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[253] $1^{134} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 9 & 3 \\ 3 & 4 & 12 & 0 & 16 & 4 \\ 7 & 12 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 36 & 0 & 46 & 12 \\ 3 & 4 & 9 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[254] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 437, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[255] $1^{144} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 12 & 3 \\ 3 & 4 & 21 & 0 & 21 & 4 \\ 11 & 21 & 76 & 0 & 78 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 78 & 0 & 76 & 16 \\ 3 & 4 & 16 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[256] $1^{14} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 446, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 5 & 0 & 26 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 26 & 24 & 0 & 118 & 20 \\ 3 & 4 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[257] $1^{13} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 449, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 5 & 0 & 22 & 4 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 22 & 21 & 0 & 85 & 17 \\ 3 & 4 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[258] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 452, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 3 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 3 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[259] $1^{13} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 459, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 5 & 0 & 27 & 4 \\ 3 & 5 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[260] $1^{13} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 461, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 5 & 0 & 22 & 4 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 22 & 21 & 0 & 85 & 17 \\ 3 & 4 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[261] $1^{113} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 32 & 21 & 0 & 193 & 25 \\ 3 & 4 & 3 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[262] $1^{113} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 469, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 3 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[263] $1^{123} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 10 & 3 \\ 3 & 4 & 13 & 0 & 17 & 4 \\ 7 & 13 & 28 & 0 & 39 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 39 & 0 & 49 & 13 \\ 3 & 4 & 10 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[264] $1^{113} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 477, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[265] $1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 17 & 3 \\ 3 & 4 & 9 & 0 & 28 & 4 \\ 6 & 9 & 13 & 0 & 48 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 28 & 48 & 0 & 154 & 22 \\ 3 & 4 & 7 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[266] $1^{122} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 11 & 3 \\ 3 & 4 & 18 & 0 & 18 & 4 \\ 10 & 18 & 58 & 0 & 60 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 60 & 0 & 58 & 14 \\ 3 & 4 & 14 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[267] $1^{14} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 494, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 5 & 0 & 26 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 26 & 24 & 0 & 118 & 20 \\ 3 & 4 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[268] $1^{134} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 497, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 12 & 3 \\ 3 & 4 & 12 & 0 & 21 & 4 \\ 7 & 12 & 25 & 0 & 45 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 45 & 0 & 76 & 16 \\ 3 & 4 & 9 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[269] $1^{112} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 499, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 13 & 3 \\ 3 & 4 & 9 & 0 & 23 & 4 \\ 5 & 9 & 13 & 0 & 39 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 23 & 39 & 0 & 100 & 18 \\ 3 & 4 & 7 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[271] $1^{12} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 500, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 5 & 0 & 23 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 23 & 24 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[270] $21^{1111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 500, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 20 & 15 & 15 & 7 \\ 20 & 25 & 20 & 20 & 9 \\ 15 & 20 & 13 & 15 & 7 \\ 15 & 20 & 15 & 13 & 7 \\ 7 & 9 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[272] $1^{14}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 506, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 5 & 0 & 26 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 26 & 24 & 0 & 118 & 20 \\ 3 & 4 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[273] $1^{133}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 509, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 12 & 3 \\ 3 & 4 & 17 & 0 & 22 & 4 \\ 9 & 17 & 49 & 0 & 66 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 66 & 0 & 85 & 17 \\ 3 & 4 & 13 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[274] $1^{112}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 514, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 16 & 3 \\ 3 & 4 & 9 & 0 & 28 & 4 \\ 5 & 9 & 13 & 0 & 48 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 28 & 48 & 0 & 154 & 22 \\ 3 & 4 & 7 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[275] $1^{112}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 522, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 18 & 4 \\ 5 & 9 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 30 & 0 & 58 & 14 \\ 3 & 4 & 7 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[276] $1^{14}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 524, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 5 & 0 & 31 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 31 & 30 & 0 & 172 & 24 \\ 3 & 4 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[277] $1^{133}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 524, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 9 & 3 \\ 3 & 4 & 17 & 0 & 17 & 4 \\ 9 & 17 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 51 & 0 & 49 & 13 \\ 3 & 4 & 13 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[278] $1^{124}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 526, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 12 & 3 \\ 3 & 4 & 8 & 0 & 21 & 4 \\ 5 & 8 & 10 & 0 & 30 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 30 & 0 & 76 & 16 \\ 3 & 4 & 6 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[279] $1^{124}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 15 & 3 \\ 3 & 4 & 18 & 0 & 26 & 4 \\ 10 & 18 & 58 & 0 & 84 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 84 & 0 & 118 & 20 \\ 3 & 4 & 14 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[280] $1^{114}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 534, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 15 & 3 \\ 3 & 4 & 9 & 0 & 26 & 4 \\ 6 & 9 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 42 & 0 & 118 & 20 \\ 3 & 4 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[281] $1^{12} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 5 & 0 & 23 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 23 & 24 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[282] $1^{134} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 539, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 17 & 3 \\ 3 & 4 & 17 & 0 & 31 & 4 \\ 9 & 17 & 49 & 0 & 93 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 31 & 93 & 0 & 172 & 24 \\ 3 & 4 & 13 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[283] $1^{124} 3^1 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 14 & 3 \\ 3 & 4 & 18 & 0 & 26 & 4 \\ 10 & 18 & 58 & 0 & 84 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 84 & 0 & 118 & 20 \\ 3 & 4 & 14 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[284] $1^{113} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 549, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 12 & 3 \\ 3 & 4 & 9 & 0 & 22 & 4 \\ 5 & 9 & 13 & 0 & 36 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 36 & 0 & 85 & 17 \\ 3 & 4 & 7 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[285] $1^{111} 3^1 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 560, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 9 & 3 \\ 3 & 4 & 14 & 0 & 14 & 4 \\ 9 & 14 & 37 & 0 & 39 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 39 & 0 & 37 & 11 \\ 3 & 4 & 11 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[286] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 3 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[287] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 3 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[288] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 565, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 3 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[289] $1^{144} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 574, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 14 & 3 \\ 3 & 4 & 21 & 0 & 26 & 4 \\ 11 & 21 & 76 & 0 & 96 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 96 & 0 & 118 & 20 \\ 3 & 4 & 16 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[291] $1^{134} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 579, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 11 & 3 \\ 3 & 4 & 17 & 0 & 21 & 4 \\ 9 & 17 & 49 & 0 & 63 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 63 & 0 & 76 & 16 \\ 3 & 4 & 13 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[290] $1^{112} 3^1 1^{12} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 579, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 15 & 3 \\ 3 & 4 & 9 & 0 & 23 & 4 \\ 5 & 9 & 13 & 0 & 39 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 23 & 39 & 0 & 100 & 18 \\ 3 & 4 & 7 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[292] $1^{124} 3^1 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 12 & 3 \\ 3 & 4 & 13 & 0 & 21 & 4 \\ 7 & 13 & 28 & 0 & 48 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 48 & 0 & 76 & 16 \\ 3 & 4 & 10 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[293] $1^{12} 3^1 1^{13} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 584, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 5 & 0 & 23 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 23 & 24 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[294] $1^{112} 3^1 1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 11 & 3 \\ 3 & 4 & 14 & 0 & 18 & 4 \\ 9 & 14 & 37 & 0 & 48 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 48 & 0 & 58 & 14 \\ 3 & 4 & 11 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[295] $1^{134} 3^1 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 591, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 12 & 3 \\ 3 & 4 & 17 & 0 & 21 & 4 \\ 9 & 17 & 49 & 0 & 63 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 63 & 0 & 76 & 16 \\ 3 & 4 & 13 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[297] $1^{12} 3^1 1^{13} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 5 & 0 & 23 & 4 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 23 & 24 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[296] $1^{113} 3^1 1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 3 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[298] $1^{124} 3^1 1^{11} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 598, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 22 & 3 \\ 3 & 4 & 13 & 0 & 36 & 4 \\ 7 & 13 & 28 & 0 & 84 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 36 & 84 & 0 & 238 & 28 \\ 3 & 4 & 10 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[299] $1^{112} 3^1 1^{112} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 602, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 18 & 4 \\ 6 & 9 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 30 & 0 & 58 & 14 \\ 3 & 4 & 7 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[300] $1^{123} 3^1 1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 604, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 13 & 3 \\ 3 & 4 & 18 & 0 & 22 & 4 \\ 10 & 18 & 58 & 0 & 72 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 72 & 0 & 85 & 17 \\ 3 & 4 & 14 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[301] $1^{124} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 606, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 16 & 3 \\ 3 & 4 & 13 & 0 & 26 & 4 \\ 7 & 13 & 28 & 0 & 60 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 26 & 60 & 0 & 118 & 20 \\ 3 & 4 & 10 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[302] $1^{123} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 607, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 16 & 3 \\ 3 & 4 & 13 & 0 & 27 & 4 \\ 7 & 13 & 28 & 0 & 63 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 63 & 0 & 133 & 21 \\ 3 & 4 & 10 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[303] $1^{134} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 614, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 15 & 3 \\ 3 & 4 & 17 & 0 & 26 & 4 \\ 9 & 17 & 49 & 0 & 78 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 78 & 0 & 118 & 20 \\ 3 & 4 & 13 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[304] $1^{124} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 616, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 12 & 3 \\ 3 & 4 & 13 & 0 & 21 & 4 \\ 8 & 13 & 28 & 0 & 48 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 48 & 0 & 76 & 16 \\ 3 & 4 & 10 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[306] $1^{123} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 619, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 15 & 3 \\ 3 & 4 & 13 & 0 & 27 & 4 \\ 7 & 13 & 28 & 0 & 63 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 27 & 63 & 0 & 133 & 21 \\ 3 & 4 & 10 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[305] $1^{112} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 619, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 14 & 3 \\ 3 & 4 & 9 & 0 & 23 & 4 \\ 6 & 9 & 13 & 0 & 39 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 23 & 39 & 0 & 100 & 18 \\ 3 & 4 & 7 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[307] $1^{122} 3^1 1^{22} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 624, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 10 & 3 \\ 3 & 4 & 18 & 0 & 18 & 4 \\ 10 & 18 & 58 & 0 & 60 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 60 & 0 & 58 & 14 \\ 3 & 4 & 14 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[308] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 628, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 3 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[309] $1^{13} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 5 & 0 & 32 & 4 \\ 3 & 5 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 32 & 33 & 0 & 193 & 25 \\ 3 & 4 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[310] $1^{123} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 12 & 3 \\ 3 & 4 & 13 & 0 & 22 & 4 \\ 7 & 13 & 28 & 0 & 51 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 51 & 0 & 85 & 17 \\ 3 & 4 & 10 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[311] $1^{114} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 634, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 14 & 3 \\ 3 & 4 & 9 & 0 & 26 & 4 \\ 5 & 9 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 42 & 0 & 118 & 20 \\ 3 & 4 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[312] $1^{113} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 3 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[313] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 3 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[314] $1^{12} 3^1 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 638, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 5 & 0 & 28 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 28 & 30 & 0 & 154 & 22 \\ 3 & 4 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[315] $1^{13} 3^1 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 639, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 5 & 0 & 27 & 4 \\ 3 & 5 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[316] $1^{113} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 640, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 19 & 3 \\ 3 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 32 & 21 & 0 & 193 & 25 \\ 3 & 4 & 3 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[317] $1^{114} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 17 & 3 \\ 3 & 4 & 9 & 0 & 26 & 4 \\ 5 & 9 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 26 & 42 & 0 & 118 & 20 \\ 3 & 4 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[318] $1^{122} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 11 & 3 \\ 3 & 4 & 13 & 0 & 18 & 4 \\ 8 & 13 & 28 & 0 & 42 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 42 & 0 & 58 & 14 \\ 3 & 4 & 10 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[319] $1^{123} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 649, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 13 & 3 \\ 3 & 4 & 13 & 0 & 22 & 4 \\ 7 & 13 & 28 & 0 & 51 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 51 & 0 & 85 & 17 \\ 3 & 4 & 10 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[320] $1^{113} 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 649, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 13 & 3 \\ 3 & 4 & 9 & 0 & 22 & 4 \\ 6 & 9 & 13 & 0 & 36 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 36 & 0 & 85 & 17 \\ 3 & 4 & 7 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[321] $1^{122} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 652, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 8 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 8 & 13 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 30 & 0 & 28 & 10 \\ 3 & 4 & 10 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[322] $1^{122} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 655, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 14 & 3 \\ 3 & 4 & 9 & 0 & 23 & 4 \\ 5 & 9 & 13 & 0 & 39 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 23 & 39 & 0 & 100 & 18 \\ 3 & 4 & 7 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[323] $1^{133} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 13 & 3 \\ 3 & 4 & 22 & 0 & 22 & 4 \\ 12 & 22 & 85 & 0 & 87 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 87 & 0 & 85 & 17 \\ 3 & 4 & 17 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[324] $1^{113} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 661, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 14 & 3 \\ 3 & 4 & 9 & 0 & 22 & 4 \\ 5 & 9 & 13 & 0 & 36 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 22 & 36 & 0 & 85 & 17 \\ 3 & 4 & 7 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[325] $1^{124} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 15 & 3 \\ 3 & 4 & 18 & 0 & 26 & 4 \\ 11 & 18 & 58 & 0 & 84 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 84 & 0 & 118 & 20 \\ 3 & 4 & 14 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[327] $1^{144} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 684, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 11 & 3 \\ 3 & 4 & 21 & 0 & 21 & 4 \\ 11 & 21 & 76 & 0 & 78 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 78 & 0 & 76 & 16 \\ 3 & 4 & 16 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[326] $1^{123} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 684, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 12 & 3 \\ 3 & 4 & 18 & 0 & 22 & 4 \\ 10 & 18 & 58 & 0 & 72 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 72 & 0 & 85 & 17 \\ 3 & 4 & 14 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[329] $1^{113} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 685, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 3 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[328] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 685, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 3 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[330] $1^{112} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 690, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 3 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[351] \ 1^{123} 1^{111} 3 \ 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 787, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 16 & 3 \\ 3 & 4 & 13 & 0 & 27 & 4 \\ 8 & 13 & 28 & 0 & 63 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 63 & 0 & 133 & 21 \\ 3 & 4 & 10 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[352] \ 1^{114} 1^{11} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 788, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 3 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[353] \ 1^{134} 1^{11} 3 \ 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 794, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 14 & 3 \\ 3 & 4 & 17 & 0 & 26 & 4 \\ 9 & 17 & 49 & 0 & 78 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 78 & 0 & 118 & 20 \\ 3 & 4 & 13 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[354] \ 1^{144} 1^{11} 3 \ 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 15 & 3 \\ 3 & 4 & 26 & 0 & 26 & 4 \\ 14 & 26 & 118 & 0 & 120 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 120 & 0 & 118 & 20 \\ 3 & 4 & 20 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[355] \ 1^{113} 1^{111} 3 \ 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 812, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 16 & 3 \\ 3 & 4 & 9 & 0 & 27 & 4 \\ 6 & 9 & 13 & 0 & 45 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 45 & 0 & 133 & 21 \\ 3 & 4 & 7 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[356] \ 1^{124} 1^{11} 3 \ 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 814, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 14 & 3 \\ 3 & 4 & 13 & 0 & 26 & 4 \\ 7 & 13 & 28 & 0 & 60 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 60 & 0 & 118 & 20 \\ 3 & 4 & 10 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[357] \ 1^{113} 1^{111} 3 \ 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 817, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 19 & 3 \\ 3 & 4 & 9 & 0 & 32 & 4 \\ 6 & 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 32 & 54 & 0 & 193 & 25 \\ 3 & 4 & 7 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[358] \ 1^{12} 1^{14} 3 \ 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 818, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 5 & 0 & 28 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 28 & 30 & 0 & 154 & 22 \\ 3 & 4 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[359] \ 1^{12} 1^{13} 3 \ 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 830, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 19 & 3 \\ 3 & 4 & 5 & 0 & 28 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 28 & 30 & 0 & 154 & 22 \\ 3 & 4 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[360] \ 1^{114} 1^{111} 3 \ 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 843, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} =$$

10.0

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 18 & 3 \\ 3 & 4 & 9 & 0 & 31 & 4 \\ 6 & 9 & 13 & 0 & 51 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 31 & 51 & 0 & 172 & 24 \\ 3 & 4 & 7 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[361] \ 1^{133} 1^{111} 3 \ 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 849, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 13 & 3 \\ 3 & 4 & 17 & 0 & 22 & 4 \\ 10 & 17 & 49 & 0 & 66 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 66 & 0 & 85 & 17 \\ 3 & 4 & 13 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[362] \ 1^{124} 1^{111} 3 \ 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 850, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 21 & 3 \\ 3 & 4 & 13 & 0 & 36 & 4 \\ 8 & 13 & 28 & 0 & 84 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 36 & 84 & 0 & 238 & 28 \\ 3 & 4 & 10 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[363] \ 1^{13} 1^{11} 3 \ 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 857, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 19 & 3 \\ 3 & 4 & 5 & 0 & 32 & 4 \\ 3 & 5 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 32 & 33 & 0 & 193 & 25 \\ 3 & 4 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[365] \ 1^{134} 1^{11} 3 \ 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 860, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 15 & 3 \\ 3 & 4 & 22 & 0 & 26 & 4 \\ 12 & 22 & 85 & 0 & 102 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 102 & 0 & 118 & 20 \\ 3 & 4 & 17 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[364] \ 1^{14} 1^{12} 3 \ 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 860, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 19 & 3 \\ 3 & 4 & 5 & 0 & 31 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 31 & 30 & 0 & 172 & 24 \\ 3 & 4 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[366] \ 1^{114} 1^{11} 3 \ 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 870, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 15 & 3 \\ 3 & 4 & 9 & 0 & 26 & 4 \\ 5 & 9 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 42 & 0 & 118 & 20 \\ 3 & 4 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[367] \ 1^{114} 1^{12} 3 \ 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 874, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 16 & 3 \\ 3 & 4 & 9 & 0 & 26 & 4 \\ 5 & 9 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 26 & 42 & 0 & 118 & 20 \\ 3 & 4 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[368] \ 1^{114} 1^{14} 3 \ 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 884, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 3 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[369] \ 1^{14} 1^{13} 3 \ 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 908, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 21 & 3 \\ 3 & 4 & 5 & 0 & 31 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 31 & 30 & 0 & 172 & 24 \\ 3 & 4 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[370] $1^{113} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 909, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 18 & 3 \\ 3 & 4 & 9 & 0 & 32 & 4 \\ 5 & 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 32 & 54 & 0 & 193 & 25 \\ 3 & 4 & 7 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[371] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 916, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 3 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[372] $1^{113} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 920, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 5 & 0 & 31 & 4 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 31 & 30 & 0 & 172 & 24 \\ 3 & 4 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[373] $1^{124} 3 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 934, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 20 & 3 \\ 3 & 4 & 13 & 0 & 36 & 4 \\ 7 & 13 & 28 & 0 & 84 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 36 & 84 & 0 & 238 & 28 \\ 3 & 4 & 10 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[374] $1^{124} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 938, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 15 & 3 \\ 3 & 4 & 13 & 0 & 26 & 4 \\ 7 & 13 & 28 & 0 & 60 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 60 & 0 & 118 & 20 \\ 3 & 4 & 10 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[375] $1^{133} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 940, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 10 & 3 \\ 3 & 4 & 17 & 0 & 17 & 4 \\ 10 & 17 & 49 & 0 & 51 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 51 & 0 & 49 & 13 \\ 3 & 4 & 13 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[376] $1^{114} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 941, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 3 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[377] $1^{133} 3 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 944, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 12 & 3 \\ 3 & 4 & 22 & 0 & 22 & 4 \\ 12 & 22 & 85 & 0 & 87 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 87 & 0 & 85 & 17 \\ 3 & 4 & 17 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[378] $1^{123} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 945, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 13 & 3 \\ 3 & 4 & 13 & 0 & 22 & 4 \\ 8 & 13 & 28 & 0 & 51 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 51 & 0 & 85 & 17 \\ 3 & 4 & 10 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[379] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 948, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 3 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[380] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 972, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 17 & 3 \\ 3 & 4 & 9 & 0 & 27 & 4 \\ 5 & 9 & 13 & 0 & 45 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 27 & 45 & 0 & 133 & 21 \\ 3 & 4 & 7 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[381] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 979, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 17 & 3 \\ 3 & 4 & 9 & 0 & 31 & 4 \\ 5 & 9 & 13 & 0 & 51 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 31 & 51 & 0 & 172 & 24 \\ 3 & 4 & 7 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[382] $1^{113} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 984, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 16 & 3 \\ 3 & 4 & 9 & 0 & 27 & 4 \\ 5 & 9 & 13 & 0 & 45 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 45 & 0 & 133 & 21 \\ 3 & 4 & 7 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[383] $1^{124} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 984, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 17 & 3 \\ 3 & 4 & 13 & 0 & 31 & 4 \\ 7 & 13 & 28 & 0 & 72 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 31 & 72 & 0 & 172 & 24 \\ 3 & 4 & 10 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[384] $1^{124} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 994, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 21 & 3 \\ 3 & 4 & 13 & 0 & 36 & 4 \\ 7 & 13 & 28 & 0 & 84 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 36 & 84 & 0 & 238 & 28 \\ 3 & 4 & 10 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[385] $1^{114} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 994, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 21 & 3 \\ 3 & 4 & 9 & 0 & 36 & 4 \\ 6 & 9 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 36 & 60 & 0 & 238 & 28 \\ 3 & 4 & 7 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[386] $1^{144} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1002, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 15 & 3 \\ 3 & 4 & 21 & 0 & 26 & 4 \\ 12 & 21 & 76 & 0 & 96 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 96 & 0 & 118 & 20 \\ 3 & 4 & 16 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[387] $1^{134} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1003, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 12 & 3 \\ 3 & 4 & 17 & 0 & 21 & 4 \\ 10 & 17 & 49 & 0 & 63 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 63 & 0 & 76 & 16 \\ 3 & 4 & 13 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[388] $1^{134} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1004, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 14 & 3 \\ 3 & 4 & 22 & 0 & 26 & 4 \\ 12 & 22 & 85 & 0 & 102 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 102 & 0 & 118 & 20 \\ 3 & 4 & 17 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[389] $1^{122} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1008, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 11 & 3 \\ 3 & 4 & 18 & 0 & 18 & 4 \\ 11 & 18 & 58 & 0 & 60 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 60 & 0 & 58 & 14 \\ 3 & 4 & 14 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[390] $1^{13} 3 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1013, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 5 & 0 & 32 & 4 \\ 3 & 5 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 32 & 33 & 0 & 193 & 25 \\ 3 & 4 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[391] $1^{13} 3 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1017, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 21 & 3 \\ 3 & 4 & 9 & 0 & 32 & 4 \\ 5 & 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 32 & 54 & 0 & 193 & 25 \\ 3 & 4 & 7 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[392] $1^{114} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1024, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 21 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 3 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[393] $1^{14} 3 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1026, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 21 & 3 \\ 3 & 4 & 5 & 0 & 36 & 4 \\ 3 & 5 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[394] $1^{123} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1036, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 13 & 3 \\ 3 & 4 & 18 & 0 & 22 & 4 \\ 11 & 18 & 58 & 0 & 72 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 72 & 0 & 85 & 17 \\ 3 & 4 & 14 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[395] $1^{114} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1061, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 21 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 3 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[396] $1^{124} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1076, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 18 & 3 \\ 3 & 4 & 13 & 0 & 31 & 4 \\ 7 & 13 & 28 & 0 & 72 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 31 & 72 & 0 & 172 & 24 \\ 3 & 4 & 10 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[397] $1^{114} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1077, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 19 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 3 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[398] $1^{124} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1082, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 15 & 3 \\ 3 & 4 & 13 & 0 & 26 & 4 \\ 8 & 13 & 28 & 0 & 60 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 60 & 0 & 118 & 20 \\ 3 & 4 & 10 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[399] $1^{13} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1097, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 21 & 3 \\ 3 & 4 & 5 & 0 & 32 & 4 \\ 3 & 5 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 32 & 33 & 0 & 193 & 25 \\ 3 & 4 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[400] $1^{13} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 22 & 3 \\ 3 & 4 & 5 & 0 & 32 & 4 \\ 3 & 5 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 32 & 33 & 0 & 193 & 25 \\ 3 & 4 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[401] $1^{14} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 3 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[402] $1^{14} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 20 & 3 \\ 3 & 4 & 9 & 0 & 31 & 4 \\ 5 & 9 & 13 & 0 & 51 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 31 & 51 & 0 & 172 & 24 \\ 3 & 4 & 7 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[403] $1^{124} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 18 & 3 \\ 3 & 4 & 13 & 0 & 31 & 4 \\ 8 & 13 & 28 & 0 & 72 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 31 & 72 & 0 & 172 & 24 \\ 3 & 4 & 10 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[404] $1^{14} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 22 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 3 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[405] $1^{14} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 24 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 24 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 3 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[406] $1^{113} 3 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1177, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 19 & 3 \\ 3 & 4 & 9 & 0 & 32 & 4 \\ 5 & 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 32 & 54 & 0 & 193 & 25 \\ 3 & 4 & 7 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[407] $1^{14} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1190, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 24 & 3 \\ 3 & 4 & 9 & 0 & 36 & 4 \\ 5 & 9 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 24 & 36 & 60 & 0 & 238 & 28 \\ 3 & 4 & 7 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[408] $1^{14} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 22 & 3 \\ 3 & 4 & 5 & 0 & 36 & 4 \\ 3 & 5 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[409] $1^{13} 3 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 20 & 3 \\ 3 & 4 & 9 & 0 & 32 & 4 \\ 5 & 9 & 13 & 0 & 54 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 32 & 54 & 0 & 193 & 25 \\ 3 & 4 & 7 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[410] $1^{14} 3 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 23 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 23 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 3 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[411] $1^{14} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 12 & 3 \\ 3 & 4 & 21 & 0 & 21 & 4 \\ 12 & 21 & 76 & 0 & 78 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 78 & 0 & 76 & 16 \\ 3 & 4 & 16 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[412] $1^{13} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 15 & 3 \\ 3 & 4 & 17 & 0 & 26 & 4 \\ 10 & 17 & 49 & 0 & 78 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 78 & 0 & 118 & 20 \\ 3 & 4 & 13 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[413] $1^{14} 3 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 20 & 3 \\ 3 & 4 & 9 & 0 & 36 & 4 \\ 5 & 9 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 36 & 60 & 0 & 238 & 28 \\ 3 & 4 & 7 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[414] $1^{14} 3 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1263, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 18 & 3 \\ 3 & 4 & 9 & 0 & 31 & 4 \\ 5 & 9 & 13 & 0 & 51 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 31 & 51 & 0 & 172 & 24 \\ 3 & 4 & 7 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[415] $1^{14} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 14 & 3 \\ 3 & 4 & 26 & 0 & 26 & 4 \\ 14 & 26 & 118 & 0 & 120 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 120 & 0 & 118 & 20 \\ 3 & 4 & 20 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[416] $1^{14} 3 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1314, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 23 & 3 \\ 3 & 4 & 5 & 0 & 36 & 4 \\ 3 & 5 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 23 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[417] $1^{14} 3 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1315, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 19 & 3 \\ 3 & 4 & 9 & 0 & 31 & 4 \\ 5 & 9 & 13 & 0 & 51 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 31 & 51 & 0 & 172 & 24 \\ 3 & 4 & 7 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[418] \ 1^{14} 3 \ 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1350, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 24 & 3 \\ 3 & 4 & 5 & 0 & 36 & 4 \\ 3 & 5 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 24 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[419] \ 1^{13} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 13 & 3 \\ 3 & 4 & 22 & 0 & 22 & 4 \\ 13 & 22 & 85 & 0 & 87 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 87 & 0 & 85 & 17 \\ 3 & 4 & 17 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[420] \ 1^{13} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1484, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 15 & 3 \\ 3 & 4 & 22 & 0 & 26 & 4 \\ 13 & 22 & 85 & 0 & 102 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 102 & 0 & 118 & 20 \\ 3 & 4 & 17 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[421] \ 1^{14} 3 \ 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1554, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 23 & 3 \\ 3 & 4 & 9 & 0 & 36 & 4 \\ 5 & 9 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 23 & 36 & 60 & 0 & 238 & 28 \\ 3 & 4 & 7 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[422] \ 1^{14} 3 \ 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1586, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 21 & 3 \\ 3 & 4 & 9 & 0 & 36 & 4 \\ 5 & 9 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 36 & 60 & 0 & 238 & 28 \\ 3 & 4 & 7 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[423] \ 1^{14} 3 \ 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1686, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 22 & 3 \\ 3 & 4 & 9 & 0 & 36 & 4 \\ 5 & 9 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 36 & 60 & 0 & 238 & 28 \\ 3 & 4 & 7 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[424] \ 1^{14} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1904, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 15 & 3 \\ 3 & 4 & 26 & 0 & 26 & 4 \\ 15 & 26 & 118 & 0 & 120 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 120 & 0 & 118 & 20 \\ 3 & 4 & 20 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [425] \ 1^{11} 31^1 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 1 & 3 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & 3 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

4.862 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 0 & 13 & 7 \\ 5 & 8 & 0 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 10 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 7 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 16 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 0 & 0 & 10 & 6 \\ 5 & 8 & 0 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 18 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 10 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 36 & 5 \\ 11 & 20 & 0 & 0 & 67 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 36 & 67 & 0 & 0 & 217 & 27 \\ 5 & 8 & 0 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 19 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 18 & 5 \\ 11 & 20 & 0 & 0 & 33 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 33 & 0 & 0 & 49 & 13 \\ 5 & 8 & 0 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{111} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -296, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 27 & 5 \\ 11 & 20 & 0 & 0 & 50 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 27 & 50 & 0 & 0 & 118 & 20 \\ 5 & 8 & 0 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{111} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -342, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 27 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 10 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 27 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{111} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -346, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 26 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 26 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -388, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.86 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 18 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 7 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -522, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 28 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 28 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{111} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -550, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.83 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 27 & 5 \\ 11 & 20 & 17 & 0 & 50 & 8 \\ 9 & 17 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 27 & 50 & 42 & 0 & 118 & 20 \\ 5 & 8 & 7 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

4.863 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -62, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 6 & 0 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -62, \quad \det = -32, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 0 & 3 & 0 & 2 \\ 3 & 4 & 0 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 6 & 0 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

4.864 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 22, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 16 & 23 & 8 & 6 \\ 16 & 25 & 36 & 12 & 9 \\ 23 & 36 & 49 & 17 & 13 \\ 8 & 12 & 17 & 4 & 4 \\ 6 & 9 & 13 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 0 & 0 & 0 & 2 \\ 3 & 4 & 0 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 3 & 4 \\ 4 & 8 & 10 & 0 & 4 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 6 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 49, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 8 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 12 & 17 & 13 & 9 & 7 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 54, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.70 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 59, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 1 & 8 & 6 \\ 15 & 20 & 0 & 11 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 11 & 0 & 4 & 4 \\ 6 & 8 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 61, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.62 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 66, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 \\ 4 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 13 & 8 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 13 & 17 & 13 & 9 & 7 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 3 & 4 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 3 & 4 \\ 5 & 9 & 13 & 0 & 5 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 7 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 3 & 4 \\ 5 & 8 & 10 & 0 & 4 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 6 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 4 \\ 8 & 14 & 37 & 0 & 8 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 3 & 4 & 11 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{13} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 3 & 4 \\ 7 & 12 & 25 & 0 & 6 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 9 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{14} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 16 & 0 & 3 & 4 \\ 9 & 16 & 46 & 0 & 8 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 3 & 4 & 12 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 3 & 4 \\ 7 & 13 & 28 & 0 & 7 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 7 & 0 & 0 & 2 \\ 3 & 4 & 10 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{13} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 3 & 4 \\ 9 & 17 & 49 & 0 & 9 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 9 & 0 & 0 & 2 \\ 3 & 4 & 13 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 3 & 4 \\ 6 & 9 & 13 & 0 & 5 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 7 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{14} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 3 & 4 \\ 11 & 21 & 76 & 0 & 11 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 11 & 0 & 0 & 2 \\ 3 & 4 & 16 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[19] \quad 21^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 20 & 27 & 9 & 7 \\ 20 & 25 & 36 & 12 & 9 \\ 27 & 36 & 49 & 17 & 13 \\ 9 & 12 & 17 & 4 & 4 \\ 7 & 9 & 13 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 3 & 4 \\ 10 & 18 & 58 & 0 & 10 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 3 & 4 & 14 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 4 \\ 9 & 14 & 37 & 0 & 8 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 3 & 4 & 11 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 306, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 3 & 4 \\ 8 & 13 & 28 & 0 & 7 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 7 & 0 & 0 & 2 \\ 3 & 4 & 10 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 311, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 19 & 0 & 3 & 4 \\ 12 & 19 & 73 & 0 & 11 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 11 & 0 & 0 & 2 \\ 3 & 4 & 15 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[24] \quad 21^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 20 & 15 & 9 & 7 \\ 20 & 25 & 20 & 12 & 9 \\ 15 & 20 & 13 & 9 & 7 \\ 9 & 12 & 9 & 4 & 4 \\ 7 & 9 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{12} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 23 & 0 & 3 & 4 \\ 13 & 23 & 100 & 0 & 13 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 13 & 0 & 0 & 2 \\ 3 & 4 & 18 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{13} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 3 & 4 \\ 12 & 22 & 85 & 0 & 12 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 12 & 0 & 0 & 2 \\ 3 & 4 & 17 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{12} 3^{11} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 3 & 4 \\ 16 & 28 & 154 & 0 & 16 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 16 & 0 & 0 & 2 \\ 3 & 4 & 22 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 399, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 3 & 4 \\ 10 & 17 & 49 & 0 & 9 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 9 & 0 & 0 & 2 \\ 3 & 4 & 13 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{14} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 26 & 0 & 3 & 4 \\ 14 & 26 & 118 & 0 & 14 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 14 & 0 & 0 & 2 \\ 3 & 4 & 20 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 3 & 4 \\ 11 & 18 & 58 & 0 & 10 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 3 & 4 & 14 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{14} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 474, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 3 & 4 \\ 12 & 21 & 76 & 0 & 11 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 11 & 0 & 0 & 2 \\ 3 & 4 & 16 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{13} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 511, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 27 & 0 & 3 & 4 \\ 15 & 27 & 133 & 0 & 15 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 15 & 0 & 0 & 2 \\ 3 & 4 & 21 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{12} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 570, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 23 & 0 & 3 & 4 \\ 14 & 23 & 100 & 0 & 13 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 13 & 0 & 0 & 2 \\ 3 & 4 & 18 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{13}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 621, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 2 & 3 \\ 3 & 4 & 32 & 0 & 3 & 4 \\ 18 & 32 & 193 & 0 & 18 & 25 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 18 & 0 & 0 & 2 \\ 3 & 4 & 25 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 621, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 3 & 4 \\ 13 & 22 & 85 & 0 & 12 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 12 & 0 & 0 & 2 \\ 3 & 4 & 17 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{12}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 654, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 3 & 4 \\ 17 & 28 & 154 & 0 & 16 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 16 & 0 & 0 & 2 \\ 3 & 4 & 22 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{14}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 670, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 3 \\ 3 & 4 & 31 & 0 & 3 & 4 \\ 17 & 31 & 172 & 0 & 17 & 24 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 17 & 0 & 0 & 2 \\ 3 & 4 & 24 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[38] $1^{14}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 770, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 26 & 0 & 3 & 4 \\ 15 & 26 & 118 & 0 & 14 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 14 & 0 & 0 & 2 \\ 3 & 4 & 20 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{13}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 811, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 3 \\ 3 & 4 & 27 & 0 & 3 & 4 \\ 16 & 27 & 133 & 0 & 15 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 15 & 0 & 0 & 2 \\ 3 & 4 & 21 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{14}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 854, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 2 & 3 \\ 3 & 4 & 36 & 0 & 3 & 4 \\ 20 & 36 & 238 & 0 & 20 & 28 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 20 & 0 & 0 & 2 \\ 3 & 4 & 28 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{14}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 969, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 2 & 3 \\ 3 & 4 & 32 & 0 & 3 & 4 \\ 19 & 32 & 193 & 0 & 18 & 25 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 18 & 0 & 0 & 2 \\ 3 & 4 & 25 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{14}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 1034, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 2 & 3 \\ 3 & 4 & 31 & 0 & 3 & 4 \\ 18 & 31 & 172 & 0 & 17 & 24 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 17 & 0 & 0 & 2 \\ 3 & 4 & 24 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{11}3^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 1266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 2 & 3 \\ 3 & 4 & 36 & 0 & 3 & 4 \\ 21 & 36 & 238 & 0 & 20 & 28 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 20 & 0 & 0 & 2 \\ 3 & 4 & 28 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [44] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 5 & 3 & 5 \\ 3 & 5 & 4 & 3 & 4 \\ 3 & 5 & 3 & 0 & 2 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 5 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[45] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 2 & 3 \\ 2 & 1 & 1 & 4 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 4 & 0 & 10 & 4 & 6 \\ 2 & 2 & 0 & 4 & 0 & 2 \\ 3 & 3 & -1 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[46] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 2 & 0 & 3 \\ 2 & 3 & 1 & 4 & 0 & 3 \\ 2 & 2 & 4 & 2 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 3 & 4 & -2 & 3 \end{pmatrix}_{b_0}$$

[47] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 2 & 3 \\ 2 & 1 & 1 & 7 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 6 & 9 \\ 2 & 2 & 0 & 6 & 0 & 2 \\ 3 & 3 & -1 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[48] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 & 3 & 5 \\ 2 & 5 & 2 & 4 & 2 & 4 \\ 3 & 5 & 4 & 0 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 2 \\ 3 & 5 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[49] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -37, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 2 & 3 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 3 & 3 & -2 & 3 \end{pmatrix}_{b_0}$$

[50] $1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 5 & 0 & 3 \\ 2 & 3 & 1 & 6 & 0 & 3 \\ 4 & 5 & 6 & 9 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 3 & 7 & -2 & 3 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{11}3^{11}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -83, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 3 \\ 3 & 5 & 10 & 5 & 3 & 5 \\ 5 & 10 & 14 & 7 & 5 & 8 \\ 3 & 5 & 7 & 0 & 2 & 2 \\ 2 & 3 & 5 & 2 & 0 & 2 \\ 3 & 5 & 8 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

4.865 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -98, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 6 & 5 \\ 11 & 20 & 0 & 0 & 11 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 0 & 0 & 4 & 4 \\ 5 & 8 & 0 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -163, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 7 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

4.866 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{12}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -8, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 2 & 3 \\ 3 & 4 & 8 & 0 & 3 & 3 & 4 \\ 4 & 8 & 10 & 0 & 4 & 4 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 2 & 0 & 2 \\ 3 & 4 & 6 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{14}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -76, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.89 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 2 & 3 \\ 3 & 4 & 21 & 0 & 3 & 3 & 4 \\ 11 & 21 & 76 & 0 & 11 & 11 & 16 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 11 & 0 & 0 & 2 & 2 \\ 2 & 3 & 11 & 0 & 2 & 0 & 2 \\ 3 & 4 & 16 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{13}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -108, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 3 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 3 & 4 \\ 2 & 4 & 1 & 0 & 2 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{13}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -192, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 2 & 3 \\ 3 & 4 & 17 & 0 & 3 & 3 & 4 \\ 9 & 17 & 49 & 0 & 9 & 9 & 13 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 9 & 0 & 0 & 2 & 2 \\ 2 & 3 & 9 & 0 & 2 & 0 & 2 \\ 3 & 4 & 13 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -208, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 3 & 3 & 4 \\ 5 & 9 & 13 & 0 & 5 & 5 & 7 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 5 & 0 & 0 & 2 & 2 \\ 2 & 3 & 5 & 0 & 2 & 0 & 2 \\ 3 & 4 & 7 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -212, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 3 & 4 \\ 8 & 14 & 37 & 0 & 8 & 8 & 11 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 8 & 0 & 0 & 2 & 2 \\ 2 & 3 & 8 & 0 & 2 & 0 & 2 \\ 3 & 4 & 11 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -236, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 2 & 3 \\ 3 & 4 & 13 & 0 & 3 & 3 & 4 \\ 7 & 13 & 28 & 0 & 7 & 7 & 10 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 7 & 0 & 0 & 2 & 2 \\ 2 & 3 & 7 & 0 & 2 & 0 & 2 \\ 3 & 4 & 10 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{12}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -376, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 2 & 3 \\ 3 & 4 & 18 & 0 & 3 & 3 & 4 \\ 10 & 18 & 58 & 0 & 10 & 10 & 14 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 10 & 0 & 0 & 2 & 2 \\ 2 & 3 & 10 & 0 & 2 & 0 & 2 \\ 3 & 4 & 14 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{13}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -468, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 2 & 3 \\ 3 & 4 & 22 & 0 & 3 & 3 & 4 \\ 12 & 22 & 85 & 0 & 12 & 12 & 17 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 12 & 0 & 0 & 2 & 2 \\ 2 & 3 & 12 & 0 & 2 & 0 & 2 \\ 3 & 4 & 17 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{14}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -488, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 2 & 3 \\ 3 & 4 & 26 & 0 & 3 & 3 & 4 \\ 14 & 26 & 118 & 0 & 14 & 14 & 20 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 14 & 0 & 0 & 2 & 2 \\ 2 & 3 & 14 & 0 & 2 & 0 & 2 \\ 3 & 4 & 20 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

4.867 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -30, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 4 & 3 \\ 3 & 4 & 14 & 0 & 3 & 7 & 4 \\ 8 & 14 & 37 & 0 & 8 & 17 & 11 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 8 & 0 & 0 & 4 & 2 \\ 4 & 7 & 17 & 0 & 4 & 7 & 5 \\ 3 & 4 & 11 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{122}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -84, \ (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 9 & 3 \\ 3 & 4 & 8 & 0 & 3 & 16 & 4 \\ 4 & 8 & 10 & 0 & 4 & 22 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 9 & 16 & 22 & 0 & 9 & 44 & 12 \\ 3 & 4 & 6 & -2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 4 & 3 \\ 3 & 4 & 8 & 0 & 3 & 7 & 4 \\ 4 & 8 & 10 & 0 & 4 & 10 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 4 & 2 \\ 4 & 7 & 10 & 0 & 4 & 7 & 5 \\ 3 & 4 & 6 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 5 & 0 & 3 & 7 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 4 & 2 \\ 4 & 7 & 6 & 0 & 4 & 7 & 5 \\ 3 & 4 & 4 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 4 & 4 & 0 & 3 & 7 & 4 \\ 2 & 4 & 1 & 0 & 2 & 5 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 4 & 2 \\ 4 & 7 & 5 & 0 & 4 & 7 & 5 \\ 3 & 4 & 3 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{112}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 6 & 3 \\ 3 & 4 & 4 & 0 & 3 & 11 & 4 \\ 2 & 4 & 1 & 0 & 2 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 6 & 2 \\ 6 & 11 & 8 & 0 & 6 & 20 & 8 \\ 3 & 4 & 3 & -2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{123}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 11 & 3 \\ 3 & 4 & 8 & 0 & 3 & 20 & 4 \\ 4 & 8 & 10 & 0 & 4 & 28 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 11 & 2 \\ 11 & 20 & 28 & 0 & 11 & 69 & 15 \\ 3 & 4 & 6 & -2 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 4 & 3 \\ 3 & 4 & 9 & 0 & 3 & 7 & 4 \\ 5 & 9 & 13 & 0 & 5 & 11 & 7 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 5 & 0 & 0 & 4 & 2 \\ 4 & 7 & 11 & 0 & 4 & 7 & 5 \\ 3 & 4 & 7 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{124}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 16 & 3 \\ 3 & 4 & 8 & 0 & 3 & 29 & 4 \\ 4 & 8 & 10 & 0 & 4 & 40 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 16 & 2 \\ 16 & 29 & 40 & 0 & 16 & 148 & 22 \\ 3 & 4 & 6 & -2 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -287, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 3 & 12 & 4 \\ 2 & 4 & 1 & 0 & 2 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 7 & 2 \\ 7 & 12 & 8 & 0 & 7 & 25 & 9 \\ 3 & 4 & 3 & -2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{141}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 4 & 3 \\ 3 & 4 & 21 & 0 & 3 & 7 & 4 \\ 11 & 21 & 76 & 0 & 11 & 26 & 16 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 11 & 0 & 0 & 4 & 2 \\ 4 & 7 & 26 & 0 & 4 & 7 & 5 \\ 3 & 4 & 16 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{121}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 4 & 3 \\ 3 & 4 & 18 & 0 & 3 & 7 & 4 \\ 10 & 18 & 58 & 0 & 10 & 22 & 14 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 10 & 0 & 0 & 4 & 2 \\ 4 & 7 & 22 & 0 & 4 & 7 & 5 \\ 3 & 4 & 14 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{122}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 6 & 3 \\ 3 & 4 & 13 & 0 & 3 & 11 & 4 \\ 7 & 13 & 28 & 0 & 7 & 25 & 10 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 7 & 0 & 0 & 6 & 2 \\ 6 & 11 & 25 & 0 & 6 & 20 & 8 \\ 3 & 4 & 10 & -2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{121}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 4 & 3 \\ 3 & 4 & 13 & 0 & 3 & 7 & 4 \\ 7 & 13 & 28 & 0 & 7 & 16 & 10 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 7 & 0 & 0 & 4 & 2 \\ 4 & 7 & 16 & 0 & 4 & 7 & 5 \\ 3 & 4 & 10 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{131}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 4 & 3 \\ 3 & 4 & 17 & 0 & 3 & 7 & 4 \\ 9 & 17 & 49 & 0 & 9 & 21 & 13 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 9 & 0 & 0 & 4 & 2 \\ 4 & 7 & 21 & 0 & 4 & 7 & 5 \\ 3 & 4 & 13 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{124}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 13 & 3 \\ 3 & 4 & 8 & 0 & 3 & 24 & 4 \\ 4 & 8 & 10 & 0 & 4 & 34 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 13 & 2 \\ 13 & 24 & 34 & 0 & 13 & 100 & 18 \\ 3 & 4 & 6 & -2 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{132}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 6 & 3 \\ 3 & 4 & 17 & 0 & 3 & 11 & 4 \\ 9 & 17 & 49 & 0 & 9 & 33 & 13 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 9 & 0 & 0 & 6 & 2 \\ 6 & 11 & 33 & 0 & 6 & 20 & 8 \\ 3 & 4 & 13 & -2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{142}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 6 & 3 \\ 3 & 4 & 21 & 0 & 3 & 11 & 4 \\ 11 & 21 & 76 & 0 & 11 & 41 & 16 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 11 & 0 & 0 & 6 & 2 \\ 6 & 11 & 41 & 0 & 6 & 20 & 8 \\ 3 & 4 & 16 & -2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -477, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 4 & 4 & 0 & 3 & 16 & 4 \\ 2 & 4 & 1 & 0 & 2 & 11 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 9 & 2 \\ 9 & 16 & 11 & 0 & 9 & 44 & 12 \\ 3 & 4 & 3 & -2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11} 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -526, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 4 & 3 \\ 3 & 4 & 22 & 0 & 3 & 7 & 4 \\ 12 & 22 & 85 & 0 & 12 & 27 & 17 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 12 & 0 & 0 & 4 & 2 \\ 4 & 7 & 27 & 0 & 4 & 7 & 5 \\ 3 & 4 & 17 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{14} 2^3 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 6 & 3 \\ 3 & 4 & 26 & 0 & 3 & 11 & 4 \\ 14 & 26 & 118 & 0 & 14 & 50 & 20 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 14 & 0 & 0 & 6 & 2 \\ 6 & 11 & 50 & 0 & 6 & 20 & 8 \\ 3 & 4 & 20 & -2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11} 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 4 & 0 & 3 & 21 & 4 \\ 2 & 4 & 1 & 0 & 2 & 14 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 12 & 2 \\ 12 & 21 & 14 & 0 & 12 & 80 & 16 \\ 3 & 4 & 3 & -2 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -631, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 11 & 3 \\ 3 & 4 & 4 & 0 & 3 & 20 & 4 \\ 2 & 4 & 1 & 0 & 2 & 14 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 11 & 2 \\ 11 & 20 & 14 & 0 & 11 & 69 & 15 \\ 3 & 4 & 3 & -2 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{14} 1^3 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -666, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 4 & 3 \\ 3 & 4 & 26 & 0 & 3 & 7 & 4 \\ 14 & 26 & 118 & 0 & 14 & 32 & 20 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 14 & 0 & 0 & 4 & 2 \\ 4 & 7 & 32 & 0 & 4 & 7 & 5 \\ 3 & 4 & 20 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -749, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 13 & 3 \\ 3 & 4 & 4 & 0 & 3 & 24 & 4 \\ 2 & 4 & 1 & 0 & 2 & 17 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 13 & 2 \\ 13 & 24 & 17 & 0 & 13 & 100 & 18 \\ 3 & 4 & 3 & -2 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{11} 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -886, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 4 & 4 & 0 & 3 & 25 & 4 \\ 2 & 4 & 1 & 0 & 2 & 17 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 14 & 2 \\ 14 & 25 & 17 & 0 & 14 & 111 & 19 \\ 3 & 4 & 3 & -2 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{11} 3^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -1152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 16 & 3 \\ 3 & 4 & 4 & 0 & 3 & 29 & 4 \\ 2 & 4 & 1 & 0 & 2 & 20 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 16 & 2 \\ 16 & 29 & 20 & 0 & 16 & 148 & 22 \\ 3 & 4 & 3 & -2 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

4.868 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_{2n3\text{mix}}$

$$T = 8 \quad [0] \ 1^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 20 & 6 \\ 15 & 20 & 17 & 28 & 8 \\ 12 & 17 & 13 & 24 & 7 \\ 20 & 28 & 24 & 37 & 11 \\ 6 & 8 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 1 & 20 & 6 \\ 15 & 20 & 0 & 28 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 20 & 28 & 0 & 37 & 11 \\ 6 & 8 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 0 & 0 & 12 & 6 \\ 3 & 4 & 0 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{144} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 13 & 3 \\ 3 & 4 & 16 & 0 & 24 & 4 \\ 9 & 16 & 46 & 0 & 68 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 68 & 0 & 100 & 18 \\ 3 & 4 & 12 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{12} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 9 & 3 \\ 3 & 4 & 0 & 0 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 0 & 0 & 44 & 12 \\ 3 & 4 & 0 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 4 & 3 \\ 3 & 4 & 14 & 0 & 7 & 4 \\ 8 & 14 & 37 & 0 & 17 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 17 & 0 & 7 & 5 \\ 3 & 4 & 11 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 4 & 3 \\ 3 & 4 & 0 & 0 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 0 & 0 & 7 & 5 \\ 3 & 4 & 0 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 \\ 3 & 5 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 6 & 0 & 7 & 5 \\ 3 & 4 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 7 & 3 \\ 3 & 4 & 0 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 0 & 0 & 25 & 9 \\ 3 & 4 & 0 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 12 & 4 \\ 4 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 13 & 20 & 6 \\ 15 & 20 & 17 & 28 & 8 \\ 13 & 17 & 13 & 24 & 7 \\ 20 & 28 & 24 & 37 & 11 \\ 6 & 8 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{12} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 14 & 3 \\ 3 & 4 & 8 & 0 & 25 & 4 \\ 4 & 8 & 10 & 0 & 34 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 25 & 34 & 0 & 111 & 19 \\ 3 & 4 & 6 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{14} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 14 & 3 \\ 3 & 4 & 21 & 0 & 24 & 4 \\ 11 & 21 & 76 & 0 & 89 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 24 & 89 & 0 & 100 & 18 \\ 3 & 4 & 16 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 21 & 6 \\ 15 & 20 & 17 & 28 & 8 \\ 12 & 17 & 13 & 24 & 7 \\ 21 & 28 & 24 & 37 & 11 \\ 6 & 8 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{12} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 4 & 3 \\ 3 & 4 & 23 & 0 & 7 & 4 \\ 13 & 23 & 100 & 0 & 28 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 28 & 0 & 7 & 5 \\ 3 & 4 & 18 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 4 & 3 \\ 3 & 4 & 14 & 0 & 7 & 4 \\ 9 & 14 & 37 & 0 & 17 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 17 & 0 & 7 & 5 \\ 3 & 4 & 11 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 21^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 17 & 15 & 24 & 7 \\ 17 & 20 & 17 & 28 & 8 \\ 15 & 17 & 13 & 24 & 7 \\ 24 & 28 & 24 & 37 & 11 \\ 7 & 8 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 3 & 4 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 8 & 0 & 25 & 9 \\ 3 & 4 & 3 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{11} 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 11 & 3 \\ 3 & 4 & 12 & 0 & 20 & 4 \\ 7 & 12 & 25 & 0 & 42 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 42 & 0 & 69 & 15 \\ 3 & 4 & 9 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{12} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 11 & 4 \\ 4 & 8 & 10 & 0 & 16 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 11 & 16 & 0 & 20 & 8 \\ 3 & 4 & 6 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{12} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 7 & 4 \\ 4 & 8 & 10 & 0 & 10 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 10 & 0 & 7 & 5 \\ 3 & 4 & 6 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 16 & 4 \\ 5 & 8 & 10 & 0 & 22 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 22 & 0 & 44 & 12 \\ 3 & 4 & 6 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[22] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 16 & 4 \\ 4 & 8 & 10 & 0 & 22 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 22 & 0 & 44 & 12 \\ 3 & 4 & 6 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 11 & 4 \\ 3 & 4 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 8 & 0 & 20 & 8 \\ 3 & 4 & 3 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[24] $1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 \\ 2 & 4 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 5 & 0 & 7 & 5 \\ 3 & 4 & 3 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 \\ 3 & 4 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 5 & 0 & 7 & 5 \\ 3 & 4 & 3 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 17 & 3 \\ 3 & 4 & 8 & 0 & 29 & 4 \\ 4 & 8 & 10 & 0 & 40 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 29 & 40 & 0 & 148 & 22 \\ 3 & 4 & 6 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 3 \\ 3 & 4 & 9 & 0 & 11 & 4 \\ 5 & 9 & 13 & 0 & 17 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 17 & 0 & 20 & 8 \\ 3 & 4 & 7 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[28] $1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 \\ 2 & 4 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 5 & 0 & 7 & 5 \\ 3 & 4 & 3 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 13 & 0 & 14 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 14 & 0 & 12 & 6 \\ 3 & 4 & 7 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 3 & 4 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 11 & 0 & 44 & 12 \\ 3 & 4 & 3 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 3 \\ 3 & 4 & 8 & 0 & 20 & 4 \\ 4 & 8 & 10 & 0 & 28 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 20 & 28 & 0 & 69 & 15 \\ 3 & 4 & 6 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 7 & 3 \\ 3 & 4 & 17 & 0 & 12 & 4 \\ 9 & 17 & 49 & 0 & 36 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 36 & 0 & 25 & 9 \\ 3 & 4 & 13 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 7 & 4 \\ 5 & 9 & 13 & 0 & 11 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 11 & 0 & 7 & 5 \\ 3 & 4 & 7 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 20 & 4 \\ 3 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 14 & 0 & 69 & 15 \\ 3 & 4 & 3 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 7 & 3 \\ 3 & 4 & 21 & 0 & 12 & 4 \\ 12 & 21 & 76 & 0 & 44 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 44 & 0 & 25 & 9 \\ 3 & 4 & 16 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 20 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 18 & 0 & 69 & 15 \\ 3 & 4 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{13} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 4 & 3 \\ 3 & 4 & 32 & 0 & 7 & 4 \\ 18 & 32 & 193 & 0 & 39 & 25 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 39 & 0 & 7 & 5 \\ 3 & 4 & 25 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[38] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 6 & 9 & 13 & 0 & 14 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 14 & 0 & 12 & 6 \\ 3 & 4 & 7 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 12 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{11}2^31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 157, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 11 & 4 \\ 2 & 4 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 8 & 0 & 20 & 8 \\ 3 & 4 & 3 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 8 & 0 & 25 & 9 \\ 3 & 4 & 3 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 7 & 4 \\ 6 & 9 & 13 & 0 & 11 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 11 & 0 & 7 & 5 \\ 3 & 4 & 7 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 7 & 4 \\ 5 & 8 & 10 & 0 & 10 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 10 & 0 & 7 & 5 \\ 3 & 4 & 6 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[46] $1^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 16 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 15 & 0 & 44 & 12 \\ 3 & 4 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[44] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 8 & 0 & 25 & 9 \\ 3 & 4 & 3 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[45] $1^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 8 & 0 & 20 & 4 \\ 5 & 8 & 10 & 0 & 28 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 28 & 0 & 69 & 15 \\ 3 & 4 & 6 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[47] $1^{14}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 5 & 0 & 24 & 4 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 24 & 21 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[48] $1^{132}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 185$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 6 & 3 \\ 3 & 4 & 22 & 0 & 11 & 4 \\ 12 & 22 & 85 & 0 & 42 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 42 & 0 & 20 & 8 \\ 3 & 4 & 17 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[49] $1^{122}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 186$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 6 & 3 \\ 3 & 4 & 13 & 0 & 11 & 4 \\ 8 & 13 & 28 & 0 & 25 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 25 & 0 & 20 & 8 \\ 3 & 4 & 10 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[50] $1^{113}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 189$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 5 & 0 & 12 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{122}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 190$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 3 \\ 3 & 4 & 8 & 0 & 11 & 4 \\ 5 & 8 & 10 & 0 & 16 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 16 & 0 & 20 & 8 \\ 3 & 4 & 6 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[52] $1^{134}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 192$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 13 & 3 \\ 3 & 4 & 12 & 0 & 24 & 4 \\ 7 & 12 & 25 & 0 & 51 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 51 & 0 & 100 & 18 \\ 3 & 4 & 9 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{123}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 194$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 15 & 4 \\ 4 & 8 & 10 & 0 & 22 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 22 & 0 & 39 & 11 \\ 3 & 4 & 6 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[54] $1^{121}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 207$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 4 & 3 \\ 3 & 4 & 23 & 0 & 7 & 4 \\ 14 & 23 & 100 & 0 & 28 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 28 & 0 & 7 & 5 \\ 3 & 4 & 18 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[55] $1^{142}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 208$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.85 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 9 & 3 \\ 3 & 4 & 21 & 0 & 16 & 4 \\ 11 & 21 & 76 & 0 & 59 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 59 & 0 & 44 & 12 \\ 3 & 4 & 16 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[56] $1^{133}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 209$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 12 & 3 \\ 3 & 4 & 17 & 0 & 20 & 4 \\ 9 & 17 & 49 & 0 & 60 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 20 & 60 & 0 & 69 & 15 \\ 3 & 4 & 13 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[57] $1^{121}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{blue}{su}_2$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 214$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 4 & 3 \\ 3 & 4 & 18 & 0 & 7 & 4 \\ 10 & 18 & 58 & 0 & 22 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 22 & 0 & 7 & 5 \\ 3 & 4 & 14 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[58] $1^{112}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 16 & 4 \\ 6 & 9 & 13 & 0 & 26 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 26 & 0 & 44 & 12 \\ 3 & 4 & 7 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[59] $1^{131}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 12 & 0 & 7 & 4 \\ 7 & 12 & 25 & 0 & 15 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 15 & 0 & 7 & 5 \\ 3 & 4 & 9 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[60] $1^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 219, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 7 & 4 \\ 7 & 13 & 28 & 0 & 16 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 16 & 0 & 7 & 5 \\ 3 & 4 & 10 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[61] $1^{122}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 222, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 6 & 3 \\ 3 & 4 & 13 & 0 & 11 & 4 \\ 7 & 13 & 28 & 0 & 25 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 25 & 0 & 20 & 8 \\ 3 & 4 & 10 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{124}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 13 & 3 \\ 3 & 4 & 8 & 0 & 24 & 4 \\ 5 & 8 & 10 & 0 & 34 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 34 & 0 & 100 & 18 \\ 3 & 4 & 6 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{123}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 11 & 3 \\ 3 & 4 & 8 & 0 & 20 & 4 \\ 4 & 8 & 10 & 0 & 28 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 28 & 0 & 69 & 15 \\ 3 & 4 & 6 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[64] $1^{111}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 3 \\ 3 & 4 & 9 & 0 & 12 & 4 \\ 5 & 9 & 13 & 0 & 20 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 20 & 0 & 25 & 9 \\ 3 & 4 & 7 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 11 & 4 \\ 2 & 4 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 11 & 8 & 0 & 20 & 8 \\ 3 & 4 & 3 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{131}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 233, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 5 & 0 & 20 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 20 & 18 & 0 & 69 & 15 \\ 3 & 4 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{123}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 5 & 0 & 16 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 16 & 15 & 0 & 44 & 12 \\ 3 & 4 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{1\bar{1}} 1^{1\bar{1}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 242, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 4 & 3 \\ 3 & 4 & 16 & 0 & 7 & 4 \\ 9 & 16 & 46 & 0 & 20 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 20 & 0 & 7 & 5 \\ 3 & 4 & 12 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{1\bar{1}} 1^{1\bar{1}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 6 & 3 \\ 3 & 4 & 12 & 0 & 11 & 4 \\ 7 & 12 & 25 & 0 & 24 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 24 & 0 & 20 & 8 \\ 3 & 4 & 9 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[70] $1^{1\bar{1}} 1^{1\bar{1}} 3^1 1^{1\bar{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 258, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 16 & 3 \\ 3 & 4 & 8 & 0 & 29 & 4 \\ 4 & 8 & 10 & 0 & 40 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 29 & 40 & 0 & 148 & 22 \\ 3 & 4 & 6 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[71] $1^{1\bar{1}} 1^{1\bar{1}} 3^1 1^{1\bar{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 12 & 4 \\ 6 & 9 & 13 & 0 & 20 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 20 & 0 & 25 & 9 \\ 3 & 4 & 7 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{1\bar{1}} 1^{1\bar{1}} 3^1 1^{1\bar{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 12 & 4 \\ 7 & 13 & 28 & 0 & 28 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 28 & 0 & 25 & 9 \\ 3 & 4 & 10 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{1\bar{1}} 1^{1\bar{1}} 3^1 1^{1\bar{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 10 & 3 \\ 3 & 4 & 13 & 0 & 16 & 4 \\ 7 & 13 & 28 & 0 & 37 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 16 & 37 & 0 & 44 & 12 \\ 3 & 4 & 10 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[74] $1^{1\bar{1}} 1^{1\bar{1}} 3^1 1^{1\bar{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 273, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 12 & 4 \\ 5 & 9 & 13 & 0 & 20 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 20 & 0 & 25 & 9 \\ 3 & 4 & 7 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[75] $1^{1\bar{1}} 1^{1\bar{1}} 3^1 1^{1\bar{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 278, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 4 & 3 \\ 3 & 4 & 17 & 0 & 7 & 4 \\ 9 & 17 & 49 & 0 & 21 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 21 & 0 & 7 & 5 \\ 3 & 4 & 13 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[76] $1^{1\bar{1}} 1^{1\bar{1}} 3^1 1^{1\bar{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 11 & 0 & 44 & 12 \\ 3 & 4 & 3 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[77] $1^{1\bar{1}} 1^{1\bar{1}} 3^1 1^{1\bar{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 311, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 6 & 3 \\ 3 & 4 & 17 & 0 & 11 & 4 \\ 9 & 17 & 49 & 0 & 33 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 33 & 0 & 20 & 8 \\ 3 & 4 & 13 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[78] $1^{1\bar{1}1} 1^{1\bar{1}1} 1^{1\bar{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 313, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 7 & 3 \\ 3 & 4 & 17 & 0 & 12 & 4 \\ 10 & 17 & 49 & 0 & 36 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 36 & 0 & 25 & 9 \\ 3 & 4 & 13 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[79] $1^{1\bar{1}1} 1^{1\bar{1}1} 1^{1\bar{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 4 & 3 \\ 3 & 4 & 21 & 0 & 7 & 4 \\ 11 & 21 & 76 & 0 & 26 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 26 & 0 & 7 & 5 \\ 3 & 4 & 16 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{1\bar{1}1} 1^{1\bar{1}1} 1^{1\bar{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 14 & 3 \\ 3 & 4 & 8 & 0 & 24 & 4 \\ 4 & 8 & 10 & 0 & 34 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 24 & 34 & 0 & 100 & 18 \\ 3 & 4 & 6 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[81] $1^{1\bar{1}1} 1^{1\bar{1}1} 1^{1\bar{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 327, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 7 & 4 \\ 8 & 13 & 28 & 0 & 16 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 16 & 0 & 7 & 5 \\ 3 & 4 & 10 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[82] $1^{1\bar{1}1} 1^{1\bar{1}1} 1^{1\bar{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 4 & 3 \\ 3 & 4 & 27 & 0 & 7 & 4 \\ 15 & 27 & 133 & 0 & 33 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 33 & 0 & 7 & 5 \\ 3 & 4 & 21 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{1\bar{1}1} 1^{1\bar{1}1} 1^{1\bar{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 338, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 4 & 3 \\ 3 & 4 & 18 & 0 & 7 & 4 \\ 11 & 18 & 58 & 0 & 22 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 22 & 0 & 7 & 5 \\ 3 & 4 & 14 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[84] $1^{1\bar{1}1} 1^{1\bar{1}1} 1^{1\bar{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 341, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 15 & 4 \\ 2 & 4 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 11 & 0 & 39 & 11 \\ 3 & 4 & 3 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[85] $1^{1\bar{1}1} 1^{1\bar{1}1} 1^{1\bar{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 349, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 14 & 0 & 80 & 16 \\ 3 & 4 & 3 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[86] $1^{1\bar{1}1} 1^{1\bar{1}1} 1^{1\bar{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 13 & 3 \\ 3 & 4 & 8 & 0 & 24 & 4 \\ 4 & 8 & 10 & 0 & 34 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 34 & 0 & 100 & 18 \\ 3 & 4 & 6 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[87] $1^{1\bar{1}1} 1^{1\bar{1}1} 1^{1\bar{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 353, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 12 & 4 \\ 8 & 13 & 28 & 0 & 28 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 28 & 0 & 25 & 9 \\ 3 & 4 & 10 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[88] $1^{11} 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 16 & 11 & 0 & 44 & 12 \\ 3 & 4 & 3 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[89] $1^{11} 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 16 & 4 \\ 5 & 9 & 13 & 0 & 26 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 26 & 0 & 44 & 12 \\ 3 & 4 & 7 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[90] $1^{11} 1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 377, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 4 & 3 \\ 3 & 4 & 22 & 0 & 7 & 4 \\ 12 & 22 & 85 & 0 & 27 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 27 & 0 & 7 & 5 \\ 3 & 4 & 17 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[91] $1^{11} 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 11 & 3 \\ 3 & 4 & 13 & 0 & 20 & 4 \\ 8 & 13 & 28 & 0 & 46 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 46 & 0 & 69 & 15 \\ 3 & 4 & 10 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[92] $1^{11} 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 6 & 3 \\ 3 & 4 & 21 & 0 & 11 & 4 \\ 11 & 21 & 76 & 0 & 41 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 41 & 0 & 20 & 8 \\ 3 & 4 & 16 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[93] $1^{11} 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 387, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 6 & 3 \\ 3 & 4 & 17 & 0 & 11 & 4 \\ 10 & 17 & 49 & 0 & 33 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 33 & 0 & 20 & 8 \\ 3 & 4 & 13 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[94] $1^{11} 1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 6 & 3 \\ 3 & 4 & 26 & 0 & 11 & 4 \\ 14 & 26 & 118 & 0 & 50 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 50 & 0 & 20 & 8 \\ 3 & 4 & 20 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[95] $1^{11} 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 9 & 3 \\ 3 & 4 & 17 & 0 & 16 & 4 \\ 9 & 17 & 49 & 0 & 48 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 48 & 0 & 44 & 12 \\ 3 & 4 & 13 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{11} 1^{13} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 393, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 4 & 3 \\ 3 & 4 & 32 & 0 & 7 & 4 \\ 19 & 32 & 193 & 0 & 39 & 25 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 39 & 0 & 7 & 5 \\ 3 & 4 & 25 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[97] $1^{11} 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 396, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 6 & 3 \\ 3 & 4 & 26 & 0 & 11 & 4 \\ 15 & 26 & 118 & 0 & 50 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 50 & 0 & 20 & 8 \\ 3 & 4 & 20 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[108] $1^{143}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 501, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 11 & 3 \\ 3 & 4 & 21 & 0 & 20 & 4 \\ 11 & 21 & 76 & 0 & 74 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 74 & 0 & 69 & 15 \\ 3 & 4 & 16 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[109] $1^{143}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 502, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 4 & 3 \\ 3 & 4 & 36 & 0 & 7 & 4 \\ 20 & 36 & 238 & 0 & 44 & 28 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 44 & 0 & 7 & 5 \\ 3 & 4 & 28 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[110] $1^{123}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 505, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 12 & 3 \\ 3 & 4 & 13 & 0 & 20 & 4 \\ 7 & 13 & 28 & 0 & 46 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 20 & 46 & 0 & 69 & 15 \\ 3 & 4 & 10 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[111] $1^{113}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 513, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 12 & 3 \\ 3 & 4 & 9 & 0 & 20 & 4 \\ 5 & 9 & 13 & 0 & 32 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 20 & 32 & 0 & 69 & 15 \\ 3 & 4 & 7 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[112] $1^{134}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 516, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 14 & 3 \\ 3 & 4 & 17 & 0 & 24 & 4 \\ 9 & 17 & 49 & 0 & 72 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 24 & 72 & 0 & 100 & 18 \\ 3 & 4 & 13 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{141}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 522, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 4 & 3 \\ 3 & 4 & 26 & 0 & 7 & 4 \\ 14 & 26 & 118 & 0 & 32 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 32 & 0 & 7 & 5 \\ 3 & 4 & 20 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[114] $1^{142}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 9 & 3 \\ 3 & 4 & 21 & 0 & 16 & 4 \\ 12 & 21 & 76 & 0 & 59 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 59 & 0 & 44 & 12 \\ 3 & 4 & 16 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[115] $1^{114}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 17 & 0 & 100 & 18 \\ 3 & 4 & 3 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[116] $1^{134}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 13 & 3 \\ 3 & 4 & 17 & 0 & 24 & 4 \\ 10 & 17 & 49 & 0 & 72 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 72 & 0 & 100 & 18 \\ 3 & 4 & 13 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[117] $1^{132}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 552, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 9 & 3 \\ 3 & 4 & 17 & 0 & 16 & 4 \\ 10 & 17 & 49 & 0 & 48 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 48 & 0 & 44 & 12 \\ 3 & 4 & 13 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[118] $1^{113}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 557, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 25 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 25 & 17 & 0 & 111 & 19 \\ 3 & 4 & 3 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

[119] $1^{113}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 560, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 20 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 20 & 14 & 0 & 69 & 15 \\ 3 & 4 & 3 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[120] $1^{131}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 562, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 4 & 3 \\ 3 & 4 & 27 & 0 & 7 & 4 \\ 16 & 27 & 133 & 0 & 33 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 33 & 0 & 7 & 5 \\ 3 & 4 & 21 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[121] $1^{142}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 570, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 6 & 3 \\ 3 & 4 & 21 & 0 & 11 & 4 \\ 12 & 21 & 76 & 0 & 41 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 41 & 0 & 20 & 8 \\ 3 & 4 & 16 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[122] $1^{114}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 14 & 3 \\ 3 & 4 & 9 & 0 & 24 & 4 \\ 5 & 9 & 13 & 0 & 38 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 24 & 38 & 0 & 100 & 18 \\ 3 & 4 & 7 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[123] $1^{123}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 581, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 11 & 3 \\ 3 & 4 & 13 & 0 & 20 & 4 \\ 7 & 13 & 28 & 0 & 46 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 46 & 0 & 69 & 15 \\ 3 & 4 & 10 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[124] $1^{141}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 583, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 4 & 3 \\ 3 & 4 & 31 & 0 & 7 & 4 \\ 17 & 31 & 172 & 0 & 38 & 24 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 38 & 0 & 7 & 5 \\ 3 & 4 & 24 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[125] $1^{131}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 589, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 4 & 3 \\ 3 & 4 & 22 & 0 & 7 & 4 \\ 13 & 22 & 85 & 0 & 27 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 27 & 0 & 7 & 5 \\ 3 & 4 & 17 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[126] $1^{141}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 4 & 3 \\ 3 & 4 & 21 & 0 & 7 & 4 \\ 12 & 21 & 76 & 0 & 26 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 26 & 0 & 7 & 5 \\ 3 & 4 & 16 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[127] $1^{133}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 605, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 11 & 3 \\ 3 & 4 & 17 & 0 & 20 & 4 \\ 9 & 17 & 49 & 0 & 60 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 60 & 0 & 69 & 15 \\ 3 & 4 & 13 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[128] $1^{\overline{113}3^{\overline{113}}1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2}$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 628, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 30 & 4 \\ 2 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 30 & 20 & 0 & 165 & 23 \\ 3 & 4 & 3 & -2 & 23 & 2 \end{pmatrix}_{b_0}$$

[129] $1^{\overline{113}3^{\overline{113}}1^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2}$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 35 & 4 \\ 2 & 4 & 1 & 0 & 23 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 35 & 23 & 0 & 231 & 27 \\ 3 & 4 & 3 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

[130] $1^{\overline{113}3^{\overline{113}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2}$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 629, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 11 & 3 \\ 3 & 4 & 17 & 0 & 20 & 4 \\ 10 & 17 & 49 & 0 & 60 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 60 & 0 & 69 & 15 \\ 3 & 4 & 13 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[131] $1^{\overline{124}3^{\overline{113}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2}$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 664, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 13 & 3 \\ 3 & 4 & 13 & 0 & 24 & 4 \\ 7 & 13 & 28 & 0 & 55 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 55 & 0 & 100 & 18 \\ 3 & 4 & 10 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[132] $1^{\overline{124}3^{\overline{113}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2}$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 692, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 14 & 3 \\ 3 & 4 & 13 & 0 & 24 & 4 \\ 7 & 13 & 28 & 0 & 55 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 24 & 55 & 0 & 100 & 18 \\ 3 & 4 & 10 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[133] $1^{\overline{113}3^{\overline{113}}1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2}$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 709, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 25 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 25 & 17 & 0 & 111 & 19 \\ 3 & 4 & 3 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

[134] $1^{\overline{144}3^{\overline{113}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2}$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 744, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 13 & 3 \\ 3 & 4 & 21 & 0 & 24 & 4 \\ 11 & 21 & 76 & 0 & 89 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 89 & 0 & 100 & 18 \\ 3 & 4 & 16 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[135] $1^{\overline{143}3^{\overline{113}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2}$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 749, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 11 & 3 \\ 3 & 4 & 21 & 0 & 20 & 4 \\ 12 & 21 & 76 & 0 & 74 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 74 & 0 & 69 & 15 \\ 3 & 4 & 16 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[136] $1^{\overline{114}3^{\overline{113}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2}$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 752, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 24 & 17 & 0 & 100 & 18 \\ 3 & 4 & 3 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[137] $1^{\overline{114}3^{\overline{113}}1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2}$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 765, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 29 & 4 \\ 2 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 29 & 20 & 0 & 148 & 22 \\ 3 & 4 & 3 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[138] $1^{134}3^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 13 & 3 \\ 3 & 4 & 17 & 0 & 24 & 4 \\ 9 & 17 & 49 & 0 & 72 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 72 & 0 & 100 & 18 \\ 3 & 4 & 13 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[139] $1^{113}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 788, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 30 & 4 \\ 2 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 30 & 20 & 0 & 165 & 23 \\ 3 & 4 & 3 & -2 & 23 & 2 \end{pmatrix}_{b_0}$$

[140] $1^{113}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 797, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 21 & 3 \\ 3 & 4 & 4 & 0 & 35 & 4 \\ 2 & 4 & 1 & 0 & 23 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 35 & 23 & 0 & 231 & 27 \\ 3 & 4 & 3 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

[141] $1^{143}4^11^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 808, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 13 & 3 \\ 3 & 4 & 21 & 0 & 24 & 4 \\ 12 & 21 & 76 & 0 & 89 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 89 & 0 & 100 & 18 \\ 3 & 4 & 16 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[142] $1^{141}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 822, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 4 & 3 \\ 3 & 4 & 26 & 0 & 7 & 4 \\ 15 & 26 & 118 & 0 & 32 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 32 & 0 & 7 & 5 \\ 3 & 4 & 20 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[143] $1^{141}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 834, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 4 & 3 \\ 3 & 4 & 36 & 0 & 7 & 4 \\ 21 & 36 & 238 & 0 & 44 & 28 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 44 & 0 & 7 & 5 \\ 3 & 4 & 28 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[144] $1^{141}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 899, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 4 & 3 \\ 3 & 4 & 31 & 0 & 7 & 4 \\ 18 & 31 & 172 & 0 & 38 & 24 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 38 & 0 & 7 & 5 \\ 3 & 4 & 24 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[145] $1^{114}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 916, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 19 & 3 \\ 3 & 4 & 4 & 0 & 34 & 4 \\ 2 & 4 & 1 & 0 & 23 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 34 & 23 & 0 & 208 & 26 \\ 3 & 4 & 3 & -2 & 26 & 2 \end{pmatrix}_{b_0}$$

[146] $1^{114}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 981, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 29 & 4 \\ 2 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 29 & 20 & 0 & 148 & 22 \\ 3 & 4 & 3 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[147] $1^{114}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 997, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 22 & 3 \\ 3 & 4 & 4 & 0 & 39 & 4 \\ 2 & 4 & 1 & 0 & 26 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 39 & 26 & 0 & 280 & 30 \\ 3 & 4 & 3 & -2 & 30 & 2 \end{pmatrix}_{b_0}$$

[148] $1^{11} 1^{14} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 1140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 34 & 4 \\ 2 & 4 & 1 & 0 & 23 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 34 & 23 & 0 & 208 & 26 \\ 3 & 4 & 3 & -2 & 26 & 2 \end{pmatrix}_{b_0}$$

[149] $1^{11} 1^{14} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 1229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 23 & 3 \\ 3 & 4 & 4 & 0 & 39 & 4 \\ 2 & 4 & 1 & 0 & 26 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 23 & 39 & 26 & 0 & 280 & 30 \\ 3 & 4 & 3 & -2 & 30 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [150] $1^{11} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 8 & 3 \\ 2 & 1 & 1 & 4 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 4 & 0 & 10 & 20 & 6 \\ 8 & 9 & 0 & 20 & 35 & 11 \\ 3 & 3 & -1 & 6 & 11 & 3 \end{pmatrix}_{b_0}$$

[151] $1^{11} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 7 & 3 \\ 3 & 5 & 5 & 5 & 13 & 5 \\ 2 & 5 & 2 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 2 \\ 7 & 13 & 9 & 9 & 26 & 10 \\ 3 & 5 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[152] $1^{11} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 3 \\ 3 & 5 & 10 & 0 & 13 & 5 \\ 5 & 10 & 14 & 0 & 19 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 13 & 19 & 0 & 24 & 10 \\ 3 & 5 & 8 & -2 & 10 & 3 \end{pmatrix}_{b_0}$$

[153] $1^{11} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 1 & 2 & 0 & 3 & 0 & 5 & 2 \\ 1 & 1 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 5 & 5 & 0 & 10 & 6 \\ 2 & 2 & 2 & 3 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[154] $1^1 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 5 & 5 & 0 & 8 & 5 \\ 2 & 5 & 2 & 0 & 6 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 6 & 0 & 8 & 6 \\ 3 & 5 & 4 & -2 & 6 & 3 \end{pmatrix}_{b_0}$$

[155] $1^1 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 5 & 5 & 0 & 13 & 5 \\ 2 & 5 & 2 & 0 & 10 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 13 & 10 & 0 & 24 & 10 \\ 3 & 5 & 4 & -2 & 10 & 3 \end{pmatrix}_{b_0}$$

[156] $1^{11} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 8 & 3 \\ 2 & 1 & 1 & 7 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 29 & 9 \\ 8 & 9 & 0 & 29 & 35 & 11 \\ 3 & 3 & -1 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[157] \ 1^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 9 & 3 \\ 2 & 1 & 1 & 7 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 29 & 9 \\ 9 & 9 & 0 & 29 & 35 & 11 \\ 3 & 3 & -1 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[159] \ 1^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 & 2 \\ 1 & 2 & 0 & 4 & 0 & 4 & 2 \\ 2 & 3 & 4 & 5 & 0 & 6 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 2 & 4 & 6 & 0 & 4 & 4 \\ 2 & 2 & 2 & 5 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[158] \ 1^{\overline{11}} 1^{\overline{12}} 3^{\overline{11}} 21^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 7 & 2 \\ 1 & 2 & 0 & 3 & 0 & 9 & 2 \\ 1 & 1 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 9 & 9 & 0 & 35 & 11 \\ 2 & 2 & 2 & 3 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[161] \ 1^{\overline{11}} 1^{\overline{12}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 10 & 3 \\ 2 & 1 & 1 & 7 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 36 & 9 \\ 10 & 12 & 0 & 36 & 56 & 14 \\ 3 & 3 & -1 & 9 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[160] \ 1^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 1 & 2 & 1 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 5 & 4 & 0 & 10 & 6 \\ 2 & 2 & 2 & 2 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[162] \ 1^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 5 & 10 & 0 & 8 & 5 \\ 5 & 10 & 14 & 0 & 12 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 12 & 0 & 8 & 6 \\ 3 & 5 & 8 & -2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[163] \ 1^{\overline{11}} 1^{\overline{12}} 3^{\overline{11}} 21^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 3 & 0 & 12 & 2 \\ 1 & 1 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 12 & 12 & 0 & 56 & 14 \\ 2 & 2 & 2 & 3 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[164] \ 1^{\overline{12}} 1^{\overline{12}} 3^{\overline{11}} 21^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 16 & 3 \\ 2 & 1 & 1 & 7 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 52 & 9 \\ 16 & 16 & 0 & 52 & 116 & 20 \\ 3 & 3 & -1 & 9 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[165] \ 1^{\overline{11}} 1^{\overline{13}} 3^{\overline{11}} 21^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 2 \\ 1 & 2 & 0 & 3 & 0 & 13 & 2 \\ 1 & 1 & 3 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 7 & 10 & 13 & 13 & 0 & 74 & 16 \\ 2 & 2 & 2 & 3 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[166] \ 1^{\overline{113}} 3^{\overline{121}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -167, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 5 & 5 & 0 & 23 & 5 \\ 2 & 5 & 2 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 23 & 17 & 0 & 78 & 18 \\ 3 & 5 & 4 & -2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[167] \ 1^{\overline{113}} 3^{\overline{121}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -170, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.99 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 16 & 3 \\ 2 & 1 & 1 & 7 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 59 & 9 \\ 16 & 19 & 0 & 59 & 151 & 23 \\ 3 & 3 & -1 & 9 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[168] \ 1^{\overline{112}} 3^{\overline{121}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -173, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 1 & 2 & 1 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 9 & 7 & 0 & 35 & 11 \\ 2 & 2 & 2 & 2 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[169] \ 1^{\overline{112}} 3^{\overline{121}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -181, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 15 & 3 \\ 2 & 1 & 1 & 7 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 52 & 9 \\ 15 & 16 & 0 & 52 & 116 & 20 \\ 3 & 3 & -1 & 9 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[170] \ 1^{\overline{111}} 3^{\overline{121}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -189, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 0 & 4 & 2 \\ 1 & 2 & 0 & 4 & 0 & 5 & 2 \\ 2 & 3 & 4 & 5 & 0 & 9 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 5 & 9 & 0 & 10 & 6 \\ 2 & 2 & 2 & 5 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[171] \ 1^{\overline{113}} 3^{\overline{121}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -189, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 5 & 5 & 0 & 28 & 5 \\ 2 & 5 & 2 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 28 & 21 & 0 & 116 & 22 \\ 3 & 5 & 4 & -2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{\overline{112}} 3^{\overline{121}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -190, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.99 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 18 & 3 \\ 2 & 1 & 1 & 7 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 59 & 9 \\ 18 & 19 & 0 & 59 & 151 & 23 \\ 3 & 3 & -1 & 9 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[173] \ 1^{\overline{114}} 3^{\overline{121}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -216, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 18 & 3 \\ 2 & 1 & 1 & 7 & 22 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 66 & 9 \\ 18 & 22 & 0 & 66 & 192 & 26 \\ 3 & 3 & -1 & 9 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[174] \ 1^{\overline{112}} 3^{\overline{121}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -216, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 20 & 3 \\ 2 & 1 & 1 & 7 & 20 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 68 & 9 \\ 20 & 20 & 0 & 68 & 200 & 26 \\ 3 & 3 & -1 & 9 & 26 & 3 \end{pmatrix}_{b_0}$$

[175] $1^{11}3^{11}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 0 & 5 & 2 \\ 1 & 2 & 0 & 4 & 0 & 8 & 2 \\ 2 & 3 & 4 & 5 & 0 & 13 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 8 & 13 & 0 & 23 & 9 \\ 2 & 2 & 2 & 5 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{11}3^{11}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 17 & 3 \\ 2 & 1 & 1 & 7 & 19 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 59 & 9 \\ 17 & 19 & 0 & 59 & 151 & 23 \\ 3 & 3 & -1 & 9 & 23 & 3 \end{pmatrix}_{b_0}$$

[177] $1^{11}3^{11}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 11 & 2 \\ 1 & 2 & 0 & 3 & 0 & 16 & 2 \\ 1 & 1 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 8 & 11 & 16 & 16 & 0 & 103 & 19 \\ 2 & 2 & 2 & 3 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

[178] $1^{14}3^{11}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 13 & 2 \\ 1 & 2 & 0 & 3 & 0 & 17 & 2 \\ 1 & 1 & 3 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 9 & 13 & 17 & 17 & 0 & 127 & 21 \\ 2 & 2 & 2 & 3 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[179] $1^{11}3^{11}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -261, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 19 & 3 \\ 2 & 1 & 1 & 7 & 22 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 66 & 9 \\ 19 & 22 & 0 & 66 & 192 & 26 \\ 3 & 3 & -1 & 9 & 26 & 3 \end{pmatrix}_{b_0}$$

[180] $1^{11}3^{11}21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -279, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 21 & 3 \\ 2 & 1 & 1 & 7 & 23 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 75 & 9 \\ 21 & 23 & 0 & 75 & 243 & 29 \\ 3 & 3 & -1 & 9 & 29 & 3 \end{pmatrix}_{b_0}$$

[181] $1^{11}3^{11}21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 0 & 13 & 2 \\ 1 & 2 & 1 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 7 & 10 & 13 & 10 & 0 & 74 & 16 \\ 2 & 2 & 2 & 2 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[182] $1^{12}3^{11}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -310, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 5 & 5 & 0 & 33 & 5 \\ 2 & 5 & 2 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 33 & 24 & 0 & 164 & 26 \\ 3 & 5 & 4 & -2 & 26 & 3 \end{pmatrix}_{b_0}$$

[183] $1^{113}3^{121}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 2 & 0 & 9 \\ 1 & 2 & 0 & 1 & 0 & 10 \\ 1 & 2 & 1 & 0 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & 0 \\ 6 & 9 & 10 & 9 & 0 & 51 \\ 2 & 2 & 2 & 2 & -2 & 13 \end{pmatrix}_{b_0}$$

[184] $1^{112}3^{121}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 26 & 3 \\ 2 & 1 & 1 & 7 & 27 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 91 & 9 \\ 26 & 27 & 0 & 91 & 359 & 35 \\ 3 & 3 & -1 & 9 & 35 & 3 \end{pmatrix}_{b_0}$$

[185] $1^{112}3^{121}21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 22 & 3 \\ 2 & 1 & 1 & 7 & 23 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 75 & 9 \\ 22 & 23 & 0 & 75 & 243 & 29 \\ 3 & 3 & -1 & 9 & 29 & 3 \end{pmatrix}_{b_0}$$

[186] $1^{111}3^{121}21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 23 & 3 \\ 2 & 1 & 1 & 7 & 26 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 82 & 9 \\ 23 & 26 & 0 & 82 & 292 & 32 \\ 3 & 3 & -1 & 9 & 32 & 3 \end{pmatrix}_{b_0}$$

[187] $1^{112}3^{121}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -405, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 & 0 & 7 & 2 \\ 1 & 2 & 0 & 4 & 0 & 9 & 2 \\ 2 & 3 & 4 & 5 & 0 & 16 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 9 & 16 & 0 & 35 & 11 \\ 2 & 2 & 2 & 5 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

[188] $1^{114}3^{121}21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -413, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 2 & 0 & 13 & 2 \\ 1 & 2 & 0 & 1 & 0 & 17 & 2 \\ 1 & 2 & 1 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 9 & 13 & 17 & 13 & 0 & 127 & 21 \\ 2 & 2 & 2 & 2 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[189] $1^{112}3^{121}21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -448, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 24 & 3 \\ 2 & 1 & 1 & 7 & 26 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 82 & 9 \\ 24 & 26 & 0 & 82 & 292 & 32 \\ 3 & 3 & -1 & 9 & 32 & 3 \end{pmatrix}_{b_0}$$

[190] $1^{112}3^{121}21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 0 & 2 & 3 & 0 & 8 & 2 \\ 1 & 2 & 0 & 4 & 0 & 12 & 2 \\ 2 & 3 & 4 & 5 & 0 & 20 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 12 & 20 & 0 & 56 & 14 \\ 2 & 2 & 2 & 5 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[191] $1^{114}3^1 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 0 & 2 & 3 & 0 & 9 & 2 \\ 1 & 2 & 0 & 4 & 0 & 10 & 2 \\ 2 & 3 & 4 & 5 & 0 & 19 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 10 & 19 & 0 & 51 & 13 \\ 2 & 2 & 2 & 5 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[192] $1^{114}3^1 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 0 & 14 & 2 \\ 1 & 2 & 1 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 14 & 12 & 0 & 96 & 18 \\ 2 & 2 & 2 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[193] $1^{112}3^1 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -532, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 28 & 3 \\ 2 & 1 & 1 & 7 & 30 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 21 & 98 & 9 \\ 28 & 30 & 0 & 98 & 416 & 38 \\ 3 & 3 & -1 & 9 & 38 & 3 \end{pmatrix}_{b_0}$$

[194] $1^{113}3^1 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -717, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 7 & 2 \\ 1 & 0 & 2 & 3 & 0 & 10 & 2 \\ 1 & 2 & 0 & 4 & 0 & 13 & 2 \\ 2 & 3 & 4 & 5 & 0 & 23 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 7 & 10 & 13 & 23 & 0 & 74 & 16 \\ 2 & 2 & 2 & 5 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[195] $1^{113}3^1 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -800, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 8 & 2 \\ 1 & 0 & 2 & 3 & 0 & 11 & 2 \\ 1 & 2 & 0 & 4 & 0 & 16 & 2 \\ 2 & 3 & 4 & 5 & 0 & 27 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 8 & 11 & 16 & 27 & 0 & 103 & 19 \\ 2 & 2 & 2 & 5 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

[196] $1^{114}3^1 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -884, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 8 & 2 \\ 1 & 0 & 2 & 3 & 0 & 12 & 2 \\ 1 & 2 & 0 & 4 & 0 & 14 & 2 \\ 2 & 3 & 4 & 5 & 0 & 26 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 14 & 26 & 0 & 96 & 18 \\ 2 & 2 & 2 & 5 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[197] $1^{114}3^1 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -1125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 9 & 2 \\ 1 & 0 & 2 & 3 & 0 & 13 & 2 \\ 1 & 2 & 0 & 4 & 0 & 17 & 2 \\ 2 & 3 & 4 & 5 & 0 & 30 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 9 & 13 & 17 & 30 & 0 & 127 & 21 \\ 2 & 2 & 2 & 5 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

4.869 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_{2n3\text{mix}} + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 15 & 5 \\ 11 & 20 & 0 & 0 & 28 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 28 & 0 & 0 & 37 & 11 \\ 5 & 8 & 0 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

4.870 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_{2n3\text{mix}} + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 4 & 2 & 3 \\ 3 & 4 & 14 & 0 & 7 & 3 & 4 \\ 8 & 14 & 37 & 0 & 17 & 8 & 11 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 17 & 0 & 7 & 4 & 5 \\ 2 & 3 & 8 & 0 & 4 & 0 & 2 \\ 3 & 4 & 11 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{122}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 2 & 3 \\ 3 & 4 & 8 & 0 & 16 & 3 & 4 \\ 4 & 8 & 10 & 0 & 22 & 4 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 22 & 0 & 44 & 9 & 12 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 6 & -2 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{121}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 2 & 3 \\ 3 & 4 & 8 & 0 & 7 & 3 & 4 \\ 4 & 8 & 10 & 0 & 10 & 4 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 10 & 0 & 7 & 4 & 5 \\ 2 & 3 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & 6 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 5 & 0 & 7 & 3 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 6 & 0 & 7 & 4 & 5 \\ 2 & 3 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 0 & 7 & 3 & 4 \\ 2 & 4 & 1 & 0 & 5 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 5 & 0 & 7 & 4 & 5 \\ 2 & 3 & 2 & 0 & 4 & 0 & 2 \\ 3 & 4 & 3 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 2 & 3 \\ 3 & 4 & 4 & 0 & 11 & 3 & 4 \\ 2 & 4 & 1 & 0 & 8 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 8 & 0 & 20 & 6 & 8 \\ 2 & 3 & 2 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & -2 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{123}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -238, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 11 & 2 & 3 \\ 3 & 4 & 8 & 0 & 20 & 3 & 4 \\ 4 & 8 & 10 & 0 & 28 & 4 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 28 & 0 & 69 & 11 & 15 \\ 2 & 3 & 4 & 0 & 11 & 0 & 2 \\ 3 & 4 & 6 & -2 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 2 & 3 \\ 3 & 4 & 9 & 0 & 7 & 3 & 4 \\ 5 & 9 & 13 & 0 & 11 & 5 & 7 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 11 & 0 & 7 & 4 & 5 \\ 2 & 3 & 5 & 0 & 4 & 0 & 2 \\ 3 & 4 & 7 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{124}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 16 & 2 & 3 \\ 3 & 4 & 8 & 0 & 29 & 3 & 4 \\ 4 & 8 & 10 & 0 & 40 & 4 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 16 & 29 & 40 & 0 & 148 & 16 & 22 \\ 2 & 3 & 4 & 0 & 16 & 0 & 2 \\ 3 & 4 & 6 & -2 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 4 & 0 & 12 & 3 & 4 \\ 2 & 4 & 1 & 0 & 8 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 8 & 0 & 25 & 7 & 9 \\ 2 & 3 & 2 & 0 & 7 & 0 & 2 \\ 3 & 4 & 3 & -2 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{141}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 4 & 2 & 3 \\ 3 & 4 & 21 & 0 & 7 & 3 & 4 \\ 11 & 21 & 76 & 0 & 26 & 11 & 16 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 26 & 0 & 7 & 4 & 5 \\ 2 & 3 & 11 & 0 & 4 & 0 & 2 \\ 3 & 4 & 16 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{121}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 4 & 2 & 3 \\ 3 & 4 & 18 & 0 & 7 & 3 & 4 \\ 10 & 18 & 58 & 0 & 22 & 10 & 14 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 22 & 0 & 7 & 4 & 5 \\ 2 & 3 & 10 & 0 & 4 & 0 & 2 \\ 3 & 4 & 14 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{122}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 6 & 2 & 3 \\ 3 & 4 & 13 & 0 & 11 & 3 & 4 \\ 7 & 13 & 28 & 0 & 25 & 7 & 10 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 25 & 0 & 20 & 6 & 8 \\ 2 & 3 & 7 & 0 & 6 & 0 & 2 \\ 3 & 4 & 10 & -2 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 2 & 3 \\ 3 & 4 & 13 & 0 & 7 & 3 & 4 \\ 7 & 13 & 28 & 0 & 16 & 7 & 10 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 16 & 0 & 7 & 4 & 5 \\ 2 & 3 & 7 & 0 & 4 & 0 & 2 \\ 3 & 4 & 10 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{131}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 4 & 2 & 3 \\ 3 & 4 & 17 & 0 & 7 & 3 & 4 \\ 9 & 17 & 49 & 0 & 21 & 9 & 13 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 21 & 0 & 7 & 4 & 5 \\ 2 & 3 & 9 & 0 & 4 & 0 & 2 \\ 3 & 4 & 13 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{124}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 13 & 2 & 3 \\ 3 & 4 & 8 & 0 & 24 & 3 & 4 \\ 4 & 8 & 10 & 0 & 34 & 4 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 34 & 0 & 100 & 13 & 18 \\ 2 & 3 & 4 & 0 & 13 & 0 & 2 \\ 3 & 4 & 6 & -2 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{132}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 6 & 2 & 3 \\ 3 & 4 & 17 & 0 & 11 & 3 & 4 \\ 9 & 17 & 49 & 0 & 33 & 9 & 13 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 33 & 0 & 20 & 6 & 8 \\ 2 & 3 & 9 & 0 & 6 & 0 & 2 \\ 3 & 4 & 13 & -2 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{142}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 6 & 2 & 3 \\ 3 & 4 & 21 & 0 & 11 & 3 & 4 \\ 11 & 21 & 76 & 0 & 41 & 11 & 16 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 41 & 0 & 20 & 6 & 8 \\ 2 & 3 & 11 & 0 & 6 & 0 & 2 \\ 3 & 4 & 16 & -2 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -477, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 2 & 3 \\ 3 & 4 & 4 & 0 & 16 & 3 & 4 \\ 2 & 4 & 1 & 0 & 11 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 11 & 0 & 44 & 9 & 12 \\ 2 & 3 & 2 & 0 & 9 & 0 & 2 \\ 3 & 4 & 3 & -2 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{131}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -526, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 4 & 2 & 3 \\ 3 & 4 & 22 & 0 & 7 & 3 & 4 \\ 12 & 22 & 85 & 0 & 27 & 12 & 17 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 27 & 0 & 7 & 4 & 5 \\ 2 & 3 & 12 & 0 & 4 & 0 & 2 \\ 3 & 4 & 17 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{142}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 6 & 2 & 3 \\ 3 & 4 & 26 & 0 & 11 & 3 & 4 \\ 14 & 26 & 118 & 0 & 50 & 14 & 20 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 50 & 0 & 20 & 6 & 8 \\ 2 & 3 & 14 & 0 & 6 & 0 & 2 \\ 3 & 4 & 20 & -2 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{112}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 2 & 3 \\ 3 & 4 & 4 & 0 & 21 & 3 & 4 \\ 2 & 4 & 1 & 0 & 14 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 12 & 21 & 14 & 0 & 80 & 12 & 16 \\ 2 & 3 & 2 & 0 & 12 & 0 & 2 \\ 3 & 4 & 3 & -2 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{113}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -631, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 2 & 3 \\ 3 & 4 & 4 & 0 & 20 & 3 & 4 \\ 2 & 4 & 1 & 0 & 14 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 14 & 0 & 69 & 11 & 15 \\ 2 & 3 & 2 & 0 & 11 & 0 & 2 \\ 3 & 4 & 3 & -2 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{141}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -666, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 4 & 2 & 3 \\ 3 & 4 & 26 & 0 & 7 & 3 & 4 \\ 14 & 26 & 118 & 0 & 32 & 14 & 20 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 32 & 0 & 7 & 4 & 5 \\ 2 & 3 & 14 & 0 & 4 & 0 & 2 \\ 3 & 4 & 20 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{114}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -749, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 2 & 3 \\ 3 & 4 & 4 & 0 & 24 & 3 & 4 \\ 2 & 4 & 1 & 0 & 17 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 13 & 24 & 17 & 0 & 100 & 13 & 18 \\ 2 & 3 & 2 & 0 & 13 & 0 & 2 \\ 3 & 4 & 3 & -2 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{113}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -886, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 2 & 3 \\ 3 & 4 & 4 & 0 & 25 & 3 & 4 \\ 2 & 4 & 1 & 0 & 17 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 14 & 25 & 17 & 0 & 111 & 14 & 19 \\ 2 & 3 & 2 & 0 & 14 & 0 & 2 \\ 3 & 4 & 3 & -2 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{114}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -1152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 2 & 3 \\ 3 & 4 & 4 & 0 & 29 & 3 & 4 \\ 2 & 4 & 1 & 0 & 20 & 2 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 16 & 29 & 20 & 0 & 148 & 16 & 22 \\ 2 & 3 & 2 & 0 & 16 & 0 & 2 \\ 3 & 4 & 3 & -2 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

4.871 NHC(-2)(-3) + $\textcolor{red}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$

$$T = 9 \quad [0] \quad 1^{1121}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 7 & 3 \\ 3 & 4 & 4 & 0 & 11 & 12 & 4 \\ 2 & 4 & 1 & 0 & 8 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 6 & 11 & 8 & 0 & 20 & 23 & 8 \\ 7 & 12 & 8 & 0 & 23 & 25 & 9 \\ 3 & 4 & 3 & -2 & 8 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1211}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 4 & 3 \\ 3 & 4 & 8 & 0 & 7 & 7 & 4 \\ 4 & 8 & 10 & 0 & 10 & 10 & 6 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 10 & 0 & 7 & 9 & 5 \\ 4 & 7 & 10 & 0 & 9 & 7 & 5 \\ 3 & 4 & 6 & -2 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 4 & 3 \\ 3 & 4 & 9 & 0 & 7 & 7 & 4 \\ 5 & 9 & 13 & 0 & 11 & 11 & 7 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 11 & 0 & 7 & 9 & 5 \\ 4 & 7 & 11 & 0 & 9 & 7 & 5 \\ 3 & 4 & 7 & -2 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -303, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 7 & 3 \\ 3 & 4 & 4 & 0 & 7 & 12 & 4 \\ 2 & 4 & 1 & 0 & 5 & 8 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 5 & 0 & 7 & 15 & 5 \\ 7 & 12 & 8 & 0 & 15 & 25 & 9 \\ 3 & 4 & 3 & -2 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}3^{\overline{11}1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2} \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -308, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 4 & 3 \\ 3 & 4 & 4 & 0 & 7 & 7 & 4 \\ 2 & 4 & 1 & 0 & 5 & 5 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 5 & 0 & 7 & 9 & 5 \\ 4 & 7 & 5 & 0 & 9 & 7 & 5 \\ 3 & 4 & 3 & -2 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1112}3^{\overline{11}1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2} \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -438, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 12 & 3 \\ 3 & 4 & 4 & 0 & 7 & 21 & 4 \\ 2 & 4 & 1 & 0 & 5 & 14 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 5 & 0 & 7 & 26 & 5 \\ 12 & 21 & 14 & 0 & 26 & 80 & 16 \\ 3 & 4 & 3 & -2 & 5 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1211}3^{\overline{11}1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2} \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -448, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 4 & 3 \\ 3 & 4 & 13 & 0 & 7 & 7 & 4 \\ 7 & 13 & 28 & 0 & 16 & 16 & 10 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 16 & 0 & 7 & 9 & 5 \\ 4 & 7 & 16 & 0 & 9 & 7 & 5 \\ 3 & 4 & 10 & -2 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1112}3^{\overline{11}1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2} \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -589, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 9 & 3 \\ 3 & 4 & 4 & 0 & 7 & 16 & 4 \\ 2 & 4 & 1 & 0 & 5 & 11 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 5 & 0 & 7 & 20 & 5 \\ 9 & 16 & 11 & 0 & 20 & 44 & 12 \\ 3 & 4 & 3 & -2 & 5 & 12 & 2 \end{pmatrix}_{b_0}$$

4.872 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{\overline{11}1^{\overline{121}}3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3} \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 14 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 14 & 17 & 13 & 15 & 7 \\ 12 & 17 & 15 & 12 & 6 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}1^{\overline{11}}3^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3} \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 19 & 5 \\ 11 & 20 & 0 & 34 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 19 & 34 & 0 & 57 & 13 \\ 5 & 8 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{141}1^{\overline{11}}3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3} \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 5 & 3 \\ 3 & 4 & 16 & 0 & 9 & 4 \\ 9 & 16 & 46 & 0 & 24 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 24 & 0 & 12 & 6 \\ 3 & 4 & 12 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{132}1^{\overline{11}}3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3} \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 7 & 3 \\ 3 & 4 & 12 & 0 & 13 & 4 \\ 6 & 12 & 25 & 0 & 27 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 27 & 0 & 27 & 9 \\ 3 & 4 & 9 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 4 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 13 & 15 & 7 \\ 12 & 17 & 15 & 12 & 6 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 10 & 17 & 13 & 15 & 7 \\ 10 & 17 & 15 & 12 & 6 \\ 5 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 0 & -1 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 20 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 20 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 4 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 5 \\ 12 & 25 & 20 & 16 & 9 \\ 9 & 20 & 13 & 12 & 7 \\ 8 & 16 & 12 & 9 & 5 \\ 5 & 9 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 3 & 4 & 0 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 3 \\ 3 & 4 & 24 & 0 & 4 & 4 \\ 17 & 24 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 19 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 0 & 0 & 12 & 6 \\ 3 & 4 & 0 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

- [14] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 0 & 0 & 3 & 3 \\ 3 & 4 & 0 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [15] $1^{111} 3 1^{21} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 10 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$
- [17] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 10 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$
- [16] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 17 & 0 & 12 & 6 \\ 5 & 8 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$
- [18] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 11 & 13 & 6 \\ 15 & 20 & 16 & 17 & 8 \\ 11 & 16 & 10 & 12 & 6 \\ 13 & 17 & 12 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$
- [19] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [21] $1^{113} 3 1^{41} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 59 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 59 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 45 & -1 & 2 \end{pmatrix}_{b_0}$$
- [20] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 5 & 0 & 0 & 4 \\ 3 & 5 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$
- [22] $1^{11} 3 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 1 & 3 \\ 3 & 4 & 19 & 0 & 0 & 4 \\ 11 & 19 & 73 & 0 & 0 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$
- [24] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 1 & 6 \\ 15 & 20 & 17 & 0 & 8 \\ 12 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$
- [23] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 8 & 0 & 5 & 4 \\ 5 & 8 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 6 & 0 & 3 & 3 \\ 3 & 4 & 6 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 10 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 10 & 16 & 10 & 12 & 6 \\ 10 & 17 & 12 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[28] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 2 & 3 \\ 4 & 9 & 28 & 0 & 4 & 5 \\ 15 & 28 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 5 & 15 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{132} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 5 & 3 \\ 3 & 4 & 12 & 0 & 8 & 4 \\ 6 & 12 & 25 & 0 & 18 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 18 & 0 & 9 & 5 \\ 3 & 4 & 9 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 3 \\ 3 & 4 & 4 & 0 & 0 & 4 \\ 2 & 4 & 1 & 0 & 0 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 10 & 17 & 13 & 15 & 7 \\ 9 & 17 & 15 & 12 & 6 \\ 5 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 1 & 3 \\ 4 & 9 & 12 & 0 & 0 & 5 \\ 5 & 12 & 13 & 0 & 0 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 7 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 17 & 0 & 12 & 6 \\ 6 & 8 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 1 & 3 \\ 4 & 9 & 36 & 0 & 0 & 5 \\ 14 & 36 & 121 & 0 & 0 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 19 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 18 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

- [34] $1^{11} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 9 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 13 & 15 & 7 \\ 10 & 17 & 15 & 12 & 6 \\ 5 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$
- [37] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$
- [38] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 13 & 15 & 7 \\ 9 & 17 & 15 & 12 & 6 \\ 5 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$
- [36] $1^{11} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 12 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 13 & 15 & 7 \\ 13 & 17 & 15 & 12 & 6 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$
- [39] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 3 & 4 & 3 & -2 & 3 & 1 \end{pmatrix}_{b_0}$
- [41] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 4 & 8 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 6 & -2 & 2 & 1 \end{pmatrix}_{b_0}$
- [40] $21^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 13 & 17 & 15 & 15 & 7 \\ 17 & 20 & 17 & 17 & 8 \\ 15 & 17 & 13 & 15 & 7 \\ 15 & 17 & 15 & 12 & 6 \\ 7 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$
- [42] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 3 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 3 & 4 & 3 & -2 & 3 & 1 \end{pmatrix}_{b_0}$
- [44] $1^{11} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 13 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 13 & 17 & 13 & 15 & 7 \\ 12 & 17 & 15 & 12 & 6 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$
- [46] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 12 & 0 & 12 & 6 \\ 3 & 4 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$
- [43] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 18 & 0 & 27 & 9 \\ 3 & 4 & 6 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 4 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 1 & 3 \\ 3 & 4 & 17 & 0 & 0 & 4 \\ 9 & 17 & 49 & 0 & 0 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

[49] $1^{11} 1^3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 8 & 16 & 10 & 12 & 6 \\ 9 & 17 & 12 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[50] $1^{11} 1^{12} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 1 & 3 \\ 4 & 9 & 20 & 0 & 0 & 5 \\ 8 & 20 & 37 & 0 & 0 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 11 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 11 & 1 & 1 & 3 \\ 4 & 9 & 28 & 0 & 0 & 5 \\ 11 & 28 & 73 & 0 & 0 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 15 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 6 & 0 & 3 & 3 \\ 3 & 4 & 4 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[53] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 9 & 16 & 10 & 12 & 6 \\ 10 & 17 & 12 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[54] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 11 & 12 & 6 \\ 15 & 20 & 16 & 17 & 8 \\ 11 & 16 & 10 & 12 & 6 \\ 12 & 17 & 12 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[55] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 16 & 17 & 8 \\ 12 & 16 & 10 & 12 & 6 \\ 12 & 17 & 12 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

- [56] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 9 & 16 & 10 & 12 & 6 \\ 9 & 17 & 12 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$
- [57] $1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 1 & 3 \\ 4 & 9 & 24 & 0 & 0 & 5 \\ 10 & 24 & 58 & 0 & 0 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 14 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$
- [58] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 7 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 15 & 0 & 12 & 6 \\ 3 & 4 & 7 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$
- [59] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 \\ 11 & 14 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 11 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [60] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 3 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$
- [62] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 8 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$
- [61] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 8 & 9 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 9 & 0 & 3 & 3 \\ 3 & 4 & 7 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [63] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 6 & 0 & 3 & 3 \\ 3 & 4 & 4 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [64] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 1 & 3 \\ 3 & 4 & 14 & 0 & 0 & 4 \\ 8 & 14 & 37 & 0 & 0 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 11 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$
- [65] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 3 \\ 4 & 9 & 12 & 0 & 4 & 5 \\ 5 & 12 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 5 & 7 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [66] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 1 & 3 \\ 3 & 4 & 9 & 0 & 0 & 4 \\ 5 & 9 & 13 & 0 & 0 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[67] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 48 & 12 \\ 3 & 4 & 4 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 3 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 12 & 0 & 48 & 12 \\ 3 & 4 & 3 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 4 & 5 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 6 & 0 & 3 & 3 \\ 3 & 4 & 4 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 1 & 3 \\ 3 & 4 & 13 & 0 & 0 & 4 \\ 7 & 13 & 28 & 0 & 0 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 20 & 10 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 20 & 33 & 49 & 27 & 13 \\ 10 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 11 & 1 & 6 \\ 15 & 20 & 16 & 0 & 8 \\ 11 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[79] \ 1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 19 & 0 & 4 & 4 \\ 14 & 19 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 15 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{112} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{111} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 24 & 0 & 4 & 4 \\ 15 & 24 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 19 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[80] $1^{121} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 4 & 8 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 6 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{121} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 7 & 8 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 6 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[76] $1^{111} 3^1 1^3 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 19 & 0 & 4 & 4 \\ 11 & 19 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 15 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{131} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 97$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.18 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 12 & 0 & 9 & 4 \\ 7 & 12 & 25 & 0 & 18 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 18 & 0 & 12 & 6 \\ 3 & 4 & 9 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{111} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 98$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 48 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 48 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 98$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 5 & 9 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 9 & 0 & 3 & 3 \\ 3 & 4 & 7 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[84] $1^{14} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 99$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.6$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 18 & 0 & 75 & 15 \\ 3 & 4 & 4 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 101$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 12 & 0 & 9 & 5 \\ 3 & 4 & 7 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 105$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 3 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 3 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{121} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 & 3 \\ 3 & 4 & 18 & 0 & 4 & 4 \\ 10 & 18 & 58 & 0 & 12 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 14 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 3 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 9 & 0 & 27 & 9 \\ 3 & 4 & 3 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 12 & 0 & 9 & 5 \\ 3 & 4 & 7 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 17 & 10 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 17 & 33 & 49 & 27 & 13 \\ 10 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[92] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 \\ 11 & 13 & 28 & 0 & 12 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 12 & 0 & 3 & 3 \\ 3 & 4 & 10 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 35 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 35 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[95] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 9 & 0 & 36 & 10 \\ 3 & 4 & 3 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 23 & 1 & 6 \\ 15 & 20 & 33 & 0 & 8 \\ 23 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[97] $1^{141} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 5 & 3 \\ 3 & 4 & 21 & 0 & 9 & 4 \\ 15 & 21 & 76 & 0 & 33 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 33 & 0 & 12 & 6 \\ 3 & 4 & 16 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[96] $1^{111} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 9 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[98] $1^{113} 3 1^3 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 1 & 3 \\ 4 & 9 & 36 & 0 & 0 & 5 \\ 15 & 36 & 133 & 0 & 0 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 21 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[99] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 \\ 5 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 7 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[100] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 5 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 12 & 0 & 12 & 6 \\ 3 & 4 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[101] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[102] $1^{122} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 4 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 12 & 0 & 9 & 5 \\ 3 & 4 & 6 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[105] $1^{111} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 3 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[104] $1^{111} 3 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 3 \\ 3 & 4 & 24 & 0 & 4 & 4 \\ 16 & 24 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 19 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[103] $1^{113} 3 1^2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 1 & 3 \\ 4 & 9 & 32 & 0 & 0 & 5 \\ 13 & 32 & 100 & 0 & 0 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 18 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[106] $1^{111} 3 1^{31} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 26 & 12 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 26 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[107] $1^{12}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 12 & 0 & 27 & 9 \\ 3 & 4 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[108] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 23 & 13 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 23 & 33 & 49 & 27 & 13 \\ 13 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[109] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 3 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{11}3^11^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 28 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 28 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[111] $1^{12}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 1 & 3 \\ 3 & 4 & 28 & 0 & 0 & 4 \\ 16 & 28 & 154 & 0 & 0 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[110] $1^{11}3^11^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 2 & 3 \\ 4 & 9 & 24 & 0 & 4 & 5 \\ 10 & 24 & 58 & 0 & 12 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 5 & 14 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[113] $1^{12}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 18 & 0 & 27 & 9 \\ 3 & 4 & 6 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[116] $1^{11}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 9 & 0 & 36 & 10 \\ 3 & 4 & 3 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[115] $21^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 16 & 12 & 12 & 6 \\ 16 & 20 & 16 & 17 & 8 \\ 12 & 16 & 10 & 12 & 6 \\ 12 & 17 & 12 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[117] $1^{11}3^11^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 5 \\ 12 & 25 & 72 & 0 & 9 \\ 34 & 72 & 202 & 0 & 26 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 26 & -1 & 2 \end{pmatrix}_{b_0}$$

[114] $1^{1^{11}} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 1 & 3 \\ 3 & 4 & 18 & 0 & 0 & 4 \\ 10 & 18 & 58 & 0 & 0 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 14 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[118] $1^{1^{11}} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 23 & 12 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 23 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[119] $1^{1^{11}} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 & 3 \\ 3 & 4 & 21 & 0 & 5 & 4 \\ 11 & 21 & 76 & 0 & 18 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 18 & 0 & 3 & 3 \\ 3 & 4 & 16 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[120] $1^{1^{11}} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 17 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[121] $1^{1^{12}} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 157, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 2 & 3 \\ 4 & 9 & 20 & 0 & 4 & 5 \\ 10 & 20 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 5 & 11 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[122] $1^{1^{14}} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 158, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 1 & 3 \\ 4 & 9 & 40 & 0 & 0 & 5 \\ 16 & 40 & 154 & 0 & 0 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 22 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[123] $1^{1^{11}} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[124] $1^{1^{11}} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 3 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[125] $1^{1^{12}} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 1 & 3 \\ 3 & 4 & 23 & 0 & 0 & 4 \\ 13 & 23 & 100 & 0 & 0 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[128] $1^{1^{11}} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 29 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 29 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[126] $1^{1^{11}} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 26 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 5 & 9 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 9 & 0 & 3 & 3 \\ 3 & 4 & 7 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 11 & 13 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 10 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[130] \ 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 170, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 15 & 0 & 48 & 12 \\ 3 & 4 & 4 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 & 3 \\ 3 & 4 & 21 & 0 & 4 & 4 \\ 11 & 21 & 76 & 0 & 15 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 16 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 7 & 13 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 10 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 5 & 8 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 18 & 0 & 27 & 9 \\ 3 & 4 & 6 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[138] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 \\ 7 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 7 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{11} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 \\ 8 & 14 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 11 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[136] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[135] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 \\ 5 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 7 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{122} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 5 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 12 & 0 & 9 & 5 \\ 3 & 4 & 6 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[142] $1^{112} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 26 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[141] $1^{14} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 1 & 3 \\ 3 & 4 & 26 & 0 & 0 & 4 \\ 14 & 26 & 118 & 0 & 0 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 20 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{13} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 1 & 3 \\ 3 & 4 & 22 & 0 & 0 & 4 \\ 12 & 22 & 85 & 0 & 0 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 17 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{113} 3 1^{21} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 46 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 46 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 33 & -1 & 2 \end{pmatrix}_{b_0}$$

[143] $21^{111} 31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 15 & 12 & 7 \\ 20 & 25 & 20 & 16 & 9 \\ 15 & 20 & 13 & 12 & 7 \\ 12 & 16 & 12 & 9 & 5 \\ 7 & 9 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[144] $1^{132} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 12 & 0 & 13 & 4 \\ 7 & 12 & 25 & 0 & 27 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 27 & 0 & 27 & 9 \\ 3 & 4 & 9 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[145] $1^{114} 3 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 1 & 3 \\ 4 & 9 & 48 & 0 & 0 & 5 \\ 20 & 48 & 238 & 0 & 0 & 28 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 28 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[146] $1^{131} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 9 & 17 & 49 & 0 & 12 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 13 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{112} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 2 & 3 \\ 4 & 9 & 20 & 0 & 4 & 5 \\ 8 & 20 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 5 & 11 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

- [148] $1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 12 & 0 & 27 & 9 \\ 3 & 4 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$
- [149] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 19 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$
- [150] $1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 19 & 9 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 19 & 33 & 49 & 27 & 13 \\ 9 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$
- [151] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 17 & 9 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 17 & 33 & 49 & 27 & 13 \\ 9 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$
- [152] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 19 & 10 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 19 & 33 & 49 & 27 & 13 \\ 10 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$
- [153] $1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 8 & 4 \\ 7 & 13 & 28 & 0 & 18 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 18 & 0 & 9 & 5 \\ 3 & 4 & 10 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$
- [154] $1^{14} 3 1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 1 & 3 \\ 4 & 9 & 44 & 0 & 0 & 5 \\ 18 & 44 & 193 & 0 & 0 & 25 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 25 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$
- [155] $1^{13} 3 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 205, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 2 & 3 \\ 4 & 9 & 32 & 0 & 4 & 5 \\ 16 & 32 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 5 & 18 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [156] $1^{13} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 \\ 9 & 17 & 49 & 0 & 15 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 15 & 0 & 3 & 3 \\ 3 & 4 & 13 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [157] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 18 & 10 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 18 & 33 & 49 & 27 & 13 \\ 10 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$
- [158] $1^{11} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 45 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 45 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[159] $1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 6 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 7 & 13 & 28 & 0 & 21 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 21 & 0 & 12 & 6 \\ 3 & 4 & 10 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[160] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 6 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 21 & 0 & 27 & 9 \\ 3 & 4 & 7 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[161] $1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 \\ 7 & 13 & 28 & 0 & 12 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 12 & 0 & 3 & 3 \\ 3 & 4 & 10 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[162] $1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 8 & 4 \\ 8 & 13 & 28 & 0 & 18 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 18 & 0 & 9 & 5 \\ 3 & 4 & 10 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[163] $1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 7 & 9 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 9 & 0 & 3 & 3 \\ 3 & 4 & 7 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[164] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 9 & 0 & 27 & 9 \\ 3 & 4 & 3 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[165] $1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 8 & 4 \\ 7 & 13 & 28 & 0 & 18 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 18 & 0 & 9 & 5 \\ 3 & 4 & 10 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[172] $1^{111} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 19 & 0 & 4 & 4 \\ 13 & 19 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 15 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[166] $1^{111} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 19 & 0 & 4 & 4 \\ 12 & 19 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 15 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[167] $1^{131} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 4 & 4 \\ 10 & 12 & 25 & 0 & 9 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 9 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[168] $1^{132} 3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 4 & 3 \\ 3 & 4 & 12 & 0 & 8 & 4 \\ 8 & 12 & 25 & 0 & 18 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 18 & 0 & 9 & 5 \\ 3 & 4 & 9 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[169] $1^{121} 3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 6 & 8 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 6 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[170] $1^{131} 3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 & 3 \\ 3 & 4 & 22 & 0 & 4 & 4 \\ 12 & 22 & 85 & 0 & 15 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 17 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[171] $1^{121} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 5 & 8 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 6 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[173] $1^{114} 3^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 22 & 1 & 2 & 3 \\ 4 & 9 & 44 & 0 & 4 & 5 \\ 22 & 44 & 193 & 0 & 21 & 25 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 3 & 5 & 25 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[174] $1^{111} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 233, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 18 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[175] $1^{111} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 25 & 12 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 25 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{112} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 237, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 27 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 27 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[179] $1^{113} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 9 & 0 & 24 & 8 \\ 3 & 4 & 3 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

[177] $1^{111} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 \\ 6 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 3 & 4 & 7 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{\overline{111}} 3^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 240$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 \\ 10 & 14 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 11 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[181] $1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 241$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 24 & 12 & 6 & 6 \\ 15 & 20 & 33 & 17 & 8 & 8 \\ 24 & 33 & 49 & 27 & 13 & 13 \\ 12 & 17 & 27 & 12 & 6 & 6 \\ 6 & 8 & 13 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[180] $1^{\overline{111}} 3^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 241$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 38 & 1 & 5 & 5 \\ 11 & 20 & 67 & 0 & 8 & 8 \\ 38 & 67 & 217 & 0 & 27 & 27 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 5 & 8 & 27 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[182] $1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 242$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 6 & 9 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 9 & 0 & 3 & 3 \\ 3 & 4 & 7 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[184] $1^{\overline{113}} 3^{\overline{131}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 244$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 2 & 3 \\ 4 & 9 & 36 & 0 & 4 & 5 \\ 15 & 36 & 133 & 0 & 18 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 5 & 21 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[183] $21^{\overline{111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 244$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 13 & 17 & 27 & 15 & 7 & 7 \\ 17 & 20 & 33 & 17 & 8 & 8 \\ 27 & 33 & 49 & 27 & 13 & 13 \\ 15 & 17 & 27 & 12 & 6 & 6 \\ 7 & 8 & 13 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[185] $1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 245$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 18 & 9 & 5 & 5 \\ 11 & 20 & 33 & 17 & 8 & 8 \\ 18 & 33 & 49 & 27 & 13 & 13 \\ 9 & 17 & 27 & 12 & 6 & 6 \\ 5 & 8 & 13 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[187] $1^{\overline{13}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 253$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 1 & 3 \\ 3 & 4 & 27 & 0 & 0 & 4 \\ 15 & 27 & 133 & 0 & 0 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[186] $1^{\overline{113}} 3^{\overline{131}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 253$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 5 & 5 \\ 12 & 25 & 108 & 0 & 9 & 9 \\ 51 & 108 & 457 & 0 & 39 & 39 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 5 & 9 & 39 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[188] $1^{\overline{111}} 3^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 254$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 47 & 1 & 5 & 5 \\ 11 & 20 & 84 & 0 & 8 & 8 \\ 47 & 84 & 346 & 0 & 34 & 34 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 5 & 8 & 34 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[189] $1^{\overline{132}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 257$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 5 & 3 \\ 3 & 4 & 17 & 0 & 8 & 4 \\ 9 & 17 & 49 & 0 & 24 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 24 & 0 & 9 & 5 \\ 3 & 4 & 13 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[190] \ 1^{111} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 15 & 0 & 12 & 6 \\ 3 & 4 & 7 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[191] \ 1^{141} 3 \ 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 & 3 \\ 3 & 4 & 31 & 0 & 4 & 4 \\ 17 & 31 & 172 & 0 & 21 & 24 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 21 & 0 & 0 & 2 \\ 3 & 4 & 24 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[192] \ 1^{111} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 15 & 0 & 12 & 6 \\ 3 & 4 & 7 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[193] \ 1^{132} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 12 & 0 & 8 & 4 \\ 7 & 12 & 25 & 0 & 18 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 18 & 0 & 9 & 5 \\ 3 & 4 & 9 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[194] \ 1^{111} 3 \ 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 \\ 9 & 14 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 11 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[195] \ 21^{112} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 20 & 30 & 12 & 7 \\ 20 & 25 & 40 & 16 & 9 \\ 30 & 40 & 58 & 24 & 14 \\ 12 & 16 & 24 & 9 & 5 \\ 7 & 9 & 14 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[196] \ 1^{113} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 38 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 38 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[197] \ 1^{112} 3 \ 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 12 & 0 & 57 & 13 \\ 3 & 4 & 3 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[198] \ 1^{111} 3 \ 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 277, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 36 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 36 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[199] \ 1^{113} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 285, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 35 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[200] $21^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 286$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 16 & 24 & 12 & 6 \\ 16 & 20 & 33 & 17 & 8 \\ 24 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[201] $1^{111}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 286$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 28 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 28 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[202] $1^{111}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 286$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 27 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 27 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{13}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 288$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 1 & 3 \\ 3 & 4 & 32 & 0 & 0 & 4 \\ 18 & 32 & 193 & 0 & 0 & 25 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[204] $1^{111}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 290$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 46 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 46 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[205] $1^{132}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 290$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 8 & 3 \\ 3 & 4 & 17 & 0 & 13 & 4 \\ 9 & 17 & 49 & 0 & 39 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 39 & 0 & 27 & 9 \\ 3 & 4 & 13 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[206] $1^{132}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 293$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 4 & 3 \\ 3 & 4 & 17 & 0 & 8 & 4 \\ 11 & 17 & 49 & 0 & 24 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 24 & 0 & 9 & 5 \\ 3 & 4 & 13 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{12}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 295$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 \\ 10 & 13 & 28 & 0 & 12 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 12 & 0 & 3 & 3 \\ 3 & 4 & 10 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[208] $1^{121}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 296$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 7 & 13 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 10 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[209] $1^{113}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 301$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.22 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 12 & 0 & 48 & 12 \\ 3 & 4 & 3 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[210] $1^{1\bar{3}1} 3 1^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 303, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 5 & 3 \\ 3 & 4 & 17 & 0 & 9 & 4 \\ 12 & 17 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 27 & 0 & 12 & 6 \\ 3 & 4 & 13 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[212] $1^{1\bar{4}1} 3 1^{1\bar{2}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 & 3 \\ 3 & 4 & 26 & 0 & 4 & 4 \\ 14 & 26 & 118 & 0 & 18 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 20 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[213] $1^{1\bar{3}1} 3 1^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 2 & 3 \\ 4 & 9 & 28 & 0 & 4 & 5 \\ 14 & 28 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 5 & 15 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[211] $1^{1\bar{3}1} 3 1^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 11 & 1 & 2 & 3 \\ 4 & 9 & 28 & 0 & 4 & 5 \\ 11 & 28 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 5 & 15 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[214] $1^{1\bar{1}1} 3 1^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 6 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 15 & 0 & 12 & 6 \\ 3 & 4 & 7 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[216] $1^{1\bar{4}1} 3 1^{1\bar{3}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 309, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 1 & 3 \\ 3 & 4 & 31 & 0 & 0 & 4 \\ 17 & 31 & 172 & 0 & 0 & 24 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[215] $1^{1\bar{1}2} 3 1^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 309, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 9 & 0 & 27 & 9 \\ 3 & 4 & 3 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[217] $1^{1\bar{1}1} 3 1^{1\bar{3}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 37 & 1 & 5 & 8 \\ 11 & 20 & 67 & 0 & 8 & 8 \\ 37 & 67 & 217 & 0 & 27 & 27 \\ 1 & 0 & 0 & -3 & -1 & 1 \\ 5 & 8 & 27 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{1\bar{3}1} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 4 & 4 \\ 7 & 12 & 25 & 0 & 9 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 9 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[219] $1^{1\bar{1}2} 3 1^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 326, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 21 & 0 & 27 & 9 \\ 3 & 4 & 7 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[220] $1^{1\bar{1}3} 3 1^{1\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 27 & 0 & 48 & 12 \\ 3 & 4 & 7 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[221] \ 1^{113} 3 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 333, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 43 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 43 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 33 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[222] \ 1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 \\ 13 & 17 & 49 & 0 & 15 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 15 & 0 & 3 & 3 \\ 3 & 4 & 13 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[223] \ 1^{132} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 341, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 4 & 3 \\ 3 & 4 & 17 & 0 & 8 & 4 \\ 9 & 17 & 49 & 0 & 24 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 24 & 0 & 9 & 5 \\ 3 & 4 & 13 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[224] \ 1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 \\ 8 & 13 & 28 & 0 & 12 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 12 & 0 & 3 & 3 \\ 3 & 4 & 10 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[225] \ 1^{113} 3 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 349, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 2 & 3 \\ 4 & 9 & 32 & 0 & 4 & 5 \\ 13 & 32 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 5 & 18 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[227] \ 1^{132} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 4 & 3 \\ 3 & 4 & 22 & 0 & 8 & 4 \\ 12 & 22 & 85 & 0 & 30 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 30 & 0 & 9 & 5 \\ 3 & 4 & 17 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[226] \ 1^{132} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 4 & 3 \\ 3 & 4 & 22 & 0 & 8 & 4 \\ 13 & 22 & 85 & 0 & 30 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 30 & 0 & 9 & 5 \\ 3 & 4 & 17 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[229] \ 1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 7 & 13 & 28 & 0 & 21 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 21 & 0 & 12 & 6 \\ 3 & 4 & 10 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[228] \ 1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 9 & 13 & 28 & 0 & 21 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 21 & 0 & 12 & 6 \\ 3 & 4 & 10 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[230] \ 1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 371, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 \\ 9 & 13 & 28 & 0 & 12 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 12 & 0 & 3 & 3 \\ 3 & 4 & 10 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[231] $1^{112} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 12 & 0 & 57 & 13 \\ 3 & 4 & 3 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[232] $1^{113} 3 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 45 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 33 & -1 & 2 \end{pmatrix}_{b_0}$$

[233] $1^{114} 3 1^{41} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 2 & 3 \\ 4 & 9 & 48 & 0 & 4 & 5 \\ 20 & 48 & 238 & 0 & 24 & 28 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 24 & 0 & 0 & 2 \\ 3 & 5 & 28 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[234] $1^{114} 3 1^{41} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 56 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 56 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -1 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{131} 3 1^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 4 & 4 \\ 9 & 12 & 25 & 0 & 9 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 9 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[236] $1^{112} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 386, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 21 & 0 & 27 & 9 \\ 3 & 4 & 7 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[237] $1^{132} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 7 & 3 \\ 3 & 4 & 17 & 0 & 13 & 4 \\ 11 & 17 & 49 & 0 & 39 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 39 & 0 & 27 & 9 \\ 3 & 4 & 13 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[238] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 10 & 13 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 10 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[239] $1^{13} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 394, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 \\ 10 & 17 & 49 & 0 & 15 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 15 & 0 & 3 & 3 \\ 3 & 4 & 13 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[240] $1^{14} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 395, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 & 3 \\ 3 & 4 & 21 & 0 & 5 & 4 \\ 12 & 21 & 76 & 0 & 18 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 18 & 0 & 3 & 3 \\ 3 & 4 & 16 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[241] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 44 & 1 & 5 & 9 \\ 12 & 25 & 96 & 0 & 9 & 9 \\ 44 & 96 & 346 & 0 & 34 & 34 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 5 & 9 & 34 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[242] $1^{14} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 1 & 3 \\ 3 & 4 & 36 & 0 & 0 & 4 \\ 20 & 36 & 238 & 0 & 0 & 28 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[243] $1^{114} 3^1 1^{41} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 5 & 9 \\ 12 & 25 & 144 & 0 & 9 & 9 \\ 68 & 144 & 814 & 0 & 52 & 52 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 5 & 9 & 52 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[244] $1^{141} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 5 & 3 \\ 3 & 4 & 21 & 0 & 9 & 4 \\ 11 & 21 & 76 & 0 & 33 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 33 & 0 & 12 & 6 \\ 3 & 4 & 16 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[245] $1^{131} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 411, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 5 & 3 \\ 3 & 4 & 17 & 0 & 9 & 4 \\ 9 & 17 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 27 & 0 & 12 & 6 \\ 3 & 4 & 13 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[246] $1^{121} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 4 & 4 \\ 14 & 18 & 58 & 0 & 12 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 14 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[247] $1^{131} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 9 & 17 & 49 & 0 & 12 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 13 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[248] $1^{131} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 4 & 4 \\ 8 & 12 & 25 & 0 & 9 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 9 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[249] $1^{141} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 16 & 0 & 4 & 4 \\ 9 & 16 & 46 & 0 & 12 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 12 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[250] $1^{121} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 4 & 4 \\ 10 & 18 & 58 & 0 & 12 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 14 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[251] $1^{121} 3^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 4 & 4 \\ 16 & 28 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 22 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[252] $1^{122} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 8 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 30 & 0 & 27 & 9 \\ 3 & 4 & 10 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[253] $21^{113} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 45 & 12 & 7 \\ 20 & 25 & 60 & 16 & 9 \\ 45 & 60 & 133 & 36 & 21 \\ 12 & 16 & 36 & 9 & 5 \\ 7 & 9 & 21 & 5 & 2 \end{pmatrix}_{b_0}$$

[254] $1^{114} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 2 & 3 \\ 4 & 9 & 36 & 0 & 4 & 5 \\ 14 & 36 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 5 & 19 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[255] $1^{132} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 4 & 3 \\ 3 & 4 & 17 & 0 & 8 & 4 \\ 10 & 17 & 49 & 0 & 24 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 24 & 0 & 9 & 5 \\ 3 & 4 & 13 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[256] $1^{113} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 37 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 37 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[257] $1^{113} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 36 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 36 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[258] $1^{113} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 2 & 3 \\ 4 & 9 & 28 & 0 & 4 & 5 \\ 13 & 28 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 5 & 15 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[259] $1^{113} 3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 15 & 0 & 84 & 16 \\ 3 & 4 & 3 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[260] $1^{114} 3^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 2 & 3 \\ 4 & 9 & 40 & 0 & 4 & 5 \\ 20 & 40 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 5 & 22 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

- [261] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 48 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 48 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$
- [262] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 453, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 12 & 0 & 48 & 12 \\ 3 & 4 & 3 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$
- [263] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 455, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 10 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 27 & 0 & 48 & 12 \\ 3 & 4 & 7 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$
- [264] $1^{121} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 23 & 0 & 4 & 4 \\ 13 & 23 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 18 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [265] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 8 & 13 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 10 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [266] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 458, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 \\ 12 & 17 & 49 & 0 & 15 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 15 & 0 & 3 & 3 \\ 3 & 4 & 13 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [267] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 30 & 0 & 27 & 9 \\ 3 & 4 & 10 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$
- [268] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 \\ 11 & 17 & 49 & 0 & 15 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 15 & 0 & 3 & 3 \\ 3 & 4 & 13 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [269] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 488, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 9 & 13 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 3 & 4 & 10 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [270] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 491, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.19 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 8 & 13 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 30 & 0 & 27 & 9 \\ 3 & 4 & 10 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$
- [271] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 493, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 2 & 3 \\ 4 & 9 & 36 & 0 & 4 & 5 \\ 18 & 36 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 5 & 19 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[272] 1^{121} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 498, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 8 & 13 & 28 & 0 & 21 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 21 & 0 & 12 & 6 \\ 3 & 4 & 10 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[273] 1^{114} 3 1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 501, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.47 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 63 & 1 & 5 & 9 \\ 12 & 25 & 128 & 0 & 9 & 46 \\ 63 & 128 & 634 & 0 & 46 & -1 \\ 1 & 0 & 0 & -3 & -1 & 2 \\ 5 & 9 & 46 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[274] 1^{141} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 503, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 & 3 \\ 3 & 4 & 21 & 0 & 5 & 4 \\ 15 & 21 & 76 & 0 & 18 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 18 & 0 & 3 & 3 \\ 3 & 4 & 16 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[275] 1^{114} 3 1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.47 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 5 & 9 \\ 12 & 25 & 128 & 0 & 9 & 46 \\ 60 & 128 & 634 & 0 & 46 & -1 \\ 1 & 0 & 0 & -3 & -1 & 2 \\ 5 & 9 & 46 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[276] 1^{114} 3 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 5 & 9 \\ 12 & 25 & 112 & 0 & 9 & 40 \\ 52 & 112 & 478 & 0 & 40 & -1 \\ 1 & 0 & 0 & -3 & -1 & 2 \\ 5 & 9 & 40 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[277] 1^{141} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 535, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 & 3 \\ 3 & 4 & 21 & 0 & 5 & 4 \\ 13 & 21 & 76 & 0 & 18 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 18 & 0 & 3 & 3 \\ 3 & 4 & 16 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[278] 1^{141} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 4 & 4 \\ 11 & 21 & 76 & 0 & 15 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 16 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[279] 1^{114} 3 1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 549, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 2 & 3 \\ 4 & 9 & 44 & 0 & 4 & 5 \\ 18 & 44 & 193 & 0 & 21 & 25 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 3 & 5 & 25 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[280] 1^{114} 3 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 2 & 3 \\ 4 & 9 & 40 & 0 & 4 & 5 \\ 16 & 40 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 5 & 22 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[281] 1^{132} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 566, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 7 & 3 \\ 3 & 4 & 17 & 0 & 13 & 4 \\ 9 & 17 & 49 & 0 & 39 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 39 & 0 & 27 & 9 \\ 3 & 4 & 13 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[282] $1^{141} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 571, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 & 3 \\ 3 & 4 & 21 & 0 & 5 & 4 \\ 14 & 21 & 76 & 0 & 18 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 18 & 0 & 3 & 3 \\ 3 & 4 & 16 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[283] $1^{121} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 584, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 3 \\ 3 & 4 & 23 & 0 & 4 & 4 \\ 17 & 23 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 18 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[284] $1^{141} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 586, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 5 & 3 \\ 3 & 4 & 21 & 0 & 9 & 4 \\ 14 & 21 & 76 & 0 & 33 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 33 & 0 & 12 & 6 \\ 3 & 4 & 16 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[285] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 15 & 0 & 84 & 16 \\ 3 & 4 & 3 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[286] $1^{131} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 5 & 3 \\ 3 & 4 & 17 & 0 & 9 & 4 \\ 11 & 17 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 27 & 0 & 12 & 6 \\ 3 & 4 & 13 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[287] $1^{121} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 4 & 4 \\ 13 & 18 & 58 & 0 & 12 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 14 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[288] $1^{141} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 16 & 0 & 4 & 4 \\ 12 & 16 & 46 & 0 & 12 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 12 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[289] $1^{121} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 4 & 4 \\ 11 & 18 & 58 & 0 & 12 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 14 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[290] $1^{141} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 16 & 0 & 4 & 4 \\ 10 & 16 & 46 & 0 & 12 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 12 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[291] $21^{114} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 634, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 60 & 12 & 7 \\ 20 & 25 & 80 & 16 & 9 \\ 60 & 80 & 238 & 48 & 28 \\ 12 & 16 & 48 & 9 & 5 \\ 7 & 9 & 28 & 5 & 2 \end{pmatrix}_{b_0}$$

[292] $1^{114} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 637$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 45 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 45 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[293] $1^{131} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 639$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 5 & 3 \\ 3 & 4 & 17 & 0 & 9 & 4 \\ 10 & 17 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 27 & 0 & 12 & 6 \\ 3 & 4 & 13 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[294] $1^{114} 3^1 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 645$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 55 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 55 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -1 & 2 \end{pmatrix}_{b_0}$$

[295] $1^{131} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 656$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 4 & 4 \\ 12 & 22 & 85 & 0 & 15 & 17 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 17 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[296] $1^{114} 3^1 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 660$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 19 & 1 & 2 & 3 \\ 4 & 9 & 40 & 0 & 4 & 5 \\ 19 & 40 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 5 & 22 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[297] $1^{114} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 661$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 47 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 47 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[298] $1^{114} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 669$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 17 & 1 & 2 & 3 \\ 4 & 9 & 36 & 0 & 4 & 5 \\ 17 & 36 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 5 & 19 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[303] $1^{131} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 672$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 13 & 17 & 49 & 0 & 12 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 13 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[304] $1^{141} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 672$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 16 & 0 & 4 & 4 \\ 11 & 16 & 46 & 0 & 12 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 12 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[299] $1^{131} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 672$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 10 & 17 & 49 & 0 & 12 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 13 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

- [300] $1^{121} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 4 & 4 \\ 17 & 28 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 22 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [301] $1^{121} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 4 & 4 \\ 12 & 18 & 58 & 0 & 12 & 14 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 14 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [302] $1^{121} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 4 & 4 \\ 20 & 28 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 22 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [305] $1^{121} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 680, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 23 & 0 & 4 & 4 \\ 14 & 23 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 18 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [306] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 717, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 2 & 3 \\ 4 & 9 & 36 & 0 & 4 & 5 \\ 16 & 36 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 5 & 19 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [307] $1^{132} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 722, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 7 & 3 \\ 3 & 4 & 17 & 0 & 13 & 4 \\ 10 & 17 & 49 & 0 & 39 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 39 & 0 & 27 & 9 \\ 3 & 4 & 13 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$
- [308] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 46 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 46 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$
- [309] $1^{141} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 730, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 5 & 3 \\ 3 & 4 & 21 & 0 & 9 & 4 \\ 12 & 21 & 76 & 0 & 33 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 33 & 0 & 12 & 6 \\ 3 & 4 & 16 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$
- [310] $1^{121} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 744, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 3 \\ 3 & 4 & 23 & 0 & 4 & 4 \\ 16 & 23 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 18 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [311] $1^{114} 3 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 756, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 54 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 54 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -1 & 2 \end{pmatrix}_{b_0}$$
- [312] $1^{121} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 776, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 23 & 0 & 4 & 4 \\ 15 & 23 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 18 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[313] $1^{141} 3^{1111} 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 790, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 5 & 3 \\ 3 & 4 & 21 & 0 & 9 & 4 \\ 13 & 21 & 76 & 0 & 33 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 33 & 0 & 12 & 6 \\ 3 & 4 & 16 & -2 & 6 & 1 \end{pmatrix} b_0$$

[317] $1^{121}3^{114}+[\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{s}\mathfrak{u}}_2+\textcolor{orange}{\mathfrak{s}\mathfrak{o}}_8+\textcolor{red}{\mathfrak{s}\mathfrak{u}}_3$ $T=8$, $T_{\max}=12$, $\Delta=-92$, $\det=800$, $(n_+, n_-, n_0)=(1, 8, 0)$, $c_{\text{ext}}=10.5$

$$T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 4 & 4 \\ 18 & 28 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 22 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[314] $1^{131} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 12 & 17 & 49 & 0 & 12 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 13 & -2 & 2 & 1 \end{pmatrix} b_0$$

$$[315] \ 1^{121} 3^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 800, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 4 & 4 \\ 19 & 28 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 22 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[316] $1^{131}3^{111}+[\textcolor{red}{\mathfrak{s}u}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{s}u}_2+\textcolor{orange}{\mathfrak{s}o}_8+\textcolor{red}{\mathfrak{s}u}_3$ $T=8$, $T_{\max}=12$, $\Delta=-92$, $\det=800$, $(n_+,n_-,n_0)=(1,8,0)$, $c_{\text{ext}}=10.8$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 11 & 17 & 49 & 0 & 12 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 3 & 4 & 13 & -2 & 2 & 1 \end{pmatrix} b_0$$

$$[318] \ 1^{131} 3^1 1^{13} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{green}{\mathbf{su}}_2 + \textcolor{orange}{\mathbf{so}}_8 + \textcolor{red}{\mathbf{su}}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 816, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 27 & 0 & 4 & 4 \\ 15 & 27 & 133 & 0 & 18 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 21 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[319] $1^{141} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 888, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 4 & 4 \\ 12 & 21 & 76 & 0 & 15 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 3 & 4 & 16 & -2 & 2 & 1 \end{pmatrix} b_0$$

$$[321] \ 1^{131} 3^1 1^{14} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{green}{\mathbf{su}}_2 + \textcolor{orange}{\mathbf{so}}_8 + \textcolor{red}{\mathbf{su}}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 896, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 2 & 3 \\ 3 & 4 & 32 & 0 & 4 & 4 \\ 18 & 32 & 193 & 0 & 21 & 25 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 3 & 4 & 25 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[320] \ 1^{141} 3^1 1^{12} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{green}{\mathbf{su}}_2 + \textcolor{orange}{\mathbf{so}}_8 + \textcolor{red}{\mathbf{su}}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 896, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 26 & 0 & 4 & 4 \\ 14 & 26 & 118 & 0 & 18 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 3 & 4 & 20 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[322] $21^{115}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 904$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.1$
 $T_{\text{ext}} = 11.38 \rightarrow 12$, $T_{\text{ext}} = 12$, $T \in [12, 12]$

$$x_{\min} = 11.999999999999999, x_{\max} = 12, x \in [12, 12]$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 4 & 3 \\ 3 & 4 & 21 & 0 & 7 & 4 \\ 13 & 21 & 101 & 0 & 31 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 31 & 0 & 7 & 5 \\ 3 & 4 & 19 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[353] \ 1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[354] \ 1^{111} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 8 & 8 & 6 \\ 12 & 13 & 10 & 10 & 7 \\ 8 & 10 & 5 & 7 & 5 \\ 8 & 10 & 7 & 4 & 4 \\ 6 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[358] \ 1^{12} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 12 & 1 & 5 \\ 9 & 13 & 20 & 0 & 7 \\ 12 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[357] \ 1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 12 & 8 & 6 \\ 12 & 13 & 14 & 10 & 7 \\ 12 & 14 & 13 & 9 & 7 \\ 8 & 10 & 9 & 4 & 4 \\ 6 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[355] \ 21^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 8 & 8 & 6 \\ 12 & 13 & 10 & 10 & 7 \\ 8 & 10 & 5 & 7 & 5 \\ 8 & 10 & 7 & 4 & 4 \\ 6 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[356] \ 21^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 12 & 8 & 6 \\ 12 & 13 & 14 & 10 & 7 \\ 12 & 14 & 13 & 9 & 7 \\ 8 & 10 & 9 & 4 & 4 \\ 6 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[359] \ 21^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 14 & 15 & 9 & 7 \\ 14 & 13 & 14 & 10 & 7 \\ 15 & 14 & 13 & 9 & 7 \\ 9 & 10 & 9 & 4 & 4 \\ 7 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[360] \ 1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 9 & 7 & 5 \\ 9 & 13 & 14 & 10 & 7 \\ 9 & 14 & 13 & 9 & 7 \\ 7 & 10 & 9 & 4 & 4 \\ 5 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[361] \ 1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -49, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 10 & 0 & 3 & 4 \\ 6 & 10 & 21 & 0 & 8 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 3 & 4 & 9 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[363] \ 1^{111} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 6 & 8 & 5 \\ 9 & 13 & 10 & 10 & 7 \\ 6 & 10 & 5 & 7 & 5 \\ 8 & 10 & 7 & 4 & 4 \\ 5 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[362] $21^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -52$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 13 & 14 & 9 & 9 & 7 \\ 14 & 13 & 10 & 10 & 7 \\ 9 & 10 & 5 & 7 & 5 \\ 9 & 10 & 7 & 4 & 4 \\ 7 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[364] $1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -54$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 5 \\ 9 & 13 & 14 & 0 & 7 \\ 9 & 14 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[365] $1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 9 & 6 & 5 \\ 9 & 13 & 14 & 10 & 7 \\ 9 & 14 & 13 & 9 & 7 \\ 6 & 10 & 9 & 4 & 4 \\ 5 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[366] $1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -61$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 5 \\ 9 & 13 & 10 & 0 & 7 \\ 6 & 10 & 5 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

[368] $1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 20 & 8 & 6 \\ 12 & 13 & 24 & 10 & 7 \\ 20 & 24 & 38 & 16 & 12 \\ 8 & 10 & 16 & 4 & 4 \\ 6 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

[367] $21^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -64$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.75 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 20 & 8 & 6 \\ 12 & 13 & 24 & 10 & 7 \\ 20 & 24 & 38 & 16 & 12 \\ 8 & 10 & 16 & 4 & 4 \\ 6 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

[370] $1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 3 & 4 \\ 5 & 8 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 7 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[369] $1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 7 & 4 \\ 5 & 8 & 13 & 0 & 11 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 11 & 0 & 7 & 5 \\ 3 & 4 & 7 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[371] $1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -70$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 3 & 0 & 7 & 4 \\ 2 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 5 & 0 & 7 & 5 \\ 3 & 4 & 3 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[372] $1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -94$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.66 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 7 & 0 & 3 & 4 \\ 4 & 7 & 8 & 0 & 5 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 6 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[373] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -97, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 4 & 3 \\ 3 & 4 & 10 & 0 & 7 & 4 \\ 6 & 10 & 21 & 0 & 15 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 15 & 0 & 7 & 5 \\ 3 & 4 & 9 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[374] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -109, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.79 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 6 & 7 & 5 \\ 9 & 13 & 10 & 10 & 7 \\ 6 & 10 & 5 & 7 & 5 \\ 7 & 10 & 7 & 4 & 4 \\ 5 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[375] \ 1^{\frac{1}{12}} 4^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -115, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 3 & 4 \\ 10 & 17 & 61 & 0 & 13 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 13 & 0 & 0 & 2 \\ 3 & 4 & 15 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[376] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -119, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 15 & 1 & 5 \\ 9 & 13 & 24 & 0 & 7 \\ 15 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[377] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -120, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 7 & 0 & 4 & 4 \\ 4 & 7 & 8 & 0 & 5 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 5 & 0 & 0 & 2 \\ 3 & 4 & 6 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[378] \ 1^{\frac{1}{12}} 4^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -139, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.69 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 11 & 0 & 3 & 4 \\ 6 & 11 & 21 & 0 & 7 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 7 & 0 & 0 & 2 \\ 3 & 4 & 9 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[379] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -142, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 15 & 7 & 5 \\ 9 & 13 & 24 & 10 & 7 \\ 15 & 24 & 38 & 16 & 12 \\ 7 & 10 & 16 & 4 & 4 \\ 5 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[380] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -146, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 3 & 4 \\ 11 & 18 & 70 & 0 & 14 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 14 & 0 & 0 & 2 \\ 3 & 4 & 16 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[381] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -151, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 7 & 0 & 7 & 4 \\ 4 & 7 & 8 & 0 & 10 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 10 & 0 & 7 & 5 \\ 3 & 4 & 6 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[382] \ 1^{\frac{1}{12}} 4^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -166, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 4 \\ 8 & 14 & 38 & 0 & 10 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 3 & 4 & 12 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[383] \ 1^{13} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 15 & 0 & 3 & 4 \\ 8 & 15 & 40 & 0 & 9 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 9 & 0 & 0 & 2 \\ 3 & 4 & 12 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[384] \ 1^{12} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 21 & 1 & 5 \\ 9 & 13 & 34 & 0 & 7 \\ 21 & 34 & 77 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[385] \ 1^{14} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 19 & 0 & 3 & 4 \\ 10 & 19 & 65 & 0 & 11 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 11 & 0 & 0 & 2 \\ 3 & 4 & 15 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[386] \ 1^{111} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 4 & 3 \\ 3 & 4 & 18 & 0 & 7 & 4 \\ 11 & 18 & 70 & 0 & 26 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 26 & 0 & 7 & 5 \\ 3 & 4 & 16 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[387] \ 1^{13} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 27 & 1 & 5 \\ 9 & 13 & 44 & 0 & 7 \\ 27 & 44 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[388] \ 1^{14} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 33 & 1 & 5 \\ 9 & 13 & 54 & 0 & 7 \\ 33 & 54 & 197 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[389] \ 1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 3 & 4 \\ 7 & 12 & 26 & 0 & 8 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 3 & 4 & 10 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[390] \ 1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 15 & 0 & 3 & 4 \\ 9 & 15 & 45 & 0 & 11 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 11 & 0 & 0 & 2 \\ 3 & 4 & 13 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[391] \ 1^{111} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 4 & 4 \\ 7 & 12 & 26 & 0 & 8 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 8 & 0 & 0 & 2 \\ 3 & 4 & 10 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[392] $1^{13} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 24 & 0 & 3 & 4 \\ 14 & 24 & 121 & 0 & 18 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 18 & 0 & 0 & 2 \\ 3 & 4 & 21 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[393] $1^{111} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 12 & 0 & 7 & 4 \\ 7 & 12 & 26 & 0 & 16 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 16 & 0 & 7 & 5 \\ 3 & 4 & 10 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[394] $21^{11} 41^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 14 & 24 & 9 & 7 & \\ 14 & 13 & 24 & 10 & 7 & \\ 24 & 24 & 38 & 16 & 12 & \\ 9 & 10 & 16 & 4 & 4 & \\ 7 & 7 & 12 & 4 & 3 & \end{pmatrix}_{b_0}$$

[395] $1^{13} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -265, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 3 & 4 \\ 10 & 18 & 61 & 0 & 12 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 12 & 0 & 0 & 2 \\ 3 & 4 & 15 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[396] $1^{111} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -270, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 4 & 3 \\ 3 & 4 & 15 & 0 & 7 & 4 \\ 9 & 15 & 45 & 0 & 21 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 21 & 0 & 7 & 5 \\ 3 & 4 & 13 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[397] $1^{13} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -286, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 3 & 4 \\ 12 & 21 & 88 & 0 & 15 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 15 & 0 & 0 & 2 \\ 3 & 4 & 18 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[398] $21^{111} 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 10 & 7 & 7 & 5 & \\ 10 & 13 & 10 & 10 & 7 & \\ 7 & 10 & 5 & 7 & 5 & \\ 7 & 10 & 7 & 4 & 4 & \\ 5 & 7 & 5 & 4 & 3 & \end{pmatrix}_{b_0}$$

[399] $1^{12} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -325, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 16 & 0 & 3 & 4 \\ 9 & 16 & 45 & 0 & 10 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 3 & 4 & 13 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[400] $1^{12} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 25 & 0 & 3 & 4 \\ 15 & 25 & 132 & 0 & 19 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 19 & 0 & 0 & 2 \\ 3 & 4 & 22 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[401] $1^{14} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 3 & 4 \\ 12 & 22 & 90 & 0 & 14 & 18 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 14 & 0 & 0 & 2 \\ 3 & 4 & 18 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[402] \ 21 \overset{1}{1^1} 41 \overset{1}{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -382, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 16 & 7 & 5 \\ 10 & 13 & 24 & 10 & 7 \\ 16 & 24 & 38 & 16 & 12 \\ 7 & 10 & 16 & 4 & 4 \\ 5 & 7 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[403] \ 1 \overset{1}{1^1} 2 \overset{1}{4} 1 \overset{1}{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -406, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 19 & 0 & 3 & 4 \\ 11 & 19 & 68 & 0 & 13 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 13 & 0 & 0 & 2 \\ 3 & 4 & 16 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[404] \ 1 \overset{1}{1^1} 2 \overset{1}{4} 1 \overset{1}{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -409, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 3 & 4 \\ 13 & 22 & 97 & 0 & 16 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 16 & 0 & 0 & 2 \\ 3 & 4 & 19 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[405] \ 1 \overset{1}{1^1} 3 \overset{1}{4} 1 \overset{1}{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 20 & 0 & 3 & 4 \\ 11 & 20 & 70 & 0 & 12 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 12 & 0 & 0 & 2 \\ 3 & 4 & 16 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[406] \ 1 \overset{1}{1^1} 4 \overset{1}{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -439, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 4 & 25 & 0 & 3 & 4 \\ 14 & 25 & 121 & 0 & 17 & 21 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 17 & 0 & 0 & 2 \\ 3 & 4 & 21 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[407] \ 1 \overset{1}{1^1} 4 \overset{1}{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -454, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 3 \\ 3 & 4 & 28 & 0 & 3 & 4 \\ 16 & 28 & 158 & 0 & 20 & 24 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 20 & 0 & 0 & 2 \\ 3 & 4 & 24 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[408] \ 1 \overset{1}{1^1} 4 \overset{1}{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 24 & 0 & 3 & 4 \\ 13 & 24 & 101 & 0 & 14 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 14 & 0 & 0 & 2 \\ 3 & 4 & 19 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[409] \ 1 \overset{1}{1^1} 3 \overset{1}{4} 1 \overset{1}{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -579, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 23 & 0 & 3 & 4 \\ 13 & 23 & 97 & 0 & 15 & 19 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 15 & 0 & 0 & 2 \\ 3 & 4 & 19 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[410] \ 1 \overset{1}{1^1} 3 \overset{1}{4} 1 \overset{1}{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -651, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 3 \\ 3 & 4 & 29 & 0 & 3 & 4 \\ 17 & 29 & 169 & 0 & 21 & 25 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 21 & 0 & 0 & 2 \\ 3 & 4 & 25 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[411] \ 1^{13} \overset{1}{4} \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -654, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.74 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 26 & 0 & 3 & 4 \\ 15 & 26 & 130 & 0 & 18 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 18 & 0 & 0 & 2 \\ 3 & 4 & 22 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[412] \ 1^{14} \overset{1}{4} \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -734, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.72 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 27 & 0 & 3 & 4 \\ 15 & 27 & 132 & 0 & 17 & 22 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 17 & 0 & 0 & 2 \\ 3 & 4 & 22 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[413] \ 1^{14} \overset{1}{4} \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -881, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.73 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 2 & 3 \\ 3 & 4 & 30 & 0 & 3 & 4 \\ 17 & 30 & 169 & 0 & 20 & 25 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 20 & 0 & 0 & 2 \\ 3 & 4 & 25 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[414] \ 1^{14} \overset{1}{4} \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -950, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.75 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 2 & 3 \\ 3 & 4 & 33 & 0 & 3 & 4 \\ 19 & 33 & 212 & 0 & 23 & 28 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 23 & 0 & 0 & 2 \\ 3 & 4 & 28 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [415] \ 1^{11} \overset{1}{4} \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -62, \ \det = 12, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 4 & 2 & 1 & 5 \\ 4 & 1 & 1 & 1 & 3 \\ 2 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 5 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

4.873 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{e}_6$

$$T = 9 \quad [0] \ 1^{11} \overset{1}{3} \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{e}_6 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -115, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, \ T_{\max} = 99, \ T \in [15, 99]$$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 9 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 9 & 7 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} \overset{1}{3} \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{e}_6 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -271, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 7 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} \overset{1}{3} \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{e}_6 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -334, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 12 & 0 & 6 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} \overset{1}{3} \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{e}_6 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -931, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 17 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 17 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 13 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

4.874 NHC(-2)(-3) + su₂ + so₈ + su₃ + f₄

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 9 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 9 & 7 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 7 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 12 & 0 & 6 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 17 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 17 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 13 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

4.875 NHC(-2)(-3) + su₂ + so₈ + su₃ + so₈

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 9 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 7 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 1 & 5 \\ 12 & 25 & 76 & 0 & 0 & 9 \\ 35 & 76 & 217 & 0 & 0 & 27 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & 0 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 1 & 5 \\ 12 & 25 & 36 & 0 & 0 & 9 \\ 17 & 36 & 49 & 0 & 0 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 13 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 10 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -133, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 1 & 5 \\ 11 & 20 & 17 & 0 & 0 & 8 \\ 9 & 17 & 13 & 0 & 0 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & 7 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 10 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 1 & 5 \\ 11 & 20 & 16 & 0 & 0 & 8 \\ 8 & 16 & 10 & 0 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & 6 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 1 & 5 \\ 12 & 25 & 56 & 0 & 0 & 9 \\ 26 & 56 & 118 & 0 & 0 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 20 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 1 & 5 \\ 12 & 25 & 72 & 0 & 0 & 9 \\ 34 & 72 & 202 & 0 & 0 & 26 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 26 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 8 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{so}_8 + \text{su}_3 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -215, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 10 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 16 & 0 & 17 & 8 \\ 9 & 16 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 5 & 8 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3 \ 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 1 & 5 \\ 11 & 20 & 50 & 0 & 0 & 8 \\ 26 & 50 & 118 & 0 & 0 & 20 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 1 & 5 \\ 11 & 20 & 33 & 0 & 0 & 8 \\ 17 & 33 & 49 & 0 & 0 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{113} 3 \ 1^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 1 & 5 \\ 12 & 25 & 108 & 0 & 0 & 9 \\ 51 & 108 & 457 & 0 & 0 & 39 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 39 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{113} 3 \ 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 43 & 1 & 1 & 5 \\ 12 & 25 & 92 & 0 & 0 & 9 \\ 43 & 92 & 325 & 0 & 0 & 33 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 33 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{114} 3 \ 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 1 & 5 \\ 12 & 25 & 112 & 0 & 0 & 9 \\ 52 & 112 & 478 & 0 & 0 & 40 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{114} 3 \ 1^{41} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 1 & 5 \\ 12 & 25 & 144 & 0 & 0 & 9 \\ 68 & 144 & 814 & 0 & 0 & 52 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 52 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{114} 3 \ 1^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 1 & 5 \\ 12 & 25 & 128 & 0 & 0 & 9 \\ 60 & 128 & 634 & 0 & 0 & 46 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 46 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -539, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 19 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 19 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -539, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 17 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -735, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 9 & 5 \\ 11 & 20 & 33 & 0 & 17 & 8 \\ 18 & 33 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 27 & 0 & 12 & 6 \\ 5 & 8 & 13 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \ 1^{\overline{111}} 4^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 165, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 9 & 6 & 1 & 5 \\ 9 & 13 & 14 & 10 & 0 & 7 \\ 9 & 14 & 13 & 9 & 0 & 7 \\ 6 & 10 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 7 & 7 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

4.876 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -16, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 \\ 4 & 5 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & 0 & -2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

4.877 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{so}_8 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 12 & 1 & 7 \\ 17 & 20 & 16 & 0 & 8 \\ 12 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 27 & 1 & 7 \\ 17 & 20 & 33 & 0 & 8 \\ 27 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 325, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 27 & 1 & 7 \\ 20 & 25 & 36 & 0 & 9 \\ 27 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{11}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 434, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 72 & 1 & 7 \\ 17 & 20 & 84 & 0 & 8 \\ 72 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{11}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 42 & 1 & 7 \\ 17 & 20 & 50 & 0 & 8 \\ 42 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{11}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 499, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 57 & 1 & 7 \\ 17 & 20 & 67 & 0 & 8 \\ 57 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[7] $1^{13}1^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 525, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 15 & 1 & 7 \\ 20 & 25 & 20 & 0 & 9 \\ 15 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [8] $1^{11}4^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 7 & 1 & 5 \\ 10 & 13 & 10 & 0 & 7 \\ 7 & 10 & 5 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

[9] $1^{11}4^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 16 & 1 & 5 \\ 10 & 13 & 24 & 0 & 7 \\ 16 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[10] $1^{11}4^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 14 & 1 & 5 \\ 10 & 13 & 20 & 0 & 7 \\ 14 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 4 \ 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -589, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 21 & 1 & 5 \\ 10 & 13 & 30 & 0 & 7 \\ 21 & 30 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 4 \ 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -709, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 23 & 1 & 5 \\ 10 & 13 & 34 & 0 & 7 \\ 23 & 34 & 77 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 4 \ 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -904, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 28 & 1 & 5 \\ 10 & 13 & 40 & 0 & 7 \\ 28 & 40 & 110 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 20 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 4 \ 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -1164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 30 & 1 & 5 \\ 10 & 13 & 44 & 0 & 7 \\ 30 & 44 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11} 4 \ 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -1709, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 37 & 1 & 5 \\ 10 & 13 & 54 & 0 & 7 \\ 37 & 54 & 197 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

4.878 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 7 \quad [3] \ 1^{11} 3 \ 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9, T_{\max} = 10, T \in [9, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112} 3 \ 1^{113} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 3 \\ 3 & 4 & 9 & 23 & 4 \\ 5 & 9 & 13 & 39 & 7 \\ 17 & 23 & 39 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{13} 3 \ 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 13 & 17 & 15 & 49 & 13 \\ 3 & 4 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{12} 3 \ 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 5 & 5 & 4 & 6 & 4 \\ 3 & 5 & 6 & 4 & 4 \\ 3 & 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

- [4] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 5 & 5 & 4 & 3 & 4 \\ 4 & 4 & 3 & 1 & 3 \\ 3 & 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [5] $1^{123} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 16 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 5 & 8 & 10 & 30 & 6 \\ 16 & 22 & 30 & 85 & 17 \\ 3 & 4 & 6 & 17 & 3 \end{pmatrix}_{b_0}$
- [6] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 3 & 5 & 4 & 3 & 4 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [9] $1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.50 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 4 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$
- [8] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.50 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 4 & 5 & 4 & 3 & 4 \\ 4 & 4 & 3 & 1 & 3 \\ 3 & 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [7] $21^1 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.50 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 3 & 1 & 3 \\ 4 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$
- [12] $1^1 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 4 & 4 \\ 3 & 4 & 0 & 4 & 3 \end{pmatrix}_{b_0}$
- [11] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 4 & 9 & 4 \\ 8 & 9 & 9 & 13 & 7 \\ 3 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$
- [10] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 8 & 9 & 13 & 21 & 7 \\ 9 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$
- [15] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -10, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 6 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$
- [13] $1^1 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -10, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 3 & 5 & 4 & 3 & 4 \\ 2 & 4 & 3 & 1 & 3 \\ 3 & 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$

[14] $1^{11} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -10, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 4 & 0 & 3 & 3 \end{pmatrix}_{b_0}$

[16] $1^{11} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -10, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 4 & 4 & 0 & 1 & 3 \\ 3 & 4 & 0 & 3 & 3 \end{pmatrix}_{b_0}$

[19] $1^{11} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -13, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.46 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 4 & 5 & 0 & 4 & 4 \\ 3 & 4 & 0 & 4 & 3 \end{pmatrix}_{b_0}$

[18] $1^{11} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -13, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.46 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 4 & 5 & 4 & 3 & 4 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$

[17] $1^{124} 3^1 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -13, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 21 & 3 \\ 3 & 4 & 8 & 31 & 4 \\ 5 & 8 & 10 & 42 & 6 \\ 21 & 31 & 42 & 172 & 24 \\ 3 & 4 & 6 & 24 & 3 \end{pmatrix}_{b_0}$

[20] $1^{11} 3^1 1^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 5 & 14 & 4 \\ 5 & 5 & 4 & 15 & 4 \\ 10 & 14 & 15 & 37 & 11 \\ 3 & 4 & 4 & 11 & 3 \end{pmatrix}_{b_0}$

[22] $1^{111} 3^1 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 5 & 9 & 13 & 15 & 7 \\ 7 & 9 & 15 & 13 & 7 \\ 3 & 4 & 7 & 7 & 3 \end{pmatrix}_{b_0}$

[21] $1^{11} 3^1 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 5 & 5 & 4 & 9 & 4 \\ 7 & 9 & 9 & 13 & 7 \\ 3 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$

[26] $1^{112} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 5 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 3 \end{pmatrix}_{b_0}$

[25] $1^{112} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 11 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$

[24] $1^{112} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 7 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 3 \end{pmatrix}_{b_0}$

- [23] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.33 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 1 & 3 \\ 3 & 4 & 0 & 3 & 3 \end{pmatrix}_{b_0}$
- [32] $1^{133} 3 1^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 18 & 3 \\ 3 & 4 & 17 & 27 & 4 \\ 11 & 17 & 49 & 81 & 13 \\ 18 & 27 & 81 & 133 & 21 \\ 3 & 4 & 13 & 21 & 3 \end{pmatrix}_{b_0}$
- [31] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 7 & 8 & 10 & 12 & 6 \\ 7 & 9 & 12 & 13 & 7 \\ 3 & 4 & 6 & 7 & 3 \end{pmatrix}_{b_0}$
- [30] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 4 & 9 & 4 \\ 5 & 9 & 9 & 13 & 7 \\ 3 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$
- [29] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 3 & 5 & 4 & 6 & 4 \\ 4 & 5 & 6 & 4 & 4 \\ 3 & 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [27] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 5 & 9 & 13 & 12 & 7 \\ 5 & 8 & 12 & 10 & 6 \\ 3 & 4 & 7 & 6 & 3 \end{pmatrix}_{b_0}$
- [28] $1^{133} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 11 & 3 \\ 3 & 4 & 12 & 17 & 4 \\ 9 & 12 & 25 & 36 & 9 \\ 11 & 17 & 36 & 49 & 13 \\ 3 & 4 & 9 & 13 & 3 \end{pmatrix}_{b_0}$
- [34] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 5 & 8 & 10 & 12 & 6 \\ 7 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 3 \end{pmatrix}_{b_0}$
- [33] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 5 & 5 & 4 & 12 & 4 \\ 11 & 13 & 12 & 28 & 10 \\ 3 & 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$
- [43] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 4 & 5 & 4 & 6 & 4 \\ 4 & 5 & 6 & 4 & 4 \\ 3 & 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [35] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 8 & 9 & 13 & 15 & 7 \\ 6 & 9 & 15 & 13 & 7 \\ 3 & 4 & 7 & 7 & 3 \end{pmatrix}_{b_0}$

- [44] $1^{111} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 10 & 14 & 9 & 37 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$
- [36] $1^{11} 13 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 2 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$
- [37] $1^{121} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 6 & 9 & 12 & 13 & 7 \\ 3 & 4 & 6 & 7 & 3 \end{pmatrix}_{b_0}$
- [38] $1^{12} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 4 & 12 & 4 \\ 8 & 13 & 12 & 28 & 10 \\ 3 & 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$
- [39] $1^{111} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 11 & 14 & 9 & 37 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$
- [40] $1^{113} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 14 & 3 \\ 3 & 4 & 14 & 22 & 4 \\ 10 & 14 & 34 & 57 & 11 \\ 14 & 22 & 57 & 85 & 17 \\ 3 & 4 & 11 & 17 & 3 \end{pmatrix}_{b_0}$
- [41] $1^{111} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 4 & 4 & 1 & 3 & 3 \\ 4 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$
- [45] $1^{113} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 1 & 12 & 3 \\ 11 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$
- [42] $1^{122} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 16 & 3 \\ 3 & 4 & 13 & 23 & 4 \\ 8 & 13 & 28 & 54 & 10 \\ 16 & 23 & 54 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0}$
- [46] $1^{12} 3 1^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 20 & 3 \\ 3 & 4 & 5 & 28 & 4 \\ 5 & 5 & 4 & 30 & 4 \\ 20 & 28 & 30 & 154 & 22 \\ 3 & 4 & 4 & 22 & 3 \end{pmatrix}_{b_0}$
- [47] $1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 5 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$

- [48] $1^{113} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 5 & 9 & 13 & 27 & 7 \\ 12 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$
- [49] $1^{14} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 3 & 5 & 4 & 18 & 4 \\ 14 & 21 & 18 & 76 & 16 \\ 3 & 4 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$
- [50] $1^{133} 3 1^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 19 & 3 \\ 3 & 4 & 17 & 27 & 4 \\ 12 & 17 & 49 & 81 & 13 \\ 19 & 27 & 81 & 133 & 21 \\ 3 & 4 & 13 & 21 & 3 \end{pmatrix}_{b_0}$$
- [51] $1^{111} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$
- [52] $1^{111} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 4 & 4 & 1 & 3 & 3 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$
- [53] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 7 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$
- [54] $1^{12} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 5 & 8 & 0 & 10 & 6 \\ 3 & 4 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$
- [55] $1^{12} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 8 & 0 & 10 & 6 \\ 3 & 4 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$
- [60] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 5 & 8 & 10 & 12 & 6 \\ 6 & 9 & 12 & 13 & 7 \\ 3 & 4 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$
- [56] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 7 & 8 & 10 & 24 & 6 \\ 12 & 17 & 24 & 49 & 13 \\ 3 & 4 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$
- [57] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 4 & 12 & 4 \\ 10 & 13 & 12 & 28 & 10 \\ 3 & 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[58] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 7 & 8 & 10 & 12 & 6 \\ 8 & 9 & 12 & 13 & 7 \\ 3 & 4 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[59] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 9 & 0 & 13 & 7 \\ 3 & 4 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

[61] $21^{113} 31^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 15 & 3 \\ 4 & 4 & 14 & 22 & 4 \\ 9 & 14 & 37 & 57 & 11 \\ 15 & 22 & 57 & 85 & 17 \\ 3 & 4 & 11 & 17 & 3 \end{pmatrix}_{b_0}$$

[62] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 5 & 8 & 10 & 18 & 6 \\ 10 & 13 & 18 & 28 & 10 \\ 3 & 4 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[68] $1^{12} 3 1^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 5 & 18 & 4 \\ 5 & 5 & 4 & 18 & 4 \\ 14 & 18 & 18 & 58 & 14 \\ 3 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[63] $1^{123} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 10 & 18 & 6 \\ 10 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[69] $1^{111} 3 1^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 8 & 9 & 13 & 24 & 7 \\ 10 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[67] $1^{11} 3 1^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 5 & 14 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 12 & 14 & 15 & 37 & 11 \\ 3 & 4 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$

[64] $1^{123} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 7 & 8 & 10 & 18 & 6 \\ 8 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[65] $1^{112} 3 1^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 12 & 3 \\ 3 & 4 & 14 & 18 & 4 \\ 11 & 14 & 37 & 48 & 11 \\ 12 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

[66] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 8 & 13 & 18 & 28 & 10 \\ 3 & 4 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[71] $1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 3 & 5 & 4 & 18 & 4 \\ 13 & 21 & 18 & 76 & 16 \\ 3 & 4 & 4 & 16 & 3 \end{pmatrix}_{b_0}$

[72] $1^{111} 1^3 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 4 & 9 & 4 \\ 5 & 9 & 9 & 13 & 7 \\ 3 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$

[70] $1^{121} 1^{22} 3^1 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 7 & 9 & 12 & 13 & 7 \\ 3 & 4 & 6 & 7 & 3 \end{pmatrix}_{b_0}$

[73] $1^{124} 1^{12} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 17 & 3 \\ 3 & 4 & 5 & 23 & 4 \\ 5 & 5 & 4 & 24 & 4 \\ 17 & 23 & 24 & 100 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[74] $1^{11} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 4 & 9 & 4 \\ 7 & 9 & 9 & 13 & 7 \\ 3 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$

[75] $1^{12} 1^{12} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 6 & 8 & 0 & 10 & 6 \\ 3 & 4 & 0 & 6 & 3 \end{pmatrix}_{b_0}$

[79] $1^{13} 1^3 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 12 & 17 & 15 & 49 & 13 \\ 3 & 4 & 4 & 13 & 3 \end{pmatrix}_{b_0}$

[76] $1^{122} 1^{43} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 13 & 3 \\ 3 & 4 & 13 & 18 & 4 \\ 11 & 13 & 28 & 42 & 10 \\ 13 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$

[77] $1^{123} 1^{43} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 12 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 11 & 13 & 28 & 39 & 10 \\ 12 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0}$

[78] $1^{112} 1^{14} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 13 & 21 & 7 \\ 11 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$

[81] $1^{132} 1^{32} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7, T_{\max} = 10, \Delta = -84, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 9 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 9 & 12 & 25 & 27 & 9 \\ 9 & 13 & 27 & 28 & 10 \\ 3 & 4 & 9 & 10 & 3 \end{pmatrix}_{b_0}$

[80] $1^{11} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.62 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 13 & 7 \\ 3 & 4 & 0 & 7 & 3 \end{pmatrix}_{b_0}$

[93] $1^{111} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 7 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$

[89] $1^{113} 3 1^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 16 & 3 \\ 3 & 4 & 9 & 22 & 4 \\ 5 & 9 & 13 & 36 & 7 \\ 16 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$

[83] $1^{111} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.50 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$

[84] $21^{111} 31^{113} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 9 & 15 & 4 \\ 5 & 4 & 9 & 17 & 4 \\ 9 & 9 & 13 & 27 & 7 \\ 15 & 17 & 27 & 49 & 13 \\ 4 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$

[85] $1^{113} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 8 & 12 & 9 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$

[86] $1^{111} 3 1^{122} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 12 & 10 & 3 \\ 3 & 4 & 14 & 14 & 4 \\ 12 & 14 & 37 & 39 & 11 \\ 10 & 14 & 39 & 37 & 11 \\ 3 & 4 & 11 & 11 & 3 \end{pmatrix}_{b_0}$

[87] $1^{133} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 12 & 12 & 4 \\ 10 & 12 & 25 & 27 & 9 \\ 8 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 3 \end{pmatrix}_{b_0}$

[94] $1^{111} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 12 & 14 & 9 & 37 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$

[95] $1^{111} 3 1^{122} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 11 & 3 \\ 3 & 4 & 14 & 14 & 4 \\ 9 & 14 & 37 & 39 & 11 \\ 11 & 14 & 39 & 37 & 11 \\ 3 & 4 & 11 & 11 & 3 \end{pmatrix}_{b_0}$

[88] $1^{111} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 9 & 14 & 9 & 37 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$

[92] $1^{132} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 8 & 12 & 25 & 27 & 9 \\ 8 & 13 & 27 & 28 & 10 \\ 3 & 4 & 9 & 10 & 3 \end{pmatrix}_{b_0}$

[91] $1^{113} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 4 & 4 & 1 & 15 & 3 \\ 15 & 22 & 15 & 85 & 17 \\ 3 & 4 & 3 & 17 & 3 \end{pmatrix}_{b_0}$

[82] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 5 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$

[90] $1^{113} 3 1^4 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 10 & 12 & 9 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$

[96] $1^{124} 3 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 20 & 3 \\ 3 & 4 & 8 & 31 & 4 \\ 5 & 8 & 10 & 42 & 6 \\ 20 & 31 & 42 & 172 & 24 \\ 3 & 4 & 6 & 24 & 3 \end{pmatrix}_{b_0}$

[97] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.56 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 9 & 0 & 13 & 7 \\ 3 & 4 & 0 & 7 & 3 \end{pmatrix}_{b_0}$

[100] $21^{112} 31^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 6 & 21 & 4 \\ 5 & 4 & 5 & 22 & 4 \\ 6 & 5 & 4 & 21 & 4 \\ 21 & 22 & 21 & 85 & 17 \\ 4 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$

[99] $1^{112} 3 1^{121} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 14 & 13 & 4 \\ 10 & 14 & 37 & 33 & 11 \\ 10 & 13 & 33 & 28 & 10 \\ 3 & 4 & 11 & 10 & 3 \end{pmatrix}_{b_0}$

[98] $1^{122} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 8 & 13 & 28 & 30 & 10 \\ 10 & 13 & 30 & 28 & 10 \\ 3 & 4 & 10 & 10 & 3 \end{pmatrix}_{b_0}$

[102] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 8 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$

[101] $1^{11} 31^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 4 & 9 & 4 \\ 6 & 9 & 9 & 13 & 7 \\ 3 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$

[103] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 4 & 9 & 4 \\ 8 & 9 & 9 & 13 & 7 \\ 3 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[105] $1^{112} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 6 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

[106] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 6 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[107] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 8 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[108] $1^{112} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 4 & 4 & 1 & 12 & 3 \\ 13 & 18 & 12 & 58 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[109] $1^{112} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 6 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

[110] $1^{123} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 15 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 6 & 8 & 10 & 30 & 6 \\ 15 & 22 & 30 & 85 & 17 \\ 3 & 4 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

[111] $1^{112} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 14 & 18 & 12 & 58 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[112] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 4 & 9 & 4 \\ 6 & 9 & 9 & 13 & 7 \\ 3 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[113] $1^{113} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 12 & 17 & 15 & 49 & 13 \\ 3 & 4 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[114] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 9 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

- [115] $21^{11}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 3 & 9 & 4 \\ 5 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 9 & 9 & 6 & 13 & 7 \\ 4 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$
- [104] $21^{112}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 4 & 9 & 8 & 4 \\ 6 & 9 & 13 & 12 & 7 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 7 & 6 & 3 \end{pmatrix}_{b_0}$
- [116] $1^{112}31^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 11 & 3 \\ 3 & 4 & 14 & 13 & 4 \\ 11 & 14 & 37 & 33 & 11 \\ 11 & 13 & 33 & 28 & 10 \\ 3 & 4 & 11 & 10 & 3 \end{pmatrix}_{b_0}$
- [117] $1^{144}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 13 & 3 \\ 3 & 4 & 16 & 21 & 4 \\ 11 & 16 & 46 & 60 & 12 \\ 13 & 21 & 60 & 76 & 16 \\ 3 & 4 & 12 & 16 & 3 \end{pmatrix}_{b_0}$
- [118] $1^{13}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 10 & 17 & 15 & 49 & 13 \\ 3 & 4 & 4 & 13 & 3 \end{pmatrix}_{b_0}$
- [120] $1^{112}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 7 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 3 \end{pmatrix}_{b_0}$
- [121] $1^{112}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 17 & 23 & 15 & 100 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0}$
- [119] $1^{112}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 5 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 3 \end{pmatrix}_{b_0}$
- [122] $1^{12}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.69 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 4 & 12 & 4 \\ 9 & 13 & 12 & 28 & 10 \\ 3 & 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$
- [123] $1^{13}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 11 & 17 & 15 & 49 & 13 \\ 3 & 4 & 4 & 13 & 3 \end{pmatrix}_{b_0}$
- [124] $1^{122}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.69 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 9 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 11 & 13 & 28 & 30 & 10 \\ 9 & 13 & 30 & 28 & 10 \\ 3 & 4 & 10 & 10 & 3 \end{pmatrix}_{b_0}$
- [125] $1^{111}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 5 & 9 & 13 & 24 & 7 \\ 10 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 11 & 14 & 9 & 37 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 10 & 14 & 9 & 37 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 14 & 0 & 37 & 11 \\ 3 & 4 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{132} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 8 & 12 & 25 & 27 & 9 \\ 9 & 13 & 27 & 28 & 10 \\ 3 & 4 & 9 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[130] \ 1^{123} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 5 & 8 & 10 & 30 & 6 \\ 13 & 22 & 30 & 85 & 17 \\ 3 & 4 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{114} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 14 & 0 & 37 & 11 \\ 3 & 4 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{112} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 4 & 9 & 4 \\ 7 & 9 & 9 & 13 & 7 \\ 3 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{112} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 7 & 9 & 13 & 21 & 7 \\ 8 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{123} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 4 & 12 & 4 \\ 8 & 13 & 12 & 28 & 10 \\ 3 & 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[135] \ 1^{123} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 4 & 12 & 4 \\ 9 & 13 & 12 & 28 & 10 \\ 3 & 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[136] \ 21^{123} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 15 & 3 \\ 4 & 4 & 8 & 22 & 4 \\ 6 & 8 & 10 & 30 & 6 \\ 15 & 22 & 30 & 85 & 17 \\ 3 & 4 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

[137] $1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 6 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[144] $1^{134} 3 1^{43} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 12 & 16 & 4 \\ 10 & 12 & 25 & 36 & 9 \\ 11 & 16 & 36 & 46 & 12 \\ 3 & 4 & 9 & 12 & 3 \end{pmatrix}_{b_0}$$

[143] $1^{112} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 7 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

[142] $1^{112} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 5 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

[141] $1^{134} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 12 & 0 & 25 & 9 \\ 3 & 4 & 0 & 9 & 3 \end{pmatrix}_{b_0}$$

[140] $1^{133} 3 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 12 & 17 & 4 \\ 8 & 12 & 25 & 36 & 9 \\ 10 & 17 & 36 & 49 & 13 \\ 3 & 4 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

[139] $1^{112} 3 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 11 & 3 \\ 3 & 4 & 14 & 13 & 4 \\ 12 & 14 & 37 & 33 & 11 \\ 11 & 13 & 33 & 28 & 10 \\ 3 & 4 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

[145] $1^{132} 3 1^{144} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 10 & 12 & 25 & 27 & 9 \\ 11 & 13 & 27 & 28 & 10 \\ 3 & 4 & 9 & 10 & 3 \end{pmatrix}_{b_0}$$

[138] $1^{132} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 12 & 0 & 25 & 9 \\ 3 & 4 & 0 & 9 & 3 \end{pmatrix}_{b_0}$$

[146] $1^{124} 3 1^{33} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 7 & 8 & 10 & 24 & 6 \\ 11 & 16 & 24 & 46 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[147] $1^{132} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 10 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 9 & 12 & 25 & 27 & 9 \\ 10 & 13 & 27 & 28 & 10 \\ 3 & 4 & 9 & 10 & 3 \end{pmatrix}_{b_0}$$

[148] $1^{112} 3 1^{121} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 4 & 14 & 13 & 4 \\ 9 & 14 & 37 & 33 & 11 \\ 8 & 13 & 33 & 28 & 10 \\ 3 & 4 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

[149] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 13 & 0 & 28 & 10 \\ 3 & 4 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[150] $21^{11} 31^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -62, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 15 & 17 & 12 & 49 & 13 \\ 4 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[151] $1^{12} 3 1^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 4 & 12 & 4 \\ 11 & 13 & 12 & 28 & 10 \\ 3 & 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[152] $1^{111} 3 1^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 7 & 9 & 13 & 24 & 7 \\ 9 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[153] $1^{13} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 13 & 17 & 15 & 49 & 13 \\ 3 & 4 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[154] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 7 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[155] $1^{112} 3 1^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 20 & 3 \\ 3 & 4 & 9 & 28 & 4 \\ 5 & 9 & 13 & 48 & 7 \\ 20 & 28 & 48 & 154 & 22 \\ 3 & 4 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$

[156] $1^{111} 3 1^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 5 & 9 & 13 & 33 & 7 \\ 13 & 19 & 33 & 73 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[157] $21^{11} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 6 & 9 & 4 \\ 5 & 4 & 5 & 9 & 4 \\ 6 & 5 & 4 & 9 & 4 \\ 9 & 9 & 9 & 13 & 7 \\ 4 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[158] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 7 & 8 & 10 & 18 & 6 \\ 10 & 13 & 18 & 28 & 10 \\ 3 & 4 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[159] $1^{13} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 12 & 0 & 25 & 9 \\ 3 & 4 & 0 & 9 & 3 \end{pmatrix}_{b_0}$$

[160] $1^{13} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 5 & 14 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 11 & 14 & 15 & 37 & 11 \\ 3 & 4 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$

[163] $1^{12} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 5 & 9 & 13 & 30 & 7 \\ 13 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

[162] $1^{12} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 11 & 3 \\ 3 & 4 & 18 & 17 & 4 \\ 11 & 18 & 58 & 54 & 14 \\ 11 & 17 & 54 & 49 & 13 \\ 3 & 4 & 14 & 13 & 3 \end{pmatrix}_{b_0}$$

[161] $1^{14} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 5 & 9 & 13 & 33 & 7 \\ 14 & 21 & 33 & 76 & 16 \\ 3 & 4 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[166] $2 1^{13} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 12 & 4 \\ 5 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 12 & 13 & 9 & 28 & 10 \\ 4 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[165] $1^{12} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 4 & 12 & 4 \\ 10 & 13 & 12 & 28 & 10 \\ 3 & 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[164] $1^{12} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 10 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[167] $1^{12} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -69, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 10 & 3 \\ 3 & 4 & 14 & 13 & 4 \\ 11 & 14 & 37 & 33 & 11 \\ 10 & 13 & 33 & 28 & 10 \\ 3 & 4 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

[173] $1^{12} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 5 & 8 & 10 & 30 & 6 \\ 15 & 22 & 30 & 85 & 17 \\ 3 & 4 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

[172] $1^{11} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 14 & 0 & 37 & 11 \\ 3 & 4 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

[171] $1^{11} 3^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 14 & 0 & 37 & 11 \\ 3 & 4 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

[170] $1^{12} 3^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 9 & 3 \\ 3 & 4 & 14 & 13 & 4 \\ 10 & 14 & 37 & 33 & 11 \\ 9 & 13 & 33 & 28 & 10 \\ 3 & 4 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

[169] $1^{12} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 8 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 6 & 8 & 10 & 6 \\ 3 & 4 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

[168] $1^{12} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 11 & 13 & 18 & 28 & 10 \\ 3 & 4 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[175] $1^{12} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 5 & 9 & 13 & 21 & 7 \\ 9 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[174] $1^{13} 3^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 16 & 3 \\ 3 & 4 & 14 & 22 & 4 \\ 11 & 14 & 37 & 57 & 11 \\ 16 & 22 & 57 & 85 & 17 \\ 3 & 4 & 11 & 17 & 3 \end{pmatrix}_{b_0}$$

[178] $1^{12} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 13 & 0 & 28 & 10 \\ 3 & 4 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[177] $1^{11} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 5 & 14 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 9 & 14 & 15 & 37 & 11 \\ 3 & 4 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$

[176] $1^{12} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 5 & 18 & 4 \\ 4 & 5 & 4 & 18 & 4 \\ 11 & 18 & 18 & 58 & 14 \\ 3 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[179] $1^{14} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 4 & 4 & 1 & 12 & 3 \\ 11 & 16 & 12 & 46 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

[180] $1^{11} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 5 & 9 & 13 & 15 & 7 \\ 6 & 9 & 15 & 13 & 7 \\ 3 & 4 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[182] \ 1^{113} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -76, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.84 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 7 & 9 & 13 & 27 & 7 \\ 11 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[181] \ 1^{122} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -76, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.84 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 5 & 8 & 10 & 12 & 6 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[183] \ 1^{113} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -76, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.84 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 9 & 12 & 9 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[184] \ 2 1^{122} 3 1^{124} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -76, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.74 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 18 & 24 & 4 \\ 5 & 4 & 18 & 26 & 4 \\ 18 & 18 & 58 & 84 & 14 \\ 24 & 26 & 84 & 118 & 20 \\ 4 & 4 & 14 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[185] \ 1^{122} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -76, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.84 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 7 & 8 & 10 & 12 & 6 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[186] \ 1^{114} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -76, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.84 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 4 & 4 & 1 & 15 & 3 \\ 14 & 21 & 15 & 76 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[187] \ 1^{113} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -76, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.84 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 9 & 12 & 9 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[189] \ 1^{114} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -79, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 14 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 7 & 9 & 13 & 33 & 7 \\ 14 & 21 & 33 & 76 & 16 \\ 3 & 4 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[188] \ 1^{112} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -79, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.71 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 13 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 8 & 9 & 13 & 30 & 7 \\ 13 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[191] \ 2 1^{111} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -80, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.60 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[190] \ 1^{133} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -80, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.6$$

$$T_{\min} = 9.80 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 13 & 3 \\ 3 & 4 & 17 & 17 & 4 \\ 11 & 17 & 49 & 51 & 13 \\ 13 & 17 & 51 & 49 & 13 \\ 3 & 4 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$

[192] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 13 & 0 & 28 & 10 \\ 3 & 4 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[194] $1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -82, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 5 & 14 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 10 & 14 & 15 & 37 & 11 \\ 3 & 4 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$

[193] $1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -82, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 14 & 18 & 4 \\ 10 & 14 & 37 & 48 & 11 \\ 11 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

[195] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 5 & 8 & 10 & 24 & 6 \\ 13 & 17 & 24 & 49 & 13 \\ 3 & 4 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[199] $1^{12} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 7 & 8 & 10 & 12 & 6 \\ 7 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 3 \end{pmatrix}_{b_0}$$

[197] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 10 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[200] $1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 14 & 3 \\ 3 & 4 & 18 & 18 & 4 \\ 12 & 18 & 58 & 60 & 14 \\ 14 & 18 & 60 & 58 & 14 \\ 3 & 4 & 14 & 14 & 3 \end{pmatrix}_{b_0}$$

[198] $1^{12} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 5 & 8 & 10 & 12 & 6 \\ 5 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 3 \end{pmatrix}_{b_0}$$

[196] $1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 12 & 3 \\ 3 & 4 & 18 & 17 & 4 \\ 12 & 18 & 58 & 54 & 14 \\ 12 & 17 & 54 & 49 & 13 \\ 3 & 4 & 14 & 13 & 3 \end{pmatrix}_{b_0}$$

[201] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.64 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 13 & 0 & 28 & 10 \\ 3 & 4 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[202] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 15 & 3 \\ 3 & 4 & 8 & 21 & 4 \\ 7 & 8 & 10 & 30 & 6 \\ 15 & 21 & 30 & 76 & 16 \\ 3 & 4 & 6 & 16 & 3 \end{pmatrix}_{b_0}$$

[203] $1^{122} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 5 & 8 & 10 & 18 & 6 \\ 9 & 13 & 18 & 28 & 10 \\ 3 & 4 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[204] $1^{123} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 5 & 8 & 10 & 30 & 6 \\ 14 & 22 & 30 & 85 & 17 \\ 3 & 4 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

[212] $1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 10 & 12 & 9 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[205] $21^{111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.59 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 6 & 9 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[206] $1^{113} 3 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 15 & 3 \\ 3 & 4 & 14 & 22 & 4 \\ 9 & 14 & 37 & 57 & 11 \\ 15 & 22 & 57 & 85 & 17 \\ 3 & 4 & 11 & 17 & 3 \end{pmatrix}_{b_0}$$

[207] $1^{113} 3 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 13 & 3 \\ 3 & 4 & 14 & 22 & 4 \\ 9 & 14 & 37 & 57 & 11 \\ 13 & 22 & 57 & 85 & 17 \\ 3 & 4 & 11 & 17 & 3 \end{pmatrix}_{b_0}$$

[208] $1^{133} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 13 & 3 \\ 3 & 4 & 12 & 17 & 4 \\ 10 & 12 & 25 & 36 & 9 \\ 13 & 17 & 36 & 49 & 13 \\ 3 & 4 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

[209] $1^{111} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 5 & 14 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 12 & 14 & 15 & 37 & 11 \\ 3 & 4 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$

[210] $21^{132} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 9 & 3 \\ 4 & 4 & 12 & 13 & 4 \\ 9 & 12 & 25 & 27 & 9 \\ 9 & 13 & 27 & 28 & 10 \\ 3 & 4 & 9 & 10 & 3 \end{pmatrix}_{b_0}$$

[211] $1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 8 & 12 & 9 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[213] $1^{122} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 5 & 8 & 10 & 18 & 6 \\ 8 & 13 & 18 & 28 & 10 \\ 3 & 4 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[214] $1^{112} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 11 & 18 & 12 & 58 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[216] $1^{123} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 15 & 3 \\ 3 & 4 & 18 & 22 & 4 \\ 14 & 18 & 58 & 72 & 14 \\ 15 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0}$$

[215] $1^{12} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 5 & 18 & 4 \\ 3 & 5 & 4 & 18 & 4 \\ 14 & 18 & 18 & 58 & 14 \\ 3 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[217] $1^{111} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 8 & 9 & 13 & 15 & 7 \\ 7 & 9 & 15 & 13 & 7 \\ 3 & 4 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[218] $1^{123} 3 1^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 13 & 3 \\ 3 & 4 & 18 & 17 & 4 \\ 13 & 18 & 58 & 54 & 14 \\ 13 & 17 & 54 & 49 & 13 \\ 3 & 4 & 14 & 13 & 3 \end{pmatrix}_{b_0}$$

[219] $1^{113} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 13 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 8 & 9 & 13 & 27 & 7 \\ 13 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[220] $1^{122} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -93, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 7 & 8 & 10 & 18 & 6 \\ 11 & 13 & 18 & 28 & 10 \\ 3 & 4 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[221] $1^{122} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 12 & 3 \\ 3 & 4 & 13 & 18 & 4 \\ 10 & 13 & 28 & 42 & 10 \\ 12 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[222] $1^{113} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 16 & 3 \\ 3 & 4 & 9 & 22 & 4 \\ 8 & 9 & 13 & 36 & 7 \\ 16 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

[223] $1^{122} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 9 & 13 & 18 & 28 & 10 \\ 3 & 4 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[224] $2 1^{123} 3 1^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 12 & 3 \\ 4 & 4 & 18 & 17 & 4 \\ 12 & 18 & 58 & 54 & 14 \\ 12 & 17 & 54 & 49 & 13 \\ 3 & 4 & 14 & 13 & 3 \end{pmatrix}_{b_0}$$

[225] $1^{112} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 9 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[226] $1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 1 & 12 & 3 \\ 12 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[227] $1^{112} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 4 & 4 & 1 & 12 & 3 \\ 14 & 18 & 12 & 58 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[228] $1^{112} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 13 & 18 & 12 & 58 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[229] $1^{123} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 16 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 6 & 8 & 10 & 30 & 6 \\ 16 & 22 & 30 & 85 & 17 \\ 3 & 4 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

[230] $1^{13} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 5 & 22 & 4 \\ 4 & 5 & 4 & 21 & 4 \\ 14 & 22 & 21 & 85 & 17 \\ 3 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

[231] $1^{123} 3 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 22 & 3 \\ 3 & 4 & 13 & 32 & 4 \\ 8 & 13 & 28 & 75 & 10 \\ 22 & 32 & 75 & 193 & 25 \\ 3 & 4 & 10 & 25 & 3 \end{pmatrix}_{b_0}$$

[232] $1^{112} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 8 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[233] $1^{112} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 16 & 23 & 15 & 100 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

[234] $1^{112} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 5 & 14 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 10 & 14 & 15 & 37 & 11 \\ 3 & 4 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$

[237] $1^{122} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 13 & 3 \\ 3 & 4 & 13 & 18 & 4 \\ 8 & 13 & 28 & 42 & 10 \\ 13 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[235] $1^{111} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -97$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.76 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 6 & 9 & 13 & 24 & 7 \\ 11 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[236] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -97$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.76 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 8 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[238] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -98$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.76 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 11 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[239] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -99$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.65 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 6 & 9 & 13 & 15 & 7 \\ 7 & 9 & 15 & 13 & 7 \\ 3 & 4 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[240] $1^{11} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -99$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 15 & 19 & 0 & 73 & 15 \\ 3 & 4 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[241] $1^{123} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -99$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.3$
 $T_{\min} = 9.81 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 10 & 13 & 28 & 39 & 10 \\ 11 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

[242] $1^{111} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -99$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 15 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 8 & 9 & 13 & 33 & 7 \\ 15 & 19 & 33 & 73 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[243] $1^{111} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -99$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 5 & 9 & 13 & 24 & 7 \\ 9 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[249] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -100$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 8 & 9 & 13 & 15 & 7 \\ 8 & 9 & 15 & 13 & 7 \\ 3 & 4 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[244] $1^{11} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -100$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 5 & 19 & 4 \\ 3 & 5 & 4 & 21 & 4 \\ 15 & 19 & 21 & 73 & 15 \\ 3 & 4 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[245] $1^{124} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -100$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 17 & 3 \\ 3 & 4 & 8 & 26 & 4 \\ 6 & 8 & 10 & 36 & 6 \\ 17 & 26 & 36 & 118 & 20 \\ 3 & 4 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

[246] $1^{122} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 17 & 3 \\ 3 & 4 & 13 & 23 & 4 \\ 9 & 13 & 28 & 54 & 10 \\ 17 & 23 & 54 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[247] $1^{122} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 3 \end{pmatrix}_{b_0}$$

[248] $1^{14} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 16 & 0 & 46 & 12 \\ 3 & 4 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[250] $1^{14} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 17 & 3 \\ 3 & 4 & 5 & 26 & 4 \\ 4 & 5 & 4 & 24 & 4 \\ 17 & 26 & 24 & 118 & 20 \\ 3 & 4 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[251] $1^{11} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 5 & 14 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 11 & 14 & 15 & 37 & 11 \\ 3 & 4 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$

[252] $1^{134} 3 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 13 & 3 \\ 3 & 4 & 22 & 21 & 4 \\ 13 & 22 & 85 & 81 & 17 \\ 13 & 21 & 81 & 76 & 16 \\ 3 & 4 & 17 & 16 & 3 \end{pmatrix}_{b_0}$$

[253] $1^{11} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 19 & 0 & 73 & 15 \\ 3 & 4 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[254] $1^{133} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 15 & 3 \\ 3 & 4 & 17 & 22 & 4 \\ 13 & 17 & 49 & 66 & 13 \\ 15 & 22 & 66 & 85 & 17 \\ 3 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

[255] $1^{132} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 17 & 18 & 4 \\ 10 & 17 & 49 & 54 & 13 \\ 11 & 18 & 54 & 58 & 14 \\ 3 & 4 & 13 & 14 & 3 \end{pmatrix}_{b_0}$$

[256] $1^{133} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 16 & 3 \\ 3 & 4 & 22 & 22 & 4 \\ 14 & 22 & 85 & 87 & 17 \\ 16 & 22 & 87 & 85 & 17 \\ 3 & 4 & 17 & 17 & 3 \end{pmatrix}_{b_0}$$

[260] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 10 & 13 & 18 & 28 & 10 \\ 3 & 4 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[259] $1^{123} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 10 & 18 & 6 \\ 9 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[258] $1^{112} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 8 & 9 & 13 & 21 & 7 \\ 10 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[257] $1^{123} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 7 & 8 & 10 & 18 & 6 \\ 9 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[262] $1^{14} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 16 & 0 & 46 & 12 \\ 3 & 4 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[263] $1^{113} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 10 & 12 & 9 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[264] $1^{111} 3 1^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 7 & 9 & 13 & 24 & 7 \\ 12 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[261] $1^{113} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 8 & 12 & 9 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[265] $1^{111} 3 1^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 8 & 9 & 13 & 24 & 7 \\ 11 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[266] $21^{112} 31^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -110, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 15 & 21 & 4 \\ 5 & 4 & 17 & 22 & 4 \\ 15 & 17 & 49 & 66 & 13 \\ 21 & 22 & 66 & 85 & 17 \\ 4 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

[267] $21^{111} 31^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 6 & 3 \\ 4 & 4 & 14 & 9 & 4 \\ 9 & 14 & 37 & 24 & 11 \\ 6 & 9 & 24 & 13 & 7 \\ 3 & 4 & 11 & 7 & 3 \end{pmatrix}_{b_0}$$

[268] $21^{111} 31^{143} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 18 & 15 & 4 \\ 5 & 4 & 21 & 17 & 4 \\ 18 & 21 & 76 & 63 & 16 \\ 15 & 17 & 63 & 49 & 13 \\ 4 & 4 & 16 & 13 & 3 \end{pmatrix}_{b_0}$$

[269] $1^{113}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 5 & 9 & 13 & 27 & 7 \\ 11 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[271] $1^{112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 17 & 0 & 49 & 13 \\ 3 & 4 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[273] $1^{122}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 15 & 3 \\ 3 & 4 & 13 & 23 & 4 \\ 8 & 13 & 28 & 54 & 10 \\ 15 & 23 & 54 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[272] $1^{112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 11 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[270] $1^{112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 5 & 9 & 13 & 21 & 7 \\ 8 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[274] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.69 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 6 & 9 & 13 & 15 & 7 \\ 6 & 9 & 15 & 13 & 7 \\ 3 & 4 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[277] $1^{123}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 5 & 18 & 4 \\ 3 & 5 & 4 & 18 & 4 \\ 11 & 18 & 18 & 58 & 14 \\ 3 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[276] $1^{122}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 11 & 3 \\ 3 & 4 & 13 & 18 & 4 \\ 9 & 13 & 28 & 42 & 10 \\ 11 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[275] $1^{123}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 5 & 18 & 4 \\ 3 & 5 & 4 & 18 & 4 \\ 13 & 18 & 18 & 58 & 14 \\ 3 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[284] $1^{114}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 4 & 4 & 1 & 12 & 3 \\ 12 & 16 & 12 & 46 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

[283] $21^{123}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 18 & 4 \\ 5 & 4 & 4 & 18 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 18 & 18 & 12 & 58 & 14 \\ 4 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[282] $1^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 12 & 16 & 12 & 46 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

[281] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 11 & 18 & 12 & 58 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[280] $1^{112}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 15 & 23 & 15 & 100 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

[279] $1^{114}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 4 & 4 & 1 & 18 & 3 \\ 18 & 26 & 18 & 118 & 20 \\ 3 & 4 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

[278] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 12 & 18 & 12 & 58 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[285] $1^{12}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 5 & 18 & 4 \\ 4 & 5 & 4 & 18 & 4 \\ 12 & 18 & 18 & 58 & 14 \\ 3 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[286] $1^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 16 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 4 & 4 & 1 & 15 & 3 \\ 16 & 22 & 15 & 85 & 17 \\ 3 & 4 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

[288] $1^{12}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 18 & 0 & 58 & 14 \\ 3 & 4 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[289] $1^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 9 & 12 & 9 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[287] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 13 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[290] $1^{112}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -121, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 17 & 3 \\ 3 & 4 & 9 & 23 & 4 \\ 8 & 9 & 13 & 39 & 7 \\ 17 & 23 & 39 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[291] $1^{123} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -122, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 16 & 3 \\ 3 & 4 & 13 & 22 & 4 \\ 11 & 13 & 28 & 51 & 10 \\ 16 & 22 & 51 & 85 & 17 \\ 3 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0}$$

[292] $1^{123} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 13 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 7 & 8 & 10 & 24 & 6 \\ 13 & 17 & 24 & 49 & 13 \\ 3 & 4 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[293] $1^{122} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 17 & 3 \\ 3 & 4 & 13 & 23 & 4 \\ 10 & 13 & 28 & 54 & 10 \\ 17 & 23 & 54 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[295] $1^{124} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 17 & 3 \\ 3 & 4 & 18 & 26 & 4 \\ 13 & 18 & 58 & 84 & 14 \\ 17 & 26 & 84 & 118 & 20 \\ 3 & 4 & 14 & 20 & 3 \end{pmatrix}_{b_0}$$

[296] $1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 7 & 9 & 13 & 15 & 7 \\ 9 & 13 & 15 & 13 & 7 \\ 3 & 4 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[297] $1^{134} 3 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 14 & 3 \\ 3 & 4 & 22 & 21 & 4 \\ 14 & 22 & 85 & 81 & 17 \\ 14 & 21 & 81 & 76 & 16 \\ 3 & 4 & 17 & 16 & 3 \end{pmatrix}_{b_0}$$

[298] $1^{132} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 12 & 3 \\ 3 & 4 & 17 & 18 & 4 \\ 11 & 17 & 49 & 54 & 13 \\ 12 & 18 & 54 & 58 & 14 \\ 3 & 4 & 13 & 14 & 3 \end{pmatrix}_{b_0}$$

[299] $1^{111} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 6 & 9 & 13 & 33 & 7 \\ 14 & 19 & 33 & 73 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[300] $1^{123} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 6 & 8 & 10 & 24 & 6 \\ 11 & 17 & 24 & 49 & 13 \\ 3 & 4 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[294] $1^{113} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 13 & 27 & 7 \\ 10 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[307] $1^{112} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 13 & 3 \\ 3 & 4 & 14 & 18 & 4 \\ 9 & 14 & 37 & 48 & 11 \\ 13 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

- [303] $1^{123} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -126$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.3$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 8 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$
- [304] $1^{112} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -126$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 12 & 14 & 3 \\ 3 & 4 & 14 & 18 & 4 \\ 12 & 14 & 37 & 48 & 11 \\ 14 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$
- [305] $1^{12} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -126$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 5 & 18 & 4 \\ 3 & 5 & 4 & 18 & 4 \\ 12 & 18 & 18 & 58 & 14 \\ 3 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$
- [301] $1^{133} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -126$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.6$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 12 & 3 \\ 3 & 4 & 12 & 17 & 4 \\ 8 & 12 & 25 & 36 & 9 \\ 12 & 17 & 36 & 49 & 13 \\ 3 & 4 & 9 & 13 & 3 \end{pmatrix}_{b_0}$
- [302] $1^{123} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -126$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.3$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 10 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$
- [306] $1^{111} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -126$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 12 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 8 & 9 & 13 & 24 & 7 \\ 12 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$
- [308] $1^{124} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -126$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 8 & 21 & 4 \\ 6 & 8 & 10 & 30 & 6 \\ 13 & 21 & 30 & 76 & 16 \\ 3 & 4 & 6 & 16 & 3 \end{pmatrix}_{b_0}$
- [309] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -127$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.66 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 9 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$
- [311] $21^{111} 31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -128$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 9 & 9 & 3 \\ 4 & 4 & 14 & 14 & 4 \\ 9 & 14 & 37 & 39 & 11 \\ 9 & 14 & 39 & 37 & 11 \\ 3 & 4 & 11 & 11 & 3 \end{pmatrix}_{b_0}$
- [310] $21^{112} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -128$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.75 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 4 & 4 & 8 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 6 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 3 \end{pmatrix}_{b_0}$
- [313] $1^{134} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -129$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 9.82 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 14 & 3 \\ 3 & 4 & 12 & 21 & 4 \\ 9 & 12 & 25 & 45 & 9 \\ 14 & 21 & 45 & 76 & 16 \\ 3 & 4 & 9 & 16 & 3 \end{pmatrix}_{b_0}$
- [314] $1^{133} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -129$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.6$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 12 & 3 \\ 3 & 4 & 12 & 17 & 4 \\ 9 & 12 & 25 & 36 & 9 \\ 12 & 17 & 36 & 49 & 13 \\ 3 & 4 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[312] \ 1^{111} 3 \ 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 6 & 9 & 13 & 24 & 7 \\ 9 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[315] \ 1^{112} 3 \ 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.66 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 13 & 21 & 7 \\ 10 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[316] \ 1^{123} 3 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 10 & 18 & 6 \\ 8 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[317] \ 1^{13} 3 \ 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 5 & 22 & 4 \\ 3 & 5 & 4 & 21 & 4 \\ 16 & 22 & 21 & 85 & 17 \\ 3 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[318] \ 1^{123} 3 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 7 & 8 & 10 & 18 & 6 \\ 10 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[319] \ 1^{133} 3 \ 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 16 & 3 \\ 3 & 4 & 17 & 22 & 4 \\ 11 & 17 & 49 & 66 & 13 \\ 16 & 22 & 66 & 85 & 17 \\ 3 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[320] \ 1^{113} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -133, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 10 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[321] \ 1^{112} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -133, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 10 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[322] \ 1^{143} 3 \ 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 15 & 3 \\ 3 & 4 & 21 & 22 & 4 \\ 15 & 21 & 76 & 81 & 16 \\ 15 & 22 & 81 & 85 & 17 \\ 3 & 4 & 16 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[324] \ 1^{13} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 17 & 0 & 49 & 13 \\ 3 & 4 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[323] \ 1^{132} 3 \ 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 13 & 3 \\ 3 & 4 & 17 & 18 & 4 \\ 12 & 17 & 49 & 54 & 13 \\ 13 & 18 & 54 & 58 & 14 \\ 3 & 4 & 13 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[326] \ 1^{113} 3 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 15 & 3 \\ 3 & 4 & 14 & 22 & 4 \\ 10 & 14 & 37 & 57 & 11 \\ 15 & 22 & 57 & 85 & 17 \\ 3 & 4 & 11 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[325] \ 1^{114} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 5 & 9 & 13 & 33 & 7 \\ 13 & 21 & 33 & 76 & 16 \\ 3 & 4 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[327] \ 21^{112} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 6 & 18 & 4 \\ 5 & 4 & 5 & 18 & 4 \\ 6 & 5 & 4 & 18 & 4 \\ 18 & 18 & 18 & 58 & 14 \\ 4 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[328] \ 1^{12} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 18 & 0 & 58 & 14 \\ 3 & 4 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[329] \ 1^{113} 3 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 14 & 3 \\ 3 & 4 & 14 & 22 & 4 \\ 9 & 14 & 37 & 57 & 11 \\ 14 & 22 & 57 & 85 & 17 \\ 3 & 4 & 11 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[336] \ 1^{124} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 14 & 3 \\ 3 & 4 & 13 & 21 & 4 \\ 10 & 13 & 28 & 48 & 10 \\ 14 & 21 & 48 & 76 & 16 \\ 3 & 4 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[335] \ 1^{133} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 3 \\ 3 & 4 & 12 & 17 & 4 \\ 8 & 12 & 25 & 36 & 9 \\ 11 & 17 & 36 & 49 & 13 \\ 3 & 4 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[334] \ 1^{123} 3 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 13 & 3 \\ 3 & 4 & 18 & 17 & 4 \\ 14 & 18 & 58 & 54 & 14 \\ 13 & 17 & 54 & 49 & 13 \\ 3 & 4 & 14 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[333] \ 1^{12} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 18 & 0 & 58 & 14 \\ 3 & 4 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[332] \ 21^{111} 3 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.68 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 9 & 12 & 4 \\ 5 & 4 & 9 & 13 & 4 \\ 9 & 9 & 13 & 21 & 7 \\ 12 & 13 & 21 & 28 & 10 \\ 4 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[331] \ 1^{111} 3 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 7 & 9 & 13 & 24 & 7 \\ 10 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[330] $1^{113} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 14 & 3 \\ 3 & 4 & 9 & 22 & 4 \\ 7 & 9 & 13 & 36 & 7 \\ 14 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

[338] $1^{112} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -138, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 12 & 18 & 12 & 58 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[337] $1^{114} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -138, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 11 & 16 & 12 & 46 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

[344] $1^{13} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 5 & 22 & 4 \\ 3 & 5 & 4 & 21 & 4 \\ 13 & 22 & 21 & 85 & 17 \\ 3 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

[343] $1^{113} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.69 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 13 & 27 & 7 \\ 13 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[339] $1^{111} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 14 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 7 & 9 & 13 & 33 & 7 \\ 14 & 19 & 33 & 73 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[340] $1^{112} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 16 & 3 \\ 3 & 4 & 9 & 23 & 4 \\ 5 & 9 & 13 & 39 & 7 \\ 16 & 23 & 39 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[341] $1^{122} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 16 & 3 \\ 3 & 4 & 13 & 23 & 4 \\ 9 & 13 & 28 & 54 & 10 \\ 16 & 23 & 54 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[345] $1^{114} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.69 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 17 & 0 & 49 & 13 \\ 3 & 4 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[342] $1^{113} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 5 & 9 & 13 & 27 & 7 \\ 10 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[348] $21^{112} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 6 & 15 & 4 \\ 5 & 4 & 5 & 14 & 4 \\ 6 & 5 & 4 & 15 & 4 \\ 15 & 14 & 15 & 37 & 11 \\ 4 & 4 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$

[347] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 1 & 12 & 3 \\ 13 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[346] $21^{113} 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 6 & 30 & 4 \\ 5 & 4 & 5 & 31 & 4 \\ 6 & 5 & 4 & 30 & 4 \\ 30 & 31 & 30 & 172 & 24 \\ 4 & 4 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[349] $1^{124} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 17 & 3 \\ 3 & 4 & 13 & 26 & 4 \\ 10 & 13 & 28 & 60 & 10 \\ 17 & 26 & 60 & 118 & 20 \\ 3 & 4 & 10 & 20 & 3 \end{pmatrix}_{b_0}$$

[351] $1^{112} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 13 & 21 & 7 \\ 8 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[350] $1^{111} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 6 & 9 & 13 & 24 & 7 \\ 10 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[352] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -143, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 5 & 8 & 10 & 24 & 6 \\ 10 & 17 & 24 & 49 & 13 \\ 3 & 4 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[357] $1^{112} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 15 & 3 \\ 3 & 4 & 9 & 23 & 4 \\ 7 & 9 & 13 & 39 & 7 \\ 15 & 23 & 39 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[353] $1^{114} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 18 & 3 \\ 3 & 4 & 9 & 26 & 4 \\ 5 & 9 & 13 & 42 & 7 \\ 18 & 26 & 42 & 118 & 20 \\ 3 & 4 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[354] $1^{112} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 7 & 9 & 13 & 21 & 7 \\ 9 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[355] $21^{111} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.69 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 9 & 9 & 4 \\ 5 & 4 & 9 & 9 & 4 \\ 9 & 9 & 13 & 15 & 7 \\ 9 & 9 & 15 & 13 & 7 \\ 4 & 4 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[356] $1^{111} 3 1^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 15 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 7 & 9 & 13 & 33 & 7 \\ 15 & 19 & 33 & 73 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[358] $1^{112} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 8 & 9 & 13 & 21 & 7 \\ 11 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[359] $1^{112} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 16 & 23 & 15 & 100 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

[360] $1^{12} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 5 & 18 & 4 \\ 4 & 5 & 4 & 18 & 4 \\ 13 & 18 & 18 & 58 & 14 \\ 3 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[361] $1^{13} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 5 & 22 & 4 \\ 4 & 5 & 4 & 21 & 4 \\ 15 & 22 & 21 & 85 & 17 \\ 3 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

[362] $1^{124} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 7 & 8 & 10 & 24 & 6 \\ 12 & 16 & 24 & 46 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[364] $1^{124} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 5 & 8 & 10 & 24 & 6 \\ 12 & 16 & 24 & 46 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[363] $1^{112} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 7 & 9 & 13 & 21 & 7 \\ 11 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[368] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 10 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[367] $1^{123} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 9 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[366] $1^{113} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 5 & 19 & 4 \\ 4 & 5 & 4 & 21 & 4 \\ 14 & 19 & 21 & 73 & 15 \\ 3 & 4 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[365] $1^{134} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 20 & 3 \\ 3 & 4 & 17 & 31 & 4 \\ 10 & 17 & 49 & 93 & 13 \\ 20 & 31 & 93 & 172 & 24 \\ 3 & 4 & 13 & 24 & 3 \end{pmatrix}_{b_0}$$

[373] $1^{111} 3^1 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 9 & 14 & 4 \\ 7 & 9 & 13 & 24 & 7 \\ 11 & 14 & 24 & 37 & 11 \\ 3 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[374] $1^{134} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 14 & 3 \\ 3 & 4 & 17 & 21 & 4 \\ 13 & 17 & 49 & 63 & 13 \\ 14 & 21 & 63 & 76 & 16 \\ 3 & 4 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

[369] $1^{123} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 5 & 8 & 10 & 24 & 6 \\ 12 & 17 & 24 & 49 & 13 \\ 3 & 4 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[370] $1^{13} 3^1 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 5 & 22 & 4 \\ 3 & 5 & 4 & 21 & 4 \\ 15 & 22 & 21 & 85 & 17 \\ 3 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

[371] $1^{133} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 13 & 3 \\ 3 & 4 & 12 & 17 & 4 \\ 9 & 12 & 25 & 36 & 9 \\ 13 & 17 & 36 & 49 & 13 \\ 3 & 4 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

[372] $1^{124} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 15 & 3 \\ 3 & 4 & 13 & 21 & 4 \\ 8 & 13 & 28 & 48 & 10 \\ 15 & 21 & 48 & 76 & 16 \\ 3 & 4 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[377] $1^{123} 3^1 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 16 & 3 \\ 3 & 4 & 13 & 22 & 4 \\ 8 & 13 & 28 & 51 & 10 \\ 16 & 22 & 51 & 85 & 17 \\ 3 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0}$$

[376] $21^{111} 31^{132} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 15 & 12 & 4 \\ 5 & 4 & 17 & 13 & 4 \\ 15 & 17 & 49 & 39 & 13 \\ 12 & 13 & 39 & 28 & 10 \\ 4 & 4 & 13 & 10 & 3 \end{pmatrix}_{b_0}$$

[375] $1^{113} 3^1 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 16 & 3 \\ 3 & 4 & 14 & 22 & 4 \\ 10 & 14 & 37 & 57 & 11 \\ 16 & 22 & 57 & 85 & 17 \\ 3 & 4 & 11 & 17 & 3 \end{pmatrix}_{b_0}$$

[378] $1^{12} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -153, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 15 & 23 & 0 & 100 & 18 \\ 3 & 4 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[379] $1^{12} 3^1 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 5 & 18 & 4 \\ 4 & 5 & 4 & 18 & 4 \\ 14 & 18 & 18 & 58 & 14 \\ 3 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[380] $1^{14}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 5 & 26 & 4 \\ 3 & 5 & 4 & 24 & 4 \\ 18 & 26 & 24 & 118 & 20 \\ 3 & 4 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[385] $1^{113}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 12 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[386] $1^{123}3^11^{22} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 14 & 3 \\ 3 & 4 & 18 & 22 & 4 \\ 13 & 18 & 58 & 72 & 14 \\ 14 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0}$$

[381] $1^{12}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 5 & 22 & 4 \\ 3 & 5 & 4 & 21 & 4 \\ 14 & 22 & 21 & 85 & 17 \\ 3 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

[383] $1^{14}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 4 & 5 & 26 & 4 \\ 4 & 5 & 4 & 24 & 4 \\ 18 & 26 & 24 & 118 & 20 \\ 3 & 4 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[384] $1^{114}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 4 & 4 & 1 & 15 & 3 \\ 15 & 21 & 15 & 76 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[382] $1^{133}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 14 & 3 \\ 3 & 4 & 17 & 22 & 4 \\ 12 & 17 & 49 & 66 & 13 \\ 14 & 22 & 66 & 85 & 17 \\ 3 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

[389] $1^{133}3^11^{43} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 9 & 3 \\ 3 & 4 & 12 & 12 & 4 \\ 10 & 12 & 25 & 27 & 9 \\ 9 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 3 \end{pmatrix}_{b_0}$$

[390] $1^{133}3^11^{23} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 12 & 12 & 4 \\ 8 & 12 & 25 & 27 & 9 \\ 9 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 3 \end{pmatrix}_{b_0}$$

[387] $1^{112}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 5 & 9 & 13 & 30 & 7 \\ 12 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

[388] $21^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.64 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 4 & 9 & 9 & 4 \\ 6 & 9 & 13 & 15 & 7 \\ 6 & 9 & 15 & 13 & 7 \\ 3 & 4 & 7 & 7 & 3 \end{pmatrix}_{b_0} \\
[391] \quad & 21^{122}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -158, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 6 & 9 & 3 \\ 4 & 4 & 8 & 13 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 9 & 13 & 18 & 28 & 10 \\ 3 & 4 & 6 & 10 & 3 \end{pmatrix}_{b_0} \\
[392] \quad & 1^{111}31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 11 & 3 \\ 3 & 4 & 14 & 14 & 4 \\ 12 & 14 & 37 & 39 & 11 \\ 11 & 14 & 39 & 37 & 11 \\ 3 & 4 & 11 & 11 & 3 \end{pmatrix}_{b_0} \\
[394] \quad & 21^{112}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 4 & 4 & 18 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 12 & 18 & 12 & 58 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0} \\
[393] \quad & 1^{111}31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 9 & 10 & 3 \\ 3 & 4 & 14 & 14 & 4 \\ 9 & 14 & 37 & 39 & 11 \\ 10 & 14 & 39 & 37 & 11 \\ 3 & 4 & 11 & 11 & 3 \end{pmatrix}_{b_0} \\
[395] \quad & 1^{122}31^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -162, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 9 & 14 & 3 \\ 3 & 4 & 13 & 18 & 4 \\ 9 & 13 & 28 & 42 & 10 \\ 14 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0} \\
[396] \quad & 1^{112}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -162, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 14 & 18 & 12 & 58 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0} \\
[397] \quad & 1^{144}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -162, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 15 & 3 \\ 3 & 4 & 16 & 21 & 4 \\ 12 & 16 & 46 & 60 & 12 \\ 15 & 21 & 60 & 76 & 16 \\ 3 & 4 & 12 & 16 & 3 \end{pmatrix}_{b_0} \\
[398] \quad & 1^{112}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -162, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 13 & 18 & 12 & 58 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0} \\
[399] \quad & 1^{112}31^{134} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 17 & 23 & 15 & 100 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0} \\
[400] \quad & 1^{113}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 11 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0} \\
[402] \quad & 1^{114}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -165, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 6 & 9 & 13 & 33 & 7 \\ 13 & 21 & 33 & 76 & 16 \\ 3 & 4 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[401] $1^{112} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -165, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.69 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 13 & 21 & 7 \\ 9 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[403] $1^{143} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -166, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 14 & 3 \\ 3 & 4 & 21 & 22 & 4 \\ 13 & 21 & 76 & 81 & 16 \\ 14 & 22 & 81 & 85 & 17 \\ 3 & 4 & 16 & 17 & 3 \end{pmatrix}_{b_0}$$

[405] $1^{14} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -166, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 5 & 26 & 4 \\ 3 & 5 & 4 & 24 & 4 \\ 16 & 26 & 24 & 118 & 20 \\ 3 & 4 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[404] $1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -166, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 22 & 0 & 85 & 17 \\ 3 & 4 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[407] $1^{14} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -169, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 5 & 26 & 4 \\ 3 & 5 & 4 & 24 & 4 \\ 17 & 26 & 24 & 118 & 20 \\ 3 & 4 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[406] $1^{144} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -169, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 17 & 3 \\ 3 & 4 & 21 & 26 & 4 \\ 15 & 21 & 76 & 96 & 16 \\ 17 & 26 & 96 & 118 & 20 \\ 3 & 4 & 16 & 20 & 3 \end{pmatrix}_{b_0}$$

[410] $1^{123} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 5 & 8 & 10 & 24 & 6 \\ 11 & 17 & 24 & 49 & 13 \\ 3 & 4 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[409] $1^{113} 3^1 1^{23} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 7 & 9 & 13 & 27 & 7 \\ 12 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[408] $1^{113} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 3 \\ 3 & 4 & 9 & 22 & 4 \\ 5 & 9 & 13 & 36 & 7 \\ 15 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

[411] $1^{113} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 19 & 3 \\ 3 & 4 & 9 & 27 & 4 \\ 5 & 9 & 13 & 45 & 7 \\ 19 & 27 & 45 & 133 & 21 \\ 3 & 4 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[414] $1^{13} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 17 & 3 \\ 3 & 4 & 5 & 27 & 4 \\ 4 & 5 & 4 & 27 & 4 \\ 17 & 27 & 27 & 133 & 21 \\ 3 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[413] \ 1^{112} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -175, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.66 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 7 & 9 & 13 & 21 & 7 \\ 10 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[412] \ 1^{12} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -175, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 5 & 23 & 4 \\ 4 & 5 & 4 & 24 & 4 \\ 15 & 23 & 24 & 100 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[420] \ 21^{122} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -176, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.82 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 4 & 8 & 8 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[419] \ 1^{111} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -176, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.82 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 14 & 14 & 4 \\ 10 & 14 & 37 & 39 & 11 \\ 11 & 14 & 39 & 37 & 11 \\ 3 & 4 & 11 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[417] \ 21^{111} 31^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -176, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.75 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 15 & 15 & 4 \\ 5 & 4 & 17 & 17 & 4 \\ 15 & 17 & 49 & 51 & 13 \\ 15 & 17 & 51 & 49 & 13 \\ 4 & 4 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[416] \ 1^{13} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -176, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.70 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 16 & 3 \\ 3 & 4 & 5 & 22 & 4 \\ 4 & 5 & 4 & 21 & 4 \\ 16 & 22 & 21 & 85 & 17 \\ 3 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[415] \ 1^{113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -176, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.70 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 16 & 22 & 15 & 85 & 17 \\ 3 & 4 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[418] \ 21^{12} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -176, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.70 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 21 & 4 \\ 5 & 4 & 4 & 22 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 21 & 22 & 15 & 85 & 17 \\ 4 & 4 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[421] \ 1^{114} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -178, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 11 & 16 & 12 & 46 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[422] \ 1^{114} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -180, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.76 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 15 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 7 & 9 & 13 & 33 & 7 \\ 15 & 21 & 33 & 76 & 16 \\ 3 & 4 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[423] \ 1^{112} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -180, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.76 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 7 & 9 & 13 & 30 & 7 \\ 12 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

[425] $1^{14} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 21 & 0 & 76 & 16 \\ 3 & 4 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[426] $1^{144} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 14 & 3 \\ 3 & 4 & 16 & 21 & 4 \\ 11 & 16 & 46 & 60 & 12 \\ 14 & 21 & 60 & 76 & 16 \\ 3 & 4 & 12 & 16 & 3 \end{pmatrix}_{b_0}$$

[424] $1^{12} 3 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 16 & 23 & 0 & 100 & 18 \\ 3 & 4 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[427] $1^{133} 3 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 12 & 12 & 4 \\ 10 & 12 & 25 & 27 & 9 \\ 10 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 3 \end{pmatrix}_{b_0}$$

[428] $1^{112} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 14 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 8 & 9 & 13 & 30 & 7 \\ 14 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

[429] $1^{111} 3 1^{122} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 12 & 3 \\ 3 & 4 & 14 & 14 & 4 \\ 12 & 14 & 37 & 39 & 11 \\ 12 & 14 & 39 & 37 & 11 \\ 3 & 4 & 11 & 11 & 3 \end{pmatrix}_{b_0}$$

[430] $21^{112} 31^{122} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 12 & 3 \\ 4 & 4 & 14 & 18 & 4 \\ 9 & 14 & 37 & 48 & 11 \\ 12 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

[431] $1^{124} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 18 & 3 \\ 3 & 4 & 8 & 26 & 4 \\ 5 & 8 & 10 & 36 & 6 \\ 18 & 26 & 36 & 118 & 20 \\ 3 & 4 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

[432] $1^{133} 3 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 12 & 12 & 4 \\ 8 & 12 & 25 & 27 & 9 \\ 8 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 3 \end{pmatrix}_{b_0}$$

[433] $1^{12} 3 1^{113} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 5 & 23 & 4 \\ 3 & 5 & 4 & 24 & 4 \\ 17 & 23 & 24 & 100 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[434] $1^{123} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 6 & 8 & 10 & 24 & 6 \\ 12 & 17 & 24 & 49 & 13 \\ 3 & 4 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[435] $1^{111} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 9 & 3 \\ 3 & 4 & 14 & 14 & 4 \\ 9 & 14 & 37 & 39 & 11 \\ 9 & 14 & 39 & 37 & 11 \\ 3 & 4 & 11 & 11 & 3 \end{pmatrix}_{b_0}$$

[438] $1^{122} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 8 & 13 & 28 & 30 & 10 \\ 9 & 13 & 30 & 28 & 10 \\ 3 & 4 & 10 & 10 & 3 \end{pmatrix}_{b_0}$$

[439] $1^{113} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 11 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[440] $1^{112} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 5 & 9 & 13 & 30 & 7 \\ 11 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

[441] $1^{113} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 13 & 27 & 7 \\ 11 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[437] $1^{112} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 19 & 3 \\ 3 & 4 & 9 & 28 & 4 \\ 5 & 9 & 13 & 48 & 7 \\ 19 & 28 & 48 & 154 & 22 \\ 3 & 4 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$

[436] $1^{123} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 19 & 3 \\ 3 & 4 & 13 & 27 & 4 \\ 8 & 13 & 28 & 63 & 10 \\ 19 & 27 & 63 & 133 & 21 \\ 3 & 4 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

[443] $21^{111} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -188, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.68 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 12 & 12 & 4 \\ 5 & 4 & 13 & 13 & 4 \\ 12 & 13 & 28 & 30 & 10 \\ 12 & 13 & 30 & 28 & 10 \\ 4 & 4 & 10 & 10 & 3 \end{pmatrix}_{b_0}$$

[442] $1^{124} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -188, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 5 & 8 & 10 & 24 & 6 \\ 11 & 16 & 24 & 46 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[445] $1^{143} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -189, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 15 & 3 \\ 3 & 4 & 21 & 22 & 4 \\ 14 & 21 & 76 & 81 & 16 \\ 15 & 22 & 81 & 85 & 17 \\ 3 & 4 & 16 & 17 & 3 \end{pmatrix}_{b_0}$$

[444] $1^{134} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -189, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 13 & 3 \\ 3 & 4 & 17 & 21 & 4 \\ 12 & 17 & 49 & 63 & 13 \\ 13 & 21 & 63 & 76 & 16 \\ 3 & 4 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

[451] $21^{112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -190, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 9 & 15 & 4 \\ 5 & 4 & 9 & 14 & 4 \\ 9 & 9 & 13 & 24 & 7 \\ 15 & 14 & 24 & 37 & 11 \\ 4 & 4 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[449] \ 1^{123} 3 \ 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -190, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 12 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 8 & 13 & 28 & 39 & 10 \\ 12 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[448] \ 1^{134} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -190, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 13 & 3 \\ 3 & 4 & 12 & 21 & 4 \\ 8 & 12 & 25 & 45 & 9 \\ 13 & 21 & 45 & 76 & 16 \\ 3 & 4 & 9 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[447] \ 1^{112} 3 \ 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -190, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 13 & 3 \\ 3 & 4 & 14 & 18 & 4 \\ 11 & 14 & 37 & 48 & 11 \\ 13 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[446] \ 1^{114} 3 \ 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -190, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 12 & 16 & 12 & 46 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[450] \ 1^{112} 3 \ 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -190, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 14 & 3 \\ 3 & 4 & 14 & 18 & 4 \\ 10 & 14 & 37 & 48 & 11 \\ 14 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[453] \ 1^{112} 3 \ 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -192, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 6 & 9 & 13 & 30 & 7 \\ 14 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[452] \ 1^{113} 3 \ 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -192, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 13 & 27 & 7 \\ 12 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[454] \ 1^{12} 3 \ 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -193, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 17 & 23 & 0 & 100 & 18 \\ 3 & 4 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[457] \ 1^{12} 3 \ 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -196, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 5 & 23 & 4 \\ 3 & 5 & 4 & 24 & 4 \\ 15 & 23 & 24 & 100 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[458] \ 1^{124} 3 \ 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -196, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 18 & 3 \\ 3 & 4 & 8 & 26 & 4 \\ 6 & 8 & 10 & 36 & 6 \\ 18 & 26 & 36 & 118 & 20 \\ 3 & 4 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[456] \ 1^{124} 3 \ 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -196, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 6 & 8 & 10 & 24 & 6 \\ 11 & 16 & 24 & 46 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[455] \ 1^{123} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -196, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 6 & 8 & 10 & 24 & 6 \\ 13 & 17 & 24 & 49 & 13 \\ 3 & 4 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[459] \ 1^{114} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -197, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 15 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 6 & 9 & 13 & 33 & 7 \\ 15 & 21 & 33 & 76 & 16 \\ 3 & 4 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[464] \ 1^{123} 3 \ 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 15 & 3 \\ 3 & 4 & 18 & 22 & 4 \\ 11 & 18 & 58 & 72 & 14 \\ 15 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[463] \ 1^{134} 3 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 12 & 3 \\ 3 & 4 & 12 & 16 & 4 \\ 8 & 12 & 25 & 36 & 9 \\ 12 & 16 & 36 & 46 & 12 \\ 3 & 4 & 9 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[461] \ 1^{13} 3 \ 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 15 & 22 & 0 & 85 & 17 \\ 3 & 4 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[460] \ 1^{134} 3 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 12 & 3 \\ 3 & 4 & 12 & 16 & 4 \\ 10 & 12 & 25 & 36 & 9 \\ 12 & 16 & 36 & 46 & 12 \\ 3 & 4 & 9 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[462] \ 1^{112} 3 \ 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 11 & 3 \\ 3 & 4 & 14 & 18 & 4 \\ 9 & 14 & 37 & 48 & 11 \\ 11 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[465] \ 1^{12} 3 \ 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -199, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 5 & 23 & 4 \\ 3 & 5 & 4 & 24 & 4 \\ 16 & 23 & 24 & 100 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[466] \ 1^{124} 3 \ 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 16 & 3 \\ 3 & 4 & 8 & 26 & 4 \\ 5 & 8 & 10 & 36 & 6 \\ 16 & 26 & 36 & 118 & 20 \\ 3 & 4 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[467] \ 21^{112} 3 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.66 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 9 & 13 & 9 & 28 & 10 \\ 3 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[468] \ 1^{113} 3 \ 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 13 & 22 & 15 & 85 & 17 \\ 3 & 4 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

[469] $1^{133}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 9 & 3 \\ 3 & 4 & 12 & 12 & 4 \\ 9 & 12 & 25 & 27 & 9 \\ 9 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 3 \end{pmatrix}_{b_0}$$

[470] $1^{14}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -201, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 15 & 21 & 0 & 76 & 16 \\ 3 & 4 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[472] $1^{122}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -202, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 14 & 3 \\ 3 & 4 & 13 & 18 & 4 \\ 11 & 13 & 28 & 42 & 10 \\ 14 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[471] $1^{143}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -202, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 16 & 3 \\ 3 & 4 & 21 & 22 & 4 \\ 15 & 21 & 76 & 81 & 16 \\ 16 & 22 & 81 & 85 & 17 \\ 3 & 4 & 16 & 17 & 3 \end{pmatrix}_{b_0}$$

[473] $1^{122}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -203, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 10 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 11 & 13 & 28 & 30 & 10 \\ 10 & 13 & 30 & 28 & 10 \\ 3 & 4 & 10 & 10 & 3 \end{pmatrix}_{b_0}$$

[474] $21^{123}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -206, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 24 & 4 \\ 5 & 4 & 4 & 26 & 4 \\ 3 & 4 & 1 & 18 & 3 \\ 24 & 26 & 18 & 118 & 20 \\ 4 & 4 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

[475] $1^{112}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 6 & 9 & 13 & 30 & 7 \\ 11 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

[476] $1^{113}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 1 & 18 & 3 \\ 17 & 27 & 18 & 133 & 21 \\ 3 & 4 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

[477] $1^{113}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 15 & 22 & 15 & 85 & 17 \\ 3 & 4 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

[478] $21^{112}31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 12 & 18 & 4 \\ 5 & 4 & 13 & 18 & 4 \\ 12 & 13 & 28 & 42 & 10 \\ 18 & 18 & 42 & 58 & 14 \\ 4 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[479] $21^{113}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 9 & 12 & 9 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

- [480] $21^{134}31^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 15 & 3 \\ 4 & 4 & 22 & 21 & 4 \\ 15 & 22 & 85 & 81 & 17 \\ 15 & 21 & 81 & 76 & 16 \\ 3 & 4 & 17 & 16 & 3 \end{pmatrix}_{b_0}$$
- [481] $1^{113}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.69 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 13 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 7 & 9 & 13 & 27 & 7 \\ 13 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$
- [482] $1^{113}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 1 & 18 & 3 \\ 19 & 27 & 18 & 133 & 21 \\ 3 & 4 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$
- [483] $1^{134}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -210, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 15 & 3 \\ 3 & 4 & 12 & 21 & 4 \\ 8 & 12 & 25 & 45 & 9 \\ 15 & 21 & 45 & 76 & 16 \\ 3 & 4 & 9 & 16 & 3 \end{pmatrix}_{b_0}$$
- [484] $1^{114}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -210, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 6 & 9 & 13 & 33 & 7 \\ 14 & 21 & 33 & 76 & 16 \\ 3 & 4 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$
- [485] $1^{112}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -210, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 12 & 3 \\ 3 & 4 & 14 & 18 & 4 \\ 9 & 14 & 37 & 48 & 11 \\ 12 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$
- [486] $1^{114}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 17 & 3 \\ 3 & 4 & 9 & 26 & 4 \\ 7 & 9 & 13 & 42 & 7 \\ 17 & 26 & 42 & 118 & 20 \\ 3 & 4 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$
- [487] $1^{122}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.68 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 10 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 9 & 13 & 28 & 30 & 10 \\ 10 & 13 & 30 & 28 & 10 \\ 3 & 4 & 10 & 10 & 3 \end{pmatrix}_{b_0}$$
- [488] $1^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 16 & 22 & 0 & 85 & 17 \\ 3 & 4 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$
- [489] $1^{112}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 12 & 3 \\ 3 & 4 & 14 & 18 & 4 \\ 10 & 14 & 37 & 48 & 11 \\ 12 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$
- [490] $1^{122}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 12 & 3 \\ 3 & 4 & 13 & 18 & 4 \\ 8 & 13 & 28 & 42 & 10 \\ 12 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[491] $1^{144} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -214$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 16 & 3 \\ 3 & 4 & 21 & 26 & 4 \\ 14 & 21 & 76 & 96 & 16 \\ 16 & 26 & 96 & 118 & 20 \\ 3 & 4 & 16 & 20 & 3 \end{pmatrix}_{b_0}$$

[501] $1^{134} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -216$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 19 & 3 \\ 3 & 4 & 17 & 31 & 4 \\ 11 & 17 & 49 & 93 & 13 \\ 19 & 31 & 93 & 172 & 24 \\ 3 & 4 & 13 & 24 & 3 \end{pmatrix}_{b_0}$$

[499] $1^{123} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -216$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.3$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 17 & 3 \\ 3 & 4 & 13 & 27 & 4 \\ 9 & 13 & 28 & 63 & 10 \\ 17 & 27 & 63 & 133 & 21 \\ 3 & 4 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

[497] $1^{113} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -216$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 22 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 2 & 4 & 1 & 21 & 3 \\ 22 & 32 & 21 & 193 & 25 \\ 3 & 4 & 3 & 25 & 3 \end{pmatrix}_{b_0}$$

[496] $1^{111} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -216$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 14 & 14 & 4 \\ 10 & 14 & 37 & 39 & 11 \\ 10 & 14 & 39 & 37 & 11 \\ 3 & 4 & 11 & 11 & 3 \end{pmatrix}_{b_0}$$

[495] $1^{124} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -216$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.81 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 3 \\ 3 & 4 & 8 & 26 & 4 \\ 5 & 8 & 10 & 36 & 6 \\ 17 & 26 & 36 & 118 & 20 \\ 3 & 4 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

[503] $1^{113} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -216$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 14 & 22 & 15 & 85 & 17 \\ 3 & 4 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

[502] $1^{122} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -216$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 8 & 13 & 28 & 30 & 10 \\ 8 & 13 & 30 & 28 & 10 \\ 3 & 4 & 10 & 10 & 3 \end{pmatrix}_{b_0}$$

[494] $1^{111} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -216$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 11 & 3 \\ 3 & 4 & 14 & 14 & 4 \\ 11 & 14 & 37 & 39 & 11 \\ 11 & 14 & 39 & 37 & 11 \\ 3 & 4 & 11 & 11 & 3 \end{pmatrix}_{b_0}$$

[493] $1^{14} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -216$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 20 & 3 \\ 3 & 4 & 5 & 31 & 4 \\ 4 & 5 & 4 & 30 & 4 \\ 20 & 31 & 30 & 172 & 24 \\ 3 & 4 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[492] $1^{13} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -216$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 16 & 3 \\ 3 & 4 & 5 & 23 & 4 \\ 4 & 5 & 4 & 24 & 4 \\ 16 & 23 & 24 & 100 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[500] $1^{112} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 3 \\ 3 & 4 & 9 & 23 & 4 \\ 5 & 9 & 13 & 39 & 7 \\ 15 & 23 & 39 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[498] $1^{124} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 13 & 3 \\ 3 & 4 & 13 & 21 & 4 \\ 9 & 13 & 28 & 48 & 10 \\ 13 & 21 & 48 & 76 & 16 \\ 3 & 4 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[504] $1^{122} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -217, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 3 \\ 3 & 4 & 13 & 18 & 4 \\ 8 & 13 & 28 & 42 & 10 \\ 11 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[505] $1^{113} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -217, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 12 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[506] $1^{123} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -217, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 21 & 3 \\ 3 & 4 & 13 & 32 & 4 \\ 8 & 13 & 28 & 75 & 10 \\ 21 & 32 & 75 & 193 & 25 \\ 3 & 4 & 10 & 25 & 3 \end{pmatrix}_{b_0}$$

[510] $1^{124} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -220, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 17 & 3 \\ 3 & 4 & 18 & 26 & 4 \\ 11 & 18 & 58 & 84 & 14 \\ 17 & 26 & 84 & 118 & 20 \\ 3 & 4 & 14 & 20 & 3 \end{pmatrix}_{b_0}$$

[511] $1^{113} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -220, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 14 & 22 & 15 & 85 & 17 \\ 3 & 4 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

[507] $1^{124} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -220, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 6 & 8 & 10 & 24 & 6 \\ 12 & 16 & 24 & 46 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[508] $1^{12} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -220, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 19 & 28 & 0 & 154 & 22 \\ 3 & 4 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

[509] $1^{124} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -220, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 16 & 3 \\ 3 & 4 & 18 & 26 & 4 \\ 12 & 18 & 58 & 84 & 14 \\ 16 & 26 & 84 & 118 & 20 \\ 3 & 4 & 14 & 20 & 3 \end{pmatrix}_{b_0}$$

[512] $1^{113} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -223, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 13 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[513] $1^{134} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 14 & 3 \\ 3 & 4 & 17 & 21 & 4 \\ 10 & 17 & 49 & 63 & 13 \\ 14 & 21 & 63 & 76 & 16 \\ 3 & 4 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

[514] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 15 & 21 & 15 & 76 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[515] $1^{134} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 11 & 3 \\ 3 & 4 & 12 & 16 & 4 \\ 9 & 12 & 25 & 36 & 9 \\ 11 & 16 & 36 & 46 & 12 \\ 3 & 4 & 9 & 12 & 3 \end{pmatrix}_{b_0}$$

[516] $1^{123} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -228, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 13 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 11 & 13 & 28 & 39 & 10 \\ 13 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

[517] $1^{124} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -229, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 3 \\ 3 & 4 & 8 & 21 & 4 \\ 5 & 8 & 10 & 30 & 6 \\ 15 & 21 & 30 & 76 & 16 \\ 3 & 4 & 6 & 16 & 3 \end{pmatrix}_{b_0}$$

[518] $21^{133} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 12 & 3 \\ 4 & 4 & 12 & 17 & 4 \\ 9 & 12 & 25 & 36 & 9 \\ 12 & 17 & 36 & 49 & 13 \\ 3 & 4 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

[520] $1^{124} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -234, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 8 & 21 & 4 \\ 6 & 8 & 10 & 30 & 6 \\ 14 & 21 & 30 & 76 & 16 \\ 3 & 4 & 6 & 16 & 3 \end{pmatrix}_{b_0}$$

[519] $1^{134} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -234, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 17 & 3 \\ 3 & 4 & 22 & 26 & 4 \\ 16 & 22 & 85 & 102 & 17 \\ 17 & 26 & 102 & 118 & 20 \\ 3 & 4 & 17 & 20 & 3 \end{pmatrix}_{b_0}$$

[522] $21^{112} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.66 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 9 & 3 \\ 4 & 4 & 9 & 13 & 4 \\ 6 & 9 & 13 & 21 & 7 \\ 9 & 13 & 21 & 28 & 10 \\ 3 & 4 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

[521] $1^{113} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 1 & 18 & 3 \\ 18 & 27 & 18 & 133 & 21 \\ 3 & 4 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

[523] $21^{13} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 27 & 4 \\ 5 & 4 & 4 & 27 & 4 \\ 3 & 4 & 1 & 18 & 3 \\ 27 & 27 & 18 & 133 & 21 \\ 4 & 4 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

[524] $1^{113} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 4 & 5 & 27 & 4 \\ 4 & 5 & 4 & 27 & 4 \\ 18 & 27 & 27 & 133 & 21 \\ 3 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[525] \ 1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 13 & 21 & 15 & 76 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[526] \ 1^{134} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -239, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 15 & 3 \\ 3 & 4 & 12 & 21 & 4 \\ 9 & 12 & 25 & 45 & 9 \\ 15 & 21 & 45 & 76 & 16 \\ 3 & 4 & 9 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[527] \ 1^{12} 3 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -239, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 17 & 3 \\ 3 & 4 & 5 & 23 & 4 \\ 4 & 5 & 4 & 24 & 4 \\ 17 & 23 & 24 & 100 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[528] \ 1^{122} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -240, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 11 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 11 & 13 & 28 & 30 & 10 \\ 11 & 13 & 30 & 28 & 10 \\ 3 & 4 & 10 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[529] \ 1^{113} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -241, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 9 & 22 & 4 \\ 6 & 9 & 13 & 36 & 7 \\ 13 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[530] \ 1^{122} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -241, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 13 & 3 \\ 3 & 4 & 13 & 18 & 4 \\ 10 & 13 & 28 & 42 & 10 \\ 13 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[531] \ 1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -243, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 13 & 21 & 15 & 76 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[536] \ 1^{134} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 14 & 3 \\ 3 & 4 & 12 & 21 & 4 \\ 8 & 12 & 25 & 45 & 9 \\ 14 & 21 & 45 & 76 & 16 \\ 3 & 4 & 9 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[537] \ 1^{12} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 20 & 28 & 0 & 154 & 22 \\ 3 & 4 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[532] \ 1^{113} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 2 & 4 & 1 & 21 & 3 \\ 21 & 32 & 21 & 193 & 25 \\ 3 & 4 & 3 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[533] \ 1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 14 & 21 & 15 & 76 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[534] \ 1^{113} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 9 & 22 & 4 \\ 5 & 9 & 13 & 36 & 7 \\ 14 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[535] \ 1^{113} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 1 & 18 & 3 \\ 17 & 27 & 18 & 133 & 21 \\ 3 & 4 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[538] \ 1^{122} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -246, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 12 & 3 \\ 3 & 4 & 13 & 18 & 4 \\ 9 & 13 & 28 & 42 & 10 \\ 12 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[539] \ 1^{123} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -246, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 8 & 13 & 28 & 39 & 10 \\ 10 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[540] \ 1^{133} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -246, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 14 & 3 \\ 3 & 4 & 17 & 22 & 4 \\ 10 & 17 & 49 & 66 & 13 \\ 14 & 22 & 66 & 85 & 17 \\ 3 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[542] \ 1^{13} 3 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -248, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 20 & 3 \\ 3 & 4 & 5 & 32 & 4 \\ 4 & 5 & 4 & 33 & 4 \\ 20 & 32 & 33 & 193 & 25 \\ 3 & 4 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[541] \ 21^{14} 3 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -248, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 33 & 4 \\ 5 & 4 & 4 & 32 & 4 \\ 3 & 4 & 1 & 21 & 3 \\ 33 & 32 & 21 & 193 & 25 \\ 4 & 4 & 3 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[543] \ 1^{113} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -248, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 2 & 4 & 1 & 21 & 3 \\ 20 & 32 & 21 & 193 & 25 \\ 3 & 4 & 3 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[546] \ 1^{112} 3 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -250, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 14 & 3 \\ 3 & 4 & 14 & 18 & 4 \\ 11 & 14 & 37 & 48 & 11 \\ 14 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[545] \ 1^{112} 3 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -250, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 13 & 3 \\ 3 & 4 & 14 & 18 & 4 \\ 10 & 14 & 37 & 48 & 11 \\ 13 & 18 & 48 & 58 & 14 \\ 3 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[544] \ 1^{134} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -250, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 3 \\ 3 & 4 & 12 & 16 & 4 \\ 8 & 12 & 25 & 36 & 9 \\ 11 & 16 & 36 & 46 & 12 \\ 3 & 4 & 9 & 12 & 3 \end{pmatrix}_{b_0}$$

[548] $1^{13}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -252, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 5 & 27 & 4 \\ 3 & 5 & 4 & 27 & 4 \\ 17 & 27 & 27 & 133 & 21 \\ 3 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[547] $1^{13}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -252, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 5 & 27 & 4 \\ 3 & 5 & 4 & 27 & 4 \\ 19 & 27 & 27 & 133 & 21 \\ 3 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[549] $1^{124}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -253, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 8 & 21 & 4 \\ 5 & 8 & 10 & 30 & 6 \\ 13 & 21 & 30 & 76 & 16 \\ 3 & 4 & 6 & 16 & 3 \end{pmatrix}_{b_0}$$

[551] $1^{112}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -255, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 6 & 9 & 13 & 30 & 7 \\ 13 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

[550] $1^{113}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -255, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 18 & 27 & 0 & 133 & 21 \\ 3 & 4 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

[554] $21^{113}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 6 & 27 & 4 \\ 5 & 4 & 5 & 27 & 4 \\ 6 & 5 & 4 & 27 & 4 \\ 27 & 27 & 27 & 133 & 21 \\ 4 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[555] $1^{14}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 17 & 26 & 0 & 118 & 20 \\ 3 & 4 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[556] $21^{123}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 9 & 3 \\ 4 & 4 & 8 & 12 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 9 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[552] $1^{123}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 13 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 9 & 13 & 28 & 39 & 10 \\ 13 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

[553] $1^{122}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 9 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 9 & 13 & 28 & 30 & 10 \\ 9 & 13 & 30 & 28 & 10 \\ 3 & 4 & 10 & 10 & 3 \end{pmatrix}_{b_0}$$

[558] $1^{114}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -259, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 3 \\ 3 & 4 & 9 & 26 & 4 \\ 5 & 9 & 13 & 42 & 7 \\ 17 & 26 & 42 & 118 & 20 \\ 3 & 4 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[557] $1^{112} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -259, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 13 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 7 & 9 & 13 & 30 & 7 \\ 13 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

[559] $1^{112} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -260, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 6 & 9 & 13 & 30 & 7 \\ 12 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

[561] $1^{122} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 13 & 3 \\ 3 & 4 & 13 & 18 & 4 \\ 9 & 13 & 28 & 42 & 10 \\ 13 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[560] $1^{13} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 5 & 27 & 4 \\ 3 & 5 & 4 & 27 & 4 \\ 18 & 27 & 27 & 133 & 21 \\ 3 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[563] $21^{122} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -262, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 15 & 18 & 4 \\ 5 & 4 & 14 & 18 & 4 \\ 15 & 14 & 37 & 48 & 11 \\ 18 & 18 & 48 & 58 & 14 \\ 4 & 4 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

[562] $1^{123} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -262, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 22 & 3 \\ 3 & 4 & 13 & 32 & 4 \\ 9 & 13 & 28 & 75 & 10 \\ 22 & 32 & 75 & 193 & 25 \\ 3 & 4 & 10 & 25 & 3 \end{pmatrix}_{b_0}$$

[564] $1^{122} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.68 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 10 & 13 & 28 & 30 & 10 \\ 10 & 13 & 30 & 28 & 10 \\ 3 & 4 & 10 & 10 & 3 \end{pmatrix}_{b_0}$$

[566] $1^{123} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -265, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 15 & 3 \\ 3 & 4 & 13 & 22 & 4 \\ 10 & 13 & 28 & 51 & 10 \\ 15 & 22 & 51 & 85 & 17 \\ 3 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0}$$

[565] $1^{124} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -265, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 8 & 21 & 4 \\ 5 & 8 & 10 & 30 & 6 \\ 14 & 21 & 30 & 76 & 16 \\ 3 & 4 & 6 & 16 & 3 \end{pmatrix}_{b_0}$$

[568] $1^{144} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 12 & 3 \\ 3 & 4 & 16 & 16 & 4 \\ 11 & 16 & 46 & 48 & 12 \\ 12 & 16 & 48 & 46 & 12 \\ 3 & 4 & 12 & 12 & 3 \end{pmatrix}_{b_0}$$

[567] $1^{113} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 15 & 22 & 15 & 85 & 17 \\ 3 & 4 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

[569] $1^{134} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -270, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 12 & 3 \\ 3 & 4 & 12 & 16 & 4 \\ 9 & 12 & 25 & 36 & 9 \\ 12 & 16 & 36 & 46 & 12 \\ 3 & 4 & 9 & 12 & 3 \end{pmatrix}_{b_0}$$

[571] $1^{112} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 16 & 3 \\ 3 & 4 & 9 & 23 & 4 \\ 7 & 9 & 13 & 39 & 7 \\ 16 & 23 & 39 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[570] $1^{114} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 20 & 3 \\ 3 & 4 & 9 & 31 & 4 \\ 7 & 9 & 13 & 51 & 7 \\ 20 & 31 & 51 & 172 & 24 \\ 3 & 4 & 7 & 24 & 3 \end{pmatrix}_{b_0}$$

[572] $1^{113} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -273, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 1 & 18 & 3 \\ 18 & 27 & 18 & 133 & 21 \\ 3 & 4 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

[573] $1^{134} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -274, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 16 & 3 \\ 3 & 4 & 22 & 26 & 4 \\ 15 & 22 & 85 & 102 & 17 \\ 16 & 26 & 102 & 118 & 20 \\ 3 & 4 & 17 & 20 & 3 \end{pmatrix}_{b_0}$$

[574] $1^{12} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -274, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 5 & 28 & 4 \\ 3 & 5 & 4 & 30 & 4 \\ 20 & 28 & 30 & 154 & 22 \\ 3 & 4 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

[575] $21^{112} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -274, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 9 & 24 & 4 \\ 5 & 4 & 9 & 26 & 4 \\ 9 & 9 & 13 & 42 & 7 \\ 24 & 26 & 42 & 118 & 20 \\ 4 & 4 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[576] $1^{123} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -274, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 12 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 10 & 13 & 28 & 39 & 10 \\ 12 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

[577] $1^{122} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -274, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 14 & 3 \\ 3 & 4 & 13 & 18 & 4 \\ 10 & 13 & 28 & 42 & 10 \\ 14 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[579] $1^{133} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -275, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 17 & 17 & 4 \\ 10 & 17 & 49 & 51 & 13 \\ 11 & 17 & 51 & 49 & 13 \\ 3 & 4 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$

[578] $1^{123} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -275, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 8 & 13 & 28 & 39 & 10 \\ 11 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

[580] $1^{113} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -279, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 21 & 3 \\ 3 & 4 & 9 & 32 & 4 \\ 7 & 9 & 13 & 54 & 7 \\ 21 & 32 & 54 & 193 & 25 \\ 3 & 4 & 7 & 25 & 3 \end{pmatrix}_{b_0}$$

[581] $1^{113} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -279, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 15 & 3 \\ 3 & 4 & 9 & 22 & 4 \\ 7 & 9 & 13 & 36 & 7 \\ 15 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

[582] $1^{123} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 11 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 9 & 13 & 28 & 39 & 10 \\ 11 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

[583] $1^{112} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 14 & 3 \\ 3 & 4 & 9 & 18 & 4 \\ 7 & 9 & 13 & 30 & 7 \\ 14 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

[584] $21^{113} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 18 & 3 \\ 4 & 4 & 4 & 27 & 4 \\ 3 & 4 & 1 & 18 & 3 \\ 18 & 27 & 18 & 133 & 21 \\ 3 & 4 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

[585] $1^{113} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 22 & 3 \\ 3 & 4 & 9 & 32 & 4 \\ 5 & 9 & 13 & 54 & 7 \\ 22 & 32 & 54 & 193 & 25 \\ 3 & 4 & 7 & 25 & 3 \end{pmatrix}_{b_0}$$

[586] $1^{12} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -281, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 19 & 3 \\ 3 & 4 & 5 & 28 & 4 \\ 4 & 5 & 4 & 30 & 4 \\ 19 & 28 & 30 & 154 & 22 \\ 3 & 4 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

[588] $1^{13} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 19 & 3 \\ 3 & 4 & 5 & 27 & 4 \\ 4 & 5 & 4 & 27 & 4 \\ 19 & 27 & 27 & 133 & 21 \\ 3 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[589] $1^{123} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 19 & 3 \\ 3 & 4 & 13 & 27 & 4 \\ 10 & 13 & 28 & 63 & 10 \\ 19 & 27 & 63 & 133 & 21 \\ 3 & 4 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

[590] $1^{13} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 19 & 27 & 0 & 133 & 21 \\ 3 & 4 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

[587] $1^{14} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 4 & 21 & 3 \\ 3 & 4 & 5 & 31 & 4 \\ 4 & 5 & 4 & 30 & 4 \\ 21 & 31 & 30 & 172 & 24 \\ 3 & 4 & 4 & 24 & 3 \end{pmatrix}_{b_0} \\
[591] \quad & 21^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -284, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 6 & 24 & 4 \\ 5 & 4 & 5 & 23 & 4 \\ 6 & 5 & 4 & 24 & 4 \\ 24 & 23 & 24 & 100 & 18 \\ 4 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0} \\
[592] \quad & 21^{112}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -286, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 6 & 15 & 3 \\ 4 & 4 & 9 & 23 & 4 \\ 6 & 9 & 13 & 39 & 7 \\ 15 & 23 & 39 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0} \\
[594] \quad & 1^{14}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -288, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 18 & 26 & 0 & 118 & 20 \\ 3 & 4 & 0 & 20 & 3 \end{pmatrix}_{b_0} \\
[593] \quad & 1^{122}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -288, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 11 & 12 & 3 \\ 3 & 4 & 18 & 18 & 4 \\ 11 & 18 & 58 & 60 & 14 \\ 12 & 18 & 60 & 58 & 14 \\ 3 & 4 & 14 & 14 & 3 \end{pmatrix}_{b_0} \\
[595] \quad & 21^{112}31^{134} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -290, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 15 & 24 & 4 \\ 5 & 4 & 17 & 26 & 4 \\ 15 & 17 & 49 & 78 & 13 \\ 24 & 26 & 78 & 118 & 20 \\ 4 & 4 & 13 & 20 & 3 \end{pmatrix}_{b_0} \\
[596] \quad & 1^{113}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -291, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 18 & 3 \\ 3 & 4 & 9 & 27 & 4 \\ 5 & 9 & 13 & 45 & 7 \\ 18 & 27 & 45 & 133 & 21 \\ 3 & 4 & 7 & 21 & 3 \end{pmatrix}_{b_0} \\
[597] \quad & 21^{113}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 6 & 12 & 3 \\ 4 & 4 & 9 & 17 & 4 \\ 6 & 9 & 13 & 27 & 7 \\ 12 & 17 & 27 & 49 & 13 \\ 3 & 4 & 7 & 13 & 3 \end{pmatrix}_{b_0} \\
[601] \quad & 1^{113}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 22 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 3 & 4 & 1 & 21 & 3 \\ 22 & 32 & 21 & 193 & 25 \\ 3 & 4 & 3 & 25 & 3 \end{pmatrix}_{b_0} \\
[598] \quad & 1^{124}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 8 & 14 & 3 \\ 3 & 4 & 13 & 21 & 4 \\ 8 & 13 & 28 & 48 & 10 \\ 14 & 21 & 48 & 76 & 16 \\ 3 & 4 & 10 & 16 & 3 \end{pmatrix}_{b_0} \\
[600] \quad & 1^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 16 & 22 & 15 & 85 & 17 \\ 3 & 4 & 3 & 17 & 3 \end{pmatrix}_{b_0} \\
[599] \quad & 1^{14}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 5 & 31 & 4 \\ 3 & 5 & 4 & 30 & 4 \\ 19 & 31 & 30 & 172 & 24 \\ 3 & 4 & 4 & 24 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [602] $1^{133} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -293, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 13 & 3 \\ 3 & 4 & 17 & 22 & 4 \\ 10 & 17 & 49 & 66 & 13 \\ 13 & 22 & 66 & 85 & 17 \\ 3 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$
- [603] $1^{124} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -294, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 15 & 3 \\ 3 & 4 & 8 & 21 & 4 \\ 6 & 8 & 10 & 30 & 6 \\ 15 & 21 & 30 & 76 & 16 \\ 3 & 4 & 6 & 16 & 3 \end{pmatrix}_{b_0}$$
- [604] $1^{124} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -295, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 21 & 3 \\ 3 & 4 & 13 & 31 & 4 \\ 10 & 13 & 28 & 72 & 10 \\ 21 & 31 & 72 & 172 & 24 \\ 3 & 4 & 10 & 24 & 3 \end{pmatrix}_{b_0}$$
- [606] $21^{122} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -296, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 15 & 15 & 4 \\ 5 & 4 & 14 & 14 & 4 \\ 15 & 14 & 37 & 39 & 11 \\ 15 & 14 & 39 & 37 & 11 \\ 4 & 4 & 11 & 11 & 3 \end{pmatrix}_{b_0}$$
- [605] $21^{124} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -296, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 18 & 3 \\ 4 & 4 & 8 & 26 & 4 \\ 6 & 8 & 10 & 36 & 6 \\ 18 & 26 & 36 & 118 & 20 \\ 3 & 4 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$
- [607] $1^{114} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -297, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 14 & 21 & 15 & 76 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$
- [608] $1^{114} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -298, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 1 & 18 & 3 \\ 16 & 26 & 18 & 118 & 20 \\ 3 & 4 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$
- [609] $1^{112} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -301, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 17 & 3 \\ 3 & 4 & 9 & 23 & 4 \\ 6 & 9 & 13 & 39 & 7 \\ 17 & 23 & 39 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$
- [610] $21^{123} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -302, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 12 & 3 \\ 4 & 4 & 8 & 17 & 4 \\ 6 & 8 & 10 & 24 & 6 \\ 12 & 17 & 24 & 49 & 13 \\ 3 & 4 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$
- [611] $1^{14} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 5 & 31 & 4 \\ 3 & 5 & 4 & 30 & 4 \\ 21 & 31 & 30 & 172 & 24 \\ 3 & 4 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$
- [612] $1^{134} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -305, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 19 & 3 \\ 3 & 4 & 17 & 31 & 4 \\ 10 & 17 & 49 & 93 & 13 \\ 19 & 31 & 93 & 172 & 24 \\ 3 & 4 & 13 & 24 & 3 \end{pmatrix}_{b_0}$$
- [614] $1^{13} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -306, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$

$$\begin{aligned}
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 21 & 32 & 0 & 193 & 25 \\ 3 & 4 & 0 & 25 & 3 \end{pmatrix}_{b_0} \\
[613] & 21^{112} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -306, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 9 & 18 & 4 \\ 5 & 4 & 9 & 18 & 4 \\ 9 & 9 & 13 & 30 & 7 \\ 18 & 18 & 30 & 58 & 14 \\ 4 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0} \\
[616] & 1^{13} 1^{14} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -307, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 5 & 31 & 4 \\ 3 & 5 & 4 & 30 & 4 \\ 20 & 31 & 30 & 172 & 24 \\ 3 & 4 & 4 & 24 & 3 \end{pmatrix}_{b_0} \\
[615] & 1^{112} 1^{14} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -307, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 3 & 4 & 1 & 21 & 3 \\ 19 & 31 & 21 & 172 & 24 \\ 3 & 4 & 3 & 24 & 3 \end{pmatrix}_{b_0} \\
[617] & 1^{112} 1^{13} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -309, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 15 & 3 \\ 3 & 4 & 9 & 23 & 4 \\ 6 & 9 & 13 & 39 & 7 \\ 15 & 23 & 39 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0} \\
[619] & 21^{144} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -310, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 12 & 15 & 3 \\ 4 & 4 & 16 & 21 & 4 \\ 12 & 16 & 46 & 60 & 12 \\ 15 & 21 & 60 & 76 & 16 \\ 3 & 4 & 12 & 16 & 3 \end{pmatrix}_{b_0} \\
[621] & 21^{122} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -310, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 9 & 12 & 3 \\ 4 & 4 & 13 & 18 & 4 \\ 9 & 13 & 28 & 42 & 10 \\ 12 & 18 & 42 & 58 & 14 \\ 3 & 4 & 10 & 14 & 3 \end{pmatrix}_{b_0} \\
[620] & 1^{114} 1^{14} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -310, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 1 & 18 & 3 \\ 18 & 26 & 18 & 118 & 20 \\ 3 & 4 & 3 & 20 & 3 \end{pmatrix}_{b_0} \\
[618] & 1^{123} 1^{24} 31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -310, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 16 & 3 \\ 3 & 4 & 18 & 22 & 4 \\ 12 & 18 & 58 & 72 & 14 \\ 16 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0} \\
[622] & 1^{134} 1^{13} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -311, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 17 & 3 \\ 3 & 4 & 17 & 26 & 4 \\ 10 & 17 & 49 & 78 & 13 \\ 17 & 26 & 78 & 118 & 20 \\ 3 & 4 & 13 & 20 & 3 \end{pmatrix}_{b_0} \\
[624] & 1^{123} 1^{13} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -315, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 7 & 18 & 3 \\ 3 & 4 & 9 & 27 & 4 \\ 7 & 9 & 13 & 45 & 7 \\ 18 & 27 & 45 & 133 & 21 \\ 3 & 4 & 7 & 21 & 3 \end{pmatrix}_{b_0} \\
[623] & 1^{113} 1^{14} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -315, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 1 & 18 & 3 \\ 19 & 27 & 18 & 133 & 21 \\ 3 & 4 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

[625] $1^{114} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -316, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 16 & 3 \\ 3 & 4 & 9 & 26 & 4 \\ 5 & 9 & 13 & 42 & 7 \\ 16 & 26 & 42 & 118 & 20 \\ 3 & 4 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[627] $21^{114} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -318, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 15 & 3 \\ 4 & 4 & 9 & 21 & 4 \\ 6 & 9 & 13 & 33 & 7 \\ 15 & 21 & 33 & 76 & 16 \\ 3 & 4 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[628] $1^{114} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -318, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 20 & 3 \\ 3 & 4 & 5 & 28 & 4 \\ 4 & 5 & 4 & 30 & 4 \\ 20 & 28 & 30 & 154 & 22 \\ 3 & 4 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

[626] $21^{112} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -318, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 12 & 3 \\ 4 & 4 & 9 & 18 & 4 \\ 6 & 9 & 13 & 30 & 7 \\ 12 & 18 & 30 & 58 & 14 \\ 3 & 4 & 7 & 14 & 3 \end{pmatrix}_{b_0}$$

[629] $1^{114} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -318, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 1 & 18 & 3 \\ 16 & 26 & 18 & 118 & 20 \\ 3 & 4 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

[631] $1^{124} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -319, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 13 & 3 \\ 3 & 4 & 13 & 21 & 4 \\ 8 & 13 & 28 & 48 & 10 \\ 13 & 21 & 48 & 76 & 16 \\ 3 & 4 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[630] $1^{124} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -319, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 15 & 3 \\ 3 & 4 & 13 & 21 & 4 \\ 10 & 13 & 28 & 48 & 10 \\ 15 & 21 & 48 & 76 & 16 \\ 3 & 4 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[632] $21^{114} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -320, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 4 & 4 & 16 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 12 & 16 & 12 & 46 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

[634] $1^{114} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -324, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 21 & 3 \\ 3 & 4 & 9 & 31 & 4 \\ 5 & 9 & 13 & 51 & 7 \\ 21 & 31 & 51 & 172 & 24 \\ 3 & 4 & 7 & 24 & 3 \end{pmatrix}_{b_0}$$

[635] $1^{144} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -324, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 11 & 3 \\ 3 & 4 & 16 & 16 & 4 \\ 11 & 16 & 46 & 48 & 12 \\ 11 & 16 & 48 & 46 & 12 \\ 3 & 4 & 12 & 12 & 3 \end{pmatrix}_{b_0}$$

[633] $1^{133} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -324, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 17 & 17 & 4 \\ 10 & 17 & 49 & 51 & 13 \\ 10 & 17 & 51 & 49 & 13 \\ 3 & 4 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$

[639] $1^{14} 3 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 20 & 31 & 0 & 172 & 24 \\ 3 & 4 & 0 & 24 & 3 \end{pmatrix}_{b_0}$$

[636] $1^{14} 3 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 23 & 3 \\ 3 & 4 & 5 & 36 & 4 \\ 4 & 5 & 4 & 36 & 4 \\ 23 & 36 & 36 & 238 & 28 \\ 3 & 4 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

[637] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 12 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 9 & 13 & 28 & 39 & 10 \\ 12 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

[638] $1^{13} 3 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 21 & 3 \\ 3 & 4 & 5 & 32 & 4 \\ 4 & 5 & 4 & 33 & 4 \\ 21 & 32 & 33 & 193 & 25 \\ 3 & 4 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

[640] $1^{123} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 15 & 3 \\ 3 & 4 & 13 & 22 & 4 \\ 8 & 13 & 28 & 51 & 10 \\ 15 & 22 & 51 & 85 & 17 \\ 3 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0}$$

[641] $1^{134} 3 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 21 & 3 \\ 3 & 4 & 17 & 31 & 4 \\ 12 & 17 & 49 & 93 & 13 \\ 21 & 31 & 93 & 172 & 24 \\ 3 & 4 & 13 & 24 & 3 \end{pmatrix}_{b_0}$$

[642] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -326, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 1 & 18 & 3 \\ 17 & 26 & 18 & 118 & 20 \\ 3 & 4 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

[643] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -327, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 15 & 21 & 15 & 76 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[647] $1^{124} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -328, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 16 & 3 \\ 3 & 4 & 18 & 26 & 4 \\ 11 & 18 & 58 & 84 & 14 \\ 16 & 26 & 84 & 118 & 20 \\ 3 & 4 & 14 & 20 & 3 \end{pmatrix}_{b_0}$$

[646] $1^{112} 3 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -328, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 20 & 3 \\ 3 & 4 & 9 & 28 & 4 \\ 7 & 9 & 13 & 48 & 7 \\ 20 & 28 & 48 & 154 & 22 \\ 3 & 4 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$

[645] $21^{13} 31^{123} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -328, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 12 & 27 & 4 \\ 5 & 4 & 13 & 27 & 4 \\ 12 & 13 & 28 & 63 & 10 \\ 27 & 27 & 63 & 133 & 21 \\ 4 & 4 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

[644] $1^{134} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -328, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 18 & 3 \\ 3 & 4 & 17 & 26 & 4 \\ 13 & 17 & 49 & 78 & 13 \\ 18 & 26 & 78 & 118 & 20 \\ 3 & 4 & 13 & 20 & 3 \end{pmatrix}_{b_0}$$

[648] $1^{123} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -329, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 13 & 3 \\ 3 & 4 & 13 & 22 & 4 \\ 8 & 13 & 28 & 51 & 10 \\ 13 & 22 & 51 & 85 & 17 \\ 3 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0}$$

[650] $1^{133} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 12 & 3 \\ 3 & 4 & 17 & 17 & 4 \\ 11 & 17 & 49 & 51 & 13 \\ 12 & 17 & 51 & 49 & 13 \\ 3 & 4 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$

[649] $1^{133} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 15 & 3 \\ 3 & 4 & 17 & 22 & 4 \\ 11 & 17 & 49 & 66 & 13 \\ 15 & 22 & 66 & 85 & 17 \\ 3 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

[651] $21^{13} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -332, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 30 & 4 \\ 5 & 4 & 4 & 31 & 4 \\ 3 & 4 & 1 & 21 & 3 \\ 30 & 31 & 21 & 172 & 24 \\ 4 & 4 & 3 & 24 & 3 \end{pmatrix}_{b_0}$$

[653] $1^{123} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -334, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 14 & 3 \\ 3 & 4 & 13 & 22 & 4 \\ 9 & 13 & 28 & 51 & 10 \\ 14 & 22 & 51 & 85 & 17 \\ 3 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0}$$

[654] $1^{112} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -334, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 16 & 3 \\ 3 & 4 & 9 & 23 & 4 \\ 6 & 9 & 13 & 39 & 7 \\ 16 & 23 & 39 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[655] $1^{123} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -334, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 18 & 3 \\ 3 & 4 & 13 & 27 & 4 \\ 8 & 13 & 28 & 63 & 10 \\ 18 & 27 & 63 & 133 & 21 \\ 3 & 4 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

[656] $1^{113} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -334, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 9 & 22 & 4 \\ 6 & 9 & 13 & 36 & 7 \\ 14 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

[657] $21^{112} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -334, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 9 & 21 & 4 \\ 5 & 4 & 9 & 22 & 4 \\ 9 & 9 & 13 & 36 & 7 \\ 21 & 22 & 36 & 85 & 17 \\ 4 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

[652] $1^{124} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -334, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 16 & 3 \\ 3 & 4 & 13 & 26 & 4 \\ 9 & 13 & 28 & 60 & 10 \\ 16 & 26 & 60 & 118 & 20 \\ 3 & 4 & 10 & 20 & 3 \end{pmatrix}_{b_0}$$

[658] $1^{123} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -335, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$

$$T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 13 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 10 & 13 & 28 & 39 & 10 \\ 13 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[659] \ 1^{122} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -336, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 13 & 3 \\ 3 & 4 & 18 & 18 & 4 \\ 12 & 18 & 58 & 60 & 14 \\ 13 & 18 & 60 & 58 & 14 \\ 3 & 4 & 14 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[660] \ 21^{133} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -336, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 9 & 3 \\ 4 & 4 & 12 & 12 & 4 \\ 9 & 12 & 25 & 27 & 9 \\ 9 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[662] \ 1^{144} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -340, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 12 & 3 \\ 3 & 4 & 16 & 16 & 4 \\ 12 & 16 & 46 & 48 & 12 \\ 12 & 16 & 48 & 46 & 12 \\ 3 & 4 & 12 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[661] \ 1^{134} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -340, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 15 & 3 \\ 3 & 4 & 17 & 21 & 4 \\ 11 & 17 & 49 & 63 & 13 \\ 15 & 21 & 63 & 76 & 16 \\ 3 & 4 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[663] \ 1^{134} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -340, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 21 & 3 \\ 3 & 4 & 17 & 31 & 4 \\ 11 & 17 & 49 & 93 & 13 \\ 21 & 31 & 93 & 172 & 24 \\ 3 & 4 & 13 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[664] \ 1^{112} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -340, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 17 & 3 \\ 3 & 4 & 9 & 23 & 4 \\ 7 & 9 & 13 & 39 & 7 \\ 17 & 23 & 39 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[665] \ 1^{122} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -340, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 11 & 3 \\ 3 & 4 & 18 & 18 & 4 \\ 11 & 18 & 58 & 60 & 14 \\ 11 & 18 & 60 & 58 & 14 \\ 3 & 4 & 14 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[668] \ 1^{133} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -344, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$$

$$T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 16 & 3 \\ 3 & 4 & 17 & 22 & 4 \\ 13 & 17 & 49 & 66 & 13 \\ 16 & 22 & 66 & 85 & 17 \\ 3 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[667] \ 1^{133} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -344, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 14 & 3 \\ 3 & 4 & 17 & 22 & 4 \\ 11 & 17 & 49 & 66 & 13 \\ 14 & 22 & 66 & 85 & 17 \\ 3 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[666] \ 21^{113} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -344, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 12 & 17 & 12 & 49 & 13 \\ 3 & 4 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[670] \ 1^{13} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -346, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 22 & 32 & 0 & 193 & 25 \\ 3 & 4 & 0 & 25 & 3 \end{pmatrix}_{b_0}$$

[669] $1^{113} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -346, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 16 & 3 \\ 3 & 4 & 9 & 22 & 4 \\ 6 & 9 & 13 & 36 & 7 \\ 16 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

[671] $1^{124} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -348, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 18 & 3 \\ 3 & 4 & 18 & 26 & 4 \\ 12 & 18 & 58 & 84 & 14 \\ 18 & 26 & 84 & 118 & 20 \\ 3 & 4 & 14 & 20 & 3 \end{pmatrix}_{b_0}$$

[672] $1^{124} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -351, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 14 & 3 \\ 3 & 4 & 13 & 21 & 4 \\ 9 & 13 & 28 & 48 & 10 \\ 14 & 21 & 48 & 76 & 16 \\ 3 & 4 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[673] $1^{122} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -352, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 13 & 3 \\ 3 & 4 & 18 & 18 & 4 \\ 14 & 18 & 58 & 60 & 14 \\ 13 & 18 & 60 & 58 & 14 \\ 3 & 4 & 14 & 14 & 3 \end{pmatrix}_{b_0}$$

[674] $1^{113} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -352, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 3 \\ 3 & 4 & 9 & 27 & 4 \\ 5 & 9 & 13 & 45 & 7 \\ 17 & 27 & 45 & 133 & 21 \\ 3 & 4 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[675] $1^{133} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -355, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 13 & 3 \\ 3 & 4 & 17 & 17 & 4 \\ 12 & 17 & 49 & 51 & 13 \\ 13 & 17 & 51 & 49 & 13 \\ 3 & 4 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$

[676] $21^{112} 31^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -356, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 12 & 21 & 4 \\ 5 & 4 & 13 & 22 & 4 \\ 12 & 13 & 28 & 51 & 10 \\ 21 & 22 & 51 & 85 & 17 \\ 4 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0}$$

[677] $1^{123} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -358, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 13 & 3 \\ 3 & 4 & 18 & 22 & 4 \\ 11 & 18 & 58 & 72 & 14 \\ 13 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0}$$

[679] $1^{124} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -360, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 18 & 3 \\ 3 & 4 & 18 & 26 & 4 \\ 13 & 18 & 58 & 84 & 14 \\ 18 & 26 & 84 & 118 & 20 \\ 3 & 4 & 14 & 20 & 3 \end{pmatrix}_{b_0}$$

[678] $1^{114} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -360, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 18 & 3 \\ 3 & 4 & 9 & 26 & 4 \\ 7 & 9 & 13 & 42 & 7 \\ 18 & 26 & 42 & 118 & 20 \\ 3 & 4 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[681] $1^{133} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -361, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 15 & 3 \\ 3 & 4 & 17 & 22 & 4 \\ 12 & 17 & 49 & 66 & 13 \\ 15 & 22 & 66 & 85 & 17 \\ 3 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

[680] $1^{144} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -361, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 17 & 3 \\ 3 & 4 & 21 & 26 & 4 \\ 13 & 21 & 76 & 96 & 16 \\ 17 & 26 & 96 & 118 & 20 \\ 3 & 4 & 16 & 20 & 3 \end{pmatrix}_{b_0}$$

[684] $1^{113} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 16 & 3 \\ 3 & 4 & 9 & 22 & 4 \\ 7 & 9 & 13 & 36 & 7 \\ 16 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

[683] $1^{114} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 16 & 3 \\ 3 & 4 & 9 & 26 & 4 \\ 6 & 9 & 13 & 42 & 7 \\ 16 & 26 & 42 & 118 & 20 \\ 3 & 4 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[682] $1^{112} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 20 & 3 \\ 3 & 4 & 9 & 28 & 4 \\ 6 & 9 & 13 & 48 & 7 \\ 20 & 28 & 48 & 154 & 22 \\ 3 & 4 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$

[685] $1^{112} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -367, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 19 & 3 \\ 3 & 4 & 9 & 28 & 4 \\ 6 & 9 & 13 & 48 & 7 \\ 19 & 28 & 48 & 154 & 22 \\ 3 & 4 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$

[687] $1^{13} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -367, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 5 & 32 & 4 \\ 3 & 5 & 4 & 33 & 4 \\ 21 & 32 & 33 & 193 & 25 \\ 3 & 4 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

[686] $1^{123} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -367, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 17 & 3 \\ 3 & 4 & 13 & 27 & 4 \\ 8 & 13 & 28 & 63 & 10 \\ 17 & 27 & 63 & 133 & 21 \\ 3 & 4 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

[688] $21^{124} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -368, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 12 & 3 \\ 4 & 4 & 8 & 16 & 4 \\ 6 & 8 & 10 & 24 & 6 \\ 12 & 16 & 24 & 46 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[689] $21^{113} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -368, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 4 & 4 & 22 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 15 & 22 & 15 & 85 & 17 \\ 3 & 4 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

[691] $1^{113} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -369, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 15 & 3 \\ 3 & 4 & 9 & 22 & 4 \\ 6 & 9 & 13 & 36 & 7 \\ 15 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0}$$

[692] $1^{124} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -369, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 23 & 3 \\ 3 & 4 & 13 & 36 & 4 \\ 9 & 13 & 28 & 84 & 10 \\ 23 & 36 & 84 & 238 & 28 \\ 3 & 4 & 10 & 28 & 3 \end{pmatrix}_{b_0}$$

[690] $1^{14} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -369, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 21 & 31 & 0 & 172 & 24 \\ 3 & 4 & 0 & 24 & 3 \end{pmatrix}_{b_0}$$

[693] $1^{13} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -370, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 22 & 3 \\ 3 & 4 & 5 & 32 & 4 \\ 3 & 5 & 4 & 33 & 4 \\ 22 & 32 & 33 & 193 & 25 \\ 3 & 4 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

[694] $1^{14} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -370, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 2 & 4 & 1 & 21 & 3 \\ 21 & 31 & 21 & 172 & 24 \\ 3 & 4 & 3 & 24 & 3 \end{pmatrix}_{b_0}$$

[695] $1^{123} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -370, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 14 & 3 \\ 3 & 4 & 18 & 22 & 4 \\ 11 & 18 & 58 & 72 & 14 \\ 14 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0}$$

[696] $1^{14} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -370, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 1 & 18 & 3 \\ 17 & 26 & 18 & 118 & 20 \\ 3 & 4 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

[697] $1^{134} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -371, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 20 & 3 \\ 3 & 4 & 17 & 31 & 4 \\ 11 & 17 & 49 & 93 & 13 \\ 20 & 31 & 93 & 172 & 24 \\ 3 & 4 & 13 & 24 & 3 \end{pmatrix}_{b_0}$$

[698] $1^{124} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -372, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 15 & 3 \\ 3 & 4 & 13 & 21 & 4 \\ 9 & 13 & 28 & 48 & 10 \\ 15 & 21 & 48 & 76 & 16 \\ 3 & 4 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[699] $1^{14} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -374, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 2 & 4 & 1 & 21 & 3 \\ 19 & 31 & 21 & 172 & 24 \\ 3 & 4 & 3 & 24 & 3 \end{pmatrix}_{b_0}$$

[700] $1^{124} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 17 & 3 \\ 3 & 4 & 18 & 26 & 4 \\ 12 & 18 & 58 & 84 & 14 \\ 17 & 26 & 84 & 118 & 20 \\ 3 & 4 & 14 & 20 & 3 \end{pmatrix}_{b_0}$$

[701] $1^{14} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -378, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 23 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 3 & 4 & 1 & 24 & 3 \\ 23 & 36 & 24 & 238 & 28 \\ 3 & 4 & 3 & 28 & 3 \end{pmatrix}_{b_0}$$

[702] $21^{122} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -380, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.68 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 9 & 3 \\ 4 & 4 & 13 & 13 & 4 \\ 9 & 13 & 28 & 30 & 10 \\ 9 & 13 & 30 & 28 & 10 \\ 3 & 4 & 10 & 10 & 3 \end{pmatrix}_{b_0}$$

[703] $1^{123} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -381, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 18 & 3 \\ 3 & 4 & 13 & 27 & 4 \\ 9 & 13 & 28 & 63 & 10 \\ 18 & 27 & 63 & 133 & 21 \\ 3 & 4 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

[704] $1^{123} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -382, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 16 & 3 \\ 3 & 4 & 18 & 22 & 4 \\ 14 & 18 & 58 & 72 & 14 \\ 16 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0}$$

[705] $1^{133} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -382, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 16 & 3 \\ 3 & 4 & 17 & 22 & 4 \\ 12 & 17 & 49 & 66 & 13 \\ 16 & 22 & 66 & 85 & 17 \\ 3 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

[706] $1^{13} 3 1^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 22 & 3 \\ 3 & 4 & 5 & 32 & 4 \\ 4 & 5 & 4 & 33 & 4 \\ 22 & 32 & 33 & 193 & 25 \\ 3 & 4 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

[707] $1^{114} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 2 & 4 & 1 & 21 & 3 \\ 20 & 31 & 21 & 172 & 24 \\ 3 & 4 & 3 & 24 & 3 \end{pmatrix}_{b_0}$$

[708] $1^{123} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 14 & 3 \\ 3 & 4 & 13 & 22 & 4 \\ 8 & 13 & 28 & 51 & 10 \\ 14 & 22 & 51 & 85 & 17 \\ 3 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0}$$

[709] $1^{134} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -385, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 13 & 3 \\ 3 & 4 & 17 & 21 & 4 \\ 10 & 17 & 49 & 63 & 13 \\ 13 & 21 & 63 & 76 & 16 \\ 3 & 4 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

[710] $1^{113} 3 1^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -387, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 21 & 3 \\ 3 & 4 & 9 & 32 & 4 \\ 5 & 9 & 13 & 54 & 7 \\ 21 & 32 & 54 & 193 & 25 \\ 3 & 4 & 7 & 25 & 3 \end{pmatrix}_{b_0}$$

[711] $1^{113} 3 1^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -388, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 23 & 36 & 0 & 238 & 28 \\ 3 & 4 & 0 & 28 & 3 \end{pmatrix}_{b_0}$$

[712] $1^{124} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -394, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 18 & 3 \\ 3 & 4 & 13 & 26 & 4 \\ 8 & 13 & 28 & 60 & 10 \\ 18 & 26 & 60 & 118 & 20 \\ 3 & 4 & 10 & 20 & 3 \end{pmatrix}_{b_0}$$

[713] $1^{133} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -396, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 11 & 3 \\ 3 & 4 & 17 & 17 & 4 \\ 11 & 17 & 49 & 51 & 13 \\ 11 & 17 & 51 & 49 & 13 \\ 3 & 4 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[714] \ 1^{114} 3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -397, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.84 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 3 & 4 & 1 & 21 & 3 \\ 20 & 31 & 21 & 172 & 24 \\ 3 & 4 & 3 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[715] \ 21^{134} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -398, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.72 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 21 & 3 \\ 4 & 4 & 17 & 31 & 4 \\ 12 & 17 & 49 & 93 & 13 \\ 21 & 31 & 93 & 172 & 24 \\ 3 & 4 & 13 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[716] \ 1^{122} 3^{1122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -404, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.79 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 12 & 3 \\ 3 & 4 & 18 & 18 & 4 \\ 12 & 18 & 58 & 60 & 14 \\ 12 & 18 & 60 & 58 & 14 \\ 3 & 4 & 14 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[717] \ 1^{114} 3^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -406, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 22 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 2 & 4 & 1 & 24 & 3 \\ 22 & 36 & 24 & 238 & 28 \\ 3 & 4 & 3 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[718] \ 1^{114} 3^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -406, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.81 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 24 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 2 & 4 & 1 & 24 & 3 \\ 24 & 36 & 24 & 238 & 28 \\ 3 & 4 & 3 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[719] \ 1^{123} 3^{1122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -406, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.81 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 14 & 3 \\ 3 & 4 & 18 & 22 & 4 \\ 12 & 18 & 58 & 72 & 14 \\ 14 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[720] \ 21^{14} 31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -406, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.81 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 36 & 4 \\ 5 & 4 & 4 & 36 & 4 \\ 3 & 4 & 1 & 24 & 3 \\ 36 & 36 & 24 & 238 & 28 \\ 4 & 4 & 3 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[721] \ 1^{134} 3^{1122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -406, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.81 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 18 & 3 \\ 3 & 4 & 22 & 26 & 4 \\ 14 & 22 & 85 & 102 & 17 \\ 18 & 26 & 102 & 118 & 20 \\ 3 & 4 & 17 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[722] \ 21^{122} 31^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -406, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.73 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 18 & 21 & 4 \\ 5 & 4 & 18 & 22 & 4 \\ 18 & 18 & 58 & 72 & 14 \\ 21 & 22 & 72 & 85 & 17 \\ 4 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[723] \ 1^{114} 3^{1114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -406, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.81 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 24 & 3 \\ 3 & 4 & 5 & 36 & 4 \\ 4 & 5 & 4 & 36 & 4 \\ 24 & 36 & 36 & 238 & 28 \\ 3 & 4 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[724] \ 21^{134} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -408, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.83 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 15 & 3 \\ 4 & 4 & 12 & 21 & 4 \\ 9 & 12 & 25 & 45 & 9 \\ 15 & 21 & 45 & 76 & 16 \\ 3 & 4 & 9 & 16 & 3 \end{pmatrix}_{b_0}$$

[725] $1^{123} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -414, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 13 & 15 & 3 \\ 3 & 4 & 18 & 22 & 4 \\ 13 & 18 & 58 & 72 & 14 \\ 15 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0}$

[726] $21^{112} 31^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -416, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 12 & 24 & 4 \\ 5 & 4 & 13 & 26 & 4 \\ 12 & 13 & 28 & 60 & 10 \\ 24 & 26 & 60 & 118 & 20 \\ 4 & 4 & 10 & 20 & 3 \end{pmatrix}_{b_0}$

[727] $1^{133} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -416, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 13 & 14 & 3 \\ 3 & 4 & 22 & 22 & 4 \\ 13 & 22 & 85 & 87 & 17 \\ 14 & 22 & 87 & 85 & 17 \\ 3 & 4 & 17 & 17 & 3 \end{pmatrix}_{b_0}$

[728] $1^{124} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -418, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 18 & 3 \\ 3 & 4 & 13 & 26 & 4 \\ 10 & 13 & 28 & 60 & 10 \\ 18 & 26 & 60 & 118 & 20 \\ 3 & 4 & 10 & 20 & 3 \end{pmatrix}_{b_0}$

[729] $1^{114} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -418, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 1 & 18 & 3 \\ 18 & 26 & 18 & 118 & 20 \\ 3 & 4 & 3 & 20 & 3 \end{pmatrix}_{b_0}$

[730] $1^{114} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -418, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 23 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 2 & 4 & 1 & 24 & 3 \\ 23 & 36 & 24 & 238 & 28 \\ 3 & 4 & 3 & 28 & 3 \end{pmatrix}_{b_0}$

[731] $1^{123} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -420, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 16 & 3 \\ 3 & 4 & 13 & 22 & 4 \\ 10 & 13 & 28 & 51 & 10 \\ 16 & 22 & 51 & 85 & 17 \\ 3 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0}$

[734] $21^{113} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 9 & 24 & 4 \\ 5 & 4 & 9 & 23 & 4 \\ 9 & 9 & 13 & 39 & 7 \\ 24 & 23 & 39 & 100 & 18 \\ 4 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$

[733] $21^{114} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 6 & 36 & 4 \\ 5 & 4 & 5 & 36 & 4 \\ 6 & 5 & 4 & 36 & 4 \\ 36 & 36 & 36 & 238 & 28 \\ 4 & 4 & 4 & 28 & 3 \end{pmatrix}_{b_0}$

[732] $1^{134} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 13 & 3 \\ 3 & 4 & 17 & 21 & 4 \\ 11 & 17 & 49 & 63 & 13 \\ 13 & 21 & 63 & 76 & 16 \\ 3 & 4 & 13 & 16 & 3 \end{pmatrix}_{b_0}$

[735] $1^{114} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -427, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 20 & 3 \\ 3 & 4 & 9 & 31 & 4 \\ 5 & 9 & 13 & 51 & 7 \\ 20 & 31 & 51 & 172 & 24 \\ 3 & 4 & 7 & 24 & 3 \end{pmatrix}_{b_0}$

[737] $1^{123} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -430, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$

$$\begin{aligned}
& T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 9 & 16 & 3 \\ 3 & 4 & 13 & 22 & 4 \\ 9 & 13 & 28 & 51 & 10 \\ 16 & 22 & 51 & 85 & 17 \\ 3 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0} \\
[736] & 21^{123} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -430, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 9 & 18 & 3 \\ 4 & 4 & 13 & 27 & 4 \\ 9 & 13 & 28 & 63 & 10 \\ 18 & 27 & 63 & 133 & 21 \\ 3 & 4 & 10 & 21 & 3 \end{pmatrix}_{b_0} \\
[738] & 1^{144} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -430, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 13 & 16 & 3 \\ 3 & 4 & 21 & 26 & 4 \\ 13 & 21 & 76 & 96 & 16 \\ 16 & 26 & 96 & 118 & 20 \\ 3 & 4 & 16 & 20 & 3 \end{pmatrix}_{b_0} \\
[740] & 221^{1111} 3 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -432, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 3 & 1 & 3 & 3 \\ 3 & 3 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[739] & 1^{123} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -432, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 9 & 19 & 3 \\ 3 & 4 & 13 & 27 & 4 \\ 9 & 13 & 28 & 63 & 10 \\ 19 & 27 & 63 & 133 & 21 \\ 3 & 4 & 10 & 21 & 3 \end{pmatrix}_{b_0} \\
[741] & 1^{113} 3 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -435, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 17 & 3 \\ 3 & 4 & 9 & 27 & 4 \\ 6 & 9 & 13 & 45 & 7 \\ 17 & 27 & 45 & 133 & 21 \\ 3 & 4 & 7 & 21 & 3 \end{pmatrix}_{b_0} \\
[745] & 1^{134} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -436, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 13 & 15 & 3 \\ 3 & 4 & 17 & 21 & 4 \\ 13 & 17 & 49 & 63 & 13 \\ 15 & 21 & 63 & 76 & 16 \\ 3 & 4 & 13 & 16 & 3 \end{pmatrix}_{b_0} \\
[744] & 1^{122} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -436, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 14 & 3 \\ 3 & 4 & 18 & 18 & 4 \\ 14 & 18 & 58 & 60 & 14 \\ 14 & 18 & 60 & 58 & 14 \\ 3 & 4 & 14 & 14 & 3 \end{pmatrix}_{b_0} \\
[743] & 1^{133} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -436, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6 \\
& T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 12 & 3 \\ 3 & 4 & 17 & 17 & 4 \\ 12 & 17 & 49 & 51 & 13 \\ 12 & 17 & 51 & 49 & 13 \\ 3 & 4 & 13 & 13 & 3 \end{pmatrix}_{b_0} \\
[742] & 1^{122} 3 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -436, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 13 & 13 & 3 \\ 3 & 4 & 18 & 18 & 4 \\ 13 & 18 & 58 & 60 & 14 \\ 13 & 18 & 60 & 58 & 14 \\ 3 & 4 & 14 & 14 & 3 \end{pmatrix}_{b_0} \\
[748] & 1^{114} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 17 & 3 \\ 3 & 4 & 9 & 26 & 4 \\ 6 & 9 & 13 & 42 & 7 \\ 17 & 26 & 42 & 118 & 20 \\ 3 & 4 & 7 & 20 & 3 \end{pmatrix}_{b_0} \\
[746] & 1^{123} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 9 & 15 & 3 \\ 3 & 4 & 13 & 22 & 4 \\ 9 & 13 & 28 & 51 & 10 \\ 15 & 22 & 51 & 85 & 17 \\ 3 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0}$$

[747] $1^{134} 3 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 17 & 3 \\ 3 & 4 & 17 & 26 & 4 \\ 12 & 17 & 49 & 78 & 13 \\ 17 & 26 & 78 & 118 & 20 \\ 3 & 4 & 13 & 20 & 3 \end{pmatrix}_{b_0}$$

[749] $21^{114} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -440, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 6 & 30 & 4 \\ 5 & 4 & 5 & 28 & 4 \\ 6 & 5 & 4 & 30 & 4 \\ 30 & 28 & 30 & 154 & 22 \\ 4 & 4 & 4 & 22 & 3 \end{pmatrix}_{b_0}$$

[750] $1^{13} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -441, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 23 & 3 \\ 3 & 4 & 5 & 36 & 4 \\ 3 & 5 & 4 & 36 & 4 \\ 23 & 36 & 36 & 238 & 28 \\ 3 & 4 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

[751] $1^{14} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -444, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 24 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 24 & 36 & 0 & 238 & 28 \\ 3 & 4 & 0 & 28 & 3 \end{pmatrix}_{b_0}$$

[752] $1^{133} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -444, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 13 & 3 \\ 3 & 4 & 17 & 17 & 4 \\ 13 & 17 & 49 & 51 & 13 \\ 13 & 17 & 51 & 49 & 13 \\ 3 & 4 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$

[753] $21^{124} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -448, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 18 & 3 \\ 4 & 4 & 18 & 26 & 4 \\ 12 & 18 & 58 & 84 & 14 \\ 18 & 26 & 84 & 118 & 20 \\ 3 & 4 & 14 & 20 & 3 \end{pmatrix}_{b_0}$$

[754] $1^{113} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -448, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 19 & 3 \\ 3 & 4 & 9 & 27 & 4 \\ 7 & 9 & 13 & 45 & 7 \\ 19 & 27 & 45 & 133 & 21 \\ 3 & 4 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[755] $21^{114} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -448, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 24 & 3 \\ 4 & 4 & 4 & 36 & 4 \\ 3 & 4 & 1 & 24 & 3 \\ 24 & 36 & 24 & 238 & 28 \\ 3 & 4 & 3 & 28 & 3 \end{pmatrix}_{b_0}$$

[756] $21^{134} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -448, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 12 & 3 \\ 4 & 4 & 12 & 16 & 4 \\ 9 & 12 & 25 & 36 & 9 \\ 12 & 16 & 36 & 46 & 12 \\ 3 & 4 & 9 & 12 & 3 \end{pmatrix}_{b_0}$$

[759] $1^{14} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -450, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 24 & 3 \\ 3 & 4 & 5 & 36 & 4 \\ 3 & 5 & 4 & 36 & 4 \\ 24 & 36 & 36 & 238 & 28 \\ 3 & 4 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

[760] $1^{134} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -450, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 14 & 3 \\ 3 & 4 & 17 & 21 & 4 \\ 12 & 17 & 49 & 63 & 13 \\ 14 & 21 & 63 & 76 & 16 \\ 3 & 4 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

[757] $1^{123} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -450, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 12 & 15 & 3 \\ 3 & 4 & 18 & 22 & 4 \\ 12 & 18 & 58 & 72 & 14 \\ 15 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0}$

[758] $1^{144} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -450, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 14 & 18 & 3 \\ 3 & 4 & 21 & 26 & 4 \\ 14 & 21 & 76 & 96 & 16 \\ 18 & 26 & 96 & 118 & 20 \\ 3 & 4 & 16 & 20 & 3 \end{pmatrix}_{b_0}$

[761] $1^{144} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -451, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 13 & 14 & 3 \\ 3 & 4 & 21 & 21 & 4 \\ 13 & 21 & 76 & 78 & 16 \\ 14 & 21 & 78 & 76 & 16 \\ 3 & 4 & 16 & 16 & 3 \end{pmatrix}_{b_0}$

[762] $1^{114} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -456, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 18 & 3 \\ 3 & 4 & 9 & 26 & 4 \\ 6 & 9 & 13 & 42 & 7 \\ 18 & 26 & 42 & 118 & 20 \\ 3 & 4 & 7 & 20 & 3 \end{pmatrix}_{b_0}$

[763] $1^{113} 3 1^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -460, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 22 & 3 \\ 3 & 4 & 9 & 32 & 4 \\ 7 & 9 & 13 & 54 & 7 \\ 22 & 32 & 54 & 193 & 25 \\ 3 & 4 & 7 & 25 & 3 \end{pmatrix}_{b_0}$

[764] $1^{114} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -462, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 24 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 3 & 4 & 1 & 24 & 3 \\ 24 & 36 & 24 & 238 & 28 \\ 3 & 4 & 3 & 28 & 3 \end{pmatrix}_{b_0}$

[765] $1^{114} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -463, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 3 & 4 & 1 & 21 & 3 \\ 21 & 31 & 21 & 172 & 24 \\ 3 & 4 & 3 & 24 & 3 \end{pmatrix}_{b_0}$

[766] $21^{115} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -464, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.1$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 4 & 4 & 20 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 15 & 20 & 15 & 73 & 15 \\ 3 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$

[768] $21^{122} 31^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -468, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 18 & 18 & 4 \\ 5 & 4 & 18 & 18 & 4 \\ 18 & 18 & 58 & 60 & 14 \\ 18 & 18 & 60 & 58 & 14 \\ 4 & 4 & 14 & 14 & 3 \end{pmatrix}_{b_0}$

[769] $1^{114} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -468, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 21 & 3 \\ 3 & 4 & 9 & 31 & 4 \\ 7 & 9 & 13 & 51 & 7 \\ 21 & 31 & 51 & 172 & 24 \\ 3 & 4 & 7 & 24 & 3 \end{pmatrix}_{b_0}$

[767] $1^{134} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -468, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 16 & 3 \\ 3 & 4 & 17 & 26 & 4 \\ 10 & 17 & 49 & 78 & 13 \\ 16 & 26 & 78 & 118 & 20 \\ 3 & 4 & 13 & 20 & 3 \end{pmatrix}_{b_0}$

[770] $1^{144} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -469, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 17 & 3 \\ 3 & 4 & 21 & 26 & 4 \\ 14 & 21 & 76 & 96 & 16 \\ 17 & 26 & 96 & 118 & 20 \\ 3 & 4 & 16 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[771] \ 1^{14} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -472, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 19 & 3 \\ 3 & 4 & 9 & 31 & 4 \\ 5 & 9 & 13 & 51 & 7 \\ 19 & 31 & 51 & 172 & 24 \\ 3 & 4 & 7 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[772] \ 1^{144} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -474, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 18 & 3 \\ 3 & 4 & 21 & 26 & 4 \\ 15 & 21 & 76 & 96 & 16 \\ 18 & 26 & 96 & 118 & 20 \\ 3 & 4 & 16 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[773] \ 1^{134} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -475, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 14 & 3 \\ 3 & 4 & 17 & 21 & 4 \\ 11 & 17 & 49 & 63 & 13 \\ 14 & 21 & 63 & 76 & 16 \\ 3 & 4 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[774] \ 1^{134} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -476, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 16 & 3 \\ 3 & 4 & 17 & 26 & 4 \\ 11 & 17 & 49 & 78 & 13 \\ 16 & 26 & 78 & 118 & 20 \\ 3 & 4 & 13 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[775] \ 21^{14} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -476, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 6 & 33 & 4 \\ 5 & 4 & 5 & 32 & 4 \\ 6 & 5 & 4 & 33 & 4 \\ 33 & 32 & 33 & 193 & 25 \\ 4 & 4 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[776] \ 21^{124} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -478, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 15 & 3 \\ 4 & 4 & 8 & 21 & 4 \\ 6 & 8 & 10 & 30 & 6 \\ 15 & 21 & 30 & 76 & 16 \\ 3 & 4 & 6 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[777] \ 21^{122} 31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -480, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 12 & 3 \\ 4 & 4 & 18 & 18 & 4 \\ 12 & 18 & 58 & 60 & 14 \\ 12 & 18 & 60 & 58 & 14 \\ 3 & 4 & 14 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[778] \ 1^{114} 3^1 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -480, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 24 & 3 \\ 3 & 4 & 9 & 36 & 4 \\ 5 & 9 & 13 & 60 & 7 \\ 24 & 36 & 60 & 238 & 28 \\ 3 & 4 & 7 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[781] \ 1^{113} 3^1 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -490, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 20 & 3 \\ 3 & 4 & 9 & 32 & 4 \\ 6 & 9 & 13 & 54 & 7 \\ 20 & 32 & 54 & 193 & 25 \\ 3 & 4 & 7 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[780] \ 21^{14} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -490, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 9 & 30 & 4 \\ 5 & 4 & 9 & 28 & 4 \\ 9 & 9 & 13 & 48 & 7 \\ 30 & 28 & 48 & 154 & 22 \\ 4 & 4 & 7 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[779] \ 1^{123} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -490, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 13 & 16 & 3 \\ 3 & 4 & 18 & 22 & 4 \\ 13 & 18 & 58 & 72 & 14 \\ 16 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0} \\
[782] & 21^{123}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -494, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 9 & 12 & 3 \\ 4 & 4 & 13 & 17 & 4 \\ 9 & 13 & 28 & 39 & 10 \\ 12 & 17 & 39 & 49 & 13 \\ 3 & 4 & 10 & 13 & 3 \end{pmatrix}_{b_0} \\
[783] & 21^{122}31^{134} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -494, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 21 & 24 & 4 \\ 5 & 4 & 22 & 26 & 4 \\ 21 & 22 & 85 & 102 & 17 \\ 24 & 26 & 102 & 118 & 20 \\ 4 & 4 & 17 & 20 & 3 \end{pmatrix}_{b_0} \\
[786] & 1^{133}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -496, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6 \\
& T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 15 & 3 \\ 3 & 4 & 22 & 22 & 4 \\ 14 & 22 & 85 & 87 & 17 \\ 15 & 22 & 87 & 85 & 17 \\ 3 & 4 & 17 & 17 & 3 \end{pmatrix}_{b_0} \\
[787] & 21^{114}31^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -496, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 12 & 36 & 4 \\ 5 & 4 & 13 & 36 & 4 \\ 12 & 13 & 28 & 84 & 10 \\ 36 & 36 & 84 & 238 & 28 \\ 4 & 4 & 10 & 28 & 3 \end{pmatrix}_{b_0} \\
[785] & 21^{113}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -496, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 6 & 15 & 3 \\ 4 & 4 & 9 & 22 & 4 \\ 6 & 9 & 13 & 36 & 7 \\ 15 & 22 & 36 & 85 & 17 \\ 3 & 4 & 7 & 17 & 3 \end{pmatrix}_{b_0} \\
[784] & 1^{134}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -496, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 11 & 18 & 3 \\ 3 & 4 & 17 & 26 & 4 \\ 11 & 17 & 49 & 78 & 13 \\ 18 & 26 & 78 & 118 & 20 \\ 3 & 4 & 13 & 20 & 3 \end{pmatrix}_{b_0} \\
[788] & 1^{113}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -500, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 18 & 3 \\ 3 & 4 & 9 & 27 & 4 \\ 6 & 9 & 13 & 45 & 7 \\ 18 & 27 & 45 & 133 & 21 \\ 3 & 4 & 7 & 21 & 3 \end{pmatrix}_{b_0} \\
[789] & 1^{124}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -502, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 8 & 24 & 3 \\ 3 & 4 & 13 & 36 & 4 \\ 8 & 13 & 28 & 84 & 10 \\ 24 & 36 & 84 & 238 & 28 \\ 3 & 4 & 10 & 28 & 3 \end{pmatrix}_{b_0} \\
[790] & 1^{133}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -504, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 13 & 13 & 3 \\ 3 & 4 & 22 & 22 & 4 \\ 13 & 22 & 85 & 87 & 17 \\ 13 & 22 & 87 & 85 & 17 \\ 3 & 4 & 17 & 17 & 3 \end{pmatrix}_{b_0} \\
[791] & 1^{114}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -504, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 7 & 24 & 3 \\ 3 & 4 & 9 & 36 & 4 \\ 7 & 9 & 13 & 60 & 7 \\ 24 & 36 & 60 & 238 & 28 \\ 3 & 4 & 7 & 28 & 3 \end{pmatrix}_{b_0} \\
[792] & 1^{113}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -507, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 19 & 3 \\ 3 & 4 & 9 & 27 & 4 \\ 6 & 9 & 13 & 45 & 7 \\ 19 & 27 & 45 & 133 & 21 \\ 3 & 4 & 7 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[793] $1^{144} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -507, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 14 & 15 & 3 \\ 3 & 4 & 21 & 21 & 4 \\ 14 & 21 & 76 & 78 & 16 \\ 15 & 21 & 78 & 76 & 16 \\ 3 & 4 & 16 & 16 & 3 \end{pmatrix}_{b_0}$

[794] $1^{124} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -509, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 17 & 3 \\ 3 & 4 & 13 & 26 & 4 \\ 8 & 13 & 28 & 60 & 10 \\ 17 & 26 & 60 & 118 & 20 \\ 3 & 4 & 10 & 20 & 3 \end{pmatrix}_{b_0}$

[795] $1^{124} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -510, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 16 & 3 \\ 3 & 4 & 13 & 26 & 4 \\ 8 & 13 & 28 & 60 & 10 \\ 16 & 26 & 60 & 118 & 20 \\ 3 & 4 & 10 & 20 & 3 \end{pmatrix}_{b_0}$

[796] $1^{124} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -511, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 21 & 3 \\ 3 & 4 & 13 & 31 & 4 \\ 8 & 13 & 28 & 72 & 10 \\ 21 & 31 & 72 & 172 & 24 \\ 3 & 4 & 10 & 24 & 3 \end{pmatrix}_{b_0}$

[797] $21^{125} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -512, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 6 & 15 & 3 \\ 4 & 4 & 8 & 20 & 4 \\ 6 & 8 & 10 & 30 & 6 \\ 15 & 20 & 30 & 73 & 15 \\ 3 & 4 & 6 & 15 & 3 \end{pmatrix}_{b_0}$

[798] $21^{133} 31^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -512, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 12 & 15 & 3 \\ 4 & 4 & 17 & 22 & 4 \\ 12 & 17 & 49 & 66 & 13 \\ 15 & 22 & 66 & 85 & 17 \\ 3 & 4 & 13 & 17 & 3 \end{pmatrix}_{b_0}$

[799] $1^{124} 3 1^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -514, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 22 & 3 \\ 3 & 4 & 13 & 36 & 4 \\ 8 & 13 & 28 & 84 & 10 \\ 22 & 36 & 84 & 238 & 28 \\ 3 & 4 & 10 & 28 & 3 \end{pmatrix}_{b_0}$

[800] $1^{114} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -517, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 19 & 3 \\ 3 & 4 & 9 & 31 & 4 \\ 6 & 9 & 13 & 51 & 7 \\ 19 & 31 & 51 & 172 & 24 \\ 3 & 4 & 7 & 24 & 3 \end{pmatrix}_{b_0}$

[801] $21^{114} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -520, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 4 & 4 & 21 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 15 & 21 & 15 & 76 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$

[802] $1^{134} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -525, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 12 & 15 & 3 \\ 3 & 4 & 17 & 21 & 4 \\ 12 & 17 & 49 & 63 & 13 \\ 15 & 21 & 63 & 76 & 16 \\ 3 & 4 & 13 & 16 & 3 \end{pmatrix}_{b_0}$

[803] $1^{124} 3 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -529, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 17 & 3 \\ 3 & 4 & 13 & 26 & 4 \\ 9 & 13 & 28 & 60 & 10 \\ 17 & 26 & 60 & 118 & 20 \\ 3 & 4 & 10 & 20 & 3 \end{pmatrix}_{b_0}$

[804] $1^{134} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -530, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 16 & 3 \\ 3 & 4 & 22 & 26 & 4 \\ 13 & 22 & 85 & 102 & 17 \\ 16 & 26 & 102 & 118 & 20 \\ 3 & 4 & 17 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[805] \ 1^{124} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -535, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 20 & 3 \\ 3 & 4 & 13 & 31 & 4 \\ 9 & 13 & 28 & 72 & 10 \\ 20 & 31 & 72 & 172 & 24 \\ 3 & 4 & 10 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[807] \ 1^{133} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -544, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$$

$$T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 16 & 3 \\ 3 & 4 & 22 & 22 & 4 \\ 15 & 22 & 85 & 87 & 17 \\ 16 & 22 & 87 & 85 & 17 \\ 3 & 4 & 17 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[806] \ 1^{144} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -544, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 13 & 3 \\ 3 & 4 & 21 & 21 & 4 \\ 13 & 21 & 76 & 78 & 16 \\ 13 & 21 & 78 & 76 & 16 \\ 3 & 4 & 16 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[808] \ 21^{144} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -560, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 12 & 3 \\ 4 & 4 & 16 & 16 & 4 \\ 12 & 16 & 46 & 48 & 12 \\ 12 & 16 & 48 & 46 & 12 \\ 3 & 4 & 12 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[809] \ 1^{124} 3 1^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -565, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 23 & 3 \\ 3 & 4 & 13 & 36 & 4 \\ 8 & 13 & 28 & 84 & 10 \\ 23 & 36 & 84 & 238 & 28 \\ 3 & 4 & 10 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[810] \ 21^{113} 31^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -568, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 21 & 3 \\ 4 & 4 & 9 & 32 & 4 \\ 6 & 9 & 13 & 54 & 7 \\ 21 & 32 & 54 & 193 & 25 \\ 3 & 4 & 7 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[811] \ 21^{123} 31^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -568, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 15 & 3 \\ 4 & 4 & 18 & 22 & 4 \\ 12 & 18 & 58 & 72 & 14 \\ 15 & 22 & 72 & 85 & 17 \\ 3 & 4 & 14 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[812] \ 1^{124} 3 1^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -570, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 24 & 3 \\ 3 & 4 & 13 & 36 & 4 \\ 9 & 13 & 28 & 84 & 10 \\ 24 & 36 & 84 & 238 & 28 \\ 3 & 4 & 10 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[813] \ 1^{114} 31^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -571, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 23 & 3 \\ 3 & 4 & 9 & 36 & 4 \\ 5 & 9 & 13 & 60 & 7 \\ 23 & 36 & 60 & 238 & 28 \\ 3 & 4 & 7 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[814] \ 21^{113} 31^{124} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -572, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 12 & 30 & 4 \\ 5 & 4 & 13 & 31 & 4 \\ 12 & 13 & 28 & 72 & 10 \\ 30 & 31 & 72 & 172 & 24 \\ 4 & 4 & 10 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[815] \ 1^{124} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -575, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 8 & 19 & 3 \\ 3 & 4 & 13 & 31 & 4 \\ 8 & 13 & 28 & 72 & 10 \\ 19 & 31 & 72 & 172 & 24 \\ 3 & 4 & 10 & 24 & 3 \end{pmatrix}_{b_0} \\
[816] \quad & 21^{113}31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -576, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 9 & 27 & 4 \\ 5 & 4 & 9 & 27 & 4 \\ 9 & 9 & 13 & 45 & 7 \\ 27 & 27 & 45 & 133 & 21 \\ 4 & 4 & 7 & 21 & 3 \end{pmatrix}_{b_0} \\
[817] \quad & 1^{134}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -579, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 11 & 17 & 3 \\ 3 & 4 & 17 & 26 & 4 \\ 11 & 17 & 49 & 78 & 13 \\ 17 & 26 & 78 & 118 & 20 \\ 3 & 4 & 13 & 20 & 3 \end{pmatrix}_{b_0} \\
[818] \quad & 21^{124}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -580, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 9 & 15 & 3 \\ 4 & 4 & 13 & 21 & 4 \\ 9 & 13 & 28 & 48 & 10 \\ 15 & 21 & 48 & 76 & 16 \\ 3 & 4 & 10 & 16 & 3 \end{pmatrix}_{b_0} \\
[819] \quad & 21^{114}31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -584, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 21 & 3 \\ 4 & 4 & 4 & 31 & 4 \\ 3 & 4 & 1 & 21 & 3 \\ 21 & 31 & 21 & 172 & 24 \\ 3 & 4 & 3 & 24 & 3 \end{pmatrix}_{b_0} \\
[820] \quad & 21^{122}31^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -584, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 21 & 21 & 4 \\ 5 & 4 & 22 & 22 & 4 \\ 21 & 22 & 85 & 87 & 17 \\ 21 & 22 & 87 & 85 & 17 \\ 4 & 4 & 17 & 17 & 3 \end{pmatrix}_{b_0} \\
[821] \quad & 1^{113}31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -585, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 21 & 3 \\ 3 & 4 & 9 & 32 & 4 \\ 6 & 9 & 13 & 54 & 7 \\ 21 & 32 & 54 & 193 & 25 \\ 3 & 4 & 7 & 25 & 3 \end{pmatrix}_{b_0} \\
[822] \quad & 21^{113}31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -588, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 6 & 18 & 3 \\ 4 & 4 & 9 & 27 & 4 \\ 6 & 9 & 13 & 45 & 7 \\ 18 & 27 & 45 & 133 & 21 \\ 3 & 4 & 7 & 21 & 3 \end{pmatrix}_{b_0} \\
[823] \quad & 21^{135}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -592, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.9 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 9 & 15 & 3 \\ 4 & 4 & 12 & 20 & 4 \\ 9 & 12 & 25 & 45 & 9 \\ 15 & 20 & 45 & 73 & 15 \\ 3 & 4 & 9 & 15 & 3 \end{pmatrix}_{b_0} \\
[824] \quad & 21^{124}31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -598, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 9 & 24 & 3 \\ 4 & 4 & 13 & 36 & 4 \\ 9 & 13 & 28 & 84 & 10 \\ 24 & 36 & 84 & 238 & 28 \\ 3 & 4 & 10 & 28 & 3 \end{pmatrix}_{b_0} \\
[826] \quad & 1^{133}31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 14 & 3 \\ 3 & 4 & 22 & 22 & 4 \\ 14 & 22 & 85 & 87 & 17 \\ 14 & 22 & 87 & 85 & 17 \\ 3 & 4 & 17 & 17 & 3 \end{pmatrix}_{b_0} \\
[825] \quad & 1^{124}31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 8 & 20 & 3 \\ 3 & 4 & 13 & 31 & 4 \\ 8 & 13 & 28 & 72 & 10 \\ 20 & 31 & 72 & 172 & 24 \\ 3 & 4 & 10 & 24 & 3 \end{pmatrix}_{b_0} \\
[827] \quad & 1^{134}31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -606, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8
\end{aligned}$$

$$T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 16 & 3 \\ 3 & 4 & 22 & 26 & 4 \\ 14 & 22 & 85 & 102 & 17 \\ 16 & 26 & 102 & 118 & 20 \\ 3 & 4 & 17 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[829] \ 21^{123}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -608, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 15 & 3 \\ 4 & 4 & 13 & 22 & 4 \\ 9 & 13 & 28 & 51 & 10 \\ 15 & 22 & 51 & 85 & 17 \\ 3 & 4 & 10 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[828] \ 21^{114}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -608, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 18 & 3 \\ 4 & 4 & 4 & 26 & 4 \\ 3 & 4 & 1 & 18 & 3 \\ 18 & 26 & 18 & 118 & 20 \\ 3 & 4 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[830] \ 1^{124}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -610, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 18 & 3 \\ 3 & 4 & 13 & 26 & 4 \\ 9 & 13 & 28 & 60 & 10 \\ 18 & 26 & 60 & 118 & 20 \\ 3 & 4 & 10 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[831] \ 1^{134}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -616, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 18 & 3 \\ 3 & 4 & 17 & 26 & 4 \\ 12 & 17 & 49 & 78 & 13 \\ 18 & 26 & 78 & 118 & 20 \\ 3 & 4 & 13 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[832] \ 1^{144}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -616, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 14 & 3 \\ 3 & 4 & 21 & 21 & 4 \\ 14 & 21 & 76 & 78 & 16 \\ 14 & 21 & 78 & 76 & 16 \\ 3 & 4 & 16 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[834] \ 1^{114}31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -622, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 20 & 3 \\ 3 & 4 & 9 & 31 & 4 \\ 6 & 9 & 13 & 51 & 7 \\ 20 & 31 & 51 & 172 & 24 \\ 3 & 4 & 7 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[833] \ 1^{113}31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -622, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 22 & 3 \\ 3 & 4 & 9 & 32 & 4 \\ 6 & 9 & 13 & 54 & 7 \\ 22 & 32 & 54 & 193 & 25 \\ 3 & 4 & 7 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[835] \ 1^{114}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -624, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 22 & 3 \\ 3 & 4 & 9 & 36 & 4 \\ 6 & 9 & 13 & 60 & 7 \\ 22 & 36 & 60 & 238 & 28 \\ 3 & 4 & 7 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[837] \ 21^{116}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -640, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 18 & 3 \\ 4 & 4 & 4 & 24 & 4 \\ 3 & 4 & 1 & 18 & 3 \\ 18 & 24 & 18 & 106 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[836] \ 21^{113}31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -640, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 9 & 30 & 4 \\ 5 & 4 & 9 & 31 & 4 \\ 9 & 9 & 13 & 51 & 7 \\ 30 & 31 & 51 & 172 & 24 \\ 4 & 4 & 7 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[838] \ 21^{122}31^{144} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -644, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 24 & 24 & 4 \\ 5 & 4 & 26 & 26 & 4 \\ 24 & 26 & 118 & 120 & 20 \\ 24 & 26 & 120 & 118 & 20 \\ 4 & 4 & 20 & 20 & 3 \end{pmatrix}_{b_0}$$

[840] $1^{134} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -646, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 17 & 3 \\ 3 & 4 & 22 & 26 & 4 \\ 15 & 22 & 85 & 102 & 17 \\ 17 & 26 & 102 & 118 & 20 \\ 3 & 4 & 17 & 20 & 3 \end{pmatrix}_{b_0}$$

[839] $1^{134} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -646, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 18 & 3 \\ 3 & 4 & 22 & 26 & 4 \\ 16 & 22 & 85 & 102 & 17 \\ 18 & 26 & 102 & 118 & 20 \\ 3 & 4 & 17 & 20 & 3 \end{pmatrix}_{b_0}$$

[841] $21^{114} 31^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -646, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 18 & 3 \\ 4 & 4 & 9 & 26 & 4 \\ 6 & 9 & 13 & 42 & 7 \\ 18 & 26 & 42 & 118 & 20 \\ 3 & 4 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[842] $1^{134} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -650, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 17 & 3 \\ 3 & 4 & 22 & 26 & 4 \\ 14 & 22 & 85 & 102 & 17 \\ 17 & 26 & 102 & 118 & 20 \\ 3 & 4 & 17 & 20 & 3 \end{pmatrix}_{b_0}$$

[843] $1^{144} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -656, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 17 & 3 \\ 3 & 4 & 26 & 26 & 4 \\ 16 & 26 & 118 & 120 & 20 \\ 17 & 26 & 120 & 118 & 20 \\ 3 & 4 & 20 & 20 & 3 \end{pmatrix}_{b_0}$$

[844] $1^{144} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -656, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 15 & 3 \\ 3 & 4 & 21 & 21 & 4 \\ 15 & 21 & 76 & 78 & 16 \\ 15 & 21 & 78 & 76 & 16 \\ 3 & 4 & 16 & 16 & 3 \end{pmatrix}_{b_0}$$

[845] $1^{133} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -664, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 15 & 3 \\ 3 & 4 & 22 & 22 & 4 \\ 15 & 22 & 85 & 87 & 17 \\ 15 & 22 & 87 & 85 & 17 \\ 3 & 4 & 17 & 17 & 3 \end{pmatrix}_{b_0}$$

[846] $21^{133} 31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -668, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 12 & 3 \\ 4 & 4 & 17 & 17 & 4 \\ 12 & 17 & 49 & 51 & 13 \\ 12 & 17 & 51 & 49 & 13 \\ 3 & 4 & 13 & 13 & 3 \end{pmatrix}_{b_0}$$

[847] $1^{114} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -669, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 21 & 3 \\ 3 & 4 & 9 & 31 & 4 \\ 6 & 9 & 13 & 51 & 7 \\ 21 & 31 & 51 & 172 & 24 \\ 3 & 4 & 7 & 24 & 3 \end{pmatrix}_{b_0}$$

[848] $1^{124} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -676, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 21 & 3 \\ 3 & 4 & 13 & 31 & 4 \\ 9 & 13 & 28 & 72 & 10 \\ 21 & 31 & 72 & 172 & 24 \\ 3 & 4 & 10 & 24 & 3 \end{pmatrix}_{b_0}$$

[849] $21^{126} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -688, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 18 & 3 \\ 4 & 4 & 8 & 24 & 4 \\ 6 & 8 & 10 & 36 & 6 \\ 18 & 24 & 36 & 106 & 18 \\ 3 & 4 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

[850] $1^{133} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -696, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$

$$\begin{aligned}
& T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 16 & 16 & 3 \\ 3 & 4 & 22 & 22 & 4 \\ 16 & 22 & 85 & 87 & 17 \\ 16 & 22 & 87 & 85 & 17 \\ 3 & 4 & 17 & 17 & 3 \end{pmatrix}_{b_0} \\
[851] & 21^{145} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -704, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.1 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 12 & 15 & 3 \\ 4 & 4 & 16 & 20 & 4 \\ 12 & 16 & 46 & 60 & 12 \\ 15 & 20 & 60 & 73 & 15 \\ 3 & 4 & 12 & 15 & 3 \end{pmatrix}_{b_0} \\
[852] & 21^{144} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -718, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 15 & 18 & 3 \\ 4 & 4 & 21 & 26 & 4 \\ 15 & 21 & 76 & 96 & 16 \\ 18 & 26 & 96 & 118 & 20 \\ 3 & 4 & 16 & 20 & 3 \end{pmatrix}_{b_0} \\
[853] & 1^{134} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -730, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 15 & 18 & 3 \\ 3 & 4 & 22 & 26 & 4 \\ 15 & 22 & 85 & 102 & 17 \\ 18 & 26 & 102 & 118 & 20 \\ 3 & 4 & 17 & 20 & 3 \end{pmatrix}_{b_0} \\
[854] & 1^{144} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -736, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 17 & 18 & 3 \\ 3 & 4 & 26 & 26 & 4 \\ 17 & 26 & 118 & 120 & 20 \\ 18 & 26 & 120 & 118 & 20 \\ 3 & 4 & 20 & 20 & 3 \end{pmatrix}_{b_0} \\
[855] & 1^{114} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -759, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 23 & 3 \\ 3 & 4 & 9 & 36 & 4 \\ 6 & 9 & 13 & 60 & 7 \\ 23 & 36 & 60 & 238 & 28 \\ 3 & 4 & 7 & 28 & 3 \end{pmatrix}_{b_0} \\
[856] & 21^{114} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -766, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 9 & 33 & 4 \\ 5 & 4 & 9 & 32 & 4 \\ 9 & 9 & 13 & 54 & 7 \\ 33 & 32 & 54 & 193 & 25 \\ 4 & 4 & 7 & 25 & 3 \end{pmatrix}_{b_0} \\
[857] & 21^{136} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -768, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 9 & 18 & 3 \\ 4 & 4 & 12 & 24 & 4 \\ 9 & 12 & 25 & 54 & 9 \\ 18 & 24 & 54 & 106 & 18 \\ 3 & 4 & 9 & 18 & 3 \end{pmatrix}_{b_0} \\
[858] & 1^{144} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -804, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 16 & 16 & 3 \\ 3 & 4 & 26 & 26 & 4 \\ 16 & 26 & 118 & 120 & 20 \\ 16 & 26 & 120 & 118 & 20 \\ 3 & 4 & 20 & 20 & 3 \end{pmatrix}_{b_0} \\
[859] & 21^{134} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -814, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 12 & 15 & 3 \\ 4 & 4 & 17 & 21 & 4 \\ 12 & 17 & 49 & 63 & 13 \\ 15 & 21 & 63 & 76 & 16 \\ 3 & 4 & 13 & 16 & 3 \end{pmatrix}_{b_0} \\
[860] & 1^{114} 3 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -836, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 24 & 3 \\ 3 & 4 & 9 & 36 & 4 \\ 6 & 9 & 13 & 60 & 7 \\ 24 & 36 & 60 & 238 & 28 \\ 3 & 4 & 7 & 28 & 3 \end{pmatrix}_{b_0} \\
[862] & 21^{117} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -848, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 21 & 3 \\ 4 & 4 & 4 & 28 & 4 \\ 3 & 4 & 1 & 21 & 3 \\ 21 & 28 & 21 & 145 & 21 \\ 3 & 4 & 3 & 21 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [861] $21^{155}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -848, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 15 & 15 & 3 \\ 4 & 4 & 20 & 20 & 4 \\ 15 & 20 & 73 & 75 & 15 \\ 15 & 20 & 75 & 73 & 15 \\ 3 & 4 & 15 & 15 & 3 \end{pmatrix}_{b_0}$
- [863] $21^{114}31^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -862, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 6 & 21 & 3 \\ 4 & 4 & 9 & 31 & 4 \\ 6 & 9 & 13 & 51 & 7 \\ 21 & 31 & 51 & 172 & 24 \\ 3 & 4 & 7 & 24 & 3 \end{pmatrix}_{b_0}$
- [864] $21^{124}31^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -878, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 9 & 18 & 3 \\ 4 & 4 & 13 & 26 & 4 \\ 9 & 13 & 28 & 60 & 10 \\ 18 & 26 & 60 & 118 & 20 \\ 3 & 4 & 10 & 20 & 3 \end{pmatrix}_{b_0}$
- [865] $21^{146}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -880, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 12 & 18 & 3 \\ 4 & 4 & 16 & 24 & 4 \\ 12 & 16 & 46 & 72 & 12 \\ 18 & 24 & 72 & 106 & 18 \\ 3 & 4 & 12 & 18 & 3 \end{pmatrix}_{b_0}$
- [866] $21^{124}31^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -884, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 9 & 21 & 3 \\ 4 & 4 & 13 & 31 & 4 \\ 9 & 13 & 28 & 72 & 10 \\ 21 & 31 & 72 & 172 & 24 \\ 3 & 4 & 10 & 24 & 3 \end{pmatrix}_{b_0}$
- [868] $21^{127}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -896, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 6 & 21 & 3 \\ 4 & 4 & 8 & 28 & 4 \\ 6 & 8 & 10 & 42 & 6 \\ 21 & 28 & 42 & 145 & 21 \\ 3 & 4 & 6 & 21 & 3 \end{pmatrix}_{b_0}$
- [867] $21^{133}31^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -896, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.71 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 15 & 15 & 3 \\ 4 & 4 & 22 & 22 & 4 \\ 15 & 22 & 85 & 87 & 17 \\ 15 & 22 & 87 & 85 & 17 \\ 3 & 4 & 17 & 17 & 3 \end{pmatrix}_{b_0}$
- [869] $1^{144}31^{133} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -900, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 17 & 17 & 3 \\ 3 & 4 & 26 & 26 & 4 \\ 17 & 26 & 118 & 120 & 20 \\ 17 & 26 & 120 & 118 & 20 \\ 3 & 4 & 20 & 20 & 3 \end{pmatrix}_{b_0}$
- [870] $21^{134}31^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -902, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 12 & 18 & 3 \\ 4 & 4 & 17 & 26 & 4 \\ 12 & 17 & 49 & 78 & 13 \\ 18 & 26 & 78 & 118 & 20 \\ 3 & 4 & 13 & 20 & 3 \end{pmatrix}_{b_0}$
- [871] $21^{114}31^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -954, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 9 & 36 & 4 \\ 5 & 4 & 9 & 36 & 4 \\ 9 & 9 & 13 & 60 & 7 \\ 36 & 36 & 60 & 238 & 28 \\ 4 & 4 & 7 & 28 & 3 \end{pmatrix}_{b_0}$
- [872] $1^{144}31^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -964, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 18 & 18 & 3 \\ 3 & 4 & 26 & 26 & 4 \\ 18 & 26 & 118 & 120 & 20 \\ 18 & 26 & 120 & 118 & 20 \\ 3 & 4 & 20 & 20 & 3 \end{pmatrix}_{b_0}$
- [873] $21^{114}31^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -966, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 6 & 24 & 3 \\ 4 & 4 & 9 & 36 & 4 \\ 6 & 9 & 13 & 60 & 7 \\ 24 & 36 & 60 & 238 & 28 \\ 3 & 4 & 7 & 28 & 3 \end{pmatrix}_{b_0}$
- [874] $21^{137}31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -976, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.1$

$$\begin{aligned}
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 9 & 21 & 3 \\ 4 & 4 & 12 & 28 & 4 \\ 9 & 12 & 25 & 63 & 9 \\ 21 & 28 & 63 & 145 & 21 \\ 3 & 4 & 9 & 21 & 3 \end{pmatrix}_{b_0} \\
[875] & 21^{134}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -1016, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 15 & 18 & 3 \\ 4 & 4 & 22 & 26 & 4 \\ 15 & 22 & 85 & 102 & 17 \\ 18 & 26 & 102 & 118 & 20 \\ 3 & 4 & 17 & 20 & 3 \end{pmatrix}_{b_0} \\
[876] & 21^{144}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -1020, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 15 & 15 & 3 \\ 4 & 4 & 21 & 21 & 4 \\ 15 & 21 & 76 & 78 & 16 \\ 15 & 21 & 78 & 76 & 16 \\ 3 & 4 & 16 & 16 & 3 \end{pmatrix}_{b_0} \\
[877] & 21^{156}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -1024, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.4 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 15 & 18 & 3 \\ 4 & 4 & 20 & 24 & 4 \\ 15 & 20 & 73 & 90 & 15 \\ 18 & 24 & 90 & 106 & 18 \\ 3 & 4 & 15 & 18 & 3 \end{pmatrix}_{b_0} \\
[878] & 21^{147}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -1088, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 12 & 21 & 3 \\ 4 & 4 & 16 & 28 & 4 \\ 12 & 16 & 46 & 84 & 12 \\ 21 & 28 & 84 & 145 & 21 \\ 3 & 4 & 12 & 21 & 3 \end{pmatrix}_{b_0} \\
[879] & 21^{166}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -1200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 18 & 18 & 3 \\ 4 & 4 & 24 & 24 & 4 \\ 18 & 24 & 106 & 108 & 18 \\ 18 & 24 & 108 & 106 & 18 \\ 3 & 4 & 18 & 18 & 3 \end{pmatrix}_{b_0} \\
[880] & 21^{157}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -1232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 15 & 21 & 3 \\ 4 & 4 & 20 & 28 & 4 \\ 15 & 20 & 73 & 105 & 15 \\ 21 & 28 & 105 & 145 & 21 \\ 3 & 4 & 15 & 21 & 3 \end{pmatrix}_{b_0} \\
[881] & 21^{144}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -1376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.72 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 18 & 18 & 3 \\ 4 & 4 & 26 & 26 & 4 \\ 18 & 26 & 118 & 120 & 20 \\ 18 & 26 & 120 & 118 & 20 \\ 3 & 4 & 20 & 20 & 3 \end{pmatrix}_{b_0} \\
[882] & 21^{167}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -1408, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.6 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 18 & 21 & 3 \\ 4 & 4 & 24 & 28 & 4 \\ 18 & 24 & 106 & 126 & 18 \\ 21 & 28 & 126 & 145 & 21 \\ 3 & 4 & 18 & 21 & 3 \end{pmatrix}_{b_0} \\
[883] & 21^{177}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -1616, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 21 & 21 & 3 \\ 4 & 4 & 28 & 28 & 4 \\ 21 & 28 & 145 & 147 & 21 \\ 21 & 28 & 147 & 145 & 21 \\ 3 & 4 & 21 & 21 & 3 \end{pmatrix}_{b_0} \\
T = 8 & [887] 1^{12}31^131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 0 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 0 & 28 & 10 \\ 2 & 3 & -1 & 0 & 10 & 3 \end{pmatrix}_{b_0} \\
[886] & 1^{11}31^131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 1 & 3 \\ 2 & 3 & -1 & 4 & 3 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[885] \ 1^{\textcolor{blue}{1}}31^{\textcolor{blue}{1}}31^{\textcolor{blue}{1}\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{s}\textcolor{blue}{u}_2 + \textcolor{red}{s}\textcolor{red}{u}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 3, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 4 \\ 2 & 3 & -1 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[884] \quad \begin{array}{c} 1^1 \\ 1^1 4^1 1^1 \\ 1^1 1^1 \end{array} + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{5}\textcolor{teal}{u}_2 + \textcolor{orange}{5}\textcolor{orange}{u}_2 \quad T = 8, \, T_{\max} = 10, \, \Delta = -75, \, \det = 3, \, (n_+, n_-, n_0) = (1, 8, 0), \, c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \, T_{\max} = 10, \, T \in [10, 10]$$

$$\left(\begin{array}{c} 1 \, 3 \, 3 \, 2 \, 3 \\ 3 \, 4 \, 3 \, 3 \, 4 \\ 3 \, 3 \, 1 \, 3 \, 3 \\ 2 \, 3 \, 3 \, 1 \, 3 \\ 3 \, 4 \, 3 \, 3 \, 4 \end{array} \right)_{b_0}$$

$$[888] \begin{array}{c} 1^{11} \\ 1 \ 4 \ 1^1 \\ 1^1 \end{array} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -75, \ \det = 4, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\left(\begin{array}{cccc} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{array} \right)_{b_0}$$

$$[889] \quad \begin{array}{c} 1^1 \\ 1^1 4^1 1^1 \end{array} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \, T_{\max} = 10, \, \Delta = -75, \, \det = 7, \, (n_+, n_-, n_0) = (1, 8, 0), \, c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \, T_{\max} = 10, \, T \in [10, 10]$$

$$\left(\begin{array}{cc|cc} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 0 & 4 \end{array} \right)_{b_0}$$

$$[890] \ 1^{\textcolor{red}{1}}1^{\textcolor{red}{1}}31^{\textcolor{red}{1}}31^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{green}{u}_2 + \textcolor{red}{s}\textcolor{red}{u}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 14, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.86 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 3 \\ 3 & 6 & 0 & 6 & 13 & 7 \\ 2 & 3 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

[892] $1^{11}1131^{13}1^{11} + [\textcolor{brown}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{brown}{su}_2 + \textcolor{brown}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 6 & 3 \\ 4 & 6 & 0 & 6 & 13 & 7 \\ 2 & 3 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$

$$[891] \ 1^{111}3131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 3 & 1 & 3 \\ 2 & 3 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

[894] $1^{111}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -63, \quad \det = 26, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 6 & 3 \\ 3 & 6 & 0 & 6 & 13 & 7 \\ 2 & 3 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$

$$[893] \quad 1^{11}31^{11}31^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -63, \quad \det = 26, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 9 & 4 \\ 3 & 6 & 0 & 9 & 13 & 7 \\ 2 & 3 & -1 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[896] \begin{array}{c} 1^{11} \\ 2^3 2 + [\textcolor{red}{5}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{5}u_2 + \textcolor{orange}{5}u_2 \\ 1^{11} \end{array} \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -58, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 2 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 & 2 \\ 2 & 2 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 2 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

[895] $2 \overset{11}{\underset{132}{3}} 2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 6 & 6 & 2 \\ 2 & 1 & 3 & 9 & 8 & 3 \\ 2 & 3 & 0 & 6 & 7 & 2 \\ 6 & 9 & 6 & 25 & 24 & 9 \\ 6 & 8 & 7 & 24 & 21 & 9 \\ 2 & 3 & 2 & 9 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[897] \ 1^{11}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 45, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 9 & 4 \\ 4 & 6 & 0 & 9 & 13 & 7 \\ 2 & 3 & -1 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[898] \ 1^{12}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 45, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 12 & 4 \\ 5 & 9 & 0 & 12 & 28 & 10 \\ 2 & 3 & -1 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[899] \ 1^{11}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 56, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & 1 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 13 & 15 & 7 \\ 4 & 6 & 0 & 15 & 13 & 7 \\ 2 & 3 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[901] \ 2 \overset{11}{\underset{1^{12}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 64, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 4 & 2 \\ 2 & 1 & 3 & 6 & 5 & 3 \\ 2 & 3 & 0 & 4 & 5 & 2 \\ 4 & 6 & 4 & 10 & 10 & 6 \\ 4 & 5 & 5 & 10 & 8 & 6 \\ 2 & 3 & 2 & 6 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[900] \ 1^{11}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 64, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & 1 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 13 & 15 & 7 \\ 3 & 6 & 0 & 15 & 13 & 7 \\ 2 & 3 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[902] \ 1^{11}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 69, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 3 \\ 4 & 6 & 0 & 6 & 13 & 7 \\ 2 & 3 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[904] \ 1^{12}31^{23}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 12 & 3 \\ 7 & 12 & 0 & 12 & 58 & 14 \\ 2 & 3 & -1 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[903] \ 1^{12}31^{23}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 12 & 3 \\ 7 & 12 & 0 & 12 & 58 & 14 \\ 2 & 3 & -1 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[905] \ 1^{12}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 9 & 3 \\ 5 & 9 & 0 & 9 & 28 & 10 \\ 2 & 3 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[906] \ 2 \overset{1}{\underset{1^{11}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 96, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.83 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 2 & 3 & 2 & 1 & 3 & 3 \\ 2 & 3 & 2 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[907] \ 1^{12}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 108, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 9 & 3 \\ 5 & 9 & 0 & 9 & 28 & 10 \\ 2 & 3 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0} \\
[908] \quad & 1^{11231} 1^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & 1 & 6 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 13 & 21 & 7 \\ 5 & 9 & 0 & 21 & 28 & 10 \\ 2 & 3 & -1 & 7 & 10 & 3 \end{pmatrix}_{b_0} \\
[909] \quad & 2 \overset{1}{\underset{1^{112}}{3}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 3 & 6 & 3 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 1 & 6 & 3 \\ 4 & 6 & 4 & 6 & 10 & 6 \\ 2 & 3 & 2 & 3 & 6 & 3 \end{pmatrix}_{b_0} \\
[910] \quad & 2 \overset{1}{\underset{1^{111}}{3}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 3 & 7 & 3 \\ 2 & 3 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 1 & 7 & 3 \\ 6 & 7 & 8 & 7 & 21 & 9 \\ 2 & 3 & 2 & 3 & 9 & 3 \end{pmatrix}_{b_0} \\
[911] \quad & 1^{11231} 1^{11} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 1 & 6 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 13 & 21 & 7 \\ 5 & 9 & 0 & 21 & 28 & 10 \\ 2 & 3 & -1 & 7 & 10 & 3 \end{pmatrix}_{b_0} \\
[912] \quad & 2 \overset{1}{\underset{1^{122}}{3}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 4 & 4 & 2 \\ 2 & 1 & 3 & 6 & 6 & 3 \\ 2 & 3 & 0 & 4 & 4 & 2 \\ 4 & 6 & 4 & 10 & 12 & 6 \\ 4 & 6 & 4 & 12 & 10 & 6 \\ 2 & 3 & 2 & 6 & 6 & 3 \end{pmatrix}_{b_0} \\
[913] \quad & 1^{111} 31^{11} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 1 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 13 & 15 & 7 \\ 4 & 6 & 0 & 15 & 13 & 7 \\ 2 & 3 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0} \\
[914] \quad & 2 \overset{1}{\underset{1^{111}}{3}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 3 & 5 & 3 \\ 2 & 3 & 0 & 2 & 5 & 2 \\ 2 & 3 & 2 & 1 & 5 & 3 \\ 4 & 5 & 5 & 5 & 8 & 6 \\ 2 & 3 & 2 & 3 & 6 & 3 \end{pmatrix}_{b_0} \\
[915] \quad & 2 \overset{1}{\underset{1^{113}}{3}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 3 & 9 & 3 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 1 & 9 & 3 \\ 6 & 9 & 6 & 9 & 25 & 9 \\ 2 & 3 & 2 & 3 & 9 & 3 \end{pmatrix}_{b_0} \\
[917] \quad & 2 \overset{1}{\underset{1^{123}}{3}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 4 & 10 & 2 \\ 2 & 1 & 3 & 6 & 13 & 3 \\ 2 & 3 & 0 & 4 & 12 & 2 \\ 4 & 6 & 4 & 10 & 26 & 6 \\ 10 & 13 & 12 & 26 & 61 & 15 \\ 2 & 3 & 2 & 6 & 15 & 3 \end{pmatrix}_{b_0} \\
[916] \quad & 2 \overset{1}{\underset{1^{123}}{3}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 6 & 2 \\ 2 & 1 & 3 & 6 & 9 & 3 \\ 2 & 3 & 0 & 4 & 6 & 2 \\ 4 & 6 & 4 & 10 & 18 & 6 \\ 6 & 9 & 6 & 18 & 25 & 9 \\ 2 & 3 & 2 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[918] $2 \overset{1^{11}}{\underset{1^{122}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 6 & 2 \\ 2 & 1 & 3 & 6 & 8 & 3 \\ 2 & 3 & 0 & 4 & 7 & 2 \\ 4 & 6 & 4 & 10 & 16 & 6 \\ 6 & 8 & 7 & 16 & 21 & 9 \\ 2 & 3 & 2 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[919] $2 \overset{1^1}{\underset{1^{114}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 3 & 12 & 3 \\ 2 & 3 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 1 & 12 & 3 \\ 8 & 12 & 8 & 12 & 46 & 12 \\ 2 & 3 & 2 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

[920] $2 \overset{1^1}{\underset{1^{124}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 8 & 2 \\ 2 & 1 & 3 & 6 & 12 & 3 \\ 2 & 3 & 0 & 4 & 8 & 2 \\ 4 & 6 & 4 & 10 & 24 & 6 \\ 8 & 12 & 8 & 24 & 46 & 12 \\ 2 & 3 & 2 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[921] $2 \overset{1^1}{\underset{1^{115}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.1$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 3 & 15 & 3 \\ 2 & 3 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 1 & 15 & 3 \\ 10 & 15 & 10 & 15 & 73 & 15 \\ 2 & 3 & 2 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

[922] $2 \overset{1^{11}}{\underset{1^{112}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 3 & 8 & 3 \\ 2 & 3 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 1 & 8 & 3 \\ 6 & 8 & 7 & 8 & 21 & 9 \\ 2 & 3 & 2 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[923] $2 \overset{1^{11}}{\underset{1^{123}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 8 & 2 \\ 2 & 1 & 3 & 6 & 11 & 3 \\ 2 & 3 & 0 & 4 & 9 & 2 \\ 4 & 6 & 4 & 10 & 22 & 6 \\ 8 & 11 & 9 & 22 & 40 & 12 \\ 2 & 3 & 2 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[925] $2 \overset{1^{12}}{\underset{1^{112}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 3 & 10 & 3 \\ 2 & 3 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 1 & 10 & 3 \\ 8 & 10 & 10 & 10 & 38 & 12 \\ 2 & 3 & 2 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

[924] $2 \overset{1^1}{\underset{1^{116}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 12 & 2 \\ 2 & 1 & 3 & 3 & 18 & 3 \\ 2 & 3 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 1 & 18 & 3 \\ 12 & 18 & 12 & 18 & 106 & 18 \\ 2 & 3 & 2 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

[926] $2 \overset{1^{11}}{\underset{1^{113}}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 3 & 11 & 3 \\ 2 & 3 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 1 & 11 & 3 \\ 8 & 11 & 9 & 11 & 40 & 12 \\ 2 & 3 & 2 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[927] \quad 2 \overset{1^{12}}{\underset{1^{124}}{3}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 12 & 2 \\ 2 & 1 & 3 & 6 & 16 & 3 \\ 2 & 3 & 0 & 4 & 14 & 2 \\ 4 & 6 & 4 & 10 & 32 & 6 \\ 12 & 16 & 14 & 32 & 90 & 18 \\ 2 & 3 & 2 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[928] \quad 2 \overset{1^1}{\underset{1^{117}}{3}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 14 & 2 \\ 2 & 1 & 3 & 3 & 21 & 3 \\ 2 & 3 & 0 & 2 & 14 & 2 \\ 2 & 3 & 2 & 1 & 21 & 3 \\ 14 & 21 & 14 & 21 & 145 & 21 \\ 2 & 3 & 2 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[929] \quad 2 \overset{1^{11}}{\underset{1^{124}}{3}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 520, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 10 & 2 \\ 2 & 1 & 3 & 6 & 14 & 3 \\ 2 & 3 & 0 & 4 & 11 & 2 \\ 4 & 6 & 4 & 10 & 28 & 6 \\ 10 & 14 & 11 & 28 & 65 & 15 \\ 2 & 3 & 2 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[930] \quad 2 \overset{1^{11}}{\underset{1^{114}}{3}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 616, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 3 & 14 & 3 \\ 2 & 3 & 0 & 2 & 11 & 2 \\ 2 & 3 & 2 & 1 & 14 & 3 \\ 10 & 14 & 11 & 14 & 65 & 15 \\ 2 & 3 & 2 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[931] \quad 2 \overset{1^{12}}{\underset{1^{113}}{3}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 640, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 3 & 13 & 3 \\ 2 & 3 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 1 & 13 & 3 \\ 10 & 13 & 12 & 13 & 61 & 15 \\ 2 & 3 & 2 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[932] \quad 2 \overset{1^{12}}{\underset{1^{114}}{3}} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 920, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 12 & 2 \\ 2 & 1 & 3 & 3 & 16 & 3 \\ 2 & 3 & 0 & 2 & 14 & 2 \\ 2 & 3 & 2 & 1 & 16 & 3 \\ 12 & 16 & 14 & 16 & 90 & 18 \\ 2 & 3 & 2 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [933] \quad 1^{\textcolor{red}{11}} 23131^{\textcolor{red}{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 4 & 0 & 2 & 4 \\ 2 & 0 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[934] \quad 1^{\textcolor{red}{111}} 231^{\textcolor{red}{1}} 31^{\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 2 & 1 & 4 & 7 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 4 & 0 & 2 & 7 & 4 \\ 3 & 7 & 0 & 7 & 14 & 8 \\ 2 & 4 & -1 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[935] \quad 1^{\textcolor{red}{11}} 231^{\textcolor{red}{1}} 31^{\textcolor{red}{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 0 & 7 & 0 & 14 & 8 \\ 2 & 0 & 3 & 0 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[936] \quad 1^{\textcolor{red}{133}} 2^{\textcolor{red}{3}} 21^{\textcolor{red}{123}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 2 & 11 & 12 & 3 \\ 1 & 2 & 0 & 9 & 12 & 2 \\ 6 & 11 & 9 & 49 & 60 & 15 \\ 8 & 12 & 12 & 60 & 70 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[937] $1^{11} 1^2 3 21^{21} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 1 & 5 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & 17 & 12 & 9 \\ 2 & 4 & 0 & 12 & 6 & 6 \\ 2 & 3 & -1 & 9 & 6 & 4 \end{pmatrix}_{b_0}$$

[938] $1^{11} 1^2 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 3 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 2 & 3 & 1 & 1 & 4 & 3 \\ 4 & 4 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 3 & 6 & 4 \end{pmatrix}_{b_0}$$

[939] $1^{12} 2^3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 2 & 4 & 4 & 6 & 12 & 6 \\ 5 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

[940] $1^{12} 3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 6 & 2 \\ 2 & 0 & 1 & 1 & 10 & 3 \\ 6 & 0 & 6 & 10 & 34 & 12 \\ 2 & 0 & 2 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[941] $1^{11} 2 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 3 & 1 & 1 & 4 & 3 \\ 2 & 4 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 3 & 6 & 4 \end{pmatrix}_{b_0}$$

[942] $1^{13} 2^3 21^{34} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 1 & 12 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 84 & 18 \\ 8 & 13 & 0 & 84 & 97 & 21 \\ 2 & 3 & -1 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

[943] $1^{14} 2^3 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 4 & 4 & 4 & 6 & 28 & 6 \\ 10 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[944] $1^{11} 5 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -70, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 4 & 6 & 8 & 5 & 6 \\ 2 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

[945] $1^{11} 2^3 21^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

[946] $1^{11} 1^2 3 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 2 & 3 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 2 & 3 & 1 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 & 6 & 6 \\ 2 & 3 & 2 & 3 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[947] \quad 1 \overset{1}{\underset{1^1, 1^{11}}{5}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -70, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 4 & 6 & 8 & 5 & 6 \\ 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[949] \quad 1 \overset{1^{11}}{\underset{1^1, 1^1}{5}} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -70, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[948] \quad 21 \overset{1}{\underset{1^1}{5}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -70, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 3 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[950] \quad 1 \overset{1}{\underset{1^1, 1^{11}}{5}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -70, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 5 & 6 & 8 & 5 & 6 \\ 2 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[951] \quad 1 \overset{1}{\underset{1^1, 1^{11}}{5}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -70, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 4 & 6 & 8 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[952] \quad 21 \overset{1^{11}}{\underset{1^1}{5}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -70, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 5 & 6 & 8 & 5 & 6 \\ 2 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[953] \quad 1 \overset{1^{12}}{\underset{1^{11}}{22}} 3 \overset{1^{12}}{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 2 & 7 & 8 & 3 \\ 1 & 2 & 0 & 5 & 8 & 2 \\ 4 & 7 & 5 & 17 & 24 & 9 \\ 6 & 8 & 8 & 24 & 30 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[954] \quad 1 \overset{1^{11}}{\underset{1^{12}}{22}} 3 \overset{1^{11}}{21} \overset{23}{23} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 1 & 8 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 40 & 12 \\ 6 & 9 & 0 & 40 & 49 & 15 \\ 2 & 3 & -1 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[955] \quad 1 \overset{1}{\underset{1^1, 1^1}{5}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -70, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[957] \quad 1 \overset{1^{11}}{\underset{1^{32}}{32}} 3 \overset{1^{11}}{21} \overset{43}{43} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 13 & 9 & 2 \\ 8 & 0 & 13 & 97 & 71 & 21 \\ 6 & 0 & 9 & 71 & 49 & 15 \\ 2 & 0 & 2 & 21 & 15 & 4 \end{pmatrix}_{b_0}$$

[959] $1^{114}2321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 3 & 15 & 3 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 1 & 3 & 1 & 1 & 15 & 3 \\ 9 & 15 & 13 & 15 & 97 & 21 \\ 2 & 3 & 2 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[958] $1^{112}2321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 3 & 7 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 3 & 1 & 1 & 7 & 3 \\ 5 & 7 & 5 & 7 & 17 & 9 \\ 2 & 3 & 2 & 3 & 9 & 4 \end{pmatrix}_{b_0}$$

[956] $1^{111}2321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 5 & 2 \\ 2 & 0 & 1 & 1 & 7 & 3 \\ 4 & 0 & 5 & 7 & 17 & 9 \\ 2 & 0 & 2 & 3 & 9 & 4 \end{pmatrix}_{b_0}$$

[960] $1^{111}231^{13}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 2 & 1 & 7 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 7 & 0 & 14 & 7 & 8 \\ 2 & 4 & 0 & 7 & 2 & 4 \\ 2 & 4 & -1 & 8 & 4 & 4 \end{pmatrix}_{b_0}$$

[962] $1^{112}2321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 9 & 2 \\ 2 & 0 & 4 & 6 & 20 & 6 \\ 6 & 0 & 9 & 20 & 49 & 15 \\ 2 & 0 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[961] $1^{111}2321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 5 & 2 \\ 2 & 0 & 4 & 6 & 12 & 6 \\ 4 & 0 & 5 & 12 & 17 & 9 \\ 2 & 0 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

[964] $1^{113}2321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 3 & 1 & 1 & 11 & 3 \\ 7 & 11 & 9 & 11 & 49 & 15 \\ 2 & 3 & 2 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[963] $1^{112}2321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & 0 & 6 & 4 \end{pmatrix}_{b_0}$$

[966] $1^{114}2321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 3 & 14 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 3 & 1 & 1 & 14 & 3 \\ 10 & 14 & 10 & 14 & 74 & 18 \\ 2 & 3 & 2 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[965] $1^{112}2321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 10 & 2 \\ 2 & 0 & 1 & 1 & 14 & 3 \\ 8 & 0 & 10 & 14 & 74 & 18 \\ 2 & 0 & 2 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[974] $1^{143}2321^{43} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & 1 & 1 & 16 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 96 & 24 \\ 6 & 12 & 0 & 96 & 70 & 18 \\ 2 & 3 & -1 & 24 & 18 & 4 \end{pmatrix}_{b_0}$$

[973] $1^{112}2321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 10 & 2 \\ 2 & 0 & 4 & 6 & 24 & 6 \\ 8 & 0 & 10 & 24 & 74 & 18 \\ 2 & 0 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[972] $1^{144}2321^{44} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & 1 & 16 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 128 & 24 \\ 8 & 16 & 0 & 128 & 126 & 24 \\ 2 & 3 & -1 & 24 & 24 & 4 \end{pmatrix}_{b_0}$$

[971] $1^{122}2321^{22} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 1 & 8 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 32 & 12 \\ 4 & 8 & 0 & 32 & 30 & 12 \\ 2 & 3 & -1 & 12 & 12 & 4 \end{pmatrix}_{b_0}$$

[970] $1^{121}2321^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 1 & 8 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 16 & 12 \\ 2 & 4 & 0 & 16 & 6 & 6 \\ 2 & 3 & -1 & 12 & 6 & 4 \end{pmatrix}_{b_0}$$

[969] $1^{131}2321^{31} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 1 & 12 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 24 & 18 \\ 2 & 4 & 0 & 24 & 6 & 6 \\ 2 & 3 & -1 & 18 & 6 & 4 \end{pmatrix}_{b_0}$$

[968] $1^{111}2321^{33} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 6 & 6 & 2 \\ 6 & 0 & 6 & 34 & 36 & 12 \\ 6 & 0 & 6 & 36 & 34 & 12 \\ 2 & 0 & 2 & 12 & 12 & 4 \end{pmatrix}_{b_0}$$

[975] $1^{142}2321^{42} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & 1 & 16 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 64 & 24 \\ 4 & 8 & 0 & 64 & 30 & 12 \\ 2 & 3 & -1 & 24 & 12 & 4 \end{pmatrix}_{b_0}$$

[976] $1^{141}2321^{41} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & 1 & 16 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 32 & 24 \\ 2 & 4 & 0 & 32 & 6 & 6 \\ 2 & 3 & -1 & 24 & 6 & 4 \end{pmatrix}_{b_0}$$

[977] $1^{111}2321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 4 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 8 & 6 \\ 2 & 4 & 0 & 8 & 6 & 6 \\ 2 & 3 & -1 & 6 & 6 & 4 \end{pmatrix}_{b_0}$$

[978] $1^{132}2^13^121^{32} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & 1 & 12 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 48 & 18 \\ 4 & 8 & 0 & 48 & 30 & 12 \\ 2 & 3 & -1 & 18 & 12 & 4 \end{pmatrix}_{b_0}$$

[967] $1^{133}2^13^121^{33} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & 1 & 12 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 72 & 18 \\ 6 & 12 & 0 & 72 & 70 & 18 \\ 2 & 3 & -1 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[980] $1^{111}2^13^121^{32} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 6 & 5 & 2 \\ 6 & 0 & 6 & 34 & 26 & 12 \\ 4 & 0 & 5 & 26 & 17 & 9 \\ 2 & 0 & 2 & 12 & 9 & 4 \end{pmatrix}_{b_0}$$

[979] $1^{112}2^13^121^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 4 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 20 & 6 \\ 6 & 9 & 0 & 20 & 49 & 15 \\ 2 & 3 & -1 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[981] $1^{122}2^13^121^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 7 & 5 & 0 & 17 & 9 \\ 2 & 3 & 2 & 0 & 9 & 4 \end{pmatrix}_{b_0}$$

[982] $1^{123}2^13^121^{24} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -22, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 1 & 8 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 56 & 12 \\ 8 & 13 & 0 & 56 & 97 & 21 \\ 2 & 3 & -1 & 12 & 21 & 4 \end{pmatrix}_{b_0}$$

[983] $1^{122}2^13^121^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 7 & 5 & 0 & 17 & 9 \\ 2 & 3 & 2 & 0 & 9 & 4 \end{pmatrix}_{b_0}$$

[984] $1^{121}2^13^121^{32} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 9 & 5 & 2 \\ 6 & 0 & 9 & 49 & 31 & 15 \\ 4 & 0 & 5 & 31 & 17 & 9 \\ 2 & 0 & 2 & 15 & 9 & 4 \end{pmatrix}_{b_0}$$

[985] $1^{122}2^13^121^{34} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 6 & 10 & 2 \\ 6 & 0 & 6 & 34 & 52 & 12 \\ 8 & 0 & 10 & 52 & 74 & 18 \\ 2 & 0 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[987] $1^{112}2^13^121^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 4 & 4 & 6 & 12 & 6 \\ 6 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

[986] $1^{122}2^3 3 21^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & 2 & 7 & 8 & 3 \\ 1 & 2 & 0 & 5 & 8 & 2 \\ 6 & 7 & 5 & 17 & 24 & 9 \\ 7 & 8 & 8 & 24 & 30 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

[990] $1^{111}2^3 3 21^{22} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 5 & 5 & 2 \\ 4 & 0 & 5 & 17 & 19 & 9 \\ 4 & 0 & 5 & 19 & 17 & 9 \\ 2 & 0 & 2 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

[989] $1^{112}2^3 3 21^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 3 & 7 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 3 & 1 & 1 & 7 & 3 \\ 4 & 7 & 5 & 7 & 17 & 9 \\ 2 & 3 & 2 & 3 & 9 & 4 \end{pmatrix}_{b_0}$$

[988] $1^{122}2^3 3 21^{33} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 9 & 9 & 2 \\ 6 & 0 & 9 & 49 & 51 & 15 \\ 6 & 0 & 9 & 51 & 49 & 15 \\ 2 & 0 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

[993] $1^{122}2^3 3 21^{34} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 9 & 10 & 2 \\ 6 & 0 & 9 & 49 & 62 & 15 \\ 8 & 0 & 10 & 62 & 74 & 18 \\ 2 & 0 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[992] $1^{112}2^3 3 21^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 4 & 4 & 4 & 6 & 12 & 6 \\ 5 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

[991] $1^{113}2^3 3 21^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 2 & 4 & 4 & 6 & 20 & 6 \\ 8 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[994] $1^{112}2^3 3 21^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 2 & 3 & 7 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 3 & 1 & 1 & 7 & 3 \\ 6 & 7 & 5 & 7 & 17 & 9 \\ 2 & 3 & 2 & 3 & 9 & 4 \end{pmatrix}_{b_0}$$

[997] $1^{114}2^3 3 21^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 3 & 15 & 3 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 2 & 3 & 1 & 1 & 15 & 3 \\ 10 & 15 & 13 & 15 & 97 & 21 \\ 2 & 3 & 2 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[995] $1^{13}2^3 3 21^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 10 & 6 & 0 & 34 & 12 \\ 2 & 3 & 2 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

[996] $1^{13}2^3 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 10 & 6 & 0 & 34 & 12 \\ 2 & 3 & 2 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

[998] $1^{11}2^3 3 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 1 & 4 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 28 & 6 \\ 8 & 13 & 0 & 28 & 97 & 21 \\ 2 & 3 & -1 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[999] $1^{12}2^3 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 8 & 3 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 8 & 8 & 0 & 30 & 12 \\ 2 & 3 & 2 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

[1000] $1^{14}2^3 3 21^{11}2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 4 & 4 & 4 & 6 & 24 & 6 \\ 10 & 14 & 10 & 24 & 74 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[1001] $1^{11}2^3 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 2 & 4 & 4 & 6 & 8 & 6 \\ 3 & 4 & 4 & 8 & 6 & 6 \\ 2 & 3 & 2 & 6 & 6 & 4 \end{pmatrix}_{b_0}$$

[1002] $1^{11}2^3 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 4 & 4 & 4 & 6 & 8 & 6 \\ 3 & 4 & 4 & 8 & 6 & 6 \\ 2 & 3 & 2 & 6 & 6 & 4 \end{pmatrix}_{b_0}$$

[1003] $1^{14}2^3 3 21^{11}2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 2 & 4 & 4 & 6 & 24 & 6 \\ 9 & 14 & 10 & 24 & 74 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[1004] $1^{12}2^3 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -37, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 2 & 3 & 7 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 3 & 1 & 1 & 7 & 3 \\ 5 & 7 & 5 & 7 & 17 & 9 \\ 2 & 3 & 2 & 3 & 9 & 4 \end{pmatrix}_{b_0}$$

[1005] $1^{13}2^3 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -37, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 2 & 3 & 1 & 1 & 11 & 3 \\ 7 & 11 & 9 & 11 & 49 & 15 \\ 2 & 3 & 2 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[1006] $1^{11}2^3 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 7 & 3 \\ 2 & 0 & 4 & 2 & 7 & 4 \\ 4 & 0 & 7 & 7 & 14 & 8 \\ 2 & 0 & 3 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

[1008] $1^{122}2^13^121^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -42$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & 2 & 7 & 8 & 3 \\ 1 & 2 & 0 & 5 & 8 & 2 \\ 5 & 7 & 5 & 17 & 24 & 9 \\ 6 & 8 & 8 & 24 & 30 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

[1007] $1^{123}2^13^121^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -42$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.3$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 6 & 8 & 8 & 30 & 40 & 12 \\ 9 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

[1010] $1^{122}2^13^121^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -43$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 2 & 7 & 8 & 3 \\ 1 & 2 & 0 & 5 & 8 & 2 \\ 4 & 7 & 5 & 17 & 24 & 9 \\ 5 & 8 & 8 & 24 & 30 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

[1009] $1^{123}2^13^121^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -43$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.3$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 2 & 7 & 10 & 3 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 4 & 7 & 5 & 17 & 26 & 9 \\ 7 & 10 & 6 & 26 & 34 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

[1012] $1^{112}2^13^121^{24} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -46$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 5 & 10 & 2 \\ 4 & 0 & 5 & 17 & 38 & 9 \\ 8 & 0 & 10 & 38 & 74 & 18 \\ 2 & 0 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

[1011] $1^{112}2^13^121^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -46$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 2 & 4 & 4 & 6 & 12 & 6 \\ 4 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

[1013] $1^{112}2^13^121^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 4 & 4 & 4 & 6 & 12 & 6 \\ 6 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

[1014] $1^{112}2^13^121^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.82 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 4 & 4 & 6 & 12 & 6 \\ 4 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

[1015] $1^{113}2^13^121^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -52$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.85 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 2 & 3 & 1 & 1 & 11 & 3 \\ 8 & 11 & 9 & 11 & 49 & 15 \\ 2 & 3 & 2 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[1017] $1^{114}2^13^121^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -53$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & 2 & 3 & 15 & 3 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 2 & 3 & 1 & 1 & 15 & 3 \\ 11 & 15 & 13 & 15 & 97 & 21 \\ 2 & 3 & 2 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[1016] \ 1^{13} 2^1 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -53, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 2 & 3 & 1 & 1 & 11 & 3 \\ 9 & 11 & 9 & 11 & 49 & 15 \\ 2 & 3 & 2 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1018] \ 1^{13} 2^1 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -55, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 11 & 9 & 0 & 49 & 15 \\ 2 & 3 & 2 & 0 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1019] \ 1^{13} 2^1 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -59, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.85 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 11 & 9 & 0 & 49 & 15 \\ 2 & 3 & 2 & 0 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1021] \ 1^{12} 2^1 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 2 & 4 & 4 & 6 & 16 & 6 \\ 6 & 8 & 8 & 16 & 30 & 12 \\ 2 & 3 & 2 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1022] \ 1^{12} 2^1 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 6 & 7 & 5 & 17 & 31 & 9 \\ 8 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1023] \ 1^{12} 2^1 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 4 & 4 & 4 & 6 & 16 & 6 \\ 6 & 8 & 8 & 16 & 30 & 12 \\ 2 & 3 & 2 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1020] \ 1^{11} 2^1 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.80 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 3 & 4 & 4 & 6 & 8 & 6 \\ 3 & 4 & 4 & 8 & 6 & 6 \\ 2 & 3 & 2 & 6 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1024] \ 1^{12} 2^1 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -63, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.79 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 4 & 4 & 6 & 12 & 6 \\ 5 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[1025] \ 1^{12} 2^1 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -64, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 10 & 10 & 2 \\ 8 & 0 & 10 & 74 & 76 & 18 \\ 8 & 0 & 10 & 76 & 74 & 18 \\ 2 & 0 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1027] \ 1^{14} 2^1 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -70, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 11 & 2 \\ 1 & 1 & 2 & 14 & 15 & 3 \\ 1 & 2 & 0 & 10 & 13 & 2 \\ 10 & 14 & 10 & 74 & 86 & 18 \\ 11 & 15 & 13 & 86 & 97 & 21 \\ 2 & 3 & 2 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[1026] \ 1^{134} 2^3 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -70, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 10 & 2 \\ 1 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 7 & 11 & 9 & 49 & 62 & 15 \\ 10 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1028] \ 1^{114} 2^3 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -74, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 2 & 4 & 4 & 6 & 28 & 6 \\ 11 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[1029] \ 1^{113} 2^3 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -74, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 4 & 4 & 4 & 6 & 20 & 6 \\ 8 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1031] \ 1^{123} 2^3 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -75, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 8 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 7 & 8 & 8 & 30 & 40 & 12 \\ 8 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1030] \ 1^{123} 2^3 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -75, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & 2 & 7 & 10 & 3 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 6 & 7 & 5 & 17 & 26 & 9 \\ 7 & 10 & 6 & 26 & 34 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1032] \ 1^{132} 2^3 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -76, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 12 & 3 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 12 & 12 & 0 & 70 & 18 \\ 2 & 3 & 2 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1033] \ 1^{122} 2^3 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -77, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 1 & 2 & 7 & 7 & 3 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 6 & 7 & 5 & 17 & 19 & 9 \\ 5 & 7 & 5 & 19 & 17 & 9 \\ 2 & 3 & 2 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[1034] \ 1^{142} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -80, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 14 & 3 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 14 & 10 & 0 & 74 & 18 \\ 2 & 3 & 2 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1035] \ 1^{142} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -80, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 14 & 3 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 14 & 10 & 0 & 74 & 18 \\ 2 & 3 & 2 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1037] \ 1^{114} 2^3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -82, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 3 & 14 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 3 & 1 & 1 & 14 & 3 \\ 8 & 14 & 10 & 14 & 74 & 18 \\ 2 & 3 & 2 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[1036] $1^{144}2^13^{22}21^{123} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 1 & 2 & 14 & 15 & 3 \\ 1 & 2 & 0 & 10 & 13 & 2 \\ 8 & 14 & 10 & 14 & 74 & 18 \\ 9 & 15 & 13 & 86 & 97 & 21 \\ 2 & 3 & 2 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

[1038] $1^{144}2^13^{33}21^{123} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 10 & 2 \\ 1 & 1 & 2 & 14 & 15 & 3 \\ 1 & 2 & 0 & 10 & 13 & 2 \\ 9 & 14 & 10 & 74 & 86 & 18 \\ 10 & 15 & 13 & 86 & 97 & 21 \\ 2 & 3 & 2 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

[1039] $1^{122}2^13^{12}21^{111} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 2 & 7 & 7 & 3 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 4 & 7 & 5 & 17 & 19 & 9 \\ 5 & 7 & 5 & 19 & 17 & 9 \\ 2 & 3 & 2 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

[1040] $1^{124}2^13^{22}21^{123} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 6 & 8 & 8 & 30 & 56 & 12 \\ 9 & 15 & 13 & 56 & 97 & 21 \\ 2 & 3 & 2 & 12 & 21 & 4 \end{pmatrix}_{b_0}$$

[1041] $1^{114}2^13^{12}21^{112} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 8 & 14 & 10 & 24 & 74 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[1042] $1^{133}2^13^{44}21^{123} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 10 & 2 \\ 1 & 1 & 2 & 11 & 12 & 3 \\ 1 & 2 & 0 & 9 & 12 & 2 \\ 9 & 11 & 9 & 49 & 60 & 15 \\ 10 & 12 & 12 & 60 & 70 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[1043] $1^{113}2^13^{11}21^{112} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 4 & 4 & 6 & 20 & 6 \\ 6 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[1046] $1^{123}2^13^{23}21^{113} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 4 & 4 & 4 & 6 & 24 & 6 \\ 9 & 12 & 12 & 24 & 70 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[1045] $1^{114}2^13^{12}21^{112} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 2 & 4 & 4 & 6 & 24 & 6 \\ 8 & 14 & 10 & 24 & 74 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[1044] $1^{113}2^13^{13}21^{113} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 2 & 4 & 4 & 6 & 24 & 6 \\ 9 & 12 & 12 & 24 & 70 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[1047] $1^{123}2^321^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -102, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & 2 & 7 & 10 & 3 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 5 & 7 & 0 & 5 & 17 & 26 \\ 6 & 10 & 6 & 26 & 34 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

[1048] $1^{113}2^321^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 2 & 4 & 4 & 6 & 20 & 6 \\ 7 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[1049] $1^{142}2^321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 15 & 3 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 15 & 13 & 0 & 97 & 21 \\ 2 & 3 & 2 & 0 & 21 & 4 \end{pmatrix}_{b_0}$$

[1050] $1^{114}2^321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 3 & 14 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 3 & 1 & 1 & 14 & 3 \\ 9 & 14 & 10 & 14 & 74 & 18 \\ 2 & 3 & 2 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[1054] $1^{112}2^321^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 4 & 4 & 6 & 16 & 6 \\ 5 & 8 & 8 & 16 & 30 & 12 \\ 2 & 3 & 2 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[1052] $1^{112}2^321^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 4 & 4 & 6 & 16 & 6 \\ 7 & 8 & 8 & 16 & 30 & 12 \\ 2 & 3 & 2 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[1053] $1^{134}2^321^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & 2 & 10 & 14 & 3 \\ 1 & 2 & 0 & 6 & 10 & 2 \\ 6 & 10 & 6 & 34 & 52 & 12 \\ 10 & 14 & 10 & 52 & 74 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[1051] $1^{134}2^321^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 1 & 2 & 10 & 14 & 3 \\ 1 & 2 & 0 & 6 & 10 & 2 \\ 7 & 10 & 6 & 34 & 52 & 12 \\ 9 & 14 & 10 & 52 & 74 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[1055] $1^{124}2^321^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 2 \\ 1 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 4 & 7 & 5 & 17 & 43 & 9 \\ 11 & 15 & 13 & 43 & 97 & 21 \\ 2 & 3 & 2 & 9 & 21 & 4 \end{pmatrix}_{b_0}$$

[1056] $1^{142}2^321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 0 & 15 & 3 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 15 & 13 & 0 & 97 & 21 \\ 2 & 3 & 2 & 0 & 21 & 4 \end{pmatrix}_{b_0}$$

[1057] $1^{123}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 4 & 7 & 5 & 17 & 31 & 9 \\ 8 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[1058] $1^{123}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 6 & 8 & 8 & 30 & 40 & 12 \\ 7 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

[1059] $1^{133}2^3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 1 & 2 & 11 & 12 & 3 \\ 1 & 2 & 0 & 9 & 12 & 2 \\ 8 & 11 & 9 & 49 & 60 & 15 \\ 9 & 12 & 12 & 60 & 70 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[1061] $1^{114}2^3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 3 & 14 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 3 & 1 & 1 & 14 & 3 \\ 10 & 14 & 10 & 14 & 74 & 18 \\ 2 & 3 & 2 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[1060] $1^{123}2^3 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 2 & 7 & 10 & 3 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 4 & 7 & 5 & 17 & 26 & 9 \\ 6 & 10 & 6 & 26 & 34 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

[1062] $1^{122}2^3 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 2 & 7 & 7 & 3 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 4 & 7 & 5 & 17 & 19 & 9 \\ 4 & 7 & 5 & 19 & 17 & 9 \\ 2 & 3 & 2 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

[1063] $1^{123}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 5 & 8 & 8 & 30 & 40 & 12 \\ 6 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

[1064] $1^{113}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 4 & 4 & 4 & 6 & 20 & 6 \\ 9 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[1065] $1^{123}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 5 & 8 & 8 & 30 & 40 & 12 \\ 7 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

[1066] $1^{123}2^3 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 1 & 2 & 7 & 10 & 3 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 5 & 7 & 5 & 17 & 26 & 9 \\ 7 & 10 & 6 & 26 & 34 & 12 \\ 2 & 3 & 2 & 9 & 12 & 4 \end{pmatrix}_{b_0}$$

[1069] $1^{134}2^3 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 6 & 11 & 9 & 49 & 71 & 15 \\ 10 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

[1068] $1^{122}2^3 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 5 & 2 \\ 1 & 1 & 2 & 7 & 7 & 3 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 5 & 7 & 5 & 17 & 19 & 9 \\ 5 & 7 & 5 & 19 & 17 & 9 \\ 2 & 3 & 2 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

[1067] $1^{133}2^3 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & 2 & 10 & 10 & 3 \\ 1 & 2 & 0 & 6 & 6 & 2 \\ 6 & 10 & 6 & 34 & 36 & 12 \\ 7 & 10 & 6 & 36 & 34 & 12 \\ 2 & 3 & 2 & 12 & 12 & 4 \end{pmatrix}_{b_0}$$

[1070] $1^{124}2^3 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 5 & 7 & 5 & 17 & 38 & 9 \\ 8 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

[1074] $1^{122}2^3 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & 2 & 8 & 8 & 3 \\ 1 & 2 & 0 & 8 & 8 & 2 \\ 5 & 8 & 8 & 30 & 32 & 12 \\ 6 & 8 & 8 & 32 & 30 & 12 \\ 2 & 3 & 2 & 12 & 12 & 4 \end{pmatrix}_{b_0}$$

[1071] $1^{112}2^3 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 4 & 4 & 6 & 16 & 6 \\ 6 & 8 & 8 & 16 & 30 & 12 \\ 2 & 3 & 2 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[1072] $1^{122}2^3 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 6 & 2 \\ 1 & 1 & 2 & 8 & 8 & 3 \\ 1 & 2 & 0 & 8 & 8 & 2 \\ 7 & 8 & 8 & 30 & 32 & 12 \\ 6 & 8 & 8 & 32 & 30 & 12 \\ 2 & 3 & 2 & 12 & 12 & 4 \end{pmatrix}_{b_0}$$

[1073] $1^{142}2^3 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 1 & 2 & 0 & 16 & 3 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 16 & 16 & 0 & 126 & 24 \\ 2 & 3 & 2 & 0 & 24 & 4 \end{pmatrix}_{b_0}$$

[1075] $1^{124}2^3 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 6 & 7 & 5 & 17 & 38 & 9 \\ 10 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

[1076] $1^{114}2^3 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 4 & 4 & 4 & 6 & 28 & 6 \\ 11 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[1077] $1^{13}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 4 & 4 & 6 & 20 & 6 \\ 9 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[1078] $1^{124}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -142, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 7 & 5 & 17 & 38 & 9 \\ 10 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

[1079] $1^{123}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -143, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 7 & 8 & 8 & 30 & 40 & 12 \\ 9 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

[1080] $1^{124}2^3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -143, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 11 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 7 & 8 & 8 & 30 & 56 & 12 \\ 11 & 15 & 13 & 56 & 97 & 21 \\ 2 & 3 & 2 & 12 & 21 & 4 \end{pmatrix}_{b_0}$$

[1081] $1^{114}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 9 & 14 & 10 & 24 & 74 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[1082] $1^{114}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 10 & 14 & 10 & 24 & 74 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[1083] $1^{123}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 5 & 7 & 5 & 17 & 31 & 9 \\ 7 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[1084] $1^{123}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 6 & 8 & 8 & 30 & 40 & 12 \\ 8 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

[1086] $1^{114}2^3 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & 2 & 4 & 16 & 3 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 2 & 4 & 4 & 6 & 32 & 6 \\ 12 & 16 & 16 & 32 & 126 & 24 \\ 2 & 3 & 2 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[1085] $1^{114}2^3 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 1 & 2 & 4 & 16 & 3 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 4 & 4 & 4 & 6 & 32 & 6 \\ 12 & 16 & 16 & 32 & 126 & 24 \\ 2 & 3 & 2 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[1087] $1^{123}2^321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 6 & 7 & 5 & 17 & 31 & 9 \\ 9 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[1088] $1^{113}2^321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 4 & 4 & 6 & 20 & 6 \\ 7 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[1089] $1^{123}2^321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 5 & 7 & 5 & 17 & 31 & 9 \\ 9 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[1090] $1^{113}2^321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 4 & 4 & 6 & 20 & 6 \\ 8 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

[1092] $1^{123}2^321^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & 2 & 8 & 12 & 3 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 5 & 8 & 8 & 30 & 48 & 12 \\ 9 & 12 & 12 & 48 & 70 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[1091] $1^{134}2^321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 2 & 10 & 14 & 3 \\ 1 & 2 & 0 & 6 & 10 & 2 \\ 6 & 10 & 6 & 34 & 52 & 12 \\ 8 & 14 & 10 & 52 & 74 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[1094] $1^{123}2^321^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 1 & 2 & 8 & 12 & 3 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 7 & 8 & 8 & 30 & 48 & 12 \\ 9 & 12 & 12 & 48 & 70 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[1093] $1^{123}2^321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 4 & 7 & 5 & 17 & 31 & 9 \\ 6 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[1095] $1^{134}2^321^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 10 & 2 \\ 1 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 9 & 11 & 9 & 49 & 62 & 15 \\ 10 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[1096] $1^{114}2^321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 2 & 4 & 4 & 6 & 28 & 6 \\ 10 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[1097] $1^{123}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 4 & 7 & 5 & 17 & 31 & 9 \\ 7 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[1098] $1^{134}2^3 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 10 & 2 \\ 1 & 1 & 2 & 12 & 15 & 3 \\ 1 & 2 & 0 & 12 & 13 & 2 \\ 9 & 12 & 12 & 70 & 84 & 18 \\ 10 & 15 & 13 & 84 & 97 & 21 \\ 2 & 3 & 2 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

[1099] $1^{124}2^3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -197, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 5 & 7 & 5 & 17 & 43 & 9 \\ 10 & 15 & 13 & 43 & 97 & 21 \\ 2 & 3 & 2 & 9 & 21 & 4 \end{pmatrix}_{b_0}$$

[1100] $1^{122}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & 2 & 8 & 8 & 3 \\ 1 & 2 & 0 & 8 & 8 & 2 \\ 6 & 8 & 8 & 30 & 32 & 12 \\ 6 & 8 & 8 & 32 & 30 & 12 \\ 2 & 3 & 2 & 12 & 12 & 4 \end{pmatrix}_{b_0}$$

[1102] $1^{123}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 5 & 7 & 5 & 17 & 31 & 9 \\ 8 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[1101] $1^{133}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 6 & 11 & 9 & 49 & 51 & 15 \\ 7 & 11 & 9 & 51 & 49 & 15 \\ 2 & 3 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

[1103] $1^{134}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & 2 & 10 & 14 & 3 \\ 1 & 2 & 0 & 6 & 10 & 2 \\ 6 & 10 & 6 & 34 & 52 & 12 \\ 9 & 14 & 10 & 52 & 74 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[1104] $1^{134}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 10 & 2 \\ 1 & 1 & 2 & 10 & 14 & 3 \\ 1 & 2 & 0 & 6 & 10 & 2 \\ 7 & 10 & 6 & 34 & 52 & 12 \\ 10 & 14 & 10 & 52 & 74 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[1105] $1^{134}2^3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 11 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 9 & 11 & 9 & 49 & 71 & 15 \\ 11 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

[1106] $1^{134}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 8 & 11 & 9 & 49 & 62 & 15 \\ 9 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[1107] $1^{133}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 8 & 2 \\ 1 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 9 & 11 & 9 & 49 & 51 & 15 \\ 8 & 11 & 9 & 51 & 49 & 15 \\ 2 & 3 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

[1108] $1^{134}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 1 & 2 & 12 & 15 & 3 \\ 1 & 2 & 0 & 12 & 13 & 2 \\ 8 & 12 & 12 & 70 & 84 & 18 \\ 9 & 15 & 13 & 84 & 97 & 21 \\ 2 & 3 & 2 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

[1109] $1^{134}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 6 & 11 & 9 & 49 & 62 & 15 \\ 8 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[1110] $1^{124}2^3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 5 & 8 & 8 & 30 & 56 & 12 \\ 10 & 15 & 13 & 56 & 97 & 21 \\ 2 & 3 & 2 & 12 & 21 & 4 \end{pmatrix}_{b_0}$$

[1115] $1^{124}2^3 21^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 12 & 2 \\ 1 & 1 & 2 & 8 & 16 & 3 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 7 & 8 & 8 & 30 & 64 & 12 \\ 12 & 16 & 16 & 64 & 126 & 24 \\ 2 & 3 & 2 & 12 & 24 & 4 \end{pmatrix}_{b_0}$$

[1114] $1^{124}2^3 21^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 12 & 2 \\ 1 & 1 & 2 & 8 & 16 & 3 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 5 & 8 & 8 & 30 & 64 & 12 \\ 12 & 16 & 16 & 64 & 126 & 24 \\ 2 & 3 & 2 & 12 & 24 & 4 \end{pmatrix}_{b_0}$$

[1113] $1^{113}2^3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 10 & 12 & 12 & 24 & 70 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[1112] $1^{113}2^3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 8 & 12 & 12 & 24 & 70 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[1111] $1^{124}2^3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 4 & 7 & 5 & 17 & 43 & 9 \\ 10 & 15 & 13 & 43 & 97 & 21 \\ 2 & 3 & 2 & 9 & 21 & 4 \end{pmatrix}_{b_0}$$

[1116] $1^{133}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 8 & 2 \\ 1 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 7 & 11 & 9 & 49 & 51 & 15 \\ 8 & 11 & 9 & 51 & 49 & 15 \\ 2 & 3 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

[1117] $1^{134}2^3 3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -234$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 8 & 2 \\ 1 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 7 & 11 & 9 & 49 & 62 & 15 \\ 8 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[1119] $1^{134}2^3 3 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -242$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 1 & 2 & 12 & 15 & 3 \\ 1 & 2 & 0 & 12 & 13 & 2 \\ 8 & 12 & 12 & 70 & 84 & 18 \\ 10 & 15 & 13 & 84 & 97 & 21 \\ 2 & 3 & 2 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

[1118] $1^{114}2^3 3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -242$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 2 & 4 & 4 & 6 & 28 & 6 \\ 9 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[1122] $1^{123}2^3 3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -252$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.3$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 2 & 8 & 12 & 3 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 6 & 8 & 8 & 30 & 48 & 12 \\ 8 & 12 & 12 & 48 & 70 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[1121] $1^{123}2^3 3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -252$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.3$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & 2 & 8 & 12 & 3 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 6 & 8 & 8 & 30 & 48 & 12 \\ 10 & 12 & 12 & 48 & 70 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[1120] $1^{113}2^3 3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -252$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 9 & 12 & 12 & 24 & 70 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[1123] $1^{124}2^3 3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -253$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 4 & 7 & 5 & 17 & 43 & 9 \\ 9 & 15 & 13 & 43 & 97 & 21 \\ 2 & 3 & 2 & 9 & 21 & 4 \end{pmatrix}_{b_0}$$

[1124] $1^{114}2^3 3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -255$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 3 & 4 & 4 & 6 & 28 & 6 \\ 9 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[1125] $1^{124}2^3 3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -259$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 7 & 5 & 17 & 38 & 9 \\ 9 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

[1126] $1^{124}2^3 3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -263$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 5 & 7 & 5 & 17 & 38 & 9 \\ 9 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

[1127] $1^{124}2^3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 11 & 2 \\ 1 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 5 & 7 & 5 & 17 & 43 & 9 \\ 11 & 15 & 13 & 43 & 97 & 21 \\ 2 & 3 & 2 & 9 & 21 & 4 \end{pmatrix}_{b_0}$$

[1128] $1^{124}2^3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -270, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 6 & 8 & 8 & 30 & 56 & 12 \\ 10 & 15 & 13 & 56 & 97 & 21 \\ 2 & 3 & 2 & 12 & 21 & 4 \end{pmatrix}_{b_0}$$

[1129] $1^{134}2^3 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 11 & 2 \\ 1 & 1 & 2 & 12 & 15 & 3 \\ 1 & 2 & 0 & 12 & 13 & 2 \\ 9 & 12 & 12 & 70 & 84 & 18 \\ 11 & 15 & 13 & 84 & 97 & 21 \\ 2 & 3 & 2 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

[1130] $1^{134}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 7 & 11 & 9 & 49 & 62 & 15 \\ 9 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[1131] $1^{134}2^3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 11 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 7 & 11 & 9 & 49 & 71 & 15 \\ 11 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

[1132] $1^{124}2^3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -282, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 7 & 5 & 17 & 38 & 9 \\ 8 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

[1133] $1^{134}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -286, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 8 & 11 & 9 & 49 & 62 & 15 \\ 10 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[1134] $1^{124}2^3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 5 & 8 & 8 & 30 & 56 & 12 \\ 9 & 15 & 13 & 56 & 97 & 21 \\ 2 & 3 & 2 & 12 & 21 & 4 \end{pmatrix}_{b_0}$$

[1135] $1^{133}2^3 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 1 & 2 & 12 & 12 & 3 \\ 1 & 2 & 0 & 12 & 12 & 2 \\ 8 & 12 & 12 & 70 & 72 & 18 \\ 9 & 12 & 12 & 72 & 70 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[1136] $1^{133}2^3 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 9 & 2 \\ 1 & 1 & 2 & 12 & 12 & 3 \\ 1 & 2 & 0 & 12 & 12 & 2 \\ 10 & 12 & 12 & 70 & 72 & 18 \\ 9 & 12 & 12 & 72 & 70 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[1137] $1^{124}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 5 & 7 & 5 & 17 & 38 & 9 \\ 10 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

[1138] $1^{144}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 1 & 2 & 14 & 14 & 3 \\ 1 & 2 & 0 & 10 & 10 & 2 \\ 8 & 14 & 10 & 74 & 76 & 18 \\ 9 & 14 & 10 & 76 & 74 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[1139] $1^{124}2^3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -306, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 11 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 6 & 8 & 8 & 30 & 56 & 12 \\ 11 & 15 & 13 & 56 & 97 & 21 \\ 2 & 3 & 2 & 12 & 21 & 4 \end{pmatrix}_{b_0}$$

[1140] $1^{114}2^3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 3 & 4 & 4 & 6 & 28 & 6 \\ 10 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[1142] $1^{133}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 9 & 2 \\ 1 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 9 & 11 & 9 & 49 & 51 & 15 \\ 9 & 11 & 9 & 51 & 49 & 15 \\ 2 & 3 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

[1141] $1^{144}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 9 & 2 \\ 1 & 1 & 2 & 14 & 14 & 3 \\ 1 & 2 & 0 & 10 & 10 & 2 \\ 10 & 14 & 10 & 74 & 76 & 18 \\ 9 & 14 & 10 & 76 & 74 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[1143] $1^{123}2^3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & 2 & 8 & 12 & 3 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 6 & 8 & 8 & 30 & 48 & 12 \\ 9 & 12 & 12 & 48 & 70 & 18 \\ 2 & 3 & 2 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[1144] $1^{114}2^3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -327, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 11 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 3 & 4 & 4 & 6 & 28 & 6 \\ 11 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[1145] $1^{133}2^3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -332, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 8 & 11 & 9 & 49 & 71 & 15 \\ 10 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

[1146] $1^{122}2^3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 7 & 2 \\ 1 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 7 & 11 & 9 & 49 & 51 & 15 \\ 7 & 11 & 9 & 51 & 49 & 15 \\ 2 & 3 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

[1147] $1^{133}2^3 3^{21}1^{22} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 8 & 11 & 9 & 49 & 51 & 15 \\ 8 & 11 & 9 & 51 & 49 & 15 \\ 2 & 3 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

[1149] $1^{134}2^3 3^{21}1^{34} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -348, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 12 & 2 \\ 1 & 1 & 2 & 12 & 16 & 3 \\ 1 & 2 & 0 & 12 & 16 & 2 \\ 10 & 12 & 12 & 70 & 96 & 18 \\ 12 & 16 & 16 & 96 & 126 & 24 \\ 2 & 3 & 2 & 18 & 24 & 4 \end{pmatrix}_{b_0}$$

[1148] $1^{134}2^3 3^{21}1^{34} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -348, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 12 & 2 \\ 1 & 1 & 2 & 12 & 16 & 3 \\ 1 & 2 & 0 & 12 & 16 & 2 \\ 8 & 12 & 12 & 70 & 96 & 18 \\ 12 & 16 & 16 & 96 & 126 & 24 \\ 2 & 3 & 2 & 18 & 24 & 4 \end{pmatrix}_{b_0}$$

[1150] $1^{134}2^3 3^{21}1^{23} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -373, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 6 & 11 & 9 & 49 & 71 & 15 \\ 9 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

[1151] $1^{134}2^3 3^{21}1^{23} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 7 & 11 & 9 & 49 & 71 & 15 \\ 9 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

[1152] $1^{114}2^3 3^{21}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -396, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 11 & 2 \\ 1 & 1 & 2 & 4 & 16 & 3 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 3 & 4 & 4 & 6 & 32 & 6 \\ 11 & 16 & 16 & 32 & 126 & 24 \\ 2 & 3 & 2 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[1153] $1^{144}2^3 3^{21}1^{33} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -413, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 10 & 2 \\ 1 & 1 & 2 & 15 & 15 & 3 \\ 1 & 2 & 0 & 13 & 13 & 2 \\ 9 & 15 & 13 & 97 & 99 & 21 \\ 10 & 15 & 13 & 99 & 97 & 21 \\ 2 & 3 & 2 & 21 & 21 & 4 \end{pmatrix}_{b_0}$$

[1155] $1^{124}2^3 3^{21}1^{24} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -420, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 11 & 2 \\ 1 & 1 & 2 & 8 & 16 & 3 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 6 & 8 & 8 & 30 & 64 & 12 \\ 11 & 16 & 16 & 64 & 126 & 24 \\ 2 & 3 & 2 & 12 & 24 & 4 \end{pmatrix}_{b_0}$$

[1154] $1^{114}2^3 3^{21}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -420, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 12 & 2 \\ 1 & 1 & 2 & 4 & 16 & 3 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 3 & 4 & 4 & 6 & 32 & 6 \\ 12 & 16 & 16 & 32 & 126 & 24 \\ 2 & 3 & 2 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[1156] $1^{144}2^3 3^{21}1^{33} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -437, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 11 & 2 \\ 1 & 1 & 2 & 15 & 15 & 3 \\ 1 & 2 & 0 & 13 & 13 & 2 \\ 10 & 15 & 13 & 97 & 99 & 21 \\ 11 & 15 & 13 & 99 & 97 & 21 \\ 2 & 3 & 2 & 21 & 21 & 4 \end{pmatrix}_{b_0}$$

[1158] $1^{133}2^3 3 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 9 & 2 \\ 1 & 1 & 2 & 12 & 12 & 3 \\ 1 & 2 & 0 & 12 & 12 & 2 \\ 9 & 12 & 12 & 70 & 72 & 18 \\ 9 & 12 & 12 & 72 & 70 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[1157] $1^{144}2^3 3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & 2 & 14 & 14 & 3 \\ 1 & 2 & 0 & 10 & 10 & 2 \\ 8 & 14 & 10 & 74 & 76 & 18 \\ 8 & 14 & 10 & 76 & 74 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[1159] $1^{134}2^3 3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -453, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 10 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 7 & 11 & 9 & 49 & 71 & 15 \\ 10 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

[1160] $1^{134}2^3 3 21^{134} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -460, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 11 & 2 \\ 1 & 1 & 2 & 12 & 16 & 3 \\ 1 & 2 & 0 & 12 & 16 & 2 \\ 9 & 12 & 12 & 70 & 96 & 18 \\ 11 & 16 & 16 & 96 & 126 & 24 \\ 2 & 3 & 2 & 18 & 24 & 4 \end{pmatrix}_{b_0}$$

[1161] $1^{134}2^3 3 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -469, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 11 & 2 \\ 1 & 1 & 2 & 11 & 15 & 3 \\ 1 & 2 & 0 & 9 & 13 & 2 \\ 8 & 11 & 9 & 49 & 71 & 15 \\ 11 & 15 & 13 & 71 & 97 & 21 \\ 2 & 3 & 2 & 15 & 21 & 4 \end{pmatrix}_{b_0}$$

[1163] $1^{144}2^3 3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 9 & 2 \\ 1 & 1 & 2 & 14 & 14 & 3 \\ 1 & 2 & 0 & 10 & 10 & 2 \\ 9 & 14 & 10 & 74 & 76 & 18 \\ 9 & 14 & 10 & 76 & 74 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[1162] $1^{144}2^3 3 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 10 & 2 \\ 1 & 1 & 2 & 14 & 14 & 3 \\ 1 & 2 & 0 & 10 & 10 & 2 \\ 10 & 14 & 10 & 74 & 76 & 18 \\ 10 & 14 & 10 & 76 & 74 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

[1164] $1^{124}2^3 3 21^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -492, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 12 & 2 \\ 1 & 1 & 2 & 8 & 16 & 3 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 6 & 8 & 8 & 30 & 64 & 12 \\ 12 & 16 & 16 & 64 & 126 & 24 \\ 2 & 3 & 2 & 12 & 24 & 4 \end{pmatrix}_{b_0}$$

[1165] $1^{144}2^3 3 21^{144} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 12 & 2 \\ 1 & 1 & 2 & 16 & 16 & 3 \\ 1 & 2 & 0 & 16 & 16 & 2 \\ 11 & 16 & 16 & 126 & 128 & 24 \\ 12 & 16 & 16 & 128 & 126 & 24 \\ 2 & 3 & 2 & 24 & 24 & 4 \end{pmatrix}_{b_0}$$

[1166] $1^{144}2^3 3 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 9 & 2 \\ 1 & 1 & 2 & 15 & 15 & 3 \\ 1 & 2 & 0 & 13 & 13 & 2 \\ 9 & 15 & 13 & 97 & 99 & 21 \\ 9 & 15 & 13 & 99 & 97 & 21 \\ 2 & 3 & 2 & 21 & 21 & 4 \end{pmatrix}_{b_0}$$

[1167] $1^{134}2^3 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -612, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 12 & 2 \\ 1 & 1 & 2 & 12 & 16 & 3 \\ 1 & 2 & 0 & 12 & 16 & 2 \\ 9 & 12 & 12 & 70 & 96 & 18 \\ 12 & 16 & 16 & 96 & 126 & 24 \\ 2 & 3 & 2 & 18 & 24 & 4 \end{pmatrix}_{b_0}$$

[1168] $1^{144}2^3 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -648, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 10 & 2 \\ 1 & 1 & 2 & 15 & 15 & 3 \\ 1 & 2 & 0 & 13 & 13 & 2 \\ 10 & 15 & 13 & 97 & 99 & 21 \\ 10 & 15 & 13 & 99 & 97 & 21 \\ 2 & 3 & 2 & 21 & 21 & 4 \end{pmatrix}_{b_0}$$

[1169] $1^{144}2^3 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 11 & 2 \\ 1 & 1 & 2 & 15 & 15 & 3 \\ 1 & 2 & 0 & 13 & 13 & 2 \\ 11 & 15 & 13 & 97 & 99 & 21 \\ 11 & 15 & 13 & 99 & 97 & 21 \\ 2 & 3 & 2 & 21 & 21 & 4 \end{pmatrix}_{b_0}$$

[1170] $1^{144}2^3 21^{144} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -780, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 12 & 2 \\ 1 & 1 & 2 & 16 & 16 & 3 \\ 1 & 2 & 0 & 16 & 16 & 2 \\ 12 & 16 & 16 & 126 & 128 & 24 \\ 12 & 16 & 16 & 128 & 126 & 24 \\ 2 & 3 & 2 & 24 & 24 & 4 \end{pmatrix}_{b_0}$$

$T = 10$ [1176] $1^{111}3141^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 3 & 0 & 1 & 4 & 3 \\ 3 & 4 & 0 & 4 & 5 & 5 \\ 2 & 3 & 0 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

[1175] $1^{11}31^1 41^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 4 & 5 & 4 \\ 3 & 4 & 0 & 5 & 5 & 5 \\ 2 & 3 & 0 & 4 & 5 & 4 \end{pmatrix}_{b_0}$$

[1174] $1^{111}314131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 & 2 & 4 \\ 2 & 3 & 0 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

[1173] $1^{11}5131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -49, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 5 & 3 & 0 & 5 & 5 \\ 2 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -2 & 0 & 5 \\ 3 & 5 & 3 & 0 & 5 & 5 \end{pmatrix}_{b_0}$$

[1172] $1^{11}4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -45, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 3 \\ 3 & 5 & 4 & 4 & 5 & 5 \\ 3 & 4 & 2 & 4 & 4 & 4 \\ 2 & 4 & 4 & 2 & 4 & 4 \\ 3 & 5 & 4 & 4 & 5 & 5 \end{pmatrix}_{b_0}$$

[1171] $1^{11}4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -45, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 3 & 5 & 0 & 4 & 5 \end{pmatrix}_{b_0} \\
[1177] & 1^{11} 31^{11} 4131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 3 & 0 & 1 & 6 & 3 \\ 4 & 6 & 1 & 6 & 14 & 8 \\ 2 & 3 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0} \\
[1179] & 1^{11} 2231^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -6, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 3 & 1 & 8 & 5 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 8 & 0 & 15 & 8 & 9 \\ 2 & 5 & 0 & 8 & 3 & 5 \\ 2 & 5 & -1 & 9 & 5 & 5 \end{pmatrix}_{b_0} \\
[1178] & 1^{11} 5131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -49, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1181] & 1^{11} 2231^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -6, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 5 & 8 & 4 \\ 2 & 0 & 5 & 3 & 8 & 5 \\ 4 & 0 & 8 & 8 & 15 & 9 \\ 2 & 0 & 4 & 5 & 9 & 5 \end{pmatrix}_{b_0} \\
[1180] & 1^{11} 3141^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 4 & 0 & 5 & 4 & 5 \\ 2 & 3 & 1 & 4 & 2 & 4 \\ 2 & 3 & 0 & 5 & 4 & 4 \end{pmatrix}_{b_0} \\
[1182] & 1^{11} 31^{11} 41^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & 1 & 4 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 4 & 0 & 5 & 9 & 5 \\ 4 & 6 & 1 & 9 & 14 & 8 \\ 2 & 3 & 0 & 5 & 8 & 4 \end{pmatrix}_{b_0} \\
[1183] & 1^{11} 31^{11} 41^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 1 & 6 & 7 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 4 & 6 & 1 & 14 & 16 & 8 \\ 5 & 7 & 1 & 16 & 17 & 9 \\ 2 & 3 & 0 & 8 & 9 & 4 \end{pmatrix}_{b_0} \\
[1184] & 1^{11} 31^{11} 4131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 2 & 3 & 1 & 2 & 7 & 4 \\ 4 & 6 & 1 & 7 & 14 & 8 \\ 2 & 3 & 0 & 4 & 8 & 4 \end{pmatrix}_{b_0} \\
[1185] & 1^{11} 31^{11} 41^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 1 & 3 & 10 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 2 & 3 & 1 & 2 & 11 & 4 \\ 7 & 10 & 1 & 11 & 39 & 13 \\ 2 & 3 & 0 & 4 & 13 & 4 \end{pmatrix}_{b_0} \\
[1186] & 231^{11} 3151^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 428, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 5 & 1 & 5 & 5 & 2 \\ 5 & 33 & 1 & 35 & 35 & 11 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 35 & 0 & 33 & 35 & 11 \\ 5 & 35 & 0 & 35 & 33 & 11 \\ 2 & 11 & -1 & 11 & 11 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
& T = 11 \quad [1188] \quad 1^{111}3151231^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 4 & 0 & 2 & 4 & 4 \\ 2 & 4 & 0 & 4 & 2 & 4 \\ 2 & 4 & -1 & 4 & 4 & 4 \end{pmatrix}_{b_0} \\
& [1189] \quad 1^{11}3151231^{111} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 0 & 4 & 0 & 2 & 4 \\ 2 & 0 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0} \\
& [1190] \quad 1^{11}61231^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 12, \Delta = -47, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
& [1191] \quad 1^{11}31^{11}41321^{111} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 0 & 3 & 3 & 2 \\ 3 & 2 & 3 & 6 & 5 & 6 \\ 2 & 2 & 3 & 5 & 3 & 5 \\ 2 & 2 & 2 & 6 & 5 & 5 \end{pmatrix}_{b_0} \\
& [1192] \quad 1^{111}3151^{11}231^{111} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 9 & 3 \\ 2 & 0 & 4 & 2 & 9 & 4 \\ 5 & 0 & 9 & 9 & 27 & 11 \\ 2 & 0 & 3 & 4 & 11 & 4 \end{pmatrix}_{b_0} \\
& [1193] \quad 1^{111}3151^{11}231^{112} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 9 & 3 \\ 2 & 0 & 4 & 2 & 9 & 4 \\ 6 & 0 & 9 & 9 & 27 & 11 \\ 2 & 0 & 3 & 4 & 11 & 4 \end{pmatrix}_{b_0} \\
& [1194] \quad 1^{111}31^{11}41321^{111} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 4 & 6 & 3 \\ 2 & 2 & 5 & 8 & 4 \\ 4 & 5 & 7 & 12 & 7 \\ 6 & 8 & 12 & 19 & 11 \\ 3 & 4 & 7 & 11 & 6 \end{pmatrix}_{b_0} \\
& [1195] \quad 1^{114}3151^{21}231^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 2 & 1 & 4 & 26 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 4 & 0 & 2 & 26 & 4 \\ 11 & 26 & 0 & 26 & 202 & 30 \\ 2 & 4 & -1 & 4 & 30 & 4 \end{pmatrix}_{b_0} \\
& [1196] \quad 1^{12}3151^{22}231^{113} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 0 & 18 & 0 & 114 & 22 \\ 2 & 0 & 3 & 0 & 22 & 4 \end{pmatrix}_{b_0} \\
& [1197] \quad 1^{13}3151^{22}231^{114} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 22 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 0 & 22 & 0 & 154 & 26 \\ 2 & 0 & 3 & 0 & 26 & 4 \end{pmatrix}_{b_0}$$

[1198] $2^1 3^{111} 15131^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & 2 & 7 & 7 & 3 \\ 2 & 2 & 0 & 5 & 5 & 2 \\ 4 & 7 & 5 & 17 & 19 & 9 \\ 4 & 7 & 5 & 19 & 17 & 9 \\ 2 & 3 & 2 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

[1199] $2^1 3^{113} 15131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -48, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 8 & 2 \\ 2 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 2 & 3 & 1 & 1 & 11 & 3 \\ 8 & 11 & 9 & 11 & 49 & 15 \\ 2 & 3 & 2 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[1200] $1^{12} 3151^1 231^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -53, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 2 & 1 & 4 & 13 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 13 & 4 \\ 6 & 13 & 0 & 13 & 49 & 15 \\ 2 & 4 & -1 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

[1201] $1^{113} 3151^1 231^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 2 & 1 & 4 & 17 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 17 & 4 \\ 7 & 17 & 0 & 17 & 79 & 19 \\ 2 & 4 & -1 & 4 & 19 & 4 \end{pmatrix}_{b_0}$$

[1202] $2^1 3^{111} 15131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 3 & 2 \\ 2 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 3 & 4 & 4 & 6 & 8 & 6 \\ 3 & 4 & 4 & 8 & 6 & 6 \\ 2 & 3 & 2 & 6 & 6 & 4 \end{pmatrix}_{b_0}$$

[1203] $2^1 3^{112} 15131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -93, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 5 & 2 \\ 2 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 4 & 4 & 6 & 12 & 6 \\ 5 & 7 & 5 & 12 & 17 & 9 \\ 2 & 3 & 2 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

[1204] $1^{114} 3151^3 231^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -107, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 17 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 31 & 3 \\ 2 & 0 & 4 & 2 & 31 & 4 \\ 17 & 0 & 31 & 31 & 317 & 37 \\ 2 & 0 & 3 & 4 & 37 & 4 \end{pmatrix}_{b_0}$$

[1205] $1^{112} 3151^1 231^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -107, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 13 & 3 \\ 2 & 0 & 4 & 2 & 13 & 4 \\ 7 & 0 & 13 & 13 & 49 & 15 \\ 2 & 0 & 3 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

[1206] $1^{113} 3151^2 231^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 22 & 3 \\ 2 & 0 & 4 & 2 & 22 & 4 \\ 12 & 0 & 22 & 22 & 154 & 26 \\ 2 & 0 & 3 & 4 & 26 & 4 \end{pmatrix}_{b_0}$$

$$[1207] \quad 1^{112}3151^{1231^{113}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -133, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 13 & 3 \\ 2 & 0 & 4 & 2 & 13 & 4 \\ 8 & 0 & 13 & 13 & 49 & 15 \\ 2 & 0 & 3 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1208] \quad 2^1 3^{123} 15131^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -148, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.3$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 8 & 2 \\ 2 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 6 & 8 & 8 & 30 & 40 & 12 \\ 8 & 11 & 9 & 40 & 49 & 15 \\ 2 & 3 & 2 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1209] \quad 2^1 3^{112} 15131^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -148, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 6 & 2 \\ 2 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 4 & 4 & 6 & 16 & 6 \\ 6 & 8 & 8 & 16 & 30 & 12 \\ 2 & 3 & 2 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1210] \quad 1^{113}3151^{2231^{114}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -182, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 13 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 22 & 3 \\ 2 & 0 & 4 & 2 & 22 & 4 \\ 13 & 0 & 22 & 22 & 154 & 26 \\ 2 & 0 & 3 & 4 & 26 & 4 \end{pmatrix}_{b_0}$$

$$[1211] \quad 2^1 3^{114} 15131^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -216, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 10 & 2 \\ 2 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 10 & 14 & 10 & 24 & 74 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1213] \quad 2^1 3^{122} 15131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -220, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 5 & 2 \\ 2 & 1 & 2 & 7 & 7 & 3 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 5 & 7 & 5 & 17 & 19 & 9 \\ 5 & 7 & 5 & 19 & 17 & 9 \\ 2 & 3 & 2 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[1212] \quad 2^1 3^{122} 15131^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -220, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 6 & 2 \\ 2 & 1 & 2 & 8 & 8 & 3 \\ 1 & 2 & 0 & 8 & 8 & 2 \\ 6 & 8 & 8 & 30 & 32 & 12 \\ 6 & 8 & 8 & 32 & 30 & 12 \\ 2 & 3 & 2 & 12 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1214] \quad 2^1 3^{113} 15131^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -241, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 8 & 2 \\ 2 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 4 & 4 & 6 & 20 & 6 \\ 8 & 11 & 9 & 20 & 49 & 15 \\ 2 & 3 & 2 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[1215] \quad 2^1 3^{113} 15131^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -29, \quad \det = -268, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 9 & 2 \\ 2 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 3 & 4 & 4 & 6 & 24 & 6 \\ 9 & 12 & 12 & 24 & 70 & 18 \\ 2 & 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[1216] $2^1 3^{123} 15131^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -313$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.3$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 8 & 2 \\ 2 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 5 & 7 & 5 & 17 & 31 & 9 \\ 8 & 11 & 9 & 31 & 49 & 15 \\ 2 & 3 & 2 & 9 & 15 & 4 \end{pmatrix}_{b_0}$$

[1217] $1^{114} 3151^{2231} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -344$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 26 & 3 \\ 2 & 0 & 4 & 2 & 26 & 4 \\ 14 & 0 & 26 & 26 & 202 & 30 \\ 2 & 0 & 3 & 4 & 30 & 4 \end{pmatrix}_{b_0}$$

[1218] $2^1 3^{124} 15131^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -344$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 11 & 2 \\ 2 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 5 & 7 & 5 & 17 & 43 & 9 \\ 11 & 15 & 13 & 43 & 97 & 21 \\ 2 & 3 & 2 & 9 & 21 & 4 \end{pmatrix}_{b_0}$$

[1219] $2^1 3^{134} 15131^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -436$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 10 & 2 \\ 2 & 1 & 2 & 11 & 14 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 8 & 11 & 9 & 49 & 62 & 15 \\ 10 & 14 & 10 & 62 & 74 & 18 \\ 2 & 3 & 2 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[1220] $2^1 3^{114} 15131^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -436$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 12 & 2 \\ 2 & 1 & 2 & 4 & 16 & 3 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 3 & 4 & 4 & 6 & 32 & 6 \\ 12 & 16 & 16 & 32 & 126 & 24 \\ 2 & 3 & 2 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[1221] $2^1 3^{114} 15131^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -437$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 11 & 2 \\ 2 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 3 & 4 & 4 & 6 & 28 & 6 \\ 11 & 15 & 13 & 28 & 97 & 21 \\ 2 & 3 & 2 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[1222] $2^1 3^{124} 15131^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -508$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 10 & 2 \\ 2 & 1 & 2 & 7 & 14 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 5 & 7 & 5 & 17 & 38 & 9 \\ 10 & 14 & 10 & 38 & 74 & 18 \\ 2 & 3 & 2 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

[1223] $2^1 3^{133} 15131^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -516$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.6$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 8 & 2 \\ 2 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 8 & 11 & 9 & 49 & 51 & 15 \\ 8 & 11 & 9 & 51 & 49 & 15 \\ 2 & 3 & 2 & 15 & 15 & 4 \end{pmatrix}_{b_0}$$

[1224] $2^1 3^{144} 15131^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -796$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 10 & 2 \\ 2 & 1 & 2 & 14 & 14 & 3 \\ 1 & 2 & 0 & 10 & 10 & 2 \\ 10 & 14 & 10 & 74 & 76 & 18 \\ 10 & 14 & 10 & 76 & 74 & 18 \\ 2 & 3 & 2 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1227] \quad 1^{11} 31 \quad 5 \quad 13221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 12, 1), \quad c_{\text{ext}} = 5.0$$

$$[1226] \quad 1^{12} 31 \quad 5 \quad 13221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 12, 1), \quad c_{\text{ext}} = 5.5$$

$$[1225] \quad 1^{13} 31 \quad 5 \quad 13221^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 12, 1), \quad c_{\text{ext}} = 5.8$$

$$[1228] \quad 1^{12} 231 \quad 5 \quad 1321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 12, 1), \quad c_{\text{ext}} = 5.8$$

$$[1229] \quad 1^{12} 1431 \quad 5 \quad 13221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 12, 1), \quad c_{\text{ext}} = 6.0$$

$$[1230] \quad 1^{12} 1131 \quad 5 \quad 13221^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 12, 1), \quad c_{\text{ext}} = 5.0$$

$$[1231] \quad 1^{11} 31 \quad 5 \quad 13221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 12, 1), \quad c_{\text{ext}} = 5.8$$

$$[1240] \quad 1^{11} 5 \quad 131 \quad 5 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -43, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 3 & 5 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[1239] \quad 1^{11} 5 \quad 131 \quad 5 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -43, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 4 & 4 & 5 \\ 3 & 4 & 2 & 4 & 4 \\ 2 & 4 & 4 & 2 & 4 \\ 3 & 5 & 4 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[1243] \quad 1^{11} 314141 \quad 1^{11} 31 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -3, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 2 & 1 & 5 & 4 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 5 & 0 & 6 & 5 & 6 \\ 2 & 4 & 1 & 5 & 3 & 5 \\ 2 & 4 & 0 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[1241] \quad 1^{12} 2321 \quad 1^{11} 5131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 3 & 1 & 5 & 17 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 3 & 17 & 5 \\ 6 & 17 & 0 & 17 & 67 & 19 \\ 2 & 5 & -1 & 5 & 19 & 5 \end{pmatrix}_{b_0}$$

$$[1242] \quad 1^{11} 7 \quad 12231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -45, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[1244] \quad 1^{11} 1331 \quad 5 \quad 1321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 5, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 5 & 11 & 3 \\ 2 & 0 & 5 & 7 & 22 & 7 \\ 6 & 0 & 11 & 22 & 53 & 17 \\ 2 & 0 & 3 & 7 & 17 & 5 \end{pmatrix}_{b_0}$$

$$[1245] \quad 1^{11} 1131 \quad 5 \quad 1321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 6, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.67 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 5 & 0 & 7 & 7 \\ 2 & 3 & 3 & 0 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[1246] \quad 1^{11} 1231 \quad 5 \quad 1321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 7 & 6 & 0 & 18 & 10 \\ 2 & 3 & 3 & 0 & 10 & 5 \end{pmatrix}_{b_0}$$

[1247] $1^{1\bar{1}2}31\bar{5}1321^{1\bar{1}3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 2 & 2 & 5 & 14 & 4 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 2 & 5 & 4 & 7 & 23 & 7 \\ 8 & 14 & 9 & 23 & 58 & 18 \\ 2 & 4 & 2 & 7 & 18 & 5 \end{pmatrix}_{b_0}$$

[1248] $1^{1\bar{1}1}31^{1\bar{1}1}414131^{1\bar{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -3, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 2 & 1 & 4 & 9 & 4 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 2 & 4 & 1 & 3 & 10 & 5 \\ 4 & 9 & 1 & 10 & 23 & 11 \\ 2 & 4 & 0 & 5 & 11 & 5 \end{pmatrix}_{b_0}$$

[1249] $1^{1\bar{1}2}31\bar{5}1321^{1\bar{1}3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 2 & 2 & 5 & 14 & 4 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 5 & 4 & 7 & 23 & 7 \\ 6 & 14 & 9 & 23 & 58 & 18 \\ 2 & 4 & 2 & 7 & 18 & 5 \end{pmatrix}_{b_0}$$

[1257] $1^{1\bar{1}2}31\bar{5}1321^{1\bar{1}2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 2 & 1 & 5 & 10 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 7 & 18 & 7 \\ 4 & 10 & 0 & 18 & 34 & 14 \\ 2 & 4 & -1 & 7 & 14 & 5 \end{pmatrix}_{b_0}$$

[1251] $1^{1\bar{1}1}31\bar{5}1321^{1\bar{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 7 & 9 & 7 \\ 2 & 5 & 0 & 9 & 7 & 7 \\ 2 & 4 & -1 & 7 & 7 & 5 \end{pmatrix}_{b_0}$$

[1252] $1^{1\bar{1}4}31\bar{5}1321^{1\bar{1}4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 2 & 1 & 20 & 20 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 20 & 0 & 142 & 144 & 28 \\ 8 & 20 & 0 & 144 & 142 & 28 \\ 2 & 4 & -1 & 28 & 28 & 5 \end{pmatrix}_{b_0}$$

[1253] $1^{1\bar{1}4}31\bar{5}1321^{1\bar{1}4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 2 & 1 & 5 & 20 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 7 & 36 & 7 \\ 8 & 20 & 0 & 36 & 142 & 28 \\ 2 & 4 & -1 & 7 & 28 & 5 \end{pmatrix}_{b_0}$$

[1254] $1^{1\bar{1}3}31\bar{5}1321^{1\bar{1}3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 2 & 1 & 15 & 20 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 15 & 0 & 79 & 108 & 21 \\ 8 & 20 & 0 & 108 & 142 & 28 \\ 2 & 4 & -1 & 21 & 28 & 5 \end{pmatrix}_{b_0}$$

[1255] $1^{1\bar{1}3}31\bar{5}1321^{1\bar{1}3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 2 & 1 & 5 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 7 & 27 & 7 \\ 6 & 15 & 0 & 27 & 79 & 21 \\ 2 & 4 & -1 & 7 & 21 & 5 \end{pmatrix}_{b_0}$$

[1256] $2\bar{3}151321^{1\bar{1}2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 6 & 2 \\ 2 & 1 & 2 & 8 & 8 & 3 \\ 1 & 2 & 1 & 10 & 10 & 3 \\ 6 & 8 & 10 & 34 & 36 & 14 \\ 6 & 8 & 10 & 36 & 34 & 14 \\ 2 & 3 & 3 & 14 & 14 & 5 \end{pmatrix}_{b_0}$$

[1250] $2^1 3^{112} 151321^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 6 & 2 \\ 2 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 1 & 5 & 10 & 3 \\ 3 & 4 & 5 & 7 & 18 & 7 \\ 6 & 8 & 10 & 18 & 34 & 14 \\ 2 & 3 & 3 & 7 & 14 & 5 \end{pmatrix}_{b_0}$$

[1258] $2^1 3^{141} 151321^{141} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 3 & 2 \\ 2 & 1 & 2 & 16 & 4 & 3 \\ 1 & 2 & 1 & 20 & 5 & 3 \\ 12 & 16 & 20 & 142 & 36 & 28 \\ 3 & 4 & 5 & 36 & 7 & 7 \\ 2 & 3 & 3 & 28 & 7 & 5 \end{pmatrix}_{b_0}$$

[1259] $2^1 3^{142} 151321^{142} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 6 & 2 \\ 2 & 1 & 2 & 16 & 8 & 3 \\ 1 & 2 & 1 & 20 & 10 & 3 \\ 12 & 16 & 20 & 142 & 72 & 28 \\ 6 & 8 & 10 & 72 & 34 & 14 \\ 2 & 3 & 3 & 28 & 14 & 5 \end{pmatrix}_{b_0}$$

[1260] $1^{24} 31^1 51321^{124} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 2 & 1 & 10 & 20 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 10 & 0 & 34 & 72 & 14 \\ 8 & 20 & 0 & 72 & 142 & 28 \\ 2 & 4 & -1 & 14 & 28 & 5 \end{pmatrix}_{b_0}$$

[1261] $2^1 3^{143} 151321^{143} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 9 & 2 \\ 2 & 1 & 2 & 16 & 12 & 3 \\ 1 & 2 & 1 & 20 & 15 & 3 \\ 12 & 16 & 20 & 142 & 108 & 28 \\ 9 & 12 & 15 & 108 & 79 & 21 \\ 2 & 3 & 3 & 28 & 21 & 5 \end{pmatrix}_{b_0}$$

[1262] $2^1 3^{132} 151321^{132} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 6 & 2 \\ 2 & 1 & 2 & 12 & 8 & 3 \\ 1 & 2 & 1 & 15 & 10 & 3 \\ 9 & 12 & 15 & 79 & 54 & 21 \\ 6 & 8 & 10 & 54 & 34 & 14 \\ 2 & 3 & 3 & 21 & 14 & 5 \end{pmatrix}_{b_0}$$

[1263] $2^1 3^{133} 151321^{133} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 9 & 2 \\ 2 & 1 & 2 & 12 & 12 & 3 \\ 1 & 2 & 1 & 15 & 15 & 3 \\ 9 & 12 & 15 & 79 & 81 & 21 \\ 9 & 12 & 15 & 81 & 79 & 21 \\ 2 & 3 & 3 & 21 & 21 & 5 \end{pmatrix}_{b_0}$$

[1264] $1^{23} 31^1 51321^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 2 & 1 & 10 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 10 & 0 & 34 & 54 & 14 \\ 6 & 15 & 0 & 54 & 79 & 21 \\ 2 & 4 & -1 & 14 & 21 & 5 \end{pmatrix}_{b_0}$$

[1265] $2^1 3^{144} 151321^{144} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 12 & 2 \\ 2 & 1 & 2 & 16 & 16 & 3 \\ 1 & 2 & 1 & 20 & 20 & 3 \\ 12 & 16 & 20 & 142 & 144 & 28 \\ 12 & 16 & 20 & 144 & 142 & 28 \\ 2 & 3 & 3 & 28 & 28 & 5 \end{pmatrix}_{b_0}$$

[1266] $1^{33} 31^1 51321^{133} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 2 & 1 & 15 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 15 & 0 & 79 & 81 & 21 \\ 6 & 15 & 0 & 81 & 79 & 21 \\ 2 & 4 & -1 & 21 & 21 & 5 \end{pmatrix}_{b_0}$$

[1267] $2^1 3^{131} 151321^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 3 & 2 \\ 2 & 1 & 2 & 12 & 4 & 3 \\ 1 & 2 & 1 & 15 & 5 & 3 \\ 9 & 12 & 15 & 79 & 27 & 21 \\ 3 & 4 & 5 & 27 & 7 & 7 \\ 2 & 3 & 3 & 21 & 7 & 5 \end{pmatrix}_{b_0}$$

[1268] $1^{22} 31^1 51321^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 2 & 1 & 10 & 10 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 10 & 0 & 34 & 36 & 14 \\ 4 & 10 & 0 & 36 & 34 & 14 \\ 2 & 4 & -1 & 14 & 14 & 5 \end{pmatrix}_{b_0}$$

[1269] $2^1 3^{111} 151321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 3 & 2 \\ 2 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 5 & 7 & 9 & 7 \\ 3 & 4 & 5 & 9 & 7 & 7 \\ 2 & 3 & 3 & 7 & 7 & 5 \end{pmatrix}_{b_0}$$

[1271] $1^{112} 31^1 51321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 2 & 2 & 4 & 9 & 4 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 4 & 1 & 2 & 9 & 4 \\ 5 & 9 & 5 & 9 & 21 & 11 \\ 2 & 4 & 2 & 4 & 11 & 5 \end{pmatrix}_{b_0}$$

[1270] $1^{111} 31^1 51321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 2 & 2 & 4 & 9 & 4 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 4 & 1 & 2 & 9 & 4 \\ 4 & 9 & 5 & 9 & 21 & 11 \\ 2 & 4 & 2 & 4 & 11 & 5 \end{pmatrix}_{b_0}$$

[1273] $1^{111} 31^1 51321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 2 & 2 & 5 & 5 & 4 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 2 & 5 & 4 & 7 & 9 & 7 \\ 3 & 5 & 4 & 9 & 7 & 7 \\ 2 & 4 & 2 & 7 & 7 & 5 \end{pmatrix}_{b_0}$$

[1272] $1^{112} 31^1 51321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & 2 & 4 & 7 & 3 \\ 1 & 2 & 1 & 5 & 6 & 3 \\ 3 & 4 & 5 & 7 & 13 & 7 \\ 4 & 7 & 6 & 13 & 18 & 10 \\ 2 & 3 & 3 & 7 & 10 & 5 \end{pmatrix}_{b_0}$$

[1274] $1^{12} 31^1 51321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 8 & 3 \\ 1 & 2 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 8 & 10 & 0 & 34 & 14 \\ 2 & 3 & 3 & 0 & 14 & 5 \end{pmatrix}_{b_0}$$

[1275] $1^{21} 31^1 51321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 4 & 4 & 5 & 7 & 9 & 7 \\ 3 & 4 & 5 & 9 & 7 & 7 \\ 2 & 3 & 3 & 7 & 7 & 5 \end{pmatrix}_{b_0}$$

[1276] $1^{111} 31^1 51321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 2 & 4 & 5 & 7 & 9 & 7 \\ 3 & 4 & 5 & 9 & 7 & 7 \\ 2 & 3 & 3 & 7 & 7 & 5 \end{pmatrix}_{b_0}$$

[1277] $1^{12}31^151321^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 22$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 2 & 2 & 5 & 18 & 4 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 5 & 4 & 7 & 28 & 7 \\ 9 & 18 & 10 & 28 & 90 & 22 \\ 2 & 4 & 2 & 7 & 22 & 5 \end{pmatrix}_{b_0}$$

[1278] $1^{12}31^151321^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 22$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 2 & 2 & 5 & 18 & 4 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 2 & 5 & 4 & 7 & 28 & 7 \\ 9 & 18 & 10 & 28 & 90 & 22 \\ 2 & 4 & 2 & 7 & 22 & 5 \end{pmatrix}_{b_0}$$

[1279] $1^{124}31^151321^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 1 & 10 & 16 & 3 \\ 5 & 8 & 10 & 34 & 62 & 14 \\ 10 & 15 & 16 & 62 & 106 & 24 \\ 2 & 3 & 3 & 14 & 24 & 5 \end{pmatrix}_{b_0}$$

[1280] $1^{12}31^14141^131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -3$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 2 & 1 & 4 & 14 & 4 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 2 & 4 & 1 & 3 & 15 & 5 \\ 7 & 14 & 1 & 15 & 55 & 17 \\ 2 & 4 & 0 & 5 & 17 & 5 \end{pmatrix}_{b_0}$$

[1281] $1^{111}31^151321^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 2 & 2 & 9 & 13 & 4 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 4 & 9 & 5 & 21 & 32 & 11 \\ 7 & 13 & 6 & 32 & 43 & 15 \\ 2 & 4 & 2 & 11 & 15 & 5 \end{pmatrix}_{b_0}$$

[1282] $1^{13}31^151321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 11 & 11 & 0 & 53 & 17 \\ 2 & 3 & 3 & 0 & 17 & 5 \end{pmatrix}_{b_0}$$

[1283] $1^{123}31^151321^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.3$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & 2 & 8 & 11 & 3 \\ 1 & 2 & 1 & 10 & 11 & 3 \\ 5 & 8 & 10 & 34 & 44 & 14 \\ 6 & 11 & 11 & 44 & 53 & 17 \\ 2 & 3 & 3 & 14 & 17 & 5 \end{pmatrix}_{b_0}$$

[1284] $1^{13}31^151321^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 27$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 1 & 5 & 11 & 3 \\ 2 & 4 & 5 & 7 & 22 & 7 \\ 7 & 11 & 11 & 22 & 53 & 17 \\ 2 & 3 & 3 & 7 & 17 & 5 \end{pmatrix}_{b_0}$$

[1285] $1^{13}31^151321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 27$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 11 & 11 & 0 & 53 & 17 \\ 2 & 3 & 3 & 0 & 17 & 5 \end{pmatrix}_{b_0}$$

[1286] $1^{111}31^151321^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 27$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 2 & 2 & 5 & 9 & 4 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 2 & 5 & 4 & 7 & 14 & 7 \\ 4 & 9 & 5 & 14 & 21 & 11 \\ 2 & 4 & 2 & 7 & 11 & 5 \end{pmatrix}_{b_0}$$

[1287] $1^{111}31\ 5\ 1321^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 27$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 2 & 2 & 5 & 9 & 4 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 5 & 4 & 7 & 14 & 7 \\ 5 & 9 & 5 & 14 & 21 & 11 \\ 2 & 4 & 2 & 7 & 11 & 5 \end{pmatrix}_{b_0}$$

[1288] $1^{112}31\ 5\ 1321^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 34$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 1 & 5 & 10 & 3 \\ 2 & 4 & 5 & 7 & 18 & 7 \\ 6 & 8 & 10 & 18 & 34 & 14 \\ 2 & 3 & 3 & 7 & 14 & 5 \end{pmatrix}_{b_0}$$

[1289] $1^{13}31\ 5\ 1321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 38$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 12 & 3 \\ 1 & 2 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 12 & 15 & 0 & 79 & 21 \\ 2 & 3 & 3 & 0 & 21 & 5 \end{pmatrix}_{b_0}$$

[1290] $1^{111}31\ 5\ 1321^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 40$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 5 & 7 & 9 & 7 \\ 3 & 4 & 5 & 9 & 7 & 7 \\ 2 & 3 & 3 & 7 & 7 & 5 \end{pmatrix}_{b_0}$$

[1291] $1^{12}31\ 5\ 1321^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 42$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 2 & 2 & 4 & 18 & 4 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 4 & 1 & 2 & 18 & 4 \\ 8 & 18 & 10 & 18 & 90 & 22 \\ 2 & 4 & 2 & 4 & 22 & 5 \end{pmatrix}_{b_0}$$

[1292] $1^{111}31\ 5\ 1321^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 42$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 2 & 2 & 9 & 9 & 4 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 5 & 9 & 5 & 21 & 23 & 11 \\ 4 & 9 & 5 & 23 & 21 & 11 \\ 2 & 4 & 2 & 11 & 11 & 5 \end{pmatrix}_{b_0}$$

[1294] $1^{112}23151^{1231} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -1$, $\det = 43$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$

$T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 5 & 15 & 4 \\ 2 & 0 & 5 & 3 & 15 & 5 \\ 7 & 0 & 15 & 15 & 53 & 17 \\ 2 & 0 & 4 & 5 & 17 & 5 \end{pmatrix}_{b_0}$$

[1293] $1^{113}23151^{1231} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -1$, $\det = 43$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$

$T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 3 & 1 & 5 & 20 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 3 & 20 & 5 \\ 7 & 20 & 0 & 20 & 88 & 22 \\ 2 & 5 & -1 & 5 & 22 & 5 \end{pmatrix}_{b_0}$$

[1295] $1^{12}31\ 5\ 1321^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 48$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 2 & 2 & 4 & 18 & 4 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 4 & 1 & 2 & 18 & 4 \\ 9 & 18 & 10 & 18 & 90 & 22 \\ 2 & 4 & 2 & 4 & 22 & 5 \end{pmatrix}_{b_0}$$

[1296] $1^{14}31\ 5\ 1321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 53$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 15 & 3 \\ 1 & 2 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 15 & 16 & 0 & 106 & 24 \\ 2 & 3 & 3 & 0 & 24 & 5 \end{pmatrix}_{b_0}$$

[1297] $1^{12}3151321^{114} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 2 & 2 & 5 & 18 & 4 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 2 & 5 & 4 & 7 & 28 & 7 \\ 8 & 18 & 10 & 28 & 90 & 22 \\ 2 & 4 & 2 & 7 & 22 & 5 \end{pmatrix}_{b_0}$$

[1298] $1^{13}3151321^{113} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 1 & 5 & 15 & 3 \\ 2 & 4 & 5 & 7 & 27 & 7 \\ 9 & 12 & 15 & 27 & 79 & 21 \\ 2 & 3 & 3 & 7 & 21 & 5 \end{pmatrix}_{b_0}$$

[1299] $1^{14}3151321^{113} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 61, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 1 & 5 & 16 & 3 \\ 2 & 4 & 5 & 7 & 31 & 7 \\ 10 & 15 & 16 & 31 & 106 & 24 \\ 2 & 3 & 3 & 7 & 24 & 5 \end{pmatrix}_{b_0}$$

[1300] $1^{13}3151321^{112} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 63, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 1 & 6 & 11 & 3 \\ 4 & 7 & 6 & 18 & 33 & 10 \\ 8 & 11 & 11 & 33 & 53 & 17 \\ 2 & 3 & 3 & 10 & 17 & 5 \end{pmatrix}_{b_0}$$

[1301] $1^{12}2321^15131^{113} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 64, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 5 & 17 & 4 \\ 2 & 0 & 5 & 3 & 17 & 5 \\ 8 & 0 & 17 & 17 & 67 & 19 \\ 2 & 0 & 4 & 5 & 19 & 5 \end{pmatrix}_{b_0}$$

[1302] $1^{12}3151321^{113} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 67, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 2 & 2 & 5 & 14 & 4 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 2 & 5 & 4 & 7 & 23 & 7 \\ 7 & 14 & 9 & 23 & 58 & 18 \\ 2 & 4 & 2 & 7 & 18 & 5 \end{pmatrix}_{b_0}$$

[1303] $1^{12}3151321^{113} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 67, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 2 & 2 & 5 & 14 & 4 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 5 & 4 & 7 & 23 & 7 \\ 7 & 14 & 9 & 23 & 58 & 18 \\ 2 & 4 & 2 & 7 & 18 & 5 \end{pmatrix}_{b_0}$$

[1304] $1^{12}3151321^{112} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 70, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 1 & 5 & 10 & 3 \\ 3 & 4 & 5 & 7 & 18 & 7 \\ 5 & 8 & 10 & 18 & 34 & 14 \\ 2 & 3 & 3 & 7 & 14 & 5 \end{pmatrix}_{b_0}$$

[1305] $1^{12}3151321^{112} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 70, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 1 & 5 & 10 & 3 \\ 3 & 4 & 5 & 7 & 18 & 7 \\ 7 & 8 & 10 & 18 & 34 & 14 \\ 2 & 3 & 3 & 7 & 14 & 5 \end{pmatrix}_{b_0}$$

[1306] $1^{11}3151321^{123} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 75, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 2 & 2 & 9 & 13 & 4 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 4 & 9 & 5 & 21 & 32 & 11 \\ 6 & 13 & 6 & 32 & 43 & 15 \\ 2 & 4 & 2 & 11 & 15 & 5 \end{pmatrix}_{b_0}$$

[1307] $1^{11}31 \overset{11}{5} 1321^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 80$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 2 & 2 & 9 & 9 & 4 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 4 & 9 & 5 & 21 & 23 & 11 \\ 4 & 9 & 5 & 23 & 21 & 11 \\ 2 & 4 & 2 & 11 & 11 & 5 \end{pmatrix}_{b_0}$$

[1308] $1^{14}31 \overset{14}{5} 1321^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 82$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & 2 & 4 & 16 & 3 \\ 1 & 2 & 1 & 5 & 20 & 3 \\ 2 & 4 & 5 & 7 & 36 & 7 \\ 12 & 16 & 20 & 36 & 142 & 28 \\ 2 & 3 & 3 & 7 & 28 & 5 \end{pmatrix}_{b_0}$$

[1309] $1^{13}31 \overset{13}{5} 1321^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 82$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 1 & 5 & 11 & 3 \\ 3 & 4 & 5 & 7 & 22 & 7 \\ 8 & 11 & 11 & 22 & 53 & 17 \\ 2 & 3 & 3 & 7 & 17 & 5 \end{pmatrix}_{b_0}$$

[1310] $1^{12}31 \overset{12}{5} 1321^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 82$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 1 & 2 & 8 & 8 & 3 \\ 1 & 2 & 1 & 10 & 10 & 3 \\ 5 & 8 & 10 & 34 & 36 & 14 \\ 6 & 8 & 10 & 36 & 34 & 14 \\ 2 & 3 & 3 & 14 & 14 & 5 \end{pmatrix}_{b_0}$$

[1311] $1^{12}31 \overset{12}{5} 1321^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 87$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 1 & 6 & 16 & 3 \\ 4 & 7 & 6 & 18 & 46 & 10 \\ 9 & 15 & 16 & 46 & 106 & 24 \\ 2 & 3 & 3 & 10 & 24 & 5 \end{pmatrix}_{b_0}$$

[1312] $1^{12}31 \overset{12}{5} 1321^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 88$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 1 & 5 & 10 & 3 \\ 3 & 4 & 5 & 7 & 18 & 7 \\ 6 & 8 & 10 & 18 & 34 & 14 \\ 2 & 3 & 3 & 7 & 14 & 5 \end{pmatrix}_{b_0}$$

[1313] $1^{13}31 \overset{13}{5} 1321^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 88$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 1 & 5 & 11 & 3 \\ 3 & 4 & 5 & 7 & 22 & 7 \\ 7 & 11 & 11 & 22 & 53 & 17 \\ 2 & 3 & 3 & 7 & 17 & 5 \end{pmatrix}_{b_0}$$

[1314] $1^{13}31 \overset{13}{5} 1321^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 92$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & 2 & 7 & 15 & 3 \\ 1 & 2 & 1 & 6 & 16 & 3 \\ 4 & 7 & 6 & 18 & 46 & 10 \\ 10 & 15 & 16 & 46 & 106 & 24 \\ 2 & 3 & 3 & 10 & 24 & 5 \end{pmatrix}_{b_0}$$

[1315] $1^{12}31 \overset{12}{5} 1321^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 93$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 2 & 2 & 9 & 14 & 4 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 4 & 9 & 5 & 21 & 37 & 11 \\ 7 & 14 & 9 & 37 & 58 & 18 \\ 2 & 4 & 2 & 11 & 18 & 5 \end{pmatrix}_{b_0}$$

[1316] $1^{12}31 \overset{12}{5} 1321^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 102$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 2 & 2 & 14 & 14 & 4 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 6 & 14 & 9 & 58 & 60 & 18 \\ 7 & 14 & 9 & 60 & 58 & 18 \\ 2 & 4 & 2 & 18 & 18 & 5 \end{pmatrix}_{b_0}$$

[1317] $1^{123}31 \overset{1^{12}}{5} 1321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 1 & 6 & 11 & 3 \\ 4 & 7 & 6 & 18 & 33 & 10 \\ 7 & 11 & 11 & 33 & 53 & 17 \\ 2 & 3 & 3 & 10 & 17 & 5 \end{pmatrix}_{b_0}$$

[1318] $1^{123}31 \overset{1^{13}}{5} 1321^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 102, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & 2 & 8 & 12 & 3 \\ 1 & 2 & 1 & 10 & 15 & 3 \\ 5 & 8 & 10 & 34 & 54 & 14 \\ 9 & 12 & 15 & 54 & 79 & 21 \\ 2 & 3 & 3 & 14 & 21 & 5 \end{pmatrix}_{b_0}$$

[1319] $1^{112}31 \overset{1^{11}}{5} 1321^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 111, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 2 & 2 & 9 & 14 & 4 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 4 & 9 & 5 & 21 & 37 & 11 \\ 6 & 14 & 9 & 37 & 58 & 18 \\ 2 & 4 & 2 & 11 & 18 & 5 \end{pmatrix}_{b_0}$$

[1320] $1^{124}31 \overset{1^{12}}{5} 1321^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 115, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 1 & 2 & 8 & 15 & 3 \\ 1 & 2 & 1 & 10 & 16 & 3 \\ 5 & 8 & 10 & 34 & 62 & 14 \\ 9 & 15 & 16 & 62 & 106 & 24 \\ 2 & 3 & 3 & 14 & 24 & 5 \end{pmatrix}_{b_0}$$

[1321] $1^{114}231 \overset{1^{14}}{5} 151^2 231^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 118, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 14 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 5 & 30 & 4 \\ 2 & 0 & 5 & 3 & 30 & 5 \\ 14 & 0 & 30 & 30 & 218 & 34 \\ 2 & 0 & 4 & 5 & 34 & 5 \end{pmatrix}_{b_0}$$

[1322] $1^{114}31 \overset{1^{12}}{5} 1321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 121, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 1 & 5 & 16 & 3 \\ 2 & 4 & 5 & 7 & 31 & 7 \\ 9 & 15 & 16 & 31 & 106 & 24 \\ 2 & 3 & 3 & 7 & 24 & 5 \end{pmatrix}_{b_0}$$

[1323] $1^{133}31 \overset{1^{12}}{5} 1321^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 125, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.6$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & 2 & 11 & 11 & 3 \\ 1 & 2 & 1 & 11 & 11 & 3 \\ 6 & 11 & 11 & 53 & 55 & 17 \\ 7 & 11 & 11 & 55 & 53 & 17 \\ 2 & 3 & 3 & 17 & 17 & 5 \end{pmatrix}_{b_0}$$

[1324] $1^{122}31 \overset{1^{12}}{5} 1321^{134} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 138, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 2 & 2 & 14 & 18 & 4 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 6 & 14 & 9 & 58 & 74 & 18 \\ 8 & 18 & 10 & 74 & 90 & 22 \\ 2 & 4 & 2 & 18 & 22 & 5 \end{pmatrix}_{b_0}$$

[1325] $1^{113}31 \overset{1^{12}}{5} 1321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 150, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 1 & 5 & 15 & 3 \\ 3 & 4 & 5 & 7 & 27 & 7 \\ 8 & 12 & 15 & 27 & 79 & 21 \\ 2 & 3 & 3 & 7 & 21 & 5 \end{pmatrix}_{b_0}$$

[1326] $1^{112}31 \overset{1^{13}}{5} 1321^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 158, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 2 & 2 & 9 & 18 & 4 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 9 & 5 & 21 & 46 & 11 \\ 9 & 18 & 10 & 46 & 90 & 22 \\ 2 & 4 & 2 & 11 & 22 & 5 \end{pmatrix}_{b_0}$$

[1327] $1^{114}31^15^1321^{113}+[\textcolor{red}{\mathfrak{su}_2}\cdot\textcolor{blue}{\mathfrak{g}_2}]+\textcolor{green}{\mathfrak{su}_2}+\textcolor{red}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 162, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 1 & 5 & 16 & 3 \\ 3 & 4 & 5 & 7 & 31 & 7 \\ 9 & 15 & 16 & 31 & 106 & 24 \\ 2 & 3 & 3 & 7 & 24 & 5 \end{pmatrix}_{b_0}$$

[1328] $1^{113}31^15^1321^{113}+[\textcolor{red}{\mathfrak{su}_2}\cdot\textcolor{blue}{\mathfrak{g}_2}]+\textcolor{green}{\mathfrak{su}_2}+\textcolor{red}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 168, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 4 & 12 & 3 \\ 1 & 2 & 1 & 5 & 15 & 3 \\ 3 & 4 & 5 & 7 & 27 & 7 \\ 9 & 12 & 15 & 27 & 79 & 21 \\ 2 & 3 & 3 & 7 & 21 & 5 \end{pmatrix}_{b_0}$$

[1329] $1^{112}31^15^1321^{124}+[\textcolor{red}{\mathfrak{su}_2}\cdot\textcolor{blue}{\mathfrak{g}_2}]+\textcolor{green}{\mathfrak{su}_2}+\textcolor{red}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 188, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 2 & 2 & 9 & 18 & 4 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 9 & 5 & 21 & 46 & 11 \\ 8 & 18 & 10 & 46 & 90 & 22 \\ 2 & 4 & 2 & 11 & 22 & 5 \end{pmatrix}_{b_0}$$

[1330] $1^{114}31^15^1321^{113}+[\textcolor{red}{\mathfrak{su}_2}\cdot\textcolor{blue}{\mathfrak{g}_2}]+\textcolor{green}{\mathfrak{su}_2}+\textcolor{red}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & 2 & 4 & 15 & 3 \\ 1 & 2 & 1 & 5 & 16 & 3 \\ 3 & 4 & 5 & 7 & 31 & 7 \\ 10 & 15 & 16 & 31 & 106 & 24 \\ 2 & 3 & 3 & 7 & 24 & 5 \end{pmatrix}_{b_0}$$

[1331] $1^{114}31^15^1321^{114}+[\textcolor{red}{\mathfrak{su}_2}\cdot\textcolor{blue}{\mathfrak{g}_2}]+\textcolor{green}{\mathfrak{su}_2}+\textcolor{red}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 262, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 11 & 2 \\ 1 & 1 & 2 & 4 & 16 & 3 \\ 1 & 2 & 1 & 5 & 20 & 3 \\ 3 & 4 & 5 & 7 & 36 & 7 \\ 11 & 16 & 20 & 36 & 142 & 28 \\ 2 & 3 & 3 & 7 & 28 & 5 \end{pmatrix}_{b_0}$$

$T = 13$ [1332] $1^{112}31^1414141^{111}+[\textcolor{red}{\mathfrak{su}_2}\cdot\textcolor{blue}{\mathfrak{g}_2}]+\textcolor{green}{\mathfrak{su}_2}+\textcolor{red}{\mathfrak{su}_2} \quad T = 13, T_{\max} = 13, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 5.5$

[1333] $1^{111}4^114141^141^1+[\textcolor{red}{\mathfrak{su}_2}\cdot\textcolor{blue}{\mathfrak{g}_2}]+\textcolor{green}{\mathfrak{su}_2}+\textcolor{red}{\mathfrak{su}_2} \quad T = 13, T_{\max} = 13, \Delta = -14, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 5.0$

[1334] $1^{11}6^1131^15^11^1+[\textcolor{red}{\mathfrak{su}_2}\cdot\textcolor{blue}{\mathfrak{g}_2}]+\textcolor{green}{\mathfrak{su}_2}+\textcolor{red}{\mathfrak{su}_2} \quad T = 13, T_{\max} = 14, \Delta = -40, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

[1335] $21^{11}6^1131^15^11^1+[\textcolor{red}{\mathfrak{su}_2}\cdot\textcolor{blue}{\mathfrak{g}_2}]+\textcolor{green}{\mathfrak{su}_2}+\textcolor{red}{\mathfrak{su}_2} \quad T = 13, T_{\max} = 14, \Delta = -40, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 2 & 0 & 6 & 6 \\ 3 & 2 & 0 & 6 & 5 \end{pmatrix}_{b_0}$$

[1336] $1^{111}6^1131^15^11^1+[\textcolor{red}{\mathfrak{su}_2}\cdot\textcolor{blue}{\mathfrak{g}_2}]+\textcolor{green}{\mathfrak{su}_2}+\textcolor{red}{\mathfrak{su}_2} \quad T = 13, T_{\max} = 14, \Delta = -40, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 1 & 2 \\ 3 & 2 & 3 & 5 & 3 & 5 \\ 1 & 1 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

[1337] $1^{111}6^11231^15^11^1+[\textcolor{red}{\mathfrak{su}_2}\cdot\textcolor{blue}{\mathfrak{g}_2}]+\textcolor{green}{\mathfrak{su}_2}+\textcolor{red}{\mathfrak{su}_2} \quad T = 13, T_{\max} = 14, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 1 & 2 & 3 \\ 1 & 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & 0 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[1338] & 1^{\overline{1}} \overline{6} 131 \overline{5} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -40, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 & 2 \\ 2 & 2 & 3 & 5 & 4 & 5 \\ 2 & 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 2 & 5 & 3 & 4 \end{pmatrix}_{b_0} \\
[1339] & 1^{\overline{1}} \overline{6} 131 \overline{5} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -40, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 2 & 2 & 3 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
[1340] & 1^{\overline{1}} \overline{6} 131 \overline{5} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -40, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 1 & 2 \\ 2 & 2 & 3 & 5 & 3 & 5 \\ 1 & 1 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & 5 & 2 & 4 \end{pmatrix}_{b_0} \\
[1341] & 1^{\overline{1}} \overline{6} 131 \overline{5} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -40, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 & 2 \\ 3 & 2 & 3 & 5 & 4 & 5 \\ 1 & 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 2 & 5 & 3 & 4 \end{pmatrix}_{b_0} \\
[1342] & 1^{\overline{1}} \overline{6} 1231 \overline{5} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -41, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 3 & 2 & 0 & 2 & 5 \end{pmatrix}_{b_0} \\
[1343] & 1^{\overline{1}} 15131 \overline{6} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -40, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[1344] & 1^{\overline{1}} \overline{1} 321 \overline{6} 1^{\overline{1}} 2321^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 2 & 4 & 2 \\ 1 & 1 & 0 & 3 & 5 & 2 \\ 2 & 2 & 3 & 3 & 9 & 5 \\ 6 & 4 & 5 & 9 & 19 & 11 \\ 2 & 2 & 2 & 5 & 11 & 5 \end{pmatrix}_{b_0} \\
[1345] & 1^{\overline{1}} \overline{1} 321 \overline{6} 12321^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -43, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 1 & 2 & 6 & 2 \\ 1 & 1 & 0 & 3 & 7 & 2 \\ 2 & 2 & 3 & 3 & 13 & 5 \\ 7 & 6 & 7 & 13 & 39 & 15 \\ 2 & 2 & 2 & 5 & 15 & 5 \end{pmatrix}_{b_0} \\
[1346] & 1^{\overline{1}} \overline{1} 321 \overline{6} 12321^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -43, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 4 & 11 & 3 \\ 2 & 2 & 5 & 14 & 4 \\ 4 & 5 & 7 & 22 & 7 \\ 11 & 14 & 22 & 62 & 20 \\ 3 & 4 & 7 & 20 & 6 \end{pmatrix}_{b_0} \\
[1347] & 1^{\overline{1}} \overline{1} 321 \overline{6} 1^{\overline{1}} 2321^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -43, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 4 & 19 & 3 \\ 2 & 2 & 5 & 24 & 4 \\ 4 & 5 & 7 & 36 & 7 \\ 19 & 24 & 36 & 178 & 34 \\ 3 & 4 & 7 & 34 & 6 \end{pmatrix}_{b_0} \\
& [1348] \ 1^{\overline{113}} 321^{\overline{161}} 2321^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -54, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 4 & 18 & 3 \\ 2 & 2 & 5 & 21 & 4 \\ 4 & 5 & 7 & 33 & 7 \\ 18 & 21 & 33 & 145 & 31 \\ 3 & 4 & 7 & 31 & 6 \end{pmatrix}_{b_0} \\
& [1349] \ 1^{\overline{113}} 32161^{\overline{12321}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 4 & 12 & 3 \\ 2 & 2 & 5 & 15 & 4 \\ 4 & 5 & 7 & 23 & 7 \\ 12 & 15 & 23 & 67 & 21 \\ 3 & 4 & 7 & 21 & 6 \end{pmatrix}_{b_0} \\
& [1350] \ 1^{\overline{114}} 321^{\overline{161}} 2321^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -196, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 4 & 22 & 3 \\ 2 & 2 & 5 & 26 & 4 \\ 4 & 5 & 7 & 42 & 7 \\ 22 & 26 & 42 & 220 & 38 \\ 3 & 4 & 7 & 38 & 6 \end{pmatrix}_{b_0} \\
& T = 14 \quad [1351] \ 1^{\overline{1}} \begin{pmatrix} 8 \\ 1^{\overline{1}} 2321 \end{pmatrix} 1231^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -26, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 5.0 \\
& [1355] \ 1^{\overline{1}} 712231^{\overline{1}} 51^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -39, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 3 & 2 & 0 & 2 & 5 \end{pmatrix}_{b_0} \\
& [1356] \ 21^{\overline{1}} 71231^{\overline{1}} 51^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -39, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 2 & 0 & 6 & 6 \\ 3 & 2 & 0 & 6 & 5 \end{pmatrix}_{b_0} \\
& [1357] \ 1^{\overline{1}} 71231^{\overline{1}} 51^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -39, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 & 2 \\ 3 & 2 & 3 & 5 & 4 & 5 \\ 1 & 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 2 & 5 & 3 & 4 \end{pmatrix}_{b_0} \\
& [1358] \ 1^{\overline{1}} 71231^{\overline{1}} 51^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -39, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 & 2 \\ 2 & 2 & 3 & 5 & 4 & 5 \\ 2 & 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 2 & 5 & 3 & 4 \end{pmatrix}_{b_0} \\
& [1359] \ 1^{\overline{1}} 71231^{\overline{1}} 51^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -39, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
& [1360] \ 1^{\overline{1}} 71231^{\overline{1}} 51^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -39, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 2 & 2 & 3 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 5 & 0 & 4 \end{pmatrix}_{b_0}$$

[1361] $1^{11} 7^{11} 1232151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 14, T_{\max} = 15, \Delta = -41, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[1362] $15^{11} 13217^{11} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 14, T_{\max} = 15, \Delta = -39, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[1363] $2^1 3^{111} 21712321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 14, T_{\max} = 14, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 3 & 2 \\ 2 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 5 & 7 & 7 & 7 \\ 3 & 4 & 5 & 9 & 7 & 7 \\ 2 & 3 & 3 & 7 & 7 & 5 \end{pmatrix}_{b_0}$$

[1364] $2^1 3^{131} 21712321^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 14, T_{\max} = 14, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 3 & 2 \\ 2 & 1 & 2 & 12 & 4 & 3 \\ 1 & 2 & 1 & 15 & 5 & 3 \\ 9 & 12 & 15 & 79 & 27 & 21 \\ 3 & 4 & 5 & 27 & 7 & 7 \\ 2 & 3 & 3 & 21 & 7 & 5 \end{pmatrix}_{b_0}$$

[1365] $2^1 3^{121} 21712321^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 14, T_{\max} = 14, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 3 & 2 \\ 2 & 1 & 2 & 8 & 4 & 3 \\ 1 & 2 & 1 & 10 & 5 & 3 \\ 6 & 8 & 10 & 34 & 18 & 14 \\ 3 & 4 & 5 & 18 & 7 & 7 \\ 2 & 3 & 3 & 14 & 7 & 5 \end{pmatrix}_{b_0}$$

[1366] $2^1 3^{132} 21712321^{132} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 14, T_{\max} = 14, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 6 & 2 \\ 2 & 1 & 2 & 12 & 8 & 3 \\ 1 & 2 & 1 & 15 & 10 & 3 \\ 9 & 12 & 15 & 79 & 54 & 21 \\ 6 & 8 & 10 & 54 & 34 & 14 \\ 2 & 3 & 3 & 21 & 14 & 5 \end{pmatrix}_{b_0}$$

[1367] $2^1 3^{122} 21712321^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 14, T_{\max} = 14, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 6 & 2 \\ 2 & 1 & 2 & 8 & 8 & 3 \\ 1 & 2 & 1 & 10 & 10 & 3 \\ 6 & 8 & 10 & 34 & 36 & 14 \\ 6 & 8 & 10 & 36 & 34 & 14 \\ 2 & 3 & 3 & 14 & 14 & 5 \end{pmatrix}_{b_0}$$

[1368] $2^1 3^{142} 21712321^{142} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 14, T_{\max} = 14, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 6 & 2 \\ 2 & 1 & 2 & 16 & 8 & 3 \\ 1 & 2 & 1 & 20 & 10 & 3 \\ 12 & 16 & 20 & 142 & 72 & 28 \\ 6 & 8 & 10 & 72 & 34 & 14 \\ 2 & 3 & 3 & 28 & 14 & 5 \end{pmatrix}_{b_0}$$

[1369] $2^1 3^{141} 21712321^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 14, T_{\max} = 14, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 3 & 2 \\ 2 & 1 & 2 & 16 & 4 & 3 \\ 1 & 2 & 1 & 20 & 5 & 3 \\ 12 & 16 & 20 & 142 & 36 & 28 \\ 3 & 4 & 5 & 36 & 7 & 7 \\ 2 & 3 & 3 & 28 & 7 & 5 \end{pmatrix}_{b_0}$$

$T = 15$ [1370] $1^{11} 6^{131} 1316131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 15, T_{\max} = 15, \Delta = -24, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 5.0$

[1380] $1^{\overline{1}1}81223151^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 2 & 2 & 3 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 5 & 0 & 4 \end{pmatrix}_{b_0}$

[1371] $1^{\overline{1}1}81223151^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$

[1381] $21^{\overline{1}1}8123151^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 3 & 2 \\ 3 & 2 & 3 & 5 & 6 & 5 \\ 2 & 2 & 3 & 6 & 5 & 5 \\ 2 & 2 & 2 & 5 & 5 & 4 \end{pmatrix}_{b_0}$

[1378] $21^{\overline{1}1}8123151^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 1 & 2 \\ 3 & 2 & 3 & 5 & 2 & 5 \\ 1 & 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 5 & 2 & 4 \end{pmatrix}_{b_0}$

[1377] $21^{\overline{1}1}8123151^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 3 & 1 & 2 \\ 1 & 2 & 0 & 4 & 2 & 2 \\ 4 & 3 & 4 & 11 & 5 & 7 \\ 1 & 1 & 2 & 5 & 1 & 3 \\ 2 & 2 & 2 & 7 & 3 & 4 \end{pmatrix}_{b_0}$

[1376] $221^{\overline{1}1}8123151^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 1 & 1 & 5 & 3 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 3 & 0 & 12 & 8 \\ 3 & 2 & 0 & 8 & 5 \end{pmatrix}_{b_0}$

[1375] $1^{\overline{1}1}81223151^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 & 2 \\ 3 & 2 & 3 & 5 & 4 & 5 \\ 1 & 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 2 & 5 & 3 & 4 \end{pmatrix}_{b_0}$

[1374] $1^{\overline{1}1}81223151^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 & 2 \\ 2 & 2 & 3 & 5 & 4 & 5 \\ 2 & 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 2 & 5 & 3 & 4 \end{pmatrix}_{b_0}$

[1373] $21^{\overline{1}1}81223151^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 1 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 2 & 0 & 6 & 6 \\ 3 & 2 & 0 & 6 & 5 \end{pmatrix}_{b_0}$

[1372] $21^1 8^1 1231^1 5^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$

[1379] $1^1 8^1 1231^1 5^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 15, T_{\max} = 16, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 & 2 \\ 3 & 2 & 3 & 5 & 4 & 5 \\ 1 & 1 & 2 & 4 & 1 & 3 \\ 2 & 2 & 2 & 5 & 3 & 4 \end{pmatrix}_{b_0}$

[1382] $1^1 5^1 13221^1 8^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 15, T_{\max} = 16, \Delta = -38, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

$T = 16$ [1383] $1^1 1^1 5^1 1321^1 8^1 12321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 16, T_{\max} = 16, \Delta = -20, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 5.0$

[1388] $1^1 6^1 1316^1 13151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 16, T_{\max} = 17, \Delta = -39, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 2 & 0 & 5 & 3 & 5 \\ 1 & 1 & 0 & 3 & 0 & 2 \\ 2 & 2 & 0 & 5 & 2 & 4 \end{pmatrix}_{b_0}$

[1387] $1^1 6^1 1316^1 13151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 16, T_{\max} = 17, \Delta = -39, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 1 & 1 & 3 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & 3 & 2 & 4 \end{pmatrix}_{b_0}$

[1389] $1^1 1^1 2^1 31^1 6^1 1316131^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 16, T_{\max} = 16, \Delta = -3, \det = 64, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 5 & 2 \\ 2 & 6 & 1 & 8 & 16 & 6 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 3 & 8 & 1 & 7 & 17 & 7 \\ 5 & 16 & 1 & 17 & 31 & 13 \\ 2 & 6 & 0 & 7 & 13 & 5 \end{pmatrix}_{b_0}$

$T = 17$ [1390] $1^1 6^1 12321^1 8^1 12321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 17, T_{\max} = 17, \Delta = -19, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 5.0$

[1392] $1^1 8^1 1231513151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 17, T_{\max} = 18, \Delta = -40, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

$T = 19$ [1394] $1^1 1^1 321^1 8^1 12321^1 8^1 1231^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 19, T_{\max} = 19, \Delta = -23, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 5.0$

[1395] $1^1 1^1 321^1 8^1 12321^1 8^1 12321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 19, T_{\max} = 19, \Delta = -25, \det = 0, (n_+, n_-, n_0) = (1, 19, 2), c_{\text{ext}} = 5.0$

$$[1396] \ 1^{\overline{11}} 5 \ 1316131613151^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 19, \ T_{\max} = 19, \ \Delta = -12, \ \det = 0, \ (n_+, n_-, n_0) = (1, 19, 1), \ c_{\text{ext}} = 5.0$$

$$[1397] \ 31^{\overline{1}} 6 \ 13161316 \ 131^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 19, \ T_{\max} = 19, \ \Delta = -2, \ \det = 0, \ (n_+, n_-, n_0) = (1, 19, 1), \ c_{\text{ext}} = 5.0$$

$$[1399] \ 1^{\overline{1}} 7 \ 123218 \ 123151^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 19, \ T_{\max} = 20, \ \Delta = -38, \ \det = -2, \ (n_+, n_-, n_0) = (1, 19, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 20, \ T_{\max} = 20, \ T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 1 & 1 & 3 \\ 1 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 23 \quad [1402] \ 1^{\overline{11}} 5 \ 13218123218123151^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 23, \ T_{\max} = 23, \ \Delta = -11, \ \det = 0, \ (n_+, n_-, n_0) = (1, 23, 1), \ c_{\text{ext}} = 5.0$$

$$T = 24 \quad [1403] \ 1^{\overline{11}} 6 \ 123218123218123151^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 24, \ T_{\max} = 24, \ \Delta = -10, \ \det = 0, \ (n_+, n_-, n_0) = (1, 24, 1), \ c_{\text{ext}} = 5.0$$

$$T = 25 \quad [1405] \ 1^{\overline{1}} 8 \ 123218 \ 123218123151^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 25, \ T_{\max} = 26, \ \Delta = -36, \ \det = -2, \ (n_+, n_-, n_0) = (1, 25, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 26, \ T_{\max} = 26, \ T \in [26, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 2 & 0 & 5 & 2 & 5 \\ 1 & 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & 0 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 33 \quad [1407] \ 1^{\overline{11}} 5 \ 13221 \ 12231513221(12)1223151^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 33, \ T_{\max} = 33, \ \Delta = -10, \ \det = 0, \ (n_+, n_-, n_0) = (1, 33, 1), \ c_{\text{ext}} = 5.0$$

$$T = 36 \quad [1408] \ 1^{\overline{11}} 5 \ 13221 \ 12231513221(12)122315131^{\overline{11}} 6 \ 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 36, \ T_{\max} = 36, \ \Delta = -9, \ \det = 0, \ (n_+, n_-, n_0) = (1, 36, 1), \ c_{\text{ext}} = 5.0$$

$$T = 44 \quad [1409] \ 1^{\overline{11}} 5 \ 13221(12)12231513221 \ 12231513221(12)1223151^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 44, \ T_{\max} = 44, \ \Delta = -9, \ \det = 0, \ (n_+, n_-, n_0) = (1, 44, 1), \ c_{\text{ext}} = 5.0$$

4.879 $\text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{e}}_6$

$$T = 8 \quad [0] \ 1^{\overline{11}} 31^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{e}}_6 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 49, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 31^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{e}}_6 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 112, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 31^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{e}}_6 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 171, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.34 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6$ $T = 8$, $T_{\max} = 92$, $\Delta = -133$, $\det = 189$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 92$, $T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 5 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

4.880 $\text{NHC}(-2)(-3) + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{violet}{e}_6 + \text{NHC}(-2)(-3)$

$T = 10$ [0] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10$, $T_{\max} = 13$, $\Delta = -83$, $\det = 44$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 5 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 3 & 0 & 4 & 4 & 4 \\ 2 & 4 & 3 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10$, $T_{\max} = 13$, $\Delta = -83$, $\det = 429$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.80 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 5 & 4 & 4 \\ 3 & 5 & 4 & 9 & 0 & 6 & 3 & 4 \\ 5 & 9 & 9 & 13 & 0 & 9 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 6 & 9 & 0 & 4 & 4 & 4 \\ 2 & 4 & 3 & 6 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & 7 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

4.881 $\text{NHC}(-2)(-3) + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{so}_8$

$T = 9$ [0] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -114$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.89 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 1 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 \\ 9 & 17 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 7 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -512$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.52 \rightarrow 14$, $T_{\max} = 93$, $T \in [14, 93]$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 13 & 27 & 0 & 7 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 7 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

4.882 $\text{NHC}(-2)(-3) + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{so}_8 + \textcolor{red}{su}_2$

$T = 10$ [0] $1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{so}_8 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 22$, $\Delta = -102$, $\det = 372$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 9 & 5 \\ 11 & 20 & 17 & 33 & 0 & 17 & 8 \\ 9 & 17 & 13 & 27 & 0 & 15 & 7 \\ 17 & 33 & 27 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 27 & 0 & 13 & 7 \\ 5 & 8 & 7 & 13 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

4.883 $\text{NHC}(-2)(-3) + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{red}{su}_2$

$T = 9$ [0] $1^{111}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -16$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 18.25 \rightarrow 19$, $T_{\max} = 21$, $T \in [19, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 1 & 3 & 0 & 15 & 3 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 15 & 15 & 0 & 100 & 18 \\ 3 & 4 & 3 & 3 & -4 & 18 & -6 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 3 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{113}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 5 & 9 & 0 & 17 & 4 \\ 3 & 5 & 4 & 9 & 0 & 15 & 4 \\ 5 & 9 & 9 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 27 & 0 & 49 & 13 \\ 3 & 4 & 4 & 7 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{1112}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 4 & 3 \\ 3 & 4 & 4 & 9 & 0 & 8 & 4 \\ 2 & 4 & 1 & 6 & 0 & 6 & 3 \\ 5 & 9 & 6 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 6 & 12 & 0 & 10 & 6 \\ 3 & 4 & 3 & 7 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{113}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 5 & 4 & 0 & 17 & 4 \\ 3 & 5 & 4 & 3 & 0 & 15 & 4 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 12 & 0 & 49 & 13 \\ 3 & 4 & 4 & 3 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{112}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 17.66 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 5 & 4 & 0 & 28 & 4 \\ 3 & 5 & 4 & 3 & 0 & 30 & 4 \\ 2 & 4 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 16 & 28 & 30 & 18 & 0 & 154 & 22 \\ 3 & 4 & 4 & 3 & -4 & 22 & -5 \end{pmatrix}_{b_0}$$

[6] $1^{111}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 5 & 4 & 0 & 14 & 4 \\ 3 & 5 & 4 & 3 & 0 & 15 & 4 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 15 & 9 & 0 & 37 & 11 \\ 3 & 4 & 4 & 3 & -4 & 11 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -170, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 7 & 7 \\ 3 & 4 & 4 & 3 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{112}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{e}_6 + \text{su}_2$ $T = 9, T_{\max} = 21, \Delta = -103, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 13 & 4 \\ 3 & 5 & 4 & 3 & 0 & 12 & 4 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 12 & 9 & 0 & 28 & 10 \\ 3 & 4 & 4 & 3 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -224, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.48 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 5 & 9 & 0 & 5 & 4 \\ 3 & 5 & 4 & 9 & 0 & 6 & 4 \\ 5 & 9 & 9 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 6 & 9 & 0 & 4 & 4 \\ 3 & 4 & 4 & 7 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1113} 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -232, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 13.10 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 1 & 3 & 0 & 21 & 3 \\ 2 & 4 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 21 & 21 & 0 & 193 & 25 \\ 3 & 4 & 3 & 3 & -4 & 25 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1111} 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -232, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.28 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & 3 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1112} 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -256, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.44 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 6 & 6 & 0 & 10 & 6 \\ 3 & 4 & 3 & 3 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1113} 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -296, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.65 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 9 & 9 & 0 & 25 & 9 \\ 3 & 4 & 3 & 3 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1111} 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -296, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.28 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & 3 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1112} 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -338, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.21 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 16 & 3 \\ 3 & 4 & 4 & 9 & 0 & 28 & 4 \\ 2 & 4 & 1 & 6 & 0 & 18 & 3 \\ 5 & 9 & 6 & 13 & 0 & 48 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 16 & 28 & 18 & 48 & 0 & 154 & 22 \\ 3 & 4 & 3 & 7 & -4 & 22 & -1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{112} 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -340, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.75 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 5 & 4 & 0 & 23 & 4 \\ 3 & 5 & 4 & 3 & 0 & 24 & 4 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 24 & 15 & 0 & 100 & 18 \\ 3 & 4 & 4 & 3 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{1114}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -352, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.86 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 12 & 0 & 46 & 12 \\ 3 & 4 & 3 & 3 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1111}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -368, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.49 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 8 & 3 \\ 3 & 4 & 4 & 9 & 0 & 14 & 4 \\ 2 & 4 & 1 & 6 & 0 & 9 & 3 \\ 5 & 9 & 6 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 24 & 0 & 37 & 11 \\ 3 & 4 & 3 & 7 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{112}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -386, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.44 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 5 & 4 & 0 & 18 & 4 \\ 3 & 5 & 4 & 3 & 0 & 18 & 4 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 4 & 3 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1115}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -424, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.1$$

$$T_{\min} = 13.06 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 4 & 0 & 20 & 4 \\ 2 & 4 & 1 & 3 & 0 & 15 & 3 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 20 & 15 & 15 & 0 & 73 & 15 \\ 3 & 4 & 3 & 3 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{112}31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -440, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.35 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 5 & 9 & 0 & 13 & 4 \\ 3 & 5 & 4 & 9 & 0 & 12 & 4 \\ 5 & 9 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 12 & 21 & 0 & 28 & 10 \\ 3 & 4 & 4 & 7 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{114}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -490, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.38 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 5 & 4 & 0 & 26 & 4 \\ 3 & 5 & 4 & 3 & 0 & 24 & 4 \\ 2 & 4 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 24 & 18 & 0 & 118 & 20 \\ 3 & 4 & 4 & 3 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111}31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -504, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.37 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 5 & 9 & 0 & 9 & 4 \\ 3 & 5 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 4 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{1116}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -512, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.22 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 1 & 3 & 0 & 18 & 3 \\ 2 & 4 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 24 & 18 & 18 & 0 & 106 & 18 \\ 3 & 4 & 3 & 3 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1112} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -512, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 12 & 12 & 0 & 58 & 14 \\ 3 & 4 & 3 & 3 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{113} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -520, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 5 & 4 & 0 & 22 & 4 \\ 3 & 5 & 4 & 3 & 0 & 21 & 4 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 21 & 15 & 0 & 85 & 17 \\ 3 & 4 & 4 & 3 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -540, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 1 & 6 & 0 & 6 & 3 \\ 5 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 15 & 0 & 13 & 7 \\ 3 & 4 & 3 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -592, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & 3 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1117} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -616, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 13.35 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 4 & 0 & 28 & 4 \\ 2 & 4 & 1 & 3 & 0 & 21 & 3 \\ 2 & 4 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 28 & 21 & 21 & 0 & 145 & 21 \\ 3 & 4 & 3 & 3 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1114} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -666, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 11 & 3 \\ 3 & 4 & 4 & 9 & 0 & 21 & 4 \\ 2 & 4 & 1 & 6 & 0 & 15 & 3 \\ 5 & 9 & 6 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 15 & 33 & 0 & 76 & 16 \\ 3 & 4 & 3 & 7 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{113} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -736, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.66 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 5 & 4 & 0 & 32 & 4 \\ 3 & 5 & 4 & 3 & 0 & 33 & 4 \\ 2 & 4 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 33 & 21 & 0 & 193 & 25 \\ 3 & 4 & 4 & 3 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -746, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.34 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 7 & 3 \\ 3 & 4 & 4 & 9 & 0 & 13 & 4 \\ 2 & 4 & 1 & 6 & 0 & 9 & 3 \\ 5 & 9 & 6 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 21 & 0 & 28 & 10 \\ 3 & 4 & 3 & 7 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1113} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -746, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.47 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 5 & 4 & 0 & 27 & 4 \\ 3 & 5 & 4 & 3 & 0 & 27 & 4 \\ 2 & 4 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 4 & 3 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1113} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -788, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 9 & 3 \\ 3 & 4 & 4 & 9 & 0 & 17 & 4 \\ 2 & 4 & 1 & 6 & 0 & 12 & 3 \\ 5 & 9 & 6 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 27 & 0 & 49 & 13 \\ 3 & 4 & 3 & 7 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1113} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -872, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 3 & 0 & 18 & 3 \\ 2 & 4 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 18 & 18 & 0 & 133 & 21 \\ 3 & 4 & 3 & 3 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1112} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -890, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 13 & 3 \\ 3 & 4 & 4 & 9 & 0 & 23 & 4 \\ 2 & 4 & 1 & 6 & 0 & 15 & 3 \\ 5 & 9 & 6 & 13 & 0 & 39 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 15 & 39 & 0 & 100 & 18 \\ 3 & 4 & 3 & 7 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1113} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -904, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & 3 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1114} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -988, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.40 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 5 & 4 & 0 & 31 & 4 \\ 3 & 5 & 4 & 3 & 0 & 30 & 4 \\ 2 & 4 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 30 & 21 & 0 & 172 & 24 \\ 3 & 4 & 4 & 3 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1112} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1026, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 10 & 3 \\ 3 & 4 & 4 & 9 & 0 & 18 & 4 \\ 2 & 4 & 1 & 6 & 0 & 12 & 3 \\ 5 & 9 & 6 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 12 & 30 & 0 & 58 & 14 \\ 3 & 4 & 3 & 7 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1113} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1096, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.36 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 15 & 3 \\ 3 & 4 & 4 & 9 & 0 & 27 & 4 \\ 2 & 4 & 1 & 6 & 0 & 18 & 3 \\ 5 & 9 & 6 & 13 & 0 & 45 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 18 & 45 & 0 & 133 & 21 \\ 3 & 4 & 3 & 7 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1114} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1850, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 14 & 3 \\ 3 & 4 & 4 & 9 & 0 & 26 & 4 \\ 2 & 4 & 1 & 6 & 0 & 18 & 3 \\ 5 & 9 & 6 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 18 & 42 & 0 & 118 & 20 \\ 3 & 4 & 3 & 7 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1114} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2618, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.39 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 17 & 3 \\ 3 & 4 & 4 & 9 & 0 & 31 & 4 \\ 2 & 4 & 1 & 6 & 0 & 21 & 3 \\ 5 & 9 & 6 & 13 & 0 & 51 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 21 & 51 & 0 & 172 & 24 \\ 3 & 4 & 3 & 7 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1114} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -2970, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.42 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 20 & 3 \\ 3 & 4 & 4 & 9 & 0 & 36 & 4 \\ 2 & 4 & 1 & 6 & 0 & 24 & 3 \\ 5 & 9 & 6 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 24 & 60 & 0 & 238 & 28 \\ 3 & 4 & 3 & 7 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

4.884 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 10 \quad [0] \quad 1^{1111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -102, \quad \det = 372, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 9 & 5 \\ 11 & 20 & 17 & 33 & 0 & 17 & 8 \\ 9 & 17 & 13 & 27 & 0 & 15 & 7 \\ 17 & 33 & 27 & 49 & 0 & 27 & 13 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 27 & 0 & 13 & 7 \\ 5 & 8 & 7 & 13 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

4.885 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{1111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 148, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 5 & 4 & 4 \\ 3 & 5 & 4 & 9 & 0 & 6 & 3 & 4 \\ 5 & 9 & 9 & 13 & 0 & 9 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 6 & 9 & 0 & 4 & 3 & 4 \\ 2 & 4 & 3 & 6 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & 7 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 166, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 3 & 4 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & 3 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 220, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.85 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 8 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 14 & 4 & 4 \\ 2 & 4 & 1 & 6 & 0 & 9 & 3 & 3 \\ 5 & 9 & 6 & 13 & 0 & 24 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 24 & 0 & 37 & 9 & 11 \\ 2 & 4 & 3 & 6 & 0 & 9 & 0 & 2 \\ 3 & 4 & 3 & 7 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{1121}31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 274$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.5$

$T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 10 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 18 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 18 & 3 & 4 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 18 & 12 & 0 & 58 & 12 & 14 \\ 2 & 4 & 3 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & 3 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{1131}31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 454$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.8$

$T_{\min} = 12.98 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 15 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 27 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 27 & 3 & 4 \\ 2 & 4 & 3 & 1 & 0 & 18 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 15 & 27 & 27 & 18 & 0 & 133 & 18 & 21 \\ 2 & 4 & 3 & 3 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & 3 & -4 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{1111}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 660$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 6 & 0 & 6 & 12 & 6 \\ 3 & 4 & 3 & 3 & -4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{1131}31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 676$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.8$

$T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 12 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 22 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 21 & 3 & 4 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 21 & 15 & 0 & 85 & 15 & 17 \\ 2 & 4 & 3 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & 3 & -4 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{1141}31^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 706$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 12.99 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 20 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 36 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 36 & 3 & 4 \\ 2 & 4 & 3 & 1 & 0 & 24 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 20 & 36 & 36 & 24 & 0 & 238 & 24 & 28 \\ 2 & 4 & 3 & 3 & 0 & 24 & 0 & 2 \\ 3 & 4 & 4 & 3 & -4 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{1111}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 712$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.87 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{1111}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 724$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.82 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{1121}31^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 796$, $(n_+, n_-, n_0) =$

$$(1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 7 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 13 & 4 & 4 \\ 3 & 5 & 4 & 9 & 0 & 12 & 3 & 4 \\ 5 & 9 & 9 & 13 & 0 & 21 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 12 & 21 & 0 & 28 & 9 & 10 \\ 2 & 4 & 3 & 6 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & 7 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 9 & 4 & 4 \\ 3 & 5 & 4 & 9 & 0 & 9 & 3 & 4 \\ 5 & 9 & 9 & 13 & 0 & 15 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 15 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 6 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & 7 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1032, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 6 & 0 & 15 & 12 & 6 \\ 3 & 4 & 3 & 3 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1141} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1180, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 17 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 31 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 30 & 3 & 4 \\ 2 & 4 & 3 & 1 & 0 & 21 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 17 & 31 & 30 & 21 & 0 & 172 & 21 & 24 \\ 2 & 4 & 3 & 3 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & 3 & -4 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 & 4 \\ 2 & 4 & 1 & 6 & 0 & 6 & 3 & 3 \\ 5 & 9 & 6 & 13 & 0 & 15 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 15 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 6 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & 7 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

4.886 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{11} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.86 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 6 & 3 & 0 & 3 & 3 \\ 3 & 4 & 4 & 3 & -4 & 3 & -4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.71 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 9 & 0 & 36 & 10 \\ 3 & 4 & 3 & 3 & -4 & 10 & -4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & 3 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{113}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -107, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.32 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 5 & 4 & 0 & 17 & 4 \\ 3 & 5 & 4 & 3 & 0 & 15 & 4 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 12 & 0 & 48 & 12 \\ 3 & 4 & 4 & 3 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{113}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 3 & 4 & 3 & 3 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{113}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -219, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 4 & 3 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{112}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -245, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.74 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 13 & 4 \\ 3 & 5 & 4 & 3 & 0 & 12 & 4 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 12 & 9 & 0 & 27 & 9 \\ 3 & 4 & 4 & 3 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -341, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 4 \\ 3 & 5 & 4 & 9 & 0 & 3 & 4 \\ 5 & 9 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 4 & 7 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -344, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.81 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -359, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 & 3 \\ 3 & 4 & 4 & 9 & 0 & 5 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 3 \\ 5 & 9 & 6 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 3 & 9 & 0 & 3 & 3 \\ 3 & 4 & 3 & 7 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{111}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -408, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 12 & 6 \\ 3 & 4 & 3 & 3 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{\overline{11111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -485, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 3 \\ 5 & 9 & 6 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 3 & 7 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{\overline{11111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -729, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 1 & 6 & 0 & 6 & 3 \\ 5 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 15 & 0 & 12 & 6 \\ 3 & 4 & 3 & 7 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

4.887 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{\overline{11111}} 3^{\overline{11111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 148, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 5 & 4 \\ 3 & 5 & 4 & 9 & 0 & 3 & 6 & 4 \\ 5 & 9 & 9 & 13 & 0 & 6 & 9 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 9 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & 7 & -4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 166, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 9 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & 3 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11111}} 3^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 220, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.85 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 14 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 9 & 3 \\ 5 & 9 & 6 & 13 & 0 & 6 & 24 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 9 & 2 \\ 8 & 14 & 9 & 24 & 0 & 9 & 37 & 11 \\ 3 & 4 & 3 & 7 & -4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1112}} 3^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 274, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.97 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 18 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 18 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 12 & 2 \\ 10 & 18 & 18 & 12 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & 3 & -4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1113}} 3^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 454, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.98 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 15 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 27 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 27 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 18 & 2 \\ 15 & 27 & 27 & 18 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & 3 & -4 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

- [5] $1^{\overline{11111}}31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 12 & 6 & 6 \\ 2 & 4 & 3 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 3 & 3 & -4 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$
- [6] $1^{\overline{1113}}31^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 22 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 21 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 15 & 2 \\ 12 & 22 & 21 & 15 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & 3 & -4 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$
- [7] $1^{\overline{1114}}31^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 706, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 20 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 36 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 36 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 & 24 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 24 & 2 \\ 20 & 36 & 36 & 24 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & 3 & -4 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{\overline{11111}}31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 712, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & -4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{\overline{11111}}31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 724, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 6 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & 3 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{\overline{1112}}31^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 796, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 13 & 4 \\ 3 & 5 & 4 & 9 & 0 & 3 & 12 & 4 \\ 5 & 9 & 9 & 13 & 0 & 6 & 21 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 9 & 2 \\ 7 & 13 & 12 & 21 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & 7 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{\overline{11111}}31^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 9 & 0 & 3 & 9 & 4 \\ 5 & 9 & 9 & 13 & 0 & 6 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 15 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & 7 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{\overline{11111}}31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1032, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 12 & 15 & 6 \\ 5 & 9 & 6 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 3 & 3 & -4 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{111111}31^{1113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 1180$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 12.96 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 17 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 31 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 30 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 & 21 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 21 & 2 \\ 17 & 31 & 30 & 21 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & 3 & -4 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{111111}31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 1276$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 6 & 3 \\ 5 & 9 & 6 & 13 & 0 & 6 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 6 & 2 \\ 5 & 9 & 6 & 15 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

4.888 NHC(-2)(-3) + $\textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_3$

$T = 10$ [0] $1^{111111}31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 228$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 5 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111111}31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 231$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.13 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 9 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & 3 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{111111}31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 507$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.92 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 4 & 4 \\ 3 & 5 & 4 & 9 & 0 & 3 & 3 & 4 \\ 5 & 9 & 9 & 13 & 0 & 6 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 6 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & 7 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{111111}31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 651$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.17 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 5 & 4 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 3 & 3 \\ 5 & 9 & 6 & 13 & 0 & 9 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 9 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 6 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 7 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.889 NHC(-2)(-3) + $\textcolor{red}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{f}_4$

$T = 8$ [0] $1^{111111}31^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 20$, $\Delta = -107$, $\det = 39$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 5 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

4.890 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8$

$T = 9 \quad [0] \quad 1^{111}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 1 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 \\ 9 & 17 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 7 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 13 & 27 & 0 & 7 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 7 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

4.891 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9 \quad [0] \quad 1^{111}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 3 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 4 & 3 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 3 & 0 & 2 & 3 \\ 2 & 4 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 3 & 4 \\ 3 & 5 & 4 & 9 & 0 & 3 & 4 \\ 5 & 9 & 9 & 13 & 0 & 5 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 4 & 7 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^{111}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 3 & 4 \\ 2 & 4 & 1 & 6 & 0 & 2 & 3 \\ 5 & 9 & 6 & 13 & 0 & 5 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 5 & 0 & 0 & 2 \\ 3 & 4 & 3 & 7 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

4.892 NHC(-2)(-3) + $\mathbf{su}_2 + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$

$T = 10 \quad [0] \quad 1^{111}3^{111}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 2 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & 3 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^{111}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 3 & 4 & 4 \\ 3 & 5 & 4 & 9 & 0 & 3 & 3 & 4 \\ 5 & 9 & 9 & 13 & 0 & 5 & 6 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 5 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 6 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & 7 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^{111}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 3 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 2 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 2 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 2 & 0 & 0 & 5 & 2 \\ 5 & 9 & 6 & 6 & 0 & 5 & 12 & 6 \\ 3 & 4 & 3 & 3 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^{111}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 3 & 4 & 4 \\ 2 & 4 & 1 & 6 & 0 & 2 & 3 & 3 \\ 5 & 9 & 6 & 13 & 0 & 5 & 6 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 5 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 6 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & 7 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.893 NHC(-2)(-3) + $\mathbf{su}_2 + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$

$T = 9 \quad [0] \quad 1^{111}3^{111}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 5 & 4 & 0 & 7 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 5 & 0 & 7 & 5 \\ 3 & 4 & 4 & 3 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^{111}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 5 & 9 & 0 & 7 & 4 \\ 3 & 5 & 4 & 9 & 0 & 6 & 4 \\ 5 & 9 & 9 & 13 & 0 & 11 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 11 & 0 & 7 & 5 \\ 3 & 4 & 4 & 7 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{1113}} 3^{\overline{111}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -194$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 12 & 4 \\ 3 & 5 & 4 & 3 & 0 & 12 & 4 \\ 2 & 4 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 12 & 8 & 0 & 25 & 9 \\ 3 & 4 & 4 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{113}} 3^{\overline{111}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -194$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 5 & 4 & 0 & 20 & 4 \\ 3 & 5 & 4 & 3 & 0 & 18 & 4 \\ 2 & 4 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 18 & 14 & 0 & 69 & 15 \\ 3 & 4 & 4 & 3 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{1113}} 3^{\overline{111}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -246$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 6 & 3 \\ 3 & 4 & 4 & 9 & 0 & 11 & 4 \\ 2 & 4 & 1 & 6 & 0 & 8 & 3 \\ 5 & 9 & 6 & 13 & 0 & 17 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 8 & 17 & 0 & 20 & 8 \\ 3 & 4 & 4 & 7 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{1123}} 3^{\overline{111}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -260$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.60 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 5 & 4 & 0 & 16 & 4 \\ 3 & 5 & 4 & 3 & 0 & 15 & 4 \\ 2 & 4 & 3 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 15 & 11 & 0 & 44 & 12 \\ 3 & 4 & 4 & 3 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{1113}} 3^{\overline{111}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -264$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 3 & 0 & 8 & 3 \\ 2 & 4 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 8 & 8 & 0 & 25 & 9 \\ 3 & 4 & 3 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{\overline{1113}} 3^{\overline{111}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -276$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.61 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 & 0 & 7 & 4 \\ 2 & 4 & 1 & 3 & 0 & 5 & 3 \\ 2 & 4 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 5 & 5 & 0 & 7 & 5 \\ 3 & 4 & 3 & 3 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{\overline{1113}} 3^{\overline{111}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -316$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 4 & 3 \\ 3 & 4 & 4 & 9 & 0 & 7 & 4 \\ 2 & 4 & 1 & 6 & 0 & 5 & 3 \\ 5 & 9 & 6 & 13 & 0 & 11 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 5 & 11 & 0 & 7 & 5 \\ 3 & 4 & 3 & 7 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{\overline{1113}} 3^{\overline{111}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -416$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.71 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 4 & 0 & 11 & 4 \\ 2 & 4 & 1 & 3 & 0 & 8 & 3 \\ 2 & 4 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 8 & 8 & 0 & 20 & 8 \\ 3 & 4 & 3 & 3 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{1112}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 3 & 0 & 14 & 3 \\ 2 & 4 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 21 & 14 & 14 & 0 & 80 & 16 \\ 3 & 4 & 3 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 7 & 3 \\ 3 & 4 & 4 & 9 & 0 & 12 & 4 \\ 2 & 4 & 1 & 6 & 0 & 8 & 3 \\ 5 & 9 & 6 & 13 & 0 & 20 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 8 & 20 & 0 & 25 & 9 \\ 3 & 4 & 3 & 7 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{1113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -564, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 15 & 4 \\ 2 & 4 & 1 & 3 & 0 & 11 & 3 \\ 2 & 4 & 3 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 11 & 11 & 0 & 39 & 11 \\ 3 & 4 & 3 & 3 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -620, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 1 & 3 & 0 & 11 & 3 \\ 2 & 4 & 3 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 11 & 11 & 0 & 44 & 12 \\ 3 & 4 & 3 & 3 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1114}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -720, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 4 & 0 & 19 & 4 \\ 2 & 4 & 1 & 3 & 0 & 14 & 3 \\ 2 & 4 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 14 & 14 & 0 & 64 & 14 \\ 3 & 4 & 3 & 3 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -810, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 9 & 3 \\ 3 & 4 & 4 & 9 & 0 & 16 & 4 \\ 2 & 4 & 1 & 6 & 0 & 11 & 3 \\ 5 & 9 & 6 & 13 & 0 & 26 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 11 & 26 & 0 & 44 & 12 \\ 3 & 4 & 3 & 7 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1113}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -840, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 4 & 0 & 30 & 4 \\ 2 & 4 & 1 & 3 & 0 & 20 & 3 \\ 2 & 4 & 3 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 17 & 30 & 20 & 20 & 0 & 165 & 23 \\ 3 & 4 & 3 & 3 & -3 & 23 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{1113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -964, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 11 & 3 \\ 3 & 4 & 4 & 9 & 0 & 20 & 4 \\ 2 & 4 & 1 & 6 & 0 & 14 & 3 \\ 5 & 9 & 6 & 13 & 0 & 32 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 14 & 32 & 0 & 69 & 15 \\ 3 & 4 & 3 & 7 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11133^1 1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -984, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 4 & 0 & 20 & 4 \\ 2 & 4 & 1 & 3 & 0 & 14 & 3 \\ 2 & 4 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 14 & 14 & 0 & 69 & 15 \\ 3 & 4 & 3 & 3 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11143^1 1^{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -986, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.67 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 13 & 3 \\ 3 & 4 & 4 & 9 & 0 & 24 & 4 \\ 2 & 4 & 1 & 6 & 0 & 17 & 3 \\ 5 & 9 & 6 & 13 & 0 & 38 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 17 & 38 & 0 & 100 & 18 \\ 3 & 4 & 3 & 7 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11143^1 1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1076, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.60 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 4 & 0 & 25 & 4 \\ 2 & 4 & 1 & 3 & 0 & 17 & 3 \\ 2 & 4 & 3 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 14 & 25 & 17 & 17 & 0 & 111 & 19 \\ 3 & 4 & 3 & 3 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11143^1 1^{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1296, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 22 & 3 \\ 3 & 4 & 4 & 4 & 0 & 39 & 4 \\ 2 & 4 & 1 & 3 & 0 & 26 & 3 \\ 2 & 4 & 3 & 1 & 0 & 26 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 22 & 39 & 26 & 26 & 0 & 280 & 30 \\ 3 & 4 & 3 & 3 & -3 & 30 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11143^1 1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1356, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 1 & 3 & 0 & 17 & 3 \\ 2 & 4 & 3 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 17 & 17 & 0 & 100 & 18 \\ 3 & 4 & 3 & 3 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11143^1 1^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1644, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 19 & 3 \\ 3 & 4 & 4 & 4 & 0 & 34 & 4 \\ 2 & 4 & 1 & 3 & 0 & 23 & 3 \\ 2 & 4 & 3 & 1 & 0 & 23 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 19 & 34 & 23 & 23 & 0 & 208 & 26 \\ 3 & 4 & 3 & 3 & -3 & 26 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{11143^1 1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 4 & 0 & 29 & 4 \\ 2 & 4 & 1 & 3 & 0 & 20 & 3 \\ 2 & 4 & 3 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 16 & 29 & 20 & 20 & 0 & 148 & 22 \\ 3 & 4 & 3 & 3 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

4.894 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{blue}{1}\textcolor{blue}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 4 & 3 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{2}} 3^{\textcolor{red}{1}} 1^{\textcolor{blue}{1}\textcolor{blue}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 13 & 4 \\ 3 & 5 & 4 & 3 & 0 & 12 & 4 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 12 & 9 & 0 & 27 & 9 \\ 3 & 4 & 4 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{2}} 3^{\textcolor{red}{1}} 1^{\textcolor{blue}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{2}} 3^{\textcolor{red}{1}} 1^{\textcolor{blue}{1}\textcolor{blue}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 4 \\ 3 & 5 & 4 & 9 & 0 & 3 & 4 \\ 5 & 9 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 4 & 7 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{blue}{1}\textcolor{blue}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 6 & 0 & 12 & 6 \\ 3 & 4 & 3 & 3 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{blue}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 3 \\ 5 & 9 & 6 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 3 & 7 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{blue}{1}\textcolor{blue}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 1 & 6 & 0 & 6 & 3 \\ 5 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 15 & 0 & 12 & 6 \\ 3 & 4 & 3 & 7 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

4.895 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{blue}{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 3 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 2 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & 3 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 3 & 4 \\ 3 & 5 & 4 & 9 & 0 & 3 & 3 & 4 \\ 5 & 9 & 9 & 13 & 0 & 6 & 5 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 & 2 \\ 2 & 3 & 3 & 5 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & 7 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 3 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 2 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 6 & 0 & 12 & 5 & 6 \\ 2 & 3 & 2 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 3 & 3 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 3 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 2 & 3 \\ 5 & 9 & 6 & 13 & 0 & 6 & 5 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 3 & 7 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.896 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 10, \Delta = -34, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 0 & 5 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 5 & 0 & 4 & 3 & 0 & 4 \\ 2 & 4 & 0 & 3 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 0 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

4.897 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{11} 3^1 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 14 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 14 & 17 & 13 & 15 & 7 \\ 12 & 17 & 15 & 13 & 7 \\ 6 & 8 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 3 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 5 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 5 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 1 & 1 & 6 \\ 15 & 20 & 0 & 0 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 6 & 8 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 17 & 0 & 13 & 7 \\ 6 & 8 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.48 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 13 & 15 & 7 \\ 13 & 17 & 15 & 13 & 7 \\ 6 & 8 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 4 & 5 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{12} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 4 & 5 & 4 & 12 & 0 & 4 \\ 8 & 13 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 13 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 13 & 17 & 13 & 15 & 7 \\ 13 & 17 & 15 & 13 & 7 \\ 6 & 8 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 1 & 13 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 17 & 0 & 13 & 7 \\ 6 & 8 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 4 & 5 & 4 & 3 & 0 & 4 \\ 3 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 1 & 11 & 6 \\ 15 & 20 & 0 & 16 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 16 & 0 & 10 & 6 \\ 6 & 8 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

- [14] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 3 \\ 3 & 4 & 9 & 8 & 0 & 4 \\ 5 & 9 & 13 & 12 & 0 & 7 \\ 5 & 8 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 6 & -2 & 2 \end{pmatrix}_{b_0}$
- [15] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & 3 & -2 & 2 \end{pmatrix}_{b_0}$
- [17] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 \\ 3 & 5 & 4 & 6 & 0 & 4 \\ 4 & 5 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 4 & -2 & 2 \end{pmatrix}_{b_0}$
- [16] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 10 & 15 & 13 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 13 & 17 & 13 & 27 & 7 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$
- [18] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 3 & 4 & 1 & 6 & 0 & 3 \\ 5 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 7 & -2 & 2 \end{pmatrix}_{b_0}$
- [19] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 3 & -2 & 2 \end{pmatrix}_{b_0}$
- [20] $1^{12} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 3 \\ 3 & 4 & 0 & 8 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & 6 & -2 & 2 \end{pmatrix}_{b_0}$
- [21] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 10 & 15 & 12 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 13 & 27 & 7 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$
- [22] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 3 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 3 & -2 & 2 \end{pmatrix}_{b_0}$
- [23] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 10 & 15 & 12 & 39 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 39 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$
- [24] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 16 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 16 & 0 & 10 & 6 \\ 6 & 8 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[25] $1^{121} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 1 & 3 \\ 3 & 4 & 8 & 9 & 0 & 4 \\ 5 & 8 & 10 & 12 & 0 & 6 \\ 6 & 9 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{11} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 1 & 3 \\ 3 & 4 & 0 & 9 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 9 & 0 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{111} 3^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 25 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 13 & 27 & 7 \\ 25 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[27] $1^{11} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[28] $1^{11} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 7 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{14} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 1 & 3 \\ 3 & 4 & 5 & 21 & 0 & 4 \\ 3 & 5 & 4 & 18 & 0 & 4 \\ 13 & 21 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{111} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 5 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{111} 3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 9 & 14 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{111} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 24 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 13 & 27 & 7 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[34] $1^{11} 3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 1 & 3 \\ 3 & 4 & 0 & 14 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 9 & 14 & 0 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 36 & 6 \\ 15 & 20 & 0 & 50 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 36 & 50 & 0 & 118 & 20 \\ 6 & 8 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[36] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 3 & 0 \\ 3 & 4 & 3 & 1 & 0 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 3 & -2 \end{pmatrix}_{b_0}$$

[37] $1^{112} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 6 & 8 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

[38] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 1 & 3 \\ 3 & 4 & 0 & 13 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 133, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 6 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.68 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 6 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{123} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 1 & 3 \\ 3 & 4 & 8 & 22 & 0 & 4 \\ 5 & 8 & 10 & 30 & 0 & 6 \\ 13 & 22 & 30 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 4 & 5 & 4 & 9 & 0 & 4 \\ 6 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 13 & 24 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 13 & 17 & 13 & 27 & 7 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[44] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 1 & 24 & 6 \\ 15 & 20 & 0 & 33 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 24 & 33 & 0 & 49 & 13 \\ 6 & 8 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

[45] $1^{13} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 143, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 4 & 15 & 0 & 4 \\ 10 & 17 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

[46] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 5 & 8 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

[47] $1^{112} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 5 & 9 & 13 & 21 & 0 & 7 \\ 9 & 13 & 21 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[48] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 4 & 12 & 0 & 4 \\ 9 & 13 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[49] $1^{111} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 38 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 38 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[50] $1^{13} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 4 & 15 & 0 & 4 \\ 11 & 17 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.66 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 4 & 12 & 0 & 4 \\ 8 & 13 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{111} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 13 & 36 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 13 & 17 & 13 & 42 & 7 \\ 36 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

[52] $1^{12} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 1 & 3 \\ 3 & 4 & 5 & 18 & 0 & 4 \\ 4 & 5 & 4 & 18 & 0 & 4 \\ 11 & 18 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 14 & -2 & 2 \end{pmatrix}_{b_0}$$

[54] $1^{111} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 37 & 6 \\ 15 & 20 & 17 & 50 & 8 \\ 12 & 17 & 13 & 42 & 7 \\ 37 & 50 & 42 & 118 & 20 \\ 6 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$

- [55] $1^{111} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 171$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 10 & 15 & 12 & 51 & 6 \\ 15 & 20 & 17 & 67 & 8 \\ 12 & 17 & 13 & 57 & 7 \\ 51 & 67 & 57 & 217 & 27 \\ 6 & 8 & 7 & 27 & 3 \end{pmatrix}_{b_0}$$
- [56] $1^{112} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 3 & 4 & 1 & 6 & 0 & 3 \\ 5 & 8 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$
- [57] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 177$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.69 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 3 & 4 & 1 & 6 & 0 & 3 \\ 6 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$
- [58] $1^{112} 3 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 185$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 1 & 3 \\ 3 & 4 & 14 & 13 & 0 & 4 \\ 9 & 14 & 37 & 33 & 0 & 11 \\ 8 & 13 & 33 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$
- [59] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 200$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.66 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 5 & 9 & 13 & 15 & 0 & 7 \\ 6 & 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$
- [60] $1^{111} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 209$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 5 & 14 & 0 & 4 \\ 4 & 5 & 4 & 15 & 0 & 4 \\ 9 & 14 & 15 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$
- [61] $1^{113} 3 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 224$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.8$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 13 & 1 & 3 \\ 3 & 4 & 14 & 22 & 0 & 4 \\ 9 & 14 & 37 & 57 & 0 & 11 \\ 13 & 22 & 57 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$
- [62] $1^{112} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 228$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.70 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 3 & 4 & 1 & 12 & 0 & 3 \\ 11 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 14 & -2 & 2 \end{pmatrix}_{b_0}$$
- [63] $21^{123} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 232$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.5$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 13 & 17 & 15 & 42 & 7 \\ 17 & 20 & 17 & 50 & 8 \\ 15 & 17 & 13 & 42 & 7 \\ 42 & 50 & 42 & 118 & 20 \\ 7 & 8 & 7 & 20 & 3 \end{pmatrix}_{b_0}$$
- [64] $1^{121} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 245$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.91 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 5 & 14 & 0 & 4 \\ 3 & 5 & 4 & 15 & 0 & 4 \\ 10 & 14 & 15 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$
- [65] $1^{122} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -83$, $\det = 256$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 10.81 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 3 \\ 3 & 4 & 8 & 8 & 0 & 4 \\ 5 & 8 & 10 & 12 & 0 & 6 \\ 5 & 8 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 12 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 8 & 12 & 9 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 9 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 270, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 1 & 3 \\ 3 & 4 & 8 & 13 & 0 & 4 \\ 5 & 8 & 10 & 18 & 0 & 6 \\ 8 & 13 & 18 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{111} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 273, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 9 & 14 & 0 & 4 \\ 5 & 9 & 13 & 24 & 0 & 7 \\ 9 & 14 & 24 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$

[70] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 277, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.66 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 8 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[71] $1^{113} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 1 & 3 \\ 3 & 4 & 9 & 17 & 0 & 4 \\ 5 & 9 & 13 & 27 & 0 & 7 \\ 11 & 17 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 281, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 3 & 4 & 1 & 9 & 0 & 3 \\ 8 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{132} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 298, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 1 & 3 \\ 3 & 4 & 17 & 18 & 0 & 4 \\ 10 & 17 & 49 & 54 & 0 & 13 \\ 11 & 18 & 54 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 14 & -2 & 2 \end{pmatrix}_{b_0}$$

[74] $1^{112} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.66 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 5 & 9 & 13 & 21 & 0 & 7 \\ 8 & 13 & 21 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[75] $1^{113} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 1 & 3 \\ 3 & 4 & 9 & 22 & 0 & 4 \\ 5 & 9 & 13 & 36 & 0 & 7 \\ 15 & 22 & 36 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[86] $1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 1 & 3 \\ 3 & 4 & 5 & 18 & 0 & 4 \\ 3 & 5 & 4 & 18 & 0 & 4 \\ 12 & 18 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 14 & -2 & 2 \end{pmatrix}_{b_0}$$

[87] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 388, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 1 & 3 \\ 3 & 4 & 9 & 17 & 0 & 4 \\ 5 & 9 & 13 & 27 & 0 & 7 \\ 10 & 17 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

[88] $1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 393, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 1 & 3 \\ 3 & 4 & 9 & 14 & 0 & 4 \\ 6 & 9 & 13 & 24 & 0 & 7 \\ 9 & 14 & 24 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$

[89] $1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 398, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 1 & 3 \\ 3 & 4 & 9 & 18 & 0 & 4 \\ 5 & 9 & 13 & 30 & 0 & 7 \\ 12 & 18 & 30 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 14 & -2 & 2 \end{pmatrix}_{b_0}$$

[90] $1^{114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 16 & 0 & 4 \\ 2 & 4 & 1 & 12 & 0 & 3 \\ 11 & 16 & 12 & 46 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

[91] $1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 401, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 1 & 3 \\ 3 & 4 & 5 & 22 & 0 & 4 \\ 3 & 5 & 4 & 21 & 0 & 4 \\ 13 & 22 & 21 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[92] $1^{123} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 414, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 1 & 3 \\ 3 & 4 & 8 & 17 & 0 & 4 \\ 5 & 8 & 10 & 24 & 0 & 6 \\ 10 & 17 & 24 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

[93] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 421, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 2 & 4 & 1 & 12 & 0 & 3 \\ 10 & 17 & 12 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

[94] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 427, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 6 & 9 & 13 & 21 & 0 & 7 \\ 8 & 13 & 21 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

[95] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 437, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 1 & 15 & 0 & 3 \\ 13 & 22 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[106] $1^{112} 3^{121} 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 579, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 1 & 3 \\ 3 & 4 & 9 & 23 & 0 & 4 \\ 5 & 9 & 13 & 39 & 0 & 7 \\ 15 & 23 & 39 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

[107] $1^{12} 3^{121} 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 584, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 1 & 3 \\ 3 & 4 & 5 & 23 & 0 & 4 \\ 3 & 5 & 4 & 24 & 0 & 4 \\ 15 & 23 & 24 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

[108] $1^{112} 3^{111} 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 11 & 1 & 3 \\ 3 & 4 & 14 & 18 & 0 & 4 \\ 9 & 14 & 37 & 48 & 0 & 11 \\ 11 & 18 & 48 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & 14 & -2 & 2 \end{pmatrix}_{b_0}$$

[110] $1^{12} 3^{131} 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 1 & 3 \\ 3 & 4 & 5 & 23 & 0 & 4 \\ 3 & 5 & 4 & 24 & 0 & 4 \\ 16 & 23 & 24 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 18 & -2 & 2 \end{pmatrix}_{b_0}$$

[109] $1^{113} 3^{12} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 1 & 15 & 0 & 3 \\ 15 & 22 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[111] $1^{112} 3^{111} 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 602, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 1 & 3 \\ 3 & 4 & 9 & 18 & 0 & 4 \\ 6 & 9 & 13 & 30 & 0 & 7 \\ 11 & 18 & 30 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 14 & -2 & 2 \end{pmatrix}_{b_0}$$

[112] $1^{113} 3^{12} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 628, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 1 & 15 & 0 & 3 \\ 14 & 22 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{114} 3^{111} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 1 & 3 \\ 3 & 4 & 4 & 21 & 0 & 4 \\ 2 & 4 & 1 & 15 & 0 & 3 \\ 15 & 21 & 15 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

[114] $1^{114} 3^{112} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 1 & 3 \\ 3 & 4 & 9 & 26 & 0 & 4 \\ 5 & 9 & 13 & 42 & 0 & 7 \\ 17 & 26 & 42 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

[115] $1^{122} 3^{112} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 1 & 3 \\ 3 & 4 & 4 & 26 & 0 & 4 \\ 2 & 4 & 1 & 18 & 0 & 3 \\ 18 & 26 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[136] \ 1^{14} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 908, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 1 & 3 \\ 3 & 4 & 5 & 31 & 0 & 4 \\ 3 & 5 & 4 & 30 & 0 & 4 \\ 21 & 31 & 30 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 24 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{14} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 916, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 1 & 3 \\ 3 & 4 & 4 & 26 & 0 & 4 \\ 2 & 4 & 1 & 18 & 0 & 3 \\ 16 & 26 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[138] \ 1^{14} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 920, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 1 & 3 \\ 3 & 4 & 5 & 31 & 0 & 4 \\ 3 & 5 & 4 & 30 & 0 & 4 \\ 20 & 31 & 30 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 24 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{133} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 940, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 1 & 3 \\ 3 & 4 & 17 & 17 & 0 & 4 \\ 10 & 17 & 49 & 51 & 0 & 13 \\ 10 & 17 & 51 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{123} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 945, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.3$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 13 & 1 & 3 \\ 3 & 4 & 13 & 22 & 0 & 4 \\ 8 & 13 & 28 & 51 & 0 & 10 \\ 13 & 22 & 51 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[141] \ 1^{14} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 948, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 1 & 3 \\ 3 & 4 & 4 & 26 & 0 & 4 \\ 2 & 4 & 1 & 18 & 0 & 3 \\ 17 & 26 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 20 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{113} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 972, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 1 & 3 \\ 3 & 4 & 9 & 27 & 0 & 4 \\ 5 & 9 & 13 & 45 & 0 & 7 \\ 17 & 27 & 45 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 21 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[143] \ 1^{122} 3^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1008, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 11 & 1 & 3 \\ 3 & 4 & 18 & 18 & 0 & 4 \\ 11 & 18 & 58 & 60 & 0 & 14 \\ 11 & 18 & 60 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & 14 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[144] \ 1^{113} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 1017, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 21 & 1 & 3 \\ 3 & 4 & 9 & 32 & 0 & 4 \\ 5 & 9 & 13 & 54 & 0 & 7 \\ 21 & 32 & 54 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 25 & -2 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 5 & 24 & 1 & 3 \\ 3 & 4 & 9 & 36 & 0 & 4 \\ 5 & 9 & 13 & 60 & 0 & 7 \\ 24 & 36 & 60 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 28 & -2 & 2 \end{pmatrix}_{b_0}$$

[155] $1^{114} 3^1 1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 23 & 1 & 3 \\ 3 & 4 & 4 & 36 & 0 & 4 \\ 2 & 4 & 1 & 24 & 0 & 3 \\ 23 & 36 & 24 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 28 & -2 & 2 \end{pmatrix}_{b_0}$$

[156] $1^{14} 3^1 1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1314, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 23 & 1 & 3 \\ 3 & 4 & 5 & 36 & 0 & 4 \\ 3 & 5 & 4 & 36 & 0 & 4 \\ 23 & 36 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 28 & -2 & 2 \end{pmatrix}_{b_0}$$

[157] $1^{114} 3^1 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1315, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 19 & 1 & 3 \\ 3 & 4 & 9 & 31 & 0 & 4 \\ 5 & 9 & 13 & 51 & 0 & 7 \\ 19 & 31 & 51 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 24 & -2 & 2 \end{pmatrix}_{b_0}$$

[158] $1^{14} 3^1 1^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1350, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 24 & 1 & 3 \\ 3 & 4 & 5 & 36 & 0 & 4 \\ 3 & 5 & 4 & 36 & 0 & 4 \\ 24 & 36 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 28 & -2 & 2 \end{pmatrix}_{b_0}$$

[159] $1^{133} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.6$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 13 & 1 & 3 \\ 3 & 4 & 22 & 22 & 0 & 4 \\ 13 & 22 & 85 & 87 & 0 & 17 \\ 13 & 22 & 87 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 17 & 17 & -2 & 2 \end{pmatrix}_{b_0}$$

[160] $1^{114} 3^1 1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 1554, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 23 & 1 & 3 \\ 3 & 4 & 9 & 36 & 0 & 4 \\ 5 & 9 & 13 & 60 & 0 & 7 \\ 23 & 36 & 60 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 28 & -2 & 2 \end{pmatrix}_{b_0}$$

4.898 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$

$T = 9$ [0] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 1 & 5 \\ 11 & 20 & 0 & 17 & 0 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 9 & 17 & 0 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 0 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 10 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

4.899 NHC(-2)(-3) + su₂ + su₂ + so₈ + su₃

$$T = 9 \quad [0] \quad 1^{11} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & 6 & 0 & 3 & 4 \\ 4 & 5 & 6 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 5 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 5 & 0 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 3 & 0 & 0 & 2 \\ 3 & 4 & 0 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

4.900 NHC(-2)(-3) + su₂ + su₂ + su₂

$$T = 9 \quad [1] \quad 1^{11} 31^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 3 & 9 & 4 \\ 2 & 3 & 0 & 3 & 1 & 6 & 3 \\ 4 & 6 & 0 & 9 & 6 & 13 & 7 \\ 2 & 3 & -1 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -45, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 3 & 3 & 1 & 3 & 0 & 3 \\ 2 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12} 31^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 0 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 12 & 4 \\ 5 & 9 & 0 & 0 & 12 & 28 & 10 \\ 2 & 3 & -1 & 0 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -45, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 1 & 3 & 0 & 3 \\ 2 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 11^{11} 31^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 6 & 6 & 3 \\ 4 & 6 & 0 & 6 & 13 & 15 & 7 \\ 3 & 6 & 0 & 6 & 15 & 13 & 7 \\ 2 & 3 & -1 & 3 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 11^{11} 31^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 1 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 3 & 3 \\ 3 & 6 & 0 & 6 & 13 & 6 & 7 \\ 1 & 3 & 0 & 3 & 6 & 1 & 3 \\ 2 & 3 & -1 & 3 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 11^{11} 31^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$[16] \quad \begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 4 & 2 \\ 1 & 1 & 1 & 6 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 6 & 0 & 13 & 15 & 15 & 7 \\ 4 & 6 & 0 & 15 & 13 & 15 & 7 \\ 4 & 6 & 0 & 15 & 15 & 13 & 7 \\ 2 & 3 & -1 & 7 & 7 & 7 & 3 \end{pmatrix}_{b_0} \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -183, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

[16] $1^{1112}31^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -33$, $\det = -183$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 5 & 2 \\ 1 & 1 & 1 & 6 & 6 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 6 & 0 & 13 & 15 & 21 & 7 \\ 4 & 6 & 0 & 15 & 13 & 21 & 7 \\ 5 & 9 & 0 & 21 & 21 & 28 & 10 \\ 2 & 3 & -1 & 7 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [22] \quad 1^1 \quad 5 \quad 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 11, \quad \Delta = -40, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 0 & 4 \\ 4 & 6 & 8 & 5 & 0 & 6 \\ 2 & 3 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[21] $1^{11} \begin{smallmatrix} 1^{11} \\ 5 \\ 1^{11}, 1^{11} \end{smallmatrix} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \\ 4 & 6 & 8 & 5 & 5 & 6 \\ 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[20] $1 \begin{smallmatrix} 1^{11} \\ 1^{11}, 1^1 \end{smallmatrix} 5 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 11, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \\ 4 & 6 & 8 & 5 & 5 & 6 \\ 2 & 3 & 5 & 1 & 2 & 3 \\ 2 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[19] $1^1 \begin{smallmatrix} 1^1 \\ 1^{11} \end{smallmatrix} 5^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -40, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 3 & 3 \\ 3 & 4 & 6 & 3 & 3 & 3 & 4 \\ 4 & 6 & 8 & 5 & 5 & 6 & \\ 2 & 3 & 5 & 1 & 2 & 3 & \\ 3 & 3 & 5 & 2 & 1 & 3 & \\ 3 & 4 & 6 & 3 & 3 & 4 & \end{pmatrix}_{b_0}$$

[24] $1 \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 5 \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 1 \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} + [\textcolor{red}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s} \textcolor{green}{u}_2 + \textcolor{orange}{s} \textcolor{orange}{u}_2 + \textcolor{red}{s} \textcolor{red}{u}_2 \quad T = 10, T_{\max} = 11, \Delta = -40, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 3 \\ 2 & 4 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 3 & 2 & 3 \\ 2 & 3 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[23] $1^1 \begin{smallmatrix} 1^1 \\ 1^{11} \end{smallmatrix} 5 \begin{smallmatrix} 1^1 \\ 1^{11} \end{smallmatrix} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -40, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 3 \\ 2 & 4 & 6 & 5 & 5 & 4 \\ 3 & 6 & 8 & 5 & 5 & 6 \\ 2 & 3 & 5 & 1 & 2 & 3 \\ 2 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[25] $1^1 \begin{smallmatrix} 1^1 \\ 1^1, 1^1 \end{smallmatrix} 5 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -40, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [29] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3 1^{\textcolor{red}{1}} 4 1^{\textcolor{blue}{1}} 3 1^{\textcolor{blue}{1}\textcolor{red}{1}} + [\textcolor{red}{s} \textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{s} \textcolor{blue}{u}_2 + \textcolor{red}{s} \textcolor{red}{u}_2 + \textcolor{red}{s} \textcolor{red}{u}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 4 & 2 \\ 1 & 1 & 1 & 4 & 3 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 1 & 0 \\ 3 & 4 & 0 & 5 & 4 & 9 & 5 \\ 2 & 3 & 1 & 4 & 2 & 7 & 4 \\ 4 & 6 & 1 & 9 & 7 & 14 & 8 \\ 2 & 3 & 0 & 5 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

4.901 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11111} 31^{11} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 & 2 \\ 1 & 1 & 1 & 3 & 6 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 6 & 6 & 3 & 3 \\ 4 & 6 & 0 & 6 & 13 & 15 & 6 & 7 \\ 3 & 6 & 0 & 6 & 15 & 13 & 6 & 7 \\ 2 & 3 & 0 & 3 & 6 & 6 & 1 & 3 \\ 2 & 3 & -1 & 3 & 7 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1121} 31^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 5 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 9 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 9 & 12 & 9 & 4 \\ 3 & 6 & 0 & 9 & 13 & 21 & 15 & 7 \\ 5 & 9 & 0 & 12 & 21 & 28 & 21 & 10 \\ 4 & 6 & 0 & 9 & 15 & 21 & 13 & 7 \\ 2 & 3 & -1 & 4 & 7 & 10 & 7 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [2] \quad 1^{11111} 31^1 41^1 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 7.0$$

$$[4] \quad 1^{111} 5^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -10, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 3 & 0 & 4 \\ 5 & 6 & 8 & 5 & 5 & 0 & 6 \\ 2 & 3 & 5 & 1 & 2 & 0 & 3 \\ 2 & 3 & 5 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 5^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -10, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 3 & 0 & 4 \\ 2 & 3 & 1 & 3 & 2 & 0 & 3 \\ 2 & 3 & 3 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [6] \quad 1^{111} 6^{11} 131^1 5^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -9, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 7.0$$

$$[5] \quad 1^{111} 6^{11} 131^1 5^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 14, T_{\max} = 14, \Delta = -9, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 7.0$$

4.902 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{111} 5^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -9, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 8.0$$

4.903 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{111} 5^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -8, \det = 0, (n_+, n_-, n_0) = (1, 11, 2), c_{\text{ext}} = 9.0$$

4.904 NHC(-2)(-3) + su₂ + su₂ + su₂ + su₂ → (-3)

$$T = 10 \quad [0] \quad 1^{1111}31^{1311}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 6 & 2 & 3 \\ 2 & 2 & 1 & 4 & 9 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 2 & 9 & 4 & 4 \\ 6 & 9 & 0 & 9 & 22 & 9 & 10 \\ 2 & 4 & 0 & 4 & 9 & 2 & 4 \\ 3 & 4 & -1 & 4 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

4.905 NHC(-2)(-3) + su₂ + su₂ + su₂ → (-3)

$$T = 9 \quad [1] \quad 1^{1311}1^{1311}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 3 & 4 \\ 1 & 2 & 0 & 0 & 3 & 0 & 2 \\ 2 & 3 & -1 & 0 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{1131}1^{1311}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 3 & 2 & 0 & 2 \\ 2 & 3 & -1 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1113}1^{1311}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 1 & 2 \\ 1 & 1 & 1 & 3 & 6 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 6 & 2 & 3 \\ 3 & 6 & 0 & 6 & 13 & 5 & 7 \\ 1 & 2 & 0 & 2 & 5 & 0 & 2 \\ 2 & 3 & -1 & 3 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1113}1^{1311}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 2 & 1 & 4 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 4 & 4 \\ 3 & 4 & 0 & 4 & 2 & 4 \\ 3 & 4 & -1 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1113}1^{1311}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 6 & 3 \\ 2 & 2 & 1 & 4 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 2 & 9 & 4 \\ 6 & 9 & 0 & 9 & 22 & 10 \\ 3 & 4 & -1 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1113}1^{1311}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 2 & 1 & 4 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 9 & 4 \\ 6 & 9 & 0 & 9 & 22 & 10 \\ 3 & 4 & -1 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{1113}1^{1311}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 3 & 6 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 6 & 2 & 3 \\ 4 & 6 & 0 & 6 & 13 & 5 & 7 \\ 1 & 2 & 0 & 2 & 5 & 0 & 2 \\ 2 & 3 & -1 & 3 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{1113}1^{1311}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 3 & 2 & 3 \\ 2 & 3 & 0 & 3 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 3 & -1 & 3 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [8] \quad 1^{111}3^1 141^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 3 & 2 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 & 0 \\ 3 & 4 & 0 & 5 & 4 & 3 & 5 \\ 2 & 3 & 1 & 4 & 2 & 2 & 4 \\ 1 & 2 & 0 & 3 & 2 & 0 & 2 \\ 2 & 3 & 0 & 5 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [12] \quad 1^{111}3^1 14141^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0$$

$$[16] \quad 1^{12}3^1 1^1 15^1 1321^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.5$$

$$[17] \quad 1^{13}3^1 1^1 15^1 1321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.8$$

$$[18] \quad 1^{22}3^1 1^1 15^1 1321^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0$$

$$[15] \quad 1^{44}3^1 1^1 15^1 1321^{144} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.0$$

$$[14] \quad 1^{23}3^1 1^1 15^1 1321^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.3$$

$$[13] \quad 1^{14}3^1 1^1 15^1 1321^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0$$

$$[11] \quad 1^{34}3^1 1^1 15^1 1321^{134} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.8$$

$$[10] \quad 1^{24}3^1 1^1 15^1 1321^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.5$$

$$[9] \quad 1^{11}3^1 1^1 15^1 1321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0$$

$$[19] \quad 1^{33}3^1 1^1 15^1 1321^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.6$$

$$[20] \quad 1^{111}3^1 15^1 1^1 23^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 5 & 13 & 4 \\ 3 & 0 & 5 & 3 & 13 & 5 \\ 9 & 0 & 13 & 13 & 43 & 15 \\ 3 & 0 & 4 & 5 & 15 & 5 \end{pmatrix}_{b_0}$$

$$[21] \quad 2^1 3^1 15^1 1^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 4 & 1 & 2 \\ 1 & 1 & 2 & 7 & 7 & 2 & 3 \\ 2 & 2 & 0 & 5 & 5 & 1 & 2 \\ 4 & 7 & 5 & 17 & 19 & 5 & 9 \\ 4 & 7 & 5 & 19 & 17 & 5 & 9 \\ 1 & 2 & 1 & 5 & 5 & 0 & 2 \\ 2 & 3 & 2 & 9 & 9 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{112}3^1 15^1 1^1 23^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 55, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 1 & 2 \\ 1 & 2 & 1 & 4 & 13 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 13 & 3 & 4 \\ 6 & 13 & 0 & 13 & 49 & 11 & 15 \\ 1 & 3 & 0 & 3 & 11 & 1 & 3 \\ 2 & 4 & -1 & 4 & 15 & 3 & 4 \end{pmatrix}_{b_0}$$

4.906 NHC(-2)(-3) + su₂ + su₂ + su₂ → (-3) + su₂

$$T = 10 \quad [0] \quad 1^{111} 3^1 1^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 6 & 2 & 3 \\ 2 & 2 & 1 & 4 & 9 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 2 & 9 & 4 & 4 \\ 6 & 9 & 0 & 9 & 22 & 9 & 10 \\ 2 & 4 & 0 & 4 & 9 & 2 & 4 \\ 3 & 4 & -1 & 4 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

4.907 NHC(-2)(-3) + su₂ + su₂ + su₂n3mix

$$T = 12 \quad [17] \quad 1^{141} 3^1 1^1 5 1321^{1141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0$$

$$[7] \quad 1^{341} 3^1 1^1 5 1321^{1341} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.8$$

$$[16] \quad 1^{134} 3^1 1^1 5 1321^{1134} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.8$$

$$[15] \quad 1^{142} 3^1 1^1 5 1321^{1142} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0$$

$$[14] \quad 1^{242} 3^1 1^1 5 1321^{1242} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.5$$

$$[13] \quad 1^{143} 3^1 1^1 5 1321^{1143} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0$$

$$[12] \quad 1^{223} 3^1 1^1 5 1321^{1223} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0$$

$$[11] \quad 1^{144} 3^1 1^1 5 1321^{1144} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0$$

$$[10] \quad 1^{232} 3^1 1^1 5 1321^{1232} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.3$$

$$[23] \quad 1^{123} 3^1 1^1 5 1321^{1123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.5$$

$$[22] \quad 1^{241} 3^1 1^1 5 1321^{1241} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.5$$

$$[9] \quad 1^{222} 3^1 1^1 5 1321^{1222} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0$$

$$[21] \quad 1^{122} 3^1 1^1 5 1321^{1122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.5$$

$$[18] \quad 1^{121} 3^1 1^1 5 1321^{1121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.5$$

$$[20] \quad 1^{224} 3^1 1^1 5 1321^{1224} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0$$

$$[19] \quad 1^{133} 3^1 1^1 5 1321^{1133} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.8$$

$$[4] \quad 1^{132} 3^1 1^1 5 1321^{1132} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.8$$

$$[3] \quad 1^{113} 3^1 1^1 5 1321^{1113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0$$

$$[2] \quad 1^{114} 3^1 1^1 5 1321^{1114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0$$

$$[6] \quad 1^{131} 3^1 1^1 5 1321^{1131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.8$$

$$\begin{aligned}
[1] \ 1^{221} 3^1 1^1 5 1321^{1221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0 \\
[0] \ 1^{233} 3^1 1^1 5 1321^{1233} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.3 \\
[24] \ 1^{441} 3^1 1^1 5 1321^{1441} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.0 \\
[8] \ 1^{111} 3^1 1^1 5 1321^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0 \\
[25] \ 1^{124} 3^1 1^1 5 1321^{1124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.5 \\
[26] \ 1^{231} 3^1 1^1 5 1321^{1231} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.3 \\
[27] \ 1^{331} 3^1 1^1 5 1321^{1331} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.6 \\
[28] \ 1^{1122} 3^1 1^{12} 5 1321^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.5 \\
[29] \ 1^{112} 3^1 1^1 5 1321^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0 \\
[5] \ 1^{332} 3^1 1^1 5 1321^{1332} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.6 \\
[30] \ 2^1 3^{1111} 15 13^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0
\end{aligned}$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 3 & 3 & 2 \\ 2 & 1 & 2 & 4 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 4 & 3 & 2 \\ 3 & 4 & 4 & 6 & 8 & 7 & 6 \\ 3 & 4 & 4 & 8 & 6 & 7 & 6 \\ 3 & 4 & 3 & 7 & 5 & 5 & 5 \\ 2 & 3 & 2 & 6 & 6 & 5 & 4 \end{pmatrix}_{b_0}$$

4.908 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 0 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 5 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 5 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 0 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 3 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 4 & 5 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 6 & 9 & 13 & 15 & 0 & 7 \\ 6 & 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[35] $1^{112} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 285, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 6 & 9 & 13 & 21 & 0 & 7 \\ 8 & 13 & 21 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

4.909 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$T = 9 \quad [0] 1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & 0 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 3 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 4 \\ 4 & 5 & 4 & 9 & 0 & 3 & 4 \\ 5 & 9 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 6 & 0 & 1 & 3 \\ 3 & 4 & 4 & 7 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[4] $1^{111} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 0 & 3 & 0 & 1 & 3 \\ 3 & 4 & 0 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 5 & 0 & 4 & 3 & 6 & 4 \\ 3 & 4 & 0 & 3 & 0 & 3 & 2 \\ 4 & 5 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & 0 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[6] $1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 0 & 3 & 2 \\ 4 & 5 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & 0 & 0 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

- [7] $1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 4 & 5 & 4 & 3 & 0 & 3 & 4 \\ 3 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$
- [8] $1^{11} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 & 1 & 3 \\ 3 & 4 & 5 & 5 & 4 & 0 & 4 \\ 4 & 5 & 4 & 6 & 3 & 0 & 4 \\ 4 & 5 & 6 & 4 & 3 & 0 & 4 \\ 3 & 4 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 4 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$
- [9] $1^{121} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 1 & 3 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 & 4 \\ 4 & 5 & 4 & 12 & 0 & 3 & 4 \\ 8 & 13 & 12 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 9 & 0 & 1 & 3 \\ 3 & 4 & 4 & 10 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$
- [10] $1^{1121} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 6 & 3 \\ 3 & 4 & 9 & 8 & 0 & 9 & 4 \\ 5 & 9 & 13 & 12 & 0 & 15 & 7 \\ 5 & 8 & 12 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 15 & 12 & 0 & 13 & 7 \\ 3 & 4 & 7 & 6 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$
- [11] $1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 6 & 3 \\ 3 & 4 & 0 & 4 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 0 & 6 & 0 & 13 & 7 \\ 3 & 4 & 0 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$
- [12] $1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 4 \\ 3 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 4 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$
- [14] $1^{1211} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 1 & 6 & 3 \\ 3 & 4 & 8 & 9 & 0 & 9 & 4 \\ 5 & 8 & 10 & 12 & 0 & 12 & 6 \\ 6 & 9 & 12 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 12 & 15 & 0 & 13 & 7 \\ 3 & 4 & 6 & 7 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$
- [13] $1^{121} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 8 & 0 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 0 & 12 & 0 & 13 & 7 \\ 3 & 4 & 0 & 6 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$
- [15] $1^{11} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -33, \det = -45, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 5 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 5 & 0 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 0 & 9 & 0 & 13 & 7 \\ 3 & 4 & 0 & 4 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{112} 3 \quad 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 4 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 4 & 0 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 13 & 0 & 9 & 0 & 28 & 10 \\ 3 & 4 & 0 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111} 3 \quad 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 4 \\ 3 & 4 & 1 & 6 & 0 & 3 & 3 \\ 5 & 9 & 6 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 6 & 0 & 1 & 3 \\ 3 & 4 & 3 & 7 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{12} 3 \quad 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 5 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 5 & 0 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 13 & 0 & 12 & 0 & 28 & 10 \\ 3 & 4 & 0 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{112} 3 \quad 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 9 & 3 \\ 3 & 4 & 4 & 9 & 0 & 13 & 4 \\ 2 & 4 & 1 & 6 & 0 & 9 & 3 \\ 5 & 9 & 6 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 9 & 13 & 9 & 21 & 0 & 28 & 10 \\ 3 & 4 & 3 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{112} 3 \quad 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 5 & 4 & 0 & 13 & 4 \\ 3 & 5 & 4 & 3 & 0 & 12 & 4 \\ 3 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 13 & 12 & 9 & 0 & 28 & 10 \\ 3 & 4 & 4 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111} 3 \quad 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 6 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{111} 3 \quad 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 3 \quad 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 6 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 3 & 4 & 1 & 6 & 0 & 6 & 3 \\ 6 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 6 & 15 & 0 & 13 & 7 \\ 3 & 4 & 3 & 7 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1112} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -407, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.00 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 8 & 3 \\ 3 & 4 & 4 & 9 & 0 & 13 & 4 \\ 3 & 4 & 1 & 6 & 0 & 9 & 3 \\ 6 & 9 & 6 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 13 & 9 & 21 & 0 & 28 & 10 \\ 3 & 4 & 3 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

4.910 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{111} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -3, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 3 & 0 & 4 & 9 & 4 \\ 6 & 9 & 0 & 6 & 0 & 9 & 13 & 7 \\ 3 & 4 & 0 & 3 & -1 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1112} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -3, \quad \det = 36, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 13 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 12 & 4 \\ 3 & 4 & 3 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 3 & 0 & 1 & 9 & 3 \\ 8 & 13 & 12 & 9 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & 3 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -3, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 6 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 6 & 6 & 0 & 13 & 15 & 7 \\ 6 & 9 & 6 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 3 & 3 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 3 \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -3, \quad \det = 92, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 6 & 6 & 0 & 13 & 6 & 7 \\ 3 & 4 & 3 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 3 & 3 & -1 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1111} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -3, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 6 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 0 & 3 & 6 & 3 \\ 5 & 9 & 6 & 13 & 0 & 6 & 15 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 6 & 0 & 1 & 6 & 3 \\ 6 & 9 & 6 & 15 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & 7 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1121} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -3, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 8 & 6 & 3 \\ 3 & 4 & 0 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 3 & 4 & 0 & 1 & 0 & 9 & 6 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 8 & 13 & 0 & 9 & 0 & 28 & 21 & 10 \\ 6 & 9 & 0 & 6 & 0 & 21 & 13 & 7 \\ 3 & 4 & 0 & 3 & -1 & 10 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -3, \ \det = 144, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 3 & 3 & 0 & 1 & 3 & 3 \\ 3 & 4 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -3, \ \det = 224, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 3 \\ 3 & 4 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 3 & 3 & 0 & 1 & 6 & 3 \\ 6 & 9 & 6 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & 3 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -3, \ \det = 364, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 6 & 3 \\ 3 & 4 & 3 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 6 & 6 & 0 & 13 & 15 & 7 \\ 6 & 9 & 6 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 3 & 3 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -3, \ \det = 400, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 9 & 4 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 3 \\ 3 & 4 & 3 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 6 & 6 & 0 & 13 & 15 & 7 \\ 6 & 9 & 6 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 3 & 3 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

4.911 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 10, \ \Delta = -5, \ \det = 9, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & 0 & 3 & -1 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 10, \ \Delta = -5, \ \det = 9, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 3 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 3 & 4 \\ 3 & 4 & 3 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 3 & 2 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & 3 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

4.912 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 10, \ \Delta = -35, \ \det = -11, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 0 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 4 & 0 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 10, \ \Delta = -35, \ \det = -18, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 5 & 0 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 5 & 0 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 3 & 0 & 0 & 2 \\ 3 & 4 & 0 & 4 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 10, \ \Delta = -35, \ \det = -18, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 3 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 3 & 4 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 4 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 10, \ \Delta = -35, \ \det = -36, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 3 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 4 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 10, \ \Delta = -35, \ \det = -64, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 3 & 0 & 2 & 3 \\ 2 & 4 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 2 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 10, \ \Delta = -35, \ \det = -91, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 3 & 0 & 2 & 3 \\ 3 & 4 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 2 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

4.913 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -5, \ \det = 9, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 0 & 3 & -1 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -5, \ \det = 9, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 3 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 3 & 4 \\ 3 & 4 & 3 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

4.914 $\mathbf{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 7 \quad [0] \ 1^{\overset{1^2}{\color{red}1}\overset{\color{red}1}{\color{blue}3}}\overset{\color{blue}1}{\color{red}1} + [\textcolor{red}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{green}{u}_2 + \textcolor{brown}{s}\textcolor{brown}{u}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -6, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 5 & 5 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix} b_0$$

[1] $1^1 1^3 1^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -13, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 9.46 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 4 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[2] $1^1 1^1 3^1 1^1 + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{green}{\mathbf{su}}_2 + \textcolor{brown}{\mathbf{su}}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 9.59 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 4 & 4 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix} b_0$$

[3] $13^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T=7, T_{\max}=10, \Delta=-86, \det=-18, (n_+, n_-, n_0)=(1, 7, 0), c_{\text{ext}}=5.0$

$$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 3 & 5 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix} b_0$$

[6] $13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix} b_0$$

[5] $1^1 1^3 1^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 4 & 5 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix} b_0$$

[4] $21^{\textcolor{blue}{1}}3^{\textcolor{brown}{1}}1^{\textcolor{teal}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{teal}{\mathfrak{g}_2}] + \textcolor{teal}{\mathfrak{su}_2} + \textcolor{brown}{\mathfrak{su}_2} \rightarrow (-3)$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$

$$T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 4 & 4 & 3 & 2 & 3 \end{pmatrix} b_0$$

[7] $1^{\overline{1}}1^{\overline{1}}3^{\overline{1}}1^{\overline{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + \textcolor{orange}{s}\textcolor{blue}{u}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix} b_0$$

[8] $1^{1^3} 1^{12} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 7 & 8 & 10 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_b$$

[10] $21^{\textcolor{blue}{1}1^{\textcolor{brown}{1}}3^{\textcolor{brown}{1}}1^{\textcolor{brown}{1}} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}u_2 + \textcolor{brown}{s}u_2 \rightarrow (-3)$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -36$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 6 & 3 & 4 \\ 5 & 4 & 5 & 3 & 4 \\ 6 & 5 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 4 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[9] $21^{11}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -36$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 18 & 3 & 4 \\ 5 & 4 & 21 & 3 & 4 \\ 18 & 21 & 76 & 11 & 16 \\ 3 & 3 & 11 & 0 & 2 \\ 4 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 5 & 8 & 10 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 5 & 9 & 13 & 5 & 7 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{13}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 8 & 9 & 13 & 5 & 7 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{12}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 6 & 8 & 10 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 21^{11}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.64 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{13}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 10 & 12 & 25 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -62, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 6 & 9 & 13 & 5 & 7 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 7 & 9 & 13 & 5 & 7 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{13}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 8 & 12 & 25 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 12 & 14 & 37 & 8 & 11 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{13}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

- [22] $\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 9 & 12 & 25 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$
 $21^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
- [23] $\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 4 & 14 & 3 & 4 \\ 9 & 14 & 37 & 8 & 11 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 11 & 2 & 3 \end{pmatrix}_{b_0}$
 $21^{11}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
- [24] $\begin{pmatrix} 4 & 5 & 15 & 3 & 4 \\ 5 & 4 & 17 & 3 & 4 \\ 15 & 17 & 49 & 9 & 13 \\ 3 & 3 & 9 & 0 & 2 \\ 4 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$
 $21^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
- [25] $\begin{pmatrix} 4 & 5 & 9 & 3 & 4 \\ 5 & 4 & 9 & 3 & 4 \\ 9 & 9 & 13 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 \\ 4 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$
 $21^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
- [26] $\begin{pmatrix} 1 & 4 & 6 & 2 & 3 \\ 4 & 4 & 8 & 3 & 4 \\ 6 & 8 & 10 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
 $1^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
- [27] $\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 9 & 14 & 37 & 8 & 11 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 11 & 2 & 3 \end{pmatrix}_{b_0}$
 $21^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
- [29] $\begin{pmatrix} 1 & 4 & 6 & 2 & 3 \\ 4 & 4 & 9 & 3 & 4 \\ 6 & 9 & 13 & 5 & 7 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$
 $1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
- [28] $\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 11 & 14 & 37 & 8 & 11 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 11 & 2 & 3 \end{pmatrix}_{b_0}$
 $21^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
- [30] $\begin{pmatrix} 4 & 5 & 12 & 3 & 4 \\ 5 & 4 & 13 & 3 & 4 \\ 12 & 13 & 28 & 7 & 10 \\ 3 & 3 & 7 & 0 & 2 \\ 4 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$
 $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -98, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
- [32] $\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 11 & 13 & 28 & 7 & 10 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$
 $1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
- [31] $\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 10 & 14 & 37 & 8 & 11 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 11 & 2 & 3 \end{pmatrix}_{b_0}$
 $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
- [31] $\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 8 & 13 & 28 & 7 & 10 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$

[34] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 10 & 13 & 28 & 7 & 10 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[33] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 11 & 16 & 46 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[35] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 9 & 13 & 28 & 7 & 10 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[36] $1^{14}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 12 & 16 & 46 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[37] $1^{11}3^{11}1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 15 & 19 & 73 & 11 & 15 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[38] $21^{13}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 4 & 12 & 3 & 4 \\ 9 & 12 & 25 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[39] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -138, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 10 & 17 & 49 & 9 & 13 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[40] $1^{11}3^{11}1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -138, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 14 & 19 & 73 & 11 & 15 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[41] $21^{12}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 15 & 3 & 4 \\ 5 & 4 & 14 & 3 & 4 \\ 15 & 14 & 37 & 8 & 11 \\ 3 & 3 & 8 & 0 & 2 \\ 4 & 4 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

[42] $1^{12}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -153, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 11 & 18 & 58 & 10 & 14 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

[43] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 11 & 17 & 49 & 9 & 13 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

- [44] $21^{12}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 4 & 13 & 3 & 4 \\ 9 & 13 & 28 & 7 & 10 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$
- [45] $1^{12}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -182, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 12 & 18 & 58 & 10 & 14 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$
- [46] $1^{13}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -183, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 13 & 17 & 49 & 9 & 13 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$
- [47] $1^{13}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -186, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 12 & 17 & 49 & 9 & 13 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$
- [48] $1^{12}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -186, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 14 & 18 & 58 & 10 & 14 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$
- [49] $1^{12}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -193, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 13 & 18 & 58 & 10 & 14 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$
- [50] $21^{14}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -206, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 3 \\ 4 & 4 & 16 & 3 & 4 \\ 12 & 16 & 46 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$
- [52] $1^{13}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 13 & 22 & 85 & 12 & 17 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 17 & 2 & 3 \end{pmatrix}_{b_0}$$
- [51] $21^{12}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 18 & 3 & 4 \\ 5 & 4 & 18 & 3 & 4 \\ 18 & 18 & 58 & 10 & 14 \\ 3 & 3 & 10 & 0 & 2 \\ 4 & 4 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$
- [53] $21^{12}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -221, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 4 & 23 & 3 & 4 \\ 15 & 23 & 100 & 13 & 18 \\ 2 & 3 & 13 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$
- [54] $1^{14}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -222, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

[illegible]

5.8

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 3 \\ 4 & 4 & 17 & 3 & 4 \\ 12 & 17 & 49 & 9 & 13 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[66] 221^{11}3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -295, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 3 & 0 & 3 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[67] 21^{15}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -296, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.1$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 4 & 20 & 3 & 4 \\ 15 & 20 & 73 & 10 & 15 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[68] 1^{14}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -297, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 16 & 22 & 85 & 12 & 17 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 17 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[69] 1^{12}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -321, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 19 & 2 & 3 \\ 3 & 4 & 28 & 3 & 4 \\ 19 & 28 & 154 & 16 & 22 \\ 2 & 3 & 16 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[70] 21^{13}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -336, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 24 & 3 & 4 \\ 5 & 4 & 23 & 3 & 4 \\ 24 & 23 & 100 & 13 & 18 \\ 3 & 3 & 13 & 0 & 2 \\ 4 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[71] 1^{12}3^11^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -338, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 20 & 2 & 3 \\ 3 & 4 & 28 & 3 & 4 \\ 20 & 28 & 154 & 16 & 22 \\ 2 & 3 & 16 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[72] 1^{14}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -338, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 16 & 26 & 118 & 14 & 20 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 20 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[73] 1^{13}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -343, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 17 & 27 & 133 & 15 & 21 \\ 2 & 3 & 15 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[74] 1^{14}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -381, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 17 & 26 & 118 & 14 & 20 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 20 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[75] 1^{13}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -382, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 18 & 27 & 133 & 15 & 21 \\ 2 & 3 & 15 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

- [76] $21^{13}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -396, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 4 & 22 & 3 & 4 \\ 15 & 22 & 85 & 12 & 17 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 17 & 2 & 3 \end{pmatrix}_{b_0}$
- [77] $1^{14}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -403, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 19 & 2 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 19 & 27 & 133 & 15 & 21 \\ 2 & 3 & 15 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$
- [79] $21^{16}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -406, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 18 & 2 & 3 \\ 4 & 4 & 24 & 3 & 4 \\ 18 & 24 & 106 & 12 & 18 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$
- [78] $1^{14}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -406, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 18 & 26 & 118 & 14 & 20 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 20 & 2 & 3 \end{pmatrix}_{b_0}$
- [80] $21^{14}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -421, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 4 & 21 & 3 & 4 \\ 15 & 21 & 76 & 11 & 16 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$
- [81] $21^{13}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -441, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 27 & 3 & 4 \\ 5 & 4 & 27 & 3 & 4 \\ 27 & 27 & 133 & 15 & 21 \\ 3 & 3 & 15 & 0 & 2 \\ 4 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$
- [82] $1^{14}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -446, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 19 & 2 & 3 \\ 3 & 4 & 31 & 3 & 4 \\ 19 & 31 & 172 & 17 & 24 \\ 2 & 3 & 17 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$
- [83] $21^{13}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -451, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 18 & 2 & 3 \\ 4 & 4 & 27 & 3 & 4 \\ 18 & 27 & 133 & 15 & 21 \\ 2 & 3 & 15 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$
- [84] $21^{13}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -456, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 21 & 2 & 3 \\ 4 & 4 & 32 & 3 & 4 \\ 21 & 32 & 193 & 18 & 25 \\ 2 & 3 & 18 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$
- [85] $21^{14}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -456, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 30 & 3 & 4 \\ 5 & 4 & 28 & 3 & 4 \\ 30 & 28 & 154 & 16 & 22 \\ 3 & 3 & 16 & 0 & 2 \\ 4 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$
- [86] $1^{13}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3)$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -468, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 21 & 2 & 3 \\ 3 & 4 & 32 & 3 & 4 \\ 21 & 32 & 193 & 18 & 25 \\ 2 & 3 & 18 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[88] \ 1^{13}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -501, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 20 & 2 & 3 \\ 3 & 4 & 31 & 3 & 4 \\ 20 & 31 & 172 & 17 & 24 \\ 2 & 3 & 17 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[87] \ 1^{14}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -501, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 22 & 2 & 3 \\ 3 & 4 & 32 & 3 & 4 \\ 22 & 32 & 193 & 18 & 25 \\ 2 & 3 & 18 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[89] \ 21^{13}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -516, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.81 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 30 & 3 & 4 \\ 5 & 4 & 31 & 3 & 4 \\ 30 & 31 & 172 & 17 & 24 \\ 3 & 3 & 17 & 0 & 2 \\ 4 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[90] \ 21^{17}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -536, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 21 & 2 & 3 \\ 4 & 4 & 28 & 3 & 4 \\ 21 & 28 & 145 & 14 & 21 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[91] \ 1^{14}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -538, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.84 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 21 & 2 & 3 \\ 3 & 4 & 31 & 3 & 4 \\ 21 & 31 & 172 & 17 & 24 \\ 2 & 3 & 17 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[92] \ 21^{14}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -586, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.78 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 3 \\ 4 & 4 & 26 & 3 & 4 \\ 18 & 26 & 118 & 14 & 20 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 20 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[93] \ 1^{14}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -613, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 23 & 2 & 3 \\ 3 & 4 & 36 & 3 & 4 \\ 23 & 36 & 238 & 20 & 28 \\ 2 & 3 & 20 & 0 & 2 \\ 3 & 4 & 28 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[94] \ 21^{14}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -621, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.87 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 33 & 3 & 4 \\ 5 & 4 & 32 & 3 & 4 \\ 33 & 32 & 193 & 18 & 25 \\ 3 & 3 & 18 & 0 & 2 \\ 4 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[95] \ 1^{14}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -662, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 24 & 2 & 3 \\ 3 & 4 & 36 & 3 & 4 \\ 24 & 36 & 238 & 20 & 28 \\ 2 & 3 & 20 & 0 & 2 \\ 3 & 4 & 28 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[96] \ 21^{14}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -701, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.80 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 21 & 2 & 3 \\ 4 & 4 & 31 & 3 & 4 \\ 21 & 31 & 172 & 17 & 24 \\ 2 & 3 & 17 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[97] \ 21^{14}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -756, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$\begin{aligned}
& T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 36 & 3 & 4 \\ 5 & 4 & 36 & 3 & 4 \\ 36 & 36 & 238 & 20 & 28 \\ 3 & 3 & 20 & 0 & 2 \\ 4 & 4 & 28 & 2 & 3 \end{pmatrix}_{b_0} \\
[98] & 21^{14}3^{11}1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -766, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 24 & 2 & 3 \\ 4 & 4 & 36 & 3 & 4 \\ 24 & 36 & 238 & 20 & 28 \\ 2 & 3 & 20 & 0 & 2 \\ 3 & 4 & 28 & 2 & 3 \end{pmatrix}_{b_0} \\
& T = 8 \quad [99] \quad 1^13^{11}131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 2 & 3 & -1 & 0 & 2 & 3 \end{pmatrix}_{b_0} \\
[100] & 1^13^{11}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 \\ 4 & 2 & 4 & 2 & 4 \\ 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0} \\
[101] & 13^121^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 \\ 2 & 2 & 4 & 2 & 4 \\ 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0} \\
[102] & 1^13^{11}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 3 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 5 & 0 & 2 & 4 \end{pmatrix}_{b_0} \\
[104] & 13^121^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 5 & 5 & 3 & 5 \\ 2 & 5 & 2 & 2 & 4 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 5 & 4 & 2 & 4 \end{pmatrix}_{b_0} \\
[103] & 1^{11}3^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 2 & 1 & 4 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 4 \\ 3 & 4 & -1 & 4 & 4 \end{pmatrix}_{b_0} \\
[105] & 1^13^{11}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.50 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0} \\
[106] & 1^13^{11}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 5 & 5 & 3 & 5 \\ 4 & 5 & 2 & 2 & 4 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 5 & 4 & 2 & 4 \end{pmatrix}_{b_0} \\
[107] & 1^13^{11}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 \\ 3 & 2 & 4 & 2 & 4 \\ 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

- [108] $1^1 3^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -65, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 2 & 1 & 5 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 5 & 0 & 8 & 6 \\ 3 & 4 & -1 & 6 & 4 \end{pmatrix}_{b_0}$
- [109] $1^1 1^3 3^1 1^3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -65, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 \\ 1 & 3 & 0 & 1 & 2 \\ 1 & 2 & 0 & 2 & 0 \\ 2 & 3 & -1 & 3 & 2 \end{pmatrix}_{b_0}$
- [110] $1^1 1^3 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -65, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 3 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 3 & 4 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 2 & 3 & -1 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [111] $1^1 1^3 1^1 2^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$
- [112] $1^1 1^3 1^1 2^1 1^2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 5 & 10 & 3 & 5 \\ 5 & 10 & 14 & 4 & 8 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 5 & 8 & 2 & 4 \end{pmatrix}_{b_0}$
- [113] $1^1 1^3 1^1 2^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$
- [114] $1^1 1^3 1^1 2^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.65 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 5 & 3 & 5 \\ 3 & 5 & 2 & 2 & 4 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 5 & 4 & 2 & 4 \end{pmatrix}_{b_0}$
- [115] $1^1 1^3 1^1 2^1 1^2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 5 & 10 & 3 & 5 \\ 7 & 10 & 14 & 4 & 8 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 5 & 8 & 2 & 4 \end{pmatrix}_{b_0}$
- [116] $2^1 1^2 3^1 2^1 1^3 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 12 & 12 & 16 & 70 & 18 \\ 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$
- [118] $1^1 1^3 1^1 2^1 1^2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 5 & 10 & 3 & 5 \\ 6 & 10 & 14 & 4 & 8 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 5 & 8 & 2 & 4 \end{pmatrix}_{b_0}$
- [117] $1^1 1^3 1^1 2^1 1^3 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 5 & 15 & 3 & 5 \\ 8 & 15 & 34 & 6 & 12 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 5 & 12 & 2 & 4 \end{pmatrix}_{b_0}$
- [119] $1^1 1^3 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -65, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$

$$\begin{aligned}
& T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 6 & 3 \\ 6 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[130] & 1^{11}3^{12}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 6 & 3 \\ 5 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[132] & 21^{11}3^{12}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 6 & 3 \\ 5 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[134] & 1^{11}3^{12}21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 10 & 3 \\ 7 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0} \\
[133] & 1^{14}3^{12}21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 10 & 3 \\ 10 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0} \\
[136] & 1^43^{12}21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 5 & 20 & 3 & 5 \\ 12 & 20 & 62 & 8 & 16 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 5 & 16 & 2 & 4 \end{pmatrix}_{b_0} \\
[135] & 1^{11}3^{12}21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 5 & 15 & 3 & 5 \\ 8 & 15 & 32 & 7 & 12 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 5 & 12 & 2 & 4 \end{pmatrix}_{b_0} \\
[137] & 1^{11}3^{11}1^431^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 6 & 2 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 13 & 5 & 7 \\ 1 & 2 & 0 & 5 & 0 & 2 \\ 2 & 3 & -1 & 7 & 2 & 3 \end{pmatrix}_{b_0} \\
[139] & 1^{12}3^{12}21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 10 & 3 \\ 8 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0} \\
[138] & 1^{13}3^{12}21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 10 & 3 \\ 9 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0} \\
[141] & 1^{13}3^{12}21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 5 & 15 & 3 & 5 \\ 10 & 15 & 32 & 7 & 12 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 5 & 12 & 2 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[140] $1^{11}3^{12}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 5 & 20 & 3 & 5 \\ 10 & 20 & 58 & 9 & 16 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 5 & 16 & 2 & 4 \end{pmatrix}_{b_0}$$

[143] $1^{12}3^{12}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 1 & 8 & 3 \\ 9 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[142] $1^{12}3^{12}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 1 & 8 & 3 \\ 7 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[144] $1^{12}3^{12}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 5 & 15 & 3 & 5 \\ 9 & 15 & 32 & 7 & 12 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 5 & 12 & 2 & 4 \end{pmatrix}_{b_0}$$

[145] $1^{12}3^{12}21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 1 & 8 & 3 \\ 8 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[146] $1^{11}3^{12}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 12 & 9 & 14 & 49 & 15 \\ 3 & 3 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[147] $1^{12}3^{12}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 11 & 9 & 14 & 49 & 15 \\ 3 & 3 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[148] $1^{12}3^{12}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 5 & 20 & 3 & 5 \\ 11 & 20 & 58 & 9 & 16 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 5 & 16 & 2 & 4 \end{pmatrix}_{b_0}$$

[149] $1^{12}3^{12}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 12 & 3 \\ 9 & 10 & 12 & 42 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[150] $21^{13}3^{12}21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 1 & 3 & 17 & 3 \\ 2 & 3 & 1 & 22 & 3 \\ 17 & 17 & 22 & 137 & 25 \\ 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

[151] $1^{14}3^{12}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 5 & 25 & 3 & 5 \\ 15 & 25 & 92 & 11 & 20 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 5 & 20 & 2 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 5 & 25 & 3 & 5 \\ 15 & 25 & 94 & 12 & 20 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 5 & 20 & 2 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 13 & 12 & 16 & 70 & 18 \\ 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 12 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 14 & 12 & 16 & 70 & 18 \\ 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 15 & 14 & 20 & 106 & 22 \\ 3 & 3 & 3 & 22 & 4 \end{pmatrix}_b$$

$$[3] \text{ } 1^{13} 3^{12} 21^{12} + [5u_2]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 13 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$9] 1^{13} 3^{14} 21^{12} + [\mathfrak{su}_2, \mathfrak{su}_2]_{\text{min}} = 9.91 \rightarrow 10, 7$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 14 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$] 1^{12} 3^{14} 21^{14} + [su_2, \dots]$$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 5 & 30 & 3 & 5 \\ 16 & 30 & 134 & 14 & 24 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 5 & 24 & 2 & 4 \end{pmatrix}_b$$

$$2] 1^{12} 3^{14} 21^{14} + [5u_2, \\ \text{min} = 9.93 \rightarrow 10, T$$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 16 & 14 & 20 & 106 & 22 \\ 3 & 3 & 3 & 22 & 4 \end{pmatrix}_b$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10.11 \rightarrow 10$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 18 & 3 \\ 15 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[173] $1^{13}3^{12}21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 5 & 30 & 3 & 5 \\ 17 & 30 & 134 & 14 & 24 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 5 & 24 & 2 & 4 \end{pmatrix}_{b_0}$$

[174] $1^{12}3^{12}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 18 & 3 \\ 14 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[175] $21^{13}3^{12}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 13 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[176] $1^{12}3^{12}21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 258, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 5 & 30 & 3 & 5 \\ 18 & 30 & 134 & 14 & 24 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 5 & 24 & 2 & 4 \end{pmatrix}_{b_0}$$

[177] $1^{14}3^{12}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 16 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 15 & 16 & 16 & 94 & 20 \\ 3 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

[178] $1^{14}3^{12}21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 16 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 16 & 16 & 16 & 94 & 20 \\ 3 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

[179] $1^{13}3^{12}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 277, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 18 & 3 \\ 15 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[180] $1^{13}3^{12}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 293, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 18 & 3 \\ 16 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[181] $1^{12}3^{12}21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 324, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 18 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 16 & 18 & 20 & 130 & 24 \\ 3 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[182] $1^{13}3^{12}21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 325, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 17 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 1 & 22 & 3 \\ 17 & 17 & 22 & 137 & 25 \\ 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

[183] $1^{14}3^{12}21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 357, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 1 & 22 & 3 \\ 18 & 17 & 22 & 137 & 25 \\ 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

[184] $1^{14}3^{13}21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 364, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 17 & 3 \\ 2 & 1 & 3 & 18 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 17 & 18 & 20 & 130 & 24 \\ 3 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[185] $21^{14}3^{13}21^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 1 & 3 & 20 & 3 \\ 2 & 3 & 1 & 24 & 3 \\ 20 & 20 & 24 & 174 & 28 \\ 3 & 3 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

[186] $1^{14}3^{13}21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 388, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 18 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 18 & 18 & 20 & 130 & 24 \\ 3 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[187] $21^{14}3^{13}21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 430, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 1 & 3 & 18 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 18 & 18 & 20 & 130 & 24 \\ 3 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[188] $1^{14}3^{13}21^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 19 & 3 \\ 2 & 1 & 3 & 20 & 3 \\ 2 & 3 & 1 & 24 & 3 \\ 19 & 20 & 24 & 174 & 28 \\ 3 & 3 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

[189] $1^{14}3^{13}21^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 20 & 3 \\ 2 & 1 & 3 & 20 & 3 \\ 2 & 3 & 1 & 24 & 3 \\ 20 & 20 & 24 & 174 & 28 \\ 3 & 3 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

$T = 9$ [191] $1^{12}3^{11}131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 2 & 0 & 3 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

[190] $1^{11}23^{11}131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 4 & 2 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 4 & 0 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 4 & -1 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

[194] $23^{11}141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 5 & 5 & 4 & 5 \\ 3 & 5 & 4 & 3 & 4 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 5 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[193] $1^{14}1^{12}23^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 0 & 2 & 0 & 2 \\ 3 & 2 & 4 & 0 & 4 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 4 & 0 & 4 \end{pmatrix}_{b_0} \\
[192] \quad & 1^{11}4123^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[195] \quad & 1^{11}4123^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 2 & 0 & 3 \end{pmatrix}_{b_0} \\
[196] \quad & 1^{11}4123^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 0 & 1 & 0 & 2 \\ 2 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[197] \quad & 1^{11}2313^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -37, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 5 & 4 \\ 3 & 0 & 5 & 3 & 5 \\ 3 & 0 & 4 & 5 & 5 \end{pmatrix}_{b_0} \\
[198] \quad & 1^{11}23^1131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 2 & 3 \\ 2 & 0 & 4 & 2 & 2 & 4 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 2 & 0 & 3 & 4 & 2 & 4 \end{pmatrix}_{b_0} \\
[199] \quad & 1^{11}41^123^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 4 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 0 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[200] \quad & 1^{11}4123^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 2 & 2 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 3 & 0 & 3 \end{pmatrix}_{b_0} \\
[201] \quad & 1^{11}4123^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 0 & 3 & 0 & 2 \\ 3 & 3 & 2 & 0 & 4 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 4 & 0 & 4 \end{pmatrix}_{b_0} \\
[202] \quad & 1^{11}41^223^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 6 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 7 & 6 & 0 & 32 & 0 & 10 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 10 & 0 & 3 \end{pmatrix}_{b_0} \\
[203] \quad & 1^{11}41^223^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 5 & 1 & 3 \\ 2 & 0 & 3 & 0 & 2 \\ 5 & 3 & 6 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 6 & 0 & 4 \end{pmatrix}_{b_0} \\
[204] & 23^{\overset{1}{1}}141^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 3 & 2 & 2 \\ 1 & 0 & 2 & 3 & 1 & 2 \\ 2 & 2 & 0 & 4 & 2 & 2 \\ 3 & 3 & 4 & 5 & 3 & 5 \\ 2 & 1 & 2 & 3 & 0 & 2 \\ 2 & 2 & 2 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[205] & 23^{\overset{1}{1}}141^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 5 & 5 & 4 & 5 \\ 3 & 5 & 2 & 3 & 4 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 5 & 4 & 3 & 4 \end{pmatrix}_{b_0} \\
[206] & 1^{\overset{1}{1}}41^{\overset{2}{2}}23^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 8 & 1 & 3 \\ 2 & 0 & 5 & 0 & 2 \\ 8 & 5 & 24 & 0 & 10 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 10 & 0 & 4 \end{pmatrix}_{b_0} \\
[207] & 1^{\overset{12}{1}}41^{\overset{2}{2}}23^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 1 & 8 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 8 & 8 & 0 & 46 & 0 & 12 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 12 & 0 & 3 \end{pmatrix}_{b_0} \\
[208] & 23^{\overset{1}{1}}141^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 5 & 10 & 4 & 5 \\ 6 & 10 & 14 & 6 & 8 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 5 & 8 & 3 & 4 \end{pmatrix}_{b_0} \\
[209] & 23^{\overset{1}{1}}141^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 5 & 10 & 4 & 5 \\ 6 & 10 & 14 & 6 & 8 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 5 & 8 & 3 & 4 \end{pmatrix}_{b_0} \\
[210] & 1^{\overset{1}{1}}41^{\overset{1}{1}}23^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 6 & 1 & 3 \\ 2 & 0 & 5 & 0 & 2 \\ 6 & 5 & 12 & 0 & 8 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 8 & 0 & 4 \end{pmatrix}_{b_0} \\
[211] & 1^{\overset{1}{1}}41^{\overset{1}{1}}23^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 4 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 5 & 4 & 1 & 11 & 0 & 7 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 7 & 0 & 3 \end{pmatrix}_{b_0} \\
[212] & 1^{\overset{1}{1}}41^{\overset{1}{1}}23^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 7 & 1 & 3 \\ 2 & 0 & 4 & 0 & 2 \\ 7 & 4 & 12 & 0 & 8 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 8 & 0 & 4 \end{pmatrix}_{b_0} \\
[213] & 23^{\overset{1}{1}}141^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 5 & 15 & 4 & 5 \\ 9 & 15 & 34 & 9 & 12 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 5 & 12 & 3 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[214] \ 1^{11}41^{23}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -85, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 9 & 1 & 3 \\ 2 & 0 & 7 & 0 & 2 \\ 9 & 7 & 34 & 0 & 12 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 12 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[215] \ 1^{12}41^{323}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -90, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 0 & 1 & 10 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 11 & 10 & 0 & 84 & 0 & 16 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[216] \ 23^{112}141^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -96, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 & 2 \\ 1 & 0 & 2 & 6 & 1 & 2 \\ 2 & 2 & 0 & 8 & 2 & 2 \\ 6 & 6 & 8 & 26 & 6 & 10 \\ 2 & 1 & 2 & 6 & 0 & 2 \\ 2 & 2 & 2 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[217] \ 23^{114}141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -112, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 5 & 20 & 4 & 5 \\ 12 & 20 & 62 & 12 & 16 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 5 & 16 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[218] \ 1^{11}41^{223}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -115, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 1 & 6 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 8 & 6 & 1 & 33 & 0 & 11 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[219] \ 1^{13}41^{323}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -122, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 10.95 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 0 & 1 & 12 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 12 & 12 & 0 & 106 & 0 & 18 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 18 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[222] \ 1^{14}41^{223}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -132, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 10 & 1 & 3 \\ 2 & 0 & 6 & 0 & 2 \\ 10 & 6 & 30 & 0 & 12 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 12 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[221] \ 23^{112}141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -132, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 5 & 15 & 4 & 5 \\ 9 & 15 & 32 & 9 & 12 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 5 & 12 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[220] \ 1^{11}41^{223}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -132, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 8 & 1 & 3 \\ 2 & 0 & 6 & 0 & 2 \\ 8 & 6 & 18 & 0 & 10 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 10 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[223] \ 1^{11}41^{323}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -147, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 0 & 1 & 8 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 11 & 8 & 1 & 67 & 0 & 15 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 15 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[224] \quad 23^{1^{15}}141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.1$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 & 3 \\ 4 & 5 & 25 & 4 & 5 \\ 15 & 25 & 98 & 15 & 20 \\ 3 & 4 & 15 & 1 & 3 \\ 3 & 5 & 20 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[225] \quad 1^{14}41^{323^11^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 13 & 1 & 3 \\ 2 & 0 & 8 & 0 & 2 \\ 13 & 8 & 60 & 0 & 16 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 16 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[226] \quad 1^{13}41^{423^11^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 1 & 2 \\ 1 & 0 & 1 & 14 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 15 & 14 & 0 & 160 & 0 & 22 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 22 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[227] \quad 1^{12}41^{223^11^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 1 & 8 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 9 & 8 & 1 & 47 & 0 & 13 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 13 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[228] \quad 1^{11}41^{23^11^{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 10 & 1 & 3 \\ 2 & 0 & 7 & 0 & 2 \\ 10 & 7 & 28 & 0 & 12 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 12 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[229] \quad 23^{1^{16}}141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 18 & 3 & 3 \\ 4 & 5 & 30 & 4 & 5 \\ 18 & 30 & 142 & 18 & 24 \\ 3 & 4 & 18 & 1 & 3 \\ 3 & 5 & 24 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[230] \quad 1^{11}41^{223^11^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 11 & 1 & 3 \\ 2 & 0 & 8 & 0 & 2 \\ 11 & 8 & 40 & 0 & 14 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 14 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[231] \quad 1^{14}41^{223^11^{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 12 & 1 & 3 \\ 2 & 0 & 7 & 0 & 2 \\ 12 & 7 & 40 & 0 & 14 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 14 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[232] \quad 1^{14}41^{423^11^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 1 & 2 \\ 1 & 0 & 1 & 16 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 16 & 16 & 0 & 190 & 0 & 24 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 24 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[233] \quad 23^{1^{13}}141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

5.0

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 5 & 20 & 4 & 5 \\ 12 & 20 & 58 & 12 & 16 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 5 & 16 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[234] \quad 1^{12}41^323^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -223, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 0 & 1 & 10 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 12 & 10 & 1 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 17 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[235] \quad 1^{11}41^323^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 14 & 1 & 3 \\ 2 & 0 & 10 & 0 & 2 \\ 14 & 10 & 74 & 0 & 18 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 18 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[236] \quad 23^{1^{17}}141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 21 & 3 & 3 \\ 4 & 5 & 35 & 4 & 5 \\ 21 & 35 & 194 & 21 & 28 \\ 3 & 4 & 21 & 1 & 3 \\ 3 & 5 & 28 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[237] \quad 1^{12}41^223^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 1 & 8 & 0 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 10 & 8 & 2 & 50 & 0 & 14 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 14 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[238] \quad 1^{12}41^423^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 1 & 2 \\ 1 & 0 & 1 & 12 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 15 & 12 & 1 & 135 & 0 & 21 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 21 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[239] \quad 1^{14}41^223^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 14 & 1 & 3 \\ 2 & 0 & 8 & 0 & 2 \\ 14 & 8 & 54 & 0 & 16 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 16 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[240] \quad 1^{14}41^323^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -277, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 15 & 1 & 3 \\ 2 & 0 & 9 & 0 & 2 \\ 15 & 9 & 70 & 0 & 18 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 18 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[241] \quad 1^{13}41^323^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 1 & 2 \\ 1 & 0 & 1 & 12 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 13 & 12 & 1 & 107 & 0 & 19 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 19 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[242] \quad 23^{1^{14}}141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 & 3 \\ 4 & 5 & 25 & 4 & 5 \\ 15 & 25 & 92 & 15 & 20 \\ 3 & 4 & 15 & 1 & 3 \\ 3 & 5 & 20 & 3 & 4 \end{pmatrix}_{b_0}$$

[243] $1^{11}41^223^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 13 & 1 & 3 \\ 2 & 0 & 9 & 0 & 2 \\ 13 & 9 & 50 & 0 & 16 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 16 & 0 & 4 \end{pmatrix}_{b_0}$$

[244] $1^{14}41^423^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -325, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 18 & 1 & 3 \\ 2 & 0 & 11 & 0 & 2 \\ 18 & 11 & 112 & 0 & 22 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 22 & 0 & 4 \end{pmatrix}_{b_0}$$

[245] $1^{12}41^323^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -342, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 1 & 2 \\ 1 & 0 & 1 & 10 & 0 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 13 & 10 & 2 & 88 & 0 & 18 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 18 & 0 & 3 \end{pmatrix}_{b_0}$$

[246] $1^{13}41^423^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 1 & 2 \\ 1 & 0 & 1 & 14 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 16 & 14 & 1 & 161 & 0 & 23 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 23 & 0 & 3 \end{pmatrix}_{b_0}$$

[247] $1^{11}41^323^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -373, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 16 & 1 & 3 \\ 2 & 0 & 11 & 0 & 2 \\ 16 & 11 & 84 & 0 & 20 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 20 & 0 & 4 \end{pmatrix}_{b_0}$$

[248] $1^{14}41^323^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 17 & 1 & 3 \\ 2 & 0 & 10 & 0 & 2 \\ 17 & 10 & 84 & 0 & 20 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 20 & 0 & 4 \end{pmatrix}_{b_0}$$

[249] $1^{11}41^223^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 15 & 1 & 3 \\ 2 & 0 & 10 & 0 & 2 \\ 15 & 10 & 64 & 0 & 18 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 18 & 0 & 4 \end{pmatrix}_{b_0}$$

[250] $1^{12}41^423^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -418, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 1 & 2 \\ 1 & 0 & 1 & 12 & 0 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 16 & 12 & 2 & 138 & 0 & 22 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 22 & 0 & 3 \end{pmatrix}_{b_0}$$

[251] $1^{14}41^423^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -423, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 1 & 2 \\ 1 & 0 & 1 & 16 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 17 & 16 & 1 & 191 & 0 & 25 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 25 & 0 & 3 \end{pmatrix}_{b_0}$$

[252] $1^{11}41^423^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -429, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 19 & 1 & 3 \\ 2 & 0 & 13 & 0 & 2 \\ 19 & 13 & 130 & 0 & 24 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 24 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[253] \ 1^{13}41^323^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -430, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 1 & 2 \\ 1 & 0 & 1 & 12 & 0 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 14 & 12 & 2 & 110 & 0 & 20 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 20 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[254] \ 1^{14}41^423^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -480, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.92 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 20 & 1 & 3 \\ 2 & 0 & 12 & 0 & 2 \\ 20 & 12 & 126 & 0 & 24 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 24 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[255] \ 1^{14}41^323^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -512, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.89 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 18 & 1 & 3 \\ 2 & 0 & 12 & 0 & 2 \\ 18 & 12 & 98 & 0 & 22 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 22 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[256] \ 1^{13}41^423^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -530, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 10.92 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 1 & 2 \\ 1 & 0 & 1 & 14 & 0 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 17 & 14 & 2 & 164 & 0 & 24 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 24 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[257] \ 1^{14}41^423^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -612, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 21 & 1 & 3 \\ 2 & 0 & 14 & 0 & 2 \\ 21 & 14 & 144 & 0 & 26 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 2 & 26 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[258] \ 1^{14}41^423^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -626, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 1 & 2 \\ 1 & 0 & 1 & 16 & 0 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 18 & 16 & 2 & 194 & 0 & 26 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 26 & 0 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [259] \ 1^{11}22313^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 10, \ \Delta = -10, \ \det = 0, \ (n_+, n_-, n_0) = (1, 10, 1), \ c_{\text{ext}} = 5.0$$

$$[261] \ 1^{14}13^121^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 11, \ \Delta = -29, \ \det = 3, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.67 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[262] \ 1^{11}23^11321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 10, \ \Delta = -10, \ \det = 9, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 3 & 1 & 5 & 3 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 3 & 3 & 5 \\ 1 & 3 & 0 & 3 & 1 & 3 \\ 2 & 5 & -1 & 5 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[263] \ 31^{13}1^151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 11, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 6 & 7 & 9 & 5 & 7 \\ 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[264] \ 1^{11}5131^13^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 15, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} =$$

5.0

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 6 & 7 & 9 & 0 & 7 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 7 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[265] \quad 1^{11}3^{11}14131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 3 & 2 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 2 & 3 & 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 0 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[266] \quad 1^{11}3^{11}141^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 1 & 4 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 5 & 3 & 5 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 2 & 3 & 0 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[267] \quad 31^{23}3^{11}51^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 12 & 14 & 42 & 10 & 14 \\ 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[268] \quad 1^{12}5^{11}131^{23}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 12 & 14 & 42 & 0 & 14 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 14 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[269] \quad 31^{33}3^{11}51^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 18 & 21 & 97 & 15 & 21 \\ 3 & 3 & 15 & 1 & 3 \\ 3 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[270] \quad 1^{13}5^{11}131^{33}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 18 & 21 & 97 & 0 & 21 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 21 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[271] \quad 31^{11}5^{11}13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 65, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 2 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 3 & 2 & 4 \\ 3 & 4 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[272] \quad 31^{11}5^{12}13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 74, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 2 & 4 & 8 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 6 & 8 & 6 & 14 & 8 \\ 3 & 4 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[273] \quad 31^{11}5^{13}13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 89, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 2 & 4 & 12 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 9 & 12 & 9 & 34 & 12 \\ 3 & 4 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[274] \quad 231^{11}3^{11}151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -60, \quad \det = 275, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 5 & 1 & 5 & 3 & 2 \\ 5 & 33 & 1 & 35 & 20 & 11 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 35 & 0 & 33 & 20 & 11 \\ 3 & 20 & 0 & 20 & 10 & 6 \\ 2 & 11 & -1 & 11 & 6 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [275] \quad 1^{11}23^{11}141321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -9, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 11, 1), \quad c_{\text{ext}} = 5.0$$

$$[277] \quad 1^{13}31^5 13^{11}1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 13 & 1 & 2 \\ 8 & 0 & 13 & 97 & 11 & 21 \\ 1 & 0 & 1 & 11 & 0 & 2 \\ 2 & 0 & 2 & 21 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[278] \quad 1^{12}3151^{12}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 3 & 1 & 18 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 18 & 0 & 74 & 20 \\ 3 & 5 & -1 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[279] \quad 1^{13}1^5 13^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 1 & 3 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 2 \\ 2 & 3 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[280] \quad 1^{13}1^5 13^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 2 \\ 2 & 0 & 1 & 1 & 2 & 3 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 2 & 0 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[281] \quad 1^{12}31^5 13^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 2 & 7 & 1 & 3 \\ 1 & 2 & 0 & 5 & 2 & 2 \\ 6 & 7 & 5 & 17 & 4 & 9 \\ 1 & 1 & 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[282] \quad 1^{13}1^5 15131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.67 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 3 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[284] \quad 1^{11}31^5 13^{11}1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 6 & 1 & 2 \\ 6 & 0 & 6 & 34 & 7 & 12 \\ 1 & 0 & 1 & 7 & 0 & 2 \\ 2 & 0 & 2 & 12 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[283] \quad 1^{11}31^5 13^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 1 & 2 \\ 2 & 0 & 4 & 6 & 3 & 6 \\ 1 & 0 & 1 & 3 & 0 & 2 \\ 2 & 0 & 2 & 6 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[285] \quad 1^{14}31513^{11}12 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 1 & 2 & 14 & 1 & 3 \\ 1 & 2 & 0 & 10 & 2 & 2 \\ 8 & 14 & 10 & 74 & 8 & 18 \\ 1 & 1 & 2 & 8 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[286] \quad 1^{11}3^{11}151^{23}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 9 & 3 & 3 \\ 6 & 0 & 9 & 27 & 8 & 11 \\ 1 & 0 & 3 & 8 & 1 & 3 \\ 2 & 0 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[287] \quad 1^{11}31513^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 3 & 6 \\ 1 & 1 & 0 & 3 & 0 & 2 \\ 2 & 3 & -1 & 6 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[288] \quad 1^{12}3^{11}151^{23}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 18 & 3 & 3 \\ 12 & 0 & 18 & 114 & 16 & 22 \\ 1 & 0 & 3 & 16 & 1 & 3 \\ 2 & 0 & 3 & 22 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[289] \quad 1^{14}31513^{11}1^4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 1 & 1 & 16 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 12 & 24 \\ 1 & 1 & 0 & 12 & 0 & 2 \\ 2 & 3 & -1 & 24 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[293] \quad 2^13^{11}1513^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 8 & 1 & 2 \\ 1 & 1 & 2 & 14 & 2 & 3 \\ 2 & 2 & 0 & 10 & 1 & 2 \\ 8 & 14 & 10 & 74 & 10 & 18 \\ 1 & 2 & 1 & 10 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[292] \quad 1^{12}31513^{11}1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 1 & 8 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 6 & 12 \\ 1 & 1 & 0 & 6 & 0 & 2 \\ 2 & 3 & -1 & 12 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[291] \quad 1^{13}31513^{11}1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 12 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 9 & 18 \\ 1 & 1 & 0 & 9 & 0 & 2 \\ 2 & 3 & -1 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[290] \quad 1^{12}31513^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 2 & 7 & 1 & 3 \\ 1 & 2 & 0 & 5 & 2 & 2 \\ 4 & 7 & 5 & 17 & 4 & 9 \\ 1 & 1 & 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[294] \quad 1^{11} 2^3 3^{15} 13^{11} 12 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 1 & 2 \\ 1 & 1 & 2 & 7 & 2 & 3 \\ 2 & 2 & 0 & 5 & 1 & 2 \\ 4 & 7 & 5 & 17 & 5 & 9 \\ 1 & 2 & 1 & 5 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[295] \quad 1^{11} 3^{11} 5^{12} 13^{11} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -7, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.86 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 2 & 4 & 2 & 3 \\ 1 & 2 & 0 & 4 & 1 & 2 \\ 4 & 4 & 4 & 6 & 3 & 6 \\ 1 & 2 & 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 6 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[296] \quad 1^{12} 3^{11} 5^{13} 13^{11} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -8, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 2 & 7 & 1 & 3 \\ 1 & 2 & 0 & 5 & 2 & 2 \\ 5 & 7 & 5 & 17 & 4 & 9 \\ 1 & 1 & 2 & 4 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[297] \quad 1^{11} 3^{11} 5^{13} 13^{11} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -8, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 3 & 2 & 3 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 2 & 3 & 1 & 1 & 2 & 3 \\ 1 & 2 & 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[298] \quad 1^{11} 3^{14} 14^{13} 21^{11} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -8, \quad \det = -9, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & 0 & 3 & 0 & 2 \\ 2 & 2 & 3 & 3 & 0 & 5 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 5 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[300] \quad 1^{12} 3^{11} 5^{13} 13^{11} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -10, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 9 & 1 & 2 \\ 6 & 0 & 9 & 49 & 8 & 15 \\ 1 & 0 & 1 & 8 & 0 & 2 \\ 2 & 0 & 2 & 15 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[299] \quad 1^{11} 3^{11} 5^{13} 13^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -10, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 5 & 1 & 2 \\ 4 & 0 & 5 & 17 & 5 & 9 \\ 1 & 0 & 1 & 5 & 0 & 2 \\ 2 & 0 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[302] \quad 1^{14} 3^{15} 13^{11} 12 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -14, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 1 & 2 & 14 & 1 & 3 \\ 1 & 2 & 0 & 10 & 2 & 2 \\ 10 & 14 & 10 & 74 & 8 & 18 \\ 1 & 1 & 2 & 8 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[301] \quad 1^{14} 3^{15} 13^{11} 12 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -14, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$\begin{aligned}
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 1 & 2 & 14 & 1 & 3 \\ 1 & 2 & 0 & 10 & 2 & 2 \\ 9 & 14 & 10 & 74 & 8 & 18 \\ 1 & 1 & 2 & 8 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0} \\
[306] & 1^{11}3^{15}123^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 1 & 2 & 3 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 5 & 4 \\ 3 & 0 & 5 & 3 & 5 \\ 3 & 0 & 4 & 5 & 5 \end{pmatrix}_{b_0} \\
[305] & 1^{11}3^{15}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 4 & 2 & 3 \\ 1 & 2 & 0 & 4 & 1 & 2 \\ 2 & 4 & 4 & 6 & 3 & 6 \\ 1 & 2 & 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 6 & 2 & 4 \end{pmatrix}_{b_0} \\
[304] & 1^{11}3^{15}1^{15}1^{23}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 2 & 1 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 9 & 0 & 27 & 8 & 11 \\ 1 & 3 & 0 & 8 & 1 & 3 \\ 2 & 4 & -1 & 11 & 3 & 4 \end{pmatrix}_{b_0} \\
[303] & 1^{11}3^{15}151231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 2 & 1 & 4 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 3 & 4 \\ 1 & 3 & 0 & 3 & 1 & 3 \\ 2 & 4 & -1 & 4 & 3 & 4 \end{pmatrix}_{b_0} \\
[307] & 1^{12}3^{15}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 10 & 1 & 2 \\ 8 & 0 & 10 & 74 & 10 & 18 \\ 1 & 0 & 1 & 10 & 0 & 2 \\ 2 & 0 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0} \\
[308] & 1^{11}3^{15}15131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 2 & 4 & 2 & 3 \\ 1 & 2 & 0 & 4 & 1 & 2 \\ 3 & 4 & 4 & 6 & 3 & 6 \\ 1 & 2 & 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 6 & 2 & 4 \end{pmatrix}_{b_0} \\
[309] & 1^{12}3^{15}15131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 2 & 7 & 2 & 3 \\ 1 & 2 & 0 & 5 & 1 & 2 \\ 6 & 7 & 5 & 17 & 5 & 9 \\ 1 & 2 & 1 & 5 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0} \\
[310] & 1^{12}3^{15}15131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 1 & 2 & 8 & 2 & 3 \\ 1 & 2 & 0 & 8 & 1 & 2 \\ 7 & 8 & 8 & 30 & 6 & 12 \\ 1 & 2 & 1 & 6 & 0 & 2 \\ 2 & 3 & 2 & 12 & 2 & 4 \end{pmatrix}_{b_0} \\
[311] & 1^{12}3^{15}15131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -42, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 2 & 8 & 2 & 3 \\ 1 & 2 & 0 & 8 & 1 & 2 \\ 6 & 8 & 8 & 30 & 6 & 12 \\ 1 & 2 & 1 & 6 & 0 & 2 \\ 2 & 3 & 2 & 12 & 2 & 4 \end{pmatrix}_{b_0} \\
[312] & 1^{12}3^{11}5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -42, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5 \\
& T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 2 & 7 & 2 & 3 \\ 1 & 2 & 0 & 5 & 1 & 2 \\ 5 & 7 & 5 & 17 & 5 & 9 \\ 1 & 2 & 1 & 5 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0} \\
[314] & 1^{11}3^{15}1^{12}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -43, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 1 & 2 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 13 & 4 \\ 9 & 0 & 13 & 43 & 15 \\ 3 & 0 & 4 & 15 & 5 \end{pmatrix}_{b_0} \\
[313] & 1^{12}3^{11}5^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -43, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5 \\
& T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 2 & 8 & 2 & 3 \\ 1 & 2 & 0 & 8 & 1 & 2 \\ 5 & 8 & 8 & 30 & 6 & 12 \\ 1 & 2 & 1 & 6 & 0 & 2 \\ 2 & 3 & 2 & 12 & 2 & 4 \end{pmatrix}_{b_0} \\
[315] & 1^{12}3^{11}5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -43, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5 \\
& T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 2 & 7 & 2 & 3 \\ 1 & 2 & 0 & 5 & 1 & 2 \\ 4 & 7 & 5 & 17 & 5 & 9 \\ 1 & 2 & 1 & 5 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0} \\
[316] & 1^{12}3^{11}5^{12}1^{23}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -60, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 2 & 1 & 13 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 13 & 0 & 49 & 11 & 15 \\ 1 & 3 & 0 & 11 & 1 & 3 \\ 2 & 4 & -1 & 15 & 3 & 4 \end{pmatrix}_{b_0} \\
[317] & 1^{13}3^{11}5^{13}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -63, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.8 \\
& T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & 2 & 0 & 12 & 1 & 2 \\ 10 & 12 & 12 & 70 & 9 & 18 \\ 1 & 2 & 1 & 9 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0} \\
[318] & 1^{13}3^{11}5^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -73, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.8 \\
& T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 1 & 2 & 11 & 2 & 3 \\ 1 & 2 & 0 & 9 & 1 & 2 \\ 9 & 11 & 9 & 49 & 8 & 15 \\ 1 & 2 & 1 & 8 & 0 & 2 \\ 2 & 3 & 2 & 15 & 2 & 4 \end{pmatrix}_{b_0} \\
[319] & 3^{11}4^{13}1^{13}1^{15}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 13, \quad \Delta = -58, \quad \det = -80, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13] \\
& \begin{pmatrix} 0 & 3 & 1 & 8 & 3 & 2 \\ 3 & 4 & 1 & 12 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 8 & 12 & 0 & 26 & 8 & 10 \\ 3 & 3 & 0 & 8 & 1 & 3 \\ 2 & 4 & 0 & 10 & 3 & 3 \end{pmatrix}_{b_0} \\
[320] & 1^{13}3^{11}5^{13}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -82, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.8 \\
& T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & 2 & 0 & 12 & 1 & 2 \\ 9 & 12 & 12 & 70 & 9 & 18 \\ 1 & 2 & 1 & 9 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[321] \quad 1^{13}3^{1^2}1^5131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -87, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & 2 & 0 & 12 & 1 & 2 \\ 8 & 12 & 12 & 70 & 9 & 18 \\ 1 & 2 & 1 & 9 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[322] \quad 1^{13}3^{1^3}1^5131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -92, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 1 & 2 & 11 & 2 & 3 \\ 1 & 2 & 0 & 9 & 1 & 2 \\ 8 & 11 & 9 & 49 & 8 & 15 \\ 1 & 2 & 1 & 8 & 0 & 2 \\ 2 & 3 & 2 & 15 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[323] \quad 1^{13}3^{1^2}1^5131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -97, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 1 & 2 & 11 & 2 & 3 \\ 1 & 2 & 0 & 9 & 1 & 2 \\ 7 & 11 & 9 & 49 & 8 & 15 \\ 1 & 2 & 1 & 8 & 0 & 2 \\ 2 & 3 & 2 & 15 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[324] \quad 1^{12}3^{15}1^{23}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 1 & 2 & 12 & 3 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 2 & 18 & 4 \\ 12 & 0 & 18 & 74 & 20 \\ 3 & 0 & 4 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[325] \quad 1^{13}3^{1^5}1^{12}231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -115, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 2 & 1 & 22 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 22 & 0 & 154 & 19 & 26 \\ 1 & 3 & 0 & 19 & 1 & 3 \\ 2 & 4 & -1 & 26 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[326] \quad 1^{13}3^{1^5}1^{12}231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -115, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 2 & 1 & 17 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 17 & 0 & 79 & 14 & 19 \\ 1 & 3 & 0 & 14 & 1 & 3 \\ 2 & 4 & -1 & 19 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[327] \quad 41^{11}3^{15}1^3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 13, \quad \Delta = -58, \quad \det = -122, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 4 & 2 & 4 & 1 & 2 \\ 4 & 10 & 5 & 12 & 0 & 6 \\ 2 & 5 & 0 & 4 & 0 & 2 \\ 4 & 12 & 4 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 6 & 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[329] \quad 1^{14}3^{1^4}1^5131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -138, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 1 & 2 & 14 & 2 & 3 \\ 1 & 2 & 0 & 10 & 1 & 2 \\ 10 & 14 & 10 & 74 & 10 & 18 \\ 1 & 2 & 1 & 10 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[328] \quad 1^{14}3^{1^4}1^5131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -138, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 1 & 2 & 16 & 2 & 3 \\ 1 & 2 & 0 & 16 & 1 & 2 \\ 12 & 16 & 16 & 126 & 12 & 24 \\ 1 & 2 & 1 & 12 & 0 & 2 \\ 2 & 3 & 2 & 24 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[330] \quad 1^{14}3^{11}5^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -147, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 1 & 2 & 14 & 2 & 3 \\ 1 & 2 & 0 & 10 & 1 & 2 \\ 9 & 14 & 10 & 74 & 10 & 18 \\ 1 & 2 & 1 & 10 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[331] \quad 1^{14}3^{11}5^{13}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -147, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 1 & 2 & 16 & 2 & 3 \\ 1 & 2 & 0 & 16 & 1 & 2 \\ 11 & 16 & 16 & 126 & 12 & 24 \\ 1 & 2 & 1 & 12 & 0 & 2 \\ 2 & 3 & 2 & 24 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[332] \quad 1^{14}3^{11}5^{13}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -158, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 1 & 2 & 15 & 2 & 3 \\ 1 & 2 & 0 & 13 & 1 & 2 \\ 11 & 15 & 13 & 97 & 11 & 21 \\ 1 & 2 & 1 & 11 & 0 & 2 \\ 2 & 3 & 2 & 21 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[333] \quad 3^{11}1^{14}1^{11}3^{15}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -162, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 7 & 1 & 12 & 3 \\ 7 & 12 & 1 & 24 & 8 \\ 1 & 1 & -2 & 0 & 0 \\ 12 & 24 & 0 & 42 & 14 \\ 3 & 8 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[334] \quad 1^{14}3^{11}5^{13}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -162, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 1 & 2 & 15 & 2 & 3 \\ 1 & 2 & 0 & 13 & 1 & 2 \\ 9 & 15 & 13 & 97 & 11 & 21 \\ 1 & 2 & 1 & 11 & 0 & 2 \\ 2 & 3 & 2 & 21 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[335] \quad 1^{14}3^{11}5^{13}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -167, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 1 & 2 & 15 & 2 & 3 \\ 1 & 2 & 0 & 13 & 1 & 2 \\ 10 & 15 & 13 & 97 & 11 & 21 \\ 1 & 2 & 1 & 11 & 0 & 2 \\ 2 & 3 & 2 & 21 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[336] \quad 1^{14}3^{11}5^{12}2^{23}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -210, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 2 & 1 & 26 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 12 & 26 & 0 & 202 & 22 & 30 \\ 1 & 3 & 0 & 22 & 1 & 3 \\ 2 & 4 & -1 & 30 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[337] \quad 3^{11}1^{14}1^{11}3^{15}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -227, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 7 & 1 & 7 & 3 \\ 7 & 12 & 1 & 14 & 8 \\ 1 & 1 & -2 & 0 & 0 \\ 7 & 14 & 0 & 12 & 8 \\ 3 & 8 & 0 & 8 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [339] \quad 1^{11}3^{14}1^{14}1^{13}2^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -7, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$$

$$[338] \quad 1^{43}1^{11}5^{13}2^{21}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) =$$

$$(1, 12, 1), c_{\text{ext}} = 6.0$$

$$[354] \ 1^{\overline{1^3 1^{21}}} \ 6 \ 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\text{max}} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\text{min}} = 13, T_{\text{max}} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 0 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[355] \ 1^{\overline{1^3 1^{21}}} \ 6 \ 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\text{max}} = 13, \Delta = -29, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\text{min}} = 12.67 \rightarrow 13, T_{\text{max}} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[359] \ 1^{\overline{1^2 3^1 1^5 1321^{12}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\text{max}} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\text{min}} = 12, T_{\text{max}} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 2 & 8 & 2 & 3 \\ 1 & 2 & 1 & 10 & 1 & 3 \\ 6 & 8 & 10 & 34 & 6 & 14 \\ 1 & 2 & 1 & 6 & 0 & 2 \\ 2 & 3 & 3 & 14 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[360] \ 1^{\overline{1^4 3^1 1^5 1321^{14}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\text{max}} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\text{min}} = 12, T_{\text{max}} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 1 & 2 & 16 & 2 & 3 \\ 1 & 2 & 1 & 20 & 1 & 3 \\ 12 & 16 & 20 & 142 & 12 & 28 \\ 1 & 2 & 1 & 12 & 0 & 2 \\ 2 & 3 & 3 & 28 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[356] \ 1^{\overline{1^3 3^1 1^5 1321^{12}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\text{max}} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$$

$$T_{\text{min}} = 12, T_{\text{max}} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 11 & 1 & 3 \\ 6 & 0 & 11 & 53 & 8 & 17 \\ 1 & 0 & 1 & 8 & 0 & 2 \\ 2 & 0 & 3 & 17 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[357] \ 1^{\overline{1^1 3^1 1^5 1321^{11}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\text{max}} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\text{min}} = 12, T_{\text{max}} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 2 & 4 & 2 & 3 \\ 1 & 2 & 1 & 5 & 1 & 3 \\ 3 & 4 & 5 & 7 & 3 & 7 \\ 1 & 2 & 1 & 3 & 0 & 2 \\ 2 & 3 & 3 & 7 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[358] \ 1^{\overline{1^3 3^1 1^5 1321^{13}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, T_{\text{max}} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$$

$$T_{\text{min}} = 12, T_{\text{max}} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & 2 & 1 & 15 & 1 & 3 \\ 9 & 12 & 15 & 79 & 9 & 21 \\ 1 & 2 & 1 & 9 & 0 & 2 \\ 2 & 3 & 3 & 21 & 2 & 5 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [370] \ 1^{\overline{1^2 3^1 1^6 1^5 131^1 1^3}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 15, T_{\text{max}} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.5$$

$$T_{\text{min}} = 15, T_{\text{max}} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 12 & 9 & 9 & 61 & 21 \\ 3 & 3 & 2 & 21 & 6 \end{pmatrix}_{b_0}$$

4.915 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_6$

$$T = 9 \quad [0] \quad 1^1 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{e}_6 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 17.60 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 3 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 5 & 0 & 2 & -2 & 0 \\ 3 & 5 & 0 & 2 & 0 & -4 \end{pmatrix}_{b_0}$$

4.916 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4$

$$T = 8 \quad [0] \quad 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 5 & 9 & 13 & 5 & 0 & 7 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 7 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

4.917 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \quad 1^1 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 3 & 0 & 5 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 6 & 3 & 0 & 4 & 4 & 4 \\ 2 & 4 & 3 & 2 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 1^1 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 600, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 9 & 3 & 0 & 5 & 4 & 4 \\ 5 & 9 & 13 & 5 & 0 & 9 & 6 & 7 \\ 2 & 3 & 5 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 9 & 3 & 0 & 4 & 4 & 4 \\ 2 & 4 & 6 & 2 & 0 & 4 & 0 & 2 \\ 3 & 4 & 7 & 2 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

4.918 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^1 1^1 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 9 & 6 & 1 & 5 \\ 11 & 20 & 17 & 11 & 0 & 8 \\ 9 & 17 & 13 & 9 & 0 & 7 \\ 6 & 11 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 7 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

4.919 NHC(-2)(-3) + su₂ + su₂→(-3) + f₄ + su₂

$$T = 9 \quad [0] \quad 1^{112} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 4 & 3 \\ 3 & 4 & 9 & 3 & 0 & 8 & 4 \\ 5 & 9 & 13 & 5 & 0 & 12 & 7 \\ 2 & 3 & 5 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 12 & 4 & 0 & 10 & 6 \\ 3 & 4 & 7 & 2 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{112} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 1 & 3 \\ 3 & 4 & 4 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{13} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 5 & 3 & 0 & 17 & 4 \\ 3 & 5 & 4 & 3 & 0 & 15 & 4 \\ 2 & 3 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 9 & 0 & 49 & 13 \\ 3 & 4 & 4 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{112} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 3 & 0 & 8 & 4 \\ 2 & 4 & 1 & 2 & 0 & 6 & 3 \\ 2 & 3 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 4 & 0 & 10 & 6 \\ 3 & 4 & 3 & 2 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{13} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 3 & 0 & 5 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 6 & 3 & 0 & 4 & 4 \\ 3 & 4 & 4 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{112} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 3 & 0 & 14 & 4 \\ 2 & 4 & 1 & 2 & 0 & 9 & 3 \\ 2 & 3 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 8 & 0 & 37 & 11 \\ 3 & 4 & 3 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{12} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 3 & 0 & 13 & 4 \\ 3 & 5 & 4 & 3 & 0 & 12 & 4 \\ 2 & 3 & 3 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 12 & 7 & 0 & 28 & 10 \\ 3 & 4 & 4 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11}3^{11}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -234, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 3 & 0 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 4 \\ 2 & 3 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 5 & 0 & 13 & 7 \\ 3 & 4 & 4 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{111}3^{11}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -254, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.54 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 1 & 2 & 0 & 6 & 3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 5 & 0 & 13 & 7 \\ 3 & 4 & 3 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{111}3^{11}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -314, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.59 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 11 & 3 \\ 3 & 4 & 9 & 3 & 0 & 21 & 4 \\ 5 & 9 & 13 & 5 & 0 & 33 & 7 \\ 2 & 3 & 5 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 33 & 11 & 0 & 76 & 16 \\ 3 & 4 & 7 & 2 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{112}3^{11}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -394, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.55 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 3 & 0 & 13 & 4 \\ 2 & 4 & 1 & 2 & 0 & 9 & 3 \\ 2 & 3 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 7 & 0 & 28 & 10 \\ 3 & 4 & 3 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111}3^{11}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -464, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 5 & 3 \\ 3 & 4 & 9 & 3 & 0 & 9 & 4 \\ 5 & 9 & 13 & 5 & 0 & 15 & 7 \\ 2 & 3 & 5 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 15 & 5 & 0 & 13 & 7 \\ 3 & 4 & 7 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{112}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -474, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 3 & 0 & 23 & 4 \\ 2 & 4 & 1 & 2 & 0 & 15 & 3 \\ 2 & 3 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 15 & 13 & 0 & 100 & 18 \\ 3 & 4 & 3 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{113}3^{11}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -494, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 3 & 0 & 17 & 4 \\ 2 & 4 & 1 & 2 & 0 & 12 & 3 \\ 2 & 3 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 9 & 0 & 49 & 13 \\ 3 & 4 & 3 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{113}3^{11}1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -504, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.57 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 9 & 3 \\ 3 & 4 & 9 & 3 & 0 & 17 & 4 \\ 5 & 9 & 13 & 5 & 0 & 27 & 7 \\ 2 & 3 & 5 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 27 & 9 & 0 & 49 & 13 \\ 3 & 4 & 7 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{112}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 3 & 0 & 18 & 4 \\ 2 & 4 & 1 & 2 & 0 & 12 & 3 \\ 2 & 3 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 10 & 0 & 58 & 14 \\ 3 & 4 & 3 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{114}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 3 & 0 & 21 & 4 \\ 2 & 4 & 1 & 2 & 0 & 15 & 3 \\ 2 & 3 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 15 & 11 & 0 & 76 & 16 \\ 3 & 4 & 3 & 2 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{112}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -554, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 7 & 3 \\ 3 & 4 & 9 & 3 & 0 & 13 & 4 \\ 5 & 9 & 13 & 5 & 0 & 21 & 7 \\ 2 & 3 & 5 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 21 & 7 & 0 & 28 & 10 \\ 3 & 4 & 7 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{113}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -824, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 3 & 0 & 22 & 4 \\ 2 & 4 & 1 & 2 & 0 & 15 & 3 \\ 2 & 3 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 15 & 12 & 0 & 85 & 17 \\ 3 & 4 & 3 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{113}3^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 3 & 0 & 32 & 4 \\ 2 & 4 & 1 & 2 & 0 & 21 & 3 \\ 2 & 3 & 2 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 21 & 18 & 0 & 193 & 25 \\ 3 & 4 & 3 & 2 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{113}3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -974, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 3 & 0 & 27 & 4 \\ 2 & 4 & 1 & 2 & 0 & 18 & 3 \\ 2 & 3 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 18 & 15 & 0 & 133 & 21 \\ 3 & 4 & 3 & 2 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{114}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1084, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 3 & 0 & 26 & 4 \\ 2 & 4 & 1 & 2 & 0 & 18 & 3 \\ 2 & 3 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 18 & 14 & 0 & 118 & 20 \\ 3 & 4 & 3 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{114}3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1434, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 3 & 0 & 31 & 4 \\ 2 & 4 & 1 & 2 & 0 & 21 & 3 \\ 2 & 3 & 2 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 21 & 17 & 0 & 172 & 24 \\ 3 & 4 & 3 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{1111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -79$, $\det = -1604$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 3 & 0 & 36 & 4 \\ 2 & 4 & 1 & 2 & 0 & 24 & 3 \\ 2 & 3 & 2 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 24 & 20 & 0 & 238 & 28 \\ 3 & 4 & 3 & 2 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

4.920 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 10$ [0] $1^{1111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 72$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 3 & 0 & 5 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 6 & 3 & 0 & 4 & 3 & 4 \\ 2 & 4 & 3 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{1111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 392$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 & 3 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{1111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 462$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 3 & 0 & 9 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 5 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 2 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{1111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 512$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 & 6 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 5 & 0 & 6 & 12 & 6 \\ 3 & 4 & 3 & 2 & -3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{1111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 622$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 9 & 4 & 4 \\ 2 & 4 & 1 & 2 & 0 & 6 & 3 & 3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 5 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 2 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & 2 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{1111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 1152$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 9 & 3 & 0 & 9 & 4 & 4 \\ 5 & 9 & 13 & 5 & 0 & 15 & 6 & 7 \\ 2 & 3 & 5 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 15 & 5 & 0 & 13 & 6 & 7 \\ 2 & 4 & 6 & 2 & 0 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.921 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{f}_4$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 3 & 0 & 5 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 & 3 \\ 2 & 3 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 3 & 4 & 3 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.31 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 3 & 0 & 8 & 4 \\ 2 & 4 & 1 & 2 & 0 & 6 & 3 \\ 2 & 3 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 4 & 0 & 9 & 5 \\ 3 & 4 & 3 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 4 & 3 \\ 3 & 4 & 9 & 3 & 0 & 8 & 4 \\ 5 & 9 & 13 & 5 & 0 & 12 & 7 \\ 2 & 3 & 5 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 12 & 4 & 0 & 9 & 5 \\ 3 & 4 & 7 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 1 & 2 & 0 & 6 & 3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 5 & 0 & 12 & 6 \\ 3 & 4 & 3 & 2 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 & 4 \\ 5 & 9 & 13 & 5 & 0 & 6 & 7 \\ 2 & 3 & 5 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 2 & 0 & 0 & 2 \\ 3 & 4 & 7 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -476, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 3 & 0 & 13 & 4 \\ 2 & 4 & 1 & 2 & 0 & 9 & 3 \\ 2 & 3 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 7 & 0 & 27 & 9 \\ 3 & 4 & 3 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$4.922 \quad \text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$$

$$T = 10 \quad [0] \ 1^{113}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 + \textcolor{magenta}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 72, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 & 5 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 6 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & 2 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 + \textcolor{magenta}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 392, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 & 3 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 2 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & 2 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 + \textcolor{magenta}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 462, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 9 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 5 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & 2 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 + \textcolor{magenta}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 512, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 9 & 4 & 4 \\ 2 & 4 & 1 & 2 & 0 & 6 & 3 & 3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 5 & 0 & 12 & 6 & 6 \\ 2 & 4 & 3 & 2 & 0 & 6 & 1 & 3 \\ 3 & 4 & 3 & 2 & -3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 + \textcolor{magenta}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 622, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 & 6 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 6 & 2 \\ 5 & 9 & 6 & 5 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & 2 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 + \textcolor{magenta}{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 1152, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 & 9 & 4 \\ 5 & 9 & 13 & 5 & 0 & 6 & 15 & 7 \\ 2 & 3 & 5 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 2 & 0 & 0 & 6 & 2 \\ 5 & 9 & 15 & 5 & 0 & 6 & 13 & 7 \\ 3 & 4 & 7 & 2 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.923 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 12 & 8 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 12 & 17 & 13 & 9 & 7 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 1 & 8 & 6 \\ 15 & 20 & 0 & 11 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 11 & 0 & 4 & 4 \\ 6 & 8 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.62 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 \\ 4 & 5 & 4 & 3 & 0 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 15 & 13 & 8 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 13 & 17 & 13 & 9 & 7 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 3 & 4 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 7 \\ 5 & 9 & 13 & 5 & 0 & 7 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{12} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$\begin{aligned}
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 8 & 3 & 0 & 4 \\ 5 & 8 & 10 & 4 & 0 & 6 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 2 & -2 & 2 \end{pmatrix}_{b_0} \\
[10] & 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 6 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 6 & 9 & 13 & 5 & 0 & 7 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 2 & -2 & 2 \end{pmatrix}_{b_0} \\
[11] & 1^{11} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 14 & 3 & 0 & 4 \\ 9 & 14 & 37 & 8 & 0 & 11 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & 2 & -2 & 2 \end{pmatrix}_{b_0} \\
[12] & 1^{12} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 306, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 13 & 3 & 0 & 4 \\ 8 & 13 & 28 & 7 & 0 & 10 \\ 2 & 3 & 7 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 2 & -2 & 2 \end{pmatrix}_{b_0} \\
[13] & 1^{13} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 399, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 17 & 3 & 0 & 4 \\ 10 & 17 & 49 & 9 & 0 & 13 \\ 2 & 3 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 2 & -2 & 2 \end{pmatrix}_{b_0} \\
[14] & 1^{12} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 18 & 3 & 0 & 4 \\ 11 & 18 & 58 & 10 & 0 & 14 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & 2 & -2 & 2 \end{pmatrix}_{b_0} \\
[15] & 1^{13} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 621, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 13 & 2 & 1 & 3 \\ 3 & 4 & 22 & 3 & 0 & 4 \\ 13 & 22 & 85 & 12 & 0 & 17 \\ 2 & 3 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 17 & 2 & -2 & 2 \end{pmatrix}_{b_0} \\
T = 9 & [16] 1^{13} 3^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 2 & 0 & 3 \\ 2 & 3 & 1 & 4 & 0 & 3 \\ 2 & 2 & 4 & 2 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 3 & 4 & -2 & 3 \end{pmatrix}_{b_0} \\
[17] & 1^{13} 3^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 5 & 5 & 3 & 5 & 5 \\ 2 & 5 & 2 & 2 & 4 & 4 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 3 & 5 & 4 & 2 & 0 & 2 \\ 3 & 5 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[18] $1^{11}3^{11}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -37, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 3 & 0 & 3 \\ 2 & 3 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 3 & 3 & -2 & 3 \end{pmatrix}_{b_0}$$

[19] $1^{11}3^{11}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 1 & 3 \\ 2 & 1 & 3 & 5 & 0 & 3 \\ 2 & 3 & 1 & 6 & 0 & 3 \\ 4 & 5 & 6 & 9 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 3 & 7 & -2 & 3 \end{pmatrix}_{b_0}$$

[20] $1^{11}3^{11}21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 3 \\ 3 & 5 & 10 & 3 & 5 & 5 \\ 5 & 10 & 14 & 5 & 7 & 8 \\ 2 & 3 & 5 & 0 & 2 & 2 \\ 3 & 5 & 7 & 2 & 0 & 2 \\ 3 & 5 & 8 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$T = 12$ [21] $1^{11}3^{11}5^{11}13^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{so}}_8 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 3 & 8 & 6 \\ 1 & 1 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 8 & 3 & 4 & 4 \\ 2 & 3 & -1 & 6 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

4.924 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{so}_8$

$T = 9$ [0] $1^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 1 & 5 \\ 11 & 20 & 0 & 11 & 0 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 11 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 0 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 6 & 1 & 5 \\ 11 & 20 & 17 & 11 & 0 & 8 \\ 9 & 17 & 13 & 9 & 0 & 7 \\ 6 & 11 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

4.925 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 3 & 0 & 3 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 1 & 2 & 0 & 2 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 2 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & 2 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -208, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 3 \\ 3 & 4 & 9 & 3 & 0 & 3 & 4 \\ 5 & 9 & 13 & 5 & 0 & 5 & 7 \\ 2 & 3 & 5 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 5 & 2 & 0 & 0 & 2 \\ 3 & 4 & 7 & 2 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

4.926 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ 3mix

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -91, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 5 & 3 & 0 & 7 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 3 & 3 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 6 & 4 & 0 & 7 & 5 \\ 3 & 4 & 4 & 2 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -194, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 3 & 0 & 7 & 4 \\ 2 & 4 & 1 & 2 & 0 & 5 & 3 \\ 2 & 3 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 5 & 4 & 0 & 7 & 5 \\ 3 & 4 & 3 & 2 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -236, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 3 & 0 & 11 & 4 \\ 2 & 4 & 1 & 2 & 0 & 8 & 3 \\ 2 & 3 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 8 & 6 & 0 & 20 & 8 \\ 3 & 4 & 3 & 2 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -251, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 4 & 3 \\ 3 & 4 & 9 & 3 & 0 & 7 & 4 \\ 5 & 9 & 13 & 5 & 0 & 11 & 7 \\ 2 & 3 & 5 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 11 & 4 & 0 & 7 & 5 \\ 3 & 4 & 7 & 2 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -287, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 3 & 0 & 12 & 4 \\ 2 & 4 & 1 & 2 & 0 & 8 & 3 \\ 2 & 3 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 8 & 7 & 0 & 25 & 9 \\ 3 & 4 & 3 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -477, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 3 & 0 & 16 & 4 \\ 2 & 4 & 1 & 2 & 0 & 11 & 3 \\ 2 & 3 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 11 & 9 & 0 & 44 & 12 \\ 3 & 4 & 3 & 2 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -584, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.94 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 3 & 0 & 21 & 4 \\ 2 & 4 & 1 & 2 & 0 & 14 & 3 \\ 2 & 3 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 14 & 12 & 0 & 80 & 16 \\ 3 & 4 & 3 & 2 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -631, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 3 & 0 & 20 & 4 \\ 2 & 4 & 1 & 2 & 0 & 14 & 3 \\ 2 & 3 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 14 & 11 & 0 & 69 & 15 \\ 3 & 4 & 3 & 2 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{114}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -749, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.92 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 3 & 0 & 24 & 4 \\ 2 & 4 & 1 & 2 & 0 & 17 & 3 \\ 2 & 3 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 17 & 13 & 0 & 100 & 18 \\ 3 & 4 & 3 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{113}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -886, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.89 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 3 & 0 & 25 & 4 \\ 2 & 4 & 1 & 2 & 0 & 17 & 3 \\ 2 & 3 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 14 & 25 & 17 & 14 & 0 & 111 & 19 \\ 3 & 4 & 3 & 2 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{114}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -1152, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 3 & 0 & 29 & 4 \\ 2 & 4 & 1 & 2 & 0 & 20 & 3 \\ 2 & 3 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 16 & 29 & 20 & 16 & 0 & 148 & 22 \\ 3 & 4 & 3 & 2 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

4.927 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 9 \quad [4] \ 1^{11}3^{11}1^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -35, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 3 & 3 & 4 \\ 1 & 2 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 0 & 3 & 2 & 1 & 3 \\ 2 & 3 & -1 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -35, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 0 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 0 & 3 & 4 & 4 \\ 2 & 3 & -1 & 0 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}21^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -2, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 13 & 3 \\ 3 & 5 & 15 & 3 & 20 & 5 \\ 10 & 15 & 34 & 6 & 45 & 12 \\ 2 & 3 & 6 & 0 & 9 & 2 \\ 13 & 20 & 45 & 9 & 58 & 16 \\ 3 & 5 & 12 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{12} 3^{12} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 8 & 3 \\ 2 & 1 & 3 & 5 & 8 & 3 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 7 & 5 & 6 & 9 & 16 & 7 \\ 8 & 8 & 8 & 16 & 22 & 10 \\ 3 & 3 & 3 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{12} 3^{12} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 8 & 3 \\ 2 & 1 & 3 & 5 & 8 & 3 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 4 & 5 & 6 & 9 & 16 & 7 \\ 8 & 8 & 8 & 16 & 22 & 10 \\ 3 & 3 & 3 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12} 3^{12} 21^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 10 & 3 \\ 3 & 5 & 15 & 3 & 20 & 5 \\ 8 & 15 & 34 & 6 & 45 & 12 \\ 2 & 3 & 6 & 0 & 9 & 2 \\ 10 & 20 & 45 & 9 & 58 & 16 \\ 3 & 5 & 12 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{12} 3^{12} 21^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 14 & 3 \\ 3 & 5 & 15 & 3 & 25 & 5 \\ 8 & 15 & 34 & 6 & 57 & 12 \\ 2 & 3 & 6 & 0 & 11 & 2 \\ 14 & 25 & 57 & 11 & 92 & 20 \\ 3 & 5 & 12 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[7] \quad 21^{121} 3^{12} 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 5 & 3 \\ 3 & 1 & 3 & 12 & 5 & 3 \\ 2 & 3 & 1 & 16 & 6 & 3 \\ 12 & 12 & 16 & 70 & 28 & 18 \\ 5 & 5 & 6 & 28 & 9 & 7 \\ 3 & 3 & 3 & 18 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{12} 3^{12} 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 14 & 3 \\ 3 & 5 & 5 & 3 & 25 & 5 \\ 4 & 5 & 2 & 2 & 19 & 4 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 14 & 25 & 19 & 11 & 92 & 20 \\ 3 & 5 & 4 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{12} 3^{12} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 6 & 3 \\ 3 & 5 & 5 & 3 & 10 & 5 \\ 4 & 5 & 2 & 2 & 7 & 4 \\ 2 & 3 & 2 & 0 & 5 & 2 \\ 6 & 10 & 7 & 5 & 14 & 8 \\ 3 & 5 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12} 3^{12} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 8 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 10 & 3 \\ 4 & 2 & 4 & 2 & 10 & 4 \\ 8 & 7 & 10 & 10 & 25 & 11 \\ 3 & 3 & 3 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[12] \quad & 1^{11} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -10, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 4 & 6 & 3 \\ 2 & 2 & 4 & 2 & 6 & 4 \\ 5 & 5 & 6 & 6 & 9 & 7 \\ 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0} \\
[13] \quad & 1^{11} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -10, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 10 & 3 \\ 2 & 2 & 4 & 2 & 10 & 4 \\ 9 & 7 & 10 & 10 & 25 & 11 \\ 3 & 3 & 3 & 4 & 11 & 4 \end{pmatrix}_{b_0} \\
[10] \quad & 1^{12} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -10, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 4 & 6 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 4 & 6 & 3 \\ 4 & 2 & 4 & 2 & 6 & 4 \\ 6 & 5 & 6 & 6 & 9 & 7 \\ 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0} \\
[14] \quad & 1^{11} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 2 & 8 & 3 \\ 3 & 5 & 10 & 3 & 15 & 5 \\ 5 & 10 & 14 & 4 & 22 & 8 \\ 2 & 3 & 4 & 0 & 7 & 2 \\ 8 & 15 & 22 & 7 & 32 & 12 \\ 3 & 5 & 8 & 2 & 12 & 4 \end{pmatrix}_{b_0} \\
[15] \quad & 1^{11} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 3 & 12 & 3 \\ 2 & 3 & 1 & 2 & 16 & 3 \\ 2 & 3 & 2 & 1 & 12 & 3 \\ 13 & 12 & 16 & 12 & 70 & 18 \\ 3 & 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0} \\
[16] \quad & 1^{11} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 7 & 2 & 12 & 3 \\ 3 & 5 & 10 & 3 & 20 & 5 \\ 7 & 10 & 14 & 4 & 30 & 8 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 12 & 20 & 30 & 9 & 58 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0} \\
[17] \quad & 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 3 \\ 2 & 1 & 3 & 0 & 3 & 3 \\ 2 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 3 & 2 & 0 & 1 & 3 \\ 3 & 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0} \\
[18] \quad & 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 3 & 3 \\ 2 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 3 \\ 3 & 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0} \\
[19] \quad & 1^{13} 3^1 21^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 12 & 3 \\ 3 & 5 & 15 & 3 & 20 & 5 \\ 9 & 15 & 34 & 6 & 45 & 12 \\ 2 & 3 & 6 & 0 & 9 & 2 \\ 12 & 20 & 45 & 9 & 58 & 16 \\ 3 & 5 & 12 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11} 1^1 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -12, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.67 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 5 & 0 & 3 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 3 & 5 & 0 & 2 & 2 & 4 \\ 3 & 5 & 0 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11} 1^1 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -12, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.67 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 3 \\ 2 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 2 & 4 & 0 & 2 & 4 \\ 3 & 3 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^1 1^1 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -14, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.71 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 5 & 5 & 3 & 5 & 5 \\ 2 & 5 & 2 & 2 & 4 & 4 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 3 & 5 & 4 & 2 & 2 & 4 \\ 3 & 5 & 4 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{124} 1^{12} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 16 & 3 \\ 2 & 1 & 3 & 2 & 14 & 3 \\ 2 & 3 & 1 & 4 & 20 & 3 \\ 4 & 2 & 4 & 2 & 20 & 4 \\ 16 & 14 & 20 & 20 & 106 & 22 \\ 3 & 3 & 3 & 4 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{11} 1^1 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 4 & 4 & 3 \\ 3 & 2 & 4 & 2 & 4 & 4 \\ 3 & 2 & 4 & 2 & 4 & 4 \\ 3 & 3 & 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{124} 1^{12} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 12 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 4 & 12 & 3 \\ 4 & 2 & 4 & 2 & 12 & 4 \\ 12 & 10 & 12 & 12 & 42 & 14 \\ 3 & 3 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{13} 1^{122} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 15 & 3 \\ 2 & 1 & 3 & 10 & 14 & 3 \\ 2 & 3 & 1 & 12 & 20 & 3 \\ 9 & 10 & 12 & 42 & 68 & 14 \\ 15 & 14 & 20 & 68 & 106 & 22 \\ 3 & 3 & 3 & 14 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^2 1^{12} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 4 & 12 & 3 \\ 2 & 2 & 4 & 2 & 12 & 4 \\ 10 & 10 & 12 & 12 & 42 & 14 \\ 3 & 3 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{111} 3^1 1^1 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -35, \quad \det = -18, \quad (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 1 & 2 & 6 & 3 \\ 1 & 2 & 0 & 2 & 0 & 5 & 2 \\ 3 & 6 & 0 & 6 & 5 & 13 & 7 \\ 2 & 3 & -1 & 3 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{111}3^{11}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 & 3 \\ 2 & 2 & 1 & 9 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 9 & 0 & 22 & 9 & 10 \\ 3 & 4 & 0 & 9 & 2 & 4 \\ 3 & 4 & -1 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{11}3^{12}21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 9 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 2 & 5 & 2 & 2 & 11 & 4 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 9 & 15 & 11 & 7 & 32 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[31] \quad 13^{11}21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 5 & 5 & 3 & 10 & 5 \\ 2 & 5 & 2 & 2 & 7 & 4 \\ 2 & 3 & 2 & 0 & 5 & 2 \\ 5 & 10 & 7 & 5 & 14 & 8 \\ 3 & 5 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{12}3^{11}21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 3 & 12 & 3 \\ 2 & 3 & 1 & 2 & 16 & 3 \\ 2 & 3 & 2 & 1 & 12 & 3 \\ 12 & 12 & 16 & 12 & 70 & 18 \\ 3 & 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{111}3^{11}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 2 & 1 & 4 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 4 & 4 \\ 3 & 4 & 0 & 4 & 2 & 4 \\ 3 & 4 & -1 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{112}3^{11}21^{133} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 12 & 3 \\ 2 & 1 & 3 & 9 & 12 & 3 \\ 2 & 3 & 1 & 14 & 16 & 3 \\ 11 & 9 & 14 & 49 & 60 & 15 \\ 12 & 12 & 16 & 60 & 70 & 18 \\ 3 & 3 & 3 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{12}3^{11}21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 5 & 0 & 3 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 6 & 10 & 0 & 4 & 14 & 8 \\ 3 & 5 & 0 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{11}3^{11}21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 4 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 4 & 6 & 3 \\ 3 & 2 & 4 & 2 & 6 & 4 \\ 4 & 5 & 6 & 6 & 9 & 7 \\ 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

[37] $1^{13} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -19$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 4 & 6 & 3 \\ 3 & 2 & 4 & 2 & 6 & 4 \\ 7 & 5 & 6 & 6 & 9 & 7 \\ 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$

[38] $1^2 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -20$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 3 \\ 3 & 5 & 5 & 3 & 10 & 5 \\ 2 & 5 & 2 & 2 & 8 & 4 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 6 & 10 & 8 & 4 & 14 & 8 \\ 3 & 5 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$

[39] $1^{111} 3^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -35$, $\det = -21$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 2 & 1 & 4 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 9 & 4 \\ 6 & 9 & 0 & 9 & 22 & 10 \\ 3 & 4 & -1 & 4 & 10 & 4 \end{pmatrix}_{b_0}$

[40] $1^{21} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -22$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.82 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 3 \\ 3 & 5 & 5 & 3 & 5 & 5 \\ 4 & 5 & 2 & 2 & 4 & 4 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 3 & 5 & 4 & 2 & 2 & 4 \\ 3 & 5 & 4 & 2 & 4 & 4 \end{pmatrix}_{b_0}$

[43] $1^{24} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 4 & 14 & 3 \\ 2 & 1 & 3 & 2 & 12 & 3 \\ 2 & 3 & 1 & 4 & 16 & 3 \\ 4 & 2 & 4 & 2 & 16 & 4 \\ 14 & 12 & 16 & 16 & 70 & 18 \\ 3 & 3 & 3 & 4 & 18 & 4 \end{pmatrix}_{b_0}$

[42] $1^{21} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 2 & 8 & 3 \\ 3 & 5 & 10 & 3 & 15 & 5 \\ 6 & 10 & 14 & 4 & 22 & 8 \\ 2 & 3 & 4 & 0 & 7 & 2 \\ 8 & 15 & 22 & 7 & 32 & 12 \\ 3 & 5 & 8 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[41] $1^{14} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 2 & 12 & 3 \\ 2 & 3 & 1 & 4 & 16 & 3 \\ 2 & 2 & 4 & 2 & 16 & 4 \\ 14 & 12 & 16 & 16 & 70 & 18 \\ 3 & 3 & 3 & 4 & 18 & 4 \end{pmatrix}_{b_0}$

[44] $1^3 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -26$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 2 & 15 & 3 \\ 2 & 3 & 1 & 4 & 18 & 3 \\ 2 & 2 & 4 & 2 & 18 & 4 \\ 15 & 15 & 18 & 18 & 97 & 21 \\ 3 & 3 & 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$

[45] $1^{13} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -27$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 11 & 3 \\ 2 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 4 & 5 & 6 & 9 & 22 & 7 \\ 11 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{11} 3^{11} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -27, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 5 & 0 & 3 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 8 & 15 & 0 & 7 & 32 & 12 \\ 3 & 5 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{112} 3^{11} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -27, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 11 & 3 \\ 2 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 7 & 5 & 6 & 9 & 22 & 7 \\ 11 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{11} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -27, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 5 & 0 & 3 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 6 & 10 & 0 & 5 & 14 & 8 \\ 3 & 5 & 0 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{13} 3^{11} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -28, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.86 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 5 & 0 & 3 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 9 & 15 & 0 & 6 & 34 & 12 \\ 3 & 5 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{122} 3^{11} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -28, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.86 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 6 & 3 \\ 3 & 5 & 5 & 3 & 10 & 5 \\ 4 & 5 & 2 & 2 & 8 & 4 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 6 & 10 & 8 & 4 & 14 & 8 \\ 3 & 5 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{111} 3^{11} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -28, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.86 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 3 & 5 & 3 \\ 2 & 3 & 1 & 2 & 6 & 3 \\ 3 & 3 & 2 & 1 & 5 & 3 \\ 5 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{11} 3^{11} 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -28, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.86 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 3 \\ 3 & 5 & 10 & 3 & 5 & 5 \\ 5 & 10 & 14 & 4 & 8 & 8 \\ 2 & 3 & 4 & 0 & 2 & 2 \\ 3 & 5 & 8 & 2 & 2 & 4 \\ 3 & 5 & 8 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{12} 3^{11} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -28, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.86 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 6 & 3 \\ 2 & 1 & 3 & 3 & 5 & 3 \\ 2 & 3 & 1 & 2 & 6 & 3 \\ 2 & 3 & 2 & 1 & 5 & 3 \\ 6 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[54] $1^3 1^3 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 9 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 2 & 5 & 2 & 2 & 12 & 4 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 9 & 15 & 12 & 6 & 34 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[57] $1^{11} 1^3 1^3 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 0 & 2 & 2 \\ 2 & 3 & 0 & 3 & 2 & 1 & 3 \\ 2 & 3 & -1 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[56] $1^{11} 1^3 1^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 3 & 5 & 3 \\ 2 & 3 & 1 & 2 & 6 & 3 \\ 3 & 3 & 2 & 1 & 5 & 3 \\ 7 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[58] $1^{11} 1^3 1^3 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 3 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 2 & 6 & 3 \\ 1 & 2 & 0 & 2 & 0 & 5 & 2 \\ 4 & 6 & 0 & 6 & 5 & 13 & 7 \\ 2 & 3 & -1 & 3 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[55] $1^3 1^3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 3 & 5 & 3 \\ 2 & 3 & 1 & 2 & 6 & 3 \\ 2 & 3 & 2 & 1 & 5 & 3 \\ 4 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[60] $1^{12} 1^3 1^3 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 12 & 3 \\ 2 & 1 & 3 & 7 & 10 & 3 \\ 2 & 3 & 1 & 10 & 12 & 3 \\ 8 & 7 & 10 & 25 & 34 & 11 \\ 12 & 10 & 12 & 34 & 42 & 14 \\ 3 & 3 & 3 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

[61] $1^{12} 1^3 1^3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 10 & 3 \\ 2 & 1 & 3 & 3 & 10 & 3 \\ 2 & 3 & 1 & 2 & 12 & 3 \\ 3 & 3 & 2 & 1 & 10 & 3 \\ 10 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[62] $1^{14} 1^{12} 1^3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 3 & 10 & 3 \\ 2 & 3 & 1 & 2 & 12 & 3 \\ 2 & 3 & 2 & 1 & 10 & 3 \\ 12 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[63] $1^{12} 1^3 1^3 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 6 & 3 \\ 3 & 5 & 10 & 3 & 10 & 5 \\ 5 & 10 & 14 & 4 & 16 & 8 \\ 2 & 3 & 4 & 0 & 4 & 2 \\ 6 & 10 & 16 & 4 & 14 & 8 \\ 3 & 5 & 8 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{132} 3^{121} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -34, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 10 & 3 \\ 2 & 1 & 3 & 7 & 10 & 3 \\ 2 & 3 & 1 & 10 & 12 & 3 \\ 9 & 7 & 10 & 25 & 34 & 11 \\ 10 & 10 & 12 & 34 & 42 & 14 \\ 3 & 3 & 3 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{133} 3^{121} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -35, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 10 & 3 \\ 3 & 5 & 10 & 3 & 15 & 5 \\ 7 & 10 & 14 & 4 & 22 & 8 \\ 2 & 3 & 4 & 0 & 7 & 2 \\ 10 & 15 & 22 & 7 & 32 & 12 \\ 3 & 5 & 8 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{122} 3^{121} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -38, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 9 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 4 & 5 & 2 & 2 & 11 & 4 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 9 & 15 & 11 & 7 & 32 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{123} 3^{121} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -38, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 9 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 4 & 5 & 2 & 2 & 12 & 4 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 9 & 15 & 12 & 6 & 34 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{111} 3^{121} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -40, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.70 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 3 \\ 3 & 5 & 5 & 3 & 5 & 5 \\ 3 & 5 & 2 & 2 & 4 & 4 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 3 & 5 & 4 & 2 & 2 & 4 \\ 3 & 5 & 4 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{14} 3^{121} 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -40, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 20 & 3 \\ 2 & 1 & 3 & 2 & 20 & 3 \\ 2 & 3 & 1 & 4 & 24 & 3 \\ 2 & 2 & 4 & 2 & 24 & 4 \\ 20 & 20 & 24 & 24 & 174 & 28 \\ 3 & 3 & 3 & 4 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{14} 3^{121} 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -42, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 5 & 0 & 3 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 12 & 20 & 0 & 8 & 62 & 16 \\ 3 & 5 & 0 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{12} 3^{121} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -43, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.79 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 6 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 4 & 6 & 3 \\ 3 & 2 & 4 & 2 & 6 & 4 \\ 6 & 5 & 6 & 6 & 9 & 7 \\ 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[71] \quad & 1^{11} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -43, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.79 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 3 \\ 2 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 7 \\ 3 & 3 & 3 & 0 & 7 & 4 \end{pmatrix}_{b_0} \\
[72] \quad & 1^{12} 3^1 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -43, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.79 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 3 \\ 2 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 7 \\ 3 & 3 & 3 & 0 & 7 & 4 \end{pmatrix}_{b_0} \\
[73] \quad & 1^{11} 3^1 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -43, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.79 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 4 & 6 & 3 \\ 3 & 2 & 4 & 2 & 6 & 4 \\ 5 & 5 & 6 & 6 & 9 & 7 \\ 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0} \\
[79] \quad & 1^{13} 3^1 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -44, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 & 15 & 3 \\ 2 & 1 & 3 & 2 & 15 & 3 \\ 2 & 3 & 1 & 2 & 18 & 3 \\ 3 & 3 & 2 & 1 & 15 & 3 \\ 15 & 15 & 18 & 15 & 97 & 21 \\ 3 & 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0} \\
[78] \quad & 1^{13} 3^1 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -44, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 7 & 3 \\ 3 & 5 & 5 & 3 & 10 & 5 \\ 3 & 5 & 2 & 2 & 8 & 4 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 7 & 10 & 8 & 4 & 14 & 8 \\ 3 & 5 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0} \\
[77] \quad & 1^{11} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -44, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 2 & 5 & 2 & 2 & 11 & 4 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 8 & 15 & 11 & 7 & 32 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0} \\
[76] \quad & 1^{13} 3^1 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -44, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 12 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 2 & 5 & 2 & 2 & 15 & 4 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 12 & 20 & 15 & 9 & 58 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0} \\
[75] \quad & 1^{14} 3^1 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -44, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 12 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 2 & 5 & 2 & 2 & 16 & 4 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 12 & 20 & 16 & 8 & 62 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0} \\
[74] \quad & 1^{13} 3^1 21^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -44, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 9 & 3 \\ 3 & 5 & 10 & 3 & 15 & 5 \\ 5 & 10 & 14 & 4 & 24 & 8 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 9 & 15 & 24 & 6 & 34 & 12 \\ 3 & 5 & 8 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{12} 1^{11} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -45, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 11 & 3 \\ 3 & 5 & 10 & 3 & 20 & 5 \\ 5 & 10 & 14 & 4 & 30 & 8 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 11 & 20 & 30 & 9 & 58 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{12} 1^{111} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -45, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 6 & 3 \\ 2 & 1 & 3 & 3 & 5 & 3 \\ 2 & 3 & 1 & 2 & 6 & 3 \\ 3 & 3 & 2 & 1 & 5 & 3 \\ 6 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{11} 1^{111} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -45, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 3 & 5 & 3 \\ 2 & 3 & 1 & 2 & 6 & 3 \\ 2 & 3 & 2 & 1 & 5 & 3 \\ 5 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{12} 1^{11} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -50, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 9 & 3 \\ 3 & 5 & 10 & 3 & 15 & 5 \\ 6 & 10 & 14 & 4 & 22 & 8 \\ 2 & 3 & 4 & 0 & 7 & 2 \\ 9 & 15 & 22 & 7 & 32 & 12 \\ 3 & 5 & 8 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{32} 1^{11} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -50, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 6 & 3 \\ 3 & 5 & 10 & 3 & 10 & 5 \\ 7 & 10 & 14 & 4 & 16 & 8 \\ 2 & 3 & 4 & 0 & 4 & 2 \\ 6 & 10 & 16 & 4 & 14 & 8 \\ 3 & 5 & 8 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{11} 1^{11} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -51, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 6 & 3 \\ 3 & 5 & 5 & 3 & 10 & 5 \\ 3 & 5 & 2 & 2 & 7 & 4 \\ 2 & 3 & 2 & 0 & 5 & 2 \\ 6 & 10 & 7 & 5 & 14 & 8 \\ 3 & 5 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{24} 1^{13} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -52, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 12 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 4 & 5 & 2 & 2 & 16 & 4 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 12 & 20 & 16 & 8 & 62 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{12} 1^{11} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -53, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 5 & 0 & 3 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 11 & 20 & 0 & 9 & 58 & 16 \\ 3 & 5 & 0 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[88] $1^{134} 1^{11} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 15 & 3 \\ 3 & 5 & 10 & 3 & 25 & 5 \\ 7 & 10 & 14 & 4 & 38 & 8 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 15 & 25 & 38 & 11 & 92 & 20 \\ 3 & 5 & 8 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[89] $1^{12} 1^{11} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 5 & 0 & 3 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 9 & 15 & 0 & 7 & 32 & 12 \\ 3 & 5 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[90] $1^{12} 1^{11} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 8 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 3 & 5 & 2 & 2 & 12 & 4 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 8 & 15 & 12 & 6 & 34 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[93] $1^{12} 1^{11} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 6 & 3 \\ 3 & 5 & 5 & 3 & 10 & 5 \\ 3 & 5 & 2 & 2 & 8 & 4 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 6 & 10 & 8 & 4 & 14 & 8 \\ 3 & 5 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

[92] $1^{14} 1^{11} 4^1 3^1 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 20 & 3 \\ 2 & 1 & 3 & 3 & 20 & 3 \\ 2 & 3 & 1 & 2 & 24 & 3 \\ 3 & 3 & 2 & 1 & 20 & 3 \\ 20 & 20 & 24 & 20 & 174 & 28 \\ 3 & 3 & 3 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

[91] $1^{14} 1^{11} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 12 & 3 \\ 3 & 5 & 10 & 3 & 20 & 5 \\ 5 & 10 & 14 & 4 & 32 & 8 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 12 & 20 & 32 & 8 & 62 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[96] $1^{12} 1^{11} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 8 & 3 \\ 2 & 1 & 3 & 5 & 7 & 3 \\ 2 & 3 & 1 & 6 & 10 & 3 \\ 4 & 5 & 6 & 9 & 17 & 7 \\ 8 & 7 & 10 & 17 & 25 & 11 \\ 3 & 3 & 3 & 7 & 11 & 4 \end{pmatrix}_{b_0}$$

[97] $1^{22} 1^{11} 3^1 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 6 & 3 \\ 3 & 5 & 15 & 3 & 10 & 5 \\ 8 & 15 & 34 & 6 & 24 & 12 \\ 2 & 3 & 6 & 0 & 4 & 2 \\ 6 & 10 & 24 & 4 & 14 & 8 \\ 3 & 5 & 12 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

[95] $1^{33} 1^{11} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 9 & 3 \\ 2 & 1 & 3 & 5 & 7 & 3 \\ 2 & 3 & 1 & 6 & 10 & 3 \\ 7 & 5 & 6 & 9 & 17 & 7 \\ 9 & 7 & 10 & 17 & 25 & 11 \\ 3 & 3 & 3 & 7 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{133} 1^{13} 3^{121} 2^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -60, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 9 & 3 \\ 3 & 5 & 10 & 3 & 15 & 5 \\ 7 & 10 & 14 & 4 & 24 & 8 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 9 & 15 & 24 & 6 & 34 & 12 \\ 3 & 5 & 8 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[99] \quad 1^{124} 1^{122} 3^{121} 2^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 16 & 3 \\ 2 & 1 & 3 & 10 & 14 & 3 \\ 2 & 3 & 1 & 12 & 20 & 3 \\ 10 & 10 & 12 & 42 & 68 & 14 \\ 16 & 14 & 20 & 68 & 106 & 22 \\ 3 & 3 & 3 & 14 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[98] \quad 1^{111} 1^{12} 3^{121} 2^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 4 & 12 & 3 \\ 3 & 2 & 4 & 2 & 12 & 4 \\ 9 & 10 & 12 & 12 & 42 & 14 \\ 3 & 3 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[100] \quad 1^{111} 1^{12} 3^{121} 2^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 7 & 3 \\ 2 & 1 & 3 & 0 & 8 & 3 \\ 2 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 8 & 8 & 0 & 22 & 10 \\ 3 & 3 & 3 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[101] \quad 1^{133} 1^{12} 3^{121} 2^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 9 & 3 \\ 2 & 1 & 3 & 0 & 8 & 3 \\ 2 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 8 & 8 & 0 & 22 & 10 \\ 3 & 3 & 3 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[102] \quad 1^{133} 1^{124} 3^{121} 2^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 15 & 3 \\ 2 & 1 & 3 & 8 & 16 & 3 \\ 2 & 3 & 1 & 8 & 16 & 3 \\ 9 & 8 & 8 & 22 & 48 & 10 \\ 15 & 16 & 16 & 48 & 94 & 20 \\ 3 & 3 & 3 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[103] \quad 1^{14} 1^{112} 3^{121} 2^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -66, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 16 & 3 \\ 2 & 1 & 3 & 5 & 14 & 3 \\ 2 & 3 & 1 & 6 & 20 & 3 \\ 4 & 5 & 6 & 9 & 34 & 7 \\ 16 & 14 & 20 & 34 & 106 & 22 \\ 3 & 3 & 3 & 7 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[104] \quad 1^{13} 1^{112} 3^{121} 2^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -67, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 9 & 3 \\ 2 & 1 & 3 & 5 & 8 & 3 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 5 & 5 & 6 & 9 & 16 & 7 \\ 9 & 8 & 8 & 16 & 22 & 10 \\ 3 & 3 & 3 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& [110] \ 1^{11} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -67, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5 \\
& \quad T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10] \\
& \quad \begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 3 & 8 & 3 \\ 2 & 3 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 1 & 8 & 3 \\ 7 & 8 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0} \\
& [109] \ 1^{12} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -67, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5 \\
& \quad T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10] \\
& \quad \begin{pmatrix} 1 & 2 & 2 & 6 & 7 & 3 \\ 2 & 1 & 3 & 5 & 8 & 3 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 6 & 5 & 6 & 9 & 16 & 7 \\ 7 & 8 & 8 & 16 & 22 & 10 \\ 3 & 3 & 3 & 7 & 10 & 4 \end{pmatrix}_{b_0} \\
& [108] \ 1^{12} 3^{11} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -67, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0 \\
& \quad T_{\min} = 9.87 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10] \\
& \quad \begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 3 \\ 2 & 1 & 3 & 0 & 7 & 3 \\ 2 & 3 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 11 \\ 3 & 3 & 3 & 0 & 11 & 4 \end{pmatrix}_{b_0} \\
& [107] \ 1^{14} 3^{11} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -67, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0 \\
& \quad T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10] \\
& \quad \begin{pmatrix} 1 & 2 & 2 & 3 & 10 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 10 & 3 \\ 3 & 2 & 4 & 2 & 10 & 4 \\ 10 & 7 & 10 & 10 & 25 & 11 \\ 3 & 3 & 3 & 4 & 11 & 4 \end{pmatrix}_{b_0} \\
& [106] \ 1^{13} 3^{11} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -67, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0 \\
& \quad T_{\min} = 9.87 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10] \\
& \quad \begin{pmatrix} 1 & 2 & 2 & 1 & 9 & 3 \\ 2 & 1 & 3 & 0 & 7 & 3 \\ 2 & 3 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 7 & 10 & 0 & 25 & 11 \\ 3 & 3 & 3 & 0 & 11 & 4 \end{pmatrix}_{b_0} \\
& [105] \ 1^{11} 3^{11} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -67, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0 \\
& \quad T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10] \\
& \quad \begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 10 & 3 \\ 3 & 2 & 4 & 2 & 10 & 4 \\ 7 & 7 & 10 & 10 & 25 & 11 \\ 3 & 3 & 3 & 4 & 11 & 4 \end{pmatrix}_{b_0} \\
& [111] \ 1^{13} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -67, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5 \\
& \quad T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10] \\
& \quad \begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 1 & 8 & 3 \\ 9 & 8 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0} \\
& [112] \ 1^{11} 3^{11} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -67, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5 \\
& \quad T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10] \\
& \quad \begin{pmatrix} 1 & 3 & 5 & 2 & 10 & 3 \\ 3 & 5 & 10 & 3 & 20 & 5 \\ 5 & 10 & 14 & 4 & 30 & 8 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 10 & 20 & 30 & 9 & 58 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0} \\
& [113] \ 1^{23} 3^{11} 21^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -70, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.6 \\
& \quad T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 9 & 3 \\ 3 & 5 & 15 & 3 & 15 & 5 \\ 8 & 15 & 34 & 6 & 36 & 12 \\ 2 & 3 & 6 & 0 & 6 & 2 \\ 9 & 15 & 36 & 6 & 34 & 12 \\ 3 & 5 & 12 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[114] \quad 1^{122} 1^{11} 3^{121} 1^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -70, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 13 & 3 \\ 3 & 5 & 15 & 3 & 25 & 5 \\ 8 & 15 & 34 & 6 & 57 & 12 \\ 2 & 3 & 6 & 0 & 11 & 2 \\ 13 & 25 & 57 & 11 & 92 & 20 \\ 3 & 5 & 12 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[117] \quad 1^{23} 1^{11} 3^{121} 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -72, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 10 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 4 & 5 & 2 & 2 & 11 & 4 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 10 & 15 & 11 & 7 & 32 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[115] \quad 21^{132} 3^{121} 1^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -72, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 10 & 3 \\ 3 & 1 & 3 & 17 & 10 & 3 \\ 2 & 3 & 1 & 22 & 12 & 3 \\ 17 & 17 & 22 & 137 & 78 & 25 \\ 10 & 10 & 12 & 78 & 42 & 14 \\ 3 & 3 & 3 & 25 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[116] \quad 1^{23} 1^{11} 3^{121} 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -72, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 12 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 4 & 5 & 2 & 2 & 15 & 4 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 12 & 20 & 15 & 9 & 58 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[119] \quad 1^{34} 1^{11} 3^{121} 1^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -74, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 12 & 3 \\ 3 & 5 & 10 & 3 & 20 & 5 \\ 7 & 10 & 14 & 4 & 32 & 8 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 12 & 20 & 32 & 8 & 62 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[120] \quad 1^{12} 1^{11} 3^{121} 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -74, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 2 & 5 & 2 & 2 & 15 & 4 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 11 & 20 & 15 & 9 & 58 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[118] \quad 1^{14} 1^{11} 3^{121} 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -74, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 15 & 3 \\ 3 & 5 & 5 & 3 & 25 & 5 \\ 2 & 5 & 2 & 2 & 19 & 4 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 15 & 25 & 19 & 11 & 92 & 20 \\ 3 & 5 & 4 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[121] \quad 1^{33} 1^{11} 3^{121} 1^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -75, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 14 & 3 \\ 3 & 5 & 15 & 3 & 25 & 5 \\ 9 & 15 & 34 & 6 & 57 & 12 \\ 2 & 3 & 6 & 0 & 11 & 2 \\ 14 & 25 & 57 & 11 & 92 & 20 \\ 3 & 5 & 12 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[122] \quad 1^{13} 1^{114} 3^{121} 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -75, \quad (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 15 & 3 \\ 2 & 1 & 3 & 5 & 16 & 3 \\ 2 & 3 & 1 & 6 & 16 & 3 \\ 4 & 5 & 6 & 9 & 32 & 7 \\ 15 & 16 & 16 & 32 & 94 & 20 \\ 3 & 3 & 3 & 7 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{\overline{1}22} 3^{\overline{1}} 21^{\overline{1}22} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 6 & 3 \\ 3 & 5 & 10 & 3 & 10 & 5 \\ 6 & 10 & 14 & 4 & 16 & 8 \\ 2 & 3 & 4 & 0 & 4 & 2 \\ 6 & 10 & 16 & 4 & 14 & 8 \\ 3 & 5 & 8 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{\overline{1}34} 3^{\overline{1}} 21^{\overline{1}2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 14 & 3 \\ 2 & 1 & 3 & 5 & 13 & 3 \\ 2 & 3 & 1 & 6 & 14 & 3 \\ 7 & 5 & 6 & 9 & 27 & 7 \\ 14 & 13 & 14 & 27 & 65 & 17 \\ 3 & 3 & 3 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{\overline{1}3} 3^{\overline{1}} 21^{\overline{1}2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 13 & 3 \\ 2 & 1 & 3 & 5 & 13 & 3 \\ 2 & 3 & 1 & 6 & 14 & 3 \\ 4 & 5 & 6 & 9 & 27 & 7 \\ 13 & 13 & 14 & 27 & 65 & 17 \\ 3 & 3 & 3 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[126] \ 1^{\overline{1}4} 3^{\overline{1}} 21^{\overline{1}3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 10 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 3 & 5 & 2 & 2 & 12 & 4 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 10 & 15 & 12 & 6 & 34 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{\overline{1}2} 3^{\overline{1}} 21^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 3 \\ 2 & 1 & 3 & 0 & 8 & 3 \\ 2 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 8 & 8 & 0 & 22 & 10 \\ 3 & 3 & 3 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{\overline{1}3} 3^{\overline{1}} 21^{\overline{1}3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 11 & 3 \\ 2 & 1 & 3 & 0 & 9 & 3 \\ 2 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 9 & 14 & 0 & 49 & 15 \\ 3 & 3 & 3 & 0 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{\overline{1}13} 3^{\overline{1}} 21^{\overline{1}24} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 14 & 3 \\ 3 & 5 & 10 & 3 & 25 & 5 \\ 5 & 10 & 14 & 4 & 38 & 8 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 14 & 25 & 38 & 11 & 92 & 20 \\ 3 & 5 & 8 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{\overline{1}4} 3^{\overline{1}} 21^{\overline{1}3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 12 & 3 \\ 2 & 1 & 3 & 0 & 9 & 3 \\ 2 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 9 & 14 & 0 & 49 & 15 \\ 3 & 3 & 3 & 0 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[130] \quad 1^{13} 1^{11} 3^{12} 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -83, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 5 & 0 & 3 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 12 & 20 & 0 & 9 & 58 & 16 \\ 3 & 5 & 0 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[132] \quad 1^{13} 1^{11} 3^{12} 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -83, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 5 & 0 & 3 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 14 & 25 & 0 & 11 & 92 & 20 \\ 3 & 5 & 0 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[128] \quad 1^{13} 1^{12} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -83, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 3 & 10 & 3 \\ 2 & 3 & 1 & 2 & 12 & 3 \\ 2 & 3 & 2 & 1 & 10 & 3 \\ 11 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[129] \quad 1^{13} 1^{12} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -83, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 11 & 3 \\ 2 & 1 & 3 & 3 & 10 & 3 \\ 2 & 3 & 1 & 2 & 12 & 3 \\ 3 & 3 & 2 & 1 & 10 & 3 \\ 11 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[135] \quad 1^{14} 1^{13} 3^{12} 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -84, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 10 & 3 \\ 3 & 5 & 10 & 3 & 15 & 5 \\ 6 & 10 & 14 & 4 & 24 & 8 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 10 & 15 & 24 & 6 & 34 & 12 \\ 3 & 5 & 8 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[136] \quad 1^{14} 1^{13} 3^{12} 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -84, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 3 & 17 & 3 \\ 2 & 3 & 1 & 2 & 22 & 3 \\ 2 & 3 & 2 & 1 & 17 & 3 \\ 18 & 17 & 22 & 17 & 137 & 25 \\ 3 & 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[137] \quad 1^{14} 1^{13} 3^{12} 21^{134} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -84, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 12 & 3 \\ 3 & 5 & 15 & 3 & 20 & 5 \\ 8 & 15 & 34 & 6 & 48 & 12 \\ 2 & 3 & 6 & 0 & 8 & 2 \\ 12 & 20 & 48 & 8 & 62 & 16 \\ 3 & 5 & 12 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[138] \quad 1^{11} 1^{11} 3^{12} 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -85, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 5 & 3 \\ 2 & 1 & 3 & 5 & 5 & 3 \\ 2 & 3 & 1 & 6 & 6 & 3 \\ 4 & 5 & 6 & 9 & 11 & 7 \\ 5 & 5 & 6 & 11 & 9 & 7 \\ 3 & 3 & 3 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

[139] $1^{1^{32}} 1^{111} 3^1 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 6 & 3 \\ 2 & 1 & 3 & 5 & 5 & 3 \\ 2 & 3 & 1 & 6 & 6 & 3 \\ 7 & 5 & 6 & 9 & 11 & 7 \\ 6 & 5 & 6 & 11 & 9 & 7 \\ 3 & 3 & 3 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

[140] $1^{1^{13}} 1^{13} 3^1 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 9 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 3 & 5 & 2 & 2 & 12 & 4 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 9 & 15 & 12 & 6 & 34 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[141] $1^{1^{12}} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 3 & 10 & 3 \\ 2 & 3 & 1 & 2 & 12 & 3 \\ 2 & 3 & 2 & 1 & 10 & 3 \\ 9 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[142] $1^{1^{21}} 1^{112} 3^1 21^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 9 & 3 \\ 2 & 1 & 3 & 7 & 10 & 3 \\ 2 & 3 & 1 & 10 & 12 & 3 \\ 8 & 7 & 10 & 25 & 34 & 11 \\ 9 & 10 & 12 & 34 & 42 & 14 \\ 3 & 3 & 3 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

[143] $1^{1^{12}} 1^{113} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 10 & 3 \\ 3 & 2 & 4 & 2 & 10 & 4 \\ 8 & 7 & 10 & 10 & 25 & 11 \\ 3 & 3 & 3 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

[144] $1^{1^{13}} 1^{113} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 10 & 3 \\ 3 & 2 & 4 & 2 & 10 & 4 \\ 9 & 7 & 10 & 10 & 25 & 11 \\ 3 & 3 & 3 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

[145] $1^{1^{13}} 1^{13} 3^1 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 11 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 3 & 5 & 2 & 2 & 16 & 4 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 11 & 20 & 16 & 8 & 62 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[146] $1^{1^{24}} 1^{12} 3^1 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 18 & 3 \\ 3 & 5 & 5 & 3 & 30 & 5 \\ 4 & 5 & 2 & 2 & 22 & 4 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 18 & 30 & 22 & 14 & 134 & 24 \\ 3 & 5 & 4 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[147] $1^{1^3} 1^{113} 3^1 21^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -93, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 17 & 3 \\ 2 & 1 & 3 & 3 & 17 & 3 \\ 2 & 3 & 1 & 2 & 22 & 3 \\ 2 & 3 & 2 & 1 & 17 & 3 \\ 17 & 17 & 22 & 17 & 137 & 25 \\ 3 & 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[148] \quad 1^{134} 1^{11} 3^{121} 1^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -93, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 13 & 3 \\ 3 & 5 & 10 & 3 & 20 & 5 \\ 7 & 10 & 14 & 4 & 30 & 8 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 13 & 20 & 30 & 9 & 58 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[149] \quad 1^{43} 1^{13} 1^{21} 1^{133} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -94, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.6$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 9 & 3 \\ 3 & 5 & 15 & 3 & 15 & 5 \\ 10 & 15 & 34 & 6 & 36 & 12 \\ 2 & 3 & 6 & 0 & 6 & 2 \\ 9 & 15 & 36 & 6 & 34 & 12 \\ 3 & 5 & 12 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[151] \quad 1^{12} 1^{112} 3^{121} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 3 & 8 & 3 \\ 2 & 3 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 1 & 8 & 3 \\ 8 & 8 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[156] \quad 1^{24} 1^{11} 3^{121} 1^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 13 & 3 \\ 3 & 5 & 10 & 3 & 20 & 5 \\ 6 & 10 & 14 & 4 & 30 & 8 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 13 & 20 & 30 & 9 & 58 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[155] \quad 1^{12} 1^{11} 3^{121} 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 13 & 3 \\ 3 & 5 & 5 & 3 & 25 & 5 \\ 2 & 5 & 2 & 2 & 19 & 4 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 13 & 25 & 19 & 11 & 92 & 20 \\ 3 & 5 & 4 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[150] \quad 1^{22} 1^{11} 3^{121} 1^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 13 & 3 \\ 3 & 5 & 10 & 3 & 25 & 5 \\ 6 & 10 & 14 & 4 & 38 & 8 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 13 & 25 & 38 & 11 & 92 & 20 \\ 3 & 5 & 8 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[152] \quad 1^{23} 1^{13} 1^{21} 1^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 11 & 3 \\ 3 & 5 & 10 & 3 & 20 & 5 \\ 6 & 10 & 14 & 4 & 32 & 8 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 11 & 20 & 32 & 8 & 62 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[153] \quad 1^{12} 1^{12} 3^{121} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 5 & 0 & 3 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 14 & 25 & 0 & 12 & 94 & 20 \\ 3 & 5 & 0 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[154] $1^{1^2}1^{12}3^121^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 3 & 8 & 3 \\ 2 & 3 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 1 & 8 & 3 \\ 8 & 8 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[157] $1^{1^{33}}1^{12}3^121^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 9 & 3 \\ 2 & 1 & 3 & 5 & 8 & 3 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 7 & 5 & 6 & 9 & 16 & 7 \\ 9 & 8 & 8 & 16 & 22 & 10 \\ 3 & 3 & 3 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

[158] $1^{1^{11}}1^{11}3^121^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 8 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 3 & 5 & 2 & 2 & 11 & 4 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 8 & 15 & 11 & 7 & 32 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[159] $1^{1^1}1^{12}3^121^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 7 & 3 \\ 2 & 1 & 3 & 5 & 8 & 3 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 4 & 5 & 6 & 9 & 16 & 7 \\ 7 & 8 & 8 & 16 & 22 & 10 \\ 3 & 3 & 3 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

[160] $1^{1^{12}}1^{12}3^121^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 10 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 4 & 12 & 3 \\ 3 & 2 & 4 & 2 & 12 & 4 \\ 10 & 10 & 12 & 12 & 42 & 14 \\ 3 & 3 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

[161] $1^{1^{14}}1^{12}3^121^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 4 & 12 & 3 \\ 3 & 2 & 4 & 2 & 12 & 4 \\ 12 & 10 & 12 & 12 & 42 & 14 \\ 3 & 3 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

[162] $1^{1^{24}}1^{123}3^121^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -102, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 14 & 3 \\ 2 & 1 & 3 & 10 & 13 & 3 \\ 2 & 3 & 1 & 12 & 14 & 3 \\ 10 & 10 & 12 & 42 & 54 & 14 \\ 14 & 13 & 14 & 54 & 65 & 17 \\ 3 & 3 & 3 & 14 & 17 & 4 \end{pmatrix}_{b_0}$$

[163] $1^{1^{43}}1^{123}3^121^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -102, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 13 & 3 \\ 2 & 1 & 3 & 10 & 13 & 3 \\ 2 & 3 & 1 & 12 & 14 & 3 \\ 12 & 10 & 12 & 42 & 54 & 14 \\ 13 & 13 & 14 & 54 & 65 & 17 \\ 3 & 3 & 3 & 14 & 17 & 4 \end{pmatrix}_{b_0}$$

[164] $1^{1^2}1^{12}3^121^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -102, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 3 & 10 & 3 \\ 2 & 3 & 1 & 2 & 12 & 3 \\ 2 & 3 & 2 & 1 & 10 & 3 \\ 10 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[165] \quad 1^{112} 3^{121} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -102, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 3 & 10 & 3 \\ 2 & 3 & 1 & 2 & 12 & 3 \\ 3 & 3 & 2 & 1 & 10 & 3 \\ 12 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[166] \quad 1^{123} 3^{121} 12^{23} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -106, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 9 & 3 \\ 3 & 5 & 10 & 3 & 15 & 5 \\ 6 & 10 & 14 & 4 & 24 & 8 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 9 & 15 & 24 & 6 & 34 & 12 \\ 3 & 5 & 8 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[167] \quad 1^{122} 3^{121} 21^{223} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -106, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 11 & 3 \\ 3 & 5 & 10 & 3 & 20 & 5 \\ 6 & 10 & 14 & 4 & 30 & 8 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 11 & 20 & 30 & 9 & 58 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[168] \quad 1^{111} 3^{121} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -107, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 10 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 3 & 5 & 2 & 2 & 15 & 4 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 10 & 20 & 15 & 9 & 58 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[169] \quad 1^{113} 3^{121} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -107, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 10 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 3 & 5 & 2 & 2 & 11 & 4 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 10 & 15 & 11 & 7 & 32 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[174] \quad 1^{144} 3^{121} 13^{34} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -108, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 12 & 3 \\ 3 & 5 & 15 & 3 & 20 & 5 \\ 10 & 15 & 34 & 6 & 48 & 12 \\ 2 & 3 & 6 & 0 & 8 & 2 \\ 12 & 20 & 48 & 8 & 62 & 16 \\ 3 & 5 & 12 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[175] \quad 1^{133} 3^{121} 13^{34} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -108, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 11 & 3 \\ 3 & 5 & 15 & 3 & 20 & 5 \\ 9 & 15 & 34 & 6 & 48 & 12 \\ 2 & 3 & 6 & 0 & 8 & 2 \\ 11 & 20 & 48 & 8 & 62 & 16 \\ 3 & 5 & 12 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[176] \quad 1^{122} 3^{121} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -108, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 11 & 3 \\ 2 & 1 & 3 & 8 & 10 & 3 \\ 2 & 3 & 1 & 8 & 12 & 3 \\ 9 & 8 & 8 & 22 & 32 & 10 \\ 11 & 10 & 12 & 32 & 42 & 14 \\ 3 & 3 & 3 & 10 & 14 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[173] \quad & 1^{124} 1^{111} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 6 & 10 & 3 \\ 2 & 1 & 3 & 5 & 7 & 3 \\ 2 & 3 & 1 & 6 & 10 & 3 \\ 6 & 5 & 6 & 9 & 17 & 7 \\ 10 & 7 & 10 & 17 & 25 & 11 \\ 3 & 3 & 3 & 7 & 11 & 4 \end{pmatrix}_{b_0} \\
[170] \quad & 1^{13} 1^{11} 3^1 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 14 & 3 \\ 3 & 5 & 5 & 3 & 25 & 5 \\ 2 & 5 & 2 & 2 & 19 & 4 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 14 & 25 & 19 & 11 & 92 & 20 \\ 3 & 5 & 4 & 2 & 20 & 4 \end{pmatrix}_{b_0} \\
[171] \quad & 1^{13} 1^{122} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 7 & 11 & 3 \\ 2 & 1 & 3 & 8 & 10 & 3 \\ 2 & 3 & 1 & 8 & 12 & 3 \\ 7 & 8 & 8 & 22 & 32 & 10 \\ 11 & 10 & 12 & 32 & 42 & 14 \\ 3 & 3 & 3 & 10 & 14 & 4 \end{pmatrix}_{b_0} \\
[172] \quad & 1^{11} 1^{111} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 5 & 7 & 3 \\ 2 & 1 & 3 & 5 & 7 & 3 \\ 2 & 3 & 1 & 6 & 10 & 3 \\ 5 & 5 & 6 & 9 & 17 & 7 \\ 7 & 7 & 10 & 17 & 25 & 11 \\ 3 & 3 & 3 & 7 & 11 & 4 \end{pmatrix}_{b_0} \\
[177] \quad & 1^{34} 1^{12} 3^1 21^{134} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -109, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 9 & 2 & 15 & 3 \\ 3 & 5 & 15 & 3 & 25 & 5 \\ 9 & 15 & 34 & 6 & 57 & 12 \\ 2 & 3 & 6 & 0 & 11 & 2 \\ 15 & 25 & 57 & 11 & 92 & 20 \\ 3 & 5 & 12 & 2 & 20 & 4 \end{pmatrix}_{b_0} \\
[179] \quad & 1^{13} 1^{12} 3^1 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -110, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 17 & 3 \\ 3 & 5 & 5 & 3 & 30 & 5 \\ 2 & 5 & 2 & 2 & 22 & 4 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 17 & 30 & 22 & 14 & 134 & 24 \\ 3 & 5 & 4 & 2 & 24 & 4 \end{pmatrix}_{b_0} \\
[178] \quad & 1^{24} 1^{11} 3^1 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -110, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 2 & 15 & 3 \\ 3 & 5 & 5 & 3 & 25 & 5 \\ 4 & 5 & 2 & 2 & 19 & 4 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 15 & 25 & 19 & 11 & 92 & 20 \\ 3 & 5 & 4 & 2 & 20 & 4 \end{pmatrix}_{b_0} \\
[182] \quad & 1^{13} 1^{12} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -112, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 & 11 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 4 & 12 & 3 \\ 3 & 2 & 4 & 2 & 12 & 4 \\ 11 & 10 & 12 & 12 & 42 & 14 \\ 3 & 3 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0} \\
[181] \quad & 1^{22} 1^{111} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -112, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 8 & 3 \\ 2 & 1 & 3 & 5 & 7 & 3 \\ 2 & 3 & 1 & 6 & 10 & 3 \\ 6 & 5 & 6 & 9 & 17 & 7 \\ 8 & 7 & 10 & 17 & 25 & 11 \\ 3 & 3 & 3 & 7 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[180] \quad 1^{113} 3^{121} 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -112, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.86 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 9 & 3 \\ 2 & 1 & 3 & 5 & 7 & 3 \\ 2 & 3 & 1 & 6 & 10 & 3 \\ 5 & 5 & 6 & 9 & 17 & 7 \\ 9 & 7 & 10 & 17 & 25 & 11 \\ 3 & 3 & 3 & 7 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[184] \quad 1^{112} 3^{121} 1^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -114, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 10 & 3 \\ 2 & 1 & 3 & 7 & 10 & 3 \\ 2 & 3 & 1 & 10 & 12 & 3 \\ 7 & 7 & 10 & 25 & 34 & 11 \\ 10 & 10 & 12 & 34 & 42 & 14 \\ 3 & 3 & 3 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[183] \quad 1^{112} 3^{121} 1^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -114, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 12 & 3 \\ 2 & 1 & 3 & 7 & 10 & 3 \\ 2 & 3 & 1 & 10 & 12 & 3 \\ 10 & 7 & 10 & 25 & 34 & 11 \\ 12 & 10 & 12 & 34 & 42 & 14 \\ 3 & 3 & 3 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[185] \quad 1^{113} 3^{121} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -116, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 16 & 3 \\ 2 & 1 & 3 & 5 & 15 & 3 \\ 2 & 3 & 1 & 6 & 18 & 3 \\ 4 & 5 & 6 & 9 & 33 & 7 \\ 16 & 15 & 18 & 33 & 97 & 21 \\ 3 & 3 & 3 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[186] \quad 1^{113} 3^{121} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -117, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 3 & 13 & 3 \\ 2 & 3 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 1 & 13 & 3 \\ 12 & 13 & 14 & 13 & 65 & 17 \\ 3 & 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[187] \quad 1^{112} 3^{121} 1^{24} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -117, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 13 & 3 \\ 3 & 5 & 10 & 3 & 25 & 5 \\ 5 & 10 & 14 & 4 & 38 & 8 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 13 & 25 & 38 & 11 & 92 & 20 \\ 3 & 5 & 8 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[189] \quad 1^{114} 3^{121} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -117, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 5 & 0 & 3 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 15 & 25 & 0 & 11 & 92 & 20 \\ 3 & 5 & 0 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[188] \quad 1^{111} 3^{121} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -117, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.82 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 5 & 3 \\ 2 & 1 & 3 & 5 & 5 & 3 \\ 2 & 3 & 1 & 6 & 6 & 3 \\ 6 & 5 & 6 & 9 & 11 & 7 \\ 5 & 5 & 6 & 11 & 9 & 7 \\ 3 & 3 & 3 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

[190] $1^{124} 1^{11} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 2 & 13 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 4 & 5 & 2 & 2 & 15 & 4 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 13 & 20 & 15 & 9 & 58 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$

[191] $21^{123} 3^1 21^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 12 & 17 & 3 \\ 3 & 1 & 3 & 12 & 17 & 3 \\ 2 & 3 & 1 & 16 & 22 & 3 \\ 12 & 12 & 16 & 70 & 100 & 18 \\ 17 & 17 & 22 & 100 & 137 & 25 \\ 3 & 3 & 3 & 18 & 25 & 4 \end{pmatrix}_{b_0}$

[192] $1^{134} 1^{13} 21^{144} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -122, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 2 & 12 & 3 \\ 3 & 5 & 20 & 3 & 20 & 5 \\ 11 & 20 & 62 & 8 & 64 & 16 \\ 2 & 3 & 8 & 0 & 8 & 2 \\ 12 & 20 & 64 & 8 & 62 & 16 \\ 3 & 5 & 16 & 2 & 16 & 4 \end{pmatrix}_{b_0}$

[193] $1^{14} 1^{12} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 12 & 3 \\ 2 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 10 & 12 & 0 & 42 & 14 \\ 3 & 3 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$

[194] $1^{12} 1^{12} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 3 \\ 2 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 14 \\ 3 & 3 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$

[195] $1^{13} 1^{11} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 7 & 15 & 3 \\ 2 & 1 & 3 & 7 & 15 & 3 \\ 2 & 3 & 1 & 10 & 18 & 3 \\ 7 & 7 & 10 & 25 & 51 & 11 \\ 15 & 15 & 18 & 51 & 97 & 21 \\ 3 & 3 & 3 & 11 & 21 & 4 \end{pmatrix}_{b_0}$

[196] $1^{12} 1^{11} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 9 & 3 \\ 3 & 5 & 5 & 3 & 15 & 5 \\ 3 & 5 & 2 & 2 & 11 & 4 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 9 & 15 & 11 & 7 & 32 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[197] $1^{14} 1^{11} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 16 & 3 \\ 2 & 1 & 3 & 3 & 15 & 3 \\ 2 & 3 & 1 & 2 & 18 & 3 \\ 3 & 3 & 2 & 1 & 15 & 3 \\ 16 & 15 & 18 & 15 & 97 & 21 \\ 3 & 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$

[200] $1^{13} 1^{11} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 11 & 3 \\ 2 & 1 & 3 & 7 & 9 & 3 \\ 2 & 3 & 1 & 10 & 14 & 3 \\ 7 & 7 & 10 & 25 & 37 & 11 \\ 11 & 9 & 14 & 37 & 49 & 15 \\ 3 & 3 & 3 & 11 & 15 & 4 \end{pmatrix}_{b_0}$$

[199] $1^{\overline{144}} 1^{\overline{111}} 3^{\overline{1}} 21^{\overline{123}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 12 & 3 \\ 2 & 1 & 3 & 7 & 9 & 3 \\ 2 & 3 & 1 & 10 & 14 & 3 \\ 10 & 7 & 10 & 25 & 37 & 11 \\ 12 & 9 & 14 & 37 & 49 & 15 \\ 3 & 3 & 3 & 11 & 15 & 4 \end{pmatrix}_{b_0}$$

[198] $1^{\overline{13}} 1^{\overline{12}} 3^{\overline{1}} 21^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 5 & 0 & 3 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 15 & 25 & 0 & 12 & 94 & 20 \\ 3 & 5 & 0 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[201] $1^{\overline{14}} 1^{\overline{3}} 3^{\overline{1}} 21^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 12 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 3 & 5 & 2 & 2 & 16 & 4 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 12 & 20 & 16 & 8 & 62 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[202] $1^{\overline{23}} 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{123}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 12 & 3 \\ 3 & 5 & 10 & 3 & 20 & 5 \\ 6 & 10 & 14 & 4 & 30 & 8 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 12 & 20 & 30 & 9 & 58 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[203] $1^{\overline{33}} 1^{\overline{3}} 3^{\overline{1}} 21^{\overline{133}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 9 & 3 \\ 3 & 5 & 15 & 3 & 15 & 5 \\ 9 & 15 & 34 & 6 & 36 & 12 \\ 2 & 3 & 6 & 0 & 6 & 2 \\ 9 & 15 & 36 & 6 & 34 & 12 \\ 3 & 5 & 12 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[204] $1^{\overline{14}} 1^{\overline{114}} 3^{\overline{1}} 21^{\overline{124}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -138, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 20 & 3 \\ 2 & 1 & 3 & 7 & 20 & 3 \\ 2 & 3 & 1 & 10 & 24 & 3 \\ 7 & 7 & 10 & 25 & 68 & 11 \\ 20 & 20 & 24 & 68 & 174 & 28 \\ 3 & 3 & 3 & 11 & 28 & 4 \end{pmatrix}_{b_0}$$

[205] $1^{\overline{2}} 1^{\overline{12}} 3^{\overline{1}} 21^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 16 & 3 \\ 3 & 5 & 5 & 3 & 30 & 5 \\ 2 & 5 & 2 & 2 & 22 & 4 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 16 & 30 & 22 & 14 & 134 & 24 \\ 3 & 5 & 4 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[206] $1^{\overline{13}} 1^{\overline{12}} 3^{\overline{1}} 21^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 11 & 3 \\ 2 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 10 & 12 & 0 & 42 & 14 \\ 3 & 3 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

[207] $1^{12} 1^{13} 3^1 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 14 & 3 \\ 2 & 1 & 3 & 2 & 15 & 3 \\ 2 & 3 & 1 & 4 & 18 & 3 \\ 3 & 2 & 4 & 2 & 18 & 4 \\ 14 & 15 & 18 & 18 & 97 & 21 \\ 3 & 3 & 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$

[208] $1^{122} 1^{111} 3^1 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 6 & 6 & 3 \\ 2 & 1 & 3 & 5 & 5 & 3 \\ 2 & 3 & 1 & 6 & 6 & 3 \\ 6 & 5 & 6 & 9 & 11 & 7 \\ 6 & 5 & 6 & 11 & 9 & 7 \\ 3 & 3 & 3 & 7 & 7 & 4 \end{pmatrix}_{b_0}$

[209] $1^{124} 1^{13} 3^1 21^{124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 2 & 12 & 3 \\ 3 & 5 & 10 & 3 & 20 & 5 \\ 6 & 10 & 14 & 4 & 32 & 8 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 12 & 20 & 32 & 8 & 62 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0}$

[210] $1^{111} 1^{111} 3^1 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 5 & 5 & 3 \\ 2 & 1 & 3 & 5 & 5 & 3 \\ 2 & 3 & 1 & 6 & 6 & 3 \\ 5 & 5 & 6 & 9 & 11 & 7 \\ 5 & 5 & 6 & 11 & 9 & 7 \\ 3 & 3 & 3 & 7 & 7 & 4 \end{pmatrix}_{b_0}$

[211] $1^{13} 1^{12} 3^1 21^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 5 & 0 & 3 & 30 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 17 & 30 & 0 & 14 & 134 & 24 \\ 3 & 5 & 0 & 2 & 24 & 4 \end{pmatrix}_{b_0}$

[212] $21^{111} 1^{13} 3^1 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 5 & 3 \\ 3 & 1 & 3 & 5 & 5 & 3 \\ 2 & 3 & 1 & 6 & 6 & 3 \\ 5 & 5 & 6 & 9 & 11 & 7 \\ 5 & 5 & 6 & 11 & 9 & 7 \\ 3 & 3 & 3 & 7 & 7 & 4 \end{pmatrix}_{b_0}$

[213] $1^{143} 1^{123} 3^1 21^{133} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 14 & 15 & 3 \\ 2 & 1 & 3 & 12 & 15 & 3 \\ 2 & 3 & 1 & 16 & 18 & 3 \\ 14 & 12 & 16 & 70 & 84 & 18 \\ 15 & 15 & 18 & 84 & 97 & 21 \\ 3 & 3 & 3 & 18 & 21 & 4 \end{pmatrix}_{b_0}$

[214] $1^{143} 1^{123} 3^1 21^{144} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 16 & 17 & 3 \\ 2 & 1 & 3 & 14 & 17 & 3 \\ 2 & 3 & 1 & 20 & 22 & 3 \\ 16 & 14 & 20 & 106 & 122 & 22 \\ 17 & 17 & 22 & 122 & 137 & 25 \\ 3 & 3 & 3 & 22 & 25 & 4 \end{pmatrix}_{b_0}$

[215] $1^{13} 1^{112} 3^1 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 13 & 3 \\ 2 & 1 & 3 & 5 & 12 & 3 \\ 2 & 3 & 1 & 6 & 16 & 3 \\ 4 & 5 & 6 & 9 & 28 & 7 \\ 13 & 12 & 16 & 28 & 70 & 18 \\ 3 & 3 & 3 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

[216] $1^{123} 1^{112} 3^{121} 1^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 13 & 3 \\ 2 & 1 & 3 & 9 & 12 & 3 \\ 2 & 3 & 1 & 14 & 16 & 3 \\ 11 & 9 & 14 & 49 & 60 & 15 \\ 13 & 12 & 16 & 60 & 70 & 18 \\ 3 & 3 & 3 & 15 & 18 & 4 \end{pmatrix}_{b_0}$$

[217] $1^{123} 1^{111} 3^{121} 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 9 & 3 \\ 2 & 1 & 3 & 5 & 7 & 3 \\ 2 & 3 & 1 & 6 & 10 & 3 \\ 6 & 5 & 6 & 9 & 17 & 7 \\ 9 & 7 & 10 & 17 & 25 & 11 \\ 3 & 3 & 3 & 7 & 11 & 4 \end{pmatrix}_{b_0}$$

[218] $1^{112} 1^{111} 3^{121} 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 8 & 3 \\ 2 & 1 & 3 & 5 & 7 & 3 \\ 2 & 3 & 1 & 6 & 10 & 3 \\ 5 & 5 & 6 & 9 & 17 & 7 \\ 8 & 7 & 10 & 17 & 25 & 11 \\ 3 & 3 & 3 & 7 & 11 & 4 \end{pmatrix}_{b_0}$$

[219] $1^{143} 1^{111} 3^{121} 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 9 & 3 \\ 2 & 1 & 3 & 7 & 7 & 3 \\ 2 & 3 & 1 & 10 & 10 & 3 \\ 10 & 7 & 10 & 25 & 27 & 11 \\ 9 & 7 & 10 & 27 & 25 & 11 \\ 3 & 3 & 3 & 11 & 11 & 4 \end{pmatrix}_{b_0}$$

[220] $1^{112} 1^{111} 3^{121} 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 8 & 3 \\ 2 & 1 & 3 & 7 & 7 & 3 \\ 2 & 3 & 1 & 10 & 10 & 3 \\ 7 & 7 & 10 & 25 & 27 & 11 \\ 8 & 7 & 10 & 27 & 25 & 11 \\ 3 & 3 & 3 & 11 & 11 & 4 \end{pmatrix}_{b_0}$$

[221] $1^{122} 1^{113} 3^{121} 1^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 14 & 3 \\ 2 & 1 & 3 & 7 & 15 & 3 \\ 2 & 3 & 1 & 10 & 18 & 3 \\ 8 & 7 & 10 & 25 & 51 & 11 \\ 14 & 15 & 18 & 51 & 97 & 21 \\ 3 & 3 & 3 & 11 & 21 & 4 \end{pmatrix}_{b_0}$$

[222] $1^{122} 1^{113} 3^{121} 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 12 & 3 \\ 2 & 1 & 3 & 5 & 13 & 3 \\ 2 & 3 & 1 & 6 & 14 & 3 \\ 6 & 5 & 6 & 9 & 27 & 7 \\ 12 & 13 & 14 & 27 & 65 & 17 \\ 3 & 3 & 3 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

[223] $1^{122} 1^{112} 3^{121} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 8 & 3 \\ 2 & 1 & 3 & 5 & 8 & 3 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 6 & 5 & 6 & 9 & 16 & 7 \\ 8 & 8 & 8 & 16 & 22 & 10 \\ 3 & 3 & 3 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

[224] $1^{1^{12}}3^{1^{11}}21^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 8 & 3 \\ 2 & 1 & 3 & 5 & 8 & 3 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 5 & 5 & 6 & 9 & 16 & 7 \\ 8 & 8 & 8 & 16 & 22 & 10 \\ 3 & 3 & 3 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

[225] $1^{1^{33}}3^{1^{21}}21^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 13 & 3 \\ 2 & 1 & 3 & 8 & 13 & 3 \\ 2 & 3 & 1 & 8 & 14 & 3 \\ 9 & 8 & 8 & 22 & 40 & 10 \\ 13 & 13 & 14 & 40 & 65 & 17 \\ 3 & 3 & 3 & 10 & 17 & 4 \end{pmatrix}_{b_0}$$

[226] $1^{1^{23}}3^{1^{21}}21^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 14 & 3 \\ 2 & 1 & 3 & 8 & 13 & 3 \\ 2 & 3 & 1 & 8 & 14 & 3 \\ 7 & 8 & 8 & 22 & 40 & 10 \\ 14 & 13 & 14 & 40 & 65 & 17 \\ 3 & 3 & 3 & 10 & 17 & 4 \end{pmatrix}_{b_0}$$

[227] $1^{1^{23}}3^{1^{21}}21^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 9 & 3 \\ 2 & 1 & 3 & 5 & 8 & 3 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 6 & 5 & 6 & 9 & 16 & 7 \\ 9 & 8 & 8 & 16 & 22 & 10 \\ 3 & 3 & 3 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

[228] $1^{1^{11}}3^{1^{21}}21^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 7 & 3 \\ 2 & 1 & 3 & 5 & 8 & 3 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 5 & 5 & 6 & 9 & 16 & 7 \\ 7 & 8 & 8 & 16 & 22 & 10 \\ 3 & 3 & 3 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

[229] $1^{1^{12}}3^{1^{21}}21^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 12 & 3 \\ 2 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 7 & 5 & 6 & 9 & 22 & 7 \\ 12 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

[230] $1^{1^2}3^{1^{12}}21^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 10 & 3 \\ 2 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 4 & 5 & 6 & 9 & 22 & 7 \\ 10 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

[231] $1^{1^{12}}3^{1^{21}}21^{1^{13}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 2 & 12 & 3 \\ 2 & 3 & 1 & 4 & 16 & 3 \\ 3 & 2 & 4 & 2 & 16 & 4 \\ 12 & 12 & 16 & 16 & 70 & 18 \\ 3 & 3 & 3 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

[232] $1^{1^{24}}3^{1^{21}}21^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 2 & 8 & 12 & 3 \\ 2 & 1 & 3 & 8 & 10 & 3 \\ 2 & 3 & 1 & 8 & 12 & 3 \\ 8 & 8 & 8 & 22 & 32 & 10 \\ 12 & 10 & 12 & 32 & 42 & 14 \\ 3 & 3 & 3 & 10 & 14 & 4 \end{pmatrix}_{b_0} \\
[233] & 1^{122} 3^{121} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -168, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 8 & 10 & 3 \\ 2 & 1 & 3 & 8 & 10 & 3 \\ 2 & 3 & 1 & 8 & 12 & 3 \\ 8 & 8 & 8 & 22 & 32 & 10 \\ 10 & 10 & 12 & 32 & 42 & 14 \\ 3 & 3 & 3 & 10 & 14 & 4 \end{pmatrix}_{b_0} \\
[234] & 1^{112} 3^{121} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -171, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 7 & 9 & 3 \\ 2 & 1 & 3 & 7 & 10 & 3 \\ 2 & 3 & 1 & 10 & 12 & 3 \\ 7 & 7 & 10 & 25 & 34 & 11 \\ 9 & 10 & 12 & 34 & 42 & 14 \\ 3 & 3 & 3 & 11 & 14 & 4 \end{pmatrix}_{b_0} \\
[235] & 1^{12} 3^{121} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -171, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 12 & 3 \\ 2 & 1 & 3 & 0 & 13 & 3 \\ 2 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 13 & 14 & 0 & 65 & 17 \\ 3 & 3 & 3 & 0 & 17 & 4 \end{pmatrix}_{b_0} \\
[236] & 1^{13} 3^{121} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -171, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 & 17 & 3 \\ 2 & 1 & 3 & 3 & 18 & 3 \\ 2 & 3 & 1 & 2 & 20 & 3 \\ 3 & 3 & 2 & 1 & 18 & 3 \\ 17 & 18 & 20 & 18 & 130 & 24 \\ 3 & 3 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0} \\
[237] & 1^{124} 3^{121} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -172, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 8 & 12 & 3 \\ 2 & 1 & 3 & 7 & 9 & 3 \\ 2 & 3 & 1 & 10 & 14 & 3 \\ 8 & 7 & 10 & 25 & 37 & 11 \\ 12 & 9 & 14 & 37 & 49 & 15 \\ 3 & 3 & 3 & 11 & 15 & 4 \end{pmatrix}_{b_0} \\
[238] & 1^{13} 3^{121} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -172, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 9 & 13 & 3 \\ 2 & 1 & 3 & 10 & 12 & 3 \\ 2 & 3 & 1 & 12 & 16 & 3 \\ 9 & 10 & 12 & 42 & 56 & 14 \\ 13 & 12 & 16 & 56 & 70 & 18 \\ 3 & 3 & 3 & 14 & 18 & 4 \end{pmatrix}_{b_0} \\
[239] & 1^{11} 3^{122} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -172, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 7 & 9 & 3 \\ 2 & 1 & 3 & 8 & 10 & 3 \\ 2 & 3 & 1 & 8 & 12 & 3 \\ 7 & 8 & 8 & 22 & 32 & 10 \\ 9 & 10 & 12 & 32 & 42 & 14 \\ 3 & 3 & 3 & 10 & 14 & 4 \end{pmatrix}_{b_0} \\
[240] & 1^{133} 3^{121} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -172, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 9 & 11 & 3 \\ 2 & 1 & 3 & 7 & 9 & 3 \\ 2 & 3 & 1 & 10 & 14 & 3 \\ 9 & 7 & 10 & 25 & 37 & 11 \\ 11 & 9 & 14 & 37 & 49 & 15 \\ 3 & 3 & 3 & 11 & 15 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[241] \quad & 1^{13} 3^{12} 3^1 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 13 & 3 \\ 2 & 1 & 3 & 0 & 12 & 3 \\ 2 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 12 & 16 & 0 & 70 & 18 \\ 3 & 3 & 3 & 0 & 18 & 4 \end{pmatrix}_{b_0} \\
[242] \quad & 1^{14} 3^{11} 3^1 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 3 & 15 & 3 \\ 2 & 3 & 1 & 2 & 18 & 3 \\ 2 & 3 & 2 & 1 & 15 & 3 \\ 16 & 15 & 18 & 15 & 97 & 21 \\ 3 & 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0} \\
[243] \quad & 1^{34} 3^1 21^{134} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -178, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 9 & 2 & 12 & 3 \\ 3 & 5 & 15 & 3 & 20 & 5 \\ 9 & 15 & 34 & 6 & 48 & 12 \\ 2 & 3 & 6 & 0 & 8 & 2 \\ 12 & 20 & 48 & 8 & 62 & 16 \\ 3 & 5 & 12 & 2 & 16 & 4 \end{pmatrix}_{b_0} \\
[244] \quad & 1^{144} 3^{112} 3^1 21^{134} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -178, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 12 & 16 & 3 \\ 2 & 1 & 3 & 9 & 14 & 3 \\ 2 & 3 & 1 & 14 & 20 & 3 \\ 12 & 9 & 14 & 49 & 74 & 15 \\ 16 & 14 & 20 & 74 & 106 & 22 \\ 3 & 3 & 3 & 15 & 22 & 4 \end{pmatrix}_{b_0} \\
[245] \quad & 1^{24} 3^{112} 3^1 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -178, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 6 & 16 & 3 \\ 2 & 1 & 3 & 5 & 14 & 3 \\ 2 & 3 & 1 & 6 & 20 & 3 \\ 6 & 5 & 6 & 9 & 34 & 7 \\ 16 & 14 & 20 & 34 & 106 & 22 \\ 3 & 3 & 3 & 7 & 22 & 4 \end{pmatrix}_{b_0} \\
[246] \quad & 1^{23} 3^{123} 3^1 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -178, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 8 & 15 & 3 \\ 2 & 1 & 3 & 8 & 15 & 3 \\ 2 & 3 & 1 & 8 & 18 & 3 \\ 8 & 8 & 8 & 22 & 48 & 10 \\ 15 & 15 & 18 & 48 & 97 & 21 \\ 3 & 3 & 3 & 10 & 21 & 4 \end{pmatrix}_{b_0} \\
[247] \quad & 1^{13} 3^{113} 3^1 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 & 13 & 3 \\ 2 & 1 & 3 & 3 & 13 & 3 \\ 2 & 3 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 1 & 13 & 3 \\ 13 & 13 & 14 & 13 & 65 & 17 \\ 3 & 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0} \\
[248] \quad & 21^{113} 3^1 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 5 & 17 & 3 \\ 3 & 1 & 3 & 5 & 17 & 3 \\ 2 & 3 & 1 & 6 & 22 & 3 \\ 5 & 5 & 6 & 9 & 39 & 7 \\ 17 & 17 & 22 & 39 & 137 & 25 \\ 3 & 3 & 3 & 7 & 25 & 4 \end{pmatrix}_{b_0} \\
[249] \quad & 1^{14} 3^{113} 3^1 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 3 & 13 & 3 \\ 2 & 3 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 1 & 13 & 3 \\ 14 & 13 & 14 & 13 & 65 & 17 \\ 3 & 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0} \\
[250] & 1^{134} 3^{123} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 9 & 16 & 3 \\ 2 & 1 & 3 & 7 & 15 & 3 \\ 2 & 3 & 1 & 10 & 18 & 3 \\ 9 & 7 & 10 & 25 & 51 & 11 \\ 16 & 15 & 18 & 51 & 97 & 21 \\ 3 & 3 & 3 & 11 & 21 & 4 \end{pmatrix}_{b_0} \\
[251] & 1^{12} 3^{121} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 11 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 3 & 5 & 2 & 2 & 15 & 4 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 11 & 20 & 15 & 9 & 58 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0} \\
[252] & 1^{14} 3^{121} 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 5 & 0 & 3 & 30 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 18 & 30 & 0 & 14 & 134 & 24 \\ 3 & 5 & 0 & 2 & 24 & 4 \end{pmatrix}_{b_0} \\
[253] & 1^{32} 3^{123} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 11 & 12 & 3 \\ 2 & 1 & 3 & 10 & 13 & 3 \\ 2 & 3 & 1 & 12 & 14 & 3 \\ 11 & 10 & 12 & 42 & 54 & 14 \\ 12 & 13 & 14 & 54 & 65 & 17 \\ 3 & 3 & 3 & 14 & 17 & 4 \end{pmatrix}_{b_0} \\
[254] & 1^{14} 3^{123} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 7 & 16 & 3 \\ 2 & 1 & 3 & 8 & 15 & 3 \\ 2 & 3 & 1 & 8 & 18 & 3 \\ 7 & 8 & 8 & 22 & 48 & 10 \\ 16 & 15 & 18 & 48 & 97 & 21 \\ 3 & 3 & 3 & 10 & 21 & 4 \end{pmatrix}_{b_0} \\
[255] & 1^{12} 3^{123} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 3 & 15 & 3 \\ 2 & 3 & 1 & 2 & 18 & 3 \\ 2 & 3 & 2 & 1 & 15 & 3 \\ 14 & 15 & 18 & 15 & 97 & 21 \\ 3 & 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0} \\
[256] & 1^{32} 3^{111} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 9 & 8 & 3 \\ 2 & 1 & 3 & 7 & 7 & 3 \\ 2 & 3 & 1 & 10 & 10 & 3 \\ 9 & 7 & 10 & 25 & 27 & 11 \\ 8 & 7 & 10 & 27 & 25 & 11 \\ 3 & 3 & 3 & 11 & 11 & 4 \end{pmatrix}_{b_0} \\
[257] & 1^{14} 3^{123} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 14 & 3 \\ 2 & 1 & 3 & 0 & 12 & 3 \\ 2 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 14 & 12 & 16 & 0 & 70 & 18 \\ 3 & 3 & 3 & 0 & 18 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[258] $1^{124}3^121^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 20 & 3 \\ 2 & 1 & 3 & 8 & 20 & 3 \\ 2 & 3 & 1 & 8 & 24 & 3 \\ 8 & 8 & 8 & 22 & 64 & 10 \\ 20 & 20 & 24 & 64 & 174 & 28 \\ 3 & 3 & 3 & 10 & 28 & 4 \end{pmatrix}_{b_0}$$

[259] $1^{123}3^121^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 11 & 3 \\ 2 & 1 & 3 & 7 & 10 & 3 \\ 2 & 3 & 1 & 10 & 12 & 3 \\ 8 & 7 & 10 & 25 & 34 & 11 \\ 11 & 10 & 12 & 34 & 42 & 14 \\ 3 & 3 & 3 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

[260] $1^{13}3^121^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 15 & 3 \\ 2 & 1 & 3 & 2 & 15 & 3 \\ 2 & 3 & 1 & 4 & 18 & 3 \\ 3 & 2 & 4 & 2 & 18 & 4 \\ 15 & 15 & 18 & 18 & 97 & 21 \\ 3 & 3 & 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

[261] $1^{13}3^121^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 15 & 3 \\ 2 & 1 & 3 & 5 & 14 & 3 \\ 2 & 3 & 1 & 6 & 20 & 2 \\ 4 & 5 & 6 & 9 & 34 & 7 \\ 15 & 14 & 20 & 34 & 106 & 22 \\ 3 & 3 & 3 & 7 & 22 & 4 \end{pmatrix}_{b_0}$$

[262] $1^{133}3^121^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 11 & 3 \\ 2 & 1 & 3 & 7 & 10 & 3 \\ 2 & 3 & 1 & 10 & 12 & 3 \\ 9 & 7 & 10 & 25 & 34 & 11 \\ 11 & 10 & 12 & 34 & 42 & 14 \\ 3 & 3 & 3 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

[263] $1^{114}3^121^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -197, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 13 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 3 & 5 & 2 & 2 & 15 & 4 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 13 & 20 & 15 & 9 & 58 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[264] $1^{112}3^121^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -197, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 13 & 3 \\ 3 & 5 & 5 & 3 & 25 & 5 \\ 3 & 5 & 2 & 2 & 19 & 4 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 13 & 25 & 19 & 11 & 92 & 20 \\ 3 & 5 & 4 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[265] $1^{113}3^121^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -197, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 3 & 15 & 3 \\ 2 & 3 & 1 & 2 & 18 & 3 \\ 2 & 3 & 2 & 1 & 15 & 3 \\ 15 & 15 & 18 & 15 & 97 & 21 \\ 3 & 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[266] $1^{122}3^121^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 10 & 3 \\ 2 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 6 & 5 & 6 & 9 & 22 & 7 \\ 10 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[267] \quad 1^{14} 1^{12} 3^{12} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -198, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 12 & 3 \\ 2 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 5 & 5 & 6 & 9 & 22 & 7 \\ 12 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[268] \quad 21^{12} 3^{12} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 8 & 3 \\ 3 & 1 & 3 & 5 & 8 & 3 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 5 & 5 & 6 & 9 & 16 & 7 \\ 8 & 8 & 8 & 16 & 22 & 10 \\ 3 & 3 & 3 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[269] \quad 1^{13} 3^{12} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 14 & 3 \\ 3 & 5 & 10 & 3 & 25 & 5 \\ 6 & 10 & 14 & 4 & 38 & 8 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 14 & 25 & 38 & 11 & 92 & 20 \\ 3 & 5 & 8 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[270] \quad 1^{14} 1^{13} 3^{12} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -203, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 14 & 3 \\ 2 & 1 & 3 & 0 & 13 & 3 \\ 2 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 14 & 13 & 14 & 0 & 65 & 17 \\ 3 & 3 & 3 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[271] \quad 1^{13} 1^{13} 3^{12} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -203, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 13 & 3 \\ 2 & 1 & 3 & 0 & 13 & 3 \\ 2 & 3 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 13 & 14 & 0 & 65 & 17 \\ 3 & 3 & 3 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[272] \quad 1^{13} 1^{12} 3^{12} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -204, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 13 & 3 \\ 2 & 1 & 3 & 2 & 12 & 3 \\ 2 & 3 & 1 & 4 & 16 & 3 \\ 3 & 2 & 4 & 2 & 16 & 4 \\ 13 & 12 & 16 & 16 & 70 & 18 \\ 3 & 3 & 3 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[273] \quad 1^{14} 1^{14} 3^{12} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -206, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 18 & 3 \\ 2 & 1 & 3 & 5 & 18 & 3 \\ 2 & 3 & 1 & 6 & 20 & 3 \\ 4 & 5 & 6 & 9 & 38 & 7 \\ 18 & 18 & 20 & 38 & 130 & 24 \\ 3 & 3 & 3 & 7 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[274] \quad 1^{12} 1^{12} 3^{12} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -208, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 8 & 3 \\ 2 & 1 & 3 & 8 & 8 & 3 \\ 2 & 3 & 1 & 8 & 8 & 3 \\ 7 & 8 & 8 & 22 & 24 & 10 \\ 8 & 8 & 8 & 24 & 22 & 10 \\ 3 & 3 & 3 & 10 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[275] \quad 1^{12} 1^{22} 3^{12} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -208, \quad (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 8 & 3 \\ 2 & 1 & 3 & 8 & 8 & 3 \\ 2 & 3 & 1 & 8 & 8 & 3 \\ 9 & 8 & 8 & 22 & 24 & 10 \\ 8 & 8 & 8 & 24 & 22 & 10 \\ 3 & 3 & 3 & 10 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[276] \quad 1^{13} 3^{11} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -211, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 12 & 3 \\ 3 & 5 & 5 & 3 & 20 & 5 \\ 3 & 5 & 2 & 2 & 15 & 4 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 12 & 20 & 15 & 9 & 58 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[277] \quad 1^{13} 3^{11} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -213, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 3 & 13 & 3 \\ 2 & 3 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 1 & 13 & 3 \\ 13 & 13 & 14 & 13 & 65 & 17 \\ 3 & 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[278] \quad 1^{14} 3^{11} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -213, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 14 & 3 \\ 2 & 1 & 3 & 3 & 13 & 3 \\ 2 & 3 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 1 & 13 & 3 \\ 14 & 13 & 14 & 13 & 65 & 17 \\ 3 & 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[279] \quad 1^{122} 3^{11} 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -214, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 10 & 3 \\ 2 & 1 & 3 & 7 & 10 & 3 \\ 2 & 3 & 1 & 10 & 12 & 3 \\ 8 & 7 & 10 & 25 & 34 & 11 \\ 10 & 10 & 12 & 34 & 42 & 14 \\ 3 & 3 & 3 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[280] \quad 1^{134} 3^{11} 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -214, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 12 & 3 \\ 2 & 1 & 3 & 7 & 10 & 3 \\ 2 & 3 & 1 & 10 & 12 & 3 \\ 9 & 7 & 10 & 25 & 34 & 11 \\ 12 & 10 & 12 & 34 & 42 & 14 \\ 3 & 3 & 3 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[281] \quad 1^{14} 3^{11} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 14 & 3 \\ 2 & 1 & 3 & 2 & 12 & 3 \\ 2 & 3 & 1 & 4 & 16 & 3 \\ 3 & 2 & 4 & 2 & 16 & 4 \\ 14 & 12 & 16 & 16 & 70 & 18 \\ 3 & 3 & 3 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[282] \quad 1^{14} 3^{11} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 16 & 3 \\ 2 & 1 & 3 & 2 & 15 & 3 \\ 2 & 3 & 1 & 4 & 18 & 3 \\ 3 & 2 & 4 & 2 & 18 & 4 \\ 16 & 15 & 18 & 18 & 97 & 21 \\ 3 & 3 & 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[284] \quad 1^{144} 3^{11} 21^{144} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 12 & 3 \\ 3 & 5 & 20 & 3 & 20 & 5 \\ 12 & 20 & 62 & 8 & 64 & 16 \\ 2 & 3 & 8 & 0 & 8 & 2 \\ 12 & 20 & 64 & 8 & 62 & 16 \\ 3 & 5 & 16 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[283] $1^{1^{34}} 1^{122} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 9 & 12 & 3 \\ 2 & 1 & 3 & 8 & 10 & 3 \\ 2 & 3 & 1 & 8 & 12 & 3 \\ 9 & 8 & 8 & 22 & 32 & 10 \\ 12 & 10 & 12 & 32 & 42 & 14 \\ 3 & 3 & 3 & 10 & 14 & 4 \end{pmatrix}_{b_0}$

[285] $1^{1^2} 1^{113} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 4 & 12 & 3 \\ 2 & 1 & 3 & 5 & 13 & 3 \\ 2 & 3 & 1 & 6 & 14 & 3 \\ 4 & 5 & 6 & 9 & 27 & 7 \\ 12 & 13 & 14 & 27 & 65 & 17 \\ 3 & 3 & 3 & 7 & 17 & 4 \end{pmatrix}_{b_0}$

[286] $1^{1^{12}} 1^{122} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 7 & 10 & 3 \\ 2 & 1 & 3 & 8 & 10 & 3 \\ 2 & 3 & 1 & 8 & 12 & 3 \\ 7 & 8 & 8 & 22 & 32 & 10 \\ 10 & 10 & 12 & 32 & 42 & 14 \\ 3 & 3 & 3 & 10 & 14 & 4 \end{pmatrix}_{b_0}$

[287] $1^{1^{14}} 1^{12} 3^1 21^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 16 & 3 \\ 2 & 1 & 3 & 0 & 14 & 3 \\ 2 & 3 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 16 & 14 & 20 & 0 & 106 & 22 \\ 3 & 3 & 3 & 0 & 22 & 4 \end{pmatrix}_{b_0}$

[288] $1^{1^{32}} 1^{123} 3^1 21^{133} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 13 & 14 & 3 \\ 2 & 1 & 3 & 12 & 15 & 3 \\ 2 & 3 & 1 & 16 & 18 & 3 \\ 13 & 12 & 16 & 70 & 84 & 18 \\ 14 & 15 & 18 & 84 & 97 & 21 \\ 3 & 3 & 3 & 18 & 21 & 4 \end{pmatrix}_{b_0}$

[289] $1^{1^{34}} 1^{113} 3^1 21^{124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 9 & 18 & 3 \\ 2 & 1 & 3 & 7 & 17 & 3 \\ 2 & 3 & 1 & 10 & 22 & 3 \\ 9 & 7 & 10 & 25 & 61 & 11 \\ 18 & 17 & 22 & 61 & 137 & 25 \\ 3 & 3 & 3 & 11 & 25 & 4 \end{pmatrix}_{b_0}$

[290] $1^{1^{24}} 1^{11} 3^1 21^{124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 2 & 15 & 3 \\ 3 & 5 & 10 & 3 & 25 & 5 \\ 6 & 10 & 14 & 4 & 38 & 8 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 15 & 25 & 38 & 11 & 92 & 20 \\ 3 & 5 & 8 & 2 & 20 & 4 \end{pmatrix}_{b_0}$

[291] $1^{1^{23}} 1^{112} 3^1 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 6 & 13 & 3 \\ 2 & 1 & 3 & 5 & 12 & 3 \\ 2 & 3 & 1 & 6 & 16 & 3 \\ 6 & 5 & 6 & 9 & 28 & 7 \\ 13 & 12 & 16 & 28 & 70 & 18 \\ 3 & 3 & 3 & 7 & 18 & 4 \end{pmatrix}_{b_0}$

[292] $1^{1^{33}} 1^{112} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 13 & 3 \\ 2 & 1 & 3 & 7 & 12 & 3 \\ 2 & 3 & 1 & 10 & 16 & 3 \\ 9 & 7 & 10 & 25 & 44 & 11 \\ 13 & 12 & 16 & 44 & 70 & 18 \\ 3 & 3 & 3 & 11 & 18 & 4 \end{pmatrix}_{b_0}$$

[293] $1^{1^{34}} 1^{111} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -237, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 12 & 3 \\ 2 & 1 & 3 & 7 & 9 & 3 \\ 2 & 3 & 1 & 10 & 14 & 3 \\ 9 & 7 & 10 & 25 & 37 & 11 \\ 12 & 9 & 14 & 37 & 49 & 15 \\ 3 & 3 & 3 & 11 & 15 & 4 \end{pmatrix}_{b_0}$$

[294] $1^{1^{23}} 1^{111} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -237, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 11 & 3 \\ 2 & 1 & 3 & 7 & 9 & 3 \\ 2 & 3 & 1 & 10 & 14 & 3 \\ 8 & 7 & 10 & 25 & 37 & 11 \\ 11 & 9 & 14 & 37 & 49 & 15 \\ 3 & 3 & 3 & 11 & 15 & 4 \end{pmatrix}_{b_0}$$

[295] $1^{1^2} 1^{112} 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 12 & 3 \\ 2 & 1 & 3 & 5 & 12 & 3 \\ 2 & 3 & 1 & 6 & 16 & 3 \\ 4 & 5 & 6 & 9 & 28 & 7 \\ 12 & 12 & 16 & 28 & 70 & 18 \\ 3 & 3 & 3 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

[296] $1^{1^{11}} 1^{112} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 9 & 3 \\ 2 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 5 & 5 & 6 & 9 & 22 & 7 \\ 9 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

[297] $1^{1^{33}} 1^{111} 3^1 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 9 & 3 \\ 2 & 1 & 3 & 7 & 7 & 3 \\ 2 & 3 & 1 & 10 & 10 & 3 \\ 9 & 7 & 10 & 25 & 27 & 11 \\ 9 & 7 & 10 & 27 & 25 & 11 \\ 3 & 3 & 3 & 11 & 11 & 4 \end{pmatrix}_{b_0}$$

[298] $1^{1^{22}} 1^{111} 3^1 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 8 & 3 \\ 2 & 1 & 3 & 7 & 7 & 3 \\ 2 & 3 & 1 & 10 & 10 & 3 \\ 8 & 7 & 10 & 25 & 27 & 11 \\ 8 & 7 & 10 & 27 & 25 & 11 \\ 3 & 3 & 3 & 11 & 11 & 4 \end{pmatrix}_{b_0}$$

[299] $1^{1^{12}} 1^{113} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 14 & 3 \\ 2 & 1 & 3 & 7 & 15 & 3 \\ 2 & 3 & 1 & 10 & 18 & 3 \\ 7 & 7 & 10 & 25 & 51 & 11 \\ 14 & 15 & 18 & 51 & 97 & 21 \\ 3 & 3 & 3 & 11 & 21 & 4 \end{pmatrix}_{b_0}$$

[300] $1^{1^{23}} 1^{122} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 11 & 3 \\ 2 & 1 & 3 & 8 & 10 & 3 \\ 2 & 3 & 1 & 8 & 12 & 3 \\ 8 & 8 & 8 & 22 & 32 & 10 \\ 11 & 10 & 12 & 32 & 42 & 14 \\ 3 & 3 & 3 & 10 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[301] \quad 1^{13} 1^{12} 3^1 21^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -251, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 13 & 3 \\ 2 & 1 & 3 & 7 & 12 & 3 \\ 2 & 3 & 1 & 10 & 16 & 3 \\ 7 & 7 & 10 & 25 & 44 & 11 \\ 13 & 12 & 16 & 44 & 70 & 18 \\ 3 & 3 & 3 & 11 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[303] \quad 1^{13} 1^{12} 3^1 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 15 & 3 \\ 2 & 1 & 3 & 2 & 14 & 3 \\ 2 & 3 & 1 & 4 & 20 & 3 \\ 3 & 2 & 4 & 2 & 20 & 4 \\ 15 & 14 & 20 & 20 & 106 & 22 \\ 3 & 3 & 3 & 4 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[302] \quad 1^{13} 1^{12} 3^1 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 9 & 3 \\ 2 & 1 & 3 & 8 & 8 & 3 \\ 2 & 3 & 1 & 8 & 8 & 3 \\ 9 & 8 & 8 & 22 & 24 & 10 \\ 9 & 8 & 8 & 24 & 22 & 10 \\ 3 & 3 & 3 & 10 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[304] \quad 1^{11} 1^{12} 3^1 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 7 & 3 \\ 2 & 1 & 3 & 8 & 8 & 3 \\ 2 & 3 & 1 & 8 & 8 & 3 \\ 7 & 8 & 8 & 22 & 24 & 10 \\ 7 & 8 & 8 & 24 & 22 & 10 \\ 3 & 3 & 3 & 10 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[305] \quad 1^{13} 1^{14} 3^1 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 15 & 3 \\ 2 & 1 & 3 & 0 & 16 & 3 \\ 2 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 16 & 16 & 0 & 94 & 20 \\ 3 & 3 & 3 & 0 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[306] \quad 1^{13} 1^{13} 3^1 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -259, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 15 & 3 \\ 2 & 1 & 3 & 0 & 15 & 3 \\ 2 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 15 & 18 & 0 & 97 & 21 \\ 3 & 3 & 3 & 0 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[307] \quad 1^{14} 1^{14} 3^1 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -266, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 18 & 3 \\ 2 & 1 & 3 & 3 & 18 & 3 \\ 2 & 3 & 1 & 2 & 20 & 3 \\ 3 & 3 & 2 & 1 & 18 & 3 \\ 18 & 18 & 20 & 18 & 130 & 24 \\ 3 & 3 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[308] \quad 1^{23} 1^{14} 3^1 21^{124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -267, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.00 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 19 & 3 \\ 2 & 1 & 3 & 7 & 20 & 3 \\ 2 & 3 & 1 & 10 & 24 & 3 \\ 8 & 7 & 10 & 25 & 68 & 11 \\ 19 & 20 & 24 & 68 & 174 & 28 \\ 3 & 3 & 3 & 11 & 28 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[309] \quad & 1^{13} 1^{14} 3^1 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -268, \quad (n_+, n_-, n_0) = \\
& (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 & 19 & 3 \\ 2 & 1 & 3 & 2 & 20 & 3 \\ 2 & 3 & 1 & 4 & 24 & 3 \\ 3 & 2 & 4 & 2 & 24 & 4 \\ 19 & 20 & 24 & 24 & 174 & 28 \\ 3 & 3 & 3 & 4 & 28 & 4 \end{pmatrix}_{b_0} \\
[310] \quad & 1^{14} 3^1 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -272, \quad (n_+, n_-, n_0) = \\
& (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 16 & 3 \\ 2 & 1 & 3 & 0 & 16 & 3 \\ 2 & 3 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 16 & 16 & 16 & 0 & 94 & 20 \\ 3 & 3 & 3 & 0 & 20 & 4 \end{pmatrix}_{b_0} \\
[311] \quad & 1^{13} 1^{11} 3^1 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -275, \quad (n_+, n_-, n_0) = \\
& (1, 9, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 14 & 3 \\ 3 & 5 & 5 & 3 & 25 & 5 \\ 3 & 5 & 2 & 2 & 19 & 4 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 14 & 25 & 19 & 11 & 92 & 20 \\ 3 & 5 & 4 & 2 & 20 & 4 \end{pmatrix}_{b_0} \\
[312] \quad & 1^{124} 1^{122} 3^1 21^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -280, \quad (n_+, n_-, n_0) = \\
& (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 10 & 14 & 3 \\ 2 & 1 & 3 & 10 & 12 & 3 \\ 2 & 3 & 1 & 12 & 16 & 3 \\ 10 & 10 & 12 & 42 & 56 & 14 \\ 14 & 12 & 16 & 56 & 70 & 18 \\ 3 & 3 & 3 & 14 & 18 & 4 \end{pmatrix}_{b_0} \\
[313] \quad & 1^{144} 1^{122} 3^1 21^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -280, \quad (n_+, n_-, n_0) = \\
& (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 12 & 14 & 3 \\ 2 & 1 & 3 & 10 & 12 & 3 \\ 2 & 3 & 1 & 12 & 16 & 3 \\ 12 & 10 & 12 & 42 & 56 & 14 \\ 14 & 12 & 16 & 56 & 70 & 18 \\ 3 & 3 & 3 & 14 & 18 & 4 \end{pmatrix}_{b_0} \\
[314] \quad & 21^{1123} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -280, \quad (n_+, n_-, n_0) = \\
& (1, 9, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 5 & 10 & 3 \\ 3 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 5 & 5 & 6 & 9 & 22 & 7 \\ 10 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0} \\
[315] \quad & 1^{12} 1^{123} 3^1 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -283, \quad (n_+, n_-, n_0) = \\
& (1, 9, 0), \quad c_{\text{ext}} = 7.3 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 7 & 14 & 3 \\ 2 & 1 & 3 & 8 & 15 & 3 \\ 2 & 3 & 1 & 8 & 18 & 3 \\ 7 & 8 & 8 & 22 & 48 & 10 \\ 14 & 15 & 18 & 48 & 97 & 21 \\ 3 & 3 & 3 & 10 & 21 & 4 \end{pmatrix}_{b_0} \\
[316] \quad & 1^{22} 1^{122} 3^1 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -288, \quad (n_+, n_-, n_0) = \\
& (1, 9, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 8 & 8 & 3 \\ 2 & 1 & 3 & 8 & 8 & 3 \\ 2 & 3 & 1 & 8 & 8 & 3 \\ 8 & 8 & 8 & 22 & 24 & 10 \\ 8 & 8 & 8 & 24 & 22 & 10 \\ 3 & 3 & 3 & 10 & 10 & 4 \end{pmatrix}_{b_0} \\
[317] \quad & 1^{14} 1^{13} 3^1 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -291, \quad (n_+, n_-, n_0) = \\
& (1, 9, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 16 & 3 \\ 2 & 1 & 3 & 0 & 15 & 3 \\ 2 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 16 & 15 & 18 & 0 & 97 & 21 \\ 3 & 3 & 3 & 0 & 21 & 4 \end{pmatrix}_{b_0}$$

[318] $1^{14}3^{12}21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 16 & 3 \\ 2 & 1 & 3 & 2 & 14 & 3 \\ 2 & 3 & 1 & 4 & 20 & 3 \\ 3 & 2 & 4 & 2 & 20 & 4 \\ 16 & 14 & 20 & 20 & 106 & 22 \\ 3 & 3 & 3 & 4 & 22 & 4 \end{pmatrix}_{b_0}$$

[319] $1^{12}3^{12}21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 16 & 3 \\ 3 & 5 & 5 & 3 & 30 & 5 \\ 3 & 5 & 2 & 2 & 22 & 4 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 16 & 30 & 22 & 14 & 134 & 24 \\ 3 & 5 & 4 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[320] $1^{13}3^{13}21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -301, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 15 & 3 \\ 2 & 1 & 3 & 5 & 15 & 3 \\ 2 & 3 & 1 & 6 & 18 & 3 \\ 4 & 5 & 6 & 9 & 33 & 7 \\ 15 & 15 & 18 & 33 & 97 & 21 \\ 3 & 3 & 3 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

[321] $1^{14}3^{13}21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 18 & 3 \\ 2 & 1 & 3 & 5 & 17 & 3 \\ 2 & 3 & 1 & 6 & 22 & 3 \\ 4 & 5 & 6 & 9 & 39 & 7 \\ 18 & 17 & 22 & 39 & 137 & 25 \\ 3 & 3 & 3 & 7 & 25 & 4 \end{pmatrix}_{b_0}$$

[326] $1^{13}3^{12}21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 11 & 3 \\ 2 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 5 & 5 & 6 & 9 & 22 & 7 \\ 11 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

[323] $1^{23}3^{14}21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 17 & 3 \\ 2 & 1 & 3 & 5 & 18 & 3 \\ 2 & 3 & 1 & 6 & 20 & 3 \\ 6 & 5 & 6 & 9 & 38 & 7 \\ 17 & 18 & 20 & 38 & 130 & 24 \\ 3 & 3 & 3 & 7 & 24 & 4 \end{pmatrix}_{b_0}$$

[324] $1^{12}3^{12}21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 12 & 3 \\ 2 & 1 & 3 & 10 & 13 & 3 \\ 2 & 3 & 1 & 12 & 14 & 3 \\ 9 & 10 & 12 & 42 & 54 & 14 \\ 12 & 13 & 14 & 54 & 65 & 17 \\ 3 & 3 & 3 & 14 & 17 & 4 \end{pmatrix}_{b_0}$$

[325] $1^{13}3^{13}21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 17 & 3 \\ 2 & 1 & 3 & 2 & 17 & 3 \\ 2 & 3 & 1 & 4 & 22 & 3 \\ 3 & 2 & 4 & 2 & 22 & 4 \\ 17 & 17 & 22 & 22 & 137 & 25 \\ 3 & 3 & 3 & 4 & 25 & 4 \end{pmatrix}_{b_0}$$

[322] $1^{1^{13}} 1^{1^{12}} 3^1 21^{1^{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 15 & 3 \\ 2 & 1 & 3 & 5 & 14 & 3 \\ 2 & 3 & 1 & 6 & 20 & 3 \\ 5 & 5 & 6 & 9 & 34 & 7 \\ 15 & 14 & 20 & 34 & 106 & 22 \\ 3 & 3 & 3 & 7 & 22 & 4 \end{pmatrix}_{b_0}$$

[327] $1^{1^{14}} 1^{1^{23}} 3^1 21^{1^{24}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 18 & 3 \\ 2 & 1 & 3 & 10 & 17 & 3 \\ 2 & 3 & 1 & 12 & 22 & 3 \\ 9 & 10 & 12 & 42 & 78 & 14 \\ 18 & 17 & 22 & 78 & 137 & 25 \\ 3 & 3 & 3 & 14 & 25 & 4 \end{pmatrix}_{b_0}$$

[328] $1^{1^{33}} 1^{1^{12}} 3^1 21^{1^{34}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 15 & 3 \\ 2 & 1 & 3 & 9 & 14 & 3 \\ 2 & 3 & 1 & 14 & 20 & 3 \\ 11 & 9 & 14 & 49 & 74 & 15 \\ 15 & 14 & 20 & 74 & 106 & 22 \\ 3 & 3 & 3 & 15 & 22 & 4 \end{pmatrix}_{b_0}$$

[329] $1^{1^{23}} 1^{1^{12}} 3^1 21^{1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 11 & 3 \\ 2 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 6 & 5 & 6 & 9 & 22 & 7 \\ 11 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

[330] $1^{1^{23}} 1^{1^{13}} 3^1 21^{1^{23}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -309, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 15 & 3 \\ 2 & 1 & 3 & 7 & 15 & 3 \\ 2 & 3 & 1 & 10 & 18 & 3 \\ 8 & 7 & 10 & 25 & 51 & 11 \\ 15 & 15 & 18 & 51 & 97 & 21 \\ 3 & 3 & 3 & 11 & 21 & 4 \end{pmatrix}_{b_0}$$

[331] $1^{1^{14}} 1^{1^{11}} 3^1 21^{1^{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -309, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 15 & 3 \\ 3 & 5 & 5 & 3 & 25 & 5 \\ 3 & 5 & 2 & 2 & 19 & 4 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 15 & 25 & 19 & 11 & 92 & 20 \\ 3 & 5 & 4 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[332] $1^{1^{14}} 1^{1^{12}} 3^1 21^{1^{24}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -310, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 16 & 3 \\ 2 & 1 & 3 & 7 & 14 & 3 \\ 2 & 3 & 1 & 10 & 20 & 3 \\ 7 & 7 & 10 & 25 & 54 & 11 \\ 16 & 14 & 20 & 54 & 106 & 22 \\ 3 & 3 & 3 & 11 & 22 & 4 \end{pmatrix}_{b_0}$$

[333] $1^{1^{13}} 1^{1^{23}} 3^1 21^{1^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 15 & 3 \\ 2 & 1 & 3 & 8 & 15 & 3 \\ 2 & 3 & 1 & 8 & 18 & 3 \\ 7 & 8 & 8 & 22 & 48 & 10 \\ 15 & 15 & 18 & 48 & 97 & 21 \\ 3 & 3 & 3 & 10 & 21 & 4 \end{pmatrix}_{b_0}$$

[334] $1^{1^{24}} 1^{1^{13}} 3^1 21^{1^{23}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 16 & 3 \\ 2 & 1 & 3 & 7 & 15 & 3 \\ 2 & 3 & 1 & 10 & 18 & 3 \\ 8 & 7 & 10 & 25 & 51 & 11 \\ 16 & 15 & 18 & 51 & 97 & 21 \\ 3 & 3 & 3 & 11 & 21 & 4 \end{pmatrix}_{b_0}$$

[335] $1^{1^{24}3^121^{134}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 19 & 3 \\ 2 & 1 & 3 & 12 & 20 & 3 \\ 2 & 3 & 1 & 16 & 24 & 3 \\ 13 & 12 & 16 & 70 & 112 & 18 \\ 19 & 20 & 24 & 112 & 174 & 28 \\ 3 & 3 & 3 & 18 & 28 & 4 \end{pmatrix}_{b_0}$$

[336] $1^{1^{24}3^121^{123}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 17 & 3 \\ 2 & 1 & 3 & 10 & 18 & 3 \\ 2 & 3 & 1 & 12 & 20 & 3 \\ 11 & 10 & 12 & 42 & 76 & 14 \\ 17 & 18 & 20 & 76 & 130 & 24 \\ 3 & 3 & 3 & 14 & 24 & 4 \end{pmatrix}_{b_0}$$

[337] $1^{1^{12}3^121^{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 12 & 3 \\ 2 & 1 & 3 & 5 & 12 & 3 \\ 2 & 3 & 1 & 6 & 16 & 3 \\ 5 & 5 & 6 & 9 & 28 & 7 \\ 12 & 12 & 16 & 28 & 70 & 18 \\ 3 & 3 & 3 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

[338] $1^{1^{12}3^121^{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 10 & 3 \\ 2 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 5 & 5 & 6 & 9 & 22 & 7 \\ 10 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

[339] $1^{1^{22}3^121^{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 12 & 3 \\ 2 & 1 & 3 & 8 & 13 & 3 \\ 2 & 3 & 1 & 8 & 14 & 3 \\ 8 & 8 & 8 & 22 & 40 & 10 \\ 12 & 13 & 14 & 40 & 65 & 17 \\ 3 & 3 & 3 & 10 & 17 & 4 \end{pmatrix}_{b_0}$$

[340] $1^{1^{44}3^121^{123}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 16 & 3 \\ 2 & 1 & 3 & 10 & 15 & 3 \\ 2 & 3 & 1 & 12 & 18 & 3 \\ 12 & 10 & 12 & 42 & 66 & 14 \\ 16 & 15 & 18 & 66 & 97 & 21 \\ 3 & 3 & 3 & 14 & 21 & 4 \end{pmatrix}_{b_0}$$

[341] $1^{1^{22}3^121^{123}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 12 & 3 \\ 2 & 1 & 3 & 7 & 12 & 3 \\ 2 & 3 & 1 & 10 & 16 & 3 \\ 8 & 7 & 10 & 25 & 44 & 11 \\ 12 & 12 & 16 & 44 & 70 & 18 \\ 3 & 3 & 3 & 11 & 18 & 4 \end{pmatrix}_{b_0}$$

[342] $1^{1^{24}3^121^{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 12 & 3 \\ 2 & 1 & 3 & 5 & 10 & 3 \\ 2 & 3 & 1 & 6 & 12 & 3 \\ 6 & 5 & 6 & 9 & 22 & 7 \\ 12 & 10 & 12 & 22 & 42 & 14 \\ 3 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

[343] $1^{1^3} 1^{14} 3^1 21^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -323, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 19 & 3 \\ 2 & 1 & 3 & 3 & 20 & 3 \\ 2 & 3 & 1 & 2 & 24 & 3 \\ 2 & 3 & 2 & 1 & 20 & 3 \\ 19 & 20 & 24 & 20 & 174 & 28 \\ 3 & 3 & 3 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

[344] $1^{1^{13}} 1^{14} 3^1 21^{124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -323, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 19 & 3 \\ 2 & 1 & 3 & 7 & 20 & 3 \\ 2 & 3 & 1 & 10 & 24 & 3 \\ 7 & 7 & 10 & 25 & 68 & 11 \\ 19 & 20 & 24 & 68 & 174 & 28 \\ 3 & 3 & 3 & 11 & 28 & 4 \end{pmatrix}_{b_0}$$

[345] $1^{1^{12}} 1^{22} 3^1 21^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 10 & 3 \\ 2 & 1 & 3 & 10 & 10 & 3 \\ 2 & 3 & 1 & 12 & 12 & 3 \\ 9 & 10 & 12 & 42 & 44 & 14 \\ 10 & 10 & 12 & 44 & 42 & 14 \\ 3 & 3 & 3 & 14 & 14 & 4 \end{pmatrix}_{b_0}$$

[346] $1^{1^{22}} 1^{22} 3^1 21^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -328, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 12 & 3 \\ 2 & 1 & 3 & 10 & 12 & 3 \\ 2 & 3 & 1 & 12 & 16 & 3 \\ 10 & 10 & 12 & 42 & 56 & 14 \\ 12 & 12 & 16 & 56 & 70 & 18 \\ 3 & 3 & 3 & 14 & 18 & 4 \end{pmatrix}_{b_0}$$

[347] $1^{1^{24}} 1^{22} 3^1 21^{134} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -328, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 16 & 3 \\ 2 & 1 & 3 & 12 & 14 & 3 \\ 2 & 3 & 1 & 16 & 20 & 3 \\ 12 & 12 & 16 & 70 & 88 & 18 \\ 16 & 14 & 20 & 88 & 106 & 22 \\ 3 & 3 & 3 & 18 & 22 & 4 \end{pmatrix}_{b_0}$$

[348] $1^{1^{14}} 1^{14} 3^1 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -328, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 20 & 3 \\ 2 & 1 & 3 & 2 & 20 & 3 \\ 2 & 3 & 1 & 4 & 24 & 3 \\ 3 & 2 & 4 & 2 & 24 & 4 \\ 20 & 20 & 24 & 24 & 174 & 28 \\ 3 & 3 & 3 & 4 & 28 & 4 \end{pmatrix}_{b_0}$$

[349] $1^{1^4} 1^{14} 3^1 21^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 20 & 3 \\ 2 & 1 & 3 & 3 & 20 & 3 \\ 2 & 3 & 1 & 2 & 24 & 3 \\ 2 & 3 & 2 & 1 & 20 & 3 \\ 20 & 20 & 24 & 20 & 174 & 28 \\ 3 & 3 & 3 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

[350] $1^{1^{33}} 1^{22} 3^1 21^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -332, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 13 & 3 \\ 2 & 1 & 3 & 10 & 12 & 3 \\ 2 & 3 & 1 & 12 & 16 & 3 \\ 11 & 10 & 12 & 42 & 56 & 14 \\ 13 & 12 & 16 & 56 & 70 & 18 \\ 3 & 3 & 3 & 14 & 18 & 4 \end{pmatrix}_{b_0}$$

[351] $1^{1^{13}} 1^{14} 3^1 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -332, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 2 & 1 & 17 & 3 \\ 2 & 1 & 3 & 0 & 18 & 3 \\ 2 & 3 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 17 & 18 & 20 & 0 & 130 & 24 \\ 3 & 3 & 3 & 0 & 24 & 4 \end{pmatrix}_{b_0} \\
[352] & 1^{\overline{123}} 1^{\overline{113}} 3^{\overline{1}} 21^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 6 & 15 & 3 \\ 2 & 1 & 3 & 5 & 15 & 3 \\ 2 & 3 & 1 & 6 & 18 & 3 \\ 6 & 5 & 6 & 9 & 33 & 7 \\ 15 & 15 & 18 & 33 & 97 & 21 \\ 3 & 3 & 3 & 7 & 21 & 4 \end{pmatrix}_{b_0} \\
[353] & 1^{\overline{123}} 1^{\overline{113}} 3^{\overline{1}} 21^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 6 & 13 & 3 \\ 2 & 1 & 3 & 5 & 13 & 3 \\ 2 & 3 & 1 & 6 & 14 & 3 \\ 6 & 5 & 6 & 9 & 27 & 7 \\ 13 & 13 & 14 & 27 & 65 & 17 \\ 3 & 3 & 3 & 7 & 17 & 4 \end{pmatrix}_{b_0} \\
[354] & 1^{\overline{114}} 1^{\overline{113}} 3^{\overline{1}} 21^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 5 & 14 & 3 \\ 2 & 1 & 3 & 5 & 13 & 3 \\ 2 & 3 & 1 & 6 & 14 & 3 \\ 5 & 5 & 6 & 9 & 27 & 7 \\ 14 & 13 & 14 & 27 & 65 & 17 \\ 3 & 3 & 3 & 7 & 17 & 4 \end{pmatrix}_{b_0} \\
[355] & 1^{\overline{112}} 1^{\overline{113}} 3^{\overline{1}} 21^{\overline{123}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 7 & 12 & 3 \\ 2 & 1 & 3 & 7 & 12 & 3 \\ 2 & 3 & 1 & 10 & 16 & 3 \\ 7 & 7 & 10 & 25 & 44 & 11 \\ 12 & 12 & 16 & 44 & 70 & 18 \\ 3 & 3 & 3 & 11 & 18 & 4 \end{pmatrix}_{b_0} \\
[356] & 1^{\overline{124}} 1^{\overline{123}} 3^{\overline{1}} 21^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 8 & 16 & 3 \\ 2 & 1 & 3 & 8 & 15 & 3 \\ 2 & 3 & 1 & 8 & 18 & 3 \\ 8 & 8 & 8 & 22 & 48 & 10 \\ 16 & 15 & 18 & 48 & 97 & 21 \\ 3 & 3 & 3 & 10 & 21 & 4 \end{pmatrix}_{b_0} \\
[357] & 1^{\overline{123}} 1^{\overline{123}} 3^{\overline{1}} 21^{\overline{133}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 12 & 15 & 3 \\ 2 & 1 & 3 & 12 & 15 & 3 \\ 2 & 3 & 1 & 16 & 18 & 3 \\ 12 & 12 & 16 & 70 & 84 & 18 \\ 15 & 15 & 18 & 84 & 97 & 21 \\ 3 & 3 & 3 & 18 & 21 & 4 \end{pmatrix}_{b_0} \\
[358] & 1^{\overline{114}} 1^{\overline{113}} 3^{\overline{1}} 21^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 18 & 3 \\ 2 & 1 & 3 & 0 & 17 & 3 \\ 2 & 3 & 1 & 0 & 22 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 18 & 17 & 22 & 0 & 137 & 25 \\ 3 & 3 & 3 & 0 & 25 & 4 \end{pmatrix}_{b_0} \\
[359] & 1^{\overline{134}} 1^{\overline{112}} 3^{\overline{1}} 21^{\overline{124}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -342, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 9 & 16 & 3 \\ 2 & 1 & 3 & 7 & 14 & 3 \\ 2 & 3 & 1 & 10 & 20 & 3 \\ 9 & 7 & 10 & 25 & 54 & 11 \\ 16 & 14 & 20 & 54 & 106 & 22 \\ 3 & 3 & 3 & 11 & 22 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[360] $1^{1^{14}} 1^{113} 3^1 21^{124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -349, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 18 & 3 \\ 2 & 1 & 3 & 7 & 17 & 3 \\ 2 & 3 & 1 & 10 & 22 & 3 \\ 7 & 7 & 10 & 25 & 61 & 11 \\ 18 & 17 & 22 & 61 & 137 & 25 \\ 3 & 3 & 3 & 11 & 25 & 4 \end{pmatrix}_{b_0}$$

[361] $1^{1^{24}} 1^{124} 3^1 21^{134} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 20 & 3 \\ 2 & 1 & 3 & 12 & 20 & 3 \\ 2 & 3 & 1 & 16 & 24 & 3 \\ 12 & 12 & 16 & 70 & 112 & 18 \\ 20 & 20 & 24 & 112 & 174 & 28 \\ 3 & 3 & 3 & 18 & 28 & 4 \end{pmatrix}_{b_0}$$

[362] $1^{1^{14}} 1^{113} 3^1 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 18 & 3 \\ 2 & 1 & 3 & 2 & 17 & 3 \\ 2 & 3 & 1 & 4 & 22 & 3 \\ 3 & 2 & 4 & 2 & 22 & 4 \\ 18 & 17 & 22 & 22 & 137 & 25 \\ 3 & 3 & 3 & 4 & 25 & 4 \end{pmatrix}_{b_0}$$

[363] $1^{1^{24}} 1^{112} 3^1 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -358, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 14 & 3 \\ 2 & 1 & 3 & 5 & 12 & 3 \\ 2 & 3 & 1 & 6 & 16 & 3 \\ 6 & 5 & 6 & 9 & 28 & 7 \\ 14 & 12 & 16 & 28 & 70 & 18 \\ 3 & 3 & 3 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

[364] $1^{1^{14}} 1^{112} 3^1 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -358, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 16 & 3 \\ 2 & 1 & 3 & 5 & 14 & 3 \\ 2 & 3 & 1 & 6 & 20 & 3 \\ 5 & 5 & 6 & 9 & 34 & 7 \\ 16 & 14 & 20 & 34 & 106 & 22 \\ 3 & 3 & 3 & 7 & 22 & 4 \end{pmatrix}_{b_0}$$

[365] $1^{1^{14}} 1^{112} 3^1 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -358, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 14 & 3 \\ 2 & 1 & 3 & 5 & 12 & 3 \\ 2 & 3 & 1 & 6 & 16 & 3 \\ 5 & 5 & 6 & 9 & 28 & 7 \\ 14 & 12 & 16 & 28 & 70 & 18 \\ 3 & 3 & 3 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

[366] $1^{1^{12}} 1^{122} 3^1 21^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -360, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 12 & 3 \\ 2 & 1 & 3 & 10 & 12 & 3 \\ 2 & 3 & 1 & 12 & 16 & 3 \\ 9 & 10 & 12 & 42 & 56 & 14 \\ 12 & 12 & 16 & 56 & 70 & 18 \\ 3 & 3 & 3 & 14 & 18 & 4 \end{pmatrix}_{b_0}$$

[367] $21^{123} 3^1 21^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -360, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 13 & 3 \\ 3 & 1 & 3 & 10 & 13 & 3 \\ 2 & 3 & 1 & 12 & 14 & 3 \\ 10 & 10 & 12 & 42 & 54 & 14 \\ 13 & 13 & 14 & 54 & 65 & 17 \\ 3 & 3 & 3 & 14 & 17 & 4 \end{pmatrix}_{b_0}$$

[368] $1^{1^4} 1^{114} 3^1 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -366, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 3 & 18 & 3 \\ 2 & 3 & 1 & 2 & 20 & 3 \\ 2 & 3 & 2 & 1 & 18 & 3 \\ 18 & 18 & 20 & 18 & 130 & 24 \\ 3 & 3 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[369] $1^{13} 1^{12} 3^{12} 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -368$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 17 & 3 \\ 3 & 5 & 5 & 3 & 30 & 5 \\ 3 & 5 & 2 & 2 & 22 & 4 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 17 & 30 & 22 & 14 & 134 & 24 \\ 3 & 5 & 4 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[370] $1^{13} 1^{114} 3^{12} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -371$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 10.00 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 17 & 3 \\ 2 & 1 & 3 & 3 & 18 & 3 \\ 2 & 3 & 1 & 2 & 20 & 3 \\ 2 & 3 & 2 & 1 & 18 & 3 \\ 17 & 18 & 20 & 18 & 130 & 24 \\ 3 & 3 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[371] $1^{24} 1^{114} 3^{12} 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -378$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 16 & 3 \\ 2 & 1 & 3 & 5 & 16 & 3 \\ 2 & 3 & 1 & 6 & 16 & 3 \\ 6 & 5 & 6 & 9 & 32 & 7 \\ 16 & 16 & 16 & 32 & 94 & 20 \\ 3 & 3 & 3 & 7 & 20 & 4 \end{pmatrix}_{b_0}$$

[372] $1^{14} 1^{114} 3^{12} 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -378$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 16 & 3 \\ 2 & 1 & 3 & 5 & 16 & 3 \\ 2 & 3 & 1 & 6 & 16 & 3 \\ 5 & 5 & 6 & 9 & 32 & 7 \\ 16 & 16 & 16 & 32 & 94 & 20 \\ 3 & 3 & 3 & 7 & 20 & 4 \end{pmatrix}_{b_0}$$

[373] $1^{14} 1^{14} 3^{12} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -380$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 18 & 3 \\ 2 & 1 & 3 & 0 & 18 & 3 \\ 2 & 3 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 18 & 18 & 20 & 0 & 130 & 24 \\ 3 & 3 & 3 & 0 & 24 & 4 \end{pmatrix}_{b_0}$$

[374] $1^{34} 1^{123} 3^{12} 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -387$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.3$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 14 & 3 \\ 2 & 1 & 3 & 8 & 13 & 3 \\ 2 & 3 & 1 & 8 & 14 & 3 \\ 9 & 8 & 8 & 22 & 40 & 10 \\ 14 & 13 & 14 & 40 & 65 & 17 \\ 3 & 3 & 3 & 10 & 17 & 4 \end{pmatrix}_{b_0}$$

[375] $1^{13} 1^{123} 3^{12} 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -387$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.3$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 13 & 3 \\ 2 & 1 & 3 & 8 & 13 & 3 \\ 2 & 3 & 1 & 8 & 14 & 3 \\ 7 & 8 & 8 & 22 & 40 & 10 \\ 13 & 13 & 14 & 40 & 65 & 17 \\ 3 & 3 & 3 & 10 & 17 & 4 \end{pmatrix}_{b_0}$$

[376] $1^{13} 1^{112} 3^{12} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -387$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 13 & 3 \\ 2 & 1 & 3 & 5 & 12 & 3 \\ 2 & 3 & 1 & 6 & 16 & 3 \\ 5 & 5 & 6 & 9 & 28 & 7 \\ 13 & 12 & 16 & 28 & 70 & 18 \\ 3 & 3 & 3 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

[377] $1^{123} 1^{122} 3^1 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 11 & 3 \\ 2 & 1 & 3 & 10 & 10 & 3 \\ 2 & 3 & 1 & 12 & 12 & 3 \\ 10 & 10 & 12 & 42 & 44 & 14 \\ 11 & 10 & 12 & 44 & 42 & 14 \\ 3 & 3 & 3 & 14 & 14 & 4 \end{pmatrix}_{b_0}$$

[378] $1^{143} 1^{122} 3^1 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -388, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 11 & 3 \\ 2 & 1 & 3 & 10 & 10 & 3 \\ 2 & 3 & 1 & 12 & 12 & 3 \\ 12 & 10 & 12 & 42 & 44 & 14 \\ 11 & 10 & 12 & 44 & 42 & 14 \\ 3 & 3 & 3 & 14 & 14 & 4 \end{pmatrix}_{b_0}$$

[379] $1^{13} 1^{114} 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 17 & 3 \\ 2 & 1 & 3 & 5 & 18 & 3 \\ 2 & 3 & 1 & 6 & 20 & 3 \\ 4 & 5 & 6 & 9 & 38 & 7 \\ 17 & 18 & 20 & 38 & 130 & 24 \\ 3 & 3 & 3 & 7 & 24 & 4 \end{pmatrix}_{b_0}$$

[380] $1^{12} 1^{113} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -397, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 12 & 3 \\ 2 & 1 & 3 & 5 & 13 & 3 \\ 2 & 3 & 1 & 6 & 14 & 3 \\ 5 & 5 & 6 & 9 & 27 & 7 \\ 12 & 13 & 14 & 27 & 65 & 17 \\ 3 & 3 & 3 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

[381] $1^{14} 1^{12} 3^1 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -398, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 18 & 3 \\ 3 & 5 & 5 & 3 & 30 & 5 \\ 3 & 5 & 2 & 2 & 22 & 4 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 18 & 30 & 22 & 14 & 134 & 24 \\ 3 & 5 & 4 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[383] $1^{14} 1^{124} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 16 & 3 \\ 2 & 1 & 3 & 8 & 16 & 3 \\ 2 & 3 & 1 & 8 & 16 & 3 \\ 7 & 8 & 8 & 22 & 48 & 10 \\ 16 & 16 & 16 & 48 & 94 & 20 \\ 3 & 3 & 3 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

[382] $1^{34} 1^{124} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 16 & 3 \\ 2 & 1 & 3 & 8 & 16 & 3 \\ 2 & 3 & 1 & 8 & 16 & 3 \\ 9 & 8 & 8 & 22 & 48 & 10 \\ 16 & 16 & 16 & 48 & 94 & 20 \\ 3 & 3 & 3 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

[384] $1^{11} 1^{122} 3^1 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 9 & 3 \\ 2 & 1 & 3 & 10 & 10 & 3 \\ 2 & 3 & 1 & 12 & 12 & 3 \\ 9 & 10 & 12 & 42 & 44 & 14 \\ 9 & 10 & 12 & 44 & 42 & 14 \\ 3 & 3 & 3 & 14 & 14 & 4 \end{pmatrix}_{b_0}$$

[385] $1^{24} 1^{112} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -406, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 14 & 3 \\ 2 & 1 & 3 & 7 & 12 & 3 \\ 2 & 3 & 1 & 10 & 16 & 3 \\ 8 & 7 & 10 & 25 & 44 & 11 \\ 14 & 12 & 16 & 44 & 70 & 18 \\ 3 & 3 & 3 & 11 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[386] \quad 1^{1^{34}} 1^{12} 3^1 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -406, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 14 & 3 \\ 2 & 1 & 3 & 7 & 12 & 3 \\ 2 & 3 & 1 & 10 & 16 & 3 \\ 9 & 7 & 10 & 25 & 44 & 11 \\ 14 & 12 & 16 & 44 & 70 & 18 \\ 3 & 3 & 3 & 11 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[387] \quad 21^{122} 3^1 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -412, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 10 & 3 \\ 3 & 1 & 3 & 10 & 10 & 3 \\ 2 & 3 & 1 & 12 & 12 & 3 \\ 10 & 10 & 12 & 42 & 44 & 14 \\ 10 & 10 & 12 & 44 & 42 & 14 \\ 3 & 3 & 3 & 14 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[388] \quad 1^{1^{24}} 1^{13} 3^1 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -412, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 18 & 3 \\ 2 & 1 & 3 & 5 & 17 & 3 \\ 2 & 3 & 1 & 6 & 22 & 3 \\ 6 & 5 & 6 & 9 & 39 & 7 \\ 18 & 17 & 22 & 39 & 137 & 25 \\ 3 & 3 & 3 & 7 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[389] \quad 1^{1^{13}} 1^{13} 3^1 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -412, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 13 & 3 \\ 2 & 1 & 3 & 5 & 13 & 3 \\ 2 & 3 & 1 & 6 & 14 & 3 \\ 5 & 5 & 6 & 9 & 27 & 7 \\ 13 & 13 & 14 & 27 & 65 & 17 \\ 3 & 3 & 3 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[390] \quad 1^{1^{24}} 1^{13} 3^1 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -412, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 14 & 3 \\ 2 & 1 & 3 & 5 & 13 & 3 \\ 2 & 3 & 1 & 6 & 14 & 3 \\ 6 & 5 & 6 & 9 & 27 & 7 \\ 14 & 13 & 14 & 27 & 65 & 17 \\ 3 & 3 & 3 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[391] \quad 1^{1^{44}} 1^{23} 3^1 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -418, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 14 & 3 \\ 2 & 1 & 3 & 10 & 13 & 3 \\ 2 & 3 & 1 & 12 & 14 & 3 \\ 12 & 10 & 12 & 42 & 54 & 14 \\ 14 & 13 & 14 & 54 & 65 & 17 \\ 3 & 3 & 3 & 14 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[392] \quad 1^{1^{23}} 1^{23} 3^1 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -418, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 13 & 3 \\ 2 & 1 & 3 & 10 & 13 & 3 \\ 2 & 3 & 1 & 12 & 14 & 3 \\ 10 & 10 & 12 & 42 & 54 & 14 \\ 13 & 13 & 14 & 54 & 65 & 17 \\ 3 & 3 & 3 & 14 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[393] \quad 1^{1^{13}} 1^{14} 3^1 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -427, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.00 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 15 & 3 \\ 2 & 1 & 3 & 5 & 16 & 3 \\ 2 & 3 & 1 & 6 & 16 & 3 \\ 5 & 5 & 6 & 9 & 32 & 7 \\ 15 & 16 & 16 & 32 & 94 & 20 \\ 3 & 3 & 3 & 7 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[394] \quad 1^{1^{24}} 1^{24} 3^1 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -432, \quad (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 19 & 3 \\ 2 & 1 & 3 & 8 & 20 & 3 \\ 2 & 3 & 1 & 8 & 24 & 3 \\ 7 & 8 & 8 & 22 & 64 & 10 \\ 19 & 20 & 24 & 64 & 174 & 28 \\ 3 & 3 & 3 & 10 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[395] \quad 1^{123} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 13 & 3 \\ 2 & 1 & 3 & 7 & 12 & 3 \\ 2 & 3 & 1 & 10 & 16 & 3 \\ 8 & 7 & 10 & 25 & 44 & 11 \\ 13 & 12 & 16 & 44 & 70 & 18 \\ 3 & 3 & 3 & 11 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[396] \quad 1^{13} 3^1 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -437, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 17 & 3 \\ 2 & 1 & 3 & 5 & 17 & 3 \\ 2 & 3 & 1 & 6 & 22 & 3 \\ 4 & 5 & 6 & 9 & 39 & 7 \\ 17 & 17 & 22 & 39 & 137 & 25 \\ 3 & 3 & 3 & 7 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[397] \quad 1^{134} 3^1 21^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -438, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 16 & 3 \\ 2 & 1 & 3 & 9 & 14 & 3 \\ 2 & 3 & 1 & 14 & 20 & 3 \\ 11 & 9 & 14 & 49 & 74 & 15 \\ 16 & 14 & 20 & 74 & 106 & 22 \\ 3 & 3 & 3 & 15 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[398] \quad 1^{123} 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 17 & 3 \\ 2 & 1 & 3 & 8 & 18 & 3 \\ 2 & 3 & 1 & 8 & 20 & 3 \\ 8 & 8 & 8 & 22 & 56 & 10 \\ 17 & 18 & 20 & 56 & 130 & 24 \\ 3 & 3 & 3 & 10 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[399] \quad 1^{124} 3^1 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 20 & 3 \\ 2 & 1 & 3 & 7 & 20 & 3 \\ 2 & 3 & 1 & 10 & 24 & 3 \\ 8 & 7 & 10 & 25 & 68 & 11 \\ 20 & 20 & 24 & 68 & 174 & 28 \\ 3 & 3 & 3 & 11 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[400] \quad 1^{123} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 13 & 3 \\ 2 & 1 & 3 & 10 & 12 & 3 \\ 2 & 3 & 1 & 12 & 16 & 3 \\ 10 & 10 & 12 & 42 & 56 & 14 \\ 13 & 12 & 16 & 56 & 70 & 18 \\ 3 & 3 & 3 & 14 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[401] \quad 1^{14} 3^1 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -448, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 20 & 3 \\ 2 & 1 & 3 & 0 & 20 & 3 \\ 2 & 3 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 20 & 20 & 24 & 0 & 174 & 28 \\ 3 & 3 & 3 & 0 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[402] \quad 1^{14} 3^1 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -448, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 20 & 3 \\ 2 & 1 & 3 & 8 & 20 & 3 \\ 2 & 3 & 1 & 8 & 24 & 3 \\ 7 & 8 & 8 & 22 & 64 & 10 \\ 20 & 20 & 24 & 64 & 174 & 28 \\ 3 & 3 & 3 & 10 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[403] \quad 1^{123} 3^{121} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -451, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 12 & 3 \\ 2 & 1 & 3 & 8 & 13 & 3 \\ 2 & 3 & 1 & 8 & 14 & 3 \\ 7 & 8 & 8 & 22 & 40 & 10 \\ 12 & 13 & 14 & 40 & 65 & 17 \\ 3 & 3 & 3 & 10 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[404] \quad 1^{123} 3^{121} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -454, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 12 & 3 \\ 2 & 1 & 3 & 10 & 13 & 3 \\ 2 & 3 & 1 & 12 & 14 & 3 \\ 10 & 10 & 12 & 42 & 54 & 14 \\ 12 & 13 & 14 & 54 & 65 & 17 \\ 3 & 3 & 3 & 14 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[405] \quad 1^{123} 3^{121} 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -459, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 15 & 3 \\ 2 & 1 & 3 & 7 & 14 & 3 \\ 2 & 3 & 1 & 10 & 20 & 3 \\ 8 & 7 & 10 & 25 & 54 & 11 \\ 15 & 14 & 20 & 54 & 106 & 22 \\ 3 & 3 & 3 & 11 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[406] \quad 1^{123} 3^{121} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -459, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 15 & 3 \\ 2 & 1 & 3 & 10 & 15 & 3 \\ 2 & 3 & 1 & 12 & 18 & 3 \\ 9 & 10 & 12 & 42 & 66 & 14 \\ 15 & 15 & 18 & 66 & 97 & 21 \\ 3 & 3 & 3 & 14 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[407] \quad 1^{122} 3^{121} 21^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -460, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 15 & 3 \\ 2 & 1 & 3 & 12 & 14 & 3 \\ 2 & 3 & 1 & 16 & 20 & 3 \\ 13 & 12 & 16 & 70 & 88 & 18 \\ 15 & 14 & 20 & 88 & 106 & 22 \\ 3 & 3 & 3 & 18 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[409] \quad 1^{123} 3^{121} 21^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -467, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 16 & 3 \\ 2 & 1 & 3 & 12 & 15 & 3 \\ 2 & 3 & 1 & 16 & 18 & 3 \\ 13 & 12 & 16 & 70 & 84 & 18 \\ 16 & 15 & 18 & 84 & 97 & 21 \\ 3 & 3 & 3 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[408] \quad 1^{123} 3^{121} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -467, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 13 & 3 \\ 2 & 1 & 3 & 10 & 13 & 3 \\ 2 & 3 & 1 & 12 & 14 & 3 \\ 11 & 10 & 12 & 42 & 54 & 14 \\ 13 & 13 & 14 & 54 & 65 & 17 \\ 3 & 3 & 3 & 14 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[410] \quad 1^{123} 3^{121} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -467, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 14 & 3 \\ 2 & 1 & 3 & 10 & 13 & 3 \\ 2 & 3 & 1 & 12 & 14 & 3 \\ 11 & 10 & 12 & 42 & 54 & 14 \\ 14 & 13 & 14 & 54 & 65 & 17 \\ 3 & 3 & 3 & 14 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[411] \quad 21^{113} 3^{121} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -468, \quad (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 13 & 3 \\ 3 & 1 & 3 & 5 & 13 & 3 \\ 2 & 3 & 1 & 6 & 14 & 3 \\ 5 & 5 & 6 & 9 & 27 & 7 \\ 13 & 13 & 14 & 27 & 65 & 17 \\ 3 & 3 & 3 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[413] \ 1^{122} 3^{14} 21^{134} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -472, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 16 & 3 \\ 2 & 1 & 3 & 12 & 14 & 3 \\ 2 & 3 & 1 & 16 & 20 & 3 \\ 14 & 12 & 16 & 70 & 88 & 18 \\ 16 & 14 & 20 & 88 & 106 & 22 \\ 3 & 3 & 3 & 18 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[412] \ 1^{122} 3^{14} 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -472, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 14 & 3 \\ 2 & 1 & 3 & 10 & 12 & 3 \\ 2 & 3 & 1 & 12 & 16 & 3 \\ 11 & 10 & 12 & 42 & 56 & 14 \\ 14 & 12 & 16 & 56 & 70 & 18 \\ 3 & 3 & 3 & 14 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[414] \ 1^{123} 3^{133} 21^{144} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -475, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 17 & 3 \\ 2 & 1 & 3 & 14 & 17 & 3 \\ 2 & 3 & 1 & 20 & 22 & 3 \\ 15 & 14 & 20 & 106 & 122 & 22 \\ 17 & 17 & 22 & 122 & 137 & 25 \\ 3 & 3 & 3 & 22 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[415] \ 1^{122} 3^{123} 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -480, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 13 & 3 \\ 2 & 1 & 3 & 12 & 12 & 3 \\ 2 & 3 & 1 & 16 & 16 & 3 \\ 12 & 12 & 16 & 70 & 72 & 18 \\ 13 & 12 & 16 & 72 & 70 & 18 \\ 3 & 3 & 3 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[418] \ 1^{122} 3^{121} 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -482, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 14 & 3 \\ 2 & 1 & 3 & 12 & 15 & 3 \\ 2 & 3 & 1 & 16 & 18 & 3 \\ 12 & 12 & 16 & 70 & 84 & 18 \\ 14 & 15 & 18 & 84 & 97 & 21 \\ 3 & 3 & 3 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[416] \ 1^{123} 3^{124} 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -482, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 14 & 3 \\ 2 & 1 & 3 & 8 & 13 & 3 \\ 2 & 3 & 1 & 8 & 14 & 3 \\ 8 & 8 & 8 & 22 & 40 & 10 \\ 14 & 13 & 14 & 40 & 65 & 17 \\ 3 & 3 & 3 & 10 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[417] \ 1^{123} 3^{123} 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -482, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 13 & 3 \\ 2 & 1 & 3 & 8 & 13 & 3 \\ 2 & 3 & 1 & 8 & 14 & 3 \\ 8 & 8 & 8 & 22 & 40 & 10 \\ 13 & 13 & 14 & 40 & 65 & 17 \\ 3 & 3 & 3 & 10 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[419] \ 1^{114} 3^{14} 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -490, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 20 & 3 \\ 2 & 1 & 3 & 5 & 20 & 3 \\ 2 & 3 & 1 & 6 & 24 & 3 \\ 4 & 5 & 6 & 9 & 44 & 7 \\ 20 & 20 & 24 & 44 & 174 & 28 \\ 3 & 3 & 3 & 7 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[422] \quad 1^{13} 1^{124} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -496, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 17 & 3 \\ 2 & 1 & 3 & 10 & 18 & 3 \\ 2 & 3 & 1 & 12 & 20 & 3 \\ 9 & 10 & 12 & 42 & 76 & 14 \\ 17 & 18 & 20 & 76 & 130 & 24 \\ 3 & 3 & 3 & 14 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[420] \quad 1^{22} 1^{122} 3^1 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -496, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 10 & 3 \\ 2 & 1 & 3 & 10 & 10 & 3 \\ 2 & 3 & 1 & 12 & 12 & 3 \\ 10 & 10 & 12 & 42 & 44 & 14 \\ 10 & 10 & 12 & 44 & 42 & 14 \\ 3 & 3 & 3 & 14 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[421] \quad 1^{44} 1^{122} 3^1 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -496, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 12 & 3 \\ 2 & 1 & 3 & 10 & 10 & 3 \\ 2 & 3 & 1 & 12 & 12 & 3 \\ 12 & 10 & 12 & 42 & 44 & 14 \\ 12 & 10 & 12 & 44 & 42 & 14 \\ 3 & 3 & 3 & 14 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[424] \quad 21^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -500, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 15 & 3 \\ 3 & 1 & 3 & 5 & 15 & 3 \\ 2 & 3 & 1 & 6 & 18 & 3 \\ 5 & 5 & 6 & 9 & 33 & 7 \\ 15 & 15 & 18 & 33 & 97 & 21 \\ 3 & 3 & 3 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[423] \quad 1^{13} 1^{113} 3^1 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -500, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 17 & 3 \\ 2 & 1 & 3 & 7 & 17 & 3 \\ 2 & 3 & 1 & 10 & 22 & 3 \\ 7 & 7 & 10 & 25 & 61 & 11 \\ 17 & 17 & 22 & 61 & 137 & 25 \\ 3 & 3 & 3 & 11 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[425] \quad 1^{44} 1^{123} 3^1 21^{144} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -514, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 18 & 3 \\ 2 & 1 & 3 & 14 & 17 & 3 \\ 2 & 3 & 1 & 20 & 22 & 3 \\ 16 & 14 & 20 & 106 & 122 & 22 \\ 18 & 17 & 22 & 122 & 137 & 25 \\ 3 & 3 & 3 & 22 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[426] \quad 1^{12} 1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -517, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 14 & 3 \\ 2 & 1 & 3 & 5 & 15 & 3 \\ 2 & 3 & 1 & 6 & 18 & 3 \\ 5 & 5 & 6 & 9 & 33 & 7 \\ 14 & 15 & 18 & 33 & 97 & 21 \\ 3 & 3 & 3 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[427] \quad 1^{24} 1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -520, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 16 & 3 \\ 2 & 1 & 3 & 5 & 15 & 3 \\ 2 & 3 & 1 & 6 & 18 & 3 \\ 6 & 5 & 6 & 9 & 33 & 7 \\ 16 & 15 & 18 & 33 & 97 & 21 \\ 3 & 3 & 3 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[428] \quad 1^{24} 1^{114} 3^1 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -522, \quad (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 20 & 3 \\ 2 & 1 & 3 & 5 & 20 & 3 \\ 2 & 3 & 1 & 6 & 24 & 3 \\ 6 & 5 & 6 & 9 & 44 & 7 \\ 20 & 20 & 24 & 44 & 174 & 28 \\ 3 & 3 & 3 & 7 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[430] \quad 1^{1^{33}} 3^1 21^{1^{23}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -523, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 15 & 3 \\ 2 & 1 & 3 & 10 & 15 & 3 \\ 2 & 3 & 1 & 12 & 18 & 3 \\ 11 & 10 & 12 & 42 & 66 & 14 \\ 15 & 15 & 18 & 66 & 97 & 21 \\ 3 & 3 & 3 & 14 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[429] \quad 1^{1^{34}} 3^1 21^{1^{24}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -523, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 18 & 3 \\ 2 & 1 & 3 & 10 & 17 & 3 \\ 2 & 3 & 1 & 12 & 22 & 3 \\ 11 & 10 & 12 & 42 & 78 & 14 \\ 18 & 17 & 22 & 78 & 137 & 25 \\ 3 & 3 & 3 & 14 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[431] \quad 1^{1^{23}} 3^1 21^{1^{24}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -525, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 17 & 3 \\ 2 & 1 & 3 & 7 & 17 & 3 \\ 2 & 3 & 1 & 10 & 22 & 3 \\ 8 & 7 & 10 & 25 & 61 & 11 \\ 17 & 17 & 22 & 61 & 137 & 25 \\ 3 & 3 & 3 & 11 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[432] \quad 1^{1^{23}} 3^1 21^{1^{22}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 11 & 3 \\ 2 & 1 & 3 & 10 & 10 & 3 \\ 2 & 3 & 1 & 12 & 12 & 3 \\ 11 & 10 & 12 & 42 & 44 & 14 \\ 11 & 10 & 12 & 44 & 42 & 14 \\ 3 & 3 & 3 & 14 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[433] \quad 1^{1^{44}} 3^1 21^{1^{33}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -534, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 16 & 3 \\ 2 & 1 & 3 & 12 & 15 & 3 \\ 2 & 3 & 1 & 16 & 18 & 3 \\ 14 & 12 & 16 & 70 & 84 & 18 \\ 16 & 15 & 18 & 84 & 97 & 21 \\ 3 & 3 & 3 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[434] \quad 1^{1^{33}} 3^1 21^{1^{33}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 15 & 3 \\ 2 & 1 & 3 & 12 & 15 & 3 \\ 2 & 3 & 1 & 16 & 18 & 3 \\ 13 & 12 & 16 & 70 & 84 & 18 \\ 15 & 15 & 18 & 84 & 97 & 21 \\ 3 & 3 & 3 & 18 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[435] \quad 1^{1^{34}} 3^1 21^{1^{33}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 14 & 3 \\ 2 & 1 & 3 & 12 & 12 & 3 \\ 2 & 3 & 1 & 16 & 16 & 3 \\ 13 & 12 & 16 & 70 & 72 & 18 \\ 14 & 12 & 16 & 72 & 70 & 18 \\ 3 & 3 & 3 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[436] \quad 1^{1^{23}} 3^1 21^{1^{34}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -552, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 15 & 3 \\ 2 & 1 & 3 & 12 & 14 & 3 \\ 2 & 3 & 1 & 16 & 20 & 3 \\ 12 & 12 & 16 & 70 & 88 & 18 \\ 15 & 14 & 20 & 88 & 106 & 22 \\ 3 & 3 & 3 & 18 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[437] \quad 1^{124} 3^{121} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -558, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 18 & 3 \\ 2 & 1 & 3 & 5 & 18 & 3 \\ 2 & 3 & 1 & 6 & 20 & 3 \\ 6 & 5 & 6 & 9 & 38 & 7 \\ 18 & 18 & 20 & 38 & 130 & 24 \\ 3 & 3 & 3 & 7 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[438] \quad 1^{124} 3^{121} 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -562, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 16 & 3 \\ 2 & 1 & 3 & 7 & 14 & 3 \\ 2 & 3 & 1 & 10 & 20 & 3 \\ 8 & 7 & 10 & 25 & 54 & 11 \\ 16 & 14 & 20 & 54 & 106 & 22 \\ 3 & 3 & 3 & 11 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[439] \quad 1^{124} 3^{121} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -576, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 15 & 3 \\ 2 & 1 & 3 & 8 & 16 & 3 \\ 2 & 3 & 1 & 8 & 16 & 3 \\ 7 & 8 & 8 & 22 & 48 & 10 \\ 15 & 16 & 16 & 48 & 94 & 20 \\ 3 & 3 & 3 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[440] \quad 1^{123} 3^{121} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -592, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 15 & 3 \\ 2 & 1 & 3 & 8 & 16 & 3 \\ 2 & 3 & 1 & 8 & 16 & 3 \\ 8 & 8 & 8 & 22 & 48 & 10 \\ 15 & 16 & 16 & 48 & 94 & 20 \\ 3 & 3 & 3 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[441] \quad 1^{123} 3^{121} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -598, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 14 & 3 \\ 2 & 1 & 3 & 10 & 15 & 3 \\ 2 & 3 & 1 & 12 & 18 & 3 \\ 10 & 10 & 12 & 42 & 66 & 14 \\ 14 & 15 & 18 & 66 & 97 & 21 \\ 3 & 3 & 3 & 14 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[442] \quad 21^{123} 3^{121} 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -600, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 13 & 3 \\ 3 & 1 & 3 & 8 & 13 & 3 \\ 2 & 3 & 1 & 8 & 14 & 3 \\ 8 & 8 & 8 & 22 & 40 & 10 \\ 13 & 13 & 14 & 40 & 65 & 17 \\ 3 & 3 & 3 & 10 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[443] \quad 1^{123} 3^{121} 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -603, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 17 & 3 \\ 2 & 1 & 3 & 10 & 17 & 3 \\ 2 & 3 & 1 & 12 & 22 & 3 \\ 9 & 10 & 12 & 42 & 78 & 14 \\ 17 & 17 & 22 & 78 & 137 & 25 \\ 3 & 3 & 3 & 14 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[444] \quad 1^{124} 3^{121} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -604, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 18 & 3 \\ 2 & 1 & 3 & 8 & 18 & 3 \\ 2 & 3 & 1 & 8 & 20 & 3 \\ 7 & 8 & 8 & 22 & 56 & 10 \\ 18 & 18 & 20 & 56 & 130 & 24 \\ 3 & 3 & 3 & 10 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[445] \quad 1^{122} 3^{121} 21^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -608, \quad (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 12 & 3 \\ 2 & 1 & 3 & 12 & 12 & 3 \\ 2 & 3 & 1 & 16 & 16 & 3 \\ 12 & 12 & 16 & 70 & 72 & 18 \\ 12 & 12 & 16 & 72 & 70 & 18 \\ 3 & 3 & 3 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[446] \quad 1^{123} 3^{121} 1^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 19 & 3 \\ 2 & 1 & 3 & 12 & 20 & 3 \\ 2 & 3 & 1 & 16 & 24 & 3 \\ 12 & 12 & 16 & 70 & 112 & 18 \\ 19 & 20 & 24 & 112 & 174 & 28 \\ 3 & 3 & 3 & 18 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[447] \quad 1^{113} 3^{121} 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -612, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 15 & 3 \\ 2 & 1 & 3 & 5 & 15 & 3 \\ 2 & 3 & 1 & 6 & 18 & 3 \\ 5 & 5 & 6 & 9 & 33 & 7 \\ 15 & 15 & 18 & 33 & 97 & 21 \\ 3 & 3 & 3 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[448] \quad 1^{122} 3^{121} 1^{144} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -612, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 16 & 3 \\ 2 & 1 & 3 & 14 & 14 & 3 \\ 2 & 3 & 1 & 20 & 20 & 3 \\ 15 & 14 & 20 & 106 & 108 & 22 \\ 16 & 14 & 20 & 108 & 106 & 22 \\ 3 & 3 & 3 & 22 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[449] \quad 1^{113} 3^{121} 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -613, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 16 & 3 \\ 2 & 1 & 3 & 5 & 15 & 3 \\ 2 & 3 & 1 & 6 & 18 & 3 \\ 5 & 5 & 6 & 9 & 33 & 7 \\ 16 & 15 & 18 & 33 & 97 & 21 \\ 3 & 3 & 3 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[450] \quad 1^{114} 3^{121} 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -627, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 19 & 3 \\ 2 & 1 & 3 & 5 & 20 & 3 \\ 2 & 3 & 1 & 6 & 24 & 3 \\ 4 & 5 & 6 & 9 & 44 & 7 \\ 19 & 20 & 24 & 44 & 174 & 28 \\ 3 & 3 & 3 & 7 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[451] \quad 1^{123} 3^{121} 1^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -630, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 18 & 3 \\ 2 & 1 & 3 & 12 & 17 & 3 \\ 2 & 3 & 1 & 16 & 22 & 3 \\ 14 & 12 & 16 & 70 & 100 & 18 \\ 18 & 17 & 22 & 100 & 137 & 25 \\ 3 & 3 & 3 & 18 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[452] \quad 21^{123} 3^{121} 1^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -632, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 15 & 3 \\ 3 & 1 & 3 & 10 & 15 & 3 \\ 2 & 3 & 1 & 12 & 18 & 3 \\ 10 & 10 & 12 & 42 & 66 & 14 \\ 15 & 15 & 18 & 66 & 97 & 21 \\ 3 & 3 & 3 & 14 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[453] \quad 1^{122} 3^{121} 1^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -636, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 16 & 3 \\ 2 & 1 & 3 & 12 & 14 & 3 \\ 2 & 3 & 1 & 16 & 20 & 3 \\ 13 & 12 & 16 & 70 & 88 & 18 \\ 16 & 14 & 20 & 88 & 106 & 22 \\ 3 & 3 & 3 & 18 & 22 & 4 \end{pmatrix}_{b_0}$$

[454] $1^{124} 1^{123} 3^1 21^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -642, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 18 & 3 \\ 2 & 1 & 3 & 12 & 17 & 3 \\ 2 & 3 & 1 & 16 & 22 & 3 \\ 12 & 12 & 16 & 70 & 100 & 18 \\ 18 & 17 & 22 & 100 & 137 & 25 \\ 3 & 3 & 3 & 18 & 25 & 4 \end{pmatrix}_{b_0}$$

[455] $1^{112} 1^{123} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -643, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 14 & 3 \\ 2 & 1 & 3 & 10 & 15 & 3 \\ 2 & 3 & 1 & 12 & 18 & 3 \\ 9 & 10 & 12 & 42 & 66 & 14 \\ 14 & 15 & 18 & 66 & 97 & 21 \\ 3 & 3 & 3 & 14 & 21 & 4 \end{pmatrix}_{b_0}$$

[456] $1^{114} 1^{124} 3^1 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -648, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 20 & 3 \\ 2 & 1 & 3 & 10 & 20 & 3 \\ 2 & 3 & 1 & 12 & 24 & 3 \\ 9 & 10 & 12 & 42 & 88 & 14 \\ 20 & 20 & 24 & 88 & 174 & 28 \\ 3 & 3 & 3 & 14 & 28 & 4 \end{pmatrix}_{b_0}$$

[457] $1^{113} 1^{13} 3^1 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -652, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 17 & 3 \\ 2 & 1 & 3 & 5 & 17 & 3 \\ 2 & 3 & 1 & 6 & 22 & 3 \\ 5 & 5 & 6 & 9 & 39 & 7 \\ 17 & 17 & 22 & 39 & 137 & 25 \\ 3 & 3 & 3 & 7 & 25 & 4 \end{pmatrix}_{b_0}$$

[458] $1^{123} 1^{123} 3^1 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -658, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 17 & 3 \\ 2 & 1 & 3 & 10 & 17 & 3 \\ 2 & 3 & 1 & 12 & 22 & 3 \\ 10 & 10 & 12 & 42 & 78 & 14 \\ 17 & 17 & 22 & 78 & 137 & 25 \\ 3 & 3 & 3 & 14 & 25 & 4 \end{pmatrix}_{b_0}$$

[459] $1^{124} 1^{113} 3^1 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -664, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 18 & 3 \\ 2 & 1 & 3 & 7 & 17 & 3 \\ 2 & 3 & 1 & 10 & 22 & 3 \\ 8 & 7 & 10 & 25 & 61 & 11 \\ 18 & 17 & 22 & 61 & 137 & 25 \\ 3 & 3 & 3 & 11 & 25 & 4 \end{pmatrix}_{b_0}$$

[461] $1^{134} 1^{124} 3^1 21^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -672, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 20 & 3 \\ 2 & 1 & 3 & 12 & 20 & 3 \\ 2 & 3 & 1 & 16 & 24 & 3 \\ 13 & 12 & 16 & 70 & 112 & 18 \\ 20 & 20 & 24 & 112 & 174 & 28 \\ 3 & 3 & 3 & 18 & 28 & 4 \end{pmatrix}_{b_0}$$

[460] $1^{124} 1^{124} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -672, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 16 & 3 \\ 2 & 1 & 3 & 8 & 16 & 3 \\ 2 & 3 & 1 & 8 & 16 & 3 \\ 8 & 8 & 8 & 22 & 48 & 10 \\ 16 & 16 & 16 & 48 & 94 & 20 \\ 3 & 3 & 3 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

[462] $1^{124} 1^{124} 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -680, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 18 & 3 \\ 2 & 1 & 3 & 8 & 18 & 3 \\ 2 & 3 & 1 & 8 & 20 & 3 \\ 8 & 8 & 8 & 22 & 56 & 10 \\ 18 & 18 & 20 & 56 & 130 & 24 \\ 3 & 3 & 3 & 10 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[463] \quad 1^{13} 1^{124} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -684, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 17 & 3 \\ 2 & 1 & 3 & 8 & 18 & 3 \\ 2 & 3 & 1 & 8 & 20 & 3 \\ 7 & 8 & 8 & 22 & 56 & 10 \\ 17 & 18 & 20 & 56 & 130 & 24 \\ 3 & 3 & 3 & 10 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[464] \quad 1^{24} 1^{123} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -694, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 16 & 3 \\ 2 & 1 & 3 & 10 & 15 & 3 \\ 2 & 3 & 1 & 12 & 18 & 3 \\ 10 & 10 & 12 & 42 & 66 & 14 \\ 16 & 15 & 18 & 66 & 97 & 21 \\ 3 & 3 & 3 & 14 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[465] \quad 1^{33} 1^{122} 3^1 21^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -704, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 13 & 3 \\ 2 & 1 & 3 & 12 & 12 & 3 \\ 2 & 3 & 1 & 16 & 16 & 3 \\ 13 & 12 & 16 & 70 & 72 & 18 \\ 13 & 12 & 16 & 72 & 70 & 18 \\ 3 & 3 & 3 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[466] \quad 1^{34} 1^{124} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -712, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 20 & 3 \\ 2 & 1 & 3 & 10 & 20 & 3 \\ 2 & 3 & 1 & 12 & 24 & 3 \\ 11 & 10 & 12 & 42 & 88 & 14 \\ 20 & 20 & 24 & 88 & 174 & 28 \\ 3 & 3 & 3 & 14 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[468] \quad 21^{124} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -716, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 18 & 3 \\ 3 & 1 & 3 & 10 & 18 & 3 \\ 2 & 3 & 1 & 12 & 20 & 3 \\ 10 & 10 & 12 & 42 & 76 & 14 \\ 18 & 18 & 20 & 76 & 130 & 24 \\ 3 & 3 & 3 & 14 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[467] \quad 21^{124} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -716, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 18 & 3 \\ 3 & 1 & 3 & 8 & 18 & 3 \\ 2 & 3 & 1 & 8 & 20 & 3 \\ 8 & 8 & 8 & 22 & 56 & 10 \\ 18 & 18 & 20 & 56 & 130 & 24 \\ 3 & 3 & 3 & 10 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[469] \quad 1^{14} 1^{113} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -733, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 18 & 3 \\ 2 & 1 & 3 & 5 & 17 & 3 \\ 2 & 3 & 1 & 6 & 22 & 3 \\ 5 & 5 & 6 & 9 & 39 & 7 \\ 18 & 17 & 22 & 39 & 137 & 25 \\ 3 & 3 & 3 & 7 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[470] \quad 1^{44} 1^{122} 3^1 21^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -736, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 14 & 3 \\ 2 & 1 & 3 & 12 & 12 & 3 \\ 2 & 3 & 1 & 16 & 16 & 3 \\ 14 & 12 & 16 & 70 & 72 & 18 \\ 14 & 12 & 16 & 72 & 70 & 18 \\ 3 & 3 & 3 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[471] \quad 1^{34} 1^{124} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -740, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 18 & 3 \\ 2 & 1 & 3 & 10 & 18 & 3 \\ 2 & 3 & 1 & 12 & 20 & 3 \\ 11 & 10 & 12 & 42 & 76 & 14 \\ 18 & 18 & 20 & 76 & 130 & 24 \\ 3 & 3 & 3 & 14 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[472] \quad 1^{114} 3^{113} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -747, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 17 & 3 \\ 2 & 1 & 3 & 5 & 18 & 3 \\ 2 & 3 & 1 & 6 & 20 & 3 \\ 5 & 5 & 6 & 9 & 38 & 7 \\ 17 & 18 & 20 & 38 & 130 & 24 \\ 3 & 3 & 3 & 7 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[473] \quad 1^{123} 3^{121} 21^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -747, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 17 & 3 \\ 2 & 1 & 3 & 12 & 17 & 3 \\ 2 & 3 & 1 & 16 & 22 & 3 \\ 13 & 12 & 16 & 70 & 100 & 18 \\ 17 & 17 & 22 & 100 & 137 & 25 \\ 3 & 3 & 3 & 18 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[474] \quad 1^{114} 3^{121} 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -778, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 18 & 3 \\ 2 & 1 & 3 & 5 & 18 & 3 \\ 2 & 3 & 1 & 6 & 20 & 3 \\ 5 & 5 & 6 & 9 & 38 & 7 \\ 18 & 18 & 20 & 38 & 130 & 24 \\ 3 & 3 & 3 & 7 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[475] \quad 1^{122} 3^{121} 21^{144} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -784, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 15 & 3 \\ 2 & 1 & 3 & 14 & 14 & 3 \\ 2 & 3 & 1 & 20 & 20 & 3 \\ 15 & 14 & 20 & 106 & 108 & 22 \\ 15 & 14 & 20 & 108 & 106 & 22 \\ 3 & 3 & 3 & 22 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[476] \quad 1^{124} 3^{121} 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -784, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 18 & 3 \\ 2 & 1 & 3 & 10 & 18 & 3 \\ 2 & 3 & 1 & 12 & 20 & 3 \\ 10 & 10 & 12 & 42 & 76 & 14 \\ 18 & 18 & 20 & 76 & 130 & 24 \\ 3 & 3 & 3 & 14 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[477] \quad 1^{123} 3^{121} 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -786, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 15 & 3 \\ 2 & 1 & 3 & 10 & 15 & 3 \\ 2 & 3 & 1 & 12 & 18 & 3 \\ 10 & 10 & 12 & 42 & 66 & 14 \\ 15 & 15 & 18 & 66 & 97 & 21 \\ 3 & 3 & 3 & 14 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[478] \quad 1^{123} 3^{121} 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -790, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 18 & 3 \\ 2 & 1 & 3 & 10 & 17 & 3 \\ 2 & 3 & 1 & 12 & 22 & 3 \\ 10 & 10 & 12 & 42 & 78 & 14 \\ 18 & 17 & 22 & 78 & 137 & 25 \\ 3 & 3 & 3 & 14 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[479] \quad 1^{123} 3^{121} 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -803, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 16 & 3 \\ 2 & 1 & 3 & 10 & 15 & 3 \\ 2 & 3 & 1 & 12 & 18 & 3 \\ 11 & 10 & 12 & 42 & 66 & 14 \\ 16 & 15 & 18 & 66 & 97 & 21 \\ 3 & 3 & 3 & 14 & 21 & 4 \end{pmatrix}_{b_0}$$

[480] $1^{123}3^{123}21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -804, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 10 & 17 & 3 \\ 2 & 1 & 3 & 10 & 18 & 3 \\ 2 & 3 & 1 & 12 & 20 & 3 \\ 10 & 10 & 12 & 42 & 76 & 14 \\ 17 & 18 & 20 & 76 & 130 & 24 \\ 3 & 3 & 3 & 14 & 24 & 4 \end{pmatrix}_{b_0}$

[481] $21^{114}3^{121}1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -808, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 20 & 3 \\ 3 & 1 & 3 & 5 & 20 & 3 \\ 2 & 3 & 1 & 6 & 24 & 3 \\ 5 & 5 & 6 & 9 & 44 & 7 \\ 20 & 20 & 24 & 44 & 174 & 28 \\ 3 & 3 & 3 & 7 & 28 & 4 \end{pmatrix}_{b_0}$

[482] $21^{114}3^{121}1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -824, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 18 & 3 \\ 3 & 1 & 3 & 5 & 18 & 3 \\ 2 & 3 & 1 & 6 & 20 & 3 \\ 5 & 5 & 6 & 9 & 38 & 7 \\ 18 & 18 & 20 & 38 & 130 & 24 \\ 3 & 3 & 3 & 7 & 24 & 4 \end{pmatrix}_{b_0}$

[483] $1^{122}3^{121}1^{144} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -880, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 16 & 16 & 3 \\ 2 & 1 & 3 & 14 & 14 & 3 \\ 2 & 3 & 1 & 20 & 20 & 3 \\ 16 & 14 & 20 & 106 & 108 & 22 \\ 16 & 14 & 20 & 108 & 106 & 22 \\ 3 & 3 & 3 & 22 & 22 & 4 \end{pmatrix}_{b_0}$

[484] $1^{123}3^{121}1^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -882, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 12 & 17 & 3 \\ 2 & 1 & 3 & 12 & 17 & 3 \\ 2 & 3 & 1 & 16 & 22 & 3 \\ 12 & 12 & 16 & 70 & 100 & 18 \\ 17 & 17 & 22 & 100 & 137 & 25 \\ 3 & 3 & 3 & 18 & 25 & 4 \end{pmatrix}_{b_0}$

[485] $1^{114}3^{121}1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -907, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 5 & 19 & 3 \\ 2 & 1 & 3 & 5 & 20 & 3 \\ 2 & 3 & 1 & 6 & 24 & 3 \\ 5 & 5 & 6 & 9 & 44 & 7 \\ 19 & 20 & 24 & 44 & 174 & 28 \\ 3 & 3 & 3 & 7 & 28 & 4 \end{pmatrix}_{b_0}$

[487] $1^{124}3^{121}1^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -940, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 9 & 19 & 3 \\ 2 & 1 & 3 & 10 & 20 & 3 \\ 2 & 3 & 1 & 12 & 24 & 3 \\ 9 & 10 & 12 & 42 & 88 & 14 \\ 19 & 20 & 24 & 88 & 174 & 28 \\ 3 & 3 & 3 & 14 & 28 & 4 \end{pmatrix}_{b_0}$

[486] $21^{124}3^{121}1^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -940, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 10 & 20 & 3 \\ 3 & 1 & 3 & 10 & 20 & 3 \\ 2 & 3 & 1 & 12 & 24 & 3 \\ 10 & 10 & 12 & 42 & 88 & 14 \\ 20 & 20 & 24 & 88 & 174 & 28 \\ 3 & 3 & 3 & 14 & 28 & 4 \end{pmatrix}_{b_0}$

[488] $1^{124}3^{121}1^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -988, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 2 & 2 & 10 & 19 & 3 \\ 2 & 1 & 3 & 10 & 20 & 3 \\ 2 & 3 & 1 & 12 & 24 & 3 \\ 10 & 10 & 12 & 42 & 88 & 14 \\ 19 & 20 & 24 & 88 & 174 & 28 \\ 3 & 3 & 3 & 14 & 28 & 4 \end{pmatrix}_{b_0}$

$$[489] \quad 1^{134} 1^{123} 3^1 21^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -1011, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 18 & 3 \\ 2 & 1 & 3 & 12 & 17 & 3 \\ 2 & 3 & 1 & 16 & 22 & 3 \\ 13 & 12 & 16 & 70 & 100 & 18 \\ 18 & 17 & 22 & 100 & 137 & 25 \\ 3 & 3 & 3 & 18 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[490] \quad 1^{114} 3^1 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -1018, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 20 & 3 \\ 2 & 1 & 3 & 5 & 20 & 3 \\ 2 & 3 & 1 & 6 & 24 & 3 \\ 5 & 5 & 6 & 9 & 44 & 7 \\ 20 & 20 & 24 & 44 & 174 & 28 \\ 3 & 3 & 3 & 7 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[491] \quad 1^{124} 3^1 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -1192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 20 & 3 \\ 2 & 1 & 3 & 10 & 20 & 3 \\ 2 & 3 & 1 & 12 & 24 & 3 \\ 10 & 10 & 12 & 42 & 88 & 14 \\ 20 & 20 & 24 & 88 & 174 & 28 \\ 3 & 3 & 3 & 14 & 28 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [492] \quad 1^{111} 3^1 141^1 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 2 & 4 & 3 \\ 1 & 1 & 1 & 3 & 2 & 4 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 3 & 1 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 2 & 0 & 3 & 2 \\ 3 & 4 & 0 & 4 & 3 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [493] \quad 1^{111} 3^1 14141^1 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0$$

$$[496] \quad 1^{111} 3151^1 23^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 & 9 & 3 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 2 & 1 & 2 & 5 & 13 & 4 \\ 3 & 0 & 5 & 3 & 13 & 5 \\ 9 & 0 & 13 & 13 & 43 & 15 \\ 3 & 0 & 4 & 5 & 15 & 5 \end{pmatrix}_{b_0}$$

$$[497] \quad 1^{111} 3^1 1^1 5131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 2 & 2 \\ 2 & 3 & 1 & 0 & 2 & 1 & 3 \\ 2 & 3 & 2 & 0 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[498] \quad 1^{112} 31^1 513^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 0 & 1 & 7 & 3 \\ 1 & 2 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 4 & 2 \\ 5 & 7 & 5 & 0 & 4 & 17 & 9 \\ 2 & 3 & 2 & 0 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[499] \quad 1^{14} 31^1 513^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 1 & 14 & 3 \\ 1 & 2 & 0 & 0 & 2 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 1 & 2 & 0 & 0 & 8 & 2 \\ 9 & 14 & 10 & 0 & 8 & 74 & 18 \\ 2 & 3 & 2 & 0 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[500] \quad & 1^{112}31^1513^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 1 & 9 & 2 \\ 2 & 0 & 4 & 6 & 3 & 20 & 6 \\ 1 & 0 & 1 & 3 & 0 & 8 & 2 \\ 6 & 0 & 9 & 20 & 8 & 49 & 15 \\ 2 & 0 & 2 & 6 & 2 & 15 & 4 \end{pmatrix}_{b_0} \\
[501] \quad & 1^{111}31^1513^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 1 & 5 & 2 \\ 2 & 0 & 4 & 6 & 3 & 12 & 6 \\ 1 & 0 & 1 & 3 & 0 & 5 & 2 \\ 4 & 0 & 5 & 12 & 5 & 17 & 9 \\ 2 & 0 & 2 & 6 & 2 & 9 & 4 \end{pmatrix}_{b_0} \\
[502] \quad & 1^{113}31^15131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 10, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 0 & 2 & 4 & 3 \\ 1 & 2 & 0 & 0 & 1 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 3 & 2 \\ 3 & 4 & 4 & 0 & 3 & 6 & 6 \\ 2 & 3 & 2 & 0 & 2 & 6 & 4 \end{pmatrix}_{b_0} \\
[508] \quad & 1^{133}31^15131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.6 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 9 & 72 & 18 \\ 1 & 1 & 0 & 9 & 0 & 9 & 2 \\ 6 & 12 & 0 & 72 & 9 & 70 & 18 \\ 2 & 3 & -1 & 18 & 2 & 18 & 4 \end{pmatrix}_{b_0} \\
[509] \quad & 1^{142}31^15131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 4 & 2 \\ 1 & 1 & 1 & 16 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 12 & 64 & 24 \\ 1 & 1 & 0 & 12 & 0 & 6 & 2 \\ 4 & 8 & 0 & 64 & 6 & 30 & 12 \\ 2 & 3 & -1 & 24 & 2 & 12 & 4 \end{pmatrix}_{b_0} \\
[515] \quad & 2^13^11513^11^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 12, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 4 & 1 & 4 & 2 \\ 1 & 1 & 2 & 7 & 2 & 7 & 3 \\ 2 & 2 & 0 & 5 & 1 & 5 & 2 \\ 4 & 7 & 5 & 17 & 5 & 19 & 9 \\ 1 & 2 & 1 & 5 & 0 & 5 & 2 \\ 4 & 7 & 5 & 19 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 2 & 9 & 4 \end{pmatrix}_{b_0} \\
[514] \quad & 1^{122}31^15131^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 4 & 2 \\ 1 & 1 & 1 & 8 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 6 & 32 & 12 \\ 1 & 1 & 0 & 6 & 0 & 6 & 2 \\ 4 & 8 & 0 & 32 & 6 & 30 & 12 \\ 2 & 3 & -1 & 12 & 2 & 12 & 4 \end{pmatrix}_{b_0} \\
[513] \quad & 1^{111}31^15131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 3 & 8 & 6 \\ 1 & 1 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 0 & 8 & 3 & 6 & 6 \\ 2 & 3 & -1 & 6 & 2 & 6 & 4 \end{pmatrix}_{b_0} \\
[512] \quad & 1^{112}31^15131^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) =
\end{aligned}$$

$$(1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 1 & 10 & 2 \\ 2 & 0 & 4 & 6 & 3 & 24 & 6 \\ 1 & 0 & 1 & 3 & 0 & 10 & 2 \\ 8 & 0 & 10 & 24 & 10 & 74 & 18 \\ 2 & 0 & 2 & 6 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[511] \quad 1^{113}31513^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 1 & 1 & 4 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 3 & 24 & 6 \\ 1 & 1 & 0 & 3 & 0 & 9 & 2 \\ 6 & 12 & 0 & 24 & 9 & 70 & 18 \\ 2 & 3 & -1 & 6 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[503] \quad 1^{144}31513^11^{44} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 8 & 2 \\ 1 & 1 & 1 & 16 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 12 & 128 & 24 \\ 1 & 1 & 0 & 12 & 0 & 12 & 2 \\ 8 & 16 & 0 & 128 & 12 & 126 & 24 \\ 2 & 3 & -1 & 24 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[504] \quad 1^{143}31513^11^{43} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 6 & 2 \\ 1 & 1 & 1 & 16 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 12 & 96 & 24 \\ 1 & 1 & 0 & 12 & 0 & 9 & 2 \\ 6 & 12 & 0 & 96 & 9 & 70 & 18 \\ 2 & 3 & -1 & 24 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[505] \quad 1^{112}31513^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 1 & 1 & 4 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 3 & 16 & 6 \\ 1 & 1 & 0 & 3 & 0 & 6 & 2 \\ 4 & 8 & 0 & 16 & 6 & 30 & 12 \\ 2 & 3 & -1 & 6 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[506] \quad 1^{132}31513^11^{32} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 4 & 2 \\ 1 & 1 & 1 & 12 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 9 & 48 & 18 \\ 1 & 1 & 0 & 9 & 0 & 6 & 2 \\ 4 & 8 & 0 & 48 & 6 & 30 & 12 \\ 2 & 3 & -1 & 18 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[507] \quad 1^{114}31513^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 8 & 2 \\ 1 & 1 & 1 & 4 & 1 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 3 & 32 & 6 \\ 1 & 1 & 0 & 3 & 0 & 12 & 2 \\ 8 & 16 & 0 & 32 & 12 & 126 & 24 \\ 2 & 3 & -1 & 6 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[510] \quad 1^{122}31513^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 4 & 2 \\ 1 & 1 & 2 & 7 & 1 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 & 5 & 2 \\ 4 & 7 & 5 & 17 & 4 & 19 & 9 \\ 1 & 1 & 2 & 4 & 0 & 4 & 2 \\ 4 & 7 & 5 & 19 & 4 & 17 & 9 \\ 2 & 3 & 2 & 9 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[516] \quad 1^{122}31513^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 13, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 5 & 2 \\ 1 & 1 & 2 & 7 & 1 & 7 & 3 \\ 1 & 2 & 0 & 5 & 2 & 5 & 2 \\ 4 & 7 & 5 & 17 & 4 & 19 & 9 \\ 1 & 1 & 2 & 4 & 0 & 4 & 2 \\ 5 & 7 & 5 & 19 & 4 & 17 & 9 \\ 2 & 3 & 2 & 9 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[517] \quad 1^{1113}1^1 5 131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 1 & 4 & 2 \\ 4 & 4 & 4 & 6 & 3 & 8 & 6 \\ 1 & 2 & 1 & 3 & 0 & 3 & 2 \\ 3 & 4 & 4 & 8 & 3 & 6 & 6 \\ 2 & 3 & 2 & 6 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[518] \quad 1^{113}3^1 1^1 5 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 7 & 2 \\ 1 & 1 & 2 & 3 & 2 & 11 & 3 \\ 1 & 2 & 0 & 1 & 1 & 9 & 2 \\ 2 & 3 & 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 1 & 2 & 0 & 8 & 2 \\ 7 & 11 & 9 & 11 & 8 & 49 & 15 \\ 2 & 3 & 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[519] \quad 1^{123}1^1 5 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 2 & 8 & 3 \\ 1 & 2 & 0 & 0 & 1 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 6 & 2 \\ 6 & 8 & 8 & 0 & 6 & 30 & 12 \\ 2 & 3 & 2 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[520] \quad 1^{123}1^1 5 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 0 & 2 & 7 & 3 \\ 1 & 2 & 0 & 0 & 1 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 5 & 2 \\ 5 & 7 & 5 & 0 & 5 & 17 & 9 \\ 2 & 3 & 2 & 0 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[521] \quad 1^{1123}1^1 5 131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 6 & 2 \\ 1 & 1 & 2 & 4 & 2 & 7 & 3 \\ 1 & 2 & 0 & 4 & 1 & 5 & 2 \\ 3 & 4 & 4 & 6 & 3 & 12 & 6 \\ 1 & 2 & 1 & 3 & 0 & 5 & 2 \\ 6 & 7 & 5 & 12 & 5 & 17 & 9 \\ 2 & 3 & 2 & 6 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[522] \quad 1^{113}3^1 1^1 5 131^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 31, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 8 & 2 \\ 1 & 1 & 2 & 4 & 2 & 11 & 3 \\ 1 & 2 & 0 & 4 & 1 & 9 & 2 \\ 2 & 4 & 4 & 6 & 3 & 20 & 6 \\ 1 & 2 & 1 & 3 & 0 & 8 & 2 \\ 8 & 11 & 9 & 20 & 8 & 49 & 15 \\ 2 & 3 & 2 & 6 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[523] \quad 1^{113}3^1 1^1 5 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 8 & 2 \\ 1 & 1 & 2 & 3 & 2 & 11 & 3 \\ 1 & 2 & 0 & 1 & 1 & 9 & 2 \\ 2 & 3 & 1 & 1 & 2 & 11 & 3 \\ 1 & 2 & 1 & 2 & 0 & 8 & 2 \\ 8 & 11 & 9 & 11 & 8 & 49 & 15 \\ 2 & 3 & 2 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[524] \quad 1^{1113}1^1 5 131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 34, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 2 & 1 & 4 & 3 & 13 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 3 & 13 & 4 \\ 1 & 3 & 0 & 3 & 1 & 11 & 3 \\ 6 & 13 & 0 & 13 & 11 & 49 & 15 \\ 2 & 4 & -1 & 4 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[533] \quad 1^{112}3^1 1^5 131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 58, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 7 & 2 \\ 1 & 1 & 2 & 4 & 2 & 8 & 3 \\ 1 & 2 & 0 & 4 & 1 & 8 & 2 \\ 3 & 4 & 4 & 6 & 3 & 16 & 6 \\ 1 & 2 & 1 & 3 & 0 & 6 & 2 \\ 7 & 8 & 8 & 16 & 6 & 30 & 12 \\ 2 & 3 & 2 & 6 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[534] \quad 1^{112}3^1 1^5 131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 63, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 5 & 2 \\ 1 & 1 & 2 & 4 & 2 & 7 & 3 \\ 1 & 2 & 0 & 4 & 1 & 5 & 2 \\ 3 & 4 & 4 & 6 & 3 & 12 & 6 \\ 1 & 2 & 1 & 3 & 0 & 5 & 2 \\ 5 & 7 & 5 & 12 & 5 & 17 & 9 \\ 2 & 3 & 2 & 6 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[535] \quad 1^{113}3^1 1^5 131^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 66, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 9 & 2 \\ 1 & 1 & 2 & 4 & 2 & 12 & 3 \\ 1 & 2 & 0 & 4 & 1 & 12 & 2 \\ 2 & 4 & 4 & 6 & 3 & 24 & 6 \\ 1 & 2 & 1 & 3 & 0 & 9 & 2 \\ 9 & 12 & 12 & 24 & 9 & 70 & 18 \\ 2 & 3 & 2 & 6 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[536] \quad 1^{114}3^1 1^5 131^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 69, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 11 & 2 \\ 1 & 1 & 2 & 4 & 2 & 15 & 3 \\ 1 & 2 & 0 & 4 & 1 & 13 & 2 \\ 2 & 4 & 4 & 6 & 3 & 28 & 6 \\ 1 & 2 & 1 & 3 & 0 & 11 & 2 \\ 11 & 15 & 13 & 28 & 11 & 97 & 21 \\ 2 & 3 & 2 & 6 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[537] \quad 1^{112}3^1 1^5 131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 90, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 5 & 2 \\ 1 & 1 & 2 & 4 & 2 & 8 & 3 \\ 1 & 2 & 0 & 4 & 1 & 8 & 2 \\ 3 & 4 & 4 & 6 & 3 & 16 & 6 \\ 1 & 2 & 1 & 3 & 0 & 6 & 2 \\ 5 & 8 & 8 & 16 & 6 & 30 & 12 \\ 2 & 3 & 2 & 6 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[538] \quad 1^{123}3^1 1^5 131^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 90, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 7 & 2 \\ 1 & 1 & 2 & 8 & 2 & 11 & 3 \\ 1 & 2 & 0 & 8 & 1 & 9 & 2 \\ 5 & 8 & 8 & 30 & 6 & 40 & 12 \\ 1 & 2 & 1 & 6 & 0 & 8 & 2 \\ 7 & 11 & 9 & 40 & 8 & 49 & 15 \\ 2 & 3 & 2 & 12 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[539] \quad 1^{114}3^1 1^5 131^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 94, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 12 & 2 \\ 1 & 1 & 2 & 4 & 2 & 16 & 3 \\ 1 & 2 & 0 & 4 & 1 & 16 & 2 \\ 2 & 4 & 4 & 6 & 3 & 32 & 6 \\ 1 & 2 & 1 & 3 & 0 & 12 & 2 \\ 12 & 16 & 16 & 32 & 12 & 126 & 24 \\ 2 & 3 & 2 & 6 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[540] \quad 1^{123}3^1 1^5 131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 95, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 8 & 2 \\ 1 & 1 & 2 & 7 & 2 & 11 & 3 \\ 1 & 2 & 0 & 5 & 1 & 9 & 2 \\ 4 & 7 & 5 & 17 & 5 & 31 & 9 \\ 1 & 2 & 1 & 5 & 0 & 8 & 2 \\ 8 & 11 & 9 & 31 & 8 & 49 & 15 \\ 2 & 3 & 2 & 9 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[541] \quad 1^{112}3^11^{12}5131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 6 & 2 \\ 1 & 1 & 2 & 4 & 2 & 8 & 3 \\ 1 & 2 & 0 & 4 & 1 & 8 & 2 \\ 3 & 4 & 4 & 6 & 3 & 16 & 6 \\ 1 & 2 & 1 & 3 & 0 & 6 & 2 \\ 6 & 8 & 8 & 16 & 6 & 30 & 12 \\ 2 & 3 & 2 & 6 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[542] \quad 1^{14}3^11^{13}5131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 2 & 15 & 3 \\ 1 & 2 & 0 & 0 & 1 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 11 & 2 \\ 10 & 15 & 13 & 0 & 11 & 97 & 21 \\ 2 & 3 & 2 & 0 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[543] \quad 1^{122}3^11^{21}5131^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 102, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 5 & 2 \\ 1 & 1 & 2 & 8 & 2 & 8 & 3 \\ 1 & 2 & 0 & 8 & 1 & 8 & 2 \\ 6 & 8 & 8 & 30 & 6 & 32 & 12 \\ 1 & 2 & 1 & 6 & 0 & 6 & 2 \\ 5 & 8 & 8 & 32 & 6 & 30 & 12 \\ 2 & 3 & 2 & 12 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[544] \quad 1^{122}3^11^{21}5131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 102, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 4 & 2 \\ 1 & 1 & 2 & 7 & 2 & 7 & 3 \\ 1 & 2 & 0 & 5 & 1 & 5 & 2 \\ 5 & 7 & 5 & 17 & 5 & 19 & 9 \\ 1 & 2 & 1 & 5 & 0 & 5 & 2 \\ 4 & 7 & 5 & 19 & 5 & 17 & 9 \\ 2 & 3 & 2 & 9 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[545] \quad 1^{113}3^11^{12}5131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 111, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 7 & 2 \\ 1 & 1 & 2 & 4 & 2 & 11 & 3 \\ 1 & 2 & 0 & 4 & 1 & 9 & 2 \\ 2 & 4 & 4 & 6 & 3 & 20 & 6 \\ 1 & 2 & 1 & 3 & 0 & 8 & 2 \\ 7 & 11 & 9 & 20 & 8 & 49 & 15 \\ 2 & 3 & 2 & 6 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[546] \quad 1^{113}3^11^{12}5131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 157, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 7 & 2 \\ 1 & 1 & 2 & 4 & 2 & 11 & 3 \\ 1 & 2 & 0 & 4 & 1 & 9 & 2 \\ 3 & 4 & 4 & 6 & 3 & 20 & 6 \\ 1 & 2 & 1 & 3 & 0 & 8 & 2 \\ 7 & 11 & 9 & 20 & 8 & 49 & 15 \\ 2 & 3 & 2 & 6 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[547] \quad 1^{113}3^11^{13}5131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 163, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 8 & 2 \\ 1 & 1 & 2 & 4 & 2 & 11 & 3 \\ 1 & 2 & 0 & 4 & 1 & 9 & 2 \\ 3 & 4 & 4 & 6 & 3 & 20 & 6 \\ 1 & 2 & 1 & 3 & 0 & 8 & 2 \\ 8 & 11 & 9 & 20 & 8 & 49 & 15 \\ 2 & 3 & 2 & 6 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[548] \quad 1^{113}3^11^{12}5131^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 178, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 8 & 2 \\ 1 & 1 & 2 & 4 & 2 & 12 & 3 \\ 1 & 2 & 0 & 4 & 1 & 12 & 2 \\ 3 & 4 & 4 & 6 & 3 & 24 & 6 \\ 1 & 2 & 1 & 3 & 0 & 9 & 2 \\ 8 & 12 & 12 & 24 & 9 & 70 & 18 \\ 2 & 3 & 2 & 6 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[549] \quad 1^{113}3^1 1^5 131^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 180, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 9 & 2 \\ 1 & 1 & 2 & 4 & 2 & 12 & 3 \\ 1 & 2 & 0 & 4 & 1 & 12 & 2 \\ 3 & 4 & 4 & 6 & 3 & 24 & 6 \\ 1 & 2 & 1 & 3 & 0 & 9 & 2 \\ 9 & 12 & 12 & 24 & 9 & 70 & 18 \\ 2 & 3 & 2 & 6 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[550] \quad 1^{114}3^1 1^5 131^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 181, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 10 & 2 \\ 1 & 1 & 2 & 4 & 2 & 15 & 3 \\ 1 & 2 & 0 & 4 & 1 & 13 & 2 \\ 2 & 4 & 4 & 6 & 3 & 28 & 6 \\ 1 & 2 & 1 & 3 & 0 & 11 & 2 \\ 10 & 15 & 13 & 28 & 11 & 97 & 21 \\ 2 & 3 & 2 & 6 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[551] \quad 1^{124}3^1 1^5 131^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 188, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 10 & 2 \\ 1 & 1 & 2 & 7 & 2 & 15 & 3 \\ 1 & 2 & 0 & 5 & 1 & 13 & 2 \\ 4 & 7 & 5 & 17 & 5 & 43 & 9 \\ 1 & 2 & 1 & 5 & 0 & 11 & 2 \\ 10 & 15 & 13 & 43 & 11 & 97 & 21 \\ 2 & 3 & 2 & 9 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[552] \quad 1^{123}3^1 1^5 131^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 207, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 7 & 2 \\ 1 & 1 & 2 & 7 & 2 & 11 & 3 \\ 1 & 2 & 0 & 5 & 1 & 9 & 2 \\ 4 & 7 & 5 & 17 & 5 & 31 & 9 \\ 1 & 2 & 1 & 5 & 0 & 8 & 2 \\ 7 & 11 & 9 & 31 & 8 & 49 & 15 \\ 2 & 3 & 2 & 9 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[553] \quad 1^{114}3^1 1^5 131^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 297, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 10 & 2 \\ 1 & 1 & 2 & 4 & 2 & 15 & 3 \\ 1 & 2 & 0 & 4 & 1 & 13 & 2 \\ 3 & 4 & 4 & 6 & 3 & 28 & 6 \\ 1 & 2 & 1 & 3 & 0 & 11 & 2 \\ 10 & 15 & 13 & 28 & 11 & 97 & 21 \\ 2 & 3 & 2 & 6 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[554] \quad 1^{114}3^1 1^5 131^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 298, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 11 & 2 \\ 1 & 1 & 2 & 4 & 2 & 16 & 3 \\ 1 & 2 & 0 & 4 & 1 & 16 & 2 \\ 3 & 4 & 4 & 6 & 3 & 32 & 6 \\ 1 & 2 & 1 & 3 & 0 & 12 & 2 \\ 11 & 16 & 16 & 32 & 12 & 126 & 24 \\ 2 & 3 & 2 & 6 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

4.928 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{111}1^1 31^1 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 & 2 & 3 \\ 2 & 2 & 1 & 9 & 4 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 9 & 0 & 22 & 9 & 9 & 10 \\ 3 & 4 & 0 & 9 & 2 & 4 & 4 \\ 2 & 4 & 0 & 9 & 4 & 2 & 4 \\ 3 & 4 & -1 & 10 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

4.929 NHC(-2)(-3) + su₂ + su₂→(-3) + su₂ + su2n3mix

$$\begin{aligned}
T = 12 \quad [8] \quad & 1^{11431}1513^{11141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.0 \\
[12] \quad & 1^{13231}1513^{11321} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.3 \\
[17] \quad & 1^{112231}1513^{11123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.5 \\
[16] \quad & 1^{13331}1513^{11331} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.6 \\
[15] \quad & 1^{14231}1513^{11421} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.5 \\
[14] \quad & 1^{112131}1513^{11122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.5 \\
[13] \quad & 1^{11231}1513^{11121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.5 \\
[1] \quad & 1^{111131}1513^{11112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0 \\
[2] \quad & 1^{111131}1513^{11111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0 \\
[3] \quad & 1^{113131}1513^{11132} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.8 \\
[4] \quad & 1^{11331}1513^{11131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.8 \\
[5] \quad & 1^{114131}1513^{11142} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.0 \\
[6] \quad & 1^{111231}1513^{11113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0 \\
[7] \quad & 1^{122131}1513^{11222} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.0 \\
[0] \quad & 1^{12231}1513^{11221} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.0 \\
[9] \quad & 1^{111331}1513^{11114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0 \\
[10] \quad & 1^{14431}1513^{11441} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 9.0 \\
[11] \quad & 1^{14331}1513^{11431} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 8.8
\end{aligned}$$

4.930 NHC(-2)(-3) + su₂ + su₂→(-3) + su₂→(-3)

$$T = 8 \quad [0] \quad 1^{12}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 4 & 5 & 3 & 3 & 4 \\ 5 & 5 & 4 & 3 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{12}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$\begin{aligned}
[10] \quad & 1^{1^1} 1^{1^2} 3^{1^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 4 & 8 & 3 & 3 & 4 \\ 5 & 8 & 10 & 4 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 & 2 \\ 2 & 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[13] \quad & 21^{1^1} 3^{1^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 6 & 3 & 3 & 4 \\ 5 & 4 & 5 & 3 & 3 & 4 \\ 6 & 5 & 4 & 3 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 & 2 \\ 3 & 3 & 3 & 2 & 0 & 2 \\ 4 & 4 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[12] \quad & 1^{1^2} 1^{1^1} 3^{1^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 3 \\ 3 & 4 & 8 & 3 & 3 & 4 \\ 6 & 8 & 10 & 4 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 & 2 \\ 2 & 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[11] \quad & 1^{1^3} 1^{1^1} 3^{1^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 8 & 2 & 2 & 3 \\ 3 & 4 & 9 & 3 & 3 & 4 \\ 8 & 9 & 13 & 5 & 5 & 7 \\ 2 & 3 & 5 & 0 & 2 & 2 \\ 2 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[14] \quad & 1^{1^3} 1^{1^1} 3^{1^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 9 & 2 & 2 & 3 \\ 3 & 4 & 12 & 3 & 3 & 4 \\ 9 & 12 & 25 & 6 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 & 2 \\ 2 & 3 & 6 & 2 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[15] \quad & 1^{1^4} 1^{1^1} 3^{1^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 2 & 2 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 12 & 14 & 37 & 8 & 8 & 11 \\ 2 & 3 & 8 & 0 & 2 & 2 \\ 2 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[16] \quad & 1^{1^2} 1^{1^1} 3^{1^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 8 & 2 & 2 & 3 \\ 3 & 4 & 12 & 3 & 3 & 4 \\ 8 & 12 & 25 & 6 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 & 2 \\ 2 & 3 & 6 & 2 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[17] \quad & 21^{1^1} 3^{1^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 2 & 2 & 3 \\ 4 & 4 & 4 & 3 & 3 & 4 \\ 3 & 4 & 1 & 2 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[18] \quad & 1^{1^2} 1^{1^1} 3^{1^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 13 & 2 & 2 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 13 & 21 & 76 & 11 & 11 & 16 \\ 2 & 3 & 11 & 0 & 2 & 2 \\ 2 & 3 & 11 & 2 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[46] \quad & 21^{12}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 360, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 18 & 3 & 3 & 4 \\ 5 & 4 & 18 & 3 & 3 & 4 \\ 18 & 18 & 58 & 10 & 10 & 14 \\ 3 & 3 & 10 & 0 & 2 & 2 \\ 3 & 3 & 10 & 2 & 0 & 2 \\ 4 & 4 & 14 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[47] \quad & 21^{12}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 376, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 12 & 2 & 2 & 3 \\ 4 & 4 & 18 & 3 & 3 & 4 \\ 12 & 18 & 58 & 10 & 10 & 14 \\ 2 & 3 & 10 & 0 & 2 & 2 \\ 2 & 3 & 10 & 2 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[48] \quad & 1^{14}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 376, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 2 & 2 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 14 & 21 & 76 & 11 & 11 & 16 \\ 2 & 3 & 11 & 0 & 2 & 2 \\ 2 & 3 & 11 & 2 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[49] \quad & 21^{16}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 376, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 7.2 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 18 & 2 & 2 & 3 \\ 4 & 4 & 24 & 3 & 3 & 4 \\ 18 & 24 & 106 & 12 & 12 & 18 \\ 2 & 3 & 12 & 0 & 2 & 2 \\ 2 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[50] \quad & 1^{14}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 388, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 15 & 2 & 2 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 15 & 21 & 76 & 11 & 11 & 16 \\ 2 & 3 & 11 & 0 & 2 & 2 \\ 2 & 3 & 11 & 2 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[51] \quad & 21^{12}3^{11}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 424, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 21 & 3 & 3 & 4 \\ 5 & 4 & 22 & 3 & 3 & 4 \\ 21 & 22 & 85 & 12 & 12 & 17 \\ 3 & 3 & 12 & 0 & 2 & 2 \\ 3 & 3 & 12 & 2 & 0 & 2 \\ 4 & 4 & 17 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[52] \quad & 1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 424, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 2 & 2 & 3 \\ 3 & 4 & 22 & 3 & 3 & 4 \\ 14 & 22 & 85 & 12 & 12 & 17 \\ 2 & 3 & 12 & 0 & 2 & 2 \\ 2 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[53] \quad & 21^{12}3^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 424, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 24 & 3 & 3 & 4 \\ 5 & 4 & 26 & 3 & 3 & 4 \\ 24 & 26 & 118 & 14 & 14 & 20 \\ 3 & 3 & 14 & 0 & 2 & 2 \\ 3 & 3 & 14 & 2 & 0 & 2 \\ 4 & 4 & 20 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[54] \quad & 21^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 448, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 2 & 3 \\ 4 & 4 & 17 & 3 & 3 & 4 \\ 12 & 17 & 49 & 9 & 9 & 13 \\ 2 & 3 & 9 & 0 & 2 & 2 \\ 2 & 3 & 9 & 2 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 468, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 2 & 3 \\ 3 & 4 & 22 & 3 & 3 & 4 \\ 15 & 22 & 85 & 12 & 12 & 17 \\ 2 & 3 & 12 & 0 & 2 & 2 \\ 2 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{14}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 2 & 3 \\ 3 & 4 & 22 & 3 & 3 & 4 \\ 16 & 22 & 85 & 12 & 12 & 17 \\ 2 & 3 & 12 & 0 & 2 & 2 \\ 2 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[57] \quad 21^{17}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 21 & 2 & 2 & 3 \\ 4 & 4 & 28 & 3 & 3 & 4 \\ 21 & 28 & 145 & 14 & 14 & 21 \\ 2 & 3 & 14 & 0 & 2 & 2 \\ 2 & 3 & 14 & 2 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{14}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 604, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 2 & 3 \\ 3 & 4 & 26 & 3 & 3 & 4 \\ 17 & 26 & 118 & 14 & 14 & 20 \\ 2 & 3 & 14 & 0 & 2 & 2 \\ 2 & 3 & 14 & 2 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[59] \quad 21^{14}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 616, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 2 & 3 \\ 4 & 4 & 21 & 3 & 3 & 4 \\ 15 & 21 & 76 & 11 & 11 & 16 \\ 2 & 3 & 11 & 0 & 2 & 2 \\ 2 & 3 & 11 & 2 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 21^{13}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 640, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 2 & 3 \\ 4 & 4 & 22 & 3 & 3 & 4 \\ 15 & 22 & 85 & 12 & 12 & 17 \\ 2 & 3 & 12 & 0 & 2 & 2 \\ 2 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{14}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 640, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 2 & 3 \\ 3 & 4 & 26 & 3 & 3 & 4 \\ 18 & 26 & 118 & 14 & 14 & 20 \\ 2 & 3 & 14 & 0 & 2 & 2 \\ 2 & 3 & 14 & 2 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 21^{14}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 920, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 2 & 3 \\ 4 & 4 & 26 & 3 & 3 & 4 \\ 18 & 26 & 118 & 14 & 14 & 20 \\ 2 & 3 & 14 & 0 & 2 & 2 \\ 2 & 3 & 14 & 2 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [63] \quad 1^{11}3^113^11^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -37, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 2 & 1 & 4 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 3 & 4 \\ 2 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [64] \quad 1^{11} 5^{13} 1^1 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 1 & 3 \\ 3 & 4 & 7 & 0 & 0 & 4 \\ 6 & 7 & 9 & 0 & 0 & 7 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 7 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{12} 5^{13} 1^1 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 1 & 3 \\ 3 & 4 & 14 & 3 & 0 & 4 \\ 12 & 14 & 42 & 10 & 0 & 14 \\ 3 & 3 & 10 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 14 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{13} 5^{13} 1^1 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 1 & 3 \\ 3 & 4 & 21 & 3 & 0 & 4 \\ 18 & 21 & 97 & 15 & 0 & 21 \\ 3 & 3 & 15 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 21 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{11} 5^{13} 1^1 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 1 & 3 \\ 3 & 4 & 7 & 3 & 0 & 4 \\ 6 & 7 & 9 & 5 & 0 & 7 \\ 3 & 3 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 7 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[70] \quad 3^{11} 1^1 5^{13} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -96, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 2 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 3 & 0 & 3 \\ 3 & 4 & 3 & 2 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[68] \quad 3^{11} 1^1 5^{13} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -96, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 2 & 4 & 12 & 0 & 4 \\ 3 & 4 & 1 & 9 & 0 & 3 \\ 9 & 12 & 9 & 34 & 0 & 12 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 12 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[69] \quad 3^{11} 1^1 5^{13} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -31, \quad \det = -96, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 3 \\ 3 & 2 & 4 & 8 & 0 & 4 \\ 3 & 4 & 1 & 6 & 0 & 3 \\ 6 & 8 & 6 & 14 & 0 & 8 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 8 & 0 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [76] \quad 1^{14} 3^{11} 5^{13} 1^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 1 & 2 \\ 1 & 1 & 2 & 14 & 1 & 2 & 3 \\ 1 & 2 & 0 & 10 & 2 & 1 & 2 \\ 10 & 14 & 10 & 74 & 8 & 10 & 18 \\ 1 & 1 & 2 & 8 & 0 & 1 & 2 \\ 1 & 2 & 1 & 10 & 1 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{12}3^{11}5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 1 & 2 & 7 & 1 & 2 & 3 \\ 1 & 2 & 0 & 5 & 2 & 1 & 2 \\ 5 & 7 & 5 & 17 & 4 & 5 & 9 \\ 1 & 1 & 2 & 4 & 0 & 1 & 2 \\ 1 & 2 & 1 & 5 & 1 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

4.931 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{13}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 & 0 & 4 \\ 3 & 5 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 2 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 3 & 0 & 4 \\ 2 & 4 & 1 & 2 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

4.932 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 9 \quad [8] \ 1^{132}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 2 & 5 & 3 \\ 3 & 4 & 12 & 3 & 3 & 8 & 4 \\ 10 & 12 & 25 & 6 & 6 & 18 & 9 \\ 2 & 3 & 6 & 0 & 2 & 4 & 2 \\ 2 & 3 & 6 & 2 & 0 & 4 & 2 \\ 5 & 8 & 18 & 4 & 4 & 10 & 6 \\ 3 & 4 & 9 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{132}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 2 & 11 & 3 \\ 3 & 4 & 12 & 3 & 3 & 13 & 4 \\ 10 & 12 & 25 & 6 & 6 & 27 & 9 \\ 2 & 3 & 6 & 0 & 2 & 7 & 2 \\ 2 & 3 & 6 & 2 & 0 & 7 & 2 \\ 11 & 13 & 27 & 7 & 7 & 28 & 10 \\ 3 & 4 & 9 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 7 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 4 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 2 & 0 & 4 & 2 \\ 7 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 11 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -4, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 4 & 3 \\ 3 & 4 & 5 & 3 & 3 & 4 & 4 \\ 5 & 5 & 4 & 3 & 3 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 & 2 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 & 2 \\ 4 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -4, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 2 & 3 \\ 3 & 4 & 8 & 3 & 3 & 4 & 4 \\ 7 & 8 & 10 & 4 & 4 & 6 & 6 \\ 2 & 3 & 4 & 0 & 2 & 2 & 2 \\ 2 & 3 & 4 & 2 & 0 & 2 & 2 \\ 2 & 4 & 6 & 2 & 2 & 1 & 3 \\ 3 & 4 & 6 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{123} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -4, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 16 & 3 \\ 3 & 4 & 8 & 3 & 3 & 22 & 4 \\ 5 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 12 & 2 \\ 2 & 3 & 4 & 2 & 0 & 12 & 2 \\ 16 & 22 & 30 & 12 & 12 & 85 & 17 \\ 3 & 4 & 6 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{13} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -4, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 6 & 2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 10 & 12 & 0 & 6 & 6 & 25 & 9 \\ 3 & 4 & 0 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{13} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -4, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 3 & 0 & 4 \\ 4 & 4 & 1 & 2 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 3 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -16, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 & 4 \\ 4 & 4 & 1 & 2 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 2 & 2 \\ 2 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{13} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -16, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 4 & 3 \\ 3 & 4 & 4 & 3 & 3 & 5 & 4 \\ 4 & 4 & 1 & 2 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 2 & 3 & 2 \\ 2 & 3 & 2 & 2 & 0 & 3 & 2 \\ 4 & 5 & 3 & 3 & 3 & 4 & 4 \\ 3 & 4 & 3 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{13} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -16, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 1 & 3 \\ 3 & 4 & 0 & 3 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 0 & 2 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{124} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -16, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 2 & 0 & 6 & 2 \\ 10 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{121} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -16, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 2 & 8 & 3 \\ 3 & 4 & 12 & 3 & 3 & 13 & 4 \\ 8 & 12 & 25 & 6 & 6 & 27 & 9 \\ 2 & 3 & 6 & 0 & 2 & 7 & 2 \\ 2 & 3 & 6 & 2 & 0 & 7 & 2 \\ 8 & 13 & 27 & 7 & 7 & 28 & 10 \\ 3 & 4 & 9 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[15] \ 21^{132} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -16, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 2 & 9 & 3 \\ 4 & 4 & 12 & 3 & 3 & 13 & 4 \\ 9 & 12 & 25 & 6 & 6 & 27 & 9 \\ 2 & 3 & 6 & 0 & 2 & 7 & 2 \\ 2 & 3 & 6 & 2 & 0 & 7 & 2 \\ 9 & 13 & 27 & 7 & 7 & 28 & 10 \\ 3 & 4 & 9 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[14] \ 21^1 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -16, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 3 & 3 & 4 \\ 5 & 4 & 4 & 3 & 3 & 4 & 4 \\ 3 & 4 & 1 & 2 & 2 & 3 & 3 \\ 3 & 3 & 2 & 0 & 2 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 1 & 3 \\ 4 & 4 & 3 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{122} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -16, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 5 & 3 \\ 3 & 4 & 8 & 3 & 3 & 8 & 4 \\ 7 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 4 & 2 \\ 2 & 3 & 4 & 2 & 0 & 4 & 2 \\ 5 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 6 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{123} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -16, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 7 & 3 \\ 3 & 4 & 8 & 3 & 3 & 8 & 4 \\ 7 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 4 & 2 \\ 2 & 3 & 4 & 2 & 0 & 4 & 2 \\ 7 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 6 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{121} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 8 & 3 \\ 3 & 4 & 8 & 3 & 3 & 9 & 4 \\ 7 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 5 & 2 \\ 2 & 3 & 4 & 2 & 0 & 5 & 2 \\ 8 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{113} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 8 & 9 & 0 & 5 & 5 & 13 & 7 \\ 3 & 4 & 0 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{121} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 9 & 4 \\ 6 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 5 & 2 \\ 2 & 3 & 4 & 2 & 0 & 5 & 2 \\ 6 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[26] \quad 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 2 & 3 \\ 3 & 4 & 5 & 3 & 3 & 4 & 4 \\ 3 & 5 & 4 & 3 & 3 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 & 2 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 & 2 \\ 2 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{14} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 3 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 8 & 2 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 12 & 14 & 0 & 8 & 8 & 37 & 11 \\ 3 & 4 & 0 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{123} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 10 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 7 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 6 & 2 \\ 2 & 3 & 4 & 2 & 0 & 6 & 2 \\ 10 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 2 & 1 & 3 \\ 3 & 4 & 0 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{122} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 2 & 9 & 3 \\ 3 & 4 & 12 & 3 & 3 & 13 & 4 \\ 8 & 12 & 25 & 6 & 6 & 27 & 9 \\ 2 & 3 & 6 & 0 & 2 & 7 & 2 \\ 2 & 3 & 6 & 2 & 0 & 7 & 2 \\ 9 & 13 & 27 & 7 & 7 & 28 & 10 \\ 3 & 4 & 9 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{121} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 5 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 4 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 2 & 0 & 4 & 2 \\ 5 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{132} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 8 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 7 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 6 & 2 \\ 2 & 3 & 4 & 2 & 0 & 6 & 2 \\ 8 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{122}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 10 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 7 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 7 & 2 \\ 2 & 3 & 4 & 2 & 0 & 7 & 2 \\ 10 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 3 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 4 & 5 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & 0 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{132}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 2 & 10 & 3 \\ 3 & 4 & 12 & 3 & 3 & 13 & 4 \\ 9 & 12 & 25 & 6 & 6 & 27 & 9 \\ 2 & 3 & 6 & 0 & 2 & 7 & 2 \\ 2 & 3 & 6 & 2 & 0 & 7 & 2 \\ 10 & 13 & 27 & 7 & 7 & 28 & 10 \\ 3 & 4 & 9 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 3 & 3 \\ 3 & 4 & 5 & 3 & 3 & 4 & 4 \\ 4 & 5 & 4 & 3 & 3 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 & 2 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 & 4 \\ 4 & 4 & 1 & 2 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 3 & 3 \\ 3 & 4 & 8 & 3 & 3 & 4 & 4 \\ 7 & 8 & 10 & 4 & 4 & 6 & 6 \\ 2 & 3 & 4 & 0 & 2 & 2 & 2 \\ 2 & 3 & 4 & 2 & 0 & 2 & 2 \\ 3 & 4 & 6 & 2 & 2 & 1 & 3 \\ 3 & 4 & 6 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 3 & 4 & 0 & 2 & 2 & 1 & 3 \\ 3 & 4 & 0 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[35] \quad 21^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 2 & 6 & 3 \\ 4 & 4 & 8 & 3 & 3 & 9 & 4 \\ 6 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 5 & 2 \\ 2 & 3 & 4 & 2 & 0 & 5 & 2 \\ 6 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{122} 1^{113} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 8 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 2 & 0 & 6 & 2 \\ 8 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{122} 1^{113} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 4 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 2 & 0 & 4 & 2 \\ 6 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[38] \quad 13^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 & 4 \\ 2 & 4 & 1 & 2 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 2 & 2 \\ 2 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{123} 1^{113} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 8 & 3 \\ 3 & 4 & 4 & 3 & 3 & 9 & 4 \\ 4 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 5 & 2 \\ 2 & 3 & 2 & 2 & 0 & 5 & 2 \\ 8 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{123} 1^{113} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 9 & 4 \\ 5 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 5 & 2 \\ 2 & 3 & 4 & 2 & 0 & 5 & 2 \\ 6 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{124} 1^{113} 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 14 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 2 & 0 & 8 & 2 \\ 12 & 14 & 9 & 8 & 8 & 37 & 11 \\ 3 & 4 & 3 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{11} 1^{123} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 4 & 2 \\ 2 & 3 & 0 & 2 & 0 & 4 & 2 \\ 5 & 8 & 0 & 4 & 4 & 10 & 6 \\ 3 & 4 & 0 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[45] \quad 21^{123} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 2 & 15 & 3 \\ 4 & 4 & 8 & 3 & 3 & 22 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 12 & 2 \\ 2 & 3 & 4 & 2 & 0 & 12 & 2 \\ 15 & 22 & 30 & 12 & 12 & 85 & 17 \\ 3 & 4 & 6 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{111}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 14 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 2 & 0 & 8 & 2 \\ 10 & 14 & 9 & 8 & 8 & 37 & 11 \\ 3 & 4 & 3 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{124}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 3 & 16 & 4 \\ 7 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 8 & 2 \\ 2 & 3 & 4 & 2 & 0 & 8 & 2 \\ 11 & 16 & 24 & 8 & 8 & 46 & 12 \\ 3 & 4 & 6 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{121}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 7 & 3 \\ 3 & 4 & 8 & 3 & 3 & 9 & 4 \\ 6 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 5 & 2 \\ 2 & 3 & 4 & 2 & 0 & 5 & 2 \\ 7 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{123}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 4 & 2 \\ 2 & 3 & 0 & 2 & 0 & 4 & 2 \\ 6 & 8 & 0 & 4 & 4 & 10 & 6 \\ 3 & 4 & 0 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{122}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 8 & 4 \\ 7 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 4 & 2 \\ 2 & 3 & 4 & 2 & 0 & 4 & 2 \\ 6 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 6 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{123}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 10 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 5 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 7 & 2 \\ 2 & 3 & 4 & 2 & 0 & 7 & 2 \\ 10 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[50] \quad 21^{113}3^{11}1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 3 & 15 & 4 \\ 5 & 4 & 4 & 3 & 3 & 17 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 9 & 2 \\ 3 & 3 & 2 & 2 & 0 & 9 & 2 \\ 15 & 17 & 12 & 9 & 9 & 49 & 13 \\ 4 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{131}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 14 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{113} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 8 & 3 \\ 3 & 4 & 4 & 3 & 3 & 9 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 5 & 2 \\ 2 & 3 & 2 & 0 & 5 & 2 & 2 \\ 8 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{132} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 2 & 6 & 3 \\ 3 & 4 & 12 & 3 & 3 & 8 & 4 \\ 10 & 12 & 25 & 6 & 6 & 18 & 9 \\ 2 & 3 & 6 & 0 & 2 & 4 & 2 \\ 2 & 3 & 6 & 2 & 0 & 4 & 2 \\ 6 & 8 & 18 & 4 & 4 & 10 & 6 \\ 3 & 4 & 9 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{112} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 9 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 6 & 2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 8 & 12 & 0 & 6 & 6 & 25 & 9 \\ 3 & 4 & 0 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{14} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 7 & 2 \\ 2 & 3 & 0 & 2 & 0 & 7 & 2 \\ 11 & 13 & 0 & 7 & 7 & 28 & 10 \\ 3 & 4 & 0 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{123} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 16 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 2 & 0 & 8 & 2 \\ 11 & 16 & 12 & 8 & 8 & 46 & 12 \\ 3 & 4 & 3 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 5 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 2 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 2 & 0 & 4 & 2 \\ 5 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{123} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 7 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 7 & 2 \\ 2 & 3 & 4 & 2 & 0 & 7 & 2 \\ 11 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[69] $1^{1113}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 & 4 \\ 3 & 4 & 1 & 2 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[68] $1^{1143}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 16 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 12 & 16 & 12 & 8 & 8 & 46 & 12 \\ 3 & 4 & 3 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[70] $1^{1133}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 0 & 2 & 6 & 2 \\ 9 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{113}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 7 & 9 & 0 & 5 & 5 & 13 & 7 \\ 3 & 4 & 0 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[74] $1^{1243}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 3 & 16 & 4 \\ 7 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 8 & 2 \\ 2 & 3 & 4 & 2 & 0 & 8 & 2 \\ 12 & 16 & 24 & 8 & 8 & 46 & 12 \\ 3 & 4 & 6 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[71] $1^{1233}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 13 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 7 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 9 & 2 \\ 2 & 3 & 4 & 2 & 0 & 9 & 2 \\ 13 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[73] $1^{143}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 8 & 2 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 12 & 16 & 0 & 8 & 8 & 46 & 12 \\ 3 & 4 & 0 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[75] $1^{1123}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 16 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 2 & 0 & 8 & 2 \\ 12 & 16 & 12 & 8 & 8 & 46 & 12 \\ 3 & 4 & 3 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{14}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 8 & 2 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 11 & 16 & 0 & 8 & 8 & 46 & 12 \\ 3 & 4 & 0 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[83] \ 1^{122}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 5 & 3 \\ 3 & 4 & 8 & 3 & 3 & 8 & 4 \\ 5 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 4 & 2 \\ 2 & 3 & 4 & 2 & 0 & 4 & 2 \\ 5 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 6 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[82] \ 21^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 3 & 9 & 4 \\ 5 & 4 & 4 & 3 & 3 & 9 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 3 & 3 & 2 & 0 & 2 & 5 & 2 \\ 3 & 3 & 2 & 2 & 0 & 5 & 2 \\ 9 & 9 & 6 & 5 & 5 & 13 & 7 \\ 4 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{113}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 8 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 2 & 0 & 6 & 2 \\ 8 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[88] \ 1^{123}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -156, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 9 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 5 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 6 & 2 \\ 2 & 3 & 4 & 2 & 0 & 6 & 2 \\ 9 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[89] \ 1^{113}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -156, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 3 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 8 & 2 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 11 & 14 & 0 & 8 & 8 & 37 & 11 \\ 3 & 4 & 0 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[90] \ 1^{123}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -156, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 15 & 3 \\ 3 & 4 & 8 & 3 & 3 & 22 & 4 \\ 5 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 12 & 2 \\ 2 & 3 & 4 & 2 & 0 & 12 & 2 \\ 15 & 22 & 30 & 12 & 12 & 85 & 17 \\ 3 & 4 & 6 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[92] \ 1^{112}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -160, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 3 & 16 & 4 \\ 5 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 8 & 2 \\ 2 & 3 & 4 & 2 & 0 & 8 & 2 \\ 12 & 16 & 24 & 8 & 8 & 46 & 12 \\ 3 & 4 & 6 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[99] $1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 13 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 5 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 9 & 2 \\ 2 & 3 & 4 & 2 & 0 & 9 & 2 \\ 13 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[102] $1^{123}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 14 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[103] $1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 2 & 0 & 9 & 2 \\ 12 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[104] $1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 8 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 6 & 2 \\ 2 & 3 & 4 & 2 & 0 & 6 & 2 \\ 8 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[101] $1^{123}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 16 & 3 \\ 3 & 4 & 8 & 3 & 3 & 22 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 12 & 2 \\ 2 & 3 & 4 & 2 & 0 & 12 & 2 \\ 16 & 22 & 30 & 12 & 12 & 85 & 17 \\ 3 & 4 & 6 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[105] $1^{123}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 14 & 3 \\ 3 & 4 & 8 & 3 & 3 & 22 & 4 \\ 5 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 12 & 2 \\ 2 & 3 & 4 & 2 & 0 & 12 & 2 \\ 14 & 22 & 30 & 12 & 12 & 85 & 17 \\ 3 & 4 & 6 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[106] $1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 9 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 5 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 7 & 2 \\ 2 & 3 & 4 & 2 & 0 & 7 & 2 \\ 9 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[107] $1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 8 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 5 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 6 & 2 \\ 2 & 3 & 4 & 2 & 0 & 6 & 2 \\ 8 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[108] \ 1^{12} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -196, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 3 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 8 & 2 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 10 & 14 & 0 & 8 & 8 & 37 & 11 \\ 3 & 4 & 0 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{14} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -196, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 16 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 2 & 3 & 2 & 0 & 0 & 8 & 2 \\ 11 & 16 & 12 & 8 & 8 & 46 & 12 \\ 3 & 4 & 3 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{122} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -200, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 9 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 7 & 2 \\ 2 & 3 & 4 & 2 & 0 & 7 & 2 \\ 9 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{123} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -200, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 9 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 6 & 2 \\ 2 & 3 & 4 & 2 & 0 & 6 & 2 \\ 9 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{122} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -204, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 8 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 5 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 7 & 2 \\ 2 & 3 & 4 & 2 & 0 & 7 & 2 \\ 8 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -204, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 7 & 2 \\ 2 & 3 & 0 & 2 & 0 & 7 & 2 \\ 10 & 13 & 0 & 7 & 7 & 28 & 10 \\ 3 & 4 & 0 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[114] \ 21^{111} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -208, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.85 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 & 3 & 3 \\ 4 & 4 & 4 & 3 & 3 & 4 & 4 \\ 3 & 4 & 1 & 2 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{122} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -224, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 3 & 16 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 8 & 2 \\ 2 & 3 & 4 & 2 & 0 & 8 & 2 \\ 12 & 16 & 24 & 8 & 8 & 46 & 12 \\ 3 & 4 & 6 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{123} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -256, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 16 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 4 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 2 & 0 & 12 & 2 \\ 16 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{124} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -256, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 3 & 16 & 4 \\ 5 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 8 & 2 \\ 2 & 3 & 4 & 2 & 0 & 8 & 2 \\ 11 & 16 & 24 & 8 & 8 & 46 & 12 \\ 3 & 4 & 6 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[123] \ 21^{123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -256, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 & 6 & 3 \\ 4 & 4 & 4 & 3 & 3 & 8 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 2 & 0 & 4 & 2 \\ 6 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{123} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -260, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 13 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -260, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 9 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{124} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -264, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 3 & 16 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 8 & 2 \\ 2 & 3 & 4 & 2 & 0 & 8 & 2 \\ 11 & 16 & 24 & 8 & 8 & 46 & 12 \\ 3 & 4 & 6 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[130] \ 1^{123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -264, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 9 & 2 \\ 2 & 3 & 4 & 2 & 0 & 9 & 2 \\ 11 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -268, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 8 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 8 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{113}3^{14}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -276, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 2 & 0 & 9 & 2 \\ 13 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{114}3^{14}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -292, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 4 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 11 & 2 \\ 2 & 3 & 2 & 2 & 0 & 11 & 2 \\ 15 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{13}3^{14}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -296, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 9 & 2 \\ 2 & 3 & 0 & 2 & 0 & 9 & 2 \\ 13 & 17 & 0 & 9 & 9 & 49 & 13 \\ 3 & 4 & 0 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[135] \ 21^{122}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -304, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 2 & 6 & 3 \\ 4 & 4 & 8 & 3 & 3 & 8 & 4 \\ 6 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 4 & 2 \\ 2 & 3 & 4 & 2 & 0 & 4 & 2 \\ 6 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 6 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[136] \ 1^{123}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -312, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 3 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 10 & 2 \\ 2 & 3 & 0 & 2 & 0 & 10 & 2 \\ 14 & 18 & 0 & 10 & 10 & 58 & 14 \\ 3 & 4 & 0 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -332, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 10 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[138] \ 21^{123}3^{11}1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -336, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 3 & 18 & 4 \\ 5 & 4 & 4 & 3 & 3 & 18 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 10 & 2 \\ 3 & 3 & 2 & 2 & 0 & 10 & 2 \\ 18 & 18 & 12 & 10 & 10 & 58 & 14 \\ 4 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[139] \ 21^{122}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -336, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

[164] $1^{1^{12}} 1^{1^{13}} 3^{1^1 1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -468, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 2 & 0 & 9 & 2 \\ 11 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[165] $1^{1^{23}} 1^{1^{13}} 3^{1^1 1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -480, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 14 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 11 & 2 \\ 2 & 3 & 4 & 2 & 0 & 11 & 2 \\ 14 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[166] $1^{1^{14}} 1^{1^{23}} 3^{1^1 1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -484, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 15 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 5 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 11 & 2 \\ 2 & 3 & 4 & 2 & 0 & 11 & 2 \\ 15 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[167] $21^{1^{24}} 3^{1^1 1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 2 & 12 & 3 \\ 4 & 4 & 8 & 3 & 3 & 16 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 8 & 2 \\ 2 & 3 & 4 & 2 & 0 & 8 & 2 \\ 12 & 16 & 24 & 8 & 8 & 46 & 12 \\ 3 & 4 & 6 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[168] $1^{1^{13}} 1^{1^{24}} 3^{1^1 1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 17 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 5 & 8 & 10 & 4 & 4 & 36 & 6 \\ 2 & 3 & 4 & 0 & 2 & 14 & 2 \\ 2 & 3 & 4 & 2 & 0 & 14 & 2 \\ 17 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[169] $1^{1^{14}} 1^{1^{13}} 3^{1^1 1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 2 & 0 & 9 & 2 \\ 13 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[170] $1^{1^{14}} 1^{1^{13}} 3^{1^1 1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 11 & 2 \\ 2 & 3 & 2 & 2 & 0 & 11 & 2 \\ 15 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[172] $21^{1^{12}} 3^{1^1 1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 & 9 & 3 \\ 4 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 9 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[171] $1^{1^{13}} 1^{1^{13}} 3^{1^1 1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -528, (n_+, n_-, n_0) =$

$(1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 2 & 0 & 9 & 2 \\ 12 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[173] \text{ } 1^{13} 1^{13} 1^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -540, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 3 & 3 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 12 & 2 \\ 2 & 3 & 0 & 2 & 0 & 12 & 2 \\ 15 & 22 & 0 & 12 & 12 & 85 & 17 \\ 3 & 4 & 0 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[174] \quad 1^{13} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -548, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 16 & 3 \\ 3 & 4 & 0 & 3 & 3 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 12 & 2 \\ 2 & 3 & 0 & 2 & 0 & 12 & 2 \\ 16 & 22 & 0 & 12 & 12 & 85 & 17 \\ 3 & 4 & 0 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[175] $1^{124}3^{11}1^{11}+[\textcolor{red}{s}\textcolor{red}{u}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{s}\textcolor{green}{u}_2+\textcolor{brown}{s}\textcolor{brown}{u}_2\rightarrow(-3)+\textcolor{red}{s}\textcolor{red}{u}_2\rightarrow(-3)+\textcolor{blue}{s}\textcolor{blue}{u}_2$ $T=9$, $T_{\max}=10$, $\Delta=-29$, $\det=-556$, $(n_+, n_-, n_0)=(1, 9, 0)$, $c_{\text{ext}}=8.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 14 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 5 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 11 & 2 \\ 2 & 3 & 4 & 2 & 0 & 11 & 2 \\ 14 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[176] \ 1^{11}1^33^{11}1^{11}+[\textcolor{red}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{brown}{su}_2\rightarrow(-3)+\textcolor{red}{su}_2\rightarrow(-3)+\textcolor{teal}{su}_2 \quad T=9, \ T_{\max}=10, \ \Delta=-29, \ \det=-560, \ (n_+, n_-, n_0)=(1, 9, 0), \ c_{\text{ext}}=8.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 11 & 2 \\ 2 & 3 & 2 & 2 & 0 & 11 & 2 \\ 14 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[177] \ 1^{113}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -564, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 2 & 0 & 12 & 2 \\ 15 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix} b_0$$

$$[178] \ 1^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -576, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 2 & 0 & 12 & 2 \\ 14 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix} b_0$$

[179] $1^{124}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -584$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.5$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 15 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 11 & 2 \\ 2 & 3 & 4 & 2 & 0 & 11 & 2 \\ 15 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix} b_0$$

[180] $21^{115}3^{111}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -592$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 8.1$

$$[197] \ 21^{114}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -1168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 & 15 & 3 \\ 4 & 4 & 4 & 3 & 3 & 21 & 4 \\ 3 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 11 & 2 \\ 2 & 3 & 2 & 2 & 0 & 11 & 2 \\ 15 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[198] \ 21^{114}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -1536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 2 & 18 & 3 \\ 4 & 4 & 4 & 3 & 3 & 26 & 4 \\ 3 & 4 & 1 & 2 & 2 & 18 & 3 \\ 2 & 3 & 2 & 0 & 2 & 14 & 2 \\ 2 & 3 & 2 & 2 & 0 & 14 & 2 \\ 18 & 26 & 18 & 14 & 14 & 118 & 20 \\ 3 & 4 & 3 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

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$$T = 8 \quad [0] \ 1^{122}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 6 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 \\ 5 & 5 & 4 & 3 & 6 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 \\ 6 & 7 & 6 & 4 & 7 & 5 \\ 3 & 4 & 4 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 3 \\ 3 & 4 & 9 & 3 & 3 & 4 \\ 5 & 9 & 13 & 5 & 5 & 7 \\ 2 & 3 & 5 & 0 & 2 & 2 \\ 3 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 5 & 3 \\ 3 & 4 & 14 & 3 & 7 & 4 \\ 9 & 14 & 37 & 8 & 17 & 11 \\ 2 & 3 & 8 & 0 & 4 & 2 \\ 5 & 7 & 17 & 4 & 7 & 5 \\ 3 & 4 & 11 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{132}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 8 & 3 \\ 3 & 4 & 17 & 3 & 11 & 4 \\ 10 & 17 & 49 & 9 & 33 & 13 \\ 2 & 3 & 9 & 0 & 6 & 2 \\ 8 & 11 & 33 & 6 & 20 & 8 \\ 3 & 4 & 13 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{113}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 \\ 2 & 4 & 1 & 2 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{123}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 3 \\ 3 & 4 & 8 & 3 & 3 & 4 \\ 5 & 8 & 10 & 4 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 & 2 \\ 3 & 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 5 & 3 \\ 3 & 4 & 14 & 3 & 7 & 4 \\ 11 & 14 & 37 & 8 & 17 & 11 \\ 2 & 3 & 8 & 0 & 4 & 2 \\ 5 & 7 & 17 & 4 & 7 & 5 \\ 3 & 4 & 11 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{13}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 3 \\ 3 & 4 & 5 & 3 & 3 & 4 \\ 3 & 5 & 4 & 3 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 & 2 \\ 3 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[7] $1^{13}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 0 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[9] $1^{21}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 & 3 \\ 3 & 4 & 12 & 3 & 3 & 4 \\ 8 & 12 & 25 & 6 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 & 2 \\ 3 & 3 & 6 & 2 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[10] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 \\ 3 & 5 & 4 & 3 & 6 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 \\ 5 & 7 & 6 & 4 & 7 & 5 \\ 3 & 4 & 4 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[11] $1^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 16 & 4 \\ 5 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 12 & 16 & 22 & 9 & 44 & 12 \\ 3 & 4 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[12] $1^{13}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 6 & 3 \\ 3 & 4 & 17 & 3 & 7 & 4 \\ 11 & 17 & 49 & 9 & 21 & 13 \\ 2 & 3 & 9 & 0 & 4 & 2 \\ 6 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[14] $1^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 6 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 \\ 5 & 8 & 10 & 4 & 10 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 \\ 6 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[13] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 \\ 4 & 4 & 1 & 2 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 3 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[15] $1^{13}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 6 & 3 \\ 3 & 4 & 12 & 3 & 7 & 4 \\ 8 & 12 & 25 & 6 & 15 & 9 \\ 2 & 3 & 6 & 0 & 4 & 2 \\ 6 & 7 & 15 & 4 & 7 & 5 \\ 3 & 4 & 9 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[16] $1^{111}3^{111}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 6 & 3 \\ 3 & 4 & 14 & 3 & 7 & 4 \\ 12 & 14 & 37 & 8 & 17 & 11 \\ 2 & 3 & 8 & 0 & 4 & 2 \\ 6 & 7 & 17 & 4 & 7 & 5 \\ 3 & 4 & 11 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[22] $1^{11}3^{111}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 3 \\ 3 & 4 & 5 & 3 & 3 & 4 \\ 4 & 5 & 4 & 3 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 & 2 \\ 3 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[23] $1^{14}3^{111}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 & 3 \\ 3 & 4 & 16 & 3 & 3 & 4 \\ 11 & 16 & 46 & 8 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 & 2 \\ 3 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[21] $21^13^{111}1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 6 & 4 \\ 5 & 4 & 4 & 3 & 7 & 4 \\ 3 & 4 & 1 & 2 & 5 & 3 \\ 3 & 3 & 2 & 0 & 4 & 2 \\ 6 & 7 & 5 & 4 & 7 & 5 \\ 4 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[20] $1^{11}3^{111}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 \\ 4 & 5 & 4 & 3 & 6 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 \\ 5 & 7 & 6 & 4 & 7 & 5 \\ 3 & 4 & 4 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[19] $1^{111}3^{111}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 \\ 2 & 4 & 1 & 2 & 5 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 6 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[18] $1^{14}3^{111}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 13 & 21 & 76 & 11 & 11 & 16 \\ 2 & 3 & 11 & 0 & 2 & 2 \\ 3 & 3 & 11 & 2 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[17] $1^{112}3^{111}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 6 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 \\ 4 & 5 & 4 & 3 & 6 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 \\ 6 & 7 & 6 & 4 & 7 & 5 \\ 3 & 4 & 4 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[24] $1^{111}3^{111}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} =$

6.0

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 6 & 3 \\ 3 & 4 & 14 & 3 & 7 & 4 \\ 11 & 14 & 37 & 8 & 17 & 11 \\ 2 & 3 & 8 & 0 & 4 & 2 \\ 6 & 7 & 17 & 4 & 7 & 5 \\ 3 & 4 & 11 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{1413^{11}1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 6 & 3 \\ 3 & 4 & 16 & 3 & 7 & 4 \\ 11 & 16 & 46 & 8 & 20 & 12 \\ 2 & 3 & 8 & 0 & 4 & 2 \\ 6 & 7 & 20 & 4 & 7 & 5 \\ 3 & 4 & 12 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{1113^{11}1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 5 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 \\ 8 & 9 & 13 & 5 & 11 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 \\ 5 & 7 & 11 & 4 & 7 & 5 \\ 3 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{1113^{11}1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 \\ 3 & 4 & 1 & 2 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{1113^{11}1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 6 & 7 & 0 & 4 & 7 & 5 \\ 3 & 4 & 0 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{123^{11}1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 & 3 \\ 3 & 4 & 18 & 3 & 3 & 4 \\ 11 & 18 & 58 & 10 & 10 & 14 \\ 2 & 3 & 10 & 0 & 2 & 2 \\ 3 & 3 & 10 & 2 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{123^{11}1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 & 3 \\ 3 & 4 & 8 & 3 & 3 & 4 \\ 7 & 8 & 10 & 4 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 & 2 \\ 3 & 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{1113^{11}1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 5 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 \\ 5 & 9 & 13 & 5 & 11 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 \\ 5 & 7 & 11 & 4 & 7 & 5 \\ 3 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{123^{11}1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 3 \\ 3 & 4 & 8 & 3 & 3 & 4 \\ 6 & 8 & 10 & 4 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 & 2 \\ 3 & 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[33] $1^{11} 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 & 3 \\ 3 & 4 & 13 & 3 & 3 & 4 \\ 8 & 13 & 28 & 7 & 7 & 10 \\ 2 & 3 & 7 & 0 & 2 & 2 \\ 3 & 3 & 7 & 2 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[34] $1^{12} 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 6 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 \\ 6 & 9 & 13 & 5 & 11 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 \\ 6 & 7 & 11 & 4 & 7 & 5 \\ 3 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[35] $1^{21} 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 \\ 4 & 4 & 1 & 2 & 5 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[36] $1^{21} 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 5 & 3 \\ 3 & 4 & 14 & 3 & 7 & 4 \\ 10 & 14 & 37 & 8 & 17 & 11 \\ 2 & 3 & 8 & 0 & 4 & 2 \\ 5 & 7 & 17 & 4 & 7 & 5 \\ 3 & 4 & 11 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[37] $1^{31} 1^{21} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 5 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 \\ 7 & 8 & 10 & 4 & 10 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 \\ 5 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[38] $1^{23} 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 9 & 3 \\ 3 & 4 & 12 & 3 & 11 & 4 \\ 8 & 12 & 25 & 6 & 24 & 9 \\ 2 & 3 & 6 & 0 & 6 & 2 \\ 9 & 11 & 24 & 6 & 20 & 8 \\ 3 & 4 & 9 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[39] $1^{11} 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 9 & 14 & 37 & 8 & 8 & 11 \\ 2 & 3 & 8 & 0 & 2 & 2 \\ 3 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[40] $1^{33} 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 9 & 3 \\ 3 & 4 & 16 & 3 & 11 & 4 \\ 11 & 16 & 46 & 8 & 32 & 12 \\ 2 & 3 & 8 & 0 & 6 & 2 \\ 9 & 11 & 32 & 6 & 20 & 8 \\ 3 & 4 & 12 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

- [41] $1^{1223}1^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 58, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 9 & 3 \\ 3 & 4 & 8 & 3 & 11 & 4 \\ 5 & 8 & 10 & 4 & 16 & 6 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 9 & 11 & 16 & 6 & 20 & 8 \\ 3 & 4 & 6 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$
- [42] $1^{131}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 5 & 3 \\ 3 & 4 & 12 & 3 & 7 & 4 \\ 10 & 12 & 25 & 6 & 15 & 9 \\ 2 & 3 & 6 & 0 & 4 & 2 \\ 5 & 7 & 15 & 4 & 7 & 5 \\ 3 & 4 & 9 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$
- [43] $1^{111}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 5 & 7 & 0 & 4 & 7 & 5 \\ 3 & 4 & 0 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$
- [44] $1^{111}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 \\ 4 & 4 & 1 & 2 & 5 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 6 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$
- [45] $1^{123}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 17 & 3 \\ 3 & 4 & 8 & 3 & 25 & 4 \\ 5 & 8 & 10 & 4 & 34 & 6 \\ 2 & 3 & 4 & 0 & 14 & 2 \\ 17 & 25 & 34 & 14 & 111 & 19 \\ 3 & 4 & 6 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$
- [46] $1^{111}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 \\ 4 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 9 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$
- [47] $1^{131}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 & 3 \\ 3 & 4 & 12 & 3 & 3 & 4 \\ 9 & 12 & 25 & 6 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 & 2 \\ 3 & 3 & 6 & 2 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$
- [48] $1^{111}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 3 \\ 3 & 4 & 9 & 3 & 3 & 4 \\ 6 & 9 & 13 & 5 & 5 & 7 \\ 2 & 3 & 5 & 0 & 2 & 2 \\ 3 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$
- [49] $1^{131}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 & 3 \\ 3 & 4 & 12 & 3 & 3 & 4 \\ 10 & 12 & 25 & 6 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 & 2 \\ 3 & 3 & 6 & 2 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[50] $1^{1^4} 1^{11} 2^3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 13 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[51] $1^{1^3} 1^{11} 2^3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 9 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[52] $1^{1^{33}} 1^{11} 2^3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 9 & 3 \\ 3 & 4 & 9 & 3 & 11 & 4 \\ 8 & 9 & 13 & 5 & 17 & 7 \\ 2 & 3 & 5 & 0 & 6 & 2 \\ 9 & 11 & 17 & 6 & 20 & 8 \\ 3 & 4 & 7 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[53] $1^{1^2} 1^{11} 2^3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 9 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[54] $1^{1^1} 1^{11} 2^3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 \\ 2 & 4 & 1 & 2 & 5 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[55] $1^{1^{31}} 1^{11} 2^3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 & 3 \\ 3 & 4 & 9 & 3 & 3 & 4 \\ 8 & 9 & 13 & 5 & 5 & 7 \\ 2 & 3 & 5 & 0 & 2 & 2 \\ 3 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[56] $1^{1^{34}} 1^{11} 2^3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 15 & 3 \\ 3 & 4 & 8 & 3 & 20 & 4 \\ 7 & 8 & 10 & 4 & 28 & 6 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 15 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[57] $1^{1^{34}} 1^{11} 2^3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 13 & 3 \\ 3 & 4 & 8 & 3 & 16 & 4 \\ 7 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 13 & 16 & 22 & 9 & 44 & 12 \\ 3 & 4 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

- [58] $1^{132}3^{1112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 6 & 3 \\ 3 & 4 & 22 & 3 & 7 & 4 \\ 15 & 22 & 85 & 12 & 27 & 17 \\ 2 & 3 & 12 & 0 & 4 & 2 \\ 6 & 7 & 27 & 4 & 7 & 5 \\ 3 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$
- [59] $1^{121}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 6 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 \\ 7 & 8 & 10 & 4 & 10 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 \\ 6 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$
- [62] $1^{113}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 9 & 11 & 0 & 6 & 20 & 8 \\ 3 & 4 & 0 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$
- [60] $1^{141}3^{1112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 & 3 \\ 3 & 4 & 26 & 3 & 3 & 4 \\ 16 & 26 & 118 & 14 & 14 & 20 \\ 2 & 3 & 14 & 0 & 2 & 2 \\ 3 & 3 & 14 & 2 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$
- [61] $1^{141}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 & 3 \\ 3 & 4 & 16 & 3 & 3 & 4 \\ 12 & 16 & 46 & 8 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 & 2 \\ 3 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$
- [63] $1^{122}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 8 & 3 \\ 3 & 4 & 8 & 3 & 11 & 4 \\ 7 & 8 & 10 & 4 & 16 & 6 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 8 & 11 & 16 & 6 & 20 & 8 \\ 3 & 4 & 6 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$
- [64] $21^{113}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 9 & 3 & 6 & 4 \\ 5 & 4 & 9 & 3 & 7 & 4 \\ 9 & 9 & 13 & 5 & 11 & 7 \\ 3 & 3 & 5 & 0 & 4 & 2 \\ 6 & 7 & 11 & 4 & 7 & 5 \\ 4 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$
- [65] $1^{111}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 4 & 1 & 2 & 5 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 6 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$
- [66] $1^{111}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 & 3 \\ 3 & 4 & 9 & 3 & 3 & 4 \\ 7 & 9 & 13 & 5 & 5 & 7 \\ 2 & 3 & 5 & 0 & 2 & 2 \\ 3 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[68] $1^{1^{22}}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 8 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 \\ 4 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[67] $1^{1^{42}}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 8 & 3 \\ 3 & 4 & 12 & 3 & 11 & 4 \\ 10 & 12 & 25 & 6 & 24 & 9 \\ 2 & 3 & 6 & 0 & 6 & 2 \\ 8 & 11 & 24 & 6 & 20 & 8 \\ 3 & 4 & 9 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[69] $1^{1^{23}}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 10 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 \\ 4 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 10 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[70] $1^{1^{13}}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 10 & 12 & 0 & 7 & 25 & 9 \\ 3 & 4 & 0 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[71] $1^{1^{22}}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 6 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 \\ 6 & 8 & 10 & 4 & 10 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 \\ 6 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{1^{32}}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 6 & 3 \\ 3 & 4 & 21 & 3 & 7 & 4 \\ 14 & 21 & 76 & 11 & 26 & 16 \\ 2 & 3 & 11 & 0 & 4 & 2 \\ 6 & 7 & 26 & 4 & 7 & 5 \\ 3 & 4 & 16 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[73] $1^{1^{32}}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 6 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 \\ 8 & 9 & 13 & 5 & 11 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 \\ 6 & 7 & 11 & 4 & 7 & 5 \\ 3 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[74] $1^{1^{11}}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 5 & 3 \\ 3 & 4 & 18 & 3 & 7 & 4 \\ 11 & 18 & 58 & 10 & 22 & 14 \\ 2 & 3 & 10 & 0 & 4 & 2 \\ 5 & 7 & 22 & 4 & 7 & 5 \\ 3 & 4 & 14 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[75] $1^{11} 1^{12} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 5 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 \\ 5 & 8 & 10 & 4 & 10 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 \\ 5 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[76] $1^{11} 1^{12} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 8 & 3 \\ 3 & 4 & 9 & 3 & 11 & 4 \\ 6 & 9 & 13 & 5 & 17 & 7 \\ 2 & 3 & 5 & 0 & 6 & 2 \\ 8 & 11 & 17 & 6 & 20 & 8 \\ 3 & 4 & 7 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[77] $1^{14} 1^{12} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 15 & 4 \\ 5 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 12 & 15 & 22 & 8 & 39 & 11 \\ 3 & 4 & 6 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[78] $1^{21} 1^{13} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 & 3 \\ 3 & 4 & 17 & 3 & 3 & 4 \\ 11 & 17 & 49 & 9 & 9 & 13 \\ 2 & 3 & 9 & 0 & 2 & 2 \\ 3 & 3 & 9 & 2 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[79] $1^{12} 1^{12} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 8 & 3 \\ 3 & 4 & 13 & 3 & 11 & 4 \\ 8 & 13 & 28 & 7 & 25 & 10 \\ 2 & 3 & 7 & 0 & 6 & 2 \\ 8 & 11 & 25 & 6 & 20 & 8 \\ 3 & 4 & 10 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[80] $1^{12} 1^{12} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 8 & 11 & 0 & 6 & 20 & 8 \\ 3 & 4 & 0 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[81] $1^{11} 1^{11} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 \\ 3 & 4 & 1 & 2 & 5 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[82] $1^{22} 1^{12} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 16 & 4 \\ 6 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 11 & 16 & 22 & 9 & 44 & 12 \\ 3 & 4 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[83] $1^{23} 1^{12} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 \\ 4 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 9 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 112, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 8 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[85] \quad 21^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 112, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 12 & 4 \\ 5 & 4 & 4 & 3 & 12 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 0 & 7 & 2 \\ 12 & 12 & 8 & 7 & 25 & 9 \\ 4 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{121}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 113, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 6 & 3 \\ 3 & 4 & 18 & 3 & 7 & 4 \\ 13 & 18 & 58 & 10 & 22 & 14 \\ 2 & 3 & 10 & 0 & 4 & 2 \\ 6 & 7 & 22 & 4 & 7 & 5 \\ 3 & 4 & 14 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{131}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 113, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 6 & 3 \\ 3 & 4 & 12 & 3 & 7 & 4 \\ 9 & 12 & 25 & 6 & 15 & 9 \\ 2 & 3 & 6 & 0 & 4 & 2 \\ 6 & 7 & 15 & 4 & 7 & 5 \\ 3 & 4 & 9 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{121}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 113, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 10 & 14 & 37 & 8 & 3 & 11 \\ 2 & 3 & 8 & 0 & 2 & 2 \\ 3 & 3 & 8 & 0 & 2 & 2 \\ 3 & 4 & 11 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{142}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 114, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 6 & 3 \\ 3 & 4 & 12 & 3 & 7 & 4 \\ 10 & 12 & 25 & 6 & 15 & 9 \\ 2 & 3 & 6 & 0 & 4 & 2 \\ 6 & 7 & 15 & 4 & 7 & 5 \\ 3 & 4 & 9 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 115, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 6 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 \\ 7 & 9 & 13 & 5 & 11 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 \\ 6 & 7 & 11 & 4 & 7 & 5 \\ 3 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{122}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 116, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 16 & 4 \\ 5 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 11 & 16 & 22 & 9 & 44 & 12 \\ 3 & 4 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{124}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 116, \quad (n_+, n_-, n_0) =$$

$$(1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 13 & 3 \\ 3 & 4 & 8 & 3 & 16 & 4 \\ 6 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 13 & 16 & 22 & 9 & 44 & 12 \\ 3 & 4 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{12}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 5 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 \\ 6 & 8 & 10 & 4 & 10 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 \\ 5 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{14}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 15 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 12 & 15 & 11 & 8 & 39 & 11 \\ 3 & 4 & 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 9 & 12 & 0 & 7 & 25 & 9 \\ 3 & 4 & 0 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[96] \quad 1^{12}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 & 3 \\ 3 & 4 & 13 & 3 & 3 & 4 \\ 9 & 13 & 28 & 7 & 7 & 10 \\ 2 & 3 & 7 & 0 & 2 & 2 \\ 3 & 3 & 7 & 2 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[97] \quad 1^{122}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 6 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 \\ 9 & 13 & 28 & 7 & 16 & 10 \\ 2 & 3 & 7 & 0 & 4 & 2 \\ 6 & 7 & 16 & 4 & 7 & 5 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[99] \quad 1^{112}3^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 16 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 12 & 2 \\ 16 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[98] \quad 1^{12}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[100] \quad 1^{111}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 125, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 10 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[101] & 1^{\overset{121}{111}3^{\overset{11}{11}}1^{\overset{11}{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 126, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 7 & 2 & 5 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 \\ 7 & 9 & 13 & 5 & 11 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 \\ 5 & 7 & 11 & 4 & 7 & 5 \\ 3 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[102] & 1^{\overset{141}{121}3^{\overset{11}{11}}1^{\overset{11}{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 129, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 11 & 2 & 5 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 \\ 11 & 13 & 28 & 7 & 16 & 10 \\ 2 & 3 & 7 & 0 & 4 & 2 \\ 5 & 7 & 16 & 4 & 7 & 5 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[103] & 1^{\overset{111}{111}3^{\overset{11}{11}}1^{\overset{11}{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 129, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 2 & 5 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 \\ 6 & 9 & 13 & 5 & 11 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 \\ 5 & 7 & 11 & 4 & 7 & 5 \\ 3 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[104] & 1^{\overset{121}{131}3^{\overset{11}{11}}1^{\overset{11}{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 131, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 8 & 2 & 5 & 3 \\ 3 & 4 & 12 & 3 & 7 & 4 \\ 8 & 12 & 25 & 6 & 15 & 9 \\ 2 & 3 & 6 & 0 & 4 & 2 \\ 5 & 7 & 15 & 4 & 7 & 5 \\ 3 & 4 & 9 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[105] & 1^{\overset{131}{113}3^{\overset{11}{11}}1^{\overset{12}{12}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 133, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 11 & 2 & 3 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 11 & 14 & 37 & 8 & 8 & 11 \\ 2 & 3 & 8 & 0 & 2 & 2 \\ 3 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[106] & 1^{\overset{142}{141}3^{\overset{11}{11}}1^{\overset{11}{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 134, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 2 & 6 & 3 \\ 3 & 4 & 16 & 3 & 7 & 4 \\ 12 & 16 & 46 & 8 & 20 & 12 \\ 2 & 3 & 8 & 0 & 4 & 2 \\ 6 & 7 & 20 & 4 & 7 & 5 \\ 3 & 4 & 12 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[107] & 1^{\overset{114}{123}3^{\overset{11}{11}}1^{\overset{11}{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 135, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 2 & 15 & 3 \\ 3 & 4 & 8 & 3 & 20 & 4 \\ 5 & 8 & 10 & 4 & 28 & 6 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 15 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[108] & 21^{\overset{113}{113}3^{\overset{11}{11}}1^{\overset{131}{131}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 136, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 15 & 3 & 6 & 4 \\ 5 & 4 & 17 & 3 & 7 & 4 \\ 15 & 17 & 49 & 9 & 21 & 13 \\ 3 & 3 & 9 & 0 & 4 & 2 \\ 6 & 7 & 21 & 4 & 7 & 5 \\ 4 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[109] & 1^{\overset{131}{131}3^{\overset{11}{11}}1^{\overset{11}{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 136, \quad (n_+, n_-, n_0) =
\end{aligned}$$

$$(1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 5 & 3 \\ 3 & 4 & 12 & 3 & 7 & 4 \\ 9 & 12 & 25 & 6 & 15 & 9 \\ 2 & 3 & 6 & 0 & 4 & 2 \\ 5 & 7 & 15 & 4 & 7 & 5 \\ 3 & 4 & 9 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{122}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 138, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 9 & 3 \\ 3 & 4 & 8 & 3 & 11 & 4 \\ 7 & 8 & 10 & 4 & 16 & 6 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 9 & 11 & 16 & 6 & 20 & 8 \\ 3 & 4 & 6 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{123}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 15 & 4 \\ 7 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 11 & 15 & 22 & 8 & 39 & 11 \\ 3 & 4 & 6 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{13}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 12 & 15 & 0 & 8 & 39 & 11 \\ 3 & 4 & 0 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[114] \ 21^{11}3^{11}1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 12 & 3 & 6 & 4 \\ 5 & 4 & 13 & 3 & 7 & 4 \\ 12 & 13 & 28 & 7 & 16 & 10 \\ 3 & 3 & 7 & 0 & 4 & 2 \\ 6 & 7 & 16 & 4 & 7 & 5 \\ 4 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{131}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 5 & 3 \\ 3 & 4 & 22 & 3 & 7 & 4 \\ 13 & 22 & 85 & 12 & 27 & 17 \\ 2 & 3 & 12 & 0 & 4 & 2 \\ 5 & 7 & 27 & 4 & 7 & 5 \\ 3 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{123}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 & 3 \\ 3 & 4 & 22 & 3 & 3 & 4 \\ 14 & 22 & 85 & 12 & 12 & 17 \\ 2 & 3 & 12 & 0 & 2 & 2 \\ 3 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 4 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 12 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 & 3 \\ 3 & 4 & 13 & 3 & 3 & 4 \\ 11 & 13 & 28 & 7 & 7 & 10 \\ 2 & 3 & 7 & 0 & 2 & 2 \\ 3 & 3 & 7 & 0 & 2 & 2 \\ 3 & 4 & 10 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[118] \quad & 1^{123} 1^{113} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 147, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 15 & 4 \\ 4 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 11 & 15 & 11 & 8 & 39 & 11 \\ 3 & 4 & 3 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[119] \quad & 1^{123} 1^{122} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 150, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 16 & 4 \\ 6 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 12 & 16 & 22 & 9 & 44 & 12 \\ 3 & 4 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[121] \quad & 1^{112} 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 152, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 9 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[120] \quad & 1^{112} 1^{122} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 152, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 2 & 8 & 3 \\ 3 & 4 & 8 & 3 & 11 & 4 \\ 5 & 8 & 10 & 4 & 16 & 6 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 8 & 11 & 16 & 6 & 20 & 8 \\ 3 & 4 & 6 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[123] \quad & 21^{12} 3^{11} 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 154, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 18 & 3 & 6 & 4 \\ 5 & 4 & 18 & 3 & 7 & 4 \\ 18 & 18 & 58 & 10 & 22 & 14 \\ 3 & 3 & 10 & 0 & 4 & 2 \\ 6 & 7 & 22 & 4 & 7 & 5 \\ 4 & 4 & 14 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[122] \quad & 1^{21} 1^{12} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 154, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 2 & 3 & 3 \\ 3 & 4 & 18 & 3 & 3 & 4 \\ 12 & 18 & 58 & 10 & 10 & 14 \\ 2 & 3 & 10 & 0 & 2 & 2 \\ 3 & 3 & 10 & 2 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[124] \quad & 1^{31} 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 154, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 2 & 3 & 3 \\ 3 & 4 & 13 & 3 & 3 & 4 \\ 10 & 13 & 28 & 7 & 7 & 10 \\ 2 & 3 & 7 & 0 & 2 & 2 \\ 3 & 3 & 7 & 2 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[125] \quad & 1^{124} 1^{112} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 156, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 2 & 16 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 4 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 12 & 2 \\ 16 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[126] \quad & 1^{123} 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 156, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 4 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 4 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 14 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[127] \quad & 21^{113}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 160, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 3 & 3 & 18 & 4 \\ 5 & 4 & 4 & 3 & 20 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 11 & 2 \\ 18 & 20 & 14 & 11 & 69 & 15 \\ 4 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[128] \quad & 1^{112}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 161, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 9 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[129] \quad & 1^{143}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 162, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 2 & 3 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 14 & 21 & 76 & 11 & 11 & 16 \\ 2 & 3 & 11 & 0 & 2 & 2 \\ 3 & 3 & 11 & 2 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[130] \quad & 1^{13}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 163, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 11 & 15 & 0 & 8 & 39 & 11 \\ 3 & 4 & 0 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[131] \quad & 1^{141}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 164, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 2 & 5 & 3 \\ 3 & 4 & 16 & 3 & 7 & 4 \\ 12 & 16 & 46 & 8 & 20 & 12 \\ 2 & 3 & 8 & 0 & 4 & 2 \\ 5 & 7 & 20 & 4 & 7 & 5 \\ 3 & 4 & 12 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[133] \quad & 21^{113}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 166, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 3 & 3 & 15 & 4 \\ 5 & 4 & 4 & 3 & 16 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 9 & 2 \\ 15 & 16 & 11 & 9 & 44 & 12 \\ 4 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[134] \quad & 21^{122}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 166, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 6 & 2 & 11 & 3 \\ 4 & 4 & 8 & 3 & 16 & 4 \\ 6 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 9 & 2 \\ 11 & 16 & 22 & 9 & 44 & 12 \\ 3 & 4 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[132] \quad & 1^{112}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 166, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 12 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[135] \quad & 21^{113}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 168, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 8 & 3 \\ 4 & 4 & 4 & 3 & 12 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 7 & 2 \\ 8 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[136] \quad 1^{131}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 169, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 5 & 3 \\ 3 & 4 & 17 & 3 & 7 & 4 \\ 10 & 17 & 49 & 9 & 21 & 13 \\ 2 & 3 & 9 & 0 & 4 & 2 \\ 5 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[137] \quad 1^{122}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 170, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 9 & 3 \\ 3 & 4 & 8 & 3 & 11 & 4 \\ 6 & 8 & 10 & 4 & 16 & 6 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 9 & 11 & 16 & 6 & 20 & 8 \\ 3 & 4 & 6 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[138] \quad 1^{113}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 170, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 15 & 4 \\ 4 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 12 & 15 & 11 & 8 & 39 & 11 \\ 3 & 4 & 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[139] \quad 1^{141}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 171, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 5 & 3 \\ 3 & 4 & 16 & 3 & 7 & 4 \\ 11 & 16 & 46 & 8 & 20 & 12 \\ 2 & 3 & 8 & 0 & 4 & 2 \\ 5 & 7 & 20 & 4 & 7 & 5 \\ 3 & 4 & 12 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[140] \quad 1^{132}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 173, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 9 & 3 \\ 3 & 4 & 12 & 3 & 11 & 4 \\ 10 & 12 & 25 & 6 & 24 & 9 \\ 2 & 3 & 6 & 0 & 6 & 2 \\ 9 & 11 & 24 & 6 & 20 & 8 \\ 3 & 4 & 9 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[141] \quad 1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 174, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 5 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 \\ 8 & 13 & 28 & 7 & 16 & 10 \\ 2 & 3 & 7 & 0 & 4 & 2 \\ 5 & 7 & 16 & 4 & 7 & 5 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[142] \quad 1^{113}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 175, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 13 & 17 & 0 & 10 & 55 & 13 \\ 3 & 4 & 0 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[143] \quad 1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 176, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 13 & 16 & 0 & 9 & 44 & 12 \\ 3 & 4 & 0 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[145] \quad 21^{111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 180, \quad (n_+, n_-, n_0) =$$

$(1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 5 & 3 \\ 4 & 4 & 4 & 3 & 7 & 4 \\ 3 & 4 & 1 & 2 & 5 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$[144] \ 1^{112}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 180, (n_+, n_-, n_0) =$
 $(1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$[146] \ 1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 181, (n_+, n_-, n_0) =$
 $(1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 4 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 13 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$[147] \ 1^{131}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 183, (n_+, n_-, n_0) =$
 $(1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 & 3 \\ 3 & 4 & 17 & 3 & 3 & 4 \\ 12 & 17 & 49 & 9 & 9 & 13 \\ 2 & 3 & 9 & 0 & 2 & 2 \\ 3 & 3 & 9 & 2 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$[148] \ 1^{132}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 185, (n_+, n_-, n_0) =$
 $(1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 9 & 3 \\ 3 & 4 & 12 & 3 & 11 & 4 \\ 9 & 12 & 25 & 6 & 24 & 9 \\ 2 & 3 & 6 & 0 & 6 & 2 \\ 9 & 11 & 24 & 6 & 20 & 8 \\ 3 & 4 & 9 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$[149] \ 1^{113}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 187, (n_+, n_-, n_0) =$
 $(1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 15 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 11 & 15 & 11 & 8 & 39 & 11 \\ 3 & 4 & 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$[150] \ 1^{132}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 188, (n_+, n_-, n_0) =$
 $(1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 8 & 3 \\ 3 & 4 & 12 & 3 & 11 & 4 \\ 8 & 12 & 25 & 6 & 24 & 9 \\ 2 & 3 & 6 & 0 & 6 & 2 \\ 8 & 11 & 24 & 6 & 20 & 8 \\ 3 & 4 & 9 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$[151] \ 1^{113}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 190, (n_+, n_-, n_0) =$
 $(1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 18 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 4 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 18 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$[152] \ 1^{142}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 192, (n_+, n_-, n_0) =$
 $(1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 8 & 3 \\ 3 & 4 & 21 & 3 & 11 & 4 \\ 13 & 21 & 76 & 11 & 41 & 16 \\ 2 & 3 & 11 & 0 & 6 & 2 \\ 8 & 11 & 41 & 6 & 20 & 8 \\ 3 & 4 & 16 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[153] \quad 1^{122}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 192, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 8 & 3 \\ 3 & 4 & 8 & 3 & 11 & 4 \\ 6 & 8 & 10 & 4 & 16 & 6 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 8 & 11 & 16 & 6 & 20 & 8 \\ 3 & 4 & 6 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[154] \quad 21^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 194, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 5 & 3 \\ 4 & 4 & 9 & 3 & 7 & 4 \\ 6 & 9 & 13 & 5 & 11 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 \\ 5 & 7 & 11 & 4 & 7 & 5 \\ 3 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[155] \quad 1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 199, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 6 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 \\ 11 & 13 & 28 & 7 & 16 & 10 \\ 2 & 3 & 7 & 0 & 4 & 2 \\ 6 & 7 & 16 & 4 & 7 & 5 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[156] \quad 1^{12}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 200, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 11 & 16 & 0 & 9 & 44 & 12 \\ 3 & 4 & 0 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[157] \quad 1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 202, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 6 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 \\ 10 & 13 & 28 & 7 & 16 & 10 \\ 2 & 3 & 7 & 0 & 4 & 2 \\ 6 & 7 & 16 & 4 & 7 & 5 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[158] \quad 1^{121}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 202, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 6 & 3 \\ 3 & 4 & 18 & 3 & 7 & 4 \\ 14 & 18 & 58 & 10 & 22 & 14 \\ 2 & 3 & 10 & 0 & 4 & 2 \\ 6 & 7 & 22 & 4 & 7 & 5 \\ 3 & 4 & 14 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[159] \quad 1^{131}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 202, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 6 & 3 \\ 3 & 4 & 17 & 3 & 7 & 4 \\ 12 & 17 & 49 & 9 & 21 & 13 \\ 2 & 3 & 9 & 0 & 4 & 2 \\ 6 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[162] \quad 1^{121}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 204, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 5 & 3 \\ 3 & 4 & 18 & 3 & 7 & 4 \\ 14 & 18 & 58 & 10 & 22 & 14 \\ 2 & 3 & 10 & 0 & 4 & 2 \\ 5 & 7 & 22 & 4 & 7 & 5 \\ 3 & 4 & 14 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[161] \quad 21^{121}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 204, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 5 & 3 \\ 4 & 4 & 8 & 3 & 7 & 4 \\ 6 & 8 & 10 & 4 & 10 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 \\ 5 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[160] \quad 1^{124}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 14 & 3 \\ 3 & 4 & 8 & 3 & 19 & 4 \\ 7 & 8 & 10 & 4 & 28 & 6 \\ 2 & 3 & 4 & 0 & 10 & 2 \\ 14 & 19 & 28 & 10 & 64 & 14 \\ 3 & 4 & 6 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[163] \quad 1^{123}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 205, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 15 & 4 \\ 7 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 12 & 15 & 22 & 8 & 39 & 11 \\ 3 & 4 & 6 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[164] \quad 1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 9 & 3 \\ 3 & 4 & 13 & 3 & 11 & 4 \\ 11 & 13 & 28 & 7 & 25 & 10 \\ 2 & 3 & 7 & 0 & 6 & 2 \\ 9 & 11 & 25 & 6 & 20 & 8 \\ 3 & 4 & 10 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[165] \quad 1^{142}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 9 & 3 \\ 3 & 4 & 16 & 3 & 11 & 4 \\ 12 & 16 & 46 & 8 & 32 & 12 \\ 2 & 3 & 8 & 0 & 6 & 2 \\ 9 & 11 & 32 & 6 & 20 & 8 \\ 3 & 4 & 12 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[166] \quad 1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 2 \\ 12 & 16 & 0 & 9 & 44 & 12 \\ 3 & 4 & 0 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[167] \quad 1^{132}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 8 & 3 \\ 3 & 4 & 12 & 3 & 11 & 4 \\ 9 & 12 & 25 & 6 & 24 & 9 \\ 2 & 3 & 6 & 0 & 6 & 2 \\ 8 & 11 & 24 & 6 & 20 & 8 \\ 3 & 4 & 9 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[168] \quad 1^{122}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 9 & 3 \\ 3 & 4 & 13 & 3 & 11 & 4 \\ 10 & 13 & 28 & 7 & 25 & 10 \\ 2 & 3 & 7 & 0 & 6 & 2 \\ 9 & 11 & 25 & 6 & 20 & 8 \\ 3 & 4 & 10 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[169] \quad 1^{114}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 19 & 4 \\ 4 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 10 & 2 \\ 14 & 19 & 14 & 10 & 64 & 14 \\ 3 & 4 & 3 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[170] \quad & 1^{121}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 216, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 2 & 5 & 3 \\ 3 & 4 & 18 & 3 & 7 & 4 \\ 12 & 18 & 58 & 10 & 22 & 14 \\ 2 & 3 & 10 & 0 & 4 & 2 \\ 5 & 7 & 22 & 4 & 7 & 5 \\ 3 & 4 & 14 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[171] \quad & 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 217, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 11 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[172] \quad & 1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 217, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 12 & 2 \\ 15 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[173] \quad & 1^{123}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 219, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 15 & 4 \\ 5 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 11 & 15 & 22 & 8 & 39 & 11 \\ 3 & 4 & 6 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[174] \quad & 1^{123}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 222, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 13 & 2 & 3 & 3 \\ 3 & 4 & 18 & 3 & 3 & 4 \\ 13 & 18 & 58 & 10 & 10 & 14 \\ 2 & 3 & 10 & 0 & 2 & 2 \\ 3 & 3 & 10 & 2 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[175] \quad & 1^{13}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 223, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 13 & 2 & 3 & 3 \\ 3 & 4 & 17 & 3 & 3 & 4 \\ 13 & 17 & 49 & 9 & 9 & 13 \\ 2 & 3 & 9 & 0 & 2 & 2 \\ 3 & 3 & 9 & 2 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[176] \quad & 1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 226, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 2 & 5 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 \\ 10 & 13 & 28 & 7 & 16 & 10 \\ 2 & 3 & 7 & 0 & 4 & 2 \\ 5 & 7 & 16 & 4 & 7 & 5 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[179] \quad & 1^{142}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 232, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 11 & 2 & 8 & 3 \\ 3 & 4 & 16 & 3 & 11 & 4 \\ 11 & 16 & 46 & 8 & 32 & 12 \\ 2 & 3 & 8 & 0 & 6 & 2 \\ 8 & 11 & 32 & 6 & 20 & 8 \\ 3 & 4 & 12 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[178] \quad & 1^{122}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 232, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 8 & 3 \\ 3 & 4 & 13 & 3 & 11 & 4 \\ 10 & 13 & 28 & 7 & 25 & 10 \\ 2 & 3 & 7 & 0 & 6 & 2 \\ 8 & 11 & 25 & 6 & 20 & 8 \\ 3 & 4 & 10 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[177] \quad 1^{132} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 232, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 8 & 3 \\ 3 & 4 & 26 & 3 & 11 & 4 \\ 17 & 26 & 118 & 14 & 50 & 20 \\ 2 & 3 & 14 & 0 & 6 & 2 \\ 8 & 11 & 50 & 6 & 20 & 8 \\ 3 & 4 & 20 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[180] \quad 1^{142} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 234, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 6 & 3 \\ 3 & 4 & 26 & 3 & 7 & 4 \\ 18 & 26 & 118 & 14 & 32 & 20 \\ 2 & 3 & 14 & 0 & 4 & 2 \\ 6 & 7 & 32 & 4 & 7 & 5 \\ 3 & 4 & 20 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[181] \quad 1^{14} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 236, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 19 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 14 & 19 & 0 & 10 & 64 & 14 \\ 3 & 4 & 0 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[182] \quad 1^{122} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 240, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 8 & 3 \\ 3 & 4 & 13 & 3 & 11 & 4 \\ 9 & 13 & 28 & 7 & 25 & 10 \\ 2 & 3 & 7 & 0 & 6 & 2 \\ 8 & 11 & 25 & 6 & 20 & 8 \\ 3 & 4 & 10 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[183] \quad 1^{142} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 240, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 8 & 3 \\ 3 & 4 & 16 & 3 & 11 & 4 \\ 12 & 16 & 46 & 8 & 32 & 12 \\ 2 & 3 & 8 & 0 & 6 & 2 \\ 8 & 11 & 32 & 6 & 20 & 8 \\ 3 & 4 & 12 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[184] \quad 1^{121} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 241, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 5 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 \\ 9 & 13 & 28 & 7 & 16 & 10 \\ 2 & 3 & 7 & 0 & 4 & 2 \\ 5 & 7 & 16 & 4 & 7 & 5 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[185] \quad 1^{113} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 243, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 4 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 15 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[186] \quad 21^{131} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 244, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 5 & 3 \\ 4 & 4 & 12 & 3 & 7 & 4 \\ 9 & 12 & 25 & 6 & 15 & 9 \\ 2 & 3 & 6 & 0 & 4 & 2 \\ 5 & 7 & 15 & 4 & 7 & 5 \\ 3 & 4 & 9 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[187] \quad 1^{123} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 249, \quad (n_+, n_-, n_0) =$$

$$(1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 14 & 3 \\ 3 & 4 & 8 & 3 & 20 & 4 \\ 5 & 8 & 10 & 4 & 28 & 6 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 14 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[188] \quad 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 250, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 & 3 \\ 3 & 4 & 18 & 3 & 3 & 4 \\ 14 & 18 & 58 & 10 & 10 & 14 \\ 2 & 3 & 10 & 0 & 2 & 2 \\ 3 & 3 & 10 & 2 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[189] \quad 1^{14} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 250, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 & 3 \\ 3 & 4 & 26 & 3 & 3 & 4 \\ 17 & 26 & 118 & 14 & 14 & 20 \\ 2 & 3 & 14 & 0 & 2 & 2 \\ 3 & 3 & 14 & 2 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[190] \quad 1^{14} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 250, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 15 & 21 & 76 & 11 & 11 & 16 \\ 2 & 3 & 11 & 0 & 2 & 2 \\ 3 & 3 & 11 & 2 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[191] \quad 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 251, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 5 & 3 \\ 3 & 4 & 18 & 3 & 7 & 4 \\ 13 & 18 & 58 & 10 & 22 & 14 \\ 2 & 3 & 10 & 0 & 4 & 2 \\ 5 & 7 & 22 & 4 & 7 & 5 \\ 3 & 4 & 14 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[192] \quad 1^{124} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 20 & 3 \\ 3 & 4 & 8 & 3 & 29 & 4 \\ 6 & 8 & 10 & 4 & 40 & 6 \\ 2 & 3 & 4 & 0 & 16 & 2 \\ 20 & 29 & 40 & 16 & 148 & 22 \\ 3 & 4 & 6 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[193] \quad 1^{113} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 15 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 12 & 15 & 11 & 8 & 39 & 11 \\ 3 & 4 & 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[194] \quad 21^{121} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 5 & 3 \\ 4 & 4 & 18 & 3 & 7 & 4 \\ 12 & 18 & 58 & 10 & 22 & 14 \\ 2 & 3 & 10 & 0 & 4 & 2 \\ 5 & 7 & 22 & 4 & 7 & 5 \\ 3 & 4 & 14 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[195] \quad 1^{13} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 & 3 \\ 3 & 4 & 22 & 3 & 3 & 4 \\ 15 & 22 & 85 & 12 & 12 & 17 \\ 2 & 3 & 12 & 0 & 2 & 2 \\ 3 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[196] $1^{124}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 2 & 17 & 3 \\ 3 & 4 & 4 & 3 & 24 & 4 \\ 4 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 13 & 2 \\ 17 & 24 & 17 & 13 & 100 & 18 \\ 3 & 4 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$

[197] $1^{12}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 12 & 2 \\ 14 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[198] $1^{114}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 19 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 10 & 2 \\ 14 & 19 & 14 & 10 & 64 & 14 \\ 3 & 4 & 3 & 2 & 14 & 3 \end{pmatrix}_{b_0}$

[199] $21^{123}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 3 & 3 & 21 & 4 \\ 5 & 4 & 4 & 3 & 21 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 12 & 2 \\ 21 & 21 & 14 & 12 & 80 & 16 \\ 4 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[200] $1^{123}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 15 & 4 \\ 6 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 12 & 15 & 22 & 8 & 39 & 11 \\ 3 & 4 & 6 & 2 & 11 & 3 \end{pmatrix}_{b_0}$

[201] $1^{132}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 2 & 8 & 3 \\ 3 & 4 & 17 & 3 & 11 & 4 \\ 11 & 17 & 49 & 9 & 33 & 13 \\ 2 & 3 & 9 & 0 & 6 & 2 \\ 8 & 11 & 33 & 6 & 20 & 8 \\ 3 & 4 & 13 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[202] $1^{112}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 13 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[203] $1^{131}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 270, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 16 & 2 & 6 & 3 \\ 3 & 4 & 22 & 3 & 7 & 4 \\ 16 & 22 & 85 & 12 & 27 & 17 \\ 2 & 3 & 12 & 0 & 4 & 2 \\ 6 & 7 & 27 & 4 & 7 & 5 \\ 3 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[204] $1^{113}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 15 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[205] \quad 1^{14} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 276, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 20 & 3 \\ 3 & 4 & 8 & 3 & 29 & 4 \\ 5 & 8 & 10 & 4 & 40 & 6 \\ 2 & 3 & 4 & 0 & 16 & 2 \\ 20 & 29 & 40 & 16 & 148 & 22 \\ 3 & 4 & 6 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[206] \quad 1^{43} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 278, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 9 & 3 \\ 3 & 4 & 21 & 3 & 11 & 4 \\ 15 & 21 & 76 & 11 & 41 & 16 \\ 2 & 3 & 11 & 0 & 6 & 2 \\ 9 & 11 & 41 & 6 & 20 & 8 \\ 3 & 4 & 16 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[207] \quad 1^{13} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 280, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 15 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 11 & 15 & 11 & 8 & 39 & 11 \\ 3 & 4 & 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[208] \quad 1^{23} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 284, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 14 & 3 \\ 3 & 4 & 8 & 3 & 20 & 4 \\ 6 & 8 & 10 & 4 & 28 & 6 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 14 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[209] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 285, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 11 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[210] \quad 1^{42} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 287, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 6 & 3 \\ 3 & 4 & 21 & 3 & 7 & 4 \\ 15 & 21 & 76 & 11 & 26 & 16 \\ 2 & 3 & 11 & 0 & 4 & 2 \\ 6 & 7 & 26 & 4 & 7 & 5 \\ 3 & 4 & 16 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[211] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 287, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 13 & 3 \\ 3 & 4 & 8 & 3 & 20 & 4 \\ 5 & 8 & 10 & 4 & 28 & 6 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 13 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[212] \quad 1^{23} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 292, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 15 & 4 \\ 6 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 11 & 15 & 22 & 8 & 39 & 11 \\ 3 & 4 & 6 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[213] \quad & 1^{13}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 295, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 13 & 2 & 6 & 3 \\ 3 & 4 & 17 & 3 & 7 & 4 \\ 13 & 17 & 49 & 9 & 21 & 13 \\ 2 & 3 & 9 & 0 & 4 & 2 \\ 6 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[214] \quad & 21^{12}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 298, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 21 & 3 & 6 & 4 \\ 5 & 4 & 22 & 3 & 7 & 4 \\ 21 & 22 & 85 & 12 & 27 & 17 \\ 3 & 3 & 12 & 0 & 4 & 2 \\ 6 & 7 & 27 & 4 & 7 & 5 \\ 4 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[217] \quad & 1^{12}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 300, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 2 & 14 & 3 \\ 3 & 4 & 8 & 3 & 19 & 4 \\ 5 & 8 & 10 & 4 & 28 & 6 \\ 2 & 3 & 4 & 0 & 10 & 2 \\ 14 & 19 & 28 & 10 & 64 & 14 \\ 3 & 4 & 6 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[216] \quad & 21^{14}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 300, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 12 & 2 & 5 & 3 \\ 4 & 4 & 16 & 3 & 7 & 4 \\ 12 & 16 & 46 & 8 & 20 & 12 \\ 2 & 3 & 8 & 0 & 4 & 2 \\ 5 & 7 & 20 & 4 & 7 & 5 \\ 3 & 4 & 12 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[215] \quad & 1^{12}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 300, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 12 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[219] \quad & 1^{12}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 304, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 16 & 21 & 0 & 12 & 80 & 16 \\ 3 & 4 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[218] \quad & 21^{12}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 304, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 2 & 8 & 3 \\ 4 & 4 & 4 & 3 & 11 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[220] \quad & 1^{13}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 305, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 14 & 20 & 0 & 11 & 69 & 15 \\ 3 & 4 & 0 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[221] \quad & 1^{13}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 306, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 11 & 2 & 5 & 3 \\ 3 & 4 & 17 & 3 & 7 & 4 \\ 11 & 17 & 49 & 9 & 21 & 13 \\ 2 & 3 & 9 & 0 & 4 & 2 \\ 5 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[222] \quad & 1^{14} 1^{13} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 307, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 3 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 15 & 20 & 0 & 11 & 69 & 15 \\ 3 & 4 & 0 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[223] \quad & 1^{13} 1^{12} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 310, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 15 & 21 & 0 & 12 & 80 & 16 \\ 3 & 4 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[224] \quad & 21^{12} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 320, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 2 & 14 & 3 \\ 4 & 4 & 4 & 3 & 21 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 12 & 2 \\ 14 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[225] \quad & 1^{21} 1^{14} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 321, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 13 & 2 & 5 & 3 \\ 3 & 4 & 21 & 3 & 7 & 4 \\ 13 & 21 & 76 & 11 & 26 & 16 \\ 2 & 3 & 11 & 0 & 4 & 2 \\ 5 & 7 & 26 & 4 & 7 & 5 \\ 3 & 4 & 16 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[226] \quad & 1^{24} 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 322, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 2 & 15 & 3 \\ 3 & 4 & 8 & 3 & 20 & 4 \\ 6 & 8 & 10 & 4 & 28 & 6 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 15 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[227] \quad & 21^{12} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 328, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 6 & 2 & 8 & 3 \\ 4 & 4 & 8 & 3 & 11 & 4 \\ 6 & 8 & 10 & 4 & 16 & 6 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 8 & 11 & 16 & 6 & 20 & 8 \\ 3 & 4 & 6 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[229] \quad & 1^{43} 1^{13} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 331, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 13 & 2 & 9 & 3 \\ 3 & 4 & 17 & 3 & 11 & 4 \\ 13 & 17 & 49 & 9 & 33 & 13 \\ 2 & 3 & 9 & 0 & 6 & 2 \\ 9 & 11 & 33 & 6 & 20 & 8 \\ 3 & 4 & 13 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[228] \quad & 1^{21} 1^{13} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 331, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 2 & 5 & 3 \\ 3 & 4 & 22 & 3 & 7 & 4 \\ 14 & 22 & 85 & 12 & 27 & 17 \\ 2 & 3 & 12 & 0 & 4 & 2 \\ 5 & 7 & 27 & 4 & 7 & 5 \\ 3 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[231] \quad & 1^{13} 1^{24} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 334, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 19 & 3 \\ 3 & 4 & 8 & 3 & 29 & 4 \\ 5 & 8 & 10 & 4 & 40 & 6 \\ 2 & 3 & 4 & 0 & 16 & 2 \\ 19 & 29 & 40 & 16 & 148 & 22 \\ 3 & 4 & 6 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[230] \quad 1^{13}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 334, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 5 & 3 \\ 3 & 4 & 17 & 3 & 7 & 4 \\ 13 & 17 & 49 & 9 & 21 & 13 \\ 2 & 3 & 9 & 0 & 4 & 2 \\ 5 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[232] \quad 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 336, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 14 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[233] \quad 1^{13}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 337, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 & 3 \\ 3 & 4 & 22 & 3 & 3 & 4 \\ 16 & 22 & 85 & 12 & 12 & 17 \\ 2 & 3 & 12 & 0 & 2 & 2 \\ 3 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[234] \quad 21^{12}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 352, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 20 & 3 \\ 4 & 4 & 8 & 3 & 29 & 4 \\ 6 & 8 & 10 & 4 & 40 & 6 \\ 2 & 3 & 4 & 0 & 16 & 2 \\ 20 & 29 & 40 & 16 & 148 & 22 \\ 3 & 4 & 6 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[235] \quad 1^{11}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 353, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 12 & 2 \\ 15 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[236] \quad 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 355, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 13 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[237] \quad 1^{11}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 356, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 12 & 2 \\ 16 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[238] \quad 1^{13}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 361, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 5 & 3 \\ 3 & 4 & 17 & 3 & 7 & 4 \\ 12 & 17 & 49 & 9 & 21 & 13 \\ 2 & 3 & 9 & 0 & 4 & 2 \\ 5 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[239] \quad 1^{13}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 364, \quad (n_+, n_-, n_0) =$$

$$(1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 8 & 3 \\ 3 & 4 & 17 & 3 & 11 & 4 \\ 13 & 17 & 49 & 9 & 33 & 13 \\ 2 & 3 & 9 & 0 & 6 & 2 \\ 8 & 11 & 33 & 6 & 20 & 8 \\ 3 & 4 & 13 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[240] \quad 21^{1323^{11}1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 8 & 3 \\ 4 & 4 & 12 & 3 & 11 & 4 \\ 9 & 12 & 25 & 6 & 24 & 9 \\ 2 & 3 & 6 & 0 & 6 & 2 \\ 8 & 11 & 24 & 6 & 20 & 8 \\ 3 & 4 & 9 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[241] \quad 21^{1213^{11}1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 5 & 3 \\ 4 & 4 & 13 & 3 & 7 & 4 \\ 9 & 13 & 28 & 7 & 16 & 10 \\ 2 & 3 & 7 & 0 & 4 & 2 \\ 5 & 7 & 16 & 4 & 7 & 5 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[242] \quad 21^{1513^{11}1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.1$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 5 & 3 \\ 4 & 4 & 20 & 3 & 7 & 4 \\ 15 & 20 & 73 & 10 & 25 & 15 \\ 2 & 3 & 10 & 0 & 4 & 2 \\ 5 & 7 & 25 & 4 & 7 & 5 \\ 3 & 4 & 15 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[243] \quad 1^{23}1^{243^{11}1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 374, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 16 & 3 \\ 3 & 4 & 8 & 3 & 24 & 4 \\ 6 & 8 & 10 & 4 & 34 & 6 \\ 2 & 3 & 4 & 0 & 13 & 2 \\ 16 & 24 & 34 & 13 & 100 & 18 \\ 3 & 4 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[244] \quad 1^{41}1^{143^{11}1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 374, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 3 & 3 \\ 3 & 4 & 26 & 3 & 3 & 4 \\ 18 & 26 & 118 & 14 & 14 & 20 \\ 2 & 3 & 14 & 0 & 2 & 2 \\ 3 & 3 & 14 & 2 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[245] \quad 1^{14}1^{1133^{11}1^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 18 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 2 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 18 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[246] \quad 21^{123^{11}1^{141}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 24 & 3 & 6 & 4 \\ 5 & 4 & 26 & 3 & 7 & 4 \\ 24 & 26 & 118 & 14 & 32 & 20 \\ 3 & 3 & 14 & 0 & 4 & 2 \\ 6 & 7 & 32 & 4 & 7 & 5 \\ 4 & 4 & 20 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[247] \quad 21^{123^{11}1^{113}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 24 & 4 \\ 5 & 4 & 4 & 3 & 25 & 4 \\ 3 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 14 & 2 \\ 24 & 25 & 17 & 14 & 111 & 19 \\ 4 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[248] \quad 21^{1223^{11}1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 8 & 3 \\ 4 & 4 & 13 & 3 & 11 & 4 \\ 9 & 13 & 28 & 7 & 25 & 10 \\ 2 & 3 & 7 & 0 & 6 & 2 \\ 8 & 11 & 25 & 6 & 20 & 8 \\ 3 & 4 & 10 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[249] \quad 1^{132}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 388, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 8 & 3 \\ 3 & 4 & 17 & 3 & 11 & 4 \\ 12 & 17 & 49 & 9 & 33 & 13 \\ 2 & 3 & 9 & 0 & 6 & 2 \\ 8 & 11 & 33 & 6 & 20 & 8 \\ 3 & 4 & 13 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[250] \quad 1^{143}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 13 & 2 \\ 16 & 24 & 0 & 13 & 100 & 18 \\ 3 & 4 & 0 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[251] \quad 1^{141}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 396, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 5 & 3 \\ 3 & 4 & 26 & 3 & 7 & 4 \\ 16 & 26 & 118 & 14 & 32 & 20 \\ 2 & 3 & 14 & 0 & 4 & 2 \\ 5 & 7 & 32 & 4 & 7 & 5 \\ 3 & 4 & 20 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[252] \quad 1^{124}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 396, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 17 & 3 \\ 3 & 4 & 8 & 3 & 24 & 4 \\ 5 & 8 & 10 & 4 & 34 & 6 \\ 2 & 3 & 4 & 0 & 13 & 2 \\ 17 & 24 & 34 & 13 & 100 & 18 \\ 3 & 4 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[253] \quad 1^{114}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 19 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 10 & 2 \\ 14 & 19 & 14 & 10 & 64 & 14 \\ 3 & 4 & 3 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[254] \quad 1^{113}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 13 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[255] \quad 21^{112}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 412, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 11 & 3 \\ 4 & 4 & 4 & 3 & 16 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 \\ 11 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[256] \quad 1^{124}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 14 & 3 \\ 3 & 4 & 8 & 3 & 19 & 4 \\ 6 & 8 & 10 & 4 & 28 & 6 \\ 2 & 3 & 4 & 0 & 10 & 2 \\ 14 & 19 & 28 & 10 & 64 & 14 \\ 3 & 4 & 6 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[257] $1^{14}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 13 & 2 \\ 17 & 24 & 0 & 13 & 100 & 18 \\ 3 & 4 & 0 & 2 & 18 & 3 \end{pmatrix}_{b_0}$

[258] $21^{14}23^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 12 & 2 & 8 & 3 \\ 4 & 4 & 16 & 3 & 11 & 4 \\ 12 & 16 & 46 & 8 & 32 & 12 \\ 2 & 3 & 8 & 0 & 6 & 2 \\ 8 & 11 & 32 & 6 & 20 & 8 \\ 3 & 4 & 12 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[259] $1^{13}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 431, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 3 & 25 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 17 & 25 & 0 & 14 & 111 & 19 \\ 3 & 4 & 0 & 2 & 19 & 3 \end{pmatrix}_{b_0}$

[261] $1^{14}23^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 18 & 2 & 8 & 3 \\ 3 & 4 & 26 & 3 & 11 & 4 \\ 18 & 26 & 118 & 14 & 50 & 20 \\ 2 & 3 & 14 & 0 & 6 & 2 \\ 8 & 11 & 50 & 6 & 20 & 8 \\ 3 & 4 & 20 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[260] $21^{12}3^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 3 & 3 & 27 & 4 \\ 5 & 4 & 4 & 3 & 29 & 4 \\ 3 & 4 & 1 & 2 & 20 & 3 \\ 3 & 3 & 2 & 0 & 16 & 2 \\ 27 & 29 & 20 & 16 & 148 & 22 \\ 4 & 4 & 3 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[262] $1^{13}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 15 & 2 & 5 & 3 \\ 3 & 4 & 22 & 3 & 7 & 4 \\ 15 & 22 & 85 & 12 & 27 & 17 \\ 2 & 3 & 12 & 0 & 4 & 2 \\ 5 & 7 & 27 & 4 & 7 & 5 \\ 3 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[263] $1^{11}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 439, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 17 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 2 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 17 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$

[264] $1^{14}23^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 14 & 2 & 8 & 3 \\ 3 & 4 & 21 & 3 & 11 & 4 \\ 14 & 21 & 76 & 11 & 41 & 16 \\ 2 & 3 & 11 & 0 & 6 & 2 \\ 8 & 11 & 41 & 6 & 20 & 8 \\ 3 & 4 & 16 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[265] $1^{14}13^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 446, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 14 & 2 & 5 & 3 \\ 3 & 4 & 21 & 3 & 7 & 4 \\ 14 & 21 & 76 & 11 & 26 & 16 \\ 2 & 3 & 11 & 0 & 4 & 2 \\ 5 & 7 & 26 & 4 & 7 & 5 \\ 3 & 4 & 16 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[266] \quad & 21^{1233^{11}1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 450, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 6 & 2 & 14 & 3 \\ 4 & 4 & 8 & 3 & 20 & 4 \\ 6 & 8 & 10 & 4 & 28 & 6 \\ 2 & 3 & 4 & 0 & 11 & 2 \\ 14 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[267] \quad & 1^{1133^{11}1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 454, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.00 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 16 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 2 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 16 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[268] \quad & 1^{1143^{11}1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 457, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 0 & 3 & 25 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 18 & 25 & 0 & 14 & 111 & 19 \\ 3 & 4 & 0 & 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[269] \quad & 1^{1243^{11}1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 458, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 2 & 16 & 3 \\ 3 & 4 & 8 & 3 & 24 & 4 \\ 5 & 8 & 10 & 4 & 34 & 6 \\ 2 & 3 & 4 & 0 & 13 & 2 \\ 16 & 24 & 34 & 13 & 100 & 18 \\ 3 & 4 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[270] \quad & 1^{1313^{11}1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 459, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 16 & 2 & 5 & 3 \\ 3 & 4 & 22 & 3 & 7 & 4 \\ 16 & 22 & 85 & 12 & 27 & 17 \\ 2 & 3 & 12 & 0 & 4 & 2 \\ 5 & 7 & 27 & 4 & 7 & 5 \\ 3 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[271] \quad & 21^{1613^{11}1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 460, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.2 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 18 & 2 & 5 & 3 \\ 4 & 4 & 24 & 3 & 7 & 4 \\ 18 & 24 & 106 & 12 & 30 & 18 \\ 2 & 3 & 12 & 0 & 4 & 2 \\ 5 & 7 & 30 & 4 & 7 & 5 \\ 3 & 4 & 18 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[272] \quad & 1^{1133^{11}1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 461, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 14 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[273] \quad & 21^{1133^{11}1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 468, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 2 & 11 & 3 \\ 4 & 4 & 4 & 3 & 15 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 11 & 15 & 11 & 8 & 39 & 11 \\ 3 & 4 & 3 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[274] \quad & 1^{1143^{11}1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 469, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 17 & 3 \\ 3 & 4 & 4 & 3 & 24 & 4 \\ 2 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 13 & 2 \\ 17 & 24 & 17 & 13 & 100 & 18 \\ 3 & 4 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[275] $1^{14}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 15 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[276] $1^{24}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 484, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 17 & 3 \\ 3 & 4 & 8 & 3 & 24 & 4 \\ 6 & 8 & 10 & 4 & 34 & 6 \\ 2 & 3 & 4 & 0 & 13 & 2 \\ 17 & 24 & 34 & 13 & 100 & 18 \\ 3 & 4 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[277] $1^{41}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 489, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 5 & 3 \\ 3 & 4 & 21 & 3 & 7 & 4 \\ 15 & 21 & 76 & 11 & 26 & 16 \\ 2 & 3 & 11 & 0 & 4 & 2 \\ 5 & 7 & 26 & 4 & 7 & 5 \\ 3 & 4 & 16 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[278] $21^{23}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 492, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 11 & 3 \\ 4 & 4 & 8 & 3 & 15 & 4 \\ 6 & 8 & 10 & 4 & 22 & 6 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 11 & 15 & 22 & 8 & 39 & 11 \\ 3 & 4 & 6 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[279] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 502, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 16 & 3 \\ 3 & 4 & 4 & 3 & 24 & 4 \\ 2 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 13 & 2 \\ 16 & 24 & 17 & 13 & 100 & 18 \\ 3 & 4 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[280] $1^{42}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 8 & 3 \\ 3 & 4 & 21 & 3 & 11 & 4 \\ 15 & 21 & 76 & 11 & 41 & 16 \\ 2 & 3 & 11 & 0 & 6 & 2 \\ 8 & 11 & 41 & 6 & 20 & 8 \\ 3 & 4 & 16 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[281] $1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 556, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 3 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 17 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[282] $21^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 562, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 5 & 3 \\ 4 & 4 & 17 & 3 & 7 & 4 \\ 12 & 17 & 49 & 9 & 21 & 13 \\ 2 & 3 & 9 & 0 & 4 & 2 \\ 5 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[283] $21^{17}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.3$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 21 & 2 & 5 & 3 \\ 4 & 4 & 28 & 3 & 7 & 4 \\ 21 & 28 & 145 & 14 & 35 & 21 \\ 2 & 3 & 14 & 0 & 4 & 2 \\ 5 & 7 & 35 & 4 & 7 & 5 \\ 3 & 4 & 21 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[284] \quad 1^{141}3^{111}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 571, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 5 & 3 \\ 3 & 4 & 26 & 3 & 7 & 4 \\ 17 & 26 & 118 & 14 & 32 & 20 \\ 2 & 3 & 14 & 0 & 4 & 2 \\ 5 & 7 & 32 & 4 & 7 & 5 \\ 3 & 4 & 20 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[285] \quad 1^{14}3^{111}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 596, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 0 & 3 & 29 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 16 & 2 \\ 20 & 29 & 0 & 16 & 148 & 22 \\ 3 & 4 & 0 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[286] \quad 1^{113}3^{111}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 601, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 3 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 18 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[287] \quad 21^{131}3^{111}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 604, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 5 & 3 \\ 4 & 4 & 22 & 3 & 7 & 4 \\ 15 & 22 & 85 & 12 & 27 & 17 \\ 2 & 3 & 12 & 0 & 4 & 2 \\ 5 & 7 & 27 & 4 & 7 & 5 \\ 3 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[288] \quad 1^{114}3^{111}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 608, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 3 & 24 & 4 \\ 3 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 13 & 2 \\ 16 & 24 & 17 & 13 & 100 & 18 \\ 3 & 4 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[289] \quad 21^{132}3^{111}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 616, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 8 & 3 \\ 4 & 4 & 17 & 3 & 11 & 4 \\ 12 & 17 & 49 & 9 & 33 & 13 \\ 2 & 3 & 9 & 0 & 6 & 2 \\ 8 & 11 & 33 & 6 & 20 & 8 \\ 3 & 4 & 13 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[290] \quad 1^{114}3^{111}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 628, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 20 & 3 \\ 3 & 4 & 4 & 3 & 29 & 4 \\ 2 & 4 & 1 & 2 & 20 & 3 \\ 2 & 3 & 2 & 0 & 16 & 2 \\ 20 & 29 & 20 & 16 & 148 & 22 \\ 3 & 4 & 3 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[291] \quad 21^{142}3^{111}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 648, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 8 & 3 \\ 4 & 4 & 26 & 3 & 11 & 4 \\ 18 & 26 & 118 & 14 & 50 & 20 \\ 2 & 3 & 14 & 0 & 6 & 2 \\ 8 & 11 & 50 & 6 & 20 & 8 \\ 3 & 4 & 20 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[292] \quad 1^{114}3^{111}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 657, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 19 & 3 \\ 3 & 4 & 4 & 3 & 29 & 4 \\ 2 & 4 & 1 & 2 & 20 & 3 \\ 2 & 3 & 2 & 0 & 16 & 2 \\ 19 & 29 & 20 & 16 & 148 & 22 \\ 3 & 4 & 3 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[293] $1^{14}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 661, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 3 & 24 & 4 \\ 3 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 13 & 2 \\ 17 & 24 & 17 & 13 & 100 & 18 \\ 3 & 4 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[294] $1^{14}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 664, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 5 & 3 \\ 3 & 4 & 26 & 3 & 7 & 4 \\ 18 & 26 & 118 & 14 & 32 & 20 \\ 2 & 3 & 14 & 0 & 4 & 2 \\ 5 & 7 & 32 & 4 & 7 & 5 \\ 3 & 4 & 20 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[295] $21^{14}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 14 & 3 \\ 4 & 4 & 4 & 3 & 19 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 10 & 2 \\ 14 & 19 & 14 & 10 & 64 & 14 \\ 3 & 4 & 3 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[296] $21^{12}43^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 696, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 14 & 3 \\ 4 & 4 & 8 & 3 & 19 & 4 \\ 6 & 8 & 10 & 4 & 28 & 6 \\ 2 & 3 & 4 & 0 & 10 & 2 \\ 14 & 19 & 28 & 10 & 64 & 14 \\ 3 & 4 & 6 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[297] $21^{11}33^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 696, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 14 & 3 \\ 4 & 4 & 4 & 3 & 20 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 2 & 3 & 2 & 0 & 11 & 2 \\ 14 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[298] $21^{11}33^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 17 & 3 \\ 4 & 4 & 4 & 3 & 25 & 4 \\ 3 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 17 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[299] $1^{13}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 745, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 3 & 29 & 4 \\ 3 & 4 & 1 & 2 & 20 & 3 \\ 2 & 3 & 2 & 0 & 16 & 2 \\ 19 & 29 & 20 & 16 & 148 & 22 \\ 3 & 4 & 3 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[300] $21^{14}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 770, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 5 & 3 \\ 4 & 4 & 21 & 3 & 7 & 4 \\ 15 & 21 & 76 & 11 & 26 & 16 \\ 2 & 3 & 11 & 0 & 4 & 2 \\ 5 & 7 & 26 & 4 & 7 & 5 \\ 3 & 4 & 16 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[301] $21^{12}43^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 774, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 17 & 3 \\ 4 & 4 & 8 & 3 & 24 & 4 \\ 6 & 8 & 10 & 4 & 34 & 6 \\ 2 & 3 & 4 & 0 & 13 & 2 \\ 17 & 24 & 34 & 13 & 100 & 18 \\ 3 & 4 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[302] \quad 1^{14}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 832, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 3 & 29 & 4 \\ 3 & 4 & 1 & 2 & 20 & 3 \\ 2 & 3 & 2 & 0 & 16 & 2 \\ 20 & 29 & 20 & 16 & 148 & 22 \\ 3 & 4 & 3 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[303] \quad 21^{14}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 864, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 8 & 3 \\ 4 & 4 & 21 & 3 & 11 & 4 \\ 15 & 21 & 76 & 11 & 41 & 16 \\ 2 & 3 & 11 & 0 & 6 & 2 \\ 8 & 11 & 41 & 6 & 20 & 8 \\ 3 & 4 & 16 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[304] \quad 21^{14}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 964, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 5 & 3 \\ 4 & 4 & 26 & 3 & 7 & 4 \\ 18 & 26 & 118 & 14 & 32 & 20 \\ 2 & 3 & 14 & 0 & 4 & 2 \\ 5 & 7 & 32 & 4 & 7 & 5 \\ 3 & 4 & 20 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[305] \quad 21^{14}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 1020, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 17 & 3 \\ 4 & 4 & 4 & 3 & 24 & 4 \\ 3 & 4 & 1 & 2 & 17 & 3 \\ 2 & 3 & 2 & 0 & 13 & 2 \\ 17 & 24 & 17 & 13 & 100 & 18 \\ 3 & 4 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[306] \quad 21^{14}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 1168, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.86 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 20 & 3 \\ 4 & 4 & 4 & 3 & 29 & 4 \\ 3 & 4 & 1 & 2 & 20 & 3 \\ 2 & 3 & 2 & 0 & 16 & 2 \\ 20 & 29 & 20 & 16 & 148 & 22 \\ 3 & 4 & 3 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [311] \quad 1^{12}3^{11}5^{13}1^{41} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 10 & 1 & 3 & 2 \\ 8 & 0 & 10 & 74 & 10 & 22 & 18 \\ 1 & 0 & 1 & 10 & 0 & 3 & 2 \\ 2 & 0 & 3 & 22 & 3 & 5 & 5 \\ 2 & 0 & 2 & 18 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[312] \quad 1^{11}3^{11}5^{13}1^{21} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 5 & 1 & 3 & 2 \\ 4 & 0 & 5 & 17 & 5 & 11 & 9 \\ 1 & 0 & 1 & 5 & 0 & 3 & 2 \\ 2 & 0 & 3 & 11 & 3 & 5 & 5 \\ 2 & 0 & 2 & 9 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[319] \quad 1^{14}3^{11}5^{13}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 3 & 2 \\ 1 & 1 & 2 & 16 & 2 & 4 & 3 \\ 1 & 2 & 0 & 16 & 1 & 3 & 2 \\ 11 & 16 & 16 & 126 & 12 & 28 & 24 \\ 1 & 2 & 1 & 12 & 0 & 3 & 2 \\ 3 & 4 & 3 & 28 & 3 & 5 & 5 \\ 2 & 3 & 2 & 24 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[313] \quad 1^{11}3^{11}5^{13}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.0$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 0 & 2 & 4 & 3 \\ 1 & 2 & 0 & 0 & 1 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 3 & 2 \\ 3 & 4 & 3 & 0 & 3 & 5 & 5 \\ 2 & 3 & 2 & 0 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 3 & 2 \\ 1 & 1 & 2 & 12 & 2 & 4 & 3 \\ 1 & 2 & 0 & 12 & 1 & 3 & 2 \\ 8 & 12 & 12 & 70 & 9 & 21 & 18 \\ 1 & 2 & 1 & 9 & 0 & 3 & 2 \\ 3 & 4 & 3 & 21 & 3 & 5 & 5 \\ 2 & 3 & 2 & 18 & 2 & 5 & 4 \end{pmatrix} b_0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 3 & 2 \\ 1 & 1 & 2 & 8 & 2 & 4 & 3 \\ 1 & 2 & 0 & 8 & 1 & 3 & 2 \\ 5 & 8 & 8 & 30 & 6 & 14 & 12 \\ 1 & 2 & 1 & 6 & 0 & 3 & 2 \\ 3 & 4 & 3 & 14 & 3 & 5 & 5 \\ 2 & 3 & 2 & 12 & 2 & 5 & 4 \end{pmatrix} b_0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 3 & 7 & 6 \\ 1 & 2 & 1 & 3 & 0 & 3 & 2 \\ 3 & 4 & 3 & 7 & 3 & 5 & 5 \\ 2 & 3 & 2 & 6 & 2 & 5 & 4 \end{pmatrix} b_0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 3 & 2 \\ 1 & 1 & 2 & 8 & 2 & 4 & 3 \\ 1 & 2 & 0 & 8 & 1 & 3 & 2 \\ 7 & 8 & 8 & 30 & 6 & 14 & 12 \\ 1 & 2 & 1 & 6 & 0 & 3 & 2 \\ 3 & 4 & 3 & 14 & 3 & 5 & 5 \\ 2 & 3 & 2 & 12 & 2 & 5 & 4 \end{pmatrix} b_0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 1 & 3 & 2 \\ 4 & 4 & 4 & 6 & 3 & 7 & 6 \\ 1 & 2 & 1 & 3 & 0 & 3 & 2 \\ 3 & 4 & 3 & 7 & 3 & 5 & 5 \\ 2 & 3 & 2 & 6 & 2 & 5 & 4 \end{pmatrix} b_0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 2 & 8 & 3 \\ 1 & 2 & 0 & 0 & 1 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 6 & 2 \\ 6 & 8 & 7 & 0 & 6 & 27 & 11 \\ 2 & 3 & 2 & 0 & 2 & 11 & 4 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 1 & 2 & 4 & 2 & 8 & 3 \\ 1 & 2 & 0 & 4 & 1 & 7 & 2 \\ 2 & 4 & 4 & 6 & 3 & 15 & 6 \\ 1 & 2 & 1 & 3 & 0 & 6 & 2 \\ 6 & 8 & 7 & 15 & 6 & 27 & 11 \\ 2 & 3 & 2 & 6 & 2 & 11 & 4 \end{pmatrix} b_0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 6 & 2 \\ 1 & 1 & 2 & 7 & 2 & 8 & 3 \\ 1 & 2 & 0 & 5 & 1 & 7 & 2 \\ 4 & 7 & 5 & 17 & 5 & 23 & 9 \\ 1 & 2 & 1 & 5 & 0 & 6 & 2 \\ 6 & 8 & 7 & 23 & 6 & 27 & 11 \\ 2 & 3 & 2 & 9 & 2 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[325] \quad 1^{111}3^{11}1^{11}5^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 3 & 2 \\ 1 & 1 & 2 & 4 & 2 & 4 & 3 \\ 1 & 2 & 0 & 4 & 1 & 3 & 2 \\ 3 & 4 & 4 & 6 & 3 & 7 & 6 \\ 1 & 2 & 1 & 3 & 0 & 3 & 2 \\ 3 & 4 & 3 & 7 & 3 & 5 & 5 \\ 2 & 3 & 2 & 6 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[323] \quad 1^{13}3^{11}1^{13}5^{13}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 3 & 2 \\ 1 & 1 & 2 & 12 & 2 & 4 & 3 \\ 1 & 2 & 0 & 12 & 1 & 3 & 2 \\ 9 & 12 & 12 & 70 & 9 & 21 & 18 \\ 1 & 2 & 1 & 9 & 0 & 3 & 2 \\ 3 & 4 & 3 & 21 & 3 & 5 & 5 \\ 2 & 3 & 2 & 18 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[324] \quad 1^{12}3^{11}1^{12}5^{13}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 3 & 2 \\ 1 & 1 & 2 & 8 & 2 & 4 & 3 \\ 1 & 2 & 0 & 8 & 1 & 3 & 2 \\ 6 & 8 & 8 & 30 & 6 & 14 & 12 \\ 1 & 2 & 1 & 6 & 0 & 3 & 2 \\ 3 & 4 & 3 & 14 & 3 & 5 & 5 \\ 2 & 3 & 2 & 12 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[326] \quad 1^{13}3^{11}5^{13}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 2 & 12 & 3 \\ 1 & 2 & 0 & 0 & 1 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 9 & 2 \\ 9 & 12 & 11 & 0 & 9 & 65 & 17 \\ 2 & 3 & 2 & 0 & 2 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[327] \quad 1^{112}3^{11}1^{12}5^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 100, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 6 & 2 \\ 1 & 1 & 2 & 4 & 2 & 8 & 3 \\ 1 & 2 & 0 & 4 & 1 & 7 & 2 \\ 3 & 4 & 4 & 6 & 3 & 15 & 6 \\ 1 & 2 & 1 & 3 & 0 & 6 & 2 \\ 6 & 8 & 7 & 15 & 6 & 27 & 11 \\ 2 & 3 & 2 & 6 & 2 & 11 & 4 \end{pmatrix}_{b_0}$$

4.934 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$

$$T = 9 \quad [1] \quad 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 3 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 & 4 \\ 3 & 5 & 4 & 3 & 6 & 3 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 & 2 \\ 5 & 7 & 6 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{112}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 3 & 3 & 2 & 2 & 0 & 10 & 2 \\ 13 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 9 & 4 \\ 5 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 5 & 2 \\ 3 & 3 & 4 & 2 & 0 & 5 & 2 \\ 6 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 5 & 3 \\ 3 & 4 & 9 & 3 & 3 & 8 & 4 \\ 5 & 9 & 13 & 5 & 5 & 12 & 7 \\ 2 & 3 & 5 & 0 & 2 & 4 & 2 \\ 3 & 3 & 5 & 2 & 0 & 4 & 2 \\ 5 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 7 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 1^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 6 & 4 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 & 4 \\ 5 & 5 & 4 & 3 & 6 & 3 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 & 2 \\ 6 & 7 & 6 & 4 & 7 & 5 & 5 \\ 4 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{14} 1^{12} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 15 & 1 & 3 \\ 3 & 4 & 8 & 3 & 20 & 0 & 4 \\ 5 & 8 & 10 & 4 & 28 & 0 & 6 \\ 2 & 3 & 4 & 0 & 11 & 0 & 2 \\ 15 & 20 & 28 & 11 & 69 & 0 & 15 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 2 & 15 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 1^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 4 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 & 4 \\ 4 & 4 & 1 & 2 & 5 & 3 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 & 5 \\ 4 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{12} 1^{12} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 5 & 1 & 3 \\ 3 & 4 & 8 & 3 & 7 & 0 & 4 \\ 7 & 8 & 10 & 4 & 10 & 0 & 6 \\ 2 & 3 & 4 & 0 & 4 & 0 & 2 \\ 5 & 7 & 10 & 4 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{13} 1^{12} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 8 & 1 & 3 \\ 3 & 4 & 12 & 3 & 11 & 0 & 4 \\ 10 & 12 & 25 & 6 & 24 & 0 & 9 \\ 2 & 3 & 6 & 0 & 6 & 0 & 2 \\ 8 & 11 & 24 & 6 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 9 & 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11} 1^{12} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 8 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 3 & 3 & 2 & 2 & 0 & 7 & 2 \\ 8 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11} 1^{11} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 3 & 3 \\ 3 & 4 & 5 & 3 & 3 & 4 & 4 \\ 3 & 5 & 4 & 3 & 3 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 & 2 & 2 \\ 3 & 3 & 3 & 2 & 0 & 2 & 2 \\ 3 & 4 & 4 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{1\overline{32}3^{11}1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 & 8 & 3 \\ 3 & 4 & 12 & 3 & 3 & 13 & 4 \\ 8 & 12 & 25 & 6 & 6 & 27 & 9 \\ 2 & 3 & 6 & 0 & 2 & 7 & 2 \\ 3 & 3 & 6 & 2 & 0 & 7 & 2 \\ 8 & 13 & 27 & 7 & 7 & 28 & 10 \\ 3 & 4 & 9 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 5 & 4 \\ 2 & 4 & 1 & 2 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 2 & 3 & 2 \\ 3 & 3 & 2 & 0 & 3 & 2 \\ 3 & 5 & 3 & 3 & 3 & 4 & 4 \\ 3 & 4 & 3 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 3 & 7 & 0 & 4 \\ 4 & 4 & 1 & 2 & 5 & 0 & 3 \\ 2 & 3 & 2 & 0 & 4 & 0 & 2 \\ 5 & 7 & 5 & 4 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1121}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 8 & 4 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 4 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 & 8 \\ 4 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1211}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 5 & 2 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 & 4 \\ 7 & 8 & 10 & 4 & 10 & 6 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 & 2 \\ 5 & 7 & 10 & 4 & 7 & 5 & 5 \\ 2 & 4 & 6 & 2 & 5 & 1 & 3 \\ 3 & 4 & 6 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{122}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 8 & 1 & 3 \\ 3 & 4 & 8 & 3 & 11 & 0 & 4 \\ 7 & 8 & 10 & 4 & 16 & 0 & 6 \\ 2 & 3 & 4 & 0 & 6 & 0 & 2 \\ 8 & 11 & 16 & 6 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -17, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 5 & 3 & 7 & 0 & 4 \\ 3 & 5 & 4 & 3 & 6 & 0 & 4 \\ 2 & 3 & 3 & 0 & 4 & 0 & 2 \\ 5 & 7 & 6 & 4 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 4 & 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 3 & 3 & 2 & 2 & 0 & 10 & 2 \\ 11 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 & 7 & 3 \\ 3 & 4 & 8 & 3 & 3 & 9 & 4 \\ 7 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 5 & 2 \\ 3 & 3 & 4 & 2 & 0 & 5 & 2 \\ 7 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{111} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 6 & 1 & 3 \\ 3 & 4 & 14 & 3 & 7 & 0 & 4 \\ 12 & 14 & 37 & 8 & 17 & 0 & 11 \\ 2 & 3 & 8 & 0 & 4 & 0 & 2 \\ 6 & 7 & 17 & 4 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 11 & 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 4 & 3 \\ 3 & 4 & 4 & 3 & 3 & 5 & 4 \\ 4 & 4 & 1 & 2 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 2 & 3 & 2 \\ 3 & 3 & 2 & 2 & 0 & 3 & 2 \\ 4 & 5 & 3 & 3 & 3 & 4 & 4 \\ 3 & 4 & 3 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{132} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 & 9 & 3 \\ 3 & 4 & 12 & 3 & 3 & 13 & 4 \\ 9 & 12 & 25 & 6 & 6 & 27 & 9 \\ 2 & 3 & 6 & 0 & 2 & 7 & 2 \\ 3 & 3 & 6 & 2 & 0 & 7 & 2 \\ 9 & 13 & 27 & 7 & 7 & 28 & 10 \\ 3 & 4 & 9 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 4 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 & 4 \\ 4 & 5 & 4 & 3 & 6 & 3 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 & 2 \\ 5 & 7 & 6 & 4 & 7 & 5 & 5 \\ 4 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 5 & 7 & 3 \\ 3 & 4 & 8 & 3 & 7 & 9 & 4 \\ 7 & 8 & 10 & 4 & 10 & 12 & 6 \\ 2 & 3 & 4 & 0 & 4 & 5 & 2 \\ 5 & 7 & 10 & 4 & 7 & 11 & 5 \\ 7 & 9 & 12 & 5 & 11 & 13 & 7 \\ 3 & 4 & 6 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -25, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 4 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 5 & 7 & 5 & 4 & 7 & 11 & 5 \\ 6 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -25, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 3 & 12 & 0 & 4 \\ 2 & 4 & 1 & 2 & 8 & 0 & 3 \\ 2 & 3 & 2 & 0 & 7 & 0 & 2 \\ 9 & 12 & 8 & 7 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -25, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 2 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 & 4 \\ 3 & 5 & 4 & 3 & 6 & 3 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 & 2 \\ 5 & 7 & 6 & 4 & 7 & 5 & 5 \\ 2 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 5 & 5 & 3 \\ 3 & 4 & 9 & 3 & 7 & 8 & 4 \\ 5 & 9 & 13 & 5 & 11 & 12 & 7 \\ 2 & 3 & 5 & 0 & 4 & 4 & 2 \\ 5 & 7 & 11 & 4 & 7 & 10 & 5 \\ 5 & 8 & 12 & 4 & 10 & 10 & 6 \\ 3 & 4 & 7 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{121} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 6 & 6 & 3 \\ 3 & 4 & 8 & 3 & 7 & 9 & 4 \\ 5 & 8 & 10 & 4 & 10 & 12 & 6 \\ 2 & 3 & 4 & 0 & 4 & 5 & 2 \\ 6 & 7 & 10 & 4 & 7 & 11 & 5 \\ 6 & 9 & 12 & 5 & 11 & 13 & 7 \\ 3 & 4 & 6 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{112} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 3 & 3 & 2 & 2 & 0 & 7 & 2 \\ 9 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{111} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 & 4 \\ 2 & 4 & 1 & 2 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{121} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 3 & 3 \\ 3 & 4 & 8 & 3 & 3 & 4 & 4 \\ 5 & 8 & 10 & 4 & 4 & 6 & 6 \\ 2 & 3 & 4 & 0 & 2 & 2 & 2 \\ 3 & 3 & 4 & 2 & 0 & 2 & 2 \\ 3 & 4 & 6 & 2 & 2 & 1 & 3 \\ 3 & 4 & 6 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{111} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 6 & 1 & 3 \\ 3 & 4 & 14 & 3 & 7 & 0 & 4 \\ 11 & 14 & 37 & 8 & 17 & 0 & 11 \\ 2 & 3 & 8 & 0 & 4 & 0 & 2 \\ 6 & 7 & 17 & 4 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 11 & 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 5 & 3 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 & 4 \\ 5 & 9 & 13 & 5 & 11 & 6 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 & 2 \\ 5 & 7 & 11 & 4 & 7 & 5 & 5 \\ 3 & 4 & 6 & 2 & 5 & 1 & 3 \\ 3 & 4 & 7 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$1^{213}$$

$$[40] \ 1^{111}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 14 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 3 & 3 & 2 & 2 & 0 & 8 & 2 \\ 11 & 14 & 9 & 8 & 8 & 37 & 11 \\ 3 & 4 & 3 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$1^{111}$$

$$[39] \ 1^{13}1^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 0 & 3 & 3 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 3 & 2 \\ 3 & 3 & 0 & 2 & 0 & 3 & 2 \\ 4 & 5 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & 0 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$1^{111}$$

$$[38] \ 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 3 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 & 4 \\ 4 & 5 & 4 & 3 & 6 & 3 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 & 2 \\ 5 & 7 & 6 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$1^{21}$$

$$[37] \ 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 2 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 6 & 7 & 5 & 4 & 7 & 11 & 5 \\ 6 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[36] \ 21^{113}1^{11}1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 6 & 9 & 4 \\ 5 & 4 & 4 & 3 & 7 & 9 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 3 & 3 & 2 & 0 & 4 & 5 & 2 \\ 6 & 7 & 5 & 4 & 7 & 11 & 5 \\ 9 & 9 & 6 & 5 & 11 & 13 & 7 \\ 4 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$1^{11}$$

$$[35] \ 1^{111}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 14 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 3 & 3 & 2 & 2 & 0 & 8 & 2 \\ 9 & 14 & 9 & 8 & 8 & 37 & 11 \\ 3 & 4 & 3 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$1^{312}$$

$$[41] \ 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 5 & 4 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 & 4 \\ 8 & 9 & 13 & 5 & 11 & 6 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 & 2 \\ 5 & 7 & 11 & 4 & 7 & 5 & 5 \\ 4 & 4 & 6 & 2 & 5 & 1 & 3 \\ 3 & 4 & 7 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$1^{12}$$

$$[43] \ 1^{112}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 2 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 4 & 2 \\ 3 & 3 & 2 & 2 & 0 & 4 & 2 \\ 6 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 9 & 4 \\ 6 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 5 & 2 \\ 3 & 3 & 4 & 2 & 0 & 5 & 2 \\ 6 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 6 & 3 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 & 4 \\ 4 & 5 & 4 & 3 & 6 & 3 & 4 \\ 2 & 3 & 3 & 0 & 4 & 2 & 2 \\ 6 & 7 & 6 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{141} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 & 3 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 & 4 \\ 13 & 21 & 76 & 11 & 11 & 15 & 16 \\ 2 & 3 & 11 & 0 & 2 & 2 & 2 \\ 3 & 3 & 11 & 2 & 0 & 2 & 2 \\ 3 & 4 & 15 & 2 & 2 & 1 & 3 \\ 3 & 4 & 16 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 1 & 3 \\ 3 & 4 & 5 & 3 & 7 & 0 & 4 \\ 4 & 5 & 4 & 3 & 6 & 0 & 4 \\ 2 & 3 & 3 & 0 & 4 & 0 & 2 \\ 5 & 7 & 6 & 4 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 4 & 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 3 & 3 \\ 3 & 4 & 5 & 3 & 3 & 4 & 4 \\ 4 & 5 & 4 & 3 & 3 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 & 2 & 2 \\ 3 & 3 & 3 & 2 & 0 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 7 & 0 & 4 \\ 4 & 5 & 4 & 3 & 6 & 0 & 4 \\ 2 & 3 & 3 & 0 & 4 & 0 & 2 \\ 6 & 7 & 6 & 4 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 4 & 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{122} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 8 & 4 \\ 5 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 4 & 2 \\ 3 & 3 & 4 & 2 & 0 & 4 & 2 \\ 6 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 6 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{111} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 2 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 & 4 \\ 4 & 4 & 1 & 2 & 5 & 3 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 & 5 \\ 2 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{123} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 17 & 1 & 3 \\ 3 & 4 & 8 & 3 & 25 & 0 & 4 \\ 5 & 8 & 10 & 4 & 34 & 0 & 6 \\ 2 & 3 & 4 & 0 & 14 & 0 & 2 \\ 17 & 25 & 34 & 14 & 111 & 0 & 19 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 2 & 19 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{111} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 1 & 3 \\ 3 & 4 & 0 & 3 & 7 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 0 & 2 \\ 5 & 7 & 0 & 4 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{122} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 8 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 5 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 7 & 2 \\ 3 & 3 & 4 & 2 & 0 & 7 & 2 \\ 8 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{122} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 9 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 5 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 7 & 2 \\ 3 & 3 & 4 & 2 & 0 & 7 & 2 \\ 9 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{122} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 3 & 11 & 0 & 4 \\ 4 & 4 & 1 & 2 & 8 & 0 & 3 \\ 2 & 3 & 2 & 0 & 6 & 0 & 2 \\ 8 & 11 & 8 & 6 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 9 & 2 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 & 4 \\ 4 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 7 & 2 & 2 \\ 9 & 12 & 8 & 7 & 25 & 8 & 9 \\ 2 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -38, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 6 & 3 \\ 3 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 \\ 3 & 3 & 0 & 2 & 0 & 5 & 2 \\ 6 & 9 & 0 & 5 & 5 & 13 & 7 \\ 3 & 4 & 0 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{111} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -38, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 3 & 3 & 0 & 2 & 0 & 2 & 2 \\ 3 & 4 & 0 & 2 & 2 & 1 & 3 \\ 3 & 4 & 0 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{124}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 17 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 6 & 8 & 10 & 4 & 4 & 36 & 6 \\ 2 & 3 & 4 & 0 & 2 & 14 & 2 \\ 3 & 3 & 4 & 2 & 0 & 14 & 2 \\ 17 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{113}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 6 & 2 \\ 3 & 3 & 2 & 2 & 0 & 6 & 2 \\ 9 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 12 & 3 \\ 3 & 4 & 0 & 3 & 3 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 10 & 2 \\ 3 & 3 & 0 & 2 & 0 & 10 & 2 \\ 12 & 18 & 0 & 10 & 10 & 58 & 14 \\ 3 & 4 & 0 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{131}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 & 3 & 3 \\ 3 & 4 & 12 & 3 & 3 & 4 & 4 \\ 8 & 12 & 25 & 6 & 6 & 9 & 9 \\ 2 & 3 & 6 & 0 & 2 & 2 & 2 \\ 3 & 3 & 6 & 2 & 0 & 2 & 2 \\ 3 & 4 & 9 & 2 & 2 & 1 & 3 \\ 3 & 4 & 9 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{112}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 6 & 3 \\ 3 & 4 & 0 & 3 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 4 & 2 \\ 3 & 3 & 0 & 2 & 0 & 4 & 2 \\ 6 & 8 & 0 & 4 & 4 & 10 & 6 \\ 3 & 4 & 0 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 9 & 4 \\ 2 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 5 & 2 \\ 3 & 3 & 2 & 2 & 0 & 5 & 2 \\ 6 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 3 & 16 & 0 & 4 \\ 4 & 4 & 1 & 2 & 11 & 0 & 3 \\ 2 & 3 & 2 & 0 & 9 & 0 & 2 \\ 12 & 16 & 11 & 9 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 3 & 3 & 2 & 2 & 0 & 9 & 2 \\ 11 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{1^{12}3^{11}1^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 14 & 3 \\ 3 & 4 & 8 & 3 & 3 & 22 & 4 \\ 5 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 12 & 2 \\ 3 & 3 & 4 & 2 & 0 & 12 & 2 \\ 14 & 22 & 30 & 12 & 12 & 85 & 17 \\ 3 & 4 & 6 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{1^{13}3^{11}1^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 9 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 5 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 6 & 2 \\ 3 & 3 & 4 & 2 & 0 & 6 & 2 \\ 9 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{1^{11}1^{11}3^{11}1^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 5 & 3 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 & 4 \\ 8 & 9 & 13 & 5 & 11 & 6 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 & 2 \\ 5 & 7 & 11 & 4 & 7 & 5 & 5 \\ 3 & 4 & 6 & 2 & 5 & 1 & 3 \\ 3 & 4 & 7 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{2^{13}3^{11}1^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 15 & 3 \\ 3 & 4 & 8 & 3 & 3 & 22 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 12 & 2 \\ 3 & 3 & 4 & 2 & 0 & 12 & 2 \\ 15 & 22 & 30 & 12 & 12 & 85 & 17 \\ 3 & 4 & 6 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{2^{12}3^{11}1^{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 7 & 3 \\ 3 & 4 & 8 & 3 & 3 & 9 & 4 \\ 6 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 5 & 2 \\ 3 & 3 & 4 & 2 & 0 & 5 & 2 \\ 7 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{2^{12}3^{11}1^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 & 6 & 3 \\ 3 & 4 & 12 & 3 & 3 & 8 & 4 \\ 8 & 12 & 25 & 6 & 6 & 18 & 9 \\ 2 & 3 & 6 & 0 & 2 & 4 & 2 \\ 3 & 3 & 6 & 2 & 0 & 4 & 2 \\ 6 & 8 & 18 & 4 & 4 & 10 & 6 \\ 3 & 4 & 9 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{1^{13}3^{11}1^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 9 & 3 \\ 3 & 4 & 0 & 3 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 6 & 2 \\ 3 & 3 & 0 & 2 & 0 & 6 & 2 \\ 9 & 12 & 0 & 6 & 6 & 25 & 9 \\ 3 & 4 & 0 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{1^{14}3^{11}1^{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 16 & 3 \\ 3 & 4 & 4 & 3 & 3 & 26 & 4 \\ 3 & 4 & 1 & 2 & 2 & 18 & 3 \\ 2 & 3 & 2 & 0 & 2 & 14 & 2 \\ 3 & 3 & 2 & 2 & 0 & 14 & 2 \\ 16 & 26 & 18 & 14 & 14 & 118 & 20 \\ 3 & 4 & 3 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 11 & 3 \\ 3 & 4 & 0 & 3 & 7 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 7 & 2 \\ 5 & 7 & 0 & 4 & 7 & 16 & 5 \\ 11 & 13 & 0 & 7 & 16 & 28 & 10 \\ 3 & 4 & 0 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{114}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 16 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 3 & 3 & 2 & 0 & 8 & 2 & 2 \\ 12 & 16 & 12 & 8 & 8 & 46 & 12 \\ 3 & 4 & 3 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 3 & 12 & 0 & 4 \\ 4 & 4 & 1 & 2 & 8 & 0 & 3 \\ 2 & 3 & 2 & 0 & 7 & 0 & 2 \\ 10 & 12 & 8 & 7 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{111}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 5 & 1 & 3 \\ 3 & 4 & 14 & 3 & 7 & 0 & 4 \\ 10 & 14 & 37 & 8 & 17 & 0 & 11 \\ 2 & 3 & 8 & 0 & 4 & 0 & 2 \\ 5 & 7 & 17 & 4 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 11 & 2 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{112}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 13 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 4 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 5 & 7 & 5 & 4 & 7 & 22 & 5 \\ 13 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{124}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 12 & 3 \\ 3 & 4 & 8 & 3 & 3 & 16 & 4 \\ 5 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 8 & 2 \\ 3 & 3 & 4 & 2 & 0 & 8 & 2 \\ 12 & 16 & 24 & 8 & 8 & 46 & 12 \\ 3 & 4 & 6 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{124}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 18 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 5 & 8 & 10 & 4 & 4 & 36 & 6 \\ 2 & 3 & 4 & 0 & 2 & 14 & 2 \\ 3 & 3 & 4 & 2 & 0 & 14 & 2 \\ 18 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{111}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 & 4 \\ 4 & 4 & 1 & 2 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 & 2 \\ 3 & 3 & 2 & 2 & 0 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{124} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -60, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 16 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 5 & 8 & 10 & 4 & 4 & 36 & 6 \\ 2 & 3 & 4 & 0 & 2 & 14 & 2 \\ 3 & 3 & 4 & 2 & 0 & 14 & 2 \\ 16 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{121} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -60, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 6 & 3 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 & 4 \\ 5 & 8 & 10 & 4 & 10 & 6 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 & 2 \\ 6 & 7 & 10 & 4 & 7 & 5 & 5 \\ 3 & 4 & 6 & 2 & 5 & 1 & 3 \\ 3 & 4 & 6 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{141} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 1 & 3 \\ 3 & 4 & 0 & 3 & 16 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 0 & 2 \\ 13 & 16 & 0 & 9 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{111} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 5 & 3 \\ 3 & 4 & 4 & 3 & 7 & 8 & 4 \\ 4 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 10 & 5 \\ 5 & 8 & 6 & 4 & 10 & 10 & 6 \\ 3 & 4 & 3 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 2 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 7 & 2 & 2 \\ 8 & 12 & 8 & 7 & 25 & 8 & 9 \\ 2 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{112} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 14 & 2 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 & 3 \\ 2 & 3 & 2 & 0 & 12 & 2 & 2 \\ 14 & 21 & 14 & 12 & 80 & 14 & 16 \\ 2 & 4 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 3 & 2 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{111} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 0 & 3 & 7 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 2 & 2 \\ 5 & 7 & 0 & 4 & 7 & 5 & 5 \\ 2 & 4 & 0 & 2 & 5 & 1 & 3 \\ 3 & 4 & 0 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{123} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 17 & 3 \\ 3 & 4 & 0 & 3 & 11 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 14 & 2 \\ 8 & 11 & 0 & 6 & 20 & 50 & 8 \\ 17 & 26 & 0 & 14 & 50 & 118 & 20 \\ 3 & 4 & 0 & 2 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{123} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 1 & 3 \\ 3 & 4 & 0 & 3 & 11 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 0 & 2 \\ 8 & 11 & 0 & 6 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{113} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 3 & 15 & 0 & 4 \\ 4 & 4 & 1 & 2 & 11 & 0 & 3 \\ 2 & 3 & 2 & 0 & 8 & 0 & 2 \\ 11 & 15 & 11 & 8 & 39 & 0 & 11 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{112} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 8 & 2 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 4 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 & 8 \\ 2 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{112} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 16 & 4 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 & 3 \\ 2 & 3 & 2 & 0 & 12 & 2 & 2 \\ 16 & 21 & 14 & 12 & 80 & 14 & 16 \\ 4 & 4 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 3 & 2 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[96] \quad 1^{112} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -66, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 4 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 4 & 2 \\ 3 & 3 & 2 & 2 & 0 & 4 & 2 \\ 6 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{121} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -66, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 5 & 6 & 3 \\ 3 & 4 & 8 & 3 & 7 & 9 & 4 \\ 6 & 8 & 10 & 4 & 10 & 12 & 6 \\ 2 & 3 & 4 & 0 & 4 & 5 & 2 \\ 5 & 7 & 10 & 4 & 7 & 11 & 5 \\ 6 & 9 & 12 & 5 & 11 & 13 & 7 \\ 3 & 4 & 6 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[97] \quad 1^{113} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -67, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 3 & 3 & 2 & 2 & 0 & 7 & 2 \\ 10 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[98] \quad 1^{114} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -68, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 12 & 3 \\ 3 & 4 & 0 & 3 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 8 & 2 \\ 3 & 3 & 0 & 2 & 0 & 8 & 2 \\ 12 & 16 & 0 & 8 & 8 & 46 & 12 \\ 3 & 4 & 0 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[101] \ 1^{1^{311}} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -70, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 & 3 & 3 \\ 3 & 4 & 16 & 3 & 3 & 4 & 4 \\ 11 & 16 & 46 & 8 & 8 & 12 & 12 \\ 2 & 3 & 8 & 0 & 2 & 2 & 2 \\ 3 & 3 & 8 & 2 & 0 & 2 & 2 \\ 3 & 4 & 12 & 2 & 2 & 1 & 3 \\ 3 & 4 & 12 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[100] \ 1^{1^{111}} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -70, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 3 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 & 4 \\ 2 & 4 & 1 & 2 & 5 & 3 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 & 2 \\ 6 & 7 & 5 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[99] \ 1^{1^{111}} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -70, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 7 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 2 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 6 & 7 & 5 & 4 & 7 & 11 & 5 \\ 7 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[102] \ 1^{1^{121}} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -72, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 1 & 3 \\ 3 & 4 & 0 & 3 & 12 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 7 & 0 & 2 \\ 9 & 12 & 0 & 7 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{1^{112}} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -74, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 11 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 5 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 9 & 2 \\ 3 & 3 & 4 & 2 & 0 & 9 & 2 \\ 11 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{1^{113}} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -74, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 12 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 5 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 9 & 2 \\ 3 & 3 & 4 & 2 & 0 & 9 & 2 \\ 12 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{1^{111}} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -75, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 8 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 3 & 3 & 2 & 2 & 0 & 7 & 2 \\ 8 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{1^{121}} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -75, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 6 & 3 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 & 4 \\ 6 & 9 & 13 & 5 & 11 & 6 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 & 2 \\ 6 & 7 & 11 & 4 & 7 & 5 & 5 \\ 3 & 4 & 6 & 2 & 5 & 1 & 3 \\ 3 & 4 & 7 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[107] \quad 1^{1111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 5 & 2 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 & 4 \\ 5 & 9 & 13 & 5 & 11 & 6 & 7 \\ 2 & 3 & 5 & 0 & 4 & 2 & 2 \\ 5 & 7 & 11 & 4 & 7 & 5 & 5 \\ 2 & 4 & 6 & 2 & 5 & 1 & 3 \\ 3 & 4 & 7 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[108] \quad 1^{113} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 6 & 2 \\ 3 & 3 & 2 & 2 & 0 & 6 & 2 \\ 9 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[109] \quad 1^{1112} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 8 & 4 \\ 2 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 4 & 2 \\ 6 & 7 & 5 & 4 & 7 & 10 & 5 \\ 6 & 8 & 6 & 4 & 10 & 10 & 6 \\ 3 & 4 & 3 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[110] \quad 1^{1112} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 14 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 2 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 6 & 7 & 5 & 4 & 7 & 22 & 5 \\ 14 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[111] \quad 1^{142} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 & 6 & 3 \\ 3 & 4 & 16 & 3 & 3 & 8 & 4 \\ 11 & 16 & 46 & 8 & 8 & 24 & 12 \\ 2 & 3 & 8 & 0 & 2 & 4 & 2 \\ 3 & 3 & 8 & 2 & 0 & 4 & 2 \\ 6 & 8 & 24 & 4 & 4 & 10 & 6 \\ 3 & 4 & 12 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[112] \quad 1^{113} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 3 & 3 & 2 & 2 & 0 & 9 & 2 \\ 12 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[113] \quad 1^{112} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 9 & 3 \\ 3 & 4 & 0 & 3 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 7 & 2 \\ 3 & 3 & 0 & 2 & 0 & 7 & 2 \\ 9 & 13 & 0 & 7 & 7 & 28 & 10 \\ 3 & 4 & 0 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[114] \quad 1^{114} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 11 & 2 \\ 3 & 3 & 2 & 2 & 0 & 11 & 2 \\ 14 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[115] \quad 1^{1^{123}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 8 & 3 \\ 3 & 4 & 0 & 3 & 7 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 5 & 2 \\ 6 & 7 & 0 & 4 & 7 & 11 & 5 \\ 8 & 9 & 0 & 5 & 11 & 13 & 7 \\ 3 & 4 & 0 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[116] \quad 1^{1^{12}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 3 & 3 & 2 & 2 & 0 & 7 & 2 \\ 9 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[117] \quad 1^{1^{12}} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 12 & 2 \\ 3 & 3 & 2 & 2 & 0 & 12 & 2 \\ 14 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[118] \quad 1^{1^{31}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 3 & 16 & 0 & 4 \\ 2 & 4 & 1 & 2 & 11 & 0 & 3 \\ 2 & 3 & 2 & 0 & 9 & 0 & 2 \\ 12 & 16 & 11 & 9 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[119] \quad 1^{1^{212}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 7 & 3 \\ 3 & 4 & 4 & 3 & 3 & 9 & 4 \\ 4 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 5 & 2 \\ 3 & 3 & 2 & 2 & 0 & 5 & 2 \\ 7 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[120] \quad 1^{1^{12}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 7 & 3 \\ 3 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 \\ 3 & 3 & 0 & 2 & 0 & 5 & 2 \\ 7 & 9 & 0 & 5 & 5 & 13 & 7 \\ 3 & 4 & 0 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[121] \quad 1^{1^{23}} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 13 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 2 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 6 & 7 & 5 & 4 & 7 & 22 & 5 \\ 13 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[122] \quad 1^{1^{22}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 9 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 2 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 6 & 7 & 5 & 4 & 7 & 16 & 5 \\ 9 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -86, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 10 & 3 \\ 3 & 4 & 0 & 3 & 3 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 8 & 2 \\ 3 & 3 & 0 & 2 & 0 & 8 & 2 \\ 10 & 14 & 0 & 8 & 8 & 37 & 11 \\ 3 & 4 & 0 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -88, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.86 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 & 4 \\ 3 & 4 & 1 & 2 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 & 2 & 2 \\ 3 & 4 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{\overline{121}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -88, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 12 & 2 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 & 4 \\ 4 & 4 & 1 & 2 & 11 & 3 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 & 2 \\ 12 & 16 & 11 & 9 & 44 & 11 & 12 \\ 2 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[126] \ 21^{\overline{11}} 3^{\overline{11}} 1^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -88, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 15 & 3 & 4 \\ 5 & 4 & 4 & 3 & 16 & 4 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 & 3 \\ 3 & 3 & 2 & 0 & 9 & 2 & 2 \\ 15 & 16 & 11 & 9 & 44 & 11 & 12 \\ 3 & 4 & 3 & 2 & 11 & 1 & 3 \\ 4 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{\overline{114}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -90, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 16 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 3 & 3 & 2 & 0 & 8 & 2 & 2 \\ 12 & 16 & 12 & 8 & 46 & 12 & 12 \\ 3 & 4 & 3 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -91, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 9 & 3 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 7 & 2 & 2 \\ 9 & 12 & 8 & 7 & 25 & 8 & 9 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -91, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 7 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 2 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 5 & 7 & 5 & 4 & 7 & 11 & 5 \\ 7 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[130] \ 1^{\overline{121}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -92, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 & 3 \\ 3 & 4 & 0 & 3 & 7 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 2 & 2 \\ 6 & 7 & 0 & 4 & 7 & 5 & 5 \\ 3 & 4 & 0 & 2 & 5 & 1 & 3 \\ 3 & 4 & 0 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[131] \quad 1^{113} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -94, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 7 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 4 & 2 \\ 3 & 3 & 2 & 2 & 0 & 4 & 2 \\ 7 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[132] \quad 1^{112} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -96, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 14 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 3 & 3 & 2 & 2 & 0 & 8 & 2 \\ 10 & 14 & 9 & 8 & 8 & 37 & 11 \\ 3 & 4 & 3 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[133] \quad 1^{1123} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -96, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 6 & 13 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 3 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 6 & 7 & 5 & 4 & 7 & 22 & 5 \\ 13 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[134] \quad 1^{122} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -96, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 11 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 7 & 2 \\ 3 & 3 & 4 & 2 & 0 & 7 & 2 \\ 11 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[135] \quad 1^{113} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -97, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 15 & 1 & 3 \\ 3 & 4 & 4 & 3 & 20 & 0 & 4 \\ 4 & 4 & 1 & 2 & 14 & 0 & 3 \\ 2 & 3 & 2 & 0 & 11 & 0 & 2 \\ 15 & 20 & 14 & 11 & 69 & 0 & 15 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 15 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[136] \quad 1^{121} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -97, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 5 & 6 & 3 \\ 3 & 4 & 8 & 3 & 7 & 9 & 4 \\ 5 & 8 & 10 & 4 & 10 & 12 & 6 \\ 2 & 3 & 4 & 0 & 4 & 5 & 2 \\ 5 & 7 & 10 & 4 & 7 & 11 & 5 \\ 6 & 9 & 12 & 5 & 11 & 13 & 7 \\ 3 & 4 & 6 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[137] \quad 1^{122} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 6 & 3 \\ 3 & 4 & 0 & 3 & 7 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 4 & 2 \\ 6 & 7 & 0 & 4 & 7 & 10 & 5 \\ 6 & 8 & 0 & 4 & 10 & 10 & 6 \\ 3 & 4 & 0 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[138] \quad 1^{113} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 3 & 3 & 2 & 2 & 0 & 10 & 2 \\ 13 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{1111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 6 & 7 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 4 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 6 & 7 & 5 & 4 & 7 & 11 & 5 \\ 7 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{1211}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 6 & 7 & 3 \\ 3 & 4 & 8 & 3 & 7 & 9 & 4 \\ 6 & 8 & 10 & 4 & 10 & 12 & 6 \\ 2 & 3 & 4 & 0 & 4 & 5 & 2 \\ 6 & 7 & 10 & 4 & 7 & 11 & 5 \\ 7 & 9 & 12 & 5 & 11 & 13 & 7 \\ 3 & 4 & 6 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[144] \ 1^{1111}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 0 & 3 & 7 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 4 & 2 \\ 5 & 7 & 0 & 4 & 7 & 10 & 5 \\ 5 & 8 & 0 & 4 & 10 & 10 & 6 \\ 3 & 4 & 0 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[141] \ 1^{1211}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 8 & 1 & 3 \\ 3 & 4 & 9 & 3 & 11 & 0 & 4 \\ 6 & 9 & 13 & 5 & 17 & 0 & 7 \\ 2 & 3 & 5 & 0 & 6 & 0 & 2 \\ 8 & 11 & 17 & 6 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 7 & 2 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{1311}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 1 & 3 \\ 3 & 4 & 0 & 3 & 15 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 8 & 0 & 2 \\ 11 & 15 & 0 & 8 & 39 & 0 & 11 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[143] \ 1^{1141}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 14 & 1 & 3 \\ 3 & 4 & 4 & 3 & 19 & 0 & 4 \\ 4 & 4 & 1 & 2 & 14 & 0 & 3 \\ 2 & 3 & 2 & 0 & 10 & 0 & 2 \\ 14 & 19 & 14 & 10 & 64 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 14 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[148] \ 1^{1221}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 8 & 4 \\ 7 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 4 & 2 \\ 3 & 3 & 4 & 2 & 0 & 4 & 2 \\ 6 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 6 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[147] \ 1^{1211}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 0 & 3 & 11 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 2 & 2 \\ 8 & 11 & 0 & 6 & 20 & 8 & 8 \\ 2 & 4 & 0 & 2 & 8 & 1 & 3 \\ 3 & 4 & 0 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[146] \quad 1^{1111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 2 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 & 4 \\ 2 & 4 & 1 & 2 & 5 & 3 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 & 5 \\ 2 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[145] \quad 1^{111}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 1 & 3 \\ 3 & 4 & 0 & 3 & 17 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 10 & 0 & 2 \\ 13 & 17 & 0 & 10 & 55 & 0 & 13 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 13 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[149] \quad 1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -102, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 3 & 3 & 2 & 2 & 0 & 10 & 2 \\ 12 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[153] \quad 1^{1211}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 5 & 3 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 & 4 \\ 7 & 8 & 10 & 4 & 10 & 6 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 & 2 \\ 5 & 7 & 10 & 4 & 7 & 5 & 5 \\ 3 & 4 & 6 & 2 & 5 & 1 & 3 \\ 3 & 4 & 6 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[152] \quad 1^{112}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 4 & 2 \\ 3 & 3 & 2 & 2 & 0 & 4 & 2 \\ 6 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[151] \quad 1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 11 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 9 & 2 \\ 3 & 3 & 4 & 2 & 0 & 9 & 2 \\ 11 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[150] \quad 1^{1111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 & 4 \\ 4 & 4 & 1 & 2 & 5 & 3 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[154] \quad 1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 12 & 3 \\ 3 & 4 & 0 & 3 & 7 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 9 & 2 \\ 6 & 7 & 0 & 4 & 7 & 21 & 5 \\ 12 & 17 & 0 & 9 & 21 & 49 & 13 \\ 3 & 4 & 0 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[158] \quad 1^{1212} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 5 & 7 & 3 \\ 3 & 4 & 8 & 3 & 7 & 9 & 4 \\ 6 & 8 & 10 & 4 & 10 & 12 & 6 \\ 2 & 3 & 4 & 0 & 4 & 5 & 2 \\ 5 & 7 & 10 & 4 & 7 & 11 & 5 \\ 7 & 9 & 12 & 5 & 11 & 13 & 7 \\ 3 & 4 & 6 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[157] \quad 1^{1223} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 13 & 1 & 3 \\ 3 & 4 & 8 & 3 & 16 & 0 & 4 \\ 6 & 8 & 10 & 4 & 22 & 0 & 6 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 13 & 16 & 22 & 9 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[156] \quad 1^{1223} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 11 & 1 & 3 \\ 3 & 4 & 8 & 3 & 16 & 0 & 4 \\ 5 & 8 & 10 & 4 & 22 & 0 & 6 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 11 & 16 & 22 & 9 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[155] \quad 1^{1123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 9 & 3 \\ 3 & 4 & 0 & 3 & 7 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 6 & 2 \\ 6 & 7 & 0 & 4 & 7 & 15 & 5 \\ 9 & 12 & 0 & 6 & 15 & 25 & 9 \\ 3 & 4 & 0 & 2 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[159] \quad 1^{1111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 9 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 5 & 2 \\ 3 & 3 & 2 & 2 & 0 & 5 & 2 \\ 6 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[161] \quad 1^{1123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 4 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 2 & 2 \\ 13 & 16 & 0 & 9 & 44 & 11 & 12 \\ 4 & 4 & 0 & 2 & 11 & 1 & 3 \\ 3 & 4 & 0 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[160] \quad 1^{1111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 7 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 4 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 5 & 7 & 5 & 4 & 7 & 11 & 5 \\ 7 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[164] \quad 1^{1123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 14 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 5 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 11 & 2 \\ 3 & 3 & 4 & 2 & 0 & 11 & 2 \\ 14 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[162] \quad 1^{1^{14}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 15 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 5 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 11 & 2 \\ 3 & 3 & 4 & 2 & 0 & 11 & 2 \\ 15 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[163] \quad 1^{1^{14}} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 16 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 4 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 12 & 2 \\ 3 & 3 & 2 & 2 & 0 & 12 & 2 \\ 16 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[165] \quad 1^{1^{13}} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 17 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 5 & 8 & 10 & 4 & 4 & 36 & 6 \\ 2 & 3 & 4 & 0 & 2 & 14 & 2 \\ 3 & 3 & 4 & 2 & 0 & 14 & 2 \\ 17 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[166] \quad 1^{1^{12}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 7 & 3 \\ 3 & 4 & 0 & 3 & 7 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 5 & 2 \\ 6 & 7 & 0 & 4 & 7 & 11 & 5 \\ 7 & 9 & 0 & 5 & 11 & 13 & 7 \\ 3 & 4 & 0 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[167] \quad 1^{1^{14}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 3 & 3 & 2 & 2 & 0 & 9 & 2 \\ 13 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[168] \quad 1^{1^{12}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 8 & 4 \\ 4 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 10 & 5 \\ 6 & 8 & 6 & 4 & 10 & 10 & 6 \\ 3 & 4 & 3 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[169] \quad 1^{1^{12}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -121, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 11 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 4 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 11 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[170] \quad 1^{1^{11}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -122, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 3 \\ 3 & 4 & 0 & 3 & 7 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 2 & 2 \\ 5 & 7 & 0 & 4 & 7 & 5 & 5 \\ 3 & 4 & 0 & 2 & 5 & 1 & 3 \\ 3 & 4 & 0 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[171] \quad 1^{1^{124}} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -122, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 12 & 3 \\ 3 & 4 & 0 & 3 & 7 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 8 & 2 \\ 6 & 7 & 0 & 4 & 7 & 20 & 5 \\ 12 & 16 & 0 & 8 & 20 & 46 & 12 \\ 3 & 4 & 0 & 2 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[172] \quad 1^{1^{231}} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 10 & 3 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 & 4 \\ 4 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 7 & 2 & 2 \\ 10 & 12 & 8 & 7 & 25 & 8 & 9 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[173] \quad 1^{1^{31}} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 1 & 3 \\ 3 & 4 & 0 & 3 & 16 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 0 & 2 \\ 12 & 16 & 0 & 9 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[174] \quad 1^{1^{212}} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 8 & 4 \\ 6 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 4 & 0 & 2 & 4 & 2 \\ 3 & 3 & 4 & 2 & 0 & 4 & 2 \\ 6 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 6 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[175] \quad 1^{1^{112}} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 11 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 2 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 5 & 7 & 5 & 4 & 7 & 22 & 5 \\ 11 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[176] \quad 1^{1^{13}} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 3 & 3 & 2 & 2 & 0 & 9 & 2 \\ 12 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[177] \quad 1^{1^{221}} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 6 & 3 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 & 4 \\ 4 & 4 & 1 & 2 & 5 & 3 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 & 2 \\ 6 & 7 & 5 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[179] \quad 1^{1^{13}} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 12 & 3 \\ 3 & 4 & 0 & 3 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 9 & 2 \\ 3 & 3 & 0 & 2 & 0 & 9 & 2 \\ 12 & 17 & 0 & 9 & 9 & 49 & 13 \\ 3 & 4 & 0 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[178] \ 1^{\overline{113}} 3^{\overline{11}} 1^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -126, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 15 & 3 \\ 3 & 4 & 0 & 3 & 3 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 12 & 2 \\ 3 & 3 & 0 & 2 & 0 & 12 & 2 \\ 15 & 22 & 0 & 12 & 12 & 85 & 17 \\ 3 & 4 & 0 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[180] \ 1^{\overline{114}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -131, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 4 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 11 & 2 \\ 3 & 3 & 2 & 2 & 0 & 11 & 2 \\ 15 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[182] \ 1^{\overline{1122}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -132, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 6 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 8 & 4 \\ 4 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 4 & 2 \\ 6 & 7 & 5 & 4 & 7 & 10 & 5 \\ 6 & 8 & 6 & 4 & 10 & 10 & 6 \\ 3 & 4 & 3 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[181] \ 21^{\overline{1112}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -132, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 5 & 6 & 3 \\ 4 & 4 & 9 & 3 & 7 & 8 & 4 \\ 6 & 9 & 13 & 5 & 11 & 12 & 7 \\ 2 & 3 & 5 & 0 & 4 & 4 & 2 \\ 5 & 7 & 11 & 4 & 7 & 10 & 5 \\ 6 & 8 & 12 & 4 & 10 & 10 & 6 \\ 3 & 4 & 7 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[183] \ 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -134, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 6 & 3 \\ 3 & 4 & 0 & 3 & 7 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 4 & 2 \\ 5 & 7 & 0 & 4 & 7 & 10 & 5 \\ 6 & 8 & 0 & 4 & 10 & 10 & 6 \\ 3 & 4 & 0 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[184] \ 1^{\overline{113}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -136, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 6 & 2 \\ 3 & 3 & 2 & 2 & 0 & 6 & 2 \\ 9 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[185] \ 1^{\overline{131}} 3^{\overline{11}} 1^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -138, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 15 & 1 & 3 \\ 3 & 4 & 4 & 3 & 21 & 0 & 4 \\ 2 & 4 & 1 & 2 & 14 & 0 & 3 \\ 2 & 3 & 2 & 0 & 12 & 0 & 2 \\ 15 & 21 & 14 & 12 & 80 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[186] \ 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -139, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 7 & 3 \\ 3 & 4 & 0 & 3 & 7 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 5 & 2 \\ 5 & 7 & 0 & 4 & 7 & 11 & 5 \\ 7 & 9 & 0 & 5 & 11 & 13 & 7 \\ 3 & 4 & 0 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[188] \quad 1^{114} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 6 & 2 \\ 3 & 3 & 2 & 2 & 0 & 6 & 2 \\ 10 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[187] \quad 1^{1121} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 9 & 3 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 & 2 \\ 9 & 11 & 8 & 6 & 20 & 8 & 8 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[198] \quad 1^{1112} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 5 & 3 \\ 3 & 4 & 4 & 3 & 7 & 8 & 4 \\ 2 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 10 & 5 \\ 5 & 8 & 6 & 4 & 10 & 10 & 6 \\ 3 & 4 & 3 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[197] \quad 1^{1212} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 9 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 7 & 2 \\ 3 & 3 & 4 & 2 & 0 & 7 & 2 \\ 9 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[196] \quad 1^{113} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 10 & 3 \\ 3 & 4 & 0 & 3 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 7 & 2 \\ 3 & 3 & 0 & 2 & 0 & 7 & 2 \\ 10 & 13 & 0 & 7 & 7 & 28 & 10 \\ 3 & 4 & 0 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[195] \quad 1^{121} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 5 & 3 \\ 3 & 4 & 0 & 3 & 11 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 4 & 2 \\ 8 & 11 & 0 & 6 & 20 & 16 & 8 \\ 5 & 8 & 0 & 4 & 16 & 10 & 6 \\ 3 & 4 & 0 & 2 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[194] \quad 1^{113} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 3 & 15 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 8 & 2 & 2 \\ 11 & 15 & 0 & 8 & 39 & 11 & 11 \\ 2 & 4 & 0 & 2 & 11 & 1 & 3 \\ 3 & 4 & 0 & 2 & 11 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[193] \quad 1^{12} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 2 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 & 8 \\ 2 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[192] \ 1^{1112}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 10 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 2 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 6 & 7 & 5 & 4 & 7 & 16 & 5 \\ 10 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[191] \ 21^{113}3^{11}1^{1112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 6 & 12 & 4 \\ 5 & 4 & 4 & 3 & 7 & 13 & 4 \\ 3 & 4 & 1 & 2 & 5 & 9 & 3 \\ 3 & 3 & 2 & 0 & 4 & 7 & 2 \\ 6 & 7 & 5 & 4 & 7 & 16 & 5 \\ 12 & 13 & 9 & 7 & 16 & 28 & 10 \\ 4 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[190] \ 1^{113}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 8 & 3 \\ 3 & 4 & 0 & 3 & 7 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 6 & 2 \\ 5 & 7 & 0 & 4 & 7 & 15 & 5 \\ 8 & 12 & 0 & 6 & 15 & 25 & 9 \\ 3 & 4 & 0 & 2 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[189] \ 1^{141}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 1 & 3 \\ 3 & 4 & 0 & 3 & 19 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 10 & 0 & 2 \\ 14 & 19 & 0 & 10 & 64 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 14 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[199] \ 1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -145, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 6 & 3 \\ 3 & 4 & 0 & 3 & 7 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 5 & 2 \\ 5 & 7 & 0 & 4 & 7 & 11 & 5 \\ 6 & 9 & 0 & 5 & 11 & 13 & 7 \\ 3 & 4 & 0 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[200] \ 1^{1112}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -145, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 9 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 4 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 9 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[201] \ 1^{123}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -146, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 10 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 6 & 2 \\ 3 & 3 & 4 & 2 & 0 & 6 & 2 \\ 10 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[203] \ 1^{1112}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -147, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 8 & 3 \\ 3 & 4 & 0 & 3 & 7 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 7 & 2 \\ 5 & 7 & 0 & 4 & 7 & 16 & 5 \\ 8 & 13 & 0 & 7 & 16 & 28 & 10 \\ 3 & 4 & 0 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[202] \quad 1^{1111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 6 & 8 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 6 & 7 & 5 & 4 & 7 & 11 & 5 \\ 8 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[204] \quad 1^{1112} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -149, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 7 & 3 \\ 3 & 4 & 4 & 3 & 3 & 9 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 5 & 2 \\ 3 & 3 & 2 & 2 & 0 & 5 & 2 \\ 7 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[205] \quad 1^{113} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 15 & 1 & 3 \\ 3 & 4 & 4 & 3 & 20 & 0 & 4 \\ 2 & 4 & 1 & 2 & 14 & 0 & 3 \\ 2 & 3 & 2 & 0 & 11 & 0 & 2 \\ 15 & 20 & 14 & 11 & 69 & 0 & 15 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 15 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[211] \quad 1^{113} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 9 & 3 \\ 3 & 4 & 0 & 3 & 7 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 6 & 2 \\ 5 & 7 & 0 & 4 & 7 & 15 & 5 \\ 9 & 12 & 0 & 6 & 15 & 25 & 9 \\ 3 & 4 & 0 & 2 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[210] \quad 1^{123} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 16 & 3 \\ 3 & 4 & 8 & 3 & 3 & 22 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 12 & 2 \\ 3 & 3 & 4 & 2 & 0 & 12 & 2 \\ 16 & 22 & 30 & 12 & 12 & 85 & 17 \\ 3 & 4 & 6 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[209] \quad 1^{1112} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 14 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 4 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 5 & 7 & 5 & 4 & 7 & 22 & 5 \\ 14 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[208] \quad 1^{123} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 9 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 6 & 2 \\ 3 & 3 & 4 & 2 & 0 & 6 & 2 \\ 9 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[207] \quad 1^{114} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 17 & 3 \\ 3 & 4 & 4 & 3 & 3 & 26 & 4 \\ 2 & 4 & 1 & 2 & 2 & 18 & 3 \\ 2 & 3 & 2 & 0 & 2 & 14 & 2 \\ 3 & 3 & 2 & 2 & 0 & 14 & 2 \\ 17 & 26 & 18 & 14 & 14 & 118 & 20 \\ 3 & 4 & 3 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[206] \ 1^{1111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -154, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 & 4 \\ 2 & 4 & 1 & 2 & 5 & 3 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[212] \ 1^{1123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -155, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 3 & 3 & 2 & 2 & 0 & 7 & 2 \\ 10 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[213] \ 1^{1112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -156, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 13 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 2 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 5 & 7 & 5 & 4 & 7 & 22 & 5 \\ 13 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[214] \ 1^{1124}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -162, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 16 & 3 \\ 3 & 4 & 0 & 3 & 7 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 12 & 2 \\ 6 & 7 & 0 & 4 & 7 & 27 & 5 \\ 16 & 22 & 0 & 12 & 27 & 85 & 17 \\ 3 & 4 & 0 & 2 & 5 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[216] \ 1^{1112}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -166, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 8 & 4 \\ 2 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 10 & 5 \\ 6 & 8 & 6 & 4 & 10 & 10 & 6 \\ 3 & 4 & 3 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[215] \ 1^{1214}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -166, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 3 & 3 & 2 & 2 & 0 & 10 & 2 \\ 14 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[217] \ 1^{1111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -169, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 2 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 5 & 7 & 5 & 4 & 7 & 11 & 5 \\ 6 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[218] \ 1^{1143}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -171, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 11 & 2 \\ 3 & 3 & 2 & 2 & 0 & 11 & 2 \\ 15 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[219] \ 1^{1211} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -172, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 6 & 3 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 & 4 \\ 7 & 8 & 10 & 4 & 10 & 6 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 & 2 \\ 6 & 7 & 10 & 4 & 7 & 5 & 5 \\ 3 & 4 & 6 & 2 & 5 & 1 & 3 \\ 3 & 4 & 6 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[220] \ 1^{114} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -173, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 15 & 3 \\ 3 & 4 & 0 & 3 & 7 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 11 & 2 \\ 6 & 7 & 0 & 4 & 7 & 26 & 5 \\ 15 & 21 & 0 & 11 & 26 & 76 & 16 \\ 3 & 4 & 0 & 2 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[221] \ 1^{121} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -174, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 & 3 \\ 3 & 4 & 0 & 3 & 11 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 2 & 2 \\ 9 & 11 & 0 & 6 & 20 & 8 & 8 \\ 3 & 4 & 0 & 2 & 8 & 1 & 3 \\ 3 & 4 & 0 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[226] \ 1^{123} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 12 & 1 & 3 \\ 3 & 4 & 8 & 3 & 16 & 0 & 4 \\ 6 & 8 & 10 & 4 & 22 & 0 & 6 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 12 & 16 & 22 & 9 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 2 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[225] \ 1^{113} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 12 & 2 \\ 3 & 3 & 2 & 2 & 0 & 12 & 2 \\ 15 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[227] \ 1^{112} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 3 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 12 & 2 \\ 3 & 3 & 2 & 2 & 0 & 12 & 2 \\ 14 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[222] \ 1^{114} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 15 & 3 \\ 3 & 4 & 0 & 3 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 11 & 2 \\ 3 & 3 & 0 & 2 & 0 & 11 & 2 \\ 15 & 21 & 0 & 11 & 11 & 76 & 16 \\ 3 & 4 & 0 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[223] \ 1^{123} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 10 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 2 & 3 & 4 & 0 & 2 & 7 & 2 \\ 3 & 3 & 4 & 2 & 0 & 7 & 2 \\ 10 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[224] \quad 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 1 & 3 \\ 3 & 4 & 0 & 3 & 21 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 12 & 0 & 2 \\ 16 & 21 & 0 & 12 & 80 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[228] \quad 1^{114} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -178, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 16 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 3 & 3 & 2 & 0 & 8 & 2 & 2 \\ 12 & 16 & 12 & 8 & 8 & 46 & 12 \\ 3 & 4 & 3 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[229] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 7 & 2 & 2 \\ 9 & 12 & 0 & 7 & 25 & 8 & 9 \\ 3 & 4 & 0 & 2 & 8 & 1 & 3 \\ 3 & 4 & 0 & 2 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[233] \quad 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 2 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 & 2 \\ 11 & 16 & 11 & 9 & 44 & 11 & 12 \\ 2 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[232] \quad 1^{122} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 6 & 3 \\ 3 & 4 & 0 & 3 & 11 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 4 & 2 \\ 9 & 11 & 0 & 6 & 20 & 16 & 8 \\ 6 & 8 & 0 & 4 & 16 & 10 & 6 \\ 3 & 4 & 0 & 2 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[231] \quad 1^{113} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 13 & 3 \\ 3 & 4 & 0 & 3 & 3 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 10 & 2 \\ 3 & 3 & 0 & 2 & 0 & 10 & 2 \\ 13 & 18 & 0 & 10 & 10 & 58 & 14 \\ 3 & 4 & 0 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[230] \quad 1^{113} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 16 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 12 & 2 \\ 3 & 3 & 2 & 2 & 0 & 12 & 2 \\ 16 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[234] \quad 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 3 & 3 & 2 & 2 & 0 & 9 & 2 \\ 11 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[235] \quad 1^{114} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 12 & 3 \\ 3 & 4 & 0 & 3 & 7 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 8 & 2 \\ 5 & 7 & 0 & 4 & 7 & 20 & 5 \\ 12 & 16 & 0 & 8 & 20 & 46 & 12 \\ 3 & 4 & 0 & 2 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[237] \quad 1^{121} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 6 & 3 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 & 4 \\ 3 & 4 & 1 & 2 & 5 & 3 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 & 2 \\ 6 & 7 & 5 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[236] \quad 1^{141} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 1 & 3 \\ 3 & 4 & 0 & 3 & 20 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 11 & 0 & 2 \\ 15 & 20 & 0 & 11 & 69 & 0 & 15 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 15 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[238] \quad 1^{122} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 2 & 2 \\ 11 & 16 & 0 & 9 & 44 & 11 & 12 \\ 2 & 4 & 0 & 2 & 11 & 1 & 3 \\ 3 & 4 & 0 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[239] \quad 1^{122} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 8 & 3 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 4 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 & 8 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[240] \quad 1^{112} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 3 & 3 & 2 & 2 & 0 & 10 & 2 \\ 12 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[241] \quad 1^{214} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 3 & 3 & 2 & 2 & 0 & 7 & 2 \\ 11 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[243] \quad 1^{14} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 4 & 0 & 3 & 19 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 10 & 2 & 2 \\ 14 & 19 & 0 & 10 & 64 & 14 & 14 \\ 2 & 4 & 0 & 2 & 14 & 1 & 3 \\ 3 & 4 & 0 & 2 & 14 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[242] \quad 1^{123} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 8 & 3 \\ 3 & 4 & 0 & 3 & 11 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 6 & 2 \\ 8 & 11 & 0 & 6 & 20 & 24 & 8 \\ 8 & 12 & 0 & 6 & 24 & 25 & 9 \\ 3 & 4 & 0 & 2 & 8 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[244] \quad 1^{124} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 12 & 3 \\ 3 & 4 & 8 & 3 & 3 & 16 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 8 & 2 \\ 3 & 3 & 4 & 2 & 0 & 8 & 2 \\ 12 & 16 & 24 & 8 & 8 & 46 & 12 \\ 3 & 4 & 6 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[245] \quad 1^{114} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 11 & 3 \\ 3 & 4 & 0 & 3 & 7 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 8 & 2 \\ 5 & 7 & 0 & 4 & 7 & 20 & 5 \\ 11 & 16 & 0 & 8 & 20 & 46 & 12 \\ 3 & 4 & 0 & 2 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[246] \quad 1^{132} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 5 & 3 \\ 3 & 4 & 0 & 3 & 15 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 8 & 4 & 2 \\ 11 & 15 & 0 & 8 & 39 & 22 & 11 \\ 5 & 8 & 0 & 4 & 22 & 10 & 6 \\ 3 & 4 & 0 & 2 & 11 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[247] \quad 1^{124} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -197, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 11 & 3 \\ 3 & 4 & 0 & 3 & 7 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 7 & 2 \\ 6 & 7 & 0 & 4 & 7 & 16 & 5 \\ 11 & 13 & 0 & 7 & 16 & 28 & 10 \\ 3 & 4 & 0 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[248] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -197, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 6 & 7 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 6 & 7 & 5 & 4 & 7 & 11 & 5 \\ 7 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[249] \quad 1^{112} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 14 & 3 \\ 3 & 4 & 0 & 3 & 7 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 10 & 2 \\ 6 & 7 & 0 & 4 & 7 & 22 & 5 \\ 14 & 18 & 0 & 10 & 22 & 58 & 14 \\ 3 & 4 & 0 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[250] \quad 1^{112} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 6 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 8 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 4 & 2 \\ 6 & 7 & 5 & 4 & 7 & 10 & 5 \\ 6 & 8 & 6 & 4 & 10 & 10 & 6 \\ 3 & 4 & 3 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[251] \quad 1^{1^{114}} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 14 & 3 \\ 3 & 4 & 0 & 3 & 7 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 10 & 2 \\ 5 & 7 & 0 & 4 & 7 & 22 & 5 \\ 14 & 18 & 0 & 10 & 22 & 58 & 14 \\ 3 & 4 & 0 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[252] \quad 1^{1^{111}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 5 & 3 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 & 4 \\ 5 & 8 & 10 & 4 & 10 & 6 & 6 \\ 2 & 3 & 4 & 0 & 4 & 2 & 2 \\ 5 & 7 & 10 & 4 & 7 & 5 & 5 \\ 3 & 4 & 6 & 2 & 5 & 1 & 3 \\ 3 & 4 & 6 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[253] \quad 1^{1^{123}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 10 & 3 \\ 3 & 4 & 0 & 3 & 7 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 7 & 2 \\ 6 & 7 & 0 & 4 & 7 & 16 & 5 \\ 10 & 13 & 0 & 7 & 16 & 28 & 10 \\ 3 & 4 & 0 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[254] \quad 1^{1^{121}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 9 & 3 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 7 & 2 & 2 \\ 9 & 12 & 8 & 7 & 25 & 8 & 9 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[255] \quad 1^{1^{114}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 13 & 3 \\ 3 & 4 & 0 & 3 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 9 & 2 \\ 3 & 3 & 0 & 2 & 0 & 9 & 2 \\ 13 & 17 & 0 & 9 & 9 & 49 & 13 \\ 3 & 4 & 0 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[256] \quad 1^{1^{111}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -210, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 5 & 7 & 5 & 4 & 7 & 11 & 5 \\ 6 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[257] \quad 1^{1^{13}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -211, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 10 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 2 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 10 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[258] \quad 1^{1^{14}} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 18 & 3 \\ 3 & 4 & 0 & 3 & 3 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 14 & 2 \\ 3 & 3 & 0 & 2 & 0 & 14 & 2 \\ 18 & 26 & 0 & 14 & 14 & 118 & 20 \\ 3 & 4 & 0 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[259] $1^{1^{12}3^{11}1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 11 & 3 \\ 3 & 4 & 0 & 3 & 11 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 9 & 2 \\ 8 & 11 & 0 & 6 & 20 & 33 & 8 \\ 11 & 17 & 0 & 9 & 33 & 49 & 13 \\ 3 & 4 & 0 & 2 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

[260] $1^{1^{12}1^{11}3^{11}1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 & 3 \\ 3 & 4 & 0 & 3 & 11 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 2 & 2 \\ 8 & 11 & 0 & 6 & 20 & 8 & 8 \\ 3 & 4 & 0 & 2 & 8 & 1 & 3 \\ 3 & 4 & 0 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

[261] $21^{1^{11}1^{12}3^{11}1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 5 & 12 & 3 \\ 4 & 4 & 4 & 3 & 7 & 18 & 4 \\ 3 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 5 & 7 & 5 & 4 & 7 & 22 & 5 \\ 12 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

[262] $1^{1^{11}1^{11}3^{11}1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 & 4 \\ 3 & 4 & 1 & 2 & 5 & 3 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[263] $1^{1^{11}1^{11}3^{11}1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 7 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 5 & 7 & 5 & 4 & 7 & 11 & 5 \\ 7 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[264] $1^{1^{11}1^{12}3^{11}1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 12 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 3 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 5 & 7 & 5 & 4 & 7 & 22 & 5 \\ 12 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

[265] $1^{1^{11}1^{12}3^{11}1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 3 & 3 & 2 & 2 & 0 & 7 & 2 \\ 9 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[266] $1^{1^{11}1^{12}3^{11}1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 5 & 10 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 4 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 10 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[267] \quad 1^{1^{12}2^13^{11}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 9 & 3 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 4 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 & 2 \\ 9 & 11 & 8 & 6 & 20 & 8 & 8 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[268] \quad 1^{1^{12}2^13^{11}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 6 & 9 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 3 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 6 & 7 & 5 & 4 & 7 & 16 & 5 \\ 9 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[269] \quad 1^{1^{24}3^{11}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 14 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 11 & 2 \\ 3 & 3 & 4 & 2 & 0 & 11 & 2 \\ 14 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[270] \quad 1^{1^{12}2^13^{11}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 6 & 3 \\ 3 & 4 & 0 & 3 & 11 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 4 & 2 \\ 8 & 11 & 0 & 6 & 20 & 16 & 8 \\ 6 & 8 & 0 & 4 & 16 & 10 & 6 \\ 3 & 4 & 0 & 2 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[271] \quad 1^{1^{12}2^13^{11}} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 15 & 3 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 & 3 \\ 2 & 3 & 2 & 0 & 12 & 2 & 2 \\ 15 & 21 & 14 & 12 & 80 & 14 & 16 \\ 3 & 4 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 3 & 2 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[274] \quad 1^{1^{12}2^13^{11}} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 12 & 3 \\ 3 & 4 & 0 & 3 & 7 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 10 & 2 \\ 5 & 7 & 0 & 4 & 7 & 22 & 5 \\ 12 & 18 & 0 & 10 & 22 & 58 & 14 \\ 3 & 4 & 0 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[273] \quad 1^{1^{11}2^13^{11}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 11 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 3 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 11 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[272] \quad 1^{1^{11}2^13^{11}} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 12 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 2 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 5 & 7 & 5 & 4 & 7 & 22 & 5 \\ 12 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[275] \ 1^{1^{13}3^{11}1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -229, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 3 & 3 & 2 & 2 & 0 & 9 & 2 \\ 13 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[276] \ 1^{1^{13}3^{11}1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -232, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 14 & 1 & 3 \\ 3 & 4 & 8 & 3 & 20 & 0 & 4 \\ 5 & 8 & 10 & 4 & 28 & 0 & 6 \\ 2 & 3 & 4 & 0 & 11 & 0 & 2 \\ 14 & 20 & 28 & 11 & 69 & 0 & 15 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 2 & 15 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[277] \ 1^{1^{12}3^{11}1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -233, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 12 & 3 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 & 4 \\ 4 & 4 & 1 & 2 & 11 & 3 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 & 2 \\ 12 & 16 & 11 & 9 & 44 & 11 & 12 \\ 3 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[278] \ 1^{1^{12}3^{11}1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -236, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 14 & 3 \\ 3 & 4 & 0 & 3 & 3 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 10 & 2 \\ 3 & 3 & 0 & 2 & 0 & 10 & 2 \\ 14 & 18 & 0 & 10 & 10 & 58 & 14 \\ 3 & 4 & 0 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[281] \ 1^{1^{11}2^3 1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -240, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 14 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 3 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 5 & 7 & 5 & 4 & 7 & 22 & 5 \\ 14 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[280] \ 1^{1^{12}3^{11}1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -240, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 20 & 1 & 3 \\ 3 & 4 & 8 & 3 & 29 & 0 & 4 \\ 5 & 8 & 10 & 4 & 40 & 0 & 6 \\ 2 & 3 & 4 & 0 & 16 & 0 & 2 \\ 20 & 29 & 40 & 16 & 148 & 0 & 22 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 2 & 22 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[279] \ 1^{1^{11}2^3 1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -240, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 8 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 10 & 5 \\ 6 & 8 & 6 & 4 & 10 & 10 & 6 \\ 3 & 4 & 3 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[282] \ 1^{1^{12}3^{11}1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -243, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 3 & 3 & 2 & 2 & 0 & 7 & 2 \\ 11 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[283] $1^{123}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 9 & 3 \\ 3 & 4 & 0 & 3 & 11 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 6 & 2 \\ 8 & 11 & 0 & 6 & 20 & 24 & 8 \\ 9 & 12 & 0 & 6 & 24 & 25 & 9 \\ 3 & 4 & 0 & 2 & 8 & 9 & 3 \end{pmatrix}_{b_0}$$

[284] $1^{112}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 10 & 3 \\ 3 & 4 & 0 & 3 & 7 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 7 & 2 \\ 5 & 7 & 0 & 4 & 7 & 16 & 5 \\ 10 & 13 & 0 & 7 & 16 & 28 & 10 \\ 3 & 4 & 0 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[285] $1^{112}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -252, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 & 8 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

[286] $1^{112}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 6 & 14 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 3 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 6 & 7 & 5 & 4 & 7 & 22 & 5 \\ 14 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

[287] $1^{114}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 18 & 3 \\ 3 & 4 & 4 & 3 & 3 & 26 & 4 \\ 2 & 4 & 1 & 2 & 2 & 18 & 3 \\ 2 & 3 & 2 & 0 & 2 & 14 & 2 \\ 3 & 3 & 2 & 2 & 0 & 14 & 2 \\ 18 & 26 & 18 & 14 & 14 & 118 & 20 \\ 3 & 4 & 3 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[288] $1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 10 & 3 \\ 3 & 4 & 0 & 3 & 11 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 7 & 2 \\ 8 & 11 & 0 & 6 & 20 & 25 & 8 \\ 10 & 13 & 0 & 7 & 25 & 28 & 10 \\ 3 & 4 & 0 & 2 & 8 & 10 & 3 \end{pmatrix}_{b_0}$$

[289] $1^{124}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 11 & 3 \\ 3 & 4 & 0 & 3 & 11 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 8 & 2 \\ 8 & 11 & 0 & 6 & 20 & 32 & 8 \\ 11 & 16 & 0 & 8 & 32 & 46 & 12 \\ 3 & 4 & 0 & 2 & 8 & 12 & 3 \end{pmatrix}_{b_0}$$

[290] $1^{142}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 5 & 3 \\ 3 & 4 & 0 & 3 & 19 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 10 & 4 & 2 \\ 14 & 19 & 0 & 10 & 64 & 28 & 14 \\ 5 & 8 & 0 & 4 & 28 & 10 & 6 \\ 3 & 4 & 0 & 2 & 14 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[291] \quad 1^{14} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 1 & 3 \\ 3 & 4 & 0 & 3 & 24 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 13 & 0 & 2 \\ 17 & 24 & 0 & 13 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 18 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[292] \quad 1^{113} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 18 & 1 & 3 \\ 3 & 4 & 4 & 3 & 25 & 0 & 4 \\ 2 & 4 & 1 & 2 & 17 & 0 & 3 \\ 2 & 3 & 2 & 0 & 14 & 0 & 2 \\ 18 & 25 & 17 & 14 & 111 & 0 & 19 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & 19 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[293] \quad 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -259, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 2 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 8 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[294] \quad 1^{123} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -266, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 12 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 9 & 2 \\ 3 & 3 & 4 & 2 & 0 & 9 & 2 \\ 12 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[295] \quad 1^{122} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 9 & 3 \\ 3 & 4 & 0 & 3 & 11 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 7 & 2 \\ 8 & 11 & 0 & 6 & 20 & 25 & 8 \\ 9 & 13 & 0 & 7 & 25 & 28 & 10 \\ 3 & 4 & 0 & 2 & 8 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[296] \quad 1^{124} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 12 & 3 \\ 3 & 4 & 0 & 3 & 11 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 8 & 2 \\ 8 & 11 & 0 & 6 & 20 & 32 & 8 \\ 12 & 16 & 0 & 8 & 32 & 46 & 12 \\ 3 & 4 & 0 & 2 & 8 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[297] \quad 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -273, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 12 & 3 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 & 2 \\ 12 & 16 & 11 & 9 & 44 & 11 & 12 \\ 3 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[298] \quad 1^{113} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 2 & 0 & 2 & 7 & 2 \\ 3 & 3 & 2 & 2 & 0 & 7 & 2 \\ 10 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[299] $1^{124}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 18 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 6 & 8 & 10 & 4 & 4 & 36 & 6 \\ 2 & 3 & 4 & 0 & 2 & 14 & 2 \\ 3 & 3 & 4 & 2 & 0 & 14 & 2 \\ 18 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[300] $1^{114}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -278, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 16 & 3 \\ 3 & 4 & 0 & 3 & 3 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 & 12 & 2 \\ 3 & 3 & 0 & 2 & 0 & 12 & 2 \\ 16 & 22 & 0 & 12 & 12 & 85 & 17 \\ 3 & 4 & 0 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[302] $1^{112}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 8 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 3 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 8 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[301] $1^{13}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 1 & 3 \\ 3 & 4 & 0 & 3 & 25 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 14 & 0 & 2 \\ 18 & 25 & 0 & 14 & 111 & 0 & 19 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 19 & 0 & 3 \end{pmatrix}_{b_0}$$

[303] $1^{112}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -281, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 9 & 3 \\ 3 & 4 & 0 & 3 & 7 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 7 & 2 \\ 5 & 7 & 0 & 4 & 7 & 16 & 5 \\ 9 & 13 & 0 & 7 & 16 & 28 & 10 \\ 3 & 4 & 0 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[304] $1^{13}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 4 & 0 & 3 & 20 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 11 & 2 & 2 \\ 14 & 20 & 0 & 11 & 69 & 14 & 15 \\ 2 & 4 & 0 & 2 & 14 & 1 & 3 \\ 3 & 4 & 0 & 2 & 15 & 3 & 3 \end{pmatrix}_{b_0}$$

[306] $1^{13}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 & 3 \\ 3 & 4 & 0 & 3 & 15 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 8 & 2 & 2 \\ 12 & 15 & 0 & 8 & 39 & 11 & 11 \\ 3 & 4 & 0 & 2 & 11 & 1 & 3 \\ 3 & 4 & 0 & 2 & 11 & 3 & 3 \end{pmatrix}_{b_0}$$

[305] $1^{112}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 14 & 3 \\ 3 & 4 & 0 & 3 & 7 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 12 & 2 \\ 5 & 7 & 0 & 4 & 7 & 27 & 5 \\ 14 & 22 & 0 & 12 & 27 & 85 & 17 \\ 3 & 4 & 0 & 2 & 5 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[307] \quad 1^{132} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -290, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 6 & 3 \\ 3 & 4 & 0 & 3 & 15 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 8 & 4 & 2 \\ 12 & 15 & 0 & 8 & 39 & 22 & 11 \\ 6 & 8 & 0 & 4 & 22 & 10 & 6 \\ 3 & 4 & 0 & 2 & 11 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[308] \quad 1^{121} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -291, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 2 & 2 \\ 13 & 16 & 0 & 9 & 44 & 11 & 12 \\ 3 & 4 & 0 & 2 & 11 & 1 & 3 \\ 3 & 4 & 0 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[309] \quad 1^{113} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 13 & 3 \\ 3 & 4 & 0 & 3 & 7 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 9 & 2 \\ 6 & 7 & 0 & 4 & 7 & 21 & 5 \\ 13 & 17 & 0 & 9 & 21 & 49 & 13 \\ 3 & 4 & 0 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[310] \quad 1^{113} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 13 & 3 \\ 3 & 4 & 0 & 3 & 7 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 10 & 2 \\ 5 & 7 & 0 & 4 & 7 & 22 & 5 \\ 13 & 18 & 0 & 10 & 22 & 58 & 14 \\ 3 & 4 & 0 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[311] \quad 1^{114} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 3 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 11 & 2 \\ 3 & 3 & 2 & 2 & 0 & 11 & 2 \\ 14 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[312] \quad 1^{121} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -301, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 13 & 3 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 & 4 \\ 4 & 4 & 1 & 2 & 11 & 3 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 & 2 \\ 13 & 16 & 11 & 9 & 44 & 11 & 12 \\ 3 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[313] \quad 1^{123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 14 & 1 & 3 \\ 3 & 4 & 8 & 3 & 20 & 0 & 4 \\ 6 & 8 & 10 & 4 & 28 & 0 & 6 \\ 2 & 3 & 4 & 0 & 11 & 0 & 2 \\ 14 & 20 & 28 & 11 & 69 & 0 & 15 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 2 & 15 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[316] \quad 1^{112} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 13 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 3 & 4 & 1 & 2 & 5 & 12 & 3 \\ 2 & 3 & 2 & 0 & 4 & 10 & 2 \\ 5 & 7 & 5 & 4 & 7 & 22 & 5 \\ 13 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

[315] $1^{123}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 13 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 2 & 3 & 4 & 0 & 2 & 9 & 2 \\ 3 & 3 & 4 & 2 & 0 & 9 & 2 \\ 13 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[314] $1^{112}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 3 & 3 & 2 & 2 & 0 & 10 & 2 \\ 13 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[317] $1^{112}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 11 & 3 \\ 3 & 4 & 0 & 3 & 7 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 9 & 2 \\ 5 & 7 & 0 & 4 & 7 & 21 & 5 \\ 11 & 17 & 0 & 9 & 21 & 49 & 13 \\ 3 & 4 & 0 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[318] $1^{112}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 9 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 2 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 9 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[319] $1^{121}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -323, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 2 & 2 \\ 11 & 16 & 0 & 9 & 44 & 11 & 12 \\ 3 & 4 & 0 & 2 & 11 & 1 & 3 \\ 3 & 4 & 0 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

[320] $21^{111}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -324, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 5 & 6 & 3 \\ 4 & 4 & 4 & 3 & 7 & 9 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 5 & 2 \\ 5 & 7 & 5 & 4 & 7 & 11 & 5 \\ 6 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[321] $1^{112}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -328, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 9 & 3 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 & 2 \\ 9 & 11 & 8 & 6 & 20 & 8 & 8 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

[322] $1^{131}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 & 3 \\ 3 & 4 & 0 & 3 & 15 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 8 & 2 & 2 \\ 11 & 15 & 0 & 8 & 39 & 11 & 11 \\ 3 & 4 & 0 & 2 & 11 & 1 & 3 \\ 3 & 4 & 0 & 2 & 11 & 3 & 3 \end{pmatrix}_{b_0}$$

[323] $1^{113}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -341, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 3 & 3 & 2 & 2 & 0 & 9 & 2 \\ 12 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[324] $1^{132}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -342, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 6 & 3 \\ 3 & 4 & 0 & 3 & 15 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 8 & 4 & 2 \\ 11 & 15 & 0 & 8 & 39 & 22 & 11 \\ 6 & 8 & 0 & 4 & 22 & 10 & 6 \\ 3 & 4 & 0 & 2 & 11 & 6 & 3 \end{pmatrix}_{b_0}$$

[325] $1^{1124}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -349, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 6 & 11 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 3 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 6 & 7 & 5 & 4 & 7 & 16 & 5 \\ 11 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[326] $1^{1124}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -350, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 10 & 2 \\ 3 & 3 & 2 & 2 & 0 & 10 & 2 \\ 14 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[327] $1^{121}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -353, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 9 & 2 & 2 \\ 12 & 16 & 0 & 9 & 44 & 11 & 12 \\ 3 & 4 & 0 & 2 & 11 & 1 & 3 \\ 3 & 4 & 0 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

[328] $1^{121}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -357, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 3 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 & 2 \\ 11 & 16 & 11 & 9 & 44 & 11 & 12 \\ 3 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

[329] $1^{124}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 17 & 1 & 3 \\ 3 & 4 & 8 & 3 & 24 & 0 & 4 \\ 5 & 8 & 10 & 4 & 34 & 0 & 6 \\ 2 & 3 & 4 & 0 & 13 & 0 & 2 \\ 17 & 24 & 34 & 13 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 2 & 18 & 0 & 3 \end{pmatrix}_{b_0}$$

[330] $1^{121}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 8 & 3 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 & 8 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[332] \quad 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -371, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 13 & 3 \\ 3 & 4 & 0 & 3 & 7 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 9 & 2 \\ 5 & 7 & 0 & 4 & 7 & 21 & 5 \\ 13 & 17 & 0 & 9 & 21 & 49 & 13 \\ 3 & 4 & 0 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[331] \quad 1^{1123} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -371, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 6 & 10 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 3 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 6 & 7 & 5 & 4 & 7 & 16 & 5 \\ 10 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[333] \quad 21^{1113} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 5 & 3 & 3 \\ 4 & 4 & 4 & 3 & 7 & 4 & 4 \\ 3 & 4 & 1 & 2 & 5 & 3 & 3 \\ 2 & 3 & 2 & 0 & 4 & 2 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[334] \quad 1^{132} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -378, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 6 & 3 \\ 3 & 4 & 0 & 3 & 20 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 11 & 4 & 2 \\ 15 & 20 & 0 & 11 & 69 & 28 & 15 \\ 6 & 8 & 0 & 4 & 28 & 10 & 6 \\ 3 & 4 & 0 & 2 & 15 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[335] \quad 1^{13} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 2 & 3 \\ 3 & 4 & 0 & 3 & 25 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 14 & 2 & 2 \\ 17 & 25 & 0 & 14 & 111 & 17 & 19 \\ 2 & 4 & 0 & 2 & 17 & 1 & 3 \\ 3 & 4 & 0 & 2 & 19 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[336] \quad 1^{121} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 16 & 3 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 & 3 \\ 2 & 3 & 2 & 0 & 12 & 2 & 2 \\ 16 & 21 & 14 & 12 & 80 & 14 & 16 \\ 3 & 4 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 3 & 2 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[337] \quad 1^{114} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 17 & 3 \\ 3 & 4 & 4 & 3 & 3 & 26 & 4 \\ 3 & 4 & 1 & 2 & 2 & 18 & 3 \\ 2 & 3 & 2 & 0 & 2 & 14 & 2 \\ 3 & 3 & 2 & 2 & 0 & 14 & 2 \\ 17 & 26 & 18 & 14 & 14 & 118 & 20 \\ 3 & 4 & 3 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[338] \quad 1^{141} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -387, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 2 & 3 \\ 3 & 4 & 0 & 3 & 24 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 13 & 2 & 2 \\ 17 & 24 & 0 & 13 & 100 & 17 & 18 \\ 2 & 4 & 0 & 2 & 17 & 1 & 3 \\ 3 & 4 & 0 & 2 & 18 & 3 & 3 \end{pmatrix}_{b_0}$$

[339] $1^{1^{214}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 15 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 2 & 3 & 4 & 0 & 2 & 11 & 2 \\ 3 & 3 & 4 & 2 & 0 & 11 & 2 \\ 15 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[340] $1^{1^{113}}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 3 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 12 & 2 \\ 3 & 3 & 2 & 2 & 0 & 12 & 2 \\ 15 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[341] $1^{1^{234}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 13 & 3 \\ 3 & 4 & 0 & 3 & 11 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 9 & 2 \\ 8 & 11 & 0 & 6 & 20 & 33 & 8 \\ 13 & 17 & 0 & 9 & 33 & 49 & 13 \\ 3 & 4 & 0 & 2 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

[342] $1^{1^{213}}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -410, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 12 & 2 & 2 \\ 15 & 21 & 0 & 12 & 80 & 14 & 16 \\ 3 & 4 & 0 & 2 & 14 & 1 & 3 \\ 3 & 4 & 0 & 2 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

[343] $1^{1^{114}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -413, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 2 & 3 & 2 & 0 & 2 & 9 & 2 \\ 3 & 3 & 2 & 2 & 0 & 9 & 2 \\ 13 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[345] $1^{1^{211}}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 12 & 2 & 2 \\ 16 & 21 & 0 & 12 & 80 & 14 & 16 \\ 3 & 4 & 0 & 2 & 14 & 1 & 3 \\ 3 & 4 & 0 & 2 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

[344] $1^{1^{113}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 10 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 3 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 10 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[346] $21^{1^{112}}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 5 & 6 & 3 \\ 4 & 4 & 4 & 3 & 7 & 8 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 2 & 3 & 2 & 0 & 4 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 10 & 5 \\ 6 & 8 & 6 & 4 & 10 & 10 & 6 \\ 3 & 4 & 3 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[347] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -425, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 12 & 3 \\ 3 & 4 & 0 & 3 & 7 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 9 & 2 \\ 5 & 7 & 0 & 4 & 7 & 21 & 5 \\ 12 & 17 & 0 & 9 & 21 & 49 & 13 \\ 3 & 4 & 0 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[348] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 14 & 3 \\ 3 & 4 & 0 & 3 & 11 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 11 & 2 \\ 8 & 11 & 0 & 6 & 20 & 41 & 8 \\ 14 & 21 & 0 & 11 & 41 & 76 & 16 \\ 3 & 4 & 0 & 2 & 8 & 16 & 3 \end{pmatrix}_{b_0}$$

[349] $1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -434, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 5 & 9 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 3 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 9 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[350] $1^{141}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 13 & 3 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 & 2 \\ 13 & 16 & 11 & 9 & 44 & 11 & 12 \\ 3 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

[351] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -452, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 12 & 3 \\ 3 & 4 & 0 & 3 & 11 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 9 & 2 \\ 8 & 11 & 0 & 6 & 20 & 33 & 8 \\ 12 & 17 & 0 & 9 & 33 & 49 & 13 \\ 3 & 4 & 0 & 2 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

[352] $1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -452, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 11 & 3 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 & 2 \\ 11 & 16 & 11 & 9 & 44 & 11 & 12 \\ 3 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

[353] $1^{124}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 18 & 3 \\ 3 & 4 & 0 & 3 & 11 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 14 & 2 \\ 18 & 26 & 0 & 6 & 20 & 50 & 8 \\ 3 & 4 & 0 & 2 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

[354] $1^{114}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -475, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 3 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 11 & 2 \\ 3 & 3 & 2 & 2 & 0 & 11 & 2 \\ 15 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[355] $1^{141}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 & 3 \\ 3 & 4 & 0 & 3 & 19 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 10 & 2 & 2 \\ 14 & 19 & 0 & 10 & 64 & 14 & 14 \\ 3 & 4 & 0 & 2 & 14 & 1 & 3 \\ 3 & 4 & 0 & 2 & 14 & 3 & 3 \end{pmatrix}_{b_0}$$

[356] $1^{142}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -488, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 6 & 3 \\ 3 & 4 & 0 & 3 & 19 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 10 & 4 & 2 \\ 14 & 19 & 0 & 10 & 64 & 28 & 14 \\ 6 & 8 & 0 & 4 & 28 & 10 & 6 \\ 3 & 4 & 0 & 2 & 14 & 6 & 3 \end{pmatrix}_{b_0}$$

[357] $1^{113}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -491, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 14 & 3 \\ 3 & 4 & 0 & 3 & 7 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 11 & 2 \\ 5 & 7 & 0 & 4 & 7 & 26 & 5 \\ 14 & 21 & 0 & 11 & 26 & 76 & 16 \\ 3 & 4 & 0 & 2 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[358] $1^{113}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 15 & 3 \\ 3 & 4 & 0 & 3 & 7 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 12 & 2 \\ 5 & 7 & 0 & 4 & 7 & 27 & 5 \\ 15 & 22 & 0 & 12 & 27 & 85 & 17 \\ 3 & 4 & 0 & 2 & 5 & 17 & 3 \end{pmatrix}_{b_0}$$

[359] $1^{1121}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 12 & 3 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 & 2 \\ 12 & 16 & 11 & 9 & 44 & 11 & 12 \\ 3 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

[360] $1^{1131}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -533, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 & 3 \\ 3 & 4 & 0 & 3 & 20 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 11 & 2 & 2 \\ 14 & 20 & 0 & 11 & 69 & 14 & 15 \\ 3 & 4 & 0 & 2 & 14 & 1 & 3 \\ 3 & 4 & 0 & 2 & 15 & 3 & 3 \end{pmatrix}_{b_0}$$

[361] $1^{141}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 2 & 3 \\ 3 & 4 & 0 & 3 & 29 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 16 & 2 & 2 \\ 20 & 29 & 0 & 16 & 148 & 20 & 22 \\ 2 & 4 & 0 & 2 & 20 & 1 & 3 \\ 3 & 4 & 0 & 2 & 22 & 3 & 3 \end{pmatrix}_{b_0}$$

[362] $1^{142}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -540, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 6 & 3 \\ 3 & 4 & 0 & 3 & 24 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 13 & 4 & 2 \\ 17 & 24 & 0 & 13 & 100 & 34 & 18 \\ 6 & 8 & 0 & 4 & 34 & 10 & 6 \\ 3 & 4 & 0 & 2 & 18 & 6 & 3 \end{pmatrix}_{b_0}$$

[363] $1^{113}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -540, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 16 & 3 \\ 3 & 4 & 0 & 3 & 7 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 12 & 2 \\ 5 & 7 & 0 & 4 & 7 & 27 & 5 \\ 16 & 22 & 0 & 12 & 27 & 85 & 17 \\ 3 & 4 & 0 & 2 & 5 & 17 & 3 \end{pmatrix}_{b_0}$$

[364] $1^{113}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 16 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 3 & 4 & 1 & 2 & 2 & 15 & 3 \\ 2 & 3 & 2 & 0 & 2 & 12 & 2 \\ 3 & 3 & 2 & 2 & 0 & 12 & 2 \\ 16 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[365] $1^{131}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -555, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 & 3 \\ 3 & 4 & 0 & 3 & 20 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 11 & 2 & 2 \\ 15 & 20 & 0 & 11 & 69 & 14 & 15 \\ 3 & 4 & 0 & 2 & 14 & 1 & 3 \\ 3 & 4 & 0 & 2 & 15 & 3 & 3 \end{pmatrix}_{b_0}$$

[366] $1^{114}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -577, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 15 & 3 \\ 3 & 4 & 0 & 3 & 7 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 11 & 2 \\ 5 & 7 & 0 & 4 & 7 & 26 & 5 \\ 15 & 21 & 0 & 11 & 26 & 76 & 16 \\ 3 & 4 & 0 & 2 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[367] $1^{114}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 17 & 3 \\ 3 & 4 & 0 & 3 & 7 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 14 & 2 \\ 5 & 7 & 0 & 4 & 7 & 32 & 5 \\ 17 & 26 & 0 & 14 & 32 & 118 & 20 \\ 3 & 4 & 0 & 2 & 5 & 20 & 3 \end{pmatrix}_{b_0}$$

[368] $1^{131}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -614, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 & 3 \\ 3 & 4 & 0 & 3 & 25 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 14 & 2 & 2 \\ 17 & 25 & 0 & 14 & 111 & 17 & 19 \\ 3 & 4 & 0 & 2 & 19 & 3 & 3 \end{pmatrix}_{b_0}$$

[369] $21^{112}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 8 & 3 & 3 \\ 4 & 4 & 4 & 3 & 11 & 4 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 & 3 \\ 2 & 3 & 2 & 0 & 6 & 2 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 & 8 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

[370] $1^{114}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -630, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 18 & 3 \\ 3 & 4 & 4 & 3 & 3 & 26 & 4 \\ 3 & 4 & 1 & 2 & 2 & 18 & 3 \\ 2 & 3 & 2 & 0 & 2 & 14 & 2 \\ 3 & 3 & 2 & 2 & 0 & 14 & 2 \\ 18 & 26 & 18 & 14 & 14 & 118 & 20 \\ 3 & 4 & 3 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[371] $1^{124}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -640, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 15 & 3 \\ 3 & 4 & 0 & 3 & 11 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 6 & 11 & 2 \\ 8 & 11 & 0 & 6 & 20 & 41 & 8 \\ 15 & 21 & 0 & 11 & 41 & 76 & 16 \\ 3 & 4 & 0 & 2 & 8 & 16 & 3 \end{pmatrix}_{b_0}$$

[372] $21^{1121}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -660, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 11 & 3 & 3 \\ 4 & 4 & 4 & 3 & 16 & 4 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 & 3 \\ 2 & 3 & 2 & 0 & 9 & 2 & 2 \\ 11 & 16 & 11 & 9 & 44 & 11 & 12 \\ 3 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

[373] $21^{1112}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -676, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 5 & 9 & 3 \\ 4 & 4 & 4 & 3 & 7 & 13 & 4 \\ 3 & 4 & 1 & 2 & 5 & 9 & 3 \\ 2 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 9 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[374] $1^{131}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 & 3 \\ 3 & 4 & 0 & 3 & 25 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 14 & 2 & 2 \\ 18 & 25 & 0 & 14 & 111 & 17 & 19 \\ 3 & 4 & 0 & 2 & 17 & 1 & 3 \\ 3 & 4 & 0 & 2 & 19 & 3 & 3 \end{pmatrix}_{b_0}$$

[375] $1^{114}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -770, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 18 & 3 \\ 3 & 4 & 0 & 3 & 7 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 4 & 14 & 2 \\ 5 & 7 & 0 & 4 & 7 & 32 & 5 \\ 18 & 26 & 0 & 14 & 32 & 118 & 20 \\ 3 & 4 & 0 & 2 & 5 & 20 & 3 \end{pmatrix}_{b_0}$$

[376] $1^{141}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -771, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 & 3 \\ 3 & 4 & 0 & 3 & 24 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 13 & 2 & 2 \\ 17 & 24 & 0 & 13 & 100 & 17 & 18 \\ 3 & 4 & 0 & 2 & 17 & 1 & 3 \\ 3 & 4 & 0 & 2 & 18 & 3 & 3 \end{pmatrix}_{b_0}$$

[377] $1^{141}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 & 3 \\ 3 & 4 & 0 & 3 & 29 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 16 & 2 & 2 \\ 20 & 29 & 0 & 16 & 148 & 20 & 22 \\ 3 & 4 & 0 & 2 & 20 & 1 & 3 \\ 3 & 4 & 0 & 2 & 22 & 3 & 3 \end{pmatrix}_{b_0}$$

4.935 NHC(-2)(-3) + $\mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$

$T = 8$ [0] $1^{13}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3$ $T = 8, T_{\max} = 10, \Delta = -65, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 0 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 3 & 4 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 0 & 3 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 1 & 3 \\ 3 & 4 & 0 & 2 & -1 & 3 & 3 \end{pmatrix} b_0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & 0 & 2 & -1 & 4 & 3 \end{pmatrix} b_0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 2 & 0 & 1 & 3 \\ 3 & 4 & 4 & 2 & -1 & 3 & 3 \end{pmatrix} b_0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 2 & 0 & 1 & 3 \\ 3 & 4 & 4 & 2 & -1 & 3 & 3 \end{pmatrix} b_0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 2 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -35, \ \det = -91, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 2 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

4.937 NHC(-2)(-3) + su₂ + su₂ → (-3) + su₃ + su₂ + su₂

$$T = 10 \quad [0] \ 1^{11} 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -5, \ \det = 9, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 0 & 4 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 1 & 3 & 3 \\ 3 & 5 & 0 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 0 & 2 & -1 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -5, \ \det = 9, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 3 & 2 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 2 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & 2 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

4.938 NHC(-2)(-3) + su₂ + su₂n3mix

$$T = 7 \quad [2] \ 21^{13} 3^{11} 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -4, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 24 & 6 & 4 \\ 5 & 4 & 23 & 7 & 4 \\ 24 & 23 & 100 & 28 & 18 \\ 6 & 7 & 28 & 7 & 5 \\ 4 & 4 & 18 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -4, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 5 & 5 & 4 & 6 & 4 \\ 6 & 7 & 6 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -4, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 5 & 9 & 13 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{121} 3^{11} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -4, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 6 & 3 \\ 3 & 4 & 23 & 7 & 4 \\ 17 & 23 & 100 & 28 & 18 \\ 6 & 7 & 28 & 7 & 5 \\ 3 & 4 & 18 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 1^{11} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -6, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 14 & 7 & 4 \\ 9 & 14 & 37 & 17 & 11 \\ 5 & 7 & 17 & 7 & 5 \\ 3 & 4 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

- [4] $1^{133}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 7 & 8 & 10 & 16 & 6 \\ 10 & 12 & 16 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$
- [5] $1^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 10 & 16 & 6 \\ 8 & 12 & 16 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$
- [6] $1^{132}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 17 & 11 & 4 \\ 10 & 17 & 49 & 33 & 13 \\ 8 & 11 & 33 & 20 & 8 \\ 3 & 4 & 13 & 8 & 3 \end{pmatrix}_{b_0}$
- [8] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.57 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [9] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 0 & 2 & 3 \end{pmatrix}_{b_0}$
- [10] $1^{13}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 3 & 5 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [11] $1^{122}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -10, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 6 & 8 & 10 & 16 & 6 \\ 9 & 12 & 16 & 25 & 9 \\ 3 & 4 & 6 & 9 & 3 \end{pmatrix}_{b_0}$
- [12] $1^{113}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -10, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 5 & 12 & 4 \\ 5 & 5 & 4 & 12 & 4 \\ 9 & 12 & 12 & 25 & 9 \\ 3 & 4 & 4 & 9 & 3 \end{pmatrix}_{b_0}$
- [13] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -13, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 5 & 8 & 10 & 4 & 6 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [14] $1^{111}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -13, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 3 & 5 & 4 & 6 & 4 \\ 5 & 7 & 6 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$
- [18] $1^{111}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7, T_{\max} = 10, \Delta = -86, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 5 & 9 & 13 & 14 & 7 \\ 6 & 8 & 14 & 12 & 6 \\ 3 & 4 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 5 & 16 & 4 \\ 5 & 5 & 4 & 15 & 4 \\ 13 & 16 & 15 & 44 & 12 \\ 3 & 4 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 5 & 16 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 13 & 16 & 15 & 44 & 12 \\ 3 & 4 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 13^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 3 & 5 & 4 & 6 & 4 \\ 4 & 7 & 6 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{111} 3^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 14 & 7 & 4 \\ 11 & 14 & 37 & 17 & 11 \\ 5 & 7 & 17 & 7 & 5 \\ 3 & 4 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 4 & 5 & 4 & 6 & 4 \\ 5 & 7 & 6 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 4 & 5 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{122} 3^{11} 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 9 & 3 \\ 3 & 4 & 17 & 12 & 4 \\ 11 & 17 & 49 & 36 & 13 \\ 9 & 12 & 36 & 25 & 9 \\ 3 & 4 & 13 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{121} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 5 & 8 & 10 & 10 & 6 \\ 6 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[24] \quad 21^1 3^{11} 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 4 & 7 & 4 \\ 3 & 4 & 1 & 5 & 3 \\ 6 & 7 & 5 & 7 & 5 \\ 4 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{121} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 4 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{112} 3^{11} 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

- [27] $\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 5 & 9 & 13 & 20 & 7 \\ 9 & 12 & 20 & 25 & 9 \\ 3 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0}^{1^2}$ $1^{111}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 2 & 4 & 1 & 5 & 3 \\ 6 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$
- [28] $\begin{pmatrix} 1 & 3 & 11 & 6 & 3 \\ 3 & 4 & 17 & 7 & 4 \\ 11 & 17 & 49 & 21 & 13 \\ 6 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}^{1^{22}}$ $1^{131}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.84 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
- [29] $\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 4 & 5 & 4 & 6 & 4 \\ 6 & 7 & 6 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}^{1^{12}}$ $1^{111}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
- [30] $\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 7 & 8 & 10 & 22 & 6 \\ 12 & 16 & 22 & 44 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}^{1^{33}}$ $1^{122}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
- [31] $\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 5 & 8 & 10 & 22 & 6 \\ 12 & 16 & 22 & 44 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}^{1^{13}}$ $1^{122}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
- [32] $\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}^{1^{11}}$ $1^{111}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -22$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.64 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
- [37] $\begin{pmatrix} 1 & 3 & 11 & 16 & 3 \\ 3 & 4 & 16 & 24 & 4 \\ 11 & 16 & 46 & 68 & 12 \\ 16 & 24 & 68 & 100 & 18 \\ 3 & 4 & 12 & 18 & 3 \end{pmatrix}_{b_0}^{1^{33}}$ $1^{144}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
- [34] $\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 14 & 7 & 4 \\ 9 & 14 & 37 & 17 & 11 \\ 4 & 7 & 17 & 7 & 5 \\ 3 & 4 & 11 & 5 & 3 \end{pmatrix}_{b_0}^{1^1}$ $1^{111}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
- [33] $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 6 & 8 & 0 & 12 & 6 \\ 3 & 4 & 0 & 6 & 3 \end{pmatrix}_{b_0}^{1^{11}}$ $1^{111}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
- [35] $\begin{pmatrix} 1 & 3 & 11 & 6 & 3 \\ 3 & 4 & 14 & 7 & 4 \\ 11 & 14 & 37 & 17 & 11 \\ 6 & 7 & 17 & 7 & 5 \\ 3 & 4 & 11 & 5 & 3 \end{pmatrix}_{b_0}^{1^{32}}$ $1^{111}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
- [36] $\begin{pmatrix} 1 & 3 & 11 & 6 & 3 \\ 3 & 4 & 14 & 7 & 4 \\ 11 & 14 & 37 & 17 & 11 \\ 6 & 7 & 17 & 7 & 5 \\ 3 & 4 & 11 & 5 & 3 \end{pmatrix}_{b_0}^{1^{14}}$ $1^{123}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 15 & 3 \\ 3 & 4 & 13 & 20 & 4 \\ 8 & 13 & 28 & 46 & 10 \\ 15 & 20 & 46 & 69 & 15 \\ 3 & 4 & 10 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{12} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 6 & 7 & 0 & 7 & 5 \\ 3 & 4 & 0 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{13} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 5 & 12 & 4 \\ 3 & 5 & 4 & 12 & 4 \\ 10 & 12 & 12 & 25 & 9 \\ 3 & 4 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{13} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 8 & 12 & 25 & 6 & 9 \\ 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{14} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 8 & 3 \\ 3 & 4 & 21 & 12 & 4 \\ 13 & 21 & 76 & 44 & 16 \\ 8 & 12 & 44 & 25 & 9 \\ 3 & 4 & 16 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{131} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 19 & 6 & 3 \\ 3 & 4 & 27 & 7 & 4 \\ 19 & 27 & 133 & 33 & 21 \\ 6 & 7 & 33 & 7 & 5 \\ 3 & 4 & 21 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{131} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 8 & 12 & 25 & 15 & 9 \\ 6 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{111} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 6 & 3 \\ 3 & 4 & 14 & 7 & 4 \\ 12 & 14 & 37 & 17 & 11 \\ 6 & 7 & 17 & 7 & 5 \\ 3 & 4 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 5 & 9 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 6 & 9 & 13 & 11 & 7 \\ 6 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{124} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 17 & 3 \\ 3 & 4 & 8 & 24 & 4 \\ 7 & 8 & 10 & 34 & 6 \\ 17 & 24 & 34 & 100 & 18 \\ 3 & 4 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

- [48] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -31$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.77 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 8 & 9 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$
- [49] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -31$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.77 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 5 & 7 & 0 & 7 & 5 \\ 3 & 4 & 0 & 5 & 3 \end{pmatrix}_{b_0}$
- [50] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -33$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.79 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 14 & 7 & 4 \\ 10 & 14 & 37 & 17 & 11 \\ 5 & 7 & 17 & 7 & 5 \\ 3 & 4 & 11 & 5 & 3 \end{pmatrix}_{b_0}$
- [51] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -33$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.79 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 4 & 4 & 1 & 5 & 3 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$
- [52] $1^{12}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -34$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.76 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 6 & 8 & 10 & 4 & 6 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [54] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -34$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 9 & 14 & 37 & 8 & 11 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 11 & 2 & 3 \end{pmatrix}_{b_0}$
- [53] $1^{13}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -34$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 5 & 20 & 4 \\ 3 & 5 & 4 & 18 & 4 \\ 15 & 20 & 18 & 69 & 15 \\ 3 & 4 & 4 & 15 & 3 \end{pmatrix}_{b_0}$
- [55] $1^{12}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -34$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.76 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 8 & 13 & 28 & 7 & 10 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$
- [58] $1^{14}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -36$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 5 & 24 & 4 \\ 3 & 5 & 4 & 21 & 4 \\ 17 & 24 & 21 & 100 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$
- [57] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -36$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 5 & 9 & 13 & 17 & 7 \\ 7 & 11 & 17 & 20 & 8 \\ 3 & 4 & 7 & 8 & 3 \end{pmatrix}_{b_0}$
- [56] $1^{144}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -36$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 17 & 3 \\ 3 & 4 & 21 & 24 & 4 \\ 13 & 21 & 76 & 89 & 16 \\ 17 & 24 & 89 & 100 & 18 \\ 3 & 4 & 16 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{121} 3^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 5 & 3 \\ 3 & 4 & 23 & 7 & 4 \\ 15 & 23 & 100 & 28 & 18 \\ 5 & 7 & 28 & 7 & 5 \\ 3 & 4 & 18 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{111} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 6 & 9 & 13 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{122} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 8 & 11 & 4 \\ 5 & 8 & 10 & 16 & 6 \\ 9 & 11 & 16 & 20 & 8 \\ 3 & 4 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{123} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 11 & 18 & 58 & 10 & 14 \\ 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{123} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 7 & 8 & 10 & 4 & 6 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{123} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 12 & 11 & 4 \\ 8 & 12 & 25 & 24 & 9 \\ 9 & 11 & 24 & 20 & 8 \\ 3 & 4 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{134} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 16 & 3 \\ 3 & 4 & 12 & 24 & 4 \\ 9 & 12 & 25 & 51 & 9 \\ 16 & 24 & 51 & 100 & 18 \\ 3 & 4 & 9 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{141} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 4 & 16 & 7 & 4 \\ 12 & 16 & 46 & 20 & 12 \\ 4 & 7 & 20 & 7 & 5 \\ 3 & 4 & 12 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{111} 3^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 8 & 9 & 13 & 14 & 7 \\ 6 & 8 & 14 & 12 & 6 \\ 3 & 4 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{111} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 3 & 4 & 1 & 5 & 3 \\ 4 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{131} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 9 & 12 & 25 & 15 & 9 \\ 4 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{12} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 6 & 8 & 10 & 10 & 6 \\ 4 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{122} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 1 & 8 & 3 \\ 9 & 12 & 8 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{133} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 14 & 3 \\ 3 & 4 & 12 & 20 & 4 \\ 9 & 12 & 25 & 42 & 9 \\ 14 & 20 & 42 & 69 & 15 \\ 3 & 4 & 9 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{111} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 2 & 4 & 1 & 5 & 3 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{111} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 1 & 8 & 3 \\ 9 & 12 & 8 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{112} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 2 & 4 & 1 & 8 & 3 \\ 9 & 11 & 8 & 20 & 8 \\ 3 & 4 & 3 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{111} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 4 & 4 & 1 & 5 & 3 \\ 6 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{141} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 3 \\ 3 & 4 & 16 & 7 & 4 \\ 11 & 16 & 46 & 20 & 12 \\ 6 & 7 & 20 & 7 & 5 \\ 3 & 4 & 12 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{122} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 4 & 4 & 1 & 11 & 3 \\ 11 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{14} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 1 & 11 & 3 \\ 13 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

- [80] $1^{12} 1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 8 & 13 & 28 & 28 & 10 \\ 9 & 12 & 28 & 25 & 9 \\ 3 & 4 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$
- [81] $1^{12} 1^{31} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 7 & 8 & 10 & 10 & 6 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$
- [82] $1^{123} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 3 \\ 3 & 4 & 8 & 25 & 4 \\ 5 & 8 & 10 & 34 & 6 \\ 17 & 25 & 34 & 111 & 19 \\ 3 & 4 & 6 & 19 & 3 \end{pmatrix}_{b_0}$$
- [86] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 8 & 9 & 13 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$
- [87] $21^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 9 & 6 & 4 \\ 5 & 4 & 9 & 7 & 4 \\ 9 & 9 & 13 & 11 & 7 \\ 6 & 7 & 11 & 7 & 5 \\ 4 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$
- [85] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 8 & 9 & 13 & 17 & 7 \\ 9 & 11 & 17 & 20 & 8 \\ 3 & 4 & 7 & 8 & 3 \end{pmatrix}_{b_0}$$
- [84] $1^{142} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 9 & 3 \\ 3 & 4 & 16 & 11 & 4 \\ 11 & 16 & 46 & 32 & 12 \\ 9 & 11 & 32 & 20 & 8 \\ 3 & 4 & 12 & 8 & 3 \end{pmatrix}_{b_0}$$
- [83] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 13 & 21 & 76 & 11 & 16 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$
- [88] $1^{14} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 11 & 16 & 46 & 8 & 12 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$
- [92] $1^{111} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 3 & 4 & 1 & 5 & 3 \\ 6 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$
- [91] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 7 & 9 & 13 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

- [90] $1^{132}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -52$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.85 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 15 & 6 & 3 \\ 3 & 4 & 22 & 7 & 4 \\ 15 & 22 & 85 & 27 & 17 \\ 6 & 7 & 27 & 7 & 5 \\ 3 & 4 & 17 & 5 & 3 \end{pmatrix}_{b_0}$
- [89] $1^{13}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -52$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 9 & 20 & 4 \\ 5 & 9 & 13 & 32 & 7 \\ 14 & 20 & 32 & 69 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$
- [96] $1^{132}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -54$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 13 & 7 & 3 \\ 3 & 4 & 22 & 11 & 4 \\ 13 & 22 & 85 & 42 & 17 \\ 7 & 11 & 42 & 20 & 8 \\ 3 & 4 & 17 & 8 & 3 \end{pmatrix}_{b_0}$
- [95] $1^{131}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -54$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.85 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 9 & 12 & 25 & 6 & 9 \\ 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$
- [94] $1^{142}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -54$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.85 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 15 & 9 & 3 \\ 3 & 4 & 21 & 12 & 4 \\ 15 & 21 & 76 & 44 & 16 \\ 9 & 12 & 44 & 25 & 9 \\ 3 & 4 & 16 & 9 & 3 \end{pmatrix}_{b_0}$
- [93] $1^{11}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -54$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 5 & 12 & 4 \\ 3 & 5 & 4 & 12 & 4 \\ 9 & 12 & 12 & 25 & 9 \\ 3 & 4 & 4 & 9 & 3 \end{pmatrix}_{b_0}$
- [98] $1^{122}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -55$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 12 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 8 & 13 & 28 & 37 & 10 \\ 12 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$
- [97] $1^{142}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -55$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 14 & 12 & 3 \\ 3 & 4 & 21 & 16 & 4 \\ 14 & 21 & 76 & 59 & 16 \\ 12 & 16 & 59 & 44 & 12 \\ 3 & 4 & 16 & 12 & 3 \end{pmatrix}_{b_0}$
- [99] $1^{12}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -55$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 5 & 16 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 12 & 16 & 15 & 44 & 12 \\ 3 & 4 & 4 & 12 & 3 \end{pmatrix}_{b_0}$
- [100] $1^{11}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -56$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 6 & 9 & 13 & 14 & 7 \\ 6 & 8 & 14 & 12 & 6 \\ 3 & 4 & 7 & 6 & 3 \end{pmatrix}_{b_0}$
- [101] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -56$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 6 & 9 & 13 & 11 & 7 \\ 4 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0} \\
[102] & 1^{12} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 11 & 0 & 20 & 8 \\ 3 & 4 & 0 & 8 & 3 \end{pmatrix}_{b_0} \\
[109] & 1^{11} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 12 & 0 & 25 & 9 \\ 3 & 4 & 0 & 9 & 3 \end{pmatrix}_{b_0} \\
[107] & 1^{122} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 11 & 8 & 3 \\ 3 & 4 & 13 & 11 & 4 \\ 11 & 13 & 28 & 25 & 10 \\ 8 & 11 & 25 & 20 & 8 \\ 3 & 4 & 10 & 8 & 3 \end{pmatrix}_{b_0} \\
[106] & 1^{131} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 17 & 12 & 4 \\ 10 & 17 & 49 & 36 & 13 \\ 8 & 12 & 36 & 25 & 9 \\ 3 & 4 & 13 & 9 & 3 \end{pmatrix}_{b_0} \\
[105] & 1^{113} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 6 & 9 & 13 & 20 & 7 \\ 10 & 12 & 20 & 25 & 9 \\ 3 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0} \\
[104] & 1^{112} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 6 & 9 & 13 & 17 & 7 \\ 8 & 11 & 17 & 20 & 8 \\ 3 & 4 & 7 & 8 & 3 \end{pmatrix}_{b_0} \\
[103] & 1^{113} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 12 & 0 & 25 & 9 \\ 3 & 4 & 0 & 9 & 3 \end{pmatrix}_{b_0} \\
[108] & 1^{111} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 3 & 4 & 1 & 5 & 3 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0} \\
[110] & 21^{143} 3^{11} 1^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 33 & 6 & 4 \\ 5 & 4 & 32 & 7 & 4 \\ 33 & 32 & 193 & 39 & 25 \\ 6 & 7 & 39 & 7 & 5 \\ 4 & 4 & 25 & 5 & 3 \end{pmatrix}_{b_0} \\
[111] & 1^{111} 3^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 5 & 12 & 4 \\ 4 & 5 & 4 & 12 & 4 \\ 8 & 12 & 12 & 25 & 9 \\ 3 & 4 & 4 & 9 & 3 \end{pmatrix}_{b_0} \\
[112] & 1^{12} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 11 & 0 & 20 & 8 \\ 3 & 4 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[113] $21^{141}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 8 & 3 \\ 4 & 4 & 21 & 12 & 4 \\ 15 & 21 & 76 & 44 & 16 \\ 8 & 12 & 44 & 25 & 9 \\ 3 & 4 & 16 & 9 & 3 \end{pmatrix}_{b_0}$$

$$1^{12}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

[114] $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 13 & 11 & 4 \\ 8 & 13 & 28 & 25 & 10 \\ 8 & 11 & 25 & 20 & 8 \\ 3 & 4 & 10 & 8 & 3 \end{pmatrix}_{b_0}$$

$$1^{23}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

[115] $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 14 & 3 \\ 3 & 4 & 12 & 20 & 4 \\ 8 & 12 & 25 & 42 & 9 \\ 14 & 20 & 42 & 69 & 15 \\ 3 & 4 & 9 & 15 & 3 \end{pmatrix}_{b_0}$$

$$1^{111}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

[116] $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 1 & 8 & 3 \\ 8 & 12 & 8 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

$$1^{23}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

[117] $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 1 & 8 & 3 \\ 10 & 12 & 8 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

$$1^{22}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

[118] $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 4 & 4 & 1 & 8 & 3 \\ 8 & 11 & 8 & 20 & 8 \\ 3 & 4 & 3 & 8 & 3 \end{pmatrix}_{b_0}$$

$$21^{11}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

[119] $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 12 & 4 \\ 5 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 8 & 3 \\ 12 & 12 & 8 & 25 & 9 \\ 4 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

$$1^{22}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

[126] $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 7 & 9 & 13 & 17 & 7 \\ 8 & 11 & 17 & 20 & 8 \\ 3 & 4 & 7 & 8 & 3 \end{pmatrix}_{b_0}$$

$$1^{11}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

[120] $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 18 & 7 & 4 \\ 11 & 18 & 58 & 22 & 14 \\ 5 & 7 & 22 & 7 & 5 \\ 3 & 4 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

$$1^{24}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

[121] $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 12 & 3 \\ 3 & 4 & 12 & 15 & 4 \\ 8 & 12 & 25 & 33 & 9 \\ 12 & 15 & 33 & 39 & 11 \\ 3 & 4 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

$$1^{11}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

[122] $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 5 & 8 & 10 & 10 & 6 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[123] \quad 1^{14} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 10 & 12 & 25 & 6 & 9 \\ 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[124] \quad 1^{13} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 5 & 24 & 4 \\ 3 & 5 & 4 & 21 & 4 \\ 16 & 24 & 21 & 100 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[125] \quad 1^{11} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 5 & 12 & 4 \\ 3 & 5 & 4 & 12 & 4 \\ 8 & 12 & 12 & 25 & 9 \\ 3 & 4 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[133] \quad 1^{11} 3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 13 & 19 & 73 & 11 & 15 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[132] \quad 1^{122} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 6 & 8 & 10 & 10 & 6 \\ 6 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[131] \quad 1^{121} 3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 4 & 3 \\ 3 & 4 & 23 & 7 & 4 \\ 15 & 23 & 100 & 28 & 18 \\ 4 & 7 & 28 & 7 & 5 \\ 3 & 4 & 18 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[130] \quad 1^{121} 3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 7 & 9 & 13 & 14 & 7 \\ 6 & 8 & 14 & 12 & 6 \\ 3 & 4 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[129] \quad 1^{123} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 7 & 9 & 13 & 17 & 7 \\ 9 & 11 & 17 & 20 & 8 \\ 3 & 4 & 7 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[128] \quad 1^{124} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 18 & 3 \\ 3 & 4 & 8 & 25 & 4 \\ 6 & 8 & 10 & 34 & 6 \\ 18 & 25 & 34 & 111 & 19 \\ 3 & 4 & 6 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[127] \quad 1^{132} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 8 & 9 & 13 & 11 & 7 \\ 6 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[134] \quad 1^{112} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 12 & 0 & 25 & 9 \\ 3 & 4 & 0 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[142] \quad 1^{1^{12}} 1^{123} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 16 & 3 \\ 3 & 4 & 8 & 25 & 4 \\ 5 & 8 & 10 & 34 & 6 \\ 16 & 25 & 34 & 111 & 19 \\ 3 & 4 & 6 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[143] \quad 1^{1^{34}} 1^{132} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 13 & 3 \\ 3 & 4 & 17 & 16 & 4 \\ 12 & 17 & 49 & 48 & 13 \\ 13 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[144] \quad 1^{1^{32}} 1^{141} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 6 & 3 \\ 3 & 4 & 21 & 7 & 4 \\ 14 & 21 & 76 & 26 & 16 \\ 6 & 7 & 26 & 7 & 5 \\ 3 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[145] \quad 1^{1^{32}} 1^{121} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 6 & 3 \\ 3 & 4 & 18 & 7 & 4 \\ 13 & 18 & 58 & 22 & 14 \\ 6 & 7 & 22 & 7 & 5 \\ 3 & 4 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[140] \quad 1^{1^{21}} 1^{113} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 10 & 14 & 37 & 8 & 11 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[139] \quad 1^{1^3} 1^{133} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 5 & 20 & 4 \\ 3 & 5 & 4 & 18 & 4 \\ 14 & 20 & 18 & 69 & 15 \\ 3 & 4 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[138] \quad 1^{1^{34}} 1^{143} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 12 & 3 \\ 3 & 4 & 16 & 15 & 4 \\ 11 & 16 & 46 & 44 & 12 \\ 12 & 15 & 44 & 39 & 11 \\ 3 & 4 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[137] \quad 1^{1^{32}} 1^{121} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 7 & 8 & 10 & 10 & 6 \\ 6 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[136] \quad 1^{1^{22}} 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 7 & 9 & 13 & 11 & 7 \\ 6 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[135] \quad 1^{1^{14}} 1^{123} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 8 & 15 & 4 \\ 5 & 8 & 10 & 22 & 6 \\ 12 & 15 & 22 & 39 & 11 \\ 3 & 4 & 6 & 11 & 3 \end{pmatrix}_{b_0}$$

[141] $1^{1^{12}} 1^{132} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 17 & 16 & 4 \\ 10 & 17 & 49 & 48 & 13 \\ 11 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

[146] $1^{1^{11}} 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 6 & 9 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

[148] $1^{1^{21}} 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 11 & 17 & 49 & 9 & 13 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[147] $1^{1^{22}} 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 9 & 13 & 28 & 16 & 10 \\ 6 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

[149] $1^{1^{21}} 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 7 & 9 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

[157] $1^{1^2} 1^{123} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 2 & 4 & 1 & 8 & 3 \\ 8 & 11 & 8 & 20 & 8 \\ 3 & 4 & 3 & 8 & 3 \end{pmatrix}_{b_0}$$

[155] $1^{1^{22}} 1^{133} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 13 & 3 \\ 3 & 4 & 12 & 20 & 4 \\ 8 & 12 & 25 & 42 & 9 \\ 13 & 20 & 42 & 69 & 15 \\ 3 & 4 & 9 & 15 & 3 \end{pmatrix}_{b_0}$$

[156] $1^{1^{41}} 1^{131} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 10 & 12 & 25 & 15 & 9 \\ 5 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[150] $1^{1^4} 1^{112} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 1 & 14 & 3 \\ 16 & 21 & 14 & 80 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[151] $1^{1^{31}} 1^{141} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 8 & 3 \\ 3 & 4 & 21 & 12 & 4 \\ 14 & 21 & 76 & 44 & 16 \\ 8 & 12 & 44 & 25 & 9 \\ 3 & 4 & 16 & 9 & 3 \end{pmatrix}_{b_0}$$

[152] $1^{1^{13}} 1^{111} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 5 & 12 & 4 \\ 4 & 5 & 4 & 12 & 4 \\ 10 & 12 & 12 & 25 & 9 \\ 3 & 4 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[153] $1^{113}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 8 & 3 \\ 10 & 12 & 8 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[154] $1^{21}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 9 & 13 & 28 & 7 & 10 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[160] $1^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 8 & 3 \\ 8 & 12 & 8 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[159] $1^{132}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 8 & 11 & 4 \\ 7 & 8 & 10 & 16 & 6 \\ 8 & 11 & 16 & 20 & 8 \\ 3 & 4 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

[158] $1^{133}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 5 & 20 & 4 \\ 4 & 5 & 4 & 18 & 4 \\ 14 & 20 & 18 & 69 & 15 \\ 3 & 4 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[166] $1^{122}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 5 & 8 & 10 & 22 & 6 \\ 11 & 16 & 22 & 44 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[161] $1^{122}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 13 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 7 & 8 & 10 & 22 & 6 \\ 13 & 16 & 22 & 44 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[162] $1^{113}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 5 & 12 & 4 \\ 4 & 5 & 4 & 12 & 4 \\ 9 & 12 & 12 & 25 & 9 \\ 3 & 4 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[163] $21^{111}3^{11}1^{143} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 18 & 18 & 4 \\ 5 & 4 & 21 & 20 & 4 \\ 18 & 21 & 76 & 74 & 16 \\ 18 & 20 & 74 & 69 & 15 \\ 4 & 4 & 16 & 15 & 3 \end{pmatrix}_{b_0}$$

[164] $1^{21}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 6 & 8 & 10 & 10 & 6 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[165] $1^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 5 & 16 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 11 & 16 & 15 & 44 & 12 \\ 3 & 4 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[167] \quad 1^{1^{22}} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 6 & 8 & 10 & 22 & 6 \\ 11 & 16 & 22 & 44 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[168] \quad 1^{1^{12}} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 5 & 16 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 11 & 16 & 15 & 44 & 12 \\ 3 & 4 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[169] \quad 1^{1^{24}} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 6 & 8 & 10 & 22 & 6 \\ 13 & 16 & 22 & 44 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[170] \quad 21^{1^{11}} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 6 & 12 & 4 \\ 5 & 4 & 5 & 12 & 4 \\ 6 & 5 & 4 & 12 & 4 \\ 12 & 12 & 12 & 25 & 9 \\ 4 & 4 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[171] \quad 1^{1^{14}} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 2 & 4 & 1 & 11 & 3 \\ 12 & 15 & 11 & 39 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[172] \quad 1^{1^{23}} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 4 & 4 & 1 & 8 & 3 \\ 9 & 11 & 8 & 20 & 8 \\ 3 & 4 & 3 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[173] \quad 1^{1^{12}} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 5 & 16 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 10 & 16 & 15 & 44 & 12 \\ 3 & 4 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[174] \quad 1^{1^{23}} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 12 & 3 \\ 3 & 4 & 17 & 16 & 4 \\ 11 & 17 & 49 & 48 & 13 \\ 12 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[178] \quad 1^{1^{21}} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -82, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 16 & 26 & 118 & 14 & 20 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 20 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[176] \quad 1^{1^{41}} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -82, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 12 & 16 & 46 & 8 & 12 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[177] \quad 1^{1^2} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -82, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 5 & 20 & 4 \\ 3 & 5 & 4 & 18 & 4 \\ 13 & 20 & 18 & 69 & 15 \\ 3 & 4 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[175] $1^{131}3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -82, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 11 & 14 & 37 & 8 & 11 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

[179] $1^{132}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 9 & 12 & 25 & 15 & 9 \\ 6 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[180] $1^{122}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 5 & 8 & 10 & 22 & 6 \\ 10 & 16 & 22 & 44 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[181] $21^{113}3^{1121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 12 & 6 & 4 \\ 5 & 4 & 13 & 7 & 4 \\ 12 & 13 & 28 & 16 & 10 \\ 6 & 7 & 16 & 7 & 5 \\ 4 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

[183] $1^{111}3^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 8 & 9 & 13 & 20 & 7 \\ 9 & 12 & 20 & 25 & 9 \\ 3 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0}$$

[182] $1^{134}3^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 15 & 3 \\ 3 & 4 & 9 & 20 & 4 \\ 8 & 9 & 13 & 32 & 7 \\ 15 & 20 & 32 & 69 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[185] $1^{112}3^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 8 & 3 \\ 9 & 12 & 8 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[184] $1^{133}3^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 15 & 3 \\ 3 & 4 & 12 & 20 & 4 \\ 9 & 12 & 25 & 42 & 9 \\ 15 & 20 & 42 & 69 & 15 \\ 3 & 4 & 9 & 15 & 3 \end{pmatrix}_{b_0}$$

[187] $1^{123}3^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 15 & 3 \\ 3 & 4 & 8 & 20 & 4 \\ 7 & 8 & 10 & 28 & 6 \\ 15 & 20 & 28 & 69 & 15 \\ 3 & 4 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

[188] $1^{121}3^{1113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 5 & 3 \\ 3 & 4 & 23 & 7 & 4 \\ 16 & 23 & 100 & 28 & 18 \\ 5 & 7 & 28 & 7 & 5 \\ 3 & 4 & 18 & 5 & 3 \end{pmatrix}_{b_0}$$

[186] $1^{123}3^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 3 \\ 3 & 4 & 8 & 20 & 4 \\ 5 & 8 & 10 & 28 & 6 \\ 15 & 20 & 28 & 69 & 15 \\ 3 & 4 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[189] \quad 1^{1^{43}} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -88, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 10 & 3 \\ 3 & 4 & 17 & 12 & 4 \\ 13 & 17 & 49 & 36 & 13 \\ 10 & 12 & 36 & 25 & 9 \\ 3 & 4 & 13 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[191] \quad 1^{1^{11}} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -88, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 5 & 9 & 13 & 20 & 7 \\ 8 & 12 & 20 & 25 & 9 \\ 3 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[190] \quad 1^{1^{32}} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -88, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 8 & 3 \\ 3 & 4 & 22 & 11 & 4 \\ 15 & 22 & 85 & 42 & 17 \\ 8 & 11 & 42 & 20 & 8 \\ 3 & 4 & 17 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[192] \quad 21^{11} 3^1 1^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -89, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 15 & 6 & 4 \\ 5 & 4 & 17 & 7 & 4 \\ 15 & 17 & 49 & 21 & 13 \\ 6 & 7 & 21 & 7 & 5 \\ 4 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[193] \quad 1^{1^{21}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -90, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 8 & 11 & 4 \\ 6 & 8 & 10 & 16 & 6 \\ 7 & 11 & 16 & 20 & 8 \\ 3 & 4 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[196] \quad 1^{1^{41}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -90, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 7 & 3 \\ 3 & 4 & 16 & 11 & 4 \\ 12 & 16 & 46 & 32 & 12 \\ 7 & 11 & 32 & 20 & 8 \\ 3 & 4 & 12 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[195] \quad 1^{1^{11}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -90, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 3 & 4 & 1 & 8 & 3 \\ 7 & 11 & 8 & 20 & 8 \\ 3 & 4 & 3 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[194] \quad 1^{1^{31}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -90, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 12 & 11 & 4 \\ 9 & 12 & 25 & 24 & 9 \\ 7 & 11 & 24 & 20 & 8 \\ 3 & 4 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[197] \quad 21^{11} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -91, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 9 & 18 & 4 \\ 5 & 4 & 9 & 20 & 4 \\ 9 & 9 & 13 & 32 & 7 \\ 18 & 20 & 32 & 69 & 15 \\ 4 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[198] \quad 1^{1^{14}} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -91, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 6 & 9 & 13 & 26 & 7 \\ 13 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[199] \quad 1^{1^2} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -91, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 5 & 9 & 13 & 26 & 7 \\ 11 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[200] \ 1^{121} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -93, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.80 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 11 & 13 & 28 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[201] \ 1^{121} 3^{11} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -93, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.80 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 5 & 3 \\ 3 & 4 & 23 & 7 & 4 \\ 17 & 23 & 100 & 28 & 18 \\ 5 & 7 & 28 & 7 & 5 \\ 3 & 4 & 18 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[203] \ 1^{131} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -94, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.79 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 10 & 13 & 28 & 7 & 10 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[202] \ 21^{123} 3^{11} 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -94, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.79 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 18 & 6 & 4 \\ 5 & 4 & 18 & 7 & 4 \\ 18 & 18 & 58 & 22 & 14 \\ 6 & 7 & 22 & 7 & 5 \\ 4 & 4 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[204] \ 1^{121} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -94, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.79 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 12 & 18 & 58 & 10 & 14 \\ 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[208] \ 1^{112} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -95, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.80 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 1 & 11 & 3 \\ 12 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[207] \ 1^{123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -95, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.80 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 4 & 4 & 1 & 11 & 3 \\ 12 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[206] \ 1^{123} 3^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -95, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.80 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 5 & 16 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 12 & 16 & 15 & 44 & 12 \\ 3 & 4 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[205] \ 21^{113} 3^{11} 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -95, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.80 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 4 & 4 & 16 & 4 \\ 3 & 4 & 1 & 11 & 3 \\ 15 & 16 & 11 & 44 & 12 \\ 4 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[215] \ 1^{133} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -96, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 11 & 3 \\ 3 & 4 & 17 & 15 & 4 \\ 11 & 17 & 49 & 45 & 13 \\ 11 & 15 & 45 & 39 & 11 \\ 3 & 4 & 13 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[214] \ 1^{133} 3^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -96, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.83 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 5 & 20 & 4 \\ 4 & 5 & 4 & 18 & 4 \\ 15 & 20 & 18 & 69 & 15 \\ 3 & 4 & 4 & 15 & 3 \end{pmatrix}_{b_0} \\
[213] \quad & 21^{1^{33}3^1 1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 9 & 14 & 3 \\ 4 & 4 & 12 & 20 & 4 \\ 9 & 12 & 25 & 42 & 9 \\ 14 & 20 & 42 & 69 & 15 \\ 3 & 4 & 9 & 15 & 3 \end{pmatrix}_{b_0} \\
[212] \quad & 1^{1^{12}3^1 1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 17 & 15 & 4 \\ 10 & 17 & 49 & 45 & 13 \\ 10 & 15 & 45 & 39 & 11 \\ 3 & 4 & 13 & 11 & 3 \end{pmatrix}_{b_0} \\
[211] \quad & 1^{1^{14}2^3 1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 5 & 16 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 13 & 16 & 15 & 44 & 12 \\ 3 & 4 & 4 & 12 & 3 \end{pmatrix}_{b_0} \\
[210] \quad & 21^{1^{11}3^1 1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 8 & 3 \\ 4 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 8 & 3 \\ 8 & 12 & 8 & 25 & 9 \\ 3 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0} \\
[209] \quad & 1^{1^{12}2^1 1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 9 & 13 & 28 & 16 & 10 \\ 4 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0} \\
[223] \quad & 1^{1^{12}2^3 1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 8 & 11 & 4 \\ 5 & 8 & 10 & 16 & 6 \\ 8 & 11 & 16 & 20 & 8 \\ 3 & 4 & 6 & 8 & 3 \end{pmatrix}_{b_0} \\
[216] \quad & 1^{1^{21}3^1 1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 10 & 13 & 28 & 28 & 10 \\ 8 & 12 & 28 & 25 & 9 \\ 3 & 4 & 10 & 9 & 3 \end{pmatrix}_{b_0} \\
[217] \quad & 1^{1^{21}3^1 1^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 15 & 23 & 100 & 13 & 18 \\ 3 & 3 & 13 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0} \\
[220] \quad & 1^{1^{41}2^3 1^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 11 & 13 & 28 & 7 & 10 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0} \\
[221] \quad & 1^{1^{31}3^1 1^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 14 & 19 & 73 & 11 & 15 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0} \\
[222] \quad & 1^{1^{21}3^1 1^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 31 & 3 & 4 \\ 19 & 31 & 172 & 17 & 24 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[219] \ 1^{1^{42}} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -97, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 12 & 11 & 4 \\ 10 & 12 & 25 & 24 & 9 \\ 8 & 11 & 24 & 20 & 8 \\ 3 & 4 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[218] \ 1^{1^{21}} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -97, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 8 & 12 & 25 & 15 & 9 \\ 5 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[224] \ 1^{1^{11}} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -98, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 4 & 22 & 7 & 4 \\ 13 & 22 & 85 & 27 & 17 \\ 5 & 7 & 27 & 7 & 5 \\ 3 & 4 & 17 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[225] \ 21^{1^{12}} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -99, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 8 & 3 \\ 4 & 4 & 9 & 11 & 4 \\ 6 & 9 & 13 & 17 & 7 \\ 8 & 11 & 17 & 20 & 8 \\ 3 & 4 & 7 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[226] \ 1^{1^{24}} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -99, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.81 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 15 & 3 \\ 3 & 4 & 17 & 20 & 4 \\ 11 & 17 & 49 & 60 & 13 \\ 15 & 20 & 60 & 69 & 15 \\ 3 & 4 & 13 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[227] \ 1^{1^{42}} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -99, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 10 & 12 & 25 & 15 & 9 \\ 6 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[228] \ 1^{1^{21}} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -99, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.81 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 14 & 22 & 85 & 12 & 17 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 17 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[230] \ 1^{1^{12}} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -99, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 17 & 0 & 55 & 13 \\ 3 & 4 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[229] \ 1^{1^{14}} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -99, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 15 & 0 & 39 & 11 \\ 3 & 4 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[231] \ 1^{1^{23}} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -100, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.80 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 8 & 16 & 4 \\ 6 & 8 & 10 & 22 & 6 \\ 12 & 16 & 22 & 44 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[232] \ 1^{1^{33}} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -100, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.80 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

- [233] $21^{113}3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 12 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 8 & 9 & 13 & 26 & 7 \\ 12 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$
- [234] $1^{112}3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 18 & 4 \\ 5 & 4 & 4 & 20 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 18 & 20 & 14 & 69 & 15 \\ 4 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$
- [235] $1^{113}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 4 & 4 & 1 & 14 & 3 \\ 14 & 20 & 14 & 69 & 15 \\ 3 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$
- [236] $1^{123}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 4 & 4 & 1 & 11 & 3 \\ 11 & 15 & 11 & 39 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$
- [237] $1^{112}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 3 & 4 & 1 & 8 & 3 \\ 9 & 11 & 8 & 20 & 8 \\ 3 & 4 & 3 & 8 & 3 \end{pmatrix}_{b_0}$$
- [238] $1^{121}3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 8 & 13 & 28 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$
- [240] $1^{113}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 15 & 0 & 39 & 11 \\ 3 & 4 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$
- [239] $1^{113}3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 17 & 0 & 55 & 13 \\ 3 & 4 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$
- [241] $1^{121}3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 4 & 18 & 7 & 4 \\ 12 & 18 & 58 & 22 & 14 \\ 4 & 7 & 22 & 7 & 5 \\ 3 & 4 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$
- [242] $21^{111}3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 3 \\ 4 & 4 & 4 & 7 & 4 \\ 3 & 4 & 1 & 5 & 3 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$
- [243] $1^{131}3^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 21 & 5 & 3 \\ 3 & 4 & 32 & 7 & 4 \\ 21 & 32 & 193 & 39 & 25 \\ 5 & 7 & 39 & 7 & 5 \\ 3 & 4 & 25 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[244] \quad 1^{131} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 9 & 12 & 25 & 15 & 9 \\ 5 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[245] \quad 1^{112} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 3 & 4 & 1 & 8 & 3 \\ 8 & 11 & 8 & 20 & 8 \\ 3 & 4 & 3 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[246] \quad 1^{114} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 16 & 3 \\ 3 & 4 & 9 & 24 & 4 \\ 5 & 9 & 13 & 38 & 7 \\ 16 & 24 & 38 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[247] \quad 1^{122} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 13 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 9 & 13 & 28 & 37 & 10 \\ 13 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[248] \quad 1^{142} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 11 & 3 \\ 3 & 4 & 21 & 16 & 4 \\ 13 & 21 & 76 & 59 & 16 \\ 11 & 16 & 59 & 44 & 12 \\ 3 & 4 & 16 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[249] \quad 1^{121} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 17 & 27 & 133 & 15 & 21 \\ 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[250] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 7 & 9 & 13 & 20 & 7 \\ 8 & 12 & 20 & 25 & 9 \\ 3 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[251] \quad 21^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 5 & 3 \\ 4 & 4 & 9 & 7 & 4 \\ 6 & 9 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[252] \quad 1^{122} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 8 & 11 & 4 \\ 7 & 8 & 10 & 16 & 6 \\ 9 & 11 & 16 & 20 & 8 \\ 3 & 4 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[253] \quad 1^{132} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 9 & 3 \\ 3 & 4 & 17 & 11 & 4 \\ 12 & 17 & 49 & 33 & 13 \\ 9 & 11 & 33 & 20 & 8 \\ 3 & 4 & 13 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[254] \quad 1^{121} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -110, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 9 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 11 & 13 & 28 & 28 & 10 \\ 9 & 12 & 28 & 25 & 9 \\ 3 & 4 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

[257] $1^{1^{11}} 1^{13} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 17 & 7 & 4 \\ 10 & 17 & 49 & 21 & 13 \\ 5 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[256] $1^{1^{33}} 1^{123} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 8 & 15 & 4 \\ 7 & 8 & 10 & 22 & 6 \\ 11 & 15 & 22 & 39 & 11 \\ 3 & 4 & 6 & 11 & 3 \end{pmatrix}_{b_0}$$

[255] $1^{1^{23}} 1^{143} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 14 & 3 \\ 3 & 4 & 21 & 20 & 4 \\ 13 & 21 & 76 & 74 & 16 \\ 14 & 20 & 74 & 69 & 15 \\ 3 & 4 & 16 & 15 & 3 \end{pmatrix}_{b_0}$$

[258] $1^{1^{14}} 1^{12} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 16 & 0 & 44 & 12 \\ 3 & 4 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[259] $1^{1^{12}} 1^{12} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 16 & 0 & 44 & 12 \\ 3 & 4 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

[260] $1^{1^{42}} 1^{141} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 6 & 3 \\ 3 & 4 & 16 & 7 & 4 \\ 12 & 16 & 46 & 20 & 12 \\ 6 & 7 & 20 & 7 & 5 \\ 3 & 4 & 12 & 5 & 3 \end{pmatrix}_{b_0}$$

[261] $1^{1^{23}} 1^{122} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 8 & 11 & 4 \\ 6 & 8 & 10 & 16 & 6 \\ 9 & 11 & 16 & 20 & 8 \\ 3 & 4 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

[262] $1^{1^{22}} 1^{123} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 8 & 20 & 4 \\ 6 & 8 & 10 & 28 & 6 \\ 13 & 20 & 28 & 69 & 15 \\ 3 & 4 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

[263] $1^{1^{24}} 1^{114} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 20 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 4 & 4 & 1 & 20 & 3 \\ 20 & 29 & 20 & 148 & 22 \\ 3 & 4 & 3 & 22 & 3 \end{pmatrix}_{b_0}$$

[264] $1^{1^3} 1^{113} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 2 & 4 & 1 & 11 & 3 \\ 11 & 15 & 11 & 39 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

[265] $1^{1^{31}} 1^{13} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 12 & 17 & 49 & 9 & 13 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[266] \ 1^{14} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -118, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 14 & 21 & 76 & 11 & 16 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[267] \ 21^{122} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 11 & 3 \\ 4 & 4 & 8 & 16 & 4 \\ 6 & 8 & 10 & 22 & 6 \\ 11 & 16 & 22 & 44 & 12 \\ 3 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[268] \ 1^{12} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 16 & 0 & 44 & 12 \\ 3 & 4 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[269] \ 1^{121} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 10 & 13 & 28 & 16 & 10 \\ 6 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[270] \ 1^{112} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 4 & 4 & 1 & 11 & 3 \\ 13 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[271] \ 1^{113} 3^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 4 & 4 & 1 & 17 & 3 \\ 18 & 25 & 17 & 111 & 19 \\ 3 & 4 & 3 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[272] \ 1^{112} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 1 & 11 & 3 \\ 11 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[273] \ 1^{112} 3^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 1 & 14 & 3 \\ 15 & 21 & 14 & 80 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[275] \ 1^{131} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 6 & 3 \\ 3 & 4 & 17 & 7 & 4 \\ 12 & 17 & 49 & 21 & 13 \\ 6 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[276] \ 1^{121} 3^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 6 & 3 \\ 3 & 4 & 18 & 7 & 4 \\ 14 & 18 & 58 & 22 & 14 \\ 6 & 7 & 22 & 7 & 5 \\ 3 & 4 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[274] \ 1^{143} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 15 & 3 \\ 3 & 4 & 21 & 20 & 4 \\ 14 & 21 & 76 & 74 & 16 \\ 15 & 20 & 74 & 69 & 15 \\ 3 & 4 & 16 & 15 & 3 \end{pmatrix}_{b_0}$$

$$1^{33}$$

$$[278] \ 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 8 & 9 & 13 & 20 & 7 \\ 10 & 12 & 20 & 25 & 9 \\ 3 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0}$$

$$1^{22}$$

$$[277] \ 1^{122} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 8 & 3 \\ 3 & 4 & 21 & 11 & 4 \\ 13 & 21 & 76 & 41 & 16 \\ 8 & 11 & 41 & 20 & 8 \\ 3 & 4 & 16 & 8 & 3 \end{pmatrix}_{b_0}$$

$$1^{11}$$

$$[280] \ 1^{122} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 13 & 11 & 4 \\ 8 & 13 & 28 & 25 & 10 \\ 7 & 11 & 25 & 20 & 8 \\ 3 & 4 & 10 & 8 & 3 \end{pmatrix}_{b_0}$$

$$1^{31}$$

$$[281] \ 1^{131} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 8 & 3 \\ 3 & 4 & 17 & 12 & 4 \\ 12 & 17 & 49 & 36 & 13 \\ 8 & 12 & 36 & 25 & 9 \\ 3 & 4 & 13 & 9 & 3 \end{pmatrix}_{b_0}$$

$$1^{22}$$

$$[279] \ 1^{122} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 8 & 11 & 4 \\ 6 & 8 & 10 & 16 & 6 \\ 8 & 11 & 16 & 20 & 8 \\ 3 & 4 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

$$1^3$$

$$[282] \ 1^{112} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 1 & 17 & 3 \\ 18 & 26 & 17 & 128 & 20 \\ 3 & 4 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

$$1^1$$

$$[283] \ 1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 1 & 11 & 3 \\ 10 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$1^{24}$$

$$[284] \ 1^{113} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 4 & 4 & 1 & 11 & 3 \\ 12 & 15 & 11 & 39 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

$$1^{11}$$

$$[285] \ 1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 1 & 11 & 3 \\ 10 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$1^{21}$$

$$[286] \ 1^{121} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 4 & 18 & 7 & 4 \\ 12 & 18 & 58 & 22 & 14 \\ 5 & 7 & 22 & 7 & 5 \\ 3 & 4 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

- [287] $1^{1^{12}} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 6 & 9 & 13 & 20 & 7 \\ 9 & 12 & 20 & 25 & 9 \\ 3 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0}$$
- [288] $1^{1^{32}} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 9 & 3 \\ 3 & 4 & 17 & 12 & 4 \\ 12 & 17 & 49 & 36 & 13 \\ 9 & 12 & 36 & 25 & 9 \\ 3 & 4 & 13 & 9 & 3 \end{pmatrix}_{b_0}$$
- [289] $1^{1^{41}} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 5 & 3 \\ 3 & 4 & 18 & 7 & 4 \\ 14 & 18 & 58 & 22 & 14 \\ 5 & 7 & 22 & 7 & 5 \\ 3 & 4 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$
- [290] $1^{1^{42}} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 11 & 13 & 28 & 16 & 10 \\ 6 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$
- [293] $1^{1^{31}} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -133, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 13 & 18 & 58 & 10 & 14 \\ 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$
- [291] $1^{1^{43}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -133, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 12 & 15 & 4 \\ 10 & 12 & 25 & 33 & 9 \\ 11 & 15 & 33 & 39 & 11 \\ 3 & 4 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$
- [294] $1^{1^{22}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -133, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 12 & 11 & 4 \\ 8 & 12 & 25 & 24 & 9 \\ 8 & 11 & 24 & 20 & 8 \\ 3 & 4 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$
- [292] $1^{1^{42}} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -133, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 8 & 3 \\ 3 & 4 & 22 & 11 & 4 \\ 16 & 22 & 85 & 42 & 17 \\ 8 & 11 & 42 & 20 & 8 \\ 3 & 4 & 17 & 8 & 3 \end{pmatrix}_{b_0}$$
- [295] $1^{1^{33}} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -134, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 9 & 3 \\ 3 & 4 & 13 & 11 & 4 \\ 10 & 13 & 28 & 25 & 10 \\ 9 & 11 & 25 & 20 & 8 \\ 3 & 4 & 10 & 8 & 3 \end{pmatrix}_{b_0}$$
- [296] $1^{1^{33}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -134, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 9 & 3 \\ 3 & 4 & 12 & 11 & 4 \\ 9 & 12 & 25 & 24 & 9 \\ 9 & 11 & 24 & 20 & 8 \\ 3 & 4 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$
- [297] $21^{1^{21}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -134, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 5 & 3 \\ 4 & 4 & 8 & 7 & 4 \\ 6 & 8 & 10 & 10 & 6 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

- [298] $21^{111}3^11^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -135$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 15 & 15 & 4 \\ 5 & 4 & 17 & 16 & 4 \\ 15 & 17 & 49 & 48 & 13 \\ 15 & 16 & 48 & 44 & 12 \\ 4 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$
- [299] $1^{111}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -136$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 7 & 9 & 13 & 20 & 7 \\ 10 & 12 & 20 & 25 & 9 \\ 3 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0}$
- [300] $1^{131}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -136$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 17 & 5 & 3 \\ 3 & 4 & 27 & 7 & 4 \\ 17 & 27 & 133 & 33 & 21 \\ 5 & 7 & 33 & 7 & 5 \\ 3 & 4 & 21 & 5 & 3 \end{pmatrix}_{b_0}$
- [301] $1^{131}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -136$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 8 & 3 \\ 3 & 4 & 17 & 12 & 4 \\ 11 & 17 & 49 & 36 & 13 \\ 8 & 12 & 36 & 25 & 9 \\ 3 & 4 & 13 & 9 & 3 \end{pmatrix}_{b_0}$
- [302] $1^{131}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -136$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 4 & 17 & 7 & 4 \\ 12 & 17 & 49 & 21 & 13 \\ 4 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$
- [303] $21^{111}3^11^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -136$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 12 & 12 & 4 \\ 5 & 4 & 13 & 12 & 4 \\ 12 & 13 & 28 & 28 & 10 \\ 12 & 12 & 28 & 25 & 9 \\ 4 & 4 & 10 & 9 & 3 \end{pmatrix}_{b_0}$
- [304] $1^{131}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -136$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 21 & 4 & 3 \\ 3 & 4 & 32 & 7 & 4 \\ 21 & 32 & 193 & 39 & 25 \\ 4 & 7 & 39 & 7 & 5 \\ 3 & 4 & 25 & 5 & 3 \end{pmatrix}_{b_0}$
- [305] $1^{122}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -137$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 9 & 3 \\ 3 & 4 & 13 & 11 & 4 \\ 11 & 13 & 28 & 25 & 10 \\ 9 & 11 & 25 & 20 & 8 \\ 3 & 4 & 10 & 8 & 3 \end{pmatrix}_{b_0}$
- [306] $1^{121}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -138$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 10 & 13 & 28 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$
- [311] $1^{122}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -141$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 13 & 11 & 4 \\ 10 & 13 & 28 & 25 & 10 \\ 8 & 11 & 25 & 20 & 8 \\ 3 & 4 & 10 & 8 & 3 \end{pmatrix}_{b_0}$
- [307] $1^{121}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -141$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 8 & 13 & 28 & 28 & 10 \\ 8 & 12 & 28 & 25 & 9 \\ 3 & 4 & 10 & 9 & 3 \end{pmatrix}_{b_0}$

[310] $1^{1^{33}} 1^{1^{21}} 3^1 1^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 10 & 13 & 28 & 28 & 10 \\ 10 & 12 & 28 & 25 & 9 \\ 3 & 4 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

[308] $1^{1^{21}} 1^{1^{21}} 3^1 1^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 9 & 13 & 28 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

[309] $1^{1^{32}} 1^{1^{42}} 3^1 1^{1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 17 & 8 & 3 \\ 3 & 4 & 26 & 11 & 4 \\ 17 & 26 & 118 & 50 & 20 \\ 8 & 11 & 50 & 20 & 8 \\ 3 & 4 & 20 & 8 & 3 \end{pmatrix}_{b_0}$$

[313] $1^{1^{11}} 1^{1^{11}} 3^1 1^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 6 & 9 & 13 & 20 & 7 \\ 8 & 12 & 20 & 25 & 9 \\ 3 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0}$$

[312] $1^{1^{22}} 1^{1^{22}} 3^1 1^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 4 & 13 & 11 & 4 \\ 9 & 13 & 28 & 25 & 10 \\ 8 & 11 & 25 & 20 & 8 \\ 3 & 4 & 10 & 8 & 3 \end{pmatrix}_{b_0}$$

[314] $1^{1^{13}} 1^{1^{23}} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -143, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 8 & 15 & 4 \\ 5 & 8 & 10 & 22 & 6 \\ 11 & 15 & 22 & 39 & 11 \\ 3 & 4 & 6 & 11 & 3 \end{pmatrix}_{b_0}$$

[315] $1^{1^{31}} 1^{1^{41}} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -143, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 16 & 7 & 4 \\ 11 & 16 & 46 & 20 & 12 \\ 5 & 7 & 20 & 7 & 5 \\ 3 & 4 & 12 & 5 & 3 \end{pmatrix}_{b_0}$$

[316] $1^{1^{41}} 1^{1^{31}} 3^1 1^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 13 & 17 & 49 & 9 & 13 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[317] $1^{1^{42}} 1^{1^{41}} 3^1 1^{1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 18 & 6 & 3 \\ 3 & 4 & 26 & 7 & 4 \\ 18 & 26 & 118 & 32 & 20 \\ 6 & 7 & 32 & 7 & 5 \\ 3 & 4 & 20 & 5 & 3 \end{pmatrix}_{b_0}$$

[318] $1^1 1^{1^{12}} 3^1 1^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 5 & 9 & 13 & 26 & 7 \\ 10 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

[319] $21^{1^{11}} 3^1 1^{1^{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 18 & 21 & 4 \\ 5 & 4 & 21 & 24 & 4 \\ 18 & 21 & 76 & 89 & 16 \\ 21 & 24 & 89 & 100 & 18 \\ 4 & 4 & 16 & 18 & 3 \end{pmatrix}_{b_0}$$

- [320] $1^{1^{43}} 1^{1^{32}} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -147, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 9 & 3 \\ 3 & 4 & 12 & 11 & 4 \\ 10 & 12 & 25 & 24 & 9 \\ 9 & 11 & 24 & 20 & 8 \\ 3 & 4 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$
- [321] $1^{1^{22}} 1^{1^{34}} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -147, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 15 & 3 \\ 3 & 4 & 12 & 24 & 4 \\ 8 & 12 & 25 & 51 & 9 \\ 15 & 24 & 51 & 100 & 18 \\ 3 & 4 & 9 & 18 & 3 \end{pmatrix}_{b_0}$$
- [322] $1^{1^2} 1^{1^{13}} 3^1 1^{1^{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -147, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 9 & 20 & 4 \\ 5 & 9 & 13 & 32 & 7 \\ 13 & 20 & 32 & 69 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$
- [323] $1^{1^{41}} 1^{1^{14}} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 4 & 16 & 7 & 4 \\ 12 & 16 & 46 & 20 & 12 \\ 5 & 7 & 20 & 7 & 5 \\ 3 & 4 & 12 & 5 & 3 \end{pmatrix}_{b_0}$$
- [324] $21^{1^{12}} 3^1 1^{1^{12}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -149, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 21 & 4 \\ 5 & 4 & 4 & 21 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 21 & 21 & 14 & 80 & 16 \\ 4 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$
- [325] $1^{1^2} 1^{1^{12}} 3^1 1^{1^2} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -149, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 1 & 14 & 3 \\ 14 & 21 & 14 & 80 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$
- [328] $1^{1^{22}} 1^{1^{11}} 3^1 1^{1^{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 7 & 9 & 13 & 20 & 7 \\ 9 & 12 & 20 & 25 & 9 \\ 3 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0}$$
- [326] $1^{1^{42}} 1^{1^{31}} 3^1 1^{1^{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 9 & 3 \\ 3 & 4 & 17 & 12 & 4 \\ 13 & 17 & 49 & 36 & 13 \\ 9 & 12 & 36 & 25 & 9 \\ 3 & 4 & 13 & 9 & 3 \end{pmatrix}_{b_0}$$
- [327] $1^{1^{31}} 1^{1^{21}} 3^1 1^{1^2} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 4 & 18 & 7 & 4 \\ 13 & 18 & 58 & 22 & 14 \\ 5 & 7 & 22 & 7 & 5 \\ 3 & 4 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$
- [329] $1^{1^{24}} 1^{1^{14}} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -153, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 4 & 4 & 1 & 14 & 3 \\ 14 & 19 & 14 & 64 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$
- [330] $1^{1^{31}} 1^{1^{23}} 3^1 1^{1^3} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 16 & 23 & 100 & 13 & 18 \\ 3 & 3 & 13 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[331] $1^{1^{22}} 1^{1^{21}} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 9 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 9 & 13 & 28 & 28 & 10 \\ 9 & 12 & 28 & 25 & 9 \\ 3 & 4 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

[332] $1^{1^{12}} 1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 20 & 0 & 69 & 15 \\ 3 & 4 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[333] $1^{1^{41}} 1^{13} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 14 & 18 & 58 & 10 & 14 \\ 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

[334] $21^{1^{21}} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 5 & 3 \\ 4 & 4 & 18 & 7 & 4 \\ 12 & 18 & 58 & 22 & 14 \\ 5 & 7 & 22 & 7 & 5 \\ 3 & 4 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[335] $1^{1^{32}} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 4 & 12 & 11 & 4 \\ 9 & 12 & 25 & 24 & 9 \\ 8 & 11 & 24 & 20 & 8 \\ 3 & 4 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[336] $1^{1^{14}} 1^{14} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -157, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 19 & 0 & 64 & 14 \\ 3 & 4 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

[337] $1^{1^{12}} 1^{12} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -157, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 21 & 0 & 80 & 16 \\ 3 & 4 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[338] $1^{1^{31}} 1^{12} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -157, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 28 & 3 & 4 \\ 19 & 28 & 154 & 16 & 22 \\ 3 & 3 & 16 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[339] $1^{1^{13}} 1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -158, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 8 & 20 & 4 \\ 5 & 8 & 10 & 28 & 6 \\ 14 & 20 & 28 & 69 & 15 \\ 3 & 4 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

[340] $1^{1^{42}} 1^{14} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 10 & 3 \\ 3 & 4 & 16 & 15 & 4 \\ 12 & 16 & 46 & 44 & 12 \\ 10 & 15 & 44 & 39 & 11 \\ 3 & 4 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

[341] $1^{1^{32}} 1^{13} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 10 & 3 \\ 3 & 4 & 12 & 15 & 4 \\ 9 & 12 & 25 & 33 & 9 \\ 10 & 15 & 33 & 39 & 11 \\ 3 & 4 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

[342] $1^{1^{12}} 1^{113} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 3 & 4 & 1 & 11 & 3 \\ 10 & 15 & 11 & 39 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$

[343] $1^{1^{22}} 1^{123} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 8 & 15 & 4 \\ 6 & 8 & 10 & 22 & 6 \\ 10 & 15 & 22 & 39 & 11 \\ 3 & 4 & 6 & 11 & 3 \end{pmatrix}_{b_0}$

[344] $1^{1^{24}} 1^{113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 4 & 4 & 1 & 14 & 3 \\ 15 & 20 & 14 & 69 & 15 \\ 3 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$

[345] $1^{1^4} 1^{113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 2 & 4 & 1 & 14 & 3 \\ 15 & 20 & 14 & 69 & 15 \\ 3 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$

[346] $1^{1^{34}} 1^{124} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 14 & 3 \\ 3 & 4 & 8 & 19 & 4 \\ 7 & 8 & 10 & 28 & 6 \\ 14 & 19 & 28 & 64 & 14 \\ 3 & 4 & 6 & 14 & 3 \end{pmatrix}_{b_0}$

[347] $1^{1^{43}} 1^{142} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -162, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 12 & 9 & 3 \\ 3 & 4 & 16 & 11 & 4 \\ 12 & 16 & 46 & 32 & 12 \\ 9 & 11 & 32 & 20 & 8 \\ 3 & 4 & 12 & 8 & 3 \end{pmatrix}_{b_0}$

[348] $1^{1^{31}} 1^{13} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -162, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 15 & 22 & 85 & 12 & 17 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 17 & 2 & 3 \end{pmatrix}_{b_0}$

[349] $1^{1^{43}} 1^{121} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 10 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 11 & 13 & 28 & 28 & 10 \\ 10 & 12 & 28 & 25 & 9 \\ 3 & 4 & 10 & 9 & 3 \end{pmatrix}_{b_0}$

[350] $1^{1^{22}} 1^{132} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 8 & 3 \\ 3 & 4 & 17 & 11 & 4 \\ 11 & 17 & 49 & 33 & 13 \\ 8 & 11 & 33 & 20 & 8 \\ 3 & 4 & 13 & 8 & 3 \end{pmatrix}_{b_0}$

[351] $1^{1^{14}} 1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 1 & 11 & 3 \\ 13 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$

[352] $1^{1^{21}} 1^{121} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 9 & 13 & 28 & 28 & 10 \\ 8 & 12 & 28 & 25 & 9 \\ 3 & 4 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[353] \quad 1^{1^{12}3^11^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 1 & 11 & 3 \\ 11 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[354] \quad 1^{1^{42}3^11^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 6 & 3 \\ 3 & 4 & 22 & 7 & 4 \\ 16 & 22 & 85 & 27 & 17 \\ 6 & 7 & 27 & 7 & 5 \\ 3 & 4 & 17 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[355] \quad 1^{1^{23}3^11^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 8 & 15 & 4 \\ 7 & 8 & 10 & 22 & 6 \\ 12 & 15 & 22 & 39 & 11 \\ 3 & 4 & 6 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[356] \quad 1^{1^{34}3^11^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 17 & 3 \\ 3 & 4 & 12 & 24 & 4 \\ 8 & 12 & 25 & 51 & 9 \\ 17 & 24 & 51 & 100 & 18 \\ 3 & 4 & 9 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[357] \quad 1^{1^{43}3^11^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -169, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 17 & 26 & 118 & 14 & 20 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 20 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[358] \quad 21^{1^{33}3^11^{131}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -169, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 27 & 6 & 4 \\ 5 & 4 & 27 & 7 & 4 \\ 27 & 27 & 133 & 33 & 21 \\ 6 & 7 & 33 & 7 & 5 \\ 4 & 4 & 21 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[359] \quad 1^{1^{41}3^11^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -169, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 15 & 21 & 76 & 11 & 16 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[360] \quad 1^{1^{22}3^11^{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 7 & 9 & 13 & 26 & 7 \\ 11 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[361] \quad 1^{1^{34}3^11^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 17 & 3 \\ 3 & 4 & 12 & 24 & 4 \\ 9 & 12 & 25 & 51 & 9 \\ 17 & 24 & 51 & 100 & 18 \\ 3 & 4 & 9 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[362] \quad 21^{1^{11}3^11^{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 9 & 12 & 4 \\ 5 & 4 & 9 & 12 & 4 \\ 9 & 9 & 13 & 20 & 7 \\ 12 & 12 & 20 & 25 & 9 \\ 4 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[363] \quad 1^{1^{24}3^11^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 17 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 4 & 4 & 1 & 17 & 3 \\ 17 & 24 & 17 & 100 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[364] \quad 1^{1^{34}} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -171, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.84 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 13 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 8 & 9 & 13 & 26 & 7 \\ 13 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[365] \quad 1^{1^{12}} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -172, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 14 & 21 & 14 & 80 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[366] \quad 1^{1^{14}} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -172, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 3 & 4 & 1 & 11 & 3 \\ 12 & 15 & 11 & 39 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[367] \quad 1^{1^{14}} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -173, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 2 & 4 & 1 & 14 & 3 \\ 14 & 19 & 14 & 64 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[368] \quad 1^{1^{24}} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -174, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.84 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 20 & 3 \\ 3 & 4 & 8 & 29 & 4 \\ 6 & 8 & 10 & 40 & 6 \\ 20 & 29 & 40 & 148 & 22 \\ 3 & 4 & 6 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[369] \quad 1^{1^{43}} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -175, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.80 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 12 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 11 & 13 & 28 & 37 & 10 \\ 12 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[371] \quad 21^{1^{11}} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -175, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.80 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 9 & 15 & 4 \\ 5 & 4 & 9 & 16 & 4 \\ 9 & 9 & 13 & 26 & 7 \\ 15 & 16 & 26 & 44 & 12 \\ 4 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[370] \quad 1^{1^{13}} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -175, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.80 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 1 & 11 & 3 \\ 12 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[372] \quad 1^{1^4} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -176, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 4 & 3 \\ 3 & 4 & 21 & 7 & 4 \\ 15 & 21 & 76 & 26 & 16 \\ 4 & 7 & 26 & 7 & 5 \\ 3 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[373] \quad 1^{1^{32}} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -176, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.82 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 9 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 10 & 13 & 28 & 28 & 10 \\ 9 & 12 & 28 & 25 & 9 \\ 3 & 4 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[377] \quad 1^{1^{13}} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -177, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 15 & 21 & 0 & 80 & 16 \\ 3 & 4 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[376] $1^{142}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 9 & 3 \\ 3 & 4 & 21 & 11 & 4 \\ 15 & 21 & 76 & 41 & 16 \\ 9 & 11 & 41 & 20 & 8 \\ 3 & 4 & 16 & 8 & 3 \end{pmatrix}_{b_0}$$

[375] $1^{131}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 22 & 5 & 3 \\ 3 & 4 & 32 & 7 & 4 \\ 22 & 32 & 193 & 39 & 25 \\ 5 & 7 & 39 & 7 & 5 \\ 3 & 4 & 25 & 5 & 3 \end{pmatrix}_{b_0}$$

[374] $1^{124}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 20 & 3 \\ 3 & 4 & 8 & 29 & 4 \\ 5 & 8 & 10 & 40 & 6 \\ 20 & 29 & 40 & 148 & 22 \\ 3 & 4 & 6 & 22 & 3 \end{pmatrix}_{b_0}$$

[378] $1^{13}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -178, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 20 & 0 & 69 & 15 \\ 3 & 4 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[379] $1^{113}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -178, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 3 & 4 & 1 & 11 & 3 \\ 11 & 15 & 11 & 39 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

[380] $21^{12}3^11^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 21 & 6 & 4 \\ 5 & 4 & 22 & 7 & 4 \\ 21 & 22 & 85 & 27 & 17 \\ 6 & 7 & 27 & 7 & 5 \\ 4 & 4 & 17 & 5 & 3 \end{pmatrix}_{b_0}$$

[381] $1^{112}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 6 & 9 & 13 & 26 & 7 \\ 12 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

[382] $1^{142}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 8 & 3 \\ 3 & 4 & 16 & 11 & 4 \\ 11 & 16 & 46 & 32 & 12 \\ 8 & 11 & 32 & 20 & 8 \\ 3 & 4 & 12 & 8 & 3 \end{pmatrix}_{b_0}$$

[383] $21^{111}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 8 & 3 \\ 4 & 4 & 9 & 12 & 4 \\ 6 & 9 & 13 & 20 & 7 \\ 8 & 12 & 20 & 25 & 9 \\ 3 & 4 & 7 & 9 & 3 \end{pmatrix}_{b_0}$$

[384] $21^{131}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 8 & 3 \\ 4 & 4 & 17 & 12 & 4 \\ 12 & 17 & 49 & 36 & 13 \\ 8 & 12 & 36 & 25 & 9 \\ 3 & 4 & 13 & 9 & 3 \end{pmatrix}_{b_0}$$

[385] $1^{12}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 16 & 21 & 0 & 80 & 16 \\ 3 & 4 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[386] \quad 1^{1^{44}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 14 & 3 \\ 3 & 4 & 12 & 19 & 4 \\ 10 & 12 & 25 & 42 & 9 \\ 14 & 19 & 42 & 64 & 14 \\ 3 & 4 & 9 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[387] \quad 1^{1^{23}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 3 \\ 3 & 4 & 12 & 15 & 4 \\ 8 & 12 & 25 & 33 & 9 \\ 11 & 15 & 33 & 39 & 11 \\ 3 & 4 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[388] \quad 1^{1^{12}} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -182, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 8 & 20 & 4 \\ 5 & 8 & 10 & 28 & 6 \\ 13 & 20 & 28 & 69 & 15 \\ 3 & 4 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[389] \quad 1^{1^{24}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 8 & 15 & 4 \\ 6 & 8 & 10 & 22 & 6 \\ 12 & 15 & 22 & 39 & 11 \\ 3 & 4 & 6 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[390] \quad 21^{1^{12}} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 14 & 3 \\ 4 & 4 & 4 & 21 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 14 & 21 & 14 & 80 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[391] \quad 21^{1^{12}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 8 & 3 \\ 4 & 4 & 4 & 11 & 4 \\ 3 & 4 & 1 & 8 & 3 \\ 8 & 11 & 8 & 20 & 8 \\ 3 & 4 & 3 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[392] \quad 1^{1^3} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 4 & 3 \\ 3 & 4 & 22 & 7 & 4 \\ 15 & 22 & 85 & 27 & 17 \\ 4 & 7 & 27 & 7 & 5 \\ 3 & 4 & 17 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[393] \quad 1^{1^{11}} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 6 & 9 & 13 & 26 & 7 \\ 10 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[394] \quad 1^{1^3} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 18 & 4 & 3 \\ 3 & 4 & 27 & 7 & 4 \\ 18 & 27 & 133 & 33 & 21 \\ 4 & 7 & 33 & 7 & 5 \\ 3 & 4 & 21 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[395] \quad 1^{1^{42}} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 6 & 3 \\ 3 & 4 & 21 & 7 & 4 \\ 15 & 21 & 76 & 26 & 16 \\ 6 & 7 & 26 & 7 & 5 \\ 3 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[396] \quad 21^{1^{13}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 5 & 3 \\ 4 & 4 & 12 & 7 & 4 \\ 9 & 12 & 25 & 15 & 9 \\ 5 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[397] \ 1^{14} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -186, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.81 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 15 & 20 & 0 & 69 & 15 \\ 3 & 4 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[398] \ 1^{23} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -186, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.83 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 8 & 20 & 4 \\ 6 & 8 & 10 & 28 & 6 \\ 14 & 20 & 28 & 69 & 15 \\ 3 & 4 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[399] \ 1^{42} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -187, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.86 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 11 & 3 \\ 3 & 4 & 17 & 16 & 4 \\ 13 & 17 & 49 & 48 & 13 \\ 11 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[400] \ 1^{14} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -187, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 3 \\ 3 & 4 & 9 & 24 & 4 \\ 5 & 9 & 13 & 38 & 7 \\ 15 & 24 & 38 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[401] \ 1^{42} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -187, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.86 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 6 & 3 \\ 3 & 4 & 17 & 7 & 4 \\ 13 & 17 & 49 & 21 & 13 \\ 6 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[402] \ 1^{21} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -188, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.85 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 17 & 7 & 4 \\ 11 & 17 & 49 & 21 & 13 \\ 5 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[403] \ 1^{31} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -189, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.81 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 18 & 27 & 133 & 15 & 21 \\ 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[404] \ 1^{41} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -193, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.86 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 17 & 23 & 100 & 13 & 18 \\ 3 & 3 & 13 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[405] \ 1^{23} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -196, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 8 & 15 & 4 \\ 6 & 8 & 10 & 22 & 6 \\ 11 & 15 & 22 & 39 & 11 \\ 3 & 4 & 6 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[406] \ 1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -197, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.84 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 2 & 4 & 1 & 14 & 3 \\ 14 & 20 & 14 & 69 & 15 \\ 3 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[408] \ 1^{42} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -86, \ \det = -198, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 8 & 3 \\ 3 & 4 & 16 & 11 & 4 \\ 12 & 16 & 46 & 32 & 12 \\ 8 & 11 & 32 & 20 & 8 \\ 3 & 4 & 12 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[407] \quad 1^{131} 3^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 21 & 3 & 3 \\ 3 & 4 & 32 & 3 & 4 \\ 21 & 32 & 193 & 18 & 25 \\ 3 & 3 & 18 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[410] \quad 21^{111} 3^{1134} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -199, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 15 & 21 & 4 \\ 5 & 4 & 17 & 24 & 4 \\ 15 & 17 & 49 & 72 & 13 \\ 21 & 24 & 72 & 100 & 18 \\ 4 & 4 & 13 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[409] \quad 1^{123} 3^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -199, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 16 & 3 \\ 3 & 4 & 12 & 24 & 4 \\ 8 & 12 & 25 & 51 & 9 \\ 16 & 24 & 51 & 100 & 18 \\ 3 & 4 & 9 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[413] \quad 1^{121} 3^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -201, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 5 & 3 \\ 3 & 4 & 22 & 7 & 4 \\ 14 & 22 & 85 & 27 & 17 \\ 5 & 7 & 27 & 7 & 5 \\ 3 & 4 & 17 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[412] \quad 1^{124} 3^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -201, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 18 & 3 \\ 3 & 4 & 8 & 29 & 4 \\ 5 & 8 & 10 & 40 & 6 \\ 18 & 29 & 40 & 148 & 22 \\ 3 & 4 & 6 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[411] \quad 1^{124} 3^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -201, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 8 & 19 & 4 \\ 5 & 8 & 10 & 28 & 6 \\ 14 & 19 & 28 & 64 & 14 \\ 3 & 4 & 6 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[415] \quad 1^{131} 3^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -202, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 20 & 3 & 3 \\ 3 & 4 & 31 & 3 & 4 \\ 20 & 31 & 172 & 17 & 24 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[414] \quad 1^{112} 3^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -202, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 15 & 21 & 14 & 80 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[416] \quad 1^{133} 3^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 12 & 3 \\ 3 & 4 & 12 & 15 & 4 \\ 9 & 12 & 25 & 33 & 9 \\ 12 & 15 & 33 & 39 & 11 \\ 3 & 4 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[420] \quad 1^{141} 3^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 16 & 22 & 85 & 12 & 17 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 17 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[417] \quad 1^{122} 3^{1111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 10 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 9 & 13 & 28 & 37 & 10 \\ 10 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

$$1^{31}$$

$$[421] \ 1^{142} 3^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 10 & 3 \\ 3 & 4 & 21 & 16 & 4 \\ 14 & 21 & 76 & 59 & 16 \\ 10 & 16 & 59 & 44 & 12 \\ 3 & 4 & 16 & 12 & 3 \end{pmatrix}_{b_0}$$

$$1^{14}$$

$$[419] \ 1^{112} 3^{11} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 1 & 17 & 3 \\ 19 & 26 & 17 & 128 & 20 \\ 3 & 4 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

$$1^{44}$$

$$[418] \ 1^{133} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 12 & 3 \\ 3 & 4 & 12 & 15 & 4 \\ 10 & 12 & 25 & 33 & 9 \\ 12 & 15 & 33 & 39 & 11 \\ 3 & 4 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

$$1^{32}$$

$$[424] \ 1^{122} 3^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 10 & 13 & 28 & 37 & 10 \\ 11 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

$$1^{14}$$

$$[423] \ 1^{112} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 16 & 21 & 14 & 80 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

$$1^{12}$$

$$[422] \ 1^{122} 3^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 8 & 13 & 28 & 37 & 10 \\ 11 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

$$1^2$$

$$[426] \ 1^{113} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -209, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 2 & 4 & 1 & 14 & 3 \\ 13 & 20 & 14 & 69 & 15 \\ 3 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$1^{14}$$

$$[425] \ 1^{113} 3^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -209, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 15 & 3 \\ 3 & 4 & 9 & 20 & 4 \\ 6 & 9 & 13 & 32 & 7 \\ 15 & 20 & 32 & 69 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

$$1^{24}$$

$$[427] \ 1^{123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -210, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 15 & 3 \\ 3 & 4 & 8 & 20 & 4 \\ 6 & 8 & 10 & 28 & 6 \\ 15 & 20 & 28 & 69 & 15 \\ 3 & 4 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

$$1^{44}$$

$$[428] \ 1^{123} 3^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -210, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 15 & 3 \\ 3 & 4 & 13 & 20 & 4 \\ 11 & 13 & 28 & 46 & 10 \\ 15 & 20 & 46 & 69 & 15 \\ 3 & 4 & 10 & 15 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[429] \quad & 1^{1^{24}} 1^{12} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 7 & 13 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 7 & 9 & 13 & 26 & 7 \\ 13 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0} \\
[430] \quad & 21^{1^{11}} 3^1 1^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 15 & 18 & 4 \\ 5 & 4 & 17 & 20 & 4 \\ 15 & 17 & 49 & 60 & 13 \\ 18 & 20 & 60 & 69 & 15 \\ 4 & 4 & 13 & 15 & 3 \end{pmatrix}_{b_0} \\
[431] \quad & 1^{1^{12}} 1^{12} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 6 & 9 & 13 & 26 & 7 \\ 11 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0} \\
[432] \quad & 1^{1^{43}} 1^{32} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -212, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 13 & 9 & 3 \\ 3 & 4 & 17 & 11 & 4 \\ 13 & 17 & 49 & 33 & 13 \\ 9 & 11 & 33 & 20 & 8 \\ 3 & 4 & 13 & 8 & 3 \end{pmatrix}_{b_0} \\
[434] \quad & 1^{1^{21}} 1^{41} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -213, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 19 & 5 & 3 \\ 3 & 4 & 31 & 7 & 4 \\ 19 & 31 & 172 & 38 & 24 \\ 5 & 7 & 38 & 7 & 5 \\ 3 & 4 & 24 & 5 & 3 \end{pmatrix}_{b_0} \\
[435] \quad & 1^{1^{21}} 1^{41} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -213, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 4 & 21 & 7 & 4 \\ 13 & 21 & 76 & 26 & 16 \\ 5 & 7 & 26 & 7 & 5 \\ 3 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0} \\
[433] \quad & 1^{1^{13}} 1^{24} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -213, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 19 & 3 \\ 3 & 4 & 8 & 29 & 4 \\ 5 & 8 & 10 & 40 & 6 \\ 19 & 29 & 40 & 148 & 22 \\ 3 & 4 & 6 & 22 & 3 \end{pmatrix}_{b_0} \\
[436] \quad & 1^{1^{41}} 1^{12} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 20 & 3 & 3 \\ 3 & 4 & 28 & 3 & 4 \\ 20 & 28 & 154 & 16 & 22 \\ 3 & 3 & 16 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0} \\
[437] \quad & 21^{1^{22}} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 6 & 8 & 3 \\ 4 & 4 & 8 & 11 & 4 \\ 6 & 8 & 10 & 16 & 6 \\ 8 & 11 & 16 & 20 & 8 \\ 3 & 4 & 6 & 8 & 3 \end{pmatrix}_{b_0} \\
[438] \quad & 1^{1^{14}} 1^{24} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 8 & 17 & 3 \\ 3 & 4 & 13 & 24 & 4 \\ 8 & 13 & 28 & 55 & 10 \\ 17 & 24 & 55 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0} \\
[439] \quad & 1^{1^{41}} 1^{31} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 4 & 17 & 7 & 4 \\ 13 & 17 & 49 & 21 & 13 \\ 5 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [440] $21^{111}3^11^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -216$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 12 & 18 & 4 \\ 5 & 4 & 13 & 20 & 4 \\ 12 & 13 & 28 & 46 & 10 \\ 18 & 20 & 46 & 69 & 15 \\ 4 & 4 & 10 & 15 & 3 \end{pmatrix}_{b_0}$
- [441] $1^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -216$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 7 & 3 \\ 3 & 4 & 17 & 11 & 4 \\ 11 & 17 & 49 & 33 & 13 \\ 7 & 11 & 33 & 20 & 8 \\ 3 & 4 & 13 & 8 & 3 \end{pmatrix}_{b_0}$
- [442] $1^{123}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -216$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 14 & 3 \\ 3 & 4 & 9 & 20 & 4 \\ 7 & 9 & 13 & 32 & 7 \\ 14 & 20 & 32 & 69 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$
- [443] $1^{131}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -217$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 23 & 3 & 3 \\ 3 & 4 & 36 & 3 & 4 \\ 23 & 36 & 238 & 20 & 28 \\ 3 & 3 & 20 & 0 & 2 \\ 3 & 4 & 28 & 2 & 3 \end{pmatrix}_{b_0}$
- [446] $1^{113}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -219$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 2 & 4 & 1 & 17 & 3 \\ 18 & 25 & 17 & 111 & 19 \\ 3 & 4 & 3 & 19 & 3 \end{pmatrix}_{b_0}$
- [444] $21^{123}3^11^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -219$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 3 & 24 & 4 \\ 5 & 4 & 4 & 25 & 4 \\ 3 & 4 & 1 & 17 & 3 \\ 24 & 25 & 17 & 111 & 19 \\ 4 & 4 & 3 & 19 & 3 \end{pmatrix}_{b_0}$
- [445] $21^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -219$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 9 & 5 & 3 \\ 4 & 4 & 13 & 7 & 4 \\ 9 & 13 & 28 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$
- [447] $21^{111}3^11^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -220$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 12 & 15 & 4 \\ 5 & 4 & 13 & 16 & 4 \\ 12 & 13 & 28 & 37 & 10 \\ 15 & 16 & 37 & 44 & 12 \\ 4 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$
- [448] $1^{123}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -220$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 7 & 9 & 13 & 26 & 7 \\ 12 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0}$
- [449] $21^{121}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -221$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.82 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 9 & 8 & 3 \\ 4 & 4 & 13 & 12 & 4 \\ 9 & 13 & 28 & 28 & 10 \\ 8 & 12 & 28 & 25 & 9 \\ 3 & 4 & 10 & 9 & 3 \end{pmatrix}_{b_0}$
- [451] $1^{131}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -223$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 4 & 17 & 7 & 4 \\ 12 & 17 & 49 & 21 & 13 \\ 5 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$
- [450] $1^{131}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -223$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 18 & 5 & 3 \\ 3 & 4 & 27 & 7 & 4 \\ 18 & 27 & 133 & 33 & 21 \\ 5 & 7 & 33 & 7 & 5 \\ 3 & 4 & 21 & 5 & 3 \end{pmatrix}_{b_0}$$

[453] $1^{143}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 11 & 3 \\ 3 & 4 & 21 & 15 & 4 \\ 14 & 21 & 76 & 56 & 16 \\ 11 & 15 & 56 & 39 & 11 \\ 3 & 4 & 16 & 11 & 3 \end{pmatrix}_{b_0}$$

[452] $1^{133}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 11 & 3 \\ 3 & 4 & 12 & 15 & 4 \\ 9 & 12 & 25 & 33 & 9 \\ 11 & 15 & 33 & 39 & 11 \\ 3 & 4 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

[454] $1^{13}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -228, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 18 & 26 & 0 & 128 & 20 \\ 3 & 4 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[456] $1^{132}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -228, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 17 & 16 & 4 \\ 10 & 17 & 49 & 48 & 13 \\ 10 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

[455] $1^{143}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -228, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 11 & 3 \\ 3 & 4 & 17 & 15 & 4 \\ 13 & 17 & 49 & 45 & 13 \\ 11 & 15 & 45 & 39 & 11 \\ 3 & 4 & 13 & 11 & 3 \end{pmatrix}_{b_0}$$

[458] $21^{131}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -229, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 18 & 5 & 3 \\ 4 & 4 & 27 & 7 & 4 \\ 18 & 27 & 133 & 33 & 21 \\ 5 & 7 & 33 & 7 & 5 \\ 3 & 4 & 21 & 5 & 3 \end{pmatrix}_{b_0}$$

[457] $21^{122}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -229, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 8 & 3 \\ 4 & 4 & 13 & 11 & 4 \\ 9 & 13 & 28 & 25 & 10 \\ 8 & 11 & 25 & 20 & 8 \\ 3 & 4 & 10 & 8 & 3 \end{pmatrix}_{b_0}$$

[460] $1^{143}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -231, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 11 & 3 \\ 3 & 4 & 16 & 15 & 4 \\ 11 & 16 & 46 & 44 & 12 \\ 11 & 15 & 44 & 39 & 11 \\ 3 & 4 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

[459] $1^{142}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -231, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 8 & 3 \\ 3 & 4 & 17 & 11 & 4 \\ 13 & 17 & 49 & 33 & 13 \\ 8 & 11 & 33 & 20 & 8 \\ 3 & 4 & 13 & 8 & 3 \end{pmatrix}_{b_0}$$

[461] $1^{133}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -231, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 11 & 3 \\ 3 & 4 & 17 & 15 & 4 \\ 12 & 17 & 49 & 45 & 13 \\ 11 & 15 & 45 & 39 & 11 \\ 3 & 4 & 13 & 11 & 3 \end{pmatrix}_{b_0}$$

[464] $1^{144}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 12 & 3 \\ 3 & 4 & 16 & 15 & 4 \\ 12 & 16 & 46 & 44 & 12 \\ 12 & 15 & 44 & 39 & 11 \\ 3 & 4 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[463] \quad 1^{134} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -232, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 16 & 3 \\ 3 & 4 & 17 & 24 & 4 \\ 10 & 17 & 49 & 72 & 13 \\ 16 & 24 & 72 & 100 & 18 \\ 3 & 4 & 13 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[462] \quad 1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -232, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 16 & 24 & 0 & 100 & 18 \\ 3 & 4 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[465] \quad 1^{123} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -233, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.83 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 14 & 3 \\ 3 & 4 & 13 & 20 & 4 \\ 8 & 13 & 28 & 46 & 10 \\ 14 & 20 & 46 & 69 & 15 \\ 3 & 4 & 10 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[467] \quad 1^{121} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -234, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 7 & 3 \\ 3 & 4 & 26 & 11 & 4 \\ 16 & 26 & 118 & 50 & 20 \\ 7 & 11 & 50 & 20 & 8 \\ 3 & 4 & 20 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[466] \quad 21^{123} 3^1 1^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -234, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.83 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 24 & 6 & 4 \\ 5 & 4 & 26 & 7 & 4 \\ 24 & 26 & 118 & 32 & 20 \\ 6 & 7 & 32 & 7 & 5 \\ 4 & 4 & 20 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[468] \quad 1^{113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -237, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 13 & 20 & 14 & 69 & 15 \\ 3 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[469] \quad 1^{141} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -238, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.78 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 18 & 26 & 118 & 14 & 20 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 20 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[470] \quad 1^{132} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -238, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 8 & 3 \\ 3 & 4 & 17 & 11 & 4 \\ 12 & 17 & 49 & 33 & 13 \\ 8 & 11 & 33 & 20 & 8 \\ 3 & 4 & 13 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[471] \quad 21^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -240, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.80 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 11 & 3 \\ 4 & 4 & 4 & 16 & 4 \\ 3 & 4 & 1 & 11 & 3 \\ 11 & 16 & 11 & 44 & 12 \\ 3 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[472] \quad 1^{134} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -241, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 14 & 3 \\ 3 & 4 & 12 & 19 & 4 \\ 8 & 12 & 25 & 42 & 9 \\ 14 & 19 & 42 & 64 & 14 \\ 3 & 4 & 9 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[473] \quad 1^{114} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -243, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 17 & 3 \\ 3 & 4 & 9 & 24 & 4 \\ 7 & 9 & 13 & 38 & 7 \\ 17 & 24 & 38 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[474] $1^{14}3^11^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 19 & 26 & 0 & 128 & 20 \\ 3 & 4 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[475] $1^{23}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 16 & 3 \\ 3 & 4 & 8 & 24 & 4 \\ 6 & 8 & 10 & 34 & 6 \\ 16 & 24 & 34 & 100 & 18 \\ 3 & 4 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

[476] $21^{124}3^11^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -246, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 20 & 3 \\ 4 & 4 & 8 & 29 & 4 \\ 6 & 8 & 10 & 40 & 6 \\ 20 & 29 & 40 & 148 & 22 \\ 3 & 4 & 6 & 22 & 3 \end{pmatrix}_{b_0}$$

[477] $1^{141}3^11^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -247, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 23 & 5 & 3 \\ 3 & 4 & 36 & 7 & 4 \\ 23 & 36 & 238 & 44 & 28 \\ 5 & 7 & 44 & 7 & 5 \\ 3 & 4 & 28 & 5 & 3 \end{pmatrix}_{b_0}$$

[479] $1^{13}3^11^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -247, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 17 & 25 & 0 & 111 & 19 \\ 3 & 4 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

[478] $1^{142}3^11^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -247, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 10 & 3 \\ 3 & 4 & 21 & 16 & 4 \\ 13 & 21 & 76 & 59 & 16 \\ 10 & 16 & 59 & 44 & 12 \\ 3 & 4 & 16 & 12 & 3 \end{pmatrix}_{b_0}$$

[480] $1^{122}3^11^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -247, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 8 & 13 & 28 & 37 & 10 \\ 10 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

[481] $1^{113}3^11^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 9 & 20 & 4 \\ 6 & 9 & 13 & 32 & 7 \\ 13 & 20 & 32 & 69 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[484] $1^{124}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -250, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 8 & 19 & 4 \\ 6 & 8 & 10 & 28 & 6 \\ 13 & 19 & 28 & 64 & 14 \\ 3 & 4 & 6 & 14 & 3 \end{pmatrix}_{b_0}$$

[482] $1^{134}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -250, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 13 & 3 \\ 3 & 4 & 12 & 19 & 4 \\ 9 & 12 & 25 & 42 & 9 \\ 13 & 19 & 42 & 64 & 14 \\ 3 & 4 & 9 & 14 & 3 \end{pmatrix}_{b_0}$$

[483] $1^{114}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -250, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 13 & 19 & 14 & 64 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[485] $1^{144}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -250, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 13 & 3 \\ 3 & 4 & 16 & 19 & 4 \\ 12 & 16 & 46 & 56 & 12 \\ 13 & 19 & 56 & 64 & 14 \\ 3 & 4 & 12 & 14 & 3 \end{pmatrix}_{b_0}$$

[488] $1^{124}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -252, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 3 \\ 3 & 4 & 8 & 24 & 4 \\ 5 & 8 & 10 & 34 & 6 \\ 17 & 24 & 34 & 100 & 18 \\ 3 & 4 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

[487] $1^{141}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -252, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 19 & 27 & 133 & 15 & 21 \\ 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[486] $1^{141}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -252, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 5 & 3 \\ 3 & 4 & 26 & 7 & 4 \\ 16 & 26 & 118 & 32 & 20 \\ 5 & 7 & 32 & 7 & 5 \\ 3 & 4 & 20 & 5 & 3 \end{pmatrix}_{b_0}$$

[490] $1^{113}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -253, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 2 & 4 & 1 & 17 & 3 \\ 17 & 25 & 17 & 111 & 19 \\ 3 & 4 & 3 & 19 & 3 \end{pmatrix}_{b_0}$$

[489] $1^{113}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -253, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 2 & 4 & 1 & 20 & 3 \\ 21 & 30 & 20 & 165 & 23 \\ 3 & 4 & 3 & 23 & 3 \end{pmatrix}_{b_0}$$

[491] $21^{141}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -254, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 5 & 3 \\ 4 & 4 & 16 & 7 & 4 \\ 12 & 16 & 46 & 20 & 12 \\ 5 & 7 & 20 & 7 & 5 \\ 3 & 4 & 12 & 5 & 3 \end{pmatrix}_{b_0}$$

[493] $1^{123}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -255, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 12 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 9 & 13 & 28 & 37 & 10 \\ 12 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

[492] $1^{142}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -255, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 12 & 3 \\ 3 & 4 & 21 & 16 & 4 \\ 15 & 21 & 76 & 59 & 16 \\ 12 & 16 & 59 & 44 & 12 \\ 3 & 4 & 16 & 12 & 3 \end{pmatrix}_{b_0}$$

[495] $1^{14}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 17 & 24 & 0 & 100 & 18 \\ 3 & 4 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[494] $1^{143}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 10 & 3 \\ 3 & 4 & 21 & 15 & 4 \\ 13 & 21 & 76 & 56 & 16 \\ 10 & 15 & 56 & 39 & 11 \\ 3 & 4 & 16 & 11 & 3 \end{pmatrix}_{b_0}$$

[497] $1^{1^{13}}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -258, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 9 & 20 & 4 \\ 6 & 9 & 13 & 32 & 7 \\ 14 & 20 & 32 & 69 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[496] $1^{1^{42}}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -258, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 13 & 3 \\ 3 & 4 & 21 & 20 & 4 \\ 15 & 21 & 76 & 74 & 16 \\ 13 & 20 & 74 & 69 & 15 \\ 3 & 4 & 16 & 15 & 3 \end{pmatrix}_{b_0}$$

[498] $21^{1^{23}}3^11^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 27 & 4 \\ 5 & 4 & 4 & 29 & 4 \\ 3 & 4 & 1 & 20 & 3 \\ 27 & 29 & 20 & 148 & 22 \\ 4 & 4 & 3 & 22 & 3 \end{pmatrix}_{b_0}$$

[500] $1^{1^{31}}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -262, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 5 & 3 \\ 3 & 4 & 22 & 7 & 4 \\ 15 & 22 & 85 & 27 & 17 \\ 5 & 7 & 27 & 7 & 5 \\ 3 & 4 & 17 & 5 & 3 \end{pmatrix}_{b_0}$$

[499] $1^{1^{42}}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -262, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 18 & 8 & 3 \\ 3 & 4 & 26 & 11 & 4 \\ 18 & 26 & 118 & 50 & 20 \\ 8 & 11 & 50 & 20 & 8 \\ 3 & 4 & 20 & 8 & 3 \end{pmatrix}_{b_0}$$

[501] $1^{1^{13}}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -263, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 2 & 4 & 1 & 17 & 3 \\ 16 & 25 & 17 & 111 & 19 \\ 3 & 4 & 3 & 19 & 3 \end{pmatrix}_{b_0}$$

[502] $1^{1^{4}}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -263, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 24 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 2 & 4 & 1 & 23 & 3 \\ 24 & 35 & 23 & 231 & 27 \\ 3 & 4 & 3 & 27 & 3 \end{pmatrix}_{b_0}$$

[504] $1^{1^{41}}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 18 & 4 & 3 \\ 3 & 4 & 26 & 7 & 4 \\ 18 & 26 & 118 & 32 & 20 \\ 4 & 7 & 32 & 7 & 5 \\ 3 & 4 & 20 & 5 & 3 \end{pmatrix}_{b_0}$$

[503] $21^{1^{32}}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 8 & 3 \\ 4 & 4 & 12 & 11 & 4 \\ 9 & 12 & 25 & 24 & 9 \\ 8 & 11 & 24 & 20 & 8 \\ 3 & 4 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[507] $1^{1^{22}}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -267, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 11 & 3 \\ 3 & 4 & 17 & 16 & 4 \\ 11 & 17 & 49 & 48 & 13 \\ 11 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

[506] $1^{1^{44}}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -267, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 13 & 3 \\ 3 & 4 & 17 & 16 & 4 \\ 13 & 17 & 49 & 48 & 13 \\ 13 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[505] \quad 1^{14} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -267, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 18 & 25 & 0 & 111 & 19 \\ 3 & 4 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[516] \quad 1^{42} 3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 11 & 3 \\ 3 & 4 & 21 & 16 & 4 \\ 15 & 21 & 76 & 59 & 16 \\ 11 & 16 & 59 & 44 & 12 \\ 3 & 4 & 16 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[508] \quad 1^{34} 3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 13 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 10 & 13 & 28 & 37 & 10 \\ 13 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[509] \quad 1^{41} 3^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 19 & 5 & 3 \\ 3 & 4 & 27 & 7 & 4 \\ 19 & 27 & 133 & 33 & 21 \\ 5 & 7 & 33 & 7 & 5 \\ 3 & 4 & 21 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[510] \quad 1^{43} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 11 & 3 \\ 3 & 4 & 16 & 15 & 4 \\ 12 & 16 & 46 & 44 & 12 \\ 11 & 15 & 44 & 39 & 11 \\ 3 & 4 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[511] \quad 1^{44} 3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 13 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 11 & 13 & 28 & 37 & 10 \\ 13 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[512] \quad 1^{21} 3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 10 & 3 \\ 3 & 4 & 17 & 16 & 4 \\ 11 & 17 & 49 & 48 & 13 \\ 10 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[513] \quad 1^{14} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 14 & 19 & 14 & 64 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[514] \quad 1^{32} 3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 11 & 3 \\ 3 & 4 & 21 & 16 & 4 \\ 14 & 21 & 76 & 59 & 16 \\ 11 & 16 & 59 & 44 & 12 \\ 3 & 4 & 16 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[515] \quad 1^{12} 3^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 3 & 4 & 1 & 17 & 3 \\ 16 & 25 & 17 & 111 & 19 \\ 3 & 4 & 3 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[517] \quad 1^{12} 3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -269, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 13 & 3 \\ 3 & 4 & 17 & 20 & 4 \\ 10 & 17 & 49 & 60 & 13 \\ 13 & 20 & 60 & 69 & 15 \\ 3 & 4 & 13 & 15 & 3 \end{pmatrix}_{b_0}$$

 1^{24}

$$[518] \ 1^{113} 3^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 15 & 3 \\ 3 & 4 & 9 & 20 & 4 \\ 7 & 9 & 13 & 32 & 7 \\ 15 & 20 & 32 & 69 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

 1^{13}

$$[519] \ 1^{113} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -273, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 14 & 20 & 14 & 69 & 15 \\ 3 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[520] \ 21^{14} 3^{11} 1^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -274, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 36 & 6 & 4 \\ 5 & 4 & 36 & 7 & 4 \\ 36 & 36 & 238 & 44 & 28 \\ 6 & 7 & 44 & 7 & 5 \\ 4 & 4 & 28 & 5 & 3 \end{pmatrix}_{b_0}$$

 1^{32}

$$[521] \ 1^{142} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -277, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 8 & 3 \\ 3 & 4 & 21 & 11 & 4 \\ 14 & 21 & 76 & 41 & 16 \\ 8 & 11 & 41 & 20 & 8 \\ 3 & 4 & 16 & 8 & 3 \end{pmatrix}_{b_0}$$

 1^{41}

$$[522] \ 1^{13} 3^{11} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -279, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 22 & 3 & 3 \\ 3 & 4 & 32 & 3 & 4 \\ 22 & 32 & 193 & 18 & 25 \\ 3 & 3 & 18 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

 1^{33}

$$[523] \ 1^{132} 3^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 12 & 3 \\ 3 & 4 & 17 & 16 & 4 \\ 12 & 17 & 49 & 48 & 13 \\ 12 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

 1^{41}

$$[524] \ 1^{131} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -281, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 5 & 3 \\ 3 & 4 & 22 & 7 & 4 \\ 16 & 22 & 85 & 27 & 17 \\ 5 & 7 & 27 & 7 & 5 \\ 3 & 4 & 17 & 5 & 3 \end{pmatrix}_{b_0}$$

 1^4

$$[525] \ 1^{114} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 2 & 4 & 1 & 17 & 3 \\ 17 & 24 & 17 & 100 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

 1^{14}

$$[526] \ 1^{114} 3^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 17 & 3 \\ 3 & 4 & 9 & 24 & 4 \\ 6 & 9 & 13 & 38 & 7 \\ 17 & 24 & 38 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

 1^{12}

$$[527] \ 1^{124} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -284, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 3 \\ 3 & 4 & 8 & 24 & 4 \\ 5 & 8 & 10 & 34 & 6 \\ 15 & 24 & 34 & 100 & 18 \\ 3 & 4 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[530] \ 21^{13} 3^{11} 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -285, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 30 & 4 \\ 5 & 4 & 4 & 30 & 4 \\ 3 & 4 & 1 & 20 & 3 \\ 30 & 30 & 20 & 165 & 23 \\ 4 & 4 & 3 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[529] \ 1^{1^3} 3^1 1^{1^3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -285, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 2 & 4 & 1 & 20 & 3 \\ 20 & 30 & 20 & 165 & 23 \\ 3 & 4 & 3 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[528] \ 1^{1^{14}} 3^1 1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -285, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 15 & 20 & 14 & 69 & 15 \\ 3 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[532] \ 1^{1^{31}} 3^1 1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -286, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 5 & 3 \\ 3 & 4 & 21 & 7 & 4 \\ 14 & 21 & 76 & 26 & 16 \\ 5 & 7 & 26 & 7 & 5 \\ 3 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[531] \ 1^{1^{24}} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -286, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 8 & 19 & 4 \\ 6 & 8 & 10 & 28 & 6 \\ 14 & 19 & 28 & 64 & 14 \\ 3 & 4 & 6 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[533] \ 1^{1^{22}} 3^1 1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -288, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 11 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 9 & 13 & 28 & 37 & 10 \\ 11 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[534] \ 1^{1^{41}} 3^1 1^{1^3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -289, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 21 & 3 & 3 \\ 3 & 4 & 31 & 3 & 4 \\ 21 & 31 & 172 & 17 & 24 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[536] \ 1^{1^{13}} 3^1 1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 16 & 3 \\ 3 & 4 & 8 & 24 & 4 \\ 5 & 8 & 10 & 34 & 6 \\ 16 & 24 & 34 & 100 & 18 \\ 3 & 4 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[535] \ 1^{1^{13}} 3^1 1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 16 & 3 \\ 3 & 4 & 9 & 24 & 4 \\ 6 & 9 & 13 & 38 & 7 \\ 16 & 24 & 38 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[539] \ 1^{1^{13}} 3^1 1^{1^3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -293, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 2 & 4 & 1 & 20 & 3 \\ 19 & 30 & 20 & 165 & 23 \\ 3 & 4 & 3 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[538] \ 1^{1^{34}} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -293, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 14 & 3 \\ 3 & 4 & 16 & 19 & 4 \\ 11 & 16 & 46 & 56 & 12 \\ 14 & 19 & 56 & 64 & 14 \\ 3 & 4 & 12 & 14 & 3 \end{pmatrix}_{b_0}$$

- [537] $1^{1^3}1^{1^3}1^{1^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -293$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 23 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 2 & 4 & 1 & 23 & 3 \\ 23 & 35 & 23 & 231 & 27 \\ 3 & 4 & 3 & 27 & 3 \end{pmatrix}_{b_0}$
- [540] $21^{1^2}3^11^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -295$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 6 & 11 & 3 \\ 4 & 4 & 9 & 16 & 4 \\ 6 & 9 & 13 & 26 & 7 \\ 11 & 16 & 26 & 44 & 12 \\ 3 & 4 & 7 & 12 & 3 \end{pmatrix}_{b_0}$
- [541] $1^{1^2}3^11^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -295$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 12 & 3 \\ 3 & 4 & 13 & 16 & 4 \\ 10 & 13 & 28 & 37 & 10 \\ 12 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$
- [542] $21^{1^2}3^11^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -296$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.81 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 6 & 14 & 3 \\ 4 & 4 & 8 & 20 & 4 \\ 6 & 8 & 10 & 28 & 6 \\ 14 & 20 & 28 & 69 & 15 \\ 3 & 4 & 6 & 15 & 3 \end{pmatrix}_{b_0}$
- [543] $21^{1^2}3^11^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -296$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 3 & 11 & 3 \\ 4 & 4 & 4 & 15 & 4 \\ 3 & 4 & 1 & 11 & 3 \\ 11 & 15 & 11 & 39 & 11 \\ 3 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0}$
- [544] $1^{1^4}1^{1^4}1^{1^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -296$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 24 & 4 & 3 \\ 3 & 4 & 36 & 7 & 4 \\ 24 & 36 & 238 & 44 & 28 \\ 4 & 7 & 44 & 7 & 5 \\ 3 & 4 & 28 & 5 & 3 \end{pmatrix}_{b_0}$
- [545] $1^{1^2}3^11^{1^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -301$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 22 & 31 & 0 & 188 & 24 \\ 3 & 4 & 0 & 24 & 3 \end{pmatrix}_{b_0}$
- [546] $1^{1^2}3^11^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -301$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 14 & 3 \\ 3 & 4 & 13 & 20 & 4 \\ 10 & 13 & 28 & 46 & 10 \\ 14 & 20 & 46 & 69 & 15 \\ 3 & 4 & 10 & 15 & 3 \end{pmatrix}_{b_0}$
- [547] $1^{1^4}3^11^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -303$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 2 & 4 & 1 & 17 & 3 \\ 16 & 24 & 17 & 100 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0}$
- [548] $21^{1^2}3^11^{1^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -304$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.80 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 30 & 6 & 4 \\ 5 & 4 & 31 & 7 & 4 \\ 30 & 31 & 172 & 38 & 24 \\ 6 & 7 & 38 & 7 & 5 \\ 4 & 4 & 24 & 5 & 3 \end{pmatrix}_{b_0}$
- [549] $21^{1^4}3^11^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -304$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.84 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 9 & 17 & 3 \\ 4 & 4 & 12 & 24 & 4 \\ 9 & 12 & 25 & 51 & 9 \\ 17 & 24 & 51 & 100 & 18 \\ 3 & 4 & 9 & 18 & 3 \end{pmatrix}_{b_0}$
- [550] $1^{1^4}1^{1^4}1^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -304$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 21 & 4 & 3 \\ 3 & 4 & 31 & 7 & 4 \\ 21 & 31 & 172 & 38 & 24 \\ 4 & 7 & 38 & 7 & 5 \\ 3 & 4 & 24 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[551] \quad 1^{142} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -306, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\ T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 7 & 3 \\ 3 & 4 & 21 & 11 & 4 \\ 14 & 21 & 76 & 41 & 16 \\ 7 & 11 & 41 & 20 & 8 \\ 3 & 4 & 16 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[553] \quad 1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -307, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\ T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 3 & 4 & 1 & 17 & 3 \\ 15 & 24 & 17 & 100 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[552] \quad 1^{132} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -307, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\ T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 11 & 3 \\ 3 & 4 & 17 & 16 & 4 \\ 12 & 17 & 49 & 48 & 13 \\ 11 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[554] \quad 1^{13} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -310, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\ T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 20 & 30 & 0 & 165 & 23 \\ 3 & 4 & 0 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[555] \quad 1^{14} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -313, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\ T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 19 & 29 & 0 & 148 & 22 \\ 3 & 4 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[556] \quad 1^{124} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -316, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\ T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 17 & 3 \\ 3 & 4 & 8 & 24 & 4 \\ 6 & 8 & 10 & 34 & 6 \\ 17 & 24 & 34 & 100 & 18 \\ 3 & 4 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[557] \quad 1^{134} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -316, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\ T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 14 & 3 \\ 3 & 4 & 12 & 19 & 4 \\ 9 & 12 & 25 & 42 & 9 \\ 14 & 19 & 42 & 64 & 14 \\ 3 & 4 & 9 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[558] \quad 1^{141} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -317, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\ T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 5 & 3 \\ 3 & 4 & 21 & 7 & 4 \\ 15 & 21 & 76 & 26 & 16 \\ 5 & 7 & 26 & 7 & 5 \\ 3 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[559] \quad 1^{132} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -320, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\ T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 12 & 3 \\ 3 & 4 & 17 & 16 & 4 \\ 13 & 17 & 49 & 48 & 13 \\ 12 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[560] \quad 21^{113} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -320, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\ T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 20 & 3 \\ 4 & 4 & 4 & 30 & 4 \\ 3 & 4 & 1 & 20 & 3 \\ 20 & 30 & 20 & 165 & 23 \\ 3 & 4 & 3 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[561] \quad 1^{14} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -322, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\ T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 24 & 3 & 3 \\ 3 & 4 & 36 & 3 & 4 \\ 24 & 36 & 238 & 20 & 28 \\ 3 & 3 & 20 & 0 & 2 \\ 3 & 4 & 28 & 2 & 3 \end{pmatrix}_{b_0}$$

[562] $1^{113}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -322, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 3 & 4 & 1 & 17 & 3 \\ 17 & 25 & 17 & 111 & 19 \\ 3 & 4 & 3 & 19 & 3 \end{pmatrix}_{b_0}$

[563] $1^{113}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 3 & 4 & 1 & 20 & 3 \\ 20 & 30 & 20 & 165 & 23 \\ 3 & 4 & 3 & 23 & 3 \end{pmatrix}_{b_0}$

[564] $21^{123}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -326, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 6 & 11 & 3 \\ 4 & 4 & 8 & 15 & 4 \\ 6 & 8 & 10 & 22 & 6 \\ 11 & 15 & 22 & 39 & 11 \\ 3 & 4 & 6 & 11 & 3 \end{pmatrix}_{b_0}$

[565] $1^{133}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -328, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 13 & 14 & 3 \\ 3 & 4 & 17 & 20 & 4 \\ 13 & 17 & 49 & 60 & 13 \\ 14 & 20 & 60 & 69 & 15 \\ 3 & 4 & 13 & 15 & 3 \end{pmatrix}_{b_0}$

[566] $1^{123}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -328, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 13 & 3 \\ 3 & 4 & 13 & 20 & 4 \\ 8 & 13 & 28 & 46 & 10 \\ 13 & 20 & 46 & 69 & 15 \\ 3 & 4 & 10 & 15 & 3 \end{pmatrix}_{b_0}$

[570] $1^{123}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -334, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 13 & 3 \\ 3 & 4 & 13 & 20 & 4 \\ 9 & 13 & 28 & 46 & 10 \\ 13 & 20 & 46 & 69 & 15 \\ 3 & 4 & 10 & 15 & 3 \end{pmatrix}_{b_0}$

[569] $1^{123}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -334, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 15 & 3 \\ 3 & 4 & 13 & 20 & 4 \\ 9 & 13 & 28 & 46 & 10 \\ 15 & 20 & 46 & 69 & 15 \\ 3 & 4 & 10 & 15 & 3 \end{pmatrix}_{b_0}$

[568] $21^{142}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -334, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 12 & 8 & 3 \\ 4 & 4 & 16 & 11 & 4 \\ 12 & 16 & 46 & 32 & 12 \\ 8 & 11 & 32 & 20 & 8 \\ 3 & 4 & 12 & 8 & 3 \end{pmatrix}_{b_0}$

[567] $21^{141}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -334, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 24 & 5 & 3 \\ 4 & 4 & 36 & 7 & 4 \\ 24 & 36 & 238 & 44 & 28 \\ 5 & 7 & 44 & 7 & 5 \\ 3 & 4 & 28 & 5 & 3 \end{pmatrix}_{b_0}$

[571] $1^{133}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -337, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 12 & 13 & 3 \\ 3 & 4 & 17 & 20 & 4 \\ 12 & 17 & 49 & 60 & 13 \\ 13 & 20 & 60 & 69 & 15 \\ 3 & 4 & 13 & 15 & 3 \end{pmatrix}_{b_0}$

[572] $1^{141}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -338, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 20 & 5 & 3 \\ 3 & 4 & 31 & 7 & 4 \\ 20 & 31 & 172 & 38 & 24 \\ 5 & 7 & 38 & 7 & 5 \\ 3 & 4 & 24 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[573] \quad 1^{14} 3^{11} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -342, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 21 & 30 & 0 & 165 & 23 \\ 3 & 4 & 0 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[574] \quad 21^{151} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -344, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.1$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 15 & 5 & 3 \\ 4 & 4 & 20 & 7 & 4 \\ 15 & 20 & 73 & 25 & 15 \\ 5 & 7 & 25 & 7 & 5 \\ 3 & 4 & 15 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[575] \quad 1^{142} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -346, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 8 & 3 \\ 3 & 4 & 21 & 11 & 4 \\ 15 & 21 & 76 & 41 & 16 \\ 8 & 11 & 41 & 20 & 8 \\ 3 & 4 & 16 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[576] \quad 21^{131} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -349, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 5 & 3 \\ 4 & 4 & 17 & 7 & 4 \\ 12 & 17 & 49 & 21 & 13 \\ 5 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[577] \quad 1^{14} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -349, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 20 & 29 & 0 & 148 & 22 \\ 3 & 4 & 0 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[578] \quad 1^{141} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -351, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 5 & 3 \\ 3 & 4 & 26 & 7 & 4 \\ 17 & 26 & 118 & 32 & 20 \\ 5 & 7 & 32 & 7 & 5 \\ 3 & 4 & 20 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[579] \quad 1^{143} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -352, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 3 & 4 & 1 & 17 & 3 \\ 18 & 25 & 17 & 111 & 19 \\ 3 & 4 & 3 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[580] \quad 1^{143} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -353, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 11 & 3 \\ 3 & 4 & 21 & 15 & 4 \\ 15 & 21 & 76 & 56 & 16 \\ 11 & 15 & 56 & 39 & 11 \\ 3 & 4 & 16 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[582] \quad 1^{141} 3^{11} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -356, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 24 & 5 & 3 \\ 3 & 4 & 36 & 7 & 4 \\ 24 & 36 & 238 & 44 & 28 \\ 5 & 7 & 44 & 7 & 5 \\ 3 & 4 & 28 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[581] \quad 1^{133} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -356, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 14 & 3 \\ 3 & 4 & 17 & 20 & 4 \\ 11 & 17 & 49 & 60 & 13 \\ 14 & 20 & 60 & 69 & 15 \\ 3 & 4 & 13 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[583] \quad 1^{144} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -358, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 14 & 3 \\ 3 & 4 & 16 & 19 & 4 \\ 12 & 16 & 46 & 56 & 12 \\ 14 & 19 & 56 & 64 & 14 \\ 3 & 4 & 12 & 14 & 3 \end{pmatrix}_{b_0}$$

[584] $1^{114}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -360, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 3 & 4 & 1 & 20 & 3 \\ 18 & 29 & 20 & 148 & 22 \\ 3 & 4 & 3 & 22 & 3 \end{pmatrix}_{b_0}$$

[585] $21^{131}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 5 & 3 \\ 4 & 4 & 22 & 7 & 4 \\ 15 & 22 & 85 & 27 & 17 \\ 5 & 7 & 27 & 7 & 5 \\ 3 & 4 & 17 & 5 & 3 \end{pmatrix}_{b_0}$$

[587] $1^{114}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -367, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 3 & 4 & 1 & 17 & 3 \\ 16 & 24 & 17 & 100 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

[586] $1^{124}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -367, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 16 & 3 \\ 3 & 4 & 13 & 24 & 4 \\ 8 & 13 & 28 & 55 & 10 \\ 16 & 24 & 55 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[588] $1^{114}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -369, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 2 & 4 & 1 & 20 & 3 \\ 20 & 29 & 20 & 148 & 22 \\ 3 & 4 & 3 & 22 & 3 \end{pmatrix}_{b_0}$$

[589] $1^{113}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -373, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 3 & 4 & 1 & 20 & 3 \\ 21 & 30 & 20 & 165 & 23 \\ 3 & 4 & 3 & 23 & 3 \end{pmatrix}_{b_0}$$

[590] $1^{124}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 17 & 3 \\ 3 & 4 & 13 & 24 & 4 \\ 10 & 13 & 28 & 55 & 10 \\ 17 & 24 & 55 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[591] $21^{133}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 11 & 3 \\ 4 & 4 & 12 & 15 & 4 \\ 9 & 12 & 25 & 33 & 9 \\ 11 & 15 & 33 & 39 & 11 \\ 3 & 4 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

[592] $21^{132}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -379, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 8 & 3 \\ 4 & 4 & 17 & 11 & 4 \\ 12 & 17 & 49 & 33 & 13 \\ 8 & 11 & 33 & 20 & 8 \\ 3 & 4 & 13 & 8 & 3 \end{pmatrix}_{b_0}$$

[593] $21^{133}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -381, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 11 & 3 \\ 4 & 4 & 17 & 15 & 4 \\ 12 & 17 & 49 & 45 & 13 \\ 11 & 15 & 45 & 39 & 11 \\ 3 & 4 & 13 & 11 & 3 \end{pmatrix}_{b_0}$$

[594] $21^{113}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -381, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 14 & 3 \\ 4 & 4 & 9 & 20 & 4 \\ 6 & 9 & 13 & 32 & 7 \\ 14 & 20 & 32 & 69 & 15 \\ 3 & 4 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[595] $1^{1^2} 1^{14} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -381, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 2 & 4 & 1 & 20 & 3 \\ 18 & 29 & 20 & 148 & 22 \\ 3 & 4 & 3 & 22 & 3 \end{pmatrix}_{b_0}$$

[596] $1^{1^{22}} 1^{14} 3^1 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -382, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 13 & 3 \\ 3 & 4 & 21 & 20 & 4 \\ 13 & 21 & 76 & 74 & 16 \\ 13 & 20 & 74 & 69 & 15 \\ 3 & 4 & 16 & 15 & 3 \end{pmatrix}_{b_0}$$

[597] $1^{1^3} 1^{14} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -387, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 2 & 4 & 1 & 20 & 3 \\ 19 & 29 & 20 & 148 & 22 \\ 3 & 4 & 3 & 22 & 3 \end{pmatrix}_{b_0}$$

[598] $1^{1^3} 1^{14} 3^1 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -388, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 34 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 22 & 34 & 0 & 208 & 26 \\ 3 & 4 & 0 & 26 & 3 \end{pmatrix}_{b_0}$$

[600] $21^{1^3} 3^1 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -391, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 33 & 4 \\ 5 & 4 & 4 & 34 & 4 \\ 3 & 4 & 1 & 23 & 3 \\ 33 & 34 & 23 & 208 & 26 \\ 4 & 4 & 3 & 26 & 3 \end{pmatrix}_{b_0}$$

[599] $1^{1^{23}} 1^{12} 3^1 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -391, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 14 & 3 \\ 3 & 4 & 13 & 20 & 4 \\ 9 & 13 & 28 & 46 & 10 \\ 14 & 20 & 46 & 69 & 15 \\ 3 & 4 & 10 & 15 & 3 \end{pmatrix}_{b_0}$$

[601] $21^{1^{42}} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -394, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 18 & 8 & 3 \\ 4 & 4 & 26 & 11 & 4 \\ 18 & 26 & 118 & 50 & 20 \\ 8 & 11 & 50 & 20 & 8 \\ 3 & 4 & 20 & 8 & 3 \end{pmatrix}_{b_0}$$

[602] $1^{1^{34}} 1^{12} 3^1 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -396, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 15 & 3 \\ 3 & 4 & 13 & 20 & 4 \\ 10 & 13 & 28 & 46 & 10 \\ 15 & 20 & 46 & 69 & 15 \\ 3 & 4 & 10 & 15 & 3 \end{pmatrix}_{b_0}$$

[604] $1^{1^{14}} 1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -403, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 3 & 4 & 1 & 17 & 3 \\ 17 & 24 & 17 & 100 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

[603] $1^{1^{24}} 1^{14} 3^1 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -403, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 17 & 3 \\ 3 & 4 & 17 & 24 & 4 \\ 11 & 17 & 49 & 72 & 13 \\ 17 & 24 & 72 & 100 & 18 \\ 3 & 4 & 13 & 18 & 3 \end{pmatrix}_{b_0}$$

[605] $1^{1^{12}} 1^{14} 3^1 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -404, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 15 & 3 \\ 3 & 4 & 13 & 24 & 4 \\ 8 & 13 & 28 & 55 & 10 \\ 15 & 24 & 55 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[606] $1^{141} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -408, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 18 & 5 & 3 \\ 3 & 4 & 26 & 7 & 4 \\ 18 & 26 & 118 & 32 & 20 \\ 5 & 7 & 32 & 7 & 5 \\ 3 & 4 & 20 & 5 & 3 \end{pmatrix}_{b_0}$$

[607] $1^{133} 3^1 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -409, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 15 & 3 \\ 3 & 4 & 17 & 20 & 4 \\ 12 & 17 & 49 & 60 & 13 \\ 15 & 20 & 60 & 69 & 15 \\ 3 & 4 & 13 & 15 & 3 \end{pmatrix}_{b_0}$$

[608] $1^{13} 3^1 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -411, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 24 & 3 \\ 3 & 4 & 0 & 35 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 24 & 35 & 0 & 231 & 27 \\ 3 & 4 & 0 & 27 & 3 \end{pmatrix}_{b_0}$$

[610] $21^{142} 3^1 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -415, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 11 & 3 \\ 4 & 4 & 21 & 16 & 4 \\ 15 & 21 & 76 & 59 & 16 \\ 11 & 16 & 59 & 44 & 12 \\ 3 & 4 & 16 & 12 & 3 \end{pmatrix}_{b_0}$$

[609] $21^{122} 3^1 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -415, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 11 & 3 \\ 4 & 4 & 13 & 16 & 4 \\ 9 & 13 & 28 & 37 & 10 \\ 11 & 16 & 37 & 44 & 12 \\ 3 & 4 & 10 & 12 & 3 \end{pmatrix}_{b_0}$$

[611] $21^{113} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -416, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 14 & 3 \\ 4 & 4 & 4 & 20 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 14 & 20 & 4 & 69 & 15 \\ 3 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

[613] $1^{141} 3^1 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -421, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 21 & 5 & 3 \\ 3 & 4 & 31 & 7 & 4 \\ 21 & 31 & 172 & 38 & 24 \\ 5 & 7 & 38 & 7 & 5 \\ 3 & 4 & 24 & 5 & 3 \end{pmatrix}_{b_0}$$

[612] $1^{143} 3^1 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -421, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 14 & 3 \\ 3 & 4 & 21 & 20 & 4 \\ 14 & 21 & 76 & 74 & 16 \\ 14 & 20 & 74 & 69 & 15 \\ 3 & 4 & 16 & 15 & 3 \end{pmatrix}_{b_0}$$

[615] $21^{152} 3^1 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.1$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 8 & 3 \\ 4 & 4 & 20 & 11 & 4 \\ 15 & 20 & 73 & 40 & 15 \\ 8 & 11 & 40 & 20 & 8 \\ 3 & 4 & 15 & 8 & 3 \end{pmatrix}_{b_0}$$

[614] $21^{113} 3^1 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 17 & 3 \\ 4 & 4 & 4 & 25 & 4 \\ 3 & 4 & 1 & 17 & 3 \\ 17 & 25 & 4 & 111 & 19 \\ 3 & 4 & 3 & 19 & 3 \end{pmatrix}_{b_0}$$

[616] $1^{134} 3^1 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -427, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 15 & 3 \\ 3 & 4 & 17 & 24 & 4 \\ 10 & 17 & 49 & 72 & 13 \\ 15 & 24 & 72 & 100 & 18 \\ 3 & 4 & 13 & 18 & 3 \end{pmatrix}_{b_0}$$

[617] $1^{133} 3^1 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -427, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 13 & 3 \\ 3 & 4 & 17 & 20 & 4 \\ 11 & 17 & 49 & 60 & 13 \\ 13 & 20 & 60 & 69 & 15 \\ 3 & 4 & 13 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[618] \quad 1^{1^{14}3^1}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -431, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.84 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 23 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 2 & 4 & 1 & 23 & 3 \\ 23 & 34 & 23 & 208 & 26 \\ 3 & 4 & 3 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[619] \quad 1^{1^{44}3^1}1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -434, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.81 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 15 & 3 \\ 3 & 4 & 21 & 20 & 4 \\ 15 & 21 & 76 & 74 & 16 \\ 15 & 20 & 74 & 69 & 15 \\ 3 & 4 & 16 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[620] \quad 1^{1^43^1}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -436, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 34 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 23 & 34 & 0 & 208 & 26 \\ 3 & 4 & 0 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[621] \quad 1^{1^{14}3^1}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -438, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 3 & 4 & 1 & 20 & 3 \\ 19 & 29 & 20 & 148 & 22 \\ 3 & 4 & 3 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[624] \quad 1^{1^{14}3^1}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -439, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 2 & 4 & 1 & 23 & 3 \\ 21 & 34 & 23 & 208 & 26 \\ 3 & 4 & 3 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[622] \quad 21^{1^{14}3^1}1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -439, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 17 & 3 \\ 4 & 4 & 9 & 24 & 4 \\ 6 & 9 & 13 & 38 & 7 \\ 17 & 24 & 38 & 100 & 18 \\ 3 & 4 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[623] \quad 1^{1^{23}3^1}1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -439, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 16 & 3 \\ 3 & 4 & 13 & 24 & 4 \\ 9 & 13 & 28 & 55 & 10 \\ 16 & 24 & 55 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[625] \quad 1^{1^{44}3^1}1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -439, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 16 & 3 \\ 3 & 4 & 21 & 24 & 4 \\ 13 & 21 & 76 & 89 & 16 \\ 16 & 24 & 89 & 100 & 18 \\ 3 & 4 & 16 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[626] \quad 21^{1^{14}3^1}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -440, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 14 & 3 \\ 4 & 4 & 4 & 19 & 4 \\ 3 & 4 & 1 & 14 & 3 \\ 14 & 19 & 14 & 64 & 14 \\ 3 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[627] \quad 1^{1^{32}3^1}1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -444, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 13 & 3 \\ 3 & 4 & 21 & 20 & 4 \\ 14 & 21 & 76 & 74 & 16 \\ 13 & 20 & 74 & 69 & 15 \\ 3 & 4 & 16 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[628] \quad 21^{1^{43}3^1}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -446, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 11 & 3 \\ 4 & 4 & 16 & 15 & 4 \\ 12 & 16 & 46 & 44 & 12 \\ 11 & 15 & 44 & 39 & 11 \\ 3 & 4 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[629] \quad 1^{134} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -447, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 22 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 2 & 4 & 1 & 23 & 3 \\ 22 & 34 & 23 & 208 & 26 \\ 3 & 4 & 3 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[630] \quad 1^{134} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -451, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.85 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 17 & 3 \\ 3 & 4 & 17 & 24 & 4 \\ 13 & 17 & 49 & 72 & 13 \\ 17 & 24 & 72 & 100 & 18 \\ 3 & 4 & 13 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[631] \quad 21^{161} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -454, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.2$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 18 & 5 & 3 \\ 4 & 4 & 24 & 7 & 4 \\ 18 & 24 & 106 & 30 & 18 \\ 5 & 7 & 30 & 7 & 5 \\ 3 & 4 & 18 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[632] \quad 21^{132} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -455, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.80 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 11 & 3 \\ 4 & 4 & 17 & 16 & 4 \\ 12 & 17 & 49 & 48 & 13 \\ 11 & 16 & 48 & 44 & 12 \\ 3 & 4 & 13 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[633] \quad 1^{114} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -463, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 22 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 3 & 4 & 1 & 23 & 3 \\ 22 & 34 & 23 & 208 & 26 \\ 3 & 4 & 3 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[634] \quad 1^{133} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -466, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.83 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 14 & 3 \\ 3 & 4 & 17 & 20 & 4 \\ 12 & 17 & 49 & 60 & 13 \\ 14 & 20 & 60 & 69 & 15 \\ 3 & 4 & 13 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[636] \quad 1^{114} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -469, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 26 & 3 \\ 3 & 4 & 4 & 39 & 4 \\ 2 & 4 & 1 & 26 & 3 \\ 26 & 39 & 26 & 280 & 30 \\ 3 & 4 & 3 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[635] \quad 21^{143} 3^1 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -469, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 39 & 4 \\ 5 & 4 & 4 & 39 & 4 \\ 3 & 4 & 1 & 26 & 3 \\ 39 & 39 & 26 & 280 & 30 \\ 4 & 4 & 3 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[637] \quad 21^{124} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -470, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 14 & 3 \\ 4 & 4 & 8 & 19 & 4 \\ 6 & 8 & 10 & 28 & 6 \\ 14 & 19 & 28 & 64 & 14 \\ 3 & 4 & 6 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[638] \quad 1^{133} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -471, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.81 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 15 & 3 \\ 3 & 4 & 17 & 20 & 4 \\ 13 & 17 & 49 & 60 & 13 \\ 15 & 20 & 60 & 69 & 15 \\ 3 & 4 & 13 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[639] \quad 1^{124} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -472, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.84 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 17 & 3 \\ 3 & 4 & 13 & 24 & 4 \\ 9 & 13 & 28 & 55 & 10 \\ 17 & 24 & 55 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[641] \quad 1^{1^{14}3^1 1^{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -483, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 25 & 3 \\ 3 & 4 & 4 & 39 & 4 \\ 2 & 4 & 1 & 26 & 3 \\ 25 & 39 & 26 & 280 & 30 \\ 3 & 4 & 3 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[640] \quad 1^{1^{43}3^1 1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -483, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.83 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 14 & 3 \\ 3 & 4 & 21 & 20 & 4 \\ 15 & 21 & 76 & 74 & 16 \\ 14 & 20 & 74 & 69 & 15 \\ 3 & 4 & 16 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[642] \quad 1^{1^{43}3^1 1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -487, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 16 & 3 \\ 3 & 4 & 21 & 24 & 4 \\ 15 & 21 & 76 & 89 & 16 \\ 16 & 24 & 89 & 100 & 18 \\ 3 & 4 & 16 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[643] \quad 1^{1^{14}3^1 1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -492, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.82 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 3 & 4 & 1 & 20 & 3 \\ 20 & 29 & 20 & 148 & 22 \\ 3 & 4 & 3 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[644] \quad 21^{1^{14}3^1 1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -499, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.83 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 15 & 5 & 3 \\ 4 & 4 & 21 & 7 & 4 \\ 15 & 21 & 76 & 26 & 16 \\ 5 & 7 & 26 & 7 & 5 \\ 3 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[645] \quad 1^{1^{34}3^1 1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -499, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 15 & 3 \\ 3 & 4 & 17 & 24 & 4 \\ 11 & 17 & 49 & 72 & 13 \\ 15 & 24 & 72 & 100 & 18 \\ 3 & 4 & 13 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[647] \quad 21^{1^{14}3^1 1^{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -504, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 26 & 3 \\ 4 & 4 & 4 & 39 & 4 \\ 3 & 4 & 1 & 26 & 3 \\ 26 & 39 & 26 & 280 & 30 \\ 3 & 4 & 3 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[646] \quad 1^{1^{34}3^1 1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -504, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 16 & 3 \\ 3 & 4 & 17 & 24 & 4 \\ 12 & 17 & 49 & 72 & 13 \\ 16 & 24 & 72 & 100 & 18 \\ 3 & 4 & 13 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[648] \quad 21^{1^{24}3^1 1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -504, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.83 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 17 & 3 \\ 4 & 4 & 8 & 24 & 4 \\ 6 & 8 & 10 & 34 & 6 \\ 17 & 24 & 34 & 100 & 18 \\ 3 & 4 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[649] \quad 1^{1^{14}3^1 1^{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -517, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 26 & 3 \\ 3 & 4 & 0 & 39 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 26 & 39 & 0 & 280 & 30 \\ 3 & 4 & 0 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[650] \quad 21^{1^{34}3^1 1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 10, \quad \Delta = -86, \quad \det = -520, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 14 & 3 \\ 4 & 4 & 12 & 19 & 4 \\ 9 & 12 & 25 & 42 & 9 \\ 14 & 19 & 42 & 64 & 14 \\ 3 & 4 & 9 & 14 & 3 \end{pmatrix}_{b_0}$$

[652] $1^{14}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -532, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 26 & 3 \\ 3 & 4 & 4 & 39 & 4 \\ 3 & 4 & 1 & 26 & 3 \\ 26 & 39 & 26 & 280 & 30 \\ 3 & 4 & 3 & 30 & 3 \end{pmatrix}_{b_0}$$

[651] $1^{14}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -532, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 17 & 3 \\ 3 & 4 & 21 & 24 & 4 \\ 14 & 21 & 76 & 89 & 16 \\ 17 & 24 & 89 & 100 & 18 \\ 3 & 4 & 16 & 18 & 3 \end{pmatrix}_{b_0}$$

[653] $21^{16}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -534, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 18 & 8 & 3 \\ 4 & 4 & 24 & 11 & 4 \\ 18 & 24 & 106 & 48 & 18 \\ 8 & 11 & 48 & 20 & 8 \\ 3 & 4 & 18 & 8 & 3 \end{pmatrix}_{b_0}$$

[654] $1^{14}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -535, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 23 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 3 & 4 & 1 & 23 & 3 \\ 23 & 34 & 23 & 208 & 26 \\ 3 & 4 & 3 & 26 & 3 \end{pmatrix}_{b_0}$$

[655] $21^{15}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -536, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.1$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 11 & 3 \\ 4 & 4 & 20 & 15 & 4 \\ 15 & 20 & 73 & 55 & 15 \\ 11 & 15 & 55 & 39 & 11 \\ 3 & 4 & 15 & 11 & 3 \end{pmatrix}_{b_0}$$

[656] $21^{14}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -539, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 21 & 5 & 3 \\ 4 & 4 & 31 & 7 & 4 \\ 21 & 31 & 172 & 38 & 24 \\ 5 & 7 & 38 & 7 & 5 \\ 3 & 4 & 24 & 5 & 3 \end{pmatrix}_{b_0}$$

[657] $1^{13}4^31^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -544, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 16 & 3 \\ 3 & 4 & 17 & 24 & 4 \\ 11 & 17 & 49 & 72 & 13 \\ 16 & 24 & 72 & 100 & 18 \\ 3 & 4 & 13 & 18 & 3 \end{pmatrix}_{b_0}$$

[658] $21^{14}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -549, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 8 & 3 \\ 4 & 4 & 21 & 11 & 4 \\ 15 & 21 & 76 & 41 & 16 \\ 8 & 11 & 41 & 20 & 8 \\ 3 & 4 & 16 & 8 & 3 \end{pmatrix}_{b_0}$$

[659] $1^{14}4^31^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -568, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 15 & 3 \\ 3 & 4 & 21 & 24 & 4 \\ 13 & 21 & 76 & 89 & 16 \\ 15 & 24 & 89 & 100 & 18 \\ 3 & 4 & 16 & 18 & 3 \end{pmatrix}_{b_0}$$

[660] $21^{14}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -571, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 11 & 3 \\ 4 & 4 & 21 & 15 & 4 \\ 15 & 21 & 76 & 56 & 16 \\ 11 & 15 & 56 & 39 & 11 \\ 3 & 4 & 16 & 11 & 3 \end{pmatrix}_{b_0}$$

[661] $21^{12}3^21^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -581, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 14 & 3 \\ 4 & 4 & 13 & 20 & 4 \\ 9 & 13 & 28 & 46 & 10 \\ 14 & 20 & 46 & 69 & 15 \\ 3 & 4 & 10 & 15 & 3 \end{pmatrix}_{b_0}$$

[662] $21^{17}1^31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -584, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$

$$\begin{aligned}
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 21 & 5 & 3 \\ 4 & 4 & 28 & 7 & 4 \\ 21 & 28 & 145 & 35 & 21 \\ 5 & 7 & 35 & 7 & 5 \\ 3 & 4 & 21 & 5 & 3 \end{pmatrix}_{b_0} \\
[663] & 21^{144}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -590, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 12 & 14 & 3 \\ 4 & 4 & 16 & 19 & 4 \\ 12 & 16 & 46 & 56 & 12 \\ 14 & 19 & 56 & 64 & 14 \\ 3 & 4 & 12 & 14 & 3 \end{pmatrix}_{b_0} \\
[664] & 21^{141}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -594, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 18 & 5 & 3 \\ 4 & 4 & 26 & 7 & 4 \\ 18 & 26 & 118 & 32 & 20 \\ 5 & 7 & 32 & 7 & 5 \\ 3 & 4 & 20 & 5 & 3 \end{pmatrix}_{b_0} \\
& 1^{34} \\
[665] & 1^{134}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -603, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 17 & 3 \\ 3 & 4 & 17 & 24 & 4 \\ 12 & 17 & 49 & 72 & 13 \\ 17 & 24 & 72 & 100 & 18 \\ 3 & 4 & 13 & 18 & 3 \end{pmatrix}_{b_0} \\
[666] & 21^{144}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -624, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 17 & 3 \\ 4 & 4 & 4 & 24 & 4 \\ 3 & 4 & 1 & 17 & 3 \\ 17 & 24 & 17 & 100 & 18 \\ 3 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0} \\
& 1^{33} \\
[667] & 1^{144}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -639, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 16 & 3 \\ 3 & 4 & 21 & 24 & 4 \\ 14 & 21 & 76 & 89 & 16 \\ 16 & 24 & 89 & 100 & 18 \\ 3 & 4 & 16 & 18 & 3 \end{pmatrix}_{b_0} \\
[668] & 21^{163}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -646, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 18 & 11 & 3 \\ 4 & 4 & 24 & 15 & 4 \\ 18 & 24 & 106 & 66 & 18 \\ 11 & 15 & 66 & 39 & 11 \\ 3 & 4 & 18 & 11 & 3 \end{pmatrix}_{b_0} \\
[669] & 21^{144}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -656, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 23 & 3 \\ 4 & 4 & 4 & 34 & 4 \\ 3 & 4 & 1 & 23 & 3 \\ 23 & 34 & 23 & 208 & 26 \\ 3 & 4 & 3 & 26 & 3 \end{pmatrix}_{b_0} \\
[670] & 21^{172}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -664, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 21 & 8 & 3 \\ 4 & 4 & 28 & 11 & 4 \\ 21 & 28 & 145 & 56 & 21 \\ 8 & 11 & 56 & 20 & 8 \\ 3 & 4 & 21 & 8 & 3 \end{pmatrix}_{b_0} \\
& 1^{44} \\
[671] & 1^{144}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -676, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 15 & 17 & 3 \\ 3 & 4 & 21 & 24 & 4 \\ 15 & 21 & 76 & 89 & 16 \\ 17 & 24 & 89 & 100 & 18 \\ 3 & 4 & 16 & 18 & 3 \end{pmatrix}_{b_0} \\
[672] & 21^{154}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -680, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.1 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 15 & 14 & 3 \\ 4 & 4 & 20 & 19 & 4 \\ 15 & 20 & 73 & 70 & 15 \\ 14 & 19 & 70 & 64 & 14 \\ 3 & 4 & 15 & 14 & 3 \end{pmatrix}_{b_0} \\
[673] & 21^{144}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -696, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 20 & 3 \\ 4 & 4 & 4 & 29 & 4 \\ 3 & 4 & 1 & 20 & 3 \\ 20 & 29 & 20 & 148 & 22 \\ 3 & 4 & 3 & 22 & 3 \end{pmatrix}_{b_0} \\
[674] & 21^{133}3^11^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -701, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 12 & 14 & 3 \\ 4 & 4 & 17 & 20 & 4 \\ 12 & 17 & 49 & 60 & 13 \\ 14 & 20 & 60 & 69 & 15 \\ 3 & 4 & 13 & 15 & 3 \end{pmatrix}_{b_0} \\
[675] & 21^{124}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -719, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 9 & 17 & 3 \\ 4 & 4 & 13 & 24 & 4 \\ 9 & 13 & 28 & 55 & 10 \\ 17 & 24 & 55 & 100 & 18 \\ 3 & 4 & 10 & 18 & 3 \end{pmatrix}_{b_0} \\
[676] & 21^{143}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -741, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 15 & 14 & 3 \\ 4 & 4 & 21 & 20 & 4 \\ 15 & 21 & 76 & 74 & 16 \\ 14 & 20 & 74 & 69 & 15 \\ 3 & 4 & 16 & 15 & 3 \end{pmatrix}_{b_0} \\
[677] & 21^{173}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -776, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 21 & 11 & 3 \\ 4 & 4 & 28 & 15 & 4 \\ 21 & 28 & 145 & 77 & 21 \\ 11 & 15 & 77 & 39 & 11 \\ 3 & 4 & 21 & 11 & 3 \end{pmatrix}_{b_0} \\
[678] & 21^{134}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -919, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 12 & 17 & 3 \\ 4 & 4 & 17 & 24 & 4 \\ 12 & 17 & 49 & 72 & 13 \\ 17 & 24 & 72 & 100 & 18 \\ 3 & 4 & 13 & 18 & 3 \end{pmatrix}_{b_0} \\
[679] & 21^{144}3^11^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -1039, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 15 & 17 & 3 \\ 4 & 4 & 21 & 24 & 4 \\ 15 & 21 & 76 & 89 & 16 \\ 17 & 24 & 89 & 100 & 18 \\ 3 & 4 & 16 & 18 & 3 \end{pmatrix}_{b_0} \\
T = 8 & [692] 1^{11}3^121^{142} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 13 & 8 & 3 \\ 3 & 5 & 25 & 13 & 5 \\ 13 & 25 & 92 & 49 & 20 \\ 8 & 13 & 49 & 24 & 10 \\ 3 & 5 & 20 & 10 & 4 \end{pmatrix}_{b_0} \\
[680] & 1^{11}3^121^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 4 & 2 \\ 1 & 2 & 0 & 3 & 5 & 2 \\ 1 & 1 & 3 & 1 & 5 & 3 \\ 4 & 4 & 5 & 5 & 10 & 6 \\ 2 & 2 & 2 & 3 & 6 & 3 \end{pmatrix}_{b_0} \\
[681] & 1^{133}3^121^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 3 & 7 & 2 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 5 & 3 & 4 & 5 & 16 & 5 \\ 8 & 7 & 9 & 16 & 35 & 11 \\ 2 & 2 & 2 & 5 & 11 & 3 \end{pmatrix}_{b_0} \\
[682] & 1^{131}3^121^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 7 & 4 & 2 \\ 1 & 2 & 0 & 6 & 5 & 2 \\ 7 & 7 & 6 & 25 & 17 & 9 \\ 4 & 4 & 5 & 17 & 10 & 6 \\ 2 & 2 & 2 & 9 & 6 & 3 \end{pmatrix}_{b_0} \\
[683] & 1^{11}3^121^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 5 & 2 \\ 1 & 2 & 0 & 3 & 8 & 2 \\ 1 & 1 & 3 & 1 & 8 & 3 \\ 7 & 5 & 8 & 8 & 23 & 9 \\ 2 & 2 & 2 & 3 & 9 & 3 \end{pmatrix}_{b_0} \\
[684] & 1^{122}3^121^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8
\end{aligned}$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 0 & 2 & 7 & 4 & 2 \\ 1 & 2 & 0 & 6 & 5 & 2 \\ 6 & 7 & 6 & 25 & 17 & 9 \\ 5 & 4 & 5 & 17 & 10 & 6 \\ 2 & 2 & 2 & 9 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[685] \quad 1^{122} 1^{11} 3^{121} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 1 & 5 & 2 \\ 1 & 2 & 0 & 3 & 8 & 2 \\ 3 & 1 & 3 & 1 & 8 & 3 \\ 6 & 5 & 8 & 8 & 23 & 9 \\ 2 & 2 & 2 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[686] \quad 1^{122} 1^{11} 3^{121} 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 2 & 1 & 4 & 2 \\ 1 & 2 & 0 & 3 & 5 & 2 \\ 3 & 1 & 3 & 1 & 5 & 3 \\ 5 & 4 & 5 & 5 & 10 & 6 \\ 2 & 2 & 2 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[687] \quad 1^{122} 1^{11} 3^{121} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 5 & 10 & 13 & 5 \\ 7 & 10 & 14 & 19 & 8 \\ 8 & 13 & 19 & 24 & 10 \\ 3 & 5 & 8 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[688] \quad 1^{13} 1^{122} 3^{121} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 3 & 7 & 2 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 2 & 3 & 4 & 5 & 16 & 5 \\ 8 & 7 & 9 & 16 & 35 & 11 \\ 2 & 2 & 2 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[689] \quad 1^{11} 1^{12} 3^{121} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 5 & 5 & 8 & 5 \\ 3 & 5 & 4 & 6 & 4 \\ 5 & 8 & 6 & 8 & 6 \\ 3 & 5 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[693] \quad 1^{31} 1^{11} 3^{121} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 15 & 4 \\ 3 & 1 & 3 & 9 & 3 \\ 3 & 3 & 1 & 14 & 3 \\ 15 & 9 & 14 & 49 & 15 \\ 4 & 3 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[690] \quad 1^{11} 1^{12} 3^{121} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 & 3 \\ 3 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[691] \quad 1^{11} 1^{11} 3^{121} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 1 & 0 & 1 & 8 & 3 \\ 4 & 5 & 0 & 8 & 23 & 9 \\ 2 & 2 & -1 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[695] \quad 1^{11} 1^{11} 3^{121} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 1 & 0 & 1 & 4 & 3 \\ 2 & 2 & 0 & 4 & 4 & 4 \\ 2 & 2 & -1 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[694] \quad 1^{11} 1^{11} 3^{121} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 4 & 4 \\ 3 & 1 & 3 & 2 & 3 \\ 3 & 3 & 1 & 4 & 3 \\ 4 & 2 & 4 & 2 & 4 \\ 4 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[696] \quad 1^{1^{12}} 3^{1^{32}} 21^{32} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 1 & 5 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 5 & 0 & 33 & 29 & 11 \\ 4 & 5 & 0 & 29 & 23 & 9 \\ 2 & 2 & -1 & 11 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[697] \quad 1^{1^{11}} 3^{1^{13}} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 7, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 3 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 17 & 5 \\ 6 & 6 & 0 & 17 & 42 & 12 \\ 2 & 2 & -1 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[713] \quad 1^{1^{24}} 3^{1^{12}} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 1 & 7 & 2 \\ 1 & 2 & 0 & 3 & 9 & 2 \\ 3 & 1 & 3 & 1 & 9 & 3 \\ 9 & 7 & 9 & 35 & 11 \\ 2 & 2 & 2 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[716] \quad 1^{1^{13}} 3^{1^{14}} 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 1 & 3 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 27 & 5 \\ 8 & 11 & 0 & 27 & 103 & 19 \\ 2 & 2 & -1 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[715] \quad 1^{1^{24}} 3^{1^{14}} 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 11 & 2 \\ 1 & 0 & 2 & 1 & 9 & 2 \\ 1 & 2 & 0 & 3 & 15 & 2 \\ 3 & 1 & 3 & 1 & 15 & 3 \\ 11 & 9 & 15 & 15 & 83 & 17 \\ 2 & 2 & 2 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[714] \quad 1^{1^{22}} 3^{1^{33}} 21^{33} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & 7 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 45 & 52 & 13 \\ 6 & 8 & 0 & 52 & 56 & 14 \\ 2 & 2 & -1 & 13 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[712] \quad 1^{1^{21}} 3^{1^{22}} 21^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 6 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 26 & 10 \\ 4 & 5 & 0 & 26 & 23 & 9 \\ 2 & 2 & -1 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[717] \quad 1^{1^{22}} 3^{1^{23}} 21^{23} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 1 & 6 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 40 & 10 \\ 6 & 8 & 0 & 40 & 56 & 14 \\ 2 & 2 & -1 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[718] \quad 1^{1^{23}} 3^{1^{24}} 21^{24} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 1 & 6 & 11 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 54 & 10 \\ 8 & 11 & 0 & 54 & 103 & 19 \\ 2 & 2 & -1 & 10 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[698] \quad 1^{1^{21}} 3^{1^{12}} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 20 & 4 \\ 3 & 1 & 3 & 16 & 3 \\ 3 & 3 & 1 & 16 & 3 \\ 20 & 16 & 16 & 94 & 20 \\ 4 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[711] \ 1^{1123^1 21^{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 8, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 3 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 20 & 5 \\ 6 & 8 & 0 & 20 & 56 & 14 \\ 2 & 2 & -1 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[719] \ 1^{1423^1 21^{43}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 8, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & 0 & 1 & 12 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 12 & 0 & 110 & 80 & 20 \\ 6 & 8 & 0 & 80 & 56 & 14 \\ 2 & 2 & -1 & 20 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[699] \ 1^{133^1 21^{31}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 8, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 9 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 9 & 0 & 61 & 18 & 15 \\ 2 & 2 & 0 & 18 & 4 & 4 \\ 2 & 2 & -1 & 15 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[700] \ 1^{1423^1 21^{122}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 8, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 7 & 2 \\ 1 & 0 & 2 & 10 & 7 & 2 \\ 1 & 2 & 0 & 10 & 9 & 2 \\ 8 & 10 & 10 & 58 & 46 & 14 \\ 7 & 7 & 9 & 46 & 35 & 11 \\ 2 & 2 & 2 & 14 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[710] \ 1^{123^1 21^{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 8, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 7 & 2 \\ 1 & 2 & 0 & 3 & 9 & 2 \\ 1 & 1 & 3 & 1 & 9 & 3 \\ 7 & 7 & 9 & 9 & 35 & 11 \\ 2 & 2 & 2 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[701] \ 1^{131^1 21^{32}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 8, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 1 & 9 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 9 & 0 & 61 & 39 & 15 \\ 4 & 5 & 0 & 39 & 23 & 9 \\ 2 & 2 & -1 & 15 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[702] \ 1^{111^1 3^1 21^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 8, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 3 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 13 & 5 \\ 4 & 5 & 0 & 13 & 23 & 9 \\ 2 & 2 & -1 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[709] \ 1^{143^1 21^{41}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 8, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 1 & 12 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 12 & 0 & 110 & 24 & 20 \\ 2 & 2 & 0 & 24 & 4 & 4 \\ 2 & 2 & -1 & 20 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[708] \ 1^{112^1 3^1 21^{31}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 8, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 5 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 5 & 0 & 33 & 14 & 11 \\ 2 & 2 & 0 & 14 & 4 & 4 \\ 2 & 2 & -1 & 11 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[707] \ 1^{113^1 21^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 8, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 6 & 5 \\ 2 & 2 & 0 & 6 & 4 & 4 \\ 2 & 2 & -1 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[706] $1^{14}3^{12}21^{42} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 1 & 12 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 12 & 0 & 110 & 52 & 20 \\ 4 & 5 & 0 & 52 & 23 & 9 \\ 2 & 2 & -1 & 20 & 9 & 3 \end{pmatrix}_{b_0}$$

[705] $1^{12}3^{12}21^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 6 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 12 & 10 \\ 2 & 2 & 0 & 12 & 4 & 4 \\ 2 & 2 & -1 & 10 & 4 & 3 \end{pmatrix}_{b_0}$$

[703] $1^{132}3^{12}21^{33} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 1 & 9 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 9 & 0 & 61 & 60 & 15 \\ 6 & 8 & 0 & 60 & 56 & 14 \\ 2 & 2 & -1 & 15 & 14 & 3 \end{pmatrix}_{b_0}$$

[704] $1^{12}3^{12}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 10 & 4 \\ 3 & 1 & 3 & 8 & 3 \\ 3 & 3 & 1 & 8 & 3 \\ 10 & 8 & 8 & 22 & 10 \\ 4 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[722] $1^{111}3^{12}21^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 4 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 4 & 0 & 16 & 21 & 8 \\ 4 & 5 & 0 & 21 & 23 & 9 \\ 2 & 2 & -1 & 8 & 9 & 3 \end{pmatrix}_{b_0}$$

[721] $1^{11}3^{12}21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 11 & 4 \\ 3 & 1 & 3 & 7 & 3 \\ 3 & 3 & 1 & 10 & 3 \\ 11 & 7 & 10 & 25 & 11 \\ 4 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[720] $1^{11}3^{12}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 7 & 4 \\ 3 & 1 & 3 & 5 & 3 \\ 3 & 3 & 1 & 6 & 3 \\ 7 & 5 & 6 & 9 & 7 \\ 4 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[727] $1^{11}3^{13}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -65, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 3 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 4 & 5 & 4 \\ 2 & 4 & 0 & 5 & 4 & 4 \\ 2 & 3 & -1 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

[726] $1^{122}3^{12}21^{24} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 1 & 6 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 48 & 10 \\ 8 & 9 & 0 & 48 & 83 & 17 \\ 2 & 2 & -1 & 10 & 17 & 3 \end{pmatrix}_{b_0}$$

[725] $1^{11}3^{12}21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 & 2 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 2 & 3 & 4 & 5 & 6 & 5 \\ 3 & 2 & 4 & 6 & 4 & 4 \\ 2 & 2 & 2 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[724] \quad 1^{1^{33}} 3^{1^{21}} 21^{1^{33}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 0 & 2 & 5 & 8 & 2 \\ 1 & 2 & 0 & 10 & 12 & 2 \\ 7 & 5 & 10 & 33 & 44 & 11 \\ 9 & 8 & 12 & 44 & 56 & 14 \\ 2 & 2 & 2 & 11 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[723] \quad 1^{1^{31}} 3^{1^{21}} 21^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 & 2 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 5 & 3 & 4 & 5 & 6 & 5 \\ 3 & 2 & 4 & 6 & 4 & 4 \\ 2 & 2 & 2 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[728] \quad 1^{1^{11}} 3^{1^{21}} 21^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 4 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 4 & 0 & 16 & 10 & 8 \\ 2 & 2 & 0 & 10 & 4 & 4 \\ 2 & 2 & -1 & 8 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[729] \quad 1^1 3^{1^{21}} 21^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 5 & 8 & 5 \\ 2 & 5 & 2 & 6 & 4 \\ 5 & 8 & 6 & 8 & 6 \\ 3 & 5 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[730] \quad 1^{1^{11}} 3^{1^{21}} 21^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 5 & 0 & 8 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 5 & 8 & 0 & 8 & 6 \\ 3 & 5 & 0 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[731] \quad 1^{1^{11}} 3^{1^{21}} 21^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 4 \\ 2 & 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[732] \quad 1^{1^4} 3^{1^{21}} 21^{1^{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 1 & 8 & 2 \\ 1 & 2 & 0 & 3 & 12 & 2 \\ 1 & 1 & 3 & 1 & 12 & 3 \\ 10 & 8 & 12 & 12 & 56 & 14 \\ 2 & 2 & 2 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[733] \quad 1^{1^{24}} 3^{1^{21}} 21^{1^{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 0 & 2 & 1 & 8 & 2 \\ 1 & 2 & 0 & 3 & 12 & 2 \\ 3 & 1 & 3 & 1 & 12 & 3 \\ 10 & 8 & 12 & 12 & 56 & 14 \\ 2 & 2 & 2 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[734] \quad 21^{1^{13}} 3^{1^{21}} 21^{1^{21}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 12 & 6 & 4 \\ 5 & 5 & 15 & 8 & 5 \\ 12 & 15 & 32 & 18 & 12 \\ 6 & 8 & 18 & 8 & 6 \\ 4 & 5 & 12 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[735] \quad 21^{1^{11}} 3^{1^{21}} 21^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 4 & 2 \\ 2 & 0 & 2 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 2 & 1 & 0 & 4 & 2 \\ 4 & 4 & 5 & 4 & 10 & 6 \\ 2 & 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[736] $21^{131}3^{121}1^{41} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 4 & 2 \\ 2 & 0 & 2 & 10 & 4 & 2 \\ 1 & 2 & 0 & 15 & 5 & 2 \\ 10 & 10 & 15 & 88 & 32 & 18 \\ 4 & 4 & 5 & 32 & 10 & 6 \\ 2 & 2 & 2 & 18 & 6 & 3 \end{pmatrix}_{b_0}$$

[737] $1^{132}3^{121}1^{43} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & 0 & 1 & 10 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 88 & 72 & 18 \\ 6 & 8 & 0 & 72 & 56 & 14 \\ 2 & 2 & -1 & 18 & 14 & 3 \end{pmatrix}_{b_0}$$

[738] $1^{11}3^{121}1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 5 & 10 & 8 & 5 \\ 5 & 10 & 14 & 12 & 8 \\ 5 & 8 & 12 & 8 & 6 \\ 3 & 5 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

[739] $1^{11}3^{121}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 5 & 10 & 8 & 5 \\ 5 & 10 & 14 & 12 & 8 \\ 5 & 8 & 12 & 8 & 6 \\ 3 & 5 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

[740] $1^{11}3^{121}1^{24} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 17 & 3 \\ 3 & 5 & 10 & 28 & 5 \\ 7 & 10 & 14 & 42 & 8 \\ 17 & 28 & 42 & 116 & 22 \\ 3 & 5 & 8 & 22 & 4 \end{pmatrix}_{b_0}$$

[741] $1^{13}3^{121}1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 1 & 10 & 2 \\ 1 & 2 & 0 & 3 & 13 & 2 \\ 1 & 1 & 3 & 1 & 13 & 3 \\ 10 & 10 & 13 & 13 & 74 & 16 \\ 2 & 2 & 2 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[742] $1^{121}3^{121}1^{32} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 1 & 7 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 45 & 34 & 13 \\ 4 & 5 & 0 & 34 & 23 & 9 \\ 2 & 2 & -1 & 13 & 9 & 3 \end{pmatrix}_{b_0}$$

[743] $1^{121}3^{121}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 17 & 4 \\ 3 & 1 & 3 & 13 & 3 \\ 3 & 3 & 1 & 14 & 3 \\ 17 & 13 & 14 & 65 & 17 \\ 4 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[744] $1^{121}3^{121}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 22 & 4 \\ 3 & 1 & 3 & 14 & 3 \\ 3 & 3 & 1 & 20 & 3 \\ 22 & 14 & 20 & 106 & 22 \\ 4 & 3 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[745] $1^{121}3^{121}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 14 & 4 \\ 3 & 1 & 3 & 10 & 3 \\ 3 & 3 & 1 & 12 & 3 \\ 14 & 10 & 12 & 42 & 14 \\ 4 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[746] $1^{121}3^{121}1^{42} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 1 & 8 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 8 & 0 & 70 & 42 & 16 \\ 4 & 5 & 0 & 42 & 23 & 9 \\ 2 & 2 & -1 & 16 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[747] \quad 1^{11} 3^{12} 21^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 1 & 7 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 7 & 0 & 45 & 16 & 13 \\ 2 & 2 & 0 & 16 & 4 & 4 \\ 2 & 2 & -1 & 13 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[748] \quad 1^{12} 3^{12} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 5 & 5 & 8 & 5 \\ 4 & 5 & 2 & 6 & 4 \\ 5 & 8 & 6 & 8 & 6 \\ 3 & 5 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[749] \quad 21^{123} 3^{12} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 10 & 2 \\ 2 & 0 & 2 & 5 & 10 & 2 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 5 & 5 & 5 & 13 & 33 & 7 \\ 10 & 10 & 13 & 33 & 74 & 16 \\ 2 & 2 & 2 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[751] \quad 1^{12} 3^{12} 21^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 1 & 8 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 8 & 0 & 70 & 20 & 16 \\ 2 & 2 & 0 & 20 & 4 & 4 \\ 2 & 2 & -1 & 16 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[750] \quad 1^{12} 3^{12} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 18 & 4 \\ 3 & 1 & 3 & 12 & 3 \\ 3 & 3 & 1 & 16 & 3 \\ 18 & 12 & 16 & 70 & 18 \\ 4 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[755] \quad 1^{11} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 2 & 1 & 0 & 4 & 2 \\ 4 & 4 & 5 & 4 & 10 & 6 \\ 2 & 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[754] \quad 1^{12} 3^{12} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 2 & 1 & 0 & 4 & 2 \\ 5 & 4 & 5 & 4 & 10 & 6 \\ 2 & 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[753] \quad 1^{13} 3^{12} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 4 & 10 & 2 \\ 1 & 2 & 0 & 7 & 13 & 2 \\ 4 & 4 & 7 & 16 & 36 & 8 \\ 10 & 10 & 13 & 36 & 74 & 16 \\ 2 & 2 & 2 & 8 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[752] \quad 1^{12} 3^{12} 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 5 & 5 & 13 & 5 \\ 2 & 5 & 2 & 10 & 4 \\ 8 & 13 & 10 & 24 & 10 \\ 3 & 5 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[757] \quad 1^{13} 3^{12} 21^{42} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 1 & 10 & 5 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 88 & 47 & 18 \\ 4 & 5 & 0 & 47 & 23 & 9 \\ 2 & 2 & -1 & 18 & 9 & 3 \end{pmatrix}_{b_0}$$

[761] $1^{122} 1^{131} 3^1 21^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 6 & 2 \\ 1 & 0 & 2 & 10 & 5 & 2 \\ 1 & 2 & 0 & 15 & 8 & 2 \\ 9 & 10 & 15 & 88 & 47 & 18 \\ 6 & 5 & 8 & 47 & 23 & 9 \\ 2 & 2 & 2 & 18 & 9 & 3 \end{pmatrix}_{b_0}$$

[760] $1^{122} 1^{11} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 4 & 5 & 2 & 13 & 4 \\ 11 & 18 & 13 & 48 & 14 \\ 3 & 5 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

[759] $1^{11} 1^{12} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 5 & 10 & 13 & 5 \\ 5 & 10 & 14 & 19 & 8 \\ 7 & 13 & 19 & 24 & 10 \\ 3 & 5 & 8 & 10 & 4 \end{pmatrix}_{b_0}$$

[758] $1^{112} 1^{11} 3^1 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 1 & 3 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 24 & 5 \\ 8 & 9 & 0 & 24 & 83 & 17 \\ 2 & 2 & -1 & 5 & 17 & 3 \end{pmatrix}_{b_0}$$

[756] $1^{122} 1^{11} 3^1 21^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 15 & 3 \\ 3 & 5 & 15 & 28 & 5 \\ 8 & 15 & 34 & 63 & 12 \\ 15 & 28 & 63 & 116 & 22 \\ 3 & 5 & 12 & 22 & 4 \end{pmatrix}_{b_0}$$

[762] $1^{131} 1^{11} 3^1 21^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 0 & 1 & 10 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 10 & 0 & 88 & 22 & 18 \\ 2 & 2 & 0 & 22 & 4 & 4 \\ 2 & 2 & -1 & 18 & 4 & 3 \end{pmatrix}_{b_0}$$

[769] $1^{134} 1^{11} 3^1 21^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 17 & 3 \\ 3 & 5 & 15 & 28 & 5 \\ 9 & 15 & 34 & 63 & 12 \\ 17 & 28 & 63 & 116 & 22 \\ 3 & 5 & 12 & 22 & 4 \end{pmatrix}_{b_0}$$

[768] $1^{14} 1^{114} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 13 & 2 \\ 1 & 0 & 2 & 4 & 13 & 2 \\ 1 & 2 & 0 & 7 & 17 & 2 \\ 4 & 4 & 7 & 16 & 47 & 8 \\ 13 & 13 & 17 & 47 & 127 & 21 \\ 2 & 2 & 2 & 8 & 21 & 3 \end{pmatrix}_{b_0}$$

[767] $1^{121} 1^{31} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 5 & 4 & 2 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 6 & 5 & 5 & 13 & 13 & 7 \\ 4 & 4 & 5 & 13 & 10 & 6 \\ 2 & 2 & 2 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

[766] $1^{12} 1^{13} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 5 & 0 & 13 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 13 & 0 & 24 & 10 \\ 3 & 5 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

[765] $1^{1^{44}}3^{1^{122}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 0 & 2 & 4 & 7 & 2 \\ 1 & 2 & 0 & 7 & 9 & 2 \\ 7 & 4 & 7 & 16 & 25 & 8 \\ 9 & 7 & 9 & 25 & 35 & 11 \\ 2 & 2 & 2 & 8 & 11 & 3 \end{pmatrix}_{b_0}$

[770] $1^{1^{12}}3^{1^{122}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 4 & 7 & 2 \\ 1 & 2 & 0 & 7 & 9 & 2 \\ 4 & 4 & 7 & 16 & 25 & 8 \\ 7 & 7 & 9 & 25 & 35 & 11 \\ 2 & 2 & 2 & 8 & 11 & 3 \end{pmatrix}_{b_0}$

[763] $1^{1^{12}}3^{1^{123}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 5 & 10 & 23 & 5 \\ 5 & 10 & 14 & 34 & 8 \\ 13 & 23 & 34 & 78 & 18 \\ 3 & 5 & 8 & 18 & 4 \end{pmatrix}_{b_0}$

[771] $1^{1^{11}}3^{1^{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 1 & 4 & 2 \\ 1 & 2 & 0 & 3 & 5 & 2 \\ 2 & 1 & 3 & 1 & 5 & 3 \\ 4 & 4 & 5 & 5 & 10 & 6 \\ 2 & 2 & 2 & 3 & 6 & 3 \end{pmatrix}_{b_0}$

[772] $1^{1^{12}}3^{1^{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 5 & 4 & 2 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 4 & 5 & 5 & 13 & 13 & 7 \\ 5 & 4 & 5 & 13 & 10 & 6 \\ 2 & 2 & 2 & 7 & 6 & 3 \end{pmatrix}_{b_0}$

[773] $1^{1^{12}}3^{1^{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 1 & 4 & 2 \\ 1 & 2 & 0 & 3 & 5 & 2 \\ 2 & 1 & 3 & 1 & 5 & 3 \\ 5 & 4 & 5 & 5 & 10 & 6 \\ 2 & 2 & 2 & 3 & 6 & 3 \end{pmatrix}_{b_0}$

[774] $1^{1^{12}}3^{1^{122}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 5 & 10 & 13 & 5 \\ 5 & 10 & 14 & 20 & 8 \\ 8 & 13 & 20 & 24 & 10 \\ 3 & 5 & 8 & 10 & 4 \end{pmatrix}_{b_0}$

[764] $1^{1^{21}}3^{1^{131}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 5 & 15 & 8 & 5 \\ 8 & 15 & 34 & 18 & 12 \\ 5 & 8 & 18 & 8 & 6 \\ 3 & 5 & 12 & 6 & 4 \end{pmatrix}_{b_0}$

[782] $1^{1^2}3^{1^{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 12 & 20 & 62 & 24 & 16 \\ 4 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$

[775] $1^{1^3}3^{1^{131}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 5 & 15 & 8 & 5 \\ 9 & 15 & 34 & 18 & 12 \\ 4 & 8 & 18 & 8 & 6 \\ 3 & 5 & 12 & 6 & 4 \end{pmatrix}_{b_0}$

$$\begin{aligned}
[776] \quad & 1^1 3^1 21^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 12 & 20 & 62 & 24 & 16 \\ 4 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0} \\
[777] \quad & 1^1 3^1 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 5 & 10 & 8 & 5 \\ 6 & 10 & 14 & 12 & 8 \\ 4 & 8 & 12 & 8 & 6 \\ 3 & 5 & 8 & 6 & 4 \end{pmatrix}_{b_0} \\
[778] \quad & 1^{121} 3^1 21^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 3 & 2 \\ 1 & 0 & 2 & 12 & 2 & 2 \\ 1 & 2 & 0 & 16 & 4 & 2 \\ 10 & 12 & 16 & 110 & 24 & 20 \\ 3 & 2 & 4 & 24 & 4 & 4 \\ 2 & 2 & 2 & 20 & 4 & 3 \end{pmatrix}_{b_0} \\
[779] \quad & 1^1 3^1 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 5 & 5 & 8 & 5 \\ 3 & 5 & 2 & 6 & 4 \\ 4 & 8 & 6 & 8 & 6 \\ 3 & 5 & 4 & 6 & 4 \end{pmatrix}_{b_0} \\
[780] \quad & 1^1 3^1 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 5 & 10 & 8 & 5 \\ 6 & 10 & 14 & 12 & 8 \\ 4 & 8 & 12 & 8 & 6 \\ 3 & 5 & 8 & 6 & 4 \end{pmatrix}_{b_0} \\
[781] \quad & 1^{11} 3^1 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 2 & 6 & 2 & 2 \\ 1 & 2 & 0 & 8 & 4 & 2 \\ 5 & 6 & 8 & 26 & 12 & 10 \\ 3 & 2 & 4 & 12 & 4 & 4 \\ 2 & 2 & 2 & 10 & 4 & 3 \end{pmatrix}_{b_0} \\
[786] \quad & 1^{11} 3^1 21^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 10 & 20 & 58 & 24 & 16 \\ 5 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0} \\
[783] \quad & 1^{121} 3^1 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 5 & 10 & 8 & 5 \\ 7 & 10 & 14 & 12 & 8 \\ 5 & 8 & 12 & 8 & 6 \\ 3 & 5 & 8 & 6 & 4 \end{pmatrix}_{b_0} \\
[784] \quad & 1^{13} 3^1 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 3 & 9 & 2 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 2 & 3 & 4 & 5 & 19 & 5 \\ 9 & 9 & 10 & 19 & 51 & 13 \\ 2 & 2 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0} \\
[785] \quad & 1^{134} 3^1 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 3 & 9 & 2 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 5 & 3 & 4 & 5 & 19 & 5 \\ 10 & 9 & 10 & 19 & 51 & 13 \\ 2 & 2 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0} \\
[787] \quad & 1^{131} 3^1 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 5 & 10 & 8 & 5 \\ 7 & 10 & 14 & 12 & 8 \\ 5 & 8 & 12 & 8 & 6 \\ 3 & 5 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[788] \ 1^{131} 3^{11} 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 35, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 21 & 4 \\ 3 & 1 & 3 & 15 & 3 \\ 3 & 3 & 1 & 18 & 3 \\ 21 & 15 & 18 & 97 & 21 \\ 4 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[791] \ 1^{143} 3^{11} 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 36, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 24 & 4 \\ 3 & 1 & 3 & 18 & 3 \\ 3 & 3 & 1 & 20 & 3 \\ 24 & 18 & 20 & 130 & 24 \\ 4 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[790] \ 1^{14} 3^{11} 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 36, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 1 & 13 & 2 \\ 1 & 2 & 0 & 3 & 17 & 2 \\ 1 & 1 & 3 & 1 & 17 & 3 \\ 13 & 13 & 17 & 17 & 127 & 21 \\ 2 & 2 & 2 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[789] \ 1^{123} 3^{11} 21^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 36, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 7 & 3 \\ 3 & 5 & 25 & 13 & 5 \\ 13 & 25 & 94 & 48 & 20 \\ 7 & 13 & 48 & 24 & 10 \\ 3 & 5 & 20 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[799] \ 1^{122} 3^1 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 38, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 5 & 5 & 13 & 5 \\ 4 & 5 & 2 & 10 & 4 \\ 8 & 13 & 10 & 24 & 10 \\ 3 & 5 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[792] \ 1^{122} 3^1 21^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 38, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 5 & 15 & 13 & 5 \\ 8 & 15 & 34 & 30 & 12 \\ 8 & 13 & 30 & 24 & 10 \\ 3 & 5 & 12 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[793] \ 1^{112} 3^{11} 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 38, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 5 & 10 & 13 & 5 \\ 6 & 10 & 14 & 19 & 8 \\ 8 & 13 & 19 & 24 & 10 \\ 3 & 5 & 8 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[794] \ 1^{112} 3^{11} 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 38, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 5 & 15 & 13 & 5 \\ 8 & 15 & 32 & 29 & 12 \\ 8 & 13 & 29 & 24 & 10 \\ 3 & 5 & 12 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[795] \ 1^{111} 3^{11} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 38, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 6 \\ 2 & 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[796] \ 1^{111} 3^{11} 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 38, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 4 & 4 & 2 \\ 1 & 2 & 0 & 7 & 5 & 2 \\ 6 & 4 & 7 & 16 & 14 & 8 \\ 4 & 4 & 5 & 14 & 10 & 6 \\ 2 & 2 & 2 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[797] \quad 1^{\overline{13}} 1^{\overline{14}} 3^{\overline{21}} 21^{\overline{113}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 38, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\left(\begin{array}{cccccc} 1 & 3 & 2 & 14 & 3 & \\ 3 & 5 & 5 & 23 & 5 & \\ 2 & 5 & 2 & 17 & 4 & \\ 14 & 23 & 17 & 78 & 18 & \\ 3 & 5 & 4 & 18 & 4 & \end{array} \right)_{b_0}$$

$$[798] \quad 1^{121} 3^{121} 1^{411} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, \, T_{\max} = 10, \, \Delta = -59, \, \det = 38, \, (n_+, n_-, n_0) = (1, 8, 0), \, c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.95 \rightarrow 10, \, T_{\max} = 10, \, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 8 & 4 & 2 \\ 1 & 2 & 0 & 14 & 5 & 2 \\ 9 & 8 & 14 & 70 & 28 & 16 \\ 4 & 4 & 5 & 28 & 10 & 6 \\ 2 & 2 & 2 & 16 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[800] \ 1^{\overline{1}12} 3^{\overline{1}1} 21^{\overline{1}1} + [\textcolor{red}{\mathfrak{s}}\textcolor{red}{\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}}\textcolor{green}{\mathfrak{u}}_2 + \textcolor{orange}{\mathfrak{s}}\textcolor{orange}{\mathfrak{u}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 38, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 2 & 1 & 0 & 4 & 2 \\ 5 & 4 & 5 & 4 & 10 & 6 \\ 2 & 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[801] \ 1^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{121}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 + \textcolor{brown}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 38, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 4 & 4 & 2 \\ 1 & 2 & 0 & 7 & 5 & 2 \\ 4 & 4 & 7 & 16 & 14 & 8 \\ 4 & 4 & 5 & 14 & 10 & 6 \\ 2 & 2 & 2 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[802] \ 1^{122} 1^{131} 3^{121} 21^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 38, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 5 & 2 \\ 1 & 0 & 2 & 8 & 4 & 2 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 7 & 8 & 9 & 40 & 22 & 12 \\ 5 & 4 & 5 & 22 & 10 & 6 \\ 2 & 2 & 2 & 12 & 6 & 3 \end{pmatrix}_{b_0}$$

[803] $1^{\overline{111}1}3^{\overline{11}1}21^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 38, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 2 & 1 & 0 & 4 & 2 \\ 4 & 4 & 5 & 4 & 10 & 6 \\ 2 & 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

$$[804] \quad 1^{142} 1^{111} 3^{121} 21^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 38, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 5 & 2 \\ 1 & 0 & 2 & 4 & 4 & 2 \\ 1 & 2 & 0 & 7 & 5 & 2 \\ 7 & 4 & 7 & 16 & 14 & 8 \\ 5 & 4 & 5 & 14 & 10 & 6 \\ 2 & 2 & 2 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

[805] $1^{111}3^{122}21^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 2 & 4 & 4 & 2 \\ 1 & 2 & 0 & 7 & 5 & 2 \\ 5 & 4 & 7 & 16 & 14 & 8 \\ 5 & 4 & 5 & 14 & 10 & 6 \\ 2 & 2 & 2 & 8 & 6 & 3 \end{pmatrix}_{b_0}$

$$[806] \quad 1^{11} 3^{12} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 38, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 6 \\ 2 & 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[807] \ 1^{\overline{12} \overline{3} \overline{1} \overline{21} \overline{132}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 38, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 8 & 3 \\ 3 & 5 & 25 & 13 & 5 \\ 16 & 25 & 94 & 48 & 20 \\ 8 & 13 & 48 & 24 & 10 \\ 3 & 5 & 20 & 10 & 4 \end{pmatrix}_{b_0}$$

[808] $21^{123121}1^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 24 & 6 & 4 \\ 5 & 5 & 30 & 8 & 5 \\ 24 & 30 & 134 & 36 & 24 \\ 6 & 8 & 36 & 8 & 6 \\ 4 & 5 & 24 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[809] \ 1^{11} 3^{11} 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 5 & 5 & 8 & 5 \\ 3 & 5 & 2 & 6 & 4 \\ 5 & 8 & 6 & 8 & 6 \\ 3 & 5 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[810] \ 1^{41} 3^{11} 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 25 & 4 \\ 3 & 1 & 3 & 17 & 3 \\ 3 & 3 & 1 & 22 & 3 \\ 25 & 17 & 22 & 137 & 25 \\ 4 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[811] \ 1^{121} 3^{11} 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 & 2 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 4 & 3 & 4 & 5 & 6 & 5 \\ 3 & 2 & 4 & 6 & 4 & 4 \\ 2 & 2 & 2 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[812] \ 1^{111} 3^{11} 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 & 2 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 3 & 3 & 4 & 5 & 6 & 5 \\ 3 & 2 & 4 & 6 & 4 & 4 \\ 2 & 2 & 2 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[814] \ 1^3 3^{11} 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 2 & 5 & 2 & 14 & 4 \\ 11 & 18 & 14 & 48 & 14 \\ 3 & 5 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[813] \ 1^{1123} 3^{11} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 2 & 7 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 2 & 1 & 0 & 7 & 2 \\ 9 & 7 & 9 & 7 & 35 & 11 \\ 2 & 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[815] \ 1^{112} 3^{11} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 2 & 7 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 2 & 2 & 1 & 0 & 7 & 2 \\ 7 & 7 & 9 & 7 & 35 & 11 \\ 2 & 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[817] \ 1^{1113} 3^{11} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 3 & 5 & 2 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 5 & 3 & 4 & 5 & 13 & 5 \\ 7 & 5 & 8 & 13 & 23 & 9 \\ 2 & 2 & 2 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[818] \ 1^{113} 3^{11} 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 5 & 10 & 18 & 5 \\ 5 & 10 & 14 & 28 & 8 \\ 11 & 18 & 28 & 48 & 14 \\ 3 & 5 & 8 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[819] \ 1^{12} 3^{11} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 3 & 5 & 2 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 2 & 3 & 4 & 5 & 13 & 5 \\ 6 & 5 & 8 & 13 & 23 & 9 \\ 2 & 2 & 2 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[820] \quad 1^{11} 3^{12} 1^{21} 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 11 & 20 & 62 & 24 & 16 \\ 5 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[816] \quad 1^{31} 3^{12} 1^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 11 & 20 & 62 & 24 & 16 \\ 5 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[821] \quad 1^{11} 3^{12} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 2 & 5 & 2 & 13 & 4 \\ 10 & 18 & 13 & 48 & 14 \\ 3 & 5 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[822] \quad 1^{13} 3^{12} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 2 & 8 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 2 & 1 & 0 & 8 & 2 \\ 9 & 8 & 12 & 8 & 56 & 14 \\ 2 & 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[827] \quad 1^{11} 3^{12} 1^{21} 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 5 & 0 & 18 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 18 & 0 & 48 & 14 \\ 3 & 5 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[826] \quad 1^{13} 3^{12} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 5 & 0 & 18 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 18 & 0 & 48 & 14 \\ 3 & 5 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[823] \quad 1^{21} 3^{12} 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 5 & 10 & 8 & 5 \\ 6 & 10 & 14 & 12 & 8 \\ 5 & 8 & 12 & 8 & 6 \\ 3 & 5 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[824] \quad 1^{12} 3^{12} 1^{21} 1^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 6 & 5 & 2 \\ 1 & 2 & 0 & 8 & 8 & 2 \\ 5 & 6 & 8 & 26 & 26 & 10 \\ 6 & 5 & 8 & 26 & 23 & 9 \\ 2 & 2 & 2 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[825] \quad 1^{11} 3^{12} 1^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 5 & 10 & 8 & 5 \\ 6 & 10 & 14 & 12 & 8 \\ 5 & 8 & 12 & 8 & 6 \\ 3 & 5 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[829] \quad 1^{32} 3^{12} 1^{32} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 8 & 3 \\ 3 & 5 & 25 & 13 & 5 \\ 15 & 25 & 94 & 48 & 20 \\ 8 & 13 & 48 & 24 & 10 \\ 3 & 5 & 20 & 10 & 4 \end{pmatrix}_{b_0}$$

[830] $1^{141} 3^1 21^{131} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 5 & 15 & 8 & 5 \\ 10 & 15 & 34 & 18 & 12 \\ 5 & 8 & 18 & 8 & 6 \\ 3 & 5 & 12 & 6 & 4 \end{pmatrix}_{b_0}$

[831] $1^{132} 3^1 21^{142} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 8 & 3 \\ 3 & 5 & 20 & 13 & 5 \\ 11 & 20 & 62 & 40 & 16 \\ 8 & 13 & 40 & 24 & 10 \\ 3 & 5 & 16 & 10 & 4 \end{pmatrix}_{b_0}$

[832] $1^{134} 3^1 21^{123} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 15 & 3 \\ 3 & 5 & 10 & 23 & 5 \\ 7 & 10 & 14 & 34 & 8 \\ 15 & 23 & 34 & 78 & 18 \\ 3 & 5 & 8 & 18 & 4 \end{pmatrix}_{b_0}$

[833] $1^{132} 3^1 21^{122} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 5 & 10 & 13 & 5 \\ 7 & 10 & 14 & 20 & 8 \\ 8 & 13 & 20 & 24 & 10 \\ 3 & 5 & 8 & 10 & 4 \end{pmatrix}_{b_0}$

[834] $1^{131} 3^1 21^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 7 & 4 & 2 \\ 1 & 2 & 0 & 6 & 5 & 2 \\ 6 & 7 & 6 & 25 & 17 & 9 \\ 4 & 4 & 5 & 17 & 10 & 6 \\ 2 & 2 & 2 & 9 & 6 & 3 \end{pmatrix}_{b_0}$

[835] $1^{144} 3^1 21^{134} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 8 & 11 & 2 \\ 1 & 0 & 2 & 5 & 9 & 2 \\ 1 & 2 & 0 & 10 & 15 & 2 \\ 8 & 5 & 10 & 33 & 54 & 11 \\ 11 & 9 & 15 & 54 & 83 & 17 \\ 2 & 2 & 2 & 11 & 17 & 3 \end{pmatrix}_{b_0}$

[836] $1^{111} 3^1 21^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 3 & 4 & 2 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 2 & 3 & 4 & 5 & 9 & 5 \\ 4 & 4 & 5 & 9 & 10 & 6 \\ 2 & 2 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$

[837] $1^{111} 3^1 21^{131} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 5 & 25 & 8 & 5 \\ 13 & 25 & 94 & 30 & 20 \\ 5 & 8 & 30 & 8 & 6 \\ 3 & 5 & 20 & 6 & 4 \end{pmatrix}_{b_0}$

[838] $1^{141} 3^1 21^{131} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 5 & 2 \\ 1 & 0 & 2 & 11 & 4 & 2 \\ 1 & 2 & 0 & 13 & 5 & 2 \\ 10 & 11 & 13 & 81 & 31 & 17 \\ 5 & 4 & 5 & 31 & 10 & 6 \\ 2 & 2 & 2 & 17 & 6 & 3 \end{pmatrix}_{b_0}$

[839] $1^{132} 3^1 21^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 5 & 2 \\ 1 & 0 & 2 & 7 & 4 & 2 \\ 1 & 2 & 0 & 6 & 5 & 2 \\ 7 & 7 & 6 & 25 & 17 & 9 \\ 5 & 4 & 5 & 17 & 10 & 6 \\ 2 & 2 & 2 & 9 & 6 & 3 \end{pmatrix}_{b_0}$

[840] $1^{121} 3^1 21^{131} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 0 & 2 & 7 & 4 & 2 \\ 1 & 2 & 0 & 11 & 5 & 2 \\ 8 & 7 & 11 & 45 & 23 & 13 \\ 5 & 4 & 5 & 23 & 10 & 6 \\ 2 & 2 & 2 & 13 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[841] \ 1^{123} 3^{121} 1^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 3 \\ 3 & 5 & 15 & 18 & 5 \\ 8 & 15 & 34 & 42 & 12 \\ 11 & 18 & 42 & 48 & 14 \\ 3 & 5 & 12 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[828] \ 1^{111} 3^{121} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 2 & 3 & 4 & 2 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 5 & 3 & 4 & 5 & 9 & 5 \\ 5 & 4 & 5 & 9 & 10 & 6 \\ 2 & 2 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[842] \ 1^{141} 3^{121} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 28 & 4 \\ 3 & 1 & 3 & 20 & 3 \\ 3 & 3 & 1 & 24 & 3 \\ 28 & 20 & 24 & 174 & 28 \\ 4 & 3 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[843] \ 1^{123} 3^{121} 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 4 & 5 & 2 & 14 & 4 \\ 11 & 18 & 14 & 48 & 14 \\ 3 & 5 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[844] \ 1^{123} 3^{121} 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 4 & 5 & 2 & 17 & 4 \\ 14 & 23 & 17 & 78 & 18 \\ 3 & 5 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[846] \ 1^{12} 3^{121} 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 5 & 15 & 8 & 5 \\ 9 & 15 & 32 & 18 & 12 \\ 4 & 8 & 18 & 8 & 6 \\ 3 & 5 & 12 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[847] \ 1^{112} 3^{121} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 6 & 6 & 0 & 20 & 8 \\ 2 & 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[845] \ 1^{112} 3^{121} 1^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 5 & 20 & 18 & 5 \\ 10 & 20 & 58 & 54 & 16 \\ 10 & 18 & 54 & 48 & 14 \\ 3 & 5 & 16 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[848] \ 1^{131} 3^{121} 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 5 & 15 & 8 & 5 \\ 9 & 15 & 34 & 18 & 12 \\ 5 & 8 & 18 & 8 & 6 \\ 3 & 5 & 12 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[852] \ 1^{111} 3^{121} 1^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 5 & 5 & 2 \\ 1 & 2 & 0 & 10 & 8 & 2 \\ 7 & 5 & 10 & 33 & 29 & 11 \\ 7 & 5 & 8 & 29 & 23 & 9 \\ 2 & 2 & 2 & 11 & 9 & 3 \end{pmatrix}_{b_0}$$

[853] $1^{112} 3^{121} 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 9 \\ 2 & 2 & 2 & 0 & 9 & 3 \end{pmatrix}_{b_0}$$

[849] $1^{113} 3^{121} 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 5 & 8 & 0 & 23 & 9 \\ 2 & 2 & 2 & 0 & 9 & 3 \end{pmatrix}_{b_0}$$

[850] $1^{133} 3^{121} 21^{143} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 11 & 3 \\ 3 & 5 & 20 & 18 & 5 \\ 11 & 20 & 62 & 56 & 16 \\ 11 & 18 & 56 & 48 & 14 \\ 3 & 5 & 16 & 14 & 4 \end{pmatrix}_{b_0}$$

[851] $1^{142} 3^{121} 21^{132} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & 0 & 2 & 5 & 5 & 2 \\ 1 & 2 & 0 & 10 & 8 & 2 \\ 8 & 5 & 10 & 33 & 29 & 11 \\ 6 & 5 & 8 & 29 & 23 & 9 \\ 2 & 2 & 2 & 11 & 9 & 3 \end{pmatrix}_{b_0}$$

[855] $1^{113} 3^{121} 21^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 3 & 10 & 2 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 2 & 3 & 4 & 5 & 23 & 5 \\ 11 & 10 & 13 & 23 & 74 & 16 \\ 2 & 2 & 2 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[854] $1^{111} 3^{121} 21^{121} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 5 & 15 & 8 & 5 \\ 8 & 15 & 32 & 18 & 12 \\ 5 & 8 & 18 & 8 & 6 \\ 3 & 5 & 12 & 6 & 4 \end{pmatrix}_{b_0}$$

[857] $1^{112} 3^{121} 21^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 3 & 9 & 2 \\ 1 & 2 & 0 & 4 & 15 & 2 \\ 2 & 3 & 4 & 5 & 24 & 5 \\ 11 & 9 & 15 & 24 & 83 & 17 \\ 2 & 2 & 2 & 5 & 17 & 3 \end{pmatrix}_{b_0}$$

[856] $1^{114} 3^{121} 21^{124} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 5 & 10 & 23 & 5 \\ 5 & 10 & 14 & 36 & 8 \\ 14 & 23 & 36 & 80 & 18 \\ 3 & 5 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

[861] $1^{113} 3^{121} 21^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 2 & 2 & 1 & 0 & 10 & 2 \\ 10 & 10 & 13 & 10 & 74 & 16 \\ 2 & 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[860] $1^{124} 3^{121} 21^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 20 & 3 \\ 3 & 5 & 5 & 33 & 5 \\ 4 & 5 & 2 & 24 & 4 \\ 20 & 33 & 24 & 164 & 26 \\ 3 & 5 & 4 & 26 & 4 \end{pmatrix}_{b_0}$$

[859] $1^{13}3^{121}21^{123} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 11 & 2 \\ 1 & 0 & 2 & 4 & 10 & 2 \\ 1 & 2 & 0 & 7 & 13 & 2 \\ 6 & 4 & 7 & 16 & 36 & 8 \\ 11 & 10 & 13 & 36 & 74 & 16 \\ 2 & 2 & 2 & 8 & 16 & 3 \end{pmatrix}_{b_0}$$

[858] $1^{14}3^{121}21^{114} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 2 & 18 & 4 \\ 14 & 23 & 18 & 80 & 18 \\ 3 & 5 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

[866] $1^{12}3^{121}21^{123} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 5 & 10 & 23 & 5 \\ 6 & 10 & 14 & 34 & 8 \\ 13 & 23 & 34 & 78 & 18 \\ 3 & 5 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

[862] $1^{11}3^{121}21^{112} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 5 & 5 & 13 & 5 \\ 3 & 5 & 2 & 10 & 4 \\ 7 & 13 & 10 & 24 & 10 \\ 3 & 5 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

[863] $1^{132}3^{121}21^{122} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 7 & 2 \\ 1 & 0 & 2 & 10 & 7 & 2 \\ 1 & 2 & 0 & 10 & 9 & 2 \\ 9 & 10 & 10 & 58 & 46 & 14 \\ 7 & 7 & 9 & 46 & 35 & 11 \\ 2 & 2 & 2 & 14 & 11 & 3 \end{pmatrix}_{b_0}$$

[870] $1^{141}3^{121}21^{142} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 7 & 3 \\ 3 & 5 & 20 & 13 & 5 \\ 12 & 20 & 62 & 40 & 16 \\ 7 & 13 & 40 & 24 & 10 \\ 3 & 5 & 16 & 10 & 4 \end{pmatrix}_{b_0}$$

[864] $1^{12}3^{121}21^{112} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 1 & 7 & 2 \\ 1 & 2 & 0 & 3 & 9 & 2 \\ 2 & 1 & 3 & 1 & 9 & 3 \\ 7 & 7 & 9 & 9 & 35 & 11 \\ 2 & 2 & 2 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

[869] $1^{31}3^{121}21^{132} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 5 & 15 & 13 & 5 \\ 9 & 15 & 34 & 30 & 12 \\ 7 & 13 & 30 & 24 & 10 \\ 3 & 5 & 12 & 10 & 4 \end{pmatrix}_{b_0}$$

[868] $1^{142}3^{121}21^{142} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 7 & 2 \\ 1 & 0 & 2 & 12 & 7 & 2 \\ 1 & 2 & 0 & 16 & 9 & 2 \\ 10 & 12 & 16 & 110 & 64 & 20 \\ 7 & 7 & 9 & 64 & 35 & 11 \\ 2 & 2 & 2 & 20 & 11 & 3 \end{pmatrix}_{b_0}$$

[867] $1^{21}3^{121}21^{122} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 5 & 10 & 13 & 5 \\ 6 & 10 & 14 & 20 & 8 \\ 7 & 13 & 20 & 24 & 10 \\ 3 & 5 & 8 & 10 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[865] \quad & 1^{14} 1^{12} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 1 & 7 & 2 \\ 1 & 2 & 0 & 3 & 9 & 2 \\ 2 & 1 & 3 & 1 & 9 & 3 \\ 9 & 7 & 9 & 9 & 35 & 11 \\ 2 & 2 & 2 & 3 & 11 & 3 \end{pmatrix}_{b_0} \\
[871] \quad & 1^{42} 1^{131} 3^1 21^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 74, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 5 & 2 \\ 1 & 0 & 2 & 10 & 4 & 2 \\ 1 & 2 & 0 & 15 & 5 & 2 \\ 11 & 10 & 15 & 88 & 32 & 18 \\ 5 & 4 & 5 & 32 & 10 & 6 \\ 2 & 2 & 2 & 18 & 6 & 3 \end{pmatrix}_{b_0} \\
[872] \quad & 1^{42} 1^1 3^1 21^{132} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 74, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 5 & 15 & 13 & 5 \\ 10 & 15 & 34 & 30 & 12 \\ 8 & 13 & 30 & 24 & 10 \\ 3 & 5 & 12 & 10 & 4 \end{pmatrix}_{b_0} \\
[873] \quad & 1^{24} 1^1 3^1 21^{134} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 74, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 8 & 14 & 3 \\ 3 & 5 & 15 & 23 & 5 \\ 8 & 15 & 34 & 54 & 12 \\ 14 & 23 & 54 & 80 & 18 \\ 3 & 5 & 12 & 18 & 4 \end{pmatrix}_{b_0} \\
[874] \quad & 1^{13} 1^{112} 3^1 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 75, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 2 & 7 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 2 & 2 & 1 & 0 & 7 & 2 \\ 8 & 7 & 9 & 7 & 35 & 11 \\ 2 & 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[875] \quad & 1^4 1^{11} 3^1 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 75, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 2 & 5 & 2 & 21 & 4 \\ 17 & 28 & 21 & 116 & 22 \\ 3 & 5 & 4 & 22 & 4 \end{pmatrix}_{b_0} \\
[876] \quad & 1^{13} 1^{112} 3^1 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 75, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 2 & 7 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 2 & 1 & 0 & 7 & 2 \\ 8 & 7 & 9 & 7 & 35 & 11 \\ 2 & 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0} \\
[877] \quad & 1^{21} 1^{13} 3^1 21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 76, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.8 \\
& T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 0 & 2 & 9 & 2 & 2 \\ 1 & 2 & 0 & 12 & 4 & 2 \\ 8 & 9 & 12 & 61 & 18 & 15 \\ 3 & 2 & 4 & 18 & 4 & 4 \\ 2 & 2 & 2 & 15 & 4 & 3 \end{pmatrix}_{b_0} \\
[878] \quad & 1^{33} 1^{122} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 76, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 0 & 2 & 5 & 7 & 2 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 6 & 5 & 5 & 13 & 23 & 7 \\ 8 & 7 & 9 & 23 & 35 & 11 \\ 2 & 2 & 2 & 7 & 11 & 3 \end{pmatrix}_{b_0} \\
[879] \quad & 1^{13} 1^{122} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 76, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 2 & 5 & 7 & 2 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 4 & 5 & 5 & 13 & 23 & 7 \\ 8 & 7 & 9 & 23 & 35 & 11 \\ 2 & 2 & 2 & 7 & 11 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[881] $1^{14} 1^1 3^1 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 5 & 0 & 23 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 23 & 0 & 80 & 18 \\ 3 & 5 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

[882] $1^{12} 1^1 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 5 & 0 & 23 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 23 & 0 & 78 & 18 \\ 3 & 5 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

[880] $1^{12} 1^{12} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 0 & 2 & 8 & 7 & 2 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 6 & 8 & 9 & 40 & 39 & 12 \\ 7 & 7 & 9 & 39 & 35 & 11 \\ 2 & 2 & 2 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

[883] $1^{21} 1^{11} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 2 & 4 & 4 & 2 \\ 1 & 2 & 0 & 7 & 5 & 2 \\ 5 & 4 & 7 & 16 & 14 & 8 \\ 4 & 4 & 5 & 14 & 10 & 6 \\ 2 & 2 & 2 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

[890] $1^{31} 1^{12} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 13 & 20 & 62 & 24 & 16 \\ 5 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$$

[889] $1^{22} 1^{13} 3^1 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & 0 & 2 & 9 & 5 & 2 \\ 1 & 2 & 0 & 12 & 8 & 2 \\ 8 & 9 & 12 & 61 & 39 & 15 \\ 6 & 5 & 8 & 39 & 23 & 9 \\ 2 & 2 & 2 & 15 & 9 & 3 \end{pmatrix}_{b_0}$$

[888] $1^{32} 1^{11} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 0 & 2 & 4 & 4 & 2 \\ 1 & 2 & 0 & 7 & 5 & 2 \\ 6 & 4 & 7 & 16 & 14 & 8 \\ 5 & 4 & 5 & 14 & 10 & 6 \\ 2 & 2 & 2 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

[887] $1^{33} 1^1 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 5 & 10 & 18 & 5 \\ 7 & 10 & 14 & 28 & 8 \\ 11 & 18 & 28 & 48 & 14 \\ 3 & 5 & 8 & 14 & 4 \end{pmatrix}_{b_0}$$

[886] $1^{32} 1^{14} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 5 & 2 \\ 1 & 0 & 2 & 10 & 4 & 2 \\ 1 & 2 & 0 & 10 & 5 & 2 \\ 9 & 10 & 10 & 58 & 26 & 14 \\ 5 & 4 & 5 & 26 & 10 & 6 \\ 2 & 2 & 2 & 14 & 6 & 3 \end{pmatrix}_{b_0}$$

[885] $1^{41} 1^{12} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 8 & 4 & 2 \\ 1 & 2 & 0 & 14 & 5 & 2 \\ 10 & 8 & 14 & 70 & 28 & 16 \\ 4 & 4 & 5 & 28 & 10 & 6 \\ 2 & 2 & 2 & 16 & 6 & 3 \end{pmatrix}_{b_0}$$

[884] $1^{32} 1^{11} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 5 & 15 & 13 & 5 \\ 10 & 15 & 32 & 29 & 12 \\ 8 & 13 & 29 & 24 & 10 \\ 3 & 5 & 12 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[895] \ 1^{12} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 0 & 2 & 6 & 2 & 2 \\ 1 & 2 & 0 & 8 & 4 & 2 \\ 8 & 6 & 8 & 26 & 12 & 10 \\ 3 & 2 & 4 & 12 & 4 & 4 \\ 2 & 2 & 2 & 10 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[894] \ 1^{12} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 8 & 2 \\ 1 & 0 & 2 & 10 & 7 & 2 \\ 1 & 2 & 0 & 10 & 9 & 2 \\ 10 & 10 & 10 & 58 & 46 & 14 \\ 8 & 7 & 9 & 46 & 35 & 11 \\ 2 & 2 & 2 & 14 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[893] \ 1^{12} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 1 & 7 & 2 \\ 1 & 2 & 0 & 3 & 9 & 2 \\ 2 & 1 & 3 & 1 & 9 & 3 \\ 8 & 7 & 9 & 35 & 11 \\ 2 & 2 & 2 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[892] \ 1^{12} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 13 & 2 \\ 1 & 0 & 2 & 7 & 13 & 2 \\ 1 & 2 & 0 & 11 & 17 & 2 \\ 7 & 7 & 11 & 45 & 77 & 13 \\ 13 & 13 & 17 & 77 & 127 & 21 \\ 2 & 2 & 2 & 13 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[891] \ 1^{12} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 2 & 6 & 2 & 2 \\ 1 & 2 & 0 & 8 & 4 & 2 \\ 6 & 6 & 8 & 26 & 12 & 10 \\ 3 & 2 & 4 & 12 & 4 & 4 \\ 2 & 2 & 2 & 10 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[897] \ 1^{12} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 5 & 5 & 13 & 5 \\ 3 & 5 & 2 & 10 & 4 \\ 8 & 13 & 10 & 24 & 10 \\ 3 & 5 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[896] \ 1^{12} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 2 & 3 & 4 & 2 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 3 & 4 & 5 & 9 & 5 \\ 5 & 4 & 5 & 9 & 10 & 6 \\ 2 & 2 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[898] \ 1^{12} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 3 & 4 & 2 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 4 & 3 & 4 & 5 & 9 & 5 \\ 4 & 4 & 5 & 9 & 10 & 6 \\ 2 & 2 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[900] \ 1^{12} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 5 & 0 & 23 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 23 & 0 & 78 & 18 \\ 3 & 5 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[899] \ 1^{12} 3^{12} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 8 & 2 \\ 1 & 0 & 2 & 8 & 7 & 2 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 7 & 8 & 9 & 40 & 39 & 12 \\ 8 & 7 & 9 & 39 & 35 & 11 \\ 2 & 2 & 2 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

[901] $1^{13} 1^{11} 3^{12} 21^{124} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 16 & 3 \\ 3 & 5 & 10 & 28 & 5 \\ 5 & 10 & 14 & 42 & 8 \\ 16 & 28 & 42 & 116 & 22 \\ 3 & 5 & 8 & 22 & 4 \end{pmatrix}_{b_0}$

[904] $1^{34} 1^{13} 3^{12} 21^{144} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 14 & 3 \\ 3 & 5 & 20 & 23 & 5 \\ 11 & 20 & 62 & 72 & 16 \\ 14 & 23 & 72 & 80 & 18 \\ 3 & 5 & 16 & 18 & 4 \end{pmatrix}_{b_0}$

[903] $1^{13} 1^{11} 3^{12} 21^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 1 & 5 & 2 \\ 1 & 2 & 0 & 3 & 8 & 2 \\ 2 & 1 & 3 & 1 & 8 & 3 \\ 7 & 5 & 8 & 8 & 23 & 9 \\ 2 & 2 & 2 & 3 & 9 & 3 \end{pmatrix}_{b_0}$

[902] $1^{12} 1^{11} 3^{12} 21^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 1 & 5 & 2 \\ 1 & 2 & 0 & 3 & 8 & 2 \\ 2 & 1 & 3 & 1 & 8 & 3 \\ 6 & 5 & 8 & 8 & 23 & 9 \\ 2 & 2 & 2 & 3 & 9 & 3 \end{pmatrix}_{b_0}$

[905] $1^{41} 1^{13} 3^{12} 21^{141} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 12 & 20 & 62 & 24 & 16 \\ 5 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$

[906] $1^{21} 1^{12} 3^{12} 21^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 12 & 20 & 62 & 24 & 16 \\ 5 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$

[907] $1^{12} 1^{11} 3^{12} 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 2 & 7 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 2 & 1 & 0 & 7 & 2 \\ 7 & 7 & 9 & 7 & 35 & 11 \\ 2 & 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$

[908] $1^{12} 1^{11} 3^{12} 21^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 2 & 5 & 2 & 17 & 4 \\ 13 & 23 & 17 & 78 & 18 \\ 3 & 5 & 4 & 18 & 4 \end{pmatrix}_{b_0}$

[909] $1^{14} 1^{11} 3^{12} 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 2 & 7 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 2 & 2 & 1 & 0 & 7 & 2 \\ 9 & 7 & 9 & 7 & 35 & 11 \\ 2 & 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$

[910] $1^{41} 1^{12} 3^{12} 21^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 6 & 4 & 2 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 8 & 6 & 8 & 26 & 18 & 10 \\ 4 & 4 & 5 & 18 & 10 & 6 \\ 2 & 2 & 2 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[911] \ 1^{\overline{1}24} 3^{\overline{1}} 21^{\overline{114}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 90, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 4 & 5 & 2 & 18 & 4 \\ 14 & 23 & 18 & 80 & 18 \\ 3 & 5 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[912] \ 1^{\overline{1}22} 3^{\overline{1}} 21^{\overline{122}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 90, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 5 & 10 & 13 & 5 \\ 6 & 10 & 14 & 20 & 8 \\ 8 & 13 & 20 & 24 & 10 \\ 3 & 5 & 8 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[913] \ 1^{\overline{1}22} 3^{\overline{1}21} 21^{\overline{121}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 90, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 0 & 2 & 6 & 4 & 2 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 6 & 6 & 8 & 26 & 18 & 10 \\ 5 & 4 & 5 & 18 & 10 & 6 \\ 2 & 2 & 2 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[914] \ 1^{\overline{1}23} 3^{\overline{1}21} 21^{\overline{123}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 90, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 5 & 10 & 23 & 5 \\ 6 & 10 & 14 & 34 & 8 \\ 14 & 23 & 34 & 78 & 18 \\ 3 & 5 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[918] \ 1^{\overline{1}21} 3^{\overline{1}21} 21^{\overline{141}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 91, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 5 & 25 & 8 & 5 \\ 13 & 25 & 92 & 30 & 20 \\ 5 & 8 & 30 & 8 & 6 \\ 3 & 5 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[915] \ 1^{\overline{1}23} 3^{\overline{1}21} 21^{\overline{133}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 91, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 10 & 2 \\ 1 & 0 & 2 & 7 & 10 & 2 \\ 1 & 2 & 0 & 11 & 13 & 2 \\ 7 & 7 & 11 & 45 & 59 & 13 \\ 10 & 10 & 13 & 59 & 74 & 16 \\ 2 & 2 & 2 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[916] \ 1^{\overline{1}33} 3^{\overline{1}21} 21^{\overline{122}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 91, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 0 & 2 & 4 & 7 & 2 \\ 1 & 2 & 0 & 7 & 9 & 2 \\ 6 & 4 & 7 & 16 & 25 & 8 \\ 8 & 7 & 9 & 25 & 35 & 11 \\ 2 & 2 & 2 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[917] \ 1^{\overline{1}23} 3^{\overline{1}21} 21^{\overline{122}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 91, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 4 & 7 & 2 \\ 1 & 2 & 0 & 7 & 9 & 2 \\ 5 & 4 & 7 & 16 & 25 & 8 \\ 8 & 7 & 9 & 25 & 35 & 11 \\ 2 & 2 & 2 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[919] \ 1^{\overline{1}31} 3^{\overline{1}21} 21^{\overline{121}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 91, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 5 & 15 & 8 & 5 \\ 10 & 15 & 32 & 18 & 12 \\ 5 & 8 & 18 & 8 & 6 \\ 3 & 5 & 12 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[920] \ 21^{\overline{122}} 3^{\overline{1}21} 21^{\overline{112}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 94, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 7 & 2 \\ 2 & 0 & 2 & 5 & 7 & 2 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 5 & 5 & 5 & 13 & 23 & 7 \\ 7 & 7 & 9 & 23 & 35 & 11 \\ 2 & 2 & 2 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[924] $1^{131} 3^{121} 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 2 & 6 & 2 & 2 \\ 1 & 2 & 0 & 8 & 4 & 2 \\ 7 & 6 & 8 & 26 & 12 & 10 \\ 3 & 2 & 4 & 12 & 4 & 4 \\ 2 & 2 & 2 & 10 & 4 & 3 \end{pmatrix}_{b_0}$$

[923] $1^{133} 3^{121} 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 8 & 2 \\ 1 & 0 & 2 & 11 & 7 & 2 \\ 1 & 2 & 0 & 13 & 9 & 2 \\ 10 & 11 & 13 & 81 & 55 & 17 \\ 8 & 7 & 9 & 55 & 35 & 11 \\ 2 & 2 & 2 & 17 & 11 & 3 \end{pmatrix}_{b_0}$$

[922] $1^{13} 3^{121} 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 4 & 3 \\ 3 & 5 & 25 & 8 & 5 \\ 15 & 25 & 94 & 30 & 20 \\ 4 & 8 & 30 & 8 & 6 \\ 3 & 5 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

[921] $1^{13} 3^{121} 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 12 & 20 & 58 & 24 & 16 \\ 4 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$$

[925] $1^{111} 3^{121} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 5 & 15 & 13 & 5 \\ 8 & 15 & 32 & 29 & 12 \\ 7 & 13 & 29 & 24 & 10 \\ 3 & 5 & 12 & 10 & 4 \end{pmatrix}_{b_0}$$

[926] $1^{143} 3^{121} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 7 & 8 & 2 \\ 7 & 4 & 7 & 16 & 21 & 8 \\ 7 & 5 & 8 & 21 & 23 & 9 \\ 2 & 2 & 2 & 8 & 9 & 3 \end{pmatrix}_{b_0}$$

[927] $1^{143} 3^{121} 21^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 5 & 15 & 18 & 5 \\ 10 & 15 & 34 & 42 & 12 \\ 11 & 18 & 42 & 48 & 14 \\ 3 & 5 & 12 & 14 & 4 \end{pmatrix}_{b_0}$$

[928] $1^{112} 3^{121} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 7 & 8 & 2 \\ 4 & 4 & 7 & 16 & 21 & 8 \\ 6 & 5 & 8 & 21 & 23 & 9 \\ 2 & 2 & 2 & 8 & 9 & 3 \end{pmatrix}_{b_0}$$

[929] $1^{112} 3^{121} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 3 & 5 & 2 & 13 & 4 \\ 11 & 18 & 13 & 48 & 14 \\ 3 & 5 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

[930] $1^{124} 3^{121} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 4 & 5 & 2 & 17 & 4 \\ 15 & 23 & 17 & 78 & 18 \\ 3 & 5 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

[931] $1^{114} 3^{121} 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 2 & 13 & 2 \\ 1 & 2 & 0 & 1 & 17 & 2 \\ 2 & 2 & 1 & 0 & 13 & 2 \\ 13 & 13 & 17 & 13 & 127 & 21 \\ 2 & 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[932] $1^{1^{12}}3^{1^{12}}21^{1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 2 & 1 & 0 & 9 & 2 \\ 8 & 9 & 10 & 9 & 51 & 13 \\ 2 & 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[934] $1^{1^{24}}3^{1^{24}}21^{1^{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 17 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 4 & 5 & 2 & 21 & 4 \\ 17 & 28 & 21 & 116 & 22 \\ 3 & 5 & 4 & 22 & 4 \end{pmatrix}_{b_0}$$

[933] $1^{1^{21}}3^{1^{21}}21^{1^{21}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 5 & 15 & 8 & 5 \\ 9 & 15 & 32 & 18 & 12 \\ 5 & 8 & 18 & 8 & 6 \\ 3 & 5 & 12 & 6 & 4 \end{pmatrix}_{b_0}$$

[936] $1^{1^{14}}3^{1^{14}}21^{1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 7 & 9 & 0 & 35 & 11 \\ 2 & 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

[935] $1^{1^{13}}3^{1^{13}}21^{1^{23}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 6 & 8 & 2 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 5 & 6 & 8 & 26 & 40 & 10 \\ 9 & 8 & 12 & 40 & 56 & 14 \\ 2 & 2 & 2 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[937] $1^{1^{12}}3^{1^{12}}21^{1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 11 \\ 2 & 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

[938] $1^{1^{32}}3^{1^{32}}21^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 0 & 2 & 5 & 4 & 2 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 6 & 5 & 5 & 13 & 13 & 7 \\ 5 & 4 & 5 & 13 & 10 & 6 \\ 2 & 2 & 2 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

[940] $1^{1^{32}}3^{1^{32}}21^{1^{32}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 5 & 15 & 13 & 5 \\ 9 & 15 & 34 & 30 & 12 \\ 8 & 13 & 30 & 24 & 10 \\ 3 & 5 & 12 & 10 & 4 \end{pmatrix}_{b_0}$$

[941] $1^{1^{14}}3^{1^{14}}21^{1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 5 & 9 & 2 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 5 & 5 & 13 & 28 & 7 \\ 10 & 9 & 10 & 28 & 51 & 13 \\ 2 & 2 & 2 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[942] $1^{1^{11}}3^{1^{11}}21^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 5 & 4 & 2 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 4 & 5 & 5 & 13 & 13 & 7 \\ 4 & 4 & 5 & 13 & 10 & 6 \\ 2 & 2 & 2 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

[943] $1^{132}3^{121}1^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 17 & 8 & 3 \\ 3 & 5 & 30 & 13 & 5 \\ 17 & 30 & 134 & 58 & 24 \\ 8 & 13 & 58 & 24 & 10 \\ 3 & 5 & 24 & 10 & 4 \end{pmatrix}_{b_0}$

[944] $1^{132}3^{121}1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 9 & 5 & 2 \\ 1 & 0 & 2 & 9 & 4 & 2 \\ 1 & 2 & 0 & 12 & 5 & 2 \\ 9 & 9 & 12 & 61 & 27 & 15 \\ 5 & 4 & 5 & 27 & 10 & 6 \\ 2 & 2 & 2 & 15 & 6 & 3 \end{pmatrix}_{b_0}$

[939] $1^{123}3^{121}1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 0 & 2 & 5 & 9 & 2 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 6 & 5 & 5 & 13 & 28 & 7 \\ 9 & 9 & 10 & 28 & 51 & 13 \\ 2 & 2 & 2 & 7 & 13 & 3 \end{pmatrix}_{b_0}$

[946] $1^{113}3^{121}1^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 8 & 3 \\ 3 & 5 & 20 & 13 & 5 \\ 11 & 20 & 58 & 39 & 16 \\ 8 & 13 & 39 & 24 & 10 \\ 3 & 5 & 16 & 10 & 4 \end{pmatrix}_{b_0}$

[947] $1^{121}3^{121}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 2 & 5 & 4 & 2 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 5 & 5 & 5 & 13 & 13 & 7 \\ 4 & 4 & 5 & 13 & 10 & 6 \\ 2 & 2 & 2 & 7 & 6 & 3 \end{pmatrix}_{b_0}$

[950] $1^{122}3^{121}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 2 & 5 & 4 & 2 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 5 & 5 & 5 & 13 & 13 & 7 \\ 5 & 4 & 5 & 13 & 10 & 6 \\ 2 & 2 & 2 & 7 & 6 & 3 \end{pmatrix}_{b_0}$

[949] $1^{111}3^{121}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 3 & 4 & 2 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 4 & 3 & 4 & 5 & 9 & 5 \\ 5 & 4 & 5 & 9 & 10 & 6 \\ 2 & 2 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$

[945] $1^{111}3^{121}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 3 & 4 & 2 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 3 & 4 & 5 & 9 & 5 \\ 4 & 4 & 5 & 9 & 10 & 6 \\ 2 & 2 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$

[948] $1^{122}3^{121}1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 5 & 15 & 13 & 5 \\ 9 & 15 & 32 & 29 & 12 \\ 8 & 13 & 29 & 24 & 10 \\ 3 & 5 & 12 & 10 & 4 \end{pmatrix}_{b_0}$

[952] $1^{122}3^{121}1^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 0 & 2 & 7 & 7 & 2 \\ 1 & 2 & 0 & 11 & 9 & 2 \\ 8 & 7 & 11 & 45 & 41 & 13 \\ 8 & 7 & 9 & 41 & 35 & 11 \\ 2 & 2 & 2 & 13 & 11 & 3 \end{pmatrix}_{b_0}$$

[951] $1^{1^{22}}3^{1^{22}}21^{1^{32}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 7 & 7 & 2 \\ 1 & 2 & 0 & 11 & 9 & 2 \\ 7 & 7 & 11 & 45 & 41 & 13 \\ 7 & 7 & 9 & 41 & 35 & 11 \\ 2 & 2 & 2 & 13 & 11 & 3 \end{pmatrix}_{b_0}$$

[953] $1^{1^{34}}3^{1^{34}}21^{1^{24}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 14 & 3 \\ 3 & 5 & 10 & 23 & 5 \\ 7 & 10 & 14 & 36 & 8 \\ 14 & 23 & 36 & 80 & 18 \\ 3 & 5 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

[957] $1^{1^{13}}3^{1^{13}}21^{1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 3 & 5 & 2 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 3 & 4 & 5 & 13 & 5 \\ 7 & 5 & 8 & 13 & 23 & 9 \\ 2 & 2 & 2 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[956] $1^{1^{22}}3^{1^{22}}21^{1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 3 & 5 & 2 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 4 & 3 & 4 & 5 & 13 & 5 \\ 6 & 5 & 8 & 13 & 23 & 9 \\ 2 & 2 & 2 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[955] $21^{1^{11}}3^{1^{11}}21^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 4 & 2 \\ 2 & 0 & 2 & 3 & 4 & 2 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 3 & 4 & 5 & 9 & 5 \\ 4 & 4 & 5 & 9 & 10 & 6 \\ 2 & 2 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[954] $1^{1^{13}}3^{1^{13}}21^{1^{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 5 & 0 & 28 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 17 & 28 & 0 & 120 & 22 \\ 3 & 5 & 0 & 22 & 4 \end{pmatrix}_{b_0}$$

[958] $1^{1^{13}}3^{1^{13}}21^{1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 7 & 9 & 0 & 35 & 11 \\ 2 & 2 & 2 & 0 & 11 & 3 \end{pmatrix}_{b_0}$$

[959] $1^{1^{34}}3^{1^{34}}21^{1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 3 & 7 & 2 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 5 & 3 & 4 & 5 & 16 & 5 \\ 9 & 7 & 9 & 16 & 35 & 11 \\ 2 & 2 & 2 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

[960] $1^{1^{12}}3^{1^{12}}21^{1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 3 & 7 & 2 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 2 & 3 & 4 & 5 & 16 & 5 \\ 7 & 7 & 9 & 16 & 35 & 11 \\ 2 & 2 & 2 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

[961] $1^{1^{34}}3^{1^{34}}21^{1^{22}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 0 & 2 & 4 & 7 & 2 \\ 1 & 2 & 0 & 7 & 9 & 2 \\ 6 & 4 & 7 & 16 & 25 & 8 \\ 9 & 7 & 9 & 25 & 35 & 11 \\ 2 & 2 & 2 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

[962] $1^{1^{22}}3^{1^{21}}21^{1^{22}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 4 & 7 & 2 \\ 1 & 2 & 0 & 7 & 9 & 2 \\ 5 & 4 & 7 & 16 & 25 & 8 \\ 7 & 7 & 9 & 25 & 35 & 11 \\ 2 & 2 & 2 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

[963] $1^{1^3}3^{1^{21}}21^{1^{14}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 5 & 5 & 33 & 5 \\ 2 & 5 & 2 & 24 & 4 \\ 19 & 33 & 24 & 164 & 26 \\ 3 & 5 & 4 & 26 & 4 \end{pmatrix}_{b_0}$$

[964] $1^{1^4}3^{1^{21}}21^{1^{14}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 2 & 11 & 2 \\ 1 & 2 & 0 & 1 & 16 & 2 \\ 1 & 2 & 1 & 0 & 11 & 2 \\ 12 & 11 & 16 & 11 & 103 & 19 \\ 2 & 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[965] $1^{1^{23}}3^{1^{21}}21^{1^{33}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 0 & 2 & 9 & 8 & 2 \\ 1 & 2 & 0 & 12 & 12 & 2 \\ 8 & 9 & 12 & 61 & 60 & 15 \\ 9 & 8 & 12 & 60 & 56 & 14 \\ 2 & 2 & 2 & 15 & 14 & 3 \end{pmatrix}_{b_0}$$

[966] $1^{1^3}3^{1^{21}}21^{1^{13}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 3 & 8 & 2 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 2 & 3 & 4 & 5 & 20 & 5 \\ 9 & 8 & 12 & 20 & 56 & 14 \\ 2 & 2 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

[967] $1^{1^{13}}3^{1^{21}}21^{1^{14}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 5 & 0 & 28 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 16 & 28 & 0 & 116 & 22 \\ 3 & 5 & 0 & 22 & 4 \end{pmatrix}_{b_0}$$

[968] $1^{1^{14}}3^{1^{21}}21^{1^{13}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 2 & 2 & 1 & 0 & 10 & 2 \\ 11 & 10 & 13 & 10 & 74 & 16 \\ 2 & 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[970] $1^{1^{21}}3^{1^{21}}21^{1^{31}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 11 & 20 & 58 & 24 & 16 \\ 5 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$$

[969] $1^{1^{14}}3^{1^{21}}21^{1^{14}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 5 & 0 & 28 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 17 & 28 & 0 & 116 & 22 \\ 3 & 5 & 0 & 22 & 4 \end{pmatrix}_{b_0}$$

[971] $1^{1^{14}}3^{1^{21}}21^{1^{13}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 3 & 12 & 2 \\ 1 & 2 & 0 & 4 & 14 & 2 \\ 2 & 3 & 4 & 5 & 26 & 5 \\ 12 & 12 & 14 & 26 & 96 & 18 \\ 2 & 2 & 2 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

[972] $1^{13}3^{121}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 125, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 2 & 11 & 2 \\ 1 & 2 & 0 & 1 & 16 & 2 \\ 1 & 2 & 1 & 0 & 11 & 2 \\ 11 & 11 & 16 & 11 & 103 & 19 \\ 2 & 2 & 2 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[973] $1^{142}3^{121}1^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 5 & 2 \\ 1 & 0 & 2 & 12 & 4 & 2 \\ 1 & 2 & 0 & 16 & 5 & 2 \\ 12 & 12 & 16 & 110 & 36 & 20 \\ 5 & 4 & 5 & 36 & 10 & 6 \\ 2 & 2 & 2 & 20 & 6 & 3 \end{pmatrix}_{b_0}$$

[974] $1^{111}3^{121}1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 2 & 6 & 4 & 2 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 5 & 6 & 8 & 26 & 18 & 10 \\ 4 & 4 & 5 & 18 & 10 & 6 \\ 2 & 2 & 2 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

[975] $1^{142}3^{121}1^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 8 & 3 \\ 3 & 5 & 20 & 13 & 5 \\ 12 & 20 & 62 & 40 & 16 \\ 8 & 13 & 40 & 24 & 10 \\ 3 & 5 & 16 & 10 & 4 \end{pmatrix}_{b_0}$$

[976] $21^{121}3^{121}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 4 & 2 \\ 2 & 0 & 2 & 5 & 4 & 2 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 5 & 5 & 5 & 13 & 13 & 7 \\ 4 & 4 & 5 & 13 & 10 & 6 \\ 2 & 2 & 2 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

[985] $21^{114}3^{121}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 12 & 2 \\ 2 & 0 & 2 & 2 & 12 & 2 \\ 1 & 2 & 0 & 1 & 14 & 2 \\ 2 & 2 & 1 & 0 & 12 & 2 \\ 12 & 12 & 14 & 12 & 96 & 18 \\ 2 & 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[984] $1^{14}3^{121}1^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 4 & 3 \\ 3 & 5 & 25 & 8 & 5 \\ 15 & 25 & 92 & 30 & 20 \\ 4 & 8 & 30 & 8 & 6 \\ 3 & 5 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

[983] $1^{132}3^{121}1^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 10 & 3 \\ 3 & 5 & 15 & 18 & 5 \\ 9 & 15 & 34 & 42 & 12 \\ 10 & 18 & 42 & 48 & 14 \\ 3 & 5 & 12 & 14 & 4 \end{pmatrix}_{b_0}$$

[982] $1^{122}3^{121}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 5 & 6 & 2 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 4 & 5 & 5 & 13 & 18 & 7 \\ 6 & 6 & 6 & 18 & 20 & 8 \\ 2 & 2 & 2 & 7 & 8 & 3 \end{pmatrix}_{b_0}$$

[981] $1^{112}3^{121}1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 3 & 5 & 2 & 14 & 4 \\ 10 & 18 & 14 & 48 & 14 \\ 3 & 5 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

[980] $1^{1^{32}} 3^{1^{22}} 21^{1^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 5 & 6 & 2 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 6 & 5 & 5 & 13 & 18 & 7 \\ 6 & 6 & 6 & 18 & 20 & 8 \\ 2 & 2 & 2 & 7 & 8 & 3 \end{pmatrix}_{b_0}$$

[979] $1^{1^{22}} 3^{1^{42}} 21^{1^{21}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & 0 & 2 & 10 & 6 & 2 \\ 1 & 2 & 0 & 10 & 6 & 2 \\ 8 & 10 & 10 & 58 & 36 & 14 \\ 6 & 6 & 6 & 36 & 20 & 8 \\ 2 & 2 & 2 & 14 & 8 & 3 \end{pmatrix}_{b_0}$$

[978] $1^{1^{42}} 3^{1^{143}} 21^{1^{43}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 10 & 3 \\ 3 & 5 & 20 & 18 & 5 \\ 12 & 20 & 62 & 56 & 16 \\ 10 & 18 & 56 & 48 & 14 \\ 3 & 5 & 16 & 14 & 4 \end{pmatrix}_{b_0}$$

[977] $1^{1^{22}} 3^{1^{123}} 21^{1^{23}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 5 & 10 & 18 & 5 \\ 6 & 10 & 14 & 28 & 8 \\ 10 & 18 & 28 & 48 & 14 \\ 3 & 5 & 8 & 14 & 4 \end{pmatrix}_{b_0}$$

[988] $1^{1^{11}} 3^{1^{132}} 21^{1^{32}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 7 & 2 \\ 1 & 0 & 2 & 5 & 5 & 2 \\ 1 & 2 & 0 & 10 & 8 & 2 \\ 8 & 5 & 10 & 33 & 29 & 11 \\ 7 & 5 & 8 & 29 & 23 & 9 \\ 2 & 2 & 2 & 11 & 9 & 3 \end{pmatrix}_{b_0}$$

[989] $1^{1^{32}} 3^{1^{111}} 21^{1^{32}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 6 & 2 \\ 1 & 0 & 2 & 5 & 5 & 2 \\ 1 & 2 & 0 & 10 & 8 & 2 \\ 7 & 5 & 10 & 33 & 29 & 11 \\ 6 & 5 & 8 & 29 & 23 & 9 \\ 2 & 2 & 2 & 11 & 9 & 3 \end{pmatrix}_{b_0}$$

[987] $1^{1^{43}} 3^{1^{111}} 21^{1^{33}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 11 & 3 \\ 3 & 5 & 20 & 18 & 5 \\ 13 & 20 & 58 & 54 & 16 \\ 11 & 18 & 54 & 48 & 14 \\ 3 & 5 & 16 & 14 & 4 \end{pmatrix}_{b_0}$$

[986] $1^{1^{32}} 3^{1^{141}} 21^{1^{42}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 6 & 2 \\ 1 & 0 & 2 & 12 & 5 & 2 \\ 1 & 2 & 0 & 16 & 8 & 2 \\ 11 & 12 & 16 & 110 & 52 & 20 \\ 6 & 5 & 8 & 52 & 23 & 9 \\ 2 & 2 & 2 & 20 & 9 & 3 \end{pmatrix}_{b_0}$$

[991] $1^{1^{43}} 3^{1^{132}} 21^{1^{42}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 8 & 2 \\ 1 & 0 & 2 & 10 & 7 & 2 \\ 1 & 2 & 0 & 15 & 9 & 2 \\ 11 & 10 & 15 & 88 & 57 & 18 \\ 8 & 7 & 9 & 57 & 35 & 11 \\ 2 & 2 & 2 & 18 & 11 & 3 \end{pmatrix}_{b_0}$$

[990] $1^{1^{21}} 3^{1^{121}} 21^{1^{31}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 7 & 4 & 2 \\ 1 & 2 & 0 & 11 & 5 & 2 \\ 7 & 7 & 11 & 45 & 23 & 13 \\ 4 & 4 & 5 & 23 & 10 & 6 \\ 2 & 2 & 2 & 13 & 6 & 3 \end{pmatrix}_{b_0}$$

[992] $1^{1\overline{1}3}3^121^{1\overline{2}4} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 133, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 12 & 2 \\ 1 & 0 & 2 & 4 & 11 & 2 \\ 1 & 2 & 0 & 7 & 16 & 2 \\ 6 & 4 & 7 & 16 & 43 & 8 \\ 12 & 11 & 16 & 43 & 103 & 19 \\ 2 & 2 & 2 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$

[996] $1^{1\overline{4}4}3^121^{1\overline{3}4} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 134, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 14 & 3 \\ 3 & 5 & 15 & 23 & 5 \\ 10 & 15 & 34 & 54 & 12 \\ 14 & 23 & 54 & 80 & 18 \\ 3 & 5 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[995] $1^{1\overline{1}1}3^121^{1\overline{2}1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 134, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 8 & 4 & 2 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 6 & 8 & 9 & 40 & 22 & 12 \\ 4 & 4 & 5 & 22 & 10 & 6 \\ 2 & 2 & 2 & 12 & 6 & 3 \end{pmatrix}_{b_0}$$

[994] $1^{1\overline{3}1}3^121^{1\overline{4}1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 134, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 10 & 4 & 2 \\ 1 & 2 & 0 & 15 & 5 & 2 \\ 9 & 10 & 15 & 88 & 32 & 18 \\ 4 & 4 & 5 & 32 & 10 & 6 \\ 2 & 2 & 2 & 18 & 6 & 3 \end{pmatrix}_{b_0}$$

[993] $1^{1\overline{1}2}3^121^{1\overline{2}3} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 134, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 4 & 10 & 2 \\ 1 & 2 & 0 & 7 & 13 & 2 \\ 4 & 4 & 7 & 16 & 36 & 8 \\ 9 & 10 & 13 & 36 & 74 & 16 \\ 2 & 2 & 2 & 8 & 16 & 3 \end{pmatrix}_{b_0}$$

[997] $1^{1\overline{3}}3^121^{1\overline{1}4} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 2 & 5 & 2 & 21 & 4 \\ 16 & 28 & 21 & 116 & 22 \\ 3 & 5 & 4 & 22 & 4 \end{pmatrix}_{b_0}$$

[1000] $1^{1\overline{1}3}3^121^{1\overline{1}3} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 137, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 5 & 5 & 18 & 5 \\ 3 & 5 & 2 & 14 & 4 \\ 11 & 18 & 14 & 48 & 14 \\ 3 & 5 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

[998] $1^{1\overline{3}2}3^121^{1\overline{2}2} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 137, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 7 & 8 & 2 \\ 6 & 4 & 7 & 16 & 21 & 8 \\ 6 & 5 & 8 & 21 & 23 & 9 \\ 2 & 2 & 2 & 8 & 9 & 3 \end{pmatrix}_{b_0}$$

[999] $1^{1\overline{2}3}3^121^{1\overline{2}2} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 137, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 7 & 8 & 2 \\ 5 & 4 & 7 & 16 & 21 & 8 \\ 7 & 5 & 8 & 21 & 23 & 9 \\ 2 & 2 & 2 & 8 & 9 & 3 \end{pmatrix}_{b_0}$$

[1001] $1^{1\overline{1}3}3^121^{1\overline{2}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 2 & 1 & 0 & 9 & 2 \\ 9 & 9 & 10 & 9 & 51 & 13 \\ 2 & 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[1002] \quad & 1^{1^4} 1^{13} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 2 & 1 & 0 & 9 & 2 \\ 10 & 9 & 10 & 9 & 51 & 13 \\ 2 & 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0} \\
[1003] \quad & 1^{1^{22}} 1^{122} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 5 & 7 & 2 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 5 & 5 & 5 & 13 & 23 & 7 \\ 7 & 7 & 9 & 23 & 35 & 11 \\ 2 & 2 & 2 & 7 & 11 & 3 \end{pmatrix}_{b_0} \\
[1005] \quad & 1^{1^{31}} 1^{13} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 3 & 2 \\ 1 & 0 & 2 & 9 & 2 & 2 \\ 1 & 2 & 0 & 12 & 4 & 2 \\ 9 & 9 & 12 & 61 & 18 & 15 \\ 3 & 2 & 4 & 18 & 4 & 4 \\ 2 & 2 & 2 & 15 & 4 & 3 \end{pmatrix}_{b_0} \\
[1004] \quad & 1^{1^{24}} 1^{122} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 5 & 7 & 2 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 5 & 5 & 5 & 13 & 23 & 7 \\ 9 & 7 & 9 & 23 & 35 & 11 \\ 2 & 2 & 2 & 7 & 11 & 3 \end{pmatrix}_{b_0} \\
[1007] \quad & 1^{1^{33}} 1^{121} 3^1 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 7 & 2 \\ 1 & 0 & 2 & 7 & 5 & 2 \\ 1 & 2 & 0 & 11 & 8 & 2 \\ 8 & 7 & 11 & 45 & 34 & 13 \\ 7 & 5 & 8 & 34 & 23 & 9 \\ 2 & 2 & 2 & 13 & 9 & 3 \end{pmatrix}_{b_0} \\
[1006] \quad & 1^{1^{13}} 1^{13} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 1 & 10 & 2 \\ 1 & 2 & 0 & 3 & 13 & 2 \\ 2 & 1 & 3 & 1 & 13 & 3 \\ 10 & 10 & 13 & 13 & 74 & 16 \\ 2 & 2 & 2 & 3 & 16 & 3 \end{pmatrix}_{b_0} \\
[1009] \quad & 1^{1^{12}} 1^{12} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 1 & 8 & 2 \\ 1 & 2 & 0 & 3 & 12 & 2 \\ 2 & 1 & 3 & 1 & 12 & 3 \\ 8 & 8 & 12 & 12 & 56 & 14 \\ 2 & 2 & 2 & 3 & 14 & 3 \end{pmatrix}_{b_0} \\
[1008] \quad & 1^{1^{31}} 1^{14} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 3 & 2 \\ 1 & 0 & 2 & 12 & 2 & 2 \\ 1 & 2 & 0 & 16 & 4 & 2 \\ 11 & 12 & 16 & 110 & 24 & 20 \\ 3 & 2 & 4 & 24 & 4 & 4 \\ 2 & 2 & 2 & 20 & 4 & 3 \end{pmatrix}_{b_0} \\
[1010] \quad & 1^{1^{21}} 1^{12} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 5 & 3 \\ 3 & 5 & 25 & 8 & 5 \\ 14 & 25 & 94 & 30 & 20 \\ 5 & 8 & 30 & 8 & 6 \\ 3 & 5 & 20 & 6 & 4 \end{pmatrix}_{b_0} \\
[1015] \quad & 1^{1^{43}} 1^{121} 3^1 21^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 7 & 2 \\ 1 & 0 & 2 & 8 & 5 & 2 \\ 1 & 2 & 0 & 14 & 8 & 2 \\ 10 & 8 & 14 & 70 & 42 & 16 \\ 7 & 5 & 8 & 42 & 23 & 9 \\ 2 & 2 & 2 & 16 & 9 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[1014] \quad & 1^{123} 3^1 21^{123} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 5 & 10 & 18 & 5 \\ 6 & 10 & 14 & 28 & 8 \\ 11 & 18 & 28 & 48 & 14 \\ 3 & 5 & 8 & 14 & 4 \end{pmatrix}_{b_0} \\
[1016] \quad & 1^{123} 3^1 21^{144} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 11 & 2 \\ 1 & 0 & 2 & 8 & 11 & 2 \\ 1 & 2 & 0 & 14 & 16 & 2 \\ 9 & 8 & 14 & 70 & 86 & 16 \\ 11 & 11 & 16 & 86 & 103 & 19 \\ 2 & 2 & 2 & 16 & 19 & 3 \end{pmatrix}_{b_0} \\
[1017] \quad & 1^{112} 3^1 21^{122} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 6 & 9 & 2 \\ 1 & 2 & 0 & 8 & 10 & 2 \\ 5 & 6 & 8 & 26 & 38 & 10 \\ 8 & 9 & 10 & 38 & 51 & 13 \\ 2 & 2 & 2 & 10 & 13 & 3 \end{pmatrix}_{b_0} \\
[1013] \quad & 1^{112} 3^1 21^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 3 & 5 & 2 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 3 & 4 & 5 & 13 & 5 \\ 6 & 5 & 8 & 13 & 23 & 9 \\ 2 & 2 & 2 & 5 & 9 & 3 \end{pmatrix}_{b_0} \\
[1012] \quad & 1^{112} 3^1 21^{124} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 15 & 3 \\ 3 & 5 & 10 & 28 & 5 \\ 5 & 10 & 14 & 42 & 8 \\ 15 & 28 & 42 & 116 & 22 \\ 3 & 5 & 8 & 22 & 4 \end{pmatrix}_{b_0} \\
[1011] \quad & 1^{123} 3^1 21^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 3 & 5 & 2 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 4 & 3 & 4 & 5 & 13 & 5 \\ 7 & 5 & 8 & 13 & 23 & 9 \\ 2 & 2 & 2 & 5 & 9 & 3 \end{pmatrix}_{b_0} \\
[1018] \quad & 1^{12} 3^1 21^{112} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 3 & 9 & 2 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 2 & 3 & 4 & 5 & 19 & 5 \\ 8 & 9 & 10 & 19 & 51 & 13 \\ 2 & 2 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0} \\
[1019] \quad & 1^{133} 3^1 21^{143} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 11 & 3 \\ 3 & 5 & 25 & 18 & 5 \\ 14 & 25 & 92 & 68 & 20 \\ 11 & 18 & 68 & 48 & 14 \\ 3 & 5 & 20 & 14 & 4 \end{pmatrix}_{b_0} \\
[1020] \quad & 21^{123} 3^1 21^{134} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 1 & 7 & 11 & 2 \\ 2 & 0 & 2 & 7 & 11 & 2 \\ 1 & 2 & 0 & 11 & 16 & 2 \\ 7 & 7 & 11 & 45 & 70 & 13 \\ 11 & 11 & 16 & 70 & 103 & 19 \\ 2 & 2 & 2 & 13 & 19 & 3 \end{pmatrix}_{b_0} \\
[1021] \quad & 1^{114} 3^1 21^{113} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 5 & 10 & 2 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 4 & 5 & 5 & 13 & 33 & 7 \\ 11 & 10 & 13 & 33 & 74 & 16 \\ 2 & 2 & 2 & 7 & 16 & 3 \end{pmatrix}_{b_0} \\
[1022] \quad & 1^{113} 3^1 21^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0
\end{aligned}$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 1 & 2 & 1 & 0 & 10 & 2 \\ 11 & 10 & 13 & 10 & 74 & 16 \\ 2 & 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1028] \ 1^{1^{13}} 3^{1^{12}} 21^{1^{12} 3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 4 & 8 & 2 \\ 1 & 2 & 0 & 7 & 12 & 2 \\ 4 & 4 & 7 & 16 & 32 & 8 \\ 9 & 8 & 12 & 32 & 56 & 14 \\ 2 & 2 & 2 & 8 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1023] \ 1^{1^{13}} 3^{1^{14}} 21^{1^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 2 & 12 & 2 \\ 1 & 2 & 0 & 1 & 14 & 2 \\ 2 & 2 & 1 & 0 & 12 & 2 \\ 11 & 12 & 14 & 12 & 96 & 18 \\ 2 & 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1024] \ 1^{1^{33}} 3^{1^{12}} 21^{1^{14} 3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 9 & 2 \\ 1 & 0 & 2 & 8 & 8 & 2 \\ 1 & 2 & 0 & 14 & 12 & 2 \\ 9 & 8 & 14 & 70 & 64 & 16 \\ 9 & 8 & 12 & 64 & 56 & 14 \\ 2 & 2 & 2 & 16 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1025] \ 1^{1^{12}} 3^{1^{13}} 21^{1^{12} 3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 0 & 2 & 4 & 8 & 2 \\ 1 & 2 & 0 & 7 & 12 & 2 \\ 6 & 4 & 7 & 16 & 32 & 8 \\ 9 & 8 & 12 & 32 & 56 & 14 \\ 2 & 2 & 2 & 8 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1026] \ 1^{1^{13}} 3^{1^{12}} 21^{1^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 8 & 12 & 0 & 56 & 14 \\ 2 & 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1027] \ 1^{1^{33}} 3^{1^{14}} 21^{1^{14} 3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 9 & 2 \\ 1 & 0 & 2 & 12 & 8 & 2 \\ 1 & 2 & 0 & 16 & 12 & 2 \\ 11 & 12 & 16 & 110 & 80 & 20 \\ 9 & 8 & 12 & 80 & 56 & 14 \\ 2 & 2 & 2 & 20 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1029] \ 1^{1^{22}} 3^{1^{12}} 21^{1^{12} 2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 3 & 7 & 2 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 4 & 3 & 4 & 5 & 16 & 5 \\ 7 & 7 & 9 & 16 & 35 & 11 \\ 2 & 2 & 2 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1030] \ 1^{1^{14}} 3^{1^{12}} 21^{1^{12} 2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 3 & 7 & 2 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 3 & 4 & 5 & 16 & 5 \\ 9 & 7 & 9 & 16 & 35 & 11 \\ 2 & 2 & 2 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1031] \ 1^{1^{21}} 3^{1^{12}} 21^{1^{14} 1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 5 & 3 \\ 3 & 5 & 30 & 8 & 5 \\ 16 & 30 & 134 & 36 & 24 \\ 5 & 8 & 36 & 8 & 6 \\ 3 & 5 & 24 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1032] \ 1^{1^{41}} 3^{1^{11}} 21^{1^{13} 1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 13 & 20 & 58 & 24 & 16 \\ 5 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1033] \ 1^{13} 3^{12} 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 9 & 10 & 0 & 51 & 13 \\ 2 & 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1034] \ 1^{14} 3^{12} 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 9 & 10 & 0 & 51 & 13 \\ 2 & 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1035] \ 1^{123} 3^{12} 21^{144} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 12 & 2 \\ 1 & 0 & 2 & 8 & 11 & 2 \\ 1 & 2 & 0 & 14 & 16 & 2 \\ 10 & 8 & 14 & 70 & 86 & 16 \\ 12 & 11 & 16 & 86 & 103 & 19 \\ 2 & 2 & 2 & 16 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1036] \ 1^{13} 3^{12} 21^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 5 & 0 & 33 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 19 & 33 & 0 & 164 & 26 \\ 3 & 5 & 0 & 26 & 4 \end{pmatrix}_{b_0}$$

$$[1037] \ 1^{14} 3^{12} 21^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 157, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 1 & 10 & 2 \\ 1 & 2 & 0 & 3 & 13 & 2 \\ 2 & 1 & 3 & 1 & 13 & 3 \\ 11 & 10 & 13 & 13 & 74 & 16 \\ 2 & 2 & 2 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1040] \ 1^{121} 3^{12} 21^{131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 159, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 5 & 2 \\ 1 & 0 & 2 & 7 & 4 & 2 \\ 1 & 2 & 0 & 11 & 5 & 2 \\ 9 & 7 & 11 & 45 & 23 & 13 \\ 5 & 4 & 5 & 23 & 10 & 6 \\ 2 & 2 & 2 & 13 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1039] \ 1^{113} 3^{12} 21^{132} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 159, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 8 & 3 \\ 3 & 5 & 20 & 13 & 5 \\ 13 & 20 & 58 & 39 & 16 \\ 8 & 13 & 39 & 24 & 10 \\ 3 & 5 & 16 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1038] \ 1^{121} 3^{12} 21^{131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 159, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 7 & 4 & 2 \\ 1 & 2 & 0 & 11 & 5 & 2 \\ 9 & 7 & 11 & 45 & 23 & 13 \\ 4 & 4 & 5 & 23 & 10 & 6 \\ 2 & 2 & 2 & 13 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1042] \ 1^{123} 3^{12} 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 8 & 12 & 0 & 56 & 14 \\ 2 & 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1043] \ 1^{122} 3^{12} 21^{143} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 10 & 2 \\ 1 & 0 & 2 & 8 & 8 & 2 \\ 1 & 2 & 0 & 14 & 12 & 2 \\ 10 & 8 & 14 & 70 & 64 & 16 \\ 10 & 8 & 12 & 64 & 56 & 14 \\ 2 & 2 & 2 & 16 & 14 & 3 \end{pmatrix}_{b_0}$$

[1041] $1^{14} 1^{12} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 18 & 4 & 3 \\ 3 & 5 & 30 & 8 & 5 \\ 18 & 30 & 134 & 36 & 24 \\ 4 & 8 & 36 & 8 & 6 \\ 3 & 5 & 24 & 6 & 4 \end{pmatrix}_{b_0}$$

[1044] $1^{133} 1^{13} 3^1 21^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 11 & 3 \\ 3 & 5 & 15 & 18 & 5 \\ 9 & 15 & 34 & 42 & 12 \\ 11 & 18 & 42 & 48 & 14 \\ 3 & 5 & 12 & 14 & 4 \end{pmatrix}_{b_0}$$

[1047] $1^{114} 1^{113} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 2 & 1 & 0 & 9 & 2 \\ 10 & 9 & 10 & 9 & 51 & 13 \\ 2 & 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1046] $1^{123} 1^{12} 3^1 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 5 & 0 & 33 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 20 & 33 & 0 & 164 & 26 \\ 3 & 5 & 0 & 26 & 4 \end{pmatrix}_{b_0}$$

[1045] $1^{113} 1^{13} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 2 & 1 & 0 & 9 & 2 \\ 9 & 9 & 10 & 9 & 51 & 13 \\ 2 & 2 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1048] $1^{131} 1^{11} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 5 & 20 & 8 & 5 \\ 12 & 20 & 58 & 24 & 16 \\ 5 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$$

[1050] $1^{122} 1^{12} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 5 & 7 & 2 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 4 & 5 & 5 & 13 & 23 & 7 \\ 7 & 7 & 9 & 23 & 35 & 11 \\ 2 & 2 & 2 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[1049] $1^{122} 1^{12} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 0 & 2 & 5 & 7 & 2 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 6 & 5 & 5 & 13 & 23 & 7 \\ 9 & 7 & 9 & 23 & 35 & 11 \\ 2 & 2 & 2 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[1051] $1^{123} 1^{12} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 5 & 5 & 33 & 5 \\ 2 & 5 & 2 & 24 & 4 \\ 18 & 33 & 24 & 164 & 26 \\ 3 & 5 & 4 & 26 & 4 \end{pmatrix}_{b_0}$$

[1052] $1^{123} 1^{12} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 1 & 2 & 1 & 0 & 10 & 2 \\ 9 & 10 & 13 & 10 & 74 & 16 \\ 2 & 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1053] $1^{13}3^121^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 1 & 2 & 1 & 0 & 10 & 2 \\ 10 & 10 & 13 & 10 & 74 & 16 \\ 2 & 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1055] $1^{13}3^121^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 1 & 8 & 2 \\ 1 & 2 & 0 & 3 & 12 & 2 \\ 2 & 1 & 3 & 1 & 12 & 3 \\ 9 & 8 & 12 & 12 & 56 & 14 \\ 2 & 2 & 2 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[1054] $1^{14}3^121^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 3 & 5 & 2 & 17 & 4 \\ 15 & 23 & 17 & 78 & 18 \\ 3 & 5 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

[1056] $1^{13}3^121^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 170, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 4 & 10 & 2 \\ 1 & 2 & 0 & 7 & 13 & 2 \\ 5 & 4 & 7 & 16 & 36 & 8 \\ 11 & 10 & 13 & 36 & 74 & 16 \\ 2 & 2 & 2 & 8 & 16 & 3 \end{pmatrix}_{b_0}$$

[1057] $1^{121}3^121^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 7 & 3 \\ 3 & 5 & 20 & 13 & 5 \\ 11 & 20 & 58 & 39 & 16 \\ 7 & 13 & 39 & 24 & 10 \\ 3 & 5 & 16 & 10 & 4 \end{pmatrix}_{b_0}$$

[1058] $1^{13}3^121^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 3 & 2 \\ 1 & 0 & 2 & 9 & 2 & 2 \\ 1 & 2 & 0 & 12 & 4 & 2 \\ 10 & 9 & 12 & 61 & 18 & 15 \\ 3 & 2 & 4 & 18 & 4 & 4 \\ 2 & 2 & 2 & 15 & 4 & 3 \end{pmatrix}_{b_0}$$

[1059] $1^{113}3^121^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 4 & 10 & 2 \\ 1 & 2 & 0 & 7 & 13 & 2 \\ 5 & 4 & 7 & 16 & 36 & 8 \\ 10 & 10 & 13 & 36 & 74 & 16 \\ 2 & 2 & 2 & 8 & 16 & 3 \end{pmatrix}_{b_0}$$

[1060] $1^{113}3^121^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 16 & 3 \\ 3 & 5 & 10 & 28 & 5 \\ 6 & 10 & 14 & 42 & 8 \\ 16 & 28 & 42 & 116 & 22 \\ 3 & 5 & 8 & 22 & 4 \end{pmatrix}_{b_0}$$

[1061] $1^{123}3^121^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 1 & 8 & 2 \\ 1 & 2 & 0 & 3 & 12 & 2 \\ 2 & 1 & 3 & 1 & 12 & 3 \\ 10 & 8 & 12 & 12 & 56 & 14 \\ 2 & 2 & 2 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[1062] $21^{121}3^121^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 4 & 2 \\ 2 & 0 & 2 & 6 & 4 & 2 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 6 & 6 & 8 & 26 & 18 & 10 \\ 4 & 4 & 5 & 18 & 10 & 6 \\ 2 & 2 & 2 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1065] \quad 1^{133} 3^{121} 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 11 & 3 \\ 3 & 5 & 20 & 18 & 5 \\ 12 & 20 & 58 & 54 & 16 \\ 11 & 18 & 54 & 48 & 14 \\ 3 & 5 & 16 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1064] \quad 1^{133} 3^{121} 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 6 & 5 & 2 \\ 1 & 2 & 0 & 8 & 8 & 2 \\ 7 & 6 & 8 & 26 & 26 & 10 \\ 7 & 5 & 8 & 26 & 23 & 9 \\ 2 & 2 & 2 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1063] \quad 1^{132} 3^{121} 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 6 & 2 \\ 1 & 0 & 2 & 6 & 5 & 2 \\ 1 & 2 & 0 & 8 & 8 & 2 \\ 7 & 6 & 8 & 26 & 26 & 10 \\ 6 & 5 & 8 & 26 & 23 & 9 \\ 2 & 2 & 2 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1067] \quad 1^{113} 3^{121} 21^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 4 & 13 & 2 \\ 1 & 2 & 0 & 7 & 17 & 2 \\ 4 & 4 & 7 & 16 & 47 & 8 \\ 12 & 13 & 17 & 47 & 127 & 21 \\ 2 & 2 & 2 & 8 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1066] \quad 1^{123} 3^{121} 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 5 & 10 & 2 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 5 & 5 & 5 & 13 & 33 & 7 \\ 10 & 10 & 13 & 33 & 74 & 16 \\ 2 & 2 & 2 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1070] \quad 1^{112} 3^{121} 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 6 & 7 & 2 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 5 & 6 & 8 & 26 & 32 & 10 \\ 7 & 7 & 9 & 32 & 35 & 11 \\ 2 & 2 & 2 & 10 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1069] \quad 1^{121} 3^{121} 21^{142} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 7 & 3 \\ 3 & 5 & 30 & 13 & 5 \\ 16 & 30 & 134 & 58 & 24 \\ 7 & 13 & 58 & 24 & 10 \\ 3 & 5 & 24 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1068] \quad 1^{112} 3^{121} 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 3 & 5 & 2 & 17 & 4 \\ 13 & 23 & 17 & 78 & 18 \\ 3 & 5 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1077] \quad 1^{132} 3^{121} 21^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 182, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 8 & 3 \\ 3 & 5 & 20 & 13 & 5 \\ 12 & 20 & 58 & 39 & 16 \\ 8 & 13 & 39 & 24 & 10 \\ 3 & 5 & 16 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1079] \quad 1^{141} 3^{121} 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 182, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 8 & 4 & 2 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 9 & 8 & 9 & 40 & 22 & 12 \\ 4 & 4 & 5 & 22 & 10 & 6 \\ 2 & 2 & 2 & 12 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1075] \quad 1^{1^{32}} 3^{1^{21}} 21^{1^{21}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 182, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 5 & 2 \\ 1 & 0 & 2 & 6 & 4 & 2 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 7 & 6 & 8 & 26 & 18 & 10 \\ 5 & 4 & 5 & 18 & 10 & 6 \\ 2 & 2 & 2 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1074] \quad 1^{1^{31}} 3^{1^{21}} 21^{1^{21}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 182, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 6 & 4 & 2 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 7 & 6 & 8 & 26 & 18 & 10 \\ 4 & 4 & 5 & 18 & 10 & 6 \\ 2 & 2 & 2 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1073] \quad 1^{1^{22}} 3^{1^{21}} 21^{1^{22}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 182, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 6 & 5 & 2 \\ 1 & 2 & 0 & 8 & 8 & 2 \\ 6 & 6 & 8 & 26 & 26 & 10 \\ 6 & 5 & 8 & 26 & 23 & 9 \\ 2 & 2 & 2 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1072] \quad 1^1 3^1 21^{1^{43}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 182, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 11 & 3 \\ 3 & 5 & 20 & 18 & 5 \\ 12 & 20 & 62 & 56 & 16 \\ 11 & 18 & 56 & 48 & 14 \\ 3 & 5 & 16 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1071] \quad 1^{1^{32}} 3^1 21^{1^{42}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 182, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 8 & 3 \\ 3 & 5 & 25 & 13 & 5 \\ 14 & 25 & 92 & 49 & 20 \\ 8 & 13 & 49 & 24 & 10 \\ 3 & 5 & 20 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1076] \quad 1^{1^{32}} 3^1 21^{1^{21}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 182, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 0 & 2 & 8 & 4 & 2 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 8 & 8 & 9 & 40 & 22 & 12 \\ 5 & 4 & 5 & 22 & 10 & 6 \\ 2 & 2 & 2 & 12 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1078] \quad 1^{1^{21}} 3^1 21^{1^{22}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 182, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 7 & 2 \\ 1 & 0 & 2 & 6 & 5 & 2 \\ 1 & 2 & 0 & 8 & 8 & 2 \\ 8 & 6 & 8 & 26 & 26 & 10 \\ 7 & 5 & 8 & 26 & 23 & 9 \\ 2 & 2 & 2 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1080] \quad 1^{1^{31}} 3^1 21^{1^{41}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 5 & 3 \\ 3 & 5 & 25 & 8 & 5 \\ 14 & 25 & 92 & 30 & 20 \\ 5 & 8 & 30 & 8 & 6 \\ 3 & 5 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1081] \quad 1^{1^{21}} 3^1 21^{1^{31}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 7 & 4 & 2 \\ 1 & 2 & 0 & 11 & 5 & 2 \\ 8 & 7 & 11 & 45 & 23 & 13 \\ 4 & 4 & 5 & 23 & 10 & 6 \\ 2 & 2 & 2 & 13 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1082] \quad 1^{1^{41}} 3^1 21^{1^{31}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 5 & 3 \\ 3 & 5 & 25 & 8 & 5 \\ 16 & 25 & 94 & 30 & 20 \\ 5 & 8 & 30 & 8 & 6 \\ 3 & 5 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

[1083] $1^{141}3^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 10 & 4 & 2 \\ 1 & 2 & 0 & 10 & 5 & 2 \\ 8 & 10 & 10 & 58 & 26 & 14 \\ 4 & 4 & 5 & 26 & 10 & 6 \\ 2 & 2 & 2 & 14 & 6 & 3 \end{pmatrix}_{b_0}$$

[1085] $1^{122}3^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 7 & 8 & 2 \\ 5 & 4 & 7 & 16 & 21 & 8 \\ 6 & 5 & 8 & 21 & 23 & 9 \\ 2 & 2 & 2 & 8 & 9 & 3 \end{pmatrix}_{b_0}$$

[1084] $1^{133}3^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 0 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 7 & 8 & 2 \\ 6 & 4 & 7 & 16 & 21 & 8 \\ 7 & 5 & 8 & 21 & 23 & 9 \\ 2 & 2 & 2 & 8 & 9 & 3 \end{pmatrix}_{b_0}$$

[1086] $1^{123}3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 5 & 7 & 2 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 5 & 5 & 5 & 13 & 23 & 7 \\ 8 & 7 & 9 & 23 & 35 & 11 \\ 2 & 2 & 2 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[1088] $1^{131}3^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 5 & 3 \\ 3 & 5 & 25 & 8 & 5 \\ 15 & 25 & 94 & 30 & 20 \\ 5 & 8 & 30 & 8 & 6 \\ 3 & 5 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

[1087] $1^{134}3^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 11 & 2 \\ 1 & 0 & 2 & 7 & 10 & 2 \\ 1 & 2 & 0 & 11 & 13 & 2 \\ 8 & 7 & 11 & 45 & 59 & 13 \\ 11 & 10 & 13 & 59 & 74 & 16 \\ 2 & 2 & 2 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

[1089] $1^{113}3^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 3 & 5 & 2 & 17 & 4 \\ 14 & 23 & 17 & 78 & 18 \\ 3 & 5 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

[1090] $1^{122}3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 5 & 6 & 2 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 5 & 5 & 5 & 13 & 18 & 7 \\ 6 & 6 & 6 & 18 & 20 & 8 \\ 2 & 2 & 2 & 7 & 8 & 3 \end{pmatrix}_{b_0}$$

[1091] $1^{122}3^{143} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 10 & 3 \\ 3 & 5 & 25 & 18 & 5 \\ 13 & 25 & 92 & 68 & 20 \\ 10 & 18 & 68 & 48 & 14 \\ 3 & 5 & 20 & 14 & 4 \end{pmatrix}_{b_0}$$

[1099] $1^{142}3^{142} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 18 & 8 & 3 \\ 3 & 5 & 30 & 13 & 5 \\ 18 & 30 & 134 & 58 & 24 \\ 8 & 13 & 58 & 24 & 10 \\ 3 & 5 & 24 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1097] \ 1^{141} 3^{121} 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 194, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 5 & 2 \\ 1 & 0 & 2 & 10 & 4 & 2 \\ 1 & 2 & 0 & 10 & 5 & 2 \\ 10 & 10 & 10 & 58 & 26 & 14 \\ 5 & 4 & 5 & 26 & 10 & 6 \\ 2 & 2 & 2 & 14 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1094] \ 1^{142} 3^{121} 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 194, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 0 & 2 & 6 & 4 & 2 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 8 & 6 & 8 & 26 & 18 & 10 \\ 5 & 4 & 5 & 18 & 10 & 6 \\ 2 & 2 & 2 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1093] \ 1^{124} 3^{121} 21^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 194, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 17 & 3 \\ 3 & 5 & 10 & 28 & 5 \\ 6 & 10 & 14 & 42 & 8 \\ 17 & 28 & 42 & 116 & 22 \\ 3 & 5 & 8 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[1098] \ 1^{144} 3^{121} 21^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 194, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 12 & 2 \\ 1 & 0 & 2 & 6 & 11 & 2 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 5 & 6 & 8 & 26 & 54 & 10 \\ 12 & 11 & 16 & 54 & 103 & 19 \\ 2 & 2 & 2 & 10 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1095] \ 1^{144} 3^{121} 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 194, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 11 & 2 \\ 1 & 0 & 2 & 6 & 10 & 2 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 8 & 6 & 8 & 26 & 46 & 10 \\ 11 & 10 & 13 & 46 & 74 & 16 \\ 2 & 2 & 2 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1096] \ 1^{141} 3^{121} 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 194, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 10 & 4 & 2 \\ 1 & 2 & 0 & 10 & 5 & 2 \\ 10 & 10 & 10 & 58 & 26 & 14 \\ 4 & 4 & 5 & 26 & 10 & 6 \\ 2 & 2 & 2 & 14 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1092] \ 1^{121} 3^{121} 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 194, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 6 & 4 & 2 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 6 & 6 & 8 & 26 & 18 & 10 \\ 4 & 4 & 5 & 18 & 10 & 6 \\ 2 & 2 & 2 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1100] \ 1^{122} 3^{121} 21^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 195, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 6 & 2 \\ 1 & 0 & 2 & 7 & 5 & 2 \\ 1 & 2 & 0 & 11 & 8 & 2 \\ 7 & 7 & 11 & 45 & 34 & 13 \\ 6 & 5 & 8 & 34 & 23 & 9 \\ 2 & 2 & 2 & 13 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1101] \ 1^{13} 3^{121} 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 196, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 1 & 13 & 2 \\ 1 & 2 & 0 & 3 & 17 & 2 \\ 2 & 1 & 3 & 1 & 17 & 3 \\ 12 & 13 & 17 & 17 & 127 & 21 \\ 2 & 2 & 2 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1105] \ 1^{123} 3^{121} 21^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 200, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 5 & 10 & 23 & 5 \\ 6 & 10 & 14 & 36 & 8 \\ 13 & 23 & 36 & 80 & 18 \\ 3 & 5 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

[1107] $1^{13}3^{14}21^{144} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 13 & 3 \\ 3 & 5 & 20 & 23 & 5 \\ 12 & 20 & 62 & 72 & 16 \\ 13 & 23 & 72 & 80 & 18 \\ 3 & 5 & 16 & 18 & 4 \end{pmatrix}_{b_0}$$

[1106] $1^{12}3^{14}21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 9 & 15 & 0 & 83 & 17 \\ 2 & 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[1102] $1^{13}3^{14}21^{144} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 3 & 5 & 2 & 18 & 4 \\ 13 & 23 & 18 & 80 & 18 \\ 3 & 5 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

[1104] $1^{13}3^{14}21^{134} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 13 & 3 \\ 3 & 5 & 15 & 23 & 5 \\ 9 & 15 & 34 & 54 & 12 \\ 13 & 23 & 54 & 80 & 18 \\ 3 & 5 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

[1103] $1^{14}3^{14}21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 11 & 11 & 0 & 71 & 15 \\ 2 & 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

[1108] $1^{13}3^{14}21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 10 & 13 & 0 & 74 & 16 \\ 2 & 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

[1109] $1^{114}3^{14}21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 2 & 12 & 2 \\ 1 & 2 & 0 & 1 & 14 & 2 \\ 2 & 2 & 1 & 0 & 12 & 2 \\ 12 & 12 & 14 & 12 & 96 & 18 \\ 2 & 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1110] $1^{113}3^{14}21^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 4 & 11 & 2 \\ 1 & 2 & 0 & 7 & 16 & 2 \\ 4 & 4 & 7 & 16 & 43 & 8 \\ 12 & 11 & 16 & 43 & 103 & 19 \\ 2 & 2 & 2 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$

[1111] $1^{14}3^{14}21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 5 & 5 & 23 & 5 \\ 3 & 5 & 2 & 18 & 4 \\ 14 & 23 & 18 & 80 & 18 \\ 3 & 5 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

[1112] $1^{13}3^{14}21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 3 & 10 & 2 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 2 & 3 & 4 & 5 & 23 & 5 \\ 10 & 10 & 13 & 23 & 74 & 16 \\ 2 & 2 & 2 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1113] \ 1^{122} 3^{121} 1^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 212, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 7 & 2 \\ 1 & 0 & 2 & 7 & 7 & 2 \\ 1 & 2 & 0 & 11 & 9 & 2 \\ 8 & 7 & 11 & 45 & 41 & 13 \\ 7 & 7 & 9 & 41 & 35 & 11 \\ 2 & 2 & 2 & 13 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1114] \ 1^{123} 3^{121} 1^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 212, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 3 & 8 & 2 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 4 & 3 & 4 & 5 & 20 & 5 \\ 9 & 8 & 12 & 20 & 56 & 14 \\ 2 & 2 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1115] \ 1^{124} 3^{121} 1^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 218, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 5 & 10 & 23 & 5 \\ 6 & 10 & 14 & 36 & 8 \\ 14 & 23 & 36 & 80 & 18 \\ 3 & 5 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1116] \ 1^{122} 3^{121} 1^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 220, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 8 & 2 \\ 1 & 0 & 2 & 7 & 7 & 2 \\ 1 & 2 & 0 & 11 & 9 & 2 \\ 9 & 7 & 11 & 45 & 41 & 13 \\ 8 & 7 & 9 & 41 & 35 & 11 \\ 2 & 2 & 2 & 13 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1117] \ 1^{133} 3^{121} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 222, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 10 & 13 & 0 & 74 & 16 \\ 2 & 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1118] \ 1^{123} 3^{121} 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 222, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 3 & 11 & 2 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 2 & 3 & 4 & 5 & 27 & 5 \\ 12 & 11 & 16 & 27 & 103 & 19 \\ 2 & 2 & 2 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1119] \ 1^{122} 3^{121} 1^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 224, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 0 & 2 & 6 & 8 & 2 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 8 & 6 & 8 & 26 & 40 & 10 \\ 10 & 8 & 12 & 40 & 56 & 14 \\ 2 & 2 & 2 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1120] \ 1^{141} 3^{121} 1^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 224, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 3 & 2 \\ 1 & 0 & 2 & 12 & 2 & 2 \\ 1 & 2 & 0 & 16 & 4 & 2 \\ 12 & 12 & 16 & 110 & 24 & 20 \\ 3 & 2 & 4 & 24 & 4 & 4 \\ 2 & 2 & 2 & 20 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1121] \ 1^{124} 3^{121} 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 224, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 13 & 2 \\ 1 & 0 & 2 & 5 & 13 & 2 \\ 1 & 2 & 0 & 5 & 17 & 2 \\ 5 & 5 & 5 & 13 & 43 & 7 \\ 13 & 13 & 17 & 43 & 127 & 21 \\ 2 & 2 & 2 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1122] \ 1^{122} 3^{121} 1^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 224, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 0 & 2 & 6 & 8 & 2 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 6 & 6 & 8 & 26 & 40 & 10 \\ 10 & 8 & 12 & 40 & 56 & 14 \\ 2 & 2 & 2 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1123] \ 1^{14} 3^{12} 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 225, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 4 & 9 & 2 \\ 1 & 2 & 0 & 7 & 15 & 2 \\ 4 & 4 & 7 & 16 & 39 & 8 \\ 11 & 9 & 15 & 39 & 83 & 17 \\ 2 & 2 & 2 & 8 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1124] \ 1^{41} 3^{12} 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 227, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 5 & 3 \\ 3 & 5 & 25 & 8 & 5 \\ 15 & 25 & 92 & 30 & 20 \\ 5 & 8 & 30 & 8 & 6 \\ 3 & 5 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1126] \ 1^{23} 3^{12} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 227, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 2 & 3 & 7 & 2 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 4 & 3 & 4 & 5 & 16 & 5 \\ 8 & 7 & 9 & 16 & 35 & 11 \\ 2 & 2 & 2 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1125] \ 1^{13} 3^{12} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 227, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 0 & 2 & 3 & 7 & 2 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 3 & 4 & 5 & 16 & 5 \\ 8 & 7 & 9 & 16 & 35 & 11 \\ 2 & 2 & 2 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1127] \ 21^{12} 3^{12} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 228, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 7 & 2 \\ 2 & 0 & 2 & 3 & 7 & 2 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 3 & 4 & 5 & 16 & 5 \\ 7 & 7 & 9 & 16 & 35 & 11 \\ 2 & 2 & 2 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1128] \ 1^{21} 3^{13} 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 230, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 8 & 4 & 2 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 7 & 8 & 9 & 40 & 22 & 12 \\ 4 & 4 & 5 & 22 & 10 & 6 \\ 2 & 2 & 2 & 12 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1129] \ 1^{31} 3^{14} 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 230, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 10 & 4 & 2 \\ 1 & 2 & 0 & 10 & 5 & 2 \\ 9 & 10 & 10 & 58 & 26 & 14 \\ 4 & 4 & 5 & 26 & 10 & 6 \\ 2 & 2 & 2 & 14 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1130] \ 1^{44} 3^{12} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 230, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 0 & 2 & 6 & 9 & 2 \\ 1 & 2 & 0 & 8 & 10 & 2 \\ 8 & 6 & 8 & 26 & 38 & 10 \\ 10 & 9 & 10 & 38 & 51 & 13 \\ 2 & 2 & 2 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1131] \ 1^{23} 3^{12} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 230, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 0 & 2 & 6 & 9 & 2 \\ 1 & 2 & 0 & 8 & 10 & 2 \\ 6 & 6 & 8 & 26 & 38 & 10 \\ 9 & 9 & 10 & 38 & 51 & 13 \\ 2 & 2 & 2 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1132] \ 1^{34} 3^{12} 21^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 233, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 14 & 3 \\ 3 & 5 & 15 & 23 & 5 \\ 9 & 15 & 34 & 54 & 12 \\ 14 & 23 & 54 & 80 & 18 \\ 3 & 5 & 12 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1133] \quad 1^{123} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 3 & 9 & 2 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 4 & 3 & 4 & 5 & 19 & 5 \\ 9 & 9 & 10 & 19 & 51 & 13 \\ 2 & 2 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1134] \quad 1^{114} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 0 & 2 & 3 & 9 & 2 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 3 & 4 & 5 & 19 & 5 \\ 10 & 9 & 10 & 19 & 51 & 13 \\ 2 & 2 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1135] \quad 1^{14} 3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 1 & 13 & 2 \\ 1 & 2 & 0 & 3 & 17 & 2 \\ 2 & 1 & 3 & 1 & 17 & 3 \\ 13 & 13 & 17 & 17 & 127 & 21 \\ 2 & 2 & 2 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1136] \quad 1^{124} 3^1 21^{134} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 11 & 2 \\ 1 & 0 & 2 & 7 & 9 & 2 \\ 1 & 2 & 0 & 11 & 15 & 2 \\ 7 & 7 & 11 & 45 & 63 & 13 \\ 11 & 9 & 15 & 63 & 83 & 17 \\ 2 & 2 & 2 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1137] \quad 1^{143} 3^1 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 0 & 2 & 6 & 7 & 2 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 8 & 6 & 8 & 26 & 32 & 10 \\ 8 & 7 & 9 & 32 & 35 & 11 \\ 2 & 2 & 2 & 10 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1138] \quad 1^{123} 3^1 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 0 & 2 & 6 & 7 & 2 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 6 & 6 & 8 & 26 & 32 & 10 \\ 8 & 7 & 9 & 32 & 35 & 11 \\ 2 & 2 & 2 & 10 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1139] \quad 21^{141} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 4 & 2 \\ 2 & 0 & 2 & 10 & 4 & 2 \\ 1 & 2 & 0 & 10 & 5 & 2 \\ 10 & 10 & 10 & 58 & 26 & 14 \\ 4 & 4 & 5 & 26 & 10 & 6 \\ 2 & 2 & 2 & 14 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1140] \quad 1^{132} 3^1 21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 242, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 6 & 2 \\ 1 & 0 & 2 & 8 & 5 & 2 \\ 1 & 2 & 0 & 14 & 8 & 2 \\ 9 & 8 & 14 & 70 & 42 & 16 \\ 6 & 5 & 8 & 42 & 23 & 9 \\ 2 & 2 & 2 & 16 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1141] \quad 1^{132} 3^1 21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 242, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 6 & 2 \\ 1 & 0 & 2 & 9 & 5 & 2 \\ 1 & 2 & 0 & 12 & 8 & 2 \\ 9 & 9 & 12 & 61 & 39 & 15 \\ 6 & 5 & 8 & 39 & 23 & 9 \\ 2 & 2 & 2 & 15 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1142] \quad 1^{131} 3^1 21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 7 & 3 \\ 3 & 5 & 25 & 13 & 5 \\ 14 & 25 & 92 & 49 & 20 \\ 7 & 13 & 49 & 24 & 10 \\ 3 & 5 & 20 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1143] \quad 1^{1^{22}} 3^{1^{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 5 & 9 & 2 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 5 & 5 & 5 & 13 & 28 & 7 \\ 8 & 9 & 10 & 28 & 51 & 13 \\ 2 & 2 & 2 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1144] \quad 1^{1^{22}} 3^{1^{133}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 0 & 2 & 7 & 10 & 2 \\ 1 & 2 & 0 & 11 & 13 & 2 \\ 7 & 7 & 11 & 45 & 59 & 13 \\ 9 & 10 & 13 & 59 & 74 & 16 \\ 2 & 2 & 2 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1145] \quad 1^{1^{32}} 3^{1^{142}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 7 & 2 \\ 1 & 0 & 2 & 10 & 7 & 2 \\ 1 & 2 & 0 & 15 & 9 & 2 \\ 10 & 10 & 15 & 88 & 57 & 18 \\ 7 & 7 & 9 & 57 & 35 & 11 \\ 2 & 2 & 2 & 18 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1146] \quad 1^{1^{12}} 3^{1^{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 3 & 5 & 2 & 21 & 4 \\ 15 & 28 & 21 & 116 & 22 \\ 3 & 5 & 4 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[1147] \quad 1^{1^{24}} 3^{1^{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 3 & 7 & 2 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 4 & 3 & 4 & 5 & 16 & 5 \\ 9 & 7 & 9 & 16 & 35 & 11 \\ 2 & 2 & 2 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1148] \quad 1^{1^{13}} 3^{1^{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 1 & 11 & 2 \\ 1 & 2 & 0 & 3 & 16 & 2 \\ 2 & 1 & 3 & 1 & 16 & 3 \\ 11 & 11 & 16 & 16 & 103 & 19 \\ 2 & 2 & 2 & 3 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1149] \quad 1^{1^{12}} 3^{1^{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 3 & 7 & 2 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 3 & 4 & 5 & 16 & 5 \\ 7 & 7 & 9 & 16 & 35 & 11 \\ 2 & 2 & 2 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1151] \quad 1^{1^{31}} 3^{1^{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 246, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 8 & 4 & 2 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 8 & 8 & 9 & 40 & 22 & 12 \\ 4 & 4 & 5 & 22 & 10 & 6 \\ 2 & 2 & 2 & 12 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1150] \quad 1^{1^{42}} 3^{1^{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 246, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 5 & 2 \\ 1 & 0 & 2 & 8 & 4 & 2 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 9 & 8 & 9 & 40 & 22 & 12 \\ 5 & 4 & 5 & 22 & 10 & 6 \\ 2 & 2 & 2 & 12 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1152] \quad 1^{1^{44}} 3^{1^{133}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 11 & 2 \\ 1 & 0 & 2 & 7 & 10 & 2 \\ 1 & 2 & 0 & 11 & 13 & 2 \\ 9 & 7 & 11 & 45 & 59 & 13 \\ 11 & 10 & 13 & 59 & 74 & 16 \\ 2 & 2 & 2 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

[1153] $1^{122} 3^{121} 1^{132} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 7 & 2 \\ 1 & 0 & 2 & 11 & 7 & 2 \\ 1 & 2 & 0 & 13 & 9 & 2 \\ 9 & 11 & 13 & 81 & 55 & 17 \\ 7 & 7 & 9 & 55 & 35 & 11 \\ 2 & 2 & 2 & 17 & 11 & 3 \end{pmatrix}_{b_0}$$

[1154] $1^{133} 3^{121} 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 249, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 5 & 10 & 2 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 4 & 5 & 5 & 13 & 33 & 7 \\ 10 & 10 & 13 & 33 & 74 & 16 \\ 2 & 2 & 2 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[1155] $1^{131} 3^{121} 1^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 250, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 17 & 5 & 3 \\ 3 & 5 & 30 & 8 & 5 \\ 17 & 30 & 134 & 36 & 24 \\ 5 & 8 & 36 & 8 & 6 \\ 3 & 5 & 24 & 6 & 4 \end{pmatrix}_{b_0}$$

[1156] $1^{133} 3^{121} 1^{132} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 251, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 8 & 2 \\ 1 & 0 & 2 & 9 & 7 & 2 \\ 1 & 2 & 0 & 12 & 9 & 2 \\ 9 & 9 & 12 & 61 & 48 & 15 \\ 8 & 7 & 9 & 48 & 35 & 11 \\ 2 & 2 & 2 & 15 & 11 & 3 \end{pmatrix}_{b_0}$$

[1157] $1^{122} 3^{121} 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 5 & 10 & 2 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 4 & 5 & 5 & 13 & 33 & 7 \\ 9 & 10 & 13 & 33 & 74 & 16 \\ 2 & 2 & 2 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[1158] $1^{144} 3^{121} 1^{144} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 14 & 3 \\ 3 & 5 & 20 & 23 & 5 \\ 12 & 20 & 62 & 72 & 16 \\ 14 & 23 & 72 & 80 & 18 \\ 3 & 5 & 16 & 18 & 4 \end{pmatrix}_{b_0}$$

[1159] $21^{123} 3^{121} 1^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 11 & 2 \\ 2 & 0 & 2 & 6 & 11 & 2 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 6 & 6 & 8 & 26 & 54 & 10 \\ 11 & 11 & 16 & 54 & 103 & 19 \\ 2 & 2 & 2 & 10 & 19 & 3 \end{pmatrix}_{b_0}$$

[1160] $1^{143} 3^{121} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 12 & 14 & 0 & 96 & 18 \\ 2 & 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[1161] $21^{132} 3^{121} 1^{143} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 10 & 8 & 2 \\ 2 & 0 & 2 & 10 & 8 & 2 \\ 1 & 2 & 0 & 15 & 12 & 2 \\ 10 & 10 & 15 & 88 & 72 & 18 \\ 8 & 8 & 12 & 72 & 56 & 14 \\ 2 & 2 & 2 & 18 & 14 & 3 \end{pmatrix}_{b_0}$$

[1162] $1^{121} 3^{131} 21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 9 & 4 & 2 \\ 1 & 2 & 0 & 12 & 5 & 2 \\ 8 & 9 & 12 & 61 & 27 & 15 \\ 4 & 4 & 5 & 27 & 10 & 6 \\ 2 & 2 & 2 & 15 & 6 & 3 \end{pmatrix}_{b_0}$$

[1164] $1^{122} 3^1 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 6 & 7 & 2 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 7 & 6 & 8 & 26 & 32 & 10 \\ 7 & 7 & 9 & 32 & 35 & 11 \\ 2 & 2 & 2 & 10 & 11 & 3 \end{pmatrix}_{b_0}$$

[1163] $1^{122} 3^1 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 0 & 2 & 6 & 7 & 2 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 7 & 6 & 8 & 26 & 32 & 10 \\ 9 & 7 & 9 & 32 & 35 & 11 \\ 2 & 2 & 2 & 10 & 11 & 3 \end{pmatrix}_{b_0}$$

[1168] $1^{123} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 5 & 9 & 2 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 5 & 5 & 13 & 28 & 7 \\ 9 & 9 & 10 & 28 & 51 & 13 \\ 2 & 2 & 2 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[1167] $1^{123} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 0 & 2 & 5 & 9 & 2 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 6 & 5 & 5 & 13 & 28 & 7 \\ 10 & 9 & 10 & 28 & 51 & 13 \\ 2 & 2 & 2 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[1166] $1^{124} 3^1 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 13 & 2 \\ 1 & 0 & 2 & 4 & 13 & 2 \\ 1 & 2 & 0 & 7 & 17 & 2 \\ 5 & 4 & 7 & 16 & 47 & 8 \\ 13 & 13 & 17 & 47 & 127 & 21 \\ 2 & 2 & 2 & 8 & 21 & 3 \end{pmatrix}_{b_0}$$

[1165] $1^{123} 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 3 & 10 & 2 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 4 & 3 & 4 & 5 & 23 & 5 \\ 10 & 10 & 13 & 23 & 74 & 16 \\ 2 & 2 & 2 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1170] $1^{141} 3^1 21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 263, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 5 & 2 \\ 1 & 0 & 2 & 11 & 4 & 2 \\ 1 & 2 & 0 & 13 & 5 & 2 \\ 11 & 11 & 13 & 81 & 31 & 17 \\ 5 & 4 & 5 & 31 & 10 & 6 \\ 2 & 2 & 2 & 17 & 6 & 3 \end{pmatrix}_{b_0}$$

[1171] $1^{142} 3^1 21^{142} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 263, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 8 & 3 \\ 3 & 5 & 25 & 13 & 5 \\ 15 & 25 & 92 & 49 & 20 \\ 8 & 13 & 49 & 24 & 10 \\ 3 & 5 & 20 & 10 & 4 \end{pmatrix}_{b_0}$$

[1169] $1^{131} 3^1 21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 263, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 5 & 2 \\ 1 & 0 & 2 & 9 & 4 & 2 \\ 1 & 2 & 0 & 12 & 5 & 2 \\ 10 & 9 & 12 & 61 & 27 & 15 \\ 5 & 4 & 5 & 27 & 10 & 6 \\ 2 & 2 & 2 & 15 & 6 & 3 \end{pmatrix}_{b_0}$$

[1172] $1^{124} 3^1 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 5 & 11 & 2 \\ 1 & 2 & 0 & 5 & 11 & 2 \\ 4 & 5 & 5 & 13 & 33 & 7 \\ 11 & 11 & 11 & 33 & 71 & 15 \\ 2 & 2 & 2 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[1173] $1^{14} 3^{12} 3^1 21^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 264$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 1 & 9 & 2 \\ 1 & 2 & 0 & 3 & 15 & 2 \\ 2 & 1 & 3 & 1 & 15 & 3 \\ 11 & 9 & 15 & 15 & 83 & 17 \\ 2 & 2 & 2 & 3 & 17 & 3 \end{pmatrix}_{b_0}$$

[1174] $1^{34} 3^{12} 3^1 21^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 264$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 11 & 2 \\ 1 & 0 & 2 & 5 & 11 & 2 \\ 1 & 2 & 0 & 5 & 11 & 2 \\ 6 & 5 & 5 & 13 & 33 & 7 \\ 11 & 11 & 11 & 33 & 71 & 15 \\ 2 & 2 & 2 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[1175] $1^{33} 3^{12} 3^1 21^{133} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 265$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 0 & 2 & 7 & 10 & 2 \\ 1 & 2 & 0 & 11 & 13 & 2 \\ 8 & 7 & 11 & 45 & 59 & 13 \\ 10 & 10 & 13 & 59 & 74 & 16 \\ 2 & 2 & 2 & 13 & 16 & 3 \end{pmatrix}_{b_0}$$

[1176] $1^3 3^{14} 3^1 21^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 268$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 3 & 12 & 2 \\ 1 & 2 & 0 & 4 & 14 & 2 \\ 2 & 3 & 4 & 5 & 26 & 5 \\ 11 & 12 & 14 & 26 & 96 & 18 \\ 2 & 2 & 2 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

[1177] $21^{132} 3^1 21^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 268$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 7 & 2 \\ 2 & 0 & 2 & 8 & 7 & 2 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 8 & 8 & 9 & 40 & 39 & 12 \\ 7 & 7 & 9 & 39 & 35 & 11 \\ 2 & 2 & 2 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

[1178] $1^{14} 3^{14} 3^1 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 269$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 2 & 13 & 2 \\ 1 & 2 & 0 & 1 & 17 & 2 \\ 1 & 2 & 1 & 0 & 13 & 2 \\ 13 & 13 & 17 & 13 & 127 & 21 \\ 2 & 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1179] $1^{42} 3^{13} 3^1 21^{132} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 270$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 6 & 2 \\ 1 & 0 & 2 & 9 & 5 & 2 \\ 1 & 2 & 0 & 12 & 8 & 2 \\ 10 & 9 & 12 & 61 & 39 & 15 \\ 6 & 5 & 8 & 39 & 23 & 9 \\ 2 & 2 & 2 & 15 & 9 & 3 \end{pmatrix}_{b_0}$$

[1180] $1^{13} 3^{12} 3^1 21^{123} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 270$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 6 & 10 & 2 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 5 & 6 & 8 & 26 & 46 & 10 \\ 10 & 10 & 13 & 46 & 74 & 16 \\ 2 & 2 & 2 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[1181] $1^{21} 3^{14} 3^1 21^{141} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 270$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 12 & 4 & 2 \\ 1 & 2 & 0 & 16 & 5 & 2 \\ 10 & 12 & 16 & 110 & 36 & 20 \\ 4 & 4 & 5 & 36 & 10 & 6 \\ 2 & 2 & 2 & 20 & 6 & 3 \end{pmatrix}_{b_0}$$

[1182] $1^3 3^{14} 3^1 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -59$, $\det = 271$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 2 & 13 & 2 \\ 1 & 2 & 0 & 1 & 17 & 2 \\ 1 & 2 & 1 & 0 & 13 & 2 \\ 12 & 13 & 17 & 13 & 127 & 21 \\ 2 & 2 & 2 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1185] $1^{13} 1^{124} 3^1 21^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 6 & 12 & 2 \\ 1 & 2 & 0 & 8 & 14 & 2 \\ 5 & 6 & 8 & 26 & 52 & 10 \\ 11 & 12 & 14 & 52 & 96 & 18 \\ 2 & 2 & 2 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[1183] $1^{43} 1^{142} 3^1 21^{142} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 8 & 2 \\ 1 & 0 & 2 & 12 & 7 & 2 \\ 1 & 2 & 0 & 16 & 9 & 2 \\ 12 & 12 & 16 & 110 & 64 & 20 \\ 8 & 7 & 9 & 64 & 35 & 11 \\ 2 & 2 & 2 & 20 & 11 & 3 \end{pmatrix}_{b_0}$$

[1184] $1^{14} 1^{143} 3^1 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 2 & 12 & 2 \\ 1 & 2 & 0 & 1 & 14 & 2 \\ 1 & 2 & 1 & 0 & 12 & 2 \\ 12 & 12 & 14 & 12 & 96 & 18 \\ 2 & 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1186] $1^{23} 1^{122} 3^1 21^{133} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 0 & 2 & 7 & 8 & 2 \\ 1 & 2 & 0 & 11 & 12 & 2 \\ 7 & 7 & 11 & 45 & 52 & 13 \\ 9 & 8 & 12 & 52 & 56 & 14 \\ 2 & 2 & 2 & 13 & 14 & 3 \end{pmatrix}_{b_0}$$

[1189] $1^{14} 1^{13} 3^1 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 11 & 16 & 0 & 103 & 19 \\ 2 & 2 & 2 & 0 & 19 & 3 \end{pmatrix}_{b_0}$$

[1188] $1^{34} 1^{123} 3^1 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 10 & 2 \\ 1 & 0 & 2 & 6 & 9 & 2 \\ 1 & 2 & 0 & 8 & 10 & 2 \\ 7 & 6 & 8 & 26 & 38 & 10 \\ 10 & 9 & 10 & 38 & 51 & 13 \\ 2 & 2 & 2 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

[1187] $1^{33} 1^{123} 3^1 21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 0 & 2 & 6 & 9 & 2 \\ 1 & 2 & 0 & 8 & 10 & 2 \\ 7 & 6 & 8 & 26 & 38 & 10 \\ 9 & 9 & 10 & 38 & 51 & 13 \\ 2 & 2 & 2 & 10 & 13 & 3 \end{pmatrix}_{b_0}$$

[1191] $1^{24} 1^{123} 3^1 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 275, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 5 & 10 & 2 \\ 1 & 2 & 0 & 5 & 13 & 2 \\ 5 & 5 & 5 & 13 & 33 & 7 \\ 11 & 10 & 13 & 33 & 74 & 16 \\ 2 & 2 & 2 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[1190] $1^{43} 1^{11} 3^1 21^{143} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 275, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 11 & 3 \\ 3 & 5 & 25 & 18 & 5 \\ 15 & 25 & 92 & 68 & 20 \\ 11 & 18 & 68 & 48 & 14 \\ 3 & 5 & 20 & 14 & 4 \end{pmatrix}_{b_0}$$

[1192] $1^{14} 1^{13} 3^1 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 277, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 1 & 11 & 2 \\ 1 & 2 & 0 & 3 & 16 & 2 \\ 2 & 1 & 3 & 1 & 16 & 3 \\ 12 & 11 & 16 & 16 & 103 & 19 \\ 2 & 2 & 2 & 3 & 19 & 3 \end{pmatrix}_{b_0} \\
[1193] \quad & 21^{13}3^{121}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 278, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 1 & 3 & 11 & 2 \\ 2 & 0 & 2 & 3 & 11 & 2 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 3 & 3 & 4 & 5 & 27 & 5 \\ 11 & 11 & 16 & 27 & 103 & 19 \\ 2 & 2 & 2 & 5 & 19 & 3 \end{pmatrix}_{b_0} \\
[1194] \quad & 1^{14}3^{121}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 12 & 14 & 0 & 96 & 18 \\ 2 & 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0} \\
[1195] \quad & 1^{124}3^{121}1^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 4 & 8 & 2 \\ 1 & 2 & 0 & 7 & 12 & 2 \\ 5 & 4 & 7 & 16 & 32 & 8 \\ 10 & 8 & 12 & 32 & 56 & 14 \\ 2 & 2 & 2 & 8 & 14 & 3 \end{pmatrix}_{b_0} \\
[1196] \quad & 1^{112}3^{121}1^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 0 & 2 & 4 & 8 & 2 \\ 1 & 2 & 0 & 7 & 12 & 2 \\ 6 & 4 & 7 & 16 & 32 & 8 \\ 10 & 8 & 12 & 32 & 56 & 14 \\ 2 & 2 & 2 & 8 & 14 & 3 \end{pmatrix}_{b_0} \\
[1197] \quad & 21^{131}3^{121}1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 1 & 8 & 4 & 2 \\ 2 & 0 & 2 & 8 & 4 & 2 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 8 & 8 & 9 & 40 & 22 & 12 \\ 4 & 4 & 5 & 22 & 10 & 6 \\ 2 & 2 & 2 & 12 & 6 & 3 \end{pmatrix}_{b_0} \\
[1198] \quad & 1^{114}3^{121}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 2 & 12 & 2 \\ 1 & 2 & 0 & 1 & 14 & 2 \\ 1 & 2 & 1 & 0 & 12 & 2 \\ 11 & 12 & 14 & 12 & 96 & 18 \\ 2 & 2 & 2 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[1199] \quad & 1^{112}3^{121}1^{134} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 281, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 11 & 2 \\ 1 & 0 & 2 & 5 & 9 & 2 \\ 1 & 2 & 0 & 10 & 15 & 2 \\ 7 & 5 & 10 & 33 & 54 & 11 \\ 11 & 9 & 15 & 54 & 83 & 17 \\ 2 & 2 & 2 & 11 & 17 & 3 \end{pmatrix}_{b_0} \\
[1200] \quad & 1^{112}3^{121}1^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 281, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 11 & 2 \\ 1 & 0 & 2 & 4 & 9 & 2 \\ 1 & 2 & 0 & 7 & 15 & 2 \\ 6 & 4 & 7 & 16 & 39 & 8 \\ 11 & 9 & 15 & 39 & 83 & 17 \\ 2 & 2 & 2 & 8 & 17 & 3 \end{pmatrix}_{b_0} \\
[1201] \quad & 1^{131}3^{121}1^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 282, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 10 & 4 & 2 \\ 1 & 2 & 0 & 15 & 5 & 2 \\ 10 & 10 & 15 & 88 & 32 & 18 \\ 4 & 4 & 5 & 32 & 10 & 6 \\ 2 & 2 & 2 & 18 & 6 & 3 \end{pmatrix}_{b_0} \\
[1202] \quad & 1^{113}3^{121}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 284, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 3 & 5 & 2 & 21 & 4 \\ 16 & 28 & 21 & 116 & 22 \\ 3 & 5 & 4 & 22 & 4 \end{pmatrix}_{b_0} \\
[1203] & 21^{131} 3^{121} 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 1 & 9 & 4 & 2 \\ 2 & 0 & 2 & 9 & 4 & 2 \\ 1 & 2 & 0 & 12 & 5 & 2 \\ 9 & 9 & 12 & 61 & 27 & 15 \\ 4 & 4 & 5 & 27 & 10 & 6 \\ 2 & 2 & 2 & 15 & 6 & 3 \end{pmatrix}_{b_0} \\
[1206] & 1^{122} 3^{121} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 6 & 8 & 2 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 5 & 6 & 8 & 26 & 40 & 10 \\ 8 & 8 & 12 & 40 & 56 & 14 \\ 2 & 2 & 2 & 10 & 14 & 3 \end{pmatrix}_{b_0} \\
[1205] & 1^{142} 3^{121} 21^{143} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 8 & 2 \\ 1 & 0 & 2 & 12 & 8 & 2 \\ 1 & 2 & 0 & 16 & 12 & 2 \\ 10 & 12 & 16 & 110 & 80 & 20 \\ 8 & 8 & 12 & 80 & 56 & 14 \\ 2 & 2 & 2 & 20 & 14 & 3 \end{pmatrix}_{b_0} \\
[1204] & 1^{122} 3^{121} 21^{143} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 8 & 2 \\ 1 & 0 & 2 & 8 & 8 & 2 \\ 1 & 2 & 0 & 14 & 12 & 2 \\ 9 & 8 & 14 & 70 & 64 & 16 \\ 8 & 8 & 12 & 64 & 56 & 14 \\ 2 & 2 & 2 & 16 & 14 & 3 \end{pmatrix}_{b_0} \\
[1209] & 1^{124} 3^{121} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 3 & 9 & 2 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 4 & 3 & 4 & 5 & 19 & 5 \\ 10 & 9 & 10 & 19 & 51 & 13 \\ 2 & 2 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0} \\
[1207] & 1^{113} 3^{121} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 3 & 9 & 2 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 3 & 4 & 5 & 19 & 5 \\ 9 & 9 & 10 & 19 & 51 & 13 \\ 2 & 2 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0} \\
[1208] & 1^{112} 3^{121} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 0 & 2 & 3 & 9 & 2 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 3 & 4 & 5 & 19 & 5 \\ 8 & 9 & 10 & 19 & 51 & 13 \\ 2 & 2 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0} \\
[1212] & 1^{132} 3^{121} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 293, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 7 & 2 \\ 1 & 0 & 2 & 8 & 7 & 2 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 8 & 8 & 9 & 40 & 39 & 12 \\ 7 & 7 & 9 & 39 & 35 & 11 \\ 2 & 2 & 2 & 12 & 11 & 3 \end{pmatrix}_{b_0} \\
[1211] & 1^{144} 3^{121} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 293, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 9 & 2 \\ 1 & 0 & 2 & 8 & 7 & 2 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 9 & 8 & 9 & 40 & 39 & 12 \\ 9 & 7 & 9 & 39 & 35 & 11 \\ 2 & 2 & 2 & 12 & 11 & 3 \end{pmatrix}_{b_0} \\
[1210] & 1^{114} 3^{121} 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 293, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 5 & 5 & 28 & 5 \\ 3 & 5 & 2 & 21 & 4 \\ 17 & 28 & 21 & 116 & 22 \\ 3 & 5 & 4 & 22 & 4 \end{pmatrix}_{b_0} \\
[1213] & 21^{122} 3^{121} 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 294, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 1 & 6 & 7 & 2 \\ 2 & 0 & 2 & 6 & 7 & 2 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 6 & 6 & 8 & 26 & 32 & 10 \\ 7 & 7 & 9 & 32 & 35 & 11 \\ 2 & 2 & 2 & 10 & 11 & 3 \end{pmatrix}_{b_0} \\
[1214] & 1^{143} 3^{121} 21^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 295, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 7 & 2 \\ 1 & 0 & 2 & 7 & 5 & 2 \\ 1 & 2 & 0 & 11 & 8 & 2 \\ 9 & 7 & 11 & 45 & 34 & 13 \\ 7 & 5 & 8 & 34 & 23 & 9 \\ 2 & 2 & 2 & 13 & 9 & 3 \end{pmatrix}_{b_0} \\
[1215] & 1^{142} 3^{121} 21^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 295, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 6 & 2 \\ 1 & 0 & 2 & 7 & 5 & 2 \\ 1 & 2 & 0 & 11 & 8 & 2 \\ 9 & 7 & 11 & 45 & 34 & 13 \\ 6 & 5 & 8 & 34 & 23 & 9 \\ 2 & 2 & 2 & 13 & 9 & 3 \end{pmatrix}_{b_0} \\
[1216] & 1^{124} 3^{121} 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 3 & 8 & 2 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 4 & 3 & 4 & 5 & 20 & 5 \\ 10 & 8 & 12 & 20 & 56 & 14 \\ 2 & 2 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0} \\
[1217] & 1^{114} 3^{121} 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 0 & 2 & 3 & 8 & 2 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 3 & 3 & 4 & 5 & 20 & 5 \\ 10 & 8 & 12 & 20 & 56 & 14 \\ 2 & 2 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0} \\
[1218] & 1^{133} 3^{121} 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 0 & 2 & 6 & 8 & 2 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 7 & 6 & 8 & 26 & 40 & 10 \\ 9 & 8 & 12 & 40 & 56 & 14 \\ 2 & 2 & 2 & 10 & 14 & 3 \end{pmatrix}_{b_0} \\
[1219] & 1^{122} 3^{121} 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 297, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 8 & 7 & 2 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 7 & 8 & 9 & 40 & 39 & 12 \\ 7 & 7 & 9 & 39 & 35 & 11 \\ 2 & 2 & 2 & 12 & 11 & 3 \end{pmatrix}_{b_0} \\
[1220] & 1^{141} 3^{121} 21^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 298, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 18 & 5 & 3 \\ 3 & 5 & 30 & 8 & 5 \\ 18 & 30 & 134 & 36 & 24 \\ 5 & 8 & 36 & 8 & 6 \\ 3 & 5 & 24 & 6 & 4 \end{pmatrix}_{b_0} \\
[1221] & 1^{121} 3^{141} 21^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 303, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 11 & 4 & 2 \\ 1 & 2 & 0 & 13 & 5 & 2 \\ 9 & 11 & 13 & 81 & 31 & 17 \\ 4 & 4 & 5 & 31 & 10 & 6 \\ 2 & 2 & 2 & 17 & 6 & 3 \end{pmatrix}_{b_0} \\
[1222] & 1^{123} 3^{121} 21^{134} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 12 & 2 \\ 1 & 0 & 2 & 7 & 13 & 2 \\ 1 & 2 & 0 & 11 & 17 & 2 \\ 7 & 7 & 11 & 45 & 77 & 13 \\ 12 & 13 & 17 & 77 & 127 & 21 \\ 2 & 2 & 2 & 13 & 21 & 3 \end{pmatrix}_{b_0}$$

[1223] $1^{122} 3^{121} 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 305, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 7 & 2 \\ 1 & 0 & 2 & 9 & 7 & 2 \\ 1 & 2 & 0 & 12 & 9 & 2 \\ 8 & 9 & 12 & 61 & 48 & 15 \\ 7 & 7 & 9 & 48 & 35 & 11 \\ 2 & 2 & 2 & 15 & 11 & 3 \end{pmatrix}_{b_0}$$

[1224] $1^{132} 3^{121} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 0 & 2 & 8 & 7 & 2 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 8 & 8 & 9 & 40 & 39 & 12 \\ 8 & 7 & 9 & 39 & 35 & 11 \\ 2 & 2 & 2 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

[1225] $1^{143} 3^{121} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 8 & 2 \\ 1 & 0 & 2 & 8 & 7 & 2 \\ 1 & 2 & 0 & 9 & 9 & 2 \\ 9 & 8 & 9 & 40 & 39 & 12 \\ 8 & 7 & 9 & 39 & 35 & 11 \\ 2 & 2 & 2 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

[1226] $1^{123} 3^{121} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 4 & 8 & 2 \\ 1 & 2 & 0 & 7 & 12 & 2 \\ 5 & 4 & 7 & 16 & 32 & 8 \\ 9 & 8 & 12 & 32 & 56 & 14 \\ 2 & 2 & 2 & 8 & 14 & 3 \end{pmatrix}_{b_0}$$

[1227] $1^{143} 3^{121} 21^{143} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 9 & 2 \\ 1 & 0 & 2 & 8 & 8 & 2 \\ 1 & 2 & 0 & 14 & 12 & 2 \\ 10 & 8 & 14 & 70 & 64 & 16 \\ 9 & 8 & 12 & 64 & 56 & 14 \\ 2 & 2 & 2 & 16 & 14 & 3 \end{pmatrix}_{b_0}$$

[1228] $1^{113} 3^{121} 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 5 & 5 & 33 & 5 \\ 3 & 5 & 2 & 24 & 4 \\ 19 & 33 & 24 & 164 & 26 \\ 3 & 5 & 4 & 26 & 4 \end{pmatrix}_{b_0}$$

[1229] $1^{121} 3^{121} 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 313, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & 0 & 2 & 7 & 5 & 2 \\ 1 & 2 & 0 & 11 & 8 & 2 \\ 8 & 7 & 11 & 45 & 34 & 13 \\ 6 & 5 & 8 & 34 & 23 & 9 \\ 2 & 2 & 2 & 13 & 9 & 3 \end{pmatrix}_{b_0}$$

[1230] $1^{131} 3^{121} 21^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 313, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 6 & 2 \\ 1 & 0 & 2 & 10 & 5 & 2 \\ 1 & 2 & 0 & 15 & 8 & 2 \\ 10 & 10 & 15 & 88 & 47 & 18 \\ 6 & 5 & 8 & 47 & 23 & 9 \\ 2 & 2 & 2 & 18 & 9 & 3 \end{pmatrix}_{b_0}$$

[1231] $1^{132} 3^{121} 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 6 & 2 \\ 1 & 0 & 2 & 10 & 6 & 2 \\ 1 & 2 & 0 & 10 & 6 & 2 \\ 9 & 10 & 10 & 58 & 36 & 14 \\ 6 & 6 & 6 & 36 & 20 & 8 \\ 2 & 2 & 2 & 14 & 8 & 3 \end{pmatrix}_{b_0}$$

[1232] $1^{114} 3^{121} 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 321, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 5 & 5 & 33 & 5 \\ 3 & 5 & 2 & 24 & 4 \\ 20 & 33 & 24 & 164 & 26 \\ 3 & 5 & 4 & 26 & 4 \end{pmatrix}_{b_0}$$

$$[1233] \quad 1^{1^{12}3^1} 21^{1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 321, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 2 & 5 & 9 & 2 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 5 & 5 & 13 & 28 & 7 \\ 8 & 9 & 10 & 28 & 51 & 13 \\ 2 & 2 & 2 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1234] \quad 1^{1^{13}3^1} 21^{1^{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 324, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 3 & 8 & 2 \\ 1 & 2 & 0 & 4 & 12 & 2 \\ 3 & 3 & 4 & 5 & 20 & 5 \\ 9 & 8 & 12 & 20 & 56 & 14 \\ 2 & 2 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1235] \quad 1^{1^{42}3^1} 21^{1^{42}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 326, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 6 & 2 \\ 1 & 0 & 2 & 12 & 5 & 2 \\ 1 & 2 & 0 & 16 & 8 & 2 \\ 12 & 12 & 16 & 110 & 52 & 20 \\ 6 & 5 & 8 & 52 & 23 & 9 \\ 2 & 2 & 2 & 20 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1236] \quad 1^{1^{22}3^1} 21^{1^{22}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 0 & 2 & 6 & 7 & 2 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 8 & 6 & 8 & 26 & 32 & 10 \\ 9 & 7 & 9 & 32 & 35 & 11 \\ 2 & 2 & 2 & 10 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1237] \quad 1^{1^{22}3^1} 21^{1^{22}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 0 & 2 & 6 & 7 & 2 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 6 & 6 & 8 & 26 & 32 & 10 \\ 7 & 7 & 9 & 32 & 35 & 11 \\ 2 & 2 & 2 & 10 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1238] \quad 21^{1^{13}3^1} 21^{1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 9 & 2 \\ 2 & 0 & 2 & 3 & 9 & 2 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 3 & 4 & 5 & 19 & 5 \\ 9 & 9 & 10 & 19 & 51 & 13 \\ 2 & 2 & 2 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1239] \quad 1^{1^{22}3^1} 21^{1^{33}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 10 & 2 \\ 1 & 0 & 2 & 7 & 8 & 2 \\ 1 & 2 & 0 & 11 & 12 & 2 \\ 8 & 7 & 11 & 45 & 52 & 13 \\ 10 & 8 & 12 & 52 & 56 & 14 \\ 2 & 2 & 2 & 13 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1240] \quad 1^{1^{13}3^1} 21^{1^{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 338, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 3 & 11 & 2 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 2 & 3 & 4 & 5 & 27 & 5 \\ 11 & 11 & 16 & 27 & 103 & 19 \\ 2 & 2 & 2 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1241] \quad 1^{1^{31}3^1} 21^{1^{31}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 339, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 9 & 4 & 2 \\ 1 & 2 & 0 & 12 & 5 & 2 \\ 9 & 9 & 12 & 61 & 27 & 15 \\ 4 & 4 & 5 & 27 & 10 & 6 \\ 2 & 2 & 2 & 15 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1242] \quad 1^{1^{23}3^1} 21^{1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 339, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 5 & 9 & 2 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 5 & 5 & 5 & 13 & 28 & 7 \\ 9 & 9 & 10 & 28 & 51 & 13 \\ 2 & 2 & 2 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[1243] $1^{124} 3^{121} 21^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 339, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 5 & 9 & 2 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 5 & 5 & 5 & 13 & 28 & 7 \\ 10 & 9 & 10 & 28 & 51 & 13 \\ 2 & 2 & 2 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[1244] $1^{141} 3^{121} 21^{131} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 341, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 9 & 4 & 2 \\ 1 & 2 & 0 & 12 & 5 & 2 \\ 10 & 9 & 12 & 61 & 27 & 15 \\ 4 & 4 & 5 & 27 & 10 & 6 \\ 2 & 2 & 2 & 15 & 6 & 3 \end{pmatrix}_{b_0}$$

[1245] $1^{14} 3^{121} 21^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 341, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 3 & 13 & 2 \\ 1 & 2 & 0 & 4 & 17 & 2 \\ 2 & 3 & 4 & 5 & 30 & 5 \\ 13 & 13 & 17 & 30 & 127 & 21 \\ 2 & 2 & 2 & 5 & 21 & 3 \end{pmatrix}_{b_0}$$

[1246] $1^{134} 3^{121} 21^{134} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 344, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 13 & 2 \\ 1 & 0 & 2 & 7 & 13 & 2 \\ 1 & 2 & 0 & 11 & 17 & 2 \\ 8 & 7 & 11 & 45 & 77 & 13 \\ 13 & 13 & 17 & 77 & 127 & 21 \\ 2 & 2 & 2 & 13 & 21 & 3 \end{pmatrix}_{b_0}$$

[1247] $1^{144} 3^{121} 21^{143} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 344, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 10 & 2 \\ 1 & 0 & 2 & 10 & 8 & 2 \\ 1 & 2 & 0 & 15 & 12 & 2 \\ 11 & 10 & 15 & 88 & 72 & 18 \\ 10 & 8 & 12 & 72 & 56 & 14 \\ 2 & 2 & 2 & 18 & 14 & 3 \end{pmatrix}_{b_0}$$

[1248] $1^{14} 3^{121} 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 344, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 0 & 13 & 2 \\ 1 & 2 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 13 & 17 & 0 & 127 & 21 \\ 2 & 2 & 2 & 0 & 21 & 3 \end{pmatrix}_{b_0}$$

[1249] $1^{121} 3^{121} 21^{142} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 346, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 6 & 2 \\ 1 & 0 & 2 & 8 & 5 & 2 \\ 1 & 2 & 0 & 14 & 8 & 2 \\ 10 & 8 & 14 & 70 & 42 & 16 \\ 6 & 5 & 8 & 42 & 23 & 9 \\ 2 & 2 & 2 & 16 & 9 & 3 \end{pmatrix}_{b_0}$$

[1250] $1^{113} 3^{121} 21^{124} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 347, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 4 & 11 & 2 \\ 1 & 2 & 0 & 7 & 16 & 2 \\ 4 & 4 & 7 & 16 & 43 & 8 \\ 11 & 11 & 16 & 43 & 103 & 19 \\ 2 & 2 & 2 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$

[1251] $1^{123} 3^{121} 21^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 348, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 5 & 12 & 2 \\ 1 & 2 & 0 & 5 & 14 & 2 \\ 5 & 5 & 5 & 13 & 38 & 7 \\ 11 & 12 & 14 & 38 & 96 & 18 \\ 2 & 2 & 2 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[1252] $1^{133} 3^{121} 21^{122} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 348, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 8 & 2 \\ 1 & 0 & 2 & 6 & 7 & 2 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 7 & 6 & 8 & 26 & 32 & 10 \\ 8 & 7 & 9 & 32 & 35 & 11 \\ 2 & 2 & 2 & 10 & 11 & 3 \end{pmatrix}_{b_0}$$

[1254] $1^{131}3^{121}1^{141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 350, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 4 & 2 \\ 1 & 0 & 2 & 10 & 4 & 2 \\ 1 & 2 & 0 & 15 & 5 & 2 \\ 11 & 10 & 15 & 88 & 32 & 18 \\ 4 & 4 & 5 & 32 & 10 & 6 \\ 2 & 2 & 2 & 18 & 6 & 3 \end{pmatrix}_{b_0}$$

[1253] $1^{124}3^{121}1^{144} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 350, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 3 & 11 & 2 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 4 & 3 & 4 & 5 & 27 & 5 \\ 12 & 11 & 16 & 27 & 103 & 19 \\ 2 & 2 & 2 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[1255] $1^{114}3^{121}1^{144} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 353, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 11 & 2 \\ 1 & 0 & 2 & 3 & 9 & 2 \\ 1 & 2 & 0 & 4 & 15 & 2 \\ 3 & 3 & 4 & 5 & 24 & 5 \\ 11 & 9 & 15 & 24 & 83 & 17 \\ 2 & 2 & 2 & 5 & 17 & 3 \end{pmatrix}_{b_0}$$

[1256] $1^{123}3^{121}1^{134} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 355, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 12 & 2 \\ 1 & 0 & 2 & 7 & 11 & 2 \\ 1 & 2 & 0 & 11 & 16 & 2 \\ 7 & 7 & 11 & 45 & 70 & 13 \\ 12 & 11 & 16 & 70 & 103 & 19 \\ 2 & 2 & 2 & 13 & 19 & 3 \end{pmatrix}_{b_0}$$

[1257] $1^{122}3^{121}1^{144} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 11 & 2 \\ 1 & 0 & 2 & 8 & 9 & 2 \\ 1 & 2 & 0 & 14 & 15 & 2 \\ 9 & 8 & 14 & 70 & 78 & 16 \\ 11 & 9 & 15 & 78 & 83 & 17 \\ 2 & 2 & 2 & 16 & 17 & 3 \end{pmatrix}_{b_0}$$

[1258] $1^{124}3^{121}1^{144} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 13 & 2 \\ 1 & 0 & 2 & 5 & 13 & 2 \\ 1 & 2 & 0 & 5 & 17 & 2 \\ 4 & 5 & 5 & 13 & 43 & 7 \\ 13 & 13 & 17 & 43 & 127 & 21 \\ 2 & 2 & 2 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[1259] $1^{123}3^{121}1^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 0 & 2 & 6 & 8 & 2 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 6 & 6 & 8 & 26 & 40 & 10 \\ 9 & 8 & 12 & 40 & 56 & 14 \\ 2 & 2 & 2 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[1260] $1^{122}3^{121}1^{133} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 9 & 2 \\ 1 & 0 & 2 & 7 & 8 & 2 \\ 1 & 2 & 0 & 11 & 12 & 2 \\ 9 & 7 & 11 & 45 & 52 & 13 \\ 9 & 8 & 12 & 52 & 56 & 14 \\ 2 & 2 & 2 & 13 & 14 & 3 \end{pmatrix}_{b_0}$$

[1261] $1^{132}3^{121}1^{142} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 7 & 2 \\ 1 & 0 & 2 & 10 & 7 & 2 \\ 1 & 2 & 0 & 15 & 9 & 2 \\ 11 & 10 & 15 & 88 & 57 & 18 \\ 7 & 7 & 9 & 57 & 35 & 11 \\ 2 & 2 & 2 & 18 & 11 & 3 \end{pmatrix}_{b_0}$$

[1262] $1^{123}3^{121}1^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 374, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 10 & 2 \\ 1 & 0 & 2 & 6 & 10 & 2 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 7 & 6 & 8 & 26 & 46 & 10 \\ 10 & 10 & 13 & 46 & 74 & 16 \\ 2 & 2 & 2 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[1263] $1^{13} 1^{124} 3^1 21^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 5 & 13 & 2 \\ 1 & 2 & 0 & 5 & 17 & 2 \\ 4 & 5 & 5 & 13 & 43 & 7 \\ 12 & 13 & 17 & 43 & 127 & 21 \\ 2 & 2 & 2 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[1264] $1^{23} 1^{113} 3^1 21^{124} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 4 & 11 & 2 \\ 1 & 2 & 0 & 7 & 16 & 2 \\ 5 & 4 & 7 & 16 & 43 & 8 \\ 11 & 11 & 16 & 43 & 103 & 19 \\ 2 & 2 & 2 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$

[1265] $1^{24} 1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 383, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 3 & 10 & 2 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 4 & 3 & 4 & 5 & 23 & 5 \\ 11 & 10 & 13 & 23 & 74 & 16 \\ 2 & 2 & 2 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1267] $1^{34} 1^{122} 3^1 21^{123} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 10 & 2 \\ 1 & 0 & 2 & 6 & 8 & 2 \\ 1 & 2 & 0 & 8 & 12 & 2 \\ 7 & 6 & 8 & 26 & 40 & 10 \\ 10 & 8 & 12 & 40 & 56 & 14 \\ 2 & 2 & 2 & 10 & 14 & 3 \end{pmatrix}_{b_0}$$

[1266] $1^{42} 1^{142} 3^1 21^{121} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 6 & 2 \\ 1 & 0 & 2 & 10 & 6 & 2 \\ 1 & 2 & 0 & 10 & 6 & 2 \\ 10 & 10 & 10 & 58 & 36 & 14 \\ 6 & 6 & 6 & 36 & 20 & 8 \\ 2 & 2 & 2 & 14 & 8 & 3 \end{pmatrix}_{b_0}$$

[1268] $1^{13} 1^{124} 3^1 21^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 5 & 11 & 2 \\ 1 & 2 & 0 & 5 & 11 & 2 \\ 4 & 5 & 5 & 13 & 33 & 7 \\ 10 & 11 & 11 & 33 & 71 & 15 \\ 2 & 2 & 2 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[1269] $1^{24} 1^{114} 3^1 21^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 396, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 3 & 12 & 2 \\ 1 & 2 & 0 & 4 & 14 & 2 \\ 4 & 3 & 4 & 5 & 26 & 5 \\ 12 & 12 & 14 & 26 & 96 & 18 \\ 2 & 2 & 2 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

[1270] $1^{14} 1^{124} 3^1 21^{124} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 396, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 13 & 2 \\ 1 & 0 & 2 & 6 & 13 & 2 \\ 1 & 2 & 0 & 8 & 17 & 2 \\ 5 & 6 & 8 & 26 & 60 & 10 \\ 13 & 13 & 17 & 60 & 127 & 21 \\ 2 & 2 & 2 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

[1272] $1^{43} 1^{132} 3^1 21^{133} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 9 & 2 \\ 1 & 0 & 2 & 9 & 8 & 2 \\ 1 & 2 & 0 & 12 & 12 & 2 \\ 10 & 9 & 12 & 61 & 60 & 15 \\ 9 & 8 & 12 & 60 & 56 & 14 \\ 2 & 2 & 2 & 15 & 14 & 3 \end{pmatrix}_{b_0}$$

[1271] $21^{132} 3^1 21^{132} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 7 & 2 \\ 2 & 0 & 2 & 9 & 7 & 2 \\ 1 & 2 & 0 & 12 & 9 & 2 \\ 9 & 9 & 12 & 61 & 48 & 15 \\ 7 & 7 & 9 & 48 & 35 & 11 \\ 2 & 2 & 2 & 15 & 11 & 3 \end{pmatrix}_{b_0}$$

[1273] $1^{124} 3^{121} 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 405, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 13 & 2 \\ 1 & 0 & 2 & 3 & 13 & 2 \\ 1 & 2 & 0 & 4 & 17 & 2 \\ 4 & 3 & 4 & 5 & 30 & 5 \\ 13 & 13 & 17 & 30 & 127 & 21 \\ 2 & 2 & 2 & 5 & 21 & 3 \end{pmatrix}_{b_0}$$

[1274] $21^{113} 3^{121} 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 10 & 2 \\ 2 & 0 & 2 & 3 & 10 & 2 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 3 & 3 & 4 & 5 & 23 & 5 \\ 10 & 10 & 13 & 23 & 74 & 16 \\ 2 & 2 & 2 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1275] $1^{144} 3^{121} 1^{134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 407, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 12 & 2 \\ 1 & 0 & 2 & 7 & 11 & 2 \\ 1 & 2 & 0 & 11 & 16 & 2 \\ 9 & 7 & 11 & 45 & 70 & 13 \\ 12 & 11 & 16 & 70 & 103 & 19 \\ 2 & 2 & 2 & 13 & 19 & 3 \end{pmatrix}_{b_0}$$

[1277] $1^{112} 3^{121} 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 413, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 3 & 10 & 2 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 3 & 3 & 4 & 5 & 23 & 5 \\ 9 & 10 & 13 & 23 & 74 & 16 \\ 2 & 2 & 2 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1276] $1^{131} 3^{121} 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 413, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 11 & 4 & 2 \\ 1 & 2 & 0 & 13 & 5 & 2 \\ 10 & 11 & 13 & 81 & 31 & 17 \\ 4 & 4 & 5 & 31 & 10 & 6 \\ 2 & 2 & 2 & 17 & 6 & 3 \end{pmatrix}_{b_0}$$

[1278] $1^{133} 3^{121} 1^{143} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 9 & 2 \\ 1 & 0 & 2 & 10 & 8 & 2 \\ 1 & 2 & 0 & 15 & 12 & 2 \\ 10 & 10 & 15 & 88 & 72 & 18 \\ 9 & 8 & 12 & 72 & 56 & 14 \\ 2 & 2 & 2 & 18 & 14 & 3 \end{pmatrix}_{b_0}$$

[1279] $21^{123} 3^{121} 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 420, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 9 & 2 \\ 2 & 0 & 2 & 5 & 9 & 2 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 5 & 5 & 5 & 13 & 28 & 7 \\ 9 & 9 & 10 & 28 & 51 & 13 \\ 2 & 2 & 2 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[1280] $1^{114} 3^{121} 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 420, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 5 & 12 & 2 \\ 1 & 2 & 0 & 5 & 14 & 2 \\ 4 & 5 & 5 & 13 & 38 & 7 \\ 12 & 12 & 14 & 38 & 96 & 18 \\ 2 & 2 & 2 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[1281] $1^{133} 3^{121} 1^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 420, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 9 & 2 \\ 1 & 0 & 2 & 9 & 8 & 2 \\ 1 & 2 & 0 & 12 & 12 & 2 \\ 9 & 9 & 12 & 61 & 60 & 15 \\ 9 & 8 & 12 & 60 & 56 & 14 \\ 2 & 2 & 2 & 15 & 14 & 3 \end{pmatrix}_{b_0}$$

[1282] $1^{144} 3^{121} 1^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 10 & 2 \\ 1 & 0 & 2 & 9 & 8 & 2 \\ 1 & 2 & 0 & 12 & 12 & 2 \\ 10 & 9 & 12 & 61 & 60 & 15 \\ 10 & 8 & 12 & 60 & 56 & 14 \\ 2 & 2 & 2 & 15 & 14 & 3 \end{pmatrix}_{b_0}$$

[1283] $1^{142} 3^{121} 1^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 425, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 7 & 2 \\ 1 & 0 & 2 & 9 & 7 & 2 \\ 1 & 2 & 0 & 12 & 9 & 2 \\ 10 & 9 & 12 & 61 & 48 & 15 \\ 7 & 7 & 9 & 48 & 35 & 11 \\ 2 & 2 & 2 & 15 & 11 & 3 \end{pmatrix}_{b_0}$$

[1284] $1^{143} 3^{121} 1^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 8 & 2 \\ 1 & 0 & 2 & 11 & 7 & 2 \\ 1 & 2 & 0 & 13 & 9 & 2 \\ 11 & 11 & 13 & 81 & 55 & 17 \\ 8 & 7 & 9 & 55 & 35 & 11 \\ 2 & 2 & 2 & 17 & 11 & 3 \end{pmatrix}_{b_0}$$

[1285] $1^{122} 3^{121} 1^{133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 8 & 2 \\ 1 & 0 & 2 & 7 & 8 & 2 \\ 1 & 2 & 0 & 11 & 12 & 2 \\ 7 & 7 & 11 & 45 & 52 & 13 \\ 8 & 8 & 12 & 52 & 56 & 14 \\ 2 & 2 & 2 & 13 & 14 & 3 \end{pmatrix}_{b_0}$$

[1286] $21^{141} 3^{121} 1^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 4 & 2 \\ 2 & 0 & 2 & 12 & 4 & 2 \\ 1 & 2 & 0 & 16 & 5 & 2 \\ 12 & 12 & 16 & 110 & 36 & 20 \\ 4 & 4 & 5 & 36 & 10 & 6 \\ 2 & 2 & 2 & 20 & 6 & 3 \end{pmatrix}_{b_0}$$

[1287] $1^{141} 3^{121} 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 443, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 4 & 2 \\ 1 & 0 & 2 & 11 & 4 & 2 \\ 1 & 2 & 0 & 13 & 5 & 2 \\ 11 & 11 & 13 & 81 & 31 & 17 \\ 4 & 4 & 5 & 31 & 10 & 6 \\ 2 & 2 & 2 & 17 & 6 & 3 \end{pmatrix}_{b_0}$$

[1289] $1^{123} 3^{121} 1^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 446, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 6 & 10 & 2 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 5 & 6 & 8 & 26 & 46 & 10 \\ 9 & 10 & 13 & 46 & 74 & 16 \\ 2 & 2 & 2 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[1288] $1^{141} 3^{121} 1^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 446, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 4 & 2 \\ 1 & 0 & 2 & 12 & 4 & 2 \\ 1 & 2 & 0 & 16 & 5 & 2 \\ 11 & 12 & 16 & 110 & 36 & 20 \\ 4 & 4 & 5 & 36 & 10 & 6 \\ 2 & 2 & 2 & 20 & 6 & 3 \end{pmatrix}_{b_0}$$

[1290] $21^{124} 3^{121} 1^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 12 & 2 \\ 2 & 0 & 2 & 6 & 12 & 2 \\ 1 & 2 & 0 & 8 & 14 & 2 \\ 6 & 6 & 8 & 26 & 52 & 10 \\ 12 & 12 & 14 & 52 & 96 & 18 \\ 2 & 2 & 2 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[1291] $1^{113} 3^{121} 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 449, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 11 & 2 \\ 1 & 0 & 2 & 3 & 10 & 2 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 3 & 3 & 4 & 5 & 23 & 5 \\ 11 & 10 & 13 & 23 & 74 & 16 \\ 2 & 2 & 2 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1293] $1^{134} 3^{121} 1^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 12 & 2 \\ 1 & 0 & 2 & 6 & 11 & 2 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 7 & 6 & 8 & 26 & 54 & 10 \\ 12 & 11 & 16 & 54 & 103 & 19 \\ 2 & 2 & 2 & 10 & 19 & 3 \end{pmatrix}_{b_0}$$

[1292] $1^{1^{13}}3^{1^{12}}3^{1^{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 6 & 11 & 2 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 5 & 6 & 8 & 26 & 54 & 10 \\ 11 & 11 & 16 & 54 & 103 & 19 \\ 2 & 2 & 2 & 10 & 19 & 3 \end{pmatrix}_{b_0}$$

[1294] $1^{1^{24}}3^{1^{12}}3^{1^{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 458, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 11 & 2 \\ 1 & 0 & 2 & 6 & 10 & 2 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 6 & 6 & 8 & 26 & 46 & 16 \\ 11 & 10 & 13 & 46 & 74 & 16 \\ 2 & 2 & 2 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[1295] $1^{1^3}3^{1^{14}}3^{1^{14}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 459, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 3 & 13 & 2 \\ 1 & 2 & 0 & 4 & 17 & 2 \\ 2 & 3 & 4 & 5 & 30 & 5 \\ 12 & 13 & 17 & 30 & 127 & 21 \\ 2 & 2 & 2 & 5 & 21 & 3 \end{pmatrix}_{b_0}$$

[1296] $1^{1^{13}}3^{1^{12}}3^{1^{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 459, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 0 & 2 & 3 & 10 & 2 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 3 & 3 & 4 & 5 & 23 & 5 \\ 10 & 10 & 13 & 23 & 74 & 16 \\ 2 & 2 & 2 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1297] $1^{1^{24}}3^{1^{12}}3^{1^{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 461, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 4 & 9 & 2 \\ 1 & 2 & 0 & 7 & 15 & 2 \\ 5 & 4 & 7 & 16 & 39 & 8 \\ 11 & 9 & 15 & 39 & 83 & 17 \\ 2 & 2 & 2 & 8 & 17 & 3 \end{pmatrix}_{b_0}$$

[1299] $1^{1^{42}}3^{1^{12}}3^{1^{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 463, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 6 & 2 \\ 1 & 0 & 2 & 10 & 5 & 2 \\ 1 & 2 & 0 & 15 & 8 & 2 \\ 11 & 10 & 15 & 88 & 47 & 18 \\ 6 & 5 & 8 & 47 & 23 & 9 \\ 2 & 2 & 2 & 18 & 9 & 3 \end{pmatrix}_{b_0}$$

[1298] $1^{1^{24}}3^{1^{12}}3^{1^{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 463, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 12 & 2 \\ 1 & 0 & 2 & 4 & 11 & 2 \\ 1 & 2 & 0 & 7 & 16 & 2 \\ 5 & 4 & 7 & 16 & 43 & 8 \\ 12 & 11 & 16 & 43 & 103 & 19 \\ 2 & 2 & 2 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$

[1300] $1^{1^{24}}3^{1^{12}}3^{1^{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 5 & 11 & 2 \\ 1 & 2 & 0 & 5 & 11 & 2 \\ 5 & 5 & 5 & 13 & 33 & 7 \\ 11 & 11 & 11 & 33 & 71 & 15 \\ 2 & 2 & 2 & 7 & 15 & 3 \end{pmatrix}_{b_0}$$

[1301] $1^{1^{34}}3^{1^{12}}3^{1^{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 12 & 2 \\ 1 & 0 & 2 & 6 & 12 & 2 \\ 1 & 2 & 0 & 8 & 14 & 2 \\ 7 & 6 & 8 & 26 & 52 & 10 \\ 12 & 12 & 14 & 52 & 96 & 18 \\ 2 & 2 & 2 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[1302] $21^{1^{23}}3^{1^{12}}3^{1^{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 10 & 2 \\ 2 & 0 & 2 & 6 & 10 & 2 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 6 & 6 & 8 & 26 & 46 & 10 \\ 10 & 10 & 13 & 46 & 74 & 16 \\ 2 & 2 & 2 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1303] \quad 1^{143} 3^{121} 21^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 475, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 8 & 2 \\ 1 & 0 & 2 & 9 & 7 & 2 \\ 1 & 2 & 0 & 12 & 9 & 2 \\ 10 & 9 & 12 & 61 & 48 & 15 \\ 8 & 7 & 9 & 48 & 35 & 11 \\ 2 & 2 & 2 & 15 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1304] \quad 1^{132} 3^{121} 21^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 477, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 7 & 2 \\ 1 & 0 & 2 & 9 & 7 & 2 \\ 1 & 2 & 0 & 12 & 9 & 2 \\ 9 & 9 & 12 & 61 & 48 & 15 \\ 7 & 7 & 9 & 48 & 35 & 11 \\ 2 & 2 & 2 & 15 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1305] \quad 21^{141} 3^{121} 21^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 4 & 2 \\ 2 & 0 & 2 & 11 & 4 & 2 \\ 1 & 2 & 0 & 13 & 5 & 2 \\ 11 & 11 & 13 & 81 & 31 & 17 \\ 4 & 4 & 5 & 31 & 10 & 6 \\ 2 & 2 & 2 & 17 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1306] \quad 1^{122} 3^{121} 21^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 0 & 2 & 7 & 8 & 2 \\ 1 & 2 & 0 & 11 & 12 & 2 \\ 8 & 7 & 11 & 45 & 52 & 13 \\ 9 & 8 & 12 & 52 & 56 & 14 \\ 2 & 2 & 2 & 13 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1307] \quad 21^{142} 3^{121} 21^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 486, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 11 & 7 & 2 \\ 2 & 0 & 2 & 11 & 7 & 2 \\ 1 & 2 & 0 & 13 & 9 & 2 \\ 11 & 11 & 13 & 81 & 55 & 17 \\ 7 & 7 & 9 & 55 & 35 & 11 \\ 2 & 2 & 2 & 17 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1310] \quad 1^{142} 3^{121} 21^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 488, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 7 & 2 \\ 1 & 0 & 2 & 11 & 7 & 2 \\ 1 & 2 & 0 & 13 & 9 & 2 \\ 10 & 11 & 13 & 81 & 55 & 17 \\ 7 & 7 & 9 & 55 & 35 & 11 \\ 2 & 2 & 2 & 17 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1308] \quad 1^{124} 3^{121} 21^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 488, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 12 & 2 \\ 1 & 0 & 2 & 6 & 12 & 2 \\ 1 & 2 & 0 & 8 & 14 & 2 \\ 6 & 6 & 8 & 26 & 52 & 10 \\ 12 & 12 & 14 & 52 & 96 & 18 \\ 2 & 2 & 2 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1309] \quad 1^{142} 3^{121} 21^{142} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 488, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 7 & 2 \\ 1 & 0 & 2 & 12 & 7 & 2 \\ 1 & 2 & 0 & 16 & 9 & 2 \\ 11 & 12 & 16 & 110 & 64 & 20 \\ 7 & 7 & 9 & 64 & 35 & 11 \\ 2 & 2 & 2 & 20 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1311] \quad 1^{124} 3^{121} 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 492, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 12 & 2 \\ 1 & 0 & 2 & 5 & 12 & 2 \\ 1 & 2 & 0 & 5 & 14 & 2 \\ 5 & 5 & 5 & 13 & 38 & 7 \\ 12 & 12 & 14 & 38 & 96 & 18 \\ 2 & 2 & 2 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1312] \quad 1^{124} 3^{121} 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 492, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 5 & 12 & 2 \\ 1 & 2 & 0 & 5 & 14 & 2 \\ 4 & 5 & 5 & 13 & 38 & 7 \\ 11 & 12 & 14 & 38 & 96 & 18 \\ 2 & 2 & 2 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[1313] $1^{142} 3^{121} 21^{133} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 496, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 10 & 2 \\ 1 & 0 & 2 & 7 & 8 & 2 \\ 1 & 2 & 0 & 11 & 12 & 2 \\ 9 & 7 & 11 & 45 & 52 & 13 \\ 10 & 8 & 12 & 52 & 56 & 14 \\ 2 & 2 & 2 & 13 & 14 & 3 \end{pmatrix}_{b_0}$$

[1315] $1^{142} 3^{121} 21^{132} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 504, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 7 & 2 \\ 1 & 0 & 2 & 11 & 7 & 2 \\ 1 & 2 & 0 & 13 & 9 & 2 \\ 11 & 11 & 13 & 81 & 55 & 17 \\ 7 & 7 & 9 & 55 & 35 & 11 \\ 2 & 2 & 2 & 17 & 11 & 3 \end{pmatrix}_{b_0}$$

[1314] $1^{142} 3^{121} 21^{143} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 504, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 9 & 2 \\ 1 & 0 & 2 & 12 & 8 & 2 \\ 1 & 2 & 0 & 16 & 12 & 2 \\ 12 & 12 & 16 & 110 & 80 & 20 \\ 9 & 8 & 12 & 80 & 56 & 14 \\ 2 & 2 & 2 & 20 & 14 & 3 \end{pmatrix}_{b_0}$$

[1316] $1^{124} 3^{121} 21^{124} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 524, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 13 & 2 \\ 1 & 0 & 2 & 6 & 13 & 2 \\ 1 & 2 & 0 & 8 & 17 & 2 \\ 7 & 6 & 8 & 26 & 60 & 10 \\ 13 & 13 & 17 & 60 & 127 & 21 \\ 2 & 2 & 2 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

[1317] $1^{124} 3^{121} 21^{123} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 11 & 2 \\ 1 & 0 & 2 & 6 & 12 & 2 \\ 1 & 2 & 0 & 8 & 14 & 2 \\ 6 & 6 & 8 & 26 & 52 & 10 \\ 11 & 12 & 14 & 52 & 96 & 18 \\ 2 & 2 & 2 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[1318] $21^{124} 3^{121} 21^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 12 & 2 \\ 2 & 0 & 2 & 5 & 12 & 2 \\ 1 & 2 & 0 & 5 & 14 & 2 \\ 5 & 5 & 5 & 13 & 38 & 7 \\ 12 & 12 & 14 & 38 & 96 & 18 \\ 2 & 2 & 2 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[1319] $1^{123} 3^{121} 21^{134} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 533, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 11 & 2 \\ 1 & 0 & 2 & 7 & 11 & 2 \\ 1 & 2 & 0 & 11 & 16 & 2 \\ 8 & 7 & 11 & 45 & 70 & 13 \\ 11 & 11 & 16 & 70 & 103 & 19 \\ 2 & 2 & 2 & 13 & 19 & 3 \end{pmatrix}_{b_0}$$

[1320] $1^{114} 3^{121} 21^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 540, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 11 & 2 \\ 1 & 0 & 2 & 3 & 12 & 2 \\ 1 & 2 & 0 & 4 & 14 & 2 \\ 3 & 3 & 4 & 5 & 26 & 5 \\ 11 & 12 & 14 & 26 & 96 & 18 \\ 2 & 2 & 2 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

[1321] $1^{141} 3^{121} 21^{141} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 542, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 4 & 2 \\ 1 & 0 & 2 & 12 & 4 & 2 \\ 1 & 2 & 0 & 16 & 5 & 2 \\ 12 & 12 & 16 & 110 & 36 & 20 \\ 4 & 4 & 5 & 36 & 10 & 6 \\ 2 & 2 & 2 & 20 & 6 & 3 \end{pmatrix}_{b_0}$$

[1322] $1^{113} 3^{121} 21^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 542, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 11 & 2 \\ 1 & 0 & 2 & 3 & 11 & 2 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 3 & 3 & 4 & 5 & 27 & 5 \\ 11 & 11 & 16 & 27 & 103 & 19 \\ 2 & 2 & 2 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[1323] $1^{1^{23}} 3^1 21^{1^{23}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 546, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 10 & 2 \\ 1 & 0 & 2 & 6 & 10 & 2 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 6 & 6 & 8 & 26 & 46 & 10 \\ 10 & 10 & 13 & 46 & 74 & 16 \\ 2 & 2 & 2 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[1325] $1^{1^{34}} 3^1 21^{1^{34}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 11 & 2 \\ 1 & 0 & 2 & 7 & 9 & 2 \\ 1 & 2 & 0 & 11 & 15 & 2 \\ 8 & 7 & 11 & 45 & 63 & 13 \\ 11 & 9 & 15 & 63 & 83 & 17 \\ 2 & 2 & 2 & 13 & 17 & 3 \end{pmatrix}_{b_0}$$

[1324] $1^{1^{14}} 3^1 21^{1^{13}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 12 & 2 \\ 1 & 0 & 2 & 3 & 12 & 2 \\ 1 & 2 & 0 & 4 & 14 & 2 \\ 3 & 3 & 4 & 5 & 26 & 5 \\ 12 & 12 & 14 & 26 & 96 & 18 \\ 2 & 2 & 2 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

[1326] $1^{1^{34}} 3^1 21^{1^{23}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 550, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 11 & 2 \\ 1 & 0 & 2 & 6 & 10 & 2 \\ 1 & 2 & 0 & 8 & 13 & 2 \\ 7 & 6 & 8 & 26 & 46 & 10 \\ 11 & 10 & 13 & 46 & 74 & 16 \\ 2 & 2 & 2 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[1327] $21^{1^{42}} 3^1 21^{1^{42}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 558, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 7 & 2 \\ 2 & 0 & 2 & 12 & 7 & 2 \\ 1 & 2 & 0 & 16 & 9 & 2 \\ 12 & 12 & 16 & 110 & 64 & 20 \\ 7 & 7 & 9 & 64 & 35 & 11 \\ 2 & 2 & 2 & 20 & 11 & 3 \end{pmatrix}_{b_0}$$

[1328] $1^{1^{23}} 3^1 21^{1^{24}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 566, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 11 & 2 \\ 1 & 0 & 2 & 6 & 11 & 2 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 6 & 6 & 8 & 26 & 54 & 10 \\ 11 & 11 & 16 & 54 & 103 & 19 \\ 2 & 2 & 2 & 10 & 19 & 3 \end{pmatrix}_{b_0}$$

[1329] $1^{1^{14}} 3^1 21^{1^{14}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 578, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 12 & 2 \\ 1 & 0 & 2 & 3 & 11 & 2 \\ 1 & 2 & 0 & 4 & 16 & 2 \\ 3 & 3 & 4 & 5 & 27 & 5 \\ 12 & 11 & 16 & 27 & 103 & 19 \\ 2 & 2 & 2 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[1330] $1^{1^{23}} 3^1 21^{1^{34}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 587, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 11 & 2 \\ 1 & 0 & 2 & 7 & 11 & 2 \\ 1 & 2 & 0 & 11 & 16 & 2 \\ 7 & 7 & 11 & 45 & 70 & 13 \\ 11 & 11 & 16 & 70 & 103 & 19 \\ 2 & 2 & 2 & 13 & 19 & 3 \end{pmatrix}_{b_0}$$

[1331] $21^{1^{14}} 3^1 21^{1^{13}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 592, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 12 & 2 \\ 2 & 0 & 2 & 3 & 12 & 2 \\ 1 & 2 & 0 & 4 & 14 & 2 \\ 3 & 3 & 4 & 5 & 26 & 5 \\ 12 & 12 & 14 & 26 & 96 & 18 \\ 2 & 2 & 2 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

[1332] $1^{1^{24}} 3^1 21^{1^{24}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 614, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 12 & 2 \\ 1 & 0 & 2 & 6 & 11 & 2 \\ 1 & 2 & 0 & 8 & 16 & 2 \\ 6 & 6 & 8 & 26 & 54 & 10 \\ 12 & 11 & 16 & 54 & 103 & 19 \\ 2 & 2 & 2 & 10 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1333] \ 1^{1^{44}} 3^{1^{122}} 21^{1^{144}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 620, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 11 & 2 \\ 1 & 0 & 2 & 8 & 9 & 2 \\ 1 & 2 & 0 & 14 & 15 & 2 \\ 10 & 8 & 14 & 70 & 78 & 16 \\ 11 & 9 & 15 & 78 & 83 & 17 \\ 2 & 2 & 2 & 16 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[1334] \ 1^{1^{13}} 3^{1^{124}} 21^{1^{124}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 644, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 12 & 2 \\ 1 & 0 & 2 & 6 & 13 & 2 \\ 1 & 2 & 0 & 8 & 17 & 2 \\ 5 & 6 & 8 & 26 & 60 & 10 \\ 12 & 13 & 17 & 60 & 127 & 21 \\ 2 & 2 & 2 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1335] \ 21^{1^{143}} 3^{1^{121}} 21^{1^{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 644, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 13 & 2 \\ 2 & 0 & 2 & 3 & 13 & 2 \\ 1 & 2 & 0 & 4 & 17 & 2 \\ 3 & 3 & 4 & 5 & 30 & 5 \\ 13 & 13 & 17 & 30 & 127 & 21 \\ 2 & 2 & 2 & 5 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1336] \ 1^{1^{43}} 3^{1^{132}} 21^{1^{143}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 9 & 2 \\ 1 & 0 & 2 & 10 & 8 & 2 \\ 1 & 2 & 0 & 15 & 12 & 2 \\ 11 & 10 & 15 & 88 & 72 & 18 \\ 9 & 8 & 12 & 72 & 56 & 14 \\ 2 & 2 & 2 & 18 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1337] \ 1^{1^{34}} 3^{1^{123}} 21^{1^{134}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 673, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 12 & 2 \\ 1 & 0 & 2 & 7 & 11 & 2 \\ 1 & 2 & 0 & 11 & 16 & 2 \\ 8 & 7 & 11 & 45 & 70 & 13 \\ 12 & 11 & 16 & 70 & 103 & 19 \\ 2 & 2 & 2 & 13 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1338] \ 1^{1^{42}} 3^{1^{142}} 21^{1^{142}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 680, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 7 & 2 \\ 1 & 0 & 2 & 12 & 7 & 2 \\ 1 & 2 & 0 & 16 & 9 & 2 \\ 12 & 12 & 16 & 110 & 64 & 20 \\ 7 & 7 & 9 & 64 & 35 & 11 \\ 2 & 2 & 2 & 20 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[1339] \ 1^{1^{13}} 3^{1^{114}} 21^{1^{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 695, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 12 & 2 \\ 1 & 0 & 2 & 3 & 13 & 2 \\ 1 & 2 & 0 & 4 & 17 & 2 \\ 3 & 3 & 4 & 5 & 30 & 5 \\ 12 & 13 & 17 & 30 & 127 & 21 \\ 2 & 2 & 2 & 5 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1340] \ 21^{1^{124}} 3^{1^{124}} 21^{1^{124}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 710, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 13 & 2 \\ 2 & 0 & 2 & 6 & 13 & 2 \\ 1 & 2 & 0 & 8 & 17 & 2 \\ 6 & 6 & 8 & 26 & 60 & 10 \\ 13 & 13 & 17 & 60 & 127 & 21 \\ 2 & 2 & 2 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1341] \ 1^{1^{23}} 3^{1^{124}} 21^{1^{124}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 740, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 12 & 2 \\ 1 & 0 & 2 & 6 & 13 & 2 \\ 1 & 2 & 0 & 8 & 17 & 2 \\ 6 & 6 & 8 & 26 & 60 & 10 \\ 12 & 13 & 17 & 60 & 127 & 21 \\ 2 & 2 & 2 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1342] \ 1^{1^{14}} 3^{1^{114}} 21^{1^{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 749, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 13 & 2 \\ 1 & 0 & 2 & 3 & 13 & 2 \\ 1 & 2 & 0 & 4 & 17 & 2 \\ 3 & 3 & 4 & 5 & 30 & 5 \\ 13 & 13 & 17 & 30 & 127 & 21 \\ 2 & 2 & 2 & 5 & 21 & 3 \end{pmatrix}_{b_0}$$

[1343] $1^{124}3^{124}21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 836, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 13 & 2 \\ 1 & 0 & 2 & 6 & 13 & 2 \\ 1 & 2 & 0 & 8 & 17 & 2 \\ 6 & 6 & 8 & 26 & 60 & 10 \\ 13 & 13 & 17 & 60 & 127 & 21 \\ 2 & 2 & 2 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

$T = 9$ [1344] $1^{11}41^{11}23^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 3 \\ 2 & 0 & 2 & 5 & 2 \\ 3 & 2 & 4 & 8 & 4 \\ 7 & 5 & 8 & 12 & 8 \\ 3 & 2 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

[1345] $1^{141}1^{11}23^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 6 & 3 \\ 2 & 0 & 2 & 3 & 2 \\ 3 & 2 & 4 & 6 & 4 \\ 6 & 3 & 6 & 6 & 6 \\ 3 & 2 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

[1346] $23^{11}141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 3 & 2 \\ 1 & 0 & 2 & 1 & 3 & 2 \\ 2 & 2 & 0 & 3 & 3 & 2 \\ 2 & 1 & 3 & 1 & 3 & 3 \\ 3 & 3 & 3 & 3 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

[1347] $1^{141}1^{11}23^{11}1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 9 & 6 & 3 \\ 2 & 0 & 5 & 3 & 2 \\ 9 & 5 & 22 & 12 & 10 \\ 6 & 3 & 12 & 6 & 6 \\ 3 & 2 & 10 & 6 & 4 \end{pmatrix}_{b_0}$$

[1348] $1^{141}1^{12}23^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 11 & 3 \\ 2 & 0 & 2 & 6 & 2 \\ 3 & 2 & 4 & 12 & 4 \\ 11 & 6 & 12 & 30 & 12 \\ 3 & 2 & 4 & 12 & 4 \end{pmatrix}_{b_0}$$

[1349] $1^{11}41^{12}23^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 12 & 3 \\ 2 & 0 & 2 & 8 & 2 \\ 3 & 2 & 4 & 14 & 4 \\ 12 & 8 & 14 & 40 & 14 \\ 3 & 2 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

[1350] $1^{141}1^{13}23^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 16 & 3 \\ 2 & 0 & 2 & 9 & 2 \\ 3 & 2 & 4 & 18 & 4 \\ 16 & 9 & 18 & 70 & 18 \\ 3 & 2 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

[1351] $23^{11}141^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -22, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 3 & 2 \\ 1 & 0 & 2 & 4 & 3 & 2 \\ 2 & 2 & 0 & 7 & 3 & 2 \\ 5 & 4 & 7 & 16 & 9 & 8 \\ 3 & 3 & 3 & 9 & 4 & 4 \\ 2 & 2 & 2 & 8 & 4 & 3 \end{pmatrix}_{b_0}$$

[1352] $23^{11}141^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 2 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & 0 & 3 & 2 \\ 3 & 3 & 3 & 3 & 4 & 4 \\ 2 & 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[1353] $1^{141}1^{12}23^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 2 & 6 & 3 \\ 2 & 0 & 1 & 3 & 2 \\ 2 & 1 & 0 & 2 & 2 \\ 6 & 3 & 2 & 6 & 6 \\ 3 & 2 & 2 & 6 & 4 \end{pmatrix}_{b_0} \\
[1354] & 1^{\overline{11}} 41^{\overline{13}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 2 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 5 & 4 & 0 & 4 & 10 & 6 \\ 2 & 2 & 0 & 2 & 6 & 3 \end{pmatrix}_{b_0} \\
& 1^{\overline{11}} \\
[1355] & 23^{\overline{1}} 141^{\overline{13}} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 5 & 3 & 2 \\ 1 & 0 & 2 & 7 & 3 & 2 \\ 2 & 2 & 0 & 6 & 3 & 2 \\ 5 & 7 & 6 & 25 & 12 & 9 \\ 3 & 3 & 3 & 12 & 4 & 4 \\ 2 & 2 & 2 & 9 & 4 & 3 \end{pmatrix}_{b_0} \\
[1356] & 1^{\overline{1}} 41^{\overline{22}} 23^{\overline{1}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 8 & 11 & 3 \\ 2 & 0 & 5 & 6 & 2 \\ 8 & 5 & 24 & 28 & 10 \\ 11 & 6 & 28 & 30 & 12 \\ 3 & 2 & 10 & 12 & 4 \end{pmatrix}_{b_0} \\
[1359] & 1^{\overline{12}} 41^{\overline{11}} 23^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 4 & 7 & 2 \\ 3 & 4 & 1 & 8 & 10 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 4 & 8 & 0 & 10 & 16 & 6 \\ 7 & 10 & 1 & 16 & 21 & 9 \\ 2 & 4 & 0 & 6 & 9 & 3 \end{pmatrix}_{b_0} \\
[1357] & 1^{\overline{31}} 41^{\overline{13}} 23^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 12 & 2 & 2 \\ 3 & 4 & 1 & 24 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 12 & 24 & 0 & 106 & 12 & 18 \\ 2 & 2 & 0 & 12 & 0 & 2 \\ 2 & 4 & 0 & 18 & 2 & 3 \end{pmatrix}_{b_0} \\
[1358] & 1^{\overline{21}} 41^{\overline{12}} 23^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 8 & 2 & 2 \\ 3 & 4 & 1 & 16 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 8 & 16 & 0 & 46 & 8 & 12 \\ 2 & 2 & 0 & 8 & 0 & 2 \\ 2 & 4 & 0 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[1360] & 1^{\overline{12}} 41^{\overline{11}} 23^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 4 & 6 & 2 \\ 3 & 4 & 1 & 8 & 10 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 4 & 8 & 0 & 10 & 16 & 6 \\ 6 & 10 & 0 & 16 & 20 & 8 \\ 2 & 4 & 0 & 6 & 8 & 3 \end{pmatrix}_{b_0} \\
[1361] & 1^{\overline{11}} 41^{\overline{1}} 23^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 1 & 4 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 0 & 10 & 4 & 6 \\ 2 & 2 & 0 & 4 & 0 & 2 \\ 2 & 2 & 0 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[1362] & 1^{\overline{41}} 41^{\overline{14}} 23^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 16 & 2 & 2 \\ 3 & 4 & 1 & 32 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 16 & 32 & 0 & 190 & 16 & 24 \\ 2 & 2 & 0 & 16 & 0 & 2 \\ 2 & 4 & 0 & 24 & 2 & 3 \end{pmatrix}_{b_0} \\
[1363] & 1^{\overline{11}} 41^{\overline{11}} 23^{\overline{1}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 6 & 2 & 2 \\ 3 & 4 & 1 & 8 & 2 & 4 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 6 & 8 & 2 & 14 & 4 & 8 \\ 2 & 2 & 0 & 4 & 0 & 2 \\ 2 & 4 & 0 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[1364] & 1^{\overline{11}} 41^{\overline{13}} 23^{\overline{1}} 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 3 & 17 & 3 \\ 2 & 0 & 2 & 11 & 2 \\ 3 & 2 & 4 & 20 & 4 \\ 17 & 11 & 20 & 84 & 20 \\ 3 & 2 & 4 & 20 & 4 \end{pmatrix}_{b_0} \\
[1365] & 1^{141} 1^{14} 23^{11} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 3 & 21 & 3 \\ 2 & 0 & 2 & 12 & 2 \\ 3 & 2 & 4 & 24 & 4 \\ 21 & 12 & 24 & 126 & 24 \\ 3 & 2 & 4 & 24 & 4 \end{pmatrix}_{b_0} \\
[1366] & 1^{11} 41^{11} 23^{11} 1^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 10 & 6 & 3 \\ 2 & 0 & 7 & 3 & 2 \\ 10 & 7 & 28 & 14 & 12 \\ 6 & 3 & 14 & 6 & 6 \\ 3 & 2 & 12 & 6 & 4 \end{pmatrix}_{b_0} \\
[1367] & 1^{111} 41^{11} 23^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 5 & 2 \\ 1 & 0 & 1 & 10 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 15 & 10 & 2 & 116 & 36 & 20 \\ 5 & 4 & 0 & 36 & 10 & 6 \\ 2 & 2 & 0 & 20 & 6 & 3 \end{pmatrix}_{b_0} \\
[1368] & 1^{11} 41^{11} 23^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 6 & 6 & 3 \\ 2 & 0 & 5 & 3 & 2 \\ 6 & 5 & 12 & 10 & 8 \\ 6 & 3 & 10 & 6 & 6 \\ 3 & 2 & 8 & 6 & 4 \end{pmatrix}_{b_0} \\
[1369] & 1^{141} 2^{11} 23^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 8 & 6 & 3 \\ 2 & 0 & 5 & 3 & 2 \\ 8 & 5 & 24 & 14 & 10 \\ 6 & 3 & 14 & 6 & 6 \\ 3 & 2 & 10 & 6 & 4 \end{pmatrix}_{b_0} \\
[1370] & 1^{111} 41^{11} 23^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 5 & 2 \\ 1 & 0 & 1 & 6 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 9 & 6 & 2 & 36 & 20 & 12 \\ 5 & 4 & 0 & 20 & 10 & 6 \\ 2 & 2 & 0 & 12 & 6 & 3 \end{pmatrix}_{b_0} \\
[1371] & 1^{111} 2^3 1^{11} 141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 5 & 10 & 9 & 5 \\ 6 & 10 & 14 & 13 & 8 \\ 6 & 9 & 13 & 11 & 7 \\ 3 & 5 & 8 & 7 & 4 \end{pmatrix}_{b_0} \\
[1372] & 1^{141} 2^{11} 23^{11} 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 0 & 1 & 6 & 2 \\ 2 & 1 & 0 & 4 & 2 \\ 11 & 6 & 4 & 30 & 12 \\ 3 & 2 & 2 & 12 & 4 \end{pmatrix}_{b_0} \\
[1373] & 1^{11} 41^{14} 23^{11} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 3 & 22 & 3 \\ 2 & 0 & 2 & 14 & 2 \\ 3 & 2 & 4 & 26 & 4 \\ 22 & 14 & 26 & 144 & 26 \\ 3 & 2 & 4 & 26 & 4 \end{pmatrix}_{b_0} \\
[1374] & 1^{111} 41^{11} 23^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 1 & 4 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 5 & 4 & 1 & 11 & 4 & 7 \\ 2 & 2 & 0 & 4 & 0 & 2 \\ 2 & 2 & 0 & 7 & 2 & 3 \end{pmatrix}_{b_0} \\
[1376] & 1^{112} 41^{11} 23^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 2 & 8 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 9 & 8 & 0 & 8 & 46 & 12 \\ 2 & 2 & 0 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1375] & 1^{141} 3^{11} 23^{11} 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 0 & 1 & 9 & 2 \\ 2 & 1 & 0 & 6 & 2 \\ 16 & 9 & 6 & 70 & 18 \\ 3 & 2 & 2 & 18 & 4 \end{pmatrix}_{b_0} \\
[1377] & 1^{112} 41^{11} 23^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 1 & 4 & 6 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 4 & 4 & 0 & 10 & 16 & 6 \\ 7 & 6 & 1 & 16 & 21 & 9 \\ 2 & 2 & 0 & 6 & 9 & 3 \end{pmatrix}_{b_0} \\
[1378] & 1^{111} 41^{21} 23^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 5 & 2 \\ 1 & 0 & 1 & 6 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 7 & 6 & 0 & 32 & 20 & 10 \\ 5 & 4 & 0 & 20 & 10 & 6 \\ 2 & 2 & 0 & 10 & 6 & 3 \end{pmatrix}_{b_0} \\
[1379] & 1^{112} 41^{21} 23^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 2 & 8 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 10 & 8 & 1 & 8 & 47 & 13 \\ 2 & 2 & 0 & 2 & 13 & 3 \end{pmatrix}_{b_0} \\
& 1^{11} \\
[1380] & 23^1 141^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 6 & 2 \\ 1 & 0 & 2 & 2 & 6 & 2 \\ 2 & 2 & 0 & 1 & 7 & 2 \\ 1 & 2 & 1 & 0 & 6 & 2 \\ 6 & 6 & 7 & 6 & 23 & 9 \\ 2 & 2 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[1381] & 1^{11} 41^{31} 23^1 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 14 & 6 & 3 \\ 2 & 0 & 10 & 3 & 2 \\ 14 & 10 & 74 & 24 & 18 \\ 6 & 3 & 24 & 6 & 6 \\ 3 & 2 & 18 & 6 & 4 \end{pmatrix}_{b_0} \\
[1382] & 1^{111} 41^{31} 23^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 12 & 5 & 2 \\ 1 & 0 & 1 & 8 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 12 & 8 & 2 & 70 & 28 & 16 \\ 5 & 4 & 0 & 28 & 10 & 6 \\ 2 & 2 & 0 & 16 & 6 & 3 \end{pmatrix}_{b_0} \\
& 1^{11} \\
[1383] & 23^1 141^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 3 & 3 & 2 \\ 1 & 0 & 2 & 3 & 3 & 2 \\ 2 & 2 & 0 & 4 & 3 & 2 \\ 3 & 3 & 4 & 5 & 6 & 5 \\ 3 & 3 & 3 & 6 & 4 & 4 \\ 2 & 2 & 2 & 5 & 4 & 3 \end{pmatrix}_{b_0} \\
[1384] & 1^{112} 41^{123} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -89, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 2 & 6 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 7 & 6 & 1 & 6 & 21 & 9 \\ 2 & 2 & 0 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[1385] & 1^{11} 41^{123} 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -89, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 0 & 1 & 6 & 2 \\ 2 & 1 & 0 & 4 & 2 \\ 9 & 6 & 4 & 18 & 10 \\ 3 & 2 & 2 & 10 & 4 \end{pmatrix}_{b_0} \\
[1386] & 1^{111} 41^{11} 23^1 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & 4 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 0 & 10 & 12 & 6 \\ 5 & 4 & 0 & 12 & 10 & 6 \\ 2 & 2 & 0 & 6 & 6 & 3 \end{pmatrix}_{b_0} \\
[1387] & 1^{11} 41^{11} 23^1 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 8 & 6 & 3 \\ 2 & 0 & 6 & 3 & 2 \\ 8 & 6 & 18 & 12 & 10 \\ 6 & 3 & 12 & 6 & 6 \\ 3 & 2 & 10 & 6 & 4 \end{pmatrix}_{b_0}$$

[1388] $23^1 141^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 3 & 2 \\ 1 & 0 & 2 & 5 & 3 & 2 \\ 2 & 2 & 0 & 5 & 3 & 2 \\ 4 & 5 & 5 & 13 & 9 & 7 \\ 3 & 3 & 3 & 9 & 4 & 4 \\ 2 & 2 & 2 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[1391] $23^1 141^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 7 & 2 \\ 1 & 0 & 2 & 2 & 8 & 2 \\ 2 & 2 & 0 & 1 & 8 & 2 \\ 1 & 2 & 1 & 0 & 8 & 2 \\ 7 & 8 & 8 & 8 & 37 & 11 \\ 2 & 2 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[1390] $1^1 41^4 23^1 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 2 & 21 & 3 \\ 2 & 0 & 1 & 12 & 2 \\ 2 & 1 & 0 & 8 & 2 \\ 21 & 12 & 8 & 126 & 24 \\ 3 & 2 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[1389] $1^{113} 41^4 23^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 2 & 14 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 14 & 2 \\ 17 & 14 & 1 & 14 & 161 & 23 \\ 2 & 2 & 0 & 2 & 23 & 3 \end{pmatrix}_{b_0}$$

[1392] $23^1 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 5 & 5 & 9 & 5 \\ 3 & 5 & 2 & 7 & 4 \\ 6 & 9 & 7 & 11 & 7 \\ 3 & 5 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

[1393] $1^{111} 41^{11} 23^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & 4 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 5 & 4 & 1 & 11 & 12 & 7 \\ 5 & 4 & 0 & 12 & 10 & 6 \\ 2 & 2 & 0 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

[1394] $1^1 41^{11} 23^1 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 7 & 6 & 3 \\ 2 & 0 & 4 & 3 & 2 \\ 7 & 4 & 12 & 10 & 8 \\ 6 & 3 & 10 & 6 & 6 \\ 3 & 2 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

[1395] $1^1 41^{11} 23^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 5 & 6 & 3 \\ 2 & 0 & 3 & 3 & 2 \\ 5 & 3 & 6 & 8 & 6 \\ 6 & 3 & 8 & 6 & 6 \\ 3 & 2 & 6 & 6 & 4 \end{pmatrix}_{b_0}$$

[1396] $1^{11} 41^{11} 23^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 5 & 7 & 3 \\ 2 & 0 & 3 & 5 & 2 \\ 5 & 3 & 6 & 10 & 6 \\ 7 & 5 & 10 & 12 & 8 \\ 3 & 2 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[1398] $23^1 141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 6 & 12 & 3 \\ 4 & 5 & 10 & 19 & 5 \\ 6 & 10 & 14 & 28 & 8 \\ 12 & 19 & 28 & 53 & 15 \\ 3 & 5 & 8 & 15 & 4 \end{pmatrix}_{b_0}$$

[1397] $23^1 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 5 & 10 & 9 & 5 \\ 6 & 10 & 14 & 14 & 8 \\ 6 & 9 & 14 & 11 & 7 \\ 3 & 5 & 8 & 7 & 4 \end{pmatrix}_{b_0}$$

[1399] $2 \ 3^{111} 141^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 5 & 5 & 14 & 5 \\ 3 & 5 & 2 & 10 & 4 \\ 9 & 14 & 10 & 29 & 11 \\ 3 & 5 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

[1400] $1^{121} 41^{21} 23^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 0 & 1 & 8 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 8 & 8 & 0 & 46 & 24 & 12 \\ 5 & 4 & 0 & 24 & 10 & 6 \\ 2 & 2 & 0 & 12 & 6 & 3 \end{pmatrix}_{b_0}$$

[1404] $1^{112} 41^{11} 23^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 1 & 4 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 0 & 10 & 16 & 6 \\ 6 & 6 & 0 & 16 & 20 & 8 \\ 2 & 2 & 0 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

[1403] $1^{121} 41^{21} 23^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 0 & 1 & 8 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 9 & 8 & 1 & 47 & 8 & 13 \\ 2 & 2 & 0 & 8 & 0 & 2 \\ 2 & 2 & 0 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1402] $1^{113} 41^{12} 23^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 1 & 4 & 10 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 4 & 4 & 0 & 10 & 28 & 6 \\ 12 & 10 & 2 & 28 & 68 & 16 \\ 2 & 2 & 0 & 6 & 16 & 3 \end{pmatrix}_{b_0}$$

[1401] $1^{122} 41^{21} 23^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & 0 & 1 & 8 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 8 & 8 & 0 & 46 & 32 & 12 \\ 6 & 6 & 0 & 32 & 20 & 8 \\ 2 & 2 & 0 & 12 & 8 & 3 \end{pmatrix}_{b_0}$$

[1405] $1^{11} 41^{11} 23^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 12 & 8 & 3 \\ 2 & 0 & 8 & 4 & 2 \\ 12 & 8 & 42 & 24 & 14 \\ 8 & 4 & 24 & 12 & 8 \\ 3 & 2 & 14 & 8 & 4 \end{pmatrix}_{b_0}$$

[1406] $2 \ 3^{121} 141^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 9 & 6 & 3 \\ 4 & 5 & 15 & 9 & 5 \\ 9 & 15 & 32 & 20 & 12 \\ 6 & 9 & 20 & 11 & 7 \\ 3 & 5 & 12 & 7 & 4 \end{pmatrix}_{b_0}$$

[1407] $1^{141} 3^{11} 23^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 13 & 6 & 3 \\ 2 & 0 & 8 & 3 & 2 \\ 13 & 8 & 60 & 22 & 16 \\ 6 & 3 & 22 & 6 & 6 \\ 3 & 2 & 16 & 6 & 4 \end{pmatrix}_{b_0}$$

[1408] $1^{111} 41^{31} 23^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 5 & 2 \\ 1 & 0 & 1 & 8 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 11 & 8 & 1 & 67 & 28 & 15 \\ 5 & 4 & 0 & 28 & 10 & 6 \\ 2 & 2 & 0 & 15 & 6 & 3 \end{pmatrix}_{b_0}$$

[1409] $1^{112} 41^{22} 23^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 7 & 9 & 2 \\ 1 & 0 & 1 & 6 & 8 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 7 & 6 & 0 & 32 & 40 & 10 \\ 9 & 8 & 0 & 40 & 46 & 12 \\ 2 & 2 & 0 & 10 & 12 & 3 \end{pmatrix}_{b_0} \\
[1410] & 1^{121}41^{31}23^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 5 & 2 \\ 1 & 0 & 1 & 10 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 11 & 10 & 0 & 84 & 32 & 16 \\ 5 & 4 & 0 & 32 & 10 & 6 \\ 2 & 2 & 0 & 16 & 6 & 3 \end{pmatrix}_{b_0} \\
& 1^{131} \\
[1411] & 23^1141^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 9 & 6 & 3 \\ 4 & 5 & 15 & 9 & 5 \\ 9 & 15 & 34 & 21 & 12 \\ 6 & 9 & 21 & 11 & 7 \\ 3 & 5 & 12 & 7 & 4 \end{pmatrix}_{b_0} \\
[1412] & 1^{13}41^{32}3^11^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -138, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 2 & 12 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 13 & 12 & 0 & 12 & 106 & 18 \\ 2 & 2 & 0 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[1413] & 1^141^{23}1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 2 & 13 & 3 \\ 2 & 0 & 1 & 7 & 2 \\ 2 & 1 & 0 & 6 & 2 \\ 13 & 7 & 6 & 40 & 14 \\ 3 & 2 & 2 & 14 & 4 \end{pmatrix}_{b_0} \\
[1414] & 1^{11}41^{11}23^11^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 7 & 9 & 3 \\ 2 & 0 & 4 & 6 & 2 \\ 7 & 4 & 12 & 16 & 8 \\ 9 & 6 & 16 & 18 & 10 \\ 3 & 2 & 8 & 10 & 4 \end{pmatrix}_{b_0} \\
[1415] & 1^141^{12}23^11^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 7 & 11 & 3 \\ 2 & 0 & 4 & 6 & 2 \\ 7 & 4 & 12 & 20 & 8 \\ 11 & 6 & 20 & 30 & 12 \\ 3 & 2 & 8 & 12 & 4 \end{pmatrix}_{b_0} \\
& 1^{12} \\
[1416] & 23^1141^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 6 & 3 & 2 \\ 1 & 0 & 2 & 6 & 3 & 2 \\ 2 & 2 & 0 & 8 & 3 & 2 \\ 6 & 6 & 8 & 26 & 12 & 10 \\ 3 & 3 & 3 & 12 & 4 & 4 \\ 2 & 2 & 2 & 10 & 4 & 3 \end{pmatrix}_{b_0} \\
& 1^{12} \\
[1417] & 23^1141^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 9 & 2 \\ 1 & 0 & 2 & 2 & 9 & 2 \\ 2 & 2 & 0 & 1 & 11 & 2 \\ 1 & 2 & 1 & 0 & 9 & 2 \\ 9 & 9 & 11 & 9 & 56 & 14 \\ 2 & 2 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0} \\
[1418] & 1^{11}41^{23}1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 0 & 1 & 9 & 2 \\ 2 & 1 & 0 & 6 & 2 \\ 14 & 9 & 6 & 50 & 16 \\ 3 & 2 & 2 & 16 & 4 \end{pmatrix}_{b_0} \\
[1419] & 1^{13}41^{32}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 2 & 12 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 15 & 12 & 2 & 12 & 110 & 20 \\ 2 & 2 & 0 & 2 & 20 & 3 \end{pmatrix}_{b_0} \\
[1420] & 1^{11}41^{11}23^11^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 5 & 9 & 3 \\ 2 & 0 & 3 & 6 & 2 \\ 5 & 3 & 6 & 12 & 6 \\ 9 & 6 & 12 & 18 & 10 \\ 3 & 2 & 6 & 10 & 4 \end{pmatrix}_{b_0} \\
[1422] & 1^{11}41^{21}23^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 0 & 1 & 6 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 8 & 6 & 1 & 33 & 20 & 11 \\ 5 & 4 & 0 & 20 & 10 & 6 \\ 2 & 2 & 0 & 11 & 6 & 3 \end{pmatrix}_{b_0} \\
[1421] & 1^{11}41^{21}23^11^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 11 & 6 & 3 \\ 2 & 0 & 8 & 3 & 2 \\ 11 & 8 & 40 & 18 & 14 \\ 6 & 3 & 18 & 6 & 6 \\ 3 & 2 & 14 & 6 & 4 \end{pmatrix}_{b_0} \\
[1423] & 1^{121}41^{23}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 0 & 1 & 8 & 2 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 10 & 8 & 2 & 50 & 8 & 14 \\ 2 & 2 & 0 & 8 & 0 & 2 \\ 2 & 2 & 0 & 14 & 2 & 3 \end{pmatrix}_{b_0} \\
& 1^{141} \\
[1424] & 23^1141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 12 & 6 & 3 \\ 4 & 5 & 20 & 9 & 5 \\ 12 & 20 & 62 & 28 & 16 \\ 6 & 9 & 28 & 11 & 7 \\ 3 & 5 & 16 & 7 & 4 \end{pmatrix}_{b_0} \\
[1425] & 1^{113}41^{123}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 2 & 8 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 9 & 8 & 2 & 8 & 38 & 12 \\ 2 & 2 & 0 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[1426] & 1^{113}41^{323}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & 2 & 12 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 14 & 12 & 1 & 12 & 107 & 19 \\ 2 & 2 & 0 & 2 & 19 & 3 \end{pmatrix}_{b_0} \\
& 1^{12} \\
[1427] & 23^1141^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -178, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 2 & 8 & 3 & 2 \\ 1 & 0 & 2 & 10 & 3 & 2 \\ 2 & 2 & 0 & 10 & 3 & 2 \\ 8 & 10 & 10 & 58 & 18 & 14 \\ 3 & 3 & 3 & 18 & 4 & 4 \\ 2 & 2 & 2 & 14 & 4 & 3 \end{pmatrix}_{b_0} \\
[1428] & 1^{141}1^{323}1^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 7 & 16 & 3 \\ 2 & 0 & 4 & 9 & 2 \\ 7 & 4 & 12 & 30 & 8 \\ 16 & 9 & 30 & 70 & 18 \\ 3 & 2 & 8 & 18 & 4 \end{pmatrix}_{b_0} \\
& 1^{112} \\
[1429] & 23^1141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 5 & 5 & 14 & 5 \\ 3 & 5 & 2 & 11 & 4 \\ 9 & 14 & 11 & 29 & 11 \\ 3 & 5 & 4 & 11 & 4 \end{pmatrix}_{b_0} \\
[1430] & 1^{113}41^{223}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 2 & 10 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 12 & 10 & 2 & 10 & 68 & 16 \\ 2 & 2 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[1431] & 1^{131}41^{31}23^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 5 & 2 \\ 1 & 0 & 1 & 12 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 12 & 12 & 0 & 106 & 36 & 18 \\ 5 & 4 & 0 & 36 & 10 & 6 \\ 2 & 2 & 0 & 18 & 6 & 3 \end{pmatrix}_{b_0}$$

[1432] $1^{13}41^{23}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -189$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 1 & 2 & 10 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 11 & 10 & 1 & 10 & 65 & 15 \\ 2 & 2 & 0 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1433] $1^{31}41^{13}23^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 1 & 13 & 2 & 2 \\ 3 & 4 & 1 & 24 & 2 & 4 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 13 & 24 & 1 & 107 & 12 & 19 \\ 2 & 2 & 0 & 12 & 0 & 2 \\ 2 & 4 & 0 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1434] $1^{14}1^{11}23^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 7 & 8 & 3 \\ 2 & 0 & 4 & 4 & 2 \\ 7 & 4 & 12 & 14 & 8 \\ 8 & 4 & 14 & 12 & 8 \\ 3 & 2 & 8 & 8 & 4 \end{pmatrix}_{b_0}$$

[1435] $1^{122}23^{11}141^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 6 & 9 & 3 \\ 4 & 5 & 10 & 14 & 5 \\ 6 & 10 & 14 & 22 & 8 \\ 9 & 14 & 22 & 29 & 11 \\ 3 & 5 & 8 & 11 & 4 \end{pmatrix}_{b_0}$$

[1436] $1^{14}1^{21}23^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -194$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 10 & 6 & 3 \\ 2 & 0 & 6 & 3 & 2 \\ 10 & 6 & 30 & 16 & 12 \\ 6 & 3 & 16 & 6 & 6 \\ 3 & 2 & 12 & 6 & 4 \end{pmatrix}_{b_0}$$

[1437] $1^{11}41^{22}23^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -194$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 10 & 14 & 3 \\ 2 & 0 & 6 & 9 & 2 \\ 10 & 6 & 30 & 40 & 12 \\ 14 & 9 & 40 & 50 & 16 \\ 3 & 2 & 12 & 16 & 4 \end{pmatrix}_{b_0}$$

[1438] $1^{112}41^{12}23^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -194$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 1 & 4 & 8 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 4 & 4 & 0 & 10 & 24 & 6 \\ 10 & 8 & 1 & 24 & 47 & 13 \\ 2 & 2 & 0 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[1439] $1^{14}1^{21}23^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -194$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 14 & 6 & 3 \\ 2 & 0 & 8 & 3 & 2 \\ 14 & 8 & 54 & 20 & 16 \\ 6 & 3 & 20 & 6 & 6 \\ 3 & 2 & 16 & 6 & 4 \end{pmatrix}_{b_0}$$

[1440] $1^{121}41^{21}23^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -194$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 5 & 2 \\ 1 & 0 & 1 & 8 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 10 & 8 & 2 & 50 & 24 & 14 \\ 5 & 4 & 0 & 24 & 10 & 6 \\ 2 & 2 & 0 & 14 & 6 & 3 \end{pmatrix}_{b_0}$$

[1441] $1^{132}23^{11}141^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -196$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 12 & 9 & 3 \\ 4 & 5 & 20 & 14 & 5 \\ 12 & 20 & 58 & 42 & 16 \\ 9 & 14 & 42 & 29 & 11 \\ 3 & 5 & 16 & 11 & 4 \end{pmatrix}_{b_0}$$

[1442] $1^{12}23^{11}141^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -198$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 10.91 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 3 & 2 \\ 1 & 0 & 2 & 8 & 3 & 2 \\ 2 & 2 & 0 & 9 & 3 & 2 \\ 7 & 8 & 9 & 40 & 15 & 12 \\ 3 & 3 & 3 & 15 & 4 & 4 \\ 2 & 2 & 2 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1443] \quad 2 \overset{1}{3} \overset{151}{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.1$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 15 & 6 & 3 \\ 4 & 5 & 25 & 9 & 5 \\ 15 & 25 & 98 & 35 & 20 \\ 6 & 9 & 35 & 11 & 7 \\ 3 & 5 & 20 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[1445] \quad 1 \overset{1}{4} 1 \overset{3}{2} 3 \overset{1}{1} \overset{114}{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 18 & 3 \\ 2 & 0 & 1 & 10 & 2 \\ 2 & 1 & 0 & 8 & 2 \\ 18 & 10 & 8 & 84 & 20 \\ 3 & 2 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[1444] \quad 2 \overset{1}{3} \overset{12}{141} \overset{114}{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 10 & 2 \\ 1 & 0 & 2 & 2 & 11 & 2 \\ 2 & 2 & 0 & 1 & 12 & 2 \\ 1 & 2 & 1 & 0 & 11 & 2 \\ 10 & 11 & 12 & 11 & 76 & 16 \\ 2 & 2 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1446] \quad 2 \overset{1}{3} \overset{132}{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 9 & 9 & 3 \\ 4 & 5 & 15 & 14 & 5 \\ 9 & 15 & 34 & 33 & 12 \\ 9 & 14 & 33 & 29 & 11 \\ 3 & 5 & 12 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1447] \quad 1 \overset{1}{4} 1 \overset{3}{2} 3 \overset{1}{1} \overset{114}{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -217, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 19 & 3 \\ 2 & 0 & 1 & 12 & 2 \\ 2 & 1 & 0 & 8 & 2 \\ 19 & 12 & 8 & 98 & 22 \\ 3 & 2 & 2 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[1450] \quad 1 \overset{1}{12} 41 \overset{12}{23} 1 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 1 & 4 & 8 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 0 & 10 & 24 & 6 \\ 9 & 8 & 0 & 24 & 46 & 12 \\ 2 & 2 & 0 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1449] \quad 1 \overset{131}{41} \overset{41}{23} 1 \overset{1}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 5 & 2 \\ 1 & 0 & 1 & 14 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 15 & 14 & 0 & 160 & 44 & 22 \\ 5 & 4 & 0 & 44 & 10 & 6 \\ 2 & 2 & 0 & 22 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1448] \quad 1 \overset{114}{41} \overset{12}{23} 1 \overset{12}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 13 & 2 \\ 1 & 0 & 1 & 4 & 12 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 4 & 4 & 0 & 10 & 32 & 6 \\ 13 & 12 & 2 & 32 & 90 & 18 \\ 2 & 2 & 0 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1452] \quad 1 \overset{1}{4} 1 \overset{22}{23} 1 \overset{122}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 10 & 12 & 3 \\ 2 & 0 & 6 & 8 & 2 \\ 10 & 6 & 30 & 36 & 12 \\ 12 & 8 & 36 & 40 & 14 \\ 3 & 2 & 12 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1451] \quad 1 \overset{1}{4} 1 \overset{14}{23} 1 \overset{124}{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 7 & 21 & 3 \\ 2 & 0 & 4 & 12 & 2 \\ 7 & 4 & 12 & 40 & 8 \\ 21 & 12 & 40 & 126 & 24 \\ 3 & 2 & 8 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[1453] \quad 2 \overset{1}{3} \overset{131}{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -225, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 12 & 6 & 3 \\ 4 & 5 & 20 & 9 & 5 \\ 12 & 20 & 58 & 27 & 16 \\ 6 & 9 & 27 & 11 & 7 \\ 3 & 5 & 16 & 7 & 4 \end{pmatrix}_{b_0} \\
[1455] & 1^{14}41^{423}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -230, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & 2 & 16 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 17 & 16 & 0 & 16 & 190 & 24 \\ 2 & 2 & 0 & 2 & 24 & 3 \end{pmatrix}_{b_0} \\
[1454] & 1^{14}1^{22}23^{11}1^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -230, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 10 & 13 & 3 \\ 2 & 0 & 6 & 7 & 2 \\ 10 & 6 & 30 & 36 & 12 \\ 13 & 7 & 36 & 40 & 14 \\ 3 & 2 & 12 & 14 & 4 \end{pmatrix}_{b_0} \\
& 1^{142} \\
[1456] & 23^{11}141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 12 & 9 & 3 \\ 4 & 5 & 20 & 14 & 5 \\ 12 & 20 & 62 & 44 & 16 \\ 9 & 14 & 44 & 29 & 11 \\ 3 & 5 & 16 & 11 & 4 \end{pmatrix}_{b_0} \\
[1457] & 1^{11}41^{11}23^{11}1^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 7 & 11 & 3 \\ 2 & 0 & 4 & 7 & 2 \\ 7 & 4 & 12 & 20 & 8 \\ 11 & 7 & 20 & 28 & 12 \\ 3 & 2 & 8 & 12 & 4 \end{pmatrix}_{b_0} \\
& 1^{161} \\
[1458] & 23^{11}141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 18 & 6 & 3 \\ 4 & 5 & 30 & 9 & 5 \\ 18 & 30 & 142 & 42 & 24 \\ 6 & 9 & 42 & 11 & 7 \\ 3 & 5 & 24 & 7 & 4 \end{pmatrix}_{b_0} \\
[1459] & 1^{14}1^{12}23^{11}1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 5 & 11 & 3 \\ 2 & 0 & 3 & 6 & 2 \\ 5 & 3 & 6 & 16 & 6 \\ 11 & 6 & 16 & 30 & 12 \\ 3 & 2 & 6 & 12 & 4 \end{pmatrix}_{b_0} \\
[1460] & 1^{14}1^{21}23^{11}1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -246, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 12 & 6 & 3 \\ 2 & 0 & 7 & 3 & 2 \\ 12 & 7 & 40 & 18 & 14 \\ 6 & 3 & 18 & 6 & 6 \\ 3 & 2 & 14 & 6 & 4 \end{pmatrix}_{b_0} \\
[1461] & 1^{122}41^{22}23^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 0 & 1 & 8 & 8 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 8 & 8 & 0 & 46 & 48 & 12 \\ 9 & 8 & 0 & 48 & 46 & 12 \\ 2 & 2 & 0 & 12 & 12 & 3 \end{pmatrix}_{b_0} \\
[1462] & 1^{41}41^{14}23^{11}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 17 & 2 & 2 \\ 3 & 4 & 1 & 32 & 2 & 4 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 17 & 32 & 1 & 191 & 16 & 25 \\ 2 & 2 & 0 & 16 & 0 & 2 \\ 2 & 4 & 0 & 25 & 2 & 3 \end{pmatrix}_{b_0} \\
[1463] & 1^{11}41^{41}23^{11}1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 19 & 6 & 3 \\ 2 & 0 & 13 & 3 & 2 \\ 19 & 13 & 130 & 32 & 24 \\ 6 & 3 & 32 & 6 & 6 \\ 3 & 2 & 24 & 6 & 4 \end{pmatrix}_{b_0} \\
[1464] & 1^{11}41^{21}23^{11}1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 13 & 6 & 3 \\ 2 & 0 & 9 & 3 & 2 \\ 13 & 9 & 50 & 20 & 16 \\ 6 & 3 & 20 & 6 & 6 \\ 3 & 2 & 16 & 6 & 4 \end{pmatrix}_{b_0} \\
[1465] & 1^{11}41^{21}23^11^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 15 & 6 & 3 \\ 2 & 0 & 10 & 3 & 2 \\ 15 & 10 & 64 & 22 & 18 \\ 6 & 3 & 22 & 6 & 6 \\ 3 & 2 & 18 & 6 & 4 \end{pmatrix}_{b_0} \\
[1466] & 1^{121}41^{21}23^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 5 & 2 \\ 1 & 0 & 1 & 8 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 9 & 8 & 1 & 47 & 24 & 13 \\ 5 & 4 & 0 & 24 & 10 & 6 \\ 2 & 2 & 0 & 13 & 6 & 3 \end{pmatrix}_{b_0} \\
& 1^{112} \\
[1467] & 23^1141^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 5 & 5 & 19 & 5 \\ 3 & 5 & 2 & 14 & 4 \\ 12 & 19 & 14 & 53 & 15 \\ 3 & 5 & 4 & 15 & 4 \end{pmatrix}_{b_0} \\
[1468] & 1^{141}41^{41}23^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -270, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 5 & 2 \\ 1 & 0 & 1 & 16 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 16 & 16 & 0 & 190 & 48 & 24 \\ 5 & 4 & 0 & 48 & 10 & 6 \\ 2 & 2 & 0 & 24 & 6 & 3 \end{pmatrix}_{b_0} \\
[1469] & 1^{11}41^{22}23^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 0 & 1 & 10 & 2 \\ 2 & 1 & 0 & 8 & 2 \\ 16 & 10 & 8 & 64 & 18 \\ 3 & 2 & 2 & 18 & 4 \end{pmatrix}_{b_0} \\
[1470] & 1^{112}41^{13}23^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 13 & 2 \\ 1 & 0 & 1 & 4 & 10 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 4 & 4 & 0 & 10 & 32 & 6 \\ 13 & 10 & 1 & 32 & 85 & 17 \\ 2 & 2 & 0 & 6 & 17 & 3 \end{pmatrix}_{b_0} \\
[1471] & 1^{11}41^{12}23^11^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 7 & 14 & 3 \\ 2 & 0 & 4 & 9 & 2 \\ 7 & 4 & 12 & 26 & 8 \\ 14 & 9 & 26 & 50 & 16 \\ 3 & 2 & 8 & 16 & 4 \end{pmatrix}_{b_0} \\
& 1^{113} \\
[1472] & 23^1141^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 5 & 5 & 19 & 5 \\ 3 & 5 & 2 & 15 & 4 \\ 12 & 19 & 15 & 55 & 15 \\ 3 & 5 & 4 & 15 & 4 \end{pmatrix}_{b_0} \\
[1473] & 1^{141}41^{23}1^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -282, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 18 & 6 & 3 \\ 2 & 0 & 11 & 3 & 2 \\ 18 & 11 & 112 & 30 & 22 \\ 6 & 3 & 30 & 6 & 6 \\ 3 & 2 & 22 & 6 & 4 \end{pmatrix}_{b_0} \\
[1474] & 1^{114}41^{42}23^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 18 & 2 \\ 1 & 0 & 1 & 2 & 16 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 18 & 16 & 1 & 16 & 191 & 25 \\ 2 & 2 & 0 & 2 & 25 & 3 \end{pmatrix}_{b_0} \\
& 1^{123} \\
[1475] & 23^1141^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 6 & 15 & 3 \\ 4 & 5 & 10 & 24 & 5 \\ 6 & 10 & 14 & 36 & 8 \\ 15 & 24 & 36 & 85 & 19 \\ 3 & 5 & 8 & 19 & 4 \end{pmatrix}_{b_0} \\
[1477] & 1^{131} 41^{323} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 2 & 2 \\ 1 & 0 & 1 & 12 & 2 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 14 & 12 & 2 & 110 & 12 & 20 \\ 2 & 2 & 0 & 12 & 0 & 2 \\ 2 & 2 & 0 & 20 & 2 & 3 \end{pmatrix}_{b_0} \\
[1476] & 1^{131} 41^{123} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 1 & 4 & 10 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 0 & 10 & 28 & 6 \\ 10 & 10 & 0 & 28 & 64 & 14 \\ 2 & 2 & 0 & 6 & 14 & 3 \end{pmatrix}_{b_0} \\
[1479] & 1^{114} 41^{423} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -290, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 19 & 2 \\ 1 & 0 & 1 & 2 & 16 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 19 & 16 & 2 & 16 & 194 & 26 \\ 2 & 2 & 0 & 2 & 26 & 3 \end{pmatrix}_{b_0} \\
[1478] & 1^{141} 1^{223} 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -290, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 5 & 13 & 3 \\ 2 & 0 & 3 & 7 & 2 \\ 5 & 3 & 6 & 18 & 6 \\ 13 & 7 & 18 & 40 & 14 \\ 3 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0} \\
& 1^{123} \\
[1480] & 2 3^1 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 6 & 12 & 3 \\ 4 & 5 & 10 & 19 & 5 \\ 6 & 10 & 14 & 30 & 8 \\ 12 & 19 & 30 & 55 & 15 \\ 3 & 5 & 8 & 15 & 4 \end{pmatrix}_{b_0} \\
[1481] & 1^{141} 1^{223} 1^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 12 & 11 & 3 \\ 2 & 0 & 7 & 6 & 2 \\ 12 & 7 & 40 & 36 & 14 \\ 11 & 6 & 36 & 30 & 12 \\ 3 & 2 & 14 & 12 & 4 \end{pmatrix}_{b_0} \\
& 1^{171} \\
[1482] & 2 3^1 141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 21 & 6 & 3 \\ 4 & 5 & 35 & 9 & 5 \\ 21 & 35 & 194 & 49 & 28 \\ 6 & 9 & 49 & 11 & 7 \\ 3 & 5 & 28 & 7 & 4 \end{pmatrix}_{b_0} \\
[1483] & 1^{114} 1^{123} 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 5 & 12 & 3 \\ 2 & 0 & 3 & 8 & 2 \\ 5 & 3 & 6 & 18 & 6 \\ 12 & 8 & 18 & 40 & 14 \\ 3 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0} \\
[1484] & 1^{131} 41^{123} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 1 & 4 & 10 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 4 & 4 & 0 & 10 & 28 & 6 \\ 11 & 10 & 1 & 28 & 65 & 15 \\ 2 & 2 & 0 & 6 & 15 & 3 \end{pmatrix}_{b_0} \\
[1485] & 1^{114} 41^{223} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 2 & 12 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 12 & 12 & 1 & 12 & 87 & 17 \\ 2 & 2 & 0 & 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[1486] & 1^{112} 41^{143} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -306, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 16 & 2 \\ 1 & 0 & 1 & 4 & 12 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 4 & 4 & 0 & 10 & 40 & 6 \\ 16 & 12 & 1 & 40 & 135 & 21 \\ 2 & 2 & 0 & 6 & 21 & 3 \end{pmatrix}_{b_0} \\
[1487] & 1^{14} 41^{323} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -313, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 2 & 14 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 14 & 2 \\ 15 & 14 & 1 & 14 & 133 & 21 \\ 2 & 2 & 0 & 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1488] & 1^{13} 41^{13} 23^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 15 & 2 \\ 1 & 0 & 1 & 4 & 12 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 4 & 4 & 0 & 10 & 36 & 6 \\ 15 & 12 & 2 & 36 & 110 & 20 \\ 2 & 2 & 0 & 6 & 20 & 3 \end{pmatrix}_{b_0} \\
& 1^{142} \\
[1489] & 2 3^1 141^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 15 & 9 & 3 \\ 4 & 5 & 25 & 14 & 5 \\ 15 & 25 & 92 & 53 & 20 \\ 9 & 14 & 53 & 29 & 11 \\ 3 & 5 & 20 & 11 & 4 \end{pmatrix}_{b_0} \\
[1490] & 1^{14} 41^{223} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 2 & 12 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 13 & 12 & 2 & 12 & 90 & 18 \\ 2 & 2 & 0 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[1491] & 1^{14} 41^{323} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -328, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 2 & 14 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 1 & 2 & 0 & 0 & 14 & 2 \\ 16 & 14 & 2 & 14 & 136 & 22 \\ 2 & 2 & 0 & 2 & 22 & 3 \end{pmatrix}_{b_0} \\
& 1^{141} \\
[1492] & 2 3^1 141^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -329, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 15 & 6 & 3 \\ 4 & 5 & 25 & 9 & 5 \\ 15 & 25 & 92 & 34 & 20 \\ 6 & 9 & 34 & 11 & 7 \\ 3 & 5 & 20 & 7 & 4 \end{pmatrix}_{b_0} \\
[1493] & 1^{14} 1^{22} 23^{1122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -332, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 10 & 11 & 3 \\ 2 & 0 & 6 & 6 & 2 \\ 10 & 6 & 30 & 32 & 12 \\ 11 & 6 & 32 & 30 & 12 \\ 3 & 2 & 12 & 12 & 4 \end{pmatrix}_{b_0} \\
[1494] & 1^{121} 41^{41} 23^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -335, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 15 & 5 & 2 \\ 1 & 0 & 1 & 12 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 15 & 12 & 1 & 135 & 40 & 21 \\ 5 & 4 & 0 & 40 & 10 & 6 \\ 2 & 2 & 0 & 21 & 6 & 3 \end{pmatrix}_{b_0} \\
[1496] & 1^{14} 41^{31} 23^{1131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 16 & 6 & 3 \\ 2 & 0 & 11 & 3 & 2 \\ 16 & 11 & 84 & 26 & 20 \\ 6 & 3 & 26 & 6 & 6 \\ 3 & 2 & 20 & 6 & 4 \end{pmatrix}_{b_0} \\
[1495] & 1^{121} 41^{31} 23^{1112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 13 & 5 & 2 \\ 1 & 0 & 1 & 10 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 13 & 10 & 2 & 88 & 32 & 18 \\ 5 & 4 & 0 & 32 & 10 & 6 \\ 2 & 2 & 0 & 18 & 6 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [1499] $1^{141}3^{123}1^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -339$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 15 & 6 & 3 \\ 2 & 0 & 9 & 3 & 2 \\ 15 & 9 & 70 & 24 & 18 \\ 6 & 3 & 24 & 6 & 6 \\ 3 & 2 & 18 & 6 & 4 \end{pmatrix}_{b_0}$
- [1497] $1^{141}2^{123}1^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -339$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 7 & 13 & 3 \\ 2 & 0 & 4 & 7 & 2 \\ 7 & 4 & 12 & 24 & 8 \\ 13 & 7 & 24 & 40 & 14 \\ 3 & 2 & 8 & 14 & 4 \end{pmatrix}_{b_0}$
- [1498] $1^{121}4^{131}2^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -339$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 5 & 2 \\ 1 & 0 & 1 & 10 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 12 & 10 & 1 & 85 & 32 & 17 \\ 5 & 4 & 0 & 32 & 10 & 6 \\ 2 & 2 & 0 & 17 & 6 & 3 \end{pmatrix}_{b_0}$
- [1500] $1^{114}1^{13}2^{1112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -353$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 5 & 15 & 3 \\ 2 & 0 & 3 & 10 & 2 \\ 5 & 3 & 6 & 24 & 6 \\ 15 & 10 & 24 & 74 & 18 \\ 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$
- [1501] $1^{114}4^{12}2^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -354$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 1 & 4 & 12 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 4 & 4 & 0 & 10 & 32 & 6 \\ 12 & 12 & 1 & 32 & 87 & 17 \\ 2 & 2 & 0 & 6 & 17 & 3 \end{pmatrix}_{b_0}$
- [1502] $1^{114}1^{12}2^{1113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -383$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 5 & 14 & 3 \\ 2 & 0 & 3 & 9 & 2 \\ 5 & 3 & 6 & 20 & 6 \\ 14 & 9 & 20 & 50 & 16 \\ 3 & 2 & 6 & 16 & 4 \end{pmatrix}_{b_0}$
- [1503] $1^{141}2^{123}1^{142} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -384$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 14 & 8 & 3 \\ 2 & 0 & 8 & 4 & 2 \\ 14 & 8 & 54 & 28 & 16 \\ 8 & 4 & 28 & 12 & 8 \\ 3 & 2 & 16 & 8 & 4 \end{pmatrix}_{b_0}$
- [1504] $1^{114}1^{12}2^{1114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -396$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 5 & 16 & 3 \\ 2 & 0 & 3 & 10 & 2 \\ 5 & 3 & 6 & 22 & 6 \\ 16 & 10 & 22 & 64 & 18 \\ 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$
- [1505] $1^{121}4^{131}2^{1112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -402$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 16 & 5 & 2 \\ 1 & 0 & 1 & 12 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 16 & 12 & 2 & 138 & 40 & 22 \\ 5 & 4 & 0 & 40 & 10 & 6 \\ 2 & 2 & 0 & 22 & 6 & 3 \end{pmatrix}_{b_0}$
- [1506] $2^13^{114}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -404$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 5 & 5 & 24 & 5 \\ 3 & 5 & 2 & 19 & 4 \\ 15 & 24 & 19 & 89 & 19 \\ 3 & 5 & 4 & 19 & 4 \end{pmatrix}_{b_0}$
- [1507] $1^{113}4^{13}2^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -410$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.99 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 13 & 2 \\ 1 & 0 & 1 & 4 & 12 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 0 & 10 & 36 & 6 \\ 13 & 12 & 0 & 36 & 106 & 18 \\ 2 & 2 & 0 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

^{1¹²⁴}

$$[1508] \ 2 \ 3^1 \ 141^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 6 & 15 & 3 \\ 4 & 5 & 10 & 24 & 5 \\ 6 & 10 & 14 & 38 & 8 \\ 15 & 24 & 38 & 89 & 19 \\ 3 & 5 & 8 & 19 & 4 \end{pmatrix}_{b_0}$$

$$[1509] \ 1^{\textcolor{blue}{11} \textcolor{blue}{41} \textcolor{blue}{13} \textcolor{blue}{23} \textcolor{blue}{1}^{\textcolor{blue}{124}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 7 & 19 & 3 \\ 2 & 0 & 4 & 12 & 2 \\ 7 & 4 & 12 & 36 & 8 \\ 19 & 12 & 36 & 98 & 22 \\ 3 & 2 & 8 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[1510] \ 1^{\textcolor{blue}{141} \textcolor{blue}{41} \textcolor{blue}{42} \textcolor{blue}{23} \textcolor{blue}{1}^{\textcolor{blue}{12}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 2 & 2 \\ 1 & 0 & 1 & 16 & 2 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 18 & 16 & 2 & 194 & 16 & 26 \\ 2 & 2 & 0 & 16 & 0 & 2 \\ 2 & 2 & 0 & 26 & 2 & 3 \end{pmatrix}_{b_0}$$

^{1¹¹³}

$$[1511] \ 2 \ 3^1 \ 141^{\textcolor{blue}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -437, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 5 & 5 & 24 & 5 \\ 3 & 5 & 2 & 18 & 4 \\ 15 & 24 & 18 & 85 & 19 \\ 3 & 5 & 4 & 19 & 4 \end{pmatrix}_{b_0}$$

$$[1512] \ 1^{\textcolor{blue}{141} \textcolor{blue}{31} \textcolor{blue}{23} \textcolor{blue}{1}^{\textcolor{blue}{141}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -443, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 17 & 6 & 3 \\ 2 & 0 & 10 & 3 & 2 \\ 17 & 10 & 84 & 26 & 20 \\ 6 & 3 & 26 & 6 & 6 \\ 3 & 2 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1513] \ 1^{\textcolor{blue}{131} \textcolor{blue}{41} \textcolor{blue}{31} \textcolor{blue}{23} \textcolor{blue}{1}^{\textcolor{blue}{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -443, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 5 & 2 \\ 1 & 0 & 1 & 12 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 13 & 12 & 1 & 107 & 36 & 19 \\ 5 & 4 & 0 & 36 & 10 & 6 \\ 2 & 2 & 0 & 19 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1514] \ 1^{\textcolor{blue}{113} \textcolor{blue}{41} \textcolor{blue}{13} \textcolor{blue}{23} \textcolor{blue}{1}^{\textcolor{blue}{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -450, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 14 & 2 \\ 1 & 0 & 1 & 4 & 12 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 4 & 4 & 0 & 10 & 36 & 6 \\ 14 & 12 & 1 & 36 & 107 & 19 \\ 2 & 2 & 0 & 6 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1515] \ 1^{\textcolor{blue}{113} \textcolor{blue}{41} \textcolor{blue}{14} \textcolor{blue}{23} \textcolor{blue}{1}^{\textcolor{blue}{12}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 18 & 2 \\ 1 & 0 & 1 & 4 & 14 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 4 & 4 & 0 & 10 & 44 & 6 \\ 18 & 14 & 2 & 44 & 164 & 24 \\ 2 & 2 & 0 & 6 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[1516] \ 1^{\textcolor{blue}{141} \textcolor{blue}{13} \textcolor{blue}{23} \textcolor{blue}{1}^{\textcolor{blue}{113}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -459, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 5 & 16 & 3 \\ 2 & 0 & 3 & 9 & 2 \\ 5 & 3 & 6 & 24 & 6 \\ 16 & 9 & 24 & 70 & 18 \\ 3 & 2 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1517] \ 1^{\textcolor{blue}{141} \textcolor{blue}{12} \textcolor{blue}{23} \textcolor{blue}{1}^{\textcolor{blue}{124}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 7 & 15 & 3 \\ 2 & 0 & 4 & 8 & 2 \\ 7 & 4 & 12 & 28 & 8 \\ 15 & 8 & 28 & 54 & 16 \\ 3 & 2 & 8 & 16 & 4 \end{pmatrix}_{b_0}$$

^{1¹²⁴}

$$[1518] \ 2 \ 3^1 \ 141^{\textcolor{blue}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$\begin{aligned}
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 6 & 18 & 3 \\ 4 & 5 & 10 & 29 & 5 \\ 6 & 10 & 14 & 44 & 8 \\ 18 & 29 & 44 & 125 & 23 \\ 3 & 5 & 8 & 23 & 4 \end{pmatrix}_{b_0} \\
[1519] & 1^{114} 41^{13} 23^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -480, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 16 & 2 \\ 1 & 0 & 1 & 4 & 14 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 4 & 4 & 0 & 10 & 40 & 6 \\ 16 & 14 & 2 & 40 & 136 & 22 \\ 2 & 2 & 0 & 6 & 22 & 3 \end{pmatrix}_{b_0} \\
[1520] & 1^1 41^{13} 23^1 1^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -492, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 7 & 18 & 3 \\ 2 & 0 & 4 & 10 & 2 \\ 7 & 4 & 12 & 34 & 8 \\ 18 & 10 & 34 & 84 & 20 \\ 3 & 2 & 8 & 20 & 4 \end{pmatrix}_{b_0} \\
[1521] & 1^{11} 41^{12} 23^1 1^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -492, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 7 & 16 & 3 \\ 2 & 0 & 4 & 10 & 2 \\ 7 & 4 & 12 & 30 & 8 \\ 16 & 10 & 30 & 64 & 18 \\ 3 & 2 & 8 & 18 & 4 \end{pmatrix}_{b_0} \\
[1522] & 1^{131} 41^{31} 23^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 14 & 5 & 2 \\ 1 & 0 & 1 & 12 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 14 & 12 & 2 & 110 & 36 & 20 \\ 5 & 4 & 0 & 36 & 10 & 6 \\ 2 & 2 & 0 & 20 & 6 & 3 \end{pmatrix}_{b_0} \\
[1523] & 1^{11} 41^{31} 23^1 1^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 18 & 6 & 3 \\ 2 & 0 & 12 & 3 & 2 \\ 18 & 12 & 98 & 28 & 22 \\ 6 & 3 & 28 & 6 & 6 \\ 3 & 2 & 22 & 6 & 4 \end{pmatrix}_{b_0} \\
[1524] & 1^{114} 41^{13} 23^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 14 & 2 \\ 1 & 0 & 1 & 4 & 14 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 0 & 10 & 40 & 6 \\ 14 & 14 & 0 & 40 & 132 & 20 \\ 2 & 2 & 0 & 6 & 20 & 3 \end{pmatrix}_{b_0} \\
[1525] & 1^1 41^{14} 23^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -542, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 5 & 19 & 3 \\ 2 & 0 & 3 & 11 & 2 \\ 5 & 3 & 6 & 30 & 6 \\ 19 & 11 & 30 & 112 & 22 \\ 3 & 2 & 6 & 22 & 4 \end{pmatrix}_{b_0} \\
[1526] & 1^1 41^{41} 23^1 1^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -542, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 20 & 6 & 3 \\ 2 & 0 & 12 & 3 & 2 \\ 20 & 12 & 126 & 32 & 24 \\ 6 & 3 & 32 & 6 & 6 \\ 3 & 2 & 24 & 6 & 4 \end{pmatrix}_{b_0} \\
[1527] & 1^{131} 41^{41} 23^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -543, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 16 & 5 & 2 \\ 1 & 0 & 1 & 14 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 16 & 14 & 1 & 161 & 44 & 23 \\ 5 & 4 & 0 & 44 & 10 & 6 \\ 2 & 2 & 0 & 23 & 6 & 3 \end{pmatrix}_{b_0} \\
[1528] & 1^1 41^{13} 23^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -548, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 5 & 18 & 3 \\ 2 & 0 & 3 & 10 & 2 \\ 5 & 3 & 6 & 26 & 6 \\ 18 & 10 & 26 & 84 & 20 \\ 3 & 2 & 6 & 20 & 4 \end{pmatrix}_{b_0} \\
[1529] & 1^{11} 41^{13} 23^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -558, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =
\end{aligned}$$

5.0

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 5 & 17 & 3 \\ 2 & 0 & 3 & 11 & 2 \\ 5 & 3 & 6 & 26 & 6 \\ 17 & 11 & 26 & 84 & 20 \\ 3 & 2 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[1530] 1^{113} 41^{14} 23^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 17 & 2 \\ 1 & 0 & 1 & 4 & 14 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 4 & 4 & 0 & 10 & 44 & 6 \\ 17 & 14 & 1 & 44 & 161 & 23 \\ 2 & 2 & 0 & 6 & 23 & 3 \end{pmatrix}_{b_0}$$

$$[1531] 1^{11} 41^{41} 23^1 1^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -575, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 21 & 6 & 3 \\ 2 & 0 & 14 & 3 & 2 \\ 21 & 14 & 144 & 34 & 26 \\ 6 & 3 & 34 & 6 & 6 \\ 3 & 2 & 26 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1532] 1^{114} 41^{13} 23^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 15 & 2 \\ 1 & 0 & 1 & 4 & 14 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 4 & 4 & 0 & 10 & 40 & 6 \\ 15 & 14 & 1 & 40 & 133 & 21 \\ 2 & 2 & 0 & 6 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1533] 1^{114} 2^3 141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -629, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 18 & 3 \\ 4 & 5 & 5 & 29 & 5 \\ 3 & 5 & 2 & 22 & 4 \\ 18 & 29 & 22 & 125 & 23 \\ 3 & 5 & 4 & 23 & 4 \end{pmatrix}_{b_0}$$

$$[1534] 1^{141} 41^{41} 23^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -647, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 5 & 2 \\ 1 & 0 & 1 & 16 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 17 & 16 & 1 & 191 & 48 & 25 \\ 5 & 4 & 0 & 48 & 10 & 6 \\ 2 & 2 & 0 & 25 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1535] 1^{11} 41^{14} 23^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -647, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 5 & 20 & 3 \\ 2 & 0 & 3 & 13 & 2 \\ 5 & 3 & 6 & 32 & 6 \\ 20 & 13 & 32 & 130 & 24 \\ 3 & 2 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[1536] 1^{131} 41^{41} 23^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -662, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 17 & 5 & 2 \\ 1 & 0 & 1 & 14 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 17 & 14 & 2 & 164 & 44 & 24 \\ 5 & 4 & 0 & 44 & 10 & 6 \\ 2 & 2 & 0 & 24 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1537] 1^{114} 41^{14} 23^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -666, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 17 & 2 \\ 1 & 0 & 1 & 4 & 16 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 0 & 10 & 48 & 6 \\ 17 & 16 & 0 & 48 & 190 & 24 \\ 2 & 2 & 0 & 6 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[1538] 1^{11} 41^{13} 23^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -689, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 5 & 19 & 3 \\ 2 & 0 & 3 & 12 & 2 \\ 5 & 3 & 6 & 28 & 6 \\ 19 & 12 & 28 & 98 & 22 \\ 3 & 2 & 6 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[1539] 1^{114} 41^{14} 23^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -698, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 19 & 2 \\ 1 & 0 & 1 & 4 & 16 & 2 \\ 1 & 1 & -2 & 0 & 2 & 0 \\ 4 & 4 & 0 & 10 & 48 & 6 \\ 19 & 16 & 2 & 48 & 194 & 26 \\ 2 & 2 & 0 & 6 & 26 & 3 \end{pmatrix}_{b_0}$$

$$[1540] \ 1^{14} 4^{14} 23^{11} 1^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -749, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 5 & 21 & 3 \\ 2 & 0 & 3 & 12 & 2 \\ 5 & 3 & 6 & 32 & 6 \\ 21 & 12 & 32 & 126 & 24 \\ 3 & 2 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[1541] \ 1^{14} 4^{14} 23^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -770, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 18 & 2 \\ 1 & 0 & 1 & 4 & 16 & 2 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 4 & 4 & 0 & 10 & 48 & 6 \\ 18 & 16 & 1 & 48 & 191 & 25 \\ 2 & 2 & 0 & 6 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[1542] \ 1^{14} 4^{14} 23^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -818, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 18 & 5 & 2 \\ 1 & 0 & 1 & 16 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 18 & 16 & 2 & 194 & 48 & 26 \\ 5 & 4 & 0 & 48 & 10 & 6 \\ 2 & 2 & 0 & 26 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1543] \ 1^{11} 4^{14} 23^{11} 1^{114} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 5 & 22 & 3 \\ 2 & 0 & 3 & 14 & 2 \\ 5 & 3 & 6 & 34 & 6 \\ 22 & 14 & 34 & 144 & 26 \\ 3 & 2 & 6 & 26 & 4 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [1545] \ 1^{11} 4^{13} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -29, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 6 & 5 & 3 & 10 & 8 \\ 3 & 3 & 2 & 8 & 5 \end{pmatrix}_{b_0}$$

$$[1544] \ 1^{11} 4^{13} 21^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -29, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & 0 & 5 & 2 \\ 7 & 6 & 5 & 18 & 10 \\ 3 & 3 & 2 & 10 & 5 \end{pmatrix}_{b_0}$$

$$[1546] \ 1^{11} 4^{13} 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -29, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 7 & 2 \\ 11 & 9 & 7 & 48 & 16 \\ 3 & 3 & 2 & 16 & 5 \end{pmatrix}_{b_0}$$

$$[1548] \ 1^{11} 5^{13} 13^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 6 & 3 & 3 \\ 5 & 5 & 8 & 3 & 5 \\ 6 & 8 & 9 & 5 & 7 \\ 3 & 3 & 5 & 1 & 3 \\ 3 & 5 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1547] \ 31^{11} 3^{11} 5^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 6 & 5 & 3 \\ 5 & 5 & 8 & 7 & 5 \\ 6 & 8 & 9 & 10 & 7 \\ 5 & 7 & 10 & 8 & 6 \\ 3 & 5 & 7 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1549] \ 1^{12} 4^{13} 21^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -29, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 11 & 10 & 6 & 46 & 16 \\ 3 & 3 & 2 & 16 & 5 \end{pmatrix}_{b_0}$$

[1550] $1^{12} 4^{13} 1^{21} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -29, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & 0 & 8 & 2 \\ 12 & 11 & 8 & 60 & 18 \\ 3 & 3 & 2 & 18 & 5 \end{pmatrix}_{b_0}$$

[1551] $1^{12} 5^{13} 1^{11} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 6 & 9 & 3 \\ 5 & 5 & 8 & 11 & 5 \\ 6 & 8 & 9 & 16 & 7 \\ 9 & 11 & 16 & 22 & 10 \\ 3 & 5 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

[1552] $1^{12} 4^{13} 1^{21} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -29, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & 0 & 8 & 2 \\ 15 & 13 & 8 & 90 & 22 \\ 3 & 3 & 2 & 22 & 5 \end{pmatrix}_{b_0}$$

[1553] $1^{13} 4^{13} 1^{21} 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -29, \det = 73, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 13 & 13 & 9 & 76 & 20 \\ 3 & 3 & 2 & 20 & 5 \end{pmatrix}_{b_0}$$

[1555] $1^{121} 5^{13} 1^{12} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 12 & 3 & 3 \\ 5 & 5 & 16 & 3 & 5 \\ 12 & 16 & 42 & 10 & 14 \\ 3 & 3 & 10 & 1 & 3 \\ 3 & 5 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[1554] $1^{12} 4^{13} 1^{21} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -29, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 14 & 3 \\ 1 & 1 & 0 & 10 & 2 \\ 16 & 14 & 10 & 108 & 24 \\ 3 & 3 & 2 & 24 & 5 \end{pmatrix}_{b_0}$$

[1556] $31^{121} 3^1 1^5 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 12 & 5 & 3 \\ 5 & 5 & 16 & 7 & 5 \\ 12 & 16 & 42 & 20 & 14 \\ 5 & 7 & 20 & 8 & 6 \\ 3 & 5 & 14 & 6 & 4 \end{pmatrix}_{b_0}$$

[1557] $31^{111} 3^1 1^5 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 6 & 11 & 3 \\ 5 & 5 & 8 & 15 & 5 \\ 6 & 8 & 9 & 21 & 7 \\ 11 & 15 & 21 & 39 & 13 \\ 3 & 5 & 7 & 13 & 4 \end{pmatrix}_{b_0}$$

[1558] $1^{13} 4^{13} 1^{21} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -29, \det = 102, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 16 & 3 \\ 2 & 1 & 1 & 15 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 16 & 15 & 9 & 106 & 24 \\ 3 & 3 & 2 & 24 & 5 \end{pmatrix}_{b_0}$$

[1559] $1^{13} 4^{13} 1^{21} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -29, \det = 110, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 16 & 3 \\ 1 & 1 & 0 & 11 & 2 \\ 17 & 16 & 11 & 126 & 26 \\ 3 & 3 & 2 & 26 & 5 \end{pmatrix}_{b_0}$$

[1560] $1^{14} 4^{13} 1^{21} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -29, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 17 & 3 \\ 2 & 1 & 1 & 17 & 3 \\ 1 & 1 & 0 & 10 & 2 \\ 17 & 17 & 10 & 126 & 26 \\ 3 & 3 & 2 & 26 & 5 \end{pmatrix}_{b_0}$$

[1561] $1^{13} 4^{14} 13^{14} 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -29$, $\det = 133$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.96 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 20 & 3 \\ 2 & 1 & 1 & 18 & 3 \\ 1 & 1 & 0 & 11 & 2 \\ 20 & 18 & 11 & 168 & 30 \\ 3 & 3 & 2 & 30 & 5 \end{pmatrix}_{b_0}$$

[1562] $1^{14} 4^{13} 13^{13} 21^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -29$, $\det = 136$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.94 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 18 & 3 \\ 2 & 1 & 1 & 18 & 3 \\ 1 & 1 & 0 & 12 & 2 \\ 18 & 18 & 12 & 148 & 28 \\ 3 & 3 & 2 & 28 & 5 \end{pmatrix}_{b_0}$$

[1564] $1^{131} 5^{131} 13^{131} 3^{131} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 140$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 18 & 3 & 3 \\ 5 & 5 & 24 & 3 & 5 \\ 18 & 24 & 97 & 15 & 21 \\ 3 & 3 & 15 & 1 & 3 \\ 3 & 5 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

[1563] $31^{131} 3^{131} 1^{131} 5^{131} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 140$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 18 & 5 & 3 \\ 5 & 5 & 24 & 7 & 5 \\ 18 & 24 & 97 & 30 & 21 \\ 5 & 7 & 30 & 8 & 6 \\ 3 & 5 & 21 & 6 & 4 \end{pmatrix}_{b_0}$$

[1565] $1^{14} 4^{13} 13^{14} 21^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -29$, $\det = 171$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 21 & 3 \\ 2 & 1 & 1 & 20 & 3 \\ 1 & 1 & 0 & 12 & 2 \\ 21 & 20 & 12 & 190 & 32 \\ 3 & 3 & 2 & 32 & 5 \end{pmatrix}_{b_0}$$

$T = 11$ [1573] $1^{112} 31^{112} 5^{112} 13^{112} 1^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 10 & 2 \\ 2 & 0 & 4 & 6 & 23 & 6 \\ 7 & 0 & 10 & 23 & 67 & 17 \\ 2 & 0 & 2 & 6 & 17 & 4 \end{pmatrix}_{b_0}$$

[1566] $1^{113} 1^{11} 5^{11} 13^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 3 & 2 \\ 2 & 0 & 4 & 6 & 7 & 6 \\ 2 & 0 & 3 & 7 & 5 & 5 \\ 2 & 0 & 2 & 6 & 5 & 4 \end{pmatrix}_{b_0}$$

[1572] $1^{131} 1^{11} 5^{11} 13^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 3 & 2 \\ 2 & 0 & 1 & 1 & 4 & 3 \\ 2 & 0 & 3 & 4 & 5 & 5 \\ 2 & 0 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

[1567] $1^{111} 3^{11} 5^{11} 13^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 3 & 1 & 1 & 6 & 3 \\ 4 & 6 & 5 & 6 & 14 & 8 \\ 2 & 3 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[1568] $1^{111} 31^{11} 5^{11} 13^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 6 & 2 \\ 2 & 0 & 4 & 6 & 15 & 6 \\ 5 & 0 & 6 & 15 & 29 & 11 \\ 2 & 0 & 2 & 6 & 11 & 4 \end{pmatrix}_{b_0}$$

[1569] $1^{13}3^1 5^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 3 & 1 & 1 & 11 & 3 \\ 8 & 11 & 8 & 11 & 46 & 14 \\ 2 & 3 & 2 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[1571] $1^{11}3^1 5^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 3 & 4 & 3 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 3 & 1 & 1 & 4 & 3 \\ 3 & 4 & 3 & 4 & 5 & 5 \\ 2 & 3 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

[1570] $1^{13}3^1 1^5 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & 2 & 10 & 6 & 3 \\ 1 & 2 & 0 & 6 & 5 & 2 \\ 7 & 10 & 6 & 34 & 23 & 12 \\ 4 & 6 & 5 & 23 & 14 & 8 \\ 2 & 3 & 2 & 12 & 8 & 4 \end{pmatrix}_{b_0}$$

[1574] $1^{11}3^1 1^5 13^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 8 & 2 \\ 2 & 0 & 4 & 6 & 19 & 6 \\ 6 & 0 & 8 & 19 & 46 & 14 \\ 2 & 0 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

[1575] $1^{12}3^1 5^{13} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 11 & 2 \\ 2 & 0 & 4 & 6 & 27 & 6 \\ 9 & 0 & 11 & 27 & 96 & 20 \\ 2 & 0 & 2 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[1612] $1^{13}3^1 1^5 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 7 & 6 \\ 2 & 3 & 0 & 7 & 5 & 5 \\ 2 & 3 & -1 & 6 & 5 & 4 \end{pmatrix}_{b_0}$$

[1615] $1^{12}3^1 1^5 13^1 2^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 1 & 8 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 30 & 12 \\ 4 & 7 & 0 & 30 & 27 & 11 \\ 2 & 3 & -1 & 12 & 11 & 4 \end{pmatrix}_{b_0}$$

[1578] $1^{12}3^1 1^5 13^1 2^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 1 & 8 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 62 & 12 \\ 8 & 15 & 0 & 62 & 119 & 23 \\ 2 & 3 & -1 & 12 & 23 & 4 \end{pmatrix}_{b_0}$$

[1579] $1^{13}3^1 1^5 13^1 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 3 & 13 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 3 & 1 & 1 & 13 & 3 \\ 9 & 13 & 10 & 13 & 67 & 17 \\ 2 & 3 & 2 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[1580] $1^{13}3^{11}5^{13}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & 2 & 12 & 2 & 3 \\ 1 & 2 & 0 & 12 & 1 & 2 \\ 9 & 12 & 12 & 70 & 9 & 18 \\ 2 & 2 & 1 & 9 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

[1581] $1^{124}3^{11}5^{13}1^{24} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 1 & 8 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 70 & 12 \\ 9 & 17 & 0 & 70 & 152 & 26 \\ 2 & 3 & -1 & 12 & 26 & 4 \end{pmatrix}_{b_0}$$

[1582] $1^{13}3^{11}5^{13}1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 21 & 18 \\ 2 & 3 & 0 & 21 & 5 & 5 \\ 2 & 3 & -1 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

[1583] $1^{11}3^{11}5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 2 & 4 & 2 & 3 \\ 1 & 2 & 0 & 4 & 1 & 2 \\ 3 & 4 & 4 & 6 & 3 & 6 \\ 2 & 2 & 1 & 3 & 0 & 2 \\ 2 & 3 & 2 & 6 & 2 & 4 \end{pmatrix}_{b_0}$$

[1584] $1^{121}3^{11}5^{13}1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & 1 & 8 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 22 & 12 \\ 3 & 5 & 0 & 22 & 14 & 8 \\ 2 & 3 & -1 & 12 & 8 & 4 \end{pmatrix}_{b_0}$$

[1585] $1^{123}3^{11}5^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 2 & 8 & 2 & 3 \\ 1 & 2 & 0 & 8 & 1 & 2 \\ 6 & 8 & 8 & 30 & 6 & 12 \\ 2 & 2 & 1 & 6 & 0 & 2 \\ 2 & 3 & 2 & 12 & 2 & 4 \end{pmatrix}_{b_0}$$

[1586] $1^{131}3^{11}5^{13}1^{32} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & 1 & 12 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 45 & 18 \\ 4 & 7 & 0 & 45 & 27 & 11 \\ 2 & 3 & -1 & 18 & 11 & 4 \end{pmatrix}_{b_0}$$

[1587] $1^{123}3^{11}5^{13}1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 1 & 8 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 14 & 12 \\ 2 & 3 & 0 & 14 & 5 & 5 \\ 2 & 3 & -1 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

[1588] $1^{11}3^{11}5^{13}1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 6 & 3 & 2 \\ 6 & 0 & 6 & 34 & 15 & 12 \\ 2 & 0 & 3 & 15 & 5 & 5 \\ 2 & 0 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

[1589] $1^{123}1^{11}5131^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 9 & 3 & 2 \\ 6 & 0 & 9 & 49 & 18 & 15 \\ 2 & 0 & 3 & 18 & 5 & 5 \\ 2 & 0 & 2 & 15 & 5 & 4 \end{pmatrix}_{b_0}$$

[1590] $1^{132}3^{11}5131^{33} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 2 \\ 1 & 1 & 1 & 12 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 69 & 18 \\ 6 & 11 & 0 & 69 & 65 & 17 \\ 2 & 3 & -1 & 18 & 17 & 4 \end{pmatrix}_{b_0}$$

[1591] $1^{133}3^{11}5131^{34} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 1 & 12 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 93 & 18 \\ 8 & 15 & 0 & 93 & 119 & 23 \\ 2 & 3 & -1 & 18 & 23 & 4 \end{pmatrix}_{b_0}$$

[1592] $1^{131}31513^{11}1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & 1 & 12 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 33 & 18 \\ 3 & 5 & 0 & 33 & 14 & 8 \\ 2 & 3 & -1 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

[1593] $1^{122}31513^{11}1^{22} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 1 & 8 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 38 & 12 \\ 5 & 9 & 0 & 38 & 44 & 14 \\ 2 & 3 & -1 & 12 & 14 & 4 \end{pmatrix}_{b_0}$$

[1594] $1^{132}31513^{11}1^{32} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 5 & 2 \\ 1 & 1 & 1 & 12 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 57 & 18 \\ 5 & 9 & 0 & 57 & 44 & 14 \\ 2 & 3 & -1 & 18 & 14 & 4 \end{pmatrix}_{b_0}$$

[1595] $1^{114}31513^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 1 & 4 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 35 & 6 \\ 9 & 17 & 0 & 35 & 152 & 26 \\ 2 & 3 & -1 & 6 & 26 & 4 \end{pmatrix}_{b_0}$$

[1596] $1^{113}31513^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 1 & 4 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 27 & 6 \\ 7 & 13 & 0 & 27 & 90 & 20 \\ 2 & 3 & -1 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[1597] $1^{133}31513^{11}1^{33} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 7 & 2 \\ 1 & 1 & 1 & 12 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 81 & 18 \\ 7 & 13 & 0 & 81 & 90 & 20 \\ 2 & 3 & -1 & 18 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[1598] \ 1^{112} 31^1 513^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 4 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 19 & 6 \\ 5 & 9 & 0 & 19 & 44 & 14 \\ 2 & 3 & -1 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1599] \ 1^{134} 31^1 513^{1134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 9 & 2 \\ 1 & 1 & 1 & 12 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 105 & 18 \\ 9 & 17 & 0 & 105 & 152 & 26 \\ 2 & 3 & -1 & 18 & 26 & 4 \end{pmatrix}_{b_0}$$

$$[1600] \ 1^{143} 31^1 5131^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & 1 & 16 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 28 & 24 \\ 2 & 3 & 0 & 28 & 5 & 5 \\ 2 & 3 & -1 & 24 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1601] \ 1^{141} 31^1 5131^{42} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & 1 & 16 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 60 & 24 \\ 4 & 7 & 0 & 60 & 27 & 11 \\ 2 & 3 & -1 & 24 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1602] \ 1^{142} 31^1 5131^{43} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 6 & 2 \\ 1 & 1 & 1 & 16 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 92 & 24 \\ 6 & 11 & 0 & 92 & 65 & 17 \\ 2 & 3 & -1 & 24 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[1603] \ 1^{143} 31^1 5131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & 1 & 14 & 2 & 3 \\ 1 & 2 & 0 & 10 & 1 & 2 \\ 8 & 14 & 10 & 74 & 10 & 18 \\ 2 & 2 & 1 & 10 & 0 & 2 \\ 2 & 3 & 2 & 18 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1604] \ 1^{143} 31^1 5131^{44} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & 1 & 16 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 124 & 24 \\ 8 & 15 & 0 & 124 & 119 & 23 \\ 2 & 3 & -1 & 24 & 23 & 4 \end{pmatrix}_{b_0}$$

$$[1605] \ 1^{111} 31^1 513^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 4 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 11 & 6 \\ 3 & 5 & 0 & 11 & 14 & 8 \\ 2 & 3 & -1 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1606] \ 1^{141} 31^1 513^{1141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$$

6.0

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & 1 & 16 & 5 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 44 & 24 \\ 3 & 5 & 0 & 44 & 14 & 8 \\ 2 & 3 & -1 & 24 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1607] \ 1^{142} 31^1 513^1 1^{42} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 5 & 2 \\ 1 & 1 & 1 & 16 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 76 & 24 \\ 5 & 9 & 0 & 76 & 44 & 14 \\ 2 & 3 & -1 & 24 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1608] \ 1^{143} 31^1 513^1 1^{43} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 7 & 2 \\ 1 & 1 & 1 & 16 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 108 & 24 \\ 7 & 13 & 0 & 108 & 90 & 20 \\ 2 & 3 & -1 & 24 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[1609] \ 1^{123} 1^1 5131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 2 & 7 & 2 & 3 \\ 1 & 2 & 0 & 5 & 1 & 2 \\ 4 & 7 & 5 & 17 & 5 & 9 \\ 2 & 2 & 1 & 5 & 0 & 2 \\ 2 & 3 & 2 & 9 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1610] \ 1^{113} 31^1 15131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 1 & 4 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 31 & 6 \\ 8 & 15 & 0 & 31 & 119 & 23 \\ 2 & 3 & -1 & 6 & 23 & 4 \end{pmatrix}_{b_0}$$

$$[1611] \ 1^{144} 31^1 513^1 1^{44} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 9 & 2 \\ 1 & 1 & 1 & 16 & 17 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 140 & 24 \\ 9 & 17 & 0 & 140 & 152 & 26 \\ 2 & 3 & -1 & 24 & 26 & 4 \end{pmatrix}_{b_0}$$

$$[1576] \ 1^{123} 31^1 513^1 1^{23} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 1 & 8 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 54 & 12 \\ 7 & 13 & 0 & 54 & 90 & 20 \\ 2 & 3 & -1 & 12 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[1613] \ 1^{111} 31^1 15131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 4 & 7 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 15 & 6 \\ 4 & 7 & 0 & 15 & 27 & 11 \\ 2 & 3 & -1 & 6 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1614] \ 1^{122} 31^1 15131^{23} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 1 & 8 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 46 & 12 \\ 6 & 11 & 0 & 46 & 65 & 17 \\ 2 & 3 & -1 & 12 & 17 & 4 \end{pmatrix}_{b_0}$$

[1577] $1^{1123}1^{11}5131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 4 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 23 & 6 \\ 6 & 11 & 0 & 23 & 65 & 17 \\ 2 & 3 & -1 & 6 & 17 & 4 \end{pmatrix}_{b_0}$$

[1622] $1^{1143}1^{13}5131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 3 & 15 & 3 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 3 & 1 & 1 & 15 & 3 \\ 10 & 15 & 12 & 15 & 92 & 20 \\ 2 & 3 & 2 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

[1621] $1^{1131}1^{11}5131^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 5 & 3 & 2 \\ 4 & 0 & 5 & 17 & 11 & 9 \\ 2 & 0 & 3 & 11 & 5 & 5 \\ 2 & 0 & 2 & 9 & 5 & 4 \end{pmatrix}_{b_0}$$

[1620] $1^{111}3151^{123}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 2 & 1 & 9 & 5 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 9 & 0 & 27 & 13 & 11 \\ 2 & 5 & 0 & 13 & 5 & 5 \\ 2 & 4 & -1 & 11 & 5 & 4 \end{pmatrix}_{b_0}$$

[1619] $1^{1143}1^{13}5131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 3 & 17 & 3 \\ 1 & 2 & 0 & 1 & 14 & 2 \\ 1 & 3 & 1 & 1 & 17 & 3 \\ 11 & 17 & 14 & 17 & 121 & 23 \\ 2 & 3 & 2 & 3 & 23 & 4 \end{pmatrix}_{b_0}$$

[1618] $1^{131}315131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & 2 & 10 & 4 & 3 \\ 1 & 2 & 0 & 6 & 3 & 2 \\ 7 & 10 & 6 & 34 & 15 & 12 \\ 2 & 4 & 3 & 15 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

[1617] $1^{123}3^{11}5131^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 2 \\ 1 & 1 & 2 & 8 & 13 & 3 \\ 1 & 2 & 0 & 8 & 10 & 2 \\ 5 & 8 & 8 & 30 & 46 & 12 \\ 8 & 13 & 10 & 46 & 67 & 17 \\ 2 & 3 & 2 & 12 & 17 & 4 \end{pmatrix}_{b_0}$$

[1616] $1^{111}315131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 2 & 3 & 4 & 3 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 2 & 3 & 1 & 1 & 4 & 3 \\ 2 & 4 & 3 & 4 & 5 & 5 \\ 2 & 3 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

[1623] $1^{11}315131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 3 & 0 & 5 & 5 \\ 2 & 3 & 2 & 0 & 5 & 4 \end{pmatrix}_{b_0}$$

[1624] $1^{111}31^1513^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -9$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 3 & 2 \\ 2 & 4 & 4 & 6 & 7 & 6 \\ 3 & 4 & 3 & 7 & 5 & 5 \\ 2 & 3 & 2 & 6 & 5 & 4 \end{pmatrix}_{b_0}$$

[1625] $1^{13}3^{11}5131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -12$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 1 & 2 & 11 & 2 & 3 \\ 1 & 2 & 0 & 9 & 1 & 2 \\ 7 & 11 & 9 & 49 & 8 & 15 \\ 2 & 2 & 1 & 8 & 0 & 2 \\ 2 & 3 & 2 & 15 & 2 & 4 \end{pmatrix}_{b_0}$$

[1626] $1^{12}3^{11}5131^{41} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -14$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 10 & 3 & 2 \\ 8 & 0 & 10 & 74 & 22 & 18 \\ 2 & 0 & 3 & 22 & 5 & 5 \\ 2 & 0 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

[1627] $1^{111}31^1513^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -15$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 2 & 3 & 4 & 3 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 2 & 3 & 1 & 1 & 4 & 3 \\ 3 & 4 & 3 & 4 & 5 & 5 \\ 2 & 3 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

[1629] $1^{111}3^{11}5131^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -15$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 2 & 4 & 6 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 4 & 4 & 4 & 6 & 11 & 6 \\ 4 & 6 & 5 & 11 & 14 & 8 \\ 2 & 3 & 2 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[1628] $1^{111}3^{11}5131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -15$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 3 & 1 & 1 & 6 & 3 \\ 4 & 6 & 5 & 6 & 14 & 8 \\ 2 & 3 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[1632] $1^{111}3^{11}5131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -18$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 2 & 3 & 1 & 1 & 6 & 3 \\ 5 & 6 & 5 & 6 & 14 & 8 \\ 2 & 3 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[1631] $1^{14}3^{11}5131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -18$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 1 & 2 & 15 & 2 & 3 \\ 1 & 2 & 0 & 13 & 1 & 2 \\ 10 & 15 & 13 & 97 & 11 & 21 \\ 2 & 2 & 1 & 11 & 0 & 2 \\ 2 & 3 & 2 & 21 & 2 & 4 \end{pmatrix}_{b_0}$$

[1630] $1^{112}31^1513^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -18$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 2 & 3 & 8 & 3 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 2 & 3 & 1 & 1 & 8 & 3 \\ 6 & 8 & 7 & 8 & 27 & 11 \\ 2 & 3 & 2 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[1633] $1^{11}3^{11}1^15^{11}13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 6 & 3 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 5 & 0 & 14 & 8 \\ 2 & 3 & 2 & 0 & 8 & 4 \end{pmatrix}_{b_0}$$

[1636] $1^{11}3^{11}1^15^{11}13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 3 & 2 \\ 3 & 4 & 4 & 6 & 7 & 6 \\ 2 & 4 & 3 & 7 & 5 & 5 \\ 2 & 3 & 2 & 6 & 5 & 4 \end{pmatrix}_{b_0}$$

[1637] $1^{11}3^{11}1^15^{11}13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 9 & 3 & 3 \\ 5 & 0 & 9 & 27 & 8 & 11 \\ 2 & 0 & 3 & 8 & 1 & 3 \\ 2 & 0 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

[1640] $1^{13}3^{11}1^15^{11}13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 27 & 3 & 3 \\ 15 & 0 & 27 & 259 & 24 & 33 \\ 2 & 0 & 3 & 24 & 1 & 3 \\ 2 & 0 & 3 & 33 & 3 & 4 \end{pmatrix}_{b_0}$$

[1639] $1^{12}3^{11}1^15^{11}13^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 2 & 8 & 4 & 3 \\ 1 & 2 & 0 & 8 & 3 & 2 \\ 6 & 8 & 8 & 30 & 14 & 12 \\ 2 & 4 & 3 & 14 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

[1638] $1^{12}3^{11}1^15^{11}13^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 18 & 3 & 3 \\ 10 & 0 & 18 & 114 & 16 & 22 \\ 2 & 0 & 3 & 16 & 1 & 3 \\ 2 & 0 & 3 & 22 & 3 & 4 \end{pmatrix}_{b_0}$$

[1634] $1^{13}3^{11}1^15^{11}13^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 & 2 \\ 1 & 1 & 2 & 12 & 4 & 3 \\ 1 & 2 & 0 & 12 & 3 & 2 \\ 9 & 12 & 12 & 70 & 21 & 18 \\ 2 & 4 & 3 & 21 & 5 & 5 \\ 2 & 3 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

[1635] $1^{14}3^{11}1^15^{11}13^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 & 2 \\ 1 & 1 & 2 & 16 & 4 & 3 \\ 1 & 2 & 0 & 16 & 3 & 2 \\ 12 & 16 & 16 & 126 & 28 & 24 \\ 2 & 4 & 3 & 28 & 5 & 5 \\ 2 & 3 & 2 & 24 & 5 & 4 \end{pmatrix}_{b_0}$$

[1644] $1^{12}3^{11}1^15^{11}13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -25, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 7 & 2 \\ 2 & 4 & 4 & 6 & 15 & 6 \\ 6 & 8 & 7 & 15 & 27 & 11 \\ 2 & 3 & 2 & 6 & 11 & 4 \end{pmatrix}_{b_0}$$

[1641] $1^{121}3^1 5^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -25, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 1 & 2 & 8 & 4 & 3 \\ 1 & 2 & 0 & 8 & 3 & 2 \\ 5 & 8 & 8 & 30 & 14 & 12 \\ 3 & 4 & 3 & 14 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

[1642] $1^{111}3^1 5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -25, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 3 & 2 \\ 4 & 4 & 4 & 6 & 7 & 6 \\ 3 & 4 & 3 & 7 & 5 & 5 \\ 2 & 3 & 2 & 6 & 5 & 4 \end{pmatrix}_{b_0}$$

[1643] $1^{113}3^1 1^5 13^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -25, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 4 & 13 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 2 & 4 & 4 & 6 & 23 & 6 \\ 9 & 13 & 10 & 23 & 67 & 17 \\ 2 & 3 & 2 & 6 & 17 & 4 \end{pmatrix}_{b_0}$$

[1646] $1^{121}3^1 1^5 13^1 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & 2 & 8 & 6 & 3 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 7 & 8 & 8 & 30 & 22 & 12 \\ 4 & 6 & 5 & 22 & 14 & 8 \\ 2 & 3 & 2 & 12 & 8 & 4 \end{pmatrix}_{b_0}$$

[1645] $1^{131}3^1 1^5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & 2 & 10 & 6 & 3 \\ 1 & 2 & 0 & 6 & 5 & 2 \\ 6 & 10 & 6 & 34 & 23 & 12 \\ 4 & 6 & 5 & 23 & 14 & 8 \\ 2 & 3 & 2 & 12 & 8 & 4 \end{pmatrix}_{b_0}$$

[1648] $1^{123}1^1 5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 9 & 3 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 6 & 0 & 29 & 11 \\ 2 & 3 & 2 & 0 & 11 & 4 \end{pmatrix}_{b_0}$$

[1647] $1^{121}3^1 5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & 2 & 7 & 4 & 3 \\ 1 & 2 & 0 & 5 & 3 & 2 \\ 5 & 7 & 5 & 17 & 11 & 9 \\ 2 & 4 & 3 & 11 & 5 & 5 \\ 2 & 3 & 2 & 9 & 5 & 4 \end{pmatrix}_{b_0}$$

[1649] $1^{123}1^1 5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 8 & 3 \\ 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 8 & 7 & 0 & 27 & 11 \\ 2 & 3 & 2 & 0 & 11 & 4 \end{pmatrix}_{b_0}$$

[1650] $1^{111}3^1 5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 3 & 2 \\ 3 & 4 & 4 & 6 & 7 & 6 \\ 3 & 4 & 3 & 7 & 5 & 5 \\ 2 & 3 & 2 & 6 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1651] \ 1^{131} 31 \ 5 \ 13^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -39, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & 2 & 11 & 4 & 3 \\ 1 & 2 & 0 & 9 & 3 & 2 \\ 6 & 11 & 9 & 49 & 18 & 15 \\ 3 & 4 & 3 & 18 & 5 & 5 \\ 2 & 3 & 2 & 15 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1652] \ 1^{111} 3^1 1 \ 5 \ 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -39, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 4 & 6 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 2 & 4 & 4 & 6 & 11 & 6 \\ 4 & 6 & 5 & 11 & 14 & 8 \\ 2 & 3 & 2 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1653] \ 1^{113} 31 \ 5 \ 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -39, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 2 & 4 & 4 & 6 & 19 & 6 \\ 7 & 11 & 8 & 19 & 46 & 14 \\ 2 & 3 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1654] \ 1^{122} 31 \ 5 \ 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -43, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 2 & 7 & 8 & 3 \\ 1 & 2 & 0 & 5 & 7 & 2 \\ 4 & 7 & 5 & 17 & 23 & 9 \\ 6 & 8 & 7 & 23 & 27 & 11 \\ 2 & 3 & 2 & 9 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1655] \ 1^{121} 31 \ 5 \ 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -46, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & 2 & 7 & 4 & 3 \\ 1 & 2 & 0 & 5 & 3 & 2 \\ 4 & 7 & 5 & 17 & 11 & 9 \\ 3 & 4 & 3 & 11 & 5 & 5 \\ 2 & 3 & 2 & 9 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1656] \ 1^{112} 3^1 151^1 231^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 2 & 1 & 4 & 16 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 2 & 16 & 4 \\ 7 & 16 & 0 & 16 & 74 & 18 \\ 2 & 4 & -1 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1660] \ 1^{131} 31 \ 5 \ 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 11 & 8 & 0 & 46 & 14 \\ 2 & 3 & 2 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1659] \ 1^{13} 31 \ 5 \ 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 11 & 8 & 0 & 46 & 14 \\ 2 & 3 & 2 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1658] \ 1^{111} 3^1 151^1 231^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 4 & 12 & 3 \\ 2 & 0 & 4 & 2 & 12 & 4 \\ 7 & 0 & 12 & 12 & 46 & 14 \\ 2 & 0 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1657] \ 1^{121} 31^1 5 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & 2 & 7 & 4 & 3 \\ 1 & 2 & 0 & 5 & 3 & 2 \\ 6 & 7 & 5 & 17 & 11 & 9 \\ 3 & 4 & 3 & 11 & 5 & 5 \\ 2 & 3 & 2 & 9 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1661] \ 1^{123} 1^1 5 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -51, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 10 & 9 & 0 & 44 & 14 \\ 2 & 3 & 2 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1662] \ 2^1 3^{111} 1513^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 3 & 2 \\ 2 & 1 & 2 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 3 & 2 \\ 3 & 4 & 4 & 6 & 7 & 6 \\ 3 & 4 & 3 & 7 & 5 & 5 \\ 2 & 3 & 2 & 6 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1663] \ 2^1 3^{141} 1513^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 12 & 3 & 2 \\ 2 & 1 & 2 & 16 & 4 & 3 \\ 1 & 2 & 0 & 16 & 3 & 2 \\ 12 & 16 & 16 & 126 & 28 & 24 \\ 3 & 4 & 3 & 28 & 5 & 5 \\ 2 & 3 & 2 & 24 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1666] \ 1^{111} 3^1 1^1 5 131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & 2 & 4 & 6 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 4 & 4 & 6 & 11 & 6 \\ 5 & 6 & 5 & 11 & 14 & 8 \\ 2 & 3 & 2 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1665] \ 2^1 3^{131} 1513^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 3 & 2 \\ 2 & 1 & 2 & 12 & 4 & 3 \\ 1 & 2 & 0 & 12 & 3 & 2 \\ 9 & 12 & 12 & 70 & 21 & 18 \\ 3 & 4 & 3 & 21 & 5 & 5 \\ 2 & 3 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1664] \ 2^1 3^{121} 1513^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 3 & 2 \\ 2 & 1 & 2 & 8 & 4 & 3 \\ 1 & 2 & 0 & 8 & 3 & 2 \\ 6 & 8 & 8 & 30 & 14 & 12 \\ 3 & 4 & 3 & 14 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1668] \ 1^{131} 31^1 5 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -56, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 1 & 2 & 11 & 4 & 3 \\ 1 & 2 & 0 & 9 & 3 & 2 \\ 8 & 11 & 9 & 49 & 18 & 15 \\ 2 & 4 & 3 & 18 & 5 & 5 \\ 2 & 3 & 2 & 15 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1667] \ 1^{141} 31 5 13^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -56, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 1 & 2 & 14 & 4 & 3 \\ 1 & 2 & 0 & 10 & 3 & 2 \\ 10 & 14 & 10 & 74 & 22 & 18 \\ 2 & 4 & 3 & 22 & 5 & 5 \\ 2 & 3 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1671] \ 1^{121} 31 5 13^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -57, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 3 & 2 \\ 1 & 1 & 2 & 8 & 4 & 3 \\ 1 & 2 & 0 & 8 & 3 & 2 \\ 7 & 8 & 8 & 30 & 14 & 12 \\ 3 & 4 & 3 & 14 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1670] \ 1^{131} 31 5 13^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -57, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & 2 & 12 & 4 & 3 \\ 1 & 2 & 0 & 12 & 3 & 2 \\ 8 & 12 & 12 & 70 & 21 & 18 \\ 3 & 4 & 3 & 21 & 5 & 5 \\ 2 & 3 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1669] \ 1^{131} 31 5 13^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -57, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & 2 & 10 & 4 & 3 \\ 1 & 2 & 0 & 6 & 3 & 2 \\ 6 & 10 & 6 & 34 & 15 & 12 \\ 3 & 4 & 3 & 15 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1672] \ 1^{121} 31 5 13^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -60, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 1 & 2 & 8 & 4 & 3 \\ 1 & 2 & 0 & 8 & 3 & 2 \\ 6 & 8 & 8 & 30 & 14 & 12 \\ 3 & 4 & 3 & 14 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1674] \ 1^{113} 31 5 13^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -63, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 3 & 13 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 3 & 1 & 1 & 13 & 3 \\ 8 & 13 & 10 & 13 & 67 & 17 \\ 2 & 3 & 2 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[1673] \ 1^{113} 31 5 13^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -63, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 2 & 3 & 1 & 1 & 11 & 3 \\ 7 & 11 & 8 & 11 & 46 & 14 \\ 2 & 3 & 2 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1675] \ 1^{111} 31 5 13^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -66, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & 2 & 4 & 6 & 3 \\ 1 & 2 & 0 & 4 & 5 & 2 \\ 3 & 4 & 4 & 6 & 11 & 6 \\ 4 & 6 & 5 & 11 & 14 & 8 \\ 2 & 3 & 2 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1676] \ 1^{121} 31 5 13^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -67, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 1 & 2 & 7 & 4 & 3 \\ 1 & 2 & 0 & 5 & 3 & 2 \\ 5 & 7 & 5 & 17 & 11 & 9 \\ 3 & 4 & 3 & 11 & 5 & 5 \\ 2 & 3 & 2 & 9 & 5 & 4 \end{pmatrix}_{b_0}$$

[1677] $1^{13}3^1 5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 3 & 11 & 3 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 2 & 3 & 1 & 1 & 11 & 3 \\ 8 & 11 & 8 & 11 & 46 & 14 \\ 2 & 3 & 2 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[1678] $1^{14}3^1 5^{13}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & 2 & 3 & 17 & 3 \\ 1 & 2 & 0 & 1 & 14 & 2 \\ 2 & 3 & 1 & 1 & 17 & 3 \\ 11 & 17 & 14 & 17 & 121 & 23 \\ 2 & 3 & 2 & 3 & 23 & 4 \end{pmatrix}_{b_0}$$

[1679] $1^{13}3^1 5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 3 & 2 \\ 1 & 1 & 2 & 10 & 4 & 3 \\ 1 & 2 & 0 & 6 & 3 & 2 \\ 7 & 10 & 6 & 34 & 15 & 12 \\ 3 & 4 & 3 & 15 & 5 & 5 \\ 2 & 3 & 2 & 12 & 5 & 4 \end{pmatrix}_{b_0}$$

[1680] $1^{13}3^1 5^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -73, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 12 & 3 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 12 & 11 & 0 & 65 & 17 \\ 2 & 3 & 2 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

[1681] $1^{12}3^1 5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -75, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & 2 & 8 & 6 & 3 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 5 & 8 & 8 & 30 & 22 & 12 \\ 4 & 6 & 5 & 22 & 14 & 8 \\ 2 & 3 & 2 & 12 & 8 & 4 \end{pmatrix}_{b_0}$$

[1683] $1^{13}3^1 5^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 13 & 3 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 13 & 10 & 0 & 67 & 17 \\ 2 & 3 & 2 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

[1682] $1^{12}3^1 5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 2 & 7 & 6 & 3 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 4 & 7 & 5 & 17 & 17 & 9 \\ 4 & 6 & 5 & 17 & 14 & 8 \\ 2 & 3 & 2 & 9 & 8 & 4 \end{pmatrix}_{b_0}$$

[1684] $1^{12}3^1 5^{13}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -77, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & 2 & 7 & 6 & 3 \\ 1 & 2 & 0 & 5 & 5 & 2 \\ 5 & 7 & 5 & 17 & 17 & 9 \\ 4 & 6 & 5 & 17 & 14 & 8 \\ 2 & 3 & 2 & 9 & 8 & 4 \end{pmatrix}_{b_0}$$

[1685] $1^{12}3^1 5^{13}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -79, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 2 & 7 & 10 & 3 \\ 1 & 2 & 0 & 5 & 9 & 2 \\ 4 & 7 & 5 & 17 & 29 & 9 \\ 7 & 10 & 9 & 29 & 44 & 14 \\ 2 & 3 & 2 & 9 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1686] 1^{123} 31^1 5 13^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -79, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 8 & 2 \\ 4 & 7 & 5 & 17 & 30 & 9 \\ 8 & 11 & 8 & 30 & 46 & 14 \\ 2 & 3 & 2 & 9 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1687] 1^{112} 3^1 1^2 5 13^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -79, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 2 & 4 & 10 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 2 & 4 & 4 & 6 & 19 & 6 \\ 7 & 10 & 9 & 19 & 44 & 14 \\ 2 & 3 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1688] 1^{141} 31^1 5 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -80, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 & 2 \\ 1 & 1 & 2 & 15 & 4 & 3 \\ 1 & 2 & 0 & 13 & 3 & 2 \\ 11 & 15 & 13 & 97 & 25 & 21 \\ 2 & 4 & 3 & 25 & 5 & 5 \\ 2 & 3 & 2 & 21 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1689] 1^{114} 31^1 5 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 3 & 15 & 3 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 2 & 3 & 1 & 1 & 15 & 3 \\ 10 & 15 & 12 & 15 & 92 & 20 \\ 2 & 3 & 2 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[1690] 1^{113} 3^1 1^{13} 5 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 2 & 3 & 13 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 2 & 3 & 1 & 1 & 13 & 3 \\ 9 & 13 & 10 & 13 & 67 & 17 \\ 2 & 3 & 2 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[1691] 1^{114} 3^1 1^4 5 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -85, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & 2 & 4 & 17 & 3 \\ 1 & 2 & 0 & 4 & 14 & 2 \\ 2 & 4 & 4 & 6 & 31 & 6 \\ 12 & 17 & 14 & 31 & 121 & 23 \\ 2 & 3 & 2 & 6 & 23 & 4 \end{pmatrix}_{b_0}$$

$$[1692] 1^{121} 3^1 1^1 5 13^1 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -90, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & 2 & 8 & 6 & 3 \\ 1 & 2 & 0 & 8 & 5 & 2 \\ 6 & 8 & 8 & 30 & 22 & 12 \\ 4 & 6 & 5 & 22 & 14 & 8 \\ 2 & 3 & 2 & 12 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1693] 1^{122} 31^1 5 13^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 2 & 7 & 8 & 3 \\ 1 & 2 & 0 & 5 & 7 & 2 \\ 4 & 7 & 5 & 17 & 23 & 9 \\ 5 & 8 & 7 & 23 & 27 & 11 \\ 2 & 3 & 2 & 9 & 11 & 4 \end{pmatrix}_{b_0}$$

[1694] $1^{122}3^1 1^5 131^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 2 \\ 1 & 1 & 2 & 7 & 9 & 3 \\ 1 & 2 & 0 & 5 & 6 & 2 \\ 4 & 7 & 5 & 17 & 24 & 9 \\ 6 & 9 & 6 & 24 & 29 & 11 \\ 2 & 3 & 2 & 9 & 11 & 4 \end{pmatrix}_{b_0}$$

[1695] $1^{112}3^1 1^5 131^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 7 & 2 \\ 3 & 4 & 4 & 6 & 15 & 6 \\ 5 & 8 & 7 & 15 & 27 & 11 \\ 2 & 3 & 2 & 6 & 11 & 4 \end{pmatrix}_{b_0}$$

[1699] $1^{133}3^1 1^5 131^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -99, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 14 & 3 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 14 & 13 & 0 & 90 & 20 \\ 2 & 3 & 2 & 0 & 20 & 4 \end{pmatrix}_{b_0}$$

[1696] $1^{131}3^1 1^5 131^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -99, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & 2 & 11 & 6 & 3 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 6 & 11 & 9 & 49 & 28 & 15 \\ 4 & 6 & 5 & 28 & 14 & 8 \\ 2 & 3 & 2 & 15 & 8 & 4 \end{pmatrix}_{b_0}$$

[1697] $1^{113}3^1 1^5 131^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -99, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 4 & 4 & 6 & 19 & 6 \\ 8 & 11 & 8 & 19 & 46 & 14 \\ 2 & 3 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

[1698] $1^{112}3^1 1^5 131^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -99, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 4 & 11 & 3 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 4 & 4 & 6 & 19 & 6 \\ 7 & 11 & 8 & 19 & 46 & 14 \\ 2 & 3 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

[1701] $1^{141}3^1 1^5 131^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & 2 & 14 & 4 & 3 \\ 1 & 2 & 0 & 10 & 3 & 2 \\ 8 & 14 & 10 & 74 & 22 & 18 \\ 3 & 4 & 3 & 22 & 5 & 5 \\ 2 & 3 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

[1700] $1^{131}3^1 1^5 131^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 3 & 2 \\ 1 & 1 & 2 & 12 & 4 & 3 \\ 1 & 2 & 0 & 12 & 3 & 2 \\ 9 & 12 & 12 & 70 & 21 & 18 \\ 3 & 4 & 3 & 21 & 5 & 5 \\ 2 & 3 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0}$$

[1702] $1^{143}3^1 1^5 131^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -102, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 15 & 3 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 15 & 12 & 0 & 92 & 20 \\ 2 & 3 & 2 & 0 & 20 & 4 \end{pmatrix}_{b_0}$$

[1703] $1^{141}31^1 5 13^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 3 & 2 \\ 1 & 1 & 2 & 16 & 4 & 3 \\ 1 & 2 & 0 & 16 & 3 & 2 \\ 11 & 16 & 16 & 126 & 28 & 24 \\ 3 & 4 & 3 & 28 & 5 & 5 \\ 2 & 3 & 2 & 24 & 5 & 4 \end{pmatrix}_{b_0}$$

[1704] $1^{113}31^1 5 13^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 4 & 13 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 2 & 4 & 4 & 6 & 23 & 6 \\ 8 & 13 & 10 & 23 & 67 & 17 \\ 2 & 3 & 2 & 6 & 17 & 4 \end{pmatrix}_{b_0}$$

[1705] $1^{131}31^1 5 13^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -106, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 3 & 2 \\ 1 & 1 & 2 & 11 & 4 & 3 \\ 1 & 2 & 0 & 9 & 3 & 2 \\ 7 & 11 & 9 & 49 & 18 & 15 \\ 3 & 4 & 3 & 18 & 5 & 5 \\ 2 & 3 & 2 & 15 & 5 & 4 \end{pmatrix}_{b_0}$$

[1706] $1^{141}31^1 5 13^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -107, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 3 & 2 \\ 1 & 1 & 2 & 15 & 4 & 3 \\ 1 & 2 & 0 & 13 & 3 & 2 \\ 9 & 15 & 13 & 97 & 25 & 21 \\ 3 & 4 & 3 & 25 & 5 & 5 \\ 2 & 3 & 2 & 21 & 5 & 4 \end{pmatrix}_{b_0}$$

[1707] $1^{112}31^1 5 13^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -108, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 7 & 2 \\ 3 & 4 & 4 & 6 & 15 & 6 \\ 6 & 8 & 7 & 15 & 27 & 11 \\ 2 & 3 & 2 & 6 & 11 & 4 \end{pmatrix}_{b_0}$$

[1708] $1^{122}31^1 5 13^{12}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -123, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 1 & 2 & 8 & 10 & 3 \\ 1 & 2 & 0 & 8 & 9 & 2 \\ 5 & 8 & 8 & 30 & 38 & 12 \\ 7 & 10 & 9 & 38 & 44 & 14 \\ 2 & 3 & 2 & 12 & 14 & 4 \end{pmatrix}_{b_0}$$

[1709] $1^{131}31^1 1 5 13^{11}1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -127, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & 2 & 12 & 6 & 3 \\ 1 & 2 & 0 & 12 & 5 & 2 \\ 8 & 12 & 12 & 70 & 33 & 18 \\ 4 & 6 & 5 & 33 & 14 & 8 \\ 2 & 3 & 2 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

[1710] $1^{143}1^1 5 13^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 1 & 2 & 0 & 17 & 3 \\ 1 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 17 & 14 & 0 & 121 & 23 \\ 2 & 3 & 2 & 0 & 23 & 4 \end{pmatrix}_{b_0}$$

[1711] $1^{131}31^1 1 5 13^{11}1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -130, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & 2 & 12 & 6 & 3 \\ 1 & 2 & 0 & 12 & 5 & 2 \\ 9 & 12 & 12 & 70 & 33 & 18 \\ 4 & 6 & 5 & 33 & 14 & 8 \\ 2 & 3 & 2 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1712] \quad 1^{14} 1^3 1^{121} 5 131^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -130, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & 2 & 14 & 6 & 3 \\ 1 & 2 & 0 & 10 & 5 & 2 \\ 8 & 14 & 10 & 74 & 34 & 18 \\ 4 & 6 & 5 & 34 & 14 & 8 \\ 2 & 3 & 2 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[1713] \quad 1^{13} 1^3 1^5 13^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -135, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 1 & 2 & 11 & 4 & 3 \\ 1 & 2 & 0 & 9 & 3 & 2 \\ 8 & 11 & 9 & 49 & 18 & 15 \\ 3 & 4 & 3 & 18 & 5 & 5 \\ 2 & 3 & 2 & 15 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1714] \quad 1^{11} 1^3 1^5 131^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -135, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 2 & 4 & 4 & 6 & 27 & 6 \\ 10 & 14 & 13 & 27 & 90 & 20 \\ 2 & 3 & 2 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[1715] \quad 2^1 3^{112} 15 13^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -150, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 6 & 2 \\ 2 & 1 & 2 & 4 & 8 & 3 \\ 1 & 2 & 0 & 4 & 7 & 2 \\ 3 & 4 & 4 & 6 & 15 & 6 \\ 6 & 8 & 7 & 15 & 27 & 11 \\ 2 & 3 & 2 & 6 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1716] \quad 1^{11} 1^3 1^5 131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -150, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 10 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 4 & 4 & 6 & 19 & 6 \\ 8 & 10 & 9 & 19 & 44 & 14 \\ 2 & 3 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1717] \quad 2^1 3^{122} 15 13^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -150, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 6 & 2 \\ 2 & 1 & 2 & 7 & 8 & 3 \\ 1 & 2 & 0 & 5 & 7 & 2 \\ 5 & 7 & 5 & 17 & 23 & 9 \\ 6 & 8 & 7 & 23 & 27 & 11 \\ 2 & 3 & 2 & 9 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1718] \quad 1^{11} 1^3 1^5 131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -153, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 2 \\ 1 & 1 & 2 & 4 & 13 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 4 & 4 & 6 & 23 & 6 \\ 8 & 13 & 10 & 23 & 67 & 17 \\ 2 & 3 & 2 & 6 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[1719] \quad 1^{13} 1^3 1^5 131^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -155, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & 2 & 11 & 6 & 3 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 8 & 11 & 9 & 49 & 28 & 15 \\ 4 & 6 & 5 & 28 & 14 & 8 \\ 2 & 3 & 2 & 15 & 8 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[1720] \quad & 1^{141}31^1513^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -161, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 3 & 2 \\ 1 & 1 & 2 & 14 & 4 & 3 \\ 1 & 2 & 0 & 10 & 3 & 2 \\ 9 & 14 & 10 & 74 & 22 & 18 \\ 3 & 4 & 3 & 22 & 5 & 5 \\ 2 & 3 & 2 & 18 & 5 & 4 \end{pmatrix}_{b_0} \\
[1721] \quad & 1^{112}3^11^1513^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -162, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 4 & 10 & 3 \\ 1 & 2 & 0 & 4 & 9 & 2 \\ 3 & 4 & 4 & 6 & 19 & 6 \\ 7 & 10 & 9 & 19 & 44 & 14 \\ 2 & 3 & 2 & 6 & 14 & 4 \end{pmatrix}_{b_0} \\
[1722] \quad & 1^{131}3^11^1513^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -166, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 5.8 \\
& T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & 2 & 11 & 6 & 3 \\ 1 & 2 & 0 & 9 & 5 & 2 \\ 7 & 11 & 9 & 49 & 28 & 15 \\ 4 & 6 & 5 & 28 & 14 & 8 \\ 2 & 3 & 2 & 15 & 8 & 4 \end{pmatrix}_{b_0} \\
[1723] \quad & 1^{141}3^11^1513^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -171, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & 2 & 14 & 6 & 3 \\ 1 & 2 & 0 & 10 & 5 & 2 \\ 9 & 14 & 10 & 74 & 34 & 18 \\ 4 & 6 & 5 & 34 & 14 & 8 \\ 2 & 3 & 2 & 18 & 8 & 4 \end{pmatrix}_{b_0} \\
[1724] \quad & 1^{123}3^11^1513^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -171, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 5.5 \\
& T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 2 & 7 & 13 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 7 & 5 & 17 & 36 & 9 \\ 9 & 13 & 10 & 36 & 67 & 17 \\ 2 & 3 & 2 & 9 & 17 & 4 \end{pmatrix}_{b_0} \\
[1725] \quad & 1^{123}3^11^1513^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -172, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 5.5 \\
& T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 2 & 7 & 11 & 3 \\ 1 & 2 & 0 & 5 & 8 & 2 \\ 4 & 7 & 5 & 17 & 30 & 9 \\ 7 & 11 & 8 & 30 & 46 & 14 \\ 2 & 3 & 2 & 9 & 14 & 4 \end{pmatrix}_{b_0} \\
[1726] \quad & 1^{113}3^11^1513^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -177, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 4 & 13 & 3 \\ 1 & 2 & 0 & 4 & 10 & 2 \\ 3 & 4 & 4 & 6 & 23 & 6 \\ 9 & 13 & 10 & 23 & 67 & 17 \\ 2 & 3 & 2 & 6 & 17 & 4 \end{pmatrix}_{b_0} \\
[1727] \quad & 1^{141}31^1513^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -182, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 3 & 2 \\ 1 & 1 & 2 & 15 & 4 & 3 \\ 1 & 2 & 0 & 13 & 3 & 2 \\ 10 & 15 & 13 & 97 & 25 & 21 \\ 3 & 4 & 3 & 25 & 5 & 5 \\ 2 & 3 & 2 & 21 & 5 & 4 \end{pmatrix}_{b_0} \\
[1728] \quad & 1^{133}3^11^1513^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -183, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 5.8 \\
& T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 2 & 11 & 13 & 3 \\ 1 & 2 & 0 & 9 & 10 & 2 \\ 6 & 11 & 9 & 49 & 59 & 15 \\ 8 & 13 & 10 & 59 & 67 & 17 \\ 2 & 3 & 2 & 15 & 17 & 4 \end{pmatrix}_{b_0}$$

[1729] $1^{141}3^{11}5^{131}1^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -195, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$

$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 4 & 2 \\ 1 & 1 & 2 & 16 & 6 & 3 \\ 1 & 2 & 0 & 16 & 5 & 2 \\ 11 & 16 & 16 & 126 & 44 & 24 \\ 4 & 6 & 5 & 44 & 14 & 8 \\ 2 & 3 & 2 & 24 & 8 & 4 \end{pmatrix}_{b_0}$$

[1730] $1^{114}3^{11}5^{131}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -197, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$

$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & 2 & 4 & 17 & 3 \\ 1 & 2 & 0 & 4 & 14 & 2 \\ 2 & 4 & 4 & 6 & 31 & 6 \\ 11 & 17 & 14 & 31 & 121 & 23 \\ 2 & 3 & 2 & 6 & 23 & 4 \end{pmatrix}_{b_0}$$

[1731] $1^{114}3^{11}5^{131}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -207, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$

$T_{\min} = 12.00 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 13 & 2 \\ 1 & 1 & 2 & 4 & 18 & 3 \\ 1 & 2 & 0 & 4 & 17 & 2 \\ 2 & 4 & 4 & 6 & 35 & 6 \\ 13 & 18 & 17 & 35 & 152 & 26 \\ 2 & 3 & 2 & 6 & 26 & 4 \end{pmatrix}_{b_0}$$

[1732] $1^{141}3^{11}5^{131}1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -217, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$

$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & 2 & 15 & 6 & 3 \\ 1 & 2 & 0 & 13 & 5 & 2 \\ 9 & 15 & 13 & 97 & 39 & 21 \\ 4 & 6 & 5 & 39 & 14 & 8 \\ 2 & 3 & 2 & 21 & 8 & 4 \end{pmatrix}_{b_0}$$

[1733] $1^{123}3^{11}5^{131}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -238, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5$

$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 2 & 7 & 13 & 3 \\ 1 & 2 & 0 & 5 & 10 & 2 \\ 4 & 7 & 5 & 17 & 36 & 9 \\ 8 & 13 & 10 & 36 & 67 & 17 \\ 2 & 3 & 2 & 9 & 17 & 4 \end{pmatrix}_{b_0}$$

[1734] $1^{141}3^{11}5^{131}1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -272, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 6.0$

$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 4 & 2 \\ 1 & 1 & 2 & 15 & 6 & 3 \\ 1 & 2 & 0 & 13 & 5 & 2 \\ 10 & 15 & 13 & 97 & 39 & 21 \\ 4 & 6 & 5 & 39 & 14 & 8 \\ 2 & 3 & 2 & 21 & 8 & 4 \end{pmatrix}_{b_0}$$

[1735] $1^{124}3^{11}5^{131}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -294, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5$

$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 2 \\ 1 & 1 & 2 & 7 & 16 & 3 \\ 1 & 2 & 0 & 5 & 11 & 2 \\ 4 & 7 & 5 & 17 & 43 & 9 \\ 10 & 16 & 11 & 43 & 96 & 20 \\ 2 & 3 & 2 & 9 & 20 & 4 \end{pmatrix}_{b_0}$$

[1736] $1^{113}3^{11}5^{131}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -306, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.0$

$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 0 & 4 & 13 & 2 \\ 3 & 4 & 4 & 6 & 27 & 6 \\ 10 & 14 & 13 & 27 & 90 & 20 \\ 2 & 3 & 2 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[1737] $1^{124}3^{11}5^{131}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -325, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 5.5$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 2 \\ 1 & 1 & 2 & 7 & 17 & 3 \\ 1 & 2 & 0 & 5 & 14 & 2 \\ 4 & 7 & 5 & 17 & 48 & 9 \\ 11 & 17 & 14 & 48 & 121 & 23 \\ 2 & 3 & 2 & 9 & 23 & 4 \end{pmatrix}_{b_0}$$

$$[1738] \ 1^{1143^1} 1^5 131^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -333, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 11 & 2 \\ 1 & 1 & 2 & 4 & 17 & 3 \\ 1 & 2 & 0 & 4 & 14 & 2 \\ 3 & 4 & 4 & 6 & 31 & 6 \\ 11 & 17 & 14 & 31 & 121 & 23 \\ 2 & 3 & 2 & 6 & 23 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1739] \ 1^{113^1} 1^5 13221^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$$

$$[1740] \ 1^{113^1} 1^5 13221^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$$

$$[1741] \ 1^{13^1} 1^5 13221^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$$

$$[1742] \ 1^{213^1} 1^5 13221^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.5$$

$$[1743] \ 1^{123^1} 1^5 1321^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 2 & 2 & 14 & 5 & 4 \\ 1 & 2 & 0 & 9 & 3 & 2 \\ 6 & 14 & 9 & 58 & 21 & 18 \\ 3 & 5 & 3 & 21 & 6 & 6 \\ 2 & 4 & 2 & 18 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[1747] \ 1^{13^1} 1^5 1321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 2 & 2 & 4 & 5 & 4 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 2 & 4 & 1 & 2 & 5 & 4 \\ 3 & 5 & 3 & 5 & 6 & 6 \\ 2 & 4 & 2 & 4 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[1745] \ 1^{113^1} 1^5 1321^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 & 2 \\ 1 & 2 & 2 & 13 & 5 & 4 \\ 1 & 2 & 0 & 6 & 3 & 2 \\ 7 & 13 & 6 & 43 & 18 & 15 \\ 2 & 5 & 3 & 18 & 6 & 6 \\ 2 & 4 & 2 & 15 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[1746] \ 1^{13^1} 1^5 1321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 2 & 2 & 4 & 5 & 4 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 2 & 4 & 1 & 2 & 5 & 4 \\ 2 & 5 & 3 & 5 & 6 & 6 \\ 2 & 4 & 2 & 4 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[1744] \ 1^{113^1} 1^5 1321^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 \\ 1 & 2 & 2 & 13 & 5 & 4 \\ 1 & 2 & 0 & 6 & 3 & 2 \\ 6 & 13 & 6 & 43 & 18 & 15 \\ 3 & 5 & 3 & 18 & 6 & 6 \\ 2 & 4 & 2 & 15 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[1750] \ 1^{123^1} 1^5 1321^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 \\ 1 & 2 & 2 & 14 & 5 & 4 \\ 1 & 2 & 0 & 9 & 3 & 2 \\ 8 & 14 & 9 & 58 & 21 & 18 \\ 2 & 5 & 3 & 21 & 6 & 6 \\ 2 & 4 & 2 & 18 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[1749] \ 1^{113^1} 1^5 1321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 2 & 2 & 5 & 5 & 4 \\ 1 & 2 & 0 & 4 & 3 & 2 \\ 3 & 5 & 4 & 7 & 8 & 7 \\ 2 & 5 & 3 & 8 & 6 & 6 \\ 2 & 4 & 2 & 7 & 6 & 5 \end{pmatrix}_{b_0}$$

[1748] $1^{11}3^11^151321^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 2 & 2 & 5 & 5 & 4 \\ 1 & 2 & 0 & 4 & 3 & 2 \\ 3 & 5 & 4 & 7 & 8 & 7 \\ 2 & 5 & 3 & 8 & 6 & 6 \\ 2 & 4 & 2 & 7 & 6 & 5 \end{pmatrix}_{b_0}$$

[1758] $1^{11}3^11^151321^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 6 & 3 \\ 1 & 2 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 6 & 0 & 15 & 9 \\ 2 & 3 & 3 & 0 & 9 & 5 \end{pmatrix}_{b_0}$$

[1757] $1^{11}3^11^151321^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 2 & 4 & 6 & 3 \\ 1 & 2 & 1 & 5 & 6 & 3 \\ 4 & 4 & 5 & 7 & 12 & 7 \\ 4 & 6 & 6 & 12 & 15 & 9 \\ 2 & 3 & 3 & 7 & 9 & 5 \end{pmatrix}_{b_0}$$

[1756] $1^{121}3^11^151321^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 1 & 2 & 8 & 6 & 3 \\ 1 & 2 & 1 & 10 & 6 & 3 \\ 5 & 8 & 10 & 34 & 24 & 14 \\ 4 & 6 & 6 & 24 & 15 & 9 \\ 2 & 3 & 3 & 14 & 9 & 5 \end{pmatrix}_{b_0}$$

[1755] $1^{121}3^11^151321^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 4 & 2 \\ 1 & 1 & 2 & 8 & 6 & 3 \\ 1 & 2 & 1 & 10 & 6 & 3 \\ 7 & 8 & 10 & 34 & 24 & 14 \\ 4 & 6 & 6 & 24 & 15 & 9 \\ 2 & 3 & 3 & 14 & 9 & 5 \end{pmatrix}_{b_0}$$

[1754] $1^{11}3^11^151321^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 4 & 6 & 3 \\ 1 & 2 & 1 & 5 & 6 & 3 \\ 2 & 4 & 5 & 7 & 12 & 7 \\ 4 & 6 & 6 & 12 & 15 & 9 \\ 2 & 3 & 3 & 7 & 9 & 5 \end{pmatrix}_{b_0}$$

[1753] $1^{141}3^11^151321^{141} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 4 & 2 \\ 1 & 1 & 2 & 16 & 6 & 3 \\ 1 & 2 & 1 & 20 & 6 & 3 \\ 11 & 16 & 20 & 142 & 48 & 28 \\ 4 & 6 & 6 & 48 & 15 & 9 \\ 2 & 3 & 3 & 28 & 9 & 5 \end{pmatrix}_{b_0}$$

[1752] $1^{131}3^11^151321^{131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 1 & 2 & 12 & 6 & 3 \\ 1 & 2 & 1 & 15 & 6 & 3 \\ 8 & 12 & 15 & 79 & 36 & 21 \\ 4 & 6 & 6 & 36 & 15 & 9 \\ 2 & 3 & 3 & 21 & 9 & 5 \end{pmatrix}_{b_0}$$

[1751] $1^{11}3^{13}21^{11}6131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 3 & 2 \\ 3 & 4 & 0 & 3 & 5 & 5 \\ 2 & 3 & 0 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

[1762] $1^{12}3^1 1^1 5 1321^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 & 2 \\ 1 & 2 & 2 & 18 & 5 & 4 \\ 1 & 2 & 0 & 10 & 3 & 2 \\ 10 & 18 & 10 & 90 & 26 & 22 \\ 2 & 5 & 3 & 26 & 6 & 6 \\ 2 & 4 & 2 & 22 & 6 & 5 \end{pmatrix}_{b_0}$$

[1759] $1^{12}3^1 1^1 5 1321^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 2 & 2 & 9 & 5 & 4 \\ 1 & 2 & 0 & 5 & 3 & 2 \\ 5 & 9 & 5 & 21 & 13 & 11 \\ 2 & 5 & 3 & 13 & 6 & 6 \\ 2 & 4 & 2 & 11 & 6 & 5 \end{pmatrix}_{b_0}$$

[1761] $1^{12}3^1 1^1 5 1321^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 2 & 2 & 18 & 5 & 4 \\ 1 & 2 & 0 & 10 & 3 & 2 \\ 8 & 18 & 10 & 90 & 26 & 22 \\ 3 & 5 & 3 & 26 & 6 & 6 \\ 2 & 4 & 2 & 22 & 6 & 5 \end{pmatrix}_{b_0}$$

[1760] $1^{12}3^1 1^1 5 1321^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 \\ 1 & 2 & 2 & 9 & 5 & 4 \\ 1 & 2 & 0 & 5 & 3 & 2 \\ 4 & 9 & 5 & 21 & 13 & 11 \\ 3 & 5 & 3 & 13 & 6 & 6 \\ 2 & 4 & 2 & 11 & 6 & 5 \end{pmatrix}_{b_0}$$

[1763] $1^{11} 1^{11} 3^1 21 6 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 19, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 5 & 5 & 3 & 7 & 7 \\ 3 & 3 & 2 & 7 & 5 \end{pmatrix}_{b_0}$$

[1770] $1^{12}3^1 1^1 5 1321^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 2 & 2 & 4 & 14 & 4 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 2 & 4 & 1 & 2 & 14 & 4 \\ 7 & 14 & 8 & 14 & 55 & 17 \\ 2 & 4 & 2 & 4 & 17 & 5 \end{pmatrix}_{b_0}$$

[1769] $1^{12}3^1 1^1 5 1321^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 2 & 4 & 10 & 3 \\ 1 & 2 & 1 & 5 & 11 & 3 \\ 2 & 4 & 5 & 7 & 21 & 7 \\ 7 & 10 & 11 & 21 & 48 & 16 \\ 2 & 3 & 3 & 7 & 16 & 5 \end{pmatrix}_{b_0}$$

[1768] $1^{122}3^1 1^1 5 1321^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 2 & 7 & 10 & 3 \\ 1 & 2 & 1 & 6 & 11 & 3 \\ 4 & 7 & 6 & 18 & 31 & 10 \\ 7 & 10 & 11 & 31 & 48 & 16 \\ 2 & 3 & 3 & 10 & 16 & 5 \end{pmatrix}_{b_0}$$

[1767] $1^{111}3^1 1^1 5 1321^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 2 & 2 & 5 & 14 & 4 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 3 & 5 & 4 & 7 & 22 & 7 \\ 7 & 14 & 8 & 22 & 55 & 17 \\ 2 & 4 & 2 & 7 & 17 & 5 \end{pmatrix}_{b_0}$$

[1766] $1^{122}3^1 1^5 1321^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 2 \\ 1 & 1 & 2 & 8 & 10 & 3 \\ 1 & 2 & 1 & 10 & 11 & 3 \\ 5 & 8 & 10 & 34 & 42 & 14 \\ 7 & 10 & 11 & 42 & 48 & 16 \\ 2 & 3 & 3 & 14 & 16 & 5 \end{pmatrix}_{b_0}$$

[1765] $1^{123}3^1 1^5 1321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 10 & 11 & 0 & 48 & 16 \\ 2 & 3 & 3 & 0 & 16 & 5 \end{pmatrix}_{b_0}$$

[1764] $1^{111}3^1 1^5 1321^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 2 & 2 & 5 & 14 & 4 \\ 1 & 2 & 0 & 4 & 8 & 2 \\ 2 & 5 & 4 & 7 & 22 & 7 \\ 7 & 14 & 8 & 22 & 55 & 17 \\ 2 & 4 & 2 & 7 & 17 & 5 \end{pmatrix}_{b_0}$$

[1772] $1^{11} 1^1 3^{1211} 6^{131} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 6 & 2 \\ 7 & 9 & 0 & 6 & 34 & 12 \\ 2 & 3 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[1771] $1^{11} 1^1 3^{1211} 6^{131} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & 0 & 5 & 2 \\ 8 & 7 & 5 & 21 & 11 \\ 3 & 3 & 2 & 11 & 5 \end{pmatrix}_{b_0}$$

[1775] $1^{121}3^1 1^5 1321^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & 2 & 8 & 6 & 3 \\ 1 & 2 & 1 & 10 & 6 & 3 \\ 6 & 8 & 10 & 34 & 24 & 14 \\ 4 & 6 & 6 & 24 & 15 & 9 \\ 2 & 3 & 3 & 14 & 9 & 5 \end{pmatrix}_{b_0}$$

[1774] $1^{131}3^1 1^5 1321^{131} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 1 & 2 & 12 & 6 & 3 \\ 1 & 2 & 1 & 15 & 6 & 3 \\ 9 & 12 & 15 & 79 & 36 & 21 \\ 4 & 6 & 6 & 36 & 15 & 9 \\ 2 & 3 & 3 & 21 & 9 & 5 \end{pmatrix}_{b_0}$$

[1773] $1^{111}3^1 1^5 1321^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & 2 & 4 & 6 & 3 \\ 1 & 2 & 1 & 5 & 6 & 3 \\ 3 & 4 & 5 & 7 & 12 & 7 \\ 4 & 6 & 6 & 12 & 15 & 9 \\ 2 & 3 & 3 & 7 & 9 & 5 \end{pmatrix}_{b_0}$$

[1778] $1^{123}3^1 1^5 1321^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 2 \\ 1 & 1 & 2 & 8 & 14 & 3 \\ 1 & 2 & 1 & 10 & 16 & 3 \\ 5 & 8 & 10 & 34 & 60 & 14 \\ 10 & 14 & 16 & 60 & 99 & 23 \\ 2 & 3 & 3 & 14 & 23 & 5 \end{pmatrix}_{b_0}$$

[1776] $1^{13}3^1 1^5 1321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 14 & 3 \\ 1 & 2 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 14 & 16 & 0 & 99 & 23 \\ 2 & 3 & 3 & 0 & 23 & 5 \end{pmatrix}_{b_0}$$

[1777] $1^{13}3^1 1^5 1321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 1 & 5 & 16 & 3 \\ 2 & 4 & 5 & 7 & 30 & 7 \\ 10 & 14 & 16 & 30 & 99 & 23 \\ 2 & 3 & 3 & 7 & 23 & 5 \end{pmatrix}_{b_0}$$

[1780] $1^{11} 6^{1^{11}3^1 21^1} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 7 & 2 \\ 9 & 12 & 1 & 7 & 56 & 16 \\ 2 & 3 & 0 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1779] $1^{11} 6^{1^{12}3^1 21^1} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 11 & 3 \\ 1 & 1 & 0 & 6 & 2 \\ 11 & 11 & 6 & 42 & 16 \\ 3 & 3 & 2 & 16 & 5 \end{pmatrix}_{b_0}$$

[1781] $1^{14}3^1 1^5 1321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 81, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 13 & 2 \\ 1 & 1 & 2 & 4 & 18 & 3 \\ 1 & 2 & 1 & 5 & 21 & 3 \\ 2 & 4 & 5 & 7 & 39 & 7 \\ 13 & 18 & 21 & 39 & 168 & 30 \\ 2 & 3 & 3 & 7 & 30 & 5 \end{pmatrix}_{b_0}$$

[1782] $1^{11}3^1 1^5 1321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 100, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 \\ 1 & 2 & 2 & 9 & 14 & 4 \\ 1 & 2 & 0 & 5 & 8 & 2 \\ 4 & 9 & 5 & 21 & 36 & 11 \\ 7 & 14 & 8 & 36 & 55 & 17 \\ 2 & 4 & 2 & 11 & 17 & 5 \end{pmatrix}_{b_0}$$

[1783] $1^{12}3^1 1^5 1321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 100, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 4 & 10 & 3 \\ 1 & 2 & 1 & 5 & 11 & 3 \\ 3 & 4 & 5 & 7 & 21 & 7 \\ 7 & 10 & 11 & 21 & 48 & 16 \\ 2 & 3 & 3 & 7 & 16 & 5 \end{pmatrix}_{b_0}$$

[1784] $1^{13}3^1 1^5 1321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 196, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 2 \\ 1 & 1 & 2 & 4 & 14 & 3 \\ 1 & 2 & 1 & 5 & 16 & 3 \\ 3 & 4 & 5 & 7 & 30 & 7 \\ 10 & 14 & 16 & 30 & 99 & 23 \\ 2 & 3 & 3 & 7 & 23 & 5 \end{pmatrix}_{b_0}$$

$T = 13$ [1785] $1^{41}3^1 151321^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 8 & 4 & 3 \\ 2 & 2 & 11 & 4 & 4 \\ 8 & 11 & 35 & 14 & 15 \\ 4 & 4 & 14 & 4 & 6 \\ 3 & 4 & 15 & 6 & 6 \end{pmatrix}_{b_0} \\
[1786] & 1^{11}413^{11}151321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 4 & 4 & 3 \\ 2 & 2 & 5 & 4 & 4 \\ 4 & 5 & 7 & 6 & 7 \\ 4 & 4 & 6 & 4 & 6 \\ 3 & 4 & 7 & 6 & 6 \end{pmatrix}_{b_0} \\
[1787] & 1^{11}413^{11}151321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 2 & 3 & 2 \\ 1 & 1 & 0 & 3 & 3 & 2 \\ 2 & 2 & 3 & 3 & 6 & 5 \\ 4 & 3 & 3 & 6 & 7 & 7 \\ 2 & 2 & 2 & 5 & 7 & 5 \end{pmatrix}_{b_0} \\
[1788] & 1^{11}41^{13}151321^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 1 & 2 & 3 & 2 \\ 1 & 1 & 0 & 3 & 3 & 2 \\ 4 & 2 & 3 & 7 & 8 & 7 \\ 4 & 3 & 3 & 8 & 7 & 7 \\ 2 & 2 & 2 & 7 & 7 & 5 \end{pmatrix}_{b_0} \\
[1789] & 1^{121}41^{13}151321^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 1 & 6 & 3 & 2 \\ 1 & 1 & 0 & 9 & 3 & 2 \\ 8 & 6 & 9 & 47 & 20 & 17 \\ 4 & 3 & 3 & 20 & 7 & 7 \\ 2 & 2 & 2 & 17 & 7 & 5 \end{pmatrix}_{b_0} \\
[1790] & 1^{11}41^{13}151321^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 12 & 4 & 3 \\ 2 & 2 & 16 & 4 & 4 \\ 12 & 16 & 74 & 20 & 22 \\ 4 & 4 & 20 & 4 & 6 \\ 3 & 4 & 22 & 6 & 6 \end{pmatrix}_{b_0} \\
[1791] & 1^{111}41^{13}151321^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 1 & 4 & 3 & 2 \\ 1 & 1 & 0 & 6 & 3 & 2 \\ 6 & 4 & 6 & 22 & 14 & 12 \\ 4 & 3 & 3 & 14 & 7 & 7 \\ 2 & 2 & 2 & 12 & 7 & 5 \end{pmatrix}_{b_0} \\
[1792] & 1^{112}41^{13}151321^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -49, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 0 & 1 & 2 & 7 & 2 \\ 1 & 1 & 0 & 3 & 9 & 2 \\ 2 & 2 & 3 & 3 & 16 & 5 \\ 10 & 7 & 9 & 16 & 59 & 19 \\ 2 & 2 & 2 & 5 & 19 & 5 \end{pmatrix}_{b_0} \\
T = 15 & [1793] 1^{121}3161^{123}1513^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 12 & 3 & 3 \\ 2 & 2 & 13 & 4 & 4 \\ 12 & 13 & 58 & 15 & 20 \\ 3 & 4 & 15 & 3 & 5 \\ 3 & 4 & 20 & 5 & 6 \end{pmatrix}_{b_0} \\
[1794] & 1^{121}3161^{123}1513^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 9 & 3 & 3 \\ 2 & 2 & 11 & 4 & 4 \\ 9 & 11 & 40 & 12 & 16 \\ 3 & 4 & 12 & 3 & 5 \\ 3 & 4 & 16 & 5 & 6 \end{pmatrix}_{b_0} \\
[1795] & 1^{111}321^{16}131513^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0
\end{aligned}$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 7 & 3 & 3 \\ 2 & 2 & 9 & 4 & 4 \\ 7 & 9 & 27 & 10 & 13 \\ 3 & 4 & 10 & 3 & 5 \\ 3 & 4 & 13 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[1796] \ 1^{11} 1^{13} 1^{15} 1^{13} 6^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 2 & 1 & 5 & 4 \\ 1 & 1 & 0 & 3 & 2 \\ 4 & 5 & 3 & 7 & 7 \\ 3 & 4 & 2 & 7 & 6 \end{pmatrix}_{b_0}$$

$$[1797] \ 1^{11} 1^{12} 3161^{11} 231513^{11} 1^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -49, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 \\ 1 & 0 & 1 & 2 & 7 & 2 \\ 1 & 1 & 0 & 3 & 9 & 2 \\ 2 & 2 & 3 & 3 & 16 & 5 \\ 10 & 7 & 9 & 16 & 59 & 19 \\ 2 & 2 & 2 & 5 & 19 & 5 \end{pmatrix}_{b_0}$$

$$T = 17 \quad [1799] \ 1^{11} 1^{13} 1^{15} 1^{13} 8^{11} 12321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 5.0$$

$$[1800] \ 1^{11} 1^{13} 1^{15} 1^{13} 8^{11} 12321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 5.0$$

$$[1798] \ 1^{11} 1^{13} 1^{15} 1^{13} 8^{11} 12321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 5.0$$

4.939 NHC(-2)(-3) + su₂ + su2n3mix + f₄

$$T = 8 \quad [0] \ 1^{11} 1^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 7 & 0 & 4 \\ 3 & 5 & 4 & 6 & 0 & 4 \\ 4 & 7 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 5 & -3 & 1 \end{pmatrix}_{b_0}$$

4.940 NHC(-2)(-3) + su₂ + su2n3mix + f₄ + su₂

$$T = 9 \quad [0] \ 1^{11} 1^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 5 & 7 & 0 & 4 & 4 \\ 3 & 5 & 4 & 6 & 0 & 3 & 4 \\ 4 & 7 & 6 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 5 & 0 & 1 & 3 \\ 3 & 4 & 4 & 5 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 1^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 5 & 7 & 0 & 17 & 4 \\ 3 & 5 & 4 & 6 & 0 & 15 & 4 \\ 4 & 7 & 6 & 7 & 0 & 21 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 21 & 0 & 49 & 13 \\ 3 & 4 & 4 & 5 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 1^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 5 & 7 & 0 & 9 & 4 \\ 3 & 5 & 4 & 6 & 0 & 9 & 4 \\ 4 & 7 & 6 & 7 & 0 & 11 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 11 & 0 & 13 & 7 \\ 3 & 4 & 4 & 5 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{1123} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 5 & 7 & 0 & 13 & 4 \\ 3 & 5 & 4 & 6 & 0 & 12 & 4 \\ 4 & 7 & 6 & 7 & 0 & 16 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 12 & 16 & 0 & 28 & 10 \\ 3 & 4 & 4 & 5 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

4.941 NHC(-2)(-3) + $\mathfrak{su}_2$ + su2n3mix + $\mathfrak{f}_4$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 5 & 7 & 0 & 4 & 4 \\ 3 & 5 & 4 & 6 & 0 & 3 & 4 \\ 4 & 7 & 6 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 4 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

4.942 NHC(-2)(-3) + $\mathfrak{su}_2$ + su2n3mix + $\mathfrak{so}_8$

$$T = 8 \quad [0] 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 7 & 0 & 4 \\ 3 & 5 & 4 & 6 & 0 & 4 \\ 4 & 7 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[1] 1^{111} 3^{111} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 1 & 3 \\ 3 & 4 & 14 & 7 & 0 & 4 \\ 9 & 14 & 37 & 17 & 0 & 11 \\ 4 & 7 & 17 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[2] 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 4 & 7 & 0 & 4 \\ 3 & 4 & 1 & 5 & 0 & 3 \\ 4 & 7 & 5 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 7 & 0 & 4 \\ 2 & 4 & 1 & 5 & 0 & 3 \\ 5 & 7 & 5 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[4] 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 1 & 3 \\ 3 & 4 & 9 & 7 & 0 & 4 \\ 6 & 9 & 13 & 11 & 0 & 7 \\ 4 & 7 & 11 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[5] 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 12 & 0 & 4 \\ 2 & 4 & 1 & 8 & 0 & 3 \\ 8 & 12 & 8 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[6] 1^{111} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 1 & 3 \\ 3 & 4 & 5 & 12 & 0 & 4 \\ 3 & 5 & 4 & 12 & 0 & 4 \\ 8 & 12 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

[7] $1^{111} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 1 & 3 \\ 3 & 4 & 9 & 12 & 0 & 4 \\ 5 & 9 & 13 & 20 & 0 & 7 \\ 8 & 12 & 20 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{111} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 5 & 16 & 0 & 4 \\ 3 & 5 & 4 & 15 & 0 & 4 \\ 10 & 16 & 15 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 16 & 0 & 4 \\ 2 & 4 & 1 & 11 & 0 & 3 \\ 10 & 16 & 11 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{112} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 396, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 1 & 3 \\ 3 & 4 & 9 & 16 & 0 & 4 \\ 5 & 9 & 13 & 26 & 0 & 7 \\ 10 & 16 & 26 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

4.943 NHC(-2)(-3) + $\mathfrak{su}_2$ + su2n3mix + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$

$T = 9 \quad [0] \quad 1^{113} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 5 & 7 & 0 & 3 & 4 \\ 3 & 5 & 4 & 6 & 0 & 3 & 4 \\ 4 & 7 & 6 & 7 & 0 & 4 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 4 & 5 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

4.944 NHC(-2)(-3) + $\mathfrak{su}_2$ + su2n3mix + $\mathfrak{su}_2$

$T = 9 \quad [3] \quad 1^{111} 3^1 21^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 4 & 2 \\ 1 & 0 & 2 & 1 & 5 & 3 & 2 \\ 1 & 2 & 0 & 3 & 8 & 4 & 2 \\ 3 & 1 & 3 & 1 & 8 & 4 & 3 \\ 6 & 5 & 8 & 8 & 23 & 13 & 9 \\ 4 & 3 & 4 & 4 & 13 & 5 & 5 \\ 2 & 2 & 2 & 3 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[16] $1^{111} 3^1 21^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 & 2 \\ 1 & 0 & 1 & 1 & 5 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 1 & 0 & 1 & 8 & 4 & 3 \\ 4 & 5 & 0 & 8 & 23 & 13 & 9 \\ 2 & 3 & 0 & 4 & 13 & 5 & 5 \\ 2 & 2 & -1 & 3 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3^1 21^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 4 & 3 \\ 3 & 5 & 10 & 13 & 5 & 5 \\ 7 & 10 & 14 & 19 & 7 & 8 \\ 8 & 13 & 19 & 24 & 10 & 10 \\ 4 & 5 & 7 & 10 & 2 & 4 \\ 3 & 5 & 8 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 13 & 8 & 8 & 3 \\ 3 & 5 & 25 & 13 & 15 & 5 \\ 13 & 25 & 92 & 49 & 57 & 20 \\ 8 & 13 & 49 & 24 & 30 & 10 \\ 8 & 15 & 57 & 30 & 34 & 12 \\ 3 & 5 & 20 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{112} 3^{121} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 10 & 4 \\ 3 & 1 & 3 & 3 & 8 & 3 \\ 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 1 & 8 & 3 \\ 10 & 8 & 8 & 8 & 22 & 10 \\ 4 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{123} 3^{121} 21^{1122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 6 & 2 \\ 1 & 0 & 2 & 1 & 7 & 8 & 2 \\ 1 & 2 & 0 & 3 & 9 & 9 & 2 \\ 1 & 1 & 3 & 1 & 9 & 9 & 3 \\ 7 & 7 & 9 & 9 & 35 & 39 & 11 \\ 6 & 8 & 9 & 9 & 39 & 40 & 12 \\ 2 & 2 & 2 & 3 & 11 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{112} 3^{121} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 18 & 4 \\ 3 & 1 & 3 & 3 & 12 & 3 \\ 3 & 3 & 1 & 2 & 16 & 3 \\ 3 & 3 & 2 & 1 & 12 & 3 \\ 18 & 12 & 16 & 12 & 70 & 18 \\ 4 & 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{12} 3^{121} 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 16 & 3 \\ 3 & 5 & 0 & 13 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 48 & 10 \\ 16 & 25 & 0 & 48 & 94 & 20 \\ 3 & 5 & 0 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{132} 3^{121} 21^{313} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 6 & 2 \\ 1 & 0 & 1 & 9 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 9 & 0 & 61 & 18 & 54 & 15 \\ 2 & 2 & 0 & 18 & 4 & 16 & 4 \\ 6 & 7 & 0 & 54 & 16 & 45 & 13 \\ 2 & 2 & -1 & 15 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{111} 3^{121} 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 & 2 \\ 1 & 0 & 1 & 3 & 6 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 17 & 7 & 5 \\ 6 & 6 & 0 & 17 & 42 & 17 & 12 \\ 2 & 3 & 0 & 7 & 17 & 5 & 5 \\ 2 & 2 & -1 & 5 & 12 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11} 3^{121} 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 1 & 0 & 1 & 4 & 4 & 3 \\ 2 & 2 & 0 & 4 & 4 & 6 & 4 \\ 2 & 3 & 0 & 4 & 6 & 5 & 5 \\ 2 & 2 & -1 & 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{141} 3^{121} 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 20 & 7 & 4 \\ 3 & 1 & 3 & 16 & 5 & 3 \\ 3 & 3 & 1 & 16 & 6 & 3 \\ 20 & 16 & 16 & 94 & 32 & 20 \\ 7 & 5 & 6 & 32 & 9 & 7 \\ 4 & 3 & 3 & 20 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1211} 3^{121} 21^{1411} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 4 & 4 & 2 \\ 1 & 0 & 2 & 8 & 4 & 3 & 2 \\ 1 & 2 & 0 & 14 & 5 & 4 & 2 \\ 10 & 8 & 14 & 70 & 28 & 22 & 16 \\ 4 & 4 & 5 & 28 & 10 & 9 & 6 \\ 4 & 3 & 4 & 22 & 9 & 5 & 5 \\ 2 & 2 & 2 & 16 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[22] \quad & 1^{11} 1^{12} 1^1 3^1 21^{222} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 4 & 2 \\ 1 & 0 & 1 & 6 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 26 & 22 & 10 \\ 4 & 5 & 0 & 26 & 23 & 21 & 9 \\ 4 & 4 & 0 & 22 & 21 & 16 & 8 \\ 2 & 2 & -1 & 10 & 9 & 8 & 3 \end{pmatrix}_{b_0} \\
[37] \quad & 1^{21} 1^{11} 3^1 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 8 & 6 & 3 \\ 3 & 5 & 5 & 13 & 10 & 5 \\ 2 & 5 & 2 & 10 & 7 & 4 \\ 8 & 13 & 10 & 24 & 19 & 10 \\ 6 & 10 & 7 & 19 & 14 & 8 \\ 3 & 5 & 4 & 10 & 8 & 4 \end{pmatrix}_{b_0} \\
[23] \quad & 1^{21} 1^{11} 3^1 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 8 & 8 & 3 \\ 3 & 5 & 5 & 13 & 15 & 5 \\ 2 & 5 & 2 & 10 & 11 & 4 \\ 8 & 13 & 10 & 24 & 29 & 10 \\ 8 & 15 & 11 & 29 & 32 & 12 \\ 3 & 5 & 4 & 10 & 12 & 4 \end{pmatrix}_{b_0} \\
[24] \quad & 1^{222} 1^{11} 1^{11} 3^1 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 4 & 2 \\ 1 & 0 & 2 & 1 & 4 & 3 & 2 \\ 1 & 2 & 0 & 3 & 5 & 4 & 3 \\ 3 & 1 & 3 & 1 & 5 & 4 & 3 \\ 5 & 4 & 5 & 5 & 10 & 9 & 6 \\ 4 & 3 & 4 & 4 & 9 & 5 & 5 \\ 2 & 2 & 2 & 3 & 6 & 5 & 3 \end{pmatrix}_{b_0} \\
[25] \quad & 1^{21} 1^{111} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 & 2 \\ 1 & 0 & 2 & 2 & 4 & 2 & 2 \\ 1 & 2 & 0 & 1 & 5 & 1 & 2 \\ 1 & 2 & 1 & 0 & 4 & 2 & 2 \\ 5 & 4 & 5 & 4 & 10 & 4 & 6 \\ 2 & 2 & 1 & 2 & 4 & 0 & 2 \\ 2 & 2 & 2 & 2 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[26] \quad & 1^{11} 1^{12} 1^1 3^1 21^{212} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 4 & 2 \\ 1 & 0 & 1 & 6 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 12 & 22 & 10 \\ 2 & 2 & 0 & 12 & 4 & 10 & 4 \\ 4 & 4 & 0 & 22 & 10 & 16 & 8 \\ 2 & 2 & -1 & 10 & 4 & 8 & 3 \end{pmatrix}_{b_0} \\
[27] \quad & 1^{11} 1^{111} 3^1 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 2 & 1 & 4 & 3 & 2 \\ 1 & 2 & 0 & 3 & 5 & 4 & 2 \\ 1 & 1 & 3 & 1 & 5 & 4 & 3 \\ 4 & 4 & 5 & 5 & 10 & 9 & 6 \\ 3 & 3 & 4 & 4 & 9 & 5 & 5 \\ 2 & 2 & 2 & 3 & 6 & 5 & 3 \end{pmatrix}_{b_0} \\
[28] \quad & 1^{11} 1^{111} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 1 & 2 \\ 1 & 0 & 2 & 2 & 4 & 2 & 2 \\ 1 & 2 & 0 & 1 & 5 & 1 & 2 \\ 2 & 2 & 1 & 0 & 4 & 2 & 2 \\ 4 & 4 & 5 & 4 & 10 & 4 & 6 \\ 1 & 2 & 1 & 2 & 4 & 0 & 2 \\ 2 & 2 & 2 & 2 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[29] \quad & 1^{222} 1^{11} 1^1 3^1 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 5 & 2 \\ 1 & 0 & 2 & 1 & 5 & 4 & 2 \\ 1 & 2 & 0 & 3 & 8 & 7 & 2 \\ 3 & 1 & 3 & 1 & 8 & 7 & 3 \\ 6 & 5 & 8 & 8 & 23 & 21 & 9 \\ 5 & 4 & 7 & 7 & 21 & 16 & 8 \\ 2 & 2 & 2 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0} \\
[30] \quad & 1^{111} 1^{111} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 1 & 2 \\ 1 & 0 & 2 & 2 & 4 & 0 & 2 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & 1 & 0 & 4 & 0 & 2 \\ 4 & 4 & 5 & 4 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1^{13}} 3^1 21^{1^{11}2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 7 & 2 \\ 1 & 0 & 2 & 3 & 2 & 6 & 2 \\ 1 & 2 & 0 & 4 & 4 & 8 & 2 \\ 2 & 3 & 4 & 5 & 6 & 14 & 5 \\ 3 & 2 & 4 & 6 & 4 & 12 & 4 \\ 7 & 6 & 8 & 14 & 12 & 26 & 10 \\ 2 & 2 & 2 & 5 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1^{312}} 3^1 21^{1^{32}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 15 & 11 & 4 \\ 3 & 1 & 3 & 9 & 7 & 3 \\ 3 & 3 & 1 & 14 & 10 & 3 \\ 15 & 9 & 14 & 49 & 37 & 15 \\ 11 & 7 & 10 & 37 & 25 & 11 \\ 4 & 3 & 3 & 15 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1^{11}} 3^1 21^{1^{11}2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 5 & 2 \\ 1 & 0 & 2 & 3 & 2 & 6 & 2 \\ 1 & 2 & 0 & 4 & 4 & 8 & 2 \\ 2 & 3 & 4 & 5 & 6 & 14 & 5 \\ 3 & 2 & 4 & 6 & 4 & 12 & 4 \\ 5 & 6 & 8 & 14 & 12 & 26 & 10 \\ 2 & 2 & 2 & 5 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1^{11}} 3^1 21^{1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 7 & 4 \\ 3 & 1 & 3 & 3 & 5 & 3 \\ 3 & 3 & 1 & 2 & 6 & 3 \\ 3 & 3 & 2 & 1 & 5 & 3 \\ 7 & 5 & 6 & 5 & 9 & 7 \\ 4 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1^{33}} 3^1 21^{1^{12}2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 6 & 2 \\ 1 & 0 & 2 & 1 & 5 & 4 & 2 \\ 1 & 2 & 0 & 3 & 8 & 7 & 2 \\ 1 & 1 & 3 & 1 & 8 & 7 & 3 \\ 7 & 5 & 8 & 8 & 23 & 21 & 9 \\ 6 & 4 & 7 & 7 & 21 & 16 & 8 \\ 2 & 2 & 2 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1^{21}} 3^1 21^{1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 2 & 4 & 0 & 2 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 1 & 2 & 1 & 0 & 4 & 0 & 2 \\ 5 & 4 & 5 & 4 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1^{12}} 3^1 21^{1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 4 & 11 & 4 \\ 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 3 & 1 & 4 & 10 & 3 \\ 4 & 2 & 4 & 2 & 10 & 4 \\ 11 & 7 & 10 & 10 & 25 & 11 \\ 4 & 3 & 3 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1^{11}} 3^1 21^{1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 4 & 7 & 4 \\ 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 3 & 1 & 4 & 6 & 3 \\ 4 & 2 & 4 & 2 & 6 & 4 \\ 7 & 5 & 6 & 6 & 9 & 7 \\ 4 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1^{11}} 3^1 21^{1^{11}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 6 & 3 \\ 3 & 5 & 5 & 8 & 10 & 5 \\ 4 & 5 & 2 & 6 & 7 & 4 \\ 5 & 8 & 6 & 8 & 12 & 6 \\ 6 & 10 & 7 & 12 & 14 & 8 \\ 3 & 5 & 4 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[41] $1^{11} 1^{11} 3^{11} 21^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 5 & 3 \\ 3 & 5 & 5 & 8 & 10 & 5 \\ 2 & 5 & 2 & 6 & 7 & 4 \\ 5 & 8 & 6 & 8 & 12 & 6 \\ 5 & 10 & 7 & 12 & 14 & 8 \\ 3 & 5 & 4 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[42] $1^{123} 3^{11} 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 11 & 1 & 2 \\ 1 & 0 & 2 & 5 & 10 & 0 & 2 \\ 1 & 2 & 0 & 5 & 13 & 0 & 2 \\ 4 & 5 & 5 & 13 & 33 & 0 & 7 \\ 11 & 10 & 13 & 33 & 74 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 7 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

[43] $1^{11} 1^{11} 3^{11} 21^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 6 & 3 \\ 3 & 5 & 5 & 8 & 10 & 5 \\ 2 & 5 & 2 & 6 & 7 & 4 \\ 5 & 8 & 6 & 8 & 12 & 6 \\ 6 & 10 & 7 & 12 & 14 & 8 \\ 3 & 5 & 4 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[44] $1^{11} 1^{11} 3^{11} 21^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 & 3 \\ 3 & 5 & 10 & 8 & 5 & 5 \\ 5 & 10 & 14 & 12 & 7 & 8 \\ 5 & 8 & 12 & 8 & 6 & 6 \\ 3 & 5 & 7 & 6 & 2 & 4 \\ 3 & 5 & 8 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[45] $1^{112} 3^{11} 21^{1113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 1 & 4 & 11 & 2 \\ 1 & 2 & 0 & 3 & 5 & 13 & 2 \\ 1 & 1 & 3 & 1 & 5 & 13 & 3 \\ 4 & 4 & 5 & 5 & 10 & 31 & 6 \\ 9 & 11 & 13 & 13 & 31 & 81 & 17 \\ 2 & 2 & 2 & 3 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

[46] $1^{112} 3^{11} 21^{1113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 1 & 4 & 7 & 2 \\ 1 & 2 & 0 & 3 & 5 & 11 & 2 \\ 1 & 1 & 3 & 1 & 5 & 11 & 3 \\ 4 & 4 & 5 & 5 & 10 & 23 & 6 \\ 9 & 7 & 11 & 11 & 23 & 45 & 13 \\ 2 & 2 & 2 & 3 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[47] $1^{112} 3^{11} 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 17 & 4 \\ 3 & 1 & 3 & 3 & 13 & 3 \\ 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 1 & 13 & 3 \\ 17 & 13 & 14 & 13 & 65 & 17 \\ 4 & 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[48] $1^{114} 3^{11} 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 25 & 4 \\ 3 & 1 & 3 & 3 & 17 & 3 \\ 3 & 3 & 1 & 2 & 22 & 3 \\ 3 & 3 & 2 & 1 & 17 & 3 \\ 25 & 17 & 22 & 17 & 137 & 25 \\ 4 & 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

[50] $1^{112} 3^{11} 21^{1112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 1 & 4 & 6 & 2 \\ 1 & 2 & 0 & 3 & 5 & 8 & 2 \\ 1 & 1 & 3 & 1 & 5 & 8 & 3 \\ 4 & 4 & 5 & 5 & 10 & 18 & 6 \\ 6 & 6 & 8 & 8 & 18 & 26 & 10 \\ 2 & 2 & 2 & 3 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[49] $1^{1^{12}}3^{12}21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 14 & 4 \\ 3 & 1 & 3 & 3 & 10 & 3 \\ 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 3 & 2 & 1 & 10 & 3 \\ 14 & 10 & 12 & 10 & 42 & 14 \\ 4 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[57] $1^{1^{11}}3^{11}21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 3 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 6 & 7 \\ 2 & 2 & 0 & 6 & 4 & 6 \\ 2 & 3 & 0 & 7 & 6 & 5 \\ 2 & 2 & -1 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[51] $1^{1^{211}}3^{11}21^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 20 & 10 & 4 \\ 3 & 1 & 3 & 16 & 8 & 3 \\ 3 & 3 & 1 & 16 & 8 & 3 \\ 20 & 16 & 16 & 94 & 48 & 20 \\ 10 & 8 & 8 & 48 & 22 & 10 \\ 4 & 3 & 3 & 20 & 10 & 4 \end{pmatrix}_{b_0}$$

[52] $1^{1^{112}}3^{11}21^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 0 & 1 & 3 & 5 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 13 & 14 & 5 \\ 4 & 5 & 0 & 13 & 23 & 26 & 9 \\ 4 & 6 & 0 & 14 & 26 & 26 & 10 \\ 2 & 2 & -1 & 5 & 9 & 10 & 3 \end{pmatrix}_{b_0}$$

[53] $1^{1^{111}}3^{11}21^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 & 2 \\ 1 & 0 & 1 & 3 & 5 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 13 & 7 & 5 \\ 4 & 5 & 0 & 13 & 23 & 13 & 9 \\ 2 & 3 & 0 & 7 & 13 & 5 & 5 \\ 2 & 2 & -1 & 5 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[54] $1^{1^{32}}3^{11}21^{312} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 4 & 2 \\ 1 & 0 & 1 & 9 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 9 & 0 & 61 & 18 & 42 & 15 \\ 2 & 2 & 0 & 18 & 4 & 12 & 4 \\ 4 & 6 & 0 & 42 & 12 & 26 & 10 \\ 2 & 2 & -1 & 15 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[55] $1^{1^{31}}3^{11}21^{311} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 & 2 \\ 1 & 0 & 1 & 9 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 9 & 0 & 61 & 18 & 21 & 15 \\ 2 & 2 & 0 & 18 & 4 & 6 & 4 \\ 2 & 3 & 0 & 21 & 6 & 5 & 5 \\ 2 & 2 & -1 & 15 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[56] $1^{1^{22}}3^{11}21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 10 & 10 & 4 \\ 3 & 1 & 3 & 8 & 8 & 3 \\ 3 & 3 & 1 & 8 & 8 & 3 \\ 10 & 8 & 8 & 22 & 24 & 10 \\ 10 & 8 & 8 & 24 & 22 & 10 \\ 4 & 3 & 3 & 10 & 10 & 4 \end{pmatrix}_{b_0}$$

[64] $1^{1^{24}}3^{11}21^{214} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 12 & 56 & 10 \\ 2 & 2 & 0 & 12 & 4 & 24 & 4 \\ 8 & 12 & 0 & 56 & 24 & 110 & 20 \\ 2 & 2 & -1 & 10 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[63] $1^{1121}3^{121}1^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 & 2 \\ 1 & 0 & 1 & 3 & 8 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 20 & 7 & 5 \\ 6 & 8 & 0 & 20 & 56 & 20 & 14 \\ 2 & 3 & 0 & 7 & 20 & 5 & 5 \\ 2 & 2 & -1 & 5 & 14 & 5 & 3 \end{pmatrix}_{b_0}$

[62] $1^{112}3^{121}1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 3 & 3 & 4 & 14 & 4 \\ 3 & 1 & 3 & 2 & 10 & 3 \\ 3 & 3 & 1 & 4 & 12 & 3 \\ 4 & 2 & 4 & 2 & 12 & 4 \\ 14 & 10 & 12 & 12 & 42 & 14 \\ 4 & 3 & 3 & 4 & 14 & 4 \end{pmatrix}_{b_0}$

[61] $1^{114}3^{121}1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 3 & 3 & 4 & 22 & 4 \\ 3 & 1 & 3 & 2 & 14 & 3 \\ 3 & 3 & 1 & 4 & 20 & 3 \\ 4 & 2 & 4 & 2 & 20 & 4 \\ 22 & 14 & 20 & 20 & 106 & 22 \\ 4 & 3 & 3 & 4 & 22 & 4 \end{pmatrix}_{b_0}$

[60] $1^{114}3^{121}1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 0 & 1 & 3 & 2 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 6 & 28 & 5 \\ 2 & 2 & 0 & 6 & 4 & 24 & 4 \\ 8 & 12 & 0 & 28 & 24 & 110 & 20 \\ 2 & 2 & -1 & 5 & 4 & 20 & 3 \end{pmatrix}_{b_0}$

[59] $1^{122}3^{121}1^{212} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 4 & 2 \\ 1 & 0 & 1 & 6 & 2 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 12 & 28 & 10 \\ 2 & 2 & 0 & 12 & 4 & 12 & 4 \\ 4 & 6 & 0 & 28 & 12 & 26 & 10 \\ 2 & 2 & -1 & 10 & 4 & 10 & 3 \end{pmatrix}_{b_0}$

[58] $1^{121}3^{121}1^{211} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 & 2 \\ 1 & 0 & 1 & 6 & 2 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 12 & 14 & 10 \\ 2 & 2 & 0 & 12 & 4 & 6 & 4 \\ 2 & 3 & 0 & 14 & 6 & 5 & 5 \\ 2 & 2 & -1 & 10 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[70] $1^{121}3^{121}1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 13 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 4 & 5 & 2 & 6 & 14 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 13 & 20 & 14 & 24 & 62 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$

[73] $1^{121}3^{121}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 3 & 3 & 10 & 7 & 4 \\ 3 & 1 & 3 & 8 & 5 & 3 \\ 3 & 3 & 1 & 8 & 6 & 3 \\ 10 & 8 & 8 & 22 & 16 & 10 \\ 7 & 5 & 6 & 16 & 9 & 7 \\ 4 & 3 & 3 & 10 & 7 & 4 \end{pmatrix}_{b_0}$

[72] $1^{112}3^{121}1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 11 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 2 & 5 & 2 & 6 & 14 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 11 & 20 & 14 & 24 & 62 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$

[71] $1^{111}3^{121}1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

6.0

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 0 & 1 & 3 & 5 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 13 & 11 & 5 \\ 4 & 5 & 0 & 13 & 23 & 21 & 9 \\ 4 & 4 & 0 & 11 & 21 & 16 & 8 \\ 2 & 2 & -1 & 5 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{11} 3^{11} 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 & 3 \\ 3 & 5 & 10 & 13 & 5 & 5 \\ 5 & 10 & 14 & 19 & 7 & 8 \\ 7 & 13 & 19 & 24 & 10 & 10 \\ 3 & 5 & 7 & 10 & 2 & 4 \\ 3 & 5 & 8 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{11} 3^{12} 21^{1323} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 11 & 3 \\ 3 & 5 & 15 & 13 & 20 & 5 \\ 9 & 15 & 34 & 30 & 45 & 12 \\ 7 & 13 & 30 & 24 & 39 & 10 \\ 11 & 20 & 45 & 39 & 58 & 16 \\ 3 & 5 & 12 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[67] \ 1^{121} 3^{11} 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 2 & 1 & 7 & 3 & 2 \\ 1 & 2 & 0 & 3 & 9 & 4 & 2 \\ 1 & 1 & 3 & 1 & 9 & 4 & 3 \\ 7 & 7 & 9 & 9 & 35 & 16 & 11 \\ 3 & 3 & 4 & 4 & 16 & 5 & 5 \\ 2 & 2 & 2 & 3 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{11} 3^{12} 21^{1414} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 13 & 3 \\ 3 & 5 & 20 & 8 & 25 & 5 \\ 11 & 20 & 62 & 24 & 76 & 16 \\ 5 & 8 & 24 & 8 & 30 & 6 \\ 13 & 25 & 76 & 30 & 92 & 20 \\ 3 & 5 & 16 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{121} 3^{11} 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 & 2 \\ 1 & 0 & 2 & 3 & 5 & 1 & 2 \\ 1 & 2 & 0 & 4 & 8 & 3 & 2 \\ 2 & 3 & 4 & 5 & 13 & 4 & 5 \\ 6 & 5 & 8 & 13 & 23 & 8 & 9 \\ 2 & 1 & 3 & 4 & 8 & 1 & 3 \\ 2 & 2 & 2 & 5 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{123} 3^{11} 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 6 & 2 \\ 1 & 0 & 2 & 1 & 4 & 4 & 2 \\ 1 & 2 & 0 & 3 & 5 & 7 & 2 \\ 2 & 1 & 3 & 1 & 5 & 7 & 3 \\ 5 & 4 & 5 & 5 & 10 & 14 & 6 \\ 6 & 4 & 7 & 7 & 14 & 16 & 8 \\ 2 & 2 & 2 & 3 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{123} 3^{11} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 10 & 3 \\ 3 & 5 & 0 & 13 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 29 & 10 \\ 10 & 15 & 0 & 29 & 32 & 12 \\ 3 & 5 & 0 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{123} 3^{11} 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 8 & 2 \\ 1 & 0 & 2 & 1 & 7 & 8 & 2 \\ 1 & 2 & 0 & 3 & 9 & 9 & 2 \\ 2 & 1 & 3 & 1 & 9 & 9 & 3 \\ 7 & 7 & 9 & 9 & 35 & 39 & 11 \\ 8 & 8 & 9 & 9 & 39 & 40 & 12 \\ 2 & 2 & 2 & 3 & 11 & 12 & 3 \end{pmatrix}_{b_0}$$

[78] $1^{112} 1^{111} 3^1 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 5 & 2 \\ 1 & 0 & 2 & 1 & 4 & 4 & 2 \\ 1 & 2 & 0 & 3 & 5 & 7 & 2 \\ 2 & 1 & 3 & 1 & 5 & 7 & 3 \\ 4 & 4 & 5 & 5 & 10 & 14 & 6 \\ 5 & 4 & 7 & 7 & 14 & 16 & 8 \\ 2 & 2 & 2 & 3 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

[79] $1^{211} 1^{111} 3^1 21^{1313} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 10 & 3 \\ 3 & 5 & 15 & 8 & 20 & 5 \\ 8 & 15 & 34 & 18 & 45 & 12 \\ 5 & 8 & 18 & 8 & 24 & 6 \\ 10 & 20 & 45 & 24 & 58 & 16 \\ 3 & 5 & 12 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[77] $1^{312} 1^{111} 3^1 21^{1313} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 11 & 3 \\ 3 & 5 & 15 & 8 & 20 & 5 \\ 9 & 15 & 34 & 18 & 45 & 12 \\ 5 & 8 & 18 & 8 & 24 & 6 \\ 11 & 20 & 45 & 24 & 58 & 16 \\ 3 & 5 & 12 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[80] $1^{111} 1^{111} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 6 & 11 & 5 \\ 2 & 2 & 0 & 6 & 4 & 10 & 4 \\ 4 & 4 & 0 & 11 & 10 & 16 & 8 \\ 2 & 2 & -1 & 5 & 4 & 8 & 3 \end{pmatrix}_{b_0}$$

[87] $1^{112} 1^{111} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 7 & 11 & 4 \\ 3 & 1 & 3 & 5 & 7 & 3 \\ 3 & 3 & 1 & 6 & 10 & 3 \\ 7 & 5 & 6 & 9 & 17 & 7 \\ 11 & 7 & 10 & 17 & 25 & 11 \\ 4 & 3 & 3 & 7 & 11 & 4 \end{pmatrix}_{b_0}$$

[81] $1^{12} 1^{112} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 8 & 2 \\ 1 & 0 & 1 & 3 & 5 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 13 & 22 & 5 \\ 4 & 5 & 0 & 13 & 23 & 42 & 9 \\ 8 & 8 & 0 & 22 & 42 & 70 & 16 \\ 2 & 2 & -1 & 5 & 9 & 16 & 3 \end{pmatrix}_{b_0}$$

[86] $1^{13} 1^{113} 3^1 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 21 & 4 \\ 3 & 1 & 3 & 3 & 15 & 3 \\ 3 & 3 & 1 & 2 & 18 & 3 \\ 3 & 3 & 2 & 1 & 15 & 3 \\ 21 & 15 & 18 & 15 & 97 & 21 \\ 4 & 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[85] $1^{22} 1^{122} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 6 & 2 \\ 1 & 0 & 2 & 1 & 7 & 6 & 2 \\ 1 & 2 & 0 & 3 & 9 & 8 & 2 \\ 1 & 1 & 3 & 1 & 9 & 8 & 3 \\ 7 & 7 & 9 & 9 & 35 & 32 & 11 \\ 6 & 6 & 8 & 8 & 32 & 26 & 10 \\ 2 & 2 & 2 & 3 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

[88] $1^{1^{13}} 3^{1^{113}} 21^{1^{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 1 & 4 & 9 & 2 \\ 1 & 2 & 0 & 3 & 5 & 12 & 2 \\ 1 & 1 & 3 & 1 & 5 & 12 & 3 \\ 4 & 4 & 5 & 5 & 10 & 27 & 6 \\ 9 & 9 & 12 & 12 & 27 & 61 & 15 \\ 2 & 2 & 2 & 3 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

[89] $1^{1^{41}} 3^{1^{1411}} 21^{1^{411}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 3 & 3 & 2 \\ 1 & 0 & 2 & 12 & 2 & 3 & 2 \\ 1 & 2 & 0 & 16 & 4 & 4 & 2 \\ 10 & 12 & 16 & 110 & 24 & 28 & 20 \\ 3 & 2 & 4 & 24 & 4 & 6 & 4 \\ 3 & 3 & 4 & 28 & 6 & 5 & 5 \\ 2 & 2 & 2 & 20 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[90] $1^{1^{214}} 3^{1^{112}} 21^{1^{212}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 8 & 2 \\ 1 & 0 & 2 & 4 & 4 & 6 & 2 \\ 1 & 2 & 0 & 7 & 5 & 8 & 2 \\ 5 & 4 & 7 & 16 & 14 & 22 & 8 \\ 4 & 4 & 5 & 14 & 10 & 18 & 6 \\ 8 & 6 & 8 & 22 & 18 & 26 & 10 \\ 2 & 2 & 2 & 8 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[84] $1^{1^{13}} 3^{1^{113}} 21^{1^{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 4 & 18 & 4 \\ 3 & 1 & 3 & 2 & 12 & 3 \\ 3 & 3 & 1 & 4 & 16 & 3 \\ 4 & 2 & 4 & 2 & 16 & 4 \\ 18 & 12 & 16 & 16 & 70 & 18 \\ 4 & 3 & 3 & 4 & 18 & 4 \end{pmatrix}_{b_0}$$

[83] $1^{1^{11}} 3^{1^{213}} 21^{1^{213}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 6 & 2 \\ 1 & 0 & 1 & 6 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 12 & 36 & 10 \\ 2 & 2 & 0 & 12 & 4 & 16 & 4 \\ 6 & 7 & 0 & 36 & 16 & 45 & 13 \\ 2 & 2 & -1 & 10 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[82] $1^{1^{112}} 3^{1^{21}} 21^{1^{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 10 & 14 & 4 \\ 3 & 1 & 3 & 8 & 10 & 3 \\ 3 & 3 & 1 & 8 & 12 & 3 \\ 10 & 8 & 8 & 22 & 32 & 10 \\ 14 & 10 & 12 & 32 & 42 & 14 \\ 4 & 3 & 3 & 10 & 14 & 4 \end{pmatrix}_{b_0}$$

[91] $1^{1^{13}} 3^{1^{113}} 21^{1^{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 4 & 21 & 4 \\ 3 & 1 & 3 & 2 & 15 & 3 \\ 3 & 3 & 1 & 4 & 18 & 3 \\ 4 & 2 & 4 & 2 & 18 & 4 \\ 21 & 15 & 18 & 18 & 97 & 21 \\ 4 & 3 & 3 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

[92] $1^{1^{14}} 3^{1^{13}} 21^{1^{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 24 & 4 \\ 3 & 1 & 3 & 3 & 18 & 3 \\ 3 & 3 & 1 & 2 & 20 & 3 \\ 3 & 3 & 2 & 1 & 18 & 3 \\ 24 & 18 & 20 & 18 & 130 & 24 \\ 4 & 3 & 3 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

[96] $1^{1^{23}} 3^{1^{1}} 21^{1^{24}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 14 & 3 \\ 3 & 5 & 0 & 13 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 49 & 10 \\ 14 & 25 & 0 & 49 & 92 & 20 \\ 3 & 5 & 0 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

[95] $1^{121} 1^{11} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 6 & 3 \\ 3 & 5 & 0 & 13 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 19 & 10 \\ 6 & 10 & 0 & 19 & 14 & 8 \\ 3 & 5 & 0 & 10 & 8 & 4 \end{pmatrix}_{b_0}$

[105] $1^{123} 1^{11} 3^1 21^{1223} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 8 & 12 & 3 \\ 3 & 5 & 10 & 13 & 20 & 5 \\ 5 & 10 & 14 & 20 & 30 & 8 \\ 8 & 13 & 20 & 24 & 39 & 10 \\ 12 & 20 & 30 & 39 & 58 & 16 \\ 3 & 5 & 8 & 10 & 16 & 4 \end{pmatrix}_{b_0}$

[104] $1^{121} 1^{11} 3^1 21^{1222} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 8 & 8 & 3 \\ 3 & 5 & 10 & 13 & 15 & 5 \\ 5 & 10 & 14 & 20 & 22 & 8 \\ 8 & 13 & 20 & 24 & 29 & 10 \\ 8 & 15 & 22 & 29 & 32 & 12 \\ 3 & 5 & 8 & 10 & 12 & 4 \end{pmatrix}_{b_0}$

[103] $1^{123} 1^{113} 3^1 21^{1112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 8 & 2 \\ 1 & 0 & 2 & 1 & 4 & 8 & 2 \\ 1 & 2 & 0 & 3 & 5 & 9 & 2 \\ 2 & 1 & 3 & 1 & 5 & 9 & 3 \\ 5 & 4 & 5 & 5 & 10 & 22 & 6 \\ 8 & 8 & 9 & 9 & 22 & 40 & 12 \\ 2 & 2 & 2 & 3 & 6 & 12 & 3 \end{pmatrix}_{b_0}$

[102] $1^{121} 1^{11} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 1 & 2 \\ 1 & 0 & 2 & 1 & 4 & 0 & 2 \\ 1 & 2 & 0 & 3 & 5 & 0 & 2 \\ 2 & 1 & 3 & 1 & 5 & 0 & 3 \\ 5 & 4 & 5 & 5 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 6 & 0 & 3 \end{pmatrix}_{b_0}$

[101] $1^{121} 1^{1211} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 & 2 \\ 1 & 0 & 2 & 5 & 4 & 2 & 2 \\ 1 & 2 & 0 & 5 & 5 & 1 & 2 \\ 4 & 5 & 5 & 13 & 13 & 5 & 7 \\ 5 & 4 & 5 & 13 & 10 & 4 & 6 \\ 2 & 2 & 1 & 5 & 4 & 0 & 2 \\ 2 & 2 & 2 & 7 & 6 & 2 & 3 \end{pmatrix}_{b_0}$

[100] $1^{114} 1^{113} 3^1 21^{1112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 9 & 2 \\ 1 & 0 & 2 & 1 & 4 & 8 & 2 \\ 1 & 2 & 0 & 3 & 5 & 9 & 2 \\ 2 & 1 & 3 & 1 & 5 & 9 & 3 \\ 4 & 4 & 5 & 5 & 10 & 22 & 6 \\ 9 & 8 & 9 & 9 & 22 & 40 & 12 \\ 2 & 2 & 2 & 3 & 6 & 12 & 3 \end{pmatrix}_{b_0}$

[99] $1^{111} 1^{11} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 1 & 2 \\ 1 & 0 & 2 & 1 & 4 & 0 & 2 \\ 1 & 2 & 0 & 3 & 5 & 0 & 2 \\ 2 & 1 & 3 & 1 & 5 & 0 & 3 \\ 4 & 4 & 5 & 5 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 6 & 0 & 3 \end{pmatrix}_{b_0}$

[98] $1^{31} 1^{1211} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 1 & 2 \\ 1 & 0 & 2 & 5 & 4 & 2 & 2 \\ 1 & 2 & 0 & 5 & 5 & 1 & 2 \\ 6 & 5 & 5 & 13 & 13 & 5 & 7 \\ 4 & 4 & 5 & 13 & 10 & 4 & 6 \\ 1 & 2 & 1 & 5 & 4 & 0 & 2 \\ 2 & 2 & 2 & 7 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[94] $1^{11} 3^1 21^{1231} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 & 3 \\ 3 & 5 & 10 & 23 & 5 & 5 \\ 5 & 10 & 14 & 34 & 8 & 8 \\ 13 & 23 & 34 & 78 & 17 & 18 \\ 3 & 5 & 8 & 17 & 2 & 4 \\ 3 & 5 & 8 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[93] $1^{11} 3^1 21^{1313} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 11 & 3 \\ 3 & 5 & 15 & 8 & 20 & 5 \\ 8 & 15 & 34 & 18 & 45 & 12 \\ 5 & 8 & 18 & 8 & 24 & 6 \\ 11 & 20 & 45 & 24 & 58 & 16 \\ 3 & 5 & 12 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[97] $1^{1211} 3^1 21^{1411} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 4 & 2 & 2 \\ 1 & 0 & 2 & 8 & 4 & 3 & 2 \\ 1 & 2 & 0 & 14 & 5 & 4 & 2 \\ 9 & 8 & 14 & 70 & 28 & 22 & 16 \\ 4 & 4 & 5 & 28 & 10 & 9 & 6 \\ 2 & 3 & 4 & 22 & 9 & 5 & 5 \\ 2 & 2 & 2 & 16 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[109] $1^{111} 3^1 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 11 & 11 & 4 \\ 3 & 1 & 3 & 7 & 7 & 3 \\ 3 & 3 & 1 & 10 & 10 & 3 \\ 11 & 7 & 10 & 25 & 27 & 11 \\ 11 & 7 & 10 & 27 & 25 & 11 \\ 4 & 3 & 3 & 11 & 11 & 4 \end{pmatrix}_{b_0}$$

[108] $1^{111} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 7 & 7 & 4 \\ 3 & 1 & 3 & 5 & 5 & 3 \\ 3 & 3 & 1 & 6 & 6 & 3 \\ 7 & 5 & 6 & 9 & 11 & 7 \\ 7 & 5 & 6 & 11 & 9 & 7 \\ 4 & 3 & 3 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

[106] $1^{121} 3^1 21^{1131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 3 & 2 \\ 1 & 0 & 2 & 1 & 8 & 3 & 2 \\ 1 & 2 & 0 & 3 & 12 & 4 & 2 \\ 1 & 1 & 3 & 1 & 12 & 4 & 3 \\ 10 & 8 & 12 & 12 & 56 & 20 & 14 \\ 3 & 3 & 4 & 4 & 20 & 5 & 5 \\ 2 & 2 & 2 & 3 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[107] $1^{11} 3^1 21^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 & 3 \\ 3 & 5 & 8 & 5 & 5 & 5 \\ 2 & 5 & 2 & 6 & 4 & 4 \\ 5 & 8 & 6 & 8 & 6 & 6 \\ 3 & 5 & 4 & 6 & 2 & 4 \\ 3 & 5 & 4 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[110] $1^{112} 3^1 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 11 & 14 & 4 \\ 3 & 1 & 3 & 7 & 10 & 3 \\ 3 & 3 & 1 & 10 & 12 & 3 \\ 11 & 7 & 10 & 25 & 34 & 11 \\ 14 & 10 & 12 & 34 & 42 & 14 \\ 4 & 3 & 3 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[111] & 1^{\overline{111}} 3^{\overline{1}} 21^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 3 \\ 3 & 5 & 0 & 8 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 6 & 6 \\ 3 & 5 & 0 & 6 & 2 & 4 \\ 3 & 5 & 0 & 6 & 4 & 4 \end{pmatrix}_{b_0} \\
[112] & 1^{\overline{114}} 3^{\overline{1}} 21^{\overline{114}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 3 & 3 & 7 & 22 & 4 \\ 3 & 1 & 3 & 5 & 14 & 3 \\ 3 & 3 & 1 & 6 & 20 & 3 \\ 7 & 5 & 6 & 9 & 34 & 7 \\ 22 & 14 & 20 & 34 & 106 & 22 \\ 4 & 3 & 3 & 7 & 22 & 4 \end{pmatrix}_{b_0} \\
[117] & 1^{\overline{111}} 3^{\overline{1}} 21^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 & 2 \\ 1 & 0 & 2 & 1 & 4 & 3 & 2 \\ 1 & 2 & 0 & 3 & 5 & 4 & 2 \\ 2 & 1 & 3 & 1 & 5 & 4 & 3 \\ 4 & 4 & 5 & 5 & 10 & 9 & 6 \\ 2 & 3 & 4 & 4 & 9 & 5 & 5 \\ 2 & 2 & 2 & 3 & 6 & 5 & 3 \end{pmatrix}_{b_0} \\
[120] & 1^{\overline{113}} 3^{\overline{1}} 21^{\overline{113}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 5 & 15 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 2 & 5 & 2 & 6 & 18 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 15 & 25 & 18 & 30 & 94 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0} \\
[123] & 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{113}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 5 & 13 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 2 & 5 & 2 & 6 & 18 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 13 & 25 & 18 & 30 & 94 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0} \\
[116] & 1^{\overline{131}} 3^{\overline{1}} 21^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 1 & 12 & 0 & 2 \\ 1 & 2 & 1 & 0 & 8 & 0 & 2 \\ 9 & 8 & 12 & 8 & 56 & 0 & 14 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 14 & 0 & 3 \end{pmatrix}_{b_0} \\
[124] & 1^{\overline{123}} 3^{\overline{1}} 21^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 5 & 2 \\ 1 & 0 & 2 & 1 & 4 & 3 & 2 \\ 1 & 2 & 0 & 3 & 5 & 4 & 2 \\ 2 & 1 & 3 & 1 & 5 & 4 & 3 \\ 5 & 4 & 5 & 5 & 10 & 9 & 6 \\ 5 & 3 & 4 & 4 & 9 & 5 & 5 \\ 2 & 2 & 2 & 3 & 6 & 5 & 3 \end{pmatrix}_{b_0} \\
[122] & 1^{\overline{111}} 3^{\overline{1}} 21^{\overline{1213}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 5 & 10 & 3 \\ 3 & 5 & 10 & 8 & 20 & 5 \\ 5 & 10 & 14 & 12 & 30 & 8 \\ 5 & 8 & 12 & 8 & 24 & 6 \\ 10 & 20 & 30 & 24 & 58 & 16 \\ 3 & 5 & 8 & 6 & 16 & 4 \end{pmatrix}_{b_0} \\
[121] & 1^{\overline{112}} 3^{\overline{1}} 21^{\overline{114}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 0 & 1 & 3 & 2 & 8 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 6 & 22 & 5 \\ 2 & 2 & 0 & 6 & 4 & 20 & 4 \\ 8 & 8 & 0 & 22 & 20 & 70 & 16 \\ 2 & 2 & -1 & 5 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

[115] $1^{111} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 3 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 4 & 4 & 2 \\ 5 & 3 & 4 & 5 & 6 & 7 & 5 \\ 3 & 2 & 4 & 6 & 4 & 6 & 4 \\ 4 & 3 & 4 & 7 & 6 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[114] $1^{123} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 15 & 3 \\ 3 & 5 & 0 & 13 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 48 & 10 \\ 15 & 25 & 0 & 48 & 94 & 20 \\ 3 & 5 & 0 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

[113] $1^{121} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 1 & 2 \\ 1 & 0 & 2 & 2 & 4 & 0 & 2 \\ 1 & 2 & 0 & 1 & 5 & 0 & 2 \\ 2 & 2 & 1 & 0 & 4 & 0 & 2 \\ 5 & 4 & 5 & 4 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[118] $1^{111} 3^1 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 & 2 \\ 1 & 2 & 0 & 0 & 5 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 4 & 6 \\ 1 & 2 & 1 & 0 & 4 & 0 & 2 \\ 2 & 2 & 2 & 0 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[119] $1^{111} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 4 & 4 & 2 \\ 2 & 3 & 4 & 5 & 6 & 7 & 5 \\ 3 & 2 & 4 & 6 & 4 & 6 & 4 \\ 3 & 3 & 4 & 7 & 6 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[129] $1^{113} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.3$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 10 & 21 & 4 \\ 3 & 1 & 3 & 8 & 15 & 3 \\ 3 & 3 & 1 & 8 & 18 & 3 \\ 10 & 8 & 8 & 22 & 48 & 10 \\ 21 & 15 & 18 & 48 & 97 & 21 \\ 4 & 3 & 3 & 10 & 21 & 4 \end{pmatrix}_{b_0}$$

[128] $1^{12} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 6 & 3 \\ 3 & 5 & 5 & 8 & 10 & 5 \\ 2 & 5 & 2 & 6 & 8 & 4 \\ 5 & 8 & 6 & 8 & 12 & 6 \\ 6 & 10 & 8 & 12 & 14 & 8 \\ 3 & 5 & 4 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[127] $1^{123} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.3$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 10 & 17 & 4 \\ 3 & 1 & 3 & 8 & 13 & 3 \\ 3 & 3 & 1 & 8 & 14 & 3 \\ 10 & 8 & 8 & 22 & 40 & 10 \\ 17 & 13 & 14 & 40 & 65 & 17 \\ 4 & 3 & 3 & 10 & 17 & 4 \end{pmatrix}_{b_0}$$

[126] $1^{11} 1^{112} 3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 6 & 2 \\ 1 & 0 & 1 & 3 & 5 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 13 & 18 & 5 \\ 4 & 5 & 0 & 13 & 23 & 34 & 9 \\ 6 & 7 & 0 & 18 & 34 & 45 & 13 \\ 2 & 2 & -1 & 5 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

[125] $1^{11} 1^{211} 3^1 21^{1212} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 8 & 3 \\ 3 & 5 & 10 & 8 & 15 & 5 \\ 6 & 10 & 14 & 12 & 22 & 8 \\ 5 & 8 & 12 & 8 & 18 & 6 \\ 8 & 15 & 22 & 18 & 32 & 12 \\ 3 & 5 & 8 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[132] $1^{11} 1^{11} 3^1 21^{1311} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 & 3 \\ 3 & 5 & 20 & 8 & 5 & 5 \\ 10 & 20 & 58 & 24 & 15 & 16 \\ 5 & 8 & 24 & 8 & 6 & 6 \\ 3 & 5 & 15 & 6 & 2 & 4 \\ 3 & 5 & 16 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[130] $1^{11} 1^{212} 3^1 21^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 7 & 3 \\ 3 & 5 & 5 & 8 & 10 & 5 \\ 4 & 5 & 2 & 6 & 7 & 4 \\ 5 & 8 & 6 & 8 & 12 & 6 \\ 7 & 10 & 7 & 12 & 14 & 8 \\ 3 & 5 & 4 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[131] $1^{11} 1^{211} 3^1 21^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 & 3 \\ 3 & 5 & 10 & 8 & 5 & 5 \\ 7 & 10 & 14 & 12 & 7 & 8 \\ 5 & 8 & 12 & 8 & 6 & 6 \\ 3 & 5 & 7 & 6 & 2 & 4 \\ 3 & 5 & 8 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[136] $1^{11} 1^{112} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 5 & 0 & 8 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 12 & 6 \\ 6 & 10 & 0 & 12 & 14 & 8 \\ 3 & 5 & 0 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[135] $1^{11} 1^{112} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 6 & 2 \\ 1 & 0 & 1 & 3 & 2 & 7 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 6 & 18 & 5 \\ 2 & 2 & 0 & 6 & 4 & 16 & 4 \\ 6 & 7 & 0 & 18 & 16 & 45 & 13 \\ 2 & 2 & -1 & 5 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[134] $1^{11} 1^{111} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 5 & 0 & 8 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 12 & 6 \\ 6 & 10 & 0 & 12 & 14 & 8 \\ 3 & 5 & 0 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[133] $1^{11} 1^{112} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 5 & 0 & 8 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 24 & 6 \\ 11 & 20 & 0 & 24 & 58 & 16 \\ 3 & 5 & 0 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[139] $1^{114} 3^{114} 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 28 & 4 \\ 3 & 1 & 3 & 3 & 20 & 3 \\ 3 & 3 & 1 & 2 & 24 & 3 \\ 3 & 3 & 2 & 1 & 20 & 3 \\ 28 & 20 & 24 & 20 & 174 & 28 \\ 4 & 3 & 3 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

[140] $1^{114} 3^{114} 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 1 & 4 & 12 & 2 \\ 1 & 2 & 0 & 3 & 5 & 16 & 2 \\ 1 & 1 & 3 & 1 & 5 & 16 & 3 \\ 4 & 4 & 5 & 5 & 10 & 36 & 6 \\ 12 & 12 & 16 & 16 & 36 & 110 & 20 \\ 2 & 2 & 2 & 3 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

[141] $1^{111} 3^{111} 21^{1211} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 & 3 \\ 3 & 5 & 10 & 8 & 5 & 5 \\ 5 & 10 & 14 & 12 & 8 & 8 \\ 5 & 8 & 12 & 8 & 6 & 6 \\ 3 & 5 & 8 & 6 & 2 & 4 \\ 3 & 5 & 8 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[138] $1^{111} 3^{111} 21^{1212} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 8 & 3 \\ 3 & 5 & 10 & 8 & 15 & 5 \\ 5 & 10 & 14 & 12 & 22 & 8 \\ 5 & 8 & 12 & 8 & 18 & 6 \\ 8 & 15 & 22 & 18 & 32 & 12 \\ 3 & 5 & 8 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[137] $1^{114} 3^{114} 21^{1214} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 15 & 3 \\ 3 & 5 & 10 & 8 & 25 & 5 \\ 5 & 10 & 14 & 12 & 38 & 8 \\ 5 & 8 & 12 & 8 & 30 & 6 \\ 15 & 25 & 38 & 30 & 92 & 20 \\ 3 & 5 & 8 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[142] $1^{114} 3^{114} 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 4 & 28 & 4 \\ 3 & 1 & 3 & 2 & 20 & 3 \\ 3 & 3 & 1 & 4 & 24 & 3 \\ 4 & 2 & 4 & 2 & 24 & 4 \\ 28 & 20 & 24 & 24 & 174 & 28 \\ 4 & 3 & 3 & 4 & 28 & 4 \end{pmatrix}_{b_0}$$

[143] $1^{112} 3^{112} 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 6 & 4 \\ 4 & 3 & 4 & 0 & 6 & 5 & 5 \\ 2 & 2 & 2 & 0 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[144] $1^{111} 3^{111} 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 6 & 4 \\ 3 & 3 & 4 & 0 & 6 & 5 & 5 \\ 2 & 2 & 2 & 0 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[145] $1^{1^{11}} 1^{123} 3^1 21^{214} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 6 & 0 & 26 & 12 & 50 & 10 \\ 2 & 2 & 0 & 12 & 4 & 22 & 4 \\ 8 & 10 & 0 & 50 & 22 & 88 & 18 \\ 2 & 2 & -1 & 10 & 4 & 18 & 3 \end{pmatrix}_{b_0}$

[146] $1^{1^{31}} 1^{131} 3^1 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 3 & 2 \\ 1 & 0 & 2 & 1 & 10 & 3 & 2 \\ 1 & 2 & 0 & 3 & 13 & 4 & 2 \\ 1 & 1 & 3 & 1 & 13 & 4 & 3 \\ 10 & 10 & 13 & 13 & 74 & 23 & 16 \\ 3 & 3 & 4 & 4 & 23 & 5 & 5 \\ 2 & 2 & 2 & 3 & 16 & 5 & 3 \end{pmatrix}_{b_0}$

[147] $1^{1^{112}} 1^{113} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 3 & 3 & 7 & 17 & 4 \\ 3 & 1 & 3 & 5 & 13 & 3 \\ 3 & 3 & 1 & 6 & 14 & 3 \\ 7 & 5 & 6 & 9 & 27 & 7 \\ 17 & 13 & 14 & 27 & 65 & 17 \\ 4 & 3 & 3 & 7 & 17 & 4 \end{pmatrix}_{b_0}$

[148] $1^{1^{213}} 1^{113} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 3 & 3 & 11 & 21 & 4 \\ 3 & 1 & 3 & 7 & 15 & 3 \\ 3 & 3 & 1 & 10 & 18 & 3 \\ 11 & 7 & 10 & 25 & 51 & 11 \\ 21 & 15 & 18 & 51 & 97 & 21 \\ 4 & 3 & 3 & 11 & 21 & 4 \end{pmatrix}_{b_0}$

[149] $1^{1^{114}} 1^{13} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 3 & 3 & 4 & 25 & 4 \\ 3 & 1 & 3 & 2 & 17 & 3 \\ 3 & 3 & 1 & 4 & 22 & 3 \\ 4 & 2 & 4 & 2 & 22 & 4 \\ 25 & 17 & 22 & 22 & 137 & 25 \\ 4 & 3 & 3 & 4 & 25 & 4 \end{pmatrix}_{b_0}$

[157] $1^{1^{11}} 1^{121} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 & 2 \\ 1 & 0 & 1 & 3 & 9 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 24 & 7 & 5 \\ 8 & 9 & 0 & 24 & 83 & 24 & 17 \\ 2 & 3 & 0 & 7 & 24 & 5 & 5 \\ 2 & 2 & -1 & 5 & 17 & 5 & 3 \end{pmatrix}_{b_0}$

[156] $1^{1^{211}} 1^{11} 3^1 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 3 & 3 \\ 3 & 5 & 5 & 8 & 5 & 5 \\ 4 & 5 & 2 & 6 & 4 & 4 \\ 5 & 8 & 6 & 8 & 6 & 6 \\ 3 & 5 & 4 & 6 & 2 & 4 \\ 3 & 5 & 4 & 6 & 4 & 4 \end{pmatrix}_{b_0}$

[155] $1^{1^{32}} 1^{11} 3^1 21^{1242} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 16 & 6 & 3 \\ 3 & 5 & 10 & 28 & 10 & 5 \\ 5 & 10 & 14 & 42 & 16 & 8 \\ 16 & 28 & 42 & 116 & 42 & 22 \\ 6 & 10 & 16 & 42 & 14 & 8 \\ 3 & 5 & 8 & 22 & 8 & 4 \end{pmatrix}_{b_0}$

[154] $1^{1^{12}} 1^{12} 3^1 21^{1214} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 16 & 3 \\ 3 & 5 & 10 & 8 & 30 & 5 \\ 5 & 10 & 14 & 12 & 44 & 8 \\ 5 & 8 & 12 & 8 & 36 & 6 \\ 16 & 30 & 44 & 36 & 134 & 24 \\ 3 & 5 & 8 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[153] \ 1^{113} 3^{1141} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 1 & 2 \\ 1 & 0 & 2 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 1 & 13 & 0 & 2 \\ 2 & 2 & 1 & 0 & 10 & 0 & 2 \\ 11 & 10 & 13 & 10 & 74 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[152] \ 1^{12} 3^{1213} 21^{1214} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 17 & 3 \\ 3 & 5 & 10 & 8 & 30 & 5 \\ 6 & 10 & 14 & 12 & 44 & 8 \\ 5 & 8 & 12 & 8 & 36 & 6 \\ 17 & 30 & 44 & 36 & 134 & 24 \\ 3 & 5 & 8 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[151] \ 1^{13} 3^{1113} 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 9 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 2 & 5 & 2 & 6 & 12 & 4 \\ 5 & 8 & 6 & 8 & 18 & 6 \\ 9 & 15 & 12 & 18 & 34 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[150] \ 1^{112} 3^{1112} 21^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 6 & 3 \\ 3 & 5 & 10 & 8 & 10 & 5 \\ 5 & 10 & 14 & 12 & 16 & 8 \\ 5 & 8 & 12 & 8 & 12 & 6 \\ 6 & 10 & 16 & 12 & 14 & 8 \\ 3 & 5 & 8 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[160] \ 1^{12} 3^{1112} 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 9 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 2 & 5 & 2 & 6 & 11 & 4 \\ 5 & 8 & 6 & 8 & 18 & 6 \\ 9 & 15 & 11 & 18 & 32 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[159] \ 1^{141} 3^{113} 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 & 2 \\ 1 & 0 & 2 & 3 & 10 & 1 & 2 \\ 1 & 2 & 0 & 4 & 13 & 3 & 2 \\ 2 & 3 & 4 & 5 & 23 & 4 & 5 \\ 11 & 10 & 13 & 23 & 74 & 13 & 16 \\ 2 & 1 & 3 & 4 & 13 & 1 & 3 \\ 2 & 2 & 2 & 5 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[158] \ 1^{11} 3^{1112} 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 8 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 2 & 5 & 2 & 6 & 11 & 4 \\ 5 & 8 & 6 & 8 & 18 & 6 \\ 8 & 15 & 11 & 18 & 32 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[165] \ 1^{1214} 3^{1112} 21^{1214} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 11 & 22 & 4 \\ 3 & 1 & 3 & 7 & 14 & 3 \\ 3 & 3 & 1 & 10 & 20 & 3 \\ 11 & 7 & 10 & 25 & 54 & 11 \\ 22 & 14 & 20 & 54 & 106 & 22 \\ 4 & 3 & 3 & 11 & 22 & 4 \end{pmatrix}_{b_0}$$

[164] $1^{1^{212}} 3^{1^{122}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 3 & 3 & 17 & 14 & 4 \\ 3 & 1 & 3 & 13 & 10 & 3 \\ 3 & 3 & 1 & 14 & 12 & 3 \\ 17 & 13 & 14 & 65 & 54 & 17 \\ 14 & 10 & 12 & 54 & 42 & 14 \\ 4 & 3 & 3 & 17 & 14 & 4 \end{pmatrix}_{b_0}$

[163] $1^{1^{112}} 3^{1^{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 3 & 3 & 7 & 14 & 4 \\ 3 & 1 & 3 & 5 & 10 & 3 \\ 3 & 3 & 1 & 6 & 12 & 3 \\ 7 & 5 & 6 & 9 & 22 & 7 \\ 14 & 10 & 12 & 22 & 42 & 14 \\ 4 & 3 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$

[162] $1^{1^{113}} 3^{1^{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 5 & 0 & 8 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 18 & 6 \\ 9 & 15 & 0 & 18 & 34 & 12 \\ 3 & 5 & 0 & 6 & 12 & 4 \end{pmatrix}_{b_0}$

[161] $1^{1^{112}} 3^{1^{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 5 & 0 & 8 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 30 & 6 \\ 14 & 25 & 0 & 30 & 94 & 20 \\ 3 & 5 & 0 & 6 & 20 & 4 \end{pmatrix}_{b_0}$

[167] $1^{1^{131}} 3^{1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 1 & 2 \\ 1 & 0 & 2 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & 1 & 0 & 7 & 0 & 2 \\ 8 & 7 & 9 & 7 & 35 & 0 & 11 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 11 & 0 & 3 \end{pmatrix}_{b_0}$

[166] $1^{1^{31}} 3^{1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 1 & 2 & 1 & 0 & 7 & 0 & 2 \\ 8 & 7 & 9 & 7 & 35 & 0 & 11 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 11 & 0 & 3 \end{pmatrix}_{b_0}$

[171] $1^{1^{14}} 3^{1^{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 10 & 2 \\ 1 & 0 & 2 & 3 & 2 & 9 & 2 \\ 1 & 2 & 0 & 4 & 4 & 12 & 2 \\ 2 & 3 & 4 & 5 & 6 & 21 & 5 \\ 3 & 2 & 4 & 6 & 4 & 18 & 4 \\ 10 & 9 & 12 & 21 & 18 & 61 & 15 \\ 2 & 2 & 2 & 5 & 4 & 15 & 3 \end{pmatrix}_{b_0}$

[168] $1^{1^{31}} 3^{1^{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 & 2 \\ 1 & 0 & 2 & 1 & 8 & 1 & 2 \\ 1 & 2 & 0 & 3 & 12 & 3 & 2 \\ 2 & 1 & 3 & 1 & 12 & 3 & 3 \\ 9 & 8 & 12 & 12 & 56 & 12 & 14 \\ 2 & 1 & 3 & 3 & 12 & 1 & 3 \\ 2 & 2 & 2 & 3 & 14 & 3 & 3 \end{pmatrix}_{b_0}$

[169] $1^{1^{11}} 3^{1^{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 6 & 3 \\ 3 & 5 & 5 & 8 & 10 & 5 \\ 3 & 5 & 2 & 6 & 7 & 4 \\ 4 & 8 & 6 & 8 & 12 & 6 \\ 6 & 10 & 7 & 12 & 14 & 8 \\ 3 & 5 & 4 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[172] $1^{112} 3^{112} 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 8 & 2 \\ 1 & 0 & 2 & 3 & 2 & 9 & 2 \\ 1 & 2 & 0 & 4 & 4 & 12 & 2 \\ 2 & 3 & 4 & 5 & 6 & 21 & 5 \\ 3 & 2 & 4 & 6 & 4 & 18 & 4 \\ 8 & 9 & 12 & 21 & 18 & 61 & 15 \\ 2 & 2 & 2 & 5 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[173] $1^{133} 3^{113} 21^{1313} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 12 & 3 \\ 3 & 5 & 15 & 8 & 20 & 5 \\ 9 & 15 & 34 & 18 & 45 & 12 \\ 4 & 8 & 18 & 8 & 24 & 6 \\ 12 & 20 & 45 & 24 & 58 & 16 \\ 3 & 5 & 12 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[170] $1^{114} 3^{114} 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 10 & 28 & 4 \\ 3 & 1 & 3 & 8 & 20 & 3 \\ 3 & 3 & 1 & 8 & 24 & 3 \\ 10 & 8 & 8 & 22 & 64 & 10 \\ 28 & 20 & 24 & 64 & 174 & 28 \\ 4 & 3 & 3 & 10 & 28 & 4 \end{pmatrix}_{b_0}$$

[185] $1^{222} 3^{112} 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 9 & 3 \\ 3 & 5 & 5 & 13 & 15 & 5 \\ 4 & 5 & 2 & 10 & 11 & 4 \\ 8 & 13 & 10 & 24 & 29 & 10 \\ 9 & 15 & 11 & 29 & 32 & 12 \\ 3 & 5 & 4 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[188] $1^{213} 3^{112} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 11 & 18 & 4 \\ 3 & 1 & 3 & 7 & 12 & 3 \\ 3 & 3 & 1 & 10 & 16 & 3 \\ 11 & 7 & 10 & 25 & 44 & 11 \\ 18 & 12 & 16 & 44 & 70 & 18 \\ 4 & 3 & 3 & 11 & 18 & 4 \end{pmatrix}_{b_0}$$

[176] $1^{113} 3^{112} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 5 & 0 & 8 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 18 & 6 \\ 10 & 15 & 0 & 18 & 32 & 12 \\ 3 & 5 & 0 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[177] $1^{112} 3^{112} 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 6 & 3 \\ 3 & 5 & 5 & 8 & 10 & 5 \\ 4 & 5 & 2 & 6 & 8 & 4 \\ 5 & 8 & 6 & 8 & 12 & 6 \\ 6 & 10 & 8 & 12 & 14 & 8 \\ 3 & 5 & 4 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[178] $1^{222} 3^{112} 21^{1323} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 11 & 3 \\ 3 & 5 & 15 & 13 & 20 & 5 \\ 8 & 15 & 34 & 30 & 45 & 12 \\ 8 & 13 & 30 & 24 & 39 & 10 \\ 11 & 20 & 45 & 39 & 58 & 16 \\ 3 & 5 & 12 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[179] $1^{113} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 8 & 2 \\ 1 & 0 & 2 & 1 & 4 & 8 & 2 \\ 1 & 2 & 0 & 3 & 5 & 9 & 2 \\ 2 & 1 & 3 & 1 & 5 & 9 & 3 \\ 4 & 4 & 5 & 5 & 10 & 22 & 6 \\ 8 & 8 & 9 & 9 & 22 & 40 & 12 \\ 2 & 2 & 2 & 3 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[180] $1^{1213} 3^1 21^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 10 & 3 \\ 3 & 5 & 10 & 8 & 15 & 5 \\ 6 & 10 & 14 & 12 & 22 & 8 \\ 5 & 8 & 12 & 8 & 18 & 6 \\ 10 & 15 & 22 & 18 & 32 & 12 \\ 3 & 5 & 8 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[181] $1^{1212} 3^1 21^{1214} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 13 & 3 \\ 3 & 5 & 10 & 8 & 25 & 5 \\ 6 & 10 & 14 & 12 & 38 & 8 \\ 5 & 8 & 12 & 8 & 30 & 6 \\ 13 & 25 & 38 & 30 & 92 & 20 \\ 3 & 5 & 8 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[182] $1^{1124} 3^1 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 9 & 2 \\ 1 & 0 & 2 & 1 & 4 & 8 & 2 \\ 1 & 2 & 0 & 3 & 5 & 9 & 2 \\ 2 & 1 & 3 & 1 & 5 & 9 & 3 \\ 5 & 4 & 5 & 5 & 10 & 22 & 6 \\ 9 & 8 & 9 & 9 & 22 & 40 & 12 \\ 2 & 2 & 2 & 3 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[183] $1^{1222} 3^1 21^{1211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 5 & 4 & 2 \\ 1 & 0 & 2 & 4 & 4 & 3 & 2 \\ 1 & 2 & 0 & 7 & 5 & 4 & 2 \\ 5 & 4 & 7 & 16 & 14 & 11 & 8 \\ 5 & 4 & 5 & 14 & 10 & 9 & 6 \\ 4 & 3 & 4 & 11 & 9 & 5 & 5 \\ 2 & 2 & 2 & 8 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[184] $1^{1111} 3^1 21^{1211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 2 & 2 \\ 1 & 0 & 2 & 4 & 4 & 3 & 2 \\ 1 & 2 & 0 & 7 & 5 & 4 & 2 \\ 4 & 4 & 7 & 16 & 14 & 11 & 8 \\ 4 & 4 & 5 & 14 & 10 & 9 & 6 \\ 2 & 3 & 4 & 11 & 9 & 5 & 5 \\ 2 & 2 & 2 & 8 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[174] $1^{311} 3^1 21^{1211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 3 & 2 \\ 1 & 0 & 2 & 4 & 4 & 3 & 2 \\ 1 & 2 & 0 & 7 & 5 & 4 & 2 \\ 6 & 4 & 7 & 16 & 14 & 11 & 8 \\ 4 & 4 & 5 & 14 & 10 & 9 & 6 \\ 3 & 3 & 4 & 11 & 9 & 5 & 5 \\ 2 & 2 & 2 & 8 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[186] $1^{113} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 7 & 18 & 4 \\ 3 & 1 & 3 & 5 & 12 & 3 \\ 3 & 3 & 1 & 6 & 16 & 3 \\ 7 & 5 & 6 & 9 & 28 & 7 \\ 18 & 12 & 16 & 28 & 70 & 18 \\ 4 & 3 & 3 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

[187] $1^{142} 3^1 21^{1142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

6.5

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 6 & 3 \\ 3 & 5 & 5 & 28 & 10 & 5 \\ 2 & 5 & 2 & 21 & 8 & 4 \\ 17 & 28 & 21 & 116 & 42 & 22 \\ 6 & 10 & 8 & 42 & 14 & 8 \\ 3 & 5 & 4 & 22 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[175] \ 1^{111} 3^{121} 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 4 & 2 \\ 1 & 0 & 2 & 3 & 5 & 4 & 2 \\ 1 & 2 & 0 & 4 & 8 & 7 & 2 \\ 2 & 3 & 4 & 5 & 13 & 11 & 5 \\ 6 & 5 & 8 & 13 & 23 & 21 & 9 \\ 4 & 4 & 7 & 11 & 21 & 16 & 8 \\ 2 & 2 & 2 & 5 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[189] \ 1^{121} 3^{121} 21^{1221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 & 3 \\ 3 & 5 & 15 & 13 & 5 & 5 \\ 8 & 15 & 32 & 29 & 11 & 12 \\ 8 & 13 & 29 & 24 & 10 & 10 \\ 3 & 5 & 11 & 10 & 2 & 4 \\ 3 & 5 & 12 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[190] \ 1^{121} 3^{121} 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 & 3 \\ 3 & 5 & 10 & 13 & 5 & 5 \\ 6 & 10 & 14 & 19 & 7 & 8 \\ 8 & 13 & 19 & 24 & 10 & 10 \\ 3 & 5 & 7 & 10 & 2 & 4 \\ 3 & 5 & 8 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[191] \ 1^{121} 3^{121} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 3 & 2 \\ 1 & 2 & 0 & 0 & 5 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 9 & 6 \\ 3 & 3 & 4 & 0 & 9 & 5 & 5 \\ 2 & 2 & 2 & 0 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[192] \ 1^{123} 3^{123} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 17 & 6 \\ 7 & 7 & 6 & 0 & 17 & 25 & 9 \\ 2 & 2 & 2 & 0 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[193] \ 1^{111} 3^{111} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 2 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 5 & 4 & 2 \\ 1 & 2 & 1 & 0 & 4 & 3 & 2 \\ 4 & 4 & 5 & 4 & 10 & 9 & 6 \\ 2 & 3 & 4 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[194] \ 1^{131} 3^{131} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 & 3 \\ 3 & 5 & 5 & 23 & 5 & 5 \\ 2 & 5 & 2 & 17 & 4 & 4 \\ 14 & 23 & 17 & 78 & 17 & 18 \\ 3 & 5 & 4 & 17 & 2 & 4 \\ 3 & 5 & 4 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[195] \ 1^{112} 3^{112} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 17 & 6 \\ 6 & 7 & 6 & 0 & 17 & 25 & 9 \\ 2 & 2 & 2 & 0 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[196] $1^{1111} 3^1 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 5 & 2 \\ 1 & 0 & 2 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 5 & 4 & 2 \\ 2 & 2 & 1 & 0 & 4 & 3 & 2 \\ 5 & 4 & 5 & 4 & 10 & 9 & 6 \\ 5 & 3 & 4 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[197] $1^{1111} 3^1 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 3 & 2 \\ 1 & 0 & 2 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 5 & 4 & 2 \\ 2 & 2 & 1 & 0 & 4 & 3 & 2 \\ 5 & 4 & 5 & 4 & 10 & 9 & 6 \\ 3 & 3 & 4 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[198] $1^{1111} 3^1 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 3 & 2 \\ 1 & 2 & 0 & 0 & 5 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 9 & 6 \\ 4 & 3 & 4 & 0 & 9 & 5 & 5 \\ 2 & 2 & 2 & 0 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[199] $1^{1111} 3^1 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 5 & 4 & 2 \\ 1 & 2 & 1 & 0 & 4 & 3 & 2 \\ 4 & 4 & 5 & 4 & 10 & 9 & 6 \\ 4 & 3 & 4 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[200] $1^{1111} 3^1 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 8 & 2 \\ 1 & 0 & 1 & 3 & 2 & 10 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 6 & 25 & 5 \\ 2 & 2 & 0 & 6 & 4 & 22 & 4 \\ 8 & 10 & 0 & 25 & 22 & 88 & 18 \\ 2 & 2 & -1 & 5 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[201] $1^{1111} 3^1 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 9 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 4 & 5 & 2 & 6 & 11 & 4 \\ 5 & 8 & 6 & 8 & 18 & 6 \\ 9 & 15 & 11 & 18 & 32 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[212] $1^{1111} 3^1 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 2 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 5 & 4 & 2 \\ 1 & 2 & 1 & 0 & 4 & 3 & 2 \\ 5 & 4 & 5 & 4 & 10 & 9 & 6 \\ 4 & 3 & 4 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[208] $1^{1111} 3^1 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 12 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 3 & 5 & 2 & 6 & 14 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 12 & 20 & 14 & 24 & 62 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[213] $1^{1111} 3^{121} 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 3 & 2 \\ 1 & 0 & 2 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 5 & 4 & 2 \\ 1 & 2 & 1 & 0 & 4 & 3 & 2 \\ 5 & 4 & 5 & 4 & 10 & 9 & 6 \\ 3 & 3 & 4 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[202] $1^{1113} 3^{1213} 21^{1213} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 9 & 3 \\ 3 & 5 & 10 & 8 & 15 & 5 \\ 5 & 10 & 14 & 12 & 24 & 8 \\ 5 & 8 & 12 & 8 & 18 & 6 \\ 9 & 15 & 24 & 18 & 34 & 12 \\ 3 & 5 & 8 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[214] $1^{122} 3^{121} 21^{1122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 9 & 3 \\ 3 & 5 & 5 & 13 & 15 & 5 \\ 2 & 5 & 2 & 10 & 11 & 4 \\ 8 & 13 & 10 & 24 & 29 & 10 \\ 9 & 15 & 11 & 29 & 32 & 12 \\ 3 & 5 & 4 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[215] $1^{122} 3^{121} 21^{1123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 11 & 3 \\ 3 & 5 & 5 & 13 & 20 & 5 \\ 2 & 5 & 2 & 10 & 15 & 4 \\ 8 & 13 & 10 & 24 & 39 & 10 \\ 11 & 20 & 15 & 39 & 58 & 16 \\ 3 & 5 & 4 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[203] $1^{1214} 3^{121} 21^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 11 & 28 & 4 \\ 3 & 1 & 3 & 7 & 20 & 3 \\ 3 & 3 & 1 & 10 & 24 & 3 \\ 11 & 7 & 10 & 25 & 68 & 11 \\ 28 & 20 & 24 & 68 & 174 & 28 \\ 4 & 3 & 3 & 11 & 28 & 4 \end{pmatrix}_{b_0}$$

[204] $1^{1111} 3^{112} 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 0 & 2 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 5 & 4 & 2 \\ 2 & 2 & 1 & 0 & 4 & 3 & 2 \\ 4 & 4 & 5 & 4 & 10 & 9 & 6 \\ 4 & 3 & 4 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[205] $1^{1111} 3^{111} 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 3 & 2 \\ 1 & 0 & 2 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 5 & 4 & 2 \\ 2 & 2 & 1 & 0 & 4 & 3 & 2 \\ 4 & 4 & 5 & 4 & 10 & 9 & 6 \\ 3 & 3 & 4 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[206] $1^{121} 3^{121} 21^{1121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 & 3 \\ 3 & 5 & 5 & 13 & 5 & 5 \\ 2 & 5 & 2 & 10 & 4 & 4 \\ 8 & 13 & 10 & 24 & 10 & 10 \\ 3 & 5 & 4 & 10 & 2 & 4 \\ 3 & 5 & 4 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

[207] $1^{1211} 1^{11} 3^1 21^{1222} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 7 & 8 & 3 \\ 3 & 5 & 10 & 13 & 15 & 5 \\ 6 & 10 & 14 & 20 & 22 & 8 \\ 7 & 13 & 20 & 24 & 29 & 10 \\ 8 & 15 & 22 & 29 & 32 & 12 \\ 3 & 5 & 8 & 10 & 12 & 4 \end{pmatrix}_{b_0}$

[211] $1^{122} 1^{112} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 2 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 1 & 2 & 1 & 0 & 4 & 5 & 2 \\ 5 & 4 & 5 & 4 & 10 & 13 & 6 \\ 5 & 5 & 5 & 5 & 13 & 13 & 7 \\ 2 & 2 & 2 & 2 & 6 & 7 & 3 \end{pmatrix}_{b_0}$

[210] $1^{112} 1^{112} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 5 & 2 \\ 1 & 0 & 2 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 2 & 2 & 1 & 0 & 4 & 5 & 2 \\ 4 & 4 & 5 & 4 & 10 & 13 & 6 \\ 5 & 5 & 5 & 5 & 13 & 13 & 7 \\ 2 & 2 & 2 & 2 & 6 & 7 & 3 \end{pmatrix}_{b_0}$

[209] $1^{122} 1^{11} 3^1 21^{1231} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 13 & 2 & 3 \\ 3 & 5 & 10 & 23 & 5 & 5 \\ 6 & 10 & 14 & 34 & 8 & 8 \\ 13 & 23 & 34 & 78 & 17 & 18 \\ 2 & 5 & 8 & 17 & 2 & 4 \\ 3 & 5 & 8 & 18 & 4 & 4 \end{pmatrix}_{b_0}$

[216] $1^{112} 1^{12} 3^1 21^{1112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 6 & 2 \\ 1 & 0 & 2 & 3 & 2 & 6 & 2 \\ 1 & 2 & 0 & 4 & 4 & 8 & 2 \\ 2 & 3 & 4 & 5 & 6 & 14 & 5 \\ 3 & 2 & 4 & 6 & 4 & 12 & 4 \\ 6 & 6 & 8 & 14 & 12 & 26 & 10 \\ 2 & 2 & 2 & 5 & 4 & 10 & 3 \end{pmatrix}_{b_0}$

[217] $1^{113} 1^{124} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 3 & 3 & 10 & 24 & 4 \\ 3 & 1 & 3 & 8 & 18 & 3 \\ 3 & 3 & 1 & 8 & 20 & 3 \\ 10 & 8 & 8 & 22 & 56 & 10 \\ 24 & 18 & 20 & 56 & 130 & 24 \\ 4 & 3 & 3 & 10 & 24 & 4 \end{pmatrix}_{b_0}$

[218] $1^{213} 1^{122} 3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 3 & 3 & 14 & 18 & 4 \\ 3 & 1 & 3 & 10 & 12 & 3 \\ 3 & 3 & 1 & 12 & 16 & 3 \\ 14 & 10 & 12 & 42 & 56 & 14 \\ 18 & 12 & 16 & 56 & 70 & 18 \\ 4 & 3 & 3 & 14 & 18 & 4 \end{pmatrix}_{b_0}$

[219] $1^{14} 1^1 3^1 21^{1114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 12 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 2 & 5 & 2 & 6 & 16 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 12 & 20 & 16 & 24 & 62 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$

[220] $1^{1211} 1^{211} 3^1 21^{1411} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 4 & 3 & 2 \\ 1 & 0 & 2 & 8 & 4 & 3 & 2 \\ 1 & 2 & 0 & 14 & 5 & 4 & 2 \\ 10 & 8 & 14 & 70 & 28 & 22 & 16 \\ 4 & 4 & 5 & 28 & 10 & 9 & 6 \\ 3 & 3 & 4 & 22 & 9 & 5 & 5 \\ 2 & 2 & 2 & 16 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[236] \ 1^{\overset{1^{121}}{1}} 3^{\overset{1^{121}}{1}} 21^{\overset{1^{121}}{1221}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 & 3 \\ 3 & 5 & 10 & 13 & 5 & 5 \\ 5 & 10 & 14 & 20 & 8 & 8 \\ 8 & 13 & 20 & 24 & 10 & 10 \\ 3 & 5 & 8 & 10 & 2 & 4 \\ 3 & 5 & 8 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[224] \ 1^{\overset{1^{114}}{1}} 3^{\overset{1^{114}}{1}} 21^{\overset{1^{114}}{114}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 5 & 0 & 8 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 24 & 6 \\ 12 & 20 & 0 & 24 & 62 & 16 \\ 3 & 5 & 0 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[223] \ 1^{\overset{1^{121}}{1}} 3^{\overset{1^{121}}{1}} 21^{\overset{1^{111}}{1111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 3 & 2 \\ 1 & 0 & 2 & 5 & 4 & 3 & 2 \\ 1 & 2 & 0 & 5 & 5 & 4 & 2 \\ 4 & 5 & 5 & 13 & 13 & 10 & 7 \\ 5 & 4 & 5 & 13 & 10 & 9 & 6 \\ 3 & 3 & 4 & 10 & 9 & 5 & 5 \\ 2 & 2 & 2 & 7 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[222] \ 1^{\overset{1^{121}}{1}} 3^{\overset{1^{121}}{1}} 21^{\overset{1^{111}}{1111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 3 & 2 \\ 1 & 0 & 2 & 1 & 4 & 3 & 2 \\ 1 & 2 & 0 & 3 & 5 & 4 & 2 \\ 2 & 1 & 3 & 1 & 5 & 4 & 3 \\ 5 & 4 & 5 & 5 & 10 & 9 & 6 \\ 3 & 3 & 4 & 4 & 9 & 5 & 5 \\ 2 & 2 & 2 & 3 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[227] \ 1^{\overset{1^{112}}{1}} 3^{\overset{1^{112}}{1}} 21^{\overset{1^{111}}{1111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 0 & 2 & 1 & 4 & 3 & 2 \\ 1 & 2 & 0 & 3 & 5 & 4 & 2 \\ 2 & 1 & 3 & 1 & 5 & 4 & 3 \\ 4 & 4 & 5 & 5 & 10 & 9 & 6 \\ 4 & 3 & 4 & 4 & 9 & 5 & 5 \\ 2 & 2 & 2 & 3 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[221] \ 1^{\overset{1^{24}}{1}} 3^{\overset{1^{24}}{1}} 21^{\overset{1^{124}}{1124}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 18 & 3 \\ 3 & 5 & 5 & 13 & 30 & 5 \\ 2 & 5 & 2 & 10 & 22 & 4 \\ 8 & 13 & 10 & 24 & 58 & 10 \\ 18 & 30 & 22 & 58 & 134 & 24 \\ 3 & 5 & 4 & 10 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[226] \ 1^{\overset{1^{312}}{1}} 3^{\overset{1^{312}}{1}} 21^{\overset{1^{111}}{1111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 4 & 2 \\ 1 & 0 & 2 & 5 & 4 & 3 & 2 \\ 1 & 2 & 0 & 5 & 5 & 4 & 2 \\ 6 & 5 & 5 & 13 & 13 & 10 & 7 \\ 4 & 4 & 5 & 13 & 10 & 9 & 6 \\ 4 & 3 & 4 & 10 & 9 & 5 & 5 \\ 2 & 2 & 2 & 7 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[228] \ 1^{\overset{1^{22}}{1}} 3^{\overset{1^{22}}{1}} 21^{\overset{1^{112}}{112}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 2 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 1 & 2 & 1 & 0 & 4 & 6 & 2 \\ 5 & 4 & 5 & 4 & 10 & 18 & 6 \\ 6 & 6 & 8 & 6 & 18 & 26 & 10 \\ 2 & 2 & 2 & 2 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[225] $1^{121} 3^1 21^{1221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 3 & 2 \\ 1 & 0 & 2 & 4 & 7 & 3 & 2 \\ 1 & 2 & 0 & 7 & 9 & 4 & 2 \\ 4 & 4 & 7 & 16 & 25 & 11 & 8 \\ 7 & 7 & 9 & 25 & 35 & 16 & 11 \\ 3 & 3 & 4 & 11 & 16 & 5 & 5 \\ 2 & 2 & 2 & 8 & 11 & 5 & 3 \end{pmatrix}_{b_0}$

[229] $1^{112} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 2 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 1 & 2 & 1 & 0 & 4 & 6 & 2 \\ 5 & 4 & 5 & 4 & 10 & 18 & 6 \\ 8 & 6 & 8 & 6 & 18 & 26 & 10 \\ 2 & 2 & 2 & 2 & 6 & 10 & 3 \end{pmatrix}_{b_0}$

[230] $1^{122} 3^1 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 8 & 6 & 3 \\ 3 & 5 & 5 & 13 & 10 & 5 \\ 2 & 5 & 2 & 10 & 8 & 4 \\ 8 & 13 & 10 & 24 & 20 & 10 \\ 6 & 10 & 8 & 20 & 14 & 8 \\ 3 & 5 & 4 & 10 & 8 & 4 \end{pmatrix}_{b_0}$

[231] $1^{112} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 5 & 0 & 8 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 24 & 6 \\ 12 & 20 & 0 & 24 & 62 & 16 \\ 3 & 5 & 0 & 6 & 16 & 4 \end{pmatrix}_{b_0}$

[232] $1^{114} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 8 & 2 \\ 1 & 0 & 2 & 2 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 2 & 2 & 1 & 0 & 4 & 6 & 2 \\ 4 & 4 & 5 & 4 & 10 & 18 & 6 \\ 8 & 6 & 8 & 6 & 18 & 26 & 10 \\ 2 & 2 & 2 & 2 & 6 & 10 & 3 \end{pmatrix}_{b_0}$

[233] $1^{1213} 3^1 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 9 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 4 & 5 & 2 & 6 & 12 & 4 \\ 5 & 8 & 6 & 8 & 18 & 6 \\ 9 & 15 & 12 & 18 & 34 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0}$

[234] $1^{121} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 & 3 \\ 3 & 5 & 0 & 13 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 10 & 10 \\ 3 & 5 & 0 & 10 & 2 & 4 \\ 3 & 5 & 0 & 10 & 4 & 4 \end{pmatrix}_{b_0}$

[235] $1^{112} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 6 & 2 \\ 1 & 0 & 2 & 2 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 2 & 2 & 1 & 0 & 4 & 6 & 2 \\ 4 & 4 & 5 & 4 & 10 & 18 & 6 \\ 6 & 6 & 8 & 6 & 18 & 26 & 10 \\ 2 & 2 & 2 & 2 & 6 & 10 & 3 \end{pmatrix}_{b_0}$

[237] $1^{1211} 3^1 21^{1311} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 & 3 \\ 3 & 5 & 15 & 8 & 5 & 5 \\ 8 & 15 & 34 & 18 & 12 & 12 \\ 5 & 8 & 18 & 8 & 6 & 6 \\ 3 & 5 & 12 & 6 & 2 & 4 \\ 3 & 5 & 12 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[238] $1^{1231} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 1 & 2 \\ 1 & 0 & 2 & 4 & 7 & 0 & 2 \\ 1 & 2 & 0 & 7 & 9 & 0 & 2 \\ 5 & 4 & 7 & 16 & 25 & 0 & 8 \\ 8 & 7 & 9 & 25 & 35 & 0 & 11 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 8 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

[247] $1^{122} 3^1 21^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 9 & 3 \\ 3 & 5 & 10 & 8 & 15 & 5 \\ 6 & 10 & 14 & 12 & 22 & 8 \\ 4 & 8 & 12 & 8 & 18 & 6 \\ 9 & 15 & 22 & 18 & 32 & 12 \\ 3 & 5 & 8 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[259] $1^{114} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 8 & 2 \\ 1 & 0 & 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 0 & 4 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 12 & 4 \\ 8 & 6 & 8 & 0 & 12 & 26 & 10 \\ 2 & 2 & 2 & 0 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[240] $1^{123} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 5 & 2 \\ 1 & 2 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 6 & 6 & 0 & 20 & 18 & 8 \\ 6 & 5 & 5 & 0 & 18 & 13 & 7 \\ 2 & 2 & 2 & 0 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

[241] $1^{11} 3^1 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 & 3 \\ 3 & 5 & 5 & 8 & 5 & 5 \\ 3 & 5 & 2 & 6 & 4 & 4 \\ 4 & 8 & 6 & 8 & 6 & 6 \\ 3 & 5 & 4 & 6 & 2 & 4 \\ 3 & 5 & 4 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[242] $1^{141} 3^1 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 & 2 \\ 1 & 0 & 2 & 1 & 8 & 1 & 2 \\ 1 & 2 & 0 & 3 & 12 & 3 & 2 \\ 2 & 1 & 3 & 1 & 12 & 3 & 3 \\ 10 & 8 & 12 & 12 & 56 & 12 & 14 \\ 2 & 1 & 3 & 3 & 12 & 1 & 3 \\ 2 & 2 & 2 & 3 & 14 & 3 & 3 \end{pmatrix}_{b_0}$$

[243] $1^{131} 3^1 21^{1311} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 & 3 \\ 3 & 5 & 15 & 8 & 5 & 5 \\ 9 & 15 & 34 & 18 & 12 & 12 \\ 4 & 8 & 18 & 8 & 6 & 6 \\ 3 & 5 & 12 & 6 & 2 & 4 \\ 3 & 5 & 12 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[262] $1^{121} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 5 & 2 \\ 1 & 2 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 6 & 6 & 0 & 20 & 18 & 8 \\ 4 & 5 & 5 & 0 & 18 & 13 & 7 \\ 2 & 2 & 2 & 0 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

[263] $1^{141} 3^1 21^{1411} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 & 3 \\ 3 & 5 & 20 & 8 & 5 & 5 \\ 12 & 20 & 62 & 24 & 16 & 16 \\ 4 & 8 & 24 & 8 & 6 & 6 \\ 3 & 5 & 16 & 6 & 2 & 4 \\ 3 & 5 & 16 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[244] $1^{132} 3^1 21^{1312} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 6 & 3 \\ 3 & 5 & 15 & 8 & 10 & 5 \\ 9 & 15 & 34 & 18 & 24 & 12 \\ 4 & 8 & 18 & 8 & 12 & 6 \\ 6 & 10 & 24 & 12 & 14 & 8 \\ 3 & 5 & 12 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[245] $1^{142} 3^1 21^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 6 & 2 \\ 1 & 2 & 0 & 0 & 12 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 8 & 12 & 0 & 56 & 40 & 14 \\ 6 & 6 & 8 & 0 & 40 & 26 & 10 \\ 2 & 2 & 2 & 0 & 14 & 10 & 3 \end{pmatrix}_{b_0}$$

[260] $1^{142} 3^1 21^{1412} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 6 & 3 \\ 3 & 5 & 20 & 8 & 10 & 5 \\ 12 & 20 & 62 & 24 & 32 & 16 \\ 4 & 8 & 24 & 8 & 12 & 6 \\ 6 & 10 & 32 & 12 & 14 & 8 \\ 3 & 5 & 16 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[246] $1^{133} 3^1 21^{1313} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 9 & 3 \\ 3 & 5 & 15 & 8 & 15 & 5 \\ 9 & 15 & 34 & 18 & 36 & 12 \\ 4 & 8 & 18 & 8 & 18 & 6 \\ 9 & 15 & 36 & 18 & 34 & 12 \\ 3 & 5 & 12 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[258] $1^{144} 3^1 21^{1414} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 12 & 3 \\ 3 & 5 & 20 & 8 & 20 & 5 \\ 12 & 20 & 62 & 24 & 64 & 16 \\ 4 & 8 & 24 & 8 & 24 & 6 \\ 12 & 20 & 64 & 24 & 62 & 16 \\ 3 & 5 & 16 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[257] $1^{121} 3^1 21^{1211} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 & 3 \\ 3 & 5 & 10 & 8 & 5 & 5 \\ 6 & 10 & 14 & 12 & 8 & 8 \\ 4 & 8 & 12 & 8 & 6 & 6 \\ 3 & 5 & 8 & 6 & 2 & 4 \\ 3 & 5 & 8 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[256] $1^{122} 3^1 21^{1212} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 6 & 3 \\ 3 & 5 & 10 & 8 & 10 & 5 \\ 6 & 10 & 14 & 12 & 16 & 8 \\ 4 & 8 & 12 & 8 & 12 & 6 \\ 6 & 10 & 16 & 12 & 14 & 8 \\ 3 & 5 & 8 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[255] $1^{142} 3^1 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 7 & 1 & 2 \\ 1 & 0 & 2 & 10 & 7 & 0 & 2 \\ 1 & 2 & 0 & 10 & 9 & 0 & 2 \\ 9 & 10 & 10 & 58 & 46 & 0 & 14 \\ 7 & 7 & 9 & 46 & 35 & 0 & 11 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 14 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

[254] $1^{121} 3^1 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 1 & 2 \\ 1 & 0 & 2 & 1 & 7 & 0 & 2 \\ 1 & 2 & 0 & 3 & 9 & 0 & 2 \\ 2 & 1 & 3 & 1 & 9 & 0 & 3 \\ 7 & 7 & 9 & 9 & 35 & 0 & 11 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[253] \quad 1^{12} 3^{112} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 0 & 4 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 12 & 4 \\ 6 & 6 & 8 & 0 & 12 & 26 & 10 \\ 2 & 2 & 2 & 0 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[252] \quad 1^{12} 3^{1141} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 1 & 2 \\ 1 & 0 & 2 & 1 & 7 & 0 & 2 \\ 1 & 2 & 0 & 3 & 9 & 0 & 2 \\ 2 & 1 & 3 & 1 & 9 & 0 & 3 \\ 9 & 7 & 9 & 9 & 35 & 0 & 11 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[251] \quad 1^{122} 3^{1212} 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 14 & 14 & 4 \\ 3 & 1 & 3 & 10 & 10 & 3 \\ 3 & 3 & 1 & 12 & 12 & 3 \\ 14 & 10 & 12 & 42 & 44 & 14 \\ 14 & 10 & 12 & 44 & 42 & 14 \\ 4 & 3 & 3 & 14 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[250] \quad 1^{134} 3^{121} 21^{1314} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 12 & 3 \\ 3 & 5 & 15 & 8 & 20 & 5 \\ 9 & 15 & 34 & 18 & 48 & 12 \\ 4 & 8 & 18 & 8 & 24 & 6 \\ 12 & 20 & 48 & 24 & 62 & 16 \\ 3 & 5 & 12 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[261] \quad 1^{11} 3^{121} 21^{1331} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 & 3 \\ 3 & 5 & 20 & 18 & 5 & 5 \\ 10 & 20 & 58 & 54 & 15 & 16 \\ 10 & 18 & 54 & 48 & 14 & 14 \\ 3 & 5 & 15 & 14 & 2 & 4 \\ 3 & 5 & 16 & 14 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[249] \quad 1^{122} 3^{111} 21^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 3 & 5 & 2 \\ 1 & 0 & 2 & 6 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 4 & 8 & 2 \\ 5 & 6 & 8 & 26 & 12 & 28 & 10 \\ 3 & 2 & 4 & 12 & 4 & 12 & 4 \\ 5 & 6 & 8 & 28 & 12 & 26 & 10 \\ 2 & 2 & 2 & 10 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[248] \quad 1^{142} 3^{1141} 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 4 & 2 \\ 1 & 0 & 2 & 0 & 11 & 5 & 2 \\ 1 & 2 & 0 & 0 & 11 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 11 & 11 & 0 & 71 & 33 & 15 \\ 4 & 5 & 5 & 0 & 33 & 13 & 7 \\ 2 & 2 & 2 & 0 & 15 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[239] \quad 1^{12} 3^{124} 21^{1214} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 18 & 3 \\ 3 & 5 & 10 & 8 & 30 & 5 \\ 6 & 10 & 14 & 12 & 44 & 8 \\ 4 & 8 & 12 & 8 & 36 & 6 \\ 18 & 30 & 44 & 36 & 134 & 24 \\ 3 & 5 & 8 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[264] $1^{1^{122}} 3^{1^{122}} 21^{1^{122}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 6 & 3 \\ 3 & 5 & 0 & 13 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 20 & 10 \\ 6 & 10 & 0 & 20 & 14 & 8 \\ 3 & 5 & 0 & 10 & 8 & 4 \end{pmatrix}_{b_0}$

[269] $1^{1^{114}} 3^{1^{112}} 21^{1^{112}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 8 & 2 \\ 1 & 0 & 2 & 1 & 4 & 6 & 2 \\ 1 & 2 & 0 & 3 & 5 & 8 & 2 \\ 2 & 1 & 3 & 1 & 5 & 8 & 3 \\ 4 & 4 & 5 & 5 & 10 & 18 & 6 \\ 8 & 6 & 8 & 8 & 18 & 26 & 10 \\ 2 & 2 & 2 & 3 & 6 & 10 & 3 \end{pmatrix}_{b_0}$

[268] $1^{1^{122}} 3^{1^{122}} 21^{1^{122}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 8 & 6 & 3 \\ 3 & 5 & 10 & 13 & 10 & 5 \\ 5 & 10 & 14 & 20 & 16 & 8 \\ 8 & 13 & 20 & 24 & 20 & 10 \\ 6 & 10 & 16 & 20 & 14 & 8 \\ 3 & 5 & 8 & 10 & 8 & 4 \end{pmatrix}_{b_0}$

[267] $1^{1^{12}} 3^{1^{1312}} 21^{1^{1312}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.3$
 $T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 5 & 6 & 3 \\ 3 & 5 & 15 & 8 & 10 & 5 \\ 8 & 15 & 34 & 18 & 24 & 12 \\ 5 & 8 & 18 & 8 & 12 & 6 \\ 6 & 10 & 24 & 12 & 14 & 8 \\ 3 & 5 & 12 & 6 & 8 & 4 \end{pmatrix}_{b_0}$

[265] $1^{1^{122}} 3^{1^{112}} 21^{1^{112}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 6 & 2 \\ 1 & 0 & 2 & 1 & 4 & 6 & 2 \\ 1 & 2 & 0 & 3 & 5 & 8 & 2 \\ 2 & 1 & 3 & 1 & 5 & 8 & 3 \\ 5 & 4 & 5 & 5 & 10 & 18 & 6 \\ 6 & 6 & 8 & 8 & 18 & 26 & 10 \\ 2 & 2 & 2 & 3 & 6 & 10 & 3 \end{pmatrix}_{b_0}$

[266] $1^{1^{132}} 3^{1^{132}} 21^{1^{132}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 14 & 6 & 3 \\ 3 & 5 & 0 & 23 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 14 & 23 & 0 & 78 & 34 & 18 \\ 6 & 10 & 0 & 34 & 14 & 8 \\ 3 & 5 & 0 & 18 & 8 & 4 \end{pmatrix}_{b_0}$

[271] $1^{1^{12}} 3^{1^{1121}} 21^{1^{1121}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -67$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 & 2 \\ 1 & 0 & 2 & 1 & 7 & 3 & 2 \\ 1 & 2 & 0 & 3 & 9 & 4 & 2 \\ 2 & 1 & 3 & 1 & 9 & 4 & 3 \\ 7 & 7 & 9 & 9 & 35 & 16 & 11 \\ 2 & 3 & 4 & 4 & 16 & 5 & 5 \\ 2 & 2 & 2 & 3 & 11 & 5 & 3 \end{pmatrix}_{b_0}$

[270] $1^{1^{111}} 3^{1^{1112}} 21^{1^{1112}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -67$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 5 & 2 \\ 1 & 0 & 2 & 3 & 4 & 4 & 2 \\ 1 & 2 & 0 & 4 & 5 & 7 & 2 \\ 2 & 3 & 4 & 5 & 9 & 11 & 5 \\ 4 & 4 & 5 & 9 & 10 & 14 & 6 \\ 5 & 4 & 7 & 11 & 14 & 16 & 8 \\ 2 & 2 & 2 & 5 & 6 & 8 & 3 \end{pmatrix}_{b_0}$

[272] $1^{\overline{114}} 1^{\overline{1}} 3^{\overline{1}} 21^{\overline{1214}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 12 & 3 \\ 3 & 5 & 10 & 8 & 20 & 5 \\ 5 & 10 & 14 & 12 & 32 & 8 \\ 5 & 8 & 12 & 8 & 24 & 6 \\ 12 & 20 & 32 & 24 & 62 & 16 \\ 3 & 5 & 8 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[277] $1^{\overline{123}} 1^{\overline{1}} 3^{\overline{1}} 21^{\overline{1123}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 9 & 3 \\ 3 & 5 & 5 & 13 & 15 & 5 \\ 2 & 5 & 2 & 10 & 12 & 4 \\ 8 & 13 & 10 & 24 & 30 & 10 \\ 9 & 15 & 12 & 30 & 34 & 12 \\ 3 & 5 & 4 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[276] $1^{\overline{111}} 1^{\overline{1112}} 3^{\overline{1}} 21^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 0 & 2 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 2 & 2 & 1 & 0 & 4 & 5 & 2 \\ 4 & 4 & 5 & 4 & 10 & 13 & 6 \\ 4 & 5 & 5 & 5 & 13 & 13 & 7 \\ 2 & 2 & 2 & 2 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[275] $1^{\overline{111}} 1^{\overline{1}} 3^{\overline{1}} 21^{\overline{1211}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 & 3 \\ 3 & 5 & 10 & 8 & 5 & 5 \\ 7 & 10 & 14 & 12 & 8 & 8 \\ 5 & 8 & 12 & 8 & 6 & 6 \\ 3 & 5 & 8 & 6 & 2 & 4 \\ 3 & 5 & 8 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[274] $1^{\overline{123}} 1^{\overline{1112}} 3^{\overline{1}} 21^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 1 & 2 & 1 & 0 & 4 & 5 & 2 \\ 5 & 4 & 5 & 4 & 10 & 13 & 6 \\ 6 & 5 & 5 & 5 & 13 & 13 & 7 \\ 2 & 2 & 2 & 2 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[273] $1^{\overline{123}} 1^{\overline{1}} 3^{\overline{1}} 21^{\overline{1124}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 17 & 3 \\ 3 & 5 & 5 & 13 & 30 & 5 \\ 2 & 5 & 2 & 10 & 22 & 4 \\ 8 & 13 & 10 & 24 & 58 & 10 \\ 17 & 30 & 22 & 58 & 134 & 24 \\ 3 & 5 & 4 & 10 & 24 & 4 \end{pmatrix}_{b_0}$$

[278] $1^{\overline{111}} 1^{\overline{121}} 3^{\overline{1}} 21^{\overline{1211}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 3 & 3 & 2 \\ 1 & 0 & 2 & 6 & 2 & 3 & 2 \\ 1 & 2 & 0 & 8 & 4 & 4 & 2 \\ 5 & 6 & 8 & 26 & 12 & 14 & 10 \\ 3 & 2 & 4 & 12 & 4 & 6 & 4 \\ 3 & 3 & 4 & 14 & 6 & 5 & 5 \\ 2 & 2 & 2 & 10 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[279] $1^{\overline{31}} 1^{\overline{112}} 3^{\overline{1}} 21^{\overline{1131}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -73, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 & 2 \\ 1 & 0 & 2 & 3 & 8 & 1 & 2 \\ 1 & 2 & 0 & 4 & 12 & 3 & 2 \\ 2 & 3 & 4 & 5 & 20 & 4 & 5 \\ 9 & 8 & 12 & 20 & 56 & 12 & 14 \\ 2 & 1 & 3 & 4 & 12 & 1 & 3 \\ 2 & 2 & 2 & 5 & 14 & 3 & 3 \end{pmatrix}_{b_0}$$

[280] $1^{\overline{41}} 1^{\overline{141}} 3^{\overline{1}} 21^{\overline{1141}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 3 & 2 \\ 1 & 0 & 2 & 1 & 13 & 3 & 2 \\ 1 & 2 & 0 & 3 & 17 & 4 & 2 \\ 1 & 1 & 3 & 1 & 17 & 4 & 3 \\ 13 & 13 & 17 & 17 & 127 & 30 & 21 \\ 3 & 3 & 4 & 4 & 30 & 5 & 5 \\ 2 & 2 & 2 & 3 & 21 & 5 & 3 \end{pmatrix}_{b_0}$$

[281] $1^{\overline{1}^{214}} 3^{\overline{1}^{114}} 21^{\overline{1114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 12 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 4 & 5 & 2 & 6 & 16 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 12 & 20 & 16 & 24 & 62 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[284] $1^{\overline{1}^{113}} 3^{\overline{1}^{114}} 21^{\overline{1113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 7 & 24 & 4 \\ 3 & 1 & 3 & 5 & 18 & 3 \\ 3 & 3 & 1 & 6 & 20 & 3 \\ 7 & 5 & 6 & 9 & 38 & 7 \\ 24 & 18 & 20 & 38 & 130 & 24 \\ 4 & 3 & 3 & 7 & 24 & 4 \end{pmatrix}_{b_0}$$

[283] $1^{\overline{1}^{113}} 3^{\overline{1}^{114}} 21^{\overline{1114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 5 & 0 & 8 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 30 & 6 \\ 14 & 25 & 0 & 30 & 92 & 20 \\ 3 & 5 & 0 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[282] $1^{\overline{1}^{112}} 3^{\overline{1}^{112}} 21^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 5 & 0 & 8 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 18 & 6 \\ 9 & 15 & 0 & 18 & 32 & 12 \\ 3 & 5 & 0 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[286] $1^{\overline{1}^{111}} 3^{\overline{1}^{111}} 21^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 6 & 3 \\ 3 & 5 & 5 & 8 & 10 & 5 \\ 3 & 5 & 2 & 6 & 7 & 4 \\ 5 & 8 & 6 & 8 & 12 & 6 \\ 6 & 10 & 7 & 12 & 14 & 8 \\ 3 & 5 & 4 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[289] $1^{\overline{1}^{21}} 3^{\overline{1}^{12}} 21^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 1 & 2 & 1 & 0 & 7 & 0 & 2 \\ 7 & 7 & 9 & 7 & 35 & 0 & 11 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 2 & 2 & 2 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

[288] $1^{\overline{1}^{21}} 3^{\overline{1}^{121}} 21^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 5 & 0 & 18 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 11 & 18 & 0 & 48 & 13 & 14 \\ 3 & 5 & 0 & 13 & 2 & 4 \\ 3 & 5 & 0 & 14 & 4 & 4 \end{pmatrix}_{b_0}$$

[285] $1^{\overline{1}^{41}} 3^{\overline{1}^{12}} 21^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 1 & 2 \\ 1 & 0 & 2 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 1 & 9 & 0 & 2 \\ 2 & 2 & 1 & 0 & 7 & 0 & 2 \\ 9 & 7 & 9 & 7 & 35 & 0 & 11 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 2 & 2 & 2 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

[287] $1^{111} 1^{112} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 5 & 2 \\ 1 & 0 & 2 & 1 & 4 & 6 & 2 \\ 1 & 2 & 0 & 3 & 5 & 8 & 2 \\ 2 & 1 & 3 & 1 & 5 & 8 & 3 \\ 4 & 4 & 5 & 5 & 10 & 18 & 6 \\ 5 & 6 & 8 & 8 & 18 & 26 & 10 \\ 2 & 2 & 2 & 3 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[301] $1^{111} 3^1 21^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -76$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 5 & 2 \\ 1 & 0 & 2 & 4 & 4 & 4 & 2 \\ 1 & 2 & 0 & 7 & 5 & 7 & 2 \\ 4 & 4 & 7 & 16 & 14 & 18 & 8 \\ 4 & 4 & 5 & 14 & 10 & 14 & 6 \\ 5 & 4 & 7 & 18 & 14 & 16 & 8 \\ 2 & 2 & 2 & 8 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

[299] $1^{113} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -76$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 7 & 21 & 4 \\ 3 & 1 & 3 & 5 & 15 & 3 \\ 3 & 3 & 1 & 6 & 18 & 3 \\ 7 & 5 & 6 & 9 & 33 & 7 \\ 21 & 15 & 18 & 33 & 97 & 21 \\ 4 & 3 & 3 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

[300] $1^{312} 1^1 3^1 21^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -76$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 6 & 3 \\ 3 & 5 & 10 & 8 & 10 & 5 \\ 7 & 10 & 14 & 12 & 16 & 8 \\ 5 & 8 & 12 & 8 & 12 & 6 \\ 6 & 10 & 16 & 12 & 14 & 8 \\ 3 & 5 & 8 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[298] $1^{123} 1^1 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -76$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 9 & 3 \\ 3 & 5 & 0 & 13 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 30 & 10 \\ 9 & 15 & 0 & 30 & 34 & 12 \\ 3 & 5 & 0 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[297] $1^{112} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -76$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 5 & 14 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 28 & 6 \\ 10 & 8 & 14 & 0 & 28 & 70 & 16 \\ 2 & 2 & 2 & 0 & 6 & 16 & 3 \end{pmatrix}_{b_0}$$

[296] $1^{112} 1^{111} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -76$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 4 & 2 \\ 1 & 2 & 0 & 0 & 5 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 14 & 6 \\ 5 & 4 & 7 & 0 & 14 & 16 & 8 \\ 2 & 2 & 2 & 0 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

[295] $1^{123} 1^{113} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -76$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 9 & 2 \\ 1 & 0 & 2 & 1 & 4 & 9 & 2 \\ 1 & 2 & 0 & 3 & 5 & 12 & 2 \\ 2 & 1 & 3 & 1 & 5 & 12 & 3 \\ 5 & 4 & 5 & 5 & 10 & 27 & 6 \\ 9 & 9 & 12 & 12 & 27 & 61 & 15 \\ 2 & 2 & 2 & 3 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

[294] $1^{123} 1^{111} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -76$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$\begin{aligned}
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 4 & 2 \\ 1 & 2 & 0 & 0 & 5 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 14 & 6 \\ 6 & 4 & 7 & 0 & 14 & 16 & 8 \\ 2 & 2 & 2 & 0 & 6 & 8 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^{213} \\ [293] \end{matrix} 1^{\textcolor{blue}{1}} 3^{\textcolor{red}{1}} 21^{\textcolor{red}{1313}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.6 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 8 & 5 & 9 & 3 \\ 3 & 5 & 15 & 8 & 15 & 5 \\ 8 & 15 & 34 & 18 & 36 & 12 \\ 5 & 8 & 18 & 8 & 18 & 6 \\ 9 & 15 & 36 & 18 & 34 & 12 \\ 3 & 5 & 12 & 6 & 12 & 4 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^{123} \\ [292] \end{matrix} 1^{\textcolor{blue}{1}} 3^{\textcolor{red}{1}} 21^{\textcolor{red}{1223}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 8 & 9 & 3 \\ 3 & 5 & 10 & 13 & 15 & 5 \\ 5 & 10 & 14 & 20 & 24 & 8 \\ 8 & 13 & 20 & 24 & 30 & 10 \\ 9 & 15 & 24 & 30 & 34 & 12 \\ 3 & 5 & 8 & 10 & 12 & 4 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^{214} \\ [291] \end{matrix} 1^{\textcolor{red}{113}} 3^{\textcolor{red}{1}} 21^{\textcolor{red}{124}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 3 & 3 & 11 & 25 & 4 \\ 3 & 1 & 3 & 7 & 17 & 3 \\ 3 & 3 & 1 & 10 & 22 & 3 \\ 11 & 7 & 10 & 25 & 61 & 11 \\ 25 & 17 & 22 & 61 & 137 & 25 \\ 4 & 3 & 3 & 11 & 25 & 4 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^{312} \\ [290] \end{matrix} 1^{\textcolor{red}{1111}} 3^{\textcolor{red}{1}} 21^{\textcolor{red}{1212}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 5 & 2 \\ 1 & 0 & 2 & 4 & 4 & 4 & 2 \\ 1 & 2 & 0 & 7 & 5 & 7 & 2 \\ 6 & 4 & 7 & 16 & 14 & 18 & 8 \\ 4 & 4 & 5 & 14 & 10 & 14 & 6 \\ 5 & 4 & 7 & 18 & 14 & 16 & 8 \\ 2 & 2 & 2 & 8 & 6 & 8 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^{111} \\ [308] \end{matrix} 1^{\textcolor{red}{111}} 3^{\textcolor{red}{1}} 21^{\textcolor{red}{1111}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 3 & 2 \\ 1 & 0 & 2 & 1 & 4 & 3 & 2 \\ 1 & 2 & 0 & 3 & 5 & 4 & 2 \\ 2 & 1 & 3 & 1 & 5 & 4 & 3 \\ 4 & 4 & 5 & 5 & 10 & 9 & 6 \\ 3 & 3 & 4 & 4 & 9 & 5 & 5 \\ 2 & 2 & 2 & 3 & 6 & 5 & 3 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^{122} \\ [307] \end{matrix} 1^{\textcolor{red}{11}} 3^{\textcolor{red}{1}} 21^{\textcolor{red}{122}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 8 & 9 & 3 \\ 3 & 5 & 0 & 13 & 15 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 29 & 10 \\ 9 & 15 & 0 & 29 & 32 & 12 \\ 3 & 5 & 0 & 10 & 12 & 4 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^{213} \\ [305] \end{matrix} 1^{\textcolor{red}{11}} 3^{\textcolor{red}{1}} 21^{\textcolor{red}{1314}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 8 & 5 & 14 & 3 \\ 3 & 5 & 15 & 8 & 25 & 5 \\ 8 & 15 & 34 & 18 & 57 & 12 \\ 5 & 8 & 18 & 8 & 30 & 6 \\ 14 & 25 & 57 & 30 & 92 & 20 \\ 3 & 5 & 12 & 6 & 20 & 4 \end{pmatrix}_{b_0} \\
& \begin{matrix} 1^{124} \\ [304] \end{matrix} 1^{\textcolor{red}{114}} 3^{\textcolor{red}{1}} 21^{\textcolor{red}{1113}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 11 & 2 \\ 1 & 0 & 2 & 1 & 4 & 11 & 2 \\ 1 & 2 & 0 & 3 & 5 & 13 & 2 \\ 2 & 1 & 3 & 1 & 5 & 13 & 3 \\ 5 & 4 & 5 & 5 & 10 & 31 & 6 \\ 11 & 11 & 13 & 13 & 31 & 81 & 17 \\ 2 & 2 & 2 & 3 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

[303] $1^{11} 3^{124} 21^{1224} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -77$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 15 & 3 \\ 3 & 5 & 10 & 13 & 25 & 5 \\ 5 & 10 & 14 & 20 & 38 & 8 \\ 8 & 13 & 20 & 24 & 49 & 10 \\ 15 & 25 & 38 & 49 & 92 & 20 \\ 3 & 5 & 8 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

[302] $1^{11} 3^{122} 21^{1223} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -77$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 11 & 3 \\ 3 & 5 & 10 & 13 & 20 & 5 \\ 5 & 10 & 14 & 20 & 30 & 8 \\ 8 & 13 & 20 & 24 & 39 & 10 \\ 11 & 20 & 30 & 39 & 58 & 16 \\ 3 & 5 & 8 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[306] $1^{111} 3^{122} 21^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -77$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 4 & 2 \\ 1 & 0 & 2 & 1 & 4 & 3 & 2 \\ 1 & 2 & 0 & 3 & 5 & 4 & 2 \\ 2 & 1 & 3 & 1 & 5 & 4 & 3 \\ 5 & 4 & 5 & 5 & 10 & 9 & 6 \\ 4 & 3 & 4 & 4 & 9 & 5 & 5 \\ 2 & 2 & 2 & 3 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[309] $1^{1111} 3^{1221} 21^{1211} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -78$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 5 & 3 & 2 \\ 1 & 0 & 2 & 4 & 4 & 3 & 2 \\ 1 & 2 & 0 & 7 & 5 & 4 & 2 \\ 5 & 4 & 7 & 16 & 14 & 11 & 8 \\ 5 & 4 & 5 & 14 & 10 & 9 & 6 \\ 3 & 3 & 4 & 11 & 9 & 5 & 5 \\ 2 & 2 & 2 & 8 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[310] $1^{1111} 3^{1221} 21^{1211} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -78$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 3 & 2 \\ 1 & 0 & 2 & 4 & 4 & 3 & 2 \\ 1 & 2 & 0 & 7 & 5 & 4 & 2 \\ 4 & 4 & 7 & 16 & 14 & 11 & 8 \\ 4 & 4 & 5 & 14 & 10 & 9 & 6 \\ 3 & 3 & 4 & 11 & 9 & 5 & 5 \\ 2 & 2 & 2 & 8 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[311] $1^{1221} 3^{1321} 21^{1321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -78$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 & 3 \\ 3 & 5 & 15 & 13 & 5 & 5 \\ 8 & 15 & 34 & 30 & 12 & 12 \\ 8 & 13 & 30 & 24 & 10 & 10 \\ 3 & 5 & 12 & 10 & 2 & 4 \\ 3 & 5 & 12 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

[312] $1^{1221} 3^{1121} 21^{1121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -78$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 & 3 \\ 3 & 5 & 5 & 13 & 5 & 5 \\ 4 & 5 & 2 & 10 & 4 & 4 \\ 8 & 13 & 10 & 24 & 10 & 10 \\ 3 & 5 & 4 & 10 & 2 & 4 \\ 3 & 5 & 4 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

[313] $1^{1111} 3^{1211} 21^{1211} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -78$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 4 & 2 \\ 1 & 0 & 2 & 4 & 4 & 3 & 2 \\ 1 & 2 & 0 & 7 & 5 & 4 & 2 \\ 6 & 4 & 7 & 16 & 14 & 11 & 8 \\ 4 & 4 & 5 & 14 & 10 & 9 & 6 \\ 4 & 3 & 4 & 11 & 9 & 5 & 5 \\ 2 & 2 & 2 & 8 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[316] $1^{112} 3^{1213} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 11 & 3 \\ 3 & 5 & 10 & 8 & 20 & 5 \\ 5 & 10 & 14 & 12 & 30 & 8 \\ 5 & 8 & 12 & 8 & 24 & 6 \\ 11 & 20 & 30 & 24 & 58 & 16 \\ 3 & 5 & 8 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[315] $1^{124} 3^{1213} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 14 & 24 & 4 \\ 3 & 1 & 3 & 10 & 18 & 3 \\ 3 & 3 & 1 & 12 & 20 & 3 \\ 14 & 10 & 12 & 42 & 76 & 14 \\ 24 & 18 & 20 & 76 & 130 & 24 \\ 4 & 3 & 3 & 14 & 24 & 4 \end{pmatrix}_{b_0}$$

[314] $1^{131} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 1 & 2 \\ 1 & 0 & 2 & 1 & 7 & 0 & 2 \\ 1 & 2 & 0 & 3 & 9 & 0 & 2 \\ 2 & 1 & 3 & 1 & 9 & 0 & 3 \\ 8 & 7 & 9 & 9 & 35 & 0 & 11 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 11 & 0 & 3 \end{pmatrix}_{b_0}$$

[317] $1^{113} 3^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 12 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 2 & 5 & 2 & 6 & 15 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 12 & 20 & 15 & 24 & 58 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[318] $1^{112} 3^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 11 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 2 & 5 & 2 & 6 & 15 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 11 & 20 & 15 & 24 & 58 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[319] $1^{123} 3^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 4 & 2 \\ 1 & 2 & 0 & 0 & 8 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 21 & 9 \\ 6 & 4 & 7 & 0 & 21 & 16 & 8 \\ 2 & 2 & 2 & 0 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[320] $1^{121} 3^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 2 & 0 & 5 & 1 & 2 \\ 1 & 2 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 8 & 9 \\ 2 & 1 & 3 & 0 & 8 & 1 & 3 \\ 2 & 2 & 2 & 0 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

[321] $1^{132} 3^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 4 & 2 \\ 1 & 2 & 0 & 0 & 8 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 5 & 8 & 0 & 23 & 21 & 9 \\ 5 & 4 & 7 & 0 & 21 & 16 & 8 \\ 2 & 2 & 2 & 0 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[322] $1^{11}3^{11}21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 2 & 0 & 5 & 1 & 2 \\ 1 & 2 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 5 & 8 & 0 & 23 & 8 & 9 \\ 2 & 1 & 3 & 0 & 8 & 1 & 3 \\ 2 & 2 & 2 & 0 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

[323] $1^{124}3^{11}21^{1124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 12 & 3 \\ 3 & 5 & 5 & 13 & 20 & 5 \\ 2 & 5 & 2 & 10 & 16 & 4 \\ 8 & 13 & 10 & 24 & 40 & 10 \\ 12 & 20 & 16 & 40 & 62 & 16 \\ 3 & 5 & 4 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[328] $1^{1112}3^{11}21^{1212} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 5 & 6 & 2 \\ 1 & 0 & 2 & 4 & 4 & 6 & 2 \\ 1 & 2 & 0 & 7 & 5 & 8 & 2 \\ 5 & 4 & 7 & 16 & 14 & 22 & 8 \\ 5 & 4 & 5 & 14 & 10 & 18 & 6 \\ 6 & 6 & 8 & 22 & 18 & 26 & 10 \\ 2 & 2 & 2 & 8 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[327] $1^{122}3^{11}21^{1322} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 6 & 3 \\ 3 & 5 & 15 & 13 & 10 & 5 \\ 8 & 15 & 34 & 30 & 24 & 12 \\ 8 & 13 & 30 & 24 & 20 & 10 \\ 6 & 10 & 24 & 20 & 14 & 8 \\ 3 & 5 & 12 & 10 & 8 & 4 \end{pmatrix}_{b_0}$$

[326] $1^{112}3^{11}21^{1212} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 6 & 2 \\ 1 & 0 & 2 & 4 & 4 & 6 & 2 \\ 1 & 2 & 0 & 7 & 5 & 8 & 2 \\ 4 & 4 & 7 & 16 & 14 & 22 & 8 \\ 4 & 4 & 5 & 14 & 10 & 18 & 6 \\ 6 & 6 & 8 & 22 & 18 & 26 & 10 \\ 2 & 2 & 2 & 8 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[325] $1^{112}3^{11}21^{1113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 14 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 2 & 5 & 2 & 6 & 18 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 14 & 25 & 18 & 30 & 94 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[324] $1^{222}3^{11}21^{1122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 6 & 3 \\ 3 & 5 & 5 & 13 & 10 & 5 \\ 4 & 5 & 2 & 10 & 8 & 4 \\ 8 & 13 & 10 & 24 & 20 & 10 \\ 6 & 10 & 8 & 20 & 14 & 8 \\ 3 & 5 & 4 & 10 & 8 & 4 \end{pmatrix}_{b_0}$$

[330] $1^{124}3^{11}21^{1113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 9 & 2 \\ 1 & 0 & 2 & 1 & 4 & 7 & 2 \\ 1 & 2 & 0 & 3 & 5 & 11 & 2 \\ 2 & 1 & 3 & 1 & 5 & 11 & 3 \\ 5 & 4 & 5 & 5 & 10 & 23 & 6 \\ 9 & 7 & 11 & 11 & 23 & 45 & 13 \\ 2 & 2 & 2 & 3 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[331] $1^{124} 1^{11} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 13 & 3 \\ 3 & 5 & 0 & 13 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 39 & 10 \\ 13 & 20 & 0 & 39 & 58 & 16 \\ 3 & 5 & 0 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[332] $1^{112} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 7 & 2 \\ 1 & 0 & 2 & 1 & 4 & 7 & 2 \\ 1 & 2 & 0 & 3 & 5 & 11 & 2 \\ 2 & 1 & 3 & 1 & 5 & 11 & 3 \\ 4 & 4 & 5 & 5 & 10 & 23 & 6 \\ 7 & 7 & 11 & 11 & 23 & 45 & 13 \\ 2 & 2 & 2 & 3 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[333] $1^{114} 1^{112} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 9 & 2 \\ 1 & 0 & 2 & 1 & 4 & 7 & 2 \\ 1 & 2 & 0 & 3 & 5 & 11 & 2 \\ 2 & 1 & 3 & 1 & 5 & 11 & 3 \\ 4 & 4 & 5 & 5 & 10 & 23 & 6 \\ 9 & 7 & 11 & 11 & 23 & 45 & 13 \\ 2 & 2 & 2 & 3 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[329] $1^{212} 1^{11} 3^1 21^{1314} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 13 & 3 \\ 3 & 5 & 15 & 8 & 25 & 5 \\ 8 & 15 & 34 & 18 & 57 & 12 \\ 5 & 8 & 18 & 8 & 30 & 6 \\ 13 & 25 & 57 & 30 & 92 & 20 \\ 3 & 5 & 12 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[334] $1^{214} 1^{12} 3^1 21^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 18 & 3 \\ 3 & 5 & 5 & 8 & 30 & 5 \\ 4 & 5 & 2 & 6 & 22 & 4 \\ 5 & 8 & 6 & 8 & 36 & 6 \\ 18 & 30 & 22 & 36 & 134 & 24 \\ 3 & 5 & 4 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[335] $1^{112} 1^{11} 3^1 21^{1214} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 13 & 3 \\ 3 & 5 & 10 & 8 & 25 & 5 \\ 5 & 10 & 14 & 12 & 38 & 8 \\ 5 & 8 & 12 & 8 & 30 & 6 \\ 13 & 25 & 38 & 30 & 92 & 20 \\ 3 & 5 & 8 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[336] $1^{41} 1^{13} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 1 & 10 & 0 & 2 \\ 1 & 2 & 1 & 0 & 9 & 0 & 2 \\ 10 & 9 & 10 & 9 & 51 & 0 & 13 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 13 & 0 & 3 \end{pmatrix}_{b_0}$$

[337] $1^{41} 1^{13} 3^1 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 1 & 2 \\ 1 & 0 & 2 & 2 & 11 & 0 & 2 \\ 1 & 2 & 0 & 1 & 16 & 0 & 2 \\ 1 & 2 & 1 & 0 & 11 & 0 & 2 \\ 12 & 11 & 16 & 11 & 103 & 0 & 19 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 19 & 0 & 3 \end{pmatrix}_{b_0}$$

[338] $1^{131} 1^{113} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 1 & 2 \\ 1 & 0 & 2 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 1 & 10 & 0 & 2 \\ 2 & 2 & 1 & 0 & 9 & 0 & 2 \\ 9 & 9 & 10 & 9 & 51 & 0 & 13 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 13 & 0 & 3 \end{pmatrix}_{b_0}$$

[339] $1^{\overline{111}} 1^{\overline{1}} 3^{\overline{1}} 21^{\overline{1111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -88$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 & 3 \\ 3 & 5 & 5 & 8 & 5 & 5 \\ 3 & 5 & 2 & 6 & 4 & 4 \\ 5 & 8 & 6 & 8 & 6 & 6 \\ 3 & 5 & 4 & 6 & 2 & 4 \\ 3 & 5 & 4 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[340] $1^{\overline{114}} 1^{\overline{113}} 3^{\overline{1}} 21^{\overline{114}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -88$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 7 & 25 & 4 \\ 3 & 1 & 3 & 5 & 17 & 3 \\ 3 & 3 & 1 & 6 & 22 & 3 \\ 7 & 5 & 6 & 9 & 39 & 7 \\ 25 & 17 & 22 & 39 & 137 & 25 \\ 4 & 3 & 3 & 7 & 25 & 4 \end{pmatrix}_{b_0}$$

[344] $1^{\overline{124}} 1^{\overline{1}} 3^{\overline{1}} 21^{\overline{124}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -90$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 5 & 0 & 13 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 40 & 10 \\ 12 & 20 & 0 & 40 & 62 & 16 \\ 3 & 5 & 0 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[343] $1^{\overline{124}} 1^{\overline{1}} 3^{\overline{1}} 21^{\overline{124}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -90$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 12 & 3 \\ 3 & 5 & 10 & 13 & 20 & 5 \\ 5 & 10 & 14 & 20 & 32 & 8 \\ 8 & 13 & 20 & 24 & 40 & 10 \\ 12 & 20 & 32 & 40 & 62 & 16 \\ 3 & 5 & 8 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[341] $1^{\overline{124}} 1^{\overline{114}} 3^{\overline{1}} 21^{\overline{114}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -90$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 12 & 2 \\ 1 & 0 & 2 & 1 & 4 & 12 & 2 \\ 1 & 2 & 0 & 3 & 5 & 16 & 2 \\ 2 & 1 & 3 & 1 & 5 & 16 & 3 \\ 5 & 4 & 5 & 5 & 10 & 36 & 6 \\ 12 & 12 & 16 & 16 & 36 & 110 & 20 \\ 2 & 2 & 2 & 3 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

[342] $1^{\overline{124}} 1^{\overline{1}} 3^{\overline{1}} 21^{\overline{1314}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -90$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.8$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 12 & 3 \\ 3 & 5 & 15 & 8 & 20 & 5 \\ 8 & 15 & 34 & 18 & 48 & 12 \\ 5 & 8 & 18 & 8 & 24 & 6 \\ 12 & 20 & 48 & 24 & 62 & 16 \\ 3 & 5 & 12 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[345] $1^{\overline{111}} 1^{\overline{112}} 3^{\overline{1}} 21^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -91$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 13 & 6 \\ 4 & 5 & 5 & 0 & 13 & 13 & 7 \\ 2 & 2 & 2 & 0 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[346] $1^{\overline{123}} 1^{\overline{112}} 3^{\overline{1}} 21^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -91$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 13 & 6 \\ 6 & 5 & 5 & 0 & 13 & 13 & 7 \\ 2 & 2 & 2 & 0 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[347] $1^{123} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 6 & 2 \\ 1 & 0 & 2 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 2 & 2 & 1 & 0 & 4 & 5 & 2 \\ 5 & 4 & 5 & 4 & 10 & 13 & 6 \\ 6 & 5 & 5 & 5 & 13 & 13 & 7 \\ 2 & 2 & 2 & 2 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[348] $1^{112} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 1 & 2 & 1 & 0 & 4 & 5 & 2 \\ 4 & 4 & 5 & 4 & 10 & 13 & 6 \\ 4 & 5 & 5 & 5 & 13 & 13 & 7 \\ 2 & 2 & 2 & 2 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[349] $1^{121} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 9 & 4 & 2 \\ 1 & 2 & 1 & 0 & 7 & 3 & 2 \\ 9 & 7 & 9 & 7 & 35 & 16 & 11 \\ 4 & 3 & 4 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[350] $1^{121} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 3 & 2 \\ 1 & 0 & 2 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 9 & 4 & 2 \\ 2 & 2 & 1 & 0 & 7 & 3 & 2 \\ 7 & 7 & 9 & 7 & 35 & 16 & 11 \\ 3 & 3 & 4 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[351] $1^{12} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 & 2 \\ 1 & 0 & 2 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 9 & 4 & 2 \\ 1 & 2 & 1 & 0 & 7 & 3 & 2 \\ 7 & 7 & 9 & 7 & 35 & 16 & 11 \\ 2 & 3 & 4 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[352] $1^{31} 3^1 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 & 3 \\ 3 & 5 & 5 & 18 & 5 & 5 \\ 2 & 5 & 2 & 14 & 4 & 4 \\ 11 & 18 & 14 & 48 & 14 & 14 \\ 3 & 5 & 4 & 14 & 2 & 4 \\ 3 & 5 & 4 & 14 & 4 & 4 \end{pmatrix}_{b_0}$$

[353] $1^{213} 3^1 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -93, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 10 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 4 & 5 & 2 & 6 & 11 & 4 \\ 5 & 8 & 6 & 8 & 18 & 6 \\ 10 & 15 & 11 & 18 & 32 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[357] $1^{214} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 14 & 25 & 4 \\ 3 & 1 & 3 & 10 & 17 & 3 \\ 3 & 3 & 1 & 12 & 22 & 3 \\ 14 & 10 & 12 & 42 & 78 & 14 \\ 25 & 17 & 22 & 78 & 137 & 25 \\ 4 & 3 & 3 & 14 & 25 & 4 \end{pmatrix}_{b_0}$$

[354] $1^{113} 3^1 21^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$\begin{aligned}
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 5 & 11 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 3 & 5 & 2 & 6 & 16 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 11 & 20 & 16 & 24 & 62 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0} \\
& [355] \ 1^{\overset{131}{1}} 3^{\overset{1}{1}} 21^{\overset{1231}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 11 & 3 & 3 \\ 3 & 5 & 10 & 18 & 5 & 5 \\ 5 & 10 & 14 & 28 & 8 & 8 \\ 11 & 18 & 28 & 48 & 14 & 14 \\ 3 & 5 & 8 & 14 & 2 & 4 \\ 3 & 5 & 8 & 14 & 4 & 4 \end{pmatrix}_{b_0} \\
& [356] \ 1^{\overset{122}{1111}} 3^{\overset{1}{1}} 21^{\overset{1122}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 5 & 2 \\ 1 & 0 & 2 & 3 & 5 & 4 & 2 \\ 1 & 2 & 0 & 4 & 8 & 7 & 2 \\ 2 & 3 & 4 & 5 & 13 & 11 & 5 \\ 6 & 5 & 8 & 13 & 23 & 21 & 9 \\ 5 & 4 & 7 & 11 & 21 & 16 & 8 \\ 2 & 2 & 2 & 5 & 9 & 8 & 3 \end{pmatrix}_{b_0} \\
& [358] \ 1^{\overset{213}{123}} 3^{\overset{1}{1}} 21^{\overset{123}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 3 & 3 & 14 & 21 & 4 \\ 3 & 1 & 3 & 10 & 15 & 3 \\ 3 & 3 & 1 & 12 & 18 & 3 \\ 14 & 10 & 12 & 42 & 66 & 14 \\ 21 & 15 & 18 & 66 & 97 & 21 \\ 4 & 3 & 3 & 14 & 21 & 4 \end{pmatrix}_{b_0} \\
& [362] \ 1^{\overset{122}{112}} 3^{\overset{1}{1}} 21^{\overset{111}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 13 & 6 \\ 5 & 5 & 5 & 0 & 13 & 13 & 7 \\ 2 & 2 & 2 & 0 & 6 & 7 & 3 \end{pmatrix}_{b_0} \\
& [359] \ 1^{\overset{11}{1111}} 3^{\overset{1}{1}} 21^{\overset{111}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 2 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 5 & 4 & 2 \\ 1 & 2 & 1 & 0 & 4 & 3 & 2 \\ 4 & 4 & 5 & 4 & 10 & 9 & 6 \\ 3 & 3 & 4 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0} \\
& [360] \ 1^{\overset{122}{1111}} 3^{\overset{1}{1}} 21^{\overset{111}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 4 & 2 \\ 1 & 0 & 2 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 5 & 4 & 2 \\ 2 & 2 & 1 & 0 & 4 & 3 & 2 \\ 5 & 4 & 5 & 4 & 10 & 9 & 6 \\ 4 & 3 & 4 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 2 & 6 & 5 & 3 \end{pmatrix}_{b_0} \\
& [361] \ 1^{\overset{112}{112}} 3^{\overset{1}{1}} 21^{\overset{111}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 13 & 6 \\ 5 & 5 & 5 & 0 & 13 & 13 & 7 \\ 2 & 2 & 2 & 0 & 6 & 7 & 3 \end{pmatrix}_{b_0} \\
& [367] \ 1^{\overset{112}{122}} 3^{\overset{1}{1}} 21^{\overset{1212}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 3 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 4 & 8 & 2 \\ 5 & 6 & 8 & 26 & 12 & 28 & 10 \\ 3 & 2 & 4 & 12 & 4 & 12 & 4 \\ 6 & 6 & 8 & 28 & 12 & 26 & 10 \\ 2 & 2 & 2 & 10 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[373] $1^{1121} 3^{122} 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 9 & 4 & 2 \\ 1 & 2 & 1 & 0 & 7 & 3 & 2 \\ 7 & 7 & 9 & 7 & 35 & 16 & 11 \\ 4 & 3 & 4 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[374] $1^{1121} 3^{141} 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 3 & 2 \\ 1 & 0 & 2 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 9 & 4 & 2 \\ 2 & 2 & 1 & 0 & 7 & 3 & 2 \\ 9 & 7 & 9 & 7 & 35 & 16 & 11 \\ 3 & 3 & 4 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[364] $1^{113} 3^{113} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 9 & 2 \\ 1 & 0 & 2 & 3 & 2 & 9 & 2 \\ 1 & 2 & 0 & 4 & 4 & 12 & 2 \\ 2 & 3 & 4 & 5 & 6 & 21 & 5 \\ 3 & 2 & 4 & 6 & 4 & 18 & 4 \\ 9 & 9 & 12 & 21 & 18 & 61 & 15 \\ 2 & 2 & 2 & 5 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[372] $1^{1121} 3^{111} 21^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 5 & 2 \\ 1 & 0 & 2 & 4 & 4 & 6 & 2 \\ 1 & 2 & 0 & 7 & 5 & 8 & 2 \\ 5 & 4 & 7 & 16 & 14 & 22 & 8 \\ 4 & 4 & 5 & 14 & 10 & 18 & 6 \\ 5 & 6 & 8 & 22 & 18 & 26 & 10 \\ 2 & 2 & 2 & 8 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[371] $1^{111} 3^{121} 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 4 & 4 & 2 \\ 4 & 3 & 4 & 5 & 6 & 7 & 5 \\ 3 & 2 & 4 & 6 & 4 & 6 & 4 \\ 3 & 3 & 4 & 7 & 6 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[370] $1^{1121} 3^{122} 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 5 & 2 \\ 1 & 0 & 2 & 3 & 7 & 4 & 2 \\ 1 & 2 & 0 & 4 & 9 & 7 & 2 \\ 2 & 3 & 4 & 5 & 16 & 11 & 5 \\ 7 & 7 & 9 & 16 & 35 & 25 & 11 \\ 5 & 4 & 7 & 11 & 25 & 16 & 8 \\ 2 & 2 & 2 & 5 & 11 & 8 & 3 \end{pmatrix}_{b_0}$$

[369] $1^{113} 3^{112} 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 0 & 4 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 12 & 4 \\ 7 & 6 & 8 & 0 & 12 & 26 & 10 \\ 2 & 2 & 2 & 0 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[368] $1^{114} 3^{141} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.92 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 1 & 2 \\ 1 & 0 & 2 & 2 & 12 & 0 & 2 \\ 1 & 2 & 0 & 1 & 14 & 0 & 2 \\ 2 & 2 & 1 & 0 & 12 & 0 & 2 \\ 12 & 12 & 14 & 12 & 96 & 0 & 18 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 18 & 0 & 3 \end{pmatrix}_{b_0}$$

[363] $1^{1112} 3^1 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 6 & 2 \\ 1 & 0 & 2 & 3 & 4 & 5 & 2 \\ 1 & 2 & 0 & 4 & 5 & 5 & 2 \\ 3 & 3 & 4 & 5 & 9 & 10 & 5 \\ 5 & 4 & 5 & 9 & 10 & 13 & 6 \\ 6 & 5 & 5 & 10 & 13 & 13 & 7 \\ 2 & 2 & 2 & 5 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[366] $1^{1112} 3^1 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 4 & 2 \\ 1 & 0 & 2 & 3 & 4 & 5 & 2 \\ 1 & 2 & 0 & 4 & 5 & 5 & 2 \\ 4 & 3 & 4 & 5 & 9 & 10 & 5 \\ 4 & 4 & 5 & 9 & 10 & 13 & 6 \\ 4 & 5 & 5 & 10 & 13 & 13 & 7 \\ 2 & 2 & 2 & 5 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[365] $1^{123} 1^{12} 3^1 21^{1124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 17 & 3 \\ 3 & 5 & 5 & 13 & 30 & 5 \\ 3 & 5 & 2 & 10 & 22 & 4 \\ 8 & 13 & 10 & 24 & 58 & 10 \\ 17 & 30 & 22 & 58 & 134 & 24 \\ 3 & 5 & 4 & 10 & 24 & 4 \end{pmatrix}_{b_0}$$

[375] $1^1 3^1 21^{1132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 6 & 3 \\ 3 & 5 & 5 & 18 & 10 & 5 \\ 2 & 5 & 2 & 14 & 8 & 4 \\ 11 & 18 & 14 & 48 & 28 & 14 \\ 6 & 10 & 8 & 28 & 14 & 8 \\ 3 & 5 & 4 & 14 & 8 & 4 \end{pmatrix}_{b_0}$$

[380] $1^{141} 1^{121} 3^1 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -99$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 3 & 2 \\ 1 & 0 & 2 & 1 & 7 & 3 & 2 \\ 1 & 2 & 0 & 3 & 9 & 4 & 2 \\ 2 & 1 & 3 & 1 & 9 & 4 & 3 \\ 9 & 7 & 9 & 9 & 35 & 16 & 11 \\ 3 & 3 & 4 & 4 & 16 & 5 & 5 \\ 2 & 2 & 2 & 3 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[377] $1^{231} 1^{11} 3^1 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -99$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 & 3 \\ 3 & 5 & 5 & 23 & 5 & 5 \\ 4 & 5 & 2 & 17 & 4 & 4 \\ 14 & 23 & 17 & 78 & 17 & 18 \\ 3 & 5 & 4 & 17 & 2 & 4 \\ 3 & 5 & 4 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[378] $1^{213} 1^{11} 3^1 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -99$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 12 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 4 & 5 & 2 & 6 & 15 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 12 & 20 & 15 & 24 & 58 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[379] $1^{122} 1^{121} 3^1 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -99$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 4 & 2 \\ 1 & 0 & 2 & 1 & 7 & 3 & 2 \\ 1 & 2 & 0 & 3 & 9 & 4 & 2 \\ 2 & 1 & 3 & 1 & 9 & 4 & 3 \\ 7 & 7 & 9 & 9 & 35 & 16 & 11 \\ 4 & 3 & 4 & 4 & 16 & 5 & 5 \\ 2 & 2 & 2 & 3 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[376] $1^{112} 1^{113} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 9 & 2 \\ 1 & 0 & 2 & 1 & 4 & 10 & 2 \\ 1 & 2 & 0 & 3 & 5 & 15 & 2 \\ 2 & 1 & 3 & 1 & 5 & 15 & 3 \\ 4 & 4 & 5 & 5 & 10 & 32 & 6 \\ 9 & 10 & 15 & 15 & 32 & 88 & 18 \\ 2 & 2 & 2 & 3 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

[382] $1^{112} 1^1 3^1 21^{1412} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 6 & 3 \\ 3 & 5 & 20 & 8 & 10 & 5 \\ 11 & 20 & 62 & 24 & 32 & 16 \\ 5 & 8 & 24 & 8 & 12 & 6 \\ 6 & 10 & 32 & 12 & 14 & 8 \\ 3 & 5 & 16 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[381] $1^{132} 1^1 3^1 21^{1232} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 6 & 3 \\ 3 & 5 & 10 & 18 & 10 & 5 \\ 5 & 10 & 14 & 28 & 16 & 8 \\ 11 & 18 & 28 & 48 & 28 & 14 \\ 6 & 10 & 16 & 28 & 14 & 8 \\ 3 & 5 & 8 & 14 & 8 & 4 \end{pmatrix}_{b_0}$$

[383] $1^{111} 1^{111} 3^1 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 3 & 2 \\ 1 & 0 & 2 & 3 & 5 & 3 & 2 \\ 1 & 2 & 0 & 4 & 8 & 4 & 2 \\ 2 & 3 & 4 & 5 & 13 & 7 & 5 \\ 6 & 5 & 8 & 13 & 23 & 13 & 9 \\ 3 & 3 & 4 & 7 & 13 & 5 & 5 \\ 2 & 2 & 2 & 5 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[384] $1^{114} 1^{11} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 5 & 0 & 8 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 24 & 6 \\ 13 & 20 & 0 & 24 & 58 & 16 \\ 3 & 5 & 0 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[385] $1^{12} 1^{12} 3^1 21^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -102, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 18 & 3 \\ 3 & 5 & 5 & 8 & 30 & 5 \\ 2 & 5 & 2 & 6 & 22 & 4 \\ 5 & 8 & 6 & 8 & 36 & 6 \\ 18 & 30 & 22 & 36 & 134 & 24 \\ 3 & 5 & 4 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[386] $1^{12} 1^{12} 3^1 21^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -102, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 16 & 3 \\ 3 & 5 & 5 & 8 & 30 & 5 \\ 2 & 5 & 2 & 6 & 22 & 4 \\ 5 & 8 & 6 & 8 & 36 & 6 \\ 16 & 30 & 22 & 36 & 134 & 24 \\ 3 & 5 & 4 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[387] $1^{131} 1^1 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -102, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 & 3 \\ 3 & 5 & 0 & 18 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 11 & 18 & 0 & 48 & 14 & 14 \\ 3 & 5 & 0 & 14 & 2 & 4 \\ 3 & 5 & 0 & 14 & 4 & 4 \end{pmatrix}_{b_0}$$

[388] $1^{122} 1^{111} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 5 & 2 \\ 1 & 0 & 2 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 2 & 2 & 1 & 0 & 4 & 5 & 2 \\ 5 & 4 & 5 & 4 & 10 & 13 & 6 \\ 5 & 5 & 5 & 5 & 13 & 13 & 7 \\ 2 & 2 & 2 & 2 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[389] $1^{111} 3^1 21^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 3 & 2 \\ 1 & 2 & 0 & 0 & 5 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 9 & 6 \\ 3 & 3 & 4 & 0 & 9 & 5 & 5 \\ 2 & 2 & 2 & 0 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[390] $1^{122} 3^1 21^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 3 & 2 \\ 1 & 2 & 0 & 0 & 5 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 9 & 6 \\ 4 & 3 & 4 & 0 & 9 & 5 & 5 \\ 2 & 2 & 2 & 0 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[391] $1^{112} 3^1 21^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 1 & 2 & 1 & 0 & 4 & 5 & 2 \\ 4 & 4 & 5 & 4 & 10 & 13 & 6 \\ 5 & 5 & 5 & 5 & 13 & 13 & 7 \\ 2 & 2 & 2 & 2 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[392] $1^{23} 3^1 21^{123} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 12 & 3 \\ 3 & 5 & 5 & 13 & 20 & 5 \\ 2 & 5 & 2 & 10 & 15 & 4 \\ 8 & 13 & 10 & 24 & 39 & 10 \\ 12 & 20 & 15 & 39 & 58 & 16 \\ 3 & 5 & 4 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[393] $1^{123} 3^1 21^{124} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 14 & 3 \\ 3 & 5 & 5 & 13 & 25 & 5 \\ 2 & 5 & 2 & 10 & 19 & 4 \\ 8 & 13 & 10 & 24 & 49 & 10 \\ 14 & 25 & 19 & 49 & 92 & 20 \\ 3 & 5 & 4 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

[394] $1^{123} 3^1 21^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 2 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 1 & 2 & 1 & 0 & 4 & 6 & 2 \\ 5 & 4 & 5 & 4 & 10 & 18 & 6 \\ 7 & 6 & 8 & 6 & 18 & 26 & 10 \\ 2 & 2 & 2 & 2 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[395] $1^{113} 3^1 21^{112} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 7 & 2 \\ 1 & 0 & 2 & 2 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 2 & 2 & 1 & 0 & 4 & 6 & 2 \\ 4 & 4 & 5 & 4 & 10 & 18 & 6 \\ 7 & 6 & 8 & 6 & 18 & 26 & 10 \\ 2 & 2 & 2 & 2 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[396] $1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 0 & 4 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 18 & 4 \\ 9 & 9 & 12 & 0 & 18 & 61 & 15 \\ 2 & 2 & 2 & 0 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[397] $1^{113} 3^{114} 21^{1114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 11 & 2 \\ 1 & 0 & 2 & 3 & 2 & 12 & 2 \\ 1 & 2 & 0 & 4 & 4 & 16 & 2 \\ 2 & 3 & 4 & 5 & 6 & 28 & 5 \\ 3 & 2 & 4 & 6 & 4 & 24 & 4 \\ 11 & 12 & 16 & 28 & 24 & 110 & 20 \\ 2 & 2 & 2 & 5 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[398] $1^{112} 3^{1112} 21^{1112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 6 & 3 \\ 3 & 5 & 5 & 8 & 10 & 5 \\ 3 & 5 & 2 & 6 & 8 & 4 \\ 5 & 8 & 6 & 8 & 12 & 6 \\ 6 & 10 & 8 & 12 & 14 & 8 \\ 3 & 5 & 4 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[399] $1^{212} 3^{1112} 21^{1212} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 9 & 3 \\ 3 & 5 & 10 & 8 & 15 & 5 \\ 6 & 10 & 14 & 12 & 22 & 8 \\ 5 & 8 & 12 & 8 & 18 & 6 \\ 9 & 15 & 22 & 18 & 32 & 12 \\ 3 & 5 & 8 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[400] $1^{123} 3^{1123} 21^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 10 & 2 \\ 1 & 0 & 2 & 0 & 7 & 10 & 2 \\ 1 & 2 & 0 & 0 & 9 & 15 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 57 & 11 \\ 10 & 10 & 15 & 0 & 57 & 88 & 18 \\ 2 & 2 & 2 & 0 & 11 & 18 & 3 \end{pmatrix}_{b_0}$$

[402] $1^{133} 3^{1133} 21^{1133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 12 & 3 \\ 3 & 5 & 5 & 18 & 20 & 5 \\ 2 & 5 & 2 & 14 & 15 & 4 \\ 11 & 18 & 14 & 48 & 54 & 14 \\ 12 & 20 & 15 & 54 & 58 & 16 \\ 3 & 5 & 4 & 14 & 16 & 4 \end{pmatrix}_{b_0}$$

[401] $1^{121} 3^{1122} 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 4 & 2 \\ 1 & 2 & 0 & 0 & 9 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 25 & 11 \\ 5 & 4 & 7 & 0 & 25 & 16 & 8 \\ 2 & 2 & 2 & 0 & 11 & 8 & 3 \end{pmatrix}_{b_0}$$

[418] $1^{133} 3^{1133} 21^{1133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 9 & 3 \\ 3 & 5 & 5 & 18 & 15 & 5 \\ 2 & 5 & 2 & 14 & 12 & 4 \\ 11 & 18 & 14 & 48 & 42 & 14 \\ 9 & 15 & 12 & 42 & 34 & 12 \\ 3 & 5 & 4 & 14 & 12 & 4 \end{pmatrix}_{b_0}$$

[419] $1^{114} 3^{1114} 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 5 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 22 & 6 \\ 9 & 8 & 9 & 0 & 22 & 40 & 12 \\ 2 & 2 & 2 & 0 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[403] $1^{1^{41}} 1^{113} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 1 & 2 \\ 1 & 0 & 2 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 1 & 13 & 0 & 2 \\ 1 & 2 & 1 & 0 & 10 & 0 & 2 \\ 11 & 10 & 13 & 10 & 74 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

[404] $1^{1^{23}} 1^{112} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 7 & 2 \\ 1 & 0 & 2 & 2 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 2 & 2 & 1 & 0 & 4 & 6 & 2 \\ 5 & 4 & 5 & 4 & 10 & 18 & 6 \\ 7 & 6 & 8 & 6 & 18 & 26 & 10 \\ 2 & 2 & 2 & 2 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[405] $1^{1^{32}} 1^{11} 3^1 21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 6 & 3 \\ 3 & 5 & 0 & 28 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 16 & 28 & 0 & 116 & 42 & 22 \\ 6 & 10 & 0 & 42 & 14 & 8 \\ 3 & 5 & 0 & 22 & 8 & 4 \end{pmatrix}_{b_0}$$

[406] $1^{1^{11}} 1^{112} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 0 & 2 & 2 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 1 & 2 & 1 & 0 & 4 & 6 & 2 \\ 4 & 4 & 5 & 4 & 10 & 18 & 6 \\ 5 & 6 & 8 & 6 & 18 & 26 & 10 \\ 2 & 2 & 2 & 2 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[407] $1^{1^{11}} 1^{112} 3^1 21^{1212} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 5 & 2 \\ 1 & 0 & 2 & 4 & 4 & 6 & 2 \\ 1 & 2 & 0 & 7 & 5 & 8 & 2 \\ 4 & 4 & 7 & 16 & 14 & 22 & 8 \\ 4 & 4 & 5 & 14 & 10 & 18 & 6 \\ 5 & 6 & 8 & 22 & 18 & 26 & 10 \\ 2 & 2 & 2 & 8 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[408] $1^{1^{14}} 1^1 3^1 21^{1113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 10 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 3 & 5 & 2 & 6 & 12 & 4 \\ 5 & 8 & 6 & 8 & 18 & 6 \\ 10 & 15 & 12 & 18 & 34 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[409] $1^{1^{321}} 1^1 3^1 21^{1221} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 & 3 \\ 3 & 5 & 10 & 13 & 5 & 5 \\ 7 & 10 & 14 & 20 & 8 & 8 \\ 8 & 13 & 20 & 24 & 10 & 10 \\ 3 & 5 & 8 & 10 & 2 & 4 \\ 3 & 5 & 8 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

[410] $1^{1^{321}} 1^1 3^1 21^{1421} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -108$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 8 & 3 & 3 \\ 3 & 5 & 20 & 13 & 5 & 5 \\ 11 & 20 & 62 & 40 & 16 & 16 \\ 8 & 13 & 40 & 24 & 10 & 10 \\ 3 & 5 & 16 & 10 & 2 & 4 \\ 3 & 5 & 16 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[411] \ 1^{\overline{1}32} \ 3^{\overline{1}} \ 21^{\overline{1}32} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -108, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 6 & 3 \\ 3 & 5 & 0 & 18 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 11 & 18 & 0 & 48 & 28 & 14 \\ 6 & 10 & 0 & 28 & 14 & 8 \\ 3 & 5 & 0 & 14 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[412] \ 1^{\overline{1}112} \ 3^{\overline{1}12} \ 21^{\overline{1}12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -108, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 2 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 1 & 2 & 1 & 0 & 4 & 6 & 2 \\ 4 & 4 & 5 & 4 & 10 & 18 & 6 \\ 7 & 6 & 8 & 6 & 18 & 26 & 10 \\ 2 & 2 & 2 & 2 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[413] \ 1^{\overline{1}211} \ 3^{\overline{1}12} \ 21^{\overline{1}221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -108, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 6 & 3 & 2 \\ 1 & 0 & 2 & 6 & 5 & 3 & 2 \\ 1 & 2 & 0 & 8 & 8 & 4 & 2 \\ 5 & 6 & 8 & 26 & 26 & 14 & 10 \\ 6 & 5 & 8 & 26 & 23 & 13 & 9 \\ 3 & 3 & 4 & 14 & 13 & 5 & 5 \\ 2 & 2 & 2 & 10 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[414] \ 1^{\overline{1}211} \ 3^{\overline{1}12} \ 21^{\overline{1}311} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -108, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 5 & 3 & 2 \\ 1 & 0 & 2 & 7 & 4 & 3 & 2 \\ 1 & 2 & 0 & 11 & 5 & 4 & 2 \\ 8 & 7 & 11 & 45 & 23 & 18 & 13 \\ 5 & 4 & 5 & 23 & 10 & 9 & 6 \\ 3 & 3 & 4 & 18 & 9 & 5 & 5 \\ 2 & 2 & 2 & 13 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[415] \ 1^{\overline{1}231} \ 3^{\overline{1}} \ 21^{\overline{1}331} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -108, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 3 & 3 \\ 3 & 5 & 15 & 18 & 5 & 5 \\ 8 & 15 & 34 & 42 & 12 & 12 \\ 11 & 18 & 42 & 48 & 14 & 14 \\ 3 & 5 & 12 & 14 & 2 & 4 \\ 3 & 5 & 12 & 14 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[416] \ 1^{\overline{1}123} \ 3^{\overline{1}12} \ 21^{\overline{1}12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -108, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 5 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 22 & 6 \\ 8 & 8 & 9 & 0 & 22 & 40 & 12 \\ 2 & 2 & 2 & 0 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[417] \ 1^{\overline{1}221} \ 3^{\overline{1}12} \ 21^{\overline{1}222} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -108, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 5 & 2 \\ 1 & 0 & 2 & 1 & 7 & 6 & 2 \\ 1 & 2 & 0 & 3 & 9 & 8 & 2 \\ 2 & 1 & 3 & 1 & 9 & 8 & 3 \\ 7 & 7 & 9 & 9 & 35 & 32 & 11 \\ 5 & 6 & 8 & 8 & 32 & 26 & 10 \\ 2 & 2 & 2 & 3 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[420] \ 1^{\overline{1}11} \ 3^{\overline{1}12} \ 21^{\overline{1}112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -109, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 8 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 3 & 5 & 2 & 6 & 11 & 4 \\ 5 & 8 & 6 & 8 & 18 & 6 \\ 8 & 15 & 11 & 18 & 32 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[422] \ 1^{112} 1^{11} 3^1 21^{1112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -112, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 9 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 3 & 5 & 2 & 6 & 11 & 4 \\ 4 & 8 & 6 & 8 & 18 & 6 \\ 9 & 15 & 11 & 18 & 32 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[423] \ 1^{111} 3^1 21^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -112, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 4 & 4 & 2 \\ 3 & 3 & 4 & 5 & 6 & 7 & 5 \\ 3 & 2 & 4 & 6 & 4 & 6 & 4 \\ 3 & 3 & 4 & 7 & 6 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[424] \ 1^{134} 1^{11} 3^1 21^{1314} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -112, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 15 & 3 \\ 3 & 5 & 15 & 8 & 25 & 5 \\ 9 & 15 & 34 & 18 & 57 & 12 \\ 4 & 8 & 18 & 8 & 30 & 6 \\ 15 & 25 & 57 & 30 & 92 & 20 \\ 3 & 5 & 12 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[421] \ 1^{212} 1^{111} 3^1 21^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -112, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 4 & 4 & 2 \\ 4 & 3 & 4 & 5 & 6 & 7 & 5 \\ 3 & 2 & 4 & 6 & 4 & 6 & 4 \\ 4 & 3 & 4 & 7 & 6 & 5 & 5 \\ 2 & 2 & 2 & 5 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[428] \ 1^{1222} 1^{11} 3^1 21^{1223} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -114, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 11 & 3 \\ 3 & 5 & 10 & 13 & 20 & 5 \\ 6 & 10 & 14 & 20 & 30 & 8 \\ 8 & 13 & 20 & 24 & 39 & 10 \\ 11 & 20 & 30 & 39 & 58 & 16 \\ 3 & 5 & 8 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[429] \ 1^{1222} 1^{11} 3^1 21^{1222} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -114, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 9 & 3 \\ 3 & 5 & 10 & 13 & 15 & 5 \\ 6 & 10 & 14 & 20 & 22 & 8 \\ 8 & 13 & 20 & 24 & 29 & 10 \\ 9 & 15 & 22 & 29 & 32 & 12 \\ 3 & 5 & 8 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[430] \ 1^{1214} 1^{11} 3^1 21^{1213} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -114, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 10 & 3 \\ 3 & 5 & 10 & 8 & 15 & 5 \\ 6 & 10 & 14 & 12 & 24 & 8 \\ 5 & 8 & 12 & 8 & 18 & 6 \\ 10 & 15 & 24 & 18 & 34 & 12 \\ 3 & 5 & 8 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[425] \ 1^{1231} 1^{11} 3^1 21^{1231} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -114, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 & 3 \\ 3 & 5 & 10 & 23 & 5 & 5 \\ 6 & 10 & 14 & 34 & 8 & 8 \\ 14 & 23 & 34 & 78 & 17 & 18 \\ 3 & 5 & 8 & 17 & 2 & 4 \\ 3 & 5 & 8 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[426] $1^{1^{222}} 3^1 21^{1^{211}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 5 & 4 & 2 \\ 1 & 0 & 2 & 6 & 4 & 3 & 2 \\ 1 & 2 & 0 & 8 & 5 & 4 & 2 \\ 6 & 6 & 8 & 26 & 18 & 14 & 10 \\ 5 & 4 & 5 & 18 & 10 & 9 & 6 \\ 4 & 3 & 4 & 14 & 9 & 5 & 5 \\ 2 & 2 & 2 & 10 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[427] $1^{1^{211}} 3^1 21^{1^{211}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 3 & 2 \\ 1 & 0 & 2 & 6 & 4 & 3 & 2 \\ 1 & 2 & 0 & 8 & 5 & 4 & 2 \\ 8 & 6 & 8 & 26 & 18 & 14 & 10 \\ 4 & 4 & 5 & 18 & 10 & 9 & 6 \\ 3 & 3 & 4 & 14 & 9 & 5 & 5 \\ 2 & 2 & 2 & 10 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[432] $1^{1^{113}} 3^1 21^{1^{114}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 17 & 3 \\ 3 & 5 & 0 & 8 & 30 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 36 & 6 \\ 17 & 30 & 0 & 36 & 134 & 24 \\ 3 & 5 & 0 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[433] $1^{1^{214}} 3^1 21^{1^{113}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 16 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 4 & 5 & 2 & 6 & 18 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 16 & 25 & 18 & 30 & 94 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[434] $1^{1^{113}} 3^1 21^{1^{113}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 8 & 2 \\ 1 & 0 & 2 & 1 & 4 & 7 & 2 \\ 1 & 2 & 0 & 3 & 5 & 11 & 2 \\ 2 & 1 & 3 & 1 & 5 & 11 & 3 \\ 4 & 4 & 5 & 5 & 10 & 23 & 6 \\ 8 & 7 & 11 & 11 & 23 & 45 & 13 \\ 2 & 2 & 2 & 3 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[431] $1^{1^{113}} 3^1 21^{1^{113}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 5 & 0 & 8 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 24 & 6 \\ 12 & 20 & 0 & 24 & 58 & 16 \\ 3 & 5 & 0 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[435] $1^{1^{114}} 3^1 21^{1^{114}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 3 & 7 & 28 & 4 \\ 3 & 1 & 3 & 5 & 20 & 3 \\ 3 & 3 & 1 & 6 & 24 & 3 \\ 7 & 5 & 6 & 9 & 44 & 7 \\ 28 & 20 & 24 & 44 & 174 & 28 \\ 4 & 3 & 3 & 7 & 28 & 4 \end{pmatrix}_{b_0}$$

[437] $1^{1^{114}} 3^1 21^{1^{112}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 8 & 2 \\ 1 & 0 & 2 & 3 & 2 & 6 & 2 \\ 1 & 2 & 0 & 4 & 4 & 8 & 2 \\ 3 & 3 & 4 & 5 & 6 & 14 & 5 \\ 3 & 2 & 4 & 6 & 4 & 12 & 4 \\ 8 & 6 & 8 & 14 & 12 & 26 & 10 \\ 2 & 2 & 2 & 5 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[436] $1^{1^{212}} 3^1 21^{1^{112}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 6 & 2 \\ 1 & 0 & 2 & 3 & 2 & 6 & 2 \\ 1 & 2 & 0 & 4 & 4 & 8 & 2 \\ 4 & 3 & 0 & 5 & 6 & 14 & 5 \\ 3 & 2 & 4 & 6 & 4 & 12 & 4 \\ 6 & 6 & 8 & 14 & 12 & 26 & 10 \\ 2 & 2 & 2 & 5 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[438] \quad 1^{134} 3^{121} 21^{1134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -122, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 12 & 3 \\ 3 & 5 & 5 & 18 & 20 & 5 \\ 2 & 5 & 2 & 14 & 16 & 4 \\ 11 & 18 & 14 & 48 & 56 & 14 \\ 12 & 20 & 16 & 56 & 62 & 16 \\ 3 & 5 & 4 & 14 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[441] \quad 1^{1111} 3^{121} 21^{1211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 5 & 3 & 2 \\ 1 & 0 & 2 & 4 & 4 & 3 & 2 \\ 1 & 2 & 0 & 7 & 5 & 4 & 2 \\ 6 & 4 & 7 & 16 & 14 & 11 & 8 \\ 5 & 4 & 5 & 14 & 10 & 9 & 6 \\ 3 & 3 & 4 & 11 & 9 & 5 & 5 \\ 2 & 2 & 2 & 8 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[442] \quad 1^{121} 3^{121} 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 & 3 \\ 3 & 5 & 0 & 23 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 13 & 23 & 0 & 78 & 17 & 18 \\ 3 & 5 & 0 & 17 & 2 & 4 \\ 3 & 5 & 0 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[443] \quad 1^{131} 3^{121} 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 1 & 2 \\ 1 & 0 & 2 & 1 & 10 & 0 & 2 \\ 1 & 2 & 0 & 3 & 13 & 0 & 2 \\ 2 & 1 & 3 & 1 & 13 & 0 & 3 \\ 10 & 10 & 13 & 13 & 74 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[439] \quad 1^{321} 3^{121} 21^{1221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 & 3 \\ 3 & 5 & 15 & 13 & 5 & 5 \\ 10 & 15 & 32 & 29 & 11 & 12 \\ 8 & 13 & 29 & 24 & 10 & 10 \\ 3 & 5 & 11 & 10 & 2 & 4 \\ 3 & 5 & 12 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[440] \quad 1^{1111} 3^{121} 21^{1211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 4 & 2 \\ 1 & 0 & 2 & 4 & 4 & 3 & 2 \\ 1 & 2 & 0 & 7 & 5 & 4 & 2 \\ 5 & 4 & 7 & 16 & 14 & 11 & 8 \\ 4 & 4 & 5 & 14 & 10 & 9 & 6 \\ 4 & 3 & 4 & 11 & 9 & 5 & 5 \\ 2 & 2 & 2 & 8 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[448] \quad 1^{1212} 3^{121} 21^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 6 & 3 \\ 3 & 5 & 10 & 8 & 10 & 5 \\ 6 & 10 & 14 & 12 & 16 & 8 \\ 5 & 8 & 12 & 8 & 12 & 6 \\ 6 & 10 & 16 & 12 & 14 & 8 \\ 3 & 5 & 8 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[446] \quad 1^{1231} 3^{121} 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 & 3 \\ 3 & 5 & 5 & 18 & 5 & 5 \\ 4 & 5 & 2 & 14 & 4 & 4 \\ 11 & 18 & 14 & 48 & 14 & 14 \\ 3 & 5 & 4 & 14 & 2 & 4 \\ 3 & 5 & 4 & 14 & 4 & 4 \end{pmatrix}_{b_0}$$

[444] $1^{\overline{1112}} 3^{\overline{11}} 21^{\overline{1111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 0 & 2 & 3 & 4 & 5 & 2 \\ 1 & 2 & 0 & 4 & 5 & 5 & 2 \\ 2 & 3 & 4 & 5 & 9 & 10 & 5 \\ 4 & 4 & 5 & 9 & 10 & 13 & 6 \\ 4 & 5 & 5 & 10 & 13 & 13 & 7 \\ 2 & 2 & 2 & 5 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[450] $1^{\overline{122}} 1^{\overline{1112}} 3^{\overline{11}} 21^{\overline{1122}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 6 & 2 \\ 1 & 0 & 2 & 3 & 5 & 6 & 2 \\ 1 & 2 & 0 & 4 & 8 & 8 & 2 \\ 2 & 3 & 4 & 5 & 13 & 14 & 5 \\ 6 & 5 & 8 & 13 & 23 & 26 & 9 \\ 6 & 6 & 8 & 14 & 26 & 26 & 10 \\ 2 & 2 & 2 & 5 & 9 & 10 & 3 \end{pmatrix}_{b_0}$$

[445] $1^{\overline{214}} 1^{\overline{12}} 3^{\overline{11}} 21^{\overline{1214}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 18 & 3 \\ 3 & 5 & 10 & 8 & 30 & 5 \\ 6 & 10 & 14 & 12 & 44 & 8 \\ 5 & 8 & 12 & 8 & 36 & 6 \\ 18 & 30 & 44 & 36 & 134 & 24 \\ 3 & 5 & 8 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[449] $1^{\overline{13}} 1^{\overline{131}} 3^{\overline{11}} 21^{\overline{1131}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 2 & 2 \\ 1 & 0 & 2 & 1 & 10 & 3 & 2 \\ 1 & 2 & 0 & 3 & 13 & 4 & 2 \\ 2 & 1 & 3 & 1 & 13 & 4 & 3 \\ 10 & 10 & 13 & 13 & 74 & 23 & 16 \\ 2 & 3 & 4 & 4 & 23 & 5 & 5 \\ 2 & 2 & 2 & 3 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

[447] $1^{\overline{132}} 1^{\overline{122}} 3^{\overline{11}} 21^{\overline{122}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 6 & 2 \\ 1 & 2 & 0 & 0 & 9 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 8 & 7 & 9 & 0 & 35 & 32 & 11 \\ 6 & 6 & 8 & 0 & 32 & 26 & 10 \\ 2 & 2 & 2 & 0 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

[453] $1^{\overline{41}} 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{1141}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 & 3 \\ 3 & 5 & 5 & 28 & 5 & 5 \\ 2 & 5 & 2 & 21 & 4 & 4 \\ 17 & 28 & 21 & 116 & 21 & 22 \\ 3 & 5 & 4 & 21 & 2 & 4 \\ 3 & 5 & 4 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

[451] $1^{\overline{14}} 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{1114}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 15 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 2 & 5 & 2 & 6 & 19 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 15 & 25 & 19 & 30 & 92 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[452] $1^{\overline{132}} 1^{\overline{121}} 3^{\overline{11}} 21^{\overline{121}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 9 & 4 & 2 \\ 1 & 2 & 1 & 0 & 7 & 3 & 2 \\ 8 & 7 & 9 & 7 & 35 & 16 & 11 \\ 4 & 3 & 4 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[454] $1^{1121} 3^{1131} 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -125, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 3 & 2 \\ 1 & 0 & 2 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 9 & 4 & 2 \\ 2 & 2 & 1 & 0 & 7 & 3 & 2 \\ 8 & 7 & 9 & 7 & 35 & 16 & 11 \\ 3 & 3 & 4 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[455] $1^{113} 1^{11} 3^{11} 21^{1114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -125, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 14 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 2 & 5 & 2 & 6 & 19 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 14 & 25 & 19 & 30 & 92 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[456] $1^{113} 1^{11} 3^{11} 21^{1214} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -126, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 14 & 3 \\ 3 & 5 & 10 & 8 & 25 & 5 \\ 5 & 10 & 14 & 12 & 38 & 8 \\ 5 & 8 & 12 & 8 & 30 & 6 \\ 14 & 25 & 38 & 30 & 92 & 20 \\ 3 & 5 & 8 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[457] $1^{12} 3^{11} 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -126, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 16 & 3 \\ 3 & 5 & 0 & 8 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 30 & 6 \\ 16 & 25 & 0 & 30 & 94 & 20 \\ 3 & 5 & 0 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[458] $1^{112} 3^{11} 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -127, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 0 & 5 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 23 & 6 \\ 9 & 7 & 11 & 0 & 23 & 45 & 13 \\ 2 & 2 & 2 & 0 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[459] $1^{124} 1^{112} 3^{11} 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -127, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 0 & 5 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 23 & 6 \\ 9 & 7 & 11 & 0 & 23 & 45 & 13 \\ 2 & 2 & 2 & 0 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[460] $1^{123} 1^{111} 3^{11} 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -128, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 7 & 2 \\ 1 & 0 & 2 & 0 & 5 & 5 & 2 \\ 1 & 2 & 0 & 0 & 8 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 29 & 9 \\ 7 & 5 & 10 & 0 & 29 & 33 & 11 \\ 2 & 2 & 2 & 0 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

[461] $1^{121} 1^{111} 3^{11} 21^{1122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -128, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 4 & 2 \\ 1 & 0 & 2 & 1 & 5 & 4 & 2 \\ 1 & 2 & 0 & 3 & 8 & 7 & 2 \\ 2 & 1 & 3 & 1 & 8 & 7 & 3 \\ 6 & 5 & 8 & 8 & 23 & 21 & 9 \\ 4 & 4 & 7 & 7 & 21 & 16 & 8 \\ 2 & 2 & 2 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[462] $1^{11}3^{123}21^{1213} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 12 & 3 \\ 3 & 5 & 10 & 8 & 20 & 5 \\ 6 & 10 & 14 & 12 & 30 & 8 \\ 4 & 8 & 12 & 8 & 24 & 6 \\ 12 & 20 & 30 & 24 & 58 & 16 \\ 3 & 5 & 8 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[463] $1^{13}1^{12}3^{121}21^{1113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 15 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 3 & 5 & 2 & 6 & 18 & 4 \\ 4 & 8 & 6 & 8 & 30 & 6 \\ 15 & 25 & 18 & 30 & 94 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[464] $1^{122}1^{11}3^{121}21^{1234} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 13 & 3 \\ 3 & 5 & 10 & 18 & 25 & 5 \\ 6 & 10 & 14 & 28 & 38 & 8 \\ 10 & 18 & 28 & 48 & 68 & 14 \\ 13 & 25 & 38 & 68 & 92 & 20 \\ 3 & 5 & 8 & 14 & 20 & 4 \end{pmatrix}_{b_0}$$

[470] $1^{123}1^{112}3^{121}21^{1112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 7 & 2 \\ 1 & 0 & 2 & 1 & 4 & 6 & 2 \\ 1 & 2 & 0 & 3 & 5 & 8 & 2 \\ 2 & 1 & 3 & 1 & 5 & 8 & 3 \\ 5 & 4 & 5 & 5 & 10 & 18 & 6 \\ 7 & 6 & 8 & 8 & 18 & 26 & 10 \\ 2 & 2 & 2 & 3 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[471] $1^{123}1^{113}3^{121}21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 10 & 2 \\ 1 & 0 & 2 & 0 & 5 & 10 & 2 \\ 1 & 2 & 0 & 0 & 8 & 15 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 47 & 9 \\ 10 & 10 & 15 & 0 & 47 & 88 & 18 \\ 2 & 2 & 2 & 0 & 9 & 18 & 3 \end{pmatrix}_{b_0}$$

[465] $1^{123}1^{11}3^{121}21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 5 & 0 & 13 & 20 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 39 & 10 \\ 12 & 20 & 0 & 39 & 58 & 16 \\ 3 & 5 & 0 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[466] $1^{123}1^{11}3^{121}21^{1224} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 14 & 3 \\ 3 & 5 & 10 & 13 & 25 & 5 \\ 5 & 10 & 14 & 20 & 38 & 8 \\ 8 & 13 & 20 & 24 & 49 & 10 \\ 14 & 25 & 38 & 49 & 92 & 20 \\ 3 & 5 & 8 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

[467] $1^{141}1^{131}3^{121}21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 & 2 \\ 1 & 2 & 0 & 0 & 10 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 9 & 10 & 0 & 51 & 9 & 13 \\ 2 & 2 & 1 & 0 & 9 & 0 & 2 \\ 2 & 2 & 2 & 0 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[468] \quad 1^{113} 1^{112} 3^1 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -131, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 7 & 2 \\ 1 & 0 & 2 & 1 & 4 & 6 & 2 \\ 1 & 2 & 0 & 3 & 5 & 8 & 2 \\ 2 & 1 & 3 & 1 & 5 & 8 & 3 \\ 4 & 4 & 5 & 5 & 10 & 18 & 6 \\ 7 & 6 & 8 & 8 & 18 & 26 & 10 \\ 2 & 2 & 2 & 3 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[469] \quad 1^{13} 1^{131} 3^1 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -131, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 & 2 \\ 1 & 2 & 0 & 0 & 10 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 9 & 10 & 0 & 51 & 9 & 13 \\ 1 & 2 & 1 & 0 & 9 & 0 & 2 \\ 2 & 2 & 2 & 0 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[472] \quad 1^{11} 1^{111} 3^1 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -133, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 3 & 2 \\ 1 & 0 & 2 & 3 & 4 & 3 & 2 \\ 1 & 2 & 0 & 4 & 5 & 4 & 2 \\ 2 & 3 & 4 & 5 & 9 & 7 & 5 \\ 4 & 4 & 5 & 9 & 10 & 9 & 6 \\ 3 & 3 & 4 & 7 & 9 & 5 & 5 \\ 2 & 2 & 2 & 5 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[476] \quad 1^{13} 1^{12} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -134, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 5 & 0 & 8 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 30 & 6 \\ 15 & 25 & 0 & 30 & 94 & 20 \\ 3 & 5 & 0 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[473] \quad 1^{124} 1^{12} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -134, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 18 & 3 \\ 3 & 5 & 0 & 13 & 30 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 58 & 10 \\ 18 & 30 & 0 & 58 & 134 & 24 \\ 3 & 5 & 0 & 10 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[474] \quad 1^{124} 1^{112} 3^1 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -134, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 8 & 2 \\ 1 & 0 & 2 & 1 & 4 & 6 & 2 \\ 1 & 2 & 0 & 3 & 5 & 8 & 2 \\ 2 & 1 & 3 & 1 & 5 & 8 & 3 \\ 5 & 4 & 5 & 5 & 10 & 18 & 6 \\ 8 & 6 & 8 & 8 & 18 & 26 & 10 \\ 2 & 2 & 2 & 3 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[475] \quad 1^{112} 1^{112} 3^1 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -134, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 6 & 2 \\ 1 & 0 & 2 & 1 & 4 & 6 & 2 \\ 1 & 2 & 0 & 3 & 5 & 8 & 2 \\ 2 & 1 & 3 & 1 & 5 & 8 & 3 \\ 4 & 4 & 5 & 5 & 10 & 18 & 6 \\ 6 & 6 & 8 & 8 & 18 & 26 & 10 \\ 2 & 2 & 2 & 3 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[478] \quad 1^{214} 1^{124} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -136, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 4 & 3 & 3 & 14 & 28 & 4 \\ 3 & 1 & 3 & 10 & 20 & 3 \\ 3 & 3 & 1 & 12 & 24 & 3 \\ 14 & 10 & 12 & 42 & 88 & 14 \\ 28 & 20 & 24 & 88 & 174 & 28 \\ 4 & 3 & 3 & 14 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[477] \quad 1^{113} 1^1 3^1 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -136, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} =$$

6.8

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 9 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 3 & 5 & 2 & 6 & 12 & 4 \\ 5 & 8 & 6 & 8 & 18 & 6 \\ 9 & 15 & 12 & 18 & 34 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[479] \ 1^{11} 3^{11} 21^{1241} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -137, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 16 & 3 & 3 \\ 3 & 5 & 10 & 28 & 5 & 5 \\ 5 & 10 & 14 & 42 & 8 & 8 \\ 16 & 28 & 42 & 116 & 21 & 22 \\ 3 & 5 & 8 & 21 & 2 & 4 \\ 3 & 5 & 8 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[484] \ 1^{141} 3^{11} 21^{1241} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 & 3 \\ 3 & 5 & 10 & 23 & 5 & 5 \\ 5 & 10 & 14 & 36 & 8 & 8 \\ 14 & 23 & 36 & 80 & 18 & 18 \\ 3 & 5 & 8 & 18 & 2 & 4 \\ 3 & 5 & 8 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[480] \ 1^{111} 3^{11} 21^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 3 & 2 \\ 1 & 2 & 0 & 0 & 8 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 13 & 9 \\ 3 & 3 & 4 & 0 & 13 & 5 & 5 \\ 2 & 2 & 2 & 0 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[481] \ 1^{113} 3^{11} 21^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 1 & 13 & 0 & 2 \\ 1 & 2 & 1 & 0 & 10 & 0 & 2 \\ 10 & 10 & 13 & 10 & 74 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[482] \ 1^{122} 3^{11} 21^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 5 & 2 \\ 1 & 2 & 0 & 0 & 9 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 23 & 11 \\ 4 & 5 & 5 & 0 & 23 & 13 & 7 \\ 2 & 2 & 2 & 0 & 11 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[483] \ 1^{111} 3^{11} 21^{121} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 3 & 2 \\ 1 & 2 & 0 & 0 & 8 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 5 & 8 & 0 & 23 & 13 & 9 \\ 4 & 3 & 4 & 0 & 13 & 5 & 5 \\ 2 & 2 & 2 & 0 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[485] \ 1^{131} 3^{11} 21^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -141, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 & 3 \\ 3 & 5 & 0 & 23 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 14 & 23 & 0 & 78 & 17 & 18 \\ 3 & 5 & 0 & 17 & 2 & 4 \\ 3 & 5 & 0 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[486] \ 1^1 3^1 21^{1141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -142, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 & 3 \\ 3 & 5 & 5 & 23 & 5 & 5 \\ 2 & 5 & 2 & 18 & 4 & 4 \\ 14 & 23 & 18 & 80 & 18 & 18 \\ 3 & 5 & 4 & 18 & 2 & 4 \\ 3 & 5 & 4 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[487] $1^{1^{31}} 1^{12} 3^{121} 21^{1141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -142, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 & 3 \\ 3 & 5 & 5 & 33 & 5 & 5 \\ 2 & 5 & 2 & 24 & 4 & 4 \\ 19 & 33 & 24 & 164 & 24 & 26 \\ 3 & 5 & 4 & 24 & 2 & 4 \\ 3 & 5 & 4 & 26 & 4 & 4 \end{pmatrix}_{b_0}$$

[496] $1^{1^{42}} 1^{11} 3^{121} 21^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 6 & 3 \\ 3 & 5 & 0 & 28 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 17 & 28 & 0 & 116 & 42 & 22 \\ 6 & 10 & 0 & 42 & 14 & 8 \\ 3 & 5 & 0 & 22 & 8 & 4 \end{pmatrix}_{b_0}$$

[488] $1^{1^{214}} 1^{11} 3^{121} 21^{1224} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 12 & 3 \\ 3 & 5 & 10 & 13 & 20 & 5 \\ 6 & 10 & 14 & 20 & 32 & 8 \\ 7 & 13 & 20 & 24 & 40 & 10 \\ 12 & 20 & 32 & 40 & 62 & 16 \\ 3 & 5 & 8 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[489] $1^{1^{212}} 1^{11} 3^{121} 21^{1213} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 11 & 3 \\ 3 & 5 & 10 & 8 & 20 & 5 \\ 6 & 10 & 14 & 12 & 30 & 8 \\ 5 & 8 & 12 & 8 & 24 & 6 \\ 11 & 20 & 30 & 24 & 58 & 16 \\ 3 & 5 & 8 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[490] $1^{1^{111}} 1^{11} 3^{121} 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 & 3 \\ 3 & 5 & 5 & 13 & 5 & 5 \\ 3 & 5 & 2 & 10 & 4 & 4 \\ 7 & 13 & 10 & 24 & 10 & 10 \\ 3 & 5 & 4 & 10 & 2 & 4 \\ 3 & 5 & 4 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

[491] $1^{1^{112}} 1^{11} 3^{121} 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 6 & 3 \\ 3 & 5 & 5 & 13 & 10 & 5 \\ 3 & 5 & 2 & 10 & 8 & 4 \\ 7 & 13 & 10 & 24 & 20 & 10 \\ 6 & 10 & 8 & 20 & 14 & 8 \\ 3 & 5 & 4 & 10 & 8 & 4 \end{pmatrix}_{b_0}$$

[492] $1^{1^{113}} 1^{11} 3^{121} 21^{1123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 9 & 3 \\ 3 & 5 & 5 & 13 & 15 & 5 \\ 3 & 5 & 2 & 10 & 12 & 4 \\ 7 & 13 & 10 & 24 & 30 & 10 \\ 9 & 15 & 12 & 30 & 34 & 12 \\ 3 & 5 & 4 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[493] $1^{1^{114}} 1^{11} 3^{121} 21^{1124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 12 & 3 \\ 3 & 5 & 5 & 13 & 20 & 5 \\ 3 & 5 & 2 & 10 & 16 & 4 \\ 7 & 13 & 10 & 24 & 40 & 10 \\ 12 & 20 & 16 & 40 & 62 & 16 \\ 3 & 5 & 4 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[494] $1^{1212} 1^1 3^1 21^{1222} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 6 & 3 \\ 3 & 5 & 10 & 13 & 10 & 5 \\ 6 & 10 & 14 & 20 & 16 & 8 \\ 7 & 13 & 20 & 24 & 20 & 10 \\ 6 & 10 & 16 & 20 & 14 & 8 \\ 3 & 5 & 8 & 10 & 8 & 4 \end{pmatrix}_{b_0}$$

[495] $1^{1213} 1^1 3^1 21^{1223} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 9 & 3 \\ 3 & 5 & 10 & 13 & 15 & 5 \\ 6 & 10 & 14 & 20 & 24 & 8 \\ 7 & 13 & 20 & 24 & 30 & 10 \\ 9 & 15 & 24 & 30 & 34 & 12 \\ 3 & 5 & 8 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[497] $1^{142} 1^1 3^1 21^{1242} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 6 & 3 \\ 3 & 5 & 10 & 23 & 10 & 5 \\ 5 & 10 & 14 & 36 & 16 & 8 \\ 14 & 23 & 36 & 80 & 36 & 18 \\ 6 & 10 & 16 & 36 & 14 & 8 \\ 3 & 5 & 8 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

[498] $1^{121} 1^{11} 3^1 21^{1131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 & 3 \\ 3 & 5 & 5 & 23 & 5 & 5 \\ 2 & 5 & 2 & 17 & 4 & 4 \\ 13 & 23 & 17 & 78 & 17 & 18 \\ 3 & 5 & 4 & 17 & 2 & 4 \\ 3 & 5 & 4 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[499] $1^{142} 1^1 3^1 21^{1142} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 6 & 3 \\ 3 & 5 & 5 & 23 & 10 & 5 \\ 2 & 5 & 2 & 18 & 8 & 4 \\ 14 & 23 & 18 & 80 & 36 & 18 \\ 6 & 10 & 8 & 36 & 14 & 8 \\ 3 & 5 & 4 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

[500] $1^{132} 1^{11} 3^1 21^{1142} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 6 & 3 \\ 3 & 5 & 5 & 28 & 10 & 5 \\ 2 & 5 & 2 & 21 & 8 & 4 \\ 16 & 28 & 21 & 116 & 42 & 22 \\ 6 & 10 & 8 & 42 & 14 & 8 \\ 3 & 5 & 4 & 22 & 8 & 4 \end{pmatrix}_{b_0}$$

[503] $1^{113} 1^{11} 3^1 21^{1112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -149, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 10 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 3 & 5 & 2 & 6 & 11 & 4 \\ 5 & 8 & 6 & 8 & 18 & 6 \\ 10 & 15 & 11 & 18 & 32 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[502] $1^{112} 1^{11} 3^1 21^{1114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -149, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 13 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 3 & 5 & 2 & 6 & 19 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 13 & 25 & 19 & 30 & 92 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[501] $1^{214} 1^{11} 3^1 21^{1114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -149, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 15 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 4 & 5 & 2 & 6 & 19 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 15 & 25 & 19 & 30 & 92 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[505] $1^{\overset{1241}{1} \overset{1}{3} \overset{1}{21} \overset{1341}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 14 & 3 & 3 \\ 3 & 5 & 15 & 23 & 5 & 5 \\ 8 & 15 & 34 & 54 & 12 & 12 \\ 14 & 23 & 54 & 80 & 18 & 18 \\ 3 & 5 & 12 & 18 & 2 & 4 \\ 3 & 5 & 12 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[506] $1^{\overset{1421}{1} \overset{1}{3} \overset{1}{21} \overset{1321}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 & 3 \\ 3 & 5 & 15 & 13 & 5 & 5 \\ 10 & 15 & 34 & 30 & 12 & 12 \\ 8 & 13 & 30 & 24 & 10 & 10 \\ 3 & 5 & 12 & 10 & 2 & 4 \\ 3 & 5 & 12 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

[504] $1^{\overset{141}{1} \overset{113}{3} \overset{1}{21} \overset{1141}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 2 & 2 \\ 1 & 0 & 2 & 3 & 11 & 1 & 2 \\ 1 & 2 & 0 & 4 & 16 & 3 & 2 \\ 2 & 3 & 4 & 5 & 27 & 4 & 5 \\ 12 & 11 & 16 & 27 & 103 & 16 & 19 \\ 2 & 1 & 3 & 4 & 16 & 1 & 3 \\ 2 & 2 & 2 & 5 & 19 & 3 & 3 \end{pmatrix}_{b_0}$$

[512] $1^{\overset{114}{1} \overset{114}{3} \overset{1}{21} \overset{112}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 0 & 5 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 26 & 6 \\ 10 & 10 & 10 & 0 & 26 & 58 & 14 \\ 2 & 2 & 2 & 0 & 6 & 14 & 3 \end{pmatrix}_{b_0}$$

[514] $1^{\overset{114}{1} \overset{114}{3} \overset{1}{21} \overset{1114}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 12 & 2 \\ 1 & 0 & 2 & 3 & 2 & 12 & 2 \\ 1 & 2 & 0 & 4 & 4 & 16 & 2 \\ 2 & 3 & 4 & 5 & 6 & 28 & 5 \\ 3 & 2 & 4 & 6 & 4 & 24 & 4 \\ 12 & 12 & 16 & 28 & 24 & 110 & 20 \\ 2 & 2 & 2 & 5 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[513] $1^{\overset{1241}{1} \overset{113}{3} \overset{1}{21} \overset{123}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 11 & 1 & 2 \\ 1 & 0 & 2 & 4 & 10 & 0 & 2 \\ 1 & 2 & 0 & 7 & 13 & 0 & 2 \\ 5 & 4 & 7 & 16 & 36 & 0 & 8 \\ 11 & 10 & 13 & 36 & 74 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 8 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

[507] $1^{\overset{1121}{1} \overset{1121}{3} \overset{1}{21} \overset{121}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 2 \\ 1 & 0 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 9 & 4 & 2 \\ 1 & 2 & 1 & 0 & 7 & 3 & 2 \\ 8 & 7 & 9 & 7 & 35 & 16 & 11 \\ 3 & 3 & 4 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[508] $1^{\overset{1124}{1} \overset{1112}{3} \overset{1}{21} \overset{112}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -152, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 8 & 2 \\ 1 & 0 & 2 & 2 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 2 & 2 & 1 & 0 & 4 & 6 & 2 \\ 5 & 4 & 5 & 4 & 10 & 18 & 6 \\ 8 & 6 & 8 & 6 & 18 & 26 & 10 \\ 2 & 2 & 2 & 2 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[509] \quad 1^{1121} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -152, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 4 & 2 \\ 1 & 0 & 2 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 9 & 4 & 2 \\ 2 & 2 & 1 & 0 & 7 & 3 & 2 \\ 8 & 7 & 9 & 7 & 35 & 16 & 11 \\ 4 & 3 & 4 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[510] \quad 1^{114} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -152, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 0 & 5 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 26 & 6 \\ 10 & 10 & 10 & 0 & 26 & 58 & 14 \\ 2 & 2 & 2 & 0 & 6 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[511] \quad 1^{112} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -152, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 2 & 4 & 6 & 2 \\ 1 & 2 & 0 & 1 & 5 & 8 & 2 \\ 1 & 2 & 1 & 0 & 4 & 6 & 2 \\ 4 & 4 & 5 & 4 & 10 & 18 & 6 \\ 6 & 6 & 8 & 6 & 18 & 26 & 10 \\ 2 & 2 & 2 & 2 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[515] \quad 1^1 3^1 21^{1213} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -154, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 9 & 3 \\ 3 & 5 & 10 & 8 & 15 & 5 \\ 6 & 10 & 14 & 12 & 24 & 8 \\ 5 & 8 & 12 & 8 & 18 & 6 \\ 9 & 15 & 24 & 18 & 34 & 12 \\ 3 & 5 & 8 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[516] \quad 1^{113} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -154, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 17 & 3 \\ 3 & 5 & 5 & 8 & 30 & 5 \\ 2 & 5 & 2 & 6 & 22 & 4 \\ 5 & 8 & 6 & 8 & 36 & 6 \\ 17 & 30 & 22 & 36 & 134 & 24 \\ 3 & 5 & 4 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[517] \quad 1^{111} 3^1 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -155, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 4 & 2 \\ 1 & 0 & 2 & 1 & 5 & 3 & 2 \\ 1 & 2 & 0 & 3 & 8 & 4 & 2 \\ 2 & 1 & 3 & 1 & 8 & 4 & 3 \\ 7 & 5 & 8 & 8 & 23 & 13 & 9 \\ 4 & 3 & 4 & 4 & 13 & 5 & 5 \\ 2 & 2 & 2 & 3 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[518] \quad 1^{141} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -155, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 1 & 2 \\ 1 & 0 & 2 & 1 & 10 & 0 & 2 \\ 1 & 2 & 0 & 3 & 13 & 0 & 2 \\ 2 & 1 & 3 & 1 & 13 & 0 & 3 \\ 11 & 10 & 13 & 13 & 74 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[519] \quad 1^{111} 3^1 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -155, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 3 & 2 \\ 1 & 0 & 2 & 1 & 5 & 3 & 2 \\ 1 & 2 & 0 & 3 & 8 & 4 & 2 \\ 2 & 1 & 3 & 1 & 8 & 4 & 3 \\ 6 & 5 & 8 & 8 & 23 & 13 & 9 \\ 3 & 3 & 4 & 4 & 13 & 5 & 5 \\ 2 & 2 & 2 & 3 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[520] \quad 1^{11} 3^{1214} 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -155, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 13 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 4 & 5 & 2 & 6 & 15 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 13 & 20 & 15 & 24 & 58 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[521] \quad 1^{112} 1^{113} 3^{11} 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -156, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 8 & 2 \\ 1 & 0 & 2 & 1 & 4 & 9 & 2 \\ 1 & 2 & 0 & 3 & 5 & 12 & 2 \\ 2 & 1 & 3 & 1 & 5 & 12 & 3 \\ 4 & 4 & 5 & 5 & 10 & 27 & 6 \\ 8 & 9 & 12 & 12 & 27 & 61 & 15 \\ 2 & 2 & 2 & 3 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[522] \quad 1^{122} 1^{121} 3^{11} 21^{1221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -156, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 4 & 2 \\ 1 & 0 & 2 & 4 & 7 & 3 & 2 \\ 1 & 2 & 0 & 7 & 9 & 4 & 2 \\ 5 & 4 & 7 & 16 & 25 & 11 & 8 \\ 7 & 7 & 9 & 25 & 35 & 16 & 11 \\ 4 & 3 & 4 & 11 & 16 & 5 & 5 \\ 2 & 2 & 2 & 8 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[523] \quad 1^{141} 1^1 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -156, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 & 3 \\ 3 & 5 & 0 & 23 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 14 & 23 & 0 & 80 & 18 & 18 \\ 3 & 5 & 0 & 18 & 2 & 4 \\ 3 & 5 & 0 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[524] \quad 1^{112} 1^{111} 3^{11} 21^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -156, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 5 & 2 \\ 1 & 0 & 2 & 4 & 4 & 4 & 2 \\ 1 & 2 & 0 & 7 & 5 & 7 & 2 \\ 5 & 4 & 7 & 16 & 14 & 18 & 8 \\ 4 & 4 & 5 & 14 & 10 & 14 & 6 \\ 5 & 4 & 7 & 18 & 14 & 16 & 8 \\ 2 & 2 & 2 & 8 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[525] \quad 1^{113} 1^{1112} 3^1 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -156, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 8 & 2 \\ 1 & 0 & 2 & 3 & 4 & 7 & 2 \\ 1 & 2 & 0 & 4 & 5 & 11 & 2 \\ 2 & 3 & 4 & 5 & 9 & 18 & 5 \\ 4 & 4 & 5 & 9 & 10 & 23 & 6 \\ 8 & 7 & 11 & 18 & 23 & 45 & 13 \\ 2 & 2 & 2 & 5 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[526] \quad 1^{114} 1^{114} 3^1 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -157, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 11 & 2 \\ 1 & 0 & 2 & 1 & 4 & 11 & 2 \\ 1 & 2 & 0 & 3 & 5 & 13 & 2 \\ 2 & 1 & 3 & 1 & 5 & 13 & 3 \\ 4 & 4 & 5 & 5 & 10 & 31 & 6 \\ 11 & 11 & 13 & 13 & 31 & 81 & 17 \\ 2 & 2 & 2 & 3 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[528] \quad 1^{143} 1^1 3^1 21^{1143} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -158, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 9 & 3 \\ 3 & 5 & 5 & 23 & 15 & 5 \\ 2 & 5 & 2 & 18 & 12 & 4 \\ 14 & 23 & 18 & 80 & 54 & 18 \\ 9 & 15 & 12 & 54 & 34 & 12 \\ 3 & 5 & 4 & 18 & 12 & 4 \end{pmatrix}_{b_0}$$

[527] $1^{124} 1^{11} 3^{121} 1^{124} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -158, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 15 & 3 \\ 3 & 5 & 5 & 13 & 25 & 5 \\ 2 & 5 & 2 & 10 & 19 & 4 \\ 8 & 13 & 10 & 24 & 49 & 10 \\ 15 & 25 & 19 & 49 & 92 & 20 \\ 3 & 5 & 4 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

[538] $1^{213} 1^{113} 3^{121} 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 9 & 2 \\ 1 & 0 & 2 & 3 & 2 & 9 & 2 \\ 1 & 2 & 0 & 4 & 4 & 12 & 2 \\ 4 & 3 & 4 & 5 & 6 & 21 & 5 \\ 3 & 2 & 4 & 6 & 4 & 18 & 4 \\ 9 & 9 & 12 & 21 & 18 & 61 & 15 \\ 2 & 2 & 2 & 5 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[533] $1^{132} 1^{131} 3^{121} 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 0 & 9 & 3 & 2 \\ 1 & 2 & 0 & 0 & 10 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 9 & 10 & 0 & 51 & 19 & 13 \\ 4 & 3 & 4 & 0 & 19 & 5 & 5 \\ 2 & 2 & 2 & 0 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[532] $1^{14} 1^{14} 3^{121} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 12 & 2 \\ 1 & 0 & 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 0 & 4 & 16 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 24 & 4 \\ 12 & 12 & 16 & 0 & 24 & 110 & 20 \\ 2 & 2 & 2 & 0 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[537] $1^{113} 1^{12} 3^{121} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 6 & 2 \\ 1 & 2 & 0 & 0 & 5 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 18 & 6 \\ 7 & 6 & 8 & 0 & 18 & 26 & 10 \\ 2 & 2 & 2 & 0 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[535] $1^{131} 1^{12} 3^{121} 1^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 & 2 \\ 1 & 0 & 2 & 0 & 8 & 1 & 2 \\ 1 & 2 & 0 & 0 & 12 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 8 & 12 & 0 & 56 & 12 & 14 \\ 2 & 1 & 3 & 0 & 12 & 1 & 3 \\ 2 & 2 & 2 & 0 & 14 & 3 & 3 \end{pmatrix}_{b_0}$$

[534] $1^{231} 1^{113} 3^{121} 1^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 10 & 1 & 2 \\ 1 & 0 & 2 & 4 & 10 & 0 & 2 \\ 1 & 2 & 0 & 7 & 13 & 0 & 2 \\ 5 & 4 & 7 & 16 & 36 & 0 & 8 \\ 10 & 10 & 13 & 36 & 74 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 8 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

[530] $1^{141} 1^{131} 3^{121} 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 3 & 2 \\ 1 & 0 & 2 & 0 & 9 & 3 & 2 \\ 1 & 2 & 0 & 0 & 10 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 9 & 10 & 0 & 51 & 19 & 13 \\ 3 & 3 & 4 & 0 & 19 & 5 & 5 \\ 2 & 2 & 2 & 0 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[536] $1^{112} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 6 & 2 \\ 1 & 2 & 0 & 0 & 5 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 18 & 6 \\ 7 & 6 & 8 & 0 & 18 & 26 & 10 \\ 2 & 2 & 2 & 0 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[529] $1^{132} 1^{131} 3^1 21^{1131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 10 & 4 & 2 \\ 1 & 0 & 2 & 0 & 1 & 10 & 3 & 2 \\ 1 & 2 & 0 & 0 & 3 & 13 & 4 & 2 \\ 2 & 1 & 3 & 1 & 13 & 4 & 3 \\ 10 & 10 & 13 & 13 & 74 & 23 & 16 \\ 4 & 3 & 4 & 4 & 23 & 5 & 5 \\ 2 & 2 & 2 & 2 & 3 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

[531] $1^{112} 1^{113} 3^1 21^{1113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 8 & 2 \\ 1 & 0 & 2 & 3 & 2 & 9 & 2 \\ 1 & 2 & 0 & 4 & 4 & 12 & 2 \\ 3 & 3 & 4 & 5 & 6 & 21 & 5 \\ 3 & 2 & 4 & 6 & 4 & 18 & 4 \\ 8 & 9 & 12 & 21 & 18 & 61 & 15 \\ 2 & 2 & 2 & 5 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[542] $1^{131} 1^{121} 3^1 21^{1121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 3 & 2 \\ 1 & 0 & 2 & 1 & 7 & 3 & 2 \\ 1 & 2 & 0 & 3 & 9 & 4 & 2 \\ 2 & 1 & 3 & 1 & 9 & 4 & 3 \\ 8 & 7 & 9 & 9 & 35 & 16 & 11 \\ 3 & 3 & 4 & 4 & 16 & 5 & 5 \\ 2 & 2 & 2 & 3 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[541] $1^{142} 1^1 3^1 21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 6 & 3 \\ 3 & 5 & 0 & 23 & 10 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 14 & 23 & 0 & 80 & 36 & 18 \\ 6 & 10 & 0 & 36 & 14 & 8 \\ 3 & 5 & 0 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

[543] $1^{111} 1^{11} 3^1 21^{1122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 8 & 3 \\ 3 & 5 & 5 & 13 & 15 & 5 \\ 3 & 5 & 2 & 10 & 11 & 4 \\ 7 & 13 & 10 & 24 & 29 & 10 \\ 8 & 15 & 11 & 29 & 32 & 12 \\ 3 & 5 & 4 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[540] $1^{114} 1^{11} 3^1 21^{1114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 5 & 0 & 8 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 30 & 6 \\ 15 & 25 & 0 & 30 & 92 & 20 \\ 3 & 5 & 0 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[539] $1^{132} 1^{121} 3^1 21^{1121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 4 & 2 \\ 1 & 0 & 2 & 1 & 7 & 3 & 2 \\ 1 & 2 & 0 & 3 & 9 & 4 & 2 \\ 2 & 1 & 3 & 1 & 9 & 4 & 3 \\ 8 & 7 & 9 & 9 & 35 & 16 & 11 \\ 4 & 3 & 4 & 4 & 16 & 5 & 5 \\ 2 & 2 & 2 & 3 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[544] $1^{11} 3^{1241} 21^{1131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 & 3 \\ 3 & 5 & 5 & 23 & 5 & 5 \\ 4 & 5 & 2 & 17 & 4 & 4 \\ 15 & 23 & 17 & 78 & 17 & 18 \\ 3 & 5 & 4 & 17 & 2 & 4 \\ 3 & 5 & 4 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[546] $1^{121} 3^{1121} 21^{1121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 3 & 2 \\ 1 & 0 & 2 & 1 & 7 & 3 & 2 \\ 1 & 2 & 0 & 3 & 9 & 4 & 2 \\ 2 & 1 & 3 & 1 & 9 & 4 & 3 \\ 7 & 7 & 9 & 9 & 35 & 16 & 11 \\ 3 & 3 & 4 & 4 & 16 & 5 & 5 \\ 2 & 2 & 2 & 3 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[545] $1^{142} 3^{1121} 21^{1121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 4 & 2 \\ 1 & 0 & 2 & 1 & 7 & 3 & 2 \\ 1 & 2 & 0 & 3 & 9 & 4 & 2 \\ 2 & 1 & 3 & 1 & 9 & 4 & 3 \\ 9 & 7 & 9 & 9 & 35 & 16 & 11 \\ 4 & 3 & 4 & 4 & 16 & 5 & 5 \\ 2 & 2 & 2 & 3 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[548] $1^{122} 3^{1122} 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 6 & 2 \\ 1 & 2 & 0 & 0 & 8 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 26 & 9 \\ 6 & 6 & 8 & 0 & 26 & 26 & 10 \\ 2 & 2 & 2 & 0 & 9 & 10 & 3 \end{pmatrix}_{b_0}$$

[547] $1^{132} 3^{1122} 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 6 & 2 \\ 1 & 2 & 0 & 0 & 10 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 9 & 10 & 0 & 51 & 38 & 13 \\ 6 & 6 & 8 & 0 & 38 & 26 & 10 \\ 2 & 2 & 2 & 0 & 13 & 10 & 3 \end{pmatrix}_{b_0}$$

[549] $1^{124} 3^{1113} 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 9 & 2 \\ 1 & 2 & 0 & 0 & 5 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 27 & 6 \\ 10 & 9 & 12 & 0 & 27 & 61 & 15 \\ 2 & 2 & 2 & 0 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

[550] $1^{111} 3^{1212} 21^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 4 & 2 \\ 1 & 0 & 2 & 3 & 4 & 3 & 2 \\ 1 & 2 & 0 & 4 & 5 & 4 & 2 \\ 4 & 3 & 4 & 5 & 9 & 7 & 5 \\ 4 & 4 & 5 & 9 & 10 & 9 & 6 \\ 4 & 3 & 4 & 7 & 9 & 5 & 5 \\ 2 & 2 & 2 & 5 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[551] $1^{114} 3^{113} 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 10 & 2 \\ 1 & 0 & 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 0 & 4 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 2 & 4 & 0 & 4 & 18 & 4 \\ 10 & 9 & 12 & 0 & 18 & 61 & 15 \\ 2 & 2 & 2 & 0 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[552] \quad 1^{\overline{1}^{121}} 3^{\overline{1}} 21^{\overline{1121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -168, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 & 3 \\ 3 & 5 & 5 & 13 & 5 & 5 \\ 3 & 5 & 2 & 10 & 4 & 4 \\ 8 & 13 & 10 & 24 & 10 & 10 \\ 3 & 5 & 4 & 10 & 2 & 4 \\ 3 & 5 & 4 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[553] \quad 1^{\overline{1}^{121}} 1^{\overline{1111}} 3^{\overline{1}} 21^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -168, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 3 & 2 \\ 1 & 0 & 2 & 3 & 4 & 3 & 2 \\ 1 & 2 & 0 & 4 & 5 & 4 & 2 \\ 3 & 3 & 4 & 5 & 9 & 7 & 5 \\ 5 & 4 & 5 & 9 & 10 & 9 & 6 \\ 3 & 3 & 4 & 7 & 9 & 5 & 5 \\ 2 & 2 & 2 & 5 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[554] \quad 1^{\overline{1}^{132}} 1^{\overline{122}} 3^{\overline{1}} 21^{\overline{1122}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -168, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 6 & 2 \\ 1 & 0 & 2 & 1 & 7 & 6 & 2 \\ 1 & 2 & 0 & 3 & 9 & 8 & 2 \\ 2 & 1 & 3 & 1 & 9 & 8 & 3 \\ 8 & 7 & 9 & 9 & 35 & 32 & 11 \\ 6 & 6 & 8 & 8 & 32 & 26 & 10 \\ 2 & 2 & 2 & 3 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[555] \quad 1^{\overline{1}^{11}} 1^{\overline{1112}} 3^{\overline{1}} 21^{\overline{1112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -171, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 5 & 2 \\ 1 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 2 & 3 & 4 & 5 & 9 & 14 & 5 \\ 4 & 4 & 5 & 9 & 10 & 18 & 6 \\ 5 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[556] \quad 1^{\overline{1}^{131}} 1^{\overline{132}} 3^{\overline{1}} 21^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -171, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 0 & 9 & 5 & 2 \\ 1 & 2 & 0 & 0 & 10 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 9 & 10 & 0 & 51 & 28 & 13 \\ 4 & 5 & 5 & 0 & 28 & 13 & 7 \\ 2 & 2 & 2 & 0 & 13 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[557] \quad 1^{\overline{1}^{123}} 1^{\overline{122}} 3^{\overline{1}} 21^{\overline{1122}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -172, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 7 & 2 \\ 1 & 0 & 2 & 1 & 7 & 6 & 2 \\ 1 & 2 & 0 & 3 & 9 & 8 & 2 \\ 2 & 1 & 3 & 1 & 9 & 8 & 3 \\ 7 & 7 & 9 & 9 & 35 & 32 & 11 \\ 7 & 6 & 8 & 8 & 32 & 26 & 10 \\ 2 & 2 & 2 & 3 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[558] \quad 1^{\overline{1}^{44}} 1^{\overline{1}^{13}} 3^{\overline{1}} 21^{\overline{1144}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -172, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 12 & 3 \\ 3 & 5 & 5 & 23 & 20 & 5 \\ 2 & 5 & 2 & 18 & 16 & 4 \\ 14 & 23 & 18 & 80 & 72 & 18 \\ 12 & 20 & 16 & 72 & 62 & 16 \\ 3 & 5 & 4 & 18 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[559] \quad 1^{\overline{1}^{123}} 1^{\overline{111}} 3^{\overline{1}} 21^{\overline{1122}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -172, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 6 & 2 \\ 1 & 0 & 2 & 1 & 5 & 4 & 2 \\ 1 & 2 & 0 & 3 & 8 & 7 & 2 \\ 2 & 1 & 3 & 1 & 8 & 7 & 3 \\ 6 & 5 & 8 & 8 & 23 & 21 & 9 \\ 6 & 4 & 7 & 7 & 21 & 16 & 8 \\ 2 & 2 & 2 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[560] \quad 1^{111} 3^{121} 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -172, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 5 & 2 \\ 1 & 0 & 2 & 1 & 5 & 4 & 2 \\ 1 & 2 & 0 & 3 & 8 & 7 & 2 \\ 2 & 1 & 3 & 1 & 8 & 7 & 3 \\ 7 & 5 & 8 & 8 & 23 & 21 & 9 \\ 5 & 4 & 7 & 7 & 21 & 16 & 8 \\ 2 & 2 & 2 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[561] \quad 1^{134} 3^{121} 21^{1134} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -173, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 15 & 3 \\ 3 & 5 & 5 & 18 & 25 & 5 \\ 2 & 5 & 2 & 14 & 19 & 4 \\ 11 & 18 & 14 & 48 & 68 & 14 \\ 15 & 25 & 19 & 68 & 92 & 20 \\ 3 & 5 & 4 & 14 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[562] \quad 1^{112} 3^{121} 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -173, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 7 & 2 \\ 1 & 0 & 2 & 3 & 4 & 7 & 2 \\ 1 & 2 & 0 & 4 & 5 & 11 & 2 \\ 2 & 3 & 4 & 5 & 9 & 18 & 5 \\ 4 & 4 & 5 & 9 & 10 & 23 & 6 \\ 7 & 7 & 11 & 18 & 23 & 45 & 13 \\ 2 & 2 & 2 & 5 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[563] \quad 1^{111} 3^{121} 21^{1211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -175, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 3 & 2 \\ 1 & 0 & 2 & 4 & 4 & 3 & 2 \\ 1 & 2 & 0 & 7 & 5 & 4 & 2 \\ 5 & 4 & 7 & 16 & 14 & 11 & 8 \\ 4 & 4 & 5 & 14 & 10 & 9 & 6 \\ 3 & 3 & 4 & 11 & 9 & 5 & 5 \\ 2 & 2 & 2 & 8 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[566] \quad 1^{113} 3^{121} 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -176, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 12 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 3 & 5 & 2 & 6 & 15 & 4 \\ 4 & 8 & 6 & 8 & 24 & 6 \\ 12 & 20 & 15 & 24 & 58 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[564] \quad 1^{141} 3^{121} 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -176, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 & 2 \\ 1 & 0 & 2 & 0 & 8 & 1 & 2 \\ 1 & 2 & 0 & 0 & 12 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 8 & 12 & 0 & 56 & 12 & 14 \\ 2 & 1 & 3 & 0 & 12 & 1 & 3 \\ 2 & 2 & 2 & 0 & 14 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[565] \quad 1^{112} 3^{121} 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -176, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 14 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 3 & 5 & 2 & 6 & 18 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 14 & 25 & 18 & 30 & 94 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[567] \quad 1^{214} 3^{121} 21^{1213} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -176, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 13 & 3 \\ 3 & 5 & 10 & 8 & 20 & 5 \\ 6 & 10 & 14 & 12 & 30 & 8 \\ 5 & 8 & 12 & 8 & 24 & 6 \\ 13 & 20 & 30 & 24 & 58 & 16 \\ 3 & 5 & 8 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[568] $1^{114} 3^{114} 21^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -178, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 12 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 3 & 5 & 2 & 6 & 16 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 12 & 20 & 16 & 24 & 62 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[569] $1^{113} 3^{113} 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 0 & 5 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 23 & 6 \\ 8 & 7 & 11 & 0 & 23 & 45 & 13 \\ 2 & 2 & 2 & 0 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[570] $1^{112} 3^{112} 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 9 & 3 \\ 3 & 5 & 5 & 8 & 15 & 5 \\ 3 & 5 & 2 & 6 & 11 & 4 \\ 5 & 8 & 6 & 8 & 18 & 6 \\ 9 & 15 & 11 & 18 & 32 & 12 \\ 3 & 5 & 4 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

[571] $1^{111} 3^{111} 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 17 & 3 & 3 \\ 3 & 5 & 5 & 28 & 5 & 5 \\ 4 & 5 & 2 & 21 & 4 & 4 \\ 17 & 28 & 21 & 116 & 21 & 22 \\ 3 & 5 & 4 & 21 & 2 & 4 \\ 3 & 5 & 4 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

[572] $1^{1241} 3^{1241} 21^{1141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 & 3 \\ 3 & 5 & 5 & 23 & 5 & 5 \\ 4 & 5 & 2 & 18 & 4 & 4 \\ 14 & 23 & 18 & 80 & 18 & 18 \\ 3 & 5 & 4 & 18 & 2 & 4 \\ 3 & 5 & 4 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[573] $1^{122} 3^{122} 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -183, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 4 & 2 \\ 1 & 2 & 0 & 0 & 8 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 21 & 9 \\ 5 & 4 & 7 & 0 & 21 & 16 & 8 \\ 2 & 2 & 2 & 0 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[576] $1^{124} 3^{124} 21^{1124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 8 & 2 \\ 1 & 0 & 2 & 0 & 4 & 6 & 2 \\ 1 & 2 & 0 & 0 & 5 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 18 & 6 \\ 8 & 6 & 8 & 0 & 18 & 26 & 10 \\ 2 & 2 & 2 & 0 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[577] $1^{112} 3^{112} 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 6 & 2 \\ 1 & 0 & 2 & 3 & 2 & 6 & 2 \\ 1 & 2 & 0 & 4 & 4 & 8 & 2 \\ 3 & 3 & 4 & 5 & 6 & 14 & 5 \\ 3 & 2 & 4 & 6 & 4 & 12 & 4 \\ 6 & 6 & 8 & 14 & 12 & 26 & 10 \\ 2 & 2 & 2 & 5 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[578] \quad 1^{1^{231}} 3^1 21^{1^{221}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 8 & 3 & 2 \\ 1 & 0 & 2 & 4 & 7 & 3 & 2 \\ 1 & 2 & 0 & 7 & 9 & 4 & 2 \\ 5 & 4 & 7 & 16 & 25 & 11 & 8 \\ 8 & 7 & 9 & 25 & 35 & 16 & 11 \\ 3 & 3 & 4 & 11 & 16 & 5 & 5 \\ 2 & 2 & 2 & 8 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[575] \quad 1^{1^{214}} 3^1 21^{1^{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 8 & 2 \\ 1 & 0 & 2 & 3 & 2 & 6 & 2 \\ 1 & 2 & 0 & 4 & 4 & 8 & 2 \\ 4 & 3 & 4 & 5 & 6 & 14 & 5 \\ 3 & 2 & 4 & 6 & 4 & 12 & 4 \\ 8 & 6 & 8 & 14 & 12 & 26 & 10 \\ 2 & 2 & 2 & 5 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[574] \quad 1^{1^{112}} 3^1 21^{1^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 6 & 2 \\ 1 & 2 & 0 & 0 & 5 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 18 & 6 \\ 6 & 6 & 8 & 0 & 18 & 26 & 10 \\ 2 & 2 & 2 & 0 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[579] \quad 1^{1^{112}} 3^1 21^{1^{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -186, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 8 & 2 \\ 1 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 4 & 3 & 4 & 5 & 9 & 14 & 5 \\ 4 & 4 & 5 & 9 & 10 & 18 & 6 \\ 8 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[580] \quad 1^{1^{112}} 3^1 21^{1^{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -186, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 6 & 2 \\ 1 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 3 & 3 & 4 & 5 & 9 & 14 & 5 \\ 5 & 4 & 5 & 9 & 10 & 18 & 6 \\ 6 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[581] \quad 1^1 3^1 21^{1^{122}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -186, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 6 & 3 \\ 3 & 5 & 5 & 13 & 10 & 5 \\ 3 & 5 & 2 & 10 & 8 & 4 \\ 8 & 13 & 10 & 24 & 20 & 10 \\ 6 & 10 & 8 & 20 & 14 & 8 \\ 3 & 5 & 4 & 10 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[582] \quad 1^{1^{132}} 3^1 21^{1^{21}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -188, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 5 & 2 \\ 1 & 2 & 0 & 0 & 9 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 8 & 7 & 9 & 0 & 35 & 23 & 11 \\ 5 & 5 & 5 & 0 & 23 & 13 & 7 \\ 2 & 2 & 2 & 0 & 11 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[584] \quad 1^{1^{124}} 3^1 21^{1^{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -189, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 10 & 2 \\ 1 & 0 & 2 & 1 & 4 & 9 & 2 \\ 1 & 2 & 0 & 3 & 5 & 12 & 2 \\ 2 & 1 & 3 & 1 & 5 & 12 & 3 \\ 5 & 4 & 5 & 5 & 10 & 27 & 6 \\ 10 & 9 & 12 & 12 & 27 & 61 & 15 \\ 2 & 2 & 2 & 3 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

[585] $1^{122} 1^{11} 3^1 21^{1123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 11 & 3 \\ 3 & 5 & 5 & 13 & 20 & 5 \\ 3 & 5 & 2 & 10 & 15 & 4 \\ 8 & 13 & 10 & 24 & 39 & 10 \\ 11 & 20 & 15 & 39 & 58 & 16 \\ 3 & 5 & 4 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[586] $1^{122} 1^{11} 3^1 21^{1122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 9 & 3 \\ 3 & 5 & 5 & 13 & 15 & 5 \\ 3 & 5 & 2 & 10 & 11 & 4 \\ 8 & 13 & 10 & 24 & 29 & 10 \\ 9 & 15 & 11 & 29 & 32 & 12 \\ 3 & 5 & 4 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[587] $1^{1111} 3^1 21^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 3 & 2 \\ 1 & 0 & 2 & 3 & 4 & 3 & 2 \\ 1 & 2 & 0 & 4 & 5 & 4 & 2 \\ 4 & 3 & 4 & 5 & 9 & 7 & 5 \\ 4 & 4 & 5 & 9 & 10 & 9 & 6 \\ 3 & 3 & 4 & 7 & 9 & 5 & 5 \\ 2 & 2 & 2 & 5 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[588] $1^{112} 3^1 21^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 5 & 2 \\ 1 & 0 & 2 & 3 & 4 & 5 & 2 \\ 1 & 2 & 0 & 4 & 5 & 5 & 2 \\ 4 & 3 & 4 & 5 & 9 & 10 & 5 \\ 4 & 4 & 5 & 9 & 10 & 13 & 6 \\ 5 & 5 & 5 & 10 & 13 & 13 & 7 \\ 2 & 2 & 2 & 5 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[589] $1^{124} 1^{11} 3^1 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 15 & 3 \\ 3 & 5 & 0 & 13 & 25 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 13 & 0 & 24 & 49 & 10 \\ 15 & 25 & 0 & 49 & 92 & 20 \\ 3 & 5 & 0 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

[590] $1^{122} 1^{1111} 3^1 21^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 4 & 2 \\ 1 & 0 & 2 & 3 & 4 & 3 & 2 \\ 1 & 2 & 0 & 4 & 5 & 4 & 2 \\ 3 & 3 & 4 & 5 & 9 & 7 & 5 \\ 5 & 4 & 5 & 9 & 10 & 9 & 6 \\ 4 & 3 & 4 & 7 & 9 & 5 & 5 \\ 2 & 2 & 2 & 5 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[583] $1^{122} 1^{1112} 3^1 21^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 5 & 2 \\ 1 & 0 & 2 & 3 & 4 & 5 & 2 \\ 1 & 2 & 0 & 4 & 5 & 5 & 2 \\ 3 & 3 & 4 & 5 & 9 & 10 & 5 \\ 5 & 4 & 5 & 9 & 10 & 13 & 6 \\ 5 & 5 & 5 & 10 & 13 & 13 & 7 \\ 2 & 2 & 2 & 5 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[591] $1^{112} 1^{12} 3^1 21^{1114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 16 & 3 \\ 3 & 5 & 5 & 8 & 30 & 5 \\ 3 & 5 & 2 & 6 & 22 & 4 \\ 5 & 8 & 6 & 8 & 36 & 6 \\ 16 & 30 & 22 & 36 & 134 & 24 \\ 3 & 5 & 4 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[593] $1^{122} 1^{122} 3^1 21^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 5 & 2 \\ 1 & 0 & 2 & 0 & 6 & 5 & 2 \\ 1 & 2 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 6 & 6 & 0 & 20 & 18 & 8 \\ 5 & 5 & 5 & 0 & 18 & 13 & 7 \\ 2 & 2 & 2 & 0 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

1^{24}

$$[592] \ 1^{11} 3^{121} 21^{1214} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 15 & 3 \\ 3 & 5 & 10 & 8 & 25 & 5 \\ 6 & 10 & 14 & 12 & 38 & 8 \\ 4 & 8 & 12 & 8 & 30 & 6 \\ 15 & 25 & 38 & 30 & 92 & 20 \\ 3 & 5 & 8 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

1^{121}

$$[594] \ 1^{1111} 3^{121} 21^{1221} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 3 & 2 \\ 1 & 0 & 2 & 4 & 5 & 3 & 2 \\ 1 & 2 & 0 & 7 & 8 & 4 & 2 \\ 4 & 4 & 7 & 16 & 21 & 11 & 8 \\ 6 & 5 & 8 & 21 & 23 & 13 & 9 \\ 3 & 3 & 4 & 11 & 13 & 5 & 5 \\ 2 & 2 & 2 & 8 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

1^{214}

$$[595] \ 1^{11} 3^{121} 21^{1214} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 12 & 3 \\ 3 & 5 & 10 & 8 & 20 & 5 \\ 6 & 10 & 14 & 12 & 32 & 8 \\ 5 & 8 & 12 & 8 & 24 & 6 \\ 12 & 20 & 32 & 24 & 62 & 16 \\ 3 & 5 & 8 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

1^{21}

$$[596] \ 1^{1121} 3^{121} 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -197, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 2 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 9 & 4 & 2 \\ 1 & 2 & 1 & 0 & 7 & 3 & 2 \\ 7 & 7 & 9 & 7 & 35 & 16 & 11 \\ 3 & 3 & 4 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

1^{142}

$$[597] \ 1^{1121} 3^{121} 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -197, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 4 & 2 \\ 1 & 0 & 2 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 9 & 4 & 2 \\ 2 & 2 & 1 & 0 & 7 & 3 & 2 \\ 9 & 7 & 9 & 7 & 35 & 16 & 11 \\ 4 & 3 & 4 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

1^{212}

$$[598] \ 1^{11} 3^{121} 21^{1223} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 11 & 3 \\ 3 & 5 & 10 & 13 & 20 & 5 \\ 6 & 10 & 14 & 20 & 30 & 8 \\ 7 & 13 & 20 & 24 & 39 & 10 \\ 11 & 20 & 30 & 39 & 58 & 16 \\ 3 & 5 & 8 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

1^{132}

$$[599] \ 1^{121} 3^{121} 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 3 & 2 \\ 1 & 2 & 0 & 0 & 9 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 8 & 7 & 9 & 0 & 35 & 16 & 11 \\ 4 & 3 & 4 & 0 & 16 & 5 & 5 \\ 2 & 2 & 2 & 0 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

1^{131}

$$[600] \ 1^{121} 3^{121} 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 3 & 2 \\ 1 & 0 & 2 & 0 & 7 & 3 & 2 \\ 1 & 2 & 0 & 0 & 9 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 8 & 7 & 9 & 0 & 35 & 16 & 11 \\ 3 & 3 & 4 & 0 & 16 & 5 & 5 \\ 2 & 2 & 2 & 0 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[603] $1^{112} 3^1 21^{1112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 7 & 2 \\ 1 & 0 & 2 & 3 & 2 & 6 & 2 \\ 1 & 2 & 0 & 4 & 4 & 8 & 2 \\ 3 & 3 & 4 & 5 & 6 & 14 & 5 \\ 3 & 2 & 4 & 6 & 4 & 12 & 4 \\ 7 & 6 & 8 & 14 & 12 & 26 & 10 \\ 2 & 2 & 2 & 5 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[601] $1^{113} 3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 0 & 5 & 15 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 32 & 6 \\ 10 & 10 & 15 & 0 & 32 & 88 & 18 \\ 2 & 2 & 2 & 0 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

[602] $1^{113} 3^1 21^{1112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 7 & 2 \\ 1 & 0 & 2 & 3 & 2 & 6 & 2 \\ 1 & 2 & 0 & 4 & 4 & 8 & 2 \\ 4 & 3 & 4 & 5 & 6 & 14 & 5 \\ 3 & 2 & 4 & 6 & 4 & 12 & 4 \\ 7 & 6 & 8 & 14 & 12 & 26 & 10 \\ 2 & 2 & 2 & 5 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[604] $1^{131} 3^1 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -201, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 & 3 \\ 3 & 5 & 0 & 28 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 16 & 28 & 0 & 116 & 21 & 22 \\ 3 & 5 & 0 & 21 & 2 & 4 \\ 3 & 5 & 0 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

[605] $1^{141} 3^1 21^{1411} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -203, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 13 & 2 & 2 \\ 1 & 0 & 2 & 1 & 13 & 3 & 2 \\ 1 & 2 & 0 & 3 & 17 & 4 & 2 \\ 2 & 1 & 3 & 1 & 17 & 4 & 3 \\ 13 & 13 & 17 & 17 & 127 & 30 & 21 \\ 2 & 3 & 4 & 4 & 30 & 5 & 5 \\ 2 & 2 & 2 & 3 & 21 & 5 & 3 \end{pmatrix}_{b_0}$$

[606] $1^{123} 3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -204, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 8 & 2 \\ 1 & 0 & 2 & 0 & 7 & 7 & 2 \\ 1 & 2 & 0 & 0 & 9 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 41 & 11 \\ 8 & 7 & 11 & 0 & 41 & 45 & 13 \\ 2 & 2 & 2 & 0 & 11 & 13 & 3 \end{pmatrix}_{b_0}$$

[607] $1^{112} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -204, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 5 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 22 & 6 \\ 7 & 8 & 9 & 0 & 22 & 40 & 12 \\ 2 & 2 & 2 & 0 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[608] $1^{222} 3^1 21^{1222} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -204, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 6 & 3 \\ 3 & 5 & 10 & 13 & 10 & 5 \\ 6 & 10 & 14 & 20 & 16 & 8 \\ 8 & 13 & 20 & 24 & 20 & 10 \\ 6 & 10 & 16 & 20 & 14 & 8 \\ 3 & 5 & 8 & 10 & 8 & 4 \end{pmatrix}_{b_0}$$

[609] $1^{141} 3^{114} 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -204$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 13 & 1 & 2 \\ 1 & 0 & 2 & 1 & 13 & 0 & 2 \\ 1 & 2 & 0 & 3 & 17 & 0 & 2 \\ 2 & 1 & 3 & 1 & 17 & 0 & 3 \\ 13 & 13 & 17 & 17 & 127 & 0 & 21 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 3 & 21 & 0 & 3 \end{pmatrix}_{b_0}$$

[610] $1^{123} 3^{113} 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -204$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 9 & 2 \\ 1 & 0 & 2 & 0 & 5 & 9 & 2 \\ 1 & 2 & 0 & 0 & 8 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 39 & 9 \\ 9 & 9 & 12 & 0 & 39 & 61 & 15 \\ 2 & 2 & 2 & 0 & 9 & 15 & 3 \end{pmatrix}_{b_0}$$

[611] $1^{113} 3^{113} 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -206$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 10 & 2 \\ 1 & 0 & 2 & 1 & 4 & 10 & 2 \\ 1 & 2 & 0 & 3 & 5 & 15 & 2 \\ 2 & 1 & 3 & 1 & 5 & 15 & 3 \\ 4 & 4 & 5 & 5 & 10 & 32 & 6 \\ 10 & 10 & 15 & 15 & 32 & 88 & 18 \\ 2 & 2 & 2 & 3 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

[612] $1^{1213} 3^{112} 21^{1212} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -208$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 7 & 2 \\ 1 & 0 & 2 & 4 & 4 & 6 & 2 \\ 1 & 2 & 0 & 7 & 5 & 8 & 2 \\ 5 & 4 & 7 & 16 & 14 & 22 & 8 \\ 4 & 4 & 5 & 14 & 10 & 18 & 6 \\ 7 & 6 & 8 & 22 & 18 & 26 & 10 \\ 2 & 2 & 2 & 8 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[613] $1^{131} 3^{113} 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -211$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 0 & 10 & 5 & 2 \\ 1 & 2 & 0 & 0 & 13 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 10 & 13 & 0 & 74 & 33 & 16 \\ 4 & 5 & 5 & 0 & 33 & 13 & 7 \\ 2 & 2 & 2 & 0 & 16 & 7 & 3 \end{pmatrix}_{b_0}$$

[614] $1^{114} 3^{114} 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -212$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 18 & 3 \\ 3 & 5 & 0 & 8 & 30 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 8 & 36 & 6 \\ 18 & 30 & 0 & 36 & 134 & 24 \\ 3 & 5 & 0 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[615] $1^{12} 3^{1112} 3^{112} 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -214$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 6 & 2 \\ 1 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 2 & 3 & 4 & 5 & 9 & 14 & 5 \\ 4 & 4 & 5 & 9 & 10 & 18 & 6 \\ 6 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[616] $1^{214} 3^{114} 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -216$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 12 & 2 \\ 1 & 0 & 2 & 3 & 2 & 12 & 2 \\ 1 & 2 & 0 & 4 & 4 & 16 & 2 \\ 4 & 3 & 4 & 5 & 6 & 28 & 5 \\ 3 & 2 & 4 & 6 & 4 & 24 & 4 \\ 12 & 12 & 16 & 28 & 24 & 110 & 20 \\ 2 & 2 & 2 & 5 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[617] $1^{1112} 3^1 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 5 & 2 \\ 1 & 0 & 2 & 3 & 4 & 5 & 2 \\ 1 & 2 & 0 & 4 & 5 & 5 & 2 \\ 3 & 3 & 4 & 5 & 9 & 10 & 5 \\ 4 & 4 & 5 & 9 & 10 & 13 & 6 \\ 5 & 5 & 5 & 10 & 13 & 13 & 7 \\ 2 & 2 & 2 & 5 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[618] $1^{1112} 3^1 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 5 & 2 \\ 1 & 0 & 2 & 3 & 4 & 5 & 2 \\ 1 & 2 & 0 & 4 & 5 & 5 & 2 \\ 4 & 3 & 4 & 5 & 9 & 10 & 5 \\ 5 & 4 & 5 & 9 & 10 & 13 & 6 \\ 5 & 5 & 5 & 10 & 13 & 13 & 7 \\ 2 & 2 & 2 & 5 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[619] $21^{1112} 3^1 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 4 & 5 & 2 \\ 2 & 0 & 2 & 3 & 4 & 5 & 2 \\ 1 & 2 & 0 & 4 & 5 & 5 & 2 \\ 3 & 3 & 4 & 5 & 9 & 10 & 5 \\ 4 & 4 & 5 & 9 & 10 & 13 & 6 \\ 5 & 5 & 5 & 10 & 13 & 13 & 7 \\ 2 & 2 & 2 & 5 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[620] $1^{12} 3^1 21^{1124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 16 & 3 \\ 3 & 5 & 5 & 13 & 30 & 5 \\ 3 & 5 & 2 & 10 & 22 & 4 \\ 7 & 13 & 10 & 24 & 58 & 10 \\ 16 & 30 & 22 & 58 & 134 & 24 \\ 3 & 5 & 4 & 10 & 24 & 4 \end{pmatrix}_{b_0}$$

[621] $1^{123} 3^1 21^{1123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 9 & 3 \\ 3 & 5 & 5 & 13 & 15 & 5 \\ 3 & 5 & 2 & 10 & 12 & 4 \\ 8 & 13 & 10 & 24 & 30 & 10 \\ 9 & 15 & 12 & 30 & 34 & 12 \\ 3 & 5 & 4 & 10 & 12 & 4 \end{pmatrix}_{b_0}$$

[622] $1^{121} 3^1 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -217, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 4 & 2 \\ 1 & 0 & 2 & 1 & 8 & 3 & 2 \\ 1 & 2 & 0 & 3 & 12 & 4 & 2 \\ 2 & 1 & 3 & 1 & 12 & 4 & 3 \\ 9 & 8 & 12 & 12 & 56 & 20 & 14 \\ 4 & 3 & 4 & 4 & 20 & 5 & 5 \\ 2 & 2 & 2 & 3 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[625] $1^{112} 3^1 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 11 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 3 & 5 & 2 & 6 & 15 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 11 & 20 & 15 & 24 & 58 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[624] $1^{1111} 3^1 21^{1221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 6 & 3 & 2 \\ 1 & 0 & 2 & 4 & 5 & 3 & 2 \\ 1 & 2 & 0 & 7 & 8 & 4 & 2 \\ 6 & 4 & 7 & 16 & 21 & 11 & 8 \\ 6 & 5 & 8 & 21 & 23 & 13 & 9 \\ 3 & 3 & 4 & 11 & 13 & 5 & 5 \\ 2 & 2 & 2 & 8 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[623] $1^{141} 3^{121} 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 & 3 \\ 3 & 5 & 0 & 28 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 17 & 28 & 0 & 116 & 21 & 22 \\ 3 & 5 & 0 & 21 & 2 & 4 \\ 3 & 5 & 0 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

[626] $1^{122} 3^{121} 21^{1122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 6 & 2 \\ 1 & 0 & 2 & 1 & 7 & 6 & 2 \\ 1 & 2 & 0 & 3 & 9 & 8 & 2 \\ 2 & 1 & 3 & 1 & 9 & 8 & 3 \\ 7 & 7 & 9 & 9 & 35 & 32 & 11 \\ 6 & 6 & 8 & 8 & 32 & 26 & 10 \\ 2 & 2 & 2 & 3 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

[627] $1^{132} 3^{121} 21^{1122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 5 & 2 \\ 1 & 0 & 2 & 3 & 5 & 4 & 2 \\ 1 & 2 & 0 & 4 & 8 & 7 & 2 \\ 3 & 3 & 4 & 5 & 13 & 11 & 5 \\ 7 & 5 & 8 & 13 & 23 & 21 & 9 \\ 5 & 4 & 7 & 11 & 21 & 16 & 8 \\ 2 & 2 & 2 & 5 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[628] $1^{131} 3^{122} 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 & 3 \\ 3 & 5 & 0 & 33 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 19 & 33 & 0 & 164 & 24 & 26 \\ 3 & 5 & 0 & 24 & 2 & 4 \\ 3 & 5 & 0 & 26 & 4 & 4 \end{pmatrix}_{b_0}$$

[629] $1^{213} 3^{122} 21^{1212} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 4 & 8 & 2 \\ 6 & 6 & 8 & 26 & 12 & 28 & 10 \\ 3 & 2 & 4 & 12 & 4 & 12 & 4 \\ 7 & 6 & 8 & 28 & 12 & 26 & 10 \\ 2 & 2 & 2 & 10 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[630] $1^{113} 3^{114} 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 0 & 5 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 26 & 6 \\ 9 & 10 & 10 & 0 & 26 & 58 & 14 \\ 2 & 2 & 2 & 0 & 6 & 14 & 3 \end{pmatrix}_{b_0}$$

[631] $1^{213} 3^{111} 21^{1213} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 12 & 3 \\ 3 & 5 & 10 & 8 & 20 & 5 \\ 6 & 10 & 14 & 12 & 30 & 8 \\ 5 & 8 & 12 & 8 & 24 & 6 \\ 12 & 20 & 30 & 24 & 58 & 16 \\ 3 & 5 & 8 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[632] $1^{41} 3^{114} 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 1 & 2 \\ 1 & 0 & 2 & 2 & 12 & 0 & 2 \\ 1 & 2 & 0 & 1 & 14 & 0 & 2 \\ 1 & 2 & 1 & 0 & 12 & 0 & 2 \\ 12 & 12 & 14 & 12 & 96 & 0 & 18 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 18 & 0 & 3 \end{pmatrix}_{b_0}$$

[633] $1^{131} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -227$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10.00 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 2 & 2 \\ 1 & 0 & 2 & 1 & 11 & 3 & 2 \\ 1 & 2 & 0 & 3 & 16 & 4 & 2 \\ 2 & 1 & 3 & 1 & 16 & 4 & 3 \\ 11 & 11 & 16 & 16 & 103 & 27 & 19 \\ 2 & 3 & 4 & 4 & 27 & 5 & 5 \\ 2 & 2 & 2 & 3 & 19 & 5 & 3 \end{pmatrix}_{b_0}$$

[634] $1^{114} 3^1 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -227$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 1 & 2 \\ 1 & 0 & 2 & 2 & 13 & 0 & 2 \\ 1 & 2 & 0 & 1 & 17 & 0 & 2 \\ 1 & 2 & 1 & 0 & 13 & 0 & 2 \\ 13 & 13 & 17 & 13 & 127 & 0 & 21 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 21 & 0 & 3 \end{pmatrix}_{b_0}$$

[636] $1^{1111} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -228$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 3 & 2 \\ 1 & 0 & 2 & 3 & 5 & 3 & 2 \\ 1 & 2 & 0 & 4 & 8 & 4 & 2 \\ 4 & 3 & 4 & 5 & 13 & 7 & 5 \\ 6 & 5 & 8 & 13 & 23 & 13 & 9 \\ 3 & 3 & 4 & 7 & 13 & 5 & 5 \\ 2 & 2 & 2 & 5 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[635] $1^{1112} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -228$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 4 & 2 \\ 1 & 0 & 2 & 3 & 5 & 3 & 2 \\ 1 & 2 & 0 & 4 & 8 & 4 & 2 \\ 3 & 3 & 4 & 5 & 13 & 7 & 5 \\ 7 & 5 & 8 & 13 & 23 & 13 & 9 \\ 4 & 3 & 4 & 7 & 13 & 5 & 5 \\ 2 & 2 & 2 & 5 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[638] $1^{1111} 3^1 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -229$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 3 & 2 \\ 1 & 0 & 2 & 5 & 4 & 3 & 2 \\ 1 & 2 & 0 & 5 & 5 & 4 & 2 \\ 4 & 5 & 5 & 13 & 13 & 10 & 7 \\ 4 & 4 & 5 & 13 & 10 & 9 & 6 \\ 3 & 3 & 4 & 10 & 9 & 5 & 5 \\ 2 & 2 & 2 & 7 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[637] $1^{113} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -229$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 9 & 2 \\ 1 & 0 & 2 & 1 & 4 & 9 & 2 \\ 1 & 2 & 0 & 3 & 5 & 12 & 2 \\ 2 & 1 & 3 & 1 & 5 & 12 & 3 \\ 4 & 4 & 5 & 5 & 10 & 27 & 6 \\ 9 & 9 & 12 & 12 & 27 & 61 & 15 \\ 2 & 2 & 2 & 3 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

[639] $1^{112} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -232$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 4 & 6 & 2 \\ 1 & 0 & 2 & 4 & 4 & 6 & 2 \\ 1 & 2 & 0 & 7 & 5 & 8 & 2 \\ 5 & 4 & 7 & 16 & 14 & 22 & 8 \\ 4 & 4 & 5 & 14 & 10 & 18 & 6 \\ 6 & 6 & 8 & 22 & 18 & 26 & 10 \\ 2 & 2 & 2 & 8 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[642] $1^{142} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -29$, $\det = -235$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 3 & 2 \\ 1 & 2 & 0 & 0 & 9 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 7 & 9 & 0 & 35 & 16 & 11 \\ 4 & 3 & 4 & 0 & 16 & 5 & 5 \\ 2 & 2 & 2 & 0 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[641] $1^{114} 3^1 21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 13 & 1 & 2 \\ 1 & 0 & 2 & 0 & 4 & 13 & 0 & 2 \\ 1 & 2 & 0 & 0 & 7 & 17 & 0 & 2 \\ 5 & 4 & 7 & 16 & 47 & 0 & 8 & \\ 13 & 13 & 17 & 47 & 127 & 0 & 21 & \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & \\ 2 & 2 & 2 & 0 & 11 & 5 & 3 & \end{pmatrix}_{b_0}$$

[640] $1^{121} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 2 & 0 & 7 & 3 & 2 \\ 1 & 2 & 0 & 0 & 9 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 16 & 11 \\ 3 & 3 & 4 & 0 & 16 & 5 & 5 \\ 2 & 2 & 2 & 0 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[643] $1^{113} 3^1 21^{1241} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 3 & 3 \\ 3 & 5 & 10 & 28 & 5 & 5 \\ 5 & 10 & 14 & 42 & 8 & 8 \\ 15 & 28 & 42 & 116 & 21 & 22 \\ 3 & 5 & 8 & 21 & 2 & 4 \\ 3 & 5 & 8 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

[644] $1^{113} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 5 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 22 & 6 \\ 8 & 8 & 9 & 0 & 22 & 40 & 12 \\ 2 & 2 & 2 & 0 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[645] $1^{113} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 0 & 5 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 4 & 5 & 0 & 10 & 22 & 6 \\ 9 & 8 & 9 & 0 & 22 & 40 & 12 \\ 2 & 2 & 2 & 0 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

[646] $1^{111} 3^1 21^{1122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -237, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 5 & 2 \\ 1 & 0 & 2 & 1 & 5 & 4 & 2 \\ 1 & 2 & 0 & 3 & 8 & 7 & 2 \\ 2 & 1 & 3 & 1 & 8 & 7 & 3 \\ 6 & 5 & 8 & 8 & 23 & 21 & 9 \\ 5 & 4 & 7 & 7 & 21 & 16 & 8 \\ 2 & 2 & 2 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[650] $1^{121} 3^1 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 & 3 \\ 3 & 5 & 0 & 33 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 20 & 33 & 0 & 164 & 24 & 26 \\ 3 & 5 & 0 & 24 & 2 & 4 \\ 3 & 5 & 0 & 26 & 4 & 4 \end{pmatrix}_{b_0}$$

[647] $1^{1121} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 3 & 2 \\ 1 & 0 & 2 & 3 & 8 & 3 & 2 \\ 1 & 2 & 0 & 4 & 12 & 4 & 2 \\ 2 & 3 & 4 & 5 & 20 & 7 & 5 \\ 9 & 8 & 12 & 20 & 56 & 20 & 14 \\ 3 & 3 & 4 & 7 & 20 & 5 & 5 \\ 2 & 2 & 2 & 5 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[649] \quad 1^{113} 3^{114} 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -240, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 10 & 2 \\ 1 & 0 & 2 & 1 & 4 & 9 & 2 \\ 1 & 2 & 0 & 3 & 5 & 12 & 2 \\ 2 & 1 & 3 & 1 & 5 & 12 & 3 \\ 4 & 4 & 5 & 5 & 10 & 27 & 6 \\ 10 & 9 & 12 & 12 & 27 & 61 & 15 \\ 2 & 2 & 2 & 3 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[653] \quad 1^{12} 3^{114} 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -240, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 16 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 3 & 5 & 2 & 6 & 18 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 16 & 25 & 18 & 30 & 94 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[652] \quad 1^{1112} 3^{1213} 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -240, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 8 & 2 \\ 1 & 0 & 2 & 3 & 4 & 7 & 2 \\ 1 & 2 & 0 & 4 & 5 & 11 & 2 \\ 4 & 3 & 4 & 5 & 9 & 18 & 5 \\ 4 & 4 & 5 & 9 & 10 & 23 & 6 \\ 8 & 7 & 11 & 18 & 23 & 45 & 13 \\ 2 & 2 & 2 & 5 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[651] \quad 1^{121} 3^{121} 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -240, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 4 & 2 \\ 1 & 2 & 0 & 0 & 12 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 8 & 12 & 0 & 56 & 32 & 14 \\ 5 & 4 & 7 & 0 & 32 & 16 & 8 \\ 2 & 2 & 2 & 0 & 14 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[648] \quad 1^{113} 3^{121} 21^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -240, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 15 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 3 & 5 & 2 & 6 & 19 & 4 \\ 4 & 8 & 6 & 8 & 30 & 6 \\ 15 & 25 & 19 & 30 & 92 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[654] \quad 1^{113} 3^{121} 21^{1141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -241, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 & 3 \\ 3 & 5 & 5 & 28 & 5 & 5 \\ 2 & 5 & 2 & 21 & 4 & 4 \\ 16 & 28 & 21 & 116 & 21 & 22 \\ 3 & 5 & 4 & 21 & 2 & 4 \\ 3 & 5 & 4 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[655] \quad 1^{1111} 3^{121} 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -242, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 5 & 2 \\ 1 & 0 & 2 & 3 & 5 & 4 & 2 \\ 1 & 2 & 0 & 4 & 8 & 7 & 2 \\ 4 & 3 & 4 & 5 & 13 & 11 & 5 \\ 6 & 5 & 8 & 13 & 23 & 21 & 9 \\ 5 & 4 & 7 & 11 & 21 & 16 & 8 \\ 2 & 2 & 2 & 5 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[656] \quad 1^{141} 3^{121} 21^{1141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -243, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.00 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 13 & 4 & 2 \\ 1 & 0 & 2 & 1 & 13 & 3 & 2 \\ 1 & 2 & 0 & 3 & 17 & 4 & 2 \\ 2 & 1 & 3 & 1 & 17 & 4 & 3 \\ 13 & 13 & 17 & 17 & 127 & 30 & 21 \\ 4 & 3 & 4 & 4 & 30 & 5 & 5 \\ 2 & 2 & 2 & 3 & 21 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[657] \quad 1^{1121} 3^1 21^{1221} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -245, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 7 & 3 & 2 \\ 1 & 0 & 2 & 4 & 7 & 3 & 2 \\ 1 & 2 & 0 & 7 & 9 & 4 & 2 \\ 5 & 4 & 7 & 16 & 25 & 11 & 8 \\ 7 & 7 & 9 & 25 & 35 & 16 & 11 \\ 3 & 3 & 4 & 11 & 16 & 5 & 5 \\ 2 & 2 & 2 & 8 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[659] \quad 1^{113} 3^1 21^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -251, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 11 & 2 \\ 1 & 0 & 2 & 1 & 4 & 10 & 2 \\ 1 & 2 & 0 & 3 & 5 & 15 & 2 \\ 2 & 1 & 3 & 1 & 5 & 15 & 3 \\ 4 & 4 & 5 & 5 & 10 & 32 & 6 \\ 11 & 10 & 15 & 15 & 32 & 88 & 18 \\ 2 & 2 & 2 & 3 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[658] \quad 1^{123} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -251, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.00 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 8 & 2 \\ 1 & 0 & 2 & 0 & 7 & 8 & 2 \\ 1 & 2 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 39 & 11 \\ 8 & 8 & 9 & 0 & 39 & 40 & 12 \\ 2 & 2 & 2 & 0 & 11 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[660] \quad 1^{1112} 3^1 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -252, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 6 & 6 & 2 \\ 1 & 0 & 2 & 3 & 5 & 6 & 2 \\ 1 & 2 & 0 & 4 & 8 & 8 & 2 \\ 4 & 3 & 4 & 5 & 13 & 14 & 5 \\ 6 & 5 & 8 & 13 & 23 & 26 & 9 \\ 6 & 6 & 8 & 14 & 26 & 26 & 10 \\ 2 & 2 & 2 & 5 & 9 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[661] \quad 1^{1121} 3^1 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -253, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 3 & 2 \\ 1 & 0 & 2 & 3 & 7 & 3 & 2 \\ 1 & 2 & 0 & 4 & 9 & 4 & 2 \\ 2 & 3 & 4 & 5 & 16 & 7 & 5 \\ 7 & 7 & 9 & 16 & 35 & 16 & 11 \\ 3 & 3 & 4 & 7 & 16 & 5 & 5 \\ 2 & 2 & 2 & 5 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[668] \quad 1^{123} 3^1 21^{1133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 9 & 3 \\ 3 & 5 & 5 & 18 & 15 & 5 \\ 3 & 5 & 2 & 14 & 12 & 4 \\ 10 & 18 & 14 & 48 & 42 & 14 \\ 9 & 15 & 12 & 42 & 34 & 12 \\ 3 & 5 & 4 & 14 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[667] \quad 1^{124} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 11 & 2 \\ 1 & 0 & 2 & 0 & 7 & 12 & 2 \\ 1 & 2 & 0 & 0 & 9 & 16 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 64 & 11 \\ 11 & 12 & 16 & 0 & 64 & 110 & 20 \\ 2 & 2 & 2 & 0 & 11 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[674] \quad 1^{124} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 9 & 2 \\ 1 & 0 & 2 & 0 & 6 & 10 & 2 \\ 1 & 2 & 0 & 0 & 6 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 6 & 6 & 0 & 20 & 36 & 8 \\ 9 & 10 & 10 & 0 & 36 & 58 & 14 \\ 2 & 2 & 2 & 0 & 8 & 14 & 3 \end{pmatrix}_{b_0}$$

[673] $1^{14} 1^{12} 3^1 21^{1114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 18 & 3 \\ 3 & 5 & 5 & 8 & 30 & 5 \\ 3 & 5 & 2 & 6 & 22 & 4 \\ 4 & 8 & 6 & 8 & 36 & 6 \\ 18 & 30 & 22 & 36 & 134 & 24 \\ 3 & 5 & 4 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[672] $1^{121} 1^{12} 3^1 21^{1231} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 & 3 \\ 3 & 5 & 10 & 18 & 5 & 5 \\ 6 & 10 & 14 & 28 & 8 & 8 \\ 10 & 18 & 28 & 48 & 14 & 14 \\ 3 & 5 & 8 & 14 & 2 & 4 \\ 3 & 5 & 8 & 14 & 4 & 4 \end{pmatrix}_{b_0}$$

[671] $1^{222} 1^{12} 3^1 21^{1232} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 6 & 3 \\ 3 & 5 & 10 & 18 & 10 & 5 \\ 6 & 10 & 14 & 28 & 16 & 8 \\ 10 & 18 & 28 & 48 & 28 & 14 \\ 6 & 10 & 16 & 28 & 14 & 8 \\ 3 & 5 & 8 & 14 & 8 & 4 \end{pmatrix}_{b_0}$$

[670] $1^{421} 1^{12} 3^1 21^{1431} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 10 & 3 & 3 \\ 3 & 5 & 20 & 18 & 5 & 5 \\ 12 & 20 & 62 & 56 & 16 & 16 \\ 10 & 18 & 56 & 48 & 14 & 14 \\ 3 & 5 & 16 & 14 & 2 & 4 \\ 3 & 5 & 16 & 14 & 4 & 4 \end{pmatrix}_{b_0}$$

[669] $1^{121} 1^{12} 3^1 21^{1131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 & 3 \\ 3 & 5 & 5 & 18 & 5 & 5 \\ 3 & 5 & 2 & 14 & 4 & 4 \\ 10 & 18 & 14 & 48 & 14 & 14 \\ 3 & 5 & 4 & 14 & 2 & 4 \\ 3 & 5 & 4 & 14 & 4 & 4 \end{pmatrix}_{b_0}$$

[662] $1^{141} 1^{121} 3^1 21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 3 & 2 \\ 1 & 0 & 2 & 0 & 8 & 3 & 2 \\ 1 & 2 & 0 & 0 & 12 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 8 & 12 & 0 & 56 & 20 & 14 \\ 3 & 3 & 4 & 0 & 20 & 5 & 5 \\ 2 & 2 & 2 & 0 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[663] $1^{212} 1^{122} 3^1 21^{1212} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 4 & 8 & 2 \\ 6 & 6 & 8 & 26 & 12 & 28 & 10 \\ 3 & 2 & 4 & 12 & 4 & 12 & 4 \\ 6 & 6 & 8 & 28 & 12 & 26 & 10 \\ 2 & 2 & 2 & 10 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[664] $1^{141} 1^{11} 3^1 21^{1131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 & 3 \\ 3 & 5 & 5 & 23 & 5 & 5 \\ 3 & 5 & 2 & 17 & 4 & 4 \\ 15 & 23 & 17 & 78 & 17 & 18 \\ 3 & 5 & 4 & 17 & 2 & 4 \\ 3 & 5 & 4 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[665] $1^{12} 3^{141} 21^{141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 & 2 \\ 1 & 0 & 2 & 0 & 9 & 1 & 2 \\ 1 & 2 & 0 & 0 & 15 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 9 & 15 & 0 & 83 & 15 & 17 \\ 2 & 1 & 3 & 0 & 15 & 1 & 3 \\ 2 & 2 & 2 & 0 & 17 & 3 & 3 \end{pmatrix}_{b_0}$$

[666] $1^{121} 3^{142} 21^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 0 & 8 & 3 & 2 \\ 1 & 2 & 0 & 0 & 12 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 8 & 12 & 0 & 56 & 20 & 14 \\ 4 & 3 & 4 & 0 & 20 & 5 & 5 \\ 2 & 2 & 2 & 0 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[675] $1^{124} 3^{141} 21^{124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 12 & 3 \\ 3 & 5 & 5 & 13 & 20 & 5 \\ 3 & 5 & 2 & 10 & 16 & 4 \\ 8 & 13 & 10 & 24 & 40 & 10 \\ 12 & 20 & 16 & 40 & 62 & 16 \\ 3 & 5 & 4 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[677] $1^{13} 3^{141} 21^{141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -259, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 & 2 \\ 1 & 0 & 2 & 0 & 11 & 1 & 2 \\ 1 & 2 & 0 & 0 & 16 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 11 & 16 & 0 & 103 & 16 & 19 \\ 2 & 1 & 3 & 0 & 16 & 1 & 3 \\ 2 & 2 & 2 & 0 & 19 & 3 & 3 \end{pmatrix}_{b_0}$$

[676] $1^{123} 3^{141} 21^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -259, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 8 & 2 \\ 1 & 2 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 39 & 11 \\ 7 & 8 & 9 & 0 & 39 & 40 & 12 \\ 2 & 2 & 2 & 0 & 11 & 12 & 3 \end{pmatrix}_{b_0}$$

[679] $1^{114} 3^{141} 21^{124} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 12 & 2 \\ 1 & 0 & 2 & 0 & 5 & 12 & 2 \\ 1 & 2 & 0 & 0 & 8 & 16 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 52 & 9 \\ 12 & 12 & 16 & 0 & 52 & 110 & 20 \\ 2 & 2 & 2 & 0 & 9 & 20 & 3 \end{pmatrix}_{b_0}$$

[678] $1^{123} 3^{141} 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 15 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 3 & 5 & 2 & 6 & 18 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 15 & 25 & 18 & 30 & 94 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[681] $1^{112} 3^{141} 21^{123} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 9 & 2 \\ 1 & 0 & 2 & 0 & 5 & 7 & 2 \\ 1 & 2 & 0 & 0 & 8 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 34 & 9 \\ 9 & 7 & 11 & 0 & 34 & 45 & 13 \\ 2 & 2 & 2 & 0 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

[680] $1^{132} 3^{141} 21^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -263, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

6.5

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 5 & 2 \\ 1 & 0 & 2 & 0 & 10 & 5 & 2 \\ 1 & 2 & 0 & 0 & 13 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 10 & 13 & 0 & 74 & 33 & 16 \\ 5 & 5 & 5 & 0 & 33 & 13 & 7 \\ 2 & 2 & 2 & 0 & 16 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[682] \quad 1^{1^{112}} 3^1 21^{1^{113}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 7 & 2 \\ 1 & 0 & 2 & 3 & 4 & 7 & 2 \\ 1 & 2 & 0 & 4 & 5 & 11 & 2 \\ 3 & 3 & 4 & 5 & 9 & 18 & 5 \\ 4 & 4 & 5 & 9 & 10 & 23 & 6 \\ 7 & 7 & 11 & 18 & 23 & 45 & 13 \\ 2 & 2 & 2 & 5 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[683] \quad 21^{1^{111} 3^1} 21^{1^{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 4 & 3 & 2 \\ 2 & 0 & 2 & 3 & 4 & 3 & 2 \\ 1 & 2 & 0 & 4 & 5 & 4 & 2 \\ 3 & 3 & 4 & 5 & 9 & 7 & 5 \\ 4 & 4 & 5 & 9 & 10 & 9 & 6 \\ 3 & 3 & 4 & 7 & 9 & 5 & 5 \\ 2 & 2 & 2 & 5 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[684] \quad 1^{1^{213}} 3^1 21^{1^{214}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -266, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 14 & 3 \\ 3 & 5 & 10 & 8 & 25 & 5 \\ 6 & 10 & 14 & 12 & 38 & 8 \\ 5 & 8 & 12 & 8 & 30 & 6 \\ 14 & 25 & 38 & 30 & 92 & 20 \\ 3 & 5 & 8 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[685] \quad 1^{1^{42}} 3^1 21^{1^{31}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 4 & 2 \\ 1 & 0 & 2 & 3 & 10 & 3 & 2 \\ 1 & 2 & 0 & 3 & 13 & 4 & 2 \\ 2 & 1 & 3 & 1 & 13 & 4 & 3 \\ 11 & 10 & 13 & 13 & 74 & 23 & 16 \\ 4 & 3 & 4 & 4 & 23 & 5 & 5 \\ 2 & 2 & 2 & 3 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[686] \quad 1^{1^{124}} 3^1 21^{1^{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -270, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 8 & 2 \\ 1 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 3 & 3 & 4 & 5 & 9 & 14 & 5 \\ 5 & 4 & 5 & 9 & 10 & 18 & 6 \\ 8 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[687] \quad 1^{1^{24}} 3^1 21^{1^{124}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -270, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 18 & 3 \\ 3 & 5 & 5 & 13 & 30 & 5 \\ 3 & 5 & 2 & 10 & 22 & 4 \\ 8 & 13 & 10 & 24 & 58 & 10 \\ 18 & 30 & 22 & 58 & 134 & 24 \\ 3 & 5 & 4 & 10 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[688] \quad 1^{1^{112}} 3^1 21^{1^{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -270, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 6 & 2 \\ 1 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 4 & 3 & 4 & 5 & 9 & 14 & 5 \\ 4 & 4 & 5 & 9 & 10 & 18 & 6 \\ 6 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[689] \quad 1^{1^{222}} 3^1 21^{1^{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 4 & 2 \\ 1 & 0 & 2 & 3 & 4 & 3 & 2 \\ 1 & 2 & 0 & 4 & 5 & 4 & 2 \\ 4 & 3 & 4 & 5 & 9 & 7 & 5 \\ 5 & 4 & 5 & 9 & 10 & 9 & 6 \\ 4 & 3 & 4 & 7 & 9 & 5 & 5 \\ 2 & 2 & 2 & 5 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[690] \ 1^{131} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -272, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 0 & 9 & 3 & 2 \\ 1 & 2 & 0 & 0 & 10 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 9 & 10 & 0 & 51 & 19 & 13 \\ 4 & 3 & 4 & 0 & 19 & 5 & 5 \\ 2 & 2 & 2 & 0 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[691] \ 1^{111} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -272, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 3 & 2 \\ 1 & 0 & 2 & 3 & 4 & 3 & 2 \\ 1 & 2 & 0 & 4 & 5 & 4 & 2 \\ 3 & 3 & 4 & 5 & 9 & 7 & 5 \\ 4 & 4 & 5 & 9 & 10 & 9 & 6 \\ 3 & 3 & 4 & 7 & 9 & 5 & 5 \\ 2 & 2 & 2 & 5 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[692] \ 1^{131} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -272, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 3 & 2 \\ 1 & 0 & 2 & 0 & 9 & 3 & 2 \\ 1 & 2 & 0 & 0 & 10 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 9 & 10 & 0 & 51 & 19 & 13 \\ 3 & 3 & 4 & 0 & 19 & 5 & 5 \\ 2 & 2 & 2 & 0 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[693] \ 1^{113} 3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -275, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 11 & 2 \\ 1 & 0 & 2 & 1 & 4 & 12 & 2 \\ 1 & 2 & 0 & 3 & 5 & 16 & 2 \\ 2 & 1 & 3 & 1 & 5 & 16 & 3 \\ 4 & 4 & 5 & 5 & 10 & 36 & 6 \\ 11 & 12 & 16 & 16 & 36 & 110 & 20 \\ 2 & 2 & 2 & 3 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[694] \ 1^{114} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -275, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 13 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 3 & 5 & 2 & 6 & 15 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 13 & 20 & 15 & 24 & 58 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[695] \ 1^{112} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -279, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 5 & 2 \\ 1 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 3 & 3 & 4 & 5 & 9 & 14 & 5 \\ 4 & 4 & 5 & 9 & 10 & 18 & 6 \\ 5 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[696] \ 1^{231} 3^1 21^{1241} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -280, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 16 & 3 & 3 \\ 3 & 5 & 10 & 28 & 5 & 5 \\ 6 & 10 & 14 & 42 & 8 & 8 \\ 16 & 28 & 42 & 116 & 21 & 22 \\ 3 & 5 & 8 & 21 & 2 & 4 \\ 3 & 5 & 8 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[697] \ 1^{131} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -280, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 & 3 \\ 3 & 5 & 5 & 18 & 5 & 5 \\ 3 & 5 & 2 & 14 & 4 & 4 \\ 11 & 18 & 14 & 48 & 14 & 14 \\ 3 & 5 & 4 & 14 & 2 & 4 \\ 3 & 5 & 4 & 14 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[698] \quad 1^{1112} 3^{1214} 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -285, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 9 & 2 \\ 1 & 0 & 2 & 3 & 4 & 7 & 2 \\ 1 & 2 & 0 & 4 & 5 & 11 & 2 \\ 4 & 3 & 4 & 5 & 9 & 18 & 5 \\ 4 & 4 & 5 & 9 & 10 & 23 & 6 \\ 9 & 7 & 11 & 18 & 23 & 45 & 13 \\ 2 & 2 & 2 & 5 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[699] \quad 1^{1112} 3^{124} 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -285, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 9 & 2 \\ 1 & 0 & 2 & 3 & 4 & 7 & 2 \\ 1 & 2 & 0 & 4 & 5 & 11 & 2 \\ 3 & 3 & 4 & 5 & 9 & 18 & 5 \\ 5 & 4 & 5 & 9 & 10 & 23 & 6 \\ 9 & 7 & 11 & 18 & 23 & 45 & 13 \\ 2 & 2 & 2 & 5 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[700] \quad 1^{11} 3^{124} 21^{1123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -285, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 13 & 3 \\ 3 & 5 & 5 & 13 & 20 & 5 \\ 3 & 5 & 2 & 10 & 15 & 4 \\ 8 & 13 & 10 & 24 & 39 & 10 \\ 13 & 20 & 15 & 39 & 58 & 16 \\ 3 & 5 & 4 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[701] \quad 1^{131} 3^{142} 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -287, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 4 & 2 \\ 1 & 0 & 2 & 0 & 10 & 3 & 2 \\ 1 & 2 & 0 & 0 & 13 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 10 & 13 & 0 & 74 & 23 & 16 \\ 4 & 3 & 4 & 0 & 23 & 5 & 5 \\ 2 & 2 & 2 & 0 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[702] \quad 1^{121} 3^{121} 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -288, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 & 3 \\ 3 & 5 & 5 & 23 & 5 & 5 \\ 3 & 5 & 2 & 17 & 4 & 4 \\ 13 & 23 & 17 & 78 & 17 & 18 \\ 3 & 5 & 4 & 17 & 2 & 4 \\ 3 & 5 & 4 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[703] \quad 1^{1112} 3^{114} 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -292, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 9 & 2 \\ 1 & 0 & 2 & 3 & 4 & 7 & 2 \\ 1 & 2 & 0 & 4 & 5 & 11 & 2 \\ 3 & 3 & 4 & 5 & 9 & 18 & 5 \\ 4 & 4 & 5 & 9 & 10 & 23 & 6 \\ 9 & 7 & 11 & 18 & 23 & 45 & 13 \\ 2 & 2 & 2 & 5 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[704] \quad 1^{114} 3^{113} 21^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -296, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 11 & 2 \\ 1 & 0 & 2 & 3 & 2 & 12 & 2 \\ 1 & 2 & 0 & 4 & 4 & 16 & 2 \\ 3 & 3 & 4 & 5 & 6 & 28 & 5 \\ 3 & 2 & 4 & 6 & 4 & 24 & 4 \\ 11 & 12 & 16 & 28 & 24 & 110 & 20 \\ 2 & 2 & 2 & 5 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[705] \quad 1^{132} 3^{132} 21^{1132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -298, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 6 & 3 \\ 3 & 5 & 5 & 18 & 10 & 5 \\ 3 & 5 & 2 & 14 & 8 & 4 \\ 11 & 18 & 14 & 48 & 28 & 14 \\ 6 & 10 & 8 & 28 & 14 & 8 \\ 3 & 5 & 4 & 14 & 8 & 4 \end{pmatrix}_{b_0}$$

[707] $1^{112} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 8 & 2 \\ 1 & 0 & 2 & 0 & 5 & 7 & 2 \\ 1 & 2 & 0 & 0 & 8 & 11 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 34 & 9 \\ 8 & 7 & 11 & 0 & 34 & 45 & 13 \\ 2 & 2 & 2 & 0 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

[706] $1^{113} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 12 & 3 \\ 3 & 5 & 5 & 8 & 20 & 5 \\ 3 & 5 & 2 & 6 & 15 & 4 \\ 5 & 8 & 6 & 8 & 24 & 6 \\ 12 & 20 & 15 & 24 & 58 & 16 \\ 3 & 5 & 4 & 6 & 16 & 4 \end{pmatrix}_{b_0}$$

[710] $1^{142} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 4 & 2 \\ 1 & 0 & 2 & 0 & 12 & 5 & 2 \\ 1 & 2 & 0 & 0 & 14 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 12 & 14 & 0 & 96 & 38 & 18 \\ 4 & 5 & 5 & 0 & 38 & 13 & 7 \\ 2 & 2 & 2 & 0 & 18 & 7 & 3 \end{pmatrix}_{b_0}$$

[709] $1^{121} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 4 & 2 \\ 1 & 0 & 2 & 1 & 8 & 3 & 2 \\ 1 & 2 & 0 & 3 & 12 & 4 & 2 \\ 2 & 1 & 3 & 1 & 12 & 4 & 3 \\ 10 & 8 & 12 & 12 & 56 & 20 & 14 \\ 4 & 3 & 4 & 4 & 20 & 5 & 5 \\ 2 & 2 & 2 & 3 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[708] $1^{141} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 3 & 2 \\ 1 & 0 & 2 & 1 & 8 & 3 & 2 \\ 1 & 2 & 0 & 3 & 12 & 4 & 2 \\ 2 & 1 & 3 & 1 & 12 & 4 & 3 \\ 10 & 8 & 12 & 12 & 56 & 20 & 14 \\ 3 & 3 & 4 & 4 & 20 & 5 & 5 \\ 2 & 2 & 2 & 3 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[711] $1^{131} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -301, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 10 & 3 & 2 \\ 1 & 0 & 2 & 1 & 10 & 3 & 2 \\ 1 & 2 & 0 & 3 & 13 & 4 & 2 \\ 2 & 1 & 3 & 1 & 13 & 4 & 3 \\ 10 & 10 & 13 & 13 & 74 & 23 & 16 \\ 3 & 3 & 4 & 4 & 23 & 5 & 5 \\ 2 & 2 & 2 & 3 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

[712] $1^{113} 3^1 21^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 9 & 2 \\ 1 & 0 & 2 & 3 & 2 & 9 & 2 \\ 1 & 2 & 0 & 4 & 4 & 12 & 2 \\ 3 & 3 & 4 & 5 & 6 & 21 & 5 \\ 3 & 2 & 4 & 6 & 4 & 18 & 4 \\ 9 & 9 & 12 & 21 & 18 & 61 & 15 \\ 2 & 2 & 2 & 5 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[713] $1^{132} 3^1 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 4 & 2 \\ 1 & 2 & 0 & 0 & 12 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 8 & 12 & 0 & 56 & 32 & 14 \\ 5 & 4 & 7 & 0 & 32 & 16 & 8 \\ 2 & 2 & 2 & 0 & 14 & 8 & 3 \end{pmatrix}_{b_0}$$

[714] $1^{142} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 4 & 2 \\ 1 & 0 & 2 & 0 & 13 & 5 & 2 \\ 1 & 2 & 0 & 0 & 17 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 13 & 13 & 17 & 0 & 127 & 43 & 21 \\ 4 & 5 & 5 & 0 & 43 & 13 & 7 \\ 2 & 2 & 2 & 0 & 21 & 7 & 3 \end{pmatrix}_{b_0}$$

[715] $1^{113} 3^1 21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 10 & 2 \\ 1 & 0 & 2 & 3 & 2 & 9 & 2 \\ 1 & 2 & 0 & 4 & 4 & 12 & 2 \\ 4 & 3 & 4 & 5 & 6 & 21 & 5 \\ 3 & 2 & 4 & 6 & 4 & 18 & 4 \\ 10 & 9 & 12 & 21 & 18 & 61 & 15 \\ 2 & 2 & 2 & 5 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[716] $1^{112} 3^1 21^{1123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -306, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 11 & 3 \\ 3 & 5 & 5 & 13 & 20 & 5 \\ 3 & 5 & 2 & 10 & 15 & 4 \\ 7 & 13 & 10 & 24 & 39 & 10 \\ 11 & 20 & 15 & 39 & 58 & 16 \\ 3 & 5 & 4 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[717] $1^{131} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -312, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 3 & 2 \\ 1 & 0 & 2 & 0 & 8 & 3 & 2 \\ 1 & 2 & 0 & 0 & 12 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 8 & 12 & 0 & 56 & 20 & 14 \\ 3 & 3 & 4 & 0 & 20 & 5 & 5 \\ 2 & 2 & 2 & 0 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[718] $1^{131} 3^1 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -312, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 & 3 \\ 3 & 5 & 5 & 23 & 5 & 5 \\ 3 & 5 & 2 & 17 & 4 & 4 \\ 14 & 23 & 17 & 78 & 17 & 18 \\ 3 & 5 & 4 & 17 & 2 & 4 \\ 3 & 5 & 4 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[720] $1^{122} 3^1 21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 6 & 2 \\ 1 & 2 & 0 & 0 & 9 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 32 & 11 \\ 6 & 6 & 8 & 0 & 32 & 26 & 10 \\ 2 & 2 & 2 & 0 & 11 & 10 & 3 \end{pmatrix}_{b_0}$$

[721] $1^{141} 3^1 21^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 3 & 2 \\ 1 & 0 & 2 & 1 & 10 & 3 & 2 \\ 1 & 2 & 0 & 3 & 13 & 4 & 2 \\ 2 & 1 & 3 & 1 & 13 & 4 & 3 \\ 11 & 10 & 13 & 13 & 74 & 23 & 16 \\ 3 & 3 & 4 & 4 & 23 & 5 & 5 \\ 2 & 2 & 2 & 3 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

[719] $1^{121} 3^1 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 3 & 2 \\ 1 & 0 & 2 & 3 & 5 & 3 & 2 \\ 1 & 2 & 0 & 4 & 8 & 4 & 2 \\ 3 & 3 & 4 & 5 & 13 & 7 & 5 \\ 6 & 5 & 8 & 13 & 23 & 13 & 9 \\ 3 & 3 & 4 & 7 & 13 & 5 & 5 \\ 2 & 2 & 2 & 5 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[722] $1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -319, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 9 & 2 \\ 1 & 2 & 0 & 0 & 5 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 27 & 6 \\ 9 & 9 & 12 & 0 & 27 & 61 & 15 \\ 2 & 2 & 2 & 0 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

[723] $1^{122} 3^1 21^{1134} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 13 & 3 \\ 3 & 5 & 5 & 18 & 25 & 5 \\ 3 & 5 & 2 & 14 & 19 & 4 \\ 10 & 18 & 14 & 48 & 68 & 14 \\ 13 & 25 & 19 & 68 & 92 & 20 \\ 3 & 5 & 4 & 14 & 20 & 4 \end{pmatrix}_{b_0}$$

[724] $1^{131} 3^1 21^{1131} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 3 & 2 \\ 1 & 0 & 2 & 1 & 8 & 3 & 2 \\ 1 & 2 & 0 & 3 & 12 & 4 & 2 \\ 2 & 1 & 3 & 1 & 12 & 4 & 3 \\ 9 & 8 & 12 & 12 & 56 & 20 & 14 \\ 3 & 3 & 4 & 4 & 20 & 5 & 5 \\ 2 & 2 & 2 & 3 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[725] $1^{241} 3^1 21^{1241} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 17 & 3 & 3 \\ 3 & 5 & 10 & 28 & 5 & 5 \\ 6 & 10 & 14 & 42 & 8 & 8 \\ 17 & 28 & 42 & 116 & 21 & 22 \\ 3 & 5 & 8 & 21 & 2 & 4 \\ 3 & 5 & 8 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

[726] $1^{114} 3^1 21^{113} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -323, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 9 & 2 \\ 1 & 2 & 0 & 0 & 5 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 27 & 6 \\ 10 & 9 & 12 & 0 & 27 & 61 & 15 \\ 2 & 2 & 2 & 0 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

[727] $1^{142} 3^1 21^{121} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -327, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 5 & 2 \\ 1 & 0 & 2 & 0 & 9 & 5 & 2 \\ 1 & 2 & 0 & 0 & 10 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 9 & 10 & 0 & 51 & 28 & 13 \\ 5 & 5 & 5 & 0 & 28 & 13 & 7 \\ 2 & 2 & 2 & 0 & 13 & 7 & 3 \end{pmatrix}_{b_0}$$

[728] $1^{132} 3^1 21^{121} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -327, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 5 & 2 \\ 1 & 0 & 2 & 0 & 9 & 5 & 2 \\ 1 & 2 & 0 & 0 & 10 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 9 & 10 & 0 & 51 & 28 & 13 \\ 5 & 5 & 5 & 0 & 28 & 13 & 7 \\ 2 & 2 & 2 & 0 & 13 & 7 & 3 \end{pmatrix}_{b_0}$$

[729] $1^{142} 3^1 21^{1121} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 4 & 2 \\ 1 & 0 & 2 & 3 & 7 & 3 & 2 \\ 1 & 2 & 0 & 4 & 9 & 4 & 2 \\ 3 & 3 & 4 & 5 & 16 & 7 & 5 \\ 9 & 7 & 9 & 16 & 35 & 16 & 11 \\ 4 & 3 & 4 & 7 & 16 & 5 & 5 \\ 2 & 2 & 2 & 5 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[730] \quad 1^{1121} 3^1 21^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -333, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 3 & 2 \\ 1 & 0 & 2 & 3 & 7 & 3 & 2 \\ 1 & 2 & 0 & 4 & 9 & 4 & 2 \\ 4 & 3 & 4 & 5 & 16 & 7 & 5 \\ 7 & 7 & 9 & 16 & 35 & 16 & 11 \\ 3 & 3 & 4 & 7 & 16 & 5 & 5 \\ 2 & 2 & 2 & 5 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[731] \quad 1^{1124} 3^1 21^{124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -336, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 10 & 2 \\ 1 & 0 & 2 & 0 & 5 & 8 & 2 \\ 1 & 2 & 0 & 0 & 8 & 14 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 42 & 9 \\ 10 & 8 & 14 & 0 & 42 & 70 & 16 \\ 2 & 2 & 2 & 0 & 9 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[732] \quad 1^{1134} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -336, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 0 & 5 & 15 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 32 & 6 \\ 11 & 10 & 15 & 0 & 32 & 88 & 18 \\ 2 & 2 & 2 & 0 & 6 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[733] \quad 1^{132} 3^1 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -336, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 6 & 2 \\ 1 & 2 & 0 & 0 & 12 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 8 & 12 & 0 & 56 & 40 & 14 \\ 6 & 6 & 8 & 0 & 40 & 26 & 10 \\ 2 & 2 & 2 & 0 & 14 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[734] \quad 1^{1312} 3^1 21^{1141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -338, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 4 & 2 \\ 1 & 0 & 2 & 1 & 11 & 3 & 2 \\ 1 & 2 & 0 & 3 & 16 & 4 & 2 \\ 2 & 1 & 3 & 1 & 16 & 4 & 3 \\ 12 & 11 & 16 & 16 & 103 & 27 & 19 \\ 4 & 3 & 4 & 4 & 27 & 5 & 5 \\ 2 & 2 & 2 & 3 & 19 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[735] \quad 1^{121} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -339, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.00 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 3 & 2 \\ 1 & 0 & 2 & 0 & 9 & 3 & 2 \\ 1 & 2 & 0 & 0 & 15 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 9 & 15 & 0 & 83 & 24 & 17 \\ 3 & 3 & 4 & 0 & 24 & 5 & 5 \\ 2 & 2 & 2 & 0 & 17 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[736] \quad 1^{1123} 3^1 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -339, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 7 & 2 \\ 1 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 3 & 3 & 4 & 5 & 9 & 14 & 5 \\ 5 & 4 & 5 & 9 & 10 & 18 & 6 \\ 7 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[737] \quad 1^{1213} 3^1 21^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -339, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 7 & 2 \\ 1 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 4 & 3 & 4 & 5 & 9 & 14 & 5 \\ 4 & 4 & 5 & 9 & 10 & 18 & 6 \\ 7 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[738] $1^{11} 3^{11} 21^{1124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 14 & 3 \\ 3 & 5 & 5 & 13 & 25 & 5 \\ 3 & 5 & 2 & 10 & 19 & 4 \\ 8 & 13 & 10 & 24 & 49 & 10 \\ 14 & 25 & 19 & 49 & 92 & 20 \\ 3 & 5 & 4 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

[739] $1^{11} 3^{11} 21^{1123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -339, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 12 & 3 \\ 3 & 5 & 5 & 13 & 20 & 5 \\ 3 & 5 & 2 & 10 & 15 & 4 \\ 8 & 13 & 10 & 24 & 39 & 10 \\ 12 & 20 & 15 & 39 & 58 & 16 \\ 3 & 5 & 4 & 10 & 16 & 4 \end{pmatrix}_{b_0}$$

[740] $1^{11} 3^{11} 21^{1114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -341, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 14 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 3 & 5 & 2 & 6 & 19 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 14 & 25 & 19 & 30 & 92 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[741] $1^{11} 3^{11} 21^{1244} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -342, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 14 & 3 \\ 3 & 5 & 10 & 13 & 25 & 5 \\ 6 & 10 & 14 & 20 & 38 & 8 \\ 7 & 13 & 20 & 24 & 49 & 10 \\ 14 & 25 & 38 & 49 & 92 & 20 \\ 3 & 5 & 8 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

[742] $1^{132} 3^{11} 21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 6 & 2 \\ 1 & 0 & 2 & 0 & 10 & 6 & 2 \\ 1 & 2 & 0 & 0 & 13 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 10 & 13 & 0 & 74 & 46 & 16 \\ 6 & 6 & 8 & 0 & 46 & 26 & 10 \\ 2 & 2 & 2 & 0 & 16 & 10 & 3 \end{pmatrix}_{b_0}$$

[743] $1^{113} 3^{11} 21^{1113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 10 & 2 \\ 1 & 0 & 2 & 3 & 2 & 9 & 2 \\ 1 & 2 & 0 & 4 & 4 & 12 & 2 \\ 3 & 3 & 4 & 5 & 6 & 21 & 5 \\ 3 & 2 & 4 & 6 & 4 & 18 & 4 \\ 10 & 9 & 12 & 21 & 18 & 61 & 15 \\ 2 & 2 & 2 & 5 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[744] $1^{114} 3^{11} 21^{1214} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 15 & 3 \\ 3 & 5 & 10 & 8 & 25 & 5 \\ 6 & 10 & 14 & 12 & 38 & 8 \\ 5 & 8 & 12 & 8 & 30 & 6 \\ 15 & 25 & 38 & 30 & 92 & 20 \\ 3 & 5 & 8 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[745] $1^{123} 3^{11} 21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 11 & 2 \\ 1 & 0 & 2 & 0 & 7 & 10 & 2 \\ 1 & 2 & 0 & 0 & 9 & 15 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 57 & 11 \\ 11 & 10 & 15 & 0 & 57 & 88 & 18 \\ 2 & 2 & 2 & 0 & 11 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[746] \quad 1^{1^{122}} 1^{111} 3^1 21^{1122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -359, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 5 & 2 \\ 1 & 0 & 2 & 3 & 5 & 4 & 2 \\ 1 & 2 & 0 & 4 & 8 & 7 & 2 \\ 3 & 3 & 4 & 5 & 13 & 11 & 5 \\ 6 & 5 & 8 & 13 & 23 & 21 & 9 \\ 5 & 4 & 7 & 11 & 21 & 16 & 8 \\ 2 & 2 & 2 & 5 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[747] \quad 1^{1^{114}} 1^{114} 3^1 21^{1114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -362, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 12 & 2 \\ 1 & 0 & 2 & 1 & 4 & 12 & 2 \\ 1 & 2 & 0 & 3 & 5 & 16 & 2 \\ 2 & 1 & 3 & 1 & 5 & 16 & 3 \\ 4 & 4 & 5 & 5 & 10 & 36 & 6 \\ 12 & 12 & 16 & 16 & 36 & 110 & 20 \\ 2 & 2 & 2 & 3 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[748] \quad 1^{1^{123}} 1^{112} 3^1 21^{1122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -364, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 7 & 2 \\ 1 & 0 & 2 & 3 & 5 & 6 & 2 \\ 1 & 2 & 0 & 4 & 8 & 8 & 2 \\ 3 & 3 & 4 & 5 & 13 & 14 & 5 \\ 6 & 5 & 8 & 13 & 23 & 26 & 9 \\ 7 & 6 & 8 & 14 & 26 & 26 & 10 \\ 2 & 2 & 2 & 5 & 9 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[749] \quad 1^{1^{113}} 1^{114} 3^1 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -371, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 11 & 2 \\ 1 & 2 & 0 & 0 & 5 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 31 & 6 \\ 10 & 11 & 13 & 0 & 31 & 81 & 17 \\ 2 & 2 & 2 & 0 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[750] \quad 1^{1^{124}} 1^{124} 3^1 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -384, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 10 & 2 \\ 1 & 0 & 2 & 0 & 6 & 10 & 2 \\ 1 & 2 & 0 & 0 & 6 & 10 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 6 & 6 & 0 & 20 & 36 & 8 \\ 10 & 10 & 10 & 0 & 36 & 58 & 14 \\ 2 & 2 & 2 & 0 & 8 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[751] \quad 1^{1^{113}} 1^{12} 3^1 21^{1114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -386, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 17 & 3 \\ 3 & 5 & 5 & 8 & 30 & 5 \\ 3 & 5 & 2 & 6 & 22 & 4 \\ 5 & 8 & 6 & 8 & 36 & 6 \\ 17 & 30 & 22 & 36 & 134 & 24 \\ 3 & 5 & 4 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[752] \quad 1^{1^{122}} 1^{112} 3^1 21^{1122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -388, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 6 & 2 \\ 1 & 0 & 2 & 3 & 5 & 6 & 2 \\ 1 & 2 & 0 & 4 & 8 & 8 & 2 \\ 3 & 3 & 4 & 5 & 13 & 14 & 5 \\ 6 & 5 & 8 & 13 & 23 & 26 & 9 \\ 6 & 6 & 8 & 14 & 26 & 26 & 10 \\ 2 & 2 & 2 & 5 & 9 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[753] \quad 1^{1^{113}} 1^{112} 3^1 21^{1113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -389, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 8 & 2 \\ 1 & 0 & 2 & 3 & 4 & 7 & 2 \\ 1 & 2 & 0 & 4 & 5 & 11 & 2 \\ 3 & 3 & 4 & 5 & 9 & 18 & 5 \\ 4 & 4 & 5 & 9 & 10 & 23 & 6 \\ 8 & 7 & 11 & 18 & 23 & 45 & 13 \\ 2 & 2 & 2 & 5 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

[754] $1^{1113} 3^1 21^{1112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -391, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 7 & 2 \\ 1 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 3 & 3 & 4 & 5 & 9 & 14 & 5 \\ 4 & 4 & 5 & 9 & 10 & 18 & 6 \\ 7 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[755] $21^{1112} 3^1 21^{1112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -396, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 4 & 6 & 2 \\ 2 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 3 & 3 & 4 & 5 & 9 & 14 & 5 \\ 4 & 4 & 5 & 9 & 10 & 18 & 6 \\ 6 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[758] $1^{131} 3^1 21^{1141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 & 3 \\ 3 & 5 & 5 & 23 & 5 & 5 \\ 3 & 5 & 2 & 18 & 4 & 4 \\ 13 & 23 & 18 & 80 & 18 & 18 \\ 3 & 5 & 4 & 18 & 2 & 4 \\ 3 & 5 & 4 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[757] $1^{132} 3^1 21^{1142} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 6 & 3 \\ 3 & 5 & 5 & 23 & 10 & 5 \\ 3 & 5 & 2 & 18 & 8 & 4 \\ 13 & 23 & 18 & 80 & 36 & 18 \\ 6 & 10 & 8 & 36 & 14 & 8 \\ 3 & 5 & 4 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

[756] $1^{12} 3^1 21^{1141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 & 3 \\ 3 & 5 & 5 & 33 & 5 & 5 \\ 3 & 5 & 2 & 24 & 4 & 4 \\ 19 & 33 & 24 & 164 & 24 & 26 \\ 3 & 5 & 4 & 24 & 2 & 4 \\ 3 & 5 & 4 & 26 & 4 & 4 \end{pmatrix}_{b_0}$$

[759] $1^{142} 3^1 21^{132} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 6 & 2 \\ 1 & 0 & 2 & 0 & 12 & 6 & 2 \\ 1 & 2 & 0 & 0 & 14 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 12 & 14 & 0 & 96 & 52 & 18 \\ 6 & 6 & 8 & 0 & 52 & 26 & 10 \\ 2 & 2 & 2 & 0 & 18 & 10 & 3 \end{pmatrix}_{b_0}$$

[760] $1^{131} 3^1 21^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -419, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 3 & 2 \\ 1 & 0 & 2 & 0 & 10 & 3 & 2 \\ 1 & 2 & 0 & 0 & 13 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 10 & 13 & 0 & 74 & 23 & 16 \\ 3 & 3 & 4 & 0 & 23 & 5 & 5 \\ 2 & 2 & 2 & 0 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

[761] $1^{141} 3^1 21^{1141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 & 3 \\ 3 & 5 & 5 & 23 & 5 & 5 \\ 3 & 5 & 2 & 18 & 4 & 4 \\ 14 & 23 & 18 & 80 & 18 & 18 \\ 3 & 5 & 4 & 18 & 2 & 4 \\ 3 & 5 & 4 & 18 & 4 & 4 \end{pmatrix}_{b_0}$$

[762] $1^{12} 3^1 21^{1141} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 & 3 \\ 3 & 5 & 5 & 33 & 5 & 5 \\ 3 & 5 & 2 & 24 & 4 & 4 \\ 20 & 33 & 24 & 164 & 24 & 26 \\ 3 & 5 & 4 & 24 & 2 & 4 \\ 3 & 5 & 4 & 26 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[763] \ 1^{1121} 3^{1132} 21^{1121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -429, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 4 & 2 \\ 1 & 0 & 2 & 3 & 7 & 3 & 2 \\ 1 & 2 & 0 & 4 & 9 & 4 & 2 \\ 3 & 3 & 4 & 5 & 16 & 7 & 5 \\ 8 & 7 & 9 & 16 & 35 & 16 & 11 \\ 4 & 3 & 4 & 7 & 16 & 5 & 5 \\ 2 & 2 & 2 & 5 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[764] \ 1^{114} 3^{1114} 21^{1114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -429, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 15 & 3 \\ 3 & 5 & 5 & 8 & 25 & 5 \\ 3 & 5 & 2 & 6 & 19 & 4 \\ 5 & 8 & 6 & 8 & 30 & 6 \\ 15 & 25 & 19 & 30 & 92 & 20 \\ 3 & 5 & 4 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[765] \ 1^{114} 3^{1113} 21^{1113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -431, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 0 & 4 & 11 & 2 \\ 1 & 2 & 0 & 0 & 5 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 31 & 6 \\ 11 & 11 & 13 & 0 & 31 & 81 & 17 \\ 2 & 2 & 2 & 0 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[766] \ 1^{113} 3^{1124} 21^{1124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -431, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 11 & 2 \\ 1 & 0 & 2 & 0 & 5 & 10 & 2 \\ 1 & 2 & 0 & 0 & 8 & 15 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 6 & 5 & 8 & 0 & 23 & 47 & 9 \\ 11 & 10 & 15 & 0 & 47 & 88 & 18 \\ 2 & 2 & 2 & 0 & 9 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[767] \ 1^{123} 3^{1123} 21^{1123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 10 & 2 \\ 1 & 0 & 2 & 0 & 7 & 11 & 2 \\ 1 & 2 & 0 & 0 & 9 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 55 & 11 \\ 10 & 11 & 13 & 0 & 55 & 81 & 17 \\ 2 & 2 & 2 & 0 & 11 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[768] \ 1^{121} 3^{1131} 21^{1131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 3 & 2 \\ 1 & 0 & 2 & 3 & 8 & 3 & 2 \\ 1 & 2 & 0 & 4 & 12 & 4 & 2 \\ 4 & 3 & 4 & 5 & 20 & 7 & 5 \\ 9 & 8 & 12 & 20 & 56 & 20 & 14 \\ 3 & 3 & 4 & 7 & 20 & 5 & 5 \\ 2 & 2 & 2 & 5 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[769] \ 1^{131} 3^{1131} 21^{1131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -439, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 3 & 2 \\ 1 & 0 & 2 & 0 & 10 & 3 & 2 \\ 1 & 2 & 0 & 0 & 13 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 10 & 13 & 0 & 74 & 23 & 16 \\ 3 & 3 & 4 & 0 & 23 & 5 & 5 \\ 2 & 2 & 2 & 0 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[770] \ 1^{142} 3^{1142} 21^{1142} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 6 & 3 \\ 3 & 5 & 5 & 23 & 10 & 5 \\ 3 & 5 & 2 & 18 & 8 & 4 \\ 14 & 23 & 18 & 80 & 36 & 18 \\ 6 & 10 & 8 & 36 & 14 & 8 \\ 3 & 5 & 4 & 18 & 8 & 4 \end{pmatrix}_{b_0}$$

[771] $1^{142} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 5 & 2 \\ 1 & 0 & 2 & 0 & 12 & 5 & 2 \\ 1 & 2 & 0 & 0 & 14 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 12 & 14 & 0 & 96 & 38 & 18 \\ 5 & 5 & 5 & 0 & 38 & 13 & 7 \\ 2 & 2 & 2 & 0 & 18 & 7 & 3 \end{pmatrix}_{b_0}$$

[772] $1^{112} 3^1 21^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -446, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 6 & 2 \\ 1 & 0 & 2 & 3 & 4 & 6 & 2 \\ 1 & 2 & 0 & 4 & 5 & 8 & 2 \\ 3 & 3 & 4 & 5 & 9 & 14 & 5 \\ 4 & 4 & 5 & 9 & 10 & 18 & 6 \\ 6 & 6 & 8 & 14 & 18 & 26 & 10 \\ 2 & 2 & 2 & 5 & 6 & 10 & 3 \end{pmatrix}_{b_0}$$

[773] $1^{142} 3^1 21^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -447, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 5 & 2 \\ 1 & 0 & 2 & 0 & 11 & 4 & 2 \\ 1 & 2 & 0 & 0 & 16 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 11 & 16 & 0 & 103 & 43 & 19 \\ 5 & 4 & 7 & 0 & 43 & 16 & 8 \\ 2 & 2 & 2 & 0 & 19 & 8 & 3 \end{pmatrix}_{b_0}$$

[774] $1^{113} 3^1 21^{1124} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -450, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 14 & 3 \\ 3 & 5 & 5 & 13 & 25 & 5 \\ 3 & 5 & 2 & 10 & 19 & 4 \\ 7 & 13 & 10 & 24 & 49 & 10 \\ 14 & 25 & 19 & 49 & 92 & 20 \\ 3 & 5 & 4 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

[775] $1^{142} 3^1 21^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -451, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 5 & 2 \\ 1 & 0 & 2 & 0 & 9 & 4 & 2 \\ 1 & 2 & 0 & 0 & 15 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 9 & 15 & 0 & 83 & 39 & 17 \\ 5 & 4 & 7 & 0 & 39 & 16 & 8 \\ 2 & 2 & 2 & 0 & 17 & 8 & 3 \end{pmatrix}_{b_0}$$

[776] $1^{123} 3^1 21^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -451, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 9 & 2 \\ 1 & 0 & 2 & 0 & 7 & 9 & 2 \\ 1 & 2 & 0 & 0 & 9 & 12 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 48 & 11 \\ 9 & 9 & 12 & 0 & 48 & 61 & 15 \\ 2 & 2 & 2 & 0 & 11 & 15 & 3 \end{pmatrix}_{b_0}$$

[777] $1^{131} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -460, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 8 & 3 & 2 \\ 1 & 0 & 2 & 3 & 7 & 3 & 2 \\ 1 & 2 & 0 & 4 & 9 & 4 & 2 \\ 3 & 3 & 4 & 5 & 16 & 7 & 5 \\ 8 & 7 & 9 & 16 & 35 & 16 & 11 \\ 3 & 3 & 4 & 7 & 16 & 5 & 5 \\ 2 & 2 & 2 & 5 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[779] $1^{142} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 5 & 2 \\ 1 & 0 & 2 & 0 & 11 & 5 & 2 \\ 1 & 2 & 0 & 0 & 11 & 5 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 11 & 11 & 11 & 0 & 71 & 33 & 15 \\ 5 & 5 & 5 & 0 & 33 & 13 & 7 \\ 2 & 2 & 2 & 0 & 15 & 7 & 3 \end{pmatrix}_{b_0}$$

[778] $1^{124} 3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 11 & 2 \\ 1 & 0 & 2 & 0 & 7 & 11 & 2 \\ 1 & 2 & 0 & 0 & 9 & 13 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 55 & 11 \\ 11 & 11 & 13 & 0 & 55 & 81 & 17 \\ 2 & 2 & 2 & 0 & 11 & 17 & 3 \end{pmatrix}_{b_0}$$

[780] $1^{114} 3^1 21^{1114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -472, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 12 & 2 \\ 1 & 0 & 2 & 3 & 2 & 12 & 2 \\ 1 & 2 & 0 & 4 & 4 & 16 & 2 \\ 3 & 3 & 4 & 5 & 6 & 28 & 5 \\ 3 & 2 & 4 & 6 & 4 & 24 & 4 \\ 12 & 12 & 16 & 28 & 24 & 110 & 20 \\ 2 & 2 & 2 & 5 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[781] $1^{114} 3^1 21^{1114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -478, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 18 & 3 \\ 3 & 5 & 5 & 8 & 30 & 5 \\ 3 & 5 & 2 & 6 & 22 & 4 \\ 5 & 8 & 6 & 8 & 36 & 6 \\ 18 & 30 & 22 & 36 & 134 & 24 \\ 3 & 5 & 4 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[782] $1^{131} 3^1 21^{1141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -482, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 11 & 3 & 2 \\ 1 & 0 & 2 & 1 & 11 & 3 & 2 \\ 1 & 2 & 0 & 3 & 16 & 4 & 2 \\ 2 & 1 & 3 & 1 & 16 & 4 & 3 \\ 11 & 11 & 16 & 16 & 103 & 27 & 19 \\ 3 & 3 & 4 & 4 & 27 & 5 & 5 \\ 2 & 2 & 2 & 3 & 19 & 5 & 3 \end{pmatrix}_{b_0}$$

[783] $1^{141} 3^1 21^{1141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -491, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 13 & 3 & 2 \\ 1 & 0 & 2 & 1 & 13 & 3 & 2 \\ 1 & 2 & 0 & 3 & 17 & 4 & 2 \\ 2 & 1 & 3 & 1 & 17 & 4 & 3 \\ 13 & 13 & 17 & 17 & 127 & 30 & 21 \\ 3 & 3 & 4 & 4 & 30 & 5 & 5 \\ 2 & 2 & 2 & 3 & 21 & 5 & 3 \end{pmatrix}_{b_0}$$

[784] $1^{131} 3^1 21^{1141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 & 3 \\ 3 & 5 & 5 & 28 & 5 & 5 \\ 3 & 5 & 2 & 21 & 4 & 4 \\ 16 & 28 & 21 & 116 & 21 & 22 \\ 3 & 5 & 4 & 21 & 2 & 4 \\ 3 & 5 & 4 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

[785] $21^{1121} 3^1 21^{1121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 7 & 3 & 2 \\ 2 & 0 & 2 & 3 & 7 & 3 & 2 \\ 1 & 2 & 0 & 4 & 9 & 4 & 2 \\ 3 & 3 & 4 & 5 & 16 & 7 & 5 \\ 7 & 7 & 9 & 16 & 35 & 16 & 11 \\ 3 & 3 & 4 & 7 & 16 & 5 & 5 \\ 2 & 2 & 2 & 5 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[786] $1^{124} 3^1 21^{1124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -501, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 15 & 3 \\ 3 & 5 & 5 & 13 & 25 & 5 \\ 3 & 5 & 2 & 10 & 19 & 4 \\ 8 & 13 & 10 & 24 & 49 & 10 \\ 15 & 25 & 19 & 49 & 92 & 20 \\ 3 & 5 & 4 & 10 & 20 & 4 \end{pmatrix}_{b_0}$$

[787] $1^{114} 3^1 21^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -508, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 0 & 4 & 12 & 2 \\ 1 & 2 & 0 & 0 & 5 & 16 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 4 & 4 & 5 & 0 & 10 & 36 & 6 \\ 12 & 12 & 16 & 0 & 36 & 110 & 20 \\ 2 & 2 & 2 & 0 & 6 & 20 & 3 \end{pmatrix}_{b_0}$$

[788] $1^{132} 3^1 21^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -520, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 6 & 2 \\ 1 & 0 & 2 & 0 & 10 & 6 & 2 \\ 1 & 2 & 0 & 0 & 13 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 10 & 10 & 13 & 0 & 74 & 46 & 16 \\ 6 & 6 & 8 & 0 & 46 & 26 & 10 \\ 2 & 2 & 2 & 0 & 16 & 10 & 3 \end{pmatrix}_{b_0}$$

[789] $1^{141} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -520, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 & 3 \\ 3 & 5 & 5 & 28 & 5 & 5 \\ 3 & 5 & 2 & 21 & 4 & 4 \\ 17 & 28 & 21 & 116 & 21 & 22 \\ 3 & 5 & 4 & 21 & 2 & 4 \\ 3 & 5 & 4 & 22 & 4 & 4 \end{pmatrix}_{b_0}$$

[790] $1^{141} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 3 & 2 \\ 1 & 0 & 2 & 0 & 12 & 3 & 2 \\ 1 & 2 & 0 & 0 & 14 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 12 & 14 & 0 & 96 & 26 & 18 \\ 3 & 3 & 4 & 0 & 26 & 5 & 5 \\ 2 & 2 & 2 & 0 & 18 & 5 & 3 \end{pmatrix}_{b_0}$$

[791] $1^{121} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 10 & 3 & 2 \\ 1 & 0 & 2 & 3 & 8 & 3 & 2 \\ 1 & 2 & 0 & 4 & 12 & 4 & 2 \\ 4 & 3 & 4 & 5 & 20 & 7 & 5 \\ 10 & 8 & 12 & 20 & 56 & 20 & 14 \\ 3 & 3 & 4 & 7 & 20 & 5 & 5 \\ 2 & 2 & 2 & 5 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[792] $1^{141} 3^1 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 12 & 3 & 2 \\ 1 & 0 & 2 & 1 & 11 & 3 & 2 \\ 1 & 2 & 0 & 3 & 16 & 4 & 2 \\ 2 & 1 & 3 & 1 & 16 & 4 & 3 \\ 12 & 11 & 16 & 16 & 103 & 27 & 19 \\ 3 & 3 & 4 & 4 & 27 & 5 & 5 \\ 2 & 2 & 2 & 3 & 19 & 5 & 3 \end{pmatrix}_{b_0}$$

[793] $1^{121} 3^1 21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 10 & 3 & 2 \\ 1 & 0 & 2 & 3 & 8 & 3 & 2 \\ 1 & 2 & 0 & 4 & 12 & 4 & 2 \\ 3 & 3 & 4 & 5 & 20 & 7 & 5 \\ 10 & 8 & 12 & 20 & 56 & 20 & 14 \\ 3 & 3 & 4 & 7 & 20 & 5 & 5 \\ 2 & 2 & 2 & 5 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[794] $1^{121} 3^1 21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -548, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 7 & 3 & 2 \\ 1 & 0 & 2 & 3 & 7 & 3 & 2 \\ 1 & 2 & 0 & 4 & 9 & 4 & 2 \\ 3 & 3 & 4 & 5 & 16 & 7 & 5 \\ 7 & 7 & 9 & 16 & 35 & 16 & 11 \\ 3 & 3 & 4 & 7 & 16 & 5 & 5 \\ 2 & 2 & 2 & 5 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[795] $1^{131} 3^1 21^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -556, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 3 & 2 \\ 1 & 0 & 2 & 0 & 11 & 3 & 2 \\ 1 & 2 & 0 & 0 & 16 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 11 & 16 & 0 & 103 & 27 & 19 \\ 3 & 3 & 4 & 0 & 27 & 5 & 5 \\ 2 & 2 & 2 & 0 & 19 & 5 & 3 \end{pmatrix}_{b_0}$$

[796] $1^{132} 3^1 21^{142} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -580, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 6 & 2 \\ 1 & 0 & 2 & 0 & 11 & 6 & 2 \\ 1 & 2 & 0 & 0 & 16 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 12 & 11 & 16 & 0 & 103 & 54 & 19 \\ 6 & 6 & 8 & 0 & 54 & 26 & 10 \\ 2 & 2 & 2 & 0 & 19 & 10 & 3 \end{pmatrix}_{b_0}$$

[797] $1^{124} 3^1 21^{124} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -640, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 12 & 2 \\ 1 & 0 & 2 & 0 & 7 & 12 & 2 \\ 1 & 2 & 0 & 0 & 9 & 16 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 9 & 0 & 35 & 64 & 11 \\ 12 & 12 & 16 & 0 & 64 & 110 & 20 \\ 2 & 2 & 2 & 0 & 11 & 20 & 3 \end{pmatrix}_{b_0}$$

[798] $1^{121} 3^1 21^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -672, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 9 & 3 & 2 \\ 1 & 0 & 2 & 3 & 8 & 3 & 2 \\ 1 & 2 & 0 & 4 & 12 & 4 & 2 \\ 3 & 3 & 4 & 5 & 20 & 7 & 5 \\ 9 & 8 & 12 & 20 & 56 & 20 & 14 \\ 3 & 3 & 4 & 7 & 20 & 5 & 5 \\ 2 & 2 & 2 & 5 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[799] $1^{141} 3^1 21^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -715, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 3 & 2 \\ 1 & 0 & 2 & 0 & 13 & 3 & 2 \\ 1 & 2 & 0 & 0 & 17 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 13 & 13 & 17 & 0 & 127 & 30 & 21 \\ 3 & 3 & 4 & 0 & 30 & 5 & 5 \\ 2 & 2 & 2 & 0 & 21 & 5 & 3 \end{pmatrix}_{b_0}$$

[800] $1^{142} 3^1 21^{142} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -796, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 6 & 2 \\ 1 & 0 & 2 & 0 & 13 & 6 & 2 \\ 1 & 2 & 0 & 0 & 17 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 13 & 13 & 17 & 0 & 127 & 60 & 21 \\ 6 & 6 & 8 & 0 & 60 & 26 & 10 \\ 2 & 2 & 2 & 0 & 21 & 10 & 3 \end{pmatrix}_{b_0}$$

$T = 12$ [801] $1^{1221} 3^1 1^5 1321^{1121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.5$

[805] $1^{111} 31^5 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 3 & 2 \\ 1 & 1 & 2 & 3 & 4 & 4 & 3 \\ 1 & 2 & 0 & 1 & 3 & 4 & 2 \\ 2 & 3 & 1 & 1 & 4 & 4 & 3 \\ 2 & 4 & 3 & 4 & 5 & 7 & 5 \\ 3 & 4 & 4 & 4 & 7 & 6 & 6 \\ 2 & 3 & 2 & 3 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

[810] $1^{1114}31^1513^11^{13}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & 2 & 3 & 4 & 15 & 3 \\ 1 & 2 & 0 & 1 & 3 & 13 & 2 \\ 1 & 3 & 1 & 1 & 4 & 15 & 3 \\ 3 & 4 & 3 & 4 & 5 & 25 & 5 \\ 9 & 15 & 13 & 15 & 25 & 97 & 21 \\ 2 & 3 & 2 & 3 & 5 & 21 & 4 \end{pmatrix}_{b_0}$$

[809] $1^{1112}31^1513^11^{11}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & 2 & 3 & 4 & 7 & 3 \\ 1 & 2 & 0 & 1 & 3 & 5 & 2 \\ 1 & 3 & 1 & 1 & 4 & 7 & 3 \\ 3 & 4 & 3 & 4 & 5 & 11 & 5 \\ 5 & 7 & 5 & 7 & 11 & 17 & 9 \\ 2 & 3 & 2 & 3 & 5 & 9 & 4 \end{pmatrix}_{b_0}$$

[808] $1^{1114}3^11^15131^{113}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 1 & 2 & 3 & 6 & 15 & 3 \\ 1 & 2 & 0 & 1 & 5 & 13 & 2 \\ 1 & 3 & 1 & 1 & 6 & 15 & 3 \\ 4 & 6 & 5 & 6 & 14 & 39 & 8 \\ 9 & 15 & 13 & 15 & 39 & 97 & 21 \\ 2 & 3 & 2 & 3 & 8 & 21 & 4 \end{pmatrix}_{b_0}$$

[807] $1^{1112}3^11^15131^{111}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 5 & 2 \\ 1 & 1 & 2 & 3 & 6 & 7 & 3 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 1 & 3 & 1 & 1 & 6 & 7 & 3 \\ 4 & 6 & 5 & 6 & 14 & 17 & 8 \\ 5 & 7 & 5 & 7 & 17 & 17 & 9 \\ 2 & 3 & 2 & 3 & 8 & 9 & 4 \end{pmatrix}_{b_0}$$

[806] $1^{1113}3^11^15131^{112}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 4 & 3 & 5 & 2 \\ 2 & 0 & 4 & 6 & 7 & 12 & 6 \\ 2 & 0 & 3 & 7 & 5 & 11 & 5 \\ 4 & 0 & 5 & 12 & 11 & 17 & 9 \\ 2 & 0 & 2 & 6 & 5 & 9 & 4 \end{pmatrix}_{b_0}$$

[811] $1^{1141}3^11^15131^{131}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 11 & 2 & 2 \\ 1 & 1 & 2 & 3 & 17 & 4 & 3 \\ 1 & 2 & 0 & 1 & 14 & 4 & 2 \\ 1 & 3 & 1 & 1 & 17 & 4 & 3 \\ 11 & 17 & 14 & 17 & 121 & 31 & 23 \\ 2 & 4 & 4 & 4 & 31 & 6 & 6 \\ 2 & 3 & 2 & 3 & 23 & 6 & 4 \end{pmatrix}_{b_0}$$

[812] $1^{1113}3^11^15131^{112}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 1 & 2 & 3 & 6 & 11 & 3 \\ 1 & 2 & 0 & 1 & 5 & 9 & 2 \\ 1 & 3 & 1 & 1 & 6 & 11 & 3 \\ 4 & 6 & 5 & 6 & 14 & 28 & 8 \\ 7 & 11 & 9 & 11 & 28 & 49 & 15 \\ 2 & 3 & 2 & 3 & 8 & 15 & 4 \end{pmatrix}_{b_0}$$

[813] $1^{1113}31^1513^11^{12}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 1 & 2 & 3 & 4 & 11 & 3 \\ 1 & 2 & 0 & 1 & 3 & 9 & 2 \\ 1 & 3 & 1 & 1 & 4 & 11 & 3 \\ 3 & 4 & 3 & 4 & 5 & 18 & 5 \\ 7 & 11 & 9 & 11 & 18 & 49 & 15 \\ 2 & 3 & 2 & 3 & 5 & 15 & 4 \end{pmatrix}_{b_0}$$

[849] $1^{1224}31^1513^11^{224}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} =$

7.5

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 8 & 2 \\ 1 & 1 & 1 & 8 & 9 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 38 & 64 & 12 \\ 5 & 9 & 0 & 38 & 44 & 76 & 14 \\ 8 & 16 & 0 & 64 & 76 & 126 & 24 \\ 2 & 3 & -1 & 12 & 14 & 24 & 4 \end{pmatrix}_{b_0}$$

[835] $1^{1314}315131^{1314} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 8 & 2 \\ 1 & 1 & 1 & 12 & 5 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 33 & 96 & 18 \\ 3 & 5 & 0 & 33 & 14 & 44 & 8 \\ 8 & 16 & 0 & 96 & 44 & 126 & 24 \\ 2 & 3 & -1 & 18 & 8 & 24 & 4 \end{pmatrix}_{b_0}$$

[853] $1^{1231}315131^{1231} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 2 & 2 \\ 1 & 1 & 1 & 8 & 13 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 54 & 16 & 12 \\ 7 & 13 & 0 & 54 & 90 & 27 & 20 \\ 2 & 4 & 0 & 16 & 27 & 6 & 6 \\ 2 & 3 & -1 & 12 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

[852] $1^{1141}315131^{1141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 2 & 2 \\ 1 & 1 & 1 & 4 & 17 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 35 & 8 & 6 \\ 9 & 17 & 0 & 35 & 152 & 35 & 26 \\ 2 & 4 & 0 & 8 & 35 & 6 & 6 \\ 2 & 3 & -1 & 6 & 26 & 6 & 4 \end{pmatrix}_{b_0}$$

[855] $1^{1214}315131^{1214} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 8 & 2 \\ 1 & 1 & 1 & 8 & 5 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 22 & 64 & 12 \\ 3 & 5 & 0 & 22 & 14 & 44 & 8 \\ 8 & 16 & 0 & 64 & 44 & 126 & 24 \\ 2 & 3 & -1 & 12 & 8 & 24 & 4 \end{pmatrix}_{b_0}$$

[856] $1^{1131}315131^{1131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 & 2 \\ 1 & 1 & 1 & 4 & 13 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 27 & 8 & 6 \\ 7 & 13 & 0 & 27 & 90 & 27 & 20 \\ 2 & 4 & 0 & 8 & 27 & 6 & 6 \\ 2 & 3 & -1 & 6 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

[857] $1^{1212}315131^{1212} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 4 & 4 & 2 \\ 1 & 1 & 1 & 8 & 7 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 30 & 32 & 12 \\ 4 & 7 & 0 & 30 & 27 & 30 & 11 \\ 4 & 8 & 0 & 32 & 30 & 30 & 12 \\ 2 & 3 & -1 & 12 & 11 & 12 & 4 \end{pmatrix}_{b_0}$$

[858] $1^{1241}315131^{1241} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 2 & 2 \\ 1 & 1 & 1 & 8 & 17 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 70 & 16 & 12 \\ 9 & 17 & 0 & 70 & 152 & 35 & 26 \\ 2 & 4 & 0 & 16 & 35 & 6 & 6 \\ 2 & 3 & -1 & 12 & 26 & 6 & 4 \end{pmatrix}_{b_0}$$

[824] $1^{1124}315131^{1124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 8 & 2 \\ 1 & 1 & 1 & 4 & 9 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 19 & 32 & 6 \\ 5 & 9 & 0 & 19 & 44 & 76 & 14 \\ 8 & 16 & 0 & 32 & 76 & 126 & 24 \\ 2 & 3 & -1 & 6 & 14 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[826] \ 1^{121}3^11^15131^{211} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 & 2 \\ 1 & 1 & 1 & 8 & 3 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 14 & 16 & 12 \\ 2 & 3 & 0 & 14 & 5 & 7 & 5 \\ 2 & 4 & 0 & 16 & 7 & 6 & 6 \\ 2 & 3 & -1 & 12 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[818] \ 1^{1134}31^1513^11^{134} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 8 & 2 \\ 1 & 1 & 1 & 4 & 13 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 27 & 32 & 6 \\ 7 & 13 & 0 & 27 & 90 & 108 & 20 \\ 8 & 16 & 0 & 32 & 108 & 126 & 24 \\ 2 & 3 & -1 & 6 & 20 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[819] \ 1^{1211}31^1513^11^{211} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 2 & 2 \\ 1 & 1 & 1 & 8 & 5 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 22 & 16 & 12 \\ 3 & 5 & 0 & 22 & 14 & 11 & 8 \\ 2 & 4 & 0 & 16 & 11 & 6 & 6 \\ 2 & 3 & -1 & 12 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[820] \ 1^{1311}3^11^15131^{321} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 4 & 2 & 2 \\ 1 & 1 & 1 & 12 & 7 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 45 & 24 & 18 \\ 4 & 7 & 0 & 45 & 27 & 15 & 11 \\ 2 & 4 & 0 & 24 & 15 & 6 & 6 \\ 2 & 3 & -1 & 18 & 11 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[821] \ 1^{133}3^11^15131^{313} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.6$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 21 & 72 & 18 \\ 2 & 3 & 0 & 21 & 5 & 21 & 5 \\ 6 & 12 & 0 & 72 & 21 & 70 & 18 \\ 2 & 3 & -1 & 18 & 5 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[822] \ 1^{1122}3^11^15131^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 4 & 2 \\ 1 & 1 & 1 & 4 & 11 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 23 & 16 & 6 \\ 6 & 11 & 0 & 23 & 65 & 46 & 17 \\ 4 & 8 & 0 & 16 & 46 & 30 & 12 \\ 2 & 3 & -1 & 6 & 17 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[816] \ 1^{134}3^11^15131^{314} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 8 & 2 \\ 1 & 1 & 1 & 12 & 3 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 21 & 96 & 18 \\ 2 & 3 & 0 & 21 & 5 & 28 & 5 \\ 8 & 16 & 0 & 96 & 28 & 126 & 24 \\ 2 & 3 & -1 & 18 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[815] \ 1^{1143}31^1513^11^{143} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 6 & 2 \\ 1 & 1 & 1 & 4 & 17 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 35 & 24 & 6 \\ 9 & 17 & 0 & 35 & 152 & 105 & 26 \\ 6 & 12 & 0 & 24 & 105 & 70 & 18 \\ 2 & 3 & -1 & 6 & 26 & 18 & 4 \end{pmatrix}_{b_0}$$

[851] $1^{\overline{1121}} 31^{\overline{1}} 513^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 & 2 \\ 1 & 1 & 1 & 4 & 9 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 19 & 8 & 6 \\ 5 & 9 & 0 & 19 & 44 & 19 & 14 \\ 2 & 4 & 0 & 8 & 19 & 6 & 6 \\ 2 & 3 & -1 & 6 & 14 & 6 & 4 \end{pmatrix}_{b_0}$$

[850] $1^{\overline{1321}} 31^{\overline{1}} 513^{\overline{1321}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 5 & 2 & 2 \\ 1 & 1 & 1 & 12 & 9 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 57 & 24 & 18 \\ 5 & 9 & 0 & 57 & 44 & 19 & 14 \\ 2 & 4 & 0 & 24 & 19 & 6 & 6 \\ 2 & 3 & -1 & 18 & 14 & 6 & 4 \end{pmatrix}_{b_0}$$

[817] $1^{\overline{1111}} 31^{\overline{1}} 515131^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 & 2 \\ 1 & 1 & 1 & 4 & 7 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 15 & 8 & 6 \\ 4 & 7 & 0 & 15 & 27 & 15 & 11 \\ 2 & 4 & 0 & 8 & 15 & 6 & 6 \\ 2 & 3 & -1 & 6 & 11 & 6 & 4 \end{pmatrix}_{b_0}$$

[848] $1^{\overline{1144}} 31^{\overline{1}} 5131^{\overline{144}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 9 & 8 & 2 \\ 1 & 1 & 1 & 4 & 17 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 35 & 32 & 6 \\ 9 & 17 & 0 & 35 & 152 & 140 & 26 \\ 8 & 16 & 0 & 32 & 140 & 126 & 24 \\ 2 & 3 & -1 & 6 & 26 & 24 & 4 \end{pmatrix}_{b_0}$$

[847] $1^{\overline{1232}} 31^{\overline{1}} 5131^{\overline{232}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 7 & 4 & 2 \\ 1 & 1 & 1 & 8 & 13 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 54 & 32 & 12 \\ 7 & 13 & 0 & 54 & 90 & 54 & 20 \\ 4 & 8 & 0 & 32 & 54 & 30 & 12 \\ 2 & 3 & -1 & 12 & 20 & 12 & 4 \end{pmatrix}_{b_0}$$

[846] $1^{\overline{1223}} 31^{\overline{1}} 5131^{\overline{223}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.3$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 6 & 2 \\ 1 & 1 & 1 & 8 & 9 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 38 & 48 & 12 \\ 5 & 9 & 0 & 38 & 44 & 57 & 14 \\ 6 & 12 & 0 & 48 & 57 & 70 & 18 \\ 2 & 3 & -1 & 12 & 14 & 18 & 4 \end{pmatrix}_{b_0}$$

[845] $1^{\overline{1213}} 31^{\overline{1}} 5131^{\overline{213}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.3$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 6 & 2 \\ 1 & 1 & 1 & 8 & 5 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 22 & 48 & 12 \\ 3 & 5 & 0 & 22 & 14 & 33 & 8 \\ 6 & 12 & 0 & 48 & 33 & 70 & 18 \\ 2 & 3 & -1 & 12 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

[844] $1^{\overline{1222}} 31^{\overline{1}} 5131^{\overline{222}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 4 & 2 \\ 1 & 1 & 1 & 8 & 9 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 38 & 32 & 12 \\ 5 & 9 & 0 & 38 & 44 & 38 & 14 \\ 4 & 8 & 0 & 32 & 38 & 30 & 12 \\ 2 & 3 & -1 & 12 & 14 & 12 & 4 \end{pmatrix}_{b_0}$$

[843] $1^{1123}3^115131^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 4 & 2 \\ 1 & 1 & 1 & 4 & 7 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 15 & 16 & 6 \\ 4 & 7 & 0 & 15 & 27 & 30 & 11 \\ 4 & 8 & 0 & 16 & 30 & 30 & 12 \\ 2 & 3 & -1 & 6 & 11 & 12 & 4 \end{pmatrix}_{b_0}$$

[842] $1^{1121}3^115131^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 & 2 \\ 1 & 1 & 1 & 4 & 11 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 23 & 8 & 6 \\ 6 & 11 & 0 & 23 & 65 & 23 & 17 \\ 2 & 4 & 0 & 8 & 23 & 6 & 6 \\ 2 & 3 & -1 & 6 & 17 & 6 & 4 \end{pmatrix}_{b_0}$$

[841] $1^{1223}3^115131^{212} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 4 & 2 \\ 1 & 1 & 1 & 8 & 3 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 14 & 32 & 12 \\ 2 & 3 & 0 & 14 & 5 & 14 & 5 \\ 4 & 8 & 0 & 32 & 14 & 30 & 12 \\ 2 & 3 & -1 & 12 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

[840] $1^{1221}3^115131^{221} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 5 & 2 & 2 \\ 1 & 1 & 1 & 8 & 9 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 38 & 16 & 12 \\ 5 & 9 & 0 & 38 & 44 & 19 & 14 \\ 2 & 4 & 0 & 16 & 19 & 6 & 6 \\ 2 & 3 & -1 & 12 & 14 & 6 & 4 \end{pmatrix}_{b_0}$$

[839] $1^{1411}3^115131^{421} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 4 & 2 & 2 \\ 1 & 1 & 1 & 16 & 7 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 60 & 32 & 24 \\ 4 & 7 & 0 & 60 & 27 & 15 & 11 \\ 2 & 4 & 0 & 32 & 15 & 6 & 6 \\ 2 & 3 & -1 & 24 & 11 & 6 & 4 \end{pmatrix}_{b_0}$$

[838] $1^{1443}3^115131^{414} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 8 & 2 \\ 1 & 1 & 1 & 16 & 3 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 28 & 128 & 24 \\ 2 & 3 & 0 & 28 & 5 & 28 & 5 \\ 8 & 16 & 0 & 128 & 28 & 126 & 24 \\ 2 & 3 & -1 & 24 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[814] $1^{1114}3^115131^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 8 & 2 \\ 1 & 1 & 1 & 4 & 5 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 11 & 32 & 6 \\ 3 & 5 & 0 & 11 & 14 & 44 & 8 \\ 8 & 16 & 0 & 32 & 44 & 126 & 24 \\ 2 & 3 & -1 & 6 & 8 & 24 & 4 \end{pmatrix}_{b_0}$$

[823] $1^{1133}3^115131^{133} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 6 & 2 \\ 1 & 1 & 1 & 4 & 13 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 27 & 24 & 6 \\ 7 & 13 & 0 & 27 & 90 & 81 & 20 \\ 6 & 12 & 0 & 24 & 81 & 70 & 18 \\ 2 & 3 & -1 & 6 & 20 & 18 & 4 \end{pmatrix}_{b_0}$$

[827] $1^{132}3^11^15131^{312} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.3$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 4 & 2 \\ 1 & 1 & 1 & 12 & 3 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 21 & 48 & 18 \\ 2 & 3 & 0 & 21 & 5 & 14 & 5 \\ 4 & 8 & 0 & 48 & 14 & 30 & 12 \\ 2 & 3 & -1 & 18 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

[828] $1^{131}3^11^15131^{311} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 & 2 \\ 1 & 1 & 1 & 12 & 3 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 21 & 24 & 18 \\ 2 & 3 & 0 & 21 & 5 & 7 & 5 \\ 2 & 4 & 0 & 24 & 7 & 6 & 6 \\ 2 & 3 & -1 & 18 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

[829] $1^{111}3^11^15131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 7 & 8 & 6 \\ 2 & 3 & 0 & 7 & 5 & 7 & 5 \\ 2 & 4 & 0 & 8 & 7 & 6 & 6 \\ 2 & 3 & -1 & 6 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

[830] $1^{1242}3^11^1513^11^{242} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 9 & 4 & 2 \\ 1 & 1 & 1 & 8 & 17 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 70 & 32 & 12 \\ 9 & 17 & 0 & 70 & 152 & 70 & 26 \\ 4 & 8 & 0 & 32 & 70 & 30 & 12 \\ 2 & 3 & -1 & 12 & 26 & 12 & 4 \end{pmatrix}_{b_0}$$

[831] $1^{1311}3^11^1513^11^{311} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 2 & 2 \\ 1 & 1 & 1 & 12 & 5 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 33 & 24 & 18 \\ 3 & 5 & 0 & 33 & 14 & 11 & 8 \\ 2 & 4 & 0 & 24 & 11 & 6 & 6 \\ 2 & 3 & -1 & 18 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

[832] $1^{131}3^11^15131^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 & 2 \\ 1 & 1 & 1 & 4 & 15 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 31 & 8 & 6 \\ 8 & 15 & 0 & 31 & 119 & 31 & 23 \\ 2 & 4 & 0 & 8 & 31 & 6 & 6 \\ 2 & 3 & -1 & 6 & 23 & 6 & 4 \end{pmatrix}_{b_0}$$

[833] $1^{1313}3^11^1513^11^{313} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.6$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 6 & 2 \\ 1 & 1 & 1 & 12 & 5 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 6 & 12 & 0 & 70 & 33 & 72 & 18 \\ 3 & 5 & 0 & 33 & 14 & 33 & 8 \\ 6 & 12 & 0 & 72 & 33 & 70 & 18 \\ 2 & 3 & -1 & 18 & 8 & 18 & 4 \end{pmatrix}_{b_0}$$

[854] $1^{141}3^11^15131^{411} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 2 & 2 \\ 1 & 1 & 1 & 16 & 3 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 28 & 32 & 24 \\ 2 & 3 & 0 & 28 & 5 & 7 & 5 \\ 2 & 4 & 0 & 32 & 7 & 6 & 6 \\ 2 & 3 & -1 & 24 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

[834] $1^{124}3^11^1513^{1214}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 8 & 2 \\ 1 & 1 & 1 & 8 & 3 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 14 & 64 & 12 \\ 2 & 3 & 0 & 14 & 5 & 28 & 5 \\ 8 & 16 & 0 & 64 & 28 & 126 & 24 \\ 2 & 3 & -1 & 12 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[837] $1^{1212}3^11^1513^{1212}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 4 & 2 \\ 1 & 1 & 1 & 8 & 5 & 8 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & 30 & 22 & 32 & 12 \\ 3 & 5 & 0 & 22 & 14 & 22 & 8 \\ 4 & 8 & 0 & 32 & 22 & 30 & 12 \\ 2 & 3 & -1 & 12 & 8 & 12 & 4 \end{pmatrix}_{b_0}$$

[836] $1^{1414}3^11^1513^{1414}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 8 & 2 \\ 1 & 1 & 1 & 16 & 5 & 16 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 8 & 16 & 0 & 126 & 44 & 128 & 24 \\ 3 & 5 & 0 & 44 & 14 & 44 & 8 \\ 8 & 16 & 0 & 128 & 44 & 126 & 24 \\ 2 & 3 & -1 & 24 & 8 & 24 & 4 \end{pmatrix}_{b_0}$$

[825] $1^{1111}3^11^1513^{1111}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 & 2 \\ 1 & 1 & 1 & 4 & 5 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 11 & 8 & 6 \\ 3 & 5 & 0 & 11 & 14 & 11 & 8 \\ 2 & 4 & 0 & 8 & 11 & 6 & 6 \\ 2 & 3 & -1 & 6 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

[859] $1^{1112}3^11^1513^{111}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 5 & 2 \\ 1 & 1 & 2 & 3 & 4 & 7 & 3 \\ 1 & 2 & 0 & 1 & 3 & 5 & 2 \\ 2 & 3 & 1 & 1 & 4 & 7 & 3 \\ 2 & 4 & 3 & 4 & 5 & 11 & 5 \\ 5 & 7 & 5 & 7 & 11 & 17 & 9 \\ 2 & 3 & 2 & 3 & 5 & 9 & 4 \end{pmatrix}_{b_0}$$

[860] $1^{1111}3^11^1513^{111}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 22, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 3 & 2 \\ 1 & 1 & 2 & 4 & 4 & 4 & 3 \\ 1 & 2 & 0 & 4 & 3 & 4 & 2 \\ 2 & 4 & 4 & 6 & 7 & 8 & 6 \\ 3 & 4 & 3 & 7 & 5 & 7 & 5 \\ 3 & 4 & 4 & 8 & 7 & 6 & 6 \\ 2 & 3 & 2 & 6 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

[861] $1^{1112}3^11^1513^{111}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 4 & 2 \\ 1 & 1 & 2 & 4 & 4 & 7 & 3 \\ 1 & 2 & 0 & 4 & 3 & 5 & 2 \\ 2 & 4 & 4 & 6 & 7 & 12 & 6 \\ 3 & 4 & 3 & 7 & 5 & 11 & 5 \\ 4 & 7 & 5 & 12 & 11 & 17 & 9 \\ 2 & 3 & 2 & 6 & 5 & 9 & 4 \end{pmatrix}_{b_0}$$

[862] $1^{1113}3^11^1513^{112}+[\textcolor{blue}{su}_2\cdot\textcolor{blue}{g}_2]+\textcolor{green}{su}_2+\textcolor{orange}{su}_2+\textcolor{red}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

(1, 12, 1), $c_{\text{ext}} = 7.5$

4.947 NHC(-2)(-3) + su₂ + su2n3mix + su₂→(-3)

$T = 8$ [1] $1^{122}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8$, $T_{\text{max}} = 10$, $\Delta = -59$, $\det = 3$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$

$T_{\text{min}} = 10$, $T_{\text{max}} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 2 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 \\ 5 & 5 & 4 & 6 & 3 & 4 \\ 6 & 7 & 6 & 7 & 4 & 5 \\ 2 & 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[0] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8$, $T_{\text{max}} = 10$, $\Delta = -59$, $\det = 3$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$

$T_{\text{min}} = 10$, $T_{\text{max}} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 3 \\ 3 & 4 & 9 & 3 & 3 & 4 \\ 5 & 9 & 13 & 5 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 & 2 \\ 2 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[3] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8$, $T_{\text{max}} = 10$, $\Delta = -59$, $\det = 4$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.8$

$T_{\text{min}} = 10$, $T_{\text{max}} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 2 & 3 \\ 3 & 4 & 17 & 11 & 3 & 4 \\ 10 & 17 & 49 & 33 & 9 & 13 \\ 8 & 11 & 33 & 20 & 6 & 8 \\ 2 & 3 & 9 & 6 & 0 & 2 \\ 3 & 4 & 13 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8$, $T_{\text{max}} = 10$, $\Delta = -59$, $\det = 4$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$

$T_{\text{min}} = 10$, $T_{\text{max}} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 2 & 3 \\ 3 & 4 & 14 & 7 & 3 & 4 \\ 9 & 14 & 37 & 17 & 8 & 11 \\ 5 & 7 & 17 & 7 & 4 & 5 \\ 2 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 11 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[4] $1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8$, $T_{\text{max}} = 10$, $\Delta = -59$, $\det = 13$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$

$T_{\text{min}} = 9.85 \rightarrow 10$, $T_{\text{max}} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 \\ 2 & 4 & 1 & 2 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[5] $1^{12}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8$, $T_{\text{max}} = 10$, $\Delta = -59$, $\det = 14$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.5$

$T_{\text{min}} = 9.86 \rightarrow 10$, $T_{\text{max}} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 3 \\ 3 & 4 & 8 & 3 & 3 & 4 \\ 5 & 8 & 10 & 4 & 4 & 6 \\ 3 & 3 & 4 & 0 & 2 & 2 \\ 2 & 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[6] $1^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8$, $T_{\text{max}} = 10$, $\Delta = -59$, $\det = 16$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$

$T_{\text{min}} = 10$, $T_{\text{max}} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 2 & 3 \\ 3 & 4 & 14 & 7 & 3 & 4 \\ 11 & 14 & 37 & 17 & 8 & 11 \\ 5 & 7 & 17 & 7 & 4 & 5 \\ 2 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 11 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8$, $T_{\text{max}} = 10$, $\Delta = -59$, $\det = 18$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$

$T_{\text{min}} = 9.89 \rightarrow 10$, $T_{\text{max}} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 0 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[7] $1^13^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8$, $T_{\text{max}} = 10$, $\Delta = -59$, $\det = 18$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$

$T_{\text{min}} = 9.89 \rightarrow 10$, $T_{\text{max}} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 3 \\ 3 & 4 & 5 & 3 & 3 & 4 \\ 3 & 5 & 4 & 3 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[9] $1^{121}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 2 & 3 \\ 3 & 4 & 12 & 3 & 3 & 4 \\ 8 & 12 & 25 & 6 & 6 & 9 \\ 3 & 3 & 6 & 0 & 2 & 2 \\ 2 & 3 & 6 & 2 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[10] $1^{113}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 \\ 3 & 5 & 4 & 6 & 3 & 4 \\ 5 & 7 & 6 & 7 & 4 & 5 \\ 2 & 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[11] $1^{122}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 2 & 3 \\ 3 & 4 & 8 & 16 & 3 & 4 \\ 5 & 8 & 10 & 22 & 4 & 6 \\ 12 & 16 & 22 & 44 & 9 & 12 \\ 2 & 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 6 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[12] $1^{131}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 2 & 3 \\ 3 & 4 & 17 & 7 & 3 & 4 \\ 11 & 17 & 49 & 21 & 9 & 13 \\ 6 & 7 & 21 & 7 & 4 & 5 \\ 2 & 3 & 9 & 4 & 0 & 2 \\ 3 & 4 & 13 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[13] $1^{113}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 \\ 4 & 4 & 1 & 2 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[14] $1^{121}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 2 & 3 \\ 3 & 4 & 8 & 7 & 3 & 4 \\ 5 & 8 & 10 & 10 & 4 & 6 \\ 6 & 7 & 10 & 7 & 4 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 \\ 3 & 4 & 6 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[16] $1^{111}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 6 & 2 & 3 \\ 3 & 4 & 14 & 7 & 3 & 4 \\ 12 & 14 & 37 & 17 & 8 & 11 \\ 6 & 7 & 17 & 7 & 4 & 5 \\ 2 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 11 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[15] $1^{131}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 2 & 3 \\ 3 & 4 & 12 & 7 & 3 & 4 \\ 8 & 12 & 25 & 15 & 6 & 9 \\ 6 & 7 & 15 & 7 & 4 & 5 \\ 2 & 3 & 6 & 4 & 0 & 2 \\ 3 & 4 & 9 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[23] $1^{143}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} =$

7.0

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 2 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 13 & 21 & 76 & 11 & 11 & 16 \\ 3 & 3 & 11 & 0 & 2 & 2 \\ 2 & 3 & 11 & 2 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 2 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 \\ 4 & 5 & 4 & 6 & 3 & 4 \\ 6 & 7 & 6 & 7 & 4 & 5 \\ 2 & 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{14} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 2 & 3 \\ 3 & 4 & 16 & 3 & 3 & 4 \\ 11 & 16 & 46 & 8 & 8 & 12 \\ 3 & 3 & 8 & 0 & 2 & 2 \\ 2 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 \\ 4 & 5 & 4 & 6 & 3 & 4 \\ 5 & 7 & 6 & 7 & 4 & 5 \\ 2 & 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 3 \\ 3 & 4 & 5 & 3 & 3 & 4 \\ 4 & 5 & 4 & 3 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 \\ 2 & 4 & 1 & 5 & 2 & 3 \\ 6 & 7 & 5 & 7 & 4 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 \\ 3 & 4 & 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[17] \ 21^{13} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 3 & 4 \\ 5 & 4 & 4 & 7 & 3 & 4 \\ 3 & 4 & 1 & 5 & 2 & 3 \\ 6 & 7 & 5 & 7 & 4 & 5 \\ 3 & 3 & 2 & 4 & 0 & 2 \\ 4 & 4 & 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{11} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 2 & 3 \\ 3 & 4 & 14 & 7 & 3 & 4 \\ 11 & 14 & 37 & 17 & 8 & 11 \\ 6 & 7 & 17 & 7 & 4 & 5 \\ 2 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 11 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{14} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 2 & 3 \\ 3 & 4 & 16 & 7 & 3 & 4 \\ 11 & 16 & 46 & 20 & 8 & 12 \\ 6 & 7 & 20 & 7 & 4 & 5 \\ 2 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 12 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

- [26] $1^{111}3^{111}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 5 & 2 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 \\ 8 & 9 & 13 & 11 & 5 & 7 \\ 5 & 7 & 11 & 7 & 4 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 \\ 3 & 4 & 7 & 5 & 2 & 3 \end{pmatrix}_{b_0}$
- [27] $1^{111}3^{111}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 \\ 3 & 4 & 1 & 2 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [28] $1^{112}3^{111}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 0 & 7 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 5 \\ 2 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 0 & 5 & 2 & 3 \end{pmatrix}_{b_0}$
- [30] $1^{123}3^{111}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 & 2 & 3 \\ 3 & 4 & 18 & 3 & 3 & 4 \\ 11 & 18 & 58 & 10 & 10 & 14 \\ 3 & 3 & 10 & 0 & 2 & 2 \\ 2 & 3 & 10 & 2 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [29] $1^{123}3^{111}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 & 2 & 3 \\ 3 & 4 & 8 & 3 & 3 & 4 \\ 7 & 8 & 10 & 4 & 4 & 6 \\ 3 & 3 & 4 & 0 & 2 & 2 \\ 2 & 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [33] $1^{111}3^{111}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 5 & 2 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 \\ 5 & 9 & 13 & 11 & 5 & 7 \\ 5 & 7 & 11 & 7 & 4 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 \\ 3 & 4 & 7 & 5 & 2 & 3 \end{pmatrix}_{b_0}$
- [31] $1^{123}3^{111}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 & 2 & 3 \\ 3 & 4 & 13 & 3 & 3 & 4 \\ 8 & 13 & 28 & 7 & 7 & 10 \\ 3 & 3 & 7 & 0 & 2 & 2 \\ 2 & 3 & 7 & 2 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [34] $1^{121}3^{111}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 3 \\ 3 & 4 & 8 & 3 & 3 & 4 \\ 6 & 8 & 10 & 4 & 4 & 6 \\ 3 & 3 & 4 & 0 & 2 & 2 \\ 2 & 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [32] $1^{112}3^{111}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 2 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 \\ 6 & 9 & 13 & 11 & 5 & 7 \\ 6 & 7 & 11 & 7 & 4 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 \\ 3 & 4 & 7 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[35] $1^{111}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 \\ 4 & 4 & 1 & 5 & 2 & 3 \\ 5 & 7 & 5 & 7 & 4 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 \\ 3 & 4 & 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[36] $1^{121}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 2 & 3 \\ 3 & 4 & 8 & 7 & 3 & 4 \\ 7 & 8 & 10 & 10 & 4 & 6 \\ 5 & 7 & 10 & 7 & 4 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 \\ 3 & 4 & 6 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[37] $1^{121}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 2 & 3 \\ 3 & 4 & 14 & 7 & 3 & 4 \\ 10 & 14 & 37 & 17 & 8 & 11 \\ 5 & 7 & 17 & 7 & 4 & 5 \\ 2 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 11 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[38] $1^{123}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 2 & 3 \\ 3 & 4 & 12 & 11 & 3 & 4 \\ 8 & 12 & 25 & 24 & 6 & 9 \\ 9 & 11 & 24 & 20 & 6 & 8 \\ 2 & 3 & 6 & 6 & 0 & 2 \\ 3 & 4 & 9 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[39] $1^{111}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 2 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 9 & 14 & 37 & 8 & 8 & 11 \\ 3 & 3 & 8 & 0 & 2 & 2 \\ 2 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[40] $1^{142}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 9 & 2 & 3 \\ 3 & 4 & 16 & 11 & 3 & 4 \\ 11 & 16 & 46 & 32 & 8 & 12 \\ 9 & 11 & 32 & 20 & 6 & 8 \\ 2 & 3 & 8 & 6 & 0 & 2 \\ 3 & 4 & 12 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[41] $1^{123}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 58, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 2 & 3 \\ 3 & 4 & 8 & 11 & 3 & 4 \\ 5 & 8 & 10 & 16 & 4 & 6 \\ 9 & 11 & 16 & 20 & 6 & 8 \\ 2 & 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[43] $1^{131}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 2 & 3 \\ 3 & 4 & 12 & 7 & 3 & 4 \\ 10 & 12 & 25 & 15 & 6 & 9 \\ 5 & 7 & 15 & 7 & 4 & 5 \\ 2 & 3 & 6 & 4 & 0 & 2 \\ 3 & 4 & 9 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[42] $1^{11}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 0 & 7 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 5 \\ 2 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 0 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[44] $1^{11}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 2 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 \\ 4 & 4 & 1 & 5 & 2 & 3 \\ 6 & 7 & 5 & 7 & 4 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 \\ 3 & 4 & 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[46] $1^{11}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 2 & 3 \\ 3 & 4 & 4 & 12 & 3 & 4 \\ 4 & 4 & 1 & 8 & 2 & 3 \\ 9 & 12 & 8 & 25 & 7 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 \\ 3 & 4 & 3 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[45] $1^{12}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 2 & 3 \\ 3 & 4 & 8 & 25 & 3 & 4 \\ 5 & 8 & 10 & 34 & 4 & 6 \\ 17 & 25 & 34 & 111 & 14 & 19 \\ 2 & 3 & 4 & 14 & 0 & 2 \\ 3 & 4 & 6 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[47] $1^{13}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 2 & 3 \\ 3 & 4 & 12 & 3 & 3 & 4 \\ 9 & 12 & 25 & 6 & 6 & 9 \\ 3 & 3 & 6 & 0 & 2 & 2 \\ 2 & 3 & 6 & 2 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[48] $1^{11}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 3 \\ 3 & 4 & 9 & 3 & 3 & 4 \\ 6 & 9 & 13 & 5 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 & 2 \\ 2 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[51] $1^{11}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 2 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 \\ 2 & 4 & 1 & 11 & 2 & 3 \\ 13 & 16 & 11 & 44 & 9 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 \\ 3 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[50] $1^{12}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 2 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 \\ 2 & 4 & 1 & 8 & 2 & 3 \\ 9 & 11 & 8 & 20 & 6 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 \\ 3 & 4 & 3 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[49] $1^{13}3^{11}1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 2 & 3 \\ 3 & 4 & 12 & 3 & 3 & 4 \\ 10 & 12 & 25 & 6 & 6 & 9 \\ 3 & 3 & 6 & 0 & 2 & 2 \\ 2 & 3 & 6 & 2 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[52] $1^{111}3^{11}1^{11}+[\textcolor{red}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2\rightarrow(-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 \\ 2 & 4 & 1 & 5 & 2 & 3 \\ 5 & 7 & 5 & 7 & 4 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 \\ 3 & 4 & 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[53] $1^{111}3^{11}1^{11}+[\textcolor{red}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2\rightarrow(-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 2 & 3 \\ 3 & 4 & 4 & 12 & 3 & 4 \\ 2 & 4 & 1 & 8 & 2 & 3 \\ 9 & 12 & 8 & 25 & 7 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 \\ 3 & 4 & 3 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[54] $1^{112}3^{11}1^{11}+[\textcolor{red}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2\rightarrow(-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 2 & 3 \\ 3 & 4 & 9 & 11 & 3 & 4 \\ 8 & 9 & 13 & 17 & 5 & 7 \\ 9 & 11 & 17 & 20 & 6 & 8 \\ 2 & 3 & 5 & 6 & 0 & 2 \\ 3 & 4 & 7 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[55] $1^{111}3^{11}1^{11}+[\textcolor{red}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2\rightarrow(-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 2 & 3 \\ 3 & 4 & 9 & 3 & 3 & 4 \\ 8 & 9 & 13 & 5 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 & 2 \\ 2 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[56] $1^{123}3^{11}1^{11}+[\textcolor{red}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2\rightarrow(-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 15 & 2 & 3 \\ 3 & 4 & 8 & 20 & 3 & 4 \\ 7 & 8 & 10 & 28 & 4 & 6 \\ 15 & 20 & 28 & 69 & 11 & 15 \\ 2 & 3 & 4 & 11 & 0 & 2 \\ 3 & 4 & 6 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[57] $1^{122}3^{11}1^{11}+[\textcolor{red}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2\rightarrow(-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 13 & 2 & 3 \\ 3 & 4 & 8 & 16 & 3 & 4 \\ 7 & 8 & 10 & 22 & 4 & 6 \\ 13 & 16 & 22 & 44 & 9 & 12 \\ 2 & 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 6 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[58] $1^{131}3^{11}1^{12}+[\textcolor{red}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2\rightarrow(-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 6 & 2 & 3 \\ 3 & 4 & 22 & 7 & 3 & 4 \\ 15 & 22 & 85 & 27 & 12 & 17 \\ 6 & 7 & 27 & 7 & 4 & 5 \\ 2 & 3 & 12 & 4 & 0 & 2 \\ 3 & 4 & 17 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[59] $1^{121}3^{11}1^{11}+[\textcolor{red}{\mathfrak{su}}_2\cdot\textcolor{blue}{\mathfrak{g}}_2]+\textcolor{green}{\mathfrak{su}}_2+\textcolor{orange}{\mathfrak{su}}_2+\textcolor{red}{\mathfrak{su}}_2\rightarrow(-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 2 & 3 \\ 3 & 4 & 8 & 7 & 3 & 4 \\ 7 & 8 & 10 & 10 & 4 & 6 \\ 6 & 7 & 10 & 7 & 4 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 \\ 3 & 4 & 6 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

- [60] $1^{13}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 11 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 9 & 11 & 0 & 20 & 6 & 8 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 0 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$
- [61] $1^{12}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 2 & 3 \\ 3 & 4 & 26 & 3 & 3 & 4 \\ 16 & 26 & 118 & 14 & 14 & 20 \\ 3 & 3 & 14 & 0 & 2 & 2 \\ 2 & 3 & 14 & 2 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$
- [62] $1^{41}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 2 & 3 \\ 3 & 4 & 16 & 3 & 3 & 4 \\ 12 & 16 & 46 & 8 & 8 & 12 \\ 3 & 3 & 8 & 0 & 2 & 2 \\ 2 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$
- [63] $1^{132}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 2 & 3 \\ 3 & 4 & 8 & 11 & 3 & 4 \\ 7 & 8 & 10 & 16 & 4 & 6 \\ 8 & 11 & 16 & 20 & 6 & 8 \\ 2 & 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$
- [64] $21^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 9 & 6 & 3 & 4 \\ 5 & 4 & 9 & 7 & 3 & 4 \\ 9 & 9 & 13 & 11 & 5 & 7 \\ 6 & 7 & 11 & 7 & 4 & 5 \\ 3 & 3 & 5 & 4 & 0 & 2 \\ 4 & 4 & 7 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$
- [65] $1^{112}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 2 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 \\ 3 & 4 & 1 & 5 & 2 & 3 \\ 6 & 7 & 5 & 7 & 4 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 \\ 3 & 4 & 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$
- [66] $1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 2 & 3 \\ 3 & 4 & 9 & 3 & 3 & 4 \\ 7 & 9 & 13 & 5 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 & 2 \\ 2 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$
- [67] $1^{132}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 2 & 3 \\ 3 & 4 & 12 & 11 & 3 & 4 \\ 10 & 12 & 25 & 24 & 6 & 9 \\ 8 & 11 & 24 & 20 & 6 & 8 \\ 2 & 3 & 6 & 6 & 0 & 2 \\ 3 & 4 & 9 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$
- [68] $1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 2 & 3 \\ 3 & 4 & 4 & 12 & 3 & 4 \\ 4 & 4 & 1 & 8 & 2 & 3 \\ 10 & 12 & 8 & 25 & 7 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 \\ 3 & 4 & 3 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[69] $1^{122}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 2 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 \\ 4 & 4 & 1 & 8 & 2 & 3 \\ 8 & 11 & 8 & 20 & 6 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 \\ 3 & 4 & 3 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[70] $1^{13}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 10 & 12 & 0 & 25 & 7 & 9 \\ 2 & 3 & 0 & 7 & 0 & 2 \\ 3 & 4 & 0 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[73] $1^{122}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 2 & 3 \\ 3 & 4 & 8 & 7 & 3 & 4 \\ 6 & 8 & 10 & 10 & 4 & 6 \\ 6 & 7 & 10 & 7 & 4 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 \\ 3 & 4 & 6 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[71] $1^{132}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 2 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 \\ 8 & 9 & 13 & 11 & 5 & 7 \\ 6 & 7 & 11 & 7 & 4 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 \\ 3 & 4 & 7 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{141}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 6 & 2 & 3 \\ 3 & 4 & 21 & 7 & 3 & 4 \\ 14 & 21 & 76 & 26 & 11 & 16 \\ 6 & 7 & 26 & 7 & 4 & 5 \\ 2 & 3 & 11 & 4 & 0 & 2 \\ 3 & 4 & 16 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[75] $1^{121}3^{11}1^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 2 & 3 \\ 3 & 4 & 18 & 7 & 3 & 4 \\ 11 & 18 & 58 & 22 & 10 & 14 \\ 5 & 7 & 22 & 7 & 4 & 5 \\ 2 & 3 & 10 & 4 & 0 & 2 \\ 3 & 4 & 14 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[74] $1^{121}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 2 & 3 \\ 3 & 4 & 8 & 7 & 3 & 4 \\ 5 & 8 & 10 & 10 & 4 & 6 \\ 5 & 7 & 10 & 7 & 4 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 \\ 3 & 4 & 6 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[76] $1^{112}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \rightarrow (-3)$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 2 & 3 \\ 3 & 4 & 9 & 11 & 3 & 4 \\ 6 & 9 & 13 & 17 & 5 & 7 \\ 8 & 11 & 17 & 20 & 6 & 8 \\ 2 & 3 & 5 & 6 & 0 & 2 \\ 3 & 4 & 7 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[77] $1^{14} 1^{12} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 2 & 3 \\ 3 & 4 & 8 & 15 & 3 & 4 \\ 5 & 8 & 10 & 22 & 4 & 6 \\ 12 & 15 & 22 & 39 & 8 & 11 \\ 2 & 3 & 4 & 8 & 0 & 2 \\ 3 & 4 & 6 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

[78] $1^{21} 1^{13} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 2 & 3 \\ 3 & 4 & 17 & 3 & 3 & 4 \\ 11 & 17 & 49 & 9 & 9 & 13 \\ 3 & 3 & 9 & 0 & 2 & 2 \\ 2 & 3 & 9 & 2 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[81] $1^{12} 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 2 & 3 \\ 3 & 4 & 13 & 11 & 3 & 4 \\ 8 & 13 & 28 & 25 & 7 & 10 \\ 8 & 11 & 25 & 20 & 6 & 8 \\ 2 & 3 & 7 & 6 & 0 & 2 \\ 3 & 4 & 10 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[79] $1^{11} 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 \\ 3 & 4 & 1 & 5 & 2 & 3 \\ 5 & 7 & 5 & 7 & 4 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 \\ 3 & 4 & 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[80] $1^{12} 1^{12} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 0 & 11 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 8 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 0 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[82] $1^{22} 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 2 & 3 \\ 3 & 4 & 8 & 16 & 3 & 4 \\ 6 & 8 & 10 & 22 & 4 & 6 \\ 11 & 16 & 22 & 44 & 9 & 12 \\ 2 & 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 6 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[83] $1^{23} 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 2 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 \\ 4 & 4 & 1 & 8 & 2 & 3 \\ 9 & 11 & 8 & 20 & 6 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 \\ 3 & 4 & 3 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[85] $21^{11} 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 12 & 3 & 4 \\ 5 & 4 & 4 & 12 & 3 & 4 \\ 3 & 4 & 1 & 8 & 2 & 3 \\ 12 & 12 & 8 & 25 & 7 & 9 \\ 3 & 3 & 2 & 7 & 0 & 2 \\ 4 & 4 & 3 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[84] $1^1 1^{11} 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 3 \\ 3 & 4 & 4 & 12 & 3 & 4 \\ 2 & 4 & 1 & 8 & 2 & 3 \\ 8 & 12 & 8 & 25 & 7 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 \\ 3 & 4 & 3 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{132}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 113, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 2 & 3 \\ 3 & 4 & 12 & 7 & 3 & 4 \\ 9 & 12 & 25 & 15 & 6 & 9 \\ 6 & 7 & 15 & 7 & 4 & 5 \\ 2 & 3 & 6 & 4 & 0 & 2 \\ 3 & 4 & 9 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{121}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 113, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 6 & 2 & 3 \\ 3 & 4 & 18 & 7 & 3 & 4 \\ 13 & 18 & 58 & 22 & 10 & 14 \\ 6 & 7 & 22 & 7 & 4 & 5 \\ 2 & 3 & 10 & 4 & 0 & 2 \\ 3 & 4 & 14 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{111}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 113, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 2 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 10 & 14 & 37 & 8 & 8 & 11 \\ 3 & 3 & 8 & 0 & 2 & 2 \\ 2 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{142}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 114, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 2 & 3 \\ 3 & 4 & 12 & 7 & 3 & 4 \\ 10 & 12 & 25 & 15 & 6 & 9 \\ 6 & 7 & 15 & 7 & 4 & 5 \\ 2 & 3 & 6 & 4 & 0 & 2 \\ 3 & 4 & 9 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 115, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 2 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 \\ 7 & 9 & 13 & 11 & 5 & 7 \\ 6 & 7 & 11 & 7 & 4 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 \\ 3 & 4 & 7 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{122}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 116, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 2 & 3 \\ 3 & 4 & 8 & 16 & 3 & 4 \\ 5 & 8 & 10 & 22 & 4 & 6 \\ 11 & 16 & 22 & 44 & 9 & 12 \\ 2 & 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 6 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{124}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 116, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 2 & 3 \\ 3 & 4 & 8 & 16 & 3 & 4 \\ 6 & 8 & 10 & 22 & 4 & 6 \\ 13 & 16 & 22 & 44 & 9 & 12 \\ 2 & 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 6 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{121}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 116, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 2 & 3 \\ 3 & 4 & 8 & 7 & 3 & 4 \\ 6 & 8 & 10 & 10 & 4 & 6 \\ 5 & 7 & 10 & 7 & 4 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 \\ 3 & 4 & 6 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[94] $1^{14}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 118, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 2 & 3 \\ 3 & 4 & 4 & 15 & 3 & 4 \\ 2 & 4 & 1 & 11 & 2 & 3 \\ 12 & 15 & 11 & 39 & 8 & 11 \\ 2 & 3 & 2 & 8 & 0 & 2 \\ 3 & 4 & 3 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

[95] $1^{12}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 119, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 9 & 12 & 0 & 25 & 7 & 9 \\ 2 & 3 & 0 & 7 & 0 & 2 \\ 3 & 4 & 0 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[96] $1^{21}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 122, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 2 & 3 \\ 3 & 4 & 13 & 3 & 3 & 4 \\ 9 & 13 & 28 & 7 & 7 & 10 \\ 3 & 3 & 7 & 0 & 2 & 2 \\ 2 & 3 & 7 & 2 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[97] $1^{22}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 123, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 2 & 3 \\ 3 & 4 & 13 & 7 & 3 & 4 \\ 9 & 13 & 28 & 16 & 7 & 10 \\ 6 & 7 & 16 & 7 & 4 & 5 \\ 2 & 3 & 7 & 4 & 0 & 2 \\ 3 & 4 & 10 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[98] $1^{14}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 124, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 2 & 3 \\ 3 & 4 & 4 & 21 & 3 & 4 \\ 2 & 4 & 1 & 14 & 2 & 3 \\ 16 & 21 & 14 & 80 & 12 & 16 \\ 2 & 3 & 2 & 12 & 0 & 2 \\ 3 & 4 & 3 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[99] $1^{12}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 124, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 \\ 2 & 4 & 1 & 8 & 2 & 3 \\ 8 & 11 & 8 & 20 & 6 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 \\ 3 & 4 & 3 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[100] $1^{13}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 125, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 2 & 3 \\ 3 & 4 & 4 & 12 & 3 & 4 \\ 3 & 4 & 1 & 8 & 2 & 3 \\ 10 & 12 & 8 & 25 & 7 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 \\ 3 & 4 & 3 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[101] $1^{21}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 126, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 2 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 \\ 7 & 9 & 13 & 11 & 5 & 7 \\ 5 & 7 & 11 & 7 & 4 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 \\ 3 & 4 & 7 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[103] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 129, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 2 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 \\ 6 & 9 & 13 & 11 & 5 & 7 \\ 5 & 7 & 11 & 7 & 4 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 \\ 3 & 4 & 7 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[102] \quad 1^{121}3^{111}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 129, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 2 & 3 \\ 3 & 4 & 13 & 7 & 3 & 4 \\ 11 & 13 & 28 & 16 & 7 & 10 \\ 5 & 7 & 16 & 7 & 4 & 5 \\ 2 & 3 & 7 & 4 & 0 & 2 \\ 3 & 4 & 10 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[104] \quad 1^{131}3^{111}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 131, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 2 & 3 \\ 3 & 4 & 12 & 7 & 3 & 4 \\ 8 & 12 & 25 & 15 & 6 & 9 \\ 5 & 7 & 15 & 7 & 4 & 5 \\ 2 & 3 & 6 & 4 & 0 & 2 \\ 3 & 4 & 9 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[105] \quad 1^{113}3^{111}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 133, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 2 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 11 & 14 & 37 & 8 & 8 & 11 \\ 3 & 3 & 8 & 0 & 2 & 2 \\ 2 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[106] \quad 1^{141}3^{111}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 134, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 6 & 2 & 3 \\ 3 & 4 & 16 & 7 & 3 & 4 \\ 12 & 16 & 46 & 20 & 8 & 12 \\ 6 & 7 & 20 & 7 & 4 & 5 \\ 2 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 12 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[107] \quad 1^{123}3^{111}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 135, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 2 & 3 \\ 3 & 4 & 8 & 20 & 3 & 4 \\ 5 & 8 & 10 & 28 & 4 & 6 \\ 15 & 20 & 28 & 69 & 11 & 15 \\ 2 & 3 & 4 & 11 & 0 & 2 \\ 3 & 4 & 6 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[109] \quad 21^{113}3^{111}1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 136, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 15 & 6 & 3 & 4 \\ 5 & 4 & 17 & 7 & 3 & 4 \\ 15 & 17 & 49 & 21 & 9 & 13 \\ 6 & 7 & 21 & 7 & 4 & 5 \\ 3 & 3 & 9 & 4 & 0 & 2 \\ 4 & 4 & 13 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[108] \quad 1^{131}3^{111}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 136, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 2 & 3 \\ 3 & 4 & 12 & 7 & 3 & 4 \\ 9 & 12 & 25 & 15 & 6 & 9 \\ 5 & 7 & 15 & 7 & 4 & 5 \\ 2 & 3 & 6 & 4 & 0 & 2 \\ 3 & 4 & 9 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[110] \quad 1^{122}3^{111}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 138, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 2 & 3 \\ 3 & 4 & 8 & 11 & 3 & 4 \\ 7 & 8 & 10 & 16 & 4 & 6 \\ 9 & 11 & 16 & 20 & 6 & 8 \\ 2 & 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[111] \quad 1^{123}3^{111}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 139, \quad (n_+, n_-, n_0) =$$

$$(1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 2 & 3 \\ 3 & 4 & 8 & 15 & 3 & 4 \\ 7 & 8 & 10 & 22 & 4 & 6 \\ 11 & 15 & 22 & 39 & 8 & 11 \\ 2 & 3 & 4 & 8 & 0 & 2 \\ 3 & 4 & 6 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{13}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 12 & 15 & 0 & 39 & 8 & 11 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & 0 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[114] \ 21^{13}3^{11}1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 12 & 6 & 3 & 4 \\ 5 & 4 & 13 & 7 & 3 & 4 \\ 12 & 13 & 28 & 16 & 7 & 10 \\ 6 & 7 & 16 & 7 & 4 & 5 \\ 3 & 3 & 7 & 4 & 0 & 2 \\ 4 & 4 & 10 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{13}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 2 & 3 \\ 3 & 4 & 22 & 7 & 3 & 4 \\ 13 & 22 & 85 & 27 & 12 & 17 \\ 5 & 7 & 27 & 7 & 4 & 5 \\ 2 & 3 & 12 & 4 & 0 & 2 \\ 3 & 4 & 17 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{13}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 2 & 3 \\ 3 & 4 & 22 & 3 & 3 & 4 \\ 14 & 22 & 85 & 12 & 12 & 17 \\ 3 & 3 & 12 & 0 & 2 & 2 \\ 2 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 2 & 3 \\ 3 & 4 & 13 & 3 & 3 & 4 \\ 11 & 13 & 28 & 7 & 7 & 10 \\ 3 & 3 & 7 & 0 & 2 & 2 \\ 2 & 3 & 7 & 2 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 2 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 \\ 4 & 4 & 1 & 11 & 2 & 3 \\ 12 & 16 & 11 & 44 & 9 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 \\ 3 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{12}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 2 & 3 \\ 3 & 4 & 4 & 15 & 3 & 4 \\ 4 & 4 & 1 & 11 & 2 & 3 \\ 11 & 15 & 11 & 39 & 8 & 11 \\ 2 & 3 & 2 & 8 & 0 & 2 \\ 3 & 4 & 3 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 2 & 3 \\ 3 & 4 & 8 & 16 & 3 & 4 \\ 6 & 8 & 10 & 22 & 4 & 6 \\ 12 & 16 & 22 & 44 & 9 & 12 \\ 2 & 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 6 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[121] \quad & 1^{12} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 152, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 8 & 2 & 3 \\ 3 & 4 & 8 & 11 & 3 & 4 \\ 5 & 8 & 10 & 16 & 4 & 6 \\ 8 & 11 & 16 & 20 & 6 & 8 \\ 2 & 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[120] \quad & 1^{11} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 152, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 9 & 2 & 3 \\ 3 & 4 & 4 & 12 & 3 & 4 \\ 3 & 4 & 1 & 8 & 2 & 3 \\ 9 & 12 & 8 & 25 & 7 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 \\ 3 & 4 & 3 & 9 & 2 & 3 \end{pmatrix}_{b_0} \\
[123] \quad & 1^{12} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 154, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 3 & 2 & 3 \\ 3 & 4 & 13 & 3 & 3 & 4 \\ 10 & 13 & 28 & 7 & 7 & 10 \\ 3 & 3 & 7 & 0 & 2 & 2 \\ 2 & 3 & 7 & 2 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[124] \quad & 1^{12} 3^{11} 1^2 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 154, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 3 & 2 & 3 \\ 3 & 4 & 18 & 3 & 3 & 4 \\ 12 & 18 & 58 & 10 & 10 & 14 \\ 3 & 3 & 10 & 0 & 2 & 2 \\ 2 & 3 & 10 & 2 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[122] \quad & 21^{12} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 154, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 18 & 6 & 3 & 4 \\ 5 & 4 & 18 & 7 & 3 & 4 \\ 18 & 18 & 58 & 22 & 10 & 14 \\ 6 & 7 & 22 & 7 & 4 & 5 \\ 3 & 3 & 10 & 4 & 0 & 2 \\ 4 & 4 & 14 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[125] \quad & 1^{12} 3^{11} 1^2 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 156, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 16 & 2 & 3 \\ 3 & 4 & 4 & 21 & 3 & 4 \\ 4 & 4 & 1 & 14 & 2 & 3 \\ 16 & 21 & 14 & 80 & 12 & 16 \\ 2 & 3 & 2 & 12 & 0 & 2 \\ 3 & 4 & 3 & 16 & 2 & 3 \end{pmatrix}_{b_0} \\
[126] \quad & 1^{13} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 156, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 14 & 2 & 3 \\ 3 & 4 & 4 & 20 & 3 & 4 \\ 4 & 4 & 1 & 14 & 2 & 3 \\ 14 & 20 & 14 & 69 & 11 & 15 \\ 2 & 3 & 2 & 11 & 0 & 2 \\ 3 & 4 & 3 & 15 & 2 & 3 \end{pmatrix}_{b_0} \\
[127] \quad & 21^{11} 3^{11} 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 160, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 3 & 18 & 3 & 4 \\ 5 & 4 & 4 & 20 & 3 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 \\ 18 & 20 & 14 & 69 & 11 & 15 \\ 3 & 3 & 2 & 11 & 0 & 2 \\ 4 & 4 & 3 & 15 & 2 & 3 \end{pmatrix}_{b_0} \\
[128] \quad & 1^{12} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 161, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 2 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 \\ 3 & 4 & 1 & 8 & 2 & 3 \\ 9 & 11 & 8 & 20 & 6 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 \\ 3 & 4 & 3 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[129] \quad 1^{14}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 162, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 2 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 14 & 21 & 76 & 11 & 11 & 16 \\ 3 & 3 & 11 & 0 & 2 & 2 \\ 2 & 3 & 11 & 2 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[130] \quad 1^{13}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 163, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 15 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 11 & 15 & 0 & 39 & 8 & 11 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & 0 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[131] \quad 1^{141}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 164, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 2 & 3 \\ 3 & 4 & 16 & 7 & 3 & 4 \\ 12 & 16 & 46 & 20 & 8 & 12 \\ 5 & 7 & 20 & 7 & 4 & 5 \\ 2 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 12 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[134] \quad 21^{122}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 166, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 11 & 2 & 3 \\ 4 & 4 & 8 & 16 & 3 & 4 \\ 6 & 8 & 10 & 22 & 4 & 6 \\ 11 & 16 & 22 & 44 & 9 & 12 \\ 2 & 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 6 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[133] \quad 21^{113}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 166, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 3 & 4 \\ 5 & 4 & 4 & 16 & 3 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 \\ 15 & 16 & 11 & 44 & 9 & 12 \\ 3 & 3 & 2 & 9 & 0 & 2 \\ 4 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[132] \quad 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 166, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 2 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 \\ 2 & 4 & 1 & 11 & 2 & 3 \\ 12 & 16 & 11 & 44 & 9 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 \\ 3 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[135] \quad 21^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 168, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 8 & 2 & 3 \\ 4 & 4 & 4 & 12 & 3 & 4 \\ 3 & 4 & 1 & 8 & 2 & 3 \\ 8 & 12 & 8 & 25 & 7 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 \\ 3 & 4 & 3 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[136] \quad 1^{131}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 169, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 2 & 3 \\ 3 & 4 & 17 & 7 & 3 & 4 \\ 10 & 17 & 49 & 21 & 9 & 13 \\ 5 & 7 & 21 & 7 & 4 & 5 \\ 2 & 3 & 9 & 4 & 0 & 2 \\ 3 & 4 & 13 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[138] \quad 1^{122}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 170, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 2 & 3 \\ 3 & 4 & 8 & 11 & 3 & 4 \\ 6 & 8 & 10 & 16 & 4 & 6 \\ 9 & 11 & 16 & 20 & 6 & 8 \\ 2 & 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[137] \quad 1^{124}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 170, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 2 & 3 \\ 3 & 4 & 4 & 15 & 3 & 4 \\ 4 & 4 & 1 & 11 & 2 & 3 \\ 12 & 15 & 11 & 39 & 8 & 11 \\ 2 & 3 & 2 & 8 & 0 & 2 \\ 3 & 4 & 3 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[139] \quad 1^{141}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 171, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 2 & 3 \\ 3 & 4 & 16 & 7 & 3 & 4 \\ 11 & 16 & 46 & 20 & 8 & 12 \\ 5 & 7 & 20 & 7 & 4 & 5 \\ 2 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 12 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[140] \quad 1^{132}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 173, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 9 & 2 & 3 \\ 3 & 4 & 12 & 11 & 3 & 4 \\ 10 & 12 & 25 & 24 & 6 & 9 \\ 9 & 11 & 24 & 20 & 6 & 8 \\ 2 & 3 & 6 & 6 & 0 & 2 \\ 3 & 4 & 9 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[141] \quad 1^{121}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 174, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 2 & 3 \\ 3 & 4 & 13 & 7 & 3 & 4 \\ 8 & 13 & 28 & 16 & 7 & 10 \\ 5 & 7 & 16 & 7 & 4 & 5 \\ 2 & 3 & 7 & 4 & 0 & 2 \\ 3 & 4 & 10 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[142] \quad 1^{113}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 175, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 17 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 13 & 17 & 0 & 55 & 10 & 13 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 3 & 4 & 0 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[143] \quad 1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 176, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 0 & 16 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 13 & 16 & 0 & 44 & 9 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 0 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[144] \quad 1^{112}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 180, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 2 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 \\ 3 & 4 & 1 & 8 & 2 & 3 \\ 8 & 11 & 8 & 20 & 6 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 \\ 3 & 4 & 3 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[145] \quad 21^{111}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 180, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 2 & 3 \\ 4 & 4 & 4 & 7 & 3 & 4 \\ 3 & 4 & 1 & 5 & 2 & 3 \\ 5 & 7 & 5 & 7 & 4 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 \\ 3 & 4 & 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[146] \quad 1^{112}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 181, \quad (n_+, n_-, n_0) =$$

$$(1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 2 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 \\ 4 & 4 & 1 & 11 & 2 & 3 \\ 13 & 16 & 11 & 44 & 9 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 \\ 3 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[147] \quad 1^{13}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 2 & 3 \\ 3 & 4 & 17 & 3 & 3 & 4 \\ 12 & 17 & 49 & 9 & 9 & 13 \\ 3 & 3 & 9 & 0 & 2 & 2 \\ 2 & 3 & 9 & 2 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[148] \quad 1^{133}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 9 & 2 & 3 \\ 3 & 4 & 12 & 11 & 3 & 4 \\ 9 & 12 & 25 & 24 & 6 & 9 \\ 9 & 11 & 24 & 20 & 6 & 8 \\ 2 & 3 & 6 & 6 & 0 & 2 \\ 3 & 4 & 9 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[149] \quad 1^{13}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 2 & 3 \\ 3 & 4 & 4 & 15 & 3 & 4 \\ 2 & 4 & 1 & 11 & 2 & 3 \\ 11 & 15 & 11 & 39 & 8 & 11 \\ 2 & 3 & 2 & 8 & 0 & 2 \\ 3 & 4 & 3 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[150] \quad 1^{122}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 2 & 3 \\ 3 & 4 & 12 & 11 & 3 & 4 \\ 8 & 12 & 25 & 24 & 6 & 9 \\ 8 & 11 & 24 & 20 & 6 & 8 \\ 2 & 3 & 6 & 6 & 0 & 2 \\ 3 & 4 & 9 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[151] \quad 1^{124}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 190, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 2 & 3 \\ 3 & 4 & 4 & 25 & 3 & 4 \\ 4 & 4 & 1 & 17 & 2 & 3 \\ 18 & 25 & 17 & 111 & 14 & 19 \\ 2 & 3 & 2 & 14 & 0 & 2 \\ 3 & 4 & 3 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[153] \quad 1^{122}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 8 & 2 & 3 \\ 3 & 4 & 21 & 11 & 3 & 4 \\ 13 & 21 & 76 & 41 & 11 & 16 \\ 8 & 11 & 41 & 20 & 6 & 8 \\ 2 & 3 & 11 & 6 & 0 & 2 \\ 3 & 4 & 16 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[152] \quad 1^{122}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 2 & 3 \\ 3 & 4 & 8 & 11 & 3 & 4 \\ 6 & 8 & 10 & 16 & 4 & 6 \\ 8 & 11 & 16 & 20 & 6 & 8 \\ 2 & 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[154] \quad 21^{111}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 5 & 2 & 3 \\ 4 & 4 & 9 & 7 & 3 & 4 \\ 6 & 9 & 13 & 11 & 5 & 7 \\ 5 & 7 & 11 & 7 & 4 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 \\ 3 & 4 & 7 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[155] \quad & 1^{142} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 199, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 11 & 6 & 2 & 3 \\ 3 & 4 & 13 & 7 & 3 & 4 \\ 11 & 13 & 28 & 16 & 7 & 10 \\ 6 & 7 & 16 & 7 & 4 & 5 \\ 2 & 3 & 7 & 4 & 0 & 2 \\ 3 & 4 & 10 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[156] \quad & 1^{12} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 200, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 0 & 16 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 11 & 16 & 0 & 44 & 9 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 0 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[157] \quad & 1^{132} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 202, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 6 & 2 & 3 \\ 3 & 4 & 17 & 7 & 3 & 4 \\ 12 & 17 & 49 & 21 & 9 & 13 \\ 6 & 7 & 21 & 7 & 4 & 5 \\ 2 & 3 & 9 & 4 & 0 & 2 \\ 3 & 4 & 13 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[158] \quad & 1^{121} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 202, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 6 & 2 & 3 \\ 3 & 4 & 13 & 7 & 3 & 4 \\ 10 & 13 & 28 & 16 & 7 & 10 \\ 6 & 7 & 16 & 7 & 4 & 5 \\ 2 & 3 & 7 & 4 & 0 & 2 \\ 3 & 4 & 10 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[159] \quad & 1^{121} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 202, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 6 & 2 & 3 \\ 3 & 4 & 18 & 7 & 3 & 4 \\ 14 & 18 & 58 & 22 & 10 & 14 \\ 6 & 7 & 22 & 7 & 4 & 5 \\ 2 & 3 & 10 & 4 & 0 & 2 \\ 3 & 4 & 14 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[160] \quad & 1^{121} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 204, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 5 & 2 & 3 \\ 3 & 4 & 18 & 7 & 3 & 4 \\ 14 & 18 & 58 & 22 & 10 & 14 \\ 5 & 7 & 22 & 7 & 4 & 5 \\ 2 & 3 & 10 & 4 & 0 & 2 \\ 3 & 4 & 14 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[161] \quad & 21^{121} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 204, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 6 & 5 & 2 & 3 \\ 4 & 4 & 8 & 7 & 3 & 4 \\ 6 & 8 & 10 & 10 & 4 & 6 \\ 5 & 7 & 10 & 7 & 4 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 \\ 3 & 4 & 6 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[162] \quad & 1^{124} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 204, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 7 & 14 & 2 & 3 \\ 3 & 4 & 8 & 19 & 3 & 4 \\ 7 & 8 & 10 & 28 & 4 & 6 \\ 14 & 19 & 28 & 64 & 10 & 14 \\ 2 & 3 & 4 & 10 & 0 & 2 \\ 3 & 4 & 6 & 14 & 2 & 3 \end{pmatrix}_{b_0} \\
[163] \quad & 1^{123} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 205, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 7 & 12 & 2 & 3 \\ 3 & 4 & 8 & 15 & 3 & 4 \\ 7 & 8 & 10 & 22 & 4 & 6 \\ 12 & 15 & 22 & 39 & 8 & 11 \\ 2 & 3 & 4 & 8 & 0 & 2 \\ 3 & 4 & 6 & 11 & 2 & 3 \end{pmatrix}_{b_0} \\
[166] & 1^{143}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 206, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 9 & 2 & 3 \\ 3 & 4 & 16 & 11 & 3 & 4 \\ 12 & 16 & 46 & 32 & 8 & 12 \\ 9 & 11 & 32 & 20 & 6 & 8 \\ 2 & 3 & 8 & 6 & 0 & 2 \\ 3 & 4 & 12 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[165] & 1^{123}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 206, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 0 & 16 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 12 & 16 & 0 & 44 & 9 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 0 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[164] & 1^{1223}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 206, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 11 & 9 & 2 & 3 \\ 3 & 4 & 13 & 11 & 3 & 4 \\ 11 & 13 & 28 & 25 & 7 & 10 \\ 9 & 11 & 25 & 20 & 6 & 8 \\ 2 & 3 & 7 & 6 & 0 & 2 \\ 3 & 4 & 10 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[167] & 1^{132}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 212, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 9 & 8 & 2 & 3 \\ 3 & 4 & 12 & 11 & 3 & 4 \\ 9 & 12 & 25 & 24 & 6 & 9 \\ 8 & 11 & 24 & 20 & 6 & 8 \\ 2 & 3 & 6 & 6 & 0 & 2 \\ 3 & 4 & 9 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[168] & 1^{1223}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 214, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 9 & 2 & 3 \\ 3 & 4 & 13 & 11 & 3 & 4 \\ 10 & 13 & 28 & 25 & 7 & 10 \\ 9 & 11 & 25 & 20 & 6 & 8 \\ 2 & 3 & 7 & 6 & 0 & 2 \\ 3 & 4 & 10 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[170] & 1^{121}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 216, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 5 & 2 & 3 \\ 3 & 4 & 18 & 7 & 3 & 4 \\ 12 & 18 & 58 & 22 & 10 & 14 \\ 5 & 7 & 22 & 7 & 4 & 5 \\ 2 & 3 & 10 & 4 & 0 & 2 \\ 3 & 4 & 14 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[169] & 1^{114}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 216, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 14 & 2 & 3 \\ 3 & 4 & 4 & 19 & 3 & 4 \\ 4 & 4 & 1 & 14 & 2 & 3 \\ 14 & 19 & 14 & 64 & 10 & 14 \\ 2 & 3 & 2 & 10 & 0 & 2 \\ 3 & 4 & 3 & 14 & 2 & 3 \end{pmatrix}_{b_0} \\
[172] & 1^{1123}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 217, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 15 & 2 & 3 \\ 3 & 4 & 4 & 21 & 3 & 4 \\ 2 & 4 & 1 & 14 & 2 & 3 \\ 15 & 21 & 14 & 80 & 12 & 16 \\ 2 & 3 & 2 & 12 & 0 & 2 \\ 3 & 4 & 3 & 16 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[171] \quad & 1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 217, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 11 & 2 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 \\ 2 & 4 & 1 & 11 & 2 & 3 \\ 11 & 16 & 11 & 44 & 9 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 \\ 3 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[173] \quad & 1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 219, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 11 & 2 & 3 \\ 3 & 4 & 8 & 15 & 3 & 4 \\ 5 & 8 & 10 & 22 & 4 & 6 \\ 11 & 15 & 22 & 39 & 8 & 11 \\ 2 & 3 & 4 & 8 & 0 & 2 \\ 3 & 4 & 6 & 11 & 2 & 3 \end{pmatrix}_{b_0} \\
[174] \quad & 1^{12}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 222, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 13 & 3 & 2 & 3 \\ 3 & 4 & 18 & 3 & 3 & 4 \\ 13 & 18 & 58 & 10 & 10 & 14 \\ 3 & 3 & 10 & 0 & 2 & 2 \\ 2 & 3 & 10 & 2 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[175] \quad & 1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 223, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 13 & 3 & 2 & 3 \\ 3 & 4 & 17 & 3 & 3 & 4 \\ 13 & 17 & 49 & 9 & 9 & 13 \\ 3 & 3 & 9 & 0 & 2 & 2 \\ 2 & 3 & 9 & 2 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[176] \quad & 1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 226, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 5 & 2 & 3 \\ 3 & 4 & 13 & 7 & 3 & 4 \\ 10 & 13 & 28 & 16 & 7 & 10 \\ 5 & 7 & 16 & 7 & 4 & 5 \\ 2 & 3 & 7 & 4 & 0 & 2 \\ 3 & 4 & 10 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[179] \quad & 1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 232, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 8 & 2 & 3 \\ 3 & 4 & 13 & 11 & 3 & 4 \\ 10 & 13 & 28 & 25 & 7 & 10 \\ 8 & 11 & 25 & 20 & 6 & 8 \\ 2 & 3 & 7 & 6 & 0 & 2 \\ 3 & 4 & 10 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[178] \quad & 1^{14}23^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 232, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 17 & 8 & 2 & 3 \\ 3 & 4 & 26 & 11 & 3 & 4 \\ 17 & 26 & 118 & 50 & 14 & 20 \\ 8 & 11 & 50 & 20 & 6 & 8 \\ 2 & 3 & 14 & 6 & 0 & 2 \\ 3 & 4 & 20 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[177] \quad & 1^{14}23^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 232, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 11 & 8 & 2 & 3 \\ 3 & 4 & 16 & 11 & 3 & 4 \\ 11 & 16 & 46 & 32 & 8 & 12 \\ 8 & 11 & 32 & 20 & 6 & 8 \\ 2 & 3 & 8 & 6 & 0 & 2 \\ 3 & 4 & 12 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[180] \quad & 1^{14}23^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 234, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 18 & 6 & 2 & 3 \\ 3 & 4 & 26 & 7 & 3 & 4 \\ 18 & 26 & 118 & 32 & 14 & 20 \\ 6 & 7 & 32 & 7 & 4 & 5 \\ 2 & 3 & 14 & 4 & 0 & 2 \\ 3 & 4 & 20 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[181] \quad 1^{14}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 236, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 19 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 14 & 19 & 0 & 64 & 10 & 14 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 3 & 4 & 0 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[183] \quad 1^{122}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 240, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 2 & 3 \\ 3 & 4 & 13 & 11 & 3 & 4 \\ 9 & 13 & 28 & 25 & 7 & 10 \\ 8 & 11 & 25 & 20 & 6 & 8 \\ 2 & 3 & 7 & 6 & 0 & 2 \\ 3 & 4 & 10 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[182] \quad 1^{142}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 240, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 8 & 2 & 3 \\ 3 & 4 & 16 & 11 & 3 & 4 \\ 12 & 16 & 46 & 32 & 8 & 12 \\ 8 & 11 & 32 & 20 & 6 & 8 \\ 2 & 3 & 8 & 6 & 0 & 2 \\ 3 & 4 & 12 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[184] \quad 1^{121}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 241, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 2 & 3 \\ 3 & 4 & 13 & 7 & 3 & 4 \\ 9 & 13 & 28 & 16 & 7 & 10 \\ 5 & 7 & 16 & 7 & 4 & 5 \\ 2 & 3 & 7 & 4 & 0 & 2 \\ 3 & 4 & 10 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[185] \quad 1^{113}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 243, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 2 & 3 \\ 3 & 4 & 4 & 20 & 3 & 4 \\ 4 & 4 & 1 & 14 & 2 & 3 \\ 15 & 20 & 14 & 69 & 11 & 15 \\ 2 & 3 & 2 & 11 & 0 & 2 \\ 3 & 4 & 3 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[186] \quad 21^{131}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 244, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 5 & 2 & 3 \\ 4 & 4 & 12 & 7 & 3 & 4 \\ 9 & 12 & 25 & 15 & 6 & 9 \\ 5 & 7 & 15 & 7 & 4 & 5 \\ 2 & 3 & 6 & 4 & 0 & 2 \\ 3 & 4 & 9 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[187] \quad 1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 249, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 2 & 3 \\ 3 & 4 & 8 & 20 & 3 & 4 \\ 5 & 8 & 10 & 28 & 4 & 6 \\ 14 & 20 & 28 & 69 & 11 & 15 \\ 2 & 3 & 4 & 11 & 0 & 2 \\ 3 & 4 & 6 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[190] \quad 1^{14}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 250, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 2 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 15 & 21 & 76 & 11 & 11 & 16 \\ 3 & 3 & 11 & 0 & 2 & 2 \\ 2 & 3 & 11 & 2 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[189] \quad 1^{14}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 250, \quad (n_+, n_-, n_0) =$$

$$(1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 2 & 3 \\ 3 & 4 & 26 & 3 & 3 & 4 \\ 17 & 26 & 118 & 14 & 14 & 20 \\ 3 & 3 & 14 & 0 & 2 & 2 \\ 2 & 3 & 14 & 2 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[188] \quad 1^{12}3^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 250, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 2 & 3 \\ 3 & 4 & 18 & 3 & 3 & 4 \\ 14 & 18 & 58 & 10 & 10 & 14 \\ 3 & 3 & 10 & 0 & 2 & 2 \\ 2 & 3 & 10 & 2 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[191] \quad 1^{121}3^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 251, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 2 & 3 \\ 3 & 4 & 18 & 7 & 3 & 4 \\ 13 & 18 & 58 & 22 & 10 & 14 \\ 5 & 7 & 22 & 7 & 4 & 5 \\ 2 & 3 & 10 & 4 & 0 & 2 \\ 3 & 4 & 14 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[192] \quad 1^{124}3^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 20 & 2 & 3 \\ 3 & 4 & 8 & 29 & 3 & 4 \\ 6 & 8 & 10 & 40 & 4 & 6 \\ 20 & 29 & 40 & 148 & 16 & 22 \\ 2 & 3 & 4 & 16 & 0 & 2 \\ 3 & 4 & 6 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[193] \quad 1^{113}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 2 & 3 \\ 3 & 4 & 4 & 15 & 3 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 \\ 12 & 15 & 11 & 39 & 8 & 11 \\ 2 & 3 & 2 & 8 & 0 & 2 \\ 3 & 4 & 3 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[194] \quad 21^{121}3^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 5 & 2 & 3 \\ 4 & 4 & 18 & 7 & 3 & 4 \\ 12 & 18 & 58 & 22 & 10 & 14 \\ 5 & 7 & 22 & 7 & 4 & 5 \\ 2 & 3 & 10 & 4 & 0 & 2 \\ 3 & 4 & 14 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[196] \quad 1^{13}3^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 2 & 3 \\ 3 & 4 & 22 & 3 & 3 & 4 \\ 15 & 22 & 85 & 12 & 12 & 17 \\ 3 & 3 & 12 & 0 & 2 & 2 \\ 2 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[195] \quad 1^{114}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 17 & 2 & 3 \\ 3 & 4 & 4 & 24 & 3 & 4 \\ 4 & 4 & 1 & 17 & 2 & 3 \\ 17 & 24 & 17 & 100 & 13 & 18 \\ 2 & 3 & 2 & 13 & 0 & 2 \\ 3 & 4 & 3 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[198] \quad 1^{112}3^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 2 & 3 \\ 3 & 4 & 4 & 21 & 3 & 4 \\ 2 & 4 & 1 & 14 & 2 & 3 \\ 14 & 21 & 14 & 80 & 12 & 16 \\ 2 & 3 & 2 & 12 & 0 & 2 \\ 3 & 4 & 3 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[197] $21^{123}3^{11}1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 21 & 3 & 4 \\ 5 & 4 & 4 & 21 & 3 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 \\ 21 & 21 & 14 & 80 & 12 & 16 \\ 3 & 3 & 2 & 12 & 0 & 2 \\ 4 & 4 & 3 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[199] $1^{114}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 2 & 3 \\ 3 & 4 & 4 & 19 & 3 & 4 \\ 2 & 4 & 1 & 14 & 2 & 3 \\ 14 & 19 & 14 & 64 & 10 & 14 \\ 2 & 3 & 2 & 10 & 0 & 2 \\ 3 & 4 & 3 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

[200] $1^{123}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 2 & 3 \\ 3 & 4 & 8 & 15 & 3 & 4 \\ 6 & 8 & 10 & 22 & 4 & 6 \\ 12 & 15 & 22 & 39 & 8 & 11 \\ 2 & 3 & 4 & 8 & 0 & 2 \\ 3 & 4 & 6 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

[201] $1^{132}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 8 & 2 & 3 \\ 3 & 4 & 17 & 11 & 3 & 4 \\ 11 & 17 & 49 & 33 & 9 & 13 \\ 8 & 11 & 33 & 20 & 6 & 8 \\ 2 & 3 & 9 & 6 & 0 & 2 \\ 3 & 4 & 13 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[202] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 2 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 \\ 13 & 16 & 11 & 44 & 9 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 \\ 3 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[203] $1^{131}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 270, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 6 & 2 & 3 \\ 3 & 4 & 22 & 7 & 3 & 4 \\ 16 & 22 & 85 & 27 & 12 & 17 \\ 6 & 7 & 27 & 7 & 4 & 5 \\ 2 & 3 & 12 & 4 & 0 & 2 \\ 3 & 4 & 17 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[204] $1^{113}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 2 & 3 \\ 3 & 4 & 4 & 20 & 3 & 4 \\ 2 & 4 & 1 & 14 & 2 & 3 \\ 15 & 20 & 14 & 69 & 11 & 15 \\ 2 & 3 & 2 & 11 & 0 & 2 \\ 3 & 4 & 3 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[205] $1^{124}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 20 & 2 & 3 \\ 3 & 4 & 8 & 29 & 3 & 4 \\ 5 & 8 & 10 & 40 & 4 & 6 \\ 20 & 29 & 40 & 148 & 16 & 22 \\ 2 & 3 & 4 & 16 & 0 & 2 \\ 3 & 4 & 6 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[206] $1^{142}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 278, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 9 & 2 & 3 \\ 3 & 4 & 21 & 11 & 3 & 4 \\ 15 & 21 & 76 & 41 & 11 & 16 \\ 9 & 11 & 41 & 20 & 6 & 8 \\ 2 & 3 & 11 & 6 & 0 & 2 \\ 3 & 4 & 16 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[207] \quad 1^{13}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 280, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 2 & 3 \\ 3 & 4 & 4 & 15 & 3 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 \\ 11 & 15 & 11 & 39 & 8 & 11 \\ 2 & 3 & 2 & 8 & 0 & 2 \\ 3 & 4 & 3 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[208] \quad 1^{23}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 284, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 2 & 3 \\ 3 & 4 & 8 & 20 & 3 & 4 \\ 6 & 8 & 10 & 28 & 4 & 6 \\ 14 & 20 & 28 & 69 & 11 & 15 \\ 2 & 3 & 4 & 11 & 0 & 2 \\ 3 & 4 & 6 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[209] \quad 1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 285, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 2 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 \\ 11 & 16 & 11 & 44 & 9 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 \\ 3 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[211] \quad 1^{42}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 287, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 6 & 2 & 3 \\ 3 & 4 & 21 & 7 & 3 & 4 \\ 15 & 21 & 76 & 26 & 11 & 16 \\ 6 & 7 & 26 & 7 & 4 & 5 \\ 2 & 3 & 11 & 4 & 0 & 2 \\ 3 & 4 & 16 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[210] \quad 1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 287, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 2 & 3 \\ 3 & 4 & 8 & 20 & 3 & 4 \\ 5 & 8 & 10 & 28 & 4 & 6 \\ 13 & 20 & 28 & 69 & 11 & 15 \\ 2 & 3 & 4 & 11 & 0 & 2 \\ 3 & 4 & 6 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[212] \quad 1^{23}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 292, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 2 & 3 \\ 3 & 4 & 8 & 15 & 3 & 4 \\ 6 & 8 & 10 & 22 & 4 & 6 \\ 11 & 15 & 22 & 39 & 8 & 11 \\ 2 & 3 & 4 & 8 & 0 & 2 \\ 3 & 4 & 6 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[213] \quad 1^{42}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 295, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 6 & 2 & 3 \\ 3 & 4 & 17 & 7 & 3 & 4 \\ 13 & 17 & 49 & 21 & 9 & 13 \\ 6 & 7 & 21 & 7 & 4 & 5 \\ 2 & 3 & 9 & 4 & 0 & 2 \\ 3 & 4 & 13 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[214] \quad 21^{12}3^{11}1^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 298, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 21 & 6 & 3 & 4 \\ 5 & 4 & 22 & 7 & 3 & 4 \\ 21 & 22 & 85 & 27 & 12 & 17 \\ 6 & 7 & 27 & 7 & 4 & 5 \\ 3 & 3 & 12 & 4 & 0 & 2 \\ 4 & 4 & 17 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[215] \quad 21^{141}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 300, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 5 & 2 & 3 \\ 4 & 4 & 16 & 7 & 3 & 4 \\ 12 & 16 & 46 & 20 & 8 & 12 \\ 5 & 7 & 20 & 7 & 4 & 5 \\ 2 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 12 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[216] \quad 1^{124}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 300, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 2 & 3 \\ 3 & 4 & 8 & 19 & 3 & 4 \\ 5 & 8 & 10 & 28 & 4 & 6 \\ 14 & 19 & 28 & 64 & 10 & 14 \\ 2 & 3 & 4 & 10 & 0 & 2 \\ 3 & 4 & 6 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[217] \quad 1^{1123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 300, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 2 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 \\ 12 & 16 & 11 & 44 & 9 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 \\ 3 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[218] \quad 1^{123}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 16 & 21 & 0 & 80 & 12 & 16 \\ 2 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 0 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[219] \quad 21^{1123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 8 & 2 & 3 \\ 4 & 4 & 4 & 11 & 3 & 4 \\ 3 & 4 & 1 & 8 & 2 & 3 \\ 8 & 11 & 8 & 20 & 6 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 \\ 3 & 4 & 3 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[220] \quad 1^{13}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 305, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 0 & 20 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 14 & 20 & 0 & 69 & 11 & 15 \\ 2 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & 0 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[221] \quad 1^{131}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 306, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 2 & 3 \\ 3 & 4 & 17 & 7 & 3 & 4 \\ 11 & 17 & 49 & 21 & 9 & 13 \\ 5 & 7 & 21 & 7 & 4 & 5 \\ 2 & 3 & 9 & 4 & 0 & 2 \\ 3 & 4 & 13 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[222] \quad 1^{133}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 20 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 15 & 20 & 0 & 69 & 11 & 15 \\ 2 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & 0 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[223] \quad 1^{123}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 15 & 21 & 0 & 80 & 12 & 16 \\ 2 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 0 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[224] \quad 21^{1123}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 320, (n_+, n_-, n_0) =$$

$$(1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 14 & 2 & 3 \\ 4 & 4 & 4 & 21 & 3 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 \\ 14 & 21 & 14 & 80 & 12 & 16 \\ 2 & 3 & 2 & 12 & 0 & 2 \\ 3 & 4 & 3 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[225] \ 1^{141}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 321, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 2 & 3 \\ 3 & 4 & 21 & 7 & 3 & 4 \\ 13 & 21 & 76 & 26 & 11 & 16 \\ 5 & 7 & 26 & 7 & 4 & 5 \\ 2 & 3 & 11 & 4 & 0 & 2 \\ 3 & 4 & 16 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[226] \ 1^{124}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 322, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 15 & 2 & 3 \\ 3 & 4 & 8 & 20 & 3 & 4 \\ 6 & 8 & 10 & 28 & 4 & 6 \\ 15 & 20 & 28 & 69 & 11 & 15 \\ 2 & 3 & 4 & 11 & 0 & 2 \\ 3 & 4 & 6 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[227] \ 21^{122}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 8 & 2 & 3 \\ 4 & 4 & 8 & 11 & 3 & 4 \\ 6 & 8 & 10 & 16 & 4 & 6 \\ 8 & 11 & 16 & 20 & 6 & 8 \\ 2 & 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[229] \ 1^{143}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 9 & 2 & 3 \\ 3 & 4 & 17 & 11 & 3 & 4 \\ 13 & 17 & 49 & 33 & 9 & 13 \\ 9 & 11 & 33 & 20 & 6 & 8 \\ 2 & 3 & 9 & 6 & 0 & 2 \\ 3 & 4 & 13 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[228] \ 1^{131}3^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 14 & 5 & 2 & 3 \\ 3 & 4 & 22 & 7 & 3 & 4 \\ 14 & 22 & 85 & 27 & 12 & 17 \\ 5 & 7 & 27 & 7 & 4 & 5 \\ 2 & 3 & 12 & 4 & 0 & 2 \\ 3 & 4 & 17 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[230] \ 1^{131}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 2 & 3 \\ 3 & 4 & 17 & 7 & 3 & 4 \\ 13 & 17 & 49 & 21 & 9 & 13 \\ 5 & 7 & 21 & 7 & 4 & 5 \\ 2 & 3 & 9 & 4 & 0 & 2 \\ 3 & 4 & 13 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[231] \ 1^{124}3^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 19 & 2 & 3 \\ 3 & 4 & 8 & 29 & 3 & 4 \\ 5 & 8 & 10 & 40 & 4 & 6 \\ 19 & 29 & 40 & 148 & 16 & 22 \\ 2 & 3 & 4 & 16 & 0 & 2 \\ 3 & 4 & 6 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[232] \ 1^{113}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 2 & 3 \\ 3 & 4 & 4 & 20 & 3 & 4 \\ 2 & 4 & 1 & 14 & 2 & 3 \\ 14 & 20 & 14 & 69 & 11 & 15 \\ 2 & 3 & 2 & 11 & 0 & 2 \\ 3 & 4 & 3 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[233] \quad & 1^{141} 1^{13} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 337, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 16 & 3 & 2 & 3 \\ 3 & 4 & 22 & 3 & 3 & 4 \\ 16 & 22 & 85 & 12 & 12 & 17 \\ 3 & 3 & 12 & 0 & 2 & 2 \\ 2 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[234] \quad & 21^{124} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 352, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 6 & 20 & 2 & 3 \\ 4 & 4 & 8 & 29 & 3 & 4 \\ 6 & 8 & 10 & 40 & 4 & 6 \\ 20 & 29 & 40 & 148 & 16 & 22 \\ 2 & 3 & 4 & 16 & 0 & 2 \\ 3 & 4 & 6 & 22 & 2 & 3 \end{pmatrix}_{b_0} \\
[235] \quad & 1^{13} 1^{12} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 353, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 15 & 2 & 3 \\ 3 & 4 & 4 & 21 & 3 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 \\ 15 & 21 & 14 & 80 & 12 & 16 \\ 2 & 3 & 2 & 12 & 0 & 2 \\ 3 & 4 & 3 & 16 & 2 & 3 \end{pmatrix}_{b_0} \\
[236] \quad & 1^{12} 1^{13} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 355, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 13 & 2 & 3 \\ 3 & 4 & 4 & 20 & 3 & 4 \\ 2 & 4 & 1 & 14 & 2 & 3 \\ 13 & 20 & 14 & 69 & 11 & 15 \\ 2 & 3 & 2 & 11 & 0 & 2 \\ 3 & 4 & 3 & 15 & 2 & 3 \end{pmatrix}_{b_0} \\
[237] \quad & 1^{14} 1^{12} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 356, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 16 & 2 & 3 \\ 3 & 4 & 4 & 21 & 3 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 \\ 16 & 21 & 14 & 80 & 12 & 16 \\ 2 & 3 & 2 & 12 & 0 & 2 \\ 3 & 4 & 3 & 16 & 2 & 3 \end{pmatrix}_{b_0} \\
[238] \quad & 1^{31} 1^{13} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 361, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 5 & 2 & 3 \\ 3 & 4 & 17 & 7 & 3 & 4 \\ 12 & 17 & 49 & 21 & 9 & 13 \\ 5 & 7 & 21 & 7 & 4 & 5 \\ 2 & 3 & 9 & 4 & 0 & 2 \\ 3 & 4 & 13 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[239] \quad & 1^{42} 1^{13} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 364, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 13 & 8 & 2 & 3 \\ 3 & 4 & 17 & 11 & 3 & 4 \\ 13 & 17 & 49 & 33 & 9 & 13 \\ 8 & 11 & 33 & 20 & 6 & 8 \\ 2 & 3 & 9 & 6 & 0 & 2 \\ 3 & 4 & 13 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[240] \quad & 21^{132} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 368, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 9 & 8 & 2 & 3 \\ 4 & 4 & 12 & 11 & 3 & 4 \\ 9 & 12 & 25 & 24 & 6 & 9 \\ 8 & 11 & 24 & 20 & 6 & 8 \\ 2 & 3 & 6 & 6 & 0 & 2 \\ 3 & 4 & 9 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[241] \quad & 21^{121} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 370, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 9 & 5 & 2 & 3 \\ 4 & 4 & 13 & 7 & 3 & 4 \\ 9 & 13 & 28 & 16 & 7 & 10 \\ 5 & 7 & 16 & 7 & 4 & 5 \\ 2 & 3 & 7 & 4 & 0 & 2 \\ 3 & 4 & 10 & 5 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[242] \quad 21^{151}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 372, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.1$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 15 & 5 & 2 & 3 \\ 4 & 4 & 20 & 7 & 3 & 4 \\ 15 & 20 & 73 & 25 & 10 & 15 \\ 5 & 7 & 25 & 7 & 4 & 5 \\ 2 & 3 & 10 & 4 & 0 & 2 \\ 3 & 4 & 15 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[243] \quad 1^{14}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 374, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 2 & 3 \\ 3 & 4 & 26 & 3 & 3 & 4 \\ 18 & 26 & 118 & 14 & 14 & 20 \\ 3 & 3 & 14 & 0 & 2 & 2 \\ 2 & 3 & 14 & 2 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[244] \quad 1^{123}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 374, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 16 & 2 & 3 \\ 3 & 4 & 8 & 24 & 3 & 4 \\ 6 & 8 & 10 & 34 & 4 & 6 \\ 16 & 24 & 34 & 100 & 13 & 18 \\ 2 & 3 & 4 & 13 & 0 & 2 \\ 3 & 4 & 6 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[245] \quad 21^{12}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 378, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 24 & 6 & 3 & 4 \\ 5 & 4 & 26 & 7 & 3 & 4 \\ 24 & 26 & 118 & 32 & 14 & 20 \\ 6 & 7 & 32 & 7 & 4 & 5 \\ 3 & 3 & 14 & 4 & 0 & 2 \\ 4 & 4 & 20 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[246] \quad 1^{14}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 378, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 2 & 3 \\ 3 & 4 & 4 & 25 & 3 & 4 \\ 2 & 4 & 1 & 17 & 2 & 3 \\ 18 & 25 & 17 & 111 & 14 & 19 \\ 2 & 3 & 2 & 14 & 0 & 2 \\ 3 & 4 & 3 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[247] \quad 21^{123}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 378, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 24 & 3 & 4 \\ 5 & 4 & 4 & 25 & 3 & 4 \\ 3 & 4 & 1 & 17 & 2 & 3 \\ 24 & 25 & 17 & 111 & 14 & 19 \\ 3 & 3 & 2 & 14 & 0 & 2 \\ 4 & 4 & 3 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[248] \quad 21^{122}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 384, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 9 & 8 & 2 & 3 \\ 4 & 4 & 13 & 11 & 3 & 4 \\ 9 & 13 & 28 & 25 & 7 & 10 \\ 8 & 11 & 25 & 20 & 6 & 8 \\ 2 & 3 & 7 & 6 & 0 & 2 \\ 3 & 4 & 10 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[249] \quad 1^{132}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 388, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 8 & 2 & 3 \\ 3 & 4 & 17 & 11 & 3 & 4 \\ 12 & 17 & 49 & 33 & 9 & 13 \\ 8 & 11 & 33 & 20 & 6 & 8 \\ 2 & 3 & 9 & 6 & 0 & 2 \\ 3 & 4 & 13 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[250] \quad 1^{14}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 390, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 0 & 24 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 16 & 24 & 0 & 100 & 13 & 18 \\ 2 & 3 & 0 & 13 & 0 & 2 \\ 3 & 4 & 0 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[251] \quad & 1^{14} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 396, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 17 & 2 & 3 \\ 3 & 4 & 8 & 24 & 3 & 4 \\ 5 & 8 & 10 & 34 & 4 & 6 \\ 17 & 24 & 34 & 100 & 13 & 18 \\ 2 & 3 & 4 & 13 & 0 & 2 \\ 3 & 4 & 6 & 18 & 2 & 3 \end{pmatrix}_{b_0} \\
[252] \quad & 1^{14} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 396, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 16 & 5 & 2 & 3 \\ 3 & 4 & 26 & 7 & 3 & 4 \\ 16 & 26 & 118 & 32 & 14 & 20 \\ 5 & 7 & 32 & 7 & 4 & 5 \\ 2 & 3 & 14 & 4 & 0 & 2 \\ 3 & 4 & 20 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[253] \quad & 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 404, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 13 & 2 & 3 \\ 3 & 4 & 4 & 20 & 3 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 \\ 13 & 20 & 14 & 69 & 11 & 15 \\ 2 & 3 & 2 & 11 & 0 & 2 \\ 3 & 4 & 3 & 15 & 2 & 3 \end{pmatrix}_{b_0} \\
[254] \quad & 1^{14} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 404, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 14 & 2 & 3 \\ 3 & 4 & 4 & 19 & 3 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 \\ 14 & 19 & 14 & 64 & 10 & 14 \\ 2 & 3 & 2 & 10 & 0 & 2 \\ 3 & 4 & 3 & 14 & 2 & 3 \end{pmatrix}_{b_0} \\
[255] \quad & 21^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 412, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 11 & 2 & 3 \\ 4 & 4 & 4 & 16 & 3 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 \\ 11 & 16 & 11 & 44 & 9 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 \\ 3 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[256] \quad & 1^{24} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 416, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 14 & 2 & 3 \\ 3 & 4 & 8 & 19 & 3 & 4 \\ 6 & 8 & 10 & 28 & 4 & 6 \\ 14 & 19 & 28 & 64 & 10 & 14 \\ 2 & 3 & 4 & 10 & 0 & 2 \\ 3 & 4 & 6 & 14 & 2 & 3 \end{pmatrix}_{b_0} \\
[257] \quad & 1^{14} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 424, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 24 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 17 & 24 & 0 & 100 & 13 & 18 \\ 2 & 3 & 0 & 13 & 0 & 2 \\ 3 & 4 & 0 & 18 & 2 & 3 \end{pmatrix}_{b_0} \\
[258] \quad & 21^{14} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 424, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 12 & 8 & 2 & 3 \\ 4 & 4 & 16 & 11 & 3 & 4 \\ 12 & 16 & 46 & 32 & 8 & 12 \\ 8 & 11 & 32 & 20 & 6 & 8 \\ 2 & 3 & 8 & 6 & 0 & 2 \\ 3 & 4 & 12 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[259] \quad & 1^{13} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 431, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 0 & 25 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 17 & 25 & 0 & 111 & 14 & 19 \\ 2 & 3 & 0 & 14 & 0 & 2 \\ 3 & 4 & 0 & 19 & 2 & 3 \end{pmatrix}_{b_0} \\
[260] \quad & 21^{123}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 432, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.86 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 3 & 27 & 3 & 4 \\ 5 & 4 & 4 & 29 & 3 & 4 \\ 3 & 4 & 1 & 20 & 2 & 3 \\ 27 & 29 & 20 & 148 & 16 & 22 \\ 3 & 3 & 2 & 16 & 0 & 2 \\ 4 & 4 & 3 & 22 & 2 & 3 \end{pmatrix}_{b_0} \\
[261] \quad & 1^{142}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 432, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 18 & 8 & 2 & 3 \\ 3 & 4 & 26 & 11 & 3 & 4 \\ 18 & 26 & 118 & 50 & 14 & 20 \\ 8 & 11 & 50 & 20 & 6 & 8 \\ 2 & 3 & 14 & 6 & 0 & 2 \\ 3 & 4 & 20 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[262] \quad & 1^{131}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 436, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 15 & 5 & 2 & 3 \\ 3 & 4 & 22 & 7 & 3 & 4 \\ 15 & 22 & 85 & 27 & 12 & 17 \\ 5 & 7 & 27 & 7 & 4 & 5 \\ 2 & 3 & 12 & 4 & 0 & 2 \\ 3 & 4 & 17 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[263] \quad & 1^{113}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 439, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 17 & 2 & 3 \\ 3 & 4 & 4 & 25 & 3 & 4 \\ 2 & 4 & 1 & 17 & 2 & 3 \\ 17 & 25 & 17 & 111 & 14 & 19 \\ 2 & 3 & 2 & 14 & 0 & 2 \\ 3 & 4 & 3 & 19 & 2 & 3 \end{pmatrix}_{b_0} \\
[264] \quad & 1^{142}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 440, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 8 & 2 & 3 \\ 3 & 4 & 21 & 11 & 3 & 4 \\ 14 & 21 & 76 & 41 & 11 & 16 \\ 8 & 11 & 41 & 20 & 6 & 8 \\ 2 & 3 & 11 & 6 & 0 & 2 \\ 3 & 4 & 16 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[265] \quad & 1^{141}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 446, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 14 & 5 & 2 & 3 \\ 3 & 4 & 21 & 7 & 3 & 4 \\ 14 & 21 & 76 & 26 & 11 & 16 \\ 5 & 7 & 26 & 7 & 4 & 5 \\ 2 & 3 & 11 & 4 & 0 & 2 \\ 3 & 4 & 16 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[266] \quad & 21^{123}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 450, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.88 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 6 & 14 & 2 & 3 \\ 4 & 4 & 8 & 20 & 3 & 4 \\ 6 & 8 & 10 & 28 & 4 & 6 \\ 14 & 20 & 28 & 69 & 11 & 15 \\ 2 & 3 & 4 & 11 & 0 & 2 \\ 3 & 4 & 6 & 15 & 2 & 3 \end{pmatrix}_{b_0} \\
[267] \quad & 1^{113}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 454, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.00 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 16 & 2 & 3 \\ 3 & 4 & 4 & 25 & 3 & 4 \\ 2 & 4 & 1 & 17 & 2 & 3 \\ 16 & 25 & 17 & 111 & 14 & 19 \\ 2 & 3 & 2 & 14 & 0 & 2 \\ 3 & 4 & 3 & 19 & 2 & 3 \end{pmatrix}_{b_0} \\
[268] \quad & 1^{134}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 457, \quad (n_+, n_-, n_0) = \\
& (1, 8, 0), \quad c_{\text{ext}} = 6.0
\end{aligned}$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 0 & 25 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 18 & 25 & 0 & 111 & 14 & 19 \\ 2 & 3 & 0 & 14 & 0 & 2 \\ 3 & 4 & 0 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[269] \quad 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 458, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 16 & 2 & 3 \\ 3 & 4 & 8 & 24 & 3 & 4 \\ 5 & 8 & 10 & 34 & 4 & 6 \\ 16 & 24 & 34 & 100 & 13 & 18 \\ 2 & 3 & 4 & 13 & 0 & 2 \\ 3 & 4 & 6 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[270] \quad 1^{41} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 459, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 16 & 5 & 2 & 3 \\ 3 & 4 & 22 & 7 & 3 & 4 \\ 16 & 22 & 85 & 27 & 12 & 17 \\ 5 & 7 & 27 & 7 & 4 & 5 \\ 2 & 3 & 12 & 4 & 0 & 2 \\ 3 & 4 & 17 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[271] \quad 21^{161} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 460, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 18 & 5 & 2 & 3 \\ 4 & 4 & 24 & 7 & 3 & 4 \\ 18 & 24 & 106 & 30 & 12 & 18 \\ 5 & 7 & 30 & 7 & 4 & 5 \\ 2 & 3 & 12 & 4 & 0 & 2 \\ 3 & 4 & 18 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[272] \quad 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 461, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 2 & 3 \\ 3 & 4 & 4 & 20 & 3 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 \\ 14 & 20 & 14 & 69 & 11 & 15 \\ 2 & 3 & 2 & 11 & 0 & 2 \\ 3 & 4 & 3 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[273] \quad 21^{113} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 468, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 11 & 2 & 3 \\ 4 & 4 & 4 & 15 & 3 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 \\ 11 & 15 & 11 & 39 & 8 & 11 \\ 2 & 3 & 2 & 8 & 0 & 2 \\ 3 & 4 & 3 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[274] \quad 1^{14} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 469, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 2 & 3 \\ 3 & 4 & 4 & 24 & 3 & 4 \\ 2 & 4 & 1 & 17 & 2 & 3 \\ 17 & 24 & 17 & 100 & 13 & 18 \\ 2 & 3 & 2 & 13 & 0 & 2 \\ 3 & 4 & 3 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[275] \quad 1^{14} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 2 & 3 \\ 3 & 4 & 4 & 20 & 3 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 \\ 15 & 20 & 14 & 69 & 11 & 15 \\ 2 & 3 & 2 & 11 & 0 & 2 \\ 3 & 4 & 3 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[276] \quad 1^{24} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 484, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 17 & 2 & 3 \\ 3 & 4 & 8 & 24 & 3 & 4 \\ 6 & 8 & 10 & 34 & 4 & 6 \\ 17 & 24 & 34 & 100 & 13 & 18 \\ 2 & 3 & 4 & 13 & 0 & 2 \\ 3 & 4 & 6 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[277] \quad 1^{41} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 489, (n_+, n_-, n_0) =$$

$$(1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 5 & 2 & 3 \\ 3 & 4 & 21 & 7 & 3 & 4 \\ 15 & 21 & 76 & 26 & 11 & 16 \\ 5 & 7 & 26 & 7 & 4 & 5 \\ 2 & 3 & 11 & 4 & 0 & 2 \\ 3 & 4 & 16 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[278] \ 21^{123}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 492, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 6 & 11 & 2 & 3 \\ 4 & 4 & 8 & 15 & 3 & 4 \\ 6 & 8 & 10 & 22 & 4 & 6 \\ 11 & 15 & 22 & 39 & 8 & 11 \\ 2 & 3 & 4 & 8 & 0 & 2 \\ 3 & 4 & 6 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[279] \ 1^{13}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 502, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 2 & 3 \\ 3 & 4 & 4 & 24 & 3 & 4 \\ 2 & 4 & 1 & 17 & 2 & 3 \\ 16 & 24 & 17 & 100 & 13 & 18 \\ 2 & 3 & 2 & 13 & 0 & 2 \\ 3 & 4 & 3 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[280] \ 1^{142}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 15 & 8 & 2 & 3 \\ 3 & 4 & 21 & 11 & 3 & 4 \\ 15 & 21 & 76 & 41 & 11 & 16 \\ 8 & 11 & 41 & 20 & 6 & 8 \\ 2 & 3 & 11 & 6 & 0 & 2 \\ 3 & 4 & 16 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[281] \ 1^{13}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 556, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 2 & 3 \\ 3 & 4 & 4 & 25 & 3 & 4 \\ 3 & 4 & 1 & 17 & 2 & 3 \\ 17 & 25 & 17 & 111 & 14 & 19 \\ 2 & 3 & 2 & 14 & 0 & 2 \\ 3 & 4 & 3 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[282] \ 21^{131}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 562, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 12 & 5 & 2 & 3 \\ 4 & 4 & 17 & 7 & 3 & 4 \\ 12 & 17 & 49 & 21 & 9 & 13 \\ 5 & 7 & 21 & 7 & 4 & 5 \\ 2 & 3 & 9 & 4 & 0 & 2 \\ 3 & 4 & 13 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[283] \ 21^{171}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 21 & 5 & 2 & 3 \\ 4 & 4 & 28 & 7 & 3 & 4 \\ 21 & 28 & 145 & 35 & 14 & 21 \\ 5 & 7 & 35 & 7 & 4 & 5 \\ 2 & 3 & 14 & 4 & 0 & 2 \\ 3 & 4 & 21 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[284] \ 1^{141}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 571, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 17 & 5 & 2 & 3 \\ 3 & 4 & 26 & 7 & 3 & 4 \\ 17 & 26 & 118 & 32 & 14 & 20 \\ 5 & 7 & 32 & 7 & 4 & 5 \\ 2 & 3 & 14 & 4 & 0 & 2 \\ 3 & 4 & 20 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[285] \ 1^{14}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 0 & 29 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 20 & 29 & 0 & 148 & 16 & 22 \\ 2 & 3 & 0 & 16 & 0 & 2 \\ 3 & 4 & 0 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[286] $1^{14} 1^{13} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 601, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 2 & 3 \\ 3 & 4 & 4 & 25 & 3 & 4 \\ 3 & 4 & 1 & 17 & 2 & 3 \\ 18 & 25 & 17 & 111 & 14 & 19 \\ 2 & 3 & 2 & 14 & 0 & 2 \\ 3 & 4 & 3 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[287] $21^{13} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 604, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$
 $T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 15 & 5 & 2 & 3 \\ 4 & 4 & 22 & 7 & 3 & 4 \\ 15 & 22 & 85 & 27 & 12 & 17 \\ 5 & 7 & 27 & 7 & 4 & 5 \\ 2 & 3 & 12 & 4 & 0 & 2 \\ 3 & 4 & 17 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[288] $1^{13} 1^{14} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 608, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 2 & 3 \\ 3 & 4 & 4 & 24 & 3 & 4 \\ 3 & 4 & 1 & 17 & 2 & 3 \\ 16 & 24 & 17 & 100 & 13 & 18 \\ 2 & 3 & 2 & 13 & 0 & 2 \\ 3 & 4 & 3 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[289] $21^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 616, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 12 & 8 & 2 & 3 \\ 4 & 4 & 17 & 11 & 3 & 4 \\ 12 & 17 & 49 & 33 & 9 & 13 \\ 8 & 11 & 33 & 20 & 6 & 8 \\ 2 & 3 & 9 & 6 & 0 & 2 \\ 3 & 4 & 13 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[290] $1^{14} 1^{13} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 628, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 2 & 3 \\ 3 & 4 & 4 & 29 & 3 & 4 \\ 2 & 4 & 1 & 20 & 2 & 3 \\ 20 & 29 & 20 & 148 & 16 & 22 \\ 2 & 3 & 2 & 16 & 0 & 2 \\ 3 & 4 & 3 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[291] $21^{14} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 648, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$
 $T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 18 & 8 & 2 & 3 \\ 4 & 4 & 26 & 11 & 3 & 4 \\ 18 & 26 & 118 & 50 & 14 & 20 \\ 8 & 11 & 50 & 20 & 6 & 8 \\ 2 & 3 & 14 & 6 & 0 & 2 \\ 3 & 4 & 20 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[292] $1^{13} 1^{14} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 657, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 2 & 3 \\ 3 & 4 & 4 & 29 & 3 & 4 \\ 2 & 4 & 1 & 20 & 2 & 3 \\ 19 & 29 & 20 & 148 & 16 & 22 \\ 2 & 3 & 2 & 16 & 0 & 2 \\ 3 & 4 & 3 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[293] $1^{14} 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 661, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 2 & 3 \\ 3 & 4 & 4 & 24 & 3 & 4 \\ 3 & 4 & 1 & 17 & 2 & 3 \\ 17 & 24 & 17 & 100 & 13 & 18 \\ 2 & 3 & 2 & 13 & 0 & 2 \\ 3 & 4 & 3 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[294] $1^{41} 1^{14} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -59, \quad \det = 664, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 18 & 5 & 2 & 3 \\ 3 & 4 & 26 & 7 & 3 & 4 \\ 18 & 26 & 118 & 32 & 14 & 20 \\ 5 & 7 & 32 & 7 & 4 & 5 \\ 2 & 3 & 14 & 4 & 0 & 2 \\ 3 & 4 & 20 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[295] $21^{114}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 3 & 14 & 2 & 3 \\ 4 & 4 & 4 & 19 & 3 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 \\ 14 & 19 & 14 & 64 & 10 & 14 \\ 2 & 3 & 2 & 10 & 0 & 2 \\ 3 & 4 & 3 & 14 & 2 & 3 \end{pmatrix}_{b_0}$

[296] $21^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 696, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 3 & 14 & 2 & 3 \\ 4 & 4 & 4 & 20 & 3 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 \\ 14 & 20 & 14 & 69 & 11 & 15 \\ 2 & 3 & 2 & 11 & 0 & 2 \\ 3 & 4 & 3 & 15 & 2 & 3 \end{pmatrix}_{b_0}$

[297] $21^{124}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 696, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 6 & 14 & 2 & 3 \\ 4 & 4 & 8 & 19 & 3 & 4 \\ 6 & 8 & 10 & 28 & 4 & 6 \\ 14 & 19 & 28 & 64 & 10 & 14 \\ 2 & 3 & 4 & 10 & 0 & 2 \\ 3 & 4 & 6 & 14 & 2 & 3 \end{pmatrix}_{b_0}$

[298] $21^{113}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 3 & 17 & 2 & 3 \\ 4 & 4 & 4 & 25 & 3 & 4 \\ 3 & 4 & 1 & 17 & 2 & 3 \\ 17 & 25 & 17 & 111 & 14 & 19 \\ 2 & 3 & 2 & 14 & 0 & 2 \\ 3 & 4 & 3 & 19 & 2 & 3 \end{pmatrix}_{b_0}$

[299] $1^{114}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 745, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 19 & 2 & 3 \\ 3 & 4 & 4 & 29 & 3 & 4 \\ 3 & 4 & 1 & 20 & 2 & 3 \\ 19 & 29 & 20 & 148 & 16 & 22 \\ 2 & 3 & 2 & 16 & 0 & 2 \\ 3 & 4 & 3 & 22 & 2 & 3 \end{pmatrix}_{b_0}$

[300] $21^{141}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 770, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 15 & 5 & 2 & 3 \\ 4 & 4 & 21 & 7 & 3 & 4 \\ 15 & 21 & 76 & 26 & 11 & 16 \\ 5 & 7 & 26 & 7 & 4 & 5 \\ 2 & 3 & 11 & 4 & 0 & 2 \\ 3 & 4 & 16 & 5 & 2 & 3 \end{pmatrix}_{b_0}$

[301] $21^{124}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 774, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 6 & 17 & 2 & 3 \\ 4 & 4 & 8 & 24 & 3 & 4 \\ 6 & 8 & 10 & 34 & 4 & 6 \\ 17 & 24 & 34 & 100 & 13 & 18 \\ 2 & 3 & 4 & 13 & 0 & 2 \\ 3 & 4 & 6 & 18 & 2 & 3 \end{pmatrix}_{b_0}$

[302] $1^{114}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 832, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 20 & 2 & 3 \\ 3 & 4 & 4 & 29 & 3 & 4 \\ 3 & 4 & 1 & 20 & 2 & 3 \\ 20 & 29 & 20 & 148 & 16 & 22 \\ 2 & 3 & 2 & 16 & 0 & 2 \\ 3 & 4 & 3 & 22 & 2 & 3 \end{pmatrix}_{b_0}$

[303] $21^{142}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 864, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 15 & 8 & 2 & 3 \\ 4 & 4 & 21 & 11 & 3 & 4 \\ 15 & 21 & 76 & 41 & 11 & 16 \\ 8 & 11 & 41 & 20 & 6 & 8 \\ 2 & 3 & 11 & 6 & 0 & 2 \\ 3 & 4 & 16 & 8 & 2 & 3 \end{pmatrix}_{b_0}$

[304] $21^{141}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 964, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 18 & 5 & 2 & 3 \\ 4 & 4 & 26 & 7 & 3 & 4 \\ 18 & 26 & 118 & 32 & 14 & 20 \\ 5 & 7 & 32 & 7 & 4 & 5 \\ 2 & 3 & 14 & 4 & 0 & 2 \\ 3 & 4 & 20 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[305] \quad 21^{114}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 1020, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 17 & 2 & 3 \\ 4 & 4 & 4 & 24 & 3 & 4 \\ 3 & 4 & 1 & 17 & 2 & 3 \\ 17 & 24 & 17 & 100 & 13 & 18 \\ 2 & 3 & 2 & 13 & 0 & 2 \\ 3 & 4 & 3 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[306] \quad 21^{114}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 1168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 20 & 2 & 3 \\ 4 & 4 & 4 & 29 & 3 & 4 \\ 3 & 4 & 1 & 20 & 2 & 3 \\ 20 & 29 & 20 & 148 & 16 & 22 \\ 2 & 3 & 2 & 16 & 0 & 2 \\ 3 & 4 & 3 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [311] \quad 1^{11}3^{11}1^{11}5^{13}1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 5 & 3 & 1 & 2 \\ 4 & 0 & 5 & 17 & 11 & 5 & 9 \\ 2 & 0 & 3 & 11 & 5 & 3 & 5 \\ 1 & 0 & 1 & 5 & 3 & 0 & 2 \\ 2 & 0 & 2 & 9 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[312] \quad 1^{12}3^{11}1^{12}5^{13}1^{41} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 10 & 3 & 1 & 2 \\ 8 & 0 & 10 & 74 & 22 & 10 & 18 \\ 2 & 0 & 3 & 22 & 5 & 3 & 5 \\ 1 & 0 & 1 & 10 & 3 & 0 & 2 \\ 2 & 0 & 2 & 18 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[313] \quad 1^{141}3^{11}1^{13}5^{13}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 3 & 1 & 2 \\ 1 & 1 & 2 & 16 & 4 & 2 & 3 \\ 1 & 2 & 0 & 16 & 3 & 1 & 2 \\ 11 & 16 & 16 & 126 & 28 & 12 & 24 \\ 3 & 4 & 3 & 28 & 5 & 3 & 5 \\ 1 & 2 & 1 & 12 & 3 & 0 & 2 \\ 2 & 3 & 2 & 24 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[314] \quad 1^{111}3^{11}1^{121}5^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 3 & 1 & 2 \\ 1 & 1 & 2 & 4 & 4 & 2 & 3 \\ 1 & 2 & 0 & 4 & 3 & 1 & 2 \\ 4 & 4 & 4 & 6 & 7 & 3 & 6 \\ 3 & 4 & 3 & 7 & 5 & 3 & 5 \\ 1 & 2 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & 2 & 6 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[315] \quad 1^{131}3^{11}1^{121}5^{13}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 3 & 1 & 2 \\ 1 & 1 & 2 & 12 & 4 & 2 & 3 \\ 1 & 2 & 0 & 12 & 3 & 1 & 2 \\ 8 & 12 & 12 & 70 & 21 & 9 & 18 \\ 3 & 4 & 3 & 21 & 5 & 3 & 5 \\ 1 & 2 & 1 & 9 & 3 & 0 & 2 \\ 2 & 3 & 2 & 18 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[316] \quad 1^{11}3^{11}1^{11}5^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 1 & 2 \\ 1 & 1 & 2 & 4 & 4 & 2 & 3 \\ 1 & 2 & 0 & 4 & 3 & 1 & 2 \\ 3 & 4 & 4 & 6 & 7 & 3 & 6 \\ 3 & 4 & 3 & 7 & 5 & 3 & 5 \\ 1 & 2 & 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & 2 & 6 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[325] \quad 1^{12} 3^{11} 1^{121} 5^{13} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 3 & 1 & 2 \\ 1 & 1 & 2 & 8 & 4 & 2 & 3 \\ 1 & 2 & 0 & 8 & 3 & 1 & 2 \\ 6 & 8 & 8 & 30 & 14 & 6 & 12 \\ 3 & 4 & 3 & 14 & 5 & 3 & 5 \\ 1 & 2 & 1 & 6 & 3 & 0 & 2 \\ 2 & 3 & 2 & 12 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[326] \quad 1^{13} 3^{11} 5^{13} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 1 & 2 & 0 & 12 & 2 & 3 \\ 1 & 2 & 0 & 0 & 11 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 9 & 12 & 11 & 0 & 65 & 9 & 17 \\ 1 & 2 & 1 & 0 & 9 & 0 & 2 \\ 2 & 3 & 2 & 0 & 17 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[327] \quad 1^{12} 3^{11} 5^{12} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 100, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 1 & 2 \\ 1 & 1 & 2 & 4 & 8 & 2 & 3 \\ 1 & 2 & 0 & 4 & 7 & 1 & 2 \\ 3 & 4 & 4 & 6 & 15 & 3 & 6 \\ 6 & 8 & 7 & 15 & 27 & 6 & 11 \\ 1 & 2 & 1 & 3 & 6 & 0 & 2 \\ 2 & 3 & 2 & 6 & 11 & 2 & 4 \end{pmatrix}_{b_0}$$

4.948 NHC(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{12} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 13 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 3 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 & 4 \\ 3 & 5 & 4 & 6 & 3 & 3 & 4 \\ 5 & 7 & 6 & 7 & 4 & 5 & 5 \\ 2 & 3 & 3 & 4 & 0 & 2 & 2 \\ 3 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 4 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 5 & 3 \\ 3 & 4 & 9 & 3 & 3 & 8 & 4 \\ 5 & 9 & 13 & 5 & 5 & 12 & 7 \\ 3 & 3 & 5 & 0 & 2 & 4 & 2 \\ 2 & 3 & 5 & 2 & 0 & 4 & 2 \\ 5 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 7 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 2 & 4 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 & 4 \\ 5 & 5 & 4 & 6 & 3 & 3 & 4 \\ 6 & 7 & 6 & 7 & 4 & 5 & 5 \\ 2 & 3 & 3 & 4 & 0 & 2 & 2 \\ 4 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 4 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 9 & 4 \\ 5 & 8 & 10 & 4 & 4 & 12 & 6 \\ 3 & 3 & 4 & 0 & 2 & 5 & 2 \\ 2 & 3 & 4 & 2 & 0 & 5 & 2 \\ 6 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1311} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 2 & 1 & 3 \\ 3 & 4 & 8 & 7 & 3 & 0 & 4 \\ 7 & 8 & 10 & 10 & 4 & 0 & 6 \\ 5 & 7 & 10 & 7 & 4 & 0 & 5 \\ 2 & 3 & 4 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{141} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 2 & 1 & 3 \\ 3 & 4 & 8 & 20 & 3 & 0 & 4 \\ 5 & 8 & 10 & 28 & 4 & 0 & 6 \\ 15 & 20 & 28 & 69 & 11 & 0 & 15 \\ 2 & 3 & 4 & 11 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 15 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{1111} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 4 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 & 4 \\ 4 & 4 & 1 & 5 & 2 & 3 & 3 \\ 5 & 7 & 5 & 7 & 4 & 5 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 & 2 \\ 4 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{132} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 2 & 1 & 3 \\ 3 & 4 & 12 & 11 & 3 & 0 & 4 \\ 10 & 12 & 25 & 24 & 6 & 0 & 9 \\ 8 & 11 & 24 & 20 & 6 & 0 & 8 \\ 2 & 3 & 6 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 9 & 8 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{132} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 2 & 8 & 3 \\ 3 & 4 & 12 & 3 & 3 & 13 & 4 \\ 8 & 12 & 25 & 6 & 6 & 27 & 9 \\ 3 & 3 & 6 & 0 & 2 & 7 & 2 \\ 2 & 3 & 6 & 2 & 0 & 7 & 2 \\ 8 & 13 & 27 & 7 & 7 & 28 & 10 \\ 3 & 4 & 9 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 3 & 3 \\ 3 & 4 & 5 & 3 & 3 & 4 & 4 \\ 3 & 5 & 4 & 3 & 3 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 & 2 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 2 & 4 & 1 & 3 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 8 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -11, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 5 & 4 \\ 2 & 4 & 1 & 2 & 2 & 3 & 3 \\ 3 & 3 & 2 & 0 & 2 & 3 & 2 \\ 2 & 3 & 2 & 2 & 0 & 3 & 2 \\ 3 & 5 & 3 & 3 & 3 & 4 & 4 \\ 3 & 4 & 3 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{122} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 2 & 1 & 3 \\ 3 & 4 & 8 & 11 & 3 & 0 & 4 \\ 7 & 8 & 10 & 16 & 4 & 0 & 6 \\ 8 & 11 & 16 & 20 & 6 & 0 & 8 \\ 2 & 3 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 8 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{122} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 2 & 4 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 & 4 \\ 4 & 4 & 1 & 8 & 2 & 3 & 3 \\ 8 & 11 & 8 & 20 & 6 & 8 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 & 2 \\ 4 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 1 & 3 \\ 3 & 4 & 4 & 7 & 3 & 0 & 4 \\ 4 & 4 & 1 & 5 & 2 & 0 & 3 \\ 5 & 7 & 5 & 7 & 4 & 0 & 5 \\ 2 & 3 & 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{121} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 2 & 2 & 3 \\ 3 & 4 & 8 & 7 & 3 & 4 & 4 \\ 7 & 8 & 10 & 10 & 4 & 6 & 6 \\ 5 & 7 & 10 & 7 & 4 & 5 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 & 2 \\ 2 & 4 & 6 & 5 & 2 & 1 & 3 \\ 3 & 4 & 6 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -17, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 5 & 7 & 3 & 0 & 4 \\ 3 & 5 & 4 & 6 & 3 & 0 & 4 \\ 5 & 7 & 6 & 7 & 4 & 0 & 5 \\ 2 & 3 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 4 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{112} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 11 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 2 & 7 & 3 \\ 3 & 4 & 8 & 3 & 3 & 9 & 4 \\ 7 & 8 & 10 & 4 & 4 & 12 & 6 \\ 3 & 3 & 4 & 0 & 2 & 5 & 2 \\ 2 & 3 & 4 & 2 & 0 & 5 & 2 \\ 7 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{111} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 6 & 2 & 1 & 3 \\ 3 & 4 & 14 & 7 & 3 & 0 & 4 \\ 12 & 14 & 37 & 17 & 8 & 0 & 11 \\ 6 & 7 & 17 & 7 & 4 & 0 & 5 \\ 2 & 3 & 8 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 11 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{132}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 2 & 9 & 3 \\ 3 & 4 & 12 & 3 & 3 & 13 & 4 \\ 9 & 12 & 25 & 6 & 6 & 27 & 9 \\ 3 & 3 & 6 & 0 & 2 & 7 & 2 \\ 2 & 3 & 6 & 2 & 0 & 7 & 2 \\ 9 & 13 & 27 & 7 & 7 & 28 & 10 \\ 3 & 4 & 9 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 4 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 & 4 \\ 4 & 5 & 4 & 6 & 3 & 3 & 4 \\ 5 & 7 & 6 & 7 & 4 & 5 & 5 \\ 2 & 3 & 3 & 4 & 0 & 2 & 2 \\ 4 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 4 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 4 & 3 \\ 3 & 4 & 4 & 3 & 3 & 5 & 4 \\ 4 & 4 & 1 & 2 & 2 & 3 & 3 \\ 3 & 3 & 2 & 0 & 2 & 3 & 2 \\ 2 & 3 & 2 & 2 & 0 & 3 & 2 \\ 4 & 5 & 3 & 3 & 3 & 4 & 4 \\ 3 & 4 & 3 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 2 & 7 & 3 \\ 3 & 4 & 8 & 7 & 3 & 9 & 4 \\ 7 & 8 & 10 & 10 & 4 & 12 & 6 \\ 5 & 7 & 10 & 7 & 4 & 11 & 5 \\ 2 & 3 & 4 & 4 & 0 & 5 & 2 \\ 7 & 9 & 12 & 11 & 5 & 13 & 7 \\ 3 & 4 & 6 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -25, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 2 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 & 4 \\ 3 & 5 & 4 & 6 & 3 & 3 & 4 \\ 5 & 7 & 6 & 7 & 4 & 5 & 5 \\ 2 & 3 & 3 & 4 & 0 & 2 & 2 \\ 2 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 4 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -25, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 3 & 9 & 4 \\ 4 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 6 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -25, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 2 & 1 & 3 \\ 3 & 4 & 4 & 12 & 3 & 0 & 4 \\ 2 & 4 & 1 & 8 & 2 & 0 & 3 \\ 9 & 12 & 8 & 25 & 7 & 0 & 9 \\ 2 & 3 & 2 & 7 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 9 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 5 & 3 \\ 3 & 4 & 9 & 7 & 3 & 8 & 4 \\ 5 & 9 & 13 & 11 & 5 & 12 & 7 \\ 5 & 7 & 11 & 7 & 4 & 10 & 5 \\ 2 & 3 & 5 & 4 & 0 & 4 & 2 \\ 5 & 8 & 12 & 10 & 4 & 10 & 6 \\ 3 & 4 & 7 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1^{12}3^{11}1^{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 9 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1^{12}3^{11}1^{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 2 & 6 & 3 \\ 3 & 4 & 8 & 7 & 3 & 9 & 4 \\ 5 & 8 & 10 & 10 & 4 & 12 & 6 \\ 6 & 7 & 10 & 7 & 4 & 11 & 5 \\ 2 & 3 & 4 & 4 & 0 & 5 & 2 \\ 6 & 9 & 12 & 11 & 5 & 13 & 7 \\ 3 & 4 & 6 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1^{11}3^{11}1^{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 & 4 \\ 2 & 4 & 1 & 2 & 2 & 3 & 3 \\ 3 & 3 & 2 & 0 & 2 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1^{12}3^{11}1^{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 3 & 3 \\ 3 & 4 & 8 & 3 & 3 & 4 & 4 \\ 5 & 8 & 10 & 4 & 4 & 6 & 6 \\ 3 & 3 & 4 & 0 & 2 & 2 & 2 \\ 2 & 3 & 4 & 2 & 0 & 2 & 2 \\ 3 & 4 & 6 & 2 & 2 & 1 & 3 \\ 3 & 4 & 6 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1^{11}3^{11}1^{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 14 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 2 & 0 & 8 & 2 \\ 9 & 14 & 9 & 8 & 8 & 37 & 11 \\ 3 & 4 & 3 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1^{11}3^{11}1^{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 3 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 & 4 \\ 4 & 5 & 4 & 6 & 3 & 3 & 4 \\ 5 & 7 & 6 & 7 & 4 & 5 & 5 \\ 2 & 3 & 3 & 4 & 0 & 2 & 2 \\ 3 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 4 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1^{11}3^{11}1^{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 3 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 3 & 2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 4 & 5 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & 0 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1^{11}3^{11}1^{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 3 & 9 & 4 \\ 2 & 4 & 1 & 5 & 2 & 6 & 3 \\ 6 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 6 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{113} 3^{11} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 3 & 9 & 4 \\ 5 & 4 & 4 & 7 & 3 & 9 & 4 \\ 3 & 4 & 1 & 5 & 2 & 6 & 3 \\ 6 & 7 & 5 & 7 & 4 & 11 & 5 \\ 3 & 3 & 2 & 4 & 0 & 5 & 2 \\ 9 & 9 & 6 & 11 & 5 & 13 & 7 \\ 4 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{113} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 14 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 2 & 0 & 8 & 2 \\ 11 & 14 & 9 & 8 & 8 & 37 & 11 \\ 3 & 4 & 3 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{1321} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 2 & 1 & 3 \\ 3 & 4 & 14 & 7 & 3 & 0 & 4 \\ 11 & 14 & 37 & 17 & 8 & 0 & 11 \\ 6 & 7 & 17 & 7 & 4 & 0 & 5 \\ 2 & 3 & 8 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 11 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 2 & 3 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 & 4 \\ 5 & 9 & 13 & 11 & 5 & 6 & 7 \\ 5 & 7 & 11 & 7 & 4 & 5 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 & 2 \\ 3 & 4 & 6 & 5 & 2 & 1 & 3 \\ 3 & 4 & 7 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 2 & 4 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 & 4 \\ 8 & 9 & 13 & 11 & 5 & 6 & 7 \\ 5 & 7 & 11 & 7 & 4 & 5 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 & 2 \\ 4 & 4 & 6 & 5 & 2 & 1 & 3 \\ 3 & 4 & 7 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{112} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 2 & 4 & 1 & 2 & 2 & 6 & 3 \\ 3 & 3 & 2 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 2 & 0 & 4 & 2 \\ 6 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 9 & 4 \\ 6 & 8 & 10 & 4 & 4 & 12 & 6 \\ 3 & 3 & 4 & 0 & 2 & 5 & 2 \\ 2 & 3 & 4 & 2 & 0 & 5 & 2 \\ 6 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 1 & 3 \\ 3 & 4 & 5 & 7 & 3 & 0 & 4 \\ 4 & 5 & 4 & 6 & 3 & 0 & 4 \\ 5 & 7 & 6 & 7 & 4 & 0 & 5 \\ 2 & 3 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 4 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 2 & 1 & 3 \\ 3 & 4 & 5 & 7 & 3 & 0 & 4 \\ 4 & 5 & 4 & 6 & 3 & 0 & 4 \\ 6 & 7 & 6 & 7 & 4 & 0 & 5 \\ 2 & 3 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 4 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{141} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 2 & 3 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 & 4 \\ 13 & 21 & 76 & 11 & 11 & 15 & 16 \\ 3 & 3 & 11 & 0 & 2 & 2 & 2 \\ 2 & 3 & 11 & 2 & 0 & 2 & 2 \\ 3 & 4 & 15 & 2 & 2 & 1 & 3 \\ 3 & 4 & 16 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 2 & 3 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 & 4 \\ 4 & 5 & 4 & 6 & 3 & 3 & 4 \\ 6 & 7 & 6 & 7 & 4 & 5 & 5 \\ 2 & 3 & 3 & 4 & 0 & 2 & 2 \\ 3 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 4 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 3 & 3 \\ 3 & 4 & 5 & 3 & 3 & 4 & 4 \\ 4 & 5 & 4 & 3 & 3 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 & 2 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 2 & 2 & 3 \\ 3 & 4 & 4 & 12 & 3 & 4 & 4 \\ 4 & 4 & 1 & 8 & 2 & 3 & 3 \\ 9 & 12 & 8 & 25 & 7 & 8 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 & 2 \\ 2 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 9 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{122} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 2 & 1 & 3 \\ 3 & 4 & 4 & 11 & 3 & 0 & 4 \\ 4 & 4 & 1 & 8 & 2 & 0 & 3 \\ 8 & 11 & 8 & 20 & 6 & 0 & 8 \\ 2 & 3 & 2 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 8 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{122} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 9 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 5 & 8 & 10 & 4 & 4 & 18 & 6 \\ 3 & 3 & 4 & 0 & 2 & 7 & 2 \\ 2 & 3 & 4 & 2 & 0 & 7 & 2 \\ 9 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{122} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 8 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 5 & 8 & 10 & 4 & 4 & 18 & 6 \\ 3 & 3 & 4 & 0 & 2 & 7 & 2 \\ 2 & 3 & 4 & 2 & 0 & 7 & 2 \\ 8 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{112} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 8 & 4 \\ 5 & 8 & 10 & 4 & 4 & 12 & 6 \\ 3 & 3 & 4 & 0 & 2 & 4 & 2 \\ 2 & 3 & 4 & 2 & 0 & 4 & 2 \\ 6 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 6 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 2 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 & 4 \\ 4 & 4 & 1 & 5 & 2 & 3 & 3 \\ 5 & 7 & 5 & 7 & 4 & 5 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 & 2 \\ 2 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 1 & 3 \\ 3 & 4 & 0 & 7 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 0 & 5 \\ 2 & 3 & 0 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{123} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 2 & 1 & 3 \\ 3 & 4 & 8 & 25 & 3 & 0 & 4 \\ 5 & 8 & 10 & 34 & 4 & 0 & 6 \\ 17 & 25 & 34 & 111 & 14 & 0 & 19 \\ 2 & 3 & 4 & 14 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 19 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 2 & 2 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 3 & 4 & 0 & 2 & 2 & 1 & 3 \\ 3 & 4 & 0 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -38, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 5 & 2 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 6 & 9 & 0 & 5 & 5 & 13 & 7 \\ 3 & 4 & 0 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{113} 3^{11} 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 2 & 0 & 6 & 2 \\ 9 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{213} 3^{11} 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 17 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 6 & 8 & 10 & 4 & 4 & 36 & 6 \\ 3 & 3 & 4 & 0 & 2 & 14 & 2 \\ 2 & 3 & 4 & 2 & 0 & 14 & 2 \\ 17 & 26 & 36 & 14 & 14 & 20 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{121} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 2 & 3 & 3 \\ 3 & 4 & 12 & 3 & 3 & 4 & 4 \\ 8 & 12 & 25 & 6 & 6 & 9 & 9 \\ 3 & 3 & 6 & 0 & 2 & 2 & 2 \\ 2 & 3 & 6 & 2 & 0 & 2 & 2 \\ 3 & 4 & 9 & 2 & 2 & 1 & 3 \\ 3 & 4 & 9 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{123} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 4 & 2 \\ 2 & 3 & 0 & 2 & 0 & 4 & 2 \\ 6 & 8 & 0 & 4 & 4 & 10 & 6 \\ 3 & 4 & 0 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{112} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 3 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 10 & 2 \\ 2 & 3 & 0 & 2 & 0 & 10 & 2 \\ 12 & 18 & 0 & 10 & 10 & 58 & 14 \\ 3 & 4 & 0 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{113} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 2 & 0 & 9 & 2 \\ 11 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{111} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 9 & 4 \\ 2 & 4 & 1 & 2 & 2 & 6 & 3 \\ 3 & 3 & 2 & 0 & 2 & 5 & 2 \\ 2 & 3 & 2 & 2 & 0 & 5 & 2 \\ 6 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{123} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 2 & 1 & 3 \\ 3 & 4 & 4 & 16 & 3 & 0 & 4 \\ 4 & 4 & 1 & 11 & 2 & 0 & 3 \\ 12 & 16 & 11 & 44 & 9 & 0 & 12 \\ 2 & 3 & 2 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 12 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{123} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 9 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 5 & 8 & 10 & 4 & 4 & 18 & 6 \\ 3 & 3 & 4 & 0 & 2 & 6 & 2 \\ 2 & 3 & 4 & 2 & 0 & 6 & 2 \\ 9 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{112} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 14 & 3 \\ 3 & 4 & 8 & 3 & 3 & 22 & 4 \\ 5 & 8 & 10 & 4 & 4 & 30 & 6 \\ 3 & 3 & 4 & 0 & 2 & 12 & 2 \\ 2 & 3 & 4 & 2 & 0 & 12 & 2 \\ 14 & 22 & 30 & 12 & 12 & 85 & 17 \\ 3 & 4 & 6 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{123} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 15 & 3 \\ 3 & 4 & 8 & 3 & 3 & 22 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 3 & 3 & 4 & 0 & 2 & 12 & 2 \\ 2 & 3 & 4 & 2 & 0 & 12 & 2 \\ 15 & 22 & 30 & 12 & 12 & 85 & 17 \\ 3 & 4 & 6 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{111} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 2 & 3 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 & 4 \\ 8 & 9 & 13 & 11 & 5 & 6 & 7 \\ 5 & 7 & 11 & 7 & 4 & 5 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 & 2 \\ 3 & 4 & 6 & 5 & 2 & 1 & 3 \\ 3 & 4 & 7 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{123} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 2 & 6 & 3 \\ 3 & 4 & 12 & 3 & 3 & 8 & 4 \\ 8 & 12 & 25 & 6 & 6 & 18 & 9 \\ 3 & 3 & 6 & 0 & 2 & 4 & 2 \\ 2 & 3 & 6 & 2 & 0 & 4 & 2 \\ 6 & 8 & 18 & 4 & 4 & 10 & 6 \\ 3 & 4 & 9 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{123} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 7 & 3 \\ 3 & 4 & 8 & 3 & 3 & 9 & 4 \\ 6 & 8 & 10 & 4 & 4 & 12 & 6 \\ 3 & 3 & 4 & 0 & 2 & 5 & 2 \\ 2 & 3 & 4 & 2 & 0 & 5 & 2 \\ 7 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{13} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 6 & 2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 9 & 12 & 0 & 6 & 6 & 25 & 9 \\ 3 & 4 & 0 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{114} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 3 & 3 & 26 & 4 \\ 3 & 4 & 1 & 2 & 2 & 18 & 3 \\ 3 & 3 & 2 & 0 & 2 & 14 & 2 \\ 2 & 3 & 2 & 2 & 0 & 14 & 2 \\ 16 & 26 & 18 & 14 & 14 & 118 & 20 \\ 3 & 4 & 3 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{112} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 11 & 3 \\ 3 & 4 & 0 & 7 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 16 & 5 \\ 2 & 3 & 0 & 4 & 0 & 7 & 2 \\ 11 & 13 & 0 & 16 & 7 & 28 & 10 \\ 3 & 4 & 0 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{114} 3^{11} 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 16 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 12 & 16 & 12 & 8 & 8 & 46 & 12 \\ 3 & 4 & 3 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 2 & 1 & 3 \\ 3 & 4 & 4 & 12 & 3 & 0 & 4 \\ 4 & 4 & 1 & 8 & 2 & 0 & 3 \\ 10 & 12 & 8 & 25 & 7 & 0 & 9 \\ 2 & 3 & 2 & 7 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 9 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{124} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 18 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 5 & 8 & 10 & 4 & 4 & 36 & 6 \\ 3 & 3 & 4 & 0 & 2 & 14 & 2 \\ 2 & 3 & 4 & 2 & 0 & 14 & 2 \\ 18 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{124} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 16 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 5 & 8 & 10 & 4 & 4 & 36 & 6 \\ 3 & 3 & 4 & 0 & 2 & 14 & 2 \\ 2 & 3 & 4 & 2 & 0 & 14 & 2 \\ 16 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{124} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 3 & 16 & 4 \\ 5 & 8 & 10 & 4 & 4 & 24 & 6 \\ 3 & 3 & 4 & 0 & 2 & 8 & 2 \\ 2 & 3 & 4 & 2 & 0 & 8 & 2 \\ 12 & 16 & 24 & 8 & 8 & 46 & 12 \\ 3 & 4 & 6 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{121} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 2 & 3 & 3 \\ 3 & 4 & 8 & 7 & 3 & 4 & 4 \\ 5 & 8 & 10 & 10 & 4 & 6 & 6 \\ 6 & 7 & 10 & 7 & 4 & 5 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 & 2 \\ 3 & 4 & 6 & 5 & 2 & 1 & 3 \\ 3 & 4 & 6 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{111} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 13 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 4 & 4 & 1 & 5 & 2 & 12 & 3 \\ 5 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 13 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{111} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 2 & 1 & 3 \\ 3 & 4 & 14 & 7 & 3 & 0 & 4 \\ 10 & 14 & 37 & 17 & 8 & 0 & 11 \\ 5 & 7 & 17 & 7 & 4 & 0 & 5 \\ 2 & 3 & 8 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 11 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 & 4 \\ 4 & 4 & 1 & 2 & 2 & 3 & 3 \\ 3 & 3 & 2 & 0 & 2 & 2 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 & 2 \\ 3 & 4 & 3 & 2 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{1111}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 2 & 3 \\ 3 & 4 & 4 & 12 & 3 & 4 & 4 \\ 2 & 4 & 1 & 8 & 2 & 3 & 3 \\ 8 & 12 & 8 & 25 & 7 & 8 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 & 2 \\ 2 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 9 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{1121}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 2 & 4 & 3 \\ 3 & 4 & 4 & 21 & 3 & 4 & 4 \\ 2 & 4 & 1 & 14 & 2 & 3 & 3 \\ 16 & 21 & 14 & 80 & 12 & 14 & 16 \\ 2 & 3 & 2 & 12 & 0 & 2 & 2 \\ 4 & 4 & 3 & 14 & 2 & 1 & 3 \\ 3 & 4 & 3 & 16 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{1121}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 2 & 2 & 3 \\ 3 & 4 & 4 & 21 & 3 & 4 & 4 \\ 2 & 4 & 1 & 14 & 2 & 3 & 3 \\ 14 & 21 & 14 & 80 & 12 & 14 & 16 \\ 2 & 3 & 2 & 12 & 0 & 2 & 2 \\ 2 & 4 & 3 & 14 & 2 & 1 & 3 \\ 3 & 4 & 3 & 16 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{111}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 0 & 7 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 5 & 5 \\ 2 & 3 & 0 & 4 & 0 & 2 & 2 \\ 2 & 4 & 0 & 5 & 2 & 1 & 3 \\ 3 & 4 & 0 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{123}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 17 & 3 \\ 3 & 4 & 0 & 11 & 3 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 50 & 8 \\ 2 & 3 & 0 & 6 & 0 & 14 & 2 \\ 17 & 26 & 0 & 50 & 14 & 118 & 20 \\ 3 & 4 & 0 & 8 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{1112}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 5 & 3 \\ 3 & 4 & 4 & 7 & 3 & 8 & 4 \\ 4 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 10 & 5 \\ 2 & 3 & 2 & 4 & 0 & 4 & 2 \\ 5 & 8 & 6 & 10 & 4 & 10 & 6 \\ 3 & 4 & 3 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{113}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 2 & 1 & 3 \\ 3 & 4 & 4 & 15 & 3 & 0 & 4 \\ 4 & 4 & 1 & 11 & 2 & 0 & 3 \\ 11 & 15 & 11 & 39 & 8 & 0 & 11 \\ 2 & 3 & 2 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 11 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{1122}3^{11}1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 2 & 2 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 & 4 \\ 4 & 4 & 1 & 8 & 2 & 3 & 3 \\ 8 & 11 & 8 & 20 & 6 & 8 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 & 2 \\ 2 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{141} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 1 & 3 \\ 3 & 4 & 0 & 16 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 13 & 16 & 0 & 44 & 9 & 0 & 12 \\ 2 & 3 & 0 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 12 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{121} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 1 & 3 \\ 3 & 4 & 0 & 11 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 0 & 8 \\ 2 & 3 & 0 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 8 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{121} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 2 & 6 & 3 \\ 3 & 4 & 8 & 7 & 3 & 9 & 4 \\ 6 & 8 & 10 & 10 & 4 & 12 & 6 \\ 5 & 7 & 10 & 7 & 4 & 11 & 5 \\ 2 & 3 & 4 & 4 & 0 & 5 & 2 \\ 6 & 9 & 12 & 11 & 5 & 13 & 7 \\ 3 & 4 & 6 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[96] \quad 1^{121} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 4 & 4 & 1 & 2 & 2 & 6 & 3 \\ 3 & 3 & 2 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 2 & 0 & 4 & 2 \\ 6 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[97] \quad 1^{113} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 10 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[98] \quad 1^{141} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 8 & 2 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 12 & 16 & 0 & 8 & 8 & 46 & 12 \\ 3 & 4 & 0 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[100] \quad 1^{111} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 7 & 3 \\ 3 & 4 & 4 & 7 & 3 & 9 & 4 \\ 2 & 4 & 1 & 5 & 2 & 6 & 3 \\ 6 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 7 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[101] \quad 1^{141} 3^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 2 & 3 & 3 \\ 3 & 4 & 16 & 3 & 3 & 4 & 4 \\ 11 & 16 & 46 & 8 & 8 & 12 & 12 \\ 3 & 3 & 8 & 0 & 2 & 2 & 2 \\ 2 & 3 & 8 & 0 & 2 & 2 & 2 \\ 3 & 3 & 12 & 2 & 2 & 1 & 3 \\ 3 & 4 & 12 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[99] \ 1^{1111} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 3 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 & 4 \\ 2 & 4 & 1 & 5 & 2 & 3 & 3 \\ 6 & 7 & 5 & 7 & 4 & 5 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 & 2 \\ 3 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[102] \ 1^{111} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 1 & 3 \\ 3 & 4 & 0 & 12 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 9 & 12 & 0 & 25 & 7 & 0 & 9 \\ 2 & 3 & 0 & 7 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 9 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{112} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 5 & 8 & 10 & 4 & 4 & 24 & 6 \\ 3 & 3 & 4 & 0 & 2 & 9 & 2 \\ 2 & 3 & 4 & 2 & 0 & 9 & 2 \\ 11 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{113} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 5 & 8 & 10 & 4 & 4 & 24 & 6 \\ 3 & 3 & 4 & 0 & 2 & 9 & 2 \\ 2 & 3 & 4 & 2 & 0 & 9 & 2 \\ 12 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{111} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 2 & 3 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 & 4 \\ 6 & 9 & 13 & 11 & 5 & 6 & 7 \\ 6 & 7 & 11 & 7 & 4 & 5 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 & 2 \\ 3 & 4 & 6 & 5 & 2 & 1 & 3 \\ 3 & 4 & 7 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{111} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 2 & 2 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 & 4 \\ 5 & 9 & 13 & 11 & 5 & 6 & 7 \\ 5 & 7 & 11 & 7 & 4 & 5 & 5 \\ 2 & 3 & 5 & 4 & 0 & 2 & 2 \\ 2 & 4 & 6 & 5 & 2 & 1 & 3 \\ 3 & 4 & 7 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[107] \ 1^{112} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 8 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[108] \ 1^{121} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 2 & 0 & 6 & 2 \\ 9 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{1123} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 14 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 2 & 4 & 1 & 5 & 2 & 12 & 3 \\ 6 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 14 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{1423} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 2 & 6 & 3 \\ 3 & 4 & 16 & 3 & 3 & 8 & 4 \\ 11 & 16 & 46 & 8 & 8 & 24 & 12 \\ 3 & 3 & 8 & 0 & 2 & 4 & 2 \\ 2 & 3 & 8 & 2 & 0 & 4 & 2 \\ 6 & 8 & 24 & 4 & 4 & 10 & 6 \\ 3 & 4 & 12 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{1112} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 3 & 8 & 4 \\ 2 & 4 & 1 & 5 & 2 & 6 & 3 \\ 6 & 7 & 5 & 7 & 4 & 10 & 5 \\ 2 & 3 & 2 & 4 & 0 & 4 & 2 \\ 6 & 8 & 6 & 10 & 4 & 10 & 6 \\ 3 & 4 & 3 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 2 & 0 & 9 & 2 \\ 12 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 0 & 3 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 7 & 2 \\ 2 & 3 & 0 & 2 & 0 & 7 & 2 \\ 9 & 13 & 0 & 7 & 7 & 28 & 10 \\ 3 & 4 & 0 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[114] \ 1^{1113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 8 & 3 \\ 3 & 4 & 0 & 7 & 3 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 11 & 5 \\ 2 & 3 & 0 & 4 & 0 & 5 & 2 \\ 8 & 9 & 0 & 11 & 5 & 13 & 7 \\ 3 & 4 & 0 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{1143} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 11 & 2 \\ 2 & 3 & 2 & 2 & 0 & 11 & 2 \\ 14 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 2 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 9 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{112} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 2 & 0 & 12 & 2 \\ 14 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 3 & 3 & 9 & 4 \\ 4 & 4 & 1 & 2 & 2 & 6 & 3 \\ 3 & 3 & 2 & 0 & 2 & 5 & 2 \\ 2 & 3 & 2 & 2 & 0 & 5 & 2 \\ 7 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 2 & 1 & 3 \\ 3 & 4 & 4 & 16 & 3 & 0 & 4 \\ 2 & 4 & 1 & 11 & 2 & 0 & 3 \\ 12 & 16 & 11 & 44 & 9 & 0 & 12 \\ 2 & 3 & 2 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 12 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{112} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 13 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 2 & 4 & 1 & 5 & 2 & 12 & 3 \\ 6 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 13 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[123] \ 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 3 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 5 & 2 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 7 & 9 & 0 & 5 & 5 & 13 & 7 \\ 3 & 4 & 0 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 9 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 2 & 4 & 1 & 5 & 2 & 9 & 3 \\ 6 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 9 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{112} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 3 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 8 & 2 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 10 & 14 & 0 & 8 & 8 & 37 & 11 \\ 3 & 4 & 0 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[126] \ 21^{11} 3^{11} 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 3 & 3 & 4 \\ 5 & 4 & 4 & 16 & 3 & 4 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 & 3 \\ 15 & 16 & 11 & 44 & 9 & 11 & 12 \\ 3 & 3 & 2 & 9 & 0 & 2 & 2 \\ 3 & 4 & 3 & 11 & 2 & 1 & 3 \\ 4 & 4 & 3 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{\overline{1121}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 2 & 2 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 & 4 \\ 4 & 4 & 1 & 11 & 2 & 3 & 3 \\ 12 & 16 & 11 & 44 & 9 & 11 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 & 2 \\ 2 & 4 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 3 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 & 4 \\ 3 & 4 & 1 & 2 & 2 & 3 & 3 \\ 3 & 3 & 2 & 0 & 2 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 & 2 & 2 \\ 3 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{\overline{114}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 16 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 2 & 0 & 8 & 2 \\ 12 & 16 & 12 & 8 & 8 & 46 & 12 \\ 3 & 4 & 3 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 7 & 3 \\ 3 & 4 & 4 & 7 & 3 & 9 & 4 \\ 2 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 7 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 2 & 3 & 3 \\ 3 & 4 & 4 & 12 & 3 & 4 & 4 \\ 2 & 4 & 1 & 8 & 2 & 3 & 3 \\ 9 & 12 & 8 & 25 & 7 & 8 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 & 2 \\ 3 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 9 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[130] \ 1^{\overline{121}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 3 & 3 \\ 3 & 4 & 0 & 7 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 5 & 5 \\ 2 & 3 & 0 & 4 & 0 & 2 & 2 \\ 3 & 4 & 0 & 5 & 2 & 1 & 3 \\ 3 & 4 & 0 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{\overline{113}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 3 & 3 & 2 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 2 & 0 & 4 & 2 \\ 7 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{\overline{121}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 3 & 3 & 4 & 0 & 2 & 7 & 2 \\ 2 & 3 & 4 & 2 & 0 & 7 & 2 \\ 11 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[133] \quad 1^{112} 3^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 14 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 2 & 0 & 8 & 2 \\ 10 & 14 & 9 & 8 & 8 & 37 & 11 \\ 3 & 4 & 3 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[132] \quad 1^{123} 3^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 2 & 13 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 3 & 4 & 1 & 5 & 2 & 12 & 3 \\ 6 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 13 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[136] \quad 1^{113} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -97, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 2 & 1 & 3 \\ 3 & 4 & 4 & 20 & 3 & 0 & 4 \\ 4 & 4 & 1 & 14 & 2 & 0 & 3 \\ 15 & 20 & 14 & 69 & 11 & 0 & 15 \\ 2 & 3 & 2 & 11 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 15 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[135] \quad 1^{111} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -97, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 2 & 6 & 3 \\ 3 & 4 & 8 & 7 & 3 & 9 & 4 \\ 5 & 8 & 10 & 10 & 4 & 12 & 6 \\ 5 & 7 & 10 & 7 & 4 & 11 & 5 \\ 2 & 3 & 4 & 4 & 0 & 5 & 2 \\ 6 & 9 & 12 & 11 & 5 & 13 & 7 \\ 3 & 4 & 6 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[139] \quad 1^{122} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 6 & 3 \\ 3 & 4 & 0 & 7 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 10 & 5 \\ 2 & 3 & 0 & 4 & 0 & 4 & 2 \\ 6 & 8 & 0 & 10 & 4 & 10 & 6 \\ 3 & 4 & 0 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[137] \quad 1^{222} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 2 & 7 & 3 \\ 3 & 4 & 8 & 7 & 3 & 9 & 4 \\ 6 & 8 & 10 & 10 & 4 & 12 & 6 \\ 6 & 7 & 10 & 7 & 4 & 11 & 5 \\ 2 & 3 & 4 & 4 & 0 & 5 & 2 \\ 7 & 9 & 12 & 11 & 5 & 13 & 7 \\ 3 & 4 & 6 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[138] \quad 1^{113} 3^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 13 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[140] \quad 1^{122} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 2 & 7 & 3 \\ 3 & 4 & 4 & 7 & 3 & 9 & 4 \\ 4 & 4 & 1 & 5 & 2 & 6 & 3 \\ 6 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 7 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[146] \quad 1^{13} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 1 & 3 \\ 3 & 4 & 0 & 15 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 11 & 15 & 0 & 39 & 8 & 0 & 11 \\ 2 & 3 & 0 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 11 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[144] \quad 1^{112} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 0 & 7 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 10 & 5 \\ 2 & 3 & 0 & 4 & 0 & 4 & 2 \\ 5 & 8 & 0 & 10 & 4 & 10 & 6 \\ 3 & 4 & 0 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[143] \quad 1^{131} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 1 & 3 \\ 3 & 4 & 0 & 17 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 13 & 17 & 0 & 55 & 10 & 0 & 13 \\ 2 & 3 & 0 & 10 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 13 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[147] \quad 1^{121} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 2 & 3 \\ 3 & 4 & 0 & 11 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 8 & 8 \\ 2 & 3 & 0 & 6 & 0 & 2 & 2 \\ 2 & 4 & 0 & 8 & 2 & 1 & 3 \\ 3 & 4 & 0 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[142] \quad 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 2 & 1 & 3 \\ 3 & 4 & 9 & 11 & 3 & 0 & 4 \\ 6 & 9 & 13 & 17 & 5 & 0 & 7 \\ 8 & 11 & 17 & 20 & 6 & 0 & 8 \\ 2 & 3 & 5 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 7 & 8 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[141] \quad 1^{111} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 2 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 & 4 \\ 2 & 4 & 1 & 5 & 2 & 3 & 3 \\ 5 & 7 & 5 & 7 & 4 & 5 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 & 2 \\ 2 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[145] \quad 1^{122} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 2 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 8 & 4 \\ 7 & 8 & 10 & 4 & 4 & 12 & 6 \\ 3 & 3 & 4 & 0 & 2 & 4 & 2 \\ 2 & 3 & 4 & 2 & 0 & 4 & 2 \\ 6 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 6 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[148] \quad 1^{141} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 2 & 1 & 3 \\ 3 & 4 & 4 & 19 & 3 & 0 & 4 \\ 4 & 4 & 1 & 14 & 2 & 0 & 3 \\ 14 & 19 & 14 & 64 & 10 & 0 & 14 \\ 2 & 3 & 2 & 10 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 14 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[149] \quad 1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -102, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 12 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[153] \quad 1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 11 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 3 & 3 & 4 & 0 & 2 & 9 & 2 \\ 11 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[152] \quad 1^{1211}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 2 & 3 & 3 \\ 3 & 4 & 8 & 7 & 3 & 4 & 4 \\ 7 & 8 & 10 & 10 & 4 & 6 & 6 \\ 5 & 7 & 10 & 7 & 4 & 5 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 & 2 \\ 3 & 4 & 6 & 5 & 2 & 1 & 3 \\ 3 & 4 & 6 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[150] \quad 1^{112}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 8 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 3 & 3 & 2 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 2 & 0 & 4 & 2 \\ 6 & 8 & 6 & 4 & 4 & 10 & 6 \\ 3 & 4 & 3 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[151] \quad 1^{1111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 3 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 & 4 \\ 4 & 4 & 1 & 5 & 2 & 3 & 3 \\ 5 & 7 & 5 & 7 & 4 & 5 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 & 2 \\ 3 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[154] \quad 1^{1123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 12 & 3 \\ 3 & 4 & 0 & 7 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 21 & 5 \\ 2 & 3 & 0 & 4 & 0 & 9 & 2 \\ 12 & 17 & 0 & 21 & 9 & 49 & 13 \\ 3 & 4 & 0 & 5 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[158] \quad 1^{1123}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 9 & 3 \\ 3 & 4 & 0 & 7 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 15 & 5 \\ 2 & 3 & 0 & 4 & 0 & 6 & 2 \\ 9 & 12 & 0 & 15 & 6 & 25 & 9 \\ 3 & 4 & 0 & 5 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[157] \quad 1^{1221}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 2 & 1 & 3 \\ 3 & 4 & 8 & 16 & 3 & 0 & 4 \\ 5 & 8 & 10 & 22 & 4 & 0 & 6 \\ 11 & 16 & 22 & 44 & 9 & 0 & 12 \\ 2 & 3 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 12 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[156] \quad 1^{1^{241}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 2 & 1 & 3 \\ 3 & 4 & 8 & 16 & 3 & 0 & 4 \\ 6 & 8 & 10 & 22 & 4 & 0 & 6 \\ 13 & 16 & 22 & 44 & 9 & 0 & 12 \\ 2 & 3 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 12 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[155] \quad 1^{1^{212}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 2 & 7 & 3 \\ 3 & 4 & 8 & 7 & 3 & 9 & 4 \\ 6 & 8 & 10 & 10 & 4 & 12 & 6 \\ 5 & 7 & 10 & 7 & 4 & 11 & 5 \\ 2 & 3 & 4 & 4 & 0 & 5 & 2 \\ 7 & 9 & 12 & 11 & 5 & 13 & 7 \\ 3 & 4 & 6 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[159] \quad 1^{1^{111}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 3 & 3 & 9 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 3 & 3 & 2 & 0 & 2 & 5 & 2 \\ 2 & 3 & 2 & 2 & 0 & 5 & 2 \\ 6 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[161] \quad 1^{1^{212}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 7 & 3 \\ 3 & 4 & 4 & 7 & 3 & 9 & 4 \\ 4 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 7 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[160] \quad 1^{1^{142}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 4 & 3 \\ 3 & 4 & 0 & 16 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 13 & 16 & 0 & 44 & 9 & 11 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 & 2 \\ 4 & 4 & 0 & 11 & 2 & 1 & 3 \\ 3 & 4 & 0 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[162] \quad 1^{1^{14}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 15 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 5 & 8 & 10 & 4 & 4 & 30 & 6 \\ 3 & 3 & 4 & 0 & 2 & 11 & 2 \\ 2 & 3 & 4 & 2 & 0 & 11 & 2 \\ 15 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[165] \quad 1^{1^{13}} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 17 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 5 & 8 & 10 & 4 & 4 & 36 & 6 \\ 3 & 3 & 4 & 0 & 2 & 14 & 2 \\ 2 & 3 & 4 & 2 & 0 & 14 & 2 \\ 17 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[164] \quad 1^{1^{214}} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 4 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 2 & 0 & 12 & 2 \\ 16 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[163] $1^{1^{13}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 2 & 14 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 5 & 8 & 10 & 4 & 4 & 30 & 6 \\ 3 & 3 & 4 & 0 & 2 & 11 & 2 \\ 2 & 3 & 4 & 2 & 0 & 11 & 2 \\ 14 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[167] $1^{1^{14}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 0 & 9 & 2 & 2 \\ 13 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[166] $1^{1^{12}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 7 & 3 \\ 3 & 4 & 0 & 7 & 3 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 11 & 5 \\ 2 & 3 & 0 & 4 & 0 & 5 & 2 \\ 7 & 9 & 0 & 11 & 5 & 13 & 7 \\ 3 & 4 & 0 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[168] $1^{1^{12}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 3 & 8 & 4 \\ 4 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 10 & 5 \\ 2 & 3 & 2 & 4 & 0 & 4 & 2 \\ 6 & 8 & 6 & 10 & 4 & 10 & 6 \\ 3 & 4 & 3 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[169] $1^{1^{14}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -121, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 11 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 4 & 4 & 1 & 5 & 2 & 9 & 3 \\ 5 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 11 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[170] $1^{1^{11}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -122, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 & 3 \\ 3 & 4 & 0 & 7 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 5 & 5 \\ 2 & 3 & 0 & 4 & 0 & 2 & 2 \\ 3 & 4 & 0 & 5 & 2 & 1 & 3 \\ 3 & 4 & 0 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[171] $1^{1^{14}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -122, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 12 & 3 \\ 3 & 4 & 0 & 7 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 20 & 5 \\ 2 & 3 & 0 & 4 & 0 & 8 & 2 \\ 12 & 16 & 0 & 20 & 8 & 46 & 12 \\ 3 & 4 & 0 & 5 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[172] $1^{1^{23}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 2 & 3 & 3 \\ 3 & 4 & 4 & 12 & 3 & 4 & 4 \\ 4 & 4 & 1 & 8 & 2 & 3 & 3 \\ 10 & 12 & 8 & 25 & 7 & 8 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 & 2 \\ 3 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 9 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[174] \quad 1^{1^{12}2}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 6 & 3 \\ 3 & 4 & 8 & 3 & 3 & 8 & 4 \\ 6 & 8 & 10 & 4 & 4 & 12 & 6 \\ 3 & 3 & 4 & 0 & 2 & 4 & 2 \\ 2 & 3 & 4 & 2 & 0 & 4 & 2 \\ 6 & 8 & 12 & 4 & 4 & 10 & 6 \\ 3 & 4 & 6 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[175] \quad 1^{1^{13}}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 1 & 3 \\ 3 & 4 & 0 & 16 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 12 & 16 & 0 & 44 & 9 & 0 & 12 \\ 2 & 3 & 0 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 12 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[173] \quad 1^{1^{11}2}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 11 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 2 & 4 & 1 & 5 & 2 & 12 & 3 \\ 5 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 11 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[176] \quad 1^{1^{13}}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 2 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 2 & 0 & 9 & 2 \\ 12 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[178] \quad 1^{1^{13}}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 9 & 2 \\ 2 & 3 & 0 & 2 & 0 & 9 & 2 \\ 12 & 17 & 0 & 9 & 9 & 49 & 13 \\ 3 & 4 & 0 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[177] \quad 1^{1^{13}}3^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 0 & 3 & 3 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 12 & 2 \\ 2 & 3 & 0 & 2 & 0 & 12 & 2 \\ 15 & 22 & 0 & 12 & 12 & 85 & 17 \\ 3 & 4 & 0 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[179] \quad 1^{1^{22}1}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 2 & 3 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 & 4 \\ 4 & 4 & 1 & 5 & 2 & 3 & 3 \\ 6 & 7 & 5 & 7 & 4 & 5 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 & 2 \\ 3 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[180] \quad 1^{1^{214}}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 4 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 11 & 2 \\ 2 & 3 & 2 & 2 & 0 & 11 & 2 \\ 15 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[182] $21^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 5 & 2 & 6 & 3 \\ 4 & 4 & 9 & 7 & 3 & 8 & 4 \\ 6 & 9 & 13 & 11 & 5 & 12 & 7 \\ 5 & 7 & 11 & 7 & 4 & 10 & 5 \\ 2 & 3 & 5 & 4 & 0 & 4 & 2 \\ 6 & 8 & 12 & 10 & 4 & 10 & 6 \\ 3 & 4 & 7 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 3 & 8 & 4 \\ 4 & 4 & 1 & 5 & 2 & 6 & 3 \\ 6 & 7 & 5 & 7 & 4 & 10 & 5 \\ 2 & 3 & 2 & 4 & 0 & 4 & 2 \\ 6 & 8 & 6 & 10 & 4 & 10 & 6 \\ 3 & 4 & 3 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[181] $1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 3 & 8 & 4 \\ 4 & 4 & 1 & 5 & 2 & 6 & 3 \\ 6 & 7 & 5 & 7 & 4 & 10 & 5 \\ 2 & 3 & 2 & 4 & 0 & 4 & 2 \\ 6 & 8 & 6 & 10 & 4 & 10 & 6 \\ 3 & 4 & 3 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[183] $1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 10 & 5 \\ 2 & 3 & 0 & 4 & 0 & 4 & 2 \\ 6 & 8 & 0 & 10 & 4 & 10 & 6 \\ 3 & 4 & 0 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[184] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 2 & 0 & 6 & 2 \\ 9 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[185] $1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -138, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 2 & 1 & 3 \\ 3 & 4 & 4 & 21 & 3 & 0 & 4 \\ 2 & 4 & 1 & 14 & 2 & 0 & 3 \\ 15 & 21 & 14 & 80 & 12 & 0 & 16 \\ 2 & 3 & 2 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 16 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[186] $1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 7 & 3 \\ 3 & 4 & 0 & 7 & 3 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 11 & 5 \\ 2 & 3 & 0 & 4 & 0 & 5 & 2 \\ 7 & 9 & 0 & 11 & 5 & 13 & 7 \\ 3 & 4 & 0 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[188] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 12 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 2 & 0 & 6 & 2 \\ 10 & 12 & 9 & 6 & 6 & 25 & 9 \\ 3 & 4 & 3 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[187] $1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 2 & 3 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 & 4 \\ 2 & 4 & 1 & 8 & 2 & 3 & 3 \\ 9 & 11 & 8 & 20 & 6 & 8 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 & 2 \\ 3 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[191] \ 1^{1112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 10 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 2 & 4 & 1 & 5 & 2 & 9 & 3 \\ 6 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 10 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[196] \ 1^{13}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 2 & 3 \\ 3 & 4 & 0 & 15 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 11 & 15 & 0 & 39 & 8 & 11 & 11 \\ 2 & 3 & 0 & 8 & 0 & 2 & 2 \\ 2 & 4 & 0 & 11 & 2 & 1 & 3 \\ 3 & 4 & 0 & 11 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[195] \ 1^{122}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 9 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 3 & 3 & 4 & 0 & 2 & 7 & 2 \\ 2 & 3 & 4 & 2 & 0 & 7 & 2 \\ 9 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[194] \ 1^{123}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 7 & 2 \\ 2 & 3 & 0 & 2 & 0 & 7 & 2 \\ 10 & 13 & 0 & 7 & 7 & 28 & 10 \\ 3 & 4 & 0 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[193] \ 1^{1112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 5 & 3 \\ 3 & 4 & 4 & 7 & 3 & 8 & 4 \\ 2 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 10 & 5 \\ 2 & 3 & 2 & 4 & 0 & 4 & 2 \\ 5 & 8 & 6 & 10 & 4 & 10 & 6 \\ 3 & 4 & 3 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[197] \ 1^{113}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 8 & 3 \\ 3 & 4 & 0 & 7 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 15 & 5 \\ 2 & 3 & 0 & 4 & 0 & 6 & 2 \\ 8 & 12 & 0 & 15 & 6 & 25 & 9 \\ 3 & 4 & 0 & 5 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[198] \ 1^{1121}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 2 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 & 4 \\ 2 & 4 & 1 & 8 & 2 & 3 & 3 \\ 8 & 11 & 8 & 20 & 6 & 8 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 & 2 \\ 2 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[192] \ 1^{14}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -144, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 1 & 3 \\ 3 & 4 & 0 & 19 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 14 & 19 & 0 & 64 & 10 & 0 & 14 \\ 2 & 3 & 0 & 10 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 14 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[190] $1^{113}3^{11}1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 3 & 12 & 4 \\ 5 & 4 & 4 & 7 & 3 & 13 & 4 \\ 3 & 4 & 1 & 5 & 2 & 9 & 3 \\ 6 & 7 & 5 & 7 & 4 & 16 & 5 \\ 3 & 3 & 2 & 4 & 0 & 7 & 2 \\ 12 & 13 & 9 & 16 & 7 & 28 & 10 \\ 4 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[189] $1^{122}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 5 & 3 \\ 3 & 4 & 0 & 11 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 16 & 8 \\ 2 & 3 & 0 & 6 & 0 & 4 & 2 \\ 5 & 8 & 0 & 16 & 4 & 10 & 6 \\ 3 & 4 & 0 & 8 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[199] $1^{113}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 6 & 3 \\ 3 & 4 & 0 & 7 & 3 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 11 & 5 \\ 2 & 3 & 0 & 4 & 0 & 5 & 2 \\ 6 & 9 & 0 & 11 & 5 & 13 & 7 \\ 3 & 4 & 0 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[200] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 9 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 4 & 4 & 1 & 5 & 2 & 9 & 3 \\ 5 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 9 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[201] $1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 10 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 6 & 8 & 10 & 4 & 18 & 6 & 6 \\ 3 & 3 & 4 & 0 & 2 & 6 & 2 \\ 2 & 3 & 4 & 2 & 0 & 6 & 2 \\ 10 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[203] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 8 & 3 \\ 3 & 4 & 7 & 3 & 9 & 4 \\ 3 & 4 & 1 & 5 & 2 & 6 & 3 \\ 6 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 8 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[202] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 8 & 3 \\ 3 & 4 & 0 & 7 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 16 & 5 \\ 2 & 3 & 0 & 4 & 0 & 7 & 2 \\ 8 & 13 & 0 & 16 & 7 & 28 & 10 \\ 3 & 4 & 0 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[204] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -149, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 3 & 3 & 9 & 4 \\ 3 & 4 & 1 & 2 & 2 & 6 & 3 \\ 3 & 3 & 2 & 0 & 2 & 5 & 2 \\ 2 & 3 & 2 & 2 & 0 & 5 & 2 \\ 7 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[205] \ 1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -153, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 2 & 1 & 3 \\ 3 & 4 & 4 & 20 & 3 & 0 & 4 \\ 2 & 4 & 1 & 14 & 2 & 0 & 3 \\ 15 & 20 & 14 & 69 & 11 & 0 & 15 \\ 2 & 3 & 2 & 11 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 15 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[211] \ 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -154, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 & 4 \\ 2 & 4 & 1 & 5 & 2 & 3 & 3 \\ 5 & 7 & 5 & 7 & 4 & 5 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 & 2 \\ 3 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[206] \ 1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -154, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 14 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 4 & 4 & 1 & 5 & 2 & 12 & 3 \\ 5 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 14 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[207] \ 1^{123}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -154, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 16 & 3 \\ 3 & 4 & 8 & 3 & 3 & 22 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 3 & 3 & 4 & 0 & 2 & 12 & 2 \\ 2 & 3 & 4 & 2 & 0 & 12 & 2 \\ 16 & 22 & 30 & 12 & 12 & 85 & 17 \\ 3 & 4 & 6 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[208] \ 1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -154, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 9 & 3 \\ 3 & 4 & 0 & 7 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 15 & 5 \\ 2 & 3 & 0 & 4 & 0 & 6 & 2 \\ 9 & 12 & 0 & 15 & 6 & 25 & 9 \\ 3 & 4 & 0 & 5 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[210] \ 1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -154, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 9 & 3 \\ 3 & 4 & 8 & 3 & 3 & 12 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 3 & 3 & 4 & 0 & 2 & 6 & 2 \\ 2 & 3 & 4 & 2 & 0 & 6 & 2 \\ 9 & 12 & 18 & 6 & 6 & 25 & 9 \\ 3 & 4 & 6 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[209] \ 1^{114}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -154, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 3 & 3 & 26 & 4 \\ 2 & 4 & 1 & 2 & 2 & 18 & 3 \\ 3 & 3 & 2 & 0 & 2 & 14 & 2 \\ 2 & 3 & 2 & 2 & 0 & 14 & 2 \\ 17 & 26 & 18 & 14 & 14 & 118 & 20 \\ 3 & 4 & 3 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[212] \ 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -155, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 10 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[213] \ 1^{1112} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -156, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 13 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 2 & 4 & 1 & 5 & 2 & 12 & 3 \\ 5 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 13 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[214] \ 1^{1124} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -162, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 16 & 3 \\ 3 & 4 & 0 & 7 & 3 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 27 & 5 \\ 2 & 3 & 0 & 4 & 0 & 12 & 2 \\ 16 & 22 & 0 & 27 & 12 & 85 & 17 \\ 3 & 4 & 0 & 5 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[215] \ 1^{1112} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -166, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 3 & 8 & 4 \\ 2 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 10 & 5 \\ 2 & 3 & 2 & 4 & 0 & 4 & 2 \\ 6 & 8 & 6 & 10 & 4 & 10 & 6 \\ 3 & 4 & 3 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[216] \ 1^{1124} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -166, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 14 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[217] \ 1^{1111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -169, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 3 & 9 & 4 \\ 2 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 6 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[218] \ 1^{114} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -171, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 11 & 2 \\ 2 & 3 & 2 & 2 & 0 & 11 & 2 \\ 15 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[219] \ 1^{1211} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -172, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 2 & 3 & 3 \\ 3 & 4 & 8 & 7 & 3 & 4 & 4 \\ 7 & 8 & 10 & 10 & 4 & 6 & 6 \\ 6 & 7 & 10 & 7 & 4 & 5 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 & 2 \\ 3 & 4 & 6 & 5 & 2 & 1 & 3 \\ 3 & 4 & 6 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[220] \ 1^{1124} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -173, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 15 & 3 \\ 3 & 4 & 0 & 7 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 26 & 5 \\ 2 & 3 & 0 & 4 & 0 & 11 & 2 \\ 15 & 21 & 0 & 26 & 11 & 76 & 16 \\ 3 & 4 & 0 & 5 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[221] \ 1^{1^{131}} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -174, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 & 3 \\ 3 & 4 & 0 & 11 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 9 & 11 & 0 & 20 & 6 & 8 & 8 \\ 2 & 3 & 0 & 6 & 0 & 2 & 2 \\ 3 & 4 & 0 & 8 & 2 & 1 & 3 \\ 3 & 4 & 0 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[223] \ 1^{1^{112}} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 3 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 2 & 0 & 12 & 2 \\ 14 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[222] \ 1^{1^{114}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 0 & 3 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 11 & 2 \\ 2 & 3 & 0 & 2 & 0 & 11 & 2 \\ 15 & 21 & 0 & 11 & 11 & 76 & 16 \\ 3 & 4 & 0 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[224] \ 1^{1^{141}} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 1 & 3 \\ 3 & 4 & 0 & 21 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 16 & 21 & 0 & 80 & 12 & 0 & 16 \\ 2 & 3 & 0 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 16 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[225] \ 1^{1^{213}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 10 & 3 \\ 3 & 4 & 8 & 3 & 3 & 13 & 4 \\ 6 & 8 & 10 & 4 & 4 & 18 & 6 \\ 3 & 3 & 4 & 0 & 2 & 7 & 2 \\ 2 & 3 & 4 & 2 & 0 & 7 & 2 \\ 10 & 13 & 18 & 7 & 7 & 28 & 10 \\ 3 & 4 & 6 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[226] \ 1^{1^{231}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 2 & 1 & 3 \\ 3 & 4 & 8 & 16 & 3 & 0 & 4 \\ 6 & 8 & 10 & 22 & 4 & 0 & 6 \\ 12 & 16 & 22 & 44 & 9 & 0 & 12 \\ 2 & 3 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 12 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[227] \ 1^{1^{13}} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 2 & 0 & 12 & 2 \\ 15 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[228] \ 1^{1^{114}} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -178, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 16 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 3 & 2 & 2 & 0 & 8 & 2 \\ 12 & 16 & 12 & 8 & 8 & 46 & 12 \\ 3 & 4 & 3 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[229] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 3 & 3 \\ 3 & 4 & 0 & 12 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 9 & 12 & 0 & 25 & 7 & 8 & 9 \\ 2 & 3 & 0 & 7 & 0 & 2 & 2 \\ 3 & 4 & 0 & 8 & 2 & 1 & 3 \\ 3 & 4 & 0 & 9 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[230] \ 1^{122} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 2 & 6 & 3 \\ 3 & 4 & 0 & 11 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 9 & 11 & 0 & 20 & 6 & 16 & 8 \\ 2 & 3 & 0 & 6 & 0 & 4 & 2 \\ 6 & 8 & 0 & 16 & 4 & 10 & 6 \\ 3 & 4 & 0 & 8 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[233] \ 1^{113} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 3 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 10 & 2 \\ 2 & 3 & 0 & 2 & 0 & 10 & 2 \\ 13 & 18 & 0 & 10 & 10 & 58 & 14 \\ 3 & 4 & 0 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[232] \ 1^{113} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 2 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 2 & 0 & 12 & 2 \\ 16 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[231] \ 1^{1121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 2 & 2 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 & 4 \\ 2 & 4 & 1 & 11 & 2 & 3 & 3 \\ 11 & 16 & 11 & 44 & 9 & 11 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 & 2 \\ 2 & 4 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 3 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[234] \ 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 2 & 0 & 9 & 2 \\ 11 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[235] \ 1^{114} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 12 & 3 \\ 3 & 4 & 0 & 7 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 20 & 5 \\ 2 & 3 & 0 & 4 & 0 & 8 & 2 \\ 12 & 16 & 0 & 20 & 8 & 46 & 12 \\ 3 & 4 & 0 & 5 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[237] \ 1^{133} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 1 & 3 \\ 3 & 4 & 0 & 20 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 15 & 20 & 0 & 69 & 11 & 0 & 15 \\ 2 & 3 & 0 & 11 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 15 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[236] $1^{1111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 2 & 3 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 & 4 \\ 3 & 4 & 1 & 5 & 2 & 3 & 3 \\ 6 & 7 & 5 & 7 & 4 & 5 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 & 2 \\ 3 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[238] $1^{121}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 2 & 3 \\ 3 & 4 & 0 & 16 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 11 & 16 & 0 & 44 & 9 & 11 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 & 2 \\ 2 & 4 & 0 & 11 & 2 & 1 & 3 \\ 3 & 4 & 0 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[239] $1^{121}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 2 & 3 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 & 4 \\ 4 & 4 & 1 & 8 & 2 & 3 & 3 \\ 8 & 11 & 8 & 20 & 6 & 8 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 & 2 \\ 3 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[240] $1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 12 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[241] $1^{1214}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 11 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[245] $1^{141}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 2 & 3 \\ 3 & 4 & 0 & 19 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 14 & 19 & 0 & 64 & 10 & 14 & 14 \\ 2 & 3 & 0 & 10 & 0 & 2 & 2 \\ 2 & 4 & 0 & 14 & 2 & 1 & 3 \\ 3 & 4 & 0 & 14 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[242] $1^{122}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 8 & 3 \\ 3 & 4 & 0 & 11 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 24 & 8 \\ 2 & 3 & 0 & 6 & 0 & 6 & 2 \\ 8 & 12 & 0 & 24 & 6 & 25 & 9 \\ 3 & 4 & 0 & 8 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[243] $1^{1214}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 3 & 16 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 3 & 3 & 4 & 0 & 2 & 8 & 2 \\ 2 & 3 & 4 & 2 & 0 & 8 & 2 \\ 12 & 16 & 24 & 8 & 8 & 46 & 12 \\ 3 & 4 & 6 & 2 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[244] $1^{131}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 5 & 3 \\ 3 & 4 & 0 & 15 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 11 & 15 & 0 & 39 & 8 & 22 & 11 \\ 2 & 3 & 0 & 8 & 0 & 4 & 2 \\ 5 & 8 & 0 & 22 & 4 & 10 & 6 \\ 3 & 4 & 0 & 11 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[246] $1^{113}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 11 & 3 \\ 3 & 4 & 0 & 7 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 20 & 5 \\ 2 & 3 & 0 & 4 & 0 & 8 & 2 \\ 11 & 16 & 0 & 20 & 8 & 46 & 12 \\ 3 & 4 & 0 & 5 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[247] $1^{1124}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -197, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 11 & 3 \\ 3 & 4 & 0 & 7 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 16 & 5 \\ 2 & 3 & 0 & 4 & 0 & 7 & 2 \\ 11 & 13 & 0 & 16 & 7 & 28 & 10 \\ 3 & 4 & 0 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[248] $1^{1122}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -197, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 2 & 7 & 3 \\ 3 & 4 & 4 & 7 & 3 & 9 & 4 \\ 3 & 4 & 1 & 5 & 2 & 6 & 3 \\ 6 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 7 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[249] $1^{1124}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 14 & 3 \\ 3 & 4 & 0 & 7 & 3 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 22 & 5 \\ 2 & 3 & 0 & 4 & 0 & 10 & 2 \\ 14 & 18 & 0 & 22 & 10 & 58 & 14 \\ 3 & 4 & 0 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[250] $1^{1114}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 14 & 3 \\ 3 & 4 & 0 & 7 & 3 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 22 & 5 \\ 2 & 3 & 0 & 4 & 0 & 10 & 2 \\ 14 & 18 & 0 & 22 & 10 & 58 & 14 \\ 3 & 4 & 0 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[251] $1^{1111}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 2 & 3 & 3 \\ 3 & 4 & 8 & 7 & 3 & 4 & 4 \\ 5 & 8 & 10 & 10 & 4 & 6 & 6 \\ 5 & 7 & 10 & 7 & 4 & 5 & 5 \\ 2 & 3 & 4 & 4 & 0 & 2 & 2 \\ 3 & 4 & 6 & 5 & 2 & 1 & 3 \\ 3 & 4 & 6 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[252] $1^{1122}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 3 & 8 & 4 \\ 3 & 4 & 1 & 5 & 2 & 6 & 3 \\ 6 & 7 & 5 & 7 & 4 & 10 & 5 \\ 2 & 3 & 2 & 4 & 0 & 4 & 2 \\ 6 & 8 & 6 & 10 & 4 & 10 & 6 \\ 3 & 4 & 3 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[253] \quad 1^{1^{123}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 10 & 3 \\ 3 & 4 & 0 & 7 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 16 & 5 \\ 2 & 3 & 0 & 4 & 0 & 7 & 2 \\ 10 & 13 & 0 & 16 & 7 & 28 & 10 \\ 3 & 4 & 0 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[254] \quad 1^{1^{121}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 2 & 3 & 3 \\ 3 & 4 & 4 & 12 & 3 & 4 & 4 \\ 3 & 4 & 1 & 8 & 2 & 3 & 3 \\ 9 & 12 & 8 & 25 & 7 & 8 & 9 \\ 2 & 3 & 2 & 7 & 0 & 2 & 2 \\ 3 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 9 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[255] \quad 1^{1^{114}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 9 & 2 \\ 2 & 3 & 0 & 2 & 0 & 9 & 2 \\ 13 & 17 & 0 & 9 & 9 & 49 & 13 \\ 3 & 4 & 0 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[256] \quad 1^{1^{111}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -210, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 3 & 9 & 4 \\ 3 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 6 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[257] \quad 1^{1^{112}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -211, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 10 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 2 & 4 & 1 & 5 & 2 & 9 & 3 \\ 5 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 10 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[258] \quad 1^{1^{121}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 & 3 \\ 3 & 4 & 0 & 11 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 8 & 8 \\ 2 & 3 & 0 & 6 & 0 & 2 & 2 \\ 3 & 4 & 0 & 8 & 2 & 1 & 3 \\ 3 & 4 & 0 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[259] \quad 1^{1^{122}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 11 & 3 \\ 3 & 4 & 0 & 11 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 33 & 8 \\ 2 & 3 & 0 & 6 & 0 & 9 & 2 \\ 11 & 17 & 0 & 33 & 9 & 49 & 13 \\ 3 & 4 & 0 & 8 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[260] \quad 1^{1^{114}} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 0 & 3 & 3 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 14 & 2 \\ 2 & 3 & 0 & 2 & 0 & 14 & 2 \\ 18 & 26 & 0 & 14 & 14 & 118 & 20 \\ 3 & 4 & 0 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[261] $1^{1112}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 12 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 3 & 4 & 1 & 5 & 2 & 12 & 3 \\ 5 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 12 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[262] $21^{1112}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 2 & 12 & 3 \\ 4 & 4 & 4 & 7 & 3 & 18 & 4 \\ 3 & 4 & 1 & 5 & 2 & 12 & 3 \\ 5 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 12 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[263] $1^{1111}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 3 & 3 \\ 3 & 4 & 4 & 7 & 3 & 4 & 4 \\ 3 & 4 & 1 & 5 & 2 & 3 & 3 \\ 5 & 7 & 5 & 7 & 4 & 5 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 & 2 \\ 3 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[264] $1^{1111}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 7 & 3 \\ 3 & 4 & 4 & 7 & 3 & 9 & 4 \\ 3 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 7 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[265] $1^{1112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 2 & 10 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 4 & 4 & 1 & 5 & 2 & 9 & 3 \\ 5 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 10 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[266] $1^{1122}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -219, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 9 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[267] $1^{1231}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 2 & 3 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 & 4 \\ 4 & 4 & 1 & 8 & 2 & 3 & 3 \\ 9 & 11 & 8 & 20 & 6 & 8 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 & 2 \\ 3 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[268] $1^{1122}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 2 & 9 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 3 & 4 & 1 & 5 & 2 & 9 & 3 \\ 6 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 9 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[271] \quad 1^{1^{122}} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 6 & 3 \\ 3 & 4 & 0 & 11 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 16 & 8 \\ 2 & 3 & 0 & 6 & 0 & 4 & 2 \\ 6 & 8 & 0 & 16 & 4 & 10 & 6 \\ 3 & 4 & 0 & 8 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[270] \quad 1^{1^{121}} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 2 & 3 & 3 \\ 3 & 4 & 4 & 21 & 3 & 4 & 4 \\ 2 & 4 & 1 & 14 & 2 & 3 & 3 \\ 15 & 21 & 14 & 80 & 12 & 14 & 16 \\ 2 & 3 & 2 & 12 & 0 & 2 & 2 \\ 3 & 4 & 3 & 14 & 2 & 1 & 3 \\ 3 & 4 & 3 & 16 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[269] \quad 1^{1^{213}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 14 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 3 & 3 & 4 & 0 & 2 & 11 & 2 \\ 2 & 3 & 4 & 2 & 0 & 11 & 2 \\ 14 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[273] \quad 1^{1^{114}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 11 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 3 & 4 & 1 & 5 & 2 & 9 & 3 \\ 5 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 11 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[274] \quad 1^{1^{112}} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 12 & 3 \\ 3 & 4 & 0 & 7 & 3 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 22 & 5 \\ 2 & 3 & 0 & 4 & 0 & 10 & 2 \\ 12 & 18 & 0 & 22 & 10 & 58 & 14 \\ 3 & 4 & 0 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[272] \quad 1^{1^{112}} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 12 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 2 & 4 & 1 & 5 & 2 & 12 & 3 \\ 5 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 12 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[275] \quad 1^{1^{214}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 4 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 2 & 0 & 9 & 2 \\ 13 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[276] \quad 1^{1^{231}} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 2 & 1 & 3 \\ 3 & 4 & 8 & 20 & 3 & 0 & 4 \\ 5 & 8 & 10 & 28 & 4 & 0 & 6 \\ 14 & 20 & 28 & 69 & 11 & 0 & 15 \\ 2 & 3 & 4 & 11 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 15 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[277] \ 1^{1121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -233, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 2 & 3 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 & 4 \\ 4 & 4 & 1 & 11 & 2 & 3 & 3 \\ 12 & 16 & 11 & 44 & 9 & 11 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 & 2 \\ 3 & 4 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 3 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[278] \ 1^{114} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -236, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 0 & 3 & 3 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 10 & 2 \\ 2 & 3 & 0 & 2 & 0 & 10 & 2 \\ 14 & 18 & 0 & 10 & 10 & 58 & 14 \\ 3 & 4 & 0 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[279] \ 1^{1123} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -240, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 14 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 3 & 4 & 1 & 5 & 2 & 12 & 3 \\ 5 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 14 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[280] \ 1^{1243} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -240, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 20 & 2 & 1 & 3 \\ 3 & 4 & 8 & 29 & 3 & 0 & 4 \\ 5 & 8 & 10 & 40 & 4 & 0 & 6 \\ 20 & 29 & 40 & 148 & 16 & 0 & 22 \\ 2 & 3 & 4 & 16 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 22 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[281] \ 1^{1112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -240, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 3 & 8 & 4 \\ 3 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 10 & 5 \\ 2 & 3 & 2 & 4 & 0 & 4 & 2 \\ 6 & 8 & 6 & 10 & 4 & 10 & 6 \\ 3 & 4 & 3 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[282] \ 1^{1114} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -243, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 11 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[283] \ 1^{123} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -244, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 9 & 3 \\ 3 & 4 & 0 & 11 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 24 & 8 \\ 2 & 3 & 0 & 6 & 0 & 6 & 2 \\ 9 & 12 & 0 & 24 & 6 & 25 & 9 \\ 3 & 4 & 0 & 8 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[284] \ 1^{1113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -251, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 10 & 3 \\ 3 & 4 & 0 & 7 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 16 & 5 \\ 2 & 3 & 0 & 4 & 0 & 7 & 2 \\ 10 & 13 & 0 & 16 & 7 & 28 & 10 \\ 3 & 4 & 0 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[285] \quad 1^{1121} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -252, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 3 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 & 4 \\ 2 & 4 & 1 & 8 & 2 & 3 & 3 \\ 8 & 11 & 8 & 20 & 6 & 8 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 & 2 \\ 3 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[286] \quad 1^{1143} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 3 & 3 & 26 & 4 \\ 2 & 4 & 1 & 2 & 2 & 18 & 3 \\ 3 & 3 & 2 & 0 & 2 & 14 & 2 \\ 2 & 3 & 2 & 2 & 0 & 14 & 2 \\ 18 & 26 & 18 & 14 & 14 & 118 & 20 \\ 3 & 4 & 3 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[287] \quad 1^{11123} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 2 & 14 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 3 & 4 & 1 & 5 & 2 & 12 & 3 \\ 6 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 14 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[289] \quad 1^{123} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 10 & 3 \\ 3 & 4 & 0 & 11 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 25 & 8 \\ 2 & 3 & 0 & 6 & 0 & 7 & 2 \\ 10 & 13 & 0 & 25 & 7 & 28 & 10 \\ 3 & 4 & 0 & 8 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[290] \quad 1^{1243} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 11 & 3 \\ 3 & 4 & 0 & 11 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 32 & 8 \\ 2 & 3 & 0 & 6 & 0 & 8 & 2 \\ 11 & 16 & 0 & 32 & 8 & 46 & 12 \\ 3 & 4 & 0 & 8 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[291] \quad 1^{141} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 1 & 3 \\ 3 & 4 & 0 & 24 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 17 & 24 & 0 & 100 & 13 & 0 & 18 \\ 2 & 3 & 0 & 13 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 18 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[288] \quad 1^{1423} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 5 & 3 \\ 3 & 4 & 0 & 19 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 14 & 19 & 0 & 64 & 10 & 28 & 14 \\ 2 & 3 & 0 & 10 & 0 & 4 & 2 \\ 5 & 8 & 0 & 28 & 4 & 10 & 6 \\ 3 & 4 & 0 & 14 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[292] \quad 1^{1133} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 2 & 1 & 3 \\ 3 & 4 & 4 & 25 & 3 & 0 & 4 \\ 2 & 4 & 1 & 17 & 2 & 0 & 3 \\ 18 & 25 & 17 & 111 & 14 & 0 & 19 \\ 2 & 3 & 2 & 14 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 19 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[293] $1^{1112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -259, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 8 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 2 & 4 & 1 & 5 & 2 & 9 & 3 \\ 5 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 8 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[294] $1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -266, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$

$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 12 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 3 & 3 & 4 & 0 & 2 & 9 & 2 \\ 2 & 3 & 4 & 2 & 0 & 9 & 2 \\ 12 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[296] $1^{124}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 12 & 3 \\ 3 & 4 & 0 & 11 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 32 & 8 \\ 2 & 3 & 0 & 6 & 0 & 8 & 2 \\ 12 & 16 & 0 & 32 & 8 & 46 & 12 \\ 3 & 4 & 0 & 8 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[295] $1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 9 & 3 \\ 3 & 4 & 0 & 11 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 25 & 8 \\ 2 & 3 & 0 & 6 & 0 & 7 & 2 \\ 9 & 13 & 0 & 25 & 7 & 28 & 10 \\ 3 & 4 & 0 & 8 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[297] $1^{1121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -273, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 2 & 3 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 & 4 \\ 2 & 4 & 1 & 11 & 2 & 3 & 3 \\ 12 & 16 & 11 & 44 & 9 & 11 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 & 2 \\ 3 & 4 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 3 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[298] $1^{113}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 1 & 2 & 2 & 9 & 3 \\ 3 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 2 & 2 & 0 & 7 & 2 \\ 10 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[299] $1^{124}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 18 & 3 \\ 3 & 4 & 8 & 3 & 3 & 26 & 4 \\ 6 & 8 & 10 & 4 & 4 & 36 & 6 \\ 3 & 3 & 4 & 0 & 2 & 14 & 2 \\ 2 & 3 & 4 & 2 & 0 & 14 & 2 \\ 18 & 26 & 36 & 14 & 14 & 118 & 20 \\ 3 & 4 & 6 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[300] $1^{13}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -278, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 0 & 3 & 3 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 2 & 12 & 2 \\ 2 & 3 & 0 & 2 & 0 & 12 & 2 \\ 16 & 22 & 0 & 12 & 12 & 85 & 17 \\ 3 & 4 & 0 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[302] \ 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -280, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 8 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 3 & 4 & 1 & 5 & 2 & 9 & 3 \\ 5 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 8 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[301] \ 1^{13} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -280, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 1 & 3 \\ 3 & 4 & 0 & 25 & 3 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 18 & 25 & 0 & 111 & 14 & 0 & 19 \\ 2 & 3 & 0 & 14 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 19 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[303] \ 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -281, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 9 & 3 \\ 3 & 4 & 0 & 7 & 3 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 16 & 5 \\ 2 & 3 & 0 & 4 & 0 & 7 & 2 \\ 9 & 13 & 0 & 16 & 7 & 28 & 10 \\ 3 & 4 & 0 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[304] \ 1^{13} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -283, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 2 & 3 \\ 3 & 4 & 0 & 20 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 14 & 20 & 0 & 69 & 11 & 14 & 15 \\ 2 & 3 & 0 & 11 & 0 & 2 & 2 \\ 2 & 4 & 0 & 14 & 2 & 1 & 3 \\ 3 & 4 & 0 & 15 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[306] \ 1^{13} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -284, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 & 3 \\ 3 & 4 & 0 & 15 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 12 & 15 & 0 & 39 & 8 & 11 & 11 \\ 2 & 3 & 0 & 8 & 0 & 2 & 2 \\ 3 & 4 & 0 & 11 & 2 & 1 & 3 \\ 3 & 4 & 0 & 11 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[305] \ 1^{112} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -284, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 14 & 3 \\ 3 & 4 & 0 & 7 & 3 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 27 & 5 \\ 2 & 3 & 0 & 4 & 0 & 12 & 2 \\ 14 & 22 & 0 & 27 & 12 & 85 & 17 \\ 3 & 4 & 0 & 5 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[307] \ 1^{132} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -290, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 6 & 3 \\ 3 & 4 & 0 & 15 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 12 & 15 & 0 & 39 & 8 & 22 & 11 \\ 2 & 3 & 0 & 8 & 0 & 4 & 2 \\ 6 & 8 & 0 & 22 & 4 & 10 & 6 \\ 3 & 4 & 0 & 11 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[308] \ 1^{121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -291, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 2 & 3 & 3 \\ 3 & 4 & 0 & 16 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 13 & 16 & 0 & 44 & 9 & 11 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 & 2 \\ 3 & 4 & 0 & 11 & 2 & 1 & 3 \\ 3 & 4 & 0 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[309] \quad 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 2 & 13 & 3 \\ 3 & 4 & 0 & 7 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 4 & 21 & 5 \\ 2 & 3 & 0 & 4 & 0 & 9 & 2 \\ 13 & 17 & 0 & 21 & 9 & 49 & 13 \\ 3 & 4 & 0 & 5 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[310] \quad 1^{113} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 13 & 3 \\ 3 & 4 & 0 & 7 & 3 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 22 & 5 \\ 2 & 3 & 0 & 4 & 0 & 10 & 2 \\ 13 & 18 & 0 & 22 & 10 & 58 & 14 \\ 3 & 4 & 0 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[311] \quad 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 3 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 11 & 2 \\ 2 & 3 & 2 & 2 & 0 & 11 & 2 \\ 14 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[312] \quad 1^{1241} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -301, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 2 & 3 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 & 4 \\ 4 & 4 & 1 & 11 & 2 & 3 & 3 \\ 13 & 16 & 11 & 44 & 9 & 11 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 & 2 \\ 3 & 4 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 3 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[313] \quad 1^{1231} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 2 & 1 & 3 \\ 3 & 4 & 8 & 20 & 3 & 0 & 4 \\ 6 & 8 & 10 & 28 & 4 & 0 & 6 \\ 14 & 20 & 28 & 69 & 11 & 0 & 15 \\ 2 & 3 & 4 & 11 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 15 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[314] \quad 1^{214} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 13 & 3 \\ 3 & 4 & 8 & 3 & 3 & 17 & 4 \\ 6 & 8 & 10 & 4 & 4 & 24 & 6 \\ 3 & 3 & 4 & 0 & 2 & 9 & 2 \\ 2 & 3 & 4 & 2 & 0 & 9 & 2 \\ 13 & 17 & 24 & 9 & 9 & 49 & 13 \\ 3 & 4 & 6 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[315] \quad 1^{113} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 13 & 3 \\ 3 & 4 & 4 & 7 & 3 & 18 & 4 \\ 3 & 4 & 1 & 5 & 2 & 12 & 3 \\ 5 & 7 & 5 & 7 & 4 & 22 & 5 \\ 2 & 3 & 2 & 4 & 0 & 10 & 2 \\ 13 & 18 & 12 & 22 & 10 & 58 & 14 \\ 3 & 4 & 3 & 5 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[316] \quad 1^{113} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 13 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[317] \ 1^{112} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -315, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 11 & 3 \\ 3 & 4 & 0 & 7 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 21 & 5 \\ 2 & 3 & 0 & 4 & 0 & 9 & 2 \\ 11 & 17 & 0 & 21 & 9 & 49 & 13 \\ 3 & 4 & 0 & 5 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[318] \ 1^{112} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -321, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 9 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 2 & 4 & 1 & 5 & 2 & 9 & 3 \\ 5 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 9 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[319] \ 1^{121} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -323, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 & 3 \\ 3 & 4 & 0 & 16 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 11 & 16 & 0 & 44 & 9 & 11 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 & 2 \\ 3 & 4 & 0 & 11 & 2 & 1 & 3 \\ 3 & 4 & 0 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[320] \ 21^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -324, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 2 & 6 & 3 \\ 4 & 4 & 4 & 7 & 3 & 9 & 4 \\ 3 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 11 & 5 \\ 2 & 3 & 2 & 4 & 0 & 5 & 2 \\ 6 & 9 & 6 & 11 & 5 & 13 & 7 \\ 3 & 4 & 3 & 5 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[321] \ 1^{121} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -328, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 2 & 3 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 & 4 \\ 3 & 4 & 1 & 8 & 2 & 3 & 3 \\ 9 & 11 & 8 & 20 & 6 & 8 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 & 2 \\ 3 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[322] \ 1^{131} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -330, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 3 & 3 \\ 3 & 4 & 0 & 15 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 11 & 15 & 0 & 39 & 8 & 11 & 11 \\ 2 & 3 & 0 & 8 & 0 & 2 & 2 \\ 3 & 4 & 0 & 11 & 2 & 1 & 3 \\ 3 & 4 & 0 & 11 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[323] \ 1^{113} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -341, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 2 & 0 & 9 & 2 \\ 12 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[324] \ 1^{132} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -342, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 2 & 6 & 3 \\ 3 & 4 & 0 & 15 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 11 & 15 & 0 & 39 & 8 & 22 & 11 \\ 2 & 3 & 0 & 8 & 0 & 4 & 2 \\ 6 & 8 & 0 & 22 & 4 & 10 & 6 \\ 3 & 4 & 0 & 11 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[325] $1^{1112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -349, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 2 & 11 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 3 & 4 & 1 & 5 & 2 & 9 & 3 \\ 6 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 11 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[326] $1^{1114}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -350, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 3 & 3 & 18 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 10 & 2 \\ 2 & 3 & 2 & 2 & 0 & 10 & 2 \\ 14 & 18 & 12 & 10 & 10 & 58 & 14 \\ 3 & 4 & 3 & 2 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[327] $1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -353, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 2 & 3 & 3 \\ 3 & 4 & 0 & 16 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 12 & 16 & 0 & 44 & 9 & 11 & 12 \\ 2 & 3 & 0 & 9 & 0 & 2 & 2 \\ 3 & 4 & 0 & 11 & 2 & 1 & 3 \\ 3 & 4 & 0 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[328] $1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -357, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 2 & 3 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 & 4 \\ 2 & 4 & 1 & 11 & 2 & 3 & 3 \\ 11 & 16 & 11 & 44 & 9 & 11 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 & 2 \\ 3 & 4 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 3 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[329] $1^{124}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 2 & 1 & 3 \\ 3 & 4 & 8 & 24 & 3 & 0 & 4 \\ 5 & 8 & 10 & 34 & 4 & 0 & 6 \\ 17 & 24 & 34 & 100 & 13 & 0 & 18 \\ 2 & 3 & 4 & 13 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 18 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[330] $1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 2 & 3 & 3 \\ 3 & 4 & 4 & 11 & 3 & 4 & 4 \\ 3 & 4 & 1 & 8 & 2 & 3 & 3 \\ 8 & 11 & 8 & 20 & 6 & 8 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 & 2 \\ 3 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[331] $1^{1123}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -371, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 2 & 10 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 3 & 4 & 1 & 5 & 2 & 9 & 3 \\ 6 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 10 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[332] $1^{1134}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -371, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 13 & 3 \\ 3 & 4 & 0 & 7 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 21 & 5 \\ 2 & 3 & 0 & 4 & 0 & 9 & 2 \\ 13 & 17 & 0 & 21 & 9 & 49 & 13 \\ 3 & 4 & 0 & 5 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[333] $21^{1111}3^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 2 & 3 & 3 \\ 4 & 4 & 4 & 7 & 3 & 4 & 4 \\ 3 & 4 & 1 & 5 & 2 & 3 & 3 \\ 5 & 7 & 5 & 7 & 4 & 5 & 5 \\ 2 & 3 & 2 & 4 & 0 & 2 & 2 \\ 3 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 3 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[334] $1^{132}3^{111}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -378, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 6 & 3 \\ 3 & 4 & 0 & 20 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 15 & 20 & 0 & 69 & 11 & 28 & 15 \\ 2 & 3 & 0 & 11 & 0 & 4 & 2 \\ 6 & 8 & 0 & 28 & 4 & 10 & 6 \\ 3 & 4 & 0 & 15 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[335] $1^{131}3^{111}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 2 & 3 \\ 3 & 4 & 0 & 25 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 17 & 25 & 0 & 111 & 14 & 17 & 19 \\ 2 & 3 & 0 & 14 & 0 & 2 & 2 \\ 2 & 4 & 0 & 17 & 2 & 1 & 3 \\ 3 & 4 & 0 & 19 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[336] $1^{121}3^{111}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 2 & 3 & 3 \\ 3 & 4 & 4 & 21 & 3 & 4 & 4 \\ 3 & 4 & 1 & 14 & 2 & 3 & 3 \\ 16 & 21 & 14 & 80 & 12 & 14 & 16 \\ 2 & 3 & 2 & 12 & 0 & 2 & 2 \\ 3 & 4 & 3 & 14 & 2 & 1 & 3 \\ 3 & 4 & 3 & 16 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[337] $1^{114}3^{111}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 3 & 3 & 26 & 4 \\ 3 & 4 & 1 & 2 & 2 & 18 & 3 \\ 3 & 3 & 2 & 0 & 2 & 14 & 2 \\ 2 & 3 & 2 & 2 & 0 & 14 & 2 \\ 17 & 26 & 18 & 14 & 14 & 118 & 20 \\ 3 & 4 & 3 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[338] $1^{141}3^{111}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -387, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 2 & 3 \\ 3 & 4 & 0 & 24 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 17 & 24 & 0 & 100 & 13 & 17 & 18 \\ 2 & 3 & 0 & 13 & 0 & 2 & 2 \\ 2 & 4 & 0 & 17 & 2 & 1 & 3 \\ 3 & 4 & 0 & 18 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[339] $1^{124}3^{111}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 2 & 15 & 3 \\ 3 & 4 & 8 & 3 & 3 & 21 & 4 \\ 6 & 8 & 10 & 4 & 4 & 30 & 6 \\ 3 & 3 & 4 & 0 & 2 & 11 & 2 \\ 2 & 3 & 4 & 2 & 0 & 11 & 2 \\ 15 & 21 & 30 & 11 & 11 & 76 & 16 \\ 3 & 4 & 6 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[341] $1^{113}3^{111}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 3 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 2 & 0 & 12 & 2 \\ 15 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[340] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 13 & 3 \\ 3 & 4 & 0 & 11 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 33 & 8 \\ 2 & 3 & 0 & 6 & 0 & 9 & 2 \\ 13 & 17 & 0 & 33 & 9 & 49 & 13 \\ 3 & 4 & 0 & 8 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[342] $1^{121}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -410, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 & 3 \\ 3 & 4 & 0 & 21 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 15 & 21 & 0 & 80 & 12 & 14 & 16 \\ 2 & 3 & 0 & 12 & 0 & 2 & 2 \\ 3 & 4 & 0 & 14 & 2 & 1 & 3 \\ 3 & 4 & 0 & 16 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[343] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -413, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 3 & 3 & 17 & 4 \\ 3 & 4 & 1 & 2 & 2 & 12 & 3 \\ 3 & 3 & 2 & 0 & 2 & 9 & 2 \\ 2 & 3 & 2 & 2 & 0 & 9 & 2 \\ 13 & 17 & 12 & 9 & 9 & 49 & 13 \\ 3 & 4 & 3 & 2 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[345] $1^{121}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 2 & 3 & 3 \\ 3 & 4 & 0 & 21 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 16 & 21 & 0 & 80 & 12 & 14 & 16 \\ 2 & 3 & 0 & 12 & 0 & 2 & 2 \\ 3 & 4 & 0 & 14 & 2 & 1 & 3 \\ 3 & 4 & 0 & 16 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[344] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 10 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 3 & 4 & 1 & 5 & 2 & 9 & 3 \\ 5 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 10 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[346] $21^{112}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 2 & 6 & 3 \\ 4 & 4 & 4 & 7 & 3 & 8 & 4 \\ 3 & 4 & 1 & 5 & 2 & 6 & 3 \\ 5 & 7 & 5 & 7 & 4 & 10 & 5 \\ 2 & 3 & 2 & 4 & 0 & 4 & 2 \\ 6 & 8 & 6 & 10 & 4 & 10 & 6 \\ 3 & 4 & 3 & 5 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[347] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -425, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 12 & 3 \\ 3 & 4 & 0 & 7 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 21 & 5 \\ 2 & 3 & 0 & 4 & 0 & 9 & 2 \\ 12 & 17 & 0 & 21 & 9 & 49 & 13 \\ 3 & 4 & 0 & 5 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[348] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 14 & 3 \\ 3 & 4 & 0 & 11 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 41 & 8 \\ 2 & 3 & 0 & 6 & 0 & 11 & 2 \\ 14 & 21 & 0 & 41 & 11 & 76 & 16 \\ 3 & 4 & 0 & 8 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[349] $1^{1112}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -434, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 2 & 9 & 3 \\ 3 & 4 & 4 & 7 & 3 & 13 & 4 \\ 3 & 4 & 1 & 5 & 2 & 9 & 3 \\ 5 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 9 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[350] $1^{1121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 2 & 3 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 & 3 \\ 13 & 16 & 11 & 44 & 9 & 11 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 & 2 \\ 3 & 4 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 3 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[352] $1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -452, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 12 & 3 \\ 3 & 4 & 0 & 11 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 33 & 8 \\ 2 & 3 & 0 & 6 & 0 & 9 & 2 \\ 12 & 17 & 0 & 33 & 9 & 49 & 13 \\ 3 & 4 & 0 & 8 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[351] $1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -452, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 2 & 3 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 & 3 \\ 11 & 16 & 11 & 44 & 9 & 11 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 & 2 \\ 3 & 4 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 3 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[353] $1^{124}3^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 18 & 3 \\ 3 & 4 & 0 & 11 & 3 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 50 & 8 \\ 2 & 3 & 0 & 6 & 0 & 14 & 2 \\ 18 & 26 & 0 & 50 & 14 & 118 & 20 \\ 3 & 4 & 0 & 8 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[354] $1^{114}3^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -475, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 3 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 11 & 2 \\ 2 & 3 & 2 & 2 & 0 & 11 & 2 \\ 15 & 21 & 15 & 11 & 11 & 76 & 16 \\ 3 & 4 & 3 & 2 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[355] $1^{141}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 & 3 \\ 3 & 4 & 0 & 19 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 14 & 19 & 0 & 64 & 10 & 14 & 14 \\ 2 & 3 & 0 & 10 & 0 & 2 & 2 \\ 3 & 4 & 0 & 14 & 2 & 1 & 3 \\ 3 & 4 & 0 & 14 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[356] $1^{142}3^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -488, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 6 & 3 \\ 3 & 4 & 0 & 19 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 14 & 19 & 0 & 64 & 10 & 28 & 14 \\ 2 & 3 & 0 & 10 & 0 & 4 & 2 \\ 6 & 8 & 0 & 28 & 4 & 10 & 6 \\ 3 & 4 & 0 & 14 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[357] $1^{114}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -491, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 14 & 3 \\ 3 & 4 & 0 & 7 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 26 & 5 \\ 2 & 3 & 0 & 4 & 0 & 11 & 2 \\ 14 & 21 & 0 & 26 & 11 & 76 & 16 \\ 3 & 4 & 0 & 5 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[358] $1^{113}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -494, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 15 & 3 \\ 3 & 4 & 0 & 7 & 3 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 27 & 5 \\ 2 & 3 & 0 & 4 & 0 & 12 & 2 \\ 15 & 22 & 0 & 27 & 12 & 85 & 17 \\ 3 & 4 & 0 & 5 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[359] $1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 2 & 3 & 3 \\ 3 & 4 & 4 & 16 & 3 & 4 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 & 3 \\ 12 & 16 & 11 & 44 & 9 & 11 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 & 2 \\ 3 & 4 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 3 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[360] $1^{131}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -533, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 2 & 3 & 3 \\ 3 & 4 & 0 & 20 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 14 & 20 & 0 & 69 & 11 & 14 & 15 \\ 2 & 3 & 0 & 11 & 0 & 2 & 2 \\ 3 & 4 & 0 & 14 & 2 & 1 & 3 \\ 3 & 4 & 0 & 15 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[361] $1^{141}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 2 & 3 \\ 3 & 4 & 0 & 29 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 20 & 29 & 0 & 148 & 16 & 20 & 22 \\ 2 & 3 & 0 & 16 & 0 & 2 & 2 \\ 2 & 4 & 0 & 20 & 2 & 1 & 3 \\ 3 & 4 & 0 & 22 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[362] $1^{114}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -540, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 16 & 3 \\ 3 & 4 & 0 & 7 & 3 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 27 & 5 \\ 2 & 3 & 0 & 4 & 0 & 12 & 2 \\ 16 & 22 & 0 & 27 & 12 & 85 & 17 \\ 3 & 4 & 0 & 5 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[363] $1^{142}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -540, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 6 & 3 \\ 3 & 4 & 0 & 24 & 3 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 17 & 24 & 0 & 100 & 13 & 34 & 18 \\ 2 & 3 & 0 & 13 & 0 & 4 & 2 \\ 6 & 8 & 0 & 34 & 4 & 10 & 6 \\ 3 & 4 & 0 & 18 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[364] $1^{113}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 3 & 3 & 22 & 4 \\ 3 & 4 & 1 & 2 & 2 & 15 & 3 \\ 3 & 3 & 2 & 0 & 2 & 12 & 2 \\ 2 & 3 & 2 & 2 & 0 & 12 & 2 \\ 16 & 22 & 15 & 12 & 12 & 85 & 17 \\ 3 & 4 & 3 & 2 & 2 & 17 & 3 \end{pmatrix}_{b_0}$$

[365] $1^{1^{141}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -555, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 2 & 3 & 3 \\ 3 & 4 & 0 & 20 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 15 & 20 & 0 & 69 & 11 & 14 & 15 \\ 2 & 3 & 0 & 11 & 0 & 2 & 2 \\ 3 & 4 & 0 & 14 & 2 & 1 & 3 \\ 3 & 4 & 0 & 15 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[366] $1^{1^{144}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -577, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 15 & 3 \\ 3 & 4 & 0 & 7 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 26 & 5 \\ 2 & 3 & 0 & 4 & 0 & 11 & 2 \\ 15 & 21 & 0 & 26 & 11 & 76 & 16 \\ 3 & 4 & 0 & 5 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[367] $1^{1^{133}}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 17 & 3 \\ 3 & 4 & 0 & 7 & 3 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 32 & 5 \\ 2 & 3 & 0 & 4 & 0 & 14 & 2 \\ 17 & 26 & 0 & 32 & 14 & 118 & 20 \\ 3 & 4 & 0 & 5 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[368] $1^{1^{131}}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -614, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 & 3 \\ 3 & 4 & 0 & 25 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 17 & 25 & 0 & 111 & 14 & 17 & 19 \\ 2 & 3 & 0 & 14 & 0 & 2 & 2 \\ 3 & 4 & 0 & 17 & 2 & 1 & 3 \\ 3 & 4 & 0 & 19 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[369] $21^{1^{121}}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 8 & 2 & 3 & 3 \\ 4 & 4 & 4 & 11 & 3 & 4 & 4 \\ 3 & 4 & 1 & 8 & 2 & 3 & 3 \\ 8 & 11 & 8 & 20 & 6 & 8 & 8 \\ 2 & 3 & 2 & 6 & 0 & 2 & 2 \\ 3 & 4 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 8 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[370] $1^{1^{144}}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -630, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 3 & 3 & 26 & 4 \\ 3 & 4 & 1 & 2 & 2 & 18 & 3 \\ 3 & 3 & 2 & 0 & 2 & 14 & 2 \\ 2 & 3 & 2 & 2 & 0 & 14 & 2 \\ 18 & 26 & 18 & 14 & 14 & 118 & 20 \\ 3 & 4 & 3 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[371] $1^{1^{124}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -640, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 15 & 3 \\ 3 & 4 & 0 & 11 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 6 & 41 & 8 \\ 2 & 3 & 0 & 6 & 0 & 11 & 2 \\ 15 & 21 & 0 & 41 & 11 & 76 & 16 \\ 3 & 4 & 0 & 8 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[372] $21^{1^{121}}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -660, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 11 & 2 & 3 & 3 \\ 4 & 4 & 4 & 16 & 3 & 4 & 4 \\ 3 & 4 & 1 & 11 & 2 & 3 & 3 \\ 11 & 16 & 11 & 44 & 9 & 11 & 12 \\ 2 & 3 & 2 & 9 & 0 & 2 & 2 \\ 3 & 4 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 3 & 12 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[373] $21^{11123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -676, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 2 & 9 & 3 \\ 4 & 4 & 4 & 7 & 3 & 13 & 4 \\ 3 & 4 & 1 & 5 & 2 & 9 & 3 \\ 5 & 7 & 5 & 7 & 4 & 16 & 5 \\ 2 & 3 & 2 & 4 & 0 & 7 & 2 \\ 9 & 13 & 9 & 16 & 7 & 28 & 10 \\ 3 & 4 & 3 & 5 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[374] $1^{131}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 2 & 3 & 3 \\ 3 & 4 & 0 & 25 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 18 & 25 & 0 & 111 & 14 & 17 & 19 \\ 2 & 3 & 0 & 14 & 0 & 2 & 2 \\ 3 & 4 & 0 & 17 & 2 & 1 & 3 \\ 3 & 4 & 0 & 19 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[375] $1^{114}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -770, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 18 & 3 \\ 3 & 4 & 0 & 7 & 3 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 4 & 32 & 5 \\ 2 & 3 & 0 & 4 & 0 & 14 & 2 \\ 18 & 26 & 0 & 32 & 14 & 118 & 20 \\ 3 & 4 & 0 & 5 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[376] $1^{141}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -771, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 2 & 3 & 3 \\ 3 & 4 & 0 & 24 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 17 & 24 & 0 & 100 & 13 & 17 & 18 \\ 2 & 3 & 0 & 13 & 0 & 2 & 2 \\ 3 & 4 & 0 & 17 & 2 & 1 & 3 \\ 3 & 4 & 0 & 18 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[377] $1^{141}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 2 & 3 & 3 \\ 3 & 4 & 0 & 29 & 3 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 20 & 29 & 0 & 148 & 16 & 20 & 22 \\ 2 & 3 & 0 & 16 & 0 & 2 & 2 \\ 3 & 4 & 0 & 20 & 2 & 1 & 3 \\ 3 & 4 & 0 & 22 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

4.949 NHC(-2)(-3) + $\mathfrak{su}_2$ + su2n3mix + su2n3mix

$T = 8$ [0] $1^{123}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 15 & 3 \\ 3 & 4 & 0 & 7 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 26 & 5 \\ 15 & 21 & 0 & 26 & 80 & 16 \\ 3 & 4 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1] $1^{1211}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 6 & 3 \\ 3 & 4 & 8 & 12 & 7 & 4 \\ 5 & 8 & 10 & 16 & 10 & 6 \\ 8 & 12 & 16 & 25 & 15 & 9 \\ 6 & 7 & 10 & 15 & 7 & 5 \\ 3 & 4 & 6 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[2] $1^{1212}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 4 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 9 & 2 \\ 11 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}2}3^{\overline{11}12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 7, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 16 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 2 & 4 & 1 & 5 & 14 & 3 \\ 5 & 7 & 5 & 7 & 26 & 5 \\ 16 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}2}3^{\overline{11}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 7, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 8 & 3 \\ 3 & 4 & 4 & 12 & 11 & 4 \\ 2 & 4 & 1 & 8 & 8 & 3 \\ 9 & 12 & 8 & 25 & 23 & 9 \\ 8 & 11 & 8 & 23 & 20 & 8 \\ 3 & 4 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1211}3}1^{\overline{11}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 7, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 5 & 3 \\ 3 & 4 & 8 & 12 & 7 & 4 \\ 5 & 8 & 10 & 16 & 10 & 6 \\ 8 & 12 & 16 & 25 & 15 & 9 \\ 5 & 7 & 10 & 15 & 7 & 5 \\ 3 & 4 & 6 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1211}3}1^{\overline{11}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 8, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 6 & 3 \\ 3 & 4 & 8 & 12 & 7 & 4 \\ 6 & 8 & 10 & 16 & 10 & 6 \\ 9 & 12 & 16 & 25 & 15 & 9 \\ 6 & 7 & 10 & 15 & 7 & 5 \\ 3 & 4 & 6 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}2}3^{\overline{11}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 12, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 8 & 0 & 12 & 10 & 6 \\ 6 & 7 & 0 & 10 & 7 & 5 \\ 3 & 4 & 0 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1211}3}1^{\overline{11}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 12, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 5 & 3 \\ 3 & 4 & 8 & 12 & 7 & 4 \\ 6 & 8 & 10 & 16 & 10 & 6 \\ 9 & 12 & 16 & 25 & 15 & 9 \\ 5 & 7 & 10 & 15 & 7 & 5 \\ 3 & 4 & 6 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{111}2}3^{\overline{11}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 12, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 4 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 \\ 3 & 5 & 4 & 3 & 6 & 4 \\ 3 & 3 & 3 & 0 & 4 & 2 \\ 4 & 7 & 6 & 4 & 7 & 5 \\ 3 & 4 & 4 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}2}3^{\overline{11}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 12, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 11 & 3 \\ 3 & 4 & 8 & 3 & 16 & 4 \\ 5 & 8 & 10 & 4 & 22 & 6 \\ 3 & 3 & 4 & 0 & 9 & 2 \\ 11 & 16 & 22 & 9 & 44 & 12 \\ 3 & 4 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}2}3^{\overline{11}11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 12, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 \\ 3 & 5 & 4 & 3 & 6 & 4 \\ 3 & 3 & 3 & 0 & 4 & 2 \\ 5 & 7 & 6 & 4 & 7 & 5 \\ 3 & 4 & 4 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{111}2}3^{\overline{11}12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 16, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

- [13] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 4 & 3 & 3 \\ 3 & 4 & 14 & 7 & 3 & 4 \\ 9 & 14 & 37 & 17 & 8 & 11 \\ 4 & 7 & 17 & 7 & 4 & 5 \\ 3 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 11 & 5 & 2 & 3 \end{pmatrix}_{b_0}$
- [14] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 5 & 5 & 3 \\ 3 & 4 & 9 & 7 & 7 & 4 \\ 5 & 9 & 13 & 11 & 11 & 7 \\ 5 & 7 & 11 & 7 & 9 & 5 \\ 5 & 7 & 11 & 9 & 7 & 5 \\ 3 & 4 & 7 & 5 & 5 & 3 \end{pmatrix}_{b_0}$
- [16] $21^{11}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 3 & 6 & 12 & 4 \\ 5 & 4 & 4 & 7 & 12 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 6 & 7 & 5 & 7 & 15 & 5 \\ 12 & 12 & 8 & 15 & 25 & 9 \\ 4 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$
- [15] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 8 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 2 & 4 & 1 & 5 & 8 & 3 \\ 6 & 7 & 5 & 7 & 15 & 5 \\ 8 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$
- [17] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 11 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 20 & 5 \\ 11 & 16 & 0 & 20 & 44 & 12 \\ 3 & 4 & 0 & 5 & 12 & 3 \end{pmatrix}_{b_0}$
- [18] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 6 & 8 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 6 & 7 & 5 & 7 & 15 & 5 \\ 8 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$
- [19] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 8 & 8 & 3 \\ 3 & 4 & 4 & 12 & 11 & 4 \\ 3 & 4 & 1 & 8 & 8 & 3 \\ 8 & 12 & 8 & 25 & 23 & 9 \\ 8 & 11 & 8 & 23 & 20 & 8 \\ 3 & 4 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$
- [20] $1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 & 5 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 \\ 5 & 8 & 10 & 4 & 10 & 6 \\ 3 & 3 & 4 & 0 & 4 & 2 \\ 5 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$
- [21] $1^{111}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 5 & 3 & 3 \\ 3 & 4 & 14 & 7 & 3 & 4 \\ 11 & 14 & 37 & 17 & 8 & 11 \\ 5 & 7 & 17 & 7 & 4 & 5 \\ 3 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 11 & 5 & 2 & 3 \end{pmatrix}_{b_0}$
- [22] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 \\ 4 & 5 & 4 & 6 & 3 & 4 \\ 5 & 7 & 6 & 7 & 4 & 5 \\ 3 & 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[23] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 5 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 \\ 2 & 4 & 1 & 2 & 5 & 3 \\ 3 & 3 & 2 & 0 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[25] $1^{123}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 & 16 & 3 \\ 3 & 4 & 8 & 3 & 25 & 4 \\ 5 & 8 & 10 & 4 & 34 & 6 \\ 3 & 3 & 4 & 0 & 14 & 2 \\ 16 & 25 & 34 & 14 & 111 & 19 \\ 3 & 4 & 6 & 2 & 19 & 3 \end{pmatrix}_{b_0}$

[24] $1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 0 & 11 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 9 & 11 & 0 & 20 & 23 & 8 \\ 9 & 12 & 0 & 23 & 25 & 9 \\ 3 & 4 & 0 & 8 & 9 & 3 \end{pmatrix}_{b_0}$

[26] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 8 & 0 & 12 & 10 & 6 \\ 5 & 7 & 0 & 10 & 7 & 5 \\ 3 & 4 & 0 & 6 & 5 & 3 \end{pmatrix}_{b_0}$

[27] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 13 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 2 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 13 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$

[28] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 9 & 6 & 3 \\ 3 & 4 & 4 & 12 & 7 & 4 \\ 4 & 4 & 1 & 8 & 5 & 3 \\ 9 & 12 & 8 & 25 & 15 & 9 \\ 6 & 7 & 5 & 15 & 7 & 5 \\ 3 & 4 & 3 & 9 & 5 & 3 \end{pmatrix}_{b_0}$

[29] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 3 \\ 3 & 4 & 0 & 12 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 12 & 0 & 25 & 23 & 9 \\ 8 & 11 & 0 & 23 & 20 & 8 \\ 3 & 4 & 0 & 9 & 8 & 3 \end{pmatrix}_{b_0}$

[30] $1^{123}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 9 & 9 & 3 \\ 3 & 4 & 4 & 12 & 11 & 4 \\ 2 & 4 & 1 & 8 & 8 & 3 \\ 9 & 12 & 8 & 25 & 23 & 9 \\ 9 & 11 & 8 & 23 & 20 & 8 \\ 3 & 4 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$

[31] $1^{111}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 13 & 3 \\ 3 & 4 & 0 & 7 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 21 & 5 \\ 13 & 17 & 0 & 21 & 55 & 13 \\ 3 & 4 & 0 & 5 & 13 & 3 \end{pmatrix}_{b_0}$

[32] $1^{111}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 & 3 \\ 3 & 4 & 14 & 7 & 3 & 4 \\ 10 & 14 & 37 & 17 & 8 & 11 \\ 5 & 7 & 17 & 7 & 4 & 5 \\ 3 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 11 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{131} 3^{111} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 5 & 3 \\ 3 & 4 & 12 & 3 & 7 & 4 \\ 8 & 12 & 25 & 6 & 15 & 9 \\ 3 & 3 & 6 & 0 & 4 & 2 \\ 5 & 7 & 15 & 4 & 7 & 5 \\ 3 & 4 & 9 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{124} 3^{111} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 20 & 3 \\ 3 & 4 & 8 & 3 & 29 & 4 \\ 5 & 8 & 10 & 4 & 40 & 6 \\ 3 & 3 & 4 & 0 & 16 & 2 \\ 20 & 29 & 40 & 16 & 148 & 22 \\ 3 & 4 & 6 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{122} 3^{111} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 6 & 3 \\ 3 & 4 & 8 & 7 & 7 & 4 \\ 6 & 8 & 10 & 10 & 10 & 6 \\ 4 & 7 & 10 & 7 & 9 & 5 \\ 6 & 7 & 10 & 9 & 7 & 5 \\ 3 & 4 & 6 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{131} 3^{111} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 & 3 \\ 3 & 4 & 12 & 7 & 3 & 4 \\ 9 & 12 & 25 & 15 & 6 & 9 \\ 4 & 7 & 15 & 7 & 4 & 5 \\ 3 & 3 & 6 & 4 & 0 & 2 \\ 3 & 4 & 9 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{111} 3^{111} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 \\ 3 & 4 & 1 & 2 & 5 & 3 \\ 3 & 3 & 2 & 0 & 4 & 2 \\ 4 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{121} 3^{111} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 4 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 \\ 6 & 8 & 10 & 4 & 10 & 6 \\ 3 & 3 & 4 & 0 & 4 & 2 \\ 4 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{111} 3^{111} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 7 & 7 & 4 \\ 3 & 4 & 1 & 5 & 5 & 3 \\ 4 & 7 & 5 & 7 & 9 & 5 \\ 6 & 7 & 5 & 9 & 7 & 5 \\ 3 & 4 & 3 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{141} 3^{111} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 & 3 \\ 3 & 4 & 16 & 7 & 3 & 4 \\ 12 & 16 & 46 & 20 & 8 & 12 \\ 4 & 7 & 20 & 7 & 4 & 5 \\ 3 & 3 & 8 & 4 & 0 & 2 \\ 3 & 4 & 12 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{112} 3^{111} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 7 & 11 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 4 & 7 & 5 & 7 & 14 & 5 \\ 9 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{111} 3^{111} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 2 \\ 5 & 7 & 0 & 4 & 7 & 5 \\ 3 & 4 & 0 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[43] & 1^{121} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 8 & 3 & 5 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 \\ 8 & 13 & 28 & 7 & 16 & 10 \\ 3 & 3 & 7 & 0 & 4 & 2 \\ 5 & 7 & 16 & 4 & 7 & 5 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[42] & 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 7 & 2 \\ 8 & 12 & 0 & 7 & 25 & 9 \\ 3 & 4 & 0 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[45] & 1^{1122} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 9 & 8 & 3 \\ 3 & 4 & 4 & 12 & 11 & 4 \\ 4 & 4 & 1 & 8 & 8 & 3 \\ 9 & 12 & 8 & 25 & 23 & 9 \\ 8 & 11 & 8 & 23 & 20 & 8 \\ 3 & 4 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0} \\
[47] & 1^{123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 9 & 9 & 3 \\ 3 & 4 & 4 & 12 & 11 & 4 \\ 3 & 4 & 1 & 8 & 8 & 3 \\ 9 & 12 & 8 & 25 & 23 & 9 \\ 9 & 11 & 8 & 23 & 20 & 8 \\ 3 & 4 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0} \\
[46] & 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 & 8 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 0 & 7 & 2 \\ 8 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[48] & 21^{112} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 3 & 8 & 8 & 3 \\ 4 & 4 & 4 & 12 & 11 & 4 \\ 3 & 4 & 1 & 8 & 8 & 3 \\ 8 & 12 & 8 & 25 & 23 & 9 \\ 8 & 11 & 8 & 23 & 20 & 8 \\ 3 & 4 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0} \\
[49] & 1^{141} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 11 & 3 & 5 & 3 \\ 3 & 4 & 16 & 3 & 7 & 4 \\ 11 & 16 & 46 & 8 & 20 & 12 \\ 3 & 3 & 8 & 0 & 4 & 2 \\ 5 & 7 & 20 & 4 & 7 & 5 \\ 3 & 4 & 12 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[50] & 1^{311} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 8 & 5 & 3 & 3 \\ 3 & 4 & 9 & 7 & 3 & 4 \\ 8 & 9 & 13 & 11 & 5 & 7 \\ 5 & 7 & 11 & 7 & 4 & 5 \\ 3 & 3 & 5 & 4 & 0 & 2 \\ 3 & 4 & 7 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[51] & 1^{122} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 42, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 & 10 & 3 \\ 3 & 4 & 8 & 3 & 16 & 4 \\ 5 & 8 & 10 & 4 & 22 & 6 \\ 3 & 3 & 4 & 0 & 9 & 2 \\ 10 & 16 & 22 & 9 & 44 & 12 \\ 3 & 4 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[52] & 1^{112} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 42, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 8 & 3 \\ 3 & 4 & 8 & 3 & 11 & 4 \\ 5 & 8 & 10 & 4 & 16 & 6 \\ 3 & 3 & 4 & 0 & 6 & 2 \\ 8 & 11 & 16 & 6 & 20 & 8 \\ 3 & 4 & 6 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[53] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 8 & 8 & 3 \\ 3 & 4 & 4 & 12 & 11 & 4 \\ 2 & 4 & 1 & 8 & 8 & 3 \\ 8 & 12 & 8 & 25 & 23 & 9 \\ 8 & 11 & 8 & 23 & 20 & 8 \\ 3 & 4 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$

[54] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 & 4 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 \\ 6 & 9 & 13 & 5 & 11 & 7 \\ 3 & 3 & 5 & 0 & 4 & 2 \\ 4 & 7 & 11 & 4 & 7 & 5 \\ 3 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[56] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 8 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 0 & 6 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[55] $1^{132}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 & 8 & 3 \\ 3 & 4 & 12 & 3 & 11 & 4 \\ 8 & 12 & 25 & 6 & 24 & 9 \\ 3 & 3 & 6 & 0 & 6 & 2 \\ 8 & 11 & 24 & 6 & 20 & 8 \\ 3 & 4 & 9 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[57] $1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 5 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 \\ 4 & 4 & 1 & 2 & 5 & 3 \\ 3 & 3 & 2 & 0 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[58] $1^{131}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 & 5 & 3 \\ 3 & 4 & 22 & 3 & 7 & 4 \\ 14 & 22 & 85 & 12 & 27 & 17 \\ 3 & 3 & 12 & 0 & 4 & 2 \\ 5 & 7 & 27 & 4 & 7 & 5 \\ 3 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[59] $1^{121}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 6 & 5 & 3 \\ 3 & 4 & 8 & 7 & 7 & 4 \\ 5 & 8 & 10 & 10 & 10 & 6 \\ 6 & 7 & 10 & 7 & 9 & 5 \\ 5 & 7 & 10 & 9 & 7 & 5 \\ 3 & 4 & 6 & 5 & 5 & 3 \end{pmatrix}_{b_0}$

[61] $1^{122}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 6 & 4 & 3 \\ 3 & 4 & 13 & 7 & 7 & 4 \\ 9 & 13 & 28 & 16 & 16 & 10 \\ 6 & 7 & 16 & 7 & 9 & 5 \\ 4 & 7 & 16 & 9 & 7 & 5 \\ 3 & 4 & 10 & 5 & 5 & 3 \end{pmatrix}_{b_0}$

[60] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 10 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 9 & 2 \\ 10 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[63] $1^{113}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 4 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 11 & 2 \\ 14 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[62] $1^{1^{132}}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 11 & 3 \\ 3 & 4 & 0 & 15 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 11 & 15 & 0 & 39 & 42 & 11 \\ 11 & 16 & 0 & 42 & 44 & 12 \\ 3 & 4 & 0 & 11 & 12 & 3 \end{pmatrix}_{b_0}$$

[65] $1^{1^{132}}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 14 & 3 \\ 3 & 4 & 8 & 3 & 20 & 4 \\ 5 & 8 & 10 & 4 & 28 & 6 \\ 3 & 3 & 4 & 0 & 11 & 2 \\ 14 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[67] $1^{1^{113}}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 9 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 \\ 4 & 4 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 0 & 7 & 2 \\ 9 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[66] $1^{1^{111}}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 5 & 3 \\ 3 & 4 & 4 & 7 & 7 & 4 \\ 2 & 4 & 1 & 5 & 5 & 3 \\ 6 & 7 & 5 & 7 & 9 & 5 \\ 5 & 7 & 5 & 9 & 7 & 5 \\ 3 & 4 & 3 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[64] $1^{1^{111}}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 9 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 2 & 4 & 1 & 5 & 8 & 3 \\ 6 & 7 & 5 & 7 & 15 & 5 \\ 9 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[73] $1^{1^{123}}3^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 19 & 3 \\ 3 & 4 & 0 & 11 & 29 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 56 & 8 \\ 19 & 29 & 0 & 56 & 148 & 22 \\ 3 & 4 & 0 & 8 & 22 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{1^{112}}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 3 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 6 & 2 \\ 8 & 11 & 0 & 6 & 20 & 8 \\ 3 & 4 & 0 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[71] $1^{1^{312}}3^{11}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 8 & 3 \\ 3 & 4 & 16 & 3 & 11 & 4 \\ 11 & 16 & 46 & 8 & 32 & 12 \\ 3 & 3 & 8 & 0 & 6 & 2 \\ 8 & 11 & 32 & 6 & 20 & 8 \\ 3 & 4 & 12 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[70] $1^{1^{113}}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 16 & 3 \\ 3 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 13 & 2 \\ 16 & 24 & 0 & 13 & 100 & 18 \\ 3 & 4 & 0 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[69] $1^{1^{112}}3^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 12 & 2 \\ 14 & 21 & 0 & 12 & 80 & 16 \\ 3 & 4 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[68] $1^{122}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 11 & 8 & 3 & 3 \\ 3 & 4 & 13 & 11 & 3 & 4 \\ 11 & 13 & 28 & 25 & 7 & 10 \\ 8 & 11 & 25 & 20 & 6 & 8 \\ 3 & 3 & 7 & 6 & 0 & 2 \\ 3 & 4 & 10 & 8 & 2 & 3 \end{pmatrix}_{b_0}$

[74] $1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 15 & 5 \\ 10 & 12 & 0 & 15 & 25 & 9 \\ 3 & 4 & 0 & 5 & 9 & 3 \end{pmatrix}_{b_0}$

[76] $1^{123}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 6 & 15 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 3 & 4 & 1 & 5 & 14 & 3 \\ 6 & 7 & 5 & 7 & 26 & 5 \\ 15 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$

[75] $1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 4 & 7 & 5 & 7 & 15 & 5 \\ 9 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$

[77] $1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 8 & 7 & 3 \\ 3 & 4 & 4 & 12 & 11 & 4 \\ 3 & 4 & 1 & 8 & 8 & 3 \\ 8 & 12 & 8 & 25 & 23 & 9 \\ 7 & 11 & 8 & 23 & 20 & 8 \\ 3 & 4 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$

[78] $1^{111}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 7 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 21 & 5 \\ 13 & 17 & 0 & 21 & 55 & 13 \\ 3 & 4 & 0 & 5 & 13 & 3 \end{pmatrix}_{b_0}$

[80] $1^{123}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 & 11 & 3 \\ 3 & 4 & 8 & 3 & 15 & 4 \\ 5 & 8 & 10 & 4 & 22 & 6 \\ 3 & 3 & 4 & 0 & 8 & 2 \\ 11 & 15 & 22 & 8 & 39 & 11 \\ 3 & 4 & 6 & 2 & 11 & 3 \end{pmatrix}_{b_0}$

[79] $1^{121}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 & 5 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 \\ 7 & 8 & 10 & 4 & 10 & 6 \\ 3 & 3 & 4 & 0 & 4 & 2 \\ 5 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[81] $1^{111}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 0 & 7 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 9 & 5 \\ 5 & 7 & 0 & 9 & 7 & 5 \\ 3 & 4 & 0 & 5 & 5 & 3 \end{pmatrix}_{b_0}$

[82] $1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 6 & 3 \\ 3 & 4 & 9 & 7 & 7 & 4 \\ 8 & 9 & 13 & 11 & 11 & 7 \\ 5 & 7 & 11 & 7 & 9 & 5 \\ 6 & 7 & 11 & 9 & 7 & 5 \\ 3 & 4 & 7 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[83] $1^{1^{12}3^{11}1^{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 11 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 2 & 4 & 1 & 5 & 11 & 3 \\ 6 & 7 & 5 & 7 & 20 & 5 \\ 11 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[85] $1^{1^{14}3^{11}1^{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 18 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 4 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 14 & 2 \\ 18 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[84] $1^{1^{12}3^{11}1^{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 15 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 2 & 4 & 1 & 5 & 14 & 3 \\ 6 & 7 & 5 & 7 & 26 & 5 \\ 15 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[86] $1^{1^{12}3^{11}1^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 8 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 \\ 4 & 4 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 0 & 6 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[87] $1^{1^{13}3^{11}1^1} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 15 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 8 & 2 \\ 11 & 15 & 11 & 8 & 39 & 11 \\ 3 & 4 & 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[90] $1^{1^{23}3^{11}1^{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 17 & 3 \\ 3 & 4 & 0 & 11 & 25 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 48 & 8 \\ 17 & 25 & 0 & 48 & 111 & 19 \\ 3 & 4 & 0 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$

[89] $1^{1^{22}3^{11}1^{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 9 & 3 \\ 3 & 4 & 0 & 11 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 23 & 8 \\ 9 & 12 & 0 & 23 & 25 & 9 \\ 3 & 4 & 0 & 8 & 9 & 3 \end{pmatrix}_{b_0}$$

[88] $1^{1^{12}3^{11}1^{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 9 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 2 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 15 & 5 \\ 9 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[94] $1^{1^{42}3^{11}1^{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 7 & 3 \\ 3 & 4 & 26 & 3 & 11 & 4 \\ 16 & 26 & 118 & 14 & 50 & 20 \\ 3 & 3 & 14 & 0 & 6 & 2 \\ 7 & 11 & 50 & 6 & 20 & 8 \\ 3 & 4 & 20 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[100] $1^{1^{12}3^{11}1^{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 9 & 2 \\ 11 & 16 & 0 & 9 & 44 & 12 \\ 3 & 4 & 0 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[101] $1^{1122}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 8 & 3 \\ 3 & 4 & 4 & 12 & 11 & 4 \\ 3 & 4 & 1 & 8 & 8 & 3 \\ 9 & 12 & 8 & 25 & 23 & 9 \\ 8 & 11 & 8 & 23 & 20 & 8 \\ 3 & 4 & 3 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

[99] $1^{1122}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 7 & 3 \\ 3 & 4 & 4 & 7 & 11 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 6 & 7 & 5 & 7 & 14 & 5 \\ 7 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[98] $1^{1222}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 7 & 3 \\ 3 & 4 & 8 & 3 & 11 & 4 \\ 6 & 8 & 10 & 4 & 16 & 6 \\ 3 & 3 & 4 & 0 & 6 & 2 \\ 7 & 11 & 16 & 6 & 20 & 8 \\ 3 & 4 & 6 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[97] $1^{1233}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 18 & 3 \\ 3 & 4 & 8 & 3 & 25 & 4 \\ 6 & 8 & 10 & 4 & 34 & 6 \\ 3 & 3 & 4 & 0 & 14 & 2 \\ 18 & 25 & 34 & 14 & 111 & 19 \\ 3 & 4 & 6 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[96] $1^{1222}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 7 & 3 \\ 3 & 4 & 13 & 3 & 11 & 4 \\ 8 & 13 & 28 & 7 & 25 & 10 \\ 3 & 3 & 7 & 0 & 6 & 2 \\ 7 & 11 & 25 & 6 & 20 & 8 \\ 3 & 4 & 10 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[95] $1^{1222}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 13 & 3 \\ 3 & 4 & 0 & 11 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 31 & 8 \\ 13 & 16 & 0 & 31 & 44 & 12 \\ 3 & 4 & 0 & 8 & 12 & 3 \end{pmatrix}_{b_0}$$

[93] $1^{1122}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 0 & 6 & 2 \\ 7 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[92] $1^{1322}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 7 & 3 \\ 3 & 4 & 12 & 3 & 11 & 4 \\ 9 & 12 & 25 & 6 & 24 & 9 \\ 3 & 3 & 6 & 0 & 6 & 2 \\ 7 & 11 & 24 & 6 & 20 & 8 \\ 3 & 4 & 9 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[91] $1^{1422}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 7 & 3 \\ 3 & 4 & 16 & 3 & 11 & 4 \\ 12 & 16 & 46 & 8 & 32 & 12 \\ 3 & 3 & 8 & 0 & 6 & 2 \\ 7 & 11 & 32 & 6 & 20 & 8 \\ 3 & 4 & 12 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[102] $1^{1123}3^{11}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 14 & 3 \\ 3 & 4 & 0 & 7 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 25 & 5 \\ 14 & 20 & 0 & 25 & 69 & 15 \\ 3 & 4 & 0 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

[103] $1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 & 8 & 3 \\ 3 & 4 & 9 & 3 & 11 & 4 \\ 7 & 9 & 13 & 5 & 17 & 7 \\ 3 & 3 & 5 & 0 & 6 & 2 \\ 8 & 11 & 17 & 6 & 20 & 8 \\ 3 & 4 & 7 & 2 & 8 & 3 \end{pmatrix}_{b_0}$

[104] $1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 9 & 2 \\ 11 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[105] $1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 7 & 2 \\ 9 & 12 & 0 & 7 & 25 & 9 \\ 3 & 4 & 0 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[106] $1^{\overline{112}} 3^{\overline{11}} 1^{\overline{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 10 & 2 \\ 12 & 17 & 0 & 10 & 55 & 13 \\ 3 & 4 & 0 & 2 & 13 & 3 \end{pmatrix}_{b_0}$

[107] $1^{\overline{121}} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 & 4 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 \\ 9 & 13 & 28 & 7 & 16 & 10 \\ 3 & 3 & 7 & 0 & 4 & 2 \\ 4 & 7 & 16 & 4 & 7 & 5 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[108] $1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 \\ 3 & 4 & 1 & 2 & 5 & 3 \\ 3 & 3 & 2 & 0 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[109] $1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 & 5 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 \\ 6 & 9 & 13 & 5 & 11 & 7 \\ 3 & 3 & 5 & 0 & 4 & 2 \\ 5 & 7 & 11 & 4 & 7 & 5 \\ 3 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[110] $1^{\overline{124}} 3^{\overline{11}} 1^{\overline{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 16 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 2 & 4 & 1 & 5 & 14 & 3 \\ 6 & 7 & 5 & 7 & 26 & 5 \\ 16 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$

[111] $1^{\overline{122}} 3^{\overline{11}} 1^{\overline{1}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 8 & 3 \\ 3 & 4 & 4 & 7 & 11 & 4 \\ 2 & 4 & 1 & 5 & 8 & 3 \\ 6 & 7 & 5 & 7 & 14 & 5 \\ 8 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$

[112] $1^{\overline{112}} 3^{\overline{11}} 1^{\overline{12}} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 7 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 21 & 5 \\ 12 & 17 & 0 & 21 & 55 & 13 \\ 3 & 4 & 0 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[113] $1^{\overline{1111}}3^{\overline{11}}1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 9 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 4 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 15 & 5 \\ 9 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$

[114] $1^{\overline{121}}3^{\overline{11}}1^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 & 4 & 3 \\ 3 & 4 & 18 & 3 & 7 & 4 \\ 12 & 18 & 58 & 10 & 22 & 14 \\ 3 & 3 & 10 & 0 & 4 & 2 \\ 4 & 7 & 22 & 4 & 7 & 5 \\ 3 & 4 & 14 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[115] $1^{\overline{122}}3^{\overline{11}}1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 9 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 15 & 5 \\ 9 & 12 & 0 & 15 & 25 & 9 \\ 3 & 4 & 0 & 5 & 9 & 3 \end{pmatrix}_{b_0}$

[116] $1^{\overline{113}}3^{\overline{11}}1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 \\ 3 & 4 & 0 & 3 & 15 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 8 & 2 \\ 11 & 15 & 0 & 8 & 39 & 11 \\ 3 & 4 & 0 & 2 & 11 & 3 \end{pmatrix}_{b_0}$

[117] $1^{\overline{111}}3^{\overline{11}}1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 0 & 7 & 2 \\ 8 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[118] $1^{\overline{1311}}3^{\overline{11}}1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 7 & 5 & 5 & 3 \\ 3 & 4 & 8 & 7 & 7 & 4 \\ 7 & 8 & 10 & 10 & 10 & 6 \\ 5 & 7 & 10 & 7 & 9 & 5 \\ 5 & 7 & 10 & 9 & 7 & 5 \\ 3 & 4 & 6 & 5 & 5 & 3 \end{pmatrix}_{b_0}$

[119] $1^{\overline{1212}}3^{\overline{11}}1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 & 11 & 3 \\ 3 & 4 & 8 & 3 & 16 & 4 \\ 6 & 8 & 10 & 4 & 22 & 6 \\ 3 & 3 & 4 & 0 & 9 & 2 \\ 11 & 16 & 22 & 9 & 44 & 12 \\ 3 & 4 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[120] $1^{\overline{1113}}3^{\overline{11}}1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 10 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 4 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 15 & 5 \\ 10 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$

[121] $1^{\overline{1112}}3^{\overline{11}}1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 11 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 4 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 11 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$

[122] $1^{\overline{1111}}3^{\overline{11}}1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 4 & 5 & 6 & 3 \\ 3 & 4 & 4 & 7 & 7 & 4 \\ 4 & 4 & 1 & 5 & 5 & 3 \\ 5 & 7 & 5 & 7 & 9 & 5 \\ 6 & 7 & 5 & 9 & 7 & 5 \\ 3 & 4 & 3 & 5 & 5 & 3 \end{pmatrix}_{b_0} \\
[123] & 1^{123} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 3 & 13 & 3 \\ 3 & 4 & 8 & 3 & 20 & 4 \\ 6 & 8 & 10 & 4 & 28 & 6 \\ 3 & 3 & 4 & 0 & 11 & 2 \\ 13 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[124] & 1^{121} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 6 & 3 & 5 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 \\ 6 & 8 & 10 & 4 & 10 & 6 \\ 3 & 3 & 4 & 0 & 4 & 2 \\ 5 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[125] & 1^{121} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 12 & 3 & 5 & 3 \\ 3 & 4 & 18 & 3 & 7 & 4 \\ 12 & 18 & 58 & 10 & 22 & 14 \\ 3 & 3 & 10 & 0 & 4 & 2 \\ 5 & 7 & 22 & 4 & 7 & 5 \\ 3 & 4 & 14 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[126] & 1^{111} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 5 & 5 & 3 \\ 3 & 4 & 4 & 7 & 7 & 4 \\ 4 & 4 & 1 & 5 & 5 & 3 \\ 5 & 7 & 5 & 7 & 9 & 5 \\ 5 & 7 & 5 & 9 & 7 & 5 \\ 3 & 4 & 3 & 5 & 5 & 3 \end{pmatrix}_{b_0} \\
[127] & 1^{113} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 & 13 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 11 & 2 \\ 13 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[128] & 1^{112} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 12 & 2 \\ 14 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0} \\
[129] & 1^{111} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 9 & 2 \\ 10 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[130] & 1^{122} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 7 & 3 & 8 & 3 \\ 3 & 4 & 8 & 3 & 11 & 4 \\ 7 & 8 & 10 & 4 & 16 & 6 \\ 3 & 3 & 4 & 0 & 6 & 2 \\ 8 & 11 & 16 & 6 & 20 & 8 \\ 3 & 4 & 6 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[131] & 1^{131} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 10 & 3 & 5 & 3 \\ 3 & 4 & 12 & 3 & 7 & 4 \\ 10 & 12 & 25 & 6 & 15 & 9 \\ 3 & 3 & 6 & 0 & 4 & 2 \\ 5 & 7 & 15 & 4 & 7 & 5 \\ 3 & 4 & 9 & 2 & 5 & 3 \end{pmatrix}_{b_0} \\
[132] & 1^{123} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 13 & 3 \\ 3 & 4 & 8 & 3 & 20 & 4 \\ 5 & 8 & 10 & 4 & 28 & 6 \\ 3 & 3 & 4 & 0 & 11 & 2 \\ 13 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{\overline{1}^{122}} 3^{\overline{1}^{11}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 93, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 7 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 14 & 5 \\ 8 & 11 & 0 & 14 & 20 & 8 \\ 3 & 4 & 0 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[136] \ 1^{\overline{1}^{124}} 3^{\overline{1}^{11}} 1^{\overline{1}^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 94, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 18 & 3 \\ 3 & 4 & 8 & 3 & 29 & 4 \\ 5 & 8 & 10 & 4 & 40 & 6 \\ 3 & 3 & 4 & 0 & 16 & 2 \\ 18 & 29 & 40 & 16 & 148 & 22 \\ 3 & 4 & 6 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{\overline{1}^{114}} 3^{\overline{1}^{11}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 94, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 14 & 3 \\ 3 & 4 & 8 & 3 & 19 & 4 \\ 5 & 8 & 10 & 4 & 28 & 6 \\ 3 & 3 & 4 & 0 & 10 & 2 \\ 14 & 19 & 28 & 10 & 64 & 14 \\ 3 & 4 & 6 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[135] \ 1^{\overline{1}^{13}} 3^{\overline{1}^{11}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 94, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 9 & 3 \\ 3 & 4 & 4 & 7 & 11 & 4 \\ 2 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 14 & 5 \\ 9 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{\overline{1}^{113}} 3^{\overline{1}^{11}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 95, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 10 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 15 & 5 \\ 10 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[138] \ 1^{\overline{1}^{213}} 3^{\overline{1}^{11}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 98, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 15 & 4 \\ 4 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 8 & 2 \\ 11 & 15 & 11 & 8 & 39 & 11 \\ 3 & 4 & 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{\overline{1}^{211}} 3^{\overline{1}^{11}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 100, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 5 & 3 \\ 3 & 4 & 9 & 3 & 7 & 4 \\ 7 & 9 & 13 & 5 & 11 & 7 \\ 3 & 3 & 5 & 0 & 4 & 2 \\ 5 & 7 & 11 & 4 & 7 & 5 \\ 3 & 4 & 7 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[141] \ 1^{\overline{1}^{111}} 3^{\overline{1}^{11}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 100, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 7 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 9 & 5 \\ 5 & 7 & 0 & 9 & 7 & 5 \\ 3 & 4 & 0 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{\overline{1}^{114}} 3^{\overline{1}^{11}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 100, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 17 & 3 \\ 3 & 4 & 8 & 3 & 24 & 4 \\ 5 & 8 & 10 & 4 & 34 & 6 \\ 3 & 3 & 4 & 0 & 13 & 2 \\ 17 & 24 & 34 & 13 & 100 & 18 \\ 3 & 4 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[143] \ 1^{\overline{1}^{211}} 3^{\overline{1}^{11}} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 100, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 5 & 3 \\ 3 & 4 & 17 & 3 & 7 & 4 \\ 11 & 17 & 49 & 9 & 21 & 13 \\ 3 & 3 & 9 & 0 & 4 & 2 \\ 5 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{1^{13}} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 100, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 19 & 3 \\ 3 & 4 & 8 & 3 & 29 & 4 \\ 5 & 8 & 10 & 4 & 40 & 6 \\ 3 & 3 & 4 & 0 & 16 & 2 \\ 19 & 29 & 40 & 16 & 148 & 22 \\ 3 & 4 & 6 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[144] \ 1^{1^{311}} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 101, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 5 & 3 \\ 3 & 4 & 12 & 3 & 7 & 4 \\ 9 & 12 & 25 & 6 & 15 & 9 \\ 3 & 3 & 6 & 0 & 4 & 2 \\ 5 & 7 & 15 & 4 & 7 & 5 \\ 3 & 4 & 9 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[145] \ 1^{1^{14}} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 101, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 19 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 10 & 2 \\ 14 & 19 & 14 & 10 & 64 & 14 \\ 3 & 4 & 3 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[146] \ 1^{1^{13}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 101, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 15 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 11 & 2 \\ 15 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[147] \ 1^{1^{214}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 103, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 17 & 3 \\ 3 & 4 & 4 & 3 & 24 & 4 \\ 4 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 13 & 2 \\ 17 & 24 & 17 & 13 & 100 & 18 \\ 3 & 4 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[148] \ 1^{1^{111}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 107, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 15 & 5 \\ 8 & 12 & 0 & 15 & 25 & 9 \\ 3 & 4 & 0 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[149] \ 1^{1^{24}} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 108, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 18 & 3 \\ 3 & 4 & 0 & 7 & 25 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 31 & 5 \\ 18 & 25 & 0 & 31 & 111 & 19 \\ 3 & 4 & 0 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[150] \ 1^{1^{31}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 108, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 4 & 3 \\ 3 & 4 & 17 & 3 & 7 & 4 \\ 12 & 17 & 49 & 9 & 21 & 13 \\ 3 & 3 & 9 & 0 & 4 & 2 \\ 4 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[151] \ 1^{1^{212}} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 110, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 6 & 3 \\ 3 & 4 & 8 & 7 & 7 & 4 \\ 7 & 8 & 10 & 10 & 10 & 6 \\ 5 & 7 & 10 & 7 & 9 & 5 \\ 6 & 7 & 10 & 9 & 7 & 5 \\ 3 & 4 & 6 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[152] \ 1^{1^{213}} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 110, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 12 & 3 \\ 3 & 4 & 8 & 3 & 16 & 4 \\ 6 & 8 & 10 & 4 & 22 & 6 \\ 3 & 3 & 4 & 0 & 9 & 2 \\ 12 & 16 & 22 & 9 & 44 & 12 \\ 3 & 4 & 6 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[153] $1^{112}3^{113}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 15 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 3 & 4 & 1 & 5 & 14 & 3 \\ 4 & 7 & 5 & 7 & 26 & 5 \\ 15 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[154] $1^{112}3^{113}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 7 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 14 & 5 \\ 9 & 11 & 0 & 14 & 20 & 8 \\ 3 & 4 & 0 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[155] $1^{112}3^{114}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 16 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 4 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 12 & 2 \\ 16 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[156] $1^{14}3^{114}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 3 & 19 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 10 & 2 \\ 14 & 19 & 0 & 10 & 64 & 14 \\ 3 & 4 & 0 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[157] $1^{112}3^{122}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 8 & 3 \\ 3 & 4 & 4 & 7 & 11 & 4 \\ 4 & 4 & 1 & 5 & 8 & 3 \\ 6 & 7 & 5 & 7 & 14 & 5 \\ 8 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[158] $1^{112}3^{112}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 16 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 2 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 14 & 2 \\ 16 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[162] $21^{113}3^{111}1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 15 & 4 \\ 5 & 4 & 4 & 7 & 16 & 4 \\ 3 & 4 & 1 & 5 & 11 & 3 \\ 6 & 7 & 5 & 7 & 20 & 5 \\ 15 & 16 & 11 & 20 & 44 & 12 \\ 4 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[161] $1^{112}3^{123}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 12 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 2 & 4 & 1 & 5 & 11 & 3 \\ 6 & 7 & 5 & 7 & 20 & 5 \\ 12 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[160] $1^{112}3^{213}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 12 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 4 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 9 & 2 \\ 12 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[159] $1^{111}3^{111}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 8 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 2 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 15 & 5 \\ 8 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[163] $1^{132}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 8 & 3 \\ 3 & 4 & 12 & 3 & 11 & 4 \\ 10 & 12 & 25 & 6 & 24 & 9 \\ 3 & 3 & 6 & 0 & 6 & 2 \\ 8 & 11 & 24 & 6 & 20 & 8 \\ 3 & 4 & 9 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[164] $1^{113}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 0 & 3 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 11 & 2 \\ 14 & 20 & 0 & 11 & 69 & 15 \\ 3 & 4 & 0 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[165] $1^{113}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 3 \\ 3 & 4 & 0 & 3 & 25 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 14 & 2 \\ 17 & 25 & 0 & 14 & 111 & 19 \\ 3 & 4 & 0 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[166] $1^{113}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 11 & 2 \\ 14 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[167] $1^{111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 0 & 7 & 2 \\ 9 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[168] $1^{111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 5 & 3 \\ 3 & 4 & 4 & 7 & 7 & 4 \\ 2 & 4 & 1 & 5 & 5 & 3 \\ 5 & 7 & 5 & 7 & 9 & 5 \\ 5 & 7 & 5 & 9 & 7 & 5 \\ 3 & 4 & 3 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[169] $1^{411}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 5 & 3 \\ 3 & 4 & 16 & 3 & 7 & 4 \\ 12 & 16 & 46 & 8 & 20 & 12 \\ 3 & 3 & 8 & 0 & 4 & 2 \\ 5 & 7 & 20 & 4 & 7 & 5 \\ 3 & 4 & 12 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[170] $1^{111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 125, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 15 & 5 \\ 9 & 12 & 0 & 15 & 25 & 9 \\ 3 & 4 & 0 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[171] $1^{123}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 9 & 2 \\ 12 & 16 & 0 & 9 & 44 & 12 \\ 3 & 4 & 0 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[172] $1^{123}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 11 & 3 \\ 3 & 4 & 0 & 7 & 15 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 19 & 5 \\ 11 & 15 & 0 & 19 & 39 & 11 \\ 3 & 4 & 0 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

[173] $1^{\overline{1112}1^{214}}3^{\overline{11}1^{12}}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 16 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 4 & 4 & 1 & 5 & 14 & 3 \\ 5 & 7 & 5 & 7 & 26 & 5 \\ 16 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[174] $1^{\overline{1111}1^{122}}3^{\overline{11}1^{11}}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 9 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 6 & 7 & 5 & 7 & 15 & 5 \\ 9 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[175] $1^{\overline{123}3^{\overline{11}1^1}}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 10 & 3 \\ 3 & 4 & 8 & 3 & 15 & 4 \\ 6 & 8 & 10 & 4 & 22 & 6 \\ 3 & 3 & 4 & 0 & 8 & 2 \\ 10 & 15 & 22 & 8 & 39 & 11 \\ 3 & 4 & 6 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[176] $1^{\overline{113}3^{\overline{11}1^1}}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 3 & 15 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 8 & 2 \\ 10 & 15 & 11 & 8 & 39 & 11 \\ 3 & 4 & 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[177] $1^{\overline{112}3^{\overline{11}1^1}}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 \\ 3 & 3 & 2 & 0 & 6 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[178] $1^{\overline{1221}3^{\overline{11}1^{11}}}}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 5 & 3 \\ 3 & 4 & 9 & 7 & 7 & 4 \\ 7 & 9 & 13 & 11 & 11 & 7 \\ 6 & 7 & 11 & 7 & 9 & 5 \\ 5 & 7 & 11 & 9 & 7 & 5 \\ 3 & 4 & 7 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[179] $1^{\overline{121}3^{\overline{11}1^{11}}}}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 & 3 \\ 3 & 4 & 13 & 7 & 3 & 4 \\ 11 & 13 & 28 & 16 & 7 & 10 \\ 5 & 7 & 16 & 7 & 4 & 5 \\ 3 & 3 & 7 & 4 & 0 & 2 \\ 3 & 4 & 10 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[180] $1^{\overline{114}3^{\overline{11}1^1}}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 133, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 19 & 4 \\ 4 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 10 & 2 \\ 14 & 19 & 14 & 10 & 64 & 14 \\ 3 & 4 & 3 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[181] $1^{\overline{1111}3^{\overline{11}1^{11}}}}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 8 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 15 & 5 \\ 8 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[182] $1^{\overline{114}3^{\overline{11}1^{11}}}}+[\mathfrak{su}_2 \cdot \mathfrak{g}_2]+\mathfrak{su}_2+\mathfrak{su}_2+\mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 16 & 3 \\ 3 & 4 & 4 & 3 & 24 & 4 \\ 2 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 13 & 2 \\ 16 & 24 & 17 & 13 & 100 & 18 \\ 3 & 4 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[183] $1^{1^{121}}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 137, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 5 & 3 \\ 3 & 4 & 4 & 7 & 7 & 4 \\ 3 & 4 & 1 & 5 & 5 & 3 \\ 6 & 7 & 5 & 7 & 9 & 5 \\ 5 & 7 & 5 & 9 & 7 & 5 \\ 3 & 4 & 3 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[184] $1^{1^{122}}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 138, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 8 & 3 \\ 3 & 4 & 8 & 3 & 11 & 4 \\ 6 & 8 & 10 & 4 & 16 & 6 \\ 3 & 3 & 4 & 0 & 6 & 2 \\ 8 & 11 & 16 & 6 & 20 & 8 \\ 3 & 4 & 6 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[185] $1^{1^{124}}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 138, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 17 & 3 \\ 3 & 4 & 0 & 7 & 24 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 30 & 5 \\ 17 & 24 & 0 & 30 & 100 & 18 \\ 3 & 4 & 0 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

[186] $1^{1^{113}}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 138, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 9 & 3 \\ 3 & 4 & 4 & 7 & 11 & 4 \\ 4 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 14 & 5 \\ 9 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[187] $1^{1^{114}}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 18 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 2 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 14 & 2 \\ 18 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[188] $1^{1^{112}}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 8 & 3 \\ 3 & 4 & 4 & 7 & 11 & 4 \\ 4 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 14 & 5 \\ 8 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[189] $1^{1^{112}}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 3 & 4 & 1 & 5 & 11 & 3 \\ 4 & 7 & 5 & 7 & 20 & 5 \\ 12 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[190] $1^{1^{41}}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 4 & 3 \\ 3 & 4 & 21 & 3 & 7 & 4 \\ 15 & 21 & 76 & 11 & 26 & 16 \\ 3 & 3 & 11 & 0 & 4 & 2 \\ 4 & 7 & 26 & 4 & 7 & 5 \\ 3 & 4 & 16 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[191] $1^{1^{111}}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 10 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 6 & 7 & 5 & 7 & 15 & 5 \\ 10 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[192] $1^{1^{112}}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 15 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 2 & 4 & 1 & 5 & 14 & 3 \\ 5 & 7 & 5 & 7 & 26 & 5 \\ 15 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[193] $1^{131}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 4 & 3 \\ 3 & 4 & 22 & 3 & 7 & 4 \\ 15 & 22 & 85 & 12 & 27 & 17 \\ 3 & 3 & 12 & 0 & 4 & 2 \\ 4 & 7 & 27 & 4 & 7 & 5 \\ 3 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[194] $1^{113}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 12 & 2 \\ 15 & 21 & 0 & 12 & 80 & 16 \\ 3 & 4 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[195] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 5 & 3 \\ 3 & 4 & 9 & 7 & 7 & 4 \\ 7 & 9 & 13 & 11 & 11 & 7 \\ 5 & 7 & 11 & 7 & 9 & 5 \\ 5 & 7 & 11 & 9 & 7 & 5 \\ 3 & 4 & 7 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[196] $1^{132}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 7 & 3 \\ 3 & 4 & 17 & 3 & 11 & 4 \\ 11 & 17 & 49 & 9 & 33 & 13 \\ 3 & 3 & 9 & 0 & 6 & 2 \\ 7 & 11 & 33 & 6 & 20 & 8 \\ 3 & 4 & 13 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[197] $1^{122}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 8 & 3 \\ 3 & 4 & 0 & 11 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 9 & 11 & 0 & 20 & 22 & 8 \\ 8 & 11 & 0 & 22 & 20 & 8 \\ 3 & 4 & 0 & 8 & 8 & 3 \end{pmatrix}_{b_0}$$

[198] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 16 & 3 \\ 3 & 4 & 8 & 3 & 24 & 4 \\ 5 & 8 & 10 & 4 & 34 & 6 \\ 3 & 3 & 4 & 0 & 13 & 2 \\ 16 & 24 & 34 & 13 & 100 & 18 \\ 3 & 4 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[199] $1^{121}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 5 & 3 \\ 3 & 4 & 8 & 7 & 7 & 4 \\ 6 & 8 & 10 & 10 & 10 & 6 \\ 6 & 7 & 10 & 7 & 9 & 5 \\ 5 & 7 & 10 & 9 & 7 & 5 \\ 3 & 4 & 6 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[200] $1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 5 & 3 \\ 3 & 4 & 13 & 7 & 7 & 4 \\ 9 & 13 & 28 & 16 & 16 & 10 \\ 6 & 7 & 16 & 7 & 9 & 5 \\ 5 & 7 & 16 & 9 & 7 & 5 \\ 3 & 4 & 10 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[204] $1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 5 & 3 \\ 3 & 4 & 9 & 7 & 7 & 4 \\ 6 & 9 & 13 & 11 & 11 & 7 \\ 5 & 7 & 11 & 7 & 9 & 5 \\ 5 & 7 & 11 & 9 & 7 & 5 \\ 3 & 4 & 7 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[202] $1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 5 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 \\ 9 & 13 & 28 & 7 & 16 & 10 \\ 3 & 3 & 7 & 0 & 4 & 2 \\ 5 & 7 & 16 & 4 & 7 & 5 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[203] $1^{1^{11}1^1}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 5 & 3 \\ 3 & 4 & 8 & 7 & 7 & 4 \\ 5 & 8 & 10 & 10 & 10 & 6 \\ 5 & 7 & 10 & 7 & 9 & 5 \\ 5 & 7 & 10 & 9 & 7 & 5 \\ 3 & 4 & 6 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[201] $21^{1^{11}2^3}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 14 & 3 \\ 4 & 4 & 4 & 7 & 21 & 4 \\ 3 & 4 & 1 & 5 & 14 & 3 \\ 5 & 7 & 5 & 7 & 26 & 5 \\ 14 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[205] $1^{1^{12}2^3}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 149, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 11 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 3 & 4 & 1 & 5 & 11 & 3 \\ 6 & 7 & 5 & 7 & 20 & 5 \\ 11 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[206] $1^{1^{12}2^3}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 8 & 3 \\ 3 & 4 & 21 & 3 & 11 & 4 \\ 14 & 21 & 76 & 11 & 41 & 16 \\ 3 & 3 & 11 & 0 & 6 & 2 \\ 8 & 11 & 41 & 6 & 20 & 8 \\ 3 & 4 & 16 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[207] $1^{1^{13}2^3}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 17 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 2 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 14 & 2 \\ 17 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[208] $1^{1^{12}2^3}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 8 & 3 \\ 3 & 4 & 13 & 3 & 11 & 4 \\ 9 & 13 & 28 & 7 & 25 & 10 \\ 3 & 3 & 7 & 0 & 6 & 2 \\ 8 & 11 & 25 & 6 & 20 & 8 \\ 3 & 4 & 10 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[210] $1^{1^{12}2^4}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 16 & 3 \\ 3 & 4 & 0 & 7 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 26 & 5 \\ 16 & 21 & 0 & 26 & 80 & 16 \\ 3 & 4 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[209] $1^{1^{12}2^3}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 8 & 3 \\ 3 & 4 & 12 & 3 & 11 & 4 \\ 9 & 12 & 25 & 6 & 24 & 9 \\ 3 & 3 & 6 & 0 & 6 & 2 \\ 8 & 11 & 24 & 6 & 20 & 8 \\ 3 & 4 & 9 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[211] $1^{1^{12}2^3}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 13 & 3 \\ 3 & 4 & 0 & 11 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 39 & 8 \\ 13 & 20 & 0 & 39 & 69 & 15 \\ 3 & 4 & 0 & 8 & 15 & 3 \end{pmatrix}_{b_0}$$

[212] $1^{1^{11}2^3}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 \\ 3 & 4 & 0 & 7 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 14 & 5 \\ 8 & 11 & 0 & 14 & 20 & 8 \\ 3 & 4 & 0 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[213] \ 1^{\overset{123}{112}3} 1^{\overset{11}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 156, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 12 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 20 & 5 \\ 12 & 16 & 0 & 20 & 44 & 12 \\ 3 & 4 & 0 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[214] \ 1^{\overset{112}{112}3} 1^{\overset{12}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 161, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 12 & 2 \\ 14 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[215] \ 1^{\overset{112}{111}3} 1^{\overset{11}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 162, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 9 & 3 \\ 3 & 4 & 4 & 7 & 12 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 15 & 5 \\ 9 & 12 & 8 & 15 & 25 & 9 \\ 3 & 4 & 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[216] \ 1^{\overset{114}{112}3} 1^{\overset{11}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 162, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 20 & 5 \\ 13 & 16 & 0 & 20 & 44 & 12 \\ 3 & 4 & 0 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[217] \ 1^{\overset{124}{114}3} 1^{\overset{11}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 165, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 14 & 3 \\ 3 & 4 & 0 & 7 & 19 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 24 & 5 \\ 14 & 19 & 0 & 24 & 64 & 14 \\ 3 & 4 & 0 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[218] \ 1^{\overset{114}{114}3} 1^{\overset{11}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 167, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 17 & 3 \\ 3 & 4 & 4 & 3 & 24 & 4 \\ 2 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 13 & 2 \\ 17 & 24 & 17 & 13 & 100 & 18 \\ 3 & 4 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[219] \ 1^{\overset{114}{114}3} 1^{\overset{11}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 168, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 17 & 3 \\ 3 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 13 & 2 \\ 17 & 24 & 0 & 13 & 100 & 18 \\ 3 & 4 & 0 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[220] \ 1^{\overset{112}{1112}3} 1^{\overset{12}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 172, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 14 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 3 & 4 & 1 & 5 & 14 & 3 \\ 5 & 7 & 5 & 7 & 26 & 5 \\ 14 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[221] \ 1^{\overset{111}{1211}3} 1^{\overset{11}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 172, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 5 & 3 \\ 3 & 4 & 13 & 7 & 7 & 4 \\ 8 & 13 & 28 & 16 & 16 & 10 \\ 5 & 7 & 16 & 7 & 9 & 5 \\ 5 & 7 & 16 & 9 & 7 & 5 \\ 3 & 4 & 10 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[222] \ 1^{\overset{111}{1111}3} 1^{\overset{11}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -59, \ \det = 172, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.95 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 5 & 3 \\ 3 & 4 & 4 & 7 & 7 & 4 \\ 3 & 4 & 1 & 5 & 5 & 3 \\ 5 & 7 & 5 & 7 & 9 & 5 \\ 5 & 7 & 5 & 9 & 7 & 5 \\ 3 & 4 & 3 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[223] $1^{121}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 173, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 5 & 3 \\ 3 & 4 & 18 & 3 & 7 & 4 \\ 13 & 18 & 58 & 10 & 22 & 14 \\ 3 & 3 & 10 & 0 & 4 & 2 \\ 5 & 7 & 22 & 4 & 7 & 5 \\ 3 & 4 & 14 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[224] $1^{141}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 173, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 5 & 3 \\ 3 & 4 & 21 & 3 & 7 & 4 \\ 14 & 21 & 76 & 11 & 26 & 16 \\ 3 & 3 & 11 & 0 & 4 & 2 \\ 5 & 7 & 26 & 4 & 7 & 5 \\ 3 & 4 & 16 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[227] $1^{142}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 8 & 3 \\ 3 & 4 & 16 & 3 & 11 & 4 \\ 12 & 16 & 46 & 8 & 32 & 12 \\ 3 & 3 & 8 & 0 & 6 & 2 \\ 8 & 11 & 32 & 6 & 20 & 8 \\ 3 & 4 & 12 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[226] $1^{122}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 8 & 3 \\ 3 & 4 & 13 & 3 & 11 & 4 \\ 10 & 13 & 28 & 7 & 25 & 10 \\ 3 & 3 & 7 & 0 & 6 & 2 \\ 8 & 11 & 25 & 6 & 20 & 8 \\ 3 & 4 & 10 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[225] $1^{113}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 7 & 15 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 19 & 5 \\ 12 & 15 & 0 & 19 & 39 & 11 \\ 3 & 4 & 0 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

[228] $1^{122}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 11 & 3 \\ 3 & 4 & 0 & 11 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 31 & 8 \\ 11 & 16 & 0 & 31 & 44 & 12 \\ 3 & 4 & 0 & 8 & 12 & 3 \end{pmatrix}_{b_0}$$

[229] $1^{121}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 5 & 3 & 3 \\ 3 & 4 & 18 & 7 & 3 & 4 \\ 14 & 18 & 58 & 22 & 10 & 14 \\ 5 & 7 & 22 & 7 & 4 & 5 \\ 3 & 3 & 10 & 4 & 0 & 2 \\ 3 & 4 & 14 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[230] $1^{124}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 20 & 3 \\ 3 & 4 & 8 & 3 & 29 & 4 \\ 6 & 8 & 10 & 4 & 40 & 6 \\ 3 & 3 & 4 & 0 & 16 & 2 \\ 20 & 29 & 40 & 16 & 148 & 22 \\ 3 & 4 & 6 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[231] $1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 8 & 3 \\ 3 & 4 & 4 & 7 & 11 & 4 \\ 2 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 14 & 5 \\ 8 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[232] $1^{114}3^{11}1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 19 & 3 \\ 3 & 4 & 4 & 3 & 29 & 4 \\ 2 & 4 & 1 & 2 & 20 & 3 \\ 3 & 3 & 2 & 0 & 16 & 2 \\ 19 & 29 & 20 & 16 & 148 & 22 \\ 3 & 4 & 3 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[233] $1^{1211}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 5 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 \\ 10 & 13 & 28 & 7 & 16 & 10 \\ 3 & 3 & 7 & 0 & 4 & 2 \\ 5 & 7 & 16 & 4 & 7 & 5 \\ 3 & 4 & 10 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[234] $1^{1411}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 5 & 3 \\ 3 & 4 & 26 & 3 & 7 & 4 \\ 17 & 26 & 118 & 14 & 32 & 20 \\ 3 & 3 & 14 & 0 & 4 & 2 \\ 5 & 7 & 32 & 4 & 7 & 5 \\ 3 & 4 & 20 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[235] $1^{114}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 15 & 3 \\ 3 & 4 & 0 & 3 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 11 & 2 \\ 15 & 20 & 0 & 11 & 69 & 15 \\ 3 & 4 & 0 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[236] $1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 9 & 2 \\ 11 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[237] $1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 14 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 2 & 4 & 1 & 5 & 14 & 3 \\ 5 & 7 & 5 & 7 & 26 & 5 \\ 14 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[238] $1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 3 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 14 & 2 \\ 16 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[239] $1^{1211}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 5 & 3 \\ 3 & 4 & 8 & 7 & 7 & 4 \\ 6 & 8 & 10 & 10 & 10 & 6 \\ 5 & 7 & 10 & 7 & 9 & 5 \\ 5 & 7 & 10 & 9 & 7 & 5 \\ 3 & 4 & 6 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[240] $21^{1111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 8 & 5 & 3 \\ 4 & 4 & 4 & 12 & 7 & 4 \\ 3 & 4 & 1 & 8 & 5 & 3 \\ 8 & 12 & 8 & 25 & 15 & 9 \\ 5 & 7 & 5 & 15 & 7 & 5 \\ 3 & 4 & 3 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[241] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 15 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 4 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 11 & 2 \\ 15 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[242] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 12 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 2 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 12 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[243] $1^{114}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 20 & 3 \\ 3 & 4 & 0 & 3 & 29 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 16 & 2 \\ 20 & 29 & 0 & 16 & 148 & 22 \\ 3 & 4 & 0 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[244] $1^{1112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 12 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 4 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 12 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[245] $1^{1112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 6 & 3 \\ 3 & 4 & 13 & 7 & 7 & 4 \\ 11 & 13 & 28 & 16 & 16 & 10 \\ 5 & 7 & 16 & 7 & 9 & 5 \\ 6 & 7 & 16 & 9 & 7 & 5 \\ 3 & 4 & 10 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[246] $1^{112}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 7 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 26 & 5 \\ 14 & 21 & 0 & 26 & 80 & 16 \\ 3 & 4 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[247] $1^{1112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 13 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 4 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 13 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[249] $1^{123}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 18 & 3 \\ 3 & 4 & 0 & 11 & 25 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 48 & 8 \\ 18 & 25 & 0 & 48 & 111 & 19 \\ 3 & 4 & 0 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$

[248] $1^{1112}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 16 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 3 & 4 & 1 & 5 & 14 & 3 \\ 6 & 7 & 5 & 7 & 26 & 5 \\ 16 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[250] $1^{1112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 11 & 2 \\ 13 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[251] $1^{113}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 197, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 3 & 15 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 8 & 2 \\ 11 & 15 & 11 & 8 & 39 & 11 \\ 3 & 4 & 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[252] $1^{112}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 16 & 3 \\ 3 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 12 & 2 \\ 16 & 21 & 0 & 12 & 80 & 16 \\ 3 & 4 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[253] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 14 & 3 \\ 3 & 4 & 8 & 3 & 20 & 4 \\ 6 & 8 & 10 & 4 & 28 & 6 \\ 3 & 3 & 4 & 0 & 11 & 2 \\ 14 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[254] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 13 & 3 \\ 3 & 4 & 8 & 3 & 19 & 4 \\ 6 & 8 & 10 & 4 & 28 & 6 \\ 3 & 3 & 4 & 0 & 10 & 2 \\ 13 & 19 & 28 & 10 & 64 & 14 \\ 3 & 4 & 6 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[255] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 3 & 19 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 10 & 2 \\ 13 & 19 & 14 & 10 & 64 & 14 \\ 3 & 4 & 3 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[256] $1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 8 & 3 \\ 3 & 4 & 4 & 7 & 11 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 6 & 7 & 5 & 7 & 14 & 5 \\ 8 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[257] $1^{124}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 16 & 3 \\ 3 & 4 & 8 & 3 & 24 & 4 \\ 6 & 8 & 10 & 4 & 34 & 6 \\ 3 & 3 & 4 & 0 & 13 & 2 \\ 16 & 24 & 34 & 13 & 100 & 18 \\ 3 & 4 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[258] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 4 & 0 & 11 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 31 & 8 \\ 12 & 16 & 0 & 31 & 44 & 12 \\ 3 & 4 & 0 & 8 & 12 & 3 \end{pmatrix}_{b_0}$$

[259] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 11 & 3 \\ 3 & 4 & 8 & 3 & 15 & 4 \\ 6 & 8 & 10 & 4 & 22 & 6 \\ 3 & 3 & 4 & 0 & 8 & 2 \\ 11 & 15 & 22 & 8 & 39 & 11 \\ 3 & 4 & 6 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[260] $1^{141}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 4 & 3 \\ 3 & 4 & 26 & 3 & 7 & 4 \\ 18 & 26 & 118 & 14 & 32 & 20 \\ 3 & 3 & 14 & 0 & 4 & 2 \\ 4 & 7 & 32 & 4 & 7 & 5 \\ 3 & 4 & 20 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[261] $1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 13 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 20 & 5 \\ 13 & 16 & 0 & 20 & 44 & 12 \\ 3 & 4 & 0 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[262] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 12 & 3 \\ 3 & 4 & 0 & 11 & 15 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 30 & 8 \\ 12 & 15 & 0 & 30 & 39 & 11 \\ 3 & 4 & 0 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

[263] $1^{142}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 7 & 3 \\ 3 & 4 & 21 & 3 & 11 & 4 \\ 14 & 21 & 76 & 11 & 41 & 16 \\ 3 & 3 & 11 & 0 & 6 & 2 \\ 7 & 11 & 41 & 6 & 20 & 8 \\ 3 & 4 & 16 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[264] $1^{114}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 20 & 3 \\ 3 & 4 & 4 & 3 & 29 & 4 \\ 2 & 4 & 1 & 2 & 20 & 3 \\ 3 & 3 & 2 & 0 & 16 & 2 \\ 20 & 29 & 20 & 16 & 148 & 22 \\ 3 & 4 & 3 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[265] $1^{1112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 218, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 10 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 3 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 10 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[266] $1^{1112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 218, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 10 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 2 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 10 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[267] $1^{13}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 219, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 18 & 3 \\ 3 & 4 & 0 & 3 & 25 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 14 & 2 \\ 18 & 25 & 0 & 14 & 111 & 19 \\ 3 & 4 & 0 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[268] $1^{122}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 3 \\ 3 & 4 & 0 & 11 & 11 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 22 & 8 \\ 8 & 11 & 0 & 22 & 20 & 8 \\ 3 & 4 & 0 & 8 & 8 & 3 \end{pmatrix}_{b_0}$$

[269] $1^{1112}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 9 & 3 \\ 3 & 4 & 4 & 7 & 11 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 14 & 5 \\ 9 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[270] $1^{113}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 7 & 15 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 19 & 5 \\ 11 & 15 & 0 & 19 & 39 & 11 \\ 3 & 4 & 0 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

[271] $1^{131}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 225, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 5 & 3 \\ 3 & 4 & 17 & 3 & 7 & 4 \\ 12 & 17 & 49 & 9 & 21 & 13 \\ 3 & 3 & 9 & 0 & 4 & 2 \\ 5 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[272] $1^{131}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 225, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 5 & 3 \\ 3 & 4 & 22 & 3 & 7 & 4 \\ 15 & 22 & 85 & 12 & 27 & 17 \\ 3 & 3 & 12 & 0 & 4 & 2 \\ 5 & 7 & 27 & 4 & 7 & 5 \\ 3 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[273] $1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 20 & 5 \\ 11 & 16 & 0 & 20 & 44 & 12 \\ 3 & 4 & 0 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[274] $1^{113}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 230, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 2 & 0 & 9 & 2 \\ 12 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[275] $1^{114}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 231, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 16 & 3 \\ 3 & 4 & 0 & 7 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 26 & 5 \\ 16 & 21 & 0 & 26 & 80 & 16 \\ 3 & 4 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[276] $21^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 5 & 5 & 3 \\ 4 & 4 & 9 & 7 & 7 & 4 \\ 6 & 9 & 13 & 11 & 11 & 7 \\ 5 & 7 & 11 & 7 & 9 & 5 \\ 5 & 7 & 11 & 9 & 7 & 5 \\ 3 & 4 & 7 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[277] $1^{121}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 235, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 5 & 3 \\ 3 & 4 & 13 & 7 & 7 & 4 \\ 10 & 13 & 28 & 16 & 16 & 10 \\ 6 & 7 & 16 & 7 & 9 & 5 \\ 5 & 7 & 16 & 9 & 7 & 5 \\ 3 & 4 & 10 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[278] $1^{113}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 20 & 5 \\ 12 & 16 & 0 & 20 & 44 & 12 \\ 3 & 4 & 0 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[279] $1^{124}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 237, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 15 & 3 \\ 3 & 4 & 0 & 7 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 25 & 5 \\ 15 & 20 & 0 & 25 & 69 & 15 \\ 3 & 4 & 0 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

[280] $1^{123}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 15 & 3 \\ 3 & 4 & 8 & 3 & 20 & 4 \\ 6 & 8 & 10 & 4 & 28 & 6 \\ 3 & 3 & 4 & 0 & 11 & 2 \\ 15 & 20 & 28 & 11 & 69 & 15 \\ 3 & 4 & 6 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[281] $1^{112}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 16 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 3 & 4 & 1 & 5 & 14 & 3 \\ 5 & 7 & 5 & 7 & 26 & 5 \\ 16 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[282] $1^{132}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 8 & 3 \\ 3 & 4 & 17 & 3 & 11 & 4 \\ 12 & 17 & 49 & 9 & 33 & 13 \\ 3 & 3 & 9 & 0 & 6 & 2 \\ 8 & 11 & 33 & 6 & 20 & 8 \\ 3 & 4 & 13 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[283] $1^{\overline{112}} 3^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 12 & 2 \\ 15 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[284] $1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 249, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 11 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 2 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 11 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[285] $1^{\overline{112}} 3^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 15 & 3 \\ 3 & 4 & 4 & 7 & 21 & 4 \\ 3 & 4 & 1 & 5 & 14 & 3 \\ 5 & 7 & 5 & 7 & 26 & 5 \\ 15 & 21 & 14 & 26 & 80 & 16 \\ 3 & 4 & 3 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[286] $1^{\overline{112}} 3^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 4 & 0 & 7 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 26 & 5 \\ 15 & 21 & 0 & 26 & 80 & 16 \\ 3 & 4 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[287] $1^{\overline{112}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 8 & 3 \\ 3 & 4 & 4 & 7 & 11 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 14 & 5 \\ 8 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[289] $1^{\overline{411}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 5 & 3 \\ 3 & 4 & 17 & 3 & 7 & 4 \\ 13 & 17 & 49 & 9 & 21 & 13 \\ 3 & 3 & 9 & 0 & 4 & 2 \\ 5 & 7 & 21 & 4 & 7 & 5 \\ 3 & 4 & 13 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[288] $1^{\overline{1211}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 5 & 3 \\ 3 & 4 & 13 & 7 & 7 & 4 \\ 10 & 13 & 28 & 16 & 16 & 10 \\ 5 & 7 & 16 & 7 & 9 & 5 \\ 5 & 7 & 16 & 9 & 7 & 5 \\ 3 & 4 & 10 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[290] $1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 273, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 13 & 3 \\ 3 & 4 & 0 & 7 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 25 & 5 \\ 13 & 20 & 0 & 25 & 69 & 15 \\ 3 & 4 & 0 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

[291] $1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 275, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 13 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 3 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 13 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[292] $1^{\overline{132}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 279, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 8 & 3 \\ 3 & 4 & 17 & 3 & 11 & 4 \\ 13 & 17 & 49 & 9 & 33 & 13 \\ 3 & 3 & 9 & 0 & 6 & 2 \\ 8 & 11 & 33 & 6 & 20 & 8 \\ 3 & 4 & 13 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[293] $1^{1^{123}}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 12 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 3 & 4 & 1 & 5 & 11 & 3 \\ 6 & 7 & 5 & 7 & 20 & 5 \\ 12 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[294] $1^{1^{114}}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 281, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 19 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 10 & 2 \\ 14 & 19 & 14 & 10 & 64 & 14 \\ 3 & 4 & 3 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[295] $1^{1^{114}}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 281, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 12 & 2 \\ 16 & 21 & 14 & 12 & 80 & 16 \\ 3 & 4 & 3 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[296] $1^{1^{211}}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 5 & 3 \\ 3 & 4 & 13 & 7 & 7 & 4 \\ 9 & 13 & 28 & 16 & 16 & 10 \\ 5 & 7 & 16 & 7 & 9 & 5 \\ 5 & 7 & 16 & 9 & 7 & 5 \\ 3 & 4 & 10 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[297] $21^{1^{111}}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 5 & 3 \\ 4 & 4 & 4 & 7 & 7 & 4 \\ 3 & 4 & 1 & 5 & 5 & 3 \\ 5 & 7 & 5 & 7 & 9 & 5 \\ 5 & 7 & 5 & 9 & 7 & 5 \\ 3 & 4 & 3 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[298] $1^{1^{214}}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 14 & 3 \\ 3 & 4 & 8 & 3 & 19 & 4 \\ 6 & 8 & 10 & 4 & 28 & 6 \\ 3 & 3 & 4 & 0 & 10 & 2 \\ 14 & 19 & 28 & 10 & 64 & 14 \\ 3 & 4 & 6 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[299] $1^{1^{113}}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 11 & 2 \\ 14 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[300] $1^{1^{123}}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 299, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 11 & 3 \\ 3 & 4 & 0 & 11 & 15 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 30 & 8 \\ 11 & 15 & 0 & 30 & 39 & 11 \\ 3 & 4 & 0 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

[301] $1^{1^{114}}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 7 & 19 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 24 & 5 \\ 14 & 19 & 0 & 24 & 64 & 14 \\ 3 & 4 & 0 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

[302] $1^{1^{411}}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 308, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 5 & 3 \\ 3 & 4 & 21 & 3 & 7 & 4 \\ 15 & 21 & 76 & 11 & 26 & 16 \\ 3 & 3 & 11 & 0 & 4 & 2 \\ 5 & 7 & 26 & 4 & 7 & 5 \\ 3 & 4 & 16 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[303] $21^{1^{12}1^3}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 6 & 5 & 5 & 3 \\ 4 & 4 & 8 & 7 & 7 & 4 \\ 6 & 8 & 10 & 10 & 10 & 6 \\ 5 & 7 & 10 & 7 & 9 & 5 \\ 5 & 7 & 10 & 9 & 7 & 5 \\ 3 & 4 & 6 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[304] $1^{1^{14}3}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 314, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 3 & 24 & 4 \\ 3 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 13 & 2 \\ 16 & 24 & 17 & 13 & 100 & 18 \\ 3 & 4 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[305] $1^{1^{12}4}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 317, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 13 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 3 & 4 & 1 & 5 & 11 & 3 \\ 6 & 7 & 5 & 7 & 20 & 5 \\ 13 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[306] $1^{1^{411}}3^{11}1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 322, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 5 & 3 \\ 3 & 4 & 22 & 3 & 7 & 4 \\ 16 & 22 & 85 & 12 & 27 & 17 \\ 3 & 3 & 12 & 0 & 4 & 2 \\ 5 & 7 & 27 & 4 & 7 & 5 \\ 3 & 4 & 17 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[307] $1^{1^{13}3}3^{11}1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 325, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 3 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 14 & 2 \\ 17 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[308] $1^{1^{24}4}3^{11}1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 20 & 3 \\ 3 & 4 & 0 & 11 & 29 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 56 & 8 \\ 20 & 29 & 0 & 56 & 148 & 22 \\ 3 & 4 & 0 & 8 & 22 & 3 \end{pmatrix}_{b_0}$$

[309] $1^{1^{12}2}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 11 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 3 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 11 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[310] $1^{1^{24}4}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 17 & 3 \\ 3 & 4 & 8 & 3 & 24 & 4 \\ 6 & 8 & 10 & 4 & 34 & 6 \\ 3 & 3 & 4 & 0 & 13 & 2 \\ 17 & 24 & 34 & 13 & 100 & 18 \\ 3 & 4 & 6 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[311] $1^{1^{23}3}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 337, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 14 & 3 \\ 3 & 4 & 0 & 11 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 39 & 8 \\ 14 & 20 & 0 & 39 & 69 & 15 \\ 3 & 4 & 0 & 8 & 15 & 3 \end{pmatrix}_{b_0}$$

[313] $1^{1^{412}}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 342, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 8 & 3 \\ 3 & 4 & 21 & 3 & 11 & 4 \\ 15 & 21 & 76 & 11 & 41 & 16 \\ 3 & 3 & 11 & 0 & 6 & 2 \\ 8 & 11 & 41 & 6 & 20 & 8 \\ 3 & 4 & 16 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[312] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 342, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 16 & 3 \\ 3 & 4 & 0 & 11 & 24 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 47 & 8 \\ 16 & 24 & 0 & 47 & 100 & 18 \\ 3 & 4 & 0 & 8 & 18 & 3 \end{pmatrix}_{b_0}$$

[314] $1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 350, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 12 & 3 \\ 3 & 4 & 4 & 7 & 16 & 4 \\ 3 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 12 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[315] $1^{113}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 3 & 29 & 4 \\ 3 & 4 & 1 & 2 & 20 & 3 \\ 3 & 3 & 2 & 0 & 16 & 2 \\ 19 & 29 & 20 & 16 & 148 & 22 \\ 3 & 4 & 3 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[316] $1^{123}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 355, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 15 & 3 \\ 3 & 4 & 0 & 11 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 39 & 8 \\ 15 & 20 & 0 & 39 & 69 & 15 \\ 3 & 4 & 0 & 8 & 15 & 3 \end{pmatrix}_{b_0}$$

[317] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 358, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 3 & 20 & 4 \\ 3 & 4 & 1 & 2 & 14 & 3 \\ 3 & 3 & 2 & 0 & 11 & 2 \\ 15 & 20 & 14 & 11 & 69 & 15 \\ 3 & 4 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[318] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 359, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 0 & 7 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 25 & 5 \\ 14 & 20 & 0 & 25 & 69 & 15 \\ 3 & 4 & 0 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

[319] $1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 361, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 15 & 3 \\ 3 & 4 & 0 & 7 & 20 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 25 & 5 \\ 15 & 20 & 0 & 25 & 69 & 15 \\ 3 & 4 & 0 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

[320] $1^{411}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 5 & 3 \\ 3 & 4 & 26 & 3 & 7 & 4 \\ 18 & 26 & 118 & 14 & 32 & 20 \\ 3 & 3 & 14 & 0 & 4 & 2 \\ 5 & 7 & 32 & 4 & 7 & 5 \\ 3 & 4 & 20 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[321] $1^{113}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 388, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 17 & 3 \\ 3 & 4 & 0 & 7 & 25 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 31 & 5 \\ 17 & 25 & 0 & 31 & 111 & 19 \\ 3 & 4 & 0 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[322] $1^{124}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 14 & 3 \\ 3 & 4 & 0 & 11 & 19 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 38 & 8 \\ 14 & 19 & 0 & 38 & 64 & 14 \\ 3 & 4 & 0 & 8 & 14 & 3 \end{pmatrix}_{b_0}$$

[323] $1^{114}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 420, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 3 & 25 & 4 \\ 3 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 14 & 2 \\ 18 & 25 & 17 & 14 & 111 & 19 \\ 3 & 4 & 3 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[324] $1^{113}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 428, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 16 & 3 \\ 3 & 4 & 0 & 7 & 24 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 30 & 5 \\ 16 & 24 & 0 & 30 & 100 & 18 \\ 3 & 4 & 0 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

[325] $1^{114}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 431, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 3 & 24 & 4 \\ 3 & 4 & 1 & 2 & 17 & 3 \\ 3 & 3 & 2 & 0 & 13 & 2 \\ 17 & 24 & 17 & 13 & 100 & 18 \\ 3 & 4 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[326] $21^{112}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 8 & 3 \\ 4 & 4 & 4 & 7 & 11 & 4 \\ 3 & 4 & 1 & 5 & 8 & 3 \\ 5 & 7 & 5 & 7 & 14 & 5 \\ 8 & 11 & 8 & 14 & 20 & 8 \\ 3 & 4 & 3 & 5 & 8 & 3 \end{pmatrix}_{b_0}$$

[327] $1^{113}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 442, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 18 & 3 \\ 3 & 4 & 0 & 7 & 25 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 31 & 5 \\ 18 & 25 & 0 & 31 & 111 & 19 \\ 3 & 4 & 0 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[328] $21^{121}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 9 & 5 & 5 & 3 \\ 4 & 4 & 13 & 7 & 7 & 4 \\ 9 & 13 & 28 & 16 & 16 & 10 \\ 5 & 7 & 16 & 7 & 9 & 5 \\ 5 & 7 & 16 & 9 & 7 & 5 \\ 3 & 4 & 10 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[329] $1^{113}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 469, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 19 & 3 \\ 3 & 4 & 0 & 7 & 29 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 36 & 5 \\ 19 & 29 & 0 & 36 & 148 & 22 \\ 3 & 4 & 0 & 5 & 22 & 3 \end{pmatrix}_{b_0}$$

[330] $21^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 492, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 11 & 3 \\ 4 & 4 & 4 & 7 & 16 & 4 \\ 3 & 4 & 1 & 5 & 11 & 3 \\ 5 & 7 & 5 & 7 & 20 & 5 \\ 11 & 16 & 11 & 20 & 44 & 12 \\ 3 & 4 & 3 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

[331] $1^{114}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 505, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 3 & 29 & 4 \\ 3 & 4 & 1 & 2 & 20 & 3 \\ 3 & 3 & 2 & 0 & 16 & 2 \\ 20 & 29 & 20 & 16 & 148 & 22 \\ 3 & 4 & 3 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[332] $1^{114}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 506, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 17 & 3 \\ 3 & 4 & 0 & 7 & 24 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 30 & 5 \\ 17 & 24 & 0 & 30 & 100 & 18 \\ 3 & 4 & 0 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

[333] $1^{124}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 512, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 17 & 3 \\ 3 & 4 & 0 & 11 & 24 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 8 & 11 & 0 & 20 & 47 & 8 \\ 17 & 24 & 0 & 47 & 100 & 18 \\ 3 & 4 & 0 & 8 & 18 & 3 \end{pmatrix}_{b_0}$$

[334] $1^{114}3^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 10, \Delta = -59, \det = 599, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 20 & 3 \\ 3 & 4 & 0 & 7 & 29 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 36 & 5 \\ 20 & 29 & 0 & 36 & 148 & 22 \\ 3 & 4 & 0 & 5 & 22 & 3 \end{pmatrix}_{b_0}$$

$T = 12$ [337] $1^{1114}3^{11}5^{13}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 11 & 2 \\ 1 & 1 & 2 & 3 & 4 & 17 & 3 \\ 1 & 2 & 0 & 1 & 3 & 14 & 2 \\ 1 & 3 & 1 & 1 & 4 & 17 & 3 \\ 3 & 4 & 3 & 4 & 5 & 28 & 5 \\ 11 & 17 & 14 & 17 & 28 & 121 & 23 \\ 2 & 3 & 2 & 3 & 5 & 23 & 4 \end{pmatrix}_{b_0}$$

[338] $1^{1111}3^{11}5^{13}1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 4 & 2 \\ 1 & 1 & 2 & 4 & 4 & 6 & 3 \\ 1 & 2 & 0 & 4 & 3 & 5 & 2 \\ 2 & 4 & 4 & 6 & 7 & 11 & 6 \\ 3 & 4 & 3 & 7 & 5 & 10 & 5 \\ 4 & 6 & 5 & 11 & 10 & 14 & 8 \\ 2 & 3 & 2 & 6 & 5 & 8 & 4 \end{pmatrix}_{b_0}$$

4.950 NHC(-2)(-3) + $\mathfrak{su}_2$ + su2n3mix + su2n3mix + $\mathfrak{su}_2$

$T = 9$ [0] $1^{1211}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 5 & 1 & 3 \\ 3 & 4 & 8 & 12 & 7 & 0 & 4 \\ 5 & 8 & 10 & 16 & 10 & 0 & 6 \\ 8 & 12 & 16 & 25 & 15 & 0 & 9 \\ 5 & 7 & 10 & 15 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 9 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 5 & 2 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 & 4 \\ 3 & 5 & 4 & 3 & 6 & 3 & 4 \\ 3 & 3 & 3 & 0 & 4 & 2 & 2 \\ 5 & 7 & 6 & 4 & 7 & 5 & 5 \\ 2 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 4 & 3 & 3 \\ 3 & 4 & 5 & 3 & 7 & 4 & 4 \\ 3 & 5 & 4 & 3 & 6 & 3 & 4 \\ 3 & 3 & 3 & 0 & 4 & 2 & 2 \\ 4 & 7 & 6 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 5 & 7 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 2 & 4 & 1 & 2 & 5 & 6 & 3 \\ 3 & 3 & 2 & 0 & 4 & 5 & 2 \\ 5 & 7 & 5 & 4 & 7 & 11 & 5 \\ 7 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 9 & 1 & 3 \\ 3 & 4 & 4 & 7 & 12 & 0 & 4 \\ 2 & 4 & 1 & 5 & 8 & 0 & 3 \\ 5 & 7 & 5 & 7 & 15 & 0 & 5 \\ 9 & 12 & 8 & 15 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 5 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 1 & 3 \\ 3 & 4 & 0 & 8 & 7 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 8 & 0 & 12 & 10 & 0 & 6 \\ 5 & 7 & 0 & 10 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -17, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 5 & 6 & 3 \\ 3 & 4 & 8 & 3 & 7 & 9 & 4 \\ 5 & 8 & 10 & 4 & 10 & 12 & 6 \\ 3 & 3 & 4 & 0 & 4 & 5 & 2 \\ 5 & 7 & 10 & 4 & 7 & 11 & 5 \\ 6 & 9 & 12 & 5 & 11 & 13 & 7 \\ 3 & 4 & 6 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -17, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 & 1 & 3 \\ 3 & 4 & 5 & 7 & 3 & 0 & 4 \\ 4 & 5 & 4 & 6 & 3 & 0 & 4 \\ 5 & 7 & 6 & 7 & 4 & 0 & 5 \\ 3 & 3 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 4 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -17, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 & 3 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 & 4 \\ 4 & 5 & 4 & 6 & 3 & 3 & 4 \\ 5 & 7 & 6 & 7 & 4 & 5 & 5 \\ 3 & 3 & 3 & 4 & 0 & 2 & 2 \\ 3 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 4 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{1112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 & 4 & 3 \\ 3 & 4 & 5 & 7 & 3 & 4 & 4 \\ 4 & 5 & 4 & 6 & 3 & 3 & 4 \\ 5 & 7 & 6 & 7 & 4 & 5 & 5 \\ 3 & 3 & 3 & 4 & 0 & 2 & 2 \\ 4 & 4 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 4 & 5 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{1111} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 & 1 & 3 \\ 3 & 4 & 14 & 7 & 3 & 0 & 4 \\ 10 & 14 & 37 & 17 & 8 & 0 & 11 \\ 5 & 7 & 17 & 7 & 4 & 0 & 5 \\ 3 & 3 & 8 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 11 & 5 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1211} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 4 & 6 & 3 \\ 3 & 4 & 8 & 3 & 7 & 9 & 4 \\ 6 & 8 & 10 & 4 & 10 & 12 & 6 \\ 3 & 3 & 4 & 0 & 4 & 5 & 2 \\ 4 & 7 & 10 & 4 & 7 & 11 & 5 \\ 6 & 9 & 12 & 5 & 11 & 13 & 7 \\ 3 & 4 & 6 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -27, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 8 & 2 & 3 \\ 3 & 4 & 0 & 12 & 11 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 8 & 12 & 0 & 25 & 23 & 8 & 9 \\ 8 & 11 & 0 & 23 & 20 & 8 & 8 \\ 2 & 4 & 0 & 8 & 8 & 1 & 3 \\ 3 & 4 & 0 & 9 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 5 & 1 & 3 \\ 3 & 4 & 4 & 7 & 7 & 0 & 4 \\ 4 & 4 & 1 & 5 & 5 & 0 & 3 \\ 5 & 7 & 5 & 7 & 9 & 0 & 5 \\ 5 & 7 & 5 & 9 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 3 & 5 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -41, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 7 & 2 & 2 \\ 8 & 12 & 0 & 7 & 25 & 8 & 9 \\ 3 & 4 & 0 & 2 & 8 & 1 & 3 \\ 3 & 4 & 0 & 2 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -41, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 5 & 3 & 3 \\ 3 & 4 & 13 & 3 & 7 & 4 & 4 \\ 8 & 13 & 28 & 7 & 16 & 9 & 10 \\ 3 & 3 & 7 & 0 & 4 & 2 & 2 \\ 5 & 7 & 16 & 4 & 7 & 5 & 5 \\ 3 & 4 & 9 & 2 & 5 & 1 & 3 \\ 3 & 4 & 10 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -41, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 0 & 3 & 7 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 5 & 2 \\ 5 & 7 & 0 & 4 & 7 & 11 & 5 \\ 6 & 9 & 0 & 5 & 11 & 13 & 7 \\ 3 & 4 & 0 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -51, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 8 & 3 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 & 3 \\ 3 & 3 & 2 & 0 & 7 & 2 & 2 \\ 8 & 12 & 8 & 7 & 25 & 8 & 9 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -52, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 5 & 3 & 3 \\ 3 & 4 & 8 & 3 & 7 & 4 & 4 \\ 5 & 8 & 10 & 4 & 10 & 6 & 6 \\ 3 & 3 & 4 & 0 & 4 & 2 & 2 \\ 5 & 7 & 10 & 4 & 7 & 5 & 5 \\ 3 & 4 & 6 & 2 & 5 & 1 & 3 \\ 3 & 4 & 6 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -54, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 1 & 3 \\ 3 & 4 & 0 & 7 & 7 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 9 & 0 & 5 \\ 5 & 7 & 0 & 9 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 5 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{\overline{1112}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 8 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 3 & 3 & 2 & 0 & 4 & 4 & 2 \\ 4 & 7 & 5 & 4 & 7 & 10 & 5 \\ 6 & 8 & 6 & 4 & 10 & 10 & 6 \\ 3 & 4 & 3 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -68, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 0 & 3 & 7 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 2 & 2 \\ 5 & 7 & 0 & 4 & 7 & 5 & 5 \\ 3 & 4 & 0 & 2 & 5 & 1 & 3 \\ 3 & 4 & 0 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{\overline{1112}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -70, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 5 & 10 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 2 & 4 & 1 & 2 & 5 & 9 & 3 \\ 3 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 10 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -72, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 3 & 3 & 2 & 0 & 4 & 5 & 2 \\ 4 & 7 & 5 & 4 & 7 & 11 & 5 \\ 6 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{\overline{1112}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -74, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 0 & 3 & 7 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 4 & 2 \\ 5 & 7 & 0 & 4 & 7 & 10 & 5 \\ 6 & 8 & 0 & 4 & 10 & 10 & 6 \\ 3 & 4 & 0 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{\overline{131}} 3^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -74, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 15 & 1 & 3 \\ 3 & 4 & 4 & 7 & 21 & 0 & 4 \\ 2 & 4 & 1 & 5 & 14 & 0 & 3 \\ 5 & 7 & 5 & 7 & 26 & 0 & 5 \\ 15 & 21 & 14 & 26 & 80 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 5 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{\overline{1121}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -76, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 1 & 3 \\ 3 & 4 & 0 & 7 & 12 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 15 & 0 & 5 \\ 9 & 12 & 0 & 15 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 5 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{\overline{131}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -85, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 12 & 1 & 3 \\ 3 & 4 & 4 & 7 & 16 & 0 & 4 \\ 2 & 4 & 1 & 5 & 11 & 0 & 3 \\ 5 & 7 & 5 & 7 & 20 & 0 & 5 \\ 12 & 16 & 11 & 20 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 5 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

[36] $1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 0 & 3 & 7 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 5 & 2 \\ 5 & 7 & 0 & 4 & 7 & 11 & 5 \\ 7 & 9 & 0 & 5 & 11 & 13 & 7 \\ 3 & 4 & 0 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[37] $1^{112} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 8 & 3 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 2 & 4 & 1 & 2 & 8 & 3 & 3 \\ 3 & 3 & 2 & 0 & 6 & 2 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 & 8 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

[38] $1^{111} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 & 4 \\ 4 & 4 & 1 & 2 & 5 & 3 & 3 \\ 3 & 3 & 2 & 0 & 4 & 2 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[39] $1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 0 & 3 & 7 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 7 & 2 \\ 5 & 7 & 0 & 4 & 7 & 16 & 5 \\ 9 & 13 & 0 & 7 & 16 & 28 & 10 \\ 3 & 4 & 0 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[41] $1^{112} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 14 & 3 & 3 \\ 3 & 4 & 4 & 3 & 21 & 4 & 4 \\ 2 & 4 & 1 & 2 & 14 & 3 & 3 \\ 3 & 3 & 2 & 0 & 12 & 2 & 2 \\ 14 & 21 & 14 & 12 & 80 & 14 & 16 \\ 3 & 4 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 3 & 2 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

[40] $1^{112} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 3 & 7 & 18 & 4 \\ 3 & 4 & 1 & 2 & 5 & 12 & 3 \\ 3 & 3 & 2 & 0 & 4 & 10 & 2 \\ 4 & 7 & 5 & 4 & 7 & 22 & 5 \\ 12 & 18 & 12 & 10 & 22 & 58 & 14 \\ 3 & 4 & 3 & 2 & 5 & 14 & 3 \end{pmatrix}_{b_0}$$

[42] $1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 2 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 15 & 5 & 5 \\ 8 & 12 & 0 & 15 & 25 & 8 & 9 \\ 2 & 4 & 0 & 5 & 8 & 1 & 3 \\ 3 & 4 & 0 & 5 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

[43] $1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 8 & 3 & 3 \\ 3 & 4 & 4 & 3 & 12 & 4 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 & 3 \\ 3 & 3 & 2 & 0 & 7 & 2 & 2 \\ 8 & 12 & 8 & 7 & 25 & 8 & 9 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{112} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -100, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 1 & 3 \\ 3 & 4 & 0 & 7 & 11 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 14 & 0 & 5 \\ 8 & 11 & 0 & 14 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 5 & 8 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{111} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -100, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 2 & 3 \\ 3 & 4 & 0 & 7 & 7 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 9 & 5 & 5 \\ 5 & 7 & 0 & 9 & 7 & 5 & 5 \\ 2 & 4 & 0 & 5 & 5 & 1 & 3 \\ 3 & 4 & 0 & 5 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -104, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 5 & 9 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 2 & 4 & 1 & 2 & 5 & 9 & 3 \\ 3 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 9 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -104, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 8 & 3 \\ 3 & 4 & 0 & 7 & 7 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 9 & 16 & 5 \\ 5 & 7 & 0 & 9 & 7 & 16 & 5 \\ 8 & 13 & 0 & 16 & 16 & 28 & 10 \\ 3 & 4 & 0 & 5 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{121} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -110, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 0 & 3 & 11 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 6 & 2 & 2 \\ 8 & 11 & 0 & 6 & 20 & 8 & 8 \\ 3 & 4 & 0 & 2 & 8 & 1 & 3 \\ 3 & 4 & 0 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -119, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 0 & 3 & 16 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 9 & 2 & 2 \\ 11 & 16 & 0 & 9 & 44 & 11 & 12 \\ 3 & 4 & 0 & 2 & 11 & 1 & 3 \\ 3 & 4 & 0 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -125, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 11 & 3 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 & 4 \\ 2 & 4 & 1 & 2 & 11 & 3 & 3 \\ 3 & 3 & 2 & 0 & 9 & 2 & 2 \\ 11 & 16 & 11 & 9 & 44 & 11 & 12 \\ 3 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -129, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 5 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 3 & 3 & 2 & 0 & 4 & 5 & 2 \\ 5 & 7 & 5 & 4 & 7 & 11 & 5 \\ 6 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[52] $1^{1111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 5 & 3 & 3 \\ 3 & 4 & 0 & 7 & 7 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 6 & 7 & 0 & 7 & 9 & 5 & 5 \\ 5 & 7 & 0 & 9 & 7 & 5 & 5 \\ 3 & 4 & 0 & 5 & 5 & 1 & 3 \\ 3 & 4 & 0 & 5 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[53] $1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 3 & 4 & 1 & 2 & 5 & 9 & 3 \\ 3 & 3 & 2 & 0 & 4 & 7 & 2 \\ 4 & 7 & 5 & 4 & 7 & 16 & 5 \\ 9 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[54] $1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -137, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 15 & 5 & 5 \\ 8 & 12 & 0 & 15 & 25 & 8 & 9 \\ 3 & 4 & 0 & 5 & 8 & 1 & 3 \\ 3 & 4 & 0 & 5 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

[57] $1^{1112} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 2 & 3 \\ 3 & 4 & 0 & 7 & 11 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 14 & 5 & 5 \\ 8 & 11 & 0 & 14 & 20 & 8 & 8 \\ 2 & 4 & 0 & 5 & 8 & 1 & 3 \\ 3 & 4 & 0 & 5 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

[56] $1^{1111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 7 & 3 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 & 3 \\ 3 & 3 & 2 & 0 & 6 & 2 & 2 \\ 7 & 11 & 8 & 6 & 20 & 8 & 8 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

[55] $1^{1112} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 5 & 3 \\ 3 & 4 & 0 & 7 & 7 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 9 & 10 & 5 \\ 5 & 7 & 0 & 9 & 7 & 10 & 5 \\ 5 & 8 & 0 & 10 & 10 & 10 & 6 \\ 3 & 4 & 0 & 5 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[58] $1^{1112} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 14 & 1 & 3 \\ 3 & 4 & 4 & 7 & 21 & 0 & 4 \\ 2 & 4 & 1 & 5 & 14 & 0 & 3 \\ 5 & 7 & 5 & 7 & 26 & 0 & 5 \\ 14 & 21 & 14 & 26 & 80 & 0 & 16 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 5 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

[59] $1^{1111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 7 & 4 & 4 \\ 3 & 4 & 1 & 2 & 5 & 3 & 3 \\ 3 & 3 & 2 & 0 & 4 & 2 & 2 \\ 5 & 7 & 5 & 4 & 7 & 5 & 5 \\ 3 & 4 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 2 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -164, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 6 & 3 \\ 3 & 4 & 0 & 7 & 7 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 9 & 11 & 5 \\ 5 & 7 & 0 & 9 & 7 & 11 & 5 \\ 6 & 9 & 0 & 11 & 11 & 13 & 7 \\ 3 & 4 & 0 & 5 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{\overline{1112}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -167, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 5 & 7 & 3 \\ 3 & 4 & 4 & 3 & 7 & 9 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 3 & 3 & 2 & 0 & 4 & 5 & 2 \\ 5 & 7 & 5 & 4 & 7 & 11 & 5 \\ 7 & 9 & 6 & 5 & 11 & 13 & 7 \\ 3 & 4 & 3 & 2 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{\overline{1112}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -178, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 5 & 6 & 3 \\ 3 & 4 & 4 & 3 & 7 & 8 & 4 \\ 3 & 4 & 1 & 2 & 5 & 6 & 3 \\ 3 & 3 & 2 & 0 & 4 & 4 & 2 \\ 5 & 7 & 5 & 4 & 7 & 10 & 5 \\ 6 & 8 & 6 & 4 & 10 & 10 & 6 \\ 3 & 4 & 3 & 2 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{\overline{1121}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -191, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 & 3 \\ 3 & 4 & 0 & 7 & 12 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 15 & 5 & 5 \\ 9 & 12 & 0 & 15 & 25 & 8 & 9 \\ 3 & 4 & 0 & 5 & 8 & 1 & 3 \\ 3 & 4 & 0 & 5 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -204, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 & 3 \\ 3 & 4 & 0 & 7 & 7 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 9 & 5 & 5 \\ 5 & 7 & 0 & 9 & 7 & 5 & 5 \\ 3 & 4 & 0 & 5 & 5 & 1 & 3 \\ 3 & 4 & 0 & 5 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -211, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 2 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 20 & 5 & 5 \\ 11 & 16 & 0 & 20 & 44 & 11 & 12 \\ 2 & 4 & 0 & 5 & 11 & 1 & 3 \\ 3 & 4 & 0 & 5 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{\overline{1112}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -216, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 6 & 3 \\ 3 & 4 & 0 & 7 & 7 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 9 & 10 & 5 \\ 5 & 7 & 0 & 9 & 7 & 10 & 5 \\ 6 & 8 & 0 & 10 & 10 & 10 & 6 \\ 3 & 4 & 0 & 5 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{\overline{1121}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -264, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 8 & 3 & 3 \\ 3 & 4 & 4 & 3 & 11 & 4 & 4 \\ 3 & 4 & 1 & 2 & 8 & 3 & 3 \\ 3 & 3 & 2 & 0 & 6 & 2 & 2 \\ 8 & 11 & 8 & 6 & 20 & 8 & 8 \\ 3 & 4 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 2 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{1112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -271, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 5 & 9 & 3 \\ 3 & 4 & 4 & 3 & 7 & 13 & 4 \\ 3 & 4 & 1 & 2 & 5 & 9 & 3 \\ 3 & 3 & 2 & 0 & 4 & 7 & 2 \\ 5 & 7 & 5 & 4 & 7 & 16 & 5 \\ 9 & 13 & 9 & 7 & 16 & 28 & 10 \\ 3 & 4 & 3 & 2 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{1121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -300, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 11 & 3 & 3 \\ 3 & 4 & 4 & 3 & 16 & 4 & 4 \\ 3 & 4 & 1 & 2 & 11 & 3 & 3 \\ 3 & 3 & 2 & 0 & 9 & 2 & 2 \\ 11 & 16 & 11 & 9 & 44 & 11 & 12 \\ 3 & 4 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 2 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{1121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -314, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 8 & 3 & 3 \\ 3 & 4 & 0 & 7 & 11 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 14 & 5 & 5 \\ 8 & 11 & 0 & 14 & 20 & 8 & 8 \\ 3 & 4 & 0 & 5 & 8 & 1 & 3 \\ 3 & 4 & 0 & 5 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{1112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -336, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 9 & 3 \\ 3 & 4 & 0 & 7 & 7 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 9 & 16 & 5 \\ 5 & 7 & 0 & 9 & 7 & 16 & 5 \\ 9 & 13 & 0 & 16 & 16 & 28 & 10 \\ 3 & 4 & 0 & 5 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{1121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -375, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 11 & 3 & 3 \\ 3 & 4 & 0 & 7 & 16 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 5 & 7 & 0 & 7 & 20 & 5 & 5 \\ 11 & 16 & 0 & 20 & 44 & 11 & 12 \\ 3 & 4 & 0 & 5 & 11 & 1 & 3 \\ 3 & 4 & 0 & 5 & 12 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [73] \quad 1^{1111} 3^1 1^{11} 5^{11} 13^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -1, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 12, 1), \quad c_{\text{ext}} = 7.0$$

4.951 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 7 \quad [0] \quad 1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -8, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 8 & 5 & 4 \\ 8 & 8 & 10 & 6 & 6 \\ 5 & 5 & 6 & 3 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -9, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.56 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 5 & 5 & 4 & 6 & 4 \\ 3 & 5 & 6 & 3 & 3 \\ 3 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{131} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -93, \quad \det = -9, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 7 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 10 & 17 & 49 & 27 & 13 \\ 7 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{141} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 16 & 9 & 4 \\ 9 & 16 & 46 & 24 & 12 \\ 5 & 9 & 24 & 12 & 6 \\ 3 & 4 & 12 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -10, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.60 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 0 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{112} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -10, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 5 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 4 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{12} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 8 & 5 & 4 \\ 7 & 8 & 10 & 6 & 6 \\ 4 & 5 & 6 & 3 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{112} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 5 & 5 & 4 & 3 & 4 \\ 4 & 4 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 4 & 5 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{112} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 9 & 9 & 13 & 12 & 7 \\ 7 & 8 & 12 & 9 & 5 \\ 3 & 4 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{12} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -13, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 3 & 5 & 4 & 6 & 4 \\ 5 & 5 & 6 & 3 & 3 \\ 3 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{13} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -13, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 13 & 17 & 15 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{122} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 7 & 8 & 10 & 18 & 6 \\ 9 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 3 & 3 \\ 3 & 4 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 8 & 5 & 4 \\ 5 & 8 & 10 & 6 & 6 \\ 3 & 5 & 6 & 3 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 8 & 14 & 37 & 9 & 11 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{131} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 10 & 17 & 49 & 12 & 13 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 3 & 5 & 4 & 3 & 4 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 9 & 9 & 13 & 15 & 7 \\ 7 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 8 & 9 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 8 & 5 & 4 \\ 6 & 8 & 10 & 6 & 6 \\ 4 & 5 & 6 & 3 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 21^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 6 & 12 & 4 \\ 5 & 4 & 5 & 13 & 4 \\ 6 & 5 & 4 & 12 & 4 \\ 12 & 13 & 12 & 27 & 9 \\ 4 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 21^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$$

$$\begin{pmatrix} 4 & 5 & 6 & 3 & 4 \\ 5 & 4 & 5 & 4 & 4 \\ 6 & 5 & 4 & 3 & 4 \\ 3 & 4 & 3 & 0 & 2 \\ 4 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 4 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[24] $1^{1^{21}} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 5 & 5 & 4 & 6 & 4 \\ 4 & 5 & 6 & 3 & 3 \\ 3 & 4 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{111} 3 1^{132} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 4 & 3 \\ 3 & 4 & 19 & 4 & 4 \\ 14 & 19 & 73 & 12 & 15 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 4 & 4 & 0 & 0 & 2 \\ 3 & 4 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{12} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 7 & 13 & 28 & 12 & 10 \\ 4 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 4 & 4 & 1 & 3 & 3 \\ 5 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{141} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 7 & 3 \\ 3 & 4 & 16 & 9 & 4 \\ 12 & 16 & 46 & 24 & 12 \\ 7 & 9 & 24 & 12 & 6 \\ 3 & 4 & 12 & 6 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

[30] $1^{122} 3 1^{142} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 8 & 8 & 10 & 12 & 6 \\ 6 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{121} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 8 & 8 & 10 & 12 & 6 \\ 8 & 9 & 12 & 12 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 4 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{11} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 4 & 9 & 4 \\ 8 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{122} 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 4 & 8 & 10 & 12 & 6 \\ 6 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[44] $1^{131} 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 7 & 12 & 25 & 9 & 9 \\ 4 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{111} 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 9 & 9 & 13 & 6 & 7 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[38] $1^{11} 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 4 & 4 & 1 & 3 & 3 \\ 4 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{111} 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 4 & 4 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[40] $21^{1431} 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 27 & 3 & 4 \\ 5 & 4 & 24 & 4 & 4 \\ 27 & 24 & 121 & 15 & 19 \\ 3 & 4 & 15 & 0 & 2 \\ 4 & 4 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

[41] $1^1 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 3 & 5 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{111} 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 4 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{111} 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 24 & 4 & 4 \\ 17 & 24 & 121 & 15 & 19 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

[45] $1^{111} 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 19 & 4 & 4 \\ 12 & 19 & 73 & 12 & 15 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[46] $1^1 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 4 & 5 & 0 & 3 & 3 \\ 3 & 4 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$

[47] $1^{11} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 4 & 5 & 4 & 3 & 4 \\ 4 & 4 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

[48] $21^{131} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 2 \\ 4 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[49] $1^{112} 3^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 12 & 28 & 85 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 17 & -1 & 1 \end{pmatrix}_{b_0}$$

[55] $1^{141} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 3 \\ 3 & 4 & 16 & 9 & 4 \\ 11 & 16 & 46 & 24 & 12 \\ 6 & 9 & 24 & 12 & 6 \\ 3 & 4 & 12 & 6 & 2 \end{pmatrix}_{b_0}$$

[56] $1^{11} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 5 & 5 & 4 & 9 & 4 \\ 7 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[50] $1^{11} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 4 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{11} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[52] $1^{11} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{121} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 5 & 8 & 10 & 12 & 6 \\ 7 & 9 & 12 & 12 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[54] $1^{111} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 6 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{11} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[57] $1^{13} 3^1 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 11 & 17 & 15 & 48 & 12 \\ 3 & 4 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{13} 3^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 10 & 0 & 21 & 7 \\ 3 & 4 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{13} 3^1 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 4 & 5 & 4 & 6 & 4 \\ 5 & 5 & 6 & 3 & 3 \\ 3 & 4 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{13} 3^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 10 & 0 & 21 & 7 \\ 3 & 4 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

[61] $21^{12} 31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 4 & 8 & 5 & 4 \\ 6 & 8 & 10 & 6 & 6 \\ 3 & 5 & 6 & 3 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{11} 3^1 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 6 & 9 & 13 & 9 & 7 \\ 5 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$

[68] $1^{11} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 4 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

[64] $1^{11} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[65] $1^{11} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 4 & 5 & 4 & 3 & 4 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[66] $1^{13} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 3 \\ 3 & 4 & 12 & 9 & 4 \\ 9 & 12 & 25 & 18 & 9 \\ 6 & 9 & 18 & 12 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{11} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 3 \\ 4 & 9 & 12 & 4 & 5 \\ 6 & 12 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 2 \\ 3 & 5 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

[70] $1^1 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -25, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 3 & 5 & 4 & 6 & 4 \\ 3 & 5 & 6 & 3 & 3 \\ 3 & 4 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -25, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 12 & 6 \\ 3 & 4 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[73] $21^{11} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 4 & 14 & 4 & 4 \\ 9 & 14 & 37 & 9 & 11 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[74] $1^1 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 10 & 15 & 52 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[71] $1^1 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 3 & 5 & 4 & 6 & 4 \\ 4 & 5 & 6 & 3 & 3 \\ 3 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 4 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[79] $1^{112} 3^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 7 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[77] $1^{112} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 5 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[75] $1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 11 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[76] $1^1 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 2 \\ 3 & 5 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[78] $1^1 3^1 1^{132} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 6 & 12 & 25 & 27 & 9 \\ 7 & 13 & 27 & 27 & 9 \\ 3 & 4 & 9 & 9 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{11} 3^{14} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 4 & 9 & 4 \\ 9 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 4 & 9 & 4 \\ 5 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[80] $21^{14} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 6 & 3 \\ 4 & 4 & 16 & 9 & 4 \\ 12 & 16 & 46 & 24 & 12 \\ 6 & 9 & 24 & 12 & 6 \\ 3 & 4 & 12 & 6 & 2 \end{pmatrix}_{b_0}$$

[81] $1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 4 & 5 & 4 & 6 & 4 \\ 3 & 5 & 6 & 3 & 3 \\ 3 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[82] $1^{11} 3^{12} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 7 & 10 & 22 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{12} 3^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 8 & 9 & 13 & 12 & 7 \\ 6 & 8 & 12 & 9 & 5 \\ 3 & 4 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[84] $1^{14} 3^{14} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 4 & 5 & 4 & 18 & 4 \\ 15 & 21 & 18 & 75 & 15 \\ 3 & 4 & 4 & 15 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{21} 3^{12} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 8 & 10 & 22 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{11} 3^{12} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 5 & 5 & 4 & 9 & 4 \\ 8 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[87] $1^{12} 3^{12} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 8 & 9 & 12 & 12 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[88] $1^{11} 3^{12} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 4 & 8 & 10 & 12 & 6 \\ 5 & 9 & 12 & 12 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[92] $1^{131} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 7 & 3 \\ 3 & 4 & 12 & 9 & 4 \\ 10 & 12 & 25 & 18 & 9 \\ 7 & 9 & 18 & 12 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[91] $1^{131} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 6 & 12 & 25 & 9 & 9 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{112} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 7 & 9 & 13 & 12 & 7 \\ 5 & 8 & 12 & 9 & 5 \\ 3 & 4 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{132} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 8 & 12 & 25 & 27 & 9 \\ 10 & 13 & 27 & 27 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[95] $1^{11} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 3 \\ 4 & 9 & 12 & 4 & 5 \\ 5 & 12 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 2 \\ 3 & 5 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 9 & 12 & 0 & 5 \\ 5 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

[97] $1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[98] $1^{12} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 5 & 8 & 0 & 9 & 5 \\ 3 & 4 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[99] $1^{131} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 12 & 9 & 4 \\ 7 & 12 & 25 & 18 & 9 \\ 5 & 9 & 18 & 12 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[100] $1^{12} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 8 & 0 & 9 & 5 \\ 3 & 4 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[101] $1^{12} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 4 & 12 & 4 \\ 8 & 13 & 12 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[102] $1^{111} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 24 & 4 & 4 \\ 15 & 24 & 121 & 15 & 19 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

[103] $1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 4 & 5 & 4 & 6 & 4 \\ 4 & 5 & 6 & 3 & 3 \\ 3 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[104] $1^{113} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 1 & 12 & 3 \\ 11 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[105] $1^{111} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 4 & 4 & 1 & 3 & 3 \\ 4 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[106] $1^1 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[107] $1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 9 & 9 & 13 & 9 & 7 \\ 4 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[108] $1^{111} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 11 & 14 & 9 & 36 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[109] $1^{111} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 10 & 14 & 9 & 36 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{14} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 3 & 5 & 4 & 18 & 4 \\ 14 & 21 & 18 & 75 & 15 \\ 3 & 4 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$

[114] $1^{121} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 13 & 23 & 100 & 15 & 18 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[110] $1^{121} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 4 & 8 & 10 & 6 & 6 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{121} 3^{141} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -36, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.56 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 8 & 8 & 10 & 6 & 6 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{14} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -36, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.6$$

$$T_{\min} = 11.33 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 3 & 5 & 4 & 18 & 4 \\ 12 & 21 & 18 & 75 & 15 \\ 3 & 4 & 4 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -37, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.46 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 9 & 0 & 12 & 6 \\ 3 & 4 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{111} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -37, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 9 & 17 & 15 & 48 & 12 \\ 3 & 4 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{131} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -37, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.46 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 12 & 9 & 4 \\ 10 & 12 & 25 & 18 & 9 \\ 8 & 9 & 18 & 12 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{112} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -37, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.2$$

$$T_{\min} = 11.54 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 4 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{112} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -37, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.2$$

$$T_{\min} = 11.54 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 8 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{13} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -39, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.44 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 8 & 9 & 13 & 9 & 7 \\ 3 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{111} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -39, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.13 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{112} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -39, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.44 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 8 & 9 & 13 & 21 & 7 \\ 9 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{111} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -39, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.13 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 24 & 4 & 4 \\ 17 & 24 & 121 & 15 & 19 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

[122] $1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 7 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[123] $1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 4 & 4 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[130] $1^{111}31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 5 & 9 & 13 & 15 & 7 \\ 7 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[126] $21^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 4 & 5 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[127] $1^131^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 5 & 9 & 13 & 12 & 7 \\ 4 & 8 & 12 & 9 & 5 \\ 3 & 4 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

[128] $1^{13}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 24 & 8 \\ 3 & 4 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

[129] $1^{12}31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 4 & 12 & 4 \\ 10 & 13 & 12 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[131] $1^131^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 4 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{12}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -42, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 6 & 8 & 0 & 9 & 5 \\ 3 & 4 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[134] $1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -42, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 5 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[132] $1^{121}31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -42, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 10 & 13 & 28 & 21 & 10 \\ 5 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[146] \ 1^{111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -43, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[147] \ 1^{111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -43, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 10 & 14 & 37 & 9 & 11 \\ 4 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[135] \ 1^{121} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -43, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.30 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 8 & 13 & 28 & 9 & 10 \\ 4 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[136] \ 1^{111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -43, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.23 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 12 & 14 & 9 & 36 & 10 \\ 3 & 4 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -43, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 5 & 9 & 13 & 9 & 7 \\ 4 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[138] \ 1^{111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -43, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 11 & 14 & 37 & 9 & 11 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -43, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.49 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 5 & 9 & 13 & 12 & 7 \\ 5 & 8 & 12 & 9 & 5 \\ 3 & 4 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -43, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 4 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[141] \ 21^{12} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -43, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.2$$

$$T_{\min} = 11.23 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 4 & 4 & 14 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 15 & 14 & 9 & 36 & 10 \\ 4 & 4 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{111} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -43, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 24 & 4 & 4 \\ 16 & 24 & 121 & 15 & 19 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[143] \ 1^{111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -43, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.23 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 9 & 14 & 9 & 36 & 10 \\ 3 & 4 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[144] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 5 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[145] $1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 8 & 14 & 9 & 36 & 10 \\ 3 & 4 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[157] $1^1 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 4 & 8 & 10 & 6 & 6 \\ 2 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[155] $1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 4 & 8 & 10 & 18 & 6 \\ 8 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[154] $1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 19 & 4 & 4 \\ 11 & 19 & 73 & 12 & 15 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[153] $1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 10 & 17 & 49 & 15 & 13 \\ 4 & 5 & 15 & 3 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[152] $1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 5 & 8 & 10 & 6 & 6 \\ 4 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[151] $1^{11} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 6 & 9 & 13 & 6 & 7 \\ 4 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[150] $1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 8 & 8 & 10 & 6 & 6 \\ 4 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

[156] $1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 7 & 8 & 10 & 6 & 6 \\ 2 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[149] $1^{11} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 19 & 4 & 4 \\ 14 & 19 & 73 & 12 & 15 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[148] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 9 & 9 & 13 & 6 & 7 \\ 4 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[158] $1^{14} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 3 & 5 & 4 & 18 & 4 \\ 13 & 21 & 18 & 75 & 15 \\ 3 & 4 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$

[160] $1^{111} 1 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 4 & 9 & 4 \\ 5 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[165] $1^{121} 1^{32} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 7 & 8 & 10 & 12 & 6 \\ 7 & 9 & 12 & 12 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[164] $1^{131} 1^{21} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 12 & 9 & 4 \\ 8 & 12 & 25 & 18 & 9 \\ 6 & 9 & 18 & 12 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[163] $1^{11} 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 12 & 6 \\ 3 & 4 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[162] $1^{121} 1^1 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 5 & 8 & 10 & 12 & 6 \\ 5 & 9 & 12 & 12 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[161] $1^{122} 1^2 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 7 & 13 & 28 & 30 & 10 \\ 9 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[159] $1^{11} 1^2 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 4 & 9 & 4 \\ 7 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[166] $1^{11} 1^{21} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 11 & 15 & 52 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 1 \end{pmatrix}_{b_0}$$

[167] $1^{131} 1^{32} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 12 & 9 & 4 \\ 9 & 12 & 25 & 18 & 9 \\ 7 & 9 & 18 & 12 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[168] $1^{122} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 7 & 8 & 10 & 12 & 6 \\ 5 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[169] $1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 12 & 17 & 15 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

[170] $1^{12} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 18 & 8 & 4 \\ 10 & 18 & 58 & 24 & 14 \\ 4 & 8 & 24 & 9 & 5 \\ 3 & 4 & 14 & 5 & 1 \end{pmatrix}_{b_0}$$

[171] $1^{122} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 5 & 8 & 10 & 12 & 6 \\ 7 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[172] $1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 4 & 12 & 4 \\ 7 & 13 & 12 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[173] $1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 9 & 0 & 12 & 6 \\ 3 & 4 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[174] $1^{113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 5 & 9 & 13 & 27 & 7 \\ 12 & 17 & 27 & 48 & 12 \\ 3 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[175] $1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 5 & 9 & 13 & 6 & 7 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[181] $1^{113} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 8 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[180] $1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 9 & 14 & 9 & 36 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[179] $1^{113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 4 & 4 & 1 & 15 & 3 \\ 15 & 22 & 15 & 84 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[178] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -51, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.10 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 7 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[177] \ 1^{11} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -51, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.57 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 5 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[176] \ 1^{113} 3^1 1^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -51, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.57 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 10 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[185] \ 1^{13} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -52, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 13 & 17 & 49 & 15 & 13 \\ 5 & 5 & 15 & 3 & 3 \\ 3 & 4 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[184] \ 1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -52, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 7 & 9 & 13 & 15 & 7 \\ 5 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[183] \ 21^{111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -52, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.23 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 4 & 5 & 12 & 6 & 4 \\ 5 & 4 & 13 & 5 & 4 \\ 12 & 13 & 28 & 12 & 10 \\ 6 & 5 & 12 & 3 & 3 \\ 4 & 4 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[182] \ 1^{11} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -52, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 7 & 9 & 13 & 9 & 7 \\ 5 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[187] \ 1^{113} 3^1 1^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -53, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.23 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 17 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 17 & 40 & 172 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 24 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[186] \ 1^{132} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -53, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 6 & 12 & 25 & 18 & 9 \\ 5 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[188] \ 1^{11} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -54, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.22 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 4 & 9 & 4 \\ 6 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[190] \ 1^{121} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -54, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.22 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 6 & 9 & 12 & 12 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[189] $1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 8 & 9 & 13 & 15 & 7 \\ 6 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[192] $1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 5 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

[191] $1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 7 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[195] $1^{111} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 24 & 4 & 4 \\ 18 & 24 & 121 & 15 & 19 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

[196] $1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 7 & 9 & 13 & 12 & 7 \\ 7 & 8 & 12 & 9 & 5 \\ 3 & 4 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[194] $1^{132} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 8 & 9 & 13 & 9 & 7 \\ 5 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 1 \end{pmatrix}_{b_0}$$

[193] $1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[198] $1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 3 \\ 4 & 9 & 20 & 0 & 5 \\ 10 & 20 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 11 & -1 & 1 \end{pmatrix}_{b_0}$$

[197] $1^1 3^1 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 10 & 2 & 3 \\ 4 & 9 & 24 & 4 & 5 \\ 10 & 24 & 58 & 12 & 14 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 5 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[202] $1^{112} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 4 & 4 & 1 & 12 & 3 \\ 13 & 18 & 12 & 57 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[201] $1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 9 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[199] $1^{112}3^{14}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 14 & 18 & 12 & 57 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 6 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[204] $1^{11}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$

[205] $1^{121}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 5 & 8 & 10 & 12 & 6 \\ 6 & 9 & 12 & 12 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[206] $1^{112}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 6 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[207] $1^{114}3^{111}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.51 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 11 & 13 & 28 & 12 & 10 \\ 3 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[208] $21^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 9 & 4 \\ 5 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 9 & 9 & 6 & 12 & 6 \\ 4 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[209] $1^{121}3^{133}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 7 & 8 & 10 & 12 & 6 \\ 8 & 9 & 12 & 12 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[210] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 4 & 9 & 4 \\ 6 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[211] $1^{113}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 4 & 9 & 4 \\ 8 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[212] $1^{13}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 12 & 17 & 15 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

[213] $1^{112} 3 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 6 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[200] $1^{111} 3 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 8 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[216] $1^{112} 3 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 7 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[217] $1^{112} 3 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 6 & 9 & 13 & 12 & 7 \\ 6 & 8 & 12 & 9 & 5 \\ 3 & 4 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{122} 3 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 8 & 8 & 10 & 12 & 6 \\ 7 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[219] $1^{13} 3 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 10 & 17 & 15 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

[220] $1^{122} 3 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 18 & 8 & 4 \\ 11 & 18 & 58 & 24 & 14 \\ 5 & 8 & 24 & 9 & 5 \\ 3 & 4 & 14 & 5 & 2 \end{pmatrix}_{b_0}$$

[214] $1^{122} 3 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 4 & 8 & 10 & 12 & 6 \\ 5 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[221] $1^{111} 3 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 6 & 9 & 13 & 15 & 7 \\ 8 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[215] $1^{122} 3 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 9 & 13 & 28 & 18 & 10 \\ 7 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

[222] $1^{141} 3 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 8 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 15 & 21 & 76 & 33 & 16 \\ 8 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 1 \end{pmatrix}_{b_0}$$

[223] $1^{122} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 5 & 8 & 10 & 18 & 6 \\ 10 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[226] $1^{114} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 14 & 0 & 36 & 10 \\ 3 & 4 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[224] $1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 5 & 9 & 13 & 9 & 7 \\ 3 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[225] $1^{111} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 14 & 0 & 36 & 10 \\ 3 & 4 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[229] $1^{132} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 6 & 3 \\ 3 & 4 & 27 & 8 & 4 \\ 19 & 27 & 133 & 36 & 21 \\ 6 & 8 & 36 & 9 & 5 \\ 3 & 4 & 21 & 5 & 1 \end{pmatrix}_{b_0}$$

[228] $1^{111} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 8 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[227] $1^{112} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 3 \\ 4 & 9 & 24 & 0 & 5 \\ 10 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 14 & -1 & 2 \end{pmatrix}_{b_0}$$

[233] $1^{12} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 4 & 12 & 4 \\ 9 & 13 & 12 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{112} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 8 & 9 & 13 & 12 & 7 \\ 7 & 8 & 12 & 9 & 5 \\ 3 & 4 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[234] $1^{131} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 8 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

[236] $1^{111} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 1 \end{pmatrix}_{b_0}$$

[237] $1^{112} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 1 & 6 & 3 \\ 7 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[230] $1^{141} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 7 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 13 & 21 & 76 & 33 & 16 \\ 7 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$

[231] $1^{112} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 1 & 6 & 3 \\ 5 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[232] $1^{13} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 4 & 15 & 4 \\ 11 & 17 & 15 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

[238] $1^{132} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -62, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 10 & 13 & 28 & 12 & 10 \\ 5 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

[239] $1^{11} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 11 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 2 \end{pmatrix}_{b_0}$$

[240] $1^{131} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 9 & 17 & 49 & 27 & 13 \\ 6 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[243] $1^{13} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 12 & 0 & 24 & 8 \\ 3 & 4 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

[251] $1^{12} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 4 & 12 & 4 \\ 8 & 13 & 12 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[246] $1^1 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 5 & 9 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[247] $1^{122} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 8 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0} \\
[248] & 21^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.81 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 6 & 6 & 4 \\ 5 & 4 & 5 & 5 & 4 \\ 6 & 5 & 4 & 6 & 4 \\ 6 & 5 & 6 & 3 & 3 \\ 4 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0} \\
[245] & 21^{11} 31^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 5 & 18 & 3 & 4 \\ 5 & 4 & 21 & 4 & 4 \\ 18 & 21 & 76 & 15 & 16 \\ 3 & 4 & 15 & 0 & 2 \\ 4 & 4 & 16 & 2 & 1 \end{pmatrix}_{b_0} \\
[244] & 1^{13} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 12 & 0 & 24 & 8 \\ 3 & 4 & 0 & 8 & 2 \end{pmatrix}_{b_0} \\
[242] & 1^{11} 3 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 4 & 8 & 10 & 18 & 6 \\ 7 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0} \\
[241] & 1^{111} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 15 & 4 & 3 \\ 3 & 4 & 19 & 4 & 4 \\ 15 & 19 & 73 & 12 & 15 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0} \\
[249] & 1^{14} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 13 & 4 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 13 & 21 & 76 & 18 & 16 \\ 4 & 5 & 18 & 3 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0} \\
[250] & 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 7 & 9 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0} \\
[252] & 1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2 \\
& T_{\min} = 10.63 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 6 & 9 & 13 & 12 & 7 \\ 5 & 8 & 12 & 9 & 5 \\ 3 & 4 & 7 & 5 & 2 \end{pmatrix}_{b_0} \\
[257] & 1^{121} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 7 & 9 & 12 & 12 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0} \\
[256] & 1^{112} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 4 & 9 & 4 \\ 7 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0} \\
[255] & 1^{12} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 4 & 12 & 4 \\ 9 & 13 & 12 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[254] $1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 7 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 2 \end{pmatrix}_{b_0}$$

[253] $1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 6 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[258] $1^{111} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 9 & 9 & 13 & 15 & 7 \\ 8 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[262] $1^{111} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 10 & 14 & 9 & 36 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[261] $1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 13 & 0 & 27 & 9 \\ 3 & 4 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[260] $1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 8 & 14 & 37 & 9 & 11 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[259] $1^{111} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 11 & 14 & 9 & 36 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[264] $1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 3 \\ 4 & 9 & 20 & 0 & 5 \\ 8 & 20 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 11 & -1 & 1 \end{pmatrix}_{b_0}$$

[263] $1^{113} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 9 & 28 & 4 & 5 \\ 15 & 28 & 73 & 12 & 15 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 5 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[265] $1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -69, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 10 & 2 & 3 \\ 4 & 9 & 20 & 4 & 5 \\ 10 & 20 & 37 & 9 & 11 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 5 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

[267] $1^{132} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 7 & 12 & 25 & 27 & 9 \\ 7 & 13 & 27 & 27 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[266] $1^{113}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 9 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[268] $1^{122}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 5 & 8 & 10 & 12 & 6 \\ 4 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

[270] $1^{122}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 5 & 8 & 10 & 18 & 6 \\ 7 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[269] $1^{112}31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 13 & 21 & 7 \\ 11 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[274] $21^{131}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 6 & 3 \\ 4 & 4 & 12 & 9 & 4 \\ 9 & 12 & 25 & 18 & 9 \\ 6 & 9 & 18 & 12 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[273] $1^{113}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 12 & 0 & 24 & 8 \\ 3 & 4 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

[275] $1^{112}31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 7 & 9 & 13 & 9 & 7 \\ 3 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[271] $1^{111}31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 6 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[272] $1^{112}31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 7 & 9 & 13 & 12 & 7 \\ 6 & 8 & 12 & 9 & 5 \\ 3 & 4 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[280] $1^{112}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 7 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[277] $21^{11}31^{113} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 15 & 17 & 12 & 48 & 12 \\ 4 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[281] $1^{132} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 4 & 22 & 8 & 4 \\ 12 & 22 & 85 & 30 & 17 \\ 5 & 8 & 30 & 9 & 5 \\ 3 & 4 & 17 & 5 & 2 \end{pmatrix}_{b_0}$$

[282] $1^{13} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 7 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

[279] $1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 6 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[276] $21^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 9 & 3 & 4 \\ 8 & 9 & 12 & 4 & 5 \\ 9 & 12 & 13 & 6 & 7 \\ 3 & 4 & 6 & 0 & 2 \\ 4 & 5 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[278] $1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 5 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[283] $1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 4 & 12 & 4 \\ 11 & 13 & 12 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[284] $1^{122} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 4 & 18 & 8 & 4 \\ 12 & 18 & 58 & 24 & 14 \\ 5 & 8 & 24 & 9 & 5 \\ 3 & 4 & 14 & 5 & 1 \end{pmatrix}_{b_0}$$

[285] $1^{13} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 4 & 5 & 4 & 15 & 4 \\ 13 & 17 & 15 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

[286] $1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 8 & 9 & 13 & 6 & 7 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[287] $1^{111} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 19 & 4 & 4 \\ 13 & 19 & 73 & 12 & 15 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[291] $1^{121} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 19 & 4 & 4 \\ 13 & 19 & 73 & 12 & 15 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[288] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 19 & 4 & 4 \\ 12 & 19 & 73 & 12 & 15 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[289] $1^{121} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 6 & 8 & 10 & 6 & 6 \\ 4 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[302] $1^{11} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -77, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 8 & 2 & 3 \\ 4 & 9 & 20 & 4 & 5 \\ 8 & 20 & 37 & 9 & 11 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 5 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

[306] $1^{131} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 8 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 12 & 17 & 49 & 27 & 13 \\ 8 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[303] $1^{122} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 6 & 3 \\ 3 & 4 & 18 & 8 & 4 \\ 14 & 18 & 58 & 24 & 14 \\ 6 & 8 & 24 & 9 & 5 \\ 3 & 4 & 14 & 5 & 2 \end{pmatrix}_{b_0}$$

[307] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 7 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[305] $1^{121} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 9 & 13 & 28 & 21 & 10 \\ 8 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[304] $1^{122} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 6 & 3 \\ 3 & 4 & 18 & 8 & 4 \\ 13 & 18 & 58 & 24 & 14 \\ 6 & 8 & 24 & 9 & 5 \\ 3 & 4 & 14 & 5 & 2 \end{pmatrix}_{b_0}$$

[311] $1^{132} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 5 & 3 \\ 3 & 4 & 27 & 8 & 4 \\ 17 & 27 & 133 & 36 & 21 \\ 5 & 8 & 36 & 9 & 5 \\ 3 & 4 & 21 & 5 & 1 \end{pmatrix}_{b_0}$$

[308] $1^{132} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 10 & 12 & 25 & 27 & 9 \\ 10 & 13 & 27 & 27 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[309] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 6 & 9 & 13 & 9 & 7 \\ 3 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[310] $1^{131} 3^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 32 & 4 & 4 \\ 18 & 32 & 193 & 21 & 25 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$

[312] $1^{111} 3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 12 & 14 & 37 & 9 & 11 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[313] $1^{111} 3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 9 & 14 & 37 & 9 & 11 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[314] $1^{112} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 10 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[315] $1^{123} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 4 & 12 & 4 \\ 10 & 13 & 12 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[316] $21^{11} 3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 12 & 4 \\ 5 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 12 & 13 & 9 & 27 & 9 \\ 4 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[317] $1^{121} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 7 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

[318] $1^{123} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 13 & 0 & 27 & 9 \\ 3 & 4 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[319] $1^{113} 3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 3 \\ 4 & 9 & 32 & 0 & 5 \\ 16 & 32 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 18 & -1 & 1 \end{pmatrix}_{b_0}$$

[320] $1^{141} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 16 & 4 & 4 \\ 10 & 16 & 46 & 12 & 12 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[321] $1^{111} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 6 & 9 & 13 & 6 & 7 \\ 2 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[322] $1^{11} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 6 & 9 & 13 & 9 & 7 \\ 4 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[323] $1^{121} 3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 12 & 18 & 58 & 12 & 14 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[324] $1^{111} 3^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 11 & 21 & 76 & 18 & 16 \\ 3 & 5 & 18 & 3 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[325] $1^{111} 3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 7 & 13 & 28 & 12 & 10 \\ 3 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[326] $1^{14} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -82, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 16 & 0 & 45 & 11 \\ 3 & 4 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[327] $1^{11} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 8 & 13 & 28 & 12 & 10 \\ 4 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[330] $1^{121} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 11 & 13 & 28 & 9 & 10 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[329] $1^{111} 3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 10 & 14 & 37 & 9 & 11 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[328] $1^{111} 3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 11 & 14 & 37 & 9 & 11 \\ 4 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[336] $21^{111} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 6 & 9 & 4 \\ 5 & 4 & 5 & 9 & 4 \\ 6 & 5 & 4 & 9 & 4 \\ 9 & 9 & 9 & 12 & 6 \\ 4 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[331] $1^{121} 3^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 28 & 4 & 4 \\ 17 & 28 & 154 & 18 & 22 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[332] $1^{121} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 7 & 13 & 28 & 9 & 10 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[333] $1^{121} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 7 & 8 & 10 & 6 & 6 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[334] $1^{121} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 10 & 18 & 58 & 12 & 14 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[335] $1^{121} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 5 & 8 & 10 & 6 & 6 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[338] $21^{111} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 9 & 15 & 4 \\ 5 & 4 & 9 & 17 & 4 \\ 9 & 9 & 13 & 27 & 7 \\ 15 & 17 & 27 & 48 & 12 \\ 4 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[343] $1^{132} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 10 & 12 & 25 & 18 & 9 \\ 5 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[342] $1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 6 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[341] $1^{13} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 17 & 0 & 48 & 12 \\ 3 & 4 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[340] $1^{11} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 7 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[339] $1^{12} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 11 & 13 & 28 & 12 & 10 \\ 5 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[337] $1^{132} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 8 & 12 & 25 & 18 & 9 \\ 7 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[344] $1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.47 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 7 & 13 & 28 & 18 & 10 \\ 5 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[345] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 9 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[346] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 13 & 0 & 27 & 9 \\ 3 & 4 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[347] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 9 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[348] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 7 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[349] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 7 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[350] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 9 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[351] $1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 9 & 17 & 49 & 15 & 13 \\ 3 & 5 & 15 & 3 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[354] $1^{132} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 5 & 3 \\ 3 & 4 & 27 & 8 & 4 \\ 18 & 27 & 133 & 36 & 21 \\ 5 & 8 & 36 & 9 & 5 \\ 3 & 4 & 21 & 5 & 1 \end{pmatrix}_{b_0}$$

[353] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 9 & 20 & 4 & 5 \\ 9 & 20 & 37 & 9 & 11 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 5 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[352] $1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 8 & 13 & 28 & 18 & 10 \\ 4 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

[355] $1^{14} 3 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 9 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 2 \end{pmatrix}_{b_0}$$

[361] $21^{11} 31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 9 & 3 & 4 \\ 5 & 4 & 9 & 4 & 4 \\ 9 & 9 & 13 & 6 & 7 \\ 3 & 4 & 6 & 0 & 2 \\ 4 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[367] $1^{11} 3 1^{122} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 7 & 13 & 28 & 18 & 10 \\ 4 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

[370] $1^{121} 3 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 4 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 16 & 23 & 100 & 15 & 18 \\ 4 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[369] $1^{131} 3 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 8 & 12 & 25 & 9 & 9 \\ 4 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[368] $1^{131} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 4 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 14 & 22 & 85 & 15 & 17 \\ 4 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[371] $1^{111} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 6 & 9 & 13 & 6 & 7 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[365] $1^{111} 3 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 19 & 4 & 4 \\ 15 & 19 & 73 & 12 & 15 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[357] $21^{13} 31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 21 & 3 & 4 \\ 5 & 4 & 19 & 4 & 4 \\ 21 & 19 & 73 & 12 & 15 \\ 3 & 4 & 12 & 0 & 2 \\ 4 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[356] $1^{131} 3 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 10 & 12 & 25 & 9 & 9 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[362] $1^{111} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 12 & 14 & 37 & 9 & 11 \\ 4 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[363] $1^{11} 3^{121} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 7 & 9 & 13 & 9 & 7 \\ 4 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[364] $1^{111} 3^{112} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 9 & 14 & 37 & 9 & 11 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[358] $1^{112} 3^{121} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.58 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 7 & 9 & 13 & 21 & 7 \\ 8 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[366] $1^{132} 3^{121} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 8 & 12 & 25 & 27 & 9 \\ 8 & 13 & 27 & 27 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[360] $1^{123} 3^{111} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 13 & 0 & 27 & 9 \\ 3 & 4 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[359] $1^{122} 3^{133} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 7 & 8 & 10 & 18 & 6 \\ 10 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[374] $1^{122} 3^{112} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -93, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 8 & 13 & 28 & 18 & 10 \\ 6 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[373] $1^{132} 3^{112} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -93, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 7 & 12 & 25 & 18 & 9 \\ 6 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[372] $21^{111} 31^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -93, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 15 & 9 & 4 \\ 5 & 4 & 17 & 9 & 4 \\ 15 & 17 & 49 & 27 & 13 \\ 9 & 9 & 27 & 12 & 6 \\ 4 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[376] $1^{132} 3^{112} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 9 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 9 & 12 & 25 & 27 & 9 \\ 9 & 13 & 27 & 27 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[375] $1^{123} 3^{112} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 18 & 0 & 57 & 13 \\ 3 & 4 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

[378] $1^{112}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -95, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 4 & 4 & 1 & 9 & 3 \\ 10 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[377] $1^{112}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -95, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 13 & 21 & 7 \\ 7 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

[379] $1^{31}31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 9 & 36 & 4 & 5 \\ 15 & 36 & 133 & 18 & 21 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 5 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[380] $1^{112}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 5 & 9 & 13 & 21 & 7 \\ 9 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[381] $21^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.42 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 4 & 9 & 5 & 4 \\ 6 & 9 & 13 & 9 & 7 \\ 3 & 5 & 9 & 3 & 3 \\ 3 & 4 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[382] $21^{112}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 4 & 9 & 8 & 4 \\ 6 & 9 & 13 & 12 & 7 \\ 6 & 8 & 12 & 9 & 5 \\ 3 & 4 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[383] $1^{132}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 8 & 12 & 25 & 18 & 9 \\ 4 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

[387] $1^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 5 & 9 & 13 & 15 & 7 \\ 6 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[386] $1^{111}31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 7 & 9 & 13 & 6 & 7 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[385] $1^{113}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 8 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[384] $1^{14}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 21 & 0 & 75 & 15 \\ 3 & 4 & 0 & 15 & 1 \end{pmatrix}_{b_0}$$

[388] $1^{111} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 19 & 4 & 4 \\ 14 & 19 & 73 & 12 & 15 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[389] $1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 5 & 9 & 13 & 15 & 7 \\ 5 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[390] $1^{121} 3^1 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 6 & 8 & 10 & 6 & 6 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[397] $21^{121} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 4 & 8 & 9 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 6 & 9 & 12 & 12 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[396] $1^{121} 3^1 1^{41} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 11 & 13 & 28 & 21 & 10 \\ 6 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[395] $1^{122} 3^1 1^{32} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 7 & 8 & 10 & 12 & 6 \\ 6 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[391] $1^{132} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 10 & 17 & 49 & 24 & 13 \\ 6 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[394] $1^{121} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 7 & 13 & 28 & 21 & 10 \\ 6 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[393] $1^{12} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 10 & 13 & 28 & 12 & 10 \\ 3 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[392] $1^{122} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 5 & 8 & 10 & 12 & 6 \\ 6 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[398] $1^{122} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 8 & 13 & 28 & 30 & 10 \\ 10 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[399] $1^{122} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 10 & 13 & 28 & 18 & 10 \\ 5 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[400] $1^{113} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 14 & 28 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 15 & -1 & 1 \end{pmatrix}_{b_0}$$

[409] $1^{132} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 8 & 12 & 25 & 27 & 9 \\ 9 & 13 & 27 & 27 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[410] $1^{132} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 6 & 3 \\ 3 & 4 & 22 & 8 & 4 \\ 14 & 22 & 85 & 30 & 17 \\ 6 & 8 & 30 & 9 & 5 \\ 3 & 4 & 17 & 5 & 2 \end{pmatrix}_{b_0}$$

[411] $1^{112} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 11 & 18 & 12 & 57 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[401] $1^{113} 3 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 3 \\ 4 & 9 & 32 & 4 & 5 \\ 16 & 32 & 100 & 15 & 18 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 5 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

[402] $1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 1 & 12 & 3 \\ 12 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[408] $1^{12} 3 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 10 & 18 & 12 & 57 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[403] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 9 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$

[404] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 12 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$

[405] $1^{113} 3 1^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 15 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 21 & -1 & 2 \end{pmatrix}_{b_0}$$

[406] $1^{112}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 13 & 18 & 12 & 57 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[407] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 9 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[415] $1^{12}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 10 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 14 & -1 & 2 \end{pmatrix}_{b_0}$$

[416] $1^{14}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 11 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 2 \end{pmatrix}_{b_0}$$

[414] $1^{14}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.42 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 16 & 0 & 45 & 11 \\ 3 & 4 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[413] $1^{122}3^11^{23} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 7 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[412] $1^{122}3^11^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 5 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[418] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 1 & 9 & 3 \\ 8 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[417] $1^{132}3^11^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 8 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 13 & 17 & 49 & 39 & 13 \\ 8 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 1 \end{pmatrix}_{b_0}$$

[421] $1^{13}3^11^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 10 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

[423] $1^{121}3^11^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 8 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

[424] $1^{141}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 5 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 15 & 21 & 76 & 33 & 16 \\ 5 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 1 \end{pmatrix}_{b_0}$$

[420] $1^{132}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 7 & 12 & 25 & 18 & 9 \\ 4 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

[419] $1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 6 & 9 & 13 & 15 & 7 \\ 5 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[422] $21^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 9 & 6 & 4 \\ 5 & 4 & 9 & 5 & 4 \\ 9 & 9 & 13 & 9 & 7 \\ 6 & 5 & 9 & 3 & 3 \\ 4 & 4 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[425] $1^{122}31^{133} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -110, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.51 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 7 & 8 & 10 & 12 & 6 \\ 7 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[426] $1^{122}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -110, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.51 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 5 & 8 & 10 & 12 & 6 \\ 5 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[430] $1^{122}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 5 & 8 & 10 & 18 & 6 \\ 9 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[427] $1^{111}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 11 & 14 & 37 & 9 & 11 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[428] $1^{111}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 10 & 14 & 37 & 9 & 11 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

[429] $1^{131}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 7 & 12 & 25 & 9 & 9 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[431] $21^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[432] $1^{112} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 8 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[433] $1^{113} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 7 & 9 & 13 & 27 & 7 \\ 11 & 17 & 27 & 48 & 12 \\ 3 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[434] $1^{113} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -113, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 11 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 11 & 28 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 15 & -1 & 1 \end{pmatrix}_{b_0}$$

[437] $1^{14} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 16 & 0 & 45 & 11 \\ 3 & 4 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

[436] $1^{121} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 8 & 13 & 28 & 21 & 10 \\ 7 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[435] $1^{12} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 8 & 13 & 28 & 12 & 10 \\ 3 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[445] $1^{141} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 4 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 13 & 21 & 76 & 15 & 16 \\ 4 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[446] $1^{131} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 9 & 17 & 49 & 12 & 13 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[444] $1^{131} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 7 & 12 & 25 & 9 & 9 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[443] $1^{13} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 11 & 17 & 49 & 15 & 13 \\ 4 & 5 & 15 & 3 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[442] $1^{122} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 5 & 8 & 10 & 18 & 6 \\ 8 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[441] \ 1 \overset{1^{11}}{3} 1^{\overset{1^{21}}{121}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -115, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.74 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 7 & 13 & 28 & 9 & 10 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[440] \ 21 \overset{1^{11}}{3} 1^{\overset{1^{31}}{131}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -115, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.39 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 15 & 3 & 4 \\ 5 & 4 & 17 & 4 & 4 \\ 15 & 17 & 49 & 12 & 13 \\ 3 & 4 & 12 & 0 & 2 \\ 4 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[439] \ 1^{\overset{1^{22}}{1^{21}}} 3 \overset{1^{11}}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -115, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.25 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 9 & 13 & 28 & 9 & 10 \\ 4 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[438] \ 21 \overset{1^{23}}{1^2} 31^{\overset{1^{11}}{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -115, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.04 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 15 & 3 & 4 \\ 5 & 4 & 14 & 4 & 4 \\ 15 & 14 & 37 & 9 & 11 \\ 3 & 4 & 9 & 0 & 2 \\ 4 & 4 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[447] \ 1^{\overset{1^{31}}{1^{22}}} 3 \overset{1^{11}}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -115, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.7$$

$$T_{\min} = 10.74 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 10 & 13 & 28 & 30 & 10 \\ 8 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[448] \ 1^{\overset{1^{33}}{1^{32}}} 3 \overset{1^{11}}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -116, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.07 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 10 & 3 \\ 3 & 4 & 12 & 13 & 4 \\ 9 & 12 & 25 & 27 & 9 \\ 10 & 13 & 27 & 27 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[449] \ 1^{\overset{1^{42}}{1^{22}}} 3 \overset{1^{11}}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -117, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.7$$

$$T_{\min} = 10.25 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 9 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 11 & 13 & 28 & 30 & 10 \\ 9 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[450] \ 21 \overset{1^{11}}{1^2} 31^{\overset{1^{11}}{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -120, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.10 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 6 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[451] \ 1^{\overset{1^{11}}{1^{12}}} 3 \overset{1^{11}}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -120, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 12, \ T \in [10, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 8 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[452] \ 1^{\overset{1^4}{1^{13}}} 31^{\overset{1^{11}}{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -120, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.47 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 13 & 17 & 49 & 15 & 13 \\ 3 & 5 & 15 & 3 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[456] \ 1^{\overset{1^2}{1^{12}}} 3 \overset{1^{11}}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -121, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.44 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 9 & 13 & 28 & 12 & 10 \\ 3 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

- [453] $1^{13} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -121$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10.27 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 9 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$
- [454] $1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -121$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.11 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 8 & 9 & 13 & 15 & 7 \\ 7 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$
- [455] $1^{13} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -121$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.24 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 17 & 0 & 48 & 12 \\ 3 & 4 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$
- [459] $1^{11} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -123$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.95 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 9 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$
- [457] $1^{113} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -123$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 8 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$
- [458] $1^{113} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -123$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 10 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$
- [462] $1^{13} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -124$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.74 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 22 & 0 & 84 & 16 \\ 3 & 4 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$
- [460] $21^{11} 31^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -124$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 12 & 3 & 4 \\ 5 & 4 & 13 & 4 & 4 \\ 12 & 13 & 28 & 9 & 10 \\ 3 & 4 & 9 & 0 & 2 \\ 4 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$
- [461] $21^{111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -124$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.32 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 4 \\ 8 & 9 & 8 & 4 & 5 \\ 6 & 8 & 4 & 3 & 4 \\ 3 & 4 & 3 & 0 & 2 \\ 4 & 5 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$
- [463] $1^{122} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -124$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.7$
 $T_{\min} = 10.10 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 9 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$
- [465] $1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -124$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.2$
 $T_{\min} = 10.74 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 5 & 9 & 13 & 21 & 7 \\ 7 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$
- [464] $1^{12} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -124$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 9 & 13 & 28 & 12 & 10 \\ 4 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[466] \ 1^{113} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -125, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 3 \\ 4 & 9 & 32 & 0 & 5 \\ 13 & 32 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 18 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[467] \ 1^{12} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 18 & 0 & 57 & 13 \\ 3 & 4 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[468] \ 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 7 & 9 & 13 & 15 & 7 \\ 8 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[469] \ 1^{132} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 9 & 17 & 49 & 24 & 13 \\ 5 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[470] \ 1^{12} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 11 & 13 & 28 & 12 & 10 \\ 4 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[471] \ 1^{131} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 11 & 17 & 49 & 12 & 13 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[472] \ 1^{121} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 10 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[473] \ 1^{111} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 11 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[474] \ 21^{123} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 18 & 4 \\ 5 & 4 & 4 & 18 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 18 & 18 & 12 & 57 & 13 \\ 4 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[475] \ 1^{111} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 12 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[476] \ 1^{112} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 12 & 18 & 12 & 57 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[477] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 11 & 18 & 12 & 57 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[478] $1^{112}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 11 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[479] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 8 & 9 & 13 & 15 & 7 \\ 8 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[480] $1^{122}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 6 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[483] $1^{131}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 9 & 12 & 25 & 9 & 9 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[482] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 13 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[481] $1^{131}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 9 & 12 & 25 & 9 & 9 \\ 4 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[484] $1^{111}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 5 & 9 & 13 & 27 & 7 \\ 9 & 17 & 27 & 48 & 12 \\ 3 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[485] $1^131^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 11 & 2 & 3 \\ 4 & 9 & 28 & 4 & 5 \\ 11 & 28 & 73 & 12 & 15 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 5 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[486] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 14 & 2 & 3 \\ 4 & 9 & 28 & 4 & 5 \\ 14 & 28 & 73 & 12 & 15 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 5 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[487] $1^{111}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -133, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 6 & 9 & 13 & 15 & 7 \\ 7 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[488] $1^{13}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 10 & 17 & 49 & 15 & 13 \\ 3 & 5 & 15 & 3 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[489] $1^{13}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 9 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[490] $1^{11}3^11^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 7 & 9 & 13 & 15 & 7 \\ 6 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[491] $1^{122}3^11^{42} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 11 & 13 & 28 & 18 & 10 \\ 6 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[492] $1^{122}3^11^{23} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 10 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[493] $1^{132}3^11^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 11 & 17 & 49 & 24 & 13 \\ 4 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

[494] $1^{122}3^11^{33} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 7 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 10 & 13 & 28 & 18 & 10 \\ 7 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[495] $1^{141}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 16 & 4 & 4 \\ 9 & 16 & 46 & 12 & 12 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[496] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 5 & 9 & 13 & 27 & 7 \\ 11 & 17 & 27 & 48 & 12 \\ 3 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[497] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 8 & 9 & 13 & 21 & 7 \\ 10 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[498] $1^{12}3^11^{41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -138, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 11 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 2 \end{pmatrix}_{b_0}$$

[499] $1^{12} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -138, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 13 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 2 \end{pmatrix}_{b_0}$$

[506] $21^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 4 & 9 & 4 & 4 \\ 6 & 9 & 13 & 6 & 7 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[501] $1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 5 & 9 & 13 & 21 & 7 \\ 8 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[502] $1^{13} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 15 & 27 & 133 & 18 & 21 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[503] $1^{14} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 4 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 14 & 21 & 76 & 18 & 16 \\ 4 & 5 & 18 & 3 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[504] $1^{13} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 8 & 12 & 25 & 9 & 9 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[505] $1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 10 & 13 & 28 & 12 & 10 \\ 4 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[500] $1^{12} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 10 & 13 & 28 & 9 & 10 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[507] $21^{12} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 & 3 \\ 4 & 4 & 23 & 4 & 4 \\ 15 & 23 & 100 & 15 & 18 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[508] $1^{13} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 10 & 12 & 25 & 9 & 9 \\ 4 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[509] $1^{13} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 4 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 18 & 27 & 133 & 18 & 21 \\ 4 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[510] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 17 & 0 & 48 & 12 \\ 3 & 4 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[511] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 12 & 17 & 49 & 27 & 13 \\ 5 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[512] $1^{11} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.66 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 7 & 13 & 28 & 21 & 10 \\ 5 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[513] $1^{21} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 13 & 2 & 3 \\ 4 & 9 & 32 & 4 & 5 \\ 13 & 32 & 100 & 15 & 18 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 5 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

[514] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.66 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 9 & 13 & 28 & 21 & 10 \\ 5 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[515] $1^{14} 3 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 22 & 2 & 3 \\ 4 & 9 & 44 & 4 & 5 \\ 22 & 44 & 193 & 21 & 25 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 5 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

[516] $1^{14} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 12 & 21 & 76 & 18 & 16 \\ 3 & 5 & 18 & 3 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[517] $1^{12} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 18 & 0 & 57 & 13 \\ 3 & 4 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

[518] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 9 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[519] $1^{12} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 18 & 0 & 57 & 13 \\ 3 & 4 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

[520] $1^{14} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 11 & 21 & 76 & 15 & 16 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[523] $1^{11} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.61 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 15 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 2 \end{pmatrix}_{b_0}$$

[522] $1^{12} 3 1^{132} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 4 & 22 & 8 & 4 \\ 12 & 22 & 85 & 30 & 17 \\ 4 & 8 & 30 & 9 & 5 \\ 3 & 4 & 17 & 5 & 1 \end{pmatrix}_{b_0}$$

[521] $1^{132} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 7 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 12 & 17 & 49 & 24 & 13 \\ 7 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[524] $1^{132} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 4 & 3 \\ 3 & 4 & 22 & 8 & 4 \\ 13 & 22 & 85 & 30 & 17 \\ 4 & 8 & 30 & 9 & 5 \\ 3 & 4 & 17 & 5 & 1 \end{pmatrix}_{b_0}$$

[525] $1^{132} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.61 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 7 & 12 & 25 & 18 & 9 \\ 5 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[526] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.03 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 17 & 0 & 48 & 12 \\ 3 & 4 & 0 & 12 & 2 \end{pmatrix}_{b_0}$$

[527] $1^{11} 3 1^{132} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 9 & 17 & 49 & 24 & 13 \\ 4 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

[530] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 10 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[529] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 8 & 13 & 28 & 18 & 10 \\ 5 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[528] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 10 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[531] $1^{113} 3 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -149, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 9 & 32 & 4 & 5 \\ 15 & 32 & 100 & 15 & 18 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 5 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

[535] $1^{14} 3^{121} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 10 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 2 \end{pmatrix}_{b_0}$$

[534] $1^{111} 3^{111} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 6 & 9 & 13 & 15 & 7 \\ 6 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[533] $1^{12} 3^{111} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 9 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[532] $1^{14} 3^{112} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 15 & 26 & 0 & 117 & 19 \\ 3 & 4 & 0 & 19 & 2 \end{pmatrix}_{b_0}$$

[537] $1^{13} 3^{111} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 12 & 17 & 49 & 15 & 13 \\ 3 & 5 & 15 & 3 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[536] $1^{14} 3^{112} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 21 & 0 & 75 & 15 \\ 3 & 4 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[538] $1^{131} 3^{112} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 12 & 22 & 85 & 15 & 17 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[540] $1^{41} 3^{114} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 20 & 2 & 3 \\ 4 & 9 & 48 & 4 & 5 \\ 20 & 48 & 238 & 24 & 28 \\ 2 & 4 & 24 & 0 & 2 \\ 3 & 5 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

[539] $1^{114} 3^{12} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 20 & 40 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 22 & -1 & 1 \end{pmatrix}_{b_0}$$

[542] $1^{112} 3^{12} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -153, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 12 & 18 & 12 & 57 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[541] $1^{13} 3^{12} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -153, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 12 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 17 & -1 & 2 \end{pmatrix}_{b_0}$$

[543] $1^{122} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 9 & 13 & 28 & 18 & 10 \\ 5 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[544] $1^{121} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -155, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 8 & 13 & 28 & 9 & 10 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[545] $1^{121} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -155, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 10 & 13 & 28 & 9 & 10 \\ 4 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[547] $1^{13} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 11 & 17 & 49 & 15 & 13 \\ 3 & 5 & 15 & 3 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[548] $1^{12} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 10 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 2 \end{pmatrix}_{b_0}$$

[546] $1^{121} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 8 & 13 & 28 & 9 & 10 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[549] $1^{121} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 14 & 23 & 100 & 15 & 18 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[551] $1^{122} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -158, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 7 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 11 & 13 & 28 & 18 & 10 \\ 7 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[550] $1^{113} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -158, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.61 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 13 & 27 & 7 \\ 10 & 17 & 27 & 48 & 12 \\ 3 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[552] $1^{131} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 8 & 12 & 25 & 9 & 9 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[553] $1^{131} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 10 & 12 & 25 & 9 & 9 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[554] $1^{141} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 16 & 4 & 4 \\ 9 & 16 & 46 & 12 & 12 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

[555] $1^{12} 3^1 1^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 10 & 18 & 58 & 12 & 14 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[556] $1^{121} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 14 & 18 & 58 & 12 & 14 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[557] $21^{112} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 4 & 4 & 8 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 6 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[558] $1^{121} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 4 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 13 & 18 & 58 & 12 & 14 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[559] $1^{141} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 3 \\ 3 & 4 & 16 & 4 & 4 \\ 11 & 16 & 46 & 12 & 12 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[560] $1^{111} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 7 & 9 & 13 & 15 & 7 \\ 7 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[561] $1^{13} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 12 & 17 & 49 & 15 & 13 \\ 4 & 5 & 15 & 3 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[565] $1^{113} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 10 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[562] $1^{132} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 9 & 12 & 25 & 18 & 9 \\ 5 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[563] $1^{132} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 9 & 12 & 25 & 18 & 9 \\ 7 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[564] $1^{113} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 13 & 28 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 15 & -1 & 2 \end{pmatrix}_{b_0}$$

[566] $1^{12} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -162, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 16 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 2 \end{pmatrix}_{b_0}$$

[573] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 11 & 13 & 28 & 9 & 10 \\ 4 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[574] $1^{113} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 13 & 22 & 15 & 84 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[567] $1^{113} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 5 & 9 & 13 & 27 & 7 \\ 10 & 17 & 27 & 48 & 12 \\ 3 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[568] $1^{131} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 22 & 4 & 3 \\ 3 & 4 & 32 & 4 & 4 \\ 22 & 32 & 193 & 21 & 25 \\ 4 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$

[569] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 9 & 13 & 28 & 9 & 10 \\ 2 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[570] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 9 & 13 & 28 & 18 & 10 \\ 6 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[571] $1^{132} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 9 & 17 & 49 & 39 & 13 \\ 8 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[572] $1^{141} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 4 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 20 & 31 & 172 & 21 & 24 \\ 4 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

[577] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 3 \\ 4 & 9 & 28 & 4 & 5 \\ 12 & 28 & 73 & 12 & 15 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 5 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

[575] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 13 & 21 & 7 \\ 10 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[576] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 13 & 2 & 3 \\ 4 & 9 & 28 & 4 & 5 \\ 13 & 28 & 73 & 12 & 15 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 5 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[579] $1^{13}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -165, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 10 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[578] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -165, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 12 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[583] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 14 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 19 & -1 & 1 \end{pmatrix}_{b_0}$$

[580] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 18 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 19 & -1 & 1 \end{pmatrix}_{b_0}$$

[581] $1^{114}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 3 \\ 4 & 9 & 48 & 0 & 5 \\ 20 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 28 & -1 & 2 \end{pmatrix}_{b_0}$$

[582] $1^{111}31^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 9 & 17 & 49 & 27 & 13 \\ 5 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[584] $1^{14}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -169, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 15 & 21 & 76 & 18 & 16 \\ 3 & 5 & 18 & 3 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[585] $1^{132}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -170, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 10 & 17 & 49 & 24 & 13 \\ 4 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

[586] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 13 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 2 \end{pmatrix}_{b_0}$$

[587] $1^{12} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 12 & 22 & 15 & 84 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[590] $1^{12} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 22 & 0 & 84 & 16 \\ 3 & 4 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

[588] $1^{122} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 7 & 13 & 28 & 30 & 10 \\ 8 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[589] $1^{11} 3 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 9 & 17 & 49 & 12 & 13 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[591] $1^{14} 3 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 28 & 4 & 4 \\ 16 & 28 & 154 & 18 & 22 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[592] $1^{13} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -173, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 13 & 27 & 7 \\ 13 & 17 & 27 & 48 & 12 \\ 3 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[593] $1^{121} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -174, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 8 & 13 & 28 & 21 & 10 \\ 5 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[594] $1^{13} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 1 & 12 & 3 \\ 11 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[595] $21^{12} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 4 & 13 & 5 & 4 \\ 9 & 13 & 28 & 12 & 10 \\ 3 & 5 & 12 & 3 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[596] $1^{131} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 9 & 12 & 25 & 9 & 9 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[597] $1^{112} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 13 & 21 & 7 \\ 8 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[598] $1^{122} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 10 & 13 & 28 & 18 & 10 \\ 6 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[599] $1^{112} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 14 & 18 & 12 & 57 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[600] $1^{132} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 8 & 12 & 25 & 18 & 9 \\ 6 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[601] $1^{12} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 13 & 21 & 76 & 18 & 16 \\ 3 & 5 & 18 & 3 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[602] $1^{121} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 10 & 13 & 28 & 21 & 10 \\ 8 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[603] $1^{112} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 13 & 18 & 12 & 57 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[604] $1^{141} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 6 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 12 & 21 & 76 & 33 & 16 \\ 6 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$

[605] $1^{132} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 10 & 12 & 25 & 18 & 9 \\ 6 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[606] $1^{12} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -178, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 11 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 14 & -1 & 2 \end{pmatrix}_{b_0}$$

[607] $1^{13} 3 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 13 & 23 & 100 & 15 & 18 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[608] $1^{13} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 4 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 13 & 17 & 49 & 15 & 13 \\ 4 & 5 & 15 & 3 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[609] $1^{121} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 4 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 17 & 23 & 100 & 15 & 18 \\ 4 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[610] $1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 7 & 9 & 13 & 21 & 7 \\ 9 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[611] $1^{121} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 11 & 18 & 58 & 12 & 14 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[612] $1^{131} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 7 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 11 & 17 & 49 & 27 & 13 \\ 7 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[613] $1^{121} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 11 & 13 & 28 & 9 & 10 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[614] $1^{132} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 7 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 10 & 12 & 25 & 18 & 9 \\ 7 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[615] $1^{132} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 8 & 12 & 25 & 18 & 9 \\ 5 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[616] $21^{111} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 9 & 12 & 4 \\ 5 & 4 & 9 & 13 & 4 \\ 9 & 9 & 13 & 21 & 7 \\ 12 & 13 & 21 & 27 & 9 \\ 4 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[617] $1^1 3^1 1^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 11 & 21 & 76 & 33 & 16 \\ 5 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$

[618] $1^{111} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -183, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 15 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 2 \end{pmatrix}_{b_0}$$

[619] $21^{121} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 4 & 8 & 4 & 4 \\ 6 & 8 & 10 & 6 & 6 \\ 3 & 4 & 6 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0} \\
[620] \quad & 21^{132}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 9 & 9 & 3 \\ 4 & 4 & 12 & 13 & 4 \\ 9 & 12 & 25 & 27 & 9 \\ 9 & 13 & 27 & 27 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0} \\
[621] \quad & 1^{111}31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -185, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7 \\
& T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 7 & 13 & 28 & 30 & 10 \\ 7 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0} \\
[622] \quad & 1^{132}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -185, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 4 & 22 & 8 & 4 \\ 13 & 22 & 85 & 30 & 17 \\ 5 & 8 & 30 & 9 & 5 \\ 3 & 4 & 17 & 5 & 2 \end{pmatrix}_{b_0} \\
[623] \quad & 1^{132}31^{41} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -185, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.87 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 16 & 5 & 3 \\ 3 & 4 & 22 & 8 & 4 \\ 16 & 22 & 85 & 30 & 17 \\ 5 & 8 & 30 & 9 & 5 \\ 3 & 4 & 17 & 5 & 2 \end{pmatrix}_{b_0} \\
[624] \quad & 1^{14}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -186, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 11 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 2 \end{pmatrix}_{b_0} \\
[625] \quad & 1^{14}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -186, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 14 & 21 & 76 & 18 & 16 \\ 3 & 5 & 18 & 3 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0} \\
[628] \quad & 1^{14}31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6 \\
& T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 14 & 21 & 0 & 75 & 15 \\ 3 & 4 & 0 & 15 & 2 \end{pmatrix}_{b_0} \\
[627] \quad & 21^{123}31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 12, T \in [10, 12] \\
& \begin{pmatrix} 4 & 5 & 3 & 21 & 4 \\ 5 & 4 & 4 & 22 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 21 & 22 & 15 & 84 & 16 \\ 4 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0} \\
[626] \quad & 1^{113}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 12, T \in [10, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 16 & 22 & 15 & 84 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0} \\
[629] \quad & 1^{14}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -188, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 15 & 4 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 15 & 21 & 76 & 18 & 16 \\ 4 & 5 & 18 & 3 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0} \\
[630] \quad & 1^{132}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -188, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 11 & 7 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 11 & 17 & 49 & 39 & 13 \\ 7 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 1 \end{pmatrix}_{b_0}$$

[632] $1^{11} 3^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -189, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 14 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 2 \end{pmatrix}_{b_0}$$

[631] $21^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -189, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 9 & 9 & 4 \\ 5 & 4 & 9 & 9 & 4 \\ 9 & 9 & 13 & 15 & 7 \\ 9 & 9 & 15 & 12 & 6 \\ 4 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[633] $1^{122} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -190, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 8 & 13 & 28 & 30 & 10 \\ 7 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[634] $21^{12} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -192, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 4 & 4 & 18 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 12 & 18 & 12 & 57 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[635] $21^{11} 31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -192, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 12 & 9 & 4 \\ 5 & 4 & 13 & 9 & 4 \\ 12 & 13 & 28 & 21 & 10 \\ 9 & 9 & 21 & 12 & 6 \\ 4 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[636] $1^{121} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -196, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 9 & 13 & 28 & 9 & 10 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[637] $1^{121} 3^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -196, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 28 & 4 & 4 \\ 18 & 28 & 154 & 18 & 22 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[638] $1^{11} 3^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -197, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 14 & 2 & 3 \\ 4 & 9 & 36 & 4 & 5 \\ 14 & 36 & 121 & 15 & 19 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 5 & 19 & 2 & 1 \end{pmatrix}_{b_0}$$

[639] $1^{14} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 12 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 2 \end{pmatrix}_{b_0}$$

[640] $1^{14} 3^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 14 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[641] $1^{132} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -199, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 13 & 17 & 49 & 24 & 13 \\ 5 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[642] $1^{112} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -199, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 13 & 21 & 7 \\ 9 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[644] $1^{114} 3 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 9 & 44 & 0 & 5 \\ 18 & 44 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 25 & -1 & 1 \end{pmatrix}_{b_0}$$

[643] $1^{114} 3 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 16 & 40 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 22 & -1 & 1 \end{pmatrix}_{b_0}$$

[645] $1^{121} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -201, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 10 & 13 & 28 & 21 & 10 \\ 6 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[646] $1^{121} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -201, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 8 & 13 & 28 & 21 & 10 \\ 6 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[647] $1^{113} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -202, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 11 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[648] $1^{132} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -203, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 10 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 11 & 17 & 49 & 39 & 13 \\ 10 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[649] $1^{121} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 10 & 13 & 28 & 9 & 10 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[650] $1^{13} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 15 & 22 & 0 & 84 & 16 \\ 3 & 4 & 0 & 16 & 2 \end{pmatrix}_{b_0}$$

[651] $1^{114} 3 1^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 20 & 2 & 3 \\ 4 & 9 & 40 & 4 & 5 \\ 20 & 40 & 154 & 18 & 22 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 5 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

[652] $21^{111} 31^{132} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -205, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 15 & 12 & 4 \\ 5 & 4 & 17 & 13 & 4 \\ 15 & 17 & 49 & 39 & 13 \\ 12 & 13 & 39 & 27 & 9 \\ 4 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[653] $1^{11} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -205, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 16 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 2 \end{pmatrix}_{b_0}$$

[654] $1^{132} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -205, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 3 \\ 3 & 4 & 12 & 8 & 4 \\ 9 & 12 & 25 & 18 & 9 \\ 6 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[659] $1^{113} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 7 & 9 & 13 & 27 & 7 \\ 12 & 17 & 27 & 48 & 12 \\ 3 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[655] $1^{131} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 8 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 13 & 17 & 49 & 27 & 13 \\ 8 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[656] $1^{132} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 9 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 10 & 17 & 49 & 39 & 13 \\ 9 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[657] $1^{14} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 15 & 21 & 0 & 75 & 15 \\ 3 & 4 & 0 & 15 & 2 \end{pmatrix}_{b_0}$$

[658] $21^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 4 & 9 & 9 & 4 \\ 6 & 9 & 13 & 15 & 7 \\ 6 & 9 & 15 & 12 & 6 \\ 3 & 4 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[667] $1^{121} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.42 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 11 & 18 & 58 & 12 & 14 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[660] $1^{121} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 4 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 14 & 18 & 58 & 12 & 14 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[661] $1^{131} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 12 & 17 & 49 & 12 & 13 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[662] $1^{121} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 13 & 18 & 58 & 12 & 14 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[663] $1^{141} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 16 & 4 & 4 \\ 10 & 16 & 46 & 12 & 12 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[664] $1^{14} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 12 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 2 \end{pmatrix}_{b_0}$$

[665] $1^{141} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.42 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 16 & 4 & 4 \\ 12 & 16 & 46 & 12 & 12 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[666] $1^{141} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.42 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 4 & 16 & 4 & 4 \\ 12 & 16 & 46 & 12 & 12 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[668] $1^{114} 3 1^{31} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -209, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 21 & 2 & 3 \\ 4 & 9 & 44 & 4 & 5 \\ 21 & 44 & 193 & 21 & 25 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 5 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

[670] $1^{131} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -210, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 10 & 17 & 49 & 27 & 13 \\ 6 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[669] $1^{121} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -210, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 8 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 11 & 13 & 28 & 21 & 10 \\ 8 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[671] $1^{113} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 13 & 22 & 15 & 84 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[672] $1^{131} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 4 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 15 & 22 & 85 & 15 & 17 \\ 4 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[673] $1^{112} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 7 & 9 & 13 & 21 & 7 \\ 10 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$$

[674] $1^{121} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 17 & 23 & 100 & 15 & 18 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[675] $1^{13}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 15 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 2 \end{pmatrix}_{b_0}$$

[676] $1^{121}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -213, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 9 & 13 & 28 & 21 & 10 \\ 7 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[677] $1^{114}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -213, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 3 \\ 4 & 9 & 36 & 4 & 5 \\ 18 & 36 & 121 & 15 & 19 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 5 & 19 & 2 & 1 \end{pmatrix}_{b_0}$$

[678] $1^{121}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -213, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 7 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 11 & 13 & 28 & 21 & 10 \\ 7 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[679] $21^{122}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 9 & 3 \\ 4 & 4 & 8 & 13 & 4 \\ 6 & 8 & 10 & 18 & 6 \\ 9 & 13 & 18 & 27 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[680] $21^{112}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 12 & 3 & 4 \\ 8 & 9 & 16 & 4 & 5 \\ 12 & 16 & 22 & 6 & 8 \\ 3 & 4 & 6 & 0 & 2 \\ 4 & 5 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

[681] $1^{14}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 16 & 26 & 0 & 117 & 19 \\ 3 & 4 & 0 & 19 & 2 \end{pmatrix}_{b_0}$$

[682] $1^{141}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 & 3 \\ 3 & 4 & 36 & 4 & 4 \\ 20 & 36 & 238 & 24 & 28 \\ 3 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

[683] $1^{131}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 11 & 17 & 49 & 27 & 13 \\ 5 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[684] $1^{141}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 16 & 4 & 4 \\ 10 & 16 & 46 & 12 & 12 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[685] $1^{141}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 14 & 26 & 118 & 18 & 20 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

[686] $1^{132}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -217, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 6 & 3 \\ 3 & 4 & 22 & 8 & 4 \\ 15 & 22 & 85 & 30 & 17 \\ 6 & 8 & 30 & 9 & 5 \\ 3 & 4 & 17 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[689] \ 1^{13} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.02 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 15 & 22 & 15 & 84 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[687] \ 1^{12} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.02 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 11 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[688] \ 1^{13} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 10 & 17 & 49 & 12 & 13 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[690] \ 1^{13} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -220, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 16 & 22 & 0 & 84 & 16 \\ 3 & 4 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[691] \ 1^{13} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -221, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 13 & 27 & 7 \\ 11 & 17 & 27 & 48 & 12 \\ 3 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[692] \ 1^3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -221, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 3 \\ 4 & 9 & 44 & 4 & 5 \\ 18 & 44 & 193 & 21 & 25 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 5 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[693] \ 21^{12} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -223, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 8 & 15 & 3 & 4 \\ 8 & 9 & 20 & 4 & 5 \\ 15 & 20 & 37 & 9 & 11 \\ 3 & 4 & 9 & 0 & 2 \\ 4 & 5 & 11 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[694] \ 1^{14} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 24 & 4 & 3 \\ 3 & 4 & 36 & 4 & 4 \\ 24 & 36 & 238 & 24 & 28 \\ 4 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[695] \ 1^{12} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 12 & 18 & 58 & 12 & 14 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[696] \ 1^{14} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 16 & 4 & 4 \\ 11 & 16 & 46 & 12 & 12 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[697] \ 1^{13} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -225, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 10 & 17 & 49 & 27 & 13 \\ 5 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[698] $1^{113} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 13 & 27 & 7 \\ 12 & 17 & 27 & 48 & 12 \\ 3 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$

[699] $1^{141} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 5 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 14 & 21 & 76 & 33 & 16 \\ 5 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$

[700] $1^{12} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 12 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 14 & -1 & 2 \end{pmatrix}_{b_0}$$

[701] $1^{113} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -227, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 1 & 15 & 3 \\ 14 & 22 & 15 & 84 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[702] $1^{132} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -228, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 11 & 17 & 49 & 24 & 13 \\ 6 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[703] $1^{11} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -228, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 14 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 2 \end{pmatrix}_{b_0}$$

[704] $1^{132} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -229, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 10 & 17 & 49 & 24 & 13 \\ 5 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[705] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 12 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[706] $21^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 4 & 4 & 13 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 9 & 13 & 9 & 27 & 9 \\ 3 & 4 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

[707] $21^{122} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 4 & 8 & 8 & 4 \\ 6 & 8 & 10 & 12 & 6 \\ 6 & 8 & 12 & 9 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[708] $1^{111} 3 1^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 9 & 17 & 49 & 39 & 13 \\ 7 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[709] $1^{121} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -234, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 9 & 13 & 28 & 21 & 10 \\ 6 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[713] $1^{113} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 14 & 22 & 15 & 84 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[710] $1^{121} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 14 & 23 & 100 & 15 & 18 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[711] $1^{141} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 4 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 14 & 21 & 76 & 15 & 16 \\ 4 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[712] $1^{11} 3 1^{141} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 11 & 21 & 76 & 15 & 16 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

[716] $1^{131} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 10 & 17 & 49 & 12 & 13 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[719] $1^{131} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 13 & 17 & 49 & 12 & 13 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[714] $1^{141} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 4 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 17 & 26 & 118 & 18 & 20 \\ 4 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

[715] $1^{121} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 2 & 3 \\ 3 & 4 & 28 & 4 & 4 \\ 20 & 28 & 154 & 18 & 22 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[718] $1^{121} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 28 & 4 & 4 \\ 17 & 28 & 154 & 18 & 22 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[717] $1^{121} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 3 \\ 4 & 9 & 40 & 4 & 5 \\ 16 & 40 & 154 & 18 & 22 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 5 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

[721] $1^{141} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 7 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 14 & 21 & 76 & 33 & 16 \\ 7 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$

[720] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.03 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 13 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[722] $1^{132} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -239, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 7 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 13 & 17 & 49 & 24 & 13 \\ 7 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[724] $21^{113} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -240, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 4 & 4 & 12 & 4 \\ 3 & 4 & 1 & 9 & 3 \\ 9 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[723] $1^{141} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -240, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 17 & 31 & 172 & 21 & 24 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

[725] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -241, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 17 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 2 \end{pmatrix}_{b_0}$$

[728] $1^{132} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -241, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 5 & 3 \\ 3 & 4 & 22 & 8 & 4 \\ 14 & 22 & 85 & 30 & 17 \\ 5 & 8 & 30 & 9 & 5 \\ 3 & 4 & 17 & 5 & 2 \end{pmatrix}_{b_0}$$

[727] $21^{13} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -241, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 4 & 17 & 5 & 4 \\ 12 & 17 & 49 & 15 & 13 \\ 3 & 5 & 15 & 3 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[726] $1^{132} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -241, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.63 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 5 & 3 \\ 3 & 4 & 22 & 8 & 4 \\ 15 & 22 & 85 & 30 & 17 \\ 5 & 8 & 30 & 9 & 5 \\ 3 & 4 & 17 & 5 & 2 \end{pmatrix}_{b_0}$$

[732] $21^{111} 31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 12 & 12 & 4 \\ 5 & 4 & 13 & 13 & 4 \\ 12 & 13 & 28 & 30 & 10 \\ 12 & 13 & 30 & 27 & 9 \\ 4 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[731] $1^{121} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 12 & 18 & 58 & 12 & 14 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[730] $21^{12}31^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 18 & 3 & 4 \\ 5 & 4 & 18 & 4 & 4 \\ 18 & 18 & 58 & 12 & 14 \\ 3 & 4 & 12 & 0 & 2 \\ 4 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[729] $1^{121}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 15 & 23 & 100 & 15 & 18 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[733] $1^{114}31^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -245, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 19 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 19 & 40 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 22 & -1 & 1 \end{pmatrix}_{b_0}$$

[735] $1^{121}31^{132} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -246, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 7 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 10 & 13 & 28 & 21 & 10 \\ 7 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$

[734] $1^{122}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -246, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 8 & 13 & 28 & 30 & 10 \\ 9 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[737] $1^{13}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 13 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 17 & -1 & 2 \end{pmatrix}_{b_0}$$

[739] $1^{114}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 17 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 17 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 19 & -1 & 2 \end{pmatrix}_{b_0}$$

[738] $1^{114}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.19 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 9 & 36 & 4 & 5 \\ 15 & 36 & 121 & 15 & 19 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 5 & 19 & 2 & 1 \end{pmatrix}_{b_0}$$

[736] $1^{13}31^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 12 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[740] $1^{12}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -250, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 14 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 14 & -1 & 2 \end{pmatrix}_{b_0}$$

[741] $1^{12}31^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -250, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 13 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 14 & -1 & 2 \end{pmatrix}_{b_0}$$

[742] $1^{121}31^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -251$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 16 & 23 & 100 & 15 & 18 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[743] $1^{141}31^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -253$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.54 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 18 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 2 \end{pmatrix}_{b_0}$$

[746] $1^{131}31^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -255$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10.18 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 32 & 4 & 4 \\ 19 & 32 & 193 & 21 & 25 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$

[745] $1^{141}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -255$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10.18 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 13 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[744] $1^{131}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -255$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10.18 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 6 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 13 & 17 & 49 & 27 & 13 \\ 6 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[748] $1^{122}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -256$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.7$
 $T_{\min} = 10.31 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 9 & 13 & 28 & 30 & 10 \\ 8 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[747] $1^{131}31^{142} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -256$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10.31 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 4 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 13 & 17 & 49 & 12 & 13 \\ 4 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[749] $1^{114}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -257$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 17 & 2 & 3 \\ 4 & 9 & 36 & 4 & 5 \\ 17 & 36 & 121 & 15 & 19 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 5 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

[750] $1^{121}31^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -259$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 15 & 23 & 100 & 15 & 18 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[751] $1^{12}31^{131} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -259$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10.76 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 12 & 22 & 85 & 15 & 17 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[752] $1^{114}31^{21} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -260$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.05 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 19 & 2 & 3 \\ 4 & 9 & 40 & 4 & 5 \\ 19 & 40 & 154 & 18 & 22 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 5 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

[753] $1^{13} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 18 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 2 \end{pmatrix}_{b_0}$$

[755] $1^{141} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -262, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 12 & 21 & 76 & 33 & 16 \\ 5 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$

[754] $1^{14} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -262, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 17 & 26 & 0 & 117 & 19 \\ 3 & 4 & 0 & 19 & 2 \end{pmatrix}_{b_0}$$

[756] $1^{141} 3 1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 16 & 4 & 4 \\ 11 & 16 & 46 & 12 & 12 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[757] $1^{132} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -265, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 12 & 17 & 49 & 24 & 13 \\ 5 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[758] $21^{122} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -265, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 6 & 3 \\ 4 & 4 & 13 & 8 & 4 \\ 9 & 13 & 28 & 18 & 10 \\ 6 & 8 & 18 & 9 & 5 \\ 3 & 4 & 10 & 5 & 2 \end{pmatrix}_{b_0}$$

[764] $1^{121} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 28 & 4 & 4 \\ 18 & 28 & 154 & 18 & 22 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[762] $1^{121} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 2 & 3 \\ 3 & 4 & 28 & 4 & 4 \\ 19 & 28 & 154 & 18 & 22 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[761] $1^{122} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 10 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 11 & 13 & 28 & 30 & 10 \\ 10 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[760] $1^{131} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 4 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 19 & 27 & 133 & 18 & 21 \\ 4 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[759] $1^{131} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 11 & 17 & 49 & 12 & 13 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[763] $1^{131} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 12 & 17 & 49 & 12 & 13 \\ 2 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[765] $1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -269, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 3 \\ 4 & 9 & 36 & 4 & 5 \\ 16 & 36 & 121 & 15 & 19 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 5 & 19 & 2 & 1 \end{pmatrix}_{b_0}$$

[766] $1^{12} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -270, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 17 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 2 \end{pmatrix}_{b_0}$$

[767] $1^{132} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 7 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 10 & 17 & 49 & 39 & 13 \\ 7 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[768] $1^{132} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -273, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 6 & 3 \\ 3 & 4 & 22 & 8 & 4 \\ 16 & 22 & 85 & 30 & 17 \\ 6 & 8 & 30 & 9 & 5 \\ 3 & 4 & 17 & 5 & 2 \end{pmatrix}_{b_0}$$

[769] $1^{132} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -275, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 11 & 17 & 49 & 24 & 13 \\ 5 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[770] $1^{122} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -275, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 8 & 13 & 28 & 30 & 10 \\ 8 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[771] $1^{114} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 16 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 19 & -1 & 1 \end{pmatrix}_{b_0}$$

[774] $1^{121} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 14 & 18 & 58 & 12 & 14 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[773] $1^{121} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 13 & 18 & 58 & 12 & 14 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[772] $1^{121} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 28 & 4 & 4 \\ 19 & 28 & 154 & 18 & 22 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[775] $1^{141} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -277, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 13 & 21 & 76 & 33 & 16 \\ 5 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$

[776] $1^{131} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -279, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 13 & 22 & 85 & 15 & 17 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[777] $1^{141} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 16 & 4 & 4 \\ 12 & 16 & 46 & 12 & 12 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[780] $1^{122} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 10 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 9 & 13 & 28 & 30 & 10 \\ 10 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[779] $1^{141} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 12 & 21 & 76 & 15 & 16 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[778] $21^{113} 31^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 24 & 3 & 4 \\ 8 & 9 & 32 & 4 & 5 \\ 24 & 32 & 100 & 15 & 18 \\ 3 & 4 & 15 & 0 & 2 \\ 4 & 5 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[781] $1^{14} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -282, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 31 & 0 & 4 \\ 17 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 2 \end{pmatrix}_{b_0}$$

[782] $1^{122} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -282, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 9 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 10 & 13 & 28 & 30 & 10 \\ 9 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[783] $1^{113} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 15 & 22 & 15 & 84 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[784] $1^{114} 3 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -284, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 3 \\ 4 & 9 & 40 & 4 & 5 \\ 18 & 40 & 154 & 18 & 22 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 5 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[785] $1^{131} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -285, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 7 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 12 & 17 & 49 & 27 & 13 \\ 7 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

- [786] $1^{14} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -286$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.07 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 13 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 2 \end{pmatrix}_{b_0}$
- [787] $21^{12} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -289$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 6 & 9 & 3 \\ 4 & 4 & 9 & 13 & 4 \\ 6 & 9 & 13 & 21 & 7 \\ 9 & 13 & 21 & 27 & 9 \\ 3 & 4 & 7 & 9 & 2 \end{pmatrix}_{b_0}$
- [788] $1^{13} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -291$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10.15 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 16 & 27 & 133 & 18 & 21 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$
- [789] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -291$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10.15 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 11 & 17 & 49 & 12 & 13 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$
- [790] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -291$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10.15 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 11 & 6 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 11 & 17 & 49 & 27 & 13 \\ 6 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$
- [791] $21^{12} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -292$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 4 & 18 & 4 & 4 \\ 12 & 18 & 58 & 12 & 14 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$
- [792] $21^{13} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -292$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.62 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 15 & 6 & 3 \\ 4 & 4 & 22 & 8 & 4 \\ 15 & 22 & 85 & 30 & 17 \\ 6 & 8 & 30 & 9 & 5 \\ 3 & 4 & 17 & 5 & 2 \end{pmatrix}_{b_0}$
- [794] $1^{14} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -294$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.6$
 $T_{\min} = 10$, $T_{\max} = 12$, $T \in [10, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 18 & 26 & 0 & 117 & 19 \\ 3 & 4 & 0 & 19 & 3 \end{pmatrix}_{b_0}$
- [793] $1^{12} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -294$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10.11 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 15 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 2 \end{pmatrix}_{b_0}$
- [796] $21^{14} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -295$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.26 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 & 3 \\ 4 & 4 & 21 & 5 & 4 \\ 15 & 21 & 76 & 18 & 16 \\ 3 & 5 & 18 & 3 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$
- [795] $21^{14} 31^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -295$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.53 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 4 & 8 & 33 & 3 & 4 \\ 8 & 9 & 44 & 4 & 5 \\ 33 & 44 & 193 & 21 & 25 \\ 3 & 4 & 21 & 0 & 2 \\ 4 & 5 & 25 & 2 & 2 \end{pmatrix}_{b_0}$
- [797] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -298$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.30 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 6 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 12 & 17 & 49 & 24 & 13 \\ 6 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[798] $1^{131} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -299, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 4 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 16 & 22 & 85 & 15 & 17 \\ 4 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[799] $1^{121} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -300, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 16 & 23 & 100 & 15 & 18 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[800] $1^{132} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 8 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 12 & 17 & 49 & 39 & 13 \\ 8 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[801] $21^{131} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 4 & 12 & 4 & 4 \\ 9 & 12 & 25 & 9 & 9 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[803] $1^{141} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -306, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 6 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 13 & 21 & 76 & 33 & 16 \\ 6 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$

[802] $1^{131} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -306, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 6 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 12 & 17 & 49 & 27 & 13 \\ 6 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[804] $21^{121} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -307, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 4 & 13 & 4 & 4 \\ 9 & 13 & 28 & 9 & 10 \\ 3 & 4 & 9 & 0 & 2 \\ 3 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$$

[805] $1^{113} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -307, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 16 & 22 & 15 & 84 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[806] $1^{132} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -312, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 6 & 3 \\ 3 & 4 & 17 & 8 & 4 \\ 13 & 17 & 49 & 24 & 13 \\ 6 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 2 \end{pmatrix}_{b_0}$$

[807] $21^{13} 31^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -316, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 24 & 3 & 4 \\ 5 & 4 & 23 & 4 & 4 \\ 24 & 23 & 100 & 15 & 18 \\ 3 & 4 & 15 & 0 & 2 \\ 4 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[808] $1^{14} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -318, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 15 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[809] $1^{13} 1^3 1^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -320, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 15 & 27 & 133 & 18 & 21 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[810] $1^{13} 1^{21} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -321, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.03 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 14 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 17 & -1 & 2 \end{pmatrix}_{b_0}$$

[812] $1^{141} 1^1 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -323, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 12 & 21 & 76 & 15 & 16 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[811] $1^{141} 1^{42} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -323, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 4 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 15 & 21 & 76 & 15 & 16 \\ 4 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[815] $1^{131} 1^{42} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -324, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 7 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 13 & 17 & 49 & 27 & 13 \\ 7 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[814] $1^{121} 1^{41} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -324, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 23 & 4 & 4 \\ 17 & 23 & 100 & 15 & 18 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[813] $1^{121} 1^{41} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -324, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 & 3 \\ 3 & 4 & 28 & 4 & 4 \\ 20 & 28 & 154 & 18 & 22 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

[817] $1^{122} 1^{22} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 9 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 9 & 13 & 28 & 30 & 10 \\ 9 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$

[816] $1^{13} 1^{11} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 16 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 2 \end{pmatrix}_{b_0}$$

[821] $1^{131} 1^{31} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 12 & 17 & 49 & 12 & 13 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

[820] $1^{141} 1^{42} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 21 & 4 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 21 & 31 & 172 & 21 & 24 \\ 4 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

- [819] $1^{131} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 13 & 22 & 85 & 15 & 17 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$
- [818] $21^{123} 1^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 21 & 3 & 4 \\ 5 & 4 & 22 & 4 & 4 \\ 21 & 22 & 85 & 15 & 17 \\ 3 & 4 & 15 & 0 & 2 \\ 4 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$
- [822] $1^{132} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -332, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 10 & 17 & 49 & 39 & 13 \\ 8 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$
- [823] $1^{122} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -335, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 10 & 13 & 28 & 30 & 10 \\ 10 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0}$$
- [824] $1^{12} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -336, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 16 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 2 \end{pmatrix}_{b_0}$$
- [825] $1^{131} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -339, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 13 & 17 & 49 & 12 & 13 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$
- [827] $1^{14} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -340, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 14 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 2 \end{pmatrix}_{b_0}$$
- [826] $21^{143} 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -340, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 30 & 3 & 4 \\ 5 & 4 & 28 & 4 & 4 \\ 30 & 28 & 154 & 18 & 22 \\ 3 & 4 & 18 & 0 & 2 \\ 4 & 4 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$
- [828] $21^{113} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -343, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 12 & 3 \\ 4 & 4 & 9 & 17 & 4 \\ 6 & 9 & 13 & 27 & 7 \\ 12 & 17 & 27 & 48 & 12 \\ 3 & 4 & 7 & 12 & 2 \end{pmatrix}_{b_0}$$
- [829] $21^{121} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -345, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 6 & 3 \\ 4 & 4 & 13 & 9 & 4 \\ 9 & 13 & 28 & 21 & 10 \\ 6 & 9 & 21 & 12 & 6 \\ 3 & 4 & 10 & 6 & 2 \end{pmatrix}_{b_0}$$
- [830] $1^{141} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -349, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 7 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 15 & 21 & 76 & 33 & 16 \\ 7 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$
- [832] $21^{131} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -352, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 21 & 3 & 3 \\ 4 & 4 & 32 & 4 & 4 \\ 21 & 32 & 193 & 21 & 25 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$

[831] $21^{132}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -352, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 6 & 3 \\ 4 & 4 & 12 & 8 & 4 \\ 9 & 12 & 25 & 18 & 9 \\ 6 & 8 & 18 & 9 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[833] $1^{12}3^{21}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -354, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 18 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 2 \end{pmatrix}_{b_0}$$

[834] $1^{12}3^{41}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -354, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 17 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 2 \end{pmatrix}_{b_0}$$

[836] $1^{131}3^{14}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -355, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 16 & 22 & 85 & 15 & 17 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[835] $1^{14}1^{131}3^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -355, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 32 & 4 & 4 \\ 18 & 32 & 193 & 21 & 25 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$

[837] $1^{14}3^{14}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -358, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 20 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 2 \end{pmatrix}_{b_0}$$

[838] $1^{141}3^{142}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 4 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 18 & 26 & 118 & 18 & 20 \\ 4 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

[840] $21^{113}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 18 & 3 & 4 \\ 8 & 9 & 24 & 4 & 5 \\ 18 & 24 & 52 & 9 & 12 \\ 3 & 4 & 9 & 0 & 2 \\ 4 & 5 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[839] $1^{12}1^{141}3^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 14 & 26 & 118 & 18 & 20 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

[841] $1^{141}3^{141}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -366, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 6 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 15 & 21 & 76 & 33 & 16 \\ 6 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$

[843] $1^{13}3^{131}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -369, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 15 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 17 & -1 & 2 \end{pmatrix}_{b_0}$$

[842] $1^{141} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -369, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 6 & 3 \\ 3 & 4 & 21 & 9 & 4 \\ 14 & 21 & 76 & 33 & 16 \\ 6 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$

[844] $1^{141} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -370, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 15 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 2 \end{pmatrix}_{b_0}$$

[845] $1^{131} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -371, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 14 & 22 & 85 & 15 & 17 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[846] $1^{132} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -372, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 9 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 13 & 17 & 49 & 39 & 13 \\ 9 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[847] $1^{131} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -375, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 14 & 22 & 85 & 15 & 17 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[848] $21^{113} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.03 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 4 & 4 & 17 & 4 \\ 3 & 4 & 1 & 12 & 3 \\ 12 & 17 & 12 & 48 & 12 \\ 3 & 4 & 3 & 12 & 2 \end{pmatrix}_{b_0}$$

[849] $21^{12} 3 1^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 24 & 3 & 4 \\ 5 & 4 & 26 & 4 & 4 \\ 24 & 26 & 118 & 18 & 20 \\ 3 & 4 & 18 & 0 & 2 \\ 4 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

[850] $1^{141} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 15 & 26 & 118 & 18 & 20 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

[851] $1^{131} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -379, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 15 & 22 & 85 & 15 & 17 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[852] $1^{141} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -379, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 13 & 21 & 76 & 15 & 16 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[853] $1^{132} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -379, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 8 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 11 & 17 & 49 & 39 & 13 \\ 8 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[854] $1^{132} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 9 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 11 & 17 & 49 & 39 & 13 \\ 9 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[855] $1^{141} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 13 & 21 & 76 & 15 & 16 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[856] $1^{132} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -388, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 10 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 12 & 17 & 49 & 39 & 13 \\ 10 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[857] $1^{13} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -393, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 16 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 17 & -1 & 2 \end{pmatrix}_{b_0}$$

[858] $1^{13} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -393, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 19 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 2 \end{pmatrix}_{b_0}$$

[859] $1^{141} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -395, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 15 & 21 & 76 & 15 & 16 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[860] $1^{131} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -399, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 & 3 \\ 3 & 4 & 32 & 4 & 4 \\ 20 & 32 & 193 & 21 & 25 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$

[861] $1^{131} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -400, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 16 & 27 & 133 & 18 & 21 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[862] $21^{113} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -400, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 4 & 4 & 22 & 4 \\ 3 & 4 & 1 & 15 & 3 \\ 15 & 22 & 15 & 84 & 16 \\ 3 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[863] $1^{141} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -403, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 14 & 21 & 76 & 15 & 16 \\ 2 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[864] $1^{131} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -411, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 17 & 27 & 133 & 18 & 21 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[866] $1^{14}3^{121}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -414, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 16 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[865] $1^{12}3^{131}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -414, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 19 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 2 \end{pmatrix}_{b_0}$$

[867] $1^{13}3^{121}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -415, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 17 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 2 \end{pmatrix}_{b_0}$$

[868] $1^{14}3^{111}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -420, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 31 & 0 & 4 \\ 18 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 2 \end{pmatrix}_{b_0}$$

[870] $1^{141}3^{111}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 21 & 3 & 3 \\ 3 & 4 & 36 & 4 & 4 \\ 21 & 36 & 238 & 24 & 28 \\ 3 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

[869] $1^{141}3^{111}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 18 & 31 & 172 & 21 & 24 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

[871] $21^{113}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -433, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 21 & 3 & 4 \\ 8 & 9 & 28 & 4 & 5 \\ 21 & 28 & 73 & 12 & 15 \\ 3 & 4 & 12 & 0 & 2 \\ 4 & 5 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

[872] $1^{131}3^{131}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 15 & 22 & 85 & 15 & 17 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[873] $1^{132}3^{132}1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 9 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 12 & 17 & 49 & 39 & 13 \\ 9 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[874] $1^{131}3^{11}1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -443, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 2 & 3 \\ 3 & 4 & 32 & 4 & 4 \\ 19 & 32 & 193 & 21 & 25 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$

[875] $1^{131}3^{12}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -448, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 17 & 27 & 133 & 18 & 21 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[876] $1^{131} 3^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -448, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 2 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 19 & 27 & 133 & 18 & 21 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[877] $1^{12} 3^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -450, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 20 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 2 \end{pmatrix}_{b_0}$$

[878] $1^{132} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -451, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 10 & 3 \\ 3 & 4 & 17 & 13 & 4 \\ 13 & 17 & 49 & 39 & 13 \\ 10 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$

[879] $1^{141} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -456, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 14 & 21 & 76 & 15 & 16 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[880] $1^{141} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -460, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 15 & 26 & 118 & 18 & 20 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

[881] $1^{131} 3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -464, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 18 & 27 & 133 & 18 & 21 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[882] $1^{13} 3^{141} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -467, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 17 & 31 & 172 & 21 & 24 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

[883] $21^{131} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -471, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 6 & 3 \\ 4 & 4 & 17 & 9 & 4 \\ 12 & 17 & 49 & 27 & 13 \\ 6 & 9 & 27 & 12 & 6 \\ 3 & 4 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[884] $1^{131} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -471, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 16 & 22 & 85 & 15 & 17 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

[885] $21^{141} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -472, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 4 & 16 & 4 & 4 \\ 12 & 16 & 46 & 12 & 12 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[886] $1^{13} 3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -481, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 18 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 2 \end{pmatrix}_{b_0} \\
[887] & 21^{132} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -481, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 12 & 6 & 3 \\ 4 & 4 & 17 & 8 & 4 \\ 12 & 17 & 49 & 24 & 13 \\ 6 & 8 & 24 & 9 & 5 \\ 3 & 4 & 13 & 5 & 2 \end{pmatrix}_{b_0} \\
[888] & 1^{14} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -486, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 17 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 20 & -1 & 2 \end{pmatrix}_{b_0} \\
[889] & 21^{122} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -487, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7 \\
& T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 9 & 9 & 3 \\ 4 & 4 & 13 & 13 & 4 \\ 9 & 13 & 28 & 30 & 10 \\ 9 & 13 & 30 & 27 & 9 \\ 3 & 4 & 10 & 9 & 2 \end{pmatrix}_{b_0} \\
[890] & 1^{141} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -496, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 21 & 4 & 4 \\ 15 & 21 & 76 & 15 & 16 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0} \\
[891] & 1^{131} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -499, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 18 & 27 & 133 & 18 & 21 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0} \\
[892] & 1^{131} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -499, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 20 & 2 & 3 \\ 3 & 4 & 32 & 4 & 4 \\ 20 & 32 & 193 & 21 & 25 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0} \\
[893] & 21^{133} 31^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -499, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 27 & 3 & 4 \\ 5 & 4 & 27 & 4 & 4 \\ 27 & 27 & 133 & 18 & 21 \\ 3 & 4 & 18 & 0 & 2 \\ 4 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0} \\
[894] & 1^{13} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -501, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 20 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 2 \end{pmatrix}_{b_0} \\
[895] & 1^{141} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -504, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 16 & 26 & 118 & 18 & 20 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0} \\
[896] & 1^{131} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -511, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 21 & 3 & 3 \\ 3 & 4 & 32 & 4 & 4 \\ 21 & 32 & 193 & 21 & 25 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0} \\
[897] & 1^{14} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -514, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 21 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [898] $1^{131} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -515$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10.29 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 22 & 2 & 3 \\ 3 & 4 & 32 & 4 & 4 \\ 22 & 32 & 193 & 21 & 25 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$
- [899] $1^{131} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -523$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 21 & 2 & 3 \\ 3 & 4 & 32 & 4 & 4 \\ 21 & 32 & 193 & 21 & 25 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$
- [900] $1^{141} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -523$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 19 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 2 \end{pmatrix}_{b_0}$$
- [901] $21^{131} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -523$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10.15 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 4 & 17 & 4 & 4 \\ 12 & 17 & 49 & 12 & 13 \\ 3 & 4 & 12 & 0 & 2 \\ 3 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$
- [902] $1^{141} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -524$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.48 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 16 & 26 & 118 & 18 & 20 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$
- [903] $1^{141} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -534$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.07 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 18 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$
- [904] $1^{141} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -534$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.07 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 31 & 0 & 4 \\ 19 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 2 \end{pmatrix}_{b_0}$$
- [905] $1^{141} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -544$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.72 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 2 & 3 \\ 3 & 4 & 36 & 4 & 4 \\ 20 & 36 & 238 & 24 & 28 \\ 2 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$
- [906] $21^{131} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -547$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 3 & 3 \\ 4 & 4 & 27 & 4 & 4 \\ 18 & 27 & 133 & 18 & 21 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$
- [907] $1^{131} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -555$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10.08 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 27 & 4 & 4 \\ 19 & 27 & 133 & 18 & 21 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$
- [908] $1^{141} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -556$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.40 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 17 & 26 & 118 & 18 & 20 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$
- [909] $1^{141} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -556$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 18 & 26 & 118 & 18 & 20 \\ 2 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[910] \ 1^{141} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -571, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 18 & 31 & 172 & 21 & 24 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[911] \ 21^{143} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -574, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 8 & 24 & 3 & 4 \\ 8 & 9 & 32 & 4 & 5 \\ 24 & 32 & 94 & 12 & 16 \\ 3 & 4 & 12 & 0 & 2 \\ 4 & 5 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[912] \ 1^{141} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -576, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 19 & 31 & 172 & 21 & 24 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[913] \ 1^{131} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -585, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 21 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[914] \ 21^{141} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -585, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 15 & 6 & 3 \\ 4 & 4 & 21 & 9 & 4 \\ 15 & 21 & 76 & 33 & 16 \\ 6 & 9 & 33 & 12 & 6 \\ 3 & 4 & 16 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[915] \ 1^{131} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -591, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 22 & 3 & 3 \\ 3 & 4 & 32 & 4 & 4 \\ 22 & 32 & 193 & 21 & 25 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[916] \ 1^{141} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 22 & 3 & 3 \\ 3 & 4 & 36 & 4 & 4 \\ 22 & 36 & 238 & 24 & 28 \\ 3 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[917] \ 1^{141} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 17 & 26 & 118 & 18 & 20 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[918] \ 21^{131} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -604, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 & 3 \\ 4 & 4 & 22 & 4 & 4 \\ 15 & 22 & 85 & 15 & 17 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[919] \ 21^{143} 3 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -610, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 8 & 30 & 3 & 4 \\ 8 & 9 & 40 & 4 & 5 \\ 30 & 40 & 154 & 18 & 22 \\ 3 & 4 & 18 & 0 & 2 \\ 4 & 5 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[920] \ 21^{143} 3 1^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -619, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 33 & 3 & 4 \\ 5 & 4 & 32 & 4 & 4 \\ 33 & 32 & 193 & 21 & 25 \\ 3 & 4 & 21 & 0 & 2 \\ 4 & 4 & 25 & 2 & 2 \end{pmatrix}_{b_0}$$

- [921] $1^{14} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -624, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 31 & 0 & 4 \\ 20 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 2 \end{pmatrix}_{b_0}$$
- [922] $21^{13} 31^{141} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -640, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 30 & 3 & 4 \\ 5 & 4 & 31 & 4 & 4 \\ 30 & 31 & 172 & 21 & 24 \\ 3 & 4 & 21 & 0 & 2 \\ 4 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$
- [923] $1^{141} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -643, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 2 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 19 & 31 & 172 & 21 & 24 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$
- [924] $1^{13} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -645, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 22 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 22 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 2 \end{pmatrix}_{b_0}$$
- [925] $1^{14} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -646, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 22 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 22 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 2 \end{pmatrix}_{b_0}$$
- [926] $1^{141} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -656, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 21 & 2 & 3 \\ 3 & 4 & 36 & 4 & 4 \\ 21 & 36 & 238 & 24 & 28 \\ 2 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$
- [927] $1^{141} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -664, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 26 & 4 & 4 \\ 18 & 26 & 118 & 18 & 20 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$
- [928] $21^{132} 31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -673, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 9 & 3 \\ 4 & 4 & 17 & 13 & 4 \\ 12 & 17 & 49 & 39 & 13 \\ 9 & 13 & 39 & 27 & 9 \\ 3 & 4 & 13 & 9 & 2 \end{pmatrix}_{b_0}$$
- [929] $1^{141} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -683, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 2 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 20 & 31 & 172 & 21 & 24 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$
- [930] $21^{151} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -688, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.1$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 & 3 \\ 4 & 4 & 20 & 4 & 4 \\ 15 & 20 & 73 & 15 & 15 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$
- [931] $1^{14} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -690, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 31 & 0 & 4 \\ 21 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 2 \end{pmatrix}_{b_0}$$
- [932] $1^{141} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -691, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 21 & 2 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 21 & 31 & 172 & 21 & 24 \\ 2 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[933] \ 1^{141} 3^1 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -696, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 20 & 3 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 20 & 31 & 172 & 21 & 24 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[934] \ 21^{114} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -703, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 8 & 27 & 3 & 4 \\ 8 & 9 & 36 & 4 & 5 \\ 27 & 36 & 121 & 15 & 19 \\ 3 & 4 & 15 & 0 & 2 \\ 4 & 5 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[935] \ 1^{141} 3^1 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -736, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 22 & 2 & 3 \\ 3 & 4 & 36 & 4 & 4 \\ 22 & 36 & 238 & 24 & 28 \\ 2 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[936] \ 1^{141} 3^1 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -744, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 23 & 3 & 3 \\ 3 & 4 & 36 & 4 & 4 \\ 23 & 36 & 238 & 24 & 28 \\ 3 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[937] \ 1^{14} 3^1 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -754, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 23 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 23 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[939] \ 1^{141} 3^1 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -784, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 23 & 2 & 3 \\ 3 & 4 & 36 & 4 & 4 \\ 23 & 36 & 238 & 24 & 28 \\ 2 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[938] \ 1^{141} 3^1 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -784, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 21 & 3 & 3 \\ 3 & 4 & 31 & 4 & 4 \\ 21 & 31 & 172 & 21 & 24 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[940] \ 21^{141} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -787, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 & 3 \\ 4 & 4 & 21 & 4 & 4 \\ 15 & 21 & 76 & 15 & 16 \\ 3 & 4 & 15 & 0 & 2 \\ 3 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[941] \ 1^{141} 3^1 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -800, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 24 & 2 & 3 \\ 3 & 4 & 36 & 4 & 4 \\ 24 & 36 & 238 & 24 & 28 \\ 2 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[942] \ 1^{141} 3^1 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -838, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 24 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 24 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[943] \ 21^{115} 31^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -844, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.1$$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 8 & 30 & 3 & 4 \\ 8 & 9 & 40 & 4 & 5 \\ 30 & 40 & 148 & 15 & 20 \\ 3 & 4 & 15 & 0 & 2 \\ 4 & 5 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

[944] $21^{141}31^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -856, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 36 & 3 & 4 \\ 5 & 4 & 36 & 4 & 4 \\ 36 & 36 & 238 & 24 & 28 \\ 3 & 4 & 24 & 0 & 2 \\ 4 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

[945] $1^{141}31^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -856, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 24 & 3 & 3 \\ 3 & 4 & 36 & 4 & 4 \\ 24 & 36 & 238 & 24 & 28 \\ 3 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

[946] $21^{141}31^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -904, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 24 & 3 & 3 \\ 4 & 4 & 36 & 4 & 4 \\ 24 & 36 & 238 & 24 & 28 \\ 3 & 4 & 24 & 0 & 2 \\ 3 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

[947] $21^{161}31^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -952, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 3 & 3 \\ 4 & 4 & 24 & 4 & 4 \\ 18 & 24 & 106 & 18 & 18 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

[948] $21^{141}31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -964, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 3 & 3 \\ 4 & 4 & 26 & 4 & 4 \\ 18 & 26 & 118 & 18 & 20 \\ 3 & 4 & 18 & 0 & 2 \\ 3 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$$

[949] $21^{141}31^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -1003, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 21 & 3 & 3 \\ 4 & 4 & 31 & 4 & 4 \\ 21 & 31 & 172 & 21 & 24 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$$

[950] $21^{171}31^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -1264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 21 & 3 & 3 \\ 4 & 4 & 28 & 4 & 4 \\ 21 & 28 & 145 & 21 & 21 \\ 3 & 4 & 21 & 0 & 2 \\ 3 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

$T = 8$ [951] $1^{111}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 0 & -1 & 3 \end{pmatrix}_{b_0}$$

[952] $21^{121}41^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 12 & 6 & 4 \\ 5 & 4 & 11 & 7 & 4 \\ 12 & 11 & 28 & 16 & 10 \\ 6 & 7 & 16 & 7 & 5 \\ 4 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

[953] $21^{111}41^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 5 & 9 & 13 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

[954] $1^{111}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 1 & 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[955] $21^{\overline{1}^{11}}41^{\overline{1}^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 6 & 6 & 4 \\ 5 & 4 & 7 & 7 & 4 \\ 6 & 7 & 8 & 10 & 6 \\ 6 & 7 & 10 & 7 & 5 \\ 4 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[956] $1^{\overline{1}^{11}}41^{\overline{1}^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 5 & 9 & 13 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

[957] $1^{\overline{1}^{11}}321^{\overline{1}^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 2 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

[958] $1^{\overline{1}^{11}}41^{\overline{1}^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 11 & 0 & 4 \\ 7 & 11 & 28 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[959] $1^{\overline{1}^{11}}41^{\overline{1}^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 7 & 8 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

[963] $1^{\overline{1}^{11}}41^{\overline{1}^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 3 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[962] $1^{\overline{1}^{11}}321^{\overline{1}^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 4 & 5 \\ 1 & 1 & 0 & 4 & 0 & 2 \\ 2 & 2 & -1 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[961] $21^{\overline{1}^{11}}41^{\overline{1}^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[960] $1^{\overline{1}^{11}}41^{\overline{1}^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[966] $1^{\overline{1}^{11}}41^{\overline{1}^{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 3 & 5 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[965] \ 1^1 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 9, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.67 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 3 & 3 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[964] \ 21^1 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 9, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.67 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 3 & 5 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[967] \ 1^1 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 10, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.60 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[969] \ 21^1 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 12, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.67 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 6 & 3 & 4 \\ 5 & 4 & 7 & 3 & 4 \\ 6 & 7 & 8 & 5 & 6 \\ 3 & 3 & 5 & 0 & 2 \\ 4 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[968] \ 1^1 3^1 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 10, \ \Delta = -65, \ \det = 12, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 1 & 3 \\ 3 & 5 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[975] \ 1^1 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 13, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 7 & 7 & 8 & 10 & 6 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[974] \ 1^1 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 13, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.46 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[973] \ 1^1 3^1 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 10, \ \Delta = -65, \ \det = 13, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.85 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[972] \ 21^1 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 13, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 3 & 5 & 4 & 6 & 4 \\ 5 & 7 & 6 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[971] \ 1^1 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 13, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 3 & 5 & 4 & 6 & 4 \\ 5 & 7 & 6 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[970] \ 1^1 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 13, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 11 & 3 & 4 \\ 6 & 11 & 21 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[976] $21^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 9 & 6 & 4 \\ 5 & 4 & 8 & 7 & 4 \\ 9 & 8 & 13 & 11 & 7 \\ 6 & 7 & 11 & 7 & 5 \\ 4 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

[982] $21^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 6 & 3 & 4 \\ 5 & 4 & 7 & 4 & 4 \\ 6 & 7 & 8 & 5 & 6 \\ 3 & 4 & 5 & 0 & 2 \\ 4 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[981] $1 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 6 & 3 & 4 \\ 5 & 4 & 5 & 4 & 4 \\ 6 & 5 & 4 & 3 & 4 \\ 3 & 4 & 3 & 0 & 2 \\ 4 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[980] $21^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 7 & 3 \\ 4 & 4 & 4 & 10 & 4 \\ 3 & 4 & 1 & 7 & 3 \\ 7 & 10 & 7 & 20 & 8 \\ 3 & 4 & 3 & 8 & 3 \end{pmatrix}_{b_0}$$

[979] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 7 & 4 & 4 \\ 7 & 7 & 8 & 5 & 6 \\ 3 & 4 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[978] $1 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 3 & 5 & 4 & 6 & 4 \\ 4 & 7 & 6 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[977] $1^{11} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 2 & 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[983] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 6 & 10 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[988] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 7 & 7 & 8 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[987] $1 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 3 & 5 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[986] $1^{111} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 10 & 7 & 4 \\ 9 & 10 & 21 & 15 & 9 \\ 5 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$

[985] $1^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 5 & 8 & 13 & 6 & 7 \\ 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$

[984] $1^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 5 & 8 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$

[989] $21^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 4 & 7 & 4 \\ 3 & 4 & 1 & 5 & 3 \\ 6 & 7 & 5 & 7 & 5 \\ 4 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$

[991] $1^{111} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 2 & 4 & 1 & 5 & 3 \\ 6 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$

[990] $1^{11} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 10 & 3 & 4 \\ 6 & 10 & 21 & 8 & 9 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$

[992] $221^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 7 & 6 & 4 \\ 5 & 7 & 7 & 10 & 5 \\ 5 & 6 & 10 & 8 & 6 \\ 3 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0}$

[993] $21^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 3 & 1 & 5 & 3 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$

[994] $1^{11} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 3 & 1 & 5 & 3 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$

[995] $21^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 4 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$

[996] $1^1 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[997] \ 1^{\frac{1}{1}} 4^{\frac{1}{1}} 1^{\frac{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.67 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 5 & 8 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[998] \ 1^{\frac{1}{1}} 3^{\frac{1}{1}} 21^{\frac{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 10, \ \Delta = -65, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 3 & 5 & 2 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[999] \ 1^{\frac{1}{1}} 4^{\frac{1}{1}} 1^{\frac{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 22, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.64 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 4 & 7 & 8 & 5 & 6 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1000] \ 1^{\frac{1}{1}} 4^{\frac{1}{1}} 1^{\frac{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 24, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 10 & 3 & 4 \\ 9 & 10 & 21 & 8 & 9 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1001] \ 1^{\frac{1}{1}} 4^{\frac{1}{1}} 1^{\frac{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 24, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 1 & 5 & 3 \\ 4 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1002] \ 1^{\frac{1}{1}} 4^{\frac{1}{1}} 1^{\frac{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 24, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 5 & 8 & 13 & 6 & 7 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1003] \ 1^{\frac{1}{1}} 4^{\frac{1}{1}} 1^{\frac{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 24, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 5 & 8 & 13 & 11 & 7 \\ 4 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1004] \ 1^{\frac{1}{1}} 4^{\frac{1}{1}} 1^{\frac{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 24, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.83 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 7 & 8 & 13 & 6 & 7 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1011] \ 221^{\frac{1}{1}} 41^{\frac{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 27, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.89 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 3 & 5 & 3 \\ 6 & 13 & 8 & 12 & 7 \\ 3 & 8 & 4 & 6 & 4 \\ 5 & 12 & 6 & 7 & 5 \\ 3 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1005] \ 1^{\frac{1}{1}} 3^{\frac{1}{1}} 21^{\frac{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 10, \ \Delta = -65, \ \det = 27, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 5 & 10 & 0 & 5 \\ 5 & 10 & 14 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 8 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[1006] \ 1^{\frac{1}{1}} 3^{\frac{1}{1}} 21^{\frac{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 10, \ \Delta = -65, \ \det = 27, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 5 & 10 & 0 & 5 \\ 5 & 10 & 14 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 8 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[1007] \ 1^{\overline{13}} 4^{\overline{11}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 10 & 18 & 61 & 12 & 15 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1008] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 6 & 8 & 13 & 6 & 7 \\ 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1009] \ 21^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 6 & 8 & 13 & 6 & 7 \\ 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1010] \ 1^{\overline{11}} 4^{\overline{12}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 4 & 7 & 8 & 10 & 6 \\ 6 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1013] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 17 & 5 & 3 \\ 3 & 4 & 21 & 7 & 4 \\ 17 & 21 & 101 & 31 & 19 \\ 5 & 7 & 31 & 7 & 5 \\ 3 & 4 & 19 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1017] \ 21^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 5 & 9 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1014] \ 21^{\overline{41}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 17 & 5 & 3 \\ 3 & 4 & 21 & 7 & 4 \\ 17 & 21 & 101 & 31 & 19 \\ 5 & 7 & 31 & 7 & 5 \\ 3 & 4 & 19 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1012] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 4 & 7 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1016] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 5 & 9 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1015] \ 1^{\overline{11}} 4^{\overline{12}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 1 & 5 & 3 \\ 6 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1018] \ 1^{11} 1^1 3 21^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}u_2 + \textcolor{orange}{s}u_3 \quad T = 8, \ T_{\max} = 10, \ \Delta = -65, \ \det = 29, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 3 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1020] \ 1 \overset{141}{11} \ 4 \ 1 \overset{11}{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}u_2 + \textcolor{red}{s}u_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 31, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 11 & 12 & 26 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1021] \begin{matrix} 1^{11} & 4 & 1^{11} \\ & 1^{31} & \end{matrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_3 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 31, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 18 & 7 & 4 \\ 11 & 18 & 70 & 26 & 16 \\ 5 & 7 & 26 & 7 & 5 \\ 3 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1019] \begin{matrix} 1 \\ 11 \end{matrix} \begin{matrix} 1 \\ 4 \end{matrix} \begin{matrix} 1 \\ 1 \end{matrix} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 31, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 1 & 5 & 3 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1025] \ 1^{111} 4 \ 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{su}}_2 + \textcolor{teal}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 10 & 7 & 4 \\ 6 & 10 & 21 & 15 & 9 \\ 5 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1022] \begin{array}{c} 1^{11} \\ 1 \ 4 \ 1^{11} \\ 1^1 \end{array} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 4 & 7 & 8 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1026] \ 1^{12} 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 10 & 17 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1027] \begin{matrix} 1^1 \\ 1^1 4^1 1^{11} \end{matrix} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 33, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.79 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 6 & 8 & 13 & 6 & 7 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1024] \ 21 \overset{1}{\textcolor{brown}{1}} \overset{1}{\textcolor{brown}{1}} 4 \ 1 \overset{1}{\textcolor{brown}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 6 & 8 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1023] \begin{smallmatrix} 1^{11} \\ 1^{11} \end{smallmatrix} 4 \begin{smallmatrix} 1^{11} \\ 1^{11} \end{smallmatrix} + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{s}\textcolor{red}{u}_2 + \textcolor{teal}{s}\textcolor{red}{u}_3 \quad T = 8, \, T_{\max} = 11, \, \Delta = -84, \, \det = 33, \, (n_+, n_-, n_0) = (1, 8, 0), \, c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.79 \rightarrow 11, \, T_{\max} = 11, \, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 6 & 8 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1031] \ 21^{1\overline{1}1^{1\overline{3}1}}4 \ 1^{1\overline{1}1} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{s}u_2 + \textcolor{orange}{s}u_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 18 & 6 & 4 \\ 5 & 4 & 18 & 7 & 4 \\ 18 & 18 & 70 & 26 & 16 \\ 6 & 7 & 26 & 7 & 5 \\ 4 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

[1030] $1^{11} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 6 & 7 & 8 & 5 & 6 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[1029] $1^{12} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 8 & 14 & 38 & 10 & 12 \\ 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[1028] $1^{12} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 9 & 16 & 45 & 10 & 13 \\ 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1032] $21^{11} 4 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 30 & 3 & 4 \\ 5 & 4 & 33 & 3 & 4 \\ 30 & 33 & 212 & 23 & 28 \\ 3 & 3 & 23 & 0 & 2 \\ 4 & 4 & 28 & 2 & 3 \end{pmatrix}_{b_0}$$

[1033] $21^{11} 4 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 6 & 3 & 4 \\ 5 & 4 & 5 & 3 & 4 \\ 6 & 5 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 4 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[1034] $21^{11} 4 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 18 & 3 & 4 \\ 5 & 4 & 21 & 3 & 4 \\ 18 & 21 & 76 & 11 & 16 \\ 3 & 3 & 11 & 0 & 2 \\ 4 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1035] $21^{11} 4 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 8 & 5 & 6 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[1036] $1^{11} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 8 & 5 & 6 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[1037] $1^{12} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 10 & 17 & 61 & 13 & 15 \\ 3 & 3 & 13 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1038] $1^{11} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 7 & 12 & 26 & 8 & 10 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1040] $1^{111}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 7 & 4 & 4 \\ 4 & 7 & 8 & 5 & 6 \\ 3 & 4 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$

[1039] $1^{121}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 6 & 7 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0}$

[1042] $21^{11}41^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 2 & 4 & 1 & 5 & 3 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$

[1045] $1^{112}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 10 & 3 & 4 \\ 8 & 10 & 21 & 8 & 9 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$

[1044] $1^{112}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.63 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 6 & 7 & 8 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$

[1043] $1^{111}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 2 & 4 & 1 & 5 & 3 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$

[1041] $1^{121}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 11 & 0 & 4 \\ 6 & 11 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$

[1047] $21^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 10 & 2 & 3 \\ 4 & 4 & 15 & 3 & 4 \\ 10 & 15 & 45 & 11 & 13 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$

[1046] $1^141^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 5 & 9 & 13 & 5 & 7 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$

[1048] $21^{111}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 7 & 5 & 3 \\ 4 & 4 & 10 & 7 & 4 \\ 7 & 10 & 21 & 15 & 9 \\ 5 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$

[1049] $21^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 9 & 3 & 4 \\ 5 & 4 & 8 & 3 & 4 \\ 9 & 8 & 13 & 6 & 7 \\ 3 & 3 & 6 & 0 & 2 \\ 4 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1050] \quad 1^{13} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 9 & 15 & 40 & 9 & 12 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1051] \quad 21^{11} 4^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 9 & 15 & 40 & 9 & 12 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1052] \quad 1^{12} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1053] \quad 1^{11} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 11 & 18 & 70 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1054] \quad 1^{11} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 8 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1055] \quad 1^{14} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 15 & 27 & 132 & 17 & 22 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1056] \quad 1^{11} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 11 & 18 & 70 & 14 & 16 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1057] \quad 1^{14} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 25 & 3 & 4 \\ 14 & 25 & 121 & 17 & 21 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1058] \quad 21^{11} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 9 & 6 & 4 \\ 5 & 4 & 9 & 7 & 4 \\ 9 & 9 & 13 & 11 & 7 \\ 6 & 7 & 11 & 7 & 5 \\ 4 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1059] \quad 1^{11} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 9 & 15 & 45 & 11 & 13 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1060] $21^{11}41^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 11 & 3 & 4 \\ 7 & 11 & 21 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$

[1061] $1^{12}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 11 & 3 & 4 \\ 7 & 11 & 21 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$

[1062] $1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 3 & 3 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 5 & -1 & 3 \end{pmatrix}_{b_0}$

[1063] $1^141^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 11 & 3 & 4 \\ 6 & 11 & 21 & 7 & 9 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$

[1064] $1^{13}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 12 & 21 & 88 & 15 & 18 \\ 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$

[1068] $1^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 7 & 12 & 26 & 8 & 10 \\ 3 & 4 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$

[1065] $1^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 7 & 4 & 4 \\ 6 & 7 & 8 & 5 & 6 \\ 3 & 4 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$

[1066] $21^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$

[1067] $1^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 11 & 12 & 26 & 8 & 10 \\ 3 & 4 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$

[1069] $1^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 58, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 4 & 7 & 8 & 10 & 6 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$

[1070] $1^{13}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 121 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1072] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 62, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.74 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 7 & 12 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1071] \ 1^{\overline{12}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 62, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.74 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 11 & 3 & 4 \\ 10 & 11 & 21 & 7 & 9 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1074] \ 1^{\overline{13}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 63, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.95 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 14 & 24 & 121 & 18 & 21 \\ 3 & 3 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1073] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 63, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.70 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 9 & 15 & 45 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1076] \ 21^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 64, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 4 & 8 & 12 & 6 & 4 \\ 8 & 13 & 20 & 12 & 7 \\ 12 & 20 & 26 & 16 & 10 \\ 6 & 12 & 16 & 7 & 5 \\ 4 & 7 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1075] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 64, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 5 & 7 & 8 & 10 & 6 \\ 4 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1078] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 65, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 10 & 7 & 4 \\ 8 & 10 & 21 & 15 & 9 \\ 5 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1077] \ 21^{\overline{21}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 65, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 10 & 7 & 4 \\ 8 & 10 & 21 & 15 & 9 \\ 5 & 7 & 15 & 7 & 5 \\ 3 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1079] \ 21^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 65, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.65 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 3 \\ 4 & 4 & 7 & 3 & 4 \\ 5 & 7 & 8 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1080] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 65, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 11 & 12 & 26 & 8 & 10 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1081] $1^{14}_{1^{11}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 14 & 24 & 101 & 14 & 19 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1082] $21^{11}_{1^{11}} 4 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 14 & 24 & 101 & 14 & 19 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1083] $21^{11}_{1^{31}} 4 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 21 & 3 & 4 \\ 5 & 4 & 22 & 3 & 4 \\ 21 & 22 & 97 & 16 & 19 \\ 3 & 3 & 16 & 0 & 2 \\ 4 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1084] $1^{11}_{1^{11}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 8 & 12 & 26 & 8 & 10 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1085] $21^{11}_{1^{11}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 8 & 12 & 26 & 8 & 10 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1086] $1^{11}_{1^{11}} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 6 & 7 & 8 & 10 & 6 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[1087] $1^{11}_{1^{31}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 5 & 3 \\ 3 & 4 & 18 & 7 & 4 \\ 15 & 18 & 70 & 26 & 16 \\ 5 & 7 & 26 & 7 & 5 \\ 3 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

[1088] $1^{11}_{1^{11}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 7 & 12 & 26 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

[1091] $1^{11}_{1^{21}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 3 \\ 3 & 4 & 15 & 7 & 4 \\ 13 & 15 & 45 & 21 & 13 \\ 5 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[1090] $1^{13}_{1^{21}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 13 & 23 & 97 & 15 & 19 \\ 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1089] $1^{11}_{1^{21}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 15 & 7 & 4 \\ 9 & 15 & 45 & 21 & 13 \\ 5 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[1093] $1^{12} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 11 & 3 & 4 \\ 8 & 11 & 21 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[1092] $1^{12} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 11 & 19 & 68 & 13 & 16 \\ 3 & 3 & 13 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1095] $1^{11} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 7 & 12 & 26 & 8 & 10 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1094] $1^{13} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 10 & 18 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[1096] $1^{13} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 12 & 21 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[1097] $1^{12} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 11 & 3 & 4 \\ 9 & 11 & 21 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[1099] $21^{11} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 15 & 6 & 4 \\ 5 & 4 & 15 & 7 & 4 \\ 15 & 15 & 45 & 21 & 13 \\ 6 & 7 & 21 & 7 & 5 \\ 4 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[1100] $21^{11} 4^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 15 & 3 & 4 \\ 5 & 4 & 17 & 3 & 4 \\ 15 & 17 & 49 & 9 & 13 \\ 3 & 3 & 9 & 0 & 2 \\ 4 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1101] $21^{11} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 15 & 3 & 4 \\ 5 & 4 & 15 & 3 & 4 \\ 15 & 15 & 45 & 11 & 13 \\ 3 & 3 & 11 & 0 & 2 \\ 4 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1098] $21^{11} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 9 & 3 & 4 \\ 5 & 4 & 9 & 3 & 4 \\ 9 & 9 & 13 & 5 & 7 \\ 3 & 3 & 5 & 0 & 2 \\ 4 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

[1102] $1^{11} \overset{1^{31}}{4} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 10 & 12 & 26 & 8 & 10 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$

[1103] $1^{14} \overset{1^1}{4} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 28 & 3 & 4 \\ 16 & 28 & 158 & 20 & 24 \\ 3 & 3 & 20 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$

[1104] $1^{12} \overset{1^1}{4} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 11 & 3 & 4 \\ 7 & 11 & 21 & 7 & 9 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$

[1108] $1^{11} \overset{1^{21}}{4} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 9 & 12 & 26 & 8 & 10 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$

[1107] $21^{11} \overset{1^1}{4} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 5 & 7 & 8 & 10 & 6 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$

[1105] $1^{11} \overset{1^{11}}{4} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 5 & 7 & 8 & 10 & 6 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$

[1106] $1^{14} \overset{1^{21}}{4} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 12 & 19 & 65 & 11 & 15 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$

[1110] $21^{11} \overset{1^{11}}{4} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 4 & 5 & 12 & 6 & 4 \\ 5 & 4 & 12 & 7 & 4 \\ 12 & 12 & 26 & 16 & 10 \\ 6 & 7 & 16 & 7 & 5 \\ 4 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$

[1109] $21^{12} \overset{1^1}{4} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 6 & 2 & 3 \\ 4 & 4 & 8 & 3 & 4 \\ 6 & 8 & 10 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$

[1111] $1^{12} \overset{1^3}{4} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 11 & 3 & 4 \\ 9 & 11 & 21 & 7 & 9 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$

[1114] $21^{11} \overset{1^{11}}{4} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{aligned}
[1119] \quad & 1^{12} \overset{1^1}{4} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 13 & 22 & 97 & 16 & 19 \\ 3 & 3 & 16 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0} \\
[1124] \quad & 1^{11} \overset{1^4}{4} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 13 & 15 & 45 & 11 & 13 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0} \\
[1125] \quad & 21^{11} \overset{1^1}{4} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 4 & 5 & 12 & 3 & 4 \\ 5 & 4 & 13 & 3 & 4 \\ 12 & 13 & 28 & 7 & 10 \\ 3 & 3 & 7 & 0 & 2 \\ 4 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0} \\
[1127] \quad & 1^{12} \overset{1^1}{4} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 25 & 3 & 4 \\ 15 & 25 & 132 & 19 & 22 \\ 3 & 3 & 19 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0} \\
[1128] \quad & 1^{11} \overset{1^{31}}{4} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 12 & 15 & 45 & 11 & 13 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0} \\
[1126] \quad & 1^{12} \overset{1^{31}}{4} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 11 & 14 & 38 & 10 & 12 \\ 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[1129] \quad & 1^{12} \overset{1^1}{4} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 25 & 0 & 4 \\ 15 & 25 & 132 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0} \\
[1131] \quad & 1^{12} \overset{1^4}{4} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 12 & 14 & 38 & 10 & 12 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[1130] \quad & 1^{11} \overset{1^1}{4} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 8 & 12 & 26 & 8 & 10 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0} \\
[1134] \quad & 1^{11} \overset{1^{21}}{4} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 11 & 15 & 45 & 11 & 13 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0} \\
[1135] \quad & 1^{12} \overset{1^{31}}{4} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 13 & 17 & 61 & 13 & 15 \\ 3 & 3 & 13 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1133] $21^{21} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 11 & 15 & 45 & 11 & 13 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1132] $21^{31} 4 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 13 & 17 & 61 & 13 & 15 \\ 3 & 3 & 13 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1136] $1^{11} 4 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 15 & 18 & 70 & 14 & 16 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1137] $1^{12} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 9 & 16 & 45 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 3 \end{pmatrix}_{b_0}$$

[1138] $1^{12} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 11 & 0 & 4 \\ 8 & 11 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[1140] $1^{111} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 10 & 12 & 26 & 8 & 10 \\ 3 & 4 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1141] $21^{111} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 3 \\ 4 & 4 & 4 & 7 & 4 \\ 3 & 4 & 1 & 5 & 3 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

[1142] $1^{111} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 10 & 12 & 26 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

[1143] $21^{11} 4 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 12 & 3 & 4 \\ 5 & 4 & 12 & 4 & 4 \\ 12 & 12 & 26 & 8 & 10 \\ 3 & 4 & 8 & 0 & 2 \\ 4 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1139] $1^{111} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 8 & 12 & 26 & 16 & 10 \\ 4 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

[1144] $1^{11} \overset{1^2}{4} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 9 & 12 & 26 & 8 & 10 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1145] $21^{11} \overset{1^{11}}{4} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 8 & 2 & 3 \\ 4 & 4 & 12 & 3 & 4 \\ 8 & 12 & 26 & 8 & 10 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1146] $21^{11} \overset{1^{31}}{4} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 24 & 3 & 4 \\ 5 & 4 & 26 & 3 & 4 \\ 24 & 26 & 130 & 18 & 22 \\ 3 & 3 & 18 & 0 & 2 \\ 4 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1147] $21^{11} \overset{1^{11}}{4} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 12 & 3 & 4 \\ 5 & 4 & 12 & 3 & 4 \\ 12 & 12 & 26 & 8 & 10 \\ 3 & 3 & 8 & 0 & 2 \\ 4 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[1148] $21^{11} \overset{1^1}{4} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 6 & 5 & 3 \\ 4 & 4 & 9 & 7 & 4 \\ 6 & 9 & 13 & 11 & 7 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

[1149] $1^{11} \overset{1^{21}}{4} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 9 & 12 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[1150] $1^{12} \overset{1^4}{4} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 14 & 17 & 61 & 13 & 15 \\ 2 & 3 & 13 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1154] $1^{13} \overset{1^{31}}{4} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 11 & 15 & 40 & 9 & 12 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[1152] $21^{11} \overset{1^{11}}{4} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 21 & 3 & 4 \\ 5 & 4 & 24 & 3 & 4 \\ 21 & 24 & 101 & 14 & 19 \\ 3 & 3 & 14 & 0 & 2 \\ 4 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1153] $1^{14} \overset{1^1}{4} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 25 & 0 & 4 \\ 14 & 25 & 121 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[1155] $1^{12} \overset{1^3}{4} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 11 & 14 & 38 & 10 & 12 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1156] \ 1^{\overline{11}} \overset{1^{31}}{4} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 117, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.94 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 12 & 15 & 45 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1157] \ 1^{\overline{13}} \overset{1^{31}}{4} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 117, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 15 & 26 & 130 & 18 & 22 \\ 3 & 3 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1158] \ 1^{\overline{12}} \overset{1^{31}}{4} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 117, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.74 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 13 & 22 & 97 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1151] \ 21^{\overline{11}} \overset{1^{31}}{4} 1^{\overline{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 117, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.74 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 27 & 3 & 4 \\ 5 & 4 & 30 & 3 & 4 \\ 27 & 30 & 169 & 20 & 25 \\ 3 & 3 & 20 & 0 & 2 \\ 4 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1159] \ 1^{\overline{12}} \overset{1^{31}}{4} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 118, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 11 & 19 & 68 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1160] \ 1^{\overline{14}} \overset{1^{31}}{4} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 118, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.80 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 30 & 3 & 4 \\ 17 & 30 & 169 & 20 & 25 \\ 3 & 3 & 20 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1161] \ 1^{\overline{11}} \overset{1^{31}}{4} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 118, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 12 & 15 & 45 & 11 & 13 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1162] \ 1^{\overline{12}} \overset{1^{31}}{4} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 118, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 11 & 14 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1163] \ 1^{\overline{13}} \overset{1^{31}}{4} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 118, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 9 & 15 & 40 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1164] \ 1^{\overline{14}} \overset{1^{31}}{4} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 118, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 16 & 28 & 158 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[1165] $1^{12} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 10 & 14 & 38 & 10 & 12 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$

[1166] $1^{13} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 14 & 18 & 70 & 14 & 16 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$

[1168] $1^{121} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 9 & 12 & 26 & 8 & 10 \\ 3 & 4 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$

[1167] $1^{11} 4 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 9 & 16 & 45 & 10 & 13 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$

[1169] $21^{12} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 10 & 2 & 3 \\ 4 & 4 & 14 & 3 & 4 \\ 10 & 14 & 38 & 10 & 12 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$

[1170] $1^{12} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 13 & 17 & 61 & 13 & 15 \\ 2 & 3 & 13 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$

[1171] $1^{11} 4 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 11 & 15 & 45 & 11 & 13 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$

[1172] $21^{31} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 14 & 5 & 3 \\ 3 & 4 & 18 & 7 & 4 \\ 14 & 18 & 70 & 26 & 16 \\ 5 & 7 & 26 & 7 & 5 \\ 3 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$

[1173] $221^{11} 41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 7 & 12 & 4 \\ 5 & 7 & 7 & 16 & 5 \\ 8 & 12 & 16 & 26 & 10 \\ 3 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$

[1174] $21^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 8 & 12 & 26 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$

[1175] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 5 & 3 \\ 3 & 4 & 18 & 7 & 4 \\ 14 & 18 & 70 & 26 & 16 \\ 5 & 7 & 26 & 7 & 5 \\ 3 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

[1176] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 8 & 12 & 26 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

[1177] $1^{13} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 12 & 15 & 40 & 9 & 12 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[1178] $21^{111} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 3 \\ 4 & 4 & 7 & 7 & 4 \\ 5 & 7 & 8 & 10 & 6 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[1179] $1^{12} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 11 & 16 & 45 & 10 & 13 \\ 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1180] $21^{11} 4 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 18 & 3 & 4 \\ 5 & 4 & 19 & 3 & 4 \\ 18 & 19 & 68 & 13 & 16 \\ 3 & 3 & 13 & 0 & 2 \\ 4 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1181] $1^{13} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 12 & 18 & 61 & 12 & 15 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1182] $21^{21} 4 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 12 & 18 & 61 & 12 & 15 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1183] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 12 & 7 & 4 \\ 9 & 12 & 26 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

[1184] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 4 & 15 & 7 & 4 \\ 12 & 15 & 45 & 21 & 13 \\ 5 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[1186] $21^{12} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 4 & 22 & 3 & 4 \\ 15 & 22 & 97 & 16 & 19 \\ 2 & 3 & 16 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1185] $1^{14}_{1^{11}} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 13 & 19 & 65 & 11 & 15 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1187] $21^{13}_{1^{11}} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 4 & 12 & 3 & 4 \\ 9 & 12 & 25 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[1188] $1^{13}_{1^{11}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 138, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 11 & 20 & 70 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

[1189] $21^{11}_{1^{11}} 4 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 15 & 3 & 4 \\ 5 & 4 & 16 & 3 & 4 \\ 15 & 16 & 45 & 10 & 13 \\ 3 & 3 & 10 & 0 & 2 \\ 4 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1190] $21^{12}_{1^{11}} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 143, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 8 & 2 & 3 \\ 4 & 4 & 11 & 3 & 4 \\ 8 & 11 & 21 & 7 & 9 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[1191] $1^{13}_{1^{41}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 29 & 3 & 4 \\ 17 & 29 & 169 & 21 & 25 \\ 3 & 3 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

[1192] $21^{11}_{1^{21}} 4 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 24 & 3 & 4 \\ 5 & 4 & 27 & 3 & 4 \\ 24 & 27 & 132 & 17 & 22 \\ 3 & 3 & 17 & 0 & 2 \\ 4 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1193] $21^{11}_{1^{11}} 4 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 18 & 3 & 4 \\ 5 & 4 & 20 & 3 & 4 \\ 18 & 20 & 70 & 12 & 16 \\ 3 & 3 & 12 & 0 & 2 \\ 4 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1194] $1^{13}_{1^{11}} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 10 & 15 & 40 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[1195] $1^{13}_{1^{11}} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 12 & 15 & 40 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[1196] $1^1_{1^{11}} 4 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 11 & 19 & 68 & 13 & 16 \\ 2 & 3 & 13 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1197] \quad 1^{12} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 12 & 16 & 45 & 10 & 13 \\ 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1198] \quad 1^{13} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 20 & 3 & 4 \\ 13 & 20 & 70 & 12 & 16 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1199] \quad 21^{11} 4^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 21 & 3 & 4 \\ 5 & 4 & 23 & 3 & 4 \\ 21 & 23 & 97 & 15 & 19 \\ 3 & 3 & 15 & 0 & 2 \\ 4 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1204] \quad 1^{14} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 14 & 22 & 90 & 14 & 18 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1201] \quad 21^{11} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 8 & 5 & 3 \\ 4 & 4 & 12 & 7 & 4 \\ 8 & 12 & 26 & 16 & 10 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1202] \quad 1^{13} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 11 & 15 & 40 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1203] \quad 1^{12} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 13 & 16 & 45 & 10 & 13 \\ 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1200] \quad 1^{14} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 15 & 24 & 101 & 14 & 19 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1205] \quad 21^{11} 4^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 14 & 22 & 90 & 14 & 18 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1206] \quad 21^{11} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 15 & 7 & 4 \\ 11 & 15 & 45 & 21 & 13 \\ 5 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[1207] $1^{111} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 15 & 7 & 4 \\ 11 & 15 & 45 & 21 & 13 \\ 5 & 7 & 21 & 7 & 5 \\ 3 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[1208] $1^{13} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 10 & 15 & 40 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[1209] $1^{12} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 10 & 16 & 45 & 10 & 13 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1210] $1^{13} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 13 & 18 & 61 & 12 & 15 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1211] $1^{14} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 14 & 19 & 65 & 11 & 15 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1212] $1^{14} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 33 & 3 & 4 \\ 19 & 33 & 212 & 23 & 28 \\ 3 & 3 & 23 & 0 & 2 \\ 3 & 4 & 28 & 2 & 3 \end{pmatrix}_{b_0}$$

[1213] $1^{12} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 13 & 19 & 68 & 13 & 16 \\ 3 & 3 & 13 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1214] $21^{21} 4 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 13 & 19 & 68 & 13 & 16 \\ 3 & 3 & 13 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1215] $31^{11} 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -80, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 6 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 6 & 3 & 4 & 6 \\ 3 & 6 & 3 & 6 & 3 \\ 2 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[1216] $1^{13} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 14 & 18 & 61 & 12 & 15 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1217] $1^{13} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 13 & 23 & 97 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1218] \quad 1^{12} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 13 & 16 & 45 & 10 & 13 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1219] \quad 1^{13} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 11 & 15 & 40 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1220] \quad 21^{12} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 4 & 13 & 3 & 4 \\ 9 & 13 & 28 & 7 & 10 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1221] \quad 1^{13} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 29 & 0 & 4 \\ 17 & 29 & 169 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1222] \quad 1^{12} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 11 & 16 & 45 & 10 & 13 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1223] \quad 1^{13} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 15 & 26 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1226] \quad 21^{31} 4^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 15 & 21 & 88 & 15 & 18 \\ 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1225] \quad 1^{12} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 12 & 16 & 45 & 10 & 13 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1224] \quad 1^{13} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 15 & 21 & 88 & 15 & 18 \\ 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1227] \quad 1^{12} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 190, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 14 & 19 & 68 & 13 & 16 \\ 3 & 3 & 13 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1229] $1^{13}_{1^{21}} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 12 & 18 & 61 & 12 & 15 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$

[1228] $1^{12}_{1^{21}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 15 & 19 & 68 & 13 & 16 \\ 3 & 3 & 13 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$

[1230] $1^{14}_{1^{11}} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 12 & 19 & 65 & 11 & 15 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$

[1231] $1^{12}_{1^1} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 197, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 11 & 16 & 45 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 3 \end{pmatrix}_{b_0}$

[1232] $1^{13}_{1^{11}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 20 & 3 & 4 \\ 14 & 20 & 70 & 12 & 16 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$

[1233] $1^{13}_{1^{31}} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 16 & 21 & 88 & 15 & 18 \\ 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$

[1234] $1^{14}_{1^{21}} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 205, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 15 & 22 & 90 & 14 & 18 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$

[1235] $1^{13}_{1^{21}} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 205, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 14 & 18 & 61 & 12 & 15 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$

[1236] $21^{14}_{1^1} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 12 & 2 & 3 \\ 4 & 4 & 16 & 3 & 4 \\ 12 & 16 & 46 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$

[1237] $1^{13}_{1^{41}} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 18 & 24 & 121 & 18 & 21 \\ 3 & 3 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$

[1238] $21^{41}_{1^1} 4 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 18 & 24 & 121 & 18 & 21 \\ 3 & 3 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[1239] $1^{13} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 13 & 18 & 61 & 12 & 15 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1240] $1^{12} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 12 & 16 & 45 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 3 \end{pmatrix}_{b_0}$$

[1241] $1^{12} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 16 & 22 & 97 & 16 & 19 \\ 3 & 3 & 16 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1242] $21^{31} 4 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 16 & 22 & 97 & 16 & 19 \\ 3 & 3 & 16 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1243] $1^{12} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 17 & 22 & 97 & 16 & 19 \\ 3 & 3 & 16 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1244] $21^{21} 4 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 15 & 23 & 97 & 15 & 19 \\ 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1245] $1^{13} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 15 & 23 & 97 & 15 & 19 \\ 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1247] $1^{14} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 12 & 19 & 65 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[1246] $21^{13} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 4 & 21 & 3 & 4 \\ 15 & 21 & 88 & 15 & 18 \\ 2 & 3 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[1248] $21^{41} 4 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 25 & 3 & 4 \\ 19 & 25 & 132 & 19 & 22 \\ 3 & 3 & 19 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1249] $1^{12}_{1^{41}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 25 & 3 & 4 \\ 19 & 25 & 132 & 19 & 22 \\ 3 & 3 & 19 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$

[1250] $1^{13}_{1^2} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 14 & 18 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$

[1251] $1^{13}_{1^2} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 13 & 18 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$

[1252] $1^{14}_{1^{11}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 219, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 13 & 19 & 65 & 11 & 15 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$

[1254] $1^{13}_{1^{11}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 222, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 20 & 3 & 4 \\ 12 & 20 & 70 & 12 & 16 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$

[1253] $1^{14}_{1^2} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 222, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 15 & 27 & 132 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0}$

[1255] $1^{14}_{1^{11}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 223, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 16 & 24 & 101 & 14 & 19 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$

[1256] $1^{13}_{1^{11}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 225, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 20 & 3 & 4 \\ 15 & 20 & 70 & 12 & 16 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$

[1258] $1^{12}_{1^{21}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 13 & 19 & 68 & 13 & 16 \\ 2 & 3 & 13 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$

[1257] $1^{14}_{1^{11}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 14 & 19 & 65 & 11 & 15 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$

[1259] $21^{12}_{1^{21}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 13 & 2 & 3 \\ 4 & 4 & 19 & 3 & 4 \\ 13 & 19 & 68 & 13 & 16 \\ 2 & 3 & 13 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1260] \quad 1^{12} 4^{14} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 230, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.72 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 15 & 19 & 68 & 13 & 16 \\ 2 & 3 & 13 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1261] \quad 1^{12} 4^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 237, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.73 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 19 & 3 & 4 \\ 14 & 19 & 68 & 13 & 16 \\ 2 & 3 & 13 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1264] \quad 1^{13} 4^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 238, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.80 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 15 & 21 & 88 & 15 & 18 \\ 2 & 3 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1263] \quad 1^{14} 4^{14} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 238, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.78 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 16 & 22 & 90 & 14 & 18 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1262] \quad 1^{12} 4^{14} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 238, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.95 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 15 & 19 & 68 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1265] \quad 21^{13} 4^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 241, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.74 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 11 & 2 & 3 \\ 4 & 4 & 15 & 3 & 4 \\ 11 & 15 & 40 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1266] \quad 1^{13} 4^{14} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 241, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.77 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 16 & 21 & 88 & 15 & 18 \\ 2 & 3 & 15 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1267] \quad 1^{14} 4^{12} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 241, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.72 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 17 & 27 & 132 & 17 & 22 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1268] \quad 21^{21} 4^{11} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 241, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.72 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 17 & 27 & 132 & 17 & 22 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1269] \quad 221^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -84, \quad \det = 243, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.76 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 5 & 5 & 3 \\ 6 & 13 & 12 & 12 & 7 \\ 5 & 12 & 8 & 10 & 6 \\ 5 & 12 & 10 & 7 & 5 \\ 3 & 7 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[1272] $1^{12} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 14 & 19 & 68 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$

[1271] $1^{13} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 16 & 21 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$

[1270] $21^{12} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 11 & 2 & 3 \\ 4 & 4 & 16 & 3 & 4 \\ 11 & 16 & 45 & 10 & 13 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$

[1273] $221^{11} 41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 6 & 8 & 5 & 3 \\ 6 & 13 & 20 & 12 & 7 \\ 8 & 20 & 26 & 16 & 10 \\ 5 & 12 & 16 & 7 & 5 \\ 3 & 7 & 10 & 5 & 3 \end{pmatrix}_{b_0}$

[1274] $1^{13} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 18 & 24 & 121 & 18 & 21 \\ 2 & 3 & 18 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$

[1275] $1^{14} 41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 17 & 30 & 169 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 3 \end{pmatrix}_{b_0}$

[1276] $1^{14} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 25 & 3 & 4 \\ 17 & 25 & 121 & 17 & 21 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$

[1277] $21^{31} 41^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 25 & 3 & 4 \\ 17 & 25 & 121 & 17 & 21 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$

[1278] $1^{14} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 259, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 13 & 19 & 65 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$

[1279] $1^{12} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 17 & 22 & 97 & 16 & 19 \\ 2 & 3 & 16 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$

[1283] $1^{12} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 16 & 22 & 97 & 16 & 19 \\ 2 & 3 & 16 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1282] \ 1^{12}_{1^3} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 261, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 17 & 22 & 97 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1281] \ 1^{13}_{1^{21}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 261, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.74 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 16 & 23 & 97 & 15 & 19 \\ 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1280] \ 1^{13}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 261, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 20 & 3 & 4 \\ 13 & 20 & 70 & 12 & 16 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1285] \ 1^{12}_{1^{21}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 266, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.85 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 14 & 22 & 90 & 14 & 18 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1284] \ 1^{14}_{1^4} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 266, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.80 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 33 & 0 & 4 \\ 19 & 33 & 212 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1286] \ 1^{12}_{1^{41}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 266, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.85 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 19 & 2 & 3 \\ 3 & 4 & 25 & 3 & 4 \\ 19 & 25 & 132 & 19 & 22 \\ 2 & 3 & 19 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1287] \ 21^{13}_{1^4} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 267, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 20 & 2 & 3 \\ 4 & 4 & 29 & 3 & 4 \\ 20 & 29 & 169 & 21 & 25 \\ 2 & 3 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1288] \ 21^{13}_{1^{21}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 268, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.73 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 13 & 2 & 3 \\ 4 & 4 & 18 & 3 & 4 \\ 13 & 18 & 61 & 12 & 15 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1289] \ 1^{14}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 274, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.75 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 17 & 24 & 101 & 14 & 19 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1290] \ 1^{14}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 281, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.95 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 14 & 24 & 101 & 14 & 19 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1291] $1^{13}_{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 282, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 13 & 20 & 70 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

[1292] $1^{13}_{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 282, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 20 & 3 & 4 \\ 14 & 20 & 70 & 12 & 16 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1293] $1^{13}_{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 285, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 20 & 3 & 4 \\ 15 & 20 & 70 & 12 & 16 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1294] $1^{13}_{121} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 17 & 23 & 97 & 15 & 19 \\ 3 & 3 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1295] $1^{14}_{131} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 289, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 25 & 3 & 4 \\ 18 & 25 & 121 & 17 & 21 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[1296] $21^{13}_{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 291, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 3 \\ 4 & 4 & 17 & 3 & 4 \\ 12 & 17 & 49 & 9 & 13 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}$$

[1297] $21^{15}_{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.1$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 4 & 20 & 3 & 4 \\ 15 & 20 & 73 & 10 & 15 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1298] $1^{14}_{121} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 297, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 15 & 22 & 90 & 14 & 18 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[1299] $1^{13}_{131} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 306, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 18 & 26 & 130 & 18 & 22 \\ 3 & 3 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1300] $21^{31}_{11} 4 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 306, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 18 & 26 & 130 & 18 & 22 \\ 3 & 3 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1301] $1^{14}_{121} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 3 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 18 & 27 & 132 & 17 & 22 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1302] \ 1^{14}_{1^{21}} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 310, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.75 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 16 & 22 & 90 & 14 & 18 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1303] \ 1^{14}_{1^2} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 314, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.87 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 15 & 22 & 90 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1304] \ 1^{13}_{1^1} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 318, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 14 & 20 & 70 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1306] \ 1^{14}_{1^{41}} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 322, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.83 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 20 & 3 & 3 \\ 3 & 4 & 28 & 3 & 4 \\ 20 & 28 & 158 & 20 & 24 \\ 3 & 3 & 20 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1305] \ 21^{1}_{41} 4^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 322, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.83 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 20 & 3 & 3 \\ 3 & 4 & 28 & 3 & 4 \\ 20 & 28 & 158 & 20 & 24 \\ 3 & 3 & 20 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1307] \ 1^{13}_{1^{21}} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 327, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.80 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 15 & 23 & 97 & 15 & 19 \\ 2 & 3 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1308] \ 1^{13}_{1^1} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 330, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.99 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 15 & 20 & 70 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1309] \ 1^{13}_{1^{31}} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 333, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.80 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 19 & 26 & 130 & 18 & 22 \\ 3 & 3 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1310] \ 1^{14}_{1^{11}} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 336, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.85 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 15 & 24 & 101 & 14 & 19 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1311] \ 1^{14}_{1^2} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 338, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 16 & 22 & 90 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[1312] $21^{14}_{1^4} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 350, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 20 & 2 & 3 \\ 4 & 4 & 28 & 3 & 4 \\ 20 & 28 & 158 & 20 & 24 \\ 2 & 3 & 20 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

[1313] $1^{13}_{1^3} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 351, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 25 & 3 & 4 \\ 17 & 25 & 121 & 17 & 21 \\ 2 & 3 & 17 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[1314] $1^{13}_{1^2} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 354, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 16 & 23 & 97 & 15 & 19 \\ 2 & 3 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1315] $21^{14}_{1^4} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 359, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 14 & 2 & 3 \\ 4 & 4 & 19 & 3 & 4 \\ 14 & 19 & 65 & 11 & 15 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1318] $21^{41}_{1^4} 4 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 21 & 3 & 3 \\ 3 & 4 & 29 & 3 & 4 \\ 21 & 29 & 169 & 21 & 25 \\ 3 & 3 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

[1317] $1^{13}_{1^4} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 21 & 3 & 3 \\ 3 & 4 & 29 & 3 & 4 \\ 21 & 29 & 169 & 21 & 25 \\ 3 & 3 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

[1316] $31^{111}_{1^3} 31^{11} 3 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -80, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 3 & 6 & 3 & 2 \\ 3 & 4 & 4 & 9 & 6 & 4 \\ 3 & 4 & 0 & 6 & 3 & 2 \\ 6 & 9 & 6 & 13 & 9 & 7 \\ 3 & 6 & 3 & 9 & 3 & 3 \\ 2 & 4 & 2 & 7 & 3 & 2 \end{pmatrix}_{b_0}$$

[1319] $1^{14}_{1^2} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 361, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 19 & 27 & 132 & 17 & 22 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1320] $1^{13}_{1^2} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 363, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 23 & 3 & 4 \\ 17 & 23 & 97 & 15 & 19 \\ 2 & 3 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1321] $1^{14}_{1^1} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 365, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 15 & 24 & 101 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

[1322] $1^{13}_{1^2} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 16 & 23 & 97 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

[1323] $1^{14}_{1^{31}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 25 & 3 & 4 \\ 18 & 25 & 121 & 17 & 21 \\ 2 & 3 & 17 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[1324] $1^{14}_{1^{11}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 373, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 16 & 24 & 101 & 14 & 19 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1325] $1^{14}_{1^{31}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 379, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 & 3 \\ 3 & 4 & 30 & 3 & 4 \\ 20 & 30 & 169 & 20 & 25 \\ 3 & 3 & 20 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

[1326] $21^{31}_{1^{11}} 4 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 379, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 20 & 3 & 3 \\ 3 & 4 & 30 & 3 & 4 \\ 20 & 30 & 169 & 20 & 25 \\ 3 & 3 & 20 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

[1327] $1^{14}_{1^{13}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 25 & 0 & 4 \\ 18 & 25 & 121 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[1328] $1^{13}_{1^2} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 387, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 17 & 23 & 97 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

[1329] $21^{13}_{1^{31}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 3 \\ 4 & 4 & 26 & 3 & 4 \\ 18 & 26 & 130 & 18 & 22 \\ 2 & 3 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1330] $1^{14}_{1^{11}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 17 & 24 & 101 & 14 & 19 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1332] $21^{13}_{1^{11}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 402, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 14 & 2 & 3 \\ 4 & 4 & 20 & 3 & 4 \\ 14 & 20 & 70 & 12 & 16 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1331] $1^{13}_{1^{31}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 402, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 18 & 26 & 130 & 18 & 22 \\ 2 & 3 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1334] $21^{16}41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 3 \\ 4 & 4 & 24 & 3 & 4 \\ 18 & 24 & 106 & 12 & 18 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[1333] $1^{14}41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 20 & 2 & 3 \\ 3 & 4 & 28 & 3 & 4 \\ 20 & 28 & 158 & 20 & 24 \\ 2 & 3 & 20 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

[1335] $1^{13}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 417, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 19 & 2 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 19 & 26 & 130 & 18 & 22 \\ 2 & 3 & 18 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1336] $21^{14}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 421, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 4 & 21 & 3 & 4 \\ 15 & 21 & 76 & 11 & 16 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1337] $1^{14}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 425, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 16 & 24 & 101 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

[1338] $1^{13}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 426, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 19 & 26 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

[1339] $1^{14}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 426, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 17 & 27 & 132 & 17 & 22 \\ 2 & 3 & 17 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1340] $1^{14}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 430, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 21 & 3 & 3 \\ 3 & 4 & 30 & 3 & 4 \\ 21 & 30 & 169 & 20 & 25 \\ 3 & 3 & 20 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

[1341] $21^{14}41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 431, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 3 \\ 4 & 4 & 25 & 3 & 4 \\ 18 & 25 & 121 & 17 & 21 \\ 2 & 3 & 17 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[1342] $21^{14}41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 434, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 3 \\ 4 & 4 & 22 & 3 & 4 \\ 16 & 22 & 90 & 14 & 18 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[1343] $21^{13}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 435, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 3 \\ 4 & 4 & 23 & 3 & 4 \\ 16 & 23 & 97 & 15 & 19 \\ 2 & 3 & 15 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

[1344] $1^{13}_{1^{41}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 447, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 21 & 2 & 3 \\ 3 & 4 & 29 & 3 & 4 \\ 21 & 29 & 169 & 21 & 25 \\ 2 & 3 & 21 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

[1345] $1^{14}_{1^{41}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 461, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 17 & 24 & 101 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

[1346] $1^{14}_{1^{21}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 469, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 18 & 27 & 132 & 17 & 22 \\ 2 & 3 & 17 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1347] $1^{14}_{1^{41}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 481, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 23 & 3 & 3 \\ 3 & 4 & 33 & 3 & 4 \\ 23 & 33 & 212 & 23 & 28 \\ 3 & 3 & 23 & 0 & 2 \\ 3 & 4 & 28 & 2 & 3 \end{pmatrix}_{b_0}$$

[1348] $21^{11}_{1^{41}} 4 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 481, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 23 & 3 & 3 \\ 3 & 4 & 33 & 3 & 4 \\ 23 & 33 & 212 & 23 & 28 \\ 3 & 3 & 23 & 0 & 2 \\ 3 & 4 & 28 & 2 & 3 \end{pmatrix}_{b_0}$$

[1349] $1^{14}_{1^{31}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 492, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 18 & 27 & 132 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

[1350] $1^{14}_{1^{21}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 494, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 19 & 2 & 3 \\ 3 & 4 & 27 & 3 & 4 \\ 19 & 27 & 132 & 17 & 22 \\ 2 & 3 & 17 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[1351] $1^{14}_{1^{41}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 534, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 19 & 27 & 132 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

[1352] $21^{17}_{1^1} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 21 & 2 & 3 \\ 4 & 4 & 28 & 3 & 4 \\ 21 & 28 & 145 & 14 & 21 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[1353] $1^{14}_{1^{31}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 541, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 20 & 2 & 3 \\ 3 & 4 & 30 & 3 & 4 \\ 20 & 30 & 169 & 20 & 25 \\ 2 & 3 & 20 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$$

[1354] $1^{14}_{1^{31}} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 572, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 21 & 2 & 3 \\ 3 & 4 & 30 & 3 & 4 \\ 21 & 30 & 169 & 20 & 25 \\ 2 & 3 & 20 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$

[1355] $21^{14}_{1^{11}} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 17 & 2 & 3 \\ 4 & 4 & 24 & 3 & 4 \\ 17 & 24 & 101 & 14 & 19 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$

[1357] $21^{14}_{1^{41}} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 589, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 23 & 2 & 3 \\ 4 & 4 & 33 & 3 & 4 \\ 23 & 33 & 212 & 23 & 28 \\ 2 & 3 & 23 & 0 & 2 \\ 3 & 4 & 28 & 2 & 3 \end{pmatrix}_{b_0}$

[1356] $1^{14}_{1^3} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 589, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 21 & 30 & 169 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 3 \end{pmatrix}_{b_0}$

[1358] $1^{14}_{1^{41}} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 626, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 23 & 2 & 3 \\ 3 & 4 & 33 & 3 & 4 \\ 23 & 33 & 212 & 23 & 28 \\ 2 & 3 & 23 & 0 & 2 \\ 3 & 4 & 28 & 2 & 3 \end{pmatrix}_{b_0}$

[1359] $21^{14}_{1^{21}} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 661, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 19 & 2 & 3 \\ 4 & 4 & 27 & 3 & 4 \\ 19 & 27 & 132 & 17 & 22 \\ 2 & 3 & 17 & 0 & 2 \\ 3 & 4 & 22 & 2 & 3 \end{pmatrix}_{b_0}$

[1360] $21^{14}_{1^{31}} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 664, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 21 & 2 & 3 \\ 4 & 4 & 30 & 3 & 4 \\ 21 & 30 & 169 & 20 & 25 \\ 2 & 3 & 20 & 0 & 2 \\ 3 & 4 & 25 & 2 & 3 \end{pmatrix}_{b_0}$

$T = 9$ [1361] $1^{11} 2^3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 4 & 6 \\ 1 & 1 & 0 & 4 & 0 & 2 \\ 2 & 3 & -1 & 6 & 2 & 4 \end{pmatrix}_{b_0}$

[1362] $1^{11} 2^3 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$

[1363] $1^{11} 31^1 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 0 & 4 \end{pmatrix}_{b_0}$

[1364] $1^{11} 31^1 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 6 & 7 & 9 & 1 & 7 \\ 1 & 0 & 1 & -2 & 0 \\ 3 & 4 & 7 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[1365] \quad 1^{\overline{11}} 4 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 3 & 0 & 4 & 4 \\ 3 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1366] \quad 1^{\overline{11}} 4 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 6 & 4 & 6 \\ 4 & 2 & 4 \\ 6 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[1367] \quad 1^{\overline{11}} 4 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 4 \\ 3 & 1 & 1 & 3 \\ 2 & 1 & 0 & 2 \\ 4 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[1373] \quad 21^{\overline{11}} 5 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 4 & 6 & 4 \\ 3 & 4 & 1 & 4 & 3 \\ 6 & 6 & 4 & 7 & 5 \\ 4 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1369] \quad 1^{\overline{11}} 4 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 3 & 4 & 2 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 3 & 4 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[1370] \quad 1^{\overline{11}} 5^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 10 & 16 & 59 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1371] \quad 1^{\overline{11}} 5^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 4 & 7 & 8 & 4 & 6 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1372] \quad 21^{\overline{11}} 5 1^{\overline{11}} 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 4 & 6 & 4 \\ 3 & 4 & 1 & 4 & 3 \\ 6 & 6 & 4 & 7 & 5 \\ 4 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1368] \quad 1^{\overline{11}} 5^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 4 & 6 & 4 \\ 2 & 4 & 1 & 4 & 3 \\ 4 & 6 & 4 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1374] \quad 21^{\overline{11}} 5 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 4 & 7 & 8 & 4 & 6 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[1375] \quad & 1 \overset{1}{\underset{1^1}{5}} 1^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[1376] \quad & 1 \overset{1}{\underset{1^3, 1^2}{5}} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 12 & 19 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0} \\
[1377] \quad & 21 \overset{1}{\underset{1^1}{5}} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 3 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[1378] \quad & 1 \overset{1}{\underset{1^1, 1^1}{5}} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 4 & 6 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0} \\
[1380] \quad & 1 \overset{1}{\underset{1^1, 1^1}{5}} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[1379] \quad & 21 \overset{1}{\underset{1^1}{5}} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[1382] \quad & 1 \overset{1}{\underset{1^3 1^1}{4}} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 2 & 3 & 3 & 4 \\ 3 & 1 & 2 & 3 \\ 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 5 \end{pmatrix}_{b_0} \\
[1381] \quad & 1 \overset{1}{\underset{1^{11}}{4}} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 2 \\ 3 & 2 & 2 & 1 & 3 \\ 3 & 2 & 2 & 3 & 4 \end{pmatrix}_{b_0} \\
[1385] \quad & 1 \overset{1}{\underset{1^1}{4}} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 \\ 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0} \\
[1384] \quad & 1 \overset{1}{\underset{1^3 1^1}{4}} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 1 & 3 \\ 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0} \\
[1383] \quad & 1 \overset{1}{\underset{1^1, 1^1}{5}} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[1386] $21 \overset{1}{\underset{1}{5}} 1^{\overset{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 6 & 3 & 4 \\ 5 & 4 & 7 & 3 & 4 \\ 6 & 7 & 8 & 4 & 6 \\ 3 & 3 & 4 & 0 & 2 \\ 4 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[1387] $1^{\overset{1}{11}} 4 131^{\overset{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 3 & 1 & 5 & 5 \\ 3 & 3 & 0 & 5 & 4 \end{pmatrix}_{b_0}$$

[1388] $1 \overset{1}{\underset{1^2, 1^1}{5}} 1^{\overset{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -17, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 10 & 3 & 4 \\ 6 & 10 & 21 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[1389] $1^{\overset{1}{11}} 2321^{\overset{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 4 & 0 & 3 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & 6 & -1 & 4 \end{pmatrix}_{b_0}$$

[1390] $21 \overset{1}{\underset{1^1}{5}} 1^{\overset{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 3 & 3 & 4 \\ 3 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 4 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[1392] $21 \overset{1}{\underset{1^2}{5}} 1^{\overset{1}{11}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 1 & 5 & 3 & 3 \\ 4 & 5 & 7 & 4 & 5 \\ 3 & 3 & 4 & 1 & 3 \\ 3 & 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[1391] $21 \overset{1}{\underset{1^2}{5}} 1^{\overset{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 7 & 2 & 3 \\ 4 & 4 & 10 & 3 & 4 \\ 7 & 10 & 21 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[1395] $1 \overset{1}{\underset{1^1}{5}} 1^{\overset{1}{11}} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 4 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[1394] $21 \overset{1}{\underset{1^1}{5}} 1^{\overset{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[1396] $221 \overset{1}{\underset{1^1}{5}} 1^{\overset{1}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 5 & 2 & 3 \\ 2 & 1 & 5 & 3 & 3 \\ 5 & 5 & 7 & 4 & 5 \\ 2 & 3 & 4 & 1 & 3 \\ 3 & 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[1393] $1 \overset{1}{\underset{1^1, 1^1}{5}} 1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$

[1397] $1 \overset{1}{\underset{1^1, 1^1}{4}} 131 \overset{11}{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 1 & 4 & 3 \\ 5 & 4 & 6 & 6 \\ 4 & 3 & 6 & 5 \end{pmatrix}_{b_0}$

[1398] $1 \overset{1}{\underset{1^1, 1^1}{4}} 1 \overset{11}{31} \overset{11}{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 1 & 3 & 3 \\ 5 & 3 & 3 & 5 \\ 4 & 3 & 5 & 5 \end{pmatrix}_{b_0}$

[1399] $1 \overset{1}{\underset{1^1, 1^1}{4}} 131 \overset{11}{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 4 & 4 \\ 3 & 1 & 4 & 3 \\ 4 & 4 & 3 & 5 \\ 4 & 3 & 5 & 5 \end{pmatrix}_{b_0}$

[1400] $1 \overset{1}{\underset{1^1, 1^1}{4}} 131 \overset{12}{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 7 & 4 \\ 3 & 1 & 5 & 3 \\ 7 & 5 & 12 & 8 \\ 4 & 3 & 8 & 5 \end{pmatrix}_{b_0}$

[1401] $1 \overset{1}{\underset{1^1, 1^1}{5}} 1 \overset{11}{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 4 & 7 & 8 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$

[1402] $1 \overset{1}{\underset{1^1, 1^1}{4}} 131 \overset{11}{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 3 & 1 & 2 & 4 \\ 3 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$

[1404] $21 \overset{1}{\underset{1^1, 1^1}{5}} 1 \overset{11}{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 10 & 3 & 4 \\ 8 & 10 & 21 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$

[1403] $1 \overset{1}{\underset{1^2, 1^2}{11}} 5 1 \overset{1}{1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 10 & 16 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$

[1405] $1 \overset{1}{\underset{1^2, 1^1}{11}} 5 1 \overset{1}{1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 8 & 13 & 36 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$

[1406] $1 \overset{1}{\underset{1^3, 1^1}{5}} 1 \overset{12}{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 10 & 17 & 61 & 10 & 15 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$

[1407] $1^{11} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 5 & 0 & 8 & 6 \\ 3 & 3 & 0 & 6 & 4 \end{pmatrix}_{b_0}$$

[1409] $1^{11} 5 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 6 & 10 & 19 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[1408] $21^{11} 5 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 8 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[1410] $1^{11} 5 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 11 & 3 & 4 \\ 6 & 11 & 21 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[1411] $1^{12} 5 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 14 & 23 & 117 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[1412] $1^{12} 5 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 8 & 14 & 38 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[1414] $1^{12} 5 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 20 & 32 & 234 & 0 & 30 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 30 & -1 & 3 \end{pmatrix}_{b_0}$$

[1413] $21^4 5 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 20 & 3 & 4 \\ 16 & 20 & 90 & 12 & 18 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[1415] $1^{11} 4 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 6 & 1 & 14 & 8 \\ 3 & 3 & 0 & 8 & 4 \end{pmatrix}_{b_0}$$

[1416] $21^{12} 5 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 3 \\ 4 & 4 & 17 & 3 & 4 \\ 12 & 17 & 61 & 10 & 15 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1417] $15^{11} 1^1 2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1418] \quad 1^{1^1} 5^{1^1} 1^{1^3} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 14 & 24 & 121 & 14 & 21 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1419] \quad 1^{1^2} 5^{1^1} 1^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 16 & 26 & 150 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1420] \quad 21^{1^1} 5^{1^1} 1^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 3 \\ 4 & 4 & 7 & 3 & 4 \\ 5 & 7 & 8 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1421] \quad 1^{1^1} 1^{1^1} 31^{1^1} 4^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 5 & 1 & 9 & 7 \\ 3 & 3 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[1422] \quad 1^{1^2} 5^{1^1} 1^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 29 & 0 & 4 \\ 18 & 29 & 189 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1423] \quad 1^{1^2} 5^{1^1} 1^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 12 & 20 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1424] \quad 21^{1^1} 5^{1^1} 1^{1^2} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 11 & 3 & 4 \\ 7 & 11 & 21 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1425] \quad 1^{1^2} 5^{1^1} 1^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 36 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1426] \quad 1^{1^1} 5^{1^1} 1^{1^3} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -73, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 10 & 18 & 61 & 10 & 15 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1427] \quad 1^{1^1} 1^{1^2} 31^{1^1} 4^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 10 & 4 \\ 3 & 1 & 7 & 3 \\ 10 & 7 & 21 & 11 \\ 4 & 3 & 11 & 5 \end{pmatrix}_{b_0}$$

$$[1428] \quad 1 \overset{1}{\underset{1^3, 1^1}{5}} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 12 & 21 & 88 & 12 & 18 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1429] \quad 1 \overset{1}{\underset{1^2, 1^1}{12}} 5 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 10 & 17 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1430] \quad 21 \overset{1}{\underset{1^1}{3}} 5 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 13 & 17 & 61 & 10 & 15 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1431] \quad 1 \overset{1}{\underset{1^3, 1^3}{12}} 5 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 16 & 26 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1432] \quad 1 \overset{1}{\underset{1^3, 1^2}{12}} 5 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 14 & 23 & 113 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1433] \quad 1 \overset{1}{\underset{1^1}{5}} 1^{12} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -86, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 3 \\ 4 & 4 & 8 & 3 & 4 \\ 6 & 8 & 10 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1435] \quad 21 \overset{1}{\underset{1^1}{2}} 5 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 10 & 14 & 38 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1434] \quad 1 \overset{1}{\underset{1^2, 1^2}{12}} 5 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 12 & 20 & 82 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1437] \quad 21 \overset{1}{\underset{1^1}{1}} 5 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 15 & 3 & 4 \\ 9 & 15 & 40 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1436] \quad 1 \overset{1}{\underset{1^1, 1^1}{13}} 5 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 10 & 18 & 59 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[1438] $1^{11} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 11, \Delta = -63, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 8 & 1 & 27 & 11 \\ 3 & 3 & 0 & 11 & 4 \end{pmatrix}_{b_0}$

[1439] $21^{12} 5^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 10 & 2 & 3 \\ 4 & 4 & 14 & 3 & 4 \\ 10 & 14 & 38 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$

[1440] $1^{13} 5^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 16 & 27 & 152 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$

[1441] $221^{11} 5^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 5 & 2 & 3 \\ 6 & 13 & 12 & 5 & 7 \\ 5 & 12 & 8 & 4 & 6 \\ 2 & 5 & 4 & 0 & 2 \\ 3 & 7 & 6 & 2 & 3 \end{pmatrix}_{b_0}$

[1443] $1^{13} 5^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 12 & 21 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$

[1442] $21^{13} 5^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 17 & 2 & 3 \\ 4 & 4 & 24 & 3 & 4 \\ 17 & 24 & 121 & 14 & 21 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$

[1444] $21^{12} 5^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 8 & 2 & 3 \\ 4 & 4 & 11 & 3 & 4 \\ 8 & 11 & 21 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$

[1445] $1^{13} 5^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 115 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$

[1446] $15^{11} 1^3 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 4 & 12 & 3 & 4 \\ 9 & 12 & 25 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$

[1447] $21^2 5^{11} 1^3 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -149, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 12 & 18 & 61 & 10 & 15 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$

[1448] $1^{13} 5^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -149, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 113 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1449] \ 1^{13} \overset{1^1}{5} \overset{1^1}{1^3, 1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -153, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 18 & 30 & 189 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1450] \ 1^{13} \overset{1^1}{5} \overset{1^1}{1^3, 1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -160, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 16 & 27 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1451] \ 1^{13} \overset{1^1}{5} \overset{1^1}{1^4, 1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -172, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.56 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 33 & 0 & 4 \\ 20 & 33 & 232 & 0 & 30 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 30 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1452] \ 1^{13} \overset{1^1}{5} \overset{1^1}{1^4, 1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -173, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.54 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 22 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 22 & 36 & 281 & 0 & 33 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 33 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1453] \ 21^4 \overset{1^1}{5} \overset{1^1}{1^1} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -173, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.54 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 18 & 24 & 121 & 14 & 21 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1454] \ 21^3 \overset{1^1}{5} \overset{1^1}{1^1} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -175, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 15 & 21 & 88 & 12 & 18 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1455] \ 1^{13} \overset{1^1}{5} \overset{1^1}{1^3, 1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -175, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 18 & 30 & 187 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1456] \ 21^{13} \overset{1^1}{5} \overset{1^1}{1^3} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -199, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.58 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 4 & 21 & 3 & 4 \\ 15 & 21 & 88 & 12 & 18 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1457] \ 21^{13} \overset{1^1}{5} \overset{1^1}{1^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -207, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.68 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 11 & 2 & 3 \\ 4 & 4 & 15 & 3 & 4 \\ 11 & 15 & 40 & 8 & 12 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1458] \ 221^{12} \overset{1^1}{5} \overset{1^1}{1^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -218, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.52 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 10 & 2 & 3 \\ 6 & 13 & 24 & 5 & 7 \\ 10 & 24 & 38 & 8 & 12 \\ 2 & 5 & 8 & 0 & 2 \\ 3 & 7 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[1459] $21^{1\bar{3}}51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 13 & 2 & 3 \\ 4 & 4 & 18 & 3 & 4 \\ 13 & 18 & 61 & 10 & 15 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[1460] $221^{1\bar{3}}51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -383, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.53 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 15 & 2 & 3 \\ 6 & 13 & 36 & 5 & 7 \\ 15 & 36 & 88 & 12 & 18 \\ 2 & 5 & 12 & 0 & 2 \\ 3 & 7 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$T = 10$ [1462] $21^{1\bar{1}}51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 & 3 \\ 3 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[1461] $1^{1\bar{1}}51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 & 3 \\ 3 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[1467] $1^{1\bar{1}}51^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & 10 & 4 & 6 \\ 1 & 1 & 0 & 4 & 0 & 2 \\ 2 & 2 & -1 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[1468] $1^{1\bar{1}}51^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 1 & 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[1464] $1^{1\bar{1}}51^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 \\ 3 & 3 & 5 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[1465] $1^{1\bar{3}}51^{31} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 13 & 21 & 97 & 18 & 21 \\ 2 & 3 & 18 & 1 & 3 \\ 3 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

[1466] $1^{1\bar{1}}51^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 7 & 7 & 9 & 6 & 7 \\ 2 & 3 & 6 & 1 & 3 \\ 3 & 4 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

[1463] $1^{1\bar{1}}51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 \\ 2 & 2 & 4 & 2 & 4 \\ 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[1469] $1^{\overline{1}1} \overline{5} 131^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 3 & 0 & 5 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 3 & -1 & 4 \end{pmatrix}_{b_0}$

[1470] $1^{\overline{1}1} \overline{5} 131^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 5 & 5 & 3 & 5 \\ 2 & 5 & 2 & 2 & 4 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 5 & 4 & 2 & 4 \end{pmatrix}_{b_0}$

[1471] $1^{\overline{1}1} \overline{5} 1231^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 2 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & 4 \end{pmatrix}_{b_0}$

[1472] $1^{\overline{1}1} \overline{5} 131^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 4 & 7 & 9 & 6 & 7 \\ 2 & 3 & 6 & 1 & 3 \\ 3 & 4 & 7 & 3 & 4 \end{pmatrix}_{b_0}$

[1473] $1^{\overline{1}1} \overline{5} 1231^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 4 & 5 \\ 1 & 1 & 0 & 4 & 0 & 2 \\ 2 & 2 & -1 & 5 & 2 & 3 \end{pmatrix}_{b_0}$

[1474] $1^{\overline{1}1} \overline{5} 131^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 \\ 3 & 4 & 0 & 3 & 4 \end{pmatrix}_{b_0}$

[1475] $1^{\overline{1}2} \overline{5} 131^{\overline{2}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 8 & 14 & 42 & 12 & 14 \\ 3 & 3 & 12 & 1 & 3 \\ 3 & 4 & 14 & 3 & 4 \end{pmatrix}_{b_0}$

[1476] $1^{\overline{1}1} \overline{5} 131^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$

[1477] $1^{\overline{1}1} \overline{5} 131^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 4 & 7 & 9 & 6 & 7 \\ 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & 7 & 3 & 4 \end{pmatrix}_{b_0}$

[1478] $1^{\overline{1}3} \overline{5} 131^{\overline{3}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 12 & 21 & 97 & 18 & 21 \\ 3 & 3 & 18 & 1 & 3 \\ 3 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$

[1479] $1^{\overline{1}2} \overline{5} 131^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 10, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 5 & 11 & 0 & 5 \\ 6 & 11 & 22 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 10 & -1 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 9 & 6 & 7 \\ 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & 7 & 3 & 4 \end{pmatrix} b_0$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 2 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 & 2 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 2 & 3 & 4 & 5 & 6 & 5 \\ 3 & 2 & 4 & 6 & 4 & 4 \\ 2 & 2 & 2 & 5 & 4 & 3 \end{pmatrix} b_0$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 & 2 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 2 & 3 & 4 & 5 & 6 & 5 \\ 3 & 2 & 4 & 6 & 4 & 4 \\ 2 & 2 & 2 & 5 & 4 & 3 \end{pmatrix} b_0$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 5 & 13 & 0 & 5 \\ 6 & 13 & 25 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 11 & -1 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 5 & 5 & 8 & 5 \\ 2 & 5 & 2 & 6 & 4 \\ 5 & 8 & 6 & 8 & 6 \\ 3 & 5 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 5 & 5 & 8 & 5 \\ 2 & 5 & 2 & 6 & 4 \\ 5 & 8 & 6 & 8 & 6 \\ 3 & 5 & 4 & 6 & 4 \end{pmatrix} b_0$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 2 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 3 \\ 2 & 1 & 3 & 3 & 3 \\ 3 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix} b_0$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[1484] $1^{11} 5^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[1491] $1^{12} 5^{12} 131^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 9 & 14 & 42 & 12 & 14 \\ 2 & 3 & 12 & 1 & 3 \\ 3 & 4 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[1492] $1^{11} 5^{12} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 6 & 7 & 9 & 6 & 7 \\ 2 & 3 & 6 & 1 & 3 \\ 3 & 4 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

[1493] $1^{11} 5^{12} 231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 1 & 9 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 8 & 9 & 0 & 81 & 12 & 17 \\ 1 & 1 & 0 & 12 & 0 & 2 \\ 2 & 2 & -1 & 17 & 2 & 3 \end{pmatrix}_{b_0}$$

[1494] $1^{11} 5^{12} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 9 & 6 & 7 \\ 2 & 3 & 6 & 1 & 3 \\ 3 & 4 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

[1495] $1^{11} 5^{12} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 17, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 5 & 8 & 0 & 5 \\ 4 & 8 & 9 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 7 & -1 & 4 \end{pmatrix}_{b_0}$$

[1496] $1^{11} 5^{12} 231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 6 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 6 & 0 & 31 & 8 & 11 \\ 1 & 1 & 0 & 8 & 0 & 2 \\ 2 & 2 & -1 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

[1497] $1^{13} 5^{12} 231^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 1 & 12 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 9 & 12 & 0 & 115 & 16 & 21 \\ 1 & 1 & 0 & 16 & 0 & 2 \\ 2 & 2 & -1 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[1498] $2^3 3^1 5^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 10 & 14 & 42 & 12 & 14 \\ 3 & 3 & 12 & 1 & 3 \\ 3 & 4 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[1499] $1^{13} 5^{12} 131^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 5 & 19 & 0 & 5 \\ 10 & 19 & 65 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 17 & -1 & 4 \end{pmatrix}_{b_0}$$

[1500] $21^{12}5^{11}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 22, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 12 & 12 & 16 & 70 & 18 \\ 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[1501] $23^{11}151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 22, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 3 & 3 & 1 & 4 & 3 \\ 3 & 2 & 4 & 2 & 4 \\ 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[1502] $1^{12}5^{14}131^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 12 & 14 & 42 & 12 & 14 \\ 2 & 3 & 12 & 1 & 3 \\ 3 & 4 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[1503] $1^{12}5^{11}1231^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 9 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 7 & 9 & 0 & 66 & 12 & 16 \\ 1 & 1 & 0 & 12 & 0 & 2 \\ 2 & 2 & -1 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

[1504] $1^{11}5^{12}131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 5 & 21 & 0 & 5 \\ 10 & 21 & 70 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 18 & -1 & 4 \end{pmatrix}_{b_0}$$

[1505] $1^{11}5^{12}131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 5 & 16 & 0 & 5 \\ 8 & 16 & 42 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 14 & -1 & 4 \end{pmatrix}_{b_0}$$

[1506] $1^{12}5^{12}131^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 10 & 14 & 42 & 12 & 14 \\ 2 & 3 & 12 & 1 & 3 \\ 3 & 4 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[1508] $1^{11}5^{13}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 \\ 1 & 2 & 0 & 7 & 0 & 2 \\ 5 & 6 & 7 & 35 & 0 & 11 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 11 & -1 & 3 \end{pmatrix}_{b_0}$$

[1507] $1^{11}5^{11}131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 6 & 3 \\ 4 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1509] $1^{11}5^{12}1231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 3 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

[1511] $1^{13} \overset{1^2}{5} 131^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 14 & 21 & 97 & 18 & 21 \\ 2 & 3 & 18 & 1 & 3 \\ 3 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$

[1510] $1^{12} \overset{1^3}{5} 131^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 11 & 14 & 42 & 12 & 14 \\ 2 & 3 & 12 & 1 & 3 \\ 3 & 4 & 14 & 3 & 4 \end{pmatrix}_{b_0}$

[1515] $1^{11} \overset{1^1}{5} 131^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 5 & 8 & 5 & 5 \\ 4 & 8 & 9 & 6 & 7 \\ 3 & 5 & 6 & 1 & 3 \\ 3 & 5 & 7 & 3 & 4 \end{pmatrix}_{b_0}$

[1513] $1^{11} \overset{1^3}{5} 131^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 5 & 24 & 5 & 5 \\ 12 & 24 & 97 & 18 & 21 \\ 3 & 5 & 18 & 1 & 3 \\ 3 & 5 & 21 & 3 & 4 \end{pmatrix}_{b_0}$

[1514] $1^{11} \overset{1^2}{5} 131^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 5 & 16 & 5 & 5 \\ 8 & 16 & 42 & 12 & 14 \\ 3 & 5 & 12 & 1 & 3 \\ 3 & 5 & 14 & 3 & 4 \end{pmatrix}_{b_0}$

[1512] $1^{11} \overset{1^4}{5} 131^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 5 & 32 & 5 & 5 \\ 16 & 32 & 174 & 24 & 28 \\ 3 & 5 & 24 & 1 & 3 \\ 3 & 5 & 28 & 3 & 4 \end{pmatrix}_{b_0}$

[1516] $2^3 3^1 15^1 \overset{1^3}{1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 15 & 21 & 97 & 18 & 21 \\ 3 & 3 & 18 & 1 & 3 \\ 3 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$

[1517] $1^{11} \overset{1^4}{5} 131^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 38, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 5 & 27 & 0 & 5 \\ 14 & 27 & 130 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 24 & -1 & 4 \end{pmatrix}_{b_0}$

[1518] $1^{12} \overset{1^1}{5} 1^{23} 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 1 & 12 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 10 & 12 & 0 & 130 & 16 & 22 \\ 1 & 1 & 0 & 16 & 0 & 2 \\ 2 & 2 & -1 & 22 & 2 & 3 \end{pmatrix}_{b_0}$

[1519] $1^{12} \overset{1^1}{5} 1^3 231^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 13 & 1 & 2 \\ 1 & 0 & 1 & 15 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 13 & 15 & 0 & 218 & 20 & 28 \\ 1 & 1 & 0 & 20 & 0 & 2 \\ 2 & 2 & -1 & 28 & 2 & 3 \end{pmatrix}_{b_0}$

[1520] $1^{11} \overset{1^1}{5} 1^1 231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 5 & 15 & 0 & 5 \\ 8 & 15 & 37 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 13 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[1521] \quad 2 \overset{1^1}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 5 & 5 & 3 & 5 \\ 3 & 5 & 2 & 2 & 4 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 5 & 4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1522] \quad 1^1 \overset{1^1}{5} 131^{\textcolor{red}{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 5 & 24 & 0 & 5 \\ 12 & 24 & 97 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 21 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[1523] \quad 1^1 \overset{1^1}{5} 1^1 231^{\textcolor{red}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 5 & 15 & 5 & 5 \\ 8 & 15 & 37 & 11 & 13 \\ 3 & 5 & 11 & 1 & 3 \\ 3 & 5 & 13 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1524] \quad 1^1 \overset{1^1}{5} 1^1 231^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 5 & 0 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 \\ 4 & 5 & 5 & 18 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 8 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1527] \quad 1^1 \overset{1^1}{5} 131^{\textcolor{red}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 4 & 0 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 5 & 11 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1526] \quad 1^1 \overset{1^1}{5} 131^{\textcolor{red}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 5 & 0 & 2 \\ 1 & 2 & 0 & 6 & 0 & 2 \\ 4 & 5 & 6 & 21 & 0 & 9 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1525] \quad 1^1 \overset{1^1}{5} 131^{\textcolor{red}{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 5 & 29 & 0 & 5 \\ 14 & 29 & 137 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 25 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[1528] \quad 1^{\textcolor{red}{13}} \overset{1^3}{5} 131^{\textcolor{red}{31}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 15 & 21 & 97 & 18 & 21 \\ 2 & 3 & 18 & 1 & 3 \\ 3 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1529] \quad 2 \overset{1^{12}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 3 \\ 4 & 5 & 10 & 3 & 5 \\ 6 & 10 & 14 & 4 & 8 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 5 & 8 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1530] \quad 21^{\textcolor{red}{11}} \overset{1^1}{5} 131^{\textcolor{red}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 6 & 3 \\ 5 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1532] $1^{11} \overset{1}{5} 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 53, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 \\ 7 & 8 & 11 & 72 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 16 & -1 & 3 \end{pmatrix}_{b_0}$

[1531] $1^{11} \overset{1}{5} 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 53, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 5 & 0 & 2 \\ 1 & 2 & 0 & 8 & 0 & 2 \\ 4 & 5 & 8 & 24 & 0 & 10 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 10 & -1 & 3 \end{pmatrix}_{b_0}$

[1534] $1^{13} \overset{1}{5} 1^{23} 31^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 0 & 1 & 15 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 12 & 15 & 0 & 193 & 20 & 27 \\ 1 & 1 & 0 & 20 & 0 & 2 \\ 2 & 2 & -1 & 27 & 2 & 3 \end{pmatrix}_{b_0}$

[1533] $1^{13} \overset{1}{5} 131^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 16 & 21 & 97 & 18 & 21 \\ 2 & 3 & 18 & 1 & 3 \\ 3 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$

[1535] $1^{11} \overset{1}{5} 1^{12} 231^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 59, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 5 & 20 & 0 & 5 \\ 10 & 20 & 63 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 17 & -1 & 4 \end{pmatrix}_{b_0}$

[1536] $1^{112} 3221^{151} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 3 & 11 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 11 & 3 \\ 6 & 11 & 0 & 11 & 49 & 13 \\ 2 & 3 & -1 & 3 & 13 & 3 \end{pmatrix}_{b_0}$

[1537] $1^{11} \overset{1}{5} 1^{12} 231^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 5 & 25 & 0 & 5 \\ 12 & 25 & 97 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 21 & -1 & 4 \end{pmatrix}_{b_0}$

[1538] $1^{11} \overset{1}{5} 131^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 62, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 5 & 32 & 0 & 5 \\ 16 & 32 & 174 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 28 & -1 & 4 \end{pmatrix}_{b_0}$

[1539] $2^3 151^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 65, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 5 & 15 & 3 & 5 \\ 9 & 15 & 34 & 6 & 12 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 5 & 12 & 2 & 4 \end{pmatrix}_{b_0}$

[1540] $2^3 151^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 67, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 3 & 3 & 1 & 6 & 3 \\ 5 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$

[1541] $1^{11} \overset{1}{5} 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 \\ 5 & 6 & 9 & 36 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[1542] $1^{11} 5^{11} 131^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 10 & 0 & 2 \\ 6 & 7 & 10 & 52 & 0 & 14 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 14 & -1 & 3 \end{pmatrix}_{b_0}$$

[1543] $1^{12} 5^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 \\ 1 & 2 & 0 & 6 & 0 & 2 \\ 4 & 6 & 6 & 21 & 0 & 9 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[1544] $1^{12} 5^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 8 & 0 & 2 \\ 6 & 8 & 8 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 13 & -1 & 3 \end{pmatrix}_{b_0}$$

[1545] $1^{13} 5^{11} 1^{32} 231^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 1 & 2 \\ 1 & 0 & 1 & 18 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 15 & 18 & 0 & 295 & 24 & 33 \\ 1 & 1 & 0 & 24 & 0 & 2 \\ 2 & 2 & -1 & 33 & 2 & 3 \end{pmatrix}_{b_0}$$

[1546] $2^3 1^5 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 78, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 3 & 3 & 1 & 8 & 3 \\ 7 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[1547] $1^{12} 5^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 7 & 0 & 2 \\ 5 & 7 & 7 & 33 & 0 & 11 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 11 & -1 & 3 \end{pmatrix}_{b_0}$$

[1548] $1^{11} 5^{12} 1^{23} 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 80, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 5 & 30 & 5 & 5 \\ 16 & 30 & 154 & 22 & 26 \\ 3 & 5 & 22 & 1 & 3 \\ 3 & 5 & 26 & 3 & 4 \end{pmatrix}_{b_0}$$

[1550] $1^{11} 5^{11} 131^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 85, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 14 & 0 & 2 \\ 8 & 9 & 14 & 97 & 0 & 19 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

[1549] $1^{11} 5^{11} 131^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 85, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 13 & 0 & 2 \\ 7 & 8 & 13 & 75 & 0 & 17 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

[1551] $2^3 1^5 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 3 \\ 4 & 5 & 20 & 3 & 5 \\ 12 & 20 & 62 & 8 & 16 \\ 2 & 3 & 8 & 0 & 2 \\ 3 & 5 & 16 & 2 & 4 \end{pmatrix}_{b_0}$$

[1552] $1^{11} 5^1 1^{12} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 88$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 \\ 1 & 2 & 0 & 8 & 0 & 2 \\ 5 & 6 & 8 & 31 & 0 & 11 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 11 & -1 & 3 \end{pmatrix}_{b_0}$$

[1553] $2^1 3^1 15^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 89$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 3 \\ 4 & 5 & 5 & 8 & 5 \\ 3 & 5 & 2 & 6 & 4 \\ 5 & 8 & 6 & 8 & 6 \\ 3 & 5 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

[1554] $2^1 3^1 15^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 98$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 5 & 3 \\ 4 & 5 & 10 & 8 & 5 \\ 6 & 10 & 14 & 12 & 8 \\ 5 & 8 & 12 & 8 & 6 \\ 3 & 5 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

[1555] $2^1 3^1 15^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 105$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 3 & 3 & 1 & 10 & 3 \\ 8 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[1556] $1^{11} 5^1 1^{12} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 108$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 \\ 6 & 7 & 11 & 50 & 0 & 14 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 14 & -1 & 3 \end{pmatrix}_{b_0}$$

[1557] $1^{12} 5^1 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 108$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 \\ 5 & 7 & 9 & 36 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[1558] $21^{13} 5^1 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 108$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 1 & 3 & 17 & 3 \\ 2 & 3 & 1 & 22 & 3 \\ 17 & 17 & 22 & 137 & 25 \\ 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

[1559] $2^1 3^1 15^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 113$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 7.1$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 3 \\ 4 & 5 & 25 & 3 & 5 \\ 15 & 25 & 98 & 10 & 20 \\ 2 & 3 & 10 & 0 & 2 \\ 3 & 5 & 20 & 2 & 4 \end{pmatrix}_{b_0}$$

[1560] $2^1 3^1 15^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 113$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 5 & 3 \\ 4 & 5 & 15 & 8 & 5 \\ 9 & 15 & 34 & 18 & 12 \\ 5 & 8 & 18 & 8 & 6 \\ 3 & 5 & 12 & 6 & 4 \end{pmatrix}_{b_0}$$

[1561] $21^{12} 5^1 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 118$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 12 & 3 \\ 10 & 10 & 12 & 42 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[1562] $21^{12} 5^1 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 118$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 1 & 3 & 8 & 3 \\ 2 & 3 & 1 & 8 & 3 \\ 8 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1563] \ 1^{\frac{1}{1}} 5^{\frac{1}{2}} 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 122, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 5 & 30 & 0 & 5 \\ 16 & 30 & 154 & 0 & 26 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 26 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[1564] \ 1^{\frac{1}{1}} 5^{\frac{1}{2}} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 132, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 12 & 0 & 2 \\ 8 & 10 & 12 & 90 & 0 & 18 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1565] \ 1^{\frac{1}{1}} 5^{\frac{1}{2}} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 132, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 10 & 0 & 2 \\ 6 & 8 & 10 & 50 & 0 & 14 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 14 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1566] \ 2^{\frac{1}{1}} 3^{\frac{1}{1}} 151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 134, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 12 & 5 & 3 \\ 4 & 5 & 20 & 8 & 5 \\ 12 & 20 & 62 & 24 & 16 \\ 5 & 8 & 24 & 8 & 6 \\ 3 & 5 & 16 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1568] \ 1^{\frac{1}{1}} 5^{\frac{1}{3}} 1^{\frac{1}{3}} 231^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 24 & 3 & 3 \\ 3 & 5 & 45 & 5 & 5 \\ 24 & 45 & 349 & 33 & 39 \\ 3 & 5 & 33 & 1 & 3 \\ 3 & 5 & 39 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1567] \ 1^{\frac{1}{1}} 5^{\frac{1}{3}} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 \\ 7 & 9 & 11 & 68 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1569] \ 1^{\frac{1}{1}} 5^{\frac{1}{2}} 1^{\frac{1}{2}} 231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 141, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 12 & 0 & 2 \\ 8 & 9 & 12 & 81 & 0 & 17 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1570] \ 2^{\frac{1}{1}} 3^{\frac{1}{1}} 151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 3 \\ 4 & 5 & 30 & 3 & 5 \\ 18 & 30 & 142 & 12 & 24 \\ 2 & 3 & 12 & 0 & 2 \\ 3 & 5 & 24 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1571] \ 1^{\frac{1}{1}} 5^{\frac{1}{2}} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 13 & 0 & 2 \\ 7 & 9 & 13 & 73 & 0 & 17 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1572] \ 2^{\frac{1}{1}} 3^{\frac{1}{1}} 151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 161, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.1$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 15 & 5 & 3 \\ 4 & 5 & 25 & 8 & 5 \\ 15 & 25 & 98 & 30 & 20 \\ 5 & 8 & 30 & 8 & 6 \\ 3 & 5 & 20 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1573] \ 2 \ 3 \ 15 \ 1^{\overset{1}{12} \overset{1}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 161, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 8 & 3 \\ 4 & 5 & 5 & 13 & 5 \\ 3 & 5 & 2 & 10 & 4 \\ 8 & 13 & 10 & 24 & 10 \\ 3 & 5 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1574] \ 1^{\overset{1}{1}} 5 \ 1^{\overset{1}{2}} 231^{\overset{1}{13}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 166, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 5 & 35 & 0 & 5 \\ 18 & 35 & 202 & 0 & 30 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 30 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[1575] \ 1^{\overset{1}{12}} 3221^{\overset{1}{15}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 171, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 5 & 1 & 2 \\ 4 & 18 & 6 & 25 & 0 & 8 \\ 1 & 6 & 0 & 8 & 0 & 2 \\ 5 & 25 & 8 & 31 & 0 & 11 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 8 & 2 & 11 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1576] \ 2 \ 3 \ 15 \ 1^{\overset{1}{12} \overset{1}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 178, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 10 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 3 & 3 & 1 & 12 & 3 \\ 10 & 10 & 12 & 42 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1577] \ 1^{\overset{1}{12} \overset{1}{1}} 5 \ 131^{\overset{1}{13}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 184, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.89 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 14 & 0 & 2 \\ 8 & 10 & 14 & 93 & 0 & 19 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1578] \ 2 \ 3 \ 15 \ 1^{\overset{1}{17} \overset{1}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 185, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 7.3$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 21 & 2 & 3 \\ 4 & 5 & 35 & 3 & 5 \\ 21 & 35 & 194 & 14 & 28 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 5 & 28 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1579] \ 1^{\overset{1}{13} \overset{1}{12}} 5 \ 131^{\overset{1}{12}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 189, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 \\ 7 & 10 & 11 & 68 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1580] \ 1^{\overset{1}{11} \overset{1}{1}} 5 \ 1^{\overset{1}{2}} 231^{\overset{1}{12}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 189, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 15 & 0 & 2 \\ 9 & 10 & 15 & 108 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 20 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1581] \ 1^{\overset{1}{12} \overset{1}{1}} 5 \ 131^{\overset{1}{13}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 192, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 \\ 1 & 2 & 0 & 15 & 0 & 2 \\ 9 & 11 & 15 & 117 & 0 & 21 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1582] \ 1^{\overset{1}{11} \overset{1}{1}} 5 \ 1^{\overset{1}{2}} 231^{\overset{1}{14}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 194, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 5 & 40 & 0 & 5 \\ 20 & 40 & 258 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 34 & -1 & 4 \end{pmatrix}_{b_0}$$

[1583] $1^{12} \overset{1}{5} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 204, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 \\ 1 & 2 & 0 & 17 & 0 & 2 \\ 9 & 11 & 17 & 124 & 0 & 22 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

[1584] $1^{13} \overset{1}{5} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 205, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 \\ 1 & 2 & 0 & 12 & 0 & 2 \\ 8 & 11 & 12 & 88 & 0 & 18 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[1585] $1^{13} \overset{1}{5} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 205, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 2 \\ 1 & 2 & 0 & 13 & 0 & 2 \\ 9 & 12 & 13 & 112 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 20 & -1 & 3 \end{pmatrix}_{b_0}$$

[1586] $2 \overset{12}{3} 1 \overset{1}{5} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 217, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 3 & 3 & 1 & 14 & 3 \\ 12 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[1587] $22 \overset{111}{3} 151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 227, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 5 & 5 & 3 \\ 6 & 14 & 13 & 13 & 8 \\ 5 & 13 & 9 & 11 & 7 \\ 5 & 13 & 11 & 8 & 6 \\ 3 & 8 & 7 & 6 & 4 \end{pmatrix}_{b_0}$$

[1589] $1^{12} \overset{1}{5} 1^{22} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 \\ 1 & 2 & 0 & 13 & 0 & 2 \\ 9 & 11 & 13 & 101 & 0 & 19 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

[1590] $1^{12} \overset{1}{5} 131^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 2 \\ 1 & 2 & 0 & 18 & 0 & 2 \\ 10 & 12 & 18 & 150 & 0 & 24 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[1588] $21^{13} \overset{1}{5} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 18 & 3 \\ 15 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[1591] $1^{13} \overset{1}{5} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 245, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 \\ 1 & 2 & 0 & 14 & 0 & 2 \\ 8 & 11 & 14 & 93 & 0 & 19 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

[1592] $21^{13} \overset{1}{5} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 252, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 13 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[1593] $1^1 \overset{1}{5} 1^3 231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 257, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 24 & 1 & 3 \\ 3 & 5 & 45 & 0 & 5 \\ 24 & 45 & 349 & 0 & 39 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 39 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[1594] \quad 1^{13} 2^3 15^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 262, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 13 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 3 & 3 & 1 & 16 & 3 \\ 13 & 12 & 16 & 70 & 18 \\ 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[1595] \quad 1^{11} 5^1 1^3 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 264, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 0 & 2 & 13 & 0 & 2 \\ 1 & 2 & 0 & 19 & 0 & 2 \\ 12 & 13 & 19 & 190 & 0 & 26 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 26 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1596] \quad 22 \quad 3 \quad 151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 5 & 5 & 3 \\ 5 & 9 & 12 & 11 & 7 \\ 5 & 12 & 8 & 11 & 6 \\ 5 & 11 & 11 & 9 & 7 \\ 3 & 7 & 6 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[1597] \quad 1^{13} 5^1 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 277, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 2 \\ 1 & 2 & 0 & 15 & 0 & 2 \\ 9 & 12 & 15 & 115 & 0 & 21 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1598] \quad 1^{13} 5^1 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 293, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 13 & 0 & 2 \\ 1 & 2 & 0 & 16 & 0 & 2 \\ 10 & 13 & 16 & 141 & 0 & 23 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 23 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1599] \quad 1^{12} 5^1 1^3 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 304, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 2 \\ 1 & 2 & 0 & 16 & 0 & 2 \\ 10 & 12 & 16 & 130 & 0 & 22 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1600] \quad 1^{12} 5^1 1^3 231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 309, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 0 & 2 & 14 & 0 & 2 \\ 1 & 2 & 0 & 17 & 0 & 2 \\ 12 & 14 & 17 & 181 & 0 & 25 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 25 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1601] \quad 1^{13} 5^1 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 325, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 13 & 0 & 2 \\ 1 & 2 & 0 & 18 & 0 & 2 \\ 10 & 13 & 18 & 148 & 0 & 24 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1602] \quad 1^{15} 5^1 1^3 231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 327, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 26 & 1 & 3 \\ 3 & 5 & 50 & 0 & 5 \\ 26 & 50 & 419 & 0 & 43 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 43 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[1603] \quad 2^3 3^1 15^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 330, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 16 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 3 & 3 & 1 & 20 & 3 \\ 16 & 14 & 20 & 106 & 22 \\ 3 & 3 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[1604] $22 \ 3 \ 151^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 341, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 5 & 10 & 3 \\ 6 & 14 & 13 & 26 & 8 \\ 5 & 13 & 9 & 22 & 7 \\ 10 & 26 & 22 & 41 & 13 \\ 3 & 8 & 7 & 13 & 4 \end{pmatrix}_{b_0}$$

[1605] $1^{\overline{1}} 5^{\overline{12}} 1^{\overline{2}} 231^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 0 & 2 & 13 & 0 & 2 \\ 1 & 2 & 0 & 19 & 0 & 2 \\ 11 & 13 & 19 & 165 & 0 & 25 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 25 & -1 & 3 \end{pmatrix}_{b_0}$$

[1606] $1^{\overline{13}} 5^{\overline{14}} 131^{\overline{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 357, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 0 & 2 & 14 & 0 & 2 \\ 1 & 2 & 0 & 19 & 0 & 2 \\ 11 & 14 & 19 & 176 & 0 & 26 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 26 & -1 & 3 \end{pmatrix}_{b_0}$$

[1607] $1^{\overline{12}} 3221^{\overline{12}} 51^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 10 & 1 & 2 \\ 4 & 18 & 6 & 50 & 0 & 8 \\ 1 & 6 & 0 & 16 & 0 & 2 \\ 10 & 50 & 16 & 130 & 0 & 22 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 8 & 2 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

[1608] $2 \ 3 \ 151^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 363, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 15 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 3 & 3 & 1 & 18 & 3 \\ 15 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[1609] $22 \ 3 \ 151^{\overline{121}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 398, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 10 & 5 & 3 \\ 6 & 14 & 26 & 13 & 8 \\ 10 & 26 & 42 & 22 & 14 \\ 5 & 13 & 22 & 8 & 6 \\ 3 & 8 & 14 & 6 & 4 \end{pmatrix}_{b_0}$$

[1610] $1^{\overline{12}} 5^{\overline{1}} 3^{\overline{2}} 231^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 413, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 1 & 2 \\ 1 & 0 & 2 & 15 & 0 & 2 \\ 1 & 2 & 0 & 20 & 0 & 2 \\ 13 & 15 & 20 & 218 & 0 & 28 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 28 & -1 & 3 \end{pmatrix}_{b_0}$$

[1611] $1^{\overline{13}} 5^{\overline{1}} 3^{\overline{2}} 231^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 424, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 1 & 2 \\ 1 & 0 & 2 & 16 & 0 & 2 \\ 1 & 2 & 0 & 18 & 0 & 2 \\ 13 & 16 & 18 & 211 & 0 & 27 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

[1612] $2 \ 3 \ 151^{\overline{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 18 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 3 & 3 & 1 & 22 & 3 \\ 18 & 17 & 22 & 137 & 25 \\ 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

[1613] $1^{\overline{12}} 5^{\overline{1}} 3^{\overline{2}} 231^{\overline{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 1 & 2 \\ 1 & 0 & 2 & 16 & 0 & 2 \\ 1 & 2 & 0 & 23 & 0 & 2 \\ 14 & 16 & 23 & 261 & 0 & 31 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 31 & -1 & 3 \end{pmatrix}_{b_0}$$

[1614] $1^{12}5^1 1^3 231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 549$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 1 & 2 \\ 1 & 0 & 2 & 17 & 0 & 2 \\ 1 & 2 & 0 & 26 & 0 & 2 \\ 15 & 17 & 26 & 310 & 0 & 34 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 34 & -1 & 3 \end{pmatrix}_{b_0}$$

[1615] $1^{13}5^1 1^3 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 556$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 14 & 1 & 2 \\ 1 & 0 & 2 & 17 & 0 & 2 \\ 1 & 2 & 0 & 21 & 0 & 2 \\ 14 & 17 & 21 & 250 & 0 & 30 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 30 & -1 & 3 \end{pmatrix}_{b_0}$$

[1616] $1^{13}5^1 1^3 231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 664$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 15 & 1 & 2 \\ 1 & 0 & 2 & 18 & 0 & 2 \\ 1 & 2 & 0 & 24 & 0 & 2 \\ 15 & 18 & 24 & 295 & 0 & 33 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 33 & -1 & 3 \end{pmatrix}_{b_0}$$

[1617] $1^{13}3221^{13}5^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 675$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 15 & 1 & 2 \\ 4 & 18 & 6 & 75 & 0 & 8 \\ 1 & 6 & 0 & 24 & 0 & 2 \\ 15 & 75 & 24 & 295 & 0 & 33 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 8 & 2 & 33 & -1 & 3 \end{pmatrix}_{b_0}$$

[1618] $22^1 3^1 151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 683$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 15 & 5 & 3 \\ 6 & 14 & 39 & 13 & 8 \\ 15 & 39 & 97 & 33 & 21 \\ 5 & 13 & 33 & 8 & 6 \\ 3 & 8 & 21 & 6 & 4 \end{pmatrix}_{b_0}$$

[1619] $1^{13}5^1 1^3 231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 748$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 16 & 1 & 2 \\ 1 & 0 & 2 & 19 & 0 & 2 \\ 1 & 2 & 0 & 27 & 0 & 2 \\ 16 & 19 & 27 & 346 & 0 & 36 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 36 & -1 & 3 \end{pmatrix}_{b_0}$$

[1620] $22^1 3^1 151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 1082$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 20 & 5 & 3 \\ 6 & 14 & 52 & 13 & 8 \\ 20 & 52 & 174 & 44 & 28 \\ 5 & 13 & 44 & 8 & 6 \\ 3 & 8 & 28 & 6 & 4 \end{pmatrix}_{b_0}$$

[1621] $22^1 3^1 151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 1595$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 7.1$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 25 & 5 & 3 \\ 6 & 14 & 65 & 13 & 8 \\ 25 & 65 & 273 & 55 & 35 \\ 5 & 13 & 55 & 8 & 6 \\ 3 & 8 & 35 & 6 & 4 \end{pmatrix}_{b_0}$$

$T = 11$ [1622] $1^{11}31^1 5^1 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 12$, $\Delta = -37$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 3 & 1 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

[1623] $1^{11}31^1 5^1 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 11$, $T_{\max} = 12$, $\Delta = -37$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 4 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 6 & 4 & 6 \\ 1 & 1 & 0 & 4 & 0 & 2 \\ 2 & 3 & -1 & 6 & 2 & 4 \end{pmatrix}_{b_0}$$

[1624] $1^{11}5^{11}1321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

[1625] $2^{11}3^{11}2161^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -13, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 3 \\ 2 & 1 & 3 & 3 & 3 \\ 3 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[1626] $1^{11}3^{11}5^{11}131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 12, \Delta = -37, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 4 & 0 & 3 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & 6 & -1 & 4 \end{pmatrix}_{b_0}$$

[1627] $2^{11}3^{11}2161^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 6 & 3 \\ 3 & 3 & 1 & 7 & 3 \\ 7 & 6 & 7 & 19 & 9 \\ 3 & 3 & 3 & 9 & 4 \end{pmatrix}_{b_0}$$

[1628] $2^{11}3^{11}2161^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 3 & 3 & 1 & 4 & 3 \\ 3 & 2 & 4 & 2 & 4 \\ 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[1629] $2^{11}3^{11}2161^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -57, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 3 & 3 & 1 & 9 & 3 \\ 9 & 9 & 9 & 34 & 12 \\ 3 & 3 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[1630] $2^{11}3^{11}2161^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 3 & 3 & 1 & 6 & 3 \\ 5 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1631] $2^{11}3^{11}2161^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -78, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 10 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 3 & 3 & 1 & 11 & 3 \\ 10 & 8 & 11 & 37 & 13 \\ 3 & 3 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

[1632] $41^{11}23151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -90, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 8 & 1 & 3 \\ 9 & 51 & 48 & 0 & 15 \\ 8 & 48 & 42 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 15 & 14 & -1 & 4 \end{pmatrix}_{b_0}$$

[1633] $2^{12}3^{11}2161^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 3 & 3 & 1 & 10 & 3 \\ 8 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[1634] $2^{12}3^{11}2161^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -118, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 13 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 3 & 3 & 1 & 15 & 3 \\ 13 & 10 & 15 & 63 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[1635] $41^{12}23151^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -126, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 16 & 1 & 3 \\ 9 & 51 & 96 & 0 & 15 \\ 16 & 96 & 174 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 15 & 28 & -1 & 4 \end{pmatrix}_{b_0}$$

[1636] $23^{11}21^{11}61^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -175, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 11 & 3 \\ 3 & 3 & 1 & 13 & 3 \\ 12 & 11 & 13 & 56 & 16 \\ 3 & 3 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

[1637] $41^{13}23151^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -186, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 24 & 1 & 3 \\ 9 & 51 & 144 & 0 & 15 \\ 24 & 144 & 394 & 0 & 42 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 15 & 42 & -1 & 4 \end{pmatrix}_{b_0}$$

[1638] $41^{14}23151^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -270, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 32 & 1 & 3 \\ 9 & 51 & 192 & 0 & 15 \\ 32 & 192 & 702 & 0 & 56 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 15 & 56 & -1 & 4 \end{pmatrix}_{b_0}$$

[1639] $23^{12}21^{11}61^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -277, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 15 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 3 & 3 & 1 & 17 & 3 \\ 15 & 13 & 17 & 86 & 20 \\ 3 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

$T = 12$ [1640] $1^{11}315^{11}13221^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 12, T_{\max} = 12, \Delta = -9, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0$

[1641] $1^{11}6^{11}12321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & 2 & -1 & 4 \end{pmatrix}_{b_0}$$

[1642] $1^{11}6^{11}131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 1 & 0 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & 2 & -1 & 4 \end{pmatrix}_{b_0}$$

[1643] $1^{11}315^{11}1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 12, T_{\max} = 12, \Delta = -9, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 4 & 0 & 3 \\ 1 & 2 & 1 & 5 & 0 & 3 \\ 2 & 4 & 5 & 7 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 3 & 7 & -1 & 5 \end{pmatrix}_{b_0}$$

$T = 13$ [1644] $1^{11}6^{11}131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 \quad T = 13, T_{\max} = 13, \Delta = -8, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 3 & 6 & 5 \\ 5 & 6 & 8 & 8 \\ 4 & 5 & 8 & 7 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [1645] \ 1^{\overset{1}{1}\overset{1}{1}2315131} 6^{11} 131^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 15, \ T_{\max} = 15, \ \Delta = -8, \ \det = 0, \ (n_+, n_-, n_0) = (1, 15, 1), \ c_{\text{ext}} = 6.0$$

4.952 NHC(-2)(-3) + $\textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{violet}{e}_6$

$$T = 8 \quad [0] \ 1^{\overset{1}{1}\overset{1}{1}32} 31^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 79, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 1 & 3 \\ 3 & 4 & 12 & 13 & 0 & 4 \\ 6 & 12 & 25 & 27 & 0 & 9 \\ 7 & 13 & 27 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 9 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overset{1}{1}} 1^{\overset{1}{1}} 31^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 94, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.70 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overset{1}{1}} 1^{\overset{1}{1}} 31^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 103, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.03 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overset{1}{1}} 1^{\overset{1}{1}} 31^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 121, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.22 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 \\ 3 & 5 & 4 & 6 & 0 & 4 \\ 3 & 5 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overset{1}{1}\overset{1}{1}2} 31^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 126, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.81 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 8 & 9 & 0 & 4 \\ 4 & 8 & 10 & 12 & 0 & 6 \\ 5 & 9 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overset{1}{1}\overset{1}{1}} 1^{\overset{1}{1}} 31^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 127, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.02 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 4 & 2 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overset{1}{1}\overset{1}{1}} 1^{\overset{1}{1}} 31^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 148, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 1 & 3 \\ 4 & 9 & 12 & 4 & 0 & 5 \\ 5 & 12 & 13 & 6 & 0 & 7 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 7 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overset{1}{1}\overset{1}{1}3} 31^{\overset{1}{1}\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 160, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 4 & 15 & 0 & 4 \\ 9 & 17 & 15 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 12 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overset{1}{1}\overset{1}{1}} 1^{\overset{1}{1}} 31^{\overset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 166, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

- [9] $1^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 178$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.07 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 4 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$
- [10] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 181$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.17 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 9 & 8 & 0 & 4 \\ 5 & 9 & 13 & 12 & 0 & 7 \\ 4 & 8 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 7 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$
- [11] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 184$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 19 & 4 & 0 & 4 \\ 11 & 19 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 15 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{121}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 184$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.91 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 4 & 8 & 10 & 6 & 0 & 6 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$
- [13] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 202$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.18 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 8 & 14 & 9 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 10 & -4 & -1 \end{pmatrix}_{b_0}$$
- [14] $1^{122}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 208$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 13.54 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 1 & 3 \\ 3 & 4 & 18 & 8 & 0 & 4 \\ 10 & 18 & 58 & 24 & 0 & 14 \\ 4 & 8 & 24 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 14 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$
- [15] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 216$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$
- [16] $1^{12}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 229$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 4 & 12 & 0 & 4 \\ 7 & 13 & 12 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$
- [17] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 243$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 5 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$
- [18] $1^{112}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 256$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 10 & 2 & 1 & 3 \\ 4 & 9 & 24 & 4 & 0 & 5 \\ 10 & 24 & 58 & 12 & 0 & 14 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 14 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

- [19] $1^{122}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 274$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 1 & 3 \\ 3 & 4 & 8 & 13 & 0 & 4 \\ 4 & 8 & 10 & 18 & 0 & 6 \\ 7 & 13 & 18 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$
- [20] $1^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 286$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 5 & 9 & 13 & 9 & 0 & 7 \\ 3 & 5 & 9 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 7 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$
- [21] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 304$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 \\ 5 & 9 & 13 & 6 & 0 & 7 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 7 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$
- [22] $1^{111}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 310$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 \\ 8 & 14 & 37 & 9 & 0 & 11 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 11 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$
- [23] $1^{14}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 313$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.06 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 1 & 3 \\ 3 & 4 & 21 & 5 & 0 & 4 \\ 11 & 21 & 76 & 18 & 0 & 16 \\ 3 & 5 & 18 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 16 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$
- [24] $1^{112}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 343$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 8 & 2 & 1 & 3 \\ 4 & 9 & 20 & 4 & 0 & 5 \\ 8 & 20 & 37 & 9 & 0 & 11 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 11 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$
- [25] $1^{12}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 373$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 \\ 7 & 13 & 28 & 12 & 0 & 10 \\ 3 & 5 & 12 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 10 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$
- [26] $1^{13}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 382$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 \\ 9 & 17 & 49 & 15 & 0 & 13 \\ 3 & 5 & 15 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 13 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$
- [27] $1^{112}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 391$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 7 & 13 & 9 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$
- [28] $1^{122}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 403$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 13.04 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 1 & 3 \\ 3 & 4 & 13 & 8 & 0 & 4 \\ 7 & 13 & 28 & 18 & 0 & 10 \\ 4 & 8 & 18 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 10 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

- [29] $1^{113}31^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 436$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 1 & 4 & 15 & 2 & 1 & 3 \\ 4 & 9 & 36 & 4 & 0 & 5 \\ 15 & 36 & 133 & 18 & 0 & 21 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 21 & 2 & -4 & 0 \end{pmatrix}_{b_0}$
- [30] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 471$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 5 & 9 & 13 & 15 & 0 & 7 \\ 5 & 9 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 7 & 6 & -4 & 0 \end{pmatrix}_{b_0}$
- [31] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 486$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 2 & 4 & 1 & 12 & 0 & 3 \\ 10 & 18 & 12 & 57 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 13 & -4 & 0 \end{pmatrix}_{b_0}$
- [32] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 526$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 7 & 13 & 28 & 9 & 0 & 10 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 10 & 2 & -4 & 0 \end{pmatrix}_{b_0}$
- [33] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 547$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 2 & 4 & 1 & 12 & 0 & 3 \\ 9 & 17 & 12 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 12 & -4 & 0 \end{pmatrix}_{b_0}$
- [34] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 568$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 1 & 4 & 11 & 2 & 1 & 3 \\ 4 & 9 & 28 & 4 & 0 & 5 \\ 11 & 28 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 15 & 2 & -4 & 0 \end{pmatrix}_{b_0}$
- [35] $1^{112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 574$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 1 & 3 & 5 & 7 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 5 & 9 & 13 & 21 & 0 & 7 \\ 7 & 13 & 21 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 7 & 9 & -4 & 0 \end{pmatrix}_{b_0}$
- [36] $1^{113}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 595$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 9 & 17 & 0 & 4 \\ 5 & 9 & 13 & 27 & 0 & 7 \\ 9 & 17 & 27 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 7 & 12 & -4 & 0 \end{pmatrix}_{b_0}$
- [37] $1^{113}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 631$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$
 $\begin{pmatrix} 1 & 4 & 13 & 2 & 1 & 3 \\ 4 & 9 & 32 & 4 & 0 & 5 \\ 13 & 32 & 100 & 15 & 0 & 18 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 18 & 2 & -4 & 0 \end{pmatrix}_{b_0}$
- [38] $1^{132}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 637$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.02 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$
 $\begin{pmatrix} 1 & 3 & 9 & 4 & 1 & 3 \\ 3 & 4 & 17 & 8 & 0 & 4 \\ 9 & 17 & 49 & 24 & 0 & 13 \\ 4 & 8 & 24 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 13 & 5 & -4 & -1 \end{pmatrix}_{b_0}$

- [39] $1^{132}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 646$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.11 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 1 & 3 \\ 3 & 4 & 22 & 8 & 0 & 4 \\ 12 & 22 & 85 & 30 & 0 & 17 \\ 4 & 8 & 30 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 17 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$
- [40] $1^{121}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 648$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 1 & 3 \\ 3 & 4 & 13 & 9 & 0 & 4 \\ 7 & 13 & 28 & 21 & 0 & 10 \\ 5 & 9 & 21 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 10 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$
- [41] $1^{114}31^{41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 688$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.99 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 20 & 2 & 1 & 3 \\ 4 & 9 & 48 & 4 & 0 & 5 \\ 20 & 48 & 238 & 24 & 0 & 28 \\ 2 & 4 & 24 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 28 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$
- [42] $1^{121}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 736$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 10 & 18 & 58 & 12 & 0 & 14 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 14 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$
- [43] $1^{131}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 747$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 1 & 3 \\ 3 & 4 & 17 & 9 & 0 & 4 \\ 9 & 17 & 49 & 27 & 0 & 13 \\ 5 & 9 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 13 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$
- [44] $1^{131}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 760$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 9 & 17 & 49 & 12 & 0 & 13 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 13 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$
- [45] $1^{121}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 760$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 1 & 3 \\ 3 & 4 & 28 & 4 & 0 & 4 \\ 16 & 28 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 22 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$
- [46] $1^{141}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 768$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 1 & 3 \\ 3 & 4 & 21 & 9 & 0 & 4 \\ 11 & 21 & 76 & 33 & 0 & 16 \\ 5 & 9 & 33 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 16 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$
- [47] $1^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 778$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.66 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 1 & 15 & 0 & 3 \\ 12 & 22 & 15 & 84 & 0 & 16 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 16 & -4 & 0 \end{pmatrix}_{b_0}$$
- [48] $1^{121}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 814$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 1 & 3 \\ 3 & 4 & 23 & 4 & 0 & 4 \\ 13 & 23 & 100 & 15 & 0 & 18 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 18 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

- [49] $1^{114}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 823$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.02 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 14 & 2 & 1 & 3 \\ 4 & 9 & 36 & 4 & 0 & 5 \\ 14 & 36 & 121 & 15 & 0 & 19 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 19 & 2 & -4 & -1 \end{pmatrix}_{b_0}$$
- [50] $1^{122}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 841$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 1 & 3 \\ 3 & 4 & 13 & 13 & 0 & 4 \\ 7 & 13 & 28 & 30 & 0 & 10 \\ 7 & 13 & 30 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 10 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$
- [51] $1^{114}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 991$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 1 & 3 \\ 4 & 9 & 44 & 4 & 0 & 5 \\ 18 & 44 & 193 & 21 & 0 & 25 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 25 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$
- [52] $1^{141}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1006$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.84 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 21 & 4 & 0 & 4 \\ 11 & 21 & 76 & 15 & 0 & 16 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 16 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$
- [53] $1^{132}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1030$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 1 & 3 \\ 3 & 4 & 17 & 13 & 0 & 4 \\ 9 & 17 & 49 & 39 & 0 & 13 \\ 7 & 13 & 39 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 13 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$
- [54] $1^{114}31^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1036$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 1 & 3 \\ 4 & 9 & 40 & 4 & 0 & 5 \\ 16 & 40 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 22 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$
- [55] $1^{131}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1174$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 22 & 4 & 0 & 4 \\ 12 & 22 & 85 & 15 & 0 & 17 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 17 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$
- [56] $1^{131}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1456$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 1 & 3 \\ 3 & 4 & 27 & 4 & 0 & 4 \\ 15 & 27 & 133 & 18 & 0 & 21 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 21 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$
- [57] $1^{131}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1606$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 1 & 3 \\ 3 & 4 & 32 & 4 & 0 & 4 \\ 18 & 32 & 193 & 21 & 0 & 25 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 25 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$
- [58] $1^{141}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 1624$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 1 & 3 \\ 3 & 4 & 26 & 4 & 0 & 4 \\ 14 & 26 & 118 & 18 & 0 & 20 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 20 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{141} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 2110$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 1 & 3 \\ 3 & 4 & 31 & 4 & 0 & 4 \\ 17 & 31 & 172 & 21 & 0 & 24 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 24 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{141} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 2464$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 20 & 2 & 1 & 3 \\ 3 & 4 & 36 & 4 & 0 & 4 \\ 20 & 36 & 238 & 24 & 0 & 28 \\ 2 & 4 & 24 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 28 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

4.953 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \text{NHC}(-2)(-3)$

$T = 10$ $[0] 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2]$ $T = 10$, $T_{\max} = 14$, $\Delta = -92$, $\det = 470$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.17 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 4 & 3 \\ 3 & 5 & 3 & 3 & 0 & 4 & 3 & 3 \\ 3 & 4 & 3 & 2 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

4.954 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{so}_8$

$T = 9$ $[0] 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.23 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 9 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 9 & 7 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 7 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -334$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 12 & 0 & 6 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{so}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -931$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 17 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 17 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 13 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

4.955 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$

$T = 9$ $[0] 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 16.86 \rightarrow 17$, $T_{\max} = 23$, $T \in [17, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 4 & 5 & 0 & 5 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & 3 & 3 & -4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -56, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.71 \rightarrow 17, \ T_{\max} = 23, \ T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 2 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 & 3 \\ 8 & 14 & 9 & 36 & 0 & 9 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 9 & 0 & 1 & 3 \\ 3 & 4 & 3 & 10 & -4 & 3 & -4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -56, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.71 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 6 & 3 & 0 & 4 & 4 \\ 3 & 4 & 4 & 2 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -65, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 2 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1211} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -90, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.07 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 5 & 3 \\ 3 & 4 & 8 & 9 & 0 & 9 & 4 \\ 4 & 8 & 10 & 12 & 0 & 12 & 6 \\ 5 & 9 & 12 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 12 & 15 & 0 & 13 & 7 \\ 3 & 4 & 6 & 6 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -92, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 9 & 0 & 37 & 11 \\ 3 & 4 & 3 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -101, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 5 & 4 & 0 & 19 & 4 \\ 3 & 5 & 4 & 3 & 0 & 21 & 4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 19 & 21 & 12 & 0 & 73 & 15 \\ 3 & 4 & 4 & 2 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{131} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -107, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.32 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 & 4 \\ 3 & 5 & 4 & 15 & 0 & 3 & 4 \\ 9 & 17 & 15 & 48 & 0 & 12 & 12 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 12 & 0 & 1 & 3 \\ 3 & 4 & 4 & 12 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1211} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{violet}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.34 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 5 & 3 \\ 3 & 4 & 8 & 4 & 0 & 9 & 4 \\ 4 & 8 & 10 & 6 & 0 & 12 & 6 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 12 & 6 & 0 & 13 & 7 \\ 3 & 4 & 6 & 2 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{\overline{1111}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -116, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 5 & 3 \\ 3 & 4 & 19 & 4 & 0 & 9 & 4 \\ 11 & 19 & 73 & 12 & 0 & 33 & 15 \\ 2 & 4 & 12 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 33 & 6 & 0 & 13 & 7 \\ 3 & 4 & 15 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{\overline{1111}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -128, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & 3 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{\overline{1112}} 3^{\overline{121}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -137, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 7 & 3 \\ 3 & 4 & 14 & 4 & 0 & 13 & 4 \\ 8 & 14 & 37 & 9 & 0 & 33 & 11 \\ 2 & 4 & 9 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 33 & 9 & 0 & 28 & 10 \\ 3 & 4 & 11 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{\overline{113}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -173, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.39 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 5 & 4 & 0 & 17 & 4 \\ 3 & 5 & 4 & 3 & 0 & 15 & 4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 12 & 0 & 49 & 13 \\ 3 & 4 & 4 & 2 & -4 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{\overline{1211}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -174, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.85 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 2 & 3 \\ 3 & 4 & 8 & 9 & 0 & 4 & 4 \\ 4 & 8 & 10 & 12 & 0 & 6 & 6 \\ 5 & 9 & 12 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 6 & 0 & 1 & 3 \\ 3 & 4 & 6 & 6 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{\overline{1111}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -191, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 5 & 3 \\ 4 & 9 & 8 & 4 & 0 & 12 & 5 \\ 3 & 8 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 9 & 6 & 0 & 13 & 7 \\ 3 & 5 & 4 & 2 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{\overline{1212}} 3^{\overline{113}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 13 & 3 \\ 3 & 4 & 13 & 4 & 0 & 23 & 4 \\ 7 & 13 & 28 & 9 & 0 & 54 & 10 \\ 2 & 4 & 9 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 54 & 15 & 0 & 100 & 18 \\ 3 & 4 & 10 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{\overline{111}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -219, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 4 \\ 3 & 5 & 4 & 9 & 0 & 3 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 1 & 3 \\ 3 & 4 & 4 & 6 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{121} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -245, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 & 4 \\ 3 & 5 & 4 & 12 & 0 & 3 & 4 \\ 7 & 13 & 12 & 27 & 0 & 9 & 9 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 9 & 0 & 1 & 3 \\ 3 & 4 & 4 & 9 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{111} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -284, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 & 3 \\ 4 & 9 & 8 & 4 & 0 & 8 & 5 \\ 3 & 8 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 6 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & 2 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{111} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -341, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{111} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -344, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.81 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111} 31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -353, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 5 & 4 & 0 & 14 & 4 \\ 3 & 5 & 4 & 3 & 0 & 15 & 4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 15 & 9 & 0 & 37 & 11 \\ 3 & 4 & 4 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{111} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -359, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 \\ 3 & 5 & 3 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 9 & 0 & 13 & 7 \\ 3 & 4 & 3 & 3 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -368, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 1 & 5 & 3 \\ 4 & 9 & 12 & 4 & 0 & 12 & 5 \\ 5 & 12 & 13 & 6 & 0 & 15 & 7 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 15 & 6 & 0 & 13 & 7 \\ 3 & 5 & 7 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{112} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -380, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 13 & 4 \\ 3 & 5 & 4 & 3 & 0 & 12 & 4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 12 & 9 & 0 & 28 & 10 \\ 3 & 4 & 4 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{1112}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 6 & 6 & 0 & 10 & 6 \\ 3 & 4 & 3 & 2 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{1111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -408, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 3 \\ 5 & 9 & 6 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 1 & 3 \\ 3 & 4 & 3 & 6 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{1114}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -413, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 11 & 3 \\ 3 & 4 & 4 & 5 & 0 & 21 & 4 \\ 2 & 4 & 1 & 3 & 0 & 15 & 3 \\ 3 & 5 & 3 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 15 & 18 & 0 & 76 & 16 \\ 3 & 4 & 3 & 3 & -4 & 16 & -1 \end{pmatrix}_{b_0}$$

[28] $1^{1212}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 4 & 3 \\ 3 & 4 & 8 & 4 & 0 & 8 & 4 \\ 4 & 8 & 10 & 6 & 0 & 12 & 6 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 12 & 6 & 0 & 10 & 6 \\ 3 & 4 & 6 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

[29] $1^{1112}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -425, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 1 & 3 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 15 & 15 & 0 & 100 & 18 \\ 3 & 4 & 3 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{1113}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 9 & 9 & 0 & 25 & 9 \\ 3 & 4 & 3 & 2 & -4 & 9 & -1 \end{pmatrix}_{b_0}$$

[31] $1^{1212}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 7 & 3 \\ 3 & 4 & 8 & 9 & 0 & 13 & 4 \\ 4 & 8 & 10 & 12 & 0 & 18 & 6 \\ 5 & 9 & 12 & 12 & 0 & 21 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 18 & 21 & 0 & 28 & 10 \\ 3 & 4 & 6 & 6 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{1111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -485, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{1113}31^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -488, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 12 & 3 \\ 3 & 4 & 14 & 4 & 0 & 22 & 4 \\ 8 & 14 & 37 & 9 & 0 & 57 & 11 \\ 2 & 4 & 9 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 57 & 15 & 0 & 85 & 17 \\ 3 & 4 & 11 & 2 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{1114}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 12 & 0 & 46 & 12 \\ 3 & 4 & 3 & 2 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$

[35] $1^{112}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -533, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 13.03 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 7 & 3 \\ 3 & 4 & 4 & 5 & 0 & 13 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 \\ 3 & 5 & 3 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 12 & 0 & 28 & 10 \\ 3 & 4 & 3 & 3 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

[36] $1^{113}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -551, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 9 & 3 \\ 3 & 4 & 4 & 5 & 0 & 17 & 4 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 \\ 3 & 5 & 3 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 15 & 0 & 49 & 13 \\ 3 & 4 & 3 & 3 & -4 & 13 & -1 \end{pmatrix}_{b_0}$$

[37] $1^{1212}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 7 & 3 \\ 3 & 4 & 8 & 4 & 0 & 13 & 4 \\ 4 & 8 & 10 & 6 & 0 & 18 & 6 \\ 2 & 4 & 6 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 18 & 9 & 0 & 28 & 10 \\ 3 & 4 & 6 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{1111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -729, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 1 & 6 & 0 & 6 & 3 \\ 5 & 9 & 6 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 15 & 0 & 13 & 7 \\ 3 & 4 & 3 & 6 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{1111}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -773, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 8 & 3 \\ 3 & 4 & 9 & 4 & 0 & 14 & 4 \\ 5 & 9 & 13 & 6 & 0 & 24 & 7 \\ 2 & 4 & 6 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 24 & 9 & 0 & 37 & 11 \\ 3 & 4 & 7 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{112}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -800, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 5 & 4 & 0 & 18 & 4 \\ 3 & 5 & 4 & 3 & 0 & 18 & 4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 4 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{112} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -848, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 5 & 4 & 0 & 28 & 4 \\ 3 & 5 & 4 & 3 & 0 & 30 & 4 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 16 & 28 & 30 & 18 & 0 & 154 & 22 \\ 3 & 4 & 4 & 2 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 5 & 3 \\ 3 & 4 & 9 & 4 & 0 & 9 & 4 \\ 5 & 9 & 13 & 6 & 0 & 15 & 7 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 15 & 6 & 0 & 13 & 7 \\ 3 & 4 & 7 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -920, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 8 & 3 \\ 3 & 4 & 14 & 4 & 0 & 14 & 4 \\ 8 & 14 & 37 & 9 & 0 & 39 & 11 \\ 2 & 4 & 9 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 39 & 9 & 0 & 37 & 11 \\ 3 & 4 & 11 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -929, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 12 & 12 & 0 & 58 & 14 \\ 3 & 4 & 3 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.91 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 5 & 4 & 0 & 26 & 4 \\ 3 & 5 & 4 & 3 & 0 & 24 & 4 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 24 & 18 & 0 & 118 & 20 \\ 3 & 4 & 4 & 2 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{111} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 5 & 4 & 0 & 23 & 4 \\ 3 & 5 & 4 & 3 & 0 & 24 & 4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 24 & 15 & 0 & 100 & 18 \\ 3 & 4 & 4 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1001, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 5 & 4 & 0 & 22 & 4 \\ 3 & 5 & 4 & 3 & 0 & 21 & 4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 21 & 15 & 0 & 85 & 17 \\ 3 & 4 & 4 & 2 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1113} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 1 & 3 & 0 & 21 & 3 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 21 & 21 & 0 & 193 & 25 \\ 3 & 4 & 3 & 2 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1112} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1083, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 7 & 3 \\ 3 & 4 & 4 & 9 & 0 & 13 & 4 \\ 2 & 4 & 1 & 6 & 0 & 9 & 3 \\ 5 & 9 & 6 & 12 & 0 & 21 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 21 & 0 & 28 & 10 \\ 3 & 4 & 3 & 6 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1114} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1085, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 11 & 3 \\ 3 & 4 & 9 & 4 & 0 & 21 & 4 \\ 5 & 9 & 13 & 6 & 0 & 33 & 7 \\ 2 & 4 & 6 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 33 & 15 & 0 & 76 & 16 \\ 3 & 4 & 7 & 2 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1112} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1205, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 7 & 3 \\ 3 & 4 & 9 & 4 & 0 & 13 & 4 \\ 5 & 9 & 13 & 6 & 0 & 21 & 7 \\ 2 & 4 & 6 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 21 & 9 & 0 & 28 & 10 \\ 3 & 4 & 7 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1112} 31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 10 & 3 \\ 3 & 4 & 14 & 4 & 0 & 18 & 4 \\ 8 & 14 & 37 & 9 & 0 & 48 & 11 \\ 2 & 4 & 9 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 48 & 12 & 0 & 58 & 14 \\ 3 & 4 & 11 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1113} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 9 & 3 \\ 3 & 4 & 9 & 4 & 0 & 17 & 4 \\ 5 & 9 & 13 & 6 & 0 & 27 & 7 \\ 2 & 4 & 6 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 27 & 12 & 0 & 49 & 13 \\ 3 & 4 & 7 & 2 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1112} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 16 & 3 \\ 3 & 4 & 9 & 4 & 0 & 28 & 4 \\ 5 & 9 & 13 & 6 & 0 & 48 & 7 \\ 2 & 4 & 6 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 16 & 28 & 48 & 18 & 0 & 154 & 22 \\ 3 & 4 & 7 & 2 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1113} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1281, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 9 & 3 \\ 3 & 4 & 4 & 9 & 0 & 17 & 4 \\ 2 & 4 & 1 & 6 & 0 & 12 & 3 \\ 5 & 9 & 6 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 27 & 0 & 49 & 13 \\ 3 & 4 & 3 & 6 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{111\bar{4}} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1323, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 11 & 3 \\ 3 & 4 & 4 & 9 & 0 & 21 & 4 \\ 2 & 4 & 1 & 6 & 0 & 15 & 3 \\ 5 & 9 & 6 & 12 & 0 & 33 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 15 & 33 & 0 & 76 & 16 \\ 3 & 4 & 3 & 6 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{111\bar{3}} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1397, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & 2 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{111\bar{3}} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1565, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 5 & 4 & 0 & 27 & 4 \\ 3 & 5 & 4 & 3 & 0 & 27 & 4 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 4 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{121\bar{2}} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 10 & 3 \\ 3 & 4 & 13 & 4 & 0 & 18 & 4 \\ 7 & 13 & 28 & 9 & 0 & 42 & 10 \\ 2 & 4 & 9 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 42 & 12 & 0 & 58 & 14 \\ 3 & 4 & 10 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{111\bar{3}} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 3 & 0 & 18 & 3 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 18 & 18 & 0 & 133 & 21 \\ 3 & 4 & 3 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{111\bar{2}} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1760, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 10 & 3 \\ 3 & 4 & 9 & 4 & 0 & 18 & 4 \\ 5 & 9 & 13 & 6 & 0 & 30 & 7 \\ 2 & 4 & 6 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 30 & 12 & 0 & 58 & 14 \\ 3 & 4 & 7 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{111\bar{2}} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1781, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 13 & 3 \\ 3 & 4 & 9 & 4 & 0 & 23 & 4 \\ 5 & 9 & 13 & 6 & 0 & 39 & 7 \\ 2 & 4 & 6 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 39 & 15 & 0 & 100 & 18 \\ 3 & 4 & 7 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{121\bar{2}} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1784, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 7 & 3 \\ 3 & 4 & 13 & 4 & 0 & 13 & 4 \\ 7 & 13 & 28 & 9 & 0 & 30 & 10 \\ 2 & 4 & 9 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 30 & 9 & 0 & 28 & 10 \\ 3 & 4 & 10 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{1113}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{c}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1820, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 3 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 15 & 15 & 0 & 85 & 17 \\ 3 & 4 & 3 & 2 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[66] $1^{113}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{c}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1865, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 5 & 4 & 0 & 32 & 4 \\ 3 & 5 & 4 & 3 & 0 & 33 & 4 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 33 & 21 & 0 & 193 & 25 \\ 3 & 4 & 4 & 2 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{1114}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{c}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1889, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 3 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 15 & 15 & 0 & 76 & 16 \\ 3 & 4 & 3 & 2 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{114}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{c}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1928, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 5 & 4 & 0 & 31 & 4 \\ 3 & 5 & 4 & 3 & 0 & 30 & 4 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 30 & 21 & 0 & 172 & 24 \\ 3 & 4 & 4 & 2 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{1113}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{c}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2501, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 12 & 3 \\ 3 & 4 & 9 & 4 & 0 & 22 & 4 \\ 5 & 9 & 13 & 6 & 0 & 36 & 7 \\ 2 & 4 & 6 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 36 & 15 & 0 & 85 & 17 \\ 3 & 4 & 7 & 2 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{1212}31^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{c}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 10 & 3 \\ 3 & 4 & 18 & 4 & 0 & 18 & 4 \\ 10 & 18 & 58 & 12 & 0 & 60 & 14 \\ 2 & 4 & 12 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 60 & 12 & 0 & 58 & 14 \\ 3 & 4 & 14 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{114}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{c}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2636, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 5 & 4 & 0 & 36 & 4 \\ 3 & 5 & 4 & 3 & 0 & 36 & 4 \\ 2 & 4 & 3 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 4 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{1114}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{c}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -2720, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 1 & 3 & 0 & 18 & 3 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 18 & 18 & 0 & 118 & 20 \\ 3 & 4 & 3 & 2 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{1114}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -2780$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 1 & 3 & 0 & 24 & 3 \\ 2 & 4 & 3 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 24 & 24 & 0 & 238 & 28 \\ 3 & 4 & 3 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{1114}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -2996$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 14 & 3 \\ 3 & 4 & 9 & 4 & 0 & 26 & 4 \\ 5 & 9 & 13 & 6 & 0 & 42 & 7 \\ 2 & 4 & 6 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 42 & 18 & 0 & 118 & 20 \\ 3 & 4 & 7 & 2 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{1114}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -3017$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 1 & 3 & 0 & 21 & 3 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 21 & 21 & 0 & 172 & 24 \\ 3 & 4 & 3 & 2 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{1114}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -3200$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 15 & 3 \\ 3 & 4 & 9 & 4 & 0 & 27 & 4 \\ 5 & 9 & 13 & 6 & 0 & 45 & 7 \\ 2 & 4 & 6 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 45 & 18 & 0 & 133 & 21 \\ 3 & 4 & 7 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{1114}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -3365$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 18 & 3 \\ 3 & 4 & 9 & 4 & 0 & 32 & 4 \\ 5 & 9 & 13 & 6 & 0 & 54 & 7 \\ 2 & 4 & 6 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 54 & 21 & 0 & 193 & 25 \\ 3 & 4 & 7 & 2 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[78] $1^{1114}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -4373$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 17 & 3 \\ 3 & 4 & 9 & 4 & 0 & 31 & 4 \\ 5 & 9 & 13 & 6 & 0 & 51 & 7 \\ 2 & 4 & 6 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 51 & 21 & 0 & 172 & 24 \\ 3 & 4 & 7 & 2 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{1114}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -5216$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 20 & 3 \\ 3 & 4 & 9 & 4 & 0 & 36 & 4 \\ 5 & 9 & 13 & 6 & 0 & 60 & 7 \\ 2 & 4 & 6 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 60 & 24 & 0 & 238 & 28 \\ 3 & 4 & 7 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

4.956 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$

$T = 10$ [0] $1^{1114}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 148$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 6 & 3 & 0 & 4 & 9 & 4 \\ 5 & 9 & 9 & 6 & 0 & 9 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1111} 31^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 9 & 0 & 37 & 24 & 11 \\ 5 & 9 & 6 & 6 & 0 & 24 & 13 & 7 \\ 3 & 4 & 3 & 2 & -4 & 11 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1112} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 274, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 18 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 18 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 12 & 3 \\ 10 & 18 & 18 & 12 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & 2 & -4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1113} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 454, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 15 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 27 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 27 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 18 & 3 \\ 15 & 27 & 27 & 18 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & 2 & -4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 660, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 3 & 3 \\ 5 & 9 & 6 & 12 & 0 & 6 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 6 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & 6 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1113} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 22 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 21 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 15 & 3 \\ 12 & 22 & 21 & 15 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & 2 & -4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1114} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 706, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 20 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 36 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 36 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 24 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 24 & 3 \\ 20 & 36 & 36 & 24 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & 2 & -4 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

- [8] $1^{\overline{11111}}31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 712, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & 2 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{\overline{11111}}31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 724, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & 2 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{\overline{1112}}31^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 796, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 13 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 12 & 4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 21 & 7 \\ 7 & 13 & 12 & 9 & 0 & 21 & 28 & 10 \\ 3 & 4 & 4 & 2 & -4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{\overline{11111}}31^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{\overline{11111}}31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1032, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 6 & 3 \\ 5 & 9 & 6 & 12 & 0 & 6 & 15 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 15 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & 6 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [13] $1^{\overline{1112}}31^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 13 & 13 & 4 \\ 3 & 5 & 4 & 3 & 0 & 12 & 12 & 4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 12 & 9 & 0 & 28 & 30 & 10 \\ 7 & 13 & 12 & 9 & 0 & 30 & 28 & 10 \\ 3 & 4 & 4 & 2 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$
- [14] $1^{\overline{1114}}31^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1180, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 17 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 31 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 30 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 21 & 3 \\ 17 & 31 & 30 & 21 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & 2 & -4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$
- [15] $1^{\overline{11111}}31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 3 & 2 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{11111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 2368, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 9 & 4 & 0 & 9 & 9 & 4 \\ 5 & 9 & 13 & 6 & 0 & 15 & 15 & 7 \\ 2 & 4 & 6 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 15 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 15 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 7 & 2 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

4.957 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 10$ [0] $1^{11111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 1 & 5 & 2 & 3 \\ 4 & 9 & 12 & 4 & 0 & 12 & 4 & 5 \\ 5 & 12 & 13 & 6 & 0 & 15 & 6 & 7 \\ 2 & 4 & 6 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 12 & 15 & 6 & 0 & 13 & 6 & 7 \\ 2 & 4 & 6 & 3 & 0 & 6 & 0 & 2 \\ 3 & 5 & 7 & 2 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{11111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 231, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 6 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & 2 & -4 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{11111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 507, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 3 & 4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & 2 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{11111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 651, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 9 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{11411}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 813, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 21 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 15 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 18 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 15 & 18 & 0 & 76 & 15 & 16 \\ 2 & 4 & 3 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 16 & 2 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{112}1}31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 1173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$

$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 13 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 12 & 0 & 28 & 9 & 10 \\ 2 & 4 & 3 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{\overline{113}1}31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 1227, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.8$

$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 17 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 12 & 15 & 0 & 49 & 12 & 13 \\ 2 & 4 & 3 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

4.958 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3$

$T = 9$ [0] $1^{\overline{112}1}31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 & 3 \\ 3 & 4 & 9 & 8 & 0 & 5 & 4 \\ 5 & 9 & 13 & 12 & 0 & 9 & 7 \\ 4 & 8 & 12 & 9 & 0 & 6 & 5 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 9 & 6 & 0 & 3 & 3 \\ 3 & 4 & 7 & 5 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{111}1}31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 1 & 2 & 3 \\ 4 & 9 & 12 & 4 & 0 & 4 & 5 \\ 5 & 12 & 13 & 6 & 0 & 6 & 7 \\ 2 & 4 & 6 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 3 & 0 & 0 & 2 \\ 3 & 5 & 7 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{114}1}31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 1 & 4 & 14 & 2 & 1 & 2 & 3 \\ 4 & 9 & 36 & 4 & 0 & 4 & 5 \\ 14 & 36 & 121 & 15 & 0 & 15 & 19 \\ 2 & 4 & 15 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 15 & 3 & 0 & 0 & 2 \\ 3 & 5 & 19 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{11}1}31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 6 & 3 & 0 & 3 & 3 \\ 3 & 4 & 4 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{111}1}31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{\overline{112}1}31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

14.5

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 8 & 2 & 1 & 2 & 3 \\ 4 & 9 & 20 & 4 & 0 & 4 & 5 \\ 8 & 20 & 37 & 9 & 0 & 9 & 11 \\ 2 & 4 & 9 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 3 & 0 & 0 & 2 \\ 3 & 5 & 11 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{113} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 11 & 2 & 1 & 2 & 3 \\ 4 & 9 & 28 & 4 & 0 & 4 & 5 \\ 11 & 28 & 73 & 12 & 0 & 12 & 15 \\ 2 & 4 & 12 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 12 & 3 & 0 & 0 & 2 \\ 3 & 5 & 15 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 9 & 0 & 36 & 10 \\ 3 & 4 & 3 & 2 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{122} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -267, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.7$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 4 & 3 \\ 3 & 4 & 13 & 5 & 0 & 8 & 4 \\ 7 & 13 & 28 & 12 & 0 & 18 & 10 \\ 3 & 5 & 12 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 18 & 6 & 0 & 9 & 5 \\ 3 & 4 & 10 & 3 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{132} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 4 & 3 \\ 3 & 4 & 17 & 5 & 0 & 8 & 4 \\ 9 & 17 & 49 & 15 & 0 & 24 & 13 \\ 3 & 5 & 15 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 24 & 6 & 0 & 9 & 5 \\ 3 & 4 & 13 & 3 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -402, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 2 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 & 4 \\ 8 & 14 & 37 & 9 & 0 & 9 & 11 \\ 2 & 4 & 9 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 3 & 0 & 0 & 2 \\ 3 & 4 & 11 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{113} 3^1 1^{211} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -420, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 13 & 2 & 1 & 2 & 3 \\ 4 & 9 & 32 & 4 & 0 & 4 & 5 \\ 13 & 32 & 100 & 15 & 0 & 15 & 18 \\ 2 & 4 & 15 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 15 & 3 & 0 & 0 & 2 \\ 3 & 5 & 18 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 4 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{114} 31^{311} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -555, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 1 & 2 & 3 \\ 4 & 9 & 44 & 4 & 0 & 4 & 5 \\ 18 & 44 & 193 & 21 & 0 & 21 & 25 \\ 2 & 4 & 21 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 21 & 3 & 0 & 0 & 2 \\ 3 & 5 & 25 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -645, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 2 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 & 4 \\ 5 & 9 & 13 & 9 & 0 & 6 & 7 \\ 3 & 5 & 9 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 3 & 0 & 0 & 2 \\ 3 & 4 & 7 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{1111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -651, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 12 & 6 \\ 3 & 4 & 3 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{114} 31^{211} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -690, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 1 & 2 & 3 \\ 4 & 9 & 40 & 4 & 0 & 4 & 5 \\ 16 & 40 & 154 & 18 & 0 & 18 & 22 \\ 2 & 4 & 18 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 18 & 3 & 0 & 0 & 2 \\ 3 & 5 & 22 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{141} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -726, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 1 & 2 & 3 \\ 3 & 4 & 21 & 5 & 0 & 4 & 4 \\ 11 & 21 & 76 & 18 & 0 & 15 & 16 \\ 3 & 5 & 18 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 15 & 3 & 0 & 0 & 2 \\ 3 & 4 & 16 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -789, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 & 4 \\ 5 & 9 & 13 & 6 & 0 & 6 & 7 \\ 2 & 4 & 6 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 3 & 0 & 0 & 2 \\ 3 & 4 & 7 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{121} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -906, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 2 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 & 4 \\ 7 & 13 & 28 & 12 & 0 & 9 & 10 \\ 3 & 5 & 12 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 3 & 0 & 0 & 2 \\ 3 & 4 & 10 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{131} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -933, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 2 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 & 4 \\ 9 & 17 & 49 & 15 & 0 & 12 & 13 \\ 3 & 5 & 15 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 12 & 3 & 0 & 0 & 2 \\ 3 & 4 & 13 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -1200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 5 & 3 \\ 3 & 4 & 9 & 4 & 0 & 9 & 4 \\ 5 & 9 & 13 & 6 & 0 & 15 & 7 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 15 & 6 & 0 & 12 & 6 \\ 3 & 4 & 7 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$4.959 \quad \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$$

$$T = 10 \quad [0] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 48, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 1 & 2 & 5 & 3 \\ 4 & 9 & 12 & 4 & 0 & 4 & 12 & 5 \\ 5 & 12 & 13 & 6 & 0 & 6 & 15 & 7 \\ 2 & 4 & 6 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 3 & 0 & 0 & 6 & 2 \\ 5 & 12 & 15 & 6 & 0 & 6 & 13 & 7 \\ 3 & 5 & 7 & 2 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 228, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 231, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 3 & 4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 12 & 6 & 6 \\ 2 & 4 & 3 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 4 & 2 & -4 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 507, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 651, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 6 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & 3 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1114}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 813, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 21 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 15 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 15 & 2 \\ 11 & 21 & 15 & 18 & 0 & 15 & 76 & 16 \\ 3 & 4 & 3 & 3 & -4 & 2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1112}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1173, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 13 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 9 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 9 & 2 \\ 7 & 13 & 9 & 12 & 0 & 9 & 28 & 10 \\ 3 & 4 & 3 & 3 & -4 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1113}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1227, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 17 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 12 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 12 & 2 \\ 9 & 17 & 12 & 15 & 0 & 12 & 49 & 13 \\ 3 & 4 & 3 & 3 & -4 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

4.960 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 459, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 1296, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 & 4 & 4 \\ 5 & 9 & 13 & 9 & 0 & 6 & 6 & 7 \\ 3 & 5 & 9 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 6 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 7 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.961 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4$

$$T = 8 \quad [0] \ 1^{\overline{132}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 50, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 28, \ T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 1 & 3 \\ 3 & 4 & 12 & 13 & 0 & 4 \\ 6 & 12 & 25 & 27 & 0 & 9 \\ 7 & 13 & 27 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.81 \rightarrow 12, \ T_{\max} = 28, \ T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 82, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 28, \ T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{121}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 8 & 9 & 0 & 4 \\ 4 & 8 & 10 & 12 & 0 & 6 \\ 5 & 9 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{11}3^11^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 \\ 3 & 5 & 4 & 6 & 0 & 4 \\ 3 & 5 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 4 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 1 & 3 \\ 4 & 9 & 12 & 4 & 0 & 5 \\ 5 & 12 & 13 & 6 & 0 & 7 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 7 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{13}3^11^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.15 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 4 & 15 & 0 & 4 \\ 9 & 17 & 15 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 12 & -3 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{112}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 137, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 4 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{111}3^11^13 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 19 & 4 & 0 & 4 \\ 11 & 19 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 15 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{121}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 4 & 8 & 10 & 6 & 0 & 6 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{112}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 9 & 8 & 0 & 4 \\ 5 & 9 & 13 & 12 & 0 & 7 \\ 4 & 8 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 7 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 159, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.34 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 8 & 14 & 9 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 10 & -3 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{122}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 1 & 3 \\ 3 & 4 & 18 & 8 & 0 & 4 \\ 10 & 18 & 58 & 24 & 0 & 14 \\ 4 & 8 & 24 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 14 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{12}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 4 & 12 & 0 & 4 \\ 7 & 13 & 12 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 5 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{112}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 10 & 2 & 1 & 3 \\ 4 & 9 & 24 & 4 & 0 & 5 \\ 10 & 24 & 58 & 12 & 0 & 14 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 14 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{122}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 1 & 3 \\ 3 & 4 & 8 & 13 & 0 & 4 \\ 4 & 8 & 10 & 18 & 0 & 6 \\ 7 & 13 & 18 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 5 & 9 & 13 & 9 & 0 & 7 \\ 3 & 5 & 9 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 7 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{14}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 1 & 3 \\ 3 & 4 & 21 & 5 & 0 & 4 \\ 11 & 21 & 76 & 18 & 0 & 16 \\ 3 & 5 & 18 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 16 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 \\ 5 & 9 & 13 & 6 & 0 & 7 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 7 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 \\ 8 & 14 & 37 & 9 & 0 & 11 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 11 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 8 & 2 & 1 & 3 \\ 4 & 9 & 20 & 4 & 0 & 5 \\ 8 & 20 & 37 & 9 & 0 & 11 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 11 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{123}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 \\ 7 & 13 & 28 & 12 & 0 & 10 \\ 3 & 5 & 12 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 10 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{13}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 294, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 \\ 9 & 17 & 49 & 15 & 0 & 13 \\ 3 & 5 & 15 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 13 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 306, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 7 & 13 & 9 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{122}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 1 & 3 \\ 3 & 4 & 13 & 8 & 0 & 4 \\ 7 & 13 & 28 & 18 & 0 & 10 \\ 4 & 8 & 18 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 10 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{113}31^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 1 & 3 \\ 4 & 9 & 36 & 4 & 0 & 5 \\ 15 & 36 & 133 & 18 & 0 & 21 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 21 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 371, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 5 & 9 & 13 & 15 & 0 & 7 \\ 5 & 9 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 7 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 2 & 4 & 1 & 12 & 0 & 3 \\ 10 & 18 & 12 & 57 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 411, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 7 & 13 & 28 & 9 & 0 & 10 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 10 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 2 & 4 & 1 & 12 & 0 & 3 \\ 9 & 17 & 12 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 11 & 2 & 1 & 3 \\ 4 & 9 & 28 & 4 & 0 & 5 \\ 11 & 28 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 15 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 5 & 9 & 13 & 21 & 0 & 7 \\ 7 & 13 & 21 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 7 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{113}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 463, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 9 & 17 & 0 & 4 \\ 5 & 9 & 13 & 27 & 0 & 7 \\ 9 & 17 & 27 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 7 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{132}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 489, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 1 & 3 \\ 3 & 4 & 17 & 8 & 0 & 4 \\ 9 & 17 & 49 & 24 & 0 & 13 \\ 4 & 8 & 24 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 13 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{113}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 490, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 13 & 2 & 1 & 3 \\ 4 & 9 & 32 & 4 & 0 & 5 \\ 13 & 32 & 100 & 15 & 0 & 18 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 18 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{132}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 501, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 1 & 3 \\ 3 & 4 & 22 & 8 & 0 & 4 \\ 12 & 22 & 85 & 30 & 0 & 17 \\ 4 & 8 & 30 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 17 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{121}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 507, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 1 & 3 \\ 3 & 4 & 13 & 9 & 0 & 4 \\ 7 & 13 & 28 & 21 & 0 & 10 \\ 5 & 9 & 21 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 10 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{114}31^{41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 20 & 2 & 1 & 3 \\ 4 & 9 & 48 & 4 & 0 & 5 \\ 20 & 48 & 238 & 24 & 0 & 28 \\ 2 & 4 & 24 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 28 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{121}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 576, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 10 & 18 & 58 & 12 & 0 & 14 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 14 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{131} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 579, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 1 & 3 \\ 3 & 4 & 17 & 9 & 0 & 4 \\ 9 & 17 & 49 & 27 & 0 & 13 \\ 5 & 9 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 13 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{141} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 587, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 1 & 3 \\ 3 & 4 & 21 & 9 & 0 & 4 \\ 11 & 21 & 76 & 33 & 0 & 16 \\ 5 & 9 & 33 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 16 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{131} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 9 & 17 & 49 & 12 & 0 & 13 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 13 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{121} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 1 & 3 \\ 3 & 4 & 28 & 4 & 0 & 4 \\ 16 & 28 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 22 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 607, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 1 & 15 & 0 & 3 \\ 12 & 22 & 15 & 84 & 0 & 16 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 16 & -3 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 626, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 14 & 2 & 1 & 3 \\ 4 & 9 & 36 & 4 & 0 & 5 \\ 14 & 36 & 121 & 15 & 0 & 19 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 19 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{121} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 635, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 1 & 3 \\ 3 & 4 & 23 & 4 & 0 & 4 \\ 13 & 23 & 100 & 15 & 0 & 18 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 18 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{122} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 1 & 3 \\ 3 & 4 & 13 & 13 & 0 & 4 \\ 7 & 13 & 28 & 30 & 0 & 10 \\ 7 & 13 & 30 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 10 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{114} 3^1 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 770, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 1 & 3 \\ 4 & 9 & 44 & 4 & 0 & 5 \\ 18 & 44 & 193 & 21 & 0 & 25 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 25 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{141} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 771, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 21 & 4 & 0 & 4 \\ 11 & 21 & 76 & 15 & 0 & 16 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 16 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{132}31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 798, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 1 & 3 \\ 3 & 4 & 17 & 13 & 0 & 4 \\ 9 & 17 & 49 & 39 & 0 & 13 \\ 7 & 13 & 39 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 13 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[54] $1^{114}31^{21} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 1 & 3 \\ 4 & 9 & 40 & 4 & 0 & 5 \\ 16 & 40 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 22 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{131}31^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 915, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 22 & 4 & 0 & 4 \\ 12 & 22 & 85 & 15 & 0 & 17 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 17 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{131}31^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 1 & 3 \\ 3 & 4 & 27 & 4 & 0 & 4 \\ 15 & 27 & 133 & 18 & 0 & 21 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 21 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[57] $1^{131}31^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1251, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 1 & 3 \\ 3 & 4 & 32 & 4 & 0 & 4 \\ 18 & 32 & 193 & 21 & 0 & 25 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 25 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{141}31^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 1 & 3 \\ 3 & 4 & 26 & 4 & 0 & 4 \\ 14 & 26 & 118 & 18 & 0 & 20 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 20 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{141}31^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1643, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 1 & 3 \\ 3 & 4 & 31 & 4 & 0 & 4 \\ 17 & 31 & 172 & 21 & 0 & 24 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 24 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{141}31^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1920, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 20 & 2 & 1 & 3 \\ 3 & 4 & 36 & 4 & 0 & 4 \\ 20 & 36 & 238 & 24 & 0 & 28 \\ 2 & 4 & 24 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 28 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

4.962 $\text{NHC}(-2)(-3) + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{so}}_8$

$T = 9 \quad [0] \quad 1^{111}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 9 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 9 & 7 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[1] 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 7 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 12 & 0 & 6 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 17 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 17 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 13 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

4.963 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$

$$T = 9 \quad [0] 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 6 & 3 & 0 & 4 & 4 \\ 3 & 4 & 4 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] 1^{111} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 5 & 4 & 0 & 19 & 4 \\ 3 & 5 & 4 & 3 & 0 & 21 & 4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 19 & 21 & 12 & 0 & 73 & 15 \\ 3 & 4 & 4 & 2 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[2] 1^{111} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 9 & 0 & 37 & 11 \\ 3 & 4 & 3 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 3 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 5 & 3 \\ 3 & 4 & 19 & 4 & 0 & 9 & 4 \\ 11 & 19 & 73 & 12 & 0 & 33 & 15 \\ 2 & 4 & 12 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 33 & 6 & 0 & 13 & 7 \\ 3 & 4 & 15 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[4] 1^{121} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 2 & 3 \\ 3 & 4 & 8 & 9 & 0 & 4 & 4 \\ 4 & 8 & 10 & 12 & 0 & 6 & 6 \\ 5 & 9 & 12 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 6 & 0 & 1 & 3 \\ 3 & 4 & 6 & 6 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

- [5] $1^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 4 \\ 3 & 5 & 4 & 9 & 0 & 3 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 6 & 0 & 1 & 3 \\ 3 & 4 & 4 & 6 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$
- [6] $1^{121}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 & 4 \\ 3 & 5 & 4 & 12 & 0 & 3 & 4 \\ 7 & 13 & 12 & 27 & 0 & 9 & 9 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 9 & 0 & 1 & 3 \\ 3 & 4 & 4 & 9 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$
- [8] $1^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 4 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{111}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 5 & 4 & 0 & 14 & 4 \\ 3 & 5 & 4 & 3 & 0 & 15 & 4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 15 & 9 & 0 & 37 & 11 \\ 3 & 4 & 4 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{112}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 13 & 4 \\ 3 & 5 & 4 & 3 & 0 & 12 & 4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 12 & 9 & 0 & 28 & 10 \\ 3 & 4 & 4 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 3 \\ 5 & 9 & 6 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 6 & 0 & 1 & 3 \\ 3 & 4 & 3 & 6 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{112}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 1 & 3 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 15 & 15 & 0 & 100 & 18 \\ 3 & 4 & 3 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$
- [13] $1^{1212}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -338, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 7 & 3 \\ 3 & 4 & 8 & 9 & 0 & 13 & 4 \\ 4 & 8 & 10 & 12 & 0 & 18 & 6 \\ 5 & 9 & 12 & 12 & 0 & 21 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 18 & 21 & 0 & 28 & 10 \\ 3 & 4 & 6 & 6 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 1 & 6 & 0 & 6 & 3 \\ 5 & 9 & 6 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 15 & 0 & 13 & 7 \\ 3 & 4 & 3 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -605, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 8 & 3 \\ 3 & 4 & 9 & 4 & 0 & 14 & 4 \\ 5 & 9 & 13 & 6 & 0 & 24 & 7 \\ 2 & 4 & 6 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 24 & 9 & 0 & 37 & 11 \\ 3 & 4 & 7 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 5 & 4 & 0 & 18 & 4 \\ 3 & 5 & 4 & 3 & 0 & 18 & 4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 4 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -650, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 5 & 4 & 0 & 28 & 4 \\ 3 & 5 & 4 & 3 & 0 & 30 & 4 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 16 & 28 & 30 & 18 & 0 & 154 & 22 \\ 3 & 4 & 4 & 2 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 5 & 3 \\ 3 & 4 & 9 & 4 & 0 & 9 & 4 \\ 5 & 9 & 13 & 6 & 0 & 15 & 7 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 15 & 6 & 0 & 13 & 7 \\ 3 & 4 & 7 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -716, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 8 & 3 \\ 3 & 4 & 14 & 4 & 0 & 14 & 4 \\ 8 & 14 & 37 & 9 & 0 & 39 & 11 \\ 2 & 4 & 9 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 39 & 9 & 0 & 37 & 11 \\ 3 & 4 & 11 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -723, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{1112} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -738, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 12 & 0 & 58 & 14 \\ 3 & 4 & 3 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{112} 3^1 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 5 & 4 & 0 & 23 & 4 \\ 3 & 5 & 4 & 3 & 0 & 24 & 4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 24 & 15 & 0 & 100 & 18 \\ 3 & 4 & 4 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{113} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -779, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 5 & 4 & 0 & 22 & 4 \\ 3 & 5 & 4 & 3 & 0 & 21 & 4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 21 & 15 & 0 & 85 & 17 \\ 3 & 4 & 4 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{1113} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -828, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 1 & 3 & 0 & 21 & 3 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 21 & 21 & 0 & 193 & 25 \\ 3 & 4 & 3 & 2 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{1112} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -845, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 7 & 3 \\ 3 & 4 & 4 & 9 & 0 & 13 & 4 \\ 2 & 4 & 1 & 6 & 0 & 9 & 3 \\ 5 & 9 & 6 & 12 & 0 & 21 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 21 & 0 & 28 & 10 \\ 3 & 4 & 3 & 6 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{1112} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -941, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 7 & 3 \\ 3 & 4 & 9 & 4 & 0 & 13 & 4 \\ 5 & 9 & 13 & 6 & 0 & 21 & 7 \\ 2 & 4 & 6 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 21 & 9 & 0 & 28 & 10 \\ 3 & 4 & 7 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{1112} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -962, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 10 & 3 \\ 3 & 4 & 14 & 4 & 0 & 18 & 4 \\ 8 & 14 & 37 & 9 & 0 & 48 & 11 \\ 2 & 4 & 9 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 48 & 12 & 0 & 58 & 14 \\ 3 & 4 & 11 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{1113} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -980, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 9 & 3 \\ 3 & 4 & 9 & 4 & 0 & 17 & 4 \\ 5 & 9 & 13 & 6 & 0 & 27 & 7 \\ 2 & 4 & 6 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 27 & 12 & 0 & 49 & 13 \\ 3 & 4 & 7 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1112} 3^1 1^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -980, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 16 & 3 \\ 3 & 4 & 9 & 4 & 0 & 28 & 4 \\ 5 & 9 & 13 & 6 & 0 & 48 & 7 \\ 2 & 4 & 6 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 16 & 28 & 48 & 18 & 0 & 154 & 22 \\ 3 & 4 & 7 & 2 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1113} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -989, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 9 & 3 \\ 3 & 4 & 4 & 9 & 0 & 17 & 4 \\ 2 & 4 & 1 & 6 & 0 & 12 & 3 \\ 5 & 9 & 6 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 27 & 0 & 49 & 13 \\ 3 & 4 & 3 & 6 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1114} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1005, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 11 & 3 \\ 3 & 4 & 4 & 9 & 0 & 21 & 4 \\ 2 & 4 & 1 & 6 & 0 & 15 & 3 \\ 5 & 9 & 6 & 12 & 0 & 33 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 15 & 33 & 0 & 76 & 16 \\ 3 & 4 & 3 & 6 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1113} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1077, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{113} 3^1 1^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1221, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 5 & 4 & 0 & 27 & 4 \\ 3 & 5 & 4 & 3 & 0 & 27 & 4 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 4 & 2 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1212} 3^1 1^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1298, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 10 & 3 \\ 3 & 4 & 13 & 4 & 0 & 18 & 4 \\ 7 & 13 & 28 & 9 & 0 & 42 & 10 \\ 2 & 4 & 9 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 42 & 12 & 0 & 58 & 14 \\ 3 & 4 & 10 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1113} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1333, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 3 & 0 & 18 & 3 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 18 & 18 & 0 & 133 & 21 \\ 3 & 4 & 3 & 2 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1112} 3^1 1^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 10 & 3 \\ 3 & 4 & 9 & 4 & 0 & 18 & 4 \\ 5 & 9 & 13 & 6 & 0 & 30 & 7 \\ 2 & 4 & 6 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 30 & 12 & 0 & 58 & 14 \\ 3 & 4 & 7 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1212} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1388, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 7 & 3 \\ 3 & 4 & 13 & 4 & 0 & 13 & 4 \\ 7 & 13 & 28 & 9 & 0 & 30 & 10 \\ 2 & 4 & 9 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 30 & 9 & 0 & 28 & 10 \\ 3 & 4 & 10 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1112} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1389, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 13 & 3 \\ 3 & 4 & 9 & 4 & 0 & 23 & 4 \\ 5 & 9 & 13 & 6 & 0 & 39 & 7 \\ 2 & 4 & 6 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 39 & 15 & 0 & 100 & 18 \\ 3 & 4 & 7 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1113} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 1 & 3 & 0 & 15 & 2 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 15 & 15 & 0 & 85 & 17 \\ 3 & 4 & 3 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{113} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1451, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 5 & 4 & 0 & 32 & 4 \\ 3 & 5 & 4 & 3 & 0 & 33 & 4 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 33 & 21 & 0 & 193 & 25 \\ 3 & 4 & 4 & 2 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{114} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1500, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 5 & 4 & 0 & 31 & 4 \\ 3 & 5 & 4 & 3 & 0 & 30 & 4 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 30 & 21 & 0 & 172 & 24 \\ 3 & 4 & 4 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1113} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1949, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 12 & 3 \\ 3 & 4 & 9 & 4 & 0 & 22 & 4 \\ 5 & 9 & 13 & 6 & 0 & 36 & 7 \\ 2 & 4 & 6 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 36 & 15 & 0 & 85 & 17 \\ 3 & 4 & 7 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1212} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2048, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 10 & 3 \\ 3 & 4 & 18 & 4 & 0 & 18 & 4 \\ 10 & 18 & 58 & 12 & 0 & 60 & 14 \\ 2 & 4 & 12 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 60 & 12 & 0 & 58 & 14 \\ 3 & 4 & 14 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{114} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2054, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 20 & 3 \\ 3 & 4 & 9 & 4 & 0 & 36 & 4 \\ 5 & 9 & 13 & 6 & 0 & 60 & 7 \\ 2 & 4 & 6 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 60 & 24 & 0 & 238 & 28 \\ 3 & 4 & 7 & 2 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

4.964 NHC(-2)(-3) + su₂ + su₃ + f₄ + su₂ + su₂→(-3)

$$T = 10 \quad [0] \quad 1^{11}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 3 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 3 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 6 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 3 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 392, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 3 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 2 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 2 & 2 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 462, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 3 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 3 & 4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 6 & 0 & 13 & 5 & 7 \\ 2 & 3 & 3 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 512, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 3 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 & 2 & 3 \\ 5 & 9 & 6 & 12 & 0 & 6 & 5 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 6 & 0 & 1 & 2 & 3 \\ 2 & 3 & 2 & 5 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & 6 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1111}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 622, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 3 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 2 & 3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 6 & 0 & 13 & 5 & 7 \\ 2 & 3 & 2 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 3 & 2 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1111}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 1152, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 9 & 4 & 0 & 9 & 3 & 4 \\ 5 & 9 & 13 & 6 & 0 & 15 & 5 & 7 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 15 & 6 & 0 & 13 & 5 & 7 \\ 2 & 3 & 5 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 7 & 2 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.965 NHC(-2)(-3) + su₂ + su₃ + f₄ + su₂→(-3)

$$T = 9 \quad [0] \quad 1^{121}3^{1111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -96, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 2 & 3 \\ 3 & 4 & 8 & 9 & 0 & 3 & 4 \\ 4 & 8 & 10 & 12 & 0 & 4 & 6 \\ 5 & 9 & 12 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 5 & 0 & 0 & 2 \\ 3 & 4 & 6 & 6 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -124, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 2 & 3 \\ 4 & 9 & 8 & 4 & 0 & 4 & 5 \\ 3 & 8 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 3 & 5 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -136, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 2 & 1 & 2 & 3 \\ 3 & 4 & 19 & 4 & 0 & 3 & 4 \\ 11 & 19 & 73 & 12 & 0 & 11 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 11 & 2 & 0 & 0 & 2 \\ 3 & 4 & 15 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{121}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -136, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 2 & 3 \\ 3 & 4 & 8 & 4 & 0 & 3 & 4 \\ 4 & 8 & 10 & 6 & 0 & 4 & 6 \\ 2 & 4 & 6 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 2 & 0 & 0 & 2 \\ 3 & 4 & 6 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -146, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 3 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -156, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 3 & 4 \\ 2 & 4 & 1 & 3 & 0 & 2 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{112}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.31 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 2 & 3 \\ 3 & 4 & 4 & 8 & 0 & 3 & 4 \\ 2 & 4 & 1 & 6 & 0 & 2 & 3 \\ 4 & 8 & 6 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 4 & 0 & 0 & 2 \\ 3 & 4 & 3 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.13 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 1 & 2 & 3 \\ 4 & 9 & 12 & 4 & 0 & 4 & 5 \\ 5 & 12 & 13 & 6 & 0 & 5 & 7 \\ 2 & 4 & 6 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 2 & 0 & 0 & 2 \\ 3 & 5 & 7 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -206, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 1 & 2 & 3 \\ 3 & 4 & 28 & 4 & 0 & 3 & 4 \\ 16 & 28 & 154 & 18 & 0 & 16 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 16 & 2 & 0 & 0 & 2 \\ 3 & 4 & 22 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{141}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 2 & 3 \\ 3 & 4 & 21 & 4 & 0 & 3 & 4 \\ 11 & 21 & 76 & 15 & 0 & 11 & 16 \\ 2 & 4 & 15 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 11 & 2 & 0 & 0 & 2 \\ 3 & 4 & 16 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{121}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 2 & 3 \\ 3 & 4 & 18 & 4 & 0 & 3 & 4 \\ 10 & 18 & 58 & 12 & 0 & 10 & 14 \\ 2 & 4 & 12 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 10 & 2 & 0 & 0 & 2 \\ 3 & 4 & 14 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{121}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 1 & 2 & 3 \\ 3 & 4 & 23 & 4 & 0 & 3 & 4 \\ 13 & 23 & 100 & 15 & 0 & 13 & 18 \\ 2 & 4 & 15 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 13 & 2 & 0 & 0 & 2 \\ 3 & 4 & 18 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{131}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 2 & 3 \\ 3 & 4 & 22 & 4 & 0 & 3 & 4 \\ 12 & 22 & 85 & 15 & 0 & 12 & 17 \\ 2 & 4 & 15 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 12 & 2 & 0 & 0 & 2 \\ 3 & 4 & 17 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{131}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 1 & 2 & 3 \\ 3 & 4 & 27 & 4 & 0 & 3 & 4 \\ 15 & 27 & 133 & 18 & 0 & 15 & 21 \\ 2 & 4 & 18 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 15 & 2 & 0 & 0 & 2 \\ 3 & 4 & 21 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{141}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 1 & 2 & 3 \\ 3 & 4 & 26 & 4 & 0 & 3 & 4 \\ 14 & 26 & 118 & 18 & 0 & 14 & 20 \\ 2 & 4 & 18 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 14 & 2 & 0 & 0 & 2 \\ 3 & 4 & 20 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{131}3^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1826, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 1 & 2 & 3 \\ 3 & 4 & 32 & 4 & 0 & 3 & 4 \\ 18 & 32 & 193 & 21 & 0 & 18 & 25 \\ 2 & 4 & 21 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 18 & 2 & 0 & 0 & 2 \\ 3 & 4 & 25 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{141}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 1 & 2 & 3 \\ 3 & 4 & 31 & 4 & 0 & 3 & 4 \\ 17 & 31 & 172 & 21 & 0 & 17 & 24 \\ 2 & 4 & 21 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 17 & 2 & 0 & 0 & 2 \\ 3 & 4 & 24 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{1111}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -2816$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 20 & 2 & 1 & 2 & 3 \\ 3 & 4 & 36 & 4 & 0 & 3 & 4 \\ 20 & 36 & 238 & 24 & 0 & 20 & 28 \\ 2 & 4 & 24 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 20 & 2 & 0 & 0 & 2 \\ 3 & 4 & 28 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

4.966 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$T = 10$ [0] $1^{1111}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 72$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 3 & 5 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 6 & 4 \\ 2 & 4 & 3 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & 2 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{1111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 392$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 2 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 3 & 2 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{1111}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 462$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 3 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 5 & 2 \\ 5 & 9 & 9 & 6 & 0 & 5 & 13 & 7 \\ 3 & 4 & 4 & 2 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{1111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 512$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 3 & 4 & 4 \\ 2 & 4 & 1 & 6 & 0 & 2 & 3 & 3 \\ 5 & 9 & 6 & 12 & 0 & 5 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 5 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 6 & 0 & 2 & 1 & 3 \\ 3 & 4 & 3 & 6 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{1111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 622$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 3 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 2 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 2 & 0 & 0 & 5 & 2 \\ 5 & 9 & 6 & 6 & 0 & 5 & 13 & 7 \\ 3 & 4 & 3 & 2 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{1111}3^11^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 1152$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 9 & 4 & 0 & 3 & 9 & 4 \\ 5 & 9 & 13 & 6 & 0 & 5 & 15 & 7 \\ 2 & 4 & 6 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 2 & 0 & 0 & 5 & 2 \\ 5 & 9 & 15 & 6 & 0 & 5 & 13 & 7 \\ 3 & 4 & 7 & 2 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.967 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_{2n3\text{mix}}$

$$T = 9 \quad [0] \quad 1^{\overset{1}{111}} 3^{\overset{1}{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 8 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 9 & 5 & 0 & 12 & 6 \\ 3 & 4 & 4 & 2 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overset{1}{111}} 3^{\overset{1}{111}} 1^{\overset{1}{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 1 & 3 & 0 & 11 & 3 \\ 2 & 4 & 3 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 17 & 11 & 11 & 0 & 55 & 13 \\ 3 & 4 & 3 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overset{1}{111}} 3^{\overset{1}{111}} 1^{\overset{1}{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 5 & 4 & 0 & 7 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 5 & 0 & 7 & 5 \\ 3 & 4 & 4 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overset{1}{112}} 3^{\overset{1}{111}} 1^{\overset{1}{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 6 & 3 \\ 3 & 4 & 4 & 5 & 0 & 11 & 4 \\ 2 & 4 & 1 & 3 & 0 & 8 & 3 \\ 3 & 5 & 3 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 8 & 9 & 0 & 20 & 8 \\ 3 & 4 & 3 & 3 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overset{1}{111}} 3^{\overset{1}{111}} 1^{\overset{1}{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 5 & 0 & 7 & 4 \\ 2 & 4 & 1 & 3 & 0 & 5 & 3 \\ 3 & 5 & 3 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 5 & 6 & 0 & 7 & 5 \\ 3 & 4 & 3 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overset{1}{111}} 3^{\overset{1}{111}} 1^{\overset{1}{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 5 & 3 \\ 3 & 4 & 9 & 4 & 0 & 8 & 4 \\ 5 & 9 & 13 & 6 & 0 & 14 & 7 \\ 2 & 4 & 6 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 14 & 5 & 0 & 12 & 6 \\ 3 & 4 & 7 & 2 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\overset{1}{111}} 3^{\overset{1}{111}} 1^{\overset{1}{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 12 & 4 \\ 3 & 5 & 4 & 3 & 0 & 12 & 4 \\ 2 & 4 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 12 & 8 & 0 & 25 & 9 \\ 3 & 4 & 4 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1113}} 3^{\overline{111}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 5 & 4 & 0 & 20 & 4 \\ 3 & 5 & 4 & 3 & 0 & 18 & 4 \\ 2 & 4 & 3 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 18 & 14 & 0 & 69 & 15 \\ 3 & 4 & 4 & 2 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -424, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.07 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 & 0 & 7 & 4 \\ 2 & 4 & 1 & 3 & 0 & 5 & 3 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 5 & 5 & 0 & 7 & 5 \\ 3 & 4 & 3 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 4 & 3 \\ 3 & 4 & 4 & 9 & 0 & 7 & 4 \\ 2 & 4 & 1 & 6 & 0 & 5 & 3 \\ 5 & 9 & 6 & 12 & 0 & 11 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 5 & 11 & 0 & 7 & 5 \\ 3 & 4 & 3 & 6 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -481, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 3 & 0 & 8 & 3 \\ 2 & 4 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 8 & 8 & 0 & 25 & 9 \\ 3 & 4 & 3 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -500, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 5 & 4 & 0 & 16 & 4 \\ 3 & 5 & 4 & 3 & 0 & 15 & 4 \\ 2 & 4 & 3 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 15 & 11 & 0 & 44 & 12 \\ 3 & 4 & 4 & 2 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -509, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.06 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 4 & 3 \\ 3 & 4 & 9 & 4 & 0 & 7 & 4 \\ 5 & 9 & 13 & 6 & 0 & 11 & 7 \\ 2 & 4 & 6 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 11 & 5 & 0 & 7 & 5 \\ 3 & 4 & 7 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -896, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 7 & 3 \\ 3 & 4 & 9 & 4 & 0 & 12 & 4 \\ 5 & 9 & 13 & 6 & 0 & 20 & 7 \\ 2 & 4 & 6 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 20 & 8 & 0 & 25 & 9 \\ 3 & 4 & 7 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$4.968 \quad \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3$$

$$T = 9 \quad [0] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -39, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 1 & 2 & 3 \\ 4 & 9 & 12 & 4 & 0 & 4 & 5 \\ 5 & 12 & 13 & 6 & 0 & 6 & 7 \\ 2 & 4 & 6 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 3 & 0 & 0 & 2 \\ 3 & 5 & 7 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 6 & 3 & 0 & 3 & 3 \\ 3 & 4 & 4 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 11 & 2 & 1 & 2 & 3 \\ 4 & 9 & 28 & 4 & 0 & 4 & 5 \\ 11 & 28 & 73 & 12 & 0 & 12 & 15 \\ 2 & 4 & 12 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 12 & 3 & 0 & 0 & 2 \\ 3 & 5 & 15 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 8 & 2 & 1 & 2 & 3 \\ 4 & 9 & 20 & 4 & 0 & 4 & 5 \\ 8 & 20 & 37 & 9 & 0 & 9 & 11 \\ 2 & 4 & 9 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 3 & 0 & 0 & 2 \\ 3 & 5 & 11 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{12}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.9$

$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 4 & 3 \\ 3 & 4 & 13 & 5 & 0 & 8 & 4 \\ 7 & 13 & 28 & 12 & 0 & 18 & 10 \\ 3 & 5 & 12 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 18 & 6 & 0 & 9 & 5 \\ 3 & 4 & 10 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.34 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 1 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 9 & 0 & 36 & 10 \\ 3 & 4 & 3 & 2 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{13}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 4 & 3 \\ 3 & 4 & 17 & 5 & 0 & 8 & 4 \\ 9 & 17 & 49 & 15 & 0 & 24 & 13 \\ 3 & 5 & 15 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 24 & 6 & 0 & 9 & 5 \\ 3 & 4 & 13 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{1111}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 2 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 & 4 \\ 8 & 14 & 37 & 9 & 0 & 9 & 11 \\ 2 & 4 & 9 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 3 & 0 & 0 & 2 \\ 3 & 4 & 11 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{113}31^{211} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 13 & 2 & 1 & 2 & 3 \\ 4 & 9 & 32 & 4 & 0 & 4 & 5 \\ 13 & 32 & 100 & 15 & 0 & 15 & 18 \\ 2 & 4 & 15 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 15 & 3 & 0 & 0 & 2 \\ 3 & 5 & 18 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -399, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{114}31^{311} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 1 & 2 & 3 \\ 4 & 9 & 44 & 4 & 0 & 4 & 5 \\ 18 & 44 & 193 & 21 & 0 & 21 & 25 \\ 2 & 4 & 21 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 21 & 3 & 0 & 0 & 2 \\ 3 & 5 & 25 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 2 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 & 4 \\ 5 & 9 & 13 & 9 & 0 & 6 & 7 \\ 3 & 5 & 9 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 3 & 0 & 0 & 2 \\ 3 & 4 & 7 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 6 & 0 & 12 & 6 \\ 3 & 4 & 3 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{114}31^{211} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{f}_4 + \text{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 1 & 2 & 3 \\ 4 & 9 & 40 & 4 & 0 & 4 & 5 \\ 16 & 40 & 154 & 18 & 0 & 18 & 22 \\ 2 & 4 & 18 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 18 & 3 & 0 & 0 & 2 \\ 3 & 5 & 22 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{141} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -527, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.27 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 1 & 2 & 3 \\ 3 & 4 & 21 & 5 & 0 & 4 & 4 \\ 11 & 21 & 76 & 18 & 0 & 15 & 16 \\ 3 & 5 & 18 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 15 & 3 & 0 & 0 & 2 \\ 3 & 4 & 16 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -621, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.48 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 & 4 \\ 5 & 9 & 13 & 6 & 0 & 6 & 7 \\ 2 & 4 & 6 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 3 & 0 & 0 & 2 \\ 3 & 4 & 7 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{121} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -707, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 2 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 & 4 \\ 7 & 13 & 28 & 12 & 0 & 9 & 10 \\ 3 & 5 & 12 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 3 & 0 & 0 & 2 \\ 3 & 4 & 10 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{131} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -713, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.25 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 2 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 & 4 \\ 9 & 17 & 49 & 15 & 0 & 12 & 13 \\ 3 & 5 & 15 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 12 & 3 & 0 & 0 & 2 \\ 3 & 4 & 13 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -944, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.22 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 5 & 3 \\ 3 & 4 & 9 & 4 & 0 & 9 & 4 \\ 5 & 9 & 13 & 6 & 0 & 15 & 7 \\ 2 & 4 & 6 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 15 & 6 & 0 & 12 & 6 \\ 3 & 4 & 7 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

4.969 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$

$$T = 11 \quad [0] \ 1^{11} 5^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -11, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 2 & 3 & 3 & 2 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 3 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 5^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -15, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 1 & 3 \\ 3 & 4 & 10 & 3 & 3 & 0 & 4 \\ 6 & 10 & 21 & 6 & 8 & 0 & 9 \\ 2 & 3 & 6 & 0 & 2 & 0 & 2 \\ 2 & 3 & 8 & 2 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 9 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12} 5^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -23, \ (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 2 & 1 & 3 \\ 3 & 4 & 17 & 3 & 3 & 0 & 4 \\ 10 & 17 & 61 & 10 & 13 & 0 & 15 \\ 2 & 3 & 10 & 0 & 2 & 0 & 2 \\ 2 & 3 & 13 & 2 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 15 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} \ 5^{\overline{1}} \ 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -44, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 1 & 3 \\ 3 & 4 & 7 & 3 & 3 & 0 & 4 \\ 4 & 7 & 8 & 4 & 5 & 0 & 6 \\ 2 & 3 & 4 & 0 & 2 & 0 & 2 \\ 2 & 3 & 5 & 2 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 6 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{12}} \ 5^{\overline{1}} \ 1^{\overline{121}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -50, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 2 & 1 & 3 \\ 3 & 4 & 14 & 3 & 3 & 0 & 4 \\ 8 & 14 & 38 & 8 & 10 & 0 & 12 \\ 2 & 3 & 8 & 0 & 2 & 0 & 2 \\ 2 & 3 & 10 & 2 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 12 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{13}} \ 5^{\overline{1}} \ 1^{\overline{131}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -60, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 2 & 1 & 3 \\ 3 & 4 & 21 & 3 & 3 & 0 & 4 \\ 12 & 21 & 88 & 12 & 15 & 0 & 18 \\ 2 & 3 & 12 & 0 & 2 & 0 & 2 \\ 2 & 3 & 15 & 2 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 18 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

4.970 NHC(-2)(-3) + $\textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{so}_8$

$$T = 8 \quad [0] \ 1^{\overline{11}} \ 3 \ 1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 14 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 14 & 17 & 13 & 15 & 7 \\ 12 & 17 & 15 & 12 & 6 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} \ 3 \ 1^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 19 & 5 \\ 11 & 20 & 0 & 34 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 19 & 34 & 0 & 57 & 13 \\ 5 & 8 & 0 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{141}} \ 3 \ 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 1 & 3 \\ 3 & 4 & 16 & 9 & 0 & 4 \\ 9 & 16 & 46 & 24 & 0 & 12 \\ 5 & 9 & 24 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{132}} \ 3 \ 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 1 & 3 \\ 3 & 4 & 12 & 13 & 0 & 4 \\ 6 & 12 & 25 & 27 & 0 & 9 \\ 7 & 13 & 27 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}} \ 3 \ 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 4 & 5 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 13 & 15 & 7 \\ 12 & 17 & 15 & 12 & 6 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 10 & 17 & 13 & 15 & 7 \\ 10 & 17 & 15 & 12 & 6 \\ 5 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 0 & -1 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 20 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 20 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{11} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 1 & 3 \\ 3 & 4 & 24 & 4 & 0 & 4 \\ 17 & 24 & 121 & 15 & 0 & 19 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[12] $1^1 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 5 \\ 12 & 25 & 20 & 16 & 9 \\ 9 & 20 & 13 & 12 & 7 \\ 8 & 16 & 12 & 9 & 5 \\ 5 & 9 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{11} 3^1 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 4 & 4 & 1 & 3 & 0 & 3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 3 \\ 3 & 4 & 0 & 9 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 5 & 9 & 0 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[14] $1^1 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 5 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 5 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 0 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

- [15] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 34$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 10 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$
- [16] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 37$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 10 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$
- [17] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 37$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 17 & 0 & 12 & 6 \\ 5 & 8 & 0 & 6 & 2 \end{pmatrix}_{b_0}$
- [18] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 39$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 11 & 13 & 6 \\ 15 & 20 & 16 & 17 & 8 \\ 11 & 16 & 10 & 12 & 6 \\ 13 & 17 & 12 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [19] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 \\ 3 & 5 & 4 & 3 & 0 \\ 3 & 4 & 3 & 0 & 0 \\ 1 & 0 & 0 & -4 & -2 \end{pmatrix}_{b_0}$
- [22] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 11 & 1 & 1 & 3 \\ 3 & 4 & 19 & 0 & 0 & 4 \\ 11 & 19 & 73 & 0 & 0 & 15 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & -1 & -2 & 1 \end{pmatrix}_{b_0}$
- [21] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 5 & 0 & 0 & 4 \\ 3 & 5 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & -1 & -2 & 1 \end{pmatrix}_{b_0}$
- [20] $1^{113} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 59 & 1 & 5 \\ 12 & 25 & 124 & 0 & 9 \\ 59 & 124 & 613 & 0 & 45 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 45 & -1 & 2 \end{pmatrix}_{b_0}$
- [25] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 10 & 10 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 10 & 16 & 10 & 12 & 6 \\ 10 & 17 & 12 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [24] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.52 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 12 & 1 & 6 \\ 15 & 20 & 17 & 0 & 8 \\ 12 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$
- [23] $1^{12} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 46$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 8 & 5 & 0 & 4 \\ 5 & 8 & 10 & 6 & 0 & 6 \\ 3 & 5 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 3 \\ 3 & 4 & 4 & 0 & 0 & 4 \\ 2 & 4 & 1 & 0 & 0 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 1 & 3 \\ 4 & 9 & 28 & 4 & 0 & 5 \\ 15 & 28 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 15 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{132} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 1 & 3 \\ 3 & 4 & 12 & 8 & 0 & 4 \\ 6 & 12 & 25 & 18 & 0 & 9 \\ 5 & 8 & 18 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 10 & 17 & 13 & 15 & 7 \\ 9 & 17 & 15 & 12 & 6 \\ 5 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 17 & 0 & 12 & 6 \\ 6 & 8 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 1 & 3 \\ 4 & 9 & 12 & 0 & 0 & 5 \\ 5 & 12 & 13 & 0 & 0 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 7 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 14 & 1 & 1 & 3 \\ 4 & 9 & 36 & 0 & 0 & 5 \\ 14 & 36 & 121 & 0 & 0 & 19 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 19 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 1 & 3 \\ 4 & 9 & 12 & 4 & 0 & 5 \\ 6 & 12 & 13 & 6 & 0 & 7 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 7 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 13 & 15 & 7 \\ 10 & 17 & 15 & 12 & 6 \\ 5 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

- [37] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 2 & -2 & 2 \end{pmatrix}_{b_0}$
- [38] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 13 & 15 & 7 \\ 9 & 17 & 15 & 12 & 6 \\ 5 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$
- [36] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 12 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 13 & 15 & 7 \\ 13 & 17 & 15 & 12 & 6 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$
- [39] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 3 & -2 & 1 \end{pmatrix}_{b_0}$
- [40] $21^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 13 & 17 & 15 & 15 & 7 \\ 17 & 20 & 17 & 17 & 8 \\ 15 & 17 & 13 & 15 & 7 \\ 15 & 17 & 15 & 12 & 6 \\ 7 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$
- [41] $1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 4 & 8 & 10 & 6 & 0 & 6 \\ 3 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 2 & -2 & 1 \end{pmatrix}_{b_0}$
- [42] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 3 & 4 & 1 & 3 & 0 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 3 & -2 & 1 \end{pmatrix}_{b_0}$
- [43] $1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 8 & 1 & 3 \\ 3 & 4 & 8 & 13 & 0 & 4 \\ 4 & 8 & 10 & 18 & 0 & 6 \\ 8 & 13 & 18 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 9 & -2 & 1 \end{pmatrix}_{b_0}$
- [44] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 4 & 5 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 6 & -2 & 1 \end{pmatrix}_{b_0}$
- [45] $1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 8 & 9 & 0 & 4 \\ 4 & 8 & 10 & 12 & 0 & 6 \\ 5 & 9 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$
- [46] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 13 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 13 & 17 & 13 & 15 & 7 \\ 12 & 17 & 15 & 12 & 6 \\ 6 & 8 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$

[47] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 1 & 3 \\ 3 & 4 & 17 & 0 & 0 & 4 \\ 9 & 17 & 49 & 0 & 0 & 13 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 4 & 3 & 0 & 4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 8 & 16 & 10 & 12 & 6 \\ 9 & 17 & 12 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[50] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 11 & 1 & 1 & 3 \\ 4 & 9 & 28 & 0 & 0 & 5 \\ 11 & 28 & 73 & 0 & 0 & 15 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 15 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 \\ 3 & 5 & 4 & 6 & 0 & 4 \\ 3 & 5 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 1 & 3 \\ 4 & 9 & 20 & 0 & 0 & 5 \\ 8 & 20 & 37 & 0 & 0 & 11 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 11 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[53] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 9 & 16 & 10 & 12 & 6 \\ 10 & 17 & 12 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[54] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 11 & 12 & 6 \\ 15 & 20 & 16 & 17 & 8 \\ 11 & 16 & 10 & 12 & 6 \\ 12 & 17 & 12 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[55] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 16 & 17 & 8 \\ 12 & 16 & 10 & 12 & 6 \\ 12 & 17 & 12 & 12 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[56] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 16 & 17 & 8 \\ 9 & 16 & 10 & 12 & 6 \\ 9 & 17 & 12 & 12 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

- [57] $1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 1 & 3 \\ 4 & 9 & 24 & 0 & 0 & 5 \\ 10 & 24 & 58 & 0 & 0 & 14 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 14 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$
- [58] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 7 & 9 & 13 & 15 & 0 & 7 \\ 5 & 9 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [60] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 3 & 4 & 1 & 6 & 0 & 3 \\ 4 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$
- [59] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 \\ 11 & 14 & 37 & 9 & 0 & 11 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [62] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 8 & 9 & 13 & 9 & 0 & 7 \\ 3 & 5 & 9 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$
- [61] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 5 & 5 \\ 11 & 20 & 16 & 0 & 8 & 8 \\ 8 & 16 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 5 & 8 & 6 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$
- [63] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 \\ 3 & 5 & 4 & 6 & 0 & 4 \\ 4 & 5 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$
- [65] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 5 & 2 & 1 & 3 \\ 4 & 9 & 12 & 4 & 0 & 5 \\ 5 & 12 & 13 & 6 & 0 & 7 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 7 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [64] $1^{11} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 1 & 3 \\ 3 & 4 & 14 & 0 & 0 & 4 \\ 8 & 14 & 37 & 0 & 0 & 11 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$
- [67] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 & 5 \\ 12 & 25 & 36 & 0 & 9 & 9 \\ 17 & 36 & 49 & 0 & 13 & 13 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 5 & 9 & 13 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$
- [66] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 1 & 3 \\ 3 & 4 & 9 & 0 & 0 & 4 \\ 5 & 9 & 13 & 0 & 0 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 4 & 15 & 0 & 4 \\ 9 & 17 & 15 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 3 & 4 & 1 & 12 & 0 & 3 \\ 9 & 17 & 12 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 \\ 4 & 5 & 4 & 6 & 0 & 4 \\ 3 & 5 & 6 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 1 & 3 \\ 3 & 4 & 13 & 0 & 0 & 4 \\ 7 & 13 & 28 & 0 & 0 & 10 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 20 & 10 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 20 & 33 & 49 & 27 & 13 \\ 10 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 11 & 1 & 6 \\ 15 & 20 & 16 & 0 & 8 \\ 11 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{111} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 1 & 3 \\ 3 & 4 & 24 & 4 & 0 & 4 \\ 15 & 24 & 121 & 15 & 0 & 19 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[80] \ 1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 1 & 3 \\ 3 & 4 & 19 & 4 & 0 & 4 \\ 14 & 19 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[79] \ 1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 19 & 4 & 0 & 4 \\ 11 & 19 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[78] \ 1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 4 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{121} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 7 & 8 & 10 & 6 & 0 & 6 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[76] $1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{121} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 4 & 8 & 10 & 6 & 0 & 6 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{131} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 1 & 3 \\ 3 & 4 & 12 & 9 & 0 & 4 \\ 7 & 12 & 25 & 18 & 0 & 9 \\ 5 & 9 & 18 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[83] $1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 5 & 9 & 13 & 9 & 0 & 7 \\ 4 & 5 & 9 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{111} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 48 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 48 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[84] $1^{14} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 1 & 3 \\ 3 & 4 & 5 & 21 & 0 & 4 \\ 3 & 5 & 4 & 18 & 0 & 4 \\ 12 & 21 & 18 & 75 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 9 & 8 & 0 & 4 \\ 5 & 9 & 13 & 12 & 0 & 7 \\ 4 & 8 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 3 & 4 & 1 & 6 & 0 & 3 \\ 5 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{121} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 10 & 18 & 58 & 12 & 0 & 14 \\ 3 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 3 & 0 & 3 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 3 \\ 3 & 4 & 9 & 8 & 0 & 4 \\ 5 & 9 & 13 & 12 & 0 & 7 \\ 5 & 8 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 3 & 4 & 1 & 9 & 0 & 3 \\ 7 & 13 & 9 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[92] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 1 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 \\ 11 & 13 & 28 & 12 & 0 & 10 \\ 3 & 5 & 12 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 17 & 10 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 17 & 33 & 49 & 27 & 13 \\ 10 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[93] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 35 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 35 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[95] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 23 & 1 & 6 \\ 15 & 20 & 33 & 0 & 8 \\ 23 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[94] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 8 & 14 & 9 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 16 & 0 & 8 \\ 9 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{141} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 5 & 1 & 3 \\ 3 & 4 & 21 & 9 & 0 & 4 \\ 15 & 21 & 76 & 33 & 0 & 16 \\ 5 & 9 & 33 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

- [98] $1^{113} 3 1^3 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 1 & 1 & 3 \\ 4 & 9 & 36 & 0 & 0 & 5 \\ 15 & 36 & 133 & 0 & 0 & 21 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 21 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$
- [99] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 5 & 9 & 13 & 0 & 7 \\ 3 & 4 & 6 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [100] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 8 & 9 & 0 & 4 \\ 5 & 8 & 10 & 12 & 0 & 6 \\ 5 & 9 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [101] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [102] $1^{122} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 8 & 8 & 0 & 4 \\ 4 & 8 & 10 & 12 & 0 & 6 \\ 5 & 8 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$
- [105] $1^{113} 3 1^2 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 13 & 1 & 1 & 3 \\ 4 & 9 & 32 & 0 & 0 & 5 \\ 13 & 32 & 100 & 0 & 0 & 18 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 18 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$
- [104] $1^{111} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 3 & 0 & 3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [103] $1^{111} 3 1^{14} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 1 & 3 \\ 3 & 4 & 24 & 4 & 0 & 4 \\ 16 & 24 & 121 & 15 & 0 & 19 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [106] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 26 & 12 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 26 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$
- [108] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 4 & 12 & 0 & 4 \\ 7 & 13 & 12 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$
- [107] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 23 & 13 & 6 & 6 \\ 15 & 20 & 33 & 17 & 8 & 8 \\ 23 & 33 & 49 & 27 & 13 & 13 \\ 13 & 17 & 27 & 12 & 6 & 6 \\ 6 & 8 & 13 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[109] 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 5 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[112] 1^{112} 3 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 10 & 2 & 1 & 3 \\ 4 & 9 & 24 & 4 & 0 & 5 \\ 10 & 24 & 58 & 12 & 0 & 14 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 14 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[111] 1^{112} 3 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 28 & 1 & 5 & 5 \\ 12 & 25 & 56 & 0 & 9 & 9 \\ 28 & 56 & 118 & 0 & 20 & 20 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 5 & 9 & 20 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[110] 1^{12} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 1 & 3 \\ 3 & 4 & 28 & 0 & 0 & 4 \\ 16 & 28 & 154 & 0 & 0 & 22 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[113] 1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 1 & 3 \\ 3 & 4 & 8 & 13 & 0 & 4 \\ 4 & 8 & 10 & 18 & 0 & 6 \\ 7 & 13 & 18 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[114] 1^{112} 3 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 5 & 5 \\ 12 & 25 & 72 & 0 & 9 & 9 \\ 34 & 72 & 202 & 0 & 26 & 26 \\ 1 & 0 & 0 & -3 & -1 & -1 \\ 5 & 9 & 26 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[115] 21^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 16 & 12 & 12 & 6 & 6 \\ 16 & 20 & 16 & 17 & 8 & 8 \\ 12 & 16 & 10 & 12 & 6 & 6 \\ 12 & 17 & 12 & 12 & 6 & 6 \\ 6 & 8 & 6 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[116] 1^{12} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 1 & 3 \\ 3 & 4 & 18 & 0 & 0 & 4 \\ 10 & 18 & 58 & 0 & 0 & 14 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[117] 1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 9 & 14 & 9 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[118] 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 23 & 12 & 6 & 6 \\ 15 & 20 & 33 & 17 & 8 & 8 \\ 23 & 33 & 49 & 27 & 13 & 13 \\ 12 & 17 & 27 & 12 & 6 & 6 \\ 6 & 8 & 13 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

- [119] $1^{14}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 1 & 3 \\ 3 & 4 & 21 & 5 & 0 & 4 \\ 11 & 21 & 76 & 18 & 0 & 16 \\ 3 & 5 & 18 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$
- [120] $1^{111}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 17 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$
- [121] $1^{112}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 157, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 10 & 2 & 1 & 3 \\ 4 & 9 & 20 & 4 & 0 & 5 \\ 10 & 20 & 37 & 9 & 0 & 11 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 11 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [122] $1^{114}3^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 158, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 16 & 1 & 1 & 3 \\ 4 & 9 & 40 & 0 & 0 & 5 \\ 16 & 40 & 154 & 0 & 0 & 22 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 22 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$
- [123] $1^{111}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 6 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [124] $1^{111}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 6 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [125] $1^{12}3^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 1 & 3 \\ 3 & 4 & 23 & 0 & 0 & 4 \\ 13 & 23 & 100 & 0 & 0 & 18 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$
- [126] $1^{111}3^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 26 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$
- [127] $1^{111}3^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 29 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 29 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$
- [128] $1^{111}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 5 & 9 & 13 & 9 & 0 & 7 \\ 3 & 5 & 9 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$
- [129] $1^{121}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 11 & 13 & 28 & 9 & 0 & 10 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[130] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 170$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 4 & 15 & 0 & 4 \\ 10 & 17 & 15 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 172$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 1 & 3 \\ 3 & 4 & 21 & 4 & 0 & 4 \\ 11 & 21 & 76 & 15 & 0 & 16 \\ 3 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[131] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 172$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 7 & 13 & 28 & 9 & 0 & 10 \\ 3 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 174$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.21 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 1 & 3 \\ 3 & 4 & 8 & 13 & 0 & 4 \\ 5 & 8 & 10 & 18 & 0 & 6 \\ 7 & 13 & 18 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[138] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 \\ 7 & 9 & 13 & 6 & 0 & 7 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 \\ 5 & 9 & 13 & 6 & 0 & 7 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[136] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 1 & 6 & 0 & 3 \\ 5 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[135] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 8 & 8 & 0 & 4 \\ 5 & 8 & 10 & 12 & 0 & 6 \\ 4 & 8 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 \\ 8 & 14 & 37 & 9 & 0 & 11 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[141] $1^{11} 3^1 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 180$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.71 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 46 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 46 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 33 & -1 & 2 \end{pmatrix}_{b_0}$$

[142] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 5 \\ 12 & 25 & 56 & 0 & 9 \\ 26 & 56 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

[140] $1^{14}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 1 & 3 \\ 3 & 4 & 26 & 0 & 0 & 4 \\ 14 & 26 & 118 & 0 & 0 & 20 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 20 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{13}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 1 & 3 \\ 3 & 4 & 22 & 0 & 0 & 4 \\ 12 & 22 & 85 & 0 & 0 & 17 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 17 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[143] $21^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 15 & 12 & 7 \\ 20 & 25 & 20 & 16 & 9 \\ 15 & 20 & 13 & 12 & 7 \\ 12 & 16 & 12 & 9 & 5 \\ 7 & 9 & 7 & 5 & 2 \end{pmatrix}_{b_0}$$

[144] $1^{132}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 1 & 3 \\ 3 & 4 & 12 & 13 & 0 & 4 \\ 7 & 12 & 25 & 27 & 0 & 9 \\ 7 & 13 & 27 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[145] $1^{114}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 20 & 1 & 1 & 3 \\ 4 & 9 & 48 & 0 & 0 & 5 \\ 20 & 48 & 238 & 0 & 0 & 28 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 28 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[146] $1^{131}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 9 & 17 & 49 & 12 & 0 & 13 \\ 3 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 8 & 2 & 1 & 3 \\ 4 & 9 & 20 & 4 & 0 & 5 \\ 8 & 20 & 37 & 9 & 0 & 11 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 11 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[149] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 19 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[148] $1^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 4 & 12 & 0 & 4 \\ 8 & 13 & 12 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

- [151] $1^{111} 1^3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 17 & 9 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 17 & 33 & 49 & 27 & 13 \\ 9 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$
- [150] $1^{111} 1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 19 & 9 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 19 & 33 & 49 & 27 & 13 \\ 9 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$
- [152] $1^{111} 1^{21} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 19 & 10 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 19 & 33 & 49 & 27 & 13 \\ 10 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$
- [153] $1^{122} 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 1 & 3 \\ 3 & 4 & 13 & 8 & 0 & 4 \\ 7 & 13 & 28 & 18 & 0 & 10 \\ 5 & 8 & 18 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$
- [154] $1^{114} 1^{11} 3 1^3 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 1 & 3 \\ 4 & 9 & 44 & 0 & 0 & 5 \\ 18 & 44 & 193 & 0 & 0 & 25 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 25 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$
- [155] $1^{113} 1^{31} 3 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 205, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 1 & 3 \\ 4 & 9 & 32 & 4 & 0 & 5 \\ 16 & 32 & 100 & 15 & 0 & 18 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 18 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [157] $1^{111} 1^{11} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 45 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 45 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$
- [158] $1^{111} 1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 18 & 10 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 18 & 33 & 49 & 27 & 13 \\ 10 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$
- [156] $1^{13} 1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 \\ 9 & 17 & 49 & 15 & 0 & 13 \\ 3 & 5 & 15 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$
- [159] $1^{121} 1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 1 & 3 \\ 3 & 4 & 13 & 9 & 0 & 4 \\ 7 & 13 & 28 & 21 & 0 & 10 \\ 6 & 9 & 21 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [160] $1^{112} 1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 6 & 9 & 13 & 21 & 0 & 7 \\ 7 & 13 & 21 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[161] $1^{12\bar{3}}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 \\ 7 & 13 & 28 & 12 & 0 & 10 \\ 3 & 5 & 12 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[162] $1^{12\bar{2}}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 1 & 3 \\ 3 & 4 & 13 & 8 & 0 & 4 \\ 8 & 13 & 28 & 18 & 0 & 10 \\ 4 & 8 & 18 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[163] $1^{11}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 7 & 9 & 13 & 9 & 0 & 7 \\ 3 & 5 & 9 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[164] $1^{12\bar{2}}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 1 & 3 \\ 3 & 4 & 13 & 8 & 0 & 4 \\ 7 & 13 & 28 & 18 & 0 & 10 \\ 4 & 8 & 18 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[165] $1^{11\bar{2}}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 7 & 13 & 9 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[169] $1^{13\bar{1}}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 1 & 3 \\ 3 & 4 & 22 & 4 & 0 & 4 \\ 12 & 22 & 85 & 15 & 0 & 17 \\ 3 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 17 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[166] $1^{12\bar{1}}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 5 & 8 & 10 & 6 & 0 & 6 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[170] $1^{11\bar{1}}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 19 & 4 & 0 & 4 \\ 12 & 19 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[171] $1^{12\bar{1}}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 6 & 8 & 10 & 6 & 0 & 6 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[172] $1^{11\bar{1}}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 1 & 3 \\ 3 & 4 & 19 & 4 & 0 & 4 \\ 13 & 19 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[168] $1^{131} 3^{141} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 12 & 4 & 0 & 4 \\ 10 & 12 & 25 & 9 & 0 & 9 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[167] $1^{132} 3^{121} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 1 & 3 \\ 3 & 4 & 12 & 8 & 0 & 4 \\ 8 & 12 & 25 & 18 & 0 & 9 \\ 4 & 8 & 18 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[173] $1^{114} 3^{131} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 22 & 2 & 1 & 3 \\ 4 & 9 & 44 & 4 & 0 & 5 \\ 22 & 44 & 193 & 21 & 0 & 25 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 25 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[174] $1^{111} 3^{111} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 233, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 5 \\ 11 & 20 & 33 & 0 & 8 \\ 18 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

[175] $1^{111} 3^{111} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 25 & 12 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 25 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{112} 3^{111} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 237, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 1 & 3 \\ 4 & 9 & 20 & 4 & 0 & 5 \\ 9 & 20 & 37 & 9 & 0 & 11 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 11 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[177] $1^{111} 3^{111} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 1 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 \\ 6 & 9 & 13 & 6 & 0 & 7 \\ 2 & 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{113} 3^{111} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 12 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 7 & 12 & 9 & 24 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

[179] $1^{111} 3^{112} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 \\ 10 & 14 & 37 & 9 & 0 & 11 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[180] $1^{111} 3^{111} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 24 & 12 & 6 \\ 15 & 20 & 33 & 17 & 8 \\ 24 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[181] $1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 241$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 38 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 38 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[182] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 242$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 6 & 9 & 13 & 9 & 0 & 7 \\ 3 & 5 & 9 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[184] $1^{113} 3^1 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 244$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 1 & 3 \\ 4 & 9 & 36 & 4 & 0 & 5 \\ 15 & 36 & 133 & 18 & 0 & 21 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 21 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[183] $21^{111} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 244$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 13 & 17 & 27 & 15 & 7 \\ 17 & 20 & 33 & 17 & 8 \\ 27 & 33 & 49 & 27 & 13 \\ 15 & 17 & 27 & 12 & 6 \\ 7 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[185] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 245$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 18 & 9 & 5 \\ 11 & 20 & 33 & 17 & 8 \\ 18 & 33 & 49 & 27 & 13 \\ 9 & 17 & 27 & 12 & 6 \\ 5 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$

[186] $1^{113} 3^1 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 253$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 5 \\ 12 & 25 & 108 & 0 & 9 \\ 51 & 108 & 457 & 0 & 39 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 39 & -1 & 2 \end{pmatrix}_{b_0}$$

[187] $1^{13} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 253$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 1 & 3 \\ 3 & 4 & 27 & 0 & 0 & 4 \\ 15 & 27 & 133 & 0 & 0 & 21 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 21 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[188] $1^{111} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 254$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 47 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 47 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[189] $1^{132} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 257$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 1 & 3 \\ 3 & 4 & 17 & 8 & 0 & 4 \\ 9 & 17 & 49 & 24 & 0 & 13 \\ 5 & 8 & 24 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[190] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 267$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.17 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 5 & 9 & 13 & 15 & 0 & 7 \\ 6 & 9 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[191] $1^{141} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 268$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 3 & 1 & 3 \\ 3 & 4 & 31 & 4 & 0 & 4 \\ 17 & 31 & 172 & 21 & 0 & 24 \\ 3 & 4 & 21 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 24 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[192] \ 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 5 & 9 & 13 & 15 & 0 & 7 \\ 5 & 9 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[193] \ 1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 \\ 9 & 14 & 37 & 9 & 0 & 11 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[194] \ 1^{132} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 1 & 3 \\ 3 & 4 & 12 & 8 & 0 & 4 \\ 7 & 12 & 25 & 18 & 0 & 9 \\ 4 & 8 & 18 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[195] \ 21^{112} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 20 & 30 & 12 & 7 \\ 20 & 25 & 40 & 16 & 9 \\ 30 & 40 & 58 & 24 & 14 \\ 12 & 16 & 24 & 9 & 5 \\ 7 & 9 & 14 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[197] \ 1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 2 & 4 & 1 & 12 & 0 & 3 \\ 10 & 18 & 12 & 57 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[196] \ 1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 38 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 38 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[198] \ 1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 277, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 36 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 36 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[199] \ 1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 285, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 35 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[200] \ 1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 28 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 28 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[201] \ 1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 27 & 1 & 5 \\ 11 & 20 & 50 & 0 & 8 \\ 27 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

- [202] $21^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 16 & 24 & 12 & 6 \\ 16 & 20 & 33 & 17 & 8 \\ 24 & 33 & 49 & 27 & 13 \\ 12 & 17 & 27 & 12 & 6 \\ 6 & 8 & 13 & 6 & 2 \end{pmatrix}_{b_0}$$
- [203] $1^{13}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 1 & 3 \\ 3 & 4 & 32 & 0 & 0 & 4 \\ 18 & 32 & 193 & 0 & 0 & 25 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 25 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$
- [204] $1^{111}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 46 & 1 & 5 \\ 11 & 20 & 84 & 0 & 8 \\ 46 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$
- [205] $1^{132}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 1 & 3 \\ 3 & 4 & 17 & 13 & 0 & 4 \\ 9 & 17 & 49 & 39 & 0 & 13 \\ 8 & 13 & 39 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$
- [206] $1^{132}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 293, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 1 & 3 \\ 3 & 4 & 17 & 8 & 0 & 4 \\ 11 & 17 & 49 & 24 & 0 & 13 \\ 4 & 8 & 24 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$
- [207] $1^{12}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 295, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 \\ 10 & 13 & 28 & 12 & 0 & 10 \\ 3 & 5 & 12 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$
- [208] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 7 & 13 & 28 & 9 & 0 & 10 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [209] $1^{133}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 301, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 2 & 4 & 1 & 12 & 0 & 3 \\ 9 & 17 & 12 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$
- [210] $1^{131}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 303, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 1 & 3 \\ 3 & 4 & 17 & 9 & 0 & 4 \\ 12 & 17 & 49 & 27 & 0 & 13 \\ 5 & 9 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [213] $1^{141}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 1 & 3 \\ 3 & 4 & 26 & 4 & 0 & 4 \\ 14 & 26 & 118 & 18 & 0 & 20 \\ 3 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 20 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [212] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 14 & 2 & 1 & 3 \\ 4 & 9 & 28 & 4 & 0 & 5 \\ 14 & 28 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 15 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[211] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 11 & 2 & 1 & 3 \\ 4 & 9 & 28 & 4 & 0 & 5 \\ 11 & 28 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 15 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[214] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 6 & 9 & 13 & 15 & 0 & 7 \\ 5 & 9 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[215] $1^{14} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 309, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 1 & 3 \\ 3 & 4 & 31 & 0 & 0 & 4 \\ 17 & 31 & 172 & 0 & 0 & 24 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 24 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[216] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 309, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 1 & 9 & 0 & 3 \\ 8 & 13 & 9 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[217] $1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 37 & 1 & 5 \\ 11 & 20 & 67 & 0 & 8 \\ 37 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{131} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 12 & 4 & 0 & 4 \\ 7 & 12 & 25 & 9 & 0 & 9 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[219] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 326, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 5 & 9 & 13 & 21 & 0 & 7 \\ 7 & 13 & 21 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[220] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 9 & 17 & 0 & 4 \\ 5 & 9 & 13 & 27 & 0 & 7 \\ 9 & 17 & 27 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[221] $1^{113} 3^1 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 333, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 43 & 1 & 5 \\ 12 & 25 & 92 & 0 & 9 \\ 43 & 92 & 325 & 0 & 33 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 33 & -1 & 2 \end{pmatrix}_{b_0}$$

[222] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 1 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 \\ 13 & 17 & 49 & 15 & 0 & 13 \\ 3 & 5 & 15 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[223] $1^{132} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 341, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 1 & 3 \\ 3 & 4 & 17 & 8 & 0 & 4 \\ 9 & 17 & 49 & 24 & 0 & 13 \\ 4 & 8 & 24 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[224] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 1 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 \\ 8 & 13 & 28 & 12 & 0 & 10 \\ 3 & 5 & 12 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[225] $1^{113} 3 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 349, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 13 & 2 & 1 & 3 \\ 4 & 9 & 32 & 4 & 0 & 5 \\ 13 & 32 & 100 & 15 & 0 & 18 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 18 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[227] $1^{132} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 4 & 1 & 3 \\ 3 & 4 & 22 & 8 & 0 & 4 \\ 13 & 22 & 85 & 30 & 0 & 17 \\ 4 & 8 & 30 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 17 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[226] $1^{132} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 1 & 3 \\ 3 & 4 & 22 & 8 & 0 & 4 \\ 12 & 22 & 85 & 30 & 0 & 17 \\ 4 & 8 & 30 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 17 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[228] $1^{121} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 1 & 3 \\ 3 & 4 & 13 & 9 & 0 & 4 \\ 9 & 13 & 28 & 21 & 0 & 10 \\ 5 & 9 & 21 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[229] $1^{121} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 1 & 3 \\ 3 & 4 & 13 & 9 & 0 & 4 \\ 7 & 13 & 28 & 21 & 0 & 10 \\ 5 & 9 & 21 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[230] $1^{121} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 371, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 \\ 9 & 13 & 28 & 12 & 0 & 10 \\ 3 & 5 & 12 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[231] $1^{121} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 2 & 4 & 1 & 12 & 0 & 3 \\ 11 & 18 & 12 & 57 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[232] $1^{113} 3 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 1 & 3 \\ 4 & 9 & 32 & 4 & 0 & 5 \\ 15 & 32 & 100 & 15 & 0 & 18 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 18 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[233] $1^{131} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 12 & 4 & 0 & 4 \\ 9 & 12 & 25 & 9 & 0 & 9 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[234] $1^{114} 3^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 56 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 56 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -1 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{114} 3^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 20 & 2 & 1 & 3 \\ 4 & 9 & 48 & 4 & 0 & 5 \\ 20 & 48 & 238 & 24 & 0 & 28 \\ 2 & 4 & 24 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 28 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[236] $1^{112} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 386, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 5 & 9 & 13 & 21 & 0 & 7 \\ 8 & 13 & 21 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[237] $1^{132} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 7 & 1 & 3 \\ 3 & 4 & 17 & 13 & 0 & 4 \\ 11 & 17 & 49 & 39 & 0 & 13 \\ 7 & 13 & 39 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[238] $1^{121} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 10 & 13 & 28 & 9 & 0 & 10 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[239] $1^{13} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 394, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 \\ 10 & 17 & 49 & 15 & 0 & 13 \\ 3 & 5 & 15 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[240] $1^{14} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 395, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 1 & 3 \\ 3 & 4 & 21 & 5 & 0 & 4 \\ 12 & 21 & 76 & 18 & 0 & 16 \\ 3 & 5 & 18 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[242] $1^{114} 3^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 5 \\ 12 & 25 & 144 & 0 & 9 \\ 68 & 144 & 814 & 0 & 52 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 52 & -1 & 2 \end{pmatrix}_{b_0}$$

[241] $1^{114} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 44 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 44 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[243] $1^{141} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 1 & 3 \\ 3 & 4 & 36 & 0 & 0 & 4 \\ 20 & 36 & 238 & 0 & 0 & 28 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 28 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[244] $1^{141} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 1 & 3 \\ 3 & 4 & 21 & 9 & 0 & 4 \\ 11 & 21 & 76 & 33 & 0 & 16 \\ 5 & 9 & 33 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[245] $1^{131} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 411, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 1 & 3 \\ 3 & 4 & 17 & 9 & 0 & 4 \\ 9 & 17 & 49 & 27 & 0 & 13 \\ 5 & 9 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[246] $1^{121} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 14 & 18 & 58 & 12 & 0 & 14 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[247] $1^{131} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 9 & 17 & 49 & 12 & 0 & 13 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[248] $1^{121} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 10 & 18 & 58 & 12 & 0 & 14 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[249] $1^{131} 3 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 12 & 4 & 0 & 4 \\ 8 & 12 & 25 & 9 & 0 & 9 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[250] $1^{121} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 1 & 3 \\ 3 & 4 & 28 & 4 & 0 & 4 \\ 16 & 28 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[251] $1^{141} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 16 & 4 & 0 & 4 \\ 9 & 16 & 46 & 12 & 0 & 12 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[252] $1^{122} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 1 & 3 \\ 3 & 4 & 13 & 13 & 0 & 4 \\ 7 & 13 & 28 & 30 & 0 & 10 \\ 8 & 13 & 30 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[253] $21^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 45 & 12 & 7 \\ 20 & 25 & 60 & 16 & 9 \\ 45 & 60 & 133 & 36 & 21 \\ 12 & 16 & 36 & 9 & 5 \\ 7 & 9 & 21 & 5 & 2 \end{pmatrix}_{b_0}$$

[254] $1^{113}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 37 & 1 & 5 \\ 12 & 25 & 76 & 0 & 9 \\ 37 & 76 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

[255] $1^{132}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 1 & 3 \\ 3 & 4 & 17 & 8 & 0 & 4 \\ 10 & 17 & 49 & 24 & 0 & 13 \\ 4 & 8 & 24 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[256] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 14 & 2 & 1 & 3 \\ 4 & 9 & 36 & 4 & 0 & 5 \\ 14 & 36 & 121 & 15 & 0 & 19 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 19 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[258] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 1 & 3 \\ 4 & 9 & 28 & 4 & 0 & 5 \\ 12 & 28 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 15 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[257] $1^{113}31^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 13 & 2 & 1 & 3 \\ 4 & 9 & 28 & 4 & 0 & 5 \\ 13 & 28 & 73 & 12 & 0 & 15 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 15 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[259] $1^{114}31^{41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 20 & 2 & 1 & 3 \\ 4 & 9 & 40 & 4 & 0 & 5 \\ 20 & 40 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 22 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[260] $1^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 1 & 15 & 0 & 3 \\ 12 & 22 & 15 & 84 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[261] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 48 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 48 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[262] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 453, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 2 & 4 & 1 & 12 & 0 & 3 \\ 10 & 17 & 12 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[263] $1^{113} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 455, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 1 & 3 \\ 3 & 4 & 9 & 17 & 0 & 4 \\ 5 & 9 & 13 & 27 & 0 & 7 \\ 10 & 17 & 27 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[264] $1^{121} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 8 & 13 & 28 & 9 & 0 & 10 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[265] $1^{121} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 1 & 3 \\ 3 & 4 & 23 & 4 & 0 & 4 \\ 13 & 23 & 100 & 15 & 0 & 18 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[266] $1^{13} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 458, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 1 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 \\ 12 & 17 & 49 & 15 & 0 & 13 \\ 3 & 5 & 15 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[267] $1^{122} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 1 & 3 \\ 3 & 4 & 13 & 13 & 0 & 4 \\ 7 & 13 & 28 & 30 & 0 & 10 \\ 7 & 13 & 30 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[268] $1^{13} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 1 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 \\ 11 & 17 & 49 & 15 & 0 & 13 \\ 3 & 5 & 15 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[269] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 488, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 9 & 13 & 28 & 9 & 0 & 10 \\ 2 & 4 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[270] $1^{122} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 491, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.19 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 1 & 3 \\ 3 & 4 & 13 & 13 & 0 & 4 \\ 8 & 13 & 28 & 30 & 0 & 10 \\ 7 & 13 & 30 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[271] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 493, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 1 & 3 \\ 4 & 9 & 36 & 4 & 0 & 5 \\ 18 & 36 & 121 & 15 & 0 & 19 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 19 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[272] $1^{121} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 498, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 1 & 3 \\ 3 & 4 & 13 & 9 & 0 & 4 \\ 8 & 13 & 28 & 21 & 0 & 10 \\ 5 & 9 & 21 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

- [273] $1^{114} 3^1 1^{31} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 501, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.47 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 21 & 2 & 1 & 3 \\ 4 & 9 & 44 & 4 & 0 & 5 \\ 21 & 44 & 193 & 21 & 0 & 25 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 25 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [274] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 503, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 1 & 3 \\ 3 & 4 & 21 & 5 & 0 & 4 \\ 15 & 21 & 76 & 18 & 0 & 16 \\ 3 & 5 & 18 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$
- [275] $1^{114} 3^1 1^{21} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 5 & 5 \\ 12 & 25 & 112 & 0 & 9 & 9 \\ 52 & 112 & 478 & 0 & 40 & 40 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$
- [276] $1^{114} 3^1 1^{31} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.47 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 5 & 5 \\ 12 & 25 & 128 & 0 & 9 & 9 \\ 60 & 128 & 634 & 0 & 46 & 46 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 46 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$
- [277] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 535, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 1 & 3 \\ 3 & 4 & 21 & 5 & 0 & 4 \\ 13 & 21 & 76 & 18 & 0 & 16 \\ 3 & 5 & 18 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$
- [278] $1^{141} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 21 & 4 & 0 & 4 \\ 11 & 21 & 76 & 15 & 0 & 16 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [279] $1^{114} 3^1 1^{31} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 549, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 1 & 3 \\ 4 & 9 & 44 & 4 & 0 & 5 \\ 18 & 44 & 193 & 21 & 0 & 25 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 25 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [280] $1^{114} 3^1 1^{21} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 1 & 3 \\ 4 & 9 & 40 & 4 & 0 & 5 \\ 16 & 40 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 22 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [281] $1^{132} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 566, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 1 & 3 \\ 3 & 4 & 17 & 13 & 0 & 4 \\ 9 & 17 & 49 & 39 & 0 & 13 \\ 7 & 13 & 39 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$
- [282] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 571, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 1 & 3 \\ 3 & 4 & 21 & 5 & 0 & 4 \\ 14 & 21 & 76 & 18 & 0 & 16 \\ 3 & 5 & 18 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$
- [283] $1^{121} 3^1 1^{13} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 584, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$

$$T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 1 & 3 \\ 3 & 4 & 23 & 4 & 0 & 4 \\ 17 & 23 & 100 & 15 & 0 & 18 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[284] \ 1^{141} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 586, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 5 & 1 & 3 \\ 3 & 4 & 21 & 9 & 0 & 4 \\ 14 & 21 & 76 & 33 & 0 & 16 \\ 5 & 9 & 33 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[285] \ 1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 1 & 15 & 0 & 3 \\ 13 & 22 & 15 & 84 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[286] \ 1^{131} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 1 & 3 \\ 3 & 4 & 17 & 9 & 0 & 4 \\ 11 & 17 & 49 & 27 & 0 & 13 \\ 5 & 9 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[290] \ 1^{121} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 11 & 18 & 58 & 12 & 0 & 14 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[287] \ 1^{141} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 16 & 4 & 0 & 4 \\ 10 & 16 & 46 & 12 & 0 & 12 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[289] \ 1^{141} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 16 & 4 & 0 & 4 \\ 12 & 16 & 46 & 12 & 0 & 12 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[288] \ 1^{121} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 13 & 18 & 58 & 12 & 0 & 14 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[291] \ 21^{114} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 634, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 20 & 60 & 12 & 7 \\ 20 & 25 & 80 & 16 & 9 \\ 60 & 80 & 238 & 48 & 28 \\ 12 & 16 & 48 & 9 & 5 \\ 7 & 9 & 28 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[292] \ 1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 15 & 2 & 1 & 3 \\ 4 & 9 & 36 & 4 & 0 & 5 \\ 15 & 36 & 121 & 15 & 0 & 19 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 19 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[293] \ 1^{131} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 639, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 1 & 3 \\ 3 & 4 & 17 & 9 & 0 & 4 \\ 10 & 17 & 49 & 27 & 0 & 13 \\ 5 & 9 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[294] $1^{114} 3^{1^{31}} 3^{1^{21}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 645, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 55 & 1 & 5 \\ 12 & 25 & 112 & 0 & 9 \\ 55 & 112 & 478 & 0 & 40 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 40 & -1 & 2 \end{pmatrix}_{b_0}$$

[295] $1^{131} 3^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 22 & 4 & 0 & 4 \\ 12 & 22 & 85 & 15 & 0 & 17 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 17 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[296] $1^{114} 3^{1^{31}} 3^{1^{21}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 660, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 19 & 2 & 1 & 3 \\ 4 & 9 & 40 & 4 & 0 & 5 \\ 19 & 40 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 22 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[297] $1^{114} 3^{1^{31}} 3^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 661, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 47 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 47 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[298] $1^{114} 3^{1^{31}} 3^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 669, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 17 & 2 & 1 & 3 \\ 4 & 9 & 36 & 4 & 0 & 5 \\ 17 & 36 & 121 & 15 & 0 & 19 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 19 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[299] $1^{121} 3^{1^{14}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 20 & 2 & 1 & 3 \\ 3 & 4 & 28 & 4 & 0 & 4 \\ 20 & 28 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[302] $1^{121} 3^{1^{14}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 2 & 1 & 3 \\ 3 & 4 & 28 & 4 & 0 & 4 \\ 17 & 28 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[303] $1^{131} 3^{1^{41}} 3^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 13 & 17 & 49 & 12 & 0 & 13 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[304] $1^{141} 3^{1^{31}} 3^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 16 & 4 & 0 & 4 \\ 11 & 16 & 46 & 12 & 0 & 12 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[300] $1^{121} 3^{1^{21}} 3^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 12 & 18 & 58 & 12 & 0 & 14 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[301] $1^{131} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 10 & 17 & 49 & 12 & 0 & 13 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[305] $1^{121} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 680, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 1 & 3 \\ 3 & 4 & 23 & 4 & 0 & 4 \\ 14 & 23 & 100 & 15 & 0 & 18 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[306] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 717, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 16 & 2 & 1 & 3 \\ 4 & 9 & 36 & 4 & 0 & 5 \\ 16 & 36 & 121 & 15 & 0 & 19 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 19 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[307] $1^{132} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 722, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 7 & 1 & 3 \\ 3 & 4 & 17 & 13 & 0 & 4 \\ 10 & 17 & 49 & 39 & 0 & 13 \\ 7 & 13 & 39 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[308] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 46 & 1 & 5 \\ 12 & 25 & 96 & 0 & 9 \\ 46 & 96 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

[309] $1^{141} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 730, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 1 & 3 \\ 3 & 4 & 21 & 9 & 0 & 4 \\ 12 & 21 & 76 & 33 & 0 & 16 \\ 5 & 9 & 33 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[310] $1^{121} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 744, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 1 & 3 \\ 3 & 4 & 23 & 4 & 0 & 4 \\ 16 & 23 & 100 & 15 & 0 & 18 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[311] $1^{114} 3 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 756, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 2 & 1 & 3 \\ 4 & 9 & 40 & 4 & 0 & 5 \\ 18 & 40 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 22 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[312] $1^{121} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 776, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 1 & 3 \\ 3 & 4 & 23 & 4 & 0 & 4 \\ 15 & 23 & 100 & 15 & 0 & 18 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 18 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[313] $1^{141} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 790, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 5 & 1 & 3 \\ 3 & 4 & 21 & 9 & 0 & 4 \\ 13 & 21 & 76 & 33 & 0 & 16 \\ 5 & 9 & 33 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[315] $1^{121} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 800$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 1 & 3 \\ 3 & 4 & 28 & 4 & 0 & 4 \\ 18 & 28 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[316] $1^{131} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 800$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 12 & 17 & 49 & 12 & 0 & 13 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[317] $1^{121} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 800$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 2 & 1 & 3 \\ 3 & 4 & 28 & 4 & 0 & 4 \\ 19 & 28 & 154 & 18 & 0 & 22 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 22 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[314] $1^{131} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 800$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 11 & 17 & 49 & 12 & 0 & 13 \\ 2 & 4 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[318] $1^{131} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 816$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 1 & 3 \\ 3 & 4 & 27 & 4 & 0 & 4 \\ 15 & 27 & 133 & 18 & 0 & 21 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 21 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[319] $1^{141} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 888$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.22 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 21 & 4 & 0 & 4 \\ 12 & 21 & 76 & 15 & 0 & 16 \\ 2 & 4 & 15 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[320] $1^{131} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 896$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 18 & 2 & 1 & 3 \\ 3 & 4 & 32 & 4 & 0 & 4 \\ 18 & 32 & 193 & 21 & 0 & 25 \\ 2 & 4 & 21 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 25 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[321] $1^{141} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 896$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.21 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 1 & 3 \\ 3 & 4 & 26 & 4 & 0 & 4 \\ 14 & 26 & 118 & 18 & 0 & 20 \\ 2 & 4 & 18 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 20 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[322] $21^{115} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 904$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.1$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 75 & 12 & 7 \\ 20 & 25 & 100 & 16 & 9 \\ 75 & 100 & 373 & 60 & 35 \\ 12 & 16 & 60 & 9 & 5 \\ 7 & 9 & 35 & 5 & 2 \end{pmatrix}_{b_0}$$

[323] $1^{131} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 944$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.11 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

[363] $1^{11} 1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -206$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.71 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 12 & 3 & 0 & 4 \\ 7 & 12 & 26 & 8 & 0 & 10 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

[364] $1^{12} 1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -325$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.71 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 16 & 3 & 0 & 4 \\ 9 & 16 & 45 & 10 & 0 & 13 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

[365] $1^{12} 1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -406$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 19 & 3 & 0 & 4 \\ 11 & 19 & 68 & 13 & 0 & 16 \\ 2 & 3 & 13 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

4.971 $\text{NHC}(-2)(-3) + \text{su}_2 + \text{su}_3 + \text{so}_8 + \mathfrak{e}_6$

$T = 9$ $[0] 1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 + \mathfrak{e}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.23 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 9 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 9 & 7 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 + \mathfrak{e}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 7 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 + \mathfrak{e}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -334$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 12 & 0 & 6 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 + \mathfrak{e}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -931$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 17 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 17 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 13 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

4.972 $\text{NHC}(-2)(-3) + \text{su}_2 + \text{su}_3 + \text{so}_8 + \mathfrak{f}_4$

$T = 9$ $[0] 1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 13.27 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 9 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 9 & 7 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -215$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

15.2

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 7 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 12 & 0 & 6 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 17 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 17 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 13 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

4.973 NHC(-2)(-3) + $\textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 + \textcolor{red}{so}_8$

$$T = 9 \quad [0] 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 9 & 8 & 1 & 5 \\ 12 & 25 & 20 & 16 & 0 & 9 \\ 9 & 20 & 13 & 12 & 0 & 7 \\ 8 & 16 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 7 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[1] 1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 35 & 1 & 1 & 5 \\ 12 & 25 & 76 & 0 & 0 & 9 \\ 35 & 76 & 217 & 0 & 0 & 27 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 27 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 1 & 5 \\ 11 & 20 & 0 & 17 & 0 & 8 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 9 & 17 & 0 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 0 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 1 & 5 \\ 12 & 25 & 36 & 0 & 0 & 9 \\ 17 & 36 & 49 & 0 & 0 & 13 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 13 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[4] 1^{11} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 10 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[5] 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -133, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 1 & 5 \\ 11 & 20 & 17 & 0 & 0 & 8 \\ 9 & 17 & 13 & 0 & 0 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 10 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 13 & 15 & 0 & 7 \\ 9 & 17 & 15 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{11} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 8 & 1 & 1 & 5 \\ 11 & 20 & 16 & 0 & 0 & 8 \\ 8 & 16 & 10 & 0 & 0 & 6 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 26 & 1 & 1 & 5 \\ 12 & 25 & 56 & 0 & 0 & 9 \\ 26 & 56 & 118 & 0 & 0 & 20 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 20 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 34 & 1 & 1 & 5 \\ 12 & 25 & 72 & 0 & 0 & 9 \\ 34 & 72 & 202 & 0 & 0 & 26 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 26 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 8 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 8 & 16 & 10 & 12 & 0 & 6 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 17 & 10 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 17 & 33 & 49 & 27 & 0 & 13 \\ 10 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 16 & 17 & 0 & 8 \\ 9 & 16 & 10 & 12 & 0 & 6 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{11} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -244, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 26 & 1 & 1 & 5 \\ 11 & 20 & 50 & 0 & 0 & 8 \\ 26 & 50 & 118 & 0 & 0 & 20 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 20 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{111} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 17 & 1 & 1 & 5 \\ 11 & 20 & 33 & 0 & 0 & 8 \\ 17 & 33 & 49 & 0 & 0 & 13 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{113} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 51 & 1 & 1 & 5 \\ 12 & 25 & 108 & 0 & 0 & 9 \\ 51 & 108 & 457 & 0 & 0 & 39 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 39 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{113} 3^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 43 & 1 & 1 & 5 \\ 12 & 25 & 92 & 0 & 0 & 9 \\ 43 & 92 & 325 & 0 & 0 & 33 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 33 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{114} 3^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 52 & 1 & 1 & 5 \\ 12 & 25 & 112 & 0 & 0 & 9 \\ 52 & 112 & 478 & 0 & 0 & 40 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 40 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{114} 3^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 68 & 1 & 1 & 5 \\ 12 & 25 & 144 & 0 & 0 & 9 \\ 68 & 144 & 814 & 0 & 0 & 52 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 52 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{114} 3^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 60 & 1 & 1 & 5 \\ 12 & 25 & 128 & 0 & 0 & 9 \\ 60 & 128 & 634 & 0 & 0 & 46 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 46 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{111} 3^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 19 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 19 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{so}_8 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 17 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 17 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -735, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 18 & 9 & 1 & 5 \\ 11 & 20 & 33 & 17 & 0 & 8 \\ 18 & 33 & 49 & 27 & 0 & 13 \\ 9 & 17 & 27 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

4.974 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -16, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 4 \\ 4 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 6 & 3 & 0 & 4 & 4 \\ 3 & 4 & 4 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 0 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & 0 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 6 & 3 & 0 & 4 & 4 \\ 3 & 4 & 4 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

4.975 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 6, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.67 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 & -1 \\ 2 & 4 & 0 & 0 & 1 & 3 & 3 \\ 3 & 4 & 0 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 9, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.78 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 & -1 \\ 3 & 5 & 0 & 0 & 4 & 4 & 4 \\ 3 & 4 & 0 & -1 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 10, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.80 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 & -1 \\ 3 & 4 & 3 & 0 & 1 & 3 & 3 \\ 3 & 4 & 4 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 12, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 4 & 3 & 3 \\ 3 & 4 & 5 & 4 & 5 & 4 & 4 \\ 3 & 5 & 4 & 3 & 6 & 4 & 4 \\ 3 & 4 & 3 & 0 & 3 & 2 & 2 \\ 4 & 5 & 6 & 3 & 4 & 4 & 4 \\ 3 & 4 & 4 & 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 12, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 4 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[5] $1^{\textcolor{brown}{11}} 1^{\textcolor{blue}{11}} 3^{\textcolor{blue}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 4 & 2 & 0 & 3 \end{pmatrix} b_0$$

[6] $1^{\text{11}} 3^{\text{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[7] $1^{111} 3^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 4 & 3 \\ 3 & 4 & 5 & 4 & 5 & 4 \\ 4 & 5 & 4 & 3 & 6 & 4 \\ 3 & 4 & 3 & 0 & 3 & 2 \\ 4 & 5 & 6 & 3 & 4 & 4 \\ 3 & 4 & 4 & 2 & 4 & 3 \end{pmatrix} b_0$$

[8] $1^{12} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{su}_2 + \textcolor{brown}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 4 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[9] $1^{\textcolor{red}{1}}1^{\textcolor{red}{1}}31^{\textcolor{blue}{1}\textcolor{blue}{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[10] $1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 7 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}} 3 \ 1^{\textcolor{blue}{1}\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 28, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 4 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[12] $1^{11} 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 3 & 2 \\ 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & 0 & 2 & 4 & 3 \end{pmatrix} b_0$$

[13] $1^{11} 1^{11} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & 0 & -1 & 3 & 3 \end{pmatrix} b_0$$

[14] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 3 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[15] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 8 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 8 & 0 & 0 & 10 & 6 \\ 3 & 4 & 0 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[16] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[17] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[18] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 6 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 0 & 0 & 13 & 7 \\ 3 & 4 & 0 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[19] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 5 & 8 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 12 & 0 & 13 & 7 \\ 3 & 4 & 6 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[21] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[20] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 3 \\ 3 & 4 & 0 & 0 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 13 & 0 & 0 & 28 & 10 \\ 3 & 4 & 0 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[22] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[23] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[24] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 3 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[25] $1^{11} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[26] $1^{12} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[27] $1^{12} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[28] $1^{111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 3 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[29] $1^{111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 125, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 7 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[30] $1^{113} 3^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 1 & 15 & 0 & 3 \\ 16 & 22 & 15 & 84 & 0 & 16 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

[31] $1^{112} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[32] $1^{112} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[33] $1^{112} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 3 & 4 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[34] $1^{111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 6 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 6 & 9 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 7 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[35] $1^{112} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 285, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 8 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 6 & 9 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$T = 9$ [36] $1^{11} 3 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 2 \\ 1 & 2 & 0 & 3 & 0 & 4 & 2 \\ 1 & 1 & 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 3 & 4 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & 3 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

[37] $1^{11} 3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & 0 & 0 & 2 \\ 2 & 2 & 2 & 0 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[38] $1^{111} 3 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 1 & 3 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 4 & 7 & 5 \\ 1 & 1 & 0 & 4 & 0 & 4 & 2 \\ 2 & 3 & 0 & 7 & 4 & 5 & 5 \\ 2 & 2 & -1 & 5 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[39] $1^{11} 3 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 2 \\ 1 & 2 & 0 & 3 & 0 & 4 & 2 \\ 2 & 1 & 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 4 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & 3 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

[40] $1^1 3 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & 2 & 0 & 4 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 4 & 0 & 2 & 4 \\ 3 & 5 & 4 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

[42] $1^{11} 3 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 5 & 5 & 0 & 10 & 5 \\ 3 & 5 & 2 & 0 & 7 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 10 & 7 & 0 & 14 & 8 \\ 3 & 5 & 4 & -1 & 8 & 4 \end{pmatrix}_{b_0}$$

[41] $1^{11} 3 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 5 & 5 & 0 & 10 & 5 \\ 2 & 5 & 2 & 0 & 7 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 10 & 7 & 0 & 14 & 8 \\ 3 & 5 & 4 & -1 & 8 & 4 \end{pmatrix}_{b_0}$$

[43] $1^{111} 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 1 & 2 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 4 & 3 & 0 & 5 & 5 \\ 2 & 2 & 2 & 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

[44] $1^{111} 3^{211} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 1 & 2 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 3 & 4 & 3 & 0 & 5 & 5 \\ 2 & 2 & 2 & 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

[45] $1^{111} 3^{211} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 0 & 3 & 2 \\ 1 & 2 & 0 & 4 & 0 & 4 & 2 \\ 2 & 3 & 4 & 5 & 0 & 7 & 5 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 3 & 4 & 7 & 0 & 5 & 5 \\ 2 & 2 & 2 & 5 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$T = 10$ [51] $1^{1311} 4^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 & 3 \\ 2 & 1 & 1 & 3 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 3 & 4 \\ 3 & 2 & 0 & 3 & 1 & 3 \\ 3 & 3 & 0 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[50] $1^{11} 4^{1311} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 2 & 3 \\ 2 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 1 & 3 \\ 3 & 3 & 0 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[49] $1^{11} 4^{1311} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 6 & 4 & 3 & 6 \\ 4 & 2 & 2 & 4 \\ 3 & 2 & 1 & 3 \\ 6 & 4 & 3 & 6 \end{pmatrix}_{b_0}$$

[48] $1^{11} 4^{1311} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 6 & 4 & 11 & 6 \\ 4 & 2 & 8 & 4 \\ 11 & 8 & 19 & 11 \\ 6 & 4 & 11 & 6 \end{pmatrix}_{b_0}$$

[47] $1^{11} 4^{1311} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -2 & 1 & 0 \\ 3 & 4 & 2 & 1 & 2 & 4 \\ 3 & 4 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

[46] $1^{11} 4^{1311} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 1 & 2 & 3 \\ 3 & 3 & 1 & 2 & 2 & 4 \\ 3 & 3 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[53] $1^{111} 4^{1311} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 & 3 \\ 3 & 4 & 7 & 0 & 4 & 4 \\ 6 & 7 & 9 & 1 & 6 & 7 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 3 & 4 & 6 & 1 & 2 & 4 \\ 3 & 4 & 7 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

[52] $1^{111} 4^{1311} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 3 & 4 & 2 & 1 & 4 & 4 \\ 1 & 0 & 1 & -2 & 1 & 0 \\ 3 & 4 & 4 & 1 & 2 & 4 \\ 3 & 4 & 4 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

- [56] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 & 6 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 1 & 5 & 3 \\ 6 & 6 & 1 & 5 & 14 & 8 \\ 3 & 3 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0}$
- [57] $1^{11} 4^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 4 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 4 & 3 & 1 & 5 & 4 & 5 \\ 3 & 3 & 1 & 4 & 2 & 4 \\ 3 & 3 & 0 & 5 & 4 & 4 \end{pmatrix}_{b_0}$
- [58] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 & 2 & 3 \\ 2 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 3 & 2 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 1 & 0 & 2 \\ 3 & 2 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$
- [54] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 3 & 5 & 4 \\ 3 & 1 & 2 & 4 & 3 \\ 3 & 2 & 1 & 3 & 3 \\ 5 & 4 & 3 & 6 & 6 \\ 4 & 3 & 3 & 6 & 5 \end{pmatrix}_{b_0}$
- [55] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 & 4 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 1 & 3 & 3 \\ 4 & 3 & 1 & 3 & 5 & 5 \\ 3 & 3 & 0 & 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [59] $1^{11} 4^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 2 & 5 & 4 \\ 3 & 1 & 1 & 4 & 3 \\ 2 & 1 & 0 & 2 & 2 \\ 5 & 4 & 2 & 6 & 6 \\ 4 & 3 & 2 & 6 & 5 \end{pmatrix}_{b_0}$
- [60] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 5 & 5 & 4 \\ 3 & 1 & 4 & 3 & 3 \\ 5 & 4 & 6 & 5 & 6 \\ 5 & 3 & 5 & 3 & 5 \\ 4 & 3 & 6 & 5 & 5 \end{pmatrix}_{b_0}$
- [61] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 5 & 4 & 4 \\ 3 & 1 & 3 & 4 & 3 \\ 5 & 3 & 3 & 4 & 5 \\ 4 & 4 & 4 & 3 & 5 \\ 4 & 3 & 5 & 5 & 5 \end{pmatrix}_{b_0}$
- [62] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 5 & 6 & 3 \\ 2 & 1 & 1 & 5 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 5 & 5 & 0 & 8 & 11 & 6 \\ 6 & 6 & 1 & 11 & 14 & 8 \\ 3 & 3 & 0 & 6 & 8 & 4 \end{pmatrix}_{b_0}$
- [63] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 5 & 7 & 4 \\ 3 & 1 & 3 & 5 & 3 \\ 5 & 3 & 3 & 7 & 5 \\ 7 & 5 & 7 & 12 & 8 \\ 4 & 3 & 5 & 8 & 5 \end{pmatrix}_{b_0}$
- [64] $1^{11} 4^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 5 & 4 & 4 \\ 3 & 1 & 4 & 4 & 3 \\ 5 & 4 & 6 & 5 & 6 \\ 4 & 4 & 5 & 3 & 5 \\ 4 & 3 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

[65] $1^{111} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 5 & 3 \\ 2 & 1 & 1 & 3 & 5 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 3 & 3 & 1 & 2 & 5 & 4 \\ 5 & 5 & 0 & 5 & 8 & 6 \\ 3 & 3 & 0 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

[66] $1^{11} 4^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 2 & 4 & 4 \\ 3 & 1 & 1 & 4 & 3 \\ 2 & 1 & 0 & 2 & 2 \\ 4 & 4 & 2 & 3 & 5 \\ 4 & 3 & 2 & 5 & 5 \end{pmatrix}_{b_0}$$

[67] $1^{111} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 2 & 5 & 4 \\ 3 & 1 & 1 & 3 & 3 \\ 2 & 1 & 0 & 2 & 2 \\ 5 & 3 & 2 & 3 & 5 \\ 4 & 3 & 2 & 5 & 5 \end{pmatrix}_{b_0}$$

[68] $1^{111} 4^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 3 & 0 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 & 2 & 4 \\ 3 & 3 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[69] $1^{111} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 6 & 3 \\ 2 & 1 & 1 & 6 & 5 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 6 & 6 & 1 & 14 & 12 & 8 \\ 6 & 5 & 1 & 12 & 9 & 7 \\ 3 & 3 & 0 & 8 & 7 & 4 \end{pmatrix}_{b_0}$$

[70] $1^{111} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 3 & 5 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 3 & 0 & 1 & 5 & 3 \\ 6 & 5 & 1 & 5 & 9 & 7 \\ 3 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[74] $1^{112} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 2 & 10 & 4 \\ 3 & 1 & 1 & 7 & 3 \\ 2 & 1 & 0 & 4 & 2 \\ 10 & 7 & 4 & 21 & 11 \\ 4 & 3 & 2 & 11 & 5 \end{pmatrix}_{b_0}$$

[71] $1^{111} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 9 & 3 \\ 2 & 1 & 1 & 3 & 8 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 3 & 3 & 1 & 2 & 9 & 4 \\ 9 & 8 & 1 & 9 & 27 & 11 \\ 3 & 3 & 0 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

[72] $1^{111} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 3 & 8 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 3 & 0 & 1 & 8 & 3 \\ 9 & 8 & 1 & 8 & 27 & 11 \\ 3 & 3 & 0 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[73] $1^{111} 4^{11} 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 3 & 0 & 1 & 6 & 3 \\ 6 & 6 & 1 & 6 & 14 & 8 \\ 3 & 3 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[75] $1^{111} 1^{11} 4 131^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -33$, $\det = 72$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 6 & 3 \\ 2 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 3 & 3 & 1 & 2 & 7 & 4 \\ 6 & 6 & 1 & 7 & 14 & 8 \\ 3 & 3 & 0 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$T = 11$ [76] $1^{11} 5 1^1 231^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -11$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 5 & 5 & 0 & 15 & 5 \\ 2 & 5 & 2 & 0 & 11 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 15 & 11 & 0 & 37 & 13 \\ 3 & 5 & 4 & -1 & 13 & 4 \end{pmatrix}_{b_0}$$

[77] $1^{111} 5 1^1 231^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -12$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 2 \\ 1 & 2 & 0 & 3 & 0 & 8 & 2 \\ 1 & 1 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 6 & 8 & 8 & 0 & 31 & 11 \\ 2 & 2 & 2 & 3 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[78] $1^{11} 5 1^1 231^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -47$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 5 & 5 & 0 & 20 & 5 \\ 2 & 5 & 2 & 0 & 15 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 20 & 15 & 0 & 63 & 17 \\ 3 & 5 & 4 & -1 & 17 & 4 \end{pmatrix}_{b_0}$$

[79] $2 3 15 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -63$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 3 & 3 & 1 & 4 & 6 & 3 \\ 3 & 2 & 4 & 2 & 6 & 4 \\ 5 & 5 & 6 & 6 & 9 & 7 \\ 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

[80] $2 3 15 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -88$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 & 3 \\ 4 & 5 & 5 & 3 & 5 & 5 \\ 3 & 5 & 2 & 2 & 4 & 4 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 3 & 5 & 4 & 2 & 2 & 4 \\ 3 & 5 & 4 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

[81] $1^{11} 5 1^1 231^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -96$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.5$

$T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 5 & 5 & 0 & 35 & 5 \\ 2 & 5 & 2 & 0 & 26 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 18 & 35 & 26 & 0 & 202 & 30 \\ 3 & 5 & 4 & -1 & 30 & 4 \end{pmatrix}_{b_0}$$

[82] $2 3 15 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -106$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.5$

$T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 6 & 3 \\ 4 & 5 & 5 & 3 & 10 & 5 \\ 3 & 5 & 2 & 2 & 8 & 4 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 6 & 10 & 8 & 4 & 14 & 8 \\ 3 & 5 & 4 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

[83] $2 3 15 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -29$, $\det = -124$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} =$

8.0

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 6 & 3 \\ 4 & 5 & 10 & 3 & 10 & 5 \\ 6 & 10 & 14 & 4 & 16 & 8 \\ 2 & 3 & 4 & 0 & 4 & 2 \\ 6 & 10 & 16 & 4 & 14 & 8 \\ 3 & 5 & 8 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[84] \ 2 \ 3 \ 15 \ 1^{\overset{1}{1} \overset{13}{3}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -136, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 9 & 3 \\ 4 & 5 & 5 & 3 & 15 & 5 \\ 3 & 5 & 2 & 2 & 12 & 4 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 9 & 15 & 12 & 6 & 34 & 12 \\ 3 & 5 & 4 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{\overset{1}{1}} \ 5 \ 1^{\overset{3}{3}} \ 231^{\overset{11}{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -143, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 26 & 3 \\ 3 & 5 & 5 & 0 & 50 & 5 \\ 2 & 5 & 2 & 0 & 37 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 26 & 50 & 37 & 0 & 419 & 43 \\ 3 & 5 & 4 & -1 & 43 & 4 \end{pmatrix}_{b_0}$$

$$[86] \ 1^{\overset{1}{1}} \ 5 \ 1^{\overset{2}{2}} \ 231^{\overset{11}{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -152, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 5 & 5 & 0 & 40 & 5 \\ 2 & 5 & 2 & 0 & 30 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 20 & 40 & 30 & 0 & 258 & 34 \\ 3 & 5 & 4 & -1 & 34 & 4 \end{pmatrix}_{b_0}$$

$$[87] \ 2 \ 3 \ 15 \ 1^{\overset{1}{1} \overset{13}{3}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -154, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.3$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 9 & 3 \\ 4 & 5 & 10 & 3 & 15 & 5 \\ 6 & 10 & 14 & 4 & 24 & 8 \\ 2 & 3 & 4 & 0 & 6 & 2 \\ 9 & 15 & 24 & 6 & 34 & 12 \\ 3 & 5 & 8 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[88] \ 2 \ 3 \ 15 \ 1^{\overset{1}{1} \overset{14}{4}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -178, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 12 & 3 \\ 4 & 5 & 5 & 3 & 20 & 5 \\ 3 & 5 & 2 & 2 & 16 & 4 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 12 & 20 & 16 & 8 & 62 & 16 \\ 3 & 5 & 4 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[90] \ 2 \ 3 \ 15 \ 1^{\overset{1}{1} \overset{13}{3}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -184, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.6$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 9 & 3 \\ 4 & 5 & 15 & 3 & 15 & 5 \\ 9 & 15 & 34 & 6 & 36 & 12 \\ 2 & 3 & 6 & 0 & 6 & 2 \\ 9 & 15 & 36 & 6 & 34 & 12 \\ 3 & 5 & 12 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[89] \ 2 \ 3 \ 15 \ 1^{\overset{1}{1} \overset{11}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -184, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 3 & 3 \\ 4 & 5 & 5 & 8 & 5 & 5 \\ 3 & 5 & 2 & 6 & 4 & 4 \\ 5 & 8 & 6 & 8 & 6 & 6 \\ 3 & 5 & 4 & 6 & 2 & 4 \\ 3 & 5 & 4 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[91] \ 2 \ 3 \ 15 \ 1^{\overset{1}{1} \overset{12}{4}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 12 & 3 \\ 4 & 5 & 10 & 3 & 20 & 5 \\ 6 & 10 & 14 & 4 & 32 & 8 \\ 2 & 3 & 4 & 0 & 8 & 2 \\ 12 & 20 & 32 & 8 & 62 & 16 \\ 3 & 5 & 8 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

- [92] $2 \overset{1^{134}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -226, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 12 & 3 \\ 4 & 5 & 15 & 3 & 20 & 5 \\ 9 & 15 & 34 & 6 & 48 & 12 \\ 2 & 3 & 6 & 0 & 8 & 2 \\ 12 & 20 & 48 & 8 & 62 & 16 \\ 3 & 5 & 12 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$
- [93] $2 \overset{1^{115}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -232, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.1$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 15 & 3 \\ 4 & 5 & 5 & 3 & 25 & 5 \\ 3 & 5 & 2 & 2 & 20 & 4 \\ 2 & 3 & 2 & 0 & 10 & 2 \\ 15 & 25 & 20 & 10 & 98 & 20 \\ 3 & 5 & 4 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$
- [94] $2 \overset{1^{125}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -250, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 15 & 3 \\ 4 & 5 & 10 & 3 & 25 & 5 \\ 6 & 10 & 14 & 4 & 40 & 8 \\ 2 & 3 & 4 & 0 & 10 & 2 \\ 15 & 25 & 40 & 10 & 98 & 20 \\ 3 & 5 & 8 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$
- [95] $2 \overset{1^{144}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -268, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 12 & 3 \\ 4 & 5 & 20 & 3 & 20 & 5 \\ 12 & 20 & 62 & 8 & 64 & 16 \\ 2 & 3 & 8 & 0 & 8 & 2 \\ 12 & 20 & 64 & 8 & 62 & 16 \\ 3 & 5 & 16 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$
- [96] $2 \overset{1^{135}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -280, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.9$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 15 & 3 \\ 4 & 5 & 15 & 3 & 25 & 5 \\ 9 & 15 & 34 & 6 & 60 & 12 \\ 2 & 3 & 6 & 0 & 10 & 2 \\ 15 & 25 & 60 & 10 & 98 & 20 \\ 3 & 5 & 12 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$
- [97] $2 \overset{1^{116}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -298, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 18 & 3 \\ 4 & 5 & 5 & 3 & 30 & 5 \\ 3 & 5 & 2 & 2 & 24 & 4 \\ 2 & 3 & 2 & 0 & 12 & 2 \\ 18 & 30 & 24 & 12 & 142 & 24 \\ 3 & 5 & 4 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$
- [98] $2 \overset{1^{126}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -316, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 18 & 3 \\ 4 & 5 & 10 & 3 & 30 & 5 \\ 6 & 10 & 14 & 4 & 48 & 8 \\ 2 & 3 & 4 & 0 & 12 & 2 \\ 18 & 30 & 48 & 12 & 142 & 24 \\ 3 & 5 & 8 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$
- [99] $2 \overset{1^{145}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -322, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.1$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 15 & 3 \\ 4 & 5 & 20 & 3 & 25 & 5 \\ 12 & 20 & 62 & 8 & 80 & 16 \\ 2 & 3 & 8 & 0 & 10 & 2 \\ 15 & 25 & 80 & 10 & 98 & 20 \\ 3 & 5 & 16 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$
- [100] $2 \overset{1^{136}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -346, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.1$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 18 & 3 \\ 4 & 5 & 15 & 3 & 30 & 5 \\ 9 & 15 & 34 & 6 & 72 & 12 \\ 2 & 3 & 6 & 0 & 12 & 2 \\ 18 & 30 & 72 & 12 & 144 & 24 \\ 3 & 5 & 12 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[101] $2 \overset{1^{117}}{3} \overset{1^1}{15} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -376, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.3$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 21 & 3 \\ 4 & 5 & 3 & 2 & 35 & 5 \\ 3 & 5 & 2 & 2 & 28 & 4 \\ 2 & 3 & 2 & 0 & 14 & 2 \\ 21 & 35 & 28 & 14 & 194 & 28 \\ 3 & 5 & 4 & 2 & 28 & 4 \end{pmatrix}_{b_0}$$

[102] $2 \overset{1^{127}}{3} \overset{1^1}{15} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -394, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.8$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 21 & 3 \\ 4 & 5 & 10 & 3 & 35 & 5 \\ 6 & 10 & 14 & 4 & 56 & 8 \\ 2 & 3 & 4 & 0 & 14 & 2 \\ 21 & 35 & 56 & 14 & 194 & 28 \\ 3 & 5 & 8 & 2 & 28 & 4 \end{pmatrix}_{b_0}$$

[103] $2 \overset{1^{137}}{3} \overset{1^1}{15} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -424, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.1$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 21 & 3 \\ 4 & 5 & 15 & 3 & 35 & 5 \\ 9 & 15 & 34 & 6 & 84 & 12 \\ 2 & 3 & 6 & 0 & 14 & 2 \\ 21 & 35 & 84 & 14 & 194 & 28 \\ 3 & 5 & 12 & 2 & 28 & 4 \end{pmatrix}_{b_0}$$

[104] $22 \overset{1^{111}}{3} \overset{1^1}{15} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -568, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 5 & 5 & 5 & 3 \\ 6 & 14 & 13 & 13 & 13 & 8 \\ 5 & 13 & 9 & 11 & 11 & 7 \\ 5 & 13 & 11 & 8 & 11 & 6 \\ 5 & 13 & 11 & 11 & 9 & 7 \\ 3 & 8 & 7 & 6 & 7 & 4 \end{pmatrix}_{b_0}$$

$T = 12$ [105] $1 \overset{1^1}{1^{11}} \overset{1^1}{5} 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 1 & 2 & 1 & 1 & 0 & 6 & 3 \\ 1 & 1 & 1 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 6 & 4 & 0 & 12 & 8 \\ 2 & 2 & 3 & 2 & -1 & 8 & 4 \end{pmatrix}_{b_0}$$

[106] $1 \overset{1^1}{1^{11}} \overset{1^1}{5} 1321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 2 \\ 1 & 2 & 1 & 1 & 0 & 11 & 3 \\ 1 & 1 & 1 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 6 & 11 & 7 & 0 & 40 & 14 \\ 2 & 2 & 3 & 2 & -1 & 14 & 4 \end{pmatrix}_{b_0}$$

[107] $1 \overset{1^1}{1^{13}} \overset{1^1}{5} 1321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 7 & 2 \\ 1 & 2 & 1 & 1 & 0 & 12 & 3 \\ 1 & 1 & 1 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 7 & 12 & 9 & 0 & 56 & 16 \\ 2 & 2 & 3 & 2 & -1 & 16 & 4 \end{pmatrix}_{b_0}$$

[108] $1 \overset{1^1}{1^{12}} \overset{1^1}{5} 1321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 1 & 1 & 0 & 12 & 3 \\ 1 & 1 & 1 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 8 & 12 & 8 & 0 & 54 & 16 \\ 2 & 2 & 3 & 2 & -1 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{12} \overset{1^1}{5} \ 1321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 48, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 2 \\ 1 & 2 & 1 & 1 & 0 & 17 & 3 \\ 1 & 1 & 1 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 10 & 17 & 11 & 0 & 102 & 22 \\ 2 & 2 & 3 & 2 & -1 & 22 & 4 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{13} \overset{1^1}{5} \ 1321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 87, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 12 & 2 \\ 1 & 2 & 1 & 1 & 0 & 18 & 3 \\ 1 & 1 & 1 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 9 & 12 & 18 & 12 & 0 & 124 & 24 \\ 2 & 2 & 3 & 2 & -1 & 24 & 4 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [111] \ 1^{11} \overset{1^1}{6} \ 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 13, \ T_{\max} = 13, \ \Delta = -4, \ \det = -4, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 1 & 0 & 4 & 3 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 1 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 3 & 0 & 5 & 5 \\ 2 & 3 & 2 & 2 & -1 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{11} \overset{1^1}{6} \ 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 13, \ T_{\max} = 13, \ \Delta = -4, \ \det = -16, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 2 & 0 & 1 & 0 & 5 & 2 \\ 1 & 1 & 1 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 5 & 4 & 0 & 12 & 8 \\ 2 & 3 & 2 & 2 & -1 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{11} \overset{1^1}{6} \ 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 13, \ T_{\max} = 13, \ \Delta = -4, \ \det = -16, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 1 & 0 & 8 & 3 \\ 1 & 2 & 0 & 1 & 0 & 6 & 2 \\ 1 & 1 & 1 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 6 & 5 & 0 & 25 & 11 \\ 2 & 3 & 2 & 2 & -1 & 11 & 4 \end{pmatrix}_{b_0}$$

$$T = 16 \quad [114] \ 1^{11} \overset{1^{12}31}{6} \ 1316131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 16, \ T_{\max} = 16, \ \Delta = -3, \ \det = 0, \ (n_+, n_-, n_0) = (1, 16, 1), \ c_{\text{ext}} = 7.0$$

4.976 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$

$$T = 9 \quad [0] \ 1^{111} \overset{1^1}{3} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -33, \ \det = -4, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} \overset{1^1}{3} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -33, \ \det = -5, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.80 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 1 & 3 & 3 \\ 3 & 5 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & 0 & -1 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} \overset{1^1}{3} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -33, \ \det = -7, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.86 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 6 & 3 \\ 3 & 4 & 9 & 0 & 8 & 9 & 4 \\ 5 & 9 & 13 & 0 & 12 & 15 & 7 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 5 & 8 & 12 & 0 & 10 & 12 & 6 \\ 6 & 9 & 15 & 0 & 12 & 13 & 7 \\ 3 & 4 & 7 & -1 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{111} 3 \quad 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -21, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 0 & 4 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 0 & 6 & 13 & 7 \\ 3 & 4 & 0 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111} 3 \quad 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -32, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 6 & 3 \\ 6 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{121} 3 \quad 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 0 & 8 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 5 & 8 & 0 & 0 & 10 & 12 & 6 \\ 6 & 9 & 0 & 0 & 12 & 13 & 7 \\ 3 & 4 & 0 & -1 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{121} 3 \quad 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 6 & 3 \\ 3 & 4 & 8 & 0 & 9 & 9 & 4 \\ 5 & 8 & 10 & 0 & 12 & 12 & 6 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 12 & 0 & 13 & 15 & 7 \\ 6 & 9 & 12 & 0 & 15 & 13 & 7 \\ 3 & 4 & 6 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11} 3 \quad 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -45, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 0 & 5 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 0 & 4 & 9 & 4 \\ 6 & 9 & 0 & 0 & 9 & 13 & 7 \\ 3 & 4 & 0 & -1 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{111} 3 \quad 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -47, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 4 \\ 3 & 4 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 5 & 9 & 6 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 3 & -1 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{112} 3 \quad 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -47, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 0 & 4 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 0 & 1 & 9 & 3 \\ 8 & 13 & 0 & 0 & 9 & 28 & 10 \\ 3 & 4 & 0 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{12} 3 \quad 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -50, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 0 & 5 & 13 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 0 & 4 & 12 & 4 \\ 8 & 13 & 0 & 0 & 12 & 28 & 10 \\ 3 & 4 & 0 & -1 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1112} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -50, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 9 & 3 \\ 3 & 4 & 4 & 0 & 9 & 13 & 4 \\ 2 & 4 & 1 & 0 & 6 & 9 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 5 & 9 & 6 & 0 & 13 & 21 & 7 \\ 9 & 13 & 9 & 0 & 21 & 28 & 10 \\ 3 & 4 & 3 & -1 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{112} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -55, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 5 & 0 & 4 & 13 & 4 \\ 3 & 5 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 9 & 3 \\ 8 & 13 & 12 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -56, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 6 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 6 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{111} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -76, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -77, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 6 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -87, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 3 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 5 & 9 & 6 & 0 & 13 & 15 & 7 \\ 6 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 3 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -87, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 0 & 4 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 0 & 6 & 13 & 7 \\ 3 & 4 & 0 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{112} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -100, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 6 & 6 & 3 \\ 3 & 4 & 0 & 0 & 9 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 0 & 0 & 13 & 15 & 7 \\ 6 & 9 & 0 & 0 & 15 & 13 & 7 \\ 3 & 4 & 0 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1112} 1^{111} 3 \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 13 & 4 \\ 3 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 9 & 3 \\ 8 & 13 & 9 & 0 & 9 & 28 & 10 \\ 3 & 4 & 3 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1111} 1^{111} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -264, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 6 & 0 & 13 & 15 & 7 \\ 6 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 3 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1112} 1^{111} 3 \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -271, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.00 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 13 & 4 \\ 2 & 4 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 9 & 3 \\ 8 & 13 & 9 & 0 & 9 & 28 & 10 \\ 3 & 4 & 3 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1112} 1^{111} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -338, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 8 & 3 \\ 3 & 4 & 4 & 0 & 9 & 13 & 4 \\ 2 & 4 & 1 & 0 & 6 & 9 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 6 & 0 & 13 & 21 & 7 \\ 8 & 13 & 9 & 0 & 21 & 28 & 10 \\ 3 & 4 & 3 & -1 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1111} 1^{111} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -348, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 3 & 4 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 6 & 0 & 13 & 15 & 7 \\ 6 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 3 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1112} 1^{111} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -407, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10.00 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 4 & 0 & 9 & 13 & 4 \\ 3 & 4 & 1 & 0 & 6 & 9 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 6 & 0 & 13 & 21 & 7 \\ 8 & 13 & 9 & 0 & 21 & 28 & 10 \\ 3 & 4 & 3 & -1 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [41] \quad 1^{11} 1^{11} 3 \quad 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -5, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 3 \\ 3 & 5 & 5 & 0 & 5 & 10 & 5 \\ 2 & 5 & 2 & 0 & 4 & 7 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 4 & 0 & 2 & 7 & 4 \\ 5 & 10 & 7 & 0 & 7 & 14 & 8 \\ 3 & 5 & 4 & -1 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [45] \quad 1^{11} 1^{11} 4 \quad 131^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -3, \quad \det = -9, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 5 & 4 & 4 \\ 3 & 1 & 1 & 4 & 4 & 3 \\ 2 & 1 & 0 & 2 & 2 & 2 \\ 5 & 4 & 2 & 6 & 5 & 6 \\ 4 & 4 & 2 & 5 & 3 & 5 \\ 4 & 3 & 2 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{11} 1^{111} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 5 & 5 & 4 & 4 \\ 3 & 1 & 4 & 3 & 4 & 3 \\ 5 & 4 & 6 & 5 & 5 & 6 \\ 5 & 3 & 5 & 3 & 4 & 5 \\ 4 & 4 & 5 & 4 & 3 & 5 \\ 4 & 3 & 6 & 5 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{11} 1^{111} 4 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 5 & 4 & 2 & 4 \\ 3 & 1 & 3 & 4 & 1 & 3 \\ 5 & 3 & 3 & 4 & 2 & 5 \\ 4 & 4 & 4 & 3 & 2 & 5 \\ 2 & 1 & 2 & 2 & 0 & 2 \\ 4 & 3 & 5 & 5 & 2 & 5 \end{pmatrix}_{b_0}$$

4.977 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{111} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 6 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 & 9 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 1 & 3 & 6 & 3 \\ 3 & 5 & 0 & 0 & 3 & 4 & 9 & 4 \\ 6 & 9 & 0 & 0 & 6 & 9 & 13 & 7 \\ 3 & 4 & 0 & -1 & 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 2 \ 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 13 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 12 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 9 & 3 \\ 8 & 13 & 12 & 0 & 9 & 9 & 28 & 10 \\ 3 & 4 & 4 & -1 & 3 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 2 \ 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 6 & 6 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 6 & 6 & 3 \\ 6 & 9 & 6 & 0 & 6 & 13 & 15 & 7 \\ 6 & 9 & 6 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & 3 & -1 & 3 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 6 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 9 & 4 \\ 2 & 4 & 1 & 0 & 3 & 6 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 & 3 \\ 6 & 9 & 6 & 0 & 6 & 13 & 6 & 7 \\ 3 & 4 & 3 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 1^{11} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 9 & 4 \\ 3 & 4 & 1 & 0 & 6 & 3 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 5 & 9 & 6 & 0 & 13 & 6 & 15 & 7 \\ 2 & 4 & 3 & 0 & 6 & 1 & 6 & 3 \\ 6 & 9 & 6 & 0 & 15 & 6 & 13 & 7 \\ 3 & 4 & 3 & -1 & 7 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 & 3 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 1 & 3 & 2 \\ 3 & 5 & 0 & 0 & 3 & 4 & 3 \\ 2 & 3 & 0 & 0 & 2 & 3 & 0 \\ 3 & 4 & 0 & -1 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$4.979 \quad \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 1 & 2 & 3 \\ 3 & 5 & 0 & 0 & 3 & 4 & 3 \\ 2 & 3 & 0 & 0 & 2 & 3 & 0 \\ 3 & 4 & 0 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 3 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 0 & 5 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 0 & 2 \\ 3 & 4 & 0 & -1 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 3 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}} 3^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 3 & 4 \\ 2 & 4 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}} 3^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 3 & 4 \\ 2 & 4 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \ 1^{\overline{4}} 4 \ 13^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 5 & 1 & 4 \\ 3 & 1 & 1 & 4 & 0 & 3 \\ 2 & 1 & 0 & 2 & 0 & 2 \\ 5 & 4 & 2 & 6 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 4 & 3 & 2 & 6 & 0 & 5 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [8] \quad 1^1 5^1 1^1 2^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 5 & 5 & 0 & 15 & 4 & 5 \\ 2 & 5 & 2 & 0 & 11 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 8 & 15 & 11 & 0 & 37 & 10 & 13 \\ 2 & 4 & 3 & 0 & 10 & 1 & 3 \\ 3 & 5 & 4 & -1 & 13 & 3 & 4 \end{pmatrix}_{b_0}$$

4.980 $\text{NHC}(-2)(-3) + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2$

$$T = 10 \quad [1] \quad 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -5, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 3 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 0 & 2 & 0 & 3 & 2 \\ 3 & 5 & 0 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & 0 & -1 & 3 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -5, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 3 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & -1 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

4.981 $\text{NHC}(-2)(-3) + \textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \quad 1^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 21, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 2 \\ 3 & 4 & 0 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 36, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 61, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 3 & 4 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [5] \quad 1^1 4^1 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 11, \quad \Delta = -35, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 1 & 3 \\ 2 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 4 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 11, \quad \Delta = -35, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 4 & 0 & 0 & 4 \\ 3 & 4 & 2 & 1 & 0 & 4 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 4 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 4 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 11, \quad \Delta = -35, \quad \det = 11, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.91 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 1 & 4 \\ 3 & 1 & 1 & 0 & 3 \\ 2 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 4 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 4 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 11, \quad \Delta = -35, \quad \det = 11, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.91 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 4 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 11, \quad \Delta = -35, \quad \det = 32, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 5 & 1 & 4 \\ 3 & 1 & 4 & 0 & 3 \\ 5 & 4 & 6 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 4 & 3 & 6 & 0 & 5 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [16] \quad 1^{11} 5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \\ 4 & 7 & 9 & 6 & 5 & 7 \\ 2 & 3 & 6 & 1 & 3 & 3 \\ 2 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11} 5 123^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 5 & 4 & 3 & 5 \\ 1 & 1 & 0 & 4 & 0 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 & 0 & 2 \\ 2 & 2 & -1 & 5 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11} 5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 1 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 & 2 \\ 3 & 3 & 5 & 13 & 0 & 4 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 4 & 0 & 0 & 2 \\ 2 & 2 & 2 & 7 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 21^{11} 5 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 2 & 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[11] \quad & 1^{11} \overset{1}{5} 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -2, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 4 & 3 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 2 & 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[10] \quad & 1^{11} \overset{1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -2, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 7 & 2 & 2 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \\ 7 & 7 & 9 & 6 & 5 & 7 \\ 2 & 3 & 6 & 1 & 3 & 3 \\ 2 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[9] \quad & 1^{11} \overset{1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -2, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 1 & 2 \\ 1 & 2 & 0 & 7 & 0 & 2 & 2 \\ 5 & 6 & 7 & 35 & 0 & 6 & 11 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 6 & 0 & 0 & 2 \\ 2 & 2 & 2 & 11 & -1 & 2 & 3 \end{pmatrix}_{b_0} \\
[14] \quad & 1^{11} \overset{1}{5} 1^{11} 23^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -2, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 0 & 1 & 6 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 5 & 6 & 0 & 31 & 8 & 7 & 11 \\ 1 & 1 & 0 & 8 & 0 & 2 & 2 \\ 1 & 1 & 0 & 7 & 2 & 0 & 2 \\ 2 & 2 & -1 & 11 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[18] \quad & 1^{12} \overset{1}{5} 13^1 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -4, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 7.5 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 9 & 2 & 2 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 9 & 14 & 42 & 12 & 10 & 14 \\ 2 & 3 & 12 & 1 & 3 & 3 \\ 2 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[17] \quad & 1^{11} \overset{1}{5} 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -4, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[21] \quad & 1^{13} \overset{1}{5} 13^1 1^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 7.8 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 12 & 3 & 2 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 12 & 21 & 97 & 18 & 15 & 21 \\ 3 & 3 & 18 & 1 & 3 & 3 \\ 2 & 3 & 15 & 3 & 1 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[19] \quad & 1^{12} \overset{1}{5} 13^1 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 7.5 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 8 & 3 & 2 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 8 & 14 & 42 & 12 & 10 & 14 \\ 3 & 3 & 12 & 1 & 3 & 3 \\ 2 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[20] \quad & 1^{11} \overset{1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) =
\end{aligned}$$

$$(1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 2 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \\ 4 & 7 & 9 & 6 & 5 & 7 \\ 3 & 3 & 6 & 1 & 3 & 3 \\ 2 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11} \overset{1}{5} 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -7, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 3 & 0 & 4 & 5 \\ 2 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 5 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11} \overset{1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -7, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 5 & 11 & 0 & 4 & 5 \\ 6 & 11 & 22 & 0 & 7 & 10 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 7 & 0 & 1 & 3 \\ 3 & 5 & 10 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{11} \overset{12}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -10, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 2 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \\ 6 & 7 & 9 & 6 & 5 & 7 \\ 2 & 3 & 6 & 1 & 3 & 3 \\ 2 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{11} \overset{11}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -10, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \\ 5 & 7 & 9 & 6 & 5 & 7 \\ 2 & 3 & 6 & 1 & 3 & 3 \\ 2 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{13} \overset{12}{5} 13^1 1^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -10, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 2 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 14 & 21 & 97 & 18 & 15 & 21 \\ 2 & 3 & 18 & 1 & 3 & 3 \\ 2 & 3 & 15 & 3 & 1 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{11} \overset{1}{5} 123^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -11, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 4 & 5 \\ 2 & 5 & 2 & 0 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 5 & 4 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{11} \overset{1}{5} 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 5 & 19 & 0 & 4 & 5 \\ 10 & 19 & 65 & 0 & 12 & 17 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 12 & 0 & 1 & 3 \\ 3 & 5 & 17 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{11} \overset{1}{5} 123^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -11, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0} \\
[32] \quad & 1 \overset{1}{5} 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -11, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 & 2 & 3 \\ 2 & 3 & 1 & 2 & 3 & 3 \\ 2 & 3 & 2 & 1 & 2 & 3 \\ 2 & 2 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[29] \quad & 1 \overset{1}{5} 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -11, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0} \\
[28] \quad & 1 \overset{1}{5} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -11, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 3 \\ 3 & 5 & 5 & 3 & 4 & 5 \\ 2 & 5 & 2 & 2 & 3 & 4 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 2 & 4 & 3 & 2 & 1 & 3 \\ 3 & 5 & 4 & 2 & 3 & 4 \end{pmatrix}_{b_0} \\
[33] \quad & 1 \overset{1}{12} 5^1 13^1 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -16, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 7.5 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 12 & 2 & 2 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 12 & 14 & 42 & 12 & 10 & 14 \\ 2 & 3 & 12 & 1 & 3 & 3 \\ 2 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[34] \quad & 1 \overset{1}{12} 5^1 13^1 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -16, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 7.5 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 10 & 2 & 2 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 10 & 14 & 42 & 12 & 10 & 14 \\ 2 & 3 & 12 & 1 & 3 & 3 \\ 2 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[35] \quad & 1 \overset{1}{15} 13^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 5 & 13 & 0 & 4 & 5 \\ 6 & 13 & 25 & 0 & 8 & 11 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 8 & 0 & 1 & 3 \\ 3 & 5 & 11 & -1 & 3 & 4 \end{pmatrix}_{b_0} \\
[36] \quad & 1 \overset{1}{15} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 & 2 \\ 2 & 2 & 4 & 5 & 0 & 3 & 5 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 3 & 0 & 0 & 2 \\ 2 & 2 & 2 & 5 & -1 & 2 & 3 \end{pmatrix}_{b_0} \\
[37] \quad & 1 \overset{1}{15} 13^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = \\
& (1, 11, 0), \quad c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 5 & 27 & 0 & 4 & 5 \\ 14 & 27 & 130 & 0 & 17 & 24 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 17 & 0 & 1 & 3 \\ 3 & 5 & 24 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^1 \overset{1}{5} 13^1 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 5 & 5 & 8 & 4 & 5 \\ 2 & 5 & 2 & 6 & 3 & 4 \\ 5 & 8 & 6 & 8 & 5 & 6 \\ 2 & 4 & 3 & 5 & 1 & 3 \\ 3 & 5 & 4 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[39] \quad 21^1 \overset{1}{5} 13^1 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 5 & 5 & 8 & 4 & 5 \\ 2 & 5 & 2 & 6 & 3 & 4 \\ 5 & 8 & 6 & 8 & 5 & 6 \\ 2 & 4 & 3 & 5 & 1 & 3 \\ 3 & 5 & 4 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^1 \overset{1}{5} 13^1 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 1 & 2 \\ 1 & 2 & 0 & 8 & 0 & 2 & 2 \\ 6 & 8 & 8 & 49 & 0 & 7 & 13 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 7 & 0 & 0 & 2 \\ 2 & 2 & 2 & 13 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^1 \overset{1}{5} 13^1 1^{\overline{21}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -20, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 2 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 11 & 14 & 42 & 12 & 10 & 14 \\ 2 & 3 & 12 & 1 & 3 & 3 \\ 2 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^1 \overset{1}{5} 13^1 1^{\overline{31}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -26, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 2 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 15 & 21 & 97 & 18 & 15 & 21 \\ 2 & 3 & 18 & 1 & 3 & 3 \\ 2 & 3 & 15 & 3 & 1 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^1 \overset{1}{5} 13^1 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -28, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.86 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 5 & 8 & 0 & 4 & 5 \\ 4 & 8 & 9 & 0 & 5 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 5 & 0 & 1 & 3 \\ 3 & 5 & 7 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^1 \overset{1}{5} 13^1 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -31, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 1 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 & 2 \\ 7 & 8 & 11 & 72 & 0 & 9 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 9 & 0 & 0 & 2 \\ 2 & 2 & 2 & 16 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^1 \overset{1}{5} 13^1 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -31, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 5 & 21 & 0 & 4 & 5 \\ 10 & 21 & 70 & 0 & 13 & 18 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 13 & 0 & 1 & 3 \\ 3 & 5 & 18 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{11}_5 5^{13} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -34, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 5 & 16 & 0 & 4 & 5 \\ 8 & 16 & 42 & 0 & 10 & 14 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 10 & 0 & 1 & 3 \\ 3 & 5 & 14 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{13}_5 5^{13} 1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -34, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 2 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 16 & 21 & 97 & 18 & 15 & 21 \\ 2 & 3 & 18 & 1 & 3 & 3 \\ 2 & 3 & 15 & 3 & 1 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{11}_3 5^{13} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -44, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 5 & 24 & 0 & 4 & 5 \\ 12 & 24 & 97 & 0 & 15 & 21 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 15 & 0 & 1 & 3 \\ 3 & 5 & 21 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[50] \quad 21^{11}_5 5^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -45, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 5 & 5 & 6 & 9 & 5 & 7 \\ 2 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{11}_5 5^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -45, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 0 & 2 & 5 & 0 & 1 & 2 \\ 1 & 2 & 0 & 6 & 0 & 2 & 2 \\ 4 & 5 & 6 & 21 & 0 & 5 & 9 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 5 & 0 & 0 & 2 \\ 2 & 2 & 2 & 9 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{11}_5 5^{11} 23^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -47, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 0 & 2 & 5 & 0 & 1 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 & 2 \\ 4 & 5 & 5 & 18 & 0 & 5 & 8 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 5 & 0 & 0 & 2 \\ 2 & 2 & 2 & 8 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{11}_3 5^{13} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -47, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 5 & 29 & 0 & 4 & 5 \\ 14 & 29 & 137 & 0 & 18 & 25 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 18 & 0 & 1 & 3 \\ 3 & 5 & 25 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{11}_4 5^{13} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -58, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 2 & 3 \\ 3 & 5 & 32 & 0 & 4 & 5 \\ 16 & 32 & 174 & 0 & 20 & 28 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 20 & 0 & 1 & 3 \\ 3 & 5 & 28 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{12} \overset{1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -58, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 1 & 2 \\ 1 & 2 & 0 & 7 & 0 & 2 & 2 \\ 5 & 7 & 7 & 33 & 0 & 6 & 11 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 6 & 0 & 0 & 2 \\ 2 & 2 & 2 & 11 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{11} \overset{1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -58, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 2 & 4 & 0 & 1 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 & 2 \\ 3 & 4 & 5 & 11 & 0 & 4 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 4 & 0 & 0 & 2 \\ 2 & 2 & 2 & 7 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{11} \overset{1}{5} 1^{12} 23^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -67, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 5 & 15 & 0 & 4 & 5 \\ 8 & 15 & 37 & 0 & 10 & 13 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 10 & 0 & 1 & 3 \\ 3 & 5 & 13 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{12} \overset{1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -67, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 1 & 2 \\ 1 & 2 & 0 & 6 & 0 & 2 & 2 \\ 4 & 6 & 6 & 21 & 0 & 5 & 9 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 5 & 0 & 0 & 2 \\ 2 & 2 & 2 & 9 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[56] \quad 21^{12} \overset{1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -67, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 3 \\ 3 & 1 & 3 & 8 & 2 & 3 \\ 2 & 3 & 1 & 8 & 3 & 3 \\ 8 & 8 & 8 & 22 & 7 & 10 \\ 2 & 2 & 3 & 7 & 1 & 3 \\ 3 & 3 & 3 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{11} \overset{1}{5} 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -76, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 1 & 2 \\ 1 & 2 & 0 & 10 & 0 & 2 & 2 \\ 6 & 7 & 10 & 52 & 0 & 8 & 14 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 8 & 0 & 0 & 2 \\ 2 & 2 & 2 & 14 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{11} \overset{1}{5} 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -91, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 1 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 & 2 \\ 5 & 6 & 9 & 36 & 0 & 7 & 12 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 7 & 0 & 0 & 2 \\ 2 & 2 & 2 & 12 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{11} \overset{1}{5} 1^{12} 23^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 5 & 20 & 0 & 4 & 5 \\ 10 & 20 & 63 & 0 & 13 & 17 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 13 & 0 & 1 & 3 \\ 3 & 5 & 17 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[63] \quad 21^{12} \overset{1}{5} 13^{11} \overset{12}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -102, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 2 & 3 \\ 3 & 1 & 3 & 10 & 2 & 3 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 10 & 10 & 12 & 42 & 10 & 14 \\ 2 & 2 & 3 & 10 & 1 & 3 \\ 3 & 3 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{12} \overset{1}{5} 13^{11} \overset{12}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -102, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 1 & 2 \\ 1 & 2 & 0 & 12 & 0 & 2 & 2 \\ 8 & 10 & 12 & 90 & 0 & 10 & 18 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 10 & 0 & 0 & 2 \\ 2 & 2 & 2 & 18 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{13} \overset{1}{5} 13^{11} \overset{12}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -143, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 1 & 2 \\ 1 & 2 & 0 & 13 & 0 & 2 & 2 \\ 9 & 12 & 13 & 112 & 0 & 11 & 20 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 11 & 0 & 0 & 2 \\ 2 & 2 & 2 & 20 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{12} \overset{1}{5} 13^{11} \overset{12}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -143, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 1 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 & 2 \\ 7 & 9 & 11 & 68 & 0 & 9 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 9 & 0 & 0 & 2 \\ 2 & 2 & 2 & 16 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{12} \overset{1}{5} 13^{11} \overset{12}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -154, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 1 & 2 \\ 1 & 2 & 0 & 10 & 0 & 2 & 2 \\ 6 & 8 & 10 & 50 & 0 & 8 & 14 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 8 & 0 & 0 & 2 \\ 2 & 2 & 2 & 14 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[68] \quad 21^{13} \overset{1}{5} 13^{11} \overset{12}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -180, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 2 & 3 \\ 3 & 1 & 3 & 13 & 2 & 3 \\ 2 & 3 & 1 & 14 & 3 & 3 \\ 13 & 13 & 14 & 65 & 12 & 17 \\ 2 & 2 & 3 & 12 & 1 & 3 \\ 3 & 3 & 3 & 17 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{13} \overset{1}{5} 13^{11} \overset{12}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -180, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 1 & 2 \\ 1 & 0 & 2 & 11 & 0 & 1 & 2 \\ 1 & 2 & 0 & 12 & 0 & 2 & 2 \\ 8 & 11 & 12 & 88 & 0 & 10 & 18 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 10 & 0 & 0 & 2 \\ 2 & 2 & 2 & 18 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{13} \overset{1}{5} 13^{11} \overset{12}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -187, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 1 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 & 2 \\ 7 & 10 & 11 & 68 & 0 & 9 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 9 & 0 & 0 & 2 \\ 2 & 2 & 2 & 16 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{13} 5^{11} 13^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -197, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 2 & 3 \\ 3 & 1 & 3 & 15 & 2 & 3 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 15 & 15 & 18 & 97 & 15 & 21 \\ 2 & 2 & 3 & 15 & 1 & 3 \\ 3 & 3 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{12} 5^{13} 13^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -214, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 1 & 2 \\ 1 & 0 & 2 & 11 & 0 & 1 & 2 \\ 1 & 2 & 0 & 15 & 0 & 2 & 2 \\ 9 & 11 & 15 & 117 & 0 & 12 & 21 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 12 & 0 & 0 & 2 \\ 2 & 2 & 2 & 21 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{12} 5^{13} 13^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -227, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 12.00 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 1 & 2 \\ 1 & 2 & 0 & 14 & 0 & 2 & 2 \\ 8 & 10 & 14 & 93 & 0 & 11 & 19 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 11 & 0 & 0 & 2 \\ 2 & 2 & 2 & 19 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{13} 5^{13} 13^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -305, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 1 & 2 \\ 1 & 0 & 2 & 13 & 0 & 1 & 2 \\ 1 & 2 & 0 & 16 & 0 & 2 & 2 \\ 10 & 13 & 16 & 141 & 0 & 13 & 23 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 13 & 0 & 0 & 2 \\ 2 & 2 & 2 & 23 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{13} 5^{13} 13^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -314, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 1 & 2 \\ 1 & 2 & 0 & 15 & 0 & 2 & 2 \\ 9 & 12 & 15 & 115 & 0 & 12 & 21 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 12 & 0 & 0 & 2 \\ 2 & 2 & 2 & 21 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{13} 5^{13} 13^{11} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -427, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 12.00 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 1 & 2 \\ 1 & 0 & 2 & 14 & 0 & 1 & 2 \\ 1 & 2 & 0 & 19 & 0 & 2 & 2 \\ 11 & 14 & 19 & 176 & 0 & 15 & 26 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 15 & 0 & 0 & 2 \\ 2 & 2 & 2 & 26 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$4.982 \quad \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -35, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 0 & 3 & 4 & 0 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 0 & 2 & 1 & 3 \\ 3 & 4 & 0 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 3 & 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 4 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 3 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & 0 & -1 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[3] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 4 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 3 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[5] $1^{111}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 3 & 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 3 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$T = 11$ [7] $1^{11}4^{13}1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 11, T_{\max} = 11, \Delta = -5, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 2 & 1 & 5 & 4 \\ 3 & 1 & 1 & 0 & 4 & 3 \\ 2 & 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 4 & 2 & 0 & 6 & 6 \\ 4 & 3 & 2 & 0 & 6 & 5 \end{pmatrix}_{b_0}$$

$T = 12$ [9] $1^{11}5^{12}13^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 3 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 3 & 1 & 5 & 3 \\ 6 & 7 & 0 & 6 & 5 & 9 & 7 \\ 3 & 4 & 0 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[10] $1^{11}5^{12}13^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 0 & 3 & 3 & 7 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 3 & 1 & 5 & 3 \\ 5 & 7 & 0 & 6 & 5 & 9 & 7 \\ 3 & 4 & 0 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 5^{11} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12} 5^{12} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 12, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 3 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 & 12 & 3 \\ 2 & 3 & 0 & 3 & 1 & 10 & 3 \\ 10 & 14 & 0 & 12 & 10 & 42 & 14 \\ 3 & 4 & 0 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11} 5^{11} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 15, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 4 & 3 \\ 3 & 5 & 11 & 0 & 4 & 8 & 5 \\ 6 & 11 & 22 & 0 & 7 & 16 & 10 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 7 & 0 & 1 & 5 & 3 \\ 4 & 8 & 16 & 0 & 5 & 9 & 7 \\ 3 & 5 & 10 & -1 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11} 5^{11} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 15, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 5 & 3 & 0 & 4 & 8 & 5 \\ 2 & 3 & 1 & 0 & 2 & 5 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 5 & 3 \\ 4 & 8 & 5 & 0 & 5 & 9 & 7 \\ 3 & 5 & 3 & -1 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 5^{11} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 4 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 3 & 2 \\ 2 & 2 & 4 & 3 & 0 & 3 & 5 & 5 \\ 2 & 2 & 2 & 2 & -1 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11} 5^{11} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 5 & 5 & 0 & 4 & 15 & 5 \\ 2 & 5 & 2 & 0 & 3 & 11 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 10 & 3 \\ 8 & 15 & 11 & 0 & 10 & 37 & 13 \\ 3 & 5 & 4 & -1 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{13} 5^{13} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 17, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 0 & 3 & 3 & 21 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 & 18 & 3 \\ 2 & 3 & 0 & 3 & 1 & 15 & 3 \\ 15 & 21 & 0 & 18 & 15 & 97 & 21 \\ 3 & 4 & 0 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 5^{11} 13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 18, \quad (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 5 & 3 & 0 & 4 & 16 & 5 \\ 2 & 3 & 1 & 0 & 2 & 10 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 10 & 3 \\ 8 & 16 & 10 & 0 & 10 & 42 & 14 \\ 3 & 5 & 3 & -1 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[18] \begin{matrix} 1^{13} \\ 1^1 \end{matrix} 5 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 20, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 0 & 3 & 3 & 14 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 & 12 & 3 \\ 2 & 3 & 0 & 3 & 1 & 10 & 3 \\ 11 & 14 & 0 & 12 & 10 & 42 & 14 \\ 3 & 4 & 0 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[20] \begin{matrix} 1^1 \\ 1^{13} \end{matrix} 5 13^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 5 & 3 & 0 & 4 & 24 & 5 \\ 2 & 3 & 1 & 0 & 2 & 15 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 15 & 3 \\ 12 & 24 & 15 & 0 & 15 & 97 & 21 \\ 3 & 5 & 3 & -1 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[19] \begin{matrix} 1^1 \\ 1^{31} \end{matrix} 5 13^1 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 4 & 3 \\ 3 & 5 & 19 & 0 & 4 & 8 & 5 \\ 10 & 19 & 65 & 0 & 12 & 27 & 17 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 12 & 0 & 1 & 5 & 3 \\ 4 & 8 & 27 & 0 & 5 & 9 & 7 \\ 3 & 5 & 17 & -1 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[21] \begin{matrix} 1^1 \\ 1^1 \end{matrix} 5 13^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 5 & 5 & 3 & 4 & 5 & 5 \\ 2 & 5 & 2 & 2 & 3 & 4 & 4 \\ 2 & 3 & 2 & 0 & 2 & 2 & 2 \\ 2 & 4 & 3 & 2 & 1 & 3 & 3 \\ 3 & 5 & 4 & 2 & 3 & 2 & 4 \\ 3 & 5 & 4 & 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[22] \begin{matrix} 1^1 \\ 1^{12} \end{matrix} 5 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 5 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 6 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 6 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 5 & 2 \\ 4 & 5 & 6 & 6 & 0 & 5 & 21 & 9 \\ 2 & 2 & 2 & 2 & -1 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[23] \begin{matrix} 1^2 \\ 1^1 \end{matrix} 5 13^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 26, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 6 & 3 \\ 3 & 5 & 5 & 3 & 4 & 10 & 5 \\ 2 & 5 & 2 & 2 & 3 & 8 & 4 \\ 2 & 3 & 2 & 0 & 2 & 4 & 2 \\ 2 & 4 & 3 & 2 & 1 & 6 & 3 \\ 6 & 10 & 8 & 4 & 6 & 14 & 8 \\ 3 & 5 & 4 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[24] \begin{matrix} 1^3 \\ 1^1 \end{matrix} 5 13^1 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 31, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 9 & 3 \\ 3 & 5 & 5 & 3 & 4 & 15 & 5 \\ 2 & 5 & 2 & 2 & 3 & 12 & 4 \\ 2 & 3 & 2 & 0 & 2 & 6 & 2 \\ 2 & 4 & 3 & 2 & 1 & 9 & 3 \\ 9 & 15 & 12 & 6 & 9 & 34 & 12 \\ 3 & 5 & 4 & 2 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[27] $1^{11}_{1^{12}} 5^{11} 13^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 0 & 1 & 5 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 & 6 & 2 \\ 2 & 2 & 4 & 5 & 0 & 3 & 12 & 5 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 3 & 0 & 0 & 5 & 2 \\ 4 & 5 & 6 & 12 & 0 & 5 & 21 & 9 \\ 2 & 2 & 2 & 5 & -1 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[25] $1^{11}_{1^4} 5^{11} 13^{11} 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 12 & 3 \\ 3 & 5 & 5 & 3 & 4 & 20 & 5 \\ 2 & 5 & 2 & 2 & 3 & 16 & 4 \\ 2 & 3 & 2 & 0 & 2 & 8 & 2 \\ 2 & 4 & 3 & 2 & 1 & 12 & 3 \\ 12 & 20 & 16 & 8 & 12 & 62 & 16 \\ 3 & 5 & 4 & 2 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

[26] $1^{12}_{1^{12}} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 7 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 7 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 7 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 6 & 2 \\ 5 & 7 & 7 & 7 & 0 & 6 & 33 & 11 \\ 2 & 2 & 2 & 2 & -1 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[28] $1^{11}_{1^{11}} 5^{11} 13^{11} 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 39, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 4 & 3 \\ 3 & 5 & 13 & 0 & 4 & 8 & 5 \\ 6 & 13 & 25 & 0 & 8 & 17 & 11 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 8 & 0 & 1 & 5 & 3 \\ 4 & 8 & 17 & 0 & 5 & 9 & 7 \\ 3 & 5 & 11 & -1 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[29] $1^{11}_{1^{12}} 5^{11} 13^{11} 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 8 & 3 \\ 3 & 5 & 13 & 0 & 4 & 16 & 5 \\ 6 & 13 & 25 & 0 & 8 & 34 & 11 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 8 & 0 & 1 & 10 & 3 \\ 8 & 16 & 34 & 0 & 10 & 42 & 14 \\ 3 & 5 & 11 & -1 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[30] $1^{11}_{1^{13}} 5^{11} 13^{11} 1^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 12 & 3 \\ 3 & 5 & 13 & 0 & 4 & 24 & 5 \\ 6 & 13 & 25 & 0 & 8 & 51 & 11 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 8 & 0 & 1 & 15 & 3 \\ 12 & 24 & 51 & 0 & 15 & 97 & 21 \\ 3 & 5 & 11 & -1 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[31] $1^{11}_{1^{13}} 5^{11} 13^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 7 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 10 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 9 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 8 & 2 \\ 6 & 7 & 10 & 9 & 0 & 8 & 52 & 14 \\ 2 & 2 & 2 & 2 & -1 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[32] $1^{11}_{1^{11}} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 4 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 5 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 4 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 4 & 2 \\ 3 & 4 & 5 & 4 & 0 & 4 & 11 & 7 \\ 2 & 2 & 2 & 2 & -1 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[33] $1^{11} \overset{1}{5} 13^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 0 & 1 & 4 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 & 5 & 2 \\ 2 & 2 & 4 & 5 & 0 & 3 & 9 & 5 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 3 & 0 & 0 & 4 & 2 \\ 3 & 4 & 5 & 9 & 0 & 4 & 11 & 7 \\ 2 & 2 & 2 & 5 & -1 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[34] $1^{11} \overset{1}{5} 13^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 60, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 4 & 3 \\ 3 & 5 & 8 & 0 & 4 & 8 & 5 \\ 4 & 8 & 9 & 0 & 5 & 11 & 7 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 5 & 0 & 1 & 5 & 3 \\ 4 & 8 & 11 & 0 & 5 & 9 & 7 \\ 3 & 5 & 7 & -1 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[35] $1^{11} \overset{1}{5} 13^{11} 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 63, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 10 & 3 \\ 3 & 5 & 8 & 0 & 4 & 21 & 5 \\ 4 & 8 & 9 & 0 & 5 & 28 & 7 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 5 & 0 & 1 & 13 & 3 \\ 10 & 21 & 28 & 0 & 13 & 70 & 18 \\ 3 & 5 & 7 & -1 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[36] $1^{11} \overset{1}{5} 13^{11} 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 8 & 3 \\ 3 & 5 & 8 & 0 & 4 & 16 & 5 \\ 4 & 8 & 9 & 0 & 5 & 22 & 7 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 5 & 0 & 1 & 10 & 3 \\ 8 & 16 & 22 & 0 & 10 & 42 & 14 \\ 3 & 5 & 7 & -1 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[37] $1^{11} \overset{1}{5} 13^{11} 1^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 92, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 12 & 3 \\ 3 & 5 & 8 & 0 & 4 & 24 & 5 \\ 4 & 8 & 9 & 0 & 5 & 33 & 7 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 5 & 0 & 1 & 15 & 3 \\ 12 & 24 & 33 & 0 & 15 & 97 & 21 \\ 3 & 5 & 7 & -1 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[38] $1^{11} \overset{1}{5} 13^{11} 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 95, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 14 & 3 \\ 3 & 5 & 8 & 0 & 4 & 29 & 5 \\ 4 & 8 & 9 & 0 & 5 & 39 & 7 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 5 & 0 & 1 & 18 & 3 \\ 14 & 29 & 39 & 0 & 18 & 137 & 25 \\ 3 & 5 & 7 & -1 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

[39] $1^{12} \overset{1}{5} 13^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 9 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 11 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 10 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 9 & 2 \\ 7 & 9 & 11 & 10 & 0 & 9 & 68 & 16 \\ 2 & 2 & 2 & 2 & -1 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[40] $1^{13} \overset{1}{5} 13^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 143, (n_+, n_-, n_0) =$

$$(1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 11 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 12 & 2 \\ 1 & 1 & 1 & 0 & 0 & 1 & 11 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 10 & 2 \\ 8 & 11 & 12 & 11 & 0 & 10 & 88 & 18 \\ 2 & 2 & 2 & 2 & -1 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$4.983 \quad \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$$

$$T = 10 \quad [1] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 3 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 & 2 \\ 3 & 4 & 3 & 0 & 2 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & -1 & 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 3 & 4 & 5 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 2 & 3 & 2 \\ 2 & 4 & 0 & 0 & 2 & 1 & 3 & 3 \\ 3 & 5 & 0 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & 0 & -1 & 2 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$4.984 \quad \text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \text{mix}$$

$$T = 11 \quad [4] \quad 1^{11} 5^1 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \\ 4 & 7 & 9 & 6 & 5 & 7 \\ 2 & 3 & 6 & 1 & 3 & 3 \\ 3 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 5^1 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 3 & 10 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 9 & 3 \\ 7 & 10 & 0 & 9 & 22 & 10 \\ 3 & 4 & 0 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 5^1 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 & 3 \\ 2 & 2 & 4 & 2 & 3 & 4 \\ 3 & 2 & 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 5^1 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 & 3 \\ 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{11} 5^1 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 & 3 \\ 2 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 4 & 5 & 6 & 9 & 5 & 7 \\ 3 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

[5] $1^{11} \overset{12}{5} 13^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 9 & 3 \\ 2 & 2 & 4 & 2 & 9 & 4 \\ 8 & 7 & 9 & 9 & 22 & 10 \\ 3 & 3 & 3 & 4 & 10 & 4 \end{pmatrix}_{b_0}$

[6] $1^{12} \overset{11}{5} 13^1 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 9 & 2 & 3 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 9 & 14 & 42 & 12 & 10 & 14 \\ 2 & 3 & 12 & 1 & 3 & 3 \\ 3 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \end{pmatrix}_{b_0}$

[7] $1^{11} \overset{121}{5} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 6 & 2 & 3 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \\ 6 & 7 & 9 & 6 & 5 & 7 \\ 2 & 3 & 6 & 1 & 3 & 3 \\ 3 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \end{pmatrix}_{b_0}$

[8] $1^{11} \overset{112}{5} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 10 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 9 & 3 \\ 8 & 10 & 0 & 9 & 22 & 10 \\ 3 & 4 & 0 & 3 & 10 & 4 \end{pmatrix}_{b_0}$

[10] $1^{12} \overset{11}{5} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 0 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 15 & 3 \\ 12 & 17 & 0 & 15 & 65 & 17 \\ 3 & 4 & 0 & 3 & 17 & 4 \end{pmatrix}_{b_0}$

[9] $1^{11} \overset{11}{5} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 2 & 3 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \\ 5 & 7 & 9 & 6 & 5 & 7 \\ 2 & 3 & 6 & 1 & 3 & 3 \\ 3 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 7 & 3 & 3 & 4 \end{pmatrix}_{b_0}$

[12] $1^1 \overset{1}{5} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 1 & 1 & 1 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 3 & 0 & 4 & 4 \\ 2 & 2 & 2 & 2 & -1 & 4 & 3 \end{pmatrix}_{b_0}$

[11] $1^{11} \overset{1}{5} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 & 0 & 3 & 2 \\ 2 & 3 & 2 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 5 & 0 & 4 & 4 \\ 2 & 2 & 2 & 4 & -1 & 4 & 3 \end{pmatrix}_{b_0}$

[13] $1^{11} \overset{1}{5} 13^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -13, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 6 & 1 & 6 & 3 \\ 3 & 5 & 11 & 0 & 12 & 5 \\ 6 & 11 & 22 & 0 & 23 & 10 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 23 & 0 & 22 & 10 \\ 3 & 5 & 10 & -1 & 10 & 4 \end{pmatrix}_{b_0}$

- [14] $1^{13}_{1^1} 5^{121} 13^1 1^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 14 & 21 & 97 & 18 & 15 & 21 \\ 2 & 3 & 18 & 1 & 3 & 3 \\ 3 & 3 & 15 & 3 & 1 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$
- [15] $1^{12}_{1^1} 5^{121} 13^1 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 10 & 14 & 42 & 12 & 10 & 14 \\ 2 & 3 & 12 & 1 & 3 & 3 \\ 3 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$
- [16] $1^{12}_{1^1} 5^{131} 13^1 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \\ 11 & 14 & 42 & 12 & 10 & 14 \\ 2 & 3 & 12 & 1 & 3 & 3 \\ 3 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 14 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$
- [17] $1^{11}_{1^{11}} 5^{131} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 5 & 3 & 0 & 12 & 5 \\ 2 & 3 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 7 & 0 & 22 & 10 \\ 3 & 5 & 3 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$
- [18] $1^{11}_{1^{31}} 5^{131} 13^1 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 6 & 3 \\ 3 & 5 & 19 & 0 & 12 & 5 \\ 10 & 19 & 65 & 0 & 39 & 17 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 39 & 0 & 22 & 10 \\ 3 & 5 & 17 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$
- [20] $1^{12}_{1^{11}} 5^{131} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 & 2 \\ 1 & 2 & 0 & 6 & 0 & 3 & 2 \\ 4 & 6 & 6 & 21 & 0 & 10 & 9 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 10 & 0 & 4 & 4 \\ 2 & 2 & 2 & 9 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$
- [19] $1^{11}_{1^{11}} 5^{131} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 4 & 0 & 3 & 2 \\ 2 & 2 & 4 & 5 & 0 & 6 & 5 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 6 & 0 & 4 & 4 \\ 2 & 2 & 2 & 5 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$
- [21] $1^{11}_{1^1} 5^{131} 13^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -21, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 5 & 5 & 3 & 9 & 5 \\ 2 & 5 & 2 & 2 & 7 & 4 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 5 & 9 & 7 & 4 & 11 & 7 \\ 3 & 5 & 4 & 2 & 7 & 4 \end{pmatrix}_{b_0}$$
- [22] $1^{12}_{1^1} 5^{113} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 0 & 3 & 17 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 15 & 3 \\ 13 & 17 & 0 & 15 & 65 & 17 \\ 3 & 4 & 0 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{13}_{1^1} 5^{13} 13^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 21 & 3 \\ 17 & 24 & 0 & 21 & 130 & 24 \\ 3 & 4 & 0 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{11}_{1^{12}} 5^{13} 13^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 8 & 3 \\ 3 & 5 & 8 & 5 & 15 & 5 \\ 4 & 8 & 9 & 6 & 21 & 7 \\ 3 & 5 & 6 & 1 & 11 & 3 \\ 8 & 15 & 21 & 11 & 39 & 13 \\ 3 & 5 & 7 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{11}_{1^{21}} 5^{13} 13^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 4 & 3 \\ 3 & 5 & 16 & 0 & 7 & 5 \\ 8 & 16 & 42 & 0 & 20 & 14 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 7 & 20 & 0 & 8 & 6 \\ 3 & 5 & 14 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{11}_{1^{21}} 5^{13} 13^{11} 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 4 & 3 \\ 3 & 5 & 16 & 5 & 7 & 5 \\ 8 & 16 & 42 & 12 & 20 & 14 \\ 3 & 5 & 12 & 1 & 5 & 3 \\ 4 & 7 & 20 & 5 & 8 & 6 \\ 3 & 5 & 14 & 3 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{11}_{1^{31}} 5^{13} 13^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 4 & 3 \\ 3 & 5 & 24 & 0 & 7 & 5 \\ 12 & 24 & 97 & 0 & 30 & 21 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 7 & 30 & 0 & 8 & 6 \\ 3 & 5 & 21 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{13}_{1^1} 5^{13} 13^{11} 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \\ 15 & 21 & 97 & 18 & 15 & 21 \\ 2 & 3 & 18 & 1 & 3 & 3 \\ 3 & 3 & 15 & 3 & 1 & 3 \\ 3 & 4 & 21 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{11}_{1^{11}} 5^{13} 13^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 4 & 3 \\ 3 & 5 & 8 & 5 & 7 & 5 \\ 4 & 8 & 9 & 6 & 10 & 7 \\ 3 & 5 & 6 & 1 & 5 & 3 \\ 4 & 7 & 10 & 5 & 8 & 6 \\ 3 & 5 & 7 & 3 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11}_{1^{11}} 5^{13} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 3 \\ 3 & 5 & 8 & 0 & 7 & 5 \\ 4 & 8 & 9 & 0 & 10 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 7 & 10 & 0 & 8 & 6 \\ 3 & 5 & 7 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

[25] $1^1 \begin{smallmatrix} 1 \\ 5 \\ 13 \end{smallmatrix} 1^1 \begin{smallmatrix} 1 \\ 13 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 4 & 3 \\ 3 & 5 & 24 & 5 & 7 & 5 \\ 12 & 24 & 97 & 18 & 30 & 21 \\ 3 & 5 & 18 & 1 & 5 & 3 \\ 4 & 7 & 30 & 5 & 8 & 6 \\ 3 & 5 & 21 & 3 & 6 & 4 \end{pmatrix}_{b_0}$$

[32] $1^1 \begin{smallmatrix} 1 \\ 5 \\ 13 \end{smallmatrix} 1^1 \begin{smallmatrix} 1 \\ 12 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 5 & 3 & 0 & 20 & 5 \\ 2 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 20 & 12 & 0 & 65 & 17 \\ 3 & 5 & 3 & -1 & 17 & 4 \end{pmatrix}_{b_0}$$

[33] $1^1 \begin{smallmatrix} 1 \\ 12 \\ 5 \end{smallmatrix} 1^1 \begin{smallmatrix} 1 \\ 11 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 & 2 \\ 1 & 2 & 0 & 9 & 0 & 3 & 2 \\ 5 & 7 & 9 & 36 & 0 & 13 & 12 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 13 & 0 & 4 & 4 \\ 2 & 2 & 2 & 12 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[34] $1^1 \begin{smallmatrix} 1 \\ 5 \\ 13 \end{smallmatrix} 1^1 \begin{smallmatrix} 1 \\ 11 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 6 & 3 \\ 3 & 5 & 13 & 0 & 12 & 5 \\ 6 & 13 & 25 & 0 & 25 & 11 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 25 & 0 & 22 & 10 \\ 3 & 5 & 11 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

[35] $1^1 \begin{smallmatrix} 1 \\ 11 \\ 5 \end{smallmatrix} 1^1 \begin{smallmatrix} 1 \\ 12 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 & 2 \\ 1 & 0 & 2 & 5 & 0 & 2 & 2 \\ 1 & 2 & 0 & 8 & 0 & 3 & 2 \\ 4 & 5 & 8 & 24 & 0 & 11 & 10 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 11 & 0 & 4 & 4 \\ 2 & 2 & 2 & 10 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[36] $1^1 \begin{smallmatrix} 1 \\ 5 \\ 13 \end{smallmatrix} 1^1 \begin{smallmatrix} 1 \\ 12 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -35, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 \\ 3 & 5 & 5 & 3 & 14 & 5 \\ 2 & 5 & 2 & 2 & 11 & 4 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 8 & 14 & 11 & 6 & 29 & 11 \\ 3 & 5 & 4 & 2 & 11 & 4 \end{pmatrix}_{b_0}$$

[37] $1^1 \begin{smallmatrix} 1 \\ 11 \\ 5 \end{smallmatrix} 1^1 \begin{smallmatrix} 1 \\ 13 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 0 & 4 & 2 \\ 1 & 1 & 1 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 4 & 4 & 0 & 10 & 6 \\ 2 & 2 & 2 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[38] $1^1 \begin{smallmatrix} 1 \\ 11 \\ 5 \end{smallmatrix} 1^1 \begin{smallmatrix} 1 \\ 13 \end{smallmatrix} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_2} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -43, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 5 & 3 & 0 & 28 & 5 \\ 2 & 3 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 28 & 17 & 0 & 130 & 24 \\ 3 & 5 & 3 & -1 & 24 & 4 \end{pmatrix}_{b_0}$$

[39] $1^{11}_{1^{21}} 5^{13} 1^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 6 & 3 \\ 3 & 5 & 21 & 0 & 12 & 5 \\ 10 & 21 & 70 & 0 & 41 & 18 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 41 & 0 & 22 & 10 \\ 3 & 5 & 18 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

[40] $1^{11}_{1^{11}} 5^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 & 2 \\ 1 & 0 & 2 & 4 & 0 & 2 & 2 \\ 1 & 2 & 0 & 5 & 0 & 3 & 2 \\ 3 & 4 & 5 & 11 & 0 & 8 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 8 & 0 & 4 & 4 \\ 2 & 2 & 2 & 7 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[41] $1^{11}_{1^{12}} 5^{13} 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 10 & 3 \\ 3 & 5 & 13 & 0 & 20 & 5 \\ 6 & 13 & 25 & 0 & 42 & 11 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 20 & 42 & 0 & 65 & 17 \\ 3 & 5 & 11 & -1 & 17 & 4 \end{pmatrix}_{b_0}$$

[42] $1^{13}_{1^1} 5^{13} 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -53, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 3 \\ 3 & 5 & 5 & 3 & 19 & 5 \\ 2 & 5 & 2 & 2 & 15 & 4 \\ 2 & 3 & 2 & 0 & 8 & 2 \\ 11 & 19 & 15 & 8 & 55 & 15 \\ 3 & 5 & 4 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

[44] $1^{11}_{1^{11}} 5^{13} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 6 & 3 \\ 3 & 5 & 8 & 0 & 12 & 5 \\ 4 & 8 & 9 & 0 & 16 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 16 & 0 & 22 & 10 \\ 3 & 5 & 7 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

[43] $1^{11}_{1^{12}} 5^{13} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 5 & 8 & 0 & 15 & 5 \\ 4 & 8 & 9 & 0 & 21 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 15 & 21 & 0 & 39 & 13 \\ 3 & 5 & 7 & -1 & 13 & 4 \end{pmatrix}_{b_0}$$

[46] $1^{11}_{1^{21}} 5^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 & 2 \\ 1 & 0 & 2 & 5 & 0 & 2 & 2 \\ 1 & 2 & 0 & 6 & 0 & 3 & 2 \\ 4 & 5 & 6 & 21 & 0 & 11 & 9 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 11 & 0 & 4 & 4 \\ 2 & 2 & 2 & 9 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[45] $1^{12}_{1^{21}} 5^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 & 2 \\ 1 & 2 & 0 & 7 & 0 & 3 & 2 \\ 5 & 7 & 7 & 33 & 0 & 13 & 11 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 13 & 0 & 4 & 4 \\ 2 & 2 & 2 & 11 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[47] $1^{11} \bar{5}^{13} 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -60, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 6 & 3 \\ 3 & 5 & 16 & 0 & 12 & 5 \\ 8 & 16 & 42 & 0 & 32 & 14 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 32 & 0 & 22 & 10 \\ 3 & 5 & 14 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

[48] $1^{11} \bar{5}^{13} 1^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -63, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 6 & 3 \\ 3 & 5 & 29 & 0 & 12 & 5 \\ 14 & 29 & 137 & 0 & 57 & 25 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 57 & 0 & 22 & 10 \\ 3 & 5 & 25 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

[50] $1^{11} \bar{5}^{13} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 4 & 0 & 7 & 2 \\ 2 & 2 & 4 & 5 & 0 & 12 & 5 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 5 & 7 & 12 & 0 & 21 & 9 \\ 2 & 2 & 2 & 5 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

[49] $1^{11} \bar{5}^{13} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 0 & 2 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 4 & 0 & 9 & 2 \\ 2 & 2 & 4 & 5 & 0 & 18 & 5 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 7 & 9 & 18 & 0 & 49 & 13 \\ 2 & 2 & 2 & 5 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

[52] $1^{11} \bar{5}^{13} 1^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 14 & 3 \\ 3 & 5 & 13 & 0 & 28 & 5 \\ 6 & 13 & 25 & 0 & 59 & 11 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 28 & 59 & 0 & 130 & 24 \\ 3 & 5 & 11 & -1 & 24 & 4 \end{pmatrix}_{b_0}$$

[51] $1^{12} \bar{5}^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 & 2 \\ 1 & 2 & 0 & 8 & 0 & 3 & 2 \\ 6 & 8 & 8 & 49 & 0 & 16 & 13 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 16 & 0 & 4 & 4 \\ 2 & 2 & 2 & 13 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[53] $1^{11} \bar{5}^{13} 1^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -70, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 6 & 3 \\ 3 & 5 & 24 & 0 & 12 & 5 \\ 12 & 24 & 97 & 0 & 48 & 21 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 48 & 0 & 22 & 10 \\ 3 & 5 & 21 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

[54] $1^{11} \bar{5}^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 0 & 8 & 2 \\ 1 & 1 & 1 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 6 & 8 & 7 & 0 & 33 & 11 \\ 2 & 2 & 2 & 2 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[55] $1^{1^4}_{1^1} 5^{13} 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -75, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 14 & 3 \\ 3 & 5 & 5 & 3 & 24 & 5 \\ 2 & 5 & 2 & 2 & 19 & 4 \\ 2 & 3 & 2 & 0 & 10 & 2 \\ 14 & 24 & 19 & 10 & 89 & 19 \\ 3 & 5 & 4 & 2 & 19 & 4 \end{pmatrix}_{b_0}$$

[57] $1^{13}_{1^{21}} 5^{13} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 & 2 \\ 1 & 2 & 0 & 11 & 0 & 3 & 2 \\ 7 & 10 & 11 & 68 & 0 & 18 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 18 & 0 & 4 & 4 \\ 2 & 2 & 2 & 16 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[58] $1^{11}_{1^{21}} 5^{13} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 & 2 \\ 1 & 2 & 0 & 9 & 0 & 3 & 2 \\ 5 & 6 & 9 & 36 & 0 & 14 & 12 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 14 & 0 & 4 & 4 \\ 2 & 2 & 2 & 12 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[56] $1^{11}_{1^{12}} 5^{13} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 5 & 2 \\ 1 & 0 & 2 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 4 & 0 & 8 & 2 \\ 2 & 2 & 4 & 5 & 0 & 15 & 5 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 6 & 8 & 15 & 0 & 33 & 11 \\ 2 & 2 & 2 & 5 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[59] $1^{11}_{1^{31}} 5^{13} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -91, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 & 2 \\ 1 & 2 & 0 & 10 & 0 & 3 & 2 \\ 6 & 7 & 10 & 52 & 0 & 17 & 14 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 17 & 0 & 4 & 4 \\ 2 & 2 & 2 & 14 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[60] $1^{11}_{1^{12}} 5^{13} 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -92, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 10 & 3 \\ 3 & 5 & 8 & 0 & 20 & 5 \\ 4 & 8 & 9 & 0 & 27 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 20 & 27 & 0 & 65 & 17 \\ 3 & 5 & 7 & -1 & 17 & 4 \end{pmatrix}_{b_0}$$

[61] $1^{11}_{1^{13}} 5^{13} 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 3 \\ 3 & 5 & 8 & 0 & 23 & 5 \\ 4 & 8 & 9 & 0 & 32 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 23 & 32 & 0 & 92 & 20 \\ 3 & 5 & 7 & -1 & 20 & 4 \end{pmatrix}_{b_0}$$

[63] $1^{13}_{1^{21}} 5^{13} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 & 2 \\ 1 & 2 & 0 & 14 & 0 & 3 & 2 \\ 8 & 11 & 14 & 93 & 0 & 21 & 19 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 21 & 0 & 4 & 4 \\ 2 & 2 & 2 & 19 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[62] $1^{12} \overset{1}{5} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 & 2 \\ 1 & 2 & 0 & 13 & 0 & 3 & 2 \\ 7 & 9 & 13 & 73 & 0 & 19 & 17 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 19 & 0 & 4 & 4 \\ 2 & 2 & 2 & 17 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[64] $1^{11} \overset{1}{5} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 0 & 12 & 2 \\ 1 & 1 & 1 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 8 & 12 & 10 & 0 & 70 & 16 \\ 2 & 2 & 2 & 2 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[65] $1^{12} \overset{1}{5} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -102, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 & 2 \\ 1 & 2 & 0 & 10 & 0 & 3 & 2 \\ 6 & 8 & 10 & 50 & 0 & 16 & 14 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 16 & 0 & 4 & 4 \\ 2 & 2 & 2 & 14 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[66] $1^{12} \overset{1}{5} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -123, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 0 & 10 & 2 \\ 1 & 1 & 1 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 9 & 10 & 10 & 0 & 65 & 15 \\ 2 & 2 & 2 & 2 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

[67] $1^{12} \overset{1}{5} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -123, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 0 & 9 & 2 \\ 1 & 1 & 1 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 8 & 9 & 8 & 0 & 47 & 13 \\ 2 & 2 & 2 & 2 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

[68] $1^1 \overset{1}{5} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -142, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 14 & 3 \\ 3 & 5 & 8 & 0 & 28 & 5 \\ 4 & 8 & 9 & 0 & 38 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 28 & 38 & 0 & 130 & 24 \\ 3 & 5 & 7 & -1 & 24 & 4 \end{pmatrix}_{b_0}$$

[69] $1^{12} \overset{1}{5} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -143, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 & 2 \\ 1 & 2 & 0 & 11 & 0 & 3 & 2 \\ 7 & 9 & 11 & 68 & 0 & 19 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 19 & 0 & 4 & 4 \\ 2 & 2 & 2 & 16 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[70] $1^{13} \overset{1}{5} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -143, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 & 2 \\ 1 & 2 & 0 & 12 & 0 & 3 & 2 \\ 8 & 11 & 12 & 88 & 0 & 21 & 18 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 21 & 0 & 4 & 4 \\ 2 & 2 & 2 & 18 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[71] $1^{12} \overset{1}{5} 13^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -151$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.5$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 & 2 \\ 1 & 2 & 0 & 17 & 0 & 3 & 2 \\ 9 & 11 & 17 & 124 & 0 & 25 & 22 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 25 & 0 & 4 & 4 \\ 2 & 2 & 2 & 22 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{12} \overset{1}{5} 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -171$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.5$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 & 2 \\ 1 & 2 & 0 & 14 & 0 & 3 & 2 \\ 8 & 10 & 14 & 93 & 0 & 22 & 19 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 22 & 0 & 4 & 4 \\ 2 & 2 & 2 & 19 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[73] $1^{12} \overset{1}{5} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -187$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 0 & 13 & 2 \\ 1 & 1 & 1 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 10 & 13 & 11 & 0 & 88 & 18 \\ 2 & 2 & 2 & 2 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[74] $1^{13} \overset{1}{5} 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -197$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.8$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 & 2 \\ 1 & 0 & 2 & 12 & 0 & 2 & 2 \\ 1 & 2 & 0 & 15 & 0 & 3 & 2 \\ 9 & 12 & 15 & 115 & 0 & 24 & 21 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 24 & 0 & 4 & 4 \\ 2 & 2 & 2 & 21 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[75] $1^{13} \overset{1}{5} 13^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -203$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.8$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 & 2 \\ 1 & 0 & 2 & 13 & 0 & 2 & 2 \\ 1 & 2 & 0 & 18 & 0 & 3 & 2 \\ 10 & 13 & 18 & 148 & 0 & 27 & 24 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 3 & 27 & 0 & 4 & 4 \\ 2 & 2 & 2 & 24 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[76] $1^{13} \overset{1}{5} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -264$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 0 & 14 & 2 \\ 1 & 1 & 1 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 9 & 12 & 14 & 12 & 0 & 110 & 20 \\ 2 & 2 & 2 & 2 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

4.985 NHC(-2)(-3) + su₂ + su₃ + su2n3mix + su₂

$T = 12$ [0] $1^{11} \overset{1}{5} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 5$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 8.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 2 & 2 & 5 & 3 \\ 2 & 3 & 1 & 4 & 3 & 6 & 3 \\ 2 & 2 & 4 & 2 & 3 & 6 & 4 \\ 3 & 2 & 3 & 3 & 1 & 5 & 3 \\ 5 & 5 & 6 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[1] \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} \begin{smallmatrix} 1^1 \\ 5 \\ 1^1 \end{smallmatrix} 13^1 1^1 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 & 3 \\ 3 & 5 & 5 & 3 & 9 & 5 & 5 \\ 2 & 5 & 2 & 2 & 7 & 4 & 4 \\ 2 & 3 & 2 & 0 & 4 & 2 & 2 \\ 5 & 9 & 7 & 4 & 11 & 7 & 7 \\ 3 & 5 & 4 & 2 & 7 & 2 & 4 \\ 3 & 5 & 4 & 2 & 7 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2] \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} \begin{smallmatrix} 1^1 \\ 5 \\ 1^1 \end{smallmatrix} 13^1 1^1 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 46, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 6 & 3 \\ 3 & 5 & 5 & 3 & 9 & 10 & 5 \\ 2 & 5 & 2 & 2 & 7 & 8 & 4 \\ 2 & 3 & 2 & 0 & 4 & 4 & 2 \\ 5 & 9 & 7 & 4 & 11 & 14 & 7 \\ 6 & 10 & 8 & 4 & 14 & 14 & 8 \\ 3 & 5 & 4 & 2 & 7 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[3] \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} \begin{smallmatrix} 1^1 \\ 5 \\ 1^1 \end{smallmatrix} 13^1 1^1 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 51, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 9 & 3 \\ 3 & 5 & 5 & 3 & 9 & 15 & 5 \\ 2 & 5 & 2 & 2 & 7 & 12 & 4 \\ 2 & 3 & 2 & 0 & 4 & 6 & 2 \\ 5 & 9 & 7 & 4 & 11 & 21 & 7 \\ 9 & 15 & 12 & 6 & 21 & 34 & 12 \\ 3 & 5 & 4 & 2 & 7 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[4] \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} \begin{smallmatrix} 1^1 \\ 5 \\ 1^1 \end{smallmatrix} 13^1 1^1 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 71, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 & 3 \\ 3 & 5 & 5 & 3 & 14 & 5 & 5 \\ 2 & 5 & 2 & 2 & 11 & 4 & 4 \\ 2 & 3 & 2 & 0 & 6 & 2 & 2 \\ 8 & 14 & 11 & 6 & 29 & 11 & 11 \\ 3 & 5 & 4 & 2 & 11 & 2 & 4 \\ 3 & 5 & 4 & 2 & 11 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[5] \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} \begin{smallmatrix} 1^1 \\ 5 \\ 1^1 \end{smallmatrix} 13^1 1^1 \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 107, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 3 & 3 \\ 3 & 5 & 5 & 3 & 19 & 5 & 5 \\ 2 & 5 & 2 & 2 & 15 & 4 & 4 \\ 2 & 3 & 2 & 0 & 8 & 2 & 2 \\ 11 & 19 & 15 & 8 & 55 & 15 & 15 \\ 3 & 5 & 4 & 2 & 15 & 2 & 4 \\ 3 & 5 & 4 & 2 & 15 & 4 & 4 \end{pmatrix}_{b_0}$$

4.986 NHC(-2)(-3) + $\textcolor{red}{su}_2 + \textcolor{red}{su}_3 + \textcolor{red}{su}_3$

$$T = 9 \quad [0] \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} \begin{smallmatrix} 1^1 \\ 4 \\ 1^1 \end{smallmatrix} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 1 & 3 & 0 & 3 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1] \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} \begin{smallmatrix} 1^1 \\ 4 \\ 1^1 \end{smallmatrix} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 \\ 3 & 5 & 4 & 3 & 0 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[2] \begin{smallmatrix} 1^1 \\ 1^1 \end{smallmatrix} \begin{smallmatrix} 1^1 \\ 4 \\ 1^1 \end{smallmatrix} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[3] $1^{11} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_3$ $T = 9, T_{\max} = 11, \Delta = -63, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 7 & 0 & 3 & 4 \\ 4 & 7 & 8 & 0 & 5 & 6 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 6 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[4] $1^{11} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_3$ $T = 9, T_{\max} = 11, \Delta = -63, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 7 & 3 & 0 & 4 \\ 5 & 7 & 8 & 5 & 0 & 6 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$T = 10$ [6] $1^{11} 5 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 6 & 0 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 \\ 4 & 6 & 4 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

[5] $21^{11} 5 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 \\ 2 & 4 & 1 & 2 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[7] $1^{11} 5 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 1 & 3 \\ 3 & 4 & 10 & 3 & 0 & 4 \\ 6 & 10 & 21 & 6 & 0 & 9 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[9] $1^{11} 5 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{12} 5 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 17 & 3 & 0 & 4 \\ 10 & 17 & 61 & 10 & 0 & 15 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[12] $1^{13} 5 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 1 & 3 \\ 3 & 4 & 24 & 3 & 0 & 4 \\ 14 & 24 & 121 & 14 & 0 & 21 \\ 2 & 3 & 14 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[11] $21^{11} 5 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 3 & 4 \\ 5 & 4 & 4 & 3 & 3 & 4 \\ 3 & 4 & 1 & 2 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 & 2 \\ 4 & 4 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

4.987 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \begin{smallmatrix} 1^{11} \\ 1^{11} \\ 4^{11} \end{smallmatrix} + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 11, \quad \Delta = -33, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 & 4 \\ 2 & 3 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 2 & 2 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[1] $1^{11} 1^{11} 4^1 1^1 + [\textcolor{red}{\mathbf{s}}\textcolor{red}{\mathbf{u}}_2 \cdot \textcolor{red}{\mathbf{g}}_2] + \textcolor{teal}{\mathbf{s}}\textcolor{teal}{\mathbf{u}}_2 + \textcolor{brown}{\mathbf{s}}\textcolor{brown}{\mathbf{u}}_3 + \textcolor{red}{\mathbf{s}}\textcolor{red}{\mathbf{u}}_3 + \textcolor{teal}{\mathbf{s}}\textcolor{teal}{\mathbf{u}}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 & 4 \\ 2 & 3 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 2 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[2] $1^{11} 4^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -33$, $\det = 15$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 0 & 4 \\ 2 & 3 & 1 & 3 & 0 & 0 & 3 \\ 2 & 3 & 3 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 2 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

[4] $1 \overset{1}{\underset{11}{4}} 1 \overset{1}{\underset{11}{4}} + [\textcolor{red}{s} \textcolor{red}{u}_2 \cdot \textcolor{red}{g}_2] + \textcolor{teal}{s} \textcolor{teal}{u}_2 + \textcolor{brown}{s} \textcolor{brown}{u}_3 + \textcolor{teal}{s} \textcolor{teal}{u}_3 + \textcolor{teal}{s} \textcolor{teal}{u}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -33$, $\det = 18$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 3 & 0 & 5 & 4 \\ 2 & 3 & 1 & 3 & 0 & 3 & 3 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 3 & 3 & 0 & 4 & 4 \\ 3 & 4 & 3 & 2 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[3] $1^{11} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 5 & 3 & 0 & 7 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 4 \\ 2 & 3 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 7 & 6 & 5 & 0 & 8 & 6 \\ 3 & 4 & 4 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[5] $1^{11} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 2 & 0 & 1 & 3 \\ 3 & 4 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[6] $1^{111} 4^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 & 3 \\ 3 & 4 & 7 & 0 & 3 & 4 & 4 \\ 4 & 7 & 8 & 0 & 5 & 5 & 6 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 5 & 0 & 0 & 2 & 2 \\ 3 & 4 & 5 & 0 & 2 & 1 & 3 \\ 3 & 4 & 6 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[7] $1^{111} \overset{1}{4} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 11$, $\Delta = -33$, $\det = 64$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 2 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}}4^{\textcolor{blue}{1}}1^{\textcolor{blue}{1}} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{g}}_2] + \textcolor{green}{\mathbf{su}}_2 + \textcolor{red}{\mathbf{su}}_2 + \textcolor{blue}{\mathbf{su}}_3 + \textcolor{blue}{\mathbf{su}}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 91, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$

9.0

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 3 & 0 & 7 & 4 \\ 2 & 4 & 1 & 2 & 0 & 5 & 3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 7 & 5 & 5 & 0 & 8 & 6 \\ 3 & 4 & 3 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}} 4^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} =$$

9.0

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 3 & 0 & 7 & 4 \\ 2 & 4 & 1 & 2 & 0 & 5 & 3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 7 & 5 & 5 & 0 & 8 & 6 \\ 3 & 4 & 3 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

4.988 NHC(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [4] \ 1^{\overline{111}} 4^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 5 & 3 & 0 & 7 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 7 & 6 & 5 & 0 & 8 & 5 & 6 \\ 3 & 4 & 3 & 2 & 0 & 5 & 1 & 3 \\ 3 & 4 & 4 & 2 & -1 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 4^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 5 & 3 & 0 & 7 & 4 & 4 \\ 3 & 5 & 4 & 3 & 0 & 6 & 3 & 4 \\ 2 & 3 & 3 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 7 & 6 & 5 & 0 & 8 & 5 & 6 \\ 2 & 4 & 3 & 2 & 0 & 5 & 1 & 3 \\ 3 & 4 & 4 & 2 & -1 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 & 5 & 4 \\ 2 & 3 & 1 & 3 & 0 & 2 & 3 & 3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 2 & 0 & 1 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 3 & 2 & -1 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 1 & 3 & 3 \\ 3 & 4 & 3 & 3 & 0 & 0 & 5 & 4 \\ 2 & 3 & 1 & 3 & 0 & 0 & 3 & 3 \\ 2 & 3 & 3 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 3 & 3 & 0 & 0 & 4 & 4 \\ 3 & 4 & 3 & 2 & -1 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 4^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -81, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 7 & 0 & 3 & 4 & 4 & 4 \\ 4 & 7 & 8 & 0 & 5 & 5 & 5 & 6 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 5 & 0 & 0 & 2 & 2 & 2 \\ 3 & 4 & 5 & 0 & 2 & 1 & 3 & 3 \\ 2 & 4 & 5 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & 6 & -1 & 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

4.989 NHC(-2)(-3) + su₂ + su₃ + su₃ + su₂ + su₂ + su₂

$$T = 11 \quad [0] \quad 1^{11} \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 4 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 11.0$$

$$[1] \quad 1^{11} \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} 4 \begin{smallmatrix} 11 \\ 11 \end{smallmatrix} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + \text{su}_3 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 11.0$$

4.990 NHC(-2)(-3) + su₂ + su₃ + (-3)→(-2)

$$T = 10 \quad [0] \quad 1^{11} \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 5 \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 3 & 3 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 5 \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 1 & 2 & 0 & 3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 5 \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 10 & 3 & 0 & 4 \\ 8 & 10 & 21 & 6 & 0 & 9 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 5 \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 7 & 3 & 0 & 4 \\ 5 & 7 & 8 & 4 & 0 & 6 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{12} \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 5 \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 141, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 11 & 3 & 0 & 4 \\ 7 & 11 & 21 & 6 & 0 & 9 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{13} \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 5 \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 15 & 3 & 0 & 4 \\ 9 & 15 & 40 & 8 & 0 & 12 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{12} \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 5 \begin{smallmatrix} 1 \\ 1 \end{smallmatrix} 3^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + \text{su}_3 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 1 & 3 \\ 3 & 4 & 17 & 3 & 0 & 4 \\ 13 & 17 & 61 & 10 & 0 & 15 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{12} \underset{1^1}{5} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 196, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 14 & 3 & 0 & 4 \\ 10 & 14 & 38 & 8 & 0 & 12 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^1 \underset{1^1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 285, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 5 & 2 & 1 & 3 \\ 6 & 13 & 12 & 5 & 0 & 7 \\ 5 & 12 & 8 & 4 & 0 & 6 \\ 2 & 5 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 6 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{13} \underset{1^1}{5} 1^2 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 316, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 18 & 3 & 0 & 4 \\ 12 & 18 & 61 & 10 & 0 & 15 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{13} \underset{1^1}{5} 1^3 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 381, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 1 & 3 \\ 3 & 4 & 21 & 3 & 0 & 4 \\ 15 & 21 & 88 & 12 & 0 & 18 \\ 2 & 3 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^1 \underset{1^1}{5} 2^{12} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 516, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 10 & 2 & 1 & 3 \\ 6 & 13 & 24 & 5 & 0 & 7 \\ 10 & 24 & 38 & 8 & 0 & 12 \\ 2 & 5 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 12 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 1^1 \underset{1^1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 901, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 15 & 2 & 1 & 3 \\ 6 & 13 & 36 & 5 & 0 & 7 \\ 15 & 36 & 88 & 12 & 0 & 18 \\ 2 & 5 & 12 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 18 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

4.991 NHC(-2)(-3) + $\textcolor{red}{su}_2 + (-3) \rightarrow (-2)$

$$T = 7 \quad [0] \ 1^{11} 31^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, \ T_{\max} = 12, \ \Delta = -94, \ \det = -21, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.43 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 4 \\ 5 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 31^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, \ T_{\max} = 12, \ \Delta = -94, \ \det = -24, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.50 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 9 & 12 & 0 & 5 \\ 6 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 7 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 312^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, \ T_{\max} = 12, \ \Delta = -94, \ \det = -29, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.41 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 9 & 8 & 0 & 5 \\ 3 & 8 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & 1 \end{pmatrix}_{b_0}$$

- [3] $1^1 31^{12} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -51$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 6 & 1 & 4 \\ 5 & 4 & 5 & 0 & 4 \\ 6 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 4 & -1 & 2 \end{pmatrix}_{b_0}$
- [5] $1^1 31^{13} 11 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -66$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 15 & 1 & 4 \\ 5 & 4 & 17 & 0 & 4 \\ 15 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 13 & -1 & 2 \end{pmatrix}_{b_0}$
- [4] $1^1 31^{12} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -66$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.21 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 4 & 5 & 12 & 1 & 4 \\ 5 & 4 & 10 & 0 & 4 \\ 12 & 10 & 22 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 8 & -1 & 1 \end{pmatrix}_{b_0}$
- [7] $1^{12} 31^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -84$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 14 & 0 & 4 \\ 9 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 11 & -1 & 2 \end{pmatrix}_{b_0}$
- [6] $1^1 31^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -84$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 2 \end{pmatrix}_{b_0}$
- [8] $1^{112} 31^{12} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -89$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 9 & 20 & 0 & 5 \\ 9 & 20 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 11 & -1 & 2 \end{pmatrix}_{b_0}$
- [9] $1^1 31^{13} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -91$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 4 & 5 & 18 & 1 & 4 \\ 5 & 4 & 15 & 0 & 4 \\ 18 & 15 & 52 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 12 & -1 & 1 \end{pmatrix}_{b_0}$
- [10] $1^{11} 31^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -106$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 9 & 1 & 4 \\ 5 & 4 & 9 & 0 & 4 \\ 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 7 & -1 & 2 \end{pmatrix}_{b_0}$
- [11] $1^{12} 31^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -111$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 12 & 1 & 4 \\ 5 & 4 & 13 & 0 & 4 \\ 12 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 10 & -1 & 2 \end{pmatrix}_{b_0}$
- [12] $1^{11} 31^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -124$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.06 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 4 & 9 & 0 & 4 \\ 6 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -1 & 2 \end{pmatrix}_{b_0}$
- [13] $1^{14} 31^{12} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -144$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10.22 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 4 & 28 & 0 & 4 \\ 18 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 2 \end{pmatrix}_{b_0}$
- [14] $1^1 31^{12} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -144$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10.22 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 4 & 8 & 0 & 4 \\ 6 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 2 \end{pmatrix}_{b_0} \\
[15] \quad & 1^{113}31^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -149, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 9 & 32 & 0 & 5 \\ 15 & 32 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 18 & -1 & 1 \end{pmatrix}_{b_0} \\
[16] \quad & 1^{131}1^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 8 & 9 & 1 & 4 \\ 8 & 9 & 12 & 0 & 5 \\ 9 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 7 & -1 & 1 \end{pmatrix}_{b_0} \\
[17] \quad & 1^{113}31^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 9 & 28 & 0 & 5 \\ 12 & 28 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 15 & -1 & 1 \end{pmatrix}_{b_0} \\
[18] \quad & 131^{111}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 8 & 6 & 1 & 4 \\ 8 & 9 & 8 & 0 & 5 \\ 6 & 8 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 4 & -1 & 2 \end{pmatrix}_{b_0} \\
[19] \quad & 1^{113}31^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -201, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 15 & 1 & 4 \\ 5 & 4 & 14 & 0 & 4 \\ 15 & 14 & 37 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 11 & -1 & 2 \end{pmatrix}_{b_0} \\
[20] \quad & 1^{114}31^{32} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -209, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 21 & 1 & 3 \\ 4 & 9 & 44 & 0 & 5 \\ 21 & 44 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 25 & -1 & 1 \end{pmatrix}_{b_0} \\
[21] \quad & 1^{131}1^{32} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 12 & 0 & 4 \\ 9 & 12 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 2 \end{pmatrix}_{b_0} \\
[22] \quad & 1^{114}31^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.19 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 9 & 36 & 0 & 5 \\ 15 & 36 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 19 & -1 & 1 \end{pmatrix}_{b_0} \\
[24] \quad & 1^{113}31^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 13 & 0 & 4 \\ 9 & 13 & 28 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 2 \end{pmatrix}_{b_0} \\
[23] \quad & 1^{13}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 23 & 0 & 4 \\ 15 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [25] $1^{114}31^{12}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -284$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 9 & 40 & 0 & 5 \\ 18 & 40 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 22 & -1 & 2 \end{pmatrix}_{b_0}$
- [26] $1^{123}31^{12}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -286$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10.32 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 18 & 1 & 4 \\ 5 & 4 & 18 & 0 & 4 \\ 18 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 14 & -1 & 2 \end{pmatrix}_{b_0}$
- [27] $1^{123}31^{12}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -304$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10.11 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 4 & 18 & 0 & 4 \\ 12 & 18 & 58 & 0 & 14 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 14 & -1 & 2 \end{pmatrix}_{b_0}$
- [29] $1^{14}31^{12}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -306$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10.26 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 24 & 1 & 4 \\ 5 & 4 & 26 & 0 & 4 \\ 24 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 20 & -1 & 2 \end{pmatrix}_{b_0}$
- [28] $1^{113}31^{13}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -306$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 10.59 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 21 & 1 & 4 \\ 5 & 4 & 19 & 0 & 4 \\ 21 & 19 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 15 & -1 & 2 \end{pmatrix}_{b_0}$
- [30] $1^{13}31^{12}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -321$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 10.28 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 21 & 1 & 4 \\ 5 & 4 & 22 & 0 & 4 \\ 21 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 17 & -1 & 2 \end{pmatrix}_{b_0}$
- [31] $1^{131}14^21 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -384$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 4 & 16 & 0 & 4 \\ 12 & 16 & 46 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 2 \end{pmatrix}_{b_0}$
- [32] $31^{111}22^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -409$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 3 & 1 \\ 3 & 4 & 0 & 3 & 1 \\ 3 & 3 & 3 & 1 & 1 \\ 1 & 1 & 1 & 1 & -2 \\ 2 & 3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$
- [33] $1^{11}31^{14}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -421$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.72 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 27 & 1 & 4 \\ 5 & 4 & 24 & 0 & 4 \\ 27 & 24 & 121 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 19 & -1 & 2 \end{pmatrix}_{b_0}$
- [34] $22^11^{11}3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -439$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 4 & 7 & 7 & 7 & 4 \\ 7 & 5 & 8 & 7 & 5 \\ 7 & 8 & 4 & 7 & 4 \\ 7 & 7 & 7 & 5 & 5 \\ 4 & 5 & 4 & 5 & 2 \end{pmatrix}_{b_0}$
- [35] $1^{11}31^{13}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 12$, $\Delta = -94$, $\det = -444$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$

$$\begin{aligned}
& T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 4 & 17 & 0 & 4 \\ 12 & 17 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 2 \end{pmatrix}_{b_0} \\
[36] \quad & 1^{14}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -471, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 24 & 1 & 4 \\ 5 & 4 & 23 & 0 & 4 \\ 24 & 23 & 100 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 18 & -1 & 2 \end{pmatrix}_{b_0} \\
[37] \quad & 1^{14}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -564, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 21 & 1 & 3 \\ 4 & 4 & 32 & 0 & 4 \\ 21 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 2 \end{pmatrix}_{b_0} \\
[38] \quad & 1^{14}31^{15}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -564, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.1 \\
& T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 20 & 0 & 4 \\ 15 & 20 & 73 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 2 \end{pmatrix}_{b_0} \\
[39] \quad & 1^{12}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -564, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 22 & 0 & 4 \\ 15 & 22 & 85 & 0 & 17 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 17 & -1 & 2 \end{pmatrix}_{b_0} \\
[40] \quad & 1^{13}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -586, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 27 & 1 & 4 \\ 5 & 4 & 27 & 0 & 4 \\ 27 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 21 & -1 & 2 \end{pmatrix}_{b_0} \\
[41] \quad & 1^{13}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -604, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 4 & 27 & 0 & 4 \\ 18 & 27 & 133 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 2 \end{pmatrix}_{b_0} \\
[42] \quad & 1^{14}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -651, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 30 & 1 & 4 \\ 5 & 4 & 31 & 0 & 4 \\ 30 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 24 & -1 & 2 \end{pmatrix}_{b_0} \\
[43] \quad & 1^{11}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -664, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 21 & 0 & 4 \\ 15 & 21 & 76 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 2 \end{pmatrix}_{b_0} \\
[44] \quad & 1^{12}31^{14}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -666, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 5 & 30 & 1 & 4 \\ 5 & 4 & 28 & 0 & 4 \\ 30 & 28 & 154 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 22 & -1 & 2 \end{pmatrix}_{b_0} \\
[45] \quad & 31^{112}22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -724, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]
\end{aligned}$$

[46] $\begin{pmatrix} 0 & 3 & 3 & 6 & 1 & 2 \\ 3 & 1 & 4 & 6 & 1 & 3 \\ 3 & 4 & 0 & 6 & 1 & 2 \\ 6 & 6 & 6 & 10 & 2 & 6 \\ 1 & 1 & 1 & 2 & -2 & 0 \\ 2 & 3 & 2 & 6 & 0 & 1 \end{pmatrix}_{b_0}$ $1^1 31^{16} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 7, T_{\max} = 12, \Delta = -94, \det = -784, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 4 & 24 & 0 & 4 \\ 18 & 24 & 106 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 2 \end{pmatrix}_{b_0}$

[47] $1^{13} 31^{14} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 7, T_{\max} = 12, \Delta = -94, \det = -861, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 33 & 1 & 4 \\ 5 & 4 & 32 & 0 & 4 \\ 33 & 32 & 193 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 25 & -1 & 2 \end{pmatrix}_{b_0}$

[48] $1^{12} 31^{14} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 7, T_{\max} = 12, \Delta = -94, \det = -864, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 4 & 26 & 0 & 4 \\ 18 & 26 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 20 & -1 & 2 \end{pmatrix}_{b_0}$

[49] $1^{13} 31^{14} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 7, T_{\max} = 12, \Delta = -94, \det = -984, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 21 & 1 & 3 \\ 4 & 4 & 31 & 0 & 4 \\ 21 & 31 & 172 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 2 \end{pmatrix}_{b_0}$

[50] $1^{14} 31^{14} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 7, T_{\max} = 12, \Delta = -94, \det = -1006, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 36 & 1 & 4 \\ 5 & 4 & 36 & 0 & 4 \\ 36 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 28 & -1 & 2 \end{pmatrix}_{b_0}$

[51] $1^{14} 31^{14} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 7, T_{\max} = 12, \Delta = -94, \det = -1024, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 24 & 1 & 3 \\ 4 & 4 & 36 & 0 & 4 \\ 24 & 36 & 238 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 2 \end{pmatrix}_{b_0}$

[52] $1^1 31^{17} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 7, T_{\max} = 12, \Delta = -94, \det = -1044, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 21 & 1 & 3 \\ 4 & 4 & 28 & 0 & 4 \\ 21 & 28 & 145 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 2 \end{pmatrix}_{b_0}$

[53] $31^{113} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 7, T_{\max} = 12, \Delta = -94, \det = -1249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 3 & 3 & 9 & 1 & 2 \\ 3 & 1 & 4 & 9 & 1 & 3 \\ 3 & 4 & 0 & 9 & 1 & 2 \\ 9 & 9 & 9 & 25 & 3 & 9 \\ 1 & 1 & 1 & 3 & -2 & 0 \\ 2 & 3 & 2 & 9 & 0 & 1 \end{pmatrix}_{b_0}$

[54] $31^{114} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 7, T_{\max} = 12, \Delta = -94, \det = -1984, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 3 & 3 & 12 & 1 & 2 \\ 3 & 1 & 4 & 12 & 1 & 3 \\ 3 & 4 & 0 & 12 & 1 & 2 \\ 12 & 12 & 12 & 46 & 4 & 12 \\ 1 & 1 & 1 & 4 & -2 & 0 \\ 2 & 3 & 2 & 12 & 0 & 1 \end{pmatrix}_{b_0}$

[55] $31^{115} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 7, T_{\max} = 12, \Delta = -94, \det = -2929, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.1$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 11 & 0 & 4 \\ 7 & 11 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[67] $1^{1^2 1^1 1^1} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 4 & 7 & 0 & 4 \\ 5 & 7 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

[68] $1^{1^2 1^1 1^1} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 8 & 12 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[69] $21^{1^1 1^1} 41^{1^1 2^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 8 & 12 & 3 & 4 \\ 8 & 13 & 20 & 6 & 7 \\ 12 & 20 & 26 & 8 & 10 \\ 3 & 6 & 8 & 0 & 2 \\ 4 & 7 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{1^3} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 9 & 15 & 40 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[70] $1^{1^2 1^3} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 14 & 18 & 70 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

[71] $1^{1^2 1^2} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 10 & 14 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[74] $1^{1^2 1^2} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 11 & 15 & 45 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 3 \end{pmatrix}_{b_0}$$

[73] $1^{1^2 1^3} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 13 & 17 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[75] $1^{1^2 1^1 1^1} 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 8 & 1 & 3 \\ 4 & 4 & 12 & 0 & 4 \\ 8 & 12 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[76] $1^{2^1 1^1 1^2} 4^1 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 10 & 1 & 3 \\ 4 & 4 & 14 & 0 & 4 \\ 10 & 14 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[77] $1^{12} 4^{211} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 10 & 16 & 45 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 3 \end{pmatrix}_{b_0}$

[78] $1^{14} 4^{11} 22^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 6 & 5 & 1 & 3 \\ 6 & 13 & 12 & 0 & 7 \\ 5 & 12 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 6 & -1 & 3 \end{pmatrix}_{b_0}$

[79] $1^{11} 4^{11} 22^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 3 & 4 & 1 & 2 \\ 1 & 0 & 3 & 4 & 0 & 2 \\ 3 & 3 & 3 & 8 & 0 & 3 \\ 4 & 4 & 8 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & 6 & -1 & 2 \end{pmatrix}_{b_0}$

[80] $22^{11} 1^{11} 4^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 6 & 5 & 3 & 3 \\ 6 & 13 & 12 & 6 & 7 \\ 5 & 12 & 8 & 5 & 6 \\ 3 & 6 & 5 & 0 & 2 \\ 3 & 7 & 6 & 2 & 3 \end{pmatrix}_{b_0}$

[81] $1^{13} 4^{211} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 12 & 18 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$

[82] $22^{11} 1^{11} 4^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 6 & 8 & 3 & 3 \\ 6 & 13 & 20 & 6 & 7 \\ 8 & 20 & 26 & 8 & 10 \\ 3 & 6 & 8 & 0 & 2 \\ 3 & 7 & 10 & 2 & 3 \end{pmatrix}_{b_0}$

[83] $21^{11} 4^{11} 1^{421} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 4 & 8 & 30 & 3 & 4 \\ 8 & 13 & 56 & 6 & 7 \\ 30 & 56 & 212 & 23 & 28 \\ 3 & 6 & 23 & 0 & 2 \\ 4 & 7 & 28 & 2 & 3 \end{pmatrix}_{b_0}$

[84] $21^{11} 4^{11} 1^{221} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 4 & 8 & 18 & 3 & 4 \\ 8 & 13 & 32 & 6 & 7 \\ 18 & 32 & 68 & 13 & 16 \\ 3 & 6 & 13 & 0 & 2 \\ 4 & 7 & 16 & 2 & 3 \end{pmatrix}_{b_0}$

[85] $1^{11} 4^{11} 22^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 6 & 8 & 1 & 3 \\ 6 & 13 & 20 & 0 & 7 \\ 8 & 20 & 26 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 10 & -1 & 3 \end{pmatrix}_{b_0}$

[86] $1^{13} 4^{211} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 222, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 12 & 20 & 70 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$

- [87] $1^{12} 2^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 13 & 19 & 68 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$
- [88] $21^{11} 41^{13} 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 4 & 8 & 24 & 3 & 4 \\ 8 & 13 & 44 & 6 & 7 \\ 24 & 44 & 130 & 18 & 22 \\ 3 & 6 & 18 & 0 & 2 \\ 4 & 7 & 22 & 2 & 3 \end{pmatrix}_{b_0}$
- [89] $1^{13} 2^{11} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 15 & 21 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$
- [90] $1^{13} 2^{11} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 18 & 24 & 121 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$
- [91] $1^{12} 2^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 16 & 22 & 97 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}$
- [92] $1^{14} 2^{11} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 14 & 22 & 90 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$
- [93] $1^{12} 2^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 25 & 0 & 4 \\ 19 & 25 & 132 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0}$
- [94] $1^{12} 2^{11} 4 1^3 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 21 & 0 & 4 \\ 15 & 21 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$
- [95] $1^{11} 2^{11} 4 1^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 280, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 13 & 1 & 3 \\ 4 & 4 & 19 & 0 & 4 \\ 13 & 19 & 68 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 3 \end{pmatrix}_{b_0}$
- [96] $1^{14} 2^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 281, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 101 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0}$
- [97] $1^{14} 2^{11} 41^{12} 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 11, \Delta = -85, \det = 308, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$\begin{aligned}
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 10 & 1 & 3 \\ 6 & 13 & 24 & 0 & 7 \\ 10 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 12 & -1 & 3 \end{pmatrix}_{b_0} \\
[98] \quad & 22^1 1^{12} 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 322, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 10 & 3 & 3 \\ 6 & 13 & 24 & 6 & 7 \\ 10 & 24 & 38 & 10 & 12 \\ 3 & 6 & 10 & 0 & 2 \\ 3 & 7 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
[99] \quad & 1^{13} 2^1 1^2 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 327, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 15 & 23 & 97 & 0 & 19 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 19 & -1 & 3 \end{pmatrix}_{b_0} \\
[100] \quad & 1^{14} 2^1 1^3 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 351, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 25 & 0 & 4 \\ 17 & 25 & 121 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0} \\
[101] \quad & 1^{13} 2^1 1^3 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 402, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 18 & 26 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0} \\
[102] \quad & 1^{14} 2^1 1^4 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 28 & 0 & 4 \\ 20 & 28 & 158 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0} \\
[103] \quad & 1^{14} 2^1 1^2 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 426, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 17 & 27 & 132 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0} \\
[104] \quad & 1^{13} 2^1 1^4 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 447, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 29 & 0 & 4 \\ 21 & 29 & 169 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 3 \end{pmatrix}_{b_0} \\
[105] \quad & 1^{13} 2^1 1^4 4 1^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 20 & 1 & 3 \\ 4 & 4 & 28 & 0 & 4 \\ 20 & 28 & 158 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0} \\
[106] \quad & 22^1 1^{12} 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 475, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 13 & 3 & 3 \\ 6 & 13 & 32 & 6 & 7 \\ 13 & 32 & 68 & 13 & 16 \\ 3 & 6 & 13 & 0 & 2 \\ 3 & 7 & 16 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[107] $1^{11} 1^{21} 1^{13} 4 \ 1^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 18 & 1 & 3 \\ 4 & 4 & 26 & 0 & 4 \\ 18 & 26 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

[108] $1^{11} 1^{41} 1^{22} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 481, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 6 & 13 & 1 & 3 \\ 6 & 13 & 32 & 0 & 7 \\ 13 & 32 & 68 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

[109] $1^{14} 1^{21} 1^{13} 4 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 541, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 20 & 30 & 169 & 0 & 25 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 25 & -1 & 3 \end{pmatrix}_{b_0}$$

[110] $1^{14} 1^{41} 1^{32} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 553, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 6 & 15 & 1 & 3 \\ 6 & 13 & 36 & 0 & 7 \\ 15 & 36 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[111] $22^1 1^{13} 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 567, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 6 & 15 & 3 & 3 \\ 6 & 13 & 36 & 6 & 7 \\ 15 & 36 & 88 & 15 & 18 \\ 3 & 6 & 15 & 0 & 2 \\ 3 & 7 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[112] $1^{14} 1^{21} 1^{14} 4 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 626, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 23 & 1 & 3 \\ 3 & 4 & 33 & 0 & 4 \\ 23 & 33 & 212 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 3 \end{pmatrix}_{b_0}$$

[113] $1^{11} 1^{21} 1^{14} 4 \ 1^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 728, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 23 & 1 & 3 \\ 4 & 4 & 33 & 0 & 4 \\ 23 & 33 & 212 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 28 & -1 & 3 \end{pmatrix}_{b_0}$$

[114] $22^1 1^{13} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 846, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 6 & 18 & 3 & 3 \\ 6 & 13 & 44 & 6 & 7 \\ 18 & 44 & 130 & 18 & 22 \\ 3 & 6 & 18 & 0 & 2 \\ 3 & 7 & 22 & 2 & 3 \end{pmatrix}_{b_0}$$

[115] $1^{11} 1^{41} 1^{32} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 852, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 6 & 18 & 1 & 3 \\ 6 & 13 & 44 & 0 & 7 \\ 18 & 44 & 130 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

[116] $1^{14} 1^{41} 1^{42} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 896, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 6 & 20 & 1 & 3 \\ 6 & 13 & 48 & 0 & 7 \\ 20 & 48 & 158 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[117] \ 22^1 1^{14} 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 910, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 20 & 3 & 3 \\ 6 & 13 & 48 & 6 & 7 \\ 20 & 48 & 158 & 20 & 24 \\ 3 & 6 & 20 & 0 & 2 \\ 3 & 7 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[118] \ 22^1 1^{14} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1315, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 23 & 3 & 3 \\ 6 & 13 & 56 & 6 & 7 \\ 23 & 56 & 212 & 23 & 28 \\ 3 & 6 & 23 & 0 & 2 \\ 3 & 7 & 28 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{11} 41^{14} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1321, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 23 & 1 & 3 \\ 6 & 13 & 56 & 0 & 7 \\ 23 & 56 & 212 & 0 & 28 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 28 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{11} 41^{15} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1337, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.1$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 25 & 1 & 3 \\ 6 & 13 & 60 & 0 & 7 \\ 25 & 60 & 248 & 0 & 30 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 30 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{11} 41^{16} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 1876, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 30 & 1 & 3 \\ 6 & 13 & 72 & 0 & 7 \\ 30 & 72 & 358 & 0 & 36 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 36 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[122] \ 1^{11} 41^{17} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 2513, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 35 & 1 & 3 \\ 6 & 13 & 84 & 0 & 7 \\ 35 & 84 & 488 & 0 & 42 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 42 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [123] \ 22^1 1^{15} 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 5 & 6 & 8 & 5 & 6 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[124] \ 21^{211} 51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 5 & 9 & 3 & 4 \\ 5 & 4 & 10 & 3 & 4 \\ 9 & 10 & 19 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 4 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[125] \ 1^{11} 51^{14} 12^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 13 & 19 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{11} 51^{12} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

6.5

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 10 & 17 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[139] \ 21^{11}51^{11}22^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

6.0

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 3 & 1 & 2 \\ 3 & 4 & 5 & 6 & 0 & 4 \\ 2 & 5 & 3 & 4 & 0 & 3 \\ 3 & 6 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 3 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{12}5^{11}2^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

6.5

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 11 & 0 & 4 \\ 6 & 11 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[138] \ 1^{11}5^{14}2^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

6.5

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 16 & 22 & 117 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{11}5^{12}2^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

6.5

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 8 & 10 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[141] \ 1^{11}5^{13}2^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

6.8

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 13 & 16 & 59 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{11}5^{13}2^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

6.0

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 11 & 16 & 59 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[143] \ 1^{12}5^{12}2^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

6.5

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[144] \ 1^{11}5^{11}2^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

6.0

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[145] \ 1^{11}5^{14}2^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

6.8

$$\begin{aligned}
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 25 & 0 & 4 \\ 19 & 25 & 152 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0} \\
[146] & 1^{\frac{1}{2} 21^1} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (\textcolor{red}{-3}) \rightarrow (\textcolor{red}{-2}) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 6 & 2 & 1 & 3 \\ 6 & 13 & 5 & 0 & 7 \\ 2 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 3 & -1 & 3 \end{pmatrix}_{b_0} \\
[147] & 22^{\frac{1}{2} 1^1} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (\textcolor{red}{-3}) \rightarrow (\textcolor{red}{-2}) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -49, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 10 & 3 & 4 \\ 7 & 10 & 19 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0} \\
[148] & 22^{\frac{1}{2} 1^1} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (\textcolor{red}{-3}) \rightarrow (\textcolor{red}{-2}) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -49, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 17 & 3 & 3 \\ 3 & 4 & 22 & 3 & 4 \\ 17 & 22 & 115 & 17 & 21 \\ 3 & 3 & 17 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0} \\
[149] & 21^{\frac{1}{2} 1^1} 5^1 12^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (\textcolor{red}{-3}) \rightarrow (\textcolor{red}{-2}) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
[150] & 1^{\frac{1}{2} 3^1} 5^1 12^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (\textcolor{red}{-3}) \rightarrow (\textcolor{red}{-2}) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 14 & 24 & 121 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0} \\
[151] & 1^{\frac{1}{2} 5^1} 1^1 3^2 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (\textcolor{red}{-3}) \rightarrow (\textcolor{red}{-2}) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 15 & 19 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0} \\
[152] & 1^{\frac{1}{2} 5^1} 1^1 3^2 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (\textcolor{red}{-3}) \rightarrow (\textcolor{red}{-2}) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 9 & 13 & 36 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0} \\
[154] & 1^{\frac{1}{2} 5^1} 1^1 3^2 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (\textcolor{red}{-3}) \rightarrow (\textcolor{red}{-2}) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 7 & 10 & 19 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0} \\
[153] & 1^{\frac{1}{2} 2^1} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (\textcolor{red}{-3}) \rightarrow (\textcolor{red}{-2}) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 17 & 22 & 115 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[157] $22^{11}1^{13}5^{11}1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 16 & 3 & 4 \\ 12 & 16 & 57 & 12 & 15 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[156] $1^{11}5^{13}1^{32}1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 17 & 22 & 115 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[155] $1^{11}5^{21}1^{21}1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -63$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 7 & 10 & 19 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[158] $21^{11}5^{21}1^{11}1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -66$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 7 & 2 & 3 \\ 4 & 4 & 10 & 3 & 4 \\ 7 & 10 & 19 & 7 & 9 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[160] $1^{11}5^{13}1^{22}1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -68$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 19 & 0 & 4 \\ 14 & 19 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[159] $1^{11}5^{11}1^{22}1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -68$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 10 & 13 & 36 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[161] $22^{11}1^{11}5^{11}1^{12} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -72$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.61 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 9 & 14 & 36 & 9 & 12 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[163] $22^{11}1^{11}5^{11}1^{13} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -73$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 18 & 3 & 4 \\ 11 & 18 & 59 & 11 & 15 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[162] $1^{13}5^{12}1^{11}1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -73$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 10 & 18 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[164] $1^{13}5^{13}12^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -76$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 12 & 21 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[166] $1^{11}5^{12}1^22^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -77$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 12 & 16 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[165] $1^{11}2^{21}2^{21}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -77$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 12 & 16 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[167] $1^{11}5^{11}1^22^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 11 & 0 & 4 \\ 7 & 11 & 21 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[168] $21^{12}5^{12}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -86$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 2 & 3 \\ 4 & 4 & 8 & 3 & 4 \\ 6 & 8 & 10 & 4 & 6 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[169] $1^{12}5^{14}1^22^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -107$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 15 & 23 & 117 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[170] $1^{12}5^{11}1^22^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 9 & 14 & 36 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[171] $1^{12}2^{21}2^{21}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 9 & 14 & 36 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[173] $1^{11}5^{13}1^22^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -118$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 15 & 0 & 4 \\ 9 & 15 & 40 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[172] $1 \overset{1}{5} 1 \overset{2}{2} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -118, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 10 & 14 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[174] $1 \overset{1}{5} 1 \overset{3}{2} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 13 & 17 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[175] $21 \overset{1}{5} 1 \overset{2}{2} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 14 & 2 & 3 \\ 4 & 4 & 20 & 3 & 4 \\ 14 & 20 & 82 & 14 & 18 \\ 2 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[176] $1 \overset{1}{5} 1 \overset{3}{2} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 13 & 20 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[177] $1 \overset{1}{5} 1 \overset{2}{2} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 11 & 17 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[178] $21 \overset{1}{5} 1 \overset{3}{2} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 4 & 12 & 3 & 4 \\ 9 & 12 & 25 & 6 & 9 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

[179] $22 \overset{1}{5} 1 \overset{2}{2} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -142, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 20 & 3 & 4 \\ 14 & 20 & 82 & 14 & 18 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

[180] $1 \overset{2}{5} 1 \overset{2}{2} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 7 & 1 & 3 \\ 6 & 13 & 17 & 0 & 7 \\ 7 & 17 & 19 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[181] $1 \overset{2}{5} 1 \overset{2}{2} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 5 & 1 & 3 \\ 6 & 13 & 12 & 0 & 7 \\ 5 & 12 & 8 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

[182] $1^{12}5^{11}1^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 12 & 17 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[183] $1^{13}5^{21}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 11 & 18 & 59 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[184] $1^{13}5^{11}1^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 11 & 18 & 59 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[185] $21^{12}5^{21}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 10 & 2 & 3 \\ 4 & 4 & 14 & 3 & 4 \\ 10 & 14 & 36 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

[186] $1^{12}5^{13}1^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 15 & 20 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[187] $1^{12}5^{14}1^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 18 & 26 & 150 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[188] $22^{13}5^{11}1^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 19 & 3 & 3 \\ 3 & 4 & 26 & 3 & 4 \\ 19 & 26 & 148 & 19 & 24 \\ 3 & 3 & 19 & 0 & 2 \\ 3 & 4 & 24 & 2 & 3 \end{pmatrix}_{b_0}$$

[189] $1^{12}5^{12}1^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 14 & 20 & 82 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[190] $1^{12}5^{21}1^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 14 & 20 & 82 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[191] $1^{12}5^{13}1^22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -193, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 16 & 23 & 113 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[192] $1^{11}5^{12}1^22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -193, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 12 & 18 & 61 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[193] $1^{13}5^{12}1^22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -198, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 13 & 21 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[194] $22^{11}1^{12}5^{13}1^22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 24 & 3 & 4 \\ 16 & 24 & 113 & 16 & 21 \\ 3 & 3 & 16 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[195] $1^{13}5^{14}1^22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 17 & 27 & 152 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[196] $1^{13}5^{13}1^22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -211, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 15 & 24 & 115 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[197] $1^{12}5^{12}1^32^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -211, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 23 & 0 & 4 \\ 17 & 23 & 113 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[198] $21^{13}5^{21}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 21 & 2 & 3 \\ 4 & 4 & 30 & 3 & 4 \\ 21 & 30 & 187 & 21 & 27 \\ 2 & 3 & 21 & 0 & 2 \\ 3 & 4 & 27 & 2 & 3 \end{pmatrix}_{b_0}$$

[199] $1^{12}5^{14}1^32^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 29 & 0 & 4 \\ 21 & 29 & 189 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

[200] $1^{12} 5^{2^1 21^3} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -228$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.61 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 19 & 26 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[201] $1^{12} 5^{1^3} 1^{32^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -228$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.61 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 19 & 26 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[202] $1^{1^1} 5^{1^3} 1^{32^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -238$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.80 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 15 & 21 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[203] $1^{12} 5^{2^1 21^4} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -242$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.64 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 24 & 1 & 3 \\ 3 & 4 & 32 & 0 & 4 \\ 24 & 32 & 234 & 0 & 30 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 30 & -1 & 3 \end{pmatrix}_{b_0}$$

[204] $1^{1^1} 5^{1^3} 1^{2^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -252$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 14 & 21 & 84 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[205] $1^{1^2} 5^{2^1 21^{12}} 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -276$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 14 & 1 & 3 \\ 6 & 13 & 34 & 0 & 7 \\ 14 & 34 & 82 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[206] $1^{1^2} 5^{1^3} 1^{2^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -293$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 16 & 24 & 113 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[207] $1^{1^3} 5^{2^1 21^2} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -293$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 16 & 24 & 113 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[208] $22^1 1^3 5^1 1^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -297$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$T_{\min} = 11.64 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 21 & 3 & 3 \\ 3 & 4 & 30 & 3 & 4 \\ 21 & 30 & 187 & 21 & 27 \\ 3 & 3 & 21 & 0 & 2 \\ 3 & 4 & 27 & 2 & 3 \end{pmatrix}_{b_0}$$

[209] $21^{13}5^{2^1 1^1}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -306$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 13 & 2 & 3 \\ 4 & 4 & 18 & 3 & 4 \\ 13 & 18 & 59 & 11 & 15 \\ 2 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

[210] $1^{13}5^{1^1}1^{3^2 1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -307$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 24 & 0 & 4 \\ 17 & 24 & 115 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

[211] $1^{2^1 21^{12}}5^{1^1}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -308$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 10 & 1 & 3 \\ 6 & 13 & 24 & 0 & 7 \\ 10 & 24 & 38 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[212] $1^{13}5^{1^1}1^{2^1 2^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -316$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 18 & 27 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[213] $1^{13}5^{1^1}1^{2^1 2^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -321$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 20 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 20 & 30 & 189 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

[214] $21^{13}5^{2^1 1^2}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -336$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 17 & 2 & 3 \\ 4 & 4 & 24 & 3 & 4 \\ 17 & 24 & 113 & 16 & 21 \\ 2 & 3 & 16 & 0 & 2 \\ 3 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0}$$

[215] $1^{2^1 21^{12}}5^{1^1}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -356$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 12 & 1 & 3 \\ 6 & 13 & 29 & 0 & 7 \\ 12 & 29 & 57 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[216] $1^{13}5^{1^1}1^{3^2 2^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -358$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 19 & 1 & 3 \\ 3 & 4 & 27 & 0 & 4 \\ 19 & 27 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[217] $1^{13}5^{1^1}1^{3^2 2^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9$, $T_{\max} = 12$, $\Delta = -80$, $\det = -391$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 21 & 30 & 187 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

[218] $1^{13} 5^{2^1 21^3} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -391, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 21 & 1 & 3 \\ 3 & 4 & 30 & 0 & 4 \\ 21 & 30 & 187 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

[219] $1^{13} 5^{1^4 3^2 1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -406, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 23 & 1 & 3 \\ 3 & 4 & 33 & 0 & 4 \\ 23 & 33 & 232 & 0 & 30 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 30 & -1 & 3 \end{pmatrix}_{b_0}$$

[220] $1^{13} 5^{2^1 21^4} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -461, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 26 & 1 & 3 \\ 3 & 4 & 36 & 0 & 4 \\ 26 & 36 & 281 & 0 & 33 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 33 & -1 & 3 \end{pmatrix}_{b_0}$$

[221] $1^{1^2 21^{13}} 5^{1^3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -481, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 21 & 1 & 3 \\ 6 & 13 & 51 & 0 & 7 \\ 21 & 51 & 187 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 27 & -1 & 3 \end{pmatrix}_{b_0}$$

[222] $1^{2^1 21^{13}} 5^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -553, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 15 & 1 & 3 \\ 6 & 13 & 36 & 0 & 7 \\ 15 & 36 & 88 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

[223] $1^{2^1 21^{13}} 5^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -633, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 19 & 1 & 3 \\ 6 & 13 & 46 & 0 & 7 \\ 19 & 46 & 148 & 0 & 24 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

[224] $1^{2^1 21^{13}} 5^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -657, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 17 & 1 & 3 \\ 6 & 13 & 41 & 0 & 7 \\ 17 & 41 & 115 & 0 & 21 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

$T = 10$ [225] $1^{1^{11} 31} 5^{12^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 \\ 1 & 2 & 0 & 3 & 0 \\ 1 & 1 & 3 & 1 & 0 \\ 1 & 0 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

[226] $1^{1^{11} 31} 5^{12^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 2 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[227] \ 1^{11} 5^{11} 12^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 13, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 \\ 1 & 2 & 0 & 1 & 0 \\ 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 \\ 2 & 2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[228] \ 1^{11} 5^{21} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 24, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 3 & 2 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[229] \ 1^{11} 5^{21} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 26, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 5 & 8 & 0 & 5 \\ 5 & 8 & 9 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 7 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[230] \ 1^{11} 5^{11} 12^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 29, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 3 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[231] \ 1^{11} 5^{21} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 52, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 5 & 13 & 0 & 5 \\ 7 & 13 & 25 & 0 & 11 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 11 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[232] \ 1^{11} 5^{21} 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 62, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 5 & 18 & 0 & 5 \\ 9 & 18 & 49 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 15 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[233] \ 1^{11} 5^{21} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 69, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 4 & 0 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 \\ 4 & 4 & 5 & 11 & 0 & 7 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[234] \ 1^{11} 5^{21} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 101, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 5 & 0 & 2 \\ 1 & 2 & 0 & 8 & 0 & 2 \\ 5 & 5 & 8 & 24 & 0 & 10 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[235] \ 1^{12} 5^{21} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 108, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 \\ 1 & 2 & 0 & 6 & 0 & 2 \\ 5 & 6 & 6 & 21 & 0 & 9 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

[236] $1^{\overline{1}^2 \overline{1}^2} 5^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 122, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 5 & 21 & 0 & 5 \\ 12 & 21 & 70 & 0 & 18 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 18 & -1 & 4 \end{pmatrix}_{b_0}$$

[237] $1^{\overline{1}^2 \overline{1}^2} 5^{13} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 141, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 \\ 7 & 6 & 9 & 36 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[238] $1^{\overline{1}^2 \overline{1}^2} 5^{13} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 5 & 26 & 0 & 5 \\ 14 & 26 & 106 & 0 & 22 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 22 & -1 & 4 \end{pmatrix}_{b_0}$$

[239] $1^{\overline{1}^2 \overline{1}^2} 5^{13} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 168, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 \\ 6 & 7 & 9 & 36 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[240] $1^{\overline{1}^2 \overline{1}^2} 5^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 189, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 12 & 0 & 2 \\ 8 & 7 & 12 & 57 & 0 & 15 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[241] $1^{\overline{1}^2 \overline{1}^2} 5^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 204, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 12 & 0 & 2 \\ 7 & 8 & 12 & 57 & 0 & 15 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

[242] $1^{\overline{1}^2 \overline{1}^2} 5^{13} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 10 & 0 & 2 \\ 8 & 8 & 10 & 50 & 0 & 14 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 14 & -1 & 3 \end{pmatrix}_{b_0}$$

[243] $1^{\overline{1}^2 \overline{1}^3} 5^{13} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 229, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 13 & 0 & 2 \\ 10 & 8 & 13 & 75 & 0 & 17 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

[244] $1^{\overline{1}^3 \overline{1}^2} 5^{13} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 229, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 10 & 0 & 2 \\ 7 & 9 & 10 & 52 & 0 & 14 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 14 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[245] \quad 1^{13} 5^{2^{11}} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 293, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 13 & 0 & 2 \\ 8 & 10 & 13 & 75 & 0 & 17 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[246] \quad 1^{11} 5^{2^{13}} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 293, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 16 & 0 & 2 \\ 11 & 9 & 16 & 104 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 20 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[247] \quad 1^{12} 5^{2^{12}} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 304, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 13 & 0 & 2 \\ 9 & 9 & 13 & 73 & 0 & 17 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 17 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[248] \quad 1^{13} 5^{2^{12}} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 309, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 \\ 9 & 10 & 11 & 68 & 0 & 16 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[249] \quad 1^{12} 5^{2^{12}} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 16 & 0 & 2 \\ 10 & 10 & 16 & 102 & 0 & 20 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 20 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[250] \quad 1^{12} 5^{2^{13}} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 364, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 2 \\ 1 & 2 & 0 & 14 & 0 & 2 \\ 11 & 10 & 14 & 93 & 0 & 19 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[251] \quad 1^{13} 5^{2^{12}} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 413, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 1 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 \\ 1 & 2 & 0 & 14 & 0 & 2 \\ 10 & 11 & 14 & 93 & 0 & 19 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 19 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[252] \quad 1^{12} 5^{2^{13}} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 456, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 0 & 2 & 11 & 0 & 2 \\ 1 & 2 & 0 & 17 & 0 & 2 \\ 12 & 11 & 17 & 124 & 0 & 22 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[253] \quad 1^{13} 5^{2112} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 493, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 2 \\ 1 & 2 & 0 & 17 & 0 & 2 \\ 11 & 12 & 17 & 124 & 0 & 22 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 22 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[254] \quad 1^{13} 5^{2113} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 493, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 2 \\ 1 & 2 & 0 & 15 & 0 & 2 \\ 12 & 12 & 15 & 115 & 0 & 21 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 21 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[255] \quad 1^{13} 5^{2113} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 613, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 13 & 1 & 2 \\ 1 & 0 & 2 & 13 & 0 & 2 \\ 1 & 2 & 0 & 18 & 0 & 2 \\ 13 & 13 & 18 & 148 & 0 & 24 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 24 & -1 & 3 \end{pmatrix}_{b_0}$$

4.992 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{e}_6$

$$T = 8 \quad [0] \quad 1^{13} 1^{111} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{e}_6 \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -143, \quad \det = 925, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 98, \quad T \in [13, 98]$$

$$\begin{pmatrix} 4 & 8 & 6 & 1 & 3 & 4 \\ 8 & 9 & 8 & 0 & 4 & 5 \\ 6 & 8 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 0 & -3 & -1 \\ 4 & 5 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

4.993 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{f}_4$

$$T = 8 \quad [0] \quad 1^{13} 1^{111} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{f}_4 \quad T = 8, \quad T_{\max} = 31, \quad \Delta = -117, \quad \det = 725, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 31, \quad T \in [12, 31]$$

$$\begin{pmatrix} 4 & 8 & 6 & 1 & 3 & 4 \\ 8 & 9 & 8 & 0 & 4 & 5 \\ 6 & 8 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 0 & -2 & 0 \\ 4 & 5 & 4 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

4.994 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8$

$$T = 8 \quad [0] \quad 1^{111} 31^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, \quad T_{\max} = 13, \quad \Delta = -93, \quad \det = 34, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 12 & 1 & 7 \\ 17 & 20 & 16 & 0 & 8 \\ 12 & 16 & 10 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, \quad T_{\max} = 13, \quad \Delta = -93, \quad \det = 79, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 31^{1121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, \quad T_{\max} = 13, \quad \Delta = -93, \quad \det = 299, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 27 & 1 & 7 \\ 17 & 20 & 33 & 0 & 8 \\ 27 & 33 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 31^{111} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, \quad T_{\max} = 13, \quad \Delta = -93, \quad \det = 325, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 27 & 1 & 7 \\ 20 & 25 & 36 & 0 & 9 \\ 27 & 36 & 49 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}31^{142^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 434, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 72 & 1 & 7 \\ 17 & 20 & 84 & 0 & 8 \\ 72 & 84 & 346 & 0 & 34 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 34 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111}31^{122^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 42 & 1 & 7 \\ 17 & 20 & 50 & 0 & 8 \\ 42 & 50 & 118 & 0 & 20 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 20 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111}31^{132^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 499, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 57 & 1 & 7 \\ 17 & 20 & 67 & 0 & 8 \\ 57 & 67 & 217 & 0 & 27 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 27 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^131^{111}2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 525, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 15 & 1 & 7 \\ 20 & 25 & 20 & 0 & 9 \\ 15 & 20 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

4.995 NHC(-2)(-3) + $\mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{2^11^1}5 \ 131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 3 & 0 & 4 & 2 \\ 1 & 1 & 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 2 & 4 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & 3 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12}5 \ 131^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 7 & 2 \\ 1 & 2 & 0 & 3 & 0 & 9 & 2 \\ 1 & 1 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 7 & 9 & 9 & 0 & 36 & 12 \\ 2 & 2 & 2 & 3 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12}5 \ 131^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 9 & 2 \\ 1 & 2 & 0 & 3 & 0 & 13 & 2 \\ 1 & 1 & 3 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 9 & 9 & 13 & 13 & 0 & 73 & 17 \\ 2 & 2 & 2 & 3 & -1 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12}5 \ 131^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 1 & 0 & 11 & 2 \\ 1 & 2 & 0 & 3 & 0 & 17 & 2 \\ 1 & 1 & 3 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 12 & 11 & 17 & 17 & 0 & 124 & 22 \\ 2 & 2 & 2 & 3 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\mathbf{2}^1\mathbf{1}^1} \ 5 \ 131^{\mathbf{11}^2} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 5 & 5 & 0 & 13 & 5 \\ 2 & 5 & 2 & 0 & 10 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 7 & 13 & 10 & 0 & 25 & 11 \\ 3 & 5 & 4 & -1 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\mathbf{2}^1\mathbf{1}^1} \ 5 \ 131^{\mathbf{11}^3} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -53, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 5 & 5 & 0 & 18 & 5 \\ 2 & 5 & 2 & 0 & 14 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 18 & 14 & 0 & 49 & 15 \\ 3 & 5 & 4 & -1 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\mathbf{2}^1\mathbf{1}^2} \ 5 \ 131^{\mathbf{11}^4} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 5 & 5 & 0 & 26 & 5 \\ 2 & 5 & 2 & 0 & 20 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 26 & 20 & 0 & 106 & 22 \\ 3 & 5 & 4 & -1 & 22 & 4 \end{pmatrix}_{b_0}$$

4.996 $\text{NHC}(-2)(-3) + \mathbf{su}_2 + (-3) \rightarrow (-2) + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3)$

$$T = 12 \quad [0] \ 1^{\mathbf{2}^1\mathbf{1}^1} \ 5 \ 13^{\mathbf{1}^1}\mathbf{1}^{\mathbf{11}^2} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 5 & 5 & 0 & 13 & 4 & 5 \\ 2 & 5 & 2 & 0 & 10 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 7 & 13 & 10 & 0 & 25 & 8 & 11 \\ 2 & 4 & 3 & 0 & 8 & 1 & 3 \\ 3 & 5 & 4 & -1 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\mathbf{2}^1\mathbf{1}^2} \ 5 \ 13^{\mathbf{1}^1}\mathbf{1}^{\mathbf{11}^4} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 122, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 2 & 3 \\ 3 & 5 & 5 & 0 & 26 & 4 & 5 \\ 2 & 5 & 2 & 0 & 20 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 14 & 26 & 20 & 0 & 106 & 16 & 22 \\ 2 & 4 & 3 & 0 & 16 & 1 & 3 \\ 3 & 5 & 4 & -1 & 22 & 3 & 4 \end{pmatrix}_{b_0}$$

4.997 $\text{NHC}(-2)(-3) + \mathbf{su}_2 + (-3) \rightarrow (-2) + \mathbf{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \ 1^{\mathbf{1}^1\mathbf{3}^1\mathbf{1}^1} \ 5 \ 12^{\mathbf{1}^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 4 & 5 \\ 2 & 5 & 2 & 0 & 3 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 5 & 4 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\mathbf{1}^1\mathbf{3}^1\mathbf{1}^1} \ 5 \ 12^{\mathbf{1}^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\mathbf{2}^1\mathbf{1}^1} \ 5 \ 13^{\mathbf{1}^1}\mathbf{1}^{\mathbf{11}^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_2 + (-3) \rightarrow (-2) + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 & 2 \\ 3 & 2 & 4 & 5 & 0 & 3 & 5 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 3 & 0 & 0 & 2 \\ 2 & 2 & 2 & 5 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^1 5^{2^1 1^1} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2) + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 5 & 8 & 0 & 4 & 5 \\ 5 & 8 & 9 & 0 & 5 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 5 & 0 & 1 & 3 \\ 3 & 5 & 7 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 5^{2^1 1^1} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2) + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 0 & 2 & 4 & 0 & 1 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 & 2 \\ 4 & 4 & 5 & 11 & 0 & 4 & 7 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 4 & 0 & 0 & 2 \\ 2 & 2 & 2 & 7 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^1 5^{2^1 1^1} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2) + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 5 & 13 & 0 & 4 & 5 \\ 7 & 13 & 25 & 0 & 8 & 11 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 8 & 0 & 1 & 3 \\ 3 & 5 & 11 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{12} 5^{2^1 1^1} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2) + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -91, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 1 & 2 \\ 1 & 2 & 0 & 6 & 0 & 2 & 2 \\ 5 & 6 & 6 & 21 & 0 & 5 & 9 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 5 & 0 & 0 & 2 \\ 2 & 2 & 2 & 9 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^1 5^{2^1 1^1} 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2) + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -91, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 5 & 18 & 0 & 4 & 5 \\ 9 & 18 & 49 & 0 & 11 & 15 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 11 & 0 & 1 & 3 \\ 3 & 5 & 15 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[9] \ 1^1 5^{2^1 1^2} 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2) + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -151, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 5 & 21 & 0 & 4 & 5 \\ 12 & 21 & 70 & 0 & 13 & 18 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 13 & 0 & 1 & 3 \\ 3 & 5 & 18 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 5^{2^1 1^2} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2) + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -151, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 1 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 1 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 & 2 \\ 7 & 6 & 9 & 36 & 0 & 7 & 12 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 7 & 0 & 0 & 2 \\ 2 & 2 & 2 & 12 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^1 5^{2^1 1^2} 13^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_2 + (-3) \rightarrow (-2) + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -226, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 2 & 3 \\ 3 & 5 & 26 & 0 & 4 & 5 \\ 14 & 26 & 106 & 0 & 16 & 22 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 16 & 0 & 1 & 3 \\ 3 & 5 & 22 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[10] 1^{12} 5^{211^2} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -226, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 1 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 1 & 2 \\ 1 & 2 & 0 & 10 & 0 & 2 & 2 \\ 8 & 8 & 10 & 50 & 0 & 8 & 14 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 8 & 0 & 0 & 2 \\ 2 & 2 & 2 & 14 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[12] 1^{13} 5^{211^2} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -271, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 1 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 1 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 & 2 \\ 9 & 10 & 11 & 68 & 0 & 9 & 16 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 9 & 0 & 0 & 2 \\ 2 & 2 & 2 & 16 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[13] 1^{12} 5^{211^3} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -371, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 1 & 1 & 2 \\ 1 & 0 & 2 & 10 & 0 & 1 & 2 \\ 1 & 2 & 0 & 14 & 0 & 2 & 2 \\ 11 & 10 & 14 & 93 & 0 & 11 & 19 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 11 & 0 & 0 & 2 \\ 2 & 2 & 2 & 19 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[14] 1^{13} 5^{211^3} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -476, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 1 & 1 & 2 \\ 1 & 0 & 2 & 12 & 0 & 1 & 2 \\ 1 & 2 & 0 & 15 & 0 & 2 & 2 \\ 12 & 12 & 15 & 115 & 0 & 12 & 21 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 12 & 0 & 0 & 2 \\ 2 & 2 & 2 & 21 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

4.998 NHC(-2)(-3) + $\textcolor{blue}{su}_2$ + (-3) $\rightarrow$ (-2) + $\textcolor{blue}{su}_2 \rightarrow$ (-3) + $\textcolor{blue}{su}_2$

$$T = 12 \quad [0] 1^{12} 5^{211^1} 13^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 5 & 5 & 0 & 4 & 13 & 5 \\ 2 & 5 & 2 & 0 & 3 & 10 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 8 & 3 \\ 7 & 13 & 10 & 0 & 8 & 25 & 11 \\ 3 & 5 & 4 & -1 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1] 1^{12} 5^{211^2} 13^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 122, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 5 & 5 & 0 & 4 & 26 & 5 \\ 2 & 5 & 2 & 0 & 3 & 20 & 4 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & 1 & 16 & 3 \\ 14 & 26 & 20 & 0 & 16 & 106 & 22 \\ 3 & 5 & 4 & -1 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

4.999 NHC(-2)(-3) + $\textcolor{blue}{su}_2$ + (-3) $\rightarrow$ (-2) + $\textcolor{blue}{su}_2$ + $\textcolor{blue}{su}_2$

$$T = 11 \quad [0] 1^{12} 5^{211^1} 13^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 5 & 5 & 0 & 17 & 5 \\ 2 & 5 & 2 & 0 & 13 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 13 & 0 & 44 & 14 \\ 3 & 5 & 4 & -1 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1] 1^{12} 5^{211^2} 13^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + (-3) \rightarrow (-2) + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -131, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 5 & 5 & 0 & 30 & 5 \\ 2 & 5 & 2 & 0 & 23 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 16 & 30 & 23 & 0 & 141 & 25 \\ 3 & 5 & 4 & -1 & 25 & 4 \end{pmatrix}_{b_0}$$

4.1000 NHC(-2)(-3) + $\mathfrak{su}_2$ + (-3)→(-2) + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{11} \overset{1}{5} 1^2 \overset{1}{2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} \overset{1}{5} 1^2 \overset{1}{2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} \overset{1}{5} 1^2 \overset{1}{2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 10 & 0 & 3 & 4 \\ 8 & 10 & 21 & 0 & 6 & 9 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 9 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} \overset{1}{5} 1^2 \overset{1}{2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 7 & 0 & 3 & 4 \\ 5 & 7 & 8 & 0 & 4 & 6 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 6 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{12} \overset{1}{5} 1^2 \overset{1}{2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 141, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 11 & 0 & 3 & 4 \\ 7 & 11 & 21 & 0 & 6 & 9 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 9 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{13} \overset{1}{5} 1^2 \overset{1}{2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 15 & 0 & 3 & 4 \\ 9 & 15 & 40 & 0 & 8 & 12 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 3 & 4 & 12 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12} \overset{1}{5} 1^3 \overset{1}{2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 3 & 4 \\ 13 & 17 & 61 & 0 & 10 & 15 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 3 & 4 & 15 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{12} \overset{1}{5} 1^2 \overset{1}{2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 196, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 3 & 4 \\ 10 & 14 & 38 & 0 & 8 & 12 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 3 & 4 & 12 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^1 2^1 2^{11} 5^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 285, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 5 & 1 & 2 & 3 \\ 6 & 13 & 12 & 0 & 5 & 7 \\ 5 & 12 & 8 & 0 & 4 & 6 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 7 & 6 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^1 1^3 5^1 2^1 2^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 316, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 3 & 4 \\ 12 & 18 & 61 & 0 & 10 & 15 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 10 & 0 & 0 & 2 \\ 3 & 4 & 15 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^1 1^3 5^1 1^3 2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 381, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 2 & 3 \\ 3 & 4 & 21 & 0 & 3 & 4 \\ 15 & 21 & 88 & 0 & 12 & 18 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 12 & 0 & 0 & 2 \\ 3 & 4 & 18 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^1 2^1 2^{11} 5^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 516, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 10 & 1 & 2 & 3 \\ 6 & 13 & 24 & 0 & 5 & 7 \\ 10 & 24 & 38 & 0 & 8 & 12 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 8 & 0 & 0 & 2 \\ 3 & 7 & 12 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^1 1^1 2^1 2^{11} 5^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{orange}{(-3)} \rightarrow \textcolor{orange}{(-2)} + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 901, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 15 & 1 & 2 & 3 \\ 6 & 13 & 36 & 0 & 5 & 7 \\ 15 & 36 & 88 & 0 & 12 & 18 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 12 & 0 & 0 & 2 \\ 3 & 7 & 18 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

4.1001 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_{3n2\text{mix}}$

$$T = 7 \quad [0] \quad 1^1 1^1 31^{12} 2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -94, \quad \det = -30, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.67 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 4 & 3 \\ 4 & 4 & 8 & 5 & 4 \\ 6 & 8 & 10 & 6 & 6 \\ 4 & 5 & 6 & 3 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 1^1 31^{12} 2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -94, \quad \det = -39, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.13 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 & 4 \\ 5 & 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 4 & 4 & 3 & 0 & 2 \\ 4 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 1^1 31^{12} 2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -94, \quad \det = -60, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.33 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 4 & 3 \\ 4 & 4 & 4 & 5 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 4 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{14} 1^1 31^{12} 2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -94, \quad \det = -79, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.16 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 18 & 4 & 4 \\ 5 & 4 & 21 & 4 & 4 \\ 18 & 21 & 76 & 15 & 16 \\ 4 & 4 & 15 & 0 & 2 \\ 4 & 4 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

- [4] $1^{11}31^{11}2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 6 & 4 & 3 \\ 4 & 4 & 9 & 4 & 4 \\ 6 & 9 & 13 & 6 & 7 \\ 4 & 4 & 6 & 0 & 2 \\ 3 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$
- [5] $1^{13}31^{11}2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 3 & 4 & 3 \\ 4 & 4 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 4 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [6] $1^{11}31^{11}2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 9 & 4 & 4 \\ 5 & 4 & 9 & 4 & 4 \\ 9 & 9 & 13 & 6 & 7 \\ 4 & 4 & 6 & 0 & 2 \\ 4 & 4 & 7 & 2 & 2 \end{pmatrix}_{b_0}$
- [7] $1^{13}31^{11}2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 15 & 4 & 4 \\ 5 & 4 & 17 & 4 & 4 \\ 15 & 17 & 49 & 12 & 13 \\ 4 & 4 & 12 & 0 & 2 \\ 4 & 4 & 13 & 2 & 2 \end{pmatrix}_{b_0}$
- [8] $1^{11}31^{11}2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 3 & 7 & 3 \\ 4 & 4 & 4 & 9 & 4 \\ 3 & 4 & 1 & 6 & 3 \\ 7 & 9 & 6 & 12 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$
- [9] $1^{12}31^{11}2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 12 & 4 & 4 \\ 5 & 4 & 13 & 4 & 4 \\ 12 & 13 & 28 & 9 & 10 \\ 4 & 4 & 9 & 0 & 2 \\ 4 & 4 & 10 & 2 & 2 \end{pmatrix}_{b_0}$
- [10] $1^{12}31^{12}2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 18 & 4 & 4 \\ 5 & 4 & 18 & 4 & 4 \\ 18 & 18 & 58 & 12 & 14 \\ 4 & 4 & 12 & 0 & 2 \\ 4 & 4 & 14 & 2 & 2 \end{pmatrix}_{b_0}$
- [11] $1^{13}31^{12}2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -399, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 21 & 4 & 4 \\ 5 & 4 & 22 & 4 & 4 \\ 21 & 22 & 85 & 15 & 17 \\ 4 & 4 & 15 & 0 & 2 \\ 4 & 4 & 17 & 2 & 2 \end{pmatrix}_{b_0}$
- [12] $1^{14}31^{12}2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -444, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 24 & 4 & 4 \\ 5 & 4 & 26 & 4 & 4 \\ 24 & 26 & 118 & 18 & 20 \\ 4 & 4 & 18 & 0 & 2 \\ 4 & 4 & 20 & 2 & 2 \end{pmatrix}_{b_0}$
- [13] $1^{13}31^{13}2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -604, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 27 & 4 & 4 \\ 5 & 4 & 27 & 4 & 4 \\ 27 & 27 & 133 & 18 & 21 \\ 4 & 4 & 18 & 0 & 2 \\ 4 & 4 & 21 & 2 & 2 \end{pmatrix}_{b_0}$
- [14] $1^{14}31^{13}2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -759, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 30 & 4 & 4 \\ 5 & 4 & 31 & 4 & 4 \\ 30 & 31 & 172 & 21 & 24 \\ 4 & 4 & 21 & 0 & 2 \\ 4 & 4 & 24 & 2 & 2 \end{pmatrix}_{b_0}$
- [15] $1^{14}31^{14}2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -94, \det = -1024, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 36 & 4 & 4 \\ 5 & 4 & 36 & 4 & 4 \\ 36 & 36 & 238 & 24 & 28 \\ 4 & 4 & 24 & 0 & 2 \\ 4 & 4 & 28 & 2 & 2 \end{pmatrix}_{b_0}$

$$\begin{aligned}
& T = 8 \quad [16] \ 1^{11} 4^{11} 12^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 3 & 5 & 4 & 3 & 4 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
& [17] \ 1^{11} 4^{11} 22^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 4 & 5 & 7 & 7 & 4 \\ 5 & 4 & 7 & 6 & 4 \\ 7 & 7 & 7 & 10 & 5 \\ 7 & 6 & 10 & 8 & 6 \\ 4 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0} \\
& [18] \ 1^{12} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
& [20] \ 1^{12} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 4 & 8 & 3 & 4 \\ 6 & 8 & 10 & 4 & 6 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
& [19] \ 1^{12} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 5 & 3 & 3 \\ 4 & 4 & 7 & 3 & 4 \\ 5 & 7 & 8 & 5 & 6 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
& [21] \ 1^{12} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 4 & 12 & 3 & 4 \\ 9 & 12 & 25 & 6 & 9 \\ 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0} \\
& [22] \ 1^{11} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 8 & 3 & 3 \\ 4 & 4 & 12 & 3 & 4 \\ 8 & 12 & 26 & 8 & 10 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0} \\
& [23] \ 1^{12} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 4 & 16 & 3 & 4 \\ 12 & 16 & 46 & 8 & 12 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
& [24] \ 1^{12} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 8 & 3 & 3 \\ 4 & 4 & 11 & 3 & 4 \\ 8 & 11 & 21 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0} \\
& [25] \ 1^{12} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 11 & 3 & 3 \\ 4 & 4 & 15 & 3 & 4 \\ 11 & 15 & 40 & 9 & 12 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0} \\
& [26] \ 1^{11} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 11 & 3 & 3 \\ 4 & 4 & 16 & 3 & 4 \\ 11 & 16 & 45 & 10 & 13 \\ 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[27] \quad 1^{\overline{1}^1} 4^{\overline{1}^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 330, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 14 & 3 & 3 \\ 4 & 4 & 19 & 3 & 4 \\ 14 & 19 & 65 & 11 & 15 \\ 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{\overline{1}^1} 4^{\overline{1}^{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 14 & 3 & 3 \\ 4 & 4 & 20 & 3 & 4 \\ 14 & 20 & 70 & 12 & 16 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{\overline{1}^1} 4^{\overline{1}^{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 17 & 3 & 3 \\ 4 & 4 & 24 & 3 & 4 \\ 17 & 24 & 101 & 14 & 19 \\ 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 19 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [30] \quad 1^{\overline{1}^1} 5^{\overline{1}^1} 1^{\overline{3}^2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 11 & 13 & 40 & 8 & 12 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{\overline{1}^1} 5^{\overline{1}^1} 1^{\overline{2}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 3 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{\overline{1}^1} 5^{\overline{1}^1} 1^{\overline{2}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 10 & 3 & 4 \\ 8 & 10 & 21 & 6 & 9 \\ 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{\overline{1}^1} 5^{\overline{1}^1} 1^{\overline{2}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 8 & 4 & 6 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{\overline{1}^2} 5^{\overline{1}^1} 1^{\overline{2}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -58, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 14 & 3 & 4 \\ 10 & 14 & 38 & 8 & 12 \\ 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{\overline{1}^2} 5^{\overline{1}^1} 1^{\overline{3}^2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 13 & 17 & 61 & 10 & 15 \\ 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{\overline{1}^2} 5^{\overline{1}^1} 1^{\overline{1}^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 5 & 3 & 3 \\ 6 & 13 & 12 & 5 & 7 \\ 5 & 12 & 8 & 4 & 6 \\ 3 & 5 & 4 & 0 & 2 \\ 3 & 7 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{\overline{1}^3} 5^{\overline{1}^1} 1^{\overline{3}^2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_2 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 21 & 3 & 4 \\ 15 & 21 & 88 & 12 & 18 \\ 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{\overline{1}^{\overline{21}^{\overline{12}}}} \ 5 \ 1^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -152, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 10 & 3 & 3 \\ 6 & 13 & 24 & 5 & 7 \\ 10 & 24 & 38 & 8 & 12 \\ 3 & 5 & 8 & 0 & 2 \\ 3 & 7 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{\overline{1}^{\overline{21}^{\overline{13}}}} \ 5 \ 1^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -80, \ \det = -247, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 15 & 3 & 3 \\ 6 & 13 & 36 & 5 & 7 \\ 15 & 36 & 88 & 12 & 18 \\ 3 & 5 & 12 & 0 & 2 \\ 3 & 7 & 18 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [40] \ 1^{\overline{1}^{\overline{21}^{\overline{11}}}} \ 5 \ 131^{\overline{111}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 44, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 5 & 8 & 5 & 5 \\ 5 & 8 & 9 & 6 & 7 \\ 3 & 5 & 6 & 1 & 3 \\ 3 & 5 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{\overline{1}^{\overline{21}^{\overline{12}}}} \ 5 \ 131^{\overline{121}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 80, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 5 & 16 & 5 & 5 \\ 10 & 16 & 42 & 12 & 14 \\ 3 & 5 & 12 & 1 & 3 \\ 3 & 5 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{\overline{1}^{\overline{21}^{\overline{13}}}} \ 5 \ 131^{\overline{131}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 140, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 5 & 24 & 5 & 5 \\ 15 & 24 & 97 & 18 & 21 \\ 3 & 5 & 18 & 1 & 3 \\ 3 & 5 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

4.1002 $\text{NHC}(-2)(-3) + \mathfrak{su}_3$

$$T = 6 \quad [0] \ 1^{\overline{111}^{\overline{12}}} \ 31^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, \ T_{\max} = 76, \ \Delta = -123, \ \det = 3, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 13.33 \rightarrow 14, \ T_{\max} = 76, \ T \in [14, 76]$$

$$\begin{pmatrix} 1 & 4 & 7 & 3 \\ 4 & 9 & 12 & 5 \\ 7 & 12 & 12 & 6 \\ 3 & 5 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}^{\overline{13}}} \ 31^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, \ T_{\max} = 76, \ \Delta = -123, \ \det = 4, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, \ T_{\max} = 76, \ T \in [14, 76]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 4 \\ 5 & 4 & 0 & 2 \\ 3 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1}^{\overline{12}}} \ 31^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, \ T_{\max} = 76, \ \Delta = -123, \ \det = 10, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, \ T_{\max} = 76, \ T \in [11, 76]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 5 & 4 \\ 5 & 5 & 3 & 3 \\ 3 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 131^{\overline{111}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, \ T_{\max} = 76, \ \Delta = -123, \ \det = 10, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, \ T_{\max} = 76, \ T \in [11, 76]$$

$$\begin{pmatrix} 1 & 4 & 7 & 3 \\ 4 & 9 & 16 & 5 \\ 7 & 16 & 27 & 9 \\ 3 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1}^{\overline{12}}} \ 31^{\overline{1}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, \ T_{\max} = 76, \ \Delta = -123, \ \det = 11, \ (n_+, n_-, n_0) = (1, 6, 0), \ c_{\text{ext}} = 6.2$$

$$T_{\min} = 13.09 \rightarrow 14, \ T_{\max} = 76, \ T \in [14, 76]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 10 & 4 \\ 6 & 10 & 21 & 7 \\ 3 & 4 & 7 & -1 \end{pmatrix}_{b_0}$$

- [5] $1^1 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 11, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.45 \rightarrow 10, T_{\max} = 76, T \in [10, 76]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -3 & -1 \\ 3 & 4 & -1 & 3 \end{pmatrix}_{b_0}$
- [6] $1^1 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 11, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 76, T \in [14, 76]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 10 & 4 \\ 9 & 10 & 21 & 7 \\ 3 & 4 & 7 & -1 \end{pmatrix}_{b_0}$
- [7] $1^1 1^1 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 12, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & 0 & 2 \\ 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$
- [8] $1^1 1^1 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 14, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 5 & 4 \\ 3 & 5 & 3 & 3 \\ 3 & 4 & 3 & 2 \end{pmatrix}_{b_0}$
- [9] $1^1 1^1 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 0 & 2 \\ 3 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [10] $1^1 1^1 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 5 & 4 \\ 4 & 5 & 3 & 3 \\ 3 & 4 & 3 & 2 \end{pmatrix}_{b_0}$
- [11] $1^1 1^1 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 4 & 4 \\ 4 & 4 & 0 & 2 \\ 3 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [12] $1^{12} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 19, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 76, T \in [12, 76]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 8 & 4 \\ 8 & 8 & 9 & 5 \\ 3 & 4 & 5 & 1 \end{pmatrix}_{b_0}$
- [13] $1^1 1^1 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 19, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 76, T \in [12, 76]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 8 & 4 \\ 4 & 8 & 9 & 5 \\ 3 & 4 & 5 & 1 \end{pmatrix}_{b_0}$
- [14] $1^1 1^1 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 19, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 76, T \in [12, 76]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 10 & 4 \\ 8 & 10 & 21 & 7 \\ 3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$
- [15] $1^1 1^1 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 19, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 76, T \in [12, 76]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 10 & 4 \\ 7 & 10 & 21 & 7 \\ 3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$
- [17] $1^1 1^1 3^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 20, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 76, T \in [10, 76]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 4 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [16] $21^1 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 20, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 10, T_{\max} = 76, T \in [10, 76]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 4 & 4 \\ 3 & 4 & 0 & 2 \\ 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 21, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 76, T \in [12, 76]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 9 & 4 \\ 9 & 9 & 12 & 6 \\ 3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 21^1 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 23, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 4 & 5 & 4 \\ 3 & 5 & 3 & 3 \\ 3 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 21^1 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 76, T \in [11, 76]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 9 & 12 & 5 \\ 6 & 12 & 12 & 6 \\ 3 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 76, T \in [11, 76]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 9 & 12 & 5 \\ 6 & 12 & 12 & 6 \\ 3 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 13^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 76, T \in [13, 76]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 12 & 4 \\ 6 & 12 & 24 & 8 \\ 3 & 4 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 2131^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 26, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 76, T \in [13, 76]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 8 & 5 \\ 3 & 8 & 3 & 3 \\ 3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^1 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 26, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 76, T \in [13, 76]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 8 & 5 \\ 3 & 8 & 3 & 3 \\ 3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 13^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 27, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 76, T \in [12, 76]$$

$$\begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 9 & 12 & 5 \\ 5 & 12 & 12 & 6 \\ 3 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[26] \ 21^{11} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 29, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.55 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$$

$$\begin{pmatrix} 4 & 5 & 6 & 4 \\ 5 & 4 & 5 & 4 \\ 6 & 5 & 3 & 3 \\ 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[27] \ 13^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 29, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.55 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 9 & 4 \\ 5 & 9 & 12 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 13^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 31, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$$

$$T_{\min} = 13.03 \rightarrow 14, T_{\max} = 76, T \in [14, 76]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 16 & 4 \\ 8 & 16 & 45 & 11 \\ 3 & 4 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 6, T_{\max} = 76, \Delta = -123, \det = 31, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 8 & 4 \\ 7 & 8 & 9 & 5 \\ 3 & 4 & 5 & 2 \end{pmatrix}_{b_0}$$

- [30] $1^{12}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 31$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 8 & 4 \\ 5 & 8 & 9 & 5 \\ 3 & 4 & 5 & 2 \end{pmatrix}_{b_0}$
- [31] $21^{12}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 35$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 76$, $T \in [12, 76]$
 $\begin{pmatrix} 4 & 5 & 12 & 4 \\ 5 & 4 & 10 & 4 \\ 12 & 10 & 21 & 7 \\ 4 & 4 & 7 & 1 \end{pmatrix}_{b_0}$
- [32] $1^{12}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 35$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 10.29 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 8 & 4 \\ 6 & 8 & 9 & 5 \\ 3 & 4 & 5 & 2 \end{pmatrix}_{b_0}$
- [33] $1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 35$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.29 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 9 & 4 \\ 8 & 9 & 12 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$
- [34] $1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 39$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.15 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 9 & 4 \\ 6 & 9 & 12 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$
- [35] $1^{12}131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 40$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 14 & 4 \\ 8 & 14 & 36 & 10 \\ 3 & 4 & 10 & 2 \end{pmatrix}_{b_0}$
- [36] $131^{15} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 40$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 13.50 \rightarrow 14$, $T_{\max} = 76$, $T \in [14, 76]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 20 & 4 \\ 10 & 20 & 72 & 14 \\ 3 & 4 & 14 & -1 \end{pmatrix}_{b_0}$
- [37] $1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 41$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.10 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 9 & 4 \\ 7 & 9 & 12 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$
- [38] $21^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 44$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.09 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 4 & 4 & 4 \\ 3 & 4 & 0 & 2 \\ 3 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [39] $1^{13}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 44$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 12 & 4 \\ 7 & 12 & 24 & 8 \\ 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$
- [40] $21^{11}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 45$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.53 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 4 & 5 & 18 & 4 \\ 5 & 4 & 21 & 4 \\ 18 & 21 & 75 & 15 \\ 4 & 4 & 15 & 2 \end{pmatrix}_{b_0}$
- [41] $131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 46$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 10.74 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 13 & 4 \\ 7 & 13 & 27 & 9 \\ 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$
- [42] $21^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 47$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.30 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$

- [43] $\begin{pmatrix} 4 & 8 & 3 & 4 \\ 8 & 9 & 4 & 5 \\ 3 & 4 & 0 & 2 \\ 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$ $21^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 54$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.11 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 4 & 5 & 9 & 4 \\ 5 & 4 & 9 & 4 \\ 9 & 9 & 12 & 6 \\ 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$
- [44] $1^{11}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 56$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 14 & 4 \\ 12 & 14 & 36 & 10 \\ 3 & 4 & 10 & 2 \end{pmatrix}_{b_0}$
- [46] $1^{13}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 56$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 12 & 4 \\ 8 & 12 & 24 & 8 \\ 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$
- [47] $1^{13}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 56$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 12 & 4 \\ 10 & 12 & 24 & 8 \\ 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$
- [48] $21^{11}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 56$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.50 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 4 & 14 & 4 \\ 9 & 14 & 36 & 10 \\ 3 & 4 & 10 & 2 \end{pmatrix}_{b_0}$
- [45] $1^{11}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 56$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 14 & 4 \\ 9 & 14 & 36 & 10 \\ 3 & 4 & 10 & 2 \end{pmatrix}_{b_0}$
- [49] $1^{14}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 59$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.6$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 76$, $T \in [12, 76]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 16 & 4 \\ 9 & 16 & 45 & 11 \\ 3 & 4 & 11 & 1 \end{pmatrix}_{b_0}$
- [50] $1^{13}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 60$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 12 & 4 \\ 9 & 12 & 24 & 8 \\ 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$
- [51] $21^{11}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 62$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.16 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 4 & 5 & 15 & 4 \\ 5 & 4 & 17 & 4 \\ 15 & 17 & 48 & 12 \\ 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$
- [52] $21^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 63$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.10 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 4 & 9 & 4 \\ 6 & 9 & 12 & 6 \\ 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$
- [53] $1^{11}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 64$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.25 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 14 & 4 \\ 10 & 14 & 36 & 10 \\ 3 & 4 & 10 & 2 \end{pmatrix}_{b_0}$
- [54] $1^{12}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 64$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 10.25 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 13 & 4 \\ 8 & 13 & 27 & 9 \\ 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$

- [55] $1^{13}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 64$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.25 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 14 & 4 \\ 11 & 14 & 36 & 10 \\ 3 & 4 & 10 & 2 \end{pmatrix}_{b_0}$
- [56] $21^{13}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 65$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.06 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 4 & 5 & 12 & 4 \\ 5 & 4 & 13 & 4 \\ 12 & 13 & 27 & 9 \\ 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$
- [57] $1^331^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 65$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 17 & 4 \\ 9 & 17 & 48 & 12 \\ 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$
- [58] $21^{12}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 68$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 10.29 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 4 & 8 & 4 \\ 6 & 8 & 9 & 5 \\ 3 & 4 & 5 & 2 \end{pmatrix}_{b_0}$
- [59] $1^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 70$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 10.14 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 13 & 4 \\ 11 & 13 & 27 & 9 \\ 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$
- [60] $1^{12}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 71$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 10.76 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 18 & 4 \\ 10 & 18 & 57 & 13 \\ 3 & 4 & 13 & 2 \end{pmatrix}_{b_0}$
- [61] $1^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 74$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 10.08 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 13 & 4 \\ 9 & 13 & 27 & 9 \\ 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$
- [62] $1^{12}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 76$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 10.05 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 13 & 4 \\ 10 & 13 & 27 & 9 \\ 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$
- [63] $1^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 79$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.6$
 $T_{\min} = 10.58 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 16 & 4 \\ 10 & 16 & 45 & 11 \\ 3 & 4 & 11 & 2 \end{pmatrix}_{b_0}$
- [64] $1^331^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 86$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.6$
 $T_{\min} = 11.09 \rightarrow 12$, $T_{\max} = 76$, $T \in [12, 76]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 21 & 4 \\ 11 & 21 & 75 & 15 \\ 3 & 4 & 15 & 1 \end{pmatrix}_{b_0}$
- [65] $21^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 87$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10.34 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 4 & 8 & 9 & 4 \\ 8 & 9 & 12 & 5 \\ 9 & 12 & 12 & 6 \\ 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$
- [67] $1^{14}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 91$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.6$
 $T_{\min} = 10.37 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 16 & 4 \\ 11 & 16 & 45 & 11 \\ 3 & 4 & 11 & 2 \end{pmatrix}_{b_0}$
- [66] $1^{13}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 91$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.37 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$

- [68] $\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 4 & 17 & 4 \\ 10 & 17 & 48 & 12 \\ 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$ $21^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 92, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 4 & 5 & 15 & 4 \\ 5 & 4 & 14 & 4 \\ 15 & 14 & 36 & 10 \\ 4 & 4 & 10 & 2 \end{pmatrix}_{b_0}$
- [69] $1^{12}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 95, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 18 & 4 \\ 11 & 18 & 57 & 13 \\ 3 & 4 & 13 & 2 \end{pmatrix}_{b_0}$
- [70] $1^{14}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 95, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 16 & 4 \\ 12 & 16 & 45 & 11 \\ 3 & 4 & 11 & 2 \end{pmatrix}_{b_0}$
- [71] $1^{12}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 104, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 22 & 4 \\ 12 & 22 & 84 & 16 \\ 3 & 4 & 16 & 2 \end{pmatrix}_{b_0}$
- [72] $131^{11}2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 104, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 4 & 8 & 6 & 4 \\ 8 & 9 & 8 & 5 \\ 6 & 8 & 3 & 3 \\ 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$
- [73] $21^{13}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 108, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 4 & 12 & 4 \\ 9 & 12 & 24 & 8 \\ 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$
- [74] $1^{13}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 109, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 4 & 17 & 4 \\ 11 & 17 & 48 & 12 \\ 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$
- [75] $1^{12}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 111, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 18 & 4 \\ 12 & 18 & 57 & 13 \\ 3 & 4 & 13 & 2 \end{pmatrix}_{b_0}$
- [76] $21^{12}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 119, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 4 & 13 & 4 \\ 9 & 13 & 27 & 9 \\ 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$
- [77] $1^{12}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 119, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 18 & 4 \\ 13 & 18 & 57 & 13 \\ 3 & 4 & 13 & 2 \end{pmatrix}_{b_0}$
- [78] $1^{12}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 119, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 18 & 4 \\ 14 & 18 & 57 & 13 \\ 3 & 4 & 13 & 2 \end{pmatrix}_{b_0}$
- [79] $1^{13}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 119, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 17 & 4 \\ 12 & 17 & 48 & 12 \\ 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$

- [80] $1^{14}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 120$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.6$
 $T_{\min} = 10.50 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 4 & 21 & 4 \\ 12 & 21 & 75 & 15 \\ 3 & 4 & 15 & 2 \end{pmatrix}_{b_0}$
- [81] $1^{13}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 121$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.03 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 17 & 4 \\ 13 & 17 & 48 & 12 \\ 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$
- [82] $21^{12}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 135$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 10.04 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 4 & 5 & 18 & 4 \\ 5 & 4 & 18 & 4 \\ 18 & 18 & 57 & 13 \\ 4 & 4 & 13 & 2 \end{pmatrix}_{b_0}$
- [83] $1^{13}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 136$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.32 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 22 & 4 \\ 13 & 22 & 84 & 16 \\ 3 & 4 & 16 & 2 \end{pmatrix}_{b_0}$
- [84] $1^{12}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 139$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.6$
 $T_{\min} = 10.76 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 26 & 4 \\ 14 & 26 & 117 & 19 \\ 3 & 4 & 19 & 2 \end{pmatrix}_{b_0}$
- [85] $21^{12}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 144$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 10.08 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 18 & 4 \\ 12 & 18 & 57 & 13 \\ 3 & 4 & 13 & 2 \end{pmatrix}_{b_0}$
- [86] $1^{14}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 146$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.6$
 $T_{\min} = 10.23 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 4 & 21 & 4 \\ 13 & 21 & 75 & 15 \\ 3 & 4 & 15 & 2 \end{pmatrix}_{b_0}$
- [87] $1^{13}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 160$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.12 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 22 & 4 \\ 14 & 22 & 84 & 16 \\ 3 & 4 & 16 & 2 \end{pmatrix}_{b_0}$
- [88] $21^{14}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 164$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.6$
 $T_{\min} = 10.32 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 16 & 4 \\ 12 & 16 & 45 & 11 \\ 3 & 4 & 11 & 2 \end{pmatrix}_{b_0}$
- [89] $21^{12}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 164$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 6.2$
 $T_{\min} = 9.98 \rightarrow 10$, $T_{\max} = 76$, $T \in [10, 76]$
 $\begin{pmatrix} 4 & 5 & 21 & 4 \\ 5 & 4 & 22 & 4 \\ 21 & 22 & 84 & 16 \\ 4 & 4 & 16 & 3 \end{pmatrix}_{b_0}$
- [90] $1^{14}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 164$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.6$
 $T_{\min} = 10.10 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 4 & 21 & 4 \\ 14 & 21 & 75 & 15 \\ 3 & 4 & 15 & 2 \end{pmatrix}_{b_0}$
- [91] $1^{14}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 174$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.6$
 $T_{\min} = 10.03 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 21 & 4 \\ 15 & 21 & 75 & 15 \\ 3 & 4 & 15 & 2 \end{pmatrix}_{b_0}$
- [92] $1^{13}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3$ $T = 6$, $T_{\max} = 76$, $\Delta = -123$, $\det = 176$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.02 \rightarrow 11$, $T_{\max} = 76$, $T \in [11, 76]$

- [93] $21^{1231^{14}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 179, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 76, T \in [10, 76]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 22 & 4 \\ 15 & 22 & 84 & 16 \\ 3 & 4 & 16 & 2 \end{pmatrix}_{b_0}$
- [94] $1^{14}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 179, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 4 & 26 & 4 \\ 15 & 26 & 117 & 19 \\ 3 & 4 & 19 & 2 \end{pmatrix}_{b_0}$
- [95] $1^{13}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 184, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 76, T \in [10, 76]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 22 & 4 \\ 16 & 22 & 84 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$
- [96] $21^{1331^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 191, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.03 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 4 & 17 & 4 \\ 12 & 17 & 48 & 12 \\ 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$
- [97] $1^{14}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 211, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 16 & 3 \\ 3 & 4 & 26 & 4 \\ 16 & 26 & 117 & 19 \\ 3 & 4 & 19 & 2 \end{pmatrix}_{b_0}$
- [98] $221^{111}3 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 234, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.03 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 3 & 3 & 0 & 2 \\ 2 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [99] $1^{14}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 235, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 3 & 17 & 3 \\ 3 & 4 & 26 & 4 \\ 17 & 26 & 117 & 19 \\ 3 & 4 & 19 & 2 \end{pmatrix}_{b_0}$
- [100] $21^{15}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 236, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 20 & 4 \\ 15 & 20 & 72 & 14 \\ 3 & 4 & 14 & 2 \end{pmatrix}_{b_0}$
- [101] $21^{1331^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 248, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 76, T \in [10, 76]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 22 & 4 \\ 15 & 22 & 84 & 16 \\ 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$
- [102] $1^{14}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 251, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 76, T \in [10, 76]$
 $\begin{pmatrix} 1 & 3 & 18 & 3 \\ 3 & 4 & 26 & 4 \\ 18 & 26 & 117 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$
- [103] $21^{14}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 279, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 10.02 \rightarrow 11, T_{\max} = 76, T \in [11, 76]$
 $\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 4 & 21 & 4 \\ 15 & 21 & 75 & 15 \\ 3 & 4 & 15 & 2 \end{pmatrix}_{b_0}$
- [104] $21^{14}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 6, T_{\max} = 76, \Delta = -123, \det = 368, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 76, T \in [10, 76]$
 $\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 4 & 26 & 4 \\ 18 & 26 & 117 & 19 \\ 3 & 4 & 19 & 3 \end{pmatrix}_{b_0}$

$$\begin{aligned}
T = 7 \quad & 1^3 141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 15, T_{\max} = 23, T \in [15, 23] \\
& \begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 13 & 4 \\ 8 & 13 & 39 & 11 \\ 3 & 4 & 11 & -1 \end{pmatrix}_{b_0} \\
[106] \quad & 1^{11} 3131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13.40 \rightarrow 14, T_{\max} = 19, T \in [14, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & -1 & -1 \end{pmatrix}_{b_0} \\
[107] \quad & 1^1 321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 12, T \in [10, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0} \\
[108] \quad & 1^1 321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.57 \rightarrow 10, T_{\max} = 12, T \in [10, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 \end{pmatrix}_{b_0} \\
[109] \quad & 1^1 31^2 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 19, T \in [14, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 6 & 0 & 21 & 7 \\ 2 & 3 & -1 & 7 & -1 \end{pmatrix}_{b_0} \\
[110] \quad & 1^1 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 1 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[111] \quad & 1^1 321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 3 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[112] \quad & 1^{13} 31^1 31^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 1 & 4 & 9 & 3 \\ 2 & 4 & 3 & 9 & 3 \\ 5 & 9 & 9 & 24 & 8 \\ 2 & 3 & 3 & 8 & 2 \end{pmatrix}_{b_0} \\
[113] \quad & 1^1 321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.50 \rightarrow 10, T_{\max} = 12, T \in [10, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0} \\
[114] \quad & 1^{11} 1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 \\ 3 & 5 & 3 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0} \\
[115] \quad & 1^{12} 3131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 19, T \in [14, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 6 & 0 & 9 & 5 \\ 2 & 3 & -1 & 5 & -1 \end{pmatrix}_{b_0} \\
[116] \quad & 1^1 3131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 19, T \in [11, 19]
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 10.45 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0} \\
[125] & 31^1 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, \quad T_{\max} = 19, \quad \Delta = -103, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.45 \rightarrow 11, \quad T_{\max} = 19, \quad T \in [11, 19] \\
& \begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 4 & 4 \\ 4 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[130] & 1^1 231^1 31^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, \quad T_{\max} = 19, \quad \Delta = -102, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.2 \\
& T_{\min} = 10.45 \rightarrow 11, \quad T_{\max} = 19, \quad T \in [11, 19] \\
& \begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 1 & 4 & 6 & 3 \\ 2 & 4 & 3 & 6 & 3 \\ 4 & 6 & 6 & 9 & 5 \\ 2 & 3 & 3 & 5 & 2 \end{pmatrix}_{b_0} \\
[131] & 1^1 321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.45 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[137] & 1^1 1231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -12, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 9 & 0 & 5 \\ 1 & 0 & -3 & -1 \\ 3 & 5 & -1 & 2 \end{pmatrix}_{b_0} \\
[136] & 21^1 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, \quad T_{\max} = 23, \quad \Delta = -114, \quad \det = -12, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, \quad T_{\max} = 23, \quad T \in [12, 23] \\
& \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & 0 & 2 \\ 3 & 5 & 2 & 2 \end{pmatrix}_{b_0} \\
[135] & 1^1 1^3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -12, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.33 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[134] & 1^1 1^2 321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -12, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.2 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 3 & 4 & 2 & 5 & 3 \\ 2 & 2 & 2 & 3 & 1 \end{pmatrix}_{b_0} \\
[133] & 1^1 4^1 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, \quad T_{\max} = 23, \quad \Delta = -114, \quad \det = -12, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, \quad T_{\max} = 23, \quad T \in [12, 23] \\
& \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & 0 & 2 \\ 3 & 5 & 2 & 2 \end{pmatrix}_{b_0} \\
[132] & 1^1 1^3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -12, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.2 \\
& T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 4 & 2 & 6 & 9 & 5 \\ 2 & 2 & 2 & 5 & 1 \end{pmatrix}_{b_0} \\
[138] & 1^1 12^1 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -14, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.43 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -1 & 1 \end{pmatrix}_{b_0} \\
[139] & 1^1 1^3 1^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, \quad T_{\max} = 19, \quad \Delta = -102, \quad \det = -14, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.57 \rightarrow 11, \quad T_{\max} = 19, \quad T \in [11, 19] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 \\ 2 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 3 & -1 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[140] \ 1^{11} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 23, \ \Delta = -114, \ \det = -14, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 23, \ T \in [12, 23]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 4 & 7 & 4 \\ 7 & 7 & 7 & 5 \\ 3 & 4 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[144] \ 1^{13} 3^1 2^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -15, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 60 & 14 \\ 2 & 2 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[143] \ 1^{13} 3^1 1^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -102, \ \det = -15, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.40 \rightarrow 12, \ T_{\max} = 19, \ T \in [12, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & 3 & 3 \\ 2 & 3 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{12} 3^1 2^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -15, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 6 & 0 & 25 & 9 \\ 2 & 2 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[141] \ 2^1 2^1 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -15, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[145] \ 1^{11} 3^1 2^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -15, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 4 & 4 \\ 2 & 2 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[147] \ 1^2 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 23, \ \Delta = -114, \ \det = -16, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.25 \rightarrow 12, \ T_{\max} = 23, \ T \in [12, 23]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 10 & 4 \\ 6 & 10 & 20 & 8 \\ 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[150] \ 1^{11} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 23, \ \Delta = -114, \ \det = -16, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.25 \rightarrow 12, \ T_{\max} = 23, \ T \in [12, 23]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 10 & 4 \\ 9 & 10 & 20 & 8 \\ 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[151] \ 2^1 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 23, \ \Delta = -114, \ \det = -16, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.25 \rightarrow 12, \ T_{\max} = 23, \ T \in [12, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 0 & 2 \\ 3 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[148] \ 1^1 2^1 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -16, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.25 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 3 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[146] \ 1^1 4^1 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 23, \ \Delta = -114, \ \det = -16, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.25 \rightarrow 12, \ T_{\max} = 23, \ T \in [12, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 0 & 2 \\ 3 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[149] \ 1^1 3^1 2^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -16, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 12, \ T \in [10, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 \\ 3 & 5 & 1 & 3 \\ 3 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

[153] $21^1 1^1 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 23, T \in [11, 23]$

$$\begin{pmatrix} 4 & 5 & 6 & 4 \\ 5 & 4 & 7 & 4 \\ 6 & 7 & 7 & 5 \\ 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[152] $2231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 3 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[155] $21^1 1^1 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 23, T \in [11, 23]$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 3 & 4 \\ 3 & 3 & 0 & 2 \\ 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[154] $1^1 231^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 1 & 4 & 6 & 3 \\ 2 & 4 & 3 & 6 & 3 \\ 3 & 6 & 6 & 9 & 5 \\ 2 & 3 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[157] $1^1 31^1 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[159] $131^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 2 & 4 & 3 & 3 & 3 \\ 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[160] $31^1 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 5 & 3 & 3 \\ 5 & 5 & 5 & 5 \\ 3 & 5 & 3 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[161] $1^1 31^1 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 12.20 \rightarrow 13, T_{\max} = 19, T \in [13, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 6 & 0 & 21 & 7 \\ 2 & 3 & -1 & 7 & 0 \end{pmatrix}_{b_0}$$

[162] $2^1 2^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 4 & 2 & 6 & 9 & 5 \\ 2 & 2 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

[163] $1^1 1^1 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 23, T \in [11, 23]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 7 & 4 \\ 4 & 7 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[164] $1^1 321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 4 & 3 \\ 1 & 4 & 0 & 4 & 2 \\ 2 & 4 & 4 & 4 & 4 \\ 2 & 3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

[156] $1^1 1^1 3131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 1 & 4 & 0 & 3 & 2 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[158] & 1^{\overline{21}} 4^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 23, T \in [11, 23] \\
& \begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 4 & 4 \\ 3 & 4 & 0 & 2 \\ 4 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[165] & 1^{\overline{11}} 3^{\overline{12}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 4 & 0 & 15 & 7 \\ 2 & 2 & -1 & 7 & 2 \end{pmatrix}_{b_0} \\
[175] & 1^{\overline{11}} 3^{\overline{11}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 10 & 5 \\ 5 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0} \\
[170] & 1^{\overline{13}} 3^{\overline{11}} 3^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 13.35 \rightarrow 14, T_{\max} = 19, T \in [14, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 51 & 11 \\ 2 & 3 & -1 & 11 & -1 \end{pmatrix}_{b_0} \\
[171] & 1^{\overline{12}} 3^{\overline{11}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 10 & 5 \\ 7 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0} \\
[172] & 2^{\overline{1}} 1^{\overline{1}} 4^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.57 \rightarrow 11, T_{\max} = 23, T \in [11, 23] \\
& \begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[173] & 1^{\overline{13}} 3^{\overline{11}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 5 & 3 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[174] & 2^{\overline{11}} 2^{\overline{1}} 3^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[169] & 1^{\overline{13}} 3^{\overline{12}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 10 & 5 \\ 5 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0} \\
[176] & 1^{\overline{13}} 1^{\overline{1}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.26 \rightarrow 12, T_{\max} = 19, T \in [12, 19] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 4 \\ 1 & 4 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 2 \\ 2 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0} \\
[168] & 1^{\overline{11}} 1^{\overline{1}} 3^{\overline{13}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.57 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 \\ 2 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0} \\
[166] & 1^{\overline{11}} 3^{\overline{11}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 3 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[167] & 1^1 3 2 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 10 & 5 \\ 7 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0} \\
[183] & 2 1^{11} 4 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 23, T \in [11, 23] \\
& \begin{pmatrix} 1 & 4 & 7 & 3 \\ 4 & 4 & 10 & 4 \\ 7 & 10 & 20 & 8 \\ 3 & 4 & 8 & 3 \end{pmatrix}_{b_0} \\
[181] & 1^1 3 1^2 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 13.17 \rightarrow 14, T_{\max} = 19, T \in [14, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 6 & 0 & 21 & 7 \\ 2 & 3 & -1 & 7 & -1 \end{pmatrix}_{b_0} \\
[182] & 2 1^2 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 23, T \in [11, 23] \\
& \begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 10 & 4 \\ 8 & 10 & 20 & 8 \\ 3 & 4 & 8 & 3 \end{pmatrix}_{b_0} \\
[180] & 2 3 1^{11} 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 2 & 2 & 3 & 2 \\ 2 & 1 & 4 & 2 & 3 \\ 2 & 4 & 0 & 2 & 2 \\ 3 & 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[184] & 1^{11} 4 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 23, T \in [11, 23] \\
& \begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 7 & 4 \\ 6 & 7 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0} \\
[185] & 1^{11} 4 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 23, T \in [11, 23] \\
& \begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 10 & 4 \\ 8 & 10 & 20 & 8 \\ 3 & 4 & 8 & 3 \end{pmatrix}_{b_0} \\
[186] & 2 1^{11} 4 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 23, T \in [11, 23] \\
& \begin{pmatrix} 4 & 8 & 6 & 4 \\ 8 & 13 & 12 & 7 \\ 6 & 12 & 7 & 5 \\ 4 & 7 & 5 & 3 \end{pmatrix}_{b_0} \\
[179] & 1 3 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 19, T \in [12, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 6 & 3 \\ 1 & 4 & 0 & 6 & 2 \\ 3 & 6 & 6 & 12 & 6 \\ 2 & 3 & 2 & 6 & 1 \end{pmatrix}_{b_0} \\
[178] & 1^{11} 3 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 6 & 0 & 12 & 6 \\ 2 & 3 & -1 & 6 & 2 \end{pmatrix}_{b_0} \\
[177] & 2 1^{11} 3 2 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 6 & 3 & 4 \\ 6 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 4 & 4 & 3 & 3 \end{pmatrix}_{b_0} \\
[187] & 2 1^{11} 3 2 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -25, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]
\end{aligned}$$

$$\begin{pmatrix} 4 & 5 & 12 & 4 \\ 5 & 5 & 15 & 5 \\ 12 & 15 & 31 & 11 \\ 4 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

[188] $1^{12}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 5 & 0 & 12 & 6 \\ 2 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[189] $221^{14}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 23, T \in [11, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 7 & 4 \\ 5 & 7 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[190] $13131^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.08 \rightarrow 14, T_{\max} = 19, T \in [14, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 1 & 4 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 2 \\ 2 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[191] $21^{14}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 23, T \in [11, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 7 & 4 \\ 5 & 7 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[192] $1^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -26, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 23, T \in [11, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 7 & 4 \\ 5 & 7 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[194] $1^1321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 3 & 4 & 7 & 15 & 7 \\ 2 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

[193] $1^{11}3131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[196] $1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 5 & 10 & 5 \\ 6 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[195] $1^{11}321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 15 & 5 \\ 7 & 15 & 33 & 11 \\ 3 & 5 & 11 & 2 \end{pmatrix}_{b_0}$$

[197] $1^{12}321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 5 & 10 & 5 \\ 6 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[198] $12^131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 1 & 4 \\ 5 & 4 & 1 & 4 \\ 1 & 1 & -3 & -1 \\ 4 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

[199] $21^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 9 & 4 \\ 5 & 5 & 10 & 5 \\ 9 & 10 & 13 & 7 \\ 4 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[200] $1^{12}3^{11}21^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 7 & 0 & 44 & 12 \\ 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[201] $1^{11}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[202] $1^{12}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 4 & 3 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{12}31^{11}31^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 12 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 7 & 12 & 9 & 27 & 9 \\ 2 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[204] $1^{11}3^{11}21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & 4 & 4 \\ 2 & 2 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

[205] $1^{13}31^{12}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 19, T \in [14, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 9 & 0 & 51 & 11 \\ 2 & 3 & -1 & 11 & -1 \end{pmatrix}_{b_0}$$

[206] $1^{12}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 9 & 0 & 27 & 9 \\ 2 & 3 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[207] $1^{11}31^{23}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 9 & 0 & 36 & 10 \\ 2 & 3 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[208] $1^{13}3^{11}21^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 8 & 0 & 39 & 11 \\ 2 & 2 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[209] $1^{11}3^{12}21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 15 & 5 \\ 7 & 15 & 31 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

[210] $1^{12}3^{14}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 7 & 5 & 5 & 12 & 6 \\ 2 & 2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[211] \ 1^{\overline{1}^2} 3^{\overline{1}^1} 21^{\overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 3 & 5 & 5 & 12 & 6 \\ 2 & 2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[212] \ 1^{\overline{1}^1} 31^{\overline{1}^1} 31^{\overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 4 \\ 1 & 4 & 0 & 3 & 2 \\ 3 & 6 & 3 & 3 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[213] \ 31^{\overline{1}^1} 231^{\overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 3 & 2 & 0 & 3 & 2 \\ 3 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[214] \ 1^{\overline{1}^2} 3131^{\overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 6 & 0 & 9 & 5 \\ 2 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[215] \ 1^{\overline{1}^4} 1^{\overline{1}^1} 321^{\overline{1}^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 15 & 5 \\ 11 & 15 & 31 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[218] \ 2^{\overline{1}^1} 2^{\overline{1}^1} 31^{\overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[219] \ 1^{\overline{1}^3} 321^{\overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 7 & 0 & 24 & 8 \\ 2 & 2 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[216] \ 21^{\overline{1}^2} 321^{\overline{1}^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 7 & 11 & 44 & 12 \\ 2 & 2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[217] \ 21^{\overline{1}^1} 321^{\overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[220] \ 2^{\overline{1}^1} 2^{\overline{1}^1} 31^{\overline{1}^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 2 & 5 & 4 \\ 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[221] \ 1^{\overline{1}^3} 3^{\overline{1}^1} 21^{\overline{1}^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 10 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 10 & 0 & 87 & 17 \\ 2 & 2 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[222] \ 2^{\overline{1}^2} 3^{\overline{1}^1} 1^{\overline{1}^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 2 & 3 & 4 & 2 \\ 2 & 1 & 3 & 6 & 3 \\ 3 & 3 & 0 & 4 & 2 \\ 4 & 6 & 4 & 9 & 5 \\ 2 & 3 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[223] \ 1^{11}3^{11}21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -43, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.49 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 4 & 4 & 7 & 15 & 7 \\ 2 & 2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[224] \ 1^{12}3^{12}21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -43, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.49 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 5 & 15 & 5 \\ 8 & 15 & 33 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[225] \ 1^{14}3^{14}21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -43, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.49 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 7 & 4 & 7 & 15 & 7 \\ 2 & 2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[226] \ 1^{14}3^{14}21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -43, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.49 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 15 & 5 \\ 10 & 15 & 33 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[227] \ 1^{11}3^{11}3^{13}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -102, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.27 \rightarrow 12, \ T_{\max} = 19, \ T \in [12, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 12 & 0 & 72 & 14 \\ 2 & 3 & -1 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[228] \ 1^14^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, \ T_{\max} = 23, \ \Delta = -114, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.09 \rightarrow 12, \ T_{\max} = 23, \ T \in [12, 23]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 4 & 4 & 4 \\ 3 & 4 & 0 & 2 \\ 3 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[229] \ 2^12^13^{11}3^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.64 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 6 & 2 \\ 5 & 7 & 6 & 24 & 8 \\ 2 & 2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[230] \ 13^11^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -102, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.09 \rightarrow 12, \ T_{\max} = 19, \ T \in [12, 19]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 4 \\ 1 & 4 & 0 & 3 & 2 \\ 2 & 6 & 3 & 3 & 3 \\ 2 & 4 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[231] \ 1^{13}3^{13}1^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -102, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.27 \rightarrow 12, \ T_{\max} = 19, \ T \in [12, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 12 & 0 & 48 & 12 \\ 2 & 3 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[232] \ 1^{11}3^{11}1^131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -102, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 6 & 0 & 12 & 6 \\ 2 & 3 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[233] \ 31^{11}3^{11}1^13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -110, \ \det = -45, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.73 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 7 & 3 & 4 \\ 3 & 7 & 3 & 3 & 3 \\ 3 & 3 & 3 & 0 & 2 \\ 2 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[234] \ 12^13^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -46, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.2$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 4 & 0 & 5 & 3 \\ 2 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[240] \ 1^{11} 3 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 5 & 5 & 10 & 32 & 10 \\ 2 & 2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[239] \ 1^{11} 31^{11} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 9 & 4 \\ 1 & 4 & 0 & 6 & 2 \\ 5 & 9 & 6 & 12 & 6 \\ 2 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[235] \ 1^{11} 3 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 20 & 5 \\ 9 & 20 & 57 & 15 \\ 3 & 5 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[236] \ 131^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 4 & 7 & 3 & 4 \\ 2 & 7 & 3 & 3 & 3 \\ 1 & 3 & 3 & 0 & 2 \\ 2 & 4 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[238] \ 1^{13} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 4 & 7 & 6 & 24 & 8 \\ 2 & 2 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[242] \ 21^{11} 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 23, T \in [11, 23]$$

$$\begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 4 & 7 & 4 \\ 5 & 7 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[241] \ 1^{13} 3131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 9 & 0 & 24 & 8 \\ 2 & 3 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[237] \ 21^{11} 1^4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$$

$$\begin{pmatrix} 4 & 8 & 3 & 4 \\ 8 & 9 & 4 & 5 \\ 3 & 4 & 0 & 2 \\ 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[247] \ 2^{11} 3^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 13, \Delta = -97, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 2 & 3 \\ 2 & 4 & 0 & 2 & 2 \\ 2 & 2 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[246] \ 1^{13} 3 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 15 & 5 \\ 9 & 15 & 33 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[245] \ 2^2 2 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[244] \ 1^{12}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 6 & 5 & 5 & 12 & 6 \\ 2 & 2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[243] \ 1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 5 & 5 & 12 & 6 \\ 2 & 2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[250] \ 1^{13}321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 15 & 5 \\ 10 & 15 & 31 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[248] \ 1^{12}321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 5 & 4 & 7 & 15 & 7 \\ 2 & 2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[249] \ 1^{11}321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 5 & 15 & 5 \\ 8 & 15 & 31 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[252] \ 1^{14}31^{13}1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 15 & 0 & 75 & 15 \\ 2 & 3 & -1 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[251] \ 1^{13}321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 6 & 4 & 7 & 15 & 7 \\ 2 & 2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[253] \ 1^{12}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 12 & 6 \\ 2 & 2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[254] \ 1^{12}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 6 & 0 & 25 & 9 \\ 2 & 2 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[255] \ 2^2 2^3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 5 & 5 & 12 & 6 \\ 2 & 2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[259] \ 1^{11}31^{13}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 6 & 0 & 12 & 6 \\ 2 & 3 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[256] \ 131^{11}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 12 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 4 & 12 & 9 & 27 & 9 \\ 2 & 4 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

[258] $1^{1^2}321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 20 & 5 \\ 10 & 20 & 61 & 15 \\ 3 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

[261] $31^{11}231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 3 & 5 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 3 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 12 & 6 \\ 2 & 2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[260] $1^{12}3131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 6 & 0 & 9 & 5 \\ 2 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[263] $1^{14}3131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 12 & 0 & 45 & 11 \\ 2 & 3 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

[264] $1^{11}321^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 25 & 5 \\ 11 & 25 & 91 & 19 \\ 3 & 5 & 19 & 2 \end{pmatrix}_{b_0}$$

[257] $1^{1^2}321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 15 & 5 \\ 9 & 15 & 31 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

[262] $1^{12}321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 4 & 6 & 8 & 25 & 9 \\ 2 & 2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[266] $2^2 2 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[265] $2^2 2 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[267] $1^{1^2}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 5 & 0 & 12 & 6 \\ 2 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[268] $1^{12}31^2 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 12 & 0 & 57 & 13 \\ 2 & 3 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[269] $131^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 4 \\ 1 & 4 & 0 & 6 & 2 \\ 3 & 9 & 6 & 12 & 6 \\ 2 & 4 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[270] \ 1^{13} 3^{14} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -63, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.59 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 8 & 7 & 6 & 24 & 8 \\ 2 & 2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[271] \ 1^{11} 31^{23} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -102, \ \det = -63, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.59 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 36 & 10 \\ 2 & 3 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[272] \ 1^{11} 3^{12} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -63, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.59 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 6 & 5 & 10 & 32 & 10 \\ 2 & 2 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[273] \ 1^{13} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -63, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.59 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 5 & 7 & 6 & 24 & 8 \\ 2 & 2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[274] \ 2^1 3^1 2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 13, \ \Delta = -97, \ \det = -64, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.25 \rightarrow 11, \ T_{\max} = 13, \ T \in [11, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[275] \ 12^1 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -65, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 6 & 0 & 15 & 5 \\ 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[276] \ 1^{12} 3^{13} 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -68, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.2$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 5 & 7 & 11 & 44 & 12 \\ 2 & 2 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[277] \ 2^1 2^1 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -68, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.29 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 6 & 2 \\ 2 & 1 & 4 & 8 & 3 \\ 2 & 4 & 0 & 8 & 2 \\ 6 & 8 & 8 & 25 & 9 \\ 2 & 3 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[279] \ 31^{11} 231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -103, \ \det = -71, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.63 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 3 & 2 & 0 & 4 & 2 \\ 4 & 3 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[278] \ 1^{12} 31^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -102, \ \det = -71, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.41 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 9 & 0 & 27 & 9 \\ 2 & 3 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[281] \ 1^1 3^{13} 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -71, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.6$$

$$T_{\min} = 10.41 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 20 & 5 \\ 11 & 20 & 61 & 15 \\ 3 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[282] \ 1^{11} 3^{13} 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -71, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.41 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 7 & 5 & 10 & 32 & 10 \\ 2 & 2 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[283] \ 1^{11} 3^{14} 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -71, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.41 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 10 & 2 \\ 8 & 5 & 10 & 32 & 10 \\ 2 & 2 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[284] \ 1^{11} 31^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -102, \ \det = -71, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.63 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 9 & 4 \\ 1 & 4 & 0 & 6 & 2 \\ 4 & 9 & 6 & 12 & 6 \\ 2 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[285] \ 2^3 3^{11} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -103, \ \det = -71, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.52 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 0 & 2 & 3 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[286] \ 1^{13} 3^{11} 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -71, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.41 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 7 & 6 & 24 & 8 \\ 2 & 2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[280] \ 1^{13} 3^{12} 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -71, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.41 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 6 & 7 & 6 & 24 & 8 \\ 2 & 2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[287] \ 1^{12} 31^{11} 31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -102, \ \det = -72, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.50 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 12 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 6 & 12 & 9 & 27 & 9 \\ 2 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[288] \ 1^{11} 3^{11} 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -72, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.39 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 \\ 3 & 5 & 20 & 5 \\ 10 & 20 & 57 & 15 \\ 3 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[289] \ 1^{13} 3^{11} 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -74, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.27 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 60 & 14 \\ 2 & 2 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[290] \ 31^{11} 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -103, \ \det = -75, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.60 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 5 & 4 & 3 \\ 5 & 5 & 5 & 5 \\ 4 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[293] \ 2^3 3^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 13, \ \Delta = -97, \ \det = -76, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.32 \rightarrow 11, \ T_{\max} = 13, \ T \in [11, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 6 & 3 \\ 2 & 3 & 0 & 4 & 2 \\ 4 & 6 & 4 & 9 & 5 \\ 2 & 3 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[292] \ 1^{13} 3^{12} 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -76, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.32 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 5 & 6 & 8 & 25 & 9 \\ 2 & 2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[291] $1^{\overline{1}^4}321^{\overline{1}^4} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 20 & 5 \\ 12 & 20 & 61 & 15 \\ 3 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

[295] $21^{\overline{1}^2}321^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 5 & 5 & 5 & 12 & 6 \\ 2 & 2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[294] $2^{\overline{1}^2}231^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 2 & 2 & 0 & 7 & 2 \\ 5 & 4 & 7 & 15 & 7 \\ 2 & 2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[296] $1^{\overline{1}^2}31^{\overline{1}^1}31^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 12 & 4 \\ 1 & 4 & 0 & 9 & 2 \\ 5 & 12 & 9 & 27 & 9 \\ 2 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[297] $1^{\overline{1}^1}31^{\overline{1}^1}31^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 12 & 0 & 72 & 14 \\ 2 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[298] $2^{\overline{1}^1}231^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 3 & 2 \\ 2 & 1 & 4 & 4 & 3 \\ 2 & 4 & 0 & 4 & 2 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 3 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[303] $1^{\overline{1}^2}31^{\overline{1}^1}31^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 15 & 0 & 99 & 17 \\ 2 & 3 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[302] $1^{\overline{1}^3}3131^{\overline{1}^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 9 & 0 & 24 & 8 \\ 2 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[301] $1^{\overline{1}^3}31^{\overline{1}^2}31^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 15 & 0 & 84 & 16 \\ 2 & 3 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[299] $1^{\overline{1}^1}31^{\overline{1}^2}31^{\overline{1}^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 36 & 10 \\ 2 & 3 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[300] $1^{\overline{1}^3}321^{\overline{1}^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 5 & 8 & 9 & 39 & 11 \\ 2 & 2 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[304] $21^{\overline{1}^2}321^{\overline{1}^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 6 & 2 \\ 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 25 & 9 \\ 2 & 2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[305] $1^{13}3^{11}21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 7 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 7 & 0 & 24 & 8 \\ 2 & 2 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[306] $2231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 5 & 15 & 5 \\ 9 & 15 & 33 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

[307] $1^{14}321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 20 & 5 \\ 13 & 20 & 57 & 15 \\ 3 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

[308] $1^{12}321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 20 & 5 \\ 11 & 20 & 57 & 15 \\ 3 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

[309] $1^{12}321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 8 & 6 & 8 & 25 & 9 \\ 2 & 2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[310] $1^{12}321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 25 & 9 \\ 2 & 2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[311] $1^{13}321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.51 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 8 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 8 & 0 & 39 & 11 \\ 2 & 2 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[312] $31^{11}231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 2 & 5 & 4 \\ 4 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

[313] $221^{11}41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 23, T \in [11, 23]$

$$\begin{pmatrix} 1 & 6 & 5 & 3 \\ 6 & 13 & 12 & 7 \\ 5 & 12 & 7 & 5 \\ 3 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

[314] $1^{11}321^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.58 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 25 & 5 \\ 12 & 25 & 91 & 19 \\ 3 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[315] $1^{13}321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 \\ 3 & 5 & 20 & 5 \\ 12 & 20 & 57 & 15 \\ 3 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

[316] $1^{12}321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$

$$\begin{aligned}
& T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 8 & 2 \\ 7 & 6 & 8 & 25 & 9 \\ 2 & 2 & 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[317] & 31^{11}231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -95, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.47 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 5 & 7 & 3 \\ 5 & 5 & 10 & 5 \\ 7 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0} \\
[318] & 1^{11}321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 6 & 7 & 11 & 44 & 12 \\ 2 & 2 & 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[319] & 1^{13}31^{13}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 12 & 0 & 48 & 12 \\ 2 & 3 & -1 & 12 & 2 \end{pmatrix}_{b_0} \\
[321] & 31^{11}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -110, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.70 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 6 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 6 & 3 & 3 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[320] & 1^{12}31^{13}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 27 & 9 \\ 2 & 3 & -1 & 9 & 2 \end{pmatrix}_{b_0} \\
[323] & 1^{14}31^{23}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 10.88 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 18 & 0 & 117 & 19 \\ 2 & 3 & -1 & 19 & 2 \end{pmatrix}_{b_0} \\
[324] & 1^{111}13 \ 2 \ 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 3 & 2 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[322] & 1^{12}31^{23}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 12 & 0 & 57 & 13 \\ 2 & 3 & -1 & 13 & 2 \end{pmatrix}_{b_0} \\
[325] & 1^{13}321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 6 & 8 & 9 & 39 & 11 \\ 2 & 2 & 2 & 11 & 2 \end{pmatrix}_{b_0} \\
[328] & 1^{14}3131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.6 \\
& T_{\min} = 10.81 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 12 & 0 & 45 & 11 \\ 2 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0} \\
[327] & 1^{11}31^{33}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.81 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 12 & 0 & 72 & 14 \\ 2 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0} \\
[326] & 2 \ 2 \ 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0
\end{aligned}$$

$$T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 5 & 2 \\ 2 & 2 & 0 & 10 & 2 \\ 7 & 5 & 10 & 32 & 10 \\ 2 & 2 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[329] \quad 1^{12} 2^2 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 5 & 15 & 5 \\ 9 & 15 & 31 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[330] \quad 221^{11} 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 23, \Delta = -114, \det = -110, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 23, T \in [11, 23]$$

$$\begin{pmatrix} 1 & 5 & 5 & 3 \\ 5 & 8 & 11 & 6 \\ 5 & 11 & 7 & 5 \\ 3 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[331] \quad 231^{11} 3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.62 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[332] \quad 1^{13} 1^3 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 6 & 9 & 12 & 60 & 14 \\ 2 & 2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[333] \quad 1^{12} 321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 7 & 7 & 11 & 44 & 12 \\ 2 & 2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[334] \quad 1^{11} 321^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 \\ 3 & 5 & 25 & 5 \\ 13 & 25 & 91 & 19 \\ 3 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[336] \quad 1^{14} 31^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -119, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.65 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 15 & 0 & 75 & 15 \\ 2 & 3 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[335] \quad 21^{13} 321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -119, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 7 & 2 \\ 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 7 & 7 & 6 & 24 & 8 \\ 2 & 2 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[337] \quad 2^2 2^2 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 6 & 2 \\ 1 & 0 & 2 & 6 & 2 \\ 2 & 2 & 0 & 8 & 2 \\ 6 & 6 & 8 & 25 & 9 \\ 2 & 2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[338] \quad 31^{12} 231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 5 & 8 & 3 \\ 5 & 5 & 10 & 5 \\ 8 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[339] \quad 1^{13} 31^3 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.63 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 18 & 0 & 132 & 20 \\ 2 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[340] \quad 1^{13} 321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 7 & 8 & 9 & 39 & 11 \\ 2 & 2 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[341] \ 1^{1^3} 3^{1^3} 21^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 8 & 7 & 11 & 44 & 12 \\ 2 & 2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[342] \ 2 \ 2 \ 31^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.6$$

$$T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 5 & 20 & 5 \\ 12 & 20 & 61 & 15 \\ 3 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[343] \ 1^{1^3} 3^{1^3} 21^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 8 & 9 & 39 & 11 \\ 2 & 2 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[344] \ 1^{1^4} 3^{1^4} 21^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 9 & 8 & 9 & 39 & 11 \\ 2 & 2 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[345] \ 1^{1^3} 3^{1^3} 21^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 14 & 3 \\ 3 & 5 & 25 & 5 \\ 14 & 25 & 91 & 19 \\ 3 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[346] \ 1^{1^4} 3^{1^4} 21^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 11 & 2 \\ 9 & 7 & 11 & 44 & 12 \\ 2 & 2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[347] \ 1^{1^2} 31^{1^3} 31^{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 99 & 17 \\ 2 & 3 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[348] \ 1^{1^4} 3^{1^4} 21^{1^4} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 15 & 3 \\ 3 & 5 & 25 & 5 \\ 15 & 25 & 91 & 19 \\ 3 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[350] \ 1^{1^2} 31^{1^2} 31^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 12 & 0 & 57 & 13 \\ 2 & 3 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[349] \ 1^{1^3} 31^{1^3} 31^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 12 & 0 & 48 & 12 \\ 2 & 3 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[351] \ 1^{1^3} 3^{1^3} 21^{1^3} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -143, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 7 & 9 & 12 & 60 & 14 \\ 2 & 2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

- [352] $1^{13}31^231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -143, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 15 & 0 & 84 & 16 \\ 2 & 3 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$
- [353] $2^12^131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -143, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 7 & 2 \\ 1 & 0 & 2 & 8 & 2 \\ 2 & 2 & 0 & 9 & 2 \\ 7 & 8 & 9 & 39 & 11 \\ 2 & 2 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$
- [354] $1^{14}31^331^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -155, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 21 & 0 & 171 & 23 \\ 2 & 3 & -1 & 23 & 2 \end{pmatrix}_{b_0}$$
- [355] $21^{13}321^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.03 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 9 & 2 \\ 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 60 & 14 \\ 2 & 2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$
- [356] $2^12^131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 5 & 2 \\ 2 & 1 & 4 & 7 & 3 \\ 2 & 4 & 0 & 7 & 2 \\ 5 & 7 & 7 & 15 & 7 \\ 2 & 3 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$
- [357] $13^12^12 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 2 \\ 2 & 1 & 4 & 6 & 3 \\ 2 & 4 & 0 & 6 & 2 \\ 4 & 6 & 6 & 9 & 5 \\ 2 & 3 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$
- [358] $21^{13}321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 8 & 2 \\ 2 & 0 & 2 & 8 & 2 \\ 1 & 2 & 0 & 9 & 2 \\ 8 & 8 & 9 & 39 & 11 \\ 2 & 2 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$
- [359] $1^{13}3^121^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -167, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 8 & 9 & 12 & 60 & 14 \\ 2 & 2 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$
- [360] $2^12^131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -173, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 \\ 4 & 5 & 20 & 5 \\ 12 & 20 & 57 & 15 \\ 3 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$
- [361] $2^13^11^{11}3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 3 & 2 & 5 & 2 \\ 3 & 1 & 3 & 5 & 3 \\ 2 & 3 & 0 & 5 & 2 \\ 5 & 5 & 5 & 7 & 5 \\ 2 & 3 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$
- [362] $1^{14}31^31^{12}3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 15 & 0 & 75 & 15 \\ 2 & 3 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$
- [363] $1^{12}31^331^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 99 & 17 \\ 2 & 3 & -1 & 17 & 2 \end{pmatrix}_{b_0} \\
[364] & 1^{14}31^{23}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 10.42 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 18 & 0 & 117 & 19 \\ 2 & 3 & -1 & 19 & 2 \end{pmatrix}_{b_0} \\
[367] & 1^{13}321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -183, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 60 & 14 \\ 2 & 2 & 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[366] & 2^2 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -183, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 7 & 2 \\ 2 & 2 & 0 & 11 & 2 \\ 8 & 7 & 11 & 44 & 12 \\ 2 & 2 & 2 & 12 & 2 \end{pmatrix}_{b_0} \\
[365] & 2^2 31^{15} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -183, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 5 & 25 & 5 \\ 15 & 25 & 97 & 19 \\ 3 & 5 & 19 & 3 \end{pmatrix}_{b_0} \\
[368] & 1^{13}31^{33}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -188, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.36 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 18 & 0 & 132 & 20 \\ 2 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0} \\
[369] & 1^{13}321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -191, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.03 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 12 & 2 \\ 10 & 9 & 12 & 60 & 14 \\ 2 & 2 & 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[370] & 1^{13}31^{23}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -195, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 10.31 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & 1 & 15 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 84 & 16 \\ 2 & 3 & -1 & 16 & 2 \end{pmatrix}_{b_0} \\
[371] & 31^{13}231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -195, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.64 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 5 & 12 & 3 \\ 5 & 5 & 15 & 5 \\ 12 & 15 & 33 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0} \\
[372] & 31^{12}231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 5 & 11 & 3 \\ 5 & 5 & 15 & 5 \\ 11 & 15 & 31 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0} \\
[373] & 1^{14}31^{33}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -239, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.36 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 21 & 0 & 171 & 23 \\ 2 & 3 & -1 & 23 & 2 \end{pmatrix}_{b_0} \\
[374] & 2^2 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -240, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.03 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 9 & 2 \\ 2 & 2 & 0 & 12 & 2 \\ 9 & 9 & 12 & 60 & 14 \\ 2 & 2 & 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[375] & 1^{14}31^{23}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -248, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 10.31 \rightarrow 11, T_{\max} = 19, T \in [11, 19]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 18 & 0 & 117 & 19 \\ 2 & 3 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[376] $1^{13}31^{331^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -248, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 11 & 2 \\ 1 & 1 & 1 & 18 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 18 & 0 & 132 & 20 \\ 2 & 3 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[377] $1^{16}2^231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 18 & 3 \\ 4 & 5 & 30 & 5 \\ 18 & 30 & 141 & 23 \\ 3 & 5 & 23 & 3 \end{pmatrix}_{b_0}$$

[378] $1^{14}2^231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 \\ 4 & 5 & 25 & 5 \\ 15 & 25 & 91 & 19 \\ 3 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[379] $1^{14}31^{331^{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -315, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & 1 & 21 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 21 & 0 & 171 & 23 \\ 2 & 3 & -1 & 23 & 2 \end{pmatrix}_{b_0}$$

[380] $1^{17}2^231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -327, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 21 & 3 \\ 4 & 5 & 35 & 5 \\ 21 & 35 & 193 & 27 \\ 3 & 5 & 27 & 3 \end{pmatrix}_{b_0}$$

[381] $31^{13}231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -335, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 5 & 15 & 3 \\ 5 & 5 & 20 & 5 \\ 15 & 20 & 57 & 15 \\ 3 & 5 & 15 & 3 \end{pmatrix}_{b_0}$$

$T = 8$ [382] $1^{13}221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -66, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 6 & 0 & 6 \\ 1 & 0 & -3 & -1 \\ 3 & 6 & -1 & 5 \end{pmatrix}_{b_0}$$

[383] $1^{12}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -68, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 3 & -1 & 2 & 4 \end{pmatrix}_{b_0}$$

[384] $1^{12}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -68, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 2 & 0 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

[385] $1^{13}221^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -66, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

[386] $1^{12}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -68, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & -1 & 4 \end{pmatrix}_{b_0}$$

[387] $1^{\overline{1}}4^{11}131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 13, T \in [11, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 4 \end{pmatrix}_{b_0}$

[388] $1^{\overline{1}}4^{11}131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$
 $\begin{pmatrix} 2 & 3 & 4 \\ 3 & 1 & 3 \\ 4 & 3 & 5 \end{pmatrix}_{b_0}$

[391] $1^{\overline{1}}4^{12}131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 2 & 1 & 6 & 4 \\ 2 & 2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$

[393] $1^{\overline{1}}4^{11}131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$

[392] $1^{\overline{1}}4^{12}31^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 4 & 2 & 2 & 7 & 5 \\ 2 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$

[394] $21^{\overline{1}}41^{\overline{1}}31^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 4 & 2 & 2 & 7 & 5 \\ 2 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$

[395] $1^{\overline{1}}41^{\overline{1}}231^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 0 & 2 & 2 \\ 3 & 2 & 3 & 3 \\ 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[396] $1^{\overline{1}}4^{11}131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$

[397] $21^{\overline{1}}5^{11}1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 4 & 6 & 4 \\ 4 & 6 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[398] $21^{\overline{1}}5^{11}1^{\overline{1}}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 4 & 6 & 4 \\ 4 & 6 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[399] $1^{\overline{1}}4^{13}1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 0 & 2 & 2 \\ 3 & 2 & 3 & 3 \\ 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[390] $1^{\overline{1}}41231^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$

[411] $1 \overset{1}{\underset{1,1}{5}} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$

[412] $41 \overset{1}{\underset{1}{2}} 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 26 & 1 & 0 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 10 & -1 & -1 & 1 \end{pmatrix}_{b_0}$

[413] $14 \overset{1}{\underset{1}{4}} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 5 & 4 & 5 \\ 4 & 1 & 3 \\ 5 & 3 & 4 \end{pmatrix}_{b_0}$

[414] $1 \overset{1}{\underset{1}{4}} 1 \overset{1}{\underset{1}{4}} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 3 & 3 \\ 2 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$

[415] $141 \overset{1}{\underset{1}{2}} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 5 & 6 & 5 \\ 6 & 5 & 5 \\ 5 & 5 & 4 \end{pmatrix}_{b_0}$

[416] $21 \overset{1}{\underset{1}{5}} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$

[417] $1 \overset{1}{\underset{1}{4}} 1 \overset{1}{\underset{1}{4}} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 3 & 3 \\ 2 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$

[418] $1 \overset{1}{\underset{1}{4}} 1 \overset{1}{\underset{1}{4}} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 4 & 3 & 1 & 4 \\ 3 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 4 & 3 & 0 & 3 \end{pmatrix}_{b_0}$

[419] $1 \overset{1}{\underset{1}{4}} 1 \overset{1}{\underset{1}{4}} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 5 & 4 & 5 \\ 4 & 1 & 3 \\ 5 & 3 & 4 \end{pmatrix}_{b_0}$

[420] $1 \overset{1}{\underset{1}{4}} 1 \overset{1}{\underset{1}{4}} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 0 & 3 & 2 \\ 4 & 3 & 7 & 5 \\ 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[421] $1 \overset{1}{\underset{1}{4}} 1231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 5 & 4 & 5 \\ 4 & 1 & 3 \\ 5 & 3 & 4 \end{pmatrix}_{b_0}$

[422] $1 \overset{1}{\underset{1}{4}} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 3 & 4 & 0 & 3 \end{pmatrix}_{b_0}$

[423] $1 \overset{1}{\underset{1}{4}} 1 \overset{1}{\underset{1}{4}} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 6 & 6 & 0 & 33 & 9 \\ 2 & 2 & 0 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[424] \quad 1^{1^1} 13141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 9 & 4 & 5 \\ 2 & 4 & 0 & 2 \\ 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[425] \quad 1^{1^1} 1^3 3221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -66, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 6 & 6 & 6 \\ 3 & 6 & 2 & 4 \\ 3 & 6 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[426] \quad 1^{1^2} 1^4 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 4 & 2 & 2 & 9 & 5 \\ 2 & 2 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[427] \quad 1^{1^1} 1^4 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[428] \quad 21^{1^1} 1^4 1^3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[429] \quad 2^1 3141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[430] \quad 1^{1^1} 1^{11} 4131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 4 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[431] \quad 1^{1^1} 1^{11} 31^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[432] \quad 1^{1^1} 1^4 1^3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 4 & 3 & 4 \\ 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[433] \quad 1^{1^1} 1^{11} 31^1 1^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 5 & 2 \\ 7 & 5 & 25 & 9 \\ 3 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[434] \quad 1^{1^1} 1^{11} 31^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 1 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

- [435] $21^{\overline{1}1}41^{\overline{1}31^{\overline{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 0 & 2 \end{pmatrix}_{b_0}$
- [436] $1^{\overline{1}1}4^{\overline{1}31^{\overline{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 4 & 1 & 24 & 8 \\ 2 & 2 & 0 & 8 & 1 \end{pmatrix}_{b_0}$
- [437] $1^{\overline{1}1}41^{\overline{1}231^{\overline{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 3 & 1 & 2 & 2 \\ 3 & 4 & 1 & 2 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 2 & 1 & 0 & 2 \\ 2 & 4 & 0 & 2 & 2 \end{pmatrix}_{b_0}$
- [438] $131^{\overline{1}}41^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 \\ 2 & 4 & 0 & 2 \\ 3 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [439] $1^{\overline{1}1}4131^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 13, T \in [11, 13]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 0 & 2 & 2 \\ 3 & 2 & 1 & 3 \\ 3 & 2 & 3 & 4 \end{pmatrix}_{b_0}$
- [440] $1^{\overline{1}1}5^{\overline{1}1}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 3 & 4 \\ 3 & 3 & 0 & 2 \\ 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [441] $1^{\overline{1}1}4^{\overline{1}31^{\overline{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 7 & 4 \\ 6 & 7 & 8 & 6 \\ 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$
- [442] $1^{\overline{1}1}4131^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 2 & 3 & 2 \\ 3 & 3 & 1 & 3 \\ 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [443] $141^{\overline{1}231^{\overline{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 4 & 1 & 0 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 0 & 0 & 2 \end{pmatrix}_{b_0}$
- [444] $1^{\overline{1}31^{\overline{1}}}4^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 3 \end{pmatrix}_{b_0}$
- [445] $1^{\overline{1}31^{\overline{1}}}4^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 4 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [446] $1^{\overline{1}1}41^{\overline{1}231^{\overline{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 3 & 2 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 3 & 2 & 3 \\ 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$

$$[447] \ 1^{\overline{1}41^{\overline{1}}231^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 2 & 6 & 4 \\ 6 & 8 & 6 \\ 4 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[448] \ 31^{\overline{1}}31^{\overline{1}41^{\overline{1}}1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 3 & 2 & 0 & 4 & 2 \\ 4 & 4 & 4 & 9 & 5 \\ 2 & 2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[449] \ 1^{\overline{1}41^{\overline{1}}231^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 3 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[450] \ 31^{\overline{1}}31^{\overline{1}41^{\overline{1}}1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 6 & 4 & 3 \\ 6 & 13 & 6 & 7 \\ 4 & 6 & 1 & 3 \\ 3 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[451] \ 1^{\overline{1}1^{\overline{1}}31^{\overline{1}}4^{\overline{1}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 3 & 1 & 12 & 6 \\ 2 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[452] \ 1^{\overline{1}}51^{\overline{1}1^{\overline{1}}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[453] \ 1^{\overline{1}1^{\overline{1}}31^{\overline{1}}4^{\overline{1}1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[454] \ 2^{\overline{1}1^{\overline{1}}3^{\overline{1}}141^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[455] \ 1^{\overline{1}4^{\overline{1}3}131^{\overline{1}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 5 & 2 \\ 7 & 5 & 23 & 9 \\ 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[456] \ 1^{\overline{1}4^{\overline{1}3}131^{\overline{1}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 0 & 3 & 2 \\ 5 & 3 & 7 & 5 \\ 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[457] \ 1^{\overline{1}4^{\overline{1}3}131^{\overline{1}1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 2 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[458] \ 21^{\overline{1}1^{\overline{1}}5^{\overline{1}}1^{\overline{1}}2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 1 & 5 & 3 \\ 4 & 5 & 7 & 5 \\ 3 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

- [459] $1^{11}41231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 2 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$
- [460] $1^{11}4131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 5 & 1 & 24 & 8 \\ 2 & 2 & 0 & 8 & 2 \end{pmatrix}_{b_0}$
- [461] $2141^{131^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 0 & 1 \end{pmatrix}_{b_0}$
- [462] $1^{11}41^{131^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 0 & 1 \end{pmatrix}_{b_0}$
- [463] $131^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [464] $1^{111}41^{131^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 2 & 3 & 2 \\ 4 & 3 & 1 & 3 \\ 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [465] $1^{131^{11}}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 3 & 0 & 3 & 3 \\ 2 & 2 & 0 & 3 & 2 \end{pmatrix}_{b_0}$
- [466] $2141^{131^{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 2 & 3 & 2 \\ 4 & 3 & 1 & 3 \\ 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [467] $1^{11}41^{131^{11}}41^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 5 & 0 & 19 & 7 \\ 2 & 2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$
- [468] $21^{11}4131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 4 & 1 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 6 & 0 & 3 & 3 \\ 2 & 4 & 0 & 3 & 2 \end{pmatrix}_{b_0}$
- [469] $1^{11}41^{131^{11}}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 0 & 2 & 2 \\ 4 & 2 & 1 & 3 \\ 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$
- [470] $31^{11}413 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 2 & 3 & 2 \\ 2 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 3 & 2 \\ 3 & 3 & 3 & 3 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[471] \quad & 1321^1 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 7 & 4 \\ 5 & 7 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0} \\
[472] \quad & 131^1 41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.77 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 7 & 4 \\ 5 & 7 & 7 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0} \\
[473] \quad & 1^{11} 4^{12} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 5 & 2 & 2 & 7 & 5 \\ 2 & 2 & 0 & 5 & 1 \end{pmatrix}_{b_0} \\
[474] \quad & 31^1 31^1 41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.78 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 3 & 2 & 0 & 3 & 2 \\ 3 & 2 & 3 & 3 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[475] \quad & 1^{11} 4^{12} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 3 & 2 & 2 & 3 & 3 \\ 2 & 2 & 0 & 3 & 1 \end{pmatrix}_{b_0} \\
[478] \quad & 1^{12} 4^{13} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 0 & 4 & 2 \\ 6 & 4 & 13 & 7 \\ 3 & 2 & 7 & 3 \end{pmatrix}_{b_0} \\
[479] \quad & 1^{11} 4^{13} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 4 & 0 & 9 & 5 \\ 2 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0} \\
[480] \quad & 1^{11} 4^{13} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 4 & 2 \\ 7 & 4 & 17 & 7 \\ 3 & 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[476] \quad & 1^{11} 4^{13} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 0 & 4 & 2 \\ 6 & 4 & 13 & 7 \\ 3 & 2 & 7 & 3 \end{pmatrix}_{b_0} \\
[477] \quad & 1^{11} 4^{13} 1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 4 & 0 & 9 & 5 \\ 2 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0} \\
[481] \quad & 221^1 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 3 \\ 5 & 5 & 7 & 5 \\ 3 & 3 & 5 & 3 \end{pmatrix}_{b_0} \\
[482] \quad & 1^1 41^1 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 0 & 3 & 2 \\ 5 & 3 & 5 & 5 \\ 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[483] $1^{\overline{1}1131^{\overline{1}}4}1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 0 & 3 & 2 \\ 5 & 3 & 5 & 5 \\ 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[484] $1^{\overline{1}114131^{\overline{1}}1^{\overline{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 3 & 1 & 4 & 4 \\ 2 & 2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

[485] $1^{\overline{1}114131^{\overline{1}}1^{\overline{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 4 & 1 & 12 & 6 \\ 2 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[486] $1^{\overline{1}41^{\overline{1}}231^{\overline{1}}1^{\overline{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 13 & 0 & 7 \\ 1 & 0 & -2 & 0 \\ 3 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

[487] $1^{\overline{1}1^{12}31^{\overline{1}}4}1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 7 & 4 & 2 & 27 & 9 \\ 2 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

[489] $1^{\overline{1}1^{12}31^{\overline{1}}4}1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 3 & 2 & 15 & 7 \\ 2 & 2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

[490] $1^{\overline{1}41^{\overline{1}}231^{\overline{1}}1^{\overline{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 1 & 3 \\ 5 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 3 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[488] $1^{\overline{1}1^{12}31^{\overline{1}}4}1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 0 & 3 & 2 \\ 6 & 3 & 7 & 5 \\ 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[491] $31^{\overline{1}131^{\overline{1}}41^{\overline{1}}1^{\overline{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 5 & 5 & 3 \\ 5 & 5 & 7 & 5 \\ 5 & 7 & 7 & 5 \\ 3 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[492] $21^{\overline{1}141^{\overline{1}}131^{\overline{1}}1^{\overline{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 4 & 1 & 8 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 8 & 1 & 10 & 6 \\ 2 & 4 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

[493] $21^{\overline{1}14131^{\overline{1}}1^{\overline{1}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 4 & 1 & 6 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 6 & 1 & 4 & 4 \\ 2 & 4 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

[494] $2^{\overline{1}11}3141^{\overline{1}}1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

- [495] $2 \overset{11}{3} 141^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 2 & 5 & 4 \\ 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [496] $2 \overset{11}{3} 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 6 & 3 & 3 \\ 6 & 13 & 6 & 7 \\ 3 & 6 & 1 & 3 \\ 3 & 7 & 3 & 3 \end{pmatrix}_{b_0}$
- [497] $21^{\overset{11}{1}} 41^{\overset{11}{1}} 31^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 4 & 1 & 8 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 8 & 0 & 9 & 5 \\ 2 & 4 & 0 & 5 & 2 \end{pmatrix}_{b_0}$
- [498] $41^{\overset{11}{1}} 2 \overset{11}{3} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 14, \Delta = -96, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$
 $\begin{pmatrix} 0 & 2 & 4 & 2 & 2 \\ 2 & 1 & 5 & 3 & 3 \\ 4 & 5 & 9 & 4 & 5 \\ 2 & 3 & 4 & 0 & 2 \\ 2 & 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$
- [499] $1^{\overset{12}{1}} 4 131^{\overset{13}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 5 & 2 \\ 8 & 5 & 23 & 9 \\ 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [500] $221^{\overset{11}{1}} 5 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 6 & 2 & 3 \\ 6 & 13 & 5 & 7 \\ 2 & 5 & 0 & 2 \\ 3 & 7 & 2 & 3 \end{pmatrix}_{b_0}$
- [501] $1^{\overset{11}{1}} 4 131^{\overset{12}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 4 & 3 & 2 & 7 & 5 \\ 2 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$
- [502] $1^{\overset{11}{1}} 4^{\overset{11}{1}} 1^{\overset{11}{1}} 31^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 3 & 1 & 4 & 4 \\ 2 & 2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$
- [503] $1^{\overset{13}{1}} 4^{\overset{13}{1}} 1^{\overset{13}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 4 & 2 \\ 8 & 4 & 17 & 7 \\ 3 & 2 & 7 & 2 \end{pmatrix}_{b_0}$
- [504] $141^{\overset{13}{1}} 31^{\overset{13}{1}} 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 3 & 3 & 3 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [505] $31^{\overset{31}{1}} 4^{\overset{31}{1}} 1^{\overset{31}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 2 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [506] $1^{\overset{13}{1}} 4 131^{\overset{13}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 6 & 2 \\ 9 & 6 & 33 & 11 \\ 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[507] \ 1^{11} 2^3 141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 3 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[508] \ 41^{11} 231^1 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 4 & 3 & 4 & 2 \\ 4 & 10 & 5 & 12 & 6 \\ 3 & 5 & 0 & 4 & 2 \\ 4 & 12 & 4 & 9 & 5 \\ 2 & 6 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[509] \ 1^{1^{14}31^1} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.6$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 5 & 2 \\ 10 & 5 & 31 & 9 \\ 3 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[510] \ 1^{11} 4^{13} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 5 & 2 & 27 & 9 \\ 2 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[511] \ 1^{11} 4^{12} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 5 & 4 & 2 & 15 & 7 \\ 2 & 2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[512] \ 1^1 41^{12} 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 4 & 2 \\ 7 & 4 & 11 & 7 \\ 3 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[513] \ 1^{1^{12}31^1} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 7 & 3 \\ 2 & 0 & 4 & 2 \\ 7 & 4 & 11 & 7 \\ 3 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[514] \ 1^2 41^{11} 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 6 & 2 \\ 3 & 4 & 1 & 10 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 10 & 1 & 20 & 8 \\ 2 & 4 & 0 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[515] \ 2^3 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 5 & 3 & 3 \\ 5 & 8 & 6 & 6 \\ 3 & 6 & 1 & 3 \\ 3 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[516] \ 1^{11} 4^{11} 31^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 6 & 1 & 34 & 10 \\ 2 & 2 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[517] \ 1^{1^{12}31^1} 4^1 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 6 & 2 \\ 9 & 6 & 31 & 11 \\ 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[518] $1^1 4 131^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 8, T_{\max} = 13, \Delta = -93, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 6 & 2 \\ 10 & 6 & 37 & 11 \\ 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$

[519] $1^{\textcolor{red}{1}\textcolor{blue}{1}} 1^{\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{blue}{1}\textcolor{red}{1}} 1^{\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 4 & 1 & 10 & 6 \\ 2 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$

$$[520] \ 1^{\overline{1}231^1} 4^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -93, \ \det = 56, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.79 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 5 & 3 & 2 & 7 & 5 \\ 2 & 2 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[521] $1^{14}14^{12}231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 4 & 1 & 10 & 6 \\ 2 & 2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$

[522] $2\ 3\ 141\overset{1\ 12}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 57, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.2$
 $T_{\min} = 11.30 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$

[523] $1^{\textcolor{blue}{1}} 1^{\textcolor{brown}{1}2} 3^{\textcolor{brown}{1}1} 4^{\textcolor{brown}{1}2} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{s}\textcolor{blue}{u}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 0 & 5 & 2 \\ 8 & 5 & 19 & 9 \\ 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[524] $31^{\textcolor{blue}{1}}31^{\textcolor{red}{1}}41^{\textcolor{green}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 5 & 4 & 3 \\ 5 & 8 & 6 & 6 \\ 4 & 6 & 1 & 3 \\ 3 & 6 & 3 & 3 \end{pmatrix}_{b_0}$

[525] $1^{11} 1^{11} 31^1 4^1 1^2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 5 & 1 & 20 & 8 \\ 2 & 2 & 0 & 8 & 2 \end{pmatrix}_{b_0}$

[526] $2^{\overset{1}{\text{1}}\overset{1}{\text{4}}\overset{1}{\text{1}}} 3^{\overset{1}{\text{1}}\overset{1}{\text{1}}} + [\textcolor{red}{\mathfrak{s}}\textcolor{red}{\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}}\textcolor{green}{\mathfrak{u}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 4 & 4 & 5 & 4 \\ 2 & 4 & 0 & 4 & 2 \\ 3 & 5 & 4 & 4 & 4 \\ 2 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$

[527] $1^{\textcolor{teal}{4}}1^{\textcolor{teal}{1}}231^{\textcolor{teal}{1}} + [\textcolor{teal}{5}\textcolor{teal}{u}_2 \cdot \textcolor{teal}{g}_2] + \textcolor{teal}{5}\textcolor{teal}{u}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 3 & 1 & 4 & 2 \\ 3 & 4 & 1 & 8 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 4 & 8 & 0 & 9 & 5 \\ 2 & 4 & 0 & 5 & 2 \end{pmatrix}_{b_0}$

[528] $1^{41}1^{11}231^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}u_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 6 & 6 & 3 \\ 6 & 13 & 12 & 7 \\ 6 & 12 & 8 & 6 \\ 3 & 7 & 6 & 3 \end{pmatrix}_{b_0}$

[529] $141^{11}32 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 3 & 2 & 2 & 2 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$

- [530] $1^{\overline{1}13}4^{\overline{1}}1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 5 & 2 \\ 9 & 5 & 21 & 9 \\ 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [531] $1^{\overline{1}14}1^{\overline{1}231}1^{\overline{1}3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 9 & 3 \\ 2 & 0 & 5 & 2 \\ 9 & 5 & 21 & 9 \\ 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$
- [532] $1^{\overline{1}14}131^{\overline{1}4} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 7 & 2 \\ 11 & 7 & 47 & 13 \\ 3 & 2 & 13 & 3 \end{pmatrix}_{b_0}$
- [533] $31^{\overline{1}13}141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 5 & 4 & 3 \\ 5 & 5 & 5 & 5 \\ 4 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$
- [534] $141^{\overline{1}1231}1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 4 & 1 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 \\ 2 & 4 & 0 & 3 & 2 \end{pmatrix}_{b_0}$
- [535] $1^{\overline{1}12}1^{\overline{1}3}4^{\overline{1}}1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 6 & 4 & 2 & 13 & 7 \\ 2 & 2 & 0 & 7 & 2 \end{pmatrix}_{b_0}$
- [536] $2^{\overline{1}13}3^{\overline{1}4}141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 5 & 15 & 5 \\ 9 & 15 & 33 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$
- [537] $1^{\overline{1}14}1^{\overline{1}3}4^{\overline{1}}1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 6 & 2 \\ 11 & 6 & 35 & 11 \\ 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [538] $1^{\overline{1}12}1^{\overline{1}3}4^{\overline{1}}1^{\overline{1}2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 5 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 7 & 5 & 2 & 23 & 9 \\ 2 & 2 & 0 & 9 & 2 \end{pmatrix}_{b_0}$
- [539] $1^{\overline{1}13}1^{\overline{1}3}4^{\overline{1}}1^{\overline{1}2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 10 & 3 \\ 2 & 0 & 6 & 2 \\ 10 & 6 & 29 & 11 \\ 3 & 2 & 11 & 3 \end{pmatrix}_{b_0}$
- [540] $31^{\overline{1}13}141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 2 & 5 & 4 \\ 4 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [541] $1321^{\overline{1}14}1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 6 & 5 & 3 \\ 6 & 13 & 12 & 7 \\ 5 & 12 & 7 & 5 \\ 3 & 7 & 5 & 3 \end{pmatrix}_{b_0}$
- [542] $1^{\overline{1}12}1^{\overline{1}3}4^{\overline{1}}1^{\overline{1}3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 6 & 2 \\ 1 & 1 & -2 & 2 & 0 \\ 8 & 6 & 2 & 37 & 11 \\ 2 & 2 & 0 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[543] \quad 141^{11}231^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.47 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 4 & 2 \\ 3 & 4 & 1 & 6 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 6 & 1 & 4 & 4 \\ 2 & 4 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[544] \quad 1^{11}4^{13}31^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.05 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 11 & 3 \\ 2 & 0 & 7 & 2 \\ 11 & 7 & 41 & 13 \\ 3 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[545] \quad 1^{11}41^{11}231^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 5 & 2 \\ 3 & 4 & 1 & 8 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 8 & 1 & 10 & 6 \\ 2 & 4 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[546] \quad 31^{11}31^{14}41^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 5 & 9 & 3 \\ 5 & 5 & 12 & 5 \\ 9 & 12 & 21 & 9 \\ 3 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[547] \quad 41^{11}2231^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 28 & 0 & 25 & 9 \\ 2 & 10 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[548] \quad 1^{11}4^{14}31^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 12 & 3 \\ 2 & 0 & 7 & 2 \\ 12 & 7 & 43 & 13 \\ 3 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[549] \quad 41^{11}2231^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 4 & 1 & 5 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 28 & 0 & 25 & 9 \\ 2 & 10 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[550] \quad 1^{11}4^{14}31^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 13 & 3 \\ 2 & 0 & 8 & 2 \\ 13 & 8 & 55 & 15 \\ 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[551] \quad 41^{11}232^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 14, \Delta = -96, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$$

$$\begin{pmatrix} 0 & 4 & 2 & 4 & 2 \\ 4 & 10 & 5 & 12 & 6 \\ 2 & 5 & 0 & 4 & 2 \\ 4 & 12 & 4 & 9 & 5 \\ 2 & 6 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[552] \quad 41^{11}2231^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 26 & 29 & 0 & 10 \\ 4 & 29 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 10 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[553] \quad 41^{11}3^{11}22 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 4 & 3 & 3 & 2 \\ 4 & 10 & 9 & 8 & 6 \\ 3 & 9 & 4 & 7 & 4 \\ 3 & 8 & 7 & 4 & 4 \\ 2 & 6 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& T = 9 \quad [554] \quad 1^{\overline{1}} 4 1321^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 11, \Delta = -65, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 3 \\ 2 & 1 & 0 & 2 \\ 3 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
& [555] \quad 1^{\overline{1}} \overline{5} 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 4 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 4 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
& [556] \quad 21^{\overline{1}} \overline{5} 1 3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 3 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 3 & 2 & 3 \end{pmatrix}_{b_0} \\
& [557] \quad 21^{\overline{1}} \overline{4} 1 41^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 4 & 2 \\ 2 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 4 & 5 & 15 & 7 \\ 2 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0} \\
& [558] \quad 231^{\overline{1}} \overline{5} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 \\ 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0} \\
& [559] \quad 21^{\overline{1}} \overline{5} 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 4 & 5 & 6 & 4 \\ 5 & 5 & 8 & 5 \\ 6 & 8 & 8 & 6 \\ 4 & 5 & 6 & 4 \end{pmatrix}_{b_0} \\
& [560] \quad 21^{\overline{1}} \overline{5} 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 4 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 4 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
& [561] \quad 1^{\overline{1}} \overline{5} 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 \\ 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0} \\
& [562] \quad 1^{\overline{1}} 4 1321^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 11, \Delta = -65, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 2 & 3 & 4 \\ 3 & 2 & 4 \\ 4 & 4 & 6 \end{pmatrix}_{b_0} \\
& [563] \quad 1^{\overline{1}} \overline{4} 1 41^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 3 & 4 & 3 \\ 1 & 3 & 0 & 3 & 2 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & 2 & 4 & 3 \end{pmatrix}_{b_0} \\
& [564] \quad 14141^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0} \\
& [565] \quad 21^{\overline{1}} \overline{4} 1 41^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 3 & 4 & 3 \\ 1 & 3 & 0 & 3 & 2 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[566] $1^{\overline{1}} \overline{5} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[567] $21^{\overline{1}} 4141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 5 & 4 & 5 \\ 3 & 4 & 1 & 3 \\ 4 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

[568] $1^{\overline{1}} \overline{5} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 \\ 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[569] $1^{\overline{1}} \overline{5} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 3 & 5 \\ 2 & 3 & 0 & 2 \\ 3 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

[570] $1^{\overline{1}} \overline{4} 14141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 4 & 5 \\ 3 & 4 & 1 & 3 \\ 3 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

[571] $1^{\overline{1}} \overline{4} 14141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 \end{pmatrix}_{b_0}$$

[572] $21^{\overline{1}} \overline{4} 14141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[573] $1^{\overline{1}} \overline{5} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 \end{pmatrix}_{b_0}$$

[574] $1^{\overline{1}} \overline{4} 131^{\overline{1}} 4141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 4 & 0 & 9 & 5 \\ 2 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[575] $1^{\overline{1}} \overline{5} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[576] $21^{\overline{1}} \overline{5} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 1 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[577] \quad 1^1 5 1231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 1 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[578] \quad 1^1 14141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 4 & 5 \\ 2 & 4 & 1 & 3 \\ 3 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[579] \quad 1^{11} 322151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 3 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 2 & 0 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[580] \quad 1^1 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[581] \quad 1^1 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 1 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[582] \quad 21^1 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 5 & 3 & 5 \\ 3 & 3 & 0 & 2 \\ 4 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[583] \quad 21^1 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -9, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 4 & 3 & 4 \\ 3 & 3 & 1 & 3 \\ 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[585] \quad 21^1 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 4 & 3 & 4 \\ 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[586] \quad 21^1 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[587] \quad 1^{11} 322151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 0 & 2 \\ 2 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[588] \quad 1^1 5 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[589] \quad 1^1 3221^1 51^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 5 & 2 \\ 4 & 0 & 5 & 17 & 7 \\ 2 & 0 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[590] \quad 1^1 1^1 5^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 \\ 1 & 0 & -3 & -1 \\ 3 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[591] \quad 1^1 4^1 1^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 5 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 5 & 0 & 12 & 6 \\ 2 & 3 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[592] \quad 1^1 1^1 4^1 1^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 0 & 2 \\ 2 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[584] \quad 1^1 5^1 1^1 2^1 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 \\ 2 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[593] \quad 1^1 5^1 1^1 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 2 & 2 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[594] \quad 3^1 1^1 4^1 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 9 & 4 \\ 3 & 0 & 9 & 12 & 6 \\ 2 & 0 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[595] \quad 1^1 4^1 1^1 3^1 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 11, \Delta = -65, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 3 & 3 \\ 3 & 3 & 2 & 4 \\ 3 & 3 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[596] \quad 1^1 5^1 1^1 2^1 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[600] \quad 3^1 1^1 4^1 1^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 0 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[597] \quad 1^1 5^1 3^1 2^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 & 2 \\ 3 & 10 & 5 & 7 & 6 \\ 1 & 5 & 0 & 4 & 2 \\ 2 & 7 & 4 & 4 & 4 \\ 2 & 6 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[598] \quad 1^1 5^1 1^1 2^1 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 7 & 3 & 3 \\ 7 & 26 & 8 & 10 \\ 3 & 8 & 1 & 3 \\ 3 & 10 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[599] \quad 23151^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[601] \quad 21^{\overline{1}1}5131^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[602] \quad 14141^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 \\ 3 & 5 & 1 & 3 \\ 3 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[603] \quad 1^{\overline{1}1}51231^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 \\ 3 & 5 & 1 & 3 \\ 3 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[604] \quad 21^{\overline{1}1}5131^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 2 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[605] \quad 1^{\overline{1}1}4141^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 13 & 5 \\ 7 & 13 & 29 & 11 \\ 3 & 5 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[606] \quad 15131^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 5 & 8 & 5 \\ 4 & 8 & 8 & 6 \\ 3 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[607] \quad 1^{\overline{1}1}5131^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 2 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[608] \quad 1^{\overline{1}1}4141^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 5 & 13 & 5 \\ 8 & 13 & 29 & 11 \\ 3 & 5 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[609] \quad 1^{\overline{1}1}4141^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 5 & 9 & 5 \\ 6 & 9 & 11 & 7 \\ 3 & 5 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[610] \quad 23151^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 5 & 3 & 5 \\ 2 & 3 & 0 & 2 \\ 3 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[611] \quad 1^{\overline{1}1}5131^{\overline{1}1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 8 & 5 \\ 5 & 8 & 8 & 6 \\ 3 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

[612] $21^1 1^1 5 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 8 & 5 \\ 5 & 8 & 8 & 6 \\ 3 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

[617] $1^{11} 4 1 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 9 & 5 \\ 5 & 9 & 11 & 7 \\ 3 & 5 & 7 & 4 \end{pmatrix}_{b_0}$$

[618] $1^{11} 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 10 & 5 & 4 & 6 \\ 1 & 5 & 0 & 3 & 2 \\ 1 & 4 & 3 & 0 & 2 \\ 2 & 6 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[613] $21^1 1^1 5 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 4 & 3 & 7 & 15 & 7 \\ 2 & 2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[614] $31^1 1 3 1 5 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 3 \\ 4 & 3 & 1 & 3 \\ 3 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

[615] $1^{11} 1^1 5 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 7 & 2 \\ 4 & 3 & 7 & 15 & 7 \\ 2 & 2 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[616] $2151^1 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 10 & 5 & 4 & 6 \\ 1 & 5 & 0 & 3 & 2 \\ 1 & 4 & 3 & 0 & 2 \\ 2 & 6 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[620] $141^1 1 3 1^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 13 & 6 & 12 & 7 \\ 1 & 6 & 0 & 4 & 2 \\ 3 & 12 & 4 & 9 & 5 \\ 2 & 7 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[621] $21^1 1^1 5 1^{11} 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 4 & 11 & 3 & 4 \\ 11 & 26 & 8 & 10 \\ 3 & 8 & 1 & 3 \\ 4 & 10 & 3 & 3 \end{pmatrix}_{b_0}$$

[619] $31^1 1^1 4 1 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 5 & 1 & 9 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 9 & 0 & 12 & 6 \\ 2 & 5 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[624] $1^1 1^1 1^1 3 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 3 & 2 \\ 2 & 1 & 3 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[622] $1^1 1^1 5 1231^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -23, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.2$

$T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 5 & 10 & 5 \\ 5 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[623] $1\overset{1}{5}131\overset{12}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -23$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 5 & 13 & 5 \\ 6 & 13 & 24 & 10 \\ 3 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[625] $31\overset{11}{1}4141 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 3 & 2 \\ 2 & 5 & 8 & 9 & 5 \\ 2 & 8 & 4 & 9 & 4 \\ 3 & 9 & 9 & 12 & 6 \\ 2 & 5 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[626] $1\overset{1}{3}221\overset{15}{1}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -89$, $\det = -26$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & 1 & 6 & 0 & 3 \\ 4 & 6 & 17 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

[628] $1\overset{11}{1}41\overset{1}{3}141\overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -28$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12.29 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 8 & 0 & 25 & 9 \\ 2 & 3 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[627] $2\overset{11}{3}1\overset{1}{5}1\overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -89$, $\det = -28$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12.14 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 2 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[630] $1\overset{1}{3}221\overset{15}{1}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -89$, $\det = -30$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12.27 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 17 & 5 & 7 \\ 1 & 0 & 5 & 0 & 2 \\ 2 & 0 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[629] $41\overset{1}{1}231\overset{1}{4}1\overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -30$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12.27 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 3 & 4 & 1 & 2 \\ 3 & 10 & 17 & 0 & 6 \\ 4 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 6 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

[633] $1\overset{11}{1}3221\overset{15}{1}1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -89$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 8 & 2 \\ 5 & 0 & 8 & 30 & 10 \\ 2 & 0 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[634] $21\overset{11}{1}4141\overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 3 & 3 & 3 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[631] $1\overset{11}{1}4131\overset{1}{4}1\overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 7 & 0 & 20 & 8 \\ 2 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[632] $31\overset{1}{1}3151\overset{1}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -96$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 5.0$

$$[635] \ 1^{\overline{1}321^{\overline{1}}} \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 7 & 2 & 3 \\ 7 & 26 & 8 & 10 \\ 2 & 8 & 1 & 3 \\ 3 & 10 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[636] \ 1^{\overline{1}41^{\overline{1}}3141^{\overline{1}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 13 & 6 & 0 & 7 \\ 1 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 7 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[637] \ 1^{\overline{1}321^{\overline{1}}} \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 6 & 2 & 3 \\ 6 & 19 & 8 & 9 \\ 2 & 8 & 1 & 3 \\ 3 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[638] \ 1^{\overline{1}41^{\overline{1}}3141^{\overline{1}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 13 & 1 & 5 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 5 & 0 & 0 & 2 \\ 2 & 7 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[639] \ 141^{\overline{1}}31^{\overline{1}}41^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 13 & 1 & 12 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 12 & 0 & 9 & 5 \\ 2 & 7 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[640] \ 21^{\overline{1}}1^{\overline{1}}5131^{\overline{1}2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 13 & 5 \\ 7 & 13 & 24 & 10 \\ 3 & 5 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[641] \ 1^{\overline{1}13221^{\overline{1}}51^{\overline{1}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 8 & 0 & 30 & 10 \\ 2 & 3 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[642] \ 1^{\overline{1}}1^{\overline{1}}5131^{\overline{1}2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 5 & 13 & 5 \\ 7 & 13 & 24 & 10 \\ 3 & 5 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[643] \ 2^{\overline{1}1}32151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[644] \ 1^{\overline{1}}1^{\overline{1}}1^{\overline{1}}32 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 5 & 5 & 5 \\ 3 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[645] \ 31^{\overline{1}}4141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 9 & 4 \\ 4 & 0 & 9 & 12 & 6 \\ 2 & 0 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[646] \ 31^{\overline{1}}4141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 7 & 0 & 4 & 4 \\ 2 & 5 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

[647] $2^1 3^1 2151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 2 & 5 & 4 \\ 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$

[648] $1^1 5^1 1^1 3^1 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 2 & 5 & 4 \\ 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$

[649] $31^1 4^1 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 5 & 8 & 0 & 5 \\ 2 & 8 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 5 & 4 & -1 & 2 \end{pmatrix}_{b_0}$

[650] $2^1 3^1 1^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 5 & 8 & 5 \\ 5 & 8 & 8 & 6 \\ 3 & 5 & 6 & 4 \end{pmatrix}_{b_0}$

[651] $1321^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 7 & 1 & 3 \\ 7 & 26 & 0 & 10 \\ 1 & 0 & -3 & -1 \\ 3 & 10 & -1 & 3 \end{pmatrix}_{b_0}$

[652] $22^1 3^1 151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 4 & 7 & 4 \\ 3 & 2 & 7 & 4 & 4 \\ 2 & 2 & 4 & 4 & 2 \end{pmatrix}_{b_0}$

[653] $2321^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 8 & 3 & 3 \\ 8 & 26 & 8 & 10 \\ 3 & 8 & 1 & 3 \\ 3 & 10 & 3 & 3 \end{pmatrix}_{b_0}$

[654] $31^1 4^1 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 3 & 2 & 1 & 2 \\ 3 & 13 & 10 & 0 & 7 \\ 2 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$

[655] $1^1 3221^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 18 & 1 & 5 & 8 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 5 & 0 & 0 & 2 \\ 2 & 8 & -1 & 2 & 2 \end{pmatrix}_{b_0}$

[656] $31^1 321^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$
 $\begin{pmatrix} 1 & 7 & 4 & 3 \\ 7 & 19 & 8 & 9 \\ 4 & 8 & 1 & 3 \\ 3 & 9 & 3 & 3 \end{pmatrix}_{b_0}$

[658] $1^1 5^1 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 18 & 5 \\ 9 & 18 & 48 & 14 \\ 3 & 5 & 14 & 3 \end{pmatrix}_{b_0}$

[657] $21^1 5^1 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 5 & 18 & 5 \\ 9 & 18 & 48 & 14 \\ 3 & 5 & 14 & 3 \end{pmatrix}_{b_0}$

[659] $1^1 3221^1 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 18 & 6 & 0 & 8 \\ 1 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 8 & 2 & -1 & 3 \end{pmatrix}_{b_0}$

[660] $1^1 141^1 31^1 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 12 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 12 & 0 & 69 & 15 \\ 2 & 3 & -1 & 15 & 2 \end{pmatrix}_{b_0}$

[662] $1^1 51^1 32 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$

[663] $1^1 232151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 10 & 5 \\ 6 & 10 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$

[661] $31^1 4141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 5 & 2 \\ 2 & 5 & 1 & 16 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 16 & 0 & 37 & 11 \\ 2 & 5 & -1 & 11 & 2 \end{pmatrix}_{b_0}$

[665] $1^1 5131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 23 & 5 \\ 11 & 23 & 80 & 18 \\ 3 & 5 & 18 & 3 \end{pmatrix}_{b_0}$

[664] $21^1 5131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 11 & 3 \\ 3 & 5 & 23 & 5 \\ 11 & 23 & 80 & 18 \\ 3 & 5 & 18 & 3 \end{pmatrix}_{b_0}$

[666] $1^1 3221^1 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -67, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 11 & 0 & 49 & 13 \\ 2 & 3 & -1 & 13 & 3 \end{pmatrix}_{b_0}$

[667] $2321^1 51^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 7 & 3 & 3 \\ 7 & 19 & 8 & 9 \\ 3 & 8 & 1 & 3 \\ 3 & 9 & 3 & 3 \end{pmatrix}_{b_0}$

[669] $1^1 5131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$
 $\begin{pmatrix} 1 & 5 & 4 & 3 \\ 5 & 5 & 5 & 5 \\ 4 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$

[668] $31^1 32151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$
 $\begin{pmatrix} 1 & 5 & 4 & 3 \\ 5 & 5 & 5 & 5 \\ 4 & 5 & 1 & 3 \\ 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$

[670] $31^1 3151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$
 $\begin{pmatrix} 1 & 5 & 6 & 3 \\ 5 & 5 & 8 & 5 \\ 6 & 8 & 8 & 6 \\ 3 & 5 & 6 & 3 \end{pmatrix}_{b_0}$

$$[671] \quad 2 \overset{1^{12}}{3} \overset{1^1}{15} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -79, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 8 & 3 \\ 4 & 5 & 13 & 5 \\ 8 & 13 & 24 & 10 \\ 3 & 5 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[672] \quad 22 \overset{1^{11}}{3} 15 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 6 & 5 & 3 \\ 6 & 14 & 13 & 8 \\ 5 & 13 & 8 & 6 \\ 3 & 8 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[673] \quad 1 \overset{1^1}{5} 1^1 3 \overset{1^{13}}{2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 9 & 3 \\ 4 & 5 & 15 & 5 \\ 9 & 15 & 33 & 11 \\ 3 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[674] \quad 1 \overset{1^1}{5} 13 1^1 3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -89, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 2 & 5 & 4 \\ 4 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[675] \quad 1 \overset{1^{13}}{3} 22 1^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 14 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 14 & 0 & 74 & 16 \\ 2 & 3 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[676] \quad 22 \overset{1^{11}}{3} 15 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -103, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 5 & 5 & 3 \\ 5 & 9 & 12 & 7 \\ 5 & 12 & 8 & 6 \\ 3 & 7 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[677] \quad 41 \overset{1^{11}}{2} 3 14 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 28 & 0 & 25 & 9 \\ 2 & 10 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[678] \quad 2 \overset{1^{13}}{3} \overset{1^1}{15} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 11 & 3 \\ 4 & 5 & 18 & 5 \\ 11 & 18 & 48 & 14 \\ 3 & 5 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[679] \quad 22 \overset{1^{12}}{3} 15 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -142, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 6 & 10 & 3 \\ 6 & 14 & 26 & 8 \\ 10 & 26 & 41 & 13 \\ 3 & 8 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[680] \quad 32 1^1 4 14 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 8 & 7 & 3 \\ 8 & 17 & 16 & 9 \\ 7 & 16 & 11 & 7 \\ 3 & 9 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[681] \quad 23 1^1 3 15 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -211, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 5 & 1 & 5 & 2 \\ 5 & 33 & 1 & 35 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 35 & 0 & 32 & 10 \\ 2 & 11 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[682] \quad 22 \overset{1^{13}}{3} 15 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -237, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 6 & 15 & 3 \\ 6 & 14 & 39 & 8 \\ 15 & 39 & 96 & 20 \\ 3 & 8 & 20 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& T = 10 \quad [683] \quad 1^{\overline{1}} 4 1 3 2 2 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 11, \Delta = -36, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0} \\
& [684] \quad 1^{\overline{1}} 5 1 2 3 2 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 6 & 0 & 6 \\ 1 & 0 & -3 & -1 \\ 3 & 6 & -1 & 5 \end{pmatrix}_{b_0} \\
& [685] \quad 1^{\overline{1}} 5 1 3 2 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 6 & 3 & 6 \\ 2 & 3 & 0 & 2 \\ 3 & 6 & 2 & 5 \end{pmatrix}_{b_0} \\
& [686] \quad 1^{\overline{1}} 4 1 4 1 3 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -63, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 3 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
& [687] \quad 1^{\overline{1}} 3 1 5 1 3 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -67, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 3 & -1 & 2 & 4 \end{pmatrix}_{b_0} \\
& [688] \quad 2 1^{\overline{1}} 5 1 3 1^{\overline{1}} 4 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 5 & 7 & 5 \\ 7 & 9 & 7 \\ 5 & 7 & 5 \end{pmatrix}_{b_0} \\
& [689] \quad 1^{\overline{1}} 6 1 3 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 4 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 4 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
& [690] \quad 1^{\overline{1}} 6 1 2 3 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 4 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 4 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
& [691] \quad 2 3 2 1^{\overline{1}} 6 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 14, T \in [13, 14] \\
& \begin{pmatrix} 0 & 4 & 2 & 2 & 2 \\ 4 & 10 & 5 & 6 & 6 \\ 2 & 5 & 0 & 2 & 2 \\ 2 & 6 & 2 & 3 & 3 \\ 2 & 6 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
& [692] \quad 2 1^{\overline{1}} 6 1^{\overline{1}} 2 3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 14, T \in [13, 14] \\
& \begin{pmatrix} 0 & 4 & 2 & 2 & 2 \\ 4 & 10 & 5 & 6 & 6 \\ 2 & 5 & 0 & 2 & 2 \\ 2 & 6 & 2 & 3 & 3 \\ 2 & 6 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
& [693] \quad 2 1^{\overline{1}} 6 1 3 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 4 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 \end{pmatrix}_{b_0} \\
& [694] \quad 1^{\overline{1}} 3 1 5 1 3 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -67, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 2 & 0 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

[695] $21^{\overline{1}1} \overline{6} 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 2 & 2 & 3 & 3 \\ 2 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[696] $21^{\overline{1}1} \overline{6} 1231^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 3 & 3 \\ 2 & 2 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

[699] $1\overline{5} 1321^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 2 & 4 \\ 3 & 3 & 4 & 5 \end{pmatrix}_{b_0}$$

[700] $1^{\overline{1}1} \overline{4} 14131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -63, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

[697] $1^{\overline{1}1} \overline{5} 12321^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

[698] $1^{\overline{1}1} \overline{3} 1\overline{5} 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 12, \Delta = -67, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 2 & -1 & 4 \end{pmatrix}_{b_0}$$

[703] $21^{\overline{1}1} \overline{6} 1^{\overline{1}1} 2231^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 & 2 \\ 4 & 10 & 1 & 6 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 6 & 0 & 3 & 3 \\ 2 & 6 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

[701] $1^{\overline{1}1} \overline{3} 221^{\overline{1}} \overline{6} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 & 2 \\ 4 & 10 & 1 & 6 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 6 & 0 & 3 & 3 \\ 2 & 6 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

[702] $1^{\overline{1}1} \overline{4} 131\overline{5} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 6 & 6 \\ 6 & 6 \end{pmatrix}_{b_0}$$

[705] $1^{\overline{1}1} \overline{4} 131\overline{5} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

[704] $1^{\overline{1}1} \overline{4} 141\overline{4} 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[710] $1^1 4 13221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 11, \Delta = -36, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 2 & 4 & 4 \\ 3 & 4 & 3 & 5 \\ 3 & 4 & 5 & 6 \end{pmatrix}_{b_0}$

[706] $21^1 6 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$

[707] $1^1 413151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 0 & 0 & 2 \\ 1 & 0 & -2 & 0 \\ 3 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[708] $1^1 6 12231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$

[709] $1^1 6 1231^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$

[711] $1^{11} 4132151^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.45 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 4 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 3 & 4 & 0 & 3 \end{pmatrix}_{b_0}$

[718] $1^1 5 1321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 6 & 9 & 6 \\ 5 & 9 & 9 & 7 \\ 3 & 6 & 7 & 5 \end{pmatrix}_{b_0}$

[719] $1^1 414141^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0}$

[712] $1^1 5 1321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 2 & 5 & 5 & 5 \\ 2 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$

[713] $21^1 5 1321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 6 & 9 & 6 \\ 5 & 9 & 9 & 7 \\ 3 & 6 & 7 & 5 \end{pmatrix}_{b_0}$

[714] $1^1 5 12321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 6 & 6 & 6 \\ 3 & 6 & 2 & 4 \\ 3 & 6 & 4 & 5 \end{pmatrix}_{b_0}$

[715] $21^1 5 1321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 2 & 5 & 5 & 5 \\ 2 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0} \\
[716] & 1 \overset{1}{5} 1321 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 6 & 9 & 6 \\ 4 & 9 & 9 & 7 \\ 3 & 6 & 7 & 5 \end{pmatrix}_{b_0} \\
[717] & 1 \overset{1}{4} 14131 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 3 & 3 \\ 3 & 3 & 2 & 4 \\ 3 & 3 & 4 & 5 \end{pmatrix}_{b_0} \\
[722] & 21 \overset{1}{6} 1231 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 3 & 6 & 0 & 4 \end{pmatrix}_{b_0} \\
[723] & 1 \overset{1}{1} 414141 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 2 & 2 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[720] & 232161 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.80 \rightarrow 13, T_{\max} = 14, T \in [13, 14] \\
& \begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[721] & 1 \overset{21}{6} 131 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[726] & 1 \overset{21}{6} 131 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 2 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0} \\
[727] & 1 \overset{1}{6} 1231 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 2 \\ 3 & 2 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0} \\
[724] & 1 \overset{1}{1} 413151 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0} \\
[725] & 21 \overset{1}{6} 1231 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0} \\
[728] & 1 \overset{1}{1} 414141 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 17, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]
\end{aligned}$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 5 & 0 & 5 \\ 1 & 0 & -2 & 0 \\ 3 & 5 & 0 & 4 \end{pmatrix}_{b_0}$$

[730] $31^{14}1^{13}151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.26 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 3 & 1 & 5 & 2 \\ 3 & 4 & 1 & 6 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 6 & 0 & 7 & 5 \\ 2 & 4 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

[729] $1^{11}41^14141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 4 & 0 & 15 & 7 \\ 2 & 2 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

[736] $221^{11}6131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 3 & 1 & 10 & 6 \\ 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[731] $31^{11}3161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 1 & 7 & 3 \\ 6 & 7 & 21 & 9 \\ 3 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[732] $31^{11}32161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 3 \\ 4 & 3 & 1 & 3 \\ 3 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

[733] $21^{11}6131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 4 & 3 & 5 & 10 & 6 \\ 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[734] $41^{11}23151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & 0 & 5 & 0 & 2 \\ 4 & 5 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

[735] $1^{11}41^14141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 3 & 0 & 4 & 4 \\ 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

[737] $221^{11}6131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 19 & 1 & 9 \\ 1 & 1 & -2 & 0 \\ 3 & 9 & 0 & 4 \end{pmatrix}_{b_0}$$

[739] $1^{11}51321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 6 & 15 & 6 \\ 7 & 15 & 28 & 12 \\ 3 & 6 & 12 & 5 \end{pmatrix}_{b_0}$$

[738] $21^{11}51321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 6 & 15 & 6 \\ 7 & 15 & 28 & 12 \\ 3 & 6 & 12 & 5 \end{pmatrix}_{b_0}$$

[740] $31^1 3221^1 61^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.59 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 1 & 9 & 4 & 3 \\ 9 & 34 & 10 & 12 \\ 4 & 10 & 1 & 3 \\ 3 & 12 & 3 & 3 \end{pmatrix}_{b_0}$

[742] $1^1 41^1 32151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.27 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 13 & 0 & 7 \\ 1 & 0 & -2 & 0 \\ 3 & 7 & 0 & 3 \end{pmatrix}_{b_0}$

[741] $1^1 41^1 4141^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 33, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 5 & 5 & 5 \\ 4 & 5 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$

[743] $1^1 41^1 32151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 5 & 1 & 3 \\ 5 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 3 & 6 & 0 & 3 \end{pmatrix}_{b_0}$

[746] $1^1 6^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 4 & 1 & 18 & 8 \\ 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$

[744] $1^1 41^1 3151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 4 & 0 & 10 & 6 \\ 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$

[745] $41^1 23151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 1 & 9 & 1 & 3 \\ 9 & 51 & 0 & 15 \\ 1 & 0 & -3 & -1 \\ 3 & 15 & -1 & 4 \end{pmatrix}_{b_0}$

[747] $1^1 161232 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.30 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 2 & 5 & 4 \\ 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$

[748] $231^1 3161^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.25 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 1 & 2 & 8 & 3 \\ 2 & 1 & 7 & 3 \\ 8 & 7 & 21 & 9 \\ 3 & 3 & 9 & 3 \end{pmatrix}_{b_0}$

[750] $1^1 41321^1 51^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.16 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 5 & 1 & 3 \\ 5 & 3 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 3 & 5 & 0 & 3 \end{pmatrix}_{b_0}$

[749] $1^1 6^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 1 & 7 & 4 & 3 \\ 7 & 19 & 8 & 9 \\ 4 & 8 & 1 & 3 \\ 3 & 9 & 3 & 3 \end{pmatrix}_{b_0}$

[752] $31^1 41^1 3151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.41 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$
 $\begin{pmatrix} 0 & 5 & 1 & 3 & 2 \\ 5 & 8 & 1 & 6 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 \\ 2 & 6 & 0 & 3 & 2 \end{pmatrix}_{b_0}$

[751] $41^1 31^1 4141^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 4 & 1 & 5 & 2 \\ 4 & 26 & 1 & 32 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 32 & 0 & 37 & 11 \\ 2 & 10 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[754] $1^{\overline{1}3221^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 7 & 1 & 3 \\ 7 & 19 & 1 & 9 \\ 1 & 1 & -2 & 0 \\ 3 & 9 & 0 & 4 \end{pmatrix}_{b_0}$$

[753] $1^1 41321^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 6 & 1 & 3 \\ 6 & 6 & 0 & 6 \\ 1 & 0 & -2 & 0 \\ 3 & 6 & 0 & 4 \end{pmatrix}_{b_0}$$

[755] $41^1 31^1 4141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 57, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.30 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 5 & 1 & 4 & 2 \\ 5 & 38 & 1 & 32 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 32 & 0 & 25 & 9 \\ 2 & 12 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[756] $1^{\overline{2}321^1} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 71, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.07 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 7 & 3 & 3 \\ 7 & 19 & 8 & 9 \\ 3 & 8 & 1 & 3 \\ 3 & 9 & 3 & 3 \end{pmatrix}_{b_0}$$

[757] $1^1 216131^1 13 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 89, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 2 & 5 & 4 \\ 4 & 5 & 1 & 3 \\ 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

[758] $1^1 6131^1 13 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 89, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 6 & 6 & 3 \\ 6 & 22 & 25 & 10 \\ 6 & 25 & 21 & 9 \\ 3 & 10 & 9 & 3 \end{pmatrix}_{b_0}$$

[759] $41^1 1^1 23151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 111, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 9 & 9 & 3 \\ 9 & 51 & 48 & 15 \\ 9 & 48 & 41 & 13 \\ 3 & 15 & 13 & 4 \end{pmatrix}_{b_0}$$

[760] $41^1 1^1 232151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 14, \Delta = -90, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.32 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 4 & 28 & 0 & 25 & 9 \\ 2 & 10 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[761] $41^1 1^1 23151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.32 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 9 & 8 & 3 \\ 9 & 51 & 48 & 15 \\ 8 & 48 & 41 & 13 \\ 3 & 15 & 13 & 3 \end{pmatrix}_{b_0}$$

$T = 11$ [765] $1^1 3151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 12, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 2 & 0 & 4 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 4 & 2 & -1 & 5 \end{pmatrix}_{b_0}$$

[762] $1^1 3151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 12, \Delta = -39, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & 0 & 2 \\ 2 & 0 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

[763] $1^{\overline{1}} 4 14 13 21^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 12, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 3 & 4 & 2 & 6 \end{pmatrix}_{b_0}$

[764] $1^{\overline{1}} 5 13 22 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 3 & 5 \\ 3 & 3 & 5 & 6 \end{pmatrix}_{b_0}$

[766] $1^{\overline{1}} 3 15 13 21^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 12, \Delta = -39, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 2 & 0 & -3 & -1 \\ 1 & 0 & 0 & -1 & 5 \end{pmatrix}_{b_0}$

[768] $1^{\overline{1}} 6 13 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 13, \Delta = -65, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$

[769] $2 1^{\overline{1}} 6 13 14 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 4 & 0 & 7 & 5 \\ 2 & 3 & -1 & 5 & 3 \end{pmatrix}_{b_0}$

[770] $1^{\overline{1}} 6 13 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 13, \Delta = -64, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[767] $1^{\overline{1}} 6 13 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 13, \Delta = -65, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[772] $1^{\overline{1}} 6 13 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 13, \Delta = -64, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 5 \end{pmatrix}_{b_0}$

[773] $1^{\overline{1}} 6 13 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 13, \Delta = -65, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 3 & 4 & 0 & 5 \end{pmatrix}_{b_0}$

[774] $1^{\overline{1}} 6 12 32 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 2 & 4 \\ 3 & 3 & 4 & 5 \end{pmatrix}_{b_0}$

[771] $1^{\overline{1}} 6 13 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 13, \Delta = -65, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 2 & 3 & 4 \\ 3 & 2 & 4 \\ 4 & 4 & 6 \end{pmatrix}_{b_0}$

[775] $2 1^{\overline{1}} 5 13 15 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 4 & 5 & 3 & 4 \\ 5 & 5 & 4 & 5 \\ 3 & 4 & 1 & 3 \\ 4 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

[776] $1^1 41414141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 4 & 4 \\ 4 & 4 & 2 & 4 \\ 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$

[777] $1^1 5132151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$

[778] $21^1 61^1 23141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -85, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 4 & 12 & 3 & 4 \\ 12 & 34 & 10 & 12 \\ 3 & 10 & 1 & 3 \\ 4 & 12 & 3 & 4 \end{pmatrix}_{b_0}$

[779] $1^1 61^1 23141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -85, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 & 2 \\ 4 & 26 & 1 & 14 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 14 & 0 & 7 & 5 \\ 2 & 10 & -1 & 5 & 3 \end{pmatrix}_{b_0}$

[780] $1^1 513151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 4 & 4 \\ 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$

[783] $1^1 513151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 4 & 5 \\ 3 & 4 & 1 & 3 \\ 3 & 5 & 3 & 4 \end{pmatrix}_{b_0}$

[781] $21^1 513151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$

[784] $1^1 5132151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 \end{pmatrix}_{b_0}$

[782] $1^1 513151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 \end{pmatrix}_{b_0}$

[786] $1^1 6123141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 11, T_{\max} = 14, \Delta = -85, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$

[787] $1^1 513221^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 2 & 6 & 4 \\ 3 & 2 & 6 & 6 & 6 \\ 2 & 2 & 4 & 6 & 5 \end{pmatrix}_{b_0}$

[788] $1\overset{1}{5}13221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 7 & 10 & 7 \\ 4 & 10 & 10 & 8 \\ 3 & 7 & 8 & 6 \end{pmatrix}_{b_0}$

[789] $1\overset{1}{4}141321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 12, \Delta = -35, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 2 \\ 2 & 2 & 2 & 2 & 4 \\ 2 & 2 & 2 & 4 & 5 \end{pmatrix}_{b_0}$

[790] $21\overset{1}{5}13221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 2 & 6 & 4 \\ 3 & 2 & 6 & 6 & 6 \\ 2 & 2 & 4 & 6 & 5 \end{pmatrix}_{b_0}$

[785] $1\overset{1}{5}132151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$

[791] $1\overset{1}{5}13151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.50 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 5 & 4 & 5 \\ 2 & 4 & 1 & 3 \\ 3 & 5 & 3 & 4 \end{pmatrix}_{b_0}$

[792] $21\overset{1}{6}12321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 1 & 4 & 3 \\ 5 & 4 & 9 & 7 \\ 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$

[793] $1\overset{1}{6}12321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 2 & 5 & 5 & 5 \\ 2 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$

[794] $1\overset{1}{5}1321^{11}51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 9 & 3 & 3 \\ 9 & 51 & 12 & 15 \\ 3 & 12 & 1 & 3 \\ 3 & 15 & 3 & 4 \end{pmatrix}_{b_0}$

[795] $21\overset{1}{6}13141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 8 & 1 & 4 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 4 & 0 & 0 & 2 \\ 2 & 6 & -1 & 2 & 3 \end{pmatrix}_{b_0}$

[796] $21\overset{1}{6}131^{11}41^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 8 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 8 & 0 & 35 & 11 \\ 2 & 3 & -1 & 11 & 3 \end{pmatrix}_{b_0}$

[797] $1\overset{1}{5}13221^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 7 & 17 & 7 \\ 7 & 17 & 32 & 14 \\ 3 & 7 & 14 & 6 \end{pmatrix}_{b_0}$

[798] $21\overset{1}{5}13221^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 12, \Delta = -31, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 \\ 3 & 7 & 17 & 7 \\ 7 & 17 & 32 & 14 \\ 3 & 7 & 14 & 6 \end{pmatrix}_{b_0}$$

$$[799] \ 151^{11}23151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -39, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 9 & 1 & 3 \\ 9 & 51 & 0 & 15 \\ 1 & 0 & -3 & -1 \\ 3 & 15 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[800] \ 1^{11}4141^{11}3151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -86, \det = -40, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 6 & 4 & 0 & 13 & 7 \\ 2 & 2 & 0 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[801] \ 51^{11}223151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 7 & 2 & 3 \\ 7 & 7 & 3 & 7 \\ 2 & 3 & 0 & 2 \\ 3 & 7 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[802] \ 161^{11}23141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -85, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 26 & 1 & 7 & 10 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 7 & 0 & 0 & 2 \\ 2 & 10 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[803] \ 151321^{11}51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 9 & 2 & 3 \\ 9 & 51 & 12 & 15 \\ 2 & 12 & 1 & 3 \\ 3 & 15 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[804] \ 1^{11}512321^{11}51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -85, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 18 & 6 & 0 & 8 \\ 1 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 8 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[805] \ 41^1414141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -70, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 5 & 4 & 3 \\ 5 & 6 & 6 & 6 \\ 4 & 6 & 2 & 4 \\ 3 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[806] \ 1^{11}4141^{11}3151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -86, \det = -75, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 7 & 1 & 3 \\ 7 & 12 & 1 & 8 \\ 1 & 1 & -2 & 0 \\ 3 & 8 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[807] \ 1^{11}413221^{11}61^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -85, \det = -77, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 5 & 1 & 1 & 2 \\ 5 & 28 & 7 & 0 & 10 \\ 1 & 7 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 10 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[808] \ 1^{11}4141^{11}3151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -86, \det = -77, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 8 & 1 & 3 \\ 8 & 17 & 0 & 9 \\ 1 & 0 & -2 & 0 \\ 3 & 9 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[809] \ 41^1414141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 3 & 6 & 5 \\ 4 & 6 & 2 & 4 \\ 3 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [810] \ 1^{11}231513221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 12, \Delta = -11, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$$

[813] $1^{\overline{1}}41414131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 12, T_{\max} = 13, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 3 & 0 & 1 & 3 \\ 2 & 4 & 0 & 3 & 5 \end{pmatrix}_{b_0}$

[814] $1^{\overline{1}}6^{12321}131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 12, T_{\max} = 13, \Delta = -38, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 3 & 2 & 3 & 6 \end{pmatrix}_{b_0}$

[815] $1^{\overline{1}}31^{1131}6^{131}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 12, T_{\max} = 13, \Delta = -43, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$

[816] $1^{\overline{1}}6^{12321}131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 12, T_{\max} = 13, \Delta = -38, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$

[817] $1^{\overline{1}}5^{1315131}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 12, T_{\max} = 14, \Delta = -61, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 6 & 0 & 6 \\ 1 & 0 & -3 & -1 \\ 3 & 6 & -1 & 5 \end{pmatrix}_{b_0}$

[818] $1^{\overline{1}}6^{12321}131^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 12, T_{\max} = 13, \Delta = -38, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 2 & 3 & 4 \\ 3 & 3 & 5 \\ 4 & 5 & 7 \end{pmatrix}_{b_0}$

[819] $1^{\overline{1}}5^{13161}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 12, T_{\max} = 15, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 4 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$

[821] $1^{\overline{1}}7^{1321}1231^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 12, T_{\max} = 14, \Delta = -64, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$

[822] $21^{\overline{1}}6^{13151}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 12, T_{\max} = 15, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 3 & 3 \\ 2 & 2 & 0 & 3 & 3 \end{pmatrix}_{b_0}$

[823] $1^{\overline{1}}4^{13221}1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 12, T_{\max} = 12, \Delta = -6, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 2 & 2 \\ 2 & 3 & 2 & 3 & 5 \\ 2 & 3 & 2 & 5 & 6 \end{pmatrix}_{b_0}$

[824] $1^{\overline{1}}6^{131}13141^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 12, T_{\max} = 14, \Delta = -63, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 0 & 3 \\ 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$

[820] $1^{\overline{1}}6^{131}13141^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 12, T_{\max} = 14, \Delta = -63, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 5 & 3 \\ 4 & 5 & 11 & 7 \\ 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 6 & 6 \\ 6 & 6 \end{pmatrix}_{b_0}$$

$$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 0 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 3 & 2 & 3 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix} b_c$$

$$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 2 & 3 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix} b_c$$

$$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 1 & 0 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix} b_0$$

$$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 4 & 4 \\ 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 10 & 1 & 0 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 6 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

[847] $1\overset{1}{6}1\overset{1}{3}1\overset{1}{5}1\overset{1}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[853] $1\overset{1}{4}1414131\overset{1}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 12, T_{\max} = 13, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 2 & 4 & 4 \\ 3 & 4 & 3 & 5 \\ 3 & 4 & 5 & 6 \end{pmatrix}_{b_0}$$

[849] $1\overset{1}{5}1\overset{1}{3}2161\overset{1}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.50 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[850] $1\overset{1}{5}1\overset{1}{3}161\overset{1}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.50 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[851] $1\overset{1}{6}1\overset{1}{3}14141\overset{1}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -83, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.50 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[852] $1\overset{1}{6}1\overset{1}{2}32151\overset{1}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.50 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[855] $1\overset{1}{5}1\overset{1}{3}15131\overset{1}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 12, T_{\max} = 14, \Delta = -61, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 6 & 6 & 6 \\ 3 & 6 & 2 & 4 \\ 3 & 6 & 4 & 5 \end{pmatrix}_{b_0}$$

[854] $1\overset{1}{6}1\overset{1}{2}231\overset{1}{5}1\overset{1}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 5 & 1 & 2 & 2 \\ 5 & 18 & 1 & 4 & 8 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 0 & 2 \\ 2 & 8 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[856] $1\overset{1}{5}1\overset{1}{3}21\overset{1}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 4 & 1 & 2 & 2 \\ 4 & 10 & 1 & 3 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 2 \\ 2 & 6 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[857] $31\overset{1}{5}1\overset{1}{3}161\overset{1}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -87, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[858] $21\overset{1}{6}1\overset{1}{3}1\overset{1}{5}1\overset{1}{1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 12, T_{\max} = 15, \Delta = -79, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 8 & 1 & 6 \\ 1 & 1 & -2 & 0 \\ 3 & 6 & 0 & 4 \end{pmatrix}_{b_0}$$

[859] $1^1 61^1 2231^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 20, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.80 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 6 & 4 & 0 & 18 & 8 \\ 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[860] $31^1 613151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -87, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.83 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 2 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

[862] $2161^1 231^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 39, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.92 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 10 & 1 & 0 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 6 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

[861] $1^1 5^{11} 1321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 39, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.92 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 10 & 1 & 0 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 6 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

[863] $161^1 2231^1 51^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.77 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 5 & 1 & 1 & 2 \\ 5 & 18 & 1 & 0 & 8 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 8 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

[864] $1^1 5^{11} 1321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 7 & 1 & 3 \\ 7 & 19 & 1 & 9 \\ 1 & 1 & -2 & 0 \\ 3 & 9 & 0 & 4 \end{pmatrix}_{b_0}$$

[865] $1^1 5^{11} 1321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 7 & 1 & 3 \\ 7 & 19 & 1 & 9 \\ 1 & 1 & -2 & 0 \\ 3 & 9 & 0 & 4 \end{pmatrix}_{b_0}$$

[866] $1^1 513221^1 61^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 77, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 9 & 1 & 3 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 3 & 12 & 0 & 4 \end{pmatrix}_{b_0}$$

[869] $1^1 4141321^1 61^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -84, \det = 77, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 5 & 1 & 1 & 2 \\ 5 & 28 & 7 & 0 & 10 \\ 1 & 7 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 10 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[868] $1^1 6131^1 4141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -83, \det = 77, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 5 & 1 & 1 & 2 \\ 5 & 38 & 1 & 8 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 8 & 0 & 0 & 2 \\ 2 & 12 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[867] $1^1 4141^1 32161^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 15, \Delta = -84, \det = 77, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 5 & 1 & 1 & 2 \\ 5 & 33 & 8 & 0 & 11 \\ 1 & 8 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 11 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[871] $31^1 51321^1 61^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 87, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 14.97 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 1 & 9 & 3 & 3 \\ 9 & 34 & 10 & 12 \\ 3 & 10 & 1 & 3 \\ 3 & 12 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[870] \ 321^{\overset{1}{1}}613151^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 87, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.97 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 5 & 1 & 2 & 2 \\ 5 & 13 & 1 & 5 & 7 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 0 & 2 \\ 2 & 7 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[872] \ 31^{\overset{1}{1}51321^{\overset{1}{1}}}6^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 12, T_{\max} = 16, \Delta = -88, \det = 168, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 0 & 7 & 1 & 2 & 2 \\ 7 & 19 & 1 & 6 & 9 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 6 & 0 & 0 & 2 \\ 2 & 9 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [873] \ 1^{\overset{1}{1}}4141413221^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 13, \Delta = -5, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 5.0$$

$$[875] \ 1^{\overset{1}{1}}513151321^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 2 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 4 & -1 & 5 \end{pmatrix}_{b_0}$$

$$[874] \ 1^{\overset{1}{1}}7^{\overset{1}{1}2321}1231^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 14, \Delta = -37, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[881] \ 1^{\overset{1}{1}}6^{\overset{1}{1}5131}131^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 3 \\ 2 & 1 & 0 & 2 \\ 3 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[880] \ 21^{\overset{1}{1}}51316131^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 0 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[879] \ 1^{\overset{1}{1}}321613151^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[878] \ 1^{\overset{1}{1}}6^{\overset{1}{1}5131}131^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[877] \ 1^{\overset{1}{1}}6^{\overset{1}{1}5131}131^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[876] \ 1^{\overset{1}{1}}316123151^{\overset{1}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

[887] $1^{\overline{1}1^13221} 8^{1^1} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -63, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

[882] $15^{\overline{1}1} 1316^{1^1} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[883] $1^{\overline{1}1^13221} 8^{1^1} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -63, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[886] $1^{\overline{1}1^1321} 8^{1^1} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -63, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

[885] $15^{\overline{1}1} 1316^{1^1} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

[884] $1^{\overline{1}1^1} 4^{1^1} 14141321^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 13, \Delta = -5, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 3 & 5 & 6 \end{pmatrix}_{b_0}$$

[894] $1^{\overline{1}1^1} 31612315^{1^1} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 0 & 3 \\ 1 & 0 & -2 & 0 \\ 3 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

[893] $1^{\overline{1}1^1} 32161315^{1^1} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 0 & 3 \\ 1 & 0 & -2 & 0 \\ 3 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

[892] $15^{\overline{1}1} 1316^{1^1} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 1 & 4 \\ 1 & 1 & -2 & 0 \\ 3 & 4 & 0 & 5 \end{pmatrix}_{b_0}$$

[891] $15^{\overline{1}1} 1316^{1^1} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 2 & 3 & 4 \\ 3 & 2 & 4 \\ 4 & 4 & 6 \end{pmatrix}_{b_0}$$

[890] $16^{1^1} 1315131^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 2 & 4 \\ 3 & 3 & 4 & 5 \end{pmatrix}_{b_0}$$

[889] $1^{\overline{1}} 6^{11} 12315131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 2 & 3 & 4 \\ 3 & 2 & 4 \\ 4 & 4 & 6 \end{pmatrix}_{b_0}$

[888] $1^{\overline{1}} 5^{11} 131^{\overline{1}} 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.60 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 0 & 3 \\ 1 & 0 & -2 & 0 \\ 3 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[895] $1^{\overline{1}} 5^{11} 13151321^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 7 & 7 & 7 \\ 3 & 7 & 3 & 5 \\ 3 & 7 & 5 & 6 \end{pmatrix}_{b_0}$

[897] $1^{\overline{1}} 5^{11} 131^{\overline{1}} 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$

[900] $1^{\overline{1}} 1^{\overline{1}} 3216131^{\overline{1}} 51^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$

[896] $1^{\overline{1}} 1^{\overline{1}} 3161231^{\overline{1}} 51^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$

[898] $1^{\overline{1}} 6^{11} 1315131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 1 & 5 & 3 \\ 3 & 2 & 5 & 5 & 5 \\ 2 & 2 & 3 & 5 & 4 \end{pmatrix}_{b_0}$

[899] $21^{\overline{1}} 6^{11} 1315131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -12, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 1 & 4 & 3 \\ 5 & 4 & 9 & 7 \\ 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$

[901] $1^{\overline{1}} 5^{11} 131^{\overline{1}} 131^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -14, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 14.86 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 4 & 0 & 5 & 5 \\ 2 & 3 & 0 & 5 & 4 \end{pmatrix}_{b_0}$

[902] $41^{\overline{1}} 1^{\overline{1}} 31613151^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 16, \Delta = -86, \det = -44, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15.91 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 & 2 \\ 4 & 10 & 1 & 4 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 0 & 2 \\ 2 & 6 & 0 & 2 & 3 \end{pmatrix}_{b_0}$

[903] $1^{\overline{1}} 414141^{\overline{1}} 3161^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 16, \Delta = -82, \det = -137, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15.77 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$
 $\begin{pmatrix} 0 & 7 & 1 & 1 & 2 \\ 7 & 61 & 10 & 0 & 15 \\ 1 & 10 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 15 & 2 & -1 & 3 \end{pmatrix}_{b_0}$

$$\begin{aligned}
& T = 14 \quad [904] \quad 1^1 41414141321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 14, T_{\max} = 14, \Delta = -4, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 5.0 \\
& [905] \quad 1^1 6123151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 14, T_{\max} = 15, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \quad \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0} \\
& [906] \quad 1^1 1613151321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 14, T_{\max} = 15, \Delta = -30, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \quad \begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 \\ 2 & 3 & 3 & 5 \\ 3 & 3 & 5 & 6 \end{pmatrix}_{b_0} \\
& [907] \quad 1^1 8^{13221} 12321^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 14, T_{\max} = 15, \Delta = -36, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
& [908] \quad 1^1 6^{1315131} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 14, T_{\max} = 15, \Delta = -37, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \quad \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0} \\
& [909] \quad 1^1 8^{12321} 1231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 14, T_{\max} = 15, \Delta = -36, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 5 \end{pmatrix}_{b_0} \\
& [910] \quad 1^1 6^{1315131} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 14, T_{\max} = 15, \Delta = -37, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \quad \begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 1 & 3 \\ 3 & 2 & 3 & 6 \end{pmatrix}_{b_0} \\
& [911] \quad 1^1 2321613151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \quad \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
& [913] \quad 1^1 6^{1315131} 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 14, T_{\max} = 15, \Delta = -37, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \quad \begin{pmatrix} 2 & 3 & 4 \\ 3 & 3 & 5 \\ 4 & 5 & 7 \end{pmatrix}_{b_0} \\
& [912] \quad 1^1 3217123151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0} \\
& [915] \quad 1^1 4141414131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 14, T_{\max} = 14, \Delta = -3, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \quad \begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 3 & 5 & 5 \\ 3 & 5 & 4 & 6 \\ 3 & 5 & 6 & 7 \end{pmatrix}_{b_0} \\
& [914] \quad 1^1 5131513221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 14, T_{\max} = 14, \Delta = -4, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \quad \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 8 & 8 & 8 \\ 3 & 8 & 4 & 6 \\ 3 & 8 & 6 & 7 \end{pmatrix}_{b_0} \\
& [916] \quad 1^1 8^{1321} 123141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 14, T_{\max} = 16, \Delta = -62, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0
\end{aligned}$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 2 & 0 & 1 & 3 \\ 2 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[917] \ 1^{\overline{1}1321} \ 8^{\overline{1}} \ 123141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 16, \Delta = -62, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 0 & 3 \\ 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[918] \ 1^{\overline{1}1} \ 6^{\overline{1}} \ 123216131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 5, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15.60 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 2 & 3 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[919] \ 1^{\overline{1}1} \ 321712315^{\overline{1}} \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 5, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15.60 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 0 & 3 \\ 1 & 0 & -2 & 0 \\ 3 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[920] \ 1^{\overline{1}15131} \ 6^{\overline{1}} \ 13151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 17, \Delta = -81, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[921] \ 21^{\overline{1}1} \ 6^{\overline{1}} \ 13151321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 15, \Delta = -30, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 2 & 3 & 6 & 6 \\ 2 & 2 & 2 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[922] \ 1^{\overline{1}1} \ 232161315^{\overline{1}} \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[923] \ 1^{\overline{1}1} \ 6^{\overline{1}} \ 13151321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 15, \Delta = -30, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 2 & 6 & 4 \\ 3 & 2 & 6 & 6 & 6 \\ 2 & 2 & 4 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[924] \ 1^{\overline{1}1} \ 6^{\overline{1}} \ 131513151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 17, \Delta = -81, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 16.50 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[925] \ 1^{\overline{1}1} \ 321712315^{\overline{1}} \ 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 \\ 2 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[926] \ 31^{\overline{1}131} \ 6^{\overline{1}} \ 13161^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 16, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 2 & 3 & 5 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 5 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[927] \ 1^{\overline{1}15131} \ 6^{\overline{1}} \ 13151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 17, \Delta = -81, \det = 17, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 16.59 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 5 & 0 & 5 \\ 1 & 0 & -2 & 0 \\ 3 & 5 & 0 & 4 \end{pmatrix}_{b_0}$$

[928] $1^1 6^1 1^1 231513151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 17, \Delta = -82, \det = 77, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16.91 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 1 & 9 & 1 & 3 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 3 & 12 & 0 & 4 \end{pmatrix}_{b_0}$$

[929] $1^1 51321^1 613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 17, \Delta = -82, \det = 135, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16.80 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 1 & 12 & 1 & 3 \\ 12 & 51 & 0 & 15 \\ 1 & 0 & -2 & 0 \\ 3 & 15 & 0 & 4 \end{pmatrix}_{b_0}$$

[930] $1^1 51321^1 613151^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 14, T_{\max} = 17, \Delta = -82, \det = 137, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16.77 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 1 & 11 & 1 & 3 \\ 11 & 42 & 1 & 14 \\ 1 & 1 & -2 & 0 \\ 3 & 14 & 0 & 4 \end{pmatrix}_{b_0}$$

$T = 15$ [931] $1^1 414141414131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 15, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 5.0$

[933] $1^{131} 6^{11} 1316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 16, \Delta = -34, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 1 & 0 & 0 & -3 \\ 2 & 4 & 2 & -1 & 5 \end{pmatrix}_{b_0}$$

[935] $1^{131} 6^{11} 13161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 16, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 3 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

[934] $1^1 232171231^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 16, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

[936] $2^1 32181231^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 17, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 \\ 2 & 5 & 1 & 0 \\ 1 & 1 & -2 & 0 \\ 1 & 0 & 0 & -2 \\ 2 & 5 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

[937] $1^{131} 6^{11} 1316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 16, \Delta = -34, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15.67 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 \\ 2 & 4 & 2 & 2 & 5 \end{pmatrix}_{b_0}$$

[938] $1^{131} 6^{11} 13161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 16, \Delta = -35, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15.67 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 0 \\ 1 & 2 & 0 & 2 \\ 2 & 4 & 0 & 2 & 5 \end{pmatrix}_{b_0}$$

[939] $2^1 32181231^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 17, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 \\ 3 & 5 & 0 & 5 \\ 1 & 0 & -2 & 0 \\ 3 & 5 & 0 & 5 \end{pmatrix}_{b_0}$$

[940] $1^{1512321} 8^{11} 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 17, \Delta = -59, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

[941] $1^{\overline{1}} \overline{1}^1 6 131513221^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 3 & 7 & 5 \\ 3 & 2 & 7 & 7 & 7 \\ 2 & 2 & 5 & 7 & 6 \end{pmatrix}_{b_0}$$

[942] $21^{\overline{1}} \overline{1}^1 6 131513221^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 3 & 2 & 4 & 7 & 7 \\ 2 & 2 & 3 & 7 & 6 \end{pmatrix}_{b_0}$$

[943] $1^{\overline{1}} \overline{1}^1 512321 \quad 8 \quad 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 17, \Delta = -59, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[944] $1^{\overline{1}1} 232171231 \overline{5} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 16, \Delta = -31, \det = -8, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

[945] $21^{\overline{1}} \overline{1}^1 6 131612321^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 16, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 2 & 6 & 4 \\ 5 & 6 & 13 & 9 \\ 3 & 4 & 9 & 6 \end{pmatrix}_{b_0}$$

[946] $1^{\overline{1}} \overline{1}^1 6 1316123151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 18, \Delta = -79, \det = -8, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17.50 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[947] $2 \quad 3 \quad 2181231 \overline{5} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 17, \Delta = -59, \det = -27, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 5 & 1 & 7 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 7 & 0 & 5 & 5 \\ 2 & 5 & 0 & 5 & 4 \end{pmatrix}_{b_0}$$

[948] $1^{\overline{1}} \overline{1}^1 61^{\overline{1}} 2321613151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 18, \Delta = -81, \det = -77, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17.91 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$

$$\begin{pmatrix} 1 & 9 & 1 & 3 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 3 & 12 & 0 & 4 \end{pmatrix}_{b_0}$$

[949] $16 13161^{\overline{1}} 23151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 18, \Delta = -79, \det = -135, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17.80 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$

$$\begin{pmatrix} 1 & 10 & 1 & 3 \\ 10 & 28 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 3 & 12 & 0 & 4 \end{pmatrix}_{b_0}$$

[950] $1^{\overline{1}} \overline{1}^1 612321^{\overline{1}} 613151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 15, T_{\max} = 18, \Delta = -81, \det = -137, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 17.77 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 1 & 11 & 1 & 3 \\ 11 & 42 & 1 & 14 \\ 1 & 1 & -2 & 0 \\ 3 & 14 & 0 & 4 \end{pmatrix}_{b_0}$$

$T = 16$ [953] $1^{\overline{1}} \overline{1}^1 3151316131 \overline{5} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 17, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

[957] $1^{\overline{1}} 6^{11} 12321712321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 17, \Delta = -30, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[955] $1^{\overline{1}} 2321812231^{\overline{1}} 51^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 17, \Delta = -30, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[952] $1^{\overline{1}} \textcolor{blue}{3} 151321^{\overline{1}} 8^{11} 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 17, \Delta = -36, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & 0 & 2 \\ 3 & 3 & 2 & 6 \end{pmatrix}_{b_0}$

[954] $1^{\overline{1}} \textcolor{blue}{6}^{11} 131613141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 17, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 3 & 4 \\ 1 & 0 & 3 & 1 & 3 \\ 2 & 0 & 4 & 3 & 5 \end{pmatrix}_{b_0}$

[956] $1^{\overline{1}} \textcolor{blue}{5}^{11} 1321^{\overline{1}} 8^{11} 12321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 17, \Delta = -30, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[959] $1^{\overline{1}} \textcolor{blue}{6}^{11} 131613141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 17, \Delta = -33, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 16.67 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 3 & 1 & 3 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 1 & 3 \\ 2 & 5 & -1 & 3 & 5 \end{pmatrix}_{b_0}$

[958] $2321812321^{\overline{1}} 61^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 18, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 3 & 2 & 2 & 4 \end{pmatrix}_{b_0}$

[962] $1^{\overline{1}} 2321812231^{\overline{1}} 51^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 17, \Delta = -30, \det = 8, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$

[961] $1^{\overline{1}} \textcolor{blue}{5}^{11} 1321^{\overline{1}} 8^{11} 12321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 17, \Delta = -30, \det = 8, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$

[960] $1^{\overline{1}} \textcolor{blue}{3} 151316131^{\overline{1}} 51^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 17, \Delta = -31, \det = 8, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$

[963] $1^{\overline{1}} \textcolor{blue}{8}^{11} 123151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 \quad T = 16, T_{\max} = 19, \Delta = -80, \det = 17, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 18.59 \rightarrow 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 5 & 0 & 5 \\ 1 & 0 & -2 & 0 \\ 3 & 5 & 0 & 4 \end{pmatrix}_{b_0}$

[964] $1^1 61^1 23217123151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 16, T_{\max} = 19, \Delta = -80, \det = 77, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 18.91 \rightarrow 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 1 & 9 & 1 & 3 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 3 & 12 & 0 & 4 \end{pmatrix}_{b_0}$

[965] $1^1 513221^1 8123151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 16, T_{\max} = 19, \Delta = -80, \det = 213, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 18.70 \rightarrow 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 1 & 15 & 1 & 3 \\ 15 & 79 & 0 & 19 \\ 1 & 0 & -2 & 0 \\ 3 & 19 & 0 & 4 \end{pmatrix}_{b_0}$

[966] $1^1 513221^1 8123151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 16, T_{\max} = 19, \Delta = -80, \det = 215, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 18.69 \rightarrow 19, T_{\max} = 19, T \in [19, 19]$
 $\begin{pmatrix} 1 & 14 & 1 & 3 \\ 14 & 68 & 1 & 18 \\ 1 & 1 & -2 & 0 \\ 3 & 18 & 0 & 4 \end{pmatrix}_{b_0}$

$T = 17$ [967] $1^1 223151316131^1 51^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 17, T_{\max} = 17, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 5.0$

[968] $1^1 32161316131^1 51^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 17, T_{\max} = 18, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$

[969] $1^1 2321^1 1232161^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 17, T_{\max} = 18, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 3 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[970] $1^1 23151316131^1 51^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 17, T_{\max} = 17, \Delta = -3, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0}$

[971] $1^1 32161316131^1 51^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 17, T_{\max} = 18, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$

[972] $1^1 51321^1 81232161^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 17, T_{\max} = 20, \Delta = -79, \det = -17, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 19.59 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 5 & 4 & 0 & 5 \\ 1 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 5 & 2 & -1 & 3 \end{pmatrix}_{b_0}$

[973] $1^1 61^1 232181223151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 17, T_{\max} = 20, \Delta = -79, \det = -77, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 19.91 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 1 & 9 & 1 & 3 \\ 9 & 34 & 1 & 12 \\ 1 & 1 & -2 & 0 \\ 3 & 12 & 0 & 4 \end{pmatrix}_{b_0}$

[974] $1^1 51321^1 81232161^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 17, T_{\max} = 20, \Delta = -79, \det = -77, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 19.91 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 0 & 5 & 1 & 1 & 2 \\ 5 & 28 & 7 & 0 & 10 \\ 1 & 7 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 10 & 2 & -1 & 3 \end{pmatrix}_{b_0}$

[975] $1^1 61232181^1 223151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3$ $T = 17, T_{\max} = 20, \Delta = -79, \det = -213, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 19.70 \rightarrow 20, T_{\max} = 20, T \in [20, 20]$
 $\begin{pmatrix} 1 & 13 & 1 & 3 \\ 13 & 50 & 1 & 16 \\ 1 & 1 & -2 & 0 \\ 3 & 16 & 0 & 4 \end{pmatrix}_{b_0}$

$$\begin{aligned}
& T = 18 \quad [976] \quad 21^1 6 1316131513221^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 18, \quad T_{\max} = 18, \quad \Delta = 0, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 18, 1), \quad c_{\text{ext}} = 5.0 \\
& [978] \quad 1^1 3221^{11} (12) 12231^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 18, \quad T_{\max} = 19, \quad \Delta = -42, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 19, \quad T_{\max} = 19, \quad T \in [19, 19] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0} \\
& [979] \quad 1^1 315132181231^1 51^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 18, \quad T_{\max} = 19, \quad \Delta = -30, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 19, \quad T_{\max} = 19, \quad T \in [19, 19] \\
& \quad \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
& [980] \quad 21^1 6 131613151321^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 18, \quad T_{\max} = 18, \quad \Delta = 0, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 18, \quad T_{\max} = 18, \quad T \in [18, 18] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & 0 & 3 & 2 \\ 3 & 4 & 3 & 10 & 8 \\ 2 & 3 & 2 & 8 & 6 \end{pmatrix}_{b_0} \\
& [981] \quad 1^1 1^{11} 232161316131^1 51^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 18, \quad T_{\max} = 18, \quad \Delta = -2, \quad \det = 4, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 18, \quad T_{\max} = 18, \quad T \in [18, 18] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0} \\
& [982] \quad 1^1 1^{11} 315132181231^1 51^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 18, \quad T_{\max} = 19, \quad \Delta = -30, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 18, 0), \quad c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 19, \quad T_{\max} = 19, \quad T \in [19, 19] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0} \\
& T = 19 \quad [983] \quad 1^1 1^{11} 22315132181231^1 51^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 19, \quad T_{\max} = 19, \quad \Delta = -2, \quad \det = 0, \quad (n_+, n_-, n_0) = (1, 19, 1), \quad c_{\text{ext}} = 5.0 \\
& [986] \quad 1^1 3161232181231^1 51^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 19, \quad T_{\max} = 20, \quad \Delta = -29, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 20, \quad T_{\max} = 20, \quad T \in [20, 20] \\
& \quad \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0} \\
& [987] \quad 1^1 6 12321812315131^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 19, \quad T_{\max} = 20, \quad \Delta = -29, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 20, \quad T_{\max} = 20, \quad T \in [20, 20] \\
& \quad \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 3 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
& [988] \quad 1^1 6 13151321812321^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 19, \quad T_{\max} = 20, \quad \Delta = -29, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 20, \quad T_{\max} = 20, \quad T \in [20, 20] \\
& \quad \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 4 \\ 1 & 2 & 0 & 2 \\ 3 & 4 & 2 & 6 \end{pmatrix}_{b_0} \\
& [989] \quad 1^1 1^{11} 2315132181231^1 51^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 19, \quad T_{\max} = 19, \quad \Delta = -2, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 5.0 \\
& \quad T_{\min} = 19, \quad T_{\max} = 19, \quad T \in [19, 19] \\
& \quad \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0} \\
& [990] \quad 1^1 1^{11} 3161232181231^1 51^1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 \quad T = 19, \quad T_{\max} = 20, \quad \Delta = -29, \quad \det = -8, \quad (n_+, n_-, n_0) = (1, 19, 0), \quad c_{\text{ext}} = 5.0
\end{aligned}$$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$T = 20 \quad [991] \quad 1^{1\bar{1}} 231513161316131 \bar{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 20, T_{\max} = 20, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 5.0$$

$$[993] \quad 1^{1\bar{1}} 31513161316131 \bar{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 20, T_{\max} = 20, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 20, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 20, T_{\max} = 20, T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0}$$

$$T = 21 \quad [994] \quad 1^{1\bar{1}} 2321613161316131 \bar{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 21, T_{\max} = 21, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 5.0$$

$$[996] \quad 1^{31} \bar{5}^{13221} (12) \quad 12231 \bar{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 21, T_{\max} = 23, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 3 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[995] \quad 231 \bar{5}^{13221} (12) 12231 \bar{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 21, T_{\max} = 23, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 5 & 3 & 2 & 5 \\ 1 & 3 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 5 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[997] \quad 1^{1\bar{1}} 232171232181231 \bar{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 21, T_{\max} = 21, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 21, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0}$$

$$T = 22 \quad [999] \quad 1^{1\bar{1}} (12) \quad 122315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 22, T_{\max} = 23, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 3 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[998] \quad 21^{1\bar{1}} \bar{5}^{13221} (12) \quad 122315131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 22, T_{\max} = 23, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 3 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[1000] \quad 1^{1\bar{1}} \bar{8}^{2321} 1232181231 \bar{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 22, T_{\max} = 22, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 22, T_{\max} = 22, T \in [22, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0}$$

$$[1002] \quad 1^{1\bar{1}} (12) \quad 122315131^{1\bar{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 \quad T = 22, T_{\max} = 23, \Delta = -29, \det = 8, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 5 & 4 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 5 & 3 & 6 & 6 \\ 2 & 4 & 2 & 6 & 5 \end{pmatrix}_{b_0}$$

$$T = 22, T_{\max} = 23, \Delta = -29, \det = 8, (n_+, n_-, n_0) = (1, 22, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 2 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$T = 23, T_{\max} = 23, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 5.0$$

$$T = 23, T_{\max} = 23, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 5.0$$

$$T = 23, T_{\max} = 23, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 5.0$$

$$T = 23, T_{\max} = 23, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0}$$

$$T = 23, T_{\max} = 23, \Delta = -1, \det = -4, (n_+, n_-, n_0) = (1, 23, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 23, T_{\max} = 23, T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 2 & 6 & 5 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 6 & 3 & 7 & 7 \\ 2 & 5 & 2 & 7 & 6 \end{pmatrix}_{b_0}$$

$$T = 24, T_{\max} = 24, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 5.0$$

$$T = 24, T_{\max} = 24, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 24, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 24, T_{\max} = 24, T \in [24, 24]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 3 & 1 & 5 & 5 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 5 & 0 & 3 & 5 \\ 2 & 5 & 0 & 5 & 6 \end{pmatrix}_{b_0}$$

$$T = 26, T_{\max} = 26, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 26, 1), c_{\text{ext}} = 5.0$$

$$T = 28, T_{\max} = 28, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 28, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 28, T_{\max} = 28, T \in [28, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 2 & 2 & 3 & 3 & 5 \\ 2 & 2 & 3 & 5 & 6 \end{pmatrix}_{b_0}$$

$$T = 34, T_{\max} = 34, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 34, 1), c_{\text{ext}} = 5.0$$

4.1003 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6$

- $T = 7$ [0] $1^{111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -47$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.91 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 4 & 7 & 1 & 3 \\ 4 & 9 & 16 & 0 & 5 \\ 7 & 16 & 27 & 0 & 9 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 9 & -4 & 0 \end{pmatrix}_{b_0}$
- [1] $1^{11}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -50$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 14.32 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$
 $\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 6 & 10 & 21 & 0 & 7 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 7 & -4 & -2 \end{pmatrix}_{b_0}$
- [2] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -56$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & -4 & 0 \end{pmatrix}_{b_0}$
- [4] $21^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -60$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 3 & 3 & 3 \\ 3 & 4 & 3 & -3 & -1 \\ 3 & 5 & 3 & -1 & 0 \end{pmatrix}_{b_0}$
- [3] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -60$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 3 & 3 & 3 \\ 3 & 4 & 3 & -3 & -1 \\ 3 & 5 & 3 & -1 & 0 \end{pmatrix}_{b_0}$
- [5] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -71$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & -4 & 0 \end{pmatrix}_{b_0}$
- [6] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -80$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.70 \rightarrow 13$, $T_{\max} = 100$, $T \in [13, 100]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & -4 & 0 \end{pmatrix}_{b_0}$
- [7] $1^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -86$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 13.02 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$
 $\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 4 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 5 & -4 & -1 \end{pmatrix}_{b_0}$
- [8] $1^{13}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -96$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.33 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$
 $\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 6 & 12 & 24 & 0 & 8 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 8 & -4 & -1 \end{pmatrix}_{b_0}$
- [9] $1^{14}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -110$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.6$
 $T_{\min} = 13.60 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 8 & 16 & 45 & 0 & 11 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 11 & -4 & -1 \end{pmatrix}_{b_0}$
- [10] $131^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6$ $T = 7$, $T_{\max} = 100$, $\Delta = -172$, $\det = -119$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.57 \rightarrow 14$, $T_{\max} = 100$, $T \in [14, 100]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 3 & 3 & 3 \\ 2 & 4 & 3 & -3 & -1 \\ 3 & 5 & 3 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -172, \ \det = -126, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.05 \rightarrow 14, \ T_{\max} = 100, \ T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 9 & 12 & 0 & 5 \\ 5 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 6 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{15} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -172, \ \det = -128, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.81 \rightarrow 14, \ T_{\max} = 100, \ T \in [14, 100]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 10 & 20 & 72 & 0 & 14 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 14 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -172, \ \det = -141, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 5 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 3^1 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -172, \ \det = -188, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.91 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 36 & 0 & 10 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 10 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{12} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -172, \ \det = -215, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 7 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{13} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -172, \ \det = -293, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 9 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 12 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{12} 3^1 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -172, \ \det = -330, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 10 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{14} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -172, \ \det = -375, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 11 & 21 & 75 & 0 & 15 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 15 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{13} 3^1 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -172, \ \det = -476, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.65 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 12 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 16 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 21^{11} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -172, \ \det = -483, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.69 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 4 & 8 & 6 & 3 & 4 \\ 8 & 9 & 8 & 4 & 5 \\ 6 & 8 & 3 & 3 & 3 \\ 3 & 4 & 3 & -3 & -1 \\ 4 & 5 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{14} 3^1 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -172, \ \det = -626, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 14 & 26 & 117 & 0 & 19 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 19 & -4 & 0 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [22] \quad 1^{\overline{11}} 3 21^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0 \\ T_{\min} = 13.40 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 3 & 5 & 3 & 3 & 3 \\ 2 & 5 & 3 & -2 & 0 \\ 3 & 5 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{\overline{11}} 3 21^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0 \\ T_{\min} = 13.35 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 5 & 0 & -2 & 0 \\ 3 & 5 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{\overline{11}} 3 21^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0 \\ T_{\min} = 13.09 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 \\ 3 & 5 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{\overline{11}} 3 21^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0 \\ T_{\min} = 13.05 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{\overline{11}} 3 21^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0 \\ T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{\overline{11}} 3 1^{\overline{11}} 3 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0 \\ T_{\min} = 13.14 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 3 & 2 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{\overline{11}} 3 1^{\overline{11}} 3 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0 \\ T_{\min} = 13.14 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 3 & 6 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 2 & 6 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 2 2 3 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0 \\ T_{\min} = 12.80 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 5 & 5 & 5 & 5 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 5 & 3 & -2 & 0 \\ 3 & 5 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{\overline{11}} 3 1^{\overline{11}} 3 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0 \\ T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 2 & 2 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{\overline{11}} 3 1^{\overline{11}} 3 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 102, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0 \\ T_{\min} = 13.12 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 2 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{\overline{11}} 3 21^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 4 \\ 5 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & -2 & 0 \\ 4 & 4 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{12} 3 \ 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 4 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 5 & 2 & 3 \\ 1 & 1 & 0 & 2 & -3 & -1 \\ 2 & 2 & -1 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{13} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 2 & 6 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{13} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 219, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 3 & 2 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{13} 3 \ 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 6 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 15 & 3 & 5 \\ 1 & 1 & 0 & 3 & -3 & -1 \\ 2 & 2 & -1 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{12} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.04 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 4 & 12 & 9 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 9 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{11} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 282, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.04 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 3 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 6 & -4 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [39] \ 1^{13} 3221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 19, T_{\max} = 26, T \in [19, 26]$$

$$\begin{pmatrix} 5 & 15 & 1 & 5 \\ 15 & 42 & 0 & 12 \\ 1 & 0 & -3 & -1 \\ 5 & 12 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{11} 2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 16.82 \rightarrow 17, T_{\max} = 32, T \in [17, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 1 & 2 \\ 1 & 3 & 0 & 1 & -3 & -1 \\ 2 & 3 & -1 & 2 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{11} 2321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -22, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 1 & 0 & -3 & -1 \\ 2 & 3 & 2 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{13} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{11}4^{11}131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.09 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{11}4^{11}131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{11}4^{11}131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{11}4^{11}131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 \quad T = 9, T_{\max} = 99, \Delta = -142, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.81 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & -3 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [47] $1^{11}4^{11}1321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 20, T_{\max} = 26, T \in [20, 26]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 1 & 2 & 3 \\ 2 & 1 & 0 & 0 & 2 \\ 1 & 2 & 0 & -4 & -2 \\ 3 & 3 & 2 & -2 & -4 \end{pmatrix}_{b_0}$$

[48] $1^{11}5^{11}131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 \quad T = 10, T_{\max} = 100, \Delta = -137, \det = 46, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.48 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$T = 11$ [49] $1^{11}5^{11}1321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 21, T_{\max} = 23, T \in [21, 23]$

$$\begin{pmatrix} 5 & 15 & 6 & 5 \\ 15 & 42 & 15 & 12 \\ 6 & 15 & 4 & 4 \\ 5 & 12 & 4 & -3 \end{pmatrix}_{b_0}$$

[50] $1^{11}5^{11}12321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 \quad T = 11, T_{\max} = 24, \Delta = -111, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 21, T_{\max} = 24, T \in [21, 24]$

$$\begin{pmatrix} 5 & 15 & 1 & 5 \\ 15 & 42 & 0 & 12 \\ 1 & 0 & -3 & -1 \\ 5 & 12 & -1 & -3 \end{pmatrix}_{b_0}$$

4.1004 $\text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \text{NHC}(-2)(-3)$

$T = 9$ [0] $1^{111}3^{111}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.12 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 2 & 3 \\ 4 & 9 & 4 & 0 & 8 & 4 & 5 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 8 & 3 & 0 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 2 \\ 3 & 5 & 2 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{121}3^{111}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 & 3 \\ 3 & 4 & 8 & 0 & 4 & 5 & 4 \\ 4 & 8 & 9 & 0 & 6 & 6 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 1 & 4 & 3 \\ 3 & 5 & 6 & 0 & 4 & 3 & 3 \\ 3 & 4 & 5 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 3 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -355, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 4 & 3 \\ 3 & 5 & 3 & 0 & 4 & 3 & 3 \\ 3 & 4 & 2 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 73, \Delta = -122, \det = -588, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 73, T \in [14, 73]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 2 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 & 4 \\ 5 & 9 & 12 & 0 & 9 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 9 & 0 & 4 & 4 & 4 \\ 2 & 4 & 6 & 0 & 4 & 0 & 2 \\ 3 & 4 & 6 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [5] \ 1^{111} 3 2 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.28 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 1 & 3 \\ 4 & 9 & 0 & 4 & 8 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 3 & 0 & 0 \\ 3 & 8 & 0 & 3 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & -1 & 0 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 705, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 6 & 3 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 3 & 0 & 4 & 0 & 2 \\ 2 & 4 & 2 & 3 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1005 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 3 & 0 & 4 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & 2 & -4 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

4.1006 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{1111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{su}_3$ $T = 10, T_{\max} = 22, \Delta = -101, \det = 825, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$

$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 3 & 0 & 4 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1007 NHC(-2)(-3) + $\textcolor{violet}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{brown}{so}_8$

$T = 8$ [1] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{brown}{so}_8$ $T = 8, T_{\max} = 101, \Delta = -171, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 5 & -4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{brown}{so}_8$ $T = 8, T_{\max} = 101, \Delta = -171, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{brown}{so}_8$ $T = 8, T_{\max} = 101, \Delta = -171, \det = 302, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 3 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{brown}{so}_8$ $T = 8, T_{\max} = 101, \Delta = -171, \det = 330, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.68 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 48 & 0 & 12 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 12 & -4 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [5] $1^{111}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{brown}{so}_8$ $T = 9, T_{\max} = 99, \Delta = -143, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 5 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 3 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{brown}{so}_8$ $T = 9, T_{\max} = 99, \Delta = -143, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 13.88 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -3 & 1 & -1 \\ 1 & 2 & 0 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{111}31^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{brown}{so}_8$ $T = 9, T_{\max} = 100, \Delta = -150, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 14.07 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 1 & 3 \\ 5 & 13 & 16 & 12 & 0 & 7 \\ 5 & 16 & 12 & 12 & 0 & 6 \\ 4 & 12 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 7 & 6 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

4.1008 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

4.1009 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^{14} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.6$$

$$T_{\min} = 24, T_{\max} = 98, T \in [24, 98]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 & 3 \\ 3 & 4 & 21 & 0 & 5 & 4 \\ 11 & 21 & 75 & 0 & 18 & 15 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 18 & 0 & 4 & 4 \\ 3 & 4 & 15 & -4 & 4 & -11 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 16, T_{\max} = 98, T \in [16, 98]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 3 & -1 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 3 \\ 4 & 9 & 12 & 0 & 4 & 5 \\ 5 & 12 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 0 & 1 & 3 \\ 3 & 5 & 6 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{123} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 6 & 3 \\ 3 & 4 & 13 & 0 & 12 & 4 \\ 7 & 13 & 27 & 0 & 27 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 27 & 0 & 25 & 9 \\ 3 & 4 & 9 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 103, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.03 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 121, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.22 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 4 \\ 3 & 4 & 3 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 126, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.81 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 127, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.02 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 2 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 148, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 6 & 0 & 13 & 7 \\ 3 & 5 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{13} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 160, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 \\ 9 & 17 & 48 & 0 & 15 & 12 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 15 & 0 & 4 & 4 \\ 3 & 4 & 12 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 166, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 172, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 13.21 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 2 & -4 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{121} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 178, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.07 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 4 & 8 & 9 & 0 & 6 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 5 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{121} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 181, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 4 & 8 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 7 \\ 3 & 4 & 5 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{112}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 2 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 19 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 19 & 12 & 0 & 73 & 15 \\ 3 & 4 & 2 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{122}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 4 & 8 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 5 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

[18] $1^{131}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.36 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 4 & 4 \\ 6 & 12 & 24 & 0 & 9 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 8 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[19] $1^{111}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.18 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 \\ 8 & 14 & 36 & 0 & 9 & 10 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 10 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[20] $1^{122}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.54 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 10 & 3 \\ 3 & 4 & 8 & 0 & 18 & 4 \\ 4 & 8 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 24 & 0 & 58 & 14 \\ 3 & 4 & 5 & -4 & 14 & -1 \end{pmatrix}_{b_0}$$

[21] $1^{113}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 2 & -4 & 9 & -1 \end{pmatrix}_{b_0}$$

[22] $1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 5 & 9 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 9 & 0 & 4 & 4 \\ 3 & 4 & 6 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{123}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.34 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 6 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 4 & 8 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 18 & 0 & 25 & 9 \\ 3 & 4 & 5 & -4 & 9 & -1 \end{pmatrix}_{b_0}$$

[24] $1^{123}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 \\ 7 & 13 & 27 & 0 & 12 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 12 & 0 & 4 & 4 \\ 3 & 4 & 9 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 \\ 5 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 6 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 247, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 9 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 2 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$

[27] $1^{123}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.98 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 15 & 3 \\ 3 & 4 & 8 & 0 & 27 & 4 \\ 4 & 8 & 9 & 0 & 36 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 36 & 0 & 133 & 21 \\ 3 & 4 & 5 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

[28] $1^{114}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.31 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 2 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$

[29] $1^{112}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 24 & 12 & 0 & 58 & 14 \\ 3 & 5 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{122}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 8 & 4 \\ 7 & 13 & 27 & 0 & 18 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 18 & 0 & 10 & 6 \\ 3 & 4 & 9 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[31] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 3 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{115}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 20 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 20 & 15 & 0 & 73 & 15 \\ 3 & 4 & 2 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

[34] $1^{111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{14}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 313, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 \\ 3 & 5 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 18 & 0 & 76 & 16 \\ 3 & 4 & 3 & -4 & 16 & -1 \end{pmatrix}_{b_0}$$

[36] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 20 & 5 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 20 & 9 & 0 & 37 & 11 \\ 3 & 5 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{114}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.48 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 2 & -4 & 16 & -1 \end{pmatrix}_{b_0}$$

[38] $1^{12}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 373, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 3 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{116}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.64 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 24 & 18 & 0 & 106 & 18 \\ 3 & 4 & 2 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

[40] $1^{13}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 49 & 13 \\ 3 & 4 & 3 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 391, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 7 & 13 & 27 & 0 & 9 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 9 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 403, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 18 & 0 & 28 & 10 \\ 3 & 4 & 5 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

[43] $1^{113}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 36 & 18 & 0 & 133 & 21 \\ 3 & 5 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{117}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 454, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.3$
 $T_{\min} = 13.76 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 28 & 4 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 28 & 21 & 0 & 145 & 21 \\ 3 & 4 & 2 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

[45] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 6 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[46] $1^{121}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 486, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 4 & 4 \\ 10 & 18 & 57 & 0 & 12 & 13 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 12 & 0 & 1 & 3 \\ 3 & 4 & 13 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[47] $1^{115}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 487, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.1$
 $T_{\min} = 13.55 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 40 & 15 & 0 & 148 & 20 \\ 3 & 5 & 2 & -4 & 20 & -1 \end{pmatrix}_{b_0}$$

[48] $1^{112}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 526, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{131}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 547, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 9 & 17 & 48 & 0 & 12 & 12 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 12 & 0 & 1 & 3 \\ 3 & 4 & 12 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 568, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 28 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 28 & 12 & 0 & 73 & 15 \\ 3 & 5 & 2 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{121}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 574, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 7 & 13 & 27 & 0 & 21 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 21 & 0 & 13 & 7 \\ 3 & 4 & 9 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{131}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 595, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 5 & 3 \\ 3 & 4 & 17 & 0 & 9 & 4 \\ 9 & 17 & 48 & 0 & 27 & 12 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 27 & 0 & 13 & 7 \\ 3 & 4 & 12 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{113}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 631, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 13 & 32 & 15 & 0 & 100 & 18 \\ 3 & 5 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{123}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 17 & 4 \\ 4 & 8 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 24 & 0 & 49 & 13 \\ 3 & 4 & 5 & -4 & 13 & -1 \end{pmatrix}_{b_0}$$

[55] $1^{123}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 646, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 3 \\ 3 & 4 & 8 & 0 & 22 & 4 \\ 4 & 8 & 9 & 0 & 30 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 30 & 0 & 85 & 17 \\ 3 & 4 & 5 & -4 & 17 & -1 \end{pmatrix}_{b_0}$$

[56] $1^{112}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 648, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 12 & 0 & 21 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 6 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{114}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 688, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 9 & 4 & 0 & 48 & 5 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 20 & 48 & 24 & 0 & 238 & 28 \\ 3 & 5 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

[58] $1^{112}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 736, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{113}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 747, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 27 & 0 & 49 & 13 \\ 3 & 4 & 6 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{112}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 760, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 28 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 16 & 28 & 18 & 0 & 154 & 22 \\ 3 & 4 & 2 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

[61] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 760, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 2 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[62] $1^{114}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 21 & 4 \\ 5 & 9 & 12 & 0 & 33 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 33 & 0 & 76 & 16 \\ 3 & 4 & 6 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[63] $1^{131}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 778, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.66 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 4 & 4 \\ 12 & 22 & 84 & 0 & 15 & 16 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 15 & 0 & 1 & 3 \\ 3 & 4 & 16 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{112}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 814, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{114}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 823, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 36 & 15 & 0 & 121 & 19 \\ 3 & 5 & 2 & -4 & 19 & -1 \end{pmatrix}_{b_0}$$

[66] $1^{122}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 841, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 27 & 0 & 30 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 30 & 0 & 28 & 10 \\ 3 & 4 & 9 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{114}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 991, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 3 \\ 4 & 9 & 4 & 0 & 44 & 5 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 18 & 44 & 21 & 0 & 193 & 25 \\ 3 & 5 & 2 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{114}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1006, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 2 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{123}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1030, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 9 & 3 \\ 3 & 4 & 13 & 0 & 17 & 4 \\ 7 & 13 & 27 & 0 & 39 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 39 & 0 & 49 & 13 \\ 3 & 4 & 9 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{114}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1036, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 16 & 40 & 18 & 0 & 154 & 22 \\ 3 & 5 & 2 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{113}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 2 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{113}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{113}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1606, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 21 & 0 & 193 & 25 \\ 3 & 4 & 2 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{114}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1624, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 2 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{114}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 2110$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 2 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{114}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 2464$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [77] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -4$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 17$, $T_{\max} = 25$, $T \in [17, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -3 & 4 & -1 \\ 2 & 3 & 0 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & -1 & -1 & 5 & -4 \end{pmatrix}_{b_0}$$

[78] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -6$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 16.67 \rightarrow 17$, $T_{\max} = 25$, $T \in [17, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 3 & 0 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & -1 & 0 & 4 & -3 \end{pmatrix}_{b_0}$$

[79] $1^{111}31^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -6$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 16$, $T_{\max} = 68$, $T \in [16, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & 2 & -4 & 3 & -3 \end{pmatrix}_{b_0}$$

[80] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -10$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 16.40 \rightarrow 17$, $T_{\max} = 25$, $T \in [17, 25]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 5 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 3 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

[81] $1^{112}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -12$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 16.33 \rightarrow 17$, $T_{\max} = 25$, $T \in [17, 25]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 13 & 4 \\ 5 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 10 & 0 \\ 13 & 15 & 0 & 10 & 38 & 12 \\ 4 & 4 & -1 & 0 & 12 & -3 \end{pmatrix}_{b_0}$$

[82] $1^{12}31^{111}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -24$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 15.17 \rightarrow 16$, $T_{\max} = 68$, $T \in [16, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 & 2 \\ 2 & 4 & 4 & 12 & 0 & 6 & 4 \\ 1 & 4 & 0 & 9 & 0 & 3 & 2 \\ 4 & 12 & 9 & 27 & 0 & 12 & 9 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 12 & 0 & 4 & 4 \\ 2 & 4 & 2 & 9 & -4 & 4 & -3 \end{pmatrix}_{b_0}$$

[83] $1^{12}31^{111}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 18.58 \rightarrow 19$, $T_{\max} = 68$, $T \in [19, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 12 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 6 & 3 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 9 & 12 & 0 & 28 & 10 \\ 2 & 4 & 2 & 3 & -4 & 10 & -6 \end{pmatrix}_{b_0}$$

- [84] $1^{111} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -50$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.12 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$
 $\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 3 \\ 4 & 9 & 0 & 4 & 12 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 7 & 0 \\ 5 & 12 & 0 & 7 & 14 & 8 \\ 3 & 5 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$
- [85] $1^{111} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.35 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 \\ 1 & 3 & 3 & 3 & 0 & 1 \\ 2 & 3 & 3 & 2 & -4 & 3 \end{pmatrix}_{b_0}$
- [86] $1^{111} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -60$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 \\ 2 & 3 & 6 & 3 & 0 & 4 \\ 2 & 3 & 3 & 2 & -4 & 4 \end{pmatrix}_{b_0}$
- [88] $1^{111} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 1 \end{pmatrix}_{b_0}$
- [89] $1^{111} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 9 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$
- [87] $1^{111} 31^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 14.41 \rightarrow 15$, $T_{\max} = 68$, $T \in [15, 68]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & 3 & -4 & 3 & -2 \end{pmatrix}_{b_0}$
- [90] $1^{111} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -82$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.12 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -3 & 1 & -1 \\ 1 & 2 & 0 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$
- [91] $1^{111} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 5 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$
- [92] $1^{112} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$
 $\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 6 & 0 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$
- [93] $1^{111} 31^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 3 & 0 & 12 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 6 & 6 & 0 & 22 & 8 \\ 2 & 4 & 2 & 2 & -4 & 8 & -1 \end{pmatrix}_{b_0}$

- [94] $1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.22 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 3 & 0 \\ 3 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$
- [95] $1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.10 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 3 & 0 \\ 4 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$
- [96] $1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -131$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [97] $1^{112} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 13.11 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$
- [98] $1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -140$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [99] $1^{113} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -155$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 0 \end{pmatrix}_{b_0}$$
- [100] $1^{114} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -176$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.77 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$
- [101] $1^{11} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -177$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 4 & 4 \\ 2 & 4 & 2 & 2 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$
- [102] $1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$
- [103] $1^{115} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -203$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.1$
 $T_{\min} = 14.11 \rightarrow 15$, $T_{\max} = 25$, $T \in [15, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 15 & 0 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & 0 & 20 & -1 \end{pmatrix}_{b_0}$$

$$[104] \ 1^{111} 31^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -231, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.08 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 2 \\ 3 & 9 & 6 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 6 & 0 & 1 & 3 \\ 2 & 4 & 2 & 6 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{111} 3 \ 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -236, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.81 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 8 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{112} 3 \ 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -236, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.90 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 8 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[108] \ 1^{111} 3 \ 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -240, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.82 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 6 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[107] \ 1^{111} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -240, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.85 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[109] \ 1^{111} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -243, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.84 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[110] \ 1^{111} 3 \ 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -291, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.81 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 0 \\ 7 & 9 & 0 & 7 & 14 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{113} 31^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -303, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.26 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 3 & 0 & 15 & 3 \\ 1 & 3 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 15 & 12 & 0 & 49 & 13 \\ 2 & 3 & 3 & 2 & -4 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{121} 3 \ 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -306, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} =$$

13.2

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 1 & 4 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 5 & 2 & 4 & 3 \\ 1 & 1 & 0 & 2 & -3 & 1 & -1 \\ 1 & 2 & 0 & 4 & 1 & 0 & 2 \\ 2 & 2 & -1 & 3 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{\overline{11}1} 31^{\overline{11}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -375, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 3 & 0 & 9 & 3 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 6 & 9 & 6 & 0 & 13 & 7 \\ 2 & 3 & 3 & 2 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[114] \ 1^{\overline{11}3} 321^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -395, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 & 0 \\ 11 & 21 & 0 & 13 & 74 & 16 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{\overline{11}2} 321^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -411, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 & 0 \\ 7 & 12 & 0 & 6 & 22 & 8 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{\overline{11}1} 31^{\overline{11}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -411, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & 2 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{\overline{11}2} 31^{\overline{11}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 3 & 0 & 12 & 3 \\ 1 & 3 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 9 & 12 & 9 & 0 & 28 & 10 \\ 2 & 3 & 3 & 2 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{\overline{11}1} 31^{\overline{11}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 4 \\ 2 & 4 & 2 & 3 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{\overline{11}1} 31^{\overline{11}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -537, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.04 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 & 2 \\ 2 & 4 & 4 & 9 & 0 & 6 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 2 \\ 3 & 9 & 6 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 9 & 0 & 4 & 4 \\ 2 & 4 & 2 & 6 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{\overline{11}2} 321^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -555, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 & 0 \\ 9 & 15 & 0 & 10 & 38 & 12 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 & 1 \end{pmatrix}_{b_0}$$

- [121] $1^{114} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -560$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.03 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 14 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 0 \end{pmatrix}_{b_0}$$
- [122] $1^{112} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -576$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.87 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [124] $1^{113} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -588$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$
- [123] $1^{112} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -588$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [125] $1^{113} 31^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -615$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.35 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 6 & 3 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 9 & 0 & 13 & 7 \\ 2 & 4 & 2 & 3 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$
- [126] $1^{112} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -635$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 9 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$
- [127] $1^{111} 31^{12} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -762$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 13.12 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 3 & 0 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 \\ 1 & 3 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 9 & 9 & 0 & 37 & 11 \\ 2 & 4 & 2 & 2 & -4 & 11 & -1 \end{pmatrix}_{b_0}$$
- [128] $1^{112} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -768$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 10 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$
- [129] $1^{114} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -771$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.02 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 13 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 0 \end{pmatrix}_{b_0}$$

[130] $1^{112} 3^{113} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -875$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 11 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[131] $1^{112} 3^{114} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -876$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.5$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 0 \\ 12 & 15 & 0 & 10 & 38 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{113} 3^{111} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -876$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 12 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{111} 31^{111} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -906$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 9 & 4 \\ 1 & 4 & 0 & 6 & 0 & 6 & 2 \\ 3 & 9 & 6 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 15 & 0 & 13 & 7 \\ 2 & 4 & 2 & 6 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

[134] $1^{113} 3^{112} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -915$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 11 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[135] $1^{111} 31^{13} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -1059$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 3 & 0 & 21 & 4 \\ 1 & 4 & 0 & 3 & 0 & 12 & 2 \\ 1 & 3 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 12 & 12 & 0 & 73 & 15 \\ 2 & 4 & 2 & 2 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

[136] $1^{113} 3^{113} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.8$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 12 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{114} 3^{111} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1203$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 15 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{113} 3^{114} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -1251$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

12.8

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[138] $1^{113} 3^{12} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -1251, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 13 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{114} 3^{12} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -1260, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[141] $1^{111} 31^{14} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9, T_{\max} = 68, \Delta = -121, \det = -1302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 3 & 0 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 15 & 2 \\ 1 & 3 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 15 & 15 & 0 & 121 & 19 \\ 2 & 4 & 2 & 2 & -4 & 19 & -1 \end{pmatrix}_{b_0}$$

[142] $1^{113} 3^{13} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -1520, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 14 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[143] $1^{114} 3^{13} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -1643, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[144] $1^{113} 3^{14} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -1683, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 21 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 13 & 0 \\ 15 & 21 & 0 & 13 & 74 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[145] $1^{114} 3^{12} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -1740, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 16 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[146] $1^{114} 3^{14} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_2$ $T = 9, T_{\max} = 25, \Delta = -114, \det = -1920, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & 0 & 16 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{114} 3^{21^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -2171, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 17 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{114} 3^{21^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -2496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 1 & 3 & 27 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 16 & 0 \\ 18 & 27 & 0 & 16 & 122 & 20 \\ 3 & 4 & -1 & 0 & 20 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [149] $1^{11} 4^{1^{31}1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 17.33 \rightarrow 18, T_{\max} = 24, T \in [18, 24]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 8 & 5 & 0 & 13 & 7 \\ 3 & 4 & 2 & -3 & 7 & -3 \end{pmatrix}_{b_0}$$

[150] $1^{11} 4^{1^{31}1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 17.33 \rightarrow 18, T_{\max} = 24, T \in [18, 24]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & -3 & 3 & -3 \end{pmatrix}_{b_0}$$

[151] $1^{11} 4^{1^{31}1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[152] $1^{11} 4^{1^{31}1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 5 & 4 \\ 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[153] $1^{11} 4^{1^{31}1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.08 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 7 & 0 & 5 & 4 \\ 5 & 7 & 7 & 0 & 6 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 6 & 0 & 4 & 4 \\ 3 & 4 & 5 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[154] $1^{11} 4^{1^{31}1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[155] $1^{11} 4^{1^{31}1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 78, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 7 & 0 & 9 & 4 \\ 5 & 7 & 7 & 0 & 11 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 11 & 0 & 13 & 7 \\ 3 & 4 & 5 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[156] $1^{\overline{11}} 1^{\overline{31}^1} 4^1 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 87, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 4 & 3 & 0 & 5 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[157] $1^{\overline{11}} 1^{\overline{31}^1} 4^1 1^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 114, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 24, T \in [15, 24]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 4 & 3 & 0 & 15 & 4 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 15 & 11 & 0 & 45 & 13 \\ 3 & 4 & 2 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$

[158] $1^{\overline{11}} 1^{\overline{1}} 4^1 131^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.89 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[159] $1^{\overline{11}} 1^{\overline{1}} 1^{\overline{31}^1} 4^1 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 158, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 7 & 0 & 4 & 4 \\ 5 & 7 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 0 & 1 & 3 \\ 3 & 4 & 5 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[160] $1^{\overline{11}} 1^{\overline{1}} 1^{\overline{31}^1} 4^1 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 203, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 7 & 5 & 0 & 8 & 6 \\ 3 & 4 & 2 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[161] $1^{\overline{11}} 1^{\overline{1}} 1^{\overline{1}} 4^1 131^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[162] $1^{\overline{11}} 1^{\overline{1}} 1^{\overline{31}^2} 4^1 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 15 & 4 \\ 2 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 15 & 10 & 0 & 45 & 13 \\ 3 & 4 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[163] $1^{\overline{1}} 1^{\overline{31}^1} 4^1 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 257, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 4 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 4 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[164] $1^{\overline{1}} 1^{\overline{31}^1} 4^1 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 287, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 4 & 3 & 0 & 7 & 4 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 7 & 5 & 0 & 8 & 6 \\ 3 & 4 & 2 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[165] \ 1^{\overline{11}} 1^{\overline{131}^1} 4 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 384, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 8 & 0 & 26 & 10 \\ 3 & 4 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[166] \ 1^{\overline{11}} 1^{\overline{131}^1} 4 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 402, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 4 & 3 & 0 & 9 & 4 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 5 & 0 & 13 & 7 \\ 3 & 4 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[167] \ 1^{\overline{11}} 1^{\overline{131}^1} 4 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 444, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 4 & 3 & 0 & 12 & 4 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 8 & 0 & 26 & 10 \\ 3 & 4 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [168] \ 1^{\overline{13}} 5^{\overline{11}} 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 18, T_{\max} = 23, T \in [18, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 0 \\ 6 & 12 & 0 & 6 & 25 & 9 \\ 3 & 4 & -1 & 0 & 9 & -2 \end{pmatrix}_{b_0}$$

$$[169] \ 1^{\overline{12}} 5^{\overline{11}} 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -52, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 14.54 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 4 & 0 \\ 4 & 8 & 0 & 4 & 10 & 6 \\ 3 & 4 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[170] \ 1^{\overline{11}} 5^{\overline{11}} 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -107, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.49 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

4.1010 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 5 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 4 \\ 3 & 5 & 3 & 0 & 3 & 4 & 3 \\ 3 & 4 & 2 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

4.1011 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

4.1012 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.86 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 \\ 3 & 5 & 3 & 0 & 3 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 5 & 6 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.71 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 & 4 \\ 8 & 14 & 36 & 0 & 9 & 9 & 10 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 9 & 0 & 1 & 3 & 3 \\ 2 & 4 & 9 & 0 & 3 & 1 & 3 \\ 3 & 4 & 10 & -4 & 3 & 3 & -4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 6 & 4 \\ 3 & 5 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & 2 & -4 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 2 & -4 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{1121}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -90$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.07 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 5 & 3 \\ 3 & 4 & 9 & 0 & 8 & 9 & 4 \\ 5 & 9 & 12 & 0 & 12 & 15 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 12 & 6 \\ 5 & 9 & 15 & 0 & 12 & 13 & 7 \\ 3 & 4 & 6 & -4 & 6 & 7 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{1111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -92$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 8 & 14 & 9 & 0 & 9 & 37 & 11 \\ 3 & 4 & 2 & -4 & 3 & 11 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{1111}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -101$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 11 & 3 \\ 3 & 4 & 4 & 0 & 5 & 19 & 4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 21 & 4 \\ 11 & 19 & 12 & 0 & 21 & 73 & 15 \\ 3 & 4 & 2 & -4 & 4 & 15 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{1311}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -107$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.32 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 2 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 & 4 \\ 9 & 17 & 48 & 0 & 15 & 12 & 12 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 15 & 0 & 4 & 3 & 4 \\ 2 & 4 & 12 & 0 & 3 & 1 & 3 \\ 3 & 4 & 12 & -4 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{1111}31^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 5 & 3 \\ 3 & 4 & 4 & 0 & 19 & 9 & 4 \\ 2 & 4 & 0 & 0 & 12 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 19 & 12 & 0 & 73 & 33 & 15 \\ 5 & 9 & 6 & 0 & 33 & 13 & 7 \\ 3 & 4 & 2 & -4 & 15 & 7 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{1121}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.34 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 5 & 3 \\ 3 & 4 & 4 & 0 & 8 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 12 & 6 \\ 5 & 9 & 6 & 0 & 12 & 13 & 7 \\ 3 & 4 & 2 & -4 & 6 & 7 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{1111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -128$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 13.22 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{1112}31^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -137$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 7 & 3 \\ 3 & 4 & 4 & 0 & 14 & 13 & 4 \\ 2 & 4 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 33 & 11 \\ 7 & 13 & 9 & 0 & 33 & 28 & 10 \\ 3 & 4 & 2 & -4 & 11 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -173, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.39 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 9 & 3 \\ 3 & 4 & 4 & 0 & 5 & 17 & 4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 15 & 4 \\ 9 & 17 & 12 & 0 & 15 & 49 & 13 \\ 3 & 4 & 2 & -4 & 4 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1121} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -174, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.85 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 2 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 & 4 \\ 5 & 9 & 12 & 0 & 12 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 6 & 6 \\ 2 & 4 & 6 & 0 & 6 & 1 & 3 \\ 3 & 4 & 6 & -4 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -191, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 5 & 3 \\ 4 & 9 & 4 & 0 & 8 & 12 & 5 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 8 & 3 & 0 & 4 & 9 & 4 \\ 5 & 12 & 6 & 0 & 9 & 13 & 7 \\ 3 & 5 & 2 & -4 & 4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1122} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 13 & 3 \\ 3 & 4 & 4 & 0 & 13 & 23 & 4 \\ 2 & 4 & 0 & 0 & 9 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 54 & 10 \\ 13 & 23 & 15 & 0 & 54 & 100 & 18 \\ 3 & 4 & 2 & -4 & 10 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -219, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 2 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 & 4 \\ 5 & 9 & 12 & 0 & 9 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 9 & 0 & 4 & 3 & 4 \\ 2 & 4 & 6 & 0 & 3 & 1 & 3 \\ 3 & 4 & 6 & -4 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{121} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -245, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 2 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 & 4 \\ 7 & 13 & 27 & 0 & 12 & 9 & 9 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 12 & 0 & 4 & 3 & 4 \\ 2 & 4 & 9 & 0 & 3 & 1 & 3 \\ 3 & 4 & 9 & -4 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -284, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 8 & 3 & 0 & 4 & 6 & 4 \\ 3 & 8 & 3 & 0 & 6 & 4 & 4 \\ 3 & 5 & 2 & -4 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -341, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 9 & 4 \\ 5 & 9 & 6 & 0 & 9 & 13 & 7 \\ 3 & 4 & 2 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1111}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -344, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.81 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 2 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111}31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -353, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 8 & 3 \\ 3 & 4 & 4 & 0 & 5 & 14 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 15 & 4 \\ 8 & 14 & 9 & 0 & 15 & 37 & 11 \\ 3 & 4 & 2 & -4 & 4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -359, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 \\ 3 & 5 & 3 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{1111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -368, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 12 & 5 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 12 & 6 & 0 & 13 & 15 & 7 \\ 5 & 12 & 6 & 0 & 15 & 13 & 7 \\ 3 & 5 & 2 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{1112}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -380, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 4 & 4 & 0 & 4 & 8 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 4 & 8 & 6 & 0 & 6 & 10 & 6 \\ 3 & 4 & 2 & -4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{112}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -380, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 7 & 3 \\ 3 & 4 & 4 & 0 & 5 & 13 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 12 & 4 \\ 7 & 13 & 9 & 0 & 12 & 28 & 10 \\ 3 & 4 & 2 & -4 & 4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{1111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -408, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 & 4 \\ 5 & 9 & 12 & 0 & 6 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 1 & 3 & 3 \\ 2 & 4 & 6 & 0 & 3 & 1 & 3 \\ 3 & 4 & 6 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{114}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -413, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 5 & 0 & 4 & 21 & 4 \\ 3 & 5 & 3 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 11 & 21 & 18 & 0 & 15 & 76 & 16 \\ 3 & 4 & 3 & -4 & 3 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1123} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 8 & 4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 12 & 6 \\ 4 & 8 & 6 & 0 & 12 & 10 & 6 \\ 3 & 4 & 2 & -4 & 6 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1123} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -425, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 13 & 3 \\ 3 & 4 & 4 & 0 & 4 & 23 & 4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 13 & 23 & 15 & 0 & 15 & 100 & 18 \\ 3 & 4 & 2 & -4 & 3 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1113} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 6 & 3 \\ 3 & 4 & 4 & 0 & 4 & 12 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 6 & 12 & 9 & 0 & 9 & 25 & 9 \\ 3 & 4 & 2 & -4 & 3 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1122} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 7 & 3 \\ 3 & 4 & 9 & 0 & 8 & 13 & 4 \\ 5 & 9 & 12 & 0 & 12 & 21 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 0 & 10 & 18 & 6 \\ 7 & 13 & 21 & 0 & 18 & 28 & 10 \\ 3 & 4 & 6 & -4 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -485, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 2 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1113} 31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -488, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 12 & 3 \\ 3 & 4 & 4 & 0 & 14 & 22 & 4 \\ 2 & 4 & 0 & 0 & 9 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 57 & 11 \\ 12 & 22 & 15 & 0 & 57 & 85 & 17 \\ 3 & 4 & 2 & -4 & 11 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1114} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 16 & 4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 8 & 16 & 12 & 0 & 12 & 46 & 12 \\ 3 & 4 & 2 & -4 & 3 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1112} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{c}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -533, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.03 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 13 & 4 \\ 3 & 5 & 3 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 7 & 13 & 12 & 0 & 9 & 28 & 10 \\ 3 & 4 & 3 & -4 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -551, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 5 & 0 & 4 & 17 & 4 \\ 3 & 5 & 3 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 9 & 17 & 15 & 0 & 12 & 49 & 13 \\ 3 & 4 & 3 & -4 & 3 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1122} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -560, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 7 & 3 \\ 3 & 4 & 4 & 0 & 8 & 13 & 4 \\ 2 & 4 & 0 & 0 & 6 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 10 & 18 & 6 \\ 7 & 13 & 9 & 0 & 18 & 28 & 10 \\ 3 & 4 & 2 & -4 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -729, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 4 & 9 & 4 \\ 5 & 9 & 12 & 0 & 6 & 15 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 1 & 6 & 3 \\ 5 & 9 & 15 & 0 & 6 & 13 & 7 \\ 3 & 4 & 6 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{111} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -773, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 8 & 3 \\ 3 & 4 & 4 & 0 & 9 & 14 & 4 \\ 2 & 4 & 0 & 0 & 6 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 24 & 7 \\ 8 & 14 & 9 & 0 & 24 & 37 & 11 \\ 3 & 4 & 2 & -4 & 7 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{112} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -800, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.75 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 10 & 3 \\ 3 & 4 & 4 & 0 & 5 & 18 & 4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 18 & 4 \\ 10 & 18 & 12 & 0 & 18 & 58 & 14 \\ 3 & 4 & 2 & -4 & 4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{112} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -848, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 16 & 3 \\ 3 & 4 & 4 & 0 & 5 & 28 & 4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 30 & 4 \\ 16 & 28 & 18 & 0 & 30 & 154 & 22 \\ 3 & 4 & 2 & -4 & 4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 2 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{111} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -920, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 39 & 11 \\ 8 & 14 & 9 & 0 & 39 & 37 & 11 \\ 3 & 4 & 2 & -4 & 11 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -929, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 4 & 13 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 7 & 13 & 9 & 0 & 9 & 28 & 10 \\ 3 & 4 & 2 & -4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1112} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 4 & 18 & 4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 10 & 18 & 12 & 0 & 12 & 58 & 14 \\ 3 & 4 & 2 & -4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1112} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.91 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 14 & 3 \\ 3 & 4 & 4 & 0 & 5 & 26 & 4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 24 & 4 \\ 14 & 26 & 18 & 0 & 24 & 118 & 20 \\ 3 & 4 & 2 & -4 & 4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1112} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 13 & 3 \\ 3 & 4 & 4 & 0 & 5 & 23 & 4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 24 & 4 \\ 13 & 23 & 15 & 0 & 24 & 100 & 18 \\ 3 & 4 & 2 & -4 & 4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1112} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1001, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 12 & 3 \\ 3 & 4 & 4 & 0 & 5 & 22 & 4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 21 & 4 \\ 12 & 22 & 15 & 0 & 21 & 85 & 17 \\ 3 & 4 & 2 & -4 & 4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1113} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 18 & 3 \\ 3 & 4 & 4 & 0 & 4 & 32 & 4 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 21 & 3 \\ 18 & 32 & 21 & 0 & 21 & 193 & 25 \\ 3 & 4 & 2 & -4 & 3 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1083, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 7 & 3 \\ 3 & 4 & 9 & 0 & 4 & 13 & 4 \\ 5 & 9 & 12 & 0 & 6 & 21 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 1 & 9 & 3 \\ 7 & 13 & 21 & 0 & 9 & 28 & 10 \\ 3 & 4 & 6 & -4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1114} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1085, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 11 & 3 \\ 3 & 4 & 4 & 0 & 9 & 21 & 4 \\ 2 & 4 & 0 & 0 & 6 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 33 & 7 \\ 11 & 21 & 15 & 0 & 33 & 76 & 16 \\ 3 & 4 & 2 & -4 & 7 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1205, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 7 & 3 \\ 3 & 4 & 4 & 0 & 9 & 13 & 4 \\ 2 & 4 & 0 & 0 & 6 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 21 & 7 \\ 7 & 13 & 9 & 0 & 21 & 28 & 10 \\ 3 & 4 & 2 & -4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1112} 31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 10 & 3 \\ 3 & 4 & 4 & 0 & 14 & 18 & 4 \\ 2 & 4 & 0 & 0 & 9 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 48 & 11 \\ 10 & 18 & 12 & 0 & 48 & 58 & 14 \\ 3 & 4 & 2 & -4 & 11 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 9 & 3 \\ 3 & 4 & 4 & 0 & 9 & 17 & 4 \\ 2 & 4 & 0 & 0 & 6 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 27 & 7 \\ 9 & 17 & 12 & 0 & 27 & 49 & 13 \\ 3 & 4 & 2 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1112} 31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 16 & 3 \\ 3 & 4 & 4 & 0 & 9 & 28 & 4 \\ 2 & 4 & 0 & 0 & 6 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 48 & 7 \\ 16 & 28 & 18 & 0 & 48 & 154 & 22 \\ 3 & 4 & 2 & -4 & 7 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1281, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 9 & 3 \\ 3 & 4 & 9 & 0 & 4 & 17 & 4 \\ 5 & 9 & 12 & 0 & 6 & 27 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 1 & 12 & 3 \\ 9 & 17 & 27 & 0 & 12 & 49 & 13 \\ 3 & 4 & 6 & -4 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1323, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 11 & 3 \\ 3 & 4 & 9 & 0 & 4 & 21 & 4 \\ 5 & 9 & 12 & 0 & 6 & 33 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 1 & 15 & 3 \\ 11 & 21 & 33 & 0 & 15 & 76 & 16 \\ 3 & 4 & 6 & -4 & 3 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1113} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1397, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 4 & 4 & 0 & 4 & 17 & 4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 9 & 17 & 12 & 0 & 12 & 49 & 13 \\ 3 & 4 & 2 & -4 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{113} 31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1565, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 15 & 3 \\ 3 & 4 & 4 & 0 & 5 & 27 & 4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 27 & 4 \\ 15 & 27 & 18 & 0 & 27 & 133 & 21 \\ 3 & 4 & 2 & -4 & 4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{1122} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 10 & 3 \\ 3 & 4 & 4 & 0 & 13 & 18 & 4 \\ 2 & 4 & 0 & 0 & 9 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 42 & 10 \\ 10 & 18 & 12 & 0 & 42 & 58 & 14 \\ 3 & 4 & 2 & -4 & 10 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{1113} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 15 & 3 \\ 3 & 4 & 4 & 0 & 4 & 27 & 4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 18 & 3 \\ 15 & 27 & 18 & 0 & 18 & 133 & 21 \\ 3 & 4 & 2 & -4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{1112} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1760, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 10 & 3 \\ 3 & 4 & 4 & 0 & 9 & 18 & 4 \\ 2 & 4 & 0 & 0 & 6 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 30 & 7 \\ 10 & 18 & 12 & 0 & 30 & 58 & 14 \\ 3 & 4 & 2 & -4 & 7 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{1112} 3^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1781, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 13 & 3 \\ 3 & 4 & 4 & 0 & 9 & 23 & 4 \\ 2 & 4 & 0 & 0 & 6 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 39 & 7 \\ 13 & 23 & 15 & 0 & 39 & 100 & 18 \\ 3 & 4 & 2 & -4 & 7 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{1122} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1784, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 13 & 4 \\ 2 & 4 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 0 & 28 & 30 & 10 \\ 7 & 13 & 9 & 0 & 30 & 28 & 10 \\ 3 & 4 & 2 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{1113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1820, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 4 & 0 & 4 & 22 & 4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 12 & 22 & 15 & 0 & 15 & 85 & 17 \\ 3 & 4 & 2 & -4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{113} 3^1 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1865, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 18 & 3 \\ 3 & 4 & 4 & 0 & 5 & 32 & 4 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 33 & 4 \\ 18 & 32 & 21 & 0 & 33 & 193 & 25 \\ 3 & 4 & 2 & -4 & 4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{1114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1889, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.86 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 11 & 3 \\ 3 & 4 & 4 & 0 & 4 & 21 & 4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 11 & 21 & 15 & 0 & 15 & 76 & 16 \\ 3 & 4 & 2 & -4 & 3 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{114} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1928, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 17 & 3 \\ 3 & 4 & 4 & 0 & 5 & 31 & 4 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 30 & 4 \\ 17 & 31 & 21 & 0 & 30 & 172 & 24 \\ 3 & 4 & 2 & -4 & 4 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{1113} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2501, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 12 & 3 \\ 3 & 4 & 4 & 0 & 9 & 22 & 4 \\ 2 & 4 & 0 & 0 & 6 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 36 & 7 \\ 12 & 22 & 15 & 0 & 36 & 85 & 17 \\ 3 & 4 & 2 & -4 & 7 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{1122} 31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2624, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 18 & 4 \\ 2 & 4 & 0 & 0 & 12 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 12 & 0 & 58 & 60 & 14 \\ 10 & 18 & 12 & 0 & 60 & 58 & 14 \\ 3 & 4 & 2 & -4 & 14 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{114} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2636, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 20 & 3 \\ 3 & 4 & 4 & 0 & 5 & 36 & 4 \\ 2 & 4 & 0 & 0 & 3 & 24 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 36 & 4 \\ 20 & 36 & 24 & 0 & 36 & 238 & 28 \\ 3 & 4 & 2 & -4 & 4 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{1114} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2720, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 4 & 4 & 0 & 4 & 26 & 4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 18 & 3 \\ 14 & 26 & 18 & 0 & 18 & 118 & 20 \\ 3 & 4 & 2 & -4 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{1114} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2780, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 20 & 3 \\ 3 & 4 & 4 & 0 & 4 & 36 & 4 \\ 2 & 4 & 0 & 0 & 3 & 24 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 24 & 3 \\ 20 & 36 & 24 & 0 & 24 & 238 & 28 \\ 3 & 4 & 2 & -4 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{1114} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2996, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 14 & 3 \\ 3 & 4 & 4 & 0 & 9 & 26 & 4 \\ 2 & 4 & 0 & 0 & 6 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 42 & 7 \\ 14 & 26 & 18 & 0 & 42 & 118 & 20 \\ 3 & 4 & 2 & -4 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{1114} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -3017, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 17 & 3 \\ 3 & 4 & 4 & 0 & 4 & 31 & 4 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 21 & 3 \\ 17 & 31 & 21 & 0 & 21 & 172 & 24 \\ 3 & 4 & 2 & -4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{1113}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -3200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 15 & 3 \\ 3 & 4 & 4 & 0 & 9 & 27 & 4 \\ 2 & 4 & 0 & 0 & 6 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 45 & 7 \\ 15 & 27 & 18 & 0 & 45 & 133 & 21 \\ 3 & 4 & 2 & -4 & 7 & 21 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{1113}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -3365, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 18 & 3 \\ 3 & 4 & 4 & 0 & 9 & 32 & 4 \\ 2 & 4 & 0 & 0 & 6 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 54 & 7 \\ 18 & 32 & 21 & 0 & 54 & 193 & 25 \\ 3 & 4 & 2 & -4 & 7 & 25 & 0 \end{pmatrix}_{b_0}$$

[78] $1^{1114}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -4373, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 17 & 3 \\ 3 & 4 & 4 & 0 & 9 & 31 & 4 \\ 2 & 4 & 0 & 0 & 6 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 51 & 7 \\ 17 & 31 & 21 & 0 & 51 & 172 & 24 \\ 3 & 4 & 2 & -4 & 7 & 24 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{1114}31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -5216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 20 & 3 \\ 3 & 4 & 4 & 0 & 9 & 36 & 4 \\ 2 & 4 & 0 & 0 & 6 & 24 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 60 & 7 \\ 20 & 36 & 24 & 0 & 60 & 238 & 28 \\ 3 & 4 & 2 & -4 & 7 & 28 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [80] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{1111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{1111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 82, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 3 \\ 3 & 4 & 1 & 3 & 9 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 3 & 0 \\ 5 & 9 & 0 & 7 & 14 & 9 & 8 \\ 3 & 6 & 0 & 3 & 9 & 4 & 4 \\ 3 & 4 & -1 & 0 & 8 & 4 & 1 \end{pmatrix}_{b_0}$$

[83] $1^{1111}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 111, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 6 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 9 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 1 & 6 & 3 \\ 3 & 6 & 9 & 6 & 0 & 6 & 13 & 7 \\ 2 & 3 & 3 & 2 & -4 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{11}31^{121}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 138, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.35 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 12 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 6 & 6 & 0 & 22 & 12 & 8 \\ 2 & 6 & 3 & 3 & 0 & 12 & 4 & 4 \\ 2 & 4 & 2 & 2 & -4 & 8 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{1111}3 \quad 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 224, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.89 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 7 & 0 \\ 4 & 6 & 0 & 3 & 4 & 9 & 4 \\ 6 & 9 & 0 & 7 & 9 & 14 & 8 \\ 3 & 4 & -1 & 0 & 4 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{111}3 \quad 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 239, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 6 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 7 & 0 & 14 & 8 \\ 3 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{1111}3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 268, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.97 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{111}3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 269, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.86 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & 0 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{1112}31^{11}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 273, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.38 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 9 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 12 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 1 & 9 & 3 \\ 4 & 9 & 12 & 9 & 0 & 9 & 28 & 10 \\ 2 & 3 & 3 & 2 & -4 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{112}3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -84, \quad \det = 296, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.99 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & 0 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{11}31^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{c}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 300, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 4 & 6 & 4 \\ 2 & 6 & 3 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & 2 & 2 & -4 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

[92] $1^{1111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 320, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$

$T_{\min} = 12.90 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -2 & 3 & 3 & 0 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{111} 31^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 363, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$

$T_{\min} = 13.22 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 4 & 9 & 4 \\ 3 & 9 & 6 & 6 & 0 & 9 & 13 & 7 \\ 2 & 4 & 2 & 2 & -4 & 4 & 7 & -1 \end{pmatrix}_{b_0}$$

[94] $1^{112} 3 \ 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 374, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$

$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 9 & 3 \\ 3 & 4 & 1 & 3 & 0 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 10 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 0 & 10 & 0 & 38 & 12 \\ 3 & 4 & -1 & 0 & 0 & 12 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 381, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$

$T_{\min} = 12.90 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[96] $1^{111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 704, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$

$T_{\min} = 12.86 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 3 & 0 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{111} 31^{12} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1065, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.5$

$T_{\min} = 13.14 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 5 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 9 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 4 & 15 & 4 \\ 5 & 15 & 9 & 9 & 0 & 15 & 37 & 11 \\ 2 & 4 & 2 & 2 & -4 & 4 & 11 & -1 \end{pmatrix}_{b_0}$$

[98] $1^{111} 31^{13} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1659, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.8$

$T_{\min} = 13.14 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 7 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 21 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 12 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 4 & 21 & 4 \\ 7 & 21 & 12 & 12 & 0 & 21 & 73 & 15 \\ 2 & 4 & 2 & 2 & -4 & 4 & 15 & -1 \end{pmatrix}_{b_0}$$

[99] $1^{111} 31^{14} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 2145, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$

$T_{\min} = 13.14 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 9 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 15 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 4 & 27 & 4 \\ 9 & 27 & 15 & 15 & 0 & 27 & 121 & 19 \\ 2 & 4 & 2 & 2 & -4 & 4 & 19 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [100] \quad 1^{\overline{111}} 4^{\overline{131}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 7 & 4 \\ 2 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 6 & 4 \\ 5 & 7 & 5 & 0 & 6 & 8 & 6 \\ 3 & 4 & 2 & -3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[101] \quad 1^{\overline{111}} 4^{\overline{131}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -48, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 5 & 3 \\ 4 & 4 & 3 & 0 & 5 & 7 & 4 \\ 2 & 3 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 6 & 4 \\ 5 & 7 & 5 & 0 & 6 & 8 & 6 \\ 3 & 4 & 2 & -3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

4.1013 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{\overline{111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 148, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 5 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 6 & 9 & 4 \\ 3 & 5 & 3 & 0 & 6 & 4 & 9 & 4 \\ 5 & 9 & 6 & 0 & 9 & 9 & 13 & 7 \\ 3 & 4 & 2 & -4 & 4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 3 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & -4 & 4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1111}} 3^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 6 & 3 \\ 8 & 14 & 9 & 0 & 9 & 37 & 24 & 11 \\ 5 & 9 & 6 & 0 & 6 & 24 & 13 & 7 \\ 3 & 4 & 2 & -4 & 3 & 11 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1112}} 3^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 274, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 18 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 3 & 18 & 4 \\ 2 & 4 & 3 & 0 & 3 & 1 & 12 & 3 \\ 10 & 18 & 12 & 0 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & -4 & 4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1113}} 3^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 454, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 27 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 3 & 27 & 4 \\ 2 & 4 & 3 & 0 & 3 & 1 & 18 & 3 \\ 15 & 27 & 18 & 0 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & -4 & 4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{11111}} 3 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 660$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 & 4 \\ 5 & 9 & 12 & 0 & 6 & 6 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 1 & 3 & 3 \\ 2 & 4 & 6 & 0 & 3 & 1 & 3 \\ 2 & 4 & 6 & 0 & 3 & 3 & 1 \\ 3 & 4 & 6 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{1113}} 3 1^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 676$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.8$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 22 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 3 & 21 & 4 \\ 2 & 4 & 3 & 0 & 3 & 1 & 15 & 3 \\ 12 & 22 & 15 & 0 & 21 & 15 & 85 & 17 \\ 3 & 4 & 2 & -4 & 4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{\overline{1114}} 3 1^{\overline{114}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 706$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 36 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 24 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 3 & 36 & 4 \\ 2 & 4 & 3 & 0 & 3 & 1 & 24 & 3 \\ 20 & 36 & 24 & 0 & 36 & 24 & 238 & 28 \\ 3 & 4 & 2 & -4 & 4 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{\overline{1111}} 3 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 712$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 1 \\ 3 & 4 & 2 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{\overline{1111}} 3 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 724$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 2 & -4 & 3 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{\overline{1112}} 3 1^{\overline{1111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 796$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.5$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 4 & 0 & 5 & 9 & 13 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 9 & 12 & 4 \\ 5 & 9 & 6 & 0 & 9 & 13 & 21 & 7 \\ 7 & 13 & 9 & 0 & 12 & 21 & 28 & 10 \\ 3 & 4 & 2 & -4 & 4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{\overline{1111}} 3 1^{\overline{1111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 988$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 9 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 9 & 9 & 4 \\ 5 & 9 & 6 & 0 & 9 & 13 & 15 & 7 \\ 5 & 9 & 6 & 0 & 9 & 15 & 13 & 7 \\ 3 & 4 & 2 & -4 & 4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{\overline{11111}} 3 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 1032$, $(n_+, n_-, n_0) =$

$$(1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 & 9 & 4 \\ 5 & 9 & 12 & 0 & 6 & 6 & 15 & 6 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 6 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 15 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 6 & -4 & 3 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1122} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 4 & 0 & 5 & 13 & 13 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 12 & 12 & 4 \\ 7 & 13 & 9 & 0 & 12 & 28 & 30 & 10 \\ 7 & 13 & 9 & 0 & 12 & 30 & 28 & 10 \\ 3 & 4 & 2 & -4 & 4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1114} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1180, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 31 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 3 & 30 & 4 \\ 2 & 4 & 3 & 0 & 3 & 1 & 21 & 3 \\ 17 & 31 & 21 & 0 & 30 & 21 & 172 & 24 \\ 3 & 4 & 2 & -4 & 4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 6 & 3 \\ 5 & 9 & 6 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 6 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & 2 & -4 & 3 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{11111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 2368, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 15 & 15 & 7 \\ 5 & 9 & 6 & 0 & 15 & 13 & 15 & 7 \\ 5 & 9 & 6 & 0 & 15 & 15 & 13 & 7 \\ 3 & 4 & 2 & -4 & 7 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

4.1014 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{1111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 5 & 2 & 3 \\ 4 & 9 & 4 & 0 & 12 & 12 & 4 & 5 \\ 2 & 4 & 0 & 0 & 6 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 12 & 6 & 0 & 13 & 15 & 6 & 7 \\ 5 & 12 & 6 & 0 & 15 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 6 & 0 & 2 \\ 3 & 5 & 2 & -4 & 7 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 228, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 231, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 3 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 9 & 6 & 12 & 6 \\ 3 & 4 & 2 & -4 & 4 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3^{\overline{111}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 507, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 9 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 6 & 0 & 9 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & 2 & -4 & 4 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} 3^{\overline{111}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 651, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3^{\overline{111}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 813, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 21 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 18 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 & 3 \\ 11 & 21 & 18 & 0 & 15 & 76 & 15 & 16 \\ 2 & 4 & 3 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 16 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1121}} 3^{\overline{111}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1173, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 13 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 12 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 & 3 \\ 7 & 13 & 12 & 0 & 9 & 28 & 9 & 10 \\ 2 & 4 & 3 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1131}} 3^{\overline{111}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1227, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 17 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 15 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 & 3 \\ 9 & 17 & 15 & 0 & 12 & 49 & 12 & 13 \\ 2 & 4 & 3 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1015 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{111}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 15, \ \Delta = -93, \ \det = 86, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.56 \rightarrow 14, \ T_{\max} = 15, \ T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 2 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 3 & 2 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 15, \ \Delta = -93, \ \det = 126, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.43 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 & 2 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 1 & 2 & 3 \\ 1 & 3 & 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 3 & -4 & 3 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{112} 3^{121} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 482, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 1 & 3 & 9 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 7 & 2 & 0 \\ 7 & 9 & 0 & 7 & 14 & 5 & 8 \\ 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{112} 3^{121} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 626, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 2 & 0 \\ 7 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{112} 3^{121} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 896, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 6 & 2 & 0 \\ 8 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & 0 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{112} 3^{121} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 914, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 2 & 0 \\ 9 & 15 & 0 & 10 & 38 & 8 & 12 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{112} 3^{121} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 1280, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 1 & 3 & 15 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 10 & 2 & 0 \\ 10 & 15 & 0 & 10 & 38 & 8 & 12 \\ 2 & 3 & 0 & 2 & 8 & 0 & 2 \\ 3 & 4 & -1 & 0 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

4.1016 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{su2n3mix}$

$$T = 10 \quad [0] \quad 1^{113} 1^{111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 494, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 5 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 4 & 9 & 4 \\ 3 & 9 & 5 & 5 & 0 & 9 & 12 & 6 \\ 2 & 4 & 2 & 2 & -4 & 4 & 6 & -1 \end{pmatrix}_{b_0}$$

4.1017 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{121} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 & 3 \\ 3 & 4 & 8 & 0 & 9 & 5 & 4 \\ 4 & 8 & 9 & 0 & 12 & 6 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 12 & 0 & 13 & 9 & 7 \\ 3 & 5 & 6 & 0 & 9 & 3 & 3 \\ 3 & 4 & 5 & -4 & 7 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 2 & 3 \\ 4 & 9 & 4 & 0 & 12 & 4 & 5 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 12 & 6 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 5 & 2 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] 1^{114} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 2 & 3 \\ 4 & 9 & 4 & 0 & 36 & 4 & 5 \\ 2 & 4 & 0 & 0 & 15 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 14 & 36 & 15 & 0 & 121 & 15 & 19 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 5 & 2 & -4 & 19 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 6 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 & 3 \\ 3 & 4 & 2 & -4 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 2 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] 1^{111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] 1^{113} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 2 & 3 \\ 4 & 9 & 4 & 0 & 28 & 4 & 5 \\ 2 & 4 & 0 & 0 & 12 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 28 & 12 & 0 & 73 & 12 & 15 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 5 & 2 & -4 & 15 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] 1^{112} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 2 & 3 \\ 4 & 9 & 4 & 0 & 20 & 4 & 5 \\ 2 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 20 & 9 & 0 & 37 & 9 & 11 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 5 & 2 & -4 & 11 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[8] 1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 8 & 14 & 9 & 0 & 9 & 36 & 10 \\ 3 & 4 & 2 & -4 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[9] 1^{122} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -267, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.7$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 5 & 0 & 13 & 8 & 4 \\ 3 & 5 & 3 & 0 & 12 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 12 & 0 & 28 & 18 & 10 \\ 4 & 8 & 6 & 0 & 18 & 9 & 5 \\ 3 & 4 & 3 & -4 & 10 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{132} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -294, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 5 & 0 & 17 & 8 & 4 \\ 3 & 5 & 3 & 0 & 15 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 49 & 24 & 13 \\ 4 & 8 & 6 & 0 & 24 & 9 & 5 \\ 3 & 4 & 3 & -4 & 13 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{111} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -402, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.79 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 & 4 \\ 2 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 37 & 9 & 11 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 2 & -4 & 11 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{113} 31^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -420, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.09 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 2 & 3 \\ 4 & 9 & 4 & 0 & 32 & 4 & 5 \\ 2 & 4 & 0 & 0 & 15 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 13 & 32 & 15 & 0 & 100 & 15 & 18 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 5 & 2 & -4 & 18 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -435, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 9 & 4 \\ 5 & 9 & 6 & 0 & 9 & 12 & 6 \\ 3 & 4 & 2 & -4 & 4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{111} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -510, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 2 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{114} 31^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -555, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.10 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 2 & 3 \\ 4 & 9 & 4 & 0 & 44 & 4 & 5 \\ 2 & 4 & 0 & 0 & 21 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 18 & 44 & 21 & 0 & 193 & 21 & 25 \\ 2 & 4 & 3 & 0 & 21 & 0 & 2 \\ 3 & 5 & 2 & -4 & 25 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -645, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -651, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.05 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 6 & 12 & 6 \\ 3 & 4 & 2 & -4 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{114} 31^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -690, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.08 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 2 & 3 \\ 4 & 9 & 4 & 0 & 40 & 4 & 5 \\ 2 & 4 & 0 & 0 & 18 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 16 & 40 & 18 & 0 & 154 & 18 & 22 \\ 2 & 4 & 3 & 0 & 18 & 0 & 2 \\ 3 & 5 & 2 & -4 & 22 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{141} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -726, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 & 4 \\ 3 & 5 & 3 & 0 & 18 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 18 & 0 & 76 & 15 & 16 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 3 & -4 & 16 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -789, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 2 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{121} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -906, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 & 4 \\ 3 & 5 & 3 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 12 & 0 & 28 & 9 & 10 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 3 & -4 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{131} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -933, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 & 4 \\ 3 & 5 & 3 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 0 & 49 & 12 & 13 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 3 & -4 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -1200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 0 & 15 & 12 & 6 \\ 3 & 4 & 2 & -4 & 7 & 6 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \quad 1^{111} 3 \quad 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 21, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.43 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 5 & 1 & 3 \\ 4 & 9 & 0 & 4 & 12 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 7 & 0 & 0 \\ 5 & 12 & 0 & 7 & 14 & 0 & 8 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 0 & 8 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{111} 3 \quad 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 39, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 1 & 3 \\ 4 & 9 & 0 & 4 & 8 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 3 & 0 & 0 \\ 4 & 8 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 0 & 4 & -1 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{111} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & 0 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

[27] $1^{111} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 182, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.26 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{11} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 477, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 4 & 6 & 4 \\ 2 & 6 & 3 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 2 & 2 & -4 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

[29] $1^{111} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 486, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 9 & 3 & 3 \\ 1 & 3 & 3 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 9 & 6 & 0 & 13 & 6 & 7 \\ 1 & 3 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 3 & 3 & 2 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{111} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 774, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 6 & 0 & 13 & 9 & 7 \\ 2 & 6 & 3 & 3 & 0 & 9 & 3 & 3 \\ 2 & 4 & 2 & 2 & -4 & 7 & 3 & -1 \end{pmatrix}_{b_0}$$

$T = 11$ [31] $1^{11} 4^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 0 & 5 & 4 & 4 \\ 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 3 & 4 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & 2 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{11} 4^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -108, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 2 & 3 \\ 4 & 4 & 3 & 0 & 5 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 2 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{11} 4^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -210, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 2 & 3 \\ 4 & 4 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 4 & 2 & 0 & 1 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 2 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{11} 1^{11} 31^{11} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, \quad T_{\max} = 15, \quad \Delta = -91, \quad \det = -378, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.25 \rightarrow 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 2 & 3 \\ 4 & 4 & 3 & 0 & 9 & 3 & 4 \\ 2 & 3 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 9 & 5 & 0 & 13 & 5 & 7 \\ 2 & 3 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & 2 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

4.1018 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{1111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 48, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 2 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 4 & 12 & 5 \\ 2 & 4 & 0 & 0 & 6 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 12 & 6 & 0 & 13 & 6 & 15 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 6 & 2 \\ 5 & 12 & 6 & 0 & 15 & 6 & 13 & 7 \\ 3 & 5 & 2 & -4 & 7 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 228, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & -4 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 231, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 9 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 6 & 0 & 9 & 12 & 6 & 6 \\ 2 & 4 & 3 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & 2 & -4 & 4 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 507, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 3 & 0 & 4 & 3 & 9 & 4 \\ 5 & 9 & 6 & 0 & 9 & 6 & 13 & 7 \\ 2 & 4 & 3 & 0 & 3 & 0 & 6 & 2 \\ 3 & 4 & 2 & -4 & 4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 651, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 9 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 3 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1114} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 813, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 21 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 15 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 15 & 2 \\ 11 & 21 & 18 & 0 & 15 & 15 & 76 & 16 \\ 3 & 4 & 3 & -4 & 3 & 2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1112}} 3 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 1173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 13 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 9 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 9 & 2 \\ 7 & 13 & 12 & 0 & 9 & 9 & 28 & 10 \\ 3 & 4 & 3 & -4 & 3 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1113}} 3 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 1227, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 17 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 12 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 12 & 2 \\ 9 & 17 & 15 & 0 & 12 & 12 & 49 & 13 \\ 3 & 4 & 3 & -4 & 3 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

4.1019 $\text{NHC}(-2)(-3) + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 459, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3 1^{\overline{111}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 1296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 6 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 6 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 7 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1020 $\text{NHC}(-2)(-3) + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{\overline{13}} 3^{\overline{1}} 2 1^{\overline{11}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.36 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 3 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & -2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 5 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{13}} 3^{\overline{1}} 2 1^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.11 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 9 & 0 & 4 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 2 & 0 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 5 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{13}} 3^{\overline{1}} 2 1^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 0 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{13}} 3^{\overline{1}} 1^{\overline{1}} 3 1^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -163, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.21 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 2 & 3 & 2 & 0 & 0 & 2 \\ 2 & 3 & 3 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -193, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.21 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 2 \\ 1 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 2 & 0 & 0 & 2 \\ 2 & 4 & 2 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{13}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -363, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 2 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -508, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 & 2 \\ 3 & 9 & 6 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 5 & 0 & 0 & 2 \\ 2 & 4 & 2 & 6 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1021 NHC(-2)(-3) + $\textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$

$$T = 10 \quad [0] \ 1^{111}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 3 & 3 & 3 & 0 & 2 & 1 \\ 2 & 3 & 3 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.43 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 3 & 2 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 3 & 3 & 3 & 0 & 2 & 1 & 3 \\ 2 & 4 & 2 & 3 & -4 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 130, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 7 & 0 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 5 & 9 & 0 & 7 & 5 & 14 & 8 \\ 3 & 4 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 146, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 226, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 3 & 0 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 0 & 2 & 3 \\ 1 & 3 & 3 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 2 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & 2 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 10 & 0 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 8 & 15 & 0 & 10 & 8 & 38 & 12 \\ 3 & 4 & -1 & 0 & 2 & 12 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 3 & 0 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 5 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 2 & 4 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 3 & 0 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 4 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 7 & 0 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 6 & 9 & 0 & 7 & 5 & 14 & 8 \\ 3 & 4 & -1 & 0 & 2 & 8 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 7 & 0 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 7 & 9 & 0 & 7 & 5 & 14 & 8 \\ 3 & 4 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_b$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 6 & 0 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 7 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 2 & 8 & 1 \end{pmatrix} b_0$$

1¹²

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 6 & 0 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 8 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & 0 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 914, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 10 & 0 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 9 & 15 & 0 & 10 & 8 & 38 & 12 \\ 3 & 4 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{12} 3^{12} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 1280, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 10 & 0 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 10 & 15 & 0 & 10 & 8 & 38 & 12 \\ 3 & 4 & -1 & 0 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

4.1022 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 1 & 3 \\ 4 & 9 & 0 & 4 & 4 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 0 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 0 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 \\ 1 & 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 3 & 3 & 3 & 0 & 2 & 0 \\ 2 & 3 & 3 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 489, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 6 \\ 1 & 4 & 0 & 3 & 0 & 2 & 3 \\ 1 & 3 & 3 & 0 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 \\ 1 & 3 & 2 & 2 & 0 & 0 & 3 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1023 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \text{n3mix}$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 2 & 0 \\ 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 5 & 3 \\ 2 & 4 & 3 & 3 & 0 & 6 & 3 \\ 1 & 3 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 5 & 6 & 5 & 0 & 7 & 5 \\ 2 & 3 & 3 & 2 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 7 & 12 & 0 & 9 & 26 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 5 & 0 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & 0 & 6 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{11}3^{11}14^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -265, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 3 & 0 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 14 & 2 \\ 1 & 3 & 3 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 14 & 14 & 0 & 120 & 18 \\ 2 & 4 & 2 & 2 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{11}3^{11}11^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 5 & 2 \\ 1 & 3 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 5 & 5 & 0 & 12 & 6 \\ 2 & 4 & 2 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{11}3^{11}13^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -328, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 3 & 0 & 21 & 4 \\ 1 & 4 & 0 & 3 & 0 & 11 & 2 \\ 1 & 3 & 3 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 11 & 11 & 0 & 72 & 14 \\ 2 & 4 & 2 & 2 & -4 & 14 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -332, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 8 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & 0 & 10 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{11}3^{11}12^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -337, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 3 & 0 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 8 & 2 \\ 1 & 3 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 8 & 8 & 0 & 36 & 10 \\ 2 & 4 & 2 & 2 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 8 & 12 & 0 & 9 & 26 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{11}3^{11}21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 5 & 0 \\ 6 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{1^{12}} 3^{1^1} 21^{1^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -403, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 5 & 0 \\ 7 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{1^{12}} 3^{1^1} 21^{1^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -428, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 10 & 18 & 0 & 12 & 56 & 14 \\ 3 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{1^{12}} 3^{1^1} 21^{1^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -451, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 9 & 0 \\ 9 & 12 & 0 & 9 & 26 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{1^{11}} 3^{1^1} 21^{1^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -635, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 9 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{1^{11}} 3^{1^1} 21^{1^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -779, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 11 & 18 & 0 & 12 & 56 & 14 \\ 3 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{1^{12}} 3^{1^1} 21^{1^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -832, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 8 & 0 \\ 10 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{1^{12}} 3^{1^1} 21^{1^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -1024, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 12 & 0 \\ 12 & 18 & 0 & 12 & 56 & 14 \\ 3 & 4 & -1 & 0 & 14 & 1 \end{pmatrix}_{b_0}$$

4.1024 $\text{NHC}(-2)(-3) + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2\textcolor{blue}{\mathfrak{n}}_3\textcolor{blue}{\mathfrak{mix}} + \textcolor{blue}{\mathfrak{su}}_2$

$T = 10 \quad [0] \quad 1^{1^1} 3^{1^1} 1^{1^1} 31^{1^1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 494, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 5 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 5 & 5 & 0 & 12 & 9 & 6 \\ 2 & 6 & 3 & 3 & 0 & 9 & 4 & 4 \\ 2 & 4 & 2 & 2 & -4 & 6 & 4 & -1 \end{pmatrix}_{b_0}$$

4.1025 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 5 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 5 & 0 & 8 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.11 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 6 & 0 & 12 & 6 \\ 3 & 5 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 246, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.02 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 0 & 36 & 10 \\ 3 & 4 & 2 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [8] \ 1^{11} 3^{11} 2^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.20 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 9 & 0 & 4 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^{11} 2^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 5 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 3 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{12} 3^{11} 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -43, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -3 & 2 & -1 \\ 2 & 4 & 0 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 31^{13} 1^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 31^{13} 1^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -135, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 3 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3^{11} 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.24 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 0 & 3 & 4 & 1 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3^{11} 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -195, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11} 31^{13} 1^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -279, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111} 3^{11} 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.21 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 \\ 4 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [17] \ 1^{11} 4^{13} 1^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_3} \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 30, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 5 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{11} 4^{13} 1^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{e}_6} + \textcolor{red}{\mathfrak{su}_3} \quad T = 10, T_{\max} = 69, \Delta = -121, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 69, T \in [15, 69]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{111} 1^{111} 4^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_3$ $T = 10$, $T_{\max} = 69$, $\Delta = -121$, $\det = 132$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 14.18 \rightarrow 15$, $T_{\max} = 69$, $T \in [15, 69]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

4.1026 $\text{NHC}(-2)(-3) + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{NHC}(-2)(-3)$

$T = 10$ [0] $1^{111} 3^{111} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_3 + [\text{su}_2 \cdot \text{g}_2]$ $T = 10$, $T_{\max} = 22$, $\Delta = -101$, $\det = 249$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.39 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 3 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & 3 & -4 & 2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111} 1^{111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_3 + [\text{su}_2 \cdot \text{g}_2]$ $T = 10$, $T_{\max} = 22$, $\Delta = -101$, $\det = 825$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.42 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 4 & 3 \\ 3 & 5 & 3 & 0 & 3 & 4 & 3 & 3 \\ 3 & 4 & 2 & -4 & 2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

4.1027 $\text{NHC}(-2)(-3) + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_2$

$T = 9$ [0] $1^{121} 3^{111} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -6$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 5 & 0 & 8 & 9 & 4 \\ 3 & 5 & 3 & 0 & 6 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 9 & 12 & 5 \\ 5 & 9 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & 3 & -4 & 5 & 7 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111} 1^{111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.12 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 5 & 3 \\ 4 & 9 & 4 & 0 & 4 & 12 & 5 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 12 & 6 & 0 & 6 & 13 & 7 \\ 3 & 5 & 2 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{114} 3^{111} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.12 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 14 & 3 \\ 4 & 9 & 4 & 0 & 4 & 36 & 5 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 14 & 36 & 15 & 0 & 15 & 121 & 19 \\ 3 & 5 & 2 & -4 & 2 & 19 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{111} 1^{111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_6 + \text{su}_3 + \text{su}_2$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -150$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.23 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 \\ 3 & 5 & 3 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & 3 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -159, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.87 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 2 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -231, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{113} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -231, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.08 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 11 & 3 \\ 4 & 9 & 4 & 0 & 4 & 28 & 5 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 11 & 28 & 12 & 0 & 12 & 73 & 15 \\ 3 & 5 & 2 & -4 & 2 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -231, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.08 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 8 & 3 \\ 4 & 9 & 4 & 0 & 4 & 20 & 5 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 8 & 20 & 9 & 0 & 9 & 37 & 11 \\ 3 & 5 & 2 & -4 & 2 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -258, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.18 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 & 4 \\ 2 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 0 & 36 & 9 & 10 \\ 2 & 4 & 3 & 0 & 9 & 1 & 3 \\ 3 & 4 & 2 & -4 & 10 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{122} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -267, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.7$$

$$T_{\min} = 13.07 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 5 & 0 & 8 & 13 & 4 \\ 3 & 5 & 3 & 0 & 6 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 9 & 18 & 5 \\ 7 & 13 & 12 & 0 & 18 & 28 & 10 \\ 3 & 4 & 3 & -4 & 5 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{123} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -294, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.06 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 5 & 0 & 8 & 17 & 4 \\ 3 & 5 & 3 & 0 & 6 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 6 & 0 & 9 & 24 & 5 \\ 9 & 17 & 15 & 0 & 24 & 49 & 13 \\ 3 & 4 & 3 & -4 & 5 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -402, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.79 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 8 & 14 & 9 & 0 & 9 & 37 & 11 \\ 3 & 4 & 2 & -4 & 2 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -420, \ (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 13 & 3 \\ 4 & 9 & 4 & 0 & 4 & 32 & 5 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 13 & 32 & 15 & 0 & 15 & 100 & 18 \\ 3 & 5 & 2 & -4 & 2 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 & 3 \\ 3 & 4 & 4 & 0 & 9 & 5 & 4 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 9 & 6 \\ 3 & 5 & 3 & 0 & 9 & 4 & 4 \\ 3 & 4 & 2 & -4 & 6 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 2 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -555, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 18 & 3 \\ 4 & 9 & 4 & 0 & 4 & 44 & 5 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 18 & 44 & 21 & 0 & 21 & 193 & 25 \\ 3 & 5 & 2 & -4 & 2 & 25 & -1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -645, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 \\ 3 & 5 & 3 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -651, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 6 & 6 \\ 2 & 4 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 2 & -4 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{114} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -690, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 16 & 3 \\ 4 & 9 & 4 & 0 & 4 & 40 & 5 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 16 & 40 & 18 & 0 & 18 & 154 & 22 \\ 3 & 5 & 2 & -4 & 2 & 22 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{114} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -726, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 5 & 0 & 4 & 21 & 4 \\ 3 & 5 & 3 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 11 & 21 & 18 & 0 & 15 & 76 & 16 \\ 3 & 4 & 3 & -4 & 2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -789, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 2 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{112}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -906, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 13 & 4 \\ 3 & 5 & 3 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 7 & 13 & 12 & 0 & 9 & 28 & 10 \\ 3 & 4 & 3 & -4 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{113}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -933, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 5 & 0 & 4 & 17 & 4 \\ 3 & 5 & 3 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 9 & 17 & 15 & 0 & 12 & 49 & 13 \\ 3 & 4 & 3 & -4 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -1200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 15 & 6 \\ 5 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 2 & -4 & 6 & 7 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \ 1^{111}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 5 & 3 \\ 4 & 9 & 0 & 4 & 0 & 12 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 & 7 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 0 & 7 & 0 & 14 & 8 \\ 3 & 5 & -1 & 0 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{111}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 4 & 3 \\ 4 & 9 & 0 & 4 & 0 & 8 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 0 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & 0 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{111}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 126, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -2 & 3 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & -1 & 0 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{111}321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 182, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.26 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{11}31^{111}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 477, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 3 & 6 & 3 \\ 2 & 6 & 3 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & 2 & 2 & -4 & 3 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{\overline{1111}} 31^{\overline{11} 31^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 486, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 6 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 9 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 6 & 2 \\ 3 & 6 & 9 & 6 & 0 & 6 & 13 & 7 \\ 2 & 3 & 3 & 2 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{\overline{1111}} 31^{\overline{11} 31^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 774, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 3 & 9 & 3 \\ 3 & 9 & 6 & 6 & 0 & 9 & 13 & 7 \\ 2 & 4 & 2 & 2 & -4 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [31] \ 1^{\overline{11}} 4^{\overline{1} 31^1} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 3 & 0 & 4 & 5 & 4 \\ 3 & 3 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 3 & 5 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 2 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{\overline{11}} 4^{\overline{1} 31^1} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -108, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 & 3 \\ 4 & 4 & 3 & 0 & 3 & 5 & 4 \\ 2 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 2 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{\overline{1} 31^{\overline{11}}} 4^{\overline{1} 11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -210, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 & 3 \\ 4 & 4 & 3 & 0 & 3 & 4 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 3 & 4 & 2 & 0 & 2 & 1 & 3 \\ 3 & 4 & 2 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{\overline{11}} 4^{\overline{1} 31^{\overline{11}}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -378, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 6 & 3 \\ 4 & 4 & 3 & 0 & 3 & 9 & 4 \\ 2 & 3 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 5 & 2 \\ 6 & 9 & 5 & 0 & 5 & 13 & 7 \\ 3 & 4 & 2 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

4.1028 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 31^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 5 & 5 & 3 \\ 4 & 9 & 4 & 0 & 4 & 12 & 12 & 5 \\ 2 & 4 & 0 & 0 & 3 & 6 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 6 & 2 \\ 5 & 12 & 6 & 0 & 6 & 13 & 15 & 7 \\ 5 & 12 & 6 & 0 & 6 & 15 & 13 & 7 \\ 3 & 5 & 2 & -4 & 2 & 7 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 228, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 1 \\ 3 & 4 & 3 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 231, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 5 & 4 & 4 \\ 2 & 4 & 0 & 0 & 6 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 6 & 0 & 12 & 9 & 6 & 6 \\ 3 & 5 & 3 & 0 & 9 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 6 & 3 & 1 & 3 \\ 3 & 4 & 2 & -4 & 6 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 507, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 3 & 5 & 3 & 0 & 3 & 4 & 9 & 4 \\ 5 & 9 & 6 & 0 & 6 & 9 & 13 & 7 \\ 3 & 4 & 2 & -4 & 2 & 4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 651, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 9 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 2 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1114}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 813, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 21 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 18 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 15 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 15 & 3 \\ 11 & 21 & 18 & 0 & 15 & 15 & 76 & 16 \\ 3 & 4 & 3 & -4 & 2 & 3 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1112}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1173, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 13 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 12 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 9 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 9 & 3 \\ 7 & 13 & 12 & 0 & 9 & 9 & 28 & 10 \\ 3 & 4 & 3 & -4 & 2 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1113}} 31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1227, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 9 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 17 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 15 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 12 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 12 & 3 \\ 9 & 17 & 15 & 0 & 12 & 12 & 49 & 13 \\ 3 & 4 & 3 & -4 & 2 & 3 & 13 & -1 \end{pmatrix}_{b_0}$$

4.1029 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 459, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 & 2 \\ 3 & 4 & 3 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 1296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 9 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 3 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & 3 & -4 & 2 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1030 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{11} 3^1 2^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 16, \Delta = -96, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 16, T \in [14, 16]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 0 & 2 \\ 3 & 5 & -1 & 0 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^1 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 2 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 3 & 2 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 489, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 & 3 \\ 1 & 3 & 2 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & 2 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1031 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.11 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 & 3 \\ 4 & 9 & 4 & 0 & 4 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 8 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 2 & -4 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 5 & 3 \\ 4 & 9 & 4 & 0 & 4 & 12 & 5 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 12 & 6 & 0 & 6 & 12 & 6 \\ 3 & 5 & 2 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -405, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [3] \quad 1^{111} 3^{11} 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 321, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -2 & 3 & 0 \\ 3 & 6 & 0 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

4.1032 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{1111} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 459, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 0 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 1296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 9 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 9 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 3 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 3 & -4 & 2 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

4.1033 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_7$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 3^{11} 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_7 \quad T = 8, T_{\max} = 102, \Delta = -199, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.82 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & -4 & -2 \\ 3 & 5 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 3^{11} 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_7 \quad T = 8, T_{\max} = 102, \Delta = -199, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 0 & 0 & -5 & -3 \\ 2 & 2 & -1 & -1 & -3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 3^{11} 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_7 \quad T = 8, T_{\max} = 102, \Delta = -199, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.62 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & -4 & -2 \\ 3 & 4 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 3^{11} 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_7 \quad T = 8, T_{\max} = 102, \Delta = -199, \det = 170, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 4 \\ 5 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & -4 & -2 \\ 4 & 4 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{12} 3^{11} 3^{11} 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_7 \quad T = 8, T_{\max} = 102, \Delta = -199, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 14.62 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 4 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 5 & 2 & 3 \\ 1 & 1 & 0 & 2 & -5 & -3 \\ 2 & 2 & -1 & 3 & -3 & -2 \end{pmatrix}_{b_0}$$

[5] $1^{11} 1^{13} 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_7$ $T = 8$, $T_{\max} = 102$, $\Delta = -199$, $\det = 358$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 14.61 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 6 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 15 & 3 & 5 \\ 1 & 1 & 0 & 3 & -5 & -3 \\ 2 & 2 & -1 & 5 & -3 & -2 \end{pmatrix}_{b_0}$$

$T = 9$ [6] $1^{11} 1^{13} 221^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_7$ $T = 9$, $T_{\max} = 102$, $\Delta = -170$, $\det = -9$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 18.33 \rightarrow 19$, $T_{\max} = 102$, $T \in [19, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 6 & 0 & 6 & 6 & 6 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 6 & 0 & -3 & -1 & -1 \\ 3 & 6 & -1 & -1 & -4 & -4 \end{pmatrix}_{b_0}$$

4.1034 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_7 + \text{NHC}(-2)(-3)$

$T = 10$ [0] $1^{111} 1^{11} 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_7 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10$, $T_{\max} = 101$, $\Delta = -149$, $\det = 103$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 15.03 \rightarrow 16$, $T_{\max} = 101$, $T \in [16, 101]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 1 & 3 \\ 4 & 9 & 0 & 4 & 8 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 3 & 0 & -2 \\ 3 & 8 & 0 & 3 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & -1 & -2 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

4.1035 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{so}_8$

$T = 9$ [0] $1^{11} 1^{11} 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 103$, $\Delta = -198$, $\det = -294$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 15.63 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -5 & 1 & -3 \\ 1 & 2 & 0 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & -1 & -3 & 0 & -3 \end{pmatrix}_{b_0}$$

4.1036 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2$

$T = 9$ [0] $1^{11} 1^{11} 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -113$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 14.63 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & -1 & -2 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11} 1^{11} 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -124$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 14.97 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -5 & 1 & -3 \\ 1 & 2 & 0 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & -3 & 2 & -2 \end{pmatrix}_{b_0}$$

[3] $1^{111} 1^{11} 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -156$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 14.90 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 & 3 \\ 4 & 9 & 0 & 4 & 8 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 3 & 0 & -2 \\ 3 & 8 & 0 & 3 & 4 & 4 & 4 \\ 3 & 5 & -1 & -2 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{112} 1^{11} 3 21^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -156$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -4 & 6 & -2 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_7 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.84 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 3 & -2 \\ 4 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_7 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -201, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_7 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.63 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_7 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 14.79 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & -2 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{113} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_7 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -273, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 9 & -2 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_7 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.68 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -4 & 3 & -2 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{114} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_7 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_7 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.64 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111} 3 \ 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_7 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -377, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.63 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

- [13] $1^{115} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -417$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.1$
 $T_{\min} = 15.19 \rightarrow 16$, $T_{\max} = 102$, $T \in [16, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 30 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 15 & -2 \\ 15 & 30 & 0 & 15 & 148 & 20 \\ 3 & 4 & -1 & -2 & 20 & -2 \end{pmatrix}_{b_0}$$
- [14] $1^{121} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -476$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 14.62 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 4 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 5 & 2 & 4 \\ 1 & 1 & 0 & 2 & -5 & 1 \\ 1 & 2 & 0 & 4 & 1 & 0 \\ 2 & 2 & -1 & 3 & -3 & 2 \end{pmatrix}_{b_0}$$
- [15] $1^{112} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -641$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & -2 \\ 7 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$
- [16] $1^{112} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -896$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 14.65 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & -2 \\ 8 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$
- [17] $1^{112} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -932$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 14.64 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & -2 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$
- [18] $1^{113} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -932$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 14.70 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 9 & -2 \\ 10 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$
- [19] $1^{112} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -993$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 14.64 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & -2 \\ 9 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 8 & -1 \end{pmatrix}_{b_0}$$
- [20] $1^{114} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -1241$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 14.73 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 13 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$
- [21] $1^{113} 3 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_7 + \text{su}_2$ $T = 9$, $T_{\max} = 102$, $\Delta = -169$, $\det = -1433$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 9 & -2 \\ 11 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[22] $1^{113} 3 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{e}_7} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -1776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.66 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 9 & -2 \\ 12 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[23] $1^{113} 3 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{e}_7} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -1961, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 9 & -2 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -2 & 12 & -1 \end{pmatrix}_{b_0}$$

[24] $1^{114} 3 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{e}_7} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -1988, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.69 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

[25] $1^{114} 3 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{e}_7} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -2577, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.67 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

[26] $1^{114} 3 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{e}_7} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 102, \Delta = -169, \det = -3008, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.66 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 12 & -2 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -2 & 16 & -1 \end{pmatrix}_{b_0}$$

4.1037 $\text{NHC}(-2)(-3) + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{e}_7} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2}$

$T = 10 \quad [0] 1^{111} 3 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{e}_7} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 68, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.65 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{e}_7} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 79, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.84 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3 21^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{e}_7} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 100, \Delta = -139, \det = 112, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 3 & -2 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

4.1038 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{\overline{1111}} 3^{\overline{11}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 16, T_{\max} = 23, T \in [16, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -5 & 1 & 1 & -3 \\ 1 & 2 & 0 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 2 & 0 & 2 \\ 1 & 2 & 0 & 0 & 1 & 2 & 2 & 0 \\ 2 & 2 & -1 & -1 & -3 & 2 & 2 & -3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1111}} 3^{\overline{11}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 23, \Delta = -109, \det = -202, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.85 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

4.1039 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \quad 1^{\overline{1111}} 3^{\overline{11}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 0 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1040 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 11 \quad [0] \quad 1^{\overline{1111}} 3^{\overline{11}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 38, \Delta = -118, \det = -284, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 38, T \in [16, 38]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & -2 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

4.1041 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \quad 1^{\overline{1111}} 3^{\overline{11}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 4 & -1 & -2 & 0 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{112}} 3^{\overline{11}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.80 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & 2 & -2 \\ 6 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 348, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 2 & -2 \\ 3 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{112}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 604, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 2 & -2 \\ 5 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 2 & -2 \\ 4 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{112}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & 2 & -2 \\ 7 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{112}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 1408, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 6 & 2 & -2 \\ 8 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -2 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$4.1042 \quad \text{NHC}(-2)(-3) + \textcolor{blue}{su}_3 + \textcolor{red}{e}_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$$

$$T = 11 \quad [0] \quad 1^{111}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 2 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 0 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$4.1043 \quad \text{NHC}(-2)(-3) + \textcolor{blue}{su}_3 + \textcolor{red}{e}_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}2n3\text{mix}$$

$$T = 10 \quad [0] \quad 1^{111}3^121^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{e}_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.13 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 5 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 5 & 0 & 12 & 6 \\ 3 & 4 & -1 & -2 & 0 & 6 & -2 \end{pmatrix}_{b_0}$$

4.1044 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 10 \quad [1] \quad 1^{111} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.05 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 1 & 3 \\ 4 & 9 & 0 & 4 & 8 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 3 & 0 & -2 \\ 4 & 8 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -2 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.75 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -2 & 0 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 280, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & -2 & 4 & -2 \end{pmatrix}_{b_0}$$

4.1045 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}_7$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{1111} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 38, \Delta = -118, \det = -284, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 38, T \in [16, 38]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -4 & 3 & 3 & -2 \\ 3 & 6 & 0 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & -2 & 4 & 4 & -2 \end{pmatrix}_{b_0}$$

4.1046 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}_7$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 9 & 0 & 4 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 2 & -2 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 5 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & -2 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & -2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1047 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_{2 \rightarrow (-3)} + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11} 3^1 21^1 + [\textcolor{red}{\mathfrak{s}}\textcolor{blue}{\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}}\textcolor{blue}{\mathfrak{u}}_3 + \textcolor{orange}{\mathfrak{c}}_7 + \textcolor{red}{\mathfrak{s}}\textcolor{blue}{\mathfrak{u}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{s}}\textcolor{blue}{\mathfrak{u}}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 220, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 0 & -2 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 2 & 0 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{112}3^121^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 240, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 14.80 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix}_{b_C}$$

[2] $1^{111}3^121^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_7 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 348, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.76 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 2 & 4 & -1 \end{pmatrix} b_0$$

$$[3] \quad 1^{12} 3^{11} 3^1 21^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}u_3 + \textcolor{brown}{e}7 + \textcolor{red}{s}u_2 \rightarrow (-3) + \textcolor{blue}{s}u_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 604, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 5 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 2 & 4 & -1 \end{pmatrix} b_0$$

[4] $1^{11}3^{11}21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{e}_7 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 4 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -2 & 2 & 4 & -1 \end{pmatrix} b_0$$

[5] $1^{112}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{e}_7 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 7 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix} b_{\text{r}}$$

[6] $1^{12}3^{12}21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}_7 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 10, T_{\max} = 100, \Delta = -142, \det = 1408, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 14.75 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 6 & -2 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 8 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -2 & 2 & 8 & -1 \end{pmatrix} b_r$$

4.1048 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_{2 \rightarrow (-3)} + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11, [0] 1^{11} 3^{11} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{e}_7 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 11, T_{\max} = 25, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.88 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & 0 & 3 & -2 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & 2 & 0 & 4 & -1 \end{pmatrix}_{b_0}$$

$$4.1049 \quad \text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$$

$$T = 10 \quad [0] \ 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 101, \Delta = -151, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 1 & 3 & 3 \\ 4 & 9 & 0 & 4 & 4 & 0 & 5 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 2 & 0 & -2 & -2 \\ 2 & 4 & 0 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 & -1 \\ 3 & 5 & -1 & -2 & 2 & -1 & -2 & -2 \end{pmatrix}_{b_0}$$

$$4.1050 \quad \text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \text{mix}$$

$$T = 9 \quad [0] \ 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -161, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 2 & -2 & -2 \\ 3 & 3 & 0 & 2 & 0 & 2 & 2 \\ 3 & 4 & -1 & -2 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -345, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.81 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & -2 & -2 \\ 5 & 9 & 0 & 5 & 12 & 6 & 6 \\ 3 & 4 & -1 & -2 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.86 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 8 & -2 & -2 \\ 8 & 15 & 0 & 8 & 36 & 10 & 10 \\ 3 & 4 & -1 & -2 & 10 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.72 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & -2 & -2 \\ 6 & 9 & 0 & 5 & 12 & 6 & 6 \\ 3 & 4 & -1 & -2 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -617, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & -2 & -2 \\ 7 & 9 & 0 & 5 & 12 & 6 & 6 \\ 3 & 4 & -1 & -2 & 6 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{12} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -977, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.74 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 8 & -2 & -2 \\ 9 & 15 & 0 & 8 & 36 & 10 & 10 \\ 3 & 4 & -1 & -2 & 10 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{12} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 102, \Delta = -172, \det = -1280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.70 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 8 & -2 \\ 10 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -2 & 10 & -1 \end{pmatrix}_{b_0}$$

4.1051 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}_7$ + su2n3mix + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.13 \rightarrow 16, T_{\max} = 100, T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 1 & 3 \\ 3 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 5 & 0 & -2 \\ 5 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 6 & 0 & -2 \end{pmatrix}_{b_0}$$

4.1052 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}_7$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -178, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 9 & 0 & 4 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{12} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -178, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.94 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -5 & 2 & -3 \\ 2 & 4 & 0 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -1 & -3 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -178, \det = -209, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.03 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 1 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & -1 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -178, \det = -297, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.18 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & -2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 102, \Delta = -178, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.95 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & -2 \\ 4 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & -2 & 3 & -1 \end{pmatrix}_{b_0}$$

4.1053 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}_7$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 10 \quad [1] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 63, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.05 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 4 & 3 \\ 4 & 9 & 0 & 4 & 0 & 8 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 0 & 3 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 0 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -2 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.75 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 3 & 0 & -2 \\ 3 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -2 & 3 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3 \ 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 101, \Delta = -148, \det = 280, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.07 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & -2 & 4 & -2 \end{pmatrix}_{b_0}$$

4.1054 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{111} 3 \ 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 38, \Delta = -118, \det = -284, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 38, T \in [16, 38]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & -2 & 4 & -2 \end{pmatrix}_{b_0}$$

4.1055 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{11} 3^1 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 101, \Delta = -151, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 15.20 \rightarrow 16, T_{\max} = 101, T \in [16, 101]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -4 & 0 & 2 & -2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 0 & 2 \\ 3 & 5 & -1 & -2 & -1 & 2 & -2 \end{pmatrix}_{b_0}$$

4.1056 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_7 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{111} 3 \ 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}_7 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 10, T_{\max} = 102, \Delta = -157, \det = 489, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.28 \rightarrow 16, T_{\max} = 102, T \in [16, 102]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -4 & 3 & -2 \\ 3 & 6 & 0 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & -1 & -2 & 3 & -2 \end{pmatrix}_{b_0}$$

4.1057 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7$

$$T = 8 \quad [0] \ 1^{11} 3 \ 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}'_7 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.93 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}'_7 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 0 & 0 & -4 & -2 \\ 2 & 2 & -1 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 \ 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{e}}'_7 \quad T = 8, T_{\max} = 101, \Delta = -171, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.70 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & -3 & -1 \\ 3 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{e}}'_7 \quad T = 8, \ T_{\max} = 101, \ \Delta = -171, \ \det = 141, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0 \\ T_{\min} = 13.76 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 4 \\ 5 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & -3 & -1 \\ 4 & 4 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{12} 3^{11} 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{e}}'_7 \quad T = 8, \ T_{\max} = 101, \ \Delta = -171, \ \det = 215, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2 \\ T_{\min} = 13.72 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 4 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 5 & 2 & 3 \\ 1 & 1 & 0 & 2 & -4 & -2 \\ 2 & 2 & -1 & 3 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{13} 3^{11} 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{e}}'_7 \quad T = 8, \ T_{\max} = 101, \ \Delta = -171, \ \det = 293, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0 \\ T_{\min} = 13.72 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 6 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 15 & 3 & 5 \\ 1 & 1 & 0 & 3 & -4 & -2 \\ 2 & 2 & -1 & 5 & -2 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [6] \ 1^{13} 3^{11} 221^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{e}}'_7 \quad T = 9, \ T_{\max} = 99, \ \Delta = -142, \ \det = -6, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0 \\ T_{\min} = 18, \ T_{\max} = 99, \ T \in [18, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 6 & 6 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 6 & 0 & -2 & 0 \\ 3 & 6 & -1 & 0 & -3 \end{pmatrix}_{b_0}$$

4.1058 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \ 1^{11} 3^{11} 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{e}}'_7 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 10, \ T_{\max} = 69, \ \Delta = -121, \ \det = 82, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.12 \rightarrow 15, \ T_{\max} = 69, \ T \in [15, 69]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 1 & 3 \\ 4 & 9 & 0 & 4 & 8 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 3 & 0 & -1 \\ 3 & 8 & 0 & 3 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & -1 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

4.1059 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{11} 3^{11} 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{e}}'_7 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 102, \ \Delta = -170, \ \det = -245, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.73 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -4 & 1 & -2 \\ 1 & 2 & 0 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & -1 & -2 & 0 & -2 \end{pmatrix}_{b_0}$$

4.1060 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{11} 3^{11} 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{e}}'_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -94, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.70 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{e}}'_7 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -103, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0 \\ T_{\min} = 14.03 \rightarrow 15, \ T_{\max} = 99, \ T \in [15, 99]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -4 & 1 & -2 \\ 1 & 2 & 0 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & -2 & 2 & -2 \end{pmatrix}_{b_0}$$

[2] $1^{112} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -126$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 13.81 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 10 & 4 \\ 5 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 6 & -1 \\ 10 & 12 & 0 & 6 & 22 & 8 \\ 4 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -127$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 14.02 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 3 & -1 \\ 3 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{111} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -148$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.95 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 3 \\ 4 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 3 & -1 \\ 4 & 8 & 0 & 3 & 4 & 4 \\ 3 & 5 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -166$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.76 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{112} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -184$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$
 $T_{\min} = 13.91 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & -1 \\ 6 & 12 & 0 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 8 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -184$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.72 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 6 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{113} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -214$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$
 $T_{\min} = 14.11 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 9 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & -1 \end{pmatrix}_{b_0}$$

[9] $1^{111} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -243$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.77 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 4 \\ 4 & 4 & -1 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{114} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 14.31 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 12 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{111} 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -304$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.72 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

13.8

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 11 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{113} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -1456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 12 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{114} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -1606, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 1 & 3 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 9 & -1 \\ 13 & 18 & 0 & 9 & 52 & 12 \\ 3 & 4 & -1 & -1 & 12 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{112} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -1624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 14 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{113} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -2110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 15 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{114} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -2464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 1 & 3 & 24 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 12 & -1 \\ 16 & 24 & 0 & 12 & 94 & 16 \\ 3 & 4 & -1 & -1 & 16 & 0 \end{pmatrix}_{b_0}$$

4.1061 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_3 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$

$T = 10 \quad [0] 1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.71 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 65, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 \end{pmatrix}_{b_0}$$

[2] $1^{112} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3 \ 21^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.22 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -4 & 1 & 1 \\ 1 & 2 & 0 & 0 & 1 & 0 & 2 \\ 1 & 2 & 0 & 0 & 1 & 2 & 0 \\ 2 & 2 & -1 & -1 & -2 & 2 & -2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{113}} 3 \ 21^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 10 & 3 \\ 3 & 4 & 1 & 3 & 0 & 18 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 9 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 0 & 9 & 0 & 52 & 12 \\ 3 & 4 & -1 & -1 & 0 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3 \ 21^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 341, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3 \ 21^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 344, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.81 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{112}} 3 \ 21^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 380, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 7 & 3 \\ 3 & 4 & 1 & 3 & 0 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 7 & 12 & 0 & 6 & 0 & 22 & 8 \\ 3 & 4 & -1 & -1 & 0 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3 \ 21^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 408, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 5 & 5 & 4 \\ 5 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 0 & -3 & 3 & 3 & -1 \\ 5 & 6 & 0 & 3 & 4 & 6 & 4 \\ 5 & 6 & 0 & 3 & 6 & 4 & 4 \\ 4 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}} 3 \ 21^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 485, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{111}} 3 \ 21^{\overline{1}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}'_7 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 896, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$4.1062 \quad \text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$$

$$T = 11 \quad [0] \quad 1^{\overline{1111}} 3^{\overline{11}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -166, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.95 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 6 & 4 \\ 4 & 6 & 0 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$4.1063 \quad \text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 11 \quad [0] \quad 1^{\overline{1111}} 3^{\overline{11}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & -1 & 0 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$4.1064 \quad \text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$$

$$T = 11 \quad [0] \quad 1^{\overline{1111}} 3^{\overline{11}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 15, \Delta = -90, \det = -228, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$4.1065 \quad \text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 10 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 183, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 3 & 4 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{112}} 3^{\overline{11}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 188, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & 2 & -1 \\ 6 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{11}} 21^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 287, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 2 & -1 \\ 3 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -1 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{112}3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 495, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 2 & -1 \\ 5 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -1 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 528, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 2 & 3 \\ 3 & 4 & 1 & 3 & 6 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 2 & -1 \\ 4 & 6 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & -1 & -1 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{112}3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 807, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & 2 & -1 \\ 7 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{112}3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 1 & 3 & 12 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 6 & 2 & -1 \\ 8 & 12 & 0 & 6 & 22 & 6 & 8 \\ 2 & 3 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & -1 & -1 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

4.1066 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_3 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$

$T = 11 \quad [0] \quad 1^{111}3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 2 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 0 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

4.1067 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_3 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_{2n3\text{mix}}$

$T = 10 \quad [0] \quad 1^{111}3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 26, T \in [15, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 1 & 3 & 0 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 5 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 0 & 5 & 0 & 12 & 6 \\ 3 & 4 & -1 & -1 & 0 & 6 & -1 \end{pmatrix}_{b_0}$$

4.1068 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_3 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3$

$T = 10 \quad [0] \quad 1^{111}3^1 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{e}'_7 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 4 & 1 & 3 \\ 4 & 9 & 0 & 4 & 8 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 3 & 0 & -1 \\ 4 & 8 & 0 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] 1^{111} 3^{11} 21^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{e}'_7} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 159, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 0 & 3 & 3 \\ 3 & 4 & -1 & -1 & 0 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] 1^{111} 3^{11} 21^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{e}'_7} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 231, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.15 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$\mathbf{4.1069} \quad \text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$$

$$T = 11 \quad [0] 1^{111} 3^{11} 21^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{e}'_7} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 15, \Delta = -90, \det = -228, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 0 & 3 & 4 & 4 \\ 3 & 6 & 0 & 0 & 3 & 6 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$\mathbf{4.1070} \quad \text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3)$$

$$T = 9 \quad [0] 1^{11} 3^{11} 21^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{e}'_7} + \textcolor{red}{\mathfrak{su}_2} \rightarrow (-3) \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -92, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 3 \\ 4 & 9 & 0 & 4 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 2 & -1 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 5 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] 1^{11} 3^{11} 21^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{e}'_7} + \textcolor{red}{\mathfrak{su}_2} \rightarrow (-3) \quad T = 9, T_{\max} = 99, \Delta = -144, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.82 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & -1 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$\mathbf{4.1071} \quad \text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$$

$$T = 10 \quad [0] 1^{11} 3^{11} 21^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{e}'_7} + \textcolor{red}{\mathfrak{su}_2} \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 183, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 1 & 3 \\ 3 & 4 & 1 & 3 & 3 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 0 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] 1^{112} 3^{11} 21^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{e}'_7} + \textcolor{red}{\mathfrak{su}_2} \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 188, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 6 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{12} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 287, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 3 & -1 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{12} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 495, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 3 & -1 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 5 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{12} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 528, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 3 & -1 \\ 2 & 3 & 0 & 2 & 0 & 3 & 2 \\ 4 & 6 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{12} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 807, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 7 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12} 3^{11} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 26, \Delta = -114, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 3 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 6 & -1 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 8 & 12 & 0 & 6 & 6 & 22 & 8 \\ 3 & 4 & -1 & -1 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

4.1072 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{11} 3^{12} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 0 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & 0 & 3 & -1 \\ 2 & 3 & 0 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 3 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & -1 & 2 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

4.1073 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{11} 3^{12} 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 2 & 1 & 3 \\ 4 & 9 & 0 & 4 & 4 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 2 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & 2 & -1 & -1 \end{pmatrix}_{b_0}$$

4.1074 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}'_7$ + su2n3mix

$$T = 9 \quad [0] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -144, \quad \det = -134, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.82 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 2 & -1 \\ 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -144, \quad \det = -286, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.90 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & -1 \\ 5 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -144, \quad \det = -424, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.97 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 8 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -144, \quad \det = -464, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.81 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & -1 \\ 6 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -144, \quad \det = -510, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.79 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 1 & 3 & 9 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & -1 \\ 7 & 9 & 0 & 5 & 12 & 6 \\ 3 & 4 & -1 & -1 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -144, \quad \det = -806, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.83 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 9 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -144, \quad \det = -1056, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.79 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 1 & 3 & 15 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 8 & -1 \\ 10 & 15 & 0 & 8 & 36 & 10 \\ 3 & 4 & -1 & -1 & 10 & 0 \end{pmatrix}_{b_0}$$

4.1075 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{e}'_7$ + su2n3mix + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11} 3^1 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 26, \quad \Delta = -114, \quad \det = 150, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.23 \rightarrow 15, \quad T_{\max} = 26, \quad T \in [15, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 1 & 3 \\ 3 & 4 & 1 & 3 & 9 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 5 & 0 & -1 \\ 5 & 9 & 0 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 6 & 0 & -1 \end{pmatrix}_{b_0}$$

4.1076 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 100, \Delta = -150, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.08 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 3 \\ 4 & 9 & 0 & 4 & 0 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{12} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 100, \Delta = -150, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -4 & 2 & -2 \\ 2 & 4 & 0 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -1 & -2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 100, \Delta = -150, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.11 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 1 & 0 & 3 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 100, \Delta = -150, \det = -246, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 100, \Delta = -150, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 1 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & -1 \\ 4 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

4.1077 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.12 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 4 & 3 \\ 4 & 9 & 0 & 4 & 0 & 8 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 8 & 0 & 3 & 0 & 4 & 4 \\ 3 & 5 & -1 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 159, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 1 & 3 \\ 3 & 4 & 1 & 3 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 3 & 0 & -1 \\ 3 & 6 & 0 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & -1 & 3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3 \quad 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 58, \Delta = -120, \det = 231, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.15 \rightarrow 15, T_{\max} = 58, T \in [15, 58]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

4.1078 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{1111} 3^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 15, \Delta = -90, \det = -228, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -3 & 3 & 3 & -1 \\ 3 & 6 & 0 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 0 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & -1 & -1 & -1 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

4.1079 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{111} 3^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 80, \Delta = -123, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 80, T \in [15, 80]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & -3 & 0 & 2 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 2 & 0 & 0 & 2 \\ 3 & 5 & -1 & -1 & -1 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1080 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{111} 3^{211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}'_7 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 91, \Delta = -129, \det = 405, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.39 \rightarrow 15, T_{\max} = 91, T \in [15, 91]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 1 & 0 & 3 & 6 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & -3 & 3 & -1 \\ 3 & 6 & 0 & 0 & 3 & 3 & 3 \\ 3 & 4 & -1 & -1 & -1 & 3 & -1 \end{pmatrix}_{b_0}$$

4.1081 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{f}_4$

$$T = 7 \quad [1] \ 21^1 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 3 & 3 & 3 \\ 3 & 4 & 3 & -2 & 0 \\ 3 & 5 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 3 & 3 & 3 \\ 3 & 4 & 3 & -2 & 0 \\ 3 & 5 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 31^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 7 & 1 & 3 \\ 4 & 9 & 16 & 0 & 5 \\ 7 & 16 & 27 & 0 & 9 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 9 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.4$$

$$T_{\min} = 13.62 \rightarrow 14, T_{\max} = 97, T \in [14, 97]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 6 & 10 & 21 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 7 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -146, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

- [5] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -57$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.05 \rightarrow 13$, $T_{\max} = 97$, $T \in [13, 97]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & -3 & 0 \end{pmatrix}_{b_0}$
- [6] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -64$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & -3 & 1 \end{pmatrix}_{b_0}$
- [7] $1^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -67$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.4$
 $T_{\min} = 12.22 \rightarrow 13$, $T_{\max} = 97$, $T \in [13, 97]$
 $\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 4 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & -3 & 0 \end{pmatrix}_{b_0}$
- [8] $1^{13}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -72$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.67 \rightarrow 13$, $T_{\max} = 97$, $T \in [13, 97]$
 $\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 6 & 12 & 24 & 0 & 8 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 8 & -3 & 0 \end{pmatrix}_{b_0}$
- [9] $1^{14}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -79$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 13.10 \rightarrow 14$, $T_{\max} = 97$, $T \in [14, 97]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 8 & 16 & 45 & 0 & 11 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 11 & -3 & -1 \end{pmatrix}_{b_0}$
- [10] $1^{15}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -88$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.50 \rightarrow 14$, $T_{\max} = 97$, $T \in [14, 97]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 10 & 20 & 72 & 0 & 14 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 14 & -3 & -1 \end{pmatrix}_{b_0}$
- [11] $1^{11}131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -93$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 97$, $T \in [13, 97]$
 $\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 9 & 8 & 4 & 5 \\ 3 & 8 & 3 & 3 & 3 \\ 2 & 4 & 3 & -2 & 0 \\ 3 & 5 & 3 & 0 & 0 \end{pmatrix}_{b_0}$
- [12] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -99$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 97$, $T \in [13, 97]$
 $\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 9 & 12 & 0 & 5 \\ 5 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 6 & -3 & 0 \end{pmatrix}_{b_0}$
- [13] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -112$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 5 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & -3 & 1 \end{pmatrix}_{b_0}$
- [14] $1^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -148$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.08 \rightarrow 13$, $T_{\max} = 97$, $T \in [13, 97]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 36 & 0 & 10 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 10 & -3 & 0 \end{pmatrix}_{b_0}$
- [15] $1^{12}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -169$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.4$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$
 $\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 7 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & -3 & 1 \end{pmatrix}_{b_0}$

[16] $1^{13}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -228$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$
 $\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 9 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 12 & -3 & 1 \end{pmatrix}_{b_0}$

[17] $1^{12}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -259$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.4$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 10 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 13 & -3 & 1 \end{pmatrix}_{b_0}$

[18] $1^{14}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -289$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$
 $\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 11 & 21 & 75 & 0 & 15 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 15 & -3 & 1 \end{pmatrix}_{b_0}$

[19] $1^{13}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -372$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$
 $\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 12 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 16 & -3 & 1 \end{pmatrix}_{b_0}$

[20] $21^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -379$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$
 $\begin{pmatrix} 4 & 8 & 6 & 3 & 4 \\ 8 & 9 & 8 & 4 & 5 \\ 6 & 8 & 3 & 3 & 3 \\ 3 & 4 & 3 & -2 & 0 \\ 4 & 5 & 3 & 0 & 1 \end{pmatrix}_{b_0}$

[21] $1^{14}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4$ $T = 7$, $T_{\max} = 97$, $\Delta = -146$, $\det = -487$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 97$, $T \in [12, 97]$
 $\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 14 & 26 & 117 & 0 & 19 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 19 & -3 & 1 \end{pmatrix}_{b_0}$

$T = 8$ [22] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 22$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$
 $\begin{pmatrix} 5 & 13 & 9 & 5 \\ 13 & 30 & 20 & 10 \\ 9 & 20 & 12 & 6 \\ 5 & 10 & 6 & 2 \end{pmatrix}_{b_0}$

[23] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 27$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.37 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$
 $\begin{pmatrix} 5 & 13 & 1 & 5 \\ 13 & 30 & 0 & 10 \\ 1 & 0 & -3 & -1 \\ 5 & 10 & -1 & 2 \end{pmatrix}_{b_0}$

[24] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 28$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.14 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 \\ 1 & 2 & 0 & 3 & 0 \\ 1 & 1 & 3 & 0 & 0 \\ 1 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & -3 & 0 \end{pmatrix}_{b_0}$

[25] $1^{11}321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 30$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 35$, $T \in [14, 35]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 3 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 3 & -3 & 0 \end{pmatrix}_{b_0}$

[26] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12$, $T_{\max} = 35$, $T \in [12, 35]$
 $\begin{pmatrix} 10 & 18 & 12 & 6 \\ 18 & 30 & 20 & 10 \\ 12 & 20 & 12 & 6 \\ 6 & 10 & 6 & 3 \end{pmatrix}_{b_0}$

- [27] $1^{11}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$
 $\begin{pmatrix} 5 & 12 & 1 & 5 \\ 12 & 25 & 0 & 9 \\ 1 & 0 & -3 & -1 \\ 5 & 9 & -1 & 2 \end{pmatrix}_{b_0}$
- [28] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 35$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 3 & -3 & 2 \end{pmatrix}_{b_0}$
- [29] $1^{11}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 43$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & -3 & 1 \end{pmatrix}_{b_0}$
- [30] $1^{11}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 43$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$
- [31] $1^{11}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12$, $T_{\max} = 35$, $T \in [12, 35]$
 $\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$
- [32] $1^{11}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 52$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 4 & 0 & 3 \\ 1 & 4 & 0 & 4 & 0 & 2 \\ 2 & 4 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & 4 & -3 & 0 \end{pmatrix}_{b_0}$
- [33] $1^{111}31^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 83$, $\Delta = -125$, $\det = 60$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 83$, $T \in [13, 83]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 3 & 6 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & 6 & -3 & 0 \end{pmatrix}_{b_0}$
- [34] $1^{11}31^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 83$, $\Delta = -125$, $\det = 64$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.25 \rightarrow 13$, $T_{\max} = 83$, $T \in [13, 83]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 3 & 2 & -3 & 0 \end{pmatrix}_{b_0}$
- [35] $1^{111}3131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 83$, $\Delta = -125$, $\det = 67$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 13.22 \rightarrow 14$, $T_{\max} = 83$, $T \in [14, 83]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & 2 & -3 & -1 \end{pmatrix}_{b_0}$
- [36] $2^2231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 68$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12$, $T_{\max} = 35$, $T \in [12, 35]$
 $\begin{pmatrix} 10 & 19 & 12 & 6 \\ 19 & 30 & 20 & 10 \\ 12 & 20 & 12 & 6 \\ 6 & 10 & 6 & 3 \end{pmatrix}_{b_0}$
- [37] $1^{11}31^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 83$, $\Delta = -125$, $\det = 79$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 83$, $T \in [13, 83]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 2 & -3 & 0 \end{pmatrix}_{b_0}$

- [38] $1^{11} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 80$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 5 & 10 & 0 & 5 \\ 5 & 10 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 7 & -3 & 1 \end{pmatrix}_{b_0}$
- [41] $1^{11} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 83$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$
 $\begin{pmatrix} 13 & 14 & 1 & 7 \\ 14 & 13 & 1 & 7 \\ 1 & 1 & -3 & -1 \\ 7 & 7 & -1 & 2 \end{pmatrix}_{b_0}$
- [40] $1^{11} 3^{21} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 83$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 4 & 0 & 2 \\ 1 & 2 & 0 & 7 & 0 & 2 \\ 3 & 4 & 7 & 15 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 7 & -3 & 0 \end{pmatrix}_{b_0}$
- [39] $1^{11} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 83$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 3 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 4 & -3 & 1 \end{pmatrix}_{b_0}$
- [42] $1^{11} 3^{21} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 110$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$
 $\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 5 & 15 & 0 & 5 \\ 7 & 15 & 31 & 0 & 11 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 11 & -3 & 2 \end{pmatrix}_{b_0}$
- [43] $1^{11} 3^{21} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 110$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$
 $\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 5 & 25 & 0 & 5 \\ 11 & 25 & 91 & 0 & 19 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 19 & -3 & 2 \end{pmatrix}_{b_0}$
- [44] $1^{11} 3^{21} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 120$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 35$, $T \in [12, 35]$
 $\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 5 & 20 & 0 & 5 \\ 9 & 20 & 57 & 0 & 15 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 15 & -3 & 2 \end{pmatrix}_{b_0}$
- [45] $1^{12} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 123$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.4$
 $T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 4 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 5 & 2 & 3 \\ 1 & 1 & 0 & 2 & -2 & 0 \\ 2 & 2 & -1 & 3 & 0 & 0 \end{pmatrix}_{b_0}$
- [46] $1^{12} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 128$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.4$
 $T_{\min} = 12.03 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 5 & 0 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 \\ 3 & 5 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 6 & -3 & 0 \end{pmatrix}_{b_0}$
- [47] $1^{13} 3^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 83$, $\Delta = -125$, $\det = 160$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 83$, $T \in [13, 83]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 2 & 6 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 3 & -3 & 0 \end{pmatrix}_{b_0}$
- [48] $1^{13} 3^{21} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 163$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.21 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 6 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 15 & 3 & 5 \\ 1 & 1 & 0 & 3 & -2 & 0 \\ 2 & 2 & -1 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{13}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 163, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 6 & 0 & 2 \\ 4 & 7 & 6 & 24 & 0 & 8 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 8 & -3 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{12}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$

$$\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 4 & 12 & 9 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 9 & -3 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{13}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.53 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 3 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{12}321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 \\ 1 & 2 & 0 & 8 & 0 & 2 \\ 4 & 6 & 8 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[53] $1^{12}321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 \\ 5 & 7 & 11 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$

[54] $1^{11}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 219, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.19 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 3 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{13}321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 283, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 \\ 5 & 8 & 9 & 39 & 0 & 11 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 11 & -3 & 1 \end{pmatrix}_{b_0}$$

[56] $2^12^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 353, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 3 & 1 & 3 \\ 2 & 4 & 0 & 3 & 1 & 2 \\ 2 & 3 & 3 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & -3 & -1 \\ 2 & 3 & 2 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

[57] $1^{13}321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 363, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 12 & 0 & 2 \\ 6 & 9 & 12 & 60 & 0 & 14 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 14 & -3 & 1 \end{pmatrix}_{b_0}$$

[58] $2^12^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 548, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 0 & 2 & 2 & 4 & 1 & 2 \\ 2 & 1 & 4 & 6 & 1 & 3 \\ 2 & 4 & 0 & 6 & 1 & 2 \\ 4 & 6 & 6 & 9 & 2 & 5 \\ 1 & 1 & 1 & 2 & -3 & -1 \\ 2 & 3 & 2 & 5 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [59] \quad 1^{11} 2^1 3^1 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -17, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 1 & 0 & -2 & 0 \\ 2 & 3 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{11} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{11} 4^1 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 29, T \in [14, 29]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{11} 4^1 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13, T_{\max} = 29, T \in [13, 29]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{11} 4^1 1^1 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{11} 4^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{11} 4^1 1^1 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{11} 4^1 1^1 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -307, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 1 & 6 & 5 & 1 & 3 \\ 6 & 13 & 12 & 0 & 7 \\ 5 & 12 & 7 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 7 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [67] \quad 1^{11} 5^1 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 10, T_{\max} = 23, \Delta = -111, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.60 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 0 & 7 \\ 1 & 0 & -3 & -1 \\ 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{11} 5^1 1^1 231^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 10, T_{\max} = 24, \Delta = -112, \det = 147, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 5 & 23 & 1 & 5 \\ 23 & 90 & 0 & 18 \\ 1 & 0 & -3 & -1 \\ 5 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

4.1082 $\text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{f}}_4$

$$T = 8 \quad [0] \quad 1^{111} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 101, \Delta = -169, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 3 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [1] \quad 1^{11} 3^{11} 2^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 13 & 9 & 1 & 5 \\ 13 & 30 & 20 & 0 & 10 \\ 9 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 10 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

4.1083 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{f}_4 + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 2 & 3 \\ 4 & 9 & 4 & 0 & 8 & 4 & 5 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 8 & 3 & 0 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 2 \\ 3 & 5 & 2 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -85, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 & 3 \\ 3 & 4 & 8 & 0 & 4 & 5 & 4 \\ 4 & 8 & 9 & 0 & 6 & 6 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 4 & 3 \\ 3 & 5 & 6 & 0 & 4 & 3 & 3 \\ 3 & 4 & 5 & -3 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 6 & 0 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 3 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 4 & 3 \\ 3 & 5 & 3 & 0 & 4 & 3 & 3 \\ 3 & 4 & 2 & -3 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 15, \Delta = -96, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 2 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 & 4 \\ 5 & 9 & 12 & 0 & 9 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 9 & 0 & 4 & 4 & 4 \\ 2 & 4 & 6 & 0 & 4 & 0 & 2 \\ 3 & 4 & 6 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

4.1084 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 98, T \in [15, 98]$$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 18 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 13 & -3 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.05 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 235, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 3 & 0 & -2 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 259, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 48 & 0 & 12 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [5] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 3 & 0 & 0 & 2 \\ 3 & 5 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.13 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 1 \\ 1 & 1 & 3 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 1 & 2 & 1 & 1 & 0 & -2 \\ 2 & 2 & 2 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 5 & 13 & 9 & 1 & 5 \\ 13 & 30 & 20 & 0 & 10 \\ 9 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 10 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.10 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -110, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.49 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 0 & 2 \\ 3 & 5 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$
 $T_{\min} = 13.12 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -2 & 1 & 0 \\ 1 & 2 & 0 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[11] $1^{11}3^{11}21^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -240$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 5 & 10 & 0 & 5 & 5 \\ 5 & 10 & 13 & 0 & 7 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 7 & 0 & 0 & 2 \\ 3 & 5 & 7 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{11}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 82$, $\Delta = -124$, $\det = -277$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 82$, $T \in [14, 82]$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 1 & 3 \\ 5 & 13 & 16 & 12 & 0 & 7 \\ 5 & 16 & 12 & 12 & 0 & 6 \\ 4 & 12 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 7 & 6 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

4.1085 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$

$T = 9$ [0] $1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 13.27 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -215$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -264$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -735$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

4.1086 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -392$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 6 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

4.1087 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$

$T = 8$ [0] $1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 49$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 15.51 \rightarrow 16$, $T_{\max} = 28$, $T \in [16, 28]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 8 & 4 & 0 & 5 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 3 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

[1] $1^{123}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 50$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 6 & 3 \\ 3 & 4 & 13 & 0 & 12 & 4 \\ 7 & 13 & 27 & 0 & 27 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 27 & 0 & 25 & 9 \\ 3 & 4 & 9 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 61$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 13.11 \rightarrow 14$, $T_{\max} = 28$, $T \in [14, 28]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 2 & 3 \\ 4 & 9 & 12 & 0 & 4 & 5 \\ 5 & 12 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 3 \\ 3 & 5 & 6 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 75$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 82$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{12}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 6 & 0 & 4 & 4 \\ 3 & 4 & 3 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 98$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.22 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 116$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.10 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 6 & 0 & 13 & 7 \\ 3 & 5 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{13}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 123$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 \\ 9 & 17 & 48 & 0 & 15 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 15 & 0 & 4 & 4 \\ 3 & 4 & 12 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{12}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 128$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 28$, $T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 2 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 131, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.89 \rightarrow 12, \ T_{\max} = 28, \ T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{121} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 137, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.29 \rightarrow 13, \ T_{\max} = 28, \ T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 4 & 8 & 9 & 0 & 6 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 5 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{112} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 140, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.11 \rightarrow 13, \ T_{\max} = 28, \ T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3^1 1^3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 140, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 28, \ T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 19 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 19 & 12 & 0 & 73 & 15 \\ 3 & 4 & 2 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{121} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 141, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.37 \rightarrow 13, \ T_{\max} = 28, \ T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 4 & 8 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 12 & 0 & 13 & 7 \\ 3 & 4 & 5 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{122} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 146, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.9$$

$$T_{\min} = 12.48 \rightarrow 13, \ T_{\max} = 28, \ T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 4 & 8 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 5 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{131} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 147, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 28, \ T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 4 & 4 \\ 6 & 12 & 24 & 0 & 9 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 8 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{113} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 155, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.43 \rightarrow 13, \ T_{\max} = 28, \ T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 2 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{111} 3^1 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 159, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.34 \rightarrow 13, \ T_{\max} = 28, \ T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 \\ 8 & 14 & 36 & 0 & 9 & 10 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 10 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{123} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 161, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.75 \rightarrow 13, \ T_{\max} = 28, \ T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 6 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 4 & 8 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 18 & 0 & 25 & 9 \\ 3 & 4 & 5 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{122}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 10 & 3 \\ 3 & 4 & 8 & 0 & 18 & 4 \\ 4 & 8 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 24 & 0 & 58 & 14 \\ 3 & 4 & 5 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 5 & 9 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 9 & 0 & 4 & 4 \\ 3 & 4 & 6 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{114}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 2 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{113}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 9 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 2 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{123}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 \\ 7 & 13 & 27 & 0 & 12 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 12 & 0 & 4 & 4 \\ 3 & 4 & 9 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 \\ 5 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 6 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{123}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 197, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.16 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 15 & 3 \\ 3 & 4 & 8 & 0 & 27 & 4 \\ 4 & 8 & 9 & 0 & 36 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 36 & 0 & 133 & 21 \\ 3 & 4 & 5 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[28] $1^{112}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 24 & 12 & 0 & 58 & 14 \\ 3 & 5 & 2 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{115}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 13.11 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 20 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 20 & 15 & 0 & 73 & 15 \\ 3 & 4 & 2 & -3 & 15 & -1 \end{pmatrix}_{b_0}$$

[30] $1^{122}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 8 & 4 \\ 7 & 13 & 27 & 0 & 18 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 18 & 0 & 10 & 6 \\ 3 & 4 & 9 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 3 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{14}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 \\ 3 & 5 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 18 & 0 & 76 & 16 \\ 3 & 4 & 3 & -3 & 16 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{116}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 13.42 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 24 & 18 & 0 & 106 & 18 \\ 3 & 4 & 2 & -3 & 18 & -1 \end{pmatrix}_{b_0}$$

[34] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{111}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{114}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 2 & -3 & 16 & 0 \end{pmatrix}_{b_0}$$

[37] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 20 & 5 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 9 & 0 & 37 & 11 \\ 3 & 5 & 2 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{117}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 275, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 13.69 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 28 & 4 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 28 & 21 & 0 & 145 & 21 \\ 3 & 4 & 2 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[39] $1^{12}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 3 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{13}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 294, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 49 & 13 \\ 3 & 4 & 3 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$

[41] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 306, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 7 & 13 & 27 & 0 & 9 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 9 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{122}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 18 & 0 & 28 & 10 \\ 3 & 4 & 5 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

[43] $1^{115}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 338, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.3$
 $T_{\min} = 13.06 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 40 & 15 & 0 & 148 & 20 \\ 3 & 5 & 2 & -3 & 20 & -1 \end{pmatrix}_{b_0}$$

[44] $1^{113}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 36 & 18 & 0 & 133 & 21 \\ 3 & 5 & 2 & -3 & 21 & 0 \end{pmatrix}_{b_0}$$

[45] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 371, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{121}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 4 & 4 \\ 10 & 18 & 57 & 0 & 12 & 13 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 12 & 0 & 1 & 3 \\ 3 & 4 & 13 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 411, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{131}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 9 & 17 & 48 & 0 & 12 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 12 & 0 & 1 & 3 \\ 3 & 4 & 12 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 28 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 28 & 12 & 0 & 73 & 15 \\ 3 & 5 & 2 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{121}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 7 & 13 & 27 & 0 & 21 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 21 & 0 & 13 & 7 \\ 3 & 4 & 9 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{131}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 463, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 5 & 3 \\ 3 & 4 & 17 & 0 & 9 & 4 \\ 9 & 17 & 48 & 0 & 27 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 27 & 0 & 13 & 7 \\ 3 & 4 & 12 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{123}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 489, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 17 & 4 \\ 4 & 8 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 24 & 0 & 49 & 13 \\ 3 & 4 & 5 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{113}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 490, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 32 & 2 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 32 & 15 & 0 & 100 & 18 \\ 3 & 5 & 2 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{123}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 501, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 3 \\ 3 & 4 & 8 & 0 & 22 & 4 \\ 4 & 8 & 9 & 0 & 30 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 30 & 0 & 85 & 17 \\ 3 & 4 & 5 & -3 & 17 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{112}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 507, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 12 & 0 & 21 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 6 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{114}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 9 & 4 & 0 & 48 & 5 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 48 & 24 & 0 & 238 & 28 \\ 3 & 5 & 2 & -3 & 28 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{112}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 576, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{113}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 579, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 27 & 0 & 49 & 13 \\ 3 & 4 & 6 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{114}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 587, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 21 & 4 \\ 5 & 9 & 12 & 0 & 33 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 33 & 0 & 76 & 16 \\ 3 & 4 & 6 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{112}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 28 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 28 & 18 & 0 & 154 & 22 \\ 3 & 4 & 2 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[61] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{131}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 607, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 4 & 4 \\ 12 & 22 & 84 & 0 & 15 & 16 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 15 & 0 & 1 & 3 \\ 3 & 4 & 16 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[63] $1^{114}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 626, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 36 & 15 & 0 & 121 & 19 \\ 3 & 5 & 2 & -3 & 19 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{112}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 635, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{122}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 27 & 0 & 30 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 30 & 0 & 28 & 10 \\ 3 & 4 & 9 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{114}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 770, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 3 \\ 4 & 9 & 4 & 0 & 44 & 5 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 44 & 21 & 0 & 193 & 25 \\ 3 & 5 & 2 & -3 & 25 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{114}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 771, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 2 & -3 & 16 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{123}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 798, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 9 & 3 \\ 3 & 4 & 13 & 0 & 17 & 4 \\ 7 & 13 & 27 & 0 & 39 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 39 & 0 & 49 & 13 \\ 3 & 4 & 9 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{114}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 40 & 18 & 0 & 154 & 22 \\ 3 & 5 & 2 & -3 & 22 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{113}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 915, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{113}3^{113}3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1136$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 2 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{113}3^{113}3^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1251$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 21 & 0 & 193 & 25 \\ 3 & 4 & 2 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{114}3^{112}3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1260$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{114}3^{113}3^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1643$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{114}3^{114}3^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 1920$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 2 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [76] $1^{12}3^{12}2^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -20$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 5 & 5 & 0 & 15 & 5 \\ 2 & 5 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 10 & 0 & 38 & 12 \\ 3 & 5 & 3 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{111}3^{121}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 19 & 5 \\ 12 & 25 & 0 & 40 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 40 & 0 & 63 & 15 \\ 5 & 9 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[78] $1^{111}3^{111}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -32$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 1 \\ 2 & 3 & 3 & 2 & -3 & 3 \end{pmatrix}_{b_0}$$

[79] $1^{11}3^{121}3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -40$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 2 & 4 \\ 3 & 5 & 3 & -3 & 4 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{11}3^{11}3^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 6 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 6 & 3 & 0 & 4 & 4 \\ 2 & 3 & 3 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 1 & 2 \\ 1 & 2 & 0 & 4 & 0 & 3 & 2 \\ 2 & 3 & 4 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 1 & 3 & 4 & 0 & 1 & 3 \\ 2 & 2 & 2 & 4 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 19 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{11} 3^{14} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 23 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 23 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 5 \\ 9 & 13 & 1 & 0 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 7 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{13} 3^{11} 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 1 & 2 \\ 1 & 2 & 0 & 9 & 0 & 3 & 2 \\ 5 & 8 & 9 & 39 & 0 & 9 & 11 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 1 & 3 & 9 & 0 & 1 & 3 \\ 2 & 2 & 2 & 11 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{11} 3^{11} 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 2 & 4 & 0 & 1 & 2 \\ 1 & 2 & 0 & 7 & 0 & 3 & 2 \\ 3 & 4 & 7 & 15 & 0 & 7 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 1 & 3 & 7 & 0 & 1 & 3 \\ 2 & 2 & 2 & 7 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{11} 3^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & 3 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{11} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -2 & 1 & 0 \\ 1 & 2 & 0 & 0 & 1 & 0 & 2 \\ 2 & 2 & -1 & -1 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{112} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 13 & 14 & 1 & 28 & 7 \\ 14 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 30 & 0 & 58 & 14 \\ 7 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{111} 3^{11} 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 9 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 20 & 0 & 13 & 7 \\ 5 & 9 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[91] $1^{111}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.63 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 6 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 3 & 0 & 4 & 4 \\ 3 & 4 & 3 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[92] $1^{11}31^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 3 & 0 & 12 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 6 & 6 & 0 & 22 & 8 \\ 2 & 4 & 2 & 2 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

[93] $1^{11}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 5 & 10 & 0 & 5 & 5 \\ 5 & 10 & 13 & 0 & 7 & 7 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 7 & 0 & 2 & 4 \\ 3 & 5 & 7 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[94] $1^{111}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 1 & 10 & 5 \\ 12 & 25 & 0 & 20 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 20 & 0 & 13 & 7 \\ 5 & 9 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[95] $1^{11}321^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 4 & 0 & 3 & 3 \\ 1 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 4 & 4 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 4 & 0 & 1 & 3 \\ 2 & 3 & 2 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[96] $1^{11}321^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -89, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 1 & 4 & 4 & 0 & 4 & 3 \\ 1 & 4 & 0 & 4 & 0 & 4 & 2 \\ 2 & 4 & 4 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 7 & 0 & 5 & 5 \\ 2 & 3 & 2 & 4 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[97] $1^{112}321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 18 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 18 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[98] $1^{114}321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 66 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 66 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[99] $1^{112}321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 51 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 51 & 75 & 0 & 418 & 38 \\ 5 & 7 & -1 & 38 & 2 \end{pmatrix}_{b_0}$$

[100] $1^{111}321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 12 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 12 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[101] $1^{12}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.17 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 5 & 2 \\ 1 & 2 & 0 & 3 & 0 & 5 & 2 \\ 1 & 1 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 5 & 0 & 13 & 7 \\ 2 & 2 & 2 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[102] $1^{111}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -96$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 9 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[103] $1^{111}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -101$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 & 2 \\ 1 & 2 & 0 & 4 & 0 & 1 & 2 \\ 2 & 3 & 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 1 & 3 & 0 & 0 & 2 \\ 2 & 2 & 2 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[104] $1^{11}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -111$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.19 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 2 \\ 1 & 2 & 0 & 3 & 0 & 4 & 2 \\ 1 & 1 & 3 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[105] $1^{11}321^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -111$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 1 & 2 & 0 & 3 & 0 & 7 & 2 \\ 1 & 1 & 3 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 4 & 7 & 7 & 0 & 16 & 8 \\ 2 & 2 & 2 & 2 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

[106] $1^{112}321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 21 & 18 & 0 & 94 & 20 \\ 3 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[107] $1^{12}321^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 1 & 2 \\ 1 & 2 & 0 & 8 & 0 & 3 & 2 \\ 4 & 6 & 8 & 25 & 0 & 8 & 9 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 1 & 3 & 8 & 0 & 1 & 3 \\ 2 & 2 & 2 & 9 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[108] $1^{11}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -125$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 5 & 5 & 0 & 10 & 5 \\ 2 & 5 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 10 & 7 & 0 & 14 & 8 \\ 3 & 5 & 3 & -3 & 8 & 2 \end{pmatrix}_{b_0}$$

[109] $1^{114}321^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -136$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 & 0 & 9 & 2 \\ 1 & 2 & 0 & 4 & 0 & 7 & 2 \\ 2 & 3 & 4 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 7 & 16 & 0 & 41 & 11 \\ 2 & 2 & 2 & 4 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[110] $1^{11}31^{11}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -137$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.30 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 3 & 0 & 4 & 4 \\ 2 & 4 & 2 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[111] \ 1^{\overline{11}} 3 21^{\overline{121}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -140, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 5 & 15 & 0 & 5 & 5 \\ 7 & 15 & 31 & 0 & 11 & 11 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 11 & 0 & 2 & 4 \\ 3 & 5 & 11 & -3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[112] \ 1^{\overline{11}} 3 21^{\overline{141}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -140, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 5 & 25 & 0 & 5 & 5 \\ 11 & 25 & 91 & 0 & 19 & 19 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 19 & 0 & 2 & 4 \\ 3 & 5 & 19 & -3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{\overline{111}} 3 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -141, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 7 & 0 & 14 & 8 \\ 3 & 4 & 3 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[114] \ 1^{\overline{111}} 3 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -141, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 13 & 14 & 1 & 14 & 7 \\ 14 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 14 & 15 & 0 & 13 & 7 \\ 7 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[115] \ 1^{\overline{11}} 3 21^{\overline{131}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -160, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 5 & 20 & 0 & 5 & 5 \\ 9 & 20 & 57 & 0 & 15 & 15 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 15 & 0 & 2 & 4 \\ 3 & 5 & 15 & -3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[116] \ 1^{\overline{121}} 3 21^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -161, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 & 2 \\ 1 & 2 & 0 & 8 & 0 & 1 & 2 \\ 4 & 6 & 8 & 25 & 0 & 6 & 9 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 1 & 6 & 0 & 0 & 2 \\ 2 & 2 & 2 & 9 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{\overline{111}} 31^{\overline{1}} 31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -174, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 2 \\ 3 & 9 & 6 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 6 & 0 & 1 & 3 \\ 2 & 4 & 2 & 6 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{\overline{13}} 3 21^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -175, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.03 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 5 & 5 & 0 & 25 & 5 \\ 2 & 5 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 25 & 17 & 0 & 104 & 20 \\ 3 & 5 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{\overline{111}} 3 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 20 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[120] \ 1^{\overline{112}} 3 21^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 28 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[121] 1^{113} 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 47 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 47 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

$$[122] 1^{111} 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 11 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[123] 1^{112} 3 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 52 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 52 & 75 & 0 & 418 & 38 \\ 5 & 7 & -1 & 38 & 2 \end{pmatrix}_{b_0}$$

$$[124] 1^{111} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 22 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[125] 1^{113} 3 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 58 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 58 & 90 & 0 & 583 & 45 \\ 5 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[126] 1^{111} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 10 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 13 & 7 \\ 5 & 7 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[127] 1^{113} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -188, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 3 & 0 & 15 & 3 \\ 1 & 3 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 15 & 12 & 0 & 49 & 13 \\ 2 & 3 & 3 & 2 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[128] 1^{121} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 2 & 5 & 0 & 2 & 2 \\ 1 & 2 & 0 & 5 & 0 & 1 & 2 \\ 3 & 5 & 5 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 1 & 5 & 0 & 0 & 2 \\ 2 & 2 & 2 & 6 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[129] 1^{111} 3 21^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 5 & 5 & 0 & 25 & 5 \\ 2 & 5 & 1 & 0 & 19 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 25 & 19 & 0 & 92 & 20 \\ 3 & 5 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[130] 1^{111} 3 21^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 5 & 5 & 0 & 15 & 5 \\ 2 & 5 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 15 & 11 & 0 & 32 & 12 \\ 3 & 5 & 3 & -3 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[131] \ 1^{112} 3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 39 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 60 & 0 & 258 & 30 \\ 5 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[132] \ 1^{14} 3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 2 \\ 1 & 2 & 0 & 3 & 0 & 10 & 2 \\ 1 & 1 & 3 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 10 & 10 & 10 & 0 & 58 & 14 \\ 2 & 2 & 2 & 2 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{111} 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 21 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 30 & 0 & 63 & 15 \\ 5 & 7 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{121} 3 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 1 & 4 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 5 & 2 & 4 & 3 \\ 1 & 1 & 0 & 2 & -2 & 1 & 0 \\ 1 & 2 & 0 & 4 & 1 & 0 & 2 \\ 2 & 2 & -1 & 3 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[135] \ 1^{111} 3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 0 & 4 & 2 \\ 1 & 2 & 0 & 4 & 0 & 7 & 2 \\ 2 & 3 & 4 & 4 & 0 & 11 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 4 & 7 & 11 & 0 & 16 & 8 \\ 2 & 2 & 2 & 4 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[136] \ 1^{111} 3 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 & 2 \\ 1 & 0 & 2 & 4 & 0 & 3 & 2 \\ 1 & 2 & 0 & 7 & 0 & 4 & 2 \\ 3 & 4 & 7 & 15 & 0 & 11 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 11 & 0 & 5 & 5 \\ 2 & 2 & 2 & 7 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{11} 3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 5 & 5 & 0 & 20 & 5 \\ 2 & 5 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 15 & 0 & 58 & 16 \\ 3 & 5 & 3 & -3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[138] \ 1^{111} 3 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 0 & 3 & 2 \\ 1 & 2 & 0 & 4 & 0 & 4 & 2 \\ 2 & 3 & 4 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 7 & 0 & 5 & 5 \\ 2 & 2 & 2 & 4 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{12} 3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 2 \\ 1 & 2 & 0 & 3 & 0 & 8 & 2 \\ 1 & 1 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 8 & 0 & 26 & 10 \\ 2 & 2 & 2 & 2 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

[140] $1^{112}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -276$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.32 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 15 & 4 \\ 2 & 4 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 10 & 0 & 38 & 12 \\ 3 & 4 & 3 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[141] $1^{111}31^{11}31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -284$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.25 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 3 & 0 & 9 & 3 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 9 & 6 & 0 & 13 & 7 \\ 2 & 3 & 3 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[142] $1^{12}321^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -290$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 5 & 5 & 0 & 20 & 5 \\ 2 & 5 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 20 & 14 & 0 & 62 & 16 \\ 3 & 5 & 3 & -3 & 16 & 2 \end{pmatrix}_{b_0}$$

[144] $1^{113}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -296$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 37 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 37 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[143] $1^{112}321^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -296$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.16 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 19 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 19 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[145] $1^{112}321^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -306$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 1 & 0 & 14 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 14 & 0 & 62 & 16 \\ 3 & 4 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[146] $1^{113}321^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -311$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 0 & 7 & 2 \\ 1 & 2 & 0 & 4 & 0 & 6 & 2 \\ 2 & 3 & 4 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 13 & 0 & 25 & 9 \\ 2 & 2 & 2 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{13}321^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -311$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.14 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 3 & 0 & 9 & 2 \\ 1 & 1 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 9 & 9 & 0 & 40 & 12 \\ 2 & 2 & 2 & 2 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

[148] $1^{121}321^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -311$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.4$

$T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 & 2 \\ 1 & 0 & 2 & 5 & 0 & 3 & 2 \\ 1 & 2 & 0 & 5 & 0 & 4 & 2 \\ 3 & 5 & 5 & 12 & 0 & 10 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 10 & 0 & 5 & 5 \\ 2 & 2 & 2 & 6 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[149] \ 1^{\overline{11}1}31^{\overline{11}1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.31 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 1 & 3 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 6 & 0 & 13 & 7 \\ 2 & 4 & 2 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[150] \ 1^{\overline{11}2}31^{\overline{11}1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -332, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 3 & 0 & 12 & 3 \\ 1 & 3 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 9 & 12 & 9 & 0 & 28 & 10 \\ 2 & 3 & 3 & 2 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[151] \ 1^{\overline{12}3}21^{\overline{11}3} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 7 & 2 \\ 1 & 2 & 0 & 3 & 0 & 11 & 2 \\ 1 & 1 & 3 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 7 & 11 & 11 & 0 & 45 & 13 \\ 2 & 2 & 2 & 2 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[152] \ 1^{\overline{11}3}321^{\overline{14}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -351, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 30 & 4 \\ 2 & 4 & 1 & 0 & 25 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 30 & 25 & 0 & 188 & 28 \\ 3 & 4 & 3 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[153] \ 1^{\overline{11}2}321^{\overline{11}1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 0 & 5 & 2 \\ 1 & 2 & 0 & 4 & 0 & 5 & 2 \\ 2 & 3 & 4 & 4 & 0 & 10 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 10 & 0 & 13 & 7 \\ 2 & 2 & 2 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[154] \ 1^{\overline{11}3}321^{\overline{11}1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 13 & 0 & 74 & 16 \\ 3 & 4 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[155] \ 1^{\overline{11}2}321^{\overline{11}4} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 0 & 2 & 3 & 0 & 8 & 2 \\ 1 & 2 & 0 & 4 & 0 & 14 & 2 \\ 2 & 3 & 4 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 8 & 14 & 22 & 0 & 70 & 16 \\ 2 & 2 & 2 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[156] \ 1^{\overline{1}31}1^{\overline{11}1}31^{\overline{11}1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 6 & 0 & 4 & 4 \\ 2 & 4 & 2 & 3 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[160] \ 1^{\overline{11}4}321^{\overline{12}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 56 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 56 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

$$[161] \ 1^{\overline{11}3}321^{\overline{11}1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 28 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 28 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[158] 1^{112} 3 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 22 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 22 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[157] 1^{112} 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 20 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 20 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[162] 1^{112} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 29 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[163] 1^{112} 3 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 40 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 40 & 60 & 0 & 258 & 30 \\ 5 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

$$[159] 1^{114} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 46 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 46 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

$$[164] 1^{11} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -417, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 & 2 \\ 2 & 4 & 4 & 9 & 0 & 6 & 4 \\ 1 & 4 & 0 & 6 & 0 & 3 & 2 \\ 3 & 9 & 6 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 9 & 0 & 4 & 4 \\ 2 & 4 & 2 & 6 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[165] 1^{14} 3 21^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 5 & 5 & 0 & 35 & 5 \\ 2 & 5 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 35 & 24 & 0 & 202 & 28 \\ 3 & 5 & 3 & -3 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[166] 1^{112} 3 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.11 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 21 & 5 \\ 9 & 13 & 1 & 30 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 21 & 30 & 0 & 58 & 14 \\ 5 & 7 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[167] 1^{114} 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 0 & 16 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 27 & 16 & 0 & 122 & 20 \\ 3 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[168] 1^{11} 31^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 6 & 3 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 9 & 0 & 13 & 7 \\ 2 & 4 & 2 & 3 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[169] $1^{121} 3^{21} 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -481, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 & 2 \\ 1 & 0 & 2 & 6 & 0 & 3 & 2 \\ 1 & 2 & 0 & 8 & 0 & 4 & 2 \\ 4 & 6 & 8 & 25 & 0 & 14 & 9 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 14 & 0 & 5 & 5 \\ 2 & 2 & 2 & 9 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[170] $1^{12} 3^{21} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 5 & 5 & 0 & 25 & 5 \\ 2 & 5 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 25 & 18 & 0 & 94 & 20 \\ 3 & 5 & 3 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$

[171] $1^{113} 3^{21} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 59 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 59 & 90 & 0 & 583 & 45 \\ 5 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

[172] $1^{114} 3^{21} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 37 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 37 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[173] $1^{112} 3^{21} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 41 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 41 & 60 & 0 & 258 & 30 \\ 5 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[174] $1^{13} 3^{21} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -551, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 9 & 2 \\ 1 & 2 & 0 & 3 & 0 & 12 & 2 \\ 1 & 1 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 12 & 12 & 0 & 61 & 15 \\ 2 & 2 & 2 & 2 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

[175] $1^{13} 3^{21} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -565, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 5 & 5 & 0 & 30 & 5 \\ 2 & 5 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 30 & 21 & 0 & 142 & 24 \\ 3 & 5 & 3 & -3 & 24 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{112} 3^{21} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 42 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 42 & 60 & 0 & 258 & 30 \\ 5 & 7 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[177] $1^{112} 3^{21} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 30 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[178] $1^{113} 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 48 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 48 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[179] $1^{113} 3 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -581, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 21 & 0 & 142 & 24 \\ 3 & 4 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[180] $1^{112} 3 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -586, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 0 & 2 & 3 & 0 & 6 & 2 \\ 1 & 2 & 0 & 4 & 0 & 8 & 2 \\ 2 & 3 & 4 & 4 & 0 & 14 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 14 & 0 & 26 & 10 \\ 2 & 2 & 2 & 4 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[181] $1^{111} 31^{12} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -587, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 3 & 0 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 9 & 2 \\ 1 & 3 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 9 & 9 & 0 & 37 & 11 \\ 2 & 4 & 2 & 2 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

[182] $1^{113} 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -591, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 17 & 0 & 104 & 20 \\ 3 & 4 & 3 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[183] $1^{112} 3 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -596, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 & 0 & 7 & 2 \\ 1 & 2 & 0 & 4 & 0 & 11 & 2 \\ 2 & 3 & 4 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 7 & 11 & 18 & 0 & 45 & 13 \\ 2 & 2 & 2 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[184] $1^{14} 3 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -636, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 11 & 2 \\ 1 & 2 & 0 & 3 & 0 & 13 & 2 \\ 1 & 1 & 3 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 13 & 13 & 0 & 81 & 17 \\ 2 & 2 & 2 & 2 & -3 & 17 & 0 \end{pmatrix}_{b_0}$$

[185] $1^{12} 3 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -650, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 5 & 5 & 0 & 30 & 5 \\ 2 & 5 & 1 & 0 & 22 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 30 & 22 & 0 & 134 & 24 \\ 3 & 5 & 3 & -3 & 24 & 2 \end{pmatrix}_{b_0}$$

[190] $1^{113} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 38 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 38 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[189] $1^{113} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 29 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 29 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[188] \ 1^{112} 3 \ 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 32 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 32 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[187] \ 1^{114} 3 \ 21^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 67 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 67 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

$$[186] \ 1^{112} 3 \ 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 31 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 45 & 0 & 138 & 22 \\ 5 & 7 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[191] \ 1^{13} 3 \ 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -671, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.39 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 2 \\ 1 & 2 & 0 & 3 & 0 & 15 & 2 \\ 1 & 1 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 10 & 15 & 15 & 0 & 88 & 18 \\ 2 & 2 & 2 & 2 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[192] \ 1^{111} 31^{111} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -699, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 3 & 2 \\ 2 & 4 & 4 & 9 & 0 & 9 & 4 \\ 1 & 4 & 0 & 6 & 0 & 6 & 2 \\ 3 & 9 & 6 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 15 & 0 & 13 & 7 \\ 2 & 4 & 2 & 6 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[194] \ 1^{113} 3 \ 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 30 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 30 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[193] \ 1^{113} 3 \ 21^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 60 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 60 & 90 & 0 & 583 & 45 \\ 5 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

$$[195] \ 1^{111} 31^{13} 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -818, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 3 & 0 & 21 & 4 \\ 1 & 4 & 0 & 3 & 0 & 12 & 2 \\ 1 & 3 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 12 & 12 & 0 & 73 & 15 \\ 2 & 4 & 2 & 2 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[196] \ 1^{113} 3 \ 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -821, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 5 & 2 \\ 1 & 0 & 2 & 3 & 0 & 8 & 2 \\ 1 & 2 & 0 & 4 & 0 & 9 & 2 \\ 2 & 3 & 4 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 9 & 17 & 0 & 40 & 12 \\ 2 & 2 & 2 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[197] $1^{114} 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -846$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.35 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 30 & 4 \\ 2 & 4 & 1 & 0 & 20 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 30 & 20 & 0 & 158 & 24 \\ 3 & 4 & 3 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[198] $1^{13} 3 21^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -895$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 5 & 5 & 0 & 35 & 5 \\ 2 & 5 & 1 & 0 & 25 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 35 & 25 & 0 & 188 & 28 \\ 3 & 5 & 3 & -3 & 28 & 2 \end{pmatrix}_{b_0}$$

[199] $1^{113} 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -896$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.01 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 49 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 49 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[201] $1^{114} 3 21^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -896$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 47 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 47 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

[202] $1^{114} 3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -896$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.21 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 38 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 38 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[200] $1^{113} 3 21^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -896$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.12 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 31 & 5 \\ 9 & 13 & 1 & 45 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 31 & 45 & 0 & 133 & 21 \\ 5 & 7 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[203] $2 \ 2 \ 31^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -901$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 & 2 \\ 2 & 1 & 4 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 2 \\ 2 & 3 & 3 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 3 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[204] $1^{114} 3 21^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -926$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 0 & 2 & 3 & 0 & 10 & 2 \\ 1 & 2 & 0 & 4 & 0 & 10 & 2 \\ 2 & 3 & 4 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 10 & 10 & 20 & 0 & 58 & 14 \\ 2 & 2 & 2 & 4 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[205] $1^{114} 3 21^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -936$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.05 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 57 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 57 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

[207] $1^{113} 3 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -936, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 39 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[206] $1^{14} 3 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -936, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 12 \\ 1 & 2 & 0 & 3 & 0 & 16 \\ 1 & 1 & 3 & 0 & 0 & 16 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 8 & 12 & 16 & 16 & 0 & 110 \\ 2 & 2 & 2 & 2 & -3 & 20 \end{pmatrix}_{b_0}$$

[208] $1^{14} 3 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -950, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 5 & 5 & 0 & 40 & 5 \\ 2 & 5 & 1 & 0 & 28 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 40 & 28 & 0 & 254 & 32 \\ 3 & 5 & 3 & -3 & 32 & 2 \end{pmatrix}_{b_0}$$

[209] $1^{114} 3 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -966, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 1 & 0 & 28 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 28 & 0 & 254 & 32 \\ 3 & 4 & 3 & -3 & 32 & 1 \end{pmatrix}_{b_0}$$

[210] $1^{111} 31^{14} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -1007, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 3 & 0 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 15 & 2 \\ 1 & 3 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 15 & 15 & 0 & 121 & 19 \\ 2 & 4 & 2 & 2 & -3 & 19 & 0 \end{pmatrix}_{b_0}$$

[211] $1^{113} 3 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1016, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 61 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 61 & 90 & 0 & 583 & 45 \\ 5 & 7 & -1 & 45 & 3 \end{pmatrix}_{b_0}$$

[212] $1^{114} 3 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1016, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 33 & 4 \\ 2 & 4 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 33 & 24 & 0 & 202 & 28 \\ 3 & 4 & 3 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[213] $1^{113} 3 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 6 & 2 \\ 1 & 0 & 2 & 3 & 0 & 9 & 2 \\ 1 & 2 & 0 & 4 & 0 & 12 & 2 \\ 2 & 3 & 4 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 12 & 21 & 0 & 61 & 15 \\ 2 & 2 & 2 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[215] $1^{113} 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 40 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 40 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[216] $1^{113} 3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

12.0

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 50 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 50 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[214] $1^{114} 3^{12} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 68 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 68 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

[217] $1^{114} 3^{13} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 39 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 39 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{113} 3^{11} 21^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 7 & 2 \\ 1 & 0 & 2 & 3 & 0 & 10 & 2 \\ 1 & 2 & 0 & 4 & 0 & 15 & 2 \\ 2 & 3 & 4 & 4 & 0 & 25 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 10 & 15 & 25 & 0 & 88 & 18 \\ 2 & 2 & 2 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[219] $1^{113} 3^{14} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 41 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 41 & 60 & 0 & 243 & 29 \\ 5 & 7 & -1 & 29 & 2 \end{pmatrix}_{b_0}$$

[220] $1^{114} 3^{12} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 48 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 48 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

[221] $1^{113} 3^{14} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 51 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 51 & 75 & 0 & 393 & 37 \\ 5 & 7 & -1 & 37 & 2 \end{pmatrix}_{b_0}$$

[222] $1^{114} 3^{12} 21^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 58 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 58 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

[223] $1^{114} 3^{14} 21^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 40 & 5 \\ 9 & 13 & 1 & 60 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 40 & 60 & 0 & 238 & 28 \\ 5 & 7 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[224] $1^{114} 3^{11} 21^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 7 & 2 \\ 1 & 0 & 2 & 3 & 0 & 11 & 2 \\ 1 & 2 & 0 & 4 & 0 & 13 & 2 \\ 2 & 3 & 4 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 13 & 24 & 0 & 81 & 17 \\ 2 & 2 & 2 & 4 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[225] \ 1^{114} 3 \ 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 69 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 69 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

$$[226] \ 1^{114} 3 \ 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1616, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 49 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 49 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

$$[227] \ 1^{114} 3 \ 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 59 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 59 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

$$[228] \ 1^{114} 3 \ 21^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1846, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 8 & 2 \\ 1 & 0 & 2 & 3 & 0 & 12 & 2 \\ 1 & 2 & 0 & 4 & 0 & 16 & 2 \\ 2 & 3 & 4 & 4 & 0 & 28 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 16 & 28 & 0 & 110 & 20 \\ 2 & 2 & 2 & 4 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[229] \ 1^{114} 3 \ 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1856, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.01 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 70 & 5 \\ 9 & 13 & 1 & 105 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 70 & 105 & 0 & 778 & 52 \\ 5 & 7 & -1 & 52 & 2 \end{pmatrix}_{b_0}$$

$$[230] \ 1^{114} 3 \ 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1856, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 50 & 5 \\ 9 & 13 & 1 & 75 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 50 & 75 & 0 & 378 & 36 \\ 5 & 7 & -1 & 36 & 2 \end{pmatrix}_{b_0}$$

$$[231] \ 1^{114} 3 \ 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2016, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 60 & 5 \\ 9 & 13 & 1 & 90 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 60 & 90 & 0 & 558 & 44 \\ 5 & 7 & -1 & 44 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [232] \ 1^{11} 4 \ 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[233] \ 1^{11} 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 4 & 3 & 0 & 5 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[234] \ 1^{\overline{111}} 1^{\overline{131}} 4 \ 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 117, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 7 & 0 & 4 & 4 \\ 5 & 7 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 5 & 0 & 1 & 3 \\ 3 & 4 & 5 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[235] \ 1^{\overline{111}} 1^{\overline{131}} 4 \ 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 5 & 0 & 8 & 6 \\ 3 & 4 & 2 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[236] \ 1^{\overline{1}} 1^{\overline{131}} 4 \ 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 201, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 4 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[237] \ 1^{\overline{1}} 1^{\overline{131}} 4 \ 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 222, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 4 & 3 & 0 & 7 & 4 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 5 & 0 & 8 & 6 \\ 3 & 4 & 2 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[238] \ 1^{\overline{111}} 1^{\overline{131}} 4 \ 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 288, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 8 & 0 & 26 & 10 \\ 3 & 4 & 2 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[239] \ 1^{\overline{11}} 1^{\overline{131}} 4 \ 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 311, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 4 & 3 & 0 & 9 & 4 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 5 & 0 & 13 & 7 \\ 3 & 4 & 2 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[240] \ 1^{\overline{11}} 1^{\overline{131}} 4 \ 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 338, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 4 & 3 & 0 & 12 & 4 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 8 & 0 & 26 & 10 \\ 3 & 4 & 2 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [241] \ 1^{\overline{12}} 1^{\overline{131}} 5 \ 1^{\overline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -34, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 1 & 12 & 5 \\ 9 & 13 & 0 & 20 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 20 & 0 & 26 & 10 \\ 5 & 7 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[242] \ 1^{11} \overset{1}{5} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 14, \ \Delta = -81, \ \det = -61, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.62 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 1 & 6 & 5 \\ 9 & 13 & 0 & 10 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 10 & 0 & 5 & 5 \\ 5 & 7 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

4.1088 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8$

$$T = 9 \quad [0] \ 1^{11} \overset{1}{3} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, \ T_{\max} = 26, \ \Delta = -115, \ \det = -83, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} \overset{1}{3} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, \ T_{\max} = 26, \ \Delta = -115, \ \det = -215, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 26, \ T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} \overset{1}{3} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, \ T_{\max} = 26, \ \Delta = -115, \ \det = -264, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 26, \ T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} \overset{1}{3} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, \ T_{\max} = 26, \ \Delta = -115, \ \det = -735, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 26, \ T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

4.1089 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$

$$T = 9 \quad [0] \ 1^{11} \overset{1}{3} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -44, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.82 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 6 & 4 \\ 3 & 5 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & 2 & -3 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} \overset{1}{3} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -69, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 11 & 3 \\ 3 & 4 & 4 & 0 & 5 & 19 & 4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 21 & 4 \\ 11 & 19 & 12 & 0 & 21 & 73 & 15 \\ 3 & 4 & 2 & -3 & 4 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} \overset{1}{3} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -72, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 8 & 14 & 9 & 0 & 9 & 37 & 11 \\ 3 & 4 & 2 & -3 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1111} 3^1 1^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -84, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 5 & 3 \\ 3 & 4 & 4 & 0 & 19 & 9 & 4 \\ 2 & 4 & 0 & 0 & 12 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 19 & 12 & 0 & 73 & 33 & 15 \\ 5 & 9 & 6 & 0 & 33 & 13 & 7 \\ 3 & 4 & 2 & -3 & 15 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 2 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 & 4 \\ 5 & 9 & 12 & 0 & 12 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 12 & 0 & 10 & 6 & 6 \\ 2 & 4 & 6 & 0 & 6 & 1 & 3 \\ 3 & 4 & 6 & -3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -173, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 2 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 & 4 \\ 5 & 9 & 12 & 0 & 9 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 9 & 0 & 4 & 3 & 4 \\ 2 & 4 & 6 & 0 & 3 & 1 & 3 \\ 3 & 4 & 6 & -3 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1211} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -191, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 2 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 & 4 \\ 7 & 13 & 27 & 0 & 12 & 9 & 9 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 12 & 0 & 4 & 3 & 4 \\ 2 & 4 & 9 & 0 & 3 & 1 & 3 \\ 3 & 4 & 9 & -3 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -268, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 2 & -3 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -269, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 9 & 4 \\ 5 & 9 & 6 & 0 & 9 & 13 & 7 \\ 3 & 4 & 2 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1111} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -275, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.84 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 8 & 3 \\ 3 & 4 & 4 & 0 & 5 & 14 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 15 & 4 \\ 8 & 14 & 9 & 0 & 15 & 37 & 11 \\ 3 & 4 & 2 & -3 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1112} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -296, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.99 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 7 & 3 \\ 3 & 4 & 4 & 0 & 5 & 13 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 12 & 4 \\ 7 & 13 & 9 & 0 & 12 & 28 & 10 \\ 3 & 4 & 2 & -3 & 4 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -320, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 & 4 \\ 5 & 9 & 12 & 0 & 6 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 3 & 3 \\ 2 & 4 & 6 & 0 & 3 & 1 & 3 \\ 3 & 4 & 6 & -3 & 3 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1112} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -331, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.89 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 13 & 3 \\ 3 & 4 & 4 & 0 & 4 & 23 & 4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 13 & 23 & 15 & 0 & 15 & 100 & 18 \\ 3 & 4 & 2 & -3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1122} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -338, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 7 & 3 \\ 3 & 4 & 9 & 0 & 8 & 13 & 4 \\ 5 & 9 & 12 & 0 & 12 & 21 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 12 & 0 & 10 & 18 & 6 \\ 7 & 13 & 21 & 0 & 18 & 28 & 10 \\ 3 & 4 & 6 & -3 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -381, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 2 & -3 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -573, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 9 & 0 & 4 & 9 & 4 \\ 5 & 9 & 12 & 0 & 6 & 15 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 6 & 3 \\ 5 & 9 & 15 & 0 & 6 & 13 & 7 \\ 3 & 4 & 6 & -3 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1111} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -605, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 8 & 3 \\ 3 & 4 & 4 & 0 & 9 & 14 & 4 \\ 2 & 4 & 0 & 0 & 6 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 24 & 7 \\ 8 & 14 & 9 & 0 & 24 & 37 & 11 \\ 3 & 4 & 2 & -3 & 7 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{112} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -626, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 10 & 3 \\ 3 & 4 & 4 & 0 & 5 & 18 & 4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 18 & 4 \\ 10 & 18 & 12 & 0 & 18 & 58 & 14 \\ 3 & 4 & 2 & -3 & 4 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{112} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -86, \ \det = -650, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 16 & 3 \\ 3 & 4 & 4 & 0 & 5 & 28 & 4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 30 & 4 \\ 16 & 28 & 18 & 0 & 30 & 154 & 22 \\ 3 & 4 & 2 & -3 & 4 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1111} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -704, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 2 & -3 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1111} 31^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -716, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.85 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 9 & 0 & 37 & 39 & 11 \\ 8 & 14 & 9 & 0 & 39 & 37 & 11 \\ 3 & 4 & 2 & -3 & 11 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1112} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -723, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 4 & 13 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 7 & 13 & 9 & 0 & 9 & 28 & 10 \\ 3 & 4 & 2 & -3 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1112} 31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -738, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 4 & 18 & 4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 10 & 18 & 12 & 0 & 12 & 58 & 14 \\ 3 & 4 & 2 & -3 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1112} 31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -744, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 13 & 3 \\ 3 & 4 & 4 & 0 & 5 & 23 & 4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 24 & 4 \\ 13 & 23 & 15 & 0 & 24 & 100 & 18 \\ 3 & 4 & 2 & -3 & 4 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1113} 31^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -779, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 12 & 3 \\ 3 & 4 & 4 & 0 & 5 & 22 & 4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 21 & 4 \\ 12 & 22 & 15 & 0 & 21 & 85 & 17 \\ 3 & 4 & 2 & -3 & 4 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1113} 31^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -828, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 18 & 3 \\ 3 & 4 & 4 & 0 & 4 & 32 & 4 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 21 & 3 \\ 18 & 32 & 21 & 0 & 21 & 193 & 25 \\ 3 & 4 & 2 & -3 & 3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1112} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -845, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 7 & 3 \\ 3 & 4 & 9 & 0 & 4 & 13 & 4 \\ 5 & 9 & 12 & 0 & 6 & 21 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 9 & 3 \\ 7 & 13 & 21 & 0 & 9 & 28 & 10 \\ 3 & 4 & 6 & -3 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1112} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -941, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 7 & 3 \\ 3 & 4 & 4 & 0 & 9 & 13 & 4 \\ 2 & 4 & 0 & 0 & 6 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 21 & 7 \\ 7 & 13 & 9 & 0 & 21 & 28 & 10 \\ 3 & 4 & 2 & -3 & 7 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1112} 31^{122} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -962, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 10 & 3 \\ 3 & 4 & 4 & 0 & 14 & 18 & 4 \\ 2 & 4 & 0 & 0 & 9 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 9 & 0 & 37 & 48 & 11 \\ 10 & 18 & 12 & 0 & 48 & 58 & 14 \\ 3 & 4 & 2 & -3 & 11 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1112} 31^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -980, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 16 & 3 \\ 3 & 4 & 4 & 0 & 9 & 28 & 4 \\ 2 & 4 & 0 & 0 & 6 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 48 & 7 \\ 16 & 28 & 18 & 0 & 48 & 154 & 22 \\ 3 & 4 & 2 & -3 & 7 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1113} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -980, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 9 & 3 \\ 3 & 4 & 4 & 0 & 9 & 17 & 4 \\ 2 & 4 & 0 & 0 & 6 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 27 & 7 \\ 9 & 17 & 12 & 0 & 27 & 49 & 13 \\ 3 & 4 & 2 & -3 & 7 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1113} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -989, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 9 & 3 \\ 3 & 4 & 9 & 0 & 4 & 17 & 4 \\ 5 & 9 & 12 & 0 & 6 & 27 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 12 & 3 \\ 9 & 17 & 27 & 0 & 12 & 49 & 13 \\ 3 & 4 & 6 & -3 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1114} 31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1005, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 11 & 3 \\ 3 & 4 & 9 & 0 & 4 & 21 & 4 \\ 5 & 9 & 12 & 0 & 6 & 33 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 15 & 3 \\ 11 & 21 & 33 & 0 & 15 & 76 & 16 \\ 3 & 4 & 6 & -3 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1113} 31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1077, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 4 & 4 & 0 & 4 & 17 & 4 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 12 & 3 \\ 9 & 17 & 12 & 0 & 12 & 49 & 13 \\ 3 & 4 & 2 & -3 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{113} 31^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1221, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 15 & 3 \\ 3 & 4 & 4 & 0 & 5 & 27 & 4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 27 & 4 \\ 15 & 27 & 18 & 0 & 27 & 133 & 21 \\ 3 & 4 & 2 & -3 & 4 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1122} 3^1 1^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1298, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 10 & 3 \\ 3 & 4 & 4 & 0 & 13 & 18 & 4 \\ 2 & 4 & 0 & 0 & 9 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 42 & 10 \\ 10 & 18 & 12 & 0 & 42 & 58 & 14 \\ 3 & 4 & 2 & -3 & 10 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1113} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1333, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 15 & 3 \\ 3 & 4 & 4 & 0 & 4 & 27 & 4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 18 & 3 \\ 15 & 27 & 18 & 0 & 18 & 133 & 21 \\ 3 & 4 & 2 & -3 & 3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1112} 3^1 1^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 10 & 3 \\ 3 & 4 & 4 & 0 & 9 & 18 & 4 \\ 2 & 4 & 0 & 0 & 6 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 30 & 7 \\ 10 & 18 & 12 & 0 & 30 & 58 & 14 \\ 3 & 4 & 2 & -3 & 7 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1122} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1388, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 13 & 4 \\ 2 & 4 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 30 & 10 \\ 7 & 13 & 9 & 0 & 30 & 28 & 10 \\ 3 & 4 & 2 & -3 & 10 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1112} 3^1 1^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1389, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 13 & 3 \\ 3 & 4 & 4 & 0 & 9 & 23 & 4 \\ 2 & 4 & 0 & 0 & 6 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 39 & 7 \\ 13 & 23 & 15 & 0 & 39 & 100 & 18 \\ 3 & 4 & 2 & -3 & 7 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1113} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 4 & 0 & 4 & 22 & 4 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 15 & 3 \\ 12 & 22 & 15 & 0 & 15 & 85 & 17 \\ 3 & 4 & 2 & -3 & 3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{113} 3^1 1^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1451, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 18 & 3 \\ 3 & 4 & 4 & 0 & 5 & 32 & 4 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 33 & 4 \\ 18 & 32 & 21 & 0 & 33 & 193 & 25 \\ 3 & 4 & 2 & -3 & 4 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{114} 3^1 1^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1500, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 17 & 3 \\ 3 & 4 & 4 & 0 & 5 & 31 & 4 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 30 & 4 \\ 17 & 31 & 21 & 0 & 30 & 172 & 24 \\ 3 & 4 & 2 & -3 & 4 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1113} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1949, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 12 & 3 \\ 3 & 4 & 4 & 0 & 9 & 22 & 4 \\ 2 & 4 & 0 & 0 & 6 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 36 & 7 \\ 12 & 22 & 15 & 0 & 36 & 85 & 17 \\ 3 & 4 & 2 & -3 & 7 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1122} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2048, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 18 & 4 \\ 2 & 4 & 0 & 0 & 12 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 12 & 0 & 58 & 60 & 14 \\ 10 & 18 & 12 & 0 & 60 & 58 & 14 \\ 3 & 4 & 2 & -3 & 14 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{114} 3^1 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2054, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 20 & 3 \\ 3 & 4 & 4 & 0 & 5 & 36 & 4 \\ 2 & 4 & 0 & 0 & 3 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 36 & 4 \\ 20 & 36 & 24 & 0 & 36 & 238 & 28 \\ 3 & 4 & 2 & -3 & 4 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1114} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2106, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 4 & 4 & 0 & 4 & 26 & 4 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 18 & 3 \\ 14 & 26 & 18 & 0 & 18 & 118 & 20 \\ 3 & 4 & 2 & -3 & 3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1114} 3^1 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 20 & 3 \\ 3 & 4 & 4 & 0 & 4 & 36 & 4 \\ 2 & 4 & 0 & 0 & 3 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 24 & 3 \\ 20 & 36 & 24 & 0 & 24 & 238 & 28 \\ 3 & 4 & 2 & -3 & 3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1114} 3^1 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2324, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 14 & 3 \\ 3 & 4 & 4 & 0 & 9 & 26 & 4 \\ 2 & 4 & 0 & 0 & 6 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 42 & 7 \\ 14 & 26 & 18 & 0 & 42 & 118 & 20 \\ 3 & 4 & 2 & -3 & 7 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1114} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2347, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 17 & 3 \\ 3 & 4 & 4 & 0 & 4 & 31 & 4 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 21 & 3 \\ 17 & 31 & 21 & 0 & 21 & 172 & 24 \\ 3 & 4 & 2 & -3 & 3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1113} 3^1 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2496, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 15 & 3 \\ 3 & 4 & 4 & 0 & 9 & 27 & 4 \\ 2 & 4 & 0 & 0 & 6 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 45 & 7 \\ 15 & 27 & 18 & 0 & 45 & 133 & 21 \\ 3 & 4 & 2 & -3 & 7 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{\overline{1113}1^1} 3^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2621, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 18 & 3 \\ 3 & 4 & 4 & 0 & 9 & 32 & 4 \\ 2 & 4 & 0 & 0 & 6 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 54 & 7 \\ 18 & 32 & 21 & 0 & 54 & 193 & 25 \\ 3 & 4 & 2 & -3 & 7 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{\overline{1114}1^1} 3^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -3405, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 17 & 3 \\ 3 & 4 & 4 & 0 & 9 & 31 & 4 \\ 2 & 4 & 0 & 0 & 6 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 51 & 7 \\ 17 & 31 & 21 & 0 & 51 & 172 & 24 \\ 3 & 4 & 2 & -3 & 7 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{\overline{1114}1^1} 3^{\overline{114}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -4064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 20 & 3 \\ 3 & 4 & 4 & 0 & 9 & 36 & 4 \\ 2 & 4 & 0 & 0 & 6 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 60 & 7 \\ 20 & 36 & 24 & 0 & 60 & 238 & 28 \\ 3 & 4 & 2 & -3 & 7 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [54] \quad 1^{\overline{111}} 3^{\overline{111}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 32, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 1 & 5 \\ 9 & 13 & 1 & 0 & 0 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 7 & -1 & 0 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{\overline{11}} 3^{\overline{111}} 2^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 5 & 5 & 0 & 5 & 10 & 5 \\ 2 & 5 & 1 & 0 & 4 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 2 & 7 & 4 \\ 5 & 10 & 7 & 0 & 7 & 14 & 8 \\ 3 & 5 & 3 & -3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{\overline{111}} 3^{\overline{111}} 2^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 4 & 0 & 1 & 1 & 2 \\ 2 & 3 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 3 & 0 & 0 & 2 & 2 \\ 1 & 2 & 1 & 3 & 0 & 2 & 0 & 2 \\ 2 & 2 & 2 & 4 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{\overline{11}} 3^{\overline{11}} 2^{\overline{1311}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 160, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 2 & 3 \\ 3 & 5 & 20 & 0 & 5 & 5 & 5 \\ 9 & 20 & 57 & 0 & 15 & 15 & 15 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 15 & 0 & 2 & 4 & 4 \\ 2 & 5 & 15 & 0 & 4 & 2 & 4 \\ 3 & 5 & 15 & -3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{\overline{111}} 3^{\overline{111}} 2^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 172, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 20 & 5 \\ 9 & 13 & 1 & 0 & 30 & 7 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 20 & 30 & 0 & 0 & 63 & 15 \\ 5 & 7 & -1 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{111} 321^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 172, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 0 & 1 & 3 & 2 \\ 1 & 2 & 0 & 4 & 0 & 3 & 4 & 2 \\ 2 & 3 & 4 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 1 & 3 & 4 & 0 & 1 & 4 & 3 \\ 2 & 3 & 4 & 7 & 0 & 4 & 5 & 5 \\ 2 & 2 & 2 & 4 & -3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{111} 321^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 220, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 5 & 5 & 0 & 5 & 15 & 5 \\ 2 & 5 & 1 & 0 & 4 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 2 & 11 & 4 \\ 7 & 15 & 11 & 0 & 11 & 32 & 12 \\ 3 & 5 & 3 & -3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{111} 321^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 262, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 0 & 1 & 4 & 2 \\ 2 & 3 & 4 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 1 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 7 & 0 & 3 & 5 & 5 \\ 2 & 2 & 2 & 4 & -3 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{12} 321^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 340, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 5 & 5 & 0 & 5 & 25 & 5 \\ 2 & 5 & 1 & 0 & 4 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 2 & 18 & 4 \\ 12 & 25 & 18 & 0 & 18 & 94 & 20 \\ 3 & 5 & 3 & -3 & 4 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{111} 321^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 360, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 5 & 5 & 0 & 10 & 10 & 5 \\ 2 & 5 & 1 & 0 & 7 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 10 & 7 & 0 & 14 & 16 & 8 \\ 5 & 10 & 7 & 0 & 16 & 14 & 8 \\ 3 & 5 & 3 & -3 & 8 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{13} 321^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 380, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 17 & 3 \\ 3 & 5 & 5 & 0 & 5 & 35 & 5 \\ 2 & 5 & 1 & 0 & 4 & 25 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 2 & 25 & 4 \\ 17 & 35 & 25 & 0 & 25 & 188 & 28 \\ 3 & 5 & 3 & -3 & 4 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{111} 321^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 390, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 7 & 3 \\ 3 & 5 & 5 & 0 & 10 & 15 & 5 \\ 2 & 5 & 1 & 0 & 7 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 10 & 7 & 0 & 14 & 23 & 8 \\ 7 & 15 & 11 & 0 & 23 & 32 & 12 \\ 3 & 5 & 3 & -3 & 8 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{111} 321^{1123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 630, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 9 & 3 \\ 3 & 5 & 5 & 0 & 15 & 20 & 5 \\ 2 & 5 & 1 & 0 & 11 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 15 & 11 & 0 & 32 & 45 & 12 \\ 9 & 20 & 15 & 0 & 45 & 58 & 16 \\ 3 & 5 & 3 & -3 & 12 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{12} 3^{11} 21^{1114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 640, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 5 & 5 & 0 & 5 & 30 & 5 \\ 2 & 5 & 1 & 0 & 4 & 22 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 4 & 0 & 2 & 22 & 4 \\ 14 & 30 & 22 & 0 & 22 & 134 & 24 \\ 3 & 5 & 3 & -3 & 4 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{111} 3^{11} 21^{1122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 720, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 7 & 3 \\ 3 & 5 & 5 & 0 & 15 & 15 & 5 \\ 2 & 5 & 1 & 0 & 11 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 15 & 11 & 0 & 32 & 34 & 12 \\ 7 & 15 & 11 & 0 & 34 & 32 & 12 \\ 3 & 5 & 3 & -3 & 12 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{111} 3^{11} 21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 752, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 3 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 4 & 0 & 4 & 4 & 2 \\ 2 & 3 & 4 & 4 & 0 & 7 & 7 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 7 & 0 & 5 & 7 & 5 \\ 2 & 3 & 4 & 7 & 0 & 7 & 5 & 5 \\ 2 & 2 & 2 & 4 & -3 & 5 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{111} 3^{11} 21^{1133} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 840, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.00 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 9 & 3 \\ 3 & 5 & 5 & 0 & 20 & 20 & 5 \\ 2 & 5 & 1 & 0 & 15 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 20 & 15 & 0 & 58 & 60 & 16 \\ 9 & 20 & 15 & 0 & 60 & 58 & 16 \\ 3 & 5 & 3 & -3 & 16 & 16 & 2 \end{pmatrix}_{b_0}$$

4.1090 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \quad 1^{113} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 5 & 3 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 6 & 3 & 4 \\ 3 & 5 & 3 & 0 & 6 & 4 & 3 & 4 \\ 2 & 3 & 2 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & -3 & 4 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 392, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 3 & 2 & 3 \\ 2 & 4 & 3 & 0 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & 2 & -3 & 3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 462, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 9 & 3 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 6 & 0 & 9 & 13 & 5 & 7 \\ 2 & 3 & 2 & 0 & 3 & 5 & 0 & 2 \\ 3 & 4 & 2 & -3 & 4 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 & 3 & 4 \\ 5 & 9 & 12 & 0 & 6 & 6 & 5 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 3 & 2 & 3 \\ 2 & 4 & 6 & 0 & 3 & 1 & 2 & 3 \\ 2 & 3 & 5 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & 6 & -3 & 3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 3 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 6 & 2 & 3 \\ 5 & 9 & 6 & 0 & 6 & 13 & 5 & 7 \\ 2 & 3 & 2 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & 2 & -3 & 3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 3 & 4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 15 & 5 & 7 \\ 5 & 9 & 6 & 0 & 15 & 13 & 5 & 7 \\ 2 & 3 & 2 & 0 & 5 & 5 & 0 & 2 \\ 3 & 4 & 2 & -3 & 7 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.1091 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{\overline{1123}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 2 & 3 \\ 3 & 4 & 9 & 0 & 8 & 3 & 4 \\ 5 & 9 & 12 & 0 & 12 & 5 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 12 & 0 & 10 & 4 & 6 \\ 2 & 3 & 5 & 0 & 4 & 0 & 2 \\ 3 & 4 & 6 & -3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 2 & 3 \\ 4 & 9 & 4 & 0 & 8 & 4 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 8 & 3 & 0 & 4 & 3 & 4 \\ 2 & 4 & 2 & 0 & 3 & 0 & 2 \\ 3 & 5 & 2 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3^{\overline{11}} 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 2 & 3 \\ 3 & 4 & 4 & 0 & 19 & 3 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 19 & 12 & 0 & 73 & 11 & 15 \\ 2 & 3 & 2 & 0 & 11 & 0 & 2 \\ 3 & 4 & 2 & -3 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1123}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 0 & 8 & 3 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 10 & 4 & 6 \\ 2 & 3 & 2 & 0 & 4 & 0 & 2 \\ 3 & 4 & 2 & -3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1123}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 3 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 5 & 7 \\ 2 & 3 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 2 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{12}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 3 & 4 \\ 7 & 13 & 27 & 0 & 9 & 7 & 9 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 9 & 0 & 1 & 2 & 3 \\ 2 & 3 & 7 & 0 & 2 & 0 & 2 \\ 3 & 4 & 9 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 4 & 0 & 13 & 3 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 9 & 0 & 28 & 7 & 10 \\ 2 & 3 & 2 & 0 & 7 & 0 & 2 \\ 3 & 4 & 2 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 2 & 3 \\ 3 & 4 & 4 & 0 & 17 & 3 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 12 & 0 & 49 & 9 & 13 \\ 2 & 3 & 2 & 0 & 9 & 0 & 2 \\ 3 & 4 & 2 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{112}3^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 2 & 3 \\ 3 & 4 & 4 & 0 & 28 & 3 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 16 & 28 & 18 & 0 & 154 & 16 & 22 \\ 2 & 3 & 2 & 0 & 16 & 0 & 2 \\ 3 & 4 & 2 & -3 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{114}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 2 & 3 \\ 3 & 4 & 4 & 0 & 21 & 3 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 15 & 0 & 76 & 11 & 16 \\ 2 & 3 & 2 & 0 & 11 & 0 & 2 \\ 3 & 4 & 2 & -3 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 2 & 3 \\ 3 & 4 & 4 & 0 & 18 & 3 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 12 & 0 & 58 & 10 & 14 \\ 2 & 3 & 2 & 0 & 10 & 0 & 2 \\ 3 & 4 & 2 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{112}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 2 & 3 \\ 3 & 4 & 4 & 0 & 23 & 3 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 23 & 15 & 0 & 100 & 13 & 18 \\ 2 & 3 & 2 & 0 & 13 & 0 & 2 \\ 3 & 4 & 2 & -3 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{113}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 2 & 3 \\ 3 & 4 & 4 & 0 & 22 & 3 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 15 & 0 & 85 & 12 & 17 \\ 2 & 3 & 2 & 0 & 12 & 0 & 2 \\ 3 & 4 & 2 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11} 3^{11} 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 2 & 3 \\ 3 & 4 & 4 & 0 & 27 & 3 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 18 & 0 & 133 & 15 & 21 \\ 2 & 3 & 2 & 0 & 15 & 0 & 2 \\ 3 & 4 & 2 & -3 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 2 & 3 \\ 3 & 4 & 4 & 0 & 26 & 3 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 18 & 0 & 118 & 14 & 20 \\ 2 & 3 & 2 & 0 & 14 & 0 & 2 \\ 3 & 4 & 2 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11} 3^{11} 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1826, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 2 & 3 \\ 3 & 4 & 4 & 0 & 32 & 3 & 4 \\ 2 & 4 & 0 & 0 & 21 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 18 & 32 & 21 & 0 & 193 & 18 & 25 \\ 2 & 3 & 2 & 0 & 18 & 0 & 2 \\ 3 & 4 & 2 & -3 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{11} 3^{11} 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 2 & 3 \\ 3 & 4 & 4 & 0 & 31 & 3 & 4 \\ 2 & 4 & 0 & 0 & 21 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 31 & 21 & 0 & 172 & 17 & 24 \\ 2 & 3 & 2 & 0 & 17 & 0 & 2 \\ 3 & 4 & 2 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{11} 3^{11} 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 2 & 3 \\ 3 & 4 & 4 & 0 & 36 & 3 & 4 \\ 2 & 4 & 0 & 0 & 24 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 20 & 36 & 24 & 0 & 238 & 20 & 28 \\ 2 & 3 & 2 & 0 & 20 & 0 & 2 \\ 3 & 4 & 2 & -3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [26] \quad 1^{11} 4^{13} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 1 & 3 \\ 3 & 4 & 4 & 0 & 5 & 0 & 4 \\ 2 & 4 & 0 & 0 & 3 & 0 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & -2 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{11} 1^{11} 4^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 1 & 3 \\ 3 & 4 & 4 & 0 & 7 & 0 & 4 \\ 2 & 4 & 0 & 0 & 5 & 0 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 5 & 0 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & -2 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11} 1^{11} 4^{11} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 1 & 3 \\ 3 & 4 & 4 & 0 & 12 & 0 & 4 \\ 2 & 4 & 0 & 0 & 8 & 0 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 8 & 12 & 8 & 0 & 26 & 0 & 10 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & -2 & 10 & 0 & 2 \end{pmatrix}_{b_0}$$

4.1092 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 3 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 3 & 6 & 4 \\ 2 & 3 & 2 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & 2 & -3 & 4 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 392, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 3 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 2 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & 2 & -3 & 3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 462, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 3 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 3 & 9 & 4 \\ 2 & 3 & 2 & 0 & 3 & 0 & 5 & 2 \\ 5 & 9 & 6 & 0 & 9 & 5 & 13 & 7 \\ 3 & 4 & 2 & -3 & 4 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 512, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 3 & 4 & 4 \\ 5 & 9 & 12 & 0 & 6 & 5 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 5 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 6 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & 6 & -3 & 3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 622, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 2 & 6 & 3 \\ 2 & 3 & 2 & 0 & 2 & 0 & 5 & 2 \\ 5 & 9 & 6 & 0 & 6 & 5 & 13 & 7 \\ 3 & 4 & 2 & -3 & 3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 1152, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 3 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 5 & 15 & 7 \\ 2 & 3 & 2 & 0 & 5 & 0 & 5 & 2 \\ 5 & 9 & 6 & 0 & 15 & 5 & 13 & 7 \\ 3 & 4 & 2 & -3 & 7 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.1093 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \text{n3mix}$

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -104, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 8 & 4 \\ 2 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 9 & 4 \\ 5 & 8 & 5 & 0 & 9 & 12 & 6 \\ 3 & 4 & 2 & -3 & 4 & 6 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{1111}1}3^{\overline{112}}1^{\overline{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 4 & 17 & 4 \\ 2 & 4 & 0 & 0 & 3 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 11 & 3 \\ 10 & 17 & 11 & 0 & 11 & 55 & 13 \\ 3 & 4 & 2 & -3 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{1111}1}3^{\overline{111}}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -119$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 4 & 0 & 5 & 7 & 4 \\ 2 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 6 & 4 \\ 4 & 7 & 5 & 0 & 6 & 7 & 5 \\ 3 & 4 & 2 & -3 & 4 & 5 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{1111}1}3^{\overline{111}}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -124$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 5 & 0 & 4 & 11 & 4 \\ 3 & 5 & 3 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 8 & 3 \\ 6 & 11 & 9 & 0 & 8 & 20 & 8 \\ 3 & 4 & 3 & -3 & 3 & 8 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{1111}1}3^{\overline{111}}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -204$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 5 & 0 & 4 & 7 & 4 \\ 3 & 5 & 3 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 5 & 3 \\ 4 & 7 & 6 & 0 & 5 & 7 & 5 \\ 3 & 4 & 3 & -3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{1111}1}3^{\overline{111}}1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -269$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 8 & 4 \\ 2 & 4 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 14 & 7 \\ 5 & 8 & 5 & 0 & 14 & 12 & 6 \\ 3 & 4 & 2 & -3 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{\overline{113}1}3^{\overline{111}}1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -401$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 11 & 3 \\ 3 & 4 & 4 & 0 & 5 & 20 & 4 \\ 2 & 4 & 0 & 0 & 3 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 18 & 4 \\ 11 & 20 & 14 & 0 & 18 & 69 & 15 \\ 3 & 4 & 2 & -3 & 4 & 15 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{111}1}3^{\overline{111}}1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -401$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 7 & 3 \\ 3 & 4 & 4 & 0 & 5 & 12 & 4 \\ 2 & 4 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 12 & 4 \\ 7 & 12 & 8 & 0 & 12 & 25 & 9 \\ 3 & 4 & 2 & -3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{\overline{111}1}3^{\overline{111}}1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -424$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 4 & 4 & 0 & 4 & 7 & 4 \\ 2 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 5 & 3 \\ 4 & 7 & 5 & 0 & 5 & 7 & 5 \\ 3 & 4 & 2 & -3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{\overline{1111}3^{\overline{111}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2}$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 4 & 3 \\ 3 & 4 & 9 & 0 & 4 & 7 & 4 \\ 5 & 9 & 12 & 0 & 6 & 11 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 0 & 1 & 5 & 3 \\ 4 & 7 & 11 & 0 & 5 & 7 & 5 \\ 3 & 4 & 6 & -3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{\overline{1111}3^{\overline{111}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2}$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -481, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 4 & 12 & 4 \\ 2 & 4 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 8 & 3 \\ 7 & 12 & 8 & 0 & 8 & 25 & 9 \\ 3 & 4 & 2 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{\overline{112}3^{\overline{11}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2}$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -500, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 9 & 3 \\ 3 & 4 & 4 & 0 & 5 & 16 & 4 \\ 2 & 4 & 0 & 0 & 3 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 15 & 4 \\ 9 & 16 & 11 & 0 & 15 & 44 & 12 \\ 3 & 4 & 2 & -3 & 4 & 12 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{\overline{111}3^{\overline{11}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2}$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 4 & 3 \\ 3 & 4 & 4 & 0 & 9 & 7 & 4 \\ 2 & 4 & 0 & 0 & 6 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 11 & 7 \\ 4 & 7 & 5 & 0 & 11 & 7 & 5 \\ 3 & 4 & 2 & -3 & 7 & 5 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{\overline{111}3^{\overline{11}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2}$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 7 & 3 \\ 3 & 4 & 4 & 0 & 9 & 12 & 4 \\ 2 & 4 & 0 & 0 & 6 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 20 & 7 \\ 7 & 12 & 8 & 0 & 20 & 25 & 9 \\ 3 & 4 & 2 & -3 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

4.1094 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 9$ [0] $1^{\overline{111}3^{\overline{11}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3}$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 2 & 3 \\ 4 & 9 & 4 & 0 & 12 & 4 & 5 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 12 & 6 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 5 & 2 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{11}3^{\overline{111}}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3}$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 6 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 & 3 \\ 3 & 4 & 2 & -3 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{111}3^{\overline{11}}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3}$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

13.2

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 2 & -3 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{113} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 2 & 3 \\ 4 & 9 & 4 & 0 & 28 & 4 & 5 \\ 2 & 4 & 0 & 0 & 12 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 28 & 12 & 0 & 73 & 12 & 15 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 5 & 2 & -3 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 2 & 3 \\ 4 & 9 & 4 & 0 & 20 & 4 & 5 \\ 2 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 20 & 9 & 0 & 37 & 9 & 11 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 5 & 2 & -3 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{122} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.9$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 5 & 0 & 13 & 8 & 4 \\ 3 & 5 & 3 & 0 & 12 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 12 & 0 & 28 & 18 & 10 \\ 4 & 8 & 6 & 0 & 18 & 9 & 5 \\ 3 & 4 & 3 & -3 & 10 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.34 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 9 & 3 \\ 8 & 14 & 9 & 0 & 9 & 36 & 10 \\ 3 & 4 & 2 & -3 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{132} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 5 & 0 & 17 & 8 & 4 \\ 3 & 5 & 3 & 0 & 15 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 15 & 0 & 49 & 24 & 13 \\ 4 & 8 & 6 & 0 & 24 & 9 & 5 \\ 3 & 4 & 3 & -3 & 13 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 & 4 \\ 2 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 9 & 0 & 37 & 9 & 11 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 2 & -3 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{113} 3^1 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 2 & 3 \\ 4 & 9 & 4 & 0 & 32 & 4 & 5 \\ 2 & 4 & 0 & 0 & 15 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 32 & 15 & 0 & 100 & 15 & 18 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 5 & 2 & -3 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 0 & 5 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 0 & 4 & 9 & 4 \\ 5 & 9 & 6 & 0 & 9 & 12 & 6 \\ 3 & 4 & 2 & -3 & 4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -399, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 2 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{114} 3^1 1^{31} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 2 & 3 \\ 4 & 9 & 4 & 0 & 44 & 4 & 5 \\ 2 & 4 & 0 & 0 & 21 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 18 & 44 & 21 & 0 & 193 & 21 & 25 \\ 2 & 4 & 3 & 0 & 21 & 0 & 2 \\ 3 & 5 & 2 & -3 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 6 & 12 & 6 \\ 3 & 4 & 2 & -3 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{114} 3^1 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 2 & 3 \\ 4 & 9 & 4 & 0 & 40 & 4 & 5 \\ 2 & 4 & 0 & 0 & 18 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 16 & 40 & 18 & 0 & 154 & 18 & 22 \\ 2 & 4 & 3 & 0 & 18 & 0 & 2 \\ 3 & 5 & 2 & -3 & 22 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{141} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -527, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 & 4 \\ 3 & 5 & 3 & 0 & 18 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 18 & 0 & 76 & 15 & 16 \\ 2 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 3 & -3 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -621, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 2 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{121} 3^1 1^{111} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -707, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 & 4 \\ 3 & 5 & 3 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 12 & 0 & 28 & 9 & 10 \\ 2 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 3 & -3 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{131} 3^1 1^{111} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -713, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.25 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 & 4 \\ 3 & 5 & 3 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 15 & 0 & 49 & 12 & 13 \\ 2 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 3 & -3 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1111} 3^1 1^{111} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -944, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.22 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 0 & 15 & 12 & 6 \\ 3 & 4 & 2 & -3 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

4.1095 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \ 1^{13} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} \quad T = 8, \ T_{\max} = 35, \ \Delta = -118, \ \det = 48, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.33 \rightarrow 14, \ T_{\max} = 35, \ T \in [14, 35]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 3 & 4 \\ 6 & 12 & 24 & 0 & 6 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 8 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} \quad T = 8, \ T_{\max} = 35, \ \Delta = -118, \ \det = 72, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.11 \rightarrow 13, \ T_{\max} = 35, \ T \in [13, 35]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 5 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} \quad T = 8, \ T_{\max} = 35, \ \Delta = -118, \ \det = 98, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.22 \rightarrow 13, \ T_{\max} = 35, \ T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 3 & 4 \\ 4 & 8 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{13} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} \quad T = 8, \ T_{\max} = 35, \ \Delta = -118, \ \det = 118, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.19 \rightarrow 13, \ T_{\max} = 35, \ T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} \quad T = 8, \ T_{\max} = 35, \ \Delta = -118, \ \det = 128, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 35, \ T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 35, \Delta = -118, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 3 & 4 \\ 5 & 9 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 6 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 35, \Delta = -118, \det = 278, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 3 & 4 \\ 7 & 13 & 27 & 0 & 7 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 7 & 0 & 0 & 2 \\ 3 & 4 & 9 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{13}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 8, T_{\max} = 35, \Delta = -118, \det = 328, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 3 & 4 \\ 9 & 17 & 48 & 0 & 9 & 12 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 9 & 0 & 0 & 2 \\ 3 & 4 & 12 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [8] $1^{11}3^{12}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 13 & 1 & 6 & 5 \\ 13 & 30 & 0 & 13 & 10 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 13 & 0 & 4 & 4 \\ 5 & 10 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{11}3^{12}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 1 & 6 & 5 \\ 12 & 25 & 0 & 12 & 9 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 12 & 0 & 4 & 4 \\ 5 & 9 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{11}3^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 15, \Delta = -97, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 3 \\ 2 & 2 & 6 & 4 & 0 & 4 \\ 4 & 6 & 7 & 5 & 0 & 5 \\ 2 & 4 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11}3^{12}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 6 & 5 \\ 9 & 13 & 1 & 9 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 9 & 0 & 4 & 4 \\ 5 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

[12] $1^{11}3^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 15, \Delta = -97, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 8 & 6 & 5 & 0 & 6 \\ 2 & 6 & 1 & 4 & 0 & 3 \\ 2 & 5 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{11}3^{11}1^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 15, \Delta = -97, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 3 & 2 & 0 & 0 & 2 \\ 2 & 3 & 3 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{12}3^1 1^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 15, \quad \Delta = -97, \quad \det = -136, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.24 \rightarrow 13, \quad T_{\max} = 15, \quad T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 3 \\ 4 & 8 & 6 & 20 & 0 & 6 \\ 2 & 6 & 1 & 13 & 0 & 3 \\ 8 & 20 & 13 & 43 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11}3^1 1^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 15, \quad \Delta = -97, \quad \det = -149, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 15, \quad T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 2 \\ 1 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 2 & 0 & 0 & 2 \\ 2 & 4 & 2 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{13}1^{11}3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 15, \quad \Delta = -97, \quad \det = -256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 15, \quad T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 8 & 6 & 10 & 0 & 6 \\ 2 & 6 & 1 & 5 & 0 & 3 \\ 4 & 10 & 5 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 5 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{13}1^{11}3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 15, \quad \Delta = -97, \quad \det = -284, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.42 \rightarrow 13, \quad T_{\max} = 15, \quad T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 & 2 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11}3^1 1^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 15, \quad \Delta = -97, \quad \det = -306, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.24 \rightarrow 13, \quad T_{\max} = 15, \quad T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 8 & 6 & 15 & 0 & 6 \\ 2 & 6 & 1 & 9 & 0 & 3 \\ 6 & 15 & 9 & 21 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11}3^1 1^{13}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 15, \quad \Delta = -97, \quad \det = -394, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.23 \rightarrow 13, \quad T_{\max} = 15, \quad T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 3 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 & 2 \\ 3 & 9 & 6 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 5 & 0 & 0 & 2 \\ 2 & 4 & 2 & 6 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [20] \quad 1^{11}4^{11}13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -88, \quad \det = 32, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 13, \quad T_{\max} = 14, \quad T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 3 \\ 3 & 4 & 4 & 0 & 0 & 4 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11}4^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -88, \quad \det = 130, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 1 & 3 \\ 4 & 4 & 3 & 0 & 0 & 4 \\ 2 & 3 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & -2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [22] \quad 1^{11} 5^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 1 & 8 & 5 \\ 9 & 13 & 0 & 12 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 12 & 0 & 10 & 6 \\ 5 & 7 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

4.1096 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 6 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

4.1097 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 4 & 3 \\ 3 & 4 & 9 & 0 & 3 & 8 & 4 \\ 5 & 9 & 12 & 0 & 5 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 4 & 2 \\ 4 & 8 & 12 & 0 & 4 & 10 & 6 \\ 3 & 4 & 6 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 & 3 \\ 4 & 9 & 4 & 0 & 4 & 8 & 5 \\ 2 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 3 & 8 & 3 & 0 & 3 & 4 & 4 \\ 3 & 5 & 2 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 4 & 4 & 0 & 3 & 8 & 4 \\ 2 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 4 & 2 \\ 4 & 8 & 6 & 0 & 4 & 10 & 6 \\ 3 & 4 & 2 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 11 & 3 \\ 3 & 4 & 4 & 0 & 3 & 19 & 4 \\ 2 & 4 & 0 & 0 & 2 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 11 & 2 \\ 11 & 19 & 12 & 0 & 11 & 73 & 15 \\ 3 & 4 & 2 & -3 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 3 & 5 & 4 \\ 2 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 3 & 2 \\ 3 & 5 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 2 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 2 & 3 \\ 3 & 4 & 13 & 0 & 3 & 4 & 4 \\ 7 & 13 & 27 & 0 & 7 & 9 & 9 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 7 & 0 & 0 & 2 & 2 \\ 2 & 4 & 9 & 0 & 2 & 1 & 3 \\ 3 & 4 & 9 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 4 & 0 & 3 & 13 & 4 \\ 2 & 4 & 0 & 0 & 2 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 7 & 2 \\ 7 & 13 & 9 & 0 & 7 & 28 & 10 \\ 3 & 4 & 2 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 4 & 4 & 0 & 3 & 17 & 4 \\ 2 & 4 & 0 & 0 & 2 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 9 & 2 \\ 9 & 17 & 12 & 0 & 9 & 49 & 13 \\ 3 & 4 & 2 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{112}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 16 & 3 \\ 3 & 4 & 4 & 0 & 3 & 28 & 4 \\ 2 & 4 & 0 & 0 & 2 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 16 & 2 \\ 16 & 28 & 18 & 0 & 16 & 154 & 22 \\ 3 & 4 & 2 & -3 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{114}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 11 & 3 \\ 3 & 4 & 4 & 0 & 3 & 21 & 4 \\ 2 & 4 & 0 & 0 & 2 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 11 & 2 \\ 11 & 21 & 15 & 0 & 11 & 76 & 16 \\ 3 & 4 & 2 & -3 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{112}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 0 & 3 & 18 & 4 \\ 2 & 4 & 0 & 0 & 2 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 10 & 2 \\ 10 & 18 & 12 & 0 & 10 & 58 & 14 \\ 3 & 4 & 2 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{112}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 13 & 3 \\ 3 & 4 & 4 & 0 & 3 & 23 & 4 \\ 2 & 4 & 0 & 0 & 2 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 13 & 2 \\ 13 & 23 & 15 & 0 & 13 & 100 & 18 \\ 3 & 4 & 2 & -3 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{113}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 4 & 0 & 3 & 22 & 4 \\ 2 & 4 & 0 & 0 & 2 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 12 & 2 \\ 12 & 22 & 15 & 0 & 12 & 85 & 17 \\ 3 & 4 & 2 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{113}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -88, \det = -1696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 15 & 3 \\ 3 & 4 & 4 & 0 & 3 & 27 & 4 \\ 2 & 4 & 0 & 0 & 2 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 15 & 2 \\ 15 & 27 & 18 & 0 & 15 & 133 & 21 \\ 3 & 4 & 2 & -3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{114} 3^{11} 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 4 & 4 & 0 & 3 & 26 & 4 \\ 2 & 4 & 0 & 0 & 2 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 14 & 2 \\ 14 & 26 & 18 & 0 & 14 & 118 & 20 \\ 3 & 4 & 2 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{113} 3^{11} 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1826, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 18 & 3 \\ 3 & 4 & 4 & 0 & 3 & 32 & 4 \\ 2 & 4 & 0 & 0 & 2 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 18 & 2 \\ 18 & 32 & 21 & 0 & 18 & 193 & 25 \\ 3 & 4 & 2 & -3 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{114} 3^{11} 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 17 & 3 \\ 3 & 4 & 4 & 0 & 3 & 31 & 4 \\ 2 & 4 & 0 & 0 & 2 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 17 & 2 \\ 17 & 31 & 21 & 0 & 17 & 172 & 24 \\ 3 & 4 & 2 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{114} 3^{11} 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 20 & 3 \\ 3 & 4 & 4 & 0 & 3 & 36 & 4 \\ 2 & 4 & 0 & 0 & 2 & 24 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 20 & 2 \\ 20 & 36 & 24 & 0 & 20 & 238 & 28 \\ 3 & 4 & 2 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [26] \quad 1^{111} 4^{13} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 0 & 5 & 4 \\ 2 & 4 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 3 & 0 & 0 & 4 & 4 \\ 3 & 4 & 2 & -2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{111} 4^{13} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 0 & 7 & 4 \\ 2 & 4 & 0 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 7 & 5 & 0 & 0 & 8 & 6 \\ 3 & 4 & 2 & -2 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{111} 4^{13} 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -192, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 0 & 12 & 4 \\ 2 & 4 & 0 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 12 & 8 & 0 & 0 & 26 & 10 \\ 3 & 4 & 2 & -2 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

4.1098 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_{2 \rightarrow (-3)} + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{\textcolor{brown}{1}} 1^{\textcolor{red}{1}} 3^{\textcolor{blue}{1}} 1^{\textcolor{teal}{1}} 1^{\textcolor{violet}{1}} + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{teal}{g}_2] + \textcolor{teal}{s} \textcolor{blue}{u}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{s} \textcolor{blue}{u}_2 \rightarrow (-3) + \textcolor{blue}{s} \textcolor{teal}{u}_2 + \textcolor{violet}{s} \textcolor{blue}{u}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2 \\ T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix}
1 & 3 & 2 & 1 & 2 & 3 & 3 \\
3 & 4 & 4 & 0 & 3 & 5 & 5 \\
2 & 4 & 0 & 0 & 2 & 3 & 2 \\
1 & 0 & 0 & -5 & 0 & 0 & -3 \\
2 & 3 & 2 & 0 & 0 & 3 & 2 \\
3 & 5 & 3 & 0 & 3 & 4 & 4 \\
3 & 5 & 3 & 0 & 3 & 6 & 4 \\
3 & 4 & 2 & -3 & 2 & 4 & 1
\end{pmatrix}_{b_0}$$

$$[1] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{blue}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 392, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix}
1 & 3 & 2 & 1 & 2 & 2 & 2 & 3 \\
3 & 4 & 4 & 0 & 3 & 4 & 4 & 4 \\
2 & 4 & 0 & 0 & 2 & 3 & 3 & 2 \\
1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\
2 & 3 & 2 & 0 & 0 & 2 & 2 & 2 \\
2 & 4 & 3 & 0 & 2 & 1 & 3 & 3 \\
2 & 4 & 3 & 0 & 2 & 3 & 1 & 3 \\
3 & 4 & 2 & -3 & 2 & 3 & 3 & 1
\end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}u_3 + \textcolor{brown}{f}_4 + \textcolor{red}{s}u_2 \rightarrow (-3) + \textcolor{blue}{s}u_2 + \textcolor{violet}{s}u_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 4 & 0 & 3 & 5 & 9 & 4 \\ 2 & 4 & 0 & 0 & 2 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 3 & 5 & 2 \\ 3 & 5 & 3 & 0 & 3 & 4 & 9 & 4 \\ 5 & 9 & 6 & 0 & 5 & 9 & 13 & 7 \\ 3 & 4 & 2 & -3 & 2 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \overset{1}{1} \overset{1}{1} \overset{1}{1} \overset{1}{3} \overset{1}{1} \overset{1}{1} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}u_3 + \textcolor{brown}{f}_4 + \textcolor{red}{s}u_2 \rightarrow (-3) + \textcolor{blue}{s}u_2 + \textcolor{violet}{s}u_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 9 & 0 & 3 & 4 & 4 & 4 \\ 5 & 9 & 12 & 0 & 5 & 6 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 2 & 2 & 2 \\ 2 & 4 & 6 & 0 & 2 & 1 & 3 & 3 \\ 2 & 4 & 6 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & 6 & -3 & 2 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1111}} 3^{\overline{111}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{violet}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 & 9 & 4 \\ 2 & 4 & 0 & 0 & 2 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 5 & 2 \\ 2 & 4 & 3 & 0 & 2 & 1 & 6 & 3 \\ 5 & 9 & 6 & 0 & 5 & 6 & 13 & 7 \\ 3 & 4 & 2 & -3 & 2 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{1111}} 3^{\overline{111}} 1^{\overline{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 3 & 9 & 9 & 4 \\ 2 & 4 & 0 & 0 & 2 & 6 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 5 & 2 \\ 5 & 9 & 6 & 0 & 5 & 13 & 15 & 7 \\ 5 & 9 & 6 & 0 & 5 & 15 & 13 & 7 \\ 3 & 4 & 2 & -3 & 2 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

4.1099 $\mathbf{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_{2 \rightarrow (-3)} + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overset{1}{\color{red}2}} \overset{1}{\color{red}3} 1^{\overset{1}{\color{blue}1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{s}\textcolor{blue}{u}_2 \rightarrow (-3) + \textcolor{blue}{s}\textcolor{blue}{u}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -34, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.4$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 & 3 \\ 3 & 4 & 8 & 0 & 3 & 5 & 4 \\ 4 & 8 & 9 & 0 & 4 & 6 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 3 & 3 & 3 \\ 3 & 4 & 5 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^11^{11} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{u}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{s}\mathfrak{u}}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -36, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.33 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 4 & 5 \\ 2 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 0 & 2 \\ 3 & 5 & 2 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^11^{11} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{u}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{s}\mathfrak{u}}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -274, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.32 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^11^{11} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{u}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{s}\mathfrak{u}}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -354, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.47 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 2 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^11^{11} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{u}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{s}\mathfrak{u}}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -544, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 3 & 9 & 4 \\ 2 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 \\ 5 & 9 & 6 & 0 & 5 & 12 & 6 \\ 3 & 4 & 2 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

4.1100 NHC(-2)(-3) + $\mathfrak{s}\mathfrak{u}_3 + \mathfrak{f}_4 + \mathfrak{s}\mathfrak{u}_2\mathfrak{n}3\mathfrak{mix}$

$$T = 8 \quad [0] \ 1^{11}4^31^{11} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{u}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \quad T = 8, \ T_{\max} = 35, \ \Delta = -118, \ \det = 80, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 35, \ T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 10 & 3 \\ 3 & 4 & 9 & 0 & 19 & 4 \\ 5 & 9 & 12 & 0 & 29 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 29 & 0 & 64 & 14 \\ 3 & 4 & 6 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12}3^11^{11} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{u}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \quad T = 8, \ T_{\max} = 35, \ \Delta = -118, \ \det = 102, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.06 \rightarrow 13, \ T_{\max} = 35, \ T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 5 & 0 & 11 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 9 & 0 & 20 & 8 \\ 3 & 4 & 3 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^11^{11} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{u}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \quad T = 8, \ T_{\max} = 35, \ \Delta = -118, \ \det = 107, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 35, \ T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 5 & 0 & 12 & 6 \\ 3 & 4 & 2 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12}3^11^{11} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{u}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \quad T = 8, \ T_{\max} = 35, \ \Delta = -118, \ \det = 113, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.42 \rightarrow 13, \ T_{\max} = 35, \ T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 4 & 8 & 9 & 0 & 16 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 16 & 0 & 25 & 9 \\ 3 & 4 & 5 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^11^{11} + [\textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{s}\mathfrak{u}}_3 + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{s}\mathfrak{u}}_2 \quad T = 8, \ T_{\max} = 35, \ \Delta = -118, \ \det = 132, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.09 \rightarrow 13, \ T_{\max} = 35, \ T \in [13, 35]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 16 & 8 & 0 & 25 & 9 \\ 3 & 5 & 2 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{113}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.07 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 0 & 7 & 5 \\ 3 & 4 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}3^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 27 & 17 & 0 & 151 & 21 \\ 3 & 4 & 2 & -3 & 21 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 5 & 0 & 12 & 6 \\ 3 & 5 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{121}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 197, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 7 & 4 \\ 4 & 8 & 9 & 0 & 10 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 10 & 0 & 7 & 5 \\ 3 & 4 & 5 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{113}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 3 \\ 3 & 4 & 9 & 0 & 15 & 4 \\ 5 & 9 & 12 & 0 & 23 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 23 & 0 & 39 & 11 \\ 3 & 4 & 6 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{111}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 227, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.04 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 5 \\ 3 & 4 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{111}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 7 & 4 \\ 5 & 9 & 12 & 0 & 11 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 11 & 0 & 7 & 5 \\ 3 & 4 & 6 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{112}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 3 \\ 3 & 4 & 9 & 0 & 11 & 4 \\ 5 & 9 & 12 & 0 & 17 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 17 & 0 & 20 & 8 \\ 3 & 4 & 6 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{111}3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 22 & 14 & 0 & 97 & 17 \\ 3 & 4 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{112}3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.14 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 11 & 4 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 8 & 0 & 20 & 8 \\ 3 & 4 & 2 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111311^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 8 & 0 & 25 & 9 \\ 3 & 4 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{121311^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 7 & 4 \\ 7 & 13 & 27 & 0 & 16 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 16 & 0 & 7 & 5 \\ 3 & 4 & 9 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{111311^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 403, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 17 & 11 & 0 & 55 & 13 \\ 3 & 4 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{113311^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 443, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 15 & 4 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 11 & 0 & 39 & 11 \\ 3 & 4 & 2 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{114311^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 560, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 19 & 4 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 14 & 0 & 64 & 14 \\ 3 & 4 & 2 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{112311^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 635, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 11 & 0 & 44 & 12 \\ 3 & 4 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{112311^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 832, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 21 & 14 & 0 & 80 & 16 \\ 3 & 4 & 2 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{112311^{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 908, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 31 & 20 & 0 & 188 & 24 \\ 3 & 4 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{113311^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 908, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.01 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 20 & 4 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 14 & 0 & 69 & 15 \\ 3 & 4 & 2 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{112311^{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 923, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 26 & 17 & 0 & 128 & 20 \\ 3 & 4 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{114}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1187, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 17 & 0 & 100 & 18 \\ 3 & 4 & 2 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{114}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 25 & 4 \\ 2 & 4 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 25 & 17 & 0 & 111 & 19 \\ 3 & 4 & 2 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{114}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1520, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 30 & 4 \\ 2 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 30 & 20 & 0 & 165 & 23 \\ 3 & 4 & 2 & -3 & 23 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{114}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1667, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 35 & 4 \\ 2 & 4 & 0 & 0 & 23 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 35 & 23 & 0 & 231 & 27 \\ 3 & 4 & 2 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{114}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1708, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 29 & 4 \\ 2 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 29 & 20 & 0 & 148 & 22 \\ 3 & 4 & 2 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{114}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 2123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 19 & 3 \\ 3 & 4 & 4 & 0 & 34 & 4 \\ 2 & 4 & 0 & 0 & 23 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 19 & 34 & 23 & 0 & 208 & 26 \\ 3 & 4 & 2 & -3 & 26 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{114}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 2432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 22 & 3 \\ 3 & 4 & 4 & 0 & 39 & 4 \\ 2 & 4 & 0 & 0 & 26 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 22 & 39 & 26 & 0 & 280 & 30 \\ 3 & 4 & 2 & -3 & 30 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [32] $1^{111}3^121^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 37 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 37 & 54 & 0 & 217 & 27 \\ 5 & 7 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

[33] $1^{11}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 8 & 4 \\ 1 & 4 & 0 & 3 & 0 & 4 & 2 \\ 1 & 3 & 3 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 4 & 4 & 0 & 9 & 5 \\ 2 & 4 & 2 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{112}31^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 7 & 3 \\ 2 & 4 & 3 & 3 & 0 & 8 & 3 \\ 1 & 3 & 3 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 7 & 8 & 7 & 0 & 15 & 7 \\ 2 & 3 & 3 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{11} 3^1 1^1 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 5 & 3 \\ 2 & 4 & 3 & 3 & 0 & 6 & 3 \\ 1 & 3 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 6 & 5 & 0 & 7 & 5 \\ 2 & 3 & 3 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{11} 3^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 7 & 5 \\ 9 & 13 & 1 & 9 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 4 & 4 \\ 5 & 7 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{11} 3^1 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 44 & 5 \\ 9 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 44 & 69 & 0 & 340 & 34 \\ 5 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{11} 3^1 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 25 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 39 & 0 & 107 & 19 \\ 5 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{112} 3^1 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 56 & 5 \\ 9 & 13 & 1 & 84 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 56 & 84 & 0 & 520 & 42 \\ 5 & 7 & -1 & 42 & 3 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{111} 31^1 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 5 & 4 \\ 1 & 4 & 0 & 3 & 0 & 4 & 2 \\ 1 & 3 & 3 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 2 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{111} 31^1 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 4 & 3 \\ 2 & 4 & 3 & 3 & 0 & 5 & 3 \\ 1 & 3 & 3 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 4 & 0 & 4 & 4 \\ 2 & 3 & 3 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{111} 3^1 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 15 & 5 \\ 9 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 15 & 24 & 0 & 37 & 11 \\ 5 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{11} 31^{11} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 5 & 4 \\ 1 & 4 & 0 & 3 & 0 & 4 & 2 \\ 2 & 6 & 3 & 3 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 5 & 0 & 4 & 4 \\ 2 & 4 & 2 & 3 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{11} 3^1 1^{14} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -205, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

11.2

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 4 & 3 & 0 & 27 & 4 \\ 1 & 4 & 0 & 3 & 0 & 14 & 2 \\ 1 & 3 & 3 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 14 & 14 & 0 & 120 & 18 \\ 2 & 4 & 2 & 2 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{11}3^{11}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 9 & 4 \\ 1 & 4 & 0 & 3 & 0 & 5 & 2 \\ 1 & 3 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 5 & 5 & 0 & 12 & 6 \\ 2 & 4 & 2 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{11}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 24 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 24 & 39 & 0 & 100 & 18 \\ 5 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{11}3^{11}1^{13}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -254, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 4 & 3 & 0 & 21 & 4 \\ 1 & 4 & 0 & 3 & 0 & 11 & 2 \\ 1 & 3 & 3 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 11 & 11 & 0 & 72 & 14 \\ 2 & 4 & 2 & 2 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{11}3^{11}1^{12}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -261, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 3 & 0 & 15 & 4 \\ 1 & 4 & 0 & 3 & 0 & 8 & 2 \\ 1 & 3 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 8 & 8 & 0 & 36 & 10 \\ 2 & 4 & 2 & 2 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{111}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 26 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 39 & 0 & 107 & 19 \\ 5 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{111}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 16 & 5 \\ 9 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 16 & 24 & 0 & 37 & 11 \\ 5 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{112}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 17 & 5 \\ 9 & 13 & 1 & 24 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 17 & 24 & 0 & 37 & 11 \\ 5 & 7 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{112}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 34 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 34 & 54 & 0 & 200 & 26 \\ 5 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{111}3^{11}21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 1 & 27 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 27 & 39 & 0 & 107 & 19 \\ 5 & 7 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[54] $1^{111}31^{11}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 & 2 \\ 2 & 4 & 4 & 9 & 0 & 5 & 4 \\ 1 & 4 & 0 & 6 & 0 & 4 & 2 \\ 3 & 9 & 6 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 9 & 0 & 4 & 4 \\ 2 & 4 & 2 & 6 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{111}3^121^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 45 & 5 \\ 9 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 45 & 69 & 0 & 340 & 34 \\ 5 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[56] $1^{111}31^{11}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -430, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 3 & 0 & 11 & 4 \\ 1 & 4 & 0 & 3 & 0 & 7 & 2 \\ 1 & 3 & 3 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 11 & 7 & 7 & 0 & 20 & 8 \\ 2 & 4 & 2 & 2 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{111}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 25 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 25 & 39 & 0 & 100 & 18 \\ 5 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[58] $1^{111}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 35 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 35 & 54 & 0 & 200 & 26 \\ 5 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

[59] $1^{112}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 26 & 5 \\ 9 & 13 & 1 & 39 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 26 & 39 & 0 & 100 & 18 \\ 5 & 7 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[60] $1^{112}3^121^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.02 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 46 & 5 \\ 9 & 13 & 1 & 69 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 46 & 69 & 0 & 340 & 34 \\ 5 & 7 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[61] $1^{111}31^{12}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -679, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 3 & 0 & 17 & 4 \\ 1 & 4 & 0 & 3 & 0 & 10 & 2 \\ 1 & 3 & 3 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 17 & 10 & 10 & 0 & 48 & 12 \\ 2 & 4 & 2 & 2 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

[62] $1^{112}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -768, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 1 & 36 & 5 \\ 9 & 13 & 1 & 54 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 36 & 54 & 0 & 200 & 26 \\ 5 & 7 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{111}31^{13}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -886, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 8 & 2 \\ 2 & 4 & 4 & 3 & 0 & 23 & 4 \\ 1 & 4 & 0 & 3 & 0 & 13 & 2 \\ 1 & 3 & 3 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 23 & 13 & 13 & 0 & 88 & 16 \\ 2 & 4 & 2 & 2 & -3 & 16 & 0 \end{pmatrix}_{b_0}$$

[64] $1^{111} 31^{14} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -1051$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 10 & 2 \\ 2 & 4 & 4 & 3 & 0 & 29 & 4 \\ 1 & 4 & 0 & 3 & 0 & 16 & 2 \\ 1 & 3 & 3 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 29 & 16 & 16 & 0 & 140 & 20 \\ 2 & 4 & 2 & 2 & -3 & 20 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [65] $1^{11} 4^{13} 1^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -88$, $\det = 24$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{111} 4^{13} 1^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -88$, $\det = 56$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 7 & 5 & 0 & 8 & 6 \\ 3 & 4 & 2 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{13} 1^1 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -88$, $\det = 79$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.96 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{11} 4^{13} 1^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -88$, $\det = 105$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 4 & 3 & 0 & 8 & 4 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{111} 4^{13} 1^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -88$, $\det = 128$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 13.09 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{11} 4^{13} 1^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -88$, $\det = 150$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 4 & 3 & 0 & 5 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[71] $1^{111} 4^{13} 1^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -88$, $\det = 224$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 12 & 8 & 0 & 26 & 10 \\ 3 & 4 & 2 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{111} 4^{13} 1^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{f}_4 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 14$, $\Delta = -88$, $\det = 240$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 13.05 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [73] \quad 1^{11} 5^{13} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -89, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 1 & 14 & 5 \\ 9 & 13 & 0 & 22 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 22 & 0 & 33 & 11 \\ 5 & 7 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{12} 5^{13} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -126, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.78 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 9 & 1 & 20 & 5 \\ 9 & 13 & 0 & 32 & 7 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 32 & 0 & 70 & 16 \\ 5 & 7 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

4.1101 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{f}_4$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 & 3 \\ 3 & 4 & 4 & 0 & 8 & 5 & 4 \\ 2 & 4 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 5 & 0 & 12 & 9 & 6 \\ 3 & 5 & 3 & 0 & 9 & 4 & 4 \\ 3 & 4 & 2 & -3 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 2 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 & 4 \\ 2 & 4 & 0 & 0 & 11 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 17 & 11 & 0 & 55 & 11 & 13 \\ 2 & 4 & 3 & 0 & 11 & 1 & 3 \\ 3 & 4 & 2 & -3 & 13 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 4 & 0 & 7 & 5 & 4 \\ 2 & 4 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 6 & 5 \\ 3 & 5 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & 2 & -3 & 5 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 5 & 0 & 11 & 4 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 9 & 0 & 20 & 8 & 8 \\ 2 & 4 & 3 & 0 & 8 & 1 & 3 \\ 3 & 4 & 3 & -3 & 8 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 7 & 5 & 5 \\ 2 & 4 & 3 & 0 & 5 & 1 & 3 \\ 3 & 4 & 3 & -3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 8 & 9 & 4 \\ 2 & 4 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 5 & 0 & 12 & 14 & 6 \\ 5 & 9 & 6 & 0 & 14 & 13 & 7 \\ 3 & 4 & 2 & -3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1133}1^{\overline{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 & 3 \\ 3 & 4 & 4 & 0 & 20 & 5 & 4 \\ 2 & 4 & 0 & 0 & 14 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 14 & 0 & 69 & 18 & 15 \\ 3 & 5 & 3 & 0 & 18 & 4 & 4 \\ 3 & 4 & 2 & -3 & 15 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1113}1^{\overline{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 & 3 \\ 3 & 4 & 4 & 0 & 12 & 5 & 4 \\ 2 & 4 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 8 & 0 & 25 & 12 & 9 \\ 3 & 5 & 3 & 0 & 12 & 4 & 4 \\ 3 & 4 & 2 & -3 & 9 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1111}3^{\overline{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -424, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.07 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 & 4 \\ 2 & 4 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 5 & 5 \\ 2 & 4 & 3 & 0 & 5 & 1 & 3 \\ 3 & 4 & 2 & -3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1111}3^{\overline{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 2 & 3 \\ 3 & 4 & 9 & 0 & 7 & 4 & 4 \\ 5 & 9 & 12 & 0 & 11 & 6 & 6 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 11 & 0 & 7 & 5 & 5 \\ 2 & 4 & 6 & 0 & 5 & 1 & 3 \\ 3 & 4 & 6 & -3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1111}3^{\overline{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -481, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 & 4 \\ 2 & 4 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 8 & 0 & 25 & 8 & 9 \\ 2 & 4 & 3 & 0 & 8 & 1 & 3 \\ 3 & 4 & 2 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1123}1^{\overline{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -500, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 & 3 \\ 3 & 4 & 4 & 0 & 16 & 5 & 4 \\ 2 & 4 & 0 & 0 & 11 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 11 & 0 & 44 & 15 & 12 \\ 3 & 5 & 3 & 0 & 15 & 4 & 4 \\ 3 & 4 & 2 & -3 & 12 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1111}3^{\overline{11}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -509, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.06 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 5 & 3 \\ 3 & 4 & 4 & 0 & 7 & 9 & 4 \\ 2 & 4 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 5 & 0 & 7 & 11 & 5 \\ 5 & 9 & 6 & 0 & 11 & 13 & 7 \\ 3 & 4 & 2 & -3 & 5 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{1111}3^{\overline{111}}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -896, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 5 & 3 \\ 3 & 4 & 4 & 0 & 12 & 9 & 4 \\ 2 & 4 & 0 & 0 & 8 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 8 & 0 & 25 & 20 & 9 \\ 5 & 9 & 6 & 0 & 20 & 13 & 7 \\ 3 & 4 & 2 & -3 & 9 & 7 & 1 \end{pmatrix}_{b_0}$$

4.1102 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{f}_4 + \text{su2n3mix} + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 & 4 \\ 2 & 4 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 8 & 5 & 0 & 12 & 5 & 6 \\ 2 & 4 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & 2 & -3 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -81, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 2 & 3 \\ 4 & 9 & 4 & 0 & 12 & 4 & 5 \\ 2 & 4 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 12 & 5 & 0 & 12 & 5 & 6 \\ 2 & 4 & 3 & 0 & 5 & 0 & 2 \\ 3 & 5 & 2 & -3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 5 & 0 & 7 & 8 & 4 \\ 3 & 5 & 3 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 7 & 10 & 5 \\ 4 & 8 & 6 & 0 & 10 & 9 & 5 \\ 3 & 4 & 3 & -3 & 5 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 5 & 0 & 11 & 4 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 9 & 0 & 20 & 8 & 8 \\ 2 & 4 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & 3 & -3 & 8 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 6 & 0 & 7 & 5 & 5 \\ 2 & 4 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & 3 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

4.1103 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 5 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 5 & 0 & 8 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 0 & 9 & 5 \\ 2 & 4 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

- [3] $1^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 8$, $T_{\max} = 83$, $\Delta = -125$, $\det = 142$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 83$, $T \in [13, 83]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$
- [4] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 8$, $T_{\max} = 83$, $\Delta = -125$, $\det = 160$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 83$, $T \in [13, 83]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 6 & 0 & 12 & 6 \\ 3 & 5 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$
- [5] $1^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 8$, $T_{\max} = 83$, $\Delta = -125$, $\det = 195$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.46 \rightarrow 13$, $T_{\max} = 83$, $T \in [13, 83]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$
- [6] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 8$, $T_{\max} = 83$, $\Delta = -125$, $\det = 304$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.21 \rightarrow 13$, $T_{\max} = 83$, $T \in [13, 83]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$
- [7] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 8$, $T_{\max} = 83$, $\Delta = -125$, $\det = 307$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.19 \rightarrow 13$, $T_{\max} = 83$, $T \in [13, 83]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 0 & 36 & 10 \\ 3 & 4 & 2 & -3 & 10 & 0 \end{pmatrix}_{b_0}$
- $T = 9$ [8] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -19$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.26 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 \\ 2 & 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 3 & -3 & 1 \end{pmatrix}_{b_0}$
- [9] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -21$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$
 $\begin{pmatrix} 5 & 12 & 1 & 1 & 5 & 9 \\ 12 & 25 & 0 & 0 & 9 & 9 \\ 1 & 0 & -3 & 0 & -1 & 1 \\ 1 & 0 & 0 & -3 & -1 & 1 \\ 5 & 9 & -1 & -1 & 2 & 2 \end{pmatrix}_{b_0}$
- [10] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -26$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.46 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$
 $\begin{pmatrix} 5 & 13 & 9 & 1 & 5 & 5 \\ 13 & 30 & 20 & 0 & 10 & 10 \\ 9 & 20 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & -3 & -1 & 1 \\ 5 & 10 & 6 & -1 & 2 & 2 \end{pmatrix}_{b_0}$
- [11] $1^{12}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.4$
 $T_{\min} = 12.59 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & -2 & 2 & 0 \\ 2 & 4 & 0 & 0 & 2 & 5 & 3 \\ 2 & 2 & -1 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$
- [12] $1^{111}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 11 & 0 & 31 & 11 \\ 3 & 4 & 3 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{\overline{11}} 3 21^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 0 & 1 & 3 \\ 3 & 5 & 3 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{\overline{111}} 31^{\overline{1}} 31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{\overline{11}} 3 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 5 & 9 & 1 & 1 & 5 \\ 9 & 13 & 1 & 0 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{\overline{1}} 31^{\overline{11}} 31^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -105, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 3 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{\overline{111}} 3 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -114, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.47 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 6 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 3 & 0 & 3 & 3 \\ 3 & 4 & 3 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{\overline{12}} 3 21^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 5 & 2 \\ 1 & 2 & 0 & 3 & 0 & 5 & 2 \\ 1 & 1 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 5 & 0 & 12 & 6 \\ 2 & 2 & 2 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{\overline{11}} 3 21^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 2 \\ 1 & 2 & 0 & 3 & 0 & 4 & 2 \\ 1 & 1 & 3 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 2 & 2 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{\overline{111}} 3 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 5 & 9 & 1 & 9 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 15 & 0 & 12 & 6 \\ 5 & 7 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{\overline{11}} 3 21^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.28 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 5 & 5 & 0 & 10 & 5 \\ 2 & 5 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 10 & 7 & 0 & 13 & 7 \\ 3 & 5 & 3 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{\overline{111}} 3 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{f}_4 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 7 & 0 & 13 & 7 \\ 3 & 4 & 3 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{11}31^{11}31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.52 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{11}321^{11}321^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 5 & 9 & 1 & 10 & 5 \\ 9 & 13 & 1 & 15 & 7 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 15 & 0 & 12 & 6 \\ 5 & 7 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{11}321^{11}321^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -250, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 5 & 5 & 0 & 15 & 5 \\ 2 & 5 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 15 & 11 & 0 & 31 & 11 \\ 3 & 5 & 3 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{11}321^{11}321^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 5 & 5 & 0 & 20 & 5 \\ 2 & 5 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 15 & 0 & 57 & 15 \\ 3 & 5 & 3 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{11}321^{11}321^{111} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 2 & 3 & 0 & 3 & 2 \\ 1 & 2 & 0 & 4 & 0 & 4 & 2 \\ 2 & 3 & 4 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 7 & 0 & 4 & 4 \\ 2 & 2 & 2 & 4 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [28] $1^{11}4131^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_3} \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 5 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{11}41^{11}31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_3} \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{11}41^{11}31^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_3} \quad T = 10, T_{\max} = 15, \Delta = -95, \det = 102, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

4.1104 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$

$T = 9$ [0] $1^{11}31^{11}31^{111} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{f}_4} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 5 & 3 \\ 4 & 9 & 4 & 0 & 4 & 12 & 5 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 12 & 6 & 0 & 6 & 13 & 7 \\ 3 & 5 & 2 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 \\ 3 & 5 & 3 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & 3 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 2 & -3 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{113}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 11 & 3 \\ 4 & 9 & 4 & 0 & 4 & 28 & 5 \\ 2 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 11 & 28 & 12 & 0 & 12 & 73 & 15 \\ 3 & 5 & 2 & -3 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{112}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 8 & 3 \\ 4 & 9 & 4 & 0 & 4 & 20 & 5 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 8 & 20 & 9 & 0 & 9 & 37 & 11 \\ 3 & 5 & 2 & -3 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111}3^11^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{122}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.9$

$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 5 & 0 & 8 & 13 & 4 \\ 3 & 5 & 3 & 0 & 6 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 9 & 18 & 5 \\ 7 & 13 & 12 & 0 & 18 & 28 & 10 \\ 3 & 4 & 3 & -3 & 5 & 10 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^11^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.34 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 & 4 \\ 2 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 9 & 0 & 36 & 9 & 10 \\ 2 & 4 & 3 & 0 & 9 & 1 & 3 \\ 3 & 4 & 2 & -3 & 10 & 3 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{123}3^11^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 5 & 0 & 8 & 17 & 4 \\ 3 & 5 & 3 & 0 & 6 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 9 & 24 & 5 \\ 9 & 17 & 15 & 0 & 24 & 49 & 13 \\ 3 & 4 & 3 & -3 & 5 & 13 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{1111}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 8 & 14 & 9 & 0 & 9 & 37 & 11 \\ 3 & 4 & 2 & -3 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{113}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 13 & 3 \\ 4 & 9 & 4 & 0 & 4 & 32 & 5 \\ 2 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 13 & 32 & 15 & 0 & 15 & 100 & 18 \\ 3 & 5 & 2 & -3 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 & 3 \\ 3 & 4 & 4 & 0 & 9 & 5 & 4 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 9 & 6 \\ 3 & 5 & 3 & 0 & 9 & 4 & 4 \\ 3 & 4 & 2 & -3 & 6 & 4 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -399, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 2 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{114}3^11^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 18 & 3 \\ 4 & 9 & 4 & 0 & 4 & 44 & 5 \\ 2 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 18 & 44 & 21 & 0 & 21 & 193 & 25 \\ 3 & 5 & 2 & -3 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 9 & 4 \\ 3 & 5 & 3 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}3^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 6 & 6 \\ 2 & 4 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 2 & -3 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{114}3^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 16 & 3 \\ 4 & 9 & 4 & 0 & 4 & 40 & 5 \\ 2 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 16 & 40 & 18 & 0 & 18 & 154 & 22 \\ 3 & 5 & 2 & -3 & 2 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{114} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -527, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 5 & 0 & 4 & 21 & 4 \\ 3 & 5 & 3 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 11 & 21 & 18 & 0 & 15 & 76 & 16 \\ 3 & 4 & 3 & -3 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{114} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -621, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 2 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{112} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -707, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 0 & 4 & 13 & 4 \\ 3 & 5 & 3 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 7 & 13 & 12 & 0 & 9 & 28 & 10 \\ 3 & 4 & 3 & -3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{113} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -713, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 5 & 0 & 4 & 17 & 4 \\ 3 & 5 & 3 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 9 & 17 & 15 & 0 & 12 & 49 & 13 \\ 3 & 4 & 3 & -3 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 15 & 6 \\ 5 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 2 & -3 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

4.1105 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 5 & 0 & 8 & 3 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 9 & 4 & 5 \\ 2 & 3 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 3 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 4 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 2 & 0 & 2 & 0 & 2 \\ 3 & 5 & 2 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 3 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.47 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 3 & 4 & 2 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 0 & 9 & 3 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 0 & 12 & 5 & 6 \\ 2 & 3 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 2 & -3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

4.1106 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}2\mathfrak{n}3\mathfrak{mix}$

$$T = 9 \quad [0] \ 1^{111}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 0 & 4 & 8 & 4 \\ 2 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 5 & 8 & 5 & 0 & 5 & 12 & 6 \\ 3 & 4 & 2 & -3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -81, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 5 & 3 \\ 4 & 9 & 4 & 0 & 4 & 12 & 5 \\ 2 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 5 & 12 & 5 & 0 & 5 & 12 & 6 \\ 3 & 5 & 2 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{121}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 5 & 0 & 8 & 7 & 4 \\ 3 & 5 & 3 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 6 & 0 & 9 & 10 & 5 \\ 4 & 7 & 6 & 0 & 10 & 7 & 5 \\ 3 & 4 & 3 & -3 & 5 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{121}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 5 & 0 & 4 & 11 & 4 \\ 3 & 5 & 3 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 8 & 2 \\ 6 & 11 & 9 & 0 & 8 & 20 & 8 \\ 3 & 4 & 3 & -3 & 2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 5 & 0 & 4 & 7 & 4 \\ 3 & 5 & 3 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 4 & 7 & 6 & 0 & 5 & 7 & 5 \\ 3 & 4 & 3 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

4.1107 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.20 \rightarrow 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 & 3 \\ 4 & 9 & 4 & 0 & 4 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 8 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 2 & -3 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 5 & 3 \\ 4 & 9 & 4 & 0 & 4 & 12 & 5 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 12 & 6 & 0 & 6 & 12 & 6 \\ 3 & 5 & 2 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -104, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

4.1108 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \quad 1^{\overline{11}} 5^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 6 & 4 & 0 & 4 \\ 4 & 6 & 7 & 4 & 0 & 5 \\ 2 & 4 & 4 & 1 & 1 & 3 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 5 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 21^{\overline{1}} 5^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 4 & 3 & 4 \\ 3 & 3 & 0 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 & 3 \\ 2 & 3 & 2 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1}} 5^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 3 & 2 & 1 & 1 & 3 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 21^{\overline{1}} 5^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 34, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 4 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 & 3 \\ 3 & 3 & 2 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 21^{\overline{11}} 5^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 105, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 2 & 3 \\ 4 & 4 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 3 & 4 & 2 & 1 & 3 & 3 \\ 2 & 3 & 2 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [5] \quad 1^{11} 6^{11} 13 15 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -2 & 1 & 0 \\ 2 & 2 & 3 & 1 & 1 & 3 \\ 3 & 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

4.1109 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{11} 5^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 1 & 1 & 2 & 3 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 2 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 5^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 3 \\ 3 & 4 & 3 & 3 & 0 & 10 & 4 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 2 & 3 & 2 & 1 & 1 & 8 & 3 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 6 & 10 & 6 & 8 & 0 & 21 & 9 \\ 3 & 4 & 2 & 3 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{12} 5^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -23, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 3 & 3 & 0 & 17 & 4 \\ 2 & 3 & 0 & 2 & 0 & 10 & 2 \\ 2 & 3 & 2 & 1 & 1 & 13 & 3 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 10 & 17 & 10 & 13 & 0 & 61 & 15 \\ 3 & 4 & 2 & 3 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 5^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -44, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 3 & 3 & 0 & 7 & 4 \\ 2 & 3 & 0 & 2 & 0 & 4 & 2 \\ 2 & 3 & 2 & 1 & 1 & 5 & 3 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 4 & 7 & 4 & 5 & 0 & 8 & 6 \\ 3 & 4 & 2 & 3 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{12} 5^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 3 & 3 & 0 & 14 & 4 \\ 2 & 3 & 0 & 2 & 0 & 8 & 2 \\ 2 & 3 & 2 & 1 & 1 & 10 & 3 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 8 & 14 & 8 & 10 & 0 & 38 & 12 \\ 3 & 4 & 2 & 3 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{13} 5^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -60, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 12 & 3 \\ 3 & 4 & 3 & 3 & 0 & 21 & 4 \\ 2 & 3 & 0 & 2 & 0 & 12 & 2 \\ 2 & 3 & 2 & 1 & 1 & 15 & 3 \\ 1 & 0 & 0 & 1 & -3 & 0 & -1 \\ 12 & 21 & 12 & 15 & 0 & 88 & 18 \\ 3 & 4 & 2 & 3 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

4.1110 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8$

- $T = 7$ [0] $1^{\overline{1}12} 3^{\overline{1}1} 1^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -7$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 14.14 \rightarrow 15$, $T_{\max} = 71$, $T \in [15, 71]$
 $\begin{pmatrix} 5 & 11 & 18 & 5 \\ 11 & 20 & 34 & 8 \\ 18 & 34 & 57 & 13 \\ 5 & 8 & 13 & -1 \end{pmatrix}_{b_0}$
- [1] $21^{\overline{1}1} 31^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 14$, $T_{\max} = 71$, $T \in [14, 71]$
 $\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 25 & 20 & 9 \\ 12 & 20 & 12 & 6 \\ 6 & 9 & 6 & 0 \end{pmatrix}_{b_0}$
- [2] $1^{\overline{1}1} 31^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 14$, $T_{\max} = 71$, $T \in [14, 71]$
 $\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 25 & 20 & 9 \\ 12 & 20 & 12 & 6 \\ 6 & 9 & 6 & 0 \end{pmatrix}_{b_0}$
- [3] $1^{\overline{1}1} 31^{\overline{1}12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -15$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 5 & 11 & 19 & 5 \\ 11 & 20 & 34 & 8 \\ 19 & 34 & 57 & 13 \\ 5 & 8 & 13 & 2 \end{pmatrix}_{b_0}$
- [4] $1^{\overline{1}1} 3^{\overline{1}12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -15$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 10 & 15 & 25 & 6 \\ 15 & 20 & 34 & 8 \\ 25 & 34 & 57 & 13 \\ 6 & 8 & 13 & 2 \end{pmatrix}_{b_0}$
- [5] $1^{\overline{1}1} 3^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 5 & 12 & 8 & 5 \\ 12 & 25 & 16 & 9 \\ 8 & 16 & 9 & 5 \\ 5 & 9 & 5 & 2 \end{pmatrix}_{b_0}$
- [6] $1^{\overline{1}1} 31^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -23$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 10 & 16 & 23 & 6 \\ 16 & 25 & 36 & 9 \\ 23 & 36 & 48 & 12 \\ 6 & 9 & 12 & 2 \end{pmatrix}_{b_0}$
- [7] $1^{\overline{1}1} 31^{\overline{1}12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -27$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 4 & 7 & 1 & 3 \\ 4 & 9 & 16 & 0 & 5 \\ 7 & 16 & 27 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 9 & -2 & 1 \end{pmatrix}_{b_0}$
- [8] $1^{\overline{1}1} 31^{\overline{1}12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 71$, $T \in [14, 71]$
 $\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 6 & 10 & 21 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & -2 & -1 \end{pmatrix}_{b_0}$
- [9] $1^{\overline{1}1} 3^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & -2 & 1 \end{pmatrix}_{b_0}$
- [10] $1^{\overline{1}1} 31^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & -2 & 1 \end{pmatrix}_{b_0}$
- [13] $1^{\overline{1}1} 31^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -43$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$

$$\begin{pmatrix} 5 & 11 & 10 & 5 \\ 11 & 20 & 17 & 8 \\ 10 & 17 & 12 & 6 \\ 5 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, \ T_{\max} = 71, \ \Delta = -122, \ \det = -43, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.23 \rightarrow 12, \ T_{\max} = 71, \ T \in [12, 71]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, \ T_{\max} = 71, \ \Delta = -122, \ \det = -43, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.23 \rightarrow 12, \ T_{\max} = 71, \ T \in [12, 71]$$

$$\begin{pmatrix} 5 & 11 & 1 & 5 \\ 11 & 20 & 0 & 8 \\ 1 & 0 & -3 & -1 \\ 5 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{12}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, \ T_{\max} = 71, \ \Delta = -122, \ \det = -48, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.58 \rightarrow 12, \ T_{\max} = 71, \ T \in [12, 71]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 4 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, \ T_{\max} = 71, \ \Delta = -122, \ \det = -48, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 71, \ T \in [11, 71]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{13}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, \ T_{\max} = 71, \ \Delta = -122, \ \det = -48, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.33 \rightarrow 13, \ T_{\max} = 71, \ T \in [13, 71]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 6 & 12 & 24 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 8 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{14}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, \ T_{\max} = 71, \ \Delta = -122, \ \det = -48, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 11.6$$

$$T_{\min} = 13.25 \rightarrow 14, \ T_{\max} = 71, \ T \in [14, 71]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 8 & 16 & 45 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{15}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, \ T_{\max} = 71, \ \Delta = -122, \ \det = -48, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 14.33 \rightarrow 15, \ T_{\max} = 71, \ T \in [15, 71]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 20 & 0 & 4 \\ 10 & 20 & 72 & 0 & 14 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 14 & -2 & -2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, \ T_{\max} = 71, \ \Delta = -122, \ \det = -49, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 71, \ T \in [12, 71]$$

$$\begin{pmatrix} 5 & 11 & 9 & 5 \\ 11 & 20 & 17 & 8 \\ 9 & 17 & 12 & 6 \\ 5 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, \ T_{\max} = 71, \ \Delta = -122, \ \det = -50, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 71, \ T \in [12, 71]$$

$$\begin{pmatrix} 10 & 15 & 12 & 6 \\ 15 & 20 & 17 & 8 \\ 12 & 17 & 12 & 6 \\ 6 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, \ T_{\max} = 71, \ \Delta = -122, \ \det = -51, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.67 \rightarrow 12, \ T_{\max} = 71, \ T \in [12, 71]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 4 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 7, \ T_{\max} = 71, \ \Delta = -122, \ \det = -55, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.25 \rightarrow 12, \ T_{\max} = 71, \ T \in [12, 71]$$

$$\begin{pmatrix} 10 & 15 & 1 & 6 \\ 15 & 20 & 0 & 8 \\ 1 & 0 & -3 & -1 \\ 6 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

- [23] $1^{11} 3 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -60$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 5 & 12 & 18 & 5 \\ 12 & 25 & 36 & 9 \\ 18 & 36 & 48 & 12 \\ 5 & 9 & 12 & 2 \end{pmatrix}_{b_0}$
- [24] $1^{11} 3 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -60$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 10 & 15 & 13 & 6 \\ 15 & 20 & 17 & 8 \\ 13 & 17 & 12 & 6 \\ 6 & 8 & 6 & 2 \end{pmatrix}_{b_0}$
- [25] $1^{11} 3 1^{12} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -60$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.27 \rightarrow 13$, $T_{\max} = 71$, $T \in [13, 71]$
 $\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 7 & 10 & 21 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & -2 & 0 \end{pmatrix}_{b_0}$
- [26] $1^{11} 3 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -64$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & -2 & 1 \end{pmatrix}_{b_0}$
- [27] $21^{11} 31^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -65$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 13 & 17 & 15 & 7 \\ 17 & 20 & 17 & 8 \\ 15 & 17 & 12 & 6 \\ 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$
- [28] $13 1^{11} 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -67$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.22 \rightarrow 13$, $T_{\max} = 71$, $T \in [13, 71]$
 $\begin{pmatrix} 5 & 12 & 9 & 5 \\ 12 & 25 & 20 & 9 \\ 9 & 20 & 12 & 6 \\ 5 & 9 & 6 & 1 \end{pmatrix}_{b_0}$
- [29] $13 1^{11} 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -71$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 5 & 12 & 17 & 5 \\ 12 & 25 & 36 & 9 \\ 17 & 36 & 48 & 12 \\ 5 & 9 & 12 & 2 \end{pmatrix}_{b_0}$
- [30] $1^{11} 3 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -72$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 9 & 12 & 0 & 5 \\ 5 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 6 & -2 & 1 \end{pmatrix}_{b_0}$
- [31] $21^{11} 31^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -77$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 13 & 20 & 12 & 7 \\ 20 & 25 & 16 & 9 \\ 12 & 16 & 9 & 5 \\ 7 & 9 & 5 & 2 \end{pmatrix}_{b_0}$
- [32] $1^{11} 3 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -83$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 5 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & -2 & 1 \end{pmatrix}_{b_0}$
- [33] $21^{11} 31^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -86$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 10 & 16 & 12 & 6 \\ 16 & 20 & 17 & 8 \\ 12 & 17 & 12 & 6 \\ 6 & 8 & 6 & 2 \end{pmatrix}_{b_0}$
- [34] $1^{12} 3 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{orange}{s}\textcolor{orange}{o}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -96$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 5 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & -2 & 1 \end{pmatrix}_{b_0}$

- [35] $1^{11}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -108$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.37 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 8 & 14 & 36 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & -2 & 1 \end{pmatrix}_{b_0}$
- [36] $1^{12}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -123$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 7 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$
- [37] $1^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -123$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 6 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & -2 & 1 \end{pmatrix}_{b_0}$
- [38] $1^{13}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -128$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 7 & 12 & 24 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 8 & -2 & 1 \end{pmatrix}_{b_0}$
- [39] $1^{14}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -160$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.6$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 9 & 16 & 45 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & -2 & 1 \end{pmatrix}_{b_0}$
- [40] $1^{13}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -163$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.21 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 9 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & -2 & 1 \end{pmatrix}_{b_0}$
- [41] $1^{11}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -172$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 9 & 14 & 36 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & -2 & 1 \end{pmatrix}_{b_0}$
- [42] $1^{12}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -188$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 10 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & -2 & 1 \end{pmatrix}_{b_0}$
- [43] $1^{12}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -195$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.05 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 8 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$
- [44] $1^{14}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -203$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.6$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 11 & 21 & 75 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & -2 & 1 \end{pmatrix}_{b_0}$
- [45] $21^{11}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -207$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.03 \rightarrow 12$, $T_{\max} = 71$, $T \in [12, 71]$
 $\begin{pmatrix} 13 & 20 & 27 & 7 \\ 20 & 25 & 36 & 9 \\ 27 & 36 & 48 & 12 \\ 7 & 9 & 12 & 2 \end{pmatrix}_{b_0}$
- [46] $1^{13}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8$ $T = 7$, $T_{\max} = 71$, $\Delta = -122$, $\det = -267$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 11.0$

$$T_{\min} = 11.04 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 10 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 12 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[48] \ 21^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -275, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 13 & 20 & 15 & 7 \\ 20 & 25 & 20 & 9 \\ 15 & 20 & 12 & 6 \\ 7 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -284, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.04 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 11 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -339, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.6$$

$$T_{\min} = 11.05 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 12 & 21 & 75 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{14} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -348, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.6$$

$$T_{\min} = 11.08 \rightarrow 12, T_{\max} = 71, T \in [12, 71]$$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 14 & 26 & 117 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -396, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 71, T \in [11, 71]$$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 13 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{14} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 7, T_{\max} = 71, \Delta = -122, \det = -508, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 11.6$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 71, T \in [11, 71]$$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 15 & 26 & 117 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 19 & -2 & 2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [54] \ 1^{11} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 18, T_{\max} = 23, T \in [18, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 8 & 13 & 39 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 11 & -2 & -4 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{11} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 23, T \in [12, 23]$$

$$\begin{pmatrix} 10 & 16 & 8 & 6 \\ 16 & 25 & 12 & 9 \\ 8 & 12 & 4 & 4 \\ 6 & 9 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[56] \ 21^{11} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 23, T \in [12, 23]$$

$$\begin{pmatrix} 10 & 16 & 8 & 6 \\ 16 & 25 & 12 & 9 \\ 8 & 12 & 4 & 4 \\ 6 & 9 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[55] \ 13^1 1^1 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 3 & 3 \\ 1 & 4 & 0 & 3 & 3 & 2 \\ 1 & 3 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & 3 & 2 & 2 \\ 2 & 3 & 2 & 2 & 2 & -2 \end{pmatrix}_{b_0}$$

[61] $1^{11} 1^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 3 & 5 & 3 & 3 & 3 \\ 2 & 5 & 3 & 0 & 2 \\ 3 & 5 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[60] $1^{11} 3131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 1 & 1 & 3 \\ 4 & 10 & 1 & 0 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 6 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

[59] $1^{11} 1^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 1 & 2 & 0 & 3 & 1 & 2 \\ 1 & 1 & 3 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{11} 321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 5 & 10 & 5 & 5 \\ 7 & 10 & 13 & 7 & 7 \\ 2 & 5 & 7 & 0 & 2 \\ 3 & 5 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{11} 1^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 4 & 0 & 9 & 5 \\ 3 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[64] $1^{11} 31^1 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 4 & 3 & 3 & 3 \\ 1 & 4 & 0 & 3 & 3 & 2 \\ 2 & 3 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & 3 & 2 & 2 \\ 2 & 3 & 2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[63] $1^{11} 1^3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{11} 31^1 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 17, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.41 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 0 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

[67] $1^{11} 321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 4 & 2 \\ 2 & 3 & 0 & 4 & 3 & 3 \\ 2 & 2 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{11} 321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 4 & 3 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{13} 31^1 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 9 & 0 & 24 & 9 & 8 \\ 2 & 3 & 0 & 9 & 2 & 2 \\ 2 & 3 & -1 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{11} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 20, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.40 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 0 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 0 & 11 & 5 \\ 2 & 3 & -1 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{11} 321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 20, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 4 & 5 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{11} 321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 20, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.40 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 1 & 2 & 0 & 3 & 1 & 2 \\ 2 & 1 & 3 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{11} 321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 20, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.40 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 5 & 0 & 0 & 2 \\ 3 & 5 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{11} 321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 20, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 2 & 3 \\ 4 & 9 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 5 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{11} 321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 20, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.40 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 1 & 1 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{12} 321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 4 & 5 & 1 & 4 & 3 \\ 2 & 5 & 4 & 0 & 2 \\ 3 & 5 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[79] \ 2^2 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 22, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.55 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[78] \ 1^{11} 321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 22, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.45 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 3 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[77] \ 1^{11} 3^2 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 22, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.45 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 1 & 0 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

- [76] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 22$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$
- [80] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 24$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 13$, $T \in [11, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 2 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 3 & -2 & 3 \end{pmatrix}_{b_0}$$
- [81] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 24$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 5 & 3 & 0 & 2 \\ 3 & 5 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$
- [82] $21^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 25$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.20 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 4 & 5 & 9 & 3 & 4 \\ 5 & 5 & 10 & 5 & 5 \\ 9 & 10 & 13 & 7 & 7 \\ 3 & 5 & 7 & 0 & 2 \\ 4 & 5 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$
- [83] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 26$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$
- [84] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 28$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 6 & 7 & 0 & 20 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$
- [85] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 29$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 2 & 1 \\ 2 & 3 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [86] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 29$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 12 & 0 & 12 & 6 \\ 3 & 6 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$
- [87] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 29$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$
- [88] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 29$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$
- [89] $2231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 30$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.27 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 2 & 2 & 0 & 3 & 1 & 2 \\ 2 & 1 & 3 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{\overline{1}} 1^{\overline{1}} 3 2 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 4 & 0 & 3 \\ 1 & 4 & 0 & 4 & 0 & 2 \\ 2 & 4 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

[92] $2 1^{\overline{1}} 4 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 10 & 12 & 8 & 6 \\ 12 & 13 & 10 & 7 \\ 8 & 10 & 4 & 4 \\ 6 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[91] $1^{\overline{1}} 3 1^{\overline{1}} 3 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 0 & 4 \\ 1 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[96] $1^{\overline{1}} 1^{\overline{1}} 4 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 5 & 12 & 6 & 5 \\ 12 & 25 & 12 & 9 \\ 6 & 12 & 4 & 4 \\ 5 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

[95] $1^{\overline{1}} 1^{\overline{1}} 3 2 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

[94] $1^{\overline{1}} 1^{\overline{1}} 4 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 10 & 12 & 8 & 6 \\ 12 & 13 & 10 & 7 \\ 8 & 10 & 4 & 4 \\ 6 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[93] $1^{\overline{1}} 1^{\overline{1}} 3 2 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 4 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[97] $1^{\overline{1}} 1^{\overline{1}} 4 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

[99] $1^{\overline{1}} 1^{\overline{1}} 3 2 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 3 \\ 2 & 2 & -1 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[98] $1^{\overline{1}} 1^{\overline{1}} 3 1^{\overline{1}} 3 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 3 & 6 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[100] $1^{\overline{1}} 1^{\overline{1}} 3 1^{\overline{1}} 3 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 10 & 13 & 0 & 6 \\ 5 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 6 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[101] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 1 & 0 & 0 & 4 & 2 \\ 2 & 3 & 0 & 4 & 3 & 3 \\ 2 & 2 & -1 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

[102] $1^{11}3131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.17 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 2 & -2 & -1 \end{pmatrix}_{b_0}$$

[105] $1^{11}31^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[104] $1^{11}41^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 & 4 \\ 3 & 4 & 10 & 0 & 0 & 4 \\ 6 & 10 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 & 2 \\ 3 & 4 & 8 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[103] $1^{11}31^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 4 & 8 & 8 & 7 & 4 & 4 \\ 8 & 13 & 16 & 12 & 7 & 7 \\ 8 & 16 & 12 & 12 & 6 & 6 \\ 7 & 12 & 12 & 9 & 5 & 5 \\ 4 & 7 & 6 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

[108] $1^{11}321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 4 & 0 & 3 & 3 \\ 2 & 2 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[107] $1^{11}31^131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 2 & 2 \\ 2 & 3 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[106] $1^{12}31^131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 2 & 3 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 \\ 2 & 4 & 3 & 6 & 0 & 3 \\ 3 & 6 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[109] $1^{11}31^131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 4 & 7 & 8 & 1 & 4 & 4 \\ 7 & 10 & 13 & 0 & 0 & 6 \\ 8 & 13 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 & 2 \\ 4 & 6 & 6 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[110] $141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 5 & 9 & 6 & 5 \\ 9 & 13 & 10 & 7 \\ 6 & 10 & 4 & 4 \\ 5 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[111] $1231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 0 & 0 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

[112] $31^{11}231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 5 & 3 & 4 & 3 \\ 5 & 5 & 5 & 5 & 5 \\ 3 & 5 & 3 & 3 & 3 \\ 4 & 5 & 3 & 0 & 2 \\ 3 & 5 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{11}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 2 & 5 & 1 & 4 & 3 \\ 3 & 5 & 4 & 0 & 2 \\ 3 & 5 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[114] $2^12^131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 5 & 5 & 5 & 5 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 5 & 3 & 0 & 2 \\ 3 & 5 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[115] $1^{11}3^121^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 4 & 5 & 1 & 3 & 4 \\ 5 & 4 & 1 & 3 & 4 \\ 1 & 1 & -3 & 0 & -1 \\ 3 & 3 & 0 & 0 & 2 \\ 4 & 4 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

[116] $1^{11}321^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 5 & 25 & 0 & 5 \\ 11 & 25 & 91 & 0 & 19 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 19 & -2 & 2 \end{pmatrix}_{b_0}$$

[117] $1^{12}31^131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 9 & 6 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 2 & 3 & -1 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$

[118] $1^{11}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 4 & 0 & 2 \\ 1 & 2 & 0 & 7 & 0 & 2 \\ 3 & 4 & 7 & 15 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[119] $1^{11}31^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 3 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[120] $1^{11}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 5 & 5 & 5 \\ 3 & 5 & 1 & 4 & 3 \\ 2 & 5 & 4 & 0 & 2 \\ 3 & 5 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[121] $1^{11}31^131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 2 & 2 \\ 2 & 3 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[122] $1^{12}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 8 & 10 & 0 & 37 & 11 \\ 3 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[123] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 5 & 10 & 0 & 5 \\ 5 & 10 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

[125] $21^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 13 & 14 & 9 & 7 \\ 14 & 13 & 10 & 7 \\ 9 & 10 & 4 & 4 \\ 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[124] $1^{11}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 23, \Delta = -113, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 23, T \in [12, 23]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 4 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & -2 & 2 \end{pmatrix}_{b_0}$$

[126] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 3 & 0 & 2 \\ 1 & 2 & 0 & 4 & 0 & 2 \\ 2 & 3 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

[127] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 5 & 10 & 5 & 5 \\ 6 & 10 & 13 & 7 & 7 \\ 2 & 5 & 7 & 0 & 2 \\ 3 & 5 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[128] $1^{11}321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 5 & 20 & 0 & 5 \\ 9 & 20 & 57 & 0 & 15 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 15 & -2 & 2 \end{pmatrix}_{b_0}$$

[131] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 7 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 7 & 0 & 20 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[130] $1^{11}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 5 & 15 & 0 & 5 \\ 7 & 15 & 31 & 0 & 11 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 11 & -2 & 2 \end{pmatrix}_{b_0}$$

[129] $1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 5 & 10 & 5 & 5 \\ 5 & 10 & 13 & 7 & 7 \\ 2 & 5 & 7 & 0 & 2 \\ 3 & 5 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[132] $1^{12}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 5 & 6 & 0 & 9 & 5 \\ 3 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[133] $131^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.26 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 3 \\ 5 & 13 & 16 & 12 & 7 \\ 5 & 16 & 12 & 12 & 6 \\ 4 & 12 & 12 & 9 & 5 \\ 3 & 7 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{11}321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 5 & 10 & 5 & 5 \\ 5 & 10 & 13 & 7 & 7 \\ 3 & 5 & 7 & 0 & 2 \\ 3 & 5 & 7 & 2 & 2 \end{pmatrix}_{b_0}$

[135] $1^{14}31^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.6$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 2 & 2 \\ 1 & 1 & 1 & 12 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 12 & 0 & 45 & 12 & 11 \\ 2 & 3 & 0 & 12 & 2 & 2 \\ 2 & 3 & -1 & 11 & 2 & 1 \end{pmatrix}_{b_0}$

[136] $1^{13}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 10 & 13 & 0 & 60 & 14 \\ 3 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0}$

[137] $1^{131}1^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 3 & 6 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 3 & -2 & 1 \end{pmatrix}_{b_0}$

[138] $1^{12}31^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 1 & 6 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 6 & 0 & 9 & 6 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 2 & 3 & -1 & 5 & 2 & 1 \end{pmatrix}_{b_0}$

[139] $1^{12}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 92, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 5 & 0 & 2 \\ 1 & 2 & 0 & 5 & 0 & 2 \\ 3 & 5 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$

[140] $1^{13}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 9 & 0 & 24 & 8 \\ 3 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$

[141] $1^{13}31^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 2 & 2 \\ 1 & 1 & 1 & 9 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 5 & 9 & 0 & 24 & 9 & 8 \\ 2 & 3 & 0 & 9 & 2 & 2 \\ 2 & 3 & -1 & 8 & 2 & 1 \end{pmatrix}_{b_0}$

[142] $1^{131}1^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 0 & 3 & 0 & 2 \\ 2 & 6 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 3 & -2 & 1 \end{pmatrix}_{b_0}$

[144] $1^{12}31^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 4 & 12 & 9 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 9 & -2 & 1 \end{pmatrix}_{b_0}$

[143] $1^{13}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 6 & 0 & 2 \\ 4 & 7 & 6 & 24 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 8 & -2 & 1 \end{pmatrix}_{b_0}$

- [146] $21^{11}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{so}_8$ $T = 8$, $T_{\max} = 23$, $\Delta = -113$, $\det = 123$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 23$, $T \in [12, 23]$
 $\begin{pmatrix} 5 & 10 & 7 & 5 \\ 10 & 13 & 10 & 7 \\ 7 & 10 & 4 & 4 \\ 5 & 7 & 4 & 3 \end{pmatrix}_{b_0}$
- [145] $21^{11}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{so}_8$ $T = 8$, $T_{\max} = 23$, $\Delta = -113$, $\det = 123$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 23$, $T \in [12, 23]$
 $\begin{pmatrix} 13 & 20 & 9 & 7 \\ 20 & 25 & 12 & 9 \\ 9 & 12 & 4 & 4 \\ 7 & 9 & 4 & 3 \end{pmatrix}_{b_0}$
- [147] $1^131^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 125$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$
- [148] $1^{12}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 129$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 10 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 9 & 10 & 0 & 37 & 11 \\ 3 & 3 & -1 & 11 & 2 \end{pmatrix}_{b_0}$
- [149] $1^{12}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 132$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 6 & 0 & 2 \\ 1 & 2 & 0 & 8 & 0 & 2 \\ 4 & 6 & 8 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 9 & -2 & 1 \end{pmatrix}_{b_0}$
- [150] $1^{12}321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 140$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 7 & 0 & 2 \\ 1 & 2 & 0 & 11 & 0 & 2 \\ 5 & 7 & 11 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$
- [151] $1^{11}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 156$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 3 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$
- [152] $2^12^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 174$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.32 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 5 & 5 & 5 & 5 \\ 3 & 5 & 1 & 4 & 3 \\ 3 & 5 & 4 & 0 & 2 \\ 3 & 5 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [153] $1^{13}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 182$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.18 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 1 & 13 & 3 \\ 1 & 1 & -3 & 0 & -1 \\ 11 & 13 & 0 & 60 & 14 \\ 3 & 3 & -1 & 14 & 2 \end{pmatrix}_{b_0}$
- [154] $1^{11}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 188$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 9 & 0 & 4 \\ 1 & 4 & 0 & 6 & 0 & 2 \\ 4 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$
- [155] $1^{13}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 200$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 8 & 0 & 2 \\ 1 & 2 & 0 & 9 & 0 & 2 \\ 5 & 8 & 9 & 39 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0}$

- [156] $2^1 2^1 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 211$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 5 & 10 & 5 & 5 \\ 6 & 10 & 13 & 7 & 7 \\ 3 & 5 & 7 & 0 & 2 \\ 3 & 5 & 7 & 2 & 2 \end{pmatrix}_{b_0}$
- [157] $1^{12} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -101$, $\det = 212$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.34 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 0 & 2 & 1 & 5 & 1 & 2 \\ 2 & 4 & 4 & 12 & 0 & 4 \\ 1 & 4 & 0 & 9 & 0 & 2 \\ 5 & 12 & 9 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 9 & -2 & 1 \end{pmatrix}_{b_0}$
- [158] $2^1 2^1 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 249$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 3 & 1 & 3 \\ 2 & 4 & 0 & 3 & 1 & 2 \\ 2 & 3 & 3 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$
- [159] $1^{13} 321^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 252$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 9 & 0 & 2 \\ 1 & 2 & 0 & 12 & 0 & 2 \\ 6 & 9 & 12 & 60 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$
- [160] $2^1 2^1 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 384$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.09 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 0 & 2 & 2 & 4 & 1 & 2 \\ 2 & 1 & 4 & 6 & 1 & 3 \\ 2 & 4 & 0 & 6 & 1 & 2 \\ 4 & 6 & 6 & 9 & 2 & 5 \\ 1 & 1 & 1 & 2 & -2 & 0 \\ 2 & 3 & 2 & 5 & 0 & 1 \end{pmatrix}_{b_0}$
- $T = 9$ [161] $21^1 41^1 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -9$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.56 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 5 & 7 & 2 & 5 \\ 7 & 8 & 1 & 6 \\ 2 & 1 & -2 & 0 \\ 5 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [162] $1^1 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -18$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.44 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$
- [164] $1^1 4131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -20$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$
- [163] $1^1 4131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -20$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0}$
- [165] $1^1 4131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -92$, $\det = -20$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 2 & 3 & 2 & 4 \\ 3 & 1 & 1 & 3 \\ 2 & 1 & -2 & 0 \\ 4 & 3 & 0 & 3 \end{pmatrix}_{b_0}$
- [166] $1^1 41^{12} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -21$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 5 & 6 & 1 & 5 \\ 6 & 5 & 2 & 5 \\ 1 & 2 & -2 & 0 \\ 5 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

[167] $1^{\overline{1}1}4131^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 5 & 4 & 1 & 5 \\ 4 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 5 & 3 & 0 & 4 \end{pmatrix}_{b_0}$

[168] $1^{\overline{1}1}4131^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 4 & 3 & -1 & 2 \end{pmatrix}_{b_0}$

[169] $1^{\overline{1}1}4131^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 6 & 7 & 8 & 1 & 6 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 4 & 6 & -1 & 2 \end{pmatrix}_{b_0}$

[170] $1^{\overline{1}1}4131^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 5 & 4 & 2 & 5 \\ 4 & 1 & 1 & 3 \\ 2 & 1 & -2 & 0 \\ 5 & 3 & 0 & 3 \end{pmatrix}_{b_0}$

[172] $1^{\overline{1}1}4131^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 14 & 10 & 8 \\ 10 & 5 & 5 \\ 8 & 5 & 4 \end{pmatrix}_{b_0}$

[171] $1^{\overline{1}1}41231^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 2 & 6 & 2 & 4 \\ 6 & 8 & 1 & 6 \\ 2 & 1 & -2 & 0 \\ 4 & 6 & 0 & 3 \end{pmatrix}_{b_0}$

[173] $1^{\overline{1}1}4131^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 5 & 9 & 1 & 5 \\ 9 & 13 & 1 & 7 \\ 1 & 1 & -2 & 0 \\ 5 & 7 & 0 & 3 \end{pmatrix}_{b_0}$

[174] $1^{\overline{1}1}4131^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 13 & 9 & 1 & 7 \\ 9 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 7 & 5 & 0 & 3 \end{pmatrix}_{b_0}$

[176] $1^{\overline{1}1}4131^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$

[177] $1^{\overline{1}1}4131^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 2 & 3 & 0 & 2 \\ 3 & 3 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 3 & 2 & 3 & 0 & 2 \end{pmatrix}_{b_0}$

[175] $1^{\overline{1}1}4131^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 1 & 1 & 3 \\ 4 & 4 & 0 & 0 & 4 \\ 1 & 0 & -2 & 1 & 0 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 4 & 0 & -1 & 2 \end{pmatrix}_{b_0}$

[178] $1^{\overline{1}1}4131^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 \\ 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & -2 & 1 & 0 \\ 3 & 3 & 1 & 0 & 2 \\ 3 & 4 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[180] \ 1^{\overline{11}31^1} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 3 & 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[179] \ 1^{\overline{11}31^1} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 2 & 3 & 0 & 2 \\ 4 & 3 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 3 & 2 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[181] \ 1^{\overline{11}31^1} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 4 & 5 & 5 \\ 4 & 1 & 3 & 3 \\ 5 & 3 & 1 & 3 \\ 5 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[182] \ 1^{\overline{11}41^1231^{\overline{11}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.40 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 1 & 2 \\ 3 & 4 & 1 & 0 & 2 & 4 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & -2 & 0 \\ 2 & 4 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[183] \ 1^{\overline{11}31^1} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 7 & 1 & 5 \\ 7 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 5 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[184] \ 1^{\overline{11}31^1} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 0 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[185] \ 1^{\overline{11}4131^{\overline{12}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 3 & 2 & 2 & 3 & 1 & 3 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[189] \ 1^{\overline{11}31^1} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 10 & 1 & 5 \\ 10 & 13 & 0 & 7 \\ 1 & 0 & -2 & 0 \\ 5 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[190] \ 1^{\overline{11}31^1} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 8 & 1 & 3 \\ 2 & 0 & 4 & 1 & 2 \\ 8 & 4 & 17 & 1 & 7 \\ 1 & 1 & 1 & -2 & 0 \\ 3 & 2 & 7 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[186] \ 1^{\overline{11}31^1} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 0 & 2 & 1 & 2 \\ 4 & 2 & 1 & 1 & 3 \\ 1 & 1 & 1 & -2 & 0 \\ 3 & 2 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[187] \ 1^{\overline{11}41^1231^{\overline{11}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[188] \ 1^{\overline{11}} 4^{\overline{131}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -92, \ \det = -57, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.16 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[192] \ 1^{\overline{11}} 4^{\overline{131}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -92, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 2 & 2 & 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[191] \ 1^{\overline{11}} 4^{\overline{131}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -92, \ \det = -60, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[193] \ 1^{\overline{11}} 4^{\overline{1231}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -92, \ \det = -66, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 6 & 1 & 3 \\ 2 & 0 & 3 & 1 & 2 \\ 6 & 3 & 7 & 1 & 5 \\ 1 & 1 & 1 & -2 & 0 \\ 3 & 2 & 5 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[194] \ 1^{\overline{11}} 4^{\overline{131}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -92, \ \det = -68, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.29 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 3 & 2 & 3 \\ 1 & 1 & 0 & 2 & -2 & 0 \\ 2 & 2 & 0 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[195] \ 31^{\overline{1131}} 41^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -100, \ \det = -72, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.44 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 3 & 1 & 3 \\ 3 & 4 & 0 & 3 & 1 & 2 \\ 3 & 3 & 3 & 3 & 2 & 3 \\ 1 & 1 & 1 & 2 & -2 & 0 \\ 2 & 3 & 2 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[196] \ 1^{\overline{11}} 4^{\overline{1131}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -92, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.07 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 4 & 3 & 1 & 4 & 2 & 4 \\ 1 & 1 & 0 & 2 & -2 & 0 \\ 2 & 2 & 0 & 4 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[197] \ 1^{\overline{11}} 4^{\overline{1231}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -92, \ \det = -132, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.15 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & -2 & 2 & 0 & 0 \\ 5 & 3 & 2 & 7 & 2 & 5 \\ 1 & 1 & 0 & 2 & -2 & 0 \\ 2 & 2 & 0 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[198] \ 2^{\overline{141}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 14, \ \Delta = -93, \ \det = -186, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.11 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 0 & 3 & 2 & 2 & 1 & 2 \\ 3 & 4 & 4 & 3 & 2 & 4 \\ 2 & 4 & 0 & 3 & 1 & 2 \\ 2 & 3 & 3 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & -2 & 0 \\ 2 & 4 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[199] \ 1^{\overline{141}} 1^{\overline{1231}} 231^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 14, \ \Delta = -93, \ \det = -205, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 12.32 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 1 & 2 \\ 3 & 4 & 1 & 6 & 2 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 6 & 0 & 3 & 2 & 3 \\ 1 & 2 & 0 & 2 & -2 & 0 \\ 2 & 4 & 0 & 3 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[200] \ 1^{\overline{141}} 1^{\overline{1231}} 231^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 14, \ \Delta = -93, \ \det = -217, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$\begin{aligned}
& T_{\min} = 12.19 \rightarrow 13, T_{\max} = 14, T \in [13, 14] \\
& \begin{pmatrix} 1 & 6 & 5 & 1 & 3 \\ 6 & 13 & 12 & 0 & 7 \\ 5 & 12 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 5 & -1 & 2 \end{pmatrix}_{b_0} \\
[201] & 1^{\textcolor{red}{1}} 4^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} 231^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -250, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12.08 \rightarrow 13, T_{\max} = 14, T \in [13, 14] \\
& \begin{pmatrix} 0 & 3 & 1 & 4 & 1 & 2 \\ 3 & 4 & 1 & 6 & 2 & 4 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 4 & 6 & 1 & 4 & 2 & 4 \\ 1 & 2 & 0 & 2 & -2 & 0 \\ 2 & 4 & 0 & 4 & 0 & 1 \end{pmatrix}_{b_0} \\
T = 10 & \quad [202] \quad 21^{\textcolor{blue}{1}} 5^{\textcolor{blue}{1}} 131^{\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 5 & 3 & 5 & 5 \\ 3 & 3 & 0 & 2 & 2 \\ 3 & 5 & 2 & 0 & 2 \\ 4 & 5 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[203] & 1^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} 3221^{\textcolor{blue}{1}} 51^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 13, T_{\max} = 14, T \in [13, 14] \\
& \begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 10 & 21 & 0 & 6 \\ 9 & 21 & 42 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 6 & 12 & -1 & 3 \end{pmatrix}_{b_0} \\
[204] & 1^{\textcolor{red}{1}} 4^{\textcolor{red}{1}} 141^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 10, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 2 & 4 & 0 & 3 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & -2 & 3 \end{pmatrix}_{b_0} \\
[206] & 1^{\textcolor{red}{1}} 4^{\textcolor{red}{1}} 141^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 0 & 2 \\ 3 & 5 & -1 & 2 & 3 \end{pmatrix}_{b_0} \\
[205] & 1^{\textcolor{red}{1}} 5^{\textcolor{red}{1}} 1231^{\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 13.40 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 5 & 0 & 0 & 2 \\ 3 & 5 & -1 & 2 & 2 \end{pmatrix}_{b_0} \\
[207] & 1^{\textcolor{red}{1}} 4^{\textcolor{red}{1}} 141^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 4 & 0 & 5 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 3 & -2 & 3 \end{pmatrix}_{b_0} \\
[208] & 1^{\textcolor{red}{1}} 5^{\textcolor{red}{1}} 131^{\textcolor{red}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 4 & -1 & 2 & 3 \end{pmatrix}_{b_0} \\
[209] & 1^{\textcolor{red}{1}} 4^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} 31^{\textcolor{blue}{1}} 41^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 13.50 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 3 & 1 & 3 & 1 & 2 \\ 3 & 13 & 6 & 12 & 0 & 7 \\ 1 & 6 & 0 & 4 & 0 & 2 \\ 3 & 12 & 4 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 7 & 2 & 5 & -2 & 1 \end{pmatrix}_{b_0} \\
[210] & 1^{\textcolor{red}{1}} 1^{\textcolor{red}{1}} 4^{\textcolor{red}{1}} 141^{\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 5 & 5 & 4 & 5 \\ 3 & 5 & 1 & 3 & 3 \\ 2 & 4 & 3 & 0 & 2 \\ 3 & 5 & 3 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[211] $31^{\overline{11}} 4 141^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 57, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.16 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 2 & 3 & 1 & 2 \\ 2 & 5 & 8 & 9 & 0 & 5 \\ 2 & 8 & 4 & 9 & 0 & 4 \\ 3 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 5 & 4 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[212] $2 3^{\overline{11}} 151^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 79, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 5 & 3 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 5 & 2 & 0 & 2 \\ 3 & 5 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[213] $1^{\overline{1}} 41^{\overline{1}} 31^{\overline{1}} 41^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8$ $T = 10, T_{\max} = 14, \Delta = -91, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 8 & 1 & 7 & 3 \\ 8 & 38 & 1 & 32 & 12 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 32 & 0 & 25 & 9 \\ 3 & 12 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[214] $1^{\overline{1}} 51^{\overline{1}} 231^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 104, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 7 & 1 & 2 & 3 \\ 7 & 26 & 0 & 8 & 10 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 8 & 0 & 0 & 2 \\ 3 & 10 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

4.1111 NHC(-2)(-3) + $\textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6$

$T = 8$ [1] $1^{\overline{11}} 31^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6$ $T = 8, T_{\max} = 101, \Delta = -171, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.96 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 5 & -4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{11}} 31^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6$ $T = 8, T_{\max} = 101, \Delta = -171, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.73 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{11}} 31^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6$ $T = 8, T_{\max} = 101, \Delta = -171, \det = 302, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.86 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{11}} 31^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6$ $T = 8, T_{\max} = 101, \Delta = -171, \det = 330, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.68 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 48 & 0 & 12 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 12 & -4 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [5] $1^{\overline{11}} 321^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6$ $T = 9, T_{\max} = 99, \Delta = -143, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.83 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 5 & 3 & 0 & -2 & 0 \\ 3 & 5 & 3 & -2 & 0 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{11}} 3 21^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{violet}{e}_6$ $T = 9, T_{\max} = 99, \Delta = -143, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.88 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 & -3 & -1 \\ 2 & 2 & -1 & -1 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{11}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{violet}{\mathfrak{e}}_6$ $T = 9$, $T_{\max} = 100$, $\Delta = -150$, $\det = -354$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 14.07 \rightarrow 15$, $T_{\max} = 100$, $T \in [15, 100]$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 1 & 3 \\ 5 & 13 & 16 & 12 & 0 & 7 \\ 5 & 16 & 12 & 12 & 0 & 6 \\ 4 & 12 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 7 & 6 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

4.1112 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2$

$T = 9$ [0] $1^{11}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 14.23 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -334$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -931$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

4.1113 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{e}_7$

$T = 9$ [0] $1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{violet}{\mathfrak{e}}_7$ $T = 9$, $T_{\max} = 103$, $\Delta = -198$, $\det = -294$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 15.63 \rightarrow 16$, $T_{\max} = 103$, $T \in [16, 103]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 & -5 & -3 \\ 2 & 2 & -1 & -1 & 0 & -3 & -3 \end{pmatrix}_{b_0}$$

4.1114 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{e}'_7$

$T = 9$ [0] $1^{11}321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{violet}{\mathfrak{e}}'_7$ $T = 9$, $T_{\max} = 102$, $\Delta = -170$, $\det = -245$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 14.73 \rightarrow 15$, $T_{\max} = 102$, $T \in [15, 102]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 & -4 & -2 \\ 2 & 2 & -1 & -1 & 0 & -2 & -2 \end{pmatrix}_{b_0}$$

4.1115 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4$

$$T = 8 \quad [0] \ 1^{11} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.4$$

$$T_{\min} = 14.21 \rightarrow 15, T_{\max} = 98, T \in [15, 98]$$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 18 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 13 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 235, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -145, \det = 259, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 48 & 0 & 12 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \ 1^{11} 3^1 2^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 5 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 3 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 2^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 & 0 & 2 \\ 1 & 2 & 0 & 3 & 1 & 0 & 2 \\ 1 & 1 & 3 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 2 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^1 2^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 13 & 9 & 1 & 5 \\ 13 & 30 & 20 & 0 & 10 \\ 9 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 10 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^1 2^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^1 2^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 32, T \in [14, 32]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & -2 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 & -2 & 0 \\ 2 & 2 & -1 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{\overline{111}} 3^{\overline{111}} 1^{\overline{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -240$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 5 & 10 & 5 & 0 & 5 \\ 5 & 10 & 13 & 7 & 0 & 7 \\ 2 & 5 & 7 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 7 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4$ $T = 9$, $T_{\max} = 82$, $\Delta = -124$, $\det = -277$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 82$, $T \in [14, 82]$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 1 & 3 \\ 5 & 13 & 16 & 12 & 0 & 7 \\ 5 & 16 & 12 & 12 & 0 & 6 \\ 4 & 12 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 7 & 6 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

4.1116 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2$

$T = 9$ [0] $1^{\overline{111}} 3^{\overline{111}} 1^{\overline{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -83$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 13.27 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{11}} 3^{\overline{11}} 1^{\overline{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -215$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -264$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{111}} 3^{\overline{111}} 1^{\overline{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -735$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.88 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

4.1117 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{\overline{11}} 3^{\overline{11}} 1^{\overline{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -392$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 6 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

4.1118 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8$

$T = 8$ [0] $1^{11}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 67$, $\Delta = -121$, $\det = 12$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 14.33 \rightarrow 15, T_{\max} = 67, T \in [15, 67]$$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 18 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & -2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 67$, $\Delta = -121$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 19 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 67$, $\Delta = -121$, $\det = 53$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 67$, $\Delta = -121$, $\det = 112$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 67$, $\Delta = -121$, $\det = 123$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 10 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 67$, $\Delta = -121$, $\det = 144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 12, T_{\max} = 67, T \in [12, 67]$$

$$\begin{pmatrix} 5 & 12 & 18 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 18 & 36 & 48 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 12 & -2 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 67$, $\Delta = -121$, $\det = 147$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 9 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 67$, $\Delta = -121$, $\det = 168$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 12 & 9 & 1 & 5 \\ 12 & 25 & 20 & 0 & 9 \\ 9 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 67$, $\Delta = -121$, $\det = 188$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 67, T \in [13, 67]$$

$$\begin{pmatrix} 5 & 12 & 17 & 1 & 5 \\ 12 & 25 & 36 & 0 & 9 \\ 17 & 36 & 48 & 0 & 12 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [9] $1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -100$, $\det = -5$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$$T_{\min} = 15.40 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 4 & 5 & 1 & 1 & 3 \\ 4 & 10 & 13 & 0 & 0 & 6 \\ 5 & 13 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 6 & 6 & -1 & -2 & -2 \end{pmatrix}_{b_0}$$

- [10] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -26$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.46 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 5 & 5 & 5 & 0 & 5 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 5 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$
- [11] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -33$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.27 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 & 0 \\ 1 & 2 & 0 & 3 & 1 & 0 \\ 1 & 1 & 3 & 0 & 1 & 0 \\ 1 & 2 & 1 & 1 & -2 & 0 \\ 1 & 0 & 0 & 0 & 0 & -4 \\ 2 & 2 & 2 & 2 & 0 & -2 \end{pmatrix}_{b_0}$
- [12] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -100$, $\det = -41$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.41 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$
 $\begin{pmatrix} 1 & 4 & 1 & 1 & 5 & 3 \\ 4 & 10 & 1 & 0 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 0 & 0 & 11 & 5 \\ 3 & 6 & -1 & -1 & 5 & 1 \end{pmatrix}_{b_0}$
- [13] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -45$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$
- [14] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.42 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$
 $\begin{pmatrix} 5 & 8 & 12 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 12 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 6 & 9 & -1 & 2 \end{pmatrix}_{b_0}$
- [15] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.17 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 5 & 0 & 5 & 5 \\ 2 & 5 & 1 & 0 & 4 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 5 & 4 & 0 & 0 & 2 \\ 3 & 5 & 3 & -2 & 2 & 0 \end{pmatrix}_{b_0}$
- [16] $1^{11}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -84$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$
 $\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 4 & -2 & 2 \end{pmatrix}_{b_0}$
- [17] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.61 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$
 $\begin{pmatrix} 5 & 8 & 1 & 1 & 5 \\ 8 & 10 & 1 & 0 & 6 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 6 & -1 & -1 & 2 \end{pmatrix}_{b_0}$
- [18] $1^{11}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -144$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$
 $\begin{pmatrix} 5 & 9 & 6 & 1 & 5 \\ 9 & 13 & 10 & 0 & 7 \\ 6 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 4 & -2 & 2 \end{pmatrix}_{b_0}$
- [19] $1^{11}321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8$ $T = 9$, $T_{\max} = 14$, $\Delta = -93$, $\det = -166$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.19 \rightarrow 13$, $T_{\max} = 14$, $T \in [13, 14]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 5 & 10 & 0 & 5 & 5 \\ 5 & 10 & 13 & 0 & 7 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 5 & 7 & 0 & 0 & 2 \\ 3 & 5 & 7 & -2 & 2 & 1 \end{pmatrix}_{b_0}$
- [20] $1^{11}31^{11}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -100$, $\det = -200$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

13.0

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 1 & 3 \\ 5 & 13 & 16 & 12 & 0 & 7 \\ 5 & 16 & 12 & 12 & 0 & 6 \\ 4 & 12 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 7 & 6 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [21] \quad 1^{111} 4 \quad 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 66, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 14 & 10 & 2 & 8 \\ 10 & 5 & 1 & 5 \\ 2 & 1 & -2 & 0 \\ 8 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{111} 4 \quad 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 10 & 1 & 1 & 5 \\ 10 & 13 & 0 & 0 & 7 \\ 1 & 0 & -2 & 1 & 0 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 7 & 0 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 4 \quad 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 129, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 0 & 1 & 0 & 1 & 1 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 1 & 0 & 0 & -2 & 1 \\ 1 & 1 & 0 & 0 & 1 & -2 \\ 2 & 2 & 0 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

4.1119 NHC(-2)(-3) + $\textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$

$$T = 9 \quad [0] \quad 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 28 & 5 \\ 12 & 25 & 16 & 0 & 60 & 9 \\ 8 & 16 & 9 & 0 & 36 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 28 & 60 & 36 & 0 & 133 & 21 \\ 5 & 9 & 5 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 1 & 5 \\ 11 & 20 & 17 & 0 & 0 & 8 \\ 9 & 17 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 8 & 6 & -2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 10 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 20 & 12 & 0 & 13 & 7 \\ 5 & 9 & 5 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 10 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -133, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 0 & 13 & 7 \\ 5 & 8 & -1 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 10 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 17 & 8 \\ 9 & 17 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 13 & 7 \\ 5 & 8 & 6 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 8 & 5 \\ 11 & 20 & 0 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 0 & 0 & 10 & 6 \\ 5 & 8 & -1 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 19 & 5 \\ 12 & 25 & 16 & 0 & 40 & 9 \\ 8 & 16 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 40 & 24 & 0 & 58 & 14 \\ 5 & 9 & 5 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 20 & 5 \\ 12 & 25 & 16 & 0 & 40 & 9 \\ 8 & 16 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 40 & 24 & 0 & 58 & 14 \\ 5 & 9 & 5 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 10 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -222, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3 \ 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 26 & 5 \\ 11 & 20 & 0 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 26 & 50 & 0 & 0 & 118 & 20 \\ 5 & 8 & -1 & -2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 17 & 5 \\ 11 & 20 & 0 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 33 & 0 & 0 & 49 & 13 \\ 5 & 8 & -1 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{113} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 30 & 5 \\ 12 & 25 & 16 & 0 & 60 & 9 \\ 8 & 16 & 9 & 0 & 36 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 30 & 60 & 36 & 0 & 133 & 21 \\ 5 & 9 & 5 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{113} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 29 & 5 \\ 12 & 25 & 16 & 0 & 60 & 9 \\ 8 & 16 & 9 & 0 & 36 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 29 & 60 & 36 & 0 & 133 & 21 \\ 5 & 9 & 5 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{114} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 38 & 5 \\ 12 & 25 & 16 & 0 & 80 & 9 \\ 8 & 16 & 9 & 0 & 48 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 38 & 80 & 48 & 0 & 238 & 28 \\ 5 & 9 & 5 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{114} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 40 & 5 \\ 12 & 25 & 16 & 0 & 80 & 9 \\ 8 & 16 & 9 & 0 & 48 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 40 & 80 & 48 & 0 & 238 & 28 \\ 5 & 9 & 5 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{114} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 39 & 5 \\ 12 & 25 & 16 & 0 & 80 & 9 \\ 8 & 16 & 9 & 0 & 48 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 39 & 80 & 48 & 0 & 238 & 28 \\ 5 & 9 & 5 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111} 3 \ 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 19 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{\overline{111}} 3 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{\overline{111}} 3 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 18 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \quad 1^{\overline{11}} 4 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 165, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 9 & 5 \\ 9 & 13 & 10 & 0 & 14 & 7 \\ 6 & 10 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 9 & 0 & 13 & 7 \\ 5 & 7 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{\overline{111}} 4 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 261, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 6 & 5 \\ 9 & 13 & 10 & 0 & 10 & 7 \\ 6 & 10 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 7 & 0 & 5 & 5 \\ 5 & 7 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{\overline{111}} 4 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 480, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 6 & 1 & 15 & 5 \\ 9 & 13 & 10 & 0 & 24 & 7 \\ 6 & 10 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 24 & 16 & 0 & 38 & 12 \\ 5 & 7 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

4.1120 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 6 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

4.1121 $\text{NHC}(-2)(-3) + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 7 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 6 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

4.1122 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 1 & 5 \\ 12 & 25 & 16 & 0 & 0 & 9 \\ 8 & 16 & 9 & 0 & 0 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 5 & -2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 9 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 20 & 12 & 0 & 12 & 6 \\ 5 & 9 & 5 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 10 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 20 & 12 & 0 & 12 & 6 \\ 5 & 9 & 5 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -1 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \ 1^{11} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.20 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 6 & 5 \\ 12 & 25 & 12 & 0 & 12 & 9 \\ 6 & 12 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 7 & 0 & 4 & 4 \\ 5 & 9 & 4 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 168, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 1 & 1 & 3 \\ 5 & 13 & 16 & 12 & 0 & 0 & 7 \\ 5 & 16 & 12 & 12 & 0 & 0 & 6 \\ 4 & 12 & 12 & 9 & 0 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 6 & 5 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

4.1123 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 12 & 14 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 12 & 15 & 6 \\ 14 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 19 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 8 & 13 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 16 & 4 \\ 5 & 9 & 12 & 0 & 24 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 24 & 0 & 46 & 12 \\ 3 & 4 & 6 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{123} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 6 & 3 \\ 3 & 4 & 13 & 0 & 12 & 4 \\ 7 & 13 & 27 & 0 & 27 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 27 & 0 & 25 & 9 \\ 3 & 4 & 9 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 25, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 12 & 15 & 6 \\ 12 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 8 & -1 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 10 & 17 & 12 & 15 & 6 \\ 10 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 20 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 24 & 15 & 0 & 121 & 19 \\ 3 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^1 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 9 & 20 & 12 & 13 & 7 \\ 5 & 9 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 3 \\ 3 & 4 & 4 & 0 & 0 & 4 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 34$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 5 & 0 & 0 & 4 \\ 3 & 5 & 3 & 0 & 0 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & -2 & 0 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 34$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 16 & 0 & 10 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[14] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 34$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 1 & 3 \\ 3 & 4 & 9 & 0 & 0 & 4 \\ 5 & 9 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & -2 & 0 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 37$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 10 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 8 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 37$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 17 & 0 & 13 & 7 \\ 5 & 8 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 39$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 13 & 11 & 6 \\ 15 & 20 & 17 & 16 & 8 \\ 13 & 17 & 12 & 12 & 6 \\ 11 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.45 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 0 & 0 & 4 & 4 \\ 3 & 4 & -1 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{11} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 11 & 3 \\ 3 & 4 & 0 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 19 & 0 & 0 & 73 & 15 \\ 3 & 4 & -1 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{113} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

- [24] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 17 & 0 & 13 & 7 \\ 6 & 8 & -1 & 7 & 2 \end{pmatrix}_{b_0}$
- [25] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 10 & 10 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 10 & 17 & 12 & 12 & 6 \\ 10 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [23] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 8 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 3 & -2 & 6 & 1 \end{pmatrix}_{b_0}$
- [26] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & -1 & -2 & 3 & 1 \end{pmatrix}_{b_0}$
- [27] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 28 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 28 & 12 & 0 & 73 & 15 \\ 3 & 5 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$
- [28] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 5 & 8 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 18 & 0 & 25 & 9 \\ 3 & 4 & 5 & -2 & 9 & 1 \end{pmatrix}_{b_0}$
- [29] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 9 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 12 & 15 & 6 \\ 10 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$
- [30] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 8 & 37 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 37 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$
- [31] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 12 & 1 & 6 \\ 15 & 20 & 17 & 0 & 8 \\ 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 6 & 8 & 6 & 0 & 2 \end{pmatrix}_{b_0}$
- [32] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 8 & 10 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 10 & 20 & 12 & 13 & 7 \\ 5 & 9 & 5 & 7 & 2 \end{pmatrix}_{b_0}$
- [33] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 6 & 0 & 13 & 7 \\ 3 & 5 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 13 & 7 \\ 5 & 8 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 10 & 17 & 12 & 15 & 6 \\ 9 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 13 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 13 & 17 & 12 & 15 & 6 \\ 12 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[38] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 12 & 15 & 6 \\ 9 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[40] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[41] $21^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 17 & 15 & 15 & 7 \\ 17 & 20 & 17 & 17 & 8 \\ 15 & 17 & 12 & 15 & 6 \\ 15 & 17 & 15 & 13 & 7 \\ 7 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 5 & 9 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 9 & 0 & 4 & 4 \\ 3 & 4 & 6 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

- [44] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$
- [45] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 8 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 8 & 4 \\ 8 & 13 & 27 & 0 & 18 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 18 & 0 & 10 & 6 \\ 3 & 4 & 9 & -2 & 6 & 1 \end{pmatrix}_{b_0}$
- [46] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 12 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 12 & 15 & 6 \\ 13 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$
- [48] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 9 & 3 \\ 3 & 4 & 0 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 0 & 49 & 13 \\ 3 & 4 & -1 & -2 & 13 & 1 \end{pmatrix}_{b_0}$
- [47] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 2 & -2 & 4 & 1 \end{pmatrix}_{b_0}$
- [49] $1^1 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 9 & 8 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 9 & 17 & 12 & 12 & 6 \\ 8 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [52] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 8 & 19 & 5 \\ 12 & 25 & 16 & 40 & 9 \\ 8 & 16 & 9 & 24 & 5 \\ 19 & 40 & 24 & 58 & 14 \\ 5 & 9 & 5 & 14 & 2 \end{pmatrix}_{b_0}$
- [50] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 12 & 8 & 28 & 5 \\ 12 & 25 & 16 & 60 & 9 \\ 8 & 16 & 9 & 36 & 5 \\ 28 & 60 & 36 & 133 & 21 \\ 5 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$
- [51] $1^1 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 6 & 0 & 4 & 4 \\ 3 & 4 & 3 & -2 & 4 & 1 \end{pmatrix}_{b_0}$
- [53] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 10 & 9 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 10 & 17 & 12 & 12 & 6 \\ 9 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [54] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 11 & 6 \\ 15 & 20 & 17 & 16 & 8 \\ 12 & 17 & 12 & 12 & 6 \\ 11 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[55] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 17 & 16 & 8 \\ 12 & 17 & 12 & 12 & 6 \\ 12 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[56] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 9 & 17 & 12 & 12 & 6 \\ 9 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[57] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 20 & 5 \\ 12 & 25 & 16 & 40 & 9 \\ 8 & 16 & 9 & 24 & 5 \\ 20 & 40 & 24 & 58 & 14 \\ 5 & 9 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$

[58] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 6 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 4 & 8 & 9 & 0 & 6 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 5 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

[61] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 3 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[63] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 4 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 6 & 0 & 4 & 4 \\ 3 & 4 & 3 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

[64] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 3 \\ 3 & 4 & 0 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 0 & 0 & 37 & 11 \\ 3 & 4 & -1 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

- [65] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 84$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 6 & 0 & 13 & 7 \\ 3 & 5 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$
- [66] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 85$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 0 & 4 & 4 \\ 3 & 5 & 2 & -2 & 4 & 1 \end{pmatrix}_{b_0}$
- [67] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 85$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 0 & 0 & 13 & 7 \\ 3 & 4 & -1 & -2 & 7 & 1 \end{pmatrix}_{b_0}$
- [68] $1^{131} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 86$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 \\ 9 & 17 & 48 & 0 & 15 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 15 & 0 & 4 & 4 \\ 3 & 4 & 12 & -2 & 4 & 1 \end{pmatrix}_{b_0}$
- [69] $1^{131} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 89$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 9 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 9 & 17 & 48 & 0 & 12 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 12 & 0 & 1 & 3 \\ 3 & 4 & 12 & -2 & 3 & 1 \end{pmatrix}_{b_0}$
- [70] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 91$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 6 & 0 & 4 & 4 \\ 3 & 4 & 3 & -2 & 4 & 1 \end{pmatrix}_{b_0}$
- [71] $1^{121} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 93$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.44 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 7 & 3 \\ 3 & 4 & 0 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 0 & 0 & 28 & 10 \\ 3 & 4 & -1 & -2 & 10 & 1 \end{pmatrix}_{b_0}$
- [72] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 94$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 1 & 11 & 6 \\ 15 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 16 & 0 & 10 & 6 \\ 6 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$
- [73] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 94$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.55 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 10 & 20 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 20 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$
- [77] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 19 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 19 & 12 & 0 & 73 & 15 \\ 3 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$
- [80] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 24 & 15 & 0 & 121 & 19 \\ 3 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[79] \ 1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[78] \ 1^{121} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 4 & 8 & 9 & 0 & 6 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 5 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 19 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 19 & 12 & 0 & 73 & 15 \\ 3 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{112} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 12 & 4 \\ 5 & 9 & 12 & 0 & 18 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 18 & 0 & 25 & 9 \\ 3 & 4 & 6 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{11} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 4 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 3 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[83] \ 1^{111} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 48 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 48 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{14} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.6$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 & 3 \\ 3 & 4 & 21 & 0 & 5 & 4 \\ 12 & 21 & 75 & 0 & 18 & 15 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 18 & 0 & 4 & 4 \\ 3 & 4 & 15 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 4 & 8 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 12 & 0 & 13 & 7 \\ 3 & 4 & 5 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 \\ 5 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 6 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 7 & 13 & 27 & 0 & 9 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 9 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 9 & 4 \\ 5 & 8 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 12 & 0 & 13 & 7 \\ 3 & 4 & 5 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[93] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 35 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 35 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[91] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 17 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 17 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[92] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 3 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 1 & 23 & 6 \\ 15 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 33 & 0 & 49 & 13 \\ 6 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[94] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 \\ 8 & 14 & 36 & 0 & 9 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 10 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 16 & 0 & 10 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{114} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 15 & 3 \\ 3 & 4 & 9 & 0 & 21 & 4 \\ 5 & 9 & 12 & 0 & 33 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 21 & 33 & 0 & 76 & 16 \\ 3 & 4 & 6 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[98] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 30 & 5 \\ 12 & 25 & 16 & 60 & 9 \\ 8 & 16 & 9 & 36 & 5 \\ 30 & 60 & 36 & 133 & 21 \\ 5 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$

[99] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[100] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[101] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 5 & 9 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 9 & 0 & 4 & 4 \\ 3 & 4 & 6 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

[102] $1^{122} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 5 & 8 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 5 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[103] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 24 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 24 & 15 & 0 & 121 & 19 \\ 3 & 4 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[104] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 29 & 5 \\ 12 & 25 & 16 & 60 & 9 \\ 8 & 16 & 9 & 36 & 5 \\ 29 & 60 & 36 & 133 & 21 \\ 5 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$

[105] $1^{111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[106] $1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 129$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.22 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 26 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 26 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[107] $1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 131$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 \\ 7 & 13 & 27 & 0 & 12 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 12 & 0 & 4 & 4 \\ 3 & 4 & 9 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

[108] $1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 131$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 13 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 13 & 17 & 12 & 27 & 6 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[109] $1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 141$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 \\ 5 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 6 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[110] $1^{112} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 24 & 12 & 0 & 58 & 14 \\ 3 & 5 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[111] $1^{112} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 24 & 12 & 0 & 58 & 14 \\ 3 & 5 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{12} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 16 & 3 \\ 3 & 4 & 0 & 0 & 28 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 28 & 0 & 0 & 154 & 22 \\ 3 & 4 & -1 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

[113] $1^{122} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 146$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 8 & 4 \\ 7 & 13 & 27 & 0 & 18 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 18 & 0 & 10 & 6 \\ 3 & 4 & 9 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[114] $21^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 16 & 12 & 12 & 6 \\ 16 & 20 & 17 & 16 & 8 \\ 12 & 17 & 12 & 12 & 6 \\ 12 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[115] $1^{111} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 4 \\ 9 & 14 & 36 & 0 & 9 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 10 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

- [116] $1^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 2 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$
- [117] $1^{12}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 10 & 3 \\ 3 & 4 & 0 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 0 & 0 & 58 & 14 \\ 3 & 4 & -1 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$
- [118] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [119] $1^{14}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 \\ 3 & 5 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 18 & 0 & 76 & 16 \\ 3 & 4 & 3 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$
- [120] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$
- [121] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 157, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 20 & 5 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 20 & 9 & 0 & 37 & 11 \\ 3 & 5 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$
- [122] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 158, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 38 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 38 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$
- [123] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 \\ 6 & 9 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 9 & 0 & 4 & 4 \\ 3 & 4 & 6 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$
- [124] $1^{12}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 13 & 3 \\ 3 & 4 & 0 & 0 & 23 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 23 & 0 & 0 & 100 & 18 \\ 3 & 4 & -1 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$
- [125] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 4 \\ 6 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 6 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [126] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 26 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[128] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 3 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[127] $1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 29 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 29 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[129] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[130] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 170, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 & 3 \\ 3 & 4 & 17 & 0 & 5 & 4 \\ 10 & 17 & 48 & 0 & 15 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 15 & 0 & 4 & 4 \\ 3 & 4 & 12 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[131] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 8 & 4 \\ 7 & 13 & 27 & 0 & 18 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 18 & 0 & 10 & 6 \\ 3 & 4 & 9 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[138] $1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 8 & 4 \\ 4 & 8 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 5 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 8 & 0 & 4 & 4 \\ 5 & 8 & 9 & 0 & 6 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 5 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[136] $1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

[135] $1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[142] $1^{\overline{113}} 3^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 180$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.71 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 17 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 36 & 18 & 0 & 133 & 21 \\ 3 & 5 & 2 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

[141] $1^{\overline{112}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 180$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.51 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 16 & 6 & 0 & 22 & 8 \\ 3 & 5 & 2 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{\overline{13}} 3^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 180$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.44 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 12 & 3 \\ 3 & 4 & 0 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 0 & 0 & 85 & 17 \\ 3 & 4 & -1 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{\overline{14}} 3^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 180$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.51 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 14 & 3 \\ 3 & 4 & 0 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 0 & 0 & 118 & 20 \\ 3 & 4 & -1 & -2 & 20 & 1 \end{pmatrix}_{b_0}$$

[143] $21^{\overline{111}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 184$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 15 & 7 \\ 20 & 25 & 16 & 20 & 9 \\ 12 & 16 & 9 & 12 & 5 \\ 15 & 20 & 12 & 13 & 7 \\ 7 & 9 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

[144] $1^{\overline{123}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 185$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.18 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 12 & 4 \\ 7 & 13 & 27 & 0 & 27 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 27 & 0 & 25 & 9 \\ 3 & 4 & 9 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[145] $1^{\overline{114}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 186$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 40 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 40 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$

[146] $1^{\overline{113}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 188$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 20 & 5 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 20 & 9 & 0 & 37 & 11 \\ 3 & 5 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 19 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[149] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 & 3 \\ 3 & 4 & 13 & 0 & 5 & 4 \\ 8 & 13 & 27 & 0 & 12 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 12 & 0 & 4 & 4 \\ 3 & 4 & 9 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

[150] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 19 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 9 & 17 & 12 & 27 & 6 \\ 19 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[151] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 9 & 17 & 12 & 27 & 6 \\ 17 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[152] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 19 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 19 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[153] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 5 & 8 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 18 & 0 & 28 & 10 \\ 3 & 4 & 5 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[154] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 39 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 39 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$

[155] $1^{113} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 205, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 32 & 15 & 0 & 100 & 18 \\ 3 & 5 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

[156] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 15 & 0 & 49 & 13 \\ 3 & 4 & 3 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

- [157] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 18 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 18 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [158] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 45 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 45 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$
- [159] $1^{122} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 6 & 9 & 12 & 0 & 21 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 6 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$
- [160] $1^{121} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 6 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 7 & 13 & 27 & 0 & 21 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 21 & 0 & 13 & 7 \\ 3 & 4 & 9 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$
- [161] $1^{123} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 3 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$
- [162] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 18 & 0 & 28 & 10 \\ 3 & 4 & 5 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$
- [163] $1^{112} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 3 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$
- [164] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 13 & 4 \\ 4 & 8 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 18 & 0 & 28 & 10 \\ 3 & 4 & 5 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$
- [165] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 7 & 13 & 27 & 0 & 9 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 9 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [172] $1^{113} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$
- [166] $1^{113} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$

$$T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[167] 1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[168] 1^{111} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 19 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 19 & 12 & 0 & 73 & 15 \\ 3 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[171] 1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[169] 1^{111} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 19 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 19 & 12 & 0 & 73 & 15 \\ 3 & 4 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[170] 1^{123} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 8 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 4 & 8 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 18 & 0 & 25 & 9 \\ 3 & 4 & 5 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[173] 1^{114} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 22 & 3 \\ 4 & 9 & 4 & 0 & 44 & 5 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 44 & 21 & 0 & 193 & 25 \\ 3 & 5 & 2 & -2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[174] 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 233, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 18 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[175] 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 12 & 25 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 25 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[176] 1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 237, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 9 & 3 \\ 4 & 9 & 4 & 0 & 20 & 5 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 20 & 9 & 0 & 37 & 11 \\ 3 & 5 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[179] 1^{111} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{111} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[177] $1^{131} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 12 & 0 & 4 & 4 \\ 7 & 12 & 24 & 0 & 9 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 8 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[181] $1^{111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 24 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[180] $1^{111} 3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 38 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 38 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[182] $1^{11} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 242, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 5 & 0 & 9 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 3 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[183] $1^{113} 3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 36 & 18 & 0 & 133 & 21 \\ 3 & 5 & 2 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

[184] $21^{111} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 17 & 15 & 27 & 7 \\ 17 & 20 & 17 & 33 & 8 \\ 15 & 17 & 12 & 27 & 6 \\ 27 & 33 & 27 & 49 & 13 \\ 7 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[185] $1^{111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 18 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 9 & 17 & 12 & 27 & 6 \\ 18 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[186] $1^{13} 3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 15 & 3 \\ 3 & 4 & 0 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 27 & 0 & 0 & 133 & 21 \\ 3 & 4 & -1 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

[187] $1^{113} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

- [188] $1^{111} 3^1 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 254, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 47 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 47 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$
- [189] $1^{123} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 17 & 4 \\ 5 & 8 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 24 & 0 & 49 & 13 \\ 3 & 4 & 5 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$
- [190] $1^{111} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 6 & 9 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 6 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$
- [191] $1^{114} 3^1 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 31 & 4 \\ 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 2 & -2 & 24 & 1 \end{pmatrix}_{b_0}$$
- [192] $1^{111} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 5 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 6 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$
- [193] $1^{123} 3^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 4 & 8 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 18 & 0 & 25 & 9 \\ 3 & 4 & 5 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$
- [194] $1^{111} 3^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$
- [195] $21^{112} 31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 30 & 7 \\ 20 & 25 & 16 & 40 & 9 \\ 12 & 16 & 9 & 24 & 5 \\ 30 & 40 & 24 & 58 & 14 \\ 7 & 9 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$
- [196] $1^{121} 3^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 4 & 4 \\ 10 & 18 & 57 & 0 & 12 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 1 & 3 \\ 3 & 4 & 13 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [197] $1^{113} 3^1 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 36 & 18 & 0 & 133 & 21 \\ 3 & 5 & 2 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$
- [198] $1^{111} 3^1 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 277, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 36 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 36 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[199] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 285, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[202] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 27 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 27 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[201] $21^{111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 16 & 12 & 24 & 6 \\ 16 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[200] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 28 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 28 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{13} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 18 & 3 \\ 3 & 4 & 0 & 0 & 32 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 32 & 0 & 0 & 193 & 25 \\ 3 & 4 & -1 & -2 & 25 & 1 \end{pmatrix}_{b_0}$$

[205] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 9 & 3 \\ 3 & 4 & 13 & 0 & 17 & 4 \\ 8 & 13 & 27 & 0 & 39 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 39 & 0 & 49 & 13 \\ 3 & 4 & 9 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[204] $1^{111} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 46 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 46 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[206] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 293, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 11 & 3 \\ 3 & 4 & 8 & 0 & 17 & 4 \\ 4 & 8 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 17 & 24 & 0 & 49 & 13 \\ 3 & 4 & 5 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 295, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 3 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[208] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

- [209] $1^{131} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 301, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 9 & 17 & 48 & 0 & 12 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 1 & 3 \\ 3 & 4 & 12 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [210] $1^{113} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 303, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 12 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 17 & 27 & 0 & 49 & 13 \\ 3 & 4 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$
- [211] $1^{114} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 2 & -2 & 20 & 1 \end{pmatrix}_{b_0}$$
- [212] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 28 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 28 & 12 & 0 & 73 & 15 \\ 3 & 5 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$
- [213] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 28 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 28 & 12 & 0 & 73 & 15 \\ 3 & 5 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$
- [214] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 6 & 3 \\ 3 & 4 & 9 & 0 & 9 & 4 \\ 5 & 9 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 15 & 0 & 13 & 7 \\ 3 & 4 & 6 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$
- [215] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 309, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 4 & 4 \\ 8 & 13 & 27 & 0 & 9 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 9 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [216] $1^{14} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 309, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 0 & 17 & 3 \\ 3 & 4 & 0 & 0 & 31 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 31 & 0 & 0 & 172 & 24 \\ 3 & 4 & -1 & -2 & 24 & 1 \end{pmatrix}_{b_0}$$
- [217] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 37 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 37 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$
- [218] $1^{113} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$
- [219] $1^{121} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 326, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 7 & 13 & 27 & 0 & 21 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 21 & 0 & 13 & 7 \\ 3 & 4 & 9 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[220] 1^{131} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 5 & 3 \\ 3 & 4 & 17 & 0 & 9 & 4 \\ 9 & 17 & 48 & 0 & 27 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 27 & 0 & 13 & 7 \\ 3 & 4 & 12 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[221] 1^{113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 333, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 0 & 24 & 5 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 24 & 9 & 0 & 52 & 12 \\ 3 & 5 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[222] 1^{131} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 17 & 15 & 0 & 49 & 13 \\ 3 & 4 & 3 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[223] 1^{123} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 341, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 8 & 0 & 17 & 4 \\ 4 & 8 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 24 & 0 & 49 & 13 \\ 3 & 4 & 5 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[224] 1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 3 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[225] 1^{113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 349, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 32 & 15 & 0 & 100 & 18 \\ 3 & 5 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[227] 1^{123} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 12 & 3 \\ 3 & 4 & 8 & 0 & 22 & 4 \\ 4 & 8 & 9 & 0 & 30 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 30 & 0 & 85 & 17 \\ 3 & 4 & 5 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[226] 1^{123} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 13 & 3 \\ 3 & 4 & 8 & 0 & 22 & 4 \\ 4 & 8 & 9 & 0 & 30 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 30 & 0 & 85 & 17 \\ 3 & 4 & 5 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[228] 1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 7 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 12 & 0 & 21 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 6 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[229] 1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 9 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 12 & 0 & 21 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 6 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[230] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 371$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 5 & 0 & 13 & 4 \\ 3 & 5 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 13 & 12 & 0 & 28 & 10 \\ 3 & 4 & 3 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[231] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 372$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.09 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 18 & 0 & 4 & 4 \\ 11 & 18 & 57 & 0 & 12 & 13 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 1 & 3 \\ 3 & 4 & 13 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[232] $1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 381$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 32 & 15 & 0 & 100 & 18 \\ 3 & 5 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

[233] $1^{11} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 384$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 22 & 3 \\ 4 & 9 & 4 & 0 & 48 & 5 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 48 & 24 & 0 & 238 & 28 \\ 3 & 5 & 2 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

[234] $1^{11} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 384$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.28 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[235] $1^{11} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 384$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 9 & 4 & 0 & 48 & 5 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 48 & 24 & 0 & 238 & 28 \\ 3 & 5 & 2 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

[236] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 386$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 9 & 4 \\ 8 & 13 & 27 & 0 & 21 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 21 & 0 & 13 & 7 \\ 3 & 4 & 9 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[237] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 390$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.34 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 11 & 3 \\ 3 & 4 & 13 & 0 & 17 & 4 \\ 7 & 13 & 27 & 0 & 39 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 17 & 39 & 0 & 49 & 13 \\ 3 & 4 & 9 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[238] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 392$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[239] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 394$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[251] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[248] $1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[252] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 8 & 13 & 27 & 0 & 30 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 30 & 0 & 28 & 10 \\ 3 & 4 & 9 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[253] $21^{113} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 45 & 7 \\ 20 & 25 & 16 & 60 & 9 \\ 12 & 16 & 9 & 36 & 5 \\ 45 & 60 & 36 & 133 & 21 \\ 7 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$

[254] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 10 & 3 \\ 3 & 4 & 8 & 0 & 17 & 4 \\ 4 & 8 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 24 & 0 & 49 & 13 \\ 3 & 4 & 5 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[255] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 36 & 15 & 0 & 121 & 19 \\ 3 & 5 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[256] $1^{113} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 32 & 15 & 0 & 100 & 18 \\ 3 & 5 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

[257] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 12 & 3 \\ 4 & 9 & 4 & 0 & 28 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 28 & 12 & 0 & 73 & 15 \\ 3 & 5 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[258] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 0 & 28 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 28 & 12 & 0 & 73 & 15 \\ 3 & 5 & 2 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$

[259] $1^{131} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 4 & 4 \\ 12 & 22 & 84 & 0 & 15 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 1 & 3 \\ 3 & 4 & 16 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[260] $1^{114} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 40 & 18 & 0 & 154 & 22 \\ 3 & 5 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

[261] $1^{114} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 21 & 3 \\ 4 & 9 & 4 & 0 & 48 & 5 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 48 & 24 & 0 & 238 & 28 \\ 3 & 5 & 2 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

[262] $1^{131} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 453, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 4 & 4 \\ 10 & 17 & 48 & 0 & 12 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 12 & 0 & 1 & 3 \\ 3 & 4 & 12 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[263] $1^{131} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 455, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 5 & 3 \\ 3 & 4 & 17 & 0 & 9 & 4 \\ 10 & 17 & 48 & 0 & 27 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 27 & 0 & 13 & 7 \\ 3 & 4 & 12 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

[264] $1^{112} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

[265] $1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[266] $1^{13} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 458, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 17 & 15 & 0 & 49 & 13 \\ 3 & 4 & 3 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[267] $1^{122} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 7 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 27 & 0 & 30 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 30 & 0 & 28 & 10 \\ 3 & 4 & 9 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[268] $1^{13} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 17 & 4 \\ 3 & 5 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 17 & 15 & 0 & 49 & 13 \\ 3 & 4 & 3 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[269] $1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 488, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[270] $1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 491$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.19 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 8 & 3 \\ 3 & 4 & 13 & 0 & 13 & 4 \\ 7 & 13 & 27 & 0 & 30 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 30 & 0 & 28 & 10 \\ 3 & 4 & 9 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[271] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 493$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 36 & 15 & 0 & 121 & 19 \\ 3 & 5 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[272] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 498$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 8 & 3 \\ 3 & 4 & 9 & 0 & 13 & 4 \\ 5 & 9 & 12 & 0 & 21 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 21 & 0 & 28 & 10 \\ 3 & 4 & 6 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[273] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 501$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.47 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 21 & 3 \\ 4 & 9 & 4 & 0 & 44 & 5 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 44 & 21 & 0 & 193 & 25 \\ 3 & 5 & 2 & -2 & 25 & 1 \end{pmatrix}_{b_0}$$

[274] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 503$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 \\ 3 & 5 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 21 & 18 & 0 & 76 & 16 \\ 3 & 4 & 3 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[276] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 528$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.47 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[275] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 528$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 32 & 12 & 0 & 94 & 16 \\ 3 & 5 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[277] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 535$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 \\ 3 & 5 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 21 & 18 & 0 & 76 & 16 \\ 3 & 4 & 3 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[278] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 536$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.36 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[279] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 549$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 3 \\ 4 & 9 & 4 & 0 & 44 & 5 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 44 & 21 & 0 & 193 & 25 \\ 3 & 5 & 2 & -2 & 25 & 1 \end{pmatrix}_{b_0}$$

[280] $1^{114} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 40 & 18 & 0 & 154 & 22 \\ 3 & 5 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

[281] $1^{123} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 566, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 9 & 3 \\ 3 & 4 & 13 & 0 & 17 & 4 \\ 7 & 13 & 27 & 0 & 39 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 39 & 0 & 49 & 13 \\ 3 & 4 & 9 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[282] $1^{14} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 571, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 5 & 0 & 21 & 4 \\ 3 & 5 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 21 & 18 & 0 & 76 & 16 \\ 3 & 4 & 3 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[283] $1^{112} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 584, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 2 & -2 & 18 & 1 \end{pmatrix}_{b_0}$$

[284] $1^{114} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 586, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 14 & 3 \\ 3 & 4 & 9 & 0 & 21 & 4 \\ 5 & 9 & 12 & 0 & 33 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 21 & 33 & 0 & 76 & 16 \\ 3 & 4 & 6 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[285] $1^{131} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 2 & 3 \\ 3 & 4 & 22 & 0 & 4 & 4 \\ 13 & 22 & 84 & 0 & 15 & 16 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 15 & 0 & 1 & 3 \\ 3 & 4 & 16 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[286] $1^{113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 11 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 17 & 27 & 0 & 49 & 13 \\ 3 & 4 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[288] $1^{114} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[289] $1^{114} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[290] $1^{112} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[287] $1^{112}3^{113}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[291] $21^{114}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 634, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 60 & 7 \\ 20 & 25 & 16 & 80 & 9 \\ 12 & 16 & 9 & 48 & 5 \\ 60 & 80 & 48 & 238 & 28 \\ 7 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$

[292] $1^{114}3^{111}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 637, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 15 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 36 & 15 & 0 & 121 & 19 \\ 3 & 5 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[293] $1^{113}3^{111}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 639, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 10 & 3 \\ 3 & 4 & 9 & 0 & 17 & 4 \\ 5 & 9 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 27 & 0 & 49 & 13 \\ 3 & 4 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[294] $1^{114}3^{112}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 645, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 20 & 3 \\ 4 & 9 & 4 & 0 & 44 & 5 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 44 & 21 & 0 & 193 & 25 \\ 3 & 5 & 2 & -2 & 25 & 1 \end{pmatrix}_{b_0}$$

[295] $1^{113}3^{112}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

[296] $1^{114}3^{112}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 660, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 19 & 3 \\ 4 & 9 & 4 & 0 & 40 & 5 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 40 & 18 & 0 & 154 & 22 \\ 3 & 5 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

[297] $1^{114}3^{111}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 661, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 19 & 3 \\ 4 & 9 & 4 & 0 & 44 & 5 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 44 & 21 & 0 & 193 & 25 \\ 3 & 5 & 2 & -2 & 25 & 1 \end{pmatrix}_{b_0}$$

[298] $1^{114}3^{111}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 669, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 17 & 3 \\ 4 & 9 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 36 & 15 & 0 & 121 & 19 \\ 3 & 5 & 2 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[303] $1^{112}3^{114}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[321] $1^{114}3^11^{12}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_3+\mathfrak{so}_8+\mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 896$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.21 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 2 & -2 & 20 & 1 \end{pmatrix}_{b_0}$$

[320] $1^{113}3^11^{14}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_3+\mathfrak{so}_8+\mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 896$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 32 & 21 & 0 & 193 & 25 \\ 3 & 4 & 2 & -2 & 25 & 1 \end{pmatrix}_{b_0}$$

[322] $21^{115}31^{11}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_3+\mathfrak{so}_8+\mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 904$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.1$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 75 & 7 \\ 20 & 25 & 16 & 100 & 9 \\ 12 & 16 & 9 & 60 & 5 \\ 75 & 100 & 60 & 373 & 35 \\ 7 & 9 & 5 & 35 & 2 \end{pmatrix}_{b_0}$$

[323] $1^{113}3^11^{12}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_3+\mathfrak{so}_8+\mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 944$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.11 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

[324] $1^{113}3^11^{14}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_3+\mathfrak{so}_8+\mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 1040$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

[325] $1^{113}3^11^{12}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_3+\mathfrak{so}_8+\mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 1104$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

[326] $1^{114}3^11^{11}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_3+\mathfrak{so}_8+\mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 1112$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.17 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 2 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[328] $1^{113}3^11^{12}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_3+\mathfrak{so}_8+\mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 1136$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

[327] $1^{113}3^11^{13}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_3+\mathfrak{so}_8+\mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 1136$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 2 & -2 & 21 & 1 \end{pmatrix}_{b_0}$$

[330] $1^{114}3^11^{13}+[\mathfrak{su}_2\cdot\mathfrak{g}_2]+\mathfrak{su}_3+\mathfrak{so}_8+\mathfrak{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 1176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 22 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 2 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

[350] $1^{114} 3^{113} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 2336$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 23 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 23 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 2 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

[351] $1^{114} 3^{114} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 2400$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 24 & 3 \\ 3 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 24 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 2 & -2 & 28 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [352] $1^{111} 4^{111} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -14$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 13 & 3 \\ 3 & 4 & 7 & 0 & 21 & 4 \\ 4 & 7 & 7 & 0 & 31 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 21 & 31 & 0 & 101 & 19 \\ 3 & 4 & 5 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

[353] $1^{11} 4^{11} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -31$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

[358] $1^{12} 4^{11} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 3 & 0 & 8 & 4 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 4 & 0 & 10 & 6 \\ 3 & 4 & 2 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[354] $1^{11} 4^{11} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 8 & 12 & 6 \\ 12 & 13 & 10 & 14 & 7 \\ 8 & 10 & 4 & 9 & 4 \\ 12 & 14 & 9 & 13 & 7 \\ 6 & 7 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[355] $1^{11} 4^{11} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 8 & 8 & 6 \\ 12 & 13 & 10 & 10 & 7 \\ 8 & 10 & 4 & 7 & 4 \\ 8 & 10 & 7 & 5 & 5 \\ 6 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[356] $21^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 8 & 8 & 6 \\ 12 & 13 & 10 & 10 & 7 \\ 8 & 10 & 4 & 7 & 4 \\ 8 & 10 & 7 & 5 & 5 \\ 6 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[357] $21^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 8 & 12 & 6 \\ 12 & 13 & 10 & 14 & 7 \\ 8 & 10 & 4 & 9 & 4 \\ 12 & 14 & 9 & 13 & 7 \\ 6 & 7 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[359] $21^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -44$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 14 & 9 & 15 & 7 \\ 14 & 13 & 10 & 14 & 7 \\ 9 & 10 & 4 & 9 & 4 \\ 15 & 14 & 9 & 13 & 7 \\ 7 & 7 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[360] \ 1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 7 & 0 & 5 & 4 \\ 4 & 7 & 7 & 0 & 6 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 6 & 0 & 4 & 4 \\ 3 & 4 & 5 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[361] \ 1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -49, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 3 & 0 & 10 & 4 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 8 & 0 & 21 & 9 \\ 3 & 4 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[363] \ 1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 10 & 0 & 4 & 4 \\ 6 & 10 & 20 & 0 & 7 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 7 & 0 & 1 & 3 \\ 3 & 4 & 8 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[362] \ 21^1 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 14 & 9 & 9 & 7 \\ 14 & 13 & 10 & 10 & 7 \\ 9 & 10 & 4 & 7 & 4 \\ 9 & 10 & 7 & 5 & 5 \\ 7 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[364] \ 1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 5 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[365] \ 1^1 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 6 & 9 & 5 \\ 9 & 13 & 10 & 14 & 7 \\ 6 & 10 & 4 & 9 & 4 \\ 9 & 14 & 9 & 13 & 7 \\ 5 & 7 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[366] \ 1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[367] \ 1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 8 & 20 & 6 \\ 12 & 13 & 10 & 24 & 7 \\ 8 & 10 & 4 & 16 & 4 \\ 20 & 24 & 16 & 38 & 12 \\ 6 & 7 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[368] \ 21^1 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 8 & 20 & 6 \\ 12 & 13 & 10 & 24 & 7 \\ 8 & 10 & 4 & 16 & 4 \\ 20 & 24 & 16 & 38 & 12 \\ 6 & 7 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[370] \ 1^1 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 3 & 0 & 8 & 4 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[369] \ 1^{11} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 7 & 0 & 8 & 4 \\ 4 & 7 & 7 & 0 & 11 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 11 & 0 & 13 & 7 \\ 3 & 4 & 5 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[371] \ 1^{11} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 7 & 0 & 3 & 4 \\ 4 & 7 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 5 & 0 & 1 & 3 \\ 3 & 4 & 5 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[372] \ 1^{11} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 3 & 0 & 7 & 4 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 5 & 0 & 8 & 6 \\ 3 & 4 & 2 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[373] \ 1^{11} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -97, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 6 & 3 \\ 3 & 4 & 7 & 0 & 10 & 4 \\ 4 & 7 & 7 & 0 & 15 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 15 & 0 & 21 & 9 \\ 3 & 4 & 5 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[374] \ 1^{11} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 7 & 0 & 4 & 4 \\ 4 & 7 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 5 & 0 & 1 & 3 \\ 3 & 4 & 5 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[375] \ 1^{12} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 3 & 0 & 17 & 4 \\ 2 & 3 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 13 & 0 & 61 & 15 \\ 3 & 4 & 2 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[376] \ 1^{11} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 3 & 0 & 9 & 4 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 5 & 0 & 13 & 7 \\ 3 & 4 & 2 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[377] \ 1^{11} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 6 & 7 & 5 \\ 9 & 13 & 10 & 10 & 7 \\ 6 & 10 & 4 & 7 & 4 \\ 7 & 10 & 7 & 5 & 5 \\ 5 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[378] \ 1^{12} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 3 & 0 & 11 & 4 \\ 2 & 3 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 7 & 0 & 21 & 9 \\ 3 & 4 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 3 & 0 & 21 & 4 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 11 & 0 & 76 & 16 \\ 3 & 4 & 2 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[389] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} 1^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -206, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 3 & 0 & 12 & 4 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 8 & 0 & 26 & 10 \\ 3 & 4 & 2 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[390] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} 1^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -215, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 3 & 0 & 15 & 4 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 11 & 0 & 45 & 13 \\ 3 & 4 & 2 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[391] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} 1^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -224, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 6 & 16 & 5 \\ 9 & 13 & 10 & 24 & 7 \\ 6 & 10 & 4 & 16 & 4 \\ 16 & 24 & 16 & 38 & 12 \\ 5 & 7 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[392] \ 1^{\frac{1}{13}} 4^{\frac{1}{14}} 1^{\frac{1}{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -229, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 3 & 0 & 24 & 4 \\ 2 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 24 & 18 & 0 & 121 & 21 \\ 3 & 4 & 2 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[393] \ 1^{\frac{1}{11}} 4^{\frac{1}{11}} 1^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -254, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 7 & 0 & 12 & 4 \\ 4 & 7 & 7 & 0 & 16 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 16 & 0 & 26 & 10 \\ 3 & 4 & 5 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[394] \ 21^{\frac{1}{11}} 4^{\frac{1}{11}} 1^{\frac{1}{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -254, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 13 & 14 & 9 & 24 & 7 \\ 14 & 13 & 10 & 24 & 7 \\ 9 & 10 & 4 & 16 & 4 \\ 24 & 24 & 16 & 38 & 12 \\ 7 & 7 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[395] \ 1^{\frac{1}{13}} 4^{\frac{1}{12}} 1^{\frac{1}{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -265, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.73 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 3 & 0 & 18 & 4 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 12 & 0 & 61 & 15 \\ 3 & 4 & 2 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[396] \ 1^{\frac{1}{11}} 4^{\frac{1}{12}} 1^{\frac{1}{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -270, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 7 & 0 & 15 & 4 \\ 4 & 7 & 7 & 0 & 21 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 21 & 0 & 45 & 13 \\ 3 & 4 & 5 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[397] \ 1^{\frac{1}{13}} 4^{\frac{1}{13}} 1^{\frac{1}{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -286, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.78 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 3 & 0 & 21 & 4 \\ 2 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 15 & 0 & 88 & 18 \\ 3 & 4 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

- [398] $21^{111}41^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 10 & 7 & 7 & 5 \\ 10 & 13 & 10 & 10 & 7 \\ 7 & 10 & 4 & 7 & 4 \\ 7 & 10 & 7 & 5 & 5 \\ 5 & 7 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$
- [399] $1^{12}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -325, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 3 & 0 & 16 & 4 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 10 & 0 & 45 & 13 \\ 3 & 4 & 2 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$
- [400] $1^{12}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 3 & 0 & 25 & 4 \\ 2 & 3 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 25 & 19 & 0 & 132 & 22 \\ 3 & 4 & 2 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$
- [401] $1^{14}41^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 3 & 0 & 22 & 4 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 14 & 0 & 90 & 18 \\ 3 & 4 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$
- [402] $21^{111}41^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -382, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 10 & 7 & 16 & 5 \\ 10 & 13 & 10 & 24 & 7 \\ 7 & 10 & 4 & 16 & 4 \\ 16 & 24 & 16 & 38 & 12 \\ 5 & 7 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$
- [403] $1^{12}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -406, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 3 & 0 & 19 & 4 \\ 2 & 3 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 19 & 13 & 0 & 68 & 16 \\ 3 & 4 & 2 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$
- [404] $1^{12}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -409, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 3 & 0 & 22 & 4 \\ 2 & 3 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 16 & 0 & 97 & 19 \\ 3 & 4 & 2 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$
- [405] $1^{13}41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 3 & 0 & 20 & 4 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 20 & 12 & 0 & 70 & 16 \\ 3 & 4 & 2 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$
- [406] $1^{14}41^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -439, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 3 & 0 & 25 & 4 \\ 2 & 3 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 25 & 17 & 0 & 121 & 21 \\ 3 & 4 & 2 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$
- [407] $1^{14}41^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -83, \det = -454, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 3 & 0 & 28 & 4 \\ 2 & 3 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 28 & 20 & 0 & 158 & 24 \\ 3 & 4 & 2 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[408] \ 1^{14} 4 \ 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -509, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.73 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 3 & 0 & 24 & 4 \\ 2 & 3 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 14 & 0 & 101 & 19 \\ 3 & 4 & 2 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[409] \ 1^{13} 4 \ 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -579, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.72 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 3 & 0 & 23 & 4 \\ 2 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 23 & 15 & 0 & 97 & 19 \\ 3 & 4 & 2 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[410] \ 1^{13} 4 \ 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -651, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 3 & 0 & 29 & 4 \\ 2 & 3 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 29 & 21 & 0 & 169 & 25 \\ 3 & 4 & 2 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[411] \ 1^{13} 4 \ 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -654, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.74 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 3 & 0 & 26 & 4 \\ 2 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 18 & 0 & 130 & 22 \\ 3 & 4 & 2 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[412] \ 1^{14} 4 \ 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -734, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.72 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 3 & 0 & 27 & 4 \\ 2 & 3 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 27 & 17 & 0 & 132 & 22 \\ 3 & 4 & 2 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[413] \ 1^{14} 4 \ 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -881, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.73 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 3 & 0 & 30 & 4 \\ 2 & 3 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 30 & 20 & 0 & 169 & 25 \\ 3 & 4 & 2 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[414] \ 1^{14} 4 \ 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -950, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.75 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 19 & 3 \\ 3 & 4 & 3 & 0 & 33 & 4 \\ 2 & 3 & 0 & 0 & 23 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 33 & 23 & 0 & 212 & 28 \\ 3 & 4 & 2 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [415] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -62, \ \det = 12, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 4 & 1 & 2 & 5 \\ 4 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 0 & 2 \\ 5 & 3 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

4.1124 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{e}_6$

$$T = 9 \quad [0] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -115, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 9 & 5 & 7 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[1] 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_6 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -271, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[2] 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_6 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 1 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 8 & 16 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{c}_6 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -931, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

4.1125 NHC(-2)(-3) + $\textcolor{red}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4$

$$T = 9 \quad [0] 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 9 & 5 & 7 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[1] 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 1 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 8 & 16 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

4.1126 NHC(-2)(-3) + $\textcolor{red}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{orange}{so}_8$

$$T = 9 \quad [0] 1^{111} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{orange}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[1] 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 28 & 1 & 5 \\ 12 & 25 & 16 & 60 & 0 & 9 \\ 8 & 16 & 9 & 36 & 0 & 5 \\ 28 & 60 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 1 & 5 \\ 11 & 20 & 17 & 0 & 0 & 8 \\ 9 & 17 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 5 & 8 & 6 & -2 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 10 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 10 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[4] 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 10 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[5] 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -133, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 0 & 13 & 7 \\ 5 & 8 & -1 & -2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[6] 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 10 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[7] 1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[8] 1^{11} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 8 & 5 \\ 11 & 20 & 0 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 0 & 0 & 10 & 6 \\ 5 & 8 & -1 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

- [9] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 8 & 19 & 1 & 5 \\ 12 & 25 & 16 & 40 & 0 & 9 \\ 8 & 16 & 9 & 24 & 0 & 5 \\ 19 & 40 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 8 & 20 & 1 & 5 \\ 12 & 25 & 16 & 40 & 0 & 9 \\ 8 & 16 & 9 & 24 & 0 & 5 \\ 20 & 40 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 8 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 10 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 10 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$
- [13] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 9 & 5 \\ 11 & 20 & 17 & 0 & 16 & 8 \\ 9 & 17 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 12 & 0 & 10 & 6 \\ 5 & 8 & 6 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$
- [14] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 26 & 5 \\ 11 & 20 & 0 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 26 & 50 & 0 & 0 & 118 & 20 \\ 5 & 8 & -1 & -2 & 20 & 1 \end{pmatrix}_{b_0}$$
- [15] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 17 & 5 \\ 11 & 20 & 0 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 33 & 0 & 0 & 49 & 13 \\ 5 & 8 & -1 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$
- [16] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 8 & 30 & 1 & 5 \\ 12 & 25 & 16 & 60 & 0 & 9 \\ 8 & 16 & 9 & 36 & 0 & 5 \\ 30 & 60 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$
- [17] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_2 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$
 $T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 8 & 29 & 1 & 5 \\ 12 & 25 & 16 & 60 & 0 & 9 \\ 8 & 16 & 9 & 36 & 0 & 5 \\ 29 & 60 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{114} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -91, \quad \det = -330, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 38 & 1 & 5 \\ 12 & 25 & 16 & 80 & 0 & 9 \\ 8 & 16 & 9 & 48 & 0 & 5 \\ 38 & 80 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{114} 3^1 1^{41} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -91, \quad \det = -442, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 40 & 1 & 5 \\ 12 & 25 & 16 & 80 & 0 & 9 \\ 8 & 16 & 9 & 48 & 0 & 5 \\ 40 & 80 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{114} 3^1 1^{31} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -91, \quad \det = -510, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 39 & 1 & 5 \\ 12 & 25 & 16 & 80 & 0 & 9 \\ 8 & 16 & 9 & 48 & 0 & 5 \\ 39 & 80 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -91, \quad \det = -539, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 17 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -91, \quad \det = -539, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 19 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 19 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -91, \quad \det = -735, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 18 & 5 \\ 11 & 20 & 17 & 0 & 33 & 8 \\ 9 & 17 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 33 & 27 & 0 & 49 & 13 \\ 5 & 8 & 6 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [24] \quad 1^{11} 4^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 165, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 6 & 9 & 1 & 5 \\ 9 & 13 & 10 & 14 & 0 & 7 \\ 6 & 10 & 4 & 9 & 0 & 4 \\ 9 & 14 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 7 & 4 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$4.1127 \quad \text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -62, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 3 & 0 & 4 & 6 & 4 \\ 3 & 5 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & 2 & -2 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 0 & 5 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 3 & 0 & 4 & 6 & 4 \\ 3 & 5 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & 2 & -2 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 0 & 5 & 4 \\ 2 & 4 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 3 & 0 & 0 & 4 & 4 \\ 3 & 4 & 2 & -2 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

4.1128 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \ 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 52, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 & 5 \\ 2 & 4 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 2 & 2 \\ 3 & 5 & 2 & -2 & 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 69, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 & 3 \\ 3 & 4 & 8 & 0 & 3 & 4 & 4 \\ 4 & 8 & 9 & 0 & 4 & 5 & 5 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 2 & 2 \\ 3 & 4 & 5 & -2 & 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 85, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 6 & 5 & 5 \\ 11 & 20 & 17 & 11 & 8 & 8 \\ 10 & 17 & 12 & 9 & 6 & 6 \\ 6 & 11 & 9 & 4 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 88, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.32 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 5 & 5 \\ 11 & 20 & 0 & 11 & 8 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 6 & 11 & 0 & 4 & 4 & 4 \\ 5 & 8 & -1 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 89, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.37 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 3 & 4 & 3 & -2 & 2 & 1 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 96, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 6 & 5 \\ 11 & 20 & 17 & 11 & 8 \\ 9 & 17 & 12 & 9 & 6 \\ 6 & 11 & 9 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 12 & 8 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 12 & 17 & 12 & 9 & 6 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[8] \quad 21^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 121, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.11 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 15 & 9 & 7 \\ 17 & 20 & 17 & 11 & 8 \\ 15 & 17 & 12 & 9 & 6 \\ 9 & 11 & 9 & 4 & 4 \\ 7 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 9 & 0 & 3 & 4 \\ 5 & 9 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 6 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 21^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 16 & 12 & 8 & 6 \\ 16 & 20 & 17 & 11 & 8 \\ 12 & 17 & 12 & 9 & 6 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 13 & 0 & 3 & 4 \\ 7 & 13 & 27 & 0 & 7 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 7 & 0 & 0 & 2 \\ 3 & 4 & 9 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 230, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 17 & 0 & 3 & 4 \\ 9 & 17 & 48 & 0 & 9 & 12 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 9 & 0 & 0 & 2 \\ 3 & 4 & 12 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [13] \quad 1^{15} 5^{13} 13^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -21, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

4.1129 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 9 & 6 & 1 & 5 \\ 11 & 20 & 17 & 11 & 0 & 8 \\ 9 & 17 & 12 & 9 & 0 & 6 \\ 6 & 11 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

4.1130 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 6 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 6 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

4.1131 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 12 \quad [0] \quad 1^{11} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 1 & 2 \\ 2 & 4 & 0 & 2 & 2 & 1 \\ 3 & 4 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 3 & 7 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 6 \\ 2 & 3 & 0 & 3 & 1 & 5 \\ 4 & 7 & 0 & 6 & 5 & 9 \\ 3 & 4 & -1 & 2 & 3 & 7 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{13} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 3 & 15 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 10 & 2 \\ 2 & 3 & 0 & 3 & 1 & 9 & 3 \\ 8 & 15 & 0 & 10 & 9 & 41 & 13 \\ 3 & 4 & -1 & 2 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{12} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 50, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 0 & 3 & 3 & 11 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 8 & 2 \\ 2 & 3 & 0 & 3 & 1 & 7 & 3 \\ 6 & 11 & 0 & 8 & 7 & 22 & 10 \\ 3 & 4 & -1 & 2 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

4.1132 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 24 & 5 \\ 11 & 20 & 0 & 45 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 45 & 0 & 100 & 18 \\ 5 & 8 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.62 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 8 & 0 & 25 & 9 \\ 3 & 5 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 7 & 5 \\ 11 & 20 & 0 & 11 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 11 & 0 & 4 & 4 \\ 5 & 8 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 37, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 5 & 0 & 19 & 4 \\ 3 & 5 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 19 & 15 & 0 & 64 & 14 \\ 3 & 4 & 3 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 42, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 10 & 3 \\ 3 & 4 & 9 & 0 & 19 & 4 \\ 5 & 9 & 12 & 0 & 29 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 19 & 29 & 0 & 64 & 14 \\ 3 & 4 & 6 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 45, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.56 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 48, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.25 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 5 & 0 & 12 & 6 \\ 3 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 49, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 7 & 5 \\ 11 & 20 & 17 & 11 & 8 \\ 9 & 17 & 12 & 9 & 6 \\ 7 & 11 & 9 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 50, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 7 & 5 \\ 11 & 20 & 17 & 11 & 8 \\ 10 & 17 & 12 & 9 & 6 \\ 7 & 11 & 9 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 50, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.24 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 5 & 3 \\ 3 & 4 & 13 & 0 & 7 & 4 \\ 7 & 13 & 27 & 0 & 16 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 16 & 0 & 7 & 5 \\ 3 & 4 & 9 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 25 & 5 \\ 11 & 20 & 0 & 45 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 25 & 45 & 0 & 100 & 18 \\ 5 & 8 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{12} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 8 & 0 & 7 & 4 \\ 4 & 8 & 9 & 0 & 10 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 10 & 0 & 7 & 5 \\ 3 & 4 & 5 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 12 & 9 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 12 & 17 & 12 & 9 & 6 \\ 9 & 11 & 9 & 4 & 4 \\ 6 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 16 & 5 \\ 11 & 20 & 0 & 28 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 28 & 0 & 37 & 11 \\ 5 & 8 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 5 & 0 & 11 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 11 & 9 & 0 & 20 & 8 \\ 3 & 4 & 3 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 8 & 0 & 12 & 4 \\ 4 & 8 & 9 & 0 & 16 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 16 & 0 & 25 & 9 \\ 3 & 4 & 5 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 5 & 0 & 12 & 6 \\ 3 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 15 & 5 \\ 11 & 20 & 0 & 28 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 28 & 0 & 37 & 11 \\ 5 & 8 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 7 & 3 \\ 4 & 9 & 4 & 0 & 16 & 5 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 16 & 8 & 0 & 25 & 9 \\ 3 & 5 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 8 & 0 & 25 & 9 \\ 3 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 278, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 13 & 0 & 7 & 4 \\ 7 & 13 & 27 & 0 & 16 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 16 & 0 & 7 & 5 \\ 3 & 4 & 9 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{11}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 11 & 0 & 55 & 13 \\ 3 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{11}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 15 & 4 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 15 & 11 & 0 & 39 & 11 \\ 3 & 4 & 2 & -2 & 11 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 8 & 0 & 25 & 9 \\ 3 & 4 & 2 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{11}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 22 & 14 & 0 & 97 & 17 \\ 3 & 4 & 2 & -2 & 17 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{11}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 0 & 11 & 4 \\ 2 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 11 & 8 & 0 & 20 & 8 \\ 3 & 4 & 2 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{11}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 0 & 19 & 4 \\ 2 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 19 & 14 & 0 & 64 & 14 \\ 3 & 4 & 2 & -2 & 14 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{11}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 17 & 11 & 0 & 55 & 13 \\ 3 & 4 & 2 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{11}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 464, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 11 & 0 & 44 & 12 \\ 3 & 4 & 2 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{11}3^11^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 21 & 3 \\ 3 & 4 & 4 & 0 & 35 & 4 \\ 2 & 4 & 0 & 0 & 23 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 35 & 23 & 0 & 231 & 27 \\ 3 & 4 & 2 & -2 & 27 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{11} 1^{14} 3^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 1616, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.19 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 0 & 29 & 4 \\ 2 & 4 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 29 & 20 & 0 & 148 & 22 \\ 3 & 4 & 2 & -2 & 22 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{11} 1^{14} 3^1 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 1760, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 22 & 3 \\ 3 & 4 & 4 & 0 & 39 & 4 \\ 2 & 4 & 0 & 0 & 26 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 39 & 26 & 0 & 280 & 30 \\ 3 & 4 & 2 & -2 & 30 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{11} 1^{14} 3^1 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 1952, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 0 & 34 & 4 \\ 2 & 4 & 0 & 0 & 23 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 34 & 23 & 0 & 208 & 26 \\ 3 & 4 & 2 & -2 & 26 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{11} 1^{14} 3^1 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 2208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 23 & 3 \\ 3 & 4 & 4 & 0 & 39 & 4 \\ 2 & 4 & 0 & 0 & 26 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 23 & 39 & 26 & 0 & 280 & 30 \\ 3 & 4 & 2 & -2 & 30 & 1 \end{pmatrix}_{b_0}$$

$T = 11$ [75] $1^{11} 5^1 13^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -61, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 4 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & -1 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[76] $1^{12} 5^1 13^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -61, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 3 & 14 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 11 & 2 \\ 8 & 14 & 0 & 11 & 39 & 13 \\ 3 & 4 & -1 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[77] $1^{13} 5^1 13^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -133, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 13 & 2 \\ 10 & 18 & 0 & 13 & 62 & 16 \\ 3 & 4 & -1 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

4.1133 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \text{su2n3mix} + \mathfrak{so}_8$

$T = 9$ [0] $1^{11} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 7 & 5 \\ 11 & 20 & 17 & 0 & 11 & 8 \\ 9 & 17 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 11 & 9 & 0 & 4 & 4 \\ 5 & 8 & 6 & -2 & 4 & 1 \end{pmatrix}_{b_0}$$

4.1134 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 8 \quad [0] 1^{11} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 17.60 \rightarrow 18, T_{\max} = 20, T \in [18, 20]$$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 18 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & -4 \end{pmatrix}_{b_0}$$

$$[2] 1^{11} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 1 & 3 \\ 3 & 4 & 10 & 0 & 0 & 4 \\ 6 & 10 & 21 & 0 & 0 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -2 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] 1^{13} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 12 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 9 & 0 & 24 & 8 \\ 3 & 4 & 3 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 0 & 4 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 5 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] 1^{11} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 19 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[5] 1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 5 & 0 & 8 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[4] 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 12 & 8 & 11 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 11 & 20 & 12 & 12 & 6 \\ 5 & 9 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[7] 1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 4 & 3 \\ 3 & 4 & 0 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 0 & 0 & 9 & 5 \\ 3 & 4 & -1 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[8] 1^{111} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 9 & 0 & 36 & 10 \\ 3 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[9] 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

- [10] $1^{14} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 11 & 3 \\ 3 & 4 & 0 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 21 & 0 & 0 & 75 & 15 \\ 3 & 4 & -1 & -2 & 15 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 9 & 20 & 12 & 12 & 6 \\ 5 & 9 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{11} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$
- [13] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 3 & 0 & 3 & 3 \\ 3 & 4 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [14] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 12 & 8 & 10 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 10 & 20 & 12 & 12 & 6 \\ 5 & 9 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$
- [15] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 0 & 0 & 3 & 3 \\ 3 & 4 & -1 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [16] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 17 & 0 & 12 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$
- [17] $1^{11} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 3 & 4 & -1 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [18] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 6 & 0 & 12 & 6 \\ 3 & 5 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$
- [19] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 8 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$
- [20] $1^{11} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{so}_8 + \text{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 12 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.28 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 2 & -2 & 3 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 15 & 12 & 1 & 6 \\ 15 & 20 & 17 & 0 & 8 \\ 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{13} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 9 & 3 \\ 3 & 4 & 0 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 0 & 48 & 12 \\ 3 & 4 & -1 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 6 & 0 & 12 & 6 \\ 3 & 5 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{11} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 3 \\ 3 & 4 & 0 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 0 & 0 & 36 & 10 \\ 3 & 4 & -1 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 0 & 0 & 12 & 6 \\ 3 & 4 & -1 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{111} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 0 & 3 & 3 \\ 3 & 5 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 125, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 7 & 3 \\ 3 & 4 & 0 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 0 & 0 & 27 & 9 \\ 3 & 4 & -1 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 10 & 3 \\ 3 & 4 & 0 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 0 & 0 & 57 & 13 \\ 3 & 4 & -1 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{14} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 14 & 3 \\ 3 & 4 & 0 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 0 & 0 & 117 & 19 \\ 3 & 4 & -1 & -2 & 19 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 12 & 3 \\ 3 & 4 & 0 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 0 & 0 & 84 & 16 \\ 3 & 4 & -1 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 9 & 0 & 36 & 10 \\ 3 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[39] $21^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 13 & 20 & 12 & 15 & 7 \\ 20 & 25 & 16 & 20 & 9 \\ 12 & 16 & 9 & 12 & 5 \\ 15 & 20 & 12 & 12 & 6 \\ 7 & 9 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{111} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 14 & 9 & 0 & 36 & 10 \\ 3 & 4 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [42] $1 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9, T_{\max} = 13, \Delta = -92, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 6 & 6 & 5 \\ 12 & 25 & 12 & 12 & 9 \\ 6 & 12 & 4 & 7 & 4 \\ 6 & 12 & 7 & 4 & 4 \\ 5 & 9 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{11} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 0 & 3 \\ 1 & 4 & 0 & 0 & 0 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & -1 & -2 & 2 \end{pmatrix}_{b_0}$$

[44] $1^{11} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 8 & 8 & 7 & 1 & 4 \\ 8 & 13 & 16 & 12 & 0 & 7 \\ 8 & 16 & 12 & 12 & 0 & 6 \\ 7 & 12 & 12 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 7 & 6 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[46] $1^{11} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 1 & 1 & 5 & 3 \\ 4 & 10 & 1 & 0 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 0 & 0 & 12 & 6 \\ 3 & 6 & -1 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[45] $1^{111} 31^{11} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & 0 & 3 & 3 & 2 \\ 2 & 3 & 0 & 3 & 2 & 3 & 2 \\ 3 & 3 & 0 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & 2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{11} 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9, T_{\max} = 13, \Delta = -92, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 12 & 6 & 1 & 5 \\ 12 & 25 & 12 & 0 & 9 \\ 6 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

[48] $21^{11} 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9, T_{\max} = 13, \Delta = -92, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 13 & 20 & 9 & 9 & 7 \\ 20 & 25 & 12 & 12 & 9 \\ 9 & 12 & 4 & 7 & 4 \\ 9 & 12 & 7 & 4 & 4 \\ 7 & 9 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

[49] $1^{11} 41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9, T_{\max} = 13, \Delta = -92, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{11} 31^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9, T_{\max} = 12, \Delta = -80, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 4 \\ 2 & 7 & 3 & 3 & 0 & 3 & 3 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 3 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{11} 3^1 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 9 & 6 & 0 & 0 & 12 & 6 \\ 2 & 4 & 2 & -1 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{11} 4^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 4 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{11} 3^1 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -105, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & 0 & 3 & 3 & 2 \\ 2 & 3 & 0 & 3 & 2 & 3 & 2 \\ 2 & 3 & 0 & 3 & 3 & 0 & 2 \\ 2 & 3 & -1 & 2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{11} 4^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 3 & 0 & 7 & 4 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 5 & 0 & 7 & 5 \\ 3 & 4 & 2 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{11} 3^1 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 3 & 0 & 6 & 4 \\ 1 & 4 & 0 & 3 & 0 & 3 & 2 \\ 1 & 3 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 6 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

4.1135 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 1 & 5 \\ 12 & 25 & 16 & 0 & 0 & 9 \\ 8 & 16 & 9 & 0 & 0 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & -1 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 10 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 10 & 20 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{11} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -1 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] 1^{11} 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.20 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 12 & 6 & 6 & 1 & 5 \\ 12 & 25 & 12 & 12 & 0 & 9 \\ 6 & 12 & 4 & 7 & 0 & 4 \\ 6 & 12 & 7 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 4 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

4.1136 NHC(-2)(-3) + $\textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3$

$$T = 9 \quad [0] 1^{11} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 5 & 3 \\ 4 & 9 & 4 & 0 & 4 & 12 & 5 \\ 2 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 12 & 6 & 0 & 6 & 12 & 6 \\ 3 & 5 & 2 & -2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] 1^{11} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 4 & 3 \\ 4 & 9 & 4 & 0 & 4 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 4 & 8 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 2 & -2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] 1^{11} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 10 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 20 & 12 & 0 & 12 & 6 \\ 5 & 9 & 5 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[3] 1^{11} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -1 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4] 1^{11} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 3 & 5 & 0 & 0 & 3 & 3 & 3 \\ 2 & 4 & 0 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] 1^{11} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 3 & 2 \\ 2 & 4 & 0 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & -2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -237, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{111} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 0 & 4 & 14 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 9 & 2 \\ 8 & 14 & 0 & 0 & 9 & 36 & 10 \\ 3 & 4 & -1 & -2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -261, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & -2 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 0 & 0 & 0 & 6 & 2 \\ 5 & 9 & 0 & 0 & 6 & 12 & 6 \\ 3 & 4 & -1 & -2 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

4.1137 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8 + (-3) \rightarrow (-2)$

$T = 8 \quad [0] 1^{11} 3 1^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3 1^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 13 & 20 & 12 & 1 & 7 \\ 20 & 25 & 16 & 0 & 9 \\ 12 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3 1^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 16 & 12 & 1 & 6 \\ 16 & 20 & 17 & 0 & 8 \\ 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$T = 9 \quad [3] 1^{11} 4^{21} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 13 & 14 & 9 & 1 & 7 \\ 14 & 13 & 10 & 0 & 7 \\ 9 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{2^1 1^{11}} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{so}}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.18 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 5 & 10 & 7 & 1 & 5 \\ 10 & 13 & 10 & 0 & 7 \\ 7 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

4.1138 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2$

$T = 7 \quad [0] 1^{12^4} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 5 & 8 & 4 \\ 5 & 5 & 3 & 6 & 3 \\ 8 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{113} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 7 & 9 & 12 & 27 & 6 \\ 10 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{114} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 5 & 9 & 12 & 24 & 6 \\ 9 & 16 & 24 & 46 & 12 \\ 3 & 4 & 6 & 12 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{13} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 5 & 5 & 6 & 4 & 4 \\ 3 & 4 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -10, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 5 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -10, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.60 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{12} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 5 & 8 & 4 \\ 4 & 5 & 3 & 6 & 3 \\ 7 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 4 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 4 & 5 & 3 & 3 & 3 \\ 2 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 4 & 4 & 0 & 3 & 2 \\ 5 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{121} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 7 & 8 & 9 & 12 & 5 \\ 9 & 9 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{13} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -13, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 13 & 17 & 48 & 15 & 12 \\ 3 & 5 & 15 & 4 & 4 \\ 3 & 4 & 12 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -13, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 5 & 5 & 3 & 6 & 3 \\ 3 & 5 & 6 & 4 & 4 \\ 3 & 4 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 8 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 5 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 8 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{122} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 9 & 13 & 27 & 18 & 9 \\ 7 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 7 & 9 & 12 & 15 & 6 \\ 9 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 4 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{112} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 4 & 5 & 3 & 6 & 3 \\ 5 & 5 & 6 & 4 & 4 \\ 3 & 4 & 3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 5 & 8 & 4 \\ 4 & 5 & 3 & 6 & 3 \\ 6 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 8 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{12} 3^1 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 3 & 12 & 3 \\ 7 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 21^{111} 3^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 12 & 6 & 4 \\ 5 & 4 & 13 & 5 & 4 \\ 12 & 13 & 27 & 12 & 9 \\ 6 & 5 & 12 & 4 & 4 \\ 4 & 4 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[27] \ 21^{11} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 4 & 5 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 6 & 5 & 3 & 4 & 4 \\ 4 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{111} 3^1 1^{23} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 14 & 19 & 12 & 73 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{113} 3^1 1^{21} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 10 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{11} 3^1 1^{22} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 5 & 5 & 3 & 3 & 3 \\ 4 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{122} 3^1 1^{24} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 6 & 8 & 9 & 12 & 5 \\ 8 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{11} 3^1 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 8 & 9 & 12 & 9 & 6 \\ 3 & 5 & 9 & 4 & 4 \\ 3 & 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{12} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -18, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.44 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 4 & 8 & 0 & 10 & 6 \\ 3 & 4 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{122} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -18, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.7$$

$$T_{\min} = 10.44 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 6 & 8 & 9 & 12 & 5 \\ 4 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{114} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -18, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.44 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 7 & 9 & 12 & 24 & 6 \\ 12 & 16 & 24 & 46 & 12 \\ 3 & 4 & 6 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[30] \ 1^1 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -18, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.78 \rightarrow 10, \ T_{\max} = 12, \ T \in [10, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 5 & 0 & 4 & 4 \\ 3 & 4 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -18, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.44 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 8 & 9 & 12 & 12 & 6 \\ 8 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{111} 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -19, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 17 & 24 & 15 & 121 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -19, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 4 & 4 & 0 & 3 & 2 \\ 4 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[41] \ 1^1 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -19, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 4 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{111} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -19, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 12 & 19 & 12 & 73 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -19, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 9 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[44] \ 21^{14} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -93, \ \det = -19, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.26 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 27 & 4 \\ 5 & 4 & 4 & 24 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 27 & 24 & 15 & 121 & 19 \\ 4 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[45] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 4 & 5 & 3 & 3 & 3 \\ 4 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[46] $1^{11} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 4 & 4 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[47] $1^{11} 3^1 1^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 0 & 9 & 2 \\ 7 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[48] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[39] $21^1 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 4 & 4 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 1 & 3 \\ 4 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{11} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 4 & 4 & 0 & 3 & 2 \\ 2 & 4 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[49] $1^{11} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & 0 & 6 & 2 \\ 9 & 16 & 6 & 22 & 8 \\ 3 & 5 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 7 & 9 & 12 & 9 & 6 \\ 5 & 5 & 9 & 4 & 4 \\ 3 & 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{11} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 4 & 4 & 0 & 1 & 3 \\ 3 & 4 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$

[52] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 6 & 9 & 12 & 6 & 6 \\ 4 & 4 & 6 & 1 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[54] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 7 & 9 & 12 & 12 & 6 \\ 5 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[55] $1^{114} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 9 & 16 & 4 \\ 6 & 9 & 12 & 24 & 6 \\ 11 & 16 & 24 & 46 & 12 \\ 3 & 4 & 6 & 12 & 2 \end{pmatrix}_{b_0}$$

[56] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 3 & 5 & 3 & 3 & 3 \\ 2 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[57] $1^{121} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 5 & 5 & 3 & 6 & 3 \\ 4 & 5 & 6 & 4 & 4 \\ 3 & 4 & 3 & 4 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[59] $1^{13} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 11 & 17 & 48 & 15 & 12 \\ 4 & 5 & 15 & 4 & 4 \\ 3 & 4 & 12 & 4 & 1 \end{pmatrix}_{b_0}$$

[61] $1^{121} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 8 & 10 & 21 & 0 & 7 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 7 & 0 & 1 \end{pmatrix}_{b_0}$$

[62] $1^{11} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 7 & 10 & 21 & 0 & 7 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 7 & 0 & 1 \end{pmatrix}_{b_0}$$

[60] $21^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 4 & 5 & 8 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 6 & 8 & 6 & 10 & 6 \\ 3 & 4 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{11} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 5 & 5 & 3 & 9 & 3 \\ 6 & 9 & 9 & 13 & 7 \\ 3 & 4 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[64] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 6 & 9 & 12 & 18 & 6 \\ 9 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{\mathbf{1}^{\mathbf{11}}} 3^{\mathbf{1}^{\mathbf{1}}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 12$, $T \in [10, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 4 & 5 & 0 & 4 & 4 \\ 3 & 4 & -1 & 4 & 3 \end{pmatrix}_{b_0}$

[66] $1^{\mathbf{1}^{\mathbf{11}}} 3^{\mathbf{1}^{\mathbf{1}}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 12$, $T \in [10, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0}$

[67] $1^{\mathbf{1}^{\mathbf{11}}} 3^{\mathbf{1}^{\mathbf{1}}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 6 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 2 & 4 & 0 & 6 & 2 \\ 6 & 12 & 6 & 13 & 7 \\ 3 & 5 & 2 & 7 & 1 \end{pmatrix}_{b_0}$

[68] $1^{\mathbf{1}^{\mathbf{11}}} 3^{\mathbf{1}^{\mathbf{1}}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 12$, $T \in [10, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 4 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$

[69] $1^{\mathbf{1}^{\mathbf{11}}} 3^{\mathbf{1}^{\mathbf{1}}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -25$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 3 & 5 & 6 & 4 & 4 \\ 3 & 4 & 3 & 4 & 1 \end{pmatrix}_{b_0}$

[70] $1^{\mathbf{1}^{\mathbf{11}}} 3^{\mathbf{1}^{\mathbf{1}}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -25$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 5 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 0 & 1 \end{pmatrix}_{b_0}$

[71] $1^{\mathbf{1}^{\mathbf{11}}} 3^{\mathbf{1}^{\mathbf{1}}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 4 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$

[72] $1^{\mathbf{1}^{\mathbf{11}}} 3^{\mathbf{1}^{\mathbf{11}}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 4 & 5 & 3 & 6 & 3 \\ 3 & 5 & 6 & 4 & 4 \\ 3 & 4 & 3 & 4 & 2 \end{pmatrix}_{b_0}$

[73] $21^{\mathbf{1}^{\mathbf{11}}} 3^{\mathbf{1}^{\mathbf{12}}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 9 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$

[74] $1^{\mathbf{1}^{\mathbf{11}}} 3^{\mathbf{1}^{\mathbf{13}}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 15 & 0 & 52 & 12 \\ 3 & 4 & -1 & 12 & 1 \end{pmatrix}_{b_0}$

[75] $1^{\mathbf{1}^{\mathbf{21}}} 3^{\mathbf{1}^{\mathbf{12}}} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -29$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.2$
 $T_{\min} = 10.59 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 5 & 8 & 9 & 6 & 5 \\ 4 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

[76] $1^1 3^1 1^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 7 & 13 & 27 & 27 & 9 \\ 6 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 11 & 13 & 27 & 9 & 9 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[78] $1^1 13^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & 0 & 3 & 2 \\ 3 & 8 & 3 & 4 & 4 \\ 3 & 5 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{121} 3^1 1^3 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 7 & 8 & 9 & 6 & 5 \\ 2 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

[84] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 8 & 9 & 12 & 12 & 6 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{14} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 4 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 15 & 21 & 75 & 18 & 15 \\ 4 & 5 & 18 & 4 & 4 \\ 3 & 4 & 15 & 4 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{112} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 10 & 0 & 22 & 8 \\ 3 & 4 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 6 & 8 & 9 & 12 & 5 \\ 8 & 9 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

[86] $1^{11} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 5 & 9 & 12 & 9 & 6 \\ 4 & 5 & 9 & 4 & 4 \\ 3 & 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[87] $1^1 3^1 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 5 & 9 & 12 & 12 & 6 \\ 4 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[88] $1^1 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

- [89] $21^{11431^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 4 & 5 & 6 & 4 & 4 \\ 3 & 4 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$
- [90] $1^1 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 10 & 0 & 22 & 8 \\ 3 & 4 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$
- [80] $1^{11} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 8 & 9 & 12 & 9 & 6 \\ 5 & 5 & 9 & 4 & 4 \\ 3 & 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$
- [81] $1^{11} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 9 & 9 & 12 & 9 & 6 \\ 4 & 5 & 9 & 4 & 4 \\ 3 & 4 & 6 & 4 & 1 \end{pmatrix}_{b_0}$$
- [92] $1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 5 & 8 & 9 & 12 & 5 \\ 7 & 9 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$
- [94] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 7 & 9 & 12 & 18 & 6 \\ 10 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$
- [91] $1^{113} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 6 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$
- [93] $1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 10 & 13 & 27 & 27 & 9 \\ 8 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$
- [95] $1^1 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 2 & 4 & 0 & 6 & 2 \\ 5 & 12 & 6 & 13 & 7 \\ 3 & 5 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$
- [96] $1^{111} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & 0 & 3 & 2 \\ 4 & 8 & 3 & 4 & 4 \\ 3 & 5 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$
- [97] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 4 & 0 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[98] $21^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 8 & 9 & 3 & 4 \\ 8 & 9 & 12 & 4 & 5 \\ 9 & 12 & 12 & 6 & 6 \\ 3 & 4 & 6 & 1 & 3 \\ 4 & 5 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

[99] $1^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 5 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 5 & 0 & 2 \end{pmatrix}_{b_0}$$

[100] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 5 & 9 & 12 & 18 & 6 \\ 7 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[101] $1^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 7 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 5 & 0 & 2 \end{pmatrix}_{b_0}$$

[106] $1^{12}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 8 & 13 & 27 & 12 & 9 \\ 4 & 5 & 12 & 4 & 4 \\ 3 & 4 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

[105] $1^{11}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 3 & 9 & 3 \\ 9 & 9 & 9 & 13 & 7 \\ 3 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[104] $1^{11}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 5 & 5 & 4 \\ 4 & 5 & 3 & 6 & 3 \\ 4 & 5 & 6 & 4 & 4 \\ 3 & 4 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

[107] $1^{131}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 11 & 17 & 48 & 12 & 12 \\ 4 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[108] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 4 & 4 & 0 & 3 & 2 \\ 4 & 4 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[109] $1^{111}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 15 & 24 & 15 & 121 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[110] $1^{111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 11 & 14 & 36 & 9 & 10 \\ 4 & 4 & 9 & 1 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[103] $1^{111} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 2 & 4 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[102] $1^{111} 1^{12} 3^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 10 & 14 & 36 & 9 & 10 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[111] $1^{112} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 4 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{14} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 14 & 21 & 75 & 18 & 15 \\ 3 & 5 & 18 & 4 & 4 \\ 3 & 4 & 15 & 4 & 2 \end{pmatrix}_{b_0}$$

[114] $1^{112} 3^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 13 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[115] $1^{14} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 12 & 21 & 75 & 18 & 15 \\ 3 & 5 & 18 & 4 & 4 \\ 3 & 4 & 15 & 4 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{112} 3^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 8 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[117] $1^{113} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 8 & 9 & 12 & 18 & 6 \\ 10 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[118] $1^{111} 3^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 9 & 17 & 48 & 15 & 12 \\ 3 & 5 & 15 & 4 & 4 \\ 3 & 4 & 12 & 4 & 1 \end{pmatrix}_{b_0}$$

[119] $1^{121} 3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 4 & 8 & 9 & 6 & 5 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[120] $1^{121} 3^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 8 & 8 & 9 & 6 & 5 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[116] $1^{11} 3^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 8 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

[121] $1^{11} 3^{13} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 17 & 24 & 15 & 121 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[122] $1^{11} 3^{12} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 7 & 9 & 12 & 6 & 6 \\ 2 & 4 & 6 & 1 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[123] $1^{11} 3^{12} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 4 & 4 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[124] $1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 2 & 4 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[125] $1^{12} 3^{123} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 9 & 13 & 27 & 21 & 9 \\ 8 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[126] $21^{14} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 15 & 3 & 3 \\ 4 & 4 & 21 & 5 & 4 \\ 15 & 21 & 75 & 18 & 15 \\ 3 & 5 & 18 & 4 & 4 \\ 3 & 4 & 15 & 4 & 1 \end{pmatrix}_{b_0}$$

[127] $1^{11} 3^{13} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 8 & 9 & 9 & 13 & 7 \\ 3 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[131] $1^{11} 3^{121} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 4 & 8 & 9 & 12 & 5 \\ 5 & 9 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[130] $1^{11} 3^{12} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 7 & 9 & 12 & 15 & 6 \\ 5 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[128] $21^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 5 & 4 & 4 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[129] $1^{13}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 7 & 12 & 24 & 0 & 8 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 8 & 0 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{12}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 10 & 13 & 27 & 12 & 9 \\ 3 & 5 & 12 & 4 & 4 \\ 3 & 4 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

[133] $1^1 13^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 4 & 8 & 9 & 6 & 5 \\ 2 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -42, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 5 & 9 & 12 & 6 & 6 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[135] $1^{12}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -42, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 8 & 0 & 4 \\ 6 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 5 & 0 & 2 \end{pmatrix}_{b_0}$$

[136] $1^{12}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -42, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 5 & 9 & 12 & 21 & 6 \\ 10 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 3 & 9 & 3 \\ 5 & 9 & 9 & 13 & 7 \\ 3 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[138] $1^{111}3^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 16 & 24 & 15 & 121 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[139] $1^{111}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 12 & 14 & 36 & 9 & 10 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{111}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 4 & 4 & 0 & 9 & 2 \\ 10 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[141] $1^{111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[142] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 11 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[143] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 8 & 14 & 36 & 9 & 10 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

[144] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 4 & 4 & 0 & 9 & 2 \\ 8 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[145] $21^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 15 & 3 & 4 \\ 5 & 4 & 14 & 4 & 4 \\ 15 & 14 & 36 & 9 & 10 \\ 3 & 4 & 9 & 1 & 3 \\ 4 & 4 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

[146] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 9 & 0 & 13 & 7 \\ 3 & 4 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[147] $1^{11} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 9 & 14 & 36 & 9 & 10 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 10 & 3 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{11} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 4 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[149] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 5 & 8 & 9 & 12 & 5 \\ 5 & 9 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

[150] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 8 & 13 & 27 & 18 & 9 \\ 4 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[151] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 0 & 6 & 2 \\ 9 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[152] $1^{12} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 7 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[153] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 11 & 19 & 12 & 73 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[154] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 0 & 6 & 2 \\ 6 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[155] $1^1 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 4 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[156] $1^{12} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 0 & 6 & 2 \\ 8 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[157] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 4 & 5 & 3 & 15 & 3 \\ 10 & 17 & 15 & 49 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[158] $1^{11} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 14 & 19 & 12 & 73 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[159] $1^{12} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 0 & 6 & 2 \\ 5 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[160] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 6 & 9 & 12 & 18 & 6 \\ 8 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[161] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 21 & 5 & 4 \\ 13 & 21 & 75 & 18 & 15 \\ 3 & 5 & 18 & 4 & 4 \\ 3 & 4 & 15 & 4 & 2 \end{pmatrix}_{b_0}$$

[162] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 7 & 9 & 12 & 9 & 6 \\ 3 & 5 & 9 & 4 & 4 \\ 3 & 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[163] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 6 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

[164] $1^{122}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 9 & 13 & 27 & 30 & 9 \\ 7 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[165] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 7 & 9 & 12 & 12 & 6 \\ 7 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[166] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 5 & 9 & 12 & 12 & 6 \\ 5 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[167] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 5 & 9 & 12 & 9 & 6 \\ 3 & 5 & 9 & 4 & 4 \\ 3 & 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[171] $1^{122}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 7 & 8 & 9 & 12 & 5 \\ 5 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[172] $1^{12}31^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 8 & 18 & 4 \\ 4 & 8 & 9 & 24 & 5 \\ 10 & 18 & 24 & 58 & 14 \\ 3 & 4 & 5 & 14 & 1 \end{pmatrix}_{b_0}$$

[168] $1^{122}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 5 & 8 & 9 & 12 & 5 \\ 7 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[169] $1^{13}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 12 & 17 & 48 & 15 & 12 \\ 3 & 5 & 15 & 4 & 4 \\ 3 & 4 & 12 & 4 & 2 \end{pmatrix}_{b_0}$$

[170] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 9 & 12 & 4 \\ 7 & 9 & 12 & 18 & 6 \\ 9 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[173] $1^{12}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 15 & 0 & 52 & 12 \\ 3 & 4 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$

[175] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 7 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

[174] $1^{111} 1^3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 7 & 13 & 27 & 12 & 9 \\ 3 & 5 & 12 & 4 & 4 \\ 3 & 4 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

[177] $1^{131} 1^{32} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 4 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 15 & 22 & 84 & 15 & 16 \\ 4 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[178] $1^{131} 1^3 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 5 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 12 & 17 & 48 & 27 & 12 \\ 5 & 9 & 27 & 13 & 7 \\ 3 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[179] $1^{131} 1^{22} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 8 & 12 & 24 & 9 & 8 \\ 4 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[180] $1^{111} 1^{22} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 7 & 9 & 12 & 6 & 6 \\ 4 & 4 & 6 & 1 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[181] $1^{111} 1^1 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 5 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[182] $1^{111} 1^1 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 9 & 14 & 36 & 9 & 10 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[183] $1^{131} 1^4 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 10 & 12 & 24 & 9 & 8 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{11} 1^1 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 5 & 9 & 12 & 6 & 6 \\ 2 & 4 & 6 & 1 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[184] $1^{111} 1^2 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 5 & 9 & 12 & 15 & 6 \\ 7 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[185] $1^{13} 1^{24} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 5 & 5 & 3 & 15 & 3 \\ 13 & 17 & 15 & 49 & 13 \\ 3 & 4 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

[186] $1^{11} 3^{122} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 5 & 5 & 3 & 9 & 3 \\ 7 & 9 & 13 & 7 & 2 \\ 3 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[187] $21^{11} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 6 & 12 & 4 \\ 5 & 4 & 5 & 13 & 4 \\ 6 & 5 & 3 & 12 & 3 \\ 12 & 13 & 12 & 28 & 10 \\ 4 & 4 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[188] $1^{123} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -53, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 9 & 18 & 5 \\ 6 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

[189] $1^{113} 3^{14} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -53, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & 0 & 9 & 2 \\ 13 & 24 & 9 & 52 & 12 \\ 3 & 5 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[193] $1^{111} 3^{113} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 6 & 9 & 12 & 15 & 6 \\ 8 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[190] $1^{113} 31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 6 & 9 & 12 & 9 & 6 \\ 3 & 5 & 9 & 4 & 4 \\ 3 & 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[191] $1^{12} 3^{113} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 8 & 0 & 10 & 6 \\ 3 & 4 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[194] $1^{112} 3^{112} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 6 & 9 & 12 & 12 & 6 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[192] $1^{12} 3^{111} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 8 & 0 & 10 & 6 \\ 3 & 4 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[196] $1^{111} 3^{111} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[197] $1^{111} 3^{114} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 18 & 24 & 15 & 121 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[198] $1^{11} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 5 & 5 & 3 & 9 & 3 \\ 8 & 9 & 13 & 7 & 1 \\ 3 & 4 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

[195] $1^{121} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 7 & 8 & 9 & 12 & 5 \\ 7 & 9 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

[199] $1^{112} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & 0 & 12 & 2 \\ 11 & 24 & 12 & 58 & 14 \\ 3 & 5 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[200] $1^{12} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & 0 & 12 & 2 \\ 10 & 24 & 12 & 58 & 14 \\ 3 & 5 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[201] $1^{121} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 9 & 13 & 27 & 9 & 9 \\ 4 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[202] $1^{121} 3^1 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 6 & 8 & 9 & 6 & 5 \\ 2 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 8 & 9 & 12 & 6 & 6 \\ 4 & 4 & 6 & 1 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[205] $1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 8 & 9 & 12 & 12 & 6 \\ 7 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[206] $1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 6 & 9 & 12 & 12 & 6 \\ 5 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[207] $1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 6 & 9 & 12 & 9 & 6 \\ 4 & 5 & 9 & 4 & 4 \\ 3 & 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[208] $1^{121} 3^1 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

- [209] $1^{13}31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 6 & 8 & 9 & 6 & 5 \\ 4 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$
- [210] $1^{13}31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 12 & 17 & 48 & 15 & 12 \\ 4 & 5 & 15 & 4 & 4 \\ 3 & 4 & 12 & 4 & 2 \end{pmatrix}_{b_0}$$
- [211] $1^{11}31^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 14 & 0 & 37 & 11 \\ 3 & 4 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$
- [212] $1^{111}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 6 & 9 & 12 & 6 & 6 \\ 2 & 4 & 6 & 1 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$
- [213] $1^{121}31^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 14 & 18 & 57 & 12 & 13 \\ 2 & 4 & 12 & 1 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$
- [214] $1^{121}31^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 4 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 13 & 18 & 57 & 12 & 13 \\ 4 & 4 & 12 & 1 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$
- [215] $1^{11}31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 8 & 9 & 12 & 9 & 6 \\ 4 & 5 & 9 & 4 & 4 \\ 3 & 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$
- [204] $1^{12}31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.51 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 3 & 12 & 3 \\ 11 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$
- [217] $1^{121}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 6 & 8 & 9 & 12 & 5 \\ 6 & 9 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$
- [216] $1^{114}31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 15 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 8 & 9 & 12 & 33 & 6 \\ 15 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$
- [219] $1^{13}31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 10 & 17 & 48 & 15 & 12 \\ 3 & 5 & 15 & 4 & 4 \\ 3 & 4 & 12 & 4 & 2 \end{pmatrix}_{b_0}$$

[220] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 7 & 8 & 9 & 18 & 5 \\ 9 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

[221] $1^{122} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 8 & 18 & 4 \\ 5 & 8 & 9 & 24 & 5 \\ 11 & 18 & 24 & 58 & 14 \\ 3 & 4 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$

[222] $1^{122} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 7 & 8 & 9 & 12 & 5 \\ 8 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[223] $1^{122} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 5 & 8 & 9 & 12 & 5 \\ 4 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[224] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 8 & 9 & 12 & 15 & 6 \\ 6 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 7 & 13 & 27 & 9 & 9 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[225] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 10 & 13 & 27 & 18 & 9 \\ 5 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[226] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 12 & 14 & 36 & 0 & 10 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 10 & 0 & 2 \end{pmatrix}_{b_0}$$

[228] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 5 & 9 & 13 & 7 & 7 \\ 3 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[229] $1^{112} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & 0 & 6 & 2 \\ 8 & 16 & 6 & 22 & 8 \\ 3 & 5 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[230] $1^{123} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 19 & 3 \\ 3 & 4 & 8 & 27 & 4 \\ 6 & 8 & 9 & 36 & 5 \\ 19 & 27 & 36 & 133 & 21 \\ 3 & 4 & 5 & 21 & 1 \end{pmatrix}_{b_0}$$

[231] $1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 9 & 14 & 36 & 0 & 10 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 10 & 0 & 2 \end{pmatrix}_{b_0}$$

[227] $1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 8 & 9 & 12 & 6 & 6 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[232] $1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 13 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 7 & 9 & 12 & 33 & 6 \\ 13 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[233] $1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 9 & 0 & 13 & 7 \\ 3 & 4 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[234] $1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 5 & 8 & 9 & 6 & 5 \\ 2 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{11} 3^{11} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 24 & 0 & 121 & 19 \\ 3 & 4 & -1 & 19 & 1 \end{pmatrix}_{b_0}$$

[236] $1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 7 & 8 & 9 & 12 & 5 \\ 8 & 9 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

[237] $1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 7 & 8 & 9 & 6 & 5 \\ 4 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

[238] $1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 9 & 13 & 27 & 12 & 9 \\ 3 & 5 & 12 & 4 & 4 \\ 3 & 4 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

[239] $1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 11 & 17 & 48 & 15 & 12 \\ 3 & 5 & 15 & 4 & 4 \\ 3 & 4 & 12 & 4 & 2 \end{pmatrix}_{b_0}$$

[240] $1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -62, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 5 & 5 & 3 & 12 & 3 \\ 10 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

[241] $1^{11}3^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 19 & 0 & 73 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[242] $1^{11}3^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 12 & 27 & 6 \\ 9 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[244] $1^{11}3^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 5 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[245] $1^{11}3^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 7 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[246] $21^{11}3^{11}4 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 18 & 4 \\ 5 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 18 & 21 & 15 & 76 & 16 \\ 4 & 4 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

[247] $21^{11}3^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 6 & 6 & 4 \\ 5 & 4 & 5 & 5 & 4 \\ 6 & 5 & 3 & 6 & 3 \\ 6 & 5 & 6 & 4 & 4 \\ 4 & 4 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

[248] $1^{14}3^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 4 & 5 & 3 & 18 & 3 \\ 13 & 21 & 18 & 76 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[249] $1^{13}3^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 8 & 12 & 24 & 0 & 8 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 8 & 0 & 2 \end{pmatrix}_{b_0}$$

[250] $1^{11}3^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 7 & 13 & 27 & 18 & 9 \\ 4 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[251] $1^{122}3^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 8 & 13 & 27 & 18 & 9 \\ 6 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[252] $1^{13}3^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 10 & 12 & 24 & 0 & 8 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 8 & 0 & 2 \end{pmatrix}_{b_0}$$

[253] $1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 8 & 13 & 27 & 12 & 9 \\ 3 & 5 & 12 & 4 & 4 \\ 3 & 4 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

[243] $1^{11} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 15 & 19 & 12 & 73 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[254] $1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.63 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 5 & 8 & 9 & 12 & 5 \\ 6 & 9 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

[255] $1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 13 & 0 & 28 & 10 \\ 3 & 4 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[256] $1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 8 & 0 & 10 & 6 \\ 3 & 4 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[257] $1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 9 & 5 & 4 \\ 7 & 9 & 12 & 9 & 6 \\ 4 & 5 & 9 & 4 & 4 \\ 3 & 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[258] $1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 9 & 8 & 4 \\ 7 & 9 & 12 & 12 & 6 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[259] $1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 9 & 13 & 27 & 12 & 9 \\ 4 & 5 & 12 & 4 & 4 \\ 3 & 4 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

[262] $1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 8 & 9 & 12 & 15 & 6 \\ 9 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[260] $1^{11} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 10 & 14 & 36 & 9 & 10 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[263] $1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 8 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 9 & 0 & 2 \end{pmatrix}_{b_0}$$

[264] $1^{111} 3^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 14 & 4 & 4 \\ 11 & 14 & 36 & 9 & 10 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 10 & 3 & 2 \end{pmatrix}_{b_0}$$

[261] $1^{12} 3^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 8 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[266] $1^{112} 3^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & 0 & 6 & 2 \\ 7 & 16 & 6 & 22 & 8 \\ 3 & 5 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

[265] $1^{113} 3^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 9 & 4 & 28 & 5 \\ 2 & 4 & 0 & 12 & 2 \\ 15 & 28 & 12 & 73 & 15 \\ 3 & 5 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[267] $1^{112} 3^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -69, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 3 \\ 4 & 9 & 4 & 20 & 5 \\ 2 & 4 & 0 & 9 & 2 \\ 10 & 20 & 9 & 37 & 11 \\ 3 & 5 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[268] $1^{131} 3^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 9 & 17 & 48 & 12 & 12 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

[269] $1^{123} 3^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 7 & 13 & 27 & 27 & 9 \\ 7 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[270] $1^{122} 3^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -70, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 4 & 8 & 9 & 12 & 5 \\ 5 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[271] $1^{122} 3^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 7 & 13 & 27 & 18 & 9 \\ 5 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[272] $1^{121} 3^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 11 & 13 & 27 & 21 & 9 \\ 6 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[273] $1^{111} 3^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 6 & 9 & 12 & 6 & 6 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[274] $1^{13}3^{1^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 12 & 0 & 4 \\ 9 & 12 & 24 & 0 & 8 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 8 & 0 & 2 \end{pmatrix}_{b_0}$$

[275] $1^{11}3^{1^2}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 7 & 9 & 9 & 13 & 7 \\ 3 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[276] $21^{113}3^{1^1} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 9 & 3 \\ 4 & 4 & 9 & 12 & 4 \\ 6 & 9 & 12 & 18 & 6 \\ 9 & 12 & 18 & 25 & 9 \\ 3 & 4 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

[277] $1^{121}3^{1^{22}}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 8 & 9 & 4 \\ 6 & 8 & 9 & 12 & 5 \\ 7 & 9 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

[284] $1^{123}3^{1^1}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 5 & 8 & 9 & 30 & 5 \\ 12 & 22 & 30 & 85 & 17 \\ 3 & 4 & 5 & 17 & 2 \end{pmatrix}_{b_0}$$

[278] $21^{111}3^{1^1}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 3 & 9 & 4 \\ 8 & 9 & 4 & 12 & 5 \\ 3 & 4 & 0 & 6 & 2 \\ 9 & 12 & 6 & 13 & 7 \\ 4 & 5 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[281] $21^{111}3^{1^1}1^{31} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 15 & 3 & 4 \\ 5 & 4 & 17 & 4 & 4 \\ 15 & 17 & 48 & 12 & 12 \\ 3 & 4 & 12 & 1 & 3 \\ 4 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[282] $1^{121}3^{1^1}1^{31} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 7 & 8 & 9 & 6 & 5 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

[283] $1^{111}3^{1^1}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 9 & 0 & 13 & 7 \\ 3 & 4 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

[279] $1^{13}3^{1^1}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 12 & 0 & 25 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[280] $1^{121}3^{1^1}1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 5 & 8 & 9 & 6 & 5 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

[286] $1^{122} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 8 & 18 & 4 \\ 5 & 8 & 9 & 24 & 5 \\ 12 & 18 & 24 & 58 & 14 \\ 3 & 4 & 5 & 14 & 1 \end{pmatrix}_{b_0}$$

[287] $1^{141} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 4 & 3 \\ 3 & 4 & 17 & 5 & 4 \\ 13 & 17 & 48 & 15 & 12 \\ 4 & 5 & 15 & 4 & 4 \\ 3 & 4 & 12 & 4 & 2 \end{pmatrix}_{b_0}$$

[285] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 4 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 11 & 13 & 27 & 12 & 9 \\ 4 & 5 & 12 & 4 & 4 \\ 3 & 4 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

[288] $1^{111} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 13 & 19 & 12 & 73 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[289] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 8 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[290] $1^{111} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 12 & 19 & 12 & 73 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[291] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 8 & 13 & 27 & 27 & 9 \\ 7 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[292] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 11 & 14 & 36 & 0 & 10 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 10 & 0 & 2 \end{pmatrix}_{b_0}$$

[294] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 16 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 4 & 4 & 0 & 18 & 2 \\ 16 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[295] $1^{112} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.58 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 0 & 6 & 2 \\ 7 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[296] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 0 & 6 & 2 \\ 8 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

- [297] $1^{111} 3 1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 13 & 19 & 12 & 73 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$
- [298] $1^{111} 3 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 0 & 6 & 2 \\ 7 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$
- [299] $1^{112} 3 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 6 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$
- [300] $1^{112} 3 1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 20 & 3 \\ 3 & 4 & 4 & 28 & 4 \\ 4 & 4 & 0 & 18 & 2 \\ 20 & 28 & 18 & 154 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$
- [301] $1^{112} 3 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.58 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 5 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$
- [302] $1^{113} 3 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 3 & 9 & 3 \\ 8 & 9 & 9 & 13 & 7 \\ 3 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$
- [303] $1^{111} 3 1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 14 & 0 & 4 \\ 10 & 14 & 36 & 0 & 10 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 10 & 0 & 2 \end{pmatrix}_{b_0}$$
- [293] $1^{112} 3 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 0 & 6 & 2 \\ 6 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$
- [304] $1^{11} 3 1^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -77, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 3 \\ 4 & 9 & 4 & 20 & 5 \\ 2 & 4 & 0 & 9 & 2 \\ 8 & 20 & 9 & 37 & 11 \\ 3 & 5 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$
- [306] $1^{112} 3 1^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 8 & 9 & 12 & 21 & 6 \\ 9 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$
- [305] $1^{111} 3 1^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 9 & 4 & 4 \\ 7 & 9 & 12 & 6 & 6 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[307] $1^{122} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 8 & 18 & 4 \\ 6 & 8 & 9 & 24 & 5 \\ 14 & 18 & 24 & 58 & 14 \\ 3 & 4 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$

[309] $1^{122} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 8 & 18 & 4 \\ 6 & 8 & 9 & 24 & 5 \\ 13 & 18 & 24 & 58 & 14 \\ 3 & 4 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$

[308] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -78, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 12 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 8 & 9 & 12 & 27 & 6 \\ 12 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[320] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 9 & 0 & 13 & 7 \\ 3 & 4 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

[310] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 11 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 9 & 0 & 2 \end{pmatrix}_{b_0}$$

[318] $1^{113} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 18 & 32 & 21 & 193 & 25 \\ 3 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[317] $21^{11} 31^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 12 & 3 & 4 \\ 5 & 4 & 13 & 4 & 4 \\ 12 & 13 & 27 & 9 & 9 \\ 3 & 4 & 9 & 1 & 3 \\ 4 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[316] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 10 & 13 & 27 & 9 & 9 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[315] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 10 & 13 & 27 & 27 & 9 \\ 10 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[314] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 13 & 5 & 4 \\ 10 & 13 & 27 & 12 & 9 \\ 4 & 5 & 12 & 4 & 4 \\ 3 & 4 & 9 & 4 & 2 \end{pmatrix}_{b_0}$$

[313] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 9 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[312] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 6 & 9 & 9 & 13 & 7 \\ 3 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[311] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 12 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[319] $1^{12} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 3 \\ 3 & 4 & 8 & 27 & 4 \\ 5 & 8 & 9 & 36 & 5 \\ 17 & 27 & 36 & 133 & 21 \\ 3 & 4 & 5 & 21 & 1 \end{pmatrix}_{b_0}$$

[321] $1^{12} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 12 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[323] $1^{13} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 17 & 3 \\ 4 & 9 & 4 & 36 & 5 \\ 2 & 4 & 0 & 18 & 2 \\ 17 & 36 & 18 & 133 & 21 \\ 3 & 5 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

[324] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 6 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[322] $1^{14} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 10 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[325] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 3 & 9 & 3 \\ 6 & 9 & 9 & 13 & 7 \\ 3 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[326] $1^1 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 3 & 5 & 3 & 18 & 3 \\ 11 & 21 & 18 & 76 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[327] $1^1 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -81, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 3 & 12 & 3 \\ 7 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[328] $1^{14} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -82, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 10 & 16 & 45 & 0 & 11 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 11 & 0 & 2 \end{pmatrix}_{b_0}$$

[332] $1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 11 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[331] $1^{111} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 4 & 4 & 0 & 9 & 2 \\ 11 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[330] $1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 3 & 12 & 3 \\ 8 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[329] $1^{111} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 10 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[333] $1^{112} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 10 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[334] $1^{112} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 5 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[335] $1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 7 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[336] $1^{112} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 28 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 17 & 28 & 18 & 154 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[337] $21^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 9 & 6 & 4 \\ 5 & 4 & 9 & 5 & 4 \\ 9 & 9 & 12 & 9 & 6 \\ 6 & 5 & 9 & 4 & 4 \\ 4 & 4 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[338] $1^{112} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 7 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[340] $1 3 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 7 & 13 & 27 & 9 & 9 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[339] $1^{13} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 10 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 12 & 0 & 2 \end{pmatrix}_{b_0}$$

[341] $1^{12} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 5 & 5 & 3 & 12 & 3 \\ 11 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[342] $1^{12} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 8 & 4 & 4 \\ 6 & 8 & 9 & 6 & 5 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

[343] $1^{12} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 7 & 8 & 9 & 18 & 5 \\ 8 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[344] $1^{12} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 9 & 18 & 5 \\ 10 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[345] $21^{11} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 15 & 9 & 4 \\ 5 & 4 & 17 & 9 & 4 \\ 15 & 17 & 48 & 27 & 12 \\ 9 & 9 & 27 & 13 & 7 \\ 4 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[346] $1^{12} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -86, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.47 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 5 & 8 & 9 & 18 & 5 \\ 7 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

[347] $1^{13} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 49 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[348] $1^{13} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 9 & 12 & 24 & 9 & 8 \\ 4 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[349] $1^{13} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 9 & 12 & 24 & 9 & 8 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[350] $1^{13} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 7 & 12 & 24 & 9 & 8 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[351] $1^{12} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 9 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 9 & 0 & 2 \end{pmatrix}_{b_0}$$

[352] $1^{13} 3 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 7 & 12 & 24 & 9 & 8 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[353] $1^{11} 3 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -88, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 3 & 15 & 3 \\ 9 & 17 & 15 & 49 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[356] $1^{112} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 3 \\ 4 & 9 & 4 & 20 & 5 \\ 2 & 4 & 0 & 9 & 2 \\ 9 & 20 & 9 & 37 & 11 \\ 3 & 5 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[354] $1^{123} 3 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 18 & 3 \\ 3 & 4 & 8 & 27 & 4 \\ 5 & 8 & 9 & 36 & 5 \\ 18 & 27 & 36 & 133 & 21 \\ 3 & 4 & 5 & 21 & 1 \end{pmatrix}_{b_0}$$

[355] $1^{122} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 4 & 8 & 9 & 18 & 5 \\ 8 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

[357] $1^{14} 3 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 16 & 0 & 46 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[365] $21^{13} 31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 21 & 4 \\ 5 & 4 & 4 & 19 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 21 & 19 & 12 & 73 & 15 \\ 4 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[359] $1^{111} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 9 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[358] $1^{111} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 4 & 4 & 0 & 9 & 2 \\ 12 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[360] $1^{113} 3 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 10 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[361] $1^{11} 3 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 5 & 9 & 4 \\ 4 & 5 & 3 & 9 & 3 \\ 7 & 9 & 9 & 13 & 7 \\ 3 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[362] $1^{121} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.58 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 8 & 13 & 27 & 21 & 9 \\ 7 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[363] $1^{111} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 15 & 19 & 12 & 73 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[364] $1^{112} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 16 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 4 & 4 & 0 & 15 & 2 \\ 16 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[366] $1^1 3 1^{122} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 4 & 8 & 9 & 18 & 5 \\ 7 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

[367] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 6 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[368] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 1 & 3 \\ 3 & 4 & 13 & 0 & 4 \\ 10 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 9 & 0 & 2 \end{pmatrix}_{b_0}$$

[369] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 7 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 10 & 13 & 27 & 18 & 9 \\ 7 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[370] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 8 & 13 & 27 & 27 & 9 \\ 8 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[371] $1^{113} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 0 & 9 & 2 \\ 8 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[372] $1^{113} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 4 & 4 & 0 & 15 & 2 \\ 14 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

- [373] $21^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 9 & 4 \\ 5 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 9 & 9 & 6 & 13 & 7 \\ 4 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$
- [374] $21^{11}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -93, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 9 & 15 & 4 \\ 5 & 4 & 9 & 17 & 4 \\ 9 & 9 & 12 & 27 & 6 \\ 15 & 17 & 27 & 49 & 13 \\ 4 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [375] $1^{122}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -93, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 6 & 8 & 9 & 18 & 5 \\ 8 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$
- [376] $1^{123}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -93, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 6 & 8 & 9 & 18 & 5 \\ 7 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$
- [377] $1^{123}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 11 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 13 & 0 & 2 \end{pmatrix}_{b_0}$$
- [378] $1^{123}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 9 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 9 & 13 & 27 & 27 & 9 \\ 9 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$
- [380] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -95, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 10 & 13 & 27 & 9 & 9 \\ 4 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$
- [379] $1^{121}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -95, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 7 & 13 & 27 & 21 & 9 \\ 6 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 1 \end{pmatrix}_{b_0}$$
- [382] $1^131^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 9 & 4 & 36 & 5 \\ 2 & 4 & 0 & 18 & 2 \\ 15 & 36 & 18 & 133 & 21 \\ 3 & 5 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$
- [381] $1^{121}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 9 & 13 & 27 & 21 & 9 \\ 5 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$
- [383] $21^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.42 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 4 & 5 & 9 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 6 & 9 & 9 & 13 & 7 \\ 3 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$
- [385] $1^{123}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$

$$\begin{aligned}
& T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 4 & 8 & 9 & 18 & 5 \\ 8 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 1 \end{pmatrix}_{b_0} \\
[384] \quad & 21^{121} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2 \\
& T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 4 & 8 & 9 & 4 \\ 6 & 8 & 9 & 12 & 5 \\ 6 & 9 & 12 & 13 & 7 \\ 3 & 4 & 5 & 7 & 2 \end{pmatrix}_{b_0} \\
[389] \quad & 1^{14} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6 \\
& T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 12 & 21 & 75 & 0 & 15 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 15 & 0 & 1 \end{pmatrix}_{b_0} \\
[390] \quad & 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 7 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0} \\
[386] \quad & 1^{131} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 8 & 12 & 24 & 9 & 8 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0} \\
[387] \quad & 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 6 & 9 & 12 & 15 & 6 \\ 5 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0} \\
[388] \quad & 1^{111} 3^{11} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 14 & 19 & 12 & 73 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0} \\
[391] \quad & 1^{111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 5 & 9 & 12 & 15 & 6 \\ 5 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0} \\
[392] \quad & 1^{112} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 6 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0} \\
[396] \quad & 1^{112} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 12 & 21 & 6 \\ 7 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0} \\
[399] \quad & 1^{122} 3^{11} 1^{21} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7 \\
& T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 6 & 8 & 9 & 12 & 5 \\ 5 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0} \\
[398] \quad & 21^{122} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 4 & 9 & 8 & 4 \\ 6 & 9 & 12 & 12 & 6 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[397] $1^{122} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 6 & 8 & 9 & 12 & 5 \\ 7 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[393] $1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 12 & 21 & 6 \\ 11 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[394] $1^{123} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 6 & 8 & 9 & 24 & 5 \\ 10 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[395] $1^{12} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 3 & 12 & 3 \\ 10 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[400] $1^{122} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 10 & 13 & 27 & 30 & 9 \\ 8 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[402] $1^{122} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 5 & 8 & 9 & 18 & 5 \\ 10 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

[401] $1^{113} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 3 \\ 4 & 9 & 4 & 36 & 5 \\ 2 & 4 & 0 & 18 & 2 \\ 16 & 36 & 18 & 133 & 21 \\ 3 & 5 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

[407] $1^{123} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 6 & 8 & 9 & 30 & 5 \\ 14 & 22 & 30 & 85 & 17 \\ 3 & 4 & 5 & 17 & 2 \end{pmatrix}_{b_0}$$

[405] $1^{113} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & 0 & 9 & 2 \\ 12 & 24 & 9 & 52 & 12 \\ 3 & 5 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[404] $1^{113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 14 & 0 & 37 & 11 \\ 3 & 4 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[403] $1^{113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & 0 & 15 & 2 \\ 16 & 32 & 15 & 100 & 18 \\ 3 & 5 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[406] $1^{131} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 4 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 12 & 17 & 48 & 12 & 12 \\ 4 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[413] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 14 & 0 & 37 & 11 \\ 3 & 4 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[412] $1^{121} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 11 & 18 & 57 & 12 & 13 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[411] $1^{12} 3 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 10 & 18 & 57 & 12 & 13 \\ 2 & 4 & 12 & 1 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[410] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 9 & 13 & 27 & 27 & 9 \\ 8 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[409] $1^{121} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 13 & 18 & 57 & 12 & 13 \\ 2 & 4 & 12 & 1 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[408] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 9 & 13 & 27 & 9 & 9 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[418] $1^{12} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 18 & 0 & 58 & 14 \\ 3 & 4 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[417] $1^{14} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.42 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 11 & 16 & 45 & 0 & 11 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 11 & 0 & 2 \end{pmatrix}_{b_0}$$

[416] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 5 & 8 & 9 & 12 & 5 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[415] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 7 & 8 & 9 & 12 & 5 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[414] $1^{14}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 21 & 0 & 76 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[420] $1^{123}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 13 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 8 & 13 & 27 & 39 & 9 \\ 13 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 1 \end{pmatrix}_{b_0}$$

[419] $1^{121}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 8 & 13 & 27 & 9 & 9 \\ 2 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[426] $1^{123}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 4 & 8 & 9 & 18 & 5 \\ 7 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

[421] $1^{13}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 12 & 0 & 25 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[422] $1^{13}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 12 & 0 & 25 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[423] $21^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 6 & 9 & 4 \\ 5 & 4 & 5 & 9 & 4 \\ 6 & 5 & 3 & 9 & 3 \\ 9 & 9 & 9 & 13 & 7 \\ 4 & 4 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

[424] $1^{114}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 5 & 9 & 12 & 33 & 6 \\ 15 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

[425] $1^{111}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 5 & 9 & 12 & 15 & 6 \\ 6 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[427] $1^{122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -110, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.51 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 7 & 8 & 9 & 12 & 5 \\ 7 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[428] $1^{122}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -110, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.51 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 5 & 8 & 9 & 12 & 5 \\ 5 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[431] $1^{113} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 7 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[430] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 10 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[429] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 11 & 14 & 9 & 37 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[432] $1^{122} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 9 & 13 & 27 & 18 & 9 \\ 5 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[435] $1^{131} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 7 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 11 & 17 & 48 & 27 & 12 \\ 7 & 9 & 27 & 13 & 7 \\ 3 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[434] $1^{121} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 8 & 13 & 27 & 9 & 9 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[433] $21^{111} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[436] $1^{113} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -113, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & 0 & 9 & 2 \\ 10 & 24 & 9 & 52 & 12 \\ 3 & 5 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[437] $1^{14} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 16 & 0 & 4 \\ 12 & 16 & 45 & 0 & 11 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 11 & 0 & 2 \end{pmatrix}_{b_0}$$

[438] $1^{12} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 3 & 12 & 3 \\ 8 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[439] $1^{112} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 7 & 9 & 12 & 21 & 6 \\ 8 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[447] $1^{13} 3^{112} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 4 & 5 & 3 & 15 & 3 \\ 11 & 17 & 15 & 49 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[449] $21^{11} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 15 & 17 & 12 & 49 & 13 \\ 4 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[440] $1^{122} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 8 & 13 & 27 & 18 & 9 \\ 5 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[448] $21^{12} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 15 & 14 & 9 & 37 & 11 \\ 4 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[441] $1^{114} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 4 & 4 & 0 & 15 & 2 \\ 13 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[442] $1^{113} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 7 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[443] $1^{112} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 4 & 4 & 0 & 9 & 2 \\ 9 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[444] $1^{122} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 8 & 13 & 27 & 30 & 9 \\ 10 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[445] $1^{113} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 9 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[446] $1^1 3^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -115, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 7 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[450] $1^{123} 3^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 9 & 3 \\ 3 & 4 & 13 & 12 & 4 \\ 10 & 13 & 27 & 27 & 9 \\ 9 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[451] $1^{122} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 11 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 9 & 13 & 27 & 30 & 9 \\ 11 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[454] $21^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 4 & 9 & 4 & 4 \\ 6 & 9 & 12 & 6 & 6 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 6 & 3 & 2 \end{pmatrix}_{b_0}$$

[453] $1^{14} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.47 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 3 & 15 & 3 \\ 13 & 17 & 15 & 49 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[452] $1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 13 & 0 & 28 & 10 \\ 3 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[458] $1^{13} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -121, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 12 & 0 & 25 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[457] $1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -121, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 3 & 5 & 3 & 12 & 3 \\ 9 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[456] $1^{13} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -121, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 11 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 12 & 0 & 2 \end{pmatrix}_{b_0}$$

[455] $1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -121, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 7 & 9 & 12 & 15 & 6 \\ 8 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[460] $1^{131} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 10 & 12 & 24 & 9 & 8 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[459] $1^{131} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 8 & 12 & 24 & 9 & 8 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[461] $1 3 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 9 & 17 & 48 & 12 & 12 \\ 2 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[463] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 3 & 12 & 3 \\ 9 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[465] $1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 9 & 13 & 27 & 18 & 9 \\ 6 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[462] $1^{111} 3^1 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 7 & 13 & 27 & 21 & 9 \\ 5 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[467] $21^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 4 \\ 8 & 9 & 4 & 8 & 5 \\ 3 & 4 & 0 & 3 & 2 \\ 6 & 8 & 3 & 4 & 4 \\ 4 & 5 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

[464] $1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 13 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 16 & 0 & 2 \end{pmatrix}_{b_0}$$

[466] $21^{11} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 12 & 4 \\ 5 & 4 & 4 & 13 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 12 & 13 & 9 & 28 & 10 \\ 4 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[468] $1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -125, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 3 \\ 4 & 9 & 4 & 24 & 5 \\ 2 & 4 & 0 & 9 & 2 \\ 11 & 24 & 9 & 52 & 12 \\ 3 & 5 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[469] $1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 12 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 13 & 0 & 2 \end{pmatrix}_{b_0}$$

[470] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 8 & 9 & 12 & 15 & 6 \\ 7 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[471] $1^{123} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 5 & 8 & 9 & 24 & 5 \\ 9 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[472] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 11 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[473] $1^{12} 3^{14} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 3 & 12 & 3 \\ 11 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[474] $1^{12} 3^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 11 & 18 & 57 & 12 & 13 \\ 2 & 4 & 12 & 1 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[475] $1^{12} 3^{12} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 12 & 18 & 57 & 12 & 13 \\ 2 & 4 & 12 & 1 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[476] $1^{11} 3^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 19 & 0 & 73 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[477] $21^{12} 31^{12} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 18 & 3 & 4 \\ 5 & 4 & 18 & 4 & 4 \\ 18 & 18 & 57 & 12 & 13 \\ 3 & 4 & 12 & 1 & 3 \\ 4 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[478] $1^{11} 3^{12} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 14 & 0 & 37 & 11 \\ 3 & 4 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[479] $1^{11} 3^{13} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 14 & 0 & 37 & 11 \\ 3 & 4 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[480] $1^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 8 & 9 & 12 & 15 & 6 \\ 8 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[482] $1^{12} 3^{14} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 11 & 13 & 27 & 9 & 9 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[481] $1^{12} 3^{12} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -130, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 8 & 8 & 4 \\ 6 & 8 & 9 & 12 & 5 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[485] $1^{13} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 13 & 17 & 48 & 12 & 12 \\ 2 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[484] $1^{113} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 0 & 9 & 2 \\ 9 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[483] $1^{113} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 9 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[487] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 3 \\ 4 & 9 & 4 & 28 & 5 \\ 2 & 4 & 0 & 12 & 2 \\ 14 & 28 & 12 & 73 & 15 \\ 3 & 5 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[486] $1^{111} 3 1^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 9 & 17 & 48 & 27 & 12 \\ 5 & 9 & 27 & 13 & 7 \\ 3 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[488] $1^{111} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 3 \\ 4 & 9 & 4 & 28 & 5 \\ 2 & 4 & 0 & 12 & 2 \\ 11 & 28 & 12 & 73 & 15 \\ 3 & 5 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[489] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -133, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 7 & 9 & 12 & 15 & 6 \\ 6 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[490] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 6 & 9 & 12 & 15 & 6 \\ 7 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[491] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 6 & 8 & 9 & 18 & 5 \\ 11 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

[492] $1^{131} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 12 & 4 & 4 \\ 9 & 12 & 24 & 9 & 8 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[493] $1^{131} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 3 & 15 & 3 \\ 10 & 17 & 15 & 49 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[495] $1^{121} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 8 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 10 & 13 & 27 & 21 & 9 \\ 8 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[499] $1^{131} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 5 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 11 & 17 & 48 & 27 & 12 \\ 5 & 9 & 27 & 13 & 7 \\ 3 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[498] $1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 7 & 8 & 9 & 18 & 5 \\ 10 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

[497] $1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 3 \\ 3 & 4 & 13 & 8 & 4 \\ 10 & 13 & 27 & 18 & 9 \\ 6 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[496] $1^{123} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 4 & 8 & 9 & 24 & 5 \\ 11 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

[494] $1^{114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 9 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[501] $1^{12} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -138, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 23 & 0 & 100 & 18 \\ 3 & 4 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[500] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -138, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 13 & 0 & 28 & 10 \\ 3 & 4 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[502] $1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 0 & 9 & 2 \\ 10 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[506] $1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 8 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[509] $1^{12} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 5 & 13 & 4 \\ 4 & 5 & 3 & 12 & 3 \\ 10 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[503] $1^{121} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 8 & 13 & 27 & 21 & 9 \\ 5 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[504] $1^{14}3^{13}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 4 & 5 & 3 & 18 & 3 \\ 14 & 21 & 18 & 76 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[510] $1^{113}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 15 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[505] $21^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 6 & 9 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 2 \end{pmatrix}_{b_0}$$

[508] $1^{112}3^{13}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 10 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[507] $21^{112}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 4 & 4 & 23 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 15 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[511] $1^{113}3^{123}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 4 & 4 & 0 & 18 & 2 \\ 18 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[512] $1^{113}3^{13}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 5 & 9 & 12 & 27 & 6 \\ 12 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[516] $1^{12}3^{113}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & 0 & 15 & 2 \\ 13 & 32 & 15 & 100 & 18 \\ 3 & 5 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[517] $1^{114}3^{14}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 22 & 3 \\ 4 & 9 & 4 & 44 & 5 \\ 2 & 4 & 0 & 21 & 2 \\ 22 & 44 & 21 & 193 & 25 \\ 3 & 5 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

[515] $1^{131}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 12 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 12 & 0 & 2 \end{pmatrix}_{b_0}$$

[514] $1^{111}3^{112}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.66 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 5 & 9 & 12 & 21 & 6 \\ 7 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[513] $1^{112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -141, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.66 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 5 & 9 & 12 & 21 & 6 \\ 7 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[518] $1^{14}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 3 & 5 & 3 & 18 & 3 \\ 12 & 21 & 18 & 76 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[520] $1^{12}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 14 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 13 & 0 & 2 \end{pmatrix}_{b_0}$$

[521] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 9 & 13 & 27 & 9 & 9 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[519] $1^{12}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 18 & 0 & 4 \\ 13 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 13 & 0 & 2 \end{pmatrix}_{b_0}$$

[522] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 11 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[523] $1^{11}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.61 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 24 & 0 & 121 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[524] $1^{123}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.61 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 9 & 18 & 5 \\ 7 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[525] $1^{12}31^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 4 & 8 & 9 & 30 & 5 \\ 12 & 22 & 30 & 85 & 17 \\ 3 & 4 & 5 & 17 & 1 \end{pmatrix}_{b_0}$$

[526] $1^{13}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.03 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 17 & 0 & 4 \\ 13 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 12 & 0 & 2 \end{pmatrix}_{b_0}$$

[527] $1^{123}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 4 & 8 & 9 & 30 & 5 \\ 13 & 22 & 30 & 85 & 17 \\ 3 & 4 & 5 & 17 & 1 \end{pmatrix}_{b_0}$$

[528] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -145, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 7 & 8 & 9 & 24 & 5 \\ 12 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[531] $1^{131} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 10 & 17 & 48 & 12 & 12 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[529] $1^{11} 3 1^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 4 & 8 & 9 & 24 & 5 \\ 9 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

[530] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 5 & 8 & 9 & 18 & 5 \\ 8 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

[532] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -148, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 13 & 4 & 4 \\ 10 & 13 & 27 & 9 & 9 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[533] $1^{113} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -149, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & 0 & 15 & 2 \\ 15 & 32 & 15 & 100 & 18 \\ 3 & 5 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[534] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 13 & 0 & 28 & 10 \\ 3 & 4 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

[535] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 6 & 9 & 12 & 15 & 6 \\ 6 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[536] $1^{14} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 16 & 0 & 46 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[537] $1^{14} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -150, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 15 & 26 & 117 & 0 & 19 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 19 & 0 & 2 \end{pmatrix}_{b_0}$$

[538] $1^{14} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 13 & 21 & 75 & 0 & 15 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 15 & 0 & 2 \end{pmatrix}_{b_0}$$

[539] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 12 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[540] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 3 & 15 & 3 \\ 12 & 17 & 15 & 49 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[541] $1^{114} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 22 & 3 \\ 4 & 9 & 4 & 48 & 5 \\ 2 & 4 & 0 & 24 & 2 \\ 22 & 48 & 24 & 238 & 28 \\ 3 & 5 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

[542] $1^{14} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 20 & 3 \\ 4 & 9 & 4 & 48 & 5 \\ 2 & 4 & 0 & 24 & 2 \\ 20 & 48 & 24 & 238 & 28 \\ 3 & 5 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

[543] $1^{121} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -153, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 12 & 18 & 57 & 12 & 13 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[544] $1^{13} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -153, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 22 & 0 & 85 & 17 \\ 3 & 4 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[545] $1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 5 & 8 & 9 & 18 & 5 \\ 9 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

[546] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -155, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 4 & 4 & 0 & 9 & 2 \\ 10 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[547] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -155, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 8 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[550] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 8 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[549] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 13 & 0 & 28 & 10 \\ 3 & 4 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[551] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 3 & 5 & 3 & 15 & 3 \\ 11 & 17 & 15 & 49 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[548] $1^{12} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 14 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[552] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -158, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.61 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 10 & 17 & 48 & 27 & 12 \\ 6 & 9 & 27 & 13 & 7 \\ 3 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[553] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -158, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 7 & 8 & 9 & 18 & 5 \\ 11 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

[554] $1^{13} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 10 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[555] $1^{13} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 8 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[556] $1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 14 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[557] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 4 & 5 & 3 & 15 & 3 \\ 12 & 17 & 15 & 49 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[558] $1^{14} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 9 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

[559] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 9 & 9 & 4 \\ 7 & 9 & 12 & 15 & 6 \\ 7 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[560] $1^{14} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 11 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[561] $1^{112} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 13 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[562] $1^{12} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 10 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[563] $21^{121} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 4 & 8 & 4 & 4 \\ 6 & 8 & 9 & 6 & 5 \\ 3 & 4 & 6 & 1 & 3 \\ 3 & 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

[564] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & 0 & 15 & 2 \\ 14 & 32 & 15 & 100 & 18 \\ 3 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[565] $1^{123} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 7 & 8 & 9 & 18 & 5 \\ 9 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[566] $1^{123} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 9 & 18 & 5 \\ 9 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[567] $1^{131} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 10 & 17 & 48 & 12 & 12 \\ 2 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[568] $1^{12} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -162, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 28 & 0 & 154 & 22 \\ 3 & 4 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[570] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 4 & 4 & 0 & 9 & 2 \\ 11 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[569] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 20 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 4 & 4 & 0 & 21 & 2 \\ 20 & 31 & 21 & 172 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[571] $1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 6 & 8 & 9 & 18 & 5 \\ 9 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

[572] $1^{113} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 22 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 4 & 4 & 0 & 21 & 2 \\ 22 & 32 & 21 & 193 & 25 \\ 3 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[573] $1^{131} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 10 & 17 & 48 & 27 & 12 \\ 5 & 9 & 27 & 13 & 7 \\ 3 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[574] $1^{131} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 13 & 22 & 84 & 15 & 16 \\ 3 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[575] $1^{123} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 8 & 13 & 27 & 39 & 9 \\ 9 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[576] $1^{112} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 9 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[577] $1^{121} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 6 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 10 & 13 & 27 & 21 & 9 \\ 6 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[578] $1^{113} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 3 \\ 4 & 9 & 4 & 28 & 5 \\ 2 & 4 & 0 & 12 & 2 \\ 13 & 28 & 12 & 73 & 15 \\ 3 & 5 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[579] $1^{113} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 3 \\ 4 & 9 & 4 & 28 & 5 \\ 2 & 4 & 0 & 12 & 2 \\ 12 & 28 & 12 & 73 & 15 \\ 3 & 5 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$

[580] $1^{114} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -165, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 17 & 0 & 49 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[581] $1^{131} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -165, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 12 & 17 & 48 & 12 & 12 \\ 2 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[582] $1^{114} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & 0 & 12 & 2 \\ 13 & 32 & 12 & 94 & 16 \\ 3 & 5 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

[583] $1^{114} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & 0 & 12 & 2 \\ 16 & 32 & 12 & 94 & 16 \\ 3 & 5 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[584] $1^{111} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 5 & 9 & 12 & 27 & 6 \\ 9 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[585] $1^{114} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -168, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 21 & 3 \\ 4 & 9 & 4 & 48 & 5 \\ 2 & 4 & 0 & 24 & 2 \\ 21 & 48 & 24 & 238 & 28 \\ 3 & 5 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

[586] $1^{14} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -169, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 3 & 5 & 3 & 18 & 3 \\ 15 & 21 & 18 & 76 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[587] $1^{123} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -170, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 4 & 8 & 9 & 24 & 5 \\ 10 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

[589] $1^{12} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 12 & 22 & 84 & 15 & 16 \\ 2 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[588] $1^{11} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 19 & 0 & 73 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[590] $1^{14} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 28 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 16 & 28 & 18 & 154 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[591] $1^{11} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 9 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[592] $1^{122} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 8 & 13 & 27 & 30 & 9 \\ 7 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[593] $1^{121} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 14 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 16 & 0 & 2 \end{pmatrix}_{b_0}$$

[594] $1^{131} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -173, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 6 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 13 & 17 & 48 & 27 & 12 \\ 6 & 9 & 27 & 13 & 7 \\ 3 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[595] $1^{112} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -174, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 5 & 9 & 12 & 21 & 6 \\ 8 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[596] $1^{131} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 11 & 17 & 48 & 12 & 12 \\ 2 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[597] $1^{113} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 9 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[598] $21^{123} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 4 & 5 & 13 & 4 \\ 3 & 5 & 3 & 12 & 3 \\ 9 & 13 & 12 & 28 & 10 \\ 3 & 4 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

[599] $1^{121} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 8 & 13 & 27 & 21 & 9 \\ 6 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[600] $1^{121} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 14 & 18 & 57 & 12 & 13 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[601] $1^{121} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 18 & 4 & 4 \\ 13 & 18 & 57 & 12 & 13 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[602] $1^{14} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 3 & 5 & 3 & 18 & 3 \\ 13 & 21 & 18 & 76 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[603] $1^{112} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 8 & 9 & 12 & 21 & 6 \\ 10 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[604] $1^{114} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 6 & 9 & 12 & 33 & 6 \\ 12 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[605] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 6 & 8 & 9 & 18 & 5 \\ 10 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[606] $1^{122} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 8 & 13 & 4 \\ 6 & 8 & 9 & 18 & 5 \\ 10 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

[607] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -177, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 6 & 8 & 9 & 18 & 5 \\ 8 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[608] $1^{12} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -178, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 18 & 0 & 58 & 14 \\ 3 & 4 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[611] $1^{13} 3 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 13 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[609] $1^{13} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 5 & 17 & 4 \\ 4 & 5 & 3 & 15 & 3 \\ 13 & 17 & 15 & 49 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[610] $1^{112} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -179, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 17 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 4 & 4 & 0 & 15 & 2 \\ 17 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[612] $1^{113} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 7 & 9 & 12 & 27 & 6 \\ 11 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[613] $1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 11 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[614] $1^{121} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 7 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 9 & 13 & 27 & 21 & 9 \\ 7 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[615] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 11 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[617] $1^{111} 3 1^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 5 & 9 & 12 & 33 & 6 \\ 11 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[619] $1^{123} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 9 & 18 & 5 \\ 8 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[616] $21^{111} 31^{121} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 12 & 9 & 4 \\ 5 & 4 & 13 & 9 & 4 \\ 12 & 13 & 27 & 21 & 9 \\ 9 & 9 & 21 & 13 & 7 \\ 4 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[618] $1^{123} 3 1^{34} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 7 & 8 & 9 & 18 & 5 \\ 10 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[620] $1^{11} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -183, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 19 & 0 & 73 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[621] $21^{123} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 9 & 3 \\ 4 & 4 & 13 & 12 & 4 \\ 9 & 13 & 27 & 27 & 9 \\ 9 & 12 & 27 & 25 & 9 \\ 3 & 4 & 9 & 9 & 2 \end{pmatrix}_{b_0}$$

[622] $21^{112} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 4 & 4 & 8 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 6 & 8 & 6 & 10 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[623] $1^{123} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -185, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 16 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 5 & 8 & 9 & 30 & 5 \\ 16 & 22 & 30 & 85 & 17 \\ 3 & 4 & 5 & 17 & 2 \end{pmatrix}_{b_0}$$

[625] $1^{123} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -185, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 5 & 8 & 9 & 30 & 5 \\ 13 & 22 & 30 & 85 & 17 \\ 3 & 4 & 5 & 17 & 2 \end{pmatrix}_{b_0}$$

[624] $1^{111} 3 1^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -185, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 7 & 13 & 27 & 30 & 9 \\ 7 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[626] $1^{14} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -186, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 3 & 5 & 3 & 18 & 3 \\ 14 & 21 & 18 & 76 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[627] $1^{14} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -186, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 16 & 0 & 46 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[629] $1^{14} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 14 & 21 & 75 & 0 & 15 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 15 & 0 & 2 \end{pmatrix}_{b_0}$$

[630] $21^{12} 31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 4 & 5 & 21 & 3 & 4 \\ 5 & 4 & 22 & 4 & 4 \\ 21 & 22 & 84 & 15 & 16 \\ 3 & 4 & 15 & 1 & 3 \\ 4 & 4 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

[628] $1^{13} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 16 & 22 & 84 & 15 & 16 \\ 2 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

[631] $1^{12} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -188, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 7 & 13 & 27 & 39 & 9 \\ 11 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 1 \end{pmatrix}_{b_0}$$

[632] $1^{14} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -188, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 5 & 21 & 4 \\ 4 & 5 & 3 & 18 & 3 \\ 15 & 21 & 18 & 76 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0}$$

[634] $21^{11} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -189, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 9 & 9 & 4 \\ 5 & 4 & 9 & 9 & 4 \\ 9 & 9 & 12 & 15 & 6 \\ 9 & 9 & 15 & 13 & 7 \\ 4 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[633] $1^{11} 3^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -189, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 19 & 0 & 73 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[635] $1^{12} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -190, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 7 & 13 & 27 & 30 & 9 \\ 8 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[636] $21^{11} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -192, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 9 & 12 & 4 \\ 5 & 4 & 9 & 13 & 4 \\ 9 & 9 & 12 & 21 & 6 \\ 12 & 13 & 21 & 28 & 10 \\ 4 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[637] $21^{12} 31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -192, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 4 & 18 & 4 & 4 \\ 12 & 18 & 57 & 12 & 13 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 13 & 3 & 2 \end{pmatrix}_{b_0}$$

[638] $1^{12} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -196, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 9 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[639] $1^{112} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -196, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 28 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 18 & 28 & 18 & 154 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[640] $1^{11} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -197, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 3 \\ 4 & 9 & 4 & 36 & 5 \\ 2 & 4 & 0 & 15 & 2 \\ 14 & 36 & 15 & 121 & 19 \\ 3 & 5 & 2 & 19 & 1 \end{pmatrix}_{b_0}$$

[642] $1^{14} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 26 & 0 & 118 & 20 \\ 3 & 4 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[641] $1^{14} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 16 & 0 & 46 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[644] $1^{121} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -199, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 9 & 13 & 27 & 21 & 9 \\ 6 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[643] $1^{123} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -199, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 5 & 8 & 9 & 24 & 5 \\ 13 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[646] $1^{114} 3 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & 0 & 12 & 2 \\ 14 & 32 & 12 & 94 & 16 \\ 3 & 5 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

[645] $1^{114} 3 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -200, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 9 & 4 & 32 & 5 \\ 2 & 4 & 0 & 12 & 2 \\ 15 & 32 & 12 & 94 & 16 \\ 3 & 5 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

[648] $1^{112} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -201, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 12 & 21 & 6 \\ 8 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[647] $1^{112} 3 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -201, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 12 & 21 & 6 \\ 10 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[649] $1^{131} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -202, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 11 & 17 & 48 & 12 & 12 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[650] $1^{123} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -203, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 10 & 13 & 27 & 39 & 9 \\ 11 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[651] $1^{131} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 15 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 16 & 0 & 2 \end{pmatrix}_{b_0}$$

[652] $1^{114} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 20 & 3 \\ 4 & 9 & 4 & 40 & 5 \\ 2 & 4 & 0 & 18 & 2 \\ 20 & 40 & 18 & 154 & 22 \\ 3 & 5 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

[653] $1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 13 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 10 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[654] $1^{123} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -205, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 6 & 8 & 9 & 18 & 5 \\ 9 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[655] $1^{112} 3 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -205, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 24 & 0 & 121 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[656] $21^{111} 31^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -205, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 12 & 15 & 4 \\ 5 & 4 & 13 & 17 & 4 \\ 12 & 13 & 27 & 39 & 9 \\ 15 & 17 & 39 & 49 & 13 \\ 4 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[657] $1^{123} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 10 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 9 & 13 & 27 & 39 & 9 \\ 10 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[658] $21^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 4 & 9 & 9 & 4 \\ 6 & 9 & 12 & 15 & 6 \\ 6 & 9 & 15 & 13 & 7 \\ 3 & 4 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[659] $1^{131} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 7 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 12 & 17 & 48 & 27 & 12 \\ 7 & 9 & 27 & 13 & 7 \\ 3 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[660] $1^{141} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 1 & 3 \\ 3 & 4 & 21 & 0 & 4 \\ 15 & 21 & 75 & 0 & 15 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 15 & 0 & 2 \end{pmatrix}_{b_0}$$

[661] $1^{113} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -207, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 13 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 8 & 9 & 12 & 27 & 6 \\ 13 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[663] $1^{114} 3 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.42 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 12 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[667] $1^{14} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 21 & 0 & 76 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[668] $1^{113} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 12 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[669] $1^{112} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 13 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[666] $1^{114} 3 1^4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.42 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 12 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[665] $1^{114} 3 1^2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 10 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[662] $1^{112} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.42 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 11 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[664] $1^{112} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 14 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[670] $1^{114} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -209, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 21 & 3 \\ 4 & 9 & 4 & 44 & 5 \\ 2 & 4 & 0 & 21 & 2 \\ 21 & 44 & 21 & 193 & 25 \\ 3 & 5 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

[671] $1^{113} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -210, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 12 & 27 & 6 \\ 10 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[672] $1^{112} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -210, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 8 & 9 & 12 & 21 & 6 \\ 11 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[673] $1^{121} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 7 & 3 \\ 3 & 4 & 13 & 9 & 4 \\ 10 & 13 & 27 & 21 & 9 \\ 7 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$

[674] $1^{112} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 17 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[675] $1^{131} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 13 & 22 & 84 & 15 & 16 \\ 2 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[676] $1^{13} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 27 & 0 & 133 & 21 \\ 3 & 4 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[677] $1^{113} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 4 & 4 & 0 & 15 & 2 \\ 15 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[678] $1^{114} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -213, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 3 \\ 4 & 9 & 4 & 36 & 5 \\ 2 & 4 & 0 & 15 & 2 \\ 18 & 36 & 15 & 121 & 19 \\ 3 & 5 & 2 & 19 & 1 \end{pmatrix}_{b_0}$$

[679] $1^{112} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -213, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 7 & 9 & 12 & 21 & 6 \\ 11 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[680] $1^{112} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -213, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 7 & 9 & 12 & 21 & 6 \\ 9 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[681] $21^{122} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 6 & 3 \\ 4 & 4 & 13 & 8 & 4 \\ 9 & 13 & 27 & 18 & 9 \\ 6 & 8 & 18 & 10 & 6 \\ 3 & 4 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[682] $21^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 3 & 12 & 4 \\ 8 & 9 & 4 & 16 & 5 \\ 3 & 4 & 0 & 6 & 2 \\ 12 & 16 & 6 & 22 & 8 \\ 4 & 5 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[683] $1^{14} 3^{1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 16 & 26 & 117 & 0 & 19 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 19 & 0 & 2 \end{pmatrix}_{b_0}$$

[686] $1^{14} 3^{1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 10 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[687] $1^{113} 3^{1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 5 & 9 & 12 & 27 & 6 \\ 11 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[685] $1^{14} 3^{1^{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 3 & 4 & 0 & 24 & 2 \\ 20 & 36 & 24 & 238 & 28 \\ 3 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

[684] $1^{14} 3^{1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 14 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[688] $1^{123} 3^{1^{23}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -217, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 15 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 6 & 8 & 9 & 30 & 5 \\ 15 & 22 & 30 & 85 & 17 \\ 3 & 4 & 5 & 17 & 2 \end{pmatrix}_{b_0}$$

[689] $1^{131} 3^{1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.02 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 2 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 15 & 22 & 84 & 15 & 16 \\ 2 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[690] $1^{113} 3^{1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 10 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[691] $1^{13} 3^{1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.02 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 17 & 0 & 49 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[692] $1^{13} 3^{1^{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -220, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 1 & 3 \\ 3 & 4 & 22 & 0 & 4 \\ 16 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 16 & 0 & 3 \end{pmatrix}_{b_0}$$

[693] $1^{131} 3^{1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -221, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 6 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 11 & 17 & 48 & 27 & 12 \\ 6 & 9 & 27 & 13 & 7 \\ 3 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[694] $1^{13} 3 1^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -221, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 3 \\ 4 & 9 & 4 & 44 & 5 \\ 2 & 4 & 0 & 21 & 2 \\ 18 & 44 & 21 & 193 & 25 \\ 3 & 5 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

[695] $21^{112} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -223, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 3 & 15 & 4 \\ 8 & 9 & 4 & 20 & 5 \\ 3 & 4 & 0 & 9 & 2 \\ 15 & 20 & 9 & 37 & 11 \\ 4 & 5 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[696] $1^{114} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 24 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 4 & 4 & 0 & 24 & 2 \\ 24 & 36 & 24 & 238 & 28 \\ 3 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

[697] $1^{114} 3 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 11 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[698] $1^{112} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 12 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[699] $1^{113} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -225, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 5 & 9 & 12 & 27 & 6 \\ 10 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[700] $1^{114} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 5 & 9 & 12 & 33 & 6 \\ 14 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[701] $1^{131} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 6 & 3 \\ 3 & 4 & 17 & 9 & 4 \\ 12 & 17 & 48 & 27 & 12 \\ 6 & 9 & 27 & 13 & 7 \\ 3 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[702] $1^{12} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 18 & 0 & 58 & 14 \\ 3 & 4 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[703] $1^{131} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -227, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 2 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 14 & 22 & 84 & 15 & 16 \\ 2 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[704] $1^{12} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -228, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 23 & 0 & 100 & 18 \\ 3 & 4 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[705] $1^{123} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -228, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 6 & 8 & 9 & 24 & 5 \\ 11 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[706] $1^{123} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -229, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 5 & 8 & 9 & 24 & 5 \\ 10 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[708] $1^{111} 3 1^{123} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 7 & 13 & 27 & 39 & 9 \\ 9 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[707] $1^{131} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 12 & 17 & 48 & 12 & 12 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[709] $21^{122} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 6 & 3 \\ 4 & 4 & 8 & 8 & 4 \\ 6 & 8 & 9 & 12 & 5 \\ 6 & 8 & 12 & 10 & 6 \\ 3 & 4 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[710] $21^{121} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -232, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 4 & 13 & 4 & 4 \\ 9 & 13 & 27 & 9 & 9 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

[711] $1^{112} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -234, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 6 & 9 & 12 & 21 & 6 \\ 9 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[712] $1^{112} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 14 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[713] $1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 4 & 4 & 0 & 15 & 2 \\ 14 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[714] $1^{11} 3 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 11 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

[715] $1^{131} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 14 & 3 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 14 & 22 & 84 & 15 & 16 \\ 3 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$

[716] $1^{12} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 3 \\ 4 & 9 & 4 & 40 & 3 \\ 2 & 4 & 0 & 18 & 2 \\ 16 & 40 & 18 & 154 & 22 \\ 3 & 5 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

[717] $1^{12} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 28 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 17 & 28 & 18 & 154 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[718] $1^{12} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 28 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 20 & 28 & 18 & 154 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[719] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 10 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[720] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 13 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[721] $1^{14} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -236, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 17 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 4 & 4 & 0 & 18 & 2 \\ 17 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[722] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.03 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 3 & 3 \\ 3 & 4 & 17 & 4 & 4 \\ 13 & 17 & 48 & 12 & 12 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[723] $1^{14} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 14 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 7 & 9 & 12 & 33 & 6 \\ 14 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[724] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -239, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 13 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 7 & 8 & 9 & 24 & 5 \\ 13 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[726] $21^{13} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -240, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 3 & 3 \\ 4 & 4 & 12 & 4 & 4 \\ 9 & 12 & 24 & 9 & 8 \\ 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

[725] $1^{14} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -240, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 17 & 31 & 21 & 172 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[727] $1^{123} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -241, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.63 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 5 & 8 & 9 & 30 & 5 \\ 15 & 22 & 30 & 85 & 17 \\ 3 & 4 & 5 & 17 & 2 \end{pmatrix}_{b_0}$$

[728] $21^{13} 31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -241, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 4 & 5 & 17 & 4 \\ 3 & 5 & 3 & 15 & 3 \\ 12 & 17 & 15 & 49 & 13 \\ 3 & 4 & 3 & 13 & 2 \end{pmatrix}_{b_0}$$

[729] $1^{11} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -241, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 24 & 0 & 121 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[730] $1^{123} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -241, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 5 & 8 & 9 & 30 & 5 \\ 14 & 22 & 30 & 85 & 17 \\ 3 & 4 & 5 & 17 & 2 \end{pmatrix}_{b_0}$$

[731] $21^{123} 31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 18 & 4 \\ 5 & 4 & 4 & 18 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 18 & 18 & 12 & 58 & 14 \\ 4 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[732] $1^{112} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 15 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[733] $21^{111} 31^{122} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 12 & 12 & 4 \\ 5 & 4 & 13 & 13 & 4 \\ 12 & 13 & 27 & 30 & 9 \\ 12 & 13 & 30 & 28 & 10 \\ 4 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[734] $1^{112} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 12 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[735] $1^{114} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -245, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 20 & 3 \\ 4 & 9 & 4 & 44 & 5 \\ 2 & 4 & 0 & 21 & 2 \\ 20 & 44 & 21 & 193 & 25 \\ 3 & 5 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

[736] $1^{122} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -246, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 8 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 9 & 13 & 27 & 30 & 9 \\ 8 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[737] $1^{112} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -246, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 9 & 13 & 4 \\ 7 & 9 & 12 & 21 & 6 \\ 10 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[738] $1^{13} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 22 & 0 & 85 & 17 \\ 3 & 4 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[739] $1^{14} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 19 & 3 \\ 4 & 9 & 4 & 44 & 5 \\ 2 & 4 & 0 & 21 & 2 \\ 19 & 44 & 21 & 193 & 25 \\ 3 & 5 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[740] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 17 & 0 & 49 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[741] $1^{14} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -249, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.19 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 9 & 4 & 36 & 5 \\ 2 & 4 & 0 & 15 & 2 \\ 15 & 36 & 15 & 121 & 19 \\ 3 & 5 & 2 & 19 & 1 \end{pmatrix}_{b_0}$$

[742] $1^{12} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -250, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 18 & 0 & 58 & 14 \\ 3 & 4 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[743] $1^{12} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -250, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 18 & 0 & 58 & 14 \\ 3 & 4 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[744] $1^{12} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -251, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 16 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[745] $1^{11} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -253, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 24 & 0 & 121 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[747] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -255, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 17 & 0 & 49 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[746] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -255, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 12 & 27 & 6 \\ 13 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[748] $1^{13} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -255, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 19 & 32 & 21 & 193 & 25 \\ 3 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[749] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 0 & 12 & 2 \\ 13 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[750] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 8 & 13 & 27 & 30 & 9 \\ 9 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[751] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -257, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 17 & 3 \\ 4 & 9 & 4 & 36 & 5 \\ 2 & 4 & 0 & 15 & 2 \\ 17 & 36 & 15 & 121 & 19 \\ 3 & 5 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[752] $1^{12} 3 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -259, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 12 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[753] $1^{112} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -259, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 15 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[754] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -260, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 19 & 3 \\ 4 & 9 & 4 & 40 & 5 \\ 2 & 4 & 0 & 18 & 2 \\ 19 & 40 & 18 & 154 & 22 \\ 3 & 5 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

[755] $1^{13} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 32 & 0 & 193 & 25 \\ 3 & 4 & -1 & 25 & 2 \end{pmatrix}_{b_0}$$

[756] $1^{14} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -262, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 17 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 17 & 26 & 117 & 0 & 19 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 19 & 0 & 2 \end{pmatrix}_{b_0}$$

[757] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -262, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 5 & 9 & 12 & 33 & 6 \\ 12 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[758] $1^{114} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 11 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[759] $21^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -265, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 9 & 3 \\ 4 & 4 & 8 & 13 & 4 \\ 6 & 8 & 9 & 18 & 5 \\ 9 & 13 & 18 & 28 & 10 \\ 3 & 4 & 5 & 10 & 2 \end{pmatrix}_{b_0}$$

[760] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -265, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 5 & 8 & 9 & 24 & 5 \\ 12 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[761] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 11 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[762] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 11 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 10 & 13 & 27 & 30 & 9 \\ 11 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[763] $1^{112} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 28 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 18 & 28 & 18 & 154 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[764] $1^{112} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 28 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 19 & 28 & 18 & 154 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[765] $1^{113} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 19 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 4 & 4 & 0 & 18 & 2 \\ 19 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[766] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -268, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 0 & 12 & 2 \\ 12 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[767] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -269, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 3 \\ 4 & 9 & 4 & 36 & 5 \\ 2 & 4 & 0 & 15 & 2 \\ 16 & 36 & 15 & 121 & 19 \\ 3 & 5 & 2 & 19 & 1 \end{pmatrix}_{b_0}$$

[768] $1^{12} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -270, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 28 & 0 & 154 & 22 \\ 3 & 4 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[769] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 7 & 13 & 27 & 39 & 9 \\ 10 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[770] $1^{123} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -273, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 16 & 3 \\ 3 & 4 & 8 & 22 & 4 \\ 6 & 8 & 9 & 30 & 5 \\ 16 & 22 & 30 & 85 & 17 \\ 3 & 4 & 5 & 17 & 2 \end{pmatrix}_{b_0}$$

[771] $1^{123} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -275, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 5 & 8 & 9 & 24 & 5 \\ 11 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[772] $1^{122} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -275, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 8 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 8 & 13 & 27 & 30 & 9 \\ 8 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[775] $1^{114} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 17 & 3 \\ 4 & 9 & 4 & 40 & 5 \\ 2 & 4 & 0 & 18 & 2 \\ 17 & 40 & 18 & 154 & 22 \\ 3 & 5 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

[776] $1^{112} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 28 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 19 & 28 & 18 & 154 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[774] $1^{112} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 14 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[773] $1^{112} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -276, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 18 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 13 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[777] $1^{114} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -277, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 5 & 9 & 12 & 33 & 6 \\ 13 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[778] $1^{113} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -279, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 13 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[779] $1^{122} 3 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 9 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 10 & 13 & 27 & 30 & 9 \\ 9 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[782] $1^{114} 3 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 12 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[781] $1^{114} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 12 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

- [780] $21^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -280, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 3 & 24 & 4 \\ 8 & 9 & 4 & 32 & 5 \\ 3 & 4 & 0 & 15 & 2 \\ 24 & 32 & 15 & 100 & 18 \\ 4 & 5 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$
- [783] $1^{14}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -282, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 31 & 0 & 172 & 24 \\ 3 & 4 & -1 & 24 & 2 \end{pmatrix}_{b_0}$$
- [784] $1^{122}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -282, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 10 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 9 & 13 & 27 & 30 & 9 \\ 10 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$
- [785] $1^{131}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 15 & 3 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 15 & 22 & 84 & 15 & 16 \\ 3 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 2 \end{pmatrix}_{b_0}$$
- [786] $1^{114}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -284, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 3 \\ 4 & 9 & 4 & 40 & 5 \\ 2 & 4 & 0 & 18 & 2 \\ 18 & 40 & 18 & 154 & 22 \\ 3 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$
- [787] $1^{113}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -285, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 7 & 9 & 12 & 27 & 6 \\ 12 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [788] $1^{14}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -286, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 21 & 0 & 76 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$
- [789] $21^{121}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -289, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 6 & 3 \\ 4 & 4 & 13 & 9 & 4 \\ 9 & 13 & 27 & 21 & 9 \\ 6 & 9 & 21 & 13 & 7 \\ 3 & 4 & 9 & 7 & 2 \end{pmatrix}_{b_0}$$
- [792] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -291, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 11 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$
- [791] $1^{113}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -291, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 12 & 27 & 6 \\ 11 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [790] $1^{113}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -291, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 16 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$
- [794] $21^{123}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$

$$\begin{aligned}
& T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 6 & 15 & 3 \\ 4 & 4 & 8 & 22 & 4 \\ 6 & 8 & 9 & 30 & 5 \\ 15 & 22 & 30 & 85 & 17 \\ 3 & 4 & 5 & 17 & 2 \end{pmatrix}_{b_0} \\
[793] & 21^{112} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 4 & 4 & 18 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 12 & 18 & 12 & 58 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0} \\
[795] & 1^{141} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -294, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6 \\
& T_{\min} = 10, T_{\max} = 12, T \in [10, 12] \\
& \begin{pmatrix} 1 & 3 & 18 & 1 & 3 \\ 3 & 4 & 26 & 0 & 4 \\ 18 & 26 & 117 & 0 & 19 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 19 & 0 & 3 \end{pmatrix}_{b_0} \\
[796] & 1^{12} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -294, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 23 & 0 & 100 & 18 \\ 3 & 4 & -1 & 18 & 2 \end{pmatrix}_{b_0} \\
[797] & 21^{143} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -295, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 4 & 5 & 21 & 4 \\ 3 & 5 & 3 & 18 & 3 \\ 15 & 21 & 18 & 76 & 16 \\ 3 & 4 & 3 & 16 & 2 \end{pmatrix}_{b_0} \\
[798] & 21^{114} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -295, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.53 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 4 & 8 & 3 & 33 & 4 \\ 8 & 9 & 4 & 44 & 5 \\ 3 & 4 & 0 & 21 & 2 \\ 33 & 44 & 21 & 193 & 25 \\ 4 & 5 & 2 & 25 & 2 \end{pmatrix}_{b_0} \\
[799] & 1^{123} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -298, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 6 & 8 & 9 & 24 & 5 \\ 12 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0} \\
[800] & 1^{113} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -299, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 16 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 4 & 4 & 0 & 15 & 2 \\ 16 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0} \\
[801] & 1^{112} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -300, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 16 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0} \\
[803] & 1^{123} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 8 & 12 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 8 & 13 & 27 & 39 & 9 \\ 12 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0} \\
[802] & 21^{113} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 9 & 12 & 9 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0} \\
[805] & 1^{114} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -306, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]
\end{aligned}$$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 6 & 9 & 12 & 33 & 6 \\ 13 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[804] $1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -306, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 6 & 9 & 12 & 27 & 6 \\ 12 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[806] $1^{131} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -307, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 16 & 3 & 3 \\ 3 & 4 & 22 & 4 & 4 \\ 16 & 22 & 84 & 15 & 16 \\ 3 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

[807] $21^{112} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -307, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 4 & 4 & 13 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 9 & 13 & 9 & 28 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[808] $1^{123} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -312, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 8 & 17 & 4 \\ 6 & 8 & 9 & 24 & 5 \\ 13 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[809] $21^{131} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -316, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 24 & 4 \\ 5 & 4 & 4 & 23 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 24 & 23 & 15 & 100 & 18 \\ 4 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[810] $1^{14} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -318, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 26 & 0 & 118 & 20 \\ 3 & 4 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[811] $1^{13} 1 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -320, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 15 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[812] $1^{13} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -321, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.03 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 22 & 0 & 85 & 17 \\ 3 & 4 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[813] $1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -323, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 4 & 4 & 0 & 15 & 2 \\ 15 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[814] $1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -323, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 12 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[816] $1^{112} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -324, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 23 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 17 & 23 & 15 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[815] $1^{113} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -324, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 13 & 3 \\ 3 & 4 & 9 & 17 & 4 \\ 7 & 9 & 12 & 27 & 6 \\ 13 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[817] $1^{112} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -324, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 28 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 20 & 28 & 18 & 154 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[818] $1^{122} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 9 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 9 & 13 & 27 & 30 & 9 \\ 9 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[819] $1^{13} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 27 & 0 & 133 & 21 \\ 3 & 4 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[823] $21^{12} 31^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 21 & 4 \\ 5 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 21 & 22 & 15 & 85 & 17 \\ 4 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[822] $1^{114} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 21 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 4 & 4 & 0 & 21 & 2 \\ 21 & 31 & 21 & 172 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[821] $1^{113} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 12 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[820] $1^{113} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 13 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[824] $1^{123} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -332, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 10 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 8 & 13 & 27 & 39 & 9 \\ 10 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[825] $1^{122} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -335, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 10 & 3 \\ 3 & 4 & 13 & 13 & 4 \\ 10 & 13 & 27 & 30 & 9 \\ 10 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[826] $1^{12} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -336, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 23 & 0 & 100 & 18 \\ 3 & 4 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[827] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -339, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 13 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[829] $21^{143} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -340, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 30 & 4 \\ 5 & 4 & 4 & 28 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 30 & 28 & 18 & 154 & 22 \\ 4 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[828] $1^{14} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -340, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 21 & 0 & 76 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[830] $21^{131} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -343, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 6 & 3 \\ 4 & 4 & 17 & 9 & 4 \\ 12 & 17 & 48 & 27 & 12 \\ 6 & 9 & 27 & 13 & 7 \\ 3 & 4 & 12 & 7 & 2 \end{pmatrix}_{b_0}$$

[831] $21^{112} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -345, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 9 & 3 \\ 4 & 4 & 9 & 13 & 4 \\ 6 & 9 & 12 & 21 & 6 \\ 9 & 13 & 21 & 28 & 10 \\ 3 & 4 & 6 & 10 & 2 \end{pmatrix}_{b_0}$$

[832] $1^{14} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -349, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 15 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 7 & 9 & 12 & 33 & 6 \\ 15 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[834] $21^{113} 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -352, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 21 & 3 \\ 4 & 4 & 4 & 32 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 21 & 32 & 21 & 193 & 25 \\ 3 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[833] $21^{123} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -352, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 9 & 3 \\ 4 & 4 & 8 & 12 & 4 \\ 6 & 8 & 9 & 18 & 5 \\ 9 & 12 & 18 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[836] $1^{12} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -354, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 23 & 0 & 100 & 18 \\ 3 & 4 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[835] $1^{12} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -354, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 28 & 0 & 154 & 22 \\ 3 & 4 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[837] $1^1 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -355, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 2 & 4 & 0 & 21 & 2 \\ 18 & 32 & 21 & 193 & 25 \\ 3 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[838] $1^{113}31^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -355, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 16 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[839] $1^{14}31^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -358, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 36 & 0 & 238 & 28 \\ 3 & 4 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[841] $1^{12}31^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 14 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[840] $1^{114}31^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 4 & 4 & 0 & 18 & 2 \\ 18 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[842] $21^{113}31^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -364, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 3 & 18 & 4 \\ 8 & 9 & 4 & 24 & 5 \\ 3 & 4 & 0 & 9 & 2 \\ 18 & 24 & 9 & 52 & 12 \\ 4 & 5 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[843] $1^{114}31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -366, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 15 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 6 & 9 & 12 & 33 & 6 \\ 15 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[844] $1^{113}31^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -369, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 22 & 0 & 85 & 17 \\ 3 & 4 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[845] $1^{114}31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -369, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 9 & 21 & 4 \\ 6 & 9 & 12 & 33 & 6 \\ 14 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[846] $1^{14}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -370, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 21 & 0 & 76 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[847] $1^{113}31^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -371, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 14 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[848] $1^{123}31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -372, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 13 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 9 & 13 & 27 & 39 & 9 \\ 13 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[849] $1^{113} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -375, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 14 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[850] $21^{123} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 24 & 4 \\ 5 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 24 & 26 & 18 & 118 & 20 \\ 4 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[851] $21^{131} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.03 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 3 & 3 \\ 4 & 4 & 17 & 4 & 4 \\ 12 & 17 & 48 & 12 & 12 \\ 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

[852] $1^{114} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 15 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[853] $1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -379, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 13 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[855] $1^{123} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -379, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 11 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 8 & 13 & 27 & 39 & 9 \\ 11 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[854] $1^{113} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -379, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 15 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[856] $1^{123} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 11 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 9 & 13 & 27 & 39 & 9 \\ 11 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[857] $1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 13 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[858] $1^{123} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -388, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 12 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 10 & 13 & 27 & 39 & 9 \\ 12 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[859] $1^{13} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -393, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 32 & 0 & 193 & 25 \\ 3 & 4 & -1 & 25 & 2 \end{pmatrix}_{b_0}$$

[860] $1^{13} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -393, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 22 & 0 & 85 & 17 \\ 3 & 4 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[861] \ 1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -395, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 15 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[862] \ 1^{113} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -399, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 20 & 32 & 21 & 193 & 25 \\ 3 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[863] \ 21^{131} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -400, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$$

$$\begin{pmatrix} 1 & 4 & 15 & 3 & 3 \\ 4 & 4 & 22 & 4 & 4 \\ 15 & 22 & 84 & 15 & 16 \\ 3 & 4 & 15 & 1 & 3 \\ 3 & 4 & 16 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[864] \ 1^{113} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -400, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 16 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[865] \ 1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -403, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 0 & 15 & 2 \\ 14 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[866] \ 1^{113} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -411, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 17 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[868] \ 1^{14} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -414, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 26 & 0 & 118 & 20 \\ 3 & 4 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[867] \ 1^{12} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -414, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 28 & 0 & 154 & 22 \\ 3 & 4 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[869] \ 1^{13} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -415, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 27 & 0 & 133 & 21 \\ 3 & 4 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[870] \ 1^{14} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -420, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 31 & 0 & 172 & 24 \\ 3 & 4 & -1 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[871] \ 1^{114} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 18 & 31 & 21 & 172 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[872] $1^{114} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 3 & 4 & 0 & 24 & 2 \\ 21 & 36 & 24 & 238 & 28 \\ 3 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

[873] $21^{113} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -433, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 3 & 21 & 4 \\ 8 & 9 & 4 & 28 & 5 \\ 3 & 4 & 0 & 12 & 2 \\ 21 & 28 & 12 & 73 & 15 \\ 4 & 5 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[874] $1^{123} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 12 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 9 & 13 & 27 & 39 & 9 \\ 12 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[875] $1^{113} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -439, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 15 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[876] $1^{113} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -443, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 2 & 4 & 0 & 21 & 2 \\ 19 & 32 & 21 & 193 & 25 \\ 3 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[877] $1^{113} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -448, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 19 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[878] $1^{113} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -448, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 17 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[879] $1^{12} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -450, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 28 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 28 & 0 & 154 & 22 \\ 3 & 4 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[880] $1^{123} 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -451, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 13 & 3 \\ 3 & 4 & 13 & 17 & 4 \\ 10 & 13 & 27 & 39 & 9 \\ 13 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$

[881] $1^{114} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -456, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 14 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[882] $1^{114} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -460, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 15 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[883] $1^{113} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -464, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 18 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[884] $1^{13} 3 1^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -467, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 2 & 4 & 0 & 21 & 2 \\ 17 & 31 & 21 & 172 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[886] $21^{113} 31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -471, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 12 & 3 \\ 4 & 4 & 9 & 17 & 4 \\ 6 & 9 & 12 & 27 & 6 \\ 12 & 17 & 27 & 49 & 13 \\ 3 & 4 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[885] $1^{113} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -471, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 16 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[887] $21^{114} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -472, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 12 & 16 & 12 & 46 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[888] $1^{13} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -481, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 27 & 0 & 133 & 21 \\ 3 & 4 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[889] $21^{123} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -481, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 12 & 3 \\ 4 & 4 & 8 & 17 & 4 \\ 6 & 8 & 9 & 24 & 5 \\ 12 & 17 & 24 & 49 & 13 \\ 3 & 4 & 5 & 13 & 2 \end{pmatrix}_{b_0}$$

[890] $1^{14} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -486, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 26 & 0 & 118 & 20 \\ 3 & 4 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[891] $21^{122} 31^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -487, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.7$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 9 & 3 \\ 4 & 4 & 13 & 13 & 4 \\ 9 & 13 & 27 & 30 & 9 \\ 9 & 13 & 30 & 28 & 10 \\ 3 & 4 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

[892] $1^{114} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -496, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 15 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[893] $1^{113} 3 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -499, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 2 & 4 & 0 & 21 & 2 \\ 20 & 32 & 21 & 193 & 25 \\ 3 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[895] $1^{113} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -499, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 18 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[894] $21^{133} 31^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -499, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 27 & 4 \\ 5 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 27 & 27 & 18 & 133 & 21 \\ 4 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[896] $1^{112} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -501, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 32 & 0 & 193 & 25 \\ 3 & 4 & -1 & 25 & 2 \end{pmatrix}_{b_0}$$

[897] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -504, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 16 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[898] $1^{113} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -511, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 21 & 32 & 21 & 193 & 25 \\ 3 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[899] $1^{114} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -514, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 36 & 0 & 238 & 28 \\ 3 & 4 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

[900] $1^{113} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -515, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 22 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 2 & 4 & 0 & 21 & 2 \\ 22 & 32 & 21 & 193 & 25 \\ 3 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[903] $1^{114} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -523, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 27 & 0 & 133 & 21 \\ 3 & 4 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[902] $21^{113} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -523, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 12 & 3 \\ 4 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 12 & 2 \\ 12 & 17 & 12 & 49 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[901] $1^{113} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -523, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 2 & 4 & 0 & 21 & 2 \\ 21 & 32 & 21 & 193 & 25 \\ 3 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[904] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -524, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 16 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[906] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -534, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 26 & 0 & 118 & 20 \\ 3 & 4 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[905] \ 1^{14} 3^{12} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -534, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 31 & 0 & 172 & 24 \\ 3 & 4 & -1 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[907] \ 1^{14} 3^{11} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -544, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 2 & 4 & 0 & 24 & 2 \\ 20 & 36 & 24 & 238 & 28 \\ 3 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[908] \ 21^{113} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -547, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 18 & 3 \\ 4 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 18 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[909] \ 1^{113} 3^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -555, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 19 & 27 & 18 & 133 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[911] \ 1^{114} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -556, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 18 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[910] \ 1^{114} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -556, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 0 & 18 & 2 \\ 17 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[912] \ 1^{114} 3^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -571, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 2 & 4 & 0 & 21 & 2 \\ 18 & 31 & 21 & 172 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[913] \ 21^{114} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -574, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 8 & 3 & 24 & 4 \\ 8 & 9 & 4 & 32 & 5 \\ 3 & 4 & 0 & 12 & 2 \\ 24 & 32 & 12 & 94 & 16 \\ 4 & 5 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[914] \ 1^{114} 3^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -576, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 19 & 31 & 21 & 172 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[915] \ 1^{113} 3^{11} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -585, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 32 & 0 & 193 & 25 \\ 3 & 4 & -1 & 25 & 2 \end{pmatrix}_{b_0}$$

$$[916] \ 21^{114} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -585, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 15 & 3 \\ 4 & 4 & 9 & 21 & 4 \\ 6 & 9 & 12 & 33 & 6 \\ 15 & 21 & 33 & 76 & 16 \\ 3 & 4 & 6 & 16 & 2 \end{pmatrix}_{b_0}$$

[917] $1^{113} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -591, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 22 & 3 \\ 3 & 4 & 4 & 32 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 22 & 32 & 21 & 193 & 25 \\ 3 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[918] $1^{114} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 17 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[919] $1^{114} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 22 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 3 & 4 & 0 & 24 & 2 \\ 22 & 36 & 24 & 238 & 28 \\ 3 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

[920] $21^{113} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -604, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 15 & 22 & 15 & 85 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[921] $21^{114} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -610, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 3 & 30 & 4 \\ 8 & 9 & 4 & 40 & 5 \\ 3 & 4 & 0 & 18 & 2 \\ 30 & 40 & 18 & 154 & 22 \\ 4 & 5 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[922] $21^{14} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -619, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 33 & 4 \\ 5 & 4 & 4 & 32 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 33 & 32 & 21 & 193 & 25 \\ 4 & 4 & 2 & 25 & 2 \end{pmatrix}_{b_0}$$

[923] $1^{114} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -624, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 31 & 0 & 172 & 24 \\ 3 & 4 & -1 & 24 & 2 \end{pmatrix}_{b_0}$$

[924] $21^{13} 31^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -640, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 30 & 4 \\ 5 & 4 & 4 & 31 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 30 & 31 & 21 & 172 & 24 \\ 4 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[925] $1^{114} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -643, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 2 & 4 & 0 & 21 & 2 \\ 19 & 31 & 21 & 172 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[926] $1^{13} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -645, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 32 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 22 & 32 & 0 & 193 & 25 \\ 3 & 4 & -1 & 25 & 2 \end{pmatrix}_{b_0}$$

[927] $1^{14} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -646, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 22 & 36 & 0 & 238 & 28 \\ 3 & 4 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

- [928] $1^{114}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -656$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.55 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 2 & 4 & 0 & 24 & 2 \\ 21 & 36 & 24 & 238 & 28 \\ 3 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$
- [929] $1^{114}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -664$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.12 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 18 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$
- [930] $21^{123}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -673$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.06 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 12 & 3 \\ 4 & 4 & 13 & 17 & 4 \\ 9 & 13 & 27 & 39 & 9 \\ 12 & 17 & 39 & 49 & 13 \\ 3 & 4 & 9 & 13 & 2 \end{pmatrix}_{b_0}$$
- [931] $1^{114}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -683$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.38 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 2 & 4 & 0 & 21 & 2 \\ 20 & 31 & 21 & 172 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$
- [932] $21^{115}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -688$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.1$
 $T_{\min} = 10.30 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 4 & 4 & 20 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 15 & 20 & 15 & 73 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$
- [933] $1^{114}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -690$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.06 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 31 & 0 & 172 & 24 \\ 3 & 4 & -1 & 24 & 2 \end{pmatrix}_{b_0}$$
- [934] $1^{114}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -691$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 2 & 4 & 0 & 21 & 2 \\ 21 & 31 & 21 & 172 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$
- [935] $1^{114}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -696$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.11 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 20 & 31 & 21 & 172 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$
- [936] $21^{114}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -703$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.30 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 3 & 27 & 4 \\ 8 & 9 & 4 & 36 & 5 \\ 3 & 4 & 0 & 15 & 2 \\ 27 & 36 & 15 & 121 & 19 \\ 4 & 5 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$
- [937] $1^{114}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -736$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 22 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 2 & 4 & 0 & 24 & 2 \\ 22 & 36 & 24 & 238 & 28 \\ 3 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$
- [938] $1^{114}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -744$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.11 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 23 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 3 & 4 & 0 & 24 & 2 \\ 23 & 36 & 24 & 238 & 28 \\ 3 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$
- [939] $1^{114}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -754$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 36 & 0 & 238 & 28 \\ 3 & 4 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[941] \ 1^{14} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -784, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 23 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 2 & 4 & 0 & 24 & 2 \\ 23 & 36 & 24 & 238 & 28 \\ 3 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[940] \ 1^{14} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -784, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 21 & 31 & 21 & 172 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[942] \ 21^{14} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -787, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 15 & 3 \\ 4 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 15 & 2 \\ 15 & 21 & 15 & 76 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[943] \ 1^{14} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -800, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 24 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 2 & 4 & 0 & 24 & 2 \\ 24 & 36 & 24 & 238 & 28 \\ 3 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[944] \ 1^{14} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -838, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 24 & 3 \\ 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 36 & 0 & 238 & 28 \\ 3 & 4 & -1 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[945] \ 21^{15} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -844, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.1$$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 8 & 3 & 30 & 4 \\ 8 & 9 & 4 & 40 & 5 \\ 3 & 4 & 0 & 15 & 2 \\ 30 & 40 & 15 & 148 & 20 \\ 4 & 5 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[946] \ 21^{14} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -856, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 36 & 4 \\ 5 & 4 & 4 & 36 & 4 \\ 3 & 4 & 0 & 24 & 2 \\ 36 & 36 & 24 & 238 & 28 \\ 4 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[947] \ 1^{14} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -856, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 24 & 3 \\ 3 & 4 & 4 & 36 & 4 \\ 3 & 4 & 0 & 24 & 2 \\ 24 & 36 & 24 & 238 & 28 \\ 3 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[948] \ 21^{14} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -904, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 24 & 3 \\ 4 & 4 & 4 & 36 & 4 \\ 3 & 4 & 0 & 24 & 2 \\ 24 & 36 & 24 & 238 & 28 \\ 3 & 4 & 2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[949] \ 21^{16} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -952, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 18 & 3 \\ 4 & 4 & 4 & 24 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 18 & 24 & 18 & 106 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[950] \ 21^{14} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -964, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 18 & 3 \\ 4 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 18 & 2 \\ 18 & 26 & 18 & 118 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[951] $21^{114}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -1003, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 21 & 3 \\ 4 & 4 & 4 & 31 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 21 & 31 & 21 & 172 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[952] $21^{117}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -1264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 21 & 3 \\ 4 & 4 & 4 & 28 & 4 \\ 3 & 4 & 0 & 21 & 2 \\ 21 & 28 & 21 & 145 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

$T = 8$ [955] $21^1 4 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 6 & 6 & 4 \\ 5 & 4 & 7 & 7 & 4 \\ 6 & 7 & 7 & 10 & 5 \\ 6 & 7 & 10 & 8 & 6 \\ 4 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[956] $1^1 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 3 & 0 & 1 & 3 \\ 2 & 2 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[954] $1^{11} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 3 & 9 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 5 & 9 & 5 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[953] $1^1 3 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

[957] $21^{11} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 6 & 12 & 4 \\ 5 & 4 & 7 & 11 & 4 \\ 6 & 7 & 7 & 16 & 5 \\ 12 & 11 & 16 & 28 & 10 \\ 4 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[958] $21^1 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 3 & 9 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 5 & 9 & 5 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[961] $1^1 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 3 & 5 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 5 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[959] $1^1 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 5 & 0 & 2 & 4 \\ 3 & 5 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

[960] $1^{11} 4 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 7 & 8 & 4 \\ 5 & 7 & 7 & 11 & 5 \\ 7 & 8 & 11 & 13 & 7 \\ 3 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[962] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 4 & 2 \\ 2 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[963] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[964] $21^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[965] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[966] $21^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 3 & 5 & 4 \\ 3 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[967] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 3 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[968] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 3 & 5 & 4 \\ 3 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[969] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[971] $21^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 3 & 7 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 6 & 7 & 5 & 8 & 6 \\ 4 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[970] $1^{11} 3^{11} 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 5 & 5 & 0 & 5 \\ 3 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[977] $1^{11} 3 2 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[976] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 7 & 5 & 4 \\ 5 & 7 & 7 & 6 & 5 \\ 3 & 5 & 6 & 4 & 4 \\ 3 & 4 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[975] $1^{11} 4 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 5 & 7 & 7 & 10 & 5 \\ 7 & 7 & 10 & 8 & 6 \\ 3 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[974] $2 1^1 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 6 & 3 & 4 \\ 5 & 4 & 7 & 3 & 4 \\ 6 & 7 & 7 & 5 & 5 \\ 3 & 3 & 5 & 1 & 3 \\ 4 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[972] $1^{12} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 3 & 3 & 0 & 7 & 2 \\ 6 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[973] $1^{11} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[978] $2 1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 6 & 9 & 4 \\ 5 & 4 & 7 & 8 & 4 \\ 6 & 7 & 7 & 11 & 5 \\ 9 & 8 & 11 & 13 & 7 \\ 4 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[982] $2 1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 4 & 5 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 6 & 5 & 3 & 4 & 4 \\ 4 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[984] $2 1^{11} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 7 & 3 & 3 \\ 4 & 4 & 10 & 4 & 4 \\ 7 & 10 & 20 & 7 & 8 \\ 3 & 4 & 7 & 1 & 3 \\ 3 & 4 & 8 & 3 & 3 \end{pmatrix}_{b_0}$$

[983] $1^{11} 3 2 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 1 & 3 & 0 & 1 & 3 \\ 2 & 2 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[981] $1^{11} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 7 & 13 & 4 \\ 4 & 7 & 7 & 20 & 5 \\ 9 & 13 & 20 & 40 & 12 \\ 3 & 4 & 5 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[980] \ 1 \overset{1}{4} \overset{1}{1} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 16, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 7 & 5 & 4 \\ 4 & 7 & 7 & 6 & 5 \\ 3 & 5 & 6 & 4 & 4 \\ 3 & 4 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[979] \ 21 \overset{1}{1} \overset{1}{4} \overset{1}{1} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 16, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 4 & 7 & 4 \\ 3 & 4 & 0 & 5 & 2 \\ 6 & 7 & 5 & 8 & 6 \\ 4 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[985] \ 1 \overset{1}{1} \overset{1}{4} \overset{1}{1} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 17, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.59 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 4 & 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[988] \ 1 \overset{1}{4} \overset{1}{1} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 18, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.78 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 3 & 5 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[986] \ 1 \overset{1}{1} \overset{1}{4} \overset{1}{1} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 18, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.78 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 7 & 8 & 4 \\ 5 & 7 & 7 & 11 & 5 \\ 5 & 8 & 11 & 13 & 7 \\ 3 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[989] \ 1 \overset{1}{1} \overset{1}{1} \overset{1}{4} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 18, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.78 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 7 & 10 & 4 \\ 5 & 7 & 7 & 15 & 5 \\ 9 & 10 & 15 & 21 & 9 \\ 3 & 4 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[987] \ 1 \overset{1}{1} \overset{1}{4} \overset{1}{1} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 18, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.78 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 3 & 3 & 0 & 6 & 2 \\ 5 & 8 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[990] \ 1 \overset{1}{1} \overset{1}{1} \overset{1}{4} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 18, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.78 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 7 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[994] \ 1 \overset{1}{1} \overset{1}{1} \overset{1}{4} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 19, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 7 & 4 & 4 \\ 6 & 7 & 7 & 5 & 5 \\ 2 & 4 & 5 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[991] \ 21 \overset{1}{1} \overset{1}{4} \overset{1}{1} \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 19, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 6 & 3 & 4 \\ 5 & 4 & 7 & 4 & 4 \\ 6 & 7 & 7 & 5 & 5 \\ 3 & 4 & 5 & 1 & 3 \\ 4 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[992] $1^{11} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 3 & 3 & 0 & 8 & 2 \\ 6 & 10 & 8 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[993] $221^{11} 41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 7 & 6 & 4 \\ 5 & 7 & 7 & 10 & 5 \\ 5 & 6 & 10 & 8 & 6 \\ 3 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0}$

[996] $1^{11} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 7 & 5 & 5 \\ 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}$

[995] $21^{11} 41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 7 & 5 & 5 \\ 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}$

[997] $21^{11} 41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 3 & 4 & 2 & 1 & 3 \\ 4 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[998] $1^{11} 41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 3 & 5 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 4 & 5 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$

[999] $1^1 3 21^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 5 & 0 & 2 & 4 \\ 3 & 5 & -1 & 4 & 4 \end{pmatrix}_{b_0}$

[1000] $141^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[1001] $1^{11} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 22, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 4 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[1003] $1^{12} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 7 & 8 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$

[1006] $1^{11} 4^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 5 & 8 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0} \\
[1005] & 1 \overset{11}{4} 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 7 & 8 & 4 \\ 4 & 7 & 7 & 11 & 5 \\ 5 & 8 & 11 & 13 & 7 \\ 3 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0} \\
[1004] & 1 \overset{11}{4} 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 4 & 7 & 7 & 5 & 5 \\ 2 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0} \\
[1002] & 1 \overset{13}{11} 4 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 9 & 10 & 8 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[1012] & 221 \overset{11}{41} 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 6 & 5 & 3 & 3 \\ 6 & 13 & 12 & 8 & 7 \\ 5 & 12 & 7 & 6 & 5 \\ 3 & 8 & 6 & 4 & 4 \\ 3 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0} \\
[1010] & 1 \overset{13}{12} 4 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 3 & 3 & 0 & 12 & 2 \\ 10 & 18 & 12 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[1011] & 1 \overset{11}{11} 4 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 3 & 3 & 0 & 6 & 2 \\ 6 & 8 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0} \\
[1013] & 21 \overset{11}{11} 4 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 3 & 3 & 0 & 6 & 2 \\ 6 & 8 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0} \\
[1007] & 1 \overset{11}{11} 4 1^{\overset{12}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 6 & 7 & 7 & 10 & 5 \\ 4 & 7 & 10 & 8 & 6 \\ 3 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0} \\
[1008] & 1 \overset{11}{11} 3 21^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 5 & 0 & 10 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 10 & 0 & 14 & 8 \\ 3 & 5 & -1 & 8 & 4 \end{pmatrix}_{b_0} \\
[1009] & 1 \overset{11}{11} 3 21^{\overset{12}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 5 & 0 & 10 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 10 & 0 & 14 & 8 \\ 3 & 5 & -1 & 8 & 4 \end{pmatrix}_{b_0} \\
[1014] & 1 \overset{12}{11} 4 1^{\overset{11}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0
\end{aligned}$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 6 & 7 & 7 & 5 & 5 \\ 2 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1015] \ 21^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 7 & 9 & 4 \\ 5 & 7 & 7 & 11 & 5 \\ 5 & 9 & 11 & 13 & 7 \\ 3 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1019] \ 1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1018] \ 1^{14} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 3 \\ 3 & 4 & 7 & 21 & 4 \\ 5 & 7 & 7 & 31 & 5 \\ 17 & 21 & 31 & 101 & 19 \\ 3 & 4 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1017] \ 1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 7 & 9 & 4 \\ 5 & 7 & 7 & 11 & 5 \\ 5 & 9 & 11 & 13 & 7 \\ 3 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1016] \ 21^{14} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 17 & 3 \\ 3 & 4 & 7 & 21 & 4 \\ 5 & 7 & 7 & 31 & 5 \\ 17 & 21 & 31 & 101 & 19 \\ 3 & 4 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1020] \ 1^{11} 3 2 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1022] \ 1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 7 & 18 & 4 \\ 5 & 7 & 7 & 26 & 5 \\ 11 & 18 & 26 & 70 & 16 \\ 3 & 4 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1021] \ 1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 7 & 5 & 5 \\ 2 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1023] \ 1^{14} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 7 & 12 & 4 \\ 5 & 7 & 7 & 16 & 5 \\ 11 & 12 & 16 & 26 & 10 \\ 3 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1026] \ 1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 7 & 10 & 4 \\ 5 & 7 & 7 & 15 & 5 \\ 6 & 10 & 15 & 21 & 9 \\ 3 & 4 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1024] \ 1 \overset{1}{\underset{1}{4}} 1 \overset{11}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 6 & 8 & 6 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1029] \ 1 \overset{11}{\underset{1}{4}} 1 \overset{11}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 4 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1028] \ 1 \overset{12}{\underset{1}{4}} 1 \overset{3}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 2 & 3 & 0 & 4 & 2 \\ 7 & 8 & 4 & 10 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1027] \ 1 \overset{11}{\underset{1}{4}} 1 \overset{11}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 7 & 8 & 4 \\ 5 & 7 & 7 & 11 & 5 \\ 6 & 8 & 11 & 13 & 7 \\ 3 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1025] \ 21 \overset{11}{\underset{1}{4}} 1 \overset{11}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 7 & 8 & 4 \\ 5 & 7 & 7 & 11 & 5 \\ 6 & 8 & 11 & 13 & 7 \\ 3 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1033] \ 1 \overset{11}{\underset{1}{4}} 1 \overset{12}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 6 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1032] \ 1 \overset{12}{\underset{1}{4}} 1 \overset{11}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 3 & 16 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 9 & 16 & 10 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1031] \ 21 \overset{11}{\underset{1}{4}} 1 \overset{13}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 6 & 18 & 4 \\ 5 & 4 & 7 & 18 & 4 \\ 6 & 7 & 7 & 26 & 5 \\ 18 & 18 & 26 & 70 & 16 \\ 4 & 4 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1030] \ 1 \overset{12}{\underset{1}{4}} 1 \overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 8 & 14 & 10 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1036] \ 21 \overset{11}{\underset{1}{4}} 1 \overset{14}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 36, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.89 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 18 & 4 \\ 5 & 4 & 3 & 21 & 4 \\ 3 & 3 & 0 & 11 & 2 \\ 18 & 21 & 11 & 76 & 16 \\ 4 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1035] $21^{11} 4 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 4 & 5 & 3 & 30 & 4 \\ 5 & 4 & 3 & 33 & 4 \\ 3 & 3 & 0 & 23 & 2 \\ 30 & 33 & 23 & 212 & 28 \\ 4 & 4 & 2 & 28 & 3 \end{pmatrix}_{b_0}$

[1034] $21^{11} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 3 & 5 & 4 \\ 3 & 3 & 0 & 3 & 2 \\ 6 & 5 & 3 & 4 & 4 \\ 4 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$

[1040] $1^{11} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 5 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[1037] $21^{11} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 5 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[1039] $1^{11} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 3 & 3 & 0 & 8 & 2 \\ 7 & 12 & 8 & 26 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$

[1038] $1^{12} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 3 & 3 & 0 & 13 & 2 \\ 10 & 17 & 13 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1041] $1^{111} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 7 & 4 & 4 \\ 4 & 7 & 7 & 5 & 5 \\ 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}$

[1042] $1^{11} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 7 & 10 & 8 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[1043] $1^{12} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 2 & 3 & 0 & 4 & 2 \\ 5 & 8 & 4 & 10 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[1044] $1^{111} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 7 & 4 & 4 \\ 5 & 7 & 7 & 5 & 5 \\ 2 & 4 & 5 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}$

[1045] $1^{11} 4 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 8 & 10 & 8 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1046] $21^{1^1} 4^{1^{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 7 & 4 & 4 \\ 5 & 7 & 7 & 5 & 5 \\ 2 & 4 & 5 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[1047] $1^{1^1} 4^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.63 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 6 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[1048] $21^{1^1} 4^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 3 \\ 4 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 10 & 15 & 11 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1049] $1^1 4^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 3 & 9 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 5 & 9 & 5 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[1050] $21^{1^1} 4^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 5 & 7 & 3 \\ 4 & 4 & 7 & 10 & 4 \\ 5 & 7 & 7 & 15 & 5 \\ 7 & 10 & 15 & 21 & 9 \\ 3 & 4 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[1051] $21^{1^1} 4^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 9 & 4 \\ 5 & 4 & 3 & 8 & 4 \\ 3 & 3 & 0 & 6 & 2 \\ 9 & 8 & 6 & 13 & 7 \\ 4 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[1052] $21^{1^1} 4^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 3 & 3 & 0 & 9 & 2 \\ 9 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1053] $1^{1^1} 4^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 3 & 3 & 0 & 9 & 2 \\ 9 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1055] $1^{1^1} 4^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 9 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 8 & 9 & 5 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[1056] $1^{1^1} 4^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 5 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[1054] $1^{1^2} 4^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 8 & 4 \\ 2 & 3 & 0 & 4 & 2 \\ 6 & 8 & 4 & 10 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1057] \ 1^{14}_4 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 3 & 27 & 4 \\ 3 & 3 & 0 & 17 & 2 \\ 15 & 27 & 17 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[1058] \ 1^{11}_4 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 3 & 3 & 0 & 14 & 2 \\ 11 & 18 & 14 & 70 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1059] \ 21^{11}_4 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 6 & 9 & 4 \\ 5 & 4 & 7 & 9 & 4 \\ 6 & 7 & 7 & 11 & 5 \\ 9 & 9 & 11 & 13 & 7 \\ 4 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1060] \ 1^{14}_4 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 3 & 25 & 4 \\ 3 & 3 & 0 & 17 & 2 \\ 14 & 25 & 17 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1061] \ 1^{12}_4 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 3 & 3 & 0 & 7 & 2 \\ 7 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1062] \ 21^{11}_4 4 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 3 & 3 & 0 & 7 & 2 \\ 7 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1063] \ 1^{11}_4 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 3 & 3 & 0 & 11 & 2 \\ 9 & 15 & 11 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1064] \ 1^{11}_3 3 21^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[1065] \ 1^4_1 4 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 6 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1066] \ 1^{13}_4 4 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 3 & 3 & 0 & 15 & 2 \\ 12 & 21 & 15 & 88 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1068] $1^{11} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 7 & 9 & 4 \\ 4 & 7 & 7 & 11 & 5 \\ 6 & 9 & 11 & 13 & 7 \\ 3 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[1069] $21^{11} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.64 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 4 & 3 & 4 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[1070] $1^{11} 4^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 7 & 10 & 4 \\ 4 & 7 & 7 & 15 & 5 \\ 7 & 10 & 15 & 21 & 9 \\ 3 & 4 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[1067] $1^{11} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 7 & 21 & 4 \\ 4 & 7 & 7 & 31 & 5 \\ 14 & 21 & 31 & 101 & 19 \\ 3 & 4 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[1071] $1^{11} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 58, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 5 & 7 & 7 & 10 & 5 \\ 4 & 7 & 10 & 8 & 6 \\ 3 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[1072] $1^{13} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 10 & 12 & 6 & 25 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1073] $1^{11} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 9 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 6 & 9 & 5 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[1074] $1^{12} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 10 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1075] $1^{13} 4^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 3 & 3 & 0 & 18 & 2 \\ 14 & 24 & 18 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1076] $1^{11} 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 9 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 7 & 9 & 5 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[1078] $1^{111} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 4 & 7 & 7 & 10 & 5 \\ 5 & 7 & 10 & 8 & 6 \\ 3 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0}$

[1077] $21^{111} 4 1^1 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 4 & 8 & 6 & 12 & 4 \\ 8 & 13 & 12 & 20 & 7 \\ 6 & 12 & 7 & 16 & 5 \\ 12 & 20 & 16 & 26 & 10 \\ 4 & 7 & 5 & 10 & 3 \end{pmatrix}_{b_0}$

[1082] $1^{111} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 7 & 10 & 4 \\ 5 & 7 & 7 & 15 & 5 \\ 8 & 10 & 15 & 21 & 9 \\ 3 & 4 & 5 & 9 & 3 \end{pmatrix}_{b_0}$

[1080] $1^{111} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 11 & 12 & 8 & 26 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$

[1079] $21^{12} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 7 & 10 & 4 \\ 5 & 7 & 7 & 15 & 5 \\ 8 & 10 & 15 & 21 & 9 \\ 3 & 4 & 5 & 9 & 3 \end{pmatrix}_{b_0}$

[1081] $21^{111} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 5 & 7 & 5 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[1083] $1^{14} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 3 & 3 & 0 & 14 & 2 \\ 14 & 24 & 14 & 101 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$

[1084] $21^{111} 4 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 3 & 3 & 0 & 14 & 2 \\ 14 & 24 & 14 & 101 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$

[1085] $21^{113} 4 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 4 & 5 & 3 & 21 & 4 \\ 5 & 4 & 3 & 22 & 4 \\ 3 & 3 & 0 & 16 & 2 \\ 21 & 22 & 16 & 97 & 19 \\ 4 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$

[1088] $1^{111} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 5 & 7 & 7 & 10 & 5 \\ 6 & 7 & 10 & 8 & 6 \\ 3 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0}$

[1086] $1^{111} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 3 & 3 & 0 & 8 & 2 \\ 8 & 12 & 8 & 26 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1087] $21^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 3 & 3 & 0 & 8 & 2 \\ 8 & 12 & 8 & 26 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1089] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 3 \\ 3 & 4 & 7 & 18 & 4 \\ 5 & 7 & 7 & 26 & 5 \\ 15 & 18 & 26 & 70 & 16 \\ 3 & 4 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1090] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 7 & 12 & 4 \\ 5 & 7 & 7 & 16 & 5 \\ 7 & 12 & 16 & 26 & 10 \\ 3 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[1091] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 13 & 3 \\ 3 & 4 & 7 & 15 & 4 \\ 5 & 7 & 7 & 21 & 5 \\ 13 & 15 & 21 & 45 & 13 \\ 3 & 4 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[1092] $1^{13} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 3 & 23 & 4 \\ 3 & 3 & 0 & 15 & 2 \\ 13 & 23 & 15 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1093] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 7 & 15 & 4 \\ 5 & 7 & 7 & 21 & 5 \\ 9 & 15 & 21 & 45 & 13 \\ 3 & 4 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[1097] $1^{12} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 3 & 19 & 4 \\ 3 & 3 & 0 & 13 & 2 \\ 11 & 19 & 13 & 68 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1094] $1 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 7 & 12 & 8 & 26 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1095] $1^{12} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 3 & 3 & 0 & 7 & 2 \\ 8 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1096] $1^{13} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 8 & 12 & 6 & 25 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1099] $1^{13}_{1^1} 4^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 9 & 12 & 6 & 25 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1098] $1^{12}_{1^{11}} 4^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 3 & 3 & 0 & 7 & 2 \\ 9 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1100] $21^{11}_{1^1} 4^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 9 & 4 \\ 5 & 4 & 3 & 9 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 9 & 9 & 5 & 13 & 7 \\ 4 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[1101] $21^{11}_{1^{12}} 4^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 6 & 15 & 4 \\ 5 & 4 & 7 & 15 & 4 \\ 6 & 7 & 7 & 21 & 5 \\ 15 & 15 & 21 & 45 & 13 \\ 4 & 4 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[1102] $21^{11}_{1^{12}} 4^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 4 & 3 & 15 & 4 \\ 3 & 3 & 0 & 11 & 2 \\ 15 & 15 & 11 & 45 & 13 \\ 4 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1103] $21^{11}_{1^{13}} 4^1 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 4 & 3 & 17 & 4 \\ 3 & 3 & 0 & 9 & 2 \\ 15 & 17 & 9 & 49 & 13 \\ 4 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1104] $1^{14}_{1^{14}} 4^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 3 & 28 & 4 \\ 3 & 3 & 0 & 20 & 2 \\ 16 & 28 & 20 & 158 & 24 \\ 3 & 4 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[1105] $1^{11}_{1^{13}} 4^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 3 & 3 & 0 & 8 & 2 \\ 10 & 12 & 8 & 26 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1106] $1^{12}_{1^{11}} 4^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 7 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1109] $21^{11}_{1^1} 4^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 5 & 7 & 7 & 10 & 5 \\ 5 & 7 & 10 & 8 & 6 \\ 3 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[1107] $1^{11}_{1^{11}} 4^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 7 & 7 & 4 \\ 5 & 7 & 7 & 10 & 5 \\ 5 & 7 & 10 & 8 & 6 \\ 3 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1108] \quad 1^{14} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 3 & 19 & 4 \\ 3 & 3 & 0 & 11 & 2 \\ 12 & 19 & 11 & 65 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1110] \quad 1^{11} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 3 & 3 & 0 & 8 & 2 \\ 9 & 12 & 8 & 26 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1112] \quad 21^{11} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 6 & 12 & 4 \\ 5 & 4 & 7 & 12 & 4 \\ 6 & 7 & 7 & 16 & 5 \\ 12 & 12 & 16 & 26 & 10 \\ 4 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1111] \quad 21^{12} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 3 \\ 4 & 4 & 3 & 8 & 4 \\ 2 & 3 & 0 & 4 & 2 \\ 6 & 8 & 4 & 10 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1113] \quad 1^{12} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 9 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1115] \quad 1^{13} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 3 & 20 & 4 \\ 3 & 3 & 0 & 12 & 2 \\ 12 & 20 & 12 & 70 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1114] \quad 1^{13} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 3 & 3 & 0 & 9 & 2 \\ 10 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1116] \quad 21^{11} 4 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 3 & 20 & 4 \\ 3 & 3 & 0 & 12 & 2 \\ 12 & 20 & 12 & 70 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1117] \quad 21^{11} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 3 \\ 4 & 4 & 3 & 9 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 6 & 9 & 5 & 13 & 7 \\ 3 & 4 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1118] \quad 21^{11} 4 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 3 & 16 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 10 & 16 & 10 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1119] \quad 1^{12} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 3 & 16 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 10 & 16 & 10 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1120] \ 21 \overset{1^{12}}{1^{11}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 5 & 10 & 3 \\ 4 & 4 & 7 & 15 & 4 \\ 5 & 7 & 7 & 21 & 5 \\ 10 & 15 & 21 & 45 & 13 \\ 3 & 4 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1125] \ 1 \overset{1^3}{1^{11}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 10 & 12 & 8 & 26 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1121] \ 1 \overset{1^1}{1^{12}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 3 & 22 & 4 \\ 3 & 3 & 0 & 16 & 2 \\ 13 & 22 & 16 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1122] \ 1 \overset{1^{12}}{1^{12}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 10 & 14 & 10 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1123] \ 1 \overset{1^2}{1^{12}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 8 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1124] \ 21 \overset{1^1}{1^{12}} 4 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 10 & 14 & 10 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1126] \ 1 \overset{1^4}{1^{11}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 13 & 15 & 11 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1127] \ 21 \overset{1^1}{1^{11}} 4 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 12 & 4 \\ 5 & 4 & 3 & 13 & 4 \\ 3 & 3 & 0 & 7 & 2 \\ 12 & 13 & 7 & 28 & 10 \\ 4 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1128] \ 1 \overset{1^{13}}{1^{12}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 11 & 14 & 10 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1129] \ 1 \overset{1^1}{1^{12}} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 3 & 25 & 4 \\ 3 & 3 & 0 & 19 & 2 \\ 15 & 25 & 19 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[illegible]

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 15 & 18 & 14 & 70 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1144] $21^{111} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 5 & 3 & 3 \\ 4 & 4 & 7 & 4 & 4 \\ 5 & 7 & 7 & 5 & 5 \\ 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

[1145] $1^{111} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 7 & 12 & 4 \\ 4 & 7 & 7 & 16 & 5 \\ 8 & 12 & 16 & 26 & 10 \\ 3 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[1143] $1^{111} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 7 & 12 & 4 \\ 5 & 7 & 7 & 16 & 5 \\ 10 & 12 & 16 & 26 & 10 \\ 3 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[1142] $21^{11} 41^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 12 & 4 \\ 5 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 12 & 12 & 8 & 26 & 10 \\ 4 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1141] $1^{111} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 7 & 18 & 4 \\ 4 & 7 & 7 & 26 & 5 \\ 12 & 18 & 26 & 70 & 16 \\ 3 & 4 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1146] $1^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 9 & 12 & 8 & 26 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1147] $21^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 3 \\ 4 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 8 & 12 & 8 & 26 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1148] $21^{11} 41^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 24 & 4 \\ 5 & 4 & 3 & 26 & 4 \\ 3 & 3 & 0 & 18 & 2 \\ 24 & 26 & 18 & 130 & 22 \\ 4 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1149] $21^{11} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 12 & 4 \\ 5 & 4 & 3 & 12 & 4 \\ 3 & 3 & 0 & 8 & 2 \\ 12 & 12 & 8 & 26 & 10 \\ 4 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1150] $21^{111} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 5 & 6 & 3 \\ 4 & 4 & 7 & 9 & 4 \\ 5 & 7 & 7 & 11 & 5 \\ 6 & 9 & 11 & 13 & 7 \\ 3 & 4 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[1151] $1^{12} 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 0 & 13 & 2 \\ 14 & 17 & 13 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1152] \quad 1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 10 & 15 & 11 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1153] \quad 1^{12} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 13 & 18 & 14 & 70 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1157] \quad 1^{13} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 3 & 26 & 4 \\ 3 & 3 & 0 & 18 & 2 \\ 15 & 26 & 18 & 130 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[1158] \quad 1^{12} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 11 & 14 & 10 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1160] \quad 21^{11} 4 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 21 & 4 \\ 5 & 4 & 3 & 24 & 4 \\ 3 & 3 & 0 & 14 & 2 \\ 21 & 24 & 14 & 101 & 19 \\ 4 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1159] \quad 1^{13} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 3 & 3 & 0 & 9 & 2 \\ 11 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1156] \quad 21^{11} 4 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 27 & 4 \\ 5 & 4 & 3 & 30 & 4 \\ 3 & 3 & 0 & 20 & 2 \\ 27 & 30 & 20 & 169 & 25 \\ 4 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[1155] \quad 1^{12} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 13 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 10 & 13 & 7 & 28 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1154] \quad 1^{14} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 16 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 11 & 16 & 8 & 46 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1162] \quad 1^{14} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 3 & 30 & 4 \\ 3 & 3 & 0 & 20 & 2 \\ 17 & 30 & 20 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[1168] \ 1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 118, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 12 & 15 & 11 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1167] \ 1^{12} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 118, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 10 & 14 & 10 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1166] \ 1^{13} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 118, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 9 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1165] \ 1^{14} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 118, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 16 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 12 & 16 & 8 & 46 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1164] \ 1^{12} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 118, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 13 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 9 & 13 & 7 & 28 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1161] \ 1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 118, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 14 & 18 & 14 & 70 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1163] \ 1^{12} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 118, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 0 & 13 & 2 \\ 12 & 17 & 13 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1169] \ 1^{11} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 120, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 7 & 15 & 4 \\ 4 & 7 & 7 & 21 & 5 \\ 10 & 15 & 21 & 45 & 13 \\ 3 & 4 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1170] \ 1^4 4^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 120, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.97 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 16 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 9 & 16 & 10 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1171] \ 21^{12} 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 122, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.74 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 3 \\ 4 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 10 & 14 & 10 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1173] $1^{12}_{1^{13}} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 0 & 13 & 2 \\ 13 & 17 & 13 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1172] $1^{11}_{1^{12}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 11 & 15 & 11 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1174] $221^{11}_{1^{12}} 41^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 7 & 12 & 4 \\ 5 & 7 & 7 & 16 & 5 \\ 8 & 12 & 16 & 26 & 10 \\ 3 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[1178] $21^{11}_{1^{11}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 7 & 12 & 4 \\ 5 & 7 & 7 & 16 & 5 \\ 8 & 12 & 16 & 26 & 10 \\ 3 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[1177] $21^{13}_{1^{11}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 7 & 18 & 4 \\ 5 & 7 & 7 & 26 & 5 \\ 14 & 18 & 26 & 70 & 16 \\ 3 & 4 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1176] $1^{11}_{1^{13}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 3 \\ 3 & 4 & 7 & 18 & 4 \\ 5 & 7 & 7 & 26 & 5 \\ 14 & 18 & 26 & 70 & 16 \\ 3 & 4 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[1175] $1^{11}_{1^{11}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 7 & 12 & 4 \\ 5 & 7 & 7 & 16 & 5 \\ 8 & 12 & 16 & 26 & 10 \\ 3 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[1179] $1^{13}_{1^{11}} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 3 & 3 & 0 & 9 & 2 \\ 12 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1180] $21^{11}_{1^{11}} 4 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 3 \\ 4 & 4 & 7 & 7 & 4 \\ 5 & 7 & 10 & 10 & 5 \\ 5 & 7 & 10 & 8 & 6 \\ 3 & 4 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[1181] $1^{12}_{1^{11}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 3 & 16 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 11 & 16 & 10 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1182] $21^{11}_{1^{12}} 4 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 18 & 4 \\ 5 & 4 & 3 & 19 & 4 \\ 3 & 3 & 0 & 13 & 2 \\ 18 & 19 & 13 & 68 & 16 \\ 4 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1184] \quad 1^{\overline{111}} 4^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 7 & 15 & 4 \\ 5 & 7 & 7 & 21 & 5 \\ 12 & 15 & 21 & 45 & 13 \\ 3 & 4 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1186] \quad 21^{\overline{12}} 41^{\overline{13}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 3 & 3 & 0 & 12 & 2 \\ 12 & 18 & 12 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1183] \quad 1^{\overline{111}} 4^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 7 & 12 & 4 \\ 5 & 7 & 7 & 16 & 5 \\ 9 & 12 & 16 & 26 & 10 \\ 3 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[1185] \quad 1^{\overline{13}} 4^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 3 & 3 & 0 & 12 & 2 \\ 12 & 18 & 12 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1189] \quad 21^{\overline{13}} 41^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 3 \\ 4 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 9 & 12 & 6 & 25 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1187] \quad 21^{\overline{12}} 4^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 4 & 3 & 22 & 4 \\ 2 & 3 & 0 & 16 & 2 \\ 15 & 22 & 16 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1188] \quad 1^{\overline{14}} 4^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 3 & 19 & 4 \\ 3 & 3 & 0 & 11 & 2 \\ 13 & 19 & 11 & 65 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1190] \quad 1^{\overline{13}} 41^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 138, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 10 & 17 & 9 & 49 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1191] \quad 21^{\overline{11}} 4^{\overline{12}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 4 & 3 & 16 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 15 & 16 & 10 & 45 & 13 \\ 4 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1192] \quad 21^{\overline{12}} 4^{\overline{11}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 143, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 3 \\ 4 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 8 & 11 & 7 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[1195] \quad 21^{\overline{11}} 4^{\overline{13}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 18 & 4 \\ 5 & 4 & 3 & 20 & 4 \\ 3 & 3 & 0 & 12 & 2 \\ 18 & 20 & 12 & 70 & 16 \\ 4 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1193] \ 21 \overset{1^{12}}{1^1} 4 \ 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 24 & 4 \\ 5 & 4 & 3 & 27 & 4 \\ 3 & 3 & 0 & 17 & 2 \\ 24 & 27 & 17 & 132 & 22 \\ 4 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[1194] \ 1 \overset{1^1}{1^{13}} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 3 & 29 & 4 \\ 3 & 3 & 0 & 21 & 2 \\ 17 & 29 & 21 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[1197] \ 1 \overset{1^2}{1^{13}} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 10 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1196] \ 1 \overset{1^4}{1^{13}} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 12 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1198] \ 1 \overset{1^{12}}{1^1} 4 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 19 & 4 \\ 2 & 3 & 0 & 13 & 2 \\ 11 & 19 & 13 & 68 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1199] \ 1 \overset{1^{13}}{1^{12}} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 3 & 16 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 12 & 16 & 10 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[1201] \ 21 \overset{1^{12}}{1^1} 4 \ 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 21 & 4 \\ 5 & 4 & 3 & 23 & 4 \\ 3 & 3 & 0 & 15 & 2 \\ 21 & 23 & 15 & 97 & 19 \\ 4 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1200] \ 1 \overset{1^{12}}{1^{13}} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 3 & 20 & 4 \\ 3 & 3 & 0 & 12 & 2 \\ 13 & 20 & 12 & 70 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[1207] \ 1 \overset{1^{12}}{1^{14}} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 3 & 3 & 0 & 14 & 2 \\ 15 & 24 & 14 & 101 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1205] \ 1 \overset{1^{12}}{1^{14}} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 3 & 22 & 4 \\ 3 & 3 & 0 & 14 & 2 \\ 14 & 22 & 14 & 90 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1204] $1^{13}_4 1^{11}_4 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 11 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1203] $21^{12}_4 1^{14}_4 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 3 & 22 & 4 \\ 3 & 3 & 0 & 14 & 2 \\ 14 & 22 & 14 & 90 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1202] $1^{12}_4 1^{14}_4 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 3 & 16 & 4 \\ 3 & 3 & 0 & 10 & 2 \\ 13 & 16 & 10 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1206] $21^{11}_4 1^{11}_4 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 5 & 8 & 3 \\ 4 & 4 & 7 & 12 & 4 \\ 5 & 7 & 7 & 16 & 5 \\ 8 & 12 & 16 & 26 & 10 \\ 3 & 4 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[1209] $1^{11}_4 1^{12}_4 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 7 & 15 & 4 \\ 5 & 7 & 7 & 21 & 5 \\ 11 & 15 & 21 & 45 & 13 \\ 3 & 4 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[1208] $21^{12}_4 1^{11}_4 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 7 & 15 & 4 \\ 5 & 7 & 7 & 21 & 5 \\ 11 & 15 & 21 & 45 & 13 \\ 3 & 4 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[1210] $1^{13}_4 1^{11}_4 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 11 & 18 & 12 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1211] $1^{12}_4 1^{11}_4 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 16 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 10 & 16 & 10 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1212] $1^{13}_4 1^{11}_4 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 3 & 3 & 0 & 12 & 2 \\ 13 & 18 & 12 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1213] $1^{14}_4 1^{11}_4 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 3 & 33 & 4 \\ 3 & 3 & 0 & 23 & 2 \\ 19 & 33 & 23 & 212 & 28 \\ 3 & 4 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1216] $1^{14}_4 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 3 & 19 & 4 \\ 3 & 3 & 0 & 11 & 2 \\ 14 & 19 & 11 & 65 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1215] $1^{12}_4 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 3 & 19 & 4 \\ 3 & 3 & 0 & 13 & 2 \\ 13 & 19 & 13 & 68 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1214] $21^{12}_4 4 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 3 & 19 & 4 \\ 3 & 3 & 0 & 13 & 2 \\ 13 & 19 & 13 & 68 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1217] $1^{13}_4 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 3 & 3 & 0 & 12 & 2 \\ 14 & 18 & 12 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1218] $31^{11}_3 31^1 3 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -80, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 6 & 6 & 4 \\ 3 & 4 & 0 & 3 & 3 & 2 \\ 3 & 6 & 3 & 3 & 6 & 3 \\ 3 & 6 & 3 & 6 & 4 & 4 \\ 2 & 4 & 2 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

[1219] $1^{13}_4 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 11 & 17 & 9 & 49 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1220] $1^{12}_4 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 16 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 13 & 16 & 10 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1221] $1^{13}_4 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 0 & 15 & 2 \\ 13 & 21 & 15 & 88 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1222] $21^{12}_4 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 3 \\ 4 & 4 & 3 & 13 & 4 \\ 2 & 3 & 0 & 7 & 2 \\ 9 & 13 & 7 & 28 & 10 \\ 3 & 4 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[1223] $1^{13}_4 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 13 & 17 & 9 & 49 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1224] $1^{12}_4 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

[1234] $1^{13}_{1^{11}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 3 & 20 & 4 \\ 3 & 3 & 0 & 12 & 2 \\ 14 & 20 & 12 & 70 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1237] $1^{14}_{1^{12}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 205, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 3 & 22 & 4 \\ 3 & 3 & 0 & 14 & 2 \\ 15 & 22 & 14 & 90 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1236] $1^{13}_{1^{12}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 205, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 14 & 18 & 12 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1238] $21^{14}_{1^1} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 3 \\ 4 & 4 & 3 & 16 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 12 & 16 & 8 & 46 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1240] $1^{13}_{1^{14}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 3 & 3 & 0 & 18 & 2 \\ 18 & 24 & 18 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1239] $21^{14}_{1^1} 4 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 207, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 3 & 3 & 0 & 18 & 2 \\ 18 & 24 & 18 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1241] $1^{13}_{1^{12}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 13 & 18 & 12 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1242] $1^{12}_{1^{13}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 22 & 4 \\ 2 & 3 & 0 & 16 & 2 \\ 14 & 22 & 16 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1243] $21^{13}_{1^{11}} 4 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 3 & 22 & 4 \\ 3 & 3 & 0 & 16 & 2 \\ 16 & 22 & 16 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1244] $1^{12}_{1^{13}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 3 & 22 & 4 \\ 3 & 3 & 0 & 16 & 2 \\ 16 & 22 & 16 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1245] $1^{12}_{1^{13}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 3 & 22 & 4 \\ 3 & 3 & 0 & 16 & 2 \\ 17 & 22 & 16 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1247] $1^{13}_{1^{12}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 3 & 23 & 4 \\ 3 & 3 & 0 & 15 & 2 \\ 15 & 23 & 15 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1246] $21^{12}_{1^{11}} 4 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 3 & 23 & 4 \\ 3 & 3 & 0 & 15 & 2 \\ 15 & 23 & 15 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1248] $1^{13}_{1^{14}} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 18 & 2 \\ 16 & 24 & 18 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1249] $21^{13}_{1^{13}} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 4 & 3 & 21 & 4 \\ 2 & 3 & 0 & 15 & 2 \\ 15 & 21 & 15 & 88 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1250] $1^{13}_{1^{13}} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 0 & 15 & 2 \\ 14 & 21 & 15 & 88 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1251] $1^{14}_{1^{12}} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 22 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 13 & 22 & 14 & 90 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1252] $1^{12}_{1^{14}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 3 & 25 & 4 \\ 3 & 3 & 0 & 19 & 2 \\ 19 & 25 & 19 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1253] $21^{14}_{1^{11}} 4 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 3 & 25 & 4 \\ 3 & 3 & 0 & 19 & 2 \\ 19 & 25 & 19 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1254] $1^{14}_{1^{11}} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 219, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 19 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 13 & 19 & 11 & 65 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1255] $1^{14}_{1^1} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 222, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 13 & 21 & 11 & 76 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[illegible]

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 3 & 22 & 4 \\ 3 & 3 & 0 & 14 & 2 \\ 16 & 22 & 14 & 90 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1269] $1^{14}_{1^{12}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 3 & 27 & 4 \\ 3 & 3 & 0 & 17 & 2 \\ 17 & 27 & 17 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1270] $21^{13}_{1^{11}} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 3 \\ 4 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 11 & 15 & 9 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1267] $1^{13}_{1^{13}} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 0 & 15 & 2 \\ 16 & 21 & 15 & 88 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1268] $21^{12}_{1^{11}} 4 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 3 & 27 & 4 \\ 3 & 3 & 0 & 17 & 2 \\ 17 & 27 & 17 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1271] $221^{11}_{1^{11}} 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 6 & 5 & 5 & 3 \\ 6 & 13 & 12 & 12 & 7 \\ 5 & 12 & 7 & 10 & 5 \\ 5 & 12 & 10 & 8 & 6 \\ 3 & 7 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[1272] $1^{12}_{1^{13}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 3 & 22 & 4 \\ 2 & 3 & 0 & 16 & 2 \\ 15 & 22 & 16 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1273] $21^{12}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 3 \\ 4 & 4 & 3 & 16 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 11 & 16 & 10 & 45 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1274] $1^{13}_{1^{14}} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 18 & 2 \\ 17 & 24 & 18 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1275] $221^{11}_{1^{11}} 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 6 & 5 & 8 & 3 \\ 6 & 13 & 12 & 20 & 7 \\ 5 & 12 & 7 & 16 & 5 \\ 8 & 20 & 16 & 26 & 10 \\ 3 & 7 & 5 & 10 & 3 \end{pmatrix}_{b_0}$$

[1276] $1^{13}_{1^{14}} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 18 & 2 \\ 18 & 24 & 18 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1277] $1^{14}_{1^3} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 253, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

[1286] $1^{14}_1 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 15 & 21 & 11 & 76 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1289] $21^{13}_4 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 267, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 20 & 3 \\ 4 & 4 & 3 & 29 & 4 \\ 2 & 3 & 0 & 21 & 2 \\ 20 & 29 & 21 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1290] $21^{13}_4 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 3 \\ 4 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 13 & 18 & 12 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1291] $1^{14}_1 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 3 & 3 & 0 & 14 & 2 \\ 17 & 24 & 14 & 101 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1292] $1^{14}_1 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 281, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 14 & 24 & 14 & 101 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1294] $1^{13}_4 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 282, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 20 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 14 & 20 & 12 & 70 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1293] $1^{13}_4 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 282, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 23 & 4 \\ 2 & 3 & 0 & 15 & 2 \\ 14 & 23 & 15 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1295] $1^{13}_4 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 285, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 3 & 20 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 15 & 20 & 12 & 70 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1296] $1^{13}_4 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 3 & 23 & 4 \\ 3 & 3 & 0 & 15 & 2 \\ 17 & 23 & 15 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1297] $1^{14}_1 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 289, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 3 & 25 & 4 \\ 3 & 3 & 0 & 17 & 2 \\ 18 & 25 & 17 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1298] $21^{13}_4 4^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 291, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 3 \\ 4 & 4 & 3 & 17 & 4 \\ 2 & 3 & 0 & 9 & 2 \\ 12 & 17 & 9 & 49 & 13 \\ 3 & 4 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[1299] $21^{15}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.1$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 4 & 3 & 20 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 15 & 20 & 10 & 73 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1300] $1^{14}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 297, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 3 & 22 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 15 & 22 & 14 & 90 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1302] $21^{13}41^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 306, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 3 & 26 & 4 \\ 3 & 3 & 0 & 18 & 2 \\ 18 & 26 & 18 & 130 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1301] $1^{13}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 306, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 3 & 26 & 4 \\ 3 & 3 & 0 & 18 & 2 \\ 18 & 26 & 18 & 130 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1304] $1^{14}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 3 & 27 & 4 \\ 3 & 3 & 0 & 17 & 2 \\ 18 & 27 & 17 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1303] $1^{14}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 310, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 3 & 22 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 16 & 22 & 14 & 90 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1305] $1^{14}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 314, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 3 & 25 & 4 \\ 2 & 3 & 0 & 17 & 2 \\ 16 & 25 & 17 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1306] $1^{13}41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 318, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 3 & 26 & 4 \\ 2 & 3 & 0 & 18 & 2 \\ 16 & 26 & 18 & 130 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1308] $21^{14}41^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 322, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 3 & 28 & 4 \\ 3 & 3 & 0 & 20 & 2 \\ 20 & 28 & 20 & 158 & 24 \\ 3 & 4 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[1307] $1^{14}41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 322, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 3 & 28 & 4 \\ 3 & 3 & 0 & 20 & 2 \\ 20 & 28 & 20 & 158 & 24 \\ 3 & 4 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[1309] $1^{13}_{1^{12}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 327, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 3 & 23 & 4 \\ 2 & 3 & 0 & 15 & 2 \\ 15 & 23 & 15 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1310] $1^{13}_{1^{14}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 330, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 3 & 29 & 4 \\ 2 & 3 & 0 & 21 & 2 \\ 18 & 29 & 21 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1311] $1^{13}_{1^{13}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 333, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 3 & 26 & 4 \\ 3 & 3 & 0 & 18 & 2 \\ 19 & 26 & 18 & 130 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1312] $1^{14}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 15 & 24 & 14 & 101 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1313] $1^{14}_{1^{14}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 338, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 3 & 28 & 4 \\ 2 & 3 & 0 & 20 & 2 \\ 18 & 28 & 20 & 158 & 24 \\ 3 & 4 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[1314] $21^{14}_{1^{14}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 350, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 20 & 3 \\ 4 & 4 & 3 & 28 & 4 \\ 2 & 3 & 0 & 20 & 2 \\ 20 & 28 & 20 & 158 & 24 \\ 3 & 4 & 2 & 24 & 3 \end{pmatrix}_{b_0}$$

[1315] $1^{14}_{1^{13}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 351, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 3 & 25 & 4 \\ 2 & 3 & 0 & 17 & 2 \\ 17 & 25 & 17 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1316] $1^{13}_{1^{12}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 354, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 3 & 23 & 4 \\ 2 & 3 & 0 & 15 & 2 \\ 16 & 23 & 15 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1317] $21^{14}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 359, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 3 \\ 4 & 4 & 3 & 19 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 14 & 19 & 11 & 65 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1318] $21^{14}_{1^{14}} 4 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 3 & 29 & 4 \\ 3 & 3 & 0 & 21 & 2 \\ 21 & 29 & 21 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1319] $1^{13}_{1^{14}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 3 & 29 & 4 \\ 3 & 3 & 0 & 21 & 2 \\ 21 & 29 & 21 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1320] \quad & 31^{\overline{11}} 31^{\overline{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -80, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 3 & 6 & 2 \\ 3 & 4 & 4 & 6 & 9 & 4 \\ 3 & 4 & 0 & 3 & 6 & 2 \\ 3 & 6 & 3 & 3 & 9 & 3 \\ 6 & 9 & 6 & 9 & 13 & 7 \\ 2 & 4 & 2 & 3 & 7 & 2 \end{pmatrix}_{b_0} \\
[1321] \quad & 1^{\overline{14}} 4 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 361, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 3 & 27 & 4 \\ 3 & 3 & 0 & 17 & 2 \\ 19 & 27 & 17 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[1322] \quad & 1^{\overline{13}} 4 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 363, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 3 & 23 & 4 \\ 2 & 3 & 0 & 15 & 2 \\ 17 & 23 & 15 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1323] \quad & 1^{\overline{14}} 4 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 365, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 3 & 27 & 4 \\ 2 & 3 & 0 & 17 & 2 \\ 16 & 27 & 17 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[1324] \quad & 1^{\overline{13}} 4 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 3 & 26 & 4 \\ 2 & 3 & 0 & 18 & 2 \\ 17 & 26 & 18 & 130 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0} \\
[1325] \quad & 1^{\overline{14}} 4 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 370, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 3 & 25 & 4 \\ 2 & 3 & 0 & 17 & 2 \\ 18 & 25 & 17 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0} \\
[1326] \quad & 1^{\overline{14}} 4 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 373, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 16 & 24 & 14 & 101 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0} \\
[1328] \quad & 1^{\overline{14}} 4 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 379, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 3 & 30 & 4 \\ 3 & 3 & 0 & 20 & 2 \\ 20 & 30 & 20 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1327] \quad & 21^{\overline{13}} 4 1^{\overline{14}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 379, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 3 & 30 & 4 \\ 3 & 3 & 0 & 20 & 2 \\ 20 & 30 & 20 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0} \\
[1329] \quad & 1^{\overline{14}} 4 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 3 & 28 & 4 \\ 2 & 3 & 0 & 20 & 2 \\ 19 & 28 & 20 & 158 & 24 \\ 3 & 4 & 2 & 24 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

[1330] $1^{13}_{1^{14}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 387, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 3 & 29 & 4 \\ 2 & 3 & 0 & 21 & 2 \\ 19 & 29 & 21 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1331] $21^{13}_{1^{13}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 2 & 18 & 3 \\ 4 & 4 & 3 & 26 & 4 \\ 2 & 3 & 0 & 18 & 2 \\ 18 & 26 & 18 & 130 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[1332] $1^{14}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 17 & 24 & 14 & 101 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$

[1333] $21^{13}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 402, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 2 & 14 & 3 \\ 4 & 4 & 3 & 20 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 14 & 20 & 12 & 70 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1334] $1^{13}_{1^{13}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 402, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 3 & 26 & 4 \\ 2 & 3 & 0 & 18 & 2 \\ 18 & 26 & 18 & 130 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[1335] $21^{16}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 2 & 18 & 3 \\ 4 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 18 & 24 & 12 & 106 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$

[1336] $1^{14}_{1^{14}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 3 & 28 & 4 \\ 2 & 3 & 0 & 20 & 2 \\ 20 & 28 & 20 & 158 & 24 \\ 3 & 4 & 2 & 24 & 3 \end{pmatrix}_{b_0}$

[1337] $1^{13}_{1^{13}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 417, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 3 & 26 & 4 \\ 2 & 3 & 0 & 18 & 2 \\ 19 & 26 & 18 & 130 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[1338] $21^{14}_{1^{11}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 421, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 4 & 3 & 21 & 4 \\ 2 & 3 & 0 & 11 & 2 \\ 15 & 21 & 11 & 76 & 16 \\ 3 & 4 & 2 & 16 & 3 \end{pmatrix}_{b_0}$

[1339] $1^{14}_{1^{13}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 425, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 3 & 30 & 4 \\ 2 & 3 & 0 & 20 & 2 \\ 18 & 30 & 20 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$

[1340] $1^{13}_{1^{14}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 426, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 3 & 29 & 4 \\ 2 & 3 & 0 & 21 & 2 \\ 20 & 29 & 21 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[1341] \quad 1^{14}_{1^{12}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 426, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 3 & 27 & 4 \\ 2 & 3 & 0 & 17 & 2 \\ 17 & 27 & 17 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[1342] \quad 1^{14}_{1^{13}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 430, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 3 & 30 & 4 \\ 3 & 3 & 0 & 20 & 2 \\ 21 & 30 & 20 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[1343] \quad 21^{14}_{1^{13}} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 431, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 3 \\ 4 & 4 & 3 & 25 & 4 \\ 2 & 3 & 0 & 17 & 2 \\ 18 & 25 & 17 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1344] \quad 21^{14}_{1^{12}} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 434, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 3 \\ 4 & 4 & 3 & 22 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 16 & 22 & 14 & 90 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1345] \quad 21^{13}_{1^{12}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 435, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 3 \\ 4 & 4 & 3 & 23 & 4 \\ 2 & 3 & 0 & 15 & 2 \\ 16 & 23 & 15 & 97 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

$$[1346] \quad 1^{13}_{1^{14}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 447, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 3 & 29 & 4 \\ 2 & 3 & 0 & 21 & 2 \\ 21 & 29 & 21 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[1347] \quad 1^{14}_{1^{14}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 461, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 3 & 33 & 4 \\ 2 & 3 & 0 & 23 & 2 \\ 20 & 33 & 23 & 212 & 28 \\ 3 & 4 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[1348] \quad 1^{14}_{1^{12}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 469, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 3 & 27 & 4 \\ 2 & 3 & 0 & 17 & 2 \\ 18 & 27 & 17 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$[1349] \quad 21^{14}_{1^{11}} 4 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 481, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 23 & 3 \\ 3 & 4 & 3 & 33 & 4 \\ 3 & 3 & 0 & 23 & 2 \\ 23 & 33 & 23 & 212 & 28 \\ 3 & 4 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

$$[1350] \quad 1^{14}_{1^{14}} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 481, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 23 & 3 \\ 3 & 4 & 3 & 33 & 4 \\ 3 & 3 & 0 & 23 & 2 \\ 23 & 33 & 23 & 212 & 28 \\ 3 & 4 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1351] $1^{14}_{1^{13}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 492, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 3 & 30 & 4 \\ 2 & 3 & 0 & 20 & 2 \\ 19 & 30 & 20 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1352] $1^{14}_{1^{12}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 494, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 3 & 27 & 4 \\ 2 & 3 & 0 & 17 & 2 \\ 19 & 27 & 17 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1353] $1^{14}_{1^{14}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 534, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 3 & 33 & 4 \\ 2 & 3 & 0 & 23 & 2 \\ 21 & 33 & 23 & 212 & 28 \\ 3 & 4 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1354] $21^{17}_4 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 21 & 3 \\ 4 & 4 & 3 & 28 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 21 & 28 & 14 & 145 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1355] $1^{14}_{1^{13}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 541, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 3 & 30 & 4 \\ 2 & 3 & 0 & 20 & 2 \\ 20 & 30 & 20 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1356] $1^{14}_{1^{13}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 572, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 3 & 30 & 4 \\ 2 & 3 & 0 & 20 & 2 \\ 21 & 30 & 20 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$$

[1357] $21^{14}_4 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 580, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 17 & 3 \\ 4 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 17 & 24 & 14 & 101 & 19 \\ 3 & 4 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[1358] $21^{14}_4 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 589, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 23 & 3 \\ 4 & 4 & 3 & 33 & 4 \\ 2 & 3 & 0 & 23 & 2 \\ 23 & 33 & 23 & 212 & 28 \\ 3 & 4 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1359] $1^{14}_{1^{14}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 589, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 22 & 3 \\ 3 & 4 & 3 & 33 & 4 \\ 2 & 3 & 0 & 23 & 2 \\ 22 & 33 & 23 & 212 & 28 \\ 3 & 4 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1360] $1^{14}_{1^{14}} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 626, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 23 & 3 \\ 3 & 4 & 3 & 33 & 4 \\ 2 & 3 & 0 & 23 & 2 \\ 23 & 33 & 23 & 212 & 28 \\ 3 & 4 & 2 & 28 & 3 \end{pmatrix}_{b_0}$$

[1361] $21^{14}_4 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 661, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 19 & 3 \\ 4 & 4 & 3 & 27 & 4 \\ 2 & 3 & 0 & 17 & 2 \\ 19 & 27 & 17 & 132 & 22 \\ 3 & 4 & 2 & 22 & 3 \end{pmatrix}_{b_0}$$

[1362] $21^{14} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2$ $T = 8, T_{\max} = 11, \Delta = -84, \det = 664, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 2 & 21 & 3 \\ 4 & 4 & 3 & 30 & 4 \\ 2 & 3 & 0 & 20 & 2 \\ 21 & 30 & 20 & 169 & 25 \\ 3 & 4 & 2 & 25 & 3 \end{pmatrix}_{b_0}$

$T = 9$ [1364] $1^{11} 2 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 4 & 2 \\ 2 & 4 & 0 & 4 & 6 & 6 \\ 2 & 3 & -1 & 2 & 6 & 4 \end{pmatrix}_{b_0}$

[1363] $1^{11} 2 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 10, \Delta = -38, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & -1 & 3 & 4 \end{pmatrix}_{b_0}$

[1365] $1^{11} 3 1^{11} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 11, \Delta = -63, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 0 & 3 & 4 \end{pmatrix}_{b_0}$

[1366] $1^{11} 4 1 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 11, \Delta = -63, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 3 & 0 & 4 & 4 \\ 3 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$

[1367] $1^{11} 4 1 3 1^{11} 4 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 11, \Delta = -63, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -2 & 1 & 0 \\ 6 & 7 & 1 & 9 & 7 \\ 3 & 4 & 0 & 7 & 4 \end{pmatrix}_{b_0}$

[1368] $1^{11} 4 1 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 11, \Delta = -63, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 6 & 4 & 6 \\ 4 & 2 & 4 \\ 6 & 4 & 6 \end{pmatrix}_{b_0}$

[1369] $1^{11} 4 1 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 11, \Delta = -63, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 2 & 4 \\ 3 & 1 & 1 & 3 \\ 2 & 1 & 0 & 2 \\ 4 & 3 & 2 & 5 \end{pmatrix}_{b_0}$

[1370] $1^{11} 5 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 6 & 4 & 4 \\ 4 & 6 & 7 & 4 & 5 \\ 2 & 4 & 4 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}$

[1371] $1^{11} 5 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 6 & 4 & 4 \\ 4 & 6 & 7 & 4 & 5 \\ 3 & 4 & 4 & 1 & 3 \\ 3 & 4 & 5 & 3 & 3 \end{pmatrix}_{b_0}$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 4 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 5 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[1383] \quad 21^{\overline{1}1} \overline{5}1^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1384] \quad 1^{\overline{1}1} \overline{4} \quad 1^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 3 & 4 \\ 3 & 1 & 2 & 3 \\ 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[1385] \quad 1^{\overline{1}1} \overline{4} \quad 131^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 0 & 2 & 1 & 2 \\ 3 & 2 & 1 & 2 & 3 \\ 1 & 1 & 2 & 0 & 2 \\ 3 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1386] \quad 1^{\overline{1}1} \overline{5} \quad 1^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1387] \quad 1^{\overline{1}1} \overline{4} \quad 131^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 3 & 0 & 1 & 3 \\ 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1388] \quad 1^{\overline{1}1} \overline{4} \quad 1^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 2 & 0 & 1 & 3 \\ 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1390] \quad 21^{\overline{1}1} \overline{5}1^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 3 & 7 & 4 \\ 3 & 3 & 0 & 4 & 2 \\ 6 & 7 & 4 & 8 & 6 \\ 4 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1389] \quad 1^{\overline{1}1} \overline{3} \quad 221^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -36, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 5 & 0 & 6 & 6 \\ 2 & 2 & 3 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1391] \quad 1^{\overline{1}1} \overline{4} \quad 131^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 3 & 1 & 5 & 5 \\ 3 & 3 & 0 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[1392] \quad 1^{\overline{1}2} \overline{5} \quad 1^{\overline{1}1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -17, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 6 & 10 & 6 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1394] $1^{11} 2^1 3 21^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -38, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & -1 & 6 & 4 \end{pmatrix}_{b_0}$

[1393] $21^{11} 5 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 3 & 3 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 3 & 3 & 2 & 1 & 3 \\ 4 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[1396] $21^{11} 5 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 7 & 3 \\ 4 & 4 & 3 & 10 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 7 & 10 & 6 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[1395] $21^{11} 5 1^{11} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 1 & 5 & 3 & 3 \\ 4 & 5 & 7 & 4 & 5 \\ 3 & 3 & 4 & 1 & 3 \\ 3 & 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$

[1397] $1^{11} 5 1^{11} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 3 & 4 & 2 & 1 & 3 \\ 4 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[1401] $21^{11} 5 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[1400] $1^{11} 5 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$

[1399] $1^{11} 4 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 1 & 4 & 3 \\ 5 & 4 & 6 & 6 \\ 4 & 3 & 6 & 5 \end{pmatrix}_{b_0}$

[1398] $221^{11} 5 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 5 & 2 & 3 \\ 2 & 1 & 5 & 3 & 3 \\ 5 & 5 & 7 & 4 & 5 \\ 2 & 3 & 4 & 1 & 3 \\ 3 & 3 & 5 & 3 & 3 \end{pmatrix}_{b_0}$

[1402] $1^{11} 4^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 1 & 3 & 3 \\ 5 & 3 & 3 & 5 \\ 4 & 3 & 5 & 5 \end{pmatrix}_{b_0}$

[1404] $1^{11} 4 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 7 & 4 \\ 3 & 1 & 5 & 3 \\ 7 & 5 & 12 & 8 \\ 4 & 3 & 8 & 5 \end{pmatrix}_{b_0}$

[1403] $1^{11} 4^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 4 & 4 \\ 3 & 1 & 4 & 3 \\ 4 & 4 & 3 & 5 \\ 4 & 3 & 5 & 5 \end{pmatrix}_{b_0}$

[1405] $1^{11} 5^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 4 & 2 \\ 4 & 7 & 4 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[1406] $21^{11} 5^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 8 & 10 & 6 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[1407] $1^{11} 5^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 8 & 10 & 6 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$

[1408] $1^{11} 4^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -29, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 3 & 1 & 2 & 4 \\ 3 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$

[1409] $1^{11} 5^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 4 & 2 \\ 6 & 7 & 4 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[1410] $1^{11} 5^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 10 & 17 & 10 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$

[1411] $1^{11} 4^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 5 & 5 & 0 & 8 & 6 \\ 3 & 3 & 0 & 6 & 4 \end{pmatrix}_{b_0}$

[1413] $21^{11} 5^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 4 & 2 \\ 5 & 7 & 4 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[1412] $1^{11} 5^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 4 & 2 \\ 5 & 7 & 4 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[1415] $1^{11} 5^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 6 & 11 & 6 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1414] $1^{12} \overset{1^4}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 10 & 11 & 6 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1416] $1^{12} \overset{1^2}{5} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -46, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 8 & 14 & 8 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1417] $1^{12} \overset{1^4}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 3 & 20 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 16 & 20 & 12 & 90 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1418] $21^4 \overset{1^1}{5} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 3 & 20 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 16 & 20 & 12 & 90 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1419] $21^{12} \overset{1^3}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 3 \\ 4 & 4 & 3 & 17 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 12 & 17 & 10 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1420] $1^{11} \overset{1^{11}}{4} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -53, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 6 & 1 & 14 & 8 \\ 3 & 3 & 0 & 8 & 4 \end{pmatrix}_{b_0}$$

[1421] $15 \overset{1^1}{1} 1^{11} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 4 & 3 & 4 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[1422] $1^{14} \overset{1^4}{5} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 14 & 24 & 14 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1423] $1^{12} \overset{1^4}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 12 & 14 & 8 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1424] $21^{11} \overset{1^1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 4 & 2 \\ 5 & 7 & 4 & 8 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[1425] $1^{11} 1^{11} 4^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -63, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 6 & 3 \\ 2 & 1 & 1 & 5 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 5 & 1 & 9 & 7 \\ 3 & 3 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

[1426] $1^{12} 5^{14} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 14 & 17 & 10 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1427] $21^{15} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 7 & 11 & 6 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1428] $1^{12} 5^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 7 & 11 & 6 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1429] $1^{12} 5^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 9 & 11 & 6 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1430] $1^{12} 5^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -73, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 10 & 18 & 10 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1431] $1^{11} 4^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -63, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 2 & 3 & 10 & 4 \\ 3 & 1 & 7 & 3 \\ 10 & 7 & 21 & 11 \\ 4 & 3 & 11 & 5 \end{pmatrix}_{b_0}$$

[1432] $1^{13} 5^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 12 & 21 & 12 & 88 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1433] $1^{12} 5^{12} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 8 & 11 & 6 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1435] $21^{35} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 13 & 17 & 10 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1434] $1^{12} 5^{13} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 13 & 17 & 10 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1436] \ 1^{12} \overset{1^3}{5} \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -85, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.54 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 11 & 14 & 8 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1437] \ 1^1 \overset{1^1}{5} 1^{12} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -86, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 3 \\ 4 & 4 & 3 & 8 & 4 \\ 2 & 3 & 0 & 4 & 2 \\ 6 & 8 & 4 & 10 & 6 \\ 3 & 4 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1439] \ 21^2 \overset{1^1}{5} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -90, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.56 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 10 & 14 & 8 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1438] \ 1^{12} \overset{1^2}{5} \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -90, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.56 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 10 & 14 & 8 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1441] \ 21^1 \overset{1^1}{5} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -95, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.80 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 9 & 15 & 8 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1440] \ 1^{13} \overset{1^1}{5} \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -95, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.80 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 9 & 15 & 8 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1442] \ 1^{11} \overset{1^{11} 31^1}{4} \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -63, \ \det = -100, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 1 & 8 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 9 & 8 & 1 & 27 & 11 \\ 3 & 3 & 0 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1443] \ 21^{12} \overset{1^2}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -114, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.58 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 3 \\ 4 & 4 & 3 & 14 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 10 & 14 & 8 & 38 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1444] \ 1^{13} \overset{1^4}{5} \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.66 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 12 & 15 & 8 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1445] \ 221^{11} \overset{1^1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -119, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.50 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 5 & 3 \\ 6 & 13 & 5 & 12 & 7 \\ 2 & 5 & 0 & 4 & 2 \\ 5 & 12 & 4 & 8 & 6 \\ 3 & 7 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[1447] $21^{13} \overset{1^4}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 17 & 3 \\ 4 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 17 & 24 & 14 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[1446] $1^{13} \overset{1^2}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 10 & 15 & 8 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1448] $21^{12} \overset{1^1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -125, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 3 \\ 4 & 4 & 3 & 11 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 8 & 11 & 6 & 21 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1449] $1^{13} \overset{1^3}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 11 & 15 & 8 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

[1450] $15^1 \overset{1^1}{1} 1^3 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 3 \\ 4 & 4 & 3 & 12 & 4 \\ 2 & 3 & 0 & 6 & 2 \\ 9 & 12 & 6 & 25 & 9 \\ 3 & 4 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[1451] $21^2 \overset{1^1}{5} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -149, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 12 & 18 & 10 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1452] $1^{13} \overset{1^2}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -149, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 12 & 18 & 10 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1453] $1^{13} \overset{1^4}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -153, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 14 & 18 & 10 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1454] $1^{13} \overset{1^3}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 13 & 18 & 10 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

[1455] $1^{13} \overset{1^4}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 16 & 21 & 12 & 88 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[1457] $21^4 \overset{1^1}{5} 1^3 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 18 & 24 & 14 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1456] \quad 1^{13} \overset{1^4}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 0 & 14 & 2 \\ 18 & 24 & 14 & 121 & 21 \\ 3 & 4 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[1458] \quad 1^{13} \overset{1^3}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 15 & 21 & 12 & 88 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1459] \quad 21^{13} \overset{1^1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 15 & 21 & 12 & 88 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1460] \quad 21^{13} \overset{1^3}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 4 & 3 & 21 & 4 \\ 2 & 3 & 0 & 12 & 2 \\ 15 & 21 & 12 & 88 & 18 \\ 3 & 4 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[1461] \quad 21^{13} \overset{1^1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -207, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 3 \\ 4 & 4 & 3 & 15 & 4 \\ 2 & 3 & 0 & 8 & 2 \\ 11 & 15 & 8 & 40 & 12 \\ 3 & 4 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1462] \quad 221^{12} \overset{1^1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 10 & 3 \\ 6 & 13 & 5 & 24 & 7 \\ 2 & 5 & 0 & 8 & 2 \\ 10 & 24 & 8 & 38 & 12 \\ 3 & 7 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1463] \quad 21^{13} \overset{1^2}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 3 \\ 4 & 4 & 3 & 18 & 4 \\ 2 & 3 & 0 & 10 & 2 \\ 13 & 18 & 10 & 61 & 15 \\ 3 & 4 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[1464] \quad 221^{13} \overset{1^1}{5} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -383, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.53 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 15 & 3 \\ 6 & 13 & 5 & 36 & 7 \\ 2 & 5 & 0 & 12 & 2 \\ 15 & 36 & 12 & 88 & 18 \\ 3 & 7 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [1466] \quad 21^{11} \overset{1^{11}}{5} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 & 3 \\ 3 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1465] \quad 1^{11} \overset{1^1}{5} 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 & 3 \\ 3 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1472] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 1^{\frac{1}{1}} 231^{\frac{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 4 & 2 \\ 3 & 3 & 0 & 4 & 10 & 6 \\ 2 & 2 & -1 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1471] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 131^{\frac{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 \\ 4 & 2 & 4 & 2 & 4 \\ 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1469] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 1231^{\frac{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 3 & 0 & 1 & 3 \\ 2 & 2 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1468] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 131^{\frac{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 1 & 6 & 3 \\ 7 & 7 & 6 & 9 & 7 \\ 3 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[1467] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 131^{\frac{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 1 & 18 & 3 \\ 13 & 21 & 18 & 97 & 21 \\ 3 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[1470] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 131^{\frac{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 \\ 2 & 2 & 4 & 2 & 4 \\ 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1473] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 131^{\frac{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 5 & 3 & 0 & 5 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 5 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[1475] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 131^{\frac{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 5 & 3 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 5 & 2 & 2 & 4 \\ 3 & 5 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1474] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 1231^{\frac{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 5 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 5 & 0 & 2 & 4 \\ 3 & 5 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1477] \quad 1^{\frac{1}{1}} 5^{\frac{1}{1}} 131^{\frac{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[1476] $1^{11} 5^{11} 13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 1 & 6 & 3 \\ 4 & 7 & 6 & 9 & 7 \\ 3 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$

[1478] $1^{11} 5^{11} 123^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 4 & 2 \\ 2 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & 2 & 5 & 3 \end{pmatrix}_{b_0}$

[1482] $1^{11} 5^{11} 13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 3 & 1 & 6 & 3 \\ 4 & 7 & 6 & 9 & 7 \\ 3 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$

[1483] $1^{11} 5^{11} 13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$

[1479] $1^{11} 4^{11} 132^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 6 & 4 \\ 3 & 2 & 5 & 4 \\ 6 & 5 & 10 & 8 \\ 4 & 4 & 8 & 6 \end{pmatrix}_{b_0}$

[1480] $1^{13} 5^{11} 13^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 3 & 3 & 1 & 18 & 3 \\ 12 & 21 & 18 & 97 & 21 \\ 3 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$

[1481] $1^{12} 5^{11} 13^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 3 & 3 & 1 & 12 & 3 \\ 8 & 14 & 12 & 42 & 14 \\ 3 & 4 & 3 & 14 & 4 \end{pmatrix}_{b_0}$

[1484] $1^{12} 5^{11} 13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 10, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 5 & 3 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 \\ 4 & 5 & 2 & 2 & 4 \\ 3 & 5 & 2 & 4 & 4 \end{pmatrix}_{b_0}$

[1485] $2^{11} 3^{11} 15^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 3 & 1 & 6 & 3 \\ 5 & 7 & 6 & 9 & 7 \\ 3 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$

[1486] $1^{11} 5^{11} 13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 3 & 2 & 4 & 4 & 6 & 4 \\ 2 & 3 & 4 & 6 & 5 & 5 \\ 2 & 2 & 2 & 4 & 5 & 3 \end{pmatrix}_{b_0}$

[1487] $1^{11} 5^{11} 13^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 \\ 3 & 2 & 4 & 2 & 4 \\ 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[1488] $21^{111} 5^{111} 131^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 0 & 4 & 4 & 2 \\ 3 & 2 & 4 & 4 & 6 & 4 \\ 2 & 3 & 4 & 6 & 5 & 5 \\ 2 & 2 & 2 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[1489] $1^{11} 5^{11} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 5 & 3 & 10 & 5 \\ 2 & 3 & 0 & 4 & 2 \\ 5 & 10 & 4 & 14 & 8 \\ 3 & 5 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

[1490] $1^{11} 5^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[1491] $21^{11} 5^{11} 131^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 5 & 8 & 5 & 5 \\ 5 & 8 & 8 & 6 & 6 \\ 2 & 5 & 6 & 2 & 4 \\ 3 & 5 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[1492] $231^{11} 5^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 3 \\ 2 & 1 & 3 & 3 & 3 \\ 3 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[1493] $1^{11} 5^{11} 1231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[1494] $1^{11} 5^{11} 131^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 5 & 8 & 5 & 5 \\ 5 & 8 & 8 & 6 & 6 \\ 2 & 5 & 6 & 2 & 4 \\ 3 & 5 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[1495] $1^{11} 5^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[1496] $1^{12} 5^{11} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 1 & 12 & 3 \\ 9 & 14 & 12 & 42 & 14 \\ 3 & 4 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[1497] $1^{11} 5^{12} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 1 & 6 & 3 \\ 6 & 7 & 6 & 9 & 7 \\ 3 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1498] $1^{11} 1^1 4 1321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 11, \Delta = -35, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 2 & 5 & 4 \\ 5 & 5 & 7 & 7 \\ 4 & 4 & 7 & 6 \end{pmatrix}_{b_0}$

[1499] $1^{11} 5 1^{23} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 12 & 2 \\ 8 & 9 & 0 & 12 & 81 & 17 \\ 2 & 2 & -1 & 2 & 17 & 3 \end{pmatrix}_{b_0}$

[1500] $1^{11} 5 1^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 1 & 6 & 3 \\ 5 & 7 & 6 & 9 & 7 \\ 3 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$

[1501] $1^{11} 5 1^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 17, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 5 & 3 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 5 & 2 & 2 & 4 \\ 3 & 5 & 2 & 4 & 4 \end{pmatrix}_{b_0}$

[1502] $1^{13} 5 1^{12} 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 16 & 2 \\ 9 & 12 & 0 & 16 & 115 & 21 \\ 2 & 2 & -1 & 2 & 21 & 3 \end{pmatrix}_{b_0}$

[1503] $1^{11} 5 1^{12} 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 8 & 2 \\ 5 & 6 & 0 & 8 & 31 & 11 \\ 2 & 2 & -1 & 2 & 11 & 3 \end{pmatrix}_{b_0}$

[1504] $2^3 3 1^5 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 3 & 3 & 1 & 12 & 3 \\ 10 & 14 & 12 & 42 & 14 \\ 3 & 4 & 3 & 14 & 4 \end{pmatrix}_{b_0}$

[1505] $1^{13} 5 1^{12} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 5 & 3 & 10 & 5 \\ 2 & 3 & 0 & 4 & 2 \\ 7 & 10 & 4 & 14 & 8 \\ 3 & 5 & 2 & 8 & 4 \end{pmatrix}_{b_0}$

[1506] $2^3 3 1^5 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 22, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 3 & 3 & 1 & 4 & 3 \\ 3 & 2 & 4 & 2 & 4 \\ 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$

[1507] $21^{12} 5 1^{13} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 22, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 12 & 12 & 16 & 70 & 18 \\ 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$

[1508] $1^{12} 5 1^{12} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 1 & 12 & 3 \\ 12 & 14 & 12 & 42 & 14 \\ 3 & 4 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[1509] $1^{12} 5^1 1^{12} 231^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 12 & 2 \\ 7 & 9 & 0 & 12 & 66 & 16 \\ 2 & 2 & -1 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[1510] $1^{12} 5^1 131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 5 & 3 & 15 & 5 \\ 2 & 3 & 0 & 6 & 2 \\ 8 & 15 & 6 & 34 & 12 \\ 3 & 5 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[1511] $1^{12} 5^1 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 5 & 3 & 10 & 5 \\ 2 & 3 & 0 & 4 & 2 \\ 6 & 10 & 4 & 14 & 8 \\ 3 & 5 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

[1512] $1^{12} 5^1 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 1 & 12 & 3 \\ 10 & 14 & 12 & 42 & 14 \\ 3 & 4 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[1514] $1^{11} 5^1 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 6 & 3 \\ 4 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1513] $1^{11} 5^1 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 6 & 3 \\ 7 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1515] $1^{11} 5^1 1231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

[1516] $1^{13} 5^1 131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 1 & 18 & 3 \\ 14 & 21 & 18 & 97 & 21 \\ 3 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[1517] $1^{12} 5^1 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 3 & 14 & 4 \\ 2 & 3 & 1 & 12 & 3 \\ 11 & 14 & 12 & 42 & 14 \\ 3 & 4 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[1518] $1^{11} 5^1 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 5 & 8 & 5 & 5 \\ 4 & 8 & 8 & 6 & 6 \\ 3 & 5 & 6 & 2 & 4 \\ 3 & 5 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[1519] $1^{\frac{1^2}{1^1}} 5 131^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 5 & 8 & 10 & 5 \\ 4 & 8 & 8 & 12 & 6 \\ 6 & 10 & 12 & 14 & 8 \\ 3 & 5 & 6 & 8 & 4 \end{pmatrix}_{b_0}$

[1520] $1^{\frac{1^3}{1^1}} 5 131^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 5 & 8 & 15 & 5 \\ 4 & 8 & 8 & 18 & 6 \\ 9 & 15 & 18 & 34 & 12 \\ 3 & 5 & 6 & 12 & 4 \end{pmatrix}_{b_0}$

[1521] $1^{\frac{1^4}{1^1}} 5 131^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 5 & 8 & 20 & 5 \\ 4 & 8 & 8 & 24 & 6 \\ 12 & 20 & 24 & 62 & 16 \\ 3 & 5 & 6 & 16 & 4 \end{pmatrix}_{b_0}$

[1522] $2^{\frac{1^3}{1^1}} 3 15 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 3 & 3 & 1 & 18 & 3 \\ 15 & 21 & 18 & 97 & 21 \\ 3 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$

[1523] $1^{\frac{1^4}{1^1}} 5 131^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 38, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 5 & 3 & 15 & 5 \\ 2 & 3 & 0 & 6 & 2 \\ 10 & 15 & 6 & 34 & 12 \\ 3 & 5 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[1525] $1^{\frac{1^2}{1^1}} 5 1^{\frac{1^3}{1^1}} 231^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 20 & 2 \\ 13 & 15 & 0 & 20 & 218 & 28 \\ 2 & 2 & -1 & 2 & 28 & 3 \end{pmatrix}_{b_0}$

[1524] $1^{\frac{1^2}{1^1}} 5 1^{\frac{1^2}{1^1}} 231^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 16 & 2 \\ 10 & 12 & 0 & 16 & 130 & 22 \\ 2 & 2 & -1 & 2 & 22 & 3 \end{pmatrix}_{b_0}$

[1526] $1^{\frac{1^1}{1^1}} 5 1^{\frac{1^1}{1^1}} 231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 5 & 0 & 15 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 15 & 0 & 37 & 13 \\ 3 & 5 & -1 & 13 & 4 \end{pmatrix}_{b_0}$

[1527] $1^{\frac{1^3}{1^1}} 5 131^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 5 & 3 & 15 & 5 \\ 2 & 3 & 0 & 6 & 2 \\ 9 & 15 & 6 & 34 & 12 \\ 3 & 5 & 2 & 12 & 4 \end{pmatrix}_{b_0}$

[1528] $2^{\frac{1^{11}}{1^1}} 3 15 1^{\frac{1^1}{1^1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 5 & 3 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 5 & 2 & 2 & 4 \\ 3 & 5 & 2 & 4 & 4 \end{pmatrix}_{b_0}$

[1529] $1^{\frac{1^1}{1^1}} 5 1^{\frac{1^1}{1^1}} 231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 5 & 5 & 15 & 5 \\ 3 & 5 & 1 & 11 & 3 \\ 8 & 15 & 11 & 37 & 13 \\ 3 & 5 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

[1530] $1^{11} 5^{11} 1^{12} 31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 5 & 5 & 0 & 18 & 8 \\ 2 & 2 & 2 & -1 & 8 & 3 \end{pmatrix}_{b_0}$$

[1532] $1^{11} 5^{11} 131^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 6 & 3 \\ 5 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1533] $1^{11} 5^{12} 131^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 6 & 3 \\ 6 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1531] $1^{13} 5^{13} 131^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 5 & 3 & 20 & 5 \\ 2 & 3 & 0 & 8 & 2 \\ 11 & 20 & 8 & 62 & 16 \\ 3 & 5 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1534] $1^{13} 5^{13} 131^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 1 & 18 & 3 \\ 15 & 21 & 18 & 97 & 21 \\ 3 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[1535] $2^3 3^1 5^1 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 50, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 3 \\ 4 & 5 & 3 & 10 & 5 \\ 2 & 3 & 0 & 4 & 2 \\ 6 & 10 & 4 & 14 & 8 \\ 3 & 5 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

[1536] $21^{11} 5^{11} 131^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 6 & 3 \\ 5 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1537] $1^{11} 5^{11} 131^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 53, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 10 & 3 \\ 7 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[1538] $1^{11} 5^{14} 131^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 53, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 10 & 3 \\ 10 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[1539] $1^{13} 5^{13} 131^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 3 & 21 & 4 \\ 2 & 3 & 1 & 18 & 3 \\ 16 & 21 & 18 & 97 & 21 \\ 3 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[1540] $1^{13} 5 1^{23} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 1 & 1 & 15 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 20 & 2 \\ 12 & 15 & 0 & 20 & 193 & 27 \\ 2 & 2 & -1 & 2 & 27 & 3 \end{pmatrix}_{b_0}$$

[1541] $1^{15} 5 1^{123} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 59, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 5 & 0 & 20 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 20 & 0 & 63 & 17 \\ 3 & 5 & -1 & 17 & 4 \end{pmatrix}_{b_0}$$

[1542] $1^{121} 3 2 2 1^{15} 1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & 1 & 11 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 6 & 11 & 0 & 49 & 11 & 13 \\ 1 & 3 & 0 & 11 & 1 & 3 \\ 2 & 3 & -1 & 13 & 3 & 3 \end{pmatrix}_{b_0}$$

[1543] $1^{15} 5 1^{123} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 5 & 0 & 25 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 25 & 0 & 97 & 21 \\ 3 & 5 & -1 & 21 & 4 \end{pmatrix}_{b_0}$$

[1544] $1^{14} 5 1^{13} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 62, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 5 & 3 & 20 & 5 \\ 2 & 3 & 0 & 8 & 2 \\ 12 & 20 & 8 & 62 & 16 \\ 3 & 5 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1545] $2 3 1 5 1^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 65, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 3 \\ 4 & 5 & 3 & 15 & 5 \\ 2 & 3 & 0 & 6 & 2 \\ 9 & 15 & 6 & 34 & 12 \\ 3 & 5 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[1546] $2 3 1 5 1^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 67, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 3 & 3 & 1 & 6 & 3 \\ 5 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1547] $1^{11} 5 1^{13} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 10 & 3 \\ 9 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[1548] $1^{11} 5 1^{12} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 10 & 3 \\ 8 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[1549] $1^{12} 5 1^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 1 & 8 & 3 \\ 9 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[1550] $1^{12} 5 1^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 1 & 8 & 3 \\ 7 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[1551] $1^{13} 5^1 1^3 231^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 75$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 15 & 2 \\ 1 & 0 & 1 & 1 & 18 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 24 & 2 \\ 15 & 18 & 0 & 24 & 295 & 33 \\ 2 & 2 & -1 & 2 & 33 & 3 \end{pmatrix}_{b_0}$$

[1552] $2^3 3^1 15^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 78$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 3 & 3 & 1 & 8 & 3 \\ 7 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[1553] $1^{12} 5^1 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 80$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 1 & 8 & 3 \\ 8 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[1554] $1^{11} 5^1 231^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 80$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 5 & 5 & 30 & 5 \\ 3 & 5 & 1 & 22 & 3 \\ 16 & 30 & 22 & 154 & 26 \\ 3 & 5 & 3 & 26 & 4 \end{pmatrix}_{b_0}$$

[1555] $1^{11} 5^1 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 85$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 11 & 9 & 14 & 49 & 15 \\ 3 & 3 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[1556] $1^{11} 5^1 131^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 85$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 12 & 9 & 14 & 49 & 15 \\ 3 & 3 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[1557] $2^3 3^1 15^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 86$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 3 \\ 4 & 5 & 3 & 20 & 5 \\ 2 & 3 & 0 & 8 & 2 \\ 12 & 20 & 8 & 62 & 16 \\ 3 & 5 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[1558] $1^{11} 5^1 1^3 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 88$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 6 & 8 & 0 & 31 & 11 \\ 2 & 2 & 2 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[1559] $2^3 3^1 15^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 89$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 5 & 3 & 3 \\ 4 & 5 & 8 & 5 & 5 \\ 5 & 8 & 8 & 6 & 6 \\ 3 & 5 & 6 & 2 & 4 \\ 3 & 5 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[1560] $2^3 3^1 15^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 98$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 5 & 6 & 3 \\ 4 & 5 & 8 & 10 & 5 \\ 5 & 8 & 8 & 12 & 6 \\ 6 & 10 & 12 & 14 & 8 \\ 3 & 5 & 6 & 8 & 4 \end{pmatrix}_{b_0}$$

[1561] $2^3 3^1 15^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 105$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 3 & 3 & 1 & 10 & 3 \\ 8 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[1562] \ 21^{13} 5^{11} 131^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 1 & 3 & 17 & 3 \\ 2 & 3 & 1 & 22 & 3 \\ 17 & 17 & 22 & 137 & 25 \\ 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[1563] \ 1^{11} 5^{11} 1^{12} 231^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 7 & 11 & 0 & 50 & 14 \\ 2 & 2 & 2 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1564] \ 1^{12} 5^{11} 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 108, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 12 & 3 \\ 9 & 10 & 12 & 42 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1565] \ 2^{113} 3^{11} 151^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 113, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 5 & 9 & 3 \\ 4 & 5 & 8 & 15 & 5 \\ 5 & 8 & 8 & 18 & 6 \\ 9 & 15 & 18 & 34 & 12 \\ 3 & 5 & 6 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[1566] \ 2^{15} 3^{11} 151^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 113, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.1$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 3 \\ 4 & 5 & 3 & 25 & 5 \\ 2 & 3 & 0 & 10 & 2 \\ 15 & 25 & 10 & 98 & 20 \\ 3 & 5 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[1567] \ 21^{12} 5^{11} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 118, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 1 & 3 & 8 & 3 \\ 2 & 3 & 1 & 8 & 3 \\ 8 & 8 & 8 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1568] \ 21^{12} 5^{11} 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 118, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 12 & 3 \\ 10 & 10 & 12 & 42 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1569] \ 1^{15} 5^{12} 231^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 122, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 5 & 0 & 30 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 30 & 0 & 154 & 26 \\ 3 & 5 & -1 & 26 & 4 \end{pmatrix}_{b_0}$$

$$[1570] \ 1^{12} 5^{11} 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 132, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 12 & 3 \\ 10 & 10 & 12 & 42 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[1571] \ 1^{12} 5^{14} 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 132, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 12 & 3 \\ 12 & 10 & 12 & 42 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[1572] \quad & 2 \overset{1^{114}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 134, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 5 & 12 & 3 \\ 4 & 5 & 8 & 20 & 5 \\ 5 & 8 & 8 & 24 & 6 \\ 12 & 20 & 24 & 62 & 16 \\ 3 & 5 & 6 & 16 & 4 \end{pmatrix}_{b_0} \\
[1574] \quad & 1^1 \overset{1^1}{5} 1^3 231^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 24 & 3 \\ 3 & 5 & 5 & 45 & 5 \\ 3 & 5 & 1 & 33 & 3 \\ 24 & 45 & 33 & 349 & 39 \\ 3 & 5 & 3 & 39 & 4 \end{pmatrix}_{b_0} \\
[1573] \quad & 1^{\overset{1^3}{12}} \overset{1^1}{5} 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 12 & 3 \\ 11 & 10 & 12 & 42 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0} \\
[1575] \quad & 1^{\overset{1^1}{11}} \overset{1^1}{5} 1^{22} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 141, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 9 & 12 & 0 & 81 & 17 \\ 2 & 2 & 2 & -1 & 17 & 3 \end{pmatrix}_{b_0} \\
[1576] \quad & 2 \overset{1^{16}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 146, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.2 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 2 & 18 & 3 \\ 4 & 5 & 3 & 30 & 5 \\ 2 & 3 & 0 & 12 & 2 \\ 18 & 30 & 12 & 142 & 24 \\ 3 & 5 & 2 & 24 & 4 \end{pmatrix}_{b_0} \\
[1577] \quad & 1^{\overset{1^2}{12}} \overset{1^1}{5} 131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 12 & 12 & 16 & 70 & 18 \\ 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0} \\
[1578] \quad & 2 \overset{1^{15}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 161, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.1 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 5 & 15 & 3 \\ 4 & 5 & 8 & 25 & 5 \\ 5 & 8 & 8 & 30 & 6 \\ 15 & 25 & 30 & 98 & 20 \\ 3 & 5 & 6 & 20 & 4 \end{pmatrix}_{b_0} \\
[1579] \quad & 2 \overset{1^{121}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 161, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 4 & 8 & 3 & 3 \\ 4 & 5 & 13 & 5 & 5 \\ 8 & 13 & 24 & 10 & 10 \\ 3 & 5 & 10 & 2 & 4 \\ 3 & 5 & 10 & 4 & 4 \end{pmatrix}_{b_0} \\
[1580] \quad & 1^1 \overset{1^1}{5} 1^{22} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 5 & 0 & 35 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 35 & 0 & 202 & 30 \\ 3 & 5 & -1 & 30 & 4 \end{pmatrix}_{b_0} \\
[1581] \quad & 1^{\overset{1^{12}}{11}} 3221^{11} \overset{1^1}{5} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 171, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 1 & 5 & 2 \\ 4 & 18 & 6 & 0 & 25 & 8 \\ 1 & 6 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 25 & 8 & 0 & 31 & 11 \\ 2 & 8 & 2 & -1 & 11 & 3 \end{pmatrix}_{b_0} \\
[1582] \quad & 2 \overset{1^{12}}{3} \overset{1^1}{15} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 178, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 10 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 3 & 3 & 1 & 12 & 3 \\ 10 & 10 & 12 & 42 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[1583] $1^{12} \overset{1^3}{5} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 13 & 12 & 16 & 70 & 18 \\ 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[1584] $2 \overset{1^{17}}{3} 151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 185, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 21 & 3 \\ 4 & 5 & 3 & 35 & 5 \\ 2 & 3 & 0 & 14 & 2 \\ 21 & 35 & 14 & 194 & 28 \\ 3 & 5 & 2 & 28 & 4 \end{pmatrix}_{b_0}$$

[1585] $1^{11} \overset{1^1}{5} 1^2 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 189, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 10 & 15 & 0 & 108 & 20 \\ 2 & 2 & 2 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

[1586] $1^{13} \overset{1^2}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 189, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 12 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[1587] $1^{12} \overset{1^4}{5} 131^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 14 & 12 & 16 & 70 & 18 \\ 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[1588] $1^{15} \overset{1^1}{5} 1^2 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 194, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 5 & 0 & 40 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 40 & 0 & 258 & 34 \\ 3 & 5 & -1 & 34 & 4 \end{pmatrix}_{b_0}$$

[1589] $1^{12} \overset{1^3}{5} 131^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 204, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 15 & 14 & 20 & 106 & 22 \\ 3 & 3 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[1590] $1^{13} \overset{1^3}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 205, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 13 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[1591] $1^{13} \overset{1^4}{5} 131^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 205, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 14 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[1592] $2 \overset{1^{12}}{3} 151^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 217, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 3 & 3 & 1 & 14 & 3 \\ 12 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[1593] $22 \overset{1^{11}}{3} 151^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 227, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 5 & 5 & 3 \\ 6 & 14 & 13 & 13 & 8 \\ 5 & 13 & 8 & 11 & 6 \\ 5 & 13 & 11 & 9 & 7 \\ 3 & 8 & 6 & 7 & 4 \end{pmatrix}_{b_0}$$

[1596] $1^{12} 5^{14} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 228$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 16 & 14 & 20 & 106 & 22 \\ 3 & 3 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[1594] $21^{13} 5^{11} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 228$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 18 & 3 \\ 15 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[1595] $1^{12} 5^{11} 231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 228$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 11 & 13 & 0 & 101 & 19 \\ 2 & 2 & 2 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

[1597] $1^{13} 5^{12} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 245$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 18 & 3 \\ 14 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[1598] $21^{13} 5^{11} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 252$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 13 & 13 & 14 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[1599] $1^{11} 5^{13} 231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 257$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 24 & 3 \\ 3 & 5 & 0 & 45 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 45 & 0 & 349 & 39 \\ 3 & 5 & -1 & 39 & 4 \end{pmatrix}_{b_0}$$

[1600] $2^3 3^{13} 15^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 262$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.5$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 13 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 3 & 3 & 1 & 16 & 3 \\ 13 & 12 & 16 & 70 & 18 \\ 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[1601] $1^{11} 5^{13} 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 264$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 13 & 2 \\ 1 & 2 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 13 & 19 & 0 & 190 & 26 \\ 2 & 2 & 2 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

[1602] $22^3 3^{11} 15^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 276$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 5 & 5 & 3 \\ 5 & 9 & 12 & 11 & 7 \\ 5 & 12 & 8 & 11 & 6 \\ 5 & 11 & 11 & 9 & 7 \\ 3 & 7 & 6 & 7 & 4 \end{pmatrix}_{b_0}$$

[1603] $1^{13} 5^{11} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -58$, $\det = 277$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.8$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 18 & 3 \\ 15 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[1604] $1^{13} 5^{14} 131^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 293, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 18 & 3 \\ 16 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[1605] $1^{12} 5^{11} 231^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 304, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 12 & 16 & 0 & 130 & 22 \\ 2 & 2 & 2 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[1606] $1^{12} 5^{13} 231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 309, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 14 & 2 \\ 1 & 2 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 14 & 17 & 0 & 181 & 25 \\ 2 & 2 & 2 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

[1607] $1^{13} 5^{13} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 325, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 17 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 1 & 22 & 3 \\ 17 & 17 & 22 & 137 & 25 \\ 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

[1608] $1^{11} 5^{13} 231^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 327, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 26 & 3 \\ 3 & 5 & 0 & 50 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 26 & 50 & 0 & 419 & 43 \\ 3 & 5 & -1 & 43 & 4 \end{pmatrix}_{b_0}$$

[1609] $2^3 3^{14} 151^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 330, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 16 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 3 & 3 & 1 & 20 & 3 \\ 16 & 14 & 20 & 106 & 22 \\ 3 & 3 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[1610] $22^3 3^{121} 151^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 341, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 10 & 5 & 3 \\ 6 & 14 & 26 & 13 & 8 \\ 10 & 26 & 41 & 22 & 13 \\ 5 & 13 & 22 & 9 & 7 \\ 3 & 8 & 13 & 7 & 4 \end{pmatrix}_{b_0}$$

[1611] $1^{12} 5^{11} 231^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 356, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 13 & 2 \\ 1 & 2 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 11 & 13 & 19 & 0 & 165 & 25 \\ 2 & 2 & 2 & -1 & 25 & 3 \end{pmatrix}_{b_0}$$

[1612] $1^{13} 5^{14} 131^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 357, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 1 & 22 & 3 \\ 18 & 17 & 22 & 137 & 25 \\ 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

[1613] $1^{12} 3221^{12} 51^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 360, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 10 & 2 \\ 4 & 18 & 6 & 0 & 50 & 8 \\ 1 & 6 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 50 & 16 & 0 & 130 & 22 \\ 2 & 8 & 2 & -1 & 22 & 3 \end{pmatrix}_{b_0}$$

[1614] $2^3 3^{13} 151^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 363, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 15 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 3 & 3 & 1 & 18 & 3 \\ 15 & 15 & 18 & 97 & 21 \\ 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[1615] $1^{112} 22 \ 3 \ 151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 398, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 5 & 10 & 3 \\ 6 & 14 & 13 & 26 & 8 \\ 5 & 13 & 8 & 22 & 6 \\ 10 & 26 & 22 & 42 & 14 \\ 3 & 8 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

[1616] $1^{12} 5^1 1^3 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 413, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 0 & 15 & 2 \\ 1 & 2 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 13 & 15 & 20 & 0 & 218 & 28 \\ 2 & 2 & 2 & -1 & 28 & 3 \end{pmatrix}_{b_0}$$

[1617] $1^{13} 5^1 1^3 231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 424, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 0 & 16 & 2 \\ 1 & 2 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 13 & 16 & 18 & 0 & 211 & 27 \\ 2 & 2 & 2 & -1 & 27 & 3 \end{pmatrix}_{b_0}$$

[1618] $1^{12} 5^1 1^3 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 0 & 16 & 2 \\ 1 & 2 & 0 & 0 & 23 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 16 & 23 & 0 & 261 & 31 \\ 2 & 2 & 2 & -1 & 31 & 3 \end{pmatrix}_{b_0}$$

[1619] $2^3 3^1 15^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 493, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 18 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 3 & 3 & 1 & 22 & 3 \\ 18 & 17 & 22 & 137 & 25 \\ 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

[1620] $1^{12} 5^1 1^3 231^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 549, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 0 & 17 & 2 \\ 1 & 2 & 0 & 0 & 26 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 17 & 26 & 0 & 310 & 34 \\ 2 & 2 & 2 & -1 & 34 & 3 \end{pmatrix}_{b_0}$$

[1621] $1^{13} 5^1 1^3 231^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 556, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 14 & 2 \\ 1 & 0 & 2 & 0 & 17 & 2 \\ 1 & 2 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 17 & 21 & 0 & 250 & 30 \\ 2 & 2 & 2 & -1 & 30 & 3 \end{pmatrix}_{b_0}$$

[1622] $1^{13} 5^1 1^3 231^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 664, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 0 & 18 & 2 \\ 1 & 2 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 18 & 24 & 0 & 295 & 33 \\ 2 & 2 & 2 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

[1623] $1^{13} 3221^{13} 5^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 675, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 15 & 2 \\ 4 & 18 & 6 & 0 & 75 & 8 \\ 1 & 6 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 75 & 24 & 0 & 295 & 33 \\ 2 & 8 & 2 & -1 & 33 & 3 \end{pmatrix}_{b_0}$$

[1624] $1^{113} 22 \ 3 \ 151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 683, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 5 & 15 & 3 \\ 6 & 14 & 13 & 39 & 8 \\ 5 & 13 & 8 & 33 & 6 \\ 15 & 39 & 33 & 97 & 21 \\ 3 & 8 & 6 & 21 & 4 \end{pmatrix}_{b_0}$$

[1625] $1^{13} 5^1 1^3 231^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 748$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 16 & 2 \\ 1 & 0 & 2 & 0 & 19 & 2 \\ 1 & 2 & 0 & 0 & 27 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 16 & 19 & 27 & 0 & 346 & 36 \\ 2 & 2 & 2 & -1 & 36 & 3 \end{pmatrix}_{b_0}$$

[1626] $22^1 3^{114} 151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 1082$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 5 & 20 & 3 \\ 6 & 14 & 13 & 52 & 8 \\ 5 & 13 & 8 & 44 & 6 \\ 20 & 52 & 44 & 174 & 28 \\ 3 & 8 & 6 & 28 & 4 \end{pmatrix}_{b_0}$$

[1627] $22^1 3^{115} 151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10$, $T_{\max} = 12$, $\Delta = -59$, $\det = 1595$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 7.1$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 5 & 25 & 3 \\ 6 & 14 & 13 & 65 & 8 \\ 5 & 13 & 8 & 55 & 6 \\ 25 & 65 & 55 & 273 & 35 \\ 3 & 8 & 6 & 35 & 4 \end{pmatrix}_{b_0}$$

$T = 11$ [1629] $1^{11} 31^1 5^1 131^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -37$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

[1630] $1^{12} 5^1 1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 2 & 5 & 4 \\ 4 & 2 & 5 & 3 & 5 \\ 3 & 3 & 4 & 5 & 5 \end{pmatrix}_{b_0}$$

[1631] $1^{11} 31^1 5^1 131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -37$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 4 & 2 \\ 2 & 4 & 0 & 4 & 6 & 6 \\ 2 & 3 & -1 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

[1632] $1^{11} 5^1 1^3 2321^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -32$, $\det = -2$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 6 & 0 & 28 & 6 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 28 & 0 & 106 & 24 \\ 3 & 6 & -1 & 24 & 5 \end{pmatrix}_{b_0}$$

[1633] $1^{11} 5^1 1321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -3$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 2 & 5 & 4 \\ 3 & 2 & 5 & 3 & 5 \\ 3 & 3 & 4 & 5 & 5 \end{pmatrix}_{b_0}$$

[1634] $1^{11} 5^1 1321^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -3$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 2 & 0 & 4 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 4 & 0 & 5 \end{pmatrix}_{b_0}$$

[1635] $1^{14} 5^1 1321^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -6$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 6 & 3 & 24 & 6 \\ 2 & 3 & 0 & 8 & 2 \\ 12 & 24 & 8 & 78 & 20 \\ 3 & 6 & 2 & 20 & 5 \end{pmatrix}_{b_0}$$

[1636] $1^{1^3} 5^{1^1} 1321^{1^3} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 6 & 3 & 18 & 6 \\ 2 & 3 & 0 & 6 & 2 \\ 9 & 18 & 6 & 43 & 15 \\ 3 & 6 & 2 & 15 & 5 \end{pmatrix}_{b_0}$$

[1637] $1^{1^2} 5^{1^1} 1321^{1^2} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 6 & 3 & 12 & 6 \\ 2 & 3 & 0 & 4 & 2 \\ 6 & 12 & 4 & 18 & 10 \\ 3 & 6 & 2 & 10 & 5 \end{pmatrix}_{b_0}$$

[1638] $1^{1^1} 5^{1^1} 1321^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 6 & 3 & 6 & 6 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 6 & 2 & 3 & 5 \\ 3 & 6 & 2 & 5 & 5 \end{pmatrix}_{b_0}$$

[1640] $1^{1^1} 5^{1^1} 1321^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 3 & 2 & 5 & 5 & 7 & 5 \\ 2 & 3 & 5 & 7 & 6 & 6 \\ 2 & 2 & 3 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

[1642] $21^{1^1} 5^{1^1} 1321^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 3 & 2 & 5 & 5 & 7 & 5 \\ 2 & 3 & 5 & 7 & 6 & 6 \\ 2 & 2 & 3 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

[1641] $1^{1^1} 5^{1^1} 1^{1^1} 2321^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 6 & 6 & 16 & 6 \\ 3 & 6 & 2 & 12 & 4 \\ 8 & 16 & 12 & 38 & 14 \\ 3 & 6 & 4 & 14 & 5 \end{pmatrix}_{b_0}$$

[1639] $1^{1^1} 4^{1^1} 14131^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 2 & 2 & 1 & 3 & 2 & 4 \\ 2 & 3 & 0 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

[1643] $2^{1^1} 3^{1^1} 21^{1^1} 61^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -13, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 3 \\ 2 & 1 & 3 & 3 & 3 \\ 3 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[1644] $1^{1^1} 5^{1^1} 1321^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 2 & 7 & 4 \\ 4 & 5 & 7 & 10 & 8 \\ 3 & 3 & 4 & 8 & 5 \end{pmatrix}_{b_0}$$

[1645] $1^{1^1} 4^{1^1} 14131^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -33, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 2 & 3 & 1 & 2 & 2 & 4 \\ 2 & 3 & 0 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

[1646] $1^{1^1} 5^{1^1} 12321^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 5 & 0 & 6 & 6 \\ 2 & 2 & 3 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

[1647] $1^{11} 31^1 5131^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 12, \Delta = -37, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

[1648] $1^{11} 51321^{12} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 7 & 7 & 12 & 29 & 13 \\ 3 & 3 & 4 & 13 & 5 \end{pmatrix}_{b_0}$$

[1649] $1^{11} 4141^1 31^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 12, \Delta = -33, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 3 & 4 & 0 & 3 & 5 & 5 \\ 2 & 3 & 0 & 2 & 5 & 4 \end{pmatrix}_{b_0}$$

[1650] $1^{11} 51^1 2321^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 4 & 8 & 0 & 22 & 10 \\ 2 & 2 & 3 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

[1651] $2^1 3^1 2161^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 3 & 3 & 1 & 4 & 3 \\ 3 & 2 & 4 & 2 & 4 \\ 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[1652] $2321^1 61^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 6 & 3 \\ 3 & 3 & 1 & 7 & 3 \\ 7 & 6 & 7 & 19 & 9 \\ 3 & 3 & 3 & 9 & 4 \end{pmatrix}_{b_0}$$

[1653] $1^{14} 51321^{114} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 6 & 9 & 24 & 6 \\ 4 & 9 & 9 & 28 & 7 \\ 12 & 24 & 28 & 78 & 20 \\ 3 & 6 & 7 & 20 & 5 \end{pmatrix}_{b_0}$$

[1654] $1^{13} 51321^{113} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 6 & 9 & 18 & 6 \\ 4 & 9 & 9 & 21 & 7 \\ 9 & 18 & 21 & 43 & 15 \\ 3 & 6 & 7 & 15 & 5 \end{pmatrix}_{b_0}$$

[1655] $1^{12} 51321^{112} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 6 & 9 & 12 & 6 \\ 4 & 9 & 9 & 14 & 7 \\ 6 & 12 & 14 & 18 & 10 \\ 3 & 6 & 7 & 10 & 5 \end{pmatrix}_{b_0}$$

[1656] $1^{11} 51321^{111} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 6 & 9 & 6 & 6 \\ 4 & 9 & 9 & 7 & 7 \\ 3 & 6 & 7 & 3 & 5 \\ 3 & 6 & 7 & 5 & 5 \end{pmatrix}_{b_0}$$

[1657] $1^{11} \overset{1^2}{5} 1321^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{red}{s}\textcolor{blue}{u}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -27$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 2 & 7 & 4 \\ 6 & 5 & 7 & 10 & 8 \\ 3 & 3 & 4 & 8 & 5 \end{pmatrix}_{b_0}$

[1658] $1^{11} \overset{1^1}{5} 1321^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{red}{s}\textcolor{blue}{u}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -28$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 2 & 7 & 4 \\ 5 & 5 & 7 & 10 & 8 \\ 3 & 3 & 4 & 8 & 5 \end{pmatrix}_{b_0}$

[1659] $1^{11} \overset{1^3}{5} 1321^{13} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{red}{s}\textcolor{blue}{u}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -32$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 2 & 17 & 4 \\ 11 & 9 & 17 & 58 & 18 \\ 3 & 3 & 4 & 18 & 5 \end{pmatrix}_{b_0}$

[1660] $1^{11} \overset{1^2}{5} 1321^{12} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{red}{s}\textcolor{blue}{u}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -37$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 8 & 7 & 12 & 29 & 13 \\ 3 & 3 & 4 & 13 & 5 \end{pmatrix}_{b_0}$

[1661] $1^{11} \overset{1^3}{5} 1321^{12} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{red}{s}\textcolor{blue}{u}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -42$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 9 & 7 & 12 & 29 & 13 \\ 3 & 3 & 4 & 13 & 5 \end{pmatrix}_{b_0}$

[1662] $1^{11} \overset{1^4}{5} 1321^{13} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{red}{s}\textcolor{blue}{u}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -43$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 2 & 17 & 4 \\ 12 & 9 & 17 & 58 & 18 \\ 3 & 3 & 4 & 18 & 5 \end{pmatrix}_{b_0}$

[1663] $1^{11} \overset{1^{11}}{4} 141^{13} 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{red}{s}\textcolor{blue}{u}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -33$, $\det = -50$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 6 & 2 \\ 5 & 6 & 1 & 6 & 15 & 9 \\ 2 & 3 & 0 & 2 & 9 & 4 \end{pmatrix}_{b_0}$

[1664] $2321^{11} \overset{1^1}{6} 1^{11} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{red}{s}\textcolor{blue}{u}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -58$, $\det = -57$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 3 & 3 & 1 & 9 & 3 \\ 9 & 9 & 9 & 34 & 12 \\ 3 & 3 & 3 & 12 & 4 \end{pmatrix}_{b_0}$

[1665] $1^{11} \overset{1^1}{5} 1^{12} 2321^{12} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{red}{s}\textcolor{blue}{u}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -32$, $\det = -63$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 7 & 13 & 0 & 54 & 16 \\ 2 & 2 & 3 & -1 & 16 & 4 \end{pmatrix}_{b_0}$

[1666] $1^{12} \overset{1^1}{5} 1321^{12} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{red}{s}\textcolor{blue}{u}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -63$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 2 & 14 & 4 \\ 9 & 10 & 14 & 46 & 16 \\ 3 & 3 & 4 & 16 & 5 \end{pmatrix}_{b_0}$

[1667] $2 \overset{1^{11}}{3} 21 \overset{1^1}{6} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 3 & 3 & 1 & 6 & 3 \\ 5 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1668] $2 \overset{1^{11}}{3} 21 \overset{1^1}{6} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -78, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 10 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 3 & 3 & 1 & 11 & 3 \\ 10 & 8 & 11 & 37 & 13 \\ 3 & 3 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

[1669] $1 \overset{1^2}{1^2} 5 1321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 2 & 19 & 4 \\ 12 & 12 & 19 & 79 & 21 \\ 3 & 3 & 4 & 21 & 5 \end{pmatrix}_{b_0}$$

[1670] $1 \overset{1^2}{1^2} 5 1321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -82, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 2 & 14 & 4 \\ 10 & 10 & 14 & 46 & 16 \\ 3 & 3 & 4 & 16 & 5 \end{pmatrix}_{b_0}$$

[1671] $1 \overset{1^3}{1^3} 5 1321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -87, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 2 & 14 & 4 \\ 11 & 10 & 14 & 46 & 16 \\ 3 & 3 & 4 & 16 & 5 \end{pmatrix}_{b_0}$$

[1672] $1 \overset{1^3}{1^3} 5 1321^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -87, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 2 & 24 & 4 \\ 15 & 14 & 24 & 122 & 26 \\ 3 & 3 & 4 & 26 & 5 \end{pmatrix}_{b_0}$$

[1673] $41 \overset{1^1}{1^1} 231 5 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -90, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 1 & 8 & 3 \\ 9 & 51 & 0 & 48 & 15 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 48 & 0 & 42 & 14 \\ 3 & 15 & -1 & 14 & 4 \end{pmatrix}_{b_0}$$

[1674] $2 \overset{1^{12}}{3} 21 \overset{1^1}{6} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -105, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 3 & 3 & 1 & 10 & 3 \\ 8 & 7 & 10 & 25 & 11 \\ 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

[1675] $1 \overset{1^3}{1^3} 5 1321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -107, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 2 & 19 & 4 \\ 13 & 12 & 19 & 79 & 21 \\ 3 & 3 & 4 & 21 & 5 \end{pmatrix}_{b_0}$$

[1677] $2 \overset{1^{12}}{3} 21 \overset{1^1}{6} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -118, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 13 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 3 & 3 & 1 & 15 & 3 \\ 13 & 10 & 15 & 63 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[1676] $1 \overset{1^4}{1^4} 5 1321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -118, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 2 & 19 & 4 \\ 14 & 12 & 19 & 79 & 21 \\ 3 & 3 & 4 & 21 & 5 \end{pmatrix}_{b_0}$$

[1678] $1^{12} \overset{1^4}{5} 1321^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -118$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 2 & 24 & 4 \\ 16 & 14 & 24 & 122 & 26 \\ 3 & 3 & 4 & 26 & 5 \end{pmatrix}_{b_0}$$

[1679] $41^{12} 231 \overset{1^1}{5} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -58$, $\det = -126$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.5$
 $T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 1 & 16 & 3 \\ 9 & 51 & 0 & 96 & 15 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 96 & 0 & 174 & 28 \\ 3 & 15 & -1 & 28 & 4 \end{pmatrix}_{b_0}$$

[1680] $1^{13} \overset{1^2}{5} 1321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -147$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 11.99 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 2 & 21 & 4 \\ 14 & 15 & 21 & 106 & 24 \\ 3 & 3 & 4 & 24 & 5 \end{pmatrix}_{b_0}$$

[1681] $1^{13} \overset{1^3}{5} 1321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -172$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 11.98 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 2 & 21 & 4 \\ 15 & 15 & 21 & 106 & 24 \\ 3 & 3 & 4 & 24 & 5 \end{pmatrix}_{b_0}$$

[1682] $2 \overset{1^{11}}{3} 21 \overset{1^1}{6} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -58$, $\det = -175$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 11 & 3 \\ 3 & 3 & 1 & 13 & 3 \\ 12 & 11 & 13 & 56 & 16 \\ 3 & 3 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

[1683] $1^{13} \overset{1^3}{5} 1321^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -182$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 17 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 2 & 26 & 4 \\ 17 & 17 & 26 & 153 & 29 \\ 3 & 3 & 4 & 29 & 5 \end{pmatrix}_{b_0}$$

[1684] $1^{13} \overset{1^4}{5} 1321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -183$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 11.99 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 2 & 21 & 4 \\ 16 & 15 & 21 & 106 & 24 \\ 3 & 3 & 4 & 24 & 5 \end{pmatrix}_{b_0}$$

[1685] $41^{13} 231 \overset{1^1}{5} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -58$, $\det = -186$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 1 & 24 & 3 \\ 9 & 51 & 0 & 144 & 15 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 144 & 0 & 394 & 42 \\ 3 & 15 & -1 & 42 & 4 \end{pmatrix}_{b_0}$$

[1686] $1^{13} \overset{1^4}{5} 1321^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -213$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 6.8$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 2 & 26 & 4 \\ 18 & 17 & 26 & 153 & 29 \\ 3 & 3 & 4 & 29 & 5 \end{pmatrix}_{b_0}$$

[1687] $41^{14} 231 \overset{1^1}{5} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 13$, $\Delta = -58$, $\det = -270$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12.98 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 9 & 1 & 32 & 3 \\ 9 & 51 & 0 & 192 & 15 \\ 1 & 0 & -3 & 0 & -1 \\ 32 & 192 & 0 & 702 & 56 \\ 3 & 15 & -1 & 56 & 4 \end{pmatrix}_{b_0}$$

[1688] $2 \overset{1^2}{3} 21 \overset{1^1}{6} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 11, T_{\max} = 13, \Delta = -58, \det = -277, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 3 & 15 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 3 & 3 & 1 & 17 & 3 \\ 15 & 13 & 17 & 86 & 20 \\ 3 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$

[1689] $1 \overset{1^1}{11} 31 \overset{1^1}{5} 13221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 12, \Delta = -9, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0$

[1695] $1 \overset{1^1}{1^1} 6 \overset{1^1}{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -35, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 4 \\ 3 & 4 & 0 & 4 & 5 \end{pmatrix}_{b_0}$

[1694] $1 \overset{1^1}{1^1} 6 \overset{1^1}{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -35, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 2 & 3 & 2 & 4 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 0 & 2 \\ 4 & 4 & 2 & 6 \end{pmatrix}_{b_0}$

[1693] $1 \overset{1^1}{1^1} 6 \overset{1^1}{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -34, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 & 4 \\ 2 & 3 & 1 & 2 & 3 \\ 2 & 4 & 2 & 2 & 4 \\ 3 & 4 & 3 & 4 & 5 \end{pmatrix}_{b_0}$

[1696] $1 \overset{1^2}{1^1} 6 \overset{1^1}{12} 321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 2 & 5 & 4 \\ 4 & 2 & 5 & 3 & 5 \\ 3 & 3 & 4 & 5 & 5 \end{pmatrix}_{b_0}$

[1697] $1 \overset{1^1}{1^1} 6 \overset{1^1}{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -34, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 2 & 3 & 0 & 4 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 0 & 5 \end{pmatrix}_{b_0}$

[1698] $1 \overset{1^1}{1^1} 6 \overset{1^1}{12} 321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 2 & 0 & 4 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 4 & 0 & 5 \end{pmatrix}_{b_0}$

[1699] $1 \overset{1^1}{1^1} 6 \overset{1^1}{12} 321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 2 & 5 & 4 \\ 3 & 2 & 5 & 3 & 5 \\ 3 & 3 & 4 & 5 & 5 \end{pmatrix}_{b_0}$

[1700] $1 \overset{1^1}{1^1} 5 \overset{1^1}{13} 221^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 12, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 2 & 6 & 6 & 4 \\ 3 & 2 & 6 & 6 & 8 & 6 \\ 2 & 3 & 6 & 8 & 7 & 7 \\ 2 & 2 & 4 & 6 & 7 & 5 \end{pmatrix}_{b_0}$

[1701] $1 \overset{1^1}{1^1} 6 \overset{1^1}{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -34, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & 0 & 4 & 2 \\ 3 & 2 & 4 & 4 & 6 & 6 \\ 2 & 2 & 2 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1702] \ 1^{\frac{1}{1}1} 6 \ 131^{\frac{1}{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -34, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[1703] \ 21^{\frac{1}{1}1} 5 \ 13221^{\frac{1}{1}11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 2 & 6 & 6 & 4 \\ 3 & 2 & 6 & 6 & 8 & 6 \\ 2 & 3 & 6 & 8 & 7 & 7 \\ 2 & 2 & 4 & 6 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[1704] \ 1^{\frac{1}{1}2} 5 \ 13221^{\frac{1}{1}13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 3 & 22 & 5 \\ 12 & 12 & 22 & 88 & 24 \\ 3 & 3 & 5 & 24 & 6 \end{pmatrix}_{b_0}$$

$$[1705] \ 1^{\frac{1}{1}2} 5 \ 13221^{\frac{1}{1}12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 5, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 3 & 14 & 5 \\ 8 & 7 & 14 & 33 & 15 \\ 3 & 3 & 5 & 15 & 6 \end{pmatrix}_{b_0}$$

$$[1706] \ 1^{\frac{1}{1}1} 6 \ 131^{\frac{1}{1}11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -34, \ \det = 7, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.86 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 2 & 3 & 4 & 4 \\ 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 2 & 2 & 4 \\ 3 & 4 & 3 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[1707] \ 1^{\frac{1}{1}1} 6 \ 1^{\frac{1}{1}11} 2321^{\frac{1}{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 8, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 2 & 4 & 2 \\ 1 & 2 & 1 & 5 & 8 & 3 \\ 3 & 2 & 5 & 5 & 12 & 5 \\ 4 & 4 & 8 & 12 & 22 & 10 \\ 2 & 2 & 3 & 5 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1708] \ 21^{\frac{1}{1}1} 6 \ 1^{\frac{1}{1}11} 2321^{\frac{1}{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 8, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 5 & 6 & 3 \\ 2 & 1 & 4 & 4 & 3 \\ 5 & 4 & 9 & 10 & 7 \\ 6 & 4 & 10 & 10 & 8 \\ 3 & 3 & 7 & 8 & 5 \end{pmatrix}_{b_0}$$

$$[1709] \ 1^{\frac{1}{1}2} 5 \ 13221^{\frac{1}{1}14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 8, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 14 & 3 \\ 2 & 3 & 3 & 28 & 5 \\ 16 & 14 & 28 & 138 & 30 \\ 3 & 3 & 5 & 30 & 6 \end{pmatrix}_{b_0}$$

$$[1710] \ 1^{\frac{1}{1}1} 6 \ 12321^{\frac{1}{1}11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 8, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 1 & 5 & 5 & 3 \\ 3 & 2 & 5 & 5 & 7 & 5 \\ 2 & 3 & 5 & 7 & 6 & 6 \\ 2 & 2 & 3 & 5 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[1711] \ 21^{\frac{1}{1}1} 6 \ 12321^{\frac{1}{1}11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 8, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 5 & 4 & 3 \\ 2 & 1 & 4 & 4 & 3 \\ 5 & 4 & 9 & 8 & 7 \\ 4 & 4 & 8 & 6 & 6 \\ 3 & 3 & 7 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[1712] \ 1^{11} 31^1 5^{11} 1321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -9, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 5 & 0 & 7 & 7 \\ 2 & 3 & 3 & -1 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[1713] \ 1^1 6^{11} 12321^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 2 & 7 & 4 \\ 4 & 5 & 7 & 10 & 8 \\ 3 & 3 & 4 & 8 & 5 \end{pmatrix}_{b_0}$$

$$[1716] \ 1^{14} 5^{11} 13221^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 7 & 10 & 28 & 7 \\ 4 & 10 & 10 & 32 & 8 \\ 12 & 28 & 32 & 94 & 24 \\ 3 & 7 & 8 & 24 & 6 \end{pmatrix}_{b_0}$$

$$[1714] \ 1^{12} 5^{11} 13221^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 7 & 10 & 14 & 7 \\ 4 & 10 & 10 & 16 & 8 \\ 6 & 14 & 16 & 22 & 12 \\ 3 & 7 & 8 & 12 & 6 \end{pmatrix}_{b_0}$$

$$[1715] \ 1^{11} 5^{11} 13221^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 7 & 10 & 7 & 7 \\ 4 & 10 & 10 & 8 & 8 \\ 3 & 7 & 8 & 4 & 6 \\ 3 & 7 & 8 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[1717] \ 1^{13} 5^{11} 13221^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 7 & 10 & 21 & 7 \\ 4 & 10 & 10 & 24 & 8 \\ 9 & 21 & 24 & 52 & 18 \\ 3 & 7 & 8 & 18 & 6 \end{pmatrix}_{b_0}$$

$$[1718] \ 1^{11} 6^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -35, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 2 & 3 & 4 & 4 \\ 3 & 2 & 5 & 4 \\ 4 & 5 & 4 & 6 \\ 4 & 4 & 6 & 6 \end{pmatrix}_{b_0}$$

$$[1719] \ 1^{11} 6^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -35, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 2 & 1 & 4 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 4 & 1 & 3 & 5 \\ 3 & 4 & 0 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[1720] \ 1^{11} 6^{11} 12321^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 7 & 7 & 12 & 29 & 13 \\ 3 & 3 & 4 & 13 & 5 \end{pmatrix}_{b_0}$$

$$[1721] \ 1^{11} 6^{11} 131^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 2 & 3 & 6 & 4 \\ 2 & 3 & 1 & 6 & 3 \\ 4 & 6 & 6 & 10 & 8 \\ 3 & 4 & 3 & 8 & 5 \end{pmatrix}_{b_0}$$

[1722] $1^{\frac{1}{1}} \frac{1^{11} 2321^1}{6} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 8 & 8 & 12 & 38 & 14 \\ 3 & 3 & 4 & 14 & 5 \end{pmatrix}_{b_0}$

[1723] $1^{\frac{1}{1}} \frac{1^{11} 2321^1}{6} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 6 & 2 \\ 1 & 0 & 2 & 2 & 7 & 2 \\ 1 & 2 & 1 & 5 & 13 & 3 \\ 3 & 2 & 5 & 5 & 19 & 5 \\ 6 & 7 & 13 & 19 & 54 & 16 \\ 2 & 2 & 3 & 5 & 16 & 4 \end{pmatrix}_{b_0}$

[1724] $21^{\frac{1}{1}} \frac{1^{11} 2321^1}{6} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 5 & 10 & 3 \\ 2 & 1 & 4 & 8 & 3 \\ 5 & 4 & 9 & 18 & 7 \\ 10 & 8 & 18 & 34 & 14 \\ 3 & 3 & 7 & 14 & 5 \end{pmatrix}_{b_0}$

[1725] $1^{\frac{1}{1}} \frac{1^{11} 2321^1}{6} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 8 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 9 & 8 & 12 & 38 & 14 \\ 3 & 3 & 4 & 14 & 5 \end{pmatrix}_{b_0}$

[1726] $1^{\frac{1}{1}} \frac{1^{11} 2321^1}{6} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 2 & 17 & 4 \\ 11 & 10 & 17 & 67 & 19 \\ 3 & 3 & 4 & 19 & 5 \end{pmatrix}_{b_0}$

[1727] $1^{\frac{1}{1}} \frac{1^2 6}{6} 12321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 2 & 7 & 4 \\ 6 & 5 & 7 & 10 & 8 \\ 3 & 3 & 4 & 8 & 5 \end{pmatrix}_{b_0}$

[1728] $1^{\frac{1}{1}} \frac{1^2 6}{6} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -34, \det = 27, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 5 & 3 \\ 2 & 2 & 3 & 6 & 4 \\ 2 & 3 & 1 & 6 & 3 \\ 5 & 6 & 6 & 10 & 8 \\ 3 & 4 & 3 & 8 & 5 \end{pmatrix}_{b_0}$

[1729] $1^{\frac{1}{1}} \frac{1^2 6}{6} 12321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 28, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 2 & 7 & 4 \\ 5 & 5 & 7 & 10 & 8 \\ 3 & 3 & 4 & 8 & 5 \end{pmatrix}_{b_0}$

[1730] $1^{\frac{1}{1}} \frac{1^3 6}{6} 12321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 32, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 2 & 17 & 4 \\ 11 & 9 & 17 & 58 & 18 \\ 3 & 3 & 4 & 18 & 5 \end{pmatrix}_{b_0}$

[1731] $1^{\frac{1}{1}} \frac{1^2 6}{6} 12321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 37, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 8 & 7 & 12 & 29 & 13 \\ 3 & 3 & 4 & 13 & 5 \end{pmatrix}_{b_0}$

[1733] $1^{\frac{1}{1}} \frac{1^2 6}{6} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 2 & 17 & 4 \\ 12 & 10 & 17 & 67 & 19 \\ 3 & 3 & 4 & 19 & 5 \end{pmatrix}_{b_0}$$

$$[1732] \ 1^{11} \overset{1^3}{6} 12321^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 12 & 4 \\ 9 & 7 & 12 & 29 & 13 \\ 3 & 3 & 4 & 13 & 5 \end{pmatrix}_{b_0}$$

$$[1737] \ 1^{11} \overset{1^4}{6} 12321^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 2 & 17 & 4 \\ 12 & 9 & 17 & 58 & 18 \\ 3 & 3 & 4 & 18 & 5 \end{pmatrix}_{b_0}$$

$$[1736] \ 1^1 \overset{1^3}{6}^{12} 2321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 2 & 17 & 4 \\ 13 & 10 & 17 & 67 & 19 \\ 3 & 3 & 4 & 19 & 5 \end{pmatrix}_{b_0}$$

$$[1735] \ 1^1 \overset{1^3}{6}^{13} 2321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 2 & 22 & 4 \\ 15 & 12 & 22 & 106 & 24 \\ 3 & 3 & 4 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[1734] \ 1^1 \overset{1^1}{6}^{11} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 2 & 3 & 10 & 4 \\ 2 & 3 & 1 & 8 & 3 \\ 7 & 10 & 8 & 26 & 12 \\ 3 & 4 & 3 & 12 & 5 \end{pmatrix}_{b_0}$$

$$[1739] \ 1^1 \overset{1^4}{6}^{13} 2321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 50, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 12 & 3 \\ 2 & 3 & 2 & 22 & 4 \\ 16 & 12 & 22 & 106 & 24 \\ 3 & 3 & 4 & 24 & 5 \end{pmatrix}_{b_0}$$

$$[1738] \ 1^1 \overset{1^2}{6}^{11} 131^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -34, \det = 50, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 2 & 3 & 10 & 4 \\ 2 & 3 & 1 & 8 & 3 \\ 8 & 10 & 8 & 26 & 12 \\ 3 & 4 & 3 & 12 & 5 \end{pmatrix}_{b_0}$$

$$[1740] \ 1^1 \overset{1^{12} 2321^1}{6}^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 63, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 2 & 19 & 4 \\ 12 & 13 & 19 & 90 & 22 \\ 3 & 3 & 4 & 22 & 5 \end{pmatrix}_{b_0}$$

$$[1741] \ 1^{11} \overset{1^1}{6}^{13} 2321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 82, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 2 & 24 & 4 \\ 15 & 15 & 24 & 133 & 27 \\ 3 & 3 & 4 & 27 & 5 \end{pmatrix}_{b_0}$$

$$[1742] \ 1^{11} \overset{1^2}{6}^{14} 2321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 87, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 2 & 29 & 4 \\ 18 & 17 & 29 & 186 & 32 \\ 3 & 3 & 4 & 32 & 5 \end{pmatrix}_{b_0}$$

[1743] $1^{11} \overset{1^1}{6} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 92, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 2 & 19 & 4 \\ 13 & 13 & 19 & 90 & 22 \\ 3 & 3 & 4 & 22 & 5 \end{pmatrix}_{b_0}$$

[1744] $1^{11} \overset{1^2}{6} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 107, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 2 & 19 & 4 \\ 14 & 13 & 19 & 90 & 22 \\ 3 & 3 & 4 & 22 & 5 \end{pmatrix}_{b_0}$$

[1745] $1^{11} \overset{1^3}{6} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 2 & 19 & 4 \\ 15 & 13 & 19 & 90 & 22 \\ 3 & 3 & 4 & 22 & 5 \end{pmatrix}_{b_0}$$

[1746] $1^{11} \overset{1^2}{6} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 117, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 2 & 24 & 4 \\ 16 & 15 & 24 & 133 & 27 \\ 3 & 3 & 4 & 27 & 5 \end{pmatrix}_{b_0}$$

[1747] $1^{11} \overset{1^3}{6} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 19 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 2 & 29 & 4 \\ 19 & 17 & 29 & 186 & 32 \\ 3 & 3 & 4 & 32 & 5 \end{pmatrix}_{b_0}$$

[1748] $1^{11} \overset{1^3}{6} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 138, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 17 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 2 & 24 & 4 \\ 17 & 15 & 24 & 133 & 27 \\ 3 & 3 & 4 & 27 & 5 \end{pmatrix}_{b_0}$$

[1749] $1^{11} \overset{1^4}{6} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 145, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 2 & 24 & 4 \\ 18 & 15 & 24 & 133 & 27 \\ 3 & 3 & 4 & 27 & 5 \end{pmatrix}_{b_0}$$

[1750] $1^{11} \overset{1^4}{6} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 155, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 20 & 3 \\ 2 & 1 & 3 & 17 & 3 \\ 2 & 3 & 2 & 29 & 4 \\ 20 & 17 & 29 & 186 & 32 \\ 3 & 3 & 4 & 32 & 5 \end{pmatrix}_{b_0}$$

$T = 13$ [1752] $1^1 \overset{1^{131}}{6} 13141^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 2 & 0 \\ 1 & 0 & 2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[1753] $1^1 \overset{1^{1131}}{6} 13141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 2 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 4 & 0 & 2 & 2 & 4 \\ 2 & 4 & -1 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

[1754] $1^{\overline{11}2321} 6^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 13, T_{\max} = 13, \Delta = -8, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 3 & 6 & 5 \\ 5 & 6 & 8 & 8 \\ 4 & 5 & 8 & 7 \end{pmatrix}_{b_0}$$

[1755] $1^{\overline{11}} 5^{11} 1315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 13, T_{\max} = 14, \Delta = -31, \det = -15, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 13.93 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 5 & 0 & 6 & 6 \\ 2 & 2 & 3 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

[1756] $1^{\overline{11}31} 6^{11} 13141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 13, T_{\max} = 14, \Delta = -33, \det = -18, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 2 & 0 & 4 & 2 & 2 & 4 \\ 2 & 0 & 3 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

[1757] $1^{\overline{11}31} 6^{11} 13141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 13, T_{\max} = 14, \Delta = -33, \det = -18, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 2 & 4 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 2 & 4 \\ 2 & 4 & -1 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

[1758] $1^{\overline{12}31} 6^{11} 131^{11} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 13, T_{\max} = 14, \Delta = -33, \det = -50, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 2 & 1 & 2 & 12 & 4 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & 8 & 2 \\ 5 & 12 & 0 & 8 & 42 & 14 \\ 2 & 4 & -1 & 2 & 14 & 4 \end{pmatrix}_{b_0}$$

[1759] $1^{\overline{11}31} 6^{11} 131^{11} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 13, T_{\max} = 14, \Delta = -33, \det = -50, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 8 & 3 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 5 & 0 & 8 & 6 & 22 & 10 \\ 2 & 0 & 3 & 2 & 10 & 4 \end{pmatrix}_{b_0}$$

$T = 14$ [1760] $1^{\overline{11}} 4^{11} 1414141^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 14, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 6.0$

[1762] $1^{\overline{11}321} 8^{11} 1231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -33, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & 1 & 0 & 4 & 2 \\ 3 & 2 & 4 & 4 & 6 & 6 \\ 2 & 2 & 2 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

[1763] $1^{\overline{11}321} 8^{11} 1231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -33, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 1 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[1764] $1^{\overline{12}3161} 1^{11} 231^{11} 51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 14, T_{\max} = 15, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} =$

6.5

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 0 & 9 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 0 & 34 & 12 \\ 2 & 3 & 0 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

[1765] $1^{11}3216131\overset{1^1}{5}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 0 & 4 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 3 & 5 \\ 3 & 3 & 0 & 5 & 5 \end{pmatrix}_{b_0}$$

[1767] $1^{11}612315131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 2 & 5 & 4 \\ 5 & 5 & 7 & 7 \\ 4 & 4 & 7 & 6 \end{pmatrix}_{b_0}$$

[1768] $1^{11}3161231\overset{1^1}{5}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 3 \\ 2 & 1 & 0 & 4 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 3 & 5 \\ 3 & 3 & 0 & 5 & 5 \end{pmatrix}_{b_0}$$

[1766] $1^1321^16131\overset{1^1}{5}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 15, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 5 & 5 & 0 & 12 & 8 \\ 3 & 3 & 0 & 8 & 5 \end{pmatrix}_{b_0}$$

[1769] $1^{112}3161^1231\overset{1^1}{5}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 9 & 3 \\ 6 & 9 & 0 & 9 & 34 & 12 \\ 2 & 3 & 0 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[1770] $1^1\overset{11}{6}^{11}3151321^11 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 27, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 14.96 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 10 & 4 \\ 3 & 2 & 8 & 4 \\ 10 & 8 & 25 & 13 \\ 4 & 4 & 13 & 6 \end{pmatrix}_{b_0}$$

[1771] $1^1\overset{112}{6}^{11}3151321^11 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 27, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 14.96 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 13 & 4 \\ 3 & 2 & 10 & 4 \\ 13 & 10 & 43 & 17 \\ 4 & 4 & 17 & 6 \end{pmatrix}_{b_0}$$

[1772] $1^{12}3161^1231\overset{1^1}{5}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 32, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 0 & 10 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 10 & 0 & 36 & 14 \\ 3 & 3 & 0 & 14 & 5 \end{pmatrix}_{b_0}$$

[1773] $1^{11}3161^1231\overset{1^1}{5}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 42, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 14.90 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 0 & 7 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 7 & 0 & 21 & 11 \\ 3 & 3 & 0 & 11 & 5 \end{pmatrix}_{b_0}$$

[1774] $1^{11}321^1 613151^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 14, T_{\max} = 15, \Delta = -29, \det = 60, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 1 & 0 & 9 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 9 & 0 & 29 & 13 \\ 3 & 3 & 0 & 13 & 5 \end{pmatrix}_{b_0}$

[1775] $1^{11}3151321^1 61^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 14, T_{\max} = 15, \Delta = -29, \det = 82, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 14.95 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 2 & 3 & 18 & 4 \\ 3 & 2 & 15 & 4 \\ 18 & 15 & 88 & 24 \\ 4 & 4 & 24 & 6 \end{pmatrix}_{b_0}$

[1776] $1^{11}3161^1 23151^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 14, T_{\max} = 15, \Delta = -29, \det = 87, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 14.99 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 1 & 0 & 8 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 8 & 0 & 27 & 13 \\ 3 & 3 & 0 & 13 & 5 \end{pmatrix}_{b_0}$

[1777] $1^{13}3161^1 23151^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 14, T_{\max} = 15, \Delta = -29, \det = 87, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 14.99 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 0 & 14 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 14 & 0 & 63 & 19 \\ 3 & 3 & 0 & 19 & 5 \end{pmatrix}_{b_0}$

[1778] $1^{11}3151321^1 61^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 14, T_{\max} = 15, \Delta = -29, \det = 92, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 2 & 3 & 15 & 4 \\ 3 & 2 & 13 & 4 \\ 15 & 13 & 58 & 20 \\ 4 & 4 & 20 & 6 \end{pmatrix}_{b_0}$

[1779] $1^{12}3161^1 23151^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 14, T_{\max} = 15, \Delta = -29, \det = 107, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 14.92 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 1 & 0 & 11 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 11 & 0 & 42 & 16 \\ 3 & 3 & 0 & 16 & 5 \end{pmatrix}_{b_0}$

[1780] $1^{12}321^1 613151^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 14, T_{\max} = 15, \Delta = -29, \det = 115, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 14.99 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 11 & 3 \\ 2 & 1 & 0 & 13 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 11 & 13 & 0 & 56 & 18 \\ 3 & 3 & 0 & 18 & 5 \end{pmatrix}_{b_0}$

[1781] $1^{12}3161^1 23151^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 14, T_{\max} = 15, \Delta = -29, \det = 172, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 14.98 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 0 & 12 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 12 & 0 & 52 & 18 \\ 3 & 3 & 0 & 18 & 5 \end{pmatrix}_{b_0}$

[1782] $1^{13}3161^1 23151^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 14, T_{\max} = 15, \Delta = -29, \det = 182, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 14.96 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 1 & 2 & 1 & 13 & 3 \\ 2 & 1 & 0 & 15 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 13 & 15 & 0 & 73 & 21 \\ 3 & 3 & 0 & 21 & 5 \end{pmatrix}_{b_0}$

$T = 15$ [1785] $1^{11}6131513221^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 15, T_{\max} = 15, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 6.0$

[1786] $1^{\overline{11}} \overline{6}^{2315131} 131^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 15$, $T_{\max} = 15$, $\Delta = -8$, $\det = 0$, $(n_+, n_-, n_0) = (1, 15, 1)$, $c_{\text{ext}} = 6.0$

[1784] $1^{\overline{11}} \overline{4}^{141414141} 131^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 15$, $T_{\max} = 15$, $\Delta = -1$, $\det = 0$, $(n_+, n_-, n_0) = (1, 15, 2)$, $c_{\text{ext}} = 6.0$

[1783] $21^{\overline{11}} \overline{6}^{131513221} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 15$, $T_{\max} = 15$, $\Delta = 0$, $\det = 0$, $(n_+, n_-, n_0) = (1, 15, 1)$, $c_{\text{ext}} = 6.0$

[1794] $1^{\overline{11}} \overline{6}^{1^{\overline{13}}23151321^{\overline{1}}}$ $1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 15$, $T_{\max} = 15$, $\Delta = -1$, $\det = -4$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 1 & 1 & 9 & 3 \\ 1 & 1 & 0 & 9 & 2 \\ 12 & 9 & 9 & 61 & 21 \\ 3 & 3 & 2 & 21 & 6 \end{pmatrix}_{b_0}$$

[1795] $1^{\overline{11}} \overline{6}^{13151321} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 15$, $T_{\max} = 15$, $\Delta = 0$, $\det = -4$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 1 & 2 & 2 & 6 & 6 & 4 \\ 3 & 2 & 6 & 6 & 8 & 6 \\ 2 & 3 & 6 & 8 & 7 & 7 \\ 2 & 2 & 4 & 6 & 7 & 5 \end{pmatrix}_{b_0}$$

[1796] $21^{\overline{11}} \overline{6}^{13151321} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 15$, $T_{\max} = 15$, $\Delta = 0$, $\det = -4$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 & 2 \\ 1 & 1 & 0 & 3 & 3 & 2 \\ 3 & 2 & 3 & 6 & 5 & 6 \\ 2 & 2 & 3 & 5 & 3 & 5 \\ 2 & 2 & 2 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

[1797] $1^{\overline{11}} \overline{6}^{13151321} 1^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 15$, $T_{\max} = 15$, $\Delta = 0$, $\det = -5$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 3 & 14 & 5 \\ 8 & 7 & 14 & 33 & 15 \\ 3 & 3 & 5 & 15 & 6 \end{pmatrix}_{b_0}$$

[1798] $31^{\overline{11}} \overline{6}^{13161} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 15$, $T_{\max} = 16$, $\Delta = -33$, $\det = -8$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 5 & 0 & 8 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 8 & 0 & 6 & 6 \\ 2 & 2 & 5 & -1 & 6 & 4 \end{pmatrix}_{b_0}$$

[1799] $1^{\overline{11}} \overline{6}^{1^{\overline{11}}315131} 131^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 15$, $T_{\max} = 15$, $\Delta = -7$, $\det = -9$, $(n_+, n_-, n_0) = (1, 15, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 2 & 3 & 5 & 4 \\ 3 & 3 & 6 & 5 \\ 5 & 6 & 8 & 8 \\ 4 & 5 & 8 & 7 \end{pmatrix}_{b_0}$$

$T = 16$ [1800] $1^{\overline{11}} \overline{6}^{1^{\overline{11}}31} 131612321^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 16$, $T_{\max} = 16$, $\Delta = -7$, $\det = 0$, $(n_+, n_-, n_0) = (1, 16, 1)$, $c_{\text{ext}} = 6.0$

[1804] $1^{\overline{11}} \overline{8}^{1^{\overline{15}}51^{\overline{1}}2321} 1231^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 16$, $T_{\max} = 17$, $\Delta = -29$, $\det = 2$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 17$, $T_{\max} = 17$, $T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 4 & 2 \\ 5 & 2 & 0 & 4 & 7 & 7 \\ 2 & 2 & 0 & 2 & 7 & 4 \end{pmatrix}_{b_0}$$

[1805] $1^{\overline{11}} \overline{6}^{1^{\overline{11}}31} 1316131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2$ $T = 16$, $T_{\max} = 16$, $\Delta = -4$, $\det = 4$, $(n_+, n_-, n_0) = (1, 16, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 2 & 1 & 0 & 2 & 2 \\ 2 & 4 & 1 & 2 & 2 & 4 \\ 2 & 4 & 2 & 2 & 4 & 5 \end{pmatrix}_{b_0}$$

[1806] $1^{1^{1^1}5^{1^1}1^{2321}}8^{1231^{1^2}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 4, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 3 & 7 & 0 & 18 & 10 \\ 2 & 2 & 2 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

[1807] $2^3 3^{1^{11}} 21812315^{1^1}1^{1^2} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 & 3 \\ 3 & 5 & 0 & 9 & 5 & 5 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 6 & 9 & 0 & 13 & 9 & 9 \\ 3 & 5 & 0 & 9 & 5 & 5 \end{pmatrix}_{b_0}$$

[1808] $1^{1^{11}31}6^{13161231^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 1 & 2 & 4 & 4 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 1 & 2 & 3 & 5 \\ 2 & 4 & 0 & 2 & 5 & 5 \end{pmatrix}_{b_0}$$

[1809] $1^{1^{11}31}6^{1316131^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = -4, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 2 & 2 & 2 & 5 & 4 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 2 & 1 & 0 & 3 & 2 \\ 2 & 5 & 4 & 3 & 7 & 7 \\ 2 & 4 & 2 & 2 & 7 & 5 \end{pmatrix}_{b_0}$$

[1810] $1^{1^{11}31}6^{1316131^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = -4, \det = 9, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 2 & 2 & 2 & 5 & 4 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 2 & 1 & 0 & 3 & 2 \\ 3 & 5 & 4 & 3 & 7 & 7 \\ 2 & 4 & 2 & 2 & 7 & 5 \end{pmatrix}_{b_0}$$

[1811] $1^{1^{12}31}6^{1^{1^2}1321^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = -4, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 2 & 2 & 2 & 9 & 4 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 2 & 1 & 0 & 5 & 2 \\ 5 & 9 & 5 & 5 & 21 & 11 \\ 2 & 4 & 2 & 2 & 11 & 5 \end{pmatrix}_{b_0}$$

[1812] $1^{1^{12}31}6^{1^{1^2}1321^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = -4, \det = 25, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 2 & 2 & 2 & 9 & 4 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 2 & 1 & 0 & 5 & 2 \\ 4 & 9 & 5 & 5 & 21 & 11 \\ 2 & 4 & 2 & 2 & 11 & 5 \end{pmatrix}_{b_0}$$

$T = 17$ [1821] $1^{1^{1^1}5^{1^1}1^{1321}}8^{12321^{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 2 & 1 & 4 & 5 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 2 & 5 & 4 \\ 2 & 5 & 0 & 5 & 6 & 6 \\ 2 & 4 & 0 & 4 & 6 & 5 \end{pmatrix}_{b_0}$$

[1822] $1^{11}6^{1^{1^1}2321712321^{13}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 14 & 3 \\ 2 & 2 & 2 & 16 & 4 \\ 1 & 2 & 0 & 9 & 2 \\ 14 & 16 & 9 & 97 & 25 \\ 3 & 4 & 2 & 25 & 6 \end{pmatrix}_{b_0}$$

$$[1823] \ 1^{1^{11} 5^{13} 21} 12321^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 2 & 0 & 6 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 4 & 6 & 0 & 8 & 8 \\ 3 & 4 & 0 & 8 & 6 \end{pmatrix}_{b_0}$$

$$[1824] \ 1^{13} 232181^{12} 223151^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -8, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 2 & 0 & 18 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 18 & 0 & 86 & 24 \\ 3 & 4 & 0 & 24 & 6 \end{pmatrix}_{b_0}$$

$$[1825] \ 1^{1^{11} 31} 6^{13} 131613141^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 12 & 2 \\ 1 & 3 & 1 & 35 & 5 \\ 1 & 1 & -3 & 0 & -1 \\ 1 & 3 & 0 & 13 & 3 \\ 2 & 5 & 0 & 33 & 5 \\ 2 & 5 & -1 & 35 & 5 \end{pmatrix}_{b_0}$$

$$[1826] \ 1^{14} 232181^{12} 223151^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -9, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 1 & 15 & 3 \\ 2 & 2 & 0 & 23 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 15 & 23 & 0 & 132 & 30 \\ 3 & 4 & 0 & 30 & 6 \end{pmatrix}_{b_0}$$

$$[1827] \ 1^{113} 232181^{12} 223151^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 2 & 1 & 4 & 16 & 4 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 2 & 16 & 4 \\ 8 & 16 & 0 & 16 & 78 & 20 \\ 2 & 4 & 0 & 4 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[1828] \ 1^{1^{12} 31} 6^{13} 1316131^{14} 1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 17, T_{\max} = 17, \Delta = -3, \det = -25, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 3 & 1 & 3 & 14 & 5 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 1 & 10 & 3 \\ 5 & 14 & 0 & 10 & 46 & 16 \\ 2 & 5 & -1 & 3 & 16 & 5 \end{pmatrix}_{b_0}$$

$$T = 18 \quad [1829] \ 1^{1^{11} 31} 6^{13} 13161315131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 18, T_{\max} = 18, \Delta = -6, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 6.0$$

$$T = 19 \quad [1830] \ 1^{1^{11} 31} 6^{13} 131613161231^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 19, T_{\max} = 19, \Delta = -4, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 6.0$$

$$[1831] \ 1^{1^{11} 31} 6^{13} 13161316131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 19, T_{\max} = 19, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 6.0$$

$$[1832] \ 1^{1^{12} 31} 6^{12} 13161316131^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 19, T_{\max} = 19, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 6.5$$

$$T = 20 \quad [1834] \ 1^{1^{11} 31} 6^{13} 1316131613141^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 20, T_{\max} = 20, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 6.0$$

$$[1835] \ 1^{\overline{1}^{12}31} \ 6 \ 13161316131^{\overline{1}41^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 20, \ T_{\max} = 20, \ \Delta = -2, \ \det = 0, \ (n_+, n_-, n_0) = (1, 20, 1), \ c_{\text{ext}} = 6.0$$

4.1139 $\text{NHC}(-2)(-3) + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6$

$$T = 8 \quad [0] \ 1^{\overline{1}^{23}31^{\overline{1}^{11}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 79, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 1 & 3 \\ 3 & 4 & 13 & 12 & 0 & 4 \\ 7 & 13 & 27 & 27 & 0 & 9 \\ 6 & 12 & 27 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 9 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1}^{11}31^{\overline{1}^{11}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 94, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.70 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 5 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1}^{11}31^{\overline{1}^{11}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 103, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.03 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1}^{11}31^{\overline{1}^{11}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 121, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.22 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 3 & 5 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 4 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1}^{12}31^{\overline{1}^{11}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 126, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.81 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 9 & 8 & 0 & 4 \\ 5 & 9 & 12 & 12 & 0 & 6 \\ 4 & 8 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1}^{11}31^{\overline{1}^{11}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 127, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.02 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 4 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1}^{11}31^{\overline{1}^{11}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 148, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 12 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1}^{13}31^{\overline{1}^{11}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 160, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 \\ 9 & 17 & 48 & 15 & 0 & 12 \\ 3 & 5 & 15 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 12 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1}^{11}31^{\overline{1}^{11}}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 166, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$

- [9] $1^{121}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 178$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.07 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 4 & 8 & 9 & 6 & 0 & 5 \\ 2 & 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 5 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$
- [10] $1^{121}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 181$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.17 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 8 & 9 & 0 & 4 \\ 4 & 8 & 9 & 12 & 0 & 5 \\ 5 & 9 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 5 & 7 & -4 & -1 \end{pmatrix}_{b_0}$$
- [11] $1^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 184$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.91 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 4 & 8 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 184$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 19 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 11 & 19 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 15 & -4 & 0 \end{pmatrix}_{b_0}$$
- [13] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 202$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.18 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 \\ 8 & 14 & 36 & 9 & 0 & 10 \\ 2 & 4 & 9 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 10 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$
- [14] $1^{122}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 208$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 13.54 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 1 & 3 \\ 3 & 4 & 8 & 18 & 0 & 4 \\ 4 & 8 & 9 & 24 & 0 & 5 \\ 10 & 18 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 5 & 14 & -4 & -1 \end{pmatrix}_{b_0}$$
- [15] $1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 216$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 5 & 9 & 12 & 9 & 0 & 6 \\ 3 & 5 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$
- [16] $1^{12}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 229$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 \\ 7 & 13 & 27 & 12 & 0 & 9 \\ 3 & 5 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 9 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$
- [17] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 243$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 \\ 5 & 9 & 12 & 6 & 0 & 6 \\ 2 & 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$
- [18] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{red}{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 256$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 10 & 24 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 14 & -4 & 0 \end{pmatrix}_{b_0}$$

- [19] $1^{122}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 274$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 1 & 3 \\ 3 & 4 & 13 & 8 & 0 & 4 \\ 7 & 13 & 27 & 18 & 0 & 9 \\ 4 & 8 & 18 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 9 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$
- [20] $1^{11}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 286$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 9 & 0 & 4 \\ 3 & 5 & 9 & 0 & 3 \\ 5 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$
- [21] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 304$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$
- [22] $1^{111}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 310$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 8 & 14 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 11 & -4 & 0 \end{pmatrix}_{b_0}$$
- [23] $1^{14}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 313$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 13.06 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 1 & 3 \\ 3 & 4 & 5 & 21 & 0 & 4 \\ 3 & 5 & 3 & 18 & 0 & 3 \\ 11 & 21 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 16 & -4 & -1 \end{pmatrix}_{b_0}$$
- [24] $1^{112}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 343$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 3 \\ 4 & 9 & 4 & 20 & 0 & 5 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 8 & 20 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 11 & -4 & 0 \end{pmatrix}_{b_0}$$
- [25] $1^{12}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 373$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 3 & 12 & 0 & 3 \\ 7 & 13 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 10 & -4 & 0 \end{pmatrix}_{b_0}$$
- [26] $1^{13}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 382$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.97 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 3 & 15 & 0 & 3 \\ 9 & 17 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$
- [27] $1^{121}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 391$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 7 & 13 & 27 & 9 & 0 & 9 \\ 2 & 4 & 9 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 9 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$
- [28] $1^{122}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 403$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.7$
 $T_{\min} = 13.04 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 1 & 3 \\ 3 & 4 & 8 & 13 & 0 & 4 \\ 4 & 8 & 9 & 18 & 0 & 5 \\ 7 & 13 & 18 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 5 & 10 & -4 & -1 \end{pmatrix}_{b_0}$$

- [29] $1^{113}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 436$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 15 & 36 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 21 & -4 & 0 \end{pmatrix}_{b_0}$$
- [30] $1^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 471$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 5 & 9 & 12 & 15 & 0 & 6 \\ 5 & 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$
- [31] $1^{121}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 486$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 10 & 18 & 57 & 12 & 0 & 13 \\ 2 & 4 & 12 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 13 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$
- [32] $1^{112}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 526$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 7 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 10 & -4 & 0 \end{pmatrix}_{b_0}$$
- [33] $1^{131}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 547$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 9 & 17 & 48 & 12 & 0 & 12 \\ 2 & 4 & 12 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 12 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$
- [34] $1^{113}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 568$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.98 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 3 \\ 4 & 9 & 4 & 28 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 11 & 28 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 15 & -4 & 0 \end{pmatrix}_{b_0}$$
- [35] $1^{121}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 574$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 1 & 3 \\ 3 & 4 & 13 & 9 & 0 & 4 \\ 7 & 13 & 27 & 21 & 0 & 9 \\ 5 & 9 & 21 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 9 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$
- [36] $1^{131}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 595$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 1 & 3 \\ 3 & 4 & 17 & 9 & 0 & 4 \\ 9 & 17 & 48 & 27 & 0 & 12 \\ 5 & 9 & 27 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 12 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$
- [37] $1^{113}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 631$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 13 & 32 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 18 & -4 & 0 \end{pmatrix}_{b_0}$$
- [38] $1^{123}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 637$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.02 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 17 & 0 & 4 \\ 4 & 8 & 9 & 24 & 0 & 5 \\ 9 & 17 & 24 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 5 & 13 & -4 & -1 \end{pmatrix}_{b_0}$$

- [39] $1^{123}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 646$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.11 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 8 & 22 & 0 & 4 \\ 4 & 8 & 9 & 30 & 0 & 5 \\ 12 & 22 & 30 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 5 & 17 & -4 & -1 \end{pmatrix}_{b_0}$$
- [40] $1^{112}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 648$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 5 & 9 & 12 & 21 & 0 & 6 \\ 7 & 13 & 21 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & 10 & -4 & 0 \end{pmatrix}_{b_0}$$
- [41] $1^{114}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 688$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.99 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 20 & 1 & 3 \\ 4 & 9 & 4 & 48 & 0 & 5 \\ 2 & 4 & 0 & 24 & 0 & 2 \\ 20 & 48 & 24 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 28 & -4 & 0 \end{pmatrix}_{b_0}$$
- [42] $1^{112}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 736$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 10 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 14 & -4 & 0 \end{pmatrix}_{b_0}$$
- [43] $1^{113}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 747$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 9 & 17 & 0 & 4 \\ 5 & 9 & 12 & 27 & 0 & 6 \\ 9 & 17 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$
- [45] $1^{112}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 760$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 1 & 3 \\ 3 & 4 & 4 & 28 & 0 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 16 & 28 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 22 & -4 & 0 \end{pmatrix}_{b_0}$$
- [44] $1^{113}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 760$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.8$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 9 & 17 & 12 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$
- [46] $1^{114}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 768$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 1 & 3 \\ 3 & 4 & 9 & 21 & 0 & 4 \\ 5 & 9 & 12 & 33 & 0 & 6 \\ 11 & 21 & 33 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 6 & 16 & -4 & 0 \end{pmatrix}_{b_0}$$
- [47] $1^{131}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 778$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.66 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 22 & 4 & 0 & 4 \\ 12 & 22 & 84 & 15 & 0 & 16 \\ 2 & 4 & 15 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 16 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$
- [48] $1^{112}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 814$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.5$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 23 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 13 & 23 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 18 & -4 & 0 \end{pmatrix}_{b_0}$$

- [49] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 823, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 13.02 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 14 & 36 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 19 & -4 & -1 \end{pmatrix}_{b_0}$$
- [50] $1^{122}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 841, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.7$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 1 & 3 \\ 3 & 4 & 13 & 13 & 0 & 4 \\ 7 & 13 & 27 & 30 & 0 & 9 \\ 7 & 13 & 30 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 9 & 10 & -4 & 0 \end{pmatrix}_{b_0}$$
- [51] $1^{114}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 991, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 1 & 3 \\ 4 & 9 & 4 & 44 & 0 & 5 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 18 & 44 & 21 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 25 & -4 & 0 \end{pmatrix}_{b_0}$$
- [52] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1006, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 21 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 11 & 21 & 15 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 16 & -4 & 0 \end{pmatrix}_{b_0}$$
- [53] $1^{123}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1030, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 1 & 3 \\ 3 & 4 & 13 & 17 & 0 & 4 \\ 7 & 13 & 27 & 39 & 0 & 9 \\ 9 & 17 & 39 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 9 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$
- [54] $1^{114}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1036, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 9 & 4 & 40 & 0 & 5 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 16 & 40 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 22 & -4 & 0 \end{pmatrix}_{b_0}$$
- [55] $1^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 12 & 22 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 17 & -4 & 0 \end{pmatrix}_{b_0}$$
- [56] $1^{113}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 1 & 3 \\ 3 & 4 & 4 & 27 & 0 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 15 & 27 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 21 & -4 & 0 \end{pmatrix}_{b_0}$$
- [57] $1^{113}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1606, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 1 & 3 \\ 3 & 4 & 4 & 32 & 0 & 4 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 18 & 32 & 21 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 25 & -4 & 0 \end{pmatrix}_{b_0}$$
- [58] $1^{114}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2 + \textcolor{red}{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 1624, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 4 & 26 & 0 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 14 & 26 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 20 & -4 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{114}31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{e}}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 2110$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 1 & 3 \\ 3 & 4 & 4 & 31 & 0 & 4 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 17 & 31 & 21 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 24 & -4 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{114}31^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{e}}_6$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 2464$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 1 & 3 \\ 3 & 4 & 4 & 36 & 0 & 4 \\ 2 & 4 & 0 & 24 & 0 & 2 \\ 20 & 36 & 24 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 28 & -4 & 0 \end{pmatrix}_{b_0}$$

4.1140 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \text{NHC}(-2)(-3)$

$T = 10$ $[0] 1^{1111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{e}}_6 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2]$ $T = 10$, $T_{\max} = 14$, $\Delta = -92$, $\det = 470$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 13.17 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 4 & 3 \\ 3 & 5 & 3 & 3 & 0 & 4 & 3 & 3 \\ 3 & 4 & 2 & 3 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

4.1141 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{so}_8$

$T = 9$ $[0] 1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -115$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 14.23 \rightarrow 15$, $T_{\max} = 99$, $T \in [15, 99]$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 9 & 5 & 7 & -4 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -271$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -334$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 1 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 8 & 16 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{so}}_8$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -931$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.73 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

4.1142 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$

$T = 9$ $[0] 1^{111}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{e}}_6 + \textcolor{blue}{\mathfrak{su}}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -29$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 16.86 \rightarrow 17$, $T_{\max} = 23$, $T \in [17, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 4 \\ 3 & 5 & 3 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 6 & 3 & 0 & 4 & 4 \\ 3 & 4 & 3 & 3 & -4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{teal}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -56, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.71 \rightarrow 17, \ T_{\max} = 23, \ T \in [17, 23]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 2 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 & 4 \\ 8 & 14 & 36 & 9 & 0 & 9 & 10 \\ 2 & 4 & 9 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 3 & 0 & 1 & 3 \\ 3 & 4 & 10 & 3 & -4 & 3 & -4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{teal}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -56, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.71 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 4 & 5 & 0 & 5 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & 2 & 4 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{teal}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -65, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & 4 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1121}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{teal}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -90, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.07 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 5 & 3 \\ 3 & 4 & 9 & 8 & 0 & 9 & 4 \\ 5 & 9 & 12 & 12 & 0 & 15 & 6 \\ 4 & 8 & 12 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 15 & 12 & 0 & 13 & 7 \\ 3 & 4 & 6 & 6 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{teal}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -92, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 9 & 0 & 37 & 11 \\ 3 & 4 & 2 & 3 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{113}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{teal}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -101, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 11 & 3 \\ 3 & 4 & 4 & 5 & 0 & 19 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 3 & 5 & 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 19 & 12 & 21 & 0 & 73 & 15 \\ 3 & 4 & 2 & 4 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{131}} 3^{\overline{1}} 1^{\overline{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{teal}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -107, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.32 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 2 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 & 4 \\ 9 & 17 & 48 & 15 & 0 & 12 & 12 \\ 3 & 5 & 15 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 12 & 3 & 0 & 1 & 3 \\ 3 & 4 & 12 & 4 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1121}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{violet}{\mathfrak{e}}_6 + \textcolor{teal}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.34 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 5 & 3 \\ 3 & 4 & 4 & 8 & 0 & 9 & 4 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 4 & 8 & 6 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 12 & 0 & 13 & 7 \\ 3 & 4 & 2 & 6 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{\overline{1111}1^1} 3^{\overline{131}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -116, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 5 & 3 \\ 3 & 4 & 4 & 19 & 0 & 9 & 4 \\ 2 & 4 & 0 & 12 & 0 & 6 & 2 \\ 11 & 19 & 12 & 73 & 0 & 33 & 15 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 33 & 0 & 13 & 7 \\ 3 & 4 & 2 & 15 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{\overline{1111}1^1} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -128, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & 3 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{\overline{1112}1^1} 3^{\overline{121}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -137, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 7 & 3 \\ 3 & 4 & 4 & 14 & 0 & 13 & 4 \\ 2 & 4 & 0 & 9 & 0 & 9 & 2 \\ 8 & 14 & 9 & 37 & 0 & 33 & 11 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 33 & 0 & 28 & 10 \\ 3 & 4 & 2 & 11 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{\overline{113}1^1} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -173, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.39 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 9 & 3 \\ 3 & 4 & 4 & 5 & 0 & 17 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 3 & 5 & 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 15 & 0 & 49 & 13 \\ 3 & 4 & 2 & 4 & -4 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{\overline{1121}1^1} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -174, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.85 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 2 & 3 \\ 3 & 4 & 9 & 8 & 0 & 4 & 4 \\ 5 & 9 & 12 & 12 & 0 & 6 & 6 \\ 4 & 8 & 12 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 6 & 0 & 1 & 3 \\ 3 & 4 & 6 & 6 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{\overline{1111}1^1} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -191, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.22 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 5 & 3 \\ 4 & 9 & 4 & 8 & 0 & 12 & 5 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 3 & 8 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 6 & 9 & 0 & 13 & 7 \\ 3 & 5 & 2 & 4 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{\overline{1122}1^1} 3^{\overline{113}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 13 & 3 \\ 3 & 4 & 4 & 13 & 0 & 23 & 4 \\ 2 & 4 & 0 & 9 & 0 & 15 & 2 \\ 7 & 13 & 9 & 28 & 0 & 54 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 15 & 54 & 0 & 100 & 18 \\ 3 & 4 & 2 & 10 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{\overline{111}1^1} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -219, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 2 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 & 4 \\ 5 & 9 & 12 & 9 & 0 & 6 & 6 \\ 3 & 5 & 9 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 3 & 0 & 1 & 3 \\ 3 & 4 & 6 & 4 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{121} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -245, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 2 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 & 4 \\ 7 & 13 & 27 & 12 & 0 & 9 & 9 \\ 3 & 5 & 12 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 9 & 3 & 0 & 1 & 3 \\ 3 & 4 & 9 & 4 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -284, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 8 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 3 & 6 & 0 & 4 & 4 \\ 3 & 5 & 2 & 4 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -341, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 3 & 5 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 9 & 0 & 13 & 7 \\ 3 & 4 & 2 & 4 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -344, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.81 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & 3 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{111} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -353, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 8 & 3 \\ 3 & 4 & 4 & 5 & 0 & 14 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 3 & 5 & 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 15 & 0 & 37 & 11 \\ 3 & 4 & 2 & 4 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -359, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 3 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 3 & 3 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -368, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 5 & 3 \\ 4 & 9 & 4 & 12 & 0 & 12 & 5 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 12 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 6 & 15 & 0 & 13 & 7 \\ 3 & 5 & 2 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{112} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -380, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 7 & 3 \\ 3 & 4 & 4 & 5 & 0 & 13 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 3 & 5 & 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 12 & 0 & 28 & 10 \\ 3 & 4 & 2 & 4 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -380, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 6 & 6 & 0 & 10 & 6 \\ 3 & 4 & 2 & 3 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -408, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 & 4 \\ 5 & 9 & 12 & 6 & 0 & 6 & 6 \\ 2 & 4 & 6 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 6 & 3 & 0 & 1 & 3 \\ 3 & 4 & 6 & 3 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1114}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -413, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 5 & 4 & 0 & 21 & 4 \\ 3 & 5 & 3 & 3 & 0 & 18 & 3 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 18 & 15 & 0 & 76 & 16 \\ 3 & 4 & 3 & 3 & -4 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1122}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 4 & 3 \\ 3 & 4 & 4 & 8 & 0 & 8 & 4 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 4 & 8 & 6 & 10 & 0 & 12 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 6 & 12 & 0 & 10 & 6 \\ 3 & 4 & 2 & 6 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1112}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -425, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 15 & 15 & 0 & 100 & 18 \\ 3 & 4 & 2 & 3 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 3 \\ 3 & 4 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 9 & 9 & 0 & 25 & 9 \\ 3 & 4 & 2 & 3 & -4 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1122}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -444, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 7 & 3 \\ 3 & 4 & 9 & 8 & 0 & 13 & 4 \\ 5 & 9 & 12 & 12 & 0 & 21 & 6 \\ 4 & 8 & 12 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 21 & 18 & 0 & 28 & 10 \\ 3 & 4 & 6 & 6 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -485, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & 3 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[33] $1^{\overline{1113}} 3^{\overline{1122}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -488$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.87 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 12 & 3 \\ 3 & 4 & 4 & 14 & 0 & 22 & 4 \\ 2 & 4 & 0 & 9 & 0 & 15 & 2 \\ 8 & 14 & 9 & 37 & 0 & 57 & 11 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 15 & 57 & 0 & 85 & 17 \\ 3 & 4 & 2 & 11 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[34] $1^{\overline{1114}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -524$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.34 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 12 & 12 & 0 & 46 & 12 \\ 3 & 4 & 2 & 3 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$

[35] $1^{\overline{112}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -533$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 13.03 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 13 & 4 \\ 3 & 5 & 3 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 12 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & 3 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

[36] $1^{\overline{113}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -551$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 13.06 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 5 & 4 & 0 & 17 & 4 \\ 3 & 5 & 3 & 3 & 0 & 15 & 3 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & 3 & -4 & 13 & -1 \end{pmatrix}_{b_0}$$

[37] $1^{\overline{1122}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -560$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.93 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 7 & 3 \\ 3 & 4 & 4 & 8 & 0 & 13 & 4 \\ 2 & 4 & 0 & 6 & 0 & 9 & 2 \\ 4 & 8 & 6 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 18 & 0 & 28 & 10 \\ 3 & 4 & 2 & 6 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{\overline{1111}} 3^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -729$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 5 & 3 \\ 3 & 4 & 9 & 4 & 0 & 9 & 4 \\ 5 & 9 & 12 & 6 & 0 & 15 & 6 \\ 2 & 4 & 6 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 15 & 6 & 0 & 13 & 7 \\ 3 & 4 & 6 & 3 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{\overline{1111}} 3^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -773$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 8 & 3 \\ 3 & 4 & 4 & 9 & 0 & 14 & 4 \\ 2 & 4 & 0 & 6 & 0 & 9 & 2 \\ 5 & 9 & 6 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 24 & 0 & 37 & 11 \\ 3 & 4 & 2 & 7 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{\overline{112}} 3^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -800$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 10 & 3 \\ 3 & 4 & 4 & 5 & 0 & 18 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 3 & 5 & 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 12 & 18 & 0 & 58 & 14 \\ 3 & 4 & 2 & 4 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{112} 3^1 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -848, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 16 & 3 \\ 3 & 4 & 4 & 5 & 0 & 28 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 3 & 5 & 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 16 & 28 & 18 & 30 & 0 & 154 & 22 \\ 3 & 4 & 2 & 4 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -896, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 15 & 0 & 13 & 7 \\ 3 & 4 & 2 & 7 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{111} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -920, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.72 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 8 & 3 \\ 3 & 4 & 4 & 14 & 0 & 14 & 4 \\ 2 & 4 & 0 & 9 & 0 & 9 & 2 \\ 8 & 14 & 9 & 37 & 0 & 39 & 11 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 39 & 0 & 37 & 11 \\ 3 & 4 & 2 & 11 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -929, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.79 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 9 & 0 & 28 & 10 \\ 3 & 4 & 2 & 3 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{111} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 12 & 12 & 0 & 58 & 14 \\ 3 & 4 & 2 & 3 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{111} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.91 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 14 & 3 \\ 3 & 4 & 4 & 5 & 0 & 26 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 3 & 5 & 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 18 & 24 & 0 & 118 & 20 \\ 3 & 4 & 2 & 4 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{112} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -956, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 13 & 3 \\ 3 & 4 & 4 & 5 & 0 & 23 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 3 & 5 & 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 15 & 24 & 0 & 100 & 18 \\ 3 & 4 & 2 & 4 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{113} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1001, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 12 & 3 \\ 3 & 4 & 4 & 5 & 0 & 22 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 3 & 5 & 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 15 & 21 & 0 & 85 & 17 \\ 3 & 4 & 2 & 4 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1113}31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 2 & 4 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 21 & 21 & 0 & 193 & 25 \\ 3 & 4 & 2 & 3 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1112}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1083, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 7 & 3 \\ 3 & 4 & 9 & 4 & 0 & 13 & 4 \\ 5 & 9 & 12 & 6 & 0 & 21 & 6 \\ 2 & 4 & 6 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 21 & 9 & 0 & 28 & 10 \\ 3 & 4 & 6 & 3 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1114}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1085, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 11 & 3 \\ 3 & 4 & 4 & 9 & 0 & 21 & 4 \\ 2 & 4 & 0 & 6 & 0 & 15 & 2 \\ 5 & 9 & 6 & 13 & 0 & 33 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 15 & 33 & 0 & 76 & 16 \\ 3 & 4 & 2 & 7 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1112}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1205, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 7 & 3 \\ 3 & 4 & 4 & 9 & 0 & 13 & 4 \\ 2 & 4 & 0 & 6 & 0 & 9 & 2 \\ 5 & 9 & 6 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 21 & 0 & 28 & 10 \\ 3 & 4 & 2 & 7 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1112}31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1232, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 10 & 3 \\ 3 & 4 & 4 & 14 & 0 & 18 & 4 \\ 2 & 4 & 0 & 9 & 0 & 12 & 2 \\ 8 & 14 & 9 & 37 & 0 & 48 & 11 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 12 & 48 & 0 & 58 & 14 \\ 3 & 4 & 2 & 11 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1113}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 9 & 3 \\ 3 & 4 & 4 & 9 & 0 & 17 & 4 \\ 2 & 4 & 0 & 6 & 0 & 12 & 2 \\ 5 & 9 & 6 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 27 & 0 & 49 & 13 \\ 3 & 4 & 2 & 7 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1112}31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 16 & 3 \\ 3 & 4 & 4 & 9 & 0 & 28 & 4 \\ 2 & 4 & 0 & 6 & 0 & 18 & 2 \\ 5 & 9 & 6 & 13 & 0 & 48 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 16 & 28 & 18 & 48 & 0 & 154 & 22 \\ 3 & 4 & 2 & 7 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1113}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{c}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1281, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 9 & 3 \\ 3 & 4 & 9 & 4 & 0 & 17 & 4 \\ 5 & 9 & 12 & 6 & 0 & 27 & 6 \\ 2 & 4 & 6 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 27 & 12 & 0 & 49 & 13 \\ 3 & 4 & 6 & 3 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1114} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1323, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 11 & 3 \\ 3 & 4 & 9 & 4 & 0 & 21 & 4 \\ 5 & 9 & 12 & 6 & 0 & 33 & 6 \\ 2 & 4 & 6 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 33 & 15 & 0 & 76 & 16 \\ 3 & 4 & 6 & 3 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1113} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1397, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 12 & 12 & 0 & 49 & 13 \\ 3 & 4 & 2 & 3 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{1113} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1565, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 15 & 3 \\ 3 & 4 & 4 & 5 & 0 & 27 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 3 & 5 & 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 18 & 27 & 0 & 133 & 21 \\ 3 & 4 & 2 & 4 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{1122} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 10 & 3 \\ 3 & 4 & 4 & 13 & 0 & 18 & 4 \\ 2 & 4 & 0 & 9 & 0 & 12 & 2 \\ 7 & 13 & 9 & 28 & 0 & 42 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 12 & 42 & 0 & 58 & 14 \\ 3 & 4 & 2 & 10 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{1113} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 2 & 4 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 18 & 18 & 0 & 133 & 21 \\ 3 & 4 & 2 & 3 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{1112} 31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1760, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.74 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 10 & 3 \\ 3 & 4 & 4 & 9 & 0 & 18 & 4 \\ 2 & 4 & 0 & 6 & 0 & 12 & 2 \\ 5 & 9 & 6 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 12 & 30 & 0 & 58 & 14 \\ 3 & 4 & 2 & 7 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{1112} 31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1781, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 13 & 3 \\ 3 & 4 & 4 & 9 & 0 & 23 & 4 \\ 2 & 4 & 0 & 6 & 0 & 15 & 2 \\ 5 & 9 & 6 & 13 & 0 & 39 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 15 & 39 & 0 & 100 & 18 \\ 3 & 4 & 2 & 7 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{1112} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1784, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 7 & 3 \\ 3 & 4 & 4 & 13 & 0 & 13 & 4 \\ 2 & 4 & 0 & 9 & 0 & 9 & 2 \\ 7 & 13 & 9 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 9 & 30 & 0 & 28 & 10 \\ 3 & 4 & 2 & 10 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{1113}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -1820$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 15 & 15 & 0 & 85 & 17 \\ 3 & 4 & 2 & 3 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[66] $1^{113}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -1865$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 18 & 3 \\ 3 & 4 & 4 & 5 & 0 & 32 & 4 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 3 & 5 & 3 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 21 & 33 & 0 & 193 & 25 \\ 3 & 4 & 2 & 4 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{1114}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -1889$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 15 & 15 & 0 & 76 & 16 \\ 3 & 4 & 2 & 3 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{114}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -1928$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 17 & 3 \\ 3 & 4 & 4 & 5 & 0 & 31 & 4 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 3 & 5 & 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 21 & 30 & 0 & 172 & 24 \\ 3 & 4 & 2 & 4 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[69] $1^{1113}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -2501$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 12 & 3 \\ 3 & 4 & 4 & 9 & 0 & 22 & 4 \\ 2 & 4 & 0 & 6 & 0 & 15 & 2 \\ 5 & 9 & 6 & 13 & 0 & 36 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 15 & 36 & 0 & 85 & 17 \\ 3 & 4 & 2 & 7 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{1122}31^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -2624$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 10 & 3 \\ 3 & 4 & 4 & 18 & 0 & 18 & 4 \\ 2 & 4 & 0 & 12 & 0 & 12 & 2 \\ 10 & 18 & 12 & 58 & 0 & 60 & 14 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 12 & 60 & 0 & 58 & 14 \\ 3 & 4 & 2 & 14 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{114}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -2636$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 20 & 3 \\ 3 & 4 & 4 & 5 & 0 & 36 & 4 \\ 2 & 4 & 0 & 3 & 0 & 24 & 2 \\ 3 & 5 & 3 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 24 & 36 & 0 & 238 & 28 \\ 3 & 4 & 2 & 4 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{1114}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -2720$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.80 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 2 & 4 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 18 & 18 & 0 & 118 & 20 \\ 3 & 4 & 2 & 3 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{1114}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -2780$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 0 & 3 & 0 & 24 & 2 \\ 2 & 4 & 3 & 1 & 0 & 24 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 24 & 24 & 0 & 238 & 28 \\ 3 & 4 & 2 & 3 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

[74] $1^{1114}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -2996$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 14 & 3 \\ 3 & 4 & 4 & 9 & 0 & 26 & 4 \\ 2 & 4 & 0 & 6 & 0 & 18 & 2 \\ 5 & 9 & 6 & 13 & 0 & 42 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 18 & 42 & 0 & 118 & 20 \\ 3 & 4 & 2 & 7 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{1114}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -3017$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 2 & 4 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 21 & 21 & 0 & 172 & 24 \\ 3 & 4 & 2 & 3 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{1114}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -3200$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 15 & 3 \\ 3 & 4 & 4 & 9 & 0 & 27 & 4 \\ 2 & 4 & 0 & 6 & 0 & 18 & 2 \\ 5 & 9 & 6 & 13 & 0 & 45 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 18 & 45 & 0 & 133 & 21 \\ 3 & 4 & 2 & 7 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{1114}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -3365$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 18 & 3 \\ 3 & 4 & 4 & 9 & 0 & 32 & 4 \\ 2 & 4 & 0 & 6 & 0 & 21 & 2 \\ 5 & 9 & 6 & 13 & 0 & 54 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 21 & 54 & 0 & 193 & 25 \\ 3 & 4 & 2 & 7 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[78] $1^{1114}31^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -4373$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 17 & 3 \\ 3 & 4 & 4 & 9 & 0 & 31 & 4 \\ 2 & 4 & 0 & 6 & 0 & 21 & 2 \\ 5 & 9 & 6 & 13 & 0 & 51 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 21 & 51 & 0 & 172 & 24 \\ 3 & 4 & 2 & 7 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{1114}31^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -5216$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.75 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 20 & 3 \\ 3 & 4 & 4 & 9 & 0 & 36 & 4 \\ 2 & 4 & 0 & 6 & 0 & 24 & 2 \\ 5 & 9 & 6 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 24 & 60 & 0 & 238 & 28 \\ 3 & 4 & 2 & 7 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

4.1143 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$

$T = 10$ [0] $1^{1114}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 148$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 5 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 3 & 5 & 3 & 4 & 0 & 6 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 6 & 0 & 4 & 9 & 4 \\ 5 & 9 & 6 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & 2 & 4 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 2 & 4 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 6 & 2 \\ 2 & 4 & 3 & 1 & 0 & 9 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 9 & 9 & 0 & 37 & 24 & 11 \\ 5 & 9 & 6 & 6 & 0 & 24 & 13 & 7 \\ 3 & 4 & 2 & 3 & -4 & 11 & 7 & 0 \end{pmatrix}$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 18 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 12 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & - \\ 2 & 4 & 3 & 3 & 0 & 1 & 12 & 3 \\ 10 & 18 & 12 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & 2 & 4 & -4 & 3 & 14 & 0 \end{pmatrix}$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 15 \\ 3 & 4 & 4 & 5 & 0 & 4 & 27 \\ 2 & 4 & 0 & 3 & 0 & 3 & 18 \\ 3 & 5 & 3 & 4 & 0 & 3 & 27 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 \\ 2 & 4 & 3 & 3 & 0 & 1 & 18 \\ 15 & 27 & 18 & 27 & 0 & 18 & 133 \\ 3 & 4 & 2 & 4 & -4 & 3 & 21 \end{pmatrix}$$

$$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 & 4 & 4 \\ 5 & 9 & 12 & 6 & 0 & 6 & 6 & 6 \\ 2 & 4 & 6 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 6 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 6 & 3 & -4 & 3 & 3 & 0 \end{pmatrix} b_0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 22 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 15 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & - \\ 2 & 4 & 3 & 3 & 0 & 1 & 15 & 3 \\ 12 & 22 & 15 & 21 & 0 & 15 & 85 & 17 \\ 3 & 4 & 2 & 4 & -4 & 3 & 17 & 0 \end{pmatrix}$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 20 \\ 3 & 4 & 4 & 5 & 0 & 4 & 36 \\ 2 & 4 & 0 & 3 & 0 & 3 & 24 \\ 3 & 5 & 3 & 4 & 0 & 3 & 36 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 \\ 2 & 4 & 3 & 3 & 0 & 1 & 24 \\ 20 & 36 & 24 & 36 & 0 & 24 & 238 \\ 3 & 4 & 2 & 4 & -4 & 3 & 28 \end{pmatrix}$$

- [8] $1^{\overline{11111}}31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 712, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 2 & 3 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{\overline{11111}}31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 724, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 2 & 3 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{\overline{1112}}31^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 796, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 4 & 5 & 0 & 9 & 13 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 9 & 2 \\ 3 & 5 & 3 & 4 & 0 & 9 & 12 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 9 & 0 & 13 & 21 & 7 \\ 7 & 13 & 9 & 12 & 0 & 21 & 28 & 10 \\ 3 & 4 & 2 & 4 & -4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{\overline{11111}}31^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 988, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.74 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 9 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 6 & 2 \\ 3 & 5 & 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & 2 & 4 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{\overline{11111}}31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1032, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 & 9 & 4 \\ 5 & 9 & 12 & 6 & 0 & 6 & 15 & 6 \\ 2 & 4 & 6 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 6 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 15 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 6 & 3 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [13] $1^{\overline{1112}}31^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 4 & 5 & 0 & 13 & 13 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 9 & 2 \\ 3 & 5 & 3 & 4 & 0 & 12 & 12 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 9 & 12 & 0 & 28 & 30 & 10 \\ 7 & 13 & 9 & 12 & 0 & 30 & 28 & 10 \\ 3 & 4 & 2 & 4 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$
- [14] $1^{\overline{1114}}31^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1180, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 31 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 21 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 30 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 21 & 3 \\ 17 & 31 & 21 & 30 & 0 & 21 & 172 & 24 \\ 3 & 4 & 2 & 4 & -4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$
- [15] $1^{\overline{11111}}31^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 1276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 6 & 2 \\ 2 & 4 & 3 & 1 & 0 & 6 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 2 & 3 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{11111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 2368$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 9 & 4 \\ 2 & 4 & 0 & 6 & 0 & 6 & 6 & 2 \\ 5 & 9 & 6 & 13 & 0 & 15 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 15 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 15 & 0 & 15 & 13 & 7 \\ 3 & 4 & 2 & 7 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

4.1144 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 10$ [0] $1^{11111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 48$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 5 & 2 & 3 \\ 4 & 9 & 4 & 12 & 0 & 12 & 4 & 5 \\ 2 & 4 & 0 & 6 & 0 & 6 & 3 & 2 \\ 5 & 12 & 6 & 13 & 0 & 15 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 12 & 6 & 15 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 6 & 0 & 6 & 0 & 2 \\ 3 & 5 & 2 & 7 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{11111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 228$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{11111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 231$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.13 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 2 & 4 & -4 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{11111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 507$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.92 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 9 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 3 & 2 \\ 3 & 5 & 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 2 & 4 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{11111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 651$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.17 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 9 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

[5] $1^{11111}31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 813$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.27 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 21 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 18 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 18 & 15 & 0 & 76 & 15 & 16 \\ 2 & 4 & 3 & 3 & 0 & 15 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 16 & 2 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{1121}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 1173$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.5$

$T_{\min} = 13.15 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 13 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 12 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 12 & 9 & 0 & 28 & 9 & 10 \\ 2 & 4 & 3 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{1131}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -91$, $\det = 1227$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.8$

$T_{\min} = 13.17 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 17 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 15 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 15 & 12 & 0 & 49 & 12 & 13 \\ 2 & 4 & 3 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1145 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3$

$T = 9$ [0] $1^{121}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -6$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 & 3 \\ 3 & 4 & 8 & 9 & 0 & 5 & 4 \\ 4 & 8 & 9 & 12 & 0 & 6 & 5 \\ 5 & 9 & 12 & 13 & 0 & 9 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 6 & 9 & 0 & 3 & 3 \\ 3 & 4 & 5 & 7 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.12 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 2 & 3 \\ 4 & 9 & 4 & 12 & 0 & 4 & 5 \\ 2 & 4 & 0 & 6 & 0 & 3 & 2 \\ 5 & 12 & 6 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 \\ 3 & 5 & 2 & 7 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{114}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.12 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 2 & 3 \\ 4 & 9 & 4 & 36 & 0 & 4 & 5 \\ 2 & 4 & 0 & 15 & 0 & 3 & 2 \\ 14 & 36 & 15 & 121 & 0 & 15 & 19 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 15 & 0 & 0 & 2 \\ 3 & 5 & 2 & 19 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{113}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -150$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.23 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 4 & 5 & 0 & 5 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 3 & 6 & 0 & 3 & 3 \\ 3 & 4 & 2 & 4 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -159$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.87 \rightarrow 14$, $T_{\max} = 68$, $T \in [14, 68]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 2 & 4 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{113}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 68$, $\Delta = -121$, $\det = -231$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} =$

14.8

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 2 & 3 \\ 4 & 9 & 4 & 28 & 0 & 4 & 5 \\ 2 & 4 & 0 & 12 & 0 & 3 & 2 \\ 11 & 28 & 12 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 12 & 0 & 0 & 2 \\ 3 & 5 & 2 & 15 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] 1^{112} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 2 & 3 \\ 4 & 9 & 4 & 20 & 0 & 4 & 5 \\ 2 & 4 & 0 & 9 & 0 & 3 & 2 \\ 8 & 20 & 9 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 9 & 0 & 0 & 2 \\ 3 & 5 & 2 & 11 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[8] 1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -258, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 9 & 0 & 36 & 10 \\ 3 & 4 & 2 & 3 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[9] 1^{122} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -267, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.7$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 5 & 13 & 0 & 8 & 4 \\ 3 & 5 & 3 & 12 & 0 & 6 & 3 \\ 7 & 13 & 12 & 28 & 0 & 18 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 6 & 18 & 0 & 9 & 5 \\ 3 & 4 & 3 & 10 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[10] 1^{132} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.06 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 4 & 3 \\ 3 & 4 & 5 & 17 & 0 & 8 & 4 \\ 3 & 5 & 3 & 15 & 0 & 6 & 3 \\ 9 & 17 & 15 & 49 & 0 & 24 & 13 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 6 & 24 & 0 & 9 & 5 \\ 3 & 4 & 3 & 13 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[11] 1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -402, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 2 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 & 4 \\ 2 & 4 & 0 & 9 & 0 & 3 & 2 \\ 8 & 14 & 9 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 9 & 0 & 0 & 2 \\ 3 & 4 & 2 & 11 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[12] 1^{113} 3^1 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -420, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 2 & 3 \\ 4 & 9 & 4 & 32 & 0 & 4 & 5 \\ 2 & 4 & 0 & 15 & 0 & 3 & 2 \\ 13 & 32 & 15 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 15 & 0 & 0 & 2 \\ 3 & 5 & 2 & 18 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[13] 1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{e}_6 + \text{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -435, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 3 & 5 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 9 & 0 & 12 & 6 \\ 3 & 4 & 2 & 4 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 2 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{114} 3 1^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -555, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 1 & 2 & 3 \\ 4 & 9 & 4 & 44 & 0 & 4 & 5 \\ 2 & 4 & 0 & 21 & 0 & 3 & 2 \\ 18 & 44 & 21 & 193 & 0 & 21 & 25 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 21 & 0 & 0 & 2 \\ 3 & 5 & 2 & 25 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -645, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 & 3 \\ 5 & 9 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 3 & 7 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{1111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -651, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 6 & 0 & 12 & 6 \\ 3 & 4 & 2 & 3 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{114} 3 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -690, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 2 & 3 \\ 4 & 9 & 4 & 40 & 0 & 4 & 5 \\ 2 & 4 & 0 & 18 & 0 & 3 & 2 \\ 16 & 40 & 18 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 18 & 0 & 0 & 2 \\ 3 & 5 & 2 & 22 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{141} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -726, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 5 & 21 & 0 & 4 & 4 \\ 3 & 5 & 3 & 18 & 0 & 3 & 3 \\ 11 & 21 & 18 & 76 & 0 & 15 & 16 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 15 & 0 & 0 & 2 \\ 3 & 4 & 3 & 16 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -789, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 4 \\ 2 & 4 & 0 & 6 & 0 & 3 & 2 \\ 5 & 9 & 6 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 2 & 7 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{121} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -906, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 & 4 \\ 3 & 5 & 3 & 12 & 0 & 3 & 3 \\ 7 & 13 & 12 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 9 & 0 & 0 & 2 \\ 3 & 4 & 3 & 10 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{131} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -933, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 & 4 \\ 3 & 5 & 3 & 15 & 0 & 3 & 3 \\ 9 & 17 & 15 & 49 & 0 & 12 & 13 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 12 & 0 & 0 & 2 \\ 3 & 4 & 3 & 13 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -1200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 15 & 0 & 12 & 6 \\ 3 & 4 & 2 & 7 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

4.1146 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{1111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 48, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 2 & 5 & 3 \\ 4 & 9 & 4 & 12 & 0 & 4 & 12 & 5 \\ 2 & 4 & 0 & 6 & 0 & 3 & 6 & 2 \\ 5 & 12 & 6 & 13 & 0 & 6 & 15 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 6 & 2 \\ 5 & 12 & 6 & 15 & 0 & 6 & 13 & 7 \\ 3 & 5 & 2 & 7 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 228, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 231, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 9 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 3 & 2 \\ 3 & 5 & 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 6 & 9 & 0 & 12 & 6 & 6 \\ 2 & 4 & 3 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 2 & 4 & -4 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 507, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 2 & 4 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 651, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 9 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & 3 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1114}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 813, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 11 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 21 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 18 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 15 & 2 \\ 11 & 21 & 18 & 15 & 0 & 15 & 76 & 16 \\ 3 & 4 & 3 & 3 & -4 & 2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1112}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1173, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 13 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 12 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 9 & 2 \\ 7 & 13 & 12 & 9 & 0 & 9 & 28 & 10 \\ 3 & 4 & 3 & 3 & -4 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1113}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1227, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 17 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 15 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 12 & 2 \\ 9 & 17 & 15 & 12 & 0 & 12 & 49 & 13 \\ 3 & 4 & 3 & 3 & -4 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

4.1147 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 459, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 1296, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 & 3 & 3 \\ 5 & 9 & 9 & 13 & 0 & 6 & 6 & 7 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 6 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 7 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1148 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4$

$$T = 8 \quad [0] \ 1^{\overline{123}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 50, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 28, \ T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 1 & 3 \\ 3 & 4 & 13 & 12 & 0 & 4 \\ 7 & 13 & 27 & 27 & 0 & 9 \\ 6 & 12 & 27 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.81 \rightarrow 12, \ T_{\max} = 28, \ T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 5 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 82, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 28, \ T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{1}1}3^{\overline{1}1}1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 3 & 5 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 4 & -3 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{1}12}3^{\overline{1}1}1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 9 & 8 & 0 & 4 \\ 5 & 9 & 12 & 12 & 0 & 6 \\ 4 & 8 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{1}11}3^{\overline{1}1}1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 4 & -3 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{1}11}3^{\overline{1}1}1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 12 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 7 & -3 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{\overline{1}3}3^{\overline{1}1}1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.15 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 \\ 9 & 17 & 48 & 15 & 0 & 12 \\ 3 & 5 & 15 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 12 & 4 & -3 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{\overline{1}11}3^{\overline{1}1}1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{\overline{1}21}3^{\overline{1}1}1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 137, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.29 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 4 & 8 & 9 & 6 & 0 & 5 \\ 2 & 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{\overline{1}12}3^{\overline{1}1}1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 4 & 8 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{\overline{1}11}3^{\overline{1}1}1^{\overline{1}3} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 19 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 11 & 19 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 15 & -3 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{\overline{1}21}3^{\overline{1}1}1^{\overline{1}1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.37 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 8 & 9 & 0 & 4 \\ 4 & 8 & 9 & 12 & 0 & 5 \\ 5 & 9 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 7 & -3 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{111}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 159, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.34 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 \\ 8 & 14 & 36 & 9 & 0 & 10 \\ 2 & 4 & 9 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 10 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{122}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 12.73 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 1 & 3 \\ 3 & 4 & 8 & 18 & 0 & 4 \\ 4 & 8 & 9 & 24 & 0 & 5 \\ 10 & 18 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 14 & -3 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 5 & 9 & 12 & 9 & 0 & 6 \\ 3 & 5 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{12}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 \\ 7 & 13 & 27 & 12 & 0 & 9 \\ 3 & 5 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{111}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 \\ 5 & 9 & 12 & 6 & 0 & 6 \\ 2 & 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{112}31^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 10 & 24 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 14 & -3 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{122}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 1 & 3 \\ 3 & 4 & 13 & 8 & 0 & 4 \\ 7 & 13 & 27 & 18 & 0 & 9 \\ 4 & 8 & 18 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{111}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 \\ 5 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 7 & -3 & 0 \end{pmatrix}_{b_0}$$

[21] $1^{14}31^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 1 & 3 \\ 3 & 4 & 5 & 21 & 0 & 4 \\ 3 & 5 & 3 & 18 & 0 & 3 \\ 11 & 21 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 16 & -3 & 0 \end{pmatrix}_{b_0}$$

[22] $1^{111}31^{11} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 8 & 14 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 11 & -3 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{112}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 266, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 3 \\ 4 & 9 & 4 & 20 & 0 & 5 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 8 & 20 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 11 & -3 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{123}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 3 & 12 & 0 & 3 \\ 7 & 13 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 10 & -3 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{13}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 294, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 3 & 15 & 0 & 3 \\ 9 & 17 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 13 & -3 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{121}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 306, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 7 & 13 & 27 & 9 & 0 & 9 \\ 2 & 4 & 9 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{122}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 312, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 1 & 3 \\ 3 & 4 & 8 & 13 & 0 & 4 \\ 4 & 8 & 9 & 18 & 0 & 5 \\ 7 & 13 & 18 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 10 & -3 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{113}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 15 & 36 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 21 & -3 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 371, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 5 & 9 & 12 & 15 & 0 & 6 \\ 5 & 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{121}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 10 & 18 & 57 & 12 & 0 & 13 \\ 2 & 4 & 12 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 13 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{112}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 411, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 7 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 10 & -3 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{131}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 9 & 17 & 48 & 12 & 0 & 12 \\ 2 & 4 & 12 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 12 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{113}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 3 \\ 4 & 9 & 4 & 28 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 11 & 28 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 15 & -3 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{121}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 1 & 3 \\ 3 & 4 & 13 & 9 & 0 & 4 \\ 7 & 13 & 27 & 21 & 0 & 9 \\ 5 & 9 & 21 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{131}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 463, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 1 & 3 \\ 3 & 4 & 17 & 9 & 0 & 4 \\ 9 & 17 & 48 & 27 & 0 & 12 \\ 5 & 9 & 27 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 12 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{123}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 489, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 17 & 0 & 4 \\ 4 & 8 & 9 & 24 & 0 & 5 \\ 9 & 17 & 24 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 13 & -3 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{113}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 490, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 13 & 32 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 18 & -3 & 0 \end{pmatrix}_{b_0}$$

[39] $1^{123}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 501, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.30 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 8 & 22 & 0 & 4 \\ 4 & 8 & 9 & 30 & 0 & 5 \\ 12 & 22 & 30 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 17 & -3 & 0 \end{pmatrix}_{b_0}$$

[40] $1^{112}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 507, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 5 & 9 & 12 & 21 & 0 & 6 \\ 7 & 13 & 21 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 10 & -3 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{114}31^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 20 & 1 & 3 \\ 4 & 9 & 4 & 48 & 0 & 5 \\ 2 & 4 & 0 & 24 & 0 & 2 \\ 20 & 48 & 24 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 28 & -3 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{112}31^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 576, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 10 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 14 & -3 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 579, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 9 & 17 & 0 & 4 \\ 5 & 9 & 12 & 27 & 0 & 6 \\ 9 & 17 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{114}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 587, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 1 & 3 \\ 3 & 4 & 9 & 21 & 0 & 4 \\ 5 & 9 & 12 & 33 & 0 & 6 \\ 11 & 21 & 33 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 16 & -3 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{112}3^11^4 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 1 & 3 \\ 3 & 4 & 4 & 28 & 0 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 16 & 28 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 22 & -3 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 588, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 9 & 17 & 12 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{131}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 607, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 22 & 4 & 0 & 4 \\ 12 & 22 & 84 & 15 & 0 & 16 \\ 2 & 4 & 15 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 16 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{114}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 626, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 14 & 36 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 19 & -3 & 0 \end{pmatrix}_{b_0}$$

[49] $1^{112}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 635, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 23 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 13 & 23 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 18 & -3 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{122}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 656, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.9$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 1 & 3 \\ 3 & 4 & 13 & 13 & 0 & 4 \\ 7 & 13 & 27 & 30 & 0 & 9 \\ 7 & 13 & 30 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & 10 & -3 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{114}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 770, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 1 & 3 \\ 4 & 9 & 4 & 44 & 0 & 5 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 18 & 44 & 21 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 25 & -3 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{114}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 771, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.02 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 21 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 11 & 21 & 15 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 16 & -3 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{123}31^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 798, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 1 & 3 \\ 3 & 4 & 13 & 17 & 0 & 4 \\ 7 & 13 & 27 & 39 & 0 & 9 \\ 9 & 17 & 39 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

[54] $1^{114}31^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 9 & 4 & 40 & 0 & 5 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 16 & 40 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 22 & -3 & 0 \end{pmatrix}_{b_0}$$

[55] $1^{113}31^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 915, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 12 & 22 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 17 & -3 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{113}31^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 1 & 3 \\ 3 & 4 & 4 & 27 & 0 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 15 & 27 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 21 & -3 & 1 \end{pmatrix}_{b_0}$$

[57] $1^{113}31^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1251, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 1 & 3 \\ 3 & 4 & 4 & 32 & 0 & 4 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 18 & 32 & 21 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 25 & -3 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{114}31^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 4 & 26 & 0 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 14 & 26 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 20 & -3 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{114}31^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1643, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 1 & 3 \\ 3 & 4 & 4 & 31 & 0 & 4 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 17 & 31 & 21 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 24 & -3 & 1 \end{pmatrix}_{b_0}$$

[60] $1^{114}31^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 1920, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 1 & 3 \\ 3 & 4 & 4 & 36 & 0 & 4 \\ 2 & 4 & 0 & 24 & 0 & 2 \\ 20 & 36 & 24 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 28 & -3 & 1 \end{pmatrix}_{b_0}$$

4.1149 $\text{NHC}(-2)(-3) + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{so}}_8$

$T = 9 \quad [0] \quad 1^{111}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 9 & 5 & 7 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[1] 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 1 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 8 & 16 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

4.1150 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2$

$$T = 9 \quad [0] 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 4 & 5 & 0 & 5 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & 2 & 4 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] 1^{111} 3 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 11 & 3 \\ 3 & 4 & 4 & 5 & 0 & 19 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 3 & 5 & 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 19 & 12 & 21 & 0 & 73 & 15 \\ 3 & 4 & 2 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[2] 1^{111} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 9 & 0 & 37 & 11 \\ 3 & 4 & 2 & 3 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 3 1^{131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 5 & 3 \\ 3 & 4 & 4 & 19 & 0 & 9 & 4 \\ 2 & 4 & 0 & 12 & 0 & 6 & 2 \\ 11 & 19 & 12 & 73 & 0 & 33 & 15 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 33 & 0 & 13 & 7 \\ 3 & 4 & 2 & 15 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[4] 1^{112} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 2 & 3 \\ 3 & 4 & 9 & 8 & 0 & 4 & 4 \\ 5 & 9 & 12 & 12 & 0 & 6 & 6 \\ 4 & 8 & 12 & 10 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 6 & 0 & 1 & 3 \\ 3 & 4 & 6 & 6 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

- [5] $1^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 2 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 & 4 \\ 5 & 9 & 12 & 9 & 0 & 6 & 6 \\ 3 & 5 & 9 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 3 & 0 & 1 & 3 \\ 3 & 4 & 6 & 4 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$
- [6] $1^{121}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 2 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 & 4 \\ 7 & 13 & 27 & 12 & 0 & 9 & 9 \\ 3 & 5 & 12 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 3 & 0 & 1 & 3 \\ 3 & 4 & 9 & 4 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{111}31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -268, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & 3 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$
- [8] $1^{111}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 3 & 5 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 9 & 0 & 13 & 7 \\ 3 & 4 & 2 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{111}31^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 8 & 3 \\ 3 & 4 & 4 & 5 & 0 & 14 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 3 & 5 & 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 15 & 0 & 37 & 11 \\ 3 & 4 & 2 & 4 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{112}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -296, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 7 & 3 \\ 3 & 4 & 4 & 5 & 0 & 13 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 3 & 5 & 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 12 & 0 & 28 & 10 \\ 3 & 4 & 2 & 4 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{111}31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 & 4 \\ 5 & 9 & 12 & 6 & 0 & 6 & 6 \\ 2 & 4 & 6 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 3 & 0 & 1 & 3 \\ 3 & 4 & 6 & 3 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{1112}31^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 15 & 15 & 0 & 100 & 18 \\ 3 & 4 & 2 & 3 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$
- [13] $1^{1122}31^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -338, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 7 & 3 \\ 3 & 4 & 9 & 8 & 0 & 13 & 4 \\ 5 & 9 & 12 & 12 & 0 & 21 & 6 \\ 4 & 8 & 12 & 10 & 0 & 18 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 21 & 18 & 0 & 28 & 10 \\ 3 & 4 & 6 & 6 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & 3 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 5 & 3 \\ 3 & 4 & 9 & 4 & 0 & 9 & 4 \\ 5 & 9 & 12 & 6 & 0 & 15 & 6 \\ 2 & 4 & 6 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 15 & 6 & 0 & 13 & 7 \\ 3 & 4 & 6 & 3 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1111} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -605, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 8 & 3 \\ 3 & 4 & 4 & 9 & 0 & 14 & 4 \\ 2 & 4 & 0 & 6 & 0 & 9 & 2 \\ 5 & 9 & 6 & 13 & 0 & 24 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 24 & 0 & 37 & 11 \\ 3 & 4 & 2 & 7 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{112} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 10 & 3 \\ 3 & 4 & 4 & 5 & 0 & 18 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 3 & 5 & 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 18 & 0 & 58 & 14 \\ 3 & 4 & 2 & 4 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{112} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -650, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 16 & 3 \\ 3 & 4 & 4 & 5 & 0 & 28 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 3 & 5 & 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 16 & 28 & 18 & 30 & 0 & 154 & 22 \\ 3 & 4 & 2 & 4 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{1111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 15 & 0 & 13 & 7 \\ 3 & 4 & 2 & 7 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1111} 31^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -716, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 8 & 3 \\ 3 & 4 & 4 & 14 & 0 & 14 & 4 \\ 2 & 4 & 0 & 9 & 0 & 9 & 2 \\ 8 & 14 & 9 & 37 & 0 & 39 & 11 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 39 & 0 & 37 & 11 \\ 3 & 4 & 2 & 11 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1112} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -723, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 9 & 0 & 28 & 10 \\ 3 & 4 & 2 & 3 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{1112} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -738, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 12 & 0 & 58 & 14 \\ 3 & 4 & 2 & 3 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{1112} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 13 & 3 \\ 3 & 4 & 4 & 5 & 0 & 23 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 3 & 5 & 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 15 & 24 & 0 & 100 & 18 \\ 3 & 4 & 2 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{1113} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -779, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 12 & 3 \\ 3 & 4 & 4 & 5 & 0 & 22 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 3 & 5 & 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 15 & 21 & 0 & 85 & 17 \\ 3 & 4 & 2 & 4 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{1113} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -828, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 2 & 4 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 21 & 21 & 0 & 193 & 25 \\ 3 & 4 & 2 & 3 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{1112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -845, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 7 & 3 \\ 3 & 4 & 9 & 4 & 0 & 13 & 4 \\ 5 & 9 & 12 & 6 & 0 & 21 & 6 \\ 2 & 4 & 6 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 21 & 9 & 0 & 28 & 10 \\ 3 & 4 & 6 & 3 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{1112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -941, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 7 & 3 \\ 3 & 4 & 4 & 9 & 0 & 13 & 4 \\ 2 & 4 & 0 & 6 & 0 & 9 & 2 \\ 5 & 9 & 6 & 13 & 0 & 21 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 21 & 0 & 28 & 10 \\ 3 & 4 & 2 & 7 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{1112} 3^1 1^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -962, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 10 & 3 \\ 3 & 4 & 4 & 14 & 0 & 18 & 4 \\ 2 & 4 & 0 & 9 & 0 & 12 & 2 \\ 8 & 14 & 9 & 37 & 0 & 48 & 11 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 48 & 0 & 58 & 14 \\ 3 & 4 & 2 & 11 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{1112} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -980, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 16 & 3 \\ 3 & 4 & 4 & 9 & 0 & 28 & 4 \\ 2 & 4 & 0 & 6 & 0 & 18 & 2 \\ 5 & 9 & 6 & 13 & 0 & 48 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 16 & 28 & 18 & 48 & 0 & 154 & 22 \\ 3 & 4 & 2 & 7 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{\overline{1113}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -980, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 9 & 3 \\ 3 & 4 & 4 & 9 & 0 & 17 & 4 \\ 2 & 4 & 0 & 6 & 0 & 12 & 2 \\ 5 & 9 & 6 & 13 & 0 & 27 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 27 & 0 & 49 & 13 \\ 3 & 4 & 2 & 7 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{\overline{1113}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -989, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 9 & 3 \\ 3 & 4 & 9 & 4 & 0 & 17 & 4 \\ 5 & 9 & 12 & 6 & 0 & 27 & 6 \\ 2 & 4 & 6 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 27 & 12 & 0 & 49 & 13 \\ 3 & 4 & 6 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{\overline{1114}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1005, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 11 & 3 \\ 3 & 4 & 9 & 4 & 0 & 21 & 4 \\ 5 & 9 & 12 & 6 & 0 & 33 & 6 \\ 2 & 4 & 6 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 33 & 15 & 0 & 76 & 16 \\ 3 & 4 & 6 & 3 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{\overline{1113}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1077, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 4 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 12 & 0 & 49 & 13 \\ 3 & 4 & 2 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{\overline{113}} 3^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1221, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 15 & 3 \\ 3 & 4 & 4 & 5 & 0 & 27 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 3 & 5 & 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 18 & 27 & 0 & 133 & 21 \\ 3 & 4 & 2 & 4 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{\overline{1122}} 3^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1298, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 10 & 3 \\ 3 & 4 & 4 & 13 & 0 & 18 & 4 \\ 2 & 4 & 0 & 9 & 0 & 12 & 2 \\ 7 & 13 & 9 & 28 & 0 & 42 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 42 & 0 & 58 & 14 \\ 3 & 4 & 2 & 10 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{\overline{1113}} 3^{\overline{113}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1333, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 2 & 4 & 3 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 18 & 18 & 0 & 133 & 21 \\ 3 & 4 & 2 & 3 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{\overline{1112}} 3^{\overline{112}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 10 & 3 \\ 3 & 4 & 4 & 9 & 0 & 18 & 4 \\ 2 & 4 & 0 & 6 & 0 & 12 & 2 \\ 5 & 9 & 6 & 13 & 0 & 30 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 30 & 0 & 58 & 14 \\ 3 & 4 & 2 & 7 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1122} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1388, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 7 & 3 \\ 3 & 4 & 4 & 13 & 0 & 13 & 4 \\ 2 & 4 & 0 & 9 & 0 & 9 & 2 \\ 7 & 13 & 9 & 28 & 0 & 30 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 30 & 0 & 28 & 10 \\ 3 & 4 & 2 & 10 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1122} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1389, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 13 & 3 \\ 3 & 4 & 4 & 9 & 0 & 23 & 4 \\ 2 & 4 & 0 & 6 & 0 & 15 & 2 \\ 5 & 9 & 6 & 13 & 0 & 39 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 15 & 39 & 0 & 100 & 18 \\ 3 & 4 & 2 & 7 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1133} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1416, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 2 & 4 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 15 & 15 & 0 & 85 & 17 \\ 3 & 4 & 2 & 3 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{1133} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1451, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 18 & 3 \\ 3 & 4 & 4 & 5 & 0 & 32 & 4 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 3 & 5 & 3 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 21 & 33 & 0 & 193 & 25 \\ 3 & 4 & 2 & 4 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1143} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1500, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 17 & 3 \\ 3 & 4 & 4 & 5 & 0 & 31 & 4 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 3 & 5 & 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 21 & 30 & 0 & 172 & 24 \\ 3 & 4 & 2 & 4 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1133} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -1949, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 12 & 3 \\ 3 & 4 & 4 & 9 & 0 & 22 & 4 \\ 2 & 4 & 0 & 6 & 0 & 15 & 2 \\ 5 & 9 & 6 & 13 & 0 & 36 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 15 & 36 & 0 & 85 & 17 \\ 3 & 4 & 2 & 7 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1122} 3^1 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2048, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 10 & 3 \\ 3 & 4 & 4 & 18 & 0 & 18 & 4 \\ 2 & 4 & 0 & 12 & 0 & 12 & 2 \\ 10 & 18 & 12 & 58 & 0 & 60 & 14 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 60 & 0 & 58 & 14 \\ 3 & 4 & 2 & 14 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1143} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -86, \quad \det = -2054, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 20 & 3 \\ 3 & 4 & 4 & 9 & 0 & 36 & 4 \\ 2 & 4 & 0 & 6 & 0 & 24 & 2 \\ 5 & 9 & 6 & 13 & 0 & 60 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 24 & 60 & 0 & 238 & 28 \\ 3 & 4 & 2 & 7 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

4.1151 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 5 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 & 2 \\ 3 & 5 & 3 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 6 & 0 & 4 & 3 & 4 \\ 2 & 3 & 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 2 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 392, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 2 & 2 & 0 & 2 & 0 & 2 \\ 3 & 4 & 2 & 3 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 462, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 9 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 & 2 \\ 3 & 5 & 3 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 9 & 0 & 13 & 5 & 7 \\ 2 & 3 & 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & 2 & 4 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 512, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 & 3 & 4 \\ 5 & 9 & 12 & 6 & 0 & 6 & 5 & 6 \\ 2 & 4 & 6 & 1 & 0 & 3 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 3 & 0 & 1 & 2 & 3 \\ 2 & 3 & 5 & 2 & 0 & 2 & 0 & 2 \\ 3 & 4 & 6 & 3 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 622, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 & 2 \\ 2 & 4 & 3 & 1 & 0 & 6 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 6 & 0 & 13 & 5 & 7 \\ 2 & 3 & 2 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 2 & 3 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 1152, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 3 & 4 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 & 2 \\ 5 & 9 & 6 & 13 & 0 & 15 & 5 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 15 & 0 & 13 & 5 & 7 \\ 2 & 3 & 2 & 5 & 0 & 5 & 0 & 2 \\ 3 & 4 & 2 & 7 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

4.1152 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -96, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 2 & 3 \\ 3 & 4 & 9 & 8 & 0 & 3 & 4 \\ 5 & 9 & 12 & 12 & 0 & 5 & 6 \\ 4 & 8 & 12 & 10 & 0 & 4 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 5 & 4 & 0 & 0 & 2 \\ 3 & 4 & 6 & 6 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -124, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 2 & 3 \\ 4 & 9 & 4 & 8 & 0 & 4 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 & 2 \\ 3 & 8 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 2 & 3 & 0 & 0 & 2 \\ 3 & 5 & 2 & 4 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -136, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 2 & 3 \\ 3 & 4 & 4 & 19 & 0 & 3 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 & 2 \\ 11 & 19 & 12 & 73 & 0 & 11 & 15 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 11 & 0 & 0 & 2 \\ 3 & 4 & 2 & 15 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^{11}1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -136, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 2 & 3 \\ 3 & 4 & 4 & 8 & 0 & 3 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 & 2 \\ 4 & 8 & 6 & 10 & 0 & 4 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 4 & 0 & 0 & 2 \\ 3 & 4 & 2 & 6 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -146, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 3 & 0 & 0 & 2 \\ 3 & 4 & 2 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -156, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 3 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -166, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.31 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 2 & 3 \\ 3 & 4 & 8 & 4 & 0 & 3 & 4 \\ 4 & 8 & 9 & 6 & 0 & 4 & 5 \\ 2 & 4 & 6 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 2 & 0 & 0 & 2 \\ 3 & 4 & 5 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.13 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 2 & 3 \\ 4 & 9 & 4 & 12 & 0 & 4 & 5 \\ 2 & 4 & 0 & 6 & 0 & 2 & 2 \\ 5 & 12 & 6 & 13 & 0 & 5 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 2 & 5 & 0 & 0 & 2 \\ 3 & 5 & 2 & 7 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{121}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -206, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.37 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 2 & 3 \\ 3 & 4 & 4 & 17 & 0 & 3 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 & 2 \\ 9 & 17 & 12 & 49 & 0 & 9 & 13 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 9 & 0 & 0 & 2 \\ 3 & 4 & 2 & 13 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{114}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 2 & 3 \\ 3 & 4 & 4 & 21 & 0 & 3 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 & 2 \\ 11 & 21 & 15 & 76 & 0 & 11 & 16 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 11 & 0 & 0 & 2 \\ 3 & 4 & 2 & 16 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 2 & 3 \\ 3 & 4 & 4 & 18 & 0 & 3 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 & 2 \\ 10 & 18 & 12 & 58 & 0 & 10 & 14 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 10 & 0 & 0 & 2 \\ 3 & 4 & 2 & 14 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{112}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 2 & 3 \\ 3 & 4 & 4 & 23 & 0 & 3 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 & 2 \\ 13 & 23 & 15 & 100 & 0 & 13 & 18 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 13 & 0 & 0 & 2 \\ 3 & 4 & 2 & 18 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{113}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 2 & 3 \\ 3 & 4 & 4 & 22 & 0 & 3 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 & 2 \\ 12 & 22 & 15 & 85 & 0 & 12 & 17 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 12 & 0 & 0 & 2 \\ 3 & 4 & 2 & 17 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{113}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 1 & 2 & 3 \\ 3 & 4 & 4 & 27 & 0 & 3 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 & 2 \\ 15 & 27 & 18 & 133 & 0 & 15 & 21 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 15 & 0 & 0 & 2 \\ 3 & 4 & 2 & 21 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{114}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 1 & 2 & 3 \\ 3 & 4 & 4 & 26 & 0 & 3 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 & 2 \\ 14 & 26 & 18 & 118 & 0 & 14 & 20 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 14 & 0 & 0 & 2 \\ 3 & 4 & 2 & 20 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{113}3^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1826, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 1 & 2 & 3 \\ 3 & 4 & 4 & 32 & 0 & 3 & 4 \\ 2 & 4 & 0 & 21 & 0 & 2 & 2 \\ 18 & 32 & 21 & 193 & 0 & 18 & 25 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 18 & 0 & 0 & 2 \\ 3 & 4 & 2 & 25 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{114}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 1 & 2 & 3 \\ 3 & 4 & 4 & 31 & 0 & 3 & 4 \\ 2 & 4 & 0 & 21 & 0 & 2 & 2 \\ 17 & 31 & 21 & 172 & 0 & 17 & 24 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 17 & 0 & 0 & 2 \\ 3 & 4 & 2 & 24 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1111} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -2816, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 1 & 2 & 3 \\ 3 & 4 & 4 & 36 & 0 & 3 & 4 \\ 2 & 4 & 0 & 24 & 0 & 2 & 2 \\ 20 & 36 & 24 & 238 & 0 & 20 & 28 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 20 & 0 & 0 & 2 \\ 3 & 4 & 2 & 28 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

4.1153 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{1111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 5 & 0 & 3 & 5 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 & 3 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 3 & 6 & 0 & 3 & 4 & 4 \\ 3 & 4 & 2 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 392, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 & 3 & 2 \\ 2 & 4 & 3 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 2 & 3 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 462, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 & 6 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 3 & 0 & 0 & 5 & 2 \\ 5 & 9 & 6 & 9 & 0 & 5 & 13 & 7 \\ 3 & 4 & 2 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 512, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 4 & 0 & 3 & 4 & 4 \\ 5 & 9 & 12 & 6 & 0 & 5 & 6 & 6 \\ 2 & 4 & 6 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 5 & 2 & 0 & 0 & 2 & 2 \\ 2 & 4 & 6 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 622, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 & 6 & 2 \\ 2 & 4 & 3 & 1 & 0 & 2 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 2 & 0 & 0 & 5 & 2 \\ 5 & 9 & 6 & 6 & 0 & 5 & 13 & 7 \\ 3 & 4 & 2 & 3 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 1152, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 3 & 9 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 & 6 & 2 \\ 5 & 9 & 6 & 13 & 0 & 5 & 15 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 2 & 5 & 0 & 0 & 5 & 2 \\ 5 & 9 & 6 & 15 & 0 & 5 & 13 & 7 \\ 3 & 4 & 2 & 7 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

4.1154 NHC(-2)(-3) + su₃ + su₂ + f₄ + su2n3mix

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 8 & 4 \\ 2 & 4 & 0 & 3 & 0 & 5 & 2 \\ 3 & 5 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 5 & 9 & 0 & 12 & 6 \\ 3 & 4 & 2 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 0 & 3 & 0 & 11 & 2 \\ 2 & 4 & 3 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 17 & 11 & 11 & 0 & 55 & 13 \\ 3 & 4 & 2 & 3 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 5 & 0 & 7 & 4 \\ 2 & 4 & 0 & 3 & 0 & 5 & 2 \\ 3 & 5 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 5 & 6 & 0 & 7 & 5 \\ 3 & 4 & 2 & 4 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 5 & 4 & 0 & 11 & 4 \\ 3 & 5 & 3 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 9 & 8 & 0 & 20 & 8 \\ 3 & 4 & 3 & 3 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 5 & 4 & 0 & 7 & 4 \\ 3 & 5 & 3 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 5 & 0 & 7 & 5 \\ 3 & 4 & 3 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -269, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 8 & 4 \\ 2 & 4 & 0 & 6 & 0 & 5 & 2 \\ 5 & 9 & 6 & 13 & 0 & 14 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 5 & 14 & 0 & 12 & 6 \\ 3 & 4 & 2 & 7 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -401, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 7 & 3 \\ 3 & 4 & 4 & 5 & 0 & 12 & 4 \\ 2 & 4 & 0 & 3 & 0 & 8 & 2 \\ 3 & 5 & 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 8 & 12 & 0 & 25 & 9 \\ 3 & 4 & 2 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1113}3^1}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -401, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 11 & 3 \\ 3 & 4 & 4 & 5 & 0 & 20 & 4 \\ 2 & 4 & 0 & 3 & 0 & 14 & 2 \\ 3 & 5 & 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 14 & 18 & 0 & 69 & 15 \\ 3 & 4 & 2 & 4 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1111}3^1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -424, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.07 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 & 0 & 7 & 4 \\ 2 & 4 & 0 & 3 & 0 & 5 & 2 \\ 2 & 4 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 5 & 5 & 0 & 7 & 5 \\ 3 & 4 & 2 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1111}3^1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 4 & 3 \\ 3 & 4 & 9 & 4 & 0 & 7 & 4 \\ 5 & 9 & 12 & 6 & 0 & 11 & 6 \\ 2 & 4 & 6 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 11 & 5 & 0 & 7 & 5 \\ 3 & 4 & 6 & 3 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1111}3^1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -481, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 0 & 3 & 0 & 8 & 2 \\ 2 & 4 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 8 & 8 & 0 & 25 & 9 \\ 3 & 4 & 2 & 3 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1111}3^1}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -500, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 9 & 3 \\ 3 & 4 & 4 & 5 & 0 & 16 & 4 \\ 2 & 4 & 0 & 3 & 0 & 11 & 2 \\ 3 & 5 & 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 11 & 15 & 0 & 44 & 12 \\ 3 & 4 & 2 & 4 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1111}3^1}1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -509, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.06 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 4 & 3 \\ 3 & 4 & 4 & 9 & 0 & 7 & 4 \\ 2 & 4 & 0 & 6 & 0 & 5 & 2 \\ 5 & 9 & 6 & 13 & 0 & 11 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 5 & 11 & 0 & 7 & 5 \\ 3 & 4 & 2 & 7 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{1111}3^1}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -896, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 7 & 3 \\ 3 & 4 & 4 & 9 & 0 & 12 & 4 \\ 2 & 4 & 0 & 6 & 0 & 8 & 2 \\ 5 & 9 & 6 & 13 & 0 & 20 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 8 & 20 & 0 & 25 & 9 \\ 3 & 4 & 2 & 7 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

4.1155 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{111}3^1}1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -39, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 2 & 3 \\ 4 & 9 & 4 & 12 & 0 & 4 & 5 \\ 2 & 4 & 0 & 6 & 0 & 3 & 2 \\ 5 & 12 & 6 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 \\ 3 & 5 & 2 & 7 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 4 & 5 & 0 & 5 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 6 & 0 & 3 & 3 \\ 3 & 4 & 2 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 2 & 4 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{113}3^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 2 & 3 \\ 4 & 9 & 4 & 28 & 0 & 4 & 5 \\ 2 & 4 & 0 & 12 & 0 & 3 & 2 \\ 11 & 28 & 12 & 73 & 0 & 12 & 15 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 12 & 0 & 0 & 2 \\ 3 & 5 & 2 & 15 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{112}3^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 2 & 3 \\ 4 & 9 & 4 & 20 & 0 & 4 & 5 \\ 2 & 4 & 0 & 9 & 0 & 3 & 2 \\ 8 & 20 & 9 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 9 & 0 & 0 & 2 \\ 3 & 5 & 2 & 11 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111}3^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{122}3^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.9$

$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 4 & 3 \\ 3 & 4 & 5 & 13 & 0 & 8 & 4 \\ 3 & 5 & 3 & 12 & 0 & 6 & 3 \\ 7 & 13 & 12 & 28 & 0 & 18 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 18 & 0 & 9 & 5 \\ 3 & 4 & 3 & 10 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.34 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 4 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 9 & 0 & 36 & 10 \\ 3 & 4 & 2 & 3 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{132}3^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{f}_4 + \mathbf{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -206, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 4 & 3 \\ 3 & 4 & 5 & 17 & 0 & 8 & 4 \\ 3 & 5 & 3 & 15 & 0 & 6 & 3 \\ 9 & 17 & 15 & 49 & 0 & 24 & 13 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 24 & 0 & 9 & 5 \\ 3 & 4 & 3 & 13 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{1111}31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 2 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 & 4 \\ 2 & 4 & 0 & 9 & 0 & 3 & 2 \\ 8 & 14 & 9 & 37 & 0 & 9 & 11 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 9 & 0 & 0 & 2 \\ 3 & 4 & 2 & 11 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{113}31^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 2 & 3 \\ 4 & 9 & 4 & 32 & 0 & 4 & 5 \\ 2 & 4 & 0 & 15 & 0 & 3 & 2 \\ 13 & 32 & 15 & 100 & 0 & 15 & 18 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 15 & 0 & 0 & 2 \\ 3 & 5 & 2 & 18 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 5 & 3 \\ 3 & 4 & 4 & 5 & 0 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 3 & 5 & 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 9 & 0 & 12 & 6 \\ 3 & 4 & 2 & 4 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -399, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 2 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{114}31^{131} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -426, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 1 & 2 & 3 \\ 4 & 9 & 4 & 44 & 0 & 4 & 5 \\ 2 & 4 & 0 & 21 & 0 & 3 & 2 \\ 18 & 44 & 21 & 193 & 0 & 21 & 25 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 21 & 0 & 0 & 2 \\ 3 & 5 & 2 & 25 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{111}31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 & 3 \\ 5 & 9 & 9 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 3 & 7 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 6 & 0 & 12 & 6 \\ 3 & 4 & 2 & 3 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[16] $1^{114}31^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3$ $T = 9, T_{\max} = 14, \Delta = -95, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 2 & 3 \\ 4 & 9 & 4 & 40 & 0 & 4 & 5 \\ 2 & 4 & 0 & 18 & 0 & 3 & 2 \\ 16 & 40 & 18 & 154 & 0 & 18 & 22 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 18 & 0 & 0 & 2 \\ 3 & 5 & 2 & 22 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{141} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -527, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.27 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 1 & 2 & 3 \\ 3 & 4 & 5 & 21 & 0 & 4 & 4 \\ 3 & 5 & 3 & 18 & 0 & 3 & 3 \\ 11 & 21 & 18 & 76 & 0 & 15 & 16 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 15 & 0 & 0 & 2 \\ 3 & 4 & 3 & 16 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -621, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.48 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 4 \\ 2 & 4 & 0 & 6 & 0 & 3 & 2 \\ 5 & 9 & 6 & 13 & 0 & 6 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 6 & 0 & 0 & 2 \\ 3 & 4 & 2 & 7 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{121} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -707, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 & 4 \\ 3 & 5 & 3 & 12 & 0 & 3 & 3 \\ 7 & 13 & 12 & 28 & 0 & 9 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 9 & 0 & 0 & 2 \\ 3 & 4 & 3 & 10 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{131} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -713, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.25 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 2 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 & 4 \\ 3 & 5 & 3 & 15 & 0 & 3 & 3 \\ 9 & 17 & 15 & 49 & 0 & 12 & 13 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 12 & 0 & 0 & 2 \\ 3 & 4 & 3 & 13 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -944, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.22 \rightarrow 13, \ T_{\max} = 14, \ T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 6 & 13 & 0 & 15 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 15 & 0 & 12 & 6 \\ 3 & 4 & 2 & 7 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

4.1156 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$

$$T = 11 \quad [0] \ 1^{11} 5^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -11, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 3 & 2 & 1 & 3 & 0 \\ 2 & 3 & 2 & 3 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 2 & 3 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 5^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -15, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 1 & 3 \\ 3 & 4 & 3 & 10 & 3 & 0 & 4 \\ 2 & 3 & 0 & 6 & 2 & 0 & 2 \\ 6 & 10 & 6 & 21 & 8 & 0 & 9 \\ 2 & 3 & 2 & 8 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 2 & 9 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12} 5^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -23, \ (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 2 & 1 & 3 \\ 3 & 4 & 3 & 17 & 3 & 0 & 4 \\ 2 & 3 & 0 & 10 & 2 & 0 & 2 \\ 10 & 17 & 10 & 61 & 13 & 0 & 15 \\ 2 & 3 & 2 & 13 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 2 & 15 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} \ 5 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -44, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 2 & 1 & 3 \\ 3 & 4 & 3 & 7 & 3 & 0 & 4 \\ 2 & 3 & 0 & 4 & 2 & 0 & 2 \\ 4 & 7 & 4 & 8 & 5 & 0 & 6 \\ 2 & 3 & 2 & 5 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 2 & 6 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{12} \ 5 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -50, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 1 & 3 \\ 3 & 4 & 3 & 14 & 3 & 0 & 4 \\ 2 & 3 & 0 & 8 & 2 & 0 & 2 \\ 8 & 14 & 8 & 38 & 10 & 0 & 12 \\ 2 & 3 & 2 & 10 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 2 & 12 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{13} \ 5 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -60, (n_+, n_-, n_0) =$$

$$(1, 11, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 2 & 1 & 3 \\ 3 & 4 & 3 & 21 & 3 & 0 & 4 \\ 2 & 3 & 0 & 12 & 2 & 0 & 2 \\ 12 & 21 & 12 & 88 & 15 & 0 & 18 \\ 2 & 3 & 2 & 15 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 2 & 18 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

4.1157 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{11} \ 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 12 & 14 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 12 & 15 & 6 \\ 14 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} \ 3 \ 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 19 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 19 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 8 & 13 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{14} \ 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 9 & 16 & 0 & 4 \\ 5 & 9 & 12 & 24 & 0 & 6 \\ 9 & 16 & 24 & 46 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{123} \ 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 1 & 3 \\ 3 & 4 & 13 & 12 & 0 & 4 \\ 7 & 13 & 27 & 27 & 0 & 9 \\ 6 & 12 & 27 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} \ 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 4 & 5 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 25, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 12 & 15 & 6 \\ 12 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 26, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 10 & 17 & 12 & 15 & 6 \\ 10 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 26, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 8 & -1 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 29, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 20 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.62 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 9 & 20 & 12 & 13 & 7 \\ 5 & 9 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 4 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 3 \\ 3 & 4 & 4 & 0 & 0 & 4 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 0 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 3 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 1 & 3 \\ 3 & 4 & 4 & 24 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 17 & 24 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 16 & 0 & 10 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 5 & 0 & 0 & 4 \\ 3 & 5 & 3 & 0 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 0 & -2 & 1 \end{pmatrix}_{b_0}$$

- [15] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 1 & 3 \\ 3 & 4 & 9 & 0 & 0 & 4 \\ 5 & 9 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 0 & -2 & 1 \end{pmatrix}_{b_0}$
- [16] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 10 & 1 & 5 \\ 11 & 20 & 17 & 0 & 8 \\ 10 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 5 & 8 & 6 & 0 & 2 \end{pmatrix}_{b_0}$
- [17] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 17 & 0 & 13 & 7 \\ 5 & 8 & -1 & 7 & 2 \end{pmatrix}_{b_0}$
- [18] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 13 & 11 & 6 \\ 15 & 20 & 17 & 16 & 8 \\ 13 & 17 & 12 & 12 & 6 \\ 11 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$
- [19] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 5 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 4 & -2 & 1 \end{pmatrix}_{b_0}$
- [20] $1^{11} 3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 11 & 1 & 3 \\ 3 & 4 & 0 & 19 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 11 & 19 & 0 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 15 & -2 & 1 \end{pmatrix}_{b_0}$
- [22] $1^{113} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 13 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$
- [21] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 5 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 4 & -2 & 1 \end{pmatrix}_{b_0}$
- [25] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.52 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 17 & 0 & 13 & 7 \\ 6 & 8 & -1 & 7 & 2 \end{pmatrix}_{b_0}$
- [24] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 8 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 5 & 8 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 6 & -2 & 1 \end{pmatrix}_{b_0}$
- [23] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 10 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 10 & 17 & 12 & 12 & 6 \\ 10 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{123} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 48, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.42 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 1 & 3 \\ 3 & 4 & 8 & 12 & 0 & 4 \\ 5 & 8 & 9 & 18 & 0 & 5 \\ 6 & 12 & 18 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{11} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 48, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.42 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{113} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 48, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.8$$

$$T_{\min} = 11.75 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 28 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 15 & 28 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{11} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 49, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 12 & 15 & 6 \\ 10 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 51, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.18 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 12 & 1 & 6 \\ 15 & 20 & 17 & 0 & 8 \\ 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 6 & 8 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 51, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.65 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 8 & 10 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 10 & 20 & 12 & 13 & 7 \\ 5 & 9 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{114} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 51, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.65 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 8 & 37 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 37 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 52, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.62 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 6 & 12 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{11} 3 \ 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 55, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 10 & 17 & 12 & 15 & 6 \\ 9 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 55, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 13 & 7 \\ 5 & 8 & -1 & 7 & 2 \end{pmatrix}_{b_0}$$

- [36] $1^{111} 1^3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.64 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 17 & 17 & 8 \\ 9 & 17 & 12 & 15 & 6 \\ 9 & 17 & 15 & 13 & 7 \\ 5 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$
- [38] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 5 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 4 & -2 & 2 \end{pmatrix}_{b_0}$
- [37] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 13 & 12 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 13 & 17 & 12 & 15 & 6 \\ 12 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$
- [39] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 61$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 3 & -2 & 1 \end{pmatrix}_{b_0}$
- [41] $21^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 64$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 13 & 17 & 15 & 15 & 7 \\ 17 & 20 & 17 & 17 & 8 \\ 15 & 17 & 12 & 15 & 6 \\ 15 & 17 & 15 & 13 & 7 \\ 7 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$
- [40] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 64$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 3 & 4 & 0 & 6 & 0 & 2 \\ 4 & 8 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$
- [42] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 65$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.26 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 3 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 3 & -2 & 1 \end{pmatrix}_{b_0}$
- [45] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 66$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 9 & 8 & 0 & 4 \\ 5 & 9 & 12 & 12 & 0 & 6 \\ 4 & 8 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$
- [44] $1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 66$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 8 & 4 & 1 & 3 \\ 3 & 4 & 13 & 8 & 0 & 4 \\ 8 & 13 & 27 & 18 & 0 & 9 \\ 4 & 8 & 18 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 6 & -2 & 1 \end{pmatrix}_{b_0}$
- [43] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 66$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 10 & 15 & 12 & 13 & 6 \\ 15 & 20 & 17 & 17 & 8 \\ 12 & 17 & 12 & 15 & 6 \\ 13 & 17 & 15 & 13 & 7 \\ 6 & 8 & 6 & 7 & 2 \end{pmatrix}_{b_0}$
- [46] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 66$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 5 & 9 & 12 & 9 & 0 & 6 \\ 4 & 5 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

[48] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

[47] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 1 & 3 \\ 3 & 4 & 0 & 17 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 9 & 17 & 0 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{11} 1^3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 9 & 17 & 12 & 12 & 6 \\ 8 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[50] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 28 & 5 \\ 12 & 25 & 16 & 60 & 9 \\ 8 & 16 & 9 & 36 & 5 \\ 28 & 60 & 36 & 133 & 21 \\ 5 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{13} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 3 & 5 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 19 & 5 \\ 12 & 25 & 16 & 40 & 9 \\ 8 & 16 & 9 & 24 & 5 \\ 19 & 40 & 24 & 58 & 14 \\ 5 & 9 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 10 & 17 & 12 & 12 & 6 \\ 9 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[54] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 11 & 6 \\ 15 & 20 & 17 & 16 & 8 \\ 12 & 17 & 12 & 12 & 6 \\ 11 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[55] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 12 & 6 \\ 15 & 20 & 17 & 16 & 8 \\ 12 & 17 & 12 & 12 & 6 \\ 12 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[56] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 5 \\ 11 & 20 & 17 & 16 & 8 \\ 9 & 17 & 12 & 12 & 6 \\ 9 & 16 & 12 & 10 & 6 \\ 5 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

- [57] $1^{112} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 78, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 20 & 5 \\ 12 & 25 & 16 & 40 & 9 \\ 8 & 16 & 9 & 24 & 5 \\ 20 & 40 & 24 & 58 & 14 \\ 5 & 9 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$
- [58] $1^{111} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 5 & 9 & 12 & 15 & 0 & 6 \\ 7 & 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [60] $1^{121} 3 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 4 & 8 & 9 & 6 & 0 & 5 \\ 3 & 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$
- [59] $1^{111} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 11 & 14 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$
- [61] $1^{11} 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 \\ 8 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [62] $1^{111} 3 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 82, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.49 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 16 & 0 & 10 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$
- [63] $1^1 3 1^{111} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 \\ 4 & 5 & 3 & 6 & 0 & 3 \\ 3 & 5 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$
- [64] $1^{111} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 12 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [65] $1^{11} 3 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 1 & 3 \\ 3 & 4 & 0 & 14 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 8 & 14 & 0 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$
- [66] $1^{111} 3 1^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 4 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$
- [67] $1^{11} 3 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 3 \\ 3 & 4 & 0 & 9 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 5 & 9 & 0 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{13} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 \\ 9 & 17 & 48 & 15 & 0 & 12 \\ 3 & 5 & 15 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{13} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 9 & 17 & 48 & 12 & 0 & 12 \\ 3 & 4 & 12 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{13} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 4 & 5 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{12} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 93, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 1 & 3 \\ 3 & 4 & 0 & 13 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 7 & 13 & 0 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 10 & 20 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 20 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 15 & 1 & 11 & 6 \\ 15 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 16 & 0 & 10 & 6 \\ 6 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{112} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 4 & 8 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{111} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 4 & 19 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 14 & 19 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{111} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 1 & 3 \\ 3 & 4 & 4 & 24 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 15 & 24 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[77] \ 1^{111} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 19 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 11 & 19 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{1121} 3^1 1^{31} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 7 & 8 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{1211} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.71 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 4 & 8 & 9 & 6 & 0 & 5 \\ 2 & 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[80] $1^{1111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 97$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.18 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 1 & 3 \\ 3 & 4 & 9 & 12 & 0 & 4 \\ 5 & 9 & 12 & 18 & 0 & 6 \\ 7 & 12 & 18 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{111} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 98$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.73 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 48 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 48 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{11} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 98$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 4 & 5 & 3 & 9 & 0 & 3 \\ 5 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[84] $1^{14} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 99$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.6$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 1 & 3 \\ 3 & 4 & 21 & 5 & 0 & 4 \\ 12 & 21 & 75 & 18 & 0 & 15 \\ 3 & 5 & 18 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 15 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

[85] $1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 101$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.72 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 8 & 9 & 0 & 4 \\ 4 & 8 & 9 & 12 & 0 & 5 \\ 5 & 9 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 105$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 \\ 5 & 9 & 12 & 6 & 0 & 6 \\ 3 & 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 112$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 3 & 4 & 0 & 12 & 0 & 2 \\ 10 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[90] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 7 & 13 & 27 & 9 & 0 & 9 \\ 3 & 4 & 9 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{121} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 3 \\ 3 & 4 & 8 & 9 & 0 & 4 \\ 5 & 8 & 9 & 12 & 0 & 5 \\ 5 & 9 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{111} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 35 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 35 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[93] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 3 & 12 & 0 & 3 \\ 11 & 13 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[92] $1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 17 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 17 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[94] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 \\ 8 & 14 & 36 & 9 & 0 & 10 \\ 2 & 4 & 9 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 1 & 23 & 6 \\ 15 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 33 & 0 & 49 & 13 \\ 6 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 16 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 16 & 0 & 10 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[97] $1^{114} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 15 & 1 & 3 \\ 3 & 4 & 9 & 21 & 0 & 4 \\ 5 & 9 & 12 & 33 & 0 & 6 \\ 15 & 21 & 33 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

- [98] $1^{113} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 30 & 5 \\ 12 & 25 & 16 & 60 & 9 \\ 8 & 16 & 9 & 36 & 5 \\ 30 & 60 & 36 & 133 & 21 \\ 5 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$
- [99] $1^{111} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 3 & 4 & 0 & 6 & 0 & 2 \\ 5 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [100] $1^{11} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 5 & 9 & 12 & 9 & 0 & 6 \\ 3 & 5 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$
- [101] $1^{112} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 3 \\ 3 & 4 & 9 & 8 & 0 & 4 \\ 5 & 9 & 12 & 12 & 0 & 6 \\ 5 & 8 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [104] $1^{111} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$
- [103] $1^{113} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 29 & 5 \\ 12 & 25 & 16 & 60 & 9 \\ 8 & 16 & 9 & 36 & 5 \\ 29 & 60 & 36 & 133 & 21 \\ 5 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$
- [102] $1^{122} 3^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 8 & 8 & 0 & 4 \\ 5 & 8 & 9 & 12 & 0 & 5 \\ 4 & 8 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [105] $1^{111} 3^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 1 & 3 \\ 3 & 4 & 4 & 24 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 16 & 24 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$
- [106] $1^{111} 3^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 26 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 26 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [107] $1^{12} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 1 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 \\ 7 & 13 & 27 & 12 & 0 & 9 \\ 3 & 5 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$
- [108] $1^{111} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 13 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 13 & 17 & 12 & 27 & 6 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[109] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 141$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 \\ 5 & 9 & 12 & 6 & 0 & 6 \\ 2 & 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[110] $1^{112} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 10 & 24 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[111] $1^{112} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 11 & 24 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[112] $1^{12} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 144$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 1 & 3 \\ 3 & 4 & 0 & 28 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 16 & 28 & 0 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$

[113] $1^{122} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 146$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 1 & 3 \\ 3 & 4 & 13 & 8 & 0 & 4 \\ 7 & 13 & 27 & 18 & 0 & 9 \\ 4 & 8 & 18 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[114] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 14 & 4 & 0 & 4 \\ 9 & 14 & 36 & 9 & 0 & 10 \\ 2 & 4 & 9 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 10 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[115] $1^{112} 3 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 8 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

[116] $1^{12} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 1 & 3 \\ 3 & 4 & 0 & 18 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 10 & 18 & 0 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[117] $21^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 148$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.08 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 16 & 12 & 12 & 6 \\ 16 & 20 & 17 & 16 & 8 \\ 12 & 17 & 12 & 12 & 6 \\ 12 & 16 & 12 & 10 & 6 \\ 6 & 8 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[118] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 150$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.13 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 23 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 23 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

- [119] $1^{14}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 151, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 1 & 3 \\ 3 & 4 & 5 & 21 & 0 & 4 \\ 3 & 5 & 3 & 18 & 0 & 3 \\ 11 & 21 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$
- [120] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$
- [121] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 157, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 1 & 3 \\ 4 & 9 & 4 & 20 & 0 & 5 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 10 & 20 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$
- [122] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 158, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 38 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 38 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$
- [123] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 1 & 3 \\ 3 & 4 & 9 & 5 & 0 & 4 \\ 6 & 9 & 12 & 9 & 0 & 6 \\ 3 & 5 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$
- [125] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 1 & 3 \\ 3 & 4 & 9 & 4 & 0 & 4 \\ 6 & 9 & 12 & 6 & 0 & 6 \\ 2 & 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$
- [124] $1^{12}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 165, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 1 & 3 \\ 3 & 4 & 0 & 23 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 13 & 23 & 0 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$
- [126] $1^{11}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 \\ 5 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [128] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 26 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$
- [127] $1^{111}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 29 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 29 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$
- [129] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 11 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[130] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 170, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 1 & 3 \\ 3 & 4 & 17 & 5 & 0 & 4 \\ 10 & 17 & 48 & 15 & 0 & 12 \\ 3 & 5 & 15 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

[131] $1^{14} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 1 & 3 \\ 3 & 4 & 4 & 21 & 0 & 4 \\ 3 & 4 & 0 & 15 & 0 & 2 \\ 11 & 21 & 15 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[132] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 3 & 4 & 0 & 9 & 0 & 2 \\ 7 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[133] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 1 & 3 \\ 3 & 4 & 13 & 8 & 0 & 4 \\ 7 & 13 & 27 & 18 & 0 & 9 \\ 5 & 8 & 18 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[134] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 8 & 4 & 0 & 4 \\ 5 & 8 & 9 & 6 & 0 & 5 \\ 2 & 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[135] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 8 & 8 & 0 & 4 \\ 4 & 8 & 9 & 12 & 0 & 5 \\ 5 & 8 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[136] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 7 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[137] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[138] $1^{11} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 8 & 14 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$

[139] $1^{13} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 17 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 17 & 36 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

[140] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 7 & 16 & 6 & 22 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

[141] $1^{13} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 1 & 3 \\ 3 & 4 & 0 & 22 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 12 & 22 & 0 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 17 & -2 & 1 \end{pmatrix}_{b_0}$$

[142] $1^{14} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.51 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 1 & 3 \\ 3 & 4 & 0 & 26 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 14 & 26 & 0 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 20 & -2 & 1 \end{pmatrix}_{b_0}$$

[143] $21^{111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 15 & 7 \\ 20 & 25 & 16 & 20 & 9 \\ 12 & 16 & 9 & 12 & 5 \\ 15 & 20 & 12 & 13 & 7 \\ 7 & 9 & 5 & 7 & 2 \end{pmatrix}_{b_0}$$

[144] $1^{123} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 185, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 1 & 3 \\ 3 & 4 & 13 & 12 & 0 & 4 \\ 7 & 13 & 27 & 27 & 0 & 9 \\ 7 & 12 & 27 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[145] $1^{114} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 40 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 40 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$

[146] $1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 3 & 4 & 0 & 12 & 0 & 2 \\ 9 & 17 & 12 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[147] $1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 3 \\ 4 & 9 & 4 & 20 & 0 & 5 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 8 & 20 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 19 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[149] $1^{12} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 1 & 3 \\ 3 & 4 & 13 & 5 & 0 & 4 \\ 8 & 13 & 27 & 12 & 0 & 9 \\ 3 & 5 & 12 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

- [150] $1^{111} 1^3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 9 & 17 & 12 & 27 & 6 \\ 17 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [151] $1^{111} 1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 19 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 9 & 17 & 12 & 27 & 6 \\ 19 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [152] $1^{111} 1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 19 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 19 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [153] $1^{122} 1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 1 & 3 \\ 3 & 4 & 8 & 13 & 0 & 4 \\ 5 & 8 & 9 & 18 & 0 & 5 \\ 7 & 13 & 18 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$
- [154] $1^{114} 1^3 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 12 & 8 & 39 & 5 \\ 12 & 25 & 16 & 80 & 9 \\ 8 & 16 & 9 & 48 & 5 \\ 39 & 80 & 48 & 238 & 28 \\ 5 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$
- [155] $1^{113} 1^{31} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 205, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 16 & 32 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$
- [158] $1^{13} 1^1 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 3 & 15 & 0 & 3 \\ 9 & 17 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$
- [156] $1^{111} 1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 10 & 18 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 10 & 17 & 12 & 27 & 6 \\ 18 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [157] $1^{111} 1^{11} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 45 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 45 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$
- [160] $1^{121} 1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 1 & 3 \\ 3 & 4 & 13 & 9 & 0 & 4 \\ 7 & 13 & 27 & 21 & 0 & 9 \\ 6 & 9 & 21 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [159] $1^{112} 1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 6 & 9 & 12 & 21 & 0 & 6 \\ 7 & 13 & 21 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[161] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 211, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 3 & 12 & 0 & 3 \\ 7 & 13 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[162] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 1 & 3 \\ 3 & 4 & 8 & 13 & 0 & 4 \\ 4 & 8 & 9 & 18 & 0 & 5 \\ 8 & 13 & 18 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[163] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 \\ 7 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[164] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 7 & 13 & 27 & 9 & 0 & 9 \\ 2 & 4 & 9 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[165] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 221, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 1 & 3 \\ 3 & 4 & 8 & 13 & 0 & 4 \\ 4 & 8 & 9 & 18 & 0 & 5 \\ 7 & 13 & 18 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[172] $1^{11} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 19 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 12 & 19 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[166] $1^{13} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 3 & 4 & 0 & 15 & 0 & 2 \\ 12 & 22 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 17 & -2 & 1 \end{pmatrix}_{b_0}$$

[167] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 1 & 3 \\ 3 & 4 & 8 & 12 & 0 & 4 \\ 4 & 8 & 9 & 18 & 0 & 5 \\ 8 & 12 & 18 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[168] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 8 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[169] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 6 & 8 & 6 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[170] $1^{111} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 19 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 13 & 19 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[171] $1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 12 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 10 & 12 & 9 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[173] $1^{114} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 229, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 22 & 1 & 3 \\ 4 & 9 & 4 & 44 & 0 & 5 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 22 & 44 & 21 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 25 & -2 & 1 \end{pmatrix}_{b_0}$$

[174] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 233, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 18 & 5 \\ 11 & 20 & 0 & 33 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 33 & 0 & 49 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[175] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 25 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 25 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 237, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 1 & 3 \\ 4 & 9 & 4 & 20 & 0 & 5 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 9 & 20 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$

[178] $1^{131} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 12 & 4 & 0 & 4 \\ 7 & 12 & 24 & 9 & 0 & 8 \\ 2 & 4 & 9 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 8 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[179] $1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 10 & 14 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$

[177] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 6 & 9 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[181] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 15 & 12 & 24 & 6 \\ 15 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[180] $1^{111} 3^1 1^{13} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{so}_8}$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 241$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 38 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 38 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[182] $1^{111} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{so}_8}$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 242$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.30 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & 5 & 9 & 0 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 \\ 6 & 9 & 9 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[183] $21^{111} 31^{111} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{so}_8}$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 244$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 13 & 17 & 15 & 27 & 7 \\ 17 & 20 & 17 & 33 & 8 \\ 15 & 17 & 12 & 27 & 6 \\ 27 & 33 & 27 & 49 & 13 \\ 7 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[184] $1^{113} 3^1 1^{13} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{so}_8}$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 244$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.39 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 15 & 36 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

[185] $1^{111} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{so}_8}$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 245$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 9 & 18 & 5 \\ 11 & 20 & 17 & 33 & 8 \\ 9 & 17 & 12 & 27 & 6 \\ 18 & 33 & 27 & 49 & 13 \\ 5 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[187] $1^{13} 3^1 1^{13} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{so}_8}$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 253$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 1 & 3 \\ 3 & 4 & 0 & 27 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 15 & 27 & 0 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

[186] $1^{113} 3^1 1^{31} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{so}_8}$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 253$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 12 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[188] $1^{111} 3^1 1^{14} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{so}_8}$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 254$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.31 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 47 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 47 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[189] $1^{123} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{so}_8}$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 257$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 8 & 17 & 0 & 4 \\ 5 & 8 & 9 & 24 & 0 & 5 \\ 9 & 17 & 24 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[190] $1^{111} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{so}_8}$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 267$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.17 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 6 & 9 & 12 & 15 & 0 & 6 \\ 5 & 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

- [191] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 268, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 1 & 3 \\ 3 & 4 & 4 & 31 & 0 & 4 \\ 3 & 4 & 0 & 21 & 0 & 2 \\ 17 & 31 & 21 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 24 & -2 & 1 \end{pmatrix}_{b_0}$$
- [192] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 5 & 9 & 12 & 15 & 0 & 6 \\ 5 & 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$
- [194] $1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 9 & 14 & 9 & 37 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$
- [193] $1^{123} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 1 & 3 \\ 3 & 4 & 8 & 12 & 0 & 4 \\ 4 & 8 & 9 & 18 & 0 & 5 \\ 7 & 12 & 18 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$
- [195] $21^{112} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 30 & 7 \\ 20 & 25 & 16 & 40 & 9 \\ 12 & 16 & 9 & 24 & 5 \\ 30 & 40 & 24 & 58 & 14 \\ 7 & 9 & 5 & 14 & 2 \end{pmatrix}_{b_0}$$
- [197] $1^{121} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 10 & 18 & 57 & 12 & 0 & 13 \\ 2 & 4 & 12 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$
- [196] $1^{113} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 16 & 36 & 18 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$
- [198] $1^{111} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 277, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 36 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 36 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$
- [199] $1^{113} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 285, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 10 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 10 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$
- [202] $21^{111} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 16 & 12 & 24 & 6 \\ 16 & 20 & 17 & 33 & 8 \\ 12 & 17 & 12 & 27 & 6 \\ 24 & 33 & 27 & 49 & 13 \\ 6 & 8 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$
- [200] $1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 28 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 28 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[201] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 27 & 5 \\ 11 & 20 & 0 & 50 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 27 & 50 & 0 & 118 & 20 \\ 5 & 8 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[203] $1^{13} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 1 & 3 \\ 3 & 4 & 0 & 32 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 18 & 32 & 0 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 25 & -2 & 1 \end{pmatrix}_{b_0}$$

[204] $1^{111} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 46 & 5 \\ 11 & 20 & 0 & 84 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 46 & 84 & 0 & 346 & 34 \\ 5 & 8 & -1 & 34 & 2 \end{pmatrix}_{b_0}$$

[205] $1^{123} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 290, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 9 & 1 & 3 \\ 3 & 4 & 13 & 17 & 0 & 4 \\ 8 & 13 & 27 & 39 & 0 & 9 \\ 9 & 17 & 39 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[206] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 293, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 1 & 3 \\ 3 & 4 & 8 & 17 & 0 & 4 \\ 4 & 8 & 9 & 24 & 0 & 5 \\ 11 & 17 & 24 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[207] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 295, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 3 & 12 & 0 & 3 \\ 10 & 13 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[208] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 7 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[209] $1^{131} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 301, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 9 & 17 & 48 & 12 & 0 & 12 \\ 2 & 4 & 12 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[210] $1^{131} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 303, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 1 & 3 \\ 3 & 4 & 9 & 17 & 0 & 4 \\ 5 & 9 & 12 & 27 & 0 & 6 \\ 12 & 17 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[211] $1^{131} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.54 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 3 \\ 4 & 9 & 4 & 28 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 11 & 28 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[212] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 3 \\ 4 & 9 & 4 & 28 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 14 & 28 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[213] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 1 & 3 \\ 3 & 4 & 4 & 26 & 0 & 4 \\ 3 & 4 & 0 & 18 & 0 & 2 \\ 14 & 26 & 18 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 20 & -2 & 1 \end{pmatrix}_{b_0}$$

[214] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 1 & 3 \\ 3 & 4 & 9 & 9 & 0 & 4 \\ 5 & 9 & 12 & 15 & 0 & 6 \\ 6 & 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[215] $1^{14} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 309, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 1 & 3 \\ 3 & 4 & 0 & 31 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 17 & 31 & 0 & 172 & 0 & 24 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 24 & -2 & 1 \end{pmatrix}_{b_0}$$

[216] $1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 309, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 1 & 3 \\ 3 & 4 & 13 & 4 & 0 & 4 \\ 8 & 13 & 27 & 9 & 0 & 9 \\ 2 & 4 & 9 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[217] $1^{111} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 319, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 11 & 1 & 37 & 5 \\ 11 & 20 & 0 & 67 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 37 & 67 & 0 & 217 & 27 \\ 5 & 8 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 12 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 7 & 12 & 9 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[219] $1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 326, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 1 & 3 \\ 3 & 4 & 13 & 9 & 0 & 4 \\ 7 & 13 & 27 & 21 & 0 & 9 \\ 5 & 9 & 21 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[220] $1^{131} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 331, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 1 & 3 \\ 3 & 4 & 17 & 9 & 0 & 4 \\ 9 & 17 & 48 & 27 & 0 & 12 \\ 5 & 9 & 27 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[221] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 333, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 11 & 1 & 3 \\ 4 & 9 & 4 & 24 & 0 & 5 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 11 & 24 & 9 & 52 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[222] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 334, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 3 & 15 & 0 & 3 \\ 13 & 17 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[223] $1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 341, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 1 & 3 \\ 3 & 4 & 8 & 17 & 0 & 4 \\ 4 & 8 & 9 & 24 & 0 & 5 \\ 9 & 17 & 24 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[224] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 3 & 12 & 0 & 3 \\ 8 & 13 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[225] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 349, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 13 & 32 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

[227] $1^{123} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 1 & 3 \\ 3 & 4 & 8 & 22 & 0 & 4 \\ 4 & 8 & 9 & 30 & 0 & 5 \\ 12 & 22 & 30 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 17 & -2 & 1 \end{pmatrix}_{b_0}$$

[226] $1^{123} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 356, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 1 & 3 \\ 3 & 4 & 8 & 22 & 0 & 4 \\ 4 & 8 & 9 & 30 & 0 & 5 \\ 13 & 22 & 30 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 17 & -2 & 1 \end{pmatrix}_{b_0}$$

[228] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 5 & 9 & 12 & 21 & 0 & 6 \\ 7 & 13 & 21 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[229] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 366, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 5 & 9 & 12 & 21 & 0 & 6 \\ 9 & 13 & 21 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[230] $1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 371, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 5 & 13 & 0 & 4 \\ 3 & 5 & 3 & 12 & 0 & 3 \\ 9 & 13 & 12 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[231] $1^{121} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 372, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 11 & 2 & 1 & 3 \\ 3 & 4 & 18 & 4 & 0 & 4 \\ 11 & 18 & 57 & 12 & 0 & 13 \\ 2 & 4 & 12 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 13 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[232] $1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 15 & 32 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

[233] $1^{114} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 22 & 1 & 3 \\ 4 & 9 & 4 & 48 & 0 & 5 \\ 2 & 4 & 0 & 24 & 0 & 2 \\ 22 & 48 & 24 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

[234] $1^{114} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 20 & 1 & 3 \\ 4 & 9 & 4 & 48 & 0 & 5 \\ 2 & 4 & 0 & 24 & 0 & 2 \\ 20 & 48 & 24 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

[235] $1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 12 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 9 & 12 & 9 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[236] $1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 386, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 1 & 3 \\ 3 & 4 & 13 & 9 & 0 & 4 \\ 8 & 13 & 27 & 21 & 0 & 9 \\ 5 & 9 & 21 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[237] $1^{123} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 1 & 3 \\ 3 & 4 & 13 & 17 & 0 & 4 \\ 7 & 13 & 27 & 39 & 0 & 9 \\ 11 & 17 & 39 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[238] $1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 10 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[239] $1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 394, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 3 & 15 & 0 & 3 \\ 10 & 17 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[240] $1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 395, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 1 & 3 \\ 3 & 4 & 5 & 21 & 0 & 4 \\ 3 & 5 & 3 & 18 & 0 & 3 \\ 12 & 21 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[241] $1^{114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 13 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[242] $1^{14} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 1 & 3 \\ 3 & 4 & 0 & 36 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 20 & 36 & 0 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

[243] $1^{14} 3^1 1^{41} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 400, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 16 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[244] $1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 406, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 1 & 3 \\ 3 & 4 & 9 & 21 & 0 & 4 \\ 5 & 9 & 12 & 33 & 0 & 6 \\ 11 & 21 & 33 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[245] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 411, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 9 & 17 & 0 & 4 \\ 5 & 9 & 12 & 27 & 0 & 6 \\ 9 & 17 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[248] $1^{14} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 16 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 9 & 16 & 12 & 46 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[246] $1^{13} 3^1 1^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 12 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 8 & 12 & 9 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[247] $1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 10 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[251] $1^{12} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 1 & 3 \\ 3 & 4 & 4 & 28 & 0 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 16 & 28 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$

[250] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 9 & 17 & 12 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[249] $1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 416, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 14 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[252] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 419, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 7 & 1 & 3 \\ 3 & 4 & 13 & 13 & 0 & 4 \\ 8 & 13 & 27 & 30 & 0 & 9 \\ 7 & 13 & 30 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[253] $21^{113} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 45 & 7 \\ 20 & 25 & 16 & 60 & 9 \\ 12 & 16 & 9 & 36 & 5 \\ 45 & 60 & 36 & 133 & 21 \\ 7 & 9 & 5 & 21 & 2 \end{pmatrix}_{b_0}$$

[256] $1^{123} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 1 & 3 \\ 3 & 4 & 8 & 17 & 0 & 4 \\ 4 & 8 & 9 & 24 & 0 & 5 \\ 10 & 17 & 24 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[255] $1^{113} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 14 & 32 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

[254] $1^{114} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 429, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 14 & 36 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

[258] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 13 & 1 & 3 \\ 4 & 9 & 4 & 28 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 13 & 28 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[257] $1^{113} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 1 & 3 \\ 4 & 9 & 4 & 28 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 12 & 28 & 12 & 73 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

[259] $1^{114} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 20 & 1 & 3 \\ 4 & 9 & 4 & 40 & 0 & 5 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 20 & 40 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$

[260] $1^{131} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 436, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.09 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 1 & 3 \\ 3 & 4 & 22 & 4 & 0 & 4 \\ 12 & 22 & 84 & 15 & 0 & 16 \\ 2 & 4 & 15 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[261] $1^{114} 3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 21 & 1 & 3 \\ 4 & 9 & 4 & 48 & 0 & 5 \\ 2 & 4 & 0 & 24 & 0 & 2 \\ 21 & 48 & 24 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

[262] $1^{131} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 453, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.04 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 1 & 3 \\ 3 & 4 & 17 & 4 & 0 & 4 \\ 10 & 17 & 48 & 12 & 0 & 12 \\ 2 & 4 & 12 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[263] $1^{131} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 455, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 1 & 3 \\ 3 & 4 & 17 & 9 & 0 & 4 \\ 10 & 17 & 48 & 27 & 0 & 12 \\ 5 & 9 & 27 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[265] $1^{112} 3^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 23 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 13 & 23 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

[264] $1^{112} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.11 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 8 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[266] $1^{131} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 458, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 3 & 15 & 0 & 3 \\ 12 & 17 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[267] $1^{122} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 471, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 1 & 3 \\ 3 & 4 & 13 & 13 & 0 & 4 \\ 7 & 13 & 27 & 30 & 0 & 9 \\ 7 & 13 & 30 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[268] $1^{131} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 478, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 1 & 3 \\ 3 & 4 & 5 & 17 & 0 & 4 \\ 3 & 5 & 3 & 15 & 0 & 3 \\ 11 & 17 & 15 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[269] $1^{112} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 488, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 13 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 9 & 13 & 9 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[270] $1^{122} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 491, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.19 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 1 & 3 \\ 3 & 4 & 13 & 13 & 0 & 4 \\ 7 & 13 & 27 & 30 & 0 & 9 \\ 8 & 13 & 30 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[271] $1^{114} 3^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 493, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 18 & 36 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

[272] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 498, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 1 & 3 \\ 3 & 4 & 9 & 13 & 0 & 4 \\ 5 & 9 & 12 & 21 & 0 & 6 \\ 8 & 13 & 21 & 28 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[273] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 501, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.47 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 21 & 1 & 3 \\ 4 & 9 & 4 & 44 & 0 & 5 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 21 & 44 & 21 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 25 & -2 & 1 \end{pmatrix}_{b_0}$$

[274] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 503, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 1 & 3 \\ 3 & 4 & 5 & 21 & 0 & 4 \\ 3 & 5 & 3 & 18 & 0 & 3 \\ 15 & 21 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[276] $1^{114} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.47 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 15 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[275] $1^{114} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 528, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 14 & 1 & 3 \\ 4 & 9 & 4 & 32 & 0 & 5 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 14 & 32 & 12 & 94 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[277] $1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 535, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 1 & 3 \\ 3 & 4 & 5 & 21 & 0 & 4 \\ 3 & 5 & 3 & 18 & 0 & 3 \\ 13 & 21 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[278] $1^{114} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.36 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 21 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 11 & 21 & 15 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[279] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 549, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 18 & 1 & 3 \\ 4 & 9 & 4 & 44 & 0 & 5 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 18 & 44 & 21 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 25 & -2 & 1 \end{pmatrix}_{b_0}$$

[280] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 9 & 4 & 40 & 0 & 5 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 16 & 40 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$

[281] $1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 566, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 1 & 3 \\ 3 & 4 & 13 & 17 & 0 & 4 \\ 7 & 13 & 27 & 39 & 0 & 9 \\ 9 & 17 & 39 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[282] $1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 571, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 1 & 3 \\ 3 & 4 & 5 & 21 & 0 & 4 \\ 3 & 5 & 3 & 18 & 0 & 3 \\ 14 & 21 & 18 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[283] $1^{12} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 584, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 1 & 3 \\ 3 & 4 & 4 & 23 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 17 & 23 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

[284] $1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 586, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 14 & 1 & 3 \\ 3 & 4 & 9 & 21 & 0 & 4 \\ 5 & 9 & 12 & 33 & 0 & 6 \\ 14 & 21 & 33 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[285] $1^{13} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 1 & 3 \\ 3 & 4 & 22 & 4 & 0 & 4 \\ 13 & 22 & 84 & 15 & 0 & 16 \\ 2 & 4 & 15 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 16 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

[286] $1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 603, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 1 & 3 \\ 3 & 4 & 9 & 17 & 0 & 4 \\ 5 & 9 & 12 & 27 & 0 & 6 \\ 11 & 17 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[290] $1^{14} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 16 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 12 & 16 & 12 & 46 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[289] $1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 11 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[288] $1^{14} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 16 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 10 & 16 & 12 & 46 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[287] $1^{12} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 608, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.08 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 13 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[291] $21^{14} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 634, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 20 & 12 & 60 & 7 \\ 20 & 25 & 16 & 80 & 9 \\ 12 & 16 & 9 & 48 & 5 \\ 60 & 80 & 48 & 238 & 28 \\ 7 & 9 & 5 & 28 & 2 \end{pmatrix}_{b_0}$$

[292] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 637$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 15 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 15 & 36 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

[293] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 639$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 1 & 3 \\ 3 & 4 & 9 & 17 & 0 & 4 \\ 5 & 9 & 12 & 27 & 0 & 6 \\ 10 & 17 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[294] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 645$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.40 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 20 & 1 & 3 \\ 4 & 9 & 4 & 44 & 0 & 5 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 20 & 44 & 21 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 25 & -2 & 1 \end{pmatrix}_{b_0}$$

[295] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 656$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 12 & 22 & 15 & 85 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 17 & -2 & 1 \end{pmatrix}_{b_0}$$

[296] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 660$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 19 & 1 & 3 \\ 4 & 9 & 4 & 40 & 0 & 5 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 19 & 40 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$

[297] $1^{114} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 661$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 19 & 1 & 3 \\ 4 & 9 & 4 & 44 & 0 & 5 \\ 2 & 4 & 0 & 21 & 0 & 2 \\ 19 & 44 & 21 & 193 & 0 & 25 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 25 & -2 & 1 \end{pmatrix}_{b_0}$$

[298] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 669$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 17 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 17 & 36 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

[299] $1^{112} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 672$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 1 & 3 \\ 3 & 4 & 4 & 28 & 0 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 17 & 28 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$

[300] $1^{112} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 672$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 1 & 3 \\ 3 & 4 & 4 & 28 & 0 & 4 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 20 & 28 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$

[301] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 672$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 10 & 17 & 12 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[302] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.8$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 13 & 17 & 12 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[303] $1^{114} 3^1 1^{31} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 16 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 11 & 16 & 12 & 46 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[304] $1^{112} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 18 & 0 & 4 \\ 2 & 4 & 0 & 12 & 0 & 2 \\ 12 & 18 & 12 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[305] $1^{112} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 680, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 1 & 3 \\ 3 & 4 & 4 & 23 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 14 & 23 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

[306] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 717, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 16 & 1 & 3 \\ 4 & 9 & 4 & 36 & 0 & 5 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 16 & 36 & 15 & 121 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

[307] $1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 722, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 1 & 3 \\ 3 & 4 & 13 & 17 & 0 & 4 \\ 7 & 13 & 27 & 39 & 0 & 9 \\ 10 & 17 & 39 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[308] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.39 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 17 & 1 & 3 \\ 4 & 9 & 4 & 40 & 0 & 5 \\ 2 & 4 & 0 & 18 & 0 & 2 \\ 17 & 40 & 18 & 154 & 0 & 22 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$

[309] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 730, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 1 & 3 \\ 3 & 4 & 9 & 21 & 0 & 4 \\ 5 & 9 & 12 & 33 & 0 & 6 \\ 12 & 21 & 33 & 76 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[310] $1^{112} 3^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 744, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.5$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 1 & 3 \\ 3 & 4 & 4 & 23 & 0 & 4 \\ 2 & 4 & 0 & 15 & 0 & 2 \\ 16 & 23 & 15 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 18 & -2 & 1 \end{pmatrix}_{b_0}$$

[311] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 756, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 24 & 1 & 3 \\ 3 & 4 & 4 & 36 & 0 & 4 \\ 2 & 4 & 0 & 24 & 0 & 2 \\ 24 & 36 & 24 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [352] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[353] \quad 21^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 14 & 9 & 15 & 7 \\ 14 & 13 & 10 & 14 & 7 \\ 9 & 10 & 4 & 9 & 4 \\ 15 & 14 & 9 & 13 & 7 \\ 7 & 7 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[354] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 7 & 5 & 0 & 4 \\ 4 & 7 & 7 & 6 & 0 & 5 \\ 3 & 5 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[355] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 5 & 0 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 5 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[356] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 4 & 0 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[358] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 8 & 0 & 4 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 5 & 8 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[357] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 7 & 8 & 0 & 4 \\ 4 & 7 & 7 & 11 & 0 & 5 \\ 5 & 8 & 11 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[359] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 7 & 3 & 0 & 4 \\ 4 & 7 & 7 & 5 & 0 & 5 \\ 2 & 3 & 5 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[360] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -94, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 7 & 0 & 4 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 4 & 7 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 6 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[361] \ 1^{11} 4^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -119, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 9 & 0 & 4 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 5 & 9 & 5 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[362] \ 1^{12} 4^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -139, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.69 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & 3 & 11 & 0 & 4 \\ 2 & 3 & 0 & 7 & 0 & 2 \\ 6 & 11 & 7 & 21 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[363] \ 1^{11} 4^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -206, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 3 & 12 & 0 & 4 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 7 & 12 & 8 & 26 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[364] \ 1^{12} 4^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -325, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 3 & 16 & 0 & 4 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 9 & 16 & 10 & 45 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 13 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[365] \ 1^{12} 4^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -83, \ \det = -406, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.5$$

$$T_{\min} = 11.72 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 3 & 19 & 0 & 4 \\ 2 & 3 & 0 & 13 & 0 & 2 \\ 11 & 19 & 13 & 68 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

4.1158 $\text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{e}}_6$

$$T = 9 \quad [0] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{e}}_6 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -115, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.23 \rightarrow 15, \ T_{\max} = 99, \ T \in [15, 99]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 9 & 5 & 7 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{e}}_6 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -271, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 7 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{e}}_6 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -334, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 1 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 8 & 16 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{e}}_6 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -931, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 6 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

4.1159 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.27 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 9 & 5 & 7 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 1 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 8 & 16 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

4.1160 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{113}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 28 & 1 & 5 \\ 12 & 25 & 16 & 60 & 0 & 9 \\ 8 & 16 & 9 & 36 & 0 & 5 \\ 28 & 60 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 1 & 1 & 5 \\ 11 & 20 & 17 & 0 & 0 & 8 \\ 9 & 17 & 12 & 0 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 0 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 10 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 10 & 20 & 12 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 10 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 10 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -133, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 1 & 5 \\ 11 & 20 & 0 & 17 & 0 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 9 & 17 & 0 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & -1 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -155, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 10 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 17 & 0 & 8 \\ 9 & 17 & 12 & 15 & 0 & 6 \\ 9 & 17 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 7 & -2 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 8 & 1 & 5 \\ 11 & 20 & 0 & 16 & 0 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 8 & 16 & 0 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & -1 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 19 & 1 & 5 \\ 12 & 25 & 16 & 40 & 0 & 9 \\ 8 & 16 & 9 & 24 & 0 & 5 \\ 19 & 40 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 20 & 1 & 5 \\ 12 & 25 & 16 & 40 & 0 & 9 \\ 8 & 16 & 9 & 24 & 0 & 5 \\ 20 & 40 & 24 & 58 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 8 & 1 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 8 & 16 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{so}_8 + \text{so}_8$ $T = 9, T_{\max} = 13, \Delta = -91, \det = -215, (n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 10 & 17 & 12 & 27 & 0 & 6 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -222, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 9 & 1 & 5 \\ 11 & 20 & 17 & 16 & 0 & 8 \\ 9 & 17 & 12 & 12 & 0 & 6 \\ 9 & 16 & 12 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3 \ 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 26 & 1 & 5 \\ 11 & 20 & 0 & 50 & 0 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 26 & 50 & 0 & 118 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & -1 & 20 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -293, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 17 & 1 & 5 \\ 11 & 20 & 0 & 33 & 0 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 17 & 33 & 0 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & -1 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{113} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -295, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 30 & 1 & 5 \\ 12 & 25 & 16 & 60 & 0 & 9 \\ 8 & 16 & 9 & 36 & 0 & 5 \\ 30 & 60 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{113} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 29 & 1 & 5 \\ 12 & 25 & 16 & 60 & 0 & 9 \\ 8 & 16 & 9 & 36 & 0 & 5 \\ 29 & 60 & 36 & 133 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{114} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 38 & 1 & 5 \\ 12 & 25 & 16 & 80 & 0 & 9 \\ 8 & 16 & 9 & 48 & 0 & 5 \\ 38 & 80 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{114} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 40 & 1 & 5 \\ 12 & 25 & 16 & 80 & 0 & 9 \\ 8 & 16 & 9 & 48 & 0 & 5 \\ 40 & 80 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{114} 3 \ 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -510, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 8 & 39 & 1 & 5 \\ 12 & 25 & 16 & 80 & 0 & 9 \\ 8 & 16 & 9 & 48 & 0 & 5 \\ 39 & 80 & 48 & 238 & 0 & 28 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 28 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -539, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.17 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 19 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 19 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -539, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 17 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 17 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -735, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 11 & 9 & 18 & 1 & 5 \\ 11 & 20 & 17 & 33 & 0 & 8 \\ 9 & 17 & 12 & 27 & 0 & 6 \\ 18 & 33 & 27 & 49 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

4.1161 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -16, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 & 3 \\ 3 & 4 & 4 & 5 & 0 & 5 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 4 & 5 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & 2 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 4 & 5 & 0 & 5 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 5 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & 2 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -62, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 0 & 5 & 4 \\ 2 & 4 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 3 & 0 & 0 & 4 & 4 \\ 3 & 4 & 2 & 0 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

4.1162 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 6, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.67 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & -1 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 9, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.78 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 0 & 4 & 4 \\ 3 & 4 & -1 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

- [2] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [3] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 4 & 3 \\ 3 & 4 & 4 & 5 & 5 & 4 \\ 3 & 4 & 0 & 3 & 3 & 2 \\ 3 & 5 & 3 & 4 & 6 & 4 \\ 4 & 5 & 3 & 6 & 4 & 4 \\ 3 & 4 & 2 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [4] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -1 & 4 & 7 & 3 \end{pmatrix}_{b_0}$
- [5] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 5 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [6] $1^{11} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 4 & 4 & 3 \\ 3 & 4 & 4 & 5 & 5 & 4 \\ 3 & 4 & 0 & 3 & 3 & 2 \\ 4 & 5 & 3 & 4 & 6 & 4 \\ 4 & 5 & 3 & 6 & 4 & 4 \\ 3 & 4 & 2 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [8] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & 4 & 12 & 4 \\ 8 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -1 & 4 & 10 & 3 \end{pmatrix}_{b_0}$
- [7] $1^{11} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 4 & 0 & 0 & 4 \\ 3 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 0 & 0 & 3 \end{pmatrix}_{b_0}$
- [9] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [10] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 5 & 9 & 0 & 13 & 12 & 7 \\ 5 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & -1 & 7 & 6 & 3 \end{pmatrix}_{b_0}$
- [11] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 5 & 4 \\ 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 4 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$

$$[12] \quad 1^{111} 3 \ 1^{11} + [\textcolor{blue}{s}\textcolor{red}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}\textcolor{blue}{u}_3 + \textcolor{orange}{s}\textcolor{red}{u}_2 + \textcolor{red}{s}\textcolor{blue}{u}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -63, \ \det = 28, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & 4 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111} \begin{smallmatrix} 1 & 1 & 1 \\ 3 & 1 & 1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -63, \quad \det = 30, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\left(\begin{array}{cccccc} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 0 & 0 & 1 & 3 \\ 3 & 4 & -1 & 0 & 3 & 3 \end{array} \right)_{b_0}$$

[14] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$

$$[15] \text{ } 1^{12} 3 \text{ } 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \text{ } T_{\max} = 10, \text{ } \Delta = -63, \text{ } \det = 36, \text{ } (n_+, n_-, n_0) = (1, 8, 0), \text{ } c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \text{ } T_{\max} = 10, \text{ } T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 8 & 0 & 0 & 10 & 6 \\ 3 & 4 & -1 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[16] $1^{111}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{red}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$

$$[17] \quad 1^{111} 3^1 1^1 + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}u_3 + \textcolor{orange}{s}u_2 + \textcolor{red}{s}u_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -63, \quad \det = 52, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.85 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

[18] $1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{teal}{\mathfrak{su}}_3 + \textcolor{brown}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 6 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 9 & 0 & 0 & 13 & 7 \\ 3 & 4 & -1 & 0 & 7 & 3 \end{pmatrix}_{b_0}$

$$[19] \quad 1^{121} 1^{111} 3 \quad 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -63, \quad \det = 63, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 5 & 8 & 0 & 10 & 12 & 6 \\ 6 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & -1 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

[20] $1^{12} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \, T_{\max} = 10, \, \Delta = -63, \, \det = 69, \, (n_+, n_-, n_0) = (1, 8, 0), \, c_{\text{ext}} = 7.5$
 $T_{\min} = 9.97 \rightarrow 10, \, T_{\max} = 10, \, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 3 \\ 3 & 4 & 0 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 13 & 0 & 0 & 28 & 10 \\ 3 & 4 & -1 & 0 & 10 & 3 \end{pmatrix}_{b_0}$

$$[21] \quad 1^{111} 3 1^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}u_3 + \textcolor{orange}{s}u_2 + \textcolor{red}{s}u_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

- [22] $1^{112} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 5 & 9 & 0 & 13 & 21 & 7 \\ 9 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -1 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$
- [23] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$
- [24] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$
- [25] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 9 & 4 \\ 6 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & -1 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$
- [26] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 9 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -1 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$
- [27] $1^{12} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 12 & 4 \\ 8 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & -1 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$
- [28] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & -1 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$
- [29] $1^{111} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 125, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 5 & 9 & 0 & 13 & 15 & 7 \\ 6 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$
- [30] $1^{131} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 16 & 2 & 1 & 3 \\ 3 & 4 & 22 & 4 & 0 & 4 \\ 16 & 22 & 84 & 15 & 0 & 16 \\ 2 & 4 & 15 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 16 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$
- [31] $1^{112} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -63, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 9 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

[32] $1^{112} 3^1 1^{11} + [\textcolor{blue}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}u_3 + \textcolor{orange}{s}u_2 + \textcolor{red}{s}u_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 9 & 3 \\ 8 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$

[33] $1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, \, T_{\max} = 10, \, \Delta = -63, \, \det = 184, \, (n_+, n_-, n_0) = (1, 8, 0), \, c_{\text{ext}} = 7.5$
 $T_{\min} = 9.98 \rightarrow 10, \, T_{\max} = 10, \, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 9 & 3 \\ 8 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$

[34] $1^{111} 3^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 0 & 13 & 15 & 7 \\ 6 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & -1 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[35] $1^{112} 3 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 285, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 8 & 3 \\ 3 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 6 & 9 & 0 & 13 & 21 & 7 \\ 8 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & -1 & 7 & 10 & 3 \end{pmatrix}_{b_0}$

$$T = 9 \quad [37] \quad 1^{\textcolor{red}{11}} 3^{21 \textcolor{blue}{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -35, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 3 & 0 & 1 & 4 & 3 \\ 3 & 3 & 4 & 0 & 4 & 5 & 5 \\ 2 & 2 & 2 & -1 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{11} 3^{11} 21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 0 & 2 \\ 2 & 2 & 2 & -1 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{111} 321^{111} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{s}u_3 + \textcolor{orange}{s}u_2 + \textcolor{red}{s}u_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 4 & 4 & 2 \\ 2 & 3 & 0 & 4 & 5 & 7 & 5 \\ 2 & 3 & 0 & 4 & 7 & 5 & 5 \\ 2 & 2 & -1 & 2 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[39] $1^{11} 1^{11} 3 21^{111} + [\text{red}_2 \cdot \text{g}_2] + \text{green}_3 + \text{orange}_2 + \text{red}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -35$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 1 & 3 & 0 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 4 & 5 & 5 \\ 2 & 2 & 2 & -1 & 3 & 5 & 3 \end{pmatrix}_{b_0}$

[40] $1^{11}3^{21}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 4 & 4 \\ 3 & 5 & 0 & 4 & 2 & 4 \\ 3 & 5 & -1 & 4 & 4 & 4 \end{pmatrix}_{b_0}$

[42] $1^{11} 1^{11} 3^{21} 1^{11} + [\text{su}_2, \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -35, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 6 & 4 & 3 & 6 \\ 4 & 2 & 2 & 4 \\ 3 & 2 & 1 & 3 \\ 6 & 4 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[52] \ 1^{111} 4^{111} 131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -2 & 1 & 1 & 0 \\ 3 & 4 & 1 & 2 & 4 & 4 \\ 3 & 4 & 1 & 4 & 2 & 4 \\ 3 & 4 & 0 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{111} 4^{111} 31^{111} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 & 3 \\ 3 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & -2 & 1 & 1 & 0 \\ 6 & 7 & 1 & 9 & 6 & 7 \\ 3 & 4 & 1 & 6 & 2 & 4 \\ 3 & 4 & 0 & 7 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[54] \ 1^{11} 3^{11} 221^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -6, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 1 & 0 & 4 & 5 & 3 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 1 & 4 & 0 & 2 & 5 & 4 \\ 2 & 3 & 5 & 0 & 5 & 6 & 6 \\ 2 & 2 & 3 & -1 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{11} 4^{11} 31^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 6 & 3 \\ 2 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 1 & 5 & 3 \\ 6 & 6 & 1 & 5 & 14 & 8 \\ 3 & 3 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[55] \ 1^{11} 4^{11} 31^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 3 & 5 & 4 \\ 3 & 1 & 2 & 4 & 3 \\ 3 & 2 & 1 & 3 & 3 \\ 5 & 4 & 3 & 6 & 6 \\ 4 & 3 & 3 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{11} 4^{11} 31^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 4 & 3 \\ 2 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 1 & 3 & 3 \\ 4 & 3 & 1 & 3 & 5 & 5 \\ 3 & 3 & 0 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{11} 4^{11} 131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 4 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 4 & 3 & 1 & 5 & 4 & 5 \\ 3 & 3 & 1 & 4 & 2 & 4 \\ 3 & 3 & 0 & 5 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{11} 4^{11} 31^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 0 & 2 & 1 & 1 & 2 \\ 3 & 2 & 1 & 2 & 1 & 3 \\ 1 & 1 & 2 & 0 & 1 & 2 \\ 2 & 1 & 1 & 1 & 0 & 2 \\ 3 & 2 & 3 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{11} 4^{11} 131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 5 & 4 \\ 3 & 1 & 1 & 4 & 3 \\ 2 & 1 & 0 & 2 & 2 \\ 5 & 4 & 2 & 6 & 6 \\ 4 & 3 & 2 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{11} 4^{11} 31^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{aligned}
& \begin{pmatrix} 2 & 3 & 5 & 5 & 4 \\ 3 & 1 & 4 & 3 & 3 \\ 5 & 4 & 6 & 5 & 6 \\ 5 & 3 & 5 & 3 & 5 \\ 4 & 3 & 6 & 5 & 5 \end{pmatrix}_{b_0} \\
[63] & 1^{11} 4^{111} 131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 2 & 3 & 2 & 4 & 4 \\ 3 & 1 & 1 & 4 & 3 \\ 2 & 1 & 0 & 2 & 2 \\ 4 & 4 & 2 & 3 & 5 \\ 4 & 3 & 2 & 5 & 5 \end{pmatrix}_{b_0} \\
[67] & 1^{11} 4^{111} 31^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 2 & 3 & 5 & 4 & 4 \\ 3 & 1 & 3 & 4 & 3 \\ 5 & 3 & 3 & 4 & 5 \\ 4 & 4 & 4 & 3 & 5 \\ 4 & 3 & 5 & 5 & 5 \end{pmatrix}_{b_0} \\
[66] & 1^{11} 4^{11} 131^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 2 & 3 & 5 & 4 & 4 \\ 3 & 1 & 4 & 4 & 3 \\ 5 & 4 & 6 & 5 & 6 \\ 4 & 4 & 5 & 3 & 5 \\ 4 & 3 & 6 & 5 & 5 \end{pmatrix}_{b_0} \\
[65] & 1^{111} 4^{11} 31^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 3 & 5 & 3 \\ 2 & 1 & 1 & 3 & 5 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 3 & 3 & 1 & 2 & 5 & 4 \\ 5 & 5 & 0 & 5 & 8 & 6 \\ 3 & 3 & 0 & 4 & 6 & 4 \end{pmatrix}_{b_0} \\
[64] & 1^{111} 4^{11} 31^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 5 & 6 & 3 \\ 2 & 1 & 1 & 5 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 5 & 5 & 0 & 8 & 11 & 6 \\ 6 & 6 & 1 & 11 & 14 & 8 \\ 3 & 3 & 0 & 6 & 8 & 4 \end{pmatrix}_{b_0} \\
[62] & 1^{112} 31^{11} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 2 & 3 & 5 & 7 & 4 \\ 3 & 1 & 3 & 5 & 3 \\ 5 & 3 & 3 & 7 & 5 \\ 7 & 5 & 7 & 12 & 8 \\ 4 & 3 & 5 & 8 & 5 \end{pmatrix}_{b_0} \\
[68] & 1^{111} 31^{11} 4^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 21, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 2 & 3 & 2 & 5 & 4 \\ 3 & 1 & 1 & 3 & 3 \\ 2 & 1 & 0 & 2 & 2 \\ 5 & 3 & 2 & 3 & 5 \\ 4 & 3 & 2 & 5 & 5 \end{pmatrix}_{b_0} \\
[71] & 1^{111} 4^{11} 131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 3 & 0 & 1 & 3 & 3 \\ 3 & 3 & 1 & 3 & 2 & 4 \\ 3 & 3 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0} \\
[70] & 1^{111} 4^{11} 31^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 1 & 6 & 6 & 3 \\ 2 & 1 & 1 & 6 & 5 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 6 & 6 & 1 & 14 & 12 & 8 \\ 6 & 5 & 1 & 12 & 9 & 7 \\ 3 & 3 & 0 & 8 & 7 & 4 \end{pmatrix}_{b_0} \\
[69] & 1^{111} 4^{11} 31^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 3 & 5 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 3 & 0 & 1 & 5 & 3 \\ 6 & 5 & 1 & 5 & 9 & 7 \\ 3 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{111} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 11, \ \Delta = -33, \ \det = 50, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 1 & 3 & 8 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 3 & 0 & 1 & 8 & 3 \\ 9 & 8 & 1 & 8 & 27 & 11 \\ 3 & 3 & 0 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{111} 4 \ 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 11, \ \Delta = -33, \ \det = 50, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 2 & 3 & 0 & 1 & 6 & 3 \\ 6 & 6 & 1 & 6 & 14 & 8 \\ 3 & 3 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{111} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 11, \ \Delta = -33, \ \det = 50, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 9 & 3 \\ 2 & 1 & 1 & 3 & 8 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 3 & 3 & 1 & 2 & 9 & 4 \\ 9 & 8 & 1 & 9 & 27 & 11 \\ 3 & 3 & 0 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{112} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 11, \ \Delta = -33, \ \det = 50, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 10 & 4 \\ 3 & 1 & 1 & 7 & 3 \\ 2 & 1 & 0 & 4 & 2 \\ 10 & 7 & 4 & 21 & 11 \\ 4 & 3 & 2 & 11 & 5 \end{pmatrix}_{b_0}$$

$$[76] \ 1^{111} 4 \ 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 10, \ T_{\max} = 11, \ \Delta = -33, \ \det = 72, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 6 & 3 \\ 2 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -2 & 1 & 1 & 0 \\ 3 & 3 & 1 & 2 & 7 & 4 \\ 6 & 6 & 1 & 7 & 14 & 8 \\ 3 & 3 & 0 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [77] \ 1^{11} 4 \ 13221^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 11, \ \Delta = -5, \ \det = 0, \ (n_+, n_-, n_0) = (1, 11, 1), \ c_{\text{ext}} = 7.0$$

$$[78] \ 1^{11} 4 \ 1321^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 11, \ \Delta = -5, \ \det = -4, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 6 & 5 & 4 \\ 3 & 2 & 5 & 5 & 4 \\ 6 & 5 & 10 & 9 & 8 \\ 5 & 5 & 9 & 7 & 7 \\ 4 & 4 & 8 & 7 & 6 \end{pmatrix}_{b_0}$$

$$[79] \ 1^{11} 5 \ 1^{11} 231^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -11, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 5 & 0 & 5 & 15 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 11 & 4 \\ 8 & 15 & 0 & 11 & 37 & 13 \\ 3 & 5 & -1 & 4 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[80] \ 1^{11} 5 \ 1^{11} 231^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -12, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 1 & 6 & 2 \\ 1 & 2 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 3 & 0 & 1 & 8 & 3 \\ 5 & 6 & 8 & 0 & 8 & 31 & 11 \\ 2 & 2 & 2 & -1 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{\overline{1}} \overline{5} \overline{1}^{\overline{1}} 231^{\overline{112}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -47, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.89 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 5 & 0 & 5 & 20 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 15 & 4 \\ 10 & 20 & 0 & 15 & 63 & 17 \\ 3 & 5 & -1 & 4 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[82] \ 2^{\overline{111}} \overline{3} \overline{15} \overline{1}^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -63, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 3 & 3 & 1 & 4 & 6 & 3 \\ 3 & 2 & 4 & 2 & 6 & 4 \\ 5 & 5 & 6 & 6 & 9 & 7 \\ 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[83] \ 2^{\overline{111}} \overline{3} \overline{15} \overline{1}^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -88, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 & 3 \\ 4 & 5 & 3 & 5 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 3 & 5 & 2 & 2 & 4 & 4 \\ 3 & 5 & 2 & 4 & 2 & 4 \\ 3 & 5 & 2 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{\overline{1}} \overline{5} \overline{1}^{\overline{2}} 231^{\overline{113}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -96, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 18 & 3 \\ 3 & 5 & 0 & 5 & 35 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 26 & 4 \\ 18 & 35 & 0 & 26 & 202 & 30 \\ 3 & 5 & -1 & 4 & 30 & 4 \end{pmatrix}_{b_0}$$

$$[85] \ 2^{\overline{112}} \overline{3} \overline{15} \overline{1}^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -106, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.89 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 6 & 3 \\ 4 & 5 & 3 & 5 & 10 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 3 & 5 & 2 & 2 & 8 & 4 \\ 6 & 10 & 4 & 8 & 14 & 8 \\ 3 & 5 & 2 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[86] \ 2^{\overline{122}} \overline{3} \overline{15} \overline{1}^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -124, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 6 & 3 \\ 4 & 5 & 3 & 10 & 10 & 5 \\ 2 & 3 & 0 & 4 & 4 & 2 \\ 6 & 10 & 4 & 14 & 16 & 8 \\ 6 & 10 & 4 & 16 & 14 & 8 \\ 3 & 5 & 2 & 8 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[87] \ 2^{\overline{113}} \overline{3} \overline{15} \overline{1}^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -136, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 9 & 3 \\ 4 & 5 & 3 & 5 & 15 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 3 & 5 & 2 & 2 & 12 & 4 \\ 9 & 15 & 6 & 12 & 34 & 12 \\ 3 & 5 & 2 & 4 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[88] \ 1^{\overline{1}} \overline{5} \overline{1}^{\overline{3}} 231^{\overline{114}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -143, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 26 & 3 \\ 3 & 5 & 0 & 5 & 50 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 37 & 4 \\ 26 & 50 & 0 & 37 & 419 & 43 \\ 3 & 5 & -1 & 4 & 43 & 4 \end{pmatrix}_{b_0}$$

$$[89] \ 1^{\overline{1}} \overline{5} \overline{1}^{\overline{2}} 231^{\overline{114}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -29, \ \det = -152, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 20 & 3 \\ 3 & 5 & 0 & 5 & 40 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 30 & 4 \\ 20 & 40 & 0 & 30 & 258 & 34 \\ 3 & 5 & -1 & 4 & 34 & 4 \end{pmatrix}_{b_0}$$

- [90] $2 \begin{smallmatrix} 1^{123} & 1^1 \\ 2 & 3 & 15 & 1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -154, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 6 & 9 & 3 \\ 4 & 5 & 3 & 10 & 15 & 5 \\ 2 & 3 & 0 & 4 & 6 & 2 \\ 6 & 10 & 4 & 14 & 24 & 8 \\ 9 & 15 & 6 & 24 & 34 & 12 \\ 3 & 5 & 2 & 8 & 12 & 4 \end{pmatrix}_{b_0}$
- [91] $2 \begin{smallmatrix} 1^{114} & 1^1 \\ 2 & 3 & 15 & 1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -178, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 3 & 12 & 3 \\ 4 & 5 & 3 & 5 & 20 & 5 \\ 2 & 3 & 0 & 2 & 8 & 2 \\ 3 & 5 & 2 & 2 & 16 & 4 \\ 12 & 20 & 8 & 16 & 62 & 16 \\ 3 & 5 & 2 & 4 & 16 & 4 \end{pmatrix}_{b_0}$
- [92] $2 \begin{smallmatrix} 1^{1111} & 1^1 \\ 2 & 3 & 15 & 1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -184, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 5 & 3 & 3 & 3 \\ 4 & 5 & 8 & 5 & 5 & 5 \\ 5 & 8 & 8 & 6 & 6 & 6 \\ 3 & 5 & 6 & 2 & 4 & 4 \\ 3 & 5 & 6 & 4 & 2 & 4 \\ 3 & 5 & 6 & 4 & 4 & 4 \end{pmatrix}_{b_0}$
- [93] $2 \begin{smallmatrix} 1^{133} & 1^1 \\ 2 & 3 & 15 & 1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -184, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 9 & 9 & 3 \\ 4 & 5 & 3 & 15 & 15 & 5 \\ 2 & 3 & 0 & 6 & 6 & 2 \\ 9 & 15 & 6 & 34 & 36 & 12 \\ 9 & 15 & 6 & 36 & 34 & 12 \\ 3 & 5 & 2 & 12 & 12 & 4 \end{pmatrix}_{b_0}$
- [94] $2 \begin{smallmatrix} 1^{124} & 1^1 \\ 2 & 3 & 15 & 1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -196, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 6 & 12 & 3 \\ 4 & 5 & 3 & 10 & 20 & 5 \\ 2 & 3 & 0 & 4 & 8 & 2 \\ 6 & 10 & 4 & 14 & 32 & 8 \\ 12 & 20 & 8 & 32 & 62 & 16 \\ 3 & 5 & 2 & 8 & 16 & 4 \end{pmatrix}_{b_0}$
- [95] $2 \begin{smallmatrix} 1^{134} & 1^1 \\ 2 & 3 & 15 & 1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -226, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 9 & 12 & 3 \\ 4 & 5 & 3 & 15 & 20 & 5 \\ 2 & 3 & 0 & 6 & 8 & 2 \\ 9 & 15 & 6 & 34 & 48 & 12 \\ 12 & 20 & 8 & 48 & 62 & 16 \\ 3 & 5 & 2 & 12 & 16 & 4 \end{pmatrix}_{b_0}$
- [96] $2 \begin{smallmatrix} 1^{115} & 1^1 \\ 2 & 3 & 15 & 1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -232, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.1$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 3 & 15 & 3 \\ 4 & 5 & 3 & 5 & 25 & 5 \\ 2 & 3 & 0 & 2 & 10 & 2 \\ 3 & 5 & 2 & 2 & 20 & 4 \\ 15 & 25 & 10 & 20 & 98 & 20 \\ 3 & 5 & 2 & 4 & 20 & 4 \end{pmatrix}_{b_0}$
- [97] $2 \begin{smallmatrix} 1^{125} & 1^1 \\ 2 & 3 & 15 & 1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -250, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 2 & 6 & 15 & 3 \\ 4 & 5 & 3 & 10 & 25 & 5 \\ 2 & 3 & 0 & 4 & 10 & 2 \\ 6 & 10 & 4 & 14 & 40 & 8 \\ 15 & 25 & 10 & 40 & 98 & 20 \\ 3 & 5 & 2 & 8 & 20 & 4 \end{pmatrix}_{b_0}$
- [98] $2 \begin{smallmatrix} 1^{144} & 1^1 \\ 2 & 3 & 15 & 1 \end{smallmatrix} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -268, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 12 & 3 \\ 4 & 5 & 3 & 20 & 20 & 5 \\ 2 & 3 & 0 & 8 & 8 & 2 \\ 12 & 20 & 8 & 62 & 64 & 16 \\ 12 & 20 & 8 & 64 & 62 & 16 \\ 3 & 5 & 2 & 16 & 16 & 4 \end{pmatrix}_{b_0}$$

[99] $2 \overset{1^{135}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -280, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.9$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 15 & 3 \\ 4 & 5 & 3 & 15 & 25 & 5 \\ 2 & 3 & 0 & 6 & 10 & 2 \\ 9 & 15 & 6 & 34 & 60 & 12 \\ 15 & 25 & 10 & 60 & 98 & 20 \\ 3 & 5 & 2 & 12 & 20 & 4 \end{pmatrix}_{b_0}$$

[100] $2 \overset{1^{116}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -298, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 18 & 3 \\ 4 & 5 & 3 & 5 & 30 & 5 \\ 2 & 3 & 0 & 2 & 12 & 2 \\ 3 & 5 & 2 & 2 & 24 & 4 \\ 18 & 30 & 12 & 24 & 142 & 24 \\ 3 & 5 & 2 & 4 & 24 & 4 \end{pmatrix}_{b_0}$$

[101] $2 \overset{1^{126}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -316, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.8$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 18 & 3 \\ 4 & 5 & 3 & 10 & 30 & 5 \\ 2 & 3 & 0 & 4 & 12 & 2 \\ 6 & 10 & 4 & 14 & 48 & 8 \\ 18 & 30 & 12 & 48 & 142 & 24 \\ 3 & 5 & 2 & 8 & 24 & 4 \end{pmatrix}_{b_0}$$

[102] $2 \overset{1^{145}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -322, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.1$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 12 & 15 & 3 \\ 4 & 5 & 3 & 20 & 25 & 5 \\ 2 & 3 & 0 & 8 & 10 & 2 \\ 12 & 20 & 8 & 62 & 80 & 16 \\ 15 & 25 & 10 & 80 & 98 & 20 \\ 3 & 5 & 2 & 16 & 20 & 4 \end{pmatrix}_{b_0}$$

[103] $2 \overset{1^{136}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -346, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.1$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 18 & 3 \\ 4 & 5 & 3 & 15 & 30 & 5 \\ 2 & 3 & 0 & 6 & 12 & 2 \\ 9 & 15 & 6 & 34 & 72 & 12 \\ 18 & 30 & 12 & 72 & 142 & 24 \\ 3 & 5 & 2 & 12 & 24 & 4 \end{pmatrix}_{b_0}$$

[104] $2 \overset{1^{117}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -376, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.3$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 21 & 3 \\ 4 & 5 & 3 & 5 & 35 & 5 \\ 2 & 3 & 0 & 2 & 14 & 2 \\ 3 & 5 & 2 & 2 & 28 & 4 \\ 21 & 35 & 14 & 28 & 194 & 28 \\ 3 & 5 & 2 & 4 & 28 & 4 \end{pmatrix}_{b_0}$$

[105] $2 \overset{1^{127}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -394, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.8$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 21 & 3 \\ 4 & 5 & 3 & 10 & 35 & 5 \\ 2 & 3 & 0 & 4 & 14 & 2 \\ 6 & 10 & 4 & 14 & 56 & 8 \\ 21 & 35 & 14 & 56 & 194 & 28 \\ 3 & 5 & 2 & 8 & 28 & 4 \end{pmatrix}_{b_0}$$

[106] $2 \overset{1^{137}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -424, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.1$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 21 & 3 \\ 4 & 5 & 3 & 15 & 35 & 5 \\ 2 & 3 & 0 & 6 & 14 & 2 \\ 9 & 15 & 6 & 34 & 84 & 12 \\ 21 & 35 & 14 & 84 & 194 & 28 \\ 3 & 5 & 2 & 12 & 28 & 4 \end{pmatrix}_{b_0}$$

[107] $22 \overset{1^{111}}{3} \overset{1^1}{15} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -568, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 5 & 5 & 5 & 3 \\ 6 & 14 & 13 & 13 & 13 & 8 \\ 5 & 13 & 8 & 11 & 11 & 6 \\ 5 & 13 & 11 & 9 & 11 & 7 \\ 5 & 13 & 11 & 11 & 9 & 7 \\ 3 & 8 & 6 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [108] \quad 1^{11} 4^{11} 141321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.0$$

$$[109] \quad 1^{11} 5^{11} 1321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 2 & 5 & 7 & 4 \\ 3 & 2 & 5 & 3 & 7 & 5 \\ 5 & 5 & 7 & 7 & 10 & 8 \\ 3 & 3 & 4 & 5 & 8 & 5 \end{pmatrix}_{b_0}$$

$$[110] \quad 1^{11} 5^{11} 1321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 4 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 2 & 5 & 7 & 4 \\ 3 & 2 & 5 & 3 & 7 & 5 \\ 4 & 5 & 7 & 7 & 10 & 8 \\ 3 & 3 & 4 & 5 & 8 & 5 \end{pmatrix}_{b_0}$$

$$[111] \quad 1^{11} 4^{11} 14131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -3, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 2 & 2 & 3 & 3 \\ 1 & 1 & -2 & 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 0 & 3 & 2 & 2 \\ 2 & 2 & 1 & 3 & 2 & 3 & 4 \\ 2 & 3 & 1 & 2 & 3 & 2 & 4 \\ 2 & 3 & 0 & 2 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[112] \quad 1^{11} 4^{11} 141^{11} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -3, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 2 & 3 & 4 & 3 \\ 1 & 1 & -2 & 0 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 & 3 & 2 \\ 2 & 3 & 1 & 2 & 2 & 4 & 4 \\ 3 & 4 & 0 & 3 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 4 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[115] \quad 1^{11} 5^{11} 1321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 & 3 \\ 3 & 6 & 3 & 6 & 6 & 6 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 3 & 6 & 2 & 3 & 5 & 5 \\ 3 & 6 & 2 & 5 & 3 & 5 \\ 3 & 6 & 2 & 5 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[114] \quad 1^{11} 5^{11} 1321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 & 3 \\ 3 & 6 & 3 & 18 & 6 & 6 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 9 & 18 & 6 & 43 & 15 & 15 \\ 3 & 6 & 2 & 15 & 3 & 5 \\ 3 & 6 & 2 & 15 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[113] \quad 1^{11} 5^{11} 1321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 & 3 \\ 3 & 6 & 3 & 12 & 6 & 6 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 6 & 12 & 4 & 18 & 10 & 10 \\ 3 & 6 & 2 & 10 & 3 & 5 \\ 3 & 6 & 2 & 10 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[116] \quad 1^{11} 5^{11} 1321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 3 \\ 2 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 2 & 0 & 7 & 4 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 5 & 7 & 0 & 10 & 8 \\ 3 & 3 & 4 & 0 & 8 & 5 \end{pmatrix}_{b_0}$$

[117] $1^{11} \overset{1^{12}}{5} 1321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 15$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.87 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 3 \\ 2 & 1 & 3 & 0 & 7 & 3 \\ 2 & 3 & 2 & 0 & 12 & 4 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 7 & 12 & 0 & 29 & 13 \\ 3 & 3 & 4 & 0 & 13 & 5 \end{pmatrix}_{b_0}$

[118] $1^{11} \overset{1^{12}}{5} 1321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 22$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 2 & 5 & 12 & 4 \\ 3 & 2 & 5 & 3 & 12 & 5 \\ 8 & 7 & 12 & 12 & 29 & 13 \\ 3 & 3 & 4 & 5 & 13 & 5 \end{pmatrix}_{b_0}$

[119] $1^{11} \overset{1^{13}}{5} 1321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 22$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 2 & 5 & 12 & 4 \\ 3 & 2 & 5 & 3 & 12 & 5 \\ 9 & 7 & 12 & 12 & 29 & 13 \\ 3 & 3 & 4 & 5 & 13 & 5 \end{pmatrix}_{b_0}$

[121] $1^{111} \overset{1^{11}}{4} 141^{131} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -3$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 2 & 3 & 6 & 3 \\ 1 & 1 & -2 & 0 & 1 & 1 & 0 \\ 1 & 2 & 0 & 0 & 2 & 6 & 2 \\ 2 & 3 & 1 & 2 & 2 & 7 & 4 \\ 5 & 6 & 1 & 6 & 7 & 15 & 9 \\ 2 & 3 & 0 & 2 & 4 & 9 & 4 \end{pmatrix}_{b_0}$

[120] $1^{11} \overset{1^{13}}{5} 1321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 25$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 9 & 3 \\ 2 & 1 & 3 & 0 & 7 & 3 \\ 2 & 3 & 2 & 0 & 12 & 4 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 7 & 12 & 0 & 29 & 13 \\ 3 & 3 & 4 & 0 & 13 & 5 \end{pmatrix}_{b_0}$

[123] $1^{12} \overset{1^{12}}{5} 1321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 42$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.5$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 3 \\ 2 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 2 & 0 & 14 & 4 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 10 & 14 & 0 & 46 & 16 \\ 3 & 3 & 4 & 0 & 16 & 5 \end{pmatrix}_{b_0}$

[122] $1^{12} \overset{1^{12}}{5} 1321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 42$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.5$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 2 & 12 & 3 \\ 2 & 3 & 2 & 5 & 19 & 4 \\ 3 & 2 & 5 & 3 & 19 & 5 \\ 12 & 12 & 19 & 19 & 79 & 21 \\ 3 & 3 & 4 & 5 & 21 & 5 \end{pmatrix}_{b_0}$

[128] $1^{12} \overset{1^{13}}{5} 1321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 48$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.5$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 13 & 3 \\ 2 & 1 & 3 & 2 & 12 & 3 \\ 2 & 3 & 2 & 5 & 19 & 4 \\ 3 & 2 & 5 & 3 & 19 & 5 \\ 13 & 12 & 19 & 19 & 79 & 21 \\ 3 & 3 & 4 & 5 & 21 & 5 \end{pmatrix}_{b_0}$

[126] $1^1 \overset{1^{21}}{5} 1321^{1121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 48$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.5$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 6 & 3 & 3 \\ 3 & 6 & 9 & 12 & 6 & 6 \\ 4 & 9 & 9 & 14 & 7 & 7 \\ 6 & 12 & 14 & 18 & 10 & 10 \\ 3 & 6 & 7 & 10 & 3 & 5 \\ 3 & 6 & 7 & 10 & 5 & 5 \end{pmatrix}_{b_0}$

[125] $1^1 \overset{1^{11}}{5} 1321^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 48$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 & 3 \\ 3 & 6 & 9 & 6 & 6 & 6 \\ 4 & 9 & 9 & 7 & 7 & 7 \\ 3 & 6 & 7 & 3 & 5 & 5 \\ 3 & 6 & 7 & 5 & 3 & 5 \\ 3 & 6 & 7 & 5 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[124] \ 1^{1^{13}} 5^{12} 1321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 48, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 13 & 3 \\ 2 & 1 & 3 & 0 & 12 & 3 \\ 2 & 3 & 2 & 0 & 19 & 4 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 12 & 19 & 0 & 79 & 21 \\ 3 & 3 & 4 & 0 & 21 & 5 \end{pmatrix}_{b_0}$$

$$[127] \ 1^{1^{31}} 5^{13} 1321^{1131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 48, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 & 3 \\ 3 & 6 & 9 & 18 & 6 & 6 \\ 4 & 9 & 9 & 21 & 7 & 7 \\ 9 & 18 & 21 & 43 & 15 & 15 \\ 3 & 6 & 7 & 15 & 3 & 5 \\ 3 & 6 & 7 & 15 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[129] \ 1^{1^{13}} 5^{13} 1321^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 70, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 17 & 3 \\ 2 & 1 & 3 & 2 & 17 & 3 \\ 2 & 3 & 2 & 5 & 26 & 4 \\ 3 & 2 & 5 & 3 & 26 & 5 \\ 17 & 17 & 26 & 26 & 153 & 29 \\ 3 & 3 & 4 & 5 & 29 & 5 \end{pmatrix}_{b_0}$$

$$[130] \ 1^{1^{13}} 5^{13} 1321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 87, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 15 & 3 \\ 2 & 1 & 3 & 0 & 15 & 3 \\ 2 & 3 & 2 & 0 & 21 & 4 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 15 & 21 & 0 & 106 & 24 \\ 3 & 3 & 4 & 0 & 24 & 5 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [133] \ 1^{1^{11}} 4^{11} 1414131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, \ T_{\max} = 13, \ \Delta = -2, \ \det = 0, \ (n_+, n_-, n_0) = (1, 13, 1), \ c_{\text{ext}} = 7.0$$

$$[132] \ 1^{1^{11}} 4^{11} 14141^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, \ T_{\max} = 13, \ \Delta = -2, \ \det = 0, \ (n_+, n_-, n_0) = (1, 13, 1), \ c_{\text{ext}} = 7.0$$

$$[131] \ 1^{1^{11}} 4^{11} 14141^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, \ T_{\max} = 13, \ \Delta = -2, \ \det = 0, \ (n_+, n_-, n_0) = (1, 13, 1), \ c_{\text{ext}} = 7.0$$

$$[136] \ 1^{1^{11}} 6^{11} 131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, \ T_{\max} = 13, \ \Delta = -4, \ \det = -4, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 5 & 3 \\ 2 & 2 & 3 & 4 & 6 & 4 \\ 2 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 2 & 2 & 6 & 4 \\ 5 & 6 & 6 & 6 & 10 & 8 \\ 3 & 4 & 3 & 4 & 8 & 5 \end{pmatrix}_{b_0}$$

$$[135] \ 1^{1^{11}} 6^{11} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, \ T_{\max} = 13, \ \Delta = -4, \ \det = -4, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 4 & 0 & 2 \\ 1 & 1 & 1 & 0 & 4 & 0 & 2 \\ 3 & 2 & 4 & 4 & 6 & 0 & 6 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & 2 & 6 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[134] \ 1^{1^{11}} 6^{11} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, \ T_{\max} = 13, \ \Delta = -4, \ \det = -4, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 3 \\ 2 & 2 & 3 & 0 & 4 & 4 \\ 2 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 2 & 0 & 2 & 4 \\ 3 & 4 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[138] \ 1^{1^{11}} 6^{11} 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 13, \ T_{\max} = 13, \ \Delta = -4, \ \det = -16, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 3 \\ 2 & 2 & 3 & 0 & 6 & 4 \\ 2 & 3 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 6 & 6 & 0 & 10 & 8 \\ 3 & 4 & 3 & 0 & 8 & 5 \end{pmatrix}_{b_0}$$

$$[137] \ 1^{\underline{1}} \overset{11}{6} \ 131^{\underline{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = -4, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 7 & 3 \\ 2 & 2 & 3 & 0 & 10 & 4 \\ 2 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 10 & 8 & 0 & 26 & 12 \\ 3 & 4 & 3 & 0 & 12 & 5 \end{pmatrix}_{b_0}$$

$$[139] \ 1^{\underline{1}} \overset{11}{6} \ 131^{\underline{112}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = -4, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 2 & 3 & 4 & 10 & 4 \\ 2 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 2 & 2 & 10 & 4 \\ 7 & 10 & 8 & 10 & 26 & 12 \\ 3 & 4 & 3 & 4 & 12 & 5 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{\underline{1}} \overset{11}{6} \ 131^{\underline{112}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, T_{\max} = 13, \Delta = -4, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 7 & 3 \\ 2 & 2 & 3 & 6 & 10 & 4 \\ 2 & 3 & 1 & 6 & 8 & 3 \\ 5 & 6 & 6 & 10 & 18 & 8 \\ 7 & 10 & 8 & 18 & 26 & 12 \\ 3 & 4 & 3 & 8 & 12 & 5 \end{pmatrix}_{b_0}$$

$$T = 16 \quad [141] \ 1^{\underline{1}} \overset{11}{6} \ 131^{\underline{112}} 6 \ 131^{\underline{1}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 7.0$$

$$[144] \ 1^{\underline{11}} \overset{12}{6} \ 1316 \ 131^{\underline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 7.0$$

$$[143] \ 1^{\underline{1}} \overset{11}{6} \ 1316 \ 131^{\underline{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 7.5$$

$$[142] \ 1^{\underline{1}} \overset{11}{6} \ 1316 \ 131^{\underline{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 16, T_{\max} = 16, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 7.5$$

$$T = 17 \quad [145] \ 1^{\underline{1}} \overset{112}{6} \ 1316131^{\underline{1}} 41^{\underline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 17, T_{\max} = 17, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 7.0$$

$$T = 19 \quad [146] \ 1^{\underline{1}} \overset{112}{6} \ 13161316 \ 131^{\underline{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 19, T_{\max} = 19, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 19, 2), c_{\text{ext}} = 7.5$$

$$T = 20 \quad [147] \ 1^{\underline{1}} \overset{112}{6} \ 13161316131^{\underline{1}} 41^{\underline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 20, T_{\max} = 20, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 20, 2), c_{\text{ext}} = 7.0$$

4.1163 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\underline{111}} \overset{1}{3} 1^{\underline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\underline{111}} \overset{1}{3} 1^{\underline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 3 & 3 \\ 3 & 5 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3 \ 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 3 & 1 & 3 \\ 3 & 4 & -1 & 0 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3 \ 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 5 & 0 & 4 & 4 \\ 3 & 4 & 0 & 3 & 0 & 2 & 2 \\ 3 & 5 & 3 & 4 & 6 & 0 & 4 \\ 4 & 5 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 4 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3 \ 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 5 & 3 & 3 \\ 3 & 4 & 0 & 5 & 9 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 0 & 9 & 13 & 6 & 7 \\ 3 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & -1 & 4 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}} 3 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & 4 & 3 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{121}} 3 \ 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 8 & 3 & 3 \\ 3 & 4 & 0 & 5 & 13 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & 4 & 12 & 3 & 4 \\ 8 & 13 & 0 & 12 & 28 & 9 & 10 \\ 3 & 4 & 0 & 3 & 9 & 1 & 3 \\ 3 & 4 & -1 & 4 & 10 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 4 & 3 \\ 3 & 4 & 4 & 0 & 0 & 5 & 4 \\ 3 & 4 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 5 & 3 & 0 & 0 & 4 & 4 \\ 3 & 4 & 2 & 0 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{11}} 3 \ 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 4 & 1 & 3 \\ 3 & 4 & 4 & 5 & 5 & 0 & 4 \\ 3 & 4 & 0 & 3 & 3 & 0 & 2 \\ 4 & 5 & 3 & 4 & 6 & 0 & 4 \\ 4 & 5 & 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 4 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11\bar{1}\bar{2}} 1^{11\bar{1}} 3 \quad 1^{11\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 5 & 6 & 3 \\ 3 & 4 & 0 & 9 & 8 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 5 & 9 & 0 & 13 & 12 & 15 & 7 \\ 5 & 8 & 0 & 12 & 10 & 12 & 6 \\ 6 & 9 & 0 & 15 & 12 & 13 & 7 \\ 3 & 4 & -1 & 7 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11\bar{1}} 1^{11\bar{1}} 3 \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -21, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 0 & 6 & 13 & 7 \\ 3 & 4 & -1 & 0 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11\bar{1}} 1^{11\bar{1}} 3 \quad 1^{11\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -32, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 3 & 4 & 0 & 3 & 1 & 6 & 3 \\ 6 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & -1 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{12\bar{1}\bar{1}} 1^{11\bar{1}} 3 \quad 1^{11\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 6 & 6 & 3 \\ 3 & 4 & 0 & 8 & 9 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 5 & 8 & 0 & 10 & 12 & 12 & 6 \\ 6 & 9 & 0 & 12 & 13 & 15 & 7 \\ 6 & 9 & 0 & 12 & 15 & 13 & 7 \\ 3 & 4 & -1 & 6 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{12\bar{1}} 1^{11\bar{1}} 3 \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 6 & 3 \\ 3 & 4 & 0 & 0 & 8 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 5 & 8 & 0 & 0 & 10 & 12 & 6 \\ 6 & 9 & 0 & 0 & 12 & 13 & 7 \\ 3 & 4 & -1 & 0 & 6 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11} 1^{11\bar{1}} 3 \quad 1^{11\bar{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -45, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 0 & 5 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 5 & 0 & 0 & 4 & 9 & 4 \\ 6 & 9 & 0 & 0 & 9 & 13 & 7 \\ 3 & 4 & -1 & 0 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11\bar{2}} 1^{11\bar{1}} 3 \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -47, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 0 & 4 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 1 & 9 & 3 \\ 8 & 13 & 0 & 0 & 9 & 28 & 10 \\ 3 & 4 & -1 & 0 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11\bar{1}\bar{1}} 1^{11\bar{1}} 3 \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -47, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & -1 & 3 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{12} 1^{11} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -50, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 8 & 3 \\ 3 & 4 & 0 & 0 & 5 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 5 & 0 & 0 & 4 & 12 & 4 \\ 8 & 13 & 0 & 0 & 12 & 28 & 10 \\ 3 & 4 & -1 & 0 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1112} 1^{12} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -50, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 9 & 3 \\ 3 & 4 & 0 & 4 & 9 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 6 & 9 & 3 \\ 5 & 9 & 0 & 6 & 13 & 21 & 7 \\ 9 & 13 & 0 & 9 & 21 & 28 & 10 \\ 3 & 4 & -1 & 3 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{112} 1^{11} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -55, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 9.98 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 12 & 4 \\ 3 & 4 & 0 & 3 & 1 & 9 & 3 \\ 8 & 13 & 0 & 12 & 9 & 28 & 10 \\ 3 & 4 & -1 & 4 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1111} 1^{11} 3 \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -56, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & -1 & 3 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1111} 1^{11} 3 \quad 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -76, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1111} 1^{11} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -77, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 9 & 4 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 \\ 6 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & -1 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1111} 1^{11} 3 \quad 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -87, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 1 & 6 & 3 \\ 6 & 9 & 0 & 0 & 6 & 13 & 7 \\ 3 & 4 & -1 & 0 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1111} 1^{11} 3 \quad 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -33, \quad \det = -87, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 5 & 0 & 5 & 5 & 10 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 4 & 7 & 4 \\ 3 & 5 & 0 & 4 & 2 & 7 & 4 \\ 5 & 10 & 0 & 7 & 7 & 14 & 8 \\ 3 & 5 & -1 & 4 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [44] \quad 1^{11} 1^{1111} 31^1 4^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 5 & 5 & 4 & 4 \\ 3 & 1 & 4 & 3 & 4 & 3 \\ 5 & 4 & 6 & 5 & 5 & 6 \\ 5 & 3 & 5 & 3 & 4 & 5 \\ 4 & 4 & 5 & 4 & 3 & 5 \\ 4 & 3 & 6 & 5 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{11} 1^{1111} 31^1 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 4 & 5 & 4 \\ 3 & 1 & 1 & 4 & 3 & 3 \\ 2 & 1 & 0 & 2 & 2 & 2 \\ 4 & 4 & 2 & 3 & 4 & 5 \\ 5 & 3 & 2 & 4 & 3 & 5 \\ 4 & 3 & 2 & 5 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{11} 1^{11} 4^1 131^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 5 & 4 & 4 \\ 3 & 1 & 1 & 4 & 4 & 3 \\ 2 & 1 & 0 & 2 & 2 & 2 \\ 5 & 4 & 2 & 6 & 5 & 6 \\ 4 & 4 & 2 & 5 & 3 & 5 \\ 4 & 3 & 2 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

4.1164 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{1111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 6 & 3 \\ 3 & 4 & 0 & 0 & 4 & 5 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 3 & 6 & 3 \\ 3 & 5 & 0 & 0 & 3 & 4 & 9 & 4 \\ 6 & 9 & 0 & 0 & 6 & 9 & 13 & 7 \\ 3 & 4 & -1 & 0 & 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1112} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 8 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 12 & 4 \\ 3 & 4 & 0 & 3 & 1 & 3 & 9 & 3 \\ 2 & 4 & 0 & 3 & 3 & 1 & 9 & 3 \\ 8 & 13 & 0 & 12 & 9 & 9 & 28 & 10 \\ 3 & 4 & -1 & 4 & 3 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11111} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 6 & 6 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 6 & 3 \\ 6 & 9 & 0 & 6 & 6 & 13 & 15 & 7 \\ 6 & 9 & 0 & 6 & 6 & 15 & 13 & 7 \\ 3 & 4 & -1 & 3 & 3 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -3, \det = 92, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 6 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 6 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 6 & 3 & 3 \\ 6 & 9 & 0 & 6 & 6 & 13 & 6 & 7 \\ 3 & 4 & 0 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11111} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -3, \ \det = 100, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 6 & 3 & 6 & 3 \\ 5 & 9 & 0 & 6 & 13 & 6 & 15 & 7 \\ 2 & 4 & 0 & 3 & 6 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 15 & 6 & 13 & 7 \\ 3 & 4 & -1 & 3 & 7 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11211} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -3, \ \det = 100, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 8 & 6 & 3 \\ 3 & 4 & 0 & 0 & 4 & 13 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 & 0 \\ 3 & 4 & 0 & 0 & 1 & 9 & 6 & 3 \\ 8 & 13 & 0 & 0 & 9 & 28 & 21 & 10 \\ 6 & 9 & 0 & 0 & 6 & 21 & 13 & 7 \\ 3 & 4 & -1 & 0 & 3 & 10 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11111} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -3, \ \det = 144, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & -1 & 3 & 3 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11111} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -3, \ \det = 224, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 9.98 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 0 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 3 & 3 & 1 & 6 & 3 \\ 6 & 9 & 0 & 6 & 6 & 13 & 7 & 7 \\ 3 & 4 & -1 & 3 & 3 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11111} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -3, \ \det = 364, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 6 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 0 & 3 & 1 & 6 & 6 & 3 \\ 6 & 9 & 0 & 6 & 6 & 13 & 15 & 7 \\ 6 & 9 & 0 & 6 & 6 & 15 & 13 & 7 \\ 3 & 4 & -1 & 3 & 3 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11111} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -3, \ \det = 400, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 6 & 6 & 3 \\ 3 & 4 & 0 & 4 & 4 & 9 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 0 & 3 & 1 & 6 & 6 & 3 \\ 6 & 9 & 0 & 6 & 6 & 13 & 15 & 7 \\ 6 & 9 & 0 & 6 & 6 & 15 & 13 & 7 \\ 3 & 4 & -1 & 3 & 3 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [10] \ 1^{11} 4 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 11, \ \Delta = -2, \ \det = 0, \ (n_+, n_-, n_0) = (1, 11, 1), \ c_{\text{ext}} = 9.0$$

4.1165 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 10, \ \Delta = -5, \ \det = 9, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$T = 11 \quad [7] \quad 1^{11} 4 13 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 11, \quad \Delta = -5, \quad \det = -16, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 5 & 1 & 4 \\ 3 & 1 & 1 & 4 & 0 & 3 \\ 2 & 1 & 0 & 2 & 0 & 2 \\ 5 & 4 & 2 & 6 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 4 & 3 & 2 & 6 & 0 & 5 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [8] \quad 1^{11} 5 1^{11} 23 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 5 & 0 & 5 & 15 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 11 & 3 & 4 \\ 8 & 15 & 0 & 11 & 37 & 10 & 13 \\ 2 & 4 & 0 & 3 & 10 & 1 & 3 \\ 3 & 5 & -1 & 4 & 13 & 3 & 4 \end{pmatrix}_{b_0}$$

4.1167 $\text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2$

$$T = 10 \quad [0] \quad 1^{111} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -5, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 3 & 4 \\ 3 & 4 & 0 & 3 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & -1 & 4 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -5, \quad \det = 9, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 3 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 4 & 0 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 0 & 2 & 0 & 3 & 2 \\ 3 & 5 & 0 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 3 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

4.1168 $\text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 21, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 0 & 3 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 0 & 2 \\ 3 & 4 & -1 & 0 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 36, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 40, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -65, \quad \det = 61, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 1 & 2 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & -1 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \quad 1^{11} 4^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 3 & 1 & 3 \\ 2 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 4^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -2 & 1 & 0 & 0 \\ 3 & 4 & 1 & 2 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 4^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 4^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 1 & 4 \\ 3 & 1 & 1 & 0 & 3 \\ 2 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 4 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 4^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 5 & 1 & 4 \\ 3 & 1 & 4 & 0 & 3 \\ 5 & 4 & 6 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 4 & 3 & 6 & 0 & 5 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [11] \quad 1^{11} 5^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 2 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 4 & 7 & 6 & 9 & 5 & 7 \\ 2 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 5^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 2 & 3 \\ 2 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 7 & 5 & 6 & 9 & 5 & 7 \\ 2 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{12} 5^{13} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 2 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 & 3 \\ 4 & 2 & 4 & 2 & 3 & 4 \\ 2 & 2 & 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[15] \quad & 1^{11} \overset{1^3}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 7 & 2 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \\ 2 & 3 & 1 & 6 & 3 & 2 \\ 7 & 7 & 6 & 9 & 5 & 7 \\ 2 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
[14] \quad & 21^{11} \overset{1^{11}}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 3 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 2 & 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[13] \quad & 1^{11} \overset{1^1}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 4 & 3 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 2 & 3 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[9] \quad & 1^{11} \overset{1^1}{5} 123^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -2, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 4 & 2 & 2 \\ 2 & 3 & 0 & 4 & 5 & 3 & 5 \\ 1 & 1 & 0 & 2 & 3 & 0 & 2 \\ 2 & 2 & -1 & 2 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[12] \quad & 1^{11} \overset{1^1}{5} 1^{12} 23^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -2, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & 1 & 6 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 8 & 2 & 2 \\ 5 & 6 & 0 & 8 & 31 & 7 & 11 \\ 1 & 1 & 0 & 2 & 7 & 0 & 2 \\ 2 & 2 & -1 & 2 & 11 & 2 & 3 \end{pmatrix}_{b_0} \\
[18] \quad & 1^{12} \overset{1^1}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -4, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 9 & 2 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 9 & 14 & 12 & 42 & 10 & 14 \\ 2 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0} \\
[17] \quad & 1^{11} \overset{1^1}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -4, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0} \\
[21] \quad & 1^{11} \overset{1^1}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 4 & 2 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \\ 3 & 3 & 1 & 6 & 3 & 3 \\ 4 & 7 & 6 & 9 & 5 & 7 \\ 2 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
[20] \quad & 1^{12} \overset{1^1}{5} 13^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.5
\end{aligned}$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 2 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \\ 3 & 3 & 1 & 12 & 3 & 3 \\ 8 & 14 & 12 & 42 & 10 & 14 \\ 2 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{13} \overset{1}{5} 13^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 2 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \\ 3 & 3 & 1 & 18 & 3 & 3 \\ 12 & 21 & 18 & 97 & 15 & 21 \\ 2 & 3 & 3 & 15 & 1 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11} \overset{1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 5 & 3 & 0 & 4 & 5 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 5 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{12} \overset{1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 2 & 3 \\ 3 & 5 & 3 & 5 & 4 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 4 & 5 & 2 & 2 & 3 & 4 \\ 2 & 4 & 2 & 3 & 1 & 3 \\ 3 & 5 & 2 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{11} \overset{1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 5 & 7 & 6 & 9 & 5 & 7 \\ 2 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{13} \overset{1}{5} 13^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 2 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 14 & 21 & 18 & 97 & 15 & 21 \\ 2 & 3 & 3 & 15 & 1 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{11} \overset{1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 6 & 7 & 6 & 9 & 5 & 7 \\ 2 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{15} \overset{1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 3 \\ 3 & 5 & 3 & 5 & 4 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 2 & 5 & 2 & 2 & 3 & 4 \\ 2 & 4 & 2 & 3 & 1 & 3 \\ 3 & 5 & 2 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{15} \overset{1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 3 & 2 & 3 \\ 2 & 3 & 1 & 2 & 3 & 3 \\ 2 & 3 & 2 & 1 & 2 & 3 \\ 2 & 2 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[29] $1^{11}5^{11}123^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 1 & 1 & 2 & 0 & 1 & 2 \\ 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[32] $1^{11}5^{11}13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 3 \\ 3 & 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

[31] $1^{11}5^{11}123^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 5 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 3 \\ 3 & 5 & -1 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[27] $1^{13}5^{13}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 2 & 3 \\ 3 & 5 & 3 & 10 & 4 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 7 & 10 & 4 & 14 & 6 & 8 \\ 2 & 4 & 2 & 6 & 1 & 3 \\ 3 & 5 & 2 & 8 & 3 & 4 \end{pmatrix}_{b_0}$$

[34] $1^{12}5^{12}13^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 2 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 10 & 14 & 12 & 42 & 10 & 14 \\ 2 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[33] $1^{12}5^{12}13^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 2 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 12 & 14 & 12 & 42 & 10 & 14 \\ 2 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[39] $1^{11}5^{11}13^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 5 & 8 & 5 & 4 & 5 \\ 5 & 8 & 8 & 6 & 5 & 6 \\ 2 & 5 & 6 & 2 & 3 & 4 \\ 2 & 4 & 5 & 3 & 1 & 3 \\ 3 & 5 & 6 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[38] $1^{14}5^{14}13^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 2 & 3 \\ 3 & 5 & 3 & 15 & 4 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 10 & 15 & 6 & 34 & 9 & 12 \\ 2 & 4 & 2 & 9 & 1 & 3 \\ 3 & 5 & 2 & 12 & 3 & 4 \end{pmatrix}_{b_0}$$

[37] $21 \overset{1}{\underset{1}{5}} 13 \overset{1}{\underset{1}{1}} \overset{11}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 5 & 8 & 5 & 4 & 5 \\ 5 & 8 & 8 & 6 & 5 & 6 \\ 2 & 5 & 6 & 2 & 3 & 4 \\ 2 & 4 & 5 & 3 & 1 & 3 \\ 3 & 5 & 6 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[36] $1 \overset{1}{\underset{1}{5}} 13 \overset{1}{\underset{1}{1}} \overset{12}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 5 & 3 & 10 & 4 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 5 & 10 & 4 & 14 & 6 & 8 \\ 2 & 4 & 2 & 6 & 1 & 3 \\ 3 & 5 & 2 & 8 & 3 & 4 \end{pmatrix}_{b_0}$$

[35] $1 \overset{1}{\underset{1}{5}} 13 \overset{1}{\underset{1}{1}} \overset{11}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 2 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 & 3 \\ 3 & 2 & 4 & 2 & 3 & 4 \\ 2 & 2 & 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[40] $1 \overset{13}{\underset{1}{5}} 13 \overset{1}{\underset{1}{1}} \overset{11}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 2 & 3 \\ 2 & 1 & 3 & 8 & 2 & 3 \\ 2 & 3 & 1 & 8 & 3 & 3 \\ 9 & 8 & 8 & 22 & 7 & 10 \\ 2 & 2 & 3 & 7 & 1 & 3 \\ 3 & 3 & 3 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$

[41] $1 \overset{13}{\underset{1}{5}} 13 \overset{1}{\underset{1}{1}} \overset{12}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 2 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 11 & 14 & 12 & 42 & 10 & 14 \\ 2 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[42] $1 \overset{13}{\underset{1}{5}} 13 \overset{1}{\underset{1}{1}} \overset{13}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -26, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 2 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 15 & 21 & 18 & 97 & 15 & 21 \\ 2 & 3 & 3 & 15 & 1 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

[43] $1 \overset{1}{\underset{1}{5}} 13 \overset{1}{\underset{1}{1}} \overset{11}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 3 \\ 3 & 5 & 3 & 5 & 4 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 3 & 5 & 2 & 2 & 3 & 4 \\ 2 & 4 & 2 & 3 & 1 & 3 \\ 3 & 5 & 2 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[45] $1 \overset{11}{\underset{1}{5}} 13 \overset{1}{\underset{1}{1}} \overset{12}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 2 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 10 & 3 & 3 \\ 10 & 7 & 10 & 25 & 8 & 11 \\ 2 & 2 & 3 & 8 & 1 & 3 \\ 3 & 2 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

[44] $1 \overset{1}{\underset{1}{5}} 13 \overset{1}{\underset{1}{1}} \overset{13}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 3 \\ 3 & 5 & 3 & 15 & 4 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 8 & 15 & 6 & 34 & 9 & 12 \\ 2 & 4 & 2 & 9 & 1 & 3 \\ 3 & 5 & 2 & 12 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{13}_{1^1} 5^{14} 13^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -34, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 2 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 16 & 21 & 18 & 97 & 15 & 21 \\ 2 & 3 & 3 & 15 & 1 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{12}_{1^1} 5^{12} 13^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -34, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 2 & 3 \\ 3 & 5 & 3 & 10 & 4 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 6 & 10 & 4 & 14 & 6 & 8 \\ 2 & 4 & 2 & 6 & 1 & 3 \\ 3 & 5 & 2 & 8 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{13}_{1^1} 5^{13} 13^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -44, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 2 & 3 \\ 3 & 5 & 3 & 15 & 4 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 9 & 15 & 6 & 34 & 9 & 12 \\ 2 & 4 & 2 & 9 & 1 & 3 \\ 3 & 5 & 2 & 12 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{12}_{1^1} 5^{12} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 2 & 3 \\ 2 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 2 \\ 6 & 5 & 6 & 9 & 5 & 7 \\ 2 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[50] \quad 21^{11}_{1^1} 5^{11} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 5 & 5 & 6 & 9 & 5 & 7 \\ 2 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{13}_{1^1} 5^{13} 13^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 2 & 3 \\ 3 & 5 & 3 & 20 & 4 & 5 \\ 2 & 3 & 0 & 8 & 2 & 2 \\ 11 & 20 & 8 & 62 & 12 & 16 \\ 2 & 4 & 2 & 12 & 1 & 3 \\ 3 & 5 & 2 & 16 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{11}_{1^1} 5^{11} 13^1 23^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 0 & 5 & 1 & 2 \\ 1 & 2 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 5 & 5 & 0 & 18 & 5 & 8 \\ 1 & 1 & 2 & 0 & 5 & 0 & 2 \\ 2 & 2 & 2 & -1 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{11}_{1^1} 5^{11} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 2 & 3 \\ 2 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 5 & 5 & 6 & 9 & 5 & 7 \\ 2 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1^4} 5^{12} 13^{11} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -58, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 2 & 3 \\ 3 & 5 & 3 & 20 & 4 & 5 \\ 2 & 3 & 0 & 8 & 2 & 2 \\ 12 & 20 & 8 & 62 & 12 & 16 \\ 2 & 4 & 2 & 12 & 1 & 3 \\ 3 & 5 & 2 & 16 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1^2} 5^{12} 13^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -58, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 2 & 3 \\ 2 & 1 & 3 & 8 & 2 & 3 \\ 2 & 3 & 1 & 8 & 3 & 3 \\ 8 & 8 & 8 & 22 & 7 & 10 \\ 2 & 2 & 3 & 7 & 1 & 3 \\ 3 & 3 & 3 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[56] \quad 21^{1^2} 5^{12} 13^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -67, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 3 \\ 3 & 1 & 3 & 8 & 2 & 3 \\ 2 & 3 & 1 & 8 & 3 & 3 \\ 8 & 8 & 8 & 22 & 7 & 10 \\ 2 & 2 & 3 & 7 & 1 & 3 \\ 3 & 3 & 3 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1^2} 5^{12} 13^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -67, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 2 & 3 \\ 2 & 1 & 3 & 8 & 2 & 3 \\ 2 & 3 & 1 & 8 & 3 & 3 \\ 7 & 8 & 8 & 22 & 7 & 10 \\ 2 & 2 & 3 & 7 & 1 & 3 \\ 3 & 3 & 3 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1^1} 5^{12} 13^{11} 23^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -67, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 \\ 3 & 5 & 0 & 15 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 8 & 15 & 0 & 37 & 10 & 13 \\ 2 & 4 & 0 & 10 & 1 & 3 \\ 3 & 5 & -1 & 13 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{1^3} 5^{12} 13^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -76, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 2 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 10 & 3 & 3 \\ 9 & 7 & 10 & 25 & 8 & 11 \\ 2 & 2 & 3 & 8 & 1 & 3 \\ 3 & 3 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{1^2} 5^{12} 13^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -91, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 2 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 10 & 3 & 3 \\ 8 & 7 & 10 & 25 & 8 & 11 \\ 2 & 2 & 3 & 8 & 1 & 3 \\ 3 & 3 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{1^1} 5^{12} 13^{11} 23^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 2 & 3 \\ 3 & 5 & 0 & 20 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 10 & 20 & 0 & 63 & 13 & 17 \\ 2 & 4 & 0 & 13 & 1 & 3 \\ 3 & 5 & -1 & 17 & 3 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 2 & 3 \\ 3 & 1 & 3 & 15 & 2 & 3 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 15 & 15 & 18 & 97 & 15 & 21 \\ 2 & 2 & 3 & 15 & 1 & 3 \\ 3 & 3 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{12} \overset{14}{5} 13^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -214, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 2 & 3 \\ 2 & 1 & 3 & 12 & 2 & 3 \\ 2 & 3 & 1 & 16 & 3 & 3 \\ 14 & 12 & 16 & 70 & 13 & 18 \\ 2 & 2 & 3 & 13 & 1 & 3 \\ 3 & 3 & 3 & 18 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{12} \overset{13}{5} 13^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -227, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 12.00 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 2 & 3 \\ 2 & 1 & 3 & 12 & 2 & 3 \\ 2 & 3 & 1 & 16 & 3 & 3 \\ 13 & 12 & 16 & 70 & 13 & 18 \\ 2 & 2 & 3 & 13 & 1 & 3 \\ 3 & 3 & 3 & 18 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{13} \overset{14}{5} 13^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -305, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 2 & 3 \\ 2 & 1 & 3 & 15 & 2 & 3 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 16 & 15 & 18 & 97 & 15 & 21 \\ 2 & 2 & 3 & 15 & 1 & 3 \\ 3 & 3 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{13} \overset{13}{5} 13^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -314, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 2 & 3 \\ 2 & 1 & 3 & 15 & 2 & 3 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 15 & 15 & 18 & 97 & 15 & 21 \\ 2 & 2 & 3 & 15 & 1 & 3 \\ 3 & 3 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{13} \overset{14}{5} 13^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -427, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 12.00 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 18 & 2 & 3 \\ 2 & 1 & 3 & 17 & 2 & 3 \\ 2 & 3 & 1 & 22 & 3 & 3 \\ 18 & 17 & 22 & 137 & 18 & 25 \\ 2 & 2 & 3 & 18 & 1 & 3 \\ 3 & 3 & 3 & 25 & 3 & 4 \end{pmatrix}_{b_0}$$

4.1169 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11} \overset{11}{3} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -35, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 0 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 0 & 2 & 1 & 3 \\ 3 & 4 & -1 & 0 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} \overset{11}{3} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -35, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 3 & 4 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & -1 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} \overset{11}{3} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -35, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 3 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -35, \ \det = -36, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & -1 & 4 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -35, \ \det = -64, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & -1 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -35, \ \det = -91, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 3 & 4 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & -1 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \ 1^{11} 4^1 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 11, \ \Delta = -5, \ \det = -16, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 2 & 1 & 5 & 4 \\ 3 & 1 & 1 & 0 & 4 & 3 \\ 2 & 1 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 4 & 2 & 0 & 6 & 6 \\ 4 & 3 & 2 & 0 & 6 & 5 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [8] \ 1^{11} 5^1 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 9, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 3 & 0 & 3 & 7 & 4 \\ 2 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 3 & 0 & 1 & 5 & 3 \\ 5 & 7 & 6 & 0 & 5 & 9 & 7 \\ 3 & 4 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 5^1 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 9, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 3 & 2 & 0 & 3 \\ 2 & 3 & 1 & 2 & 3 & 0 & 3 \\ 2 & 3 & 2 & 1 & 2 & 0 & 3 \\ 2 & 2 & 3 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 5^1 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 9, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 6 & 3 \\ 3 & 4 & 3 & 0 & 3 & 7 & 4 \\ 2 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 3 & 0 & 1 & 5 & 3 \\ 6 & 7 & 6 & 0 & 5 & 9 & 7 \\ 3 & 4 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{12} 5^1 13^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 12, \ (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 3 & 0 & 3 & 14 & 4 \\ 2 & 3 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 3 & 0 & 1 & 10 & 3 \\ 10 & 14 & 12 & 0 & 10 & 42 & 14 \\ 3 & 4 & 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[12] \begin{matrix} 1^{12} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^{11} \\ 1^1 \end{matrix} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 2 & 3 & 3 \\ 3 & 5 & 3 & 5 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 & 2 \\ 4 & 5 & 2 & 2 & 3 & 4 & 4 \\ 2 & 4 & 2 & 3 & 1 & 3 & 3 \\ 3 & 5 & 2 & 4 & 3 & 2 & 4 \\ 3 & 5 & 2 & 4 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[13] \begin{matrix} 1^{11} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^{11} \\ 1^1 \end{matrix} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 5 & 3 & 0 & 4 & 5 & 5 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 2 & 0 & 1 & 3 & 3 \\ 3 & 5 & 2 & 0 & 3 & 2 & 4 \\ 3 & 5 & 2 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[15] \begin{matrix} 1^{11} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^{11} \\ 1^1 \end{matrix} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 16, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 3 & 0 & 2 & 2 & 3 \\ 2 & 3 & 1 & 0 & 3 & 4 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 3 & 3 \\ 3 & 2 & 4 & 0 & 3 & 2 & 4 \\ 3 & 3 & 3 & 0 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[14] \begin{matrix} 1^{11} \\ 1^1 \end{matrix} 5 \begin{matrix} 1^{13} \\ 1^1 \end{matrix} 23^{11} 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 16, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 5 & 0 & 5 & 4 & 15 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 5 & 0 & 2 & 3 & 11 & 4 \\ 2 & 4 & 0 & 3 & 1 & 10 & 3 \\ 8 & 15 & 0 & 11 & 10 & 37 & 13 \\ 3 & 5 & -1 & 4 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[16] \begin{matrix} 1^{13} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^{11} \\ 1^1 \end{matrix} 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 17, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 15 & 3 \\ 3 & 4 & 3 & 0 & 3 & 21 & 4 \\ 2 & 3 & 1 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 3 & 0 & 1 & 15 & 3 \\ 15 & 21 & 18 & 0 & 15 & 97 & 21 \\ 3 & 4 & 3 & 0 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[17] \begin{matrix} 1^{12} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^{11} \\ 1^1 \end{matrix} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 18, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 6 & 3 \\ 3 & 5 & 3 & 0 & 4 & 10 & 5 \\ 2 & 3 & 0 & 0 & 2 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 2 & 0 & 1 & 6 & 3 \\ 6 & 10 & 4 & 0 & 6 & 14 & 8 \\ 3 & 5 & 2 & 0 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[18] \begin{matrix} 1^{13} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^{11} \\ 1^1 \end{matrix} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 20, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 11 & 3 \\ 3 & 4 & 3 & 0 & 3 & 14 & 4 \\ 2 & 3 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 3 & 3 & 0 & 1 & 10 & 3 \\ 11 & 14 & 12 & 0 & 10 & 42 & 14 \\ 3 & 4 & 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[19] $1^{\frac{1}{1^1}} 5 13^{\frac{1}{1^1}} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 5 & 3 & 0 & 4 & 15 & 5 \\ 2 & 3 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 2 & 0 & 1 & 9 & 3 \\ 9 & 15 & 6 & 0 & 9 & 34 & 12 \\ 3 & 5 & 2 & 0 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[20] $1^{\frac{1}{1^1}} 5 13^{\frac{1}{1^1}} 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 2 & 3 & 3 \\ 3 & 5 & 3 & 10 & 4 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 & 2 \\ 7 & 10 & 4 & 14 & 6 & 8 & 8 \\ 2 & 4 & 2 & 6 & 1 & 3 & 3 \\ 3 & 5 & 2 & 8 & 3 & 2 & 4 \\ 3 & 5 & 2 & 8 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[21] $1^{\frac{1}{1^1}} 5 13^{\frac{1}{1^1}} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 3 & 0 & 2 & 5 & 3 \\ 2 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 5 & 3 \\ 6 & 5 & 6 & 0 & 5 & 9 & 7 \\ 3 & 3 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[22] $1^{\frac{1}{1^1}} 5 13^{\frac{1}{1^1}} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 3 & 3 \\ 3 & 5 & 3 & 5 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 & 2 \\ 2 & 5 & 2 & 2 & 3 & 4 & 4 \\ 2 & 4 & 2 & 3 & 1 & 3 & 3 \\ 3 & 5 & 2 & 4 & 3 & 2 & 4 \\ 3 & 5 & 2 & 4 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[23] $1^{\frac{1}{1^1}} 5 13^{\frac{1}{1^1}} 1^{112} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 26, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 6 & 3 \\ 3 & 5 & 3 & 5 & 4 & 10 & 5 \\ 2 & 3 & 0 & 2 & 2 & 4 & 2 \\ 2 & 5 & 2 & 2 & 3 & 8 & 4 \\ 2 & 4 & 2 & 3 & 1 & 6 & 3 \\ 6 & 10 & 4 & 8 & 6 & 14 & 8 \\ 3 & 5 & 2 & 4 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[24] $1^{\frac{1}{1^1}} 5 13^{\frac{1}{1^1}} 1^{113} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 31, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 9 & 3 \\ 3 & 5 & 3 & 5 & 4 & 15 & 5 \\ 2 & 3 & 0 & 2 & 2 & 6 & 2 \\ 2 & 5 & 2 & 2 & 3 & 12 & 4 \\ 2 & 4 & 2 & 3 & 1 & 9 & 3 \\ 9 & 15 & 6 & 12 & 9 & 34 & 12 \\ 3 & 5 & 2 & 4 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[25] $1^{\frac{1}{1^1}} 5 13^{\frac{1}{1^1}} 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 12 & 3 \\ 3 & 5 & 3 & 5 & 4 & 20 & 5 \\ 2 & 3 & 0 & 2 & 2 & 8 & 2 \\ 2 & 5 & 2 & 2 & 3 & 16 & 4 \\ 2 & 4 & 2 & 3 & 1 & 12 & 3 \\ 12 & 20 & 8 & 16 & 12 & 62 & 16 \\ 3 & 5 & 2 & 4 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

[26] $1^{\frac{1}{1^1}} 5 13^{\frac{1}{1^1}} 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 2 & 6 & 3 \\ 2 & 1 & 3 & 2 & 2 & 5 & 3 \\ 2 & 3 & 1 & 4 & 3 & 6 & 3 \\ 3 & 2 & 4 & 2 & 3 & 6 & 4 \\ 2 & 2 & 3 & 3 & 1 & 5 & 3 \\ 6 & 5 & 6 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{12} \overset{11}{5} 13^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 8 & 3 \\ 2 & 1 & 3 & 0 & 2 & 8 & 3 \\ 2 & 3 & 1 & 0 & 3 & 8 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 7 & 3 \\ 8 & 8 & 8 & 0 & 7 & 22 & 10 \\ 3 & 3 & 3 & 0 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11} \overset{12}{5} 13^{11} \overset{121}{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 39, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 & 3 \\ 3 & 5 & 3 & 10 & 4 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 & 2 \\ 5 & 10 & 4 & 14 & 6 & 8 & 8 \\ 2 & 4 & 2 & 6 & 1 & 3 & 3 \\ 3 & 5 & 2 & 8 & 3 & 2 & 4 \\ 3 & 5 & 2 & 8 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{12} \overset{11}{5} 13^{11} \overset{122}{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 6 & 3 \\ 3 & 5 & 3 & 10 & 4 & 10 & 5 \\ 2 & 3 & 0 & 4 & 2 & 4 & 2 \\ 5 & 10 & 4 & 14 & 6 & 16 & 8 \\ 2 & 4 & 2 & 6 & 1 & 6 & 3 \\ 6 & 10 & 4 & 16 & 6 & 14 & 8 \\ 3 & 5 & 2 & 8 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{11} \overset{13}{5} 13^{11} \overset{12}{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 3 & 0 & 2 & 7 & 3 \\ 2 & 3 & 1 & 0 & 3 & 10 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 8 & 3 \\ 9 & 7 & 10 & 0 & 8 & 25 & 11 \\ 3 & 3 & 3 & 0 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{11} \overset{13}{5} 13^{11} \overset{123}{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.3$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 9 & 3 \\ 3 & 5 & 3 & 10 & 4 & 15 & 5 \\ 2 & 3 & 0 & 4 & 2 & 6 & 2 \\ 5 & 10 & 4 & 14 & 6 & 24 & 8 \\ 2 & 4 & 2 & 6 & 1 & 9 & 3 \\ 9 & 15 & 6 & 24 & 9 & 34 & 12 \\ 3 & 5 & 2 & 8 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{11} \overset{11}{5} 13^{11} \overset{111}{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 5 & 3 \\ 2 & 1 & 3 & 0 & 2 & 5 & 3 \\ 2 & 3 & 1 & 0 & 3 & 6 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 5 & 3 \\ 5 & 5 & 6 & 0 & 5 & 9 & 7 \\ 3 & 3 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{11} \overset{11}{5} 13^{11} \overset{111}{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 54, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 2 & 5 & 3 \\ 2 & 1 & 3 & 2 & 2 & 5 & 3 \\ 2 & 3 & 1 & 4 & 3 & 6 & 3 \\ 3 & 2 & 4 & 2 & 3 & 6 & 4 \\ 2 & 2 & 3 & 3 & 1 & 5 & 3 \\ 5 & 5 & 6 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{11} \overset{11}{5} 13^{11} \overset{111}{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 60, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 3 & 3 \\ 3 & 5 & 3 & 5 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 & 2 \\ 3 & 5 & 2 & 2 & 3 & 4 & 4 \\ 2 & 4 & 2 & 3 & 1 & 3 & 3 \\ 3 & 5 & 2 & 4 & 3 & 2 & 4 \\ 3 & 5 & 2 & 4 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[35] \begin{matrix} 1^{12} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^{11} \\ 1^1 \end{matrix} 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 63, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 8 & 3 \\ 3 & 5 & 3 & 5 & 4 & 15 & 5 \\ 2 & 3 & 0 & 2 & 2 & 6 & 2 \\ 3 & 5 & 2 & 2 & 3 & 12 & 4 \\ 2 & 4 & 2 & 3 & 1 & 9 & 3 \\ 8 & 15 & 6 & 12 & 9 & 34 & 12 \\ 3 & 5 & 2 & 4 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[36] \begin{matrix} 1^{12} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^{11} \\ 1^1 \end{matrix} 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 6 & 3 \\ 3 & 5 & 3 & 5 & 4 & 10 & 5 \\ 2 & 3 & 0 & 2 & 2 & 4 & 2 \\ 3 & 5 & 2 & 2 & 3 & 8 & 4 \\ 2 & 4 & 2 & 3 & 1 & 6 & 3 \\ 6 & 10 & 4 & 8 & 6 & 14 & 8 \\ 3 & 5 & 2 & 4 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[37] \begin{matrix} 1^{13} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^{11} \\ 1^1 \end{matrix} 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 92, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 9 & 3 \\ 3 & 5 & 3 & 5 & 4 & 15 & 5 \\ 2 & 3 & 0 & 2 & 2 & 6 & 2 \\ 3 & 5 & 2 & 2 & 3 & 12 & 4 \\ 2 & 4 & 2 & 3 & 1 & 9 & 3 \\ 9 & 15 & 6 & 12 & 9 & 34 & 12 \\ 3 & 5 & 2 & 4 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[38] \begin{matrix} 1^{13} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^{11} \\ 1^1 \end{matrix} 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 95, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 2 & 11 & 3 \\ 3 & 5 & 3 & 5 & 4 & 20 & 5 \\ 2 & 3 & 0 & 2 & 2 & 8 & 2 \\ 3 & 5 & 2 & 2 & 3 & 16 & 4 \\ 2 & 4 & 2 & 3 & 1 & 12 & 3 \\ 11 & 20 & 8 & 16 & 12 & 62 & 16 \\ 3 & 5 & 2 & 4 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[39] \begin{matrix} 1^{13} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^{11} \\ 1^1 \end{matrix} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 11 & 3 \\ 2 & 1 & 3 & 0 & 2 & 10 & 3 \\ 2 & 3 & 1 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 10 & 3 \\ 11 & 10 & 12 & 0 & 10 & 42 & 14 \\ 3 & 3 & 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[40] \begin{matrix} 1^{13} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^{11} \\ 1^1 \end{matrix} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 143, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 13 & 3 \\ 2 & 1 & 3 & 0 & 2 & 13 & 3 \\ 2 & 3 & 1 & 0 & 3 & 14 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 2 & 3 & 0 & 1 & 12 & 3 \\ 13 & 13 & 14 & 0 & 12 & 65 & 17 \\ 3 & 3 & 3 & 0 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$\mathbf{4.1170} \quad \text{NHC}(-2)(-3) + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$$

$$T = 10 \quad [1] \begin{matrix} 1^{11} \\ 1^1 \end{matrix} 1^{11} 3 \begin{matrix} 1^{11} \\ 1^1 \end{matrix} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 4 & 3 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 & 2 \\ 3 & 4 & 0 & 3 & 2 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & -1 & 4 & 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{\overline{11}} 5^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -5, \ \det = 9, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 0 & 3 & 4 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 0 & 0 & 0 & 2 & 3 & 2 \\ 2 & 4 & 0 & 0 & 2 & 1 & 3 & 3 \\ 3 & 5 & 0 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & -1 & 0 & 2 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

4.1171 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_{2n3\text{mix}}$

$$T = 11 \quad [4] \ 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -2, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 & 3 \\ 2 & 2 & 4 & 2 & 3 & 4 \\ 3 & 2 & 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -2, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 4 & 7 & 6 & 9 & 5 & 7 \\ 3 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -2, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 & 3 \\ 2 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 4 & 5 & 6 & 9 & 5 & 7 \\ 3 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -2, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 3 & 0 & 10 & 4 \\ 2 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 10 & 9 & 0 & 22 & 10 \\ 3 & 4 & 3 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -2, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -4, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 9 & 3 \\ 2 & 2 & 4 & 2 & 9 & 4 \\ 8 & 7 & 9 & 9 & 22 & 10 \\ 3 & 3 & 3 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{12}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 12, \ \Delta = -30, \ \det = -6, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 9 & 14 & 12 & 42 & 10 & 14 \\ 3 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

- [7] $1^{11}_{1^1} 5^{121} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 3 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \\ 2 & 3 & 1 & 6 & 3 & 2 \\ 6 & 7 & 6 & 9 & 5 & 7 \\ 3 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$
- [9] $1^{11}_{1^1} 5^{12} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 3 & 0 & 10 & 4 \\ 2 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 10 & 9 & 0 & 22 & 10 \\ 3 & 4 & 3 & 0 & 10 & 4 \end{pmatrix}_{b_0}$
- [8] $1^{11}_{1^1} 5^{11} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 3 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 5 & 7 & 6 & 9 & 5 & 7 \\ 3 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$
- [10] $1^{12}_{1^1} 5^{12} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 3 & 0 & 17 & 4 \\ 2 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 12 & 17 & 15 & 0 & 65 & 17 \\ 3 & 4 & 3 & 0 & 17 & 4 \end{pmatrix}_{b_0}$
- [12] $1^{11}_{1^1} 5^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 3 & 2 & 3 \\ 2 & 3 & 1 & 2 & 3 & 3 \\ 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 2 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$
- [11] $1^{11}_{1^1} 5^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 3 \\ 2 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 2 & 3 & 0 & 1 & 3 \\ 3 & 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$
- [13] $1^{11}_{1^1} 5^{121} 13^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -13, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 4 & 5 & 3 \\ 3 & 5 & 3 & 5 & 9 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 4 & 5 & 2 & 2 & 7 & 4 \\ 5 & 9 & 4 & 7 & 11 & 7 \\ 3 & 5 & 2 & 4 & 7 & 4 \end{pmatrix}_{b_0}$
- [16] $1^{12}_{1^1} 5^{131} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 11 & 3 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 11 & 14 & 12 & 42 & 10 & 14 \\ 3 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$
- [15] $1^{13}_{1^1} 5^{121} 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 14 & 21 & 18 & 97 & 15 & 21 \\ 3 & 3 & 3 & 15 & 1 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

[14] $1^{12}_{1^1} 5^{121} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 10 & 14 & 12 & 42 & 10 & 14 \\ 3 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[17] $1^{11}_{1^1} 5^{11} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 5 & 3 & 0 & 9 & 5 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 9 & 4 & 0 & 11 & 7 \\ 3 & 5 & 2 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

[18] $1^{131}_{1^1} 5^{131} 13^1 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 5 & 3 \\ 3 & 5 & 3 & 10 & 9 & 5 \\ 2 & 3 & 0 & 4 & 4 & 2 \\ 7 & 10 & 4 & 14 & 14 & 8 \\ 5 & 9 & 4 & 14 & 11 & 7 \\ 3 & 5 & 2 & 8 & 7 & 4 \end{pmatrix}_{b_0}$$

[19] $1^{11}_{1^1} 5^{11} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 & 3 \\ 3 & 2 & 4 & 2 & 3 & 4 \\ 3 & 2 & 3 & 3 & 1 & 3 \\ 3 & 3 & 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[20] $1^{12}_{1^1} 5^{11} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 & 3 \\ 2 & 1 & 3 & 8 & 2 & 3 \\ 2 & 3 & 1 & 8 & 3 & 3 \\ 7 & 8 & 8 & 22 & 7 & 10 \\ 3 & 2 & 3 & 7 & 1 & 3 \\ 3 & 3 & 3 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$

[21] $1^{11}_{1^1} 5^{131} 13^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -21, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 5 & 3 & 5 & 9 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 2 & 5 & 2 & 2 & 7 & 4 \\ 5 & 9 & 4 & 7 & 11 & 7 \\ 3 & 5 & 2 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

[22] $1^{12}_{1^1} 5^{13} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 3 & 0 & 17 & 4 \\ 2 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 17 & 15 & 0 & 65 & 17 \\ 3 & 4 & 3 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

[23] $1^{13}_{1^1} 5^{13} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 3 & 0 & 24 & 4 \\ 2 & 3 & 1 & 0 & 21 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 17 & 24 & 21 & 0 & 130 & 24 \\ 3 & 4 & 3 & 0 & 24 & 4 \end{pmatrix}_{b_0}$$

- [29] $1^{111}_{1^1} 5^{11} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 3 & 3 \\ 3 & 5 & 3 & 5 & 4 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 3 & 5 & 2 & 2 & 3 & 4 \\ 3 & 4 & 2 & 3 & 1 & 3 \\ 3 & 5 & 2 & 4 & 3 & 4 \end{pmatrix}_{b_0}$
- [31] $1^{131}_{1^1} 5^{13} 13^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 9 & 3 & 3 \\ 3 & 5 & 3 & 15 & 4 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 9 & 15 & 6 & 34 & 9 & 12 \\ 3 & 4 & 2 & 9 & 1 & 3 \\ 3 & 5 & 2 & 12 & 3 & 4 \end{pmatrix}_{b_0}$
- [30] $1^{121}_{1^1} 5^{13} 13^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 6 & 3 & 3 \\ 3 & 5 & 8 & 10 & 4 & 5 \\ 4 & 8 & 8 & 12 & 5 & 6 \\ 6 & 10 & 12 & 14 & 6 & 8 \\ 3 & 4 & 5 & 6 & 1 & 3 \\ 3 & 5 & 6 & 8 & 3 & 4 \end{pmatrix}_{b_0}$
- [27] $1^{111}_{1^1} 5^{13} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 3 & 3 \\ 3 & 5 & 8 & 5 & 4 & 5 \\ 4 & 8 & 8 & 6 & 5 & 6 \\ 3 & 5 & 6 & 2 & 3 & 4 \\ 3 & 4 & 5 & 3 & 1 & 3 \\ 3 & 5 & 6 & 4 & 3 & 4 \end{pmatrix}_{b_0}$
- [26] $1^{131}_{1^1} 5^{13} 13^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 9 & 3 & 3 \\ 3 & 5 & 8 & 15 & 4 & 5 \\ 4 & 8 & 8 & 18 & 5 & 6 \\ 9 & 15 & 18 & 34 & 9 & 12 \\ 3 & 4 & 5 & 9 & 1 & 3 \\ 3 & 5 & 6 & 12 & 3 & 4 \end{pmatrix}_{b_0}$
- [25] $1^{112}_{1^1} 5^{13} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 6 & 3 \\ 3 & 5 & 8 & 5 & 9 & 5 \\ 4 & 8 & 8 & 6 & 11 & 6 \\ 3 & 5 & 6 & 2 & 7 & 4 \\ 6 & 9 & 11 & 7 & 11 & 7 \\ 3 & 5 & 6 & 4 & 7 & 4 \end{pmatrix}_{b_0}$
- [24] $1^{131}_{1^1} 5^{13} 13^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 15 & 3 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 15 & 21 & 18 & 97 & 15 & 21 \\ 3 & 3 & 3 & 15 & 1 & 3 \\ 3 & 4 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$
- [28] $1^{121}_{1^1} 5^{13} 13^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 3 & 3 \\ 3 & 5 & 3 & 10 & 4 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 6 & 10 & 4 & 14 & 6 & 8 \\ 3 & 4 & 2 & 6 & 1 & 3 \\ 3 & 5 & 2 & 8 & 3 & 4 \end{pmatrix}_{b_0}$
- [32] $1^{112}_{1^1} 5^{13} 13^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 5 & 3 & 0 & 14 & 5 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 14 & 6 & 0 & 29 & 11 \\ 3 & 5 & 2 & 0 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{11}_{1^1} 5^{11} 13^{11} 1^{121} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 5 & 3 \\ 3 & 5 & 3 & 10 & 9 & 5 \\ 2 & 3 & 0 & 4 & 4 & 2 \\ 5 & 10 & 4 & 14 & 14 & 8 \\ 5 & 9 & 4 & 14 & 11 & 7 \\ 3 & 5 & 2 & 8 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{11}_{1^1} 5^{11} 13^{11} 1^{12} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 10 & 3 & 3 \\ 7 & 7 & 10 & 25 & 8 & 11 \\ 3 & 2 & 3 & 8 & 1 & 3 \\ 3 & 3 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{12}_{1^1} 5^{11} 13^{11} 1^{12} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 & 3 \\ 2 & 1 & 3 & 10 & 2 & 3 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 9 & 10 & 12 & 42 & 10 & 14 \\ 3 & 2 & 3 & 10 & 1 & 3 \\ 3 & 3 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{12}_{1^1} 5^{11} 13^{11} 1^{112} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -35, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 \\ 3 & 5 & 3 & 5 & 14 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 2 & 5 & 2 & 2 & 11 & 4 \\ 8 & 14 & 6 & 11 & 29 & 11 \\ 3 & 5 & 2 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{11}_{1^1} 5^{11} 13^{11} 1^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 3 \\ 2 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 5 & 5 & 0 & 8 & 6 \\ 3 & 3 & 3 & 0 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{11}_{1^1} 5^{13} 13^{11} 1^{13} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -43, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 11 & 3 \\ 3 & 5 & 3 & 0 & 19 & 5 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 19 & 8 & 0 & 55 & 15 \\ 3 & 5 & 2 & 0 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{11}_{1^1} 5^{11} 13^{11} 1^{11} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 3 & 3 \\ 2 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 5 & 5 & 6 & 9 & 5 & 7 \\ 3 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{11}_{1^1} 5^{13} 13^{11} 1^{131} + [\textcolor{blue}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 5 & 3 \\ 3 & 5 & 3 & 15 & 9 & 5 \\ 2 & 3 & 0 & 6 & 4 & 2 \\ 8 & 15 & 6 & 34 & 21 & 12 \\ 5 & 9 & 4 & 21 & 11 & 7 \\ 3 & 5 & 2 & 12 & 7 & 4 \end{pmatrix}_{b_0}$$

[41] $1^{1^{12}}_5 13^1 1^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 8 & 3 \\ 3 & 5 & 3 & 10 & 14 & 5 \\ 2 & 3 & 0 & 4 & 6 & 2 \\ 5 & 10 & 4 & 14 & 22 & 8 \\ 8 & 14 & 6 & 22 & 29 & 11 \\ 3 & 5 & 2 & 8 & 11 & 4 \end{pmatrix}_{b_0}$$

[42] $1^{1^3}_5 13^1 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -53, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 3 \\ 3 & 5 & 3 & 5 & 19 & 5 \\ 2 & 3 & 0 & 2 & 8 & 2 \\ 2 & 5 & 2 & 2 & 15 & 4 \\ 11 & 19 & 8 & 15 & 55 & 15 \\ 3 & 5 & 2 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

[44] $1^{1^{11}}_5 13^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 5 & 3 \\ 3 & 5 & 3 & 5 & 9 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 3 & 5 & 2 & 2 & 7 & 4 \\ 5 & 9 & 4 & 7 & 11 & 7 \\ 3 & 5 & 2 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

[43] $1^{1^{12}}_5 13^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 6 & 3 \\ 3 & 5 & 3 & 5 & 9 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 3 & 5 & 2 & 2 & 7 & 4 \\ 6 & 9 & 4 & 7 & 11 & 7 \\ 3 & 5 & 2 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

[46] $1^{1^{21}}_5 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 & 3 \\ 2 & 1 & 3 & 8 & 2 & 3 \\ 2 & 3 & 1 & 8 & 3 & 3 \\ 8 & 8 & 8 & 22 & 7 & 10 \\ 3 & 2 & 3 & 7 & 1 & 3 \\ 3 & 3 & 3 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$

[45] $1^{1^{21}}_5 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 3 & 3 \\ 2 & 1 & 3 & 5 & 2 & 3 \\ 2 & 3 & 1 & 6 & 3 & 3 \\ 6 & 5 & 6 & 9 & 5 & 7 \\ 3 & 2 & 3 & 5 & 1 & 3 \\ 3 & 3 & 3 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

[47] $1^{1^{21}}_5 13^1 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -60, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 5 & 3 \\ 3 & 5 & 3 & 10 & 9 & 5 \\ 2 & 3 & 0 & 4 & 4 & 2 \\ 6 & 10 & 4 & 14 & 14 & 8 \\ 5 & 9 & 4 & 14 & 11 & 7 \\ 3 & 5 & 2 & 8 & 7 & 4 \end{pmatrix}_{b_0}$$

[48] $1^{1^{31}}_5 13^1 1^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -63, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 5 & 3 \\ 3 & 5 & 3 & 20 & 9 & 5 \\ 2 & 3 & 0 & 8 & 4 & 2 \\ 11 & 20 & 8 & 62 & 28 & 16 \\ 5 & 9 & 4 & 28 & 11 & 7 \\ 3 & 5 & 2 & 16 & 7 & 4 \end{pmatrix}_{b_0}$$

- [50] $1^{11}_{1^1} 5^{13} 13^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 9 & 3 \\ 3 & 2 & 4 & 2 & 9 & 4 \\ 9 & 7 & 9 & 9 & 22 & 10 \\ 3 & 3 & 3 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$
- [49] $1^{11}_{1^1} 5^{11} 13^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 9 & 3 \\ 3 & 2 & 4 & 2 & 9 & 4 \\ 7 & 7 & 9 & 9 & 22 & 10 \\ 3 & 3 & 3 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$
- [51] $1^{12}_{1^1} 5^{31} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 & 3 \\ 2 & 1 & 3 & 8 & 2 & 3 \\ 2 & 3 & 1 & 8 & 3 & 3 \\ 9 & 8 & 8 & 22 & 7 & 10 \\ 3 & 2 & 3 & 7 & 1 & 3 \\ 3 & 3 & 3 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$
- [52] $1^{11}_{1^1} 5^{13} 13^1 1^{123} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -67, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 11 & 3 \\ 3 & 5 & 3 & 10 & 19 & 5 \\ 2 & 3 & 0 & 4 & 8 & 2 \\ 5 & 10 & 4 & 14 & 30 & 8 \\ 11 & 19 & 8 & 30 & 55 & 15 \\ 3 & 5 & 2 & 8 & 15 & 4 \end{pmatrix}_{b_0}$$
- [53] $1^{11}_{1^1} 5^{131} 13^1 1^{131} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -70, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 5 & 3 \\ 3 & 5 & 3 & 15 & 9 & 5 \\ 2 & 3 & 0 & 6 & 4 & 2 \\ 9 & 15 & 6 & 34 & 21 & 12 \\ 5 & 9 & 4 & 21 & 11 & 7 \\ 3 & 5 & 2 & 12 & 7 & 4 \end{pmatrix}_{b_0}$$
- [54] $1^{11}_{1^1} 5^{12} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 3 \\ 2 & 1 & 3 & 0 & 7 & 3 \\ 2 & 3 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 8 & 7 & 9 & 0 & 22 & 10 \\ 3 & 3 & 3 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$
- [55] $1^{14}_{1^1} 5^{131} 13^1 1^{114} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -75, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 14 & 3 \\ 3 & 5 & 3 & 5 & 24 & 5 \\ 2 & 3 & 0 & 2 & 10 & 2 \\ 2 & 5 & 2 & 2 & 19 & 4 \\ 14 & 24 & 10 & 19 & 89 & 19 \\ 3 & 5 & 2 & 4 & 19 & 4 \end{pmatrix}_{b_0}$$
- [57] $1^{13}_{1^1} 5^{21} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 & 3 \\ 2 & 1 & 3 & 13 & 2 & 3 \\ 2 & 3 & 1 & 14 & 3 & 3 \\ 12 & 13 & 14 & 65 & 12 & 17 \\ 3 & 2 & 3 & 12 & 1 & 3 \\ 3 & 3 & 3 & 17 & 3 & 4 \end{pmatrix}_{b_0}$$
- [56] $1^{11}_{1^1} 5^{21} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 10 & 3 & 3 \\ 8 & 7 & 10 & 25 & 8 & 11 \\ 3 & 2 & 3 & 8 & 1 & 3 \\ 3 & 3 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[58] \ 1 \overset{12}{\underset{11}{1}} \overset{13}{5} \overset{11}{13} \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 4 & 9 & 3 \\ 3 & 2 & 4 & 2 & 9 & 4 \\ 8 & 7 & 9 & 9 & 22 & 10 \\ 3 & 3 & 3 & 4 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[59] \ 1 \overset{131}{\underset{11}{1}} \overset{11}{5} \overset{13}{13} \overset{12}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -91, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 10 & 3 & 3 \\ 9 & 7 & 10 & 25 & 8 & 11 \\ 3 & 2 & 3 & 8 & 1 & 3 \\ 3 & 3 & 3 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[60] \ 1 \overset{12}{\underset{11}{1}} \overset{13}{5} \overset{13}{13} \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -92, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 8 & 3 \\ 3 & 5 & 3 & 5 & 14 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 3 & 5 & 2 & 2 & 11 & 4 \\ 8 & 14 & 6 & 11 & 29 & 11 \\ 3 & 5 & 2 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[61] \ 1 \overset{13}{\underset{11}{1}} \overset{13}{5} \overset{13}{13} \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 9 & 3 \\ 3 & 5 & 3 & 5 & 14 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 3 & 5 & 2 & 2 & 11 & 4 \\ 9 & 14 & 6 & 11 & 29 & 11 \\ 3 & 5 & 2 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[63] \ 1 \overset{121}{\underset{11}{1}} \overset{13}{5} \overset{13}{13} \overset{13}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 & 3 \\ 2 & 1 & 3 & 12 & 2 & 3 \\ 2 & 3 & 1 & 16 & 3 & 3 \\ 12 & 12 & 16 & 70 & 13 & 18 \\ 3 & 2 & 3 & 13 & 1 & 3 \\ 3 & 3 & 3 & 18 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[62] \ 1 \overset{121}{\underset{11}{1}} \overset{13}{5} \overset{13}{13} \overset{13}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 & 3 \\ 2 & 1 & 3 & 15 & 2 & 3 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 14 & 15 & 18 & 97 & 15 & 21 \\ 3 & 2 & 3 & 15 & 1 & 3 \\ 3 & 3 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[64] \ 1 \overset{113}{\underset{11}{1}} \overset{13}{5} \overset{13}{13} \overset{12}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 11 & 3 \\ 2 & 1 & 3 & 0 & 9 & 3 \\ 2 & 3 & 1 & 0 & 13 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 9 & 13 & 0 & 44 & 14 \\ 3 & 3 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[65] \ 1 \overset{121}{\underset{11}{1}} \overset{13}{5} \overset{13}{13} \overset{12}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -102, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 3 & 3 \\ 2 & 1 & 3 & 10 & 2 & 3 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 10 & 10 & 12 & 42 & 10 & 14 \\ 3 & 2 & 3 & 10 & 1 & 3 \\ 3 & 3 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[67] $1^{12} 5^{11} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -123$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 3 \\ 2 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 10 & 11 & 0 & 39 & 13 \\ 3 & 3 & 3 & 0 & 13 & 4 \end{pmatrix}_{b_0}$$

[66] $1^{12} 5^{13} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -123$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 11 & 3 \\ 2 & 1 & 3 & 0 & 10 & 3 \\ 2 & 3 & 1 & 0 & 11 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 11 & 10 & 11 & 0 & 39 & 13 \\ 3 & 3 & 3 & 0 & 13 & 4 \end{pmatrix}_{b_0}$$

[68] $1^{13} 5^{13} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -142$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 11 & 3 \\ 3 & 5 & 3 & 5 & 19 & 5 \\ 2 & 3 & 0 & 2 & 8 & 2 \\ 3 & 5 & 2 & 2 & 15 & 4 \\ 11 & 19 & 8 & 15 & 55 & 15 \\ 3 & 5 & 2 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

[70] $1^{13} 5^{31} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -143$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.8$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 & 3 \\ 2 & 1 & 3 & 13 & 2 & 3 \\ 2 & 3 & 1 & 14 & 3 & 3 \\ 13 & 13 & 14 & 65 & 12 & 17 \\ 3 & 2 & 3 & 12 & 1 & 3 \\ 3 & 3 & 3 & 17 & 3 & 4 \end{pmatrix}_{b_0}$$

[69] $1^{12} 5^{31} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -143$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.5$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 & 3 \\ 2 & 1 & 3 & 10 & 2 & 3 \\ 2 & 3 & 1 & 12 & 3 & 3 \\ 11 & 10 & 12 & 42 & 10 & 14 \\ 3 & 2 & 3 & 10 & 1 & 3 \\ 3 & 3 & 3 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

[71] $1^{12} 5^{31} 13^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -151$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.5$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 & 3 \\ 2 & 1 & 3 & 14 & 2 & 3 \\ 2 & 3 & 1 & 20 & 3 & 3 \\ 15 & 14 & 20 & 106 & 16 & 22 \\ 3 & 2 & 3 & 16 & 1 & 3 \\ 3 & 3 & 3 & 22 & 3 & 4 \end{pmatrix}_{b_0}$$

[72] $1^{12} 5^{31} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -171$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.5$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 & 3 \\ 2 & 1 & 3 & 12 & 2 & 3 \\ 2 & 3 & 1 & 16 & 3 & 3 \\ 13 & 12 & 16 & 70 & 13 & 18 \\ 3 & 2 & 3 & 13 & 1 & 3 \\ 3 & 3 & 3 & 18 & 3 & 4 \end{pmatrix}_{b_0}$$

[73] $1^{12} 5^{13} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -187$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 13 & 3 \\ 2 & 1 & 3 & 0 & 12 & 3 \\ 2 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 13 & 12 & 15 & 0 & 65 & 17 \\ 3 & 3 & 3 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

[74] $1^{13}_{1^1} 5^{131} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -197, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 & 3 \\ 2 & 1 & 3 & 15 & 2 & 3 \\ 2 & 3 & 1 & 18 & 3 & 3 \\ 15 & 15 & 18 & 97 & 15 & 21 \\ 3 & 2 & 3 & 15 & 1 & 3 \\ 3 & 3 & 3 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

[75] $1^{13}_{1^1} 5^{131} 13^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -203, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 17 & 3 & 3 \\ 2 & 1 & 3 & 17 & 2 & 3 \\ 2 & 3 & 1 & 22 & 3 & 3 \\ 17 & 17 & 22 & 137 & 18 & 25 \\ 3 & 2 & 3 & 18 & 1 & 3 \\ 3 & 3 & 3 & 25 & 3 & 4 \end{pmatrix}_{b_0}$$

[76] $1^{13}_{1^1} 5^{131} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -264, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 15 & 3 \\ 2 & 1 & 3 & 0 & 15 & 3 \\ 2 & 3 & 1 & 0 & 17 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 15 & 15 & 17 & 0 & 92 & 20 \\ 3 & 3 & 3 & 0 & 20 & 4 \end{pmatrix}_{b_0}$$

4.1172 NHC(-2)(-3) + $\textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}2\textcolor{blue}{n}3\textcolor{blue}{mix} + \textcolor{blue}{su}_2$

$T = 12$ [0] $1^{11}_{1^1} 5^{111} 13^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 2 & 2 & 5 & 3 \\ 2 & 3 & 1 & 4 & 3 & 6 & 3 \\ 2 & 2 & 4 & 2 & 3 & 6 & 4 \\ 3 & 2 & 3 & 3 & 1 & 5 & 3 \\ 5 & 5 & 6 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[1] $1^{11}_{1^1} 5^{111} 13^1 1^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 & 3 \\ 3 & 5 & 3 & 5 & 9 & 5 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 & 2 \\ 2 & 5 & 2 & 2 & 7 & 4 & 4 \\ 5 & 9 & 4 & 7 & 11 & 7 & 7 \\ 3 & 5 & 2 & 4 & 7 & 2 & 4 \\ 3 & 5 & 2 & 4 & 7 & 4 & 4 \end{pmatrix}_{b_0}$$

[2] $1^{11}_{1^1} 5^{111} 13^1 1^{1112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 46, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 6 & 3 \\ 3 & 5 & 3 & 5 & 9 & 10 & 5 \\ 2 & 3 & 0 & 2 & 4 & 4 & 2 \\ 2 & 5 & 2 & 2 & 7 & 8 & 4 \\ 5 & 9 & 4 & 7 & 11 & 14 & 7 \\ 6 & 10 & 4 & 8 & 14 & 14 & 8 \\ 3 & 5 & 2 & 4 & 7 & 8 & 4 \end{pmatrix}_{b_0}$$

[3] $1^{11}_{1^1} 5^{113} 13^1 1^{1113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 51, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 9 & 3 \\ 3 & 5 & 3 & 5 & 9 & 15 & 5 \\ 2 & 3 & 0 & 2 & 4 & 6 & 2 \\ 2 & 5 & 2 & 2 & 7 & 12 & 4 \\ 5 & 9 & 4 & 7 & 11 & 21 & 7 \\ 9 & 15 & 6 & 12 & 21 & 34 & 12 \\ 3 & 5 & 2 & 4 & 7 & 12 & 4 \end{pmatrix}_{b_0}$$

[4] $1^{11}_{1^1} 5^{121} 13^1 1^{1121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 71, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 & 3 \\ 3 & 5 & 3 & 5 & 14 & 5 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 & 2 \\ 2 & 5 & 2 & 2 & 11 & 4 & 4 \\ 8 & 14 & 6 & 11 & 29 & 11 & 11 \\ 3 & 5 & 2 & 4 & 11 & 2 & 4 \\ 3 & 5 & 2 & 4 & 11 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1^{31}} 5^{1^{11}} 13^{1^{11}} 1^{1131} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 107, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 3 & 3 \\ 3 & 5 & 3 & 5 & 19 & 5 & 5 \\ 2 & 3 & 0 & 2 & 8 & 2 & 2 \\ 2 & 5 & 2 & 2 & 15 & 4 & 4 \\ 11 & 19 & 8 & 15 & 55 & 15 & 15 \\ 3 & 5 & 2 & 4 & 15 & 2 & 4 \\ 3 & 5 & 2 & 4 & 15 & 4 & 4 \end{pmatrix}_{b_0}$$

4.1173 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{1^1} 4^{1^1} 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -63, \quad \det = -21, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1^1} 4^{1^1} 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -63, \quad \det = -36, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 5 & 0 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 5 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1^1} 4^{1^1} 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -63, \quad \det = -40, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 4 & 0 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1^{11}} 4^{1^1} 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -63, \quad \det = -61, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.97 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 & 0 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1^{11}} 4^{1^1} 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 11, \quad \Delta = -63, \quad \det = -100, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 7 & 0 & 4 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 5 & 7 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [6] \quad 21^{1^1} 5^{1^1} 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 13, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 4 & 3 & 4 \\ 3 & 3 & 0 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 2 & 3 \\ 2 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1^{11}} 5^{1^1} 1^{1^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 13, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 6 & 4 & 0 & 4 \\ 4 & 6 & 7 & 4 & 0 & 5 \\ 2 & 4 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 3 & 21 & 0 & 4 \\ 2 & 3 & 0 & 12 & 0 & 2 \\ 12 & 21 & 12 & 88 & 0 & 18 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{13} \quad 5 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 88, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 3 & 18 & 0 & 4 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 10 & 18 & 10 & 61 & 0 & 15 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[18] \quad 21^{11} \quad 5 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 96, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 2 & 3 \\ 4 & 4 & 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 3 & 4 & 2 & 1 & 2 & 3 \\ 2 & 3 & 2 & 2 & 0 & 2 \\ 3 & 4 & 2 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[19] \quad 21^{12} \quad 5 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 120, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 2 & 3 \\ 4 & 4 & 3 & 8 & 3 & 4 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 6 & 8 & 4 & 10 & 4 & 6 \\ 2 & 3 & 2 & 4 & 0 & 2 \\ 3 & 4 & 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[20] \quad 21^{13} \quad 5 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 9 & 2 & 3 \\ 4 & 4 & 3 & 12 & 3 & 4 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 9 & 12 & 6 & 25 & 6 & 9 \\ 2 & 3 & 2 & 6 & 0 & 2 \\ 3 & 4 & 2 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

4.1174 NHC(-2)(-3) + $\textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$

$$T = 10 \quad [0] \quad 1^{11} \quad 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 2 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} \quad 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} \quad 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 0 & 4 \\ 2 & 3 & 0 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & 1 & 0 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 3 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} \quad 4 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 3 & 0 & 5 & 4 \\ 2 & 3 & 0 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 3 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

[4] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 4 & 3 \\ 3 & 4 & 3 & 5 & 0 & 7 & 4 \\ 2 & 3 & 0 & 3 & 0 & 5 & 2 \\ 3 & 5 & 3 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 7 & 5 & 6 & 0 & 8 & 6 \\ 3 & 4 & 2 & 4 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[5] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 4 & 3 & 5 & 0 & 4 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 2 \\ 3 & 5 & 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & 4 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[6] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 4 & 3 \\ 3 & 4 & 3 & 4 & 0 & 7 & 4 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 3 & 4 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 7 & 5 & 5 & 0 & 8 & 6 \\ 3 & 4 & 2 & 3 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[7] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 4 & 0 & 4 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 91, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 3 & 4 & 0 & 7 & 4 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 2 & 4 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 7 & 5 & 5 & 0 & 8 & 6 \\ 3 & 4 & 2 & 3 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[9] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -33, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 3 & 4 & 0 & 7 & 4 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 2 & 4 & 2 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 7 & 5 & 5 & 0 & 8 & 6 \\ 3 & 4 & 2 & 3 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

4.1175 NHC(-2)(-3) + su₃ + su₂ + su₃ + su₂ + su₂

$T = 11$ [3] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 3 & 5 & 0 & 7 & 4 & 4 \\ 2 & 3 & 0 & 3 & 0 & 5 & 2 & 2 \\ 3 & 5 & 3 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 7 & 5 & 6 & 0 & 8 & 5 & 6 \\ 3 & 4 & 2 & 3 & 0 & 5 & 1 & 3 \\ 3 & 4 & 2 & 4 & -1 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

[2] $1^{11} 4 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 & 5 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 & 3 & 2 \\ 2 & 3 & 3 & 1 & 0 & 2 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 2 & 0 & 1 & 3 & 3 \\ 3 & 5 & 3 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 2 & 3 & -1 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

[1] $1^{11} 4 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{violet}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 1 & 3 & 3 \\ 3 & 4 & 3 & 3 & 0 & 0 & 5 & 4 \\ 2 & 3 & 0 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 0 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 3 & 3 & 0 & 0 & 4 & 4 \\ 3 & 4 & 2 & 3 & -1 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

[4] $1^{11} 4 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{violet}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 3 & 5 & 0 & 7 & 4 & 4 \\ 2 & 3 & 0 & 3 & 0 & 5 & 2 & 2 \\ 3 & 5 & 3 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 7 & 5 & 6 & 0 & 8 & 5 & 6 \\ 2 & 4 & 2 & 3 & 0 & 5 & 1 & 3 \\ 3 & 4 & 2 & 4 & -1 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

[5] $1^{11} 4 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{violet}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 11, \Delta = -3, \det = -81, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 3 & 4 & 0 & 7 & 4 & 4 \\ 2 & 3 & 0 & 2 & 0 & 5 & 2 & 2 \\ 3 & 4 & 2 & 1 & 0 & 5 & 3 & 3 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 7 & 5 & 5 & 0 & 8 & 5 & 6 \\ 2 & 4 & 2 & 3 & 0 & 5 & 1 & 3 \\ 3 & 4 & 2 & 3 & -1 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

4.1176 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$T = 11$ [0] $1^{11} 5 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{violet}{\mathfrak{su}_2} + \textcolor{violet}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 11, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 11.0$

[1] $1^{11} 5 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_3} + \textcolor{blue}{\mathfrak{su}_2} + \textcolor{violet}{\mathfrak{su}_2} + \textcolor{violet}{\mathfrak{su}_2}$ $T = 11, T_{\max} = 11, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 11, 1), c_{\text{ext}} = 11.0$

4.1177 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + (-3) \rightarrow (-2)$

$T = 10$ [0] $1^{11} 5 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 3 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[1] $1^{11} 5 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 4 & 0 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[2] $1^{11} 5 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 10 & 0 & 4 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 8 & 10 & 6 & 21 & 0 & 9 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} \overset{1}{5} 1^{12} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 85, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 3 & 7 & 0 & 4 \\ 2 & 3 & 0 & 4 & 0 & 2 \\ 5 & 7 & 4 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{12} \overset{1}{5} 1^{12} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 141, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 3 & 11 & 0 & 4 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 7 & 11 & 6 & 21 & 0 & 9 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 9 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{13} \overset{1}{5} 1^{12} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 181, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.99 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 3 & 15 & 0 & 4 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 9 & 15 & 8 & 40 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{12} \overset{1}{5} 1^{32} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 181, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 3 & 17 & 0 & 4 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 13 & 17 & 10 & 61 & 0 & 15 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{12} \overset{1}{5} 1^{22} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 196, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 3 & 14 & 0 & 4 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 10 & 14 & 8 & 38 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{2^1 21^{11}} \overset{1}{5} 1^{12} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 285, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 5 & 1 & 3 \\ 6 & 13 & 5 & 12 & 0 & 7 \\ 2 & 5 & 0 & 4 & 0 & 2 \\ 5 & 12 & 4 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 2 & 6 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{13} \overset{1}{5} 1^{22} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 316, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 3 & 18 & 0 & 4 \\ 2 & 3 & 0 & 10 & 0 & 2 \\ 12 & 18 & 10 & 61 & 0 & 15 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 15 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{13} \overset{1}{5} 1^{32} 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + (-3) \rightarrow (-2) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -59, \quad \det = 381, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 1 & 3 \\ 3 & 4 & 3 & 21 & 0 & 4 \\ 2 & 3 & 0 & 12 & 0 & 2 \\ 15 & 21 & 12 & 88 & 0 & 18 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^1 2^{12} 1^{12} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 516, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 10 & 1 & 3 \\ 6 & 13 & 5 & 24 & 0 & 7 \\ 2 & 5 & 0 & 8 & 0 & 2 \\ 10 & 24 & 8 & 38 & 0 & 12 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 2 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 1^1 2^{12} 1^{13} 5^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 + (-3) \rightarrow (-2) \quad T = 10, \ T_{\max} = 12, \ \Delta = -59, \ \det = 901, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 15 & 1 & 3 \\ 6 & 13 & 5 & 36 & 0 & 7 \\ 2 & 5 & 0 & 12 & 0 & 2 \\ 15 & 36 & 12 & 88 & 0 & 18 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 7 & 2 & 18 & -1 & 3 \end{pmatrix}_{b_0}$$

4.1178 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 7 \quad [0] \ 1^1 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -17, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 5 & 5 & 3 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 13^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -20, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 2 & 4 & 0 & 2 & 2 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{14} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -22, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.45 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 9 & 9 & 12 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{13} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -24, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.83 \rightarrow 10, \ T_{\max} = 12, \ T \in [10, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 \\ 3 & 4 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{13} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -28, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.29 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 4 & 4 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 13^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -29, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.59 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 3 & 5 & 3 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 13^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -29, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.2$$

$$T_{\min} = 11.41 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 4 & 8 & 9 & 4 & 5 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 5 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 13^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -32, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.38 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 7 & 12 & 24 & 6 & 8 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{13}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -72, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0 \\ T_{\min} = 10.44 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 10 & 12 & 24 & 6 & 8 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -73, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0 \\ T_{\min} = 10.23 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 2 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 6 & 9 & 12 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -74, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0 \\ T_{\min} = 10.16 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 7 & 9 & 12 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 21^{11}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -76, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0 \\ T_{\min} = 10.32 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 4 \\ 8 & 9 & 4 & 4 & 5 \\ 3 & 4 & 0 & 2 & 2 \\ 3 & 4 & 2 & 0 & 2 \\ 4 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 13^{11}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -77, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.2 \\ T_{\min} = 10.74 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 7 & 13 & 27 & 7 & 9 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 21^{11}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -79, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0 \\ T_{\min} = 10.16 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 4 & 4 & 3 & 4 \\ 3 & 4 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{12}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -84, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0 \\ T_{\min} = 10.43 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 8 & 12 & 24 & 6 & 8 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{13}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -87, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0 \\ T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 9 & 12 & 24 & 6 & 8 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[27] \ 13^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -98, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0 \\ T_{\min} = 10.94 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 9 & 17 & 48 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[29] \ 21^{11}3^{11}1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -99, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0 \\ T_{\min} = 10.17 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 15 & 3 & 4 \\ 5 & 4 & 17 & 3 & 4 \\ 15 & 17 & 48 & 9 & 12 \\ 3 & 3 & 9 & 0 & 2 \\ 4 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 21^{11}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -99, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0 \\ T_{\min} = 10.17 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

- [31] $\begin{pmatrix} 4 & 5 & 9 & 3 & 4 \\ 5 & 4 & 9 & 3 & 4 \\ 9 & 9 & 12 & 5 & 6 \\ 3 & 3 & 5 & 0 & 2 \\ 4 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$
 $21^{123}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 6 & 2 & 3 \\ 4 & 4 & 8 & 3 & 4 \\ 6 & 8 & 9 & 4 & 5 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$
- [30] $1^{123}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 11 & 13 & 27 & 7 & 9 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$
- [32] $1^{123}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 8 & 2 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 8 & 13 & 27 & 7 & 9 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$
- [34] $21^{113}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 6 & 2 & 3 \\ 4 & 4 & 9 & 3 & 4 \\ 6 & 9 & 12 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$
- [33] $21^{113}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 12 & 3 & 4 \\ 5 & 4 & 13 & 3 & 4 \\ 12 & 13 & 27 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 4 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$
- [35] $1^{123}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 10 & 13 & 27 & 7 & 9 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$
- [36] $1^{123}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 9 & 2 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 9 & 13 & 27 & 7 & 9 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$
- [37] $1^{133}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -149, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 10 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 10 & 17 & 48 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$
- [38] $21^{133}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 4 & 12 & 3 & 4 \\ 9 & 12 & 24 & 6 & 8 \\ 2 & 3 & 6 & 0 & 2 \\ 3 & 4 & 8 & 2 & 2 \end{pmatrix}_{b_0}$
- [39] $1^{133}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -182, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 11 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 11 & 17 & 48 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$
- [40] $1^{133}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -194, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 13 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 13 & 17 & 48 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{13}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -197, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 2 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 12 & 17 & 48 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[42] $21^{12}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 2 & 3 \\ 4 & 4 & 13 & 3 & 4 \\ 9 & 13 & 27 & 7 & 9 \\ 2 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[43] $21^{13}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -314, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 12 & 2 & 3 \\ 4 & 4 & 17 & 3 & 4 \\ 12 & 17 & 48 & 9 & 12 \\ 2 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

$T = 8$ [44] $1^{13}3^{11}21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -68, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

[45] $31^{11}31^{13}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 4 & 7 & 3 & 0 & 4 \\ 3 & 7 & 3 & 3 & 0 & 3 \\ 3 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 3 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

[46] $31^{11}31^{13}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 138, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 4 & 4 & 6 & 0 & 4 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 6 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 2 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

[47] $31^{11}31^{13}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 5 & 4 & 5 & 3 \\ 5 & 8 & 6 & 10 & 6 \\ 4 & 6 & 1 & 5 & 3 \\ 5 & 10 & 5 & 7 & 5 \\ 3 & 6 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

[48] $31^{11}31^{13}3^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 & 2 \\ 3 & 4 & 4 & 6 & 3 & 4 \\ 3 & 4 & 0 & 3 & 2 & 2 \\ 3 & 6 & 3 & 3 & 3 & 3 \\ 2 & 3 & 2 & 3 & 0 & 2 \\ 2 & 4 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [49] $1^{14}413^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -65, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 4 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

[50] $1^{14}413^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -65, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[51] \ 1^{\overline{1}} \overline{1}^{\overline{1}} 4 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -65, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 1 & 0 & 3 \\ 1 & 0 & -2 & 0 \\ 4 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [52] \ 1^{\overline{1}} \overline{1}^{\overline{1}} 5 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 3 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[53] \ 1^{\overline{1}} \overline{1}^{\overline{1}} 5 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 2 & 3 \\ 3 & 3 & 1 & 3 & 3 \\ 3 & 2 & 3 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[54] \ 21^{\overline{1}} \overline{1}^{\overline{1}} 5 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 2 & 3 \\ 3 & 3 & 1 & 3 & 3 \\ 3 & 2 & 3 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[55] \ 21^{\overline{1}} \overline{1}^{\overline{1}} 5 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 5 & 6 & 3 & 4 \\ 5 & 5 & 8 & 4 & 5 \\ 6 & 8 & 8 & 5 & 6 \\ 3 & 4 & 5 & 1 & 3 \\ 4 & 5 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[56] \ 1^{\overline{1}} \overline{1}^{\overline{1}} 5 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{\overline{1}} \overline{1}^{\overline{1}} 5 123^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 2 & 2 \\ 1 & 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & -1 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{\overline{1}} \overline{1}^{\overline{1}} 4 13^{\overline{1}} 21^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 11, \Delta = -38, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 4 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{\overline{1}} \overline{1}^{\overline{1}} 5 123^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 5 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{\overline{1}} \overline{1}^{\overline{1}} 5 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 5 & 3 & 4 & 5 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 \\ 3 & 5 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{\overline{1}} \overline{5} 123^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 21^{\overline{1}} \overline{5} 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{\overline{1}} \overline{5} 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{\overline{1}} \overline{5} 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 5 & 8 & 4 & 5 \\ 5 & 8 & 8 & 5 & 6 \\ 2 & 4 & 5 & 1 & 3 \\ 3 & 5 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[64] \quad 21^{\overline{1}} \overline{5} 13^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 3 \\ 3 & 5 & 8 & 4 & 5 \\ 5 & 8 & 8 & 5 & 6 \\ 2 & 4 & 5 & 1 & 3 \\ 3 & 5 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [67] \quad 1^{\overline{1}} \overline{4} 1413^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 3 & 0 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{\overline{1}} \overline{4} 1413^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -35, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 2 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{\overline{1}} \overline{6} 123^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 2 & 3 \\ 3 & 3 & 1 & 3 & 3 \\ 3 & 2 & 3 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{\overline{1}} \overline{4} 13^{\overline{1}} 1^{\overline{1}} 51^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 6 & 3 & 6 \\ 3 & 1 & 3 \\ 6 & 3 & 6 \end{pmatrix}_{b_0}$$

$$[68] \ 21^1 6 123^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 13, \ \Delta = -59, \ \det = -3, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[71] \ 1^1 3^1 15 131^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 12, \ \Delta = -39, \ \det = -8, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 3 \\ 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 3 & 2 & -1 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[73] \ 1^1 413^1 15 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 13, \ \Delta = -59, \ \det = -12, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 0 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 & 2 \\ 2 & 2 & 0 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[72] \ 1^1 1^1 413^1 15 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 13, \ \Delta = -59, \ \det = -12, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[74] \ 1^1 413^1 15 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 13, \ \Delta = -59, \ \det = -13, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.85 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 \\ 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 1 & 0 & 0 & 2 \\ 3 & 2 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[76] \ 1^1 6 123^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 13, \ \Delta = -59, \ \det = -13, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.85 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[75] \ 21^1 6 123^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 11, \ T_{\max} = 13, \ \Delta = -59, \ \det = -13, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.85 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 1 & 3 \\ 4 & 8 & 1 & 0 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 6 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [78] \ 1^1 6 1^1 23^1 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 14, \ \Delta = -58, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 & 2 & 2 \\ 4 & 26 & 1 & 14 & 12 & 10 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 14 & 0 & 7 & 6 & 5 \\ 2 & 12 & 0 & 6 & 4 & 4 \\ 2 & 10 & -1 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[77] \ 21^1 6 1^1 23^1 141^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 14, \ \Delta = -58, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 4 & 12 & 3 & 6 & 4 \\ 12 & 34 & 10 & 18 & 12 \\ 3 & 10 & 1 & 5 & 3 \\ 6 & 18 & 5 & 8 & 6 \\ 4 & 12 & 3 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[79] \ 1^{\overline{1}^1 1^3 1^{21}} 6^{\overline{1}^1} 131^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 13, \Delta = -38, \det = 8, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 4 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[80] \ 1^{\overline{1}^1} 6^{\overline{1}^1} 123^{\overline{1}^1} 141^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 4 & 4 \\ 2 & 3 & -1 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[81] \ 1^{\overline{1}^1} 4141^{\overline{1}^1} 13^{\overline{1}^1} 151^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 134, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13.99 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 8 & 1 & 4 & 3 \\ 8 & 17 & 0 & 8 & 9 \\ 1 & 0 & -2 & 0 & 0 \\ 4 & 8 & 0 & 2 & 4 \\ 3 & 9 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [83] \ 1^{\overline{1}^1} 5^{\overline{1}^1} 131513^{\overline{1}^1} 1^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 6 & 0 & 5 & 6 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 5 & 0 & 2 & 4 \\ 3 & 6 & -1 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{\overline{1}^1} 5^{\overline{1}^1} 131513^{\overline{1}^1} 1^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 14, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 3 & 0 & 1 & 3 \\ 2 & 2 & 3 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{\overline{1}^1} 6^{\overline{1}^1} 131513^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -22, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[85] \ 1^{\overline{1}^1} 6^{\overline{1}^1} 1231513^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 13, T_{\max} = 15, \Delta = -60, \det = -108, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 9 & 3 & 1 & 3 \\ 9 & 34 & 10 & 0 & 12 \\ 3 & 10 & 1 & 0 & 3 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 12 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [86] \ 1^{\overline{1}^1} 3^{\overline{1}^1} 21613151^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 2 & 3 & 0 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[87] \ 1^{\overline{1}^1} 6^{\overline{1}^1} 131513^{\overline{1}^1} 1^{\overline{1}^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) \quad T = 14, T_{\max} = 15, \Delta = -30, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 2 & 4 & 4 \\ 2 & 2 & 4 & 2 & 4 \\ 3 & 3 & 4 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^1 3^1 21613151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 14, \quad T_{\max} = 15, \quad \Delta = -32, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 0 & 2 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 3 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^1 61231513^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 14, \quad T_{\max} = 15, \quad \Delta = -31, \quad \det = 2, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 2 & 4 \\ 3 & 2 & 1 & 4 \\ 2 & 1 & 0 & 2 \\ 4 & 4 & 2 & 6 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [90] \quad 1^1 3^1 217123151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 15, \quad T_{\max} = 16, \quad \Delta = -31, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 15, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 16, \quad T_{\max} = 16, \quad T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 & 2 \\ 2 & 3 & 0 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^1 3^1 217123151^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 15, \quad T_{\max} = 16, \quad \Delta = -31, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 15, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 16, \quad T_{\max} = 16, \quad T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 0 & 2 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 0 & 2 \\ 3 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0}$$

4.1179 $\text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4$

$$T = 8 \quad [0] \quad 1^1 1^1 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 72, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.11 \rightarrow 13, \quad T_{\max} = 35, \quad T \in [13, 35]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 1^1 2^1 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 98, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.22 \rightarrow 13, \quad T_{\max} = 35, \quad T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 8 & 3 & 0 & 4 \\ 4 & 8 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 118, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.19 \rightarrow 13, \quad T_{\max} = 35, \quad T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^1 1^1 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, \quad T_{\max} = 35, \quad \Delta = -118, \quad \det = 128, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, \quad T_{\max} = 35, \quad T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, \ T_{\max} = 35, \ \Delta = -118, \ \det = 208, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 35, \ T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 5 & 9 & 12 & 5 & 0 & 6 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{12}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, \ T_{\max} = 35, \ \Delta = -118, \ \det = 278, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.91 \rightarrow 12, \ T_{\max} = 35, \ T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 13 & 3 & 0 & 4 \\ 7 & 13 & 27 & 7 & 0 & 9 \\ 2 & 3 & 7 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{13}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, \ T_{\max} = 35, \ \Delta = -118, \ \det = 328, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 35, \ T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 17 & 3 & 0 & 4 \\ 9 & 17 & 48 & 9 & 0 & 12 \\ 2 & 3 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 12 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

4.1180 $\text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{so}}_8$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -392, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.92 \rightarrow 13, \ T_{\max} = 32, \ T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 9 & 6 & 1 & 5 \\ 11 & 20 & 17 & 11 & 0 & 8 \\ 9 & 17 & 12 & 9 & 0 & 6 \\ 6 & 11 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

4.1181 $\text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -96, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.17 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 4 & 3 \\ 3 & 4 & 9 & 3 & 0 & 8 & 4 \\ 5 & 9 & 12 & 5 & 0 & 12 & 6 \\ 2 & 3 & 5 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 12 & 4 & 0 & 10 & 6 \\ 3 & 4 & 6 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 2 & 0 & 3 & 2 \\ 2 & 4 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 3 & 0 & 4 & 4 \\ 3 & 5 & 2 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -136, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 3 & 0 & 8 & 4 \\ 2 & 4 & 0 & 2 & 0 & 6 & 2 \\ 2 & 3 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 4 & 0 & 10 & 6 \\ 3 & 4 & 2 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -136, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 3 & 0 & 14 & 4 \\ 2 & 4 & 0 & 2 & 0 & 9 & 2 \\ 2 & 3 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 8 & 0 & 37 & 11 \\ 3 & 4 & 2 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 0 & 2 & 0 & 6 & 2 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 5 & 0 & 13 & 7 \\ 3 & 4 & 2 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 2 & 3 \\ 3 & 4 & 13 & 3 & 0 & 4 & 4 \\ 7 & 13 & 27 & 7 & 0 & 9 & 9 \\ 2 & 3 & 7 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 9 & 2 & 0 & 1 & 3 \\ 3 & 4 & 9 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -626, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 7 & 3 \\ 3 & 4 & 4 & 3 & 0 & 13 & 4 \\ 2 & 4 & 0 & 2 & 0 & 9 & 2 \\ 2 & 3 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 9 & 7 & 0 & 28 & 10 \\ 3 & 4 & 2 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 9 & 3 \\ 3 & 4 & 4 & 3 & 0 & 17 & 4 \\ 2 & 4 & 0 & 2 & 0 & 12 & 2 \\ 2 & 3 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 12 & 9 & 0 & 49 & 13 \\ 3 & 4 & 2 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{112} 3^{11} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 16 & 3 \\ 3 & 4 & 4 & 3 & 0 & 28 & 4 \\ 2 & 4 & 0 & 2 & 0 & 18 & 2 \\ 2 & 3 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 16 & 28 & 18 & 16 & 0 & 154 & 22 \\ 3 & 4 & 2 & 2 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{114} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -866, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.03 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 11 & 3 \\ 3 & 4 & 4 & 3 & 0 & 21 & 4 \\ 2 & 4 & 0 & 2 & 0 & 15 & 2 \\ 2 & 3 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 15 & 11 & 0 & 76 & 16 \\ 3 & 4 & 2 & 2 & -3 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{112} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 4 & 3 & 0 & 18 & 4 \\ 2 & 4 & 0 & 2 & 0 & 12 & 2 \\ 2 & 3 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 12 & 10 & 0 & 58 & 14 \\ 3 & 4 & 2 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{112} 3^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -946, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 13 & 3 \\ 3 & 4 & 4 & 3 & 0 & 23 & 4 \\ 2 & 4 & 0 & 2 & 0 & 15 & 2 \\ 2 & 3 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 15 & 13 & 0 & 100 & 18 \\ 3 & 4 & 2 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{113}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 12 & 3 \\ 3 & 4 & 4 & 3 & 0 & 22 & 4 \\ 2 & 4 & 0 & 2 & 0 & 15 & 2 \\ 2 & 3 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 15 & 12 & 0 & 85 & 17 \\ 3 & 4 & 2 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{113}3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 15 & 3 \\ 3 & 4 & 4 & 3 & 0 & 27 & 4 \\ 2 & 4 & 0 & 2 & 0 & 18 & 2 \\ 2 & 3 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 18 & 15 & 0 & 133 & 21 \\ 3 & 4 & 2 & 2 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{114}3^{112} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 14 & 3 \\ 3 & 4 & 4 & 3 & 0 & 26 & 4 \\ 2 & 4 & 0 & 2 & 0 & 18 & 2 \\ 2 & 3 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 18 & 14 & 0 & 118 & 20 \\ 3 & 4 & 2 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{113}3^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1826, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 18 & 3 \\ 3 & 4 & 4 & 3 & 0 & 32 & 4 \\ 2 & 4 & 0 & 2 & 0 & 21 & 2 \\ 2 & 3 & 2 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 21 & 18 & 0 & 193 & 25 \\ 3 & 4 & 2 & 2 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{114}3^{113} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2386, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 17 & 3 \\ 3 & 4 & 4 & 3 & 0 & 31 & 4 \\ 2 & 4 & 0 & 2 & 0 & 21 & 2 \\ 2 & 3 & 2 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 21 & 17 & 0 & 172 & 24 \\ 3 & 4 & 2 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{114}3^{114} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 20 & 3 \\ 3 & 4 & 4 & 3 & 0 & 36 & 4 \\ 2 & 4 & 0 & 2 & 0 & 24 & 2 \\ 2 & 3 & 2 & 0 & 0 & 20 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 24 & 20 & 0 & 238 & 28 \\ 3 & 4 & 2 & 2 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

4.1182 $\text{NHC}(-2)(-3) + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$

$$T = 10 \quad [0] \quad 1^{113}3^{1111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 4 & 3 & 0 & 5 & 5 & 4 \\ 2 & 4 & 0 & 2 & 0 & 3 & 3 & 2 \\ 2 & 3 & 2 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 3 & 0 & 4 & 6 & 4 \\ 3 & 5 & 3 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & 2 & 2 & -3 & 4 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 392, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 4 & 4 \\ 2 & 4 & 0 & 2 & 0 & 3 & 3 & 2 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 2 & 0 & 3 & 1 & 3 \\ 3 & 4 & 2 & 2 & -3 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 462, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 3 & 0 & 5 & 9 & 4 \\ 2 & 4 & 0 & 2 & 0 & 3 & 6 & 2 \\ 2 & 3 & 2 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 3 & 3 & 0 & 4 & 9 & 4 \\ 5 & 9 & 6 & 5 & 0 & 9 & 13 & 7 \\ 3 & 4 & 2 & 2 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 512, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 & 4 & 4 \\ 5 & 9 & 12 & 5 & 0 & 6 & 6 & 6 \\ 2 & 3 & 5 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 6 & 2 & 0 & 1 & 3 & 3 \\ 2 & 4 & 6 & 2 & 0 & 3 & 1 & 3 \\ 3 & 4 & 6 & 2 & -3 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 622, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 9 & 4 \\ 2 & 4 & 0 & 2 & 0 & 3 & 6 & 2 \\ 2 & 3 & 2 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 5 & 0 & 6 & 13 & 7 \\ 3 & 4 & 2 & 2 & -3 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 1152, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.94 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 4 & 3 & 0 & 9 & 9 & 4 \\ 2 & 4 & 0 & 2 & 0 & 6 & 6 & 2 \\ 2 & 3 & 2 & 0 & 0 & 5 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 6 & 5 & 0 & 13 & 15 & 7 \\ 5 & 9 & 6 & 5 & 0 & 15 & 13 & 7 \\ 3 & 4 & 2 & 2 & -3 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

4.1183 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{123}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -34, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 & 3 \\ 3 & 4 & 8 & 3 & 0 & 5 & 4 \\ 4 & 8 & 9 & 4 & 0 & 6 & 5 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 6 & 3 & 0 & 3 & 3 \\ 3 & 4 & 5 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -36, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.33 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 4 & 0 & 4 & 5 \\ 2 & 4 & 0 & 2 & 0 & 3 & 2 \\ 2 & 4 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 3 & 5 & 2 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -274, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.47 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 4 \\ 2 & 4 & 0 & 2 & 0 & 3 & 2 \\ 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 2 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 3 & 0 & 9 & 4 \\ 2 & 4 & 0 & 2 & 0 & 6 & 2 \\ 2 & 3 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 5 & 0 & 12 & 6 \\ 3 & 4 & 2 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

4.1184 NHC(-2)(-3) + $\mathbf{su}_3$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{so}_8$

$$T = 8 \quad [0] \ 1^{111}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 8 & 3 & 0 & 4 \\ 4 & 8 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 10 & 6 & 5 \\ 11 & 20 & 17 & 11 & 8 \\ 10 & 17 & 12 & 9 & 6 \\ 6 & 11 & 9 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 1 & 6 & 5 \\ 11 & 20 & 0 & 11 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 11 & 0 & 4 & 4 \\ 5 & 8 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{111} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{11}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 11 & 9 & 6 & 5 \\ 11 & 20 & 17 & 11 & 8 \\ 9 & 17 & 12 & 9 & 6 \\ 6 & 11 & 9 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[7] $1^{11}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 12 & 8 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 12 & 17 & 12 & 9 & 6 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[8] $21^{11}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 121, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.11 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 15 & 9 & 7 \\ 17 & 20 & 17 & 11 & 8 \\ 15 & 17 & 12 & 9 & 6 \\ 9 & 11 & 9 & 4 & 4 \\ 7 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{11}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 5 & 9 & 12 & 5 & 0 & 6 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[10] $21^{11}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 16 & 12 & 8 & 6 \\ 16 & 20 & 17 & 11 & 8 \\ 12 & 17 & 12 & 9 & 6 \\ 8 & 11 & 9 & 4 & 4 \\ 6 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$$

[11] $1^{12}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 201, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 7 & 2 & 1 & 3 \\ 3 & 4 & 13 & 3 & 0 & 4 \\ 7 & 13 & 27 & 7 & 0 & 9 \\ 2 & 3 & 7 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{12}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 230, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 9 & 2 & 1 & 3 \\ 3 & 4 & 17 & 3 & 0 & 4 \\ 9 & 17 & 48 & 9 & 0 & 12 \\ 2 & 3 & 9 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 12 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

4.1185 $\text{NHC}(-2)(-3) + \textcolor{blue}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{f}}_4$

$T = 9 \quad [0] 1^{11}3^{11}1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{f}}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 9 & 6 & 1 & 5 \\ 11 & 20 & 17 & 11 & 0 & 8 \\ 9 & 17 & 12 & 9 & 0 & 6 \\ 6 & 11 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 6 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

4.1186 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 9 & 6 & 1 & 5 \\ 11 & 20 & 17 & 11 & 0 & 8 \\ 9 & 17 & 12 & 9 & 0 & 6 \\ 6 & 11 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

4.1187 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 3 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & -1 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -65, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 3 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & -1 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \quad 1^{11} 4^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 & 3 \\ 2 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 4 \\ 3 & 3 & 0 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 4^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 4 & 1 & 0 & 2 & 4 \\ 3 & 4 & 0 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 4^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 2 & 3 \\ 2 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 3 \\ 3 & 3 & 0 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 4^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 11, (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 1 & 2 & 4 \\ 3 & 1 & 0 & 1 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 2 & 1 & 0 & 0 & 2 \\ 4 & 3 & 0 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} 1^4 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -35, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 1 & 5 & 4 \\ 3 & 1 & 0 & 4 & 3 \\ 1 & 0 & -2 & 0 & 0 \\ 5 & 4 & 0 & 6 & 6 \\ 4 & 3 & 0 & 6 & 5 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [9] \quad 1^{11} 5^{11} 123^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 1 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 2 & 4 \\ 1 & 1 & 0 & 2 & 0 & 3 \\ 2 & 3 & 0 & 4 & 3 & 5 \\ 2 & 2 & -1 & 2 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 5^{13} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 2 & 3 & 1 & 5 & 3 \\ 7 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11} 5^{13} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 21^{11} 5^{13} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{11} 5^{13} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 4 & 3 \\ 3 & 4 & 3 & 3 & 7 & 4 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 3 & 3 & 1 & 5 & 3 \\ 4 & 7 & 6 & 5 & 9 & 7 \\ 3 & 4 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 5^{13} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 7 & 3 \\ 3 & 4 & 3 & 3 & 7 & 4 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 3 & 3 & 1 & 5 & 3 \\ 7 & 7 & 6 & 5 & 9 & 7 \\ 3 & 4 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11} 5^{11} 1^{11} 23^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 2 & 8 & 2 \\ 1 & 1 & 0 & 2 & 0 & 7 & 2 \\ 5 & 6 & 0 & 8 & 7 & 31 & 11 \\ 2 & 2 & -1 & 2 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1^2}_1 5 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 3 & 4 & 3 \\ 2 & 2 & 3 & 1 & 3 & 3 \\ 4 & 2 & 4 & 3 & 2 & 4 \\ 3 & 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{1^1}_1 5 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 1 & 3 & 0 & 3 \\ 2 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1^2}_1 5 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -4, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 9 & 3 \\ 3 & 4 & 3 & 3 & 14 & 4 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 3 & 1 & 10 & 3 \\ 9 & 14 & 12 & 10 & 42 & 14 \\ 3 & 4 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1^2}_1 5 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 3 & 14 & 4 \\ 3 & 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 3 & 1 & 10 & 3 \\ 8 & 14 & 12 & 10 & 42 & 14 \\ 3 & 4 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1^3}_1 5 13^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 3 & 21 & 4 \\ 3 & 3 & 1 & 3 & 18 & 3 \\ 2 & 3 & 3 & 1 & 15 & 3 \\ 12 & 21 & 18 & 15 & 97 & 21 \\ 3 & 4 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1^1}_1 5 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -6, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 4 & 3 \\ 3 & 4 & 3 & 3 & 7 & 4 \\ 3 & 3 & 1 & 3 & 6 & 3 \\ 2 & 3 & 3 & 1 & 5 & 3 \\ 4 & 7 & 6 & 5 & 9 & 7 \\ 3 & 4 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1^2}_1 5 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -7, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.86 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 4 & 3 \\ 3 & 5 & 3 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 & 3 \\ 4 & 5 & 2 & 3 & 2 & 4 \\ 3 & 5 & 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1^1}_1 5 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -7, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.86 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 5 & 3 & 4 & 0 & 5 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 2 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[25] $1^{11} \overset{1^2}{5} 1^3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 3 \\ 3 & 4 & 3 & 3 & 7 & 4 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 3 & 3 & 1 & 5 & 3 \\ 6 & 7 & 6 & 5 & 9 & 7 \\ 3 & 4 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[26] $1^{13} \overset{1^2}{5} 1^3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 14 & 3 \\ 3 & 4 & 3 & 3 & 21 & 4 \\ 2 & 3 & 1 & 3 & 18 & 3 \\ 2 & 3 & 3 & 1 & 15 & 3 \\ 14 & 21 & 18 & 15 & 97 & 21 \\ 3 & 4 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[24] $1^{11} \overset{1^1}{5} 1^3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 4 & 3 & 3 & 7 & 4 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 3 & 3 & 1 & 5 & 3 \\ 5 & 7 & 6 & 5 & 9 & 7 \\ 3 & 4 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[32] $1^{11} \overset{1^1}{5} 1^3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 5 & 0 & 3 & 2 & 4 \\ 3 & 5 & -1 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[31] $1^{11} \overset{1^1}{5} 1^3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 3 \\ 3 & 5 & 3 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 & 3 \\ 2 & 5 & 2 & 3 & 2 & 4 \\ 3 & 5 & 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[30] $1^{11} \overset{1^3}{5} 1^3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 7 & 3 \\ 3 & 5 & 3 & 4 & 10 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 2 & 4 & 2 & 1 & 6 & 3 \\ 7 & 10 & 4 & 6 & 14 & 8 \\ 3 & 5 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[29] $1^{11} \overset{1^1}{5} 1^3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 & 2 & 1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 0 & 0 & 1 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

[28] $1^{11} \overset{1^1}{5} 1^3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 2 & 3 & 3 \\ 2 & 3 & 1 & 3 & 2 & 3 \\ 2 & 2 & 3 & 1 & 2 & 3 \\ 2 & 3 & 2 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[27] $1^{11} 5^{11} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 2 & 0 & 3 \\ 2 & 3 & 1 & 3 & 0 & 3 \\ 2 & 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[33] $1^{12} 5^{12} 13^{12} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 10 & 3 \\ 3 & 4 & 3 & 3 & 14 & 4 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 3 & 1 & 10 & 3 \\ 10 & 14 & 12 & 10 & 42 & 14 \\ 3 & 4 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[34] $1^{12} 5^{12} 13^{12} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 12 & 3 \\ 3 & 4 & 3 & 3 & 14 & 4 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 3 & 1 & 10 & 3 \\ 12 & 14 & 12 & 10 & 42 & 14 \\ 3 & 4 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[35] $1^{11} 5^{11} 13^{11} 23^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -32, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 1 & 3 & 2 \\ 1 & 2 & 0 & 0 & 2 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 0 & 0 & 4 & 2 \\ 3 & 3 & 4 & 0 & 4 & 10 & 6 \\ 2 & 2 & 2 & -1 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[40] $1^{11} 5^{11} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 5 & 8 & 4 & 5 & 5 \\ 5 & 8 & 8 & 5 & 6 & 6 \\ 2 & 4 & 5 & 1 & 3 & 3 \\ 2 & 5 & 6 & 3 & 2 & 4 \\ 3 & 5 & 6 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[36] $21^{11} 5^{11} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 2 & 2 & 3 \\ 3 & 5 & 8 & 4 & 5 & 5 \\ 5 & 8 & 8 & 5 & 6 & 6 \\ 2 & 4 & 5 & 1 & 3 & 3 \\ 2 & 5 & 6 & 3 & 2 & 4 \\ 3 & 5 & 6 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[41] $1^{14} 5^{14} 13^{14} 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 10 & 3 \\ 3 & 5 & 3 & 4 & 15 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 2 & 4 & 2 & 1 & 9 & 3 \\ 10 & 15 & 6 & 9 & 34 & 12 \\ 3 & 5 & 2 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[37] $1^{12} 5^{12} 13^{12} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 8 & 3 \\ 2 & 2 & 3 & 1 & 7 & 3 \\ 9 & 8 & 8 & 7 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[38] $1^{11} 5^{11} 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 3 & 4 & 3 \\ 2 & 2 & 3 & 1 & 3 & 3 \\ 3 & 2 & 4 & 3 & 2 & 4 \\ 3 & 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1^1} 5^{1^1} 13^{1^1} 1^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 5 & 3 & 4 & 10 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 2 & 4 & 2 & 1 & 6 & 3 \\ 5 & 10 & 4 & 6 & 14 & 8 \\ 3 & 5 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1^3} 5^{1^3} 13^{1^1} 1^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 3 \\ 3 & 4 & 3 & 3 & 14 & 4 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 3 & 1 & 10 & 3 \\ 11 & 14 & 12 & 10 & 42 & 14 \\ 3 & 4 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1^3} 5^{1^3} 13^{1^1} 1^{1^3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -26, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 15 & 3 \\ 3 & 4 & 3 & 3 & 21 & 4 \\ 2 & 3 & 1 & 3 & 18 & 3 \\ 2 & 3 & 3 & 1 & 15 & 3 \\ 15 & 21 & 18 & 15 & 97 & 21 \\ 3 & 4 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1^1} 5^{1^1} 13^{1^1} 1^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 \\ 3 & 5 & 3 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 & 3 \\ 3 & 5 & 2 & 3 & 2 & 4 \\ 3 & 5 & 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1^4} 5^{1^4} 13^{1^1} 1^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 3 & 10 & 3 \\ 2 & 2 & 3 & 1 & 8 & 3 \\ 10 & 7 & 10 & 8 & 25 & 11 \\ 3 & 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1^2} 5^{1^2} 13^{1^1} 1^{1^3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 \\ 3 & 5 & 3 & 4 & 15 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 2 & 4 & 2 & 1 & 9 & 3 \\ 8 & 15 & 6 & 9 & 34 & 12 \\ 3 & 5 & 2 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1^2} 5^{1^2} 13^{1^1} 1^{1^2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -34, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 6 & 3 \\ 3 & 5 & 3 & 4 & 10 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 2 & 4 & 2 & 1 & 6 & 3 \\ 6 & 10 & 4 & 6 & 14 & 8 \\ 3 & 5 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1^4} 5^{1^4} 13^{1^1} 1^{1^3} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -34, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 16 & 3 \\ 3 & 4 & 3 & 3 & 21 & 4 \\ 2 & 3 & 1 & 3 & 18 & 3 \\ 2 & 3 & 3 & 1 & 15 & 3 \\ 16 & 21 & 18 & 15 & 97 & 21 \\ 3 & 4 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1^3} 5^{1^3} 13^{1^3} 1^{1^3} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -44, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.91 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 9 & 3 \\ 3 & 5 & 3 & 4 & 15 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 2 & 4 & 2 & 1 & 9 & 3 \\ 9 & 15 & 6 & 9 & 34 & 12 \\ 3 & 5 & 2 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1^2} 5^{1^2} 13^{1^2} 1^{1^2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -45, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 6 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 2 & 3 & 1 & 5 & 3 \\ 6 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[50] \quad 21^{1^1} 5^{1^1} 13^{1^1} 1^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -45, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.89 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 \\ 3 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 2 & 3 & 1 & 5 & 3 \\ 5 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1^1} 5^{1^1} 1^{1^1} 23^{1^1} 1^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -32, \quad \det = -47, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 1 & 5 & 2 \\ 1 & 2 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 0 & 0 & 5 & 2 \\ 4 & 5 & 5 & 0 & 5 & 18 & 8 \\ 2 & 2 & 2 & -1 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1^3} 5^{1^3} 13^{1^3} 1^{1^3} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -47, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 11 & 3 \\ 3 & 5 & 3 & 4 & 20 & 5 \\ 2 & 3 & 0 & 2 & 8 & 2 \\ 2 & 4 & 2 & 1 & 12 & 3 \\ 11 & 20 & 8 & 12 & 62 & 16 \\ 3 & 5 & 2 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1^4} 5^{1^4} 13^{1^4} 1^{1^4} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -58, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 12 & 3 \\ 3 & 5 & 3 & 4 & 20 & 5 \\ 2 & 3 & 0 & 2 & 8 & 2 \\ 2 & 4 & 2 & 1 & 12 & 3 \\ 12 & 20 & 8 & 12 & 62 & 16 \\ 3 & 5 & 2 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1^2} 5^{1^2} 13^{1^2} 1^{1^2} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -58, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 8 & 3 \\ 2 & 2 & 3 & 1 & 7 & 3 \\ 8 & 8 & 8 & 7 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1^1} 5^{1^1} 13^{1^1} 1^{1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 11, \quad T_{\max} = 12, \quad \Delta = -30, \quad \det = -58, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.93 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 2 & 3 & 1 & 5 & 3 \\ 5 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[59] \quad & 1^1 5^1 1^1 2^3 1^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -67, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 8 & 3 \\ 3 & 5 & 0 & 4 & 15 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 10 & 3 \\ 8 & 15 & 0 & 10 & 37 & 13 \\ 3 & 5 & -1 & 3 & 13 & 4 \end{pmatrix}_{b_0} \\
[57] \quad & 21^{12} 5^1 1^1 1^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -67, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 8 & 3 \\ 3 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 8 & 3 \\ 2 & 2 & 3 & 1 & 7 & 3 \\ 8 & 8 & 8 & 7 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0} \\
[58] \quad & 1^{12} 5^1 1^1 1^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -67, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 8 & 3 \\ 2 & 2 & 3 & 1 & 7 & 3 \\ 7 & 8 & 8 & 7 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0} \\
[60] \quad & 1^{11} 5^1 1^1 1^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 3 & 10 & 3 \\ 2 & 2 & 3 & 1 & 8 & 3 \\ 9 & 7 & 10 & 8 & 25 & 11 \\ 3 & 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0} \\
[61] \quad & 1^{11} 5^1 1^1 1^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -91, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 3 & 10 & 3 \\ 2 & 2 & 3 & 1 & 8 & 3 \\ 8 & 7 & 10 & 8 & 25 & 11 \\ 3 & 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0} \\
[62] \quad & 1^1 5^1 1^1 2^3 1^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -98, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 5 & 0 & 4 & 20 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 13 & 3 \\ 10 & 20 & 0 & 13 & 63 & 17 \\ 3 & 5 & -1 & 3 & 17 & 4 \end{pmatrix}_{b_0} \\
[64] \quad & 21^{12} 5^1 1^1 1^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -102, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 2 & 10 & 3 \\ 3 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 2 & 2 & 3 & 1 & 10 & 3 \\ 10 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0} \\
[63] \quad & 1^{12} 5^1 1^1 1^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -102, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 2 & 2 & 3 & 1 & 10 & 3 \\ 12 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0} \\
[65] \quad & 1^{13} 5^1 1^1 1^1 1^1 + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -143, (n_+, n_-, n_0) = \\
& (1, 11, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

[74] $1^{13} \overset{1^4}{5} 1^3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -305, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 2 & 15 & 3 \\ 2 & 3 & 1 & 3 & 18 & 3 \\ 2 & 2 & 3 & 1 & 15 & 3 \\ 16 & 15 & 18 & 15 & 97 & 21 \\ 3 & 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[75] $1^{13} \overset{1^3}{5} 1^3 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -314, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 2 & 15 & 3 \\ 2 & 3 & 1 & 3 & 18 & 3 \\ 2 & 2 & 3 & 1 & 15 & 3 \\ 15 & 15 & 18 & 15 & 97 & 21 \\ 3 & 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[76] $1^{13} \overset{1^4}{5} 1^3 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -427, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 12.00 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 2 & 17 & 3 \\ 2 & 3 & 1 & 3 & 22 & 3 \\ 2 & 2 & 3 & 1 & 18 & 3 \\ 18 & 17 & 22 & 18 & 137 & 25 \\ 3 & 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

$T = 12$ [84] $1^{11} 4 1^3 1^5 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 3 & 3 \\ 2 & 0 & 0 & 1 & 3 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 2 \\ 3 & 3 & 0 & 1 & 2 & 4 \\ 3 & 2 & 0 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

[85] $1^{12} \overset{1^2}{6} 1^3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 4 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 3 & 4 & 3 \\ 2 & 2 & 3 & 1 & 3 & 3 \\ 4 & 2 & 4 & 3 & 2 & 4 \\ 3 & 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[78] $2 1^{\overset{1^3}{1}} 6 1^{\overset{1^1}{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 2 & 0 & 3 & 0 & 4 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 2 & 0 & 4 & 0 & 4 & 4 \\ 2 & 2 & 0 & 3 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

[79] $1^{11} \overset{1^1}{6} 1^3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 2 & 1 & 3 \\ 4 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[80] $1^{\overset{1^1}{1}} \overset{1^1}{6} 1^{\overset{1^1}{1}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 9 & 4 \\ 3 & 1 & 3 & 2 & 6 & 3 \\ 3 & 3 & 1 & 3 & 7 & 3 \\ 3 & 2 & 3 & 1 & 7 & 3 \\ 9 & 6 & 7 & 7 & 19 & 9 \\ 4 & 3 & 3 & 3 & 9 & 4 \end{pmatrix}_{b_0}$$

[81] $1^{11} 4 1^3 1^5 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) =$

$$(1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 6 & 3 & 7 & 6 \\ 3 & 1 & 3 & 3 \\ 7 & 3 & 7 & 7 \\ 6 & 3 & 7 & 6 \end{pmatrix}_{b_0}$$

$$[82] \ 1^{11}413^{11}51^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 1 & 3 \\ 2 & 0 & 0 & 1 & 2 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 3 & 2 & 0 & 2 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[83] \ 1^{11}6123^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 2 & 3 & 1 & 5 & 3 \\ 7 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[77] \ 21^{11}6123^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 2 & 0 & 3 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 0 & 2 & 0 & 0 & 2 \\ 2 & 2 & 0 & 3 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[86] \ 21^{11}3^{11}21^{11}61^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 0 & 1 & 2 & 0 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 2 & 0 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 4 & 4 & 0 & 6 & 0 & 10 & 6 \\ 2 & 2 & 0 & 3 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[88] \ 1^{11}41^{11}3^{11}151^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 6 & 3 & 9 & 6 \\ 3 & 1 & 5 & 3 \\ 9 & 5 & 11 & 9 \\ 6 & 3 & 9 & 6 \end{pmatrix}_{b_0}$$

$$[87] \ 1^{11}6^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 4 & 3 & 3 & 3 & 12 & 4 \\ 3 & 1 & 3 & 2 & 9 & 3 \\ 3 & 3 & 1 & 3 & 9 & 3 \\ 3 & 2 & 3 & 1 & 9 & 3 \\ 12 & 9 & 9 & 9 & 34 & 12 \\ 4 & 3 & 3 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[92] \ 1^{11}6^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 2 & 6 & 3 \\ 2 & 3 & 1 & 3 & 7 & 3 \\ 2 & 2 & 3 & 1 & 7 & 3 \\ 7 & 6 & 7 & 7 & 19 & 9 \\ 3 & 3 & 3 & 3 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[91] \ 1^{11}6^{11}123^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 2 & 0 & 3 \\ 2 & 3 & 1 & 3 & 0 & 3 \\ 2 & 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[90] $1 \overset{1}{\underset{11}{6}} 123 \overset{1}{1} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 2 & 3 & 3 \\ 2 & 3 & 1 & 3 & 2 & 3 \\ 2 & 2 & 3 & 1 & 2 & 3 \\ 2 & 3 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$

[89] $1 \overset{1}{\underset{13^1 21^1}{6}} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 2 & 9 & 3 \\ 2 & 3 & 1 & 3 & 9 & 3 \\ 2 & 2 & 3 & 1 & 9 & 3 \\ 10 & 9 & 9 & 9 & 34 & 12 \\ 3 & 3 & 3 & 3 & 12 & 4 \end{pmatrix}_{b_0}$

[93] $1 \overset{1}{\underset{11}{4}} 1 \overset{1}{\underset{11}{3}} 1 \overset{1}{\underset{11}{5}} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 14, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 6 & 3 & 16 & 6 \\ 3 & 1 & 8 & 3 \\ 16 & 8 & 38 & 16 \\ 6 & 3 & 16 & 6 \end{pmatrix}_{b_0}$

[94] $1 \overset{1}{\underset{11}{2}} 41 \overset{1}{\underset{11}{3}} 1 \overset{1}{\underset{11}{5}} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 14, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 6 & 3 & 23 & 6 \\ 3 & 1 & 11 & 3 \\ 23 & 11 & 83 & 23 \\ 6 & 3 & 23 & 6 \end{pmatrix}_{b_0}$

[95] $1 \overset{1}{\underset{11}{3}} 6 \overset{1}{\underset{11}{1}} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 2 & 6 & 3 \\ 2 & 1 & 3 & 2 & 6 & 3 \\ 2 & 3 & 1 & 3 & 7 & 3 \\ 2 & 2 & 3 & 1 & 7 & 3 \\ 6 & 6 & 7 & 7 & 19 & 9 \\ 3 & 3 & 3 & 3 & 9 & 4 \end{pmatrix}_{b_0}$

[97] $1 \overset{1}{\underset{11}{6}} 123 \overset{1}{1} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 19, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 3 & 4 & 3 \\ 2 & 2 & 3 & 1 & 3 & 3 \\ 3 & 2 & 4 & 3 & 2 & 4 \\ 3 & 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$

[96] $1 \overset{1}{\underset{11}{3}} 6 \overset{1}{\underset{11}{1}} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 19, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 11 & 3 \\ 2 & 2 & 3 & 1 & 10 & 3 \\ 11 & 8 & 11 & 10 & 37 & 13 \\ 3 & 3 & 3 & 3 & 13 & 4 \end{pmatrix}_{b_0}$

[98] $1 \overset{1}{\underset{11}{4}} 1 \overset{1}{\underset{11}{3}} 1 \overset{1}{\underset{11}{5}} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 4 & 2 \\ 1 & 0 & 1 & 2 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 & 1 & 4 & 2 \\ 2 & 1 & 0 & 1 & 0 & 4 & 2 \\ 4 & 4 & 0 & 4 & 4 & 10 & 6 \\ 2 & 2 & 0 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

[99] $1 \overset{1}{\underset{11}{4}} 1 \overset{1}{\underset{11}{3}} 1 \overset{1}{\underset{11}{5}} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 1 & 0 & 1 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 0 & 4 & 2 \\ 4 & 4 & 0 & 0 & 4 & 10 & 6 \\ 2 & 2 & 0 & 0 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[100] \quad 1^{11} \overset{1^4}{6} 123^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 31, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.97 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 3 & 10 & 3 \\ 2 & 2 & 3 & 1 & 8 & 3 \\ 10 & 7 & 10 & 8 & 25 & 11 \\ 3 & 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[101] \quad 21^{11} \overset{1^3 1^{21^1}}{6} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 44, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.91 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 1 & 9 & 3 \\ 4 & 8 & 1 & 0 & 18 & 6 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 18 & 0 & 0 & 34 & 12 \\ 3 & 6 & 0 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[102] \quad 1^{11} \overset{1^1}{6} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 44, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.91 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 2 & 9 & 3 \\ 2 & 3 & 1 & 3 & 9 & 3 \\ 2 & 2 & 3 & 1 & 9 & 3 \\ 9 & 9 & 9 & 3 & 12 & 12 \\ 3 & 3 & 3 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[103] \quad 1^{11} \overset{1^2}{6} 123^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 45, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.89 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 6 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 2 & 3 & 1 & 5 & 3 \\ 6 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[104] \quad 21^{11} \overset{1^1}{6} 123^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 45, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.89 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 1 & 5 & 3 \\ 4 & 8 & 1 & 0 & 10 & 6 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 5 & 10 & 1 & 0 & 9 & 7 \\ 3 & 6 & 0 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[105] \quad 21^{11} \overset{1^{11} 3^1 21^1}{6} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 45, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.89 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 1 & 1 & 10 & 3 \\ 4 & 8 & 1 & 0 & 18 & 6 \\ 1 & 1 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 10 & 18 & 1 & 0 & 35 & 13 \\ 3 & 6 & 0 & 0 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[106] \quad 1^{11} \overset{1^4}{6} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 45, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.89 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 2 & 11 & 3 \\ 2 & 3 & 1 & 3 & 13 & 3 \\ 2 & 2 & 3 & 1 & 12 & 3 \\ 14 & 11 & 13 & 12 & 56 & 16 \\ 3 & 3 & 3 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[107] \quad 1^{11} \overset{1^3 1^{21^1}}{6} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 47, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.98 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 2 & 9 & 3 \\ 2 & 3 & 1 & 3 & 9 & 3 \\ 2 & 2 & 3 & 1 & 9 & 3 \\ 8 & 9 & 9 & 9 & 34 & 12 \\ 3 & 3 & 3 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[108] \quad & 1^{\overline{1}} \overset{1^4}{6} \overset{1}{123} \overset{1}{21} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 50, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 3 & 15 & 3 \\ 2 & 2 & 3 & 1 & 13 & 3 \\ 14 & 10 & 15 & 13 & 63 & 17 \\ 3 & 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0} \\
[111] \quad & 1^{\overline{1}} 41^{\overline{1}} 3^{\overline{1}} 151^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 58, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 2 & 7 & 3 \\ 2 & 0 & 0 & 1 & 4 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 3 & 2 \\ 7 & 4 & 0 & 3 & 12 & 8 \\ 3 & 2 & 0 & 2 & 8 & 4 \end{pmatrix}_{b_0} \\
[112] \quad & 1^{\overline{1}} 41^{\overline{1}} 3^{\overline{1}} 151^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 58, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 2 & 5 & 3 \\ 2 & 0 & 0 & 1 & 3 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 3 & 2 \\ 5 & 3 & 0 & 3 & 6 & 6 \\ 3 & 2 & 0 & 2 & 6 & 4 \end{pmatrix}_{b_0} \\
[109] \quad & 1^{\overline{1}} \overset{1^1}{6} 123^{\overline{1}} 1^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 58, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 2 & 3 & 1 & 5 & 3 \\ 5 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[110] \quad & 1^{\overline{1}} \overset{1^2}{6} \overset{1}{113} \overset{1}{21} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 58, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 11 & 3 \\ 2 & 2 & 3 & 1 & 10 & 3 \\ 10 & 8 & 11 & 10 & 37 & 13 \\ 3 & 3 & 3 & 3 & 13 & 4 \end{pmatrix}_{b_0} \\
[113] \quad & 1^{\overline{1}} \overset{1^1}{6} \overset{1}{113} \overset{1}{21} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 67, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 11 & 3 \\ 2 & 2 & 3 & 1 & 10 & 3 \\ 9 & 8 & 11 & 10 & 37 & 13 \\ 3 & 3 & 3 & 3 & 13 & 4 \end{pmatrix}_{b_0} \\
[115] \quad & 1^{\overline{1}} 41^{\overline{1}} 3^{\overline{1}} 151^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 76, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 2 & 6 & 3 \\ 2 & 0 & 0 & 1 & 5 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 4 & 2 \\ 6 & 5 & 0 & 4 & 12 & 8 \\ 3 & 2 & 0 & 2 & 8 & 4 \end{pmatrix}_{b_0} \\
[116] \quad & 1^{\overline{1}} 41^{\overline{1}} 3^{\overline{1}} 151^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 76, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 1 & 2 & 10 & 3 \\ 2 & 0 & 0 & 1 & 7 & 2 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 4 & 2 \\ 10 & 7 & 0 & 4 & 28 & 12 \\ 3 & 2 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0} \\
[114] \quad & 1^{\overline{1}} \overset{1^3}{6} 123^{\overline{1}} 1^{\overline{1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 76, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]
\end{aligned}$$

$$[125] \ 1^{11} \overset{1^2}{6} \overset{1^1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 12, \ T_{\max} = 13, \ \Delta = -29, \ \det = 239, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 2 & 13 & 3 \\ 2 & 3 & 1 & 3 & 17 & 3 \\ 2 & 2 & 3 & 1 & 15 & 3 \\ 14 & 13 & 17 & 15 & 86 & 20 \\ 3 & 3 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [126] \ 1^{11} \overset{1^1}{6} \overset{1^1}{13} \overset{1^1}{5} \overset{1^1}{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 14, \ T_{\max} = 15, \ \Delta = -29, \ \det = 11, \ (n_+, n_-, n_0) = (1, 14, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 14.91 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 3 & 3 \\ 3 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 3 \\ 3 & 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [129] \ 1^{11} \overset{1^3}{6} \overset{1^1}{13} \overset{1^1}{15} \overset{1^1}{13} \overset{1^1}{1} \overset{1^1}{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 15, \ T_{\max} = 15, \ \Delta = 0, \ \det = -4, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 2 & 4 & 12 & 4 \\ 2 & 2 & 4 & 2 & 10 & 4 \\ 9 & 7 & 12 & 10 & 29 & 13 \\ 3 & 3 & 4 & 4 & 13 & 5 \end{pmatrix}_{b_0}$$

$$[128] \ 1^{11} \overset{1^1}{6} \overset{1^1}{13} \overset{1^1}{15} \overset{1^1}{13} \overset{1^1}{1} \overset{1^1}{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \quad T = 15, \ T_{\max} = 15, \ \Delta = 0, \ \det = -4, \ (n_+, n_-, n_0) = (1, 15, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 2 & 4 & 7 & 4 \\ 2 & 2 & 4 & 2 & 6 & 4 \\ 5 & 5 & 7 & 6 & 10 & 8 \\ 3 & 3 & 4 & 4 & 8 & 5 \end{pmatrix}_{b_0}$$

4.1188 $\text{NHC}(-2)(-3) + \textcolor{red}{su}_3 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{red}{su}_2$

$$T = 9 \quad [0] \ 1^{11} \overset{1^1}{3} \overset{1^1}{1} \overset{1^1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -35, \ \det = -11, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 0 & 3 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 4 & 0 & 2 & 0 & 1 & 3 \\ 3 & 4 & -1 & 2 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} \overset{1^1}{3} \overset{1^1}{1} \overset{1^1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -35, \ \det = -18, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 4 \\ 3 & 4 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & -1 & 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} \overset{1^1}{3} \overset{1^1}{1} \overset{1^1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -35, \ \det = -18, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 0 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & -1 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} \overset{1^1}{3} \overset{1^1}{1} \overset{1^1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -35, \ \det = -36, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 4 \\ 2 & 4 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & -1 & 2 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 & 3 \\ 2 & 4 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & -1 & 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -35, \det = -91, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 3 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 2 & 4 & 0 & 2 & 1 & 3 & 3 \\ 3 & 4 & 0 & 2 & 3 & 1 & 3 \\ 3 & 4 & -1 & 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [7] \quad 1^{111} 4^{11} 13^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -5, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 2 & 3 & 1 & 2 & 5 & 4 \\ 3 & 1 & 0 & 1 & 4 & 3 \\ 1 & 0 & -2 & 0 & 0 & 0 \\ 2 & 1 & 0 & 0 & 2 & 2 \\ 5 & 4 & 0 & 2 & 6 & 6 \\ 4 & 3 & 0 & 2 & 6 & 5 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [8] \quad 1^{111} 5^{11} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 3 & 3 & 0 & 7 & 4 \\ 2 & 3 & 1 & 3 & 0 & 6 & 3 \\ 2 & 3 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 7 & 6 & 5 & 0 & 9 & 7 \\ 3 & 4 & 3 & 3 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{111} 5^{11} 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 2 & 0 & 3 & 3 \\ 2 & 3 & 1 & 3 & 0 & 2 & 3 \\ 2 & 2 & 3 & 1 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 2 & 2 & 0 & 1 & 3 \\ 3 & 3 & 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{112} 5^{11} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 3 \\ 3 & 4 & 3 & 3 & 0 & 7 & 4 \\ 2 & 3 & 1 & 3 & 0 & 6 & 3 \\ 2 & 3 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 7 & 6 & 5 & 0 & 9 & 7 \\ 3 & 4 & 3 & 3 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{12} 5^{11} 13^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 12, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 10 & 3 \\ 3 & 4 & 3 & 3 & 0 & 14 & 4 \\ 2 & 3 & 1 & 3 & 0 & 12 & 3 \\ 2 & 3 & 3 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 10 & 14 & 12 & 10 & 0 & 42 & 14 \\ 3 & 4 & 3 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111} 5^{11} 13^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 5 & 3 & 4 & 0 & 5 & 5 \\ 2 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 2 & 3 & 0 & 2 & 4 \\ 3 & 5 & 2 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[13] \begin{matrix} 1^{121} \\ 1^1 \end{matrix} 5^{13} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 4 & 3 & 3 \\ 3 & 5 & 3 & 4 & 5 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 & 3 & 3 \\ 4 & 5 & 2 & 3 & 2 & 4 & 4 \\ 3 & 5 & 2 & 3 & 4 & 2 & 4 \\ 3 & 5 & 2 & 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[15] \begin{matrix} 1^1 \\ 1^1 \end{matrix} 5^{11} 23^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 16, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 5 & 0 & 4 & 5 & 15 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 10 & 3 \\ 2 & 5 & 0 & 3 & 2 & 11 & 4 \\ 8 & 15 & 0 & 10 & 11 & 37 & 13 \\ 3 & 5 & -1 & 3 & 4 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[14] \begin{matrix} 1^{11} \\ 1^1 \end{matrix} 5^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 16, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 3 & 3 \\ 2 & 1 & 3 & 2 & 0 & 2 & 3 \\ 2 & 3 & 1 & 3 & 0 & 4 & 3 \\ 2 & 2 & 3 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 2 & 4 & 3 & 0 & 2 & 4 \\ 3 & 3 & 3 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[16] \begin{matrix} 1^{13} \\ 1^1 \end{matrix} 5^{13} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 17, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 15 & 3 \\ 3 & 4 & 3 & 3 & 0 & 21 & 4 \\ 2 & 3 & 1 & 3 & 0 & 18 & 3 \\ 2 & 3 & 3 & 1 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 15 & 21 & 18 & 15 & 0 & 97 & 21 \\ 3 & 4 & 3 & 3 & 0 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[17] \begin{matrix} 1^{12} \\ 1^1 \end{matrix} 5^{13} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 18, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 6 & 3 \\ 3 & 5 & 3 & 4 & 0 & 10 & 5 \\ 2 & 3 & 0 & 2 & 0 & 4 & 2 \\ 2 & 4 & 2 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 10 & 4 & 6 & 0 & 14 & 8 \\ 3 & 5 & 2 & 3 & 0 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[18] \begin{matrix} 1^{13} \\ 1^1 \end{matrix} 5^{13} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 20, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 11 & 3 \\ 3 & 4 & 3 & 3 & 0 & 14 & 4 \\ 2 & 3 & 1 & 3 & 0 & 12 & 3 \\ 2 & 3 & 3 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 11 & 14 & 12 & 10 & 0 & 42 & 14 \\ 3 & 4 & 3 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[20] \begin{matrix} 1^{12} \\ 1^1 \end{matrix} 5^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) =$$

$$(1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 6 & 3 \\ 2 & 1 & 3 & 2 & 0 & 5 & 3 \\ 2 & 3 & 1 & 3 & 0 & 6 & 3 \\ 2 & 2 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 6 & 5 & 6 & 5 & 0 & 9 & 7 \\ 3 & 3 & 3 & 3 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

[19] $1^{\frac{1}{1^1}} 5 13^{\frac{1}{1^1}} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 9 & 3 \\ 3 & 5 & 3 & 4 & 0 & 15 & 5 \\ 2 & 3 & 0 & 2 & 0 & 6 & 2 \\ 2 & 4 & 2 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 15 & 6 & 9 & 0 & 34 & 12 \\ 3 & 5 & 2 & 3 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

[21] $1^{\frac{1}{1^1}} 5 13^{\frac{1}{1^1}} 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 7 & 3 & 3 \\ 3 & 5 & 3 & 4 & 10 & 5 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 & 2 \\ 2 & 4 & 2 & 1 & 6 & 3 & 3 \\ 7 & 10 & 4 & 6 & 14 & 8 & 8 \\ 3 & 5 & 2 & 3 & 8 & 2 & 4 \\ 3 & 5 & 2 & 3 & 8 & 4 & 4 \end{pmatrix}_{b_0}$$

[22] $1^{\frac{1}{1^1}} 5 13^{\frac{1}{1^1}} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 23, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 3 & 3 \\ 3 & 5 & 3 & 4 & 5 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 & 3 & 3 \\ 2 & 5 & 2 & 3 & 2 & 4 & 4 \\ 3 & 5 & 2 & 3 & 4 & 2 & 4 \\ 3 & 5 & 2 & 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

[23] $1^{\frac{1}{1^1}} 5 13^{\frac{1}{1^1}} 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 26, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 6 & 3 \\ 3 & 5 & 3 & 4 & 5 & 10 & 5 \\ 2 & 3 & 0 & 2 & 2 & 4 & 2 \\ 2 & 4 & 2 & 1 & 3 & 6 & 3 \\ 2 & 5 & 2 & 3 & 2 & 8 & 4 \\ 6 & 10 & 4 & 6 & 8 & 14 & 8 \\ 3 & 5 & 2 & 3 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

[24] $1^{\frac{1}{1^1}} 5 13^{\frac{1}{1^1}} 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 31, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 9 & 3 \\ 3 & 5 & 3 & 4 & 5 & 15 & 5 \\ 2 & 3 & 0 & 2 & 2 & 6 & 2 \\ 2 & 4 & 2 & 1 & 3 & 9 & 3 \\ 2 & 5 & 2 & 3 & 2 & 12 & 4 \\ 9 & 15 & 6 & 9 & 12 & 34 & 12 \\ 3 & 5 & 2 & 3 & 4 & 12 & 4 \end{pmatrix}_{b_0}$$

[25] $1^{\frac{1}{1^1}} 5 13^{\frac{1}{1^1}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 8 & 3 \\ 2 & 1 & 3 & 2 & 0 & 8 & 3 \\ 2 & 3 & 1 & 3 & 0 & 8 & 3 \\ 2 & 2 & 3 & 1 & 0 & 7 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 8 & 8 & 8 & 7 & 0 & 22 & 10 \\ 3 & 3 & 3 & 3 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

[26] $1^{\frac{1}{1^1}} 5 13^{\frac{1}{1^1}} 1^{114} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 12 & 3 \\ 3 & 5 & 3 & 4 & 5 & 20 & 5 \\ 2 & 3 & 0 & 2 & 2 & 8 & 2 \\ 2 & 4 & 2 & 1 & 3 & 12 & 3 \\ 2 & 5 & 2 & 3 & 2 & 16 & 4 \\ 12 & 20 & 8 & 12 & 16 & 62 & 16 \\ 3 & 5 & 2 & 3 & 4 & 16 & 4 \end{pmatrix}_{b_0}$$

[27] $1^{\frac{1}{1^1}} 5 13^{\frac{1}{1^1}} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 & 6 & 3 \\ 2 & 1 & 3 & 2 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 4 & 6 & 3 \\ 2 & 2 & 3 & 1 & 3 & 5 & 3 \\ 3 & 2 & 4 & 3 & 2 & 6 & 4 \\ 6 & 5 & 6 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1^{11}}_1 5^{13} 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 39, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 3 & 3 \\ 3 & 5 & 3 & 4 & 10 & 5 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 & 2 \\ 2 & 4 & 2 & 1 & 6 & 3 & 3 \\ 5 & 10 & 4 & 6 & 14 & 8 & 8 \\ 3 & 5 & 2 & 3 & 8 & 2 & 4 \\ 3 & 5 & 2 & 3 & 8 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1^{12}}_1 5^{13} 1^{122} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 42, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 6 & 3 \\ 3 & 5 & 3 & 4 & 10 & 10 & 5 \\ 2 & 3 & 0 & 2 & 4 & 4 & 2 \\ 2 & 4 & 2 & 1 & 6 & 6 & 3 \\ 5 & 10 & 4 & 6 & 14 & 16 & 8 \\ 6 & 10 & 4 & 6 & 16 & 14 & 8 \\ 3 & 5 & 2 & 3 & 8 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1^{13}}_1 5^{13} 1^{123} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 47, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 9.3$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 5 & 9 & 3 \\ 3 & 5 & 3 & 4 & 10 & 15 & 5 \\ 2 & 3 & 0 & 2 & 4 & 6 & 2 \\ 2 & 4 & 2 & 1 & 6 & 9 & 3 \\ 5 & 10 & 4 & 6 & 14 & 24 & 8 \\ 9 & 15 & 6 & 9 & 24 & 34 & 12 \\ 3 & 5 & 2 & 3 & 8 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1^{13}}_1 5^{13} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 47, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 9 & 3 \\ 2 & 1 & 3 & 2 & 0 & 7 & 3 \\ 2 & 3 & 1 & 3 & 0 & 10 & 3 \\ 2 & 2 & 3 & 1 & 0 & 8 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 7 & 10 & 8 & 0 & 25 & 11 \\ 3 & 3 & 3 & 3 & 0 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1^{11}}_1 5^{13} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 49, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 5 & 3 \\ 2 & 1 & 3 & 2 & 0 & 5 & 3 \\ 2 & 3 & 1 & 3 & 0 & 6 & 3 \\ 2 & 2 & 3 & 1 & 0 & 5 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 5 & 6 & 5 & 0 & 9 & 7 \\ 3 & 3 & 3 & 3 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1^{11}}_1 5^{13} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 54, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 2 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 4 & 6 & 3 \\ 2 & 2 & 3 & 1 & 3 & 5 & 3 \\ 3 & 2 & 4 & 3 & 2 & 6 & 4 \\ 5 & 5 & 6 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1^{11}}_1 5^{13} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 60, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 3 & 3 \\ 3 & 5 & 3 & 4 & 5 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 & 3 & 3 \\ 3 & 5 & 2 & 3 & 2 & 4 & 4 \\ 3 & 5 & 2 & 3 & 4 & 2 & 4 \\ 3 & 5 & 2 & 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1^{12}}_1 5^{13} 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = 0, \quad \det = 63, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 8 & 3 \\ 3 & 5 & 3 & 4 & 5 & 15 & 5 \\ 2 & 3 & 0 & 2 & 2 & 6 & 2 \\ 2 & 4 & 2 & 1 & 3 & 9 & 3 \\ 3 & 5 & 2 & 3 & 2 & 12 & 4 \\ 8 & 15 & 6 & 9 & 12 & 34 & 12 \\ 3 & 5 & 2 & 3 & 4 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[36] \begin{matrix} 1^{12} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^1 \\ 1^1 \end{matrix} 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 6 & 3 \\ 3 & 5 & 3 & 4 & 5 & 10 & 5 \\ 2 & 3 & 0 & 2 & 2 & 4 & 2 \\ 2 & 4 & 2 & 1 & 3 & 6 & 3 \\ 3 & 5 & 2 & 3 & 2 & 8 & 4 \\ 6 & 10 & 4 & 6 & 8 & 14 & 8 \\ 3 & 5 & 2 & 3 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[37] \begin{matrix} 1^{13} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^1 \\ 1^1 \end{matrix} 1^{113} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 92, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 9 & 3 \\ 3 & 5 & 3 & 4 & 5 & 15 & 5 \\ 2 & 3 & 0 & 2 & 2 & 6 & 2 \\ 2 & 4 & 2 & 1 & 3 & 9 & 3 \\ 3 & 5 & 2 & 3 & 2 & 12 & 4 \\ 9 & 15 & 6 & 9 & 12 & 34 & 12 \\ 3 & 5 & 2 & 3 & 4 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[38] \begin{matrix} 1^{13} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^1 \\ 1^1 \end{matrix} 1^{114} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 95, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 & 11 & 3 \\ 3 & 5 & 3 & 4 & 5 & 20 & 5 \\ 2 & 3 & 0 & 2 & 2 & 8 & 2 \\ 2 & 4 & 2 & 1 & 3 & 12 & 3 \\ 3 & 5 & 2 & 3 & 2 & 16 & 4 \\ 11 & 20 & 8 & 12 & 16 & 62 & 16 \\ 3 & 5 & 2 & 3 & 4 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[39] \begin{matrix} 1^{13} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^1 \\ 1^1 \end{matrix} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 11 & 3 \\ 2 & 1 & 3 & 2 & 0 & 10 & 3 \\ 2 & 3 & 1 & 3 & 0 & 12 & 3 \\ 2 & 2 & 3 & 1 & 0 & 10 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 11 & 10 & 12 & 10 & 0 & 42 & 14 \\ 3 & 3 & 3 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[40] \begin{matrix} 1^{13} \\ 1^1 \end{matrix} 5 \begin{matrix} 13^1 \\ 1^1 \end{matrix} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 143, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 13 & 3 \\ 2 & 1 & 3 & 2 & 0 & 13 & 3 \\ 2 & 3 & 1 & 3 & 0 & 14 & 3 \\ 2 & 2 & 3 & 1 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 13 & 13 & 14 & 12 & 0 & 65 & 17 \\ 3 & 3 & 3 & 3 & 0 & 17 & 4 \end{pmatrix}_{b_0}$$

4.1189 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [1] \begin{matrix} 1^{11} \\ 1^1 \end{matrix} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 3 & 5 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 & 2 \\ 3 & 5 & 0 & 3 & 4 & 3 & 3 & 4 \\ 3 & 4 & 0 & 2 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 2 & 3 & 3 & 1 & 3 \\ 3 & 4 & -1 & 2 & 4 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[0] \begin{matrix} 1^{11} \\ 1^1 \end{matrix} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 10, \Delta = -5, \det = 9, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 3 & 0 & 4 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 & 0 \\ 2 & 4 & 0 & 2 & 0 & 1 & 3 & 3 \\ 3 & 5 & 0 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & -1 & 2 & 0 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

4.1190 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_{2n3\text{mix}}$

$$T = 7 \quad [0] \ 1^{\overline{11}} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 5 & 15 & 4 \\ 3 & 5 & 3 & 12 & 3 \\ 8 & 15 & 12 & 39 & 11 \\ 3 & 4 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{14}} 1^{\overline{11}} 3^{\overline{1}} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 14 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{12}} 1^{\overline{13}} 3^{\overline{1}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 5 & 15 & 4 \\ 4 & 5 & 3 & 12 & 3 \\ 10 & 15 & 12 & 39 & 11 \\ 3 & 4 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1}} 1^{\overline{13}} 3^{\overline{1}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -13, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 5 & 9 & 12 & 5 & 6 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1}} 1^{\overline{13}} 3^{\overline{1}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{12}} 1^{\overline{13}} 3^{\overline{1}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 6 & 7 & 6 & 7 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{13}} 1^{\overline{13}} 3^{\overline{1}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 4 & 5 & 3 & 6 & 3 \\ 7 & 7 & 6 & 7 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1}} 1^{\overline{13}} 3^{\overline{1}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{11}} 1^{\overline{13}} 3^{\overline{1}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 10 & 17 & 48 & 9 & 12 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{23}} 1^{\overline{12}} 3^{\overline{1}} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 6 & 8 & 9 & 10 & 5 \\ 7 & 7 & 10 & 7 & 5 \\ 3 & 4 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 5 & 11 & 4 \\ 5 & 5 & 3 & 9 & 3 \\ 9 & 11 & 9 & 20 & 8 \\ 3 & 4 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 3 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 5 & 5 & 3 & 3 & 3 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{14} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 5 & 19 & 4 \\ 3 & 5 & 3 & 15 & 3 \\ 11 & 19 & 15 & 64 & 14 \\ 3 & 4 & 3 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 7 & 9 & 12 & 11 & 6 \\ 7 & 7 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11} 3^{11} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 20 & 27 & 17 & 151 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 4 & 4 & 0 & 5 & 2 \\ 4 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 0 & 5 & 2 \\ 6 & 8 & 5 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 0 & 5 & 2 \\ 6 & 8 & 5 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 0 & 5 & 2 \\ 5 & 8 & 5 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 4 & 5 & 3 & 6 & 3 \\ 4 & 7 & 6 & 7 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 0 & 5 & 2 \\ 7 & 8 & 5 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 5 & 11 & 4 \\ 4 & 5 & 3 & 9 & 3 \\ 7 & 11 & 9 & 20 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 8 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & 0 & 8 & 2 \\ 8 & 16 & 8 & 25 & 9 \\ 3 & 5 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -21, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 9 & 16 & 5 \\ 9 & 12 & 16 & 25 & 9 \\ 3 & 4 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 5 & 9 & 12 & 17 & 6 \\ 8 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 5 & 8 & 9 & 4 & 5 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 5 & 5 & 3 & 6 & 3 \\ 7 & 7 & 6 & 7 & 5 \\ 3 & 4 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -23, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 5 & 15 & 4 \\ 3 & 5 & 3 & 12 & 3 \\ 11 & 15 & 12 & 39 & 11 \\ 3 & 4 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 4 & 4 & 0 & 5 & 2 \\ 7 & 8 & 5 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[30] \quad 13^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 2 & 4 & 0 & 5 & 2 \\ 5 & 8 & 5 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

- [31] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 5 & 3 & 4 \\ 4 & 5 & 3 & 3 & 3 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$
- [38] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 4 & 4 & 0 & 14 & 2 \\ 12 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$
- [33] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 4 & 4 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [34] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 5 & 5 & 3 & 6 & 3 \\ 6 & 7 & 6 & 7 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$
- [35] $1^{11}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 17 & 22 & 14 & 97 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$
- [36] $1^{11}3^{11}1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 16 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 4 & 4 & 0 & 14 & 2 \\ 16 & 22 & 14 & 97 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$
- [37] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 9 & 15 & 4 \\ 6 & 9 & 12 & 23 & 6 \\ 12 & 15 & 23 & 39 & 11 \\ 3 & 4 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$
- [32] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 2 & 4 & 0 & 8 & 2 \\ 10 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$
- [39] $1^{11}3^{11}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 25 & 3 \\ 3 & 4 & 4 & 39 & 4 \\ 4 & 4 & 0 & 26 & 2 \\ 25 & 39 & 26 & 280 & 30 \\ 3 & 4 & 2 & 30 & 1 \end{pmatrix}_{b_0}$$
- [40] $1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 5 & 11 & 4 \\ 4 & 5 & 3 & 9 & 3 \\ 10 & 11 & 9 & 20 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$
- [41] $13^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 5 & 11 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 6 & 11 & 9 & 20 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{14} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 5 & 19 & 4 \\ 3 & 5 & 3 & 15 & 3 \\ 13 & 19 & 15 & 64 & 14 \\ 3 & 4 & 3 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 4 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[43] \quad 21^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 9 & 4 \\ 5 & 4 & 4 & 8 & 4 \\ 3 & 4 & 0 & 5 & 2 \\ 9 & 8 & 5 & 12 & 6 \\ 4 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{13} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 13 & 0 & 36 & 10 \\ 3 & 4 & -1 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{14} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 5 & 19 & 4 \\ 4 & 5 & 3 & 15 & 3 \\ 14 & 19 & 15 & 64 & 14 \\ 3 & 4 & 3 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 8 & 0 & 12 & 6 \\ 3 & 4 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{13} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 8 & 0 & 12 & 6 \\ 3 & 4 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[49] \quad 13^{11} 1^{12} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 4 & 8 & 9 & 16 & 5 \\ 7 & 12 & 16 & 25 & 9 \\ 3 & 4 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -33, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 5 & 11 & 4 \\ 5 & 5 & 3 & 9 & 3 \\ 10 & 11 & 9 & 20 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 4 & 8 & 9 & 10 & 5 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 5 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 4 & 8 & 4 \\ 3 & 4 & 0 & 5 & 2 \\ 6 & 8 & 5 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{13} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 8 & 12 & 24 & 6 & 8 \\ 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{11} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 3 \\ 4 & 9 & 4 & 16 & 5 \\ 2 & 4 & 0 & 8 & 2 \\ 7 & 16 & 8 & 25 & 9 \\ 3 & 5 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[54] \quad 21^{13} 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 \\ 5 & 4 & 4 & 7 & 4 \\ 3 & 4 & 0 & 5 & 2 \\ 6 & 7 & 5 & 7 & 5 \\ 4 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 4 & 7 & 6 & 7 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{113} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 3 & 4 & 0 & 5 & 2 \\ 7 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{121} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 9 & 16 & 5 \\ 7 & 12 & 16 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{12} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 5 & 11 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 8 & 11 & 9 & 20 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{14} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 5 & 19 & 4 \\ 3 & 5 & 3 & 15 & 3 \\ 12 & 19 & 15 & 64 & 14 \\ 3 & 4 & 3 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{121} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 6 & 8 & 9 & 16 & 5 \\ 8 & 12 & 16 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{123} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 6 & 8 & 9 & 16 & 5 \\ 10 & 12 & 16 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{112} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 6 & 9 & 12 & 17 & 6 \\ 6 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

[64] $1^{\overline{11}} 1^{\overline{114}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 5 & 9 & 12 & 29 & 6 \\ 10 & 19 & 29 & 64 & 14 \\ 3 & 4 & 6 & 14 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{\overline{11}} 1^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 5 & 7 & 6 & 7 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{\overline{11}} 1^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 4 & 7 & 0 & 7 & 5 \\ 3 & 4 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[68] $21^{\overline{11}} 1^{\overline{11}} 1^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 11 & 3 \\ 4 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 11 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{\overline{13}} 1^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 5 & 15 & 4 \\ 3 & 5 & 3 & 12 & 3 \\ 9 & 15 & 12 & 39 & 11 \\ 3 & 4 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{\overline{11}} 1^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 7 & 9 & 12 & 11 & 6 \\ 4 & 7 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[71] $21^{\overline{11}} 1^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 8 & 3 & 12 & 4 \\ 8 & 9 & 4 & 16 & 5 \\ 3 & 4 & 0 & 8 & 2 \\ 12 & 16 & 8 & 25 & 9 \\ 4 & 5 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[70] $1^{\overline{11}} 1^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 2 & 4 & 0 & 5 & 2 \\ 6 & 12 & 5 & 12 & 6 \\ 3 & 5 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{\overline{12}} 1^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 5 & 8 & 9 & 10 & 5 \\ 6 & 7 & 10 & 7 & 5 \\ 3 & 4 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{\overline{12}} 1^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 6 & 8 & 9 & 4 & 5 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

[74] $1^{\overline{13}} 1^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 4 & 4 & 0 & 11 & 2 \\ 9 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{111} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 4 & 4 & 0 & 5 & 2 \\ 7 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{112} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 4 & 5 & 3 & 6 & 3 \\ 6 & 7 & 6 & 7 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{111} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 2 & 4 & 0 & 5 & 2 \\ 6 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 8 & 13 & 27 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{13} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 5 & 15 & 4 \\ 4 & 5 & 3 & 12 & 3 \\ 11 & 15 & 12 & 39 & 11 \\ 3 & 4 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{112} 3^{11} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 18 & 27 & 17 & 151 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{111} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 10 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{13} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 5 & 15 & 4 \\ 3 & 5 & 3 & 12 & 3 \\ 10 & 15 & 12 & 39 & 11 \\ 3 & 4 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 4 & 5 & 3 & 6 & 3 \\ 5 & 7 & 6 & 7 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[82] \quad 21^{121} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 8 & 3 \\ 4 & 4 & 8 & 12 & 4 \\ 6 & 8 & 9 & 16 & 5 \\ 8 & 12 & 16 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 8 & 0 & 12 & 6 \\ 3 & 4 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{123} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 13 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 7 & 9 & 12 & 29 & 6 \\ 13 & 19 & 29 & 64 & 14 \\ 3 & 4 & 6 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[86] \quad 13^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 2 & 4 & 0 & 5 & 2 \\ 5 & 12 & 5 & 12 & 6 \\ 3 & 5 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 5 & 11 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 7 & 11 & 9 & 20 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 6 & 9 & 12 & 5 & 6 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{124} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 7 & 9 & 12 & 17 & 6 \\ 10 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{113} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 6 & 9 & 12 & 17 & 6 \\ 9 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{131} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 8 & 3 & 4 \\ 7 & 8 & 9 & 4 & 5 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{121} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 7 & 13 & 27 & 16 & 9 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 5 & 11 & 4 \\ 4 & 5 & 3 & 9 & 3 \\ 8 & 11 & 9 & 20 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{13} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 0 & 8 & 2 \\ 10 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[96] \quad 1^{121} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -48, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 0 & 8 & 2 \\ 8 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

^{1²¹}

$$[97] \ 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -49, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 7 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 7 & 9 & 12 & 17 & 6 \\ 7 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

^{1²¹}

$$[102] \ 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 4 & 4 & 0 & 11 & 2 \\ 10 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

^{1²²}

$$[104] \ 1^{111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 0 & 11 & 2 \\ 12 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

^{1¹⁴}

$$[103] \ 1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 5 & 15 & 4 \\ 4 & 5 & 3 & 12 & 3 \\ 12 & 15 & 12 & 39 & 11 \\ 3 & 4 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

^{1²²}

$$[101] \ 1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 10 & 3 \\ 3 & 4 & 9 & 15 & 4 \\ 7 & 9 & 12 & 23 & 6 \\ 10 & 15 & 23 & 39 & 11 \\ 3 & 4 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$

^{1¹³}

$$[100] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 5 & 11 & 4 \\ 4 & 5 & 3 & 9 & 3 \\ 9 & 11 & 9 & 20 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

^{1¹⁴}

$$[99] \ 13^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.35 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 16 & 27 & 17 & 151 & 21 \\ 3 & 4 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

^{1¹²}

$$[98] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 7 & 0 & 7 & 5 \\ 3 & 4 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

^{1¹²}

$$[106] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 6 & 9 & 12 & 11 & 6 \\ 6 & 7 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

^{1¹}

$$[105] \ 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 11 & 0 & 20 & 8 \\ 3 & 4 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[107] $1^{111}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 5 & 9 & 12 & 11 & 6 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[108] $13^{1121}1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 4 & 8 & 9 & 10 & 5 \\ 4 & 7 & 10 & 7 & 5 \\ 3 & 4 & 5 & 5 & 1 \end{pmatrix}_{b_0}$$

[111] $1^{131}3^{111}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 8 & 9 & 12 & 11 & 6 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[110] $1^{121}3^{111}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.58 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 5 & 8 & 9 & 16 & 5 \\ 8 & 12 & 16 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

[109] $1^{111}3^{111}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 7 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 7 & 0 & 7 & 5 \\ 3 & 4 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[112] $1^{111}3^{111}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 8 & 9 & 12 & 5 & 6 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

[116] $13^{111}1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 2 & 4 & 0 & 5 & 2 \\ 4 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[115] $1^{113}3^{111}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 3 \\ 3 & 4 & 9 & 15 & 4 \\ 5 & 9 & 12 & 23 & 6 \\ 10 & 15 & 23 & 39 & 11 \\ 3 & 4 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$

[114] $1^{111}3^{111}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 4 & 4 & 0 & 5 & 2 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{121}3^{111}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 10 & 4 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 10 & 13 & 27 & 16 & 9 \\ 4 & 7 & 16 & 7 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[118] $1^{121}3^{111}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 6 & 8 & 9 & 10 & 5 \\ 4 & 7 & 10 & 7 & 5 \\ 3 & 4 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[120] \quad 1^{11} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 14 & 22 & 14 & 97 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[119] \quad 1^{11} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 3 & 4 & 0 & 5 & 2 \\ 4 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[117] \quad 21^{14} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 14 & 3 \\ 4 & 4 & 5 & 19 & 4 \\ 3 & 5 & 3 & 15 & 3 \\ 14 & 19 & 15 & 64 & 14 \\ 3 & 4 & 3 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[121] \quad 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -61, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 7 & 9 & 12 & 5 & 6 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[122] \quad 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -62, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 13 & 0 & 36 & 10 \\ 3 & 4 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[123] \quad 13^{11} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 3 \\ 3 & 4 & 9 & 15 & 4 \\ 5 & 9 & 12 & 23 & 6 \\ 8 & 15 & 23 & 39 & 11 \\ 3 & 4 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[124] \quad 1^{13} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 12 & 3 & 4 \\ 9 & 12 & 24 & 6 & 8 \\ 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 8 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[125] \quad 1^{13} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 15 & 0 & 39 & 11 \\ 3 & 4 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[126] \quad 1^{12} 3^{11} 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 4 & 4 & 0 & 8 & 2 \\ 7 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[127] \quad 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 8 & 12 & 4 \\ 6 & 8 & 9 & 16 & 5 \\ 9 & 12 & 16 & 25 & 9 \\ 3 & 4 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[128] \quad 21^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

- [129] $21^{1131^{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 3 \\ 4 & 4 & 5 & 7 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 5 & 7 & 6 & 7 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$
- [130] $1^{11131^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 4 & 4 & 0 & 5 & 2 \\ 6 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$
- [131] $1^{11131^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 2 & 4 & 0 & 5 & 2 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$
- [132] $1^{1131^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 13 & 0 & 36 & 10 \\ 3 & 4 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$
- [133] $13^{111^{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.63 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 5 & 9 & 12 & 11 & 6 \\ 4 & 7 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$
- [135] $1^{1431^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 19 & 0 & 64 & 14 \\ 3 & 4 & -1 & 14 & 1 \end{pmatrix}_{b_0}$$
- [134] $1^{1131^{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 18 & 0 & 72 & 14 \\ 3 & 4 & -1 & 14 & 1 \end{pmatrix}_{b_0}$$
- [137] $1^{11331^{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 9 & 3 \\ 3 & 4 & 9 & 15 & 4 \\ 6 & 9 & 12 & 23 & 6 \\ 9 & 15 & 23 & 39 & 11 \\ 3 & 4 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$
- [138] $1^{12131^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 7 & 8 & 9 & 10 & 5 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$
- [139] $1^{11331^{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 12 & 0 & 25 & 9 \\ 3 & 4 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$
- [136] $1^{1131^{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -73, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 12 & 0 & 25 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[140] $21^{13}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 11 & 3 \\ 4 & 4 & 5 & 15 & 4 \\ 3 & 5 & 3 & 12 & 3 \\ 11 & 15 & 12 & 39 & 11 \\ 3 & 4 & 3 & 11 & 2 \end{pmatrix}_{b_0}$$

[141] $13^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 5 & 9 & 12 & 17 & 6 \\ 6 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

[142] $1^{13}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 19 & 27 & 17 & 151 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[143] $1^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 11 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[144] $1^{12}1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 3 & 4 & 0 & 5 & 2 \\ 6 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

[145] $1^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 4 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 6 & 9 & 12 & 11 & 6 \\ 4 & 7 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[146] $21^{12}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 8 & 3 \\ 4 & 4 & 5 & 11 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 8 & 11 & 9 & 20 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

[147] $1^{12}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 11 & 0 & 20 & 8 \\ 3 & 4 & -1 & 8 & 1 \end{pmatrix}_{b_0}$$

[148] $1^{11}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 7 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[149] $1^{12}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 5 & 8 & 9 & 10 & 5 \\ 4 & 7 & 10 & 7 & 5 \\ 3 & 4 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

[150] $1^{12}3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 6 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[151] \quad 1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -81, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.56 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 5 & 9 & 12 & 17 & 6 \\ 7 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[152] \quad 1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -81, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.16 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 11 & 3 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 11 & 17 & 48 & 9 & 12 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[154] \quad 1^{114} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -82, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.63 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 12 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 5 & 9 & 12 & 29 & 6 \\ 12 & 19 & 29 & 64 & 14 \\ 3 & 4 & 6 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[153] \quad 1^{121} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -82, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.10 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 9 & 13 & 27 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[155] \quad 1^{113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -83, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 13 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[156] \quad 1^{111} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -84, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.24 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 3 & 4 & 0 & 5 & 2 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[157] \quad 1^{122} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -88, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.32 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 6 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 6 & 8 & 9 & 10 & 5 \\ 6 & 7 & 10 & 7 & 5 \\ 3 & 4 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[158] \quad 1^{121} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -89, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.46 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 5 & 8 & 9 & 10 & 5 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[159] \quad 1^{112} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -91, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.49 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 15 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[160] \quad 1^{111} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -91, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.58 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 17 & 27 & 17 & 151 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[161] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -91, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 6 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 7 & 9 & 12 & 11 & 6 \\ 6 & 7 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

 1^{24}

$$[162] \ 1^{111} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 17 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 4 & 4 & 0 & 14 & 2 \\ 17 & 22 & 14 & 97 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

 1^3

$$[163] \ 1^{111} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 16 & 22 & 14 & 97 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

 1^3

$$[164] \ 1^{112} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 2 & 4 & 0 & 8 & 2 \\ 9 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

 1^{13}

$$[165] \ 13^1 1^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 13 & 22 & 14 & 97 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

 1^1

$$[166] \ 13^1 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 2 & 4 & 0 & 8 & 2 \\ 6 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

 1^{22}

$$[167] \ 1^{112} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 4 & 4 & 0 & 14 & 2 \\ 14 & 21 & 14 & 80 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

 1^1

$$[168] \ 1^{114} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -93, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 5 & 9 & 12 & 29 & 6 \\ 11 & 19 & 29 & 64 & 14 \\ 3 & 4 & 6 & 14 & 2 \end{pmatrix}_{b_0}$$

 1^1

$$[169] \ 1^{113} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 3 \\ 3 & 4 & 9 & 15 & 4 \\ 5 & 9 & 12 & 23 & 6 \\ 9 & 15 & 23 & 39 & 11 \\ 3 & 4 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$

 1^{11}

$$[170] \ 1^{12} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 11 & 0 & 20 & 8 \\ 3 & 4 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

 1^{22}

$$[172] \ 1^{121} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 6 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 9 & 13 & 27 & 16 & 9 \\ 6 & 7 & 16 & 7 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[173] $1^{1^{24}} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 0 & 8 & 2 \\ 11 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[174] $1^{1^{13}} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 18 & 0 & 72 & 14 \\ 3 & 4 & -1 & 14 & 1 \end{pmatrix}_{b_0}$$

[171] $1^{1^{12}} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 18 & 0 & 72 & 14 \\ 3 & 4 & -1 & 14 & 1 \end{pmatrix}_{b_0}$$

[175] $1^{1^{13}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 8 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[176] $1^{1^{22}} 1^{14} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 4 & 4 & 0 & 20 & 2 \\ 18 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[177] $1^{1^{24}} 1^{12} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 22 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 4 & 4 & 0 & 20 & 2 \\ 22 & 31 & 20 & 188 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[178] $1^{1^2} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 0 & 8 & 2 \\ 9 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[179] $1^{1^{22}} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 0 & 8 & 2 \\ 9 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[180] $1^{1^{11}} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 0 & 8 & 2 \\ 7 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[181] $1^{1^{11}} 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 6 & 9 & 12 & 11 & 6 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[182] $1^{1^{14}} 1^{14} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 14 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 6 & 9 & 12 & 29 & 6 \\ 14 & 19 & 29 & 64 & 14 \\ 3 & 4 & 6 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[183] \quad 1^{1^{12}} 3^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 12 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 6 & 9 & 12 & 29 & 6 \\ 12 & 19 & 29 & 64 & 14 \\ 3 & 4 & 6 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[184] \quad 1^{1^{21}} 3^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -98, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 5 & 3 \\ 3 & 4 & 9 & 7 & 4 \\ 7 & 9 & 12 & 11 & 6 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[186] \quad 1^{1^{12}} 3^{1^{14}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 18 & 27 & 17 & 151 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[185] \quad 1^{1^{11}} 3^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 17 & 0 & 55 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[188] \quad 21^{1^{14}} 3^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 30 & 4 \\ 5 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 30 & 27 & 17 & 151 & 21 \\ 4 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[187] \quad 1^{1^{13}} 3^{1^{14}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 19 & 27 & 17 & 151 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[189] \quad 21^{1^{11}} 3^{1^{121}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 12 & 6 & 4 \\ 5 & 4 & 13 & 7 & 4 \\ 12 & 13 & 27 & 16 & 9 \\ 6 & 7 & 16 & 7 & 5 \\ 4 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[190] \quad 1^{1^{21}} 3^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 8 & 7 & 4 \\ 6 & 8 & 9 & 10 & 5 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[191] \quad 1^{1^{31}} 3^{1^{11}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -103, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 3 & 3 \\ 3 & 4 & 13 & 3 & 4 \\ 10 & 13 & 27 & 7 & 9 \\ 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[192] \quad 13^{1^{11}} 3^{1^{121}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -104, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.77 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 7 & 13 & 27 & 16 & 9 \\ 4 & 7 & 16 & 7 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[193] \quad 1^{1^{13}} 3^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 11 & 0 & 20 & 8 \\ 3 & 4 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[194] $1^{112}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 7 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 6 & 9 & 12 & 17 & 6 \\ 7 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

[198] $1^{123}3^11^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 0 & 11 & 2 \\ 13 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[195] $13^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 10 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[197] $1^{123}3^11^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 4 & 4 & 0 & 17 & 2 \\ 18 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[196] $1^{122}3^11^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 16 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 4 & 4 & 0 & 17 & 2 \\ 16 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[202] $1^{111}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 12 & 0 & 25 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[201] $1^{113}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 12 & 0 & 25 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[200] $1^{112}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 8 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 6 & 9 & 12 & 17 & 6 \\ 8 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

[199] $1^{113}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 11 & 3 \\ 3 & 4 & 9 & 15 & 4 \\ 6 & 9 & 12 & 23 & 6 \\ 11 & 15 & 23 & 39 & 11 \\ 3 & 4 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$

[204] $1^{114}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 10 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[203] $1^{112}3^11^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 10 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$1^{24}$$

$$[210] 1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 4 & 4 & 0 & 8 & 2 \\ 10 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$1^{12}$$

$$[208] 1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 11 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 11 & 0 & 20 & 8 \\ 3 & 4 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$1^1$$

$$[207] 1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 16 & 0 & 44 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

$$1^1$$

$$[206] 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 2 & 4 & 0 & 8 & 2 \\ 8 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$1^{22}$$

$$[205] 1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 4 & 4 & 0 & 8 & 2 \\ 8 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$1^{23}$$

$$[209] 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 4 & 4 & 0 & 8 & 2 \\ 10 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$1^{23}$$

$$[211] 1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -113, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 9 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 7 & 9 & 12 & 17 & 6 \\ 9 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

$$1^{22}$$

$$[212] 1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -114, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 7 & 8 & 3 \\ 3 & 4 & 9 & 11 & 4 \\ 7 & 9 & 12 & 17 & 6 \\ 8 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

$$1^{12}$$

$$[215] 1^1 3^1 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 23 & 0 & 120 & 18 \\ 3 & 4 & -1 & 18 & 1 \end{pmatrix}_{b_0}$$

$$1^2$$

$$[214] 1^{121} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 4 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 9 & 13 & 27 & 16 & 9 \\ 4 & 7 & 16 & 7 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[213] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 11 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[216] $1^{113}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 22 & 0 & 97 & 17 \\ 3 & 4 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[217] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -121, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 12 & 0 & 25 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[218] $1^{114}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -122, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 14 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 7 & 9 & 12 & 29 & 6 \\ 14 & 19 & 29 & 64 & 14 \\ 3 & 4 & 6 & 14 & 2 \end{pmatrix}_{b_0}$$

[219] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -122, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 11 & 3 \\ 3 & 4 & 9 & 15 & 4 \\ 7 & 9 & 12 & 23 & 6 \\ 11 & 15 & 23 & 39 & 11 \\ 3 & 4 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$

[220] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -123, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 12 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[222] $1^{114}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 4 & 4 & 0 & 14 & 2 \\ 12 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[223] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 10 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[221] $1^{112}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 2 & 4 & 0 & 8 & 2 \\ 8 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[231] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 6 & 10 & 3 \\ 3 & 4 & 9 & 15 & 4 \\ 6 & 9 & 12 & 23 & 6 \\ 10 & 15 & 23 & 39 & 11 \\ 3 & 4 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$

[230] $1^{111}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 14 & 22 & 14 & 97 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[229] $1^{11}3^{12}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 15 & 22 & 14 & 97 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[228] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 8 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[227] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 2 & 4 & 0 & 8 & 2 \\ 7 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[226] $21^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 12 & 4 \\ 5 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 12 & 12 & 8 & 25 & 9 \\ 4 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

[225] $1^{11}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 27 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 20 & 27 & 17 & 151 & 21 \\ 3 & 4 & 2 & 21 & 2 \end{pmatrix}_{b_0}$$

[224] $1^{11}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -124, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 15 & 22 & 14 & 97 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[332] $1^{13}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 12 & 3 & 3 \\ 3 & 4 & 17 & 3 & 4 \\ 12 & 17 & 48 & 9 & 12 \\ 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

[234] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 7 & 12 & 3 \\ 3 & 4 & 9 & 15 & 4 \\ 7 & 9 & 12 & 23 & 6 \\ 12 & 15 & 23 & 39 & 11 \\ 3 & 4 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$

[233] $1^{11}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -127, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 27 & 0 & 151 & 21 \\ 3 & 4 & -1 & 21 & 1 \end{pmatrix}_{b_0}$$

[236] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.53 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 4 & 4 & 0 & 8 & 2 \\ 9 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[235] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 8 & 5 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 8 & 13 & 27 & 16 & 9 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[237] \ 1^{14} 3^{11} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -128, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.84 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 23 & 0 & 120 & 18 \\ 3 & 4 & -1 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[238] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -129, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.32 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 15 & 0 & 39 & 11 \\ 3 & 4 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[239] \ 1^{14} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -130, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.15 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 6 & 13 & 3 \\ 3 & 4 & 9 & 19 & 4 \\ 6 & 9 & 12 & 29 & 6 \\ 13 & 19 & 29 & 64 & 14 \\ 3 & 4 & 6 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[244] \ 1^{13} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -131, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.34 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 10 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 4 & 4 & 0 & 11 & 2 \\ 10 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[243] \ 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -131, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.34 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 9 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[245] \ 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -131, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.34 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 8 & 4 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 8 & 13 & 27 & 16 & 9 \\ 4 & 7 & 16 & 7 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[242] \ 13^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -131, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.44 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 8 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[240] \ 1^{11} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -131, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.34 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 4 & 4 & 0 & 11 & 2 \\ 14 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[241] \ 1^{11} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -131, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.40 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 11 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[246] \ 1^1 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -134, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 23 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 23 & 0 & 120 & 18 \\ 3 & 4 & -1 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[247] \ 1^{12} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -135, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 12 & 21 & 14 & 80 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[248] \ 1^{1^{11}} 1^{12} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 7 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[249] \ 1^{1^4} 1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 13 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[250] \ 1^{1^{22}} 1^{12} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -139, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 4 & 4 & 0 & 11 & 2 \\ 11 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[252] \ 1^{1^{14}} 1^{11} 3^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 14 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[251] \ 1^{1^{12}} 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 9 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[253] \ 21^{1^{11}} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -146, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 8 & 3 & 9 & 4 \\ 8 & 9 & 4 & 12 & 5 \\ 3 & 4 & 0 & 5 & 2 \\ 9 & 12 & 5 & 12 & 6 \\ 4 & 5 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[254] \ 1^{1^{22}} 1^{14} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -147, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 4 & 4 & 0 & 17 & 2 \\ 15 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[255] \ 1^{1^1} 1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -149, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 20 & 0 & 69 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[256] \ 21^{1^{11}} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 5 & 3 \\ 4 & 4 & 4 & 7 & 4 \\ 3 & 4 & 0 & 5 & 2 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[257] \ 21^{1^{11}} 3^1 1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 21 & 4 \\ 5 & 4 & 4 & 24 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 21 & 24 & 17 & 100 & 18 \\ 4 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[258] \ 1^{1^{13}} 1^{12} 3^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 9 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[259] $1^{112}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 19 & 31 & 20 & 188 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[260] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -153, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 17 & 0 & 55 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[266] $1^{111}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 17 & 22 & 14 & 97 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[264] $1^{111}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 16 & 22 & 14 & 97 & 17 \\ 3 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[263] $1^{122}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 4 & 4 & 0 & 14 & 2 \\ 13 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[262] $21^{13}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 24 & 4 \\ 5 & 4 & 4 & 22 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 24 & 22 & 14 & 97 & 17 \\ 4 & 4 & 2 & 17 & 2 \end{pmatrix}_{b_0}$$

[261] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 12 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

[265] $21^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 5 & 3 \\ 4 & 4 & 9 & 7 & 4 \\ 6 & 9 & 12 & 11 & 6 \\ 5 & 7 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

[267] $1^{112}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -159, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 11 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 8 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[268] $1^{113}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 20 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 4 & 4 & 0 & 20 & 2 \\ 20 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

[269] $1^{113}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 20 & 35 & 23 & 231 & 27 \\ 3 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[270] \ 1^{14} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -162, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.44 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 19 & 0 & 64 & 14 \\ 3 & 4 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[271] \ 1^{14} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -163, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.37 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 12 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[275] \ 21^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -164, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.15 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 8 & 3 \\ 4 & 4 & 4 & 12 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 8 & 12 & 8 & 25 & 9 \\ 3 & 4 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[276] \ 1^{12} 3^{11} 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -164, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.34 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 21 & 0 & 80 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[274] \ 1^{11} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -164, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.12 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 13 & 17 & 11 & 55 & 13 \\ 3 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[273] \ 21^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -164, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.12 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 18 & 4 \\ 5 & 4 & 4 & 17 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 18 & 17 & 11 & 55 & 13 \\ 4 & 4 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[272] \ 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -164, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.07 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 10 & 5 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 10 & 13 & 27 & 16 & 9 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[277] \ 1^{12} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -167, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.10 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 9 & 5 & 3 \\ 3 & 4 & 13 & 7 & 4 \\ 9 & 13 & 27 & 16 & 9 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[278] \ 1^{13} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -171, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 15 & 0 & 39 & 11 \\ 3 & 4 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[279] \ 13^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -171, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.78 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 9 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[280] \ 1^{11} 3^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -171, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.73 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 17 & 0 & 55 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[281] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -172, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.14 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 16 & 0 & 44 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[282] \quad 21^{11} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -175, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 15 & 16 & 11 & 44 & 12 \\ 4 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[283] \quad 13^1 1^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -176, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 10 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[284] \quad 21^{121} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -181, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.31 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 5 & 3 \\ 4 & 4 & 8 & 7 & 4 \\ 6 & 8 & 9 & 10 & 5 \\ 5 & 7 & 10 & 7 & 5 \\ 3 & 4 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[285] \quad 1^{13} 3^{11} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -183, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.55 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 15 & 0 & 39 & 11 \\ 3 & 4 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[286] \quad 1^{112} 3^{11} 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -183, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.29 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 17 & 0 & 55 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[291] \quad 1^{113} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.43 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 11 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[290] \quad 1^{14} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.50 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 24 & 0 & 100 & 18 \\ 3 & 4 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[289] \quad 1^{113} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.43 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 9 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[288] \quad 21^{112} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.11 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 17 & 3 \\ 4 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 17 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[287] \quad 21^{113} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -184, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.43 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 18 & 4 \\ 5 & 4 & 4 & 20 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 18 & 20 & 14 & 69 & 15 \\ 4 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[292] $1^{112}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -186, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 8 & 3 \\ 4 & 4 & 9 & 11 & 4 \\ 6 & 9 & 12 & 17 & 6 \\ 8 & 11 & 17 & 20 & 8 \\ 3 & 4 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

[293] $1^{123}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 11 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 4 & 4 & 0 & 11 & 2 \\ 11 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[294] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 9 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[295] $1^{124}3^11^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 24 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 4 & 4 & 0 & 23 & 2 \\ 24 & 35 & 23 & 231 & 27 \\ 3 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

[296] $1^{114}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 22 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 4 & 4 & 0 & 23 & 2 \\ 22 & 34 & 23 & 208 & 26 \\ 3 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

[297] $1^{123}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -188, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 4 & 4 & 0 & 14 & 2 \\ 15 & 21 & 14 & 80 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[298] $1^{111}3^11^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -189, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 22 & 0 & 97 & 17 \\ 3 & 4 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[299] $1^{113}3^11^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -189, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 17 & 0 & 55 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[300] $1^{113}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -189, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 15 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 15 & 0 & 39 & 11 \\ 3 & 4 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[301] $1^{123}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -195, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 4 & 4 & 0 & 11 & 2 \\ 12 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[302] $1^{13}3^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -195, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 12 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[303] $1^{113}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -203, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 11 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[304] $21^{113}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 11 & 3 \\ 4 & 4 & 9 & 15 & 4 \\ 6 & 9 & 12 & 23 & 6 \\ 11 & 15 & 23 & 39 & 11 \\ 3 & 4 & 6 & 11 & 2 \end{pmatrix}_{b_0}$$

[306] $1^{12}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 16 & 0 & 44 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[305] $1^{12}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 26 & 0 & 128 & 20 \\ 3 & 4 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

[307] $1^{12}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 16 & 0 & 44 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[308] $21^{114}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -210, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 6 & 14 & 3 \\ 4 & 4 & 9 & 19 & 4 \\ 6 & 9 & 12 & 29 & 6 \\ 14 & 19 & 29 & 64 & 14 \\ 3 & 4 & 6 & 14 & 2 \end{pmatrix}_{b_0}$$

[309] $1^{113}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 10 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[310] $1^{112}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 10 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[311] $1^{112}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 10 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[312] $1^{124}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 19 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 4 & 4 & 0 & 17 & 2 \\ 19 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[313] $1^{113}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -211, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 4 & 4 & 0 & 11 & 2 \\ 12 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[314] \quad 1^{113}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 17 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

$$[315] \quad 1^{111}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -217, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.63 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 27 & 0 & 151 & 21 \\ 3 & 4 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[318] \quad 1^{112}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 2 & 4 & 0 & 11 & 2 \\ 11 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[317] \quad 1^{112}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 4 & 4 & 0 & 11 & 2 \\ 13 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[316] \quad 1^{112}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 16 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[320] \quad 1^{114}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -220, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 13 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 4 & 4 & 0 & 14 & 2 \\ 13 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[319] \quad 1^{113}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -220, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 16 & 0 & 44 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[323] \quad 1^{112}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 16 & 21 & 14 & 80 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[322] \quad 13^11^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 12 & 21 & 14 & 80 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[321] \quad 1^{113}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -224, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 14 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[324] \quad 1^{13}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -227, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.34 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 25 & 0 & 111 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[325] \quad 1^{12} 1^{14} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -228, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 19 & 0 & 64 & 14 \\ 3 & 4 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[326] \quad 1^{11} 1^{14} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -231, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 11 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[327] \quad 1^{11} 1^{13} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -233, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 20 & 0 & 69 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[328] \quad 1^{11} 1^{14} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.53 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 13 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[329] \quad 1^{12} 1^{11} 3^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -237, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 22 & 0 & 97 & 17 \\ 3 & 4 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[330] \quad 1^{11} 1^{12} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -239, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 13 & 21 & 14 & 80 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[331] \quad 1^{12} 1^{11} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -240, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 10 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[332] \quad 1^{11} 1^{12} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -242, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 21 & 0 & 80 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[333] \quad 1^{23} 1^{11} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -243, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 17 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 4 & 4 & 0 & 17 & 2 \\ 17 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[334] \quad 1^{11} 1^{12} 3^{11} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.62 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 31 & 0 & 188 & 24 \\ 3 & 4 & -1 & 24 & 2 \end{pmatrix}_{b_0}$$

[335] $1^{13} 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -251, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 15 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[337] $1^{11} 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -252, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 11 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[336] $1^{114} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -252, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 26 & 3 \\ 3 & 4 & 4 & 39 & 4 \\ 4 & 4 & 0 & 26 & 2 \\ 26 & 39 & 26 & 280 & 30 \\ 3 & 4 & 2 & 30 & 2 \end{pmatrix}_{b_0}$$

[338] $1^{14} 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -252, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 18 & 31 & 20 & 188 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[339] $1^{123} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -252, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 16 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 4 & 4 & 0 & 14 & 2 \\ 16 & 21 & 14 & 80 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[340] $1^{113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -252, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 4 & 4 & 0 & 14 & 2 \\ 14 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[341] $1^{114} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.02 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 11 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

[342] $1^{114} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 12 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[343] $1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -259, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 13 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[344] $1^{112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -259, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 11 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[345] $1^{114} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 22 & 0 & 97 & 17 \\ 3 & 4 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[346] $1^{11}3^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 22 & 0 & 97 & 17 \\ 3 & 4 & -1 & 17 & 2 \end{pmatrix}_{b_0}$$

[347] $1^{11}3^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 15 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 11 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

[348] $21^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -266, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 9 & 5 & 3 \\ 4 & 4 & 13 & 7 & 4 \\ 9 & 13 & 27 & 16 & 9 \\ 5 & 7 & 16 & 7 & 5 \\ 3 & 4 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[349] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -270, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 19 & 0 & 64 & 14 \\ 3 & 4 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[350] $21^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 8 & 3 \\ 4 & 4 & 4 & 11 & 4 \\ 3 & 4 & 0 & 8 & 2 \\ 8 & 11 & 8 & 20 & 8 \\ 3 & 4 & 2 & 8 & 2 \end{pmatrix}_{b_0}$$

[351] $1^{11}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -271, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 20 & 31 & 20 & 188 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

[352] $1^{11}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -272, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 19 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 4 & 4 & 0 & 20 & 2 \\ 19 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[353] $1^{11}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -272, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 13 & 21 & 14 & 80 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[354] $1^{11}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -272, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 15 & 21 & 14 & 80 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

[355] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -275, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 12 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

[356] $1^{11}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 19 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[357] \quad 1^{11} 3^{11} 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 27 & 0 & 151 & 21 \\ 3 & 4 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

$$[358] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 16 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 4 & 4 & 0 & 17 & 2 \\ 16 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[359] \quad 1^{14} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -284, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.42 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 14 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 4 & 4 & 0 & 14 & 2 \\ 14 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[362] \quad 1^{14} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -288, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 19 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 19 & 0 & 64 & 14 \\ 3 & 4 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[360] \quad 1^{12} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -288, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 14 & 21 & 14 & 80 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[361] \quad 1^{14} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -288, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 29 & 0 & 148 & 22 \\ 3 & 4 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[363] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -292, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 24 & 0 & 100 & 18 \\ 3 & 4 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[364] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -293, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 20 & 0 & 69 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[365] \quad 1^{12} 3^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -296, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 21 & 0 & 80 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[366] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -296, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 12 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[367] \quad 1^{13} 3^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -297, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 30 & 0 & 165 & 23 \\ 3 & 4 & -1 & 23 & 2 \end{pmatrix}_{b_0}$$

1^{24}

$$[368] \ 1^{113} 3^{11} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 21 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 4 & 4 & 0 & 20 & 2 \\ 21 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

1^4

$$[369] \ 1^{114} 3^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 14 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

1^2

$$[370] \ 1^{114} 3^{11} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 12 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

1^{11}

$$[371] \ 1^{12} 3^{11} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 26 & 0 & 128 & 20 \\ 3 & 4 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

1^1

$$[372] \ 1^{112} 3^{11} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -307, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 16 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[373] \ 21^{123} 1^{112} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -311, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 21 & 4 \\ 5 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 21 & 21 & 14 & 80 & 16 \\ 4 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

1^1

$$[374] \ 1^{114} 3^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -311, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 16 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

1^{12}

$$[375] \ 1^{112} 3^{11} 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -311, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 21 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 14 & 21 & 14 & 80 & 16 \\ 3 & 4 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

1^{12}

$$[376] \ 1^{112} 3^{11} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -315, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 17 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

1^1

$$[380] \ 1^{113} 3^{11} 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -316, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.63 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 12 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

1^4

$$[381] \ 1^{112} 3^{11} 1^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -316, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 22 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 22 & 31 & 20 & 188 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

1^{24}

$$[377] \ 1^{113} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -316, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 4 & 4 & 0 & 14 & 2 \\ 15 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

1^4

$$[378] \ 1^{113} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -316, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 15 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

1^1

$$[379] \ 1^{112} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -316, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.51 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 19 & 31 & 20 & 188 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

1^1

$$[383] \ 1^{114} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -319, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 22 & 3 \\ 3 & 4 & 4 & 39 & 4 \\ 3 & 4 & 0 & 26 & 2 \\ 22 & 39 & 26 & 280 & 30 \\ 3 & 4 & 2 & 30 & 2 \end{pmatrix}_{b_0}$$

1^{12}

$$[382] \ 1^{114} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -319, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 12 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

1^3

$$[384] \ 1^{114} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -320, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 2 & 4 & 0 & 14 & 2 \\ 13 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

1^3

$$[385] \ 1^{112} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -323, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 18 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

1^{13}

$$[386] \ 1^{11} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 27 & 0 & 151 & 21 \\ 3 & 4 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

1^{13}

$$[387] \ 1^{12} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -326, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 21 & 0 & 80 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

1^{13}

$$[389] \ 1^{13} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -329, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 20 & 0 & 69 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[388] $1^{13}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -329, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 25 & 0 & 111 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[390] $1^{12}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 17 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

[391] $1^{14}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -332, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 21 & 0 & 80 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[393] $1^{14}3^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -339, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 19 & 34 & 23 & 208 & 26 \\ 3 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

[392] $13^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -339, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 13 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

[394] $1^{13}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -341, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 20 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 20 & 0 & 69 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[395] $1^{11}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -343, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 27 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 27 & 0 & 151 & 21 \\ 3 & 4 & -1 & 21 & 2 \end{pmatrix}_{b_0}$$

[396] $1^{11}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -344, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 21 & 35 & 23 & 231 & 27 \\ 3 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

[398] $1^{12}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -347, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 4 & 4 & 0 & 17 & 2 \\ 18 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[397] $13^{11}1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -347, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 14 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[401] $1^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -348, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 23 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 4 & 4 & 0 & 23 & 2 \\ 23 & 34 & 23 & 208 & 26 \\ 3 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[411] \ 1^{1^{13}} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -375, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 13 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[415] \ 1^{1^{12}} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -376, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.21 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 26 & 0 & 128 & 20 \\ 3 & 4 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[412] \ 1^{1^{12}} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -376, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.16 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 24 & 0 & 100 & 18 \\ 3 & 4 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[413] \ 1^{1^{11}} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -376, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.21 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 18 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

$$[414] \ 1^{1^{12}} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -376, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.21 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 13 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 13 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[417] \ 1^{1^{13}} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -379, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.12 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 18 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[416] \ 1^{1^{11}} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -379, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 14 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[418] \ 21^{1^{12}} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -380, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.17 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 11 & 3 \\ 4 & 4 & 4 & 16 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 11 & 16 & 11 & 44 & 12 \\ 3 & 4 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[419] \ 1^{1^1} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -384, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.36 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 34 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 34 & 0 & 208 & 26 \\ 3 & 4 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[420] \ 1^{1^{24}} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -387, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 17 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 4 & 4 & 0 & 17 & 2 \\ 17 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[422] \ 21^{1^{13}} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \ T_{\max} = 12, \ \Delta = -95, \ \det = -399, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.11 \rightarrow 11, \ T_{\max} = 12, \ T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 27 & 4 \\ 5 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 27 & 26 & 17 & 128 & 20 \\ 4 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[421] \quad 1^{114} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -399, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 19 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 14 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[423] \quad 1^{112} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -407, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 25 & 0 & 111 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[424] \quad 1^{112} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -411, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 26 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 19 & 26 & 17 & 128 & 20 \\ 3 & 4 & 2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[425] \quad 1^{114} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -414, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 29 & 0 & 148 & 22 \\ 3 & 4 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[426] \quad 1^{112} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -415, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 22 & 3 \\ 3 & 4 & 4 & 31 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 22 & 31 & 20 & 188 & 24 \\ 3 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[429] \quad 13^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -416, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.72 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 17 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

$$[427] \quad 21^{123} 3^1 1^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -416, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 24 & 4 \\ 5 & 4 & 4 & 25 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 24 & 25 & 17 & 111 & 19 \\ 4 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[428] \quad 1^{114} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -416, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 20 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 4 & 4 & 0 & 20 & 2 \\ 20 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[430] \quad 1^{113} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -417, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 30 & 0 & 165 & 23 \\ 3 & 4 & -1 & 23 & 2 \end{pmatrix}_{b_0}$$

$$[431] \quad 1^{113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -419, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 15 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[433] \quad 1^{113} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 26 & 0 & 128 & 20 \\ 3 & 4 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[432] \quad 1^{1^{13}}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -424, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.19 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 14 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[434] \quad 1^{1^{14}}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -427, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.74 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 14 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[435] \quad 21^{1^{13}}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -431, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.10 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 23 & 3 \\ 4 & 4 & 4 & 35 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 23 & 35 & 23 & 231 & 27 \\ 3 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[436] \quad 1^{1^{14}}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -436, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.16 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 24 & 0 & 100 & 18 \\ 3 & 4 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[438] \quad 21^{1^{13}}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -439, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 3 & 11 & 3 \\ 4 & 4 & 4 & 15 & 4 \\ 3 & 4 & 0 & 11 & 2 \\ 11 & 15 & 11 & 39 & 11 \\ 3 & 4 & 2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[437] \quad 21^{1^{14}}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -439, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.10 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 33 & 4 \\ 5 & 4 & 4 & 31 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 33 & 31 & 20 & 188 & 24 \\ 4 & 4 & 2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[439] \quad 1^{1^{13}}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -440, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.18 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 20 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 15 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[440] \quad 1^{1^{13}}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -443, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.32 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 18 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[441] \quad 1^{1^{12}}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -448, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.29 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 31 & 0 & 188 & 24 \\ 3 & 4 & -1 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[442] \quad 1^{1^{14}}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -448, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.29 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 26 & 0 & 128 & 20 \\ 3 & 4 & -1 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[443] \quad 13^11^{1^{13}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -459, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.71 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 2 & 4 & 0 & 23 & 2 \\ 20 & 35 & 23 & 231 & 27 \\ 3 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

[444] $1^{11}3^{12}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -459, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 16 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[445] $1^{13}3^{12}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -461, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 25 & 0 & 111 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[446] $1^{11}3^{12}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -464, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 16 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[447] $1^{11}3^{12}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -467, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 17 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[448] $1^{14}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -472, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 24 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 24 & 0 & 100 & 18 \\ 3 & 4 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

[449] $1^{14}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -472, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 39 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 22 & 39 & 0 & 280 & 30 \\ 3 & 4 & -1 & 30 & 2 \end{pmatrix}_{b_0}$$

[450] $13^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -476, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 16 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[451] $21^{12}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -479, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 27 & 4 \\ 5 & 4 & 4 & 29 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 27 & 29 & 20 & 148 & 22 \\ 4 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[452] $1^{11}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -479, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 17 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[453] $1^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -483, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 15 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[454] $1^{14}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -491, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 25 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 25 & 0 & 111 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[455] $1^{114}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -491, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 15 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 15 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[456] $1^{113}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -496, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 18 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

[457] $1^{113}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -496, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 22 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 22 & 35 & 23 & 231 & 27 \\ 3 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

[458] $1^{13}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -497, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 35 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 35 & 0 & 231 & 27 \\ 3 & 4 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[459] $1^{114}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -499, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 17 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[460] $1^{113}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -504, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 19 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

[461] $1^{114}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -507, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.47 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 2 & 4 & 0 & 17 & 2 \\ 16 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[462] $1^{13}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -513, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 30 & 0 & 165 & 23 \\ 3 & 4 & -1 & 23 & 2 \end{pmatrix}_{b_0}$$

[463] $1^{12}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -514, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 31 & 0 & 188 & 24 \\ 3 & 4 & -1 & 24 & 2 \end{pmatrix}_{b_0}$$

[464] $1^{14}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -516, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 18 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 18 & 29 & 0 & 148 & 22 \\ 3 & 4 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[465] $1^{14}3^11^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -528, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 34 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 34 & 0 & 208 & 26 \\ 3 & 4 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[466] \ 1^{\overline{1^{11}}} 1^{\overline{1^{14}}} 3^{\overline{1^{11}}} 1^{\overline{1^{13}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -531, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 20 & 34 & 23 & 208 & 26 \\ 3 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

$$[467] \ 1^{\overline{1^{11}}} 1^{\overline{1^{14}}} 3^{\overline{1^{11}}} 1^{\overline{1^{14}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -535, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.23 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 23 & 3 \\ 3 & 4 & 4 & 39 & 4 \\ 3 & 4 & 0 & 26 & 2 \\ 23 & 39 & 26 & 280 & 30 \\ 3 & 4 & 2 & 30 & 2 \end{pmatrix}_{b_0}$$

$$[468] \ 1^{\overline{1^{13}}} 1^{\overline{1^{13}}} 3^{\overline{1^{12}}} 1^{\overline{1^{12}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -536, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 17 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[469] \ 1^{\overline{1^2}} 1^{\overline{1^{13}}} 3^{\overline{1^{13}}} 1^{\overline{1^{13}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -544, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 19 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

$$[470] \ 1^{\overline{1^4}} 1^{\overline{1^{13}}} 3^{\overline{1^{13}}} 1^{\overline{1^{13}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -544, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 21 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

$$[471] \ 1^{\overline{1^1}} 1^{\overline{1^{13}}} 3^{\overline{1^{14}}} 1^{\overline{1^{14}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -547, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.54 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 2 & 4 & 0 & 23 & 2 \\ 21 & 35 & 23 & 231 & 27 \\ 3 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[472] \ 1^{\overline{1^{14}}} 1^{\overline{1^{12}}} 3^{\overline{1^{14}}} 1^{\overline{1^{14}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -556, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 31 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 22 & 31 & 0 & 188 & 24 \\ 3 & 4 & -1 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[473] \ 1^{\overline{1^3}} 1^{\overline{1^{13}}} 3^{\overline{1^{13}}} 1^{\overline{1^{13}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -560, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 20 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

$$[474] \ 1^{\overline{1^{13}}} 1^{\overline{1^{14}}} 3^{\overline{1^{11}}} 1^{\overline{1^{11}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -571, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.22 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 16 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 16 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[475] \ 1^{\overline{1^1}} 1^{\overline{1^{14}}} 3^{\overline{1^{12}}} 1^{\overline{1^{12}}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -572, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 17 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[476] $1^{14} 1^{13} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -576, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 25 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 18 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[477] $1^{13} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -585, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 30 & 0 & 165 & 23 \\ 3 & 4 & -1 & 23 & 2 \end{pmatrix}_{b_0}$$

[478] $1^{13} 1^{14} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -587, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 2 & 4 & 0 & 23 & 2 \\ 19 & 34 & 23 & 208 & 26 \\ 3 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

[479] $1^{14} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -594, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 19 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 29 & 0 & 148 & 22 \\ 3 & 4 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[480] $1^{13} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 20 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

[481] $21^{13} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 30 & 4 \\ 5 & 4 & 4 & 30 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 30 & 30 & 20 & 165 & 23 \\ 4 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

[482] $1^{12} 1^{13} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -603, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 22 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 2 & 4 & 0 & 23 & 2 \\ 22 & 35 & 23 & 231 & 27 \\ 3 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

[483] $1^{12} 1^{13} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -611, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 35 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 22 & 35 & 0 & 231 & 27 \\ 3 & 4 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[484] $1^{14} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -615, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 18 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 18 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[485] $1^{13} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -616, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 23 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 23 & 35 & 23 & 231 & 27 \\ 3 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

[487] $1^{14} 1^{13} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -619, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 24 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 2 & 4 & 0 & 23 & 2 \\ 24 & 35 & 23 & 231 & 27 \\ 3 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

[486] $1^{114}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -619, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 17 & 3 \\ 3 & 4 & 4 & 24 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 17 & 24 & 17 & 100 & 18 \\ 3 & 4 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

[488] $1^{113}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -627, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 23 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 2 & 4 & 0 & 23 & 2 \\ 23 & 35 & 23 & 231 & 27 \\ 3 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

[489] $1^{114}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -633, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 30 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 30 & 0 & 165 & 23 \\ 3 & 4 & -1 & 23 & 2 \end{pmatrix}_{b_0}$$

[490] $1^{114}3^11^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -634, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 39 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 39 & 0 & 280 & 30 \\ 3 & 4 & -1 & 30 & 2 \end{pmatrix}_{b_0}$$

[491] $1^{114}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -636, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.52 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 18 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[492] $21^{113}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -640, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 20 & 3 \\ 4 & 4 & 4 & 30 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 20 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

[493] $21^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -644, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 14 & 3 \\ 4 & 4 & 4 & 20 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 14 & 20 & 14 & 69 & 15 \\ 3 & 4 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

[494] $1^{114}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -648, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 21 & 3 \\ 3 & 4 & 0 & 34 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 21 & 34 & 0 & 208 & 26 \\ 3 & 4 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

[495] $1^{114}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -648, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 20 & 3 \\ 3 & 4 & 0 & 29 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 20 & 29 & 0 & 148 & 22 \\ 3 & 4 & -1 & 22 & 2 \end{pmatrix}_{b_0}$$

[496] $21^{114}3^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -655, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 14 & 3 \\ 4 & 4 & 4 & 19 & 4 \\ 3 & 4 & 0 & 14 & 2 \\ 14 & 19 & 14 & 64 & 14 \\ 3 & 4 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$

[497] $1^{113}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -664, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 4 & 30 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 21 & 30 & 20 & 165 & 23 \\ 3 & 4 & 2 & 23 & 2 \end{pmatrix}_{b_0}$$

[499] $1^{1^4} 1^{1^4} 3^1 1^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -668, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 20 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[498] $1^{1^3} 1^{1^4} 3^1 1^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -668, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 2 & 4 & 0 & 20 & 2 \\ 19 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[500] $1^{1^4} 1^{1^4} 3^1 1^{1^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -672, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 22 & 3 \\ 3 & 4 & 4 & 39 & 4 \\ 2 & 4 & 0 & 26 & 2 \\ 22 & 39 & 26 & 280 & 30 \\ 3 & 4 & 2 & 30 & 2 \end{pmatrix}_{b_0}$$

[501] $1^{1^{12}} 1^{1^4} 3^1 1^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -691, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 21 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 21 & 34 & 23 & 208 & 26 \\ 3 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

[502] $1^{1^{13}} 1^{1^4} 3^1 1^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -691, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 2 & 4 & 0 & 23 & 2 \\ 20 & 34 & 23 & 208 & 26 \\ 3 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

[503] $1^{1^{13}} 3^1 1^{1^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -701, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.18 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 35 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 35 & 0 & 231 & 27 \\ 3 & 4 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[504] $1^{1^{14}} 1^{1^4} 3^1 1^{1^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -704, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 24 & 3 \\ 3 & 4 & 4 & 35 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 24 & 35 & 23 & 231 & 27 \\ 3 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

[505] $21^{1^{13}} 3^1 1^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -711, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 17 & 3 \\ 4 & 4 & 4 & 25 & 4 \\ 3 & 4 & 0 & 17 & 2 \\ 17 & 25 & 17 & 111 & 19 \\ 3 & 4 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

[506] $1^{1^{13}} 1^{1^4} 3^1 1^{1^2} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -719, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 19 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 19 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[507] $1^{1^{12}} 1^{1^4} 3^1 1^{1^4} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -719, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 24 & 3 \\ 3 & 4 & 4 & 39 & 4 \\ 3 & 4 & 0 & 26 & 2 \\ 24 & 39 & 26 & 280 & 30 \\ 3 & 4 & 2 & 30 & 2 \end{pmatrix}_{b_0}$$

[508] $21^{1^4} 3^1 1^{1^3} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -736, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.11 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 36 & 4 \\ 5 & 4 & 4 & 35 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 36 & 35 & 23 & 231 & 27 \\ 4 & 4 & 2 & 27 & 2 \end{pmatrix}_{b_0}$$

[509] $1^{14}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -744, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 22 & 3 \\ 3 & 4 & 0 & 34 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 22 & 34 & 0 & 208 & 26 \\ 3 & 4 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

[510] $21^{13}3^{11}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -759, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 3 & 33 & 4 \\ 5 & 4 & 4 & 34 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 33 & 34 & 23 & 208 & 26 \\ 4 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

[511] $1^{14}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -763, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 21 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 2 & 4 & 0 & 23 & 2 \\ 21 & 34 & 23 & 208 & 26 \\ 3 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

[512] $1^{14}3^{11}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -767, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.19 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 24 & 3 \\ 3 & 4 & 0 & 35 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 35 & 0 & 231 & 27 \\ 3 & 4 & -1 & 27 & 2 \end{pmatrix}_{b_0}$$

[513] $1^{14}3^{11}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -772, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.17 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 24 & 3 \\ 3 & 4 & 0 & 39 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 39 & 0 & 280 & 30 \\ 3 & 4 & -1 & 30 & 2 \end{pmatrix}_{b_0}$$

[514] $1^{14}3^{11}1^{14} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -784, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.58 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 23 & 3 \\ 3 & 4 & 4 & 39 & 4 \\ 2 & 4 & 0 & 26 & 2 \\ 23 & 39 & 26 & 280 & 30 \\ 3 & 4 & 2 & 30 & 2 \end{pmatrix}_{b_0}$$

[515] $1^{14}3^{11}1^{12} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -791, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 20 & 3 \\ 3 & 4 & 4 & 29 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 20 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

[516] $1^{14}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -803, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 22 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 2 & 4 & 0 & 23 & 2 \\ 22 & 34 & 23 & 208 & 26 \\ 3 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

[517] $1^{14}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -811, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 23 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 2 & 4 & 0 & 23 & 2 \\ 23 & 34 & 23 & 208 & 26 \\ 3 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

[518] $1^{14}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -816, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 1 & 23 & 3 \\ 3 & 4 & 0 & 34 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 23 & 34 & 0 & 208 & 26 \\ 3 & 4 & -1 & 26 & 2 \end{pmatrix}_{b_0}$$

[519] $1^{14}3^{11}1^{13} + [\mathbf{su}_2 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -819, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 22 & 3 \\ 3 & 4 & 4 & 34 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 22 & 34 & 23 & 208 & 26 \\ 3 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 3 & 20 & 3 \\ 4 & 4 & 4 & 29 & 4 \\ 3 & 4 & 0 & 20 & 2 \\ 20 & 29 & 20 & 148 & 22 \\ 3 & 4 & 2 & 22 & 2 \end{pmatrix}_{b_0} \\
[532] \quad & 21^{114} 3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -1144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 4 & 3 & 23 & 3 \\ 4 & 4 & 4 & 34 & 4 \\ 3 & 4 & 0 & 23 & 2 \\ 23 & 34 & 23 & 208 & 26 \\ 3 & 4 & 2 & 26 & 2 \end{pmatrix}_{b_0} \\
& T = 8 \quad [533] \quad 31^{11} 31^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 4 & 2 \\ 3 & 4 & 4 & 6 & 5 & 4 \\ 3 & 4 & 0 & 3 & 4 & 2 \\ 3 & 6 & 3 & 3 & 5 & 3 \\ 4 & 5 & 4 & 5 & 4 & 4 \\ 2 & 4 & 2 & 3 & 4 & 2 \end{pmatrix}_{b_0} \\
[534] \quad & 31^{111} 31^{11} 3^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 255, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 4 & 2 \\ 3 & 4 & 4 & 6 & 6 & 4 \\ 3 & 4 & 0 & 3 & 3 & 2 \\ 3 & 6 & 3 & 3 & 6 & 3 \\ 4 & 6 & 3 & 6 & 4 & 4 \\ 2 & 4 & 2 & 3 & 4 & 2 \end{pmatrix}_{b_0} \\
& T = 10 \quad [535] \quad 1^{11} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 1 & 3 & 3 \\ 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[536] \quad & 1^{13} 5^{11} 13^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 14 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 3 & 3 & 1 & 21 & 3 \\ 14 & 24 & 21 & 130 & 24 \\ 3 & 4 & 3 & 24 & 4 \end{pmatrix}_{b_0} \\
[537] \quad & 1^{12} 5^{11} 13^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 3 & 3 & 1 & 15 & 3 \\ 10 & 17 & 15 & 65 & 17 \\ 3 & 4 & 3 & 17 & 4 \end{pmatrix}_{b_0} \\
[538] \quad & 1^{111} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 3 & 3 & 1 & 9 & 3 \\ 6 & 10 & 9 & 22 & 10 \\ 3 & 4 & 3 & 10 & 4 \end{pmatrix}_{b_0} \\
[539] \quad & 1^{111} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 2 & 3 & 1 & 9 & 3 \\ 7 & 10 & 9 & 22 & 10 \\ 3 & 4 & 3 & 10 & 4 \end{pmatrix}_{b_0} \\
[541] \quad & 1^{13} 5^{11} 13^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 16 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 1 & 21 & 3 \\ 16 & 24 & 21 & 130 & 24 \\ 3 & 4 & 3 & 24 & 4 \end{pmatrix}_{b_0} \\
[542] \quad & 1^{111} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0
\end{aligned}$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 3 & 10 & 4 \\ 2 & 3 & 1 & 9 & 3 \\ 8 & 10 & 9 & 22 & 10 \\ 3 & 4 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[540] \ 1^{\overline{1}} \overline{5} 1^{\overline{2}} 3^{\overline{1}} 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & 1 & 16 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 22 & 2 \\ 13 & 16 & 0 & 22 & 29 & 29 \\ 2 & 2 & -1 & 2 & 29 & 3 \end{pmatrix}_{b_0}$$

$$[543] \ 1^{\overline{12}} \overline{5} 1^{\overline{1}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 1 & 15 & 3 \\ 12 & 17 & 15 & 65 & 17 \\ 3 & 4 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[544] \ 1^{\overline{1}} \overline{5} 1^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 5 & 3 & 4 & 5 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 4 & 2 & 1 & 3 \\ 3 & 5 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[545] \ 1^{\overline{1}} \overline{5} 1^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 5 & 8 & 4 & 5 \\ 4 & 8 & 5 & 5 & 6 \\ 3 & 4 & 5 & 1 & 3 \\ 3 & 5 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[546] \ 1^{\overline{12}} \overline{5} 1^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 5 & 8 & 9 & 5 \\ 4 & 8 & 8 & 11 & 6 \\ 6 & 9 & 11 & 11 & 7 \\ 3 & 5 & 6 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[547] \ 1^{\overline{13}} \overline{5} 1^{\overline{1}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 5 & 8 & 14 & 5 \\ 4 & 8 & 8 & 17 & 6 \\ 9 & 14 & 17 & 29 & 11 \\ 3 & 5 & 6 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[548] \ 1^{\overline{14}} \overline{5} 1^{\overline{1}} 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 5 & 8 & 19 & 5 \\ 4 & 8 & 8 & 23 & 6 \\ 12 & 19 & 23 & 55 & 15 \\ 3 & 5 & 6 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[549] \ 1^{\overline{1}} \overline{5} 1^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 3 & 3 \\ 3 & 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[550] \ 1^{\overline{14}} \overline{5} 1^{\overline{1}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 1 & 15 & 3 \\ 14 & 17 & 15 & 65 & 17 \\ 3 & 4 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[551] \ 1^{\overline{13}} \overline{5} 1^{\overline{3}} 2^{\overline{1}} 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 16 & 2 \\ 1 & 0 & 1 & 1 & 19 & 2 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 0 & 0 & 26 & 2 \\ 16 & 19 & 0 & 26 & 339 & 35 \\ 2 & 2 & -1 & 2 & 35 & 3 \end{pmatrix}_{b_0}$$

$$[552] \ 1^{12} \overset{1^3}{5} 13^{11} 1^2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 3 \\ 3 & 4 & 3 & 17 & 4 \\ 2 & 3 & 1 & 15 & 3 \\ 13 & 17 & 15 & 65 & 17 \\ 3 & 4 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[553] \ 1^{15} \overset{1^1}{5} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 5 & 3 & 9 & 5 \\ 2 & 3 & 0 & 4 & 2 \\ 5 & 9 & 4 & 11 & 7 \\ 3 & 5 & 2 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[554] \ 1^{15} \overset{1^2}{5} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 27, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 5 & 3 & 9 & 5 \\ 2 & 3 & 0 & 4 & 2 \\ 6 & 9 & 4 & 11 & 7 \\ 3 & 5 & 2 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[555] \ 1^{13} \overset{1^3}{5} 13^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 1 & 21 & 3 \\ 17 & 24 & 21 & 130 & 24 \\ 3 & 4 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[556] \ 1^{11} \overset{1^1}{5} 13^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 5 & 3 \\ 5 & 5 & 5 & 8 & 6 \\ 3 & 3 & 3 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[557] \ 21^{11} \overset{1^1}{5} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 37, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 9 & 3 \\ 7 & 7 & 9 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[558] \ 1^{13} \overset{1^4}{5} 13^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 3 \\ 3 & 4 & 3 & 24 & 4 \\ 2 & 3 & 1 & 21 & 3 \\ 18 & 24 & 21 & 130 & 24 \\ 3 & 4 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[559] \ 1^{12} \overset{1^2}{5} 13^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 5 & 3 & 14 & 5 \\ 2 & 3 & 0 & 6 & 2 \\ 8 & 14 & 6 & 29 & 11 \\ 3 & 5 & 2 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[560] \ 1^{13} \overset{1^3}{5} 13^{11} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 5 & 3 & 14 & 5 \\ 2 & 3 & 0 & 6 & 2 \\ 9 & 14 & 6 & 29 & 11 \\ 3 & 5 & 2 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[561] \ 1^{11} \overset{1^1}{5} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 9 & 3 \\ 7 & 7 & 9 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[562] $1^{11} \overset{1^3}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 9 & 3 \\ 9 & 7 & 9 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[563] $1^{11} \overset{1^3}{5} 13^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 70, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 5 & 3 & 19 & 5 \\ 2 & 3 & 0 & 8 & 2 \\ 11 & 19 & 8 & 55 & 15 \\ 3 & 5 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

[564] $1^{11} \overset{1^2}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 9 & 3 \\ 8 & 7 & 9 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[565] $1^{11} \overset{1^4}{5} 13^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 75, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 5 & 3 & 19 & 5 \\ 2 & 3 & 0 & 8 & 2 \\ 12 & 19 & 8 & 55 & 15 \\ 3 & 5 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

[566] $1^{11} \overset{1^3}{5} 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 1 & 13 & 3 \\ 11 & 9 & 13 & 44 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[567] $1^{11} \overset{1^4}{5} 13^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 101, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 3 \\ 3 & 5 & 3 & 24 & 5 \\ 2 & 3 & 0 & 10 & 2 \\ 14 & 24 & 10 & 89 & 19 \\ 3 & 5 & 2 & 19 & 4 \end{pmatrix}_{b_0}$$

[568] $21^{12} \overset{1^1}{5} 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 105, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 3 \\ 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 1 & 15 & 3 \\ 12 & 12 & 15 & 65 & 17 \\ 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[570] $1^{12} \overset{1^3}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 125, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 11 & 3 \\ 11 & 10 & 11 & 39 & 13 \\ 3 & 3 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

[569] $1^{12} \overset{1^2}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 125, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 11 & 3 \\ 10 & 10 & 11 & 39 & 13 \\ 3 & 3 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

[571] $21^{12} \overset{1^1}{5} 13^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 147, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 11 & 3 \\ 10 & 10 & 11 & 39 & 13 \\ 3 & 3 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

[572] $1^{12} \overset{1^2}{5} 13^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 173, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

[583] $1^{13}5^{14}13^{11}1^{14} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -60, \det = 424, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 20 & 3 \\ 2 & 1 & 3 & 19 & 3 \\ 2 & 3 & 1 & 25 & 3 \\ 20 & 19 & 25 & 176 & 28 \\ 3 & 3 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

[584] $1^{13}5^{11}3^{12}23^{11}1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 10, T_{\max} = 12, \Delta = -62, \det = 629, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.00 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 15 & 2 \\ 1 & 0 & 2 & 0 & 18 & 2 \\ 1 & 2 & 0 & 0 & 23 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 18 & 23 & 0 & 290 & 32 \\ 2 & 2 & 2 & -1 & 32 & 3 \end{pmatrix}_{b_0}$$

$T = 11$ [586] $1^{11}6^{11}1^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 4 & 3 & 3 & 15 & 4 \\ 3 & 1 & 3 & 11 & 3 \\ 3 & 3 & 1 & 12 & 3 \\ 15 & 11 & 12 & 55 & 15 \\ 4 & 3 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[585] $21^{11}6^{11}3^{12}21^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 0 & 1 & 2 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 3 & 6 & 3 \\ 5 & 4 & 0 & 6 & 10 & 6 \\ 2 & 2 & 0 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

[587] $1^{11}413^{11}15^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 6 & 10 & 6 \\ 10 & 16 & 10 \\ 6 & 10 & 6 \end{pmatrix}_{b_0}$$

[588] $1^{11}41^{11}3^{11}15^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 0 & 1 & 2 & 5 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 0 & 5 & 2 \\ 6 & 5 & 0 & 5 & 20 & 8 \\ 2 & 2 & 0 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

[589] $1^{11}6^{11}123^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -13, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 3 & 3 \\ 3 & 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[590] $1^{11}41^{11}3^{11}15^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 6 & 19 & 6 \\ 19 & 57 & 19 \\ 6 & 19 & 6 \end{pmatrix}_{b_0}$$

[591] $21^{11}6^{11}3^{12}21^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 1 & 11 & 3 \\ 4 & 8 & 1 & 18 & 6 \\ 1 & 1 & -2 & 1 & 0 \\ 11 & 18 & 1 & 35 & 13 \\ 3 & 6 & 0 & 13 & 4 \end{pmatrix}_{b_0}$$

[592] $1^{11}413^{11}15^{11}1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 13, \Delta = -59, \det = -26, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 0 & 3 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 4 \\ 2 & 2 & 0 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

[593] $1^{12}41^13^1151^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -28, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 6 & 26 & 6 \\ 26 & 108 & 26 \\ 6 & 26 & 6 \end{pmatrix}_{b_0}$

[595] $1^1413^1151^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 4 & 3 \\ 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 4 & 2 & 0 & 2 & 4 \\ 3 & 2 & 0 & 4 & 4 \end{pmatrix}_{b_0}$

[594] $1^{11}6^{11}123^11^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 5 & 3 \\ 2 & 1 & 3 & 5 & 3 \\ 2 & 3 & 1 & 5 & 3 \\ 5 & 5 & 5 & 8 & 6 \\ 3 & 3 & 3 & 6 & 4 \end{pmatrix}_{b_0}$

[596] $21^{11}6^{11}123^11^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -37, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 1 & 6 & 3 \\ 4 & 8 & 1 & 10 & 6 \\ 1 & 1 & -2 & 1 & 0 \\ 6 & 10 & 1 & 9 & 7 \\ 3 & 6 & 0 & 7 & 4 \end{pmatrix}_{b_0}$

[597] $1^{11}41^13^1151^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -48, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 0 & 5 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 5 & 0 & 0 & 20 & 8 \\ 2 & 2 & 0 & 0 & 8 & 3 \end{pmatrix}_{b_0}$

[598] $21^{11}6^{11}123^11^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 4 & 1 & 10 & 3 \\ 4 & 8 & 1 & 18 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 10 & 18 & 0 & 34 & 12 \\ 3 & 6 & 0 & 12 & 4 \end{pmatrix}_{b_0}$

[603] $1^{11}6^{11}123^11^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 9 & 3 \\ 7 & 7 & 9 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$

[599] $1^{11}6^{11}123^11^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 9 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 9 & 3 \\ 9 & 7 & 9 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$

[600] $1^16^{12}1^13^{121}1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 10 & 3 \\ 2 & 3 & 1 & 14 & 3 \\ 12 & 10 & 14 & 60 & 16 \\ 3 & 3 & 3 & 16 & 4 \end{pmatrix}_{b_0}$

[601] $1^{11}413^1151^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 1 & 5 & 3 \\ 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 5 & 4 & 0 & 6 & 6 \\ 3 & 2 & 0 & 6 & 4 \end{pmatrix}_{b_0}$

[602] $1^141^13^1151^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_2} \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 4 & 0 & 14 & 8 \\ 3 & 2 & 0 & 8 & 4 \end{pmatrix}_{b_0}$$

[604] $1^{12}41^13^1151^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -70, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & 0 & 7 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 7 & 7 & 0 & 0 & 32 & 10 \\ 2 & 2 & 0 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

[605] $1^{12}6123^11^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -72, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 1 & 9 & 3 \\ 8 & 7 & 9 & 22 & 10 \\ 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[606] $1^{11}6^{12}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -85, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 3 & 11 & 3 \\ 2 & 3 & 1 & 12 & 3 \\ 12 & 11 & 12 & 55 & 15 \\ 3 & 3 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[607] $1^{11}6^{12}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -85, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 11 & 3 \\ 2 & 3 & 1 & 12 & 3 \\ 11 & 11 & 12 & 55 & 15 \\ 3 & 3 & 3 & 15 & 4 \end{pmatrix}_{b_0}$$

[608] $1^{11}41^13^1151^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 8 & 3 \\ 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 8 & 6 & 0 & 22 & 10 \\ 3 & 2 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

[609] $1^{13}6123^11^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 1 & 13 & 3 \\ 11 & 9 & 13 & 44 & 14 \\ 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[610] $1^{14}1^13^1151^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -125, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 9 & 3 \\ 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 9 & 5 & 0 & 20 & 10 \\ 3 & 2 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

[612] $1^{11}6^{14}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -141, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 16 & 13 & 16 & 83 & 19 \\ 3 & 3 & 3 & 19 & 4 \end{pmatrix}_{b_0}$$

[611] $1^{11}6^{11}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -141, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 13 & 13 & 16 & 83 & 19 \\ 3 & 3 & 3 & 19 & 4 \end{pmatrix}_{b_0}$$

[614] $1^{11}6^{13}1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -157, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 15 & 13 & 16 & 83 & 19 \\ 3 & 3 & 3 & 19 & 4 \end{pmatrix}_{b_0}$$

[613] $1^{11} 6^{1^2} 1^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -157, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 3 & 13 & 3 \\ 2 & 3 & 1 & 16 & 3 \\ 14 & 13 & 16 & 83 & 19 \\ 3 & 3 & 3 & 19 & 4 \end{pmatrix}_{b_0}$$

[615] $1^{11} 6^{1^1} 1^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -173, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 15 & 15 & 20 & 119 & 23 \\ 3 & 3 & 3 & 23 & 4 \end{pmatrix}_{b_0}$$

[616] $1^{11} 41^{1^1} 3^{1^1} 15^{1^2} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -189, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 10 & 3 \\ 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 10 & 7 & 0 & 28 & 12 \\ 3 & 2 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

[617] $1^{11} 6^{1^2} 1^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -205, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 16 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 16 & 15 & 20 & 119 & 23 \\ 3 & 3 & 3 & 23 & 4 \end{pmatrix}_{b_0}$$

[618] $1^{11} 6^{1^4} 1^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -221, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 18 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 18 & 15 & 20 & 119 & 23 \\ 3 & 3 & 3 & 23 & 4 \end{pmatrix}_{b_0}$$

[619] $1^{11} 6^{1^3} 1^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -221, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.88 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 17 & 3 \\ 2 & 1 & 3 & 15 & 3 \\ 2 & 3 & 1 & 20 & 3 \\ 17 & 15 & 20 & 119 & 23 \\ 3 & 3 & 3 & 23 & 4 \end{pmatrix}_{b_0}$$

[620] $1^{11} 41^{1^1} 3^{1^1} 15^{1^3} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -272, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 12 & 3 \\ 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & -2 & 0 & 0 \\ 12 & 8 & 0 & 38 & 14 \\ 3 & 2 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$T = 12$ [621] $1^{11} 6^{1^1} 123^{1^1} 141^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 3 & 6 & 0 & 5 & 13 & 7 \\ 2 & 3 & -1 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[622] $1^{11} 6^{1^1} 1^{1^1} 23^{1^1} 141^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 64, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 2 & 10 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & 10 & 2 \\ 7 & 10 & 0 & 10 & 56 & 14 \\ 2 & 3 & -1 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[623] $1^{11} 6^{1^1} 123^{1^1} 141^{1^1} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 64, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$

$$\begin{aligned}
& T_{\min} = 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 2 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & 5 & 2 \\ 4 & 6 & 0 & 5 & 13 & 7 \\ 2 & 3 & -1 & 2 & 7 & 3 \end{pmatrix}_{b_0} \\
[624] & 1^{11} 4 1 1^1 3^1 1 5^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 13.94 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 8 & 1 & 7 & 3 \\ 8 & 17 & 0 & 16 & 9 \\ 1 & 0 & -2 & 0 & 0 \\ 7 & 16 & 0 & 14 & 8 \\ 3 & 9 & 0 & 8 & 4 \end{pmatrix}_{b_0} \\
[625] & 1^{11} 6^1 1^1 2 3^1 1 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 85, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 13.98 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 1 & 2 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 7 & 13 & 0 & 12 & 79 & 17 \\ 2 & 3 & -1 & 2 & 17 & 3 \end{pmatrix}_{b_0} \\
[626] & 1^{11} 6^1 1^1 2 3^1 1 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 157, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 13.89 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 2 & 13 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 0 & 0 & 12 & 2 \\ 8 & 13 & 0 & 12 & 79 & 17 \\ 2 & 3 & -1 & 2 & 17 & 3 \end{pmatrix}_{b_0} \\
T = 13 & [627] 1^{11} 6^1 1 3 1 5 1 3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -27, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 3 & 3 \\ 3 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
T = 14 & [628] 1^{11} 6^1 1 3 1 5 1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -30, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 14.86 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 2 & 4 & 4 \\ 3 & 2 & 4 & 2 & 4 \\ 3 & 3 & 4 & 4 & 5 \end{pmatrix}_{b_0} \\
[629] & 1^{11} 6^1 1 3 1 5 1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -30, \det = 32, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 2 & 7 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 11 & 4 \\ 7 & 7 & 11 & 26 & 12 \\ 3 & 3 & 4 & 12 & 5 \end{pmatrix}_{b_0} \\
[630] & 1^{11} 6^1 1 3 1 5 1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -30, \det = 43, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 14.98 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 2 & 8 & 3 \\ 2 & 1 & 3 & 7 & 3 \\ 2 & 3 & 2 & 11 & 4 \\ 8 & 7 & 11 & 26 & 12 \\ 3 & 3 & 4 & 12 & 5 \end{pmatrix}_{b_0} \\
[631] & 1^{11} 6^1 1 3 1 5 1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -30, \det = 50, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 2 & 11 & 3 \\ 2 & 1 & 3 & 9 & 3 \\ 2 & 3 & 2 & 16 & 4 \\ 11 & 9 & 16 & 53 & 17 \\ 3 & 3 & 4 & 17 & 5 \end{pmatrix}_{b_0} \\
[632] & 1^{11} 6^1 1^2 3^1 1 5 1 3 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -31, \det = 98, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 6.0
\end{aligned}$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 2 & 3 & 17 & 4 \\ 3 & 2 & 14 & 4 \\ 17 & 14 & 72 & 22 \\ 4 & 4 & 22 & 6 \end{pmatrix}_{b_0}$$

4.1191 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{f}_4$

$$T = 8 \quad [0] \ 1^{\overline{114}3^1} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 5 & 10 & 1 & 3 \\ 3 & 4 & 9 & 19 & 0 & 4 \\ 5 & 9 & 12 & 29 & 0 & 6 \\ 10 & 19 & 29 & 64 & 0 & 14 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 14 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{12}3^1} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 102, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & 5 & 11 & 0 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 \\ 6 & 11 & 9 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 8 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}3^1} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 5 & 8 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{12}1} 3^1 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 113, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 1 & 3 \\ 3 & 4 & 8 & 12 & 0 & 4 \\ 4 & 8 & 9 & 16 & 0 & 5 \\ 7 & 12 & 16 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 9 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}1} 3^1 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 4 & 2 & 7 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 7 & 16 & 8 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 9 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}1} 3^1 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.07 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 7 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 4 & 7 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{11}1} 3^1 1^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 1 & 3 \\ 3 & 4 & 4 & 27 & 0 & 4 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 16 & 27 & 17 & 151 & 0 & 21 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 21 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}1} 3^1 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 5 & 12 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{12}1} 3^1 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 197, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 1 & 3 \\ 3 & 4 & 8 & 7 & 0 & 4 \\ 4 & 8 & 9 & 10 & 0 & 5 \\ 4 & 7 & 10 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 1 & 3 \\ 3 & 4 & 9 & 15 & 0 & 4 \\ 5 & 9 & 12 & 23 & 0 & 6 \\ 8 & 15 & 23 & 39 & 0 & 11 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 11 & -3 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 227, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.04 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 7 & 0 & 4 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 4 & 7 & 5 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 9 & 7 & 0 & 4 \\ 5 & 9 & 12 & 11 & 0 & 6 \\ 4 & 7 & 11 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 5 & -3 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 1 & 3 \\ 3 & 4 & 9 & 11 & 0 & 4 \\ 5 & 9 & 12 & 17 & 0 & 6 \\ 6 & 11 & 17 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{111}3^11^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 13 & 22 & 14 & 97 & 0 & 17 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 17 & -3 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{112}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 332, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.14 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & 4 & 11 & 0 & 4 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 6 & 11 & 8 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 8 & -3 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 368, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 12 & 0 & 4 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 7 & 12 & 8 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{121}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 382, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 7 & 4 & 1 & 3 \\ 3 & 4 & 13 & 7 & 0 & 4 \\ 7 & 13 & 27 & 16 & 0 & 9 \\ 4 & 7 & 16 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 9 & 5 & -3 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{111}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 403, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 \\ 2 & 4 & 0 & 11 & 0 & 2 \\ 10 & 17 & 11 & 55 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{113}3^11^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 443, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 1 & 3 \\ 3 & 4 & 4 & 35 & 0 & 4 \\ 2 & 4 & 0 & 23 & 0 & 2 \\ 20 & 35 & 23 & 231 & 0 & 27 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 27 & -3 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{114}3^11^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 1708, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 1 & 3 \\ 3 & 4 & 4 & 29 & 0 & 4 \\ 2 & 4 & 0 & 20 & 0 & 2 \\ 16 & 29 & 20 & 148 & 0 & 22 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 22 & -3 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{114}3^11^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 2123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 19 & 1 & 3 \\ 3 & 4 & 4 & 34 & 0 & 4 \\ 2 & 4 & 0 & 23 & 0 & 2 \\ 19 & 34 & 23 & 208 & 0 & 26 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 26 & -3 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{114}3^11^{14} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 2432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 2 & 22 & 1 & 3 \\ 3 & 4 & 4 & 39 & 0 & 4 \\ 2 & 4 & 0 & 26 & 0 & 2 \\ 22 & 39 & 26 & 280 & 0 & 30 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 30 & -3 & 1 \end{pmatrix}_{b_0}$$

4.1192 NHC(-2)(-3) + $\mathfrak{su}_3$ + su2n3mix + $\mathfrak{f}_4$ + $\mathfrak{su}_2$

$T = 9 \quad [0] \quad 1^{111}3^11^{111} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 & 3 \\ 3 & 4 & 4 & 8 & 0 & 5 & 4 \\ 2 & 4 & 0 & 5 & 0 & 3 & 2 \\ 5 & 8 & 5 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 9 & 0 & 4 & 4 \\ 3 & 4 & 2 & 6 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{1111}3^11^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 2 & 3 \\ 3 & 4 & 4 & 17 & 0 & 4 & 4 \\ 2 & 4 & 0 & 11 & 0 & 3 & 2 \\ 10 & 17 & 11 & 55 & 0 & 11 & 13 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 11 & 0 & 1 & 3 \\ 3 & 4 & 2 & 13 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{1111}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 & 3 \\ 3 & 4 & 4 & 7 & 0 & 5 & 4 \\ 2 & 4 & 0 & 5 & 0 & 3 & 2 \\ 4 & 7 & 5 & 7 & 0 & 6 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 6 & 0 & 4 & 4 \\ 3 & 4 & 2 & 5 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{1211}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 5 & 11 & 0 & 4 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 & 3 \\ 6 & 11 & 9 & 20 & 0 & 8 & 8 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 8 & 0 & 1 & 3 \\ 3 & 4 & 3 & 8 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{1111}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 5 & 7 & 0 & 4 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 & 3 \\ 4 & 7 & 6 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 5 & 0 & 1 & 3 \\ 3 & 4 & 3 & 5 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{111}3^1}1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -269$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 8 & 0 & 9 & 4 \\ 2 & 4 & 0 & 5 & 0 & 6 & 2 \\ 5 & 8 & 5 & 12 & 0 & 14 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 14 & 0 & 13 & 7 \\ 3 & 4 & 2 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{113}3^1}1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -401$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.40 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 & 3 \\ 3 & 4 & 4 & 20 & 0 & 5 & 4 \\ 2 & 4 & 0 & 14 & 0 & 3 & 2 \\ 11 & 20 & 14 & 69 & 0 & 18 & 15 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 18 & 0 & 4 & 4 \\ 3 & 4 & 2 & 15 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{\overline{111}3^1}1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -401$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.96 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 & 3 \\ 3 & 4 & 4 & 12 & 0 & 5 & 4 \\ 2 & 4 & 0 & 8 & 0 & 3 & 2 \\ 7 & 12 & 8 & 25 & 0 & 12 & 9 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 12 & 0 & 4 & 4 \\ 3 & 4 & 2 & 9 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{\overline{111}3^1}1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -424$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 2 & 3 \\ 3 & 4 & 4 & 7 & 0 & 4 & 4 \\ 2 & 4 & 0 & 5 & 0 & 3 & 2 \\ 4 & 7 & 5 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 5 & 0 & 1 & 3 \\ 3 & 4 & 2 & 5 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{\overline{111}3^1}1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -464$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 2 & 3 \\ 3 & 4 & 9 & 7 & 0 & 4 & 4 \\ 5 & 9 & 12 & 11 & 0 & 6 & 6 \\ 4 & 7 & 11 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 6 & 5 & 0 & 1 & 3 \\ 3 & 4 & 6 & 5 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{\overline{111}3^1}1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -481$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 2 & 3 \\ 3 & 4 & 4 & 12 & 0 & 4 & 4 \\ 2 & 4 & 0 & 8 & 0 & 3 & 2 \\ 7 & 12 & 8 & 25 & 0 & 8 & 9 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 8 & 0 & 1 & 3 \\ 3 & 4 & 2 & 9 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{\overline{112}3^1}1^{\overline{111}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -500$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.08 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 & 3 \\ 3 & 4 & 4 & 16 & 0 & 5 & 4 \\ 2 & 4 & 0 & 11 & 0 & 3 & 2 \\ 9 & 16 & 11 & 44 & 0 & 15 & 12 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 15 & 0 & 4 & 4 \\ 3 & 4 & 2 & 12 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{\overline{111}3^1}1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -509$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.06 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 5 & 3 \\ 3 & 4 & 4 & 7 & 0 & 9 & 4 \\ 2 & 4 & 0 & 5 & 0 & 6 & 2 \\ 4 & 7 & 5 & 7 & 0 & 11 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 11 & 0 & 13 & 7 \\ 3 & 4 & 2 & 5 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{\overline{111}3^11^{\overline{111}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -896, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 5 & 3 \\ 3 & 4 & 4 & 12 & 0 & 9 & 4 \\ 2 & 4 & 0 & 8 & 0 & 6 & 2 \\ 7 & 12 & 8 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 20 & 0 & 13 & 7 \\ 3 & 4 & 2 & 9 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

4.1193 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \text{su2n3mix} + \mathfrak{f}_4 + \mathfrak{su}_3$

$T = 9 \quad [0] \quad 1^{\overline{111}3^11^{\overline{111}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 & 4 \\ 2 & 4 & 0 & 5 & 0 & 3 & 2 \\ 5 & 8 & 5 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 2 & 6 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{111}3^11^{\overline{111}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -81, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 2 & 3 \\ 4 & 9 & 4 & 12 & 0 & 4 & 5 \\ 2 & 4 & 0 & 5 & 0 & 3 & 2 \\ 5 & 12 & 5 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 5 & 0 & 0 & 2 \\ 3 & 5 & 2 & 6 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{111}3^11^{\overline{111}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$

$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 5 & 7 & 0 & 8 & 4 \\ 3 & 5 & 3 & 6 & 0 & 6 & 3 \\ 4 & 7 & 6 & 7 & 0 & 10 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 10 & 0 & 9 & 5 \\ 3 & 4 & 3 & 5 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{121}3^11^{\overline{111}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -226, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 2 & 3 \\ 3 & 4 & 5 & 11 & 0 & 4 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 & 3 \\ 6 & 11 & 9 & 20 & 0 & 8 & 8 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 8 & 0 & 0 & 2 \\ 3 & 4 & 3 & 8 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{111}3^11^{\overline{111}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -346, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 5 & 7 & 0 & 4 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 & 3 \\ 4 & 7 & 6 & 7 & 0 & 5 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 3 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

4.1194 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \text{su2n3mix} + \mathfrak{so}_8$

$T = 8 \quad [0] \quad 1^{\overline{111}3^11^{\overline{12}}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 5 & 11 & 1 & 24 & 5 \\ 11 & 20 & 0 & 45 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 24 & 45 & 0 & 100 & 18 \\ 5 & 8 & -1 & 18 & 2 \end{pmatrix}_{b_0}$$

- [2] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 2 & 8 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 8 & 16 & 8 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 9 & -2 & 1 \end{pmatrix}_{b_0}$
- [1] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$
- [3] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 34$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 5 & 11 & 1 & 7 & 5 \\ 11 & 20 & 0 & 11 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 11 & 0 & 4 & 4 \\ 5 & 8 & -1 & 4 & 2 \end{pmatrix}_{b_0}$
- [4] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 37$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 3 & 11 & 1 & 3 \\ 3 & 4 & 5 & 19 & 0 & 4 \\ 3 & 5 & 3 & 15 & 0 & 3 \\ 11 & 19 & 15 & 64 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 14 & -2 & 1 \end{pmatrix}_{b_0}$
- [5] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 42$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 5 & 10 & 1 & 3 \\ 3 & 4 & 9 & 19 & 0 & 4 \\ 5 & 9 & 12 & 29 & 0 & 6 \\ 10 & 19 & 29 & 64 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 14 & -2 & 1 \end{pmatrix}_{b_0}$
- [6] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.56 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 3 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 3 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$
- [7] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 6 & 8 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$
- [8] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 49$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 5 & 11 & 9 & 7 & 5 \\ 11 & 20 & 17 & 11 & 8 \\ 9 & 17 & 12 & 9 & 6 \\ 7 & 11 & 9 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$
- [9] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 50$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 7 & 5 & 1 & 3 \\ 3 & 4 & 13 & 7 & 0 & 4 \\ 7 & 13 & 27 & 16 & 0 & 9 \\ 5 & 7 & 16 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 9 & 5 & -2 & 1 \end{pmatrix}_{b_0}$
- [10] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 50$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 5 & 11 & 10 & 7 & 5 \\ 11 & 20 & 17 & 11 & 8 \\ 10 & 17 & 12 & 9 & 6 \\ 7 & 11 & 9 & 4 & 4 \\ 5 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$

- [11] $1^{11}3^{11}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 50$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 5 & 11 & 1 & 25 & 5 \\ 11 & 20 & 0 & 45 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 25 & 45 & 0 & 100 & 18 \\ 5 & 8 & -1 & 18 & 2 \end{pmatrix}_{b_0}$
- [12] $1^{121}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 53$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 \\ 3 & 4 & 8 & 7 & 0 & 4 \\ 4 & 8 & 9 & 10 & 0 & 5 \\ 5 & 7 & 10 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 5 & -2 & 1 \end{pmatrix}_{b_0}$
- [13] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 64$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 10 & 15 & 12 & 9 & 6 \\ 15 & 20 & 17 & 11 & 8 \\ 12 & 17 & 12 & 9 & 6 \\ 9 & 11 & 9 & 4 & 4 \\ 6 & 8 & 6 & 4 & 2 \end{pmatrix}_{b_0}$
- [14] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 73$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 5 & 11 & 1 & 16 & 5 \\ 11 & 20 & 0 & 28 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 28 & 0 & 37 & 11 \\ 5 & 8 & -1 & 11 & 2 \end{pmatrix}_{b_0}$
- [15] $1^{12}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 73$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & 5 & 11 & 0 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 \\ 6 & 11 & 9 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 8 & -2 & 1 \end{pmatrix}_{b_0}$
- [16] $1^{121}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 80$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 4 & 7 & 1 & 3 \\ 3 & 4 & 8 & 12 & 0 & 4 \\ 4 & 8 & 9 & 16 & 0 & 5 \\ 7 & 12 & 16 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 9 & -2 & 1 \end{pmatrix}_{b_0}$
- [17] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 80$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 8 & 0 & 4 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 5 & 8 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$
- [18] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 85$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 5 & 11 & 1 & 15 & 5 \\ 11 & 20 & 0 & 28 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 28 & 0 & 37 & 11 \\ 5 & 8 & -1 & 11 & 2 \end{pmatrix}_{b_0}$
- [19] $1^{11}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 2 & 7 & 1 & 3 \\ 4 & 9 & 4 & 16 & 0 & 5 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 7 & 16 & 8 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 9 & -2 & 1 \end{pmatrix}_{b_0}$
- [20] $1^{11}3^{11}1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 101$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 6 & 12 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$
- [21] $1^{11}3^{11}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8$ $T = 8$, $T_{\max} = 13$, $\Delta = -94$, $\det = 104$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 1 & 3 \\ 3 & 4 & 4 & 27 & 0 & 4 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 16 & 27 & 17 & 151 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 7 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 4 & 7 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 5 & 15 & 0 & 4 \\ 3 & 5 & 3 & 12 & 0 & 3 \\ 9 & 15 & 12 & 39 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 5 & 12 & 5 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 7 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 5 & 7 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 133, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 1 & 3 \\ 3 & 4 & 9 & 7 & 0 & 4 \\ 5 & 9 & 12 & 11 & 0 & 6 \\ 5 & 7 & 11 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 5 & 11 & 0 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 \\ 7 & 11 & 9 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{12} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 141, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 4 & 1 & 3 \\ 3 & 4 & 8 & 7 & 0 & 4 \\ 4 & 8 & 9 & 10 & 0 & 5 \\ 4 & 7 & 10 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 8 & 1 & 3 \\ 3 & 4 & 9 & 15 & 0 & 4 \\ 5 & 9 & 12 & 23 & 0 & 6 \\ 8 & 15 & 23 & 39 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 7 & 0 & 4 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 4 & 7 & 5 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{11} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 198, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$\begin{pmatrix} 1 & 3 & 5 & 6 & 1 & 3 \\ 3 & 4 & 9 & 11 & 0 & 4 \\ 5 & 9 & 12 & 17 & 0 & 6 \\ 6 & 11 & 17 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 7 & 0 & 4 \\ 2 & 4 & 0 & 5 & 0 & 2 \\ 5 & 7 & 5 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 4 & 1 & 3 \\ 3 & 4 & 9 & 7 & 0 & 4 \\ 5 & 9 & 12 & 11 & 0 & 6 \\ 4 & 7 & 11 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 7 & 1 & 3 \\ 3 & 4 & 9 & 11 & 0 & 4 \\ 5 & 9 & 12 & 17 & 0 & 6 \\ 7 & 11 & 17 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & 4 & 11 & 0 & 4 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 6 & 11 & 8 & 20 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 22 & 0 & 4 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 13 & 22 & 14 & 97 & 0 & 17 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 17 & -2 & 1 \end{pmatrix}_b$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 11 & 1 & 3 \\ 3 & 4 & 9 & 19 & 0 & 4 \\ 5 & 9 & 12 & 29 & 0 & 6 \\ 11 & 19 & 29 & 64 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 1 & 3 \\ 3 & 4 & 4 & 27 & 0 & 4 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 17 & 27 & 17 & 151 & 0 & 21 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 21 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.12 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 9 & 1 & 3 \\ 3 & 4 & 9 & 15 & 0 & 4 \\ 5 & 9 & 12 & 23 & 0 & 6 \\ 9 & 15 & 23 & 39 & 0 & 11 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 6 & 11 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.18 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 4 & 12 & 0 & 4 \\ 2 & 4 & 0 & 8 & 0 & 2 \\ 7 & 12 & 8 & 25 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 21 & 0 & 4 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 12 & 21 & 14 & 80 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 1 & 3 \\ 3 & 4 & 4 & 16 & 0 & 4 \\ 2 & 4 & 0 & 11 & 0 & 2 \\ 10 & 16 & 11 & 44 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 20 & 0 & 4 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 11 & 20 & 14 & 69 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.16 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 18 & 1 & 3 \\ 3 & 4 & 4 & 31 & 0 & 4 \\ 2 & 4 & 0 & 20 & 0 & 2 \\ 18 & 31 & 20 & 188 & 0 & 24 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 24 & -2 & 1 \end{pmatrix}_b$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 15 & 1 & 3 \\ 3 & 4 & 4 & 26 & 0 & 4 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 15 & 26 & 17 & 128 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 20 & -2 & 1 \end{pmatrix}_b$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & 19 & 0 & 4 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 11 & 19 & 14 & 64 & 0 & 14 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 14 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 21 & 0 & 4 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 13 & 21 & 14 & 80 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 16 & -2 & 1 \end{pmatrix} b_0$$

$$T_{\min} = 11.35 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 4 & 24 & 0 & 4 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 13 & 24 & 17 & 100 & 0 & 18 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 18 & -2 & 1 \end{pmatrix}.$$

$$T_{\min} = 11 \ 12 \rightarrow 12 \quad T_{\max} = 13 \quad T \in [12 \ 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 16 & 1 & 3 \\ 3 & 4 & 4 & 26 & 0 & 4 \\ 2 & 4 & 0 & 17 & 0 & 2 \\ 16 & 26 & 17 & 128 & 0 & 20 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 20 & -2 & 1 \end{pmatrix},$$

$$T : -11\ 21 \rightarrow 12 \quad T : -13 \quad T \in [12\ 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 4 & 20 & 0 & 4 \\ 2 & 4 & 0 & 14 & 0 & 2 \\ 12 & 20 & 14 & 69 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 15 & 2 & 1 \end{pmatrix},$$

$$[1] \quad 1 \quad 3 \quad 1 \quad | \quad [2] \quad 2 \quad 3 \quad 2 \quad | \quad 3 \quad 3 \quad 3 \quad | \quad 2 \quad 2 \quad 2 \quad | \quad 1 \quad 1 \quad 1 \quad | \quad 1$$

$$T_{\min} = 11.19 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 1 & 3 \\ 3 & 4 & 4 & 29 & 0 & 4 \\ 2 & 4 & 0 & 20 & 0 & 2 \\ 17 & 29 & 20 & 148 & 0 & 22 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 22 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{11} 1^{14} 3^{11} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 1760, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 22 & 1 & 3 \\ 3 & 4 & 4 & 39 & 0 & 4 \\ 2 & 4 & 0 & 26 & 0 & 2 \\ 22 & 39 & 26 & 280 & 0 & 30 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 30 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{11} 1^{14} 3^{11} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 1952, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 20 & 1 & 3 \\ 3 & 4 & 4 & 34 & 0 & 4 \\ 2 & 4 & 0 & 23 & 0 & 2 \\ 20 & 34 & 23 & 208 & 0 & 26 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 26 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{11} 1^{14} 3^{11} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 2208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 23 & 1 & 3 \\ 3 & 4 & 4 & 39 & 0 & 4 \\ 2 & 4 & 0 & 26 & 0 & 2 \\ 23 & 39 & 26 & 280 & 0 & 30 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 30 & -2 & 1 \end{pmatrix}_{b_0}$$

4.1195 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}2\mathfrak{n}3\mathfrak{mix}$ + $\mathfrak{so}_8$ + $\mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{11} 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 9 & 7 & 1 & 5 \\ 11 & 20 & 17 & 11 & 0 & 8 \\ 9 & 17 & 12 & 9 & 0 & 6 \\ 7 & 11 & 9 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 6 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

4.1196 NHC(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}2\mathfrak{n}3\mathfrak{mix}$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{11} 1^{11} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 1 & 3 \\ 3 & 4 & 3 & 10 & 0 & 4 \\ 2 & 3 & 1 & 9 & 0 & 3 \\ 7 & 10 & 9 & 22 & 0 & 10 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 10 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 1^{11} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 4 & 3 \\ 3 & 4 & 3 & 3 & 7 & 4 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 3 & 3 & 3 & 1 & 5 & 3 \\ 4 & 7 & 6 & 5 & 9 & 7 \\ 3 & 4 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 1^{11} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 2 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 3 & 4 & 3 \\ 3 & 2 & 3 & 1 & 3 & 3 \\ 2 & 2 & 4 & 3 & 2 & 4 \\ 3 & 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 1^{11} 5^{11} 13^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 4 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 3 & 2 & 3 & 1 & 5 & 3 \\ 4 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

- [4] $1^{11}_1 5^{11} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 1 & 3 & 0 & 3 \\ 3 & 3 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$
- [5] $1^{11}_1 5^{12} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 8 & 2 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 9 & 4 & 3 \\ 8 & 7 & 9 & 22 & 9 & 10 \\ 2 & 2 & 4 & 9 & 2 & 4 \\ 3 & 3 & 3 & 10 & 4 & 4 \end{pmatrix}_{b_0}$
- [6] $1^{11}_1 5^{12} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 6 & 3 \\ 3 & 4 & 3 & 3 & 7 & 4 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 3 & 3 & 3 & 1 & 5 & 3 \\ 6 & 7 & 6 & 5 & 9 & 7 \\ 3 & 4 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$
- [7] $1^{11}_1 5^{12} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 9 & 3 \\ 3 & 4 & 3 & 3 & 14 & 4 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 3 & 3 & 3 & 1 & 10 & 3 \\ 9 & 14 & 12 & 10 & 42 & 14 \\ 3 & 4 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$
- [10] $1^{12}_1 5^{21} 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 12 & 1 & 3 \\ 3 & 4 & 3 & 17 & 0 & 4 \\ 2 & 3 & 1 & 15 & 0 & 3 \\ 12 & 17 & 15 & 65 & 0 & 17 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 17 & 0 & 4 \end{pmatrix}_{b_0}$
- [9] $1^{11}_1 5^{12} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 5 & 3 \\ 3 & 4 & 3 & 3 & 7 & 4 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 3 & 3 & 3 & 1 & 5 & 3 \\ 5 & 7 & 6 & 5 & 9 & 7 \\ 3 & 4 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$
- [8] $1^{11}_1 5^{12} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 3 & 10 & 0 & 4 \\ 2 & 3 & 1 & 9 & 0 & 3 \\ 8 & 10 & 9 & 22 & 0 & 10 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 10 & 0 & 4 \end{pmatrix}_{b_0}$
- [11] $1^{11}_1 5^{12} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 1 & 3 \\ 2 & 1 & 3 & 2 & 0 & 3 \\ 2 & 3 & 1 & 3 & 0 & 3 \\ 3 & 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$
- [12] $1^{11}_1 5^{12} 13^1 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 2 & 3 & 3 \\ 2 & 3 & 1 & 3 & 2 & 3 \\ 3 & 2 & 3 & 1 & 2 & 3 \\ 3 & 3 & 2 & 2 & 1 & 3 \\ 3 & 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$
- [13] $1^{11}_1 5^{12} 13^1 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -13, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$

7.0

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 4 & 3 \\ 3 & 5 & 3 & 9 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 5 & 9 & 4 & 11 & 7 & 7 \\ 4 & 5 & 2 & 7 & 2 & 4 \\ 3 & 5 & 2 & 7 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{12}_{1^1} 5^{13} 13^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$$

7.5

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 10 & 3 \\ 3 & 4 & 3 & 3 & 14 & 4 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 3 & 3 & 3 & 1 & 10 & 3 \\ 10 & 14 & 12 & 10 & 42 & 14 \\ 3 & 4 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{12}_{1^1} 5^{13} 13^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$$

7.5

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 11 & 3 \\ 3 & 4 & 3 & 3 & 14 & 4 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 3 & 3 & 3 & 1 & 10 & 3 \\ 11 & 14 & 12 & 10 & 42 & 14 \\ 3 & 4 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{13}_{1^1} 5^{12} 13^1 1^{13} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$$

7.8

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 14 & 3 \\ 3 & 4 & 3 & 3 & 21 & 4 \\ 2 & 3 & 1 & 3 & 18 & 3 \\ 3 & 3 & 3 & 1 & 15 & 3 \\ 14 & 21 & 18 & 15 & 97 & 21 \\ 3 & 4 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{11}_{1^1} 5^{13} 13^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$$

7.0

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 5 & 3 & 9 & 0 & 5 \\ 2 & 3 & 0 & 4 & 0 & 2 \\ 5 & 9 & 4 & 11 & 0 & 7 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 2 & 7 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{13}_{1^1} 5^{12} 13^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$$

7.5

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 7 & 3 \\ 3 & 5 & 3 & 9 & 10 & 5 \\ 2 & 3 & 0 & 4 & 4 & 2 \\ 5 & 9 & 4 & 11 & 14 & 7 \\ 7 & 10 & 4 & 14 & 14 & 8 \\ 3 & 5 & 2 & 7 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{12}_{1^1} 5^{11} 13^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$$

7.5

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 8 & 3 \\ 3 & 2 & 3 & 1 & 7 & 3 \\ 7 & 8 & 8 & 7 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{11}_{1^1} 5^{13} 13^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$$

7.0

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 3 & 4 & 3 \\ 3 & 2 & 3 & 1 & 3 & 3 \\ 3 & 2 & 4 & 3 & 2 & 4 \\ 3 & 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{11}_{1^1} 5^{13} 13^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -21, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} =$$

7.0

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 \\ 3 & 5 & 3 & 9 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 5 & 9 & 4 & 11 & 7 & 7 \\ 2 & 5 & 2 & 7 & 2 & 4 \\ 3 & 5 & 2 & 7 & 4 & 4 \end{pmatrix}_{b_0}$$

[22] $1^{13}_{1^1} 5^{131} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 17 & 1 & 3 \\ 3 & 4 & 3 & 24 & 0 & 4 \\ 2 & 3 & 1 & 21 & 0 & 3 \\ 17 & 24 & 21 & 130 & 0 & 24 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 24 & 0 & 4 \end{pmatrix}_{b_0}$$

[23] $1^{12}_{1^1} 5^{131} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -22, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 13 & 1 & 3 \\ 3 & 4 & 3 & 17 & 0 & 4 \\ 2 & 3 & 1 & 15 & 0 & 3 \\ 13 & 17 & 15 & 65 & 0 & 17 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & 17 & 0 & 4 \end{pmatrix}_{b_0}$$

[24] $1^{13}_{1^1} 5^{131} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 9 & 3 \\ 3 & 5 & 8 & 4 & 15 & 5 \\ 4 & 8 & 8 & 5 & 18 & 6 \\ 3 & 4 & 5 & 1 & 9 & 3 \\ 9 & 15 & 18 & 9 & 34 & 12 \\ 3 & 5 & 6 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[25] $1^{11}_{1^1} 5^{131} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 & 3 \\ 3 & 5 & 3 & 4 & 5 & 5 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 3 & 4 & 2 & 1 & 3 & 3 \\ 3 & 5 & 2 & 3 & 2 & 4 \\ 3 & 5 & 2 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[27] $1^{13}_{1^1} 5^{131} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 15 & 3 \\ 3 & 4 & 3 & 3 & 21 & 4 \\ 2 & 3 & 1 & 3 & 18 & 3 \\ 3 & 3 & 3 & 1 & 15 & 3 \\ 15 & 21 & 18 & 15 & 97 & 21 \\ 3 & 4 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[28] $1^{13}_{1^1} 5^{131} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 9 & 3 \\ 3 & 5 & 3 & 4 & 15 & 5 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 3 & 4 & 2 & 1 & 9 & 3 \\ 9 & 15 & 6 & 9 & 34 & 12 \\ 3 & 5 & 2 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[29] $1^{11}_{1^1} 5^{131} 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 & 3 \\ 3 & 5 & 8 & 4 & 5 & 5 \\ 4 & 8 & 8 & 5 & 6 & 6 \\ 3 & 4 & 5 & 1 & 3 & 3 \\ 3 & 5 & 6 & 3 & 2 & 4 \\ 3 & 5 & 6 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[30] $1^{12}_{1^1} 5^{131} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 6 & 3 \\ 3 & 5 & 8 & 4 & 10 & 5 \\ 4 & 8 & 8 & 5 & 12 & 6 \\ 3 & 4 & 5 & 1 & 6 & 3 \\ 6 & 10 & 12 & 6 & 14 & 8 \\ 3 & 5 & 6 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

- [31] $1^{1^{21}}_1 5^{1^{11}} 13^{1^{111}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 6 & 3 & 3 \\ 3 & 5 & 8 & 9 & 5 & 5 \\ 4 & 8 & 8 & 11 & 6 & 6 \\ 6 & 9 & 11 & 11 & 7 & 7 \\ 3 & 5 & 6 & 7 & 2 & 4 \\ 3 & 5 & 6 & 7 & 4 & 4 \end{pmatrix}_{b_0}$
- [26] $1^{1^{12}}_1 5^{1^{12}} 13^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 6 & 3 \\ 3 & 5 & 3 & 4 & 10 & 5 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 3 & 4 & 2 & 1 & 6 & 3 \\ 6 & 10 & 4 & 6 & 14 & 8 \\ 3 & 5 & 2 & 3 & 8 & 4 \end{pmatrix}_{b_0}$
- [32] $1^{1^{21}}_1 5^{1^{12}} 13^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 5 & 3 & 14 & 0 & 5 \\ 2 & 3 & 0 & 6 & 0 & 2 \\ 8 & 14 & 6 & 29 & 0 & 11 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 2 & 11 & 0 & 4 \end{pmatrix}_{b_0}$
- [35] $1^{1^{11}}_1 5^{1^{12}} 13^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 3 & 10 & 3 \\ 3 & 2 & 3 & 1 & 8 & 3 \\ 7 & 7 & 10 & 8 & 25 & 11 \\ 3 & 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$
- [34] $1^{1^{11}}_1 5^{1^{12}} 13^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 3 & 2 & 3 & 1 & 10 & 3 \\ 9 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$
- [33] $1^{1^{11}}_1 5^{1^{12}} 13^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 5 & 5 & 3 \\ 3 & 5 & 3 & 9 & 10 & 5 \\ 2 & 3 & 0 & 4 & 4 & 2 \\ 5 & 9 & 4 & 11 & 14 & 7 \\ 5 & 10 & 4 & 14 & 14 & 8 \\ 3 & 5 & 2 & 7 & 8 & 4 \end{pmatrix}_{b_0}$
- [36] $1^{1^{12}}_1 5^{1^{121}} 13^{1^{121}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -35, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 3 \\ 3 & 5 & 3 & 14 & 5 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 8 & 14 & 6 & 29 & 11 & 11 \\ 2 & 5 & 2 & 11 & 2 & 4 \\ 3 & 5 & 2 & 11 & 4 & 4 \end{pmatrix}_{b_0}$
- [37] $1^{1^{11}}_1 5^{1^{12}} 13^{1^{12}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 5 & 1 & 3 \\ 2 & 1 & 3 & 5 & 0 & 3 \\ 2 & 3 & 1 & 5 & 0 & 3 \\ 5 & 5 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 6 & 0 & 4 \end{pmatrix}_{b_0}$
- [38] $1^{1^{31}}_1 5^{1^{13}} 13^{1^{13}} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -43, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 1 & 3 \\ 3 & 5 & 3 & 19 & 0 & 5 \\ 2 & 3 & 0 & 8 & 0 & 2 \\ 11 & 19 & 8 & 55 & 0 & 15 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 2 & 15 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{12}_1 5^{11} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 8 & 3 \\ 3 & 5 & 3 & 9 & 15 & 5 \\ 2 & 3 & 0 & 4 & 6 & 2 \\ 5 & 9 & 4 & 11 & 21 & 7 \\ 8 & 15 & 6 & 21 & 34 & 12 \\ 3 & 5 & 2 & 7 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{11}_1 5^{11} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 3 & 2 & 3 & 1 & 5 & 3 \\ 5 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{21}_1 5^{11} 13^{11} 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -47, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 5 & 3 \\ 3 & 5 & 3 & 14 & 10 & 5 \\ 2 & 3 & 0 & 6 & 4 & 2 \\ 8 & 14 & 6 & 29 & 22 & 11 \\ 5 & 10 & 4 & 22 & 14 & 8 \\ 3 & 5 & 2 & 11 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{13}_1 5^{11} 13^{11} 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -53, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 2 & 3 \\ 3 & 5 & 3 & 19 & 5 & 5 \\ 2 & 3 & 0 & 8 & 2 & 2 \\ 11 & 19 & 8 & 55 & 15 & 15 \\ 2 & 5 & 2 & 15 & 2 & 4 \\ 3 & 5 & 2 & 15 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{21}_1 5^{11} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 & 3 \\ 3 & 5 & 3 & 9 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 6 & 9 & 4 & 11 & 7 & 7 \\ 3 & 5 & 2 & 7 & 2 & 4 \\ 3 & 5 & 2 & 7 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{11}_1 5^{11} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -54, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 & 3 \\ 3 & 5 & 3 & 9 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 \\ 5 & 9 & 4 & 11 & 7 & 7 \\ 3 & 5 & 2 & 7 & 2 & 4 \\ 3 & 5 & 2 & 7 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{12}_1 5^{11} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 6 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 3 & 2 & 3 & 1 & 5 & 3 \\ 6 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{12}_1 5^{11} 13^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -58, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 8 & 3 \\ 3 & 2 & 3 & 1 & 7 & 3 \\ 8 & 8 & 8 & 7 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[47] $1^{1^{12}}_1 5^{13} 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -60$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.5$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 6 & 3 \\ 3 & 5 & 3 & 9 & 10 & 5 \\ 2 & 3 & 0 & 4 & 4 & 2 \\ 5 & 9 & 4 & 11 & 14 & 7 \\ 6 & 10 & 4 & 14 & 14 & 8 \\ 3 & 5 & 2 & 7 & 8 & 4 \end{pmatrix}_{b_0}$$

[48] $1^{1^{13}}_1 5^{13} 1^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -63$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 8.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 11 & 3 \\ 3 & 5 & 3 & 9 & 20 & 5 \\ 2 & 3 & 0 & 4 & 8 & 2 \\ 5 & 9 & 4 & 11 & 28 & 7 \\ 11 & 20 & 8 & 28 & 62 & 16 \\ 3 & 5 & 2 & 7 & 16 & 4 \end{pmatrix}_{b_0}$$

[49] $1^{1^{31}}_1 5^{13} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -64$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 9 & 3 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 9 & 4 & 3 \\ 9 & 7 & 9 & 22 & 9 & 10 \\ 3 & 2 & 4 & 9 & 2 & 4 \\ 3 & 3 & 3 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

[50] $1^{1^{11}}_1 5^{13} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -64$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 9 & 4 & 3 \\ 7 & 7 & 9 & 22 & 9 & 10 \\ 3 & 2 & 4 & 9 & 2 & 4 \\ 3 & 3 & 3 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$

[51] $1^{1^{31}}_1 5^{13} 1^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -67$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.5$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 5 & 3 \\ 3 & 5 & 3 & 19 & 10 & 5 \\ 2 & 3 & 0 & 8 & 4 & 2 \\ 11 & 19 & 8 & 55 & 30 & 15 \\ 5 & 10 & 4 & 30 & 14 & 8 \\ 3 & 5 & 2 & 15 & 8 & 4 \end{pmatrix}_{b_0}$$

[52] $1^{1^{13}}_1 5^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -67$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.5$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 8 & 3 \\ 3 & 2 & 3 & 1 & 7 & 3 \\ 9 & 8 & 8 & 7 & 22 & 10 \\ 3 & 3 & 3 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

[53] $1^{1^{13}}_1 5^{13} 1^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -70$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.8$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 9 & 3 \\ 3 & 5 & 3 & 9 & 15 & 5 \\ 2 & 3 & 0 & 4 & 6 & 2 \\ 5 & 9 & 4 & 11 & 21 & 7 \\ 9 & 15 & 6 & 21 & 34 & 12 \\ 3 & 5 & 2 & 7 & 12 & 4 \end{pmatrix}_{b_0}$$

[54] $1^{1^{21}}_1 5^{13} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -72$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 1 & 3 \\ 2 & 1 & 3 & 7 & 0 & 3 \\ 2 & 3 & 1 & 9 & 0 & 3 \\ 8 & 7 & 9 & 22 & 0 & 10 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 10 & 0 & 4 \end{pmatrix}_{b_0}$$

- [55] $1^{14}_{1^1} 5 13^1 1^{141} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -75, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 14 & 2 & 3 \\ 3 & 5 & 3 & 24 & 5 & 5 \\ 2 & 3 & 0 & 10 & 2 & 2 \\ 14 & 24 & 10 & 89 & 19 & 19 \\ 2 & 5 & 2 & 19 & 2 & 4 \\ 3 & 5 & 2 & 19 & 4 & 4 \end{pmatrix}_{b_0}$$
- [58] $1^{121}_{1^1} 5 13^1 1^{111} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 8 & 3 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 1 & 9 & 4 & 3 \\ 8 & 7 & 9 & 22 & 9 & 10 \\ 3 & 2 & 4 & 9 & 2 & 4 \\ 3 & 3 & 3 & 10 & 4 & 4 \end{pmatrix}_{b_0}$$
- [57] $1^{112}_{1^1} 5 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 3 & 10 & 3 \\ 3 & 2 & 3 & 1 & 8 & 3 \\ 8 & 7 & 10 & 8 & 25 & 11 \\ 3 & 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$
- [56] $1^{112}_{1^1} 5 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -76, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 2 & 13 & 3 \\ 2 & 3 & 1 & 3 & 14 & 3 \\ 3 & 2 & 3 & 1 & 12 & 3 \\ 12 & 13 & 14 & 12 & 65 & 17 \\ 3 & 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$
- [59] $1^{113}_{1^1} 5 13^1 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -91, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 1 & 3 & 10 & 3 \\ 3 & 2 & 3 & 1 & 8 & 3 \\ 9 & 7 & 10 & 8 & 25 & 11 \\ 3 & 3 & 3 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$
- [60] $1^{121}_{1^1} 5 13^1 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -92, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 & 3 \\ 3 & 5 & 3 & 14 & 5 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 8 & 14 & 6 & 29 & 11 & 11 \\ 3 & 5 & 2 & 11 & 2 & 4 \\ 3 & 5 & 2 & 11 & 4 & 4 \end{pmatrix}_{b_0}$$
- [61] $1^{131}_{1^1} 5 13^1 1^{121} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -96, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 & 3 \\ 3 & 5 & 3 & 14 & 5 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 \\ 9 & 14 & 6 & 29 & 11 & 11 \\ 3 & 5 & 2 & 11 & 2 & 4 \\ 3 & 5 & 2 & 11 & 4 & 4 \end{pmatrix}_{b_0}$$
- [62] $1^{112}_{1^1} 5 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 14 & 3 \\ 2 & 1 & 3 & 2 & 15 & 3 \\ 2 & 3 & 1 & 3 & 18 & 3 \\ 3 & 2 & 3 & 1 & 15 & 3 \\ 14 & 15 & 18 & 15 & 97 & 21 \\ 3 & 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$
- [63] $1^{112}_{1^1} 5 13^1 1^{13} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -98, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 2 & 12 & 3 \\ 2 & 3 & 1 & 3 & 16 & 3 \\ 3 & 2 & 3 & 1 & 13 & 3 \\ 12 & 12 & 16 & 13 & 70 & 18 \\ 3 & 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{11} \overset{1}{1} \overset{31}{5} \overset{1}{13} \overset{1}{1} \overset{12}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -100, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 1 & 3 \\ 2 & 1 & 3 & 9 & 0 & 3 \\ 2 & 3 & 1 & 13 & 0 & 3 \\ 11 & 9 & 13 & 44 & 0 & 14 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 14 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{12} \overset{1}{1} \overset{12}{5} \overset{1}{13} \overset{1}{1} \overset{12}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -102, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 10 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 3 & 2 & 3 & 1 & 10 & 3 \\ 10 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[67] \ 1^{12} \overset{1}{1} \overset{21}{5} \overset{1}{13} \overset{1}{1} \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -123, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 10 & 1 & 3 \\ 2 & 1 & 3 & 10 & 0 & 3 \\ 2 & 3 & 1 & 11 & 0 & 3 \\ 10 & 10 & 11 & 39 & 0 & 13 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 13 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{12} \overset{1}{1} \overset{31}{5} \overset{1}{13} \overset{1}{1} \overset{11}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -123, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 1 & 3 \\ 2 & 1 & 3 & 10 & 0 & 3 \\ 2 & 3 & 1 & 11 & 0 & 3 \\ 11 & 10 & 11 & 39 & 0 & 13 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 13 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{11} \overset{1}{1} \overset{31}{5} \overset{1}{13} \overset{1}{1} \overset{131}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -142, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 & 3 \\ 3 & 5 & 3 & 19 & 5 & 5 \\ 2 & 3 & 0 & 8 & 2 & 2 \\ 11 & 19 & 8 & 55 & 15 & 15 \\ 3 & 5 & 2 & 15 & 2 & 4 \\ 3 & 5 & 2 & 15 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{13} \overset{1}{1} \overset{13}{5} \overset{1}{13} \overset{1}{1} \overset{12}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -143, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 13 & 3 \\ 2 & 1 & 3 & 2 & 13 & 3 \\ 2 & 3 & 1 & 3 & 14 & 3 \\ 3 & 2 & 3 & 1 & 12 & 3 \\ 13 & 13 & 14 & 12 & 65 & 17 \\ 3 & 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{12} \overset{1}{1} \overset{13}{5} \overset{1}{13} \overset{1}{1} \overset{12}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -143, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 11 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 3 & 12 & 3 \\ 3 & 2 & 3 & 1 & 10 & 3 \\ 11 & 10 & 12 & 10 & 42 & 14 \\ 3 & 3 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{12} \overset{1}{1} \overset{13}{5} \overset{1}{13} \overset{1}{1} \overset{14}{1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -151, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 15 & 3 \\ 2 & 1 & 3 & 2 & 14 & 3 \\ 2 & 3 & 1 & 3 & 20 & 3 \\ 3 & 2 & 3 & 1 & 16 & 3 \\ 15 & 14 & 20 & 16 & 106 & 22 \\ 3 & 3 & 3 & 3 & 22 & 4 \end{pmatrix}_{b_0}$$

[72] $1^{12} \overset{1^{13}}{5} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -171, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 13 & 3 \\ 2 & 1 & 3 & 2 & 12 & 3 \\ 2 & 3 & 1 & 3 & 16 & 3 \\ 3 & 2 & 3 & 1 & 13 & 3 \\ 13 & 12 & 16 & 13 & 70 & 18 \\ 3 & 3 & 3 & 3 & 18 & 4 \end{pmatrix}_{b_0}$$

[73] $1^{12} \overset{1^{31}}{5} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -187, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 1 & 3 \\ 2 & 1 & 3 & 12 & 0 & 3 \\ 2 & 3 & 1 & 15 & 0 & 3 \\ 13 & 12 & 15 & 65 & 0 & 17 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 17 & 0 & 4 \end{pmatrix}_{b_0}$$

[74] $1^{13} \overset{1^{13}}{5} 13^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -197, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 15 & 3 \\ 2 & 1 & 3 & 2 & 15 & 3 \\ 2 & 3 & 1 & 3 & 18 & 3 \\ 3 & 2 & 3 & 1 & 15 & 3 \\ 15 & 15 & 18 & 15 & 97 & 21 \\ 3 & 3 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[75] $1^{13} \overset{1^{13}}{5} 13^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -203, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 17 & 3 \\ 2 & 1 & 3 & 2 & 17 & 3 \\ 2 & 3 & 1 & 3 & 22 & 3 \\ 3 & 2 & 3 & 1 & 18 & 3 \\ 17 & 17 & 22 & 18 & 137 & 25 \\ 3 & 3 & 3 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

[76] $1^{13} \overset{1^{31}}{5} 13^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -264, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 15 & 1 & 3 \\ 2 & 1 & 3 & 15 & 0 & 3 \\ 2 & 3 & 1 & 17 & 0 & 3 \\ 15 & 15 & 17 & 92 & 0 & 20 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 20 & 0 & 4 \end{pmatrix}_{b_0}$$

$T = 12$ [78] $1^{11} \overset{1^1}{6} 123^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 4 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 6 & 3 \\ 3 & 2 & 3 & 1 & 5 & 3 \\ 4 & 5 & 6 & 5 & 9 & 7 \\ 3 & 3 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[77] $1^{11} \overset{1^1}{6} 123^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 2 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 1 & 3 & 4 & 3 \\ 3 & 2 & 3 & 1 & 3 & 3 \\ 2 & 2 & 4 & 3 & 2 & 4 \\ 3 & 3 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[80] $1^{11} \overset{1^1}{1^1 3^1 21^1} 6 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 2 & 9 & 3 \\ 2 & 3 & 1 & 3 & 9 & 3 \\ 3 & 2 & 3 & 1 & 9 & 3 \\ 8 & 9 & 9 & 9 & 34 & 12 \\ 3 & 3 & 3 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

- [79] $1^{\overline{11}} \begin{smallmatrix} 1 \\ 6 \end{smallmatrix} \begin{smallmatrix} 1 \\ 13^1 21^1 \end{smallmatrix} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 6 & 3 \\ 2 & 1 & 3 & 2 & 6 & 3 \\ 2 & 3 & 1 & 3 & 7 & 3 \\ 3 & 2 & 3 & 1 & 7 & 3 \\ 6 & 6 & 7 & 7 & 19 & 9 \\ 3 & 3 & 3 & 3 & 9 & 4 \end{pmatrix}_{b_0}$
- [81] $1^{\overline{11}} \begin{smallmatrix} 1 \\ 41^1 3^1 15^1 \end{smallmatrix} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 13, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & 0 & 3 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 5 & 4 \\ 3 & 2 & 0 & 0 & 5 & 4 & 4 \\ 2 & 2 & 0 & 0 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [82] $1^{\overline{11}} \begin{smallmatrix} 1 \\ 13^1 3^1 21^1 \end{smallmatrix} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 2 & 6 & 3 \\ 2 & 3 & 1 & 3 & 7 & 3 \\ 3 & 2 & 3 & 1 & 7 & 3 \\ 7 & 6 & 7 & 7 & 19 & 9 \\ 3 & 3 & 3 & 3 & 9 & 4 \end{pmatrix}_{b_0}$
- [83] $1^{\overline{11}} \begin{smallmatrix} 1 \\ 6 \end{smallmatrix} \begin{smallmatrix} 1 \\ 13^1 3^1 21^1 \end{smallmatrix} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 19, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 8 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 11 & 3 \\ 3 & 2 & 3 & 1 & 10 & 3 \\ 8 & 8 & 11 & 10 & 37 & 13 \\ 3 & 3 & 3 & 3 & 13 & 4 \end{pmatrix}_{b_0}$
- [84] $1^{\overline{11}} \begin{smallmatrix} 1 \\ 413^1 15^1 \end{smallmatrix} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 21, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & 0 & 3 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 3 & 4 \\ 1 & 2 & 0 & 0 & 3 & 0 & 2 \\ 2 & 2 & 0 & 0 & 4 & 2 & 3 \end{pmatrix}_{b_0}$
- [85] $1^{\overline{11}} \begin{smallmatrix} 1 \\ 6 \end{smallmatrix} \begin{smallmatrix} 1 \\ 13^1 3^1 21^1 \end{smallmatrix} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 31, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 12 & 3 \\ 2 & 1 & 3 & 2 & 13 & 3 \\ 2 & 3 & 1 & 3 & 17 & 3 \\ 3 & 2 & 3 & 1 & 15 & 3 \\ 12 & 13 & 17 & 15 & 86 & 20 \\ 3 & 3 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$
- [86] $1^{\overline{11}} \begin{smallmatrix} 1 \\ 6 \end{smallmatrix} \begin{smallmatrix} 1 \\ 123^1 1^1 \end{smallmatrix} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 5 & 1 & 3 \\ 2 & 1 & 3 & 5 & 0 & 3 \\ 2 & 3 & 1 & 5 & 0 & 3 \\ 5 & 5 & 5 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 6 & 0 & 4 \end{pmatrix}_{b_0}$
- [87] $1^{\overline{11}} \begin{smallmatrix} 1 \\ 6 \end{smallmatrix} \begin{smallmatrix} 1 \\ 13^1 3^1 21^1 \end{smallmatrix} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 44, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 2 & 9 & 3 \\ 2 & 3 & 1 & 3 & 9 & 3 \\ 3 & 2 & 3 & 1 & 9 & 3 \\ 9 & 9 & 9 & 9 & 34 & 12 \\ 3 & 3 & 3 & 3 & 12 & 4 \end{pmatrix}_{b_0}$
- [89] $1^{\overline{11}} \begin{smallmatrix} 1 \\ 41^1 3^1 15^1 \end{smallmatrix} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 45, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 5 & 2 \\ 1 & 0 & 1 & 0 & 3 & 6 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 11 & 4 \\ 5 & 6 & 0 & 0 & 11 & 20 & 8 \\ 2 & 2 & 0 & 0 & 4 & 8 & 3 \end{pmatrix}_{b_0}$$

[88] $1^{11} \begin{smallmatrix} 6 \\ 1^{11} 3^1 21^1 \end{smallmatrix} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 45, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 10 & 3 \\ 2 & 1 & 3 & 2 & 11 & 3 \\ 2 & 3 & 1 & 3 & 13 & 3 \\ 3 & 2 & 3 & 1 & 12 & 3 \\ 10 & 11 & 13 & 12 & 56 & 16 \\ 3 & 3 & 3 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

[90] $1^{11} \begin{smallmatrix} 6 \\ 1^{12} 3^1 21^1 \end{smallmatrix} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 50, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 11 & 3 \\ 2 & 1 & 3 & 2 & 10 & 3 \\ 2 & 3 & 1 & 3 & 15 & 3 \\ 3 & 2 & 3 & 1 & 13 & 3 \\ 11 & 10 & 15 & 13 & 63 & 17 \\ 3 & 3 & 3 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[91] $1^{121} 41^{13} 1^1 51^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 53, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & 0 & 7 & 2 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 7 & 7 & 0 & 0 & 32 & 7 & 10 \\ 1 & 2 & 0 & 0 & 7 & 0 & 2 \\ 2 & 2 & 0 & 0 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[92] $1^{11} \begin{smallmatrix} 6 \\ 1^{13} 3^1 21^1 \end{smallmatrix} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 58, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 9 & 3 \\ 2 & 1 & 3 & 2 & 8 & 3 \\ 2 & 3 & 1 & 3 & 11 & 3 \\ 3 & 2 & 3 & 1 & 10 & 3 \\ 9 & 8 & 11 & 10 & 37 & 13 \\ 3 & 3 & 3 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

[93] $1^{111} 41^{13} 1^1 51^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 60, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 1 & 0 & 3 & 4 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 & 0 \\ 3 & 3 & 0 & 0 & 4 & 8 & 4 \\ 4 & 4 & 0 & 0 & 8 & 10 & 6 \\ 2 & 2 & 0 & 0 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

[94] $1^{11} \begin{smallmatrix} 6 \\ 1^{13} 3^1 21^1 \end{smallmatrix} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 83, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 11 & 1 & 3 \\ 2 & 1 & 3 & 11 & 0 & 3 \\ 2 & 3 & 1 & 12 & 0 & 3 \\ 11 & 11 & 12 & 55 & 0 & 15 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 15 & 0 & 4 \end{pmatrix}_{b_0}$$

[95] $1^{11} \begin{smallmatrix} 6 \\ 1^{11} 3^1 21^1 \end{smallmatrix} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 110, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 11 & 3 \\ 2 & 1 & 3 & 2 & 11 & 3 \\ 2 & 3 & 1 & 3 & 13 & 3 \\ 3 & 2 & 3 & 1 & 12 & 3 \\ 11 & 11 & 13 & 12 & 56 & 16 \\ 3 & 3 & 3 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

[96] $1^{11} \begin{smallmatrix} 6 \\ 1^{11} 3^1 21^1 \end{smallmatrix} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 123, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 13 & 1 & 3 \\ 2 & 1 & 3 & 13 & 0 & 3 \\ 2 & 3 & 1 & 16 & 0 & 3 \\ 13 & 13 & 16 & 83 & 0 & 19 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 3 & 19 & 0 & 4 \end{pmatrix}_{b_0}$$

[97] $1^{11} \begin{smallmatrix} 6 \\ 1^{12} 3^1 21^1 \end{smallmatrix} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -29, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} =$

7.0

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 13 & 3 \\ 2 & 1 & 3 & 2 & 13 & 3 \\ 2 & 3 & 1 & 3 & 17 & 3 \\ 3 & 2 & 3 & 1 & 15 & 3 \\ 13 & 13 & 17 & 15 & 86 & 20 \\ 3 & 3 & 3 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [99] \quad 1^{11} \overset{11}{6} 131513^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 1 & 3 \\ 2 & 1 & 3 & 2 & 0 & 3 \\ 2 & 3 & 2 & 4 & 0 & 4 \\ 3 & 2 & 4 & 2 & 0 & 4 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 3 & 4 & 4 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[100] \quad 1^{11} \overset{11}{6} 131513^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -4, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 & 3 \\ 2 & 1 & 3 & 2 & 2 & 3 \\ 2 & 3 & 2 & 4 & 5 & 4 \\ 3 & 2 & 4 & 2 & 4 & 4 \\ 3 & 2 & 5 & 4 & 3 & 5 \\ 3 & 3 & 4 & 4 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[101] \quad 1^{11} \overset{11}{6} 131513^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -9, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 7 & 3 \\ 2 & 1 & 3 & 2 & 7 & 3 \\ 2 & 3 & 2 & 4 & 12 & 4 \\ 3 & 2 & 4 & 2 & 10 & 4 \\ 7 & 7 & 12 & 10 & 29 & 13 \\ 3 & 3 & 4 & 4 & 13 & 5 \end{pmatrix}_{b_0}$$

$$[102] \quad 1^{11} \overset{11}{6} 131513^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 7 & 3 & 3 \\ 2 & 1 & 3 & 7 & 2 & 3 \\ 2 & 3 & 2 & 11 & 5 & 4 \\ 7 & 7 & 11 & 26 & 11 & 12 \\ 3 & 2 & 5 & 11 & 3 & 5 \\ 3 & 3 & 4 & 12 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[103] \quad 1^{11} \overset{11}{6} 131513^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 15, T_{\max} = 15, \Delta = 0, \det = -23, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 5 & 3 \\ 2 & 1 & 3 & 2 & 5 & 3 \\ 2 & 3 & 2 & 4 & 7 & 4 \\ 3 & 2 & 4 & 2 & 6 & 4 \\ 5 & 5 & 7 & 6 & 10 & 8 \\ 3 & 3 & 4 & 4 & 8 & 5 \end{pmatrix}_{b_0}$$

4.1197 NHC(-2)(-3) + $\textcolor{red}{su}_3$ + $\textcolor{red}{su}2\textcolor{red}{n}3\textcolor{red}{mix}$ + $\textcolor{red}{su}_2$ + $\textcolor{red}{su}_2$

$$T = 12 \quad [0] \quad 1^{11} \overset{11}{5} 13^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 5, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 2 & 5 & 3 \\ 2 & 1 & 3 & 2 & 2 & 5 & 3 \\ 2 & 3 & 1 & 3 & 4 & 6 & 3 \\ 3 & 2 & 3 & 1 & 3 & 5 & 3 \\ 2 & 2 & 4 & 3 & 2 & 6 & 4 \\ 5 & 5 & 6 & 5 & 6 & 9 & 7 \\ 3 & 3 & 3 & 3 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} \overset{11}{5} 13^1 1^{1111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 43, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 3 & 3 \\ 3 & 5 & 3 & 9 & 5 & 5 & 5 \\ 2 & 3 & 0 & 4 & 2 & 2 & 2 \\ 5 & 9 & 4 & 11 & 7 & 7 & 7 \\ 2 & 5 & 2 & 7 & 2 & 4 & 4 \\ 3 & 5 & 2 & 7 & 4 & 2 & 4 \\ 3 & 5 & 2 & 7 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2] 1^1 \begin{smallmatrix} 12 \\ 5 \\ 13 \\ 11 \end{smallmatrix} 1^{1112} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 46, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 6 & 3 \\ 3 & 5 & 3 & 9 & 5 & 10 & 5 \\ 2 & 3 & 0 & 4 & 2 & 4 & 2 \\ 5 & 9 & 4 & 11 & 7 & 14 & 7 \\ 2 & 5 & 2 & 7 & 2 & 8 & 4 \\ 6 & 10 & 4 & 14 & 8 & 14 & 8 \\ 3 & 5 & 2 & 7 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[3] 1^1 \begin{smallmatrix} 13 \\ 5 \\ 13 \\ 11 \end{smallmatrix} 1^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 51, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 2 & 9 & 3 \\ 3 & 5 & 3 & 9 & 5 & 15 & 5 \\ 2 & 3 & 0 & 4 & 2 & 6 & 2 \\ 5 & 9 & 4 & 11 & 7 & 21 & 7 \\ 2 & 5 & 2 & 7 & 2 & 12 & 4 \\ 9 & 15 & 6 & 21 & 12 & 34 & 12 \\ 3 & 5 & 2 & 7 & 4 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[4] 1^1 \begin{smallmatrix} 21 \\ 5 \\ 13 \\ 11 \end{smallmatrix} 1^{1211} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 71, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 2 & 3 & 3 \\ 3 & 5 & 3 & 14 & 5 & 5 & 5 \\ 2 & 3 & 0 & 6 & 2 & 2 & 2 \\ 8 & 14 & 6 & 29 & 11 & 11 & 11 \\ 2 & 5 & 2 & 11 & 2 & 4 & 4 \\ 3 & 5 & 2 & 11 & 4 & 2 & 4 \\ 3 & 5 & 2 & 11 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[5] 1^1 \begin{smallmatrix} 31 \\ 5 \\ 13 \\ 11 \end{smallmatrix} 1^{1311} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 107, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 2 & 3 & 3 \\ 3 & 5 & 3 & 19 & 5 & 5 & 5 \\ 2 & 3 & 0 & 8 & 2 & 2 & 2 \\ 11 & 19 & 8 & 55 & 15 & 15 & 15 \\ 2 & 5 & 2 & 15 & 2 & 4 & 4 \\ 3 & 5 & 2 & 15 & 4 & 2 & 4 \\ 3 & 5 & 2 & 15 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

4.1198 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 7 \quad [0] 1^{13} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 5 & 12 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 7 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[1] 1^1 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 2 & 4 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 2 \\ 3 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] 1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 5 & 8 & 4 \\ 4 & 5 & 3 & 6 & 3 \\ 5 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4] 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 3 & 5 & 3 & 3 & 3 \\ 4 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] 1^{13} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 5 & 12 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 8 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$

- [2] $1^{111} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -11$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 14.09 \rightarrow 15$, $T_{\max} = 19$, $T \in [15, 19]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 5 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 2 & 2 & -2 \end{pmatrix}_{b_0}$
- [6] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -12$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 5 & 12 & 4 \\ 4 & 5 & 3 & 9 & 3 \\ 9 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$
- [7] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -14$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 19$, $T \in [12, 19]$
 $\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 5 & 8 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 4 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 1 \end{pmatrix}_{b_0}$
- [8] $21^{13} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -15$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 4 & 5 & 12 & 4 \\ 3 & 5 & 3 & 9 & 3 \\ 9 & 12 & 9 & 24 & 8 \\ 3 & 4 & 3 & 8 & 2 \end{pmatrix}_{b_0}$
- [13] $1^{13} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -17$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 17.59 \rightarrow 18$, $T_{\max} = 19$, $T \in [18, 19]$
 $\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 10 & 0 & 21 & 7 \\ 3 & 4 & -1 & 7 & -5 \end{pmatrix}_{b_0}$
- [12] $1^{14} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -17$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.6$
 $T_{\min} = 17.59 \rightarrow 18$, $T_{\max} = 19$, $T \in [18, 19]$
 $\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 16 & 0 & 45 & 11 \\ 3 & 4 & -1 & 11 & -5 \end{pmatrix}_{b_0}$
- [11] $1^{13} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -17$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 13.35 \rightarrow 14$, $T_{\max} = 19$, $T \in [14, 19]$
 $\begin{pmatrix} 1 & 3 & 6 & 1 & 3 \\ 3 & 4 & 10 & 0 & 4 \\ 6 & 10 & 21 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 7 & -1 & -1 \end{pmatrix}_{b_0}$
- [10] $1^{121} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -17$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 13.35 \rightarrow 14$, $T_{\max} = 19$, $T \in [14, 19]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 5 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & -1 \end{pmatrix}_{b_0}$
- [9] $1^{111} 3 1^2 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -17$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 7 & 1 & 3 \\ 4 & 9 & 16 & 0 & 5 \\ 7 & 16 & 27 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 9 & -1 & 2 \end{pmatrix}_{b_0}$
- [14] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -18$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 10.44 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 5 & 8 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 6 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [15] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 5 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$

- [16] $1^{11}31^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0}$
- [17] $1^{11}31^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 1 & 1 & 3 \\ 3 & 4 & 0 & 0 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & -1 & -1 & 2 \end{pmatrix}_{b_0}$
- [18] $1^{13}31^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 13.33 \rightarrow 14$, $T_{\max} = 19$, $T \in [14, 19]$
 $\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 12 & 0 & 24 & 8 \\ 3 & 4 & -1 & 8 & -1 \end{pmatrix}_{b_0}$
- [20] $1^{12}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -27$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 10.48 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 5 & 8 & 4 \\ 4 & 5 & 3 & 6 & 3 \\ 7 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [19] $1^{11}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -27$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.48 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 4 & 5 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 4 & 5 & 3 & 3 & 3 \\ 3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [23] $1^{11}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -29$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.59 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 5 & 0 & 3 & 3 \\ 3 & 4 & -1 & 3 & 2 \end{pmatrix}_{b_0}$
- [22] $1^{12}31^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -29$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 19$, $T \in [12, 19]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 4 & 8 & 0 & 9 & 5 \\ 3 & 4 & -1 & 5 & 1 \end{pmatrix}_{b_0}$
- [21] $1^{12}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -29$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 10.59 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 8 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 5 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$
- [24] $21^{11}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -30$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.40 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 4 & 8 & 3 & 3 & 4 \\ 8 & 9 & 4 & 4 & 5 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 0 & 2 \\ 4 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [26] $1^{11}31^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 19$, $T \in [12, 19]$
 $\begin{pmatrix} 1 & 3 & 1 & 4 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 4 & 4 & 0 & 0 & 2 \\ 3 & 4 & -1 & 2 & 1 \end{pmatrix}_{b_0}$
- [28] $1^{12}31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3$ $T = 7$, $T_{\max} = 19$, $\Delta = -102$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 10.38 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 5 & 8 & 4 \\ 4 & 5 & 3 & 6 & 3 \\ 6 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 19, T \in [13, 19]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 10 & 0 & 21 & 7 \\ 3 & 4 & -1 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{\overline{11}} 3^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 10 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 10 & 0 & 21 & 7 \\ 3 & 4 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 19, T \in [13, 19]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & 0 & 3 & 2 \\ 3 & 8 & 3 & 3 & 3 \\ 3 & 5 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{\overline{11}} 3^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 4 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{\overline{11}} 3^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{\overline{11}} 3^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 9 & 4 & 12 & 5 \\ 2 & 4 & 0 & 6 & 2 \\ 5 & 12 & 6 & 12 & 6 \\ 3 & 5 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{\overline{11}} 3^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 3 \\ 4 & 9 & 4 & 8 & 5 \\ 2 & 4 & 0 & 3 & 2 \\ 4 & 8 & 3 & 3 & 3 \\ 3 & 5 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[45] \ 21^{\overline{12}} 31^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 4 & 5 & 8 & 4 \\ 3 & 5 & 3 & 6 & 3 \\ 6 & 8 & 6 & 9 & 5 \\ 3 & 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.35 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 2 \\ 3 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{\overline{11}} 3^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 9 & 0 & 12 & 6 \\ 3 & 4 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{\overline{11}} 3^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 4 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{\overline{11}} 3^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.36 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 2 & 11 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 11 & 14 & 9 & 36 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{111} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.76 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{111} 3^1 1^{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 0 & 6 & 2 \\ 6 & 9 & 6 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[52] \quad 21^{11} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.46 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 5 & 4 & 4 \\ 3 & 5 & 3 & 3 & 3 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{12} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 10.63 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 5 & 8 & 0 & 9 & 5 \\ 3 & 4 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{12} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -65, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 8 & 0 & 9 & 5 \\ 3 & 4 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{11} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 14 & 0 & 36 & 10 \\ 3 & 4 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.63 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 5 & 9 & 6 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[57] \quad 21^{11} 31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 3 \\ 4 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 9 & 14 & 9 & 36 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{11} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.28 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 9 & 0 & 12 & 6 \\ 3 & 4 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{111} 3^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{12} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -77, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 13 & 0 & 27 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[60] $1^{12} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -77, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 8 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 8 & 0 & 9 & 5 \\ 3 & 4 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 7 & 9 & 6 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{11} 1^3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 5 & 9 & 6 & 12 & 6 \\ 3 & 4 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

[64] $1^{12} 1^3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 8 & 14 & 9 & 36 & 10 \\ 3 & 4 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[65] $1^{13} 1^3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 12 & 0 & 24 & 8 \\ 3 & 4 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{11} 1^3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 6 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 6 & 9 & 0 & 12 & 6 \\ 3 & 4 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{11} 1^3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 3 \\ 3 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 7 & 9 & 0 & 12 & 6 \\ 3 & 4 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{111} 1^{13} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -95, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.47 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 8 & 9 & 6 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{111} 1^1 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 6 & 2 \\ 6 & 9 & 6 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[70] $1^{13} 1^3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -98, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 48 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[72] $1^{111} 1^2 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 2 & 10 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 10 & 14 & 9 & 36 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[71] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 9 & 14 & 9 & 36 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{14} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -101, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.6$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 16 & 0 & 45 & 11 \\ 3 & 4 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 2 & 4 & 0 & 9 & 2 \\ 9 & 14 & 9 & 36 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[75] $21^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 3 & 9 & 4 \\ 5 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 9 & 9 & 6 & 12 & 6 \\ 4 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[76] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.41 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 6 & 9 & 6 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[77] $1^{11} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.59 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 14 & 0 & 36 & 10 \\ 3 & 4 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[78] $1^{14} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.6$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 21 & 0 & 75 & 15 \\ 3 & 4 & -1 & 15 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{12} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -117, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 18 & 0 & 57 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[80] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -119, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 7 & 9 & 6 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[81] $21^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 8 & 3 & 9 & 4 \\ 8 & 9 & 4 & 12 & 5 \\ 3 & 4 & 0 & 6 & 2 \\ 9 & 12 & 6 & 12 & 6 \\ 4 & 5 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[82] $1^{13} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 12 & 0 & 24 & 8 \\ 3 & 4 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[83] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 11 & 14 & 9 & 36 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[84] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 8 & 13 & 0 & 27 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[85] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -131, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.34 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 10 & 14 & 9 & 36 & 10 \\ 3 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[86] $1^{13} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -132, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.79 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 12 & 0 & 24 & 8 \\ 3 & 4 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[87] $21^{12} 31^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 3 & 15 & 4 \\ 5 & 4 & 4 & 14 & 4 \\ 3 & 4 & 0 & 9 & 2 \\ 15 & 14 & 9 & 36 & 10 \\ 4 & 4 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[88] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 14 & 0 & 36 & 10 \\ 3 & 4 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[89] $1^{11} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 14 & 0 & 36 & 10 \\ 3 & 4 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[90] $21^{11} 31^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.62 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[91] $1^{14} 3 1^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.6$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 16 & 0 & 45 & 11 \\ 3 & 4 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[92] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 13 & 0 & 27 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[93] $1^{13} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 22 & 0 & 84 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[94] $1^{12} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -167, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 13 & 0 & 27 & 9 \\ 3 & 4 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[95] $21^{11} 31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -171, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 8 & 3 & 6 & 4 \\ 8 & 9 & 4 & 8 & 5 \\ 3 & 4 & 0 & 3 & 2 \\ 6 & 8 & 3 & 3 & 3 \\ 4 & 5 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[96] $1^{13} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -176, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 17 & 0 & 48 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[97] $21^{11} 31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -183, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 3 \\ 4 & 4 & 4 & 9 & 4 \\ 3 & 4 & 0 & 6 & 2 \\ 6 & 9 & 6 & 12 & 6 \\ 3 & 4 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[98] $1^{12} 3^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -189, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 18 & 0 & 57 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[99] $1^{14} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -197, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.6$
 $T_{\min} = 10.65 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 16 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 16 & 0 & 45 & 11 \\ 3 & 4 & -1 & 11 & 2 \end{pmatrix}_{b_0}$$

[100] $1^{14} 3^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -209, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.6$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 26 & 0 & 117 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[101] $1^{14} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.6$
 $T_{\min} = 10.53 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 21 & 0 & 75 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[102] $1^{13} 3^1 1^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -230, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 11 & 17 & 0 & 48 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[103] $1^{12} 3^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -237, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 18 & 0 & 57 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[104] $1^{13} 3^1 1^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -260, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 22 & 0 & 84 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[105] $1^{13} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -260, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 3 \\ 3 & 4 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 17 & 0 & 48 & 12 \\ 3 & 4 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[106] $1^{12} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 18 & 0 & 57 & 13 \\ 3 & 4 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[107] $1^{14} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -297, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.6$
 $T_{\min} = 10.38 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 13 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 13 & 21 & 0 & 75 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[108] $1^{14} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -329, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.6$
 $T_{\min} = 10.42 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 26 & 0 & 117 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[109] $1^{13} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -332, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 22 & 0 & 84 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[110] $1^{14} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -351, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.6$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 3 \\ 3 & 4 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 14 & 21 & 0 & 75 & 15 \\ 3 & 4 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[111] $1^{13} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -380, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 15 & 3 \\ 3 & 4 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 15 & 22 & 0 & 84 & 16 \\ 3 & 4 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[112] $1^{14} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -425, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.6$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 16 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 16 & 26 & 0 & 117 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{14} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 7, T_{\max} = 19, \Delta = -102, \det = -497, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.6$
 $T_{\min} = 10.28 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 1 & 17 & 3 \\ 3 & 4 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & -1 \\ 17 & 26 & 0 & 117 & 19 \\ 3 & 4 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

$T = 8$ [115] $1^{12} 3 1^{21} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 4 & 1 & 2 \\ 1 & 1 & 4 & 6 & 0 & 3 \\ 2 & 4 & 3 & 6 & 0 & 3 \\ 4 & 6 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[114] $1^{11} 3 1 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

- [116] $1^{11}31^231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 0 & 0 & 0 & 2 \\ 2 & 3 & -1 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$
- [117] $21^141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -93, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 2 & 4 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 2 \\ 3 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [118] $1^141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -93, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 2 & 4 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 2 \\ 3 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [119] $1^{11}31^131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 0 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 0 & 0 & 12 & 6 \\ 2 & 3 & -1 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$
- [120] $1^131^131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 0 & 4 \\ 1 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 4 & 2 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$
- [121] $1^{11}31^{11}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 5 & 2 \\ 1 & 1 & 1 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 12 & 15 & 6 \\ 5 & 6 & 0 & 15 & 12 & 6 \\ 2 & 3 & -1 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$
- [122] $1^{13}31^131^{21} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 5 & 1 & 2 \\ 1 & 1 & 4 & 9 & 0 & 3 \\ 2 & 4 & 3 & 9 & 0 & 3 \\ 5 & 9 & 9 & 24 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 3 & 3 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$
- [123] $21^141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -93, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [124] $1^{11}41^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -93, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [129] $1^141^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 13, \Delta = -93, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 3 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [128] $1^{13}31^131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

- [127] $1^{11}31^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 9 & 3 \\ 5 & 9 & 0 & 9 & 24 & 8 \\ 2 & 3 & -1 & 3 & 8 & 2 \end{pmatrix}_{b_0}$$
- [126] $1^{11}31^{23}1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 3 & 3 \\ 2 & 3 & -1 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$
- [125] $1^{12}31^131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 6 & 3 \\ 3 & 6 & 0 & 6 & 9 & 5 \\ 2 & 3 & -1 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$
- [130] $1^1 1^1 4^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 9 & 4 & 4 & 5 \\ 2 & 4 & 0 & 2 & 2 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [131] $1^1 1^1 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [132] $1^{11}31^331^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 0 & 12 & 2 \\ 8 & 12 & 0 & 12 & 72 & 14 \\ 2 & 3 & -1 & 2 & 14 & 2 \end{pmatrix}_{b_0}$$
- [133] $1^1 1^1 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$
- [134] $1^{11}31^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$
- [135] $21^1 1^1 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 3 & 3 & 4 \\ 3 & 3 & 0 & 3 & 2 \\ 3 & 3 & 3 & 0 & 2 \\ 4 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [136] $1^{12}31^131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 6 & 0 & 9 & 5 \\ 2 & 3 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$
- [137] $1^{12} 1^1 4^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 4 & 4 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[138] \ 1 \overset{1^3}{1^1} 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 7 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[142] \ 21 \overset{1^{11}}{1^1} 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[141] \ 21 \overset{1^{11}}{1^1} 4 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 4 & 5 & 6 & 3 & 4 \\ 5 & 4 & 7 & 3 & 4 \\ 6 & 7 & 7 & 5 & 5 \\ 3 & 3 & 5 & 0 & 2 \\ 4 & 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[140] \ 31 \overset{1^{11}}{1^1} 231 \overset{1^{11}}{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 5 & 3 & 4 & 3 \\ 5 & 5 & 5 & 5 & 5 \\ 3 & 5 & 3 & 3 & 3 \\ 4 & 5 & 3 & 1 & 3 \\ 3 & 5 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[139] \ 1 \overset{21^{11}}{1^1} 4 \ 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 4 \\ 8 & 9 & 4 & 4 & 5 \\ 3 & 4 & 0 & 3 & 2 \\ 3 & 4 & 3 & 0 & 2 \\ 4 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[143] \ 1 \overset{1^1}{1^1} 4 \ 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 4 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[144] \ 1 \overset{1^1}{1^1} 4 \overset{1^1}{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 4 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[145] \ 1 \overset{1^{11}}{1^1} 31 \overset{1^{11}}{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 3 & 3 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & -1 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[146] \ 1 \overset{1^{11}}{1^1} 31 \overset{1^{12}}{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 0 & 6 & 2 \\ 5 & 6 & 0 & 6 & 12 & 6 \\ 2 & 3 & -1 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[147] \ 1 \overset{1^{12}}{1^1} 31 \overset{1^{12}}{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 6 & 3 \\ 4 & 6 & 0 & 6 & 9 & 5 \\ 2 & 3 & -1 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[148] \ 1 \overset{1^1}{1^1} 4 \overset{1^1}{1^1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[149] \ 21^{1^1} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 3 & 4 & 2 & 0 & 2 \\ 4 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[150] \ 1^1 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 4 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[151] \ 1^{11} 31^{11} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 9 & 6 & 0 & 12 & 6 \\ 2 & 4 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[152] \ 1^{11} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[153] \ 21^{11} 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 3 & 3 & 0 & 5 & 2 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[154] \ 1^1 31^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 0 & 6 & 4 \\ 1 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 3 & 0 & 3 & 3 \\ 2 & 4 & 2 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[155] \ 1^{12} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 6 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[157] \ 1^{11} 31^{12} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 0 & 9 & 2 \\ 5 & 9 & 0 & 9 & 36 & 10 \\ 2 & 3 & -1 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[156] \ 1^{12} 31^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 12 & 3 \\ 4 & 9 & 0 & 12 & 27 & 9 \\ 2 & 3 & -1 & 3 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[159] \ 1^{11} 4^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[158] \ 1^{11} 31^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

- $T_{\min} = 10.74 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 3 & 6 & 0 & 6 & 12 & 6 \\ 2 & 3 & -1 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$
- [160] $1^{11}31^{23}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3$ $T = 8$, $T_{\max} = 11$, $\Delta = -81$, $\det = 63$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.2$
 $T_{\min} = 10.70 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 0 & 9 & 2 \\ 6 & 9 & 0 & 9 & 36 & 10 \\ 2 & 3 & -1 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$
- [161] $31^{11}231^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3$ $T = 8$, $T_{\max} = 11$, $\Delta = -82$, $\det = 63$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.70 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 7 & 3 \\ 3 & 4 & 4 & 9 & 4 \\ 4 & 4 & 1 & 7 & 3 \\ 7 & 9 & 7 & 13 & 7 \\ 3 & 4 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$
- [162] $1^{11}31^{13}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3$ $T = 8$, $T_{\max} = 11$, $\Delta = -81$, $\det = 72$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 0 & 0 & 6 & 2 \\ 4 & 6 & 0 & 6 & 12 & 6 \\ 2 & 3 & -1 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$
- [163] $21^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -93$, $\det = 76$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.32 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 4 \\ 8 & 9 & 4 & 4 & 5 \\ 3 & 4 & 0 & 2 & 2 \\ 3 & 4 & 2 & 0 & 2 \\ 4 & 5 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [164] $21^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -93$, $\det = 79$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.16 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 4 & 3 & 4 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [165] $1^{14}31^{13}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3$ $T = 8$, $T_{\max} = 11$, $\Delta = -81$, $\det = 87$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.69 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 1 & 3 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 18 & 3 \\ 7 & 15 & 0 & 18 & 75 & 15 \\ 2 & 3 & -1 & 3 & 15 & 2 \end{pmatrix}_{b_0}$$
- [166] $21^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -93$, $\det = 88$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 3 \\ 4 & 4 & 3 & 7 & 4 \\ 2 & 3 & 0 & 5 & 2 \\ 5 & 7 & 5 & 7 & 5 \\ 3 & 4 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$
- [167] $221^{11}41^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3$ $T = 8$, $T_{\max} = 13$, $\Delta = -93$, $\det = 93$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11.23 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 1 & 5 & 5 & 3 & 3 \\ 5 & 8 & 11 & 6 & 6 \\ 5 & 11 & 7 & 6 & 5 \\ 3 & 6 & 6 & 3 & 3 \\ 3 & 6 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$
- [168] $1^{11}31^{13}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3$ $T = 8$, $T_{\max} = 11$, $\Delta = -81$, $\det = 96$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 9 & 3 \\ 4 & 6 & 0 & 9 & 12 & 6 \\ 2 & 3 & -1 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$
- [170] $1^{12}31^{11}31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3$ $T = 8$, $T_{\max} = 11$, $\Delta = -81$, $\det = 108$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 10.67 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 9 & 0 & 27 & 9 \\ 2 & 4 & 2 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$
- [169] $1^{11}31^{11}31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3$ $T = 8$, $T_{\max} = 11$, $\Delta = -81$, $\det = 108$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 7.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & 1 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 12 & 15 & 6 \\ 3 & 6 & 0 & 15 & 12 & 6 \\ 2 & 3 & -1 & 6 & 6 & 2 \end{pmatrix}_{b_0} \\
[171] \quad & 1^{11}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 0 & 9 & 4 \\ 1 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 9 & 6 & 0 & 12 & 6 \\ 2 & 4 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0} \\
[173] \quad & 1^{13}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 15 & 3 \\ 6 & 12 & 0 & 15 & 48 & 12 \\ 2 & 3 & -1 & 3 & 12 & 2 \end{pmatrix}_{b_0} \\
[172] \quad & 1^{111}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & 1 & 6 & 6 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 6 & 0 & 12 & 15 & 6 \\ 4 & 6 & 0 & 15 & 12 & 6 \\ 2 & 3 & -1 & 6 & 6 & 2 \end{pmatrix}_{b_0} \\
[174] \quad & 1^{12}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 12 & 3 \\ 5 & 9 & 0 & 12 & 27 & 9 \\ 2 & 3 & -1 & 3 & 9 & 2 \end{pmatrix}_{b_0} \\
[175] \quad & 1^{12}31^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 132, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 4 & 0 & 12 & 4 \\ 1 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 9 & 0 & 27 & 9 \\ 2 & 4 & 2 & -1 & 9 & 2 \end{pmatrix}_{b_0} \\
[176] \quad & 1^{12}31^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 1 & 3 & 9 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 12 & 3 \\ 6 & 9 & 0 & 12 & 27 & 9 \\ 2 & 3 & -1 & 3 & 9 & 2 \end{pmatrix}_{b_0} \\
[177] \quad & 1^{111}2^31^3 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 3 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 & 3 \\ 3 & 3 & 0 & 2 & 2 & 2 \\ 2 & 3 & 2 & 0 & 3 & 2 \\ 2 & 3 & 2 & 3 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[178] \quad & 1^{14}31^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.61 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 1 & 3 & 15 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 18 & 3 \\ 8 & 15 & 0 & 18 & 75 & 15 \\ 2 & 3 & -1 & 3 & 15 & 2 \end{pmatrix}_{b_0} \\
[179] \quad & 1^{13}31^{11}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 7 & 2 \\ 1 & 1 & 1 & 3 & 12 & 3 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 0 & 3 & 15 & 3 \\ 7 & 12 & 0 & 15 & 48 & 12 \\ 2 & 3 & -1 & 3 & 12 & 2 \end{pmatrix}_{b_0} \\
[180] \quad & 31^{111}231^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 255, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 5 & 4 & 7 & 3 \\ 5 & 5 & 5 & 10 & 5 \\ 4 & 5 & 1 & 7 & 3 \\ 7 & 10 & 7 & 13 & 7 \\ 3 & 5 & 3 & 7 & 3 \end{pmatrix}_{b_0} \\
[181] \quad & 31^{111}231^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 270, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 1 & 5 & 4 & 4 & 3 \\ 5 & 5 & 5 & 5 & 5 \\ 4 & 5 & 1 & 4 & 3 \\ 4 & 5 & 4 & 1 & 3 \\ 3 & 5 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [182] \quad 31^{11}31^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 4 & 6 & 4 \\ 4 & 4 & 1 & 5 & 3 \\ 5 & 6 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[183] \quad 21^{11}51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[184] \quad 1^{11}51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 4 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[185] \quad 1^{11}51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[186] \quad 31^{11}31^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -27, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 5 & 3 \\ 4 & 2 & 5 & 5 & 4 \\ 4 & 5 & 1 & 5 & 3 \\ 5 & 5 & 5 & 7 & 5 \\ 3 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[187] \quad 31^{11}31^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -30, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 4 & 3 & 3 \\ 5 & 8 & 6 & 6 & 6 \\ 4 & 6 & 1 & 3 & 3 \\ 3 & 6 & 3 & 3 & 3 \\ 3 & 6 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[188] \quad 21^{11}51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 4 \\ 5 & 4 & 3 & 3 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 4 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[189] \quad 21^{11}51^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 3 \\ 4 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[190] \quad 31^{11}31^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 5 & 5 & 4 & 3 \\ 5 & 5 & 7 & 5 & 5 \\ 5 & 7 & 7 & 5 & 5 \\ 4 & 5 & 5 & 1 & 3 \\ 3 & 5 & 5 & 3 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [191] \quad 1^{11}41^{11}41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -63, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 2 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

[192] $1^{\overline{1}} \overline{1}^{\overline{1}} 4 1 4 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -63, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 5 & -1 & 3 & 4 \end{pmatrix}_{b_0}$

[193] $1^{\overline{1}} \overline{1}^{\overline{1}} 5 1 3 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -67, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$

$T = 11$ [194] $1^{\overline{1}} \overline{1}^{\overline{1}} 6 1 2 3 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -65, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$

$T = 12$ [195] $1^{\overline{1}} \overline{1}^{\overline{1}} 5 1 3 1^{\overline{1}} 5 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 12, T_{\max} = 14, \Delta = -61, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 2 & 4 & 0 & 4 \\ 2 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 4 \end{pmatrix}_{b_0}$

[196] $1^{\overline{1}} \overline{1}^{\overline{1}} 5 1 3 1^{\overline{1}} 5 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 12, T_{\max} = 14, \Delta = -61, \det = 13, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 13.85 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 1 & 3 & 1 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 \\ 1 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 \\ 3 & 5 & -1 & 3 & 4 \end{pmatrix}_{b_0}$

$T = 13$ [197] $2 1^{\overline{1}} \overline{1}^{\overline{1}} 6 1^{\overline{1}} 2 3 1^{\overline{1}} 5 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 & 1 & 2 \\ 4 & 10 & 1 & 6 & 0 & 6 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 6 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 6 & 0 & 3 & 0 & 3 \end{pmatrix}_{b_0}$

[198] $1^{\overline{1}} \overline{1}^{\overline{1}} 6 1 3 1^{\overline{1}} 5 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 4 & 3 & 3 & 3 & 4 \\ 3 & 1 & 3 & 2 & 3 \\ 3 & 3 & 1 & 3 & 3 \\ 3 & 2 & 3 & 1 & 3 \\ 4 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0}$

[200] $1^{\overline{1}} \overline{1}^{\overline{1}} 5 1 3 2 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 4 & 1 & 2 & 1 & 2 \\ 4 & 10 & 1 & 6 & 0 & 6 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 6 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 6 & 0 & 3 & 0 & 3 \end{pmatrix}_{b_0}$

[199] $2 1^{\overline{1}} \overline{1}^{\overline{1}} 6 1 3 1^{\overline{1}} 5 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 2 & 0 & 2 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 0 & 3 & 0 & 3 \end{pmatrix}_{b_0}$

[203] $1^{\overline{1}} \overline{1}^{\overline{1}} 6 1 3 1^{\overline{1}} 5 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$

$$\begin{aligned}
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 2 & 4 & 2 & 4 \\ 2 & 4 & 1 & 2 & 3 \\ 2 & 2 & 2 & 0 & 2 \\ 3 & 4 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[202] & 21^{\overline{1}} 513161^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 4 & 5 & 11 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 7 & -1 & 4 \end{pmatrix}_{b_0} \\
[201] & 1^{\overline{1}} 6123151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 6 & 2 & 6 \\ 2 & 0 & 2 \\ 6 & 2 & 6 \end{pmatrix}_{b_0} \\
[205] & 1^{\overline{1}} 6123151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -13, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 14.85 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0} \\
[206] & 21^{\overline{1}} 613151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -13, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 14.85 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 4 & 1 & 1 & 3 \\ 4 & 8 & 1 & 0 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 6 & 0 & 0 & 4 \end{pmatrix}_{b_0} \\
[207] & 1613151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -13, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 14.85 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 2 & 3 & 1 & 3 \\ 3 & 3 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[204] & 1^{\overline{1}} 613151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -13, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 14.85 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 3 & 0 & 2 \\ 2 & 2 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 4 & -1 & 3 \end{pmatrix}_{b_0} \\
[208] & 1513161^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -16, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0} \\
[209] & 1^{\overline{1}} 5^{\overline{1}} 1321^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -27, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 7 & 1 & 3 & 3 \\ 7 & 19 & 1 & 6 & 9 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 6 & 0 & 1 & 3 \\ 3 & 9 & 0 & 3 & 4 \end{pmatrix}_{b_0} \\
[210] & 1^{\overline{1}} 61^{\overline{1}} 223151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -85, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 14.98 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 9 & 1 & 1 & 3 \\ 9 & 34 & 1 & 0 & 12 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 12 & 0 & 0 & 4 \end{pmatrix}_{b_0} \\
T = 14 & [211] 1^{\overline{1}} 6^{\overline{1}} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{red}{su}_3 \quad T = 14, T_{\max} = 15, \Delta = -37, \det = 8, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15]
\end{aligned}$$

$$\begin{pmatrix} 2 & 3 & 1 & 4 \\ 3 & 2 & 0 & 4 \\ 1 & 0 & -2 & 0 \\ 4 & 4 & 0 & 6 \end{pmatrix}_{b_0}$$

$$T = 16 \quad [212] \quad 1^{\overline{1}} 6 13161^{\overline{1}} 23151^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 16, T_{\max} = 18, \Delta = -58, \det = 168, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 17.93 \rightarrow 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 1 & 10 & 1 & 2 & 3 \\ 10 & 28 & 1 & 6 & 12 \\ 1 & 1 & -2 & 0 & 0 \\ 2 & 6 & 0 & 0 & 2 \\ 3 & 12 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

4.1199 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6$

$$T = 8 \quad [0] \quad 1^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 2 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 8 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 4 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.11 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 3 & 2 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 183, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.58 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 12 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 2 & 6 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 246, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 2 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 6 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\overline{1}} 1^{\overline{1}} 31^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 99, \Delta = -151, \det = 390, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.02 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 8 & 14 & 9 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 2 & 10 & -4 & -1 \end{pmatrix}_{b_0}$$

4.1200 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 249, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 6 & 3 & 0 & 4 & 4 & 4 \\ 2 & 4 & 3 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 3 & 2 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 22, \Delta = -101, \det = 825, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 3 & 2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 4 & 3 \\ 3 & 5 & 3 & 3 & 0 & 4 & 3 & 3 \\ 3 & 4 & 2 & 2 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

4.1201 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{121}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 5 & 8 & 0 & 9 & 4 \\ 3 & 5 & 3 & 6 & 0 & 9 & 3 \\ 4 & 8 & 6 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 12 & 0 & 13 & 7 \\ 3 & 4 & 3 & 5 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 6 & 6 & 0 & 13 & 7 \\ 3 & 5 & 2 & 2 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{114}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 14 & 3 \\ 4 & 9 & 4 & 4 & 0 & 36 & 5 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 36 & 15 & 15 & 0 & 121 & 19 \\ 3 & 5 & 2 & 2 & -4 & 19 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 4 \\ 3 & 5 & 3 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 6 & 3 & 0 & 4 & 4 \\ 3 & 4 & 3 & 2 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.87 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 3 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & 2 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{112}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -231, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 4 & 0 & 20 & 5 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 20 & 9 & 9 & 0 & 37 & 11 \\ 3 & 5 & 2 & 2 & -4 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -231, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{113} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -231, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.08 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 4 & 0 & 28 & 5 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 28 & 12 & 12 & 0 & 73 & 15 \\ 3 & 5 & 2 & 2 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -258, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.18 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 2 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 & 4 \\ 2 & 4 & 0 & 9 & 0 & 3 & 2 \\ 8 & 14 & 9 & 36 & 0 & 9 & 10 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 9 & 0 & 1 & 3 \\ 3 & 4 & 2 & 10 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{122} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -267, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.7$$

$$T_{\min} = 13.07 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 5 & 8 & 0 & 13 & 4 \\ 3 & 5 & 3 & 6 & 0 & 12 & 3 \\ 4 & 8 & 6 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 12 & 18 & 0 & 28 & 10 \\ 3 & 4 & 3 & 5 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{123} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -294, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.06 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 5 & 8 & 0 & 17 & 4 \\ 3 & 5 & 3 & 6 & 0 & 15 & 3 \\ 4 & 8 & 6 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 24 & 0 & 49 & 13 \\ 3 & 4 & 3 & 5 & -4 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1111} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -402, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.79 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 9 & 9 & 0 & 37 & 11 \\ 3 & 4 & 2 & 2 & -4 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -420, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.09 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 32 & 15 & 15 & 0 & 100 & 18 \\ 3 & 5 & 2 & 2 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -435, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.08 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 & 3 \\ 3 & 4 & 4 & 9 & 0 & 5 & 4 \\ 2 & 4 & 0 & 6 & 0 & 3 & 2 \\ 5 & 9 & 6 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 3 & 9 & 0 & 4 & 4 \\ 3 & 4 & 2 & 6 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -510, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.34 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{114}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -555, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.10 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 18 & 3 \\ 4 & 9 & 4 & 4 & 0 & 44 & 5 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 44 & 21 & 21 & 0 & 193 & 25 \\ 3 & 5 & 2 & 2 & -4 & 25 & -1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -645, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 3 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 7 & 7 \\ 3 & 4 & 3 & 2 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -651, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.05 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 4 \\ 2 & 4 & 0 & 6 & 0 & 3 & 2 \\ 5 & 9 & 6 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 6 & 0 & 1 & 3 \\ 3 & 4 & 2 & 6 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{114}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -690, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.08 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 4 & 0 & 40 & 5 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 16 & 40 & 18 & 18 & 0 & 154 & 22 \\ 3 & 5 & 2 & 2 & -4 & 22 & -1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{114}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -726, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 5 & 4 & 0 & 21 & 4 \\ 3 & 5 & 3 & 3 & 0 & 18 & 3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 18 & 15 & 0 & 76 & 16 \\ 3 & 4 & 3 & 2 & -4 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -789, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 0 & 13 & 7 & 7 \\ 3 & 4 & 2 & 2 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{112}31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -906, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 13 & 4 \\ 3 & 5 & 3 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 12 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & 2 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1113} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -933, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.8$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 5 & 4 & 0 & 17 & 4 \\ 3 & 5 & 3 & 3 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 15 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & 2 & -4 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 68, \quad \Delta = -121, \quad \det = -1200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.06 \rightarrow 14, \quad T_{\max} = 68, \quad T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 6 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 6 & 15 & 0 & 13 & 7 \\ 3 & 4 & 2 & 6 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

4.1202 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{1111} 31^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 48, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 5 & 5 & 3 \\ 4 & 9 & 4 & 4 & 0 & 12 & 12 & 5 \\ 2 & 4 & 0 & 3 & 0 & 6 & 6 & 2 \\ 2 & 4 & 3 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 12 & 6 & 6 & 0 & 13 & 15 & 7 \\ 5 & 12 & 6 & 6 & 0 & 15 & 13 & 7 \\ 3 & 5 & 2 & 2 & -4 & 7 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 228, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.25 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & 2 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 231, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 5 & 4 & 4 \\ 2 & 4 & 0 & 6 & 0 & 3 & 3 & 2 \\ 5 & 9 & 6 & 12 & 0 & 9 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 9 & 0 & 4 & 3 & 4 \\ 2 & 4 & 3 & 6 & 0 & 3 & 1 & 3 \\ 3 & 4 & 2 & 6 & -4 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 507, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.92 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 5 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 6 & 2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 3 & 3 & 0 & 4 & 9 & 4 \\ 5 & 9 & 6 & 6 & 0 & 9 & 13 & 7 \\ 3 & 4 & 2 & 2 & -4 & 4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 651, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.17 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 6 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & 2 & -4 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1114}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 813, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.27 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 11 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 21 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 18 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 15 & 3 \\ 11 & 21 & 18 & 15 & 0 & 15 & 76 & 16 \\ 3 & 4 & 3 & 2 & -4 & 3 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1112}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1173, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 13 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 12 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 9 & 3 \\ 7 & 13 & 12 & 9 & 0 & 9 & 28 & 10 \\ 3 & 4 & 3 & 2 & -4 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1113}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 1227, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 9 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 17 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 15 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 12 & 3 \\ 9 & 17 & 15 & 12 & 0 & 12 & 49 & 13 \\ 3 & 4 & 3 & 2 & -4 & 3 & 13 & -1 \end{pmatrix}_{b_0}$$

4.1203 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 459, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 1296, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 9 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 6 & 7 \\ 2 & 4 & 3 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 3 & 2 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1204 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 92, \ \Delta = -130, \ \det = -27, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.11 \rightarrow 15, \ T_{\max} = 92, \ T \in [15, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 3 & 3 & 0 & 3 & 3 \\ 3 & 5 & 2 & 2 & -4 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 92, \ \Delta = -130, \ \det = -90, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 92, \ T \in [14, 92]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 6 & 6 & 0 & 12 & 6 \\ 3 & 5 & 2 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 92, \ \Delta = -130, \ \det = -405, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} =$$

15.0

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

4.1205 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 459, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & 2 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 1296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 3 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 3 & 2 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

4.1206 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.20 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{12}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 8 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 4 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 142, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 12 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 2 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.21 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{\overline{111}} 3 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 83, \Delta = -125, \det = 307, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 12.19 \rightarrow 13, T_{\max} = 83, T \in [13, 83]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 8 & 14 & 9 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 2 & 10 & -3 & 0 \end{pmatrix}_{b_0}$$

4.1207 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$

$T = 9 \quad [0] 1^{\overline{111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 6 & 6 & 0 & 13 & 7 \\ 3 & 5 & 2 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{11}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 5 & 4 \\ 3 & 5 & 3 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 6 & 3 & 0 & 4 & 4 \\ 3 & 4 & 3 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & 2 & -3 & 4 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{113}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -159, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 11 & 3 \\ 4 & 9 & 4 & 4 & 0 & 28 & 5 \\ 2 & 4 & 0 & 3 & 0 & 12 & 2 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 28 & 12 & 12 & 0 & 73 & 15 \\ 3 & 5 & 2 & 2 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{112}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$

$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 8 & 3 \\ 4 & 9 & 4 & 4 & 0 & 20 & 5 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 9 & 9 & 0 & 37 & 11 \\ 3 & 5 & 2 & 2 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -182, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{122}3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -200$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.9$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 5 & 8 & 0 & 13 & 4 \\ 3 & 5 & 3 & 6 & 0 & 12 & 3 \\ 4 & 8 & 6 & 9 & 0 & 18 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 12 & 18 & 0 & 28 & 10 \\ 3 & 4 & 3 & 5 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -203$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$$T_{\min} = 12.34 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 2 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 & 4 \\ 2 & 4 & 0 & 9 & 0 & 3 & 2 \\ 8 & 14 & 9 & 36 & 0 & 9 & 10 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 9 & 0 & 1 & 3 \\ 3 & 4 & 2 & 10 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{123}3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -206$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 9 & 3 \\ 3 & 4 & 5 & 8 & 0 & 17 & 4 \\ 3 & 5 & 3 & 6 & 0 & 15 & 3 \\ 4 & 8 & 6 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 24 & 0 & 49 & 13 \\ 3 & 4 & 3 & 5 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{111}3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -315$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 8 & 3 \\ 3 & 4 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 0 & 3 & 0 & 9 & 2 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 9 & 9 & 0 & 37 & 11 \\ 3 & 4 & 2 & 2 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{113}3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -321$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 13 & 3 \\ 4 & 9 & 4 & 4 & 0 & 32 & 5 \\ 2 & 4 & 0 & 3 & 0 & 15 & 2 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 32 & 15 & 15 & 0 & 100 & 18 \\ 3 & 5 & 2 & 2 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -344$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 & 3 \\ 3 & 4 & 4 & 9 & 0 & 5 & 4 \\ 2 & 4 & 0 & 6 & 0 & 3 & 2 \\ 5 & 9 & 6 & 12 & 0 & 9 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 3 & 9 & 0 & 4 & 4 \\ 3 & 4 & 2 & 6 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{111}3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -399$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{114}3^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 14$, $\Delta = -95$, $\det = -426$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 18 & 3 \\ 4 & 9 & 4 & 4 & 0 & 44 & 5 \\ 2 & 4 & 0 & 3 & 0 & 21 & 2 \\ 2 & 4 & 3 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 44 & 21 & 21 & 0 & 193 & 25 \\ 3 & 5 & 2 & 2 & -3 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 5 & 4 & 0 & 9 & 4 \\ 3 & 5 & 3 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 0 & 13 & 7 & 0 \\ 3 & 4 & 3 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 & 4 \\ 2 & 4 & 0 & 6 & 0 & 3 & 2 \\ 5 & 9 & 6 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 6 & 0 & 1 & 3 \\ 3 & 4 & 2 & 6 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1114} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 16 & 3 \\ 4 & 9 & 4 & 4 & 0 & 40 & 5 \\ 2 & 4 & 0 & 3 & 0 & 18 & 2 \\ 2 & 4 & 3 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 16 & 40 & 18 & 18 & 0 & 154 & 22 \\ 3 & 5 & 2 & 2 & -3 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{1144} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -527, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 5 & 4 & 0 & 21 & 4 \\ 3 & 5 & 3 & 3 & 0 & 18 & 3 \\ 2 & 4 & 3 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 18 & 15 & 0 & 76 & 16 \\ 3 & 4 & 3 & 2 & -3 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -621, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 6 & 0 & 13 & 7 \\ 3 & 4 & 2 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{112} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -707, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.7$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 5 & 4 & 0 & 13 & 4 \\ 3 & 5 & 3 & 3 & 0 & 12 & 3 \\ 2 & 4 & 3 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 12 & 9 & 0 & 28 & 10 \\ 3 & 4 & 3 & 2 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{113} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -713, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 5 & 4 & 0 & 17 & 4 \\ 3 & 5 & 3 & 3 & 0 & 15 & 3 \\ 2 & 4 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 15 & 12 & 0 & 49 & 13 \\ 3 & 4 & 3 & 2 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1111} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -944, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.22 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 5 & 3 \\ 3 & 4 & 4 & 9 & 0 & 9 & 4 \\ 2 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 6 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 6 & 15 & 0 & 13 & 7 \\ 3 & 4 & 2 & 6 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

4.1208 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 5 & 8 & 0 & 3 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 & 3 \\ 4 & 8 & 6 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 3 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 2 & 3 \\ 4 & 9 & 4 & 4 & 0 & 4 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 2 & 2 & 0 & 0 & 2 \\ 3 & 5 & 2 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 3 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 3 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -354, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.47 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 2 & 0 & 0 & 2 \\ 3 & 4 & 2 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 2 & 3 \\ 3 & 4 & 4 & 9 & 0 & 3 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 & 2 \\ 5 & 9 & 6 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 2 & 5 & 0 & 0 & 2 \\ 3 & 4 & 2 & 6 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

4.1209 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_{2n3\text{mix}}$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.09 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 5 & 3 \\ 3 & 4 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 0 & 3 & 0 & 5 & 2 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 5 & 5 & 0 & 12 & 6 \\ 3 & 4 & 2 & 2 & -3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -81, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 3 & 0 & 5 & 2 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 5 & 5 & 0 & 12 & 6 \\ 3 & 5 & 2 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -106, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.26 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 4 & 3 \\ 3 & 4 & 5 & 8 & 0 & 7 & 4 \\ 3 & 5 & 3 & 6 & 0 & 6 & 3 \\ 4 & 8 & 6 & 9 & 0 & 10 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 10 & 0 & 7 & 5 \\ 3 & 4 & 3 & 5 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -226, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.25 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 5 & 4 & 0 & 11 & 4 \\ 3 & 5 & 3 & 3 & 0 & 9 & 3 \\ 2 & 4 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 9 & 8 & 0 & 20 & 8 \\ 3 & 4 & 3 & 2 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -346, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.27 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 5 & 4 & 0 & 7 & 4 \\ 3 & 5 & 3 & 3 & 0 & 6 & 3 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 6 & 5 & 0 & 7 & 5 \\ 3 & 4 & 3 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

4.1210 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 21, \ \Delta = -104, \ \det = -15, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 14.20 \rightarrow 15, \ T_{\max} = 21, \ T \in [15, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 3 & 3 & 0 & 3 & 3 \\ 3 & 5 & 2 & 2 & -3 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 21, \ \Delta = -104, \ \det = -69, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 6 & 6 & 0 & 12 & 6 \\ 3 & 5 & 2 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, \ T_{\max} = 21, \ \Delta = -104, \ \det = -321, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.55 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

4.1211 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{11} 3^1 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -101, \ \det = 5, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 17.60 \rightarrow 18, \ T_{\max} = 20, \ T \in [18, 20]$$

$$\begin{pmatrix} 5 & 11 & 18 & 1 & 5 \\ 11 & 20 & 34 & 0 & 8 \\ 18 & 34 & 57 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 13 & -1 & -4 \end{pmatrix}_{b_0}$$

- [1] $1^{13} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 5 & 12 & 0 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 \\ 7 & 12 & 9 & 24 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$
- [2] $1^{11} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 13.40 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 3 & 6 & 1 & 1 & 3 \\ 3 & 4 & 10 & 0 & 0 & 4 \\ 6 & 10 & 21 & 0 & 0 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 7 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$
- [3] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [4] $1^{12} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 8 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 4 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$
- [5] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 7 & 1 & 1 & 3 \\ 4 & 9 & 16 & 0 & 0 & 5 \\ 7 & 16 & 27 & 0 & 0 & 9 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 9 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$
- [6] $1^{11} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 11 & 1 & 19 & 5 \\ 11 & 20 & 0 & 34 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 19 & 34 & 0 & 57 & 13 \\ 5 & 8 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$
- [9] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 5 \\ 12 & 25 & 16 & 0 & 9 \\ 8 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$
- [8] $1^{12} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 1 & 3 \\ 3 & 4 & 0 & 8 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 4 & 8 & 0 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{111} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 3 & 4 & 0 & 9 & 0 & 2 \\ 8 & 14 & 9 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{14} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 11 & 1 & 3 \\ 3 & 4 & 0 & 21 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 11 & 21 & 0 & 75 & 0 & 15 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 15 & -2 & 1 \end{pmatrix}_{b_0}$$
- [11] $1 3 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8$ $T = 8, T_{\max} = 20, \Delta = -101, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 9 & 20 & 12 & 12 & 6 \\ 5 & 9 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 5 \\ 11 & 20 & 0 & 0 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & -1 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 4 & 5 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 12 & 8 & 10 & 5 \\ 12 & 25 & 16 & 20 & 9 \\ 8 & 16 & 9 & 12 & 5 \\ 10 & 20 & 12 & 12 & 6 \\ 5 & 9 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 77, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 3 \\ 3 & 4 & 0 & 5 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 11 & 1 & 10 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 10 & 17 & 0 & 12 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 6 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 6 & 12 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{12} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 8 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 5 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 5 \\ 11 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 12 & 6 \\ 5 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 3 & -2 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 15 & 1 & 12 & 6 \\ 15 & 20 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & -1 \\ 12 & 17 & 0 & 12 & 6 \\ 6 & 8 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.41 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 9 & 1 & 3 \\ 3 & 4 & 0 & 17 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 9 & 17 & 0 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 5 & 1 & 3 \\ 4 & 9 & 4 & 12 & 0 & 5 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 12 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{11} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 1 & 3 \\ 3 & 4 & 0 & 14 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 8 & 14 & 0 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 4 & 8 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 1 & 3 \\ 3 & 4 & 0 & 9 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 5 & 9 & 0 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{12} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 125, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 7 & 1 & 3 \\ 3 & 4 & 0 & 13 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 7 & 13 & 0 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.75 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{12} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.44 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 10 & 1 & 3 \\ 3 & 4 & 0 & 18 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 10 & 18 & 0 & 57 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 188, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 3 & 4 & 0 & 6 & 0 & 2 \\ 5 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{14} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.6$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 14 & 1 & 3 \\ 3 & 4 & 0 & 26 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 14 & 26 & 0 & 117 & 0 & 19 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 19 & -2 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{13} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 1 & 12 & 1 & 3 \\ 3 & 4 & 0 & 22 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 12 & 22 & 0 & 84 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 8 & 14 & 9 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

[38] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 5 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 5 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[39] $21^{111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 261, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 13 & 20 & 12 & 15 & 7 \\ 20 & 25 & 16 & 20 & 9 \\ 12 & 16 & 9 & 12 & 5 \\ 15 & 20 & 12 & 12 & 6 \\ 7 & 9 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 6 & 1 & 3 \\ 3 & 4 & 4 & 9 & 0 & 4 \\ 2 & 4 & 0 & 6 & 0 & 2 \\ 6 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{111} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -101, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.34 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 2 & 9 & 1 & 3 \\ 3 & 4 & 4 & 14 & 0 & 4 \\ 2 & 4 & 0 & 9 & 0 & 2 \\ 9 & 14 & 9 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 10 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [42] \quad 1 \overset{1}{4} 1 \overset{11}{1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 6 & 6 & 5 \\ 12 & 25 & 12 & 12 & 9 \\ 6 & 12 & 4 & 7 & 4 \\ 6 & 12 & 7 & 4 & 4 \\ 5 & 9 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[43] \quad 1 \overset{1}{11} 4 \overset{1}{1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 \\ 2 & 4 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 21 \overset{1}{11} 4 \overset{1}{1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 13 & 20 & 9 & 9 & 7 \\ 20 & 25 & 12 & 12 & 9 \\ 9 & 12 & 4 & 7 & 4 \\ 9 & 12 & 7 & 4 & 4 \\ 7 & 9 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[45] \quad 1 \overset{1}{11} 4 \overset{1}{1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -66, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.36 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1 \overset{1}{11} 4 \overset{1}{1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 4 & 0 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1 \overset{1}{11} 4 \overset{1}{1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 13, \Delta = -92, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 3 & 7 & 0 & 4 \\ 2 & 3 & 0 & 5 & 0 & 2 \\ 4 & 7 & 5 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 2 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

4.1212 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1 \overset{1}{11} 3 \overset{1}{1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 1 & 5 \\ 12 & 25 & 16 & 0 & 0 & 9 \\ 8 & 16 & 9 & 0 & 0 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & -1 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1 \overset{1}{11} 3 \overset{1}{1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 9 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 9 & 20 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1 \overset{1}{11} 3 \overset{1}{1} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{so}}_8 \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 8 & 10 & 1 & 5 \\ 12 & 25 & 16 & 20 & 0 & 9 \\ 8 & 16 & 9 & 12 & 0 & 5 \\ 10 & 20 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 5 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{so}_8} + \textcolor{blue}{\mathfrak{so}_8} \quad T = 9, T_{\max} = 21, \Delta = -100, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 11 & 1 & 9 & 1 & 5 \\ 11 & 20 & 0 & 17 & 0 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 9 & 17 & 0 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & -1 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] 1^{11} 4 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{so}_8} + \textcolor{blue}{\mathfrak{so}_8} \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 45, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.20 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 12 & 6 & 6 & 1 & 5 \\ 12 & 25 & 12 & 12 & 0 & 9 \\ 6 & 12 & 4 & 7 & 0 & 4 \\ 6 & 12 & 7 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 4 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

4.1213 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 9 \quad [0] 1^{111} 3 1^{111} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{so}_8} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 5 & 3 \\ 4 & 9 & 4 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 3 & 0 & 6 & 2 \\ 2 & 4 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 6 & 6 & 0 & 12 & 6 \\ 3 & 5 & 2 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] 1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{so}_8} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -63, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 4 & 3 \\ 4 & 9 & 4 & 4 & 0 & 8 & 5 \\ 2 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 3 & 3 & 0 & 3 & 3 \\ 3 & 5 & 2 & 2 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] 1^{111} 3 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{so}_8} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 8 & 1 & 10 & 5 \\ 12 & 25 & 16 & 0 & 20 & 9 \\ 8 & 16 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 20 & 12 & 0 & 12 & 6 \\ 5 & 9 & 5 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[3] 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{so}_8} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -1 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4] 1^{11} 3 1^{11} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{so}_8} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 1 & 2 & 3 \\ 3 & 4 & 0 & 5 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 3 & 0 & 0 & 2 \\ 3 & 4 & -1 & 3 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] 1^{111} 3 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{green}{\mathfrak{su}_3} + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{so}_8} + \textcolor{blue}{\mathfrak{su}_3} \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 2 & 3 \\ 3 & 4 & 0 & 4 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 0 & 3 & 0 & 0 & 2 \\ 3 & 4 & -1 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -237, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 8 & 3 \\ 3 & 4 & 0 & 4 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 0 & 9 & 0 & 36 & 10 \\ 3 & 4 & -1 & 2 & -2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -261, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 5 & 4 & 0 & 4 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & 2 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{so}}_8 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 5 & 3 \\ 3 & 4 & 0 & 4 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 0 & 6 & 0 & 12 & 6 \\ 3 & 4 & -1 & 2 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

4.1214 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 5 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 3 & 0 & 7 & 4 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 4 & 7 & 5 & 0 & 8 & 6 \\ 3 & 4 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1 \overset{11}{1} \overset{11}{4} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -63, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 3 & 0 & 7 & 4 \\ 2 & 3 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 7 & 5 & 0 & 8 & 6 \\ 3 & 4 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [6] \quad 1 \overset{11}{1} \overset{11}{5} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 \\ 4 & 6 & 7 & 0 & 4 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 4 & 0 & 1 & 3 \\ 3 & 4 & 5 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 21 \overset{11}{1} \overset{11}{5} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 2 & 4 & 2 & 2 & 1 & 3 \\ 3 & 4 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1 \overset{11}{1} \overset{12}{5} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 4 & 3 \\ 3 & 4 & 3 & 3 & 4 & 4 \\ 2 & 3 & 0 & 2 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 & 2 \\ 4 & 4 & 2 & 2 & 1 & 3 \\ 3 & 4 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[9] \quad 1 \overset{12}{1} \overset{13}{5} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 7 & 3 \\ 3 & 4 & 3 & 3 & 8 & 4 \\ 2 & 3 & 0 & 2 & 4 & 2 \\ 2 & 3 & 2 & 0 & 4 & 2 \\ 7 & 8 & 4 & 4 & 10 & 6 \\ 3 & 4 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1 \overset{11}{1} \overset{11}{5} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 28, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 3 & 0 & 4 \\ 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 21 \overset{11}{1} \overset{11}{5} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 5 & 3 & 3 & 3 & 4 \\ 5 & 4 & 3 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 & 2 \\ 3 & 4 & 2 & 2 & 1 & 3 \\ 4 & 4 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1 \overset{13}{1} \overset{14}{5} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 10 & 3 \\ 3 & 4 & 3 & 3 & 12 & 4 \\ 2 & 3 & 0 & 2 & 6 & 2 \\ 2 & 3 & 2 & 0 & 6 & 2 \\ 10 & 12 & 6 & 6 & 25 & 9 \\ 3 & 4 & 2 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[10] \quad 1 \overset{11}{1} \overset{11}{5} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 3 \\ 2 & 2 & 4 & 0 & 7 & 4 \\ 2 & 4 & 1 & 0 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 7 & 6 & 0 & 14 & 8 \\ 3 & 4 & 3 & -1 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{\overset{11}{1}} 4^{\overset{11}{1}} 14^{\overset{11}{1}} 1^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 4 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 4 & 3 \\ 2 & 4 & 0 & 4 & 2 & 4 \\ 3 & 5 & -1 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{\overset{11}{1}} 4^{\overset{11}{1}} 14^{\overset{11}{1}} 1^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 4 & 4 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 5 & 0 & 3 & 2 & 4 \\ 3 & 5 & -1 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{\overset{11}{1}} 4^{\overset{11}{1}} 14^{\overset{11}{1}} 1^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 5 & 0 & 4 & 5 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 3 & 5 & 0 & 3 & 2 & 4 \\ 3 & 5 & -1 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{\overset{11}{1}} 4^{\overset{11}{1}} 14^{\overset{11}{1}} 1^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -33, \det = -50, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 5 & 3 \\ 3 & 5 & 0 & 4 & 9 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 7 & 3 \\ 5 & 9 & 0 & 7 & 12 & 8 \\ 3 & 5 & -1 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [26] \ 1^{\overset{11}{1}} 5^{\overset{11}{1}} 13^{\overset{11}{1}} 1^{\overset{11}{1}} 5^{\overset{11}{1}} 1^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 13, T_{\max} = 14, \Delta = -31, \det = -11, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 0 & 4 & 5 & 5 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 5 & 0 & 3 & 2 & 4 \\ 3 & 5 & -1 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [27] \ 1^{\overset{11}{1}} 6^{\overset{11}{1}} 12^{\overset{11}{1}} 3^{\overset{11}{1}} 1^{\overset{11}{1}} 5^{\overset{11}{1}} 1^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 2 & 3 \\ 2 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 3 \\ 3 & 3 & 0 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{\overset{12}{1}} 5^{\overset{12}{1}} 13^{\overset{12}{1}} 2^{\overset{12}{1}} 1^{\overset{12}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 20, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 6 & 2 & 15 & 6 \\ 2 & 0 & 6 & 2 \\ 15 & 6 & 31 & 15 \\ 6 & 2 & 15 & 6 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{\overset{11}{1}} 5^{\overset{11}{1}} 13^{\overset{11}{1}} 2^{\overset{11}{1}} 1^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 44, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 9 & 3 \\ 2 & 1 & 1 & 0 & 9 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 9 & 0 & 0 & 34 & 12 \\ 3 & 3 & 0 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{\overset{11}{1}} 6^{\overset{11}{1}} 11^{\overset{11}{1}} 22^{\overset{11}{1}} 3^{\overset{11}{1}} 1^{\overset{11}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 188, (n_+, n_-, n_0) =$$

$$(1, 14, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14.98 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 9 & 1 & 1 & 9 & 3 \\ 9 & 34 & 1 & 0 & 36 & 12 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 9 & 36 & 0 & 0 & 34 & 12 \\ 3 & 12 & 0 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

4.1215 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 \\ 2 & 3 & 3 & 0 & 1 & 2 \\ 3 & 4 & 2 & 0 & 2 & 1 \\ 3 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 2 & 3 \\ 2 & 4 & 2 & 0 & 2 & 1 & 3 \\ 3 & 4 & 2 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 3 & 0 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 2 & -1 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 4 & 3 \\ 3 & 4 & 3 & 0 & 5 & 7 & 4 \\ 2 & 3 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 3 & 0 & 4 & 6 & 4 \\ 4 & 7 & 5 & 0 & 6 & 8 & 6 \\ 3 & 4 & 2 & -1 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 18, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 3 & 0 & 3 & 5 & 4 \\ 2 & 3 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 3 & 3 \\ 3 & 5 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 2 & -1 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 0 & 5 & 4 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 3 & 0 & 4 & 3 & 4 \\ 2 & 4 & 2 & 0 & 3 & 1 & 3 \\ 3 & 4 & 2 & -1 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 55, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 3 & 0 & 7 & 4 & 4 \\ 2 & 3 & 0 & 0 & 5 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 4 & 7 & 5 & 0 & 8 & 5 & 6 \\ 3 & 4 & 2 & 0 & 5 & 1 & 3 \\ 3 & 4 & 2 & -1 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 4 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 & 4 \\ 2 & 3 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 3 & 3 \\ 2 & 4 & 2 & 0 & 3 & 1 & 3 \\ 3 & 4 & 2 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 4^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 91, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 4 & 3 \\ 3 & 4 & 3 & 0 & 4 & 7 & 4 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 5 & 3 \\ 4 & 7 & 5 & 0 & 5 & 8 & 6 \\ 3 & 4 & 2 & -1 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 4^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -33, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 3 & 0 & 4 & 7 & 4 \\ 2 & 3 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 5 & 3 \\ 5 & 7 & 5 & 0 & 5 & 8 & 6 \\ 3 & 4 & 2 & -1 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [12] \ 1^{111} 4^{11} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -3, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 5 & 0 & 4 & 5 & 5 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 5 & 0 & 3 & 2 & 4 & 4 \\ 3 & 5 & 0 & 3 & 4 & 2 & 4 \\ 3 & 5 & -1 & 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 4^{11} 4^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -3, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 5 & 0 & 4 & 5 & 9 & 5 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 4 & 0 & 1 & 3 & 7 & 3 \\ 2 & 5 & 0 & 3 & 2 & 7 & 4 \\ 5 & 9 & 0 & 7 & 7 & 12 & 8 \\ 3 & 5 & -1 & 3 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

4.1216 NHC(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 11 \quad [4] \ 1^{111} 4^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 3 & 0 & 5 & 7 & 4 & 4 \\ 2 & 3 & 0 & 0 & 3 & 5 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 3 & 0 & 4 & 6 & 3 & 4 \\ 4 & 7 & 5 & 0 & 6 & 8 & 5 & 6 \\ 2 & 4 & 2 & 0 & 3 & 5 & 1 & 3 \\ 3 & 4 & 2 & -1 & 4 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 4^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 3 & 0 & 5 & 4 \\ 2 & 3 & 0 & 0 & 3 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 5 & 3 & 0 & 3 & 0 & 4 & 4 \\ 3 & 4 & 2 & -1 & 3 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 4^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 11, \Delta = -3, \det = -9, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 & 5 & 4 \\ 2 & 3 & 0 & 0 & 3 & 2 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 2 & 3 & 3 \\ 2 & 4 & 2 & 0 & 2 & 1 & 3 & 3 \\ 3 & 5 & 3 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & 2 & -1 & 3 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 11, \ \Delta = -3, \ \det = -9, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 3 & 0 & 5 & 7 & 4 & 4 \\ 2 & 3 & 0 & 0 & 3 & 5 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 3 & 0 & 4 & 6 & 3 & 4 \\ 4 & 7 & 5 & 0 & 6 & 8 & 5 & 6 \\ 3 & 4 & 2 & 0 & 3 & 5 & 1 & 3 \\ 3 & 4 & 2 & -1 & 4 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 11, \ \Delta = -3, \ \det = -81, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 4 & 3 & 2 & 3 \\ 3 & 4 & 3 & 0 & 7 & 4 & 4 & 4 \\ 2 & 3 & 0 & 0 & 5 & 2 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 4 & 7 & 5 & 0 & 8 & 5 & 5 & 6 \\ 3 & 4 & 2 & 0 & 5 & 1 & 3 & 3 \\ 2 & 4 & 2 & 0 & 5 & 3 & 1 & 3 \\ 3 & 4 & 2 & -1 & 6 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$4.1217 \quad \text{NHC}(-2)(-3) + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$$

$$T = 11 \quad [0] \ 1^{111} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 11, \ \Delta = -2, \ \det = 0, \ (n_+, n_-, n_0) = (1, 11, 1), \ c_{\text{ext}} = 11.0$$

$$[1] \ 1^{111} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 + \textcolor{red}{su}_2 \quad T = 11, \ T_{\max} = 11, \ \Delta = -2, \ \det = 0, \ (n_+, n_-, n_0) = (1, 11, 1), \ c_{\text{ext}} = 11.0$$

$$4.1218 \quad \text{NHC}(-2)(-3) + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_3$$

$$T = 8 \quad [0] \ 1^{12} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -81, \ \det = 6, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 1 & 3 & 3 \\ 3 & 4 & 5 & 8 & 0 & 4 & 4 \\ 4 & 5 & 3 & 6 & 0 & 3 & 3 \\ 5 & 8 & 6 & 9 & 0 & 5 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 & -1 \\ 3 & 4 & 3 & 5 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -81, \ \det = 12, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 1 & 3 & 3 \\ 4 & 9 & 4 & 4 & 0 & 5 & 5 \\ 2 & 4 & 0 & 3 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 & -1 \\ 3 & 5 & 2 & 2 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -81, \ \det = 12, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 2 & 2 & 5 & 3 & 3 \\ 4 & 9 & 4 & 4 & 8 & 5 & 5 \\ 2 & 4 & 0 & 3 & 3 & 2 & 2 \\ 2 & 4 & 3 & 0 & 3 & 2 & 2 \\ 5 & 8 & 3 & 3 & 3 & 3 & 3 \\ 3 & 5 & 2 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{13} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -81, \ \det = 12, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 & 3 \\ 3 & 4 & 5 & 12 & 0 & 4 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 & 3 \\ 7 & 12 & 9 & 24 & 0 & 8 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 & -1 \\ 3 & 4 & 3 & 8 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -81, \ \det = 18, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.78 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

- [5] $1^{121} 1^{13} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 & 4 \\ 3 & 5 & 3 & 3 & 3 & 3 \\ 4 & 4 & 3 & 0 & 3 & 2 \\ 3 & 4 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [6] $1^{111} 1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 2 & 3 \\ 3 & 4 & 5 & 4 & 4 & 4 \\ 3 & 5 & 3 & 3 & 3 & 3 \\ 4 & 4 & 3 & 0 & 3 & 2 \\ 2 & 4 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [7] $1^{121} 1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & 5 & 8 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 6 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$
- [8] $1^{111} 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 1 & 3 \\ 3 & 4 & 4 & 5 & 0 & 4 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 4 & 5 & 3 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$
- [11] $1^{111} 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 3 & 3 \\ 3 & 4 & 0 & 0 & 5 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 5 & 0 & 0 & 3 & 3 \\ 3 & 4 & -1 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$
- [10] $1^{111} 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 2 & 3 \\ 3 & 4 & 0 & 0 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 0 & 0 & 0 & 2 \\ 3 & 4 & -1 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$
- [13] $1^{121} 1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 7 & 3 \\ 3 & 4 & 0 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 7 & 13 & 0 & 0 & 27 & 9 \\ 3 & 4 & -1 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$
- [9] $1^{111} 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 6 & 3 \\ 4 & 9 & 4 & 0 & 12 & 5 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 6 & 0 & 12 & 6 \\ 3 & 5 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$
- [14] $1^{131} 1^{11} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 12 & 3 \\ 3 & 4 & 0 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 22 & 0 & 0 & 84 & 16 \\ 3 & 4 & -1 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$
- [12] $1^{111} 1^{11} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 3 \\ 3 & 4 & 0 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 14 & 0 & 0 & 36 & 10 \\ 3 & 4 & -1 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[15] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 4 & 3 \\ 3 & 4 & 4 & 5 & 4 & 4 \\ 2 & 4 & 0 & 3 & 3 & 2 \\ 4 & 5 & 3 & 3 & 3 & 3 \\ 4 & 4 & 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{12} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 5 & 8 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 5 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{12} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 10 & 3 \\ 3 & 4 & 0 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 18 & 0 & 0 & 57 & 13 \\ 3 & 4 & -1 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 5 & 3 \\ 3 & 4 & 0 & 0 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 9 & 0 & 0 & 12 & 6 \\ 3 & 4 & -1 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 11 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -1 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[22] $1^{11} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 2 & 4 & 3 & 3 \\ 3 & 4 & 4 & 5 & 4 & 4 \\ 2 & 4 & 0 & 3 & 3 & 2 \\ 4 & 5 & 3 & 3 & 3 & 3 \\ 3 & 4 & 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[23] $21^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 6 & 4 \\ 8 & 9 & 4 & 4 & 8 & 5 \\ 3 & 4 & 0 & 3 & 3 & 2 \\ 3 & 4 & 3 & 0 & 3 & 2 \\ 6 & 8 & 3 & 3 & 3 & 3 \\ 4 & 5 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[24] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 & 4 \\ 3 & 5 & 3 & 3 & 3 & 3 \\ 2 & 4 & 3 & 0 & 3 & 2 \\ 3 & 4 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 & 4 \\ 3 & 5 & 3 & 3 & 3 & 3 \\ 3 & 4 & 3 & 0 & 3 & 2 \\ 3 & 4 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[26] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{13} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 7 & 3 \\ 3 & 4 & 0 & 0 & 12 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 7 & 12 & 0 & 0 & 24 & 8 \\ 3 & 4 & -1 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{111} 3 1^1 + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 3 & 2 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & -1 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[28] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 117, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & -1 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 & 4 \\ 4 & 5 & 3 & 3 & 3 & 3 \\ 3 & 4 & 3 & 0 & 3 & 2 \\ 3 & 4 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{111} 3 1^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.72 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 7 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & -1 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[32] $21^{111} 31^{11} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 & 3 \\ 4 & 4 & 5 & 4 & 4 & 4 \\ 3 & 5 & 3 & 3 & 3 & 3 \\ 3 & 4 & 3 & 0 & 3 & 2 \\ 3 & 4 & 3 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 0 & 9 & 2 \\ 9 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -1 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{111} 3 1^{12} + [\text{su}_2 \cdot \mathfrak{g}_2] + \text{su}_3 + \text{su}_3 + \text{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -81, \det = 189, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 10 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -1 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{11} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 9 & 3 \\ 3 & 4 & 0 & 0 & 14 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 14 & 0 & 0 & 36 & 10 \\ 3 & 4 & -1 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 6 & 2 \\ 6 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & -1 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{12} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 210, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 8 & 3 \\ 3 & 4 & 0 & 0 & 13 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 13 & 0 & 0 & 27 & 9 \\ 3 & 4 & -1 & -1 & 9 & 2 \end{pmatrix}_{b_0}$$

[38] $1^{11} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 213, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 2 & 4 & 0 & 0 & 9 & 2 \\ 9 & 14 & 0 & 9 & 36 & 10 \\ 3 & 4 & -1 & 2 & 10 & 2 \end{pmatrix}_{b_0}$$

[39] $1^{14} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.6$
 $T_{\min} = 10.69 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 12 & 3 \\ 3 & 4 & 0 & 0 & 21 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 21 & 0 & 0 & 75 & 15 \\ 3 & 4 & -1 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 237, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 6 & 3 \\ 3 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 3 & 4 & 0 & 0 & 6 & 2 \\ 6 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & -1 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{13} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 237, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 10 & 3 \\ 3 & 4 & 0 & 0 & 17 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 17 & 0 & 0 & 48 & 12 \\ 3 & 4 & -1 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{12} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 279, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 11 & 3 \\ 3 & 4 & 0 & 0 & 18 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 11 & 18 & 0 & 0 & 57 & 13 \\ 3 & 4 & -1 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[43] $1^{13} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 10.68 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 13 & 3 \\ 3 & 4 & 0 & 0 & 22 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 13 & 22 & 0 & 0 & 84 & 16 \\ 3 & 4 & -1 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[44] $1^{14} 3 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$ $T = 8, T_{\max} = 11, \Delta = -81, \det = 363, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.6$
 $T_{\min} = 10.63 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 1 & 1 & 15 & 3 \\ 3 & 4 & 0 & 0 & 26 & 4 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 26 & 0 & 0 & 117 & 19 \\ 3 & 4 & -1 & -1 & 19 & 2 \end{pmatrix}_{b_0}$$

4.1219 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 11 & 1 & 1 & 9 & 5 \\ 11 & 20 & 0 & 0 & 17 & 8 \\ 1 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 17 & 0 & 0 & 12 & 6 \\ 5 & 8 & -1 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 0 & 5 & 4 & 0 & 4 \\ 1 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 5 & 0 & 3 & 3 & 0 & 3 \\ 2 & 4 & 0 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & -1 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -80, \det = -261, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 5 & 4 & 4 & 0 & 4 \\ 3 & 5 & 3 & 3 & 3 & 0 & 3 \\ 2 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 3 & 2 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

4.1220 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 1 & 3 \\ 4 & 4 & 5 & 12 & 0 & 4 \\ 3 & 5 & 3 & 9 & 0 & 3 \\ 9 & 12 & 9 & 24 & 0 & 8 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{11}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 8 & 3 & 3 & 1 & 4 \\ 8 & 9 & 4 & 4 & 0 & 5 \\ 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 5 & 2 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{12}} 3^{\overline{11}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 9 & 1 & 3 \\ 4 & 4 & 4 & 14 & 0 & 4 \\ 3 & 4 & 0 & 9 & 0 & 2 \\ 9 & 14 & 9 & 36 & 0 & 10 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 10 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{11}} 3^{\overline{12}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 10.74 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 1 & 3 \\ 4 & 4 & 5 & 8 & 0 & 4 \\ 3 & 5 & 3 & 6 & 0 & 3 \\ 6 & 8 & 6 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{11}} 3^{\overline{11}} 2^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 1 & 3 \\ 4 & 4 & 5 & 4 & 0 & 4 \\ 3 & 5 & 3 & 3 & 0 & 3 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 237, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.73 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 9 & 1 & 4 \\ 5 & 4 & 4 & 9 & 0 & 4 \\ 3 & 4 & 0 & 6 & 0 & 2 \\ 9 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 4 & 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{11}31^{111}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 453, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.75 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 6 & 1 & 3 \\ 4 & 4 & 4 & 9 & 0 & 4 \\ 3 & 4 & 0 & 6 & 0 & 2 \\ 6 & 9 & 6 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

4.1221 NHC(-2)(-3) + $\textcolor{red}{su}_3 + \textcolor{red}{su}_3 + \textcolor{red}{su}_3$ + su3n2mix

$T = 8 \quad [0] \quad 1^{111}31^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 4 & 5 & 3 & 6 & 4 & 4 \\ 5 & 4 & 4 & 5 & 4 & 4 \\ 3 & 4 & 0 & 3 & 3 & 2 \\ 6 & 5 & 3 & 3 & 3 & 3 \\ 4 & 4 & 3 & 3 & 0 & 2 \\ 4 & 4 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{111}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + \textcolor{orange}{su}_3 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 255, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 4 & 3 \\ 4 & 4 & 4 & 4 & 5 & 4 \\ 3 & 4 & 0 & 3 & 3 & 2 \\ 3 & 4 & 3 & 0 & 3 & 2 \\ 4 & 5 & 3 & 3 & 3 & 3 \\ 3 & 4 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

4.1222 NHC(-2)(-3) + $\textcolor{red}{su}_3 + (-3) \rightarrow (-2)$

$T = 7 \quad [0] \quad 1^{111}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 9 & 12 & 0 & 5 \\ 6 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{111}312^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 19, T \in [13, 19]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 9 & 8 & 0 & 5 \\ 3 & 8 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 3 & -1 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 4 \\ 5 & 4 & 4 & 0 & 4 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10.57 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 4 & 5 & 0 & 4 \\ 3 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 3 & -1 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{11}31^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 11.30 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 4 & 5 & 6 & 1 & 4 \\ 5 & 4 & 5 & 0 & 4 \\ 6 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{13}31^{11}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -89, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0$

- $T_{\min} = 10.37 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
- $$\begin{pmatrix} 4 & 5 & 15 & 1 & 4 \\ 5 & 4 & 17 & 0 & 4 \\ 15 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 12 & -1 & 2 \end{pmatrix}_{b_0}$$
- [6] $1^{13}31^{11}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 19$, $\Delta = -103$, $\det = -116$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
- $T_{\min} = 10.62 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
- $$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 4 & 4 & 0 & 4 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$
- [7] $1^{12}31^{11}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 19$, $\Delta = -103$, $\det = -116$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
- $T_{\min} = 10.62 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
- $$\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 14 & 0 & 4 \\ 9 & 14 & 36 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 10 & -1 & 2 \end{pmatrix}_{b_0}$$
- [8] $1^{13}31^{11}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 19$, $\Delta = -103$, $\det = -125$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
- $T_{\min} = 10.78 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
- $$\begin{pmatrix} 4 & 8 & 3 & 1 & 4 \\ 8 & 9 & 4 & 0 & 5 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$
- [9] $1^{11}31^{11}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 19$, $\Delta = -103$, $\det = -129$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
- $T_{\min} = 10.63 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
- $$\begin{pmatrix} 4 & 5 & 9 & 1 & 4 \\ 5 & 4 & 9 & 0 & 4 \\ 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$
- [10] $1^{12}31^{11}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 19$, $\Delta = -103$, $\det = -134$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
- $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
- $$\begin{pmatrix} 4 & 5 & 12 & 1 & 4 \\ 5 & 4 & 13 & 0 & 4 \\ 12 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$
- [11] $1^{11}31^{11}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 19$, $\Delta = -103$, $\det = -156$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.0$
- $T_{\min} = 10.46 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
- $$\begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 4 & 9 & 0 & 4 \\ 6 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$
- [12] $1^{13}31^{12}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 19$, $\Delta = -103$, $\det = -176$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
- $T_{\min} = 10.73 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
- $$\begin{pmatrix} 1 & 4 & 6 & 1 & 3 \\ 4 & 4 & 8 & 0 & 4 \\ 6 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$
- [13] $1^{13}31^{13}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 19$, $\Delta = -103$, $\det = -276$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 9.0$
- $T_{\min} = 10.72 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
- $$\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 12 & 0 & 4 \\ 9 & 12 & 24 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$
- [14] $1^{11}31^{12}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 19$, $\Delta = -103$, $\det = -296$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
- $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
- $$\begin{pmatrix} 1 & 4 & 9 & 1 & 3 \\ 4 & 4 & 13 & 0 & 4 \\ 9 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$
- [15] $1^{12}31^{12}2^1 + [\text{su}_2 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2)$ $T = 7$, $T_{\max} = 19$, $\Delta = -103$, $\det = -336$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 8.2$
- $T_{\min} = 10.38 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 4 & 18 & 0 & 4 \\ 12 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 13 & -1 & 2 \end{pmatrix}_{b_0} \\
[16] & 1^{11}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -476, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.42 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 4 & 12 & 1 & 3 \\ 4 & 4 & 17 & 0 & 4 \\ 12 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 12 & -1 & 2 \end{pmatrix}_{b_0} \\
[17] & 31^{11}1^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -561, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10.40 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 3 & 1 & 3 \\ 3 & 4 & 0 & 3 & 1 & 2 \\ 3 & 3 & 3 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0} \\
[18] & 1^{12}31^{13}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -596, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.0 \\
& T_{\min} = 10.34 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 1 & 4 & 15 & 1 & 3 \\ 4 & 4 & 22 & 0 & 4 \\ 15 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 16 & -1 & 2 \end{pmatrix}_{b_0} \\
T = 8 & [19] 1^{12}4^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.44 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0} \\
[20] & 21^{11}41^{12}2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 4 & 8 & 6 & 1 & 4 \\ 8 & 13 & 12 & 0 & 7 \\ 6 & 12 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 7 & 5 & -1 & 2 \end{pmatrix}_{b_0} \\
[21] & 1^{11}4^{12}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 9 & 4 & 0 & 5 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 2 & -1 & 2 \end{pmatrix}_{b_0} \\
[22] & 1^{11}2^{11}4^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 4 & 5 & 6 & 1 & 4 \\ 5 & 4 & 7 & 0 & 4 \\ 6 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0} \\
[23] & 1^{11}4^{12}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0} \\
[24] & 1^{21}4^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 4 & 5 & 3 & 1 & 4 \\ 5 & 4 & 4 & 0 & 4 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0} \\
[25] & 1^{12}2^{11}4^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 4 & 5 & 3 & 1 & 4 \\ 5 & 4 & 3 & 0 & 4 \\ 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 4 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [26] $22^1 1^1 41^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 & 3 \\ 3 & 4 & 7 & 3 & 4 \\ 5 & 7 & 7 & 5 & 5 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$
- [27] $21^1 41^{11} 2^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 4 & 8 & 6 & 3 & 4 \\ 8 & 13 & 12 & 6 & 7 \\ 6 & 12 & 7 & 5 & 5 \\ 3 & 6 & 5 & 0 & 2 \\ 4 & 7 & 5 & 2 & 2 \end{pmatrix}_{b_0}$
- [28] $1^1 2^1 1^1 4 1^2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 7 & 1 & 3 \\ 4 & 4 & 10 & 0 & 4 \\ 7 & 10 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 8 & -1 & 2 \end{pmatrix}_{b_0}$
- [30] $1^1 1^1 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$
- [29] $1^1 41^{11} 22^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 57, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$
- [31] $1^1 2^1 1^1 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$
- [32] $1^1 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 4 & 4 & 0 & 4 \\ 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$
- [33] $1^1 2^1 1^1 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 4 & 7 & 0 & 4 \\ 5 & 7 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0}$
- [34] $1^1 41^{11} 22^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.19 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 6 & 5 & 1 & 3 \\ 6 & 13 & 12 & 0 & 7 \\ 5 & 12 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 5 & -1 & 2 \end{pmatrix}_{b_0}$
- [35] $22^1 1^1 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.17 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 6 & 5 & 3 & 3 \\ 6 & 13 & 12 & 6 & 7 \\ 5 & 12 & 7 & 5 & 5 \\ 3 & 6 & 5 & 0 & 2 \\ 3 & 7 & 5 & 2 & 2 \end{pmatrix}_{b_0}$
- [36] $1^1 41^{11} 22^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2)$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.22 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 5 & 5 & 1 & 3 \\ 5 & 8 & 11 & 0 & 6 \\ 5 & 11 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 6 & 5 & -1 & 2 \end{pmatrix}_{b_0}$

$$\begin{aligned}
T = 9 \quad [37] \quad 21^1 5^1 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
\begin{pmatrix} 1 & 4 & 4 & 1 & 3 \\ 4 & 4 & 6 & 0 & 4 \\ 4 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 3 \end{pmatrix}_{b_0} \\
[38] \quad 1^1 5^1 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
\begin{pmatrix} 1 & 4 & 4 & 1 & 3 \\ 4 & 4 & 6 & 0 & 4 \\ 4 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 3 \end{pmatrix}_{b_0} \\
[39] \quad 1^1 5^1 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
\begin{pmatrix} 1 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 5 & 6 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 2 \end{pmatrix}_{b_0} \\
[40] \quad 1^1 5^1 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 11.83 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 3 \end{pmatrix}_{b_0} \\
[41] \quad 21^1 5^1 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 1 & 5 & 0 & 3 \\ 4 & 5 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 5 & -1 & 3 \end{pmatrix}_{b_0} \\
[42] \quad 1^1 5^1 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 2 & -1 & 2 \end{pmatrix}_{b_0} \\
[43] \quad 1^1 5^1 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
\begin{pmatrix} 1 & 2 & 5 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 5 & 5 & 7 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 5 & -1 & 2 \end{pmatrix}_{b_0} \\
[44] \quad 1^1 5^1 1^1 2^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
\begin{pmatrix} 1 & 6 & 2 & 1 & 3 \\ 6 & 13 & 5 & 0 & 7 \\ 2 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 2 & -1 & 3 \end{pmatrix}_{b_0} \\
[45] \quad 22^1 1^1 5^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
\begin{pmatrix} 1 & 6 & 2 & 3 & 3 \\ 6 & 13 & 5 & 6 & 7 \\ 2 & 5 & 0 & 2 & 2 \\ 3 & 6 & 2 & 0 & 2 \\ 3 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
T = 11 \quad [46] \quad 1^1 6^1 13^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{su}_3 + (-3) \rightarrow (-2) \quad T = 11, T_{\max} = 13, \Delta = -65, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0 \\
T_{\min} = 13, T_{\max} = 13, T \in [13, 13]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [47] \quad 1^{1^1 1^1 5^{131}} 6^{12^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 6 & 2 & 6 \\ 2 & 0 & 2 \\ 6 & 2 & 6 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1^1 1^1 5^{131}} 6^{12^1} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -13, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 14.85 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 3 \\ 2 & 1 & 1 & 0 & 3 \\ 1 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 3 & 0 & 0 & 4 \end{pmatrix}_{b_0}$$

4.1223 NHC(-2)(-3) + $\mathfrak{su}_3$ + (-3) $\rightarrow$ (-2) + $\mathfrak{so}_8$

$$T = 8 \quad [0] \quad 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.49 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 17 & 15 & 1 & 7 \\ 17 & 20 & 17 & 0 & 8 \\ 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.56 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 13 & 20 & 12 & 1 & 7 \\ 20 & 25 & 16 & 0 & 9 \\ 12 & 16 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 5 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 16 & 12 & 1 & 6 \\ 16 & 20 & 17 & 0 & 8 \\ 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \quad 1^{11} 4^{1^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 13 & 14 & 9 & 1 & 7 \\ 14 & 13 & 10 & 0 & 7 \\ 9 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{2^1 1^{11}} 4^{1^1} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 10 & 7 & 1 & 5 \\ 10 & 13 & 10 & 0 & 7 \\ 7 & 10 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

4.1224 NHC(-2)(-3) + $\mathfrak{su}_3$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{1^1} 51^{1^1} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 3 & 0 & 3 & 4 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 3 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

- [1] $1^{11} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 3 & 0 & 4 & 4 \\ 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$
- [2] $1^{11} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 60, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 3 & 0 & 10 & 4 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 8 & 10 & 6 & 0 & 21 & 9 \\ 3 & 4 & 2 & -1 & 9 & 3 \end{pmatrix}_{b_0}$
- [3] $1^{11} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 85, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 3 & 0 & 7 & 4 \\ 2 & 3 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 7 & 4 & 0 & 8 & 6 \\ 3 & 4 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$
- [4] $1^{12} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 141, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 3 & 0 & 11 & 4 \\ 2 & 3 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 7 & 11 & 6 & 0 & 21 & 9 \\ 3 & 4 & 2 & -1 & 9 & 3 \end{pmatrix}_{b_0}$
- [5] $1^{13} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 3 & 0 & 15 & 4 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 15 & 8 & 0 & 40 & 12 \\ 3 & 4 & 2 & -1 & 12 & 3 \end{pmatrix}_{b_0}$
- [6] $1^{12} \overset{1}{5} 1^3 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 181, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 3 & 0 & 17 & 4 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 13 & 17 & 10 & 0 & 61 & 15 \\ 3 & 4 & 2 & -1 & 15 & 3 \end{pmatrix}_{b_0}$
- [7] $1^{12} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 196, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 3 & 0 & 14 & 4 \\ 2 & 3 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 14 & 8 & 0 & 38 & 12 \\ 3 & 4 & 2 & -1 & 12 & 3 \end{pmatrix}_{b_0}$
- [8] $1^{2^1 21^{11}} \overset{1}{5} 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 285, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 6 & 2 & 1 & 5 & 3 \\ 6 & 13 & 5 & 0 & 12 & 7 \\ 2 & 5 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 5 & 12 & 4 & 0 & 8 & 6 \\ 3 & 7 & 2 & -1 & 6 & 3 \end{pmatrix}_{b_0}$
- [9] $1^{13} \overset{1}{5} 1^2 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_2$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 316, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 3 & 0 & 18 & 4 \\ 2 & 3 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 12 & 18 & 10 & 0 & 61 & 15 \\ 3 & 4 & 2 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{13} \overset{1}{5} 1^3 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 381, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 3 & 0 & 21 & 4 \\ 2 & 3 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 21 & 12 & 0 & 88 & 18 \\ 3 & 4 & 2 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{21} \overset{2}{5} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 516, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 1 & 10 & 3 \\ 6 & 13 & 5 & 0 & 24 & 7 \\ 2 & 5 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 24 & 8 & 0 & 38 & 12 \\ 3 & 7 & 2 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{21} \overset{2}{5} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 901, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 2 & 1 & 15 & 3 \\ 6 & 13 & 5 & 0 & 36 & 7 \\ 2 & 5 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 15 & 36 & 12 & 0 & 88 & 18 \\ 3 & 7 & 2 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

$$T = 14 \quad [13] \ 1^{21} \overset{1}{6} 1^{13} \overset{1}{5} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 10, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 6 & 2 & 12 & 6 \\ 2 & 0 & 5 & 2 \\ 12 & 5 & 20 & 12 \\ 6 & 2 & 12 & 6 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11} \overset{1}{5} 1^{13} \overset{1}{6} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_2 \quad T = 14, T_{\max} = 15, \Delta = -29, \det = 11, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 2 & 3 \\ 2 & 1 & 1 & 0 & 3 & 3 \\ 1 & 1 & -2 & 0 & 0 & 0 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 0 & 0 & 1 & 3 \\ 3 & 3 & 0 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

4.1225 NHC(-2)(-3) + $\mathfrak{su}_3$ + (-3) $\rightarrow$ (-2) + $\mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{13} 31^{13} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 9 & 3 \\ 4 & 4 & 5 & 0 & 12 & 4 \\ 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 12 & 9 & 0 & 24 & 8 \\ 3 & 4 & 3 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{13} 31^{13} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 8 & 3 & 1 & 3 & 4 \\ 8 & 9 & 4 & 0 & 4 & 5 \\ 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 4 & 5 & 2 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{12} 31^{11} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + (-3) \rightarrow (-2) + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 9 & 3 \\ 4 & 4 & 4 & 0 & 14 & 4 \\ 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 14 & 9 & 0 & 36 & 10 \\ 3 & 4 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}31^{12}2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 108, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 10.74 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 6 & 3 \\ 4 & 4 & 5 & 0 & 8 & 4 \\ 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 3 & -1 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}31^{11}2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 153, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.80 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 & 3 \\ 4 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 3 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111}31^{11}2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 237, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.73 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 1 & 9 & 4 \\ 5 & 4 & 4 & 0 & 9 & 4 \\ 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 9 & 9 & 6 & 0 & 12 & 6 \\ 4 & 4 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11}31^{111}2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + (-3) \rightarrow (-2) + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -82, \ \det = 453, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.75 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 6 & 3 \\ 4 & 4 & 4 & 0 & 9 & 4 \\ 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

4.1226 NHC(-2)(-3) + $\mathfrak{su}_3$ + su3n2mix

$$T = 7 \quad [0] \ 1^{11}31^{13}2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -103, \ \det = -15, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.33 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 9 & 4 & 3 \\ 4 & 4 & 12 & 5 & 4 \\ 9 & 12 & 24 & 9 & 8 \\ 4 & 5 & 9 & 3 & 3 \\ 3 & 4 & 8 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}31^{11}2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -103, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.45 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 3 & 4 & 3 \\ 4 & 4 & 5 & 4 & 4 \\ 3 & 5 & 3 & 3 & 3 \\ 4 & 4 & 3 & 0 & 2 \\ 3 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}31^{12}2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -103, \ \det = -65, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.2$$

$$T_{\min} = 10.38 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 6 & 4 & 3 \\ 4 & 4 & 8 & 5 & 4 \\ 6 & 8 & 9 & 6 & 5 \\ 4 & 5 & 6 & 3 & 3 \\ 3 & 4 & 5 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}31^{12}2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -103, \ \det = -71, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.63 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 & 4 \\ 5 & 4 & 4 & 4 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 4 & 4 & 3 & 0 & 2 \\ 4 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}31^{11}2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -103, \ \det = -95, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.47 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 1 & 4 & 3 & 4 & 3 \\ 4 & 4 & 4 & 5 & 4 \\ 3 & 4 & 0 & 3 & 2 \\ 4 & 5 & 3 & 3 & 3 \\ 3 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111}31^{11}2^1 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 7, \ T_{\max} = 19, \ \Delta = -103, \ \det = -156, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.38 \rightarrow 11, \ T_{\max} = 19, \ T \in [11, 19]$$

$$\begin{pmatrix} 4 & 5 & 9 & 4 & 4 \\ 5 & 4 & 9 & 4 & 4 \\ 9 & 9 & 12 & 6 & 6 \\ 4 & 4 & 6 & 0 & 2 \\ 4 & 4 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [6] \quad 1^{11} 4^{11} 4^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 4^{11} 4^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 & 4 \\ 5 & 4 & 3 & 4 & 4 \\ 3 & 3 & 0 & 2 & 2 \\ 4 & 4 & 2 & 0 & 2 \\ 4 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{211} 4^{11} 4^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 87, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 & 3 \\ 4 & 4 & 4 & 3 & 4 \\ 3 & 4 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [9] \quad 1^{11} 5^{11} 1^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -51, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 \\ 4 & 4 & 3 & 3 & 4 \\ 2 & 3 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [10] \quad 1^{211} 6^{11} 13 1^1 5 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -27, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 \\ 4 & 8 & 1 & 5 & 6 \\ 1 & 1 & -2 & 0 & 0 \\ 3 & 5 & 0 & 1 & 3 \\ 3 & 6 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

4.1227 NHC(-2)(-3) + $\mathfrak{su}_3$ + su3n2mix + $\mathfrak{su}_3$

$$T = 8 \quad [0] \quad 1^{111} 31^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.70 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 3 & 4 & 6 & 4 \\ 5 & 4 & 4 & 4 & 5 & 4 \\ 3 & 4 & 0 & 3 & 3 & 2 \\ 4 & 4 & 3 & 0 & 3 & 2 \\ 6 & 5 & 3 & 3 & 3 & 3 \\ 4 & 4 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{11} 2^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{su}}_3 + \textcolor{orange}{\mathfrak{su}}_3 + \textcolor{red}{\mathfrak{su}}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 255, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 3 & 4 & 3 & 3 \\ 4 & 4 & 4 & 5 & 4 & 4 \\ 3 & 4 & 0 & 3 & 3 & 2 \\ 4 & 5 & 3 & 3 & 3 & 3 \\ 3 & 4 & 3 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

5 NHC $(-2)(-3)(-2)$ External (163 combos, 2345 bases)

5.1 NHC $(-2)(-3)(-2)$

$$T = 12 \quad [167] \quad 1^{\overline{1}1} 4 14 13 22 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[166] \quad 1^{\overline{1}1} 3 15 13 22 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, T_{\max} = 12, \Delta = -5, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 2 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 4 & 0 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 13 \quad [178] \quad 1^{\overline{1}1} 4 14 14 13 21^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 3 & 4 & 5 \\ 2 & 4 & 3 & 5 \\ 3 & 5 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$T = 15 \quad [190] \quad 1^{\overline{1}1} 23 15 13 1^{\overline{1}} 6 13 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 15, T_{\max} = 15, \Delta = -4, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$T = 16 \quad [194] \quad 1^{\overline{1}1} 6 13 16 12 32 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 16, T_{\max} = 16, \Delta = -3, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 3 & 5 \\ 3 & 4 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$T = 17 \quad [197] \quad 1^{\overline{1}1} 23 15 13 21^{\overline{1}} 8 12 31^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 17, T_{\max} = 17, \Delta = -3, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$T = 18 \quad [200] \quad 1^{\overline{1}1} 6 13 16 13 15 13 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 18, T_{\max} = 18, \Delta = -2, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 3 & 5 \\ 3 & 4 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$T = 19 \quad [203] \quad 1^{\overline{1}1} 8 13 16 13 16 12 31^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 19, T_{\max} = 19, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 3 & 1 & 3 & 5 \\ 2 & 4 & 0 & 5 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$[204] \quad 1^{\overline{1}1} 8 12 32 17 12 32 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 19, T_{\max} = 19, \Delta = -2, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 3 & 5 \\ 3 & 4 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$$

$$T = 20 \quad [205] \quad 1^{1^{1^{2321}}} 8^{1^{1^{12321}}} 8^{1^{1^{1231}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 20, \quad T_{\max} = 20, \quad \Delta = -1, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 20, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 20, \quad T_{\max} = 20, \quad T \in [20, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 1 & 2 & 3 \\ 1 & 2 & 2 & 2 & 4 \\ 2 & 3 & 3 & 4 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$$

$$T = 22 \quad [208] \quad 1^{1^{1^{321}}} 8^{1^{1^{12321}}} 18^{1^{1^{12315131}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 22, \quad T_{\max} = 22, \quad \Delta = -1, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 22, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 22, \quad T_{\max} = 22, \quad T \in [22, 22]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 3 & 5 \\ 3 & 4 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

$$T = 23 \quad [209] \quad 1^{1^{1^{321}}} 8^{1^{1^{12321}}} 18^{1^{1^{123216131}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 23, \quad T_{\max} = 23, \quad \Delta = 0, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 23, 0), \quad c_{\text{ext}} = 4.0$$

$$T_{\min} = 23, \quad T_{\max} = 23, \quad T \in [23, 23]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 2 & 1 & 3 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 3 & 1 & 3 & 5 \\ 2 & 4 & 0 & 5 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{20}$$

5.2 NHC(-2)(-3)(-2) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{1^{1^{1^{1^{11}}}}} 3^{1^{1^{1^{1^{1^{11}}}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 & 2 & 3 \\ 4 & 9 & 7 & 0 & 8 & 4 & 5 \\ 3 & 7 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 8 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 4 & 0 & 3 & 0 & 2 \\ 3 & 5 & 4 & -4 & 4 & 2 & -12 \end{pmatrix}_{b_0} \rightarrow \text{bd}_4$$

5.3 NHC(-2)(-3)(-2) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{1^{1^{1^{1^{1^{11}}}}} 3^{1^{1^{1^{1^{1^{1^{11}}}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 25, \quad T_{\max} = 25, \quad T \in [25, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 3 & 3 \\ 4 & 9 & 7 & 0 & 4 & 8 & 5 \\ 3 & 7 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 3 & 8 & 5 & 0 & 3 & 4 & 4 \\ 3 & 5 & 4 & -4 & 2 & 4 & -12 \end{pmatrix}_{b_0} \rightarrow \text{bd}_4$$

5.4 NHC(-2)(-3)(-2) + $\mathfrak{su}_2$

$$T = 12 \quad [187] \quad 1^{1^{1^{1^{1^{1^{1^{11}}}}} 3^{1^{1^{1^{1^{1^{1^{1^{11}}}}} 5^{1^{1^{1^{1^{1^{1^{1^{11}}}}} 13^{1^{1^{1^{1^{1^{1^{1^{11}}}}} 21^{1^{1^{1^{1^{1^{1^{1^{11}}}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 1 & 1 & 5 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 4 & 5 & 0 & 7 & 7 \\ 2 & 3 & 3 & 0 & 7 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$[188] \quad 1^{1^{1^{1^{1^{1^{1^{1^{11}}}}} 3^{1^{1^{1^{1^{1^{1^{1^{11}}}}} 5^{1^{1^{1^{1^{1^{1^{1^{11}}}}} 13^{1^{1^{1^{1^{1^{1^{1^{11}}}}} 21^{1^{1^{1^{1^{1^{1^{1^{11}}}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 1 & 1 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 4 & 7 & 6 & 0 & 18 & 10 \\ 2 & 3 & 3 & 0 & 10 & 5 \end{pmatrix}_{b_0} \rightarrow \text{bd}_8$$

$$T = 13 \quad [193] \quad 1^{\overline{11}2321} \quad 6 \quad 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 13, \quad T_{\max} = 13, \quad \Delta = -1, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 13, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 2 & 3 & 3 \\ 2 & 2 & 2 & 5 & 4 \\ 4 & 3 & 5 & 7 & 7 \\ 3 & 3 & 4 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[196] \quad 1^{\overline{11}122321} \quad 6 \quad 131^{\overline{14}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 13, \quad T_{\max} = 13, \quad \Delta = -1, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 13, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 14 & 3 \\ 2 & 1 & 2 & 12 & 3 \\ 2 & 2 & 2 & 19 & 4 \\ 14 & 12 & 19 & 106 & 26 \\ 3 & 3 & 4 & 26 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[195] \quad 1^{\overline{11}122321} \quad 6 \quad 131^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 13, \quad T_{\max} = 13, \quad \Delta = -1, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 13, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 12 & 3 \\ 2 & 1 & 2 & 9 & 3 \\ 2 & 2 & 2 & 15 & 4 \\ 12 & 9 & 15 & 67 & 21 \\ 3 & 3 & 4 & 21 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$[194] \quad 1^{\overline{11}112321} \quad 6 \quad 131^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 13, \quad T_{\max} = 13, \quad \Delta = -1, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 13, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 2 & 6 & 3 \\ 2 & 2 & 2 & 9 & 4 \\ 6 & 6 & 9 & 22 & 12 \\ 3 & 3 & 4 & 12 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{10}$$

$$T = 14 \quad [197] \quad 1^{\overline{11}112321} \quad 7 \quad 1231^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 14, \quad T_{\max} = 14, \quad \Delta = 0, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 14, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 2 & 3 & 3 \\ 2 & 2 & 2 & 5 & 4 \\ 4 & 3 & 5 & 7 & 7 \\ 3 & 3 & 4 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{11}$$

$$T = 15 \quad [200] \quad 1^{\overline{11}11315131} \quad 6 \quad 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 15, \quad T_{\max} = 15, \quad \Delta = 0, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 15, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 4 & 3 \\ 2 & 1 & 2 & 3 & 3 \\ 2 & 2 & 2 & 5 & 4 \\ 4 & 3 & 5 & 7 & 7 \\ 3 & 3 & 4 & 7 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[199] \quad 1^{\overline{11}11315131} \quad 6 \quad 131^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 15, \quad T_{\max} = 15, \quad \Delta = 0, \quad \det = -1, \quad (n_+, n_-, n_0) = (1, 15, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 15, \quad T_{\max} = 15, \quad T \in [15, 15]$$

$$\begin{pmatrix} 1 & 2 & 2 & 6 & 3 \\ 2 & 1 & 2 & 6 & 3 \\ 2 & 2 & 2 & 9 & 4 \\ 6 & 6 & 9 & 22 & 12 \\ 3 & 3 & 4 & 12 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

5.5 NHC(-2)(-3)(-2) + $\mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 12 \quad [71] \quad 1^{\overline{11}3^{\overline{1}}1^{\overline{1}}5131^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 4 & 2 & 3 \\ 1 & 2 & 0 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 4 & 4 & 0 & 6 & 3 & 6 \\ 1 & 2 & 1 & 0 & 3 & 0 & 2 \\ 2 & 3 & 2 & 0 & 6 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

$$[72] \quad 1^{\overline{12}3^{\overline{1}}1^{\overline{1}}5131^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 12, \quad T_{\max} = 12, \quad \Delta = -2, \quad \det = 1, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 1 & 2 & 0 & 7 & 2 & 3 \\ 1 & 2 & 0 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 7 & 5 & 0 & 17 & 5 & 9 \\ 1 & 2 & 1 & 0 & 5 & 0 & 2 \\ 2 & 3 & 2 & 0 & 9 & 2 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

5.6 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{red}{\mathfrak{su}_2}] + \textcolor{red}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{e}_6} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 25, T_{\max} = 25, T \in [25, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 & 3 \\ 4 & 9 & 7 & 4 & 0 & 8 & 5 \\ 3 & 7 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 5 & 3 & 0 & 4 & 4 \\ 3 & 5 & 4 & 2 & -4 & 4 & -12 \end{pmatrix}_{b_0} \rightarrow \text{bd}_4$$

Non-unimodular ($|\det| > 1$) — **70731** bases

5.7 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3)(-2)$

$$T = 10 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [(-2) \cdot \textcolor{red}{\mathfrak{sp}_1} \cdot \textcolor{green}{\mathfrak{g}_2}] + \textcolor{red}{\mathfrak{e}_6} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{red}{\mathfrak{su}_2}] \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 436, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 4 & 6 & 4 \\ 2 & 3 & 4 & 0 & 4 & 0 & 4 & 2 \\ 3 & 6 & 5 & 0 & 6 & 4 & 4 & 4 \\ 3 & 4 & 4 & -4 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

5.8 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3)(-2)$

$$T = 9 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{red}{\mathfrak{e}_6} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{red}{\mathfrak{su}_2}] \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 4 & 0 & 4 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 & 4 \\ 3 & 4 & -4 & 3 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{red}{\mathfrak{e}_6} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{red}{\mathfrak{su}_2}] \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 6 & 4 \\ 2 & 4 & 0 & 4 & 0 & 4 & 2 \\ 3 & 5 & 0 & 6 & 4 & 4 & 4 \\ 3 & 4 & -4 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{red}{\mathfrak{e}_6} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{red}{\mathfrak{su}_2}] \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -3 & 3 & 0 & 3 & -1 \\ 3 & 8 & 3 & 4 & 1 & 6 & 4 \\ 1 & 0 & 0 & 1 & -3 & 1 & -1 \\ 3 & 8 & 3 & 6 & 1 & 4 & 4 \\ 3 & 5 & -1 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

5.9 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3)(-2) + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{red}{\mathfrak{e}_6} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{red}{\mathfrak{su}_2}] + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 212, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 6 & 6 & 4 \\ 2 & 4 & 0 & 4 & 0 & 4 & 3 & 2 \\ 3 & 5 & 0 & 6 & 4 & 4 & 6 & 4 \\ 3 & 5 & 0 & 6 & 3 & 6 & 4 & 4 \\ 3 & 4 & -4 & 4 & 2 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{red}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{g}_2}] + \textcolor{red}{\mathfrak{e}_6} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{red}{\mathfrak{su}_2}] + \textcolor{red}{\mathfrak{su}_2} \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 592, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 5 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 6 & 9 & 4 \\ 2 & 4 & 0 & 4 & 0 & 4 & 6 & 2 \\ 3 & 5 & 0 & 6 & 4 & 4 & 9 & 4 \\ 5 & 9 & 0 & 9 & 6 & 9 & 13 & 7 \\ 3 & 4 & -4 & 4 & 2 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

5.10 NHC(-2)(-3) + $\mathfrak{e}_6$ + NHC(-2)(-3)(-2) + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{1111} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 0 & 3 & 0 & -1 \\ 3 & 8 & 3 & 4 & 1 & 6 & 0 & 4 \\ 1 & 0 & 0 & 1 & -3 & 1 & 0 & -1 \\ 3 & 8 & 3 & 6 & 1 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & -1 & 4 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

5.11 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + NHC(-2)(-3)(-2)

$$T = 10 \quad [0] \quad 1^{11} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 212, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 6 & 3 & 6 & 4 \\ 3 & 5 & 0 & 6 & 4 & 4 & 6 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 4 & 2 \\ 3 & 5 & 0 & 6 & 6 & 4 & 4 & 4 \\ 3 & 4 & -4 & 4 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 592, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 9 & 5 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 9 & 6 & 9 & 7 \\ 3 & 5 & 0 & 9 & 4 & 4 & 6 & 4 \\ 2 & 4 & 0 & 6 & 4 & 0 & 4 & 2 \\ 3 & 5 & 0 & 9 & 6 & 4 & 4 & 4 \\ 3 & 4 & -4 & 7 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

5.12 NHC(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + NHC(-2)(-3)(-2)

$$T = 10 \quad [0] \quad 1^{1111} 3^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 1 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 0 & 8 & 5 \\ 2 & 4 & -3 & 0 & 3 & 0 & 3 & -1 \\ 1 & 0 & 0 & -3 & 0 & 0 & 0 & -1 \\ 3 & 8 & 3 & 0 & 4 & 1 & 6 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & 1 & -1 \\ 3 & 8 & 3 & 0 & 6 & 1 & 4 & 4 \\ 3 & 5 & -1 & -1 & 4 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

5.13 NHC(-2)(-3) + $\mathfrak{e}_7$ + NHC(-2)(-3)(-2)

$$T = 10 \quad [0] \quad 1^{11} 3^1 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 10, T_{\max} = 100, \Delta = -142, \det = 272, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.82 \rightarrow 15, T_{\max} = 100, T \in [15, 100]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 4 & 5 \\ 7 & 8 & 1 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -3 & 2 & -1 \\ 4 & 4 & 0 & 2 & 0 & 2 \\ 5 & 6 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \quad 1^{11} 2^1 3^1 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 11, T_{\max} = 28, \Delta = -115, \det = -44, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.91 \rightarrow 15, T_{\max} = 28, T \in [15, 28]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 0 & 2 \\ 1 & 0 & 1 & -5 & 0 & 1 & 0 & -3 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & 0 & 1 & -2 & 0 \\ 2 & 0 & 2 & -3 & 0 & 2 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{221} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_7 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{so}_7 \cdot \textcolor{blue}{su}_2] \quad T = 11, \ T_{\max} = 26, \ \Delta = -113, \ \det = -224, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.93 \rightarrow 15, \ T_{\max} = 26, \ T \in [15, 26]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 4 & 5 \\ 7 & 8 & 1 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 2 & 0 \\ 4 & 4 & 0 & 2 & 0 & 2 \\ 5 & 6 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

5.14 $\text{NHC}(-2)(-3) + \textcolor{red}{e}'_7 + \text{NHC}(-2)(-3)(-2)$

$$T = 10 \quad [0] \ 1^{11} 3^{21} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{so}_7 \cdot \textcolor{blue}{su}_2] \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 224, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.93 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 4 & 5 \\ 7 & 8 & 1 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 2 & 0 \\ 4 & 4 & 0 & 2 & 0 & 2 \\ 5 & 6 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [1] \ 1^{11} 2^3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}'_7 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{so}_7 \cdot \textcolor{blue}{su}_2] \quad T = 11, \ T_{\max} = 14, \ \Delta = -87, \ \det = -36, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 3 & 0 & 2 \\ 1 & 0 & 1 & -4 & 0 & 1 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 1 & 1 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & 0 & 1 & -2 & 0 \\ 2 & 0 & 2 & -2 & 0 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

5.15 $\text{NHC}(-2)(-3) + \textcolor{red}{e}_8 + \text{NHC}(-2)(-3)(-2)$

$$T = 11 \quad [0] \ 1^{11} 3^{221} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{e}_8 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{so}_7 \cdot \textcolor{blue}{su}_2] \quad T = 11, \ T_{\max} = 106, \ \Delta = -228, \ \det = -416, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.65 \rightarrow 19, \ T_{\max} = 106, \ T \in [19, 106]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 4 & 5 \\ 7 & 8 & 1 & 5 & 4 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -6 & 2 & -4 \\ 4 & 4 & 0 & 2 & 0 & 2 \\ 5 & 6 & -1 & -4 & 2 & -3 \end{pmatrix}_{b_0}$$

5.16 $\text{NHC}(-2)(-3) + \textcolor{red}{f}_4 + \text{NHC}(-2)(-3)(-2)$

$$T = 9 \quad [0] \ 1^{11} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{so}_7 \cdot \textcolor{blue}{su}_2] \quad T = 9, \ T_{\max} = 13, \ \Delta = -89, \ \det = -68, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.35 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 4 & 0 & 4 & 2 \\ 3 & 5 & 0 & 3 & 4 & 4 & 4 \\ 3 & 4 & -3 & 3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{so}_7 \cdot \textcolor{blue}{su}_2] \quad T = 9, \ T_{\max} = 13, \ \Delta = -89, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.09 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 6 & 4 \\ 2 & 4 & 0 & 4 & 0 & 4 & 2 \\ 3 & 5 & 0 & 6 & 4 & 4 & 4 \\ 3 & 4 & -3 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{f}_4 + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{so}_7 \cdot \textcolor{blue}{su}_2] \quad T = 9, \ T_{\max} = 13, \ \Delta = -89, \ \det = -272, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.35 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 1 & 3 & 3 \\ 4 & 9 & 4 & 8 & 0 & 8 & 5 \\ 2 & 4 & -2 & 3 & 0 & 3 & 0 \\ 3 & 8 & 3 & 4 & 1 & 6 & 4 \\ 1 & 0 & 0 & 1 & -3 & 1 & -1 \\ 3 & 8 & 3 & 6 & 1 & 4 & 4 \\ 3 & 5 & 0 & 4 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

5.17 $\text{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \text{NHC}(-2)(-3)(-2)$

$$T = 10 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 212, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 5 & 4 & 5 & 4 \\ 3 & 5 & 4 & 0 & 6 & 3 & 6 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 6 & 0 & 4 & 4 & 6 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 4 & 2 \\ 3 & 5 & 6 & 0 & 6 & 4 & 4 & 4 \\ 3 & 4 & 4 & -4 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 592, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 9 & 0 & 5 & 4 & 5 & 4 \\ 5 & 9 & 13 & 0 & 9 & 6 & 9 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 9 & 0 & 4 & 4 & 6 & 4 \\ 2 & 4 & 6 & 0 & 4 & 0 & 4 & 2 \\ 3 & 5 & 9 & 0 & 6 & 4 & 4 & 4 \\ 3 & 4 & 7 & -4 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

5.18 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \text{NHC}(-2)(-3)(-2)$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 & 2 & 3 & 3 \\ 4 & 9 & 4 & 0 & 8 & 4 & 8 & 5 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 8 & 3 & 0 & 4 & 4 & 6 & 4 \\ 2 & 4 & 3 & 0 & 4 & 0 & 4 & 2 \\ 3 & 8 & 3 & 0 & 6 & 4 & 4 & 4 \\ 3 & 5 & 2 & -4 & 4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

5.19 $\text{NHC}(-2)(-3)(-2)$

$$T = 6 \quad [0] \ 21^{\overline{1}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 6, T_{\max} = 33, \Delta = -118, \det = 3, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 33, T \in [14, 33]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 4 & 7 & 4 \\ 4 & 7 & 9 & 5 \\ 3 & 4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 6, T_{\max} = 33, \Delta = -118, \det = 9, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.56 \rightarrow 10, T_{\max} = 33, T \in [10, 33]$$

$$\begin{pmatrix} 5 & 7 & 5 \\ 7 & 8 & 6 \\ 5 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[2] \ 13^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 6, T_{\max} = 33, \Delta = -118, \det = 10, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 33, T \in [11, 33]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 9 & 7 & 5 \\ 3 & 7 & 4 & 4 \\ 3 & 5 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 13^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 6, T_{\max} = 33, \Delta = -118, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 33, T \in [11, 33]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 4 \\ 3 & 6 & 4 & 4 \\ 3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 21^{\overline{1}} 31^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 6, T_{\max} = 33, \Delta = -118, \det = 20, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 33, T \in [10, 33]$$

$$\begin{pmatrix} 4 & 5 & 5 & 4 \\ 5 & 4 & 6 & 4 \\ 5 & 6 & 4 & 4 \\ 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] \quad T = 6, T_{\max} = 33, \Delta = -118, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

- $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 33, T \in [13, 33]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 9 & 11 & 5 \\ 4 & 11 & 9 & 5 \\ 3 & 5 & 5 & 0 \end{pmatrix}_{b_0}$
- [5] $2131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 6, T_{\max} = 33, \Delta = -118, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.17 \rightarrow 13, T_{\max} = 33, T \in [13, 33]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 9 & 11 & 5 \\ 4 & 11 & 9 & 5 \\ 3 & 5 & 5 & 0 \end{pmatrix}_{b_0}$
- [7] $21^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 6, T_{\max} = 33, \Delta = -118, \det = 27, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.37 \rightarrow 11, T_{\max} = 33, T \in [11, 33]$
 $\begin{pmatrix} 4 & 8 & 5 & 4 \\ 8 & 9 & 7 & 5 \\ 5 & 7 & 4 & 4 \\ 4 & 5 & 4 & 2 \end{pmatrix}_{b_0}$
- [8] $21^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 6, T_{\max} = 33, \Delta = -118, \det = 44, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 33, T \in [10, 33]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 4 & 6 & 4 \\ 4 & 6 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [9] $131^{11}2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 6, T_{\max} = 33, \Delta = -118, \det = 96, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 33, T \in [11, 33]$
 $\begin{pmatrix} 4 & 8 & 8 & 4 \\ 8 & 9 & 11 & 5 \\ 8 & 11 & 9 & 5 \\ 4 & 5 & 5 & 2 \end{pmatrix}_{b_0}$
- [10] $221^{11}3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 6, T_{\max} = 33, \Delta = -118, \det = 216, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 33, T \in [10, 33]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 0 & 4 & 2 \\ 3 & 3 & 4 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- $T = 7 \quad [11] \quad 1^1 1^1 3 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7, T_{\max} = 11, \Delta = -90, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -3 & 1 & -1 \\ 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$
- [12] $21^1 3 2 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7, T_{\max} = 11, \Delta = -90, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$
 $\begin{pmatrix} 4 & 5 & 5 & 4 \\ 5 & 5 & 7 & 5 \\ 5 & 7 & 5 & 5 \\ 4 & 5 & 5 & 4 \end{pmatrix}_{b_0}$
- [13] $1^1 1^1 3 2 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7, T_{\max} = 11, \Delta = -90, \det = -7, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.57 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0}$
- [14] $1^1 3 1^1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7, T_{\max} = 13, \Delta = -97, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 \\ 2 & 4 & 3 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 2 \end{pmatrix}_{b_0}$
- [15] $1^1 1^1 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7, T_{\max} = 19, \Delta = -109, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 5 & 4 \\ 3 & 5 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [16] $21^1 4 1^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7, T_{\max} = 19, \Delta = -109, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 5 & 4 \\ 4 & 5 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [17] $1^1 3 1^1 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7, T_{\max} = 13, \Delta = -97, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

[18] $1^1 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 11, \Delta = -90, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 7 & 5 \\ 3 & 7 & 5 & 5 \\ 3 & 5 & 5 & 3 \end{pmatrix}_{b_0}$$

[19] $31^1 231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 14, \Delta = -98, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 14, T \in [11, 14]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 3 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{11} 1321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 11, \Delta = -90, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{11} 3131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 13, \Delta = -97, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -3 & 1 & -1 \\ 1 & 3 & 1 & 1 & 3 \\ 2 & 3 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$

[22] $21^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 19, \Delta = -109, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 5 & 4 \\ 5 & 4 & 5 & 4 \\ 5 & 5 & 4 & 4 \\ 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

[23] $2231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 11, \Delta = -90, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[25] $1^1 141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 19, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 4 \\ 3 & 6 & 4 & 4 \\ 3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

[24] $2141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 19, \Delta = -109, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 4 \\ 3 & 6 & 4 & 4 \\ 3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

[28] $131^1 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 13, \Delta = -97, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 2 & 4 & 3 & 4 & 3 \\ 1 & 3 & 4 & 1 & 3 \\ 2 & 3 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

[27] $1^1 141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 19, \Delta = -109, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 5 & 5 & 4 \\ 5 & 4 & 6 & 4 \\ 5 & 6 & 4 & 4 \\ 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

[26] $2231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 11, \Delta = -90, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 2 & 1 & 4 & 1 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

[29] $221^1 41^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 7, T_{\max} = 19, \Delta = -109, \det = -24, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$

- $T_{\min} = 10.50 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 3 & 3 & 3 \\ 1 & 2 & 3 & 0 & 2 \\ 2 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- $[31] \quad 2 \overset{1}{3} \overset{11}{1} 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 14$, $\Delta = -98$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.29 \rightarrow 11$, $T_{\max} = 14$, $T \in [11, 14]$
 $\begin{pmatrix} 0 & 3 & 2 & 2 & 2 \\ 3 & 1 & 3 & 2 & 3 \\ 2 & 3 & 0 & 4 & 2 \\ 2 & 2 & 4 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- $[30] \quad 21 \overset{11}{1} 321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 11$, $\Delta = -90$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.14 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- $[32] \quad 21 \overset{11}{1} 41^1 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 19$, $\Delta = -109$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.50 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 4 & 8 & 8 & 4 \\ 8 & 13 & 15 & 7 \\ 8 & 15 & 13 & 7 \\ 4 & 7 & 7 & 3 \end{pmatrix}_{b_0}$
- $[33] \quad 31 \overset{11}{1} 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 14$, $\Delta = -98$, $\det = -35$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 14$, $T \in [12, 14]$
 $\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 3 & 2 & 0 & 4 & 2 \\ 3 & 1 & 4 & 1 & 3 \\ 2 & 2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$
- $[34] \quad 2 \overset{1}{3} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -40$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.30 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 2 & 3 \\ 2 & 4 & 0 & 3 & 2 \\ 2 & 2 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- $[35] \quad 2 \overset{11}{2} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 11$, $\Delta = -90$, $\det = -40$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 5 & 7 & 5 \\ 4 & 7 & 5 & 5 \\ 3 & 5 & 5 & 4 \end{pmatrix}_{b_0}$
- $[36] \quad 131 \overset{11}{1} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 13$, $\Delta = -97$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 13$, $T \in [11, 13]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 4 \\ 1 & 4 & 0 & 4 & 2 \\ 2 & 6 & 4 & 4 & 4 \\ 2 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$
- $[37] \quad 1 \overset{11}{4} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 19$, $\Delta = -109$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.82 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 4 & 6 & 4 \\ 4 & 6 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- $[38] \quad 2 \overset{1}{3} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 11$, $\Delta = -92$, $\det = -64$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 3 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$
- $[39] \quad 31 \overset{11}{1} 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 14$, $\Delta = -98$, $\det = -76$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.26 \rightarrow 11$, $T_{\max} = 14$, $T \in [11, 14]$
 $\begin{pmatrix} 1 & 5 & 5 & 3 \\ 5 & 5 & 7 & 5 \\ 5 & 7 & 5 & 5 \\ 3 & 5 & 5 & 3 \end{pmatrix}_{b_0}$
- $[40] \quad 221 \overset{11}{1} 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 19$, $\Delta = -109$, $\det = -88$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.50 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 6 & 6 & 3 \\ 6 & 13 & 15 & 7 \\ 6 & 15 & 13 & 7 \\ 3 & 7 & 7 & 3 \end{pmatrix}_{b_0}$
- $[41] \quad 13 \overset{11}{2} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7$, $T_{\max} = 11$, $\Delta = -90$, $\det = -96$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$

$$\begin{aligned}
& T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 4 & 2 \\ 2 & 3 & 4 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[42] & 31^{11}31^{11}3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7, T_{\max} = 19, \Delta = -105, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.48 \rightarrow 11, T_{\max} = 19, T \in [11, 19] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 6 & 4 \\ 3 & 4 & 0 & 4 & 2 \\ 3 & 6 & 4 & 4 & 4 \\ 2 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0} \\
[43] & 231^{11}3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 7, T_{\max} = 14, \Delta = -98, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.29 \rightarrow 11, T_{\max} = 14, T \in [11, 14] \\
& \begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 1 & 3 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 \\ 3 & 3 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0} \\
T = 8 & [44] 1^1 2^1 3^1 2^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[45] & 1^1 4^1 1^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 2 & 4 \\ 2 & 2 & 0 & 2 \\ 3 & 4 & 2 & 4 \end{pmatrix}_{b_0} \\
[46] & 21^1 4^1 1^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 2 & 4 \\ 2 & 2 & 0 & 2 \\ 3 & 4 & 2 & 4 \end{pmatrix}_{b_0} \\
[47] & 1^1 3^1 2^1 2^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 10, \Delta = -61, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 3 & 0 & 4 \end{pmatrix}_{b_0} \\
[48] & 1^1 4^1 1^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0} \\
[49] & 21^1 4^1 1^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 4 & 3 & 4 \\ 2 & 3 & 0 & 2 \\ 3 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[50] & 131^1 4^1 1^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 5 & 4 \\ 4 & 5 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0} \\
[51] & 1^1 4^1 1^1 2^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 0 & 3 & 2 \\ 3 & 3 & 4 & 4 \\ 3 & 2 & 4 & 3 \end{pmatrix}_{b_0} \\
[54] & 1^1 4^1 1^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -88, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

- [55] $31^1 3141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 14, \Delta = -96, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$
 $\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 3 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [56] $1^{11} 41231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 2 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$
- [57] $1^1 41231^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [58] $1^1 14131^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [52] $1^{11} 31^1 4^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 1 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$
- [53] $1^{11} 31^1 4^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 \\ 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [59] $1^1 23141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [60] $1^{11} 41^1 231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$
- [61] $31^1 31^1 41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 14, \Delta = -96, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$
 $\begin{pmatrix} 1 & 5 & 6 & 3 \\ 5 & 5 & 9 & 5 \\ 6 & 9 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$
- [62] $1^{11} 23141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 2 & 1 & 4 & 1 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [65] $2^1 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 4 & 2 \\ 3 & 2 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [66] $21^{11} 4131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 4 & 1 & 6 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 6 & 1 & 4 & 4 \\ 2 & 4 & 0 & 4 & 2 \end{pmatrix}_{b_0}$

- [63] $141^{11}32 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 2 & 3 & 2 \\ 2 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 4 & 2 \\ 3 & 3 & 4 & 4 & 4 \\ 2 & 3 & 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [64] $2^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 5 & 9 & 5 \\ 6 & 9 & 13 & 7 \\ 3 & 5 & 7 & 3 \end{pmatrix}_{b_0}$
- [67] $31^{31}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 14, \Delta = -96, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$
 $\begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 2 \\ 3 & 3 & 3 & 1 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [69] $31^131^{41}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 14, \Delta = -96, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$
 $\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 3 & 2 & 0 & 4 & 2 \\ 3 & 2 & 4 & 4 & 4 \\ 2 & 2 & 2 & 4 & 2 \end{pmatrix}_{b_0}$
- [68] $21^{41}1^{11}31 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 0 & 3 & 2 & 2 \\ 2 & 3 & 3 & 3 & 3 \\ 2 & 2 & 3 & 0 & 2 \\ 2 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [70] $23141^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 5 & 7 & 5 \\ 4 & 7 & 5 & 5 \\ 3 & 5 & 5 & 4 \end{pmatrix}_{b_0}$
- [73] $2141^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 2 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$
- [72] $1^{11}31^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 2 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 2 \end{pmatrix}_{b_0}$
- [71] $1^{131}1^{11}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -88, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 4 & 6 & 4 \\ 4 & 6 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [74] $41^{11}322 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 4 & 3 & 3 & 2 \\ 4 & 10 & 9 & 8 & 6 \\ 3 & 9 & 4 & 8 & 4 \\ 3 & 8 & 8 & 5 & 5 \\ 2 & 6 & 4 & 5 & 3 \end{pmatrix}_{b_0}$
- [75] $31^{11}3141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 14, \Delta = -96, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$
 $\begin{pmatrix} 1 & 5 & 5 & 3 \\ 5 & 5 & 7 & 5 \\ 5 & 7 & 5 & 5 \\ 3 & 5 & 5 & 3 \end{pmatrix}_{b_0}$
- [76] $141^{11}231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 8, T_{\max} = 12, \Delta = -89, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 4 & 1 & 6 & 4 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 6 & 1 & 4 & 4 \\ 2 & 4 & 0 & 4 & 2 \end{pmatrix}_{b_0}$

$$[77] \quad 1321^{1^1}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 6 & 3 \\ 6 & 13 & 15 & 7 \\ 6 & 15 & 13 & 7 \\ 3 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[78] \quad 3131^{1^1}41^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 14, \Delta = -96, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.50 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$$

$$\begin{pmatrix} 1 & 6 & 6 & 3 \\ 6 & 13 & 15 & 7 \\ 6 & 15 & 13 & 7 \\ 3 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[79] \quad 141^{1^1}31^13 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 14, \Delta = -96, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.48 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 6 & 4 \\ 3 & 4 & 0 & 4 & 2 \\ 3 & 6 & 4 & 4 & 4 \\ 2 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[80] \quad 31^{1^1}413 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 14, \Delta = -96, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.29 \rightarrow 12, T_{\max} = 14, T \in [12, 14]$$

$$\begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 4 & 4 & 6 & 4 \\ 2 & 4 & 0 & 4 & 2 \\ 3 & 6 & 4 & 4 & 4 \\ 2 & 4 & 2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[81] \quad 41^{1^1}2231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 1 & -1 \\ 4 & 28 & 1 & 26 & 10 \\ 2 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[82] \quad 41^{1^1}232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 8, T_{\max} = 12, \Delta = -91, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 4 & 2 & 4 & 2 \\ 4 & 10 & 5 & 12 & 6 \\ 2 & 5 & 0 & 5 & 2 \\ 4 & 12 & 5 & 10 & 6 \\ 2 & 6 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [83] \quad 141^{1^1}41^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 1 & 3 \\ 1 & 3 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[84] \quad 14^{1^1}41^{1^1}2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 5 & 5 & 4 \\ 5 & 5 & 7 & 5 \\ 5 & 7 & 5 & 5 \\ 4 & 5 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{1^1}41321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 11, \Delta = -60, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 1 & 3 \\ 3 & 3 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[86] \quad 21^{1^1}4141^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 5 & 5 & 4 \\ 5 & 5 & 6 & 5 \\ 5 & 6 & 5 & 5 \\ 4 & 5 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{1^1}51231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{1^1}5131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[88] $21^{1^1}5^{1^1}131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[87] $14141^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 5 & 6 & 5 \\ 3 & 6 & 5 & 5 \\ 3 & 5 & 5 & 4 \end{pmatrix}_{b_0}$$

[92] $1^1 5^{1^1} 131^1 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 3 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[91] $231^{1^1}51^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[93] $31^{1^1}3151^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 3 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[96] $314141^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 0 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

[97] $1^1 5^{1^1} 132 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[98] $232151^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[99] $1^1 41^{1^1} 3141^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 13 & 1 & 5 & 7 \\ 1 & 1 & -3 & 1 & -1 \\ 1 & 5 & 1 & 1 & 3 \\ 2 & 7 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[94] $21^{1^1}51^{1^1}32 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 3 & 6 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[95] $1^1 5^{1^1} 1^1 321^{1^1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 3 & 3 & 5 & 10 & 6 \\ 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0} \\
[100] & 223151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 4 & 3 & 4 \\ 1 & 2 & 3 & 0 & 2 \\ 2 & 2 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[101] & 1^1 3221^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 18 & 6 & 0 & 8 \\ 1 & 6 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 8 & 2 & 0 & 3 \end{pmatrix}_{b_0} \\
[102] & 31^1 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 3 & 2 \\ 2 & 5 & 1 & 9 & 5 \\ 1 & 1 & -3 & 1 & -1 \\ 3 & 9 & 1 & 13 & 7 \\ 2 & 5 & -1 & 7 & 3 \end{pmatrix}_{b_0} \\
[103] & 2321^1 51^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 3 & 5 & 10 & 6 \\ 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0} \\
[104] & 2^1 3^{141} 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 4 & 4 & 3 \\ 4 & 6 & 8 & 6 \\ 4 & 8 & 6 & 6 \\ 3 & 6 & 6 & 5 \end{pmatrix}_{b_0} \\
[105] & 31^1 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 5 & 1 & 7 & 5 \\ 1 & 1 & -3 & 1 & -1 \\ 2 & 7 & 1 & 5 & 5 \\ 2 & 5 & -1 & 5 & 3 \end{pmatrix}_{b_0} \\
[106] & 223151^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -49, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 4 & 8 & 4 \\ 3 & 2 & 8 & 5 & 5 \\ 2 & 2 & 4 & 5 & 3 \end{pmatrix}_{b_0} \\
[107] & 221^{11} 5131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 4 & 3 & 6 & 4 \\ 1 & 3 & 0 & 3 & 2 \\ 3 & 6 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0} \\
[108] & 2^1 3^{11} 2^{141} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 11, \Delta = -62, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 & 4 \\ 2 & 4 & 1 & 4 & 3 \\ 2 & 4 & 4 & 2 & 4 \\ 2 & 4 & 3 & 4 & 4 \end{pmatrix}_{b_0} \\
[109] & 151^{11} 2231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 18 & 6 & 20 & 8 \\ 1 & 6 & 0 & 6 & 2 \\ 4 & 20 & 6 & 18 & 8 \\ 2 & 8 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[110] & 41^{11} 23141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 1 & -1 \\ 4 & 28 & 1 & 26 & 10 \\ 2 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[111] \ 321^{\overline{1}} 4141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 8 & 8 & 3 \\ 8 & 17 & 19 & 9 \\ 8 & 19 & 17 & 9 \\ 3 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [112] \ 1^{\overline{1}} 413221^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 11, \Delta = -31, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 5 \end{pmatrix}_{b_0}$$

$$[113] \ 1^{\overline{1}} 315131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 12, \Delta = -62, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[114] \ 1^{\overline{1}} 414131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & 1 & -2 & 1 & 0 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[121] \ 1^{\overline{1}} 6131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[115] \ 21^{\overline{1}} 61231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[116] \ 221^{\overline{1}} 6131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 3 & 1 & 10 & 6 \\ 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[117] \ 1^{\overline{1}} 413151^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 2 & 3 & 4 \\ 3 & 1 & 3 \\ 4 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[118] \ 1^{\overline{1}} 61231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[119] \ 1^{\overline{1}} 612231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[120] \ 21^{\overline{1}} 6131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

- [122] $21 \overset{1}{\underset{1}{6}} 131 \overset{1}{\underset{1}{3}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 3 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [123] $232161 \overset{1}{\underset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 10, T_{\max} = 13, \Delta = -83, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [124] $1 \overset{1}{\underset{1}{6}} 131 \overset{1}{\underset{1}{3}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 4 & 2 & 2 \\ 3 & 4 & 12 & 4 & 6 \\ 1 & 2 & 4 & 0 & 2 \\ 2 & 2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$
- [125] $31 \overset{1}{\underset{1}{3}} 161 \overset{1}{\underset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 15, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.33 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 4 & 1 & 2 \\ 3 & 4 & 12 & 4 & 6 \\ 1 & 1 & 4 & 0 & 2 \\ 2 & 2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$
- [126] $21 \overset{1}{\underset{1}{6}} 1 \overset{1}{\underset{1}{3}} 21 \overset{1}{\underset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$
- [127] $1 \overset{1}{\underset{1}{6}} 1 \overset{1}{\underset{1}{3}} 21 \overset{1}{\underset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 5 & 2 \\ 3 & 3 & 5 & 10 & 6 \\ 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$
- [128] $21 \overset{1}{\underset{1}{6}} 1232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 10, T_{\max} = 13, \Delta = -83, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 3 & 6 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0}$
- [129] $1 \overset{1}{\underset{1}{6}} 1232 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 10, T_{\max} = 13, \Delta = -83, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [131] $1 \overset{1}{\underset{1}{4}} 1321 \overset{1}{\underset{1}{5}} 1 \overset{1}{\underset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 10, T_{\max} = 13, \Delta = -83, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 0 & 2 & 2 \\ 4 & 2 & 2 & 4 \\ 3 & 2 & 4 & 4 \end{pmatrix}_{b_0}$
- [130] $1 \overset{1}{\underset{1}{6}} 1 \overset{1}{\underset{1}{3}} 221 \overset{1}{\underset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 3 & 1 & 10 & 6 \\ 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$
- [132] $1 \overset{1}{\underset{1}{6}} 1 \overset{1}{\underset{1}{3}} 21 \overset{1}{\underset{1}{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 10, T_{\max} = 13, \Delta = -83, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 2 & 2 & 0 & 5 & 2 \\ 4 & 3 & 5 & 10 & 6 \\ 2 & 2 & 2 & 6 & 3 \end{pmatrix}_{b_0}$

$$[133] \quad 41^1 31^1 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 41, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 5 & 1 & 4 & 2 \\ 5 & 38 & 1 & 32 & 12 \\ 1 & 1 & -3 & 1 & -1 \\ 4 & 32 & 1 & 26 & 10 \\ 2 & 12 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[134] \quad 221^{11} 61231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 10 & 1 & 12 & 6 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 12 & 1 & 10 & 6 \\ 2 & 6 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[135] \quad 1^{221^{11}} 6 \quad 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 10 & 1 & 12 & 6 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 12 & 1 & 10 & 6 \\ 2 & 6 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[136] \quad 41^1 4141^{11} 3 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 3 & 3 & 4 & 1 & 3 \\ 2 & 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[137] \quad 2221^{11} 6131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 18 & 1 & 20 & 8 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 20 & 1 & 18 & 8 \\ 2 & 8 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[138] \quad 41^{11} 314141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 26 & 1 & 28 & 10 \\ 1 & 1 & -3 & 1 & -1 \\ 4 & 28 & 1 & 26 & 10 \\ 2 & 10 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[139] \quad 2161^{11} 2231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 12 & 1 & 10 & 6 \\ 2 & 6 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[140] \quad 1^{13221^{11}} 6 \quad 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 12 & 1 & 10 & 6 \\ 2 & 6 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[141] \quad 51^{11} 223141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 164, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 7 & 7 & 3 \\ 7 & 7 & 9 & 7 \\ 7 & 9 & 7 & 7 \\ 3 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[142] \quad 4131^{11} 4141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 10, T_{\max} = 13, \Delta = -86, \det = 184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 5 & 1 & 5 & 2 \\ 5 & 38 & 1 & 40 & 12 \\ 1 & 1 & -3 & 1 & -1 \\ 5 & 40 & 1 & 38 & 12 \\ 2 & 12 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [143] \quad 1^{11} 31^1 51321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 12, \Delta = -34, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 5 \end{pmatrix}_{b_0}$$

$$[147] \quad 1^{11} 6 \quad 131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 13, \Delta = -59, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

[146] $1^{\overline{1}} \overline{6}^{1321} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 13, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 3 \\ 2 & 2 & 1 & 3 \\ 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

[145] $1^{\overline{1}} \overline{6}^{1321} 131^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 13, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

[144] $1^{\overline{1}} \overline{4}^{141321} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 2 & 4 \\ 3 & 4 & 4 & 6 \end{pmatrix}_{b_0}$$

[150] $21^{\overline{1}} \overline{6}^{13141} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -3 & 1 & -1 \\ 2 & 4 & 1 & 8 & 6 \\ 2 & 3 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

[151] $221^{\overline{1}} \overline{7}^{1231} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 3 & 3 & 1 & 10 & 6 \\ 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

[152] $21^{\overline{1}} \overline{7}^{12231} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 2 & 1 & 4 & 4 \\ 2 & 2 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

[149] $1^{\overline{1}} \overline{7}^{12231} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[153] $1^{\overline{1}} \overline{6}^{123141} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 14, \Delta = -80, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 1 & -1 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

[154] $21^{\overline{1}} \overline{7}^{1232} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 14, \Delta = -81, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 3 & 2 \\ 2 & 2 & 3 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[155] $1^{\overline{21}} \overline{7}^{1231} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[148] $21^{\overline{1}} \overline{7}^{1231} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[156] & 2321\overset{1}{7}1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.67 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 3 & 1 & 2 \\ 2 & 3 & 4 & 3 & 4 \\ 1 & 1 & 3 & 0 & 2 \\ 2 & 2 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[157] & 1\overset{1}{4}141\overset{1}{3}151\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.84 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 0 & 3 & 2 \\ 5 & 3 & 8 & 6 \\ 3 & 2 & 6 & 4 \end{pmatrix}_{b_0} \\
[159] & 1\overset{1}{6}1\overset{1}{2}3141\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 14, \Delta = -80, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 26 & 1 & 7 & 10 \\ 1 & 1 & -3 & 1 & -1 \\ 1 & 7 & 1 & 1 & 3 \\ 2 & 10 & -1 & 3 & 3 \end{pmatrix}_{b_0} \\
[160] & 21\overset{1}{7}3221\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 3 & 1 & 10 & 6 \\ 2 & 2 & 0 & 6 & 3 \end{pmatrix}_{b_0} \\
[158] & 1\overset{1}{7}3221\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -20, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 6 & 2 \\ 4 & 4 & 6 & 18 & 8 \\ 2 & 2 & 2 & 8 & 3 \end{pmatrix}_{b_0} \\
[161] & 1\overset{1}{4}13221\overset{1}{6}1\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 14, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.67 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 5 & 1 & 1 & 2 \\ 5 & 28 & 7 & 0 & 10 \\ 1 & 7 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 10 & 2 & 0 & 3 \end{pmatrix}_{b_0} \\
[162] & 221\overset{1}{6}13141\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -52, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.92 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 4 & 3 & 6 & 4 \\ 1 & 3 & 0 & 3 & 2 \\ 3 & 6 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & 3 \end{pmatrix}_{b_0} \\
[163] & 2\overset{1}{3}1\overset{1}{4}141\overset{1}{2} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 & 5 \\ 2 & 5 & 2 & 5 & 4 \\ 2 & 5 & 5 & 3 & 5 \\ 2 & 5 & 4 & 5 & 5 \end{pmatrix}_{b_0} \\
[164] & 161\overset{1}{2}223141\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 14, \Delta = -80, \det = -164, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.78 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 5 & 1 & 5 & 2 \\ 5 & 28 & 7 & 30 & 10 \\ 1 & 7 & 0 & 7 & 2 \\ 5 & 30 & 7 & 28 & 10 \\ 2 & 10 & 2 & 10 & 3 \end{pmatrix}_{b_0} \\
[165] & 15131\overset{1}{4}141\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 11, T_{\max} = 14, \Delta = -81, \det = -184, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.89 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 8 & 8 & 3 \\ 8 & 17 & 19 & 9 \\ 8 & 19 & 17 & 9 \\ 3 & 9 & 9 & 4 \end{pmatrix}_{b_0} \\
T = 12 & [168] 1\overset{1}{6}2321\overset{1}{1}31\overset{1}{1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, T_{\max} = 13, \Delta = -33, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 2 & 4 \\ 3 & 3 & 4 & 6 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[171] \ 1^{\overline{1}131} \ 6 \ 13141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, \ T_{\max} = 14, \ \Delta = -58, \ \det = 3, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -3 & 1 & -1 \\ 1 & 2 & 1 & 2 & 4 \\ 2 & 3 & -1 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[170] \ 1^{\overline{1}131} \ 6 \ 13141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, \ T_{\max} = 14, \ \Delta = -58, \ \det = 3, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 0 & 3 \\ 1 & 3 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[169] \ 1^{\overline{1}1321} \ 7 \ 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, \ T_{\max} = 14, \ \Delta = -59, \ \det = 3, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 1 & 3 \\ 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{\overline{1}1} \ 8 \ 1232^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, \ T_{\max} = 15, \ \Delta = -80, \ \det = 7, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.57 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 3 & 1 & 2 \\ 2 & 3 & 4 & 3 & 4 \\ 2 & 1 & 3 & 0 & 2 \\ 2 & 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[173] \ 2321^{\overline{1}1} \ 8 \ 12 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, \ T_{\max} = 15, \ \Delta = -80, \ \det = 12, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.67 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 3 & 1 & 2 \\ 2 & 3 & 4 & 3 & 4 \\ 1 & 1 & 3 & 0 & 2 \\ 2 & 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[174] \ 1^{\overline{1}1} \ 4141^{\overline{1}1} \ 32161^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, \ T_{\max} = 15, \ \Delta = -79, \ \det = 19, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.84 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 5 & 1 & 1 & 2 \\ 5 & 33 & 8 & 0 & 11 \\ 1 & 8 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 11 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[175] \ 1^{\overline{1}1} \ 4141321^{\overline{1}1} \ 61^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, \ T_{\max} = 15, \ \Delta = -79, \ \det = 24, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.67 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 5 & 1 & 1 & 2 \\ 5 & 28 & 7 & 0 & 10 \\ 1 & 7 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 10 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[176] \ 61^{\overline{1}1} \ 22314141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, \ T_{\max} = 15, \ \Delta = -79, \ \det = 220, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.65 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 6 & 1 & 6 & 2 \\ 6 & 64 & 1 & 66 & 16 \\ 1 & 1 & -3 & 1 & -1 \\ 6 & 66 & 1 & 64 & 16 \\ 2 & 16 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[177] \ 5131^{\overline{1}1} \ 414141^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 12, \ T_{\max} = 15, \ \Delta = -80, \ \det = 312, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 14.63 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 1 & 10 & 10 & 3 \\ 10 & 21 & 23 & 11 \\ 10 & 23 & 21 & 11 \\ 3 & 11 & 11 & 4 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [179] \ 1^{\overline{1}1} \ 7 \ 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 13, \ T_{\max} = 14, \ \Delta = -32, \ \det = -2, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 2 & 4 \\ 3 & 3 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[183] \ 1^{\overline{1}1} \ 8 \ 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 13, \ T_{\max} = 15, \ \Delta = -58, \ \det = -3, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[182] \ 1^{\overline{1}1} \ 8 \ 1231^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 13, \ T_{\max} = 15, \ \Delta = -58, \ \det = -3, \ (n_+, n_-, n_0) = (1, 13, 0), \ c_{\text{ext}} = 4.0$$

$$\begin{aligned}
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 3 & 4 \end{pmatrix}_{b_0} \\
[181] & 1^{1^1} 2^1 3^1 2^1 8^1 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 13, T_{\max} = 15, \Delta = -60, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 1 & 2 & 1 & 3 \\ 2 & 2 & 2 & 3 & 4 \end{pmatrix}_{b_0} \\
[180] & 1^{1^1} 8^1 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 1 & 3 \\ 3 & 2 & 3 & 5 \end{pmatrix}_{b_0} \\
[184] & 2^{1^1} 3^{1^1} 4^{1^1} 14141 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 13, T_{\max} = 13, \Delta = -2, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 6 & 6 & 6 \\ 2 & 6 & 3 & 6 & 5 \\ 2 & 6 & 6 & 4 & 6 \\ 2 & 6 & 5 & 6 & 6 \end{pmatrix}_{b_0} \\
T = 14 & [186] 1^{1^1} 6^{1^1} 315131 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 14, T_{\max} = 15, \Delta = -32, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 2 & 4 \\ 3 & 3 & 4 & 6 \end{pmatrix}_{b_0} \\
[185] & 1^{1^1} 8^1 12321^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 14, T_{\max} = 15, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 2 & 4 \\ 3 & 2 & 4 & 6 \end{pmatrix}_{b_0} \\
[187] & 1^{1^1} 8^1 1231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 14, T_{\max} = 15, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 2 & 3 \\ 1 & 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 3 & 5 \end{pmatrix}_{b_0} \\
[188] & 31^{1^1} 6^{1^1} 13161^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 14, T_{\max} = 16, \Delta = -58, \det = 10, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15.80 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 2 & 3 & 5 & 1 & 5 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
T = 15 & [189] 1^{1^1} 6^{1^1} 22315131 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 15, T_{\max} = 15, \Delta = -4, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 4.0 \\
[193] & 1^{1^1} 8^1 123141^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 15, T_{\max} = 16, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 0 & 3 \\ 1 & 3 & 2 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 4 & 0 & 5 \end{pmatrix}_{b_0} \\
[192] & 1^{1^1} 6^{1^1} 13161231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 15, T_{\max} = 16, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 2 & 1 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0} \\
[191] & 1^{1^1} 6^{1^1} 1316131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] \quad T = 15, T_{\max} = 16, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 2 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 2 & 2 & 4 \\ 2 & 3 & 2 & 4 & 5 \end{pmatrix}_{b_0}$$

$$T = 16 \quad [195] \quad 1^{\mathbf{1}^{13151321}} 8^{1231^1} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 16, T_{\max} = 17, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 2 & 4 \\ 3 & 3 & 4 & 6 \end{pmatrix}_{b_0}$$

$$T = 17 \quad [196] \quad 1^{\mathbf{1}^{1223151321}} 8^{1231^1} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 17, T_{\max} = 17, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 4.0$$

$$[198] \quad 1^{\mathbf{1}^{1321}} 8^{123216131^1} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 17, T_{\max} = 18, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 2 & 1 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$T = 18 \quad [199] \quad 1^{\mathbf{1}^{131}} 6^{131613151321^1} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 18, T_{\max} = 18, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 18, 1), c_{\text{ext}} = 4.0$$

$$[201] \quad 1^{\mathbf{1}^{1321}} 8^{1232171231^1} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 18, T_{\max} = 19, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 2 & 1 & 2 & 4 \\ 2 & 3 & 0 & 4 & 5 \end{pmatrix}_{b_0}$$

$$T = 19 \quad [202] \quad 1^{\mathbf{1}^{131}} 6^{1316131612321^1} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 19, T_{\max} = 19, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0$$

$$T = 21 \quad [206] \quad 1^{\mathbf{1}^{131}} 6^{131613161315131^1} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 21, T_{\max} = 21, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 21, 1), c_{\text{ext}} = 4.0$$

$$T = 22 \quad [207] \quad 1^{\mathbf{1}^{1321}} 8^{123218123151321^1} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 22, T_{\max} = 22, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 4.0$$

$$T = 24 \quad [210] \quad 1^{\mathbf{1}^{1321}} 8^{12321812321712321^1} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 24, T_{\max} = 24, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 24, 1), c_{\text{ext}} = 4.0$$

$$T = 25 \quad [211] \quad 1^{\mathbf{1}^{12321}} 8^{12321812321^1} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 25, T_{\max} = 25, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 25, 1), c_{\text{ext}} = 4.0$$

$$T = 27 \quad [212] \quad 1^{\mathbf{1}^{1321}} 8^{12321812321812315131^1} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] \quad T = 27, T_{\max} = 27, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 27, 1), c_{\text{ext}} = 4.0$$

5.20 NHC(-2)(-3)(-2) + e₆

$$T = 7 \quad [1] \quad 21^{\mathbf{1}^{131111}} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] + \mathbf{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -165, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 9 & 11 & 4 & 5 \\ 4 & 11 & 9 & 4 & 5 \\ 3 & 4 & 4 & -3 & -1 \\ 3 & 5 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{\mathbf{1}^{111}} 31^{\mathbf{1}^{11}} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] + \mathbf{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -165, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 9 & 11 & 4 & 5 \\ 4 & 11 & 9 & 4 & 5 \\ 3 & 4 & 4 & -3 & -1 \\ 3 & 5 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -165, \ \det = -47, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.91 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 9 & 7 & 0 & 5 \\ 3 & 7 & 4 & 0 & 4 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 5 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -165, \ \det = -76, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.47 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -165, \ \det = -104, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.42 \rightarrow 14, \ T_{\max} = 100, \ T \in [14, 100]$$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 3 \\ 4 & 9 & 11 & 4 & 5 \\ 4 & 11 & 9 & 4 & 5 \\ 2 & 4 & 4 & -3 & -1 \\ 3 & 5 & 5 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 21^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 \quad T = 7, \ T_{\max} = 100, \ \Delta = -165, \ \det = -440, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.46 \rightarrow 13, \ T_{\max} = 100, \ T \in [13, 100]$$

$$\begin{pmatrix} 4 & 8 & 8 & 3 & 4 \\ 8 & 9 & 11 & 4 & 5 \\ 8 & 11 & 9 & 4 & 5 \\ 3 & 4 & 4 & -3 & -1 \\ 4 & 5 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [6] \ 1^{11}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -137, \ \det = 7, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.14 \rightarrow 16, \ T_{\max} = 98, \ T \in [16, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & -3 & -1 \\ 2 & 2 & -1 & 3 & -1 & -3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -144, \ \det = 80, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.05 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 4 & 0 & 3 \\ 1 & 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 3 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -144, \ \det = 200, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.92 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 0 & 4 & 0 & 2 \\ 2 & 6 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 2 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [9] \ 1^{11}2321^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 \quad T = 9, \ T_{\max} = 21, \ \Delta = -110, \ \det = -3, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 19, \ T_{\max} = 21, \ T \in [19, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 3 & 1 & 0 & -3 & -1 \\ 2 & 3 & 2 & 0 & -1 & -5 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11}4^11^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -46, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.74 \rightarrow 14, \ T_{\max} = 95, \ T \in [14, 95]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 4 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 21^{11}41^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 \quad T = 9, \ T_{\max} = 95, \ \Delta = -135, \ \det = -136, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.62 \rightarrow 14, \ T_{\max} = 95, \ T \in [14, 95]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 \\ 2 & 0 & 3 & 2 & 0 & 2 \\ 2 & 3 & 3 & 3 & 0 & 3 \\ 2 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 3 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1}31^{\overline{1}1}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 \quad T = 9, T_{\max} = 95, \Delta = -135, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 95, T \in [14, 95]$$

$$\begin{pmatrix} 1 & 4 & 4 & 1 & 3 \\ 4 & 4 & 6 & 0 & 4 \\ 4 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

5.21 NHC(-2)(-3)(-2) + e₆ + e₆

$$T = 8 \quad [0] \ 1^{\overline{1}11^{\overline{1}1}} 31^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 103, \Delta = -214, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.14 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 1 & 3 \\ 4 & 9 & 11 & 4 & 0 & 5 \\ 4 & 11 & 9 & 4 & 0 & 5 \\ 2 & 4 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 5 & -1 & -4 & -3 \end{pmatrix}_{b_0}$$

5.22 NHC(-2)(-3)(-2) + e₆ + f₄

$$T = 8 \quad [0] \ 1^{\overline{1}11^{\overline{1}1}} 31^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{f}_4 \quad T = 8, T_{\max} = 102, \Delta = -188, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.31 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 1 & 3 \\ 4 & 9 & 11 & 4 & 0 & 5 \\ 4 & 11 & 9 & 4 & 0 & 5 \\ 2 & 4 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 5 & -1 & -3 & -2 \end{pmatrix}_{b_0}$$

5.23 NHC(-2)(-3)(-2) + e₆ + NHC(-2)(-2)(-3)

$$T = 10 \quad [0] \ 1^{\overline{1}111^{\overline{1}1}} 31^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{red}{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 436, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 5 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 5 & 0 & 4 & 4 & 4 & 4 \\ 2 & 4 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

5.24 NHC(-2)(-3)(-2) + e₆ + NHC(-2)(-3)

$$T = 9 \quad [0] \ 1^{\overline{1}11^{\overline{1}1}} 31^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 & 2 & 3 \\ 4 & 9 & 7 & 0 & 8 & 4 & 5 \\ 3 & 7 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 8 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 4 & 0 & 4 & 0 & 2 \\ 3 & 5 & 4 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1}11^{\overline{1}1}} 31^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1}11^{\overline{1}1}} 31^{\overline{1}1} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{red}{g}_2] \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 4 & 3 \\ 2 & 4 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

5.25 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 212, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 592, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 4 & 9 & 4 \\ 2 & 4 & 4 & 0 & 4 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

5.26 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 5 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 4 & 3 & 3 \\ 2 & 4 & 4 & 0 & 4 & 0 & 3 & 2 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 4 & 4 & -4 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

5.27 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -164, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 5 \\ 12 & 25 & 23 & 0 & 9 \\ 11 & 23 & 20 & 0 & 8 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 9 & 8 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -164, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 1 & 3 \\ 4 & 9 & 11 & 4 & 0 & 5 \\ 4 & 11 & 9 & 4 & 0 & 5 \\ 2 & 4 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 5 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 101, \Delta = -164, \det = 300, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.65 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 5 \\ 11 & 20 & 22 & 0 & 8 \\ 11 & 22 & 20 & 0 & 8 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 8 & 8 & -4 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 4 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 13 & 0 & 6 \\ 4 & 12 & 13 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \ 1^{\overline{1}} 3^{\overline{1111}} 4^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 95, \Delta = -134, \det = 544, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.49 \rightarrow 15, T_{\max} = 95, T \in [15, 95]$$

$$\begin{pmatrix} 5 & 10 & 10 & 1 & 5 \\ 10 & 13 & 15 & 0 & 7 \\ 10 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 7 & 7 & -3 & 1 \end{pmatrix}_{b_0}$$

5.28 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 9 & 5 \\ 12 & 25 & 23 & 0 & 20 & 9 \\ 11 & 23 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 20 & 17 & 0 & 13 & 7 \\ 5 & 9 & 8 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 17 & 0 & 13 & 7 \\ 5 & 8 & 8 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -278, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 17 & 5 \\ 12 & 25 & 23 & 0 & 36 & 9 \\ 11 & 23 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 36 & 33 & 0 & 49 & 13 \\ 5 & 9 & 8 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{112}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -442, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 26 & 5 \\ 12 & 25 & 23 & 0 & 56 & 9 \\ 11 & 23 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 56 & 50 & 0 & 118 & 20 \\ 5 & 9 & 8 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

5.29 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 17 & 0 & 12 & 6 \\ 5 & 8 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

5.30 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^{\overline{116}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 21, T_{\max} = 94, T \in [21, 94]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 24 & 24 & 0 & 106 & 18 \\ 3 & 4 & 4 & -4 & 18 & -8 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 19.56 \rightarrow 20, T_{\max} = 94, T \in [20, 94]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 7 & 3 \\ 4 & 9 & 7 & 0 & 16 & 5 \\ 3 & 7 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 16 & 13 & 0 & 28 & 10 \\ 3 & 5 & 4 & -4 & 10 & -7 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 9, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 16.89 \rightarrow 17, T_{\max} = 94, T \in [17, 94]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 3 \\ 4 & 9 & 7 & 0 & 4 & 5 \\ 3 & 7 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 3 \\ 3 & 5 & 4 & -4 & 3 & -4 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 20.83 \rightarrow 21, T_{\max} = 94, T \in [21, 94]$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 1 & 3 \\ 4 & 9 & 11 & 4 & 0 & 5 \\ 4 & 11 & 9 & 4 & 0 & 5 \\ 2 & 4 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 5 & 5 & -1 & 0 & -8 \end{pmatrix}_{b_0}$$

[4] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 42, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.62 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 6 & 3 \\ 4 & 9 & 7 & 0 & 16 & 5 \\ 3 & 7 & 4 & 0 & 10 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 16 & 10 & 0 & 22 & 8 \\ 3 & 5 & 4 & -4 & 8 & -2 \end{pmatrix}_{b_0}$$

[5] $1^{115}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.1$
 $T_{\min} = 13.38 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 20 & 4 \\ 3 & 6 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 20 & 20 & 0 & 73 & 15 \\ 3 & 4 & 4 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.47 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 4 \\ 3 & 4 & 4 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 & 3 \\ 4 & 9 & 7 & 0 & 8 & 5 \\ 3 & 7 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 8 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{112}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 14.02 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 12 & 3 \\ 4 & 9 & 7 & 0 & 28 & 5 \\ 3 & 7 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 28 & 22 & 0 & 85 & 17 \\ 3 & 5 & 4 & -4 & 17 & -2 \end{pmatrix}_{b_0}$$

[9] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.82 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 16 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 16 & 0 & 46 & 12 \\ 3 & 4 & 4 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.62 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.94 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 5 & 3 \\ 4 & 9 & 7 & 0 & 12 & 5 \\ 3 & 7 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 9 & 0 & 13 & 7 \\ 3 & 5 & 4 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{112}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 136, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.53 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 17 & 0 & 49 & 13 \\ 3 & 4 & 4 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[23] $1^{114}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 620, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.52 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 21 & 0 & 76 & 16 \\ 3 & 4 & 4 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{113}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 6 & 0 & 32 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 32 & 0 & 193 & 25 \\ 3 & 4 & 4 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[24] $1^{113}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 724, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 22 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 22 & 0 & 85 & 17 \\ 3 & 4 & 4 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[26] $1^{113}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[27] $1^{114}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 1000, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 26 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 26 & 0 & 118 & 20 \\ 3 & 4 & 4 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{114}31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 1228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 6 & 0 & 31 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 31 & 0 & 172 & 24 \\ 3 & 4 & 4 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[29] $1^{114}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 1304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [30] $1^{111}31^131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 4 & 0 & 3 \\ 1 & 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 \\ 1 & 3 & 3 & 3 & 0 & 1 \\ 2 & 3 & 3 & 3 & -4 & 3 \end{pmatrix}_{b_0}$$

[31] $1^{11}31^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 13.24 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 \\ 1 & 4 & 0 & 4 & 0 & 3 \\ 2 & 6 & 4 & 4 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 \\ 1 & 3 & 3 & 3 & 0 & 1 \\ 2 & 4 & 2 & 4 & -4 & 3 \end{pmatrix}_{b_0}$$

[32] $1^{14}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

11.8

$$T_{\min} = 14.18 \rightarrow 15, T_{\max} = 25, T \in [15, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 6 & 0 & 30 & 4 \\ 1 & 4 & 0 & 4 & 0 & 21 & 2 \\ 2 & 6 & 4 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 21 & 30 & 0 & 172 & 24 \\ 2 & 4 & 2 & 4 & -4 & 24 & -2 \end{pmatrix}_{b_0}$$

$$[33] \ 1^1 31^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 4 \\ 1 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 6 & 4 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 4 \\ 2 & 4 & 2 & 4 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[34] \ 1^1 31^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 4 & 0 & 6 & 2 \\ 2 & 6 & 4 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 9 & 0 & 13 & 7 \\ 2 & 4 & 2 & 4 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[35] \ 1^1 31^{12} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 12 & 4 \\ 1 & 4 & 0 & 4 & 0 & 6 & 2 \\ 2 & 6 & 4 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 6 & 12 & 0 & 22 & 8 \\ 2 & 4 & 2 & 4 & -4 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[36] \ 1^1 31^{13} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -724, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 6 & 0 & 18 & 4 \\ 1 & 4 & 0 & 4 & 0 & 9 & 2 \\ 2 & 6 & 4 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 9 & 18 & 0 & 52 & 12 \\ 2 & 4 & 2 & 4 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{12} 31^{12} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -784, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 6 & 0 & 18 & 4 \\ 1 & 4 & 0 & 4 & 0 & 12 & 2 \\ 2 & 6 & 4 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 12 & 18 & 0 & 58 & 14 \\ 2 & 4 & 2 & 4 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{11} 31^{12} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -964, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 6 & 0 & 15 & 4 \\ 1 & 4 & 0 & 4 & 0 & 9 & 2 \\ 2 & 6 & 4 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 9 & 15 & 0 & 37 & 11 \\ 2 & 4 & 2 & 4 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[39] \ 1^1 31^{14} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -976, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 6 & 0 & 24 & 4 \\ 1 & 4 & 0 & 4 & 0 & 12 & 2 \\ 2 & 6 & 4 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 12 & 24 & 0 & 94 & 16 \\ 2 & 4 & 2 & 4 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{13} 31^{13} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 13.03 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 6 & 0 & 27 & 4 \\ 1 & 4 & 0 & 4 & 0 & 18 & 2 \\ 2 & 6 & 4 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 18 & 27 & 0 & 133 & 21 \\ 2 & 4 & 2 & 4 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{11}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 6 & 0 & 21 & 4 \\ 1 & 4 & 0 & 4 & 0 & 12 & 2 \\ 2 & 6 & 4 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 12 & 21 & 0 & 73 & 15 \\ 2 & 4 & 2 & 4 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{12}31^{113}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1684, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 6 & 0 & 24 & 4 \\ 1 & 4 & 0 & 4 & 0 & 15 & 2 \\ 2 & 6 & 4 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 15 & 24 & 0 & 100 & 18 \\ 2 & 4 & 2 & 4 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{14}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1936, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 6 & 0 & 36 & 4 \\ 1 & 4 & 0 & 4 & 0 & 24 & 2 \\ 2 & 6 & 4 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 36 & 24 & 36 & 0 & 238 & 28 \\ 2 & 4 & 2 & 4 & -4 & 28 & -1 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{11}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -2116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 6 & 0 & 27 & 4 \\ 1 & 4 & 0 & 4 & 0 & 15 & 2 \\ 2 & 6 & 4 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 15 & 27 & 0 & 121 & 19 \\ 2 & 4 & 2 & 4 & -4 & 19 & 0 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{13}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -2596, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 6 & 0 & 33 & 4 \\ 1 & 4 & 0 & 4 & 0 & 21 & 2 \\ 2 & 6 & 4 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 33 & 21 & 33 & 0 & 193 & 25 \\ 2 & 4 & 2 & 4 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{12}31^{114}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -2656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 6 & 0 & 30 & 4 \\ 1 & 4 & 0 & 4 & 0 & 18 & 2 \\ 2 & 6 & 4 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 18 & 30 & 0 & 154 & 22 \\ 2 & 4 & 2 & 4 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [47] \ 1^{12}4 \ 1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 16.15 \rightarrow 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 7 & 3 \\ 3 & 4 & 5 & 0 & 11 & 4 \\ 4 & 5 & 4 & 0 & 10 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 10 & 0 & 21 & 9 \\ 3 & 4 & 4 & -3 & 9 & -2 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{11}4 \ 1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 29, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.41 \rightarrow 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 3 & 4 \\ 4 & 5 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 3 & 4 & 0 & 1 & 3 \\ 3 & 4 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{11}4 \ 1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 124, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

5.33 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{1132} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.3$$

$$T_{\min} = 16.50 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 13 & 4 \\ 3 & 6 & 4 & 0 & 12 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 12 & 0 & 25 & 27 & 9 \\ 7 & 13 & 13 & 0 & 27 & 28 & 10 \\ 3 & 4 & 4 & -4 & 9 & 10 & -4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -18, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 18.89 \rightarrow 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 5 & 3 \\ 4 & 9 & 7 & 0 & 4 & 12 & 5 \\ 3 & 7 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 6 & 3 \\ 5 & 12 & 9 & 0 & 6 & 13 & 7 \\ 3 & 5 & 4 & -4 & 3 & 7 & -6 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1153} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.9$$

$$T_{\min} = 16.50 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 6 & 3 \\ 3 & 4 & 6 & 0 & 20 & 12 & 4 \\ 3 & 6 & 4 & 0 & 20 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 20 & 20 & 0 & 73 & 45 & 15 \\ 6 & 12 & 12 & 0 & 45 & 25 & 9 \\ 3 & 4 & 4 & -4 & 15 & 9 & -4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1144} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 15, T_{\max} = 21, T \in [15, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 8 & 3 \\ 3 & 4 & 6 & 0 & 16 & 16 & 4 \\ 3 & 6 & 4 & 0 & 16 & 16 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 16 & 0 & 46 & 48 & 12 \\ 8 & 16 & 16 & 0 & 48 & 46 & 12 \\ 3 & 4 & 4 & -4 & 12 & 12 & -2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1152} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.6$$

$$T_{\min} = 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 4 & 3 \\ 3 & 4 & 6 & 0 & 20 & 8 & 4 \\ 3 & 6 & 4 & 0 & 20 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 20 & 20 & 0 & 73 & 30 & 15 \\ 4 & 8 & 8 & 0 & 30 & 10 & 6 \\ 3 & 4 & 4 & -4 & 15 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{1112} 3^1 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 7 & 3 \\ 3 & 4 & 6 & 0 & 14 & 13 & 4 \\ 3 & 6 & 4 & 0 & 14 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 14 & 0 & 37 & 33 & 11 \\ 7 & 13 & 13 & 0 & 33 & 28 & 10 \\ 3 & 4 & 4 & -4 & 11 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{1112} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 6 & 0 & 4 & 23 & 4 \\ 3 & 6 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 15 & 3 \\ 13 & 23 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & 4 & -4 & 3 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3 1^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 6 & 0 & 5 & 14 & 4 \\ 3 & 6 & 4 & 0 & 5 & 14 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 15 & 4 \\ 8 & 14 & 14 & 0 & 15 & 37 & 11 \\ 3 & 4 & 4 & -4 & 4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{115} 3 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.1$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 6 & 0 & 20 & 4 & 4 \\ 3 & 6 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 20 & 20 & 0 & 73 & 15 & 15 \\ 2 & 4 & 4 & 0 & 15 & 1 & 3 \\ 3 & 4 & 4 & -4 & 15 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1143} 3 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 6 & 3 \\ 3 & 4 & 6 & 0 & 16 & 12 & 4 \\ 3 & 6 & 4 & 0 & 16 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 16 & 0 & 46 & 36 & 12 \\ 6 & 12 & 12 & 0 & 36 & 25 & 9 \\ 3 & 4 & 4 & -4 & 12 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1121} 3 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 6 & 0 & 8 & 9 & 4 \\ 3 & 6 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & 4 & -4 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1142} 3 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 4 & 3 \\ 3 & 4 & 6 & 0 & 16 & 8 & 4 \\ 3 & 6 & 4 & 0 & 16 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 16 & 0 & 46 & 24 & 12 \\ 4 & 8 & 8 & 0 & 24 & 10 & 6 \\ 3 & 4 & 4 & -4 & 12 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1133} 3 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 6 & 3 \\ 3 & 4 & 6 & 0 & 12 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 12 & 0 & 25 & 27 & 9 \\ 6 & 12 & 12 & 0 & 27 & 25 & 9 \\ 3 & 4 & 4 & -4 & 9 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1141} 3 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 6 & 0 & 16 & 4 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 16 & 0 & 46 & 12 & 12 \\ 2 & 4 & 4 & 0 & 12 & 1 & 3 \\ 3 & 4 & 4 & -4 & 12 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{113} 3 1^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.56 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 6 & 0 & 5 & 17 & 4 \\ 3 & 6 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 15 & 4 \\ 9 & 17 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & 4 & -4 & 4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1132} 3 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{e}_6 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

13.3

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 4 & 3 \\ 3 & 4 & 6 & 0 & 12 & 8 & 4 \\ 3 & 6 & 4 & 0 & 12 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 12 & 0 & 25 & 18 & 9 \\ 4 & 8 & 8 & 0 & 18 & 10 & 6 \\ 3 & 4 & 4 & -4 & 9 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.48 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 0 & 5 & 9 & 4 \\ 3 & 6 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & 4 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{113} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 12 & 0 & 25 & 9 & 9 \\ 2 & 4 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & 4 & -4 & 9 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{112} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 6 & 0 & 8 & 8 & 4 \\ 3 & 6 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 8 & 0 & 10 & 12 & 6 \\ 4 & 8 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & 4 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{121} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 8 & 0 & 10 & 6 & 6 \\ 2 & 4 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & 4 & -4 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1123} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 6 & 0 & 8 & 22 & 4 \\ 3 & 6 & 4 & 0 & 8 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 8 & 0 & 10 & 30 & 6 \\ 12 & 22 & 22 & 0 & 30 & 85 & 17 \\ 3 & 4 & 4 & -4 & 6 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{1133} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 9 & 3 \\ 3 & 4 & 6 & 0 & 12 & 17 & 4 \\ 3 & 6 & 4 & 0 & 12 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 12 & 0 & 25 & 36 & 9 \\ 9 & 17 & 17 & 0 & 36 & 49 & 13 \\ 3 & 4 & 4 & -4 & 9 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{112} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 6 & 0 & 5 & 13 & 4 \\ 3 & 6 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 12 & 4 \\ 7 & 13 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & 4 & -4 & 4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{1111} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{1113} 3^1 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 6 & 0 & 4 & 32 & 4 \\ 3 & 6 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 21 & 3 \\ 18 & 32 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & 4 & -4 & 3 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{1111} 3^1 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.49 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 5 & 3 \\ 3 & 4 & 6 & 0 & 14 & 9 & 4 \\ 3 & 6 & 4 & 0 & 14 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 14 & 0 & 37 & 24 & 11 \\ 5 & 9 & 9 & 0 & 24 & 13 & 7 \\ 3 & 4 & 4 & -4 & 11 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{1111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -368, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.49 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{1122} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -408, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.53 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 6 & 0 & 8 & 13 & 4 \\ 3 & 6 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & 4 & -4 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{1123} 3^1 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.52 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 9 & 3 \\ 3 & 4 & 6 & 0 & 18 & 17 & 4 \\ 3 & 6 & 4 & 0 & 18 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 18 & 0 & 58 & 54 & 14 \\ 9 & 17 & 17 & 0 & 54 & 49 & 13 \\ 3 & 4 & 4 & -4 & 14 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{1123} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.49 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 6 & 0 & 5 & 18 & 4 \\ 3 & 6 & 4 & 0 & 5 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 18 & 4 \\ 10 & 18 & 18 & 0 & 18 & 58 & 14 \\ 3 & 4 & 4 & -4 & 4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{1134} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -520, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 11 & 3 \\ 3 & 4 & 6 & 0 & 12 & 21 & 4 \\ 3 & 6 & 4 & 0 & 12 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 12 & 0 & 25 & 45 & 9 \\ 11 & 21 & 21 & 0 & 45 & 76 & 16 \\ 3 & 4 & 4 & -4 & 9 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{1112} 3^1 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.49 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 6 & 0 & 4 & 18 & 4 \\ 3 & 6 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 12 & 3 \\ 10 & 18 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & -4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1143} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -584, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.55 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 12 & 3 \\ 3 & 4 & 6 & 0 & 21 & 22 & 4 \\ 3 & 6 & 4 & 0 & 21 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 21 & 0 & 76 & 81 & 16 \\ 12 & 22 & 22 & 0 & 81 & 85 & 17 \\ 3 & 4 & 4 & -4 & 16 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -592, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.49 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & 4 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{113} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -648, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 6 & 0 & 5 & 22 & 4 \\ 3 & 6 & 4 & 0 & 5 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 21 & 4 \\ 12 & 22 & 22 & 0 & 21 & 85 & 17 \\ 3 & 4 & 4 & -4 & 4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1112} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -648, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 4 & 13 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & -4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1123} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -672, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.54 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 6 & 0 & 8 & 17 & 4 \\ 3 & 6 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 8 & 0 & 10 & 24 & 6 \\ 9 & 17 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & 4 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1114} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -696, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.52 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 6 & 0 & 5 & 26 & 4 \\ 3 & 6 & 4 & 0 & 5 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 24 & 4 \\ 14 & 26 & 26 & 0 & 24 & 118 & 20 \\ 3 & 4 & 4 & -4 & 4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1114} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -736, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.54 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 6 & 0 & 9 & 21 & 4 \\ 3 & 6 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 33 & 7 \\ 11 & 21 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & 4 & -4 & 7 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -800, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 6 & 0 & 9 & 13 & 4 \\ 3 & 6 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 21 & 7 \\ 7 & 13 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & 4 & -4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{1124} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -816, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.53 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 6 & 0 & 8 & 26 & 4 \\ 3 & 6 & 4 & 0 & 8 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 8 & 0 & 10 & 36 & 6 \\ 14 & 26 & 26 & 0 & 36 & 118 & 20 \\ 3 & 4 & 4 & -4 & 6 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1113} 3^1 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -848, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.51 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 6 & 0 & 9 & 17 & 4 \\ 3 & 6 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 27 & 7 \\ 9 & 17 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & 4 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -912, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.51 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 6 & 0 & 4 & 17 & 4 \\ 3 & 6 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & 4 & -4 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1124} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -920, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.55 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 6 & 0 & 8 & 21 & 4 \\ 3 & 6 & 4 & 0 & 8 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 8 & 0 & 10 & 30 & 6 \\ 11 & 21 & 21 & 0 & 30 & 76 & 16 \\ 3 & 4 & 4 & -4 & 6 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1113} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 6 & 0 & 4 & 27 & 4 \\ 3 & 6 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 18 & 3 \\ 15 & 27 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & -4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1122} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1016, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 6 & 0 & 13 & 18 & 4 \\ 3 & 6 & 4 & 0 & 13 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 13 & 0 & 28 & 42 & 10 \\ 10 & 18 & 18 & 0 & 42 & 58 & 14 \\ 3 & 4 & 4 & -4 & 10 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1112} 3^1 1^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1024, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.49 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 6 & 0 & 9 & 18 & 4 \\ 3 & 6 & 4 & 0 & 9 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 30 & 7 \\ 10 & 18 & 18 & 0 & 30 & 58 & 14 \\ 3 & 4 & 4 & -4 & 7 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1113} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1152, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 6 & 0 & 4 & 22 & 4 \\ 3 & 6 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 15 & 3 \\ 12 & 22 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & -4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1123} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1152, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 6 & 0 & 13 & 13 & 4 \\ 3 & 6 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 13 & 0 & 28 & 30 & 10 \\ 7 & 13 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & 4 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1114} 3^1 1^{11} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1160, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.53 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 4 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 15 & 3 \\ 11 & 21 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & 4 & -4 & 3 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1123} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1344, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.51 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 6 & 0 & 13 & 17 & 4 \\ 3 & 6 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 13 & 0 & 28 & 39 & 10 \\ 9 & 17 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & 4 & -4 & 10 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1124} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1376, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.52 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 6 & 0 & 13 & 21 & 4 \\ 3 & 6 & 4 & 0 & 13 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 13 & 0 & 28 & 48 & 10 \\ 11 & 21 & 21 & 0 & 48 & 76 & 16 \\ 3 & 4 & 4 & -4 & 10 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1114} 3^1 1^{14} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1448, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 6 & 0 & 4 & 36 & 4 \\ 3 & 6 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 24 & 3 \\ 20 & 36 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & -4 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1133} 3^1 1^{112} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1488, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.51 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 12 & 3 \\ 3 & 4 & 6 & 0 & 17 & 22 & 4 \\ 3 & 6 & 4 & 0 & 17 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 17 & 0 & 49 & 66 & 13 \\ 12 & 22 & 22 & 0 & 66 & 85 & 17 \\ 3 & 4 & 4 & -4 & 13 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1113} 3^1 1^{112} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1520, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 6 & 0 & 9 & 22 & 4 \\ 3 & 6 & 4 & 0 & 9 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 36 & 7 \\ 12 & 22 & 22 & 0 & 36 & 85 & 17 \\ 3 & 4 & 4 & -4 & 7 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1133} 3^1 1^{111} + [\mathbf{su}_2 \cdot \mathbf{so}_7 \cdot \mathbf{su}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1680, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.51 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 6 & 0 & 17 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 17 & 0 & 49 & 51 & 13 \\ 9 & 17 & 17 & 0 & 51 & 49 & 13 \\ 3 & 4 & 4 & -4 & 13 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1114} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1704, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.51 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 6 & 0 & 4 & 26 & 4 \\ 3 & 6 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 18 & 3 \\ 14 & 26 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & 4 & -4 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1144} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1784, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.52 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 14 & 3 \\ 3 & 4 & 6 & 0 & 21 & 26 & 4 \\ 3 & 6 & 4 & 0 & 21 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 21 & 0 & 76 & 96 & 16 \\ 14 & 26 & 26 & 0 & 96 & 118 & 20 \\ 3 & 4 & 4 & -4 & 16 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{1123} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1800, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 12 & 3 \\ 3 & 4 & 6 & 0 & 13 & 22 & 4 \\ 3 & 6 & 4 & 0 & 13 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 13 & 0 & 28 & 51 & 10 \\ 12 & 22 & 22 & 0 & 51 & 85 & 17 \\ 3 & 4 & 4 & -4 & 10 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{1114} 3^{113} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1800, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 6 & 0 & 4 & 31 & 4 \\ 3 & 6 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 21 & 3 \\ 17 & 31 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & -4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{1134} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1856, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.52 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 11 & 3 \\ 3 & 4 & 6 & 0 & 17 & 21 & 4 \\ 3 & 6 & 4 & 0 & 17 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 17 & 0 & 49 & 63 & 13 \\ 11 & 21 & 21 & 0 & 63 & 76 & 16 \\ 3 & 4 & 4 & -4 & 13 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{1114} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -1856, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 6 & 0 & 9 & 26 & 4 \\ 3 & 6 & 4 & 0 & 9 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 42 & 7 \\ 14 & 26 & 26 & 0 & 42 & 118 & 20 \\ 3 & 4 & 4 & -4 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{1144} 3^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -2176, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.53 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 11 & 3 \\ 3 & 4 & 6 & 0 & 21 & 21 & 4 \\ 3 & 6 & 4 & 0 & 21 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 21 & 0 & 76 & 78 & 16 \\ 11 & 21 & 21 & 0 & 78 & 76 & 16 \\ 3 & 4 & 4 & -4 & 16 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{1134} 3^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -105, \quad \det = -2400, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.51 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 14 & 3 \\ 3 & 4 & 6 & 0 & 17 & 26 & 4 \\ 3 & 6 & 4 & 0 & 17 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 17 & 0 & 49 & 78 & 13 \\ 14 & 26 & 26 & 0 & 78 & 118 & 20 \\ 3 & 4 & 4 & -4 & 13 & 20 & 0 \end{pmatrix}_{b_0}$$

[65] $1^{1124}31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -2424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$

$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 14 & 3 \\ 3 & 4 & 6 & 0 & 13 & 26 & 4 \\ 3 & 6 & 4 & 0 & 13 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 13 & 0 & 28 & 60 & 10 \\ 14 & 26 & 26 & 0 & 60 & 118 & 20 \\ 3 & 4 & 4 & -4 & 10 & 20 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [66] $1^{1131}1^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 848, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 9 & 4 \\ 1 & 4 & 0 & 4 & 0 & 3 & 6 & 2 \\ 2 & 6 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 9 & 4 \\ 3 & 9 & 6 & 9 & 0 & 9 & 13 & 7 \\ 2 & 4 & 2 & 4 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

[67] $1^{131}1^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 6 & 4 \\ 1 & 4 & 0 & 4 & 0 & 3 & 3 & 2 \\ 2 & 6 & 4 & 4 & 0 & 6 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 6 & 4 \\ 2 & 6 & 3 & 6 & 0 & 6 & 4 & 4 \\ 2 & 4 & 2 & 4 & -4 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

[68] $1^{111}31^{111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 1184, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 9 & 4 \\ 1 & 4 & 0 & 4 & 0 & 6 & 6 & 2 \\ 2 & 6 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 9 & 0 & 13 & 15 & 7 \\ 3 & 9 & 6 & 9 & 0 & 15 & 13 & 7 \\ 2 & 4 & 2 & 4 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

5.34 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

$T = 10$ [0] $1^{11111}31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 848, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 864, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & 4 & -4 & 3 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 9 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 4 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -84, \ \det = 1472, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 9 & 4 & 4 \\ 3 & 6 & 4 & 0 & 9 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 15 & 6 & 7 \\ 5 & 9 & 9 & 0 & 15 & 13 & 6 & 7 \\ 2 & 4 & 4 & 0 & 6 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 7 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

5.35 NHC(-2)(-3)(-2) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{\overline{13}} 1^{\overline{111}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -86, \ \det = 724, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 4 & 0 & 3 & 2 & 2 \\ 2 & 6 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 3 & 4 \\ 1 & 3 & 2 & 3 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{13}} 3 1^{\overline{111}} 3 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 13, \ \Delta = -86, \ \det = 848, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.99 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 3 & 4 \\ 1 & 4 & 0 & 4 & 0 & 6 & 2 & 2 \\ 2 & 6 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 6 & 9 & 0 & 13 & 5 & 7 \\ 1 & 3 & 2 & 3 & 0 & 5 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

5.36 NHC(-2)(-3)(-2) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 9 \quad [1] \ 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -148, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -148, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.49 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 6 & 0 & 14 & 4 & 4 \\ 3 & 6 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 14 & 14 & 0 & 37 & 9 & 11 \\ 2 & 4 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -4 & 11 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -148, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 5 & 2 & 3 \\ 4 & 9 & 7 & 0 & 12 & 4 & 5 \\ 3 & 7 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 12 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 4 & 0 & 6 & 0 & 2 \\ 3 & 5 & 4 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{112}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -217, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 8 & 2 & 3 \\ 4 & 9 & 7 & 0 & 20 & 4 & 5 \\ 3 & 7 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 20 & 14 & 0 & 37 & 9 & 11 \\ 2 & 4 & 4 & 0 & 9 & 0 & 2 \\ 3 & 5 & 4 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -220, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.05 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 4 & 5 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 3 & 3 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & 4 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112} 3^1 1^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -256, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 10 & 2 & 3 \\ 4 & 9 & 7 & 0 & 24 & 4 & 5 \\ 3 & 7 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 24 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 4 & 0 & 12 & 0 & 2 \\ 3 & 5 & 4 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{114} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -256, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.37 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 6 & 0 & 16 & 4 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 16 & 16 & 0 & 46 & 12 & 12 \\ 2 & 4 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -4 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -264, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.88 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 6 & 0 & 8 & 9 & 4 \\ 3 & 6 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 9 & 0 & 12 & 12 & 6 \\ 3 & 4 & 4 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{113} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -340, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 12 & 12 & 0 & 25 & 9 & 9 \\ 2 & 4 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -4 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{112} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -400, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 8 & 0 & 10 & 6 & 6 \\ 2 & 4 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -436, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{112} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -436, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.28 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 6 & 0 & 23 & 4 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 13 & 23 & 23 & 0 & 100 & 15 & 18 \\ 2 & 4 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -4 & 18 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -516, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.82 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & -4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11111^1} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -592, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11211^1} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1012, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 13 & 4 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 13 & 13 & 0 & 28 & 9 & 10 \\ 2 & 4 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11311^1} 31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1012, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.17 \rightarrow 14, \quad T_{\max} = 25, \quad T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 6 & 0 & 32 & 4 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 18 & 32 & 32 & 0 & 193 & 21 & 25 \\ 2 & 4 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -4 & 25 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11211^1} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1024, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11311^1} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1408, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 17 & 17 & 0 & 49 & 12 & 13 \\ 2 & 4 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -4 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11311^1} 31^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1744, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.98 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 15 & 27 & 27 & 0 & 133 & 18 & 21 \\ 2 & 4 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -4 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11411^1} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1780, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 & 4 \\ 3 & 6 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 21 & 0 & 76 & 15 & 16 \\ 2 & 4 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -4 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11311^1} 31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1876, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 6 & 0 & 22 & 4 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 22 & 22 & 0 & 85 & 15 & 17 \\ 2 & 4 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -4 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1141} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -2704, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 6 & 0 & 26 & 4 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 14 & 26 & 26 & 0 & 118 & 18 & 20 \\ 2 & 4 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -4 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1141} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -2752, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.99 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 20 & 36 & 36 & 0 & 238 & 24 & 28 \\ 2 & 4 & 4 & 0 & 24 & 0 & 2 \\ 3 & 4 & 4 & -4 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1141} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -3028, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 6 & 0 & 31 & 4 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 17 & 31 & 31 & 0 & 172 & 21 & 24 \\ 2 & 4 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -4 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [25] \quad 1^{1111} 31^{131} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 60, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.40 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 4 & 0 & 3 & 3 \\ 1 & 3 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 1 & 3 & 3 \\ 1 & 3 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 3 & 3 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{11} 31^{112} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 168, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.57 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 12 & 3 & 4 \\ 1 & 4 & 0 & 4 & 0 & 6 & 3 & 2 \\ 2 & 6 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 6 & 12 & 0 & 22 & 6 & 8 \\ 1 & 3 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{11} 31^{111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 492, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.20 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 3 & 4 \\ 1 & 4 & 0 & 4 & 0 & 3 & 3 & 2 \\ 2 & 6 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 3 & 6 & 0 & 4 & 3 & 4 \\ 1 & 3 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [28] \quad 1^{111} 4^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, \quad T_{\max} = 14, \quad \Delta = -84, \quad \det = -94, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.83 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 & 4 \\ 4 & 5 & 4 & 0 & 7 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 7 & 7 & 0 & 8 & 5 & 6 \\ 2 & 4 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

5.37 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{11111}} 3^{\overline{11}} 1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 848, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11111}} 3^{\overline{11}} 1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 864, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 9 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 & 6 \\ 2 & 4 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & 4 & -4 & 3 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11111}} 3^{\overline{11}} 1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11111}} 3^{\overline{11}} 1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 1472, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 6 & 15 & 7 \\ 2 & 4 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 15 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 7 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

5.38 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{11111}} 3^{\overline{11}} 1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 492, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 5 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 1 & 3 & 3 & 3 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

5.39 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{\overline{131}} 2^{\overline{11}} 1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 16.67 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 1 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 1 & 2 & 2 & 3 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 2 & 2 & 1 & 2 & 4 & 4 \\ 2 & 2 & 0 & 4 & -2 & 0 \\ 3 & 3 & -1 & 4 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{131}} 2^{\overline{11}} 1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 29, \Delta = -116, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 0 & 4 & 0 & 2 & 2 \\ 2 & 6 & 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 2 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

5.40 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$T = 10$ [0] $1^1 3^1 1^1 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 724, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 4 & 0 & 2 & 3 & 2 \\ 2 & 6 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & -4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[1] $1^1 3^1 1^1 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 10, T_{\max} = 13, \Delta = -86, \det = 848, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 9 & 4 \\ 1 & 4 & 0 & 4 & 0 & 2 & 6 & 2 \\ 2 & 6 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 3 & 0 & 0 & 5 & 2 \\ 3 & 9 & 6 & 9 & 0 & 5 & 13 & 7 \\ 2 & 4 & 2 & 4 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

5.41 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$

$T = 10$ [0] $1^1 3^1 1^1 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$ $T = 10, T_{\max} = 14, \Delta = -89, \det = 576, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 3 & 4 \\ 1 & 4 & 0 & 4 & 0 & 2 & 2 & 2 \\ 2 & 6 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 3 & 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

5.42 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_2 \mathfrak{n}3\text{mix}$

$T = 9$ [0] $1^1 3^1 1^1 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9, T_{\max} = 29, \Delta = -116, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 12 & 4 \\ 1 & 4 & 0 & 4 & 0 & 8 & 2 \\ 2 & 6 & 4 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 8 & 12 & 0 & 25 & 9 \\ 2 & 4 & 2 & 4 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

[1] $1^1 3^1 1^1 1^1 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9, T_{\max} = 29, \Delta = -116, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 29, T \in [13, 29]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 4 & 0 & 5 & 2 \\ 2 & 6 & 4 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 5 & 9 & 0 & 12 & 6 \\ 2 & 4 & 2 & 4 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

5.43 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3$

$T = 8$ [0] $1^1 1^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 8, T_{\max} = 98, \Delta = -144, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 3 \\ 4 & 9 & 7 & 0 & 4 & 5 \\ 3 & 7 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 3 & 5 & 4 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^1 1^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 8, T_{\max} = 98, \Delta = -144, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 1 & 3 \\ 4 & 9 & 11 & 4 & 0 & 5 \\ 4 & 11 & 9 & 4 & 0 & 5 \\ 2 & 4 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 5 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

[2] $1^1 1^1 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 8, T_{\max} = 98, \Delta = -144, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 5 & 0 & 3 & 3 \\ 3 & 4 & 4 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{113} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 98, \ \Delta = -144, \ \det = 192, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.33 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 12 & 0 & 24 & 8 \\ 3 & 4 & 4 & -4 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 98, \ \Delta = -144, \ \det = 212, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.09 \rightarrow 14, \ T_{\max} = 98, \ T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 8 & 0 & 9 & 5 \\ 3 & 4 & 4 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 98, \ \Delta = -144, \ \det = 224, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 3 & 4 & 4 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 98, \ \Delta = -144, \ \det = 300, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.81 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 98, \ \Delta = -144, \ \det = 440, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 13 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 13 & 0 & 27 & 9 \\ 3 & 4 & 4 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{113} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 98, \ \Delta = -144, \ \det = 572, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 17 & 0 & 48 & 12 \\ 3 & 4 & 4 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [9] \ 1^{111} 31^1 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 79, \ \Delta = -123, \ \det = -156, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 79, \ T \in [14, 79]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 4 & 0 & 3 \\ 1 & 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 \\ 1 & 3 & 3 & 3 & 0 & 2 \\ 2 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 79, \ \Delta = -123, \ \det = -300, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.16 \rightarrow 14, \ T_{\max} = 79, \ T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 \\ 1 & 4 & 0 & 4 & 0 & 3 \\ 2 & 6 & 4 & 4 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 \\ 1 & 3 & 3 & 3 & 0 & 2 \\ 2 & 4 & 2 & 4 & -4 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 9, \ T_{\max} = 79, \ \Delta = -123, \ \det = -636, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.31 \rightarrow 14, \ T_{\max} = 79, \ T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 \\ 1 & 4 & 0 & 4 & 0 & 3 \\ 2 & 6 & 4 & 4 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 \\ 2 & 6 & 3 & 6 & 0 & 3 \\ 2 & 4 & 2 & 4 & -4 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11}31^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 79, \ \Delta = -123, \ \det = -696, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.07 \rightarrow 14, \ T_{\max} = 79, \ T \in [14, 79]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 4 & 0 & 6 & 2 \\ 2 & 6 & 4 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 6 & 9 & 0 & 12 & 6 \\ 2 & 4 & 2 & 4 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [13] \ 1^{11}4^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 35, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.83 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 4 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11}4^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 26, \ \Delta = -114, \ \det = 344, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.95 \rightarrow 14, \ T_{\max} = 26, \ T \in [14, 26]$$

$$\begin{pmatrix} 1 & 4 & 4 & 1 & 2 & 3 \\ 4 & 4 & 6 & 0 & 3 & 4 \\ 4 & 6 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

5.44 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \ 1^{1111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, \ T_{\max} = 15, \ \Delta = -94, \ \det = 172, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.16 \rightarrow 14, \ T_{\max} = 15, \ T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 1 & 4 & 3 \\ 2 & 4 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

5.45 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 99, \ \Delta = -143, \ \det = -524, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.77 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 17 & 0 & 12 & 6 \\ 5 & 8 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

5.46 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 9 \quad [1] \ 1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -148, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 4 & 5 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 3 & 5 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -148, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 5 & 3 \\ 4 & 9 & 7 & 0 & 4 & 12 & 5 \\ 3 & 7 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 5 & 12 & 9 & 0 & 6 & 13 & 7 \\ 3 & 5 & 4 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -148, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.49 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 6 & 0 & 4 & 14 & 4 \\ 3 & 6 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 8 & 14 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & 4 & -4 & 2 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[4] 1^{112} 3^1 1^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -217, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 8 & 3 \\ 4 & 9 & 7 & 0 & 4 & 20 & 5 \\ 3 & 7 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 8 & 20 & 14 & 0 & 9 & 37 & 11 \\ 3 & 5 & 4 & -4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[5] 1^{111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[7] 1^{112} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 10 & 3 \\ 4 & 9 & 7 & 0 & 4 & 24 & 5 \\ 3 & 7 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 10 & 24 & 18 & 0 & 12 & 58 & 14 \\ 3 & 5 & 4 & -4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[6] 1^{1114} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 6 & 0 & 4 & 16 & 4 \\ 3 & 6 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 8 & 16 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & 4 & -4 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[8] 1^{1112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 6 & 0 & 9 & 8 & 4 \\ 3 & 6 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 12 & 12 & 6 \\ 4 & 8 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & 4 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[9] 1^{1113} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -340, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 6 & 0 & 4 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 6 & 12 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & 4 & -4 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[10] 1^{1112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 6 & 0 & 4 & 8 & 4 \\ 3 & 6 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 4 & 8 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & 4 & -4 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[12] 1^{1111} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -436, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1112} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -436, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.28 \rightarrow 14, \quad T_{\max} = 25, \quad T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 6 & 0 & 4 & 23 & 4 \\ 3 & 6 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 13 & 23 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & 4 & -4 & 2 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -516, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.82 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 12 & 6 & 6 \\ 2 & 4 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & 4 & -4 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -592, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1113} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1012, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.17 \rightarrow 14, \quad T_{\max} = 25, \quad T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 6 & 0 & 4 & 32 & 4 \\ 3 & 6 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 21 & 2 \\ 18 & 32 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & 4 & -4 & 2 & 25 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1112} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1012, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 4 & 13 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 7 & 13 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{1112} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1024, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 6 & 0 & 4 & 18 & 4 \\ 3 & 6 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 10 & 18 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & -4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1113} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1408, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 6 & 0 & 4 & 17 & 4 \\ 3 & 6 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 9 & 17 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & 4 & -4 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1113} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1744, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.98 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 6 & 0 & 4 & 27 & 4 \\ 3 & 6 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 18 & 2 \\ 15 & 27 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & -4 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1114} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1780, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 4 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 11 & 21 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & 4 & -4 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1113} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -1876, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 6 & 0 & 4 & 22 & 4 \\ 3 & 6 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 12 & 22 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & -4 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1114} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -2704, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.93 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 6 & 0 & 4 & 26 & 4 \\ 3 & 6 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 18 & 2 \\ 14 & 26 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & 4 & -4 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1114} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -2752, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.99 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 6 & 0 & 4 & 36 & 4 \\ 3 & 6 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 24 & 2 \\ 20 & 36 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & -4 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1114} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \quad T_{\max} = 25, \quad \Delta = -114, \quad \det = -3028, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 25, \quad T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 6 & 0 & 4 & 31 & 4 \\ 3 & 6 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 21 & 2 \\ 17 & 31 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & -4 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [25] \quad 1^{1111} 31^{13} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 60, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.40 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 4 & 0 & 3 & 3 \\ 1 & 3 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 3 \\ 1 & 3 & 3 & 3 & 0 & 3 & 1 \\ 2 & 3 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{11} 31^{12} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 168, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.57 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 12 & 4 \\ 1 & 4 & 0 & 4 & 0 & 3 & 6 & 2 \\ 2 & 6 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 6 & 2 \\ 4 & 12 & 6 & 12 & 0 & 6 & 22 & 8 \\ 2 & 4 & 2 & 4 & -4 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{11} 31^{11} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 15, \quad \Delta = -93, \quad \det = 492, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.20 \rightarrow 14, \quad T_{\max} = 15, \quad T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 4 & 0 & 3 & 3 & 2 \\ 2 & 6 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 6 & 0 & 3 & 4 & 4 \\ 2 & 4 & 2 & 4 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [28] \quad 1^{\overline{1111}} 1^{\overline{1311}} 4^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 7 & 4 \\ 4 & 5 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 5 & 7 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & 4 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

5.47 NHC(-2)(-3)(-2) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{\overline{11111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 848, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 864, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 12 & 6 & 6 & 6 \\ 2 & 4 & 4 & 0 & 6 & 1 & 3 & 3 \\ 2 & 4 & 4 & 0 & 6 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 6 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 6 & 2 \\ 2 & 4 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{11111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 1472, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 9 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

5.48 NHC(-2)(-3)(-2) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{\overline{11111}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 492, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

5.49 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 6 & 0 & 5 & 8 & 4 \\ 3 & 6 & 4 & 0 & 5 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 3 & 6 & 3 \\ 4 & 8 & 8 & 0 & 6 & 9 & 5 \\ 3 & 4 & 4 & -4 & 3 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -660, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -816, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 216, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 4 & 0 & 3 & 3 \\ 1 & 3 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 3 \\ 1 & 3 & 3 & 3 & 0 & 3 & 0 \\ 2 & 3 & 3 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 792, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 6 & 4 \\ 1 & 4 & 0 & 4 & 0 & 3 & 3 & 2 \\ 2 & 6 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 3 & 6 & 0 & 3 & 3 & 3 \\ 2 & 4 & 2 & 4 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [6] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} 4^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 16, \Delta = -93, \det = -264, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.18 \rightarrow 15, T_{\max} = 16, T \in [15, 16]$$

$$\begin{pmatrix} 1 & 4 & 4 & 1 & 2 & 2 & 3 \\ 4 & 4 & 6 & 0 & 3 & 3 & 4 \\ 4 & 6 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

5.50 $\text{NHC}(-2)(-3)(-2) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 492, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

5.51 NHC(-2)(-3)(-2) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{1111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 864, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

5.52 NHC(-2)(-3)(-2) + $\mathfrak{f}_4$

$$T = 7 \quad [0] \quad 21^1 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 9 & 11 & 4 & 5 \\ 4 & 11 & 9 & 4 & 5 \\ 3 & 4 & 4 & -2 & 0 \\ 3 & 5 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -8, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 9 & 11 & 4 & 5 \\ 4 & 11 & 9 & 4 & 5 \\ 3 & 4 & 4 & -2 & 0 \\ 3 & 5 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -37, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 9 & 7 & 0 & 5 \\ 3 & 7 & 4 & 0 & 4 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 4 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^1 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.70 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 3 \\ 4 & 9 & 11 & 4 & 5 \\ 4 & 11 & 9 & 4 & 5 \\ 2 & 4 & 4 & -2 & 0 \\ 3 & 5 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 21^{111} 31^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -139, \det = -344, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 4 & 8 & 8 & 3 & 4 \\ 8 & 9 & 11 & 4 & 5 \\ 8 & 11 & 9 & 4 & 5 \\ 3 & 4 & 4 & -2 & 0 \\ 4 & 5 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [6] \quad 1^1 321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 15.33 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & 0 & 3 & -2 & 0 \\ 2 & 2 & -1 & 3 & 0 & -3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^1 321^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 5 & 7 & 0 & 5 \\ 3 & 7 & 5 & 0 & 5 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 5 & 5 & -3 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11}3^{11}21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 \quad T = 8, \ T_{\max} = 21, \ \Delta = -111, \ \det = 44, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.82 \rightarrow 12, \ T_{\max} = 21, \ T \in [12, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 1 & 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11}31^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 \quad T = 8, \ T_{\max} = 35, \ \Delta = -118, \ \det = 60, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 35, \ T \in [13, 35]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 4 & 0 & 3 \\ 1 & 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 3 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11}31^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 \quad T = 8, \ T_{\max} = 35, \ \Delta = -118, \ \det = 156, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.05 \rightarrow 13, \ T_{\max} = 35, \ T \in [13, 35]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 0 & 4 & 0 & 2 \\ 2 & 6 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 2 & 4 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 2^12^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 \quad T = 8, \ T_{\max} = 21, \ \Delta = -111, \ \det = 320, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.70 \rightarrow 12, \ T_{\max} = 21, \ T \in [12, 21]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 3 & 1 & 3 \\ 2 & 4 & 0 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 & 1 & 3 \\ 1 & 1 & 1 & 1 & -3 & -1 \\ 2 & 3 & 2 & 3 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [12] \ 1^{11}4^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 \quad T = 9, \ T_{\max} = 21, \ \Delta = -109, \ \det = -35, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 4 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 21^141^{11}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 \quad T = 9, \ T_{\max} = 21, \ \Delta = -109, \ \det = -104, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.73 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 \\ 2 & 0 & 3 & 2 & 0 & 2 \\ 2 & 3 & 3 & 3 & 0 & 3 \\ 2 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 3 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11}4^131^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 \quad T = 9, \ T_{\max} = 21, \ \Delta = -109, \ \det = -156, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 13.05 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 1 & 4 & 4 & 1 & 3 \\ 4 & 4 & 6 & 0 & 4 \\ 4 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11}41^{11}231^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 \quad T = 9, \ T_{\max} = 21, \ \Delta = -110, \ \det = -296, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.69 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 6 & 6 & 1 & 3 \\ 6 & 13 & 15 & 0 & 7 \\ 6 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 7 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

5.53 NHC(-2)(-3)(-2) + f₄ + e₆

$$T = 8 \quad [0] \ 1^{111}31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{e}_6 \quad T = 8, \ T_{\max} = 102, \ \Delta = -188, \ \det = 336, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.31 \rightarrow 15, \ T_{\max} = 102, \ T \in [15, 102]$$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 1 & 3 \\ 4 & 9 & 11 & 4 & 0 & 5 \\ 4 & 11 & 9 & 4 & 0 & 5 \\ 2 & 4 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 5 & 0 & -4 & -2 \end{pmatrix}_{b_0}$$

5.54 $\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{f}_4$

$$T = 8 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{brown}{\mathfrak{f}}_4 \quad T = 8, T_{\max} = 101, \Delta = -162, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 1 & 3 \\ 4 & 9 & 11 & 4 & 0 & 5 \\ 4 & 11 & 9 & 4 & 0 & 5 \\ 2 & 4 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 5 & 5 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

5.55 $\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \text{NHC}(-2)(-3)$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 & 2 & 3 \\ 4 & 9 & 7 & 0 & 8 & 4 & 5 \\ 3 & 7 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 8 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 4 & 0 & 4 & 0 & 2 \\ 3 & 5 & 4 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 4 & 4 \\ 2 & 4 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -272, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.35 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 4 & 3 \\ 2 & 4 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

5.56 $\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 5 \\ 12 & 25 & 23 & 0 & 9 \\ 11 & 23 & 20 & 0 & 8 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 1 & 3 \\ 4 & 9 & 11 & 4 & 0 & 5 \\ 4 & 11 & 9 & 4 & 0 & 5 \\ 2 & 4 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 5 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -138, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 5 \\ 11 & 20 & 22 & 0 & 8 \\ 11 & 22 & 20 & 0 & 8 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \ 1^{\overline{1}} 3^{\overline{1}} 2^{\overline{111}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 5 & 7 & 0 & 5 & 5 \\ 3 & 7 & 5 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 5 & 0 & 0 & 2 \\ 3 & 5 & 5 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3 1^{11} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -172, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 32, \ T \in [13, 32]$$

$$\begin{pmatrix} 1 & 4 & 5 & 4 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 13 & 0 & 6 \\ 4 & 12 & 13 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [5] \ 1^{11} 4 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 \quad T = 10, \ T_{\max} = 22, \ \Delta = -108, \ \det = 424, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.63 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 10 & 10 & 1 & 5 \\ 10 & 13 & 15 & 0 & 7 \\ 10 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

5.57 NHC(-2)(-3)(-2) + $\textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2$

$$T = 9 \quad [0] \ 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -62, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.65 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 9 & 5 \\ 12 & 25 & 23 & 0 & 20 & 9 \\ 11 & 23 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 17 & 0 & 13 & 7 \\ 5 & 9 & 8 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -176, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.25 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 17 & 0 & 13 & 7 \\ 5 & 8 & 8 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -218, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.64 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 17 & 5 \\ 12 & 25 & 23 & 0 & 36 & 9 \\ 11 & 23 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 36 & 33 & 0 & 49 & 13 \\ 5 & 9 & 8 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{112} 3 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -342, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.7$$

$$T_{\min} = 12.64 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 26 & 5 \\ 12 & 25 & 23 & 0 & 56 & 9 \\ 11 & 23 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 56 & 50 & 0 & 118 & 20 \\ 5 & 9 & 8 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \ 1^{11} 3 2 1^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -80, \ \det = 144, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 7 & 0 & 5 & 5 & 5 \\ 3 & 7 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 5 & 0 & 0 & 4 & 2 \\ 2 & 5 & 5 & 0 & 4 & 2 & 4 \\ 3 & 5 & 5 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 2 1^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -80, \ \det = 256, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.97 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 5 & 7 & 0 & 5 & 10 & 5 \\ 3 & 7 & 5 & 0 & 5 & 10 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 5 & 0 & 0 & 7 & 2 \\ 5 & 10 & 10 & 0 & 7 & 14 & 8 \\ 3 & 5 & 5 & -3 & 2 & 8 & 1 \end{pmatrix}_{b_0}$$

5.58 $\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_{2 \rightarrow (-3)}$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_{2 \rightarrow (-3)} \quad T = 9, \ T_{\max} = 21, \ \Delta = -110, \ \det = -134, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.69 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 6 & 5 \\ 12 & 25 & 23 & 0 & 12 & 9 \\ 11 & 23 & 20 & 0 & 11 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 11 & 0 & 4 & 4 \\ 5 & 9 & 8 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

5.59 $\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 32, \ \Delta = -117, \ \det = -412, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 32, \ T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 17 & 0 & 12 & 6 \\ 5 & 8 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

5.60 $\text{NHC}(-2)(-3)(-2) + \mathfrak{f}_4 + \mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 7, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 16.29 \rightarrow 17, \ T_{\max} = 20, \ T \in [17, 20]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 3 \\ 4 & 9 & 7 & 0 & 4 & 5 \\ 3 & 7 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 3 \\ 3 & 5 & 4 & -3 & 3 & -4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 7, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 18.57 \rightarrow 19, \ T_{\max} = 20, \ T \in [19, 20]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 7 & 3 \\ 4 & 9 & 7 & 0 & 16 & 5 \\ 3 & 7 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 16 & 13 & 0 & 28 & 10 \\ 3 & 5 & 4 & -3 & 10 & -6 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{15}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 20, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.3$$

$$T_{\min} = 14.60 \rightarrow 15, \ T_{\max} = 20, \ T \in [15, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 20 & 4 \\ 3 & 6 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 20 & 20 & 0 & 73 & 15 \\ 3 & 4 & 4 & -3 & 15 & -2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{12}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 30, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 14.33 \rightarrow 15, \ T_{\max} = 20, \ T \in [15, 20]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 6 & 3 \\ 4 & 9 & 7 & 0 & 16 & 5 \\ 3 & 7 & 4 & 0 & 10 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 16 & 10 & 0 & 22 & 8 \\ 3 & 5 & 4 & -3 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{14}} 3^{\overline{11}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 56, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.29 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 16 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 16 & 0 & 46 & 12 \\ 3 & 4 & 4 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 60, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 4 \\ 3 & 4 & 4 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{12}} 3^{\overline{11}} 1^{\overline{13}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -109, \ \det = 63, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 13.11 \rightarrow 14, \ T_{\max} = 20, \ T \in [14, 20]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 12 & 3 \\ 4 & 9 & 7 & 0 & 28 & 5 \\ 3 & 7 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 28 & 22 & 0 & 85 & 17 \\ 3 & 5 & 4 & -3 & 17 & -1 \end{pmatrix}_{b_0}$$

[7] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.35 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 & 3 \\ 4 & 9 & 7 & 0 & 8 & 5 \\ 3 & 7 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 102, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 5 & 3 \\ 4 & 9 & 7 & 0 & 12 & 5 \\ 3 & 7 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 9 & 0 & 13 & 7 \\ 3 & 5 & 4 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 6 \\ 3 & 4 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{112}31^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 6 & 0 & 28 & 4 \\ 3 & 6 & 4 & 0 & 28 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 28 & 28 & 0 & 154 & 22 \\ 3 & 4 & 4 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{111}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 14 & 4 \\ 3 & 6 & 4 & 0 & 14 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 14 & 0 & 37 & 11 \\ 3 & 4 & 4 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 3 \\ 3 & 4 & 4 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{112}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 8 & 3 \\ 4 & 9 & 7 & 0 & 20 & 5 \\ 3 & 7 & 4 & 0 & 14 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 14 & 0 & 37 & 11 \\ 3 & 5 & 4 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{112}31^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 186, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 10 & 3 \\ 4 & 9 & 7 & 0 & 24 & 5 \\ 3 & 7 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 24 & 18 & 0 & 58 & 14 \\ 3 & 5 & 4 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{112}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 284, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 6 & 0 & 23 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 23 & 0 & 100 & 18 \\ 3 & 4 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{112}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 284, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 13 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 13 & 0 & 28 & 10 \\ 3 & 4 & 4 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{112}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 344, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{113}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 17 & 0 & 49 & 13 \\ 3 & 4 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{114}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 476, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 21 & 0 & 76 & 16 \\ 3 & 4 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[22] $1^{113}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 22 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 22 & 0 & 85 & 17 \\ 3 & 4 & 4 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{113}31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 564, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 6 & 0 & 32 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 32 & 0 & 193 & 25 \\ 3 & 4 & 4 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{113}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 624, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[25] $1^{114}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 776, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.65 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 26 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 26 & 0 & 118 & 20 \\ 3 & 4 & 4 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{114}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 956, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 6 & 0 & 31 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 31 & 0 & 172 & 24 \\ 3 & 4 & 4 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{114}3^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 1016, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [28] $1^{11}3^{21}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 5 & 7 & 0 & 5 & 5 \\ 3 & 7 & 5 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 5 & 0 & 2 & 4 \\ 3 & 5 & 5 & -3 & 4 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{111}3^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 3 & 4 & 0 & 3 & 3 \\ 1 & 3 & 4 & 1 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 1 & 3 \\ 2 & 3 & 3 & 3 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{111}3^{21}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 2 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{11}3^{11}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 6 & 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 2 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[32] $1^{11}3^{21}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 5 & 7 & 0 & 10 & 5 \\ 3 & 7 & 5 & 0 & 10 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 10 & 10 & 0 & 14 & 8 \\ 3 & 5 & 5 & -3 & 8 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{111}3^{21}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 3 & 0 & 5 & 5 \\ 2 & 2 & 2 & 2 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{12}3^{21}1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 5 & 7 & 0 & 20 & 5 \\ 3 & 7 & 5 & 0 & 20 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 20 & 20 & 0 & 62 & 16 \\ 3 & 5 & 5 & -3 & 16 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{11}3^{21}1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 5 & 7 & 0 & 25 & 5 \\ 3 & 7 & 5 & 0 & 25 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 25 & 25 & 0 & 92 & 20 \\ 3 & 5 & 5 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$

- [34] $1^{11}321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -168$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 5 & 7 & 0 & 15 & 5 \\ 3 & 7 & 5 & 0 & 15 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 15 & 15 & 0 & 32 & 12 \\ 3 & 5 & 5 & -3 & 12 & 2 \end{pmatrix}_{b_0}$$
- [37] $1^{112}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -184$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 0 & 6 \\ 1 & 2 & 0 & 2 & 0 & 8 \\ 1 & 2 & 2 & 0 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 4 & 6 & 8 & 6 & 0 & 26 \\ 2 & 2 & 2 & 2 & -3 & 10 \end{pmatrix}_{b_0}$$
- [38] $1^{11}321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -192$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 5 & 7 & 0 & 20 & 5 \\ 3 & 7 & 5 & 0 & 20 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 20 & 0 & 58 & 16 \\ 3 & 5 & 5 & -3 & 16 & 2 \end{pmatrix}_{b_0}$$
- [39] $1^{112}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -208$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.85 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 0 & 5 \\ 1 & 2 & 0 & 2 & 0 & 5 \\ 1 & 2 & 2 & 0 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 3 & 5 & 5 & 5 & 0 & 13 \\ 2 & 2 & 2 & 2 & -3 & 7 \end{pmatrix}_{b_0}$$
- [40] $1^{113}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -272$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 0 & 7 \\ 1 & 2 & 0 & 2 & 0 & 6 \\ 1 & 2 & 2 & 0 & 0 & 7 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 4 & 7 & 6 & 7 & 0 & 25 \\ 2 & 2 & 2 & 2 & -3 & 9 \end{pmatrix}_{b_0}$$
- [41] $1^{113}321^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -304$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 2 & 0 & 9 \\ 1 & 2 & 0 & 2 & 0 & 12 \\ 1 & 2 & 2 & 0 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 6 & 9 & 12 & 9 & 0 & 61 \\ 2 & 2 & 2 & 2 & -3 & 15 \end{pmatrix}_{b_0}$$
- [42] $1^{12}321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -336$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.71 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 5 & 7 & 0 & 25 & 5 \\ 3 & 7 & 5 & 0 & 25 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 25 & 25 & 0 & 94 & 20 \\ 3 & 5 & 5 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$
- [43] $1^{131}1^{11}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -340$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.06 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 4 \\ 1 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 6 & 4 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 6 & 0 & 4 & 4 \\ 2 & 4 & 2 & 4 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$
- [44] $1^{11}31^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -384$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 4 & 0 & 6 & 2 \\ 2 & 6 & 4 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 9 & 0 & 13 & 7 \\ 2 & 4 & 2 & 4 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [45] $1^{113}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -392$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 9 & 8 & 0 & 40 & 12 \\ 2 & 2 & 2 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

[46] $1^1 31^{112} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -424$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 12.08 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 12 & 4 \\ 1 & 4 & 0 & 4 & 0 & 6 & 2 \\ 2 & 6 & 4 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 6 & 12 & 0 & 22 & 8 \\ 2 & 4 & 2 & 4 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

[47] $1^{12} 321^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -456$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 5 & 7 & 0 & 30 & 5 \\ 3 & 7 & 5 & 0 & 30 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 30 & 30 & 0 & 134 & 24 \\ 3 & 5 & 5 & -3 & 24 & 2 \end{pmatrix}_{b_0}$$

[48] $1^{114} 321^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -472$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 2 & 0 & 16 & 2 \\ 1 & 2 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 16 & 12 & 0 & 110 & 20 \\ 2 & 2 & 2 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[49] $1^1 31^{113} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -564$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.09 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 6 & 0 & 18 & 4 \\ 1 & 4 & 0 & 4 & 0 & 9 & 2 \\ 2 & 6 & 4 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 9 & 18 & 0 & 52 & 12 \\ 2 & 4 & 2 & 4 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

[50] $1^{114} 321^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -568$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 10 & 10 & 10 & 0 & 58 & 14 \\ 2 & 2 & 2 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[51] $1^{12} 31^{112} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -600$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 12.11 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 6 & 2 \\ 2 & 4 & 4 & 6 & 0 & 18 & 4 \\ 1 & 4 & 0 & 4 & 0 & 12 & 2 \\ 2 & 6 & 4 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 18 & 12 & 18 & 0 & 58 & 14 \\ 2 & 4 & 2 & 4 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{114} 321^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -624$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.82 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 2 & 0 & 13 & 2 \\ 1 & 2 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 13 & 11 & 0 & 81 & 17 \\ 2 & 2 & 2 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[53] $2^1 2^1 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -744$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 & 2 \\ 2 & 1 & 4 & 3 & 1 & 3 & 3 \\ 2 & 4 & 0 & 4 & 1 & 3 & 2 \\ 2 & 3 & 4 & 1 & 1 & 3 & 3 \\ 1 & 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 3 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[54] $1^{11} 31^{112} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -748$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.7$
 $T_{\min} = 12.05 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 6 & 0 & 15 & 4 \\ 1 & 4 & 0 & 4 & 0 & 9 & 2 \\ 2 & 6 & 4 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 15 & 9 & 15 & 0 & 37 & 11 \\ 2 & 4 & 2 & 4 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

[55] $1^1 31^{114} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -760$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.11 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 6 & 0 & 24 & 4 \\ 1 & 4 & 0 & 4 & 0 & 12 & 2 \\ 2 & 6 & 4 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 12 & 24 & 0 & 94 & 16 \\ 2 & 4 & 2 & 4 & -3 & 16 & 0 \end{pmatrix}_{b_0}$$

[56] $1^{13}31^{113}31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -960, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 6 & 0 & 27 & 4 \\ 1 & 4 & 0 & 4 & 0 & 18 & 2 \\ 2 & 6 & 4 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 18 & 27 & 0 & 133 & 21 \\ 2 & 4 & 2 & 4 & -3 & 21 & 0 \end{pmatrix}_{b_0}$$

[57] $2^1 2^1 31^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -1056, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 4 & 2 \\ 2 & 1 & 4 & 3 & 1 & 6 & 3 \\ 2 & 4 & 0 & 4 & 1 & 6 & 2 \\ 2 & 3 & 4 & 1 & 1 & 6 & 3 \\ 1 & 1 & 1 & 1 & -3 & 2 & -1 \\ 4 & 6 & 6 & 6 & 2 & 10 & 6 \\ 2 & 3 & 2 & 3 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{11}31^{113}31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1168, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 7 & 2 \\ 2 & 4 & 4 & 6 & 0 & 21 & 4 \\ 1 & 4 & 0 & 4 & 0 & 12 & 2 \\ 2 & 6 & 4 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 21 & 12 & 21 & 0 & 73 & 15 \\ 2 & 4 & 2 & 4 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{12}31^{113}31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 8 & 2 \\ 2 & 4 & 4 & 6 & 0 & 24 & 4 \\ 1 & 4 & 0 & 4 & 0 & 15 & 2 \\ 2 & 6 & 4 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 24 & 15 & 24 & 0 & 100 & 18 \\ 2 & 4 & 2 & 4 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

[60] $1^{14}31^{114}31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 12 & 2 \\ 2 & 4 & 4 & 6 & 0 & 36 & 4 \\ 1 & 4 & 0 & 4 & 0 & 24 & 2 \\ 2 & 6 & 4 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 36 & 24 & 36 & 0 & 238 & 28 \\ 2 & 4 & 2 & 4 & -3 & 28 & 0 \end{pmatrix}_{b_0}$$

[61] $1^{11}31^{114}31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1644, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 9 & 2 \\ 2 & 4 & 4 & 6 & 0 & 27 & 4 \\ 1 & 4 & 0 & 4 & 0 & 15 & 2 \\ 2 & 6 & 4 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 27 & 15 & 27 & 0 & 121 & 19 \\ 2 & 4 & 2 & 4 & -3 & 19 & 0 \end{pmatrix}_{b_0}$$

[62] $1^{13}31^{114}31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1996, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 11 & 2 \\ 2 & 4 & 4 & 6 & 0 & 33 & 4 \\ 1 & 4 & 0 & 4 & 0 & 21 & 2 \\ 2 & 6 & 4 & 4 & 0 & 33 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 33 & 21 & 33 & 0 & 193 & 25 \\ 2 & 4 & 2 & 4 & -3 & 25 & 0 \end{pmatrix}_{b_0}$$

[63] $1^{12}31^{114}31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2056, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 10 & 2 \\ 2 & 4 & 4 & 6 & 0 & 30 & 4 \\ 1 & 4 & 0 & 4 & 0 & 18 & 2 \\ 2 & 6 & 4 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 30 & 18 & 30 & 0 & 154 & 22 \\ 2 & 4 & 2 & 4 & -3 & 22 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [64] $1^{11} 4^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 \\ 4 & 5 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 7 & 0 & 8 & 6 \\ 3 & 4 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[65] 1^1 41^{111} 231^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 672, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.69 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 6 & 6 & 1 & 5 & 3 \\ 6 & 13 & 15 & 0 & 12 & 7 \\ 6 & 15 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 12 & 12 & 0 & 8 & 6 \\ 3 & 7 & 7 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

5.61 NHC(-2)(-3)(-2) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$

$$T = 9 \quad [0] 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -62, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 9 & 5 \\ 12 & 25 & 23 & 0 & 20 & 9 \\ 11 & 23 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 17 & 0 & 13 & 7 \\ 5 & 9 & 8 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[1] 1^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13.25 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 17 & 0 & 13 & 7 \\ 5 & 8 & 8 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[2] 1^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 17 & 5 \\ 12 & 25 & 23 & 0 & 36 & 9 \\ 11 & 23 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 36 & 33 & 0 & 49 & 13 \\ 5 & 9 & 8 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{112} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -342, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.7$$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 26 & 5 \\ 12 & 25 & 23 & 0 & 56 & 9 \\ 11 & 23 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 56 & 50 & 0 & 118 & 20 \\ 5 & 9 & 8 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] 1^1 321^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 5 & 7 & 0 & 5 & 5 & 5 \\ 3 & 7 & 5 & 0 & 5 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 5 & 5 & 0 & 2 & 4 & 4 \\ 2 & 5 & 5 & 0 & 4 & 0 & 2 \\ 3 & 5 & 5 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] 1^{11} 321^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 256, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 5 & 7 & 0 & 10 & 5 & 5 \\ 3 & 7 & 5 & 0 & 10 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 10 & 10 & 0 & 14 & 7 & 8 \\ 2 & 5 & 5 & 0 & 7 & 0 & 2 \\ 3 & 5 & 5 & -3 & 8 & 2 & 1 \end{pmatrix}_{b_0}$$

5.62 NHC(-2)(-3)(-2) + f₄ + su₂ + su₂

$$T = 9 \quad [2] \quad 1^{1112} 3^1 1^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 6 & 0 & 4 & 23 & 4 \\ 3 & 6 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 15 & 3 \\ 13 & 23 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & 4 & -3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -3 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 8 & 3 \\ 3 & 4 & 6 & 0 & 5 & 14 & 4 \\ 3 & 6 & 4 & 0 & 5 & 14 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 15 & 4 \\ 8 & 14 & 14 & 0 & 15 & 37 & 11 \\ 3 & 4 & 4 & -3 & 4 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1112} 3^1 1^{121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 7 & 3 \\ 3 & 4 & 6 & 0 & 14 & 13 & 4 \\ 3 & 6 & 4 & 0 & 14 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 14 & 0 & 37 & 33 & 11 \\ 7 & 13 & 13 & 0 & 33 & 28 & 10 \\ 3 & 4 & 4 & -3 & 11 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1121} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 6 & 0 & 8 & 9 & 4 \\ 3 & 6 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 9 & 0 & 12 & 13 & 7 \\ 3 & 4 & 4 & -3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{113} 3^1 1^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 9 & 3 \\ 3 & 4 & 6 & 0 & 5 & 17 & 4 \\ 3 & 6 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 15 & 4 \\ 9 & 17 & 17 & 0 & 15 & 49 & 13 \\ 3 & 4 & 4 & -3 & 4 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{1131} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 12 & 0 & 25 & 9 & 9 \\ 2 & 4 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & 4 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{1122} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 6 & 0 & 8 & 8 & 4 \\ 3 & 6 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 12 & 6 \\ 4 & 8 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & 4 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

- [8] $1^{111}31^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 0 & 5 & 9 & 4 \\ 3 & 6 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & 4 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{113}31^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.8$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 9 & 3 \\ 3 & 4 & 6 & 0 & 12 & 17 & 4 \\ 3 & 6 & 4 & 0 & 12 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 12 & 0 & 25 & 36 & 9 \\ 9 & 17 & 17 & 0 & 36 & 49 & 13 \\ 3 & 4 & 4 & -3 & 9 & 13 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{1123}31^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 6 & 0 & 8 & 22 & 4 \\ 3 & 6 & 4 & 0 & 8 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 30 & 6 \\ 12 & 22 & 22 & 0 & 30 & 85 & 17 \\ 3 & 4 & 4 & -3 & 6 & 17 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{1121}31^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 6 & 6 \\ 2 & 4 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & 4 & -3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$
- [14] $1^{1111}31^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -3 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$
- [13] $1^{112}31^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 6 & 0 & 5 & 13 & 4 \\ 3 & 6 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 12 & 4 \\ 7 & 13 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & 4 & -3 & 4 & 10 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{1113}31^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 6 & 0 & 4 & 32 & 4 \\ 3 & 6 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 21 & 3 \\ 18 & 32 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & 4 & -3 & 3 & 25 & 1 \end{pmatrix}_{b_0}$$
- [16] $1^{1111}31^{121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 5 & 3 \\ 3 & 4 & 6 & 0 & 14 & 9 & 4 \\ 3 & 6 & 4 & 0 & 14 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 14 & 0 & 37 & 24 & 11 \\ 5 & 9 & 9 & 0 & 24 & 13 & 7 \\ 3 & 4 & 4 & -3 & 11 & 7 & 1 \end{pmatrix}_{b_0}$$
- [15] $1^{1111}31^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

11.2

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & -3 & 3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{1123} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -312, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 6 & 0 & 8 & 13 & 4 \\ 3 & 6 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 18 & 6 \\ 7 & 13 & 13 & 0 & 18 & 28 & 10 \\ 3 & 4 & 4 & -3 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1123} 31^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 9 & 3 \\ 3 & 4 & 6 & 0 & 18 & 17 & 4 \\ 3 & 6 & 4 & 0 & 18 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 18 & 0 & 58 & 54 & 14 \\ 9 & 17 & 17 & 0 & 54 & 49 & 13 \\ 3 & 4 & 4 & -3 & 14 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{112} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 10 & 3 \\ 3 & 4 & 6 & 0 & 5 & 18 & 4 \\ 3 & 6 & 4 & 0 & 5 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 18 & 4 \\ 10 & 18 & 18 & 0 & 18 & 58 & 14 \\ 3 & 4 & 4 & -3 & 4 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1134} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 11 & 3 \\ 3 & 4 & 6 & 0 & 12 & 21 & 4 \\ 3 & 6 & 4 & 0 & 12 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 12 & 0 & 25 & 45 & 9 \\ 11 & 21 & 21 & 0 & 45 & 76 & 16 \\ 3 & 4 & 4 & -3 & 9 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1143} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 12 & 3 \\ 3 & 4 & 6 & 0 & 21 & 22 & 4 \\ 3 & 6 & 4 & 0 & 21 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 21 & 0 & 76 & 81 & 16 \\ 12 & 22 & 22 & 0 & 81 & 85 & 17 \\ 3 & 4 & 4 & -3 & 16 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{112} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -456, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 6 & 0 & 4 & 18 & 4 \\ 3 & 6 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 12 & 3 \\ 10 & 18 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & -3 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{1111} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & 4 & -3 & 7 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{1112} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 4 & 13 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 9 & 3 \\ 7 & 13 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & -3 & 3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{113} 3^1 1^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 12 & 3 \\ 3 & 4 & 6 & 0 & 5 & 22 & 4 \\ 3 & 6 & 4 & 0 & 5 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 21 & 4 \\ 12 & 22 & 22 & 0 & 21 & 85 & 17 \\ 3 & 4 & 4 & -3 & 4 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{1123} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 6 & 0 & 8 & 17 & 4 \\ 3 & 6 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 24 & 6 \\ 9 & 17 & 17 & 0 & 24 & 49 & 13 \\ 3 & 4 & 4 & -3 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{1114} 3^1 1^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 14 & 3 \\ 3 & 4 & 6 & 0 & 5 & 26 & 4 \\ 3 & 6 & 4 & 0 & 5 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 24 & 4 \\ 14 & 26 & 26 & 0 & 24 & 118 & 20 \\ 3 & 4 & 4 & -3 & 4 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{1114} 3^1 1^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 11 & 3 \\ 3 & 4 & 6 & 0 & 9 & 21 & 4 \\ 3 & 6 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 33 & 7 \\ 11 & 21 & 21 & 0 & 33 & 76 & 16 \\ 3 & 4 & 4 & -3 & 7 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{1112} 3^1 1^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 7 & 3 \\ 3 & 4 & 6 & 0 & 9 & 13 & 4 \\ 3 & 6 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 21 & 7 \\ 7 & 13 & 13 & 0 & 21 & 28 & 10 \\ 3 & 4 & 4 & -3 & 7 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{1124} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -624, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 6 & 0 & 8 & 26 & 4 \\ 3 & 6 & 4 & 0 & 8 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 36 & 6 \\ 14 & 26 & 26 & 0 & 36 & 118 & 20 \\ 3 & 4 & 4 & -3 & 6 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{1113} 3^1 1^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 9 & 3 \\ 3 & 4 & 6 & 0 & 9 & 17 & 4 \\ 3 & 6 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 27 & 7 \\ 9 & 17 & 17 & 0 & 27 & 49 & 13 \\ 3 & 4 & 4 & -3 & 7 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{1124} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -696, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 6 & 0 & 8 & 21 & 4 \\ 3 & 6 & 4 & 0 & 8 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 30 & 6 \\ 11 & 21 & 21 & 0 & 30 & 76 & 16 \\ 3 & 4 & 4 & -3 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{1113} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -704, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.66 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 6 & 0 & 4 & 17 & 4 \\ 3 & 6 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 12 & 3 \\ 9 & 17 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & 4 & -3 & 3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{1113} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -736, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 6 & 0 & 4 & 27 & 4 \\ 3 & 6 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 18 & 3 \\ 15 & 27 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & -3 & 3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{1122} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -792, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 10 & 3 \\ 3 & 4 & 6 & 0 & 13 & 18 & 4 \\ 3 & 6 & 4 & 0 & 13 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 13 & 0 & 28 & 42 & 10 \\ 10 & 18 & 18 & 0 & 42 & 58 & 14 \\ 3 & 4 & 4 & -3 & 10 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{1112} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -800, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 10 & 3 \\ 3 & 4 & 6 & 0 & 9 & 18 & 4 \\ 3 & 6 & 4 & 0 & 9 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 30 & 7 \\ 10 & 18 & 18 & 0 & 30 & 58 & 14 \\ 3 & 4 & 4 & -3 & 7 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{1114} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -888, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.69 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 4 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 15 & 3 \\ 11 & 21 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & 4 & -3 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{1122} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -896, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 6 & 0 & 13 & 13 & 4 \\ 3 & 6 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 13 & 0 & 28 & 30 & 10 \\ 7 & 13 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & 4 & -3 & 10 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{1113} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -896, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 6 & 0 & 4 & 22 & 4 \\ 3 & 6 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 15 & 3 \\ 12 & 22 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & -3 & 3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{1123} 31^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -1040, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.66 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 9 & 3 \\ 3 & 4 & 6 & 0 & 13 & 17 & 4 \\ 3 & 6 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 13 & 0 & 28 & 39 & 10 \\ 9 & 17 & 17 & 0 & 39 & 49 & 13 \\ 3 & 4 & 4 & -3 & 10 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{1124} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1056, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.68 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 11 & 3 \\ 3 & 4 & 6 & 0 & 13 & 21 & 4 \\ 3 & 6 & 4 & 0 & 13 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 13 & 0 & 28 & 48 & 10 \\ 11 & 21 & 21 & 0 & 48 & 76 & 16 \\ 3 & 4 & 4 & -3 & 10 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1114} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1128, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 6 & 0 & 4 & 36 & 4 \\ 3 & 6 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 24 & 3 \\ 20 & 36 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & -3 & 3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1133} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1152, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 11.66 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 12 & 3 \\ 3 & 4 & 6 & 0 & 17 & 22 & 4 \\ 3 & 6 & 4 & 0 & 17 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 17 & 0 & 49 & 66 & 13 \\ 12 & 22 & 22 & 0 & 66 & 85 & 17 \\ 3 & 4 & 4 & -3 & 13 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1113} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1184, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 12 & 3 \\ 3 & 4 & 6 & 0 & 9 & 22 & 4 \\ 3 & 6 & 4 & 0 & 9 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 36 & 7 \\ 12 & 22 & 22 & 0 & 36 & 85 & 17 \\ 3 & 4 & 4 & -3 & 7 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1133} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1296, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 11.67 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 9 & 3 \\ 3 & 4 & 6 & 0 & 17 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 17 & 0 & 49 & 51 & 13 \\ 9 & 17 & 17 & 0 & 51 & 49 & 13 \\ 3 & 4 & 4 & -3 & 13 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1114} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1320, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.65 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 6 & 0 & 4 & 26 & 4 \\ 3 & 6 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 18 & 3 \\ 14 & 26 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & 4 & -3 & 3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1144} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1368, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.68 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 14 & 3 \\ 3 & 4 & 6 & 0 & 21 & 26 & 4 \\ 3 & 6 & 4 & 0 & 21 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 21 & 0 & 76 & 96 & 16 \\ 14 & 26 & 26 & 0 & 96 & 118 & 20 \\ 3 & 4 & 4 & -3 & 16 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1114} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1400, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 6 & 0 & 4 & 31 & 4 \\ 3 & 6 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 21 & 3 \\ 17 & 31 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & -3 & 3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1123} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1400, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 12 & 3 \\ 3 & 4 & 6 & 0 & 13 & 22 & 4 \\ 3 & 6 & 4 & 0 & 13 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 13 & 0 & 28 & 51 & 10 \\ 12 & 22 & 22 & 0 & 51 & 85 & 17 \\ 3 & 4 & 4 & -3 & 10 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1134} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1424, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.68 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 11 & 3 \\ 3 & 4 & 6 & 0 & 17 & 21 & 4 \\ 3 & 6 & 4 & 0 & 17 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 17 & 0 & 49 & 63 & 13 \\ 11 & 21 & 21 & 0 & 63 & 76 & 16 \\ 3 & 4 & 4 & -3 & 13 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1114} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1440, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.65 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 14 & 3 \\ 3 & 4 & 6 & 0 & 9 & 26 & 4 \\ 3 & 6 & 4 & 0 & 9 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 42 & 7 \\ 14 & 26 & 26 & 0 & 42 & 118 & 20 \\ 3 & 4 & 4 & -3 & 7 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1144} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1664, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.69 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 11 & 3 \\ 3 & 4 & 6 & 0 & 21 & 21 & 4 \\ 3 & 6 & 4 & 0 & 21 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 21 & 0 & 76 & 78 & 16 \\ 11 & 21 & 21 & 0 & 78 & 76 & 16 \\ 3 & 4 & 4 & -3 & 16 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{1134} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1856, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.66 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 14 & 3 \\ 3 & 4 & 6 & 0 & 17 & 26 & 4 \\ 3 & 6 & 4 & 0 & 17 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 17 & 0 & 49 & 78 & 13 \\ 14 & 26 & 26 & 0 & 78 & 118 & 20 \\ 3 & 4 & 4 & -3 & 13 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{1124} 31^{112} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -1880, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.65 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 14 & 3 \\ 3 & 4 & 6 & 0 & 13 & 26 & 4 \\ 3 & 6 & 4 & 0 & 13 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 13 & 0 & 28 & 60 & 10 \\ 14 & 26 & 26 & 0 & 60 & 118 & 20 \\ 3 & 4 & 4 & -3 & 10 & 20 & 1 \end{pmatrix}_{b_0}$$

5.63 NHC(-2)(-3)(-2) + $\mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{11111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 672, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 4 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & 4 & -3 & 3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

5.64 NHC(-2)(-3)(-2) + f₄ + su₂ + su₂→(-3)

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 6 & 0 & 8 & 3 & 4 \\ 3 & 6 & 4 & 0 & 8 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 4 & 6 \\ 2 & 3 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 6 & 0 & 14 & 3 & 4 \\ 3 & 6 & 4 & 0 & 14 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 14 & 0 & 37 & 8 & 11 \\ 2 & 3 & 3 & 0 & 8 & 0 & 2 \\ 3 & 4 & 4 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 5 & 7 \\ 2 & 3 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 13 & 3 & 4 \\ 3 & 6 & 4 & 0 & 13 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 13 & 0 & 28 & 7 & 10 \\ 2 & 3 & 3 & 0 & 7 & 0 & 2 \\ 3 & 4 & 4 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 6 & 0 & 17 & 3 & 4 \\ 3 & 6 & 4 & 0 & 17 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 17 & 0 & 49 & 9 & 13 \\ 2 & 3 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -3 & 13 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11}3^{11}1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 6 & 0 & 18 & 3 & 4 \\ 3 & 6 & 4 & 0 & 18 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 18 & 0 & 58 & 10 & 14 \\ 2 & 3 & 3 & 0 & 10 & 0 & 2 \\ 3 & 4 & 4 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{114}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -552, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 6 & 0 & 21 & 3 & 4 \\ 3 & 6 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 21 & 0 & 76 & 11 & 16 \\ 2 & 3 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & 4 & -3 & 16 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{113}3^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -824, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.70 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 6 & 0 & 22 & 3 & 4 \\ 3 & 6 & 4 & 0 & 22 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 22 & 0 & 85 & 12 & 17 \\ 2 & 3 & 3 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{114}3^{11}1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -1072, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.70 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 6 & 0 & 26 & 3 & 4 \\ 3 & 6 & 4 & 0 & 26 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 26 & 0 & 118 & 14 & 20 \\ 2 & 3 & 3 & 0 & 14 & 0 & 2 \\ 3 & 4 & 4 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

5.65 NHC(-2)(-3)(-2) + $\mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_{2n3\text{mix}}$

$$T = 9 \quad [0] \ 1^{111}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -104, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.81 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 6 & 0 & 5 & 7 & 4 \\ 3 & 6 & 4 & 0 & 5 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 6 & 4 \\ 4 & 7 & 7 & 0 & 6 & 7 & 5 \\ 3 & 4 & 4 & -3 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{113}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -112, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 4 & 3 \\ 3 & 4 & 6 & 0 & 12 & 7 & 4 \\ 3 & 6 & 4 & 0 & 12 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 12 & 0 & 25 & 15 & 9 \\ 4 & 7 & 7 & 0 & 15 & 7 & 5 \\ 3 & 4 & 4 & -3 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -216, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 6 & 0 & 8 & 16 & 4 \\ 3 & 6 & 4 & 0 & 8 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 22 & 6 \\ 9 & 16 & 16 & 0 & 22 & 44 & 12 \\ 3 & 4 & 4 & -3 & 6 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{112}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -232, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.76 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 6 & 0 & 8 & 7 & 4 \\ 3 & 6 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 10 & 6 \\ 4 & 7 & 7 & 0 & 10 & 7 & 5 \\ 3 & 4 & 4 & -3 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112}3^{11}1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{f}}_4 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -288, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} =$$

11.2

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 6 & 3 \\ 3 & 4 & 6 & 0 & 9 & 11 & 4 \\ 3 & 6 & 4 & 0 & 9 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 17 & 7 \\ 6 & 11 & 11 & 0 & 17 & 20 & 8 \\ 3 & 4 & 4 & -3 & 7 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}3^1}1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 4 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 8 & 3 \\ 7 & 12 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & 4 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1111}3^1}1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 6 & 0 & 4 & 7 & 4 \\ 3 & 6 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 5 & 3 \\ 4 & 7 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & 4 & -3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1122}3^1}1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 6 & 0 & 8 & 11 & 4 \\ 3 & 6 & 4 & 0 & 8 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 16 & 6 \\ 6 & 11 & 11 & 0 & 16 & 20 & 8 \\ 3 & 4 & 4 & -3 & 6 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1111}3^1}1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 6 & 0 & 9 & 7 & 4 \\ 3 & 6 & 4 & 0 & 9 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 11 & 7 \\ 4 & 7 & 7 & 0 & 11 & 7 & 5 \\ 3 & 4 & 4 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1122}3^1}1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -408, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 6 & 0 & 4 & 11 & 4 \\ 3 & 6 & 4 & 0 & 4 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 8 & 3 \\ 6 & 11 & 11 & 0 & 8 & 20 & 8 \\ 3 & 4 & 4 & -3 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1123}3^1}1^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 4 & 3 \\ 3 & 4 & 6 & 0 & 18 & 7 & 4 \\ 3 & 6 & 4 & 0 & 18 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 18 & 0 & 58 & 22 & 14 \\ 4 & 7 & 7 & 0 & 22 & 7 & 5 \\ 3 & 4 & 4 & -3 & 14 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1113}3^1}1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 6 & 0 & 4 & 15 & 4 \\ 3 & 6 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 11 & 3 \\ 8 & 15 & 15 & 0 & 11 & 39 & 11 \\ 3 & 4 & 4 & -3 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1123}3^1}1^{\overline{12}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

11.2

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 6 & 0 & 4 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 14 & 3 \\ 12 & 21 & 21 & 0 & 14 & 80 & 16 \\ 3 & 4 & 4 & -3 & 3 & 16 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{1123}3^11^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -552, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 6 & 3 \\ 3 & 4 & 6 & 0 & 13 & 11 & 4 \\ 3 & 6 & 4 & 0 & 13 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 13 & 0 & 28 & 25 & 10 \\ 6 & 11 & 11 & 0 & 25 & 20 & 8 \\ 3 & 4 & 4 & -3 & 10 & 8 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{1123}3^11^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -552, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 6 & 0 & 13 & 7 & 4 \\ 3 & 6 & 4 & 0 & 13 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 13 & 0 & 28 & 16 & 10 \\ 4 & 7 & 7 & 0 & 16 & 7 & 5 \\ 3 & 4 & 4 & -3 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{1114}3^11^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -568, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 6 & 0 & 4 & 19 & 4 \\ 3 & 6 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 14 & 3 \\ 10 & 19 & 19 & 0 & 14 & 64 & 14 \\ 3 & 4 & 4 & -3 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{1112}3^11^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -648, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 6 & 0 & 4 & 16 & 4 \\ 3 & 6 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 11 & 3 \\ 9 & 16 & 16 & 0 & 11 & 44 & 12 \\ 3 & 4 & 4 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{1131}3^11^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 6 & 0 & 17 & 7 & 4 \\ 3 & 6 & 4 & 0 & 17 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 17 & 0 & 49 & 21 & 13 \\ 4 & 7 & 7 & 0 & 21 & 7 & 5 \\ 3 & 4 & 4 & -3 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{1141}3^11^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -808, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 4 & 3 \\ 3 & 4 & 6 & 0 & 21 & 7 & 4 \\ 3 & 6 & 4 & 0 & 21 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 21 & 0 & 76 & 26 & 16 \\ 4 & 7 & 7 & 0 & 26 & 7 & 5 \\ 3 & 4 & 4 & -3 & 16 & 5 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{1113}3^11^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -848, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 4 & 3 \\ 3 & 4 & 6 & 0 & 22 & 7 & 4 \\ 3 & 6 & 4 & 0 & 22 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 22 & 0 & 85 & 27 & 17 \\ 4 & 7 & 7 & 0 & 27 & 7 & 5 \\ 3 & 4 & 4 & -3 & 17 & 5 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{1113}3^11^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -976, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

11.2

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 4 & 20 & 4 \\ 3 & 6 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 14 & 3 \\ 11 & 20 & 20 & 0 & 14 & 69 & 15 \\ 3 & 4 & 4 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1113311^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -1104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 6 & 0 & 4 & 25 & 4 \\ 3 & 6 & 4 & 0 & 4 & 25 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 17 & 3 \\ 14 & 25 & 25 & 0 & 17 & 111 & 19 \\ 3 & 4 & 4 & -3 & 3 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{1114311^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -1224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 4 & 3 \\ 3 & 4 & 6 & 0 & 26 & 7 & 4 \\ 3 & 6 & 4 & 0 & 26 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 26 & 0 & 118 & 32 & 20 \\ 4 & 7 & 7 & 0 & 32 & 7 & 5 \\ 3 & 4 & 4 & -3 & 20 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{1114311^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -1288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 6 & 0 & 4 & 24 & 4 \\ 3 & 6 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 17 & 3 \\ 13 & 24 & 24 & 0 & 17 & 100 & 18 \\ 3 & 4 & 4 & -3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{1114311^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -1656, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 6 & 0 & 4 & 29 & 4 \\ 3 & 6 & 4 & 0 & 4 & 29 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 20 & 3 \\ 16 & 29 & 29 & 0 & 20 & 148 & 22 \\ 3 & 4 & 4 & -3 & 3 & 22 & 1 \end{pmatrix}_{b_0}$$

5.66 NHC(-2)(-3)(-2) + $\mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{111311^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.34 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 4 & 3 & 4 \\ 2 & 4 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111311^{12}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 6 & 0 & 14 & 4 & 4 \\ 3 & 6 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 14 & 14 & 0 & 37 & 9 & 11 \\ 2 & 4 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -3 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1111311^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 5 & 2 & 3 \\ 4 & 9 & 7 & 0 & 12 & 4 & 5 \\ 3 & 7 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 12 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 4 & 0 & 6 & 0 & 2 \\ 3 & 5 & 4 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1141} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -160, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 6 & 0 & 16 & 4 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 16 & 16 & 0 & 46 & 12 & 12 \\ 2 & 4 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -3 & 12 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112} 3 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -167, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 8 & 2 & 3 \\ 4 & 9 & 7 & 0 & 20 & 4 & 5 \\ 3 & 7 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 20 & 14 & 0 & 37 & 9 & 11 \\ 2 & 4 & 4 & 0 & 9 & 0 & 2 \\ 3 & 5 & 4 & -3 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -172, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 4 & 5 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 3 & 3 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & 4 & -3 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1121} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -200, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 6 & 0 & 8 & 9 & 4 \\ 3 & 6 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 12 & 6 \\ 5 & 9 & 9 & 0 & 12 & 12 & 6 \\ 3 & 4 & 4 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112} 3 1^{121} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -200, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 10 & 2 & 3 \\ 4 & 9 & 7 & 0 & 24 & 4 & 5 \\ 3 & 7 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 24 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 4 & 0 & 12 & 0 & 2 \\ 3 & 5 & 4 & -3 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1131} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -244, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.43 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 12 & 12 & 0 & 25 & 9 & 9 \\ 2 & 4 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -3 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1121} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -304, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 10 & 6 & 6 \\ 2 & 4 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & -3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1121} 3 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -340, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 6 & 0 & 23 & 4 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 23 & 23 & 0 & 100 & 15 & 18 \\ 2 & 4 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -3 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1111} 3 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -340, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} =$$

12.2

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & -3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -464, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 6 & 7 \\ 2 & 4 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & -3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1121} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -788, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 13 & 4 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 13 & 13 & 0 & 28 & 9 & 10 \\ 2 & 4 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -3 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1131} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -788, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 6 & 0 & 32 & 4 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 18 & 32 & 32 & 0 & 193 & 21 & 25 \\ 2 & 4 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -3 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1121} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -800, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 18 & 0 & 58 & 12 & 14 \\ 2 & 4 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -3 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{1131} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1088, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 17 & 17 & 0 & 49 & 12 & 13 \\ 2 & 4 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -3 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1131} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1360, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 27 & 0 & 133 & 18 & 21 \\ 2 & 4 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -3 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{1141} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 & 4 \\ 3 & 6 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 21 & 0 & 76 & 15 & 16 \\ 2 & 4 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -3 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11\bar{1}\bar{1}} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1460, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 6 & 0 & 22 & 4 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 22 & 22 & 0 & 85 & 15 & 17 \\ 2 & 4 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -3 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11\bar{1}\bar{1}} 3^1 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2096, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 6 & 0 & 26 & 4 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 26 & 26 & 0 & 118 & 18 & 20 \\ 2 & 4 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -3 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11\bar{1}\bar{1}} 3^1 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.13 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 20 & 36 & 36 & 0 & 238 & 24 & 28 \\ 2 & 4 & 4 & 0 & 24 & 0 & 2 \\ 3 & 4 & 4 & -3 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11\bar{1}\bar{1}} 3^1 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -2356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 6 & 0 & 31 & 4 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 31 & 31 & 0 & 172 & 21 & 24 \\ 2 & 4 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -3 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [24] \quad 1^{11\bar{1}\bar{1}} 4^1 1^{131^{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 5 & 2 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 & 4 \\ 4 & 5 & 4 & 0 & 7 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 7 & 7 & 0 & 8 & 5 & 6 \\ 2 & 4 & 3 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & -2 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

5.67 NHC(-2)(-3)(-2) + $\mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11\bar{1}\bar{1}\bar{1}} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 672, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 9 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 & 6 \\ 2 & 4 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & 4 & -3 & 3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

5.68 NHC(-2)(-3)(-2) + $\mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \quad 1^{11\bar{1}\bar{1}} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 3 \\ 4 & 9 & 7 & 0 & 4 & 5 \\ 3 & 7 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 5 & 4 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 21, \ \Delta = -111, \ \det = 104, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.69 \rightarrow 12, \ T_{\max} = 21, \ T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \ 1^{13}1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -90, \ \det = -232, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.14 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 2 & 4 & 1 & 3 \\ 4 & 8 & 6 & 10 & 0 & 6 \\ 2 & 6 & 1 & 6 & 0 & 3 \\ 4 & 10 & 6 & 8 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 3 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{13}1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -90, \ \det = -268, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.09 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 0 & 4 & 0 & 2 & 2 \\ 2 & 6 & 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 2 & 3 & 0 & 0 & 2 \\ 2 & 4 & 2 & 4 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

5.69 NHC(-2)(-3)(-2) + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -110, \ \det = -134, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.69 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 6 & 5 \\ 12 & 25 & 23 & 0 & 12 & 9 \\ 11 & 23 & 20 & 0 & 11 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 11 & 0 & 4 & 4 \\ 5 & 9 & 8 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

5.70 NHC(-2)(-3)(-2) + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -104, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 5 & 4 \\ 3 & 6 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 \\ 3 & 5 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -112, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 6 & 0 & 3 & 8 & 4 \\ 3 & 6 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 4 & 2 \\ 4 & 8 & 8 & 0 & 4 & 10 & 6 \\ 3 & 4 & 4 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -184, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 6 & 0 & 3 & 14 & 4 \\ 3 & 6 & 4 & 0 & 3 & 14 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 8 & 2 \\ 8 & 14 & 14 & 0 & 8 & 37 & 11 \\ 3 & 4 & 4 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -184$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & 4 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$

[4] $1^{111}3^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -288$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 5 & 2 \\ 5 & 9 & 9 & 0 & 5 & 13 & 7 \\ 3 & 4 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$

[5] $1^{112}3^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -424$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 13 & 4 \\ 3 & 6 & 4 & 0 & 3 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 7 & 2 \\ 7 & 13 & 13 & 0 & 7 & 28 & 10 \\ 3 & 4 & 4 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$

[6] $1^{113}3^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -512$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 6 & 0 & 3 & 17 & 4 \\ 3 & 6 & 4 & 0 & 3 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 9 & 2 \\ 9 & 17 & 17 & 0 & 9 & 49 & 13 \\ 3 & 4 & 4 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$

[7] $1^{112}3^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -528$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 6 & 0 & 3 & 18 & 4 \\ 3 & 6 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 10 & 2 \\ 10 & 18 & 18 & 0 & 10 & 58 & 14 \\ 3 & 4 & 4 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$

[8] $1^{114}3^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -552$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.71 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 3 & 21 & 4 \\ 3 & 6 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 11 & 2 \\ 11 & 21 & 21 & 0 & 11 & 76 & 16 \\ 3 & 4 & 4 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$

[9] $1^{113}3^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -824$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 6 & 0 & 3 & 22 & 4 \\ 3 & 6 & 4 & 0 & 3 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 12 & 2 \\ 12 & 22 & 22 & 0 & 12 & 85 & 17 \\ 3 & 4 & 4 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$

[10] $1^{114}3^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -1072$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 6 & 0 & 3 & 26 & 4 \\ 3 & 6 & 4 & 0 & 3 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 14 & 2 \\ 14 & 26 & 26 & 0 & 14 & 118 & 20 \\ 3 & 4 & 4 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$

5.71 NHC(-2)(-3)(-2) + f₄ + su₂→(-3) + su₃

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.11 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 2 & 3 \\ 4 & 9 & 7 & 0 & 4 & 4 & 5 \\ 3 & 7 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 4 & 0 & 2 & 0 & 2 \\ 3 & 5 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 6 & 0 & 3 & 8 & 4 \\ 3 & 6 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 4 & 2 \\ 4 & 8 & 8 & 0 & 4 & 9 & 5 \\ 3 & 4 & 4 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.11 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 4 & 4 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

5.72 NHC(-2)(-3)(-2) + f₄ + su₂n3mix

$$T = 8 \quad [0] \ 1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 8 & 8 & 0 & 12 & 6 \\ 3 & 4 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 0 & 7 & 4 \\ 3 & 6 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 5 \\ 3 & 4 & 4 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{11}1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 17 & 17 & 0 & 55 & 13 \\ 3 & 4 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{112}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 11 & 4 \\ 3 & 6 & 4 & 0 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 11 & 0 & 20 & 8 \\ 3 & 4 & 4 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{113}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 15 & 0 & 39 & 11 \\ 3 & 4 & 4 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{114}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 19 & 4 \\ 3 & 6 & 4 & 0 & 19 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 19 & 0 & 64 & 14 \\ 3 & 4 & 4 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 396, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 16 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 16 & 0 & 44 & 12 \\ 3 & 4 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{112}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 456, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 21 & 21 & 0 & 80 & 16 \\ 3 & 4 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{113}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 560, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 6 & 0 & 20 & 4 \\ 3 & 6 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 20 & 20 & 0 & 69 & 15 \\ 3 & 4 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{114}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 716, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.72 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 24 & 0 & 100 & 18 \\ 3 & 4 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{113}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 740, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 25 & 4 \\ 3 & 6 & 4 & 0 & 25 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 25 & 25 & 0 & 111 & 19 \\ 3 & 4 & 4 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{114}3^{11}1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 21, \Delta = -111, \det = 1016, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 21, T \in [12, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 6 & 0 & 29 & 4 \\ 3 & 6 & 4 & 0 & 29 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 29 & 29 & 0 & 148 & 22 \\ 3 & 4 & 4 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [13] \ 1^{11}31^{11}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 5 & 4 \\ 1 & 4 & 0 & 4 & 0 & 4 & 2 \\ 2 & 6 & 4 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 4 & 5 & 0 & 4 & 4 \\ 2 & 4 & 2 & 4 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{12}31^{111}3^{11}1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 5 & 2 \\ 2 & 4 & 4 & 6 & 0 & 14 & 4 \\ 1 & 4 & 0 & 4 & 0 & 10 & 2 \\ 2 & 6 & 4 & 4 & 0 & 14 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 14 & 10 & 14 & 0 & 37 & 11 \\ 2 & 4 & 2 & 4 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

[15] $1^1 3^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -292$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.18 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 8 & 4 \\ 1 & 4 & 0 & 4 & 0 & 4 & 2 \\ 2 & 6 & 4 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 4 & 8 & 0 & 9 & 5 \\ 2 & 4 & 2 & 4 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[16] $1^1 3^1 1^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -388$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 12.05 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 12 & 4 \\ 1 & 4 & 0 & 4 & 0 & 8 & 2 \\ 2 & 6 & 4 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 8 & 12 & 0 & 25 & 9 \\ 2 & 4 & 2 & 4 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

[17] $1^1 3^1 1^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -408$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.11 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 4 & 0 & 5 & 2 \\ 2 & 6 & 4 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 5 & 9 & 0 & 12 & 6 \\ 2 & 4 & 2 & 4 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[18] $1^1 3^1 1^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -496$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 12.06 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 4 & 2 \\ 2 & 4 & 4 & 6 & 0 & 11 & 4 \\ 1 & 4 & 0 & 4 & 0 & 7 & 2 \\ 2 & 6 & 4 & 4 & 0 & 11 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 11 & 7 & 11 & 0 & 20 & 8 \\ 2 & 4 & 2 & 4 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [19] $1^1 1^1 3^1 1^1 4^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -81$, $\det = 41$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 10.2$

$T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 4 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

5.73 NHC(-2)(-3)(-2) + f₄ + su₂n3mix + su₂

$T = 9$ [0] $1^1 1^1 3^1 1^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -104$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 6 & 0 & 7 & 5 & 4 \\ 3 & 6 & 4 & 0 & 7 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 15 & 5 \\ 3 & 5 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & 4 & -3 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

[1] $1^1 1^1 1^1 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -112$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 6 & 3 \\ 3 & 4 & 6 & 0 & 7 & 12 & 4 \\ 3 & 6 & 4 & 0 & 7 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 15 & 5 \\ 6 & 12 & 12 & 0 & 15 & 25 & 9 \\ 3 & 4 & 4 & -3 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

[2] $1^1 1^1 1^1 2^1 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -216$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.7$

$T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 6 & 0 & 16 & 8 & 4 \\ 3 & 6 & 4 & 0 & 16 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 16 & 0 & 44 & 22 & 12 \\ 4 & 8 & 8 & 0 & 22 & 10 & 6 \\ 3 & 4 & 4 & -3 & 12 & 6 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{1112}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 6 & 0 & 7 & 8 & 4 \\ 3 & 6 & 4 & 0 & 7 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 10 & 5 \\ 4 & 8 & 8 & 0 & 10 & 10 & 6 \\ 3 & 4 & 4 & -3 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{1121}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 5 & 3 \\ 3 & 4 & 6 & 0 & 11 & 9 & 4 \\ 3 & 6 & 4 & 0 & 11 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 11 & 0 & 20 & 17 & 8 \\ 5 & 9 & 9 & 0 & 17 & 13 & 7 \\ 3 & 4 & 4 & -3 & 8 & 7 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{1111}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 6 & 0 & 7 & 4 & 4 \\ 3 & 6 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 5 & 5 \\ 2 & 4 & 4 & 0 & 5 & 1 & 3 \\ 3 & 4 & 4 & -3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{1111}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 12 & 0 & 25 & 8 & 9 \\ 2 & 4 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & 4 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{1122}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -336, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.81 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 4 & 3 \\ 3 & 4 & 6 & 0 & 11 & 8 & 4 \\ 3 & 6 & 4 & 0 & 11 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 11 & 0 & 20 & 16 & 8 \\ 4 & 8 & 8 & 0 & 16 & 10 & 6 \\ 3 & 4 & 4 & -3 & 8 & 6 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{1111}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 6 & 0 & 7 & 9 & 4 \\ 3 & 6 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 11 & 5 \\ 5 & 9 & 9 & 0 & 11 & 13 & 7 \\ 3 & 4 & 4 & -3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{1121}3^11^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -408, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 2 & 3 \\ 3 & 4 & 6 & 0 & 11 & 4 & 4 \\ 3 & 6 & 4 & 0 & 11 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 11 & 0 & 20 & 8 & 8 \\ 2 & 4 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & 4 & -3 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{1112}3^11^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2$ $T = 9, T_{\max} = 12, \Delta = -81, \det = -424, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$

$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 10 & 3 \\ 3 & 4 & 6 & 0 & 7 & 18 & 4 \\ 3 & 6 & 4 & 0 & 7 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 22 & 5 \\ 10 & 18 & 18 & 0 & 22 & 58 & 14 \\ 3 & 4 & 4 & -3 & 5 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1113} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 & 4 \\ 3 & 6 & 4 & 0 & 15 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 15 & 0 & 39 & 11 & 11 \\ 2 & 4 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & 4 & -3 & 11 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{1121} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 & 4 \\ 3 & 6 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 21 & 21 & 0 & 80 & 14 & 16 \\ 2 & 4 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & 4 & -3 & 16 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1112} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -552, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 6 & 0 & 7 & 13 & 4 \\ 3 & 6 & 4 & 0 & 7 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 16 & 5 \\ 7 & 13 & 13 & 0 & 16 & 28 & 10 \\ 3 & 4 & 4 & -3 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1122} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -552, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 7 & 3 \\ 3 & 4 & 6 & 0 & 11 & 13 & 4 \\ 3 & 6 & 4 & 0 & 11 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 11 & 11 & 0 & 20 & 25 & 8 \\ 7 & 13 & 13 & 0 & 25 & 28 & 10 \\ 3 & 4 & 4 & -3 & 8 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1141} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -568, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 6 & 0 & 19 & 4 & 4 \\ 3 & 6 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 19 & 19 & 0 & 64 & 14 & 14 \\ 2 & 4 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & 4 & -3 & 14 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{1121} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -648, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 6 & 0 & 16 & 4 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 16 & 16 & 0 & 44 & 11 & 12 \\ 2 & 4 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & 4 & -3 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{1113} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.73 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 6 & 0 & 7 & 17 & 4 \\ 3 & 6 & 4 & 0 & 7 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 21 & 5 \\ 9 & 17 & 17 & 0 & 21 & 49 & 13 \\ 3 & 4 & 4 & -3 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{1114} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -808, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 11 & 3 \\ 3 & 4 & 6 & 0 & 7 & 21 & 4 \\ 3 & 6 & 4 & 0 & 7 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 26 & 5 \\ 11 & 21 & 21 & 0 & 26 & 76 & 16 \\ 3 & 4 & 4 & -3 & 5 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{11131} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -848, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.74 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 12 & 3 \\ 3 & 4 & 6 & 0 & 7 & 22 & 4 \\ 3 & 6 & 4 & 0 & 7 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 27 & 5 \\ 12 & 22 & 22 & 0 & 27 & 85 & 17 \\ 3 & 4 & 4 & -3 & 5 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1131} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -976, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 6 & 0 & 20 & 4 & 4 \\ 3 & 6 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 20 & 20 & 0 & 69 & 14 & 15 \\ 2 & 4 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & 4 & -3 & 15 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1131} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -1104, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.68 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 6 & 0 & 25 & 4 & 4 \\ 3 & 6 & 4 & 0 & 25 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 25 & 25 & 0 & 111 & 17 & 19 \\ 2 & 4 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & 4 & -3 & 19 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{1141} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -1224, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.73 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 14 & 3 \\ 3 & 4 & 6 & 0 & 7 & 26 & 4 \\ 3 & 6 & 4 & 0 & 7 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 32 & 5 \\ 14 & 26 & 26 & 0 & 32 & 118 & 20 \\ 3 & 4 & 4 & -3 & 5 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{1141} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -1288, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.72 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 & 4 \\ 3 & 6 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 24 & 0 & 100 & 17 & 18 \\ 2 & 4 & 4 & 0 & 17 & 1 & 3 \\ 3 & 4 & 4 & -3 & 18 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{1141} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -81, \ \det = -1656, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 6 & 0 & 29 & 4 & 4 \\ 3 & 6 & 4 & 0 & 29 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 16 & 29 & 29 & 0 & 148 & 20 & 22 \\ 2 & 4 & 4 & 0 & 20 & 1 & 3 \\ 3 & 4 & 4 & -3 & 22 & 3 & 1 \end{pmatrix}_{b_0}$$

5.74 NHC(-2)(-3)(-2) + f₄ + su₂n₃mix + su₃

$$T = 9 \quad [0] \ 1^{1111} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -90, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.34 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 17 & 17 & 0 & 55 & 11 & 13 \\ 2 & 4 & 4 & 0 & 11 & 0 & 2 \\ 3 & 4 & 4 & -3 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1111} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -90, \ \det = -468, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} =$$

12.2

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 6 & 0 & 7 & 4 & 4 \\ 3 & 6 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 7 & 0 & 7 & 5 & 5 \\ 2 & 4 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 12 & 0 & 25 & 8 & 9 \\ 2 & 4 & 4 & 0 & 8 & 0 & 2 \\ 3 & 4 & 4 & -3 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

5.75 NHC(-2)(-3)(-2) + $\mathfrak{f}_4$ + $\mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 3 \\ 4 & 9 & 7 & 0 & 4 & 5 \\ 3 & 7 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 3 & 5 & 4 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 1 & 3 \\ 4 & 9 & 11 & 4 & 0 & 5 \\ 4 & 11 & 9 & 4 & 0 & 5 \\ 2 & 4 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 5 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 0 & 3 & 3 \\ 3 & 4 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{113}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 12 & 0 & 24 & 8 \\ 3 & 4 & 4 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 8 & 0 & 9 & 5 \\ 3 & 4 & 4 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 35, T \in [13, 35]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 344, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 13 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 13 & 0 & 27 & 9 \\ 3 & 4 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{113}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 444, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 17 & 0 & 48 & 12 \\ 3 & 4 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [9] \ 1^{113}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 1 \\ 1 & 2 & 0 & 2 & 0 & 3 \\ 1 & 2 & 2 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 1 & 1 & 3 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & -3 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{113}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 5 & 7 & 0 & 5 & 5 \\ 3 & 7 & 5 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 5 & 0 & 1 & 3 \\ 3 & 5 & 5 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111}31^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 4 & 0 & 3 \\ 1 & 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 1 & 3 & 3 & 3 & 0 & 2 \\ 2 & 3 & 3 & 3 & -3 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 1 & 3 \\ 3 & 4 & 4 & 6 & 0 & 4 \\ 2 & 4 & 1 & 4 & 0 & 3 \\ 3 & 6 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 3 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -156, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 3 \\ 1 & 2 & 0 & 2 & 0 & 4 \\ 1 & 2 & 2 & 0 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 2 & 3 & 4 & 3 & 0 & 4 \\ 2 & 2 & 2 & 2 & -3 & 4 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{112}321^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -228, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 \\ 1 & 0 & 2 & 2 & 0 & 6 \\ 1 & 2 & 0 & 2 & 0 & 8 \\ 1 & 2 & 2 & 0 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 4 & 6 & 8 & 6 & 0 & 25 \\ 2 & 2 & 2 & 2 & -3 & 9 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{113}31^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -232, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 \\ 2 & 4 & 4 & 6 & 0 & 3 \\ 1 & 4 & 0 & 4 & 0 & 3 \\ 2 & 6 & 4 & 4 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 1 & 3 & 3 & 3 & 0 & 2 \\ 2 & 4 & 2 & 4 & -3 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{112}321^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -252, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 5 & 0 & 12 & 6 \\ 2 & 2 & 2 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -496$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 0 & 6 & 4 \\ 1 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 6 & 4 & 4 & 0 & 6 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 3 & 6 & 0 & 3 & 3 \\ 2 & 4 & 2 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{11}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 15$, $\Delta = -97$, $\det = -540$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 12.22 \rightarrow 13$, $T_{\max} = 15$, $T \in [13, 15]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 3 & 2 \\ 2 & 4 & 4 & 6 & 0 & 9 & 4 \\ 1 & 4 & 0 & 4 & 0 & 6 & 2 \\ 2 & 6 & 4 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 6 & 9 & 0 & 12 & 6 \\ 2 & 4 & 2 & 4 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [19] $1^{11}4^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -88$, $\det = 24$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 13$, $T_{\max} = 14$, $T \in [13, 14]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 4 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{11}4^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 10$, $T_{\max} = 14$, $\Delta = -88$, $\det = 268$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 13.09 \rightarrow 14$, $T_{\max} = 14$, $T \in [14, 14]$

$$\begin{pmatrix} 1 & 4 & 4 & 1 & 2 & 3 \\ 4 & 4 & 6 & 0 & 3 & 4 \\ 4 & 6 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

5.76 NHC(-2)(-3)(-2) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8$

$T = 9$ [0] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -412$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 17 & 0 & 12 & 6 \\ 5 & 8 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

5.77 NHC(-2)(-3)(-2) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$

$T = 9$ [2] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.10 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 5 & 3 \\ 4 & 9 & 7 & 0 & 4 & 12 & 5 \\ 3 & 7 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 5 & 12 & 9 & 0 & 6 & 13 & 7 \\ 3 & 5 & 4 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

[0] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.34 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 4 & 5 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 3 & 5 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -3 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -116$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$

$T_{\min} = 12.62 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 6 & 0 & 4 & 14 & 4 \\ 3 & 6 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 8 & 14 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & 4 & -3 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -160, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 6 & 0 & 4 & 16 & 4 \\ 3 & 6 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 8 & 16 & 16 & 0 & 12 & 46 & 12 \\ 3 & 4 & 4 & -3 & 2 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112} 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -167, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 8 & 3 \\ 4 & 9 & 7 & 0 & 4 & 20 & 5 \\ 3 & 7 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 8 & 20 & 14 & 0 & 9 & 37 & 11 \\ 3 & 5 & 4 & -3 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -172, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -3 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -200, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 10 & 3 \\ 4 & 9 & 7 & 0 & 4 & 24 & 5 \\ 3 & 7 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 10 & 24 & 18 & 0 & 12 & 58 & 14 \\ 3 & 5 & 4 & -3 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -200, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 6 & 0 & 9 & 8 & 4 \\ 3 & 6 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 12 & 12 & 6 \\ 4 & 8 & 8 & 0 & 12 & 10 & 6 \\ 3 & 4 & 4 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -244, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.43 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 6 & 3 \\ 3 & 4 & 6 & 0 & 4 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 6 & 12 & 12 & 0 & 9 & 25 & 9 \\ 3 & 4 & 4 & -3 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -304, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 6 & 0 & 4 & 8 & 4 \\ 3 & 6 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 4 & 8 & 8 & 0 & 6 & 10 & 6 \\ 3 & 4 & 4 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -340, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1112} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -340, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 6 & 0 & 4 & 23 & 4 \\ 3 & 6 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 13 & 23 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & 4 & -3 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -404, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 12 & 6 & 6 \\ 2 & 4 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & 4 & -3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & -3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1113} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -788, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.29 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 6 & 0 & 4 & 32 & 4 \\ 3 & 6 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 21 & 2 \\ 18 & 32 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & 4 & -3 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1112} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -788, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.09 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 4 & 13 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 7 & 13 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & -3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1112} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -800, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 6 & 0 & 4 & 18 & 4 \\ 3 & 6 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 10 & 18 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & -3 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{1113} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1088, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.10 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 6 & 0 & 4 & 17 & 4 \\ 3 & 6 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 9 & 17 & 17 & 0 & 12 & 49 & 13 \\ 3 & 4 & 4 & -3 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1113} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1360, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 6 & 0 & 4 & 27 & 4 \\ 3 & 6 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 18 & 2 \\ 15 & 27 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & -3 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1114} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -1364, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.12 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 4 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 11 & 21 & 21 & 0 & 15 & 76 & 16 \\ 3 & 4 & 4 & -3 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1113} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -1460, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.09 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 6 & 0 & 4 & 22 & 4 \\ 3 & 6 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 12 & 22 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & -3 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1114} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -2096, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.09 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 6 & 0 & 4 & 26 & 4 \\ 3 & 6 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 18 & 2 \\ 14 & 26 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & 4 & -3 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1114} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -2144, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.13 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 6 & 0 & 4 & 36 & 4 \\ 3 & 6 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 24 & 2 \\ 20 & 36 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & -3 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1114} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -2356, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.10 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 6 & 0 & 4 & 31 & 4 \\ 3 & 6 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 21 & 2 \\ 17 & 31 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & -3 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [24] \quad 1^{1111} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, \quad T_{\max} = 13, \quad \Delta = -58, \quad \det = -64, \quad (n_+, n_-, n_0) = (1, 11, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 7 & 4 \\ 4 & 5 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 5 & 2 \\ 5 & 7 & 7 & 0 & 5 & 8 & 6 \\ 3 & 4 & 4 & -2 & 2 & 6 & 2 \end{pmatrix}_{b_0}$$

5.78 NHC(-2)(-3)(-2) + $\mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 672, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 12 & 6 & 6 & 6 \\ 2 & 4 & 4 & 0 & 6 & 1 & 3 & 3 \\ 2 & 4 & 4 & 0 & 6 & 3 & 1 & 3 \\ 3 & 4 & 4 & -3 & 6 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

5.79 NHC(-2)(-3)(-2) + f₄ + su₃ + su₂→(-3)

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.11 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 2 & 3 \\ 4 & 9 & 7 & 0 & 4 & 4 & 5 \\ 3 & 7 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 2 & 2 \\ 2 & 4 & 3 & 0 & 2 & 0 & 2 \\ 3 & 5 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 2^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.33 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 6 & 0 & 8 & 3 & 4 \\ 3 & 6 & 4 & 0 & 8 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 8 & 0 & 9 & 4 & 5 \\ 2 & 3 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.11 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

5.80 NHC(-2)(-3)(-2) + f₄ + su₃ + su₂n3mix

$$T = 9 \quad [0] \quad 1^{11} 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.34 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 6 & 0 & 4 & 17 & 4 \\ 3 & 6 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 11 & 2 \\ 10 & 17 & 17 & 0 & 11 & 55 & 13 \\ 3 & 4 & 4 & -3 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -468, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 6 & 0 & 4 & 7 & 4 \\ 3 & 6 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 5 & 2 \\ 4 & 7 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & 4 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -528, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 4 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 8 & 2 \\ 7 & 12 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & 4 & -3 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

5.81 NHC(-2)(-3)(-2) + f₄ + su₃ + su₃

$$T = 9 \quad [0] \quad 1^{11} 2^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 4 & 3 \\ 3 & 4 & 6 & 0 & 5 & 8 & 4 \\ 3 & 6 & 4 & 0 & 5 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 3 & 6 & 3 \\ 4 & 8 & 8 & 0 & 6 & 9 & 5 \\ 3 & 4 & 4 & -3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -292, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 5 & 5 & 0 & 3 & 3 & 3 \\ 2 & 4 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 2 & 4 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_3 \quad T = 9, T_{\max} = 15, \Delta = -97, \det = -640, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 15, T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

5.82 NHC(-2)(-3)(-2) + NHC(-2)(-3)(-2)

$$T = 10 \quad [0] \ 21^{\overline{111}} 41^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] \quad T = 10, T_{\max} = 11, \Delta = -29, \det = 288, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 2 & 3 & 3 \\ 4 & 4 & 6 & 4 & 3 & 4 & 4 \\ 4 & 6 & 4 & 4 & 3 & 4 & 4 \\ 3 & 4 & 4 & 1 & 3 & 3 & 3 \\ 2 & 3 & 3 & 3 & 0 & 3 & 2 \\ 3 & 4 & 4 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

5.83 NHC(-2)(-3)(-2) + so₈

$$T = 7 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -5, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 10 & 16 & 15 & 6 \\ 16 & 25 & 23 & 9 \\ 15 & 23 & 20 & 8 \\ 6 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 12 & 11 & 5 \\ 12 & 25 & 23 & 9 \\ 11 & 23 & 20 & 8 \\ 5 & 9 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 9 & 7 & 0 & 5 \\ 3 & 7 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.14 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 7 & 4 & 5 \\ 7 & 8 & 6 & 6 \\ 4 & 6 & 0 & 2 \\ 5 & 6 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 21^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, T_{\max} = 24, \Delta = -115, \det = -38, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 24, T \in [11, 24]$$

$$\begin{pmatrix} 13 & 20 & 17 & 7 \\ 20 & 25 & 23 & 9 \\ 17 & 23 & 20 & 8 \\ 7 & 9 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -40, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 5 & 7 & 5 & 5 \\ 7 & 8 & 6 & 6 \\ 5 & 6 & 0 & 2 \\ 5 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.82 \rightarrow 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^1 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -56, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 24, \ T \in [13, 24]$$

$$\begin{pmatrix} 5 & 12 & 12 & 5 \\ 12 & 25 & 27 & 9 \\ 12 & 27 & 25 & 9 \\ 5 & 9 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -60, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.07 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 10 & 15 & 15 & 6 \\ 15 & 20 & 22 & 8 \\ 15 & 22 & 20 & 8 \\ 6 & 8 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^1 13^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -64, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.81 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 5 & 11 & 11 & 5 \\ 11 & 20 & 22 & 8 \\ 11 & 22 & 20 & 8 \\ 5 & 8 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 21^1 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -64, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.31 \rightarrow 12, \ T_{\max} = 24, \ T \in [12, 24]$$

$$\begin{pmatrix} 13 & 17 & 17 & 7 \\ 17 & 20 & 22 & 8 \\ 17 & 22 & 20 & 8 \\ 7 & 8 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 21^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -184, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.96 \rightarrow 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 10 & 16 & 16 & 6 \\ 16 & 20 & 22 & 8 \\ 16 & 22 & 20 & 8 \\ 6 & 8 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 21^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 7, \ T_{\max} = 24, \ \Delta = -115, \ \det = -248, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.89 \rightarrow 11, \ T_{\max} = 24, \ T \in [11, 24]$$

$$\begin{pmatrix} 13 & 20 & 20 & 7 \\ 20 & 25 & 27 & 9 \\ 20 & 27 & 25 & 9 \\ 7 & 9 & 9 & 3 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [13] \ 1^{11} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 8, \ T_{\max} = 11, \ \Delta = -87, \ \det = 16, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 2 & 3 & 0 & 4 & 3 & 3 \\ 2 & 2 & -1 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 21^1 4^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -106, \ \det = 16, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.50 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 10 & 12 & 12 & 6 \\ 12 & 13 & 15 & 7 \\ 12 & 15 & 13 & 7 \\ 6 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{11} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 8, \ T_{\max} = 20, \ \Delta = -106, \ \det = 16, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.50 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 10 & 12 & 12 & 6 \\ 12 & 13 & 15 & 7 \\ 12 & 15 & 13 & 7 \\ 6 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[16] \ 1^1 3^1 1^1 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 20, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.40 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

- [17] $1^{11}31^{131}1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 0 & 4 \\ 1 & 4 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 0 & -2 & 1 \end{pmatrix}_{b_0}$$
- [18] $1^{11}321^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 3 & 1 & 1 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 & 2 \\ 2 & 3 & -1 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$
- [19] $1^{11}41^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -106, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.48 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$
- [20] $1^{11}31^{131}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.14 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 4 & 7 & 8 & 1 & 4 \\ 7 & 10 & 13 & 0 & 6 \\ 8 & 13 & 12 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 4 & 6 & 6 & 0 & 2 \end{pmatrix}_{b_0}$$
- [21] $1^{11}321^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 2 & 0 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 1 & 2 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 2 & 2 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$
- [22] $1^{11}31^{131}1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 3 & 4 & 0 & 3 \\ 1 & 3 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 3 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$
- [23] $1^{11}3131^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 1 & 4 & 3 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -3 & 1 & -1 \\ 4 & 12 & 1 & 10 & 6 \\ 3 & 6 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$
- [24] $1^1 141^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -106, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 9 & 5 \\ 9 & 13 & 15 & 7 \\ 9 & 15 & 13 & 7 \\ 5 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$
- [25] $1^{11}31^{131}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 1 & 3 & 6 & 3 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 3 & 1 & 1 & 6 & 3 \\ 3 & 6 & 0 & 6 & 11 & 5 \\ 2 & 3 & -1 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$
- [26] $21^1 41^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -106, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 13 & 14 & 14 & 7 \\ 14 & 13 & 15 & 7 \\ 14 & 15 & 13 & 7 \\ 7 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$
- [27] $1^{11}31^{131}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 4 & 7 & 8 & 7 & 4 \\ 7 & 10 & 13 & 12 & 6 \\ 8 & 13 & 12 & 13 & 6 \\ 7 & 12 & 13 & 10 & 6 \\ 4 & 6 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[28] $131^1 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.31 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 4 & 3 \\ 4 & 10 & 13 & 12 & 6 \\ 5 & 13 & 12 & 13 & 6 \\ 4 & 12 & 13 & 10 & 6 \\ 3 & 6 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

[29] $1^1 31^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.29 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 0 & 4 & 0 & 2 \\ 2 & 6 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 2 & 4 & -2 & 1 \end{pmatrix}_{b_0}$$

[30] $21^{11} 41^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -106, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 10 & 10 & 5 \\ 10 & 13 & 15 & 7 \\ 10 & 15 & 13 & 7 \\ 5 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[31] $2^1 2^1 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 5 & 7 & 5 & 5 \\ 4 & 7 & 5 & 5 & 5 \\ 3 & 5 & 5 & 0 & 2 \\ 3 & 5 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[32] $1^{111} 2^1 2^1 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 3 & 1 & 3 \\ 2 & 4 & 0 & 4 & 1 & 2 \\ 2 & 3 & 4 & 1 & 1 & 3 \\ 1 & 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 2 & 3 & 0 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [33] $21^1 41^1 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 3 & 1 & 4 \\ 2 & 3 & 0 & 0 & 2 \\ 2 & 1 & 0 & -2 & 0 \\ 3 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[34] $1^1 1^1 31^{11} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 & 2 \\ 2 & 0 & 1 & 2 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 2 & 2 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[35] $21^1 41^{11} 31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 & 2 \\ 2 & 0 & 1 & 2 & 2 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 2 & 2 & 1 & 0 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

[36] $1^1 1^1 31^{11} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 & 3 \\ 2 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 1 & 2 \\ 3 & 2 & 1 & 0 & 2 \\ 3 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[37] $1^1 1^1 31^{11} 4^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 4 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

[38] $1^1 1^1 41^{11} 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 0 & 2 \\ 1 & 1 & 0 & -2 & 0 \\ 3 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{11} 4^{131} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{11} 4^{131} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 2 & 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 0 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{11} 3^{221} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 1 & 5 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 3 & 3 & 5 & 0 & 4 & 4 \\ 2 & 2 & 3 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[42] \ 2^1 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 6 & 3 & 3 \\ 4 & 5 & 9 & 5 & 5 \\ 6 & 9 & 13 & 6 & 7 \\ 3 & 5 & 6 & 0 & 2 \\ 3 & 5 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[43] \ 2^1 3^{141} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 3 & 1 & 2 \\ 2 & 1 & 4 & 3 & 1 & 3 \\ 2 & 4 & 0 & 4 & 1 & 2 \\ 3 & 3 & 4 & 4 & 2 & 4 \\ 1 & 1 & 1 & 2 & -2 & 0 \\ 2 & 3 & 2 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[44] \ 21^{141} 1^{131} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 \\ 2 & 0 & 3 & 2 & 0 & 2 \\ 2 & 3 & 3 & 3 & 0 & 3 \\ 2 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{11} 4^{131} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 4 & 6 & 3 & 4 \\ 4 & 6 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[46] \ 31^{131} 1^{141} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -112, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 6 & 6 & 4 & 3 \\ 6 & 13 & 15 & 6 & 7 \\ 6 & 15 & 13 & 6 & 7 \\ 4 & 6 & 6 & 0 & 2 \\ 3 & 7 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{141} 1^{1231} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 1 & 2 \\ 3 & 4 & 1 & 6 & 2 & 4 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 3 & 6 & 1 & 4 & 2 & 4 \\ 1 & 2 & 0 & 2 & -2 & 0 \\ 2 & 4 & 0 & 4 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{141} 1^{1231} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 6 & 1 & 3 \\ 6 & 13 & 15 & 0 & 7 \\ 6 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[49] \ 31^{131} 1^{141} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -284, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 4 & 4 & 6 & 2 & 4 \\ 3 & 4 & 0 & 4 & 1 & 2 \\ 3 & 6 & 4 & 4 & 2 & 4 \\ 1 & 2 & 1 & 2 & -2 & 0 \\ 2 & 4 & 2 & 4 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [50] \quad 2 \quad 3 \quad 21 \quad 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 & 3 \\ 4 & 6 & 8 & 6 & 6 & 6 \\ 4 & 8 & 6 & 6 & 6 & 6 \\ 3 & 6 & 6 & 1 & 3 & 3 \\ 3 & 6 & 6 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

5.84 NHC(-2)(-3)(-2) + so₈ + e₆

$$T = 8 \quad [0] \quad 1 \quad 1^{11} \quad 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -164, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 5 & 5 \\ 12 & 25 & 23 & 0 & 9 & 9 \\ 11 & 23 & 20 & 0 & 8 & 8 \\ 1 & 0 & 0 & -6 & -4 & -4 \\ 5 & 9 & 8 & -4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} \quad 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -164, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 12 & 12 & 1 & 5 & 5 \\ 12 & 25 & 27 & 0 & 9 & 9 \\ 12 & 27 & 25 & 0 & 9 & 9 \\ 1 & 0 & 0 & -6 & -4 & -4 \\ 5 & 9 & 9 & -4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} \quad 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{brown}{e}_6 \quad T = 8, T_{\max} = 101, \Delta = -164, \det = 300, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.65 \rightarrow 14, T_{\max} = 101, T \in [14, 101]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 5 & 5 \\ 11 & 20 & 22 & 0 & 8 & 8 \\ 11 & 22 & 20 & 0 & 8 & 8 \\ 1 & 0 & 0 & -6 & -4 & -4 \\ 5 & 8 & 8 & -4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [3] \quad 1^{111} \quad 31^{11} \quad 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{brown}{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.76 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 4 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 13 & 0 & 6 \\ 4 & 12 & 13 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

5.85 NHC(-2)(-3)(-2) + so₈ + e₆ + su₂

$$T = 9 \quad [0] \quad 1^{111} \quad 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 9 & 5 \\ 12 & 25 & 23 & 0 & 20 & 9 \\ 11 & 23 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 20 & 17 & 0 & 13 & 7 \\ 5 & 9 & 8 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} \quad 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.05 \rightarrow 15, T_{\max} = 94, T \in [15, 94]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 17 & 0 & 13 & 7 \\ 5 & 8 & 8 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} \quad 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{brown}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 94, \Delta = -134, \det = -278, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.50 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 17 & 5 \\ 12 & 25 & 23 & 0 & 36 & 9 \\ 11 & 23 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 36 & 33 & 0 & 49 & 13 \\ 5 & 9 & 8 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{112}31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 94$, $\Delta = -134$, $\det = -442$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.5$

$T_{\min} = 13.50 \rightarrow 14$, $T_{\max} = 94$, $T \in [14, 94]$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 26 & 5 \\ 12 & 25 & 23 & 0 & 56 & 9 \\ 11 & 23 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 56 & 50 & 0 & 118 & 20 \\ 5 & 9 & 8 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

5.86 NHC(-2)(-3)(-2) + $\mathfrak{so}_8$ + $\mathfrak{e}_6$ + $\mathfrak{su}_3$

$T = 9$ [0] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 99$, $\Delta = -143$, $\det = -524$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.77 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 17 & 0 & 12 & 6 \\ 5 & 8 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

5.87 NHC(-2)(-3)(-2) + $\mathfrak{so}_8$ + $\mathfrak{f}_4$

$T = 8$ [0] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -138$, $\det = 81$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.64 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 5 \\ 12 & 25 & 23 & 0 & 9 \\ 11 & 23 & 20 & 0 & 8 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -138$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.86 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 12 & 12 & 1 & 5 \\ 12 & 25 & 27 & 0 & 9 \\ 12 & 27 & 25 & 0 & 9 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 9 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{f}_4$ $T = 8$, $T_{\max} = 98$, $\Delta = -138$, $\det = 236$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.83 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 5 \\ 11 & 20 & 22 & 0 & 8 \\ 11 & 22 & 20 & 0 & 8 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 8 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [3] $1^{111}31^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 32$, $\Delta = -117$, $\det = -172$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.2$

$T_{\min} = 12.95 \rightarrow 13$, $T_{\max} = 32$, $T \in [13, 32]$

$$\begin{pmatrix} 1 & 4 & 5 & 4 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 13 & 0 & 6 \\ 4 & 12 & 13 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

5.88 NHC(-2)(-3)(-2) + $\mathfrak{so}_8$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$

$T = 9$ [0] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -62$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 12.65 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 9 & 5 \\ 12 & 25 & 23 & 0 & 20 & 9 \\ 11 & 23 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 20 & 17 & 0 & 13 & 7 \\ 5 & 9 & 8 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -176$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$T_{\min} = 13.25 \rightarrow 14$, $T_{\max} = 21$, $T \in [14, 21]$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 17 & 0 & 13 & 7 \\ 5 & 8 & 8 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 17 & 5 \\ 12 & 25 & 23 & 0 & 36 & 9 \\ 11 & 23 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 36 & 33 & 0 & 49 & 13 \\ 5 & 9 & 8 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{112}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -342, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.7$

$$T_{\min} = 12.64 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 26 & 5 \\ 12 & 25 & 23 & 0 & 56 & 9 \\ 11 & 23 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 56 & 50 & 0 & 118 & 20 \\ 5 & 9 & 8 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

5.89 $\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9 \quad [0] 1^{111}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -110, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$$T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 6 & 5 \\ 12 & 25 & 23 & 0 & 12 & 9 \\ 11 & 23 & 20 & 0 & 11 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 11 & 0 & 4 & 4 \\ 5 & 9 & 8 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

5.90 $\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3$

$T = 9 \quad [0] 1^{111}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{f}}_4 + \textcolor{blue}{\mathfrak{su}}_3 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 17 & 0 & 12 & 6 \\ 5 & 8 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

5.91 $\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8 + \mathfrak{so}_8$

$T = 8 \quad [0] 1^{111}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 5 \\ 12 & 25 & 23 & 0 & 9 \\ 11 & 23 & 20 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 8 & -2 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 24, T \in [12, 24]$$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 5 \\ 7 & 8 & 6 & 0 & 6 \\ 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 6 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.30 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 5 & 12 & 12 & 1 & 5 \\ 12 & 25 & 27 & 0 & 9 \\ 12 & 27 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 9 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^11^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{brown}{\mathfrak{so}}_8 \quad T = 8, T_{\max} = 24, \Delta = -114, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 5 \\ 11 & 20 & 22 & 0 & 8 \\ 11 & 22 & 20 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 8 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 4 & 5 & 3 \\ 4 & 10 & 1 & 12 & 12 & 6 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 4 & 12 & 1 & 10 & 12 & 6 \\ 5 & 12 & 0 & 12 & 11 & 5 \\ 3 & 6 & -1 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 4 & 5 & 4 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 12 & 13 & 0 & 6 \\ 4 & 12 & 13 & 10 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 6 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{11}} 4^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.50 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 5 \\ 9 & 13 & 15 & 0 & 7 \\ 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

5.92 NHC(-2)(-3)(-2) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 4 & 5 \\ 7 & 8 & 6 & 0 & 6 & 6 \\ 4 & 6 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 6 & 4 & 0 & 2 & 4 \\ 5 & 6 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -42, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 9 & 5 \\ 12 & 25 & 23 & 0 & 20 & 9 \\ 11 & 23 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 20 & 17 & 0 & 13 & 7 \\ 5 & 9 & 8 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 10 & 5 \\ 12 & 25 & 23 & 0 & 20 & 9 \\ 11 & 23 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 20 & 17 & 0 & 13 & 7 \\ 5 & 9 & 8 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 11 & 5 \\ 7 & 8 & 6 & 0 & 15 & 6 \\ 4 & 6 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 15 & 9 & 0 & 22 & 10 \\ 5 & 6 & 2 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 7 & 5 \\ 7 & 8 & 6 & 0 & 9 & 6 \\ 4 & 6 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 5 & 0 & 8 & 6 \\ 5 & 6 & 2 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{11}} 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -106, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 18 & 5 \\ 12 & 25 & 23 & 0 & 36 & 9 \\ 11 & 23 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 36 & 33 & 0 & 49 & 13 \\ 5 & 9 & 8 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -158, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 17 & 5 \\ 12 & 25 & 23 & 0 & 36 & 9 \\ 11 & 23 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 36 & 33 & 0 & 49 & 13 \\ 5 & 9 & 8 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3^{\overline{111}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -184, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 5 & 5 \\ 7 & 8 & 6 & 0 & 6 & 6 \\ 4 & 6 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 6 & 4 & 0 & 2 & 4 \\ 5 & 6 & 2 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{112}} 3^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 26 & 5 \\ 12 & 25 & 23 & 0 & 56 & 9 \\ 11 & 23 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 26 & 56 & 50 & 0 & 118 & 20 \\ 5 & 9 & 8 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}} 3^{\overline{113}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 26 & 5 \\ 7 & 8 & 6 & 0 & 33 & 6 \\ 4 & 6 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 26 & 33 & 19 & 0 & 122 & 22 \\ 5 & 6 & 2 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{112}} 3^{\overline{111}} 1^{\overline{111}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -318, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 27 & 5 \\ 12 & 25 & 23 & 0 & 56 & 9 \\ 11 & 23 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 27 & 56 & 50 & 0 & 118 & 20 \\ 5 & 9 & 8 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{111}} 3^{\overline{111}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -344, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 12 & 5 \\ 7 & 8 & 6 & 0 & 15 & 6 \\ 4 & 6 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 15 & 9 & 0 & 22 & 10 \\ 5 & 6 & 2 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{111}} 3^{\overline{112}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 19 & 5 \\ 7 & 8 & 6 & 0 & 24 & 6 \\ 4 & 6 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 24 & 14 & 0 & 62 & 16 \\ 5 & 6 & 2 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

5.93 NHC(-2)(-3)(-2) + so₈ + so₈ + su₂ → (-3)

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -97, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 6 & 5 \\ 12 & 25 & 23 & 0 & 12 & 9 \\ 11 & 23 & 20 & 0 & 11 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 11 & 0 & 4 & 4 \\ 5 & 9 & 8 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 4 & 5 \\ 7 & 8 & 6 & 0 & 5 & 6 \\ 4 & 6 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 3 & 0 & 1 & 3 \\ 5 & 6 & 2 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

5.94 NHC(-2)(-3)(-2) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_{2n3\text{mix}}$

$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 11 & 5 \\ 7 & 8 & 6 & 0 & 14 & 6 \\ 4 & 6 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 14 & 8 & 0 & 21 & 9 \\ 5 & 6 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

5.95 NHC(-2)(-3)(-2) + $\mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3$

$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 1 & 5 \\ 12 & 25 & 23 & 0 & 0 & 9 \\ 11 & 23 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 8 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 7 & 5 \\ 7 & 8 & 6 & 0 & 9 & 6 \\ 4 & 6 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 5 & 0 & 7 & 5 \\ 5 & 6 & 2 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 1 & 5 \\ 11 & 20 & 22 & 0 & 0 & 8 \\ 11 & 22 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & 8 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 17 & 0 & 12 & 6 \\ 5 & 8 & 8 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$T = 10 \quad [4] \quad 1^{\overline{11}} 4^{\overline{1}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 336, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$

$T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 6 & 5 \\ 9 & 13 & 15 & 0 & 10 & 7 \\ 9 & 15 & 13 & 0 & 10 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 10 & 0 & 4 & 4 \\ 5 & 7 & 7 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

5.96 $\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{111} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 16 & 15 & 13 & 6 \\ 16 & 25 & 23 & 20 & 9 \\ 15 & 23 & 20 & 17 & 8 \\ 13 & 20 & 17 & 13 & 7 \\ 6 & 9 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{112} 3 \ 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 6 & 0 & 28 & 4 \\ 3 & 6 & 4 & 0 & 28 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 28 & 28 & 0 & 154 & 22 \\ 3 & 4 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 6 & 0 & 0 & 4 \\ 3 & 6 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 4 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{112} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 8 & 0 & 10 & 6 \\ 3 & 4 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 16 & 15 & 23 & 6 \\ 16 & 25 & 23 & 36 & 9 \\ 15 & 23 & 20 & 33 & 8 \\ 23 & 36 & 33 & 49 & 13 \\ 6 & 9 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{112} 3 \ 1^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 16 & 15 & 36 & 6 \\ 16 & 25 & 23 & 56 & 9 \\ 15 & 23 & 20 & 50 & 8 \\ 36 & 56 & 50 & 118 & 20 \\ 6 & 9 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 21^{111} 31^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 20 & 17 & 27 & 7 \\ 20 & 25 & 23 & 36 & 9 \\ 17 & 23 & 20 & 33 & 8 \\ 27 & 36 & 33 & 49 & 13 \\ 7 & 9 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 5 & 7 & 5 \\ 7 & 8 & 6 & 9 & 6 \\ 5 & 6 & 0 & 5 & 2 \\ 7 & 9 & 5 & 8 & 6 \\ 5 & 6 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 21^{111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 20 & 17 & 15 & 7 \\ 20 & 25 & 23 & 20 & 9 \\ 17 & 23 & 20 & 17 & 8 \\ 15 & 20 & 17 & 13 & 7 \\ 7 & 9 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3 \ 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 5 & 4 & 5 \\ 7 & 8 & 6 & 6 & 6 \\ 5 & 6 & 0 & 4 & 2 \\ 4 & 6 & 4 & 2 & 4 \\ 5 & 6 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 5 & 5 & 5 \\ 7 & 8 & 6 & 6 & 6 \\ 5 & 6 & 0 & 4 & 2 \\ 5 & 6 & 4 & 2 & 4 \\ 5 & 6 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 112, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{113} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 116, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{112} 3 \ 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 120, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.87 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 6 & 0 & 28 & 4 \\ 3 & 6 & 4 & 0 & 28 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 28 & 28 & 0 & 154 & 22 \\ 3 & 4 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{112} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 120, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.87 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 8 & 0 & 10 & 6 \\ 3 & 4 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 128, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{113} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 128, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 17 & 17 & 0 & 49 & 13 \\ 3 & 4 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{112} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 140, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.83 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 13 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 13 & 13 & 0 & 28 & 10 \\ 3 & 4 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{112} 3 \ 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 140, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.83 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 6 & 0 & 23 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 23 & 23 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{111} 3 \ 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 144, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 5 & 19 & 5 \\ 7 & 8 & 6 & 24 & 6 \\ 5 & 6 & 0 & 14 & 2 \\ 19 & 24 & 14 & 62 & 16 \\ 5 & 6 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 164, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 5 & 12 & 5 \\ 7 & 8 & 6 & 15 & 6 \\ 5 & 6 & 0 & 9 & 2 \\ 12 & 15 & 9 & 22 & 10 \\ 5 & 6 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{113} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[32] \ 21^{112} 31^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 20 & 17 & 42 & 7 \\ 20 & 25 & 23 & 56 & 9 \\ 17 & 23 & 20 & 50 & 8 \\ 42 & 56 & 50 & 118 & 20 \\ 7 & 9 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{114} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 16 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 16 & 16 & 0 & 46 & 12 \\ 3 & 4 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 13 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 13 & 0 & 28 & 10 \\ 3 & 4 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{112} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 6 & 0 & 23 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 23 & 23 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{114} 3 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 16 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 16 & 16 & 0 & 46 & 12 \\ 3 & 4 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{112} 3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 13 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 13 & 0 & 28 & 10 \\ 3 & 4 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{112} 3 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 23 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 23 & 23 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[41] $1^{13} 3^1 1^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 260, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 21 & 3 \\ 3 & 4 & 6 & 0 & 32 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 32 & 32 & 0 & 193 & 25 \\ 3 & 4 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[42] $1^{11}3^{13}3^11^{12} + [\textcolor{red}{\mathbf{su}_2} \cdot \textcolor{blue}{\mathbf{so}_7} \cdot \textcolor{teal}{\mathbf{su}_2}] + \textcolor{brown}{\mathbf{so}_8} + \textcolor{red}{\mathbf{su}_2} \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 260, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$
 $\left(\begin{array}{cccccc} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 6 & 0 & 22 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 22 & 22 & 0 & 85 & 17 \\ 3 & 4 & 4 & -2 & 17 & 2 \end{array} \right)_{b_0}$

[43] $1^{11}3^{11}3^{11} + [\textcolor{red}{5u}_2 \cdot \textcolor{blue}{5o}_7 \cdot \textcolor{green}{5u}_2] + \textcolor{brown}{5o}_8 + \textcolor{red}{5u}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 272, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 17 & 0 & 49 & 13 \\ 3 & 4 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[45] $1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} \frac{1}{3} & \frac{3}{4} & \frac{3}{6} & \frac{1}{0} & \frac{11}{18} & \frac{3}{4} \\ \frac{3}{3} & \frac{4}{6} & \frac{3}{4} & \frac{0}{0} & \frac{18}{18} & \frac{4}{4} \\ \frac{3}{1} & \frac{6}{0} & \frac{4}{0} & \frac{0}{-4} & \frac{18}{0} & \frac{4}{-2} \\ \frac{11}{3} & \frac{18}{4} & \frac{18}{4} & \frac{0}{-2} & \frac{58}{14} & \frac{14}{2} \end{pmatrix}_{b_0}$$

[44] $1^{14} 3^1 1^{12} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{s}o_7 \cdot \textcolor{green}{s}u_2] + \textcolor{orange}{s}o_8 + \textcolor{red}{s}u_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 296, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 6 & 0 & 26 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 26 & 26 & 0 & 118 & 20 \\ 3 & 4 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[46] $1^{113} 3^1 1^{13} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{so}}_7 \cdot \textcolor{teal}{\mathbf{su}}_2] + \textcolor{brown}{\mathbf{so}}_8 + \textcolor{red}{\mathbf{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} \frac{1}{3} & \frac{3}{4} & \frac{3}{6} & \frac{1}{4} & \frac{18}{27} & \frac{3}{4} \\ \frac{3}{3} & \frac{4}{6} & \frac{6}{4} & \frac{0}{0} & \frac{27}{27} & \frac{4}{4} \\ \frac{3}{1} & \frac{6}{0} & \frac{4}{0} & \frac{0}{-4} & \frac{27}{0} & \frac{4}{-2} \\ \frac{1}{18} & \frac{0}{27} & \frac{0}{27} & \frac{-4}{0} & \frac{0}{133} & \frac{-2}{21} \end{pmatrix}_{b_0}$

[47] $1^{114}3^11^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 332, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$
 $T_{\min} = 10.98 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$

$$\begin{pmatrix} \frac{1}{3} & \frac{3}{4} & \frac{3}{6} & \frac{1}{0} & \frac{11}{21} & \frac{3}{4} \\ \frac{3}{3} & \frac{4}{6} & \frac{6}{4} & \frac{0}{0} & \frac{21}{21} & \frac{4}{4} \\ \frac{3}{1} & \frac{6}{0} & \frac{4}{0} & \frac{0}{-4} & \frac{21}{0} & \frac{4}{-2} \\ \frac{1}{11} & \frac{0}{21} & \frac{0}{21} & \frac{-4}{0} & \frac{0}{76} & \frac{-2}{16} \end{pmatrix}_{b_0}$$

[48] $1^{113} 1^{12} 3 1^{11} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{s}o_7 \cdot \textcolor{green}{s}u_2] + \textcolor{brown}{s}o_8 + \textcolor{red}{s}u_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 336, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} \frac{1}{3} & \frac{3}{4} & \frac{3}{6} & \frac{1}{0} & \frac{11}{17} & \frac{3}{4} \\ \frac{3}{3} & \frac{4}{6} & \frac{6}{4} & \frac{0}{0} & \frac{17}{17} & \frac{4}{4} \\ \frac{3}{3} & \frac{6}{0} & \frac{4}{0} & \frac{0}{-4} & \frac{0}{0} & \frac{4}{-2} \\ \frac{1}{11} & \frac{0}{17} & \frac{0}{17} & \frac{-4}{0} & \frac{0}{49} & \frac{-2}{13} \end{pmatrix}_{b_0}$

[49] $1^{14} 1^{13} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 380, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$
 $T_{\min} = 10.85 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 21 & 21 & 0 & 76 & 16 \\ 3 & 4 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$

[50] $1^{11} 1^{13} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 384, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 17 & 0 & 49 & 13 \\ 3 & 4 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{113}3^{11}3^{112} + [\textcolor{red}{5}u_2 \cdot \textcolor{blue}{5}07 \cdot \textcolor{green}{5}u_2] + \textcolor{brown}{5}08 + \textcolor{red}{5}u_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 22 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 22 & 22 & 0 & 85 & 17 \\ 3 & 4 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$

[52] $1^{113} 3^{11} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 404, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.8$
 $T_{\min} = 10.89 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$
 $\begin{pmatrix} \frac{1}{3} & \frac{3}{4} & \frac{3}{6} & \frac{1}{0} & \frac{18}{32} & \frac{3}{4} \\ \frac{3}{3} & \frac{4}{6} & \frac{3}{4} & \frac{0}{0} & \frac{32}{32} & \frac{4}{-2} \\ \frac{3}{1} & \frac{6}{0} & \frac{4}{0} & \frac{0}{-4} & \frac{0}{0} & \frac{-2}{25} \\ \frac{18}{3} & \frac{32}{4} & \frac{32}{4} & \frac{0}{-2} & \frac{193}{25} & \frac{25}{2} \end{pmatrix}_{b_0}$

[53] $1^{14} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 428, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$
 $T_{\min} = 10.83 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 21 & 3 \\ 3 & 4 & 6 & 0 & 31 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 31 & 31 & 0 & 172 & 24 \\ 3 & 4 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$

[54] $1^{113}3^{11}3^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$

[55] $1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{su}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 468, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 22 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 22 & 22 & 0 & 85 & 17 \\ 3 & 4 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$

[56] $1^{113} 3^1 1^{14} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{so}}_7 \cdot \textcolor{green}{\mathbf{su}}_2] + \textcolor{brown}{\mathbf{so}}_8 + \textcolor{red}{\mathbf{su}}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 468, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 6 & 0 & 32 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 32 & 32 & 0 & 193 & 25 \\ 3 & 4 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$

[57] $1^{11}1^{14}3^11^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 472, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$
 $T_{\min} = 10.83 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 24 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 24 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$

[58] $1^{11} 1^{14} 3^1 1^{11} + [\textcolor{red}{\mathbf{su}}_2 \cdot \textcolor{blue}{\mathbf{so}}_7 \cdot \textcolor{green}{\mathbf{su}}_2] + \textcolor{orange}{\mathbf{so}}_8 + \textcolor{red}{\mathbf{su}}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 508, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$
 $T_{\min} = 10.89 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$
 $\left(\begin{array}{cccccc} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 21 & 0 & 76 & 16 \\ 3 & 4 & 4 & -2 & 16 & 2 \end{array} \right)_{b_0}$

[59] $1^{113} 1^{12} 3^1 1^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 512, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$

[61] $1^{113} 3^1 1^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{su}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 516, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 19 & 3 \\ 3 & 4 & 6 & 0 & 32 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 32 & 32 & 0 & 193 & 25 \\ 3 & 4 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$

$$[60] \quad 1^{11} 3^1 1^{12} + [\textcolor{red}{5u}_2 \cdot \textcolor{blue}{507} \cdot \textcolor{green}{5u}_2] + \textcolor{orange}{508} + \textcolor{red}{5u}_2 \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 516, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.85 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 6 & 0 & 22 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 22 & 22 & 0 & 85 & 17 \\ 3 & 4 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[62] $1^{114} 3 1^{11} + [\textcolor{red}{5u2} \cdot \textcolor{blue}{507} \cdot \textcolor{green}{5u2}] + \textcolor{orange}{508} + \textcolor{red}{5u2} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 524, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 21 & 21 & 0 & 76 & 16 \\ 3 & 4 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[63] $1^{114}3^11^2 + [\textcolor{red}{5u}_2 \cdot \textcolor{blue}{507} \cdot \textcolor{green}{5u}_2] + \textcolor{brown}{508} + \textcolor{red}{5u}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 552, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 26 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 26 & 26 & 0 & 118 & 20 \\ 3 & 4 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[64] $1^{11} 1^{13} 3^1 1^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 560, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$

[65] $1^{114} 3^1 1^{12} + [\textcolor{red}{\mathbf{su}_2} \cdot \textcolor{blue}{\mathbf{so}_7} \cdot \textcolor{green}{\mathbf{su}_2}] + \textcolor{orange}{\mathbf{so}_8} + \textcolor{red}{\mathbf{su}_2} \quad T = 8, \quad T_{\max} = 11, \quad \Delta = -85, \quad \det = 600, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, \quad T_{\max} = 11, \quad T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 6 & 0 & 26 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 26 & 26 & 0 & 118 & 20 \\ 3 & 4 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[66] $1^{114}3^113 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 684, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 6 & 0 & 31 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 31 & 31 & 0 & 172 & 24 \\ 3 & 4 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$

[68] $1^{114}3^11^4 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 728, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\left(\begin{array}{cccccc} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{array} \right)_{b_0}$$

$$[67] \ 1^{114} 3 \ 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 728, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.87 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 6 & 0 & 26 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 26 & 26 & 0 & 118 & 20 \\ 3 & 4 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{114} 3^1 1^{13} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{teal}{\mathfrak{su}_2}] + \textcolor{orange}{\mathfrak{so}_8} + \textcolor{red}{\mathfrak{su}_2} \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 732, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 6 & 0 & 31 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 31 & 31 & 0 & 172 & 24 \\ 3 & 4 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$

$$[70] \ 1^{114} 3 \ 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 744, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.85 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 6 & 0 & 26 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 26 & 26 & 0 & 118 & 20 \\ 3 & 4 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{114} 3^{113} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 776, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 23 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 23 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{114} 3^{113} 1^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 860, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 6 & 0 & 31 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 31 & 31 & 0 & 172 & 24 \\ 3 & 4 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{114} 3^{113} 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 876, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 19 & 3 \\ 3 & 4 & 6 & 0 & 31 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 31 & 31 & 0 & 172 & 24 \\ 3 & 4 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{114} 3^{113} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 904, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 21 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{114} 3^{113} 1^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 920, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 22 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [76] \ 2^1 3^{141} 21^{111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -288, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 & 3 \\ 4 & 6 & 8 & 6 & 6 & 6 \\ 4 & 8 & 6 & 6 & 6 & 6 \\ 3 & 6 & 6 & 1 & 5 & 3 \\ 3 & 6 & 6 & 5 & 3 & 5 \\ 3 & 6 & 6 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

5.97 NHC(-2)(-3)(-2) + $\mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \ 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 16 & 15 & 8 & 6 \\ 16 & 25 & 23 & 12 & 9 \\ 15 & 23 & 20 & 11 & 8 \\ 8 & 12 & 11 & 4 & 4 \\ 6 & 9 & 8 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 21^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 58, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 20 & 17 & 9 & 7 \\ 20 & 25 & 23 & 12 & 9 \\ 17 & 23 & 20 & 11 & 8 \\ 9 & 12 & 11 & 4 & 4 \\ 7 & 9 & 8 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 5 & 4 & 5 \\ 7 & 8 & 6 & 5 & 6 \\ 5 & 6 & 0 & 3 & 2 \\ 4 & 5 & 3 & 1 & 3 \\ 5 & 6 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [4] \quad 1^1 41^{11} 23^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 248, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 6 & 1 & 1 & 3 \\ 6 & 13 & 15 & 0 & 0 & 7 \\ 6 & 15 & 13 & 0 & 0 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 7 & 7 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^1 41^{11} 23^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 288, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 1 & 1 & 2 \\ 3 & 4 & 1 & 6 & 2 & 0 & 4 \\ 1 & 1 & -2 & 1 & 0 & 0 & 0 \\ 3 & 6 & 1 & 4 & 2 & 0 & 4 \\ 1 & 2 & 0 & 2 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 4 & 0 & 4 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

5.98 NHC(-2)(-3)(-2) + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^1 41^{11} 23^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -560, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 6 & 1 & 1 & 5 & 3 \\ 6 & 13 & 15 & 0 & 0 & 12 & 7 \\ 6 & 15 & 13 & 0 & 0 & 12 & 7 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 5 & 12 & 12 & 0 & 0 & 8 & 6 \\ 3 & 7 & 7 & -1 & 0 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^1 41^{11} 23^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -576, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 1 & 1 & 3 & 2 \\ 3 & 4 & 1 & 6 & 2 & 0 & 6 & 4 \\ 1 & 1 & -2 & 1 & 0 & 0 & 0 & 0 \\ 3 & 6 & 1 & 4 & 2 & 0 & 6 & 4 \\ 1 & 2 & 0 & 2 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 3 & 6 & 0 & 6 & 2 & 0 & 4 & 4 \\ 2 & 4 & 0 & 4 & 0 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

5.99 NHC(-2)(-3)(-2) + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^1 1^1 3^1 1^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -60, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

5.100 NHC(-2)(-3)(-2) + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 10 \quad [1] \quad 1^1 1^1 3^1 1^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 21 & 4 \\ 3 & 6 & 4 & 0 & 3 & 3 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 & 11 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 11 & 2 \\ 11 & 21 & 21 & 0 & 11 & 11 & 76 & 16 \\ 3 & 4 & 4 & -2 & 2 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^1 1^1 3^1 1^1 1^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{so}}_8 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) + \textcolor{red}{\mathfrak{su}}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 & 9 & 4 \\ 3 & 6 & 4 & 0 & 3 & 3 & 4 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 & 5 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 & 5 & 2 \\ 2 & 4 & 4 & 0 & 2 & 2 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 5 & 5 & 6 & 13 & 7 \\ 3 & 4 & 4 & -2 & 2 & 2 & 3 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, \ T_{\max} = 11, \ \Delta = 0, \ \det = -576, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 12.5$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 & 13 & 4 \\ 3 & 6 & 4 & 0 & 3 & 3 & 4 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 & 7 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 & 7 & 2 \\ 2 & 4 & 4 & 0 & 2 & 2 & 1 & 9 & 3 \\ 7 & 13 & 13 & 0 & 7 & 7 & 9 & 28 & 10 \\ 3 & 4 & 4 & -2 & 2 & 2 & 3 & 10 & 2 \end{pmatrix}_{b_0}$$

5.102 NHC(-2)(-3)(-2) + so₈ + su₂→(-3) + su₂n3mix

$$T = 9 \quad [0] \ 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -60, \ \det = -280, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.99 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 7 & 2 \\ 7 & 12 & 12 & 0 & 7 & 25 & 9 \\ 3 & 4 & 4 & -2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -60, \ \det = -468, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 6 & 0 & 3 & 16 & 4 \\ 3 & 6 & 4 & 0 & 3 & 16 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 9 & 2 \\ 9 & 16 & 16 & 0 & 9 & 44 & 12 \\ 3 & 4 & 4 & -2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -60, \ \det = -616, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.99 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 3 & 20 & 4 \\ 3 & 6 & 4 & 0 & 3 & 20 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 11 & 2 \\ 11 & 20 & 20 & 0 & 11 & 69 & 15 \\ 3 & 4 & 4 & -2 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

5.103 NHC(-2)(-3)(-2) + so₈ + su₂→(-3) + su₂n3mix + su₂

$$T = 10 \quad [0] \ 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 11, \ \Delta = -30, \ \det = 344, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.99 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 4 & 4 \\ 3 & 6 & 4 & 0 & 3 & 12 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 7 & 2 & 2 \\ 7 & 12 & 12 & 0 & 7 & 25 & 8 & 9 \\ 2 & 4 & 4 & 0 & 2 & 8 & 1 & 3 \\ 3 & 4 & 4 & -2 & 2 & 9 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{112}3^{11}1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 11, \ \Delta = -30, \ \det = 720, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 16 & 4 & 4 \\ 3 & 6 & 4 & 0 & 3 & 16 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 9 & 2 & 2 \\ 9 & 16 & 16 & 0 & 9 & 44 & 11 & 12 \\ 2 & 4 & 4 & 0 & 2 & 11 & 1 & 3 \\ 3 & 4 & 4 & -2 & 2 & 12 & 3 & 2 \end{pmatrix}_{b_0}$$

5.104 NHC(-2)(-3)(-2) + $\mathfrak{so}_8$ + su2n3mix

$$T = 8 \quad [0] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 8 & 0 & 12 & 6 \\ 3 & 4 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 5 & 11 & 5 \\ 7 & 8 & 6 & 14 & 6 \\ 5 & 6 & 0 & 8 & 2 \\ 11 & 14 & 8 & 21 & 9 \\ 5 & 6 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 17 & 17 & 0 & 55 & 13 \\ 3 & 4 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 7 & 4 \\ 3 & 6 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 7 & 7 & 0 & 7 & 5 \\ 3 & 4 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 0 & 7 & 4 \\ 3 & 6 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 7 & 0 & 7 & 5 \\ 3 & 4 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 17 & 0 & 55 & 13 \\ 3 & 4 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 11 & 4 \\ 3 & 6 & 4 & 0 & 11 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 11 & 11 & 0 & 20 & 8 \\ 3 & 4 & 4 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 24 & 24 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[20] $1^{114}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -87, \det = 732, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 6 & 0 & 29 & 4 \\ 3 & 6 & 4 & 0 & 29 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 29 & 29 & 0 & 148 & 22 \\ 3 & 4 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[21] $1^{114}3^11^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 11, \Delta = -87, \det = 876, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 6 & 0 & 29 & 4 \\ 3 & 6 & 4 & 0 & 29 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 29 & 29 & 0 & 148 & 22 \\ 3 & 4 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [22] $1^{111}3^121^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -60, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 8 & 5 & 0 & 23 & 9 \\ 2 & 2 & 2 & 2 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

[24] $1^{112}3^121^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -60, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 12 & 8 & 0 & 56 & 14 \\ 2 & 2 & 2 & 2 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{113}3^121^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -60, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 5 & 7 & 0 & 8 & 5 \\ 3 & 7 & 5 & 0 & 8 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 8 & 0 & 8 & 6 \\ 3 & 5 & 5 & -2 & 6 & 3 \end{pmatrix}_{b_0}$$

[26] $1^{111}3^121^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -60, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 5 & 7 & 0 & 18 & 5 \\ 3 & 7 & 5 & 0 & 18 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 18 & 18 & 0 & 48 & 14 \\ 3 & 5 & 5 & -2 & 14 & 3 \end{pmatrix}_{b_0}$$

[25] $1^{113}3^121^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -60, \det = -144, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 2 & 0 & 16 & 2 \\ 1 & 2 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 8 & 11 & 16 & 11 & 0 & 103 & 19 \\ 2 & 2 & 2 & 2 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{111}3^121^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -60, \det = -220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 5 & 7 & 0 & 23 & 5 \\ 3 & 7 & 5 & 0 & 23 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 23 & 23 & 0 & 78 & 18 \\ 3 & 5 & 5 & -2 & 18 & 3 \end{pmatrix}_{b_0}$$

[28] $1^{111}3^121^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 11, \Delta = -60, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 5 & 7 & 0 & 28 & 5 \\ 3 & 7 & 5 & 0 & 28 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 28 & 28 & 0 & 116 & 22 \\ 3 & 5 & 5 & -2 & 22 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [29] \quad 1^{141}1^{11}23^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 1 & 5 & 2 \\ 3 & 4 & 1 & 6 & 2 & 6 & 4 \\ 1 & 1 & -2 & 1 & 0 & 1 & 0 \\ 3 & 6 & 1 & 4 & 2 & 6 & 4 \\ 1 & 2 & 0 & 2 & -2 & 2 & 0 \\ 5 & 6 & 1 & 6 & 2 & 5 & 5 \\ 2 & 4 & 0 & 4 & 0 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{141}1^{11}23^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 352, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 6 & 6 & 1 & 6 & 3 \\ 6 & 13 & 15 & 0 & 12 & 7 \\ 6 & 15 & 13 & 0 & 12 & 7 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 12 & 12 & 0 & 8 & 6 \\ 3 & 7 & 7 & -1 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{141}1^{11}23^11^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 1 & 4 & 2 \\ 3 & 4 & 1 & 6 & 2 & 6 & 4 \\ 1 & 1 & -2 & 1 & 0 & 0 & 0 \\ 3 & 6 & 1 & 4 & 2 & 6 & 4 \\ 1 & 2 & 0 & 2 & -2 & 2 & 0 \\ 4 & 6 & 0 & 6 & 2 & 4 & 4 \\ 2 & 4 & 0 & 4 & 0 & 4 & 2 \end{pmatrix}_{b_0}$$

5.105 NHC(-2)(-3)(-2) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{su}_2$

$$T = 10 \quad [1] \quad 1^{1112}3^{11}21^{122} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 2 & 0 & 5 & 6 & 2 \\ 1 & 2 & 0 & 2 & 0 & 8 & 8 & 2 \\ 1 & 2 & 2 & 0 & 0 & 5 & 6 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 8 & 5 & 0 & 23 & 26 & 9 \\ 4 & 6 & 8 & 6 & 0 & 26 & 26 & 10 \\ 2 & 2 & 2 & 2 & -2 & 9 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{1111}3^{11}21^{121} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 2 & 2 & 0 & 5 & 3 & 2 \\ 1 & 2 & 0 & 2 & 0 & 8 & 4 & 2 \\ 1 & 2 & 2 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 8 & 5 & 0 & 23 & 13 & 9 \\ 2 & 3 & 4 & 3 & 0 & 13 & 5 & 5 \\ 2 & 2 & 2 & 2 & -2 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}21^{1111} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 5 & 7 & 0 & 8 & 5 & 5 \\ 3 & 7 & 5 & 0 & 8 & 5 & 5 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 8 & 0 & 8 & 6 & 6 \\ 2 & 5 & 5 & 0 & 6 & 2 & 4 \\ 3 & 5 & 5 & -2 & 6 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11}3^{11}21^{1113} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 9 & 3 \\ 3 & 5 & 7 & 0 & 8 & 20 & 5 \\ 3 & 7 & 5 & 0 & 8 & 20 & 5 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 8 & 0 & 8 & 24 & 6 \\ 9 & 20 & 20 & 0 & 24 & 58 & 16 \\ 3 & 5 & 5 & -2 & 6 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1121}3^{11}21^{131} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 & 2 \\ 1 & 0 & 2 & 2 & 0 & 8 & 3 \\ 1 & 2 & 0 & 2 & 0 & 12 & 4 \\ 1 & 2 & 2 & 0 & 0 & 8 & 3 \\ 1 & 0 & 0 & 0 & -4 & 0 & 0 \\ 6 & 8 & 12 & 8 & 0 & 56 & 20 \\ 2 & 3 & 4 & 3 & 0 & 20 & 5 \\ 2 & 2 & 2 & 2 & -2 & 14 & 5 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}3^121^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 5 & 3 \\ 3 & 5 & 7 & 0 & 8 & 10 & 5 \\ 3 & 7 & 5 & 0 & 8 & 10 & 5 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 8 & 0 & 8 & 12 & 6 \\ 5 & 10 & 10 & 0 & 12 & 14 & 8 \\ 3 & 5 & 5 & -2 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11}3^121^{1112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 192, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 7 & 3 \\ 3 & 5 & 7 & 0 & 8 & 15 & 5 \\ 3 & 7 & 5 & 0 & 8 & 15 & 5 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 8 & 8 & 0 & 8 & 18 & 6 \\ 7 & 15 & 15 & 0 & 18 & 32 & 12 \\ 3 & 5 & 5 & -2 & 6 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11}3^121^{1131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 224, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 5 & 7 & 0 & 23 & 5 & 5 \\ 3 & 7 & 5 & 0 & 23 & 5 & 5 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 11 & 23 & 23 & 0 & 78 & 17 & 18 \\ 2 & 5 & 5 & 0 & 17 & 2 & 4 \\ 3 & 5 & 5 & -2 & 18 & 4 & 3 \end{pmatrix}_{b_0}$$

5.106 NHC(-2)(-3)(-2) + so₈ + su₂n₃mix + su₂ + su₂

$$T = 11 \quad [0] \quad 1^{11}3^121^{11311} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -16, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 2 & 3 \\ 3 & 5 & 7 & 0 & 23 & 5 & 5 & 5 \\ 3 & 7 & 5 & 0 & 23 & 5 & 5 & 5 \\ 1 & 0 & 0 & -4 & 0 & 0 & 0 & -2 \\ 11 & 23 & 23 & 0 & 78 & 17 & 17 & 18 \\ 2 & 5 & 5 & 0 & 17 & 2 & 4 & 4 \\ 2 & 5 & 5 & 0 & 17 & 4 & 2 & 4 \\ 3 & 5 & 5 & -2 & 18 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^121^{11111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 5 & 3 \\ 3 & 5 & 7 & 0 & 8 & 5 & 10 & 5 \\ 3 & 7 & 5 & 0 & 8 & 5 & 10 & 5 \\ 1 & 0 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 8 & 0 & 8 & 6 & 12 & 6 \\ 2 & 5 & 5 & 0 & 6 & 2 & 7 & 4 \\ 5 & 10 & 10 & 0 & 12 & 7 & 14 & 8 \\ 3 & 5 & 5 & -2 & 6 & 4 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^121^{11113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 9 & 3 \\ 3 & 5 & 7 & 0 & 8 & 5 & 20 & 5 \\ 3 & 7 & 5 & 0 & 8 & 5 & 20 & 5 \\ 1 & 0 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 8 & 0 & 8 & 6 & 24 & 6 \\ 2 & 5 & 5 & 0 & 6 & 2 & 15 & 4 \\ 9 & 20 & 20 & 0 & 24 & 15 & 58 & 16 \\ 3 & 5 & 5 & -2 & 6 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}3^121^{11112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{so}_8 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -192, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 7 & 3 \\ 3 & 5 & 7 & 0 & 8 & 5 & 15 & 5 \\ 3 & 7 & 5 & 0 & 8 & 5 & 15 & 5 \\ 1 & 0 & 0 & -4 & 0 & 0 & 0 & -2 \\ 4 & 8 & 8 & 0 & 8 & 6 & 18 & 6 \\ 2 & 5 & 5 & 0 & 6 & 2 & 11 & 4 \\ 7 & 15 & 15 & 0 & 18 & 11 & 32 & 12 \\ 3 & 5 & 5 & -2 & 6 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

5.107 NHC(-2)(-3)(-2) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -60, \det = -280, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 12 & 12 & 0 & 25 & 7 & 9 \\ 2 & 3 & 3 & 0 & 7 & 0 & 2 \\ 3 & 4 & 4 & -2 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -60, \det = -468, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 6 & 0 & 16 & 3 & 4 \\ 3 & 6 & 4 & 0 & 16 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 16 & 16 & 0 & 44 & 9 & 12 \\ 2 & 3 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -2 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -60, \det = -616, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 6 & 0 & 20 & 3 & 4 \\ 3 & 6 & 4 & 0 & 20 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 11 & 20 & 20 & 0 & 69 & 11 & 15 \\ 2 & 3 & 3 & 0 & 11 & 0 & 2 \\ 3 & 4 & 4 & -2 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

5.108 NHC(-2)(-3)(-2) + $\mathfrak{so}_8$ + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 344, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 4 & 4 \\ 3 & 6 & 4 & 0 & 12 & 3 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & 0 & -2 \\ 7 & 12 & 12 & 0 & 25 & 7 & 8 & 9 \\ 2 & 3 & 3 & 0 & 7 & 0 & 2 & 2 \\ 2 & 4 & 4 & 0 & 8 & 2 & 1 & 3 \\ 3 & 4 & 4 & -2 & 9 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 720, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 16 & 3 & 4 & 4 \\ 3 & 6 & 4 & 0 & 16 & 3 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & 0 & -2 \\ 9 & 16 & 16 & 0 & 44 & 9 & 11 & 12 \\ 2 & 3 & 3 & 0 & 9 & 0 & 2 & 2 \\ 2 & 4 & 4 & 0 & 11 & 2 & 1 & 3 \\ 3 & 4 & 4 & -2 & 12 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

5.109 NHC(-2)(-3)(-2) + $\mathfrak{so}_8$ + su2n3mix + su2n3mix

$$T = 9 \quad [0] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -60, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 6 & 0 & 8 & 7 & 4 \\ 3 & 6 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 5 & 8 & 8 & 0 & 12 & 10 & 6 \\ 4 & 7 & 7 & 0 & 10 & 7 & 5 \\ 3 & 4 & 4 & -2 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -60, \det = -316, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 6 & 0 & 7 & 12 & 4 \\ 3 & 6 & 4 & 0 & 7 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 7 & 0 & 7 & 15 & 5 \\ 7 & 12 & 12 & 0 & 15 & 25 & 9 \\ 3 & 4 & 4 & -2 & 5 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}123\overline{11}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -60, \det = -576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 6 & 0 & 7 & 16 & 4 \\ 3 & 6 & 4 & 0 & 7 & 16 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 7 & 7 & 0 & 7 & 20 & 5 \\ 9 & 16 & 16 & 0 & 20 & 44 & 12 \\ 3 & 4 & 4 & -2 & 5 & 12 & 2 \end{pmatrix}_{b_0}$$

5.110 NHC(-2)(-3)(-2) + $\mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{\overline{11}13\overline{1}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 1 & 3 \\ 4 & 9 & 7 & 0 & 0 & 5 \\ 3 & 7 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}13\overline{1}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 2 & 3 \\ 4 & 9 & 7 & 0 & 4 & 5 \\ 3 & 7 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 3 & 5 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}13\overline{1}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 5 \\ 12 & 25 & 23 & 0 & 9 \\ 11 & 23 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}13\overline{1}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 15 & 13 & 6 \\ 15 & 20 & 22 & 17 & 8 \\ 15 & 22 & 20 & 17 & 8 \\ 13 & 17 & 17 & 12 & 6 \\ 6 & 8 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}13\overline{1}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.25 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 5 \\ 7 & 8 & 6 & 0 & 6 \\ 4 & 6 & 0 & 0 & 2 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 6 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}13\overline{1}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 6 & 0 & 0 & 4 \\ 3 & 6 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{11}13\overline{1}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 55, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 7 & 4 & 8 & 5 \\ 7 & 8 & 6 & 9 & 6 \\ 4 & 6 & 0 & 5 & 2 \\ 8 & 9 & 5 & 7 & 5 \\ 5 & 6 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}13\overline{1}1^{\overline{11}}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 63, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 5 & 7 & 4 & 7 & 5 \\ 7 & 8 & 6 & 9 & 6 \\ 4 & 6 & 0 & 5 & 2 \\ 7 & 9 & 5 & 7 & 5 \\ 5 & 6 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 64, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.50 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 12 & 12 & 1 & 5 \\ 12 & 25 & 27 & 0 & 9 \\ 12 & 27 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 9 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 76, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 5 & 7 & 5 & 7 & 5 \\ 7 & 8 & 6 & 9 & 6 \\ 5 & 6 & 0 & 5 & 2 \\ 7 & 9 & 5 & 7 & 5 \\ 5 & 6 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 88, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.32 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 5 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 5 & 0 & 3 & 3 \\ 3 & 4 & 4 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 96, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.33 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 12 & 0 & 24 & 8 \\ 3 & 4 & 4 & -2 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 96, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 15 & 1 & 6 \\ 15 & 20 & 22 & 0 & 8 \\ 15 & 22 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 8 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 104, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 15 & 12 & 6 \\ 15 & 20 & 22 & 17 & 8 \\ 15 & 22 & 20 & 17 & 8 \\ 12 & 17 & 17 & 12 & 6 \\ 6 & 8 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 108, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.81 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 5 \\ 11 & 20 & 22 & 0 & 8 \\ 11 & 22 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 8 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^1 3^1 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 112, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.89 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 5 & 11 & 11 & 9 & 5 \\ 11 & 20 & 22 & 17 & 8 \\ 11 & 22 & 20 & 17 & 8 \\ 9 & 17 & 17 & 12 & 6 \\ 5 & 8 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 112, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.29 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 4 & 4 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{112} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 116, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.72 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 8 & 8 & 0 & 9 & 5 \\ 3 & 4 & 4 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 128, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 4 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{112} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.32 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 8 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 8 & 8 & 0 & 9 & 5 \\ 3 & 4 & 4 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 21^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 20 & 12 & 7 \\ 20 & 25 & 27 & 16 & 9 \\ 20 & 27 & 25 & 16 & 9 \\ 12 & 16 & 16 & 9 & 5 \\ 7 & 9 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{113} 3 \ 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 12 & 0 & 24 & 8 \\ 3 & 4 & 4 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{112} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 13 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 13 & 0 & 27 & 9 \\ 3 & 4 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[25] \ 21^{111} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 16 & 16 & 12 & 6 \\ 16 & 20 & 22 & 17 & 8 \\ 16 & 22 & 20 & 17 & 8 \\ 12 & 17 & 17 & 12 & 6 \\ 6 & 8 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{112} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 13 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 13 & 13 & 0 & 27 & 9 \\ 3 & 4 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{113} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.28 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 17 & 0 & 48 & 12 \\ 3 & 4 & 4 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{113} 3 \ 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 428, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 17 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 17 & 17 & 0 & 48 & 12 \\ 3 & 4 & 4 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [29] \quad 1^{111} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 12 & 8 & 6 \\ 12 & 13 & 15 & 10 & 7 \\ 12 & 15 & 13 & 10 & 7 \\ 8 & 10 & 10 & 4 & 4 \\ 6 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[30] \quad 21^{11} 4 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 12 & 8 & 6 \\ 12 & 13 & 15 & 10 & 7 \\ 12 & 15 & 13 & 10 & 7 \\ 8 & 10 & 10 & 4 & 4 \\ 6 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{11} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{11} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 5 \\ 9 & 13 & 15 & 0 & 7 \\ 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{11} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -89, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 5 & 0 & 7 & 4 \\ 3 & 5 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 7 & 7 & 0 & 7 & 5 \\ 3 & 4 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[34] \quad 21^{11} 4 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 14 & 14 & 9 & 7 \\ 14 & 13 & 15 & 10 & 7 \\ 14 & 15 & 13 & 10 & 7 \\ 9 & 10 & 10 & 4 & 4 \\ 7 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{111} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 9 & 7 & 5 \\ 9 & 13 & 15 & 10 & 7 \\ 9 & 15 & 13 & 10 & 7 \\ 7 & 10 & 10 & 4 & 4 \\ 5 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[36] \quad 21^{111} 4 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 10 & 7 & 5 \\ 10 & 13 & 15 & 10 & 7 \\ 10 & 15 & 13 & 10 & 7 \\ 7 & 10 & 10 & 4 & 4 \\ 5 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

5.111 NHC(-2)(-3)(-2) + so₈ + su₃ + e₆

$$T = 9 \quad [0] \quad 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 + \textcolor{blue}{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 11 & 9 & 1 & 5 \\ 11 & 20 & 22 & 17 & 0 & 8 \\ 11 & 22 & 20 & 17 & 0 & 8 \\ 9 & 17 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

5.112 $\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4$

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 11 & 9 & 1 & 5 \\ 11 & 20 & 22 & 17 & 0 & 8 \\ 11 & 22 & 20 & 17 & 0 & 8 \\ 9 & 17 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

5.113 $\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 1 & 5 \\ 12 & 25 & 23 & 0 & 0 & 9 \\ 11 & 23 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 9 & 8 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 7 & 4 & 7 & 1 & 5 \\ 7 & 8 & 6 & 9 & 0 & 6 \\ 4 & 6 & 0 & 5 & 0 & 2 \\ 7 & 9 & 5 & 7 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 6 & 2 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 1 & 5 \\ 11 & 20 & 22 & 0 & 0 & 8 \\ 11 & 22 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 8 & 8 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 9 & 5 \\ 11 & 20 & 22 & 0 & 17 & 8 \\ 11 & 22 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 17 & 0 & 12 & 6 \\ 5 & 8 & 8 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

5.114 $\text{NHC}(-2)(-3)(-2) + \mathfrak{so}_8 + (-3) \rightarrow (-2)$

$$T = 8 \quad [0] \quad 1^{11} 3^1 1^{12} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 85, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 17 & 1 & 7 \\ 20 & 25 & 23 & 0 & 9 \\ 17 & 23 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 9 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{12} 2^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 17 & 17 & 1 & 7 \\ 17 & 20 & 22 & 0 & 8 \\ 17 & 22 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 8 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [2] \quad 1^{2^1 1^{11}} 4^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -304, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 10 & 1 & 5 \\ 10 & 13 & 15 & 0 & 7 \\ 10 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

5.115 NHC(-2)(-3)(-2) + $\mathfrak{su}_2$

- $T = 7$ [0] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -10$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.60 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 4 & 5 \\ 7 & 8 & 6 & 6 \\ 4 & 6 & 2 & 4 \\ 5 & 6 & 4 & 4 \end{pmatrix}_{b_0}$
- [1] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -13$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.77 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 7 & 5 \\ 7 & 8 & 9 & 6 \\ 7 & 9 & 8 & 6 \\ 5 & 6 & 6 & 4 \end{pmatrix}_{b_0}$
- [2] $1^{11}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -18$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.78 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 6 & 5 \\ 7 & 8 & 6 & 6 \\ 6 & 6 & 2 & 4 \\ 5 & 6 & 4 & 4 \end{pmatrix}_{b_0}$
- [3] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 11 & 5 \\ 7 & 8 & 15 & 6 \\ 11 & 15 & 22 & 10 \\ 5 & 6 & 10 & 4 \end{pmatrix}_{b_0}$
- [4] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -22$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.64 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 5 & 5 \\ 7 & 8 & 6 & 6 \\ 5 & 6 & 2 & 4 \\ 5 & 6 & 4 & 4 \end{pmatrix}_{b_0}$
- [5] $21^{112}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 4 & 15 & 3 \\ 4 & 4 & 6 & 23 & 4 \\ 4 & 6 & 4 & 23 & 4 \\ 15 & 23 & 23 & 100 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$
- [6] $21^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 5 & 3 & 4 \\ 5 & 4 & 6 & 4 & 4 \\ 5 & 6 & 4 & 4 & 4 \\ 3 & 4 & 4 & 1 & 3 \\ 4 & 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$
- [7] $21^{12}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 5 & 15 & 4 \\ 5 & 4 & 6 & 14 & 4 \\ 5 & 6 & 4 & 14 & 4 \\ 15 & 14 & 14 & 37 & 11 \\ 4 & 4 & 4 & 11 & 3 \end{pmatrix}_{b_0}$
- [8] $1^{12}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -34$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 11 & 5 \\ 7 & 8 & 12 & 6 \\ 11 & 12 & 14 & 8 \\ 5 & 6 & 8 & 4 \end{pmatrix}_{b_0}$
- [9] $1^{12}31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -34$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.76 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 10 & 5 \\ 7 & 8 & 12 & 6 \\ 10 & 12 & 14 & 8 \\ 5 & 6 & 8 & 4 \end{pmatrix}_{b_0}$
- [10] $1^{11}31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -45$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.76 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 12 & 5 \\ 7 & 8 & 15 & 6 \\ 12 & 15 & 22 & 10 \\ 5 & 6 & 10 & 4 \end{pmatrix}_{b_0}$
- [11] $1^{11}31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -86$, $\det = -49$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 14 & 5 \\ 7 & 8 & 15 & 6 \\ 14 & 15 & 22 & 10 \\ 5 & 6 & 10 & 4 \end{pmatrix}_{b_0}$$

[12] $1^{13} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -54, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 15 & 5 \\ 7 & 8 & 18 & 6 \\ 15 & 18 & 34 & 12 \\ 5 & 6 & 12 & 4 \end{pmatrix}_{b_0}$

[13] $1^{11} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 13 & 5 \\ 7 & 8 & 15 & 6 \\ 13 & 15 & 22 & 10 \\ 5 & 6 & 10 & 4 \end{pmatrix}_{b_0}$

[15] $1^{13} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 16 & 5 \\ 7 & 8 & 18 & 6 \\ 16 & 18 & 34 & 12 \\ 5 & 6 & 12 & 4 \end{pmatrix}_{b_0}$

[14] $1^{11} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 19 & 5 \\ 7 & 8 & 24 & 6 \\ 19 & 24 & 62 & 16 \\ 5 & 6 & 16 & 4 \end{pmatrix}_{b_0}$

[17] $21^{11} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 5 & 18 & 4 \\ 5 & 4 & 6 & 21 & 4 \\ 5 & 6 & 4 & 21 & 4 \\ 18 & 21 & 21 & 76 & 16 \\ 4 & 4 & 4 & 16 & 3 \end{pmatrix}_{b_0}$

[16] $21^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 5 & 9 & 4 \\ 5 & 4 & 6 & 9 & 4 \\ 5 & 6 & 4 & 9 & 4 \\ 9 & 9 & 9 & 13 & 7 \\ 4 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$

[19] $1^{11} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -82, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 20 & 5 \\ 7 & 8 & 24 & 6 \\ 20 & 24 & 62 & 16 \\ 5 & 6 & 16 & 4 \end{pmatrix}_{b_0}$

[18] $1^{14} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -82, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 20 & 5 \\ 7 & 8 & 24 & 6 \\ 20 & 24 & 62 & 16 \\ 5 & 6 & 16 & 4 \end{pmatrix}_{b_0}$

[20] $1^{12} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 17 & 5 \\ 7 & 8 & 21 & 6 \\ 17 & 21 & 44 & 14 \\ 5 & 6 & 14 & 4 \end{pmatrix}_{b_0}$

[21] $1^{11} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -90, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 21 & 5 \\ 7 & 8 & 24 & 6 \\ 21 & 24 & 62 & 16 \\ 5 & 6 & 16 & 4 \end{pmatrix}_{b_0}$

[22] $21^{11} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 4 & 6 & 4 & 4 \\ 4 & 6 & 4 & 4 & 4 \\ 3 & 4 & 4 & 1 & 3 \\ 3 & 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$

[25] $21^{11} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 21 & 3 \\ 4 & 4 & 6 & 32 & 4 \\ 4 & 6 & 4 & 32 & 4 \\ 21 & 32 & 32 & 193 & 25 \\ 3 & 4 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

[23] $21^{11}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 12 & 4 \\ 5 & 4 & 6 & 13 & 4 \\ 5 & 6 & 4 & 13 & 4 \\ 12 & 13 & 13 & 28 & 10 \\ 4 & 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[26] $21^{11}31^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 15 & 4 \\ 5 & 4 & 6 & 17 & 4 \\ 5 & 6 & 4 & 17 & 4 \\ 15 & 17 & 17 & 49 & 13 \\ 4 & 4 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[24] $21^{13}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 24 & 4 \\ 5 & 4 & 6 & 23 & 4 \\ 5 & 6 & 4 & 23 & 4 \\ 24 & 23 & 23 & 100 & 18 \\ 4 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[27] $1^{14}12^3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -97, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 19 & 5 \\ 7 & 8 & 21 & 6 \\ 19 & 21 & 44 & 14 \\ 5 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

[28] $1^{13}12^3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 18 & 5 \\ 7 & 8 & 21 & 6 \\ 18 & 21 & 44 & 14 \\ 5 & 6 & 14 & 4 \end{pmatrix}_{b_0}$$

[29] $21^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 3 \\ 4 & 4 & 6 & 9 & 4 \\ 4 & 6 & 4 & 9 & 4 \\ 6 & 9 & 13 & 7 \\ 3 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

[30] $21^{11}231^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 3 \\ 4 & 4 & 6 & 8 & 4 \\ 4 & 6 & 4 & 8 & 4 \\ 6 & 8 & 8 & 10 & 6 \\ 3 & 4 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$

[31] $1^{12}12^3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -126, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 24 & 5 \\ 7 & 8 & 30 & 6 \\ 24 & 30 & 94 & 20 \\ 5 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[32] $1^{13}12^3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -133, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 22 & 5 \\ 7 & 8 & 27 & 6 \\ 22 & 27 & 74 & 18 \\ 5 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[33] $21^{12}31^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -136, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 18 & 4 \\ 5 & 4 & 6 & 18 & 4 \\ 5 & 6 & 4 & 18 & 4 \\ 18 & 18 & 18 & 58 & 14 \\ 4 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[34] $1^{14}12^3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 23 & 5 \\ 7 & 8 & 27 & 6 \\ 23 & 27 & 74 & 18 \\ 5 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

[35] $1^{12}12^3 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 25 & 5 \\ 7 & 8 & 30 & 6 \\ 25 & 30 & 94 & 20 \\ 5 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[36] $1^{12} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -157, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 31 & 5 \\ 7 & 8 & 39 & 6 \\ 31 & 39 & 164 & 26 \\ 5 & 6 & 26 & 4 \end{pmatrix}_{b_0}$$

[37] $21^{113} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 3 \\ 4 & 4 & 6 & 12 & 4 \\ 4 & 6 & 4 & 12 & 4 \\ 9 & 12 & 12 & 25 & 9 \\ 3 & 4 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[38] $1^{12} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -166, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 26 & 5 \\ 7 & 8 & 30 & 6 \\ 26 & 30 & 94 & 20 \\ 5 & 6 & 20 & 4 \end{pmatrix}_{b_0}$$

[39] $21^{112} 31^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 3 \\ 4 & 4 & 6 & 18 & 4 \\ 4 & 6 & 4 & 18 & 4 \\ 12 & 18 & 18 & 58 & 14 \\ 3 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[40] $1^{14} 3 1^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -189, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 27 & 5 \\ 7 & 8 & 33 & 6 \\ 27 & 33 & 112 & 22 \\ 5 & 6 & 22 & 4 \end{pmatrix}_{b_0}$$

[41] $1^{12} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -199, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 32 & 5 \\ 7 & 8 & 39 & 6 \\ 32 & 39 & 164 & 26 \\ 5 & 6 & 26 & 4 \end{pmatrix}_{b_0}$$

[42] $1^{13} 3 1^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -202, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 29 & 5 \\ 7 & 8 & 36 & 6 \\ 29 & 36 & 134 & 24 \\ 5 & 6 & 24 & 4 \end{pmatrix}_{b_0}$$

[46] $21^{112} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 3 \\ 4 & 4 & 6 & 13 & 4 \\ 4 & 6 & 4 & 13 & 4 \\ 9 & 13 & 13 & 28 & 10 \\ 3 & 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[45] $21^{123} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 21 & 4 \\ 5 & 4 & 6 & 22 & 4 \\ 5 & 6 & 4 & 22 & 4 \\ 21 & 22 & 22 & 85 & 17 \\ 4 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

[44] $21^{143} 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 33 & 4 \\ 5 & 4 & 6 & 32 & 4 \\ 5 & 6 & 4 & 32 & 4 \\ 33 & 32 & 32 & 193 & 25 \\ 4 & 4 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

[43] $21^{114} 31^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -216, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 3 \\ 4 & 4 & 6 & 16 & 4 \\ 4 & 6 & 4 & 16 & 4 \\ 12 & 16 & 16 & 46 & 12 \\ 3 & 4 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[47] $1^{12} 3 1^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -225, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 33 & 5 \\ 7 & 8 & 39 & 6 \\ 33 & 39 & 164 & 26 \\ 5 & 6 & 26 & 4 \end{pmatrix}_{b_0}$$

[48] $1^{12} 3 1^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -234, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 39 & 5 \\ 7 & 8 & 48 & 6 \\ 39 & 48 & 254 & 32 \\ 5 & 6 & 32 & 4 \end{pmatrix}_{b_0}$$

[49] $21^{13} 31^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -256, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 27 & 4 \\ 5 & 4 & 6 & 27 & 4 \\ 5 & 6 & 4 & 27 & 4 \\ 27 & 27 & 27 & 133 & 21 \\ 4 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[50] $1^{13} 3 1^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 36 & 5 \\ 7 & 8 & 45 & 6 \\ 36 & 45 & 214 & 30 \\ 5 & 6 & 30 & 4 \end{pmatrix}_{b_0}$$

[51] $21^{12} 31^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -264, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 24 & 4 \\ 5 & 4 & 6 & 26 & 4 \\ 5 & 6 & 4 & 26 & 4 \\ 24 & 26 & 26 & 118 & 20 \\ 4 & 4 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

[52] $1^{12} 3 1^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -274, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 40 & 5 \\ 7 & 8 & 48 & 6 \\ 40 & 48 & 254 & 32 \\ 5 & 6 & 32 & 4 \end{pmatrix}_{b_0}$$

[53] $1^{14} 3 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -286, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 34 & 5 \\ 7 & 8 & 42 & 6 \\ 34 & 42 & 182 & 28 \\ 5 & 6 & 28 & 4 \end{pmatrix}_{b_0}$$

[54] $21^{115} 31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -288, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.1$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 3 \\ 4 & 4 & 6 & 20 & 4 \\ 4 & 6 & 4 & 20 & 4 \\ 15 & 20 & 20 & 73 & 15 \\ 3 & 4 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

[55] $21^{113} 31^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -304, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 18 & 3 \\ 4 & 4 & 6 & 27 & 4 \\ 4 & 6 & 4 & 27 & 4 \\ 18 & 27 & 27 & 133 & 21 \\ 3 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[56] $1^{13} 3 1^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -307, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 37 & 5 \\ 7 & 8 & 45 & 6 \\ 37 & 45 & 214 & 30 \\ 5 & 6 & 30 & 4 \end{pmatrix}_{b_0}$$

[57] $1^{13} 3 1^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -310, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 43 & 5 \\ 7 & 8 & 54 & 6 \\ 43 & 54 & 314 & 36 \\ 5 & 6 & 36 & 4 \end{pmatrix}_{b_0}$$

[58] $21^{113} 31^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -336, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 3 \\ 4 & 4 & 6 & 17 & 4 \\ 4 & 6 & 4 & 17 & 4 \\ 12 & 17 & 17 & 49 & 13 \\ 3 & 4 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[59] $1^{13} 3 1^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -370, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 44 & 5 \\ 7 & 8 & 54 & 6 \\ 44 & 54 & 314 & 36 \\ 5 & 6 & 36 & 4 \end{pmatrix}_{b_0}$$

[60] $1^{14} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -373, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 41 & 5 \\ 7 & 8 & 51 & 6 \\ 41 & 51 & 272 & 34 \\ 5 & 6 & 34 & 4 \end{pmatrix}_{b_0}$$

[61] $21^{116} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -376, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 18 & 3 \\ 4 & 4 & 6 & 24 & 4 \\ 4 & 6 & 4 & 24 & 4 \\ 18 & 24 & 24 & 106 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[63] $21^{113} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 3 \\ 4 & 4 & 6 & 22 & 4 \\ 4 & 6 & 4 & 22 & 4 \\ 15 & 22 & 22 & 85 & 17 \\ 3 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

[62] $21^{131} 31^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -384, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 30 & 4 \\ 5 & 4 & 6 & 31 & 4 \\ 5 & 6 & 4 & 31 & 4 \\ 30 & 31 & 31 & 172 & 24 \\ 4 & 4 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[64] $21^{143} 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -424, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 36 & 4 \\ 5 & 4 & 6 & 36 & 4 \\ 5 & 6 & 4 & 36 & 4 \\ 36 & 36 & 36 & 238 & 28 \\ 4 & 4 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

[65] $1^{14} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -450, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 48 & 5 \\ 7 & 8 & 60 & 6 \\ 48 & 60 & 382 & 40 \\ 5 & 6 & 40 & 4 \end{pmatrix}_{b_0}$$

[67] $21^{114} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -472, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 3 \\ 4 & 4 & 6 & 21 & 4 \\ 4 & 6 & 4 & 21 & 4 \\ 15 & 21 & 21 & 76 & 16 \\ 3 & 4 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

[66] $21^{114} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -472, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 24 & 3 \\ 4 & 4 & 6 & 36 & 4 \\ 4 & 6 & 4 & 36 & 4 \\ 24 & 36 & 36 & 238 & 28 \\ 3 & 4 & 4 & 28 & 3 \end{pmatrix}_{b_0}$$

[68] $21^{117} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -480, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 21 & 3 \\ 4 & 4 & 6 & 28 & 4 \\ 4 & 6 & 4 & 28 & 4 \\ 21 & 28 & 28 & 145 & 21 \\ 3 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[69] $21^{114} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.83 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 21 & 3 \\ 4 & 4 & 6 & 31 & 4 \\ 4 & 6 & 4 & 31 & 4 \\ 21 & 31 & 31 & 172 & 24 \\ 3 & 4 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[70] $21^{114} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -86, \det = -600, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 18 & 3 \\ 4 & 4 & 6 & 26 & 4 \\ 4 & 6 & 4 & 26 & 4 \\ 18 & 26 & 26 & 118 & 20 \\ 3 & 4 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

$T = 8$ [72] $1^1 3^1 321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 1 & 3 & 0 & 1 & 3 \\ 2 & 2 & 2 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 2 & 1 & 0 & 3 & 1 & 3 \\ 2 & 2 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[74] \quad 2 2 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 & 3 \\ 4 & 5 & 7 & 5 & 5 & 5 \\ 4 & 7 & 5 & 5 & 5 & 5 \\ 3 & 5 & 5 & 4 & 4 & 4 \\ 3 & 5 & 5 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{11} 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 5 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 10 & 3 \\ 6 & 5 & 0 & 10 & 33 & 11 \\ 2 & 2 & -1 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{11} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.67 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{12} 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 6 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 8 & 3 \\ 4 & 6 & 0 & 8 & 26 & 10 \\ 2 & 2 & -1 & 3 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{13} 3 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 9 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 12 & 3 \\ 6 & 9 & 0 & 12 & 61 & 15 \\ 2 & 2 & -1 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{11} 3 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 2 & 3 & 0 & 4 & 5 & 5 \\ 2 & 2 & -1 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{14} 3 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 12 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 16 & 3 \\ 8 & 12 & 0 & 16 & 110 & 20 \\ 2 & 2 & -1 & 3 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{11} 3 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 10, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{11} 3 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & 1 & 4 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 7 & 3 \\ 4 & 4 & 0 & 7 & 16 & 8 \\ 2 & 2 & -1 & 3 & 8 & 3 \end{pmatrix}_{b_0}$$

[82] $1^{11} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & 0 & 5 & 3 \end{pmatrix}_{b_0}$$

[83] $21^{11} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 12 & 4 \\ 5 & 5 & 7 & 15 & 5 \\ 5 & 7 & 5 & 15 & 5 \\ 12 & 15 & 15 & 32 & 12 \\ 4 & 5 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

[84] $1^{11} 3 21^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 1 & 1 & 7 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 11 & 3 \\ 6 & 7 & 0 & 11 & 45 & 13 \\ 2 & 2 & -1 & 3 & 13 & 3 \end{pmatrix}_{b_0}$$

[85] $1^{12} 3 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 8 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 14 & 3 \\ 8 & 8 & 0 & 14 & 70 & 16 \\ 2 & 2 & -1 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

[86] $1^{13} 3 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 1 & 1 & 10 & 2 \\ 1 & 1 & -3 & 1 & 0 & -1 \\ 1 & 1 & 1 & 1 & 15 & 3 \\ 8 & 10 & 0 & 15 & 88 & 18 \\ 2 & 2 & -1 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

[87] $1^{11} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & 0 & 5 & 3 \end{pmatrix}_{b_0}$$

[88] $1^{11} 3 21^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 38, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 4 & 4 & 7 & 0 & 16 & 8 \\ 2 & 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$$

[89] $21^{12} 3 21^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 24 & 4 \\ 5 & 5 & 7 & 30 & 5 \\ 5 & 7 & 5 & 30 & 5 \\ 24 & 30 & 30 & 134 & 24 \\ 4 & 5 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[90] $1^{11} 3 21^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 5 \\ 2 & 2 & 2 & 0 & 5 & 3 \end{pmatrix}_{b_0}$$

[91] $2 2 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 3 & 2 \\ 2 & 2 & 0 & 2 & 4 & 2 \\ 1 & 2 & 2 & 0 & 3 & 2 \\ 3 & 3 & 4 & 3 & 5 & 5 \\ 2 & 2 & 2 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

- [92] $1^{12} 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 7 \\ 2 & 2 & 2 & 0 & 7 & 3 \end{pmatrix}_{b_0}$
- [93] $1^{12} 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 6 & 5 & 5 & 0 & 13 & 7 \\ 2 & 2 & 2 & 0 & 7 & 3 \end{pmatrix}_{b_0}$
- [94] $1^{11} 3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 8 \\ 2 & 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$
- [95] $2 \overset{11}{3} 2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -60, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 2 & 3 \\ 2 & 3 & 0 & 3 & 3 & 2 \\ 2 & 3 & 3 & 1 & 2 & 3 \\ 2 & 2 & 3 & 2 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$
- [96] $1^{11} 3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 6 & 4 & 7 & 0 & 16 & 8 \\ 2 & 2 & 2 & 0 & 8 & 3 \end{pmatrix}_{b_0}$
- [97] $1^{13} 3 21^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 7 & 7 & 6 & 0 & 25 & 9 \\ 2 & 2 & 2 & 0 & 9 & 3 \end{pmatrix}_{b_0}$
- [98] $2 \overset{111}{2} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 5 & 7 & 5 & 5 \\ 4 & 7 & 5 & 5 & 5 \\ 3 & 5 & 5 & 2 & 4 \\ 3 & 5 & 5 & 4 & 4 \end{pmatrix}_{b_0}$
- [100] $2 \overset{111}{2} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 4 & 6 & 3 \\ 4 & 5 & 7 & 10 & 5 \\ 4 & 7 & 5 & 10 & 5 \\ 6 & 10 & 10 & 14 & 8 \\ 3 & 5 & 5 & 8 & 4 \end{pmatrix}_{b_0}$
- [101] $2 \overset{112}{2} 31^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 6 & 2 \\ 1 & 0 & 2 & 2 & 6 & 2 \\ 2 & 2 & 0 & 2 & 8 & 2 \\ 1 & 2 & 2 & 0 & 6 & 2 \\ 6 & 6 & 8 & 6 & 26 & 10 \\ 2 & 2 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$
- [99] $2 \overset{112}{2} 31^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 1 & 4 & 4 & 6 & 3 \\ 4 & 5 & 7 & 10 & 5 \\ 4 & 7 & 5 & 10 & 5 \\ 6 & 10 & 10 & 14 & 8 \\ 3 & 5 & 5 & 8 & 4 \end{pmatrix}_{b_0}$
- [102] $1^{12} 3 21^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 10 \\ 2 & 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[103] \ 1^{12} 3^{12} 21^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 7 & 7 & 11 & 0 & 45 & 13 \\ 2 & 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[104] \ 2 \ 2 \ 31^{13} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 3 \\ 4 & 5 & 7 & 15 & 5 \\ 4 & 7 & 5 & 15 & 5 \\ 9 & 15 & 15 & 34 & 12 \\ 3 & 5 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[105] \ 1^{12} 3^{12} 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 7 & 6 & 8 & 0 & 26 & 10 \\ 2 & 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[106] \ 1^{12} 3^{14} 21^{12} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 122, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 8 & 6 & 8 & 0 & 26 & 10 \\ 2 & 2 & 2 & 0 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[113] \ 2 \ 3 \ 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 3 & 6 & 3 \\ 2 & 3 & 0 & 3 & 4 & 2 \\ 2 & 3 & 3 & 1 & 6 & 3 \\ 4 & 6 & 4 & 6 & 10 & 6 \\ 2 & 3 & 2 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[108] \ 2 \ 3 \ 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.1$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 3 & 15 & 3 \\ 2 & 3 & 0 & 3 & 10 & 2 \\ 2 & 3 & 3 & 1 & 15 & 3 \\ 10 & 15 & 10 & 15 & 73 & 15 \\ 2 & 3 & 2 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[109] \ 2 \ 3 \ 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 12 & 2 \\ 2 & 1 & 3 & 3 & 18 & 3 \\ 2 & 3 & 0 & 3 & 12 & 2 \\ 2 & 3 & 3 & 1 & 18 & 3 \\ 12 & 18 & 12 & 18 & 106 & 18 \\ 2 & 3 & 2 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[110] \ 2 \ 3 \ 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.3$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 14 & 2 \\ 2 & 1 & 3 & 3 & 21 & 3 \\ 2 & 3 & 0 & 3 & 14 & 2 \\ 2 & 3 & 3 & 1 & 21 & 3 \\ 14 & 21 & 14 & 21 & 145 & 21 \\ 2 & 3 & 2 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[111] \ 2 \ 3 \ 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 3 & 12 & 3 \\ 2 & 3 & 0 & 3 & 8 & 2 \\ 2 & 3 & 3 & 1 & 12 & 3 \\ 8 & 12 & 8 & 12 & 46 & 12 \\ 2 & 3 & 2 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[112] \ 2 \ 3 \ 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 3 & 9 & 3 \\ 2 & 3 & 0 & 3 & 6 & 2 \\ 2 & 3 & 3 & 1 & 9 & 3 \\ 6 & 9 & 6 & 9 & 25 & 9 \\ 2 & 3 & 2 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[107] \quad 2 \overset{1}{\underset{111}{3}} 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 & 2 \\ 2 & 3 & 3 & 1 & 3 & 3 \\ 2 & 3 & 2 & 3 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[114] \quad 2 \overset{1}{\underset{114}{2}} 31^{\textcolor{red}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 3 \\ 4 & 5 & 7 & 20 & 5 \\ 4 & 7 & 5 & 20 & 5 \\ 12 & 20 & 20 & 62 & 16 \\ 3 & 5 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[115] \quad 2 \overset{1}{\underset{112}{2}} 31^{\textcolor{red}{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 3 \\ 4 & 5 & 7 & 20 & 5 \\ 4 & 7 & 5 & 20 & 5 \\ 12 & 20 & 20 & 62 & 16 \\ 3 & 5 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[116] \quad 1^{\textcolor{red}{13}} \overset{1}{\underset{13}{3}} 21^{\textcolor{red}{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 150, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 8 & 8 & 9 & 0 & 40 & 12 \\ 2 & 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[117] \quad 1^{\textcolor{red}{13}} \overset{1}{\underset{13}{12}} \overset{1}{\underset{13}{3}} 21^{\textcolor{red}{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 153, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 8 & 7 & 11 & 0 & 45 & 13 \\ 2 & 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[118] \quad 1^{\textcolor{red}{14}} \overset{1}{\underset{14}{13}} \overset{1}{\underset{14}{3}} 21^{\textcolor{red}{12}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 9 & 8 & 9 & 0 & 40 & 12 \\ 2 & 2 & 2 & 0 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[119] \quad 1^{\textcolor{red}{13}} \overset{1}{\underset{13}{12}} \overset{1}{\underset{13}{3}} 21^{\textcolor{red}{14}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 9 & 8 & 14 & 0 & 70 & 16 \\ 2 & 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[120] \quad 1^{\textcolor{red}{14}} \overset{1}{\underset{14}{12}} \overset{1}{\underset{14}{3}} 21^{\textcolor{red}{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 175, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 9 & 7 & 11 & 0 & 45 & 13 \\ 2 & 2 & 2 & 0 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[121] \quad 2 \overset{1}{\underset{13}{2}} 2 31^{\textcolor{red}{13}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 9 & 2 \\ 1 & 0 & 2 & 2 & 9 & 2 \\ 2 & 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 2 & 0 & 9 & 2 \\ 9 & 9 & 12 & 9 & 61 & 15 \\ 2 & 2 & 2 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[122] \quad 2 \overset{1}{\underset{115}{2}} 2 31^{\textcolor{red}{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.1$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 3 \\ 4 & 5 & 7 & 25 & 5 \\ 4 & 7 & 5 & 25 & 5 \\ 15 & 25 & 25 & 98 & 20 \\ 3 & 5 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[123] \ 2 \overset{1^{11}}{3} \ 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -60, \ \det = 192, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 3 & 5 & 3 \\ 2 & 3 & 0 & 3 & 5 & 2 \\ 2 & 3 & 3 & 1 & 5 & 3 \\ 4 & 5 & 5 & 5 & 8 & 6 \\ 2 & 3 & 2 & 3 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[124] \ 1 \overset{1^{14}}{4} \ 3 \ 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 194, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 10 & 10 & 10 & 0 & 58 & 14 \\ 2 & 2 & 2 & 0 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[125] \ 2 \ 2 \ 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 196, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 3 \\ 4 & 5 & 7 & 15 & 5 \\ 4 & 7 & 5 & 15 & 5 \\ 9 & 15 & 15 & 32 & 12 \\ 3 & 5 & 5 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[126] \ 1 \overset{1^{13}}{3} \ 3 \ 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 203, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 9 & 9 & 12 & 0 & 61 & 15 \\ 2 & 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[127] \ 1 \overset{1^{14}}{2} \ 3 \ 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 206, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 10 & 8 & 14 & 0 & 70 & 16 \\ 2 & 2 & 2 & 0 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[128] \ 2 \ 2 \ 31^{16} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 224, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.2$$

$$T_{\min} = 9.96 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 18 & 3 \\ 4 & 5 & 7 & 30 & 5 \\ 4 & 7 & 5 & 30 & 5 \\ 18 & 30 & 30 & 142 & 24 \\ 3 & 5 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[129] \ 1 \overset{1^{13}}{3} \ 3 \ 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 234, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 15 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 10 & 10 & 15 & 0 & 88 & 18 \\ 2 & 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[130] \ 1 \overset{1^{14}}{3} \ 3 \ 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 237, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 10 & 9 & 12 & 0 & 61 & 15 \\ 2 & 2 & 2 & 0 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[131] \ 2 \ 2 \ 31^{17} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 276, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.3$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 21 & 3 \\ 4 & 5 & 7 & 35 & 5 \\ 4 & 7 & 5 & 35 & 5 \\ 21 & 35 & 35 & 194 & 28 \\ 3 & 5 & 5 & 28 & 4 \end{pmatrix}_{b_0}$$

$$[132] \ 1 \overset{1^{14}}{4} \ 3 \ 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 283, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 11 & 11 & 13 & 0 & 81 & 17 \\ 2 & 2 & 2 & 0 & 17 & 3 \end{pmatrix}_{b_0}$$

[133] $1^{13} 3^{14} 21^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 286, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 15 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 11 & 10 & 15 & 0 & 88 & 18 \\ 2 & 2 & 2 & 0 & 18 & 3 \end{pmatrix}_{b_0}$$

[134] $2^2 2 31^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 12 & 2 \\ 1 & 0 & 2 & 2 & 12 & 2 \\ 2 & 2 & 0 & 2 & 16 & 2 \\ 1 & 2 & 2 & 0 & 12 & 2 \\ 12 & 12 & 16 & 12 & 110 & 20 \\ 2 & 2 & 2 & 2 & 20 & 3 \end{pmatrix}_{b_0}$$

[135] $2^2 2 31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 3 \\ 4 & 5 & 7 & 20 & 5 \\ 4 & 7 & 5 & 20 & 5 \\ 12 & 20 & 20 & 58 & 16 \\ 3 & 5 & 5 & 16 & 4 \end{pmatrix}_{b_0}$$

[136] $2^2 3^{11} 2 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 320, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 3 & 8 & 3 \\ 2 & 3 & 0 & 3 & 7 & 2 \\ 2 & 3 & 3 & 1 & 8 & 3 \\ 6 & 8 & 7 & 8 & 21 & 9 \\ 2 & 3 & 2 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

[137] $2^2 2 31^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 3 \\ 4 & 5 & 7 & 25 & 5 \\ 4 & 7 & 5 & 25 & 5 \\ 15 & 25 & 25 & 94 & 20 \\ 3 & 5 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

[138] $1^{14} 3^{14} 21^{14} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 350, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 12 & 12 & 16 & 0 & 110 & 20 \\ 2 & 2 & 2 & 0 & 20 & 3 \end{pmatrix}_{b_0}$$

[139] $2^2 2 31^{11} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 420, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 3 \\ 4 & 5 & 7 & 25 & 5 \\ 4 & 7 & 5 & 25 & 5 \\ 15 & 25 & 25 & 92 & 20 \\ 3 & 5 & 5 & 20 & 4 \end{pmatrix}_{b_0}$$

[140] $2^2 3^{11} 2 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 3 & 3 & 11 & 3 \\ 2 & 3 & 0 & 3 & 9 & 2 \\ 2 & 3 & 3 & 1 & 11 & 3 \\ 8 & 11 & 9 & 11 & 40 & 12 \\ 2 & 3 & 2 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

[141] $2^2 2 31^{12} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 544, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 18 & 3 \\ 4 & 5 & 7 & 30 & 5 \\ 4 & 7 & 5 & 30 & 5 \\ 18 & 30 & 30 & 134 & 24 \\ 3 & 5 & 5 & 24 & 4 \end{pmatrix}_{b_0}$$

[142] $2^2 3^{11} 2 + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{blue}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 8, T_{\max} = 10, \Delta = -60, \det = 576, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 3 & 3 & 14 & 3 \\ 2 & 3 & 0 & 3 & 11 & 2 \\ 2 & 3 & 3 & 1 & 14 & 3 \\ 10 & 14 & 11 & 14 & 65 & 15 \\ 2 & 3 & 2 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 1 & 1 & 9 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 6 & 4 & 9 & 0 & 20 & 10 \\ 2 & 2 & 3 & 0 & 10 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 1 & 1 & 10 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 6 & 6 & 10 & 0 & 30 & 12 \\ 2 & 2 & 3 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 1 & 1 & 14 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 7 & 7 & 14 & 0 & 54 & 16 \\ 2 & 2 & 3 & 0 & 16 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 1 & 1 & 10 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 7 & 6 & 10 & 0 & 30 & 12 \\ 2 & 2 & 3 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 1 & 1 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 9 & 8 & 18 & 0 & 86 & 20 \\ 2 & 2 & 3 & 0 & 20 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 1 & 1 & 11 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 8 & 8 & 11 & 0 & 44 & 14 \\ 2 & 2 & 3 & 0 & 14 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 1 & 1 & 14 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 8 & 7 & 14 & 0 & 54 & 16 \\ 2 & 2 & 3 & 0 & 16 & 4 \end{pmatrix}_{b_0}$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10.04 \rightarrow 10$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10.04 \rightarrow 10$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 1 & 1 & 14 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 9 & 7 & 14 & 0 & 54 & 16 \end{pmatrix}$$

$$\begin{pmatrix} 9 & 7 & 14 & 0 & 54 & 16 \\ 2 & 2 & 3 & 0 & 16 & 4 \end{pmatrix}_{b_0}$$

$$T_{\min} = 9.97 \rightarrow 10, \quad T_{\max} = 10.03 \rightarrow 10$$

$$\begin{pmatrix} 1 & 0 & 1 & -2 & 0 & 0 \\ 10 & 8 & 18 & 0 & 86 & 20 \\ 2 & 2 & 3 & 0 & 20 & 4 \end{pmatrix}_b$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10.09 \rightarrow 10.1$$

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 15 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 9 & 9 & 15 & 0 & 70 & 18 \\ 2 & 2 & 3 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10.08 \rightarrow 10.1$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 1 & 1 & 19 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 10 & 10 & 19 & 0 & 104 & 22 \\ 2 & 2 & 3 & 0 & 22 & 4 \end{pmatrix}_{b_0}$$

[164] $1^{13} 3 221^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -167, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 1 & 1 & 15 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 10 & 9 & 15 & 0 & 70 & 18 \\ 2 & 2 & 3 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

[165] $1^{14} 3 221^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -193, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 1 & 1 & 16 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 11 & 11 & 16 & 0 & 90 & 20 \\ 2 & 2 & 3 & 0 & 20 & 4 \end{pmatrix}_{b_0}$$

[166] $1^{14} 3 221^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 1 & 1 & 19 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 11 & 10 & 19 & 0 & 104 & 22 \\ 2 & 2 & 3 & 0 & 22 & 4 \end{pmatrix}_{b_0}$$

[167] $1^{14} 3 221^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -29, \det = -250, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 1 & 1 & 20 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 12 & 12 & 20 & 0 & 126 & 24 \\ 2 & 2 & 3 & 0 & 24 & 4 \end{pmatrix}_{b_0}$$

[168] $41^{11} 2 3 2 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -59, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 2 & 4 & 4 & 2 \\ 4 & 10 & 5 & 12 & 12 & 6 \\ 2 & 5 & 0 & 5 & 4 & 2 \\ 4 & 12 & 5 & 10 & 12 & 6 \\ 4 & 12 & 4 & 12 & 10 & 6 \\ 2 & 6 & 2 & 6 & 6 & 3 \end{pmatrix}_{b_0}$$

$T = 10$ [169] $2 3 21^{141} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.5$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 3 \\ 4 & 6 & 8 & 12 & 6 \\ 4 & 8 & 6 & 12 & 6 \\ 6 & 12 & 12 & 18 & 10 \\ 3 & 6 & 6 & 10 & 5 \end{pmatrix}_{b_0}$$

[170] $2 3 21^{141} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 3 \\ 4 & 6 & 8 & 18 & 6 \\ 4 & 8 & 6 & 18 & 6 \\ 9 & 18 & 18 & 43 & 15 \\ 3 & 6 & 6 & 15 & 5 \end{pmatrix}_{b_0}$$

[171] $2 3 21^{141} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 3 \\ 4 & 6 & 8 & 24 & 6 \\ 4 & 8 & 6 & 24 & 6 \\ 12 & 24 & 24 & 78 & 20 \\ 3 & 6 & 6 & 20 & 5 \end{pmatrix}_{b_0}$$

[172] $2 3 21^{141} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.1$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 3 \\ 4 & 6 & 8 & 30 & 6 \\ 4 & 8 & 6 & 30 & 6 \\ 15 & 30 & 30 & 123 & 25 \\ 3 & 6 & 6 & 25 & 5 \end{pmatrix}_{b_0}$$

[173] $2 3 21^{141} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2$ $T = 10, T_{\max} = 11, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.2$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 4 & 18 & 3 \\ 4 & 6 & 8 & 36 & 6 \\ 4 & 8 & 6 & 36 & 6 \\ 18 & 36 & 36 & 178 & 30 \\ 3 & 6 & 6 & 30 & 5 \end{pmatrix}_{b_0}$$

[174] $2 \ 3 \ 21^{117} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.3$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 4 & 21 & 3 \\ 4 & 6 & 8 & 42 & 6 \\ 4 & 8 & 6 & 42 & 6 \\ 21 & 42 & 42 & 243 & 35 \\ 3 & 6 & 6 & 35 & 5 \end{pmatrix}_{b_0}$$

[175] $2 \ 3 \ 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 6 & 8 & 6 & 6 \\ 4 & 8 & 6 & 6 & 6 \\ 3 & 6 & 6 & 3 & 5 \\ 3 & 6 & 6 & 5 & 5 \end{pmatrix}_{b_0}$$

[178] $2 \ 3 \ 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 2 & 4 & 4 & 12 & 4 \\ 2 & 4 & 1 & 4 & 9 & 3 \\ 2 & 4 & 4 & 2 & 12 & 4 \\ 6 & 12 & 9 & 12 & 34 & 12 \\ 2 & 4 & 3 & 4 & 12 & 4 \end{pmatrix}_{b_0}$$

[177] $2 \ 3 \ 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 & 4 & 4 \\ 2 & 4 & 1 & 4 & 3 & 3 \\ 2 & 4 & 4 & 2 & 4 & 4 \\ 2 & 4 & 3 & 4 & 2 & 4 \\ 2 & 4 & 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

[179] $2 \ 3 \ 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 2 & 4 & 4 & 8 & 4 \\ 2 & 4 & 1 & 4 & 6 & 3 \\ 2 & 4 & 4 & 2 & 8 & 4 \\ 4 & 8 & 6 & 8 & 14 & 8 \\ 2 & 4 & 3 & 4 & 8 & 4 \end{pmatrix}_{b_0}$$

[176] $2 \ 3 \ 2 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 11, \Delta = -30, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 2 & 4 & 4 & 16 & 4 \\ 2 & 4 & 1 & 4 & 12 & 3 \\ 2 & 4 & 4 & 2 & 16 & 4 \\ 8 & 16 & 12 & 16 & 62 & 16 \\ 2 & 4 & 3 & 4 & 16 & 4 \end{pmatrix}_{b_0}$$

$T = 11$ [180] $1^{11} 31 \ 5 \ 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 2 & 0 & 6 & 4 \end{pmatrix}_{b_0}$$

[181] $1^{11} 31 \ 5 \ 131^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 3 & 1 & 0 & 1 & 3 \\ 2 & 3 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

[182] $1^{13} 31 \ 5 \ 131^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 10 & 3 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 6 & 10 & 6 & 0 & 34 & 12 \\ 2 & 3 & 2 & 0 & 12 & 4 \end{pmatrix}_{b_0}$$

[183] $1^{13} 31 \ 5 \ 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 6 & 11 & 9 & 0 & 49 & 15 \\ 2 & 3 & 2 & 0 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[184] \ 1^{12} 31^1 5^{11} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 4 & 7 & 5 & 0 & 17 & 9 \\ 2 & 3 & 2 & 0 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[185] \ 1^{14} 31^1 5^{12} 131^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -6, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 14 & 3 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 8 & 14 & 10 & 0 & 74 & 18 \\ 2 & 3 & 2 & 0 & 18 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [186] \ 1^{11} 31^1 5^{13} 13221^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$$

$$[192] \ 2^{1^{11} 4^{141}} 3^{11} 2 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 & 5 & 5 \\ 2 & 5 & 2 & 5 & 4 & 4 \\ 2 & 5 & 5 & 3 & 5 & 5 \\ 2 & 5 & 4 & 5 & 3 & 5 \\ 2 & 5 & 4 & 5 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[190] \ 2^{1^{13} 4^{141}} 3^{11} 2 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 3 & 5 & 5 & 15 & 5 \\ 2 & 5 & 2 & 5 & 12 & 4 \\ 2 & 5 & 5 & 3 & 15 & 5 \\ 6 & 15 & 12 & 15 & 43 & 15 \\ 2 & 5 & 4 & 5 & 15 & 5 \end{pmatrix}_{b_0}$$

$$[189] \ 2^{1^{14} 4^{141}} 3^{11} 2 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 3 & 5 & 5 & 20 & 5 \\ 2 & 5 & 2 & 5 & 16 & 4 \\ 2 & 5 & 5 & 3 & 20 & 5 \\ 8 & 20 & 16 & 20 & 78 & 20 \\ 2 & 5 & 4 & 5 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[191] \ 2^{1^{12} 4^{141}} 3^{11} 2 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 128, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 3 & 5 & 5 & 10 & 5 \\ 2 & 5 & 2 & 5 & 8 & 4 \\ 2 & 5 & 5 & 3 & 10 & 5 \\ 4 & 10 & 8 & 10 & 18 & 10 \\ 2 & 5 & 4 & 5 & 10 & 5 \end{pmatrix}_{b_0}$$

$$T = 15 \quad [198] \ 1^{11} 6^{2315131} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 15, T_{\max} = 15, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 15, 1), c_{\text{ext}} = 5.0$$

$$T = 16 \quad [201] \ 1^{11} 6^{131612321} 131^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 5.0$$

$$[202] \ 1^{11} 6^{1431} 13161^{12} 2321^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 16, T_{\max} = 16, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 16, 1), c_{\text{ext}} = 5.0$$

$$T = 17 \quad [203] \ 1^{11} 8^{23151321} 1231^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 \quad T = 17, T_{\max} = 17, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 5.0$$

5.116 NHC(-2)(-3)(-2) + $\mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 7 & 5 & 5 \\ 7 & 8 & 9 & 6 & 6 \\ 7 & 9 & 8 & 5 & 6 \\ 5 & 6 & 5 & 0 & 2 \\ 5 & 6 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 4 & 5 & 5 \\ 7 & 8 & 6 & 6 & 6 \\ 4 & 6 & 2 & 4 & 4 \\ 5 & 6 & 4 & 0 & 2 \\ 5 & 6 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 11 & 5 & 5 \\ 7 & 8 & 15 & 6 & 6 \\ 11 & 15 & 22 & 9 & 10 \\ 5 & 6 & 9 & 0 & 2 \\ 5 & 6 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 5 & 5 & 5 \\ 7 & 8 & 6 & 6 & 6 \\ 5 & 6 & 2 & 4 & 4 \\ 5 & 6 & 4 & 0 & 2 \\ 5 & 6 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 \ 1^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 19 & 5 & 5 \\ 7 & 8 & 24 & 6 & 6 \\ 19 & 24 & 62 & 14 & 16 \\ 5 & 6 & 14 & 0 & 2 \\ 5 & 6 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 \ 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 164, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 12 & 5 & 5 \\ 7 & 8 & 15 & 6 & 6 \\ 12 & 15 & 22 & 9 & 10 \\ 5 & 6 & 9 & 0 & 2 \\ 5 & 6 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [6] \ 2 \ 3 \ 21^{141} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -288, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 & 3 \\ 4 & 6 & 8 & 6 & 6 & 6 \\ 4 & 8 & 6 & 6 & 6 & 6 \\ 3 & 6 & 6 & 3 & 5 & 5 \\ 3 & 6 & 6 & 5 & 1 & 3 \\ 3 & 6 & 6 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

5.117 NHC(-2)(-3)(-2) + $\mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 2 \ 3 \ 2 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -30, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 2 & 3 & 3 \\ 2 & 3 & 0 & 3 & 3 & 2 & 2 \\ 2 & 3 & 3 & 1 & 2 & 3 & 3 \\ 2 & 2 & 3 & 2 & 1 & 2 & 3 \\ 2 & 3 & 2 & 3 & 2 & 1 & 3 \\ 2 & 3 & 2 & 3 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 2 \ 3 \ 2 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -30, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 4 & 2 \\ 2 & 1 & 3 & 3 & 6 & 5 & 3 \\ 2 & 3 & 0 & 3 & 4 & 5 & 2 \\ 2 & 3 & 3 & 1 & 6 & 5 & 3 \\ 4 & 6 & 4 & 6 & 10 & 10 & 6 \\ 4 & 5 & 5 & 5 & 10 & 8 & 6 \\ 2 & 3 & 2 & 3 & 6 & 6 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[3] \quad & 2 \overset{1}{\underset{1^{1112}}{3}} 2 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -30, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 3 & 3 & 6 & 3 \\ 2 & 3 & 0 & 3 & 2 & 4 & 2 \\ 2 & 3 & 3 & 1 & 3 & 6 & 3 \\ 2 & 3 & 2 & 3 & 1 & 6 & 3 \\ 4 & 6 & 4 & 6 & 6 & 10 & 6 \\ 2 & 3 & 2 & 3 & 3 & 6 & 3 \end{pmatrix}_{b_0} \\
[2] \quad & 2 \overset{1}{\underset{1^{1111}}{3}} 2 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -30, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 & 2 & 2 \\ 2 & 3 & 3 & 1 & 3 & 3 & 3 \\ 2 & 3 & 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 2 & 3 & 3 & 3 & 3 \end{pmatrix}_{b_0} \\
[4] \quad & 2 \overset{11}{\underset{1^{1111}}{3}} 2 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -30, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 4 & 2 \\ 2 & 1 & 3 & 3 & 3 & 5 & 3 \\ 2 & 3 & 0 & 3 & 2 & 5 & 2 \\ 2 & 3 & 3 & 1 & 3 & 5 & 3 \\ 2 & 3 & 2 & 3 & 1 & 5 & 3 \\ 4 & 5 & 5 & 5 & 5 & 8 & 6 \\ 2 & 3 & 2 & 3 & 3 & 6 & 3 \end{pmatrix}_{b_0} \\
[5] \quad & 2 \overset{11}{\underset{1^{1112}}{3}} 2 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -30, \det = -576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 6 & 2 \\ 2 & 1 & 3 & 3 & 3 & 8 & 3 \\ 2 & 3 & 0 & 3 & 2 & 7 & 2 \\ 2 & 3 & 3 & 1 & 3 & 8 & 3 \\ 2 & 3 & 2 & 3 & 1 & 8 & 3 \\ 6 & 8 & 7 & 8 & 8 & 21 & 9 \\ 2 & 3 & 2 & 3 & 3 & 9 & 3 \end{pmatrix}_{b_0} \\
T = 10 \quad [7] \quad & 1 \overset{12}{\underset{1^{114}}{2}} 3 \overset{112}{21} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 0 & 4 & 14 & 3 \\ 1 & 2 & 0 & 1 & 4 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 4 & 4 & 0 & 6 & 24 & 6 \\ 8 & 14 & 10 & 0 & 24 & 74 & 18 \\ 2 & 3 & 2 & 0 & 6 & 18 & 4 \end{pmatrix}_{b_0} \\
[6] \quad & 1 \overset{11}{\underset{1^{112}}{2}} 3 \overset{111}{21} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 0 & 4 & 7 & 3 \\ 1 & 2 & 0 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 4 & 4 & 0 & 6 & 12 & 6 \\ 4 & 7 & 5 & 0 & 12 & 17 & 9 \\ 2 & 3 & 2 & 0 & 6 & 9 & 4 \end{pmatrix}_{b_0} \\
[9] \quad & 1 \overset{122}{\underset{1^{134}}{2}} 3 \overset{112}{21} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 8 & 2 \\ 1 & 1 & 2 & 0 & 10 & 14 & 3 \\ 1 & 2 & 0 & 1 & 6 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 10 & 6 & 0 & 34 & 52 & 12 \\ 8 & 14 & 10 & 0 & 52 & 74 & 18 \\ 2 & 3 & 2 & 0 & 12 & 18 & 4 \end{pmatrix}_{b_0} \\
[8] \quad & 1 \overset{121}{\underset{1^{132}}{2}} 3 \overset{111}{21} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 4 & 2 \\ 1 & 1 & 2 & 0 & 10 & 7 & 3 \\ 1 & 2 & 0 & 1 & 6 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 10 & 6 & 0 & 34 & 26 & 12 \\ 4 & 7 & 5 & 0 & 26 & 17 & 9 \\ 2 & 3 & 2 & 0 & 12 & 9 & 4 \end{pmatrix}_{b_0} \\
[11] \quad & 1 \overset{111}{\underset{1^{122}}{2}} 3 \overset{111}{21} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 7 & 3 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 7 & 5 & 0 & 17 & 19 & 9 \\ 4 & 7 & 5 & 0 & 19 & 17 & 9 \\ 2 & 3 & 2 & 0 & 9 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{144} 2^1 3^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 8 & 2 \\ 1 & 1 & 2 & 0 & 14 & 14 & 3 \\ 1 & 2 & 0 & 1 & 10 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 14 & 10 & 0 & 74 & 76 & 18 \\ 8 & 14 & 10 & 0 & 76 & 74 & 18 \\ 2 & 3 & 2 & 0 & 18 & 18 & 4 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{124} 2^1 3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & 1 & 2 & 0 & 7 & 14 & 3 \\ 1 & 2 & 0 & 1 & 5 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 7 & 5 & 0 & 17 & 38 & 9 \\ 8 & 14 & 10 & 0 & 38 & 74 & 18 \\ 2 & 3 & 2 & 0 & 9 & 18 & 4 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [17] \ 2^1 3^{141} 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 & 3 \\ 4 & 6 & 8 & 6 & 6 & 6 \\ 4 & 8 & 6 & 6 & 6 & 6 \\ 3 & 6 & 6 & 3 & 5 & 5 \\ 3 & 6 & 6 & 5 & 3 & 5 \\ 3 & 6 & 6 & 5 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[18] \ 2^1 3^{141} 21^{1131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 3 & 3 \\ 4 & 6 & 8 & 18 & 6 & 6 \\ 4 & 8 & 6 & 18 & 6 & 6 \\ 9 & 18 & 18 & 43 & 15 & 15 \\ 3 & 6 & 6 & 15 & 3 & 5 \\ 3 & 6 & 6 & 15 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[39] \ 2^1 3^{141} 21^{1171} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.3$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 21 & 3 & 3 \\ 4 & 6 & 8 & 42 & 6 & 6 \\ 4 & 8 & 6 & 42 & 6 & 6 \\ 21 & 42 & 42 & 243 & 35 & 35 \\ 3 & 6 & 6 & 35 & 3 & 5 \\ 3 & 6 & 6 & 35 & 5 & 5 \end{pmatrix}_{b_0}$$

$$[38] \ 2^1 3^{141} 21^{1167} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.6$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 18 & 21 & 3 \\ 4 & 6 & 8 & 36 & 42 & 6 \\ 4 & 8 & 6 & 36 & 42 & 6 \\ 18 & 36 & 36 & 178 & 210 & 30 \\ 21 & 42 & 42 & 210 & 243 & 35 \\ 3 & 6 & 6 & 30 & 35 & 5 \end{pmatrix}_{b_0}$$

$$[37] \ 2^1 3^{141} 21^{1166} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 18 & 18 & 3 \\ 4 & 6 & 8 & 36 & 36 & 6 \\ 4 & 8 & 6 & 36 & 36 & 6 \\ 18 & 36 & 36 & 178 & 180 & 30 \\ 18 & 36 & 36 & 180 & 178 & 30 \\ 3 & 6 & 6 & 30 & 30 & 5 \end{pmatrix}_{b_0}$$

$$[16] \ 2^1 3^{141} 21^{1134} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 12 & 3 \\ 4 & 6 & 8 & 18 & 24 & 6 \\ 4 & 8 & 6 & 18 & 24 & 6 \\ 9 & 18 & 18 & 43 & 60 & 15 \\ 12 & 24 & 24 & 60 & 78 & 20 \\ 3 & 6 & 6 & 15 & 20 & 5 \end{pmatrix}_{b_0}$$

$$[35] \ 2^1 3^{141} 21^{1157} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

[15] $\begin{pmatrix} 1 & 4 & 4 & 15 & 21 & 3 \\ 4 & 6 & 8 & 30 & 42 & 6 \\ 4 & 8 & 6 & 30 & 42 & 6 \\ 15 & 30 & 30 & 123 & 175 & 25 \\ 21 & 42 & 42 & 175 & 243 & 35 \\ 3 & 6 & 6 & 25 & 35 & 5 \end{pmatrix}_{b_0}$ $^{1^1 41}$ $2 \ 3 \ 21^{1135} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.9$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 4 & 9 & 15 & 3 \\ 4 & 6 & 8 & 18 & 30 & 6 \\ 4 & 8 & 6 & 18 & 30 & 6 \\ 9 & 18 & 18 & 43 & 75 & 15 \\ 15 & 30 & 30 & 75 & 123 & 25 \\ 3 & 6 & 6 & 15 & 25 & 5 \end{pmatrix}_{b_0}$

[14] $\begin{pmatrix} 1 & 4 & 4 & 21 & 21 & 3 \\ 4 & 6 & 8 & 42 & 42 & 6 \\ 4 & 8 & 6 & 42 & 42 & 6 \\ 21 & 42 & 42 & 243 & 245 & 35 \\ 21 & 42 & 42 & 245 & 243 & 35 \\ 3 & 6 & 6 & 35 & 35 & 5 \end{pmatrix}_{b_0}$ $^{1^1 41}$ $2 \ 3 \ 21^{1177} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.7$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

[13] $\begin{pmatrix} 1 & 4 & 4 & 9 & 9 & 3 \\ 4 & 6 & 8 & 18 & 18 & 6 \\ 4 & 8 & 6 & 18 & 18 & 6 \\ 9 & 18 & 18 & 43 & 45 & 15 \\ 9 & 18 & 18 & 45 & 43 & 15 \\ 3 & 6 & 6 & 15 & 15 & 5 \end{pmatrix}_{b_0}$ $^{1^1 41}$ $2 \ 3 \ 21^{1133} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

[34] $\begin{pmatrix} 1 & 4 & 4 & 15 & 18 & 3 \\ 4 & 6 & 8 & 30 & 36 & 6 \\ 4 & 8 & 6 & 30 & 36 & 6 \\ 15 & 30 & 30 & 123 & 150 & 25 \\ 18 & 36 & 36 & 150 & 178 & 30 \\ 3 & 6 & 6 & 25 & 30 & 5 \end{pmatrix}_{b_0}$ $^{1^1 41}$ $2 \ 3 \ 21^{1156} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.4$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

[21] $\begin{pmatrix} 1 & 4 & 4 & 6 & 15 & 3 \\ 4 & 6 & 8 & 12 & 30 & 6 \\ 4 & 8 & 6 & 12 & 30 & 6 \\ 6 & 12 & 12 & 18 & 50 & 10 \\ 15 & 30 & 30 & 50 & 123 & 25 \\ 3 & 6 & 6 & 10 & 25 & 5 \end{pmatrix}_{b_0}$ $^{1^1 41}$ $2 \ 3 \ 21^{1125} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.6$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

[22] $\begin{pmatrix} 1 & 4 & 4 & 6 & 12 & 3 \\ 4 & 6 & 8 & 12 & 24 & 6 \\ 4 & 8 & 6 & 12 & 24 & 6 \\ 6 & 12 & 12 & 18 & 40 & 10 \\ 12 & 24 & 24 & 40 & 78 & 20 \\ 3 & 6 & 6 & 10 & 20 & 5 \end{pmatrix}_{b_0}$ $^{1^1 41}$ $2 \ 3 \ 21^{1124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

[23] $\begin{pmatrix} 1 & 4 & 4 & 6 & 9 & 3 \\ 4 & 6 & 8 & 12 & 18 & 6 \\ 4 & 8 & 6 & 12 & 18 & 6 \\ 6 & 12 & 12 & 18 & 30 & 10 \\ 9 & 18 & 18 & 30 & 43 & 15 \\ 3 & 6 & 6 & 10 & 15 & 5 \end{pmatrix}_{b_0}$ $^{1^1 41}$ $2 \ 3 \ 21^{1123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.3$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

[24] $\begin{pmatrix} 1 & 4 & 4 & 6 & 6 & 3 \\ 4 & 6 & 8 & 12 & 12 & 6 \\ 4 & 8 & 6 & 12 & 12 & 6 \\ 6 & 12 & 12 & 18 & 20 & 10 \\ 6 & 12 & 12 & 20 & 18 & 10 \\ 3 & 6 & 6 & 10 & 10 & 5 \end{pmatrix}_{b_0}$ $^{1^1 41}$ $2 \ 3 \ 21^{1122} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

[25] $\begin{pmatrix} 1 & 4 & 4 & 6 & 3 & 3 \\ 4 & 6 & 8 & 12 & 6 & 6 \\ 4 & 8 & 6 & 12 & 6 & 6 \\ 6 & 12 & 12 & 18 & 10 & 10 \\ 3 & 6 & 6 & 10 & 3 & 5 \\ 3 & 6 & 6 & 10 & 5 & 5 \end{pmatrix}_{b_0}$ $^{1^1 41}$ $2 \ 3 \ 21^{1121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

[26] $\begin{pmatrix} 1 & 4 & 4 & 15 & 21 & 3 \\ 4 & 6 & 8 & 30 & 42 & 6 \\ 4 & 8 & 6 & 30 & 42 & 6 \\ 15 & 30 & 30 & 123 & 175 & 25 \\ 21 & 42 & 42 & 175 & 243 & 35 \\ 3 & 6 & 6 & 25 & 35 & 5 \end{pmatrix}_{b_0}$ $^{1^1 41}$ $2 \ 3 \ 21^{1137} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 11, T_{\max} = 11, \Delta = 0, \det = -144, (n_+, n_-, n_0) = (1,$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 18 & 3 \\ 4 & 6 & 8 & 18 & 36 & 6 \\ 4 & 8 & 6 & 18 & 36 & 6 \\ 9 & 18 & 18 & 43 & 90 & 15 \\ 18 & 36 & 36 & 90 & 178 & 30 \\ 3 & 6 & 6 & 15 & 30 & 5 \end{pmatrix}_{b_0}$$

$$[19] \ 2 \overset{1^{141}}{3} \overset{1^{127}}{21} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 11, \ \Delta = 0, \ \det = -144, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 7.8$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 21 & 3 \\ 4 & 6 & 8 & 12 & 42 & 6 \\ 4 & 8 & 6 & 12 & 42 & 6 \\ 6 & 12 & 12 & 18 & 70 & 10 \\ 21 & 42 & 42 & 70 & 243 & 35 \\ 3 & 6 & 6 & 10 & 35 & 5 \end{pmatrix}_{b_0}$$

$$[41] \ 2 \overset{1^{1111}}{3} \overset{1^{11}}{4} 2 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, \ T_{\max} = 11, \ \Delta = 0, \ \det = -256, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 & 4 & 4 & 4 \\ 2 & 4 & 1 & 4 & 3 & 3 & 3 \\ 2 & 4 & 4 & 2 & 4 & 4 & 4 \\ 2 & 4 & 3 & 4 & 2 & 4 & 4 \\ 2 & 4 & 3 & 4 & 4 & 2 & 4 \\ 2 & 4 & 3 & 4 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [42] \ 1 \overset{1^{11}}{1^{12}} 31 \overset{1^{11}}{5} 1321 \overset{1^{11}}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 0, \ (n_+, n_-, n_0) = (1, 12, 1), \ c_{\text{ext}} = 6.5$$

$$[44] \ 1 \overset{1^{11}}{1^{12}} 31 \overset{1^{11}}{5} 131 \overset{1^{11}}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 2 & 0 & 4 & 7 & 3 \\ 1 & 2 & 0 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 4 & 4 & 0 & 6 & 12 & 6 \\ 4 & 7 & 5 & 0 & 12 & 17 & 9 \\ 2 & 3 & 2 & 0 & 6 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[43] \ 1 \overset{1^{12}}{1^{14}} 31 \overset{1^{12}}{5} 131 \overset{1^{12}}{1} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 12, \ T_{\max} = 12, \ \Delta = 0, \ \det = 4, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 2 & 0 & 4 & 14 & 3 \\ 1 & 2 & 0 & 1 & 4 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 4 & 4 & 0 & 6 & 24 & 6 \\ 8 & 14 & 10 & 0 & 24 & 74 & 18 \\ 2 & 3 & 2 & 0 & 6 & 18 & 4 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [46] \ 1 \overset{1^{11}}{1^{12}} 3 \overset{1^{11}}{6} 131 \overset{1^{11}}{1^{13}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, \ T_{\max} = 13, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 13, 1), \ c_{\text{ext}} = 6.8$$

$$[45] \ 1 \overset{1^{11}}{1^{11}} 3 \overset{1^{11}}{6} 131 \overset{1^{11}}{1^{12}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 13, \ T_{\max} = 13, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 13, 1), \ c_{\text{ext}} = 6.0$$

5.118 NHC(-2)(-3)(-2) + $\mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 12 \quad [0] \ 1 \overset{1^{11}}{1^{12}} 3 \overset{1^{11}}{1} 5 131 \overset{1^{11}}{1^{11}} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{blue}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 12, \ T_{\max} = 12, \ \Delta = -1, \ \det = 0, \ (n_+, n_-, n_0) = (1, 12, 1), \ c_{\text{ext}} = 7.5$$

5.119 NHC(-2)(-3)(-2) + $\mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 8 \quad [0] \ 1 \overset{1^1}{1^{13}} 3 \overset{1^1}{1^1} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 19, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.79 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 5 \\ 7 & 8 & 6 & 5 & 6 \\ 4 & 6 & 2 & 3 & 4 \\ 4 & 5 & 3 & 1 & 3 \\ 5 & 6 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1 \overset{1^1}{1^{13}} 3 \overset{1^1}{1^1} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 25, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.92 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 7 & 4 & 5 \\ 7 & 8 & 9 & 5 & 6 \\ 7 & 9 & 8 & 5 & 6 \\ 4 & 5 & 5 & 1 & 3 \\ 5 & 6 & 6 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[2] \ 1 \overset{1^{12}}{1^{11}} 3 \overset{1^1}{1^1} 1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{orange}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 27, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

- $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 6 & 4 & 5 \\ 7 & 8 & 6 & 5 & 6 \\ 6 & 6 & 2 & 3 & 4 \\ 4 & 5 & 3 & 1 & 3 \\ 5 & 6 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$
- [3] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 11 & 4 & 5 \\ 7 & 8 & 15 & 5 & 6 \\ 11 & 15 & 22 & 8 & 10 \\ 4 & 5 & 8 & 1 & 3 \\ 5 & 6 & 10 & 3 & 4 \end{pmatrix}_{b_0}$$
- [4] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 5 & 4 & 5 \\ 7 & 8 & 6 & 5 & 6 \\ 5 & 6 & 2 & 3 & 4 \\ 4 & 5 & 3 & 1 & 3 \\ 5 & 6 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$
- [5] $21^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 15 & 3 & 4 \\ 5 & 4 & 6 & 14 & 3 & 4 \\ 5 & 6 & 4 & 14 & 3 & 4 \\ 15 & 14 & 14 & 37 & 8 & 11 \\ 3 & 3 & 3 & 8 & 0 & 2 \\ 4 & 4 & 4 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$
- [6] $21^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 3 & 3 & 4 \\ 5 & 4 & 6 & 4 & 3 & 4 \\ 5 & 6 & 4 & 4 & 3 & 4 \\ 3 & 4 & 4 & 1 & 2 & 3 \\ 3 & 3 & 3 & 2 & 0 & 2 \\ 4 & 4 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$
- [7] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 11 & 4 & 5 \\ 7 & 8 & 12 & 5 & 6 \\ 11 & 12 & 14 & 6 & 8 \\ 4 & 5 & 6 & 1 & 3 \\ 5 & 6 & 8 & 3 & 4 \end{pmatrix}_{b_0}$$
- [8] $1^{12}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 10 & 4 & 5 \\ 7 & 8 & 12 & 5 & 6 \\ 10 & 12 & 14 & 6 & 8 \\ 4 & 5 & 6 & 1 & 3 \\ 5 & 6 & 8 & 3 & 4 \end{pmatrix}_{b_0}$$
- [9] $21^{11}3^{11}1^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 18 & 3 & 4 \\ 5 & 4 & 6 & 21 & 3 & 4 \\ 5 & 6 & 4 & 21 & 3 & 4 \\ 18 & 21 & 21 & 76 & 11 & 16 \\ 3 & 3 & 3 & 11 & 0 & 2 \\ 4 & 4 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0}$$
- [10] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 16 & 4 & 5 \\ 7 & 8 & 18 & 5 & 6 \\ 16 & 18 & 34 & 9 & 12 \\ 4 & 5 & 9 & 1 & 3 \\ 5 & 6 & 12 & 3 & 4 \end{pmatrix}_{b_0}$$
- [11] $1^{13}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 15 & 4 & 5 \\ 7 & 8 & 18 & 5 & 6 \\ 15 & 18 & 34 & 9 & 12 \\ 4 & 5 & 9 & 1 & 3 \\ 5 & 6 & 12 & 3 & 4 \end{pmatrix}_{b_0}$$
- [12] $1^{11}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 73, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$

$$\begin{aligned}
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 14 & 4 & 5 \\ 7 & 8 & 15 & 5 & 6 \\ 14 & 15 & 22 & 8 & 10 \\ 4 & 5 & 8 & 1 & 3 \\ 5 & 6 & 10 & 3 & 4 \end{pmatrix}_{b_0} \\
[13] & 1^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 81, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 12 & 4 & 5 \\ 7 & 8 & 15 & 5 & 6 \\ 12 & 15 & 22 & 8 & 10 \\ 4 & 5 & 8 & 1 & 3 \\ 5 & 6 & 10 & 3 & 4 \end{pmatrix}_{b_0} \\
[14] & 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 13 & 4 & 5 \\ 7 & 8 & 15 & 5 & 6 \\ 13 & 15 & 22 & 8 & 10 \\ 4 & 5 & 8 & 1 & 3 \\ 5 & 6 & 10 & 3 & 4 \end{pmatrix}_{b_0} \\
[15] & 1^{14} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 20 & 4 & 5 \\ 7 & 8 & 24 & 5 & 6 \\ 20 & 24 & 62 & 12 & 16 \\ 4 & 5 & 12 & 1 & 3 \\ 5 & 6 & 16 & 3 & 4 \end{pmatrix}_{b_0} \\
[16] & 1^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 19 & 4 & 5 \\ 7 & 8 & 24 & 5 & 6 \\ 19 & 24 & 62 & 13 & 16 \\ 4 & 5 & 13 & 1 & 3 \\ 5 & 6 & 16 & 3 & 4 \end{pmatrix}_{b_0} \\
[17] & 21^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 9 & 3 & 4 \\ 5 & 4 & 6 & 9 & 3 & 4 \\ 5 & 6 & 4 & 9 & 3 & 4 \\ 9 & 9 & 9 & 13 & 5 & 7 \\ 3 & 3 & 3 & 5 & 0 & 2 \\ 4 & 4 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0} \\
[18] & 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 20 & 4 & 5 \\ 7 & 8 & 24 & 5 & 6 \\ 20 & 24 & 62 & 13 & 16 \\ 4 & 5 & 13 & 1 & 3 \\ 5 & 6 & 16 & 3 & 4 \end{pmatrix}_{b_0} \\
[19] & 21^{11} 3^{11} 1^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 15 & 3 & 4 \\ 5 & 4 & 6 & 17 & 3 & 4 \\ 5 & 6 & 4 & 17 & 3 & 4 \\ 15 & 17 & 17 & 49 & 9 & 13 \\ 3 & 3 & 3 & 9 & 0 & 2 \\ 4 & 4 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0} \\
[20] & 1^{13} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 21 & 4 & 5 \\ 7 & 8 & 24 & 5 & 6 \\ 21 & 24 & 62 & 13 & 16 \\ 4 & 5 & 13 & 1 & 3 \\ 5 & 6 & 16 & 3 & 4 \end{pmatrix}_{b_0} \\
[21] & 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 5 & 7 & 17 & 4 & 5 \\ 7 & 8 & 21 & 5 & 6 \\ 17 & 21 & 44 & 11 & 14 \\ 4 & 5 & 11 & 1 & 3 \\ 5 & 6 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 145, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 19 & 4 & 5 \\ 7 & 8 & 21 & 5 & 6 \\ 19 & 21 & 44 & 11 & 14 \\ 4 & 5 & 11 & 1 & 3 \\ 5 & 6 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[23] \quad 21^{11} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 5 & 12 & 3 & 4 \\ 5 & 4 & 6 & 13 & 3 & 4 \\ 5 & 6 & 4 & 13 & 3 & 4 \\ 12 & 13 & 13 & 28 & 7 & 10 \\ 3 & 3 & 3 & 7 & 0 & 2 \\ 4 & 4 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[24] \quad 21^{11} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 2 & 3 \\ 4 & 4 & 6 & 4 & 3 & 4 \\ 4 & 6 & 4 & 4 & 3 & 4 \\ 3 & 4 & 4 & 1 & 2 & 3 \\ 2 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 18 & 4 & 5 \\ 7 & 8 & 21 & 5 & 6 \\ 18 & 21 & 44 & 11 & 14 \\ 4 & 5 & 11 & 1 & 3 \\ 5 & 6 & 14 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[26] \quad 21^{11} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 2 & 3 \\ 4 & 4 & 6 & 8 & 3 & 4 \\ 4 & 6 & 4 & 8 & 3 & 4 \\ 6 & 8 & 8 & 10 & 4 & 6 \\ 2 & 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[27] \quad 21^{11} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 192, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 2 & 3 \\ 4 & 4 & 6 & 9 & 3 & 4 \\ 4 & 6 & 4 & 9 & 3 & 4 \\ 6 & 9 & 9 & 13 & 5 & 7 \\ 2 & 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 4 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{12} 3^{11} 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 24 & 4 & 5 \\ 7 & 8 & 30 & 5 & 6 \\ 24 & 30 & 94 & 16 & 20 \\ 4 & 5 & 16 & 1 & 3 \\ 5 & 6 & 20 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 217, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 22 & 4 & 5 \\ 7 & 8 & 27 & 5 & 6 \\ 22 & 27 & 74 & 14 & 18 \\ 4 & 5 & 14 & 1 & 3 \\ 5 & 6 & 18 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[30] \quad 21^{11} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 2 & 3 \\ 4 & 4 & 6 & 12 & 3 & 4 \\ 4 & 6 & 4 & 12 & 3 & 4 \\ 9 & 12 & 12 & 25 & 6 & 9 \\ 2 & 3 & 3 & 6 & 0 & 2 \\ 3 & 4 & 4 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[31] \quad 21^{12}3^11^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -58, \quad \det = 232, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 5 & 18 & 3 & 4 \\ 5 & 4 & 6 & 18 & 3 & 4 \\ 5 & 6 & 4 & 18 & 3 & 4 \\ 18 & 18 & 18 & 58 & 10 & 14 \\ 3 & 3 & 3 & 10 & 0 & 2 \\ 4 & 4 & 4 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{13}3^11^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -58, \quad \det = 238, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 23 & 4 & 5 \\ 7 & 8 & 27 & 5 & 6 \\ 23 & 27 & 74 & 14 & 18 \\ 4 & 5 & 14 & 1 & 3 \\ 5 & 6 & 18 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{12}3^11^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -58, \quad \det = 259, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.92 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 25 & 4 & 5 \\ 7 & 8 & 30 & 5 & 6 \\ 25 & 30 & 94 & 16 & 20 \\ 4 & 5 & 16 & 1 & 3 \\ 5 & 6 & 20 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{12}3^11^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -58, \quad \det = 268, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 26 & 4 & 5 \\ 7 & 8 & 30 & 5 & 6 \\ 26 & 30 & 94 & 16 & 20 \\ 4 & 5 & 16 & 1 & 3 \\ 5 & 6 & 20 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[35] \quad 21^{114}3^11^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -58, \quad \det = 280, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 2 & 3 \\ 4 & 4 & 6 & 16 & 3 & 4 \\ 4 & 6 & 4 & 16 & 3 & 4 \\ 12 & 16 & 16 & 46 & 8 & 12 \\ 2 & 3 & 3 & 8 & 0 & 2 \\ 3 & 4 & 4 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{14}3^11^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -58, \quad \det = 297, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.89 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 27 & 4 & 5 \\ 7 & 8 & 33 & 5 & 6 \\ 27 & 33 & 112 & 17 & 22 \\ 4 & 5 & 17 & 1 & 3 \\ 5 & 6 & 22 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[37] \quad 21^{112}3^11^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -58, \quad \det = 312, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 2 & 3 \\ 4 & 4 & 6 & 18 & 3 & 4 \\ 4 & 6 & 4 & 18 & 3 & 4 \\ 12 & 18 & 18 & 58 & 10 & 14 \\ 2 & 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 4 & 14 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{13}3^11^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -58, \quad \det = 340, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 29 & 4 & 5 \\ 7 & 8 & 36 & 5 & 6 \\ 29 & 36 & 134 & 19 & 24 \\ 4 & 5 & 19 & 1 & 3 \\ 5 & 6 & 24 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[39] \quad 21^{115}3^11^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -58, \quad \det = 352, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 7.1$$

$$T_{\min} = 9.95 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 2 & 3 \\ 4 & 4 & 6 & 20 & 3 & 4 \\ 4 & 6 & 4 & 20 & 3 & 4 \\ 15 & 20 & 20 & 73 & 10 & 15 \\ 2 & 3 & 3 & 10 & 0 & 2 \\ 3 & 4 & 4 & 15 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[40] \quad 21^{112}3^11^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, \quad T_{\max} = 10, \quad \Delta = -58, \quad \det = 360, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 6.5$$

$$\begin{aligned}
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 9 & 2 & 3 \\ 4 & 4 & 6 & 13 & 3 & 4 \\ 4 & 6 & 4 & 13 & 3 & 4 \\ 9 & 13 & 13 & 28 & 7 & 10 \\ 2 & 3 & 3 & 7 & 0 & 2 \\ 3 & 4 & 4 & 10 & 2 & 3 \end{pmatrix}_{b_0} \\
[41] \quad & 21^{12}3^{11}1^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 360, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 21 & 3 & 4 \\ 5 & 4 & 6 & 22 & 3 & 4 \\ 5 & 6 & 4 & 22 & 3 & 4 \\ 21 & 22 & 22 & 85 & 12 & 17 \\ 3 & 3 & 3 & 12 & 0 & 2 \\ 4 & 4 & 4 & 17 & 2 & 3 \end{pmatrix}_{b_0} \\
[42] \quad & 21^{12}3^{11}1^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 424, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 24 & 3 & 4 \\ 5 & 4 & 6 & 26 & 3 & 4 \\ 5 & 6 & 4 & 26 & 3 & 4 \\ 24 & 26 & 26 & 118 & 14 & 20 \\ 3 & 3 & 3 & 14 & 0 & 2 \\ 4 & 4 & 4 & 20 & 2 & 3 \end{pmatrix}_{b_0} \\
[43] \quad & 21^{116}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 440, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 7.2 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 18 & 2 & 3 \\ 4 & 4 & 6 & 24 & 3 & 4 \\ 4 & 6 & 4 & 24 & 3 & 4 \\ 18 & 24 & 24 & 106 & 12 & 18 \\ 2 & 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 4 & 18 & 2 & 3 \end{pmatrix}_{b_0} \\
[44] \quad & 1^{14}3^{11}1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = \\
& 7.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 34 & 4 & 5 \\ 7 & 8 & 42 & 5 & 6 \\ 34 & 42 & 182 & 22 & 28 \\ 4 & 5 & 22 & 1 & 3 \\ 5 & 6 & 28 & 3 & 4 \end{pmatrix}_{b_0} \\
[45] \quad & 21^{117}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 544, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 7.3 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 21 & 2 & 3 \\ 4 & 4 & 6 & 28 & 3 & 4 \\ 4 & 6 & 4 & 28 & 3 & 4 \\ 21 & 28 & 28 & 145 & 14 & 21 \\ 2 & 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 4 & 21 & 2 & 3 \end{pmatrix}_{b_0} \\
[46] \quad & 21^{113}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 544, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 12 & 2 & 3 \\ 4 & 4 & 6 & 17 & 3 & 4 \\ 4 & 6 & 4 & 17 & 3 & 4 \\ 12 & 17 & 17 & 49 & 9 & 13 \\ 2 & 3 & 3 & 9 & 0 & 2 \\ 3 & 4 & 4 & 13 & 2 & 3 \end{pmatrix}_{b_0} \\
[47] \quad & 21^{113}3^{11}1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 640, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 15 & 2 & 3 \\ 4 & 4 & 6 & 22 & 3 & 4 \\ 4 & 6 & 4 & 22 & 3 & 4 \\ 15 & 22 & 22 & 85 & 12 & 17 \\ 2 & 3 & 3 & 12 & 0 & 2 \\ 3 & 4 & 4 & 17 & 2 & 3 \end{pmatrix}_{b_0} \\
[48] \quad & 21^{114}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 744, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 15 & 2 & 3 \\ 4 & 4 & 6 & 21 & 3 & 4 \\ 4 & 6 & 4 & 21 & 3 & 4 \\ 15 & 21 & 21 & 76 & 11 & 16 \\ 2 & 3 & 3 & 11 & 0 & 2 \\ 3 & 4 & 4 & 16 & 2 & 3 \end{pmatrix}_{b_0} \\
[49] \quad & 21^{114}3^{11}1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 984, (n_+, n_-, n_0) = \\
& (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 18 & 2 & 3 \\ 4 & 4 & 6 & 26 & 3 & 4 \\ 4 & 6 & 4 & 26 & 3 & 4 \\ 18 & 26 & 26 & 118 & 14 & 20 \\ 2 & 3 & 3 & 14 & 0 & 2 \\ 3 & 4 & 4 & 20 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$T = 9 \quad [50] \quad 1^1 \overset{1}{3} 1^1 21^{11} + [\text{su}_2 \cdot \text{su}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -2, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 2 & 4 & 0 & 2 & 4 \\ 3 & 3 & 3 & 0 & 4 & 4 \end{pmatrix}_{b_0}$$

[51] $1^{11}3^{1^1}21^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{teal}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 10$, $\Delta = -31$, $\det = -11$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 3 & 3 \\ 2 & 3 & 1 & 1 & 2 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 3 \\ 3 & 3 & 3 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

[52] $1^{11}3^{11}2^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 3 \\ 2 & 1 & 3 & 0 & 3 & 3 \\ 2 & 3 & 1 & 1 & 2 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 3 & 3 & 2 & 0 & 1 & 3 \\ 3 & 3 & 3 & 0 & 3 & 4 \end{pmatrix} b_0$$

[53] $1^{11}3^{12}21^{11} + [\text{su}_2 \cdot \text{su}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 3 \\ 2 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 1 & 1 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 7 \\ 3 & 3 & 3 & 0 & 7 & 4 \end{pmatrix}_{b_0}$$

[54] $1^{11}3^{11}21^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -50, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 7 & 3 \\ 2 & 1 & 3 & 0 & 7 & 3 \\ 2 & 3 & 1 & 1 & 10 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 7 & 7 & 10 & 0 & 25 & 11 \\ 3 & 3 & 3 & 0 & 11 & 4 \end{pmatrix}_{b_0}$$

[55] $1^{11}3^{11}21^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -58, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 3 \\ 2 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 1 & 1 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 7 \\ 3 & 3 & 3 & 0 & 7 & 4 \end{pmatrix} b_0$$

[56] $1^{11}3^{12}21^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -67, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 3 \\ 2 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 1 & 1 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 7 \\ 3 & 3 & 3 & 0 & 7 & 4 \end{pmatrix} b_0$$

[57] $1^{12} 1^{11} 3^{12} 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -91, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 3 \\ 2 & 1 & 3 & 0 & 7 & 3 \\ 2 & 3 & 1 & 1 & 10 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 11 \\ 3 & 3 & 3 & 0 & 11 & 4 \end{pmatrix}_{b_0}$$

[58] $1^{12}3^{13}21^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -98, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$[67] \quad 1^{14} 3^1 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -427, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 18 & 3 \\ 2 & 1 & 3 & 0 & 17 & 3 \\ 2 & 3 & 1 & 1 & 22 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 18 & 17 & 22 & 0 & 137 & 25 \\ 3 & 3 & 3 & 0 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{14} 3^1 21^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -450, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 18 & 3 \\ 2 & 1 & 3 & 0 & 18 & 3 \\ 2 & 3 & 1 & 1 & 20 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 18 & 18 & 20 & 0 & 130 & 24 \\ 3 & 3 & 3 & 0 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{14} 3^1 21^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -538, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 20 & 3 \\ 2 & 1 & 3 & 0 & 20 & 3 \\ 2 & 3 & 1 & 1 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 20 & 20 & 24 & 0 & 174 & 28 \\ 3 & 3 & 3 & 0 & 28 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [70] \quad 1^{11} 3^1 1^1 5^{12} 1321^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 12, T_{\max} = 12, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0$$

$$T = 14 \quad [73] \quad 1^{11} 7^{11} 123^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) \quad T = 14, T_{\max} = 14, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 14, 1), c_{\text{ext}} = 6.0$$

5.120 NHC(-2)(-3)(-2) + $\textcolor{red}{su}_2$ + $\textcolor{red}{su}_2 \rightarrow (-3)$ + $\textcolor{red}{su}_2$

$$T = 10 \quad [0] \quad 1^{13} 3^1 21^{113} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 15 & 3 \\ 2 & 1 & 3 & 0 & 2 & 15 & 3 \\ 2 & 3 & 1 & 1 & 4 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 4 & 0 & 2 & 18 & 4 \\ 15 & 15 & 18 & 0 & 18 & 97 & 21 \\ 3 & 3 & 3 & 0 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{12} 3^1 21^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 3 & 0 & 2 & 10 & 3 \\ 2 & 3 & 1 & 1 & 4 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 4 & 0 & 2 & 12 & 4 \\ 10 & 10 & 12 & 0 & 12 & 42 & 14 \\ 3 & 3 & 3 & 0 & 4 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 5 & 3 \\ 2 & 1 & 3 & 0 & 2 & 5 & 3 \\ 2 & 3 & 1 & 1 & 4 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 4 & 0 & 2 & 6 & 4 \\ 5 & 5 & 6 & 0 & 6 & 9 & 7 \\ 3 & 3 & 3 & 0 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{14} 3^1 21^{114} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 20 & 3 \\ 2 & 1 & 3 & 0 & 2 & 20 & 3 \\ 2 & 3 & 1 & 1 & 4 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 4 & 0 & 2 & 24 & 4 \\ 20 & 20 & 24 & 0 & 24 & 174 & 28 \\ 3 & 3 & 3 & 0 & 4 & 28 & 4 \end{pmatrix}_{b_0}$$

[7] $1^{112} 1^{111} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 8 & 3 \\ 2 & 1 & 3 & 0 & 5 & 7 & 3 \\ 2 & 3 & 1 & 1 & 6 & 10 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 17 & 7 \\ 8 & 7 & 10 & 0 & 17 & 25 & 11 \\ 3 & 3 & 3 & 0 & 7 & 11 & 4 \end{pmatrix}_{b_0}$$

[6] $1^{112} 1^{111} 3^1 21^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 6 & 3 \\ 2 & 1 & 3 & 0 & 3 & 5 & 3 \\ 2 & 3 & 1 & 1 & 2 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 5 & 3 \\ 6 & 5 & 6 & 0 & 5 & 9 & 7 \\ 3 & 3 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[5] $1^{112} 1^{111} 3^1 21^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 4 & 3 \\ 2 & 1 & 3 & 0 & 3 & 5 & 3 \\ 2 & 3 & 1 & 1 & 2 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 5 & 3 \\ 4 & 5 & 6 & 0 & 5 & 9 & 7 \\ 3 & 3 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[4] $1^{122} 1^{113} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 14 & 3 \\ 2 & 1 & 3 & 0 & 7 & 15 & 3 \\ 2 & 3 & 1 & 1 & 10 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 51 & 11 \\ 14 & 15 & 18 & 0 & 51 & 97 & 21 \\ 3 & 3 & 3 & 0 & 11 & 21 & 4 \end{pmatrix}_{b_0}$$

[8] $1^{122} 1^{113} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 12 & 3 \\ 2 & 1 & 3 & 0 & 5 & 13 & 3 \\ 2 & 3 & 1 & 1 & 6 & 14 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 27 & 7 \\ 12 & 13 & 14 & 0 & 27 & 65 & 17 \\ 3 & 3 & 3 & 0 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

[9] $1^{114} 1^{113} 3^1 21^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 20 & 3 \\ 2 & 1 & 3 & 0 & 3 & 20 & 3 \\ 2 & 3 & 1 & 1 & 2 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 2 & 0 & 1 & 20 & 3 \\ 20 & 20 & 24 & 0 & 20 & 174 & 28 \\ 3 & 3 & 3 & 0 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

[10] $1^{113} 1^{113} 3^1 21^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 15 & 3 \\ 2 & 1 & 3 & 0 & 3 & 15 & 3 \\ 2 & 3 & 1 & 1 & 2 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 2 & 0 & 1 & 15 & 3 \\ 15 & 15 & 18 & 0 & 15 & 97 & 21 \\ 3 & 3 & 3 & 0 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[11] $1^{112} 1^{112} 3^1 21^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 10 & 3 \\ 2 & 1 & 3 & 0 & 3 & 10 & 3 \\ 2 & 3 & 1 & 1 & 2 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 2 & 0 & 1 & 10 & 3 \\ 10 & 10 & 12 & 0 & 10 & 42 & 14 \\ 3 & 3 & 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[12] $1^{111} 3^1 21^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 & 5 & 3 \\ 2 & 1 & 3 & 0 & 3 & 5 & 3 \\ 2 & 3 & 1 & 1 & 2 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 2 & 0 & 1 & 5 & 3 \\ 5 & 5 & 6 & 0 & 5 & 9 & 7 \\ 3 & 3 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[13] $1^{111} 3^1 21^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 38, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 5 & 3 \\ 2 & 1 & 3 & 0 & 3 & 5 & 3 \\ 2 & 3 & 1 & 1 & 2 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 5 & 3 \\ 5 & 5 & 6 & 0 & 5 & 9 & 7 \\ 3 & 3 & 3 & 0 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[14] $1^{111} 3^1 21^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 5 & 3 \\ 2 & 1 & 3 & 0 & 5 & 5 & 3 \\ 2 & 3 & 1 & 1 & 6 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 11 & 7 \\ 5 & 5 & 6 & 0 & 11 & 9 & 7 \\ 3 & 3 & 3 & 0 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

[17] $1^{123} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 19 & 3 \\ 2 & 1 & 3 & 0 & 7 & 20 & 3 \\ 2 & 3 & 1 & 1 & 10 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 68 & 11 \\ 19 & 20 & 24 & 0 & 68 & 174 & 28 \\ 3 & 3 & 3 & 0 & 11 & 28 & 4 \end{pmatrix}_{b_0}$$

[16] $1^{111} 3^1 21^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 7 & 3 \\ 2 & 1 & 3 & 0 & 5 & 8 & 3 \\ 2 & 3 & 1 & 1 & 6 & 8 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 16 & 7 \\ 7 & 8 & 8 & 0 & 16 & 22 & 10 \\ 3 & 3 & 3 & 0 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

[20] $1^{111} 3^1 21^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 9 & 3 \\ 2 & 1 & 3 & 0 & 3 & 10 & 3 \\ 2 & 3 & 1 & 1 & 2 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 10 & 3 \\ 9 & 10 & 12 & 0 & 10 & 42 & 14 \\ 3 & 3 & 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[19] $1^{123} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 17 & 3 \\ 2 & 1 & 3 & 0 & 5 & 18 & 3 \\ 2 & 3 & 1 & 1 & 6 & 20 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 38 & 7 \\ 17 & 18 & 20 & 0 & 38 & 130 & 24 \\ 3 & 3 & 3 & 0 & 7 & 24 & 4 \end{pmatrix}_{b_0}$$

[18] $1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 13 & 3 \\ 2 & 1 & 3 & 0 & 5 & 12 & 3 \\ 2 & 3 & 1 & 1 & 6 & 16 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 28 & 7 \\ 13 & 12 & 16 & 0 & 28 & 70 & 18 \\ 3 & 3 & 3 & 0 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

[15] $1^{113} 3^{112} 21^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 11 & 3 \\ 2 & 1 & 3 & 0 & 3 & 10 & 3 \\ 2 & 3 & 1 & 1 & 2 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 10 & 3 \\ 11 & 10 & 12 & 0 & 10 & 42 & 14 \\ 3 & 3 & 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[21] $1^{112} 3^{112} 21^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 10 & 3 \\ 2 & 1 & 3 & 0 & 3 & 10 & 3 \\ 2 & 3 & 1 & 1 & 2 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 10 & 3 \\ 10 & 10 & 12 & 0 & 10 & 42 & 14 \\ 3 & 3 & 3 & 0 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

[25] $1^{114} 3^{112} 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 7 & 20 & 3 \\ 2 & 1 & 3 & 0 & 7 & 20 & 3 \\ 2 & 3 & 1 & 1 & 10 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 7 & 10 & 0 & 25 & 68 & 11 \\ 20 & 20 & 24 & 0 & 68 & 174 & 28 \\ 3 & 3 & 3 & 0 & 11 & 28 & 4 \end{pmatrix}_{b_0}$$

[24] $1^{113} 3^{112} 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 7 & 15 & 3 \\ 2 & 1 & 3 & 0 & 7 & 15 & 3 \\ 2 & 3 & 1 & 1 & 10 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 7 & 10 & 0 & 25 & 51 & 11 \\ 15 & 15 & 18 & 0 & 51 & 97 & 21 \\ 3 & 3 & 3 & 0 & 11 & 21 & 4 \end{pmatrix}_{b_0}$$

[23] $1^{112} 3^{112} 21^{122} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 7 & 10 & 3 \\ 2 & 1 & 3 & 0 & 7 & 10 & 3 \\ 2 & 3 & 1 & 1 & 10 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 7 & 10 & 0 & 25 & 34 & 11 \\ 10 & 10 & 12 & 0 & 34 & 42 & 14 \\ 3 & 3 & 3 & 0 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

[22] $1^{111} 3^{112} 21^{121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 7 & 5 & 3 \\ 2 & 1 & 3 & 0 & 7 & 5 & 3 \\ 2 & 3 & 1 & 1 & 10 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 7 & 10 & 0 & 25 & 17 & 11 \\ 5 & 5 & 6 & 0 & 17 & 9 & 7 \\ 3 & 3 & 3 & 0 & 11 & 7 & 4 \end{pmatrix}_{b_0}$$

[26] $1^{112} 3^{112} 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 102, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 10 & 3 \\ 2 & 1 & 3 & 0 & 5 & 10 & 3 \\ 2 & 3 & 1 & 1 & 6 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 22 & 7 \\ 10 & 10 & 12 & 0 & 22 & 42 & 14 \\ 3 & 3 & 3 & 0 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

[31] $1^{112} 3^{112} 21^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 14 & 3 \\ 2 & 1 & 3 & 0 & 3 & 15 & 3 \\ 2 & 3 & 1 & 1 & 2 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 15 & 3 \\ 14 & 15 & 18 & 0 & 15 & 97 & 21 \\ 3 & 3 & 3 & 0 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[27] $1^{1^{12}} 3^{1^{12}} 21^{1^{12}} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 12 & 3 \\ 2 & 1 & 3 & 0 & 5 & 13 & 3 \\ 2 & 3 & 1 & 1 & 6 & 14 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 27 & 7 \\ 12 & 13 & 14 & 0 & 27 & 65 & 17 \\ 3 & 3 & 3 & 0 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

[28] $1^{1^{14}} 3^{1^{14}} 21^{1^{14}} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 18 & 3 \\ 2 & 1 & 3 & 0 & 5 & 17 & 3 \\ 2 & 3 & 1 & 1 & 6 & 22 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 39 & 7 \\ 18 & 17 & 22 & 0 & 39 & 137 & 25 \\ 3 & 3 & 3 & 0 & 7 & 25 & 4 \end{pmatrix}_{b_0}$$

[29] $1^{1^{14}} 3^{1^{14}} 21^{1^3} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 16 & 3 \\ 2 & 1 & 3 & 0 & 3 & 15 & 3 \\ 2 & 3 & 1 & 1 & 2 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 15 & 3 \\ 16 & 15 & 18 & 0 & 15 & 97 & 21 \\ 3 & 3 & 3 & 0 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[30] $1^{1^{122}} 3^{1^{122}} 21^{1^{122}} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 8 & 3 \\ 2 & 1 & 3 & 0 & 5 & 7 & 3 \\ 2 & 3 & 1 & 1 & 6 & 10 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 17 & 7 \\ 8 & 7 & 10 & 0 & 17 & 25 & 11 \\ 3 & 3 & 3 & 0 & 7 & 11 & 4 \end{pmatrix}_{b_0}$$

[32] $1^{1^{112}} 3^{1^{112}} 21^{1^{111}} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 6 & 3 \\ 2 & 1 & 3 & 0 & 5 & 5 & 3 \\ 2 & 3 & 1 & 1 & 6 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 11 & 7 \\ 6 & 5 & 6 & 0 & 11 & 9 & 7 \\ 3 & 3 & 3 & 0 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

[33] $1^{1^{13}} 3^{1^{13}} 21^{1^{13}} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 166, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 15 & 3 \\ 2 & 1 & 3 & 0 & 3 & 15 & 3 \\ 2 & 3 & 1 & 1 & 2 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 15 & 3 \\ 15 & 15 & 18 & 0 & 15 & 97 & 21 \\ 3 & 3 & 3 & 0 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[34] $1^{1^{111}} 3^{1^{111}} 21^{1^{111}} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 180, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 5 & 3 \\ 2 & 1 & 3 & 0 & 5 & 5 & 3 \\ 2 & 3 & 1 & 1 & 6 & 6 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 11 & 7 \\ 5 & 5 & 6 & 0 & 11 & 9 & 7 \\ 3 & 3 & 3 & 0 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

[35] $1^{1^{13}} 3^{1^{13}} 21^{1^{13}} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 182, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 15 & 3 \\ 2 & 1 & 3 & 0 & 5 & 15 & 3 \\ 2 & 3 & 1 & 1 & 6 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 33 & 7 \\ 15 & 15 & 18 & 0 & 33 & 97 & 21 \\ 3 & 3 & 3 & 0 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

[36] $1^{123} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 196, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 15 & 3 \\ 2 & 1 & 3 & 0 & 8 & 15 & 3 \\ 2 & 3 & 1 & 1 & 8 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 8 & 8 & 0 & 22 & 48 & 10 \\ 15 & 15 & 18 & 0 & 48 & 97 & 21 \\ 3 & 3 & 3 & 0 & 10 & 21 & 4 \end{pmatrix}_{b_0}$$

[37] $1^{112} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 196, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 8 & 3 \\ 2 & 1 & 3 & 0 & 5 & 8 & 3 \\ 2 & 3 & 1 & 1 & 6 & 8 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 16 & 7 \\ 8 & 8 & 8 & 0 & 16 & 22 & 10 \\ 3 & 3 & 3 & 0 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

[38] $1^{124} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 196, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 20 & 3 \\ 2 & 1 & 3 & 0 & 8 & 20 & 3 \\ 2 & 3 & 1 & 1 & 8 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 8 & 8 & 0 & 22 & 64 & 10 \\ 20 & 20 & 24 & 0 & 64 & 174 & 28 \\ 3 & 3 & 3 & 0 & 10 & 28 & 4 \end{pmatrix}_{b_0}$$

[39] $1^{122} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 196, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 10 & 3 \\ 2 & 1 & 3 & 0 & 8 & 10 & 3 \\ 2 & 3 & 1 & 1 & 8 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 8 & 8 & 0 & 22 & 32 & 10 \\ 10 & 10 & 12 & 0 & 32 & 42 & 14 \\ 3 & 3 & 3 & 0 & 10 & 14 & 4 \end{pmatrix}_{b_0}$$

[40] $1^{112} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 198, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 8 & 3 \\ 2 & 1 & 3 & 0 & 5 & 7 & 3 \\ 2 & 3 & 1 & 1 & 6 & 10 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 17 & 7 \\ 8 & 7 & 10 & 0 & 17 & 25 & 11 \\ 3 & 3 & 3 & 0 & 7 & 11 & 4 \end{pmatrix}_{b_0}$$

[41] $1^{122} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 198, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 10 & 3 \\ 2 & 1 & 3 & 0 & 5 & 10 & 3 \\ 2 & 3 & 1 & 1 & 6 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 22 & 7 \\ 10 & 10 & 12 & 0 & 22 & 42 & 14 \\ 3 & 3 & 3 & 0 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

[42] $1^{122} 3^1 21^{122} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 246, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 10 & 3 \\ 2 & 1 & 3 & 0 & 7 & 10 & 3 \\ 2 & 3 & 1 & 1 & 10 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 34 & 11 \\ 10 & 10 & 12 & 0 & 34 & 42 & 14 \\ 3 & 3 & 3 & 0 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

[46] $1^{123} 3^1 21^{122} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 256, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 11 & 3 \\ 2 & 1 & 3 & 0 & 7 & 10 & 3 \\ 2 & 3 & 1 & 1 & 10 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 34 & 11 \\ 11 & 10 & 12 & 0 & 34 & 42 & 14 \\ 3 & 3 & 3 & 0 & 11 & 14 & 4 \end{pmatrix}_{b_0}$$

[43] $1^{113} 3^{114} 3^1 21^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 256, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 19 & 3 \\ 2 & 1 & 3 & 0 & 3 & 20 & 3 \\ 2 & 3 & 1 & 1 & 2 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 20 & 3 \\ 19 & 20 & 24 & 0 & 20 & 174 & 28 \\ 3 & 3 & 3 & 0 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

[44] $1^{123} 3^{112} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 256, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 13 & 3 \\ 2 & 1 & 3 & 0 & 5 & 12 & 3 \\ 2 & 3 & 1 & 1 & 6 & 16 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 28 & 7 \\ 13 & 12 & 16 & 0 & 28 & 70 & 18 \\ 3 & 3 & 3 & 0 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

[45] $1^{113} 3^{114} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 256, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 17 & 3 \\ 2 & 1 & 3 & 0 & 5 & 18 & 3 \\ 2 & 3 & 1 & 1 & 6 & 20 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 38 & 7 \\ 17 & 18 & 20 & 0 & 38 & 130 & 24 \\ 3 & 3 & 3 & 0 & 7 & 24 & 4 \end{pmatrix}_{b_0}$$

[47] $1^{114} 3^{112} 3^1 21^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 278, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 20 & 3 \\ 2 & 1 & 3 & 0 & 3 & 20 & 3 \\ 2 & 3 & 1 & 1 & 2 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 20 & 3 \\ 20 & 20 & 24 & 0 & 20 & 174 & 28 \\ 3 & 3 & 3 & 0 & 3 & 28 & 4 \end{pmatrix}_{b_0}$$

[48] $1^{123} 3^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 278, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 15 & 3 \\ 2 & 1 & 3 & 0 & 5 & 15 & 3 \\ 2 & 3 & 1 & 1 & 6 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 33 & 7 \\ 15 & 15 & 18 & 0 & 33 & 97 & 21 \\ 3 & 3 & 3 & 0 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

[49] $1^{114} 3^{112} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 294, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 20 & 3 \\ 2 & 1 & 3 & 0 & 5 & 20 & 3 \\ 2 & 3 & 1 & 1 & 6 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 5 & 6 & 0 & 9 & 44 & 7 \\ 20 & 20 & 24 & 0 & 44 & 174 & 28 \\ 3 & 3 & 3 & 0 & 7 & 28 & 4 \end{pmatrix}_{b_0}$$

[50] $1^{122} 3^{112} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 324, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 12 & 3 \\ 2 & 1 & 3 & 0 & 10 & 12 & 3 \\ 2 & 3 & 1 & 1 & 12 & 16 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 56 & 14 \\ 12 & 12 & 16 & 0 & 56 & 70 & 18 \\ 3 & 3 & 3 & 0 & 14 & 18 & 4 \end{pmatrix}_{b_0}$$

[51] $1^{112} 3^{112} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 324, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 12 & 3 \\ 2 & 1 & 3 & 0 & 5 & 12 & 3 \\ 2 & 3 & 1 & 1 & 6 & 16 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 28 & 7 \\ 12 & 12 & 16 & 0 & 28 & 70 & 18 \\ 3 & 3 & 3 & 0 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[52] \quad & 1^{123} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 326, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 15 & 3 \\ 2 & 1 & 3 & 0 & 7 & 15 & 3 \\ 2 & 3 & 1 & 1 & 10 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 51 & 11 \\ 15 & 15 & 18 & 0 & 51 & 97 & 21 \\ 3 & 3 & 3 & 0 & 11 & 21 & 4 \end{pmatrix}_{b_0} \\
[53] \quad & 1^{112} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 372, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 10 & 3 \\ 2 & 1 & 3 & 0 & 5 & 10 & 3 \\ 2 & 3 & 1 & 1 & 6 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 22 & 7 \\ 10 & 10 & 12 & 0 & 22 & 42 & 14 \\ 3 & 3 & 3 & 0 & 7 & 14 & 4 \end{pmatrix}_{b_0} \\
[55] \quad & 1^{124} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 390, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 20 & 3 \\ 2 & 1 & 3 & 0 & 5 & 20 & 3 \\ 2 & 3 & 1 & 1 & 6 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 44 & 7 \\ 20 & 20 & 24 & 0 & 44 & 174 & 28 \\ 3 & 3 & 3 & 0 & 7 & 28 & 4 \end{pmatrix}_{b_0} \\
[54] \quad & 1^{113} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 390, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 11 & 3 \\ 2 & 1 & 3 & 0 & 5 & 10 & 3 \\ 2 & 3 & 1 & 1 & 6 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 22 & 7 \\ 11 & 10 & 12 & 0 & 22 & 42 & 14 \\ 3 & 3 & 3 & 0 & 7 & 14 & 4 \end{pmatrix}_{b_0} \\
[56] \quad & 1^{124} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 400, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 6 & 18 & 3 \\ 2 & 1 & 3 & 0 & 5 & 17 & 3 \\ 2 & 3 & 1 & 1 & 6 & 22 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 5 & 6 & 0 & 9 & 39 & 7 \\ 18 & 17 & 22 & 0 & 39 & 137 & 25 \\ 3 & 3 & 3 & 0 & 7 & 25 & 4 \end{pmatrix}_{b_0} \\
[57] \quad & 1^{124} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 400, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 16 & 3 \\ 2 & 1 & 3 & 0 & 7 & 15 & 3 \\ 2 & 3 & 1 & 1 & 10 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 51 & 11 \\ 16 & 15 & 18 & 0 & 51 & 97 & 21 \\ 3 & 3 & 3 & 0 & 11 & 21 & 4 \end{pmatrix}_{b_0} \\
[58] \quad & 1^{123} 3^1 21^{122} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 438, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 11 & 3 \\ 2 & 1 & 3 & 0 & 10 & 10 & 3 \\ 2 & 3 & 1 & 1 & 12 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 44 & 14 \\ 11 & 10 & 12 & 0 & 44 & 42 & 14 \\ 3 & 3 & 3 & 0 & 14 & 14 & 4 \end{pmatrix}_{b_0} \\
[59] \quad & 1^{124} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 438, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 8 & 20 & 3 \\ 2 & 1 & 3 & 0 & 7 & 20 & 3 \\ 2 & 3 & 1 & 1 & 10 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 10 & 0 & 25 & 68 & 11 \\ 20 & 20 & 24 & 0 & 68 & 174 & 28 \\ 3 & 3 & 3 & 0 & 11 & 28 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[60] $1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 470, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 13 & 3 \\ 2 & 1 & 3 & 0 & 5 & 12 & 3 \\ 2 & 3 & 1 & 1 & 6 & 16 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 28 & 7 \\ 13 & 12 & 16 & 0 & 28 & 70 & 18 \\ 3 & 3 & 3 & 0 & 7 & 18 & 4 \end{pmatrix}_{b_0}$$

[61] $1^{113} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 484, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 13 & 3 \\ 2 & 1 & 3 & 0 & 5 & 13 & 3 \\ 2 & 3 & 1 & 1 & 6 & 14 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 27 & 7 \\ 13 & 13 & 14 & 0 & 27 & 65 & 17 \\ 3 & 3 & 3 & 0 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

[62] $1^{123} 3^1 21^{122} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 484, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 13 & 3 \\ 2 & 1 & 3 & 0 & 10 & 13 & 3 \\ 2 & 3 & 1 & 1 & 12 & 14 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 54 & 14 \\ 13 & 13 & 14 & 0 & 54 & 65 & 17 \\ 3 & 3 & 3 & 0 & 14 & 17 & 4 \end{pmatrix}_{b_0}$$

[63] $1^{123} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 518, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 13 & 3 \\ 2 & 1 & 3 & 0 & 10 & 12 & 3 \\ 2 & 3 & 1 & 1 & 12 & 16 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 56 & 14 \\ 13 & 12 & 16 & 0 & 56 & 70 & 18 \\ 3 & 3 & 3 & 0 & 14 & 18 & 4 \end{pmatrix}_{b_0}$$

[64] $1^{122} 3^1 21^{122} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 564, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 10 & 3 \\ 2 & 1 & 3 & 0 & 10 & 10 & 3 \\ 2 & 3 & 1 & 1 & 12 & 12 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 44 & 14 \\ 10 & 10 & 12 & 0 & 44 & 42 & 14 \\ 3 & 3 & 3 & 0 & 14 & 14 & 4 \end{pmatrix}_{b_0}$$

[66] $1^{123} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 17 & 3 \\ 2 & 1 & 3 & 0 & 10 & 17 & 3 \\ 2 & 3 & 1 & 1 & 12 & 22 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 78 & 14 \\ 17 & 17 & 22 & 0 & 78 & 137 & 25 \\ 3 & 3 & 3 & 0 & 14 & 25 & 4 \end{pmatrix}_{b_0}$$

[65] $1^{113} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 676, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 17 & 3 \\ 2 & 1 & 3 & 0 & 5 & 17 & 3 \\ 2 & 3 & 1 & 1 & 6 & 22 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 39 & 7 \\ 17 & 17 & 22 & 0 & 39 & 137 & 25 \\ 3 & 3 & 3 & 0 & 7 & 25 & 4 \end{pmatrix}_{b_0}$$

[67] $1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 692, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 15 & 3 \\ 2 & 1 & 3 & 0 & 5 & 15 & 3 \\ 2 & 3 & 1 & 1 & 6 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 33 & 7 \\ 15 & 15 & 18 & 0 & 33 & 97 & 21 \\ 3 & 3 & 3 & 0 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

[68] $1^{114} 1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 758, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 16 & 3 \\ 2 & 1 & 3 & 0 & 5 & 15 & 3 \\ 2 & 3 & 1 & 1 & 6 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 33 & 7 \\ 16 & 15 & 18 & 0 & 33 & 97 & 21 \\ 3 & 3 & 3 & 0 & 7 & 21 & 4 \end{pmatrix}_{b_0}$$

[69] $1^{124} 1^{123} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 806, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 16 & 3 \\ 2 & 1 & 3 & 0 & 10 & 15 & 3 \\ 2 & 3 & 1 & 1 & 12 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 66 & 14 \\ 16 & 15 & 18 & 0 & 66 & 97 & 21 \\ 3 & 3 & 3 & 0 & 14 & 21 & 4 \end{pmatrix}_{b_0}$$

[70] $1^{114} 1^{113} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 870, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 18 & 3 \\ 2 & 1 & 3 & 0 & 5 & 17 & 3 \\ 2 & 3 & 1 & 1 & 6 & 22 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 39 & 7 \\ 18 & 17 & 22 & 0 & 39 & 137 & 25 \\ 3 & 3 & 3 & 0 & 7 & 25 & 4 \end{pmatrix}_{b_0}$$

[71] $1^{123} 1^{123} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 884, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 15 & 3 \\ 2 & 1 & 3 & 0 & 10 & 15 & 3 \\ 2 & 3 & 1 & 1 & 12 & 18 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 66 & 14 \\ 15 & 15 & 18 & 0 & 66 & 97 & 21 \\ 3 & 3 & 3 & 0 & 14 & 21 & 4 \end{pmatrix}_{b_0}$$

[72] $1^{114} 1^{114} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 900, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 18 & 3 \\ 2 & 1 & 3 & 0 & 5 & 18 & 3 \\ 2 & 3 & 1 & 1 & 6 & 20 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 38 & 7 \\ 18 & 18 & 20 & 0 & 38 & 130 & 24 \\ 3 & 3 & 3 & 0 & 7 & 24 & 4 \end{pmatrix}_{b_0}$$

[73] $1^{124} 1^{124} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 900, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 18 & 3 \\ 2 & 1 & 3 & 0 & 10 & 18 & 3 \\ 2 & 3 & 1 & 1 & 12 & 20 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 76 & 14 \\ 18 & 18 & 20 & 0 & 76 & 130 & 24 \\ 3 & 3 & 3 & 0 & 14 & 24 & 4 \end{pmatrix}_{b_0}$$

[74] $1^{124} 1^{123} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 918, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 18 & 3 \\ 2 & 1 & 3 & 0 & 10 & 17 & 3 \\ 2 & 3 & 1 & 1 & 12 & 22 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 78 & 14 \\ 18 & 17 & 22 & 0 & 78 & 137 & 25 \\ 3 & 3 & 3 & 0 & 14 & 25 & 4 \end{pmatrix}_{b_0}$$

[75] $1^{114} 1^{114} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 1140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 5 & 20 & 3 \\ 2 & 1 & 3 & 0 & 5 & 20 & 3 \\ 2 & 3 & 1 & 1 & 6 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 6 & 0 & 9 & 44 & 7 \\ 20 & 20 & 24 & 0 & 44 & 174 & 28 \\ 3 & 3 & 3 & 0 & 7 & 28 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[76] \quad & 1^{124} 3^1 21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 1332, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 2 & 2 & 1 & 10 & 20 & 3 \\ 2 & 1 & 3 & 0 & 10 & 20 & 3 \\ 2 & 3 & 1 & 1 & 12 & 24 & 3 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 12 & 0 & 42 & 88 & 14 \\ 20 & 20 & 24 & 0 & 88 & 174 & 28 \\ 3 & 3 & 3 & 0 & 14 & 28 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
T = 12 \quad [77] \quad & 1^{112} 3^1 1^5 131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = \\
& 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 7.5
\end{aligned}$$

5.121 NHC(-2)(-3)(-2) + $\textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$

$$\begin{aligned}
T = 10 \quad [4] \quad & 1^{112} 3^1 21^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = \\
& 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 8.5 \\
[3] \quad & 1^{112} 3^1 21^{1121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = \\
& (1, 10, 1), c_{\text{ext}} = 8.5 \\
[2] \quad & 1^{112} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = \\
& (1, 10, 1), c_{\text{ext}} = 8.5 \\
[5] \quad & 1^{113} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = \\
& (1, 10, 1), c_{\text{ext}} = 8.8 \\
[6] \quad & 1^{114} 3^1 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = \\
& (1, 10, 1), c_{\text{ext}} = 9.0 \\
[1] \quad & 1^{111} 3^1 21^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = \\
& (1, 10, 1), c_{\text{ext}} = 8.0 \\
[0] \quad & 1^{111} 3^1 21^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = \\
& (1, 10, 1), c_{\text{ext}} = 8.0
\end{aligned}$$

5.122 NHC(-2)(-3)(-2) + $\textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3)$

$$\begin{aligned}
T = 9 \quad [0] \quad & 1^{123} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = \\
& -36, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 4 & 4 & 5 \\ 7 & 8 & 12 & 5 & 5 & 6 \\ 11 & 12 & 14 & 6 & 6 & 8 \\ 4 & 5 & 6 & 1 & 3 & 3 \\ 4 & 5 & 6 & 3 & 1 & 3 \\ 5 & 6 & 8 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[1] \quad & 1^{11} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -36, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 4 & 4 & 5 \\ 7 & 8 & 6 & 5 & 5 & 6 \\ 4 & 6 & 2 & 3 & 3 & 4 \\ 4 & 5 & 3 & 1 & 3 & 3 \\ 4 & 5 & 3 & 3 & 1 & 3 \\ 5 & 6 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[2] \quad & 1^{13} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -36, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 16 & 4 & 4 & 5 \\ 7 & 8 & 18 & 5 & 5 & 6 \\ 16 & 18 & 34 & 9 & 9 & 12 \\ 4 & 5 & 9 & 1 & 3 & 3 \\ 4 & 5 & 9 & 3 & 1 & 3 \\ 5 & 6 & 12 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[3] \quad & 1^{12} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -36, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 4 & 4 & 21 & 2 & 2 & 3 \\ 4 & 4 & 6 & 28 & 3 & 3 & 4 \\ 4 & 6 & 4 & 28 & 3 & 3 & 4 \\ 21 & 28 & 28 & 145 & 14 & 14 & 21 \\ 2 & 3 & 3 & 14 & 0 & 2 & 2 \\ 2 & 3 & 3 & 14 & 2 & 0 & 2 \\ 3 & 4 & 4 & 21 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[11] $21^{114}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 2 & 2 & 3 \\ 4 & 4 & 6 & 16 & 3 & 3 & 4 \\ 4 & 6 & 4 & 16 & 3 & 3 & 4 \\ 12 & 16 & 16 & 46 & 8 & 8 & 12 \\ 2 & 3 & 3 & 8 & 0 & 2 & 2 \\ 2 & 3 & 3 & 8 & 2 & 0 & 2 \\ 3 & 4 & 4 & 12 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[13] $21^{112}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 2 & 2 & 3 \\ 4 & 4 & 6 & 8 & 3 & 3 & 4 \\ 4 & 6 & 4 & 8 & 3 & 3 & 4 \\ 6 & 8 & 8 & 10 & 4 & 4 & 6 \\ 2 & 3 & 3 & 4 & 0 & 2 & 2 \\ 2 & 3 & 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 4 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[10] $21^{115}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.1$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 2 & 2 & 3 \\ 4 & 4 & 6 & 20 & 3 & 3 & 4 \\ 4 & 6 & 4 & 20 & 3 & 3 & 4 \\ 15 & 20 & 20 & 73 & 10 & 10 & 15 \\ 2 & 3 & 3 & 10 & 0 & 2 & 2 \\ 2 & 3 & 3 & 10 & 2 & 0 & 2 \\ 3 & 4 & 4 & 15 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[9] $21^{116}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 18 & 2 & 2 & 3 \\ 4 & 4 & 6 & 24 & 3 & 3 & 4 \\ 4 & 6 & 4 & 24 & 3 & 3 & 4 \\ 18 & 24 & 24 & 106 & 12 & 12 & 18 \\ 2 & 3 & 3 & 12 & 0 & 2 & 2 \\ 2 & 3 & 3 & 12 & 2 & 0 & 2 \\ 3 & 4 & 4 & 18 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[17] $21^{111}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -320, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 2 & 2 & 3 \\ 4 & 4 & 6 & 9 & 3 & 3 & 4 \\ 4 & 6 & 4 & 9 & 3 & 3 & 4 \\ 6 & 9 & 9 & 13 & 5 & 5 & 7 \\ 2 & 3 & 3 & 5 & 0 & 2 & 2 \\ 2 & 3 & 3 & 5 & 2 & 0 & 2 \\ 3 & 4 & 4 & 7 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[18] $21^{112}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 2 & 2 & 3 \\ 4 & 4 & 6 & 13 & 3 & 3 & 4 \\ 4 & 6 & 4 & 13 & 3 & 3 & 4 \\ 9 & 13 & 13 & 28 & 7 & 7 & 10 \\ 2 & 3 & 3 & 7 & 0 & 2 & 2 \\ 2 & 3 & 3 & 7 & 2 & 0 & 2 \\ 3 & 4 & 4 & 10 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[19] $21^{113}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -832, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 2 & 2 & 3 \\ 4 & 4 & 6 & 17 & 3 & 3 & 4 \\ 4 & 6 & 4 & 17 & 3 & 3 & 4 \\ 12 & 17 & 17 & 49 & 9 & 9 & 13 \\ 2 & 3 & 3 & 9 & 0 & 2 & 2 \\ 2 & 3 & 3 & 9 & 2 & 0 & 2 \\ 3 & 4 & 4 & 13 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[20] $21^{114}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3)$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -1088, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 2 & 2 & 3 \\ 4 & 4 & 6 & 21 & 3 & 3 & 4 \\ 4 & 6 & 4 & 21 & 3 & 3 & 4 \\ 15 & 21 & 21 & 76 & 11 & 11 & 16 \\ 2 & 3 & 3 & 11 & 0 & 2 & 2 \\ 2 & 3 & 3 & 11 & 2 & 0 & 2 \\ 3 & 4 & 4 & 16 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

5.123 NHC(-2)(-3)(-2) + su₂ + su₂→(-3) + su₂→(-3) + su₂

$$T = 10 \quad [3] \quad 1^{112}3^{11}1^1 + [\text{su}_2 \cdot \text{su}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 4 & 10 & 5 \\ 7 & 8 & 6 & 5 & 5 & 12 & 6 \\ 4 & 6 & 2 & 3 & 3 & 8 & 4 \\ 4 & 5 & 3 & 1 & 3 & 6 & 3 \\ 4 & 5 & 3 & 3 & 1 & 6 & 3 \\ 10 & 12 & 8 & 6 & 6 & 14 & 8 \\ 5 & 6 & 4 & 3 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{124}3^{11}1^1 + [\text{su}_2 \cdot \text{su}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 4 & 4 & 20 & 5 \\ 7 & 8 & 6 & 5 & 5 & 24 & 6 \\ 6 & 6 & 2 & 3 & 3 & 16 & 4 \\ 4 & 5 & 3 & 1 & 3 & 12 & 3 \\ 4 & 5 & 3 & 3 & 1 & 12 & 3 \\ 20 & 24 & 16 & 12 & 12 & 62 & 16 \\ 5 & 6 & 4 & 3 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{113}3^{11}1^1 + [\text{su}_2 \cdot \text{su}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 4 & 4 & 15 & 5 \\ 7 & 8 & 6 & 5 & 5 & 18 & 6 \\ 6 & 6 & 2 & 3 & 3 & 12 & 4 \\ 4 & 5 & 3 & 1 & 3 & 9 & 3 \\ 4 & 5 & 3 & 3 & 1 & 9 & 3 \\ 15 & 18 & 12 & 9 & 9 & 34 & 12 \\ 5 & 6 & 4 & 3 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{122}3^{11}1^1 + [\text{su}_2 \cdot \text{su}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 4 & 4 & 10 & 5 \\ 7 & 8 & 6 & 5 & 5 & 12 & 6 \\ 6 & 6 & 2 & 3 & 3 & 8 & 4 \\ 4 & 5 & 3 & 1 & 3 & 6 & 3 \\ 4 & 5 & 3 & 3 & 1 & 6 & 3 \\ 10 & 12 & 8 & 6 & 6 & 14 & 8 \\ 5 & 6 & 4 & 3 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{121}3^{11}1^1 + [\text{su}_2 \cdot \text{su}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 4 & 4 & 5 & 5 \\ 7 & 8 & 6 & 5 & 5 & 6 & 6 \\ 6 & 6 & 2 & 3 & 3 & 4 & 4 \\ 4 & 5 & 3 & 1 & 3 & 3 & 3 \\ 4 & 5 & 3 & 3 & 1 & 3 & 3 \\ 5 & 6 & 4 & 3 & 3 & 2 & 4 \\ 5 & 6 & 4 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{131}3^{11}1^1 + [\text{su}_2 \cdot \text{su}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 11 & 4 & 4 & 5 & 5 \\ 7 & 8 & 12 & 5 & 5 & 6 & 6 \\ 11 & 12 & 14 & 6 & 6 & 8 & 8 \\ 4 & 5 & 6 & 1 & 3 & 3 & 3 \\ 4 & 5 & 6 & 3 & 1 & 3 & 3 \\ 5 & 6 & 8 & 3 & 3 & 2 & 4 \\ 5 & 6 & 8 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{132}3^{11}1^1 + [\text{su}_2 \cdot \text{su}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 11 & 4 & 4 & 10 & 5 \\ 7 & 8 & 12 & 5 & 5 & 12 & 6 \\ 11 & 12 & 14 & 6 & 6 & 16 & 8 \\ 4 & 5 & 6 & 1 & 3 & 6 & 3 \\ 4 & 5 & 6 & 3 & 1 & 6 & 3 \\ 10 & 12 & 16 & 6 & 6 & 14 & 8 \\ 5 & 6 & 8 & 3 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}1^1 + [\text{su}_2 \cdot \text{su}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 72, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 4 & 5 & 5 \\ 7 & 8 & 6 & 5 & 5 & 6 & 6 \\ 4 & 6 & 2 & 3 & 3 & 4 & 4 \\ 4 & 5 & 3 & 1 & 3 & 3 & 3 \\ 4 & 5 & 3 & 3 & 1 & 3 & 3 \\ 5 & 6 & 4 & 3 & 3 & 2 & 4 \\ 5 & 6 & 4 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{142} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 72, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.3$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 16 & 4 & 4 & 10 & 5 \\ 7 & 8 & 18 & 5 & 5 & 12 & 6 \\ 16 & 18 & 34 & 9 & 9 & 24 & 12 \\ 4 & 5 & 9 & 1 & 3 & 6 & 3 \\ 4 & 5 & 9 & 3 & 1 & 6 & 3 \\ 10 & 12 & 24 & 6 & 6 & 14 & 8 \\ 5 & 6 & 12 & 3 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{131} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 72, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 16 & 4 & 4 & 5 & 5 \\ 7 & 8 & 18 & 5 & 5 & 6 & 6 \\ 16 & 18 & 34 & 9 & 9 & 12 & 12 \\ 4 & 5 & 9 & 1 & 3 & 3 & 3 \\ 4 & 5 & 9 & 3 & 1 & 3 & 3 \\ 5 & 6 & 12 & 3 & 3 & 2 & 4 \\ 5 & 6 & 12 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{114} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 72, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 4 & 20 & 5 \\ 7 & 8 & 6 & 5 & 5 & 24 & 6 \\ 4 & 6 & 2 & 3 & 3 & 16 & 4 \\ 4 & 5 & 3 & 1 & 3 & 12 & 3 \\ 4 & 5 & 3 & 3 & 1 & 12 & 3 \\ 20 & 24 & 16 & 12 & 12 & 62 & 16 \\ 5 & 6 & 4 & 3 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{113} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 72, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 4 & 15 & 5 \\ 7 & 8 & 6 & 5 & 5 & 18 & 6 \\ 4 & 6 & 2 & 3 & 3 & 12 & 4 \\ 4 & 5 & 3 & 1 & 3 & 9 & 3 \\ 4 & 5 & 3 & 3 & 1 & 9 & 3 \\ 15 & 18 & 12 & 9 & 9 & 34 & 12 \\ 5 & 6 & 4 & 3 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[12] \ 21^{11213} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 128, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 2 & 2 & 6 & 3 \\ 4 & 4 & 6 & 8 & 3 & 3 & 9 & 4 \\ 4 & 6 & 4 & 8 & 3 & 3 & 9 & 4 \\ 6 & 8 & 8 & 10 & 4 & 4 & 12 & 6 \\ 2 & 3 & 3 & 4 & 0 & 2 & 5 & 2 \\ 2 & 3 & 3 & 4 & 2 & 0 & 5 & 2 \\ 6 & 9 & 9 & 12 & 5 & 5 & 13 & 7 \\ 3 & 4 & 4 & 6 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[13] \ 21^{113} 3^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 128, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 5 & 3 & 3 & 3 & 9 & 4 \\ 5 & 4 & 6 & 4 & 3 & 3 & 9 & 4 \\ 5 & 6 & 4 & 4 & 3 & 3 & 9 & 4 \\ 3 & 4 & 4 & 1 & 2 & 2 & 6 & 3 \\ 3 & 3 & 3 & 2 & 0 & 2 & 5 & 2 \\ 3 & 3 & 3 & 2 & 0 & 2 & 5 & 2 \\ 9 & 9 & 9 & 6 & 5 & 5 & 13 & 7 \\ 4 & 4 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{141} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 144, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 20 & 4 & 4 & 5 & 5 \\ 7 & 8 & 24 & 5 & 5 & 6 & 6 \\ 20 & 24 & 62 & 12 & 12 & 16 & 16 \\ 4 & 5 & 12 & 1 & 3 & 3 & 3 \\ 4 & 5 & 12 & 3 & 1 & 3 & 3 \\ 5 & 6 & 16 & 3 & 3 & 2 & 4 \\ 5 & 6 & 16 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{121} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 144, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 10 & 4 & 4 & 5 & 5 \\ 7 & 8 & 12 & 5 & 5 & 6 & 6 \\ 10 & 12 & 14 & 6 & 6 & 8 & 8 \\ 4 & 5 & 6 & 1 & 3 & 3 & 3 \\ 4 & 5 & 6 & 3 & 1 & 3 & 3 \\ 5 & 6 & 8 & 3 & 3 & 2 & 4 \\ 5 & 6 & 8 & 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

[16] $1^{123}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 10 & 4 & 4 & 10 & 5 \\ 7 & 8 & 12 & 5 & 5 & 12 & 6 \\ 10 & 12 & 14 & 6 & 6 & 16 & 8 \\ 4 & 5 & 6 & 1 & 3 & 6 & 3 \\ 4 & 5 & 6 & 3 & 1 & 6 & 3 \\ 10 & 12 & 16 & 6 & 6 & 14 & 8 \\ 5 & 6 & 8 & 3 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

[17] $1^{123}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.3$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 10 & 4 & 4 & 15 & 5 \\ 7 & 8 & 12 & 5 & 5 & 18 & 6 \\ 10 & 12 & 14 & 6 & 6 & 24 & 8 \\ 4 & 5 & 6 & 1 & 3 & 9 & 3 \\ 4 & 5 & 6 & 3 & 1 & 9 & 3 \\ 15 & 18 & 24 & 9 & 9 & 34 & 12 \\ 5 & 6 & 8 & 3 & 3 & 12 & 4 \end{pmatrix}_{b_0}$$

[18] $1^{124}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 10 & 4 & 4 & 20 & 5 \\ 7 & 8 & 12 & 5 & 5 & 24 & 6 \\ 10 & 12 & 14 & 6 & 6 & 32 & 8 \\ 4 & 5 & 6 & 1 & 3 & 12 & 3 \\ 4 & 5 & 6 & 3 & 1 & 12 & 3 \\ 20 & 24 & 32 & 12 & 12 & 62 & 16 \\ 5 & 6 & 8 & 3 & 3 & 16 & 4 \end{pmatrix}_{b_0}$$

[19] $21^{113}3^{11}1^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 256, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 5 & 5 & 3 & 3 & 12 & 4 \\ 5 & 4 & 6 & 4 & 3 & 13 & 4 \\ 5 & 6 & 4 & 4 & 3 & 13 & 4 \\ 3 & 4 & 4 & 1 & 2 & 9 & 3 \\ 3 & 3 & 3 & 2 & 0 & 7 & 2 \\ 3 & 3 & 3 & 2 & 0 & 7 & 2 \\ 12 & 13 & 13 & 9 & 7 & 28 & 10 \\ 4 & 4 & 4 & 3 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

[20] $21^{1121}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 2 & 2 & 3 & 3 \\ 4 & 4 & 6 & 8 & 3 & 3 & 4 & 4 \\ 4 & 6 & 4 & 8 & 3 & 3 & 4 & 4 \\ 6 & 8 & 8 & 10 & 4 & 4 & 6 & 6 \\ 2 & 3 & 3 & 4 & 0 & 2 & 2 & 2 \\ 2 & 3 & 3 & 4 & 2 & 0 & 2 & 2 \\ 3 & 4 & 4 & 6 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 6 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[21] $21^{1111}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 2 & 2 & 3 & 3 \\ 4 & 4 & 6 & 4 & 3 & 3 & 4 & 4 \\ 4 & 6 & 4 & 4 & 3 & 3 & 4 & 4 \\ 3 & 4 & 4 & 1 & 2 & 2 & 3 & 3 \\ 2 & 3 & 3 & 2 & 0 & 2 & 2 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 & 2 & 2 \\ 3 & 4 & 4 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 4 & 3 & 2 & 2 & 3 & 3 \end{pmatrix}_{b_0}$$

[22] $21^{1111}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 512, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 2 & 2 & 6 & 3 \\ 4 & 4 & 6 & 4 & 3 & 3 & 9 & 4 \\ 4 & 6 & 4 & 4 & 3 & 3 & 9 & 4 \\ 3 & 4 & 4 & 1 & 2 & 2 & 6 & 3 \\ 2 & 3 & 3 & 2 & 0 & 2 & 5 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 & 5 & 2 \\ 6 & 9 & 9 & 6 & 5 & 5 & 13 & 7 \\ 3 & 4 & 4 & 3 & 2 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

[23] $21^{1112}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 1024, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 2 & 2 & 9 & 3 \\ 4 & 4 & 6 & 4 & 3 & 3 & 13 & 4 \\ 4 & 6 & 4 & 4 & 3 & 3 & 13 & 4 \\ 3 & 4 & 4 & 1 & 2 & 2 & 9 & 3 \\ 2 & 3 & 3 & 2 & 0 & 2 & 7 & 2 \\ 2 & 3 & 3 & 2 & 2 & 0 & 7 & 2 \\ 9 & 13 & 13 & 9 & 7 & 7 & 28 & 10 \\ 3 & 4 & 4 & 3 & 2 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

5.124 NHC(-2)(-3)(-2) + su₂ + su₂→(-3) + su₂→(-3) + su₂ + su₂

$$T = 10 \quad [7] \quad 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.0$$

$$[2] \quad 1^{131} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.8$$

$$[8] \quad 1^{112} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.0$$

$$[3] \quad 1^{122} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.5$$

$$[9] \quad 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.5$$

$$[1] \quad 1^{121} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.5$$

$$[10] \quad 1^{141} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 10.0$$

$$[4] \quad 1^{121} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.5$$

$$[6] \quad 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.0$$

$$[0] \quad 1^{12} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.5$$

$$[5] \quad 1^{131} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.5$$

5.125 NHC(-2)(-3)(-2) + su₂ + su₂→(-3) + su₂n3mix

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 7 & 4 & 9 & 5 \\ 7 & 8 & 9 & 5 & 11 & 6 \\ 7 & 9 & 8 & 5 & 10 & 6 \\ 4 & 5 & 5 & 1 & 6 & 3 \\ 9 & 11 & 10 & 6 & 11 & 7 \\ 5 & 6 & 6 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 11 & 4 & 9 & 5 \\ 7 & 8 & 15 & 5 & 11 & 6 \\ 11 & 15 & 22 & 8 & 17 & 10 \\ 4 & 5 & 8 & 1 & 6 & 3 \\ 9 & 11 & 17 & 6 & 11 & 7 \\ 5 & 6 & 10 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{131} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 14 & 4 & 9 & 5 \\ 7 & 8 & 15 & 5 & 11 & 6 \\ 14 & 15 & 22 & 8 & 17 & 10 \\ 4 & 5 & 8 & 1 & 6 & 3 \\ 9 & 11 & 17 & 6 & 11 & 7 \\ 5 & 6 & 10 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[7] \quad & 1^{111} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -64, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 4 & 9 & 5 \\ 7 & 8 & 6 & 5 & 11 & 6 \\ 4 & 6 & 2 & 3 & 7 & 4 \\ 4 & 5 & 3 & 1 & 6 & 3 \\ 9 & 11 & 7 & 6 & 11 & 7 \\ 5 & 6 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[4] \quad & 1^{121} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -64, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 6 & 4 & 9 & 5 \\ 7 & 8 & 6 & 5 & 11 & 6 \\ 6 & 6 & 2 & 3 & 7 & 4 \\ 4 & 5 & 3 & 1 & 6 & 3 \\ 9 & 11 & 7 & 6 & 11 & 7 \\ 5 & 6 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[3] \quad & 1^{131} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -64, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 16 & 4 & 9 & 5 \\ 7 & 8 & 18 & 5 & 11 & 6 \\ 16 & 18 & 34 & 9 & 21 & 12 \\ 4 & 5 & 9 & 1 & 6 & 3 \\ 9 & 11 & 21 & 6 & 11 & 7 \\ 5 & 6 & 12 & 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[5] \quad & 21^{113} 3^{11} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -64, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 3 & 3 & 12 & 4 \\ 5 & 4 & 6 & 4 & 3 & 12 & 4 \\ 5 & 6 & 4 & 4 & 3 & 12 & 4 \\ 3 & 4 & 4 & 1 & 2 & 8 & 3 \\ 3 & 3 & 3 & 2 & 0 & 7 & 2 \\ 12 & 12 & 12 & 8 & 7 & 25 & 9 \\ 4 & 4 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[6] \quad & 1^{121} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -64, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 4 & 9 & 5 \\ 7 & 8 & 12 & 5 & 11 & 6 \\ 11 & 12 & 14 & 6 & 14 & 8 \\ 4 & 5 & 6 & 1 & 6 & 3 \\ 9 & 11 & 14 & 6 & 11 & 7 \\ 5 & 6 & 8 & 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[8] \quad & 1^{132} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -100, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 16 & 4 & 14 & 5 \\ 7 & 8 & 18 & 5 & 17 & 6 \\ 16 & 18 & 34 & 9 & 33 & 12 \\ 4 & 5 & 9 & 1 & 9 & 3 \\ 14 & 17 & 33 & 9 & 29 & 11 \\ 5 & 6 & 12 & 3 & 11 & 4 \end{pmatrix}_{b_0} \\
[9] \quad & 1^{122} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -100, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 6 & 4 & 14 & 5 \\ 7 & 8 & 6 & 5 & 17 & 6 \\ 6 & 6 & 2 & 3 & 11 & 4 \\ 4 & 5 & 3 & 1 & 9 & 3 \\ 14 & 17 & 11 & 9 & 29 & 11 \\ 5 & 6 & 4 & 3 & 11 & 4 \end{pmatrix}_{b_0} \\
[10] \quad & 1^{122} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -100, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 4 & 14 & 5 \\ 7 & 8 & 12 & 5 & 17 & 6 \\ 11 & 12 & 14 & 6 & 22 & 8 \\ 4 & 5 & 6 & 1 & 9 & 3 \\ 14 & 17 & 22 & 9 & 29 & 11 \\ 5 & 6 & 8 & 3 & 11 & 4 \end{pmatrix}_{b_0} \\
[11] \quad & 1^{112} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -100, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 14 & 5 \\ 7 & 8 & 6 & 5 & 17 & 6 \\ 4 & 6 & 2 & 3 & 11 & 4 \\ 4 & 5 & 3 & 1 & 9 & 3 \\ 14 & 17 & 11 & 9 & 29 & 11 \\ 5 & 6 & 4 & 3 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 4 & 9 & 5 \\ 7 & 8 & 6 & 5 & 11 & 6 \\ 5 & 6 & 2 & 3 & 7 & 4 \\ 4 & 5 & 3 & 1 & 6 & 3 \\ 9 & 11 & 7 & 6 & 11 & 7 \\ 5 & 6 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{141} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 20 & 4 & 9 & 5 \\ 7 & 8 & 24 & 5 & 11 & 6 \\ 20 & 24 & 62 & 12 & 28 & 16 \\ 4 & 5 & 12 & 1 & 6 & 3 \\ 9 & 11 & 28 & 6 & 11 & 7 \\ 5 & 6 & 16 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{131} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 15 & 4 & 9 & 5 \\ 7 & 8 & 18 & 5 & 11 & 6 \\ 15 & 18 & 34 & 9 & 21 & 12 \\ 4 & 5 & 9 & 1 & 6 & 3 \\ 9 & 11 & 21 & 6 & 11 & 7 \\ 5 & 6 & 12 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{113} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 4 & 26 & 5 \\ 7 & 8 & 6 & 5 & 32 & 6 \\ 6 & 6 & 2 & 3 & 20 & 4 \\ 4 & 5 & 3 & 1 & 17 & 3 \\ 26 & 32 & 20 & 17 & 105 & 21 \\ 5 & 6 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 12 & 4 & 9 & 5 \\ 7 & 8 & 15 & 5 & 11 & 6 \\ 12 & 15 & 22 & 8 & 17 & 10 \\ 4 & 5 & 8 & 1 & 6 & 3 \\ 9 & 11 & 17 & 6 & 11 & 7 \\ 5 & 6 & 10 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 13 & 4 & 9 & 5 \\ 7 & 8 & 15 & 5 & 11 & 6 \\ 13 & 15 & 22 & 8 & 17 & 10 \\ 4 & 5 & 8 & 1 & 6 & 3 \\ 9 & 11 & 17 & 6 & 11 & 7 \\ 5 & 6 & 10 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 19 & 4 & 9 & 5 \\ 7 & 8 & 21 & 5 & 11 & 6 \\ 19 & 21 & 44 & 11 & 24 & 14 \\ 4 & 5 & 11 & 1 & 6 & 3 \\ 9 & 11 & 24 & 6 & 11 & 7 \\ 5 & 6 & 14 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -187, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 21 & 5 \\ 7 & 8 & 6 & 5 & 26 & 6 \\ 4 & 6 & 2 & 3 & 16 & 4 \\ 4 & 5 & 3 & 1 & 14 & 3 \\ 21 & 26 & 16 & 14 & 69 & 17 \\ 5 & 6 & 4 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[20] $1^{123} 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -189, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 6 & 4 & 22 & 5 \\ 7 & 8 & 6 & 5 & 26 & 6 \\ 6 & 6 & 2 & 3 & 16 & 4 \\ 4 & 5 & 3 & 1 & 14 & 3 \\ 22 & 26 & 16 & 14 & 69 & 17 \\ 5 & 6 & 4 & 3 & 17 & 4 \end{pmatrix}_{b_0}$

[21] $1^{114} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 4 & 4 & 24 & 5 \\ 7 & 8 & 6 & 5 & 29 & 6 \\ 4 & 6 & 2 & 3 & 19 & 4 \\ 4 & 5 & 3 & 1 & 15 & 3 \\ 24 & 29 & 19 & 15 & 89 & 19 \\ 5 & 6 & 4 & 3 & 19 & 4 \end{pmatrix}_{b_0}$

[23] $1^{122} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 10 & 4 & 14 & 5 \\ 7 & 8 & 12 & 5 & 17 & 6 \\ 10 & 12 & 14 & 6 & 22 & 8 \\ 4 & 5 & 6 & 1 & 9 & 3 \\ 14 & 17 & 22 & 9 & 29 & 11 \\ 5 & 6 & 8 & 3 & 11 & 4 \end{pmatrix}_{b_0}$

[22] $1^{142} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 20 & 4 & 14 & 5 \\ 7 & 8 & 24 & 5 & 17 & 6 \\ 20 & 24 & 62 & 12 & 44 & 16 \\ 4 & 5 & 12 & 1 & 9 & 3 \\ 14 & 17 & 44 & 9 & 29 & 11 \\ 5 & 6 & 16 & 3 & 11 & 4 \end{pmatrix}_{b_0}$

[24] $21^{113} 3^{11} 1^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 4 & 5 & 5 & 3 & 3 & 15 & 4 \\ 5 & 4 & 6 & 4 & 3 & 16 & 4 \\ 5 & 6 & 4 & 4 & 3 & 16 & 4 \\ 3 & 4 & 4 & 1 & 2 & 11 & 3 \\ 3 & 3 & 3 & 2 & 0 & 9 & 2 \\ 15 & 16 & 16 & 11 & 9 & 44 & 12 \\ 4 & 4 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$

[25] $1^{114} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 6 & 4 & 31 & 5 \\ 7 & 8 & 6 & 5 & 38 & 6 \\ 6 & 6 & 2 & 3 & 24 & 4 \\ 4 & 5 & 3 & 1 & 20 & 3 \\ 31 & 38 & 24 & 20 & 149 & 25 \\ 5 & 6 & 4 & 3 & 25 & 4 \end{pmatrix}_{b_0}$

[27] $1^{114} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 4 & 4 & 27 & 5 \\ 7 & 8 & 6 & 5 & 32 & 6 \\ 4 & 6 & 2 & 3 & 20 & 4 \\ 4 & 5 & 3 & 1 & 17 & 3 \\ 27 & 32 & 20 & 17 & 105 & 21 \\ 5 & 6 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$

[26] $1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 4 & 4 & 25 & 5 \\ 7 & 8 & 6 & 5 & 32 & 6 \\ 4 & 6 & 2 & 3 & 20 & 4 \\ 4 & 5 & 3 & 1 & 17 & 3 \\ 25 & 32 & 20 & 17 & 105 & 21 \\ 5 & 6 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$

[28] $21^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 2 & 8 & 3 \\ 4 & 4 & 6 & 4 & 3 & 12 & 4 \\ 4 & 6 & 4 & 4 & 3 & 12 & 4 \\ 3 & 4 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 3 & 2 & 0 & 7 & 2 \\ 8 & 12 & 12 & 8 & 7 & 25 & 9 \\ 3 & 4 & 4 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{113}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 26 & 5 \\ 7 & 8 & 6 & 5 & 32 & 6 \\ 4 & 6 & 2 & 3 & 20 & 4 \\ 4 & 5 & 3 & 1 & 17 & 3 \\ 26 & 32 & 20 & 17 & 105 & 21 \\ 5 & 6 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{121}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 17 & 4 & 9 & 5 \\ 7 & 8 & 21 & 5 & 11 & 6 \\ 17 & 21 & 44 & 11 & 24 & 14 \\ 4 & 5 & 11 & 1 & 6 & 3 \\ 9 & 11 & 24 & 6 & 11 & 7 \\ 5 & 6 & 14 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{121}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 18 & 4 & 9 & 5 \\ 7 & 8 & 21 & 5 & 11 & 6 \\ 18 & 21 & 44 & 11 & 24 & 14 \\ 4 & 5 & 11 & 1 & 6 & 3 \\ 9 & 11 & 24 & 6 & 11 & 7 \\ 5 & 6 & 14 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{113}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -301, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 4 & 27 & 5 \\ 7 & 8 & 6 & 5 & 32 & 6 \\ 6 & 6 & 2 & 3 & 20 & 4 \\ 4 & 5 & 3 & 1 & 17 & 3 \\ 27 & 32 & 20 & 17 & 105 & 21 \\ 5 & 6 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 10 & 4 & 26 & 5 \\ 7 & 8 & 12 & 5 & 32 & 6 \\ 10 & 12 & 14 & 6 & 40 & 8 \\ 4 & 5 & 6 & 1 & 17 & 3 \\ 26 & 32 & 40 & 17 & 105 & 21 \\ 5 & 6 & 8 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{112}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 4 & 21 & 5 \\ 7 & 8 & 6 & 5 & 26 & 6 \\ 5 & 6 & 2 & 3 & 16 & 4 \\ 4 & 5 & 3 & 1 & 14 & 3 \\ 21 & 26 & 16 & 14 & 69 & 17 \\ 5 & 6 & 4 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{113}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -349, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 30 & 5 \\ 7 & 8 & 6 & 5 & 38 & 6 \\ 4 & 6 & 2 & 3 & 24 & 4 \\ 4 & 5 & 3 & 1 & 20 & 3 \\ 30 & 38 & 24 & 20 & 149 & 25 \\ 5 & 6 & 4 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{123}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \rightarrow (-3) + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 10 & 4 & 27 & 5 \\ 7 & 8 & 12 & 5 & 32 & 6 \\ 10 & 12 & 14 & 6 & 40 & 8 \\ 4 & 5 & 6 & 1 & 17 & 3 \\ 27 & 32 & 40 & 17 & 105 & 21 \\ 5 & 6 & 8 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

[37] $1^{114}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -387, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 31 & 5 \\ 7 & 8 & 6 & 5 & 38 & 6 \\ 4 & 6 & 2 & 3 & 24 & 4 \\ 4 & 5 & 3 & 1 & 20 & 3 \\ 31 & 38 & 24 & 20 & 149 & 25 \\ 5 & 6 & 4 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

[38] $1^{124}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 10 & 4 & 24 & 5 \\ 7 & 8 & 12 & 5 & 29 & 6 \\ 10 & 12 & 14 & 6 & 38 & 8 \\ 4 & 5 & 6 & 1 & 15 & 3 \\ 24 & 29 & 38 & 15 & 89 & 19 \\ 5 & 6 & 8 & 3 & 19 & 4 \end{pmatrix}_{b_0}$$

[39] $1^{113}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 5 & 4 & 22 & 5 \\ 7 & 8 & 6 & 5 & 26 & 6 \\ 5 & 6 & 2 & 3 & 16 & 4 \\ 4 & 5 & 3 & 1 & 14 & 3 \\ 22 & 26 & 16 & 14 & 69 & 17 \\ 5 & 6 & 4 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

[40] $1^{114}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 5 & 4 & 24 & 5 \\ 7 & 8 & 6 & 5 & 29 & 6 \\ 5 & 6 & 2 & 3 & 19 & 4 \\ 4 & 5 & 3 & 1 & 15 & 3 \\ 24 & 29 & 19 & 15 & 89 & 19 \\ 5 & 6 & 4 & 3 & 19 & 4 \end{pmatrix}_{b_0}$$

[41] $21^{111}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 2 & 5 & 3 \\ 4 & 4 & 6 & 4 & 3 & 7 & 4 \\ 4 & 6 & 4 & 4 & 3 & 7 & 4 \\ 3 & 4 & 4 & 1 & 2 & 5 & 3 \\ 2 & 3 & 3 & 2 & 0 & 4 & 2 \\ 5 & 7 & 7 & 5 & 4 & 7 & 5 \\ 3 & 4 & 4 & 3 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[42] $21^{121}3^{11}1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 2 & 5 & 3 \\ 4 & 4 & 6 & 8 & 3 & 7 & 4 \\ 4 & 6 & 4 & 8 & 3 & 7 & 4 \\ 6 & 8 & 8 & 10 & 4 & 10 & 6 \\ 2 & 3 & 3 & 4 & 0 & 4 & 2 \\ 5 & 7 & 7 & 10 & 4 & 7 & 5 \\ 3 & 4 & 4 & 6 & 2 & 5 & 3 \end{pmatrix}_{b_0}$$

[44] $1^{131}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 23 & 4 & 9 & 5 \\ 7 & 8 & 27 & 5 & 11 & 6 \\ 23 & 27 & 74 & 14 & 31 & 18 \\ 4 & 5 & 14 & 1 & 6 & 3 \\ 9 & 11 & 31 & 6 & 11 & 7 \\ 5 & 6 & 18 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[43] $1^{131}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 22 & 4 & 9 & 5 \\ 7 & 8 & 27 & 5 & 11 & 6 \\ 22 & 27 & 74 & 14 & 31 & 18 \\ 4 & 5 & 14 & 1 & 6 & 3 \\ 9 & 11 & 31 & 6 & 11 & 7 \\ 5 & 6 & 18 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[45] $1^{124}3^{11}1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -522, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 10 & 4 & 31 & 5 \\ 7 & 8 & 12 & 5 & 38 & 6 \\ 10 & 12 & 14 & 6 & 48 & 8 \\ 4 & 5 & 6 & 1 & 20 & 3 \\ 31 & 38 & 48 & 20 & 149 & 25 \\ 5 & 6 & 8 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -530, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 4 & 26 & 5 \\ 7 & 8 & 6 & 5 & 32 & 6 \\ 5 & 6 & 2 & 3 & 20 & 4 \\ 4 & 5 & 3 & 1 & 17 & 3 \\ 26 & 32 & 20 & 17 & 105 & 21 \\ 5 & 6 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{141} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 27 & 4 & 9 & 5 \\ 7 & 8 & 33 & 5 & 11 & 6 \\ 27 & 33 & 112 & 17 & 38 & 22 \\ 4 & 5 & 17 & 1 & 6 & 3 \\ 9 & 11 & 38 & 6 & 11 & 7 \\ 5 & 6 & 22 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[48] \quad 21^{112} 3^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 2 & 8 & 3 \\ 4 & 4 & 6 & 4 & 3 & 11 & 4 \\ 4 & 6 & 4 & 4 & 3 & 11 & 4 \\ 3 & 4 & 4 & 1 & 2 & 8 & 3 \\ 2 & 3 & 3 & 2 & 0 & 6 & 2 \\ 8 & 11 & 11 & 8 & 6 & 20 & 8 \\ 3 & 4 & 4 & 3 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{113} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -592, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 4 & 27 & 5 \\ 7 & 8 & 6 & 5 & 32 & 6 \\ 5 & 6 & 2 & 3 & 20 & 4 \\ 4 & 5 & 3 & 1 & 17 & 3 \\ 27 & 32 & 20 & 17 & 105 & 21 \\ 5 & 6 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[50] \quad 21^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -720, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 2 & 11 & 3 \\ 4 & 4 & 6 & 4 & 3 & 16 & 4 \\ 4 & 6 & 4 & 4 & 3 & 16 & 4 \\ 3 & 4 & 4 & 1 & 2 & 11 & 3 \\ 2 & 3 & 3 & 2 & 0 & 9 & 2 \\ 11 & 16 & 16 & 11 & 9 & 44 & 12 \\ 3 & 4 & 4 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{114} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -738, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 4 & 31 & 5 \\ 7 & 8 & 6 & 5 & 38 & 6 \\ 5 & 6 & 2 & 3 & 24 & 4 \\ 4 & 5 & 3 & 1 & 20 & 3 \\ 31 & 38 & 24 & 20 & 149 & 25 \\ 5 & 6 & 4 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

5.126 NHC(-2)(-3)(-2) + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$ + su2n3mix + $\mathfrak{su}_2$

$$T = 10 \quad [3] \quad 1^{1211} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 4 & 9 & 12 & 5 \\ 7 & 8 & 6 & 5 & 11 & 15 & 6 \\ 6 & 6 & 2 & 3 & 7 & 9 & 4 \\ 4 & 5 & 3 & 1 & 6 & 8 & 3 \\ 9 & 11 & 7 & 6 & 11 & 17 & 7 \\ 12 & 15 & 9 & 8 & 17 & 22 & 10 \\ 5 & 6 & 4 & 3 & 7 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[0] \quad 1^{111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 4 & 9 & 17 & 5 \\ 7 & 8 & 6 & 5 & 11 & 21 & 6 \\ 4 & 6 & 2 & 3 & 7 & 13 & 4 \\ 4 & 5 & 3 & 1 & 6 & 11 & 3 \\ 9 & 11 & 7 & 6 & 11 & 24 & 7 \\ 17 & 21 & 13 & 11 & 24 & 44 & 14 \\ 5 & 6 & 4 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = -1, \ \det = 512, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 4 & 9 & 17 & 5 \\ 7 & 8 & 6 & 5 & 11 & 21 & 6 \\ 5 & 6 & 2 & 3 & 7 & 13 & 4 \\ 4 & 5 & 3 & 1 & 6 & 11 & 3 \\ 9 & 11 & 7 & 6 & 11 & 24 & 7 \\ 17 & 21 & 13 & 11 & 24 & 44 & 14 \\ 5 & 6 & 4 & 3 & 7 & 14 & 4 \end{pmatrix}_{b_0}$$

5.127 NHC(-2)(-3)(-2) + su₂ + su₂ → (-3) + su_{2n3mix} + su₂ + su₂

$$T = 10 \quad [5] \ 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 10, 1), \ c_{\text{ext}} = 9.0$$

$$[0] \ 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 10, 1), \ c_{\text{ext}} = 9.5$$

$$[1] \ 1^{1211} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 10, 1), \ c_{\text{ext}} = 9.0$$

$$[2] \ 1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 10, 1), \ c_{\text{ext}} = 9.0$$

$$[3] \ 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 10, 1), \ c_{\text{ext}} = 9.5$$

$$[4] \ 1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \ T_{\max} = 10, \ \Delta = 0, \ \det = 0, \ (n_+, n_-, n_0) = (1, 10, 1), \ c_{\text{ext}} = 9.0$$

5.128 NHC(-2)(-3)(-2) + su₂ + su_{2n3mix}

$$T = 8 \quad [0] \ 1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 4, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 19 & 9 & 5 \\ 7 & 8 & 24 & 11 & 6 \\ 19 & 24 & 62 & 27 & 16 \\ 9 & 11 & 27 & 11 & 7 \\ 5 & 6 & 16 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 12, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 7 & 8 & 5 \\ 7 & 8 & 9 & 11 & 6 \\ 7 & 9 & 8 & 10 & 6 \\ 8 & 11 & 10 & 11 & 7 \\ 5 & 6 & 6 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{131} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 16, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 21 & 9 & 5 \\ 7 & 8 & 24 & 11 & 6 \\ 21 & 24 & 62 & 27 & 16 \\ 9 & 11 & 27 & 11 & 7 \\ 5 & 6 & 16 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 16, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 19 & 8 & 5 \\ 7 & 8 & 24 & 11 & 6 \\ 19 & 24 & 62 & 27 & 16 \\ 8 & 11 & 27 & 11 & 7 \\ 5 & 6 & 16 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 17, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.88 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{aligned}
& \begin{pmatrix} 5 & 7 & 7 & 9 & 5 \\ 7 & 8 & 9 & 11 & 6 \\ 7 & 9 & 8 & 10 & 6 \\ 9 & 11 & 10 & 11 & 7 \\ 5 & 6 & 6 & 7 & 4 \end{pmatrix}_{b_0} \\
[6] & 1^{124} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 31 & 5 \\ 7 & 8 & 12 & 38 & 6 \\ 11 & 12 & 14 & 48 & 8 \\ 31 & 38 & 48 & 149 & 25 \\ 5 & 6 & 8 & 25 & 4 \end{pmatrix}_{b_0} \\
[5] & 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 9 & 5 \\ 7 & 8 & 15 & 11 & 6 \\ 11 & 15 & 22 & 17 & 10 \\ 9 & 11 & 17 & 11 & 7 \\ 5 & 6 & 10 & 7 & 4 \end{pmatrix}_{b_0} \\
[7] & 1^{133} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 15 & 26 & 5 \\ 7 & 8 & 18 & 32 & 6 \\ 15 & 18 & 34 & 60 & 12 \\ 26 & 32 & 60 & 105 & 21 \\ 5 & 6 & 12 & 21 & 4 \end{pmatrix}_{b_0} \\
[8] & 1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 22 & 5 \\ 7 & 8 & 12 & 26 & 6 \\ 11 & 12 & 14 & 32 & 8 \\ 22 & 26 & 32 & 69 & 17 \\ 5 & 6 & 8 & 17 & 4 \end{pmatrix}_{b_0} \\
[9] & 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 27, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 13 & 5 \\ 7 & 8 & 15 & 17 & 6 \\ 11 & 15 & 22 & 26 & 10 \\ 13 & 17 & 26 & 29 & 11 \\ 5 & 6 & 10 & 11 & 4 \end{pmatrix}_{b_0} \\
[13] & 1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 10 & 8 & 5 \\ 7 & 8 & 12 & 11 & 6 \\ 10 & 12 & 14 & 14 & 8 \\ 8 & 11 & 14 & 11 & 7 \\ 5 & 6 & 8 & 7 & 4 \end{pmatrix}_{b_0} \\
[12] & 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 8 & 5 \\ 7 & 8 & 6 & 11 & 6 \\ 5 & 6 & 2 & 7 & 4 \\ 8 & 11 & 7 & 11 & 7 \\ 5 & 6 & 4 & 7 & 4 \end{pmatrix}_{b_0} \\
[11] & 1^{131} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 15 & 8 & 5 \\ 7 & 8 & 18 & 11 & 6 \\ 15 & 18 & 34 & 21 & 12 \\ 8 & 11 & 21 & 11 & 7 \\ 5 & 6 & 12 & 7 & 4 \end{pmatrix}_{b_0} \\
[10] & 1^{141} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 20 & 8 & 5 \\ 7 & 8 & 24 & 11 & 6 \\ 20 & 24 & 62 & 28 & 16 \\ 8 & 11 & 28 & 11 & 7 \\ 5 & 6 & 16 & 7 & 4 \end{pmatrix}_{b_0} \\
[15] & 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 14 & 9 & 5 \\ 7 & 8 & 15 & 11 & 6 \\ 14 & 15 & 22 & 17 & 10 \\ 9 & 11 & 17 & 11 & 7 \\ 5 & 6 & 10 & 7 & 4 \end{pmatrix}_{b_0} \\
[14] & 1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5
\end{aligned}$$

- $T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
- $$\begin{pmatrix} 5 & 7 & 19 & 14 & 5 \\ 7 & 8 & 21 & 17 & 6 \\ 19 & 21 & 44 & 37 & 14 \\ 14 & 17 & 37 & 29 & 11 \\ 5 & 6 & 14 & 11 & 4 \end{pmatrix}_{b_0}$$
- [16] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -58$, $\det = 34$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
- $T_{\min} = 9.94 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
- $$\begin{pmatrix} 5 & 7 & 6 & 21 & 5 \\ 7 & 8 & 6 & 26 & 6 \\ 6 & 6 & 2 & 16 & 4 \\ 21 & 26 & 16 & 69 & 17 \\ 5 & 6 & 4 & 17 & 4 \end{pmatrix}_{b_0}$$
- [17] $1^{111} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -58$, $\det = 35$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
- $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
- $$\begin{pmatrix} 5 & 7 & 4 & 9 & 5 \\ 7 & 8 & 6 & 11 & 6 \\ 4 & 6 & 2 & 7 & 4 \\ 9 & 11 & 7 & 11 & 7 \\ 5 & 6 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$
- [18] $1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -58$, $\det = 36$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
- $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
- $$\begin{pmatrix} 5 & 7 & 21 & 10 & 5 \\ 7 & 8 & 24 & 11 & 6 \\ 21 & 24 & 62 & 27 & 16 \\ 10 & 11 & 27 & 11 & 7 \\ 5 & 6 & 16 & 7 & 4 \end{pmatrix}_{b_0}$$
- [19] $1^{133} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -58$, $\det = 38$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.8$
- $T_{\min} = 9.95 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
- $$\begin{pmatrix} 5 & 7 & 15 & 27 & 5 \\ 7 & 8 & 18 & 32 & 6 \\ 15 & 18 & 34 & 60 & 12 \\ 27 & 32 & 60 & 105 & 21 \\ 5 & 6 & 12 & 21 & 4 \end{pmatrix}_{b_0}$$
- [20] $21^{113} 3^1 1^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -58$, $\det = 40$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
- $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
- $$\begin{pmatrix} 4 & 5 & 5 & 3 & 12 & 4 \\ 5 & 4 & 6 & 4 & 12 & 4 \\ 5 & 6 & 4 & 4 & 12 & 4 \\ 3 & 4 & 4 & 1 & 8 & 3 \\ 12 & 12 & 12 & 8 & 25 & 9 \\ 4 & 4 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$
- [21] $21^{123} 3^1 1^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -58$, $\det = 40$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
- $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
- $$\begin{pmatrix} 4 & 5 & 5 & 3 & 21 & 4 \\ 5 & 4 & 6 & 4 & 21 & 4 \\ 5 & 6 & 4 & 4 & 21 & 4 \\ 3 & 4 & 4 & 1 & 14 & 3 \\ 21 & 21 & 21 & 14 & 80 & 16 \\ 4 & 4 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$
- [22] $1^{111} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -58$, $\det = 43$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
- $T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
- $$\begin{pmatrix} 5 & 7 & 6 & 9 & 5 \\ 7 & 8 & 6 & 11 & 6 \\ 6 & 6 & 2 & 7 & 4 \\ 9 & 11 & 7 & 11 & 7 \\ 5 & 6 & 4 & 7 & 4 \end{pmatrix}_{b_0}$$
- [23] $1^{111} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -58$, $\det = 43$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
- $T_{\min} = 9.91 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
- $$\begin{pmatrix} 5 & 7 & 20 & 9 & 5 \\ 7 & 8 & 24 & 11 & 6 \\ 20 & 24 & 62 & 27 & 16 \\ 9 & 11 & 27 & 11 & 7 \\ 5 & 6 & 16 & 7 & 4 \end{pmatrix}_{b_0}$$
- [24] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -58$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
- $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
- $$\begin{pmatrix} 5 & 7 & 12 & 8 & 5 \\ 7 & 8 & 15 & 11 & 6 \\ 12 & 15 & 22 & 17 & 10 \\ 8 & 11 & 17 & 11 & 7 \\ 5 & 6 & 10 & 7 & 4 \end{pmatrix}_{b_0}$$
- [25] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8$, $T_{\max} = 10$, $\Delta = -58$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 6.0$
- $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
- $$\begin{pmatrix} 5 & 7 & 14 & 15 & 5 \\ 7 & 8 & 15 & 17 & 6 \\ 14 & 15 & 22 & 26 & 10 \\ 15 & 17 & 26 & 29 & 11 \\ 5 & 6 & 10 & 11 & 4 \end{pmatrix}_{b_0}$$

- [26] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 4 & 16 & 5 \\ 7 & 8 & 6 & 20 & 6 \\ 4 & 6 & 2 & 12 & 4 \\ 16 & 20 & 12 & 41 & 13 \\ 5 & 6 & 4 & 13 & 4 \end{pmatrix}_{b_0}$
- [27] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 54, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 13 & 14 & 5 \\ 7 & 8 & 15 & 17 & 6 \\ 13 & 15 & 22 & 26 & 10 \\ 14 & 17 & 26 & 29 & 11 \\ 5 & 6 & 10 & 11 & 4 \end{pmatrix}_{b_0}$
- [28] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 58, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 6 & 17 & 5 \\ 7 & 8 & 6 & 20 & 6 \\ 6 & 6 & 2 & 12 & 4 \\ 17 & 20 & 12 & 41 & 13 \\ 5 & 6 & 4 & 13 & 4 \end{pmatrix}_{b_0}$
- [29] $1^{112} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 4 & 14 & 5 \\ 7 & 8 & 6 & 17 & 6 \\ 4 & 6 & 2 & 11 & 4 \\ 14 & 17 & 11 & 29 & 11 \\ 5 & 6 & 4 & 11 & 4 \end{pmatrix}_{b_0}$
- [30] $1^{121} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 59, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 11 & 9 & 5 \\ 7 & 8 & 12 & 11 & 6 \\ 11 & 12 & 14 & 14 & 8 \\ 9 & 11 & 14 & 11 & 7 \\ 5 & 6 & 8 & 7 & 4 \end{pmatrix}_{b_0}$
- [31] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 65, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 12 & 14 & 5 \\ 7 & 8 & 15 & 17 & 6 \\ 12 & 15 & 22 & 26 & 10 \\ 14 & 17 & 26 & 29 & 11 \\ 5 & 6 & 10 & 11 & 4 \end{pmatrix}_{b_0}$
- [32] $1^{112} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 6 & 14 & 5 \\ 7 & 8 & 6 & 17 & 6 \\ 6 & 6 & 2 & 11 & 4 \\ 14 & 17 & 11 & 29 & 11 \\ 5 & 6 & 4 & 11 & 4 \end{pmatrix}_{b_0}$
- [33] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 6 & 38 & 5 \\ 7 & 8 & 6 & 47 & 6 \\ 6 & 6 & 2 & 29 & 4 \\ 38 & 47 & 29 & 229 & 31 \\ 5 & 6 & 4 & 31 & 4 \end{pmatrix}_{b_0}$
- [38] $1^{111} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 5 & 10 & 5 \\ 7 & 8 & 6 & 11 & 6 \\ 5 & 6 & 2 & 7 & 4 \\ 10 & 11 & 7 & 11 & 7 \\ 5 & 6 & 4 & 7 & 4 \end{pmatrix}_{b_0}$
- [37] $1^{111} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 5 & 7 & 5 & 9 & 5 \\ 7 & 8 & 6 & 11 & 6 \\ 5 & 6 & 2 & 7 & 4 \\ 9 & 11 & 7 & 11 & 7 \\ 5 & 6 & 4 & 7 & 4 \end{pmatrix}_{b_0}$
- [36] $1^{141} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 5 & 7 & 20 & 10 & 5 \\ 7 & 8 & 24 & 11 & 6 \\ 20 & 24 & 62 & 28 & 16 \\ 10 & 11 & 28 & 11 & 7 \\ 5 & 6 & 16 & 7 & 4 \end{pmatrix}_{b_0} \\
[35] & 1^{122} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 10 & 10 & 5 \\ 7 & 8 & 12 & 11 & 6 \\ 10 & 12 & 14 & 14 & 8 \\ 10 & 11 & 14 & 11 & 7 \\ 5 & 6 & 8 & 7 & 4 \end{pmatrix}_{b_0} \\
[34] & 1^{132} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 15 & 10 & 5 \\ 7 & 8 & 18 & 11 & 6 \\ 15 & 18 & 34 & 21 & 12 \\ 10 & 11 & 21 & 11 & 7 \\ 5 & 6 & 12 & 7 & 4 \end{pmatrix}_{b_0} \\
[39] & 1^{12} 1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 17 & 8 & 5 \\ 7 & 8 & 21 & 11 & 6 \\ 17 & 21 & 44 & 24 & 14 \\ 8 & 11 & 24 & 11 & 7 \\ 5 & 6 & 14 & 7 & 4 \end{pmatrix}_{b_0} \\
[40] & 1^{12} 1^{121} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 80, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 24 & 8 & 5 \\ 7 & 8 & 30 & 11 & 6 \\ 24 & 30 & 94 & 34 & 20 \\ 8 & 11 & 34 & 11 & 7 \\ 5 & 6 & 20 & 7 & 4 \end{pmatrix}_{b_0} \\
[41] & 1^{141} 1^{131} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 16 & 9 & 5 \\ 7 & 8 & 18 & 11 & 6 \\ 16 & 18 & 34 & 21 & 12 \\ 9 & 11 & 21 & 11 & 7 \\ 5 & 6 & 12 & 7 & 4 \end{pmatrix}_{b_0} \\
[42] & 1^{132} 1^{122} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 14 & 5 \\ 7 & 8 & 12 & 17 & 6 \\ 11 & 12 & 14 & 22 & 8 \\ 14 & 17 & 22 & 29 & 11 \\ 5 & 6 & 8 & 11 & 4 \end{pmatrix}_{b_0} \\
[43] & 1^{121} 1^{121} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 10 & 9 & 5 \\ 7 & 8 & 12 & 11 & 6 \\ 10 & 12 & 14 & 14 & 8 \\ 9 & 11 & 14 & 11 & 7 \\ 5 & 6 & 8 & 7 & 4 \end{pmatrix}_{b_0} \\
[44] & 1^{122} 1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 10 & 21 & 5 \\ 7 & 8 & 12 & 26 & 6 \\ 10 & 12 & 14 & 32 & 8 \\ 21 & 26 & 32 & 69 & 17 \\ 5 & 6 & 8 & 17 & 4 \end{pmatrix}_{b_0} \\
[45] & 1^{11} 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 20 & 5 \\ 7 & 8 & 6 & 26 & 6 \\ 4 & 6 & 2 & 16 & 4 \\ 20 & 26 & 16 & 69 & 17 \\ 5 & 6 & 4 & 17 & 4 \end{pmatrix}_{b_0} \\
[46] & 1^{13} 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 22 & 5 \\ 7 & 8 & 6 & 26 & 6 \\ 4 & 6 & 2 & 16 & 4 \\ 22 & 26 & 16 & 69 & 17 \\ 5 & 6 & 4 & 17 & 4 \end{pmatrix}_{b_0} \\
[47] & 1^{111} 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 12 & 9 & 5 \\ 7 & 8 & 15 & 11 & 6 \\ 12 & 15 & 22 & 17 & 10 \\ 9 & 11 & 17 & 11 & 7 \\ 5 & 6 & 10 & 7 & 4 \end{pmatrix}_{b_0} \\
[48] & 1^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 16 & 5 \\ 7 & 8 & 6 & 20 & 6 \\ 5 & 6 & 2 & 12 & 4 \\ 16 & 20 & 12 & 41 & 13 \\ 5 & 6 & 4 & 13 & 4 \end{pmatrix}_{b_0} \\
[49] & 1^{113} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 19 & 5 \\ 7 & 8 & 6 & 23 & 6 \\ 4 & 6 & 2 & 15 & 4 \\ 19 & 23 & 15 & 55 & 15 \\ 5 & 6 & 4 & 15 & 4 \end{pmatrix}_{b_0} \\
[50] & 1^{123} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 6 & 26 & 5 \\ 7 & 8 & 6 & 32 & 6 \\ 6 & 6 & 2 & 20 & 4 \\ 26 & 32 & 20 & 105 & 21 \\ 5 & 6 & 4 & 21 & 4 \end{pmatrix}_{b_0} \\
[51] & 1^{121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 13 & 9 & 5 \\ 7 & 8 & 15 & 11 & 6 \\ 13 & 15 & 22 & 17 & 10 \\ 9 & 11 & 17 & 11 & 7 \\ 5 & 6 & 10 & 7 & 4 \end{pmatrix}_{b_0} \\
[52] & 1^{134} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 23 & 5 \\ 7 & 8 & 12 & 26 & 6 \\ 11 & 12 & 14 & 32 & 8 \\ 23 & 26 & 32 & 69 & 17 \\ 5 & 6 & 8 & 17 & 4 \end{pmatrix}_{b_0} \\
[54] & 1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 13 & 10 & 5 \\ 7 & 8 & 15 & 11 & 6 \\ 13 & 15 & 22 & 17 & 10 \\ 10 & 11 & 17 & 11 & 7 \\ 5 & 6 & 10 & 7 & 4 \end{pmatrix}_{b_0} \\
[53] & 1^{123} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 6 & 19 & 5 \\ 7 & 8 & 6 & 23 & 6 \\ 6 & 6 & 2 & 15 & 4 \\ 19 & 23 & 15 & 55 & 15 \\ 5 & 6 & 4 & 15 & 4 \end{pmatrix}_{b_0} \\
[55] & 1^{132} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 29 & 14 & 5 \\ 7 & 8 & 36 & 17 & 6 \\ 29 & 36 & 134 & 63 & 24 \\ 14 & 17 & 63 & 29 & 11 \\ 5 & 6 & 24 & 11 & 4 \end{pmatrix}_{b_0} \\
[56] & 1^{131} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 15 & 9 & 5 \\ 7 & 8 & 18 & 11 & 6 \\ 15 & 18 & 34 & 21 & 12 \\ 9 & 11 & 21 & 11 & 7 \\ 5 & 6 & 12 & 7 & 4 \end{pmatrix}_{b_0} \\
[57] & 1^{112} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 21 & 5 \\ 7 & 8 & 6 & 26 & 6 \\ 4 & 6 & 2 & 16 & 4 \\ 21 & 26 & 16 & 69 & 17 \\ 5 & 6 & 4 & 17 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

[58] $1^{1^{142}} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 16 & 14 & 5 \\ 7 & 8 & 18 & 17 & 6 \\ 16 & 18 & 34 & 33 & 12 \\ 14 & 17 & 33 & 29 & 11 \\ 5 & 6 & 12 & 11 & 4 \end{pmatrix}_{b_0}$$

[59] $1^{1^{13}} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 22 & 8 & 5 \\ 7 & 8 & 27 & 11 & 6 \\ 22 & 27 & 74 & 31 & 18 \\ 8 & 11 & 31 & 11 & 7 \\ 5 & 6 & 18 & 7 & 4 \end{pmatrix}_{b_0}$$

[61] $1^{1^{134}} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 11 & 27 & 5 \\ 7 & 8 & 12 & 32 & 6 \\ 11 & 12 & 14 & 40 & 8 \\ 27 & 32 & 40 & 105 & 21 \\ 5 & 6 & 8 & 21 & 4 \end{pmatrix}_{b_0}$$

[60] $1^{1^{123}} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 10 & 22 & 5 \\ 7 & 8 & 12 & 26 & 6 \\ 10 & 12 & 14 & 32 & 8 \\ 22 & 26 & 32 & 69 & 17 \\ 5 & 6 & 8 & 17 & 4 \end{pmatrix}_{b_0}$$

[62] $1^{1^{133}} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 11 & 19 & 5 \\ 7 & 8 & 12 & 23 & 6 \\ 11 & 12 & 14 & 30 & 8 \\ 19 & 23 & 30 & 55 & 15 \\ 5 & 6 & 8 & 15 & 4 \end{pmatrix}_{b_0}$$

[63] $1^{1^{112}} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 5 & 17 & 5 \\ 7 & 8 & 6 & 20 & 6 \\ 5 & 6 & 2 & 12 & 4 \\ 17 & 20 & 12 & 41 & 13 \\ 5 & 6 & 4 & 13 & 4 \end{pmatrix}_{b_0}$$

[64] $1^{1^{112}} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 5 & 14 & 5 \\ 7 & 8 & 6 & 17 & 6 \\ 5 & 6 & 2 & 11 & 4 \\ 14 & 17 & 11 & 29 & 11 \\ 5 & 6 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

[65] $1^{1^{141}} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 121, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 19 & 9 & 5 \\ 7 & 8 & 21 & 11 & 6 \\ 19 & 21 & 44 & 24 & 14 \\ 9 & 11 & 24 & 11 & 7 \\ 5 & 6 & 14 & 7 & 4 \end{pmatrix}_{b_0}$$

[66] $1^{1^{123}} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 6 & 22 & 5 \\ 7 & 8 & 6 & 26 & 6 \\ 6 & 6 & 2 & 16 & 4 \\ 22 & 26 & 16 & 69 & 17 \\ 5 & 6 & 4 & 17 & 4 \end{pmatrix}_{b_0}$$

[70] $1^{1^{123}} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 10 & 15 & 5 \\ 7 & 8 & 12 & 17 & 6 \\ 10 & 12 & 14 & 22 & 8 \\ 15 & 17 & 22 & 29 & 11 \\ 5 & 6 & 8 & 11 & 4 \end{pmatrix}_{b_0}$$

[67] $1^{1^{113}} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 5 & 15 & 5 \\ 7 & 8 & 6 & 17 & 6 \\ 5 & 6 & 2 & 11 & 4 \\ 15 & 17 & 11 & 29 & 11 \\ 5 & 6 & 4 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{132} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 128, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 15 & 15 & 5 \\ 7 & 8 & 18 & 17 & 6 \\ 15 & 18 & 34 & 33 & 12 \\ 15 & 17 & 33 & 29 & 11 \\ 5 & 6 & 12 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{142} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 128, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 20 & 15 & 5 \\ 7 & 8 & 24 & 17 & 6 \\ 20 & 24 & 62 & 44 & 16 \\ 15 & 17 & 44 & 29 & 11 \\ 5 & 6 & 16 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 131, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 28 & 5 \\ 7 & 8 & 6 & 35 & 6 \\ 4 & 6 & 2 & 21 & 4 \\ 28 & 35 & 21 & 129 & 23 \\ 5 & 6 & 4 & 23 & 4 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 131, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 24 & 5 \\ 7 & 8 & 6 & 29 & 6 \\ 4 & 6 & 2 & 19 & 4 \\ 24 & 29 & 19 & 89 & 19 \\ 5 & 6 & 4 & 19 & 4 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{122} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 132, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 10 & 14 & 5 \\ 7 & 8 & 12 & 17 & 6 \\ 10 & 12 & 14 & 22 & 8 \\ 14 & 17 & 22 & 29 & 11 \\ 5 & 6 & 8 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{141} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 132, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.94 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 20 & 9 & 5 \\ 7 & 8 & 24 & 11 & 6 \\ 20 & 24 & 62 & 28 & 16 \\ 9 & 11 & 28 & 11 & 7 \\ 5 & 6 & 16 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{124} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 135, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 34 & 5 \\ 7 & 8 & 6 & 41 & 6 \\ 6 & 6 & 2 & 25 & 4 \\ 34 & 41 & 25 & 175 & 27 \\ 5 & 6 & 4 & 27 & 4 \end{pmatrix}_{b_0}$$

$$[76] \ 21^{113} 3^1 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 136, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.91 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 5 & 3 & 15 & 4 \\ 5 & 4 & 6 & 4 & 16 & 4 \\ 5 & 6 & 4 & 4 & 16 & 4 \\ 3 & 4 & 4 & 1 & 11 & 3 \\ 15 & 16 & 16 & 11 & 44 & 12 \\ 4 & 4 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[77] \ 1^{114} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 139, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 24 & 5 \\ 7 & 8 & 6 & 29 & 6 \\ 6 & 6 & 2 & 19 & 4 \\ 24 & 29 & 19 & 89 & 19 \\ 5 & 6 & 4 & 19 & 4 \end{pmatrix}_{b_0}$$

$$[78] \ 1^{124} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 139, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 30 & 5 \\ 7 & 8 & 6 & 35 & 6 \\ 6 & 6 & 2 & 21 & 4 \\ 30 & 35 & 21 & 129 & 23 \\ 5 & 6 & 4 & 23 & 4 \end{pmatrix}_{b_0}$$

$$[79] \ 1^{133} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, \ T_{\max} = 10, \ \Delta = -58, \ \det = 139, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 16 & 19 & 5 \\ 7 & 8 & 18 & 23 & 6 \\ 16 & 18 & 34 & 45 & 12 \\ 19 & 23 & 45 & 55 & 15 \\ 5 & 6 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

 1^{14}

$$[80] \ 1^{141} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 27 & 8 & 5 \\ 7 & 8 & 33 & 11 & 6 \\ 27 & 33 & 112 & 38 & 22 \\ 8 & 11 & 38 & 11 & 7 \\ 5 & 6 & 22 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[82] \ 21^{113} 3^1 1^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 5 & 12 & 6 & 4 \\ 5 & 4 & 6 & 13 & 7 & 4 \\ 5 & 6 & 4 & 13 & 7 & 4 \\ 12 & 13 & 13 & 28 & 16 & 10 \\ 6 & 7 & 7 & 16 & 7 & 5 \\ 4 & 4 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0}$$

 1^{13}

$$[81] \ 1^{131} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 144, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 29 & 8 & 5 \\ 7 & 8 & 36 & 11 & 6 \\ 29 & 36 & 134 & 41 & 24 \\ 8 & 11 & 41 & 11 & 7 \\ 5 & 6 & 24 & 7 & 4 \end{pmatrix}_{b_0}$$

 1^{12}

$$[84] \ 1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 25 & 5 \\ 7 & 8 & 6 & 32 & 6 \\ 4 & 6 & 2 & 20 & 4 \\ 25 & 32 & 20 & 105 & 21 \\ 5 & 6 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

 1^{14}

$$[83] \ 1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 27 & 5 \\ 7 & 8 & 6 & 32 & 6 \\ 4 & 6 & 2 & 20 & 4 \\ 27 & 32 & 20 & 105 & 21 \\ 5 & 6 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

 1^{132}

$$[85] \ 1^{132} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 15 & 14 & 5 \\ 7 & 8 & 18 & 17 & 6 \\ 15 & 18 & 34 & 33 & 12 \\ 14 & 17 & 33 & 29 & 11 \\ 5 & 6 & 12 & 11 & 4 \end{pmatrix}_{b_0}$$

 1^{121}

$$[86] \ 1^{121} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 24 & 9 & 5 \\ 7 & 8 & 30 & 11 & 6 \\ 24 & 30 & 94 & 34 & 20 \\ 9 & 11 & 34 & 11 & 7 \\ 5 & 6 & 20 & 7 & 4 \end{pmatrix}_{b_0}$$

 1^{132}

$$[87] \ 1^{122} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 18 & 14 & 5 \\ 7 & 8 & 21 & 17 & 6 \\ 18 & 21 & 44 & 37 & 14 \\ 14 & 17 & 37 & 29 & 11 \\ 5 & 6 & 14 & 11 & 4 \end{pmatrix}_{b_0}$$

 1^{134}

$$[88] \ 1^{124} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 155, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 11 & 24 & 5 \\ 7 & 8 & 12 & 29 & 6 \\ 11 & 12 & 14 & 38 & 8 \\ 24 & 29 & 38 & 89 & 19 \\ 5 & 6 & 8 & 19 & 4 \end{pmatrix}_{b_0}$$

$$[89] \ 21^{111} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 8 & 3 \\ 4 & 4 & 6 & 4 & 12 & 4 \\ 4 & 6 & 4 & 4 & 12 & 4 \\ 3 & 4 & 4 & 1 & 8 & 3 \\ 8 & 12 & 12 & 8 & 25 & 9 \\ 3 & 4 & 4 & 3 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[90] \ 21^{112} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 14 & 3 \\ 4 & 4 & 6 & 4 & 21 & 4 \\ 4 & 6 & 4 & 4 & 21 & 4 \\ 3 & 4 & 4 & 1 & 14 & 3 \\ 14 & 21 & 21 & 14 & 80 & 16 \\ 3 & 4 & 4 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[91] \quad 1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 161, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 17 & 14 & 5 \\ 7 & 8 & 21 & 17 & 6 \\ 17 & 21 & 44 & 37 & 14 \\ 14 & 17 & 37 & 29 & 11 \\ 5 & 6 & 14 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[92] \quad 1^{134} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 15 & 31 & 5 \\ 7 & 8 & 18 & 38 & 6 \\ 15 & 18 & 34 & 72 & 12 \\ 31 & 38 & 72 & 149 & 25 \\ 5 & 6 & 12 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[93] \quad 1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 162, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 31 & 5 \\ 7 & 8 & 6 & 38 & 6 \\ 6 & 6 & 2 & 24 & 4 \\ 31 & 38 & 24 & 149 & 25 \\ 5 & 6 & 4 & 25 & 4 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 26 & 5 \\ 7 & 8 & 6 & 32 & 6 \\ 4 & 6 & 2 & 20 & 4 \\ 26 & 32 & 20 & 105 & 21 \\ 5 & 6 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[95] \quad 21^{113} 3^1 1^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 168, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 5 & 5 & 3 & 18 & 4 \\ 5 & 4 & 6 & 4 & 20 & 4 \\ 5 & 6 & 4 & 4 & 20 & 4 \\ 3 & 4 & 4 & 1 & 14 & 3 \\ 18 & 20 & 20 & 14 & 69 & 15 \\ 4 & 4 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[96] \quad 1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 17 & 9 & 5 \\ 7 & 8 & 21 & 11 & 6 \\ 17 & 21 & 44 & 24 & 14 \\ 9 & 11 & 24 & 11 & 7 \\ 5 & 6 & 14 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[97] \quad 1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 18 & 10 & 5 \\ 7 & 8 & 21 & 11 & 6 \\ 18 & 21 & 44 & 24 & 14 \\ 10 & 11 & 24 & 11 & 7 \\ 5 & 6 & 14 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[98] \quad 1^{121} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 26 & 9 & 5 \\ 7 & 8 & 30 & 11 & 6 \\ 26 & 30 & 94 & 34 & 20 \\ 9 & 11 & 34 & 11 & 7 \\ 5 & 6 & 20 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[99] \quad 1^{122} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 19 & 15 & 5 \\ 7 & 8 & 21 & 17 & 6 \\ 19 & 21 & 44 & 37 & 14 \\ 15 & 17 & 37 & 29 & 11 \\ 5 & 6 & 14 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[100] \quad 1^{121} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 18 & 9 & 5 \\ 7 & 8 & 21 & 11 & 6 \\ 18 & 21 & 44 & 24 & 14 \\ 9 & 11 & 24 & 11 & 7 \\ 5 & 6 & 14 & 7 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[101] \quad & 1^{1^{124}} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 178, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 6 & 23 & 5 \\ 7 & 8 & 6 & 26 & 6 \\ 6 & 6 & 2 & 16 & 4 \\ 23 & 26 & 16 & 69 & 17 \\ 5 & 6 & 4 & 17 & 4 \end{pmatrix}_{b_0} \\
[102] \quad & 1^{1^{144}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 16 & 24 & 5 \\ 7 & 8 & 18 & 29 & 6 \\ 16 & 18 & 34 & 57 & 12 \\ 24 & 29 & 57 & 89 & 19 \\ 5 & 6 & 12 & 19 & 4 \end{pmatrix}_{b_0} \\
[103] \quad & 1^{1^{142}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 20 & 14 & 5 \\ 7 & 8 & 24 & 17 & 6 \\ 20 & 24 & 62 & 44 & 16 \\ 14 & 17 & 44 & 29 & 11 \\ 5 & 6 & 16 & 11 & 4 \end{pmatrix}_{b_0} \\
[104] \quad & 1^{1^{121}} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 26 & 10 & 5 \\ 7 & 8 & 30 & 11 & 6 \\ 26 & 30 & 94 & 34 & 20 \\ 10 & 11 & 34 & 11 & 7 \\ 5 & 6 & 20 & 7 & 4 \end{pmatrix}_{b_0} \\
[105] \quad & 1^{1^{113}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 19 & 5 \\ 7 & 8 & 6 & 23 & 6 \\ 5 & 6 & 2 & 15 & 4 \\ 19 & 23 & 15 & 55 & 15 \\ 5 & 6 & 4 & 15 & 4 \end{pmatrix}_{b_0} \\
[106] \quad & 1^{1^{112}} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 21 & 5 \\ 7 & 8 & 6 & 26 & 6 \\ 5 & 6 & 2 & 16 & 4 \\ 21 & 26 & 16 & 69 & 17 \\ 5 & 6 & 4 & 17 & 4 \end{pmatrix}_{b_0} \\
[107] \quad & 1^{1^{131}} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 195, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 25 & 9 & 5 \\ 7 & 8 & 30 & 11 & 6 \\ 25 & 30 & 94 & 34 & 20 \\ 9 & 11 & 34 & 11 & 7 \\ 5 & 6 & 20 & 7 & 4 \end{pmatrix}_{b_0} \\
[109] \quad & 1^{1^{123}} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 10 & 19 & 5 \\ 7 & 8 & 12 & 23 & 6 \\ 10 & 12 & 14 & 30 & 8 \\ 19 & 23 & 30 & 55 & 15 \\ 5 & 6 & 8 & 15 & 4 \end{pmatrix}_{b_0} \\
[108] \quad & 1^{1^{124}} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 10 & 38 & 5 \\ 7 & 8 & 12 & 47 & 6 \\ 10 & 12 & 14 & 58 & 8 \\ 38 & 47 & 58 & 229 & 31 \\ 5 & 6 & 8 & 31 & 4 \end{pmatrix}_{b_0} \\
[111] \quad & 1^{1^{14}} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 34 & 5 \\ 7 & 8 & 6 & 41 & 6 \\ 4 & 6 & 2 & 25 & 4 \\ 34 & 41 & 25 & 175 & 27 \\ 5 & 6 & 4 & 27 & 4 \end{pmatrix}_{b_0} \\
[110] \quad & 1^{1^{12}} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 5 & 7 & 4 & 32 & 5 \\ 7 & 8 & 6 & 41 & 6 \\ 4 & 6 & 2 & 25 & 4 \\ 32 & 41 & 25 & 175 & 27 \\ 5 & 6 & 4 & 27 & 4 \end{pmatrix}_{b_0}$$

[114] $1^{113} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 5 & 20 & 5 \\ 7 & 8 & 6 & 23 & 6 \\ 5 & 6 & 2 & 15 & 4 \\ 20 & 23 & 15 & 55 & 15 \\ 5 & 6 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

[115] $1^{123} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 10 & 20 & 5 \\ 7 & 8 & 12 & 23 & 6 \\ 10 & 12 & 14 & 30 & 8 \\ 20 & 23 & 30 & 55 & 15 \\ 5 & 6 & 8 & 15 & 4 \end{pmatrix}_{b_0}$$

[116] $1^{112} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 5 & 28 & 5 \\ 7 & 8 & 6 & 35 & 6 \\ 5 & 6 & 2 & 21 & 4 \\ 28 & 35 & 21 & 129 & 23 \\ 5 & 6 & 4 & 23 & 4 \end{pmatrix}_{b_0}$$

[117] $1^{133} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 15 & 20 & 5 \\ 7 & 8 & 18 & 23 & 6 \\ 15 & 18 & 34 & 45 & 12 \\ 20 & 23 & 45 & 55 & 15 \\ 5 & 6 & 12 & 15 & 4 \end{pmatrix}_{b_0}$$

[112] $21^{112} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 5 & 3 \\ 4 & 4 & 6 & 18 & 7 & 4 \\ 4 & 6 & 4 & 18 & 7 & 4 \\ 12 & 18 & 18 & 58 & 22 & 14 \\ 5 & 7 & 7 & 22 & 7 & 5 \\ 3 & 4 & 4 & 14 & 5 & 3 \end{pmatrix}_{b_0}$$

[113] $1^{143} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 20 & 20 & 5 \\ 7 & 8 & 24 & 23 & 6 \\ 20 & 24 & 62 & 60 & 16 \\ 20 & 23 & 60 & 55 & 15 \\ 5 & 6 & 16 & 15 & 4 \end{pmatrix}_{b_0}$$

[118] $1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 6 & 27 & 5 \\ 7 & 8 & 6 & 32 & 6 \\ 6 & 6 & 2 & 20 & 4 \\ 27 & 32 & 20 & 105 & 21 \\ 5 & 6 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

[119] $1^{123} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 10 & 26 & 5 \\ 7 & 8 & 12 & 32 & 6 \\ 10 & 12 & 14 & 40 & 8 \\ 26 & 32 & 40 & 105 & 21 \\ 5 & 6 & 8 & 21 & 4 \end{pmatrix}_{b_0}$$

[120] $1^{141} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 34 & 8 & 5 \\ 7 & 8 & 42 & 11 & 6 \\ 34 & 42 & 182 & 48 & 28 \\ 8 & 11 & 48 & 11 & 7 \\ 5 & 6 & 28 & 7 & 4 \end{pmatrix}_{b_0}$$

[121] $1^{114} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 4 & 30 & 5 \\ 7 & 8 & 6 & 38 & 6 \\ 4 & 6 & 2 & 24 & 4 \\ 30 & 38 & 24 & 149 & 25 \\ 5 & 6 & 4 & 25 & 4 \end{pmatrix}_{b_0}$$

[122] $1^{133} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$

$$\begin{aligned}
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 23 & 19 & 5 \\ 7 & 8 & 27 & 23 & 6 \\ 23 & 27 & 74 & 65 & 18 \\ 19 & 23 & 65 & 55 & 15 \\ 5 & 6 & 18 & 15 & 4 \end{pmatrix}_{b_0} \\
[123] & 21^{1122} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 6 & 11 & 3 \\ 4 & 4 & 6 & 8 & 16 & 4 \\ 4 & 6 & 4 & 8 & 16 & 4 \\ 6 & 8 & 4 & 10 & 22 & 6 \\ 11 & 16 & 16 & 22 & 44 & 12 \\ 3 & 4 & 4 & 6 & 12 & 3 \end{pmatrix}_{b_0} \\
[124] & 1^{133} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 15 & 19 & 5 \\ 7 & 8 & 18 & 23 & 6 \\ 15 & 18 & 34 & 45 & 12 \\ 19 & 23 & 45 & 55 & 15 \\ 5 & 6 & 12 & 15 & 4 \end{pmatrix}_{b_0} \\
[125] & 1^{113} 3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 219, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 33 & 5 \\ 7 & 8 & 6 & 41 & 6 \\ 4 & 6 & 2 & 25 & 4 \\ 33 & 41 & 25 & 175 & 27 \\ 5 & 6 & 4 & 27 & 4 \end{pmatrix}_{b_0} \\
[126] & 21^{1111} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 6 & 5 & 3 \\ 4 & 4 & 6 & 9 & 7 & 4 \\ 4 & 6 & 4 & 9 & 7 & 4 \\ 6 & 9 & 13 & 11 & 7 \\ 5 & 7 & 7 & 11 & 7 & 5 \\ 3 & 4 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0} \\
[127] & 1^{112} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 226, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 23 & 5 \\ 7 & 8 & 6 & 26 & 6 \\ 5 & 6 & 2 & 16 & 4 \\ 23 & 26 & 16 & 69 & 17 \\ 5 & 6 & 4 & 17 & 4 \end{pmatrix}_{b_0} \\
[128] & 1^{112} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 230, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 22 & 5 \\ 7 & 8 & 6 & 26 & 6 \\ 5 & 6 & 2 & 16 & 4 \\ 22 & 26 & 16 & 69 & 17 \\ 5 & 6 & 4 & 17 & 4 \end{pmatrix}_{b_0} \\
[129] & 21^{123} 1^{1113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 232, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 3 & 24 & 4 \\ 5 & 4 & 6 & 4 & 25 & 4 \\ 5 & 6 & 4 & 4 & 25 & 4 \\ 3 & 4 & 4 & 1 & 17 & 3 \\ 24 & 25 & 25 & 17 & 111 & 19 \\ 4 & 4 & 4 & 3 & 19 & 3 \end{pmatrix}_{b_0} \\
[130] & 1^{114} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 31 & 5 \\ 7 & 8 & 6 & 38 & 6 \\ 4 & 6 & 2 & 24 & 4 \\ 31 & 38 & 24 & 149 & 25 \\ 5 & 6 & 4 & 25 & 4 \end{pmatrix}_{b_0} \\
[131] & 1^{112} 3^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 235, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 36 & 5 \\ 7 & 8 & 6 & 47 & 6 \\ 4 & 6 & 2 & 29 & 4 \\ 36 & 47 & 29 & 229 & 31 \\ 5 & 6 & 4 & 31 & 4 \end{pmatrix}_{b_0} \\
[132] & 1^{123} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 10 & 27 & 5 \\ 7 & 8 & 12 & 32 & 6 \\ 10 & 12 & 14 & 40 & 8 \\ 27 & 32 & 40 & 105 & 21 \\ 5 & 6 & 8 & 21 & 4 \end{pmatrix}_{b_0} \\
[133] & 21^{1111} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 240, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0
\end{aligned}$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 5 & 3 \\ 4 & 4 & 6 & 4 & 7 & 4 \\ 4 & 6 & 4 & 4 & 7 & 4 \\ 3 & 4 & 4 & 1 & 5 & 3 \\ 5 & 7 & 7 & 5 & 7 & 5 \\ 3 & 4 & 4 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[134] \quad 1^{133} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 22 & 19 & 5 \\ 7 & 8 & 27 & 23 & 6 \\ 22 & 27 & 74 & 65 & 18 \\ 19 & 23 & 65 & 55 & 15 \\ 5 & 6 & 18 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[135] \quad 1^{142} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 243, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 23 & 10 & 5 \\ 7 & 8 & 27 & 11 & 6 \\ 23 & 27 & 74 & 31 & 18 \\ 10 & 11 & 31 & 11 & 7 \\ 5 & 6 & 18 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[136] \quad 1^{143} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 20 & 19 & 5 \\ 7 & 8 & 24 & 23 & 6 \\ 20 & 24 & 62 & 60 & 16 \\ 19 & 23 & 60 & 55 & 15 \\ 5 & 6 & 16 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[137] \quad 1^{131} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 257, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 22 & 9 & 5 \\ 7 & 8 & 27 & 11 & 6 \\ 22 & 27 & 74 & 31 & 18 \\ 9 & 11 & 31 & 11 & 7 \\ 5 & 6 & 18 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[138] \quad 1^{142} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 262, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 23 & 14 & 5 \\ 7 & 8 & 27 & 17 & 6 \\ 23 & 27 & 74 & 48 & 18 \\ 14 & 17 & 48 & 29 & 11 \\ 5 & 6 & 18 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[140] \quad 1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 29 & 5 \\ 7 & 8 & 6 & 35 & 6 \\ 5 & 6 & 2 & 21 & 4 \\ 29 & 35 & 21 & 129 & 23 \\ 5 & 6 & 4 & 23 & 4 \end{pmatrix}_{b_0}$$

$$[139] \quad 1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 24 & 5 \\ 7 & 8 & 6 & 29 & 6 \\ 5 & 6 & 2 & 19 & 4 \\ 24 & 29 & 19 & 89 & 19 \\ 5 & 6 & 4 & 19 & 4 \end{pmatrix}_{b_0}$$

$$[141] \quad 21^{1121} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 5 & 3 \\ 4 & 4 & 6 & 8 & 7 & 4 \\ 4 & 6 & 4 & 8 & 7 & 4 \\ 6 & 8 & 8 & 10 & 10 & 6 \\ 5 & 7 & 7 & 10 & 7 & 5 \\ 3 & 4 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[142] \quad 1^{132} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 265, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 22 & 14 & 5 \\ 7 & 8 & 27 & 17 & 6 \\ 22 & 27 & 74 & 48 & 18 \\ 14 & 17 & 48 & 29 & 11 \\ 5 & 6 & 18 & 11 & 4 \end{pmatrix}_{b_0}$$

$$[143] \quad 1^{131} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 270, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 23 & 9 & 5 \\ 7 & 8 & 27 & 11 & 6 \\ 23 & 27 & 74 & 31 & 18 \\ 9 & 11 & 31 & 11 & 7 \\ 5 & 6 & 18 & 7 & 4 \end{pmatrix}_{b_0}$$

[144] $1^{124} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 276, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 10 & 24 & 5 \\ 7 & 8 & 12 & 29 & 6 \\ 10 & 12 & 14 & 38 & 8 \\ 24 & 29 & 38 & 89 & 19 \\ 5 & 6 & 8 & 19 & 4 \end{pmatrix}_{b_0}$$

[145] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 5 & 30 & 5 \\ 7 & 8 & 6 & 35 & 6 \\ 5 & 6 & 2 & 21 & 4 \\ 30 & 35 & 21 & 129 & 23 \\ 5 & 6 & 4 & 23 & 4 \end{pmatrix}_{b_0}$$

[146] $1^{113} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 295, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 4 & 37 & 5 \\ 7 & 8 & 6 & 47 & 6 \\ 4 & 6 & 2 & 29 & 4 \\ 37 & 47 & 29 & 229 & 31 \\ 5 & 6 & 4 & 31 & 4 \end{pmatrix}_{b_0}$$

[147] $1^{134} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 15 & 24 & 5 \\ 7 & 8 & 18 & 29 & 6 \\ 15 & 18 & 34 & 57 & 12 \\ 24 & 29 & 57 & 89 & 19 \\ 5 & 6 & 12 & 19 & 4 \end{pmatrix}_{b_0}$$

[148] $1^{142} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 34 & 14 & 5 \\ 7 & 8 & 42 & 17 & 6 \\ 34 & 42 & 182 & 74 & 28 \\ 14 & 17 & 74 & 29 & 11 \\ 5 & 6 & 28 & 11 & 4 \end{pmatrix}_{b_0}$$

[149] $21^{131} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 5 & 3 \\ 4 & 4 & 6 & 12 & 7 & 4 \\ 4 & 6 & 4 & 12 & 7 & 4 \\ 9 & 12 & 12 & 25 & 15 & 9 \\ 5 & 7 & 7 & 15 & 7 & 5 \\ 3 & 4 & 4 & 9 & 5 & 3 \end{pmatrix}_{b_0}$$

[150] $1^{131} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 308, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 29 & 9 & 5 \\ 7 & 8 & 36 & 11 & 6 \\ 29 & 36 & 134 & 41 & 24 \\ 9 & 11 & 41 & 11 & 7 \\ 5 & 6 & 24 & 7 & 4 \end{pmatrix}_{b_0}$$

[151] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 314, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 5 & 26 & 5 \\ 7 & 8 & 6 & 32 & 6 \\ 5 & 6 & 2 & 20 & 4 \\ 26 & 32 & 20 & 105 & 21 \\ 5 & 6 & 4 & 21 & 4 \end{pmatrix}_{b_0}$$

[152] $1^{114} 3^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 315, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 4 & 38 & 5 \\ 7 & 8 & 6 & 47 & 6 \\ 4 & 6 & 2 & 29 & 4 \\ 38 & 47 & 29 & 229 & 31 \\ 5 & 6 & 4 & 31 & 4 \end{pmatrix}_{b_0}$$

[153] $1^{144} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 324, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 5 & 7 & 20 & 24 & 5 \\ 7 & 8 & 24 & 29 & 6 \\ 20 & 24 & 62 & 76 & 16 \\ 24 & 29 & 76 & 89 & 19 \\ 5 & 6 & 16 & 19 & 4 \end{pmatrix}_{b_0}$$

[154] $1^{124} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 342, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{aligned}
& \begin{pmatrix} 5 & 7 & 10 & 31 & 5 \\ 7 & 8 & 12 & 38 & 6 \\ 10 & 12 & 14 & 48 & 8 \\ 31 & 38 & 48 & 149 & 25 \\ 5 & 6 & 8 & 25 & 4 \end{pmatrix}_{b_0} \\
[155] & 21^{1^{112}3^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 3 & 8 & 3 \\ 4 & 4 & 6 & 4 & 11 & 4 \\ 4 & 6 & 4 & 4 & 11 & 4 \\ 3 & 4 & 4 & 1 & 8 & 3 \\ 8 & 11 & 11 & 8 & 20 & 8 \\ 3 & 4 & 4 & 3 & 8 & 3 \end{pmatrix}_{b_0} \\
& 1^{141} \\
[156] & 1^{141} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 353, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 27 & 9 & 5 \\ 7 & 8 & 33 & 11 & 6 \\ 27 & 33 & 112 & 38 & 22 \\ 9 & 11 & 38 & 11 & 7 \\ 5 & 6 & 22 & 7 & 4 \end{pmatrix}_{b_0} \\
& 1^{114} \\
[157] & 1^{113} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 358, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 27 & 5 \\ 7 & 8 & 6 & 32 & 6 \\ 5 & 6 & 2 & 20 & 4 \\ 27 & 32 & 20 & 105 & 21 \\ 5 & 6 & 4 & 21 & 4 \end{pmatrix}_{b_0} \\
[160] & 21^{1^{141}3^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 12 & 5 & 3 \\ 4 & 4 & 6 & 16 & 7 & 4 \\ 4 & 6 & 4 & 16 & 7 & 4 \\ 12 & 16 & 16 & 46 & 20 & 12 \\ 5 & 7 & 7 & 20 & 7 & 5 \\ 3 & 4 & 4 & 12 & 5 & 3 \end{pmatrix}_{b_0} \\
[159] & 21^{1^{12}3^1 1^{114}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 3 & 27 & 4 \\ 5 & 4 & 6 & 4 & 29 & 4 \\ 5 & 6 & 4 & 4 & 29 & 4 \\ 3 & 4 & 4 & 1 & 20 & 3 \\ 27 & 29 & 29 & 20 & 148 & 22 \\ 4 & 4 & 4 & 3 & 22 & 3 \end{pmatrix}_{b_0} \\
[158] & 21^{1^{12}3^1 1^{141}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 24 & 6 & 4 \\ 5 & 4 & 6 & 26 & 7 & 4 \\ 5 & 6 & 4 & 26 & 7 & 4 \\ 24 & 26 & 26 & 118 & 32 & 20 \\ 6 & 7 & 7 & 32 & 7 & 5 \\ 4 & 4 & 4 & 20 & 5 & 3 \end{pmatrix}_{b_0} \\
& 1^{143} \\
[161] & 1^{143} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 369, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 27 & 19 & 5 \\ 7 & 8 & 33 & 23 & 6 \\ 27 & 33 & 112 & 80 & 22 \\ 19 & 23 & 80 & 55 & 15 \\ 5 & 6 & 22 & 15 & 4 \end{pmatrix}_{b_0} \\
& 1^{113} \\
[163] & 1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 33 & 5 \\ 7 & 8 & 6 & 41 & 6 \\ 5 & 6 & 2 & 25 & 4 \\ 33 & 41 & 25 & 175 & 27 \\ 5 & 6 & 4 & 27 & 4 \end{pmatrix}_{b_0} \\
[162] & 21^{1^{122}3^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 6 & 8 & 3 \\ 4 & 4 & 6 & 8 & 11 & 4 \\ 4 & 6 & 4 & 8 & 11 & 4 \\ 6 & 8 & 8 & 10 & 16 & 6 \\ 8 & 11 & 11 & 16 & 20 & 8 \\ 3 & 4 & 4 & 6 & 8 & 3 \end{pmatrix}_{b_0} \\
& 1^{142} \\
[164] & 1^{142} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 377, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 27 & 14 & 5 \\ 7 & 8 & 33 & 17 & 6 \\ 27 & 33 & 112 & 59 & 22 \\ 14 & 17 & 59 & 29 & 11 \\ 5 & 6 & 22 & 11 & 4 \end{pmatrix}_{b_0} \\
[165] & 21^{1^{122}3^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 4 & 4 & 9 & 8 & 3 \\ 4 & 4 & 6 & 13 & 11 & 4 \\ 4 & 6 & 4 & 13 & 11 & 4 \\ 9 & 13 & 13 & 28 & 25 & 10 \\ 8 & 11 & 11 & 25 & 20 & 8 \\ 3 & 4 & 4 & 10 & 8 & 3 \end{pmatrix}_{b_0} \\
[166] \quad & 21^{11213^1 1^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 424, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.5 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 9 & 5 & 3 \\ 4 & 4 & 6 & 13 & 7 & 4 \\ 4 & 6 & 4 & 13 & 7 & 4 \\ 9 & 13 & 13 & 28 & 16 & 10 \\ 5 & 7 & 7 & 16 & 7 & 5 \\ 3 & 4 & 4 & 10 & 5 & 3 \end{pmatrix}_{b_0} \\
[167] \quad & 21^{11513^1 1^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.1 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 15 & 5 & 3 \\ 4 & 4 & 6 & 20 & 7 & 4 \\ 4 & 6 & 4 & 20 & 7 & 4 \\ 15 & 20 & 20 & 73 & 25 & 15 \\ 5 & 7 & 7 & 25 & 7 & 5 \\ 3 & 4 & 4 & 15 & 5 & 3 \end{pmatrix}_{b_0} \\
[168] \quad & 21^{11123^1 1^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 432, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 3 & 11 & 3 \\ 4 & 4 & 6 & 4 & 16 & 4 \\ 4 & 6 & 4 & 4 & 16 & 4 \\ 3 & 4 & 4 & 1 & 11 & 3 \\ 11 & 16 & 16 & 11 & 44 & 12 \\ 3 & 4 & 4 & 3 & 12 & 3 \end{pmatrix}_{b_0} \\
& 1^{114} \\
[169] \quad & 1^{113} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 34 & 5 \\ 7 & 8 & 6 & 41 & 6 \\ 5 & 6 & 2 & 25 & 4 \\ 34 & 41 & 25 & 175 & 27 \\ 5 & 6 & 4 & 27 & 4 \end{pmatrix}_{b_0} \\
& 1^{114} \\
[170] \quad & 1^{114} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 450, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 31 & 5 \\ 7 & 8 & 6 & 38 & 6 \\ 5 & 6 & 2 & 24 & 4 \\ 31 & 38 & 24 & 149 & 25 \\ 5 & 6 & 4 & 25 & 4 \end{pmatrix}_{b_0} \\
& 1^{141} \\
[171] \quad & 1^{141} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 34 & 9 & 5 \\ 7 & 8 & 42 & 11 & 6 \\ 34 & 42 & 182 & 48 & 28 \\ 9 & 11 & 48 & 11 & 7 \\ 5 & 6 & 28 & 7 & 4 \end{pmatrix}_{b_0} \\
[172] \quad & 21^{11133^1 1^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 496, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 3 & 11 & 3 \\ 4 & 4 & 6 & 4 & 15 & 4 \\ 4 & 6 & 4 & 4 & 15 & 4 \\ 3 & 4 & 4 & 1 & 11 & 3 \\ 11 & 15 & 15 & 11 & 39 & 11 \\ 3 & 4 & 4 & 3 & 11 & 3 \end{pmatrix}_{b_0} \\
& 1^{114} \\
[173] \quad & 1^{114} 3^1 1^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 568, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 38 & 5 \\ 7 & 8 & 6 & 47 & 6 \\ 5 & 6 & 2 & 29 & 4 \\ 38 & 47 & 29 & 229 & 31 \\ 5 & 6 & 4 & 31 & 4 \end{pmatrix}_{b_0} \\
[174] \quad & 21^{11313^1 1^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 592, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 15 & 5 & 3 \\ 4 & 4 & 6 & 22 & 7 & 4 \\ 4 & 6 & 4 & 22 & 7 & 4 \\ 15 & 22 & 22 & 85 & 27 & 17 \\ 5 & 7 & 7 & 27 & 7 & 5 \\ 3 & 4 & 4 & 17 & 5 & 3 \end{pmatrix}_{b_0} \\
[175] \quad & 21^{11133^1 1^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 624, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 3 & 17 & 3 \\ 4 & 4 & 6 & 4 & 25 & 4 \\ 4 & 6 & 4 & 4 & 25 & 4 \\ 3 & 4 & 4 & 1 & 17 & 3 \\ 17 & 25 & 25 & 17 & 111 & 19 \\ 3 & 4 & 4 & 3 & 19 & 3 \end{pmatrix}_{b_0} \\
[176] \quad & 21^{11313^1 1^1 1^1} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 640, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.8 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 5 & 3 \\ 4 & 4 & 6 & 17 & 7 & 4 \\ 4 & 6 & 4 & 17 & 7 & 4 \\ 12 & 17 & 17 & 49 & 21 & 13 \\ 5 & 7 & 7 & 21 & 7 & 5 \\ 3 & 4 & 4 & 13 & 5 & 3 \end{pmatrix}_{b_0}$$

[177] $21^{1114}3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{s0}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 672, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 14 & 3 \\ 4 & 4 & 6 & 4 & 19 & 4 \\ 4 & 6 & 4 & 4 & 19 & 4 \\ 3 & 4 & 4 & 1 & 14 & 3 \\ 14 & 19 & 19 & 14 & 64 & 14 \\ 3 & 4 & 4 & 3 & 14 & 3 \end{pmatrix}_{b_0}$$

[178] $21^{1113}3^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{s0}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 736, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 14 & 3 \\ 4 & 4 & 6 & 4 & 20 & 4 \\ 4 & 6 & 4 & 4 & 20 & 4 \\ 3 & 4 & 4 & 1 & 14 & 3 \\ 14 & 20 & 20 & 14 & 69 & 15 \\ 3 & 4 & 4 & 3 & 15 & 3 \end{pmatrix}_{b_0}$$

[179] $21^{1141}3^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{s0}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 872, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 15 & 5 & 3 \\ 4 & 4 & 6 & 21 & 7 & 4 \\ 4 & 6 & 4 & 21 & 7 & 4 \\ 15 & 21 & 21 & 76 & 26 & 16 \\ 5 & 7 & 7 & 26 & 7 & 5 \\ 3 & 4 & 4 & 16 & 5 & 3 \end{pmatrix}_{b_0}$$

[180] $21^{1141}3^{1112} + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{s0}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 1000, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 18 & 5 & 3 \\ 4 & 4 & 6 & 26 & 7 & 4 \\ 4 & 6 & 4 & 26 & 7 & 4 \\ 18 & 26 & 26 & 118 & 32 & 20 \\ 5 & 7 & 7 & 32 & 7 & 5 \\ 3 & 4 & 4 & 20 & 5 & 3 \end{pmatrix}_{b_0}$$

[181] $21^{1114}3^{1111} + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{s0}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 1072, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 17 & 3 \\ 4 & 4 & 6 & 4 & 24 & 4 \\ 4 & 6 & 4 & 4 & 24 & 4 \\ 3 & 4 & 4 & 1 & 17 & 3 \\ 17 & 24 & 24 & 17 & 100 & 18 \\ 3 & 4 & 4 & 3 & 18 & 3 \end{pmatrix}_{b_0}$$

[182] $21^{1114}3^{1112} + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{s0}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 8, T_{\max} = 10, \Delta = -58, \det = 1120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 20 & 3 \\ 4 & 4 & 6 & 4 & 29 & 4 \\ 4 & 6 & 4 & 4 & 29 & 4 \\ 3 & 4 & 4 & 1 & 20 & 3 \\ 20 & 29 & 29 & 20 & 148 & 22 \\ 3 & 4 & 4 & 3 & 22 & 3 \end{pmatrix}_{b_0}$$

$T = 9$ [184] $1^{12}3^{121}21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{s0}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 1 & 7 & 2 \\ 1 & 2 & 0 & 1 & 3 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 3 & 0 & 1 & 9 & 3 \\ 7 & 7 & 9 & 0 & 9 & 35 & 11 \\ 2 & 2 & 2 & 0 & 3 & 11 & 3 \end{pmatrix}_{b_0}$$

[185] $1^{114}3^{121}21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{s0}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 13 & 2 \\ 1 & 0 & 2 & 0 & 4 & 13 & 2 \\ 1 & 2 & 0 & 1 & 7 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 7 & 0 & 16 & 47 & 8 \\ 13 & 13 & 17 & 0 & 47 & 127 & 21 \\ 2 & 2 & 2 & 0 & 8 & 21 & 3 \end{pmatrix}_{b_0}$$

[183] $1^{112}3^{121}21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{red}{s0}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 1 & 1 & 3 & 9 & 2 \\ 1 & 1 & -3 & 1 & 0 & 0 & -1 \\ 1 & 1 & 1 & 1 & 4 & 15 & 3 \\ 2 & 3 & 0 & 4 & 5 & 24 & 5 \\ 8 & 9 & 0 & 15 & 24 & 83 & 17 \\ 2 & 2 & -1 & 3 & 5 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[186] \quad 1^{13} 3^{12} 21^{113} + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{green}{\mathfrak{su}_2}] + \textcolor{orange}{\mathfrak{su}_2} + \textcolor{red}{\mathfrak{su}_2} \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 1 & 10 & 2 \\ 1 & 2 & 0 & 1 & 3 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 3 & 0 & 1 & 13 & 3 \\ 10 & 10 & 13 & 0 & 13 & 74 & 16 \\ 2 & 2 & 2 & 0 & 3 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[187] \ 1^{113} 3^{123} 21^{123} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{red}{\mathfrak{su}}_2 + \textcolor{red}{\mathfrak{su}}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -31, \ \det = -8, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 1 & 7 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 7 & 0 & 16 & 36 & 8 \\ 10 & 10 & 13 & 0 & 36 & 74 & 16 \\ 2 & 2 & 2 & 0 & 8 & 16 & 3 \end{pmatrix}_{b_0}$$

$$[188] \ 1^{112} 3^1 21^{122} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -31, \ \det = -18, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 1 & 7 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 7 & 0 & 16 & 25 & 8 \\ 7 & 7 & 9 & 0 & 25 & 35 & 11 \\ 2 & 2 & 2 & 0 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[189] \ 1^{14} 3^{12} 21^{14} + [\textcolor{blue}{5u}_2 \cdot \textcolor{blue}{507} \cdot \textcolor{blue}{5u}_2] + \textcolor{blue}{5u}_2 + \textcolor{red}{5u}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -31, \ \det = -18, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 0 & 1 & 13 & 2 \\ 1 & 2 & 0 & 1 & 3 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 1 & 3 & 0 & 1 & 17 & 3 \\ 13 & 13 & 17 & 0 & 17 & 127 & 21 \\ 2 & 2 & 2 & 0 & 3 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[196] \ 1^{111} 3^1 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -31, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 4 & 2 \\ 1 & 2 & 0 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 9 & 5 \\ 4 & 4 & 5 & 0 & 9 & 10 & 6 \\ 2 & 2 & 2 & 0 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[195] \quad & 1^{121} 3^1 21^{111} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \, T_{\max} = 10, \, \Delta = -31, \, \det = -32, \, (n_+, n_-, n_0) = (1, 9, 0), \, c_{\text{ext}} = 6.5 \\
& T_{\min} = 10, \, T_{\max} = 10, \, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 5 & 5 & 0 & 13 & 13 & 7 \\ 4 & 4 & 5 & 0 & 13 & 10 & 6 \\ 2 & 2 & 2 & 0 & 7 & 6 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[190] \quad & 1^{111} 3^1 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{su}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \, T_{\max} = 10, \, \Delta = -31, \, \det = -32, \, (n_+, n_-, n_0) = (1, 9, 0), \, c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, \, T_{\max} = 10, \, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 4 & 7 & 0 & 16 & 14 & 8 \\ 4 & 4 & 5 & 0 & 14 & 10 & 6 \\ 2 & 2 & 2 & 0 & 8 & 6 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[191] \ 1^{\overline{111}} \ 3^{\overline{111}} \ 21^{\overline{121}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -31, \ \det = -32, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 7 & 0 & 16 & 14 & 8 \\ 4 & 4 & 5 & 0 & 14 & 10 & 6 \\ 2 & 2 & 2 & 0 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[192] \begin{array}{c} 1^{11} \\ 1^{111} 3^1 21^{11} + [\textcolor{red}{s}u_2 \textcolor{blue}{s}o_7 \textcolor{green}{s}u_2] + \textcolor{red}{s}u_2 + \textcolor{red}{s}u_2 \\ T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \end{array} \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 0 & 8 & 4 & 2 \\ 1 & 2 & 0 & 1 & 14 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 8 & 14 & 0 & 70 & 28 & 16 \\ 4 & 4 & 5 & 0 & 28 & 10 & 6 \\ 2 & 2 & 2 & 0 & 16 & 6 & 3 \end{pmatrix}_{b_0}$$

[201] $1^{1^{121}} 3^1 21^{1^{21}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 4 & 2 \\ 1 & 2 & 0 & 1 & 7 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 14 & 8 \\ 4 & 4 & 5 & 0 & 14 & 10 & 6 \\ 2 & 2 & 2 & 0 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

[202] $1^{1^{141}} 3^1 21^{1^{21}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 4 & 2 \\ 1 & 2 & 0 & 1 & 8 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 6 & 8 & 0 & 26 & 18 & 10 \\ 4 & 4 & 5 & 0 & 18 & 10 & 6 \\ 2 & 2 & 2 & 0 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

[206] $1^{1^{114}} 3^1 21^{1^{14}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 13 & 2 \\ 13 & 13 & 17 & 0 & 13 & 127 & 21 \\ 2 & 2 & 2 & 0 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[207] $1^{1^{113}} 3^1 21^{1^{12}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 0 & 2 & 7 & 2 \\ 1 & 2 & 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 7 & 2 \\ 8 & 7 & 9 & 0 & 7 & 35 & 11 \\ 2 & 2 & 2 & 0 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[208] $1^{1^{122}} 3^1 21^{1^{32}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 7 & 2 \\ 1 & 2 & 0 & 1 & 11 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 7 & 11 & 0 & 45 & 41 & 13 \\ 7 & 7 & 9 & 0 & 41 & 35 & 11 \\ 2 & 2 & 2 & 0 & 13 & 11 & 3 \end{pmatrix}_{b_0}$$

[209] $1^{1^{14}} 3^1 21^{1^{14}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 0 & 3 & 11 & 2 \\ 1 & 2 & 0 & 1 & 4 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 27 & 5 \\ 12 & 11 & 16 & 0 & 27 & 103 & 19 \\ 2 & 2 & 2 & 0 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[205] $1^{1^{12}} 3^1 21^{1^{12}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 8 & 2 \\ 1 & 0 & 2 & 0 & 3 & 9 & 2 \\ 1 & 2 & 0 & 1 & 4 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 19 & 5 \\ 8 & 9 & 10 & 0 & 19 & 51 & 13 \\ 2 & 2 & 2 & 0 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[210] $1^{1^{122}} 3^1 21^{1^{22}} + [\textcolor{blue}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{so}}_7 \cdot \textcolor{green}{\mathfrak{su}}_2] + \textcolor{brown}{\mathfrak{su}}_2 + \textcolor{brown}{\mathfrak{su}}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 0 & 4 & 7 & 2 \\ 1 & 2 & 0 & 1 & 7 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 25 & 8 \\ 7 & 7 & 9 & 0 & 25 & 35 & 11 \\ 2 & 2 & 2 & 0 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

[211] $1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 10 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 1 & 4 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 23 & 5 \\ 10 & 10 & 13 & 0 & 23 & 74 & 16 \\ 2 & 2 & 2 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[212] $1^{112} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 10 & 2 \\ 9 & 10 & 13 & 0 & 10 & 74 & 16 \\ 2 & 2 & 2 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[213] $1^{111} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 4 & 2 \\ 1 & 2 & 0 & 1 & 4 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 9 & 5 \\ 4 & 4 & 5 & 0 & 9 & 10 & 6 \\ 2 & 2 & 2 & 0 & 5 & 6 & 3 \end{pmatrix}_{b_0}$$

[214] $1^{121} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 13 & 7 \\ 4 & 4 & 5 & 0 & 13 & 10 & 6 \\ 2 & 2 & 2 & 0 & 7 & 6 & 3 \end{pmatrix}_{b_0}$$

[215] $1^{114} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 10 & 2 \\ 11 & 10 & 13 & 0 & 10 & 74 & 16 \\ 2 & 2 & 2 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[216] $1^{121} 3^1 21^{131} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 4 & 2 \\ 1 & 2 & 0 & 1 & 11 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 7 & 11 & 0 & 45 & 23 & 13 \\ 4 & 4 & 5 & 0 & 23 & 10 & 6 \\ 2 & 2 & 2 & 0 & 13 & 6 & 3 \end{pmatrix}_{b_0}$$

[217] $1^{113} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 1 & 7 & 12 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 7 & 0 & 16 & 32 & 8 \\ 9 & 8 & 12 & 0 & 32 & 56 & 14 \\ 2 & 2 & 2 & 0 & 8 & 14 & 3 \end{pmatrix}_{b_0}$$

[218] $1^{113} 3^1 21^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 9 & 2 \\ 9 & 9 & 10 & 0 & 9 & 51 & 13 \\ 2 & 2 & 2 & 0 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[219] $1^{113} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 10 & 2 \\ 10 & 10 & 13 & 0 & 10 & 74 & 16 \\ 2 & 2 & 2 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[220] $1^{122} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 7 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 \\ 1 & 2 & 0 & 1 & 4 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 16 & 5 \\ 7 & 7 & 9 & 0 & 16 & 35 & 11 \\ 2 & 2 & 2 & 0 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

[221] $1^{114} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 0 & 2 & 10 & 2 \\ 1 & 2 & 0 & 1 & 1 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 10 & 2 \\ 11 & 10 & 13 & 0 & 10 & 74 & 16 \\ 2 & 2 & 2 & 0 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[223] $1^{123} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 9 & 2 \\ 1 & 0 & 2 & 0 & 2 & 9 & 2 \\ 1 & 2 & 0 & 1 & 1 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 9 & 2 \\ 9 & 9 & 10 & 0 & 9 & 51 & 13 \\ 2 & 2 & 2 & 0 & 2 & 13 & 3 \end{pmatrix}_{b_0}$$

[222] $1^{122} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 7 & 2 \\ 1 & 0 & 2 & 0 & 5 & 7 & 2 \\ 1 & 2 & 0 & 1 & 5 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 23 & 7 \\ 7 & 7 & 9 & 0 & 23 & 35 & 11 \\ 2 & 2 & 2 & 0 & 7 & 11 & 3 \end{pmatrix}_{b_0}$$

[224] $1^{123} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 1 & 7 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 36 & 8 \\ 10 & 10 & 13 & 0 & 36 & 74 & 16 \\ 2 & 2 & 2 & 0 & 8 & 16 & 3 \end{pmatrix}_{b_0}$$

[225] $1^{114} 3^1 21^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -174, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 0 & 4 & 11 & 2 \\ 1 & 2 & 0 & 1 & 7 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 7 & 0 & 16 & 43 & 8 \\ 12 & 11 & 16 & 0 & 43 & 103 & 19 \\ 2 & 2 & 2 & 0 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$

[226] $1^{113} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -177, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 0 & 3 & 12 & 2 \\ 1 & 2 & 0 & 1 & 4 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 26 & 5 \\ 11 & 12 & 14 & 0 & 26 & 96 & 18 \\ 2 & 2 & 2 & 0 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

[227] $1^{121} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -192, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 4 & 2 \\ 1 & 2 & 0 & 1 & 11 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 9 & 7 & 11 & 0 & 45 & 23 & 13 \\ 4 & 4 & 5 & 0 & 23 & 10 & 6 \\ 2 & 2 & 2 & 0 & 13 & 6 & 3 \end{pmatrix}_{b_0}$$

[228] $1^{123} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -200, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 10 & 2 \\ 1 & 0 & 2 & 0 & 5 & 10 & 2 \\ 1 & 2 & 0 & 1 & 5 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 33 & 7 \\ 10 & 10 & 13 & 0 & 33 & 74 & 16 \\ 2 & 2 & 2 & 0 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[229] $1^{114} 3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 13 & 2 \\ 1 & 0 & 2 & 0 & 3 & 13 & 2 \\ 1 & 2 & 0 & 1 & 4 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 30 & 5 \\ 13 & 13 & 17 & 0 & 30 & 127 & 21 \\ 2 & 2 & 2 & 0 & 5 & 21 & 3 \end{pmatrix}_{b_0}$$

[230] $1^{113} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 12 & 2 \\ 11 & 12 & 14 & 0 & 12 & 96 & 18 \\ 2 & 2 & 2 & 0 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[231] $1^{114} 3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -212, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 0 & 4 & 10 & 2 \\ 1 & 2 & 0 & 1 & 7 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 36 & 8 \\ 11 & 10 & 13 & 0 & 36 & 74 & 16 \\ 2 & 2 & 2 & 0 & 8 & 16 & 3 \end{pmatrix}_{b_0}$$

[232] $1^{113} 3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -215, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 13 & 2 \\ 12 & 13 & 17 & 0 & 13 & 127 & 21 \\ 2 & 2 & 2 & 0 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[233] $1^{123} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 10 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 1 & 4 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 23 & 5 \\ 10 & 10 & 13 & 0 & 23 & 74 & 16 \\ 2 & 2 & 2 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[236] $1^{121} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 4 & 2 \\ 1 & 2 & 0 & 1 & 8 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 18 & 10 \\ 4 & 4 & 5 & 0 & 18 & 10 & 6 \\ 2 & 2 & 2 & 0 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

[234] $1^{114} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 12 & 2 \\ 12 & 12 & 14 & 0 & 12 & 96 & 18 \\ 2 & 2 & 2 & 0 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[237] $1^{121} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 4 & 2 \\ 1 & 2 & 0 & 1 & 8 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 6 & 8 & 0 & 26 & 18 & 10 \\ 4 & 4 & 5 & 0 & 18 & 10 & 6 \\ 2 & 2 & 2 & 0 & 10 & 6 & 3 \end{pmatrix}_{b_0}$$

[235] $1^{141} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 0 & 10 & 4 & 2 \\ 1 & 2 & 0 & 1 & 10 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 10 & 0 & 58 & 26 & 14 \\ 4 & 4 & 5 & 0 & 26 & 10 & 6 \\ 2 & 2 & 2 & 0 & 14 & 6 & 3 \end{pmatrix}_{b_0}$$

[238] $1^{131} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 0 & 8 & 4 & 2 \\ 1 & 2 & 0 & 1 & 9 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 9 & 8 & 9 & 0 & 40 & 22 & 12 \\ 4 & 4 & 5 & 0 & 22 & 10 & 6 \\ 2 & 2 & 2 & 0 & 12 & 6 & 3 \end{pmatrix}_{b_0}$$

[239] $1^{121} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 0 & 7 & 4 & 2 \\ 1 & 2 & 0 & 1 & 11 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 11 & 0 & 45 & 23 & 13 \\ 4 & 4 & 5 & 0 & 23 & 10 & 6 \\ 2 & 2 & 2 & 0 & 13 & 6 & 3 \end{pmatrix}_{b_0}$$

[240] $1^{114} 3^1 21^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 13 & 2 \\ 1 & 0 & 2 & 0 & 2 & 13 & 2 \\ 1 & 2 & 0 & 1 & 1 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 13 & 2 \\ 13 & 13 & 17 & 0 & 13 & 127 & 21 \\ 2 & 2 & 2 & 0 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

[241] $1^{113} 3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -227, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 11 & 2 \\ 1 & 0 & 2 & 0 & 3 & 11 & 2 \\ 1 & 2 & 0 & 1 & 4 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 27 & 5 \\ 11 & 11 & 16 & 0 & 27 & 103 & 19 \\ 2 & 2 & 2 & 0 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[242] $1^{123} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -235, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 9 & 2 \\ 1 & 0 & 2 & 0 & 3 & 9 & 2 \\ 1 & 2 & 0 & 1 & 4 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 19 & 5 \\ 9 & 9 & 10 & 0 & 19 & 51 & 13 \\ 2 & 2 & 2 & 0 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[243] $1^{114} 3^1 21^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -240, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 0 & 2 & 12 & 2 \\ 1 & 2 & 0 & 1 & 1 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 2 & 1 & 0 & 0 & 12 & 2 \\ 12 & 12 & 14 & 0 & 12 & 96 & 18 \\ 2 & 2 & 2 & 0 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[245] $1^{132} 3^1 21^{142} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 7 & 2 \\ 1 & 2 & 0 & 1 & 15 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 15 & 0 & 88 & 57 & 18 \\ 7 & 7 & 9 & 0 & 57 & 35 & 11 \\ 2 & 2 & 2 & 0 & 18 & 11 & 3 \end{pmatrix}_{b_0}$$

[244] $1^{124} 3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 13 & 2 \\ 1 & 0 & 2 & 0 & 5 & 13 & 2 \\ 1 & 2 & 0 & 1 & 5 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 43 & 7 \\ 13 & 13 & 17 & 0 & 43 & 127 & 21 \\ 2 & 2 & 2 & 0 & 7 & 21 & 3 \end{pmatrix}_{b_0}$$

[247] $1^{132} 3^1 21^{132} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 7 & 2 \\ 1 & 2 & 0 & 1 & 11 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 7 & 11 & 0 & 45 & 41 & 13 \\ 7 & 7 & 9 & 0 & 41 & 35 & 11 \\ 2 & 2 & 2 & 0 & 13 & 11 & 3 \end{pmatrix}_{b_0}$$

[246] $1^{124} 3^1 21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 13 & 2 \\ 1 & 0 & 2 & 0 & 4 & 13 & 2 \\ 1 & 2 & 0 & 1 & 7 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 47 & 8 \\ 13 & 13 & 17 & 0 & 47 & 127 & 21 \\ 2 & 2 & 2 & 0 & 8 & 21 & 3 \end{pmatrix}_{b_0}$$

[248] $1^{112} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -282, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 7 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 \\ 1 & 2 & 0 & 1 & 4 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 16 & 5 \\ 7 & 7 & 9 & 0 & 16 & 35 & 11 \\ 2 & 2 & 2 & 0 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

[249] $1^{132} 3^1 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 7 & 2 \\ 1 & 2 & 0 & 1 & 8 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 7 & 6 & 8 & 0 & 26 & 32 & 10 \\ 7 & 7 & 9 & 0 & 32 & 35 & 11 \\ 2 & 2 & 2 & 0 & 10 & 11 & 3 \end{pmatrix}_{b_0}$$

[251] $1^{131} 3^1 21^{141} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 0 & 10 & 4 & 2 \\ 1 & 2 & 0 & 1 & 15 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 10 & 15 & 0 & 88 & 32 & 18 \\ 4 & 4 & 5 & 0 & 32 & 10 & 6 \\ 2 & 2 & 2 & 0 & 18 & 6 & 3 \end{pmatrix}_{b_0}$$

[250] $1^{131} 3^1 21^{121} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 4 & 2 \\ 1 & 0 & 2 & 0 & 8 & 4 & 2 \\ 1 & 2 & 0 & 1 & 9 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 8 & 9 & 0 & 40 & 22 & 12 \\ 4 & 4 & 5 & 0 & 22 & 10 & 6 \\ 2 & 2 & 2 & 0 & 12 & 6 & 3 \end{pmatrix}_{b_0}$$

[252] $1^{123} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 8 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 \\ 1 & 2 & 0 & 1 & 4 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 16 & 5 \\ 8 & 7 & 9 & 0 & 16 & 35 & 11 \\ 2 & 2 & 2 & 0 & 5 & 11 & 3 \end{pmatrix}_{b_0}$$

[253] $1^{113} 3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -314, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 12 & 2 \\ 1 & 0 & 2 & 0 & 3 & 13 & 2 \\ 1 & 2 & 0 & 1 & 4 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 30 & 5 \\ 12 & 13 & 17 & 0 & 30 & 127 & 21 \\ 2 & 2 & 2 & 0 & 5 & 21 & 3 \end{pmatrix}_{b_0}$$

[254] $1^{124} 3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -315, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 13 & 2 \\ 1 & 0 & 2 & 0 & 3 & 13 & 2 \\ 1 & 2 & 0 & 1 & 4 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 30 & 5 \\ 13 & 13 & 17 & 0 & 30 & 127 & 21 \\ 2 & 2 & 2 & 0 & 5 & 21 & 3 \end{pmatrix}_{b_0}$$

[255] $1^{123} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -337, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 0 & 5 & 10 & 2 \\ 1 & 2 & 0 & 1 & 5 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 33 & 7 \\ 11 & 10 & 13 & 0 & 33 & 74 & 16 \\ 2 & 2 & 2 & 0 & 7 & 16 & 3 \end{pmatrix}_{b_0}$$

[256] $1^{132} 3^1 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 7 & 2 \\ 1 & 0 & 2 & 0 & 8 & 7 & 2 \\ 1 & 2 & 0 & 1 & 9 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 8 & 8 & 9 & 0 & 40 & 39 & 12 \\ 7 & 7 & 9 & 0 & 39 & 35 & 11 \\ 2 & 2 & 2 & 0 & 12 & 11 & 3 \end{pmatrix}_{b_0}$$

[257] $1^{113} 3^1 21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 0 & 4 & 11 & 2 \\ 1 & 2 & 0 & 1 & 7 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 4 & 7 & 0 & 16 & 43 & 8 \\ 11 & 11 & 16 & 0 & 43 & 103 & 19 \\ 2 & 2 & 2 & 0 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$

[258] $1^{113} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 9 & 2 \\ 1 & 0 & 2 & 0 & 3 & 9 & 2 \\ 1 & 2 & 0 & 1 & 4 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 19 & 5 \\ 9 & 9 & 10 & 0 & 19 & 51 & 13 \\ 2 & 2 & 2 & 0 & 5 & 13 & 3 \end{pmatrix}_{b_0}$$

[259] $1^{113} 3^1 21^{114} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -350, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 0 & 3 & 11 & 2 \\ 1 & 2 & 0 & 1 & 4 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 27 & 5 \\ 12 & 11 & 16 & 0 & 27 & 103 & 19 \\ 2 & 2 & 2 & 0 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[260] $1^{123} 3^1 21^{124} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -363, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 11 & 2 \\ 1 & 0 & 2 & 0 & 4 & 11 & 2 \\ 1 & 2 & 0 & 1 & 7 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 43 & 8 \\ 11 & 11 & 16 & 0 & 43 & 103 & 19 \\ 2 & 2 & 2 & 0 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$

[261] $1^{124} 3^1 21^{113} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 12 & 2 \\ 1 & 0 & 2 & 0 & 3 & 12 & 2 \\ 1 & 2 & 0 & 1 & 4 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 26 & 5 \\ 12 & 12 & 14 & 0 & 26 & 96 & 18 \\ 2 & 2 & 2 & 0 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

[262] $1^{122} 3^1 21^{122} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -378, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 7 & 2 \\ 1 & 2 & 0 & 1 & 8 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 32 & 10 \\ 7 & 7 & 9 & 0 & 32 & 35 & 11 \\ 2 & 2 & 2 & 0 & 10 & 11 & 3 \end{pmatrix}_{b_0}$$

[263] $1^{123} 3^1 21^{123} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 0 & 4 & 8 & 2 \\ 1 & 2 & 0 & 1 & 7 & 12 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 32 & 8 \\ 9 & 8 & 12 & 0 & 32 & 56 & 14 \\ 2 & 2 & 2 & 0 & 8 & 14 & 3 \end{pmatrix}_{b_0}$$

[264] $1^{131} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -384, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 2 & 0 & 9 & 4 & 2 \\ 1 & 2 & 0 & 1 & 12 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 9 & 9 & 12 & 0 & 61 & 27 & 15 \\ 4 & 4 & 5 & 0 & 27 & 10 & 6 \\ 2 & 2 & 2 & 0 & 15 & 6 & 3 \end{pmatrix}_{b_0}$$

[265] $1^{123} 3^1 21^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -394, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 9 & 2 \\ 1 & 0 & 2 & 0 & 5 & 9 & 2 \\ 1 & 2 & 0 & 1 & 5 & 10 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 28 & 7 \\ 9 & 9 & 10 & 0 & 28 & 51 & 13 \\ 2 & 2 & 2 & 0 & 7 & 13 & 3 \end{pmatrix}_{b_0}$$

[266] $1^{141} 3^1 21^{131} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 4 & 2 \\ 1 & 0 & 2 & 0 & 9 & 4 & 2 \\ 1 & 2 & 0 & 1 & 12 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 9 & 12 & 0 & 61 & 27 & 15 \\ 4 & 4 & 5 & 0 & 27 & 10 & 6 \\ 2 & 2 & 2 & 0 & 15 & 6 & 3 \end{pmatrix}_{b_0}$$

[267] $1^{131} 3^1 21^{141} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 4 & 2 \\ 1 & 0 & 2 & 0 & 10 & 4 & 2 \\ 1 & 2 & 0 & 1 & 15 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 11 & 10 & 15 & 0 & 88 & 32 & 18 \\ 4 & 4 & 5 & 0 & 32 & 10 & 6 \\ 2 & 2 & 2 & 0 & 18 & 6 & 3 \end{pmatrix}_{b_0}$$

[268] $1^{142} 3^1 21^{142} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -427, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 7 & 2 \\ 1 & 2 & 0 & 1 & 15 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 11 & 10 & 15 & 0 & 88 & 57 & 18 \\ 7 & 7 & 9 & 0 & 57 & 35 & 11 \\ 2 & 2 & 2 & 0 & 18 & 11 & 3 \end{pmatrix}_{b_0}$$

[269] $1^{124} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -457, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 11 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 1 & 4 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 4 & 3 & 4 & 0 & 5 & 23 & 5 \\ 11 & 10 & 13 & 0 & 23 & 74 & 16 \\ 2 & 2 & 2 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[270] $1^{142} 3^1 21^{132} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -483, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 7 & 2 \\ 1 & 0 & 2 & 0 & 9 & 7 & 2 \\ 1 & 2 & 0 & 1 & 12 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 10 & 9 & 12 & 0 & 61 & 48 & 15 \\ 7 & 7 & 9 & 0 & 48 & 35 & 11 \\ 2 & 2 & 2 & 0 & 15 & 11 & 3 \end{pmatrix}_{b_0}$$

[271] $1^{141} 3^1 21^{131} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 4 & 2 \\ 1 & 0 & 2 & 0 & 11 & 4 & 2 \\ 1 & 2 & 0 & 1 & 13 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 11 & 11 & 13 & 0 & 81 & 31 & 17 \\ 4 & 4 & 5 & 0 & 31 & 10 & 6 \\ 2 & 2 & 2 & 0 & 17 & 6 & 3 \end{pmatrix}_{b_0}$$

[272] $1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -520, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 10 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 1 & 4 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 23 & 5 \\ 10 & 10 & 13 & 0 & 23 & 74 & 16 \\ 2 & 2 & 2 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[273] $1^{124} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -532, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 11 & 2 \\ 1 & 0 & 2 & 0 & 6 & 10 & 2 \\ 1 & 2 & 0 & 1 & 8 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 46 & 10 \\ 11 & 10 & 13 & 0 & 46 & 74 & 16 \\ 2 & 2 & 2 & 0 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[274] $1^{132} 3^1 21^{132} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -538, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.8$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 7 & 2 \\ 1 & 0 & 2 & 0 & 9 & 7 & 2 \\ 1 & 2 & 0 & 1 & 12 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 9 & 9 & 12 & 0 & 61 & 48 & 15 \\ 7 & 7 & 9 & 0 & 48 & 35 & 11 \\ 2 & 2 & 2 & 0 & 15 & 11 & 3 \end{pmatrix}_{b_0}$$

[275] $1^{113} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -551, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 12 & 2 \\ 1 & 0 & 2 & 0 & 4 & 11 & 2 \\ 1 & 2 & 0 & 1 & 7 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 4 & 7 & 0 & 16 & 43 & 8 \\ 12 & 11 & 16 & 0 & 43 & 103 & 19 \\ 2 & 2 & 2 & 0 & 8 & 19 & 3 \end{pmatrix}_{b_0}$$

[276] $1^{113} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -553, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 11 & 2 \\ 1 & 0 & 2 & 0 & 3 & 10 & 2 \\ 1 & 2 & 0 & 1 & 4 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 23 & 5 \\ 11 & 10 & 13 & 0 & 23 & 74 & 16 \\ 2 & 2 & 2 & 0 & 5 & 16 & 3 \end{pmatrix}_{b_0}$$

[277] $1^{124} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 12 & 2 \\ 1 & 0 & 2 & 0 & 6 & 12 & 2 \\ 1 & 2 & 0 & 1 & 8 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 52 & 10 \\ 12 & 12 & 14 & 0 & 52 & 96 & 18 \\ 2 & 2 & 2 & 0 & 10 & 18 & 3 \end{pmatrix}_{b_0}$$

[278] $1^{124} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -568, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 12 & 2 \\ 1 & 0 & 2 & 0 & 5 & 12 & 2 \\ 1 & 2 & 0 & 1 & 5 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 5 & 5 & 5 & 0 & 13 & 38 & 7 \\ 12 & 12 & 14 & 0 & 38 & 96 & 18 \\ 2 & 2 & 2 & 0 & 7 & 18 & 3 \end{pmatrix}_{b_0}$$

[280] $1^{142} 3^1 21^{132} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -578, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 7 & 2 \\ 1 & 0 & 2 & 0 & 11 & 7 & 2 \\ 1 & 2 & 0 & 1 & 13 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 11 & 11 & 13 & 0 & 81 & 55 & 17 \\ 7 & 7 & 9 & 0 & 55 & 35 & 11 \\ 2 & 2 & 2 & 0 & 17 & 11 & 3 \end{pmatrix}_{b_0}$$

[279] $1^{113} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -578, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 11 & 2 \\ 1 & 0 & 2 & 0 & 3 & 11 & 2 \\ 1 & 2 & 0 & 1 & 4 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 27 & 5 \\ 11 & 11 & 16 & 0 & 27 & 103 & 19 \\ 2 & 2 & 2 & 0 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[281] $1^{141} 3^1 21^{141} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 4 & 2 \\ 1 & 0 & 2 & 0 & 12 & 4 & 2 \\ 1 & 2 & 0 & 1 & 16 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 12 & 12 & 16 & 0 & 110 & 36 & 20 \\ 4 & 4 & 5 & 0 & 36 & 10 & 6 \\ 2 & 2 & 2 & 0 & 20 & 6 & 3 \end{pmatrix}_{b_0}$$

[282] $1^{123} 3^1 21^{123} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -616, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 10 & 2 \\ 1 & 0 & 2 & 0 & 6 & 10 & 2 \\ 1 & 2 & 0 & 1 & 8 & 13 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 46 & 10 \\ 10 & 10 & 13 & 0 & 46 & 74 & 16 \\ 2 & 2 & 2 & 0 & 10 & 16 & 3 \end{pmatrix}_{b_0}$$

[283] $1^{114} 3^1 21^{113} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -632, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 12 & 2 \\ 1 & 0 & 2 & 0 & 3 & 12 & 2 \\ 1 & 2 & 0 & 1 & 4 & 14 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 26 & 5 \\ 12 & 12 & 14 & 0 & 26 & 96 & 18 \\ 2 & 2 & 2 & 0 & 5 & 18 & 3 \end{pmatrix}_{b_0}$$

[284] $1^{113} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -687, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 12 & 2 \\ 1 & 0 & 2 & 0 & 3 & 11 & 2 \\ 1 & 2 & 0 & 1 & 4 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 27 & 5 \\ 12 & 11 & 16 & 0 & 27 & 103 & 19 \\ 2 & 2 & 2 & 0 & 5 & 19 & 3 \end{pmatrix}_{b_0}$$

[285] $1^{123} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -714, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 12 & 2 \\ 1 & 0 & 2 & 0 & 6 & 11 & 2 \\ 1 & 2 & 0 & 1 & 8 & 16 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 54 & 10 \\ 12 & 11 & 16 & 0 & 54 & 103 & 19 \\ 2 & 2 & 2 & 0 & 10 & 19 & 3 \end{pmatrix}_{b_0}$$

[286] $1^{142} 3^1 21^{142} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -762, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 7 & 2 \\ 1 & 0 & 2 & 0 & 12 & 7 & 2 \\ 1 & 2 & 0 & 1 & 16 & 9 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 12 & 12 & 16 & 0 & 110 & 64 & 20 \\ 7 & 7 & 9 & 0 & 64 & 35 & 11 \\ 2 & 2 & 2 & 0 & 20 & 11 & 3 \end{pmatrix}_{b_0}$$

[287] $1^{114} 3^1 21^{114} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -842, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 13 & 2 \\ 1 & 0 & 2 & 0 & 3 & 13 & 2 \\ 1 & 2 & 0 & 1 & 4 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 3 & 3 & 4 & 0 & 5 & 30 & 5 \\ 13 & 13 & 17 & 0 & 30 & 127 & 21 \\ 2 & 2 & 2 & 0 & 5 & 21 & 3 \end{pmatrix}_{b_0}$$

[288] $1^{124} 3^1 21^{124} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2$ $T = 9, T_{\max} = 10, \Delta = -31, \det = -938, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.5$

$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 13 & 2 \\ 1 & 0 & 2 & 0 & 6 & 13 & 2 \\ 1 & 2 & 0 & 1 & 8 & 17 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 \\ 6 & 6 & 8 & 0 & 26 & 60 & 10 \\ 13 & 13 & 17 & 0 & 60 & 127 & 21 \\ 2 & 2 & 2 & 0 & 10 & 21 & 3 \end{pmatrix}_{b_0}$$

5.129 NHC(-2)(-3)(-2) + su₂ + su_{2n3mix} + su₂

$T = 10$ [0] $1^{121} 3^1 21^{1121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 3 & 2 \\ 1 & 0 & 2 & 0 & 1 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 3 & 9 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 1 & 3 & 0 & 1 & 9 & 4 & 3 \\ 7 & 7 & 9 & 0 & 9 & 35 & 16 & 11 \\ 3 & 3 & 4 & 0 & 4 & 16 & 5 & 5 \\ 2 & 2 & 2 & 0 & 3 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

[8] $1^{112} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) =$

$(1, 10, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 7 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 7 & 2 & 2 \\ 1 & 2 & 0 & 1 & 4 & 9 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 16 & 3 & 5 \\ 7 & 7 & 9 & 0 & 16 & 35 & 7 & 11 \\ 1 & 2 & 1 & 0 & 3 & 7 & 0 & 2 \\ 2 & 2 & 2 & 0 & 5 & 11 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[1] \overset{111}{1} \overset{1112}{1} \overset{111}{3} \overset{111}{21} + [\textcolor{red}{5}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{5}\textcolor{red}{0}_7 \cdot \textcolor{green}{5}\textcolor{blue}{u}_2] + \textcolor{brown}{5}\textcolor{blue}{u}_2 + \textcolor{red}{5}\textcolor{blue}{u}_2 + \textcolor{blue}{5}\textcolor{blue}{u}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -1, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 4 & 5 & 2 \\ 4 & 4 & 5 & 0 & 4 & 10 & 13 & 6 \\ 4 & 5 & 5 & 0 & 5 & 13 & 13 & 7 \\ 2 & 2 & 2 & 0 & 2 & 6 & 7 & 3 \end{pmatrix} b_0$$

$$[2] \quad 1^{122} 3^1 21^{121} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -1, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 7 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 9 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 7 & 3 & 2 \\ 7 & 7 & 9 & 0 & 7 & 35 & 16 & 11 \\ 4 & 3 & 4 & 0 & 3 & 16 & 5 & 5 \\ 2 & 2 & 2 & 0 & 2 & 11 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 3^{21} 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -1, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 5 & 4 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 4 & 3 & 2 \\ 4 & 4 & 5 & 0 & 4 & 10 & 9 & 6 \\ 2 & 3 & 4 & 0 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 0 & 2 & 6 & 5 & 3 \end{pmatrix} b_0$$

[4] $1^{1112} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 6 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 & 5 & 2 \\ 1 & 2 & 0 & 1 & 1 & 5 & 5 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 4 & 5 & 2 \\ 4 & 4 & 5 & 0 & 4 & 10 & 13 & 6 \\ 6 & 5 & 5 & 0 & 5 & 13 & 13 & 7 \\ 2 & 2 & 2 & 0 & 2 & 6 & 7 & 3 \end{pmatrix} b_0$$

[5] $1^{1112} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2$ $T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.0$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 5 & 4 & 0 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 4 & 3 & 2 \\ 4 & 4 & 5 & 0 & 4 & 10 & 9 & 6 \\ 4 & 3 & 4 & 0 & 3 & 9 & 5 & 5 \\ 2 & 2 & 2 & 0 & 2 & 6 & 5 & 3 \end{pmatrix} b_0$$

$$[6] \quad 1^{1111} 3^1 21^{1112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -1, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 4 & 5 & 2 \\ 1 & 0 & 2 & 0 & 3 & 4 & 4 & 2 \\ 1 & 2 & 0 & 1 & 4 & 5 & 7 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 & 0 & 0 \\ 2 & 3 & 4 & 0 & 5 & 9 & 11 & 5 \\ 4 & 4 & 5 & 0 & 9 & 10 & 14 & 6 \\ 5 & 4 & 7 & 0 & 11 & 14 & 16 & 8 \\ 2 & 2 & 2 & 0 & 5 & 6 & 8 & 3 \end{pmatrix} b_0$$

[7] $1^{1111} 3^1 21^{1112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 10, \quad T_{\max} = 10, \quad \Delta = -1, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 7.5$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

5.130 NHC(-2)(-3)(-2) + su₂ + su_{2n3mix} + su₂ + su₂

$$\begin{aligned}
T=10 \quad [1] \quad & 1^{1112} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T=10, T_{\max}=10, \Delta=0, \det=0, (n_+, n_-, n_0) = \\
& (1, 10, 1), c_{\text{ext}}=8.5 \\
[2] \quad & 1^{1121} 3^1 21^{121} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T=10, T_{\max}=10, \Delta=0, \det=0, (n_+, n_-, n_0) = \\
& (1, 10, 1), c_{\text{ext}}=8.0 \\
[3] \quad & 1^{1112} 3^1 21^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T=10, T_{\max}=10, \Delta=0, \det=0, (n_+, n_-, n_0) = \\
& (1, 10, 1), c_{\text{ext}}=8.5 \\
[0] \quad & 1^{1111} 3^1 21^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T=10, T_{\max}=10, \Delta=0, \det=0, (n_+, n_-, n_0) = \\
& (1, 10, 1), c_{\text{ext}}=8.0
\end{aligned}$$

5.131 NHC(-2)(-3)(-2) + su₂ + su_{2n3mix} + su₂ → (-3)

$$\begin{aligned}
T=9 \quad [0] \quad & 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T=9, T_{\max}=10, \Delta=-31, \det=-32, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}}=7.0 \\
& T_{\min}=10, T_{\max}=10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 7 & 9 & 4 & 5 \\ 7 & 8 & 9 & 11 & 5 & 6 \\ 7 & 9 & 8 & 10 & 5 & 6 \\ 9 & 11 & 10 & 11 & 6 & 7 \\ 4 & 5 & 5 & 6 & 1 & 3 \\ 5 & 6 & 6 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
[1] \quad & 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T=9, T_{\max}=10, \Delta=-31, \det=-32, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}}=7.0 \\
& T_{\min}=10, T_{\max}=10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 14 & 9 & 4 & 5 \\ 7 & 8 & 15 & 11 & 5 & 6 \\ 14 & 15 & 22 & 17 & 8 & 10 \\ 9 & 11 & 17 & 11 & 6 & 7 \\ 4 & 5 & 8 & 6 & 1 & 3 \\ 5 & 6 & 10 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
[2] \quad & 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T=9, T_{\max}=10, \Delta=-31, \det=-32, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}}=7.0 \\
& T_{\min}=10, T_{\max}=10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 9 & 4 & 5 \\ 7 & 8 & 15 & 11 & 5 & 6 \\ 11 & 15 & 22 & 17 & 8 & 10 \\ 9 & 11 & 17 & 11 & 6 & 7 \\ 4 & 5 & 8 & 6 & 1 & 3 \\ 5 & 6 & 10 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
[5] \quad & 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T=9, T_{\max}=10, \Delta=-31, \det=-64, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}}=7.0 \\
& T_{\min}=10, T_{\max}=10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 6 & 9 & 4 & 5 \\ 7 & 8 & 6 & 11 & 5 & 6 \\ 6 & 6 & 2 & 7 & 3 & 4 \\ 9 & 11 & 7 & 11 & 6 & 7 \\ 4 & 5 & 3 & 6 & 1 & 3 \\ 5 & 6 & 4 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
[4] \quad & 21^{11} 3^{11} 1^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T=9, T_{\max}=10, \Delta=-31, \det=-64, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}}=7.0 \\
& T_{\min}=10, T_{\max}=10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 5 & 5 & 3 & 12 & 3 & 4 \\ 5 & 4 & 6 & 4 & 12 & 3 & 4 \\ 5 & 6 & 4 & 4 & 12 & 3 & 4 \\ 3 & 4 & 4 & 1 & 8 & 2 & 3 \\ 12 & 12 & 12 & 8 & 25 & 7 & 9 \\ 3 & 3 & 3 & 2 & 7 & 0 & 2 \\ 4 & 4 & 4 & 3 & 9 & 2 & 3 \end{pmatrix}_{b_0} \\
[6] \quad & 1^{121} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T=9, T_{\max}=10, \Delta=-31, \det=-64, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}}=7.5 \\
& T_{\min}=10, T_{\max}=10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 11 & 9 & 4 & 5 \\ 7 & 8 & 12 & 11 & 5 & 6 \\ 11 & 12 & 14 & 14 & 6 & 8 \\ 9 & 11 & 14 & 11 & 6 & 7 \\ 4 & 5 & 6 & 6 & 1 & 3 \\ 5 & 6 & 8 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
[3] \quad & 1^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T=9, T_{\max}=10, \Delta=-31, \det=-64, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}}=7.0
\end{aligned}$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 9 & 4 & 5 \\ 7 & 8 & 6 & 11 & 5 & 6 \\ 4 & 6 & 2 & 7 & 3 & 4 \\ 9 & 11 & 7 & 11 & 6 & 7 \\ 4 & 5 & 3 & 6 & 1 & 3 \\ 5 & 6 & 4 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{142} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 16 & 14 & 4 & 5 \\ 7 & 8 & 18 & 17 & 5 & 6 \\ 16 & 18 & 34 & 33 & 9 & 12 \\ 14 & 17 & 33 & 29 & 9 & 11 \\ 4 & 5 & 9 & 9 & 1 & 3 \\ 5 & 6 & 12 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{112} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 14 & 4 & 5 \\ 7 & 8 & 6 & 17 & 5 & 6 \\ 4 & 6 & 2 & 11 & 3 & 4 \\ 14 & 17 & 11 & 29 & 9 & 11 \\ 4 & 5 & 3 & 9 & 1 & 3 \\ 5 & 6 & 4 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{122} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 14 & 4 & 5 \\ 7 & 8 & 6 & 17 & 5 & 6 \\ 6 & 6 & 2 & 11 & 3 & 4 \\ 14 & 17 & 11 & 29 & 9 & 11 \\ 4 & 5 & 3 & 9 & 1 & 3 \\ 5 & 6 & 4 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{122} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 11 & 14 & 4 & 5 \\ 7 & 8 & 12 & 17 & 5 & 6 \\ 11 & 12 & 14 & 22 & 6 & 8 \\ 14 & 17 & 22 & 29 & 9 & 11 \\ 4 & 5 & 6 & 9 & 1 & 3 \\ 5 & 6 & 8 & 11 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{131} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 15 & 9 & 4 & 5 \\ 7 & 8 & 18 & 11 & 5 & 6 \\ 15 & 18 & 34 & 21 & 9 & 12 \\ 9 & 11 & 21 & 11 & 6 & 7 \\ 4 & 5 & 9 & 6 & 1 & 3 \\ 5 & 6 & 12 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 9 & 4 & 5 \\ 7 & 8 & 6 & 11 & 5 & 6 \\ 5 & 6 & 2 & 7 & 3 & 4 \\ 9 & 11 & 7 & 11 & 6 & 7 \\ 4 & 5 & 3 & 6 & 1 & 3 \\ 5 & 6 & 4 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{121} 3^{11} 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 10 & 9 & 4 & 5 \\ 7 & 8 & 12 & 11 & 5 & 6 \\ 10 & 12 & 14 & 14 & 6 & 8 \\ 9 & 11 & 14 & 11 & 6 & 7 \\ 4 & 5 & 6 & 6 & 1 & 3 \\ 5 & 6 & 8 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{123} 3^{11} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 26 & 4 & 5 \\ 7 & 8 & 6 & 32 & 5 & 6 \\ 6 & 6 & 2 & 20 & 3 & 4 \\ 26 & 32 & 20 & 105 & 17 & 21 \\ 4 & 5 & 3 & 17 & 1 & 3 \\ 5 & 6 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[15] \quad & 1^{113} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -144, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 19 & 4 & 5 \\ 7 & 8 & 6 & 23 & 5 & 6 \\ 4 & 6 & 2 & 15 & 3 & 4 \\ 19 & 23 & 15 & 55 & 12 & 15 \\ 4 & 5 & 3 & 12 & 1 & 3 \\ 5 & 6 & 4 & 15 & 3 & 4 \end{pmatrix}_{b_0} \\
[16] \quad & 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -160, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 13 & 9 & 4 & 5 \\ 7 & 8 & 15 & 11 & 5 & 6 \\ 13 & 15 & 22 & 17 & 8 & 10 \\ 9 & 11 & 17 & 11 & 6 & 7 \\ 4 & 5 & 8 & 6 & 1 & 3 \\ 5 & 6 & 10 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
[17] \quad & 1^{141} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -160, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 19 & 9 & 4 & 5 \\ 7 & 8 & 21 & 11 & 5 & 6 \\ 19 & 21 & 44 & 24 & 11 & 14 \\ 9 & 11 & 24 & 11 & 6 & 7 \\ 4 & 5 & 11 & 6 & 1 & 3 \\ 5 & 6 & 14 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
[18] \quad & 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -160, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 12 & 9 & 4 & 5 \\ 7 & 8 & 15 & 11 & 5 & 6 \\ 12 & 15 & 22 & 17 & 8 & 10 \\ 9 & 11 & 17 & 11 & 6 & 7 \\ 4 & 5 & 8 & 6 & 1 & 3 \\ 5 & 6 & 10 & 7 & 3 & 4 \end{pmatrix}_{b_0} \\
[19] \quad & 1^{12} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -187, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 21 & 4 & 5 \\ 7 & 8 & 6 & 26 & 5 & 6 \\ 4 & 6 & 2 & 16 & 3 & 4 \\ 21 & 26 & 16 & 69 & 14 & 17 \\ 4 & 5 & 3 & 14 & 1 & 3 \\ 5 & 6 & 4 & 17 & 3 & 4 \end{pmatrix}_{b_0} \\
[20] \quad & 1^{123} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -189, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 6 & 22 & 4 & 5 \\ 7 & 8 & 6 & 26 & 5 & 6 \\ 6 & 6 & 2 & 16 & 3 & 4 \\ 22 & 26 & 16 & 69 & 14 & 17 \\ 4 & 5 & 3 & 14 & 1 & 3 \\ 5 & 6 & 4 & 17 & 3 & 4 \end{pmatrix}_{b_0} \\
[21] \quad & 1^{122} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -200, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 10 & 14 & 4 & 5 \\ 7 & 8 & 12 & 17 & 5 & 6 \\ 10 & 12 & 14 & 22 & 6 & 8 \\ 14 & 17 & 22 & 29 & 9 & 11 \\ 4 & 5 & 6 & 9 & 1 & 3 \\ 5 & 6 & 8 & 11 & 3 & 4 \end{pmatrix}_{b_0} \\
[22] \quad & 1^{142} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -200, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 20 & 14 & 4 & 5 \\ 7 & 8 & 24 & 17 & 5 & 6 \\ 20 & 24 & 62 & 44 & 12 & 16 \\ 14 & 17 & 44 & 29 & 9 & 11 \\ 4 & 5 & 12 & 9 & 1 & 3 \\ 5 & 6 & 16 & 11 & 3 & 4 \end{pmatrix}_{b_0} \\
[23] \quad & 21^{113} 3^{11} 1^{112} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -224, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 4 & 5 & 5 & 3 & 15 & 3 & 4 \\ 5 & 4 & 6 & 4 & 16 & 3 & 4 \\ 5 & 6 & 4 & 4 & 16 & 3 & 4 \\ 3 & 4 & 4 & 1 & 11 & 2 & 3 \\ 15 & 16 & 16 & 11 & 44 & 9 & 12 \\ 3 & 3 & 3 & 2 & 9 & 0 & 2 \\ 4 & 4 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{114} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -243, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 31 & 4 & 5 \\ 7 & 8 & 6 & 38 & 5 & 6 \\ 6 & 6 & 2 & 24 & 3 & 4 \\ 31 & 38 & 24 & 149 & 20 & 25 \\ 4 & 5 & 3 & 20 & 1 & 3 \\ 5 & 6 & 4 & 25 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 25 & 4 & 5 \\ 7 & 8 & 6 & 32 & 5 & 6 \\ 4 & 6 & 2 & 20 & 3 & 4 \\ 25 & 32 & 20 & 105 & 17 & 21 \\ 4 & 5 & 3 & 17 & 1 & 3 \\ 5 & 6 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{114} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10.00 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 27 & 4 & 5 \\ 7 & 8 & 6 & 32 & 5 & 6 \\ 4 & 6 & 2 & 20 & 3 & 4 \\ 27 & 32 & 20 & 105 & 17 & 21 \\ 4 & 5 & 3 & 17 & 1 & 3 \\ 5 & 6 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[27] \quad 21^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 8 & 2 & 3 \\ 4 & 4 & 6 & 4 & 12 & 3 & 4 \\ 4 & 6 & 4 & 4 & 12 & 3 & 4 \\ 3 & 4 & 4 & 1 & 8 & 2 & 3 \\ 8 & 12 & 12 & 8 & 25 & 7 & 9 \\ 2 & 3 & 3 & 2 & 7 & 0 & 2 \\ 3 & 4 & 4 & 3 & 9 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{113} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -283, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 26 & 4 & 5 \\ 7 & 8 & 6 & 32 & 5 & 6 \\ 4 & 6 & 2 & 20 & 3 & 4 \\ 26 & 32 & 20 & 105 & 17 & 21 \\ 4 & 5 & 3 & 17 & 1 & 3 \\ 5 & 6 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{113} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 19 & 4 & 5 \\ 7 & 8 & 6 & 23 & 5 & 6 \\ 5 & 6 & 2 & 15 & 3 & 4 \\ 19 & 23 & 15 & 55 & 12 & 15 \\ 4 & 5 & 3 & 12 & 1 & 3 \\ 5 & 6 & 4 & 15 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 17 & 9 & 4 & 5 \\ 7 & 8 & 21 & 11 & 5 & 6 \\ 17 & 21 & 44 & 24 & 11 & 14 \\ 9 & 11 & 24 & 11 & 6 & 7 \\ 4 & 5 & 11 & 6 & 1 & 3 \\ 5 & 6 & 14 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{131} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 18 & 9 & 4 & 5 \\ 7 & 8 & 21 & 11 & 5 & 6 \\ 18 & 21 & 44 & 24 & 11 & 14 \\ 9 & 11 & 24 & 11 & 6 & 7 \\ 4 & 5 & 11 & 6 & 1 & 3 \\ 5 & 6 & 14 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[32] \quad & 1^{124} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -301, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 6 & 27 & 4 & 5 \\ 7 & 8 & 6 & 32 & 5 & 6 \\ 6 & 6 & 2 & 20 & 3 & 4 \\ 27 & 32 & 20 & 105 & 17 & 21 \\ 4 & 5 & 3 & 17 & 1 & 3 \\ 5 & 6 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0} \\
[33] \quad & 1^{123} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -314, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 10 & 26 & 4 & 5 \\ 7 & 8 & 12 & 32 & 5 & 6 \\ 10 & 12 & 14 & 40 & 6 & 8 \\ 26 & 32 & 40 & 105 & 17 & 21 \\ 4 & 5 & 6 & 17 & 1 & 3 \\ 5 & 6 & 8 & 21 & 3 & 4 \end{pmatrix}_{b_0} \\
[34] \quad & 1^{112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -338, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 21 & 4 & 5 \\ 7 & 8 & 6 & 26 & 5 & 6 \\ 5 & 6 & 2 & 16 & 3 & 4 \\ 21 & 26 & 16 & 69 & 14 & 17 \\ 4 & 5 & 3 & 14 & 1 & 3 \\ 5 & 6 & 4 & 17 & 3 & 4 \end{pmatrix}_{b_0} \\
[35] \quad & 1^{13} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -349, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 30 & 4 & 5 \\ 7 & 8 & 6 & 38 & 5 & 6 \\ 4 & 6 & 2 & 24 & 3 & 4 \\ 30 & 38 & 24 & 149 & 20 & 25 \\ 4 & 5 & 3 & 20 & 1 & 3 \\ 5 & 6 & 4 & 25 & 3 & 4 \end{pmatrix}_{b_0} \\
[36] \quad & 1^{124} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -364, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 10 & 27 & 4 & 5 \\ 7 & 8 & 12 & 32 & 5 & 6 \\ 10 & 12 & 14 & 40 & 6 & 8 \\ 27 & 32 & 40 & 105 & 17 & 21 \\ 4 & 5 & 6 & 17 & 1 & 3 \\ 5 & 6 & 8 & 21 & 3 & 4 \end{pmatrix}_{b_0} \\
[37] \quad & 1^{114} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -387, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 31 & 4 & 5 \\ 7 & 8 & 6 & 38 & 5 & 6 \\ 4 & 6 & 2 & 24 & 3 & 4 \\ 31 & 38 & 24 & 149 & 20 & 25 \\ 4 & 5 & 3 & 20 & 1 & 3 \\ 5 & 6 & 4 & 25 & 3 & 4 \end{pmatrix}_{b_0} \\
[38] \quad & 1^{113} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -392, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 22 & 4 & 5 \\ 7 & 8 & 6 & 26 & 5 & 6 \\ 5 & 6 & 2 & 16 & 3 & 4 \\ 22 & 26 & 16 & 69 & 14 & 17 \\ 4 & 5 & 3 & 14 & 1 & 3 \\ 5 & 6 & 4 & 17 & 3 & 4 \end{pmatrix}_{b_0} \\
[40] \quad & 21^{121} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -400, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 4 & 4 & 6 & 5 & 2 & 3 \\ 4 & 4 & 6 & 8 & 7 & 3 & 4 \\ 4 & 6 & 4 & 8 & 7 & 3 & 4 \\ 6 & 8 & 8 & 10 & 10 & 4 & 6 \\ 5 & 7 & 7 & 10 & 7 & 4 & 5 \\ 2 & 3 & 3 & 4 & 4 & 0 & 2 \\ 3 & 4 & 4 & 6 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[39] \quad & 21^{111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -400, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 5 & 2 & 3 \\ 4 & 4 & 6 & 4 & 7 & 3 & 4 \\ 4 & 6 & 4 & 4 & 7 & 3 & 4 \\ 3 & 4 & 4 & 1 & 5 & 2 & 3 \\ 5 & 7 & 7 & 5 & 7 & 4 & 5 \\ 2 & 3 & 3 & 2 & 4 & 0 & 2 \\ 3 & 4 & 4 & 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{131}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 23 & 9 & 4 & 5 \\ 7 & 8 & 27 & 11 & 5 & 6 \\ 23 & 27 & 74 & 31 & 14 & 18 \\ 9 & 11 & 31 & 11 & 6 & 7 \\ 4 & 5 & 14 & 6 & 1 & 3 \\ 5 & 6 & 18 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{131}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 22 & 9 & 4 & 5 \\ 7 & 8 & 27 & 11 & 5 & 6 \\ 22 & 27 & 74 & 31 & 14 & 18 \\ 9 & 11 & 31 & 11 & 6 & 7 \\ 4 & 5 & 14 & 6 & 1 & 3 \\ 5 & 6 & 18 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{124}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -522, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 10 & 31 & 4 & 5 \\ 7 & 8 & 12 & 38 & 5 & 6 \\ 10 & 12 & 14 & 48 & 6 & 8 \\ 31 & 38 & 48 & 149 & 20 & 25 \\ 4 & 5 & 6 & 20 & 1 & 3 \\ 5 & 6 & 8 & 25 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{113}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -530, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 26 & 4 & 5 \\ 7 & 8 & 6 & 32 & 5 & 6 \\ 5 & 6 & 2 & 20 & 3 & 4 \\ 26 & 32 & 20 & 105 & 17 & 21 \\ 4 & 5 & 3 & 17 & 1 & 3 \\ 5 & 6 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{141}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 27 & 9 & 4 & 5 \\ 7 & 8 & 33 & 11 & 5 & 6 \\ 27 & 33 & 112 & 38 & 17 & 22 \\ 9 & 11 & 38 & 11 & 6 & 7 \\ 4 & 5 & 17 & 6 & 1 & 3 \\ 5 & 6 & 22 & 7 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[46] \quad 21^{112}3^{11}1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -576, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 8 & 2 & 3 \\ 4 & 4 & 6 & 4 & 11 & 3 & 4 \\ 4 & 6 & 4 & 4 & 11 & 3 & 4 \\ 3 & 4 & 4 & 1 & 8 & 2 & 3 \\ 8 & 11 & 11 & 8 & 20 & 6 & 8 \\ 2 & 3 & 3 & 2 & 6 & 0 & 2 \\ 3 & 4 & 4 & 3 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{113}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -592, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 27 & 4 & 5 \\ 7 & 8 & 6 & 32 & 5 & 6 \\ 5 & 6 & 2 & 20 & 3 & 4 \\ 27 & 32 & 20 & 105 & 17 & 21 \\ 4 & 5 & 3 & 17 & 1 & 3 \\ 5 & 6 & 4 & 21 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[48] \quad 21^{112}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -720, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 11 & 2 & 3 \\ 4 & 4 & 6 & 4 & 16 & 3 & 4 \\ 4 & 6 & 4 & 4 & 16 & 3 & 4 \\ 3 & 4 & 4 & 1 & 11 & 2 & 3 \\ 11 & 16 & 16 & 11 & 44 & 9 & 12 \\ 2 & 3 & 3 & 2 & 9 & 0 & 2 \\ 3 & 4 & 4 & 3 & 12 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{114}3^{11}1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -738, (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 31 & 4 & 5 \\ 7 & 8 & 6 & 38 & 5 & 6 \\ 5 & 6 & 2 & 24 & 3 & 4 \\ 31 & 38 & 24 & 149 & 20 & 25 \\ 4 & 5 & 3 & 20 & 1 & 3 \\ 5 & 6 & 4 & 25 & 3 & 4 \end{pmatrix}_{b_0}$$

5.132 NHC(-2)(-3)(-2) + $\mathfrak{su}_2$ + su2n3mix + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 11 & 9 & 4 & 4 & 5 \\ 7 & 8 & 15 & 11 & 5 & 6 & 6 \\ 11 & 15 & 22 & 17 & 8 & 9 & 10 \\ 9 & 11 & 17 & 11 & 6 & 7 & 7 \\ 4 & 5 & 8 & 6 & 1 & 3 & 3 \\ 4 & 6 & 9 & 7 & 3 & 2 & 4 \\ 5 & 6 & 10 & 7 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 7 & 9 & 4 & 4 & 5 \\ 7 & 8 & 9 & 11 & 5 & 6 & 6 \\ 7 & 9 & 8 & 10 & 5 & 5 & 6 \\ 9 & 11 & 10 & 11 & 6 & 7 & 7 \\ 4 & 5 & 5 & 6 & 1 & 3 & 3 \\ 4 & 6 & 5 & 7 & 3 & 2 & 4 \\ 5 & 6 & 6 & 7 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 11 & 9 & 4 & 9 & 5 \\ 7 & 8 & 15 & 11 & 5 & 12 & 6 \\ 11 & 15 & 22 & 17 & 8 & 18 & 10 \\ 9 & 11 & 17 & 11 & 6 & 14 & 7 \\ 4 & 5 & 8 & 6 & 1 & 6 & 3 \\ 9 & 12 & 18 & 14 & 6 & 14 & 8 \\ 5 & 6 & 10 & 7 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 9 & 4 & 13 & 5 \\ 7 & 8 & 6 & 11 & 5 & 15 & 6 \\ 4 & 6 & 2 & 7 & 3 & 9 & 4 \\ 9 & 11 & 7 & 11 & 6 & 17 & 7 \\ 4 & 5 & 3 & 6 & 1 & 8 & 3 \\ 13 & 15 & 9 & 17 & 8 & 22 & 10 \\ 5 & 6 & 4 & 7 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 9 & 4 & 12 & 5 \\ 7 & 8 & 6 & 11 & 5 & 15 & 6 \\ 6 & 6 & 2 & 7 & 3 & 9 & 4 \\ 9 & 11 & 7 & 11 & 6 & 17 & 7 \\ 4 & 5 & 3 & 6 & 1 & 8 & 3 \\ 12 & 15 & 9 & 17 & 8 & 22 & 10 \\ 5 & 6 & 4 & 7 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1211} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 10 & 9 & 4 & 12 & 5 \\ 7 & 8 & 12 & 11 & 5 & 15 & 6 \\ 10 & 12 & 14 & 14 & 6 & 18 & 8 \\ 9 & 11 & 14 & 11 & 6 & 17 & 7 \\ 4 & 5 & 6 & 6 & 1 & 8 & 3 \\ 12 & 15 & 18 & 17 & 8 & 22 & 10 \\ 5 & 6 & 8 & 7 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{1111} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 6 & 9 & 4 & 5 & 5 \\ 7 & 8 & 6 & 11 & 5 & 6 & 6 \\ 6 & 6 & 2 & 7 & 3 & 4 & 4 \\ 9 & 11 & 7 & 11 & 6 & 7 & 7 \\ 4 & 5 & 3 & 6 & 1 & 3 & 3 \\ 5 & 6 & 4 & 7 & 3 & 2 & 4 \\ 5 & 6 & 4 & 7 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{violet}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 9 & 4 & 5 & 5 \\ 7 & 8 & 6 & 11 & 5 & 6 & 6 \\ 4 & 6 & 2 & 7 & 3 & 4 & 4 \\ 9 & 11 & 7 & 11 & 6 & 7 & 7 \\ 4 & 5 & 3 & 6 & 1 & 3 & 3 \\ 5 & 6 & 4 & 7 & 3 & 2 & 4 \\ 5 & 6 & 4 & 7 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{\overline{112}} 3^{\overline{11}1} 1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{su}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{violet}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 128, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 9 & 4 & 10 & 5 \\ 7 & 8 & 6 & 11 & 5 & 12 & 6 \\ 4 & 6 & 2 & 7 & 3 & 8 & 4 \\ 9 & 11 & 7 & 11 & 6 & 14 & 7 \\ 4 & 5 & 3 & 6 & 1 & 6 & 3 \\ 10 & 12 & 8 & 14 & 6 & 14 & 8 \\ 5 & 6 & 4 & 7 & 3 & 8 & 4 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{\overline{1113}} 3^{\overline{11}} 1^{\overline{11}} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{violet}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\left(\begin{array}{cccccc} 5 & 7 & 4 & 9 & 4 & 18 & 5 \\ 7 & 8 & 6 & 11 & 5 & 21 & 6 \\ 4 & 6 & 2 & 7 & 3 & 13 & 4 \\ 9 & 11 & 7 & 11 & 6 & 24 & 7 \\ 4 & 5 & 3 & 6 & 1 & 11 & 3 \\ 18 & 21 & 13 & 24 & 11 & 44 & 14 \\ 5 & 6 & 4 & 7 & 3 & 14 & 4 \end{array} \right)_{b_0}$$

$$[10] \begin{array}{l} 1^{111} \\ 1^{1111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{s0}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{violet}{su}_2 \\ (1, 10, 0), c_{\text{ext}} = 8.0 \\ T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\ \left(\begin{array}{cccccc} 5 & 7 & 4 & 9 & 4 & 12 & 5 \\ 7 & 8 & 6 & 11 & 5 & 15 & 6 \\ 4 & 6 & 2 & 7 & 3 & 9 & 4 \\ 9 & 11 & 7 & 11 & 6 & 17 & 7 \\ 4 & 5 & 3 & 6 & 1 & 8 & 3 \\ 12 & 15 & 9 & 17 & 8 & 22 & 10 \\ 5 & 6 & 4 & 7 & 3 & 10 & 4 \end{array} \right)_{b_0} \end{array} \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 144, (n_+, n_-, n_0) =$$

$$[111] \begin{matrix} 1^{111} \\ 1^{111} 2^3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{s}07 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{red}{su}_2 \end{matrix} \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 144, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.5$$

$$[12] \quad 1^{1121} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{teal}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{teal}{su}_2 \rightarrow (-3) + \textcolor{violet}{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$$

$$[14] \begin{array}{l} 1^{1111} \\ 1^{1111} 3^{11} 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{brown}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) + \textcolor{violet}{su}_2 \\ (1, 10, 0), c_{\text{ext}} = 8.0 \\ T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\ \left(\begin{array}{cccccc} 5 & 7 & 5 & 9 & 4 & 12 & 5 \\ 7 & 8 & 6 & 11 & 5 & 15 & 6 \\ 5 & 6 & 2 & 7 & 3 & 9 & 4 \\ 9 & 11 & 7 & 11 & 6 & 17 & 7 \\ 4 & 5 & 3 & 6 & 1 & 8 & 3 \\ 12 & 15 & 9 & 17 & 8 & 22 & 10 \\ 5 & 6 & 4 & 7 & 3 & 10 & 4 \end{array} \right)_{b_0} \end{array} \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 256, (n_+, n_-, n_0) =$$

$$[16] \begin{array}{l} 1^{1111} 3^{11} 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_2 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (\textcolor{blue}{-3}) + \textcolor{red}{su}_2 \\ (1, 10, 0), c_{\text{ext}} = 8.0 \\ T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\ \begin{pmatrix} 5 & 7 & 5 & 9 & 4 & 5 & 5 \\ 7 & 8 & 6 & 11 & 5 & 6 & 6 \\ 5 & 6 & 2 & 7 & 3 & 4 & 4 \\ 9 & 11 & 7 & 11 & 6 & 7 & 7 \\ 4 & 5 & 3 & 6 & 1 & 3 & 3 \\ 5 & 6 & 4 & 7 & 3 & 2 & 4 \\ 5 & 6 & 4 & 7 & 3 & 4 & 4 \end{pmatrix}_{b_0} \end{array} \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 256, (n_+, n_-, n_0) =$$

$$\begin{aligned}
[13] \quad & 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 256, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 9 & 4 & 13 & 5 \\ 7 & 8 & 6 & 11 & 5 & 15 & 6 \\ 5 & 6 & 2 & 7 & 3 & 9 & 4 \\ 9 & 11 & 7 & 11 & 6 & 17 & 7 \\ 4 & 5 & 3 & 6 & 1 & 8 & 3 \\ 13 & 15 & 9 & 17 & 8 & 22 & 10 \\ 5 & 6 & 4 & 7 & 3 & 10 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[15] \quad & 1^{1112} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 256, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 9 & 4 & 10 & 5 \\ 7 & 8 & 6 & 11 & 5 & 12 & 6 \\ 5 & 6 & 2 & 7 & 3 & 8 & 4 \\ 9 & 11 & 7 & 11 & 6 & 14 & 7 \\ 4 & 5 & 3 & 6 & 1 & 6 & 3 \\ 10 & 12 & 8 & 14 & 6 & 14 & 8 \\ 5 & 6 & 4 & 7 & 3 & 8 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[17] \quad & 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 272, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 4 & 9 & 4 & 17 & 5 \\ 7 & 8 & 6 & 11 & 5 & 21 & 6 \\ 4 & 6 & 2 & 7 & 3 & 13 & 4 \\ 9 & 11 & 7 & 11 & 6 & 24 & 7 \\ 4 & 5 & 3 & 6 & 1 & 11 & 3 \\ 17 & 21 & 13 & 24 & 11 & 44 & 14 \\ 5 & 6 & 4 & 7 & 3 & 14 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[18] \quad & 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = -1, \det = 512, (n_+, n_-, n_0) = \\
& (1, 10, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 9 & 4 & 17 & 5 \\ 7 & 8 & 6 & 11 & 5 & 21 & 6 \\ 5 & 6 & 2 & 7 & 3 & 13 & 4 \\ 9 & 11 & 7 & 11 & 6 & 24 & 7 \\ 4 & 5 & 3 & 6 & 1 & 11 & 3 \\ 17 & 21 & 13 & 24 & 11 & 44 & 14 \\ 5 & 6 & 4 & 7 & 3 & 14 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

5.133 NHC(-2)(-3)(-2) + su₂ + su₂n3mix + su₂ → (-3) + su₂ + su₂

$$\begin{aligned}
T = 10 \quad [5] \quad & 1^{1121} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = \\
& 0, (n_+, n_-, n_0) = (1, 10, 1), c_{\text{ext}} = 9.5
\end{aligned}$$

$$\begin{aligned}
[4] \quad & 1^{1211} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = \\
& (1, 10, 1), c_{\text{ext}} = 9.0
\end{aligned}$$

$$\begin{aligned}
[3] \quad & 1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = \\
& (1, 10, 1), c_{\text{ext}} = 9.0
\end{aligned}$$

$$\begin{aligned}
[2] \quad & 1^{1121} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = \\
& (1, 10, 1), c_{\text{ext}} = 9.5
\end{aligned}$$

$$\begin{aligned}
[1] \quad & 1^{1111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = \\
& (1, 10, 1), c_{\text{ext}} = 9.0
\end{aligned}$$

$$\begin{aligned}
[0] \quad & 1^{111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_2 + \text{su}_2 \quad T = 10, T_{\max} = 10, \Delta = 0, \det = 0, (n_+, n_-, n_0) = \\
& (1, 10, 1), c_{\text{ext}} = 9.0
\end{aligned}$$

5.134 NHC(-2)(-3)(-2) + su₂ + su₂n3mix + su₂n3mix

$$\begin{aligned}
T = 9 \quad [0] \quad & 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -8, (n_+, n_-, n_0) = \\
& (1, 9, 0), c_{\text{ext}} = 7.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 5 & 7 & 5 & 8 & 17 & 5 \\ 7 & 8 & 6 & 11 & 20 & 6 \\ 5 & 6 & 2 & 7 & 12 & 4 \\ 8 & 11 & 7 & 11 & 23 & 7 \\ 17 & 20 & 12 & 23 & 41 & 13 \\ 5 & 6 & 4 & 7 & 13 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[1] \quad & 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -31, \det = -18, (n_+, n_-, n_0) =
\end{aligned}$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 16 & 14 & 5 \\ 7 & 8 & 6 & 20 & 17 & 6 \\ 5 & 6 & 2 & 12 & 11 & 4 \\ 16 & 20 & 12 & 41 & 35 & 13 \\ 14 & 17 & 11 & 35 & 29 & 11 \\ 5 & 6 & 4 & 13 & 11 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 5 & 7 & 5 & 10 & 16 & 5 \\ 7 & 8 & 6 & 11 & 20 & 6 \\ 5 & 6 & 2 & 7 & 12 & 4 \\ 10 & 11 & 7 & 11 & 23 & 7 \\ 16 & 20 & 12 & 23 & 41 & 13 \\ 5 & 6 & 4 & 7 & 13 & 4 \end{pmatrix}_{b_0}$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 16 & 14 & 5 \\ 7 & 8 & 6 & 20 & 17 & 6 \\ 4 & 6 & 2 & 12 & 11 & 4 \\ 16 & 20 & 12 & 41 & 35 & 13 \\ 14 & 17 & 11 & 35 & 29 & 11 \\ 5 & 6 & 4 & 13 & 11 & 4 \end{pmatrix}_{b_0}$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 8 & 22 & 5 \\ 7 & 8 & 6 & 11 & 26 & 6 \\ 5 & 6 & 2 & 7 & 16 & 4 \\ 8 & 11 & 7 & 11 & 30 & 7 \\ 22 & 26 & 16 & 30 & 69 & 17 \\ 5 & 6 & 4 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 9 & 16 & 5 \\ 7 & 8 & 6 & 11 & 20 & 6 \\ 4 & 6 & 2 & 7 & 12 & 4 \\ 9 & 11 & 7 & 11 & 23 & 7 \\ 16 & 20 & 12 & 23 & 41 & 13 \\ 5 & 6 & 4 & 7 & 13 & 4 \end{pmatrix}_{b_0}$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 9 & 9 & 5 \\ 7 & 8 & 6 & 11 & 11 & 6 \\ 4 & 6 & 2 & 7 & 7 & 4 \\ 9 & 11 & 7 & 11 & 13 & 7 \\ 9 & 11 & 7 & 13 & 11 & 7 \\ 5 & 6 & 4 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 9 & 20 & 5 \\ 7 & 8 & 6 & 11 & 26 & 6 \\ 4 & 6 & 2 & 7 & 16 & 4 \\ 9 & 11 & 7 & 11 & 30 & 7 \\ 20 & 26 & 16 & 30 & 69 & 17 \\ 5 & 6 & 4 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 9 & 22 & 5 \\ 7 & 8 & 6 & 11 & 26 & 6 \\ 4 & 6 & 2 & 7 & 16 & 4 \\ 9 & 11 & 7 & 11 & 30 & 7 \\ 22 & 26 & 16 & 30 & 69 & 17 \\ 5 & 6 & 4 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 9 & 16 & 5 \\ 7 & 8 & 6 & 11 & 20 & 6 \\ 5 & 6 & 2 & 7 & 12 & 4 \\ 9 & 11 & 7 & 11 & 23 & 7 \\ 16 & 20 & 12 & 23 & 41 & 13 \\ 5 & 6 & 4 & 7 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{1111} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 9 & 17 & 5 \\ 7 & 8 & 6 & 11 & 20 & 6 \\ 5 & 6 & 2 & 7 & 12 & 4 \\ 9 & 11 & 7 & 11 & 23 & 7 \\ 17 & 20 & 12 & 23 & 41 & 13 \\ 5 & 6 & 4 & 7 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1111} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 9 & 9 & 5 \\ 7 & 8 & 6 & 11 & 11 & 6 \\ 5 & 6 & 2 & 7 & 7 & 4 \\ 9 & 11 & 7 & 11 & 13 & 7 \\ 9 & 11 & 7 & 13 & 11 & 7 \\ 5 & 6 & 4 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{1211} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 10 & 9 & 9 & 5 \\ 7 & 8 & 12 & 11 & 11 & 6 \\ 10 & 12 & 14 & 14 & 14 & 8 \\ 9 & 11 & 14 & 11 & 13 & 7 \\ 9 & 11 & 14 & 13 & 11 & 7 \\ 5 & 6 & 8 & 7 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -211, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 4 & 9 & 21 & 5 \\ 7 & 8 & 6 & 11 & 26 & 6 \\ 4 & 6 & 2 & 7 & 16 & 4 \\ 9 & 11 & 7 & 11 & 30 & 7 \\ 21 & 26 & 16 & 30 & 69 & 17 \\ 5 & 6 & 4 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{1121} 3^{11} 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -288, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 14 & 9 & 5 \\ 7 & 8 & 6 & 17 & 11 & 6 \\ 5 & 6 & 2 & 11 & 7 & 4 \\ 14 & 17 & 11 & 29 & 20 & 11 \\ 9 & 11 & 7 & 20 & 11 & 7 \\ 5 & 6 & 4 & 11 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{1112} 3^{11} 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_2 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -31, \quad \det = -386, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 7.0$$

$$T_{\min} = 9.96 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 5 & 7 & 5 & 9 & 21 & 5 \\ 7 & 8 & 6 & 11 & 26 & 6 \\ 5 & 6 & 2 & 7 & 16 & 4 \\ 9 & 11 & 7 & 11 & 30 & 7 \\ 21 & 26 & 16 & 30 & 69 & 17 \\ 5 & 6 & 4 & 7 & 17 & 4 \end{pmatrix}_{b_0}$$

5.135 NHC(-2)(-3)(-2) + su₃

$$T = 7 \quad [0] \quad 21^{11} 31^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -11, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.45 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 4 & 8 & 5 & 3 & 4 \\ 8 & 9 & 7 & 4 & 5 \\ 5 & 7 & 4 & 4 & 4 \\ 3 & 4 & 4 & 0 & 2 \\ 4 & 5 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 31^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -12, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.33 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 6 & 5 & 4 \\ 3 & 6 & 4 & 5 & 4 \\ 4 & 5 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{113} 31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, \quad T_{\max} = 12, \quad \Delta = -95, \quad \det = -12, \quad (n_+, n_-, n_0) = (1, 7, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.33 \rightarrow 11, \quad T_{\max} = 12, \quad T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 6 & 12 & 4 \\ 3 & 6 & 4 & 12 & 4 \\ 9 & 12 & 12 & 24 & 8 \\ 3 & 4 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 6 & 13 & 4 \\ 3 & 6 & 4 & 13 & 4 \\ 10 & 13 & 13 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 3 \\ 4 & 9 & 7 & 4 & 5 \\ 3 & 7 & 4 & 4 & 4 \\ 2 & 4 & 4 & 0 & 2 \\ 3 & 5 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3^1 1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 6 & 9 & 4 \\ 3 & 6 & 4 & 9 & 4 \\ 7 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 3 \\ 4 & 9 & 7 & 0 & 5 \\ 3 & 7 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 5 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

[7] $1^{12} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.56 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 7 & 15 & 5 \\ 7 & 8 & 21 & 6 \\ 15 & 21 & 43 & 13 \\ 5 & 6 & 13 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{11} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 5 & 7 & 4 & 5 \\ 7 & 8 & 6 & 6 \\ 4 & 6 & 1 & 3 \\ 5 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

[9] $21^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 5 & 5 & 6 & 4 \\ 5 & 4 & 6 & 5 & 4 \\ 5 & 6 & 4 & 5 & 4 \\ 6 & 5 & 5 & 3 & 3 \\ 4 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -22, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 5 & 7 & 7 & 5 \\ 7 & 8 & 9 & 6 \\ 7 & 9 & 7 & 5 \\ 5 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

[11] $1^{11} 3^1 1^{12} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 5 & 7 & 6 & 5 \\ 7 & 8 & 6 & 6 \\ 6 & 6 & 1 & 3 \\ 5 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

[12] $1^{12} 3^1 1^{14} 3^1 1^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -27, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 7 & 12 & 5 \\ 7 & 8 & 12 & 6 \\ 12 & 12 & 13 & 7 \\ 5 & 6 & 7 & 2 \end{pmatrix}_{b_0}$$

[13] $1^{11} 3^1 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 5 & 7 & 11 & 5 \\ 7 & 8 & 15 & 6 \\ 11 & 15 & 21 & 9 \\ 5 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

- [14] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & -1 & 2 \end{pmatrix}_{b_0}$
- [15] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -31$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 5 & 7 & 5 & 5 \\ 7 & 8 & 6 & 6 \\ 5 & 6 & 1 & 3 \\ 5 & 6 & 3 & 3 \end{pmatrix}_{b_0}$
- [16] $1^{11} 3^1 1^2 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.2$
 $T_{\min} = 10.38 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 6 & 8 & 4 \\ 3 & 6 & 4 & 8 & 4 \\ 6 & 8 & 8 & 9 & 5 \\ 3 & 4 & 4 & 5 & 2 \end{pmatrix}_{b_0}$
- [17] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -32$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 6 & 5 & 4 \\ 3 & 6 & 4 & 5 & 4 \\ 3 & 5 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$
- [18] $1^{11} 3^1 1^{14} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -36$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.33 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 5 & 7 & 15 & 5 \\ 7 & 8 & 15 & 6 \\ 15 & 15 & 21 & 9 \\ 5 & 6 & 9 & 2 \end{pmatrix}_{b_0}$
- [20] $1^{12} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -43$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 5 & 7 & 10 & 5 \\ 7 & 8 & 12 & 6 \\ 10 & 12 & 13 & 7 \\ 5 & 6 & 7 & 3 \end{pmatrix}_{b_0}$
- [19] $1^{12} 3^1 1^3 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -43$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.2$
 $T_{\min} = 10.49 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 5 & 7 & 11 & 5 \\ 7 & 8 & 12 & 6 \\ 11 & 12 & 13 & 7 \\ 5 & 6 & 7 & 3 \end{pmatrix}_{b_0}$
- [22] $21^1 31^{111} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 5 & 3 & 4 \\ 5 & 4 & 6 & 4 & 4 \\ 5 & 6 & 4 & 4 & 4 \\ 3 & 4 & 4 & 0 & 2 \\ 4 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [21] $1^{11} 3^1 1^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -44$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 6 & 4 & 4 \\ 3 & 6 & 4 & 4 & 4 \\ 3 & 4 & 4 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [24] $1^1 13^1 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.2$
 $T_{\min} = 11.58 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 6 & 8 & 4 \\ 3 & 6 & 4 & 8 & 4 \\ 4 & 8 & 8 & 9 & 5 \\ 3 & 4 & 4 & 5 & 1 \end{pmatrix}_{b_0}$
- [23] $1^1 13^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 6 & 4 & 4 \\ 3 & 6 & 4 & 4 & 4 \\ 2 & 4 & 4 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [25] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -54$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.22 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 5 & 7 & 12 & 5 \\ 7 & 8 & 15 & 6 \\ 12 & 15 & 21 & 9 \\ 5 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[26] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -58, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 5 & 7 & 14 & 5 \\ 7 & 8 & 15 & 6 \\ 14 & 15 & 21 & 9 \\ 5 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[28] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 6 & 8 & 4 \\ 3 & 6 & 4 & 8 & 4 \\ 5 & 8 & 8 & 9 & 5 \\ 3 & 4 & 4 & 5 & 2 \end{pmatrix}_{b_0}$$

[27] $1^{111} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 6 & 9 & 4 \\ 3 & 6 & 4 & 9 & 4 \\ 6 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 5 & 7 & 15 & 5 \\ 7 & 8 & 18 & 6 \\ 15 & 18 & 33 & 11 \\ 5 & 6 & 11 & 3 \end{pmatrix}_{b_0}$$

[32] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 5 & 7 & 13 & 5 \\ 7 & 8 & 15 & 6 \\ 13 & 15 & 21 & 9 \\ 5 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[31] $1^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 6 & 9 & 4 \\ 3 & 6 & 4 & 9 & 4 \\ 5 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[30] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 6 & 12 & 4 \\ 3 & 6 & 4 & 12 & 4 \\ 8 & 12 & 12 & 24 & 8 \\ 3 & 4 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{13} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -67, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 5 & 7 & 16 & 5 \\ 7 & 8 & 18 & 6 \\ 16 & 18 & 33 & 11 \\ 5 & 6 & 11 & 3 \end{pmatrix}_{b_0}$$

[34] $21^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -68, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.35 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 4 & 6 & 5 & 4 \\ 4 & 6 & 4 & 5 & 4 \\ 3 & 5 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{113} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 6 & 12 & 4 \\ 3 & 6 & 4 & 12 & 4 \\ 7 & 12 & 12 & 24 & 8 \\ 3 & 4 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{112} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -80, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 6 & 13 & 4 \\ 3 & 6 & 4 & 13 & 4 \\ 9 & 13 & 13 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[37] $21^{11} 3^1 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$

$$T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 5 & 9 & 4 \\ 5 & 4 & 6 & 9 & 4 \\ 5 & 6 & 4 & 9 & 4 \\ 9 & 9 & 9 & 12 & 6 \\ 4 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{14} 3 \ 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$$

$$T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 5 & 7 & 20 & 5 \\ 7 & 8 & 24 & 6 \\ 20 & 24 & 61 & 15 \\ 5 & 6 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{13} 3 \ 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 12 & 3 \\ 3 & 4 & 6 & 17 & 4 \\ 3 & 6 & 4 & 17 & 4 \\ 12 & 17 & 17 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{12} 3 \ 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 5 & 7 & 17 & 5 \\ 7 & 8 & 21 & 6 \\ 17 & 21 & 43 & 13 \\ 5 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{11} 3 \ 1^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 6 & 13 & 4 \\ 3 & 6 & 4 & 13 & 4 \\ 7 & 13 & 13 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{12} 3 \ 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -106, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 5 & 7 & 19 & 5 \\ 7 & 8 & 21 & 6 \\ 19 & 21 & 43 & 13 \\ 5 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{12} 3 \ 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 6 & 13 & 4 \\ 3 & 6 & 4 & 13 & 4 \\ 8 & 13 & 13 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{12} 3 \ 1^{11} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -108, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 5 & 7 & 18 & 5 \\ 7 & 8 & 21 & 6 \\ 18 & 21 & 43 & 13 \\ 5 & 6 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[45] \ 21^{11} 31^{12} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 5 & 12 & 4 \\ 5 & 4 & 6 & 13 & 4 \\ 5 & 6 & 4 & 13 & 4 \\ 12 & 13 & 13 & 27 & 9 \\ 4 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[46] \ 21^{11} 31^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.24 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 5 & 5 & 15 & 4 \\ 5 & 4 & 6 & 17 & 4 \\ 5 & 6 & 4 & 17 & 4 \\ 15 & 17 & 17 & 48 & 12 \\ 4 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{11} 3 \ 1^{13} + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -128, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.03 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 6 & 17 & 4 \\ 3 & 6 & 4 & 17 & 4 \\ 9 & 17 & 17 & 48 & 12 \\ 3 & 4 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[48] \ 21^{11} 31^1 + [\text{su}_2 \cdot \text{so}_7 \cdot \text{su}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -140, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 3 & 3 \\ 4 & 4 & 6 & 4 & 4 \\ 4 & 6 & 4 & 4 & 4 \\ 3 & 4 & 4 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -142, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.08 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 5 & 7 & 22 & 5 \\ 7 & 8 & 27 & 6 \\ 22 & 27 & 73 & 17 \\ 5 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -144, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 11 & 3 \\ 3 & 4 & 6 & 17 & 4 \\ 3 & 6 & 4 & 17 & 4 \\ 11 & 17 & 17 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[51] \quad 21^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -152, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 8 & 8 & 3 & 4 \\ 8 & 9 & 11 & 4 & 5 \\ 8 & 11 & 9 & 4 & 5 \\ 3 & 4 & 4 & 0 & 2 \\ 4 & 5 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[52] \quad 21^{11} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 3 \\ 4 & 4 & 6 & 9 & 4 \\ 4 & 6 & 4 & 9 & 4 \\ 6 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 6 & 17 & 4 \\ 3 & 6 & 4 & 17 & 4 \\ 10 & 17 & 17 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{13} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -160, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 5 & 7 & 23 & 5 \\ 7 & 8 & 27 & 6 \\ 23 & 27 & 73 & 17 \\ 5 & 6 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 21^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -164, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 6 & 3 \\ 4 & 4 & 6 & 8 & 4 \\ 4 & 6 & 4 & 8 & 4 \\ 6 & 8 & 8 & 9 & 5 \\ 3 & 4 & 4 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{14} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$$

$$T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 5 & 7 & 27 & 5 \\ 7 & 8 & 33 & 6 \\ 27 & 33 & 111 & 21 \\ 5 & 6 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[57] \quad 21^{13} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 3 \\ 4 & 4 & 6 & 12 & 4 \\ 4 & 6 & 4 & 12 & 4 \\ 9 & 12 & 12 & 24 & 8 \\ 3 & 4 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[58] \quad 21^{12} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -260, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 9 & 3 \\ 4 & 4 & 6 & 13 & 4 \\ 4 & 6 & 4 & 13 & 4 \\ 9 & 13 & 13 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[59] \quad 21^{13} 31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -380, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 4 & 4 & 12 & 3 \\ 4 & 4 & 6 & 17 & 4 \\ 4 & 6 & 4 & 17 & 4 \\ 12 & 17 & 17 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [60] \quad 1^{11} 4 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 5 & 4 & 4 \\ 3 & 5 & 4 & 3 & 4 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{11} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 5 & 0 & 4 \\ 3 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{11} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 3 & 5 & 4 & 7 & 4 \\ 6 & 7 & 7 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[63] \quad 21^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 5 & 3 & 4 \\ 5 & 4 & 5 & 4 & 4 \\ 5 & 5 & 4 & 3 & 4 \\ 3 & 4 & 3 & 0 & 2 \\ 4 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[64] \quad 21^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 5 & 6 & 4 \\ 5 & 4 & 5 & 7 & 4 \\ 5 & 5 & 4 & 7 & 4 \\ 6 & 7 & 7 & 7 & 5 \\ 4 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{11} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 3 & 5 & 4 & 7 & 4 \\ 5 & 7 & 7 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[66] \quad 21^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 5 & 6 & 4 \\ 5 & 4 & 6 & 7 & 4 \\ 5 & 6 & 4 & 7 & 4 \\ 6 & 7 & 7 & 7 & 5 \\ 4 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[67] \quad 21^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 5 & 5 & 3 & 4 \\ 5 & 4 & 6 & 3 & 4 \\ 5 & 6 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 4 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[68] \quad 21^{11} 4 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 5 & 7 & 4 \\ 4 & 5 & 4 & 7 & 4 \\ 5 & 7 & 7 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[69] \quad 21^{11} 4 1^2 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 8 & 8 & 6 & 4 \\ 8 & 13 & 15 & 12 & 7 \\ 8 & 15 & 13 & 12 & 7 \\ 6 & 12 & 12 & 7 & 5 \\ 4 & 7 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[70] \quad 21^{11} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 3 \\ 4 & 4 & 6 & 3 & 4 \\ 4 & 6 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[71] \quad 21^{11} 4 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 4 & 5 & 3 \\ 4 & 4 & 6 & 7 & 4 \\ 4 & 6 & 4 & 7 & 4 \\ 5 & 7 & 7 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[72] \quad 31^{11} 31^{11} 3 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -82, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.71 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 6 & 6 & 4 \\ 3 & 4 & 0 & 4 & 3 & 2 \\ 3 & 6 & 4 & 4 & 6 & 4 \\ 3 & 6 & 3 & 6 & 3 & 3 \\ 2 & 4 & 2 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[73] \quad 221^{11} 41^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 288, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.81 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 6 & 6 & 5 & 3 \\ 6 & 13 & 15 & 12 & 7 \\ 6 & 15 & 13 & 12 & 7 \\ 5 & 12 & 12 & 7 & 5 \\ 3 & 7 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [74] \quad 1^{11} 5^{11} 131^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [75] \quad 1^{11} 3221^{11} 6^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 & 2 \\ 4 & 10 & 1 & 12 & 6 & 6 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 4 & 12 & 1 & 10 & 6 & 6 \\ 2 & 6 & 0 & 6 & 3 & 3 \\ 2 & 6 & 0 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[76] \quad 21^{11} 61^{11} 2231^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 4 & 1 & 4 & 2 & 2 \\ 4 & 10 & 1 & 12 & 6 & 6 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 4 & 12 & 1 & 10 & 6 & 6 \\ 2 & 6 & 0 & 6 & 3 & 3 \\ 2 & 6 & 0 & 6 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{11} 6^{11} 1231^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -13, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{11} 3221^{11} 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 1 & 7 & 3 \\ 2 & 1 & 1 & 6 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 7 & 6 & 1 & 19 & 9 \\ 3 & 3 & 0 & 9 & 4 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [79] \quad 1^1 4141^{11} 3151^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 \quad T = 12, T_{\max} = 14, \Delta = -58, \det = 296, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 13.99 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 8 & 8 & 1 & 3 \\ 8 & 17 & 19 & 0 & 9 \\ 8 & 19 & 17 & 0 & 9 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 9 & 9 & 0 & 4 \end{pmatrix}_{b_0}$$

5.136 NHC(-2)(-3)(-2) + $\mathfrak{su}_3$ + $\mathfrak{e}_6$

$$T = 8 \quad [0] \quad 1^{11} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 7 & 4 & 0 & 5 \\ 3 & 7 & 4 & 4 & 0 & 4 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 5 & 4 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 152, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 5 & 0 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 \\ 3 & 5 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 3 & -4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{112}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 13.09 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 6 & 8 & 0 & 4 \\ 3 & 6 & 4 & 8 & 0 & 4 \\ 4 & 8 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 224, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 6 & 4 & 0 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 300, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 9 & 0 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{112}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 440, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 6 & 13 & 0 & 4 \\ 3 & 6 & 4 & 13 & 0 & 4 \\ 7 & 13 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{113}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -144, \det = 572, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 6 & 17 & 0 & 4 \\ 3 & 6 & 4 & 17 & 0 & 4 \\ 9 & 17 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 12 & -4 & 0 \end{pmatrix}_{b_0}$$

5.137 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{e}_6 + \text{NHC}(-2)(-3)$

$T = 10 \quad [0] \quad 1^{\overline{1111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 15, \Delta = -94, \det = 172, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$
 $T_{\min} = 13.16 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 5 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 3 & 0 & 1 & 4 & 3 \\ 2 & 4 & 4 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

5.138 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8$

$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$
 $T_{\min} = 13.77 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$

$$\begin{pmatrix} 5 & 11 & 11 & 9 & 1 & 5 \\ 11 & 20 & 22 & 17 & 0 & 8 \\ 11 & 22 & 20 & 17 & 0 & 8 \\ 9 & 17 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

5.139 NHC(-2)(-3)(-2) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2$

$$T = 9 \quad [3] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 6 & 4 & 0 & 5 & 4 \\ 3 & 6 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 4 & 2 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 5 & 3 \\ 4 & 9 & 7 & 4 & 0 & 12 & 5 \\ 3 & 7 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 12 & 9 & 6 & 0 & 13 & 7 \\ 3 & 5 & 4 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.49 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 6 & 4 & 0 & 14 & 4 \\ 3 & 6 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 4 & 2 & -4 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -217, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 8 & 3 \\ 4 & 9 & 7 & 4 & 0 & 20 & 5 \\ 3 & 7 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 20 & 14 & 9 & 0 & 37 & 11 \\ 3 & 5 & 4 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -220, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 5 & 0 & 4 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 3 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{114}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 6 & 4 & 0 & 16 & 4 \\ 3 & 6 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 16 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 4 & 2 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -256, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 10 & 3 \\ 4 & 9 & 7 & 4 & 0 & 24 & 5 \\ 3 & 7 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 24 & 18 & 12 & 0 & 58 & 14 \\ 3 & 5 & 4 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{\overline{112}} 3^{\overline{11}} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 6 & 9 & 0 & 8 & 4 \\ 3 & 6 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 9 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 4 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1113} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -340, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 6 & 4 & 0 & 12 & 4 \\ 3 & 6 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 12 & 9 & 0 & 25 & 9 \\ 3 & 4 & 4 & 2 & -4 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1112} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -400, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 6 & 4 & 0 & 8 & 4 \\ 3 & 6 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 4 & 2 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1111} 3 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -436, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 6 & 4 & 0 & 4 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 2 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1112} 3 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -436, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.28 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 6 & 4 & 0 & 23 & 4 \\ 3 & 6 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 23 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 4 & 2 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -516, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.82 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 6 & 9 & 0 & 4 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 4 & 6 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -592, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 6 & 4 & 0 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1113} 3 1^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -1012, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.17 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 6 & 4 & 0 & 32 & 4 \\ 3 & 6 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 32 & 32 & 21 & 0 & 193 & 25 \\ 3 & 4 & 4 & 2 & -4 & 25 & -1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1112} 3 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -1012, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 6 & 4 & 0 & 13 & 4 \\ 3 & 6 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 13 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 4 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{1112} 3 1^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -1024, \ (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 6 & 4 & 0 & 18 & 4 \\ 3 & 6 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 4 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1113} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1408, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 6 & 4 & 0 & 17 & 4 \\ 3 & 6 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 17 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 4 & 2 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{1113} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 6 & 4 & 0 & 27 & 4 \\ 3 & 6 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 4 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{1114} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1780, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 6 & 4 & 0 & 21 & 4 \\ 3 & 6 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 4 & 2 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1113} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -1876, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 6 & 4 & 0 & 22 & 4 \\ 3 & 6 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 22 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 4 & 2 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{1114} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -2704, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 6 & 4 & 0 & 26 & 4 \\ 3 & 6 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 26 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 4 & 2 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{1114} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -2752, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 6 & 4 & 0 & 36 & 4 \\ 3 & 6 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 4 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{1114} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -3028, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 6 & 4 & 0 & 31 & 4 \\ 3 & 6 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 17 & 31 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 4 & 2 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

5.140 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{11111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 848, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 4 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & 2 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 864, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 9 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 & 4 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 6 & 0 & 1 & 3 & 3 \\ 2 & 4 & 4 & 6 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & 6 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 944, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 4 & 0 & 4 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 1472, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 6 & 4 & 0 & 9 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

5.141 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{11111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 492, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 5 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 4 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

5.142 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{112}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -120, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 5 & 0 & 8 & 4 \\ 3 & 6 & 4 & 5 & 0 & 8 & 4 \\ 3 & 5 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 4 & 3 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 79, \Delta = -123, \det = -372, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 79, T \in [14, 79]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 5 & 0 & 4 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & 3 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{1111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 79$, $\Delta = -123$, $\det = -660$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 79$, $T \in [14, 79]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 6 & 4 & 0 & 4 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 4 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

[3] $1^{1111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3$ $T = 9$, $T_{\max} = 79$, $\Delta = -123$, $\det = -816$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 13.10 \rightarrow 14$, $T_{\max} = 79$, $T \in [14, 79]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 6 & 4 & 0 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 4 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

5.143 $\text{NHC}(-2)(-3)(-2) + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$

$T = 10$ [0] $1^{1111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{red}{su}_2$ $T = 10$, $T_{\max} = 15$, $\Delta = -93$, $\det = 492$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$

$T_{\min} = 13.18 \rightarrow 14$, $T_{\max} = 15$, $T \in [14, 15]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 5 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & 3 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

5.144 $\text{NHC}(-2)(-3)(-2) + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_3$

$T = 10$ [0] $1^{1111}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{e}_6 + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_3$ $T = 10$, $T_{\max} = 22$, $\Delta = -102$, $\det = 864$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.43 \rightarrow 14$, $T_{\max} = 22$, $T \in [14, 22]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 5 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 5 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 4 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 4 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

5.145 $\text{NHC}(-2)(-3)(-2) + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4$

$T = 8$ [0] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 44$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.09 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 & 5 \\ 4 & 9 & 7 & 4 & 0 & 5 & 4 \\ 3 & 7 & 4 & 4 & 0 & 4 & 2 \\ 2 & 4 & 4 & 0 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 & 0 \\ 3 & 5 & 4 & 2 & -3 & 0 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 120$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.10 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 & 4 \\ 3 & 4 & 6 & 5 & 0 & 4 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 & 0 \\ 3 & 4 & 4 & 3 & -3 & 0 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{112}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 164$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$

$T_{\min} = 12.32 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 & 4 \\ 3 & 4 & 6 & 8 & 0 & 4 & 4 \\ 3 & 6 & 4 & 8 & 0 & 4 & 4 \\ 4 & 8 & 8 & 9 & 0 & 5 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 & 0 \\ 3 & 4 & 4 & 5 & -3 & 0 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{red}{f}_4$ $T = 8$, $T_{\max} = 35$, $\Delta = -118$, $\det = 176$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$

$T_{\min} = 12.09 \rightarrow 13$, $T_{\max} = 35$, $T \in [13, 35]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 6 & 4 & 0 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 9 & 0 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{\overline{112}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 344, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 6 & 13 & 0 & 4 \\ 3 & 6 & 4 & 13 & 0 & 4 \\ 7 & 13 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{\overline{113}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 35, \Delta = -118, \det = 444, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 35, T \in [12, 35]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 6 & 17 & 0 & 4 \\ 3 & 6 & 4 & 17 & 0 & 4 \\ 9 & 17 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$

5.146 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8$

$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$

$$\begin{pmatrix} 5 & 11 & 11 & 9 & 1 & 5 \\ 11 & 20 & 22 & 17 & 0 & 8 \\ 11 & 22 & 20 & 17 & 0 & 8 \\ 9 & 17 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

5.147 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2$

$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 5 & 3 \\ 4 & 9 & 7 & 4 & 0 & 12 & 5 \\ 3 & 7 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 12 & 9 & 6 & 0 & 13 & 7 \\ 3 & 5 & 4 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.34 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 6 & 4 & 0 & 5 & 4 \\ 3 & 6 & 4 & 4 & 0 & 5 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 4 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{111}} 3^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$

$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 6 & 4 & 0 & 14 & 4 \\ 3 & 6 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 4 & 2 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 6 & 4 & 0 & 16 & 4 \\ 3 & 6 & 4 & 4 & 0 & 16 & 4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 16 & 12 & 0 & 46 & 12 \\ 3 & 4 & 4 & 2 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -167, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 8 & 3 \\ 4 & 9 & 7 & 4 & 0 & 20 & 5 \\ 3 & 7 & 4 & 4 & 0 & 14 & 4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 20 & 14 & 9 & 0 & 37 & 11 \\ 3 & 5 & 4 & 2 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -172, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.16 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 5 & 0 & 4 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 3 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{112} 3^1 1^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -200, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 4 & 3 \\ 3 & 4 & 6 & 9 & 0 & 8 & 4 \\ 3 & 6 & 4 & 9 & 0 & 8 & 4 \\ 5 & 9 & 9 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 8 & 12 & 0 & 10 & 6 \\ 3 & 4 & 4 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{112} 3^1 1^{112} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -200, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 10 & 3 \\ 4 & 9 & 7 & 4 & 0 & 24 & 5 \\ 3 & 7 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 24 & 18 & 12 & 0 & 58 & 14 \\ 3 & 5 & 4 & 2 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1113} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -244, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.43 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 1 & 6 & 3 \\ 3 & 4 & 6 & 4 & 0 & 12 & 4 & 4 \\ 3 & 6 & 4 & 4 & 0 & 12 & 4 & 4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 & 3 \\ 6 & 12 & 12 & 9 & 0 & 25 & 9 & 9 \\ 3 & 4 & 4 & 2 & -3 & 9 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{1112} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -304, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.21 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 6 & 4 & 0 & 8 & 4 \\ 3 & 6 & 4 & 4 & 0 & 8 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 8 & 6 & 0 & 10 & 6 \\ 3 & 4 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1111} 3^1 1^1 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -340, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 6 & 4 & 0 & 4 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1112} 3^1 1^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -340, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 6 & 4 & 0 & 23 & 4 \\ 3 & 6 & 4 & 4 & 0 & 23 & 4 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 23 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 4 & 2 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -404, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 6 & 9 & 0 & 4 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 4 & 6 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1111} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -464, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 6 & 4 & 0 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 4 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1113} 31^{14} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -788, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.29 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 6 & 4 & 0 & 32 & 4 \\ 3 & 6 & 4 & 4 & 0 & 32 & 4 \\ 2 & 4 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 32 & 32 & 21 & 0 & 193 & 25 \\ 3 & 4 & 4 & 2 & -3 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1112} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -788, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.09 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 6 & 4 & 0 & 13 & 4 \\ 3 & 6 & 4 & 4 & 0 & 13 & 4 \\ 2 & 4 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 4 & 2 & -3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1112} 31^{12} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -800, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.7$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 6 & 4 & 0 & 18 & 4 \\ 3 & 6 & 4 & 4 & 0 & 18 & 4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 4 & 2 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{1113} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1088, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.10 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 6 & 4 & 0 & 17 & 4 \\ 3 & 6 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 17 & 17 & 12 & 0 & 49 & 13 \\ 3 & 4 & 4 & 2 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1113} 31^{13} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1360, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 6 & 4 & 0 & 27 & 4 \\ 3 & 6 & 4 & 4 & 0 & 27 & 4 \\ 2 & 4 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 4 & 2 & -3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{1114} 31^{11} + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{green}{su}_2] + \textcolor{orange}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -1364, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 6 & 4 & 0 & 21 & 4 \\ 3 & 6 & 4 & 4 & 0 & 21 & 4 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 21 & 15 & 0 & 76 & 16 \\ 3 & 4 & 4 & 2 & -3 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{1113} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -1460, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.09 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 6 & 4 & 0 & 22 & 4 \\ 3 & 6 & 4 & 4 & 0 & 22 & 4 \\ 2 & 4 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 4 & 2 & -3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1114} 31^{12} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -2096, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.09 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 6 & 4 & 0 & 26 & 4 \\ 3 & 6 & 4 & 4 & 0 & 26 & 4 \\ 2 & 4 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 14 & 26 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 4 & 2 & -3 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1114} 31^{14} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -2144, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.13 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 6 & 4 & 0 & 36 & 4 \\ 3 & 6 & 4 & 4 & 0 & 36 & 4 \\ 2 & 4 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 4 & 2 & -3 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1114} 31^{13} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -2356, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.10 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 6 & 4 & 0 & 31 & 4 \\ 3 & 6 & 4 & 4 & 0 & 31 & 4 \\ 2 & 4 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 17 & 31 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 4 & 2 & -3 & 24 & 0 \end{pmatrix}_{b_0}$$

5.148 NHC(-2)(-3)(-2) + $\mathfrak{su}_3$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11111} 31^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 672, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 9 & 0 & 4 & 4 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 & 4 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 4 & 6 & 0 & 1 & 3 & 3 \\ 2 & 4 & 4 & 6 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & 6 & -3 & 3 & 3 & 1 \end{pmatrix}_{b_0}$$

5.149 NHC(-2)(-3)(-2) + $\mathfrak{su}_3$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -90, \quad \det = -72, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.11 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 2 & 3 \\ 4 & 9 & 7 & 4 & 0 & 4 & 5 \\ 3 & 7 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 2 & 0 & 0 & 2 \\ 3 & 5 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{112} 3^{11} 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -90, \quad \det = -216, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.33 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 2 & 3 \\ 3 & 4 & 6 & 8 & 0 & 3 & 4 \\ 3 & 6 & 4 & 8 & 0 & 3 & 4 \\ 4 & 8 & 8 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 4 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -90, \ \det = -288, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.11 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 6 & 4 & 0 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

5.150 NHC(-2)(-3)(-2) + $\mathfrak{su}_3$ + $\mathfrak{f}_4$ + su2n3mix

$$T = 9 \quad [0] \ 1^{\overline{1111}} 3^1 1^{\overline{12}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -90, \ \det = -116, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.34 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 6 & 4 & 0 & 17 & 4 \\ 3 & 6 & 4 & 4 & 0 & 17 & 4 \\ 2 & 4 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 17 & 17 & 11 & 0 & 55 & 13 \\ 3 & 4 & 4 & 2 & -3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -90, \ \det = -468, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.15 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 6 & 4 & 0 & 7 & 4 \\ 3 & 6 & 4 & 4 & 0 & 7 & 4 \\ 2 & 4 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 7 & 5 & 0 & 7 & 5 \\ 3 & 4 & 4 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^1 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -90, \ \det = -528, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.09 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 6 & 4 & 0 & 12 & 4 \\ 3 & 6 & 4 & 4 & 0 & 12 & 4 \\ 2 & 4 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 12 & 8 & 0 & 25 & 9 \\ 3 & 4 & 4 & 2 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

5.151 NHC(-2)(-3)(-2) + $\mathfrak{su}_3$ + $\mathfrak{f}_4$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{12}} 3^1 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -88, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.4$$

$$T_{\min} = 12.32 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 5 & 0 & 8 & 4 \\ 3 & 6 & 4 & 5 & 0 & 8 & 4 \\ 3 & 5 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 8 & 6 & 0 & 9 & 5 \\ 3 & 4 & 4 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^1 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -292, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.30 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 5 & 0 & 4 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 & 4 \\ 3 & 5 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -516, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.60 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 6 & 4 & 0 & 4 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 4 \\ 2 & 4 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3^{\overline{11}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{f}_4 + \textcolor{blue}{su}_3 \quad T = 9, \ T_{\max} = 15, \ \Delta = -97, \ \det = -640, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.27 \rightarrow 13, \ T_{\max} = 15, \ T \in [13, 15]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 6 & 4 & 0 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 4 \\ 2 & 4 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

5.152 NHC(-2)(-3)(-2) + $\textcolor{blue}{su}_3 + \textcolor{blue}{so}_8$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 29, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 1 & 1 & 3 \\ 4 & 9 & 7 & 0 & 0 & 5 \\ 3 & 7 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 2 & 1 & 3 \\ 4 & 9 & 7 & 4 & 0 & 5 \\ 3 & 7 & 4 & 4 & 0 & 4 \\ 2 & 4 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 5 & 4 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 37, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 5 \\ 12 & 25 & 23 & 0 & 9 \\ 11 & 23 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 9 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 44, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.09 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 10 & 15 & 15 & 13 & 6 \\ 15 & 20 & 22 & 17 & 8 \\ 15 & 22 & 20 & 17 & 8 \\ 13 & 17 & 17 & 12 & 6 \\ 6 & 8 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 48, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.25 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 5 & 7 & 4 & 1 & 5 \\ 7 & 8 & 6 & 0 & 6 \\ 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 6 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 52, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 6 & 0 & 0 & 4 \\ 3 & 6 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 55, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 5 & 7 & 11 & 1 & 5 \\ 7 & 8 & 15 & 0 & 6 \\ 11 & 15 & 21 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 6 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 63, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 5 & 7 & 7 & 4 & 5 \\ 7 & 8 & 9 & 6 & 6 \\ 7 & 9 & 7 & 5 & 5 \\ 4 & 6 & 5 & 0 & 2 \\ 5 & 6 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{1}} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{red}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 13, \ \Delta = -94, \ \det = 76, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$[19] \quad 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 9 & 0 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 21^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 200, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 20 & 20 & 12 & 7 \\ 20 & 25 & 27 & 16 & 9 \\ 20 & 27 & 25 & 16 & 9 \\ 12 & 16 & 16 & 9 & 5 \\ 7 & 9 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 208, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 6 & 12 & 0 & 4 \\ 3 & 6 & 4 & 12 & 0 & 4 \\ 7 & 12 & 12 & 24 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 21^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.10 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 16 & 16 & 12 & 6 \\ 16 & 20 & 22 & 17 & 8 \\ 16 & 22 & 20 & 17 & 8 \\ 12 & 17 & 17 & 12 & 6 \\ 6 & 8 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 6 & 13 & 0 & 4 \\ 3 & 6 & 4 & 13 & 0 & 4 \\ 7 & 13 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{112} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 1 & 3 \\ 3 & 4 & 6 & 13 & 0 & 4 \\ 3 & 6 & 4 & 13 & 0 & 4 \\ 8 & 13 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.28 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 6 & 17 & 0 & 4 \\ 3 & 6 & 4 & 17 & 0 & 4 \\ 9 & 17 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{113} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 428, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.09 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 6 & 17 & 0 & 4 \\ 3 & 6 & 4 & 17 & 0 & 4 \\ 10 & 17 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

5.153 $\text{NHC}(-2)(-3)(-2) + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8 + \textcolor{blue}{e}_6$

$$T = 9 \quad [0] \quad 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8 + \textcolor{blue}{e}_6 \quad T = 9, T_{\max} = 99, \Delta = -143, \det = -524, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 11 & 11 & 9 & 1 & 5 \\ 11 & 20 & 22 & 17 & 0 & 8 \\ 11 & 22 & 20 & 17 & 0 & 8 \\ 9 & 17 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 8 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

5.154 $\text{NHC}(-2)(-3)(-2) + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8 + \textcolor{blue}{f}_4$

$$T = 9 \quad [0] \quad 1^{111} 3 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{so}_7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{so}_8 + \textcolor{blue}{f}_4 \quad T = 9, T_{\max} = 32, \Delta = -117, \det = -412, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 32, T \in [13, 32]$$

$$\begin{pmatrix} 5 & 11 & 11 & 9 & 1 & 5 \\ 11 & 20 & 22 & 17 & 0 & 8 \\ 11 & 22 & 20 & 17 & 0 & 8 \\ 9 & 17 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 8 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

5.155 NHC(-2)(-3)(-2) + $\mathfrak{su}_3$ + $\mathfrak{so}_8$ + $\mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 12 & 11 & 1 & 1 & 5 \\ 12 & 25 & 23 & 0 & 0 & 9 \\ 11 & 23 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 9 & 8 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 7 & 7 & 4 & 1 & 5 \\ 7 & 8 & 9 & 6 & 0 & 6 \\ 7 & 9 & 7 & 5 & 0 & 5 \\ 4 & 6 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 6 & 5 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -180, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 11 & 1 & 1 & 5 \\ 11 & 20 & 22 & 0 & 0 & 8 \\ 11 & 22 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 8 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 14, \Delta = -93, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 5 & 11 & 11 & 9 & 1 & 5 \\ 11 & 20 & 22 & 17 & 0 & 8 \\ 11 & 22 & 20 & 17 & 0 & 8 \\ 9 & 17 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 8 & 8 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

5.156 NHC(-2)(-3)(-2) + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 11 \quad [0] \quad 1^{11} 5 13 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 0 & 2 & 0 & 3 & 2 \\ 1 & 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 3 & 1 & 0 & 1 & 3 \\ 2 & 2 & 2 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 5 13 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 0 & 2 & 0 & 3 & 2 \\ 1 & 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 5 13 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 2 & 3 & 0 & 4 & 4 \\ 2 & 2 & 2 & 2 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 5 13 1^{11} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 2 \\ 1 & 2 & 0 & 2 & 0 & 4 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 4 & 2 & 0 & 5 & 5 \\ 2 & 2 & 2 & 2 & -1 & 5 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 0 & 5 & 2 \\ 1 & 2 & 0 & 2 & 0 & 8 & 2 \\ 1 & 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 5 & 8 & 5 & 0 & 24 & 10 \\ 2 & 2 & 2 & 2 & -1 & 10 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 1 & 2 & 0 & 2 & 0 & 5 & 2 \\ 1 & 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 4 & 5 & 4 & 0 & 11 & 7 \\ 2 & 2 & 2 & 2 & -1 & 7 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 0 & 5 & 2 \\ 1 & 2 & 0 & 2 & 0 & 6 & 2 \\ 1 & 1 & 2 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 5 & 6 & 6 & 0 & 21 & 9 \\ 2 & 2 & 2 & 2 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 6 & 2 \\ 1 & 2 & 0 & 2 & 0 & 9 & 2 \\ 1 & 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 6 & 9 & 7 & 0 & 36 & 12 \\ 2 & 2 & 2 & 2 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 7 & 2 \\ 1 & 2 & 0 & 2 & 0 & 7 & 2 \\ 1 & 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 7 & 7 & 7 & 0 & 33 & 11 \\ 2 & 2 & 2 & 2 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 2 & 0 & 10 & 2 \\ 1 & 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 8 & 10 & 8 & 0 & 50 & 14 \\ 2 & 2 & 2 & 2 & -1 & 14 & 3 \end{pmatrix}_b$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 9 & 2 \\ 1 & 2 & 0 & 2 & 0 & 13 & 2 \\ 1 & 1 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 9 & 13 & 9 & 0 & 73 & 17 \\ 2 & 2 & 2 & 2 & -1 & 17 & 3 \end{pmatrix}_b$$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 9 & 2 \\ 1 & 2 & 0 & 2 & 0 & 11 & 2 \\ 1 & 1 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 9 & 11 & 10 & 0 & 68 & 16 \\ 2 & 2 & 2 & 2 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[12] $1^{12} \overset{1}{5} 131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -227, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$

$$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 2 \\ 1 & 2 & 0 & 2 & 0 & 14 & 2 \\ 1 & 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 10 & 14 & 11 & 0 & 93 & 19 \\ 2 & 2 & 2 & 2 & -1 & 19 & 3 \end{pmatrix}_{b_0}$$

[13] $1^{13} \overset{1}{5} 131^{12} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -242, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 11 & 2 \\ 1 & 2 & 0 & 2 & 0 & 12 & 2 \\ 1 & 1 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 11 & 12 & 11 & 0 & 88 & 18 \\ 2 & 2 & 2 & 2 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[14] $1^{13} \overset{1}{5} 131^{13} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -314, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 12 & 2 \\ 1 & 2 & 0 & 2 & 0 & 15 & 2 \\ 1 & 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 9 & 12 & 15 & 12 & 0 & 115 & 21 \\ 2 & 2 & 2 & 2 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[15] $1^{13} \overset{1}{5} 131^{14} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -338, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 1 & 0 & 13 & 2 \\ 1 & 2 & 0 & 2 & 0 & 18 & 2 \\ 1 & 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 10 & 13 & 18 & 13 & 0 & 148 & 24 \\ 2 & 2 & 2 & 2 & -1 & 24 & 3 \end{pmatrix}_{b_0}$$

5.157 NHC(-2)(-3)(-2) + $\textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$

$T = 12$ [0] $1^{11} \overset{1}{5} 131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 4 & 2 \\ 1 & 2 & 0 & 2 & 0 & 5 & 2 \\ 1 & 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 3 & 1 & 0 & 1 & 5 \\ 3 & 4 & 5 & 4 & 0 & 5 & 11 \\ 2 & 2 & 2 & 2 & -1 & 3 & 7 \end{pmatrix}_{b_0}$$

[1] $1^{12} \overset{1}{5} 131^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 2 & 0 & 3 & 10 \\ 1 & 1 & 2 & 0 & 0 & 1 & 8 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 \\ 1 & 1 & 3 & 1 & 0 & 1 & 10 \\ 6 & 8 & 10 & 8 & 0 & 10 & 50 \\ 2 & 2 & 2 & 2 & -1 & 3 & 14 \end{pmatrix}_{b_0}$$

5.158 NHC(-2)(-3)(-2) + $\textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3)$

$T = 12$ [0] $1^{11} \overset{1}{5} 131^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{s0}7 \cdot \textcolor{blue}{su}_2] + \textcolor{blue}{su}_3 + \textcolor{blue}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3)$ $T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 2 & 1 \\ 1 & 2 & 0 & 2 & 0 & 1 & 2 \\ 1 & 1 & 2 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 \\ 1 & 2 & 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 2 & 1 & 0 & 1 & 0 \\ 2 & 2 & 2 & 2 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

5.159 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 12 \quad [0] \quad 1^{11} \overset{1}{5} 13^1 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{green}{\mathfrak{su}_2}] + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2} + \textcolor{blue}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{violet}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 9.0$$

5.160 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \quad 1^1 \overset{1}{5} 13^1 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{green}{\mathfrak{su}_2}] + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1] \quad 1^{11} \overset{1}{6} 123^1 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{green}{\mathfrak{su}_2}] + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 18, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

5.161 $\text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 12 \quad [0] \quad 1^{11} \overset{1}{5} 13^1 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{green}{\mathfrak{su}_2}] + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 12, T_{\max} = 12, \Delta = -2, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 & 1 \\ 1 & 1 & 2 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 \\ 1 & 1 & 2 & 1 & 0 & 0 & 1 \\ 2 & 2 & 2 & 2 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 13 \quad [1] \quad 1^{11} \overset{1}{6} 1^{11} 6^{11} 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{green}{\mathfrak{su}_2}] + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 5 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 & 5 & 2 \\ 1 & 1 & 2 & 0 & 0 & 1 & 4 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 5 & 2 \\ 4 & 5 & 5 & 4 & 0 & 5 & 18 & 8 \\ 2 & 2 & 2 & 2 & -1 & 2 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} \overset{1}{6} 123^1 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{green}{\mathfrak{su}_2}] + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -9, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 0 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} \overset{1}{6} 1^{11} 6^{11} 1^1 + [\textcolor{red}{\mathfrak{su}_2} \cdot \textcolor{blue}{\mathfrak{so}_7} \cdot \textcolor{green}{\mathfrak{su}_2}] + \textcolor{orange}{\mathfrak{su}_3} + \textcolor{red}{\mathfrak{su}_2 \rightarrow (-3)} + \textcolor{blue}{\mathfrak{su}_2} \quad T = 13, T_{\max} = 13, \Delta = -1, \det = -36, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 6 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 & 6 & 2 \\ 1 & 1 & 2 & 0 & 0 & 1 & 6 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 1 & 0 & 0 & 6 & 2 \\ 5 & 6 & 6 & 6 & 0 & 6 & 28 & 10 \\ 2 & 2 & 2 & 2 & -1 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$$5.162 \quad \text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2$$

$$T = 12 \quad [0] \quad 1^{\overline{11}} \overset{1^1}{5} 13^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 9.0$$

$$T = 13 \quad [1] \quad 1^{\overline{11}} \overset{1^1}{6} 123^1 1^1 + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 13, T_{\max} = 13, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 13, 1), c_{\text{ext}} = 9.0$$

$$5.163 \quad \text{NHC}(-2)(-3)(-2) + \mathfrak{su}_3 + \mathfrak{su}_{2n3\text{mix}}$$

$$T = 12 \quad [0] \quad 1^{\overline{11}} \overset{1^1}{6} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 72, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 7 & 2 \\ 1 & 2 & 0 & 2 & 0 & 8 & 2 \\ 1 & 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 6 & 7 & 8 & 7 & 0 & 42 & 12 \\ 2 & 2 & 2 & 2 & -1 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} \overset{1^1}{6} 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{so}_7 \cdot \mathfrak{su}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 124, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 8 & 2 \\ 1 & 2 & 0 & 2 & 0 & 11 & 2 \\ 1 & 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 8 & 11 & 8 & 0 & 61 & 15 \\ 2 & 2 & 2 & 2 & -1 & 15 & 3 \end{pmatrix}_{b_0}$$

6 NHC $(-2)(-2)(-3)$ External (186 combos, 2947 bases)

6.1 NHC $(-2)(-2)(-3)$

$$T = 12 \quad [374] \quad 1^{\overline{1}} 231 \overline{5} 1321^{\overline{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 2 & 4 \\ 1 & 0 & 2 & 1 & 3 \\ 2 & 0 & 4 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$[375] \quad 1^{\overline{1}} 31 \overline{5} 13221^{\overline{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 12, T_{\max} = 12, \Delta = -3, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 3 & 2 & 5 \\ 1 & 0 & 2 & 0 & 2 \\ 2 & 0 & 5 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_9$$

$$T = 14 \quad [456] \quad 1^{\overline{1}} \overline{7}^{12321} 12321^{\overline{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 14, T_{\max} = 14, \Delta = -3, \det = 1, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 3 & 5 \\ 3 & 2 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$T = 15 \quad [498] \quad 1^{\overline{1}} \overline{8}^{12321} 12321^{\overline{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 15, T_{\max} = 15, \Delta = -2, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 2 & 2 & 4 \\ 1 & 1 & 2 & 1 & 3 \\ 2 & 2 & 4 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$[499] \quad 1^{\overline{1}} \overline{6}^{12315131} 131^{\overline{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 15, T_{\max} = 15, \Delta = -2, \det = -1, (n_+, n_-, n_0) = (1, 15, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{12}$$

$$T = 16 \quad [500] \quad 1^{\overline{1}} \overline{6}^{131} 131612321^{\overline{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 16, T_{\max} = 16, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 16, T_{\max} = 16, T \in [16, 16]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{13}$$

$$T = 17 \quad [504] \quad 1^{\overline{1}} \overline{8}^{123151321} 1231^{\overline{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 17, T_{\max} = 17, \Delta = -1, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 3 & 2 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$[505] \quad 1^{\overline{1}} \overline{8}^{13151321} 12321^{\overline{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 17, T_{\max} = 17, \Delta = -2, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 3 & 5 \\ 3 & 2 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$$

$$[506] \quad 1^{\overline{1}} \overline{8}^{13151321} 12321^{\overline{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] \quad T = 17, T_{\max} = 17, \Delta = -2, \det = -1, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{14}$$

$$T = 18 \quad [509] \quad 1^{\overline{1}1} 8^{\overline{1}2321} 123216131^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 18, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 3 & 5 \\ 3 & 4 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$[510] \quad 1^{\overline{1}1} 8^{\overline{1}2321} 123216131^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 18, \Delta = -1, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 0 & 3 \\ 1 & 3 & 3 & 1 & 5 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 5 & 0 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$$

$$[511] \quad 1^{\overline{1}1} 6^{\overline{1}31} 13161315131^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 18, T_{\max} = 18, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 18, T_{\max} = 18, T \in [18, 18]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{15}$$

$$T = 19 \quad [514] \quad 1^{\overline{1}1} 8^{\overline{1}2321} 1232171231^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 19, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 4 \\ 2 & 3 & 3 & 5 \\ 3 & 4 & 5 & 7 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{17}$$

$$[515] \quad 1^{\overline{1}1} 8^{\overline{1}2321} 1232171231^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 19, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 3 & 0 & 3 \\ 1 & 3 & 3 & 1 & 5 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 5 & 0 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

$$[516] \quad 1^{\overline{1}1} 8^{\overline{1}321} 12321712321^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 19, T_{\max} = 19, \Delta = 0, \det = -1, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 19, T_{\max} = 19, T \in [19, 19]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 2 & 3 & 2 & 3 & 6 \end{pmatrix}_{b_0} \rightarrow \text{bd}_{16}$$

6.2 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$

$$T = 8 \quad [0] \quad 1^{\overline{1}1} 3^{\overline{1}1} 1^{\overline{1}1} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 16, T_{\max} = 92, T \in [16, 92]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 3 & 5 & 0 & 4 & 4 \\ 3 & 3 & 4 & -4 & 4 & -3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_4$$

6.3 NHC(-2)(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 12 \quad [1] \quad 1^{\overline{1}1} 5^{\overline{1}1} 13^{\overline{1}1} 1^{\overline{1}1} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 5 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 3 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 4 & 2 \\ 3 & 5 & 3 & 5 & 0 & 4 & 13 & 7 \\ 2 & 2 & 2 & 2 & -1 & 2 & 7 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

$$[0] \quad 1^{\overline{1}1} 5^{\overline{1}1} 13^{\overline{1}1} 1^{\overline{1}1} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 1, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 3 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 4 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 3 & 2 \\ 2 & 3 & 2 & 4 & 0 & 3 & 5 & 5 \\ 2 & 2 & 2 & 2 & -1 & 2 & 5 & 3 \end{pmatrix}_{b_0} \rightarrow \text{bd}_6$$

[3] $1^{11} 5^{11} 13^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 1$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 8.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 4 & 0 & 2 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 2 & 4 & 0 & 1 & 3 & 3 \\ 2 & 2 & 5 & 0 & 3 & 2 & 4 \\ 3 & 3 & 5 & -1 & 3 & 4 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

[2] $1^{12} 5^{11} 13^{11} 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 1$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 8.5$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 10 & 3 \\ 2 & 1 & 4 & 0 & 2 & 12 & 3 \\ 3 & 4 & 5 & 0 & 4 & 21 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 2 & 4 & 0 & 1 & 13 & 3 \\ 10 & 12 & 21 & 0 & 13 & 70 & 18 \\ 3 & 3 & 5 & -1 & 3 & 18 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

6.4 NHC(-2)(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{su}_2$

$T = 12$ [2] $1^{13} 5^{11} 13^{11} 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 1$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 8.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 2 & 3 \\ 2 & 1 & 4 & 0 & 17 & 2 & 3 \\ 3 & 4 & 5 & 0 & 28 & 5 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 14 & 17 & 28 & 0 & 130 & 21 & 24 \\ 2 & 2 & 5 & 0 & 21 & 2 & 4 \\ 3 & 3 & 5 & -1 & 24 & 4 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

[1] $1^{12} 5^{11} 13^{11} 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 1$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 8.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 2 & 3 \\ 2 & 1 & 4 & 0 & 12 & 2 & 3 \\ 3 & 4 & 5 & 0 & 20 & 5 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 10 & 12 & 20 & 0 & 65 & 15 & 17 \\ 2 & 2 & 5 & 0 & 15 & 2 & 4 \\ 3 & 3 & 5 & -1 & 17 & 4 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

[0] $1^{11} 5^{11} 13^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = 0$, $\det = 1$, $(n_+, n_-, n_0) = (1, 12, 0)$, $c_{\text{ext}} = 8.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 2 & 3 \\ 2 & 1 & 4 & 0 & 7 & 2 & 3 \\ 3 & 4 & 5 & 0 & 12 & 5 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 6 & 7 & 12 & 0 & 22 & 9 & 10 \\ 2 & 2 & 5 & 0 & 9 & 2 & 4 \\ 3 & 3 & 5 & -1 & 10 & 4 & 4 \end{pmatrix}_{b_0} \rightarrow \text{bd}_7$$

6.5 NHC(-2)(-2)(-3)

$T = 6$ [0] $21^1 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 6$, $T_{\max} = 26$, $\Delta = -116$, $\det = 5$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$

$$T_{\min} = 10, T_{\max} = 26, T \in [10, 26]$$

$$\begin{pmatrix} 4 & 3 & 5 & 4 \\ 3 & 1 & 5 & 3 \\ 5 & 5 & 4 & 4 \\ 4 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

[1] $21^1 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 6$, $T_{\max} = 26$, $\Delta = -116$, $\det = 7$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$

$$T_{\min} = 12, T_{\max} = 26, T \in [12, 26]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 1 & 5 & 3 \\ 4 & 5 & 9 & 5 \\ 3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

[2] $1^1 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 6$, $T_{\max} = 26$, $\Delta = -116$, $\det = 11$, $(n_+, n_-, n_0) = (1, 6, 0)$, $c_{\text{ext}} = 4.0$

$$T_{\min} = 9.45 \rightarrow 10, T_{\max} = 26, T \in [10, 26]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 2 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{11}} 1^{\overline{1}} 3^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 6, T_{\max} = 26, \Delta = -116, \det = 12, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 26, T \in [11, 26]$$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 5 & 3 \\ 4 & 5 & 9 & 5 \\ 3 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1}} 1^{\overline{1}} 3^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 6, T_{\max} = 26, \Delta = -116, \det = 14, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.43 \rightarrow 11, T_{\max} = 26, T \in [11, 26]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 3 \\ 3 & 5 & 4 & 4 \\ 3 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{1}} 1^{\overline{1}} 3^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 6, T_{\max} = 26, \Delta = -116, \det = 16, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.25 \rightarrow 11, T_{\max} = 26, T \in [11, 26]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 4 \\ 3 & 6 & 4 & 4 \\ 3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 21^{\overline{1}} 3^{\overline{11}} 1^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 6, T_{\max} = 26, \Delta = -116, \det = 24, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 26, T \in [10, 26]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 6 & 4 \\ 4 & 6 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{\overline{11}} 1^{\overline{1}} 3^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 6, T_{\max} = 26, \Delta = -116, \det = 26, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 9 & 4 \\ 4 & 9 & 9 & 5 \\ 3 & 4 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 2131^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 6, T_{\max} = 26, \Delta = -116, \det = 26, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 9 & 4 \\ 4 & 9 & 9 & 5 \\ 3 & 4 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[9] \quad 21^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 6, T_{\max} = 26, \Delta = -116, \det = 31, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.45 \rightarrow 11, T_{\max} = 26, T \in [11, 26]$$

$$\begin{pmatrix} 4 & 3 & 8 & 4 \\ 3 & 1 & 5 & 3 \\ 8 & 5 & 9 & 5 \\ 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 21^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 6, T_{\max} = 26, \Delta = -116, \det = 36, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 26, T \in [10, 26]$$

$$\begin{pmatrix} 4 & 6 & 5 & 4 \\ 6 & 4 & 6 & 4 \\ 5 & 6 & 4 & 4 \\ 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[11] \quad 21^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 6, T_{\max} = 26, \Delta = -116, \det = 47, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 26, T \in [10, 26]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 1 & 5 & 3 \\ 4 & 5 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[12] \quad 131^{\overline{11}} 2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 6, T_{\max} = 26, \Delta = -116, \det = 104, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.42 \rightarrow 11, T_{\max} = 26, T \in [11, 26]$$

$$\begin{pmatrix} 4 & 6 & 8 & 4 \\ 6 & 4 & 9 & 4 \\ 8 & 9 & 9 & 5 \\ 4 & 4 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[13] \quad 221^{\overline{11}} 3 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 6, T_{\max} = 26, \Delta = -116, \det = 234, (n_+, n_-, n_0) = (1, 6, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.75 \rightarrow 10, T_{\max} = 26, T \in [10, 26]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 3 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 7 \quad [14] \quad 21^{\overline{1}} 1^{\overline{1}} 3^{\overline{11}} 21^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

- [15] $1 \overset{21^1}{4} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -5$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 4 & 3 & 5 & 4 \\ 3 & 1 & 5 & 3 \\ 5 & 5 & 4 & 4 \\ 4 & 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [16] $131^1 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -5$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 1 & 4 & 1 & 4 & 3 \\ 2 & 3 & 4 & 3 & 3 \\ 2 & 3 & 3 & 3 & 2 \end{pmatrix}_{b_0}$
- [17] $1^1 1^1 321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -6$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$
- [18] $1^1 1^1 321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -7$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.57 \rightarrow 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$
- [19] $1 \overset{21^1}{4} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -7$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 13$, $T_{\max} = 19$, $T \in [13, 19]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 1 & 5 & 3 \\ 4 & 5 & 9 & 5 \\ 3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$
- [20] $1^1 1^1 321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 2 & 6 & 4 \\ 3 & 6 & 5 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$
- [21] $1^{11} 1^1 3131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 \\ 1 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 3 & 3 & -1 & 2 \end{pmatrix}_{b_0}$
- [22] $21^{11} 1^1 41^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.50 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 4 & 6 & 8 & 4 \\ 6 & 8 & 13 & 6 \\ 8 & 13 & 13 & 7 \\ 4 & 6 & 7 & 3 \end{pmatrix}_{b_0}$
- [23] $1^1 1^1 1^1 321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 3 & 1 & 2 \\ 1 & 3 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [24] $1^1 1^1 321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -8$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.50 \rightarrow 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$
- [25] $1^1 1^1 1^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -11$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [26] $1^1 1^1 1^1 321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -11$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

- $$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 4 \\ 3 & 6 & 5 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$
- [27] $1^1 1321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 3 & 2 \\ 1 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [28] $1^1 131^1 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 0 & 3 & 2 \end{pmatrix}_{b_0}$$
- [29] $1^{11} 3131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 3 & 1 & 0 & 2 \\ 2 & 3 & 0 & 2 & 2 \end{pmatrix}_{b_0}$$
- [30] $21^1 41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 19, \Delta = -107, \det = -11, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 4 & 3 & 5 & 4 \\ 3 & 1 & 4 & 3 \\ 5 & 4 & 4 & 4 \\ 4 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$
- [32] $1^1 1^1 41 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 19, \Delta = -107, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 5 & 3 \\ 4 & 5 & 9 & 5 \\ 3 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$
- [31] $21^1 41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 19, \Delta = -107, \det = -12, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 5 & 3 \\ 4 & 5 & 9 & 5 \\ 3 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$
- [34] $1^1 1^1 41 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 19, \Delta = -107, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 3 \\ 3 & 5 & 4 & 4 \\ 3 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$
- [33] $21^1 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 19, \Delta = -107, \det = -14, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.43 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 3 \\ 3 & 5 & 4 & 4 \\ 3 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$
- [35] $21^1 41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 19, \Delta = -107, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$
- [36] $1^1 1^1 31^1 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 0 & 3 \\ 2 & 4 & 4 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 3 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$
- [37] $21^1 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 19, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.25 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 4 \\ 3 & 6 & 4 & 4 \\ 3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$
- [38] $1^1 1^1 41 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 19, \Delta = -107, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$

- $T_{\min} = 11.25 \rightarrow 12$, $T_{\max} = 19$, $T \in [12, 19]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 4 \\ 3 & 6 & 4 & 4 \\ 3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$
- [39] $1^1 31^1 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -16$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10$, $T_{\max} = 12$, $T \in [10, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 4 \\ 1 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$
- [40] $31^1 231^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 13$, $\Delta = -96$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.26 \rightarrow 11$, $T_{\max} = 13$, $T \in [11, 13]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 4 \\ 5 & 6 & 5 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$
- 1^1
 [41] $2231^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.26 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 6 & 4 \\ 4 & 6 & 5 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$
- 1^1
 [42] $21^1 41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.40 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 4 & 3 \\ 4 & 4 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [43] $21^{11} 321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -23$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 0 & 3 & 1 & 2 \\ 2 & 3 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [45] $21^{11} 41^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 3 & 6 & 4 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 6 & 3 & 3 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- 21^1
 [44] $1^1 4^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 6 & 4 \\ 4 & 6 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- 1^{11}
 [46] $2^1 2^1 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.50 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 1 & 4 & 3 \\ 2 & 1 & 4 & 0 & 2 \\ 2 & 2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [47] $131^1 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 2 & 4 & 4 & 4 & 4 \\ 1 & 3 & 4 & 0 & 2 \\ 2 & 3 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- 1^{111}
 [48] $1^1 4^1 12 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -26$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 13.08 \rightarrow 14$, $T_{\max} = 19$, $T \in [14, 19]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 9 & 4 \\ 4 & 9 & 9 & 5 \\ 3 & 4 & 5 & 0 \end{pmatrix}_{b_0}$
- [49] $22141^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -26$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 13.08 \rightarrow 14$, $T_{\max} = 19$, $T \in [14, 19]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 9 & 4 \\ 4 & 9 & 9 & 5 \\ 3 & 4 & 5 & 0 \end{pmatrix}_{b_0}$
- 1^1
 [50] $1^1 4^1 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -26$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 13.08 \rightarrow 14$, $T_{\max} = 19$, $T \in [14, 19]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 9 & 4 \\ 4 & 9 & 9 & 5 \\ 3 & 4 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 2 \ 2 \ 3 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[52] \quad 1 \ 4 \ 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 19, \Delta = -107, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 4 & 3 & 8 & 4 \\ 3 & 1 & 5 & 3 \\ 8 & 5 & 9 & 5 \\ 4 & 3 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[53] \quad 13 1^{11} 3 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -31, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 1 & 4 & 3 & 3 \\ 2 & 4 & 4 & 7 & 4 \\ 2 & 3 & 7 & 3 & 3 \\ 2 & 3 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[54] \quad 22 1^{11} 4 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 19, \Delta = -107, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 9 & 4 \\ 6 & 9 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[55] \quad 13 1^{11} 3 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 12, \Delta = -95, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 4 & 4 & 6 & 4 \\ 1 & 4 & 1 & 4 & 3 \\ 2 & 6 & 4 & 3 & 3 \\ 2 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[56] \quad 1 \ 4 \ 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 19, \Delta = -107, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 4 & 6 & 5 & 4 \\ 6 & 4 & 6 & 4 \\ 5 & 6 & 4 & 4 \\ 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[57] \quad 2 \ 3 \ 1^{11} 3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 13, \Delta = -96, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.30 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$$

$$\begin{pmatrix} 0 & 3 & 2 & 2 & 2 \\ 3 & 1 & 3 & 2 & 3 \\ 2 & 3 & 1 & 4 & 3 \\ 2 & 2 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[60] \quad 2 \ 3 \ 1^{11} 3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 13, \Delta = -96, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.41 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 3 & 2 \\ 2 & 1 & 4 & 2 & 3 \\ 2 & 4 & 1 & 3 & 3 \\ 3 & 2 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[58] \quad 2 \ 2 \ 3 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 11, \Delta = -88, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.83 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 2 & 6 & 4 \\ 4 & 6 & 5 & 5 \\ 3 & 4 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[59] \quad 3 1^{11} 2 3 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 13, \Delta = -96, \det = -41, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 3 & 2 & 1 & 4 & 3 \\ 3 & 1 & 4 & 0 & 2 \\ 2 & 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[61] \quad 3 1^{11} 3 1^{11} 3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 19, \Delta = -103, \det = -45, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.44 \rightarrow 12, T_{\max} = 19, T \in [12, 19]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 3 & 4 & 0 & 2 \\ 2 & 4 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[63] \quad 1 \ 4 \ 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 19, \Delta = -107, \det = -47, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

- [62] $131^{11}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -47$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.79 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 1 & 5 & 3 \\ 4 & 5 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [64] $2\overset{11}{3}2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -90$, $\det = -48$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.33 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 4 & 7 & 3 & 4 \\ 2 & 7 & 4 & 4 & 4 \\ 1 & 3 & 4 & 0 & 2 \\ 2 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [65] $31^{11}231^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 13$, $\Delta = -96$, $\det = -51$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11$, $T_{\max} = 13$, $T \in [11, 13]$
 $\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$
- [66] $2\overset{11}{3}2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -90$, $\det = -56$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 2 & 3 \\ 2 & 4 & 1 & 3 & 3 \\ 2 & 2 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [67] $221^{11}41^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -62$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.90 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 4 & 7 & 4 \\ 3 & 2 & 7 & 3 & 3 \\ 2 & 2 & 4 & 3 & 2 \end{pmatrix}_{b_0}$
- [68] $2\overset{11}{3}2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -90$, $\det = -64$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10$, $T_{\max} = 11$, $T \in [10, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 3 \end{pmatrix}_{b_0}$
- [69] $221^{11}41^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -66$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.55 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 4 & 3 & 6 & 4 \\ 1 & 3 & 0 & 3 & 2 \\ 3 & 6 & 3 & 3 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [70] $31^{11}31^13 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -103$, $\det = -72$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.44 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 7 & 3 & 4 \\ 3 & 7 & 4 & 4 & 4 \\ 3 & 3 & 4 & 0 & 2 \\ 2 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [71] $2\overset{11}{3}1^13 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 13$, $\Delta = -96$, $\det = -79$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.16 \rightarrow 11$, $T_{\max} = 13$, $T \in [11, 13]$
 $\begin{pmatrix} 0 & 2 & 3 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 3 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [72] $31^{11}231^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 13$, $\Delta = -96$, $\det = -84$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.24 \rightarrow 11$, $T_{\max} = 13$, $T \in [11, 13]$
 $\begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 2 & 6 & 4 \\ 5 & 6 & 5 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$
- [73] $31^{11}31^13 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -103$, $\det = -99$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.54 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 6 & 4 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 6 & 4 & 3 & 3 \\ 2 & 4 & 3 & 3 & 2 \end{pmatrix}_{b_0}$
- [74] $2\overset{11}{3}1^13 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2]$ $T = 7$, $T_{\max} = 13$, $\Delta = -96$, $\det = -99$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$

- $T_{\min} = 10.54 \rightarrow 11$, $T_{\max} = 13$, $T \in [11, 13]$
 $\begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 1 & 3 & 3 & 3 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [75] $221^{11}41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -102$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 1 & 5 & 6 & 3 \\ 5 & 8 & 13 & 6 \\ 6 & 13 & 13 & 7 \\ 3 & 6 & 7 & 3 \end{pmatrix}_{b_0}$
- [76] $21^{11}412 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -104$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 19$, $T \in [12, 19]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 7 & 6 & 4 \\ 2 & 7 & 4 & 7 & 4 \\ 2 & 6 & 7 & 3 & 3 \\ 2 & 4 & 4 & 3 & 1 \end{pmatrix}_{b_0}$
- [77] $1^{11}41 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -104$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 19$, $T \in [12, 19]$
 $\begin{pmatrix} 4 & 6 & 8 & 4 \\ 6 & 4 & 9 & 4 \\ 8 & 9 & 9 & 5 \\ 4 & 4 & 5 & 2 \end{pmatrix}_{b_0}$
- [78] $131^{11}31 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 12$, $\Delta = -95$, $\det = -104$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 7 & 6 & 4 \\ 2 & 7 & 4 & 7 & 4 \\ 2 & 6 & 7 & 3 & 3 \\ 2 & 4 & 4 & 3 & 1 \end{pmatrix}_{b_0}$
- [79] $13^{11}22 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -88$, $\det = -104$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.42 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 & 3 \\ 2 & 4 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [80] $2^{11}31 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 11$, $\Delta = -90$, $\det = -104$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.42 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 & 3 \\ 2 & 4 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [81] $2^{11}313 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 13$, $\Delta = -96$, $\det = -104$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.42 \rightarrow 11$, $T_{\max} = 13$, $T \in [11, 13]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 & 3 \\ 2 & 4 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [82] $231^{11}3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 13$, $\Delta = -96$, $\det = -111$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.26 \rightarrow 11$, $T_{\max} = 13$, $T \in [11, 13]$
 $\begin{pmatrix} 0 & 3 & 3 & 2 & 2 \\ 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 1 & 3 & 3 \\ 2 & 2 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [83] $31^{11}313 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -103$, $\det = -234$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 7 & 6 & 4 \\ 3 & 7 & 4 & 7 & 4 \\ 3 & 6 & 7 & 3 & 3 \\ 2 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$
- [84] $141^{11}22 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -234$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 19$, $T \in [11, 19]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 7 & 6 & 4 \\ 3 & 7 & 4 & 7 & 4 \\ 3 & 6 & 7 & 3 & 3 \\ 2 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$
- [85] $1321^{11}3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 13$, $\Delta = -96$, $\det = -234$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.75 \rightarrow 11$, $T_{\max} = 13$, $T \in [11, 13]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 3 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [86] $41^{11}222 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 7$, $T_{\max} = 19$, $\Delta = -107$, $\det = -416$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 4.0$

$$T_{\min} = 10.39 \rightarrow 11, T_{\max} = 19, T \in [11, 19]$$

$$\begin{pmatrix} 0 & 4 & 4 & 4 & 2 \\ 4 & 10 & 13 & 12 & 6 \\ 4 & 13 & 10 & 13 & 6 \\ 4 & 12 & 13 & 9 & 5 \\ 2 & 6 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[87] \quad 2^{21^{11}} 3^2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 11, \Delta = -90, \det = -416, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 11, T \in [10, 11]$$

$$\begin{pmatrix} 0 & 4 & 4 & 4 & 2 \\ 4 & 8 & 11 & 10 & 6 \\ 4 & 11 & 8 & 11 & 6 \\ 4 & 10 & 11 & 7 & 5 \\ 2 & 6 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[88] \quad 231^{11} 23 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 7, T_{\max} = 13, \Delta = -97, \det = -650, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.17 \rightarrow 11, T_{\max} = 13, T \in [11, 13]$$

$$\begin{pmatrix} 0 & 5 & 5 & 5 & 2 \\ 5 & 8 & 11 & 10 & 6 \\ 5 & 11 & 8 & 11 & 6 \\ 5 & 10 & 11 & 7 & 5 \\ 2 & 6 & 6 & 5 & 2 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [90] \quad 1^1 2^1 3^1 21^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 10, \Delta = -61, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 3 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^1 3^1 221^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 1 & 3 \\ 2 & 2 & 0 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[91] \quad 21^1 41^1 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 1 & 3 & 3 \\ 3 & 3 & 4 & 4 \\ 3 & 3 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[92] \quad 21^1 4131^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 5 & 7 & 5 \\ 7 & 9 & 7 \\ 5 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[93] \quad 21^{11} 51^{11} 22 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 0 & 2 & 3 & 2 & 2 \\ 2 & 4 & 5 & 6 & 4 \\ 3 & 5 & 4 & 5 & 4 \\ 2 & 6 & 5 & 3 & 3 \\ 2 & 4 & 4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[94] \quad 1^1 2^1 3^1 21^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 10, \Delta = -61, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[95] \quad 1^1 5^1 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 4 & 3 & 5 & 4 \\ 3 & 1 & 5 & 3 \\ 5 & 5 & 4 & 4 \\ 4 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[96] \quad 21^1 5^1 1^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 4 & 3 & 5 & 4 \\ 3 & 1 & 5 & 3 \\ 5 & 5 & 4 & 4 \\ 4 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[97] \quad 1^1 3^1 221^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 10, \Delta = -59, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 2 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 2 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

- [98] $1^1 2^1 3^1 2^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 10, \Delta = -61, \det = 5, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 0 & 2 \\ 2 & 0 & 3 & 2 & 4 \end{pmatrix}_{b_0}$
- [99] $41^1 2231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -87, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 10 & -1 & 3 \end{pmatrix}_{b_0}$
- [100] $1^1 4^1 131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -86, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 2 & 4 & 4 \\ 4 & 5 & 5 \\ 4 & 5 & 5 \end{pmatrix}_{b_0}$
- [101] $1^1 4^1 131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -86, \det = 7, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0}$
- [103] $21^1 5^1 1^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [102] $1^1 5^1 1^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 0 & 3 & 2 & 2 \\ 2 & 3 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [104] $1^1 4^1 131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -86, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 3 & 3 & 2 & 4 \end{pmatrix}_{b_0}$
- [105] $221^1 5^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 3 & 2 & 3 & 3 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$
- [106] $1^1 5^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [107] $21^1 5^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [108] $1^1 5^1 1^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 4 & 3 & 5 & 4 \\ 3 & 1 & 4 & 3 \\ 5 & 4 & 4 & 4 \\ 4 & 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [109] $41^1 231^1 3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 13, \Delta = -95, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 4 & 3 & 2 \\ 4 & 10 & 13 & 4 & 6 \\ 4 & 13 & 10 & 5 & 6 \\ 3 & 4 & 5 & 0 & 2 \\ 2 & 6 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[110] \quad 221 \overset{1}{5} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 3 \\ 3 & 5 & 4 & 4 \\ 3 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[111] \quad 1 \overset{1}{5} 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 3 \\ 3 & 5 & 4 & 4 \\ 3 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[112] \quad 1 \overset{1}{5} 12 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 3 \\ 3 & 5 & 4 & 4 \\ 3 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[113] \quad 21 \overset{1}{5} 1^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 2 & 2 & 0 & 3 & 2 \\ 2 & 2 & 3 & 3 & 3 \\ 2 & 2 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[114] \quad 21 \overset{1}{5} 1^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 1 & 6 & 3 \\ 5 & 6 & 13 & 7 \\ 3 & 3 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[115] \quad 21 \overset{1}{4} 1^1 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 0 & 3 & 0 & 2 \\ 2 & 3 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[116] \quad 1 \overset{1}{5} 1^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.33 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[120] \quad 1 \overset{1}{5} 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 4 \\ 3 & 6 & 4 & 4 \\ 3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[121] \quad 1 \overset{1}{5} 12 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.25 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 4 \\ 3 & 6 & 4 & 4 \\ 3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[119] \quad 1 \overset{1}{4} 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 2 & 3 \\ 3 & 1 & 2 & 3 \\ 2 & 2 & 0 & 2 \\ 3 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[118] \quad 1 \overset{1}{4} 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -86, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

- [117] $221\bar{5}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.25 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 4 \\ 3 & 6 & 4 & 4 \\ 3 & 4 & 4 & 2 \end{pmatrix}_{b_0}$
- [122] $1^1 41^{11} 231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -87, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 5 & 3 & 5 \\ 2 & 3 & 0 & 2 \\ 3 & 5 & 2 & 4 \end{pmatrix}_{b_0}$
- [123] $31^1 3141^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 4 \\ 5 & 6 & 5 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$
- [124] $1\bar{5}1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 4 & 3 \\ 4 & 4 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [125] $31^1 31^1 41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 13, \Delta = -94, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 3 & 1 & 2 \\ 3 & 3 & 4 & 4 & 4 \\ 3 & 1 & 4 & 0 & 2 \\ 2 & 2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [126] $1^1 41^{11} 231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -87, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.80 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$
 $\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 4 & 3 & 0 & 4 \\ 1 & 3 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0}$
- [127] $221^1 \bar{5}1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 5 & 3 & 3 \\ 5 & 8 & 7 & 6 \\ 3 & 7 & 4 & 4 \\ 3 & 6 & 4 & 3 \end{pmatrix}_{b_0}$
- [128] $21^1 \bar{5}1^{11} 22 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.24 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 3 & 3 & 2 & 2 \\ 3 & 4 & 7 & 4 & 4 \\ 3 & 7 & 4 & 5 & 4 \\ 2 & 4 & 5 & 3 & 3 \\ 2 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$
- [129] $21\bar{5}1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 6 & 4 \\ 4 & 6 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [130] $1\bar{5}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 4 & 6 & 4 \\ 4 & 6 & 4 & 4 \\ 3 & 4 & 4 & 3 \end{pmatrix}_{b_0}$
- [131] $1\bar{5}1^{11} 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 4 & 6 & 8 & 4 \\ 6 & 4 & 9 & 4 \\ 8 & 9 & 13 & 7 \\ 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$
- [132] $21^1 \bar{5}1^{11} 22 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 20, \Delta = -102, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$
 $\begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 4 & 5 & 6 & 4 \\ 2 & 5 & 4 & 5 & 4 \\ 3 & 6 & 5 & 3 & 3 \\ 2 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$
- [133] $1^1 41^{11} 231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 8, T_{\max} = 11, \Delta = -87, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$

$$\begin{aligned}
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[134] & 31^1 31^1 41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.33 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 4 & 3 & 3 & 4 \\ 1 & 3 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 2 & 4 & 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[135] & 2^1 3^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 3 & 1 & 2 & 2 \\ 3 & 4 & 3 & 3 & 4 \\ 1 & 3 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[136] & 2^1 3^1 141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[137] & 221^1 51^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.27 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 2 & 6 & 3 \\ 2 & 1 & 6 & 3 \\ 6 & 6 & 13 & 7 \\ 3 & 3 & 7 & 3 \end{pmatrix}_{b_0} \\
[138] & 31^1 31^1 41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 33, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 6 & 3 \\ 4 & 2 & 7 & 4 \\ 6 & 7 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0} \\
[139] & 41^1 1231^1 3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 4 & 3 & 4 & 2 \\ 4 & 10 & 5 & 12 & 6 \\ 3 & 5 & 1 & 5 & 3 \\ 4 & 12 & 5 & 9 & 5 \\ 2 & 6 & 3 & 5 & 2 \end{pmatrix}_{b_0} \\
[140] & 1^1 5^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 9 & 4 \\ 6 & 9 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0} \\
[141] & 3131^1 41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 9 & 4 \\ 6 & 9 & 13 & 7 \\ 3 & 4 & 7 & 3 \end{pmatrix}_{b_0} \\
[142] & 21^1 51^1 1^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 4 & 6 & 5 & 4 \\ 6 & 4 & 6 & 4 \\ 5 & 6 & 4 & 4 \\ 4 & 4 & 4 & 3 \end{pmatrix}_{b_0} \\
[143] & 1^1 5^1 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 4 & 6 & 5 & 4 \\ 6 & 4 & 6 & 4 \\ 5 & 6 & 4 & 4 \\ 4 & 4 & 4 & 3 \end{pmatrix}_{b_0} \\
[144] & 31^1 4^1 13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 40, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.30 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 3 & 2 & 2 & 2 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 4 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 2 \\ 2 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[146] & 2^1 3^1 141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 2 & 6 & 4 \\ 4 & 6 & 5 & 5 \\ 3 & 4 & 5 & 4 \end{pmatrix}_{b_0} \\
[145] & 31^{11}3141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.59 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 3 & 2 & 1 & 4 & 3 \\ 3 & 1 & 4 & 0 & 2 \\ 2 & 2 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[147] & 2^1 3^{141} 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 5 & 4 & 3 \\ 5 & 8 & 8 & 6 \\ 4 & 8 & 5 & 5 \\ 3 & 6 & 5 & 4 \end{pmatrix}_{b_0} \\
[148] & 141^{11}31^{11}3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.44 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 3 & 4 & 0 & 2 \\ 2 & 4 & 3 & 2 & 1 \end{pmatrix}_{b_0} \\
[149] & 2221^{151}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 46, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 18 & 1 & 8 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 8 & -1 & 2 \end{pmatrix}_{b_0} \\
[150] & 21^1 51^{11}2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.79 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 1 & 5 & 3 \\ 4 & 5 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0} \\
[151] & 1^1 51^{11}1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 47, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.79 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 1 & 5 & 3 \\ 4 & 5 & 4 & 4 \\ 3 & 3 & 4 & 3 \end{pmatrix}_{b_0} \\
[152] & 31^1 31^{141}1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.10 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 5 & 5 & 3 \\ 5 & 8 & 8 & 6 \\ 5 & 8 & 5 & 5 \\ 3 & 6 & 5 & 3 \end{pmatrix}_{b_0} \\
[153] & 31^{11}3141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.57 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0} \\
[154] & 41^1 2^1 3^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.93 \rightarrow 11, T_{\max} = 12, T \in [11, 12] \\
& \begin{pmatrix} 0 & 2 & 4 & 2 & 2 \\ 2 & 1 & 5 & 3 & 3 \\ 4 & 5 & 10 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 2 & 3 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[155] & 31^{11}4^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 60, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 3 & 3 & 2 & 2 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 4 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 4 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[156] & 221^{11}1^1 5^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 4 & 3 & 6 & 4 \\ 1 & 3 & 0 & 3 & 2 \\ 3 & 6 & 3 & 3 & 3 \\ 2 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[157] & 141^{11}31^{11}3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0
\end{aligned}$$

$$\begin{aligned}
& T_{\min} = 11.44 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 4 & 7 & 3 & 4 \\ 3 & 7 & 4 & 4 & 4 \\ 3 & 3 & 4 & 0 & 2 \\ 2 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0} \\
[158] & 31^{11} 4^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 3 & 2 & 2 \\ 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 2 \\ 2 & 3 & 4 & 2 & 2 \end{pmatrix}_{b_0} \\
[159] & 31^{11} 3141^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.24 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 2 & 6 & 4 \\ 5 & 6 & 5 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0} \\
[160] & 41^{11} 3^{22} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 97, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 4 & 3 & 3 & 2 \\ 4 & 10 & 9 & 8 & 6 \\ 3 & 9 & 5 & 8 & 5 \\ 3 & 8 & 8 & 4 & 4 \\ 2 & 6 & 5 & 4 & 3 \end{pmatrix}_{b_0} \\
[161] & 141^{13} 31^{11} 3 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.54 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 3 & 3 & 4 & 0 & 2 \\ 2 & 3 & 4 & 2 & 2 \end{pmatrix}_{b_0} \\
[162] & 1^{221} 5^{11} 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 102, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 1 & 5 & 6 & 3 \\ 5 & 8 & 13 & 6 \\ 6 & 13 & 13 & 7 \\ 3 & 6 & 7 & 3 \end{pmatrix}_{b_0} \\
[163] & 3131^{11} 41^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 102, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.43 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 5 & 6 & 3 \\ 5 & 8 & 13 & 6 \\ 6 & 13 & 13 & 7 \\ 3 & 6 & 7 & 3 \end{pmatrix}_{b_0} \\
[164] & 412^{111} 3^2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 & 3 \\ 2 & 4 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[165] & 41^{11} 232 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 2 & 4 & 2 \\ 4 & 10 & 5 & 12 & 6 \\ 2 & 5 & 1 & 5 & 3 \\ 4 & 12 & 5 & 9 & 5 \\ 2 & 6 & 3 & 5 & 2 \end{pmatrix}_{b_0} \\
[166] & 31^{111} 4^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.42 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 3 & 3 \\ 2 & 4 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0} \\
[167] & 2221^{11} 51^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.30 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 18 & 1 & 20 & 8 \\ 1 & 1 & -2 & 1 & 0 \\ 4 & 20 & 1 & 17 & 7 \\ 2 & 8 & 0 & 7 & 2 \end{pmatrix}_{b_0} \\
[168] & 41^{11} 31^{13} 2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 107, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 4 & 4 & 5 & 2 \\ 4 & 10 & 13 & 12 & 6 \\ 4 & 13 & 10 & 13 & 6 \\ 5 & 12 & 13 & 12 & 6 \\ 2 & 6 & 6 & 6 & 2 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[169] \ 31^{11}413 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.26 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 3 & 3 & 2 & 2 \\ 3 & 4 & 7 & 3 & 4 \\ 3 & 7 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 2 \\ 2 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[170] \ 41^{11}2231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 26 & 29 & 0 & 10 \\ 4 & 29 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 10 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[171] \ 41^{11}232 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 12, \Delta = -89, \det = 140, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 4 & 4 & 2 & 2 \\ 4 & 10 & 13 & 4 & 6 \\ 4 & 13 & 10 & 5 & 6 \\ 2 & 4 & 5 & 0 & 2 \\ 2 & 6 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[172] \ 321^{11}41^13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 154, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 5 & 3 & 5 & 2 \\ 5 & 13 & 6 & 15 & 7 \\ 3 & 6 & 1 & 6 & 3 \\ 5 & 15 & 6 & 12 & 6 \\ 2 & 7 & 3 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[173] \ 41^{11}322 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 3 & 4 & 3 & 2 \\ 3 & 5 & 9 & 7 & 5 \\ 4 & 9 & 10 & 9 & 6 \\ 3 & 7 & 9 & 4 & 4 \\ 2 & 5 & 6 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[174] \ 2221^{11}51^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 20, \Delta = -102, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.23 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 18 & 21 & 0 & 8 \\ 4 & 21 & 18 & 1 & 8 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 8 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[175] \ 41^{11}322 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 203, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.97 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 4 & 4 & 3 & 2 \\ 4 & 10 & 13 & 8 & 6 \\ 4 & 13 & 10 & 9 & 6 \\ 3 & 8 & 9 & 4 & 4 \\ 2 & 6 & 6 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[176] \ 14131^{11}3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -94, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 3 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[177] \ 41^{11}322 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 11, \Delta = -87, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 5 & 8 & 7 & 5 \\ 3 & 8 & 5 & 8 & 5 \\ 3 & 7 & 8 & 4 & 4 \\ 2 & 5 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[178] \ 41^{11}31^{11}32 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 8, T_{\max} = 13, \Delta = -95, \det = 246, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.13 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 5 & 4 & 5 & 2 \\ 5 & 13 & 13 & 15 & 7 \\ 4 & 13 & 10 & 13 & 6 \\ 5 & 15 & 13 & 12 & 6 \\ 2 & 7 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [179] \ 1^141321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 1 & 3 \\ 3 & 2 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[180] \ 21^{11}51^132 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[181] \quad 1 \overset{21}{5} 131^{\overset{1}{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[182] \quad 21^{\overset{1}{1}} 51231^{\overset{1}{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 3 & 3 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[183] \quad 1^{\overset{1}{1}} 41321^{\overset{1}{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 1 & 3 \\ 3 & 3 & 3 & 5 \end{pmatrix}_{b_0}$$

$$[184] \quad 1^{\overset{1}{1}} 5131^{\overset{1}{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 2 & 4 & 4 \\ 3 & 4 & 4 & 4 \\ 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[185] \quad 41^{\overset{1}{1}} 23141^{\overset{1}{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[186] \quad 21^{\overset{1}{1}} 5131^{\overset{1}{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 3 & 5 & 4 \\ 3 & 1 & 4 & 3 \\ 5 & 4 & 5 & 5 \\ 4 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[187] \quad 151^{\overset{1}{1}} 2231^{\overset{1}{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & 1 & 6 & 3 & 3 \\ 4 & 6 & 18 & 6 & 8 \\ 1 & 3 & 6 & 0 & 2 \\ 2 & 3 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[188] \quad 1^{\overset{1}{1}} 51231^{\overset{1}{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 3 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[189] \quad 31^{\overset{1}{1}} 13151^{\overset{1}{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 4 & 3 & 3 \\ 4 & 2 & 4 & 4 \\ 3 & 4 & 4 & 4 \\ 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$$

$$[190] \quad 21^{\overset{1}{1}} 51^{\overset{1}{1}} 231^{\overset{1}{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 4 & 9 & 5 & 4 \\ 9 & 19 & 11 & 9 \\ 5 & 11 & 5 & 5 \\ 4 & 9 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[195] \quad 21^{\overset{1}{1}} 5131^{\overset{1}{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 1 & 4 & 3 \\ 3 & 4 & 5 & 5 \\ 3 & 3 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[191] \quad 21^{\overset{1}{1}} 5131^{\overset{1}{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -81, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

- [192] $1^{\overline{1}}\overline{5}1231^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [193] $1^{\overline{1}}\overline{5}131^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 0 & 2 & 3 \end{pmatrix}_{b_0}$
- [194] $2^{\overline{1}}3^{\overline{1}}1\overline{5}1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -7$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [197] $31^{\overline{1}}321^{\overline{1}}\overline{5}1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 13$, $T_{\max} = 13$, $T \in [13, 13]$
 $\begin{pmatrix} 0 & 4 & 1 & 3 & 2 \\ 4 & 10 & 4 & 4 & 6 \\ 1 & 4 & 0 & 2 & 2 \\ 3 & 4 & 2 & 0 & 2 \\ 2 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [196] $1^{\overline{1}}\overline{5}1231^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$
- [198] $1^{\overline{1}}\overline{5}131^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.50 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 \\ 2 & 1 & 4 & 3 \\ 3 & 4 & 5 & 5 \\ 3 & 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [199] $321^{\overline{1}}4141^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -10$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 9 & 4 \\ 8 & 9 & 17 & 9 \\ 3 & 4 & 9 & 4 \end{pmatrix}_{b_0}$
- [200] $2^{\overline{1}}3^{\overline{1}}1\overline{5}1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -10$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 2 & 4 & 4 \\ 3 & 4 & 4 & 4 \\ 3 & 4 & 4 & 4 \end{pmatrix}_{b_0}$
- [201] $21^{\overline{1}}\overline{5}131^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -81$, $\det = -12$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [202] $21^{\overline{1}}\overline{5}1^{\overline{1}}231^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -12$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 2 & 1 & 2 \\ 3 & 10 & 4 & 4 & 6 \\ 2 & 4 & 0 & 2 & 2 \\ 1 & 4 & 2 & 0 & 2 \\ 2 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$
- [203] $2^{\overline{1}}3^{\overline{1}}1\overline{5}1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -17$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.59 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 4 & 3 \\ 2 & 1 & 4 & 3 \\ 4 & 4 & 5 & 5 \\ 3 & 3 & 5 & 4 \end{pmatrix}_{b_0}$
- [204] $1^{\overline{1}}\overline{4}1^{\overline{1}}3141^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -19$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$

$$\begin{aligned}
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 13 & 1 & 5 & 7 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 5 & 1 & 0 & 2 \\ 2 & 7 & 0 & 2 & 3 \end{pmatrix}_{b_0} \\
[205] & 31^1 32151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 4 \\ 5 & 6 & 5 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0} \\
[206] & 1^1 5 131^1 3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 4 \\ 5 & 6 & 5 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0} \\
[207] & 1^1 5 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 6 & 3 & 3 \\ 6 & 19 & 11 & 9 \\ 3 & 11 & 5 & 5 \\ 3 & 9 & 5 & 4 \end{pmatrix}_{b_0} \\
[208] & 1^1 41^1 3141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 13 & 6 & 0 & 7 \\ 1 & 6 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 7 & 3 & -1 & 3 \end{pmatrix}_{b_0} \\
[209] & 1^1 3221^1 51^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 18 & 6 & 0 & 8 \\ 1 & 6 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 8 & 3 & -1 & 3 \end{pmatrix}_{b_0} \\
[210] & 1^1 5 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 10 & 4 & 4 & 6 \\ 1 & 4 & 0 & 2 & 2 \\ 1 & 4 & 2 & 0 & 2 \\ 2 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[211] & 31^1 321^1 51^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 7 & 5 & 3 \\ 7 & 19 & 11 & 9 \\ 5 & 11 & 5 & 5 \\ 3 & 9 & 5 & 4 \end{pmatrix}_{b_0} \\
[212] & 1^1 3221^1 51^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 18 & 1 & 5 & 8 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 5 & 1 & 0 & 2 \\ 2 & 8 & 0 & 2 & 3 \end{pmatrix}_{b_0} \\
[213] & 1^1 5 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 4 & 1 & 2 \\ 3 & 4 & 10 & 5 & 6 \\ 1 & 1 & 5 & 0 & 2 \\ 2 & 2 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[214] & 31^1 3151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 1 & 2 & 5 & 3 \\ 2 & 1 & 4 & 3 \\ 5 & 4 & 5 & 5 \\ 3 & 3 & 5 & 4 \end{pmatrix}_{b_0} \\
[215] & 1^1 41^1 3141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 1 & 6 & 0 & 3 \\ 3 & 6 & 13 & 1 & 7 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 3 & 7 & -1 & 3 \end{pmatrix}_{b_0} \\
[216] & 31^1 4141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 3 & 1 & 2 & 2 \\ 3 & 13 & 1 & 9 & 7 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 9 & 1 & 4 & 4 \\ 2 & 7 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[217] \quad 2 \quad 3 \quad 21^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 11, \Delta = -60, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 2 & 2 & 4 & 4 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 4 & 2 & 1 & 3 \\ 2 & 4 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[218] \quad 41^{11} 231^{11} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 4 & 1 & 2 \\ 3 & 10 & 17 & 0 & 6 \\ 4 & 17 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 6 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[221] \quad 2 \quad 3 \quad 2151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[219] \quad 31 \quad 4 \quad 141^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 5 & 1 & 7 & 5 \\ 1 & 1 & -2 & 1 & 0 \\ 2 & 7 & 1 & 4 & 4 \\ 2 & 5 & 0 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[220] \quad 1^{11} 51 \quad 3 \quad 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[222] \quad 2321^{11} 51^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 & 2 \\ 4 & 10 & 4 & 4 & 6 \\ 1 & 4 & 0 & 2 & 2 \\ 2 & 4 & 2 & 0 & 2 \\ 2 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[223] \quad 31^{11} 4 \quad 141^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -85, \det = -33, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 2 & 1 & 2 \\ 3 & 13 & 10 & 0 & 7 \\ 2 & 10 & 5 & 1 & 5 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 7 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[225] \quad 1^{11} 3221^{11} 51^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 18 & 6 & 8 \\ 1 & 0 & 6 & 0 & 2 \\ 2 & 0 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[226] \quad 2 \quad 3 \quad 21^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 11, \Delta = -60, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 3 & 7 & 5 \\ 4 & 7 & 6 & 6 \\ 3 & 5 & 6 & 5 \end{pmatrix}_{b_0}$$

$$[224] \quad 231^{11} 3151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 5 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 5 & 1 & 33 & 1 & 11 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 11 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[227] \quad 2321^{11} 51^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -39, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 4 & 2 & 2 \\ 1 & 0 & 4 & 1 & 2 \\ 4 & 4 & 10 & 5 & 6 \\ 2 & 1 & 5 & 0 & 2 \\ 2 & 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

- [228] $31^{\overline{11}}4141^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -40$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 3 & 1 & 2 \\ 2 & 5 & 10 & 0 & 5 \\ 3 & 10 & 13 & 1 & 7 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$
- [229] $151^{\overline{11}}32 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -41$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 2 & 6 & 4 \\ 4 & 6 & 5 & 5 \\ 3 & 4 & 5 & 4 \end{pmatrix}_{b_0}$$
- [230] $232151^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -41$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 2 & 6 & 4 \\ 4 & 6 & 5 & 5 \\ 3 & 4 & 5 & 4 \end{pmatrix}_{b_0}$$
- [231] $31^{\overline{11}}4141^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -85$, $\det = -41$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 5 & 8 & 0 & 5 \\ 2 & 8 & 5 & 1 & 5 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 5 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$
- [232] $2321^{\overline{11}}51^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -41$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 7 & 4 & 3 \\ 7 & 19 & 11 & 9 \\ 4 & 11 & 5 & 5 \\ 3 & 9 & 5 & 4 \end{pmatrix}_{b_0}$$
- [233] $31321^{\overline{11}}51^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -46$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.43 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 12 & 4 \\ 8 & 12 & 26 & 10 \\ 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$
- [234] $15131^{\overline{11}}3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 13$, $\Delta = -89$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [235] $31^{\overline{11}}32151^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 13$, $\Delta = -90$, $\det = -51$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.57 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 3 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$
- [236] $223151^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -55$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.95 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 3 & 2 & 2 \\ 3 & 3 & 5 & 8 & 5 \\ 3 & 2 & 8 & 4 & 4 \\ 2 & 2 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$
- [237] $2^{\overline{11}}3^{\overline{11}}41^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 11$, $\Delta = -60$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 5 & 3 & 4 \\ 2 & 5 & 2 & 4 & 4 \\ 2 & 3 & 4 & 1 & 3 \\ 2 & 4 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$
- [238] $151^{\overline{11}}232 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 4 & 2 & 2 \\ 2 & 1 & 5 & 3 & 3 \\ 4 & 5 & 10 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 2 & 3 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$
- [239] $31^{\overline{11}}513 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9$, $T_{\max} = 13$, $\Delta = -89$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 11.93 \rightarrow 12$, $T_{\max} = 13$, $T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 2 & 3 \\ 2 & 4 & 1 & 3 & 3 \\ 2 & 2 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[240] $22 \overset{1}{3} \overset{1}{151} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 5 & 3 & 7 & 5 \\ 1 & 3 & 0 & 3 & 2 \\ 3 & 7 & 3 & 4 & 4 \\ 2 & 5 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[241] $2 \overset{1}{\overset{1}{3}} \overset{1}{\overset{1}{41}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 11, \Delta = -60, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 & 4 \\ 2 & 4 & 2 & 4 & 4 \\ 2 & 4 & 4 & 1 & 3 \\ 2 & 4 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[242] $31 \overset{1}{5} \overset{1}{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -64, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 3 & 2 & 3 \end{pmatrix}_{b_0}$$

[243] $41 \overset{231}{4} \overset{1}{1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 5 & 3 & 3 & 2 \\ 5 & 13 & 10 & 9 & 7 \\ 3 & 10 & 5 & 8 & 5 \\ 3 & 9 & 8 & 4 & 4 \\ 2 & 7 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[244] $15 \overset{1}{131} \overset{1}{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -89, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 2 & 6 & 4 \\ 5 & 6 & 5 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[245] $31 \overset{11}{132151} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -84, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 5 & 3 \\ 4 & 2 & 6 & 4 \\ 5 & 6 & 5 & 5 \\ 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

[246] $22 \overset{1}{3} \overset{1}{151} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -96, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 6 & 3 \\ 5 & 9 & 14 & 7 \\ 6 & 14 & 14 & 8 \\ 3 & 7 & 8 & 4 \end{pmatrix}_{b_0}$$

[247] $31 \overset{321}{5} \overset{1}{1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 2 & 5 & 2 & 2 \\ 2 & 1 & 5 & 3 & 3 \\ 5 & 5 & 8 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 2 & 3 & 6 & 2 & 3 \end{pmatrix}_{b_0}$$

[248] $41 \overset{11}{23141} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 26 & 29 & 0 & 10 \\ 4 & 29 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 10 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[249] $231 \overset{11}{13151} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 13, \Delta = -90, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 5 & 1 & 5 & 2 \\ 5 & 33 & 1 & 35 & 11 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 35 & 1 & 32 & 10 \\ 2 & 11 & 0 & 10 & 2 \end{pmatrix}_{b_0}$$

[250] $151 \overset{11}{2231} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -131, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 18 & 21 & 5 & 8 \\ 4 & 21 & 18 & 6 & 8 \\ 1 & 5 & 6 & 0 & 2 \\ 2 & 8 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

- [251] $151^{11}232 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -84, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 4 & 4 & 2 & 2 \\ 4 & 10 & 13 & 4 & 6 \\ 4 & 13 & 10 & 5 & 6 \\ 2 & 4 & 5 & 0 & 2 \\ 2 & 6 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [252] $31321^{11}51^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 13, \Delta = -90, \det = -179, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 7 & 8 & 3 \\ 7 & 19 & 25 & 9 \\ 8 & 25 & 26 & 10 \\ 3 & 9 & 10 & 3 \end{pmatrix}_{b_0}$
- [253] $41^231^{11}4 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 5 & 3 & 2 \\ 3 & 5 & 10 & 7 & 5 \\ 5 & 10 & 13 & 10 & 7 \\ 3 & 7 & 10 & 4 & 4 \\ 2 & 5 & 7 & 4 & 3 \end{pmatrix}_{b_0}$
- [254] $321^{11}4141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -218, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 7 & 8 & 3 \\ 7 & 12 & 17 & 8 \\ 8 & 17 & 17 & 9 \\ 3 & 8 & 9 & 4 \end{pmatrix}_{b_0}$
- [255] $215131^{11}13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 13, \Delta = -89, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 3 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [256] $1^{31}5^{11}1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 13, \Delta = -89, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 3 & 4 & 0 & 2 \\ 2 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$
- [257] $41^14^{11}132 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 5 & 8 & 7 & 5 \\ 3 & 8 & 5 & 8 & 5 \\ 3 & 7 & 8 & 4 & 4 \\ 2 & 5 & 5 & 4 & 3 \end{pmatrix}_{b_0}$
- [258] $151^{11}3^{11}22 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 5 & 8 & 7 & 5 \\ 3 & 8 & 5 & 8 & 5 \\ 3 & 7 & 8 & 4 & 4 \\ 2 & 5 & 5 & 4 & 3 \end{pmatrix}_{b_0}$
- [259] $231^{11}3151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 13, \Delta = -90, \det = -248, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 5 & 5 & 1 & 2 \\ 5 & 33 & 36 & 0 & 11 \\ 5 & 36 & 33 & 1 & 11 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 11 & 11 & -1 & 3 \end{pmatrix}_{b_0}$
- [260] $31^{321}5^{11}1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 13, \Delta = -90, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$
 $\begin{pmatrix} 0 & 5 & 5 & 2 & 2 \\ 5 & 8 & 11 & 4 & 6 \\ 5 & 11 & 8 & 5 & 6 \\ 2 & 4 & 5 & 0 & 2 \\ 2 & 6 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [261] $41^231^{11}4^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -302, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 5 & 5 & 3 & 2 \\ 5 & 13 & 16 & 9 & 7 \\ 5 & 16 & 13 & 10 & 7 \\ 3 & 9 & 10 & 4 & 4 \\ 2 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$
- [262] $21^{11}4132^2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 9, T_{\max} = 11, \Delta = -60, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 4 & 4 & 4 & 2 \\ 4 & 12 & 15 & 14 & 8 \\ 4 & 15 & 12 & 15 & 8 \\ 4 & 14 & 15 & 11 & 7 \\ 2 & 8 & 8 & 7 & 4 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [263] \quad 1^1 4^{11} 13221^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 11, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 2 & 4 \\ 3 & 3 & 4 & 6 \end{pmatrix}_{b_0}$$

$$[264] \quad 1^1 31^1 5^{11} 131^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 \\ 1 & 1 & -2 & 1 & 0 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 3 & 0 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[269] \quad 21^1 612^1 3^1 2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[265] \quad 21^1 612231^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 7 & 5 \\ 7 & 9 & 7 \\ 5 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[266] \quad 21^1 513141^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 4 & 3 & 4 \\ 3 & 3 & 1 & 3 \\ 3 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[267] \quad 1^1 6^{21} 1231^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 7 & 5 \\ 7 & 9 & 7 \\ 5 & 7 & 5 \end{pmatrix}_{b_0}$$

$$[268] \quad 1^1 31^1 5^{11} 131^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 3 & 3 & -1 & 4 \end{pmatrix}_{b_0}$$

$$[270] \quad 1^1 31^1 5^{11} 131^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 12, \Delta = -60, \det = 5, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 0 & 2 \\ 2 & 0 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[277] \quad 2161^1 2231^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 5 \\ 9 & 15 & 9 \\ 5 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[278] \quad 1^1 6^{11} 3221^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 5 \\ 9 & 15 & 9 \\ 5 & 9 & 5 \end{pmatrix}_{b_0}$$

$$[271] \quad 151^1 23141^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 6 & 3 \\ 3 & 4 & 6 & 4 \\ 6 & 6 & 6 & 6 \\ 3 & 4 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[272] \quad 41^1 231^1 5^{11} 1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 9 & 3 \\ 3 & 4 & 15 & 4 \\ 9 & 15 & 51 & 15 \\ 3 & 4 & 15 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned} [273] \quad & 141^1 131^1 51^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 5 & 6 & 5 \\ 6 & 6 & 6 \\ 5 & 6 & 5 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [274] \quad & 41^1 232151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 10 & -1 & 3 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [275] \quad & 1^1 413151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 5 & 4 & 5 \\ 4 & 2 & 4 \\ 5 & 4 & 5 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [276] \quad & 41^1 314141^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 10 & -1 & 3 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [284] \quad & 1^1 413151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 2 & 3 & 4 \\ 3 & 1 & 3 \\ 4 & 3 & 5 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [279] \quad & 16^1 1231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [280] \quad & 21^1 161231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 4 & 3 & 0 & 4 \\ 1 & 3 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [281] \quad & 232161^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [282] \quad & 21^1 61^{11} 2231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 0 & 4 & 2 & 1 & 2 \\ 4 & 10 & 7 & 0 & 6 \\ 2 & 7 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 4 & 0 & 3 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [283] \quad & 1^1 612231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 12.57 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 2 & 0 & 3 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [285] \quad & 1^1 413151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 12.50 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\ & \begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 1 & 3 \\ 1 & 1 & -2 & 0 \\ 3 & 3 & 0 & 4 \end{pmatrix}_{b_0} \end{aligned}$$

$$\begin{aligned} [286] \quad & 4131^1 4141^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 10, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\ & T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 5 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 5 & 1 & 38 & 1 & 12 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[288] $1^{\overline{1}321^1} 1^{\overline{1}2} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 2 & 1 & 2 \\ 3 & 4 & 5 & 0 & 4 \\ 2 & 5 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

[289] $21^{\overline{1}6} 1^{\overline{1}231^1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 2 & 3 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 4 & 0 & 3 \end{pmatrix}_{b_0}$$

[287] $21^{\overline{1}6} 1^{\overline{1}232} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 2 & 2 & 2 \\ 4 & 10 & 7 & 4 & 6 \\ 2 & 7 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 2 \\ 2 & 6 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[290] $1^{\overline{1}41^1} 1^{\overline{1}32151^1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 5 & 3 & 5 \\ 2 & 3 & 0 & 2 \\ 3 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

[292] $1^{\overline{1}5} 1^{\overline{1}31^1} 1^{\overline{1}41^1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 3 \\ 4 & 5 & 3 & 5 \\ 2 & 3 & 0 & 2 \\ 3 & 5 & 2 & 4 \end{pmatrix}_{b_0}$$

[291] $31^{\overline{1}4} 1^{\overline{1}3151^1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 19, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.26 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 5 & 8 & 5 \\ 8 & 9 & 7 \\ 5 & 7 & 4 \end{pmatrix}_{b_0}$$

[297] $1^{\overline{1}321^1} 1^{\overline{1}1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 10 & 4 & 4 & 6 \\ 1 & 4 & 0 & 2 & 2 \\ 1 & 4 & 2 & 0 & 2 \\ 2 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[296] $1^{\overline{1}5} 1^{\overline{1}31^1} 1^{\overline{1}41^1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 4 & 3 & 0 & 4 \\ 1 & 3 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[295] $1^{\overline{1}4} 1^{\overline{1}321^1} 1^{\overline{1}51^1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 2 & 2 & 0 & 4 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[294] $21^{\overline{1}321^1} 1^{\overline{1}1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 4 & 3 & 0 & 4 \\ 1 & 3 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[293] $1^{\overline{1}4} 1^{\overline{1}32151^1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 20, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 4 & 3 & 0 & 4 \\ 1 & 3 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[299] $1^{\overline{1}5} 1^{\overline{1}31^1} 1^{\overline{1}41^1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -80, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$

$$\begin{aligned}
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[301] & 141^{1^1}321^{1^1}51^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 3 & 1 & 2 \\ 4 & 2 & 4 & 0 & 4 \\ 3 & 4 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[300] & 141^{1^1}315^{1^1}1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -80, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 4 & 1 & 3 \\ 4 & 5 & 1 & 5 \\ 1 & 1 & -2 & 0 \\ 3 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
[302] & 1^{1^1}6^{1^1}1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 4 & 1 & 2 \\ 3 & 4 & 10 & 5 & 6 \\ 1 & 1 & 5 & 0 & 2 \\ 2 & 2 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[303] & 1^{1^1}41321^{1^1}51^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 2 & 3 \\ 5 & 3 & 2 & 5 \\ 2 & 2 & 0 & 2 \\ 3 & 5 & 2 & 4 \end{pmatrix}_{b_0} \\
[304] & 21^{1^1}6^{1^1}1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[305] & 1^{1^1}6^{1^1}1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 1 & 1 & 2 \\ 4 & 10 & 4 & 0 & 6 \\ 1 & 4 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 2 & 0 & 3 \end{pmatrix}_{b_0} \\
[298] & 1^{1^1}41^{1^1}32151^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 24, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 4 & 0 & 3 \end{pmatrix}_{b_0} \\
[306] & 151^{1^1}231^{1^1}41^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 6 & 3 \\ 5 & 8 & 8 & 6 \\ 6 & 8 & 6 & 6 \\ 3 & 6 & 6 & 4 \end{pmatrix}_{b_0} \\
[307] & 41^{1^1}2315^{1^1}1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 26, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & 0 & 5 & 0 & 2 \\ 4 & 5 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 2 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[308] & 1^16^{1^1}12^{1^1}3^12^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 31, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.90 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[309] & 1^{1^1}6^{1^1}1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 32, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.75 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 1 & 2 & 2 \\ 4 & 10 & 4 & 4 & 6 \\ 1 & 4 & 0 & 2 & 2 \\ 2 & 4 & 2 & 0 & 2 \\ 2 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

- [311] $1^{\overline{1}}41321^{\overline{1}}51^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 2 & 7 & 4 \\ 7 & 7 & 7 \\ 4 & 7 & 5 \end{pmatrix}_{b_0}$
- [312] $3141^{\overline{1}}3151^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$
 $\begin{pmatrix} 5 & 10 & 5 \\ 10 & 13 & 9 \\ 5 & 9 & 4 \end{pmatrix}_{b_0}$
- [310] $1^{\overline{1}}3221^{\overline{1}}6^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -79, \det = 35, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & 0 & 4 & 0 & 2 \\ 4 & 4 & 10 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [313] $1^{\overline{1}}2321^{\overline{1}}6^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 39, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 1 & 4 & 2 & 2 \\ 1 & 0 & 4 & 1 & 2 \\ 4 & 4 & 10 & 5 & 6 \\ 2 & 1 & 5 & 0 & 2 \\ 2 & 2 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [314] $31^{\overline{1}}41^{\overline{1}}3151^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 52, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 3 & 5 & 1 & 2 \\ 3 & 4 & 7 & 0 & 4 \\ 5 & 7 & 8 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [315] $41^{\overline{1}}4141^{\overline{1}}3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 5 & 4 & 7 & 5 \\ 3 & 4 & 1 & 4 & 3 \\ 3 & 7 & 4 & 4 & 4 \\ 2 & 5 & 3 & 4 & 3 \end{pmatrix}_{b_0}$
- [316] $31^{\overline{1}}41^{\overline{1}}3151^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 5 & 3 & 1 & 2 \\ 5 & 8 & 7 & 0 & 6 \\ 3 & 7 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 4 & 0 & 3 \end{pmatrix}_{b_0}$
- [317] $1^{\overline{1}}2^{\overline{1}}3^{\overline{1}}21^{\overline{1}}6^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 4 & 2 & 2 \\ 2 & 1 & 5 & 3 & 3 \\ 4 & 5 & 10 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 2 & 3 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [318] $41^{\overline{1}}31^{\overline{1}}5^{\overline{1}}13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$
 $\begin{pmatrix} 0 & 2 & 4 & 2 & 2 \\ 2 & 1 & 5 & 3 & 3 \\ 4 & 5 & 10 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 2 & 3 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [319] $2161^{\overline{1}}2^{\overline{1}}3^{\overline{1}}2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -81, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 2 & 4 & 2 & 2 \\ 2 & 1 & 5 & 3 & 3 \\ 4 & 5 & 10 & 5 & 6 \\ 2 & 3 & 5 & 0 & 2 \\ 2 & 3 & 6 & 2 & 3 \end{pmatrix}_{b_0}$
- [320] $41^{\overline{1}}31^{\overline{1}}4141^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 13, \Delta = -84, \det = 73, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 0 & 4 & 5 & 1 & 2 \\ 4 & 26 & 33 & 0 & 10 \\ 5 & 33 & 38 & 1 & 12 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 10 & 12 & -1 & 3 \end{pmatrix}_{b_0}$
- [321] $31^{\overline{1}}413151^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 10, T_{\max} = 14, \Delta = -88, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.24 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 3 & 3 & 1 & 2 \\ 3 & 4 & 7 & 0 & 4 \\ 3 & 7 & 4 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 4 & 0 & 2 \end{pmatrix}_{b_0} \\
[322] & 41^1 31^1 4141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 86, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.81 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 5 & 4 & 1 & 2 \\ 5 & 38 & 33 & 0 & 12 \\ 4 & 33 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 12 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[328] & 41^{11} 314141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 26 & 29 & 0 & 10 \\ 4 & 29 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 10 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[327] & 41^{11} 232151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 26 & 29 & 0 & 10 \\ 4 & 29 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 10 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[326] & 1^{11} 3221^{11} 6 \quad 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 10 & 13 & 0 & 6 \\ 4 & 13 & 10 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[325] & 2161^{11} 2231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 10 & 13 & 0 & 6 \\ 4 & 13 & 10 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[324] & 151^{11} 23141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 5 & 6 & 3 \\ 5 & 3 & 7 & 5 \\ 6 & 7 & 6 & 6 \\ 3 & 5 & 6 & 4 \end{pmatrix}_{b_0} \\
[323] & 41^{11} 23151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 131, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.95 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 8 & 9 & 3 \\ 8 & 42 & 49 & 14 \\ 9 & 49 & 51 & 15 \\ 3 & 14 & 15 & 4 \end{pmatrix}_{b_0} \\
[329] & 2161^{11} 232 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 4 & 2 & 2 \\ 4 & 10 & 13 & 4 & 6 \\ 4 & 13 & 10 & 5 & 6 \\ 2 & 4 & 5 & 0 & 2 \\ 2 & 6 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[330] & 1^{11} 2321^{11} 6 \quad 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 4 & 4 & 2 & 2 \\ 4 & 10 & 13 & 4 & 6 \\ 4 & 13 & 10 & 5 & 6 \\ 2 & 4 & 5 & 0 & 2 \\ 2 & 6 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[331] & 41^{11} 31513 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 140, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.97 \rightarrow 13, T_{\max} = 14, T \in [13, 14] \\
& \begin{pmatrix} 0 & 4 & 4 & 2 & 2 \\ 4 & 10 & 13 & 4 & 6 \\ 4 & 13 & 10 & 5 & 6 \\ 2 & 4 & 5 & 0 & 2 \\ 2 & 6 & 6 & 2 & 3 \end{pmatrix}_{b_0} \\
[332] & 51^{11} 223141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 188, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 6 & 7 & 3 \\ 6 & 4 & 8 & 6 \\ 7 & 8 & 7 & 7 \\ 3 & 6 & 7 & 4 \end{pmatrix}_{b_0} \\
[333] & 4131^{11} 4141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 218, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]
\end{aligned}$$

$$\begin{pmatrix} 0 & 5 & 5 & 1 & 2 \\ 5 & 38 & 41 & 0 & 12 \\ 5 & 41 & 38 & 1 & 12 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 12 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[334] $41 \overset{1}{4} \overset{11}{1413} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 234, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.86 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 2 \\ 3 & 5 & 8 & 7 & 5 \\ 3 & 8 & 5 & 8 & 5 \\ 3 & 7 & 8 & 4 & 4 \\ 2 & 5 & 5 & 4 & 3 \end{pmatrix}_{b_0}$$

[335] $3141 \overset{11}{3151} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -88, \det = 248, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.97 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$

$$\begin{pmatrix} 0 & 5 & 5 & 1 & 2 \\ 5 & 8 & 11 & 0 & 6 \\ 5 & 11 & 8 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[336] $2231 \overset{11}{3161} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 251, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 7 & 1 & 7 & 2 \\ 7 & 33 & 6 & 35 & 11 \\ 1 & 6 & 0 & 6 & 2 \\ 7 & 35 & 6 & 32 & 10 \\ 2 & 11 & 2 & 10 & 3 \end{pmatrix}_{b_0}$$

$T = 11$ [337] $1 \overset{1}{31} 5 \overset{1}{1321} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 1 & 3 \\ 2 & 0 & 3 & 3 & 5 \end{pmatrix}_{b_0}$$

[338] $1 \overset{1}{31} 5 \overset{1}{1321} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 2 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 3 & 4 & -1 & 5 \end{pmatrix}_{b_0}$$

[339] $1 \overset{1}{6} \overset{1321}{131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 1 & 3 \\ 2 & 1 & 2 & 3 \\ 1 & 2 & 0 & 2 \\ 3 & 3 & 2 & 5 \end{pmatrix}_{b_0}$$

[341] $1 \overset{1}{31} 5 \overset{1}{1321} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -32, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 2 & 4 \\ 1 & 0 & 2 & 0 & 2 \\ 2 & 0 & 4 & 2 & 5 \end{pmatrix}_{b_0}$$

[340] $1 \overset{1}{6} \overset{1321}{131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0}$$

[342] $1 \overset{21}{7} 12 \overset{11}{3} 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[343] $1 \overset{1}{6} \overset{1321}{131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 1 & 3 \\ 3 & 3 & 3 & 5 \end{pmatrix}_{b_0}$$

[344] $1 \overset{1}{6} \overset{1321}{131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -5, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.60 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 3 & 0 & 4 \end{pmatrix}_{b_0}$$

[345] $21^1 7^{11} 12 3 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 2 \\ 1 & 0 & 3 & 1 & 2 \\ 2 & 3 & 4 & 3 & 4 \\ 2 & 1 & 3 & 0 & 2 \\ 2 & 2 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[346] $23 21^1 71 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 2 & 2 & 3 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[347] $15131^1 4141^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 9 & 4 \\ 8 & 9 & 17 & 9 \\ 3 & 4 & 9 & 4 \end{pmatrix}_{b_0}$$

[349] $321^1 513151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 5 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 5 & 1 & 38 & 1 & 12 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 12 & -1 & 3 \end{pmatrix}_{b_0}$$

[348] $51231^1 4141^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -80, \det = -10, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 3 & 8 & 3 \\ 3 & 4 & 9 & 4 \\ 8 & 9 & 17 & 9 \\ 3 & 4 & 9 & 4 \end{pmatrix}_{b_0}$$

[350] $1^{23} 7^{11} 1^1 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -12, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.67 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 4 & 2 & 2 & 2 \\ 4 & 10 & 7 & 4 & 6 \\ 2 & 7 & 4 & 3 & 4 \\ 2 & 4 & 3 & 0 & 2 \\ 2 & 6 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[351] $31^1 5^{11} 1321^1 5 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.80 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 5 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 5 & 3 & 3 & 3 & 5 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[353] $3151^1 23151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -84, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 1 & 7 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 7 & 1 & 68 & 1 & 16 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 16 & -1 & 3 \end{pmatrix}_{b_0}$$

[352] $1^1 4141^1 3151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.67 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 5 & 1 & 1 & 2 \\ 5 & 8 & 3 & 0 & 6 \\ 1 & 3 & 0 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 2 & 0 & 3 \end{pmatrix}_{b_0}$$

[354] $1^1 4141^1 3151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -27, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.59 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 6 & 2 & 3 \\ 6 & 9 & 3 & 7 \\ 2 & 3 & 0 & 2 \\ 3 & 7 & 2 & 4 \end{pmatrix}_{b_0}$$

[357] $1^1 7^{11} 12 3 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -79, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 5 & 3 & 7 & 5 \\ 1 & 3 & 0 & 3 & 2 \\ 2 & 7 & 3 & 4 & 4 \\ 2 & 5 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[356] $31^1 5^{11} 13151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 11, T_{\max} = 14, \Delta = -83, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$

$$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[355] \quad 21 \overset{1}{7} 12 \overset{1}{3} 2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13.90 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[358] \quad 21 \overset{1}{7} \overset{1}{2} 1 \overset{1}{3} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 & 2 \\ 4 & 10 & 4 & 4 & 6 \\ 1 & 4 & 0 & 2 & 2 \\ 2 & 4 & 2 & 0 & 2 \\ 2 & 6 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[359] \quad 1 \overset{1}{2} 3 \overset{1}{2} 1 \overset{1}{1} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 4 & 1 & 2 & 2 \\ 4 & 18 & 5 & 10 & 8 \\ 1 & 5 & 0 & 3 & 2 \\ 2 & 10 & 3 & 4 & 4 \\ 2 & 8 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[360] \quad 51 \overset{1}{1} 231 \overset{1}{4} 141 \overset{1}{1} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -80, \det = -33, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13.67 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 7 & 7 & 3 \\ 7 & 12 & 10 & 8 \\ 7 & 10 & 7 & 7 \\ 3 & 8 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[361] \quad 31 \overset{1}{5} 13151 \overset{1}{1} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -41, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 \\ 3 & 2 & 6 & 4 \\ 4 & 6 & 5 & 5 \\ 3 & 4 & 5 & 4 \end{pmatrix}_{b_0}$$

$$[362] \quad 1 \overset{1}{1} 4141 \overset{1}{3} 151 \overset{1}{1} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 2 & 9 & 4 \\ 9 & 18 & 10 \\ 4 & 10 & 5 \end{pmatrix}_{b_0}$$

$$[363] \quad 2 \overset{1}{1} \overset{1}{1} \overset{1}{4} 141 \overset{1}{1} 2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -56, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 3 & 6 & 4 & 5 \\ 2 & 6 & 3 & 5 & 5 \\ 2 & 4 & 5 & 2 & 4 \\ 2 & 5 & 5 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[364] \quad 2 \overset{1}{1} \overset{1}{1} \overset{1}{4} 141 \overset{1}{1} 2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -64, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 & 5 \\ 2 & 5 & 3 & 5 & 5 \\ 2 & 5 & 5 & 2 & 4 \\ 2 & 5 & 5 & 4 & 5 \end{pmatrix}_{b_0}$$

$$[365] \quad 31 \overset{1}{5} 1321 \overset{1}{1} 5 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -167, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13.79 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 5 & 5 & 2 & 2 \\ 5 & 3 & 6 & 2 & 5 \\ 5 & 6 & 3 & 3 & 5 \\ 2 & 2 & 3 & 0 & 2 \\ 2 & 5 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[366] \quad 51 \overset{1}{1} 2314141 \overset{1}{1} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -80, \det = -188, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13.77 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 6 & 7 & 3 \\ 6 & 4 & 8 & 6 \\ 7 & 8 & 7 & 7 \\ 3 & 6 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[367] \quad 321 \overset{1}{1} 513151 \overset{1}{1} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -218, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 13.85 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 5 & 5 & 1 & 2 \\ 5 & 38 & 41 & 0 & 12 \\ 5 & 41 & 38 & 1 & 12 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 12 & 12 & -1 & 3 \end{pmatrix}_{b_0} \\
[368] & 51231^{11}4141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -80, \det = -218, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.85 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 7 & 8 & 3 \\ 7 & 12 & 17 & 8 \\ 8 & 17 & 17 & 9 \\ 3 & 8 & 9 & 4 \end{pmatrix}_{b_0} \\
[369] & 15131^{11}4141^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -218, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.85 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 1 & 7 & 8 & 3 \\ 7 & 12 & 17 & 8 \\ 8 & 17 & 17 & 9 \\ 3 & 8 & 9 & 4 \end{pmatrix}_{b_0} \\
[370] & 14141^{11}3151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -79, \det = -248, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 5 & 5 & 1 & 2 \\ 5 & 8 & 11 & 0 & 6 \\ 5 & 11 & 8 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 6 & 0 & 3 \end{pmatrix}_{b_0} \\
[371] & 21^{11}414132^2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -416, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 4 & 4 & 4 & 2 \\ 4 & 16 & 19 & 18 & 10 \\ 4 & 19 & 16 & 19 & 10 \\ 4 & 18 & 19 & 15 & 9 \\ 2 & 10 & 10 & 9 & 5 \end{pmatrix}_{b_0} \\
[372] & 3151^{11}23151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -84, \det = -458, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.76 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 7 & 7 & 1 & 2 \\ 7 & 68 & 71 & 0 & 16 \\ 7 & 71 & 68 & 1 & 16 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 16 & 16 & -1 & 3 \end{pmatrix}_{b_0} \\
T = 12 & [373] 1^{11}231513221^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 12, \Delta = -4, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 4.0 \\
[376] & 1^{11}6^{11}131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 2 & 4 \\ 3 & 2 & 4 & 6 \end{pmatrix}_{b_0} \\
[377] & 1^{11}316^{11}131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -36, \det = 2, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
[378] & 1^{11}6^{11}131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 13, \Delta = -31, \det = 3, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 12.67 \rightarrow 13, T_{\max} = 13, T \in [13, 13] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 5 \end{pmatrix}_{b_0} \\
[379] & 21^{11}61315^{11}13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[380] & 41^{11}31513151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 6, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 4 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 \\ 4 & 1 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 0 & 10 & -1 & 3 \end{pmatrix}_{b_0} \\
[381] & 31513161^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 7, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15]
\end{aligned}$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[382] & 1 \overset{321}{6} 13151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 10, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 5 & 10 & 5 \\ 10 & 18 & 10 \\ 5 & 10 & 5 \end{pmatrix}_{b_0} \\
[383] & 321^{11} 6123151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 10, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 5 & 10 & 5 \\ 10 & 18 & 10 \\ 5 & 10 & 5 \end{pmatrix}_{b_0} \\
[384] & 161^{11} 231 \overset{11}{5} 13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.80 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 5 & 2 & 2 \\ 2 & 1 & 6 & 3 & 3 \\ 5 & 6 & 18 & 6 & 8 \\ 2 & 3 & 6 & 0 & 2 \\ 2 & 3 & 8 & 2 & 3 \end{pmatrix}_{b_0} \\
[385] & 21^{11} \overset{11}{6} 13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 5 & 9 & 5 \\ 9 & 13 & 9 \\ 5 & 9 & 5 \end{pmatrix}_{b_0} \\
[386] & 1 \overset{11}{6} 13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 5 & 12 & 5 \\ 12 & 24 & 12 \\ 5 & 12 & 5 \end{pmatrix}_{b_0} \\
[387] & 32161^{11} 23151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 24, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 5 & 12 & 5 \\ 12 & 24 & 12 \\ 5 & 12 & 5 \end{pmatrix}_{b_0} \\
[388] & 31 \overset{11}{6} 13151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 28, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.57 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 3 \end{pmatrix}_{b_0} \\
[389] & 1 \overset{11}{6} 131 \overset{11}{5} 13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 31, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.90 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[390] & 31 \overset{11}{5} 132161^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 31, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.90 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[391] & 31 \overset{11}{5} 1321^{11} 61^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 31, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.65 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 5 & 1 & 2 & 2 \\ 5 & 18 & 5 & 5 & 8 \\ 1 & 5 & 0 & 2 & 2 \\ 2 & 5 & 2 & 0 & 2 \\ 2 & 8 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[392] & 61^{11} 231 \overset{11}{5} 132 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 36, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 3 & 6 & 3 & 2 \\ 3 & 5 & 13 & 7 & 5 \\ 6 & 13 & 28 & 13 & 10 \\ 3 & 7 & 13 & 4 & 4 \\ 2 & 5 & 10 & 4 & 3 \end{pmatrix}_{b_0} \\
[394] & 31 \overset{11}{6} 13151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 41, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.83 \rightarrow 15, T_{\max} = 15, T \in [15, 15]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 0 & 3 \\ 2 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[393] $321 \overset{11}{6} 13151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 41, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.83 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 0 & 3 \\ 2 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[395] $31 \overset{11}{6} 123151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 41, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.83 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 0 & 3 \\ 2 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[396] $321 \overset{1}{6} 13151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 46, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.65 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 5 & 2 & 1 & 2 \\ 5 & 13 & 6 & 0 & 7 \\ 2 & 6 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 7 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[397] $31 \overset{1}{5} 1321^1 61^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.77 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 5 & 2 & 2 \\ 1 & 0 & 5 & 1 & 2 \\ 5 & 5 & 18 & 6 & 8 \\ 2 & 1 & 6 & 0 & 2 \\ 2 & 2 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[398] $231 \overset{11}{5} 13161^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 61, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.69 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 5 & 3 & 7 & 5 \\ 1 & 3 & 0 & 3 & 2 \\ 3 & 7 & 3 & 4 & 4 \\ 2 & 5 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[399] $321 \overset{1}{6} 13151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 73, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.79 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 5 & 1 & 2 \\ 2 & 1 & 6 & 0 & 3 \\ 5 & 6 & 13 & 1 & 7 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

[400] $31 \overset{151321}{6} 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 85, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.64 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 7 & 2 & 1 & 2 \\ 7 & 19 & 7 & 0 & 9 \\ 2 & 7 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 9 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[401] $31 \overset{151321}{6} 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 122, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.77 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 7 & 1 & 2 \\ 2 & 1 & 7 & 0 & 3 \\ 7 & 7 & 19 & 1 & 9 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

[402] $41 \overset{11}{3} 1513151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 131, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.95 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 26 & 29 & 0 & 10 \\ 4 & 29 & 26 & 1 & 10 \\ 1 & 0 & 1 & -3 & -1 \\ 2 & 10 & 10 & -1 & 3 \end{pmatrix}_{b_0}$$

[403] $21 \overset{11}{6} 13^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -80, \det = 143, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.59 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 8 & 11 & 0 & 6 \\ 4 & 11 & 8 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[404] $161 \overset{11}{2} 31 \overset{1}{5} 13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 167, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.79 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 5 & 5 & 2 & 2 \\ 5 & 18 & 21 & 5 & 8 \\ 5 & 21 & 18 & 6 & 8 \\ 2 & 5 & 6 & 0 & 2 \\ 2 & 8 & 8 & 2 & 3 \end{pmatrix}_{b_0}$$

[405] $61^{11}22315^{11}13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 192, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 6 & 6 & 2 & 2 \\ 6 & 28 & 31 & 6 & 10 \\ 6 & 31 & 28 & 7 & 10 \\ 2 & 6 & 7 & 0 & 2 \\ 2 & 10 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[406] $61^{11}2315^{11}132 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 207, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.68 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 6 & 6 & 3 & 2 \\ 6 & 28 & 31 & 12 & 10 \\ 6 & 31 & 28 & 13 & 10 \\ 3 & 12 & 13 & 4 & 4 \\ 2 & 10 & 10 & 4 & 3 \end{pmatrix}_{b_0}$$

[407] $321^{11}6123151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 218, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.85 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 5 & 5 & 1 & 2 \\ 5 & 13 & 16 & 0 & 7 \\ 5 & 16 & 13 & 1 & 7 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 7 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

[408] $1^{32111}6^{11}13151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 218, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.85 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 5 & 5 & 1 & 2 \\ 5 & 13 & 16 & 0 & 7 \\ 5 & 16 & 13 & 1 & 7 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 7 & 7 & 0 & 3 \end{pmatrix}_{b_0}$$

[409] $3221^{11}613151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 416, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 7 & 7 & 1 & 2 \\ 7 & 26 & 29 & 0 & 10 \\ 7 & 29 & 26 & 1 & 10 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 10 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[410] $32161^{11}23151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -82, \det = 458, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.76 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 7 & 7 & 1 & 2 \\ 7 & 19 & 22 & 0 & 9 \\ 7 & 22 & 19 & 1 & 9 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 9 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

[411] $1^{115132111}6^{11}13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 458, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.76 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 7 & 7 & 1 & 2 \\ 7 & 19 & 22 & 0 & 9 \\ 7 & 22 & 19 & 1 & 9 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 9 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

[412] $3161^{11}223151^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 12, T_{\max} = 15, \Delta = -81, \det = 732, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.63 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 9 & 9 & 1 & 2 \\ 9 & 34 & 37 & 0 & 12 \\ 9 & 37 & 34 & 1 & 12 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 12 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$T = 13$ [413] $1^{112321}7^{11}1231^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$

$T_{\min} = 14, T_{\max} = 14, T \in [14, 14]$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 2 & 4 \\ 3 & 2 & 4 & 6 \end{pmatrix}_{b_0}$$

[414] $23218^{11}1231^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 2 & 3 \\ 2 & 1 & 2 & 0 & 2 \\ 2 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0}$$

[415] $23218^{11}1231^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 1 & 2 \\ 2 & 1 & 2 & 1 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 4 \end{pmatrix}_{b_0} \\
[416] & 1^{\overline{1}} \overline{7}^{12321} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 14, \Delta = -30, \det = -3, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 13.67 \rightarrow 14, T_{\max} = 14, T \in [14, 14] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 5 \end{pmatrix}_{b_0} \\
[417] & 21^{\overline{1}} 71231 \overline{5}^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -4, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 1 & 3 & 3 & 3 \\ 2 & 3 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 2 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[420] & 41^{\overline{1}} 31^{\overline{1}} \overline{6}^{115131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -6, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 5 & 9 & 5 \\ 9 & 15 & 9 \\ 5 & 9 & 5 \end{pmatrix}_{b_0} \\
[418] & 41^{\overline{1}} 316123151^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -6, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 5 & 9 & 5 \\ 9 & 15 & 9 \\ 5 & 9 & 5 \end{pmatrix}_{b_0} \\
[419] & 41^{\overline{1}} 321613151^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -6, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 5 & 9 & 5 \\ 9 & 15 & 9 \\ 5 & 9 & 5 \end{pmatrix}_{b_0} \\
[422] & 21^{\overline{1}} 71^{\overline{1}} 231 \overline{5}^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15.57 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 5 & 2 & 2 & 2 \\ 5 & 18 & 9 & 5 & 8 \\ 2 & 9 & 4 & 3 & 4 \\ 2 & 5 & 3 & 0 & 2 \\ 2 & 8 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
[421] & 31^{\overline{1}} \overline{5}^{1321} \overline{7}^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -7, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15.57 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[423] & 23^{\overline{1}} 21^{\overline{1}} 81231^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -58, \det = -10, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.80 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 2 & 2 & 1 & 2 & 3 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 3 & 2 & 4 \end{pmatrix}_{b_0} \\
[424] & 41^{\overline{1}} 31613151^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -17, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15.59 \rightarrow 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 0 & 4 & 2 & 1 & 2 \\ 4 & 10 & 5 & 0 & 6 \\ 2 & 5 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 3 & 0 & 3 \end{pmatrix}_{b_0} \\
[425] & 413161^{\overline{1}} 23151^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -24, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16, T_{\max} = 16, T \in [16, 16] \\
& \begin{pmatrix} 5 & 12 & 5 \\ 12 & 24 & 12 \\ 5 & 12 & 5 \end{pmatrix}_{b_0} \\
[426] & 31^{\overline{1}} \overline{6}^{15131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -26, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 14.92 \rightarrow 15, T_{\max} = 15, T \in [15, 15] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 3 & 5 & 5 \\ 3 & 5 & 2 & 4 \\ 3 & 5 & 4 & 5 \end{pmatrix}_{b_0} \\
[427] & 31^{\overline{1}} \overline{5}^{13221} \overline{7}^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -28, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 15.57 \rightarrow 16, T_{\max} = 16, T \in [16, 16]
\end{aligned}$$

$$\begin{pmatrix} 0 & 6 & 1 & 2 & 2 \\ 6 & 28 & 6 & 6 & 10 \\ 1 & 6 & 0 & 2 & 2 \\ 2 & 6 & 2 & 0 & 2 \\ 2 & 10 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[430] $1^1 7^{11} 1231^1 5^{11} 13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -79$, $\det = -31$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.90 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[428] $31^1 5^{11} 1322171^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -79$, $\det = -31$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.90 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 2 \\ 2 & 1 & 2 & 3 & 3 \\ 1 & 2 & 0 & 2 & 2 \\ 2 & 3 & 2 & 0 & 2 \\ 2 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[429] $1^{11} 31^1 5^{11} 1321^1 7^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -79$, $\det = -31$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.65 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 5 & 1 & 2 & 2 \\ 5 & 18 & 5 & 5 & 8 \\ 1 & 5 & 0 & 2 & 2 \\ 2 & 5 & 2 & 0 & 2 \\ 2 & 8 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[431] $31^{11} 6^{11} 131^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 15$, $\Delta = -59$, $\det = -35$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.89 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 2 & 5 & 0 & 4 \\ 2 & 5 & 2 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

[432] $231^1 5^{11} 1321^{11} 71^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -80$, $\det = -40$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.60 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 6 & 1 & 3 & 2 \\ 6 & 28 & 6 & 12 & 10 \\ 1 & 6 & 0 & 3 & 2 \\ 3 & 12 & 3 & 4 & 4 \\ 2 & 10 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[435] $41^1 31^1 6^{11} 13151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -79$, $\det = -41$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.83 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 2 & 4 & 1 & 2 \\ 2 & 1 & 5 & 0 & 3 \\ 4 & 5 & 10 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[433] $321^{11} 7^{11} 123151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -80$, $\det = -41$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.83 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 0 & 3 \\ 2 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[434] $4131^{11} 6^{11} 13151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -79$, $\det = -41$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.83 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 0 & 3 \\ 2 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[436] $41^1 321^1 613151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -80$, $\det = -43$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.65 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 4 & 7 & 1 & 2 \\ 4 & 10 & 17 & 0 & 6 \\ 7 & 17 & 26 & 1 & 10 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[437] $41^1 321^1 613151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -80$, $\det = -44$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.91 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 7 & 4 & 1 & 2 \\ 7 & 26 & 17 & 0 & 10 \\ 4 & 17 & 10 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 10 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[438] $321^{11} 7^{11} 123151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -80$, $\det = -46$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.65 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 5 & 2 & 1 & 2 \\ 5 & 13 & 6 & 0 & 7 \\ 2 & 6 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 7 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[439] $31 \overset{1}{5} 13221 \overset{1}{7} 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -55, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15.65 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 1 & 6 & 2 & 2 \\ 1 & 0 & 6 & 1 & 2 \\ 6 & 6 & 28 & 7 & 10 \\ 2 & 1 & 7 & 0 & 2 \\ 2 & 2 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[440] $2 \overset{1}{1} \overset{1}{4} 14141 \overset{1}{3} 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -56, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 7 & 5 & 6 \\ 2 & 7 & 4 & 6 & 6 \\ 2 & 5 & 6 & 3 & 5 \\ 2 & 6 & 6 & 5 & 6 \end{pmatrix}_{b_0}$$

[441] $31 \overset{1}{1} \overset{1}{5} 131 \overset{1}{6} 13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -56, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.93 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 5 & 3 & 4 \\ 2 & 5 & 2 & 4 & 4 \\ 2 & 3 & 4 & 1 & 3 \\ 2 & 4 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[442] $231 \overset{1}{5} 132171 \overset{1}{1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -61, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15.69 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 5 & 3 & 7 & 5 \\ 1 & 3 & 0 & 3 & 2 \\ 3 & 7 & 3 & 4 & 4 \\ 2 & 5 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[444] $31 \overset{1}{1} \overset{1}{5} 131 \overset{1}{6} 13 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 15, \Delta = -59, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 & 4 \\ 2 & 4 & 2 & 4 & 4 \\ 2 & 4 & 4 & 1 & 3 \\ 2 & 4 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[443] $2 \overset{1}{1} \overset{1}{4} 14141 \overset{1}{3} 2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 13, \Delta = 0, \det = -64, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 4 & 6 & 6 & 6 \\ 2 & 6 & 4 & 6 & 6 \\ 2 & 6 & 6 & 3 & 5 \\ 2 & 6 & 6 & 5 & 6 \end{pmatrix}_{b_0}$$

[445] $321 \overset{1}{5} 1321 \overset{1}{7} 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -85, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15.64 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 7 & 2 & 1 & 2 \\ 7 & 19 & 7 & 0 & 9 \\ 2 & 7 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 9 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[446] $41 \overset{1}{1} 321613151 \overset{1}{1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -131, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15.95 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 10 & 13 & 0 & 6 \\ 4 & 13 & 10 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[447] $41 \overset{1}{1} 31 \overset{1}{5} 131 \overset{1}{6} 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -79, \det = -131, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15.95 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 10 & 13 & 0 & 6 \\ 4 & 13 & 10 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[448] $41 \overset{1}{1} 316123151 \overset{1}{1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -131, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15.95 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 10 & 13 & 0 & 6 \\ 4 & 13 & 10 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[449] $71 \overset{1}{1} 2231 \overset{1}{5} 132 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 13, T_{\max} = 16, \Delta = -80, \det = -188, (n_+, n_-, n_0) = (1, 13, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 15.60 \rightarrow 16, T_{\max} = 16, T \in [16, 16]$

$$\begin{pmatrix} 0 & 7 & 7 & 3 & 2 \\ 7 & 40 & 43 & 14 & 12 \\ 7 & 43 & 40 & 15 & 12 \\ 3 & 14 & 15 & 4 & 4 \\ 2 & 12 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

[450] $171^{11}2231^1 5^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -79$, $\det = -192$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.67 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 6 & 6 & 2 & 2 \\ 6 & 28 & 31 & 6 & 10 \\ 6 & 31 & 28 & 7 & 10 \\ 2 & 6 & 7 & 0 & 2 \\ 2 & 10 & 10 & 2 & 3 \end{pmatrix}_{b_0}$$

[453] $41321^{11}613151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -80$, $\det = -416$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.67 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 7 & 7 & 1 & 2 \\ 7 & 26 & 29 & 0 & 10 \\ 7 & 29 & 26 & 1 & 10 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 10 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[452] $3221^{11}7123151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -80$, $\det = -416$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.67 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 7 & 7 & 1 & 2 \\ 7 & 26 & 29 & 0 & 10 \\ 7 & 29 & 26 & 1 & 10 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 10 & 10 & 0 & 3 \end{pmatrix}_{b_0}$$

[451] $21^{11}41414132^2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 13$, $\Delta = 0$, $\det = -416$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 12.89 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 4 & 4 & 4 & 2 \\ 4 & 20 & 23 & 22 & 12 \\ 4 & 23 & 20 & 23 & 12 \\ 4 & 22 & 23 & 19 & 11 \\ 2 & 12 & 12 & 11 & 6 \end{pmatrix}_{b_0}$$

[454] $413161^{11}23151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -80$, $\det = -458$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.76 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 7 & 7 & 1 & 2 \\ 7 & 19 & 22 & 0 & 9 \\ 7 & 22 & 19 & 1 & 9 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 9 & 9 & 0 & 3 \end{pmatrix}_{b_0}$$

[455] $32171^{11}223151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 13$, $T_{\max} = 16$, $\Delta = -80$, $\det = -732$, $(n_+, n_-, n_0) = (1, 13, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15.63 \rightarrow 16$, $T_{\max} = 16$, $T \in [16, 16]$

$$\begin{pmatrix} 0 & 9 & 9 & 1 & 2 \\ 9 & 34 & 37 & 0 & 12 \\ 9 & 37 & 34 & 1 & 12 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 12 & 12 & 0 & 3 \end{pmatrix}_{b_0}$$

$T = 14$ [459] $1^{11}8^{13221} 12321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 2$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$

$T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 2 & 2 & 3 \\ 2 & 1 & 2 & 3 \\ 2 & 2 & 2 & 4 \\ 3 & 3 & 4 & 6 \end{pmatrix}_{b_0}$$

[458] $1^{11}6^{1315131} 131^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14$, $T_{\max} = 15$, $\Delta = -30$, $\det = 2$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 2 & 4 \\ 3 & 2 & 4 & 6 \end{pmatrix}_{b_0}$$

[457] $1^{12321} 8^{1231} 1231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 2$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 1 & 2 & 3 \\ 1 & 1 & 2 & 1 & 3 \\ 2 & 2 & 3 & 3 & 5 \end{pmatrix}_{b_0}$$

[462] $1^{11}8^{13221} 12321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 3$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 \\ 1 & 2 & 2 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 4 & 0 & 5 \end{pmatrix}_{b_0}$$

[461] $1^{12321} 8^{1231} 1231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14$, $T_{\max} = 15$, $\Delta = -29$, $\det = 3$, $(n_+, n_-, n_0) = (1, 14, 0)$, $c_{\text{ext}} = 4.0$
 $T_{\min} = 14.67 \rightarrow 15$, $T_{\max} = 15$, $T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 \\ 1 & 2 & 2 & 2 & 4 \\ 1 & 1 & 2 & 0 & 2 \\ 2 & 2 & 4 & 2 & 5 \end{pmatrix}_{b_0}$$

[460] $1^{\overline{1}1315131} 6^{11} 131^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 15, \Delta = -30, \det = 3, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 14.67 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 5 \end{pmatrix}_{b_0}$$

[463] $21^{\overline{1}1} 81231^{\overline{1}1} 5^{11} 132 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 4, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 3 & 2 & 3 & 2 \\ 3 & 5 & 5 & 7 & 5 \\ 2 & 5 & 4 & 5 & 4 \\ 3 & 7 & 5 & 4 & 4 \\ 2 & 5 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

[464] $41^{\overline{1}1} 3217123151^{\overline{1}1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 6, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 5 & 9 & 5 \\ 9 & 15 & 9 \\ 5 & 9 & 5 \end{pmatrix}_{b_0}$$

[465] $231^{\overline{1}1} 5^{11} 1321^{\overline{1}1} 81^{\overline{1}1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 3 & 2 \\ 3 & 2 & 3 & 4 & 4 \\ 2 & 2 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[466] $21^{\overline{1}1} 81^{\overline{1}1} 231^{\overline{1}1} 5^{11} 132 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 7, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 6 & 2 & 3 & 2 \\ 6 & 28 & 11 & 12 & 10 \\ 2 & 11 & 4 & 5 & 4 \\ 3 & 12 & 5 & 4 & 4 \\ 2 & 10 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

[467] $21^{\overline{1}11} 8^{11} 51321^{\overline{1}1} 123 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 16, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 5 & 9 & 5 \\ 9 & 13 & 9 \\ 5 & 9 & 5 \end{pmatrix}_{b_0}$$

[471] $21^{\overline{1}1} 5^{11} 1321^{\overline{1}1} 8^{11} 123 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 28, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 7 & 4 & 1 & 2 \\ 7 & 19 & 13 & 0 & 9 \\ 4 & 13 & 8 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 9 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[468] $21^{\overline{1}1} 8^{321} 123151^{\overline{1}1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 28, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 5 & 4 & 1 & 2 \\ 5 & 13 & 11 & 0 & 7 \\ 4 & 11 & 8 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 7 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[469] $231^{\overline{1}1} 5^{11} 13221^{\overline{1}1} 81^{\overline{1}1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 28, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 7 & 1 & 3 & 2 \\ 7 & 40 & 7 & 14 & 12 \\ 1 & 7 & 0 & 3 & 2 \\ 3 & 14 & 3 & 4 & 4 \\ 2 & 12 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[470] $321^{\overline{1}1} 8^{11} 123151^{\overline{1}1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 28, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[472] $1^{\overline{1}1} 231^{\overline{1}1} 5^{11} 1321^{\overline{1}1} 8^{11} 1^{\overline{1}1} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 40, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.60 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 6 & 1 & 3 & 2 \\ 6 & 28 & 6 & 12 & 10 \\ 1 & 6 & 0 & 3 & 2 \\ 3 & 12 & 3 & 4 & 4 \\ 2 & 10 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[473] $321 \overset{11}{8} 1223151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 41, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.83 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 0 & 3 \\ 2 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[474] $3221 \overset{11}{8} 123151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 41, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.83 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 0 & 3 \\ 2 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[475] $321 \overset{11}{8} 123151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 41, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.83 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 0 & 3 \\ 2 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[476] $41^1 321^1 7123151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 44, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.91 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 7 & 4 & 1 & 2 \\ 7 & 26 & 17 & 0 & 10 \\ 4 & 17 & 10 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 10 & 6 & 0 & 3 \end{pmatrix}_{b_0}$$

[477] $321^1 \overset{11}{8} 123151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 46, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.65 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 5 & 2 & 1 & 2 \\ 5 & 13 & 6 & 0 & 7 \\ 2 & 6 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 7 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[478] $321^1 \overset{11}{8} 1223151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 46, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.65 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 5 & 2 & 1 & 2 \\ 5 & 13 & 6 & 0 & 7 \\ 2 & 6 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 7 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[479] $2231 \overset{11}{5} 1321^1 81^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 51, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.55 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 7 & 1 & 4 & 2 \\ 7 & 40 & 7 & 21 & 12 \\ 1 & 7 & 0 & 4 & 2 \\ 4 & 21 & 4 & 10 & 6 \\ 2 & 12 & 2 & 6 & 3 \end{pmatrix}_{b_0}$$

[480] $231 \overset{11}{5} 1322181^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 61, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.69 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 5 & 3 & 7 & 5 \\ 1 & 3 & 0 & 3 & 2 \\ 3 & 7 & 3 & 4 & 4 \\ 2 & 5 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[481] $181231 \overset{11}{5} 132 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 61, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.69 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 2 \\ 3 & 5 & 3 & 7 & 5 \\ 1 & 3 & 0 & 3 & 2 \\ 3 & 7 & 3 & 4 & 4 \\ 2 & 5 & 2 & 4 & 3 \end{pmatrix}_{b_0}$$

[482] $3221 \overset{11}{8} 123151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 63, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 7 & 2 & 1 & 2 \\ 7 & 26 & 8 & 0 & 10 \\ 2 & 8 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 10 & 3 & 0 & 3 \end{pmatrix}_{b_0}$$

[483] $231 \overset{11}{5} 13221^1 81^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2]$ $T = 14, T_{\max} = 17, \Delta = -79, \det = 64, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.62 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$

$$\begin{pmatrix} 0 & 1 & 7 & 3 & 2 \\ 1 & 0 & 7 & 2 & 2 \\ 7 & 7 & 40 & 15 & 12 \\ 3 & 2 & 15 & 4 & 4 \\ 2 & 2 & 12 & 4 & 3 \end{pmatrix}_{b_0}$$

- [484] $321 \begin{smallmatrix} 1^1 51321^1 \\ 8 \\ 1^1 \end{smallmatrix} 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 85, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.64 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 7 & 2 & 1 & 2 \\ 7 & 19 & 7 & 0 & 9 \\ 2 & 7 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 9 & 3 & 0 & 3 \end{pmatrix}_{b_0}$
- [485] $3221 \begin{smallmatrix} 1^1 51321^1 \\ 8 \\ 1^1 \end{smallmatrix} 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 85, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.64 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 7 & 2 & 1 & 2 \\ 7 & 19 & 7 & 0 & 9 \\ 2 & 7 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 9 & 3 & 0 & 3 \end{pmatrix}_{b_0}$
- [486] $2231 \begin{smallmatrix} 1^{11} \\ 5 \\ 132181^1 \end{smallmatrix} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 101, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 4 & 1 & 4 & 2 \\ 4 & 11 & 4 & 13 & 7 \\ 1 & 4 & 0 & 4 & 2 \\ 4 & 13 & 4 & 10 & 6 \\ 2 & 7 & 2 & 6 & 3 \end{pmatrix}_{b_0}$
- [487] $321 \begin{smallmatrix} 1^1 513221^1 \\ 8 \\ 1^1 \end{smallmatrix} 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 112, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.57 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 9 & 2 & 1 & 2 \\ 9 & 34 & 9 & 0 & 12 \\ 2 & 9 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 12 & 3 & 0 & 3 \end{pmatrix}_{b_0}$
- [488] $3221 \begin{smallmatrix} 1^1 \\ 8 \\ 123151^1 \end{smallmatrix} 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 118, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.66 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 2 & 7 & 1 & 2 \\ 2 & 1 & 8 & 0 & 3 \\ 7 & 8 & 26 & 1 & 10 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 10 & 0 & 3 \end{pmatrix}_{b_0}$
- [489] $41 \begin{smallmatrix} 1^1 3217123151^1 \\ 41 \end{smallmatrix} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 131, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.95 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 10 & 13 & 0 & 6 \\ 4 & 13 & 10 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [490] $21 \begin{smallmatrix} 1^1 51321 \\ 8 \\ 123 \end{smallmatrix} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 143, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.59 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 4 & 4 & 1 & 2 \\ 4 & 8 & 11 & 0 & 6 \\ 4 & 11 & 8 & 1 & 6 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 6 & 6 & 0 & 3 \end{pmatrix}_{b_0}$
- [491] $81 \begin{smallmatrix} 1^1 \\ 1^1 223151322 \end{smallmatrix} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 176, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.55 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 8 & 8 & 4 & 2 \\ 8 & 54 & 57 & 24 & 14 \\ 8 & 57 & 54 & 25 & 14 \\ 4 & 24 & 25 & 10 & 6 \\ 2 & 14 & 14 & 6 & 3 \end{pmatrix}_{b_0}$
- [492] $321 \begin{smallmatrix} 1^1 513221^1 \\ 8 \\ 1^1 \end{smallmatrix} 1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 181, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.65 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 2 & 9 & 1 & 2 \\ 2 & 1 & 9 & 0 & 3 \\ 9 & 9 & 34 & 1 & 12 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 12 & 0 & 3 \end{pmatrix}_{b_0}$
- [493] $181 \begin{smallmatrix} 1^1 \\ 1^1 22315132 \end{smallmatrix} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 188, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.60 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 7 & 7 & 3 & 2 \\ 7 & 40 & 43 & 14 & 12 \\ 7 & 43 & 40 & 15 & 12 \\ 3 & 14 & 15 & 4 & 4 \\ 2 & 12 & 12 & 4 & 3 \end{pmatrix}_{b_0}$
- [495] $1 \begin{smallmatrix} 3221^1 \\ 8 \\ 123151^1 \end{smallmatrix} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 416, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$
 $T_{\min} = 16.67 \rightarrow 17, T_{\max} = 17, T \in [17, 17]$
 $\begin{pmatrix} 0 & 7 & 7 & 1 & 2 \\ 7 & 26 & 29 & 0 & 10 \\ 7 & 29 & 26 & 1 & 10 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 10 & 10 & 0 & 3 \end{pmatrix}_{b_0}$
- [494] $3221 \begin{smallmatrix} 1^1 \\ 8 \\ 1223151^1 \end{smallmatrix} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 416, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0$

$$\begin{aligned}
& T_{\min} = 16.67 \rightarrow 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 7 & 7 & 1 & 2 \\ 7 & 26 & 29 & 0 & 10 \\ 7 & 29 & 26 & 1 & 10 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 10 & 10 & 0 & 3 \end{pmatrix}_{b_0} \\
[496] & 322181^{11}223151^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 732, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16.63 \rightarrow 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 9 & 9 & 1 & 2 \\ 9 & 34 & 37 & 0 & 12 \\ 9 & 37 & 34 & 1 & 12 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 12 & 12 & 0 & 3 \end{pmatrix}_{b_0} \\
[497] & 1^{11}513221^{11}8^{11}123 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 14, T_{\max} = 17, \Delta = -79, \det = 732, (n_+, n_-, n_0) = (1, 14, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16.63 \rightarrow 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 9 & 9 & 1 & 2 \\ 9 & 34 & 37 & 0 & 12 \\ 9 & 37 & 34 & 1 & 12 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 12 & 12 & 0 & 3 \end{pmatrix}_{b_0} \\
T = 16 & [501] 1^{11}3151321^{11}8^{11}1231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 2, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 2 \\ 2 & 2 & 2 & 4 \\ 3 & 2 & 4 & 6 \end{pmatrix}_{b_0} \\
[502] & 1^{11}3151321^{11}8^{11}1231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 16, T_{\max} = 17, \Delta = -29, \det = 3, (n_+, n_-, n_0) = (1, 16, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 16.67 \rightarrow 17, T_{\max} = 17, T \in [17, 17] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & 5 \end{pmatrix}_{b_0} \\
T = 17 & [503] 1^{11}23151321^{11}8^{11}12321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 17, T_{\max} = 17, \Delta = -3, \det = 0, (n_+, n_-, n_0) = (1, 17, 1), c_{\text{ext}} = 4.0 \\
[507] & 31^{31}6^{11}1316131^{11}51^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -29, \det = -56, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 17.93 \rightarrow 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 3 & 6 & 4 & 5 \\ 2 & 6 & 3 & 5 & 5 \\ 2 & 4 & 5 & 2 & 4 \\ 2 & 5 & 5 & 4 & 5 \end{pmatrix}_{b_0} \\
[508] & 31^{31}6^{11}1316131^{11}51^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 17, T_{\max} = 18, \Delta = -29, \det = -64, (n_+, n_-, n_0) = (1, 17, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 18, T_{\max} = 18, T \in [18, 18] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 3 & 5 & 5 & 5 \\ 2 & 5 & 3 & 5 & 5 \\ 2 & 5 & 5 & 2 & 4 \\ 2 & 5 & 5 & 4 & 5 \end{pmatrix}_{b_0} \\
T = 18 & [512] 1^{11}3221^{11}(12)12231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 18, T_{\max} = 19, \Delta = -35, \det = 2, (n_+, n_-, n_0) = (1, 18, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 19, T_{\max} = 19, T \in [19, 19] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 \\ 1 & 2 & 1 & 1 & 3 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 3 & 3 & 0 & 5 \end{pmatrix}_{b_0} \\
T = 19 & [513] 1^{11}2321^{11}8^{11}12321712321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 19, T_{\max} = 19, \Delta = -2, \det = 0, (n_+, n_-, n_0) = (1, 19, 1), c_{\text{ext}} = 4.0 \\
[517] & 3221^{11}(12)12231^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 19, T_{\max} = 21, \Delta = -58, \det = -26, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0 \\
& T_{\min} = 20.92 \rightarrow 21, T_{\max} = 21, T \in [21, 21] \\
& \begin{pmatrix} 1 & 3 & 3 & 3 \\ 3 & 3 & 5 & 5 \\ 3 & 5 & 2 & 4 \\ 3 & 5 & 4 & 5 \end{pmatrix}_{b_0} \\
[518] & 3221^{11}(12)12231^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 19, T_{\max} = 21, \Delta = -58, \det = -35, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} =
\end{aligned}$$

4.0

$$T_{\min} = 20.89 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 2 \\ 2 & 2 & 5 & 0 & 4 \\ 2 & 5 & 2 & 1 & 4 \\ 1 & 0 & 1 & -2 & 0 \\ 2 & 4 & 4 & 0 & 4 \end{pmatrix}_{b_0}$$

$$[519] \quad 3221 \quad (12) \quad 1223 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 19, T_{\max} = 21, \Delta = -58, \det = -56, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 20.93 \rightarrow 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 5 & 3 & 4 \\ 2 & 5 & 2 & 4 & 4 \\ 2 & 3 & 4 & 1 & 3 \\ 2 & 4 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[520] \quad 3221 \quad (12) \quad 1223 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 19, T_{\max} = 21, \Delta = -58, \det = -64, (n_+, n_-, n_0) = (1, 19, 0), c_{\text{ext}} = 4.0$$

$$T_{\min} = 21, T_{\max} = 21, T \in [21, 21]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 2 \\ 2 & 2 & 4 & 4 & 4 \\ 2 & 4 & 2 & 4 & 4 \\ 2 & 4 & 4 & 1 & 3 \\ 2 & 4 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

$$T = 20 \quad [521] \quad 1^1 8^{1^{2321}} 12321 8^{1^{2321}} 12321 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 20, T_{\max} = 20, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 20, 1), c_{\text{ext}} = 4.0$$

$$T = 22 \quad [522] \quad 1^1 8^{1^{2321}} 12321 8^{1^{2321}} 12315131 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 22, T_{\max} = 22, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 22, 1), c_{\text{ext}} = 4.0$$

$$T = 23 \quad [523] \quad 1^1 8^{1^{2321}} 12321 8^{1^{2321}} 123216131 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] \quad T = 23, T_{\max} = 23, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 23, 1), c_{\text{ext}} = 4.0$$

6.6 NHC(-2)(-2)(-3) + $\mathbf{e}_6$

$$T = 7 \quad [0] \quad 21^1 31^{1^{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 4 & 9 & 9 & 4 & 5 \\ 3 & 3 & 4 & -3 & -1 \\ 3 & 4 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1^{11}} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -28, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 4 & 9 & 9 & 4 & 5 \\ 3 & 3 & 4 & -3 & -1 \\ 3 & 4 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1^{11}} 31^{1^{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -56, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 4 & 5 & 9 & 0 & 5 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 3 & 5 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1^{11}} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -63, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 3 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 3 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1^{11}} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.47 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 131^{1^{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -111, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 100, T \in [14, 100]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 4 & 9 & 9 & 4 & 5 \\ 2 & 3 & 4 & -3 & -1 \\ 3 & 4 & 5 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 21^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{c}_6 \quad T = 7, T_{\max} = 100, \Delta = -163, \det = -475, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.42 \rightarrow 13, T_{\max} = 100, T \in [13, 100]$$

$$\begin{pmatrix} 4 & 6 & 8 & 3 & 4 \\ 6 & 4 & 9 & 3 & 4 \\ 8 & 9 & 9 & 4 & 5 \\ 3 & 3 & 4 & -3 & -1 \\ 4 & 4 & 5 & -1 & 0 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [7] \ 1^{11}31^131^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{c}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 16, T_{\max} = 98, T \in [16, 98]$$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 1 & 4 & 1 & 4 & 0 & 3 \\ 2 & 3 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 3 & 3 & -4 & -3 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11}321^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{c}_6 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13, T_{\max} = 94, T \in [13, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 3 & 1 & 0 & -3 & -1 \\ 2 & 3 & 2 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11}321^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{c}_6 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 37, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.41 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 1 & 3 & 0 & -3 & -1 \\ 2 & 2 & 3 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^1321^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{c}_6 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 43, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.35 \rightarrow 14, T_{\max} = 94, T \in [14, 94]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 3 & 6 & 5 & 5 & 5 \\ 2 & 3 & 5 & -2 & 0 \\ 3 & 4 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 2231^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{c}_6 \quad T = 8, T_{\max} = 94, \Delta = -135, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 94, T \in [13, 94]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 4 & 6 & 5 & 5 & 5 \\ 3 & 3 & 5 & -2 & 0 \\ 3 & 4 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11}31^131^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{c}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 100, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 4 & 4 & 0 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 4 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{11}31^131^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{c}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.91 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 4 & 7 & 0 & 4 \\ 2 & 3 & 7 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 3 & 4 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{11}31^131^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{c}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 13.23 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 1 & 4 & 0 & 3 \\ 2 & 6 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 3 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^131^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{c}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 214, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 4 & 4 & 0 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 2 & 4 & 4 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [16] \quad 1^{11}2^1321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 19, T_{\max} = 21, T \in [19, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 1 & 2 \\ 1 & 3 & 0 & 1 & -3 & -1 \\ 2 & 3 & 0 & 2 & -1 & -5 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{11}2^1321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 1 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 3 & 1 & 0 & -3 & -1 \\ 2 & 3 & 3 & -1 & -1 & -2 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11}3221^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -13, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 15.15 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 4 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 1 & 4 & 0 & -2 & 0 \\ 2 & 2 & 4 & -1 & 0 & -2 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11}2^1321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 17.14 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 1 & 3 \\ 1 & 2 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 1 & 3 & 0 & -3 & -1 \\ 2 & 3 & 3 & -1 & -1 & -4 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11}2^1321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -22, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 0 & 3 & 1 & -3 & -1 \\ 2 & 0 & 3 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11}2^1321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 4 & 0 & 2 & 2 \\ 2 & 3 & 3 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [22] \quad 1^{11}41321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 \quad T = 10, T_{\max} = 22, \Delta = -105, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 16.15 \rightarrow 17, T_{\max} = 22, T \in [17, 22]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 3 \\ 2 & 1 & 2 & 0 & 3 \\ 2 & 2 & 1 & 2 & 3 \\ 1 & 0 & 2 & -4 & -2 \\ 3 & 3 & 3 & -2 & -1 \end{pmatrix}_{b_0}$$

6.7 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{e}_6$

$$T = 8 \quad [0] \quad 1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{e}_6 \quad T = 8, T_{\max} = 103, \Delta = -212, \det = 460, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 15.09 \rightarrow 16, T_{\max} = 103, T \in [16, 103]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 4 & 9 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 5 & -1 & -4 & -3 \end{pmatrix}_{b_0}$$

6.8 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{f}_4$

$$T = 8 \quad [0] \quad 1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 349, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 4 & 9 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & -1 & -3 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [1] \quad 1^{\overline{111}} 3^{\overline{11}} 2^{\overline{11}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{f}_4 \quad T = 9, \quad T_{\max} = 101, \quad \Delta = -158, \quad \det = -148, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.41 \rightarrow 15, \quad T_{\max} = 101, \quad T \in [15, 101]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 0 & 4 \\ 3 & 6 & 5 & 5 & 0 & 5 \\ 2 & 3 & 5 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

6.9 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + NHC(-2)(-2)(-3)

$$T = 10 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{111}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{e}_6 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -83, \quad \det = 241, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.84 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 4 & 3 \\ 3 & 6 & 5 & 0 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 3 & 4 & 0 \\ 3 & 4 & 4 & -4 & 3 & 4 & 2 \end{pmatrix}_{b_0}$$

6.10 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + NHC(-2)(-3)

$$T = 9 \quad [0] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, \quad T_{\max} = 24, \quad \Delta = -113, \quad \det = -3, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.33 \rightarrow 17, \quad T_{\max} = 24, \quad T \in [17, 24]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 4 & 0 & 2 \\ 3 & 3 & 4 & -4 & 4 & 2 & -4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, \quad T_{\max} = 24, \quad \Delta = -113, \quad \det = -129, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.03 \rightarrow 14, \quad T_{\max} = 24, \quad T \in [14, 24]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 4 & 3 \\ 3 & 3 & 5 & 0 & 4 & 3 & 3 \\ 3 & 3 & 4 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, \quad T_{\max} = 24, \quad \Delta = -113, \quad \det = -130, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.20 \rightarrow 14, \quad T_{\max} = 24, \quad T \in [14, 24]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 4 & 5 & 9 & 0 & 8 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 8 & 0 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 4 & 0 & 2 \\ 3 & 3 & 5 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, \quad T_{\max} = 24, \quad \Delta = -113, \quad \det = -234, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.40 \rightarrow 14, \quad T_{\max} = 24, \quad T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 1 & 3 \\ 3 & 4 & 9 & 3 & 6 & 0 & 4 \\ 4 & 9 & 9 & 4 & 8 & 0 & 5 \\ 2 & 3 & 4 & -3 & 3 & 0 & -1 \\ 3 & 6 & 8 & 3 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 5 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{e}_6 + [\textcolor{red}{su}_2 \cdot \textcolor{blue}{g}_2] \quad T = 9, \quad T_{\max} = 24, \quad \Delta = -113, \quad \det = -275, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 24, \quad T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 9, \ T_{\max} = 24, \ \Delta = -113, \ \det = -322, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.18 \rightarrow 14, \ T_{\max} = 24, \ T \in [14, 24]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 4 & 3 \\ 2 & 3 & 4 & 0 & 4 & 0 & 2 \\ 3 & 3 & 4 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [6] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, \ T_{\max} = 14, \ \Delta = -92, \ \det = 428, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 6 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 6 & 3 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 3 & 0 & 4 & 0 & 2 \\ 2 & 4 & 4 & 2 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

6.11 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -83, \ \det = 51, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 4 & 3 & 4 \\ 2 & 3 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{112}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -83, \ \det = 150, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 4 & 12 & 4 \\ 2 & 3 & 4 & 0 & 4 & 0 & 9 & 2 \\ 7 & 12 & 13 & 0 & 12 & 9 & 28 & 10 \\ 3 & 4 & 4 & -4 & 4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -83, \ \det = 294, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.84 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 4 & 6 & 4 \\ 2 & 3 & 4 & 0 & 4 & 0 & 3 & 2 \\ 3 & 6 & 5 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & 4 & -4 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -83, \ \det = 721, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 4 & 9 & 4 \\ 2 & 3 & 4 & 0 & 4 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

6.12 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-3) + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -92, \ \det = 266, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.15 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 4 & 3 & 3 \\ 3 & 3 & 5 & 0 & 4 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 3 & 4 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -92, \ \det = 326, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.45 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 4 & 3 & 4 \\ 2 & 3 & 4 & 0 & 4 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

6.13 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 101, \ \Delta = -162, \ \det = 135, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.44 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 12 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 6 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 101, \ \Delta = -162, \ \det = 238, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.55 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 4 & 9 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & -1 & -2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 101, \ \Delta = -162, \ \det = 242, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.48 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 5 \\ 9 & 13 & 18 & 0 & 7 \\ 11 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 7 & 8 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 8, \ T_{\max} = 101, \ \Delta = -162, \ \det = 253, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.68 \rightarrow 14, \ T_{\max} = 101, \ T \in [14, 101]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 11 & 17 & 20 & 0 & 8 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 6 & 8 & -4 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \ 1^{11} 3^1 2^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 94, \ \Delta = -134, \ \det = -30, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.60 \rightarrow 14, \ T_{\max} = 94, \ T \in [14, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 1 & 3 \\ 1 & 2 & 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 1 & 0 & -3 & 1 & -1 \\ 1 & 1 & 2 & 0 & 1 & -2 & 0 \\ 2 & 3 & 2 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 2^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 94, \ \Delta = -134, \ \det = -76, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.68 \rightarrow 14, \ T_{\max} = 94, \ T \in [14, 94]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 1 & 3 & 1 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 3 & 0 & -3 & 1 & -1 \\ 1 & 2 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 3 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^1 2^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 94, \ \Delta = -134, \ \det = -105, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.69 \rightarrow 14, \ T_{\max} = 94, \ T \in [14, 94]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 0 & 4 \\ 3 & 6 & 5 & 5 & 0 & 5 \\ 2 & 3 & 5 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^1 1^{11} 3^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -226, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 13 & 16 & 0 & 7 \\ 5 & 12 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 7 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[8] 1^{11} 31^{11} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -391, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 1 & 5 & 4 & 5 & 1 & 3 \\ 5 & 13 & 13 & 15 & 0 & 7 \\ 4 & 13 & 10 & 13 & 0 & 6 \\ 5 & 15 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 7 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

6.14 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] 1^{111} 31^{114} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 44 & 5 \\ 8 & 10 & 17 & 0 & 60 & 6 \\ 12 & 17 & 25 & 0 & 96 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 44 & 60 & 96 & 0 & 346 & 34 \\ 5 & 6 & 9 & -4 & 34 & 0 \end{pmatrix}_{b_0}$$

$$[1] 1^{114} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 44 & 5 \\ 9 & 13 & 18 & 0 & 72 & 7 \\ 11 & 18 & 20 & 0 & 84 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 44 & 72 & 84 & 0 & 346 & 34 \\ 5 & 7 & 8 & -4 & 34 & -1 \end{pmatrix}_{b_0}$$

$$[2] 1^{11} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 20 & 0 & 13 & 7 \\ 5 & 6 & 9 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] 1^{11} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.54 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 8 & 5 \\ 9 & 13 & 18 & 0 & 12 & 7 \\ 11 & 18 & 20 & 0 & 16 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 12 & 16 & 0 & 10 & 6 \\ 5 & 7 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[4] 1^{111} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -317, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 15 & 17 & 0 & 13 & 7 \\ 5 & 7 & 8 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[5] 1^{111} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -351, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 17 & 5 \\ 8 & 10 & 17 & 0 & 24 & 6 \\ 12 & 17 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 24 & 36 & 0 & 49 & 13 \\ 5 & 6 & 9 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[6] 1^{111} 31^{113} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -439, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 35 & 5 \\ 8 & 10 & 17 & 0 & 48 & 6 \\ 12 & 17 & 25 & 0 & 76 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 35 & 48 & 76 & 0 & 217 & 27 \\ 5 & 6 & 9 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1111}} 3 1^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -525, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.56 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 & 3 \\ 3 & 4 & 9 & 3 & 0 & 6 & 4 \\ 4 & 9 & 9 & 4 & 0 & 8 & 5 \\ 2 & 3 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 8 & 3 & 0 & 4 & 4 \\ 3 & 4 & 5 & -1 & -2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3 1^{\overline{112}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -526, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.44 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 26 & 5 \\ 8 & 10 & 17 & 0 & 36 & 6 \\ 12 & 17 & 25 & 0 & 56 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 36 & 56 & 0 & 118 & 20 \\ 5 & 6 & 9 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{113}} 3 1^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -653, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.55 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 35 & 5 \\ 9 & 13 & 18 & 0 & 57 & 7 \\ 11 & 18 & 20 & 0 & 67 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 35 & 57 & 67 & 0 & 217 & 27 \\ 5 & 7 & 8 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{111}} 3 1^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -709, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.48 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 17 & 5 \\ 9 & 13 & 18 & 0 & 27 & 7 \\ 11 & 18 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 27 & 33 & 0 & 49 & 13 \\ 5 & 7 & 8 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{112}} 3 1^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 93, \ \Delta = -132, \ \det = -848, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.49 \rightarrow 14, \ T_{\max} = 93, \ T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 26 & 5 \\ 9 & 13 & 18 & 0 & 42 & 7 \\ 11 & 18 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 42 & 50 & 0 & 118 & 20 \\ 5 & 7 & 8 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [12] \ 1^{\overline{11}} 3 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -104, \ \det = 11, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 16, \ T_{\max} = 22, \ T \in [16, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 3 & 1 & 0 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 \\ 1 & 1 & 3 & 0 & -3 & 1 & 0 \\ 1 & 2 & 1 & 0 & 1 & -2 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 3 & -1 & -1 & 0 & -3 \end{pmatrix}_{b_0}$$

6.15 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{11}} 3 1^{\overline{1111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 16, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.75 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 20 & 9 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 20 & 0 & 13 & 15 & 7 \\ 9 & 12 & 20 & 0 & 15 & 13 & 7 \\ 5 & 6 & 9 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3 1^{\overline{1111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 188, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.47 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 17 & 5 \\ 8 & 10 & 17 & 0 & 12 & 24 & 6 \\ 12 & 17 & 25 & 0 & 20 & 36 & 9 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 20 & 0 & 13 & 27 & 7 \\ 17 & 24 & 36 & 0 & 27 & 49 & 13 \\ 5 & 6 & 9 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1111}} 3^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 300, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.64 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 15 & 17 & 0 & 13 & 15 & 7 \\ 9 & 15 & 17 & 0 & 15 & 13 & 7 \\ 5 & 7 & 8 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1111}} 3^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 22, \ \Delta = -102, \ \det = 832, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.52 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 17 & 5 \\ 9 & 13 & 18 & 0 & 15 & 27 & 7 \\ 11 & 18 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 15 & 17 & 0 & 13 & 27 & 7 \\ 17 & 27 & 33 & 0 & 27 & 49 & 13 \\ 5 & 7 & 8 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

6.16 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{11}} 21^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 22, \ \Delta = -107, \ \det = 138, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.70 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 4 \\ 3 & 6 & 5 & 5 & 0 & 3 & 5 \\ 2 & 3 & 5 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 5 & 0 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

6.17 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{so}_8$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -196, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 11 & 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 17 & 0 & 12 & 6 \\ 5 & 6 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{111}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -274, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.03 \rightarrow 15, \ T_{\max} = 99, \ T \in [15, 99]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 20 & 0 & 12 & 6 \\ 5 & 6 & 9 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -559, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 15 & 17 & 0 & 12 & 6 \\ 5 & 7 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [3] \ 1^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 25, \ \Delta = -113, \ \det = 87, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.84 \rightarrow 14, \ T_{\max} = 25, \ T \in [14, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 3 & 1 & 0 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 \\ 1 & 1 & 3 & 0 & -3 & 1 & 0 \\ 1 & 2 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 3 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

6.18 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$

$$T = 8 \quad [1] \ 1^{111} 3^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 25, T_{\max} = 92, T \in [25, 92]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 4 & 9 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 5 & -1 & 0 & -12 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{12} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 15.91 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 4 & 5 & 9 & 0 & 16 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 9 & 16 & 0 & 28 & 10 \\ 3 & 3 & 5 & -4 & 10 & -3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 23, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 \\ 3 & 3 & 5 & -4 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 15.27 \rightarrow 16, T_{\max} = 92, T \in [16, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 6 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 4 & 5 & 9 & 0 & 16 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 6 & 16 & 0 & 22 & 8 \\ 3 & 3 & 5 & -4 & 8 & -3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 15 & 17 & 0 & 49 & 13 \\ 3 & 4 & 4 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{112} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$$

$$T_{\min} = 13.33 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 10 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 10 & 0 & 22 & 8 \\ 3 & 4 & 4 & -4 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 & 3 \\ 3 & 4 & 9 & 3 & 6 & 4 \\ 4 & 9 & 9 & 4 & 8 & 5 \\ 3 & 3 & 4 & -3 & 3 & -1 \\ 3 & 6 & 8 & 3 & 4 & 4 \\ 3 & 4 & 5 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 21^1 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 & 3 \\ 3 & 4 & 9 & 3 & 6 & 4 \\ 4 & 9 & 9 & 4 & 8 & 5 \\ 3 & 3 & 4 & -3 & 3 & -1 \\ 3 & 6 & 8 & 3 & 4 & 4 \\ 3 & 4 & 5 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.51 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 \\ 3 & 4 & 4 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{13} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 4 & 5 & 9 & 0 & 28 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 15 & 28 & 0 & 85 & 17 \\ 3 & 3 & 5 & -4 & 17 & -1 \end{pmatrix}_{b_0}$$

[10] $1^{113}31^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 95, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 13.20 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 8 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 3 & 8 & 0 & 4 & 4 \\ 3 & 3 & 5 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

[12] $1^{1111}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 114, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 14.58 \rightarrow 15, T_{\max} = 92, T \in [15, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 4 & 3 \\ 3 & 4 & 9 & 3 & 6 & 4 \\ 4 & 9 & 9 & 4 & 8 & 5 \\ 2 & 3 & 4 & -3 & 3 & -1 \\ 4 & 6 & 8 & 3 & 4 & 4 \\ 3 & 4 & 5 & -1 & 4 & -2 \end{pmatrix}_{b_0}$$

[16] $1^{113}31^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 15 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 3 & 5 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 18 & 27 & 0 & 133 & 21 \\ 3 & 3 & 4 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{112}31^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 12 & 18 & 0 & 58 & 14 \\ 3 & 3 & 4 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{111}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.56 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 6 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 4 \\ 3 & 4 & 4 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

[15] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 7 \\ 3 & 3 & 4 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{114}31^{14} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 126, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 20 & 3 \\ 2 & 1 & 5 & 0 & 24 & 3 \\ 3 & 5 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 20 & 24 & 36 & 0 & 238 & 28 \\ 3 & 3 & 4 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{111}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 127, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 \\ 3 & 3 & 4 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

[19] $1^{112}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 130, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 12.54 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 6 \\ 3 & 3 & 4 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[20] $1^{113}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 135, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.67 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 9 & 12 & 0 & 25 & 9 \\ 3 & 3 & 4 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{114} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 142, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 8 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 12 & 16 & 0 & 46 & 12 \\ 3 & 3 & 4 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{113} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 148, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 4 & 5 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 6 & 12 & 0 & 13 & 7 \\ 3 & 3 & 5 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{115} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 151, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.1$$

$$T_{\min} = 13.02 \rightarrow 14, \ T_{\max} = 92, \ T \in [14, 92]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 15 & 20 & 0 & 73 & 15 \\ 3 & 3 & 4 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{116} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 162, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 13.22 \rightarrow 14, \ T_{\max} = 92, \ T \in [14, 92]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 18 & 24 & 0 & 106 & 18 \\ 3 & 3 & 4 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{117} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 175, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.3$$

$$T_{\min} = 13.43 \rightarrow 14, \ T_{\max} = 92, \ T \in [14, 92]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 \\ 3 & 5 & 4 & 0 & 28 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 21 & 28 & 0 & 145 & 21 \\ 3 & 3 & 4 & -4 & 21 & -1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{111} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 198, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.47 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 12 & 13 & 0 & 28 & 10 \\ 3 & 4 & 4 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 224, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.48 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{14} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 239, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.48 \rightarrow 14, \ T_{\max} = 92, \ T \in [14, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 17 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 \\ 4 & 5 & 9 & 0 & 40 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 21 & 40 & 0 & 172 & 24 \\ 3 & 3 & 5 & -4 & 24 & -1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{11} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 239, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.98 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 8 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 4 & 5 & 9 & 0 & 20 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 9 & 20 & 0 & 37 & 11 \\ 3 & 3 & 5 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[30] \ 1^1 1^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 250, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.34 \rightarrow 14, \ T_{\max} = 92, \ T \in [14, 92]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 3 \\ 3 & 4 & 9 & 3 & 6 & 4 \\ 4 & 9 & 9 & 4 & 8 & 5 \\ 2 & 3 & 4 & -3 & 3 & -1 \\ 3 & 6 & 8 & 3 & 4 & 4 \\ 3 & 4 & 5 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{112} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 253, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.44 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 9 & 13 & 0 & 28 & 10 \\ 3 & 3 & 4 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{12} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 256, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.97 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 4 & 5 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 12 & 24 & 0 & 58 & 14 \\ 3 & 3 & 5 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{11} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 296, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.18 \rightarrow 14, \ T_{\max} = 92, \ T \in [14, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 11 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 4 & 5 & 9 & 0 & 28 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 12 & 28 & 0 & 73 & 15 \\ 3 & 3 & 5 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{112} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 306, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.54 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 14 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 15 & 14 & 0 & 37 & 11 \\ 3 & 4 & 4 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{114} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 314, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.86 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 \\ 3 & 6 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 27 & 24 & 0 & 121 & 19 \\ 3 & 4 & 4 & -4 & 19 & 0 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{11} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 319, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.51 \rightarrow 14, \ T_{\max} = 92, \ T \in [14, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 14 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 4 & 5 & 9 & 0 & 36 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 15 & 36 & 0 & 121 & 19 \\ 3 & 3 & 5 & -4 & 19 & -1 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{113} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 336, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.65 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 19 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 19 & 0 & 73 & 15 \\ 3 & 4 & 4 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{112} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 336, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.48 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 24 & 26 & 0 & 118 & 20 \\ 3 & 4 & 4 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{113} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 379, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.44 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 15 & 22 & 0 & 85 & 17 \\ 3 & 3 & 4 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{113} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 382, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.47 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 17 & 0 & 49 & 13 \\ 3 & 3 & 4 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{13} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 436, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.98 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 15 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 4 & 5 & 9 & 0 & 36 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 18 & 36 & 0 & 133 & 21 \\ 3 & 3 & 5 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{112} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 440, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.49 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{112} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 450, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.48 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 21 & 22 & 0 & 85 & 17 \\ 3 & 4 & 4 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[44] \ 1^{12} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 455, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 13 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 4 & 5 & 9 & 0 & 32 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 13 & 15 & 32 & 0 & 100 & 18 \\ 3 & 3 & 5 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[45] \ 1^{114} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 505, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.44 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 17 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 \\ 3 & 5 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 21 & 31 & 0 & 172 & 24 \\ 3 & 3 & 4 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[46] \ 1^{114} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 513, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.49 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 11 & 15 & 21 & 0 & 76 & 16 \\ 3 & 3 & 4 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[47] \ 1^{12} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 620, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.5$$

$$T_{\min} = 13.01 \rightarrow 14, \ T_{\max} = 92, \ T \in [14, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 16 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 4 & 5 & 9 & 0 & 40 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 16 & 18 & 40 & 0 & 154 & 22 \\ 3 & 3 & 5 & -4 & 22 & -1 \end{pmatrix}_{b_0}$$

$$[48] \ 1^{113} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 630, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.8$$

$$T_{\min} = 12.53 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 13 & 24 & 23 & 0 & 100 & 18 \\ 3 & 4 & 4 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[49] \ 1^{114} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 634, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.45 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 3 & 5 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 14 & 18 & 26 & 0 & 118 & 20 \\ 3 & 3 & 4 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[50] \ 1^{14} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 92, \ \Delta = -133, \ \det = 688, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.99 \rightarrow 13, \ T_{\max} = 92, \ T \in [13, 92]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 20 & 3 \\ 2 & 1 & 5 & 0 & 24 & 3 \\ 4 & 5 & 9 & 0 & 48 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 20 & 24 & 48 & 0 & 238 & 28 \\ 3 & 3 & 5 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

[51] $1^{114}31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 743, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.93 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 18 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 \\ 4 & 5 & 9 & 0 & 44 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 18 & 21 & 44 & 0 & 193 & 25 \\ 3 & 3 & 5 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[52] $1^{114}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 768, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.58 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 6 & 0 & 30 & 4 \\ 3 & 6 & 4 & 0 & 28 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 16 & 30 & 28 & 0 & 154 & 22 \\ 3 & 4 & 4 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

[53] $1^{113}31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 800, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.49 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[54] $1^{113}31^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 846, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.8$
 $T_{\min} = 12.48 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 6 & 0 & 30 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 30 & 31 & 0 & 172 & 24 \\ 3 & 4 & 4 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[55] $21^{1111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1076, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.42 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 4 & 6 & 8 & 3 & 6 & 4 \\ 6 & 4 & 9 & 3 & 6 & 4 \\ 8 & 9 & 9 & 4 & 8 & 5 \\ 3 & 3 & 4 & -3 & 3 & -1 \\ 6 & 6 & 8 & 3 & 4 & 4 \\ 4 & 4 & 5 & -1 & 4 & 0 \end{pmatrix}_{b_0}$$

[56] $1^{114}31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1098, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.52 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 6 & 0 & 33 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 18 & 33 & 32 & 0 & 193 & 25 \\ 3 & 4 & 4 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[57] $1^{114}31^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 8, T_{\max} = 92, \Delta = -133, \det = 1304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 12.50 \rightarrow 13, T_{\max} = 92, T \in [13, 92]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

$T = 9$ [58] $1^{11}321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -105, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 0 \\ 1 & 2 & 1 & 1 & 3 & 0 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 1 & 1 & 3 & 0 & -3 & 0 \\ 1 & 0 & 0 & 0 & 0 & -2 \\ 2 & 2 & 3 & -1 & 1 & 0 \end{pmatrix}_{b_0}$$

[59] $1^{112}31^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -11, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 18.27 \rightarrow 19, T_{\max} = 23, T \in [19, 23]$

$$\begin{pmatrix} 0 & 1 & 1 & 2 & 1 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 3 \\ 1 & 4 & 1 & 4 & 0 & 9 & 3 \\ 2 & 3 & 4 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 9 & 9 & 12 & 0 & 28 & 10 \\ 2 & 3 & 3 & 3 & -4 & 10 & -6 \end{pmatrix}_{b_0}$$

- [60] $1^{11}321^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -105$, $\det = -19$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 16.05 \rightarrow 17$, $T_{\max} = 21$, $T \in [17, 21]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 0 & 4 \\ 3 & 6 & 5 & 5 & 0 & 5 \\ 2 & 3 & 5 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 5 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$
- [61] $1^{11}31^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 4 & 4 \\ 2 & 3 & 4 & 2 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$
- [62] $1^{11}31^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -59$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 1 & 3 \\ 2 & 3 & 4 & 2 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$
- [63] $1^{11}31^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -71$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 15$, $T_{\max} = 23$, $T \in [15, 23]$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 1 & 4 & 0 & 3 & 3 \\ 2 & 6 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 3 & 3 & -4 & 3 & -2 \end{pmatrix}_{b_0}$$
- [64] $1^{11}31^{12}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -275$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.5$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 3 \\ 2 & 4 & 4 & 4 & 0 & 15 & 4 \\ 1 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 15 & 9 & 0 & 37 & 11 \\ 2 & 3 & 4 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$
- [65] $1^{11}31^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -284$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 3 \\ 2 & 4 & 4 & 4 & 0 & 12 & 4 \\ 1 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 9 & 12 & 9 & 0 & 28 & 10 \\ 2 & 3 & 4 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$
- [66] $1^{11}31^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -311$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 6 & 9 & 6 & 0 & 13 & 7 \\ 2 & 3 & 4 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$
- [67] $1^{11}31^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -323$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 13.02 \rightarrow 14$, $T_{\max} = 23$, $T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 7 & 3 & 0 & 9 & 4 \\ 2 & 7 & 4 & 4 & 0 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 9 & 6 & 0 & 13 & 7 \\ 2 & 4 & 4 & 2 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$
- [68] $1^{11}31^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -473$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 11.0$
 $T_{\min} = 12.92 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 7 & 3 & 0 & 6 & 4 \\ 2 & 7 & 4 & 4 & 0 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 6 & 6 & 3 & 0 & 4 & 4 \\ 2 & 4 & 4 & 2 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$
- [69] $1^{11}31^{14}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2$ $T = 9$, $T_{\max} = 23$, $\Delta = -112$, $\det = -500$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 23$, $T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 10 & 2 \\ 1 & 1 & 4 & 3 & 0 & 18 & 3 \\ 2 & 4 & 4 & 4 & 0 & 30 & 4 \\ 1 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 30 & 18 & 0 & 154 & 22 \\ 2 & 3 & 4 & 2 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

[70] $1^{14}31^{12}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 3 & 0 & 18 & 3 \\ 2 & 4 & 4 & 4 & 0 & 24 & 4 \\ 1 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 18 & 24 & 18 & 0 & 118 & 20 \\ 2 & 3 & 4 & 2 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[71] $1^{13}31^{112}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 7 & 3 & 0 & 12 & 4 \\ 2 & 7 & 4 & 4 & 0 & 12 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 12 & 6 & 0 & 22 & 8 \\ 2 & 4 & 4 & 2 & -4 & 8 & 0 \end{pmatrix}_{b_0}$$

[72] $1^{112}31^{12}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -644, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 3 & 0 & 12 & 3 \\ 2 & 4 & 4 & 4 & 0 & 18 & 4 \\ 1 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 12 & 18 & 12 & 0 & 58 & 14 \\ 2 & 3 & 4 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

[73] $1^{13}31^{114}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -662, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 13.29 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 11 & 2 \\ 2 & 4 & 7 & 3 & 0 & 33 & 4 \\ 2 & 7 & 4 & 4 & 0 & 33 & 4 \\ 1 & 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 33 & 33 & 21 & 0 & 193 & 25 \\ 2 & 4 & 4 & 2 & -4 & 25 & -1 \end{pmatrix}_{b_0}$$

[74] $1^{112}31^{13}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -716, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 3 & 0 & 15 & 3 \\ 2 & 4 & 4 & 4 & 0 & 24 & 4 \\ 1 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 15 & 24 & 15 & 0 & 100 & 18 \\ 2 & 3 & 4 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

[75] $1^{113}31^{12}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -731, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 3 & 0 & 15 & 3 \\ 2 & 4 & 4 & 4 & 0 & 21 & 4 \\ 1 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 15 & 21 & 15 & 0 & 85 & 17 \\ 2 & 3 & 4 & 2 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

[76] $1^{11}31^{112}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -746, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.5$

$T_{\min} = 12.93 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 5 & 2 \\ 2 & 4 & 7 & 3 & 0 & 15 & 4 \\ 2 & 7 & 4 & 4 & 0 & 15 & 4 \\ 1 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 15 & 15 & 9 & 0 & 37 & 11 \\ 2 & 4 & 4 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

[77] $1^{12}31^{113}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -809, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 13.03 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 7 & 3 & 0 & 24 & 4 \\ 2 & 7 & 4 & 4 & 0 & 24 & 4 \\ 1 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 24 & 15 & 0 & 100 & 18 \\ 2 & 4 & 4 & 2 & -4 & 18 & -1 \end{pmatrix}_{b_0}$$

[78] $1^{13}31^{113}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -833, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 6 & 2 \\ 2 & 4 & 7 & 3 & 0 & 18 & 4 \\ 2 & 7 & 4 & 4 & 0 & 18 & 4 \\ 1 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 18 & 18 & 9 & 0 & 52 & 12 \\ 2 & 4 & 4 & 2 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$

[79] $1^{131}1^{14}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1148, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 8 & 2 \\ 2 & 4 & 7 & 3 & 0 & 24 & 4 \\ 2 & 7 & 4 & 4 & 0 & 24 & 4 \\ 1 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 24 & 24 & 12 & 0 & 94 & 16 \\ 2 & 4 & 4 & 2 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

[80] $1^{113}31^{13}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 9 & 2 \\ 1 & 1 & 4 & 3 & 0 & 18 & 3 \\ 2 & 4 & 4 & 4 & 0 & 27 & 4 \\ 1 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 18 & 27 & 18 & 0 & 133 & 21 \\ 2 & 3 & 4 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

[81] $1^{113}31^{13}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1259, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 7 & 2 \\ 2 & 4 & 7 & 3 & 0 & 21 & 4 \\ 2 & 7 & 4 & 4 & 0 & 21 & 4 \\ 1 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 21 & 21 & 12 & 0 & 73 & 15 \\ 2 & 4 & 4 & 2 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

[82] $1^{113}31^{14}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1379, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 11 & 2 \\ 1 & 1 & 4 & 3 & 0 & 21 & 3 \\ 2 & 4 & 4 & 4 & 0 & 33 & 4 \\ 1 & 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 33 & 21 & 0 & 193 & 25 \\ 2 & 3 & 4 & 2 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

[83] $1^{114}31^{13}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1400, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.8$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 10 & 2 \\ 1 & 1 & 4 & 3 & 0 & 21 & 3 \\ 2 & 4 & 4 & 4 & 0 & 30 & 4 \\ 1 & 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 21 & 30 & 21 & 0 & 172 & 24 \\ 2 & 3 & 4 & 2 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

[84] $1^{123}31^{14}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1700, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 10 & 2 \\ 2 & 4 & 7 & 3 & 0 & 30 & 4 \\ 2 & 7 & 4 & 4 & 0 & 30 & 4 \\ 1 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 30 & 30 & 18 & 0 & 154 & 22 \\ 2 & 4 & 4 & 2 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

[85] $1^{113}31^{14}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1862, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.91 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 9 & 2 \\ 2 & 4 & 7 & 3 & 0 & 27 & 4 \\ 2 & 7 & 4 & 4 & 0 & 27 & 4 \\ 1 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 27 & 27 & 15 & 0 & 121 & 19 \\ 2 & 4 & 4 & 2 & -4 & 19 & 0 \end{pmatrix}_{b_0}$$

[86] $1^{114}31^{14}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -1976, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 12 & 2 \\ 1 & 1 & 4 & 3 & 0 & 24 & 3 \\ 2 & 4 & 4 & 4 & 0 & 36 & 4 \\ 1 & 3 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 24 & 36 & 24 & 0 & 238 & 28 \\ 2 & 3 & 4 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

6.19 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \quad 1^{111} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 & 3 \\ 3 & 6 & 5 & 0 & 3 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{121} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 13 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 12 & 13 & 0 & 28 & 12 & 9 & 10 \\ 3 & 6 & 5 & 0 & 12 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 9 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 10 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 294, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 6 & 3 & 4 \\ 3 & 6 & 5 & 0 & 6 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 721, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 9 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 9 & 6 & 7 \\ 3 & 6 & 5 & 0 & 9 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 6 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 7 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

6.20 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{114} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.51 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 44 & 5 \\ 8 & 10 & 17 & 0 & 60 & 6 \\ 12 & 17 & 25 & 0 & 96 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 44 & 60 & 96 & 0 & 346 & 34 \\ 5 & 6 & 9 & -4 & 34 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{114} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.29 \rightarrow 15, T_{\max} = 93, T \in [15, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 44 & 5 \\ 9 & 13 & 18 & 0 & 72 & 7 \\ 11 & 18 & 20 & 0 & 84 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 44 & 72 & 84 & 0 & 346 & 34 \\ 5 & 7 & 8 & -4 & 34 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -139, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 20 & 0 & 13 & 7 \\ 5 & 6 & 9 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.54 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 8 & 5 \\ 9 & 13 & 18 & 0 & 12 & 7 \\ 11 & 18 & 20 & 0 & 16 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 12 & 16 & 0 & 10 & 6 \\ 5 & 7 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -317$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 15 & 17 & 0 & 13 & 7 \\ 5 & 7 & 8 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -351$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 17 & 5 \\ 8 & 10 & 17 & 0 & 24 & 6 \\ 12 & 17 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 24 & 36 & 0 & 49 & 13 \\ 5 & 6 & 9 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -439$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 35 & 5 \\ 8 & 10 & 17 & 0 & 48 & 6 \\ 12 & 17 & 25 & 0 & 76 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 35 & 48 & 76 & 0 & 217 & 27 \\ 5 & 6 & 9 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -525$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 1 & 3 \\ 3 & 4 & 9 & 3 & 6 & 0 & 4 \\ 4 & 9 & 9 & 4 & 8 & 0 & 5 \\ 2 & 3 & 4 & -3 & 3 & 0 & -1 \\ 3 & 6 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & -1 & 4 & -2 & -1 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -526$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 26 & 5 \\ 8 & 10 & 17 & 0 & 36 & 6 \\ 12 & 17 & 25 & 0 & 56 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 36 & 56 & 0 & 118 & 20 \\ 5 & 6 & 9 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{113}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -653$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.8$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 35 & 5 \\ 9 & 13 & 18 & 0 & 57 & 7 \\ 11 & 18 & 20 & 0 & 67 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 35 & 57 & 67 & 0 & 217 & 27 \\ 5 & 7 & 8 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{111}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -709$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 17 & 5 \\ 9 & 13 & 18 & 0 & 27 & 7 \\ 11 & 18 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 27 & 33 & 0 & 49 & 13 \\ 5 & 7 & 8 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{112}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -848$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.5$

$$T_{\min} = 13.49 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 26 & 5 \\ 9 & 13 & 18 & 0 & 42 & 7 \\ 11 & 18 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 42 & 50 & 0 & 118 & 20 \\ 5 & 7 & 8 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [12] \quad 1^{11} 3^{11} 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -104, \det = 11, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 16, T_{\max} = 22, T \in [16, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 3 & 0 & 1 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 3 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & 3 & -1 & -1 & 0 & 0 & -3 \end{pmatrix}_{b_0}$$

6.21 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 31^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 20 & 9 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 20 & 0 & 13 & 15 & 7 \\ 9 & 12 & 20 & 0 & 15 & 13 & 7 \\ 5 & 6 & 9 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 188, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 17 & 5 \\ 8 & 10 & 17 & 0 & 12 & 24 & 6 \\ 12 & 17 & 25 & 0 & 20 & 36 & 9 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 20 & 0 & 13 & 27 & 7 \\ 17 & 24 & 36 & 0 & 27 & 49 & 13 \\ 5 & 6 & 9 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1111} 3^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 300, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 15 & 17 & 0 & 13 & 15 & 7 \\ 9 & 15 & 17 & 0 & 15 & 13 & 7 \\ 5 & 7 & 8 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{1111} 3^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 832, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 17 & 5 \\ 9 & 13 & 18 & 0 & 15 & 27 & 7 \\ 11 & 18 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 15 & 17 & 0 & 13 & 27 & 7 \\ 17 & 27 & 33 & 0 & 27 & 49 & 13 \\ 5 & 7 & 8 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

6.22 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{1121} 3^{112} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 2 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 & 3 \\ 3 & 5 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 12 & 18 & 0 & 58 & 12 & 14 \\ 2 & 3 & 4 & 0 & 12 & 1 & 3 \\ 3 & 3 & 4 & -4 & 14 & 3 & -3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1121} 3^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 7 & 3 \\ 2 & 1 & 5 & 0 & 3 & 9 & 3 \\ 3 & 5 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 4 & 12 & 4 \\ 7 & 9 & 13 & 0 & 12 & 28 & 10 \\ 3 & 3 & 4 & -4 & 4 & 10 & -3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{1112} 3^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 12 & 7 \\ 4 & 6 & 8 & 0 & 12 & 10 & 6 \\ 3 & 3 & 4 & -4 & 7 & 6 & -3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{114} 3^1 1^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 14 & 3 \\ 2 & 1 & 5 & 0 & 3 & 18 & 3 \\ 3 & 5 & 4 & 0 & 5 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 4 & 24 & 4 \\ 14 & 18 & 26 & 0 & 24 & 118 & 20 \\ 3 & 3 & 4 & -4 & 4 & 20 & -3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -48, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 17.92 \rightarrow 18, T_{\max} = 21, T \in [18, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 8 & 3 \\ 2 & 1 & 5 & 0 & 3 & 9 & 5 \\ 4 & 5 & 9 & 0 & 4 & 20 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 9 & 3 \\ 8 & 9 & 20 & 0 & 9 & 37 & 11 \\ 3 & 3 & 5 & -4 & 3 & 11 & -5 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{114} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 8 & 11 & 3 \\ 2 & 1 & 5 & 0 & 12 & 15 & 3 \\ 3 & 5 & 4 & 0 & 16 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 12 & 16 & 0 & 46 & 60 & 12 \\ 11 & 15 & 21 & 0 & 60 & 76 & 16 \\ 3 & 3 & 4 & -4 & 12 & 16 & -3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 15.22 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 4 & 5 & 9 & 0 & 4 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 6 & 3 \\ 5 & 6 & 12 & 0 & 6 & 13 & 7 \\ 3 & 3 & 5 & -4 & 3 & 7 & -3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{121} 3^1 1^{1111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 16.09 \rightarrow 17, T_{\max} = 21, T \in [17, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 7 & 5 & 3 \\ 2 & 1 & 5 & 0 & 9 & 6 & 3 \\ 4 & 5 & 9 & 0 & 16 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 16 & 0 & 28 & 21 & 10 \\ 5 & 6 & 12 & 0 & 21 & 13 & 7 \\ 3 & 3 & 5 & -4 & 10 & 7 & -4 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.71 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 \\ 3 & 6 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -4 & 3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{112} 3^1 1^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -103, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 15.80 \rightarrow 16, T_{\max} = 21, T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 10 & 3 \\ 2 & 1 & 5 & 0 & 3 & 12 & 3 \\ 4 & 5 & 9 & 0 & 4 & 24 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 12 & 3 \\ 10 & 12 & 24 & 0 & 12 & 58 & 14 \\ 3 & 3 & 5 & -4 & 3 & 14 & -3 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{1111} 31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -92, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.83 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 7 & 3 \\ 3 & 4 & 6 & 0 & 6 & 12 & 4 \\ 3 & 6 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 12 & 4 \\ 7 & 12 & 13 & 0 & 12 & 28 & 10 \\ 3 & 4 & 4 & -4 & 4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{1111} 31^{132} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -125, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.58 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 7 & 3 \\ 3 & 4 & 6 & 0 & 15 & 12 & 4 \\ 3 & 6 & 4 & 0 & 17 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 15 & 17 & 0 & 49 & 39 & 13 \\ 7 & 12 & 13 & 0 & 39 & 28 & 10 \\ 3 & 4 & 4 & -4 & 13 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{113} 31^{113} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -127, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 16.23 \rightarrow 17, \quad T_{\max} = 21, \quad T \in [17, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 15 & 3 \\ 2 & 1 & 5 & 0 & 3 & 18 & 3 \\ 4 & 5 & 9 & 0 & 4 & 36 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 18 & 3 \\ 15 & 18 & 36 & 0 & 18 & 133 & 21 \\ 3 & 3 & 5 & -4 & 3 & 21 & -4 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} 31^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -156, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.59 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 4 & 5 & 9 & 0 & 8 & 8 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 8 & 0 & 4 & 6 & 4 \\ 3 & 3 & 8 & 0 & 6 & 4 & 4 \\ 3 & 3 & 5 & -4 & 4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{112} 31^{113} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -165, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 15.23 \rightarrow 16, \quad T_{\max} = 21, \quad T \in [16, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 13 & 3 \\ 2 & 1 & 5 & 0 & 3 & 15 & 3 \\ 4 & 5 & 9 & 0 & 4 & 32 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 15 & 3 \\ 13 & 15 & 32 & 0 & 15 & 100 & 18 \\ 3 & 3 & 5 & -4 & 3 & 18 & -3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{114} 31^{114} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -190, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.48 \rightarrow 17, \quad T_{\max} = 21, \quad T \in [17, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 20 & 3 \\ 2 & 1 & 5 & 0 & 3 & 24 & 3 \\ 4 & 5 & 9 & 0 & 4 & 48 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 24 & 3 \\ 20 & 24 & 48 & 0 & 24 & 238 & 28 \\ 3 & 3 & 5 & -4 & 3 & 28 & -4 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{1111} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -191, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.47 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 6 & 1 & 3 \\ 3 & 3 & 4 & -4 & 7 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11} 31^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -199, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.37 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 4 & 5 & 9 & 0 & 8 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 8 & 0 & 4 & 9 & 4 \\ 5 & 6 & 12 & 0 & 9 & 13 & 7 \\ 3 & 3 & 5 & -4 & 4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{1111} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -200, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.68 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 0 & 6 & 6 & 4 \\ 3 & 6 & 4 & 0 & 5 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 6 & 4 \\ 3 & 6 & 5 & 0 & 6 & 4 & 4 \\ 3 & 4 & 4 & -4 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{111} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -209, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.52 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 9 & 3 \\ 7 & 12 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & -4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{113} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -215, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.86 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 9 & 3 \\ 2 & 1 & 5 & 0 & 9 & 12 & 3 \\ 3 & 5 & 4 & 0 & 12 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 9 & 12 & 0 & 25 & 36 & 9 \\ 9 & 12 & 17 & 0 & 36 & 49 & 13 \\ 3 & 3 & 4 & -4 & 9 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{112} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -236, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 6 & 0 & 3 & 15 & 4 \\ 3 & 6 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 9 & 3 \\ 8 & 15 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & 4 & -4 & 3 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1113} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -252, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 15 & 3 \\ 2 & 1 & 5 & 0 & 6 & 18 & 3 \\ 3 & 5 & 4 & 0 & 9 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 45 & 7 \\ 15 & 18 & 27 & 0 & 45 & 133 & 21 \\ 3 & 3 & 4 & -4 & 7 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1114} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -252, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 20 & 3 \\ 2 & 1 & 5 & 0 & 6 & 24 & 3 \\ 3 & 5 & 4 & 0 & 9 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 60 & 7 \\ 20 & 24 & 36 & 0 & 60 & 238 & 28 \\ 3 & 3 & 4 & -4 & 7 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1122} 31^{122} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -252, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 10 & 3 \\ 2 & 1 & 5 & 0 & 12 & 12 & 3 \\ 3 & 5 & 4 & 0 & 18 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 12 & 18 & 0 & 58 & 60 & 14 \\ 10 & 12 & 18 & 0 & 60 & 58 & 14 \\ 3 & 3 & 4 & -4 & 14 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1111} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -252, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 15 & 7 \\ 5 & 6 & 9 & 0 & 15 & 13 & 7 \\ 3 & 3 & 4 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1121} 31^{121} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -252, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.43 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 5 & 3 \\ 2 & 1 & 5 & 0 & 12 & 6 & 3 \\ 3 & 5 & 4 & 0 & 18 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 12 & 18 & 0 & 58 & 30 & 14 \\ 5 & 6 & 9 & 0 & 30 & 13 & 7 \\ 3 & 3 & 4 & -4 & 14 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1122} 31^{121} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -254, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 7 & 3 \\ 2 & 1 & 5 & 0 & 12 & 9 & 3 \\ 3 & 5 & 4 & 0 & 18 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 12 & 18 & 0 & 58 & 42 & 14 \\ 7 & 9 & 13 & 0 & 42 & 28 & 10 \\ 3 & 3 & 4 & -4 & 14 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{1111} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -256, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.48 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 3 & 4 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{1124} 31^{122} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -260, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.54 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 14 & 3 \\ 2 & 1 & 5 & 0 & 12 & 18 & 3 \\ 3 & 5 & 4 & 0 & 18 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 12 & 18 & 0 & 58 & 84 & 14 \\ 14 & 18 & 26 & 0 & 84 & 118 & 20 \\ 3 & 3 & 4 & -4 & 14 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{1111} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -261, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.52 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{1122} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -262, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.56 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 7 & 3 \\ 2 & 1 & 5 & 0 & 6 & 9 & 3 \\ 3 & 5 & 4 & 0 & 8 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 18 & 6 \\ 7 & 9 & 13 & 0 & 18 & 28 & 10 \\ 3 & 3 & 4 & -4 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{1112} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -262, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.56 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 4 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 3 & 5 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 6 & 3 \\ 4 & 6 & 8 & 0 & 6 & 10 & 6 \\ 3 & 3 & 4 & -4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{1124} 31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.64 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 14 & 3 \\ 2 & 1 & 5 & 0 & 6 & 18 & 3 \\ 3 & 5 & 4 & 0 & 8 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 36 & 6 \\ 14 & 18 & 26 & 0 & 36 & 118 & 20 \\ 3 & 3 & 4 & -4 & 6 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{1122} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -268, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.64 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 8 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 12 & 6 \\ 4 & 6 & 8 & 0 & 12 & 10 & 6 \\ 3 & 3 & 4 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{1113} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -272, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.69 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 6 & 3 \\ 2 & 1 & 5 & 0 & 3 & 9 & 3 \\ 3 & 5 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 9 & 3 \\ 6 & 9 & 12 & 0 & 9 & 25 & 9 \\ 3 & 3 & 4 & -4 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1123} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -278, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 6 & 3 \\ 2 & 1 & 5 & 0 & 6 & 9 & 3 \\ 3 & 5 & 4 & 0 & 8 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 18 & 6 \\ 6 & 9 & 12 & 0 & 18 & 25 & 9 \\ 3 & 3 & 4 & -4 & 6 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1114} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -286, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.85 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 8 & 3 \\ 2 & 1 & 5 & 0 & 3 & 12 & 3 \\ 3 & 5 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 12 & 3 \\ 8 & 12 & 16 & 0 & 12 & 46 & 12 \\ 3 & 3 & 4 & -4 & 3 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{1133} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -288, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.88 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 6 & 3 \\ 2 & 1 & 5 & 0 & 9 & 9 & 3 \\ 3 & 5 & 4 & 0 & 12 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 9 & 12 & 0 & 25 & 27 & 9 \\ 6 & 9 & 12 & 0 & 27 & 25 & 9 \\ 3 & 3 & 4 & -4 & 9 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1124} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -292, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.92 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 8 & 3 \\ 2 & 1 & 5 & 0 & 6 & 12 & 3 \\ 3 & 5 & 4 & 0 & 8 & 16 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 24 & 6 \\ 8 & 12 & 16 & 0 & 24 & 46 & 12 \\ 3 & 3 & 4 & -4 & 6 & 12 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{1123} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -296, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.54 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 6 & 0 & 3 & 24 & 4 \\ 3 & 6 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 18 & 3 \\ 14 & 24 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & 4 & -4 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{113} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -300, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 14.91 \rightarrow 15, \quad T_{\max} = 21, \quad T \in [15, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 18 & 3 \\ 2 & 1 & 5 & 0 & 3 & 21 & 3 \\ 4 & 5 & 9 & 0 & 4 & 44 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 21 & 3 \\ 18 & 21 & 44 & 0 & 21 & 193 & 25 \\ 3 & 3 & 5 & -4 & 3 & 25 & -2 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{1134} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -302, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.02 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 8 & 3 \\ 2 & 1 & 5 & 0 & 9 & 12 & 3 \\ 3 & 5 & 4 & 0 & 12 & 16 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 9 & 12 & 0 & 25 & 36 & 9 \\ 8 & 12 & 16 & 0 & 36 & 46 & 12 \\ 3 & 3 & 4 & -4 & 9 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1115} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -304, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.1$$

$$T_{\min} = 13.04 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 10 & 3 \\ 2 & 1 & 5 & 0 & 3 & 15 & 3 \\ 3 & 5 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 15 & 3 \\ 10 & 15 & 20 & 0 & 15 & 73 & 15 \\ 3 & 3 & 4 & -4 & 3 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{1115} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -310, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 13.10 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 10 & 3 \\ 2 & 1 & 5 & 0 & 6 & 15 & 3 \\ 3 & 5 & 4 & 0 & 8 & 20 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 30 & 6 \\ 10 & 15 & 20 & 0 & 30 & 73 & 15 \\ 3 & 3 & 4 & -4 & 6 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{1114} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -316, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.15 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 8 & 8 & 3 \\ 2 & 1 & 5 & 0 & 12 & 12 & 3 \\ 3 & 5 & 4 & 0 & 16 & 16 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 12 & 16 & 0 & 46 & 48 & 12 \\ 8 & 12 & 16 & 0 & 48 & 46 & 12 \\ 3 & 3 & 4 & -4 & 12 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{1135} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -320, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.9$$

$$T_{\min} = 13.19 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 10 & 3 \\ 2 & 1 & 5 & 0 & 9 & 15 & 3 \\ 3 & 5 & 4 & 0 & 12 & 20 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 9 & 12 & 0 & 25 & 45 & 9 \\ 10 & 15 & 20 & 0 & 45 & 73 & 15 \\ 3 & 3 & 4 & -4 & 9 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1116} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -326, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.24 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 12 & 3 \\ 2 & 1 & 5 & 0 & 3 & 18 & 3 \\ 3 & 5 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 18 & 3 \\ 12 & 18 & 24 & 0 & 18 & 106 & 18 \\ 3 & 3 & 4 & -4 & 3 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1126} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -332, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.29 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 12 & 3 \\ 2 & 1 & 5 & 0 & 6 & 18 & 3 \\ 3 & 5 & 4 & 0 & 8 & 24 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 36 & 6 \\ 12 & 18 & 24 & 0 & 36 & 106 & 18 \\ 3 & 3 & 4 & -4 & 6 & 18 & -1 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{1111} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -333, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.56 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 0 & 6 & 9 & 4 \\ 3 & 6 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 9 & 4 \\ 5 & 9 & 9 & 0 & 9 & 13 & 7 \\ 3 & 4 & 4 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{1117} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -352, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 13.44 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 14 & 3 \\ 2 & 1 & 5 & 0 & 3 & 21 & 3 \\ 3 & 5 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 21 & 3 \\ 14 & 21 & 28 & 0 & 21 & 145 & 21 \\ 3 & 3 & 4 & -4 & 3 & 21 & -1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{1127} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -358, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.49 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 14 & 3 \\ 2 & 1 & 5 & 0 & 6 & 21 & 3 \\ 3 & 5 & 4 & 0 & 8 & 28 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 42 & 6 \\ 14 & 21 & 28 & 0 & 42 & 145 & 21 \\ 3 & 3 & 4 & -4 & 6 & 21 & -1 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{111} 31^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -368, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.96 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 4 & 5 & 9 & 0 & 12 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 12 & 0 & 13 & 15 & 7 \\ 5 & 6 & 12 & 0 & 15 & 13 & 7 \\ 3 & 3 & 5 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{114} 31^{112} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -404, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.02 \rightarrow 14, \quad T_{\max} = 21, \quad T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 6 & 0 & 3 & 30 & 4 \\ 3 & 6 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 18 & 3 \\ 16 & 30 & 28 & 0 & 18 & 154 & 22 \\ 3 & 4 & 4 & -4 & 3 & 22 & -1 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{1112} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -443, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.44 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 7 & 3 \\ 2 & 1 & 5 & 0 & 6 & 9 & 3 \\ 3 & 5 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 21 & 7 \\ 7 & 9 & 13 & 0 & 21 & 28 & 10 \\ 3 & 3 & 4 & -4 & 7 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{1112} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -445, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 7 & 3 \\ 2 & 1 & 5 & 0 & 3 & 9 & 3 \\ 3 & 5 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 9 & 3 \\ 7 & 9 & 13 & 0 & 9 & 28 & 10 \\ 3 & 3 & 4 & -4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{1114} 31^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -445, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 17 & 3 \\ 2 & 1 & 5 & 0 & 3 & 21 & 3 \\ 3 & 5 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 21 & 3 \\ 17 & 21 & 31 & 0 & 21 & 172 & 24 \\ 3 & 3 & 4 & -4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{1114} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -459, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.57 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 11 & 3 \\ 2 & 1 & 5 & 0 & 6 & 15 & 3 \\ 3 & 5 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 33 & 7 \\ 11 & 15 & 21 & 0 & 33 & 76 & 16 \\ 3 & 3 & 4 & -4 & 7 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{1111} 31^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -477, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.49 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 5 & 3 \\ 3 & 4 & 6 & 0 & 12 & 9 & 4 \\ 3 & 6 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 12 & 13 & 0 & 28 & 21 & 10 \\ 5 & 9 & 9 & 0 & 21 & 13 & 7 \\ 3 & 4 & 4 & -4 & 10 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{1134} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -477, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.70 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 11 & 3 \\ 2 & 1 & 5 & 0 & 9 & 15 & 3 \\ 3 & 5 & 4 & 0 & 12 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 9 & 12 & 0 & 25 & 45 & 9 \\ 11 & 15 & 21 & 0 & 45 & 76 & 16 \\ 3 & 3 & 4 & -4 & 9 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[61] \quad 1^{1111} 31^{122} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -488, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.48 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 12 & 4 \\ 3 & 6 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 12 & 13 & 0 & 28 & 30 & 10 \\ 7 & 12 & 13 & 0 & 30 & 28 & 10 \\ 3 & 4 & 4 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[62] \quad 1^{1123} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -504, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.54 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 6 & 0 & 3 & 18 & 4 \\ 3 & 6 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 12 & 3 \\ 10 & 18 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & -4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[63] \quad 1^{1123} 31^{122} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -506, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.44 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 12 & 3 \\ 2 & 1 & 5 & 0 & 12 & 15 & 3 \\ 3 & 5 & 4 & 0 & 18 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 12 & 18 & 0 & 58 & 72 & 14 \\ 12 & 15 & 22 & 0 & 72 & 85 & 17 \\ 3 & 3 & 4 & -4 & 14 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[64] \quad 1^{1113} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -508, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 12 & 3 \\ 2 & 1 & 5 & 0 & 3 & 15 & 3 \\ 3 & 5 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 15 & 3 \\ 12 & 15 & 22 & 0 & 15 & 85 & 17 \\ 3 & 3 & 4 & -4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[65] \quad 1^{1113} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -512, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.48 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 9 & 3 \\ 2 & 1 & 5 & 0 & 6 & 12 & 3 \\ 3 & 5 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 27 & 7 \\ 9 & 12 & 17 & 0 & 27 & 49 & 13 \\ 3 & 3 & 4 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[66] \quad 1^{1123} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -520, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.54 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 9 & 3 \\ 2 & 1 & 5 & 0 & 6 & 12 & 3 \\ 3 & 5 & 4 & 0 & 8 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 24 & 6 \\ 9 & 12 & 17 & 0 & 24 & 49 & 13 \\ 3 & 3 & 4 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[67] \quad 1^{112} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -524, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.51 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 6 & 0 & 3 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 15 & 3 \\ 12 & 21 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & -4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[68] \quad 1^{113} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -569, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.66 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 6 & 0 & 3 & 24 & 4 \\ 3 & 6 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 15 & 3 \\ 13 & 24 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & 4 & -4 & 3 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[69] \quad 1^{111} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -592, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.49 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 15 & 7 \\ 5 & 9 & 9 & 0 & 15 & 13 & 7 \\ 3 & 4 & 4 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[70] \quad 1^{111} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -695, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.44 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 12 & 3 \\ 2 & 1 & 5 & 0 & 6 & 15 & 3 \\ 3 & 5 & 4 & 0 & 9 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 36 & 7 \\ 12 & 15 & 22 & 0 & 36 & 85 & 17 \\ 3 & 3 & 4 & -4 & 7 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[71] \quad 1^{111} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -703, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.48 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 9 & 3 \\ 2 & 1 & 5 & 0 & 3 & 12 & 3 \\ 3 & 5 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 12 & 3 \\ 9 & 12 & 17 & 0 & 12 & 49 & 13 \\ 3 & 3 & 4 & -4 & 3 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[72] \quad 1^{113} 31^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -703, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.48 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 9 & 3 \\ 2 & 1 & 5 & 0 & 15 & 12 & 3 \\ 3 & 5 & 4 & 0 & 22 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 15 & 22 & 0 & 85 & 66 & 17 \\ 9 & 12 & 17 & 0 & 66 & 49 & 13 \\ 3 & 3 & 4 & -4 & 17 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[73] \quad 1^{112} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -760, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.45 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 9 & 3 \\ 3 & 5 & 4 & 0 & 13 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 13 & 0 & 28 & 30 & 10 \\ 7 & 9 & 13 & 0 & 30 & 28 & 10 \\ 3 & 3 & 4 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[74] \quad 1^{112} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -782, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.55 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 11 & 3 \\ 2 & 1 & 5 & 0 & 6 & 15 & 3 \\ 3 & 5 & 4 & 0 & 8 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 30 & 6 \\ 11 & 15 & 21 & 0 & 30 & 76 & 16 \\ 3 & 3 & 4 & -4 & 6 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[75] \quad 1^{113} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -909, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.55 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 6 & 0 & 3 & 27 & 4 \\ 3 & 6 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 18 & 3 \\ 15 & 27 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & -4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[76] \quad 1^{114} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -947, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.44 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 17 & 3 \\ 2 & 1 & 5 & 0 & 6 & 21 & 3 \\ 3 & 5 & 4 & 0 & 9 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 51 & 7 \\ 17 & 21 & 31 & 0 & 51 & 172 & 24 \\ 3 & 3 & 4 & -4 & 7 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[77] \quad 1^{1123} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -949, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.44 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 12 & 3 \\ 2 & 1 & 5 & 0 & 9 & 15 & 3 \\ 3 & 5 & 4 & 0 & 13 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 13 & 0 & 28 & 51 & 10 \\ 12 & 15 & 22 & 0 & 51 & 85 & 17 \\ 3 & 3 & 4 & -4 & 10 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[78] \quad 1^{1123} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -955, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.3$$

$$T_{\min} = 12.47 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 9 & 3 \\ 2 & 1 & 5 & 0 & 9 & 12 & 3 \\ 3 & 5 & 4 & 0 & 13 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 13 & 0 & 28 & 39 & 10 \\ 9 & 12 & 17 & 0 & 39 & 49 & 13 \\ 3 & 3 & 4 & -4 & 10 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[79] \quad 1^{114} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -965, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 11 & 3 \\ 2 & 1 & 5 & 0 & 3 & 15 & 3 \\ 3 & 5 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 15 & 3 \\ 11 & 15 & 21 & 0 & 15 & 76 & 16 \\ 3 & 3 & 4 & -4 & 3 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[80] \quad 1^{113} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1001, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.8$$

$$T_{\min} = 12.52 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 6 & 0 & 3 & 30 & 4 \\ 3 & 6 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 21 & 3 \\ 17 & 30 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & -4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[81] \quad 1^{1124} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1012, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.44 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 17 & 3 \\ 2 & 1 & 5 & 0 & 9 & 21 & 3 \\ 3 & 5 & 4 & 0 & 13 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 13 & 0 & 28 & 72 & 10 \\ 17 & 21 & 31 & 0 & 72 & 172 & 24 \\ 3 & 3 & 4 & -4 & 10 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[82] \quad 1^{114} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1016, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 14 & 3 \\ 2 & 1 & 5 & 0 & 6 & 18 & 3 \\ 3 & 5 & 4 & 0 & 9 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 42 & 7 \\ 14 & 18 & 26 & 0 & 42 & 118 & 20 \\ 3 & 3 & 4 & -4 & 7 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[83] \quad 1^{1114} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1018, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.46 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 14 & 3 \\ 2 & 1 & 5 & 0 & 3 & 18 & 3 \\ 3 & 5 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 18 & 3 \\ 14 & 18 & 26 & 0 & 18 & 118 & 20 \\ 3 & 3 & 4 & -4 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[84] \quad 1^{1124} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1028, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.50 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 11 & 3 \\ 2 & 1 & 5 & 0 & 9 & 15 & 3 \\ 3 & 5 & 4 & 0 & 13 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 13 & 0 & 28 & 48 & 10 \\ 11 & 15 & 21 & 0 & 48 & 76 & 16 \\ 3 & 3 & 4 & -4 & 10 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[85] \quad 1^{1144} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1034, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.52 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 14 & 3 \\ 2 & 1 & 5 & 0 & 15 & 18 & 3 \\ 3 & 5 & 4 & 0 & 21 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 15 & 21 & 0 & 76 & 96 & 16 \\ 14 & 18 & 26 & 0 & 96 & 118 & 20 \\ 3 & 3 & 4 & -4 & 16 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[86] \quad 1^{114} 3^1 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1064, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.62 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 6 & 0 & 3 & 33 & 4 \\ 3 & 6 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 21 & 3 \\ 18 & 33 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & 4 & -4 & 3 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[87] \quad 1^{1133} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1264, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.44 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 12 & 3 \\ 2 & 1 & 5 & 0 & 15 & 15 & 3 \\ 3 & 5 & 4 & 0 & 22 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 15 & 22 & 0 & 85 & 87 & 17 \\ 12 & 15 & 22 & 0 & 87 & 85 & 17 \\ 3 & 3 & 4 & -4 & 17 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[88] \quad 1^{1133} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1276, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.6$$

$$T_{\min} = 12.47 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 9 & 3 \\ 2 & 1 & 5 & 0 & 12 & 12 & 3 \\ 3 & 5 & 4 & 0 & 17 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 17 & 0 & 49 & 51 & 13 \\ 9 & 12 & 17 & 0 & 51 & 49 & 13 \\ 3 & 3 & 4 & -4 & 13 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[89] \quad 1^{1134} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1475, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.49 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 11 & 3 \\ 2 & 1 & 5 & 0 & 12 & 15 & 3 \\ 3 & 5 & 4 & 0 & 17 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 17 & 0 & 49 & 63 & 13 \\ 11 & 15 & 21 & 0 & 63 & 76 & 16 \\ 3 & 3 & 4 & -4 & 13 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[90] \quad 1^{114} 3^1 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -103, \quad \det = -1476, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.55 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 6 & 0 & 3 & 36 & 4 \\ 3 & 6 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 24 & 3 \\ 20 & 36 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & -4 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

[91] $1^{1134}31^{122} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1522$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 14 & 3 \\ 2 & 1 & 5 & 0 & 15 & 18 & 3 \\ 3 & 5 & 4 & 0 & 22 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 15 & 22 & 0 & 85 & 102 & 17 \\ 14 & 18 & 26 & 0 & 102 & 118 & 20 \\ 3 & 3 & 4 & -4 & 17 & 20 & 0 \end{pmatrix}_{b_0}$$

[92] $1^{1124}31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1522$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.5$

$T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 14 & 3 \\ 2 & 1 & 5 & 0 & 9 & 18 & 3 \\ 3 & 5 & 4 & 0 & 13 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 13 & 0 & 28 & 60 & 10 \\ 14 & 18 & 26 & 0 & 60 & 118 & 20 \\ 3 & 3 & 4 & -4 & 10 & 20 & 0 \end{pmatrix}_{b_0}$$

[93] $1^{1134}31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1528$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.8$

$T_{\min} = 12.47 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 14 & 3 \\ 2 & 1 & 5 & 0 & 12 & 18 & 3 \\ 3 & 5 & 4 & 0 & 17 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 17 & 0 & 49 & 78 & 13 \\ 14 & 18 & 26 & 0 & 78 & 118 & 20 \\ 3 & 3 & 4 & -4 & 13 & 20 & 0 \end{pmatrix}_{b_0}$$

[94] $1^{1144}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -1800$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 11 & 3 \\ 2 & 1 & 5 & 0 & 15 & 15 & 3 \\ 3 & 5 & 4 & 0 & 21 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 15 & 21 & 0 & 76 & 78 & 16 \\ 11 & 15 & 21 & 0 & 78 & 76 & 16 \\ 3 & 3 & 4 & -4 & 16 & 16 & 0 \end{pmatrix}_{b_0}$$

[95] $1^{1144}31^{122} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -103$, $\det = -2284$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.45 \rightarrow 13$, $T_{\max} = 21$, $T \in [13, 21]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 14 & 3 \\ 2 & 1 & 5 & 0 & 18 & 18 & 3 \\ 3 & 5 & 4 & 0 & 26 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 14 & 18 & 26 & 0 & 118 & 120 & 20 \\ 14 & 18 & 26 & 0 & 120 & 118 & 20 \\ 3 & 3 & 4 & -4 & 20 & 20 & 0 \end{pmatrix}_{b_0}$$

$T = 10$ [96] $1^{111}31^{112}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 73$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.5$

$T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 5 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 9 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 15 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 4 & 15 & 4 \\ 5 & 9 & 15 & 9 & 0 & 15 & 37 & 11 \\ 2 & 3 & 4 & 2 & -4 & 4 & 11 & 0 \end{pmatrix}_{b_0}$$

[97] $1^{111}31^{111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 145$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.79 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 6 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 4 & 9 & 4 \\ 3 & 6 & 9 & 6 & 0 & 9 & 13 & 7 \\ 2 & 3 & 4 & 2 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

[98] $1^{111}31^{111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 151$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 6 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 1 & 6 & 3 \\ 3 & 6 & 9 & 6 & 0 & 6 & 13 & 7 \\ 2 & 3 & 4 & 2 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

[99] $1^{112}31^{112}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 244, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 6 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 12 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 18 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 4 & 18 & 4 \\ 6 & 12 & 18 & 12 & 0 & 18 & 58 & 14 \\ 2 & 3 & 4 & 2 & -4 & 4 & 14 & 0 \end{pmatrix}_{b_0}$$

[100] $1^{112}31^{113}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 388, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 8 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 15 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 24 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 4 & 24 & 4 \\ 8 & 15 & 24 & 15 & 0 & 24 & 100 & 18 \\ 2 & 3 & 4 & 2 & -4 & 4 & 18 & 0 \end{pmatrix}_{b_0}$$

[101] $1^{113}31^{113}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 409, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.8$

$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 9 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 18 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 27 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 4 & 27 & 4 \\ 9 & 18 & 27 & 18 & 0 & 27 & 133 & 21 \\ 2 & 3 & 4 & 2 & -4 & 4 & 21 & 0 \end{pmatrix}_{b_0}$$

[102] $1^{1111}31^{121}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 583, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 6 & 3 \\ 2 & 4 & 4 & 4 & 0 & 15 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 9 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 15 & 9 & 0 & 37 & 24 & 11 \\ 3 & 6 & 9 & 6 & 0 & 24 & 13 & 7 \\ 2 & 3 & 4 & 2 & -4 & 11 & 7 & 0 \end{pmatrix}_{b_0}$$

[103] $1^{114}31^{114}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 640, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 12 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 24 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 36 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 24 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 4 & 36 & 4 \\ 12 & 24 & 36 & 24 & 0 & 36 & 238 & 28 \\ 2 & 3 & 4 & 2 & -4 & 4 & 28 & 0 \end{pmatrix}_{b_0}$$

[104] $1^{1111}31^{122}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 700, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 5 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 9 & 3 \\ 2 & 4 & 4 & 4 & 0 & 15 & 15 & 4 \\ 1 & 3 & 4 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 15 & 9 & 0 & 37 & 39 & 11 \\ 5 & 9 & 15 & 9 & 0 & 39 & 37 & 11 \\ 2 & 3 & 4 & 2 & -4 & 11 & 11 & 0 \end{pmatrix}_{b_0}$$

[105] $1^{113}31^{114}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 769, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 12.79 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 11 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 21 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 33 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 4 & 33 & 4 \\ 11 & 21 & 33 & 21 & 0 & 33 & 193 & 25 \\ 2 & 3 & 4 & 2 & -4 & 4 & 25 & 0 \end{pmatrix}_{b_0}$$

[106] $1^{1111}31^{111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 844, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 6 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 9 & 6 & 0 & 13 & 15 & 7 \\ 3 & 6 & 9 & 6 & 0 & 15 & 13 & 7 \\ 2 & 3 & 4 & 2 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[107] \quad 1^{\overline{111}} 3 1^{\overline{111}} 3 1^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 1036, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.92 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 & 2 & 2 \\ 2 & 4 & 7 & 3 & 0 & 6 & 6 & 4 \\ 2 & 7 & 4 & 4 & 0 & 6 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 6 & 3 & 0 & 4 & 6 & 4 \\ 2 & 6 & 6 & 3 & 0 & 6 & 4 & 4 \\ 2 & 4 & 4 & 2 & -4 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

6.23 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$

$$T = 10 \quad [0] \quad 1^{\overline{111}} 3 1^{\overline{111}} 3 1^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -102, \quad \det = 16, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.75 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 20 & 9 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 20 & 0 & 13 & 15 & 7 \\ 9 & 12 & 20 & 0 & 15 & 13 & 7 \\ 5 & 6 & 9 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3 1^{\overline{111}} 3 1^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -102, \quad \det = 188, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.47 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 17 & 5 \\ 8 & 10 & 17 & 0 & 12 & 24 & 6 \\ 12 & 17 & 25 & 0 & 20 & 36 & 9 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 20 & 0 & 13 & 27 & 7 \\ 17 & 24 & 36 & 0 & 27 & 49 & 13 \\ 5 & 6 & 9 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 3 1^{\overline{111}} 3 1^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -102, \quad \det = 300, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.64 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 15 & 17 & 0 & 13 & 15 & 7 \\ 9 & 15 & 17 & 0 & 15 & 13 & 7 \\ 5 & 7 & 8 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{111}} 3 1^{\overline{111}} 3 1^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 10, \quad T_{\max} = 22, \quad \Delta = -102, \quad \det = 832, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.52 \rightarrow 14, \quad T_{\max} = 22, \quad T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 17 & 5 \\ 9 & 13 & 18 & 0 & 15 & 27 & 7 \\ 11 & 18 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 15 & 17 & 0 & 13 & 27 & 7 \\ 17 & 27 & 33 & 0 & 27 & 49 & 13 \\ 5 & 7 & 8 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

6.24 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{\overline{111}} 3 1^{\overline{111}} 3 1^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 6 & 3 & 4 \\ 3 & 6 & 5 & 0 & 6 & 4 & 3 & 4 \\ 2 & 3 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{112}} 3 1^{\overline{114}} 3 1^{\overline{114}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 136, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 24 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 26 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 18 & 3 & 3 \\ 14 & 24 & 26 & 0 & 18 & 118 & 18 & 20 \\ 2 & 3 & 4 & 0 & 3 & 18 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1121}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 253, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 9 & 3 & 3 \\ 7 & 12 & 13 & 0 & 9 & 28 & 9 & 10 \\ 2 & 3 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{113}} 3^{\overline{1121}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 325, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 13 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 24 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 23 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 15 & 3 & 3 \\ 13 & 24 & 23 & 0 & 15 & 100 & 15 & 18 \\ 2 & 3 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{1111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 409, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 6 & 3 & 3 \\ 5 & 9 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}} 3^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 447, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 6 & 15 & 7 \\ 2 & 3 & 4 & 0 & 6 & 1 & 6 & 3 \\ 5 & 6 & 9 & 0 & 15 & 6 & 12 & 6 \\ 3 & 3 & 4 & -4 & 7 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1111}} 3^{\overline{1111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 496, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 & 3 \\ 4 & 5 & 9 & 0 & 12 & 12 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 6 & 12 & 0 & 13 & 15 & 6 & 7 \\ 5 & 6 & 12 & 0 & 15 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 6 & 6 & 0 & 2 \\ 3 & 3 & 5 & -4 & 7 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1111}} 3^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 499, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 9 & 3 & 4 \\ 5 & 9 & 9 & 0 & 9 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1111}} 3^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 516, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 6 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 3 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 6 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 3 & 4 & -4 & 3 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{112}} 3^{\overline{1121}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 760, \ (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 18 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 18 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 12 & 3 & 3 \\ 10 & 18 & 18 & 0 & 12 & 58 & 12 & 14 \\ 2 & 3 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11111} 3^1 1^{1131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 772, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 3 & 4 & -4 & 3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{112} 3^1 1^{1131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 820, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 21 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 22 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 15 & 3 & 3 \\ 12 & 21 & 22 & 0 & 15 & 85 & 15 & 17 \\ 2 & 3 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{114} 3^1 1^{1131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 928, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 18 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 33 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 32 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 21 & 3 & 3 \\ 18 & 33 & 32 & 0 & 21 & 193 & 21 & 25 \\ 2 & 3 & 4 & 0 & 3 & 21 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1111} 3^1 1^{1211} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 931, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 13 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 12 & 13 & 0 & 28 & 21 & 9 & 10 \\ 5 & 9 & 9 & 0 & 21 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 9 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 10 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111} 3^1 1^{1221} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 964, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 13 & 13 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 12 & 13 & 0 & 28 & 30 & 9 & 10 \\ 7 & 12 & 13 & 0 & 30 & 28 & 9 & 10 \\ 2 & 3 & 4 & 0 & 9 & 9 & 0 & 2 \\ 3 & 4 & 4 & -4 & 10 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1111} 3^1 1^{1111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 9 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 15 & 6 & 7 \\ 5 & 9 & 9 & 0 & 15 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 6 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 7 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{113} 3^1 1^{1131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1345, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 27 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 27 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 18 & 3 & 3 \\ 15 & 27 & 27 & 0 & 18 & 133 & 18 & 21 \\ 2 & 3 & 4 & 0 & 3 & 18 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

[17] $1^{113}3^11^{141} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1621, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$

$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 17 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 30 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 31 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 21 & 3 & 3 \\ 17 & 30 & 31 & 0 & 21 & 172 & 21 & 24 \\ 2 & 3 & 4 & 0 & 3 & 21 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

[18] $1^{114}3^11^{141} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 2164, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$

$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 36 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 36 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 24 & 3 & 3 \\ 20 & 36 & 36 & 0 & 24 & 238 & 24 & 28 \\ 2 & 3 & 4 & 0 & 3 & 24 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

6.25 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 10 \quad [0] 1^{113}3^11^{131}1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 97, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 4 & 3 & 4 \\ 1 & 2 & 3 & 2 & 0 & 3 & 0 & 2 \\ 2 & 3 & 4 & 2 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^11^{131}1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 102, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 4 & 2 & -4 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}3^11^{1231}1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$

$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 15 & 9 & 0 & 37 & 8 & 11 \\ 1 & 2 & 3 & 2 & 0 & 8 & 0 & 2 \\ 2 & 3 & 4 & 2 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^11^{1131}1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 543, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 9 & 6 & 0 & 13 & 5 & 7 \\ 1 & 2 & 3 & 2 & 0 & 5 & 0 & 2 \\ 2 & 3 & 4 & 2 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{13}3^11^{111}3^11 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 744, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$

$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 6 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 6 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 3 & 2 & 0 & 3 & 0 & 2 \\ 2 & 4 & 4 & 2 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

6.26 NHC(-2)(-2)(-3) + e₆ + su₂ + su2n3mix

$$T = 10 \quad [0] \quad 1^{111} 3^1 1^{111} 3 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 8 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 12 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 4 & 12 & 4 \\ 4 & 8 & 12 & 8 & 0 & 12 & 25 & 9 \\ 2 & 3 & 4 & 2 & -4 & 4 & 9 & 0 \end{pmatrix}_{b_0}$$

6.27 NHC(-2)(-2)(-3) + e₆ + su₂ + su₃

$$T = 9 \quad [0] \quad 1^{112} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 16.20 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 7 & 3 \\ 2 & 1 & 5 & 0 & 3 & 9 & 3 \\ 3 & 5 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 4 & 12 & 4 \\ 7 & 9 & 13 & 0 & 12 & 27 & 9 \\ 3 & 3 & 4 & -4 & 4 & 9 & -4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 4 & 5 & 9 & 0 & 8 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 8 & 0 & 4 & 3 & 4 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 3 & 5 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 3 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 3 \\ 3 & 5 & 4 & 0 & 9 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 9 & 7 \\ 3 & 3 & 5 & 0 & 9 & 3 & 3 \\ 3 & 3 & 4 & -4 & 7 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -68, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.24 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 \\ 3 & 3 & 5 & 0 & 3 & 3 & 3 \\ 3 & 3 & 4 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -78, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.87 \rightarrow 15, T_{\max} = 23, T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 4 & 5 & 9 & 0 & 4 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 6 & 3 \\ 5 & 6 & 12 & 0 & 6 & 12 & 6 \\ 3 & 3 & 5 & -4 & 3 & 6 & -2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{112} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -128, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 2 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 & 3 \\ 3 & 5 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 12 & 18 & 0 & 58 & 12 & 14 \\ 2 & 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 3 & 4 & -4 & 14 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{112} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -138, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 12 & 6 \\ 5 & 6 & 9 & 0 & 12 & 12 & 6 \\ 3 & 3 & 4 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -143, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.46 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 11 & 2 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 & 3 \\ 4 & 5 & 9 & 0 & 28 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 12 & 28 & 0 & 73 & 12 & 15 \\ 2 & 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 3 & 5 & -4 & 15 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -146, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -155, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.92 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 \\ 3 & 6 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & 4 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -191, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 3 & 4 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -203, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 3 \\ 4 & 5 & 9 & 0 & 12 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 12 & 0 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 6 & 0 & 2 \\ 3 & 3 & 5 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{112} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -257, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.02 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 & 3 \\ 3 & 5 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 13 & 0 & 28 & 12 & 10 \\ 3 & 3 & 5 & 0 & 12 & 3 & 3 \\ 3 & 3 & 4 & -4 & 10 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -275, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.43 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 1 & 3 \\ 3 & 4 & 9 & 3 & 6 & 0 & 4 \\ 4 & 9 & 9 & 4 & 8 & 0 & 5 \\ 2 & 3 & 4 & -3 & 3 & 0 & -1 \\ 3 & 6 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{114} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -281, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.13 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 & 3 \\ 3 & 5 & 4 & 0 & 21 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 15 & 21 & 0 & 76 & 18 & 16 \\ 3 & 3 & 5 & 0 & 18 & 3 & 3 \\ 3 & 3 & 4 & -4 & 16 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{111} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -287, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.98 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 8 & 2 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 & 3 \\ 4 & 5 & 9 & 0 & 20 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 9 & 20 & 0 & 37 & 9 & 11 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 3 & 3 & 5 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -299, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 9 & 3 \\ 7 & 12 & 13 & 0 & 9 & 27 & 9 \\ 3 & 4 & 4 & -4 & 3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{111} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -317, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.04 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 6 & 0 & 2 \\ 3 & 3 & 4 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{113} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -317, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 6 & 0 & 21 & 3 & 4 \\ 3 & 6 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 21 & 19 & 0 & 73 & 12 & 15 \\ 2 & 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -4 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{111} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -318, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 3 & 5 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 6 & 3 \\ 5 & 6 & 9 & 0 & 6 & 12 & 6 \\ 3 & 3 & 4 & -4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{113} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -329, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.01 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 3 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 & 3 \\ 3 & 5 & 4 & 0 & 17 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 17 & 0 & 49 & 15 & 13 \\ 3 & 3 & 5 & 0 & 15 & 3 & 3 \\ 3 & 3 & 4 & -4 & 13 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{121} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -338, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 10 & 2 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 & 3 \\ 4 & 5 & 9 & 0 & 24 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 12 & 24 & 0 & 58 & 12 & 14 \\ 2 & 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 3 & 5 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{111} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -351, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.88 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & -4 & 3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1111} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -378, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 13 & 15 & 7 \\ 5 & 6 & 9 & 0 & 15 & 12 & 6 \\ 3 & 3 & 4 & -4 & 7 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1111} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -383, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & -4 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1111} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -392, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.04 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 3 \\ 3 & 5 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 6 & 8 & 0 & 10 & 6 & 6 \\ 2 & 3 & 4 & 0 & 6 & 0 & 2 \\ 3 & 3 & 4 & -4 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{111} 31^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -407, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 12 & 13 & 0 & 28 & 9 & 10 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1131} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -407, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.16 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 2 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 6 & 9 & 12 & 0 & 25 & 9 & 9 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 3 & 3 & 4 & -4 & 9 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1141} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -428, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 8 & 2 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 & 3 \\ 3 & 5 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 12 & 16 & 0 & 46 & 12 & 12 \\ 2 & 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 3 & 4 & -4 & 12 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{111} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -485, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 7 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{112} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -542, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 6 & 0 & 15 & 3 & 4 \\ 3 & 6 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 15 & 14 & 0 & 37 & 9 & 11 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -4 & 11 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{131} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -563, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.98 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 15 & 2 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 & 3 \\ 4 & 5 & 9 & 0 & 36 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 15 & 18 & 36 & 0 & 133 & 18 & 21 \\ 2 & 3 & 4 & 0 & 18 & 0 & 2 \\ 3 & 3 & 5 & -4 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{121} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -620, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 13 & 2 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 & 3 \\ 4 & 5 & 9 & 0 & 32 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 13 & 15 & 32 & 0 & 100 & 15 & 18 \\ 2 & 3 & 4 & 0 & 15 & 0 & 2 \\ 3 & 3 & 5 & -4 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{112} 31^{141} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -632, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 6 & 0 & 24 & 3 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 14 & 24 & 26 & 0 & 118 & 18 & 20 \\ 2 & 3 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -4 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{121} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -674, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.01 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 16 & 2 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 & 3 \\ 4 & 5 & 9 & 0 & 40 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 16 & 18 & 40 & 0 & 154 & 18 & 22 \\ 2 & 3 & 4 & 0 & 18 & 0 & 2 \\ 3 & 3 & 5 & -4 & 22 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{112} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -696, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 5 & 3 \\ 2 & 1 & 5 & 0 & 9 & 6 & 3 \\ 3 & 5 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 13 & 0 & 28 & 21 & 10 \\ 5 & 6 & 9 & 0 & 21 & 12 & 6 \\ 3 & 3 & 4 & -4 & 10 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{112} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -698, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 2 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 & 3 \\ 3 & 5 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 13 & 0 & 28 & 9 & 10 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 3 & 3 & 4 & -4 & 10 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{141} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -878, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.99 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 20 & 2 & 3 \\ 2 & 1 & 5 & 0 & 24 & 3 & 3 \\ 4 & 5 & 9 & 0 & 48 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 20 & 24 & 48 & 0 & 238 & 24 & 28 \\ 2 & 3 & 4 & 0 & 24 & 0 & 2 \\ 3 & 3 & 5 & -4 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{113} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -887, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.01 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 2 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 & 3 \\ 3 & 5 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 15 & 22 & 0 & 85 & 15 & 17 \\ 2 & 3 & 4 & 0 & 15 & 0 & 2 \\ 3 & 3 & 4 & -4 & 17 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{113} 1^{11} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -894, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 5 & 3 \\ 2 & 1 & 5 & 0 & 12 & 6 & 3 \\ 3 & 5 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 17 & 0 & 49 & 27 & 13 \\ 5 & 6 & 9 & 0 & 27 & 12 & 6 \\ 3 & 3 & 4 & -4 & 13 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{112} 3^1 1^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 6 & 0 & 18 & 3 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 10 & 18 & 18 & 0 & 58 & 12 & 14 \\ 2 & 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -4 & 14 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{114} 1^{11} 3^1 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -950, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 17 & 2 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 & 3 \\ 3 & 5 & 4 & 0 & 31 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 17 & 21 & 31 & 0 & 172 & 21 & 24 \\ 2 & 3 & 4 & 0 & 21 & 0 & 2 \\ 3 & 3 & 4 & -4 & 24 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{114} 1^{11} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -972, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 5 & 3 \\ 2 & 1 & 5 & 0 & 15 & 6 & 3 \\ 3 & 5 & 4 & 0 & 21 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 15 & 21 & 0 & 76 & 33 & 16 \\ 5 & 6 & 9 & 0 & 33 & 12 & 6 \\ 3 & 3 & 4 & -4 & 16 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{112} 3^1 1^{131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -974, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 6 & 0 & 21 & 3 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 12 & 21 & 22 & 0 & 85 & 15 & 17 \\ 2 & 3 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -4 & 17 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{131} 3^1 1^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1043, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 18 & 2 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 & 3 \\ 4 & 5 & 9 & 0 & 44 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 18 & 21 & 44 & 0 & 193 & 21 & 25 \\ 2 & 3 & 4 & 0 & 21 & 0 & 2 \\ 3 & 3 & 5 & -4 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{113} 1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1085, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 2 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 & 3 \\ 3 & 5 & 4 & 0 & 17 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 17 & 0 & 49 & 12 & 13 \\ 2 & 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 3 & 4 & -4 & 13 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{114} 3^1 1^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1172, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 6 & 0 & 30 & 3 & 4 \\ 3 & 6 & 4 & 0 & 28 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 16 & 30 & 28 & 0 & 154 & 18 & 22 \\ 2 & 3 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -4 & 22 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{113} 31^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1199, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 6 & 0 & 24 & 3 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 13 & 24 & 23 & 0 & 100 & 15 & 18 \\ 2 & 3 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -4 & 18 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{1141} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1478, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 2 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 & 3 \\ 3 & 5 & 4 & 0 & 21 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 15 & 21 & 0 & 76 & 15 & 16 \\ 2 & 3 & 4 & 0 & 15 & 0 & 2 \\ 3 & 3 & 4 & -4 & 16 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{1141} 31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1652, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 2 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 & 3 \\ 3 & 5 & 4 & 0 & 26 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 14 & 18 & 26 & 0 & 118 & 18 & 20 \\ 2 & 3 & 4 & 0 & 18 & 0 & 2 \\ 3 & 3 & 4 & -4 & 20 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{113} 31^{131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 6 & 0 & 27 & 3 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 15 & 27 & 27 & 0 & 133 & 18 & 21 \\ 2 & 3 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -4 & 21 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{113} 31^{141} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1847, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 6 & 0 & 30 & 3 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 17 & 30 & 31 & 0 & 172 & 21 & 24 \\ 2 & 3 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -4 & 24 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{114} 31^{131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2162, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 6 & 0 & 33 & 3 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 18 & 33 & 32 & 0 & 193 & 21 & 25 \\ 2 & 3 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -4 & 25 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{114} 31^{141} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2780, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 6 & 0 & 36 & 3 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 20 & 36 & 36 & 0 & 238 & 24 & 28 \\ 2 & 3 & 4 & 0 & 24 & 0 & 2 \\ 3 & 4 & 4 & -4 & 28 & 2 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [54] \quad 1^{11} 31^{112} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 192, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.25 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 4 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 12 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 12 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 12 & 12 & 6 & 0 & 22 & 6 & 8 \\ 1 & 3 & 3 & 3 & 0 & 6 & 0 & 2 \\ 2 & 4 & 4 & 2 & -4 & 8 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{111}31^{111}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 201, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.04 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 6 & 3 & 0 & 4 & 3 & 4 \\ 3 & 6 & 9 & 6 & 0 & 9 & 12 & 6 \\ 2 & 3 & 4 & 2 & -4 & 4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{111}31^{111}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 354, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 6 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 6 & 6 & 3 & 0 & 4 & 3 & 4 \\ 1 & 3 & 3 & 3 & 0 & 3 & 0 & 2 \\ 2 & 4 & 4 & 2 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{111}31^{111}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 456, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.60 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 9 & 6 & 0 & 13 & 9 & 7 \\ 2 & 3 & 6 & 3 & 0 & 9 & 3 & 3 \\ 2 & 3 & 4 & 2 & -4 & 7 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{111}31^{112}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 606, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 5 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 15 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 15 & 15 & 9 & 0 & 37 & 9 & 11 \\ 1 & 3 & 3 & 3 & 0 & 9 & 0 & 2 \\ 2 & 4 & 4 & 2 & -4 & 11 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{111}31^{113}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 1200, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.12 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 7 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 21 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 21 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 12 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 21 & 21 & 12 & 0 & 73 & 12 & 15 \\ 1 & 3 & 3 & 3 & 0 & 12 & 0 & 2 \\ 2 & 4 & 4 & 2 & -4 & 15 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{111}31^{114}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 1686, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.13 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 9 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 27 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 27 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 15 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 27 & 27 & 15 & 0 & 121 & 15 & 19 \\ 1 & 3 & 3 & 3 & 0 & 15 & 0 & 2 \\ 2 & 4 & 4 & 2 & -4 & 19 & 2 & -1 \end{pmatrix}_{b_0}$$

6.28 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{111}31^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 100, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 3 & 6 & 4 \\ 2 & 3 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 6 & 5 & 0 & 6 & 3 & 4 & 4 \\ 3 & 4 & 4 & -4 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{112}31^{1114} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -82, \quad \det = 136, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 24 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 18 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 18 & 2 \\ 14 & 24 & 26 & 0 & 18 & 18 & 118 & 20 \\ 3 & 4 & 4 & -4 & 3 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 31^{1112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 253, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 9 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 9 & 2 \\ 7 & 12 & 13 & 0 & 9 & 9 & 28 & 10 \\ 3 & 4 & 4 & -4 & 3 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{113} 31^{1112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 325, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 24 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 23 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 15 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 15 & 2 \\ 13 & 24 & 23 & 0 & 15 & 15 & 100 & 18 \\ 3 & 4 & 4 & -4 & 3 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 31^{1111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 409, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 3 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1111} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 447, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 5 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 6 & 6 & 3 \\ 5 & 6 & 9 & 0 & 6 & 12 & 15 & 6 \\ 5 & 6 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 3 & 4 & -4 & 3 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1111} 31^{1111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 496, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 6 & 3 \\ 4 & 5 & 9 & 0 & 12 & 4 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 6 & 12 & 0 & 13 & 6 & 15 & 7 \\ 2 & 3 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 6 & 12 & 0 & 15 & 6 & 13 & 7 \\ 3 & 3 & 5 & -4 & 7 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{1111} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 499, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 4 & 3 & 9 & 4 \\ 2 & 3 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 9 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 516, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 6 & 3 & 3 \\ 5 & 6 & 9 & 0 & 6 & 12 & 6 & 6 \\ 2 & 3 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 3 & 4 & -4 & 3 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

- [9] $1^{112}31^{1112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 760$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 18 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 12 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 12 & 2 \\ 10 & 18 & 18 & 0 & 12 & 12 & 58 & 14 \\ 3 & 4 & 4 & -4 & 3 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{1111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 772$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.99 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 3 & 4 & -4 & 3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{112}31^{1113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 820$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.5$
 $T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 15 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 15 & 2 \\ 12 & 21 & 22 & 0 & 15 & 15 & 85 & 17 \\ 3 & 4 & 4 & -4 & 3 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$
- [12] $1^{114}31^{1113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 928$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 15.0$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 18 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 33 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 32 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 21 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 21 & 2 \\ 18 & 33 & 32 & 0 & 21 & 21 & 193 & 25 \\ 3 & 4 & 4 & -4 & 3 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$
- [13] $1^{1111}31^{1211} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 931$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 12 & 13 & 0 & 28 & 9 & 21 & 10 \\ 2 & 3 & 4 & 0 & 9 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 21 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 10 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$
- [14] $1^{1111}31^{1212} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 964$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.78 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 12 & 13 & 0 & 28 & 9 & 30 & 10 \\ 2 & 3 & 4 & 0 & 9 & 0 & 9 & 2 \\ 7 & 12 & 13 & 0 & 30 & 9 & 28 & 10 \\ 3 & 4 & 4 & -4 & 10 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$
- [15] $1^{1111}31^{1111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 1276$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 6 & 15 & 7 \\ 2 & 3 & 4 & 0 & 6 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 15 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 7 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$
- [16] $1^{113}31^{1113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10$, $T_{\max} = 13$, $\Delta = -82$, $\det = 1345$, $(n_+, n_-, n_0) = (1, 10, 0)$, $c_{\text{ext}} = 14.8$
 $T_{\min} = 12.77 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 27 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 18 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 18 & 2 \\ 15 & 27 & 27 & 0 & 18 & 18 & 133 & 21 \\ 3 & 4 & 4 & -4 & 3 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{113} 3^1 1^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1621, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 30 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 21 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 21 & 2 \\ 17 & 30 & 31 & 0 & 21 & 21 & 172 & 24 \\ 3 & 4 & 4 & -4 & 3 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{114} 3^1 1^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 2164, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 36 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 24 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 24 & 2 \\ 20 & 36 & 36 & 0 & 24 & 24 & 238 & 28 \\ 3 & 4 & 4 & -4 & 3 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

6.29 NHC(-2)(-2)(-3) + $\mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3$

$$T = 10 \quad [0] \ 1^{1111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 84, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.24 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 & 3 \\ 3 & 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 3 & 4 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{121} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 462, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 13 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 7 & 9 & 13 & 0 & 28 & 12 & 9 & 10 \\ 3 & 3 & 5 & 0 & 12 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 9 & 3 & 0 & 2 \\ 3 & 3 & 4 & -4 & 10 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 498, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.96 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 15 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 14 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 8 & 15 & 14 & 0 & 37 & 9 & 9 & 11 \\ 2 & 3 & 4 & 0 & 9 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 9 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 11 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1141} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 534, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.19 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 21 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 11 & 15 & 21 & 0 & 76 & 18 & 15 & 16 \\ 3 & 3 & 5 & 0 & 18 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 15 & 3 & 0 & 2 \\ 3 & 3 & 4 & -4 & 16 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 555, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.05 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 1 & 3 & 6 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & 4 & -4 & 3 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{11}3^{\overline{1}}1^{\overline{11}}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 678, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.14 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 17 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 9 & 12 & 17 & 0 & 49 & 15 & 12 & 13 \\ 3 & 3 & 5 & 0 & 15 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 12 & 3 & 0 & 2 \\ 3 & 3 & 4 & -4 & 13 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{11}1^{\overline{1}}3^{\overline{11}1^{\overline{1}}}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 894, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 13 & 6 & 6 & 7 \\ 2 & 3 & 4 & 0 & 6 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 6 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 7 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

6.30 NHC(-2)(-2)(-3) + $\mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{\overline{1}3^{\overline{1}}2^{\overline{11}}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -24, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 2 & 3 & 0 & 4 & 4 \\ 2 & 3 & 1 & 1 & 2 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & -2 & 0 \\ 3 & 4 & 3 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1}3^{\overline{1}}2^{\overline{11}}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -54, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.41 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 3 \\ 2 & 1 & 3 & 0 & 2 & 3 \\ 2 & 3 & 2 & 1 & 4 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 2 & 4 & 0 & -2 & 0 \\ 3 & 3 & 4 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1}3^{\overline{1}}2^{\overline{11}}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 21, \ \Delta = -108, \ \det = -62, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.35 \rightarrow 14, \ T_{\max} = 21, \ T \in [14, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \\ 3 & 6 & 5 & 5 & 3 & 5 \\ 2 & 3 & 5 & -2 & 2 & 0 \\ 2 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 5 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1}3^{\overline{1}}1^{\overline{1}}3^{\overline{1}1^{\overline{1}}}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -179, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.83 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 2 & 3 & 2 & 0 & 0 & 2 \\ 2 & 3 & 4 & 2 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{1}3^{\overline{1}}1^{\overline{1}}3^{\overline{1}1^{\overline{1}}}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 25, \ \Delta = -114, \ \det = -345, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.93 \rightarrow 13, \ T_{\max} = 25, \ T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 2 & 0 & 0 & 2 \\ 2 & 4 & 4 & 2 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

6.31 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8$

$$T = 10 \quad [0] \quad 1^{\overline{11}} 3^{\overline{1}} 21^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{so}_8 \quad T = 10, T_{\max} = 22, \Delta = -107, \det = 138, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.70 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 3 & 0 & 4 \\ 3 & 6 & 5 & 5 & 3 & 0 & 5 \\ 2 & 3 & 5 & -2 & 2 & 0 & 0 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 0 & 2 & -2 & 0 \end{pmatrix}_{b_0}$$

6.32 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} 31^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 97, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 6 \\ 1 & 3 & 4 & 0 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 \\ 1 & 2 & 3 & 2 & 0 & 0 & 3 \\ 2 & 3 & 6 & 3 & 0 & 3 & 4 \\ 2 & 3 & 4 & 2 & -4 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} 31^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 102, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 6 \\ 1 & 3 & 4 & 0 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 \\ 1 & 2 & 3 & 2 & 0 & 0 & 3 \\ 1 & 3 & 3 & 3 & 0 & 2 & 1 \\ 2 & 3 & 4 & 2 & -4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{12}} 31^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.5$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 9 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 2 & 3 & 2 & 0 & 0 & 8 & 2 \\ 5 & 9 & 15 & 9 & 0 & 8 & 37 & 11 \\ 2 & 3 & 4 & 2 & -4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} 31^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 543, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 6 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 2 & 3 & 2 & 0 & 0 & 5 & 2 \\ 3 & 6 & 9 & 6 & 0 & 5 & 13 & 7 \\ 2 & 3 & 4 & 2 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{11}} 31^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 744, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 6 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 2 & 0 & 0 & 3 & 2 \\ 2 & 6 & 6 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & 4 & 2 & -4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

6.33 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \quad 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{11}} 31^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 316, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 4 & 2 & -4 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{11}} 31^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 524, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 1 & 3 & 3 & 2 & 0 & 0 & 2 \\ 1 & 3 & 3 & 2 & 0 & 2 & 2 \\ 2 & 4 & 4 & 2 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

6.34 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2 \rightarrow (-3)$ + $\mathfrak{su}_3$

$$T = 10 \quad [0] \quad 1^{11}3^{12}1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 2 & 2 & 3 \\ 2 & 2 & 3 & 0 & 4 & 2 & 4 \\ 2 & 3 & 1 & 1 & 2 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 2 & 0 & -2 & 2 & 0 \\ 2 & 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 3 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^{11}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.59 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 2 & 3 & 2 & 0 & 0 & 3 & 2 \\ 2 & 3 & 6 & 3 & 0 & 3 & 3 & 3 \\ 2 & 3 & 4 & 2 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}1^{11}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 284, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.24 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 2 & 0 & 0 & 2 & 2 \\ 1 & 3 & 3 & 3 & 0 & 2 & 0 & 2 \\ 2 & 4 & 4 & 2 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

6.35 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_2 \mathfrak{n}3 \text{mix}$

$$T = 9 \quad [0] \quad 1^{11}3^{11}1^{11}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -71, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.08 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 5 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 5 & 9 & 5 & 0 & 12 & 6 \\ 2 & 3 & 4 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^{11}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 3 & 0 & 14 & 3 \\ 2 & 4 & 4 & 4 & 0 & 24 & 4 \\ 1 & 3 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 14 & 24 & 14 & 0 & 97 & 17 \\ 2 & 3 & 4 & 2 & -4 & 17 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}3^{11}1^{11}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.89 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 5 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 5 & 6 & 5 & 0 & 7 & 5 \\ 2 & 3 & 4 & 2 & -4 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}3^{11}1^{11}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -170, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 7 & 3 & 0 & 12 & 4 \\ 2 & 7 & 4 & 4 & 0 & 12 & 4 \\ 1 & 3 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 12 & 12 & 8 & 0 & 25 & 9 \\ 2 & 4 & 4 & 2 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11}3^{11}1^{11}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.85 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 3 & 0 & 11 & 3 \\ 2 & 4 & 4 & 4 & 0 & 18 & 4 \\ 1 & 3 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 11 & 18 & 11 & 0 & 55 & 13 \\ 2 & 3 & 4 & 2 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111}3^1 1^{11}31^1 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 8 & 3 \\ 2 & 4 & 4 & 4 & 0 & 12 & 4 \\ 1 & 3 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 8 & 12 & 8 & 0 & 25 & 9 \\ 2 & 3 & 4 & 2 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^1 3^1 1^{111}31^1 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 \quad T = 9, T_{\max} = 25, \Delta = -114, \det = -516, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 25, T \in [13, 25]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 7 & 3 & 0 & 9 & 4 \\ 2 & 7 & 4 & 4 & 0 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 9 & 9 & 5 & 0 & 12 & 6 \\ 2 & 4 & 4 & 2 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

6.36 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + su2n3mix + $\mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{111}3^1 1^{111}31^1 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -84, \det = 152, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.82 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 8 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 12 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 4 & 8 & 12 & 8 & 0 & 25 & 12 & 9 \\ 2 & 3 & 6 & 3 & 0 & 12 & 4 & 4 \\ 2 & 3 & 4 & 2 & -4 & 9 & 4 & 0 \end{pmatrix}_{b_0}$$

6.37 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + su2n3mix + $\mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{111}3^1 1^{111}31^1 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 482, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 3 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 9 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 5 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 9 & 9 & 5 & 0 & 12 & 5 & 6 \\ 1 & 3 & 3 & 3 & 0 & 5 & 0 & 2 \\ 2 & 4 & 4 & 2 & -4 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

6.38 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{11}3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 64, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 3 \\ 3 & 3 & 4 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12}3^1 1^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 4 & 5 & 9 & 0 & 16 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 9 & 16 & 0 & 27 & 9 \\ 3 & 3 & 5 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{e}_6 + \textcolor{red}{su}_3 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 3 & 5 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$

- [3] $1^{111}31^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 124$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 15 & 17 & 0 & 48 & 12 \\ 3 & 4 & 4 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$
- [4] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 127$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.40 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 4 & 9 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$
- [5] $1^{11}31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 151$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.97 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 8 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 3 & 8 & 0 & 3 & 3 \\ 3 & 3 & 5 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$
- [6] $1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 166$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 4 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{112}31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 189$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 10 & 12 & 18 & 0 & 57 & 13 \\ 3 & 3 & 4 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$
- [7] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 189$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.76 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 6 & 9 & 0 & 12 & 6 \\ 3 & 3 & 4 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 190$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.94 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 3 & 4 & -4 & 2 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{112}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 193$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 13.14 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 4 & 6 & 8 & 0 & 9 & 5 \\ 3 & 3 & 4 & -4 & 5 & -1 \end{pmatrix}_{b_0}$$
- [11] $1^{113}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 198$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 13.36 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 6 & 9 & 12 & 0 & 24 & 8 \\ 3 & 3 & 4 & -4 & 8 & -1 \end{pmatrix}_{b_0}$$
- [12] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 202$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.18 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 6 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 3 & 6 & 5 & 0 & 3 & 3 \\ 3 & 4 & 4 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

- [13] $1^{11}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 204$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.10 \rightarrow 14$, $T_{\max} = 98$, $T \in [14, 98]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 4 & 5 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 6 & 12 & 0 & 12 & 6 \\ 3 & 3 & 5 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$
- [14] $1^{111}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 274$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.72 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 12 & 13 & 0 & 27 & 9 \\ 3 & 4 & 4 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$
- [15] $1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 300$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 12.81 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$
- [16] $1^{112}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 316$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 12.73 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 7 & 9 & 13 & 0 & 27 & 9 \\ 3 & 3 & 4 & -4 & 9 & 0 \end{pmatrix}_{b_0}$$
- [17] $1^{113}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 442$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.65 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 12 & 15 & 22 & 0 & 84 & 16 \\ 3 & 3 & 4 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$
- [18] $1^{113}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3$ $T = 8$, $T_{\max} = 98$, $\Delta = -142$, $\det = 445$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.74 \rightarrow 13$, $T_{\max} = 98$, $T \in [13, 98]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 17 & 0 & 48 & 12 \\ 3 & 3 & 4 & -4 & 12 & 0 \end{pmatrix}_{b_0}$$
- $T = 9$ [19] $1^{11}321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -12$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13$, $T_{\max} = 25$, $T \in [13, 25]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 2 & 4 & 0 & 4 & 4 \\ 3 & 4 & 4 & 1 & 3 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 3 & 0 & -2 & 0 \\ 3 & 4 & 4 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$
- [20] $1^{11}321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -39$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.46 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 2 & 1 & 4 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 3 & 4 & 0 & -2 & 0 \\ 3 & 4 & 4 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$
- [21] $1^{11}321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -43$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.51 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 0 \\ 1 & 2 & 1 & 1 & 3 & 0 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 1 & 1 & 3 & 0 & -3 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & -1 & -1 & -1 \end{pmatrix}_{b_0}$$
- [22] $1^{11}321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3$ $T = 9$, $T_{\max} = 25$, $\Delta = -114$, $\det = -62$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 13.35 \rightarrow 14$, $T_{\max} = 25$, $T \in [14, 25]$

6.39 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + [\textcolor{blue}{su}_2 \cdot \textcolor{green}{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 266, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 4 & 3 \\ 3 & 3 & 5 & 0 & 3 & 4 & 3 & 3 \\ 3 & 3 & 4 & -4 & 2 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + [\textcolor{blue}{su}_2 \cdot \textcolor{green}{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 326, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 3 & 6 & 5 & 0 & 3 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 2 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

6.40 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 9 \quad [0] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 11 & 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 17 & 0 & 12 & 6 \\ 5 & 6 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.03 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 20 & 0 & 12 & 6 \\ 5 & 6 & 9 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -559, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 15 & 17 & 0 & 12 & 6 \\ 5 & 7 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [3] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{red}{1}} 2^{\textcolor{red}{1}\textcolor{blue}{1}} 1^{\textcolor{red}{1}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{so}_8 \quad T = 10, T_{\max} = 25, \Delta = -113, \det = 87, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 25, T \in [14, 25]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 3 & 0 & 1 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 3 & 0 & -3 & 0 & 1 & -1 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & 3 & -1 & -1 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

6.41 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{green}{1}} 3^{\textcolor{red}{1}} 1^{\textcolor{red}{1}\textcolor{blue}{1}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{red}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -5, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 16.20 \rightarrow 17, T_{\max} = 23, T \in [17, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 & 3 \\ 3 & 5 & 4 & 0 & 13 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 13 & 0 & 27 & 12 & 9 \\ 3 & 3 & 5 & 0 & 12 & 4 & 4 \\ 3 & 3 & 4 & -4 & 9 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -44, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.55 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 3 & 3 & 8 & 0 & 3 & 4 & 4 \\ 3 & 3 & 5 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -65, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.55 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 3 & 5 & 4 & 0 & 5 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 9 & 3 \\ 5 & 6 & 9 & 0 & 9 & 13 & 7 \\ 3 & 3 & 4 & -4 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -68, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.24 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 3 & 4 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -78, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.87 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 3 \\ 4 & 5 & 9 & 0 & 12 & 4 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 12 & 0 & 12 & 6 & 6 \\ 2 & 3 & 4 & 0 & 6 & 1 & 3 \\ 3 & 3 & 5 & -4 & 6 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1112} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.94 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 10 & 3 \\ 2 & 1 & 5 & 0 & 3 & 12 & 3 \\ 3 & 5 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 10 & 12 & 18 & 0 & 12 & 58 & 14 \\ 3 & 3 & 4 & -4 & 2 & 14 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1112} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -138, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.81 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 12 & 12 & 6 \\ 4 & 6 & 8 & 0 & 12 & 10 & 6 \\ 3 & 3 & 4 & -4 & 6 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -143, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.46 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 11 & 3 \\ 2 & 1 & 5 & 0 & 3 & 12 & 3 \\ 4 & 5 & 9 & 0 & 4 & 28 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 11 & 12 & 28 & 0 & 12 & 73 & 15 \\ 3 & 3 & 5 & -4 & 2 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -146, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -155, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} =$$

13.0

$$T_{\min} = 13.92 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 5 & 0 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -191, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 3 & 6 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111}31^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 4 & 5 & 9 & 0 & 4 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 5 & 6 & 12 & 0 & 6 & 13 & 7 \\ 3 & 3 & 5 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{112}31^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -257, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.02 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 7 & 3 \\ 2 & 1 & 5 & 0 & 3 & 9 & 3 \\ 3 & 5 & 4 & 0 & 5 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 12 & 3 \\ 7 & 9 & 13 & 0 & 12 & 28 & 10 \\ 3 & 3 & 4 & -4 & 3 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.43 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 & 3 \\ 3 & 4 & 9 & 3 & 0 & 6 & 4 \\ 4 & 9 & 9 & 4 & 0 & 8 & 5 \\ 2 & 3 & 4 & -3 & 0 & 3 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 3 & 6 & 8 & 3 & 0 & 4 & 4 \\ 3 & 4 & 5 & -1 & -1 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{114}31^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -281, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 11 & 3 \\ 2 & 1 & 5 & 0 & 3 & 15 & 3 \\ 3 & 5 & 4 & 0 & 5 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 18 & 3 \\ 11 & 15 & 21 & 0 & 18 & 76 & 16 \\ 3 & 3 & 4 & -4 & 3 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111}31^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 8 & 3 \\ 2 & 1 & 5 & 0 & 3 & 9 & 3 \\ 4 & 5 & 9 & 0 & 4 & 20 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 8 & 9 & 20 & 0 & 9 & 37 & 11 \\ 3 & 3 & 5 & -4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111}31^{121} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 12 & 13 & 0 & 27 & 9 & 9 \\ 2 & 3 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & 4 & -4 & 9 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{e}_6 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -317, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.04 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 3 & 5 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 5 & 6 & 9 & 0 & 6 & 13 & 7 \\ 3 & 3 & 4 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{111}31^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -317, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 3 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 11 & 21 & 19 & 0 & 12 & 73 & 15 \\ 3 & 4 & 4 & -4 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{111}31^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -318, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 12 & 6 & 6 \\ 2 & 3 & 4 & 0 & 6 & 1 & 3 \\ 3 & 3 & 4 & -4 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111}31^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -329, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.01 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 9 & 3 \\ 2 & 1 & 5 & 0 & 3 & 12 & 3 \\ 3 & 5 & 4 & 0 & 5 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 15 & 3 \\ 9 & 12 & 17 & 0 & 15 & 49 & 13 \\ 3 & 3 & 4 & -4 & 3 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111}31^{112} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 10 & 3 \\ 2 & 1 & 5 & 0 & 3 & 12 & 3 \\ 4 & 5 & 9 & 0 & 4 & 24 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 10 & 12 & 24 & 0 & 12 & 58 & 14 \\ 3 & 3 & 5 & -4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111}31^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -351, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 0 & 12 & 6 & 6 \\ 2 & 3 & 4 & 0 & 6 & 1 & 3 \\ 3 & 4 & 4 & -4 & 6 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111}31^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -378, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 12 & 15 & 6 \\ 5 & 6 & 9 & 0 & 15 & 13 & 7 \\ 3 & 3 & 4 & -4 & 6 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{111}31^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -383, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 3 & 4 & -4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{111}31^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{e}_6 + \textcolor{red}{su}_3 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.04 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 4 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 3 & 5 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 4 & 6 & 8 & 0 & 6 & 10 & 6 \\ 3 & 3 & 4 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{111}31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -407, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 7 & 12 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1113}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -407, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.16 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 6 & 3 \\ 2 & 1 & 5 & 0 & 3 & 9 & 3 \\ 3 & 5 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 6 & 9 & 12 & 0 & 9 & 25 & 9 \\ 3 & 3 & 4 & -4 & 2 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1114}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -428, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 8 & 3 \\ 2 & 1 & 5 & 0 & 3 & 12 & 3 \\ 3 & 5 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 8 & 12 & 16 & 0 & 12 & 46 & 12 \\ 3 & 3 & 4 & -4 & 2 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{111}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -485, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{112}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -542, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 6 & 0 & 3 & 15 & 4 \\ 3 & 6 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 8 & 15 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & 4 & -4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{113}31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -563, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.98 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 15 & 3 \\ 2 & 1 & 5 & 0 & 3 & 18 & 3 \\ 4 & 5 & 9 & 0 & 4 & 36 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 15 & 18 & 36 & 0 & 18 & 133 & 21 \\ 3 & 3 & 5 & -4 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{112}31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -620, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 13 & 3 \\ 2 & 1 & 5 & 0 & 3 & 15 & 3 \\ 4 & 5 & 9 & 0 & 4 & 32 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 13 & 15 & 32 & 0 & 15 & 100 & 18 \\ 3 & 3 & 5 & -4 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{112}31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -632, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 6 & 0 & 3 & 24 & 4 \\ 3 & 6 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 14 & 24 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & 4 & -4 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{112} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -674, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.01 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 16 & 3 \\ 2 & 1 & 5 & 0 & 3 & 18 & 3 \\ 4 & 5 & 9 & 0 & 4 & 40 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 16 & 18 & 40 & 0 & 18 & 154 & 22 \\ 3 & 3 & 5 & -4 & 2 & 22 & -1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{112} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -696, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 7 & 3 \\ 2 & 1 & 5 & 0 & 6 & 9 & 3 \\ 3 & 5 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 12 & 21 & 6 \\ 7 & 9 & 13 & 0 & 21 & 28 & 10 \\ 3 & 3 & 4 & -4 & 6 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{112} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -698, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 7 & 3 \\ 2 & 1 & 5 & 0 & 3 & 9 & 3 \\ 3 & 5 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 7 & 9 & 13 & 0 & 9 & 28 & 10 \\ 3 & 3 & 4 & -4 & 2 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{114} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -878, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.99 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 20 & 3 \\ 2 & 1 & 5 & 0 & 3 & 24 & 3 \\ 4 & 5 & 9 & 0 & 4 & 48 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 24 & 2 \\ 20 & 24 & 48 & 0 & 24 & 238 & 28 \\ 3 & 3 & 5 & -4 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{113} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -887, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.01 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 12 & 3 \\ 2 & 1 & 5 & 0 & 3 & 15 & 3 \\ 3 & 5 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 12 & 15 & 22 & 0 & 15 & 85 & 17 \\ 3 & 3 & 4 & -4 & 2 & 17 & -1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{113} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -894, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 9 & 3 \\ 2 & 1 & 5 & 0 & 6 & 12 & 3 \\ 3 & 5 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 12 & 27 & 6 \\ 9 & 12 & 17 & 0 & 27 & 49 & 13 \\ 3 & 3 & 4 & -4 & 6 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{112} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 6 & 0 & 3 & 18 & 4 \\ 3 & 6 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 10 & 18 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & -4 & 2 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{114} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -950, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 17 & 3 \\ 2 & 1 & 5 & 0 & 3 & 21 & 3 \\ 3 & 5 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 17 & 21 & 31 & 0 & 21 & 172 & 24 \\ 3 & 3 & 4 & -4 & 2 & 24 & -1 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{1114} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -972, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 11 & 3 \\ 2 & 1 & 5 & 0 & 6 & 15 & 3 \\ 3 & 5 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 12 & 33 & 6 \\ 11 & 15 & 21 & 0 & 33 & 76 & 16 \\ 3 & 3 & 4 & -4 & 6 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{112} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -974, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 6 & 0 & 3 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 12 & 21 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & -4 & 2 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{113} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1043, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 18 & 3 \\ 2 & 1 & 5 & 0 & 3 & 21 & 3 \\ 4 & 5 & 9 & 0 & 4 & 44 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 18 & 21 & 44 & 0 & 21 & 193 & 25 \\ 3 & 3 & 5 & -4 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{113} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1085, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 9 & 3 \\ 2 & 1 & 5 & 0 & 3 & 12 & 3 \\ 3 & 5 & 4 & 0 & 4 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 9 & 12 & 17 & 0 & 12 & 49 & 13 \\ 3 & 3 & 4 & -4 & 2 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{114} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1172, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 6 & 0 & 3 & 30 & 4 \\ 3 & 6 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 16 & 30 & 28 & 0 & 18 & 154 & 22 \\ 3 & 4 & 4 & -4 & 2 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{113} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1199, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 6 & 0 & 3 & 24 & 4 \\ 3 & 6 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 13 & 24 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & 4 & -4 & 2 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{114} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1478, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 11 & 3 \\ 2 & 1 & 5 & 0 & 3 & 15 & 3 \\ 3 & 5 & 4 & 0 & 4 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 11 & 15 & 21 & 0 & 15 & 76 & 16 \\ 3 & 3 & 4 & -4 & 2 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{114} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1652, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 14 & 3 \\ 2 & 1 & 5 & 0 & 3 & 18 & 3 \\ 3 & 5 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 14 & 18 & 26 & 0 & 18 & 118 & 20 \\ 3 & 3 & 4 & -4 & 2 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{113} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 6 & 0 & 3 & 27 & 4 \\ 3 & 6 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 15 & 27 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & -4 & 2 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{113} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1847, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 6 & 0 & 3 & 30 & 4 \\ 3 & 6 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 17 & 30 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & -4 & 2 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{114} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2162, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 6 & 0 & 3 & 33 & 4 \\ 3 & 6 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 18 & 33 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & 4 & -4 & 2 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{114} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2780, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 6 & 0 & 3 & 36 & 4 \\ 3 & 6 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 24 & 2 \\ 20 & 36 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & -4 & 2 & 28 & 0 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [54] \quad 1^{11} 31^{112} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 192, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.25 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 12 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 12 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 6 & 2 \\ 4 & 12 & 12 & 6 & 0 & 6 & 22 & 8 \\ 2 & 4 & 4 & 2 & -4 & 2 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{111} 31^{111} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 201, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.04 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 9 & 6 & 0 & 12 & 9 & 6 \\ 2 & 3 & 6 & 3 & 0 & 9 & 4 & 4 \\ 2 & 3 & 4 & 2 & -4 & 6 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{11} 31^{111} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 354, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 6 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 6 & 3 & 0 & 3 & 4 & 4 \\ 2 & 4 & 4 & 2 & -4 & 2 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[57] \quad 1^{111} 31^{111} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -91, \quad \det = 456, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.60 \rightarrow 14, \quad T_{\max} = 14, \quad T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 6 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 3 & 9 & 3 \\ 3 & 6 & 9 & 6 & 0 & 9 & 13 & 7 \\ 2 & 3 & 4 & 2 & -4 & 3 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[58] \quad 1^{111}31^{112}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 606, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 5 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 15 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 9 & 2 \\ 5 & 15 & 15 & 9 & 0 & 9 & 37 & 11 \\ 2 & 4 & 4 & 2 & -4 & 2 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[59] \quad 1^{111}31^{113}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 1200, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 7 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 21 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 21 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 12 & 2 \\ 7 & 21 & 21 & 12 & 0 & 12 & 73 & 15 \\ 2 & 4 & 4 & 2 & -4 & 2 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[60] \quad 1^{111}31^{114}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 1686, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 9 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 27 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 27 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 15 & 2 \\ 9 & 27 & 27 & 15 & 0 & 15 & 121 & 19 \\ 2 & 4 & 4 & 2 & -4 & 2 & 19 & -1 \end{pmatrix}_{b_0}$$

6.42 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{1111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 6 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 3 & 6 & 5 & 0 & 3 & 4 & 6 & 4 \\ 3 & 6 & 5 & 0 & 3 & 6 & 4 & 4 \\ 3 & 4 & 4 & -4 & 2 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{112}31^{1114} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 14 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 24 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 26 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 18 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 18 & 3 \\ 14 & 24 & 26 & 0 & 18 & 18 & 118 & 20 \\ 3 & 4 & 4 & -4 & 2 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}31^{1112} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 253, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 9 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 9 & 3 \\ 7 & 12 & 13 & 0 & 9 & 9 & 28 & 10 \\ 3 & 4 & 4 & -4 & 2 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{113}31^{1112} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 325, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 13 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 24 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 23 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 15 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 15 & 3 \\ 13 & 24 & 23 & 0 & 15 & 15 & 100 & 18 \\ 3 & 4 & 4 & -4 & 2 & 3 & 18 & 0 \end{pmatrix}_{b_0}$$

- [4] $1^{\overline{111}} 3 1^{\overline{1111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 409, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [5] $1^{\overline{1111}} 3 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 447, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 12 & 6 & 15 & 6 \\ 2 & 3 & 4 & 0 & 6 & 1 & 6 & 3 \\ 5 & 6 & 9 & 0 & 15 & 6 & 13 & 7 \\ 3 & 3 & 4 & -4 & 6 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$
- [6] $1^{\overline{1111}} 3 1^{\overline{1111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 496, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 5 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 6 & 3 \\ 4 & 5 & 9 & 0 & 4 & 12 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 6 & 2 \\ 5 & 6 & 12 & 0 & 6 & 13 & 15 & 7 \\ 5 & 6 & 12 & 0 & 6 & 15 & 13 & 7 \\ 3 & 3 & 5 & -4 & 2 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$
- [7] $1^{\overline{1111}} 3 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 499, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 3 & 6 & 5 & 0 & 3 & 4 & 9 & 4 \\ 5 & 9 & 9 & 0 & 6 & 9 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$
- [8] $1^{\overline{1111}} 3 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 516, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 12 & 6 & 6 & 6 \\ 2 & 3 & 4 & 0 & 6 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 6 & 3 & 1 & 3 \\ 3 & 3 & 4 & -4 & 6 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$
- [9] $1^{\overline{112}} 3 1^{\overline{1112}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 760, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 10 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 18 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 18 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 12 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 12 & 3 \\ 10 & 18 & 18 & 0 & 12 & 12 & 58 & 14 \\ 3 & 4 & 4 & -4 & 2 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$
- [10] $1^{\overline{1111}} 3 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 772, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$
 $T_{\min} = 12.99 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 3 & 4 & -4 & 2 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{\overline{112}} 3 1^{\overline{1113}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 10, T_{\max} = 13, \Delta = -82, \det = 820, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$
 $T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 12 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 22 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 15 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 15 & 3 \\ 12 & 21 & 22 & 0 & 15 & 15 & 85 & 17 \\ 3 & 4 & 4 & -4 & 2 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{114} 3^1 1^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 928, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 18 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 33 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 32 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 21 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 21 & 3 \\ 18 & 33 & 32 & 0 & 21 & 21 & 193 & 25 \\ 3 & 4 & 4 & -4 & 2 & 3 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1111} 3^1 1^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 931, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 6 & 2 \\ 7 & 12 & 13 & 0 & 9 & 28 & 21 & 10 \\ 5 & 9 & 9 & 0 & 6 & 21 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 10 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111} 3^1 1^{122} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 964, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 9 & 2 \\ 7 & 12 & 13 & 0 & 9 & 28 & 30 & 10 \\ 7 & 12 & 13 & 0 & 9 & 30 & 28 & 10 \\ 3 & 4 & 4 & -4 & 2 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1276, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 15 & 7 \\ 5 & 9 & 9 & 0 & 6 & 15 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{113} 3^1 1^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1345, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 15 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 27 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 27 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 18 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 18 & 3 \\ 15 & 27 & 27 & 0 & 18 & 18 & 133 & 21 \\ 3 & 4 & 4 & -4 & 2 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{113} 3^1 1^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 1621, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 17 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 30 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 31 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 21 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 21 & 3 \\ 17 & 30 & 31 & 0 & 21 & 21 & 172 & 24 \\ 3 & 4 & 4 & -4 & 2 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{114} 3^1 1^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 2164, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 20 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 36 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 36 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 24 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 24 & 3 \\ 20 & 36 & 36 & 0 & 24 & 24 & 238 & 28 \\ 3 & 4 & 4 & -4 & 2 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

6.43 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{1121} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.24 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 3 & 4 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1121} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 7 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 9 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 13 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 12 & 3 & 3 \\ 7 & 9 & 13 & 0 & 12 & 28 & 9 & 10 \\ 2 & 3 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 3 & 4 & -4 & 3 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{1121} 3^1 1^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 498, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 15 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 14 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 3 & 2 \\ 8 & 15 & 14 & 0 & 9 & 37 & 9 & 11 \\ 2 & 3 & 4 & 0 & 3 & 9 & 0 & 2 \\ 3 & 4 & 4 & -4 & 2 & 11 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{1141} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 534, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.19 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 11 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 15 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 21 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 18 & 3 & 3 \\ 11 & 15 & 21 & 0 & 18 & 76 & 15 & 16 \\ 2 & 3 & 4 & 0 & 3 & 15 & 0 & 2 \\ 3 & 3 & 4 & -4 & 3 & 16 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{1111} 3^1 1^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 555, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 6 & 3 \\ 5 & 9 & 9 & 0 & 6 & 6 & 12 & 6 \\ 3 & 4 & 4 & -4 & 2 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{1131} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 678, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 9 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 12 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 17 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 15 & 3 & 3 \\ 9 & 12 & 17 & 0 & 15 & 49 & 12 & 13 \\ 2 & 3 & 4 & 0 & 3 & 12 & 0 & 2 \\ 3 & 3 & 4 & -4 & 3 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1111} 3^1 1^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 894, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.39 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 3 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 3 & 6 & 0 & 2 \\ 3 & 4 & 4 & -4 & 2 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

6.44 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 8, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 2 & 4 & 0 & 4 & 2 & 4 \\ 3 & 4 & 4 & 1 & 3 & 3 & 4 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 4 & 3 & 0 & -2 & 2 & 0 \\ 2 & 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.59 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 6 & 3 & 0 & 3 & 3 & 3 \\ 1 & 2 & 3 & 2 & 0 & 3 & 0 & 2 \\ 2 & 3 & 4 & 2 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 284, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.24 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 & 2 \\ 1 & 3 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 4 & 4 & 2 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

6.45 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_{2n3\text{mix}}$

$$T = 10 \quad [0] \quad 1^{11} 3^{11} 1^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 482, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 3 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 9 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 5 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 5 & 2 \\ 3 & 9 & 9 & 5 & 0 & 5 & 12 & 6 \\ 2 & 4 & 4 & 2 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

6.46 $\text{NHC}(-2)(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -123, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.54 \rightarrow 15, T_{\max} = 68, T \in [15, 68]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 3 & 3 & 8 & 0 & 3 & 3 & 3 \\ 3 & 3 & 5 & -4 & 2 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -282, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 4 & 5 & 9 & 0 & 4 & 12 & 5 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 5 & 6 & 12 & 0 & 6 & 12 & 6 \\ 3 & 3 & 5 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^{11} 1^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -312, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

14.0

$$T_{\min} = 13.38 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -357, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.21 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 3 & 6 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & 4 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -507, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 9 & 0 & 12 & 6 & 6 \\ 2 & 3 & 4 & 0 & 6 & 0 & 2 \\ 3 & 3 & 4 & -4 & 6 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -573, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 68, \Delta = -121, \det = -651, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 68, T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & -4 & 2 & 6 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [8] \ 1^{11} 3^1 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 15, \Delta = -93, \det = 44, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 15, T \in [14, 15]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 3 & 0 & 4 \\ 2 & 4 & 2 & 1 & 4 & 0 & 4 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 4 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 & 1 \\ 3 & 4 & 4 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^1 1^{11} 3^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 22, \Delta = -100, \det = 558, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.37 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 6 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 1 & 3 & 3 & 3 & 0 & 0 & 3 & 2 \\ 2 & 6 & 6 & 3 & 0 & 3 & 3 & 3 \\ 2 & 4 & 4 & 2 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

6.47 NHC(-2)(-2)(-3) + $\mathfrak{e}_6$ + $\mathfrak{su}_3$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.24 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 3 & 1 & 3 \\ 3 & 3 & 4 & -4 & 3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1112}} 3^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 462, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 7 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 9 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 & 13 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 3 & 12 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 9 & 2 \\ 7 & 9 & 13 & 0 & 12 & 9 & 28 & 10 \\ 3 & 3 & 4 & -4 & 3 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{112}} 3^{\overline{1111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 498, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.96 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 8 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 15 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 14 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 9 & 2 \\ 2 & 3 & 4 & 0 & 3 & 0 & 9 & 2 \\ 8 & 15 & 14 & 0 & 9 & 9 & 37 & 11 \\ 3 & 4 & 4 & -4 & 2 & 2 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1114}} 3^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 534, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.19 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 11 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 15 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 & 21 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 3 & 18 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 15 & 2 \\ 11 & 15 & 21 & 0 & 18 & 15 & 76 & 16 \\ 3 & 3 & 4 & -4 & 3 & 2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{1111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 555, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.05 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 3 & 2 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 & 6 \\ 2 & 3 & 4 & 0 & 3 & 6 & 1 & 3 \\ 3 & 4 & 4 & -4 & 2 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1113}} 3^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 678, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.14 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 9 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 12 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 & 17 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 3 & 15 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 12 & 2 \\ 9 & 12 & 17 & 0 & 15 & 12 & 49 & 13 \\ 3 & 3 & 4 & -4 & 3 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{1111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 894, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 9 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 2 & 3 & 4 & 0 & 3 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 6 & 13 & 7 \\ 3 & 4 & 4 & -4 & 2 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

6.48 NHC(-2)(-2)(-3) + $\mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 216, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.44 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 3 & 4 & -4 & 3 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_3 + \mathbf{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 324, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 0 & 3 & 2 \\ 3 & 6 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 4 & 4 & -4 & 2 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

6.49 NHC(-2)(-2)(-3) + e₆ + (-3)→(-2)

$$T = 8 \quad [0] \ 1^{\overline{131}} 3^{\overline{111}} 2^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_6 + (-3) \rightarrow (-2) \quad T = 8, \ T_{\max} = 98, \ \Delta = -143, \ \det = 1219, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.79 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 4 & 6 & 8 & 3 & 1 & 4 & 4 \\ 6 & 4 & 9 & 3 & 0 & 4 & 4 \\ 8 & 9 & 9 & 4 & 0 & 5 & 5 \\ 3 & 3 & 4 & -3 & 0 & -1 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 & -1 \\ 4 & 4 & 5 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

6.50 NHC(-2)(-2)(-3) + e₇

$$T = 9 \quad [0] \ 1^{\overline{13}} 3^{\overline{221}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}_7 \quad T = 9, \ T_{\max} = 101, \ \Delta = -161, \ \det = -23, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 16.91 \rightarrow 17, \ T_{\max} = 101, \ T \in [17, 101]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 & 2 \\ 1 & 2 & 2 & 1 & 4 & 4 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 & -1 \\ 1 & 1 & 4 & 0 & -4 & -2 & -2 \\ 2 & 2 & 4 & -1 & -2 & -3 & -3 \end{pmatrix}_{b_0}$$

6.51 NHC(-2)(-2)(-3) + e'₇

$$T = 9 \quad [0] \ 1^{\overline{13}} 3^{\overline{221}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{e}'_7 \quad T = 9, \ T_{\max} = 93, \ \Delta = -133, \ \det = -18, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 16, \ T_{\max} = 93, \ T \in [16, 93]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 & 2 \\ 1 & 2 & 2 & 1 & 4 & 4 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 & -1 \\ 1 & 1 & 4 & 0 & -3 & -1 & -1 \\ 2 & 2 & 4 & -1 & -1 & -2 & -2 \end{pmatrix}_{b_0}$$

6.52 NHC(-2)(-2)(-3) + f₄

$$T = 7 \quad [1] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 7, \ T_{\max} = 97, \ \Delta = -137, \ \det = -2, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 14, \ T_{\max} = 97, \ T \in [14, 97]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 & 4 \\ 4 & 9 & 9 & 4 & 5 & 5 \\ 3 & 3 & 4 & -2 & 0 & 0 \\ 3 & 4 & 5 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[0] \ 21^{\overline{131}} 3^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 7, \ T_{\max} = 97, \ \Delta = -137, \ \det = -2, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 14, \ T_{\max} = 97, \ T \in [14, 97]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 & 3 \\ 3 & 4 & 9 & 3 & 4 & 4 \\ 4 & 9 & 9 & 4 & 5 & 5 \\ 3 & 3 & 4 & -2 & 0 & 0 \\ 3 & 4 & 5 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{13}} 3^{\overline{11}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 7, \ T_{\max} = 97, \ \Delta = -137, \ \det = -44, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.09 \rightarrow 13, \ T_{\max} = 97, \ T \in [13, 97]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & -3 & -3 \\ 3 & 3 & 5 & -3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 7, \ T_{\max} = 97, \ \Delta = -137, \ \det = -49, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 3 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 3 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1 3 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.60 \rightarrow 13, T_{\max} = 97, T \in [13, 97]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 9 & 3 & 4 \\ 4 & 9 & 9 & 4 & 5 \\ 2 & 3 & 4 & -2 & 0 \\ 3 & 4 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 21^{111} 3 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 7, T_{\max} = 97, \Delta = -137, \det = -371, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 97, T \in [12, 97]$$

$$\begin{pmatrix} 4 & 6 & 8 & 3 & 4 \\ 6 & 4 & 9 & 3 & 4 \\ 8 & 9 & 9 & 4 & 5 \\ 3 & 3 & 4 & -2 & 0 \\ 4 & 4 & 5 & 0 & 1 \end{pmatrix}_{b_0}$$

$$T = 8 \quad [7] \ 1^1 3 2 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 3 & 1 & 0 & -2 & 0 \\ 2 & 3 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^1 3 2 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 14, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 2 & 6 & 0 & 4 \\ 3 & 6 & 5 & 0 & 5 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & -3 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3 2 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.14 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 3 & 1 & 0 & 2 \\ 1 & 3 & 0 & 2 & 0 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 2 & 2 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^1 3 2 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 29, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.41 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 1 & 3 & 0 & -2 & 0 \\ 2 & 2 & 3 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 3 2 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 30, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 12 & 18 & 6 \\ 12 & 13 & 21 & 7 \\ 18 & 21 & 30 & 10 \\ 6 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 1^1 3 2 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 5 & 9 & 13 & 5 \\ 9 & 13 & 21 & 7 \\ 13 & 21 & 30 & 10 \\ 5 & 7 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[13] \ 1^1 3 2 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 35, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 2 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11}321^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.15 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 5 & 0 & 5 \\ 1 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 2^12^131^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.77 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 10 & 12 & 19 & 6 \\ 12 & 13 & 21 & 7 \\ 19 & 21 & 30 & 10 \\ 6 & 7 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11}31^{13}1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 \\ 2 & 4 & 4 & 4 & 0 \\ 1 & 3 & 4 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{13}1^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 4 & 7 & 0 & 4 \\ 2 & 3 & 7 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 3 & 4 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{13}1^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 124, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.32 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 1 & 4 & 0 & 3 \\ 2 & 6 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 3 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{13}1^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 167, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 4 & 4 & 0 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 2 & 4 & 4 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[20] \quad 2^12^131^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 345, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 3 & 1 & 3 \\ 2 & 4 & 1 & 4 & 1 & 3 \\ 2 & 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & -3 & -1 \\ 2 & 3 & 3 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 31^{111}231^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 787, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 31, T \in [12, 31]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 3 & 1 & 3 \\ 3 & 4 & 1 & 4 & 1 & 3 \\ 3 & 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & -3 & -1 \\ 2 & 3 & 3 & 2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [22] \quad 1^{11}2^1321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.2$$

$$T_{\min} = 11.80 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 4 & 0 & 3 & 3 \\ 2 & 3 & 3 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$

6.53 NHC(-2)(-2)(-3) + f₄ + e₆

$$T = 8 \quad [0] \quad 1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{e}_6 \quad T = 8, T_{\max} = 102, \Delta = -186, \det = 349, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.25 \rightarrow 15, T_{\max} = 102, T \in [15, 102]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 4 & 9 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 5 & 0 & -4 & -2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [1] \quad 1^{11} 3^{21} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{c}_6 \quad T = 9, \quad T_{\max} = 101, \quad \Delta = -158, \quad \det = -148, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.2$$

$$T_{\min} = 14.41 \rightarrow 15, \quad T_{\max} = 101, \quad T \in [15, 101]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 5 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 5 & 0 & -2 & 0 \\ 3 & 4 & 5 & -3 & 0 & -1 \end{pmatrix}_{b_0}$$

6.54 NHC(-2)(-2)(-3) + f₄ + f₄

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 \quad T = 8, \quad T_{\max} = 100, \quad \Delta = -160, \quad \det = 264, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.42 \rightarrow 14, \quad T_{\max} = 100, \quad T \in [14, 100]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 4 & 9 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 0 & -3 & -1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [1] \quad 1^{11} 3^{21} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -109, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.4$$

$$T_{\min} = 13.48 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 13 & 1 & 5 \\ 9 & 13 & 21 & 0 & 7 \\ 13 & 21 & 30 & 0 & 10 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 7 & 10 & -3 & 1 \end{pmatrix}_{b_0}$$

6.55 NHC(-2)(-2)(-3) + f₄ + f₄ + su₂

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 92, \quad \Delta = -130, \quad \det = -584, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.42 \rightarrow 14, \quad T_{\max} = 92, \quad T \in [14, 92]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 & 3 \\ 3 & 4 & 9 & 3 & 0 & 6 & 4 \\ 4 & 9 & 9 & 4 & 0 & 8 & 5 \\ 2 & 3 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 8 & 3 & 0 & 4 & 4 \\ 3 & 4 & 5 & 0 & -3 & 4 & -1 \end{pmatrix}_{b_0}$$

6.56 NHC(-2)(-2)(-3) + f₄ + f₄ + su₂ → (-3)

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.43 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 2 & 3 \\ 3 & 4 & 9 & 3 & 0 & 3 & 4 \\ 4 & 9 & 9 & 4 & 0 & 4 & 5 \\ 2 & 3 & 4 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 2 & 0 & 0 & 2 \\ 3 & 4 & 5 & 0 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

6.57 NHC(-2)(-2)(-3) + f₄ + NHC(-2)(-3)

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -87, \quad \det = -216, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 6 & 5 & 0 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

6.58 NHC(-2)(-2)(-3) + f₄ + so₈

$$T = 8 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 \quad T = 8, \quad T_{\max} = 98, \quad \Delta = -136, \quad \det = 106, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 12 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 6 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 4 & 9 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 0 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 5 \\ 9 & 13 & 18 & 0 & 7 \\ 11 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 7 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 11 & 17 & 20 & 0 & 8 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 6 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \ 1^{\overline{11}} 3 2 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 0 & 3 & 1 & 3 \\ 1 & 2 & 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 1 & 0 & -2 & 1 & 0 \\ 1 & 1 & 2 & 0 & 1 & -2 & 0 \\ 2 & 3 & 2 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1}} 3 2 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 2 & 6 & 0 & 4 & 4 \\ 3 & 6 & 5 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 0 & 0 & 2 \\ 3 & 4 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{11}} 3 2 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.75 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 1 & 3 & 1 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 3 & 0 & -2 & 1 & 0 \\ 1 & 2 & 1 & 0 & 1 & -2 & 0 \\ 2 & 2 & 3 & -1 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}} 3 2 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.46 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 1 & 3 \\ 1 & 2 & 0 & 2 & 0 & 2 & 2 \\ 1 & 3 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 1 & 2 & 1 & 0 & -2 & 0 \\ 2 & 3 & 2 & 2 & -3 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{11}} 3 2 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -70, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 5 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 5 & 0 & 0 & 2 \\ 3 & 4 & 5 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{11}} 3 2 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -77, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 13 & 1 & 5 \\ 9 & 13 & 21 & 0 & 7 \\ 13 & 21 & 30 & 0 & 10 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 10 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{11}} 3 1^{\overline{11}} 3 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$$

13.2

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 13 & 16 & 0 & 7 \\ 5 & 12 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 7 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11} 31^{11} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 1 & 5 & 4 & 5 & 1 & 3 \\ 5 & 13 & 13 & 15 & 0 & 7 \\ 4 & 13 & 10 & 13 & 0 & 6 \\ 5 & 15 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 7 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

6.59 NHC(-2)(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{111} 31^{114} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 44 & 5 \\ 8 & 10 & 17 & 0 & 60 & 6 \\ 12 & 17 & 25 & 0 & 96 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 44 & 60 & 96 & 0 & 346 & 34 \\ 5 & 6 & 9 & -3 & 34 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{114} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.30 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 44 & 5 \\ 9 & 13 & 18 & 0 & 72 & 7 \\ 11 & 18 & 20 & 0 & 84 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 44 & 72 & 84 & 0 & 346 & 34 \\ 5 & 7 & 8 & -3 & 34 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 20 & 0 & 13 & 7 \\ 5 & 6 & 9 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 8 & 5 \\ 9 & 13 & 18 & 0 & 12 & 7 \\ 11 & 18 & 20 & 0 & 16 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 16 & 0 & 10 & 6 \\ 5 & 7 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -249, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 17 & 0 & 13 & 7 \\ 5 & 7 & 8 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 17 & 5 \\ 8 & 10 & 17 & 0 & 24 & 6 \\ 12 & 17 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 24 & 36 & 0 & 49 & 13 \\ 5 & 6 & 9 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 31^{113} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -331, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 35 & 5 \\ 8 & 10 & 17 & 0 & 48 & 6 \\ 12 & 17 & 25 & 0 & 76 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 35 & 48 & 76 & 0 & 217 & 27 \\ 5 & 6 & 9 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{\overline{111}1^1}3^11^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -393$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 & 3 \\ 3 & 4 & 9 & 3 & 0 & 6 & 4 \\ 4 & 9 & 9 & 4 & 0 & 8 & 5 \\ 2 & 3 & 4 & -2 & 0 & 3 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 8 & 3 & 0 & 4 & 4 \\ 3 & 4 & 5 & 0 & -2 & 4 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{\overline{111}1^1}3^11^{\overline{112}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -408$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 26 & 5 \\ 8 & 10 & 17 & 0 & 36 & 6 \\ 12 & 17 & 25 & 0 & 56 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 36 & 56 & 0 & 118 & 20 \\ 5 & 6 & 9 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{\overline{113}1^1}3^11^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -501$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 35 & 5 \\ 9 & 13 & 18 & 0 & 57 & 7 \\ 11 & 18 & 20 & 0 & 67 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 35 & 57 & 67 & 0 & 217 & 27 \\ 5 & 7 & 8 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{\overline{111}1^1}3^11^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -557$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 17 & 5 \\ 9 & 13 & 18 & 0 & 27 & 7 \\ 11 & 18 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 27 & 33 & 0 & 49 & 13 \\ 5 & 7 & 8 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{\overline{112}1^1}3^11^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -662$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.7$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 26 & 5 \\ 9 & 13 & 18 & 0 & 42 & 7 \\ 11 & 18 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 42 & 50 & 0 & 118 & 20 \\ 5 & 7 & 8 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

6.60 NHC(-2)(-2)(-3) + $\mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$

$T = 9$ [0] $1^{\overline{111}1^1}3^11^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -130$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 2 & 3 \\ 3 & 4 & 9 & 3 & 0 & 3 & 4 \\ 4 & 9 & 9 & 4 & 0 & 4 & 5 \\ 2 & 3 & 4 & -2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 2 & 0 & 0 & 2 \\ 3 & 4 & 5 & 0 & -2 & 2 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{113}1^1}3^11^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -172$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 6 & 5 \\ 8 & 10 & 17 & 0 & 8 & 6 \\ 12 & 17 & 25 & 0 & 12 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 8 & 12 & 0 & 4 & 4 \\ 5 & 6 & 9 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{113}1^1}3^11^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 21$, $\Delta = -108$, $\det = -330$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.68 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 6 & 5 \\ 9 & 13 & 18 & 0 & 9 & 7 \\ 11 & 18 & 20 & 0 & 11 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 11 & 0 & 4 & 4 \\ 5 & 7 & 8 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [3] \quad 1^{11} 3^{11} 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 13, \Delta = -81, \det = 97, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 13 & 1 & 6 & 5 \\ 9 & 13 & 21 & 0 & 9 & 7 \\ 13 & 21 & 30 & 0 & 13 & 10 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 13 & 0 & 4 & 4 \\ 5 & 7 & 10 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

6.61 NHC(-2)(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + $\mathfrak{su}_{2n3\text{mix}}$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 15 & 5 \\ 8 & 10 & 17 & 0 & 20 & 6 \\ 12 & 17 & 25 & 0 & 32 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 20 & 32 & 0 & 37 & 11 \\ 5 & 6 & 9 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 15 & 5 \\ 9 & 13 & 18 & 0 & 24 & 7 \\ 11 & 18 & 20 & 0 & 28 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 24 & 28 & 0 & 37 & 11 \\ 5 & 7 & 8 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

6.62 NHC(-2)(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{so}_8$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \quad 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 11 & 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 17 & 0 & 12 & 6 \\ 5 & 6 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -213, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.24 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 20 & 0 & 12 & 6 \\ 5 & 6 & 9 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 17 & 0 & 12 & 6 \\ 5 & 7 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [3] \quad 1^{11} 3^{11} 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 & 0 & 2 \\ 1 & 2 & 1 & 1 & 3 & 1 & 0 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 3 & 0 & -2 & 1 & 0 & 0 \\ 1 & 2 & 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & -1 & 0 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

6.63 NHC(-2)(-2)(-3) + f₄ + su₂

$$T = 8 \quad [0] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 14.11 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 3 \\ 3 & 3 & 5 & -3 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 4 & 5 & 9 & 0 & 16 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 9 & 16 & 0 & 28 & 10 \\ 3 & 3 & 5 & -3 & 10 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{13} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 6 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 4 & 5 & 9 & 0 & 16 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 6 & 16 & 0 & 22 & 8 \\ 3 & 3 & 5 & -3 & 8 & -2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 17 & 0 & 49 & 13 \\ 3 & 4 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112} 3^1 1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 36, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 10 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 10 & 0 & 22 & 8 \\ 3 & 4 & 4 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 3 \\ 3 & 4 & 4 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{13} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 13.05 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 4 & 5 & 9 & 0 & 28 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 15 & 28 & 0 & 85 & 17 \\ 3 & 3 & 5 & -3 & 17 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{13} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 74, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.41 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 8 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 3 & 8 & 0 & 4 & 4 \\ 3 & 3 & 5 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{1111} 3^1 1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 86, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 14.02 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 4 & 3 \\ 3 & 4 & 9 & 3 & 6 & 4 \\ 4 & 9 & 9 & 4 & 8 & 5 \\ 2 & 3 & 4 & -2 & 3 & 0 \\ 4 & 6 & 8 & 3 & 4 & 4 \\ 3 & 4 & 5 & 0 & 4 & -2 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{113} 3^1 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 15 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 3 & 5 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 18 & 27 & 0 & 133 & 21 \\ 3 & 3 & 4 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{117}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.5$
 $T_{\min} = 13.57 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 \\ 3 & 5 & 4 & 0 & 28 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 21 & 28 & 0 & 145 & 21 \\ 3 & 3 & 4 & -3 & 21 & -1 \end{pmatrix}_{b_0}$$

[16] $1^{112}3^11^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 12 & 18 & 0 & 58 & 14 \\ 3 & 3 & 4 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{114}3^11^{14} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 20 & 3 \\ 2 & 1 & 5 & 0 & 24 & 3 \\ 3 & 5 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 24 & 36 & 0 & 238 & 28 \\ 3 & 3 & 4 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{111}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 3 \\ 3 & 3 & 4 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{113}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.94 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 12 & 0 & 25 & 9 \\ 3 & 3 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{114}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.22 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 8 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 16 & 0 & 46 & 12 \\ 3 & 3 & 4 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{112}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.73 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 0 & 10 & 6 \\ 3 & 3 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{115}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 12.59 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 15 & 20 & 0 & 73 & 15 \\ 3 & 3 & 4 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{116}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.4$
 $T_{\min} = 13.04 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 18 & 24 & 0 & 106 & 18 \\ 3 & 3 & 4 & -3 & 18 & -1 \end{pmatrix}_{b_0}$$

[17] $1^{111}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 98, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 6 & 9 & 0 & 13 & 7 \\ 3 & 3 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111} 3^1 1^1 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 99, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.70 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 6 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 5 & 0 & 4 & 4 \\ 3 & 4 & 4 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 116, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.10 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 4 & 5 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 6 & 12 & 0 & 13 & 7 \\ 3 & 3 & 5 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111} 3^1 1^{12} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 155, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 13 & 0 & 28 & 10 \\ 3 & 4 & 4 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 176, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.61 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{14} 3^1 1^{13} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 186, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.59 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 17 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 \\ 4 & 5 & 9 & 0 & 40 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 21 & 40 & 0 & 172 & 24 \\ 3 & 3 & 5 & -3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{11} 3^1 1^{12} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 186, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.16 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 8 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 4 & 5 & 9 & 0 & 20 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 9 & 20 & 0 & 37 & 11 \\ 3 & 3 & 5 & -3 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[26] \ 1^1 1^1 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 191, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.62 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 3 \\ 3 & 4 & 9 & 3 & 6 & 4 \\ 4 & 9 & 9 & 4 & 8 & 5 \\ 2 & 3 & 4 & -2 & 3 & 0 \\ 3 & 6 & 8 & 3 & 4 & 4 \\ 3 & 4 & 5 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{112} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 196, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 9 & 13 & 0 & 28 & 10 \\ 3 & 3 & 4 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{12} 3^1 1^{12} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 200, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 4 & 5 & 9 & 0 & 24 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 12 & 24 & 0 & 58 & 14 \\ 3 & 3 & 5 & -3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{11} 3^1 1^{13} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 228, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.40 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 11 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 4 & 5 & 9 & 0 & 28 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 12 & 28 & 0 & 73 & 15 \\ 3 & 3 & 5 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{114} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 239, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.09 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 \\ 3 & 6 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 27 & 24 & 0 & 121 & 19 \\ 3 & 4 & 4 & -3 & 19 & 0 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{112} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 239, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.69 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 14 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 14 & 0 & 37 & 11 \\ 3 & 4 & 4 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{11} 3^1 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 242, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.82 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 14 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 4 & 5 & 9 & 0 & 36 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 15 & 36 & 0 & 121 & 19 \\ 3 & 3 & 5 & -3 & 19 & 0 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{113} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 260, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 19 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 19 & 0 & 73 & 15 \\ 3 & 4 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{112} 3^1 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 260, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.60 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 24 & 26 & 0 & 118 & 20 \\ 3 & 4 & 4 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{113} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 294, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 17 & 0 & 49 & 13 \\ 3 & 3 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{113} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 294, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.59 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 15 & 22 & 0 & 85 & 17 \\ 3 & 3 & 4 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{13} 3^1 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 340, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.0$$

$$T_{\min} = 12.13 \rightarrow 13, \ T_{\max} = 20, \ T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 15 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 4 & 5 & 9 & 0 & 36 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 18 & 36 & 0 & 133 & 21 \\ 3 & 3 & 5 & -3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{112} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 344, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{112} 3^1 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, \ T_{\max} = 20, \ \Delta = -107, \ \det = 351, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.61 \rightarrow 12, \ T_{\max} = 20, \ T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 21 & 22 & 0 & 85 & 17 \\ 3 & 4 & 4 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[40] $1^{12}3^11^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 354, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.11 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 13 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 4 & 5 & 9 & 0 & 32 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 15 & 32 & 0 & 100 & 18 \\ 3 & 3 & 5 & -3 & 18 & 0 \end{pmatrix}_{b_0}$$

[42] $1^{114}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 15 & 21 & 0 & 76 & 16 \\ 3 & 3 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{114}3^11^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 392, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.58 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 17 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 \\ 3 & 5 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 21 & 31 & 0 & 172 & 24 \\ 3 & 3 & 4 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{12}3^11^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 480, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 12.20 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 16 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 4 & 5 & 9 & 0 & 40 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 18 & 40 & 0 & 154 & 22 \\ 3 & 3 & 5 & -3 & 22 & 0 \end{pmatrix}_{b_0}$$

[44] $1^{114}3^11^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 490, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 3 & 5 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 18 & 26 & 0 & 118 & 20 \\ 3 & 3 & 4 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{113}3^11^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 491, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.68 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 23 & 0 & 100 & 18 \\ 3 & 4 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[46] $1^{14}3^11^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 536, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.13 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 20 & 3 \\ 2 & 1 & 5 & 0 & 24 & 3 \\ 4 & 5 & 9 & 0 & 48 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 24 & 48 & 0 & 238 & 28 \\ 3 & 3 & 5 & -3 & 28 & 0 \end{pmatrix}_{b_0}$$

[47] $1^{13}3^11^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 578, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 12.09 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 18 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 \\ 4 & 5 & 9 & 0 & 44 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 21 & 44 & 0 & 193 & 25 \\ 3 & 3 & 5 & -3 & 25 & 0 \end{pmatrix}_{b_0}$$

[48] $1^{114}3^11^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 596, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.74 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 6 & 0 & 30 & 4 \\ 3 & 6 & 4 & 0 & 28 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 30 & 28 & 0 & 154 & 22 \\ 3 & 4 & 4 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[49] $1^{113}3^11^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 624, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{113}31^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 659, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 6 & 0 & 30 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 30 & 31 & 0 & 172 & 24 \\ 3 & 4 & 4 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[51] $21^{111}31^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 840, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 4 & 6 & 8 & 3 & 6 & 4 \\ 6 & 4 & 9 & 3 & 6 & 4 \\ 8 & 9 & 9 & 4 & 8 & 5 \\ 3 & 3 & 4 & -2 & 3 & 0 \\ 6 & 6 & 8 & 3 & 4 & 4 \\ 4 & 4 & 5 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

[52] $1^{114}31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 855, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 6 & 0 & 33 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 18 & 33 & 32 & 0 & 193 & 25 \\ 3 & 4 & 4 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[53] $1^{114}31^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 8, T_{\max} = 20, \Delta = -107, \det = 1016, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [54] $1^1321^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -12, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 2 & 6 & 0 & 4 & 4 \\ 3 & 6 & 5 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 0 & 2 & 4 \\ 3 & 4 & 5 & -3 & 4 & 2 \end{pmatrix}_{b_0}$$

[55] $1^{12}321^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$
 $T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 3 & 0 & 5 & 3 \\ 1 & 2 & 0 & 2 & 0 & 5 & 2 \\ 1 & 3 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 5 & 0 & 13 & 7 \\ 2 & 3 & 2 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[56] $1^1321^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 & 3 \\ 1 & 2 & 0 & 2 & 0 & 1 & 2 \\ 1 & 3 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 1 & 3 & 0 & 1 & 3 \\ 2 & 3 & 2 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[58] $1^{111}31^131^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 1 & 3 \\ 2 & 3 & 4 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[57] $1^{11}31^{11}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 6 & 3 & 0 & 4 & 4 \\ 2 & 3 & 4 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[59] $1^{11}321^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.60 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 2 & 6 & 0 & 11 & 4 \\ 3 & 6 & 5 & 0 & 15 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 15 & 0 & 32 & 12 \\ 3 & 4 & 5 & -3 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{11}321^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -83, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 3 \\ 2 & 2 & 6 & 0 & 15 & 4 \\ 3 & 6 & 5 & 0 & 20 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 20 & 0 & 58 & 16 \\ 3 & 4 & 5 & -3 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{11}321^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -87, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 \\ 2 & 2 & 6 & 0 & 19 & 4 \\ 3 & 6 & 5 & 0 & 25 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 19 & 25 & 0 & 92 & 20 \\ 3 & 4 & 5 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{11}321^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 0 & 4 & 3 \\ 1 & 2 & 0 & 2 & 0 & 3 & 2 \\ 1 & 3 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 4 & 0 & 5 & 5 \\ 2 & 3 & 2 & 2 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{12}321^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -104, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 3 \\ 2 & 2 & 6 & 0 & 22 & 4 \\ 3 & 6 & 5 & 0 & 30 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 22 & 30 & 0 & 134 & 24 \\ 3 & 4 & 5 & -3 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{13}321^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -111, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 3 & 0 & 9 & 3 \\ 1 & 2 & 0 & 2 & 0 & 8 & 2 \\ 1 & 3 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 8 & 9 & 0 & 40 & 12 \\ 2 & 3 & 2 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{11}321^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -119, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 3 & 0 & 7 & 3 \\ 1 & 2 & 0 & 2 & 0 & 4 & 2 \\ 1 & 3 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 7 & 4 & 7 & 0 & 16 & 8 \\ 2 & 3 & 2 & 2 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{12}321^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 3 & 0 & 8 & 3 \\ 1 & 2 & 0 & 2 & 0 & 6 & 2 \\ 1 & 3 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 8 & 0 & 26 & 10 \\ 2 & 3 & 2 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{112}31^{1131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 3 \\ 2 & 4 & 4 & 4 & 0 & 12 & 4 \\ 1 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 9 & 12 & 9 & 0 & 28 & 10 \\ 2 & 3 & 4 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[67] \ 1^{111}31^{1231} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -208, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 3 \\ 2 & 4 & 4 & 4 & 0 & 15 & 4 \\ 1 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 15 & 9 & 0 & 37 & 11 \\ 2 & 3 & 4 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[69] $1^{111}31^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -236, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 9 & 6 & 0 & 13 & 7 \\ 2 & 3 & 4 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{14}321^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 3 & 0 & 13 & 3 \\ 1 & 2 & 0 & 2 & 0 & 11 & 2 \\ 1 & 3 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 11 & 13 & 0 & 81 & 17 \\ 2 & 3 & 2 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[71] $1^{12}321^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -266, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 3 & 0 & 14 & 3 \\ 1 & 2 & 0 & 2 & 0 & 8 & 2 \\ 1 & 3 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 14 & 8 & 14 & 0 & 70 & 16 \\ 2 & 3 & 2 & 2 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{12}321^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.68 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 3 & 0 & 11 & 3 \\ 1 & 2 & 0 & 2 & 0 & 7 & 2 \\ 1 & 3 & 2 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 11 & 7 & 11 & 0 & 45 & 13 \\ 2 & 3 & 2 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{13}321^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -349, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 3 & 0 & 12 & 3 \\ 1 & 2 & 0 & 2 & 0 & 9 & 2 \\ 1 & 3 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 9 & 12 & 0 & 61 & 15 \\ 2 & 3 & 2 & 2 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{114}31^{12}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 3 & 0 & 18 & 3 \\ 2 & 4 & 4 & 4 & 0 & 24 & 4 \\ 1 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 18 & 24 & 18 & 0 & 118 & 20 \\ 2 & 3 & 4 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{112}31^{14}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -376, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 10 & 2 \\ 1 & 1 & 4 & 3 & 0 & 18 & 3 \\ 2 & 4 & 4 & 4 & 0 & 30 & 4 \\ 1 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 30 & 18 & 0 & 154 & 22 \\ 2 & 3 & 4 & 2 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[76] $1^{112}31^{12}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -488, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 3 & 0 & 12 & 3 \\ 2 & 4 & 4 & 4 & 0 & 18 & 4 \\ 1 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 12 & 18 & 12 & 0 & 58 & 14 \\ 2 & 3 & 4 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{13}321^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -495, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 3 & 0 & 15 & 3 \\ 1 & 2 & 0 & 2 & 0 & 10 & 2 \\ 1 & 3 & 2 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 15 & 10 & 15 & 0 & 88 & 18 \\ 2 & 3 & 2 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{11231^{13}31^1} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 3 & 0 & 15 & 3 \\ 2 & 4 & 4 & 4 & 0 & 24 & 4 \\ 1 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 24 & 15 & 0 & 100 & 18 \\ 2 & 3 & 4 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[79] $1^{11331^{12}31^1} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 3 & 0 & 15 & 3 \\ 2 & 4 & 4 & 4 & 0 & 21 & 4 \\ 1 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 15 & 21 & 15 & 0 & 85 & 17 \\ 2 & 3 & 4 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[80] $1^{14321^{114}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -566, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 3 & 0 & 16 & 3 \\ 1 & 2 & 0 & 2 & 0 & 12 & 2 \\ 1 & 3 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 12 & 16 & 0 & 110 & 20 \\ 2 & 3 & 2 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[81] $2^1 2^1 31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -781, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 & 2 \\ 2 & 1 & 4 & 3 & 1 & 3 & 3 \\ 2 & 4 & 1 & 4 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 3 & 3 & 3 & 1 & 1 & 3 \\ 2 & 3 & 3 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$

[82] $1^{11331^{13}31^1} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -908, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 9 & 2 \\ 1 & 1 & 4 & 3 & 0 & 18 & 3 \\ 2 & 4 & 4 & 4 & 0 & 27 & 4 \\ 1 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 18 & 27 & 18 & 0 & 133 & 21 \\ 2 & 3 & 4 & 2 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[84] $1^{11431^{13}31^1} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1048, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 10 & 2 \\ 1 & 1 & 4 & 3 & 0 & 21 & 3 \\ 2 & 4 & 4 & 4 & 0 & 30 & 4 \\ 1 & 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 21 & 30 & 21 & 0 & 172 & 24 \\ 2 & 3 & 4 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

[83] $1^{11331^{14}31^1} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1048, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 11 & 2 \\ 1 & 1 & 4 & 3 & 0 & 21 & 3 \\ 2 & 4 & 4 & 4 & 0 & 33 & 4 \\ 1 & 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 33 & 21 & 0 & 193 & 25 \\ 2 & 3 & 4 & 2 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[85] $2^1 2^1 31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1054, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.7$

$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 4 & 2 \\ 2 & 1 & 4 & 3 & 1 & 6 & 3 \\ 2 & 4 & 1 & 4 & 1 & 6 & 3 \\ 2 & 3 & 4 & 0 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & -3 & 2 & -1 \\ 4 & 6 & 6 & 6 & 2 & 10 & 6 \\ 2 & 3 & 3 & 2 & -1 & 6 & 1 \end{pmatrix}_{b_0}$$

[86] $1^{11431^{14}31^1} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1496, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 12 & 2 \\ 1 & 1 & 4 & 3 & 0 & 24 & 3 \\ 2 & 4 & 4 & 4 & 0 & 36 & 4 \\ 1 & 3 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 24 & 36 & 24 & 0 & 238 & 28 \\ 2 & 3 & 4 & 2 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

[87] $2 \ 2 \ 31^{\overline{1113}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -1509, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.0$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 6 & 2 \\ 2 & 1 & 4 & 3 & 1 & 9 & 3 \\ 2 & 4 & 1 & 4 & 1 & 9 & 3 \\ 2 & 3 & 4 & 0 & 1 & 9 & 2 \\ 1 & 1 & 1 & 1 & -3 & 3 & -1 \\ 6 & 9 & 9 & 3 & 25 & 9 & 9 \\ 2 & 3 & 3 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[88] $2 \ 2 \ 31^{\overline{1114}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -2146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 8 & 2 \\ 2 & 1 & 4 & 3 & 1 & 12 & 3 \\ 2 & 4 & 1 & 4 & 1 & 12 & 3 \\ 2 & 3 & 4 & 0 & 1 & 12 & 2 \\ 1 & 1 & 1 & 1 & -3 & 4 & -1 \\ 8 & 12 & 12 & 12 & 4 & 46 & 12 \\ 2 & 3 & 3 & 2 & -1 & 12 & 1 \end{pmatrix}_{b_0}$$

[89] $2 \ 2 \ 31^{\overline{1115}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -2965, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.3$
 $T_{\min} = 11.66 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 10 & 2 \\ 2 & 1 & 4 & 3 & 1 & 15 & 3 \\ 2 & 4 & 1 & 4 & 1 & 15 & 3 \\ 2 & 3 & 4 & 0 & 1 & 15 & 2 \\ 1 & 1 & 1 & 1 & -3 & 5 & -1 \\ 10 & 15 & 15 & 15 & 5 & 73 & 15 \\ 2 & 3 & 3 & 2 & -1 & 15 & 1 \end{pmatrix}_{b_0}$$

6.64 NHC(-2)(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{f}_4$

$T = 9 \quad [0] \ 1^{\overline{1111}} 31^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 92, \Delta = -130, \det = -584, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.4$

$T_{\min} = 13.42 \rightarrow 14, T_{\max} = 92, T \in [14, 92]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 1 & 3 \\ 3 & 4 & 9 & 3 & 6 & 0 & 4 \\ 4 & 9 & 9 & 4 & 8 & 0 & 5 \\ 2 & 3 & 4 & -2 & 3 & 0 & 0 \\ 3 & 6 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 0 & 4 & -3 & -1 \end{pmatrix}_{b_0}$$

6.65 NHC(-2)(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_2$ + $\mathfrak{so}_8$

$T = 9 \quad [0] \ 1^{\overline{111}} 31^{\overline{114}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 44 & 5 \\ 8 & 10 & 17 & 0 & 60 & 6 \\ 12 & 17 & 25 & 0 & 96 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 44 & 60 & 96 & 0 & 346 & 34 \\ 5 & 6 & 9 & -3 & 34 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{\overline{114}} 31^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$

$T_{\min} = 13.30 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 44 & 5 \\ 9 & 13 & 18 & 0 & 72 & 7 \\ 11 & 18 & 20 & 0 & 84 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 44 & 72 & 84 & 0 & 346 & 34 \\ 5 & 7 & 8 & -3 & 34 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{\overline{11}} 31^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -107, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 20 & 0 & 13 & 7 \\ 5 & 6 & 9 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{\overline{11}} 31^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -186, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$

$T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 8 & 5 \\ 9 & 13 & 18 & 0 & 12 & 7 \\ 11 & 18 & 20 & 0 & 16 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 16 & 0 & 10 & 6 \\ 5 & 7 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -249$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 17 & 0 & 13 & 7 \\ 5 & 7 & 8 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -275$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 17 & 5 \\ 8 & 10 & 17 & 0 & 24 & 6 \\ 12 & 17 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 24 & 36 & 0 & 49 & 13 \\ 5 & 6 & 9 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{111}31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -331$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 35 & 5 \\ 8 & 10 & 17 & 0 & 48 & 6 \\ 12 & 17 & 25 & 0 & 76 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 35 & 48 & 76 & 0 & 217 & 27 \\ 5 & 6 & 9 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -393$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 1 & 3 \\ 3 & 4 & 9 & 3 & 6 & 0 & 4 \\ 4 & 9 & 9 & 4 & 8 & 0 & 5 \\ 2 & 3 & 4 & -2 & 3 & 0 & 0 \\ 3 & 6 & 8 & 3 & 4 & 0 & 4 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 0 & 4 & -2 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -408$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 26 & 5 \\ 8 & 10 & 17 & 0 & 36 & 6 \\ 12 & 17 & 25 & 0 & 56 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 36 & 56 & 0 & 118 & 20 \\ 5 & 6 & 9 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{113}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -501$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 35 & 5 \\ 9 & 13 & 18 & 0 & 57 & 7 \\ 11 & 18 & 20 & 0 & 67 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 35 & 57 & 67 & 0 & 217 & 27 \\ 5 & 7 & 8 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{111}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -557$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 17 & 5 \\ 9 & 13 & 18 & 0 & 27 & 7 \\ 11 & 18 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 27 & 33 & 0 & 49 & 13 \\ 5 & 7 & 8 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{112}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -662$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.7$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 26 & 5 \\ 9 & 13 & 18 & 0 & 42 & 7 \\ 11 & 18 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 42 & 50 & 0 & 118 & 20 \\ 5 & 7 & 8 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

6.66 NHC(-2)(-2)(-3) + f₄ + su₂ + su₂→(-3)

$$T = 9 \quad [0] \quad 1^{112}3^{11}1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 2 & 3 \\ 3 & 5 & 4 & 0 & 8 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 10 & 4 & 6 \\ 2 & 2 & 3 & 0 & 4 & 0 & 2 \\ 3 & 3 & 4 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11}1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -122, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^{11}1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 2 & 3 \\ 3 & 5 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 2 & 3 \\ 2 & 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 3 & 4 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^{11}1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 2 & 3 \\ 3 & 5 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 13 & 5 & 7 \\ 2 & 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 3 & 4 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{112}3^{11}1^2 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 2 & 3 \\ 2 & 1 & 5 & 0 & 12 & 2 & 3 \\ 3 & 5 & 4 & 0 & 18 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 12 & 18 & 0 & 58 & 10 & 14 \\ 2 & 2 & 3 & 0 & 10 & 0 & 2 \\ 3 & 3 & 4 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{114}3^{11}1^4 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 20 & 2 & 3 \\ 2 & 1 & 5 & 0 & 24 & 2 & 3 \\ 3 & 5 & 4 & 0 & 36 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 20 & 24 & 36 & 0 & 238 & 20 & 28 \\ 2 & 2 & 3 & 0 & 20 & 0 & 2 \\ 3 & 3 & 4 & -3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{113}3^{11}1^3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 15 & 2 & 3 \\ 2 & 1 & 5 & 0 & 18 & 2 & 3 \\ 3 & 5 & 4 & 0 & 27 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 15 & 18 & 27 & 0 & 133 & 15 & 21 \\ 2 & 2 & 3 & 0 & 15 & 0 & 2 \\ 3 & 3 & 4 & -3 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111}3^{11}1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 2 & 3 \\ 2 & 1 & 5 & 0 & 18 & 2 & 3 \\ 3 & 5 & 4 & 0 & 26 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 18 & 26 & 0 & 118 & 14 & 20 \\ 2 & 2 & 3 & 0 & 14 & 0 & 2 \\ 3 & 3 & 4 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [17] \quad 1^{11}3^{11}1^{13}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 1 & 2 & 3 \\ 1 & 2 & 3 & 2 & 0 & 2 & 0 & 2 \\ 2 & 3 & 4 & 2 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{11}3^{11}1^{13}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 6 & 3 & 0 & 4 & 3 & 4 \\ 1 & 2 & 3 & 2 & 0 & 3 & 0 & 2 \\ 2 & 3 & 4 & 2 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11}1^13^{11}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 216, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 6 & 3 \\ 2 & 2 & 6 & 4 & 0 & 9 & 4 \\ 4 & 6 & 8 & 6 & 0 & 15 & 6 \\ 2 & 4 & 6 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 15 & 9 & 0 & 22 & 10 \\ 3 & 4 & 6 & 3 & -3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11}1^13^{11}1^{12}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 349, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 9 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 15 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 15 & 9 & 0 & 37 & 8 & 11 \\ 1 & 2 & 3 & 2 & 0 & 8 & 0 & 2 \\ 2 & 3 & 4 & 2 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11}1^13^{11}1^{13}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 412, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 6 & 9 & 6 & 0 & 13 & 5 & 7 \\ 1 & 2 & 3 & 2 & 0 & 5 & 0 & 2 \\ 2 & 3 & 4 & 2 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11}1^13^{11}1^{12}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 432, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 12 & 3 \\ 2 & 2 & 6 & 4 & 0 & 18 & 4 \\ 4 & 6 & 8 & 6 & 0 & 30 & 6 \\ 2 & 4 & 6 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 18 & 30 & 18 & 0 & 94 & 20 \\ 3 & 4 & 6 & 3 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11}1^13^{11}1^{13}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 792, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 18 & 3 \\ 2 & 2 & 6 & 4 & 0 & 27 & 4 \\ 4 & 6 & 8 & 6 & 0 & 45 & 6 \\ 2 & 4 & 6 & 1 & 0 & 27 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 27 & 45 & 27 & 0 & 214 & 30 \\ 3 & 4 & 6 & 3 & -3 & 30 & 2 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11}1^13^{11}1^{14}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 24 & 3 \\ 2 & 2 & 6 & 4 & 0 & 36 & 4 \\ 4 & 6 & 8 & 6 & 0 & 60 & 6 \\ 2 & 4 & 6 & 1 & 0 & 36 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 24 & 36 & 60 & 36 & 0 & 382 & 40 \\ 3 & 4 & 6 & 3 & -3 & 40 & 2 \end{pmatrix}_{b_0}$$

6.67 NHC(-2)(-2)(-3) + f₄ + su₂ + su₂→(-3) + su₃

$$T = 10 \quad [0] \quad 1^{111}3^{1111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 194, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 2 & 3 & 3 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{1111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 244, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 6 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 3 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^{1111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 302, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 2 & 6 & 3 \\ 3 & 5 & 4 & 0 & 4 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 2 & 6 & 3 \\ 2 & 2 & 3 & 0 & 2 & 0 & 5 & 2 \\ 5 & 6 & 9 & 0 & 6 & 5 & 12 & 6 \\ 3 & 3 & 4 & -3 & 3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}3^{1111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 484, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 13 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 13 & 0 & 28 & 7 & 9 & 10 \\ 2 & 3 & 3 & 0 & 7 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 9 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 10 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^{1111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \rightarrow (-3) + \text{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 622, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 9 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 5 & 6 & 7 \\ 2 & 3 & 3 & 0 & 5 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 6 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

6.68 NHC(-2)(-2)(-3) + f₄ + su₂ + su₂n3mix

$$T = 9 \quad [0] \quad 1^{111}3^{1111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -19, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 7 & 3 \\ 2 & 1 & 5 & 0 & 3 & 8 & 3 \\ 3 & 5 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 8 & 3 \\ 7 & 8 & 12 & 0 & 8 & 25 & 9 \\ 3 & 3 & 4 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{1111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -21, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.67 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 4 & 3 \\ 3 & 4 & 6 & 0 & 15 & 6 & 4 \\ 3 & 6 & 4 & 0 & 17 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 15 & 17 & 0 & 49 & 21 & 13 \\ 4 & 6 & 7 & 0 & 21 & 7 & 5 \\ 3 & 4 & 4 & -3 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1113}1^{\overline{113}}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -41, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 3 & 18 & 4 \\ 3 & 6 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 14 & 3 \\ 11 & 18 & 20 & 0 & 14 & 69 & 15 \\ 3 & 4 & 4 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{113}1^{\overline{111}}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -56, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 5 & 3 \\ 4 & 6 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & 4 & -3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{113}1^{\overline{11}}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -108, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 4 & 3 \\ 2 & 1 & 5 & 0 & 9 & 5 & 3 \\ 3 & 5 & 4 & 0 & 12 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 9 & 12 & 0 & 25 & 15 & 9 \\ 4 & 5 & 7 & 0 & 15 & 7 & 5 \\ 3 & 3 & 4 & -3 & 9 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1113}1^{\overline{111}}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -140, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 7 & 3 \\ 2 & 1 & 5 & 0 & 6 & 8 & 3 \\ 3 & 5 & 4 & 0 & 9 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 13 & 20 & 7 \\ 7 & 8 & 12 & 0 & 20 & 25 & 9 \\ 3 & 3 & 4 & -3 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1123}1^{\overline{111}}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -159, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 8 & 3 \\ 3 & 5 & 4 & 0 & 13 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 9 & 13 & 0 & 28 & 28 & 10 \\ 7 & 8 & 12 & 0 & 28 & 25 & 9 \\ 3 & 3 & 4 & -3 & 10 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1112}3^{\overline{11}}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -168, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 6 & 3 \\ 2 & 1 & 5 & 0 & 6 & 8 & 3 \\ 3 & 5 & 4 & 0 & 9 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 13 & 17 & 7 \\ 6 & 8 & 11 & 0 & 17 & 20 & 8 \\ 3 & 3 & 4 & -3 & 7 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1112}3^{\overline{111}}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -173, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 4 & 3 \\ 3 & 4 & 6 & 0 & 9 & 6 & 4 \\ 3 & 6 & 4 & 0 & 9 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 11 & 7 \\ 4 & 6 & 7 & 0 & 11 & 7 & 5 \\ 3 & 4 & 4 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{1113}1^{\overline{121}}} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{f}_4 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -183, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 4 & 3 \\ 3 & 4 & 6 & 0 & 12 & 6 & 4 \\ 3 & 6 & 4 & 0 & 13 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 13 & 0 & 28 & 16 & 10 \\ 4 & 6 & 7 & 0 & 16 & 7 & 5 \\ 3 & 4 & 4 & -3 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{1121}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -188, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 5 & 3 \\ 3 & 5 & 4 & 0 & 8 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 10 & 10 & 6 \\ 4 & 5 & 7 & 0 & 10 & 7 & 5 \\ 3 & 3 & 4 & -3 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{1113}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -203, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.73 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 5 & 3 \\ 3 & 5 & 4 & 0 & 9 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 13 & 11 & 7 \\ 4 & 5 & 7 & 0 & 11 & 7 & 5 \\ 3 & 3 & 4 & -3 & 7 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1114}3^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -215, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 19 & 3 \\ 2 & 1 & 5 & 0 & 3 & 23 & 3 \\ 3 & 5 & 4 & 0 & 4 & 34 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 23 & 3 \\ 19 & 23 & 34 & 0 & 23 & 208 & 26 \\ 3 & 3 & 4 & -3 & 3 & 26 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1111}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -236, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 4 & 3 \\ 2 & 1 & 5 & 0 & 3 & 5 & 3 \\ 3 & 5 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 5 & 3 \\ 4 & 5 & 7 & 0 & 5 & 7 & 5 \\ 3 & 3 & 4 & -3 & 3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1113}3^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -247, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.66 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 6 & 0 & 3 & 15 & 4 \\ 3 & 6 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 11 & 3 \\ 9 & 15 & 16 & 0 & 11 & 44 & 12 \\ 3 & 4 & 4 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1113}3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -257, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.75 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 8 & 3 \\ 7 & 12 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & 4 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1122}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -300, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 6 & 3 \\ 2 & 1 & 5 & 0 & 6 & 8 & 3 \\ 3 & 5 & 4 & 0 & 8 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 10 & 16 & 6 \\ 6 & 8 & 11 & 0 & 16 & 20 & 8 \\ 3 & 3 & 4 & -3 & 6 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{1113}3^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -320, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.62 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 14 & 3 \\ 2 & 1 & 5 & 0 & 3 & 17 & 3 \\ 3 & 5 & 4 & 0 & 4 & 25 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 17 & 3 \\ 14 & 17 & 25 & 0 & 17 & 111 & 19 \\ 3 & 3 & 4 & -3 & 3 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{11123} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -327, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.65 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 9 & 3 \\ 2 & 1 & 5 & 0 & 3 & 11 & 3 \\ 3 & 5 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 11 & 3 \\ 9 & 11 & 16 & 0 & 11 & 44 & 12 \\ 3 & 3 & 4 & -3 & 3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{11123} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -348, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.78 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 6 & 3 \\ 2 & 1 & 5 & 0 & 3 & 8 & 3 \\ 3 & 5 & 4 & 0 & 4 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 8 & 3 \\ 6 & 8 & 11 & 0 & 8 & 20 & 8 \\ 3 & 3 & 4 & -3 & 3 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{11143} 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -383, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.78 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 17 & 4 & 3 \\ 2 & 1 & 5 & 0 & 21 & 5 & 3 \\ 3 & 5 & 4 & 0 & 31 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 21 & 31 & 0 & 172 & 38 & 24 \\ 4 & 5 & 7 & 0 & 38 & 7 & 5 \\ 3 & 3 & 4 & -3 & 24 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11123} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -383, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 4 & 3 \\ 2 & 1 & 5 & 0 & 9 & 5 & 3 \\ 3 & 5 & 4 & 0 & 13 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 9 & 13 & 0 & 28 & 16 & 10 \\ 4 & 5 & 7 & 0 & 16 & 7 & 5 \\ 3 & 3 & 4 & -3 & 10 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11123} 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -399, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.65 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 9 & 3 \\ 2 & 1 & 5 & 0 & 6 & 11 & 3 \\ 3 & 5 & 4 & 0 & 9 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 13 & 26 & 7 \\ 9 & 11 & 16 & 0 & 26 & 44 & 12 \\ 3 & 3 & 4 & -3 & 7 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11123} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -404, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.81 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 6 & 3 \\ 2 & 1 & 5 & 0 & 9 & 8 & 3 \\ 3 & 5 & 4 & 0 & 13 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 9 & 13 & 0 & 28 & 25 & 10 \\ 6 & 8 & 11 & 0 & 25 & 20 & 8 \\ 3 & 3 & 4 & -3 & 10 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1113} 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -432, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.72 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 4 & 3 \\ 2 & 1 & 5 & 0 & 15 & 5 & 3 \\ 3 & 5 & 4 & 0 & 22 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 15 & 22 & 0 & 85 & 27 & 17 \\ 4 & 5 & 7 & 0 & 27 & 7 & 5 \\ 3 & 3 & 4 & -3 & 17 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1113} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -460, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.85 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 8 & 3 \\ 2 & 1 & 5 & 0 & 3 & 11 & 3 \\ 3 & 5 & 4 & 0 & 4 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 11 & 3 \\ 8 & 11 & 15 & 0 & 11 & 39 & 11 \\ 3 & 3 & 4 & -3 & 3 & 11 & 1 \end{pmatrix}_{b_0}$$

[26] $1^{11313^1}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -531, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 4 & 3 \\ 2 & 1 & 5 & 0 & 12 & 5 & 3 \\ 3 & 5 & 4 & 0 & 17 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 12 & 17 & 0 & 49 & 21 & 13 \\ 4 & 5 & 7 & 0 & 21 & 7 & 5 \\ 3 & 3 & 4 & -3 & 13 & 5 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{11133^1}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 11 & 3 \\ 2 & 1 & 5 & 0 & 6 & 14 & 3 \\ 3 & 5 & 4 & 0 & 9 & 20 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 13 & 32 & 7 \\ 11 & 14 & 20 & 0 & 32 & 69 & 15 \\ 3 & 3 & 4 & -3 & 7 & 15 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{11233^1}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -572, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$
 $T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 9 & 3 \\ 2 & 1 & 5 & 0 & 9 & 11 & 3 \\ 3 & 5 & 4 & 0 & 13 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 9 & 13 & 0 & 28 & 37 & 10 \\ 9 & 11 & 16 & 0 & 37 & 44 & 12 \\ 3 & 3 & 4 & -3 & 10 & 12 & 1 \end{pmatrix}_{b_0}$$

[28] $1^{11143^1}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -572, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 10 & 3 \\ 2 & 1 & 5 & 0 & 3 & 14 & 3 \\ 3 & 5 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 14 & 3 \\ 10 & 14 & 19 & 0 & 14 & 64 & 14 \\ 3 & 3 & 4 & -3 & 3 & 14 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{11143^1}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -623, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 13 & 3 \\ 2 & 1 & 5 & 0 & 6 & 17 & 3 \\ 3 & 5 & 4 & 0 & 9 & 24 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 13 & 38 & 7 \\ 13 & 17 & 24 & 0 & 38 & 100 & 18 \\ 3 & 3 & 4 & -3 & 7 & 18 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{11133^1}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -635, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 11 & 3 \\ 2 & 1 & 5 & 0 & 3 & 14 & 3 \\ 3 & 5 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 14 & 3 \\ 11 & 14 & 20 & 0 & 14 & 69 & 15 \\ 3 & 3 & 4 & -3 & 3 & 15 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{11143^1}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -647, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.71 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 4 & 3 \\ 2 & 1 & 5 & 0 & 15 & 5 & 3 \\ 3 & 5 & 4 & 0 & 21 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 15 & 21 & 0 & 76 & 26 & 16 \\ 4 & 5 & 7 & 0 & 26 & 7 & 5 \\ 3 & 3 & 4 & -3 & 16 & 5 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{11143^1}1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -79, \det = -776, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 4 & 3 \\ 2 & 1 & 5 & 0 & 18 & 5 & 3 \\ 3 & 5 & 4 & 0 & 26 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 18 & 26 & 0 & 118 & 32 & 20 \\ 4 & 5 & 7 & 0 & 32 & 7 & 5 \\ 3 & 3 & 4 & -3 & 20 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{111431^1 1^{12}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -824, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 16 & 3 \\ 2 & 1 & 5 & 0 & 3 & 20 & 3 \\ 3 & 5 & 4 & 0 & 4 & 29 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 20 & 3 \\ 16 & 20 & 29 & 0 & 20 & 148 & 22 \\ 3 & 3 & 4 & -3 & 3 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{111431^1 1^{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -943, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 13 & 3 \\ 2 & 1 & 5 & 0 & 3 & 17 & 3 \\ 3 & 5 & 4 & 0 & 4 & 24 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 17 & 3 \\ 13 & 17 & 24 & 0 & 17 & 100 & 18 \\ 3 & 3 & 4 & -3 & 3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [36] \quad 1^{1111} 31^{13} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 4 \\ 2 & 4 & 4 & 4 & 0 & 3 & 5 \\ 1 & 3 & 4 & 0 & 0 & 3 & 4 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 \\ 1 & 3 & 3 & 3 & 0 & 1 & 4 \\ 2 & 4 & 5 & 4 & 0 & 4 & 4 \\ 2 & 3 & 4 & 2 & -3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{111} 31^{11} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 101, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 7 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 11 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 6 & 3 & 0 & 4 & 11 & 4 \\ 4 & 7 & 11 & 7 & 0 & 11 & 20 & 8 \\ 2 & 3 & 4 & 2 & -3 & 4 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{111} 31^{11} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 104, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 8 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 12 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 6 & 3 & 0 & 4 & 12 & 4 \\ 4 & 8 & 12 & 8 & 0 & 12 & 25 & 9 \\ 2 & 3 & 4 & 2 & -3 & 4 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{111} 31^{11} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 365, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 4 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 5 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 6 & 9 & 6 & 0 & 13 & 9 & 7 \\ 2 & 4 & 5 & 4 & 0 & 9 & 4 & 4 \\ 2 & 3 & 4 & 2 & -3 & 7 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{111} 31^{11} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 480, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 7 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 11 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 6 & 9 & 6 & 0 & 13 & 18 & 7 \\ 4 & 7 & 11 & 7 & 0 & 18 & 20 & 8 \\ 2 & 3 & 4 & 2 & -3 & 7 & 8 & 1 \end{pmatrix}_{b_0}$$

6.69 NHC(-2)(-2)(-3) + f₄ + su₂ + su_{2n3mix} + su₃

$$T = 10 \quad [0] \quad 1^{111} 31^{11} 1^{1111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 389, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 12 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 8 & 3 & 3 \\ 7 & 12 & 12 & 0 & 8 & 25 & 8 & 9 \\ 2 & 3 & 4 & 0 & 3 & 8 & 0 & 2 \\ 3 & 4 & 4 & -3 & 3 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

6.70 NHC(-2)(-2)(-3) + f₄ + su₂ + su₃

$$T = 9 \quad [0] \quad 1^{1121} 3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 8 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 10 & 12 & 6 \\ 5 & 6 & 9 & 0 & 12 & 12 & 6 \\ 3 & 3 & 4 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 6 & 5 & 0 & 4 & 3 & 4 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 9 & 3 \\ 7 & 12 & 13 & 0 & 9 & 27 & 9 \\ 3 & 4 & 4 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{113} 3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -241, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 6 & 0 & 21 & 3 & 4 \\ 3 & 6 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 21 & 19 & 0 & 73 & 12 & 15 \\ 2 & 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -3 & 15 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{1111} 3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 3 & 5 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 6 & 3 \\ 5 & 6 & 9 & 0 & 6 & 12 & 6 \\ 3 & 3 & 4 & -3 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{1111} 3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 13 & 15 & 7 \\ 5 & 6 & 9 & 0 & 15 & 12 & 6 \\ 3 & 3 & 4 & -3 & 7 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 3^{1121} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -318, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 13 & 0 & 28 & 9 & 10 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{112} 3^1 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -423, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 2 & 3 \\ 3 & 4 & 6 & 0 & 15 & 3 & 4 \\ 3 & 6 & 4 & 0 & 14 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 15 & 14 & 0 & 37 & 9 & 11 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & -3 & 11 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{112} 3^1 1^{141} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -486, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 2 & 3 \\ 3 & 4 & 6 & 0 & 24 & 3 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 24 & 26 & 0 & 118 & 18 & 20 \\ 2 & 3 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -3 & 20 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{112} 3^1 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 5 & 3 \\ 2 & 1 & 5 & 0 & 9 & 6 & 3 \\ 3 & 5 & 4 & 0 & 13 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 9 & 13 & 0 & 28 & 21 & 10 \\ 5 & 6 & 9 & 0 & 21 & 12 & 6 \\ 3 & 3 & 4 & -3 & 10 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{113} 3^1 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -686, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 5 & 3 \\ 2 & 1 & 5 & 0 & 12 & 6 & 3 \\ 3 & 5 & 4 & 0 & 17 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 12 & 17 & 0 & 49 & 27 & 13 \\ 5 & 6 & 9 & 0 & 27 & 12 & 6 \\ 3 & 3 & 4 & -3 & 13 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{114} 3^1 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 5 & 3 \\ 2 & 1 & 5 & 0 & 15 & 6 & 3 \\ 3 & 5 & 4 & 0 & 21 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 15 & 21 & 0 & 76 & 33 & 16 \\ 5 & 6 & 9 & 0 & 33 & 12 & 6 \\ 3 & 3 & 4 & -3 & 16 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{112} 3^1 1^{121} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -738, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 2 & 3 \\ 3 & 4 & 6 & 0 & 18 & 3 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 18 & 18 & 0 & 58 & 12 & 14 \\ 2 & 3 & 4 & 0 & 12 & 0 & 2 \\ 3 & 4 & 4 & -3 & 14 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{112} 3^1 1^{131} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -759, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 2 & 3 \\ 3 & 4 & 6 & 0 & 21 & 3 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 12 & 21 & 22 & 0 & 85 & 15 & 17 \\ 2 & 3 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -3 & 17 & 2 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{114}3^11^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -906, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 2 & 3 \\ 3 & 4 & 6 & 0 & 30 & 3 & 4 \\ 3 & 6 & 4 & 0 & 28 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 16 & 30 & 28 & 0 & 154 & 18 & 22 \\ 2 & 3 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -3 & 22 & 2 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{113}3^11^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -934, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 2 & 3 \\ 3 & 4 & 6 & 0 & 24 & 3 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 24 & 23 & 0 & 100 & 15 & 18 \\ 2 & 3 & 4 & 0 & 15 & 0 & 2 \\ 3 & 4 & 4 & -3 & 18 & 2 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{113}3^11^{131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 2 & 3 \\ 3 & 4 & 6 & 0 & 27 & 3 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 15 & 27 & 27 & 0 & 133 & 18 & 21 \\ 2 & 3 & 4 & 0 & 18 & 0 & 2 \\ 3 & 4 & 4 & -3 & 21 & 2 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{113}3^11^{141} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1438, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 2 & 3 \\ 3 & 4 & 6 & 0 & 30 & 3 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 17 & 30 & 31 & 0 & 172 & 21 & 24 \\ 2 & 3 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -3 & 24 & 2 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{114}3^11^{131} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1683, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 2 & 3 \\ 3 & 4 & 6 & 0 & 33 & 3 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 18 & 33 & 32 & 0 & 193 & 21 & 25 \\ 2 & 3 & 4 & 0 & 21 & 0 & 2 \\ 3 & 4 & 4 & -3 & 25 & 2 & 1 \end{pmatrix}_{b_0}$$

[21] $1^{114}3^11^{141} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -2166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 2 & 3 \\ 3 & 4 & 6 & 0 & 36 & 3 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 20 & 36 & 36 & 0 & 238 & 24 & 28 \\ 2 & 3 & 4 & 0 & 24 & 0 & 2 \\ 3 & 4 & 4 & -3 & 28 & 2 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [22] $1^{11}3^121^{1111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 & 4 & 3 \\ 1 & 2 & 0 & 2 & 0 & 1 & 3 & 2 \\ 1 & 3 & 2 & 0 & 0 & 3 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 3 & 1 & 3 & 0 & 1 & 4 & 3 \\ 2 & 4 & 3 & 4 & 0 & 4 & 4 & 4 \\ 2 & 3 & 2 & 2 & -3 & 3 & 4 & 1 \end{pmatrix}_{b_0}$$

[23] $1^{11}3^121^{1113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 76, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 9 & 3 \\ 2 & 2 & 6 & 0 & 4 & 15 & 4 \\ 3 & 6 & 5 & 0 & 5 & 20 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 5 & 0 & 2 & 15 & 4 \\ 9 & 15 & 20 & 0 & 15 & 57 & 15 \\ 3 & 4 & 5 & -3 & 4 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{111}3^{11}2^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 365, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 2 & 3 & 0 & 4 & 4 & 3 \\ 1 & 2 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 3 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 4 & 0 & 5 & 7 & 5 \\ 2 & 4 & 3 & 4 & 0 & 7 & 4 & 4 \\ 2 & 3 & 2 & 2 & -3 & 5 & 4 & 1 \end{pmatrix}_{b_0}$$

6.71 NHC(-2)(-2)(-3) + f₄ + su₂ + su₃ + su₂→(-3)

$$T = 10 \quad [0] \quad 1^{111}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 194, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 3 & 2 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 244, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 6 & 5 & 0 & 4 & 3 & 3 & 4 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 3 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 4 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 302, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 2 & 3 \\ 3 & 5 & 4 & 0 & 4 & 9 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 6 & 2 & 3 \\ 5 & 6 & 9 & 0 & 6 & 12 & 5 & 6 \\ 2 & 2 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 3 & 4 & -3 & 3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}3^{11}1^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 484, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 13 & 0 & 28 & 9 & 7 & 10 \\ 2 & 3 & 4 & 0 & 9 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 7 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 10 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 622, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 9 & 9 & 0 & 13 & 6 & 5 & 7 \\ 2 & 3 & 4 & 0 & 6 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 5 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 7 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

6.72 NHC(-2)(-2)(-3) + f₄ + su₂ + su₃ + su₂n₃mix

$$T = 10 \quad [0] \quad 1^{111}3^{11}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 389, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 1 & 3 & 8 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 8 & 2 \\ 7 & 12 & 12 & 0 & 8 & 8 & 25 & 9 \\ 3 & 4 & 4 & -3 & 3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

6.73 NHC(-2)(-2)(-3) + f₄ + su₂→(-3)

$$T = 8 \quad [0] \quad 1^{11}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 2 & 3 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 2 \\ 3 & 3 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}3^{11}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.11 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 2 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 2 & 4 & 0 & 0 & 2 \\ 3 & 3 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 13^{11}1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 3 \\ 3 & 4 & 9 & 3 & 3 & 4 \\ 4 & 9 & 9 & 4 & 4 & 5 \\ 2 & 3 & 4 & -2 & 2 & 0 \\ 2 & 3 & 4 & 2 & 0 & 2 \\ 3 & 4 & 5 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 21^{111}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 20, \Delta = -109, \det = 562, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.61 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$$

$$\begin{pmatrix} 4 & 6 & 8 & 3 & 3 & 4 \\ 6 & 4 & 9 & 3 & 3 & 4 \\ 8 & 9 & 9 & 4 & 4 & 5 \\ 3 & 3 & 4 & -2 & 2 & 0 \\ 3 & 3 & 4 & 2 & 0 & 2 \\ 4 & 4 & 5 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [5] \quad 1^{11}3^{11}21^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 10 & 6 & 1 & 5 \\ 10 & 18 & 11 & 0 & 8 \\ 6 & 11 & 5 & 1 & 5 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 8 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11}31^{11}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 3 \\ 2 & 2 & 6 & 4 & 0 & 4 \\ 4 & 6 & 8 & 6 & 0 & 6 \\ 2 & 4 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 6 & 3 & -3 & 2 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11}3^{11}1^{11}3^{11}1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 2 & 3 & 2 & 0 & 0 & 2 \\ 2 & 3 & 4 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^1 3 1^{11} 3^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -250, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.13 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 4 & 2 & 1 & 3 \\ 4 & 8 & 11 & 5 & 0 & 6 \\ 4 & 11 & 8 & 6 & 0 & 6 \\ 2 & 5 & 6 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 6 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^1 3^1 1^{11} 3^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -269, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 10.2$$

$$T_{\min} = 12.06 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 2 & 0 & 0 & 2 \\ 2 & 4 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

6.74 $\text{NHC}(-2)(-2)(-3) + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{f}_4 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -224, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.4$$

$$T_{\min} = 13.43 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 3 & 0 & 4 \\ 4 & 9 & 9 & 4 & 4 & 0 & 5 \\ 2 & 3 & 4 & -2 & 2 & 0 & 0 \\ 2 & 3 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 5 & 0 & 2 & -3 & -1 \end{pmatrix}_{b_0}$$

6.75 $\text{NHC}(-2)(-2)(-3) + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -130, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.80 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 3 & 0 & 4 \\ 4 & 9 & 9 & 4 & 4 & 0 & 5 \\ 2 & 3 & 4 & -2 & 2 & 0 & 0 \\ 2 & 3 & 4 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & 0 & 2 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -172, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 6 & 5 \\ 8 & 10 & 17 & 0 & 8 & 6 \\ 12 & 17 & 25 & 0 & 12 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 8 & 12 & 0 & 4 & 4 \\ 5 & 6 & 9 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 9, \quad T_{\max} = 21, \quad \Delta = -108, \quad \det = -330, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.68 \rightarrow 13, \quad T_{\max} = 21, \quad T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 6 & 5 \\ 9 & 13 & 18 & 0 & 9 & 7 \\ 11 & 18 & 20 & 0 & 11 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 11 & 0 & 4 & 4 \\ 5 & 7 & 8 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

6.76 $\text{NHC}(-2)(-2)(-3) + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2$

$$T = 9 \quad [0] \quad 1^{11} 3^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -76, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.74 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 4 & 3 \\ 2 & 1 & 5 & 0 & 2 & 6 & 3 \\ 3 & 5 & 4 & 0 & 3 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 4 & 2 \\ 4 & 6 & 8 & 0 & 4 & 10 & 6 \\ 3 & 3 & 4 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -122, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.74 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 2 & 1 & 3 \\ 3 & 4 & 4 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 2 & 3 & 3 \\ 3 & 5 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 2 & 1 & 3 \\ 3 & 3 & 4 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{113}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 15 & 3 \\ 2 & 1 & 5 & 0 & 2 & 18 & 3 \\ 3 & 5 & 4 & 0 & 3 & 27 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 15 & 2 \\ 15 & 18 & 27 & 0 & 15 & 133 & 21 \\ 3 & 3 & 4 & -3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 2 & 6 & 3 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 5 & 2 \\ 5 & 6 & 9 & 0 & 5 & 13 & 7 \\ 3 & 3 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{112}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 10 & 3 \\ 2 & 1 & 5 & 0 & 2 & 12 & 3 \\ 3 & 5 & 4 & 0 & 3 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 10 & 2 \\ 10 & 12 & 18 & 0 & 10 & 58 & 14 \\ 3 & 3 & 4 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{114}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -140, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 20 & 3 \\ 2 & 1 & 5 & 0 & 2 & 24 & 3 \\ 3 & 5 & 4 & 0 & 3 & 36 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 20 & 2 \\ 20 & 24 & 36 & 0 & 20 & 238 & 28 \\ 3 & 3 & 4 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -162, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.75 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 3 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 \\ 3 & 6 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{111}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -242, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 3 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 7 & 2 \\ 7 & 12 & 13 & 0 & 7 & 28 & 10 \\ 3 & 4 & 4 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{112}3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -264, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 7 & 3 \\ 2 & 1 & 5 & 0 & 2 & 9 & 3 \\ 3 & 5 & 4 & 0 & 3 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 7 & 2 \\ 7 & 9 & 13 & 0 & 7 & 28 & 10 \\ 3 & 3 & 4 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.69 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 5 & 2 \\ 5 & 9 & 9 & 0 & 5 & 13 & 7 \\ 3 & 4 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{113}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -356, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.65 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 9 & 3 \\ 2 & 1 & 5 & 0 & 2 & 12 & 3 \\ 3 & 5 & 4 & 0 & 3 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 9 & 2 \\ 9 & 12 & 17 & 0 & 9 & 49 & 13 \\ 3 & 3 & 4 & -3 & 2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{113}3^11^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -404, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 12 & 3 \\ 2 & 1 & 5 & 0 & 2 & 15 & 3 \\ 3 & 5 & 4 & 0 & 3 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 12 & 2 \\ 12 & 15 & 22 & 0 & 12 & 85 & 17 \\ 3 & 3 & 4 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{114}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -416, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 11 & 3 \\ 2 & 1 & 5 & 0 & 2 & 15 & 3 \\ 3 & 5 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 11 & 2 \\ 11 & 15 & 21 & 0 & 11 & 76 & 16 \\ 3 & 3 & 4 & -3 & 2 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{114}3^11^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -544, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 17 & 3 \\ 2 & 1 & 5 & 0 & 2 & 21 & 3 \\ 3 & 5 & 4 & 0 & 3 & 31 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 17 & 2 \\ 17 & 21 & 31 & 0 & 17 & 172 & 24 \\ 3 & 3 & 4 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{114}3^11^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -636, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 14 & 3 \\ 2 & 1 & 5 & 0 & 2 & 18 & 3 \\ 3 & 5 & 4 & 0 & 3 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 14 & 2 \\ 14 & 18 & 26 & 0 & 14 & 118 & 20 \\ 3 & 3 & 4 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [17] \quad 1^{113}3^11^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 3 & 2 & 0 & 0 & 3 & 2 \\ 2 & 3 & 6 & 3 & 0 & 3 & 4 & 4 \\ 2 & 3 & 4 & 2 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111}3^11^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 69, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 3 \\ 1 & 3 & 4 & 0 & 0 & 2 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 \\ 1 & 2 & 3 & 2 & 0 & 0 & 2 \\ 1 & 3 & 3 & 3 & 0 & 2 & 1 \\ 2 & 3 & 4 & 2 & -3 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111}31^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 216, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 6 & 3 \\ 2 & 2 & 6 & 4 & 0 & 9 & 4 \\ 4 & 6 & 8 & 6 & 0 & 15 & 6 \\ 2 & 4 & 6 & 1 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 15 & 9 & 0 & 22 & 10 \\ 3 & 4 & 6 & 3 & -3 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{111}31^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 349, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 9 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 15 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 3 & 2 & 0 & 0 & 8 & 2 \\ 5 & 9 & 15 & 9 & 0 & 8 & 37 & 11 \\ 2 & 3 & 4 & 2 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111}31^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 412, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 2 & 6 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 1 & 2 & 3 & 2 & 0 & 0 & 5 & 2 \\ 3 & 6 & 9 & 6 & 0 & 5 & 13 & 7 \\ 2 & 3 & 4 & 2 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111}31^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 432, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 11.7$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 12 & 3 \\ 2 & 2 & 6 & 4 & 0 & 18 & 4 \\ 4 & 6 & 8 & 6 & 0 & 30 & 6 \\ 2 & 4 & 6 & 1 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 18 & 30 & 18 & 0 & 94 & 20 \\ 3 & 4 & 6 & 3 & -3 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111}31^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 792, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 18 & 3 \\ 2 & 2 & 6 & 4 & 0 & 27 & 4 \\ 4 & 6 & 8 & 6 & 0 & 45 & 6 \\ 2 & 4 & 6 & 1 & 0 & 27 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 27 & 45 & 27 & 0 & 214 & 30 \\ 3 & 4 & 6 & 3 & -3 & 30 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{114}31^{14}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 1296, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 24 & 3 \\ 2 & 2 & 6 & 4 & 0 & 36 & 4 \\ 4 & 6 & 8 & 6 & 0 & 60 & 6 \\ 2 & 4 & 6 & 1 & 0 & 36 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 24 & 36 & 60 & 36 & 0 & 382 & 40 \\ 3 & 4 & 6 & 3 & -3 & 40 & 2 \end{pmatrix}_{b_0}$$

6.77 NHC(-2)(-2)(-3) + f₄ + su₂→(-3) + su₂ + su₃

$$T = 10 \quad [0] \ 1^{111}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 194, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 2 & 1 & 3 \\ 2 & 3 & 4 & 0 & 2 & 3 & 0 \\ 3 & 4 & 4 & -3 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 244, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 6 & 5 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 4 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 302, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 2 & 3 & 6 & 3 \\ 3 & 5 & 4 & 0 & 3 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 2 & 5 & 2 \\ 2 & 3 & 4 & 0 & 2 & 1 & 6 & 3 \\ 5 & 6 & 9 & 0 & 5 & 6 & 12 & 6 \\ 3 & 3 & 4 & -3 & 2 & 3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^11^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 484, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 13 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 7 & 2 & 2 \\ 7 & 12 & 13 & 0 & 7 & 28 & 9 & 10 \\ 2 & 3 & 4 & 0 & 2 & 9 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^11^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 9 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 5 & 2 & 2 \\ 5 & 9 & 9 & 0 & 5 & 13 & 6 & 7 \\ 2 & 3 & 4 & 0 & 2 & 6 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

6.78 NHC(-2)(-2)(-3) + f₄ + su₂ → (-3) + su₃

$$T = 9 \quad [0] \ 1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.52 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 5 & 0 & 2 & 3 & 3 & 4 \\ 3 & 5 & 4 & 0 & 3 & 5 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 3 & 2 & 2 \\ 3 & 3 & 5 & 0 & 3 & 3 & 3 & 3 \\ 3 & 3 & 4 & -3 & 2 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 2 & 3 & 3 \\ 2 & 1 & 5 & 0 & 2 & 3 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 4 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 2 & 4 & 0 & 0 & 2 & 2 & 2 \\ 2 & 3 & 4 & 0 & 2 & 0 & 2 & 2 \\ 3 & 3 & 5 & -3 & 2 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.49 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 4 & 3 & 3 \\ 2 & 1 & 5 & 0 & 2 & 6 & 3 & 3 \\ 3 & 5 & 4 & 0 & 3 & 8 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 4 & 2 & 2 \\ 4 & 6 & 8 & 0 & 4 & 9 & 5 & 5 \\ 3 & 3 & 4 & -3 & 2 & 5 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 2 & 3 & 3 \\ 3 & 5 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 2 & 0 & 2 \\ 3 & 3 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -210, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 2 & 6 & 3 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 5 & 2 \\ 5 & 6 & 9 & 0 & 5 & 12 & 6 \\ 3 & 3 & 4 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -226, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{112}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -334, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 7 & 3 \\ 2 & 1 & 5 & 0 & 2 & 9 & 3 \\ 3 & 5 & 4 & 0 & 3 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 7 & 2 \\ 7 & 9 & 13 & 0 & 7 & 27 & 9 \\ 3 & 3 & 4 & -3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

6.79 NHC(-2)(-2)(-3) + $\mathbf{f}_4$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{su}_3$ + $\mathbf{su}_2$

$$T = 10 \quad [0] \quad 1^{111}3^11^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 194, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 2 & 0 & 3 \\ 2 & 3 & 4 & 0 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 244, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 6 \\ 3 & 6 & 4 & 0 & 3 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 3 \\ 2 & 3 & 4 & 0 & 2 & 0 & 3 \\ 3 & 6 & 5 & 0 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 302, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 2 & 6 & 3 & 3 \\ 3 & 5 & 4 & 0 & 3 & 9 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 2 & 3 & 0 & 0 & 5 & 2 & 2 \\ 5 & 6 & 9 & 0 & 5 & 12 & 6 & 6 \\ 2 & 3 & 4 & 0 & 2 & 6 & 1 & 3 \\ 3 & 3 & 4 & -3 & 2 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}3^11^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 484, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 7 & 2 \\ 2 & 3 & 4 & 0 & 2 & 0 & 9 & 2 \\ 7 & 12 & 13 & 0 & 7 & 9 & 28 & 10 \\ 3 & 4 & 4 & -3 & 2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{111}3^11^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_3 + \mathbf{su}_2$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 3 & 0 & 0 & 2 & 5 & 2 \\ 2 & 3 & 4 & 0 & 2 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 5 & 6 & 13 & 7 \\ 3 & 4 & 4 & -3 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

6.80 NHC(-2)(-2)(-3) + $\mathbf{f}_4$ + $\mathbf{su}_{2n3\text{mix}}$

$T = 8$ [0] $1^{123}3^11^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 14.11 \rightarrow 15, T_{\max} = 20, T \in [15, 20]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 9 & 3 \\ 2 & 1 & 5 & 0 & 11 & 3 \\ 4 & 5 & 9 & 0 & 20 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 11 & 20 & 0 & 44 & 12 \\ 3 & 3 & 5 & -3 & 12 & -2 \end{pmatrix}_{b_0}$$

[1] $1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 8 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 8 & 12 & 0 & 25 & 9 \\ 3 & 3 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{112}3^11^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 14 & 3 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 14 & 21 & 0 & 80 & 16 \\ 3 & 3 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{113}3^11^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 17 & 3 \\ 2 & 1 & 5 & 0 & 20 & 3 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 20 & 30 & 0 & 165 & 23 \\ 3 & 3 & 4 & -3 & 23 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{114}3^11^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 22 & 3 \\ 2 & 1 & 5 & 0 & 26 & 3 \\ 3 & 5 & 4 & 0 & 39 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 22 & 26 & 39 & 0 & 280 & 30 \\ 3 & 3 & 4 & -3 & 30 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{113}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 71, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 0 & 6 & 4 \\ 3 & 6 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 7 & 0 & 7 & 5 \\ 3 & 4 & 4 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{113}3^11^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 12.43 \rightarrow 13, T_{\max} = 20, T \in [13, 20]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 5 & 3 \\ 4 & 5 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 5 & 12 & 0 & 12 & 6 \\ 3 & 3 & 5 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^11^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2$ $T = 8, T_{\max} = 20, \Delta = -109, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.2$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 \\ 3 & 6 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 18 & 20 & 0 & 69 & 15 \\ 3 & 4 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

- [8] $1^{111}3^11^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 126$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 3 \\ 2 & 1 & 5 & 0 & 5 & 3 \\ 3 & 5 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 5 \\ 3 & 3 & 4 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{111}3^11^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 131$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 8 & 0 & 12 & 6 \\ 3 & 4 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{111}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 132$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.09 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 8 & 3 \\ 4 & 5 & 9 & 0 & 16 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 8 & 16 & 0 & 25 & 9 \\ 3 & 3 & 5 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$
- [11] $1^{112}3^11^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 182$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 3 \\ 2 & 1 & 5 & 0 & 8 & 3 \\ 3 & 5 & 4 & 0 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 8 & 11 & 0 & 20 & 8 \\ 3 & 3 & 4 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{111}3^11^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 219$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 16 & 0 & 44 & 12 \\ 3 & 4 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$
- [13] $1^{111}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 224$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.70 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 12 & 0 & 25 & 9 \\ 3 & 4 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$
- [14] $1^{112}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 224$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.64 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 3 \\ 2 & 1 & 5 & 0 & 11 & 3 \\ 3 & 5 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 11 & 16 & 0 & 44 & 12 \\ 3 & 3 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$
- [15] $1^{113}3^11^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 238$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 8 & 3 \\ 2 & 1 & 5 & 0 & 11 & 3 \\ 3 & 5 & 4 & 0 & 15 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 11 & 15 & 0 & 39 & 11 \\ 3 & 3 & 4 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$
- [16] $1^{114}3^11^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 294$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 14 & 3 \\ 3 & 5 & 4 & 0 & 19 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 14 & 19 & 0 & 64 & 14 \\ 3 & 3 & 4 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$
- [17] $1^{113}3^11^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 322$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.62 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 3 \\ 2 & 1 & 5 & 0 & 17 & 3 \\ 3 & 5 & 4 & 0 & 25 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 14 & 17 & 25 & 0 & 111 & 19 \\ 3 & 3 & 4 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

- [18] $1^{113}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 378$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.67 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 \\ 2 & 1 & 5 & 0 & 14 & 3 \\ 3 & 5 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 14 & 20 & 0 & 69 & 15 \\ 3 & 3 & 4 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$
- [19] $1^{114}3^11^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 420$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.61 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 19 & 3 \\ 2 & 1 & 5 & 0 & 23 & 3 \\ 3 & 5 & 4 & 0 & 34 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 19 & 23 & 34 & 0 & 208 & 26 \\ 3 & 3 & 4 & -3 & 26 & 1 \end{pmatrix}_{b_0}$$
- [20] $1^{114}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 532$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.69 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 13 & 3 \\ 2 & 1 & 5 & 0 & 17 & 3 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 13 & 17 & 24 & 0 & 100 & 18 \\ 3 & 3 & 4 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$
- [21] $1^{114}3^11^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 8$, $T_{\max} = 20$, $\Delta = -109$, $\det = 574$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 11.63 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 16 & 3 \\ 2 & 1 & 5 & 0 & 20 & 3 \\ 3 & 5 & 4 & 0 & 29 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 16 & 20 & 29 & 0 & 148 & 22 \\ 3 & 3 & 4 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$
- $T = 9$ [22] $1^{11}31^{11}3^11^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -35$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.14 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 7 & 3 & 0 & 5 & 4 \\ 2 & 7 & 4 & 4 & 0 & 5 & 4 \\ 1 & 3 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 5 & 4 & 0 & 4 & 4 \\ 2 & 4 & 4 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$
- [23] $1^{111}31^{12}3^11^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -48$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.25 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 3 & 0 & 10 & 3 \\ 2 & 4 & 4 & 4 & 0 & 17 & 4 \\ 1 & 3 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 10 & 17 & 10 & 0 & 48 & 12 \\ 2 & 3 & 4 & 2 & -3 & 12 & 0 \end{pmatrix}_{b_0}$$
- [24] $1^{111}3^11^{13}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -52$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 3 & 0 & 14 & 3 \\ 2 & 4 & 4 & 4 & 0 & 24 & 4 \\ 1 & 3 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 14 & 24 & 14 & 0 & 97 & 17 \\ 2 & 3 & 4 & 2 & -3 & 17 & 0 \end{pmatrix}_{b_0}$$
- [25] $1^{11}3^11^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -52$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 5 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 9 & 5 & 0 & 12 & 6 \\ 2 & 3 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$
- [26] $1^{111}3^11^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -52$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.15 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 5 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 5 & 6 & 5 & 0 & 7 & 5 \\ 2 & 3 & 4 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$
- [27] $1^{11}3^11^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2$ $T = 9$, $T_{\max} = 13$, $\Delta = -88$, $\det = -129$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.2$
 $T_{\min} = 12.07 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 7 & 3 & 0 & 12 & 4 \\ 2 & 7 & 4 & 4 & 0 & 12 & 4 \\ 1 & 3 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 12 & 12 & 8 & 0 & 25 & 9 \\ 2 & 4 & 4 & 2 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

[28] $1^{111}31^{13}1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 4 & 3 \\ 2 & 4 & 4 & 4 & 0 & 5 & 4 \\ 1 & 3 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 4 & 0 & 4 & 4 \\ 2 & 3 & 4 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{112}31^{14}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 11 & 2 \\ 1 & 1 & 4 & 3 & 0 & 19 & 3 \\ 2 & 4 & 4 & 4 & 0 & 32 & 4 \\ 1 & 3 & 4 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 19 & 32 & 19 & 0 & 175 & 23 \\ 2 & 3 & 4 & 2 & -3 & 23 & 0 \end{pmatrix}_{b_0}$$

[30] $1^{111}31^{113}1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -204, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 7 & 3 \\ 2 & 4 & 4 & 4 & 0 & 11 & 4 \\ 1 & 3 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 7 & 11 & 7 & 0 & 20 & 8 \\ 2 & 3 & 4 & 2 & -3 & 8 & 1 \end{pmatrix}_{b_0}$$

[33] $1^{111}31^{113}1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 \\ 1 & 1 & 4 & 3 & 0 & 8 & 3 \\ 2 & 4 & 4 & 4 & 0 & 12 & 4 \\ 1 & 3 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 12 & 8 & 0 & 25 & 9 \\ 2 & 3 & 4 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{111}31^{123}1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 3 & 0 & 11 & 3 \\ 2 & 4 & 4 & 4 & 0 & 18 & 4 \\ 1 & 3 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 11 & 18 & 11 & 0 & 55 & 13 \\ 2 & 3 & 4 & 2 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[31] $1^{113}31^{113}1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -276, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 6 & 2 \\ 1 & 1 & 4 & 3 & 0 & 13 & 3 \\ 2 & 4 & 4 & 4 & 0 & 17 & 4 \\ 1 & 3 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 13 & 17 & 13 & 0 & 60 & 14 \\ 2 & 3 & 4 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[34] $1^131^{111}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -325, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 7 & 3 & 0 & 8 & 4 \\ 2 & 7 & 4 & 4 & 0 & 8 & 4 \\ 1 & 3 & 4 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 8 & 8 & 4 & 0 & 9 & 5 \\ 2 & 4 & 4 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[35] $1^{111}31^{111}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -334, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.04 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 4 & 2 \\ 2 & 4 & 7 & 3 & 0 & 11 & 4 \\ 2 & 7 & 4 & 4 & 0 & 11 & 4 \\ 1 & 3 & 4 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 11 & 11 & 7 & 0 & 20 & 8 \\ 2 & 4 & 4 & 2 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

[36] $1^{112}31^{113}1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -352, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 5 & 2 \\ 1 & 1 & 4 & 3 & 0 & 10 & 3 \\ 2 & 4 & 4 & 4 & 0 & 14 & 4 \\ 1 & 3 & 4 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 10 & 14 & 10 & 0 & 37 & 11 \\ 2 & 3 & 4 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[37] $1^1 3^1 1^{11} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -402, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 12.09 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 7 & 3 & 0 & 9 & 4 \\ 2 & 7 & 4 & 4 & 0 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 5 & 0 & 12 & 6 & \\ 2 & 4 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[38] $1^{11} 31^{13} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -432, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 9 & 2 \\ 1 & 1 & 4 & 3 & 0 & 16 & 3 \\ 2 & 4 & 4 & 4 & 0 & 26 & 4 \\ 1 & 3 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 16 & 26 & 16 & 0 & 117 & 19 \\ 2 & 3 & 4 & 2 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

[39] $1^{11} 31^{12} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -504, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 7 & 2 \\ 1 & 1 & 4 & 3 & 0 & 13 & 3 \\ 2 & 4 & 4 & 4 & 0 & 20 & 4 \\ 1 & 3 & 4 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 13 & 20 & 13 & 0 & 71 & 15 \\ 2 & 3 & 4 & 2 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$

[40] $1^{11} 31^{12} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -736, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 8 & 2 \\ 1 & 1 & 4 & 3 & 0 & 16 & 3 \\ 2 & 4 & 4 & 4 & 0 & 23 & 4 \\ 1 & 3 & 4 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 16 & 23 & 16 & 0 & 100 & 18 \\ 2 & 3 & 4 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

[41] $1^{11} 431^{12} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -744, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 9 & 2 \\ 1 & 1 & 4 & 3 & 0 & 19 & 3 \\ 2 & 4 & 4 & 4 & 0 & 26 & 4 \\ 1 & 3 & 4 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 19 & 26 & 19 & 0 & 135 & 21 \\ 2 & 3 & 4 & 2 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

[42] $1^{11} 31^{13} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -972, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 10 & 2 \\ 1 & 1 & 4 & 3 & 0 & 19 & 3 \\ 2 & 4 & 4 & 4 & 0 & 29 & 4 \\ 1 & 3 & 4 & 0 & 0 & 19 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 19 & 29 & 19 & 0 & 152 & 22 \\ 2 & 3 & 4 & 2 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[43] $1^{11} 31^{14} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -984, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 12 & 2 \\ 1 & 1 & 4 & 3 & 0 & 22 & 3 \\ 2 & 4 & 4 & 4 & 0 & 35 & 4 \\ 1 & 3 & 4 & 0 & 0 & 22 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 22 & 35 & 22 & 0 & 216 & 26 \\ 2 & 3 & 4 & 2 & -3 & 26 & 1 \end{pmatrix}_{b_0}$$

[44] $1^{11} 431^{13} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1288, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 11 & 2 \\ 1 & 1 & 4 & 3 & 0 & 22 & 3 \\ 2 & 4 & 4 & 4 & 0 & 32 & 4 \\ 1 & 3 & 4 & 0 & 0 & 22 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 22 & 32 & 22 & 0 & 193 & 25 \\ 2 & 3 & 4 & 2 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

[45] $1^{11} 431^{14} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1608, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.2$

$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 13 & 2 \\ 1 & 1 & 4 & 3 & 0 & 25 & 3 \\ 2 & 4 & 4 & 4 & 0 & 38 & 4 \\ 1 & 3 & 4 & 0 & 0 & 25 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 25 & 38 & 25 & 0 & 263 & 29 \\ 2 & 3 & 4 & 2 & -3 & 29 & 1 \end{pmatrix}_{b_0}$$

6.81 NHC(-2)(-2)(-3) + f₄ + su2n3mix + so₈

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 15 & 5 \\ 8 & 10 & 17 & 0 & 20 & 6 \\ 12 & 17 & 25 & 0 & 32 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 20 & 32 & 0 & 37 & 11 \\ 5 & 6 & 9 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{so}_8 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 15 & 5 \\ 9 & 13 & 18 & 0 & 24 & 7 \\ 11 & 18 & 20 & 0 & 28 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 24 & 28 & 0 & 37 & 11 \\ 5 & 7 & 8 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

6.82 NHC(-2)(-2)(-3) + f₄ + su2n3mix + su₂

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -19, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 2 & 3 \\ 2 & 1 & 5 & 0 & 8 & 3 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 8 & 12 & 0 & 25 & 8 & 9 \\ 2 & 3 & 4 & 0 & 8 & 1 & 3 \\ 3 & 3 & 4 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{113} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -21, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.67 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 9 & 3 \\ 3 & 4 & 6 & 0 & 6 & 15 & 4 \\ 3 & 6 & 4 & 0 & 7 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 7 & 0 & 7 & 21 & 5 \\ 9 & 15 & 17 & 0 & 21 & 49 & 13 \\ 3 & 4 & 4 & -3 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{131} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -41, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 6 & 0 & 18 & 3 & 4 \\ 3 & 6 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 18 & 20 & 0 & 69 & 14 & 15 \\ 2 & 3 & 4 & 0 & 14 & 1 & 3 \\ 3 & 4 & 4 & -3 & 15 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -56, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 7 & 0 & 7 & 5 & 5 \\ 2 & 3 & 4 & 0 & 5 & 1 & 3 \\ 3 & 4 & 4 & -3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -108, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 6 & 3 \\ 2 & 1 & 5 & 0 & 5 & 9 & 3 \\ 3 & 5 & 4 & 0 & 7 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 15 & 5 \\ 6 & 9 & 12 & 0 & 15 & 25 & 9 \\ 3 & 3 & 4 & -3 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1111}3^11^{\overline{111}}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -140, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 5 & 3 \\ 2 & 1 & 5 & 0 & 8 & 6 & 3 \\ 3 & 5 & 4 & 0 & 12 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 8 & 12 & 0 & 25 & 20 & 9 \\ 5 & 6 & 9 & 0 & 20 & 13 & 7 \\ 3 & 3 & 4 & -3 & 9 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1112}3^11^{\overline{111}}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -159, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 7 & 3 \\ 2 & 1 & 5 & 0 & 8 & 9 & 3 \\ 3 & 5 & 4 & 0 & 12 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 8 & 12 & 0 & 25 & 28 & 9 \\ 7 & 9 & 13 & 0 & 28 & 28 & 10 \\ 3 & 3 & 4 & -3 & 9 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1112}3^11^{\overline{11}}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -168, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 5 & 3 \\ 2 & 1 & 5 & 0 & 8 & 6 & 3 \\ 3 & 5 & 4 & 0 & 11 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 8 & 11 & 0 & 20 & 17 & 8 \\ 5 & 6 & 9 & 0 & 17 & 13 & 7 \\ 3 & 3 & 4 & -3 & 8 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}3^11^{\overline{111}}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -173, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 6 & 0 & 6 & 9 & 4 \\ 3 & 6 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 7 & 0 & 7 & 11 & 5 \\ 5 & 9 & 9 & 0 & 11 & 13 & 7 \\ 3 & 4 & 4 & -3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}3^11^{\overline{112}}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -183, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 7 & 3 \\ 3 & 4 & 6 & 0 & 6 & 12 & 4 \\ 3 & 6 & 4 & 0 & 7 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 7 & 0 & 7 & 16 & 5 \\ 7 & 12 & 13 & 0 & 16 & 28 & 10 \\ 3 & 4 & 4 & -3 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1112}3^11^{\overline{1}}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -188, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.77 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 4 & 3 \\ 2 & 1 & 5 & 0 & 5 & 6 & 3 \\ 3 & 5 & 4 & 0 & 7 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 10 & 5 \\ 4 & 6 & 8 & 0 & 10 & 10 & 6 \\ 3 & 3 & 4 & -3 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{1111}3^11^{\overline{1}}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -203, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.73 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 5 & 3 \\ 2 & 1 & 5 & 0 & 5 & 6 & 3 \\ 3 & 5 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 11 & 5 \\ 5 & 6 & 9 & 0 & 11 & 13 & 7 \\ 3 & 3 & 4 & -3 & 5 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{1141}3^11^{\overline{13}}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -215, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.61 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 19 & 2 & 3 \\ 2 & 1 & 5 & 0 & 23 & 3 & 3 \\ 3 & 5 & 4 & 0 & 34 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 19 & 23 & 34 & 0 & 208 & 23 & 26 \\ 2 & 3 & 4 & 0 & 23 & 1 & 3 \\ 3 & 3 & 4 & -3 & 26 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1111}3^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -236, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.71 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 2 & 3 \\ 2 & 1 & 5 & 0 & 5 & 3 & 3 \\ 3 & 5 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 5 & 5 \\ 2 & 3 & 4 & 0 & 5 & 1 & 3 \\ 3 & 3 & 4 & -3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{1111}3^{11}1^{121} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -247, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.66 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 6 & 0 & 15 & 3 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 15 & 16 & 0 & 44 & 11 & 12 \\ 2 & 3 & 4 & 0 & 11 & 1 & 3 \\ 3 & 4 & 4 & -3 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1111}3^{11}1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -257, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.75 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 12 & 0 & 25 & 8 & 9 \\ 2 & 3 & 4 & 0 & 8 & 1 & 3 \\ 3 & 4 & 4 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{1122}3^{11}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -300, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.83 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 4 & 3 \\ 2 & 1 & 5 & 0 & 8 & 6 & 3 \\ 3 & 5 & 4 & 0 & 11 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 8 & 11 & 0 & 20 & 16 & 8 \\ 4 & 6 & 8 & 0 & 16 & 10 & 6 \\ 3 & 3 & 4 & -3 & 8 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{1131}3^{11}1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -320, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.62 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 2 & 3 \\ 2 & 1 & 5 & 0 & 17 & 3 & 3 \\ 3 & 5 & 4 & 0 & 25 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 14 & 17 & 25 & 0 & 111 & 17 & 19 \\ 2 & 3 & 4 & 0 & 17 & 1 & 3 \\ 3 & 3 & 4 & -3 & 19 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{1121}3^{11}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -327, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.65 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 2 & 3 \\ 2 & 1 & 5 & 0 & 11 & 3 & 3 \\ 3 & 5 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 11 & 16 & 0 & 44 & 11 & 12 \\ 2 & 3 & 4 & 0 & 11 & 1 & 3 \\ 3 & 3 & 4 & -3 & 12 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{1121}3^{11}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -348, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.78 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 2 & 3 \\ 2 & 1 & 5 & 0 & 8 & 3 & 3 \\ 3 & 5 & 4 & 0 & 11 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 8 & 11 & 0 & 20 & 8 & 8 \\ 2 & 3 & 4 & 0 & 8 & 1 & 3 \\ 3 & 3 & 4 & -3 & 8 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{1114}3^{11}1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_2 + \text{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -383, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.78 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 17 & 3 \\ 2 & 1 & 5 & 0 & 5 & 21 & 3 \\ 3 & 5 & 4 & 0 & 7 & 31 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 38 & 5 \\ 17 & 21 & 31 & 0 & 38 & 172 & 24 \\ 3 & 3 & 4 & -3 & 5 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[20] \quad 1^{11123}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -383, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 7 & 3 \\ 2 & 1 & 5 & 0 & 5 & 9 & 3 \\ 3 & 5 & 4 & 0 & 7 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 16 & 5 \\ 7 & 9 & 13 & 0 & 16 & 28 & 10 \\ 3 & 3 & 4 & -3 & 5 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{11123}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -399, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.65 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 5 & 3 \\ 2 & 1 & 5 & 0 & 11 & 6 & 3 \\ 3 & 5 & 4 & 0 & 16 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 11 & 16 & 0 & 44 & 26 & 12 \\ 5 & 6 & 9 & 0 & 26 & 13 & 7 \\ 3 & 3 & 4 & -3 & 12 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{11123}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -404, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.81 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 7 & 3 \\ 2 & 1 & 5 & 0 & 8 & 9 & 4 \\ 3 & 5 & 4 & 0 & 11 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 8 & 11 & 0 & 20 & 25 & 8 \\ 7 & 9 & 13 & 0 & 25 & 28 & 10 \\ 3 & 3 & 4 & -3 & 8 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1113}1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -432, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.72 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 12 & 3 \\ 2 & 1 & 5 & 0 & 5 & 15 & 3 \\ 3 & 5 & 4 & 0 & 7 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 27 & 5 \\ 12 & 15 & 22 & 0 & 27 & 85 & 17 \\ 3 & 3 & 4 & -3 & 5 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{113}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -460, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.85 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 8 & 2 & 3 \\ 2 & 1 & 5 & 0 & 11 & 3 & 3 \\ 3 & 5 & 4 & 0 & 15 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 8 & 11 & 15 & 0 & 39 & 11 & 11 \\ 2 & 3 & 4 & 0 & 11 & 1 & 3 \\ 3 & 3 & 4 & -3 & 11 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1113}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -531, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.0$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 9 & 3 \\ 2 & 1 & 5 & 0 & 5 & 12 & 3 \\ 3 & 5 & 4 & 0 & 7 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 21 & 5 \\ 9 & 12 & 17 & 0 & 21 & 49 & 13 \\ 3 & 3 & 4 & -3 & 5 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{113}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -560, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 5 & 3 \\ 2 & 1 & 5 & 0 & 14 & 6 & 3 \\ 3 & 5 & 4 & 0 & 20 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 14 & 20 & 0 & 69 & 32 & 15 \\ 5 & 6 & 9 & 0 & 32 & 13 & 7 \\ 3 & 3 & 4 & -3 & 15 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{114}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -572, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 2 & 3 \\ 2 & 1 & 5 & 0 & 14 & 3 & 3 \\ 3 & 5 & 4 & 0 & 19 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 10 & 14 & 19 & 0 & 64 & 14 & 14 \\ 2 & 3 & 4 & 0 & 14 & 1 & 3 \\ 3 & 3 & 4 & -3 & 14 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{11223^1}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -572, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.7$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 7 & 3 \\ 2 & 1 & 5 & 0 & 11 & 9 & 3 \\ 3 & 5 & 4 & 0 & 16 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 11 & 16 & 0 & 44 & 37 & 12 \\ 7 & 9 & 13 & 0 & 37 & 28 & 10 \\ 3 & 3 & 4 & -3 & 12 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{11413^1}1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -623, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.78 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 13 & 5 & 3 \\ 2 & 1 & 5 & 0 & 17 & 6 & 3 \\ 3 & 5 & 4 & 0 & 24 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 17 & 24 & 0 & 100 & 38 & 18 \\ 5 & 6 & 9 & 0 & 38 & 13 & 7 \\ 3 & 3 & 4 & -3 & 18 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{11313^1}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -635, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.67 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 2 & 3 \\ 2 & 1 & 5 & 0 & 14 & 3 & 3 \\ 3 & 5 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 14 & 20 & 0 & 69 & 14 & 15 \\ 2 & 3 & 4 & 0 & 14 & 1 & 3 \\ 3 & 3 & 4 & -3 & 15 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{11143^1}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -647, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.71 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 11 & 3 \\ 2 & 1 & 5 & 0 & 5 & 15 & 3 \\ 3 & 5 & 4 & 0 & 7 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 26 & 5 \\ 11 & 15 & 21 & 0 & 26 & 76 & 16 \\ 3 & 3 & 4 & -3 & 5 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{11143^1}1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -776, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 14 & 3 \\ 2 & 1 & 5 & 0 & 5 & 18 & 3 \\ 3 & 5 & 4 & 0 & 7 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 32 & 5 \\ 14 & 18 & 26 & 0 & 32 & 118 & 20 \\ 3 & 3 & 4 & -3 & 5 & 20 & 1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{11413^1}1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -824, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.64 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 16 & 2 & 3 \\ 2 & 1 & 5 & 0 & 20 & 3 & 3 \\ 3 & 5 & 4 & 0 & 29 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 16 & 20 & 29 & 0 & 148 & 20 & 22 \\ 2 & 3 & 4 & 0 & 20 & 1 & 3 \\ 3 & 3 & 4 & -3 & 22 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{11413^1}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -79, \quad \det = -943, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.70 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 13 & 2 & 3 \\ 2 & 1 & 5 & 0 & 17 & 3 & 3 \\ 3 & 5 & 4 & 0 & 24 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 13 & 17 & 24 & 0 & 100 & 17 & 18 \\ 2 & 3 & 4 & 0 & 17 & 1 & 3 \\ 3 & 3 & 4 & -3 & 18 & 3 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [36] \quad 1^{1111}31^13^11^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 64, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 4 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 5 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 5 & 4 & 0 & 4 & 4 & 4 \\ 1 & 3 & 3 & 3 & 0 & 4 & 1 & 3 \\ 2 & 3 & 4 & 2 & -3 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{111}31^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 101, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 7 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 11 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 7 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 11 & 7 & 0 & 20 & 11 & 8 \\ 2 & 3 & 6 & 3 & 0 & 11 & 4 & 4 \\ 2 & 3 & 4 & 2 & -3 & 8 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{111}31^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 104, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 8 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 12 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 8 & 12 & 8 & 0 & 25 & 12 & 9 \\ 2 & 3 & 6 & 3 & 0 & 12 & 4 & 4 \\ 2 & 3 & 4 & 2 & -3 & 9 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{111}31^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 365, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 4 & 6 & 3 \\ 2 & 4 & 4 & 4 & 0 & 5 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 4 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 5 & 4 & 0 & 4 & 9 & 4 \\ 3 & 6 & 9 & 6 & 0 & 9 & 13 & 7 \\ 2 & 3 & 4 & 2 & -3 & 4 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{111}31^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 480, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 4 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 7 & 6 & 3 \\ 2 & 4 & 4 & 4 & 0 & 11 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 7 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 7 & 11 & 7 & 0 & 20 & 18 & 8 \\ 3 & 6 & 9 & 6 & 0 & 18 & 13 & 7 \\ 2 & 3 & 4 & 2 & -3 & 8 & 7 & 1 \end{pmatrix}_{b_0}$$

6.83 NHC(-2)(-2)(-3) + $\mathbf{f}_4$ + $\mathbf{su}_{2n3\text{mix}}$ + $\mathbf{su}_2$ + $\mathbf{su}_3$

$$T = 10 \quad [0] \ 1^{111}31^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 389, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 12 & 0 & 25 & 8 & 8 & 9 \\ 2 & 3 & 4 & 0 & 8 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 8 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 9 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

6.84 NHC(-2)(-2)(-3) + $\mathbf{f}_4$ + $\mathbf{su}_{2n3\text{mix}}$ + $\mathbf{su}_3$

$$T = 9 \quad [0] \ 1^{112}31^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -82, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 3 & 3 \\ 2 & 1 & 5 & 0 & 8 & 3 & 3 \\ 3 & 5 & 4 & 0 & 11 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 6 & 8 & 11 & 0 & 20 & 9 & 8 \\ 3 & 3 & 5 & 0 & 9 & 3 & 3 \\ 3 & 3 & 4 & -3 & 8 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}31^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -89, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.37 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 2 & 3 \\ 2 & 1 & 5 & 0 & 8 & 3 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 8 & 12 & 0 & 25 & 8 & 9 \\ 2 & 3 & 4 & 0 & 8 & 0 & 2 \\ 3 & 3 & 4 & -3 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}31^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -97, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 5 & 3 & 3 \\ 4 & 5 & 9 & 0 & 12 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 5 & 12 & 0 & 12 & 5 & 6 \\ 2 & 3 & 4 & 0 & 5 & 0 & 2 \\ 3 & 3 & 5 & -3 & 6 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.11 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 3 & 3 \\ 2 & 1 & 5 & 0 & 5 & 3 & 3 \\ 3 & 5 & 4 & 0 & 7 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 6 & 5 \\ 3 & 3 & 5 & 0 & 6 & 3 & 3 \\ 3 & 3 & 4 & -3 & 5 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 7 & 0 & 7 & 5 & 5 \\ 2 & 3 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{131}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 2 & 3 \\ 3 & 4 & 6 & 0 & 18 & 3 & 4 \\ 3 & 6 & 4 & 0 & 20 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 11 & 18 & 20 & 0 & 69 & 14 & 15 \\ 2 & 3 & 4 & 0 & 14 & 0 & 2 \\ 3 & 4 & 4 & -3 & 15 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 7 & 2 & 3 \\ 2 & 1 & 5 & 0 & 8 & 3 & 3 \\ 4 & 5 & 9 & 0 & 16 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 8 & 16 & 0 & 25 & 8 & 9 \\ 2 & 3 & 4 & 0 & 8 & 0 & 2 \\ 3 & 3 & 5 & -3 & 9 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -202, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 8 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 8 & 0 & 12 & 5 & 6 \\ 2 & 3 & 4 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -329, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 5 & 3 \\ 2 & 1 & 5 & 0 & 5 & 6 & 3 \\ 3 & 5 & 4 & 0 & 7 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 11 & 5 \\ 5 & 6 & 9 & 0 & 11 & 12 & 6 \\ 3 & 3 & 4 & -3 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -362, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.17 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 2 & 3 \\ 2 & 1 & 5 & 0 & 5 & 3 & 3 \\ 3 & 5 & 4 & 0 & 7 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 5 & 7 & 0 & 7 & 5 & 5 \\ 2 & 3 & 4 & 0 & 5 & 0 & 2 \\ 3 & 3 & 4 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{121}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -466, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 2 & 3 \\ 3 & 4 & 6 & 0 & 15 & 3 & 4 \\ 3 & 6 & 4 & 0 & 16 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 15 & 16 & 0 & 44 & 11 & 12 \\ 2 & 3 & 4 & 0 & 11 & 0 & 2 \\ 3 & 4 & 4 & -3 & 12 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -481, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 12 & 0 & 25 & 8 & 9 \\ 2 & 3 & 4 & 0 & 8 & 0 & 2 \\ 3 & 4 & 4 & -3 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

6.85 NHC(-2)(-2)(-3) + $\mathbf{f}_4$ + $\mathbf{su}_{2n3\text{mix}}$ + $\mathbf{su}_3$ + $\mathbf{su}_2$

$$T = 10 \quad [0] \ 1^{111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 389, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 12 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 7 & 12 & 12 & 0 & 25 & 8 & 8 & 9 \\ 2 & 3 & 4 & 0 & 8 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 8 & 3 & 1 & 3 \\ 3 & 4 & 4 & -3 & 9 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

6.86 NHC(-2)(-2)(-3) + $\mathbf{f}_4$ + $\mathbf{su}_3$

$$T = 8 \quad [0] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 3 & 5 & 0 & 3 & 3 \\ 3 & 3 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 4 & 5 & 9 & 0 & 16 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 9 & 16 & 0 & 27 & 9 \\ 3 & 3 & 5 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 3 & 5 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 1 & 3 \\ 3 & 4 & 9 & 3 & 0 & 4 \\ 4 & 9 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 5 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^1 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12, T_{\max} = 28, T \in [12, 28]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 17 & 0 & 48 & 12 \\ 3 & 4 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 118, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.20 \rightarrow 14, T_{\max} = 28, T \in [14, 28]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 8 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 3 & 8 & 0 & 3 & 3 \\ 3 & 3 & 5 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{11}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{112}3^11^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 10 & 12 & 18 & 0 & 57 & 13 \\ 3 & 3 & 4 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 6 & 9 & 0 & 12 & 6 \\ 3 & 3 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{113}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 12.72 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 12 & 0 & 24 & 8 \\ 3 & 3 & 4 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 3 & 4 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{112}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 0 & 9 & 5 \\ 3 & 3 & 4 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

[12] $1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 159, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.34 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 6 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 5 & 0 & 3 & 3 \\ 3 & 4 & 4 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

[13] $1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 4 & 5 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 6 & 12 & 0 & 12 & 6 \\ 3 & 3 & 5 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[14] $1^{111}3^11^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 215, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 13 & 0 & 27 & 9 \\ 3 & 4 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 236, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{112}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 9 & 13 & 0 & 27 & 9 \\ 3 & 3 & 4 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{113}3^11^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.81 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 12 & 15 & 22 & 0 & 84 & 16 \\ 3 & 3 & 4 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{113}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 343, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 17 & 0 & 48 & 12 \\ 3 & 3 & 4 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [21] $1^{11}3^121^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 3 \\ 2 & 2 & 4 & 4 & 0 & 4 \\ 3 & 4 & 4 & 4 & 0 & 4 \\ 2 & 4 & 4 & 1 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 3 & -3 & 1 \end{pmatrix}_{b_0}$$

[20] $1^{11}3^121^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 2 & 6 & 0 & 4 & 4 \\ 3 & 6 & 5 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 5 & 0 & 1 & 3 \\ 3 & 4 & 5 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[19] $1^{11}3^121^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -26, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.46 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 5 & 9 & 10 & 1 & 5 \\ 9 & 13 & 16 & 0 & 7 \\ 10 & 16 & 18 & 1 & 8 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 7 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

[23] $1^{11}3^121^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 0 \\ 1 & 2 & 1 & 1 & 3 & 0 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 1 & 1 & 3 & 0 & -2 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 3 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[22] $1^{11}3^121^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.16 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 5 & 0 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 5 & 0 & 1 & 3 \\ 3 & 4 & 5 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$

[24] $1^{111}3^121^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -35, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.14 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 3 & 1 & 0 & 3 & 2 \\ 1 & 3 & 0 & 2 & 0 & 3 & 2 \\ 1 & 1 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 4 & 0 & 4 & 4 \\ 2 & 2 & 2 & 2 & -3 & 4 & 0 \end{pmatrix}_{b_0}$$

[25] $1^{11}3^121^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -45, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$

$$T_{\min} = 12.38 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 13 & 1 & 5 \\ 9 & 13 & 21 & 0 & 7 \\ 13 & 21 & 30 & 0 & 10 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 10 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{12}321^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 3 & 0 & 5 \\ 1 & 2 & 0 & 2 & 0 & 5 \\ 1 & 3 & 2 & 0 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 3 & 5 & 5 & 0 & 12 & 6 \\ 2 & 3 & 2 & 2 & -3 & 6 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{11}321^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -59, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 2 & 0 & 1 \\ 1 & 3 & 2 & 0 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 \\ 1 & 3 & 1 & 3 & 0 & 0 \\ 2 & 3 & 2 & 2 & -3 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{11}321^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 2 & 6 & 0 & 11 & 4 \\ 3 & 6 & 5 & 0 & 15 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 7 & 11 & 15 & 0 & 31 & 11 \\ 3 & 4 & 5 & -3 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{11}31^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -65, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 4 & 7 & 0 & 3 & 4 \\ 2 & 3 & 7 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 4 & 3 & -3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{11}321^{113} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -97, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 3 \\ 2 & 2 & 6 & 0 & 15 & 4 \\ 3 & 6 & 5 & 0 & 20 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 20 & 0 & 57 & 15 \\ 3 & 4 & 5 & -3 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{11}321^{114} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -101, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.05 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 \\ 2 & 2 & 6 & 0 & 19 & 4 \\ 3 & 6 & 5 & 0 & 25 & 5 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 11 & 19 & 25 & 0 & 91 & 19 \\ 3 & 4 & 5 & -3 & 19 & 1 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{111}31^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 3 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{11}31^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -116, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 & 3 \\ 2 & 4 & 4 & 4 & 0 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 6 & 3 & 0 & 3 & 3 \\ 2 & 3 & 4 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{112}321^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -124, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 3 & 1 & 0 & 5 & 2 \\ 1 & 3 & 0 & 2 & 0 & 5 & 2 \\ 1 & 1 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 5 & 5 & 5 & 0 & 12 & 6 \\ 2 & 2 & 2 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{11}321^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 3 & 0 & 4 & 3 \\ 1 & 2 & 0 & 2 & 0 & 3 & 2 \\ 1 & 3 & 2 & 0 & 0 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 4 & 3 & 4 & 0 & 4 & 4 \\ 2 & 3 & 2 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{13}321^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 3 & 0 & 9 & 3 \\ 1 & 2 & 0 & 2 & 0 & 8 & 2 \\ 1 & 3 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 8 & 9 & 0 & 39 & 11 \\ 2 & 3 & 2 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{11}321^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -154, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 3 & 0 & 7 & 3 \\ 1 & 2 & 0 & 2 & 0 & 4 & 2 \\ 1 & 3 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 7 & 4 & 7 & 0 & 15 & 7 \\ 2 & 3 & 2 & 2 & -3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[38] \ 1^{11}31^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -158, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 3 & 4 \\ 2 & 7 & 4 & 4 & 0 & 3 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[39] \ 1^{11}31^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -176, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.34 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 3 & 4 \\ 1 & 4 & 1 & 4 & 0 & 3 & 3 \\ 2 & 6 & 4 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 1 & 3 & 3 & 3 & 0 & 0 & 2 \\ 2 & 4 & 3 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[40] \ 1^{12}321^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -229, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.4$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 3 & 0 & 8 & 3 \\ 1 & 2 & 0 & 2 & 0 & 6 & 2 \\ 1 & 3 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 8 & 6 & 8 & 0 & 25 & 9 \\ 2 & 3 & 2 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

$$[41] \ 1^{111}31^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -312, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 3 & 0 & 6 & 3 \\ 2 & 4 & 4 & 4 & 0 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 9 & 6 & 0 & 12 & 6 \\ 2 & 3 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[42] \ 1^{11}31^{111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -417, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.19 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 3 & 2 \\ 2 & 4 & 7 & 3 & 0 & 9 & 4 \\ 2 & 7 & 4 & 4 & 0 & 9 & 4 \\ 1 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 9 & 9 & 6 & 0 & 12 & 6 \\ 2 & 4 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[43] \ 1^{13}31^{111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -536, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.45 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 7 & 3 & 0 & 6 & 4 \\ 2 & 7 & 4 & 4 & 0 & 6 & 4 \\ 1 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 6 & 6 & 3 & 0 & 3 & 3 \\ 2 & 4 & 4 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[44] \ 2^1 2^1 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -1126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 & 2 \\ 2 & 1 & 4 & 3 & 1 & 3 & 3 \\ 2 & 4 & 1 & 4 & 1 & 3 & 3 \\ 2 & 3 & 4 & 0 & 1 & 3 & 2 \\ 1 & 1 & 1 & 1 & -3 & 1 & -1 \\ 2 & 3 & 3 & 3 & 1 & 0 & 2 \\ 2 & 3 & 3 & 2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

6.87 NHC(-2)(-2)(-3) + f₄ + su₃ + so₈

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 11 & 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 17 & 0 & 12 & 6 \\ 5 & 6 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -213, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.24 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 20 & 0 & 12 & 6 \\ 5 & 6 & 9 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -440, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 17 & 0 & 12 & 6 \\ 5 & 7 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [3] \quad 1^{11} 3^1 21^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -87, \det = 56, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 1 & 1 & 3 & 0 & 1 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 3 & 0 & -2 & 0 & 1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & 1 & 0 & -2 \\ 2 & 2 & 3 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

6.88 NHC(-2)(-2)(-3) + f₄ + su₃ + su₂

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 12 & 12 & 6 \\ 4 & 6 & 8 & 0 & 12 & 10 & 6 \\ 3 & 3 & 4 & -3 & 6 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 3 \\ 3 & 4 & 4 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 3 & 6 & 5 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

- [3] $1^{111}31^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -234$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 13 & 0 & 27 & 9 & 9 \\ 2 & 3 & 4 & 0 & 9 & 1 & 3 \\ 3 & 4 & 4 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$
- [4] $1^{113}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -241$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 3 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 19 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 11 & 21 & 19 & 0 & 12 & 73 & 15 \\ 3 & 4 & 4 & -3 & 2 & 15 & 1 \end{pmatrix}_{b_0}$$
- [5] $1^{111}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -245$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 12 & 6 & 6 \\ 2 & 3 & 4 & 0 & 6 & 1 & 3 \\ 3 & 3 & 4 & -3 & 6 & 3 & 1 \end{pmatrix}_{b_0}$$
- [6] $1^{111}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -294$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 12 & 15 & 6 \\ 5 & 6 & 9 & 0 & 15 & 13 & 7 \\ 3 & 3 & 4 & -3 & 6 & 7 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{111}31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -318$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.91 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 7 & 12 & 13 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$
- [8] $1^{111}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -381$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{112}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -423$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.7$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 8 & 3 \\ 3 & 4 & 6 & 0 & 3 & 15 & 4 \\ 3 & 6 & 4 & 0 & 4 & 14 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 8 & 15 & 14 & 0 & 9 & 37 & 11 \\ 3 & 4 & 4 & -3 & 2 & 11 & 1 \end{pmatrix}_{b_0}$$
- [10] $1^{112}31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -486$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.7$
 $T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 14 & 3 \\ 3 & 4 & 6 & 0 & 3 & 24 & 4 \\ 3 & 6 & 4 & 0 & 4 & 26 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 14 & 24 & 26 & 0 & 18 & 118 & 20 \\ 3 & 4 & 4 & -3 & 2 & 20 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{111}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -539$, $(n_+, n_-, n_0) =$

$$(1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 7 & 3 \\ 2 & 1 & 5 & 0 & 6 & 9 & 3 \\ 3 & 5 & 4 & 0 & 9 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 12 & 21 & 6 \\ 7 & 9 & 13 & 0 & 21 & 28 & 10 \\ 3 & 3 & 4 & -3 & 6 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{1113} 3^1 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -686, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 9 & 3 \\ 2 & 1 & 5 & 0 & 6 & 12 & 3 \\ 3 & 5 & 4 & 0 & 9 & 17 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 12 & 27 & 6 \\ 9 & 12 & 17 & 0 & 27 & 49 & 13 \\ 3 & 3 & 4 & -3 & 6 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{1114} 3^1 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 11 & 3 \\ 2 & 1 & 5 & 0 & 6 & 15 & 3 \\ 3 & 5 & 4 & 0 & 9 & 21 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 12 & 33 & 6 \\ 11 & 15 & 21 & 0 & 33 & 76 & 16 \\ 3 & 3 & 4 & -3 & 6 & 16 & 1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{112} 3^1 1^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -738, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 10 & 3 \\ 3 & 4 & 6 & 0 & 3 & 18 & 4 \\ 3 & 6 & 4 & 0 & 4 & 18 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 10 & 18 & 18 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & -3 & 2 & 14 & 1 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{112} 3^1 1^{113} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -759, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 12 & 3 \\ 3 & 4 & 6 & 0 & 3 & 21 & 4 \\ 3 & 6 & 4 & 0 & 4 & 22 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 12 & 21 & 22 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & -3 & 2 & 17 & 1 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{114} 3^1 1^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -906, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 16 & 3 \\ 3 & 4 & 6 & 0 & 3 & 30 & 4 \\ 3 & 6 & 4 & 0 & 4 & 28 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 16 & 30 & 28 & 0 & 18 & 154 & 22 \\ 3 & 4 & 4 & -3 & 2 & 22 & 1 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{113} 3^1 1^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -934, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 13 & 3 \\ 3 & 4 & 6 & 0 & 3 & 24 & 4 \\ 3 & 6 & 4 & 0 & 4 & 23 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 13 & 24 & 23 & 0 & 15 & 100 & 18 \\ 3 & 4 & 4 & -3 & 2 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{113} 3^1 1^{113} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 15 & 3 \\ 3 & 4 & 6 & 0 & 3 & 27 & 4 \\ 3 & 6 & 4 & 0 & 4 & 27 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 15 & 27 & 27 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & -3 & 2 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{113} 3^1 1^{114} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1438, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 17 & 3 \\ 3 & 4 & 6 & 0 & 3 & 30 & 4 \\ 3 & 6 & 4 & 0 & 4 & 31 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 17 & 30 & 31 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & -3 & 2 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{114}31^{113} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1683, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 18 & 3 \\ 3 & 4 & 6 & 0 & 3 & 33 & 4 \\ 3 & 6 & 4 & 0 & 4 & 32 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 18 & 33 & 32 & 0 & 21 & 193 & 25 \\ 3 & 4 & 4 & -3 & 2 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{114}31^{114} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -2166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 20 & 3 \\ 3 & 4 & 6 & 0 & 3 & 36 & 4 \\ 3 & 6 & 4 & 0 & 4 & 36 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 24 & 2 \\ 20 & 36 & 36 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & -3 & 2 & 28 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [22] \ 1^{111}321^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 64, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 4 & 3 & 3 \\ 1 & 2 & 0 & 2 & 0 & 3 & 1 & 2 \\ 1 & 3 & 2 & 0 & 0 & 4 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 4 & 0 & 4 & 4 & 4 \\ 1 & 3 & 1 & 3 & 0 & 4 & 1 & 3 \\ 2 & 3 & 2 & 2 & -3 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{111}321^{1131} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 76, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 2 & 3 \\ 2 & 2 & 6 & 0 & 15 & 4 & 4 \\ 3 & 6 & 5 & 0 & 20 & 5 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 9 & 15 & 20 & 0 & 57 & 15 & 15 \\ 2 & 4 & 5 & 0 & 15 & 2 & 4 \\ 3 & 4 & 5 & -3 & 15 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{111}321^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 365, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 1 & 2 & 3 & 0 & 4 & 4 & 3 \\ 1 & 2 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 3 & 2 & 0 & 0 & 4 & 4 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 4 & 3 & 4 & 0 & 4 & 7 & 4 \\ 2 & 4 & 3 & 4 & 0 & 7 & 5 & 5 \\ 2 & 3 & 2 & 2 & -3 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$

6.89 NHC(-2)(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + $\mathfrak{su}_2$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 10 \quad [0] \ 1^{111}3^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 194, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^{1111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 244, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 & 2 \\ 3 & 6 & 5 & 0 & 3 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}3^1}1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 302, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 2 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 12 & 6 & 5 & 6 \\ 2 & 3 & 4 & 0 & 6 & 1 & 2 & 3 \\ 2 & 2 & 3 & 0 & 5 & 2 & 0 & 2 \\ 3 & 3 & 4 & -3 & 6 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}3^1}1^{\overline{112}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 484, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 13 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 & 2 \\ 7 & 12 & 13 & 0 & 9 & 28 & 7 & 10 \\ 2 & 3 & 3 & 0 & 2 & 7 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}3^1}1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 622, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 & 2 \\ 5 & 9 & 9 & 0 & 6 & 13 & 5 & 7 \\ 2 & 3 & 3 & 0 & 2 & 5 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

6.90 NHC(-2)(-2)(-3) + f₄ + su₃ + su₂ + su₂n3mix

$$T = 10 \quad [0] \ 1^{\overline{111}3^1}1^{\overline{1111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 389, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 3 & 8 & 2 \\ 2 & 3 & 4 & 0 & 3 & 1 & 8 & 3 \\ 7 & 12 & 12 & 0 & 8 & 8 & 25 & 9 \\ 3 & 4 & 4 & -3 & 2 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

6.91 NHC(-2)(-2)(-3) + f₄ + su₃ + su₂ → (-3)

$$T = 9 \quad [0] \ 1^{\overline{11}3^1}1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -54, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.52 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 2 & 3 \\ 3 & 5 & 4 & 0 & 5 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 2 & 3 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}3^1}1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -100, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 2 & 3 \\ 4 & 5 & 9 & 0 & 4 & 4 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 & 2 \\ 2 & 2 & 4 & 0 & 2 & 0 & 2 \\ 3 & 3 & 5 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{112}3^1}1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -146, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.49 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 2 & 3 \\ 3 & 5 & 4 & 0 & 8 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 4 & 6 & 8 & 0 & 9 & 4 & 5 \\ 2 & 2 & 3 & 0 & 4 & 0 & 2 \\ 3 & 3 & 4 & -3 & 5 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -194, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.12 \rightarrow 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 2 & 3 \\ 3 & 5 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 & 2 \\ 2 & 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 3 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -210, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 2 & 3 \\ 3 & 5 & 4 & 0 & 9 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 12 & 5 & 6 \\ 2 & 2 & 3 & 0 & 5 & 0 & 2 \\ 3 & 3 & 4 & -3 & 6 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -226, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \quad T_{\max} = 13, \quad \Delta = -88, \quad \det = -334, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.92 \rightarrow 12, \quad T_{\max} = 13, \quad T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 2 & 3 \\ 2 & 1 & 5 & 0 & 9 & 2 & 3 \\ 3 & 5 & 4 & 0 & 13 & 3 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 9 & 13 & 0 & 27 & 7 & 9 \\ 2 & 2 & 3 & 0 & 7 & 0 & 2 \\ 3 & 3 & 4 & -3 & 9 & 2 & 1 \end{pmatrix}_{b_0}$$

6.92 NHC(-2)(-2)(-3) + $\mathbf{f}_4$ + $\mathbf{su}_3$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{su}_2$

$$T = 10 \quad [0] \quad 1^{111}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 194, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 & 3 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & -3 & 2 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 244, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 4 & 3 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 & 3 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 3 & 2 \\ 3 & 6 & 5 & 0 & 3 & 3 & 4 & 4 \\ 3 & 4 & 4 & -3 & 2 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \quad T_{\max} = 12, \quad \Delta = -58, \quad \det = 302, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 6 & 2 & 3 & 3 \\ 3 & 5 & 4 & 0 & 9 & 3 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 12 & 5 & 6 & 6 \\ 2 & 2 & 3 & 0 & 5 & 0 & 2 & 2 \\ 2 & 3 & 4 & 0 & 6 & 2 & 1 & 3 \\ 3 & 3 & 4 & -3 & 6 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^11^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 484, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 3 & 13 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 & 9 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 7 & 2 \\ 7 & 12 & 13 & 0 & 9 & 7 & 28 & 10 \\ 3 & 4 & 4 & -3 & 2 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^11^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 622, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 3 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 2 & 6 & 2 \\ 2 & 3 & 3 & 0 & 2 & 0 & 5 & 2 \\ 5 & 9 & 9 & 0 & 6 & 5 & 13 & 7 \\ 3 & 4 & 4 & -3 & 2 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

6.93 NHC(-2)(-2)(-3) + f₄ + su₃ + su₂n3mix

$$T = 9 \quad [0] \ 1^{112}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -82, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 6 & 3 \\ 2 & 1 & 5 & 0 & 3 & 8 & 3 \\ 3 & 5 & 4 & 0 & 5 & 11 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 5 & 0 & 3 & 9 & 3 \\ 6 & 8 & 11 & 0 & 9 & 20 & 8 \\ 3 & 3 & 4 & -3 & 3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -89, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.37 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 7 & 3 \\ 2 & 1 & 5 & 0 & 3 & 8 & 3 \\ 3 & 5 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 8 & 2 \\ 7 & 8 & 12 & 0 & 8 & 25 & 9 \\ 3 & 3 & 4 & -3 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11}3^11^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -97, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.55 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 5 & 3 \\ 4 & 5 & 9 & 0 & 4 & 12 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 5 & 2 \\ 5 & 5 & 12 & 0 & 5 & 12 & 6 \\ 3 & 3 & 5 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -117, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.11 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 4 & 3 \\ 2 & 1 & 5 & 0 & 3 & 5 & 3 \\ 3 & 5 & 4 & 0 & 5 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 5 & 0 & 3 & 6 & 3 \\ 4 & 5 & 7 & 0 & 6 & 7 & 5 \\ 3 & 3 & 4 & -3 & 3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^11^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -127, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 5 & 2 \\ 4 & 6 & 7 & 0 & 5 & 7 & 5 \\ 3 & 4 & 4 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111}3^11^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{f}_4 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -157, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.26 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 11 & 3 \\ 3 & 4 & 6 & 0 & 3 & 18 & 4 \\ 3 & 6 & 4 & 0 & 4 & 20 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 14 & 2 \\ 11 & 18 & 20 & 0 & 14 & 69 & 15 \\ 3 & 4 & 4 & -3 & 2 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -175, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 7 & 3 \\ 2 & 1 & 5 & 0 & 3 & 8 & 3 \\ 4 & 5 & 9 & 0 & 4 & 16 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 8 & 2 \\ 7 & 8 & 16 & 0 & 8 & 25 & 9 \\ 3 & 3 & 5 & -3 & 2 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -202, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 8 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 5 & 2 \\ 5 & 9 & 8 & 0 & 5 & 12 & 6 \\ 3 & 4 & 4 & -3 & 2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -329, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 5 & 3 \\ 3 & 5 & 4 & 0 & 9 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 6 & 9 & 0 & 12 & 11 & 6 \\ 4 & 5 & 7 & 0 & 11 & 7 & 5 \\ 3 & 3 & 4 & -3 & 6 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -362, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.17 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 4 & 3 \\ 2 & 1 & 5 & 0 & 3 & 5 & 3 \\ 3 & 5 & 4 & 0 & 4 & 7 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 5 & 2 \\ 4 & 5 & 7 & 0 & 5 & 7 & 5 \\ 3 & 3 & 4 & -3 & 2 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{112}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -466, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 9 & 3 \\ 3 & 4 & 6 & 0 & 3 & 15 & 4 \\ 3 & 6 & 4 & 0 & 4 & 16 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 11 & 2 \\ 9 & 15 & 16 & 0 & 11 & 44 & 12 \\ 3 & 4 & 4 & -3 & 2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -481, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 0 & 4 & 12 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 8 & 2 \\ 7 & 12 & 12 & 0 & 8 & 25 & 9 \\ 3 & 4 & 4 & -3 & 2 & 9 & 1 \end{pmatrix}_{b_0}$$

6.94 NHC(-2)(-2)(-3) + $\mathfrak{f}_4$ + $\mathfrak{su}_3$ + $\mathfrak{su}_{2n3\text{mix}}$ + $\mathfrak{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{1111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 389, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 7 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 12 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 12 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 8 & 3 & 2 \\ 7 & 12 & 12 & 0 & 8 & 25 & 8 & 9 \\ 2 & 3 & 4 & 0 & 3 & 8 & 1 & 3 \\ 3 & 4 & 4 & -3 & 2 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

6.95 NHC(-2)(-2)(-3) + f₄ + su₃ + su₃

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -95, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.86 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 8 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 3 & 3 & 8 & 0 & 3 & 3 & 3 \\ 3 & 3 & 5 & -3 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 3 & 3 & 5 & 0 & 3 & 3 & 3 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & -3 & 3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 4 & 5 & 9 & 0 & 4 & 12 & 5 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 5 & 6 & 12 & 0 & 6 & 12 & 6 \\ 3 & 3 & 5 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -246, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 4 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -281, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 4 & 5 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 3 & 6 & 5 & 0 & 3 & 3 & 3 \\ 3 & 4 & 4 & -3 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 0 & 3 & 6 & 3 \\ 3 & 5 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 5 & 6 & 9 & 0 & 6 & 12 & 6 \\ 3 & 3 & 4 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -441, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & -3 & 2 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 3^1 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{f}_4 + \text{su}_3 + \text{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 0 & 4 & 9 & 4 \\ 1 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & -3 & 2 & 6 & 0 \end{pmatrix}_{b_0}$$

6.96 NHC(-2)(-2)(-3) + f₄ + (-3)→(-2)

$$T = 8 \quad [0] \quad 1^{11}31^{11}2^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 31, \Delta = -117, \det = 952, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 31, T \in [12, 31]$$

$$\begin{pmatrix} 4 & 6 & 8 & 3 & 1 & 4 \\ 6 & 4 & 9 & 3 & 0 & 4 \\ 8 & 9 & 9 & 4 & 0 & 5 \\ 3 & 3 & 4 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 4 & 4 & 5 & 0 & -1 & 1 \end{pmatrix}_{b_0}$$

6.97 NHC(-2)(-2)(-3) + NHC(-2)(-3)

$$T = 11 \quad [0] \quad 1^{11}5^{11}131^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 3 \\ 2 & 1 & 4 & 2 & 0 & 3 \\ 3 & 4 & 5 & 5 & 0 & 5 \\ 2 & 2 & 5 & 2 & 1 & 4 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 3 & 5 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

6.98 NHC(-2)(-2)(-3) + so₈

$$T = 7 \quad [1] \quad 1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 11 & 16 & 6 \\ 11 & 10 & 17 & 6 \\ 16 & 17 & 25 & 9 \\ 6 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[0] \quad 21^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -9, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 15, T_{\max} = 22, T \in [15, 22]$$

$$\begin{pmatrix} 13 & 12 & 17 & 7 \\ 12 & 10 & 17 & 6 \\ 17 & 17 & 20 & 8 \\ 7 & 6 & 8 & -1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^1 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 5 & 8 & 12 & 5 \\ 8 & 10 & 17 & 6 \\ 12 & 17 & 25 & 9 \\ 5 & 6 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.38 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 4 & 5 & 9 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 3 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 3 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$$

$$\begin{pmatrix} 10 & 12 & 15 & 6 \\ 12 & 13 & 18 & 7 \\ 15 & 18 & 20 & 8 \\ 6 & 7 & 8 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.15 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 1 & 2 & 0 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & -2 & 2 \end{pmatrix}_{b_0}$$

[10] $1^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 10 & 11 & 15 & 6 \\ 11 & 10 & 17 & 6 \\ 15 & 17 & 20 & 8 \\ 6 & 6 & 8 & 2 \end{pmatrix}_{b_0}$$

[9] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -50, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 3 & 3 & 0 & 2 & 2 \\ 2 & 2 & 2 & -1 & 2 & 1 \end{pmatrix}_{b_0}$$

[11] $1^131^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 22, T \in [12, 22]$

$$\begin{pmatrix} 5 & 9 & 11 & 5 \\ 9 & 13 & 18 & 7 \\ 11 & 18 & 20 & 8 \\ 5 & 7 & 8 & 2 \end{pmatrix}_{b_0}$$

[12] $21^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -53, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 12 & 20 & 7 \\ 12 & 10 & 17 & 6 \\ 20 & 17 & 25 & 9 \\ 7 & 6 & 9 & 3 \end{pmatrix}_{b_0}$$

[13] $1^131^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.05 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 5 & 8 & 11 & 5 \\ 8 & 10 & 17 & 6 \\ 11 & 17 & 20 & 8 \\ 5 & 6 & 8 & 1 \end{pmatrix}_{b_0}$$

[14] $1^131^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 22, T \in [13, 22]$

$$\begin{pmatrix} 5 & 9 & 12 & 5 \\ 9 & 13 & 21 & 7 \\ 12 & 21 & 25 & 9 \\ 5 & 7 & 9 & 1 \end{pmatrix}_{b_0}$$

[15] $21^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 12 & 16 & 6 \\ 12 & 13 & 18 & 7 \\ 16 & 18 & 20 & 8 \\ 6 & 7 & 8 & 3 \end{pmatrix}_{b_0}$$

[16] $21^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 15 & 17 & 7 \\ 15 & 13 & 18 & 7 \\ 17 & 18 & 20 & 8 \\ 7 & 7 & 8 & 3 \end{pmatrix}_{b_0}$$

[17] $21^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -198, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 10 & 12 & 16 & 6 \\ 12 & 10 & 17 & 6 \\ 16 & 17 & 20 & 8 \\ 6 & 6 & 8 & 3 \end{pmatrix}_{b_0}$$

[18] $21^{11}31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 7, T_{\max} = 22, \Delta = -113, \det = -267, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 22, T \in [11, 22]$

$$\begin{pmatrix} 13 & 15 & 20 & 7 \\ 15 & 13 & 21 & 7 \\ 20 & 21 & 25 & 9 \\ 7 & 7 & 9 & 3 \end{pmatrix}_{b_0}$$

$T = 8$ [19] $1^1321^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 2 & 6 & 0 & 4 \\ 3 & 6 & 5 & 0 & 5 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 4 & 5 & -2 & 3 \end{pmatrix}_{b_0}$$

[20] $1^1 1^1 3 2 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 10 & 5 & 0 & 6 \\ 2 & 5 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 6 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[21] $2 1^1 4 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 8 & 16 & 6 \\ 8 & 5 & 13 & 5 \\ 16 & 13 & 25 & 9 \\ 6 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[22] $1^1 4 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 8 & 16 & 6 \\ 8 & 5 & 13 & 5 \\ 16 & 13 & 25 & 9 \\ 6 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$

[23] $2 1^1 4 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 11, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 13 & 9 & 14 & 7 \\ 9 & 5 & 11 & 5 \\ 14 & 11 & 13 & 7 \\ 7 & 5 & 7 & 3 \end{pmatrix}_{b_0}$$

[24] $2 2 3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 2 & 2 & 1 & 2 \\ 1 & 0 & 2 & 1 & 2 & 2 \\ 2 & 2 & 1 & 4 & 1 & 3 \\ 2 & 1 & 4 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & -2 & 0 \\ 2 & 2 & 3 & 2 & 0 & 2 \end{pmatrix}_{b_0}$$

[25] $1^{11} 1^1 3 1^1 3 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 15, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.33 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 7 & 7 & 8 & 4 \\ 7 & 10 & 13 & 12 & 6 \\ 7 & 13 & 10 & 13 & 6 \\ 8 & 12 & 13 & 12 & 6 \\ 4 & 6 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$

[27] $2 1^1 4 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 12 & 12 & 6 \\ 12 & 13 & 15 & 7 \\ 12 & 15 & 13 & 7 \\ 6 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[26] $1^1 1^1 4 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 10 & 12 & 12 & 6 \\ 12 & 13 & 15 & 7 \\ 12 & 15 & 13 & 7 \\ 6 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$

[28] $2 1^1 4 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13.37 \rightarrow 14, T_{\max} = 20, T \in [14, 20]$

$$\begin{pmatrix} 5 & 7 & 12 & 5 \\ 7 & 5 & 13 & 5 \\ 12 & 13 & 25 & 9 \\ 5 & 5 & 9 & 1 \end{pmatrix}_{b_0}$$

[29] $1^{11} 1^1 3 2 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 3 & 6 & 5 & 5 & 5 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[30] $1^{11} 1^1 3 1^1 3 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 20, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 4 & 0 & 6 & 3 \\ 1 & 4 & 1 & 1 & 6 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 6 & 6 & 0 & 11 & 5 \\ 2 & 3 & 3 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

- [31] $1^{11}31^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 21$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 1 & 5 & 3 \\ 4 & 10 & 1 & 12 & 6 \\ 1 & 1 & -2 & 1 & 0 \\ 5 & 12 & 1 & 12 & 6 \\ 3 & 6 & 0 & 6 & 2 \end{pmatrix}_{b_0}$
- [33] $1^{11}3131^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 21$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 1 & 3 & 0 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 3 & 1 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 0 & 2 & -2 & 1 \end{pmatrix}_{b_0}$
- [32] $1^{11}31^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 21$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.43 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 3 & 3 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 3 & 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 3 & 2 & 2 \\ 2 & 3 & 0 & 2 & 2 & 1 \end{pmatrix}_{b_0}$
- [35] $1^{11}321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 24$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 3 & 0 & 3 & 3 \\ 2 & 3 & 2 & -1 & 3 & 2 \end{pmatrix}_{b_0}$
- [34] $1^{11}321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 24$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 1 & 2 & 3 & 0 & 3 \\ 1 & 2 & 0 & 2 & 0 & 2 \\ 1 & 3 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 2 & 2 & -2 & 2 \end{pmatrix}_{b_0}$
- [36] $1^{11}31^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 28$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.29 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 4 & 5 & 1 & 3 \\ 4 & 10 & 13 & 0 & 6 \\ 5 & 13 & 13 & 1 & 7 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 6 & 7 & -1 & 2 \end{pmatrix}_{b_0}$
- [38] $1^1 4^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 12.38 \rightarrow 13$, $T_{\max} = 20$, $T \in [13, 20]$
 $\begin{pmatrix} 5 & 6 & 12 & 5 \\ 6 & 5 & 13 & 5 \\ 12 & 13 & 25 & 9 \\ 5 & 5 & 9 & 2 \end{pmatrix}_{b_0}$
- [37] $1^1 4^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 32$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 1 & 4 & 0 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & -2 & 2 \end{pmatrix}_{b_0}$
- [39] $2^1 2^1 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 34$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.88 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 4 & 6 & 5 & 5 & 5 \\ 3 & 3 & 5 & 0 & 2 \\ 3 & 4 & 5 & 2 & 3 \end{pmatrix}_{b_0}$
- [40] $1^1 4^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 20$, $\Delta = -104$, $\det = 35$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 11.83 \rightarrow 12$, $T_{\max} = 20$, $T \in [12, 20]$
 $\begin{pmatrix} 5 & 6 & 9 & 5 \\ 6 & 5 & 11 & 5 \\ 9 & 11 & 13 & 7 \\ 5 & 5 & 7 & 3 \end{pmatrix}_{b_0}$
- [41] $1^{11}321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 11$, $\Delta = -85$, $\det = 36$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 8.0$
 $T_{\min} = 10.89 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 3 & 4 & 0 & 3 & 3 \\ 2 & 2 & 3 & -1 & 3 & 2 \end{pmatrix}_{b_0}$

- [42] $1^{11}31^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 39, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.13 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 7 & 1 & 8 & 4 \\ 7 & 10 & 1 & 12 & 6 \\ 1 & 1 & -2 & 1 & 0 \\ 8 & 12 & 1 & 12 & 6 \\ 4 & 6 & 0 & 6 & 2 \end{pmatrix}_{b_0}$$
- [43] $1^{11}3^{11}21^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & 1 & 3 & 2 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 4 & 2 \\ 3 & 3 & 0 & 4 & 3 & 2 \\ 2 & 2 & 0 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$
- [46] $1^{11}41^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 9 & 5 \\ 9 & 13 & 15 & 7 \\ 9 & 15 & 13 & 7 \\ 5 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$
- [45] $1^{11}31^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 7 & 8 & 8 & 4 \\ 7 & 10 & 13 & 12 & 6 \\ 8 & 13 & 13 & 16 & 7 \\ 8 & 12 & 16 & 12 & 6 \\ 4 & 6 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$
- [44] $1^{11}31^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 44, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.45 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 4 & 1 & 3 \\ 5 & 13 & 13 & 0 & 7 \\ 4 & 13 & 10 & 1 & 6 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 7 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$
- [47] $1^{11}31^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 2 & 1 & 1 & 2 \\ 1 & 1 & 4 & 3 & 0 & 3 \\ 2 & 4 & 4 & 0 & 0 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 4 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [48] $131^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 53, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.77 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 3 \\ 4 & 10 & 13 & 12 & 6 \\ 5 & 13 & 13 & 16 & 7 \\ 5 & 12 & 16 & 12 & 6 \\ 3 & 6 & 7 & 6 & 2 \end{pmatrix}_{b_0}$$
- [49] $31^{11}231^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$

$$\begin{pmatrix} 0 & 3 & 1 & 3 & 1 & 2 \\ 3 & 1 & 2 & 3 & 1 & 3 \\ 1 & 2 & 0 & 2 & 2 & 2 \\ 3 & 3 & 2 & 0 & 1 & 2 \\ 1 & 1 & 2 & 1 & -2 & 0 \\ 2 & 3 & 2 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$
- [50] $21^{11}41^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 67, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.55 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 5 & 9 & 10 & 5 \\ 9 & 13 & 15 & 7 \\ 10 & 15 & 13 & 7 \\ 5 & 7 & 7 & 3 \end{pmatrix}_{b_0}$$
- [51] $1^{11}31^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 79, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 8 & 7 & 8 & 4 \\ 8 & 13 & 13 & 15 & 7 \\ 7 & 13 & 10 & 13 & 6 \\ 8 & 15 & 13 & 12 & 6 \\ 4 & 7 & 6 & 6 & 2 \end{pmatrix}_{b_0}$$
- [52] $21^{11}41^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 83, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.93 \rightarrow 12, T_{\max} = 20, T \in [12, 20]$

$$\begin{pmatrix} 13 & 9 & 20 & 7 \\ 9 & 5 & 13 & 5 \\ 20 & 13 & 25 & 9 \\ 7 & 5 & 9 & 3 \end{pmatrix}_{b_0}$$
- [53] $1^{11}31^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 88, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.50 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{aligned}
& \begin{pmatrix} 0 & 2 & 1 & 2 & 1 & 2 \\ 2 & 4 & 4 & 6 & 0 & 4 \\ 1 & 4 & 1 & 4 & 0 & 3 \\ 2 & 6 & 4 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 3 & 3 & -2 & 1 \end{pmatrix}_{b_0} \\
[54] \quad & 131^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 91, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 5 & 4 & 5 & 3 \\ 5 & 13 & 13 & 15 & 7 \\ 4 & 13 & 10 & 13 & 6 \\ 5 & 15 & 13 & 12 & 6 \\ 3 & 7 & 6 & 6 & 2 \end{pmatrix}_{b_0} \\
[55] \quad & 21^{11}41^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 99, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.54 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 13 & 15 & 14 & 7 \\ 15 & 13 & 15 & 7 \\ 14 & 15 & 13 & 7 \\ 7 & 7 & 7 & 3 \end{pmatrix}_{b_0} \\
[56] \quad & 1^{11}31^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 2 & 2 & 1 & 1 & 2 \\ 2 & 4 & 7 & 3 & 0 & 4 \\ 2 & 7 & 4 & 4 & 0 & 4 \\ 1 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 2 & 4 & 4 & 2 & -2 & 1 \end{pmatrix}_{b_0} \\
[57] \quad & 21^{11}41^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 20, \Delta = -104, \det = 123, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.43 \rightarrow 12, T_{\max} = 20, T \in [12, 20] \\
& \begin{pmatrix} 5 & 7 & 10 & 5 \\ 7 & 5 & 11 & 5 \\ 10 & 11 & 13 & 7 \\ 5 & 5 & 7 & 3 \end{pmatrix}_{b_0} \\
[58] \quad & 1^{111}2^231^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 171, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 2 & 6 & 4 & 4 \\ 4 & 6 & 5 & 5 & 5 \\ 3 & 4 & 5 & 0 & 2 \\ 3 & 4 & 5 & 2 & 3 \end{pmatrix}_{b_0} \\
[59] \quad & 1^{111}2^231^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 1 & 2 \\ 2 & 1 & 4 & 3 & 1 & 3 \\ 2 & 4 & 1 & 4 & 1 & 3 \\ 2 & 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 3 & 2 & 0 & 2 \end{pmatrix}_{b_0} \\
[60] \quad & 31^{111}231^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 553, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 3 & 1 & 3 \\ 3 & 4 & 1 & 4 & 1 & 3 \\ 3 & 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 & -2 & 0 \\ 2 & 3 & 3 & 2 & 0 & 1 \end{pmatrix}_{b_0} \\
T = 9 \quad [61] \quad & 1^{11}4^{11}131^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -6, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 2 & 4 & 2 & 4 \\ 4 & 5 & 1 & 5 \\ 2 & 1 & -2 & 0 \\ 4 & 5 & 0 & 4 \end{pmatrix}_{b_0} \\
[62] \quad & 1^{11}2^321^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 3 & -1 & 4 & 3 \end{pmatrix}_{b_0} \\
[63] \quad & 1^{11}31^{11}4^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -40, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 1 & 2 & 1 & 3 \\ 2 & 2 & 0 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 \\ 3 & 3 & 2 & 0 & 3 \end{pmatrix}_{b_0} \\
[64] \quad & 1^{11}31^{11}4^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -2 & 0 & 0 \\ 1 & 1 & 1 & 0 & -2 & 0 \\ 2 & 2 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{1^{11}31^1} 4^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.82 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 1 & 2 & 2 & 3 \\ 2 & 2 & 0 & 2 & 2 \\ 3 & 2 & 2 & 0 & 2 \\ 3 & 3 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[66] $31^{11}31^{11}41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -88, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.55 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 4 & 4 & 3 & 2 & 4 \\ 3 & 4 & 1 & 4 & 1 & 3 \\ 3 & 3 & 4 & 0 & 1 & 2 \\ 1 & 2 & 1 & 1 & -2 & 0 \\ 2 & 4 & 3 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

[67] $31^{11}31^{11}41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.32 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 5 & 6 & 4 & 3 \\ 5 & 8 & 13 & 5 & 6 \\ 6 & 13 & 13 & 6 & 7 \\ 4 & 5 & 6 & 0 & 2 \\ 3 & 6 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[68] $2^1 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -134, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 4 & 3 & 3 \\ 5 & 8 & 8 & 5 & 6 \\ 4 & 8 & 5 & 5 & 5 \\ 3 & 5 & 5 & 0 & 2 \\ 3 & 6 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[69] $31^{11}31^{11}41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -190, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.06 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 4 & 7 & 3 & 2 & 4 \\ 3 & 7 & 4 & 4 & 2 & 4 \\ 3 & 3 & 4 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & -2 & 0 \\ 2 & 4 & 4 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

[70] $31^{11}31^{11}41^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -241, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.07 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 1 & 2 \\ 3 & 1 & 4 & 3 & 1 & 3 \\ 3 & 4 & 4 & 4 & 2 & 4 \\ 3 & 3 & 4 & 0 & 1 & 2 \\ 1 & 1 & 2 & 1 & -2 & 0 \\ 2 & 3 & 4 & 2 & 0 & 1 \end{pmatrix}_{b_0}$$

$T = 10$ [71] $21^{11}51^{11}231^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 4 & 9 & 5 & 3 & 4 \\ 9 & 19 & 11 & 7 & 9 \\ 5 & 11 & 5 & 5 & 5 \\ 3 & 7 & 5 & 0 & 2 \\ 4 & 9 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{11}51231^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 6, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 3 \\ 4 & 10 & 5 & 0 & 6 \\ 2 & 5 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 6 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

[73] $2^1 3^1 1^{11} 51^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 2 & 4 & 4 & 4 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 4 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[74] $2^1 3^1 1^{11} 51^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 \quad T = 10, T_{\max} = 13, \Delta = -79, \det = 54, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 2 & 4 & 3 & 3 \\ 2 & 1 & 4 & 2 & 3 \\ 4 & 4 & 5 & 5 & 5 \\ 3 & 2 & 5 & 0 & 2 \\ 3 & 3 & 5 & 2 & 3 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [75] \quad 1^{11}31^151321^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 \quad T = 12, \quad T_{\max} = 13, \quad \Delta = -29, \quad \det = 8, \quad (n_+, n_-, n_0) = (1, 12, 0), \quad c_{\text{ext}} = 8.0$$

$$T_{\min} = 13, \quad T_{\max} = 13, \quad T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 2 & 1 & 5 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 5 & 0 & 5 & 5 \\ 2 & 3 & 4 & -1 & 5 & 4 \end{pmatrix}_{b_0}$$

6.99 NHC(-2)(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{e}_6$

$$T = 8 \quad [0] \quad 1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -162, \quad \det = 135, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.44 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 12 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 6 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -162, \quad \det = 238, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.55 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 5 & 9 & 12 & 1 & 5 \\ 9 & 13 & 21 & 0 & 7 \\ 12 & 21 & 25 & 0 & 9 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 7 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -162, \quad \det = 242, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.48 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 5 \\ 9 & 13 & 18 & 0 & 7 \\ 11 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 7 & 8 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 8, \quad T_{\max} = 101, \quad \Delta = -162, \quad \det = 253, \quad (n_+, n_-, n_0) = (1, 8, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.68 \rightarrow 14, \quad T_{\max} = 101, \quad T \in [14, 101]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 11 & 17 & 20 & 0 & 8 \\ 1 & 0 & 0 & -6 & -4 \\ 5 & 6 & 8 & -4 & 0 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \quad 1^{11}31^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -226, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.96 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 13 & 16 & 0 & 7 \\ 5 & 12 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 6 & 7 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11}31^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 \quad T = 9, \quad T_{\max} = 99, \quad \Delta = -141, \quad \det = -391, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.73 \rightarrow 14, \quad T_{\max} = 99, \quad T \in [14, 99]$$

$$\begin{pmatrix} 1 & 5 & 4 & 5 & 1 & 3 \\ 5 & 13 & 13 & 15 & 0 & 7 \\ 4 & 13 & 10 & 13 & 0 & 6 \\ 5 & 15 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 7 & 6 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

6.100 NHC(-2)(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{e}_6$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -90, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.51 \rightarrow 14, \quad T_{\max} = 93, \quad T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 44 & 5 \\ 8 & 10 & 17 & 0 & 60 & 6 \\ 12 & 17 & 25 & 0 & 96 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 44 & 60 & 96 & 0 & 346 & 34 \\ 5 & 6 & 9 & -4 & 34 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, \quad T_{\max} = 93, \quad \Delta = -132, \quad \det = -124, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.29 \rightarrow 15, \quad T_{\max} = 93, \quad T \in [15, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 44 & 5 \\ 9 & 13 & 18 & 0 & 72 & 7 \\ 11 & 18 & 20 & 0 & 84 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 44 & 72 & 84 & 0 & 346 & 34 \\ 5 & 7 & 8 & -4 & 34 & -1 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{111}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{violet}{e}_6+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -139$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 20 & 0 & 13 & 7 \\ 5 & 6 & 9 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{111}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{violet}{e}_6+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -236$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.54 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 8 & 5 \\ 9 & 13 & 18 & 0 & 12 & 7 \\ 11 & 18 & 20 & 0 & 16 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 8 & 12 & 16 & 0 & 10 & 6 \\ 5 & 7 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{111}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{violet}{e}_6+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -317$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 15 & 17 & 0 & 13 & 7 \\ 5 & 7 & 8 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{111}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{violet}{e}_6+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -351$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 17 & 5 \\ 8 & 10 & 17 & 0 & 24 & 6 \\ 12 & 17 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 24 & 36 & 0 & 49 & 13 \\ 5 & 6 & 9 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111}31^{113}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{violet}{e}_6+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -439$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 35 & 5 \\ 8 & 10 & 17 & 0 & 48 & 6 \\ 12 & 17 & 25 & 0 & 76 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 35 & 48 & 76 & 0 & 217 & 27 \\ 5 & 6 & 9 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

[7] $1^{111}31^{111}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{violet}{e}_6+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -525$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.56 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 12 & 1 & 9 & 5 \\ 9 & 13 & 21 & 0 & 15 & 7 \\ 12 & 21 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 15 & 20 & 0 & 13 & 7 \\ 5 & 7 & 9 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{112}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{violet}{e}_6+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -526$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.44 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 26 & 5 \\ 8 & 10 & 17 & 0 & 36 & 6 \\ 12 & 17 & 25 & 0 & 56 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 36 & 56 & 0 & 118 & 20 \\ 5 & 6 & 9 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{113}31^{111}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{violet}{e}_6+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -653$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.8$

$$T_{\min} = 13.55 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 35 & 5 \\ 9 & 13 & 18 & 0 & 57 & 7 \\ 11 & 18 & 20 & 0 & 67 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 35 & 57 & 67 & 0 & 217 & 27 \\ 5 & 7 & 8 & -4 & 27 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{111}31^{111}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{violet}{e}_6+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 93$, $\Delta = -132$, $\det = -709$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 13.48 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 17 & 5 \\ 9 & 13 & 18 & 0 & 27 & 7 \\ 11 & 18 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 17 & 27 & 33 & 0 & 49 & 13 \\ 5 & 7 & 8 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{112} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, T_{\max} = 93, \Delta = -132, \det = -848, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.49 \rightarrow 14, T_{\max} = 93, T \in [14, 93]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 26 & 5 \\ 9 & 13 & 18 & 0 & 42 & 7 \\ 11 & 18 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 26 & 42 & 50 & 0 & 118 & 20 \\ 5 & 7 & 8 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

6.101 $\text{NHC}(-2)(-2)(-3) + \mathbf{so}_8 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2$

$$T = 10 \quad [0] \ 1^{111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.75 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 20 & 9 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 20 & 0 & 13 & 15 & 7 \\ 9 & 12 & 20 & 0 & 15 & 13 & 7 \\ 5 & 6 & 9 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 188, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.47 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 17 & 5 \\ 8 & 10 & 17 & 0 & 12 & 24 & 6 \\ 12 & 17 & 25 & 0 & 20 & 36 & 9 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 20 & 0 & 13 & 27 & 7 \\ 17 & 24 & 36 & 0 & 27 & 49 & 13 \\ 5 & 6 & 9 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 300, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.64 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 15 & 17 & 0 & 13 & 15 & 7 \\ 9 & 15 & 17 & 0 & 15 & 13 & 7 \\ 5 & 7 & 8 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 22, \Delta = -102, \det = 832, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.52 \rightarrow 14, T_{\max} = 22, T \in [14, 22]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 17 & 5 \\ 9 & 13 & 18 & 0 & 15 & 27 & 7 \\ 11 & 18 & 20 & 0 & 17 & 33 & 8 \\ 1 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 15 & 17 & 0 & 13 & 27 & 7 \\ 17 & 27 & 33 & 0 & 27 & 49 & 13 \\ 5 & 7 & 8 & -4 & 7 & 13 & 0 \end{pmatrix}_{b_0}$$

6.102 $\text{NHC}(-2)(-2)(-3) + \mathbf{so}_8 + \mathbf{e}_6 + \mathbf{su}_3$

$$T = 9 \quad [0] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -196, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 11 & 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 17 & 0 & 12 & 6 \\ 5 & 6 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 14.03 \rightarrow 15, T_{\max} = 99, T \in [15, 99]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 20 & 0 & 12 & 6 \\ 5 & 6 & 9 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[2] 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{c}_6 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 99, \Delta = -141, \det = -559, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, T_{\max} = 99, T \in [14, 99]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -6 & 0 & -4 \\ 9 & 15 & 17 & 0 & 12 & 6 \\ 5 & 7 & 8 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

6.103 NHC(-2)(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{f}_4$

$$T = 8 \quad [0] 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 106, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.58 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 12 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 6 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[1] 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 179, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 9 & 12 & 1 & 5 \\ 9 & 13 & 21 & 0 & 7 \\ 12 & 21 & 25 & 0 & 9 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 7 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[2] 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 191, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 5 \\ 9 & 13 & 18 & 0 & 7 \\ 11 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 7 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

$$[3] 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 8, T_{\max} = 98, \Delta = -136, \det = 196, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 11 & 17 & 20 & 0 & 8 \\ 1 & 0 & 0 & -5 & -3 \\ 5 & 6 & 8 & -3 & 1 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] 1^{111} 3^1 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -173, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.07 \rightarrow 14, T_{\max} = 26, T \in [14, 26]$$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 13 & 16 & 0 & 7 \\ 5 & 12 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 6 & 7 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

$$[5] 1^{111} 3^1 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -300, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 1 & 5 & 4 & 5 & 1 & 3 \\ 5 & 13 & 13 & 15 & 0 & 7 \\ 4 & 13 & 10 & 13 & 0 & 6 \\ 5 & 15 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 7 & 6 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

6.104 NHC(-2)(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2$

$$T = 9 \quad [0] 1^{111} 3^1 1^{114} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -44, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 44 & 5 \\ 8 & 10 & 17 & 0 & 60 & 6 \\ 12 & 17 & 25 & 0 & 96 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 44 & 60 & 96 & 0 & 346 & 34 \\ 5 & 6 & 9 & -3 & 34 & 1 \end{pmatrix}_{b_0}$$

$$[1] 1^{114} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -74, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 13.30 \rightarrow 14, T_{\max} = 21, T \in [14, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 44 & 5 \\ 9 & 13 & 18 & 0 & 72 & 7 \\ 11 & 18 & 20 & 0 & 84 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 44 & 72 & 84 & 0 & 346 & 34 \\ 5 & 7 & 8 & -3 & 34 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{111}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{red}{f}_4+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -107$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 20 & 0 & 13 & 7 \\ 5 & 6 & 9 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{11}31^{111}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{red}{f}_4+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -186$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.69 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 8 & 5 \\ 9 & 13 & 18 & 0 & 12 & 7 \\ 11 & 18 & 20 & 0 & 16 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 8 & 12 & 16 & 0 & 10 & 6 \\ 5 & 7 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{111}31^{11}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{red}{f}_4+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -249$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 17 & 0 & 13 & 7 \\ 5 & 7 & 8 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{111}31^{111}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{red}{f}_4+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -275$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 17 & 5 \\ 8 & 10 & 17 & 0 & 24 & 6 \\ 12 & 17 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 24 & 36 & 0 & 49 & 13 \\ 5 & 6 & 9 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{111}31^{113}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{red}{f}_4+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -331$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.61 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 35 & 5 \\ 8 & 10 & 17 & 0 & 48 & 6 \\ 12 & 17 & 25 & 0 & 76 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 35 & 48 & 76 & 0 & 217 & 27 \\ 5 & 6 & 9 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{111}31^1+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{red}{f}_4+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -393$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 12 & 1 & 9 & 5 \\ 9 & 13 & 21 & 0 & 15 & 7 \\ 12 & 21 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 20 & 0 & 13 & 7 \\ 5 & 7 & 9 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$

[8] $1^{111}31^{112}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{red}{f}_4+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -408$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.59 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 26 & 5 \\ 8 & 10 & 17 & 0 & 36 & 6 \\ 12 & 17 & 25 & 0 & 56 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 36 & 56 & 0 & 118 & 20 \\ 5 & 6 & 9 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[9] $1^{113}31^{111}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{red}{f}_4+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -501$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.0$

$$T_{\min} = 12.66 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 35 & 5 \\ 9 & 13 & 18 & 0 & 57 & 7 \\ 11 & 18 & 20 & 0 & 67 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 35 & 57 & 67 & 0 & 217 & 27 \\ 5 & 7 & 8 & -3 & 27 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{111}31^{111}+[(\textcolor{red}{-2})\cdot\textcolor{blue}{sp}_1\cdot\textcolor{green}{g}_2]+\textcolor{brown}{so}_8+\textcolor{red}{f}_4+\textcolor{blue}{su}_2$ $T = 9$, $T_{\max} = 21$, $\Delta = -106$, $\det = -557$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.2$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 17 & 5 \\ 9 & 13 & 18 & 0 & 27 & 7 \\ 11 & 18 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 17 & 27 & 33 & 0 & 49 & 13 \\ 5 & 7 & 8 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{112}3^11^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -106, \det = -662, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.7$$

$$T_{\min} = 12.62 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 26 & 5 \\ 9 & 13 & 18 & 0 & 42 & 7 \\ 11 & 18 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 26 & 42 & 50 & 0 & 118 & 20 \\ 5 & 7 & 8 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

6.105 NHC(-2)(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{112}3^11^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 12 & 1 & 6 & 5 \\ 9 & 13 & 21 & 0 & 9 & 7 \\ 12 & 21 & 25 & 0 & 12 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 12 & 0 & 4 & 4 \\ 5 & 7 & 9 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{112}3^11^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -172, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 6 & 5 \\ 8 & 10 & 17 & 0 & 8 & 6 \\ 12 & 17 & 25 & 0 & 12 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 8 & 12 & 0 & 4 & 4 \\ 5 & 6 & 9 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112}3^11^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -330, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.68 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 6 & 5 \\ 9 & 13 & 18 & 0 & 9 & 7 \\ 11 & 18 & 20 & 0 & 11 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 6 & 9 & 11 & 0 & 4 & 4 \\ 5 & 7 & 8 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$

6.106 NHC(-2)(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{f}_4$ + $\mathfrak{su}_2 \mathfrak{n}3\text{mix}$

$$T = 9 \quad [0] \ 1^{112}3^11^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -203, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 15 & 5 \\ 8 & 10 & 17 & 0 & 20 & 6 \\ 12 & 17 & 25 & 0 & 32 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 20 & 32 & 0 & 37 & 11 \\ 5 & 6 & 9 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{112}3^11^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 21, \Delta = -108, \det = -333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.2$$

$$T_{\min} = 12.63 \rightarrow 13, T_{\max} = 21, T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 15 & 5 \\ 9 & 13 & 18 & 0 & 24 & 7 \\ 11 & 18 & 20 & 0 & 28 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 15 & 24 & 28 & 0 & 37 & 11 \\ 5 & 7 & 8 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

6.107 NHC(-2)(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{f}_4$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{112}3^11^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 26, \Delta = -115, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 15.2$$

$$T_{\min} = 12.90 \rightarrow 13, T_{\max} = 26, T \in [13, 26]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 11 & 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 17 & 0 & 12 & 6 \\ 5 & 6 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -213$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 13.24 \rightarrow 14$, $T_{\max} = 26$, $T \in [14, 26]$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 20 & 0 & 12 & 6 \\ 5 & 6 & 9 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{f}_4 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -440$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 17 & 0 & 12 & 6 \\ 5 & 7 & 8 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

6.108 NHC(-2)(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{so}_8$

$T = 8$ [0] $1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 77$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.84 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 12 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 6 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 115$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 1 & 2 & 1 & 0 & -2 & 1 \\ 1 & 1 & 2 & 0 & 1 & -2 \\ 2 & 2 & 2 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 120$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.20 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$

$$\begin{pmatrix} 5 & 9 & 12 & 1 & 5 \\ 9 & 13 & 21 & 0 & 7 \\ 12 & 21 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 139$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 22$, $T \in [13, 22]$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 11 & 17 & 20 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 6 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 22$, $\Delta = -112$, $\det = 140$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 11.86 \rightarrow 12$, $T_{\max} = 22$, $T \in [12, 22]$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 5 \\ 9 & 13 & 18 & 0 & 7 \\ 11 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 8 & -2 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [5] $1^{11}31^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -34$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 1 & 4 & 1 & 5 & 1 & 3 \\ 4 & 10 & 1 & 12 & 0 & 6 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 5 & 12 & 1 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 6 & 0 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{11}31^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -40$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.0$

$T_{\min} = 12.50 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 1 & 1 & 3 & 3 & 0 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 3 & 1 & 0 & 3 & 0 & 2 \\ 2 & 3 & 0 & 3 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -4 & -2 \\ 2 & 3 & 0 & 2 & 2 & -2 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{11}} 4^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -84, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.43 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 6 & 9 & 1 & 5 \\ 6 & 5 & 11 & 0 & 5 \\ 9 & 11 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 5 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}} 4^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -84, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.86 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 6 & 12 & 1 & 5 \\ 6 & 5 & 13 & 0 & 5 \\ 12 & 13 & 25 & 0 & 9 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 5 & 9 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -120, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.27 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 4 & 5 & 5 & 1 & 3 \\ 4 & 10 & 13 & 12 & 0 & 6 \\ 5 & 13 & 13 & 16 & 0 & 7 \\ 5 & 12 & 16 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 6 & 7 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{11}} 4^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 \quad T = 9, \ T_{\max} = 21, \ \Delta = -103, \ \det = -120, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.50 \rightarrow 13, \ T_{\max} = 21, \ T \in [13, 21]$$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 5 \\ 9 & 13 & 15 & 0 & 7 \\ 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -4 & -2 \\ 5 & 7 & 7 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} 3^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -209, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.16 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 5 & 4 & 5 & 1 & 3 \\ 5 & 13 & 13 & 15 & 0 & 7 \\ 4 & 13 & 10 & 13 & 0 & 6 \\ 5 & 15 & 13 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 7 & 6 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [12] \ 1^{\overline{11}} 3^{\overline{1}} 4^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 101, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.85 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 1 & 2 & 1 & 1 & 3 \\ 2 & 2 & 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & -2 & 0 & 0 \\ 2 & 1 & 1 & 0 & -2 & 0 \\ 3 & 3 & 2 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

6.109 NHC(-2)(-2)(-3) + so₈ + so₈ + su₂

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -55, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.87 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 10 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 12 & 20 & 0 & 13 & 7 \\ 5 & 6 & 9 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -66, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 & 1 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & -2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 1 & -2 & 1 \\ 1 & 2 & 1 & 0 & 2 & 1 & 0 \\ 2 & 2 & 2 & -1 & 0 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -75, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} =$$

13.0

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 12 & 20 & 0 & 13 & 7 \\ 5 & 6 & 9 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3 \ 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 1 & 1 & 0 \\ 1 & 1 & 2 & 0 & 1 & -2 & 2 & 0 \\ 2 & 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & -1 & 0 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3 \ 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 18 & 5 \\ 8 & 10 & 17 & 0 & 24 & 6 \\ 12 & 17 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 24 & 36 & 0 & 49 & 13 \\ 5 & 6 & 9 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 \ 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 1 & 1 & 0 \\ 1 & 1 & 2 & 0 & 1 & -2 & 2 & 0 \\ 1 & 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 2 & 2 & 2 & -1 & 0 & 0 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3 \ 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 8 & 5 \\ 9 & 13 & 18 & 0 & 12 & 7 \\ 11 & 18 & 20 & 0 & 16 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 16 & 0 & 10 & 6 \\ 5 & 7 & 8 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3 \ 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 17 & 0 & 13 & 7 \\ 5 & 7 & 8 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3 \ 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -199, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 17 & 5 \\ 8 & 10 & 17 & 0 & 24 & 6 \\ 12 & 17 & 25 & 0 & 36 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 24 & 36 & 0 & 49 & 13 \\ 5 & 6 & 9 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111} 3 \ 1^{113} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -223, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 35 & 5 \\ 8 & 10 & 17 & 0 & 48 & 6 \\ 12 & 17 & 25 & 0 & 76 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 35 & 48 & 76 & 0 & 217 & 27 \\ 5 & 6 & 9 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 3 \ 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 1 & 3 & 0 \\ 1 & 1 & 2 & 0 & 1 & -2 & 3 & 0 \\ 4 & 3 & 3 & 0 & 3 & 3 & 4 & 4 \\ 2 & 2 & 2 & -1 & 0 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{111} 3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -286, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 1 & 3 & 0 \\ 1 & 1 & 2 & 0 & 1 & -2 & 3 & 0 \\ 2 & 3 & 3 & 0 & 3 & 3 & 4 & 4 \\ 2 & 2 & 2 & -1 & 0 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{111} 3^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -287, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.86 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 37 & 5 \\ 8 & 10 & 17 & 0 & 48 & 6 \\ 12 & 17 & 25 & 0 & 76 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 37 & 48 & 76 & 0 & 217 & 27 \\ 5 & 6 & 9 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{111} 3^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -290, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.86 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 26 & 5 \\ 8 & 10 & 17 & 0 & 36 & 6 \\ 12 & 17 & 25 & 0 & 56 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 26 & 36 & 56 & 0 & 118 & 20 \\ 5 & 6 & 9 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{111} 3^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -322, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 45 & 5 \\ 8 & 10 & 17 & 0 & 60 & 6 \\ 12 & 17 & 25 & 0 & 96 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 45 & 60 & 96 & 0 & 346 & 34 \\ 5 & 6 & 9 & -2 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{111} 3^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -334, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.84 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 46 & 5 \\ 8 & 10 & 17 & 0 & 60 & 6 \\ 12 & 17 & 25 & 0 & 96 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 46 & 60 & 96 & 0 & 346 & 34 \\ 5 & 6 & 9 & -2 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{111} 3^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -342, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.85 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 27 & 5 \\ 8 & 10 & 17 & 0 & 36 & 6 \\ 12 & 17 & 25 & 0 & 56 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 27 & 36 & 56 & 0 & 118 & 20 \\ 5 & 6 & 9 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{111} 3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -349, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 18 & 5 \\ 9 & 13 & 18 & 0 & 27 & 7 \\ 11 & 18 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 27 & 33 & 0 & 49 & 13 \\ 5 & 7 & 8 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{113} 3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 12, \quad \Delta = -82, \quad \det = -349, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 11.87 \rightarrow 12, \quad T_{\max} = 12, \quad T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 35 & 5 \\ 9 & 13 & 18 & 0 & 57 & 7 \\ 11 & 18 & 20 & 0 & 67 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 35 & 57 & 67 & 0 & 217 & 27 \\ 5 & 7 & 8 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{114} 3 \ 1^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -376, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 45 & 5 \\ 9 & 13 & 18 & 0 & 72 & 7 \\ 11 & 18 & 20 & 0 & 84 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 45 & 72 & 84 & 0 & 346 & 34 \\ 5 & 7 & 8 & -2 & 34 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{111} 3 \ 1^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -405, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 17 & 5 \\ 9 & 13 & 18 & 0 & 27 & 7 \\ 11 & 18 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 27 & 33 & 0 & 49 & 13 \\ 5 & 7 & 8 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{111} 3 \ 1^{113} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -411, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.85 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 36 & 5 \\ 8 & 10 & 17 & 0 & 48 & 6 \\ 12 & 17 & 25 & 0 & 76 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 36 & 48 & 76 & 0 & 217 & 27 \\ 5 & 6 & 9 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{111} 3 \ 1^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -431, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 1 & 3 & 0 \\ 1 & 1 & 2 & 0 & 1 & -2 & 3 & 0 \\ 3 & 3 & 3 & 0 & 3 & 3 & 4 & 4 \\ 2 & 2 & 2 & -1 & 0 & 0 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{112} 3 \ 1^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -476, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 11.86 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 26 & 5 \\ 9 & 13 & 18 & 0 & 42 & 7 \\ 11 & 18 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 26 & 42 & 50 & 0 & 118 & 20 \\ 5 & 7 & 8 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{112} 3 \ 1^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -556, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 27 & 5 \\ 9 & 13 & 18 & 0 & 42 & 7 \\ 11 & 18 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 27 & 42 & 50 & 0 & 118 & 20 \\ 5 & 7 & 8 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{113} 3 \ 1^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \quad T = 9, \ T_{\max} = 12, \ \Delta = -82, \ \det = -565, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.8$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 36 & 5 \\ 9 & 13 & 18 & 0 & 57 & 7 \\ 11 & 18 & 20 & 0 & 67 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 36 & 57 & 67 & 0 & 217 & 27 \\ 5 & 7 & 8 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

6.110 $\text{NHC}(-2)(-2)(-3) + \textcolor{brown}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{113} 3 \ 1^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -124, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.87 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 6 & 5 \\ 8 & 10 & 17 & 0 & 8 & 6 \\ 12 & 17 & 25 & 0 & 12 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 8 & 12 & 0 & 4 & 4 \\ 5 & 6 & 9 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{113} 3 \ 1^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{so}_8 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -214, \ (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 6 & 1 & 6 & 5 \\ 8 & 10 & 9 & 0 & 8 & 6 \\ 6 & 9 & 5 & 1 & 7 & 5 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 8 & 7 & 0 & 3 & 3 \\ 5 & 6 & 5 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -241, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 6 & 5 \\ 9 & 13 & 18 & 0 & 9 & 7 \\ 11 & 18 & 20 & 0 & 11 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 11 & 0 & 4 & 4 \\ 5 & 7 & 8 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

6.111 NHC(-2)(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{so}_8$ + $\mathfrak{su}_{2n3\text{mix}}$

$$T = 9 \quad [0] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -20, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 16 & 5 \\ 9 & 13 & 18 & 0 & 24 & 7 \\ 11 & 18 & 20 & 0 & 28 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 24 & 28 & 0 & 37 & 11 \\ 5 & 7 & 8 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -47, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 16 & 5 \\ 8 & 10 & 17 & 0 & 20 & 6 \\ 12 & 17 & 25 & 0 & 32 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 20 & 32 & 0 & 37 & 11 \\ 5 & 6 & 9 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 13 & 12 & 9 & 1 & 9 & 7 \\ 12 & 10 & 9 & 0 & 8 & 6 \\ 9 & 9 & 5 & 1 & 7 & 5 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 8 & 7 & 0 & 3 & 3 \\ 7 & 6 & 5 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -147, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 15 & 5 \\ 8 & 10 & 17 & 0 & 20 & 6 \\ 12 & 17 & 25 & 0 & 32 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 20 & 32 & 0 & 37 & 11 \\ 5 & 6 & 9 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -244, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 15 & 5 \\ 9 & 13 & 18 & 0 & 24 & 7 \\ 11 & 18 & 20 & 0 & 28 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 24 & 28 & 0 & 37 & 11 \\ 5 & 7 & 8 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

6.112 NHC(-2)(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{so}_8$ + $\mathfrak{su}_3$

$$T = 9 \quad [0] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 1 & 5 \\ 8 & 10 & 17 & 0 & 0 & 6 \\ 11 & 17 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 6 & 8 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -75, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.23 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 1 & 5 \\ 8 & 10 & 17 & 0 & 0 & 6 \\ 12 & 17 & 25 & 0 & 0 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 6 & 9 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3 \ 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -98, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 11 & 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 12 & 17 & 0 & 12 & 6 \\ 5 & 6 & 8 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3 \ 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -132, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.27 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 10 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 12 & 20 & 0 & 12 & 6 \\ 5 & 6 & 9 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3 \ 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -152, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.63 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 12 & 17 & 25 & 0 & 20 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 12 & 20 & 0 & 12 & 6 \\ 5 & 6 & 9 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3 \ 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -164, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.29 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 1 & 5 \\ 9 & 13 & 18 & 0 & 0 & 7 \\ 11 & 18 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 7 & 8 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3 \ 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -194, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 & 0 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 1 & 0 & 0 \\ 1 & 1 & 2 & 0 & 1 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 & -1 \\ 2 & 2 & 2 & -1 & 0 & 0 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3 \ 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -201, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 10 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 15 & 17 & 0 & 12 & 6 \\ 5 & 7 & 8 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3 \ 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 13, \ \Delta = -91, \ \det = -321, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.18 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 17 & 0 & 12 & 6 \\ 5 & 7 & 8 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [9] \ 1^{11} 4 \ 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{so}_8 + \mathbf{su}_3 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 117, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.87 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 6 & 12 & 1 & 6 & 5 \\ 6 & 5 & 13 & 0 & 7 & 5 \\ 12 & 13 & 25 & 0 & 12 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 7 & 12 & 0 & 4 & 4 \\ 5 & 5 & 9 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{111} 4 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 237, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 6 & 9 & 1 & 6 & 5 \\ 6 & 5 & 11 & 0 & 7 & 5 \\ 9 & 11 & 13 & 0 & 10 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 7 & 10 & 0 & 4 & 4 \\ 5 & 5 & 7 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 4 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 261, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 6 & 5 \\ 9 & 13 & 15 & 0 & 9 & 7 \\ 9 & 15 & 13 & 0 & 10 & 7 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 10 & 0 & 4 & 4 \\ 5 & 7 & 7 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

6.113 NHC(-2)(-2)(-3) + $\mathfrak{so}_8$ + $\mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{111} 3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 1 & 6 & 6 \\ 12 & 13 & 18 & 0 & 7 & 7 \\ 15 & 18 & 20 & 0 & 8 & 8 \\ 1 & 0 & 0 & -2 & 0 & 0 \\ 6 & 7 & 8 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 8, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 6 & 0 & 0 & 4 \\ 3 & 6 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 4 & -2 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 11 & 16 & 13 & 6 & 6 \\ 11 & 10 & 17 & 12 & 6 & 6 \\ 16 & 17 & 25 & 20 & 9 & 9 \\ 13 & 12 & 20 & 13 & 7 & 7 \\ 6 & 6 & 9 & 7 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{114} 3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 15 & 21 & 0 & 76 & 16 \\ 3 & 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{113} 3 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 21 & 3 \\ 3 & 4 & 6 & 0 & 30 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 30 & 31 & 0 & 172 & 24 \\ 3 & 4 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 11 & 16 & 23 & 6 & 6 \\ 11 & 10 & 17 & 24 & 6 & 6 \\ 16 & 17 & 25 & 36 & 9 & 9 \\ 23 & 24 & 36 & 49 & 13 & 13 \\ 6 & 6 & 9 & 13 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 17 & 0 & 49 & 13 \\ 3 & 4 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 26, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 12 & 6 \\ 12 & 13 & 18 & 15 & 7 \\ 15 & 18 & 20 & 17 & 8 \\ 12 & 15 & 17 & 13 & 7 \\ 6 & 7 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 26, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 0 \\ 1 & 2 & 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 1 & 2 & 1 & 0 & -2 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 31, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 1 & 2 & 1 & 0 & -2 & 1 \\ 2 & 1 & 2 & 0 & 1 & 0 \\ 2 & 2 & 2 & -1 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 34, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 11 & 6 \\ 12 & 13 & 18 & 12 & 7 \\ 15 & 18 & 20 & 16 & 8 \\ 11 & 12 & 16 & 10 & 6 \\ 6 & 7 & 8 & 6 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 40, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 13 & 6 \\ 12 & 13 & 18 & 15 & 7 \\ 15 & 18 & 20 & 17 & 8 \\ 13 & 15 & 17 & 13 & 7 \\ 6 & 7 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{111} 3^1 1^{13} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 40, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 15 & 17 & 0 & 49 & 13 \\ 3 & 4 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 40, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 0 & 6 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 6 & 5 & 0 & 4 & 4 \\ 3 & 4 & 4 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 41, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 3 & 4 & 0 & 1 & 3 \\ 3 & 3 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{111} 3^1 1^{13} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 42, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 11 & 16 & 62 & 6 \\ 11 & 10 & 17 & 60 & 6 \\ 16 & 17 & 25 & 96 & 9 \\ 62 & 60 & 96 & 346 & 34 \\ 6 & 6 & 9 & 34 & 3 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111} 3^1 1^{12} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 42, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 11 & 16 & 36 & 6 \\ 11 & 10 & 17 & 36 & 6 \\ 16 & 17 & 25 & 56 & 9 \\ 36 & 36 & 56 & 118 & 20 \\ 6 & 6 & 9 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{\overline{11}1} 3 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 45, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 1 & 0 \\ 1 & 1 & 2 & 0 & 1 & 0 & 2 \\ 2 & 2 & 2 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{\overline{11}1} 3 1^{\overline{13}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 48, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 11 & 16 & 49 & 6 \\ 11 & 10 & 17 & 48 & 6 \\ 16 & 17 & 25 & 76 & 9 \\ 49 & 48 & 76 & 217 & 27 \\ 6 & 6 & 9 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{\overline{11}1} 3 1^{\overline{13}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 49, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 5 & 0 \\ 4 & 5 & 7 & 0 & 5 & 24 & 8 \\ 2 & 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{\overline{11}1} 3 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 50, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.92 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 2 & 3 & 2 \\ 2 & 1 & 3 & 0 & 3 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{\overline{11}1} 3 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 52, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 0 & 1 & 3 \\ 3 & 4 & 4 & -2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{\overline{11}1} 3 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 54, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 2 & 0 \\ 2 & 2 & 1 & 0 & 2 & 0 & 2 \\ 2 & 2 & 2 & -1 & 0 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{\overline{11}1} 3 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 54, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 & 2 \\ 1 & 2 & 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 2 & 3 & 2 \\ 2 & 2 & 1 & 0 & 3 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{\overline{11}1} 3 1^{\overline{13}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 55, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.95 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 19 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 \\ 3 & 5 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 21 & 31 & 0 & 172 & 24 \\ 3 & 3 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 21^{\overline{11}1} 3 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 56, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 13 & 12 & 20 & 15 & 7 \\ 12 & 10 & 17 & 12 & 6 \\ 20 & 17 & 25 & 20 & 9 \\ 15 & 12 & 20 & 13 & 7 \\ 7 & 6 & 9 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{\overline{11}1} 3 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 59, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 & 2 \\ 1 & 2 & 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 2 & 3 & 2 \\ 1 & 2 & 1 & 0 & 3 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

[27] $1^{113} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 8 & 3 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 & 2 \\ 8 & 9 & 12 & 0 & 25 & 9 & 9 \\ 3 & 3 & 4 & -2 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[28] $1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 69, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 & 2 \\ 2 & 3 & 4 & 0 & 1 & 3 & 3 \\ 3 & 3 & 4 & -2 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{114} 3^1 1^{14} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 20 & 3 & 3 \\ 2 & 1 & 5 & 0 & 24 & 3 & 3 \\ 3 & 5 & 4 & 0 & 36 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 & 2 \\ 20 & 24 & 36 & 0 & 238 & 28 & 28 \\ 3 & 3 & 4 & -2 & 28 & 2 & 2 \end{pmatrix}_{b_0}$$

[29] $1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 3 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 & 2 \\ 5 & 6 & 9 & 0 & 13 & 7 & 7 \\ 3 & 3 & 4 & -2 & 7 & 2 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{113} 3^1 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 15 & 3 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 & 3 \\ 3 & 5 & 4 & 0 & 27 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 & 2 \\ 15 & 18 & 27 & 0 & 133 & 21 & 21 \\ 3 & 3 & 4 & -2 & 21 & 2 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{112} 3^1 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 12 & 15 & 35 & 6 & 6 \\ 12 & 13 & 18 & 42 & 7 & 7 \\ 15 & 18 & 20 & 50 & 8 & 8 \\ 35 & 42 & 50 & 118 & 20 & 20 \\ 6 & 7 & 8 & 20 & 3 & 3 \end{pmatrix}_{b_0}$$

[33] $1^{111} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 4 & 0 \\ 6 & 4 & 5 & 0 & 4 & 12 & 6 \\ 2 & 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

[30] $1^{112} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 70, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 & 3 \\ 3 & 5 & 4 & 0 & 18 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 & 2 \\ 10 & 12 & 18 & 0 & 58 & 14 & 14 \\ 3 & 3 & 4 & -2 & 14 & 2 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 72, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 0 & 6 & 4 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 & 2 \\ 3 & 6 & 5 & 0 & 4 & 4 & 4 \\ 3 & 4 & 4 & -2 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{11} 3^1 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 75, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 2 & 3 & 2 \\ 1 & 1 & 2 & 0 & 3 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{112} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 14 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 15 & 14 & 0 & 37 & 11 \\ 3 & 4 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[38] \quad 21^{111} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.79 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 12 & 20 & 27 & 7 \\ 12 & 10 & 17 & 24 & 6 \\ 20 & 17 & 25 & 36 & 9 \\ 27 & 24 & 36 & 49 & 13 \\ 7 & 6 & 9 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{111} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 3 & 0 \\ 2 & 3 & 3 & 0 & 3 & 4 & 4 \\ 2 & 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{114} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 61 & 6 \\ 12 & 13 & 18 & 72 & 7 \\ 15 & 18 & 20 & 84 & 8 \\ 61 & 72 & 84 & 346 & 34 \\ 6 & 7 & 8 & 34 & 3 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{113} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 12 & 17 & 0 & 49 & 13 \\ 3 & 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{112} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 37 & 6 \\ 12 & 13 & 18 & 42 & 7 \\ 15 & 18 & 20 & 50 & 8 \\ 37 & 42 & 50 & 118 & 20 \\ 6 & 7 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{112} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 6 & 8 & 0 & 10 & 6 \\ 3 & 3 & 4 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 24 & 6 \\ 12 & 13 & 18 & 27 & 7 \\ 15 & 18 & 20 & 33 & 8 \\ 24 & 27 & 33 & 49 & 13 \\ 6 & 7 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{111} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 6 & 0 \\ 6 & 6 & 9 & 0 & 6 & 40 & 10 \\ 2 & 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{111} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 94, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 & 2 \\ 1 & 2 & 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 4 & 0 \\ 3 & 4 & 5 & 0 & 4 & 12 & 6 \\ 2 & 2 & 2 & -1 & 0 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{111} 3 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 23 & 6 \\ 12 & 13 & 18 & 27 & 7 \\ 15 & 18 & 20 & 33 & 8 \\ 23 & 27 & 33 & 49 & 13 \\ 6 & 7 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{111} 3 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 3 & 0 \\ 4 & 3 & 3 & 0 & 3 & 4 & 4 \\ 2 & 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[49] \quad 21^{11} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 108, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 15 & 17 & 15 & 7 \\ 15 & 13 & 18 & 15 & 7 \\ 17 & 18 & 20 & 17 & 8 \\ 15 & 15 & 17 & 13 & 7 \\ 7 & 7 & 8 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{113} 3 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 48 & 6 \\ 12 & 13 & 18 & 57 & 7 \\ 15 & 18 & 20 & 67 & 8 \\ 48 & 57 & 67 & 217 & 27 \\ 6 & 7 & 8 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{111} 3 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 110, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 6 & 0 \\ 8 & 6 & 9 & 0 & 6 & 40 & 10 \\ 2 & 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{112} 3 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 111, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 8 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 9 & 13 & 0 & 28 & 10 \\ 3 & 3 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[53] \quad 1^{111} 3 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 13 & 0 & 28 & 10 \\ 3 & 4 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[54] \quad 1^{111} 3 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[55] \quad 1^{111} 3 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 13 & 0 & 28 & 10 \\ 3 & 4 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[56] \quad 1^{111} 3 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 115, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 5 & 0 \\ 7 & 5 & 7 & 0 & 5 & 24 & 8 \\ 2 & 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[57] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 124, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 2 & 6 & 2 \\ 2 & 3 & 3 & 0 & 6 & 4 & 4 \\ 2 & 2 & 2 & -1 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[58] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 128, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 9 & 0 & 13 & 7 \\ 3 & 4 & 4 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

$$[59] \ 1^{113} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 136, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.88 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 49 & 6 \\ 12 & 13 & 18 & 57 & 7 \\ 15 & 18 & 20 & 67 & 8 \\ 49 & 57 & 67 & 217 & 27 \\ 6 & 7 & 8 & 27 & 3 \end{pmatrix}_{b_0}$$

$$[60] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 139, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 3 & 0 \\ 3 & 3 & 3 & 0 & 3 & 4 & 4 \\ 2 & 2 & 2 & -1 & 0 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[61] \ 1^{112} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 139, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.86 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 13 & 0 & 28 & 10 \\ 3 & 3 & 4 & -2 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[62] \ 1^{113} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 145, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.95 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 12 & 0 & 25 & 9 \\ 3 & 3 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[63] \ 1^{111} 3^1 1^{14} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 146, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 2 & 6 & 2 \\ 1 & 2 & 0 & 1 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 6 & 0 \\ 7 & 6 & 9 & 0 & 6 & 40 & 10 \\ 2 & 2 & 2 & -1 & 0 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[64] \ 1^{112} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 146, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 36 & 6 \\ 12 & 13 & 18 & 42 & 7 \\ 15 & 18 & 20 & 50 & 8 \\ 36 & 42 & 50 & 118 & 20 \\ 6 & 7 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[65] \ 1^{111} 3^1 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 159, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 2 & 5 & 2 \\ 1 & 2 & 0 & 1 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 5 & 0 \\ 5 & 5 & 7 & 0 & 5 & 24 & 8 \\ 2 & 2 & 2 & -1 & 0 & 8 & 1 \end{pmatrix}_{b_0}$$

$$[66] \ 1^{113} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \mathfrak{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -83, \ \det = 160, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 24 & 23 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[67] \ 1^{111} 3 \ 1^{12} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 \\ 1 & 2 & 0 & 1 & 1 & 5 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 1 & 2 & 1 & 0 & -2 & 4 \\ 5 & 4 & 5 & 0 & 4 & 12 \\ 2 & 2 & 2 & -1 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[68] \ 1^{114} 3 \ 1^{12} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 12 & 16 & 0 & 46 & 12 \\ 3 & 3 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[69] \ 1^{111} 3 \ 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 3 \\ 1 & 0 & 2 & 0 & 3 & 3 \\ 1 & 2 & 0 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 2 & 3 & 3 & 0 & 2 & 6 \\ 3 & 3 & 3 & 0 & 6 & 4 \\ 2 & 2 & 2 & -1 & 2 & 4 \end{pmatrix}_{b_0}$$

$$[70] \ 1^{112} 3 \ 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 14 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 15 & 14 & 0 & 37 & 11 \\ 3 & 4 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$

$$[71] \ 1^{111} 3 \ 1^{12} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 4 \\ 1 & 2 & 0 & 1 & 1 & 5 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 1 & 2 & 1 & 0 & -2 & 4 \\ 4 & 4 & 5 & 0 & 4 & 12 \\ 2 & 2 & 2 & -1 & 0 & 6 \end{pmatrix}_{b_0}$$

$$[72] \ 1^{111} 3 \ 1^{13} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 2 & 5 \\ 1 & 2 & 0 & 1 & 1 & 7 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 1 & 2 & 1 & 0 & -2 & 5 \\ 6 & 5 & 7 & 0 & 5 & 24 \\ 2 & 2 & 2 & -1 & 0 & 8 \end{pmatrix}_{b_0}$$

$$[73] \ 1^{113} 3 \ 1^{12} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 13 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 15 & 22 & 0 & 85 & 17 \\ 3 & 3 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

$$[74] \ 1^{112} 3 \ 1^{14} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 24 & 26 & 0 & 118 & 20 \\ 3 & 4 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[75] \ 1^{112} 3 \ 1^{12} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

- [76] $1^{113} 3 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 19 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 21 & 19 & 0 & 73 & 15 \\ 3 & 4 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$
- [77] $1^{112} 3 1^{14} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 184, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 24 & 26 & 0 & 118 & 20 \\ 3 & 4 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$
- [78] $21^{111} 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 12 & 16 & 24 & 6 \\ 12 & 13 & 18 & 27 & 7 \\ 16 & 18 & 20 & 33 & 8 \\ 24 & 27 & 33 & 49 & 13 \\ 6 & 7 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$
- [79] $1^{112} 3 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 14 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 14 & 0 & 37 & 11 \\ 3 & 4 & 4 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$
- [80] $1^{113} 3 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 206, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 12 & 17 & 0 & 49 & 13 \\ 3 & 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$
- [81] $1^{113} 3 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 15 & 22 & 0 & 85 & 17 \\ 3 & 3 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$
- [82] $1^{112} 3 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 220, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 21 & 22 & 0 & 85 & 17 \\ 3 & 4 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$
- [83] $21^{123} 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 230, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 12 & 20 & 42 & 7 \\ 12 & 10 & 17 & 36 & 6 \\ 20 & 17 & 25 & 56 & 9 \\ 42 & 36 & 56 & 118 & 20 \\ 7 & 6 & 9 & 20 & 3 \end{pmatrix}_{b_0}$$
- [85] $1^{113} 3 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 12 & 17 & 0 & 49 & 13 \\ 3 & 3 & 4 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$
- [84] $1^{114} 3 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 234, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 16 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 3 & 5 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 18 & 26 & 0 & 118 & 20 \\ 3 & 3 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$
- [86] $21^{114} 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 238, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 12 & 20 & 72 & 7 \\ 12 & 10 & 17 & 60 & 6 \\ 20 & 17 & 25 & 96 & 9 \\ 72 & 60 & 96 & 346 & 34 \\ 7 & 6 & 9 & 34 & 3 \end{pmatrix}_{b_0}$$

[88] $21^{111}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 15 & 17 & 27 & 7 \\ 15 & 13 & 18 & 27 & 7 \\ 17 & 18 & 20 & 33 & 8 \\ 27 & 27 & 33 & 49 & 13 \\ 7 & 7 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

[87] $1^{112}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[89] $1^{114}31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 251, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 18 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 \\ 3 & 5 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 21 & 31 & 0 & 172 & 24 \\ 3 & 3 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[90] $1^{112}31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 252, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 22 & 0 & 85 & 17 \\ 3 & 4 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$

[91] $1^{112}31^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 24 & 26 & 0 & 118 & 20 \\ 3 & 4 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

[92] $1^{113}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 264, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 19 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 21 & 19 & 0 & 73 & 15 \\ 3 & 4 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[93] $1^{114}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 15 & 21 & 0 & 76 & 16 \\ 3 & 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[94] $1^{114}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 271, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 13 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 15 & 21 & 0 & 76 & 16 \\ 3 & 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[95] $1^{114}31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 279, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 17 & 3 \\ 2 & 1 & 5 & 0 & 21 & 3 \\ 3 & 5 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 21 & 31 & 0 & 172 & 24 \\ 3 & 3 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[96] $21^{13}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -83, \det = 284, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 12 & 20 & 57 & 7 \\ 12 & 10 & 17 & 48 & 6 \\ 20 & 17 & 25 & 76 & 9 \\ 57 & 48 & 76 & 217 & 27 \\ 7 & 6 & 9 & 27 & 3 \end{pmatrix}_{b_0}$$

- [97] $1^{114} 3 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 292, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 22 & 3 \\ 3 & 4 & 6 & 0 & 33 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 33 & 32 & 0 & 193 & 25 \\ 3 & 4 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$
- [98] $1^{112} 3 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 296, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 \\ 3 & 6 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 18 & 0 & 58 & 14 \\ 3 & 4 & 4 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$
- [99] $1^{113} 3 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 304, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$
- [100] $1^{112} 3 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 6 & 0 & 21 & 4 \\ 3 & 6 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 21 & 22 & 0 & 85 & 17 \\ 3 & 4 & 4 & -2 & 17 & 2 \end{pmatrix}_{b_0}$$
- [101] $1^{114} 3 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 346, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 3 & 5 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 18 & 26 & 0 & 118 & 20 \\ 3 & 3 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$
- [102] $1^{113} 3 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 13 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 24 & 23 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$
- [103] $1^{114} 3 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 355, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 15 & 21 & 0 & 76 & 16 \\ 3 & 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$
- [104] $21^{113} 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 360, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 15 & 17 & 57 & 7 \\ 15 & 13 & 18 & 57 & 7 \\ 17 & 18 & 20 & 67 & 8 \\ 57 & 57 & 67 & 217 & 27 \\ 7 & 7 & 8 & 27 & 3 \end{pmatrix}_{b_0}$$
- [105] $1^{114} 3 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 374, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 15 & 3 \\ 2 & 1 & 5 & 0 & 18 & 3 \\ 3 & 5 & 4 & 0 & 26 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 18 & 26 & 0 & 118 & 20 \\ 3 & 3 & 4 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$
- [106] $1^{113} 3 1^{14} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 6 & 0 & 30 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 30 & 31 & 0 & 172 & 24 \\ 3 & 4 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$
- [107] $1^{114} 3 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 19 & 3 \\ 3 & 4 & 6 & 0 & 30 & 4 \\ 3 & 6 & 4 & 0 & 28 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 30 & 28 & 0 & 154 & 22 \\ 3 & 4 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[108] $21^{112}31^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 15 & 17 & 42 & 7 \\ 15 & 13 & 18 & 42 & 7 \\ 17 & 18 & 20 & 50 & 8 \\ 42 & 42 & 50 & 118 & 20 \\ 7 & 7 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

[109] $1^{113}31^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 384, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 24 & 23 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[111] $1^{113}31^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 15 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 15 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[110] $1^{113}31^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 448, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 14 & 3 \\ 3 & 4 & 6 & 0 & 24 & 4 \\ 3 & 6 & 4 & 0 & 23 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 24 & 23 & 0 & 100 & 18 \\ 3 & 4 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[113] $1^{114}31^{14} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 24 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 24 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[112] $1^{113}31^{14} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 472, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 6 & 0 & 30 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 30 & 31 & 0 & 172 & 24 \\ 3 & 4 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[114] $1^{113}31^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 512, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[115] $1^{114}31^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 552, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 6 & 0 & 30 & 4 \\ 3 & 6 & 4 & 0 & 28 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 30 & 28 & 0 & 154 & 22 \\ 3 & 4 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[116] $1^{113}31^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 560, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 16 & 3 \\ 3 & 4 & 6 & 0 & 27 & 4 \\ 3 & 6 & 4 & 0 & 27 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 27 & 27 & 0 & 133 & 21 \\ 3 & 4 & 4 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

[117] $1^{113}31^{14} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 568, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 19 & 3 \\ 3 & 4 & 6 & 0 & 30 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 30 & 31 & 0 & 172 & 24 \\ 3 & 4 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[118] $1^{114} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 568, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 17 & 3 \\ 3 & 4 & 6 & 0 & 30 & 4 \\ 3 & 6 & 4 & 0 & 28 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 30 & 28 & 0 & 154 & 22 \\ 3 & 4 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[119] $1^{113} 3^1 1^{14} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 600, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.8$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 6 & 0 & 30 & 4 \\ 3 & 6 & 4 & 0 & 31 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 30 & 31 & 0 & 172 & 24 \\ 3 & 4 & 4 & -2 & 24 & 2 \end{pmatrix}_{b_0}$$

[120] $21^{1111} 31^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 604, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 15 & 20 & 15 & 7 \\ 15 & 13 & 21 & 15 & 7 \\ 20 & 21 & 25 & 20 & 9 \\ 15 & 15 & 20 & 13 & 7 \\ 7 & 7 & 9 & 7 & 3 \end{pmatrix}_{b_0}$$

[122] $1^{114} 3^1 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 612, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 21 & 3 \\ 3 & 4 & 6 & 0 & 33 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 33 & 32 & 0 & 193 & 25 \\ 3 & 4 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[121] $1^{114} 3^1 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 612, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 18 & 3 \\ 3 & 4 & 6 & 0 & 33 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 33 & 32 & 0 & 193 & 25 \\ 3 & 4 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[123] $1^{114} 3^1 1^{14} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 728, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[124] $1^{114} 3^1 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 772, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 20 & 3 \\ 3 & 4 & 6 & 0 & 33 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 20 & 33 & 32 & 0 & 193 & 25 \\ 3 & 4 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[125] $1^{114} 3^1 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 772, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 19 & 3 \\ 3 & 4 & 6 & 0 & 33 & 4 \\ 3 & 6 & 4 & 0 & 32 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 33 & 32 & 0 & 193 & 25 \\ 3 & 4 & 4 & -2 & 25 & 2 \end{pmatrix}_{b_0}$$

[126] $1^{114} 3^1 1^{14} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 776, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 23 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 23 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[127] $1^{114} 3^1 1^{14} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 904, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 21 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 21 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

[128] $1^{114} 3^{112} 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 920, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 22 & 3 \\ 3 & 4 & 6 & 0 & 36 & 4 \\ 3 & 6 & 4 & 0 & 36 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 36 & 36 & 0 & 238 & 28 \\ 3 & 4 & 4 & -2 & 28 & 2 \end{pmatrix}_{b_0}$$

6.114 NHC(-2)(-2)(-3) + $\mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8$

$T = 9$ [0] $1^{111} 3^{111} 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 1 & 1 & 0 \\ 2 & 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 1 & 2 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{111} 3^{111} 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 1 & 1 & 0 \\ 1 & 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 1 & 2 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{111} 3^{111} 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -136, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 8 & 5 \\ 9 & 13 & 18 & 0 & 12 & 7 \\ 11 & 18 & 20 & 0 & 16 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 16 & 0 & 10 & 6 \\ 5 & 7 & 8 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{111} 3^{111} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -181, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 17 & 0 & 13 & 7 \\ 5 & 7 & 8 & -2 & 7 & 2 \end{pmatrix}_{b_0}$$

[4] $1^{111} 3^{112} 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -274, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 1 & 2 \\ 1 & 2 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 3 & 1 & 0 \\ 4 & 3 & 3 & 0 & 3 & 4 & 3 & 4 \\ 1 & 1 & 2 & 0 & 1 & 3 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

[5] $1^{111} 3^{111} 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -286, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 1 & 2 \\ 1 & 2 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 3 & 1 & 0 \\ 2 & 3 & 3 & 0 & 3 & 4 & 3 & 4 \\ 1 & 1 & 2 & 0 & 1 & 3 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

[6] $1^{111} 3^{111} 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -349, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 18 & 5 \\ 9 & 13 & 18 & 0 & 27 & 7 \\ 11 & 18 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 18 & 27 & 33 & 0 & 49 & 13 \\ 5 & 7 & 8 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[7] 1^{111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -405, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 17 & 5 \\ 9 & 13 & 18 & 0 & 27 & 7 \\ 11 & 18 & 20 & 0 & 33 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 27 & 33 & 0 & 49 & 13 \\ 5 & 7 & 8 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$

$$[8] 1^{111} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -431, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 1 & 2 \\ 1 & 2 & 0 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 3 & 1 & 0 \\ 3 & 3 & 3 & 0 & 3 & 4 & 3 & 4 \\ 1 & 1 & 2 & 0 & 1 & 3 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 4 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[9] 1^{112} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 26 & 5 \\ 9 & 13 & 18 & 0 & 42 & 7 \\ 11 & 18 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 26 & 42 & 50 & 0 & 118 & 20 \\ 5 & 7 & 8 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[10] 1^{112} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -556, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 27 & 5 \\ 9 & 13 & 18 & 0 & 42 & 7 \\ 11 & 18 & 20 & 0 & 50 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 27 & 42 & 50 & 0 & 118 & 20 \\ 5 & 7 & 8 & -2 & 20 & 2 \end{pmatrix}_{b_0}$$

$$[11] 1^{113} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8 \quad T = 9, T_{\max} = 12, \Delta = -82, \det = -565, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 36 & 5 \\ 9 & 13 & 18 & 0 & 57 & 7 \\ 11 & 18 & 20 & 0 & 67 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 36 & 57 & 67 & 0 & 217 & 27 \\ 5 & 7 & 8 & -2 & 27 & 2 \end{pmatrix}_{b_0}$$

6.115 NHC(-2)(-2)(-3) + so₈ + su₂ → (-3)

$$T = 8 \quad [0] 1^{113} 1^{11} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 11 & 16 & 8 & 6 \\ 11 & 10 & 17 & 8 & 6 \\ 16 & 17 & 25 & 12 & 9 \\ 8 & 8 & 12 & 4 & 4 \\ 6 & 6 & 9 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[1] 1^{113} 1^{11} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 2 & 3 \\ 3 & 5 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 2 \\ 3 & 3 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[2] 1^{113} 1^{11} 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 8 & 6 \\ 12 & 13 & 18 & 9 & 7 \\ 15 & 18 & 20 & 11 & 8 \\ 8 & 9 & 11 & 4 & 4 \\ 6 & 7 & 8 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 66, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.94 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 5 & 8 & 6 & 1 & 5 \\ 8 & 10 & 9 & 0 & 6 \\ 6 & 9 & 5 & 1 & 5 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 6 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 76, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.89 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 21^{11}3^{11}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 85, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.82 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 13 & 12 & 20 & 9 & 7 \\ 12 & 10 & 17 & 8 & 6 \\ 20 & 17 & 25 & 12 & 9 \\ 9 & 8 & 12 & 4 & 4 \\ 7 & 6 & 9 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 91, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.92 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 1 & 4 & 3 \\ 2 & 1 & 3 & 0 & 5 & 3 \\ 2 & 3 & 1 & 1 & 5 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 5 & 5 & 0 & 6 & 4 \\ 3 & 3 & 3 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 21^{11}3^{11}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 114, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 16 & 8 & 6 \\ 12 & 13 & 18 & 9 & 7 \\ 16 & 18 & 20 & 11 & 8 \\ 8 & 9 & 11 & 4 & 4 \\ 6 & 7 & 8 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[8] \ 21^{11}3^{11}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 159, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 13 & 15 & 17 & 9 & 7 \\ 15 & 13 & 18 & 9 & 7 \\ 17 & 18 & 20 & 11 & 8 \\ 9 & 9 & 11 & 4 & 4 \\ 7 & 7 & 8 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[9] \ 21^{11}3^{11}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 399, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.85 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 13 & 15 & 20 & 9 & 7 \\ 15 & 13 & 21 & 9 & 7 \\ 20 & 21 & 25 & 12 & 9 \\ 9 & 9 & 12 & 4 & 4 \\ 7 & 7 & 9 & 4 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [10] \ 1^{11}3^{11}21^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -4, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 2 & 3 \\ 4 & 10 & 5 & 0 & 4 & 6 \\ 2 & 5 & 1 & 1 & 2 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 2 & 0 & 0 & 2 \\ 3 & 6 & 3 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{11}3^{11}21^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -21, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.90 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 2 & 2 & 1 & 3 \\ 2 & 2 & 3 & 4 & 0 & 4 \\ 2 & 3 & 1 & 3 & 0 & 3 \\ 2 & 4 & 3 & 1 & 0 & 3 \\ 1 & 0 & 0 & -4 & -2 & 3 \\ 3 & 4 & 3 & 3 & -2 & 3 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{11}3^{11}21^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -29, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 2 & 3 \\ 3 & 4 & 6 & 3 & 3 & 4 \\ 3 & 6 & 5 & 5 & 3 & 5 \\ 3 & 3 & 5 & 0 & 2 & 2 \\ 2 & 3 & 3 & 2 & 0 & 2 \\ 3 & 4 & 5 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

6.116 NHC(-2)(-2)(-3) + so₈ + su₂→(-3) + so₈

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -214, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 6 & 1 & 6 & 5 \\ 8 & 10 & 9 & 0 & 8 & 6 \\ 6 & 9 & 5 & 1 & 7 & 5 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 8 & 7 & 0 & 3 & 3 \\ 5 & 6 & 5 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{so}_8 \quad T = 9, \ T_{\max} = 12, \ \Delta = -84, \ \det = -241, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 6 & 5 \\ 9 & 13 & 18 & 0 & 9 & 7 \\ 11 & 18 & 20 & 0 & 11 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 11 & 0 & 4 & 4 \\ 5 & 7 & 8 & -2 & 4 & 2 \end{pmatrix}_{b_0}$$

6.117 NHC(-2)(-2)(-3) + so₈ + su₂→(-3) + su₂→(-3)

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -56, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 0 & 2 & 2 & 3 \\ 3 & 5 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 3 & 4 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 0 & 3 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 2 & 2 \\ 2 & 2 & 3 & 0 & 2 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 2 & 2 \end{pmatrix}_{b_0}$$

6.118 NHC(-2)(-2)(-3) + so₈ + su₂→(-3) + su₂n3mix

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -56, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 7 & 3 \\ 2 & 1 & 5 & 0 & 2 & 8 & 3 \\ 3 & 5 & 4 & 0 & 3 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 7 & 2 \\ 7 & 8 & 12 & 0 & 7 & 25 & 9 \\ 3 & 3 & 4 & -2 & 2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11}3^{11}1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -56, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 12 & 3 \\ 2 & 1 & 5 & 0 & 2 & 14 & 3 \\ 3 & 5 & 4 & 0 & 3 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 12 & 2 \\ 12 & 14 & 21 & 0 & 12 & 80 & 16 \\ 3 & 3 & 4 & -2 & 2 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{113}3^{111}1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -89, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 4 & 3 \\ 3 & 4 & 6 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 4 & 2 \\ 4 & 6 & 7 & 0 & 4 & 7 & 5 \\ 3 & 4 & 4 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{113}3^{111}1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -126, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 4 & 3 \\ 2 & 1 & 5 & 0 & 2 & 5 & 3 \\ 3 & 5 & 4 & 0 & 3 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 4 & 2 \\ 4 & 5 & 7 & 0 & 4 & 7 & 5 \\ 3 & 3 & 4 & -2 & 2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{113}3^{111}1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -224, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 9 & 3 \\ 2 & 1 & 5 & 0 & 2 & 11 & 3 \\ 3 & 5 & 4 & 0 & 3 & 16 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 9 & 2 \\ 9 & 11 & 16 & 0 & 9 & 44 & 12 \\ 3 & 3 & 4 & -2 & 2 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{113}3^{111}1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -322, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 14 & 3 \\ 2 & 1 & 5 & 0 & 2 & 17 & 3 \\ 3 & 5 & 4 & 0 & 3 & 25 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 14 & 2 \\ 14 & 17 & 25 & 0 & 14 & 111 & 19 \\ 3 & 3 & 4 & -2 & 2 & 19 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{113}3^{111}1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -364, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.96 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 11 & 3 \\ 2 & 1 & 5 & 0 & 2 & 14 & 3 \\ 3 & 5 & 4 & 0 & 3 & 20 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 11 & 2 \\ 11 & 14 & 20 & 0 & 11 & 69 & 15 \\ 3 & 3 & 4 & -2 & 2 & 15 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{114}3^{111}1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -476, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.99 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 13 & 3 \\ 2 & 1 & 5 & 0 & 2 & 17 & 3 \\ 3 & 5 & 4 & 0 & 3 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 13 & 2 \\ 13 & 17 & 24 & 0 & 13 & 100 & 18 \\ 3 & 3 & 4 & -2 & 2 & 18 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{114}3^{111}1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -560, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.92 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 16 & 3 \\ 2 & 1 & 5 & 0 & 2 & 20 & 3 \\ 3 & 5 & 4 & 0 & 3 & 29 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 2 & 2 & 3 & 0 & 0 & 16 & 2 \\ 16 & 20 & 29 & 0 & 16 & 148 & 22 \\ 3 & 3 & 4 & -2 & 2 & 22 & 2 \end{pmatrix}_{b_0}$$

6.119 NHC(-2)(-2)(-3) + so₈ + su_{2n3mix}

$$T = 8 \quad [0] \ 1^{113}3^{111}1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, \ T_{\max} = 11, \ \Delta = -85, \ \det = 6, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 3 & 0 \\ 3 & 3 & 5 & 0 & 3 & 11 & 5 \\ 2 & 2 & 2 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 5 & 3 \\ 3 & 5 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 5 & 7 & 0 & 7 & 5 \\ 3 & 3 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 4 & 3 & 3 & 1 & 6 & 4 \\ 3 & 1 & 3 & 0 & 5 & 3 \\ 3 & 3 & 1 & 1 & 5 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 5 & 5 & 0 & 6 & 4 \\ 4 & 3 & 3 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 11 & 16 & 21 & 6 \\ 11 & 10 & 17 & 20 & 6 \\ 16 & 17 & 25 & 32 & 9 \\ 21 & 20 & 32 & 37 & 11 \\ 6 & 6 & 9 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 2 & 3 & 2 \\ 1 & 2 & 0 & 1 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 3 & 0 \\ 4 & 3 & 5 & 0 & 3 & 11 & 5 \\ 2 & 2 & 2 & -1 & 0 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 13 & 12 & 9 & 1 & 7 \\ 12 & 10 & 9 & 0 & 6 \\ 9 & 9 & 5 & 1 & 5 \\ 1 & 0 & 1 & -3 & -1 \\ 7 & 6 & 5 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 10 & 12 & 15 & 21 & 6 \\ 12 & 13 & 18 & 24 & 7 \\ 15 & 18 & 20 & 28 & 8 \\ 21 & 24 & 28 & 37 & 11 \\ 6 & 7 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 2 & 0 \\ 3 & 2 & 3 & 0 & 2 & 3 & 3 \\ 2 & 2 & 2 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[12] \quad 1^{11} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 17 & 3 \\ 2 & 1 & 5 & 0 & 20 & 3 \\ 3 & 5 & 4 & 0 & 30 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 20 & 30 & 0 & 165 & 23 \\ 3 & 3 & 4 & -2 & 23 & 2 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 8 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 8 & 12 & 0 & 25 & 9 \\ 3 & 3 & 4 & -2 & 9 & 2 \end{pmatrix}_{b_0}$$

$$[11] \quad 1^{11} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 14 & 3 \\ 3 & 5 & 4 & 0 & 21 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 14 & 21 & 0 & 80 & 16 \\ 3 & 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{114}3^{114} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 22 & 3 \\ 2 & 1 & 5 & 0 & 26 & 3 \\ 3 & 5 & 4 & 0 & 39 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 22 & 26 & 39 & 0 & 280 & 30 \\ 3 & 3 & 4 & -2 & 30 & 2 \end{pmatrix}_{b_0}$$

[8] $1^{111}3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 49, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 12 & 15 & 20 & 6 \\ 12 & 13 & 18 & 24 & 7 \\ 15 & 18 & 20 & 28 & 8 \\ 20 & 24 & 28 & 37 & 11 \\ 6 & 7 & 8 & 11 & 3 \end{pmatrix}_{b_0}$$

[13] $1^{111}3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 0 & 6 & 4 \\ 3 & 6 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 6 & 7 & 0 & 7 & 5 \\ 3 & 4 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[14] $1^{111}3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 2 & 0 \\ 2 & 2 & 3 & 0 & 2 & 3 & 3 \\ 2 & 2 & 2 & -1 & 0 & 3 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{111}3^{113} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.82 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 12 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 \\ 3 & 6 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 18 & 20 & 0 & 69 & 15 \\ 3 & 4 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[16] $1^{112}3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 76, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 11 & 3 \\ 3 & 5 & 4 & 0 & 16 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 11 & 16 & 0 & 44 & 12 \\ 3 & 3 & 4 & -2 & 12 & 2 \end{pmatrix}_{b_0}$$

[17] $1^{111}3^{113} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 84, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 11 & 3 \\ 3 & 4 & 6 & 0 & 18 & 4 \\ 3 & 6 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 18 & 20 & 0 & 69 & 15 \\ 3 & 4 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[18] $1^{111}3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 90, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 3 \\ 2 & 1 & 5 & 0 & 5 & 3 \\ 3 & 5 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 7 & 0 & 7 & 5 \\ 3 & 3 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[19] $1^{112}3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 8 & 3 \\ 3 & 5 & 4 & 0 & 11 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 8 & 11 & 0 & 20 & 8 \\ 3 & 3 & 4 & -2 & 8 & 2 \end{pmatrix}_{b_0}$$

[20] $21^{111}3^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 103, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 3 \\ 2 & 1 & 5 & 0 & 17 & 3 \\ 3 & 5 & 4 & 0 & 25 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 17 & 25 & 0 & 111 & 19 \\ 3 & 3 & 4 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{11} 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 14 & 3 \\ 3 & 5 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 14 & 20 & 0 & 69 & 15 \\ 3 & 3 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[32] $1^{11} 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.94 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 \\ 2 & 1 & 5 & 0 & 14 & 3 \\ 3 & 5 & 4 & 0 & 20 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 14 & 20 & 0 & 69 & 15 \\ 3 & 3 & 4 & -2 & 15 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{11} 1^{11} 3^1 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 300, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.87 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 19 & 3 \\ 2 & 1 & 5 & 0 & 23 & 3 \\ 3 & 5 & 4 & 0 & 34 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 19 & 23 & 34 & 0 & 208 & 26 \\ 3 & 3 & 4 & -2 & 26 & 2 \end{pmatrix}_{b_0}$$

[34] $1^{11} 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 376, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 13 & 3 \\ 2 & 1 & 5 & 0 & 17 & 3 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 17 & 24 & 0 & 100 & 18 \\ 3 & 3 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[35] $1^{11} 1^{11} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 381, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 17 & 3 \\ 2 & 1 & 5 & 0 & 20 & 3 \\ 3 & 5 & 4 & 0 & 29 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 17 & 20 & 29 & 0 & 148 & 22 \\ 3 & 3 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

[36] $1^{11} 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 404, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 3 \\ 2 & 1 & 5 & 0 & 17 & 3 \\ 3 & 5 & 4 & 0 & 24 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 14 & 17 & 24 & 0 & 100 & 18 \\ 3 & 3 & 4 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[37] $1^{11} 1^{11} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 8, T_{\max} = 11, \Delta = -85, \det = 409, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 16 & 3 \\ 2 & 1 & 5 & 0 & 20 & 3 \\ 3 & 5 & 4 & 0 & 29 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 16 & 20 & 29 & 0 & 148 & 22 \\ 3 & 3 & 4 & -2 & 22 & 2 \end{pmatrix}_{b_0}$$

$T = 9$ [39] $1^{12} 3^1 21^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 10 & 3 \\ 4 & 10 & 5 & 0 & 24 & 6 \\ 2 & 5 & 1 & 1 & 12 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 10 & 24 & 12 & 0 & 56 & 14 \\ 3 & 6 & 3 & -1 & 14 & 3 \end{pmatrix}_{b_0}$$

[40] $1^{14} 3^1 21^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 18 & 3 \\ 4 & 10 & 5 & 0 & 44 & 6 \\ 2 & 5 & 1 & 1 & 22 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 18 & 44 & 22 & 0 & 192 & 26 \\ 3 & 6 & 3 & -1 & 26 & 3 \end{pmatrix}_{b_0}$$

- [38] $1^1 1^1 3^1 21^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -4, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 3 \\ 2 & 2 & 6 & 0 & 6 & 4 \\ 3 & 6 & 5 & 0 & 8 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 6 & 8 & 0 & 8 & 6 \\ 3 & 4 & 5 & -2 & 6 & 3 \end{pmatrix}_{b_0}$$
- [41] $1^1 1^1 3^1 21^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -7, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 9 & 3 \\ 4 & 10 & 5 & 0 & 20 & 6 \\ 2 & 5 & 1 & 1 & 9 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 20 & 9 & 0 & 35 & 11 \\ 3 & 6 & 3 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$
- [42] $1^1 1^1 3^1 21^{113} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -15, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 \\ 2 & 2 & 6 & 0 & 17 & 4 \\ 3 & 6 & 5 & 0 & 23 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 17 & 23 & 0 & 78 & 18 \\ 3 & 4 & 5 & -2 & 18 & 3 \end{pmatrix}_{b_0}$$
- [43] $1^{12} 1^1 3^1 21^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -24, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 16 & 3 \\ 4 & 10 & 5 & 0 & 36 & 6 \\ 2 & 5 & 1 & 1 & 16 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 16 & 36 & 16 & 0 & 116 & 20 \\ 3 & 6 & 3 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$
- [44] $1^1 1^1 3^1 21^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -31, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 17 & 3 \\ 4 & 10 & 5 & 0 & 40 & 6 \\ 2 & 5 & 1 & 1 & 19 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 17 & 40 & 19 & 0 & 151 & 23 \\ 3 & 6 & 3 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$
- [45] $1^1 1^1 3^1 21^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -37, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.95 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 3 & 0 & 10 & 3 \\ 1 & 2 & 0 & 2 & 0 & 9 & 2 \\ 1 & 3 & 2 & 0 & 0 & 10 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 6 & 10 & 9 & 10 & 0 & 51 & 13 \\ 2 & 3 & 2 & 2 & -2 & 13 & 2 \end{pmatrix}_{b_0}$$
- [46] $1^1 1^1 3^1 21^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -60, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 1 & 2 & 3 & 0 & 5 & 3 \\ 1 & 2 & 0 & 2 & 0 & 4 & 2 \\ 1 & 3 & 2 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 3 & 5 & 4 & 5 & 0 & 10 & 6 \\ 2 & 3 & 2 & 2 & -2 & 6 & 2 \end{pmatrix}_{b_0}$$
- [47] $1^{12} 1^1 4^1 3^1 21^{113} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -72, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 1 & 4 & 2 & 1 & 24 & 3 \\ 4 & 10 & 5 & 0 & 56 & 6 \\ 2 & 5 & 1 & 1 & 26 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 24 & 56 & 26 & 0 & 292 & 32 \\ 3 & 6 & 3 & -1 & 32 & 3 \end{pmatrix}_{b_0}$$
- [48] $1^{12} 1^1 4^1 3^1 21^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -141, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 1 & 2 & 3 & 0 & 9 & 3 \\ 1 & 2 & 0 & 2 & 0 & 7 & 2 \\ 1 & 3 & 2 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 7 & 9 & 0 & 35 & 11 \\ 2 & 3 & 2 & 2 & -2 & 11 & 2 \end{pmatrix}_{b_0}$$
- [49] $1^1 1^1 4^1 3^1 21^{113} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_2$ $T = 9, T_{\max} = 11, \Delta = -58, \det = -160, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 3 & 0 & 14 & 3 \\ 1 & 2 & 0 & 2 & 0 & 12 & 2 \\ 1 & 3 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 8 & 14 & 12 & 14 & 0 & 96 & 18 \\ 2 & 3 & 2 & 2 & -2 & 18 & 2 \end{pmatrix}_{b_0}$$

[50] $1^{11}1^{12}3^{12}1^{13}+[(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -256$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 3 & 0 & 12 & 3 \\ 1 & 2 & 0 & 2 & 0 & 8 & 2 \\ 1 & 3 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 6 & 12 & 8 & 12 & 0 & 56 & 14 \\ 2 & 3 & 2 & 2 & -2 & 14 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{11}1^{13}3^{12}1^{13}+[(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -264$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 1 & 2 & 3 & 0 & 13 & 3 \\ 1 & 2 & 0 & 2 & 0 & 10 & 2 \\ 1 & 3 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 7 & 13 & 10 & 13 & 0 & 74 & 16 \\ 2 & 3 & 2 & 2 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[52] $1^{11}1^{13}3^{12}1^{14}+[(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -421$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 3 & 0 & 16 & 3 \\ 1 & 2 & 0 & 2 & 0 & 11 & 2 \\ 1 & 3 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 8 & 16 & 11 & 16 & 0 & 103 & 19 \\ 2 & 3 & 2 & 2 & -2 & 19 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{11}1^{14}3^{12}1^{14}+[(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -429$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 1 & 2 & 3 & 0 & 17 & 3 \\ 1 & 2 & 0 & 2 & 0 & 13 & 2 \\ 1 & 3 & 2 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & 0 & -4 & 0 & -2 \\ 9 & 17 & 13 & 17 & 0 & 127 & 21 \\ 2 & 3 & 2 & 2 & -2 & 21 & 2 \end{pmatrix}_{b_0}$$

6.120 NHC(-2)(-2)(-3) + $\mathbf{so}_8$ + $\mathbf{su}_{2n3\text{mix}}$ + $\mathbf{so}_8$

$T = 9$ [0] $1^{11}1^{11}3^{11}1^{11}+[(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -84$, $\det = -61$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 11.89 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 13 & 12 & 9 & 1 & 9 & 7 \\ 12 & 10 & 9 & 0 & 8 & 6 \\ 9 & 9 & 5 & 1 & 7 & 5 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 8 & 7 & 0 & 3 & 3 \\ 7 & 6 & 5 & -1 & 3 & 2 \end{pmatrix}_{b_0}$$

6.121 NHC(-2)(-2)(-3) + $\mathbf{so}_8$ + $\mathbf{su}_{2n3\text{mix}}$ + $\mathbf{su}_2 \rightarrow (-3)$

$T = 9$ [1] $1^{11}1^{12}3^{11}1^{12}+[(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$

$T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 2 & 3 \\ 2 & 1 & 5 & 0 & 14 & 2 & 3 \\ 3 & 5 & 4 & 0 & 21 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 12 & 14 & 21 & 0 & 80 & 12 & 16 \\ 2 & 2 & 3 & 0 & 12 & 0 & 2 \\ 3 & 3 & 4 & -2 & 16 & 2 & 2 \end{pmatrix}_{b_0}$$

[0] $1^{11}1^{11}3^{11}1^{11}+[(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -56$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$

$T_{\min} = 10.93 \rightarrow 11$, $T_{\max} = 11$, $T \in [11, 11]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 2 & 3 \\ 2 & 1 & 5 & 0 & 8 & 2 & 3 \\ 3 & 5 & 4 & 0 & 12 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 8 & 12 & 0 & 25 & 7 & 9 \\ 2 & 2 & 3 & 0 & 7 & 0 & 2 \\ 3 & 3 & 4 & -2 & 9 & 2 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11}1^{11}3^{11}1^{11}+[(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{so}_8 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3)$ $T = 9$, $T_{\max} = 11$, $\Delta = -58$, $\det = -89$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 6 & 0 & 6 & 3 & 4 \\ 3 & 6 & 4 & 0 & 7 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 7 & 0 & 7 & 4 & 5 \\ 2 & 3 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^{11}1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -126, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.98 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 2 & 3 \\ 2 & 1 & 5 & 0 & 5 & 2 & 3 \\ 3 & 5 & 4 & 0 & 7 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 7 & 0 & 7 & 4 & 5 \\ 2 & 2 & 3 & 0 & 4 & 0 & 2 \\ 3 & 3 & 4 & -2 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{112}3^{11}1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -224, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.93 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 2 & 3 \\ 2 & 1 & 5 & 0 & 11 & 2 & 3 \\ 3 & 5 & 4 & 0 & 16 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 9 & 11 & 16 & 0 & 44 & 9 & 12 \\ 2 & 2 & 3 & 0 & 9 & 0 & 2 \\ 3 & 3 & 4 & -2 & 12 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{113}3^{11}1^{12} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.91 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 2 & 3 \\ 2 & 1 & 5 & 0 & 17 & 2 & 3 \\ 3 & 5 & 4 & 0 & 25 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 14 & 17 & 25 & 0 & 111 & 14 & 19 \\ 2 & 2 & 3 & 0 & 14 & 0 & 2 \\ 3 & 3 & 4 & -2 & 19 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{113}3^{11}1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -364, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.96 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 2 & 3 \\ 2 & 1 & 5 & 0 & 14 & 2 & 3 \\ 3 & 5 & 4 & 0 & 20 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 11 & 14 & 20 & 0 & 69 & 11 & 15 \\ 2 & 2 & 3 & 0 & 11 & 0 & 2 \\ 3 & 3 & 4 & -2 & 15 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{114}3^{11}1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -476, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.99 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 13 & 2 & 3 \\ 2 & 1 & 5 & 0 & 17 & 2 & 3 \\ 3 & 5 & 4 & 0 & 24 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 13 & 17 & 24 & 0 & 100 & 13 & 18 \\ 2 & 2 & 3 & 0 & 13 & 0 & 2 \\ 3 & 3 & 4 & -2 & 18 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{114}3^{11}1^{12} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \rightarrow (-3) \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -560, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 16 & 2 & 3 \\ 2 & 1 & 5 & 0 & 20 & 2 & 3 \\ 3 & 5 & 4 & 0 & 29 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 16 & 20 & 29 & 0 & 148 & 16 & 22 \\ 2 & 2 & 3 & 0 & 16 & 0 & 2 \\ 3 & 3 & 4 & -2 & 22 & 2 & 2 \end{pmatrix}_{b_0}$$

6.122 NHC(-2)(-2)(-3) + so₈ + su₂n₃mix + su₂n₃mix

$$T = 9 \quad [0] \ 1^{11}3^{11}1^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{so}_8 + \textcolor{red}{su}_2 + \textcolor{blue}{su}_2 \quad T = 9, T_{\max} = 11, \Delta = -58, \det = -32, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11, T_{\max} = 11, T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 6 & 0 & 6 & 6 & 4 \\ 3 & 6 & 4 & 0 & 7 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 6 & 7 & 0 & 7 & 9 & 5 \\ 4 & 6 & 7 & 0 & 9 & 7 & 5 \\ 3 & 4 & 4 & -2 & 5 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{1111}3^{\overline{11}1^{\overline{11}}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -91, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.96 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 4 & 3 \\ 2 & 1 & 5 & 0 & 8 & 5 & 3 \\ 3 & 5 & 4 & 0 & 12 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 7 & 8 & 12 & 0 & 25 & 15 & 9 \\ 4 & 5 & 7 & 0 & 15 & 7 & 5 \\ 3 & 3 & 4 & -2 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{1112}3^{\overline{11}1^{\overline{11}}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, \ T_{\max} = 11, \ \Delta = -58, \ \det = -325, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 10.97 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 9 & 3 \\ 2 & 1 & 5 & 0 & 5 & 11 & 3 \\ 3 & 5 & 4 & 0 & 7 & 16 & 4 \\ 1 & 0 & 0 & -4 & 0 & 0 & -2 \\ 4 & 5 & 7 & 0 & 7 & 20 & 5 \\ 9 & 11 & 16 & 0 & 20 & 44 & 12 \\ 3 & 3 & 4 & -2 & 5 & 12 & 2 \end{pmatrix}_{b_0}$$

6.123 NHC(-2)(-2)(-3) + $\mathfrak{so}_8 + \mathfrak{su}_3$

$$T = 8 \quad [4] \ 1^{\overline{111}3^{\overline{11}1^{\overline{11}}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 1 & 3 \\ 2 & 1 & 5 & 0 & 0 & 3 \\ 3 & 5 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 3 & 4 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}3^{\overline{11}1^{\overline{12}}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 2 & 0 \\ 3 & 2 & 4 & 0 & 2 & 5 & 3 \\ 2 & 2 & 2 & -1 & 0 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}3^{\overline{11}1^{\overline{11}}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 11 & 16 & 13 & 6 \\ 11 & 10 & 17 & 12 & 6 \\ 16 & 17 & 25 & 20 & 9 \\ 13 & 12 & 20 & 12 & 6 \\ 6 & 6 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{112}3^{\overline{11}1^{\overline{12}}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 11 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 11 & 12 & 18 & 0 & 57 & 13 \\ 3 & 3 & 4 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[0] \ 1^{\overline{111}3^{\overline{11}1^{\overline{11}}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 21, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 6 & 9 & 0 & 12 & 6 \\ 3 & 3 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{11}3^{\overline{11}1^{\overline{11}}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.62 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 1 & 3 \\ 2 & 1 & 5 & 0 & 0 & 3 \\ 4 & 5 & 9 & 0 & 0 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 3 & 5 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}3^{\overline{11}1^{\overline{11}}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 32, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.38 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

- [8] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 34$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 3 & 5 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 3 & 5 & 0 & 3 & 3 \\ 3 & 3 & 4 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$
- [7] $21^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 34$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.24 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 16 & 12 & 6 \\ 12 & 13 & 18 & 15 & 7 \\ 16 & 18 & 20 & 17 & 8 \\ 12 & 15 & 17 & 12 & 6 \\ 6 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$
- [9] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 41$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.41 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 15 & 1 & 6 \\ 12 & 13 & 18 & 0 & 7 \\ 15 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 7 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$
- [10] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 4 & 5 & 9 & 0 & 4 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 3 & 5 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{12}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 45$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 4 & 5 & 9 & 0 & 16 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 16 & 0 & 27 & 9 \\ 3 & 3 & 5 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 48$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.42 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 12 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 6 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$
- [13] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 51$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 11 & 15 & 12 & 6 \\ 11 & 10 & 17 & 12 & 6 \\ 15 & 17 & 20 & 17 & 8 \\ 12 & 12 & 17 & 12 & 6 \\ 6 & 6 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$
- [14] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 52$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 6 & 0 & 0 & 4 \\ 3 & 6 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$
- [15] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 56$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.57 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 8 & 12 & 10 & 5 \\ 8 & 10 & 17 & 12 & 6 \\ 12 & 17 & 25 & 20 & 9 \\ 10 & 12 & 20 & 12 & 6 \\ 5 & 6 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$
- [18] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 61$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 11 & 15 & 1 & 6 \\ 11 & 10 & 17 & 0 & 6 \\ 15 & 17 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 6 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$
- [17] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 61$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 15 & 12 & 6 \\ 12 & 13 & 18 & 15 & 7 \\ 15 & 18 & 20 & 17 & 8 \\ 12 & 15 & 17 & 12 & 6 \\ 6 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{11} 3^1 1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 61, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.28 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 0 & 2 \\ 1 & 2 & 0 & 1 & 1 & 0 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 66, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 13 & 12 & 14 & 1 & 7 \\ 12 & 10 & 13 & 0 & 6 \\ 14 & 13 & 13 & 1 & 7 \\ 1 & 0 & 1 & -3 & -1 \\ 7 & 6 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{11} 3^1 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 68, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.35 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 9 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 17 & 0 & 48 & 12 \\ 3 & 4 & 4 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 70, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 0 & 2 \\ 1 & 2 & 0 & 1 & 2 & 0 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 0 & -2 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 0 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 73, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 13 & 14 & 12 & 1 & 7 \\ 14 & 13 & 13 & 0 & 7 \\ 12 & 13 & 10 & 1 & 6 \\ 1 & 0 & 1 & -3 & -1 \\ 7 & 7 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 0 & 2 \\ 1 & 2 & 0 & 1 & 3 & 0 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 2 & 0 & 2 \\ 1 & 0 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & 2 & -1 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.07 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 15 & 13 & 6 \\ 12 & 13 & 18 & 15 & 7 \\ 15 & 18 & 20 & 17 & 8 \\ 13 & 15 & 17 & 12 & 6 \\ 6 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 76, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.26 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 8 & 1 & 5 \\ 9 & 13 & 13 & 0 & 7 \\ 8 & 13 & 10 & 1 & 6 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 7 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{11} 3^1 1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 76, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.26 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 3 & 4 & 0 & 0 & 2 \\ 3 & 3 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 80, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 9 & 1 & 5 \\ 8 & 10 & 13 & 0 & 6 \\ 9 & 13 & 13 & 1 & 7 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 6 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{111} 3^1 1^1 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 84, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.67 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 4 & 3 \\ 3 & 4 & 6 & 0 & 6 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 6 & 5 & 0 & 3 & 3 \\ 3 & 4 & 4 & -2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{111} 3^1 1^{111} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 84, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 6 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 4 & 5 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 6 & 12 & 0 & 12 & 6 \\ 3 & 3 & 5 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{111} 3^1 1^{13} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 84, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.10 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 10 & 3 \\ 3 & 4 & 6 & 0 & 15 & 4 \\ 3 & 6 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 15 & 17 & 0 & 48 & 12 \\ 3 & 4 & 4 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 89, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.37 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 5 \\ 9 & 13 & 18 & 0 & 7 \\ 11 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[31] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 89, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.37 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 11 & 10 & 5 \\ 9 & 13 & 18 & 15 & 7 \\ 11 & 18 & 20 & 17 & 8 \\ 10 & 15 & 17 & 12 & 6 \\ 5 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 96, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 4 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[34] \ 1^{112} 3^1 1^1 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 101, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.88 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 6 & 8 & 0 & 9 & 5 \\ 3 & 3 & 4 & -2 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{111} 3^1 1^1 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 104, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.46 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 1 & 5 & 0 & 3 & 3 \\ 3 & 5 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 3 & 3 & 4 & -2 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{112} 3^1 1^{12} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 105, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.16 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 18 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 12 & 18 & 0 & 57 & 13 \\ 3 & 3 & 4 & -2 & 13 & 1 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{orange}{so}_8 + \textcolor{red}{su}_3 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 105, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.16 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{aligned}
& \begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 6 & 9 & 0 & 12 & 6 \\ 3 & 3 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0} \\
[38] & 21^{11} 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 13 & 12 & 20 & 15 & 7 \\ 12 & 10 & 17 & 12 & 6 \\ 20 & 17 & 25 & 20 & 9 \\ 15 & 12 & 20 & 12 & 6 \\ 7 & 6 & 9 & 6 & 2 \end{pmatrix}_{b_0} \\
[39] & 1^{11} 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 4 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 4 & 5 & 9 & 0 & 12 & 5 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 6 & 12 & 0 & 12 & 6 \\ 3 & 3 & 5 & -2 & 6 & 1 \end{pmatrix}_{b_0} \\
[40] & 1^{11} 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & 6 & 0 & 6 & 4 \\ 3 & 6 & 4 & 0 & 5 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 3 & 6 & 5 & 0 & 3 & 3 \\ 3 & 4 & 4 & -2 & 3 & 1 \end{pmatrix}_{b_0} \\
[41] & 1^1 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 5 & 9 & 11 & 9 & 5 \\ 9 & 13 & 18 & 15 & 7 \\ 11 & 18 & 20 & 17 & 8 \\ 9 & 15 & 17 & 12 & 6 \\ 5 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0} \\
[42] & 1^{11} 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 1 & 5 & 3 \\ 2 & 1 & 5 & 0 & 6 & 3 \\ 3 & 5 & 4 & 0 & 8 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 6 & 8 & 0 & 9 & 5 \\ 3 & 3 & 4 & -2 & 5 & 1 \end{pmatrix}_{b_0} \\
[43] & 1^{11} 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2 \\
& T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 1 & 8 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 9 & 13 & 0 & 27 & 9 \\ 3 & 3 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0} \\
[44] & 1^{11} 31^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 1 & 7 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 12 & 13 & 0 & 27 & 9 \\ 3 & 4 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0} \\
[45] & 1^{11} 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 1 & 6 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 6 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0} \\
[46] & 1^{11} 31^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 1 & 8 & 3 \\ 3 & 4 & 6 & 0 & 12 & 4 \\ 3 & 6 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 8 & 12 & 13 & 0 & 27 & 9 \\ 3 & 4 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0} \\
[47] & 1^{11} 31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0 \\
& T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 3 & 3 & 1 & 5 & 3 \\ 3 & 4 & 6 & 0 & 9 & 4 \\ 3 & 6 & 4 & 0 & 9 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 5 & 9 & 9 & 0 & 12 & 6 \\ 3 & 4 & 4 & -2 & 6 & 1 \end{pmatrix}_{b_0}
\end{aligned}$$

[48] $21^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 16 & 12 & 6 \\ 12 & 10 & 17 & 12 & 6 \\ 16 & 17 & 20 & 17 & 8 \\ 12 & 12 & 17 & 12 & 6 \\ 6 & 6 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[49] $1^{112}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 13 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 13 & 0 & 27 & 9 \\ 3 & 3 & 4 & -2 & 9 & 1 \end{pmatrix}_{b_0}$$

[50] $1^{113}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 0 & 9 & 3 \\ 3 & 5 & 4 & 0 & 12 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 7 & 9 & 12 & 0 & 24 & 8 \\ 3 & 3 & 4 & -2 & 8 & 1 \end{pmatrix}_{b_0}$$

[51] $21^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 15 & 17 & 15 & 7 \\ 15 & 13 & 18 & 15 & 7 \\ 17 & 18 & 20 & 17 & 8 \\ 15 & 15 & 17 & 12 & 6 \\ 7 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[52] $21^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 205, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 15 & 20 & 12 & 7 \\ 15 & 13 & 21 & 12 & 7 \\ 20 & 21 & 25 & 16 & 9 \\ 12 & 12 & 16 & 9 & 5 \\ 7 & 7 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{113}31^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 216, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 10.98 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 13 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 13 & 15 & 22 & 0 & 84 & 16 \\ 3 & 3 & 4 & -2 & 16 & 2 \end{pmatrix}_{b_0}$$

[54] $1^{113}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 241, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 9 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 12 & 17 & 0 & 48 & 12 \\ 3 & 3 & 4 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

[55] $1^{113}31^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 244, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.10 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 12 & 3 \\ 2 & 1 & 5 & 0 & 15 & 3 \\ 3 & 5 & 4 & 0 & 22 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 12 & 15 & 22 & 0 & 84 & 16 \\ 3 & 3 & 4 & -2 & 16 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{113}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 269, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 5 & 0 & 12 & 3 \\ 3 & 5 & 4 & 0 & 17 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 12 & 17 & 0 & 48 & 12 \\ 3 & 3 & 4 & -2 & 12 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [58] $1^{11}41^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 8 & 16 & 8 & 6 \\ 8 & 5 & 13 & 7 & 5 \\ 16 & 13 & 25 & 12 & 9 \\ 8 & 7 & 12 & 4 & 4 \\ 6 & 5 & 9 & 4 & 3 \end{pmatrix}_{b_0}$$

[57] $21^{11}41^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 8 & 16 & 8 & 6 \\ 8 & 5 & 13 & 7 & 5 \\ 16 & 13 & 25 & 12 & 9 \\ 8 & 7 & 12 & 4 & 4 \\ 6 & 5 & 9 & 4 & 3 \end{pmatrix}_{b_0}$$

[60] $1^{11} 4^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 12 & 8 & 6 \\ 12 & 13 & 15 & 9 & 7 \\ 12 & 15 & 13 & 10 & 7 \\ 8 & 9 & 10 & 4 & 4 \\ 6 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[59] $1^{11} 4^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -34, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 12 & 8 & 6 \\ 12 & 13 & 15 & 9 & 7 \\ 12 & 15 & 13 & 10 & 7 \\ 8 & 9 & 10 & 4 & 4 \\ 6 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[61] $1^{11} 4^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -49, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.86 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 9 & 1 & 5 \\ 6 & 5 & 11 & 0 & 5 \\ 9 & 11 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[62] $1^{11} 4^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -52, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 9 & 7 & 5 \\ 9 & 13 & 15 & 9 & 7 \\ 9 & 15 & 13 & 10 & 7 \\ 7 & 9 & 10 & 4 & 4 \\ 5 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[63] $21^{11} 4^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -55, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 9 & 20 & 9 & 7 \\ 9 & 5 & 13 & 7 & 5 \\ 20 & 13 & 25 & 12 & 9 \\ 9 & 7 & 12 & 4 & 4 \\ 7 & 5 & 9 & 4 & 3 \end{pmatrix}_{b_0}$$

[64] $1^{11} 4^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -69, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 1 & 4 & 0 & 3 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 3 & 3 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[65] $1^{11} 4^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 9 & 1 & 5 \\ 9 & 13 & 15 & 0 & 7 \\ 9 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

[66] $1^{11} 4^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -80, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 1 & 4 & 0 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 2 & 2 & 4 & 0 & 0 & 2 \\ 3 & 3 & 4 & -2 & 2 & 2 \end{pmatrix}_{b_0}$$

[67] $1^{111} 4^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -90, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 6 & 9 & 7 & 5 \\ 6 & 5 & 11 & 7 & 5 \\ 9 & 11 & 13 & 10 & 7 \\ 7 & 7 & 10 & 4 & 4 \\ 5 & 5 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[68] $1^{11} 4^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -83, \det = -109, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.79 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 4 & 3 \\ 2 & 1 & 4 & 0 & 5 & 3 \\ 3 & 4 & 4 & 0 & 7 & 4 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 4 & 5 & 7 & 0 & 7 & 5 \\ 3 & 3 & 4 & -2 & 5 & 2 \end{pmatrix}_{b_0}$$

[69] $21^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -111$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.76 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 10 & 7 & 5 \\ 9 & 13 & 15 & 9 & 7 \\ 10 & 15 & 13 & 10 & 7 \\ 7 & 9 & 10 & 4 & 4 \\ 5 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[70] $21^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -151$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.77 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 13 & 15 & 14 & 9 & 7 \\ 15 & 13 & 15 & 9 & 7 \\ 14 & 15 & 13 & 10 & 7 \\ 9 & 9 & 10 & 4 & 4 \\ 7 & 7 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

[71] $21^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -391$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.79 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 5 & 7 & 10 & 7 & 5 \\ 7 & 5 & 11 & 7 & 5 \\ 10 & 11 & 13 & 10 & 7 \\ 7 & 7 & 10 & 4 & 4 \\ 5 & 5 & 7 & 4 & 3 \end{pmatrix}_{b_0}$$

6.124 $\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6$

$T = 9$ [0] $1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{e}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -559$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$
 $T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 9 & 11 & 9 & 1 & 5 \\ 9 & 13 & 18 & 15 & 0 & 7 \\ 11 & 18 & 20 & 17 & 0 & 8 \\ 9 & 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 7 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

6.125 $\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4$

$T = 9$ [0] $1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -440$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$
 $T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 5 & 9 & 11 & 9 & 1 & 5 \\ 9 & 13 & 18 & 15 & 0 & 7 \\ 11 & 18 & 20 & 17 & 0 & 8 \\ 9 & 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 7 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

6.126 $\text{NHC}(-2)(-2)(-3) + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8$

$T = 9$ [0] $1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -61$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$
 $T_{\min} = 12.28 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 1 & 5 \\ 8 & 10 & 17 & 0 & 0 & 6 \\ 11 & 17 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 6 & 8 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -75$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.23 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 1 & 5 \\ 8 & 10 & 17 & 0 & 0 & 6 \\ 12 & 17 & 25 & 0 & 0 & 9 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 6 & 9 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[2] $1^{11}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 13$, $\Delta = -91$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 14.0$

$T_{\min} = 12.24 \rightarrow 13$, $T_{\max} = 13$, $T \in [13, 13]$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 9 & 5 \\ 8 & 10 & 17 & 0 & 12 & 6 \\ 11 & 17 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 12 & 17 & 0 & 12 & 6 \\ 5 & 6 & 8 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 8 & 12 & 10 & 1 & 5 \\ 8 & 10 & 17 & 12 & 0 & 6 \\ 12 & 17 & 25 & 20 & 0 & 9 \\ 10 & 12 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 6 & 9 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 1 & 5 \\ 9 & 13 & 18 & 0 & 0 & 7 \\ 11 & 18 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 5 & 7 & 8 & -2 & -1 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 1 & 0 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & -2 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 0 & 1 & 0 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

[6] $1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -201, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 10 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 10 & 15 & 17 & 0 & 12 & 6 \\ 5 & 7 & 8 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{11} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + \text{su}_3 + \text{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$

$$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 9 & 5 \\ 9 & 13 & 18 & 0 & 15 & 7 \\ 11 & 18 & 20 & 0 & 17 & 8 \\ 1 & 0 & 0 & -4 & 0 & -2 \\ 9 & 15 & 17 & 0 & 12 & 6 \\ 5 & 7 & 8 & -2 & 6 & 1 \end{pmatrix}_{b_0}$$

6.127 NHC(-2)(-2)(-3) + so₈ + (-3)→(-2)

$T = 8 \quad [0] \quad 1^{11} 3^1 1^{11} 2^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 120, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.27 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 12 & 20 & 1 & 7 \\ 12 & 10 & 17 & 0 & 6 \\ 20 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 6 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

[1] $1^{11} 3^1 1^{11} 2^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.66 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 10 & 12 & 16 & 1 & 6 \\ 12 & 13 & 18 & 0 & 7 \\ 16 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 7 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

[2] $1^{11} 3^1 1^{11} 2^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 209, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$

$$T_{\min} = 11.16 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 15 & 17 & 1 & 7 \\ 15 & 13 & 18 & 0 & 7 \\ 17 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 7 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

[3] $1^{11} 3^1 1^{11} 2^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{so}_8 + (-3) \rightarrow (-2) \quad T = 8, T_{\max} = 13, \Delta = -93, \det = 685, (n_+, n_-, n_0) =$

$$(1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.21 \rightarrow 12, T_{\max} = 13, T \in [12, 13]$$

$$\begin{pmatrix} 13 & 15 & 20 & 1 & 7 \\ 15 & 13 & 21 & 0 & 7 \\ 20 & 21 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 7 & 7 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [4] \quad 1^{11} 4^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -145, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 10 & 1 & 5 \\ 9 & 13 & 15 & 0 & 7 \\ 10 & 15 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 4^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{so}_8 + (-3) \rightarrow (-2) \quad T = 9, T_{\max} = 12, \Delta = -84, \det = -313, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 5 & 7 & 10 & 1 & 5 \\ 7 & 5 & 11 & 0 & 5 \\ 10 & 11 & 13 & 0 & 7 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 5 & 7 & -1 & 3 \end{pmatrix}_{b_0}$$

6.128 NHC(-2)(-2)(-3) + $\mathfrak{su}_2$

$$T = 7 \quad [0] \quad 1^{12} 3^{11} 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -6, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 4 & 2 & 0 & 6 & 4 \\ 2 & 2 & 2 & -1 & 4 & 2 \end{pmatrix}_{b_0}$$

$$[2] \quad 21^{11} 31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -10, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 1 & 5 & 12 & 3 \\ 4 & 5 & 4 & 18 & 4 \\ 12 & 12 & 18 & 58 & 14 \\ 3 & 3 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^{11} 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -10, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.60 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & -1 & 0 & 2 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -13, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.46 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[4] \quad 21^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 4 & 6 & 4 & 4 & 4 \\ 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^{11} 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.59 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 2 & 1 & 0 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.59 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 2 \end{pmatrix}_{b_0}$$

- [7] $1^{11}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -19$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.84 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 2 & -1 & 2 & 2 \end{pmatrix}_{b_0}$
- [8] $21^{11}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -20$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 4 & 3 & 5 & 12 & 4 \\ 3 & 1 & 5 & 9 & 3 \\ 5 & 5 & 4 & 13 & 4 \\ 12 & 9 & 13 & 28 & 10 \\ 4 & 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$
- [9] $1^{12}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.67 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 5 & 4 & 0 & 12 & 6 \\ 2 & 2 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$
- [10] $1^{11}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.57 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 3 & 3 & 0 & 4 & 4 \\ 2 & 2 & 2 & -1 & 4 & 2 \end{pmatrix}_{b_0}$
- [11] $1^{11}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -33$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.79 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 4 & 5 & 0 & 12 & 6 \\ 2 & 2 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$
- [12] $1^{11}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -40$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 3 & 3 & 0 & 4 & 4 \\ 2 & 2 & 2 & -1 & 4 & 2 \end{pmatrix}_{b_0}$
- [13] $1^{13}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -41$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.83 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 8 & 7 & 0 & 36 & 10 \\ 2 & 2 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$
- [14] $1^{11}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -46$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.65 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 3 & 3 & 0 & 4 & 4 \\ 2 & 2 & 2 & -1 & 4 & 2 \end{pmatrix}_{b_0}$
- [15] $1^{12}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -46$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.74 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 6 & 6 & 0 & 22 & 8 \\ 2 & 2 & 2 & -1 & 8 & 2 \end{pmatrix}_{b_0}$
- [16] $1^{13}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -46$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.74 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 7 & 5 & 0 & 24 & 8 \\ 2 & 2 & 2 & -1 & 8 & 2 \end{pmatrix}_{b_0}$
- [17] $21^{12}31^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -50$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 4 & 3 & 5 & 24 & 4 \\ 3 & 1 & 5 & 18 & 3 \\ 5 & 5 & 4 & 26 & 4 \\ 24 & 18 & 26 & 118 & 20 \\ 4 & 3 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{12} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -55, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.65 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 5 & 4 & 0 & 12 & 6 \\ 2 & 2 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{12} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -61, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 7 & 8 & 0 & 36 & 10 \\ 2 & 2 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{14} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -61, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.89 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 9 & 6 & 0 & 40 & 10 \\ 2 & 2 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[21] \ 21^{11} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -64, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 4 & 6 & 5 & 6 & 4 \\ 6 & 4 & 6 & 6 & 4 \\ 5 & 6 & 4 & 5 & 4 \\ 6 & 6 & 5 & 4 & 4 \\ 4 & 4 & 4 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -65, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.77 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 4 & 5 & 0 & 12 & 6 \\ 2 & 2 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{12} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -65, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.77 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 5 & 4 & 0 & 12 & 6 \\ 2 & 2 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 21^{12} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -67, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.84 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 6 & 12 & 4 \\ 4 & 6 & 4 & 13 & 4 \\ 9 & 12 & 13 & 28 & 10 \\ 3 & 4 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{13} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -72, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.78 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 7 & 5 & 0 & 24 & 8 \\ 2 & 2 & 2 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{11} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -73, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 4 & 5 & 0 & 12 & 6 \\ 2 & 2 & 2 & -1 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[26] \ 21^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -73, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.79 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 1 & 5 & 6 & 3 \\ 4 & 5 & 4 & 9 & 4 \\ 6 & 6 & 9 & 13 & 7 \\ 3 & 3 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{14} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, \ T_{\max} = 10, \ \Delta = -84, \ \det = -73, \ (n_+, n_-, n_0) = (1, 7, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 11 & 10 & 0 & 72 & 14 \\ 2 & 2 & 2 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{13} 3 \ 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 7 & 5 & 0 & 24 & 8 \\ 2 & 2 & 2 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{11} 3 \ 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 5 & 7 & 0 & 24 & 8 \\ 2 & 2 & 2 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[31] \ 21^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -75, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 6 & 9 & 4 \\ 4 & 6 & 4 & 9 & 4 \\ 6 & 9 & 9 & 13 & 7 \\ 3 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{13} 3 \ 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 9 & 9 & 0 & 52 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{14} 3 \ 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -79, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 9 & 6 & 0 & 40 & 10 \\ 2 & 2 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

$$[34] \ 21^{12} 31^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 6 & 5 & 21 & 4 \\ 6 & 4 & 6 & 21 & 4 \\ 5 & 6 & 4 & 22 & 4 \\ 21 & 21 & 22 & 85 & 17 \\ 4 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

$$[35] \ 1^{12} 3 \ 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -94, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.70 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 6 & 6 & 0 & 22 & 8 \\ 2 & 2 & 2 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[36] \ 1^{11} 3 \ 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 5 & 7 & 0 & 24 & 8 \\ 2 & 2 & 2 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$

$$[37] \ 1^{14} 3 \ 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -98, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 10 & 8 & 0 & 54 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[38] \ 21^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 6 & 5 & 9 & 4 \\ 6 & 4 & 6 & 9 & 4 \\ 5 & 6 & 4 & 9 & 4 \\ 9 & 9 & 9 & 13 & 7 \\ 4 & 4 & 4 & 7 & 3 \end{pmatrix}_{b_0}$$

- [39] $1^{13}31^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -101$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.93 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 10 & 11 & 0 & 72 & 14 \\ 2 & 2 & 2 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$
- [40] $21^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -104$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.0$
 $T_{\min} = 9.81 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 1 & 5 & 3 & 3 \\ 4 & 5 & 4 & 4 & 4 \\ 3 & 3 & 4 & 1 & 3 \\ 3 & 3 & 4 & 3 & 3 \end{pmatrix}_{b_0}$$
- [41] $1^{13}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -105$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.70 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 8 & 7 & 0 & 36 & 10 \\ 2 & 2 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$
- [43] $1^{12}31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -118$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 6 & 6 & 0 & 22 & 8 \\ 2 & 2 & 2 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$
- [44] $1^{14}31^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -118$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.90 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 12 & 12 & 0 & 94 & 16 \\ 2 & 2 & 2 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$
- [42] $1^{12}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -118$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.76 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 6 & 6 & 0 & 22 & 8 \\ 2 & 2 & 2 & -1 & 8 & 2 \end{pmatrix}_{b_0}$$
- [45] $1^{12}31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -123$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.78 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 7 & 8 & 0 & 36 & 10 \\ 2 & 2 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$
- [46] $21^{14}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -124$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 9.84 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 4 & 6 & 24 & 4 \\ 4 & 6 & 4 & 26 & 4 \\ 18 & 24 & 26 & 118 & 20 \\ 3 & 4 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$
- [47] $21^{11}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -134$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.85 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 1 & 5 & 6 & 3 \\ 4 & 5 & 4 & 8 & 4 \\ 6 & 6 & 8 & 10 & 6 \\ 3 & 3 & 4 & 6 & 3 \end{pmatrix}_{b_0}$$
- [48] $1^{12}31^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -142$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 8 & 10 & 0 & 54 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$
- [49] $1^{13}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2$ $T = 7$, $T_{\max} = 10$, $\Delta = -84$, $\det = -145$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 5.8$
 $T_{\min} = 9.73 \rightarrow 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 8 & 7 & 0 & 36 & 10 \\ 2 & 2 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[50] $1^{13} 3 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -153, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 8 & 7 & 0 & 36 & 10 \\ 2 & 2 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[52] $1^{14} 3 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 10 & 8 & 0 & 54 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[51] $1^{13} 3 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -154, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 9 & 9 & 0 & 52 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[53] $21^{12} 31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.87 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 6 & 18 & 4 \\ 4 & 6 & 4 & 18 & 4 \\ 12 & 18 & 18 & 58 & 14 \\ 3 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[55] $1^{13} 3 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 8 & 7 & 0 & 36 & 10 \\ 2 & 2 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[54] $1^{12} 3 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -161, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 7 & 8 & 0 & 36 & 10 \\ 2 & 2 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[56] $21^{12} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 6 & 5 & 15 & 4 \\ 6 & 4 & 6 & 15 & 4 \\ 5 & 6 & 4 & 14 & 4 \\ 15 & 15 & 14 & 37 & 11 \\ 4 & 4 & 4 & 11 & 3 \end{pmatrix}_{b_0}$$

[58] $21^{14} 31^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -167, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 21 & 3 \\ 3 & 1 & 5 & 21 & 3 \\ 4 & 5 & 4 & 31 & 4 \\ 21 & 21 & 31 & 172 & 24 \\ 3 & 3 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[59] $1^{14} 3 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -167, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.74 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 11 & 10 & 0 & 72 & 14 \\ 2 & 2 & 2 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[57] $21^{12} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -167, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 1 & 5 & 9 & 3 \\ 4 & 5 & 4 & 13 & 4 \\ 9 & 9 & 13 & 28 & 10 \\ 3 & 3 & 4 & 10 & 3 \end{pmatrix}_{b_0}$$

[60] $21^{13} 31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.84 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 6 & 21 & 4 \\ 4 & 6 & 4 & 22 & 4 \\ 15 & 21 & 22 & 85 & 17 \\ 3 & 4 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

[61] $1^{12} 3^1 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -175, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 7 & 8 & 0 & 36 & 10 \\ 2 & 2 & 2 & -1 & 10 & 2 \end{pmatrix}_{b_0}$$

[62] $21^{112} 31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -180, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 6 & 5 & 18 & 4 \\ 6 & 4 & 6 & 18 & 4 \\ 5 & 6 & 4 & 18 & 4 \\ 18 & 18 & 18 & 58 & 14 \\ 4 & 4 & 4 & 14 & 3 \end{pmatrix}_{b_0}$$

[63] $21^{113} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -184, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 1 & 5 & 9 & 3 \\ 4 & 5 & 4 & 12 & 4 \\ 9 & 9 & 12 & 25 & 9 \\ 3 & 3 & 4 & 9 & 3 \end{pmatrix}_{b_0}$$

[64] $1^{14} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -186, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.78 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 10 & 8 & 0 & 54 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[65] $21^{123} 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -187, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 4 & 6 & 24 & 4 \\ 4 & 6 & 4 & 23 & 4 \\ 15 & 24 & 23 & 100 & 18 \\ 3 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

[66] $21^{113} 31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -188, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 1 & 5 & 15 & 3 \\ 4 & 5 & 4 & 22 & 4 \\ 15 & 15 & 22 & 85 & 17 \\ 3 & 3 & 4 & 17 & 3 \end{pmatrix}_{b_0}$$

[67] $1^{13} 3^1 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -193, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 10 & 11 & 0 & 72 & 14 \\ 2 & 2 & 2 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[68] $1^{14} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -194, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 10 & 10 & 8 & 0 & 54 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{12} 3^1 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -194, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 8 & 10 & 0 & 54 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[70] $1^{13} 3^1 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 9 & 9 & 0 & 52 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[71] $21^{113} 31^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -208, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 4 & 6 & 5 & 30 & 4 \\ 6 & 4 & 6 & 30 & 4 \\ 5 & 6 & 4 & 31 & 4 \\ 30 & 30 & 31 & 172 & 24 \\ 4 & 4 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

[72] $1^{12} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -222, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 8 & 10 & 0 & 54 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[73] $1^{14} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -226, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 12 & 12 & 0 & 94 & 16 \\ 2 & 2 & 2 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

[74] $1^{14} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -237, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.73 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 11 & 10 & 0 & 72 & 14 \\ 2 & 2 & 2 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[75] $1^{13} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -238, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 9 & 9 & 0 & 52 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[76] $1^{13} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -244, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 10 & 9 & 9 & 0 & 52 & 12 \\ 2 & 2 & 2 & -1 & 12 & 2 \end{pmatrix}_{b_0}$$

[77] $21^{14} 31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -254, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 1 & 5 & 12 & 3 \\ 4 & 5 & 4 & 16 & 4 \\ 12 & 12 & 16 & 46 & 12 \\ 3 & 3 & 4 & 12 & 3 \end{pmatrix}_{b_0}$$

[78] $1^{13} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -261, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 10 & 11 & 0 & 72 & 14 \\ 2 & 2 & 2 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[79] $21^{13} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -281, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.80 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 1 & 5 & 12 & 3 \\ 4 & 5 & 4 & 17 & 4 \\ 12 & 12 & 17 & 49 & 13 \\ 3 & 3 & 4 & 13 & 3 \end{pmatrix}_{b_0}$$

[80] $1^{14} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -283, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.76 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 10 & 11 & 10 & 0 & 72 & 14 \\ 2 & 2 & 2 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

[81] $21^{13} 31^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -291, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 4 & 6 & 27 & 4 \\ 4 & 6 & 4 & 27 & 4 \\ 18 & 27 & 27 & 133 & 21 \\ 3 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

[82] $1^{13} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -305, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 10 & 10 & 11 & 0 & 72 & 14 \\ 2 & 2 & 2 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[83] \ 1^{14} 3 \ 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -305, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 11 & 11 & 10 & 0 & 72 & 14 \\ 2 & 2 & 2 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[84] \ 1^{14} 3 \ 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -310, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 10 & 12 & 12 & 0 & 94 & 16 \\ 2 & 2 & 2 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[85] \ 21^{113} 31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -315, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 6 & 5 & 27 & 4 \\ 6 & 4 & 6 & 27 & 4 \\ 5 & 6 & 4 & 27 & 4 \\ 27 & 27 & 27 & 133 & 21 \\ 4 & 4 & 4 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[86] \ 21^{113} 31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -316, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.91 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 4 & 6 & 5 & 24 & 4 \\ 6 & 4 & 6 & 24 & 4 \\ 5 & 6 & 4 & 23 & 4 \\ 24 & 24 & 23 & 100 & 18 \\ 4 & 4 & 4 & 18 & 3 \end{pmatrix}_{b_0}$$

$$[87] \ 1^{13} 3 \ 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -325, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 11 & 10 & 11 & 0 & 72 & 14 \\ 2 & 2 & 2 & -1 & 14 & 2 \end{pmatrix}_{b_0}$$

$$[88] \ 21^{143} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -331, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.85 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 21 & 3 \\ 3 & 4 & 6 & 30 & 4 \\ 4 & 6 & 4 & 31 & 4 \\ 21 & 30 & 31 & 172 & 24 \\ 3 & 4 & 4 & 24 & 3 \end{pmatrix}_{b_0}$$

$$[89] \ 21^{115} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -344, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.1$$

$$T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 1 & 5 & 15 & 3 \\ 4 & 5 & 4 & 20 & 4 \\ 15 & 15 & 20 & 73 & 15 \\ 3 & 3 & 4 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[90] \ 21^{13} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -352, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 21 & 3 \\ 3 & 4 & 6 & 33 & 4 \\ 4 & 6 & 4 & 32 & 4 \\ 21 & 33 & 32 & 193 & 25 \\ 3 & 4 & 4 & 25 & 3 \end{pmatrix}_{b_0}$$

$$[91] \ 1^{14} 3 \ 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -370, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.77 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 11 & 12 & 12 & 0 & 94 & 16 \\ 2 & 2 & 2 & -1 & 16 & 2 \end{pmatrix}_{b_0}$$

$$[92] \ 21^{114} 31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -386, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.79 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 1 & 5 & 18 & 3 \\ 4 & 5 & 4 & 26 & 4 \\ 18 & 18 & 26 & 118 & 20 \\ 3 & 3 & 4 & 20 & 3 \end{pmatrix}_{b_0}$$

$$[93] \ 1^{14} 3 \ 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -406, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.81 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{aligned}
& \begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 12 & 12 & 12 & 0 & 94 & 16 \\ 2 & 2 & 2 & -1 & 16 & 2 \end{pmatrix}_{b_0} \\
[94] \quad & 21^{114}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -415, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.82 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 1 & 5 & 15 & 3 \\ 4 & 5 & 4 & 21 & 4 \\ 15 & 15 & 21 & 76 & 16 \\ 3 & 3 & 4 & 16 & 3 \end{pmatrix}_{b_0} \\
[95] \quad & 21^{114}31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -436, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 6 & 5 & 30 & 4 \\ 6 & 4 & 6 & 30 & 4 \\ 5 & 6 & 4 & 28 & 4 \\ 30 & 30 & 28 & 154 & 22 \\ 4 & 4 & 4 & 22 & 3 \end{pmatrix}_{b_0} \\
[96] \quad & 21^{116}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -454, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 18 & 3 \\ 3 & 1 & 5 & 18 & 3 \\ 4 & 5 & 4 & 24 & 4 \\ 18 & 18 & 24 & 106 & 18 \\ 3 & 3 & 4 & 18 & 3 \end{pmatrix}_{b_0} \\
[97] \quad & 21^{143}1^{114} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -480, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.88 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 24 & 3 \\ 3 & 4 & 6 & 36 & 4 \\ 4 & 6 & 4 & 36 & 4 \\ 24 & 36 & 36 & 238 & 28 \\ 3 & 4 & 4 & 28 & 3 \end{pmatrix}_{b_0} \\
[98] \quad & 21^{114}31^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -504, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.89 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 6 & 5 & 36 & 4 \\ 6 & 4 & 6 & 36 & 4 \\ 5 & 6 & 4 & 36 & 4 \\ 36 & 36 & 36 & 238 & 28 \\ 4 & 4 & 4 & 28 & 3 \end{pmatrix}_{b_0} \\
[99] \quad & 21^{114}31^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -523, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.90 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 4 & 6 & 5 & 33 & 4 \\ 6 & 4 & 6 & 33 & 4 \\ 5 & 6 & 4 & 32 & 4 \\ 33 & 33 & 32 & 193 & 25 \\ 4 & 4 & 4 & 25 & 3 \end{pmatrix}_{b_0} \\
[100] \quad & 221^{111}3 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -531, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.75 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 3 & 3 & 3 & 3 & 2 \\ 3 & 1 & 4 & 3 & 3 & 3 \\ 3 & 4 & 1 & 4 & 3 & 3 \\ 3 & 3 & 4 & 0 & 3 & 2 \\ 3 & 3 & 3 & 3 & 1 & 3 \\ 2 & 3 & 3 & 2 & 3 & 2 \end{pmatrix}_{b_0} \\
[101] \quad & 21^{117}31^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 7, T_{\max} = 10, \Delta = -84, \det = -584, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 1 & 3 & 4 & 21 & 3 \\ 3 & 1 & 5 & 21 & 3 \\ 4 & 5 & 4 & 28 & 4 \\ 21 & 21 & 28 & 145 & 21 \\ 3 & 3 & 4 & 21 & 3 \end{pmatrix}_{b_0} \\
T = 8 \quad & [102] \quad 2 \overset{11}{3} 2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 2 & 2 & 3 \\ 2 & 4 & 1 & 3 & 2 & 3 \\ 2 & 2 & 3 & 0 & 3 & 2 \\ 2 & 2 & 2 & 3 & 1 & 3 \\ 2 & 3 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
[103] \quad & 1^{11}31^131^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0} \\
[104] \quad & 1^{11}31^131^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 4 \\ 1 & 4 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 2 & 4 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}
\end{aligned}$$

$$\begin{aligned}
[105] \quad & \begin{smallmatrix} 1^{11} \\ 2 \ 3 \ 2 \\ 1^{111} \end{smallmatrix} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 1 & 4 & 2 & 5 & 3 \\ 2 & 4 & 1 & 3 & 5 & 3 \\ 2 & 2 & 3 & 0 & 5 & 2 \\ 4 & 5 & 5 & 5 & 8 & 6 \\ 2 & 3 & 3 & 2 & 6 & 3 \end{pmatrix}_{b_0} \\
[106] \quad & \begin{smallmatrix} 1^{12} \\ 2 \ 3 \ 2 \\ 1^{112} \end{smallmatrix} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 112, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 4 & 2 & 10 & 3 \\ 2 & 4 & 1 & 3 & 10 & 3 \\ 2 & 2 & 3 & 0 & 10 & 2 \\ 8 & 10 & 10 & 10 & 38 & 12 \\ 2 & 3 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[107] \quad & \begin{smallmatrix} 1^1 \\ 2 \ 3 \ 2 \\ 1^{111} \end{smallmatrix} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 2 & 2 \\ 2 & 1 & 4 & 2 & 3 & 3 \\ 2 & 4 & 1 & 3 & 3 & 3 \\ 2 & 2 & 3 & 0 & 2 & 2 \\ 2 & 3 & 3 & 2 & 1 & 3 \\ 2 & 3 & 3 & 2 & 3 & 3 \end{pmatrix}_{b_0} \\
[108] \quad & \begin{smallmatrix} 1^1 \\ 2 \ 3 \ 2 \\ 1^{112} \end{smallmatrix} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 128, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.94 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 4 & 2 \\ 2 & 1 & 4 & 2 & 6 & 3 \\ 2 & 4 & 1 & 3 & 6 & 3 \\ 2 & 2 & 3 & 0 & 4 & 2 \\ 4 & 6 & 6 & 4 & 10 & 6 \\ 2 & 3 & 3 & 2 & 6 & 3 \end{pmatrix}_{b_0} \\
[109] \quad & \begin{smallmatrix} 1^1 \\ 2 \ 3 \ 2 \\ 1^{113} \end{smallmatrix} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 148, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 1 & 4 & 2 & 9 & 3 \\ 2 & 4 & 1 & 3 & 9 & 3 \\ 2 & 2 & 3 & 0 & 6 & 2 \\ 6 & 9 & 9 & 6 & 25 & 9 \\ 2 & 3 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[110] \quad & \begin{smallmatrix} 1^1 \\ 2 \ 3 \ 2 \\ 1^{114} \end{smallmatrix} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 176, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 9.95 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 4 & 2 & 12 & 3 \\ 2 & 4 & 1 & 3 & 12 & 3 \\ 2 & 2 & 3 & 0 & 8 & 2 \\ 8 & 12 & 12 & 8 & 46 & 12 \\ 2 & 3 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0} \\
[111] \quad & \begin{smallmatrix} 1^1 \\ 2 \ 3 \ 2 \\ 1^{115} \end{smallmatrix} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 212, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.1 \\
& T_{\min} = 9.96 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 4 & 2 & 15 & 3 \\ 2 & 4 & 1 & 3 & 15 & 3 \\ 2 & 2 & 3 & 0 & 10 & 2 \\ 10 & 15 & 15 & 10 & 73 & 15 \\ 2 & 3 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0} \\
[112] \quad & \begin{smallmatrix} 1^{11} \\ 2 \ 3 \ 2 \\ 1^{112} \end{smallmatrix} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 228, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 6 & 2 \\ 2 & 1 & 4 & 2 & 8 & 3 \\ 2 & 4 & 1 & 3 & 8 & 3 \\ 2 & 2 & 3 & 0 & 7 & 2 \\ 6 & 8 & 8 & 7 & 21 & 9 \\ 2 & 3 & 3 & 2 & 9 & 3 \end{pmatrix}_{b_0} \\
[113] \quad & \begin{smallmatrix} 1^1 \\ 2 \ 3 \ 2 \\ 1^{116} \end{smallmatrix} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 256, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.2 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 2 & 2 & 2 & 12 & 2 \\ 2 & 1 & 4 & 2 & 18 & 3 \\ 2 & 4 & 1 & 3 & 18 & 3 \\ 2 & 2 & 3 & 0 & 12 & 2 \\ 12 & 18 & 18 & 12 & 106 & 18 \\ 2 & 3 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0} \\
[114] \quad & \begin{smallmatrix} 1^1 \\ 2 \ 3 \ 2 \\ 1^{117} \end{smallmatrix} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 308, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.3 \\
& T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]
\end{aligned}$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 14 & 2 \\ 2 & 1 & 4 & 2 & 21 & 3 \\ 2 & 4 & 1 & 3 & 21 & 3 \\ 2 & 2 & 3 & 0 & 14 & 2 \\ 14 & 21 & 21 & 14 & 145 & 21 \\ 2 & 3 & 3 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

$$[115] \quad 2 \overset{1^{12}}{\underset{1^{113}}{3}} 2 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 340, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 4 & 2 & 13 & 3 \\ 2 & 4 & 1 & 3 & 13 & 3 \\ 2 & 2 & 3 & 0 & 12 & 2 \\ 10 & 13 & 13 & 12 & 61 & 15 \\ 2 & 3 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[116] \quad 2 \overset{1^{11}}{\underset{1^{113}}{3}} 2 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 352, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 8 & 2 \\ 2 & 1 & 4 & 2 & 11 & 3 \\ 2 & 4 & 1 & 3 & 11 & 3 \\ 2 & 2 & 3 & 0 & 9 & 2 \\ 8 & 11 & 11 & 9 & 40 & 12 \\ 2 & 3 & 3 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[117] \quad 2 \overset{1^{11}}{\underset{1^{114}}{3}} 2 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 484, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 10 & 2 \\ 2 & 1 & 4 & 2 & 14 & 3 \\ 2 & 4 & 1 & 3 & 14 & 3 \\ 2 & 2 & 3 & 0 & 11 & 2 \\ 10 & 14 & 14 & 11 & 65 & 15 \\ 2 & 3 & 3 & 2 & 15 & 3 \end{pmatrix}_{b_0}$$

$$[118] \quad 2 \overset{1^{12}}{\underset{1^{114}}{3}} 2 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -58, \det = 576, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 9.93 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 2 & 2 & 2 & 12 & 2 \\ 2 & 1 & 4 & 2 & 16 & 3 \\ 2 & 4 & 1 & 3 & 16 & 3 \\ 2 & 2 & 3 & 0 & 14 & 2 \\ 12 & 16 & 16 & 14 & 90 & 18 \\ 2 & 3 & 3 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [120] \quad 1 \overset{1^{11}}{\underset{1^{12}}{2}} 3 21^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 7 & 3 \\ 1 & 2 & 1 & 1 & 5 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 7 & 5 & 0 & 17 & 9 \\ 2 & 3 & 3 & -1 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[124] \quad 1 \overset{1^1}{\underset{1^{11}}{2}} 3 21^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[123] \quad 1 \overset{1^{12}}{\underset{1^{13}}{2}} 3 21^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 10 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 6 & 2 \\ 6 & 10 & 0 & 6 & 34 & 12 \\ 2 & 3 & 0 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[122] \quad 1 \overset{1^1}{\underset{1^{11}}{2}} 3 21^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 1 & 1 & 4 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 4 & 2 \\ 2 & 4 & 0 & 4 & 6 & 6 \\ 2 & 3 & 0 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[119] \quad 1 \overset{1^1}{\underset{1^{11}}{2}} 3 21^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 2 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[121] \quad & 1^{11} 2^1 3^1 21^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -2, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 0 & 3 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 2 & 0 & 3 & 2 & 0 & 4 \end{pmatrix}_{b_0} \\
[127] \quad & 1^{12} 2^1 3^1 21^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 8 & 3 \\ 1 & 2 & 1 & 1 & 8 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 8 & 8 & 0 & 30 & 12 \\ 2 & 3 & 3 & -1 & 12 & 4 \end{pmatrix}_{b_0} \\
[128] \quad & 1^{12} 2^1 3^1 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 1 & 1 & 7 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 5 & 2 \\ 4 & 7 & 0 & 5 & 17 & 9 \\ 2 & 3 & 0 & 2 & 9 & 4 \end{pmatrix}_{b_0} \\
[129] \quad & 1^{13} 2^1 3^1 21^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 12 & 3 \\ 1 & 2 & 1 & 1 & 12 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 12 & 12 & 0 & 70 & 18 \\ 2 & 3 & 3 & -1 & 18 & 4 \end{pmatrix}_{b_0} \\
[130] \quad & 1^{14} 2^1 3^1 21^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 16 & 3 \\ 1 & 2 & 1 & 1 & 16 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 16 & 16 & 0 & 126 & 24 \\ 2 & 3 & 3 & -1 & 24 & 4 \end{pmatrix}_{b_0} \\
[131] \quad & 1^{11} 2^1 3^1 21^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 3 & 3 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 2 & 0 & 3 & 1 & 1 & 3 \\ 2 & 0 & 3 & 2 & 3 & 4 \end{pmatrix}_{b_0} \\
[125] \quad & 1^{13} 2^1 3^1 21^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 1 & 1 & 11 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 9 & 2 \\ 6 & 11 & 0 & 9 & 49 & 15 \\ 2 & 3 & 0 & 2 & 15 & 4 \end{pmatrix}_{b_0} \\
[126] \quad & 1^{11} 2^1 3^1 21^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 1 & 1 & 4 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 4 & 0 & 6 & 6 \\ 2 & 3 & 3 & -1 & 6 & 4 \end{pmatrix}_{b_0} \\
[132] \quad & 1^{13} 2^1 3^1 21^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -10, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 11 & 3 \\ 1 & 2 & 1 & 1 & 9 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 11 & 9 & 0 & 49 & 15 \\ 2 & 3 & 3 & -1 & 15 & 4 \end{pmatrix}_{b_0} \\
[133] \quad & 1^{14} 2^1 3^1 21^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -14, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10, T_{\max} = 10, T \in [10, 10] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 1 & 1 & 14 & 3 \\ 1 & 1 & -2 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 10 & 2 \\ 8 & 14 & 0 & 10 & 74 & 18 \\ 2 & 3 & 0 & 2 & 18 & 4 \end{pmatrix}_{b_0}
\end{aligned}$$

$$[134] \ 1^{\overline{11}} 2^{\overline{11}} 3^{\overline{11}} 21^{\overline{11}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -18, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.0$$

$$T_{\min} = 9.78 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 3 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[135] \quad 1^{11} 2^3 3^{21} 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 5.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 4 & 0 & 4 & 4 & 6 & 6 \\ 2 & 0 & 3 & 2 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[136] \quad 1^{14} 2^3 21^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 6.0$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 15 & 3 \\ 1 & 2 & 1 & 1 & 13 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 15 & 13 & 0 & 97 & 21 \\ 2 & 3 & 3 & -1 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[137] \quad 1^{12} 2^3 3^{21} 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -18, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 8 & 3 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 5 & 0 & 8 & 8 & 30 & 12 \\ 2 & 0 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[138] $1^{12}2^321^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 9.86 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 7 & 3 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 4 & 0 & 7 & 5 & 17 & 9 \\ 2 & 0 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$

$$[139] \quad 1^{13} 2^3 3^{21} 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -32, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 11 & 3 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 6 & 0 & 11 & 9 & 49 & 15 \\ 2 & 0 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$[141] \quad 1^{12} 2^3 3^{12} 21^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, \quad T_{\max} = 10, \quad \Delta = -29, \quad \det = -42, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.90 \rightarrow 10, \quad T_{\max} = 10, \quad T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 8 & 3 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 6 & 0 & 8 & 8 & 30 & 12 \\ 2 & 0 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[140] \ 1^{12}2^33^{21}1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -42, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 9.90 \rightarrow 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 7 & 3 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 5 & 0 & 7 & 5 & 17 & 9 \\ 2 & 0 & 3 & 2 & 9 & 4 \end{pmatrix}_{b_0}$$

$$[142] \ 1^{12} 2^3 21^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -50, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.5$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 8 & 3 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 7 & 0 & 8 & 8 & 30 & 12 \\ 2 & 0 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[143] \ 1^{13} 2^3 3^{21} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, \ T_{\max} = 10, \ \Delta = -29, \ \det = -50, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 5.8$$

$$T_{\min} = 10, \ T_{\max} = 10, \ T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 10 & 3 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 6 & 0 & 10 & 6 & 34 & 12 \\ 2 & 0 & 3 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

$$[144] \ 1^{13} 2^3 3^{21} 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -50, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 5.8$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 12 & 3 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 8 & 0 & 12 & 12 & 70 & 18 \\ 2 & 0 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 11 & 3 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 7 & 0 & 11 & 9 & 49 & 15 \\ 2 & 0 & 3 & 2 & 15 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 15 & 3 \\ 1 & 0 & 2 & 0 & 13 & 2 \\ 9 & 0 & 15 & 13 & 97 & 21 \\ 2 & 0 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 12 & 3 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 9 & 0 & 12 & 12 & 70 & 18 \\ 2 & 0 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 14 & 3 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 8 & 0 & 14 & 10 & 74 & 18 \\ 2 & 0 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_6}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 11 & 3 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 8 & 0 & 11 & 9 & 49 & 15 \\ 2 & 0 & 3 & 2 & 15 & 4 \end{pmatrix} b_i$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 11 & 3 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 9 & 0 & 11 & 9 & 49 & 15 \\ 2 & 0 & 3 & 2 & 15 & 4 \end{pmatrix}_b$$

$$[1^{14} 2^3 3^{21} 21^{13} + (-2)^{\text{min}} = 10, T_{\max} = 10,$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 12 & 3 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 10 & 0 & 12 & 12 & 70 & 18 \end{pmatrix}$$

$$\left(\begin{array}{cccccc} 10 & 0 & 12 & 12 & 70 & 18 \\ 2 & 0 & 3 & 2 & 18 & 4 \end{array} \right)_{b_0}$$

$$\left(\begin{array}{cccccc} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 16 & 3 \\ 1 & 0 & 2 & 0 & 16 & 2 \end{array} \right)$$

$$\begin{pmatrix} 1 & 0 & 2 & 0 & 16 & 2 \\ 11 & 0 & 16 & 16 & 126 & 24 \\ 2 & 0 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

$$T_{\min} = 9.93 \rightarrow 10, \quad T_{\max} = 10.07 \rightarrow 10$$

$$\begin{pmatrix} 1 & 1 & 1 & 2 & 14 & 3 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 9 & 0 & 14 & 10 & 74 & 18 \\ 2 & 0 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

$$T_{\min} = 9.91 \rightarrow 10, \quad T_{\max} = 10.09 \rightarrow 10,$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 15 & 3 \\ 1 & 0 & 2 & 0 & 13 & 2 \\ 10 & 0 & 15 & 13 & 97 & 21 \\ 2 & 0 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

[155] $1^{14}2^3 3 21^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -138, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 14 & 3 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 10 & 0 & 14 & 10 & 74 & 18 \\ 2 & 0 & 3 & 2 & 18 & 4 \end{pmatrix}_{b_0}$$

[156] $1^{14}2^3 3 21^{14} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -138, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.97 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 12 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 16 & 3 \\ 1 & 0 & 2 & 0 & 16 & 2 \\ 12 & 0 & 16 & 16 & 126 & 24 \\ 2 & 0 & 3 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[157] $1^{14}2^3 3 21^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -29, \det = -158, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 9.92 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 11 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 15 & 3 \\ 1 & 0 & 2 & 0 & 13 & 2 \\ 11 & 0 & 15 & 13 & 97 & 21 \\ 2 & 0 & 3 & 2 & 21 & 4 \end{pmatrix}_{b_0}$$

$T = 10$ [158] $231^{11} 3151^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 566, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 5.0$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 5 & 5 & 1 & 5 & 2 \\ 5 & 33 & 36 & 0 & 35 & 11 \\ 5 & 36 & 33 & 1 & 35 & 11 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 35 & 35 & 0 & 33 & 11 \\ 2 & 11 & 11 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

$T = 12$ [162] $1^{13} 231^1 5 1321^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.8$

[160] $1^{11} 31^1 5 13221^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$

[164] $1^{12} 231^1 5 13221^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 2), c_{\text{ext}} = 5.5$

[161] $1^{14} 231^1 5 1321^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 6.0$

[165] $1^{12} 31^1 5 13221^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.5$

[166] $1^{13} 31^1 5 13221^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$

[163] $1^{11} 231^1 5 1321^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = -1, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$

[159] $1^{11} 31^1 5 13221^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 0, (n_+, n_-, n_0) = (1, 12, 1), c_{\text{ext}} = 5.0$

[167] $1^{13} 31^1 5 1321^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 12 & 3 \\ 1 & 0 & 2 & 1 & 15 & 3 \\ 9 & 0 & 12 & 15 & 79 & 21 \\ 2 & 0 & 3 & 3 & 21 & 5 \end{pmatrix}_{b_0}$$

[168] $1^{12} 31^1 5 1321^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 8 & 3 \\ 1 & 2 & 2 & 1 & 10 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 8 & 10 & 0 & 34 & 14 \\ 2 & 3 & 4 & -1 & 14 & 5 \end{pmatrix}_{b_0}$$

[169] $1^{14} 31^1 5 1321^{14} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 16 & 3 \\ 1 & 2 & 2 & 1 & 20 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 16 & 20 & 0 & 142 & 28 \\ 2 & 3 & 4 & -1 & 28 & 5 \end{pmatrix}_{b_0}$$

[170] $1^{13}3151321^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 12 & 3 \\ 1 & 2 & 2 & 1 & 15 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 12 & 15 & 0 & 79 & 21 \\ 2 & 3 & 4 & -1 & 21 & 5 \end{pmatrix}_{b_0}$$

[171] $1^{14}3151321^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 15 & 3 \\ 1 & 0 & 2 & 1 & 16 & 3 \\ 9 & 0 & 15 & 16 & 106 & 24 \\ 2 & 0 & 3 & 3 & 24 & 5 \end{pmatrix}_{b_0}$$

[172] $1^{11}3151321^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 4 & 3 \\ 1 & 2 & 2 & 1 & 5 & 4 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 4 & 5 & 0 & 7 & 7 \\ 2 & 3 & 4 & -1 & 7 & 5 \end{pmatrix}_{b_0}$$

[173] $1^{11}3151321^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 4 & 4 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 2 & 0 & 4 & 1 & 2 & 4 \\ 2 & 0 & 4 & 2 & 4 & 5 \end{pmatrix}_{b_0}$$

[174] $1^{11}3151321^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 4 & 3 \\ 1 & 0 & 2 & 1 & 5 & 3 \\ 3 & 0 & 4 & 5 & 7 & 7 \\ 2 & 0 & 3 & 3 & 7 & 5 \end{pmatrix}_{b_0}$$

[175] $1^{12}3151321^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 2 & 8 & 3 \\ 1 & 0 & 2 & 1 & 10 & 3 \\ 6 & 0 & 8 & 10 & 34 & 14 \\ 2 & 0 & 3 & 3 & 14 & 5 \end{pmatrix}_{b_0}$$

[176] $1^{11}3151321^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 5 & 4 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 3 & 0 & 5 & 4 & 7 & 7 \\ 2 & 0 & 4 & 2 & 7 & 5 \end{pmatrix}_{b_0}$$

[177] $1^{11}3151321^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 15, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 11.87 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 9 & 4 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 4 & 0 & 9 & 5 & 21 & 11 \\ 2 & 0 & 4 & 2 & 11 & 5 \end{pmatrix}_{b_0}$$

[178] $1^{12}3151321^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 16, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 14 & 4 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 6 & 0 & 14 & 9 & 58 & 18 \\ 2 & 0 & 4 & 2 & 18 & 5 \end{pmatrix}_{b_0}$$

[180] $1^{11}3151321^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 9 & 4 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 5 & 0 & 9 & 5 & 21 & 11 \\ 2 & 0 & 4 & 2 & 11 & 5 \end{pmatrix}_{b_0}$$

[179] $1^{11}31^15^{12}1321^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 25, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 13 & 4 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 6 & 0 & 13 & 6 & 43 & 15 \\ 2 & 0 & 4 & 2 & 15 & 5 \end{pmatrix}_{b_0}$$

[181] $1^{12}31^15^{12}1321^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 18 & 4 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 8 & 0 & 18 & 10 & 90 & 22 \\ 2 & 0 & 4 & 2 & 22 & 5 \end{pmatrix}_{b_0}$$

[182] $1^{12}31^15^{12}1321^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 14 & 4 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 7 & 0 & 14 & 9 & 58 & 18 \\ 2 & 0 & 4 & 2 & 18 & 5 \end{pmatrix}_{b_0}$$

[183] $1^{13}31^15^{12}1321^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 19 & 4 \\ 1 & 0 & 2 & 0 & 13 & 2 \\ 9 & 0 & 19 & 13 & 113 & 25 \\ 2 & 0 & 4 & 2 & 25 & 5 \end{pmatrix}_{b_0}$$

[184] $1^{13}31^15^{12}1321^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 70, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 5.5$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 2 & 2 & 18 & 4 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 9 & 0 & 18 & 10 & 90 & 22 \\ 2 & 0 & 4 & 2 & 22 & 5 \end{pmatrix}_{b_0}$$

6.129 NHC(-2)(-2)(-3) + $\mathfrak{su}_2$ + $\mathfrak{so}_8$

$T = 8$ [0] $1^{11}3^11^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 26, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 1 & 2 & 1 & 0 & 0 & -2 \\ 2 & 2 & 2 & -1 & 0 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^11^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 31, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.90 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 & 1 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 1 & 2 & 0 & 0 & 1 \\ 1 & 2 & 1 & 0 & 1 & -2 \\ 2 & 2 & 2 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11}3^11^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 & 1 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 1 & 2 & 0 & 0 & 1 \\ 1 & 2 & 1 & 0 & 1 & -2 \\ 2 & 2 & 2 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{11}3^11^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 50, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

[13] $1^{111} 3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 139, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 & 2 \\ 1 & 2 & 0 & 1 & 3 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 3 & 0 & 4 & 3 & 4 \\ 1 & 2 & 1 & 0 & 3 & -2 & 0 \\ 2 & 2 & 2 & -1 & 4 & 0 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{111} 3 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 159, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 & 2 \\ 1 & 2 & 0 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 5 & 5 & 7 & 0 & 24 & 5 & 8 \\ 1 & 2 & 1 & 0 & 5 & -2 & 0 \\ 2 & 2 & 2 & -1 & 8 & 0 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{111} 3 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 166, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 1 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 & 2 \\ 1 & 2 & 0 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 5 & 4 & 5 & 0 & 12 & 4 & 6 \\ 1 & 2 & 1 & 0 & 4 & -2 & 0 \\ 2 & 2 & 2 & -1 & 6 & 0 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{111} 3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 169, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.89 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 3 & 2 \\ 1 & 2 & 0 & 1 & 3 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 3 & 0 & 4 & 6 & 4 \\ 2 & 3 & 3 & 0 & 6 & 2 & 2 \\ 2 & 2 & 2 & -1 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{111} 3 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 & 2 \\ 1 & 2 & 0 & 1 & 5 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 4 & 4 & 5 & 0 & 12 & 4 & 6 \\ 1 & 2 & 1 & 0 & 4 & -2 & 0 \\ 2 & 2 & 2 & -1 & 6 & 0 & 1 \end{pmatrix}_{b_0}$$

[18] $1^{111} 3 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 181, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.83 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 1 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 & 2 \\ 1 & 2 & 0 & 1 & 7 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 6 & 5 & 7 & 0 & 24 & 5 & 8 \\ 1 & 2 & 1 & 0 & 5 & -2 & 0 \\ 2 & 2 & 2 & -1 & 8 & 0 & 1 \end{pmatrix}_{b_0}$$

[19] $21^{111} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 194, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.88 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 10 & 12 & 16 & 24 & 6 \\ 12 & 13 & 18 & 27 & 7 \\ 16 & 18 & 20 & 33 & 8 \\ 24 & 27 & 33 & 49 & 13 \\ 6 & 7 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

[20] $21^{111} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 248, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.0$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 15 & 17 & 27 & 7 \\ 15 & 13 & 18 & 27 & 7 \\ 17 & 18 & 20 & 33 & 8 \\ 27 & 27 & 33 & 49 & 13 \\ 7 & 7 & 8 & 13 & 3 \end{pmatrix}_{b_0}$$

[21] $21^{123} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8$ $T = 8, T_{\max} = 11, \Delta = -83, \det = 378, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.5$
 $T_{\min} = 10.85 \rightarrow 11, T_{\max} = 11, T \in [11, 11]$

$$\begin{pmatrix} 13 & 15 & 17 & 42 & 7 \\ 15 & 13 & 18 & 42 & 7 \\ 17 & 18 & 20 & 50 & 8 \\ 42 & 42 & 50 & 118 & 20 \\ 7 & 7 & 8 & 20 & 3 \end{pmatrix}_{b_0}$$

6.130 NHC(-2)(-2)(-3) + $\mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$

$T = 9$ $[0] 1^{11} 3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{so}_8 + \mathfrak{so}_8$ $T = 9, T_{\max} = 12, \Delta = -82, \det = -99, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 1 & 2 & 0 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 0 & 2 & 1 & -2 & 0 \\ 2 & 2 & 2 & -1 & 2 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11} 1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -130$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 11.94 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 1 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 2 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 1 & 2 & 0 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & -2 & 1 & 0 \\ 1 & 1 & 2 & 0 & 2 & 1 & -2 & 0 \\ 2 & 2 & 2 & -1 & 2 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{11} 1^{12} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -274$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 3 & 3 & 0 & 4 & 3 & 3 & 4 \\ 1 & 2 & 1 & 0 & 3 & -2 & 1 & 0 \\ 1 & 1 & 2 & 0 & 3 & 1 & -2 & 0 \\ 2 & 2 & 2 & -1 & 4 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{11} 1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -286$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 4 & 3 & 3 & 4 \\ 1 & 2 & 1 & 0 & 3 & -2 & 1 & 0 \\ 1 & 1 & 2 & 0 & 3 & 1 & -2 & 0 \\ 2 & 2 & 2 & -1 & 4 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

[4] $1^{11} 1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{so}_8 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -82$, $\det = -431$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 13.0$

$T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 3 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 3 & 3 & 3 & 0 & 4 & 3 & 3 & 4 \\ 1 & 2 & 1 & 0 & 3 & -2 & 1 & 0 \\ 1 & 1 & 2 & 0 & 3 & 1 & -2 & 0 \\ 2 & 2 & 2 & -1 & 4 & 0 & 0 & 0 \end{pmatrix}_{b_0}$$

6.131 NHC(-2)(-2)(-3) + $\mathbf{su}_2$ + $\mathbf{su}_2$

$T = 9$ [0] $1^{11} 1^{11} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 10$, $\Delta = -33$, $\det = -8$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 6.0$

$T_{\min} = 10$, $T_{\max} = 10$, $T \in [10, 10]$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 1 & 2 & 2 \\ 2 & 4 & 4 & 0 & 3 & 3 & 4 \\ 1 & 4 & 1 & 1 & 3 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 3 & 1 & 3 \\ 2 & 4 & 3 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

6.132 NHC(-2)(-2)(-3) + $\mathbf{su}_2$ + $\mathbf{su}_{2n3\text{mix}}$

$T = 12$ [0] $1^{11} 1^{11} 3^1 1^{11} 5^{11} 13221^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -1$, $\det = 0$, $(n_+, n_-, n_0) = (1, 12, 2)$, $c_{\text{ext}} = 6.0$

[1] $1^{11} 1^{11} 3^1 1^{11} 5^{11} 1321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -1$, $\det = 0$, $(n_+, n_-, n_0) = (1, 12, 1)$, $c_{\text{ext}} = 6.0$

[2] $1^{11} 1^{12} 3^1 1^{11} 5^{11} 1321^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -1$, $\det = 0$, $(n_+, n_-, n_0) = (1, 12, 1)$, $c_{\text{ext}} = 6.0$

[3] $1^{11} 1^{11} 3^1 1^{11} 5^{11} 1321^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_2 + \mathbf{su}_2$ $T = 12$, $T_{\max} = 12$, $\Delta = -1$, $\det = 0$, $(n_+, n_-, n_0) = (1, 12, 1)$, $c_{\text{ext}} = 6.0$

6.133 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3$

$$T = 8 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 3 \\ 3 & 4 & 4 & 0 & 0 & 4 \\ 2 & 4 & 1 & 1 & 0 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{14}} 3^{\overline{12}} 1^{\overline{13}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.6$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 1 & 11 & 2 \\ 1 & 2 & 0 & 1 & 2 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 1 & 2 & 0 & 0 & 10 & 2 \\ 9 & 11 & 10 & 0 & 10 & 71 & 13 \\ 2 & 2 & 2 & -1 & 2 & 13 & 2 \end{pmatrix}_{b_0}$$

6.134 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 3 & 4 \\ 2 & 4 & 1 & 1 & 3 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 0 & 3 & 4 \\ 2 & 4 & 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 3 & 0 & 0 & 1 & 3 \\ 3 & 4 & 3 & -1 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 3 & 4 \\ 2 & 4 & 1 & 1 & 3 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

6.135 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3$

$$T = 7 \quad [0] \ 21^{\overline{12}} 31^{\overline{13}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -4, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10, T_{\max} = 12, T \in [10, 12]$$

$$\begin{pmatrix} 4 & 3 & 5 & 21 & 4 \\ 3 & 1 & 5 & 15 & 3 \\ 5 & 5 & 4 & 22 & 4 \\ 21 & 15 & 22 & 84 & 16 \\ 4 & 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

- [1] $1^{11} 1^3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 5 & 3 & 3 \\ 3 & 5 & 4 & 5 & 4 \\ 3 & 3 & 5 & 3 & 3 \\ 3 & 3 & 4 & 3 & 2 \end{pmatrix}_{b_0}$
- [2] $1^{112} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -15, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 9 & 3 \\ 2 & 1 & 5 & 9 & 3 \\ 3 & 5 & 4 & 13 & 4 \\ 9 & 9 & 13 & 27 & 9 \\ 3 & 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$
- [3] $1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -16, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 6 & 9 & 4 \\ 3 & 6 & 4 & 9 & 4 \\ 7 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$
- [4] $1^{12} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 4 & 2 & 0 & 5 & 3 \\ 2 & 2 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$
- [5] $1^{11} 1^3 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 5 & 3 & 3 \\ 4 & 5 & 9 & 4 & 5 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$
- [6] $1^{111} 1^3 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -17, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 4 & 7 & 3 \\ 2 & 1 & 5 & 9 & 3 \\ 4 & 5 & 9 & 16 & 5 \\ 7 & 9 & 16 & 27 & 9 \\ 3 & 3 & 5 & 9 & 2 \end{pmatrix}_{b_0}$
- [7] $1^{12} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -18, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 6 & 3 \\ 2 & 1 & 5 & 6 & 3 \\ 3 & 5 & 4 & 8 & 4 \\ 6 & 6 & 8 & 9 & 5 \\ 3 & 3 & 4 & 5 & 2 \end{pmatrix}_{b_0}$
- [8] $21^{13} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 4 & 6 & 15 & 4 \\ 4 & 6 & 4 & 17 & 4 \\ 12 & 15 & 17 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$
- [9] $1^{111} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 6 & 12 & 4 \\ 3 & 6 & 4 & 13 & 4 \\ 9 & 12 & 13 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$
- [10] $1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -19, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 3 & 6 & 4 & 4 & 4 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$
- [11] $1^1 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -20, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 4 & 5 & 9 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 5 & -1 & 2 \end{pmatrix}_{b_0}$

- [14] $1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & -1 & -1 & 1 \end{pmatrix}_{b_0}$
- [12] $1^{\overline{112}} 3^{\overline{1}} 1^{\overline{12}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 7.2$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 11 & 3 \\ 2 & 1 & 5 & 12 & 3 \\ 3 & 5 & 4 & 18 & 4 \\ 11 & 12 & 18 & 57 & 13 \\ 3 & 3 & 4 & 13 & 2 \end{pmatrix}_{b_0}$
- [13] $1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 1 & 5 & 0 & 3 \\ 3 & 5 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 4 & -1 & 2 \end{pmatrix}_{b_0}$
- [15] $1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -21$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 6 & 3 \\ 2 & 1 & 5 & 6 & 3 \\ 3 & 5 & 4 & 9 & 4 \\ 6 & 6 & 9 & 12 & 6 \\ 3 & 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$
- [16] $21^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -22$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.55 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 4 & 3 & 8 & 3 & 4 \\ 3 & 1 & 5 & 3 & 3 \\ 8 & 5 & 9 & 4 & 5 \\ 3 & 3 & 4 & 0 & 2 \\ 4 & 3 & 5 & 2 & 2 \end{pmatrix}_{b_0}$
- [18] $1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10$, $T_{\max} = 12$, $T \in [10, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 1 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 2 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 4 & 3 & -1 & 3 \end{pmatrix}_{b_0}$
- [17] $21^{\overline{11}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -24$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 1 & 5 & 6 & 3 \\ 4 & 5 & 9 & 12 & 5 \\ 6 & 6 & 12 & 12 & 6 \\ 3 & 3 & 5 & 6 & 1 \end{pmatrix}_{b_0}$
- [19] $21^{\overline{11}} 3^{\overline{1}} 1^{\overline{12}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -25$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.20 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 4 & 3 & 5 & 12 & 4 \\ 3 & 1 & 5 & 9 & 3 \\ 5 & 5 & 4 & 13 & 4 \\ 12 & 9 & 13 & 27 & 9 \\ 4 & 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$
- [20] $1^{\overline{1}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 1 & 4 & 0 & 3 \\ 3 & 4 & 4 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 3 & 3 & 4 & -1 & 2 \end{pmatrix}_{b_0}$
- [21] $1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 10.29 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$
 $\begin{pmatrix} 4 & 5 & 3 & 1 & 4 \\ 5 & 4 & 4 & 0 & 4 \\ 3 & 4 & 1 & 1 & 3 \\ 1 & 0 & 1 & -3 & -1 \\ 4 & 4 & 3 & -1 & 2 \end{pmatrix}_{b_0}$
- [22] $1^{\overline{13}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$
 $T_{\min} = 11.14 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 3 & 9 & 3 \\ 3 & 4 & 6 & 15 & 4 \\ 3 & 6 & 4 & 17 & 4 \\ 9 & 15 & 17 & 48 & 12 \\ 3 & 4 & 4 & 12 & 1 \end{pmatrix}_{b_0}$
- [23] $1^{\overline{11}} 3^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3$ $T = 7$, $T_{\max} = 12$, $\Delta = -93$, $\det = -28$, $(n_+, n_-, n_0) = (1, 7, 0)$, $c_{\text{ext}} = 6.0$

$$T_{\min} = 10.29 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 0 & 4 \\ 3 & 6 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 4 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{14} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 10 & 8 & 0 & 53 & 11 \\ 2 & 2 & 2 & -1 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{11} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 2 & 4 & 0 & 5 & 3 \\ 2 & 2 & 2 & -1 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -29, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 4 & 3 \\ 2 & 1 & 5 & 3 & 3 \\ 4 & 5 & 9 & 8 & 5 \\ 4 & 3 & 8 & 3 & 3 \\ 3 & 3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 3 & 5 & 1 & 4 \\ 3 & 1 & 4 & 0 & 3 \\ 5 & 4 & 4 & 1 & 4 \\ 1 & 0 & 1 & -3 & -1 \\ 4 & 3 & 4 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{14} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -30, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$$

$$T_{\min} = 11.20 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 9 & 6 & 0 & 39 & 9 \\ 2 & 2 & 2 & -1 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{12} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 5 & 4 & 0 & 11 & 5 \\ 2 & 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{111} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -32, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.38 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 3 \\ 3 & 4 & 6 & 15 & 4 \\ 3 & 6 & 4 & 17 & 4 \\ 10 & 15 & 17 & 48 & 12 \\ 3 & 4 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[31] \ 21^{11} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -34, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.59 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 4 & 3 & 5 & 15 & 4 \\ 3 & 1 & 5 & 12 & 3 \\ 5 & 5 & 4 & 17 & 4 \\ 15 & 12 & 17 & 48 & 12 \\ 4 & 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$[32] \ 1^{111} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 4 & 6 & 6 & 4 \\ 3 & 6 & 4 & 5 & 4 \\ 4 & 6 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[33] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -35, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 3 & 6 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$

- [34] $1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.56 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 5 & 3 & 3 \\ 3 & 5 & 4 & 4 & 4 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 3 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$
- [35] $1^{111}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -36, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 4 & 6 & 3 \\ 2 & 1 & 5 & 6 & 3 \\ 4 & 5 & 9 & 12 & 5 \\ 6 & 6 & 12 & 12 & 6 \\ 3 & 3 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$
- [37] $1^{111}31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 5 & 7 & 0 & 23 & 7 \\ 2 & 2 & 2 & -1 & 7 & 0 \end{pmatrix}_{b_0}$$
- [36] $1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -39, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 3 & 3 & 0 & 3 & 3 \\ 2 & 2 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$
- [38] $21^{11}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -40, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 4 & 6 & 4 & 4 & 4 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$
- [40] $1^{12}131^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -42, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 10 & 3 \\ 2 & 1 & 5 & 12 & 3 \\ 3 & 5 & 4 & 18 & 4 \\ 10 & 12 & 18 & 57 & 13 \\ 3 & 3 & 4 & 13 & 2 \end{pmatrix}_{b_0}$$
- [39] $1^{11}131^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -42, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 3 \\ 2 & 1 & 5 & 6 & 3 \\ 3 & 5 & 4 & 9 & 4 \\ 5 & 6 & 9 & 12 & 6 \\ 3 & 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$
- [41] $21^131^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 4 & 6 & 6 & 4 \\ 4 & 6 & 4 & 5 & 4 \\ 3 & 6 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$
- [44] $1^1131^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 3 \\ 2 & 1 & 5 & 3 & 3 \\ 3 & 5 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 2 \\ 3 & 3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$
- [43] $1^{113}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.49 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 14 & 3 \\ 2 & 1 & 5 & 15 & 3 \\ 3 & 5 & 4 & 22 & 4 \\ 14 & 15 & 22 & 84 & 16 \\ 3 & 3 & 4 & 16 & 2 \end{pmatrix}_{b_0}$$
- [42] $1^1131^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -43, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 4 & 6 & 6 & 4 \\ 3 & 6 & 4 & 5 & 4 \\ 3 & 6 & 5 & 3 & 3 \\ 3 & 4 & 4 & 3 & 1 \end{pmatrix}_{b_0}$$

- [46] $1^1 1^1 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.27 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 4 & 5 & 3 \\ 2 & 1 & 5 & 6 & 3 \\ 4 & 5 & 9 & 12 & 5 \\ 5 & 6 & 12 & 12 & 6 \\ 3 & 3 & 5 & 6 & 1 \end{pmatrix}_{b_0}$$
- [45] $1^{11} 1^1 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -44, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.45 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 4 & 5 & 0 & 11 & 5 \\ 2 & 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$
- [48] $1^1 1^1 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.83 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 3 \\ 2 & 1 & 5 & 6 & 3 \\ 3 & 5 & 4 & 8 & 4 \\ 4 & 6 & 8 & 9 & 5 \\ 3 & 3 & 4 & 5 & 1 \end{pmatrix}_{b_0}$$
- [47] $21^1 1^1 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -46, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 1 & 5 & 3 & 3 \\ 4 & 5 & 4 & 5 & 4 \\ 3 & 3 & 5 & 3 & 3 \\ 3 & 3 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$
- [50] $1^{11} 1^{12} 3^1 1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 8 & 3 \\ 2 & 1 & 5 & 9 & 3 \\ 3 & 5 & 4 & 12 & 4 \\ 8 & 9 & 12 & 24 & 8 \\ 3 & 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$
- [49] $1^{11} 1^{12} 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -51, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.57 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 3 & 3 & 0 & 3 & 3 \\ 2 & 2 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$
- [51] $1^{13} 1^1 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -52, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.46 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 8 & 7 & 0 & 35 & 9 \\ 2 & 2 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$
- [52] $1^{11} 1^{12} 3^1 1^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -53, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.77 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 3 \\ 2 & 1 & 5 & 6 & 3 \\ 3 & 5 & 4 & 8 & 4 \\ 5 & 6 & 8 & 9 & 5 \\ 3 & 3 & 4 & 5 & 2 \end{pmatrix}_{b_0}$$
- [53] $21^1 1^1 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -55, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 6 & 5 & 3 & 4 \\ 6 & 4 & 6 & 3 & 4 \\ 5 & 6 & 4 & 4 & 4 \\ 3 & 3 & 4 & 0 & 2 \\ 4 & 4 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$
- [57] $1^{13} 1^1 3^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 7 & 5 & 0 & 23 & 7 \\ 2 & 2 & 2 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$
- [56] $21^1 1^{12} 3^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 1 & 5 & 12 & 3 \\ 4 & 5 & 4 & 18 & 4 \\ 12 & 12 & 18 & 57 & 13 \\ 3 & 3 & 4 & 13 & 2 \end{pmatrix}_{b_0}$$

- [55] $1^{12}3^11^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.30 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 6 & 6 & 0 & 21 & 7 \\ 2 & 2 & 2 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$
- [54] $1^{11}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -57, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 3 & 3 & 0 & 3 & 3 \\ 2 & 2 & 2 & -1 & 3 & 1 \end{pmatrix}_{b_0}$$
- [59] $1^{111}3^11^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.36 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 3 \\ 3 & 4 & 6 & 12 & 4 \\ 3 & 6 & 4 & 13 & 4 \\ 8 & 12 & 13 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$
- [58] $1^{112}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -59, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.76 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 3 \\ 3 & 4 & 6 & 12 & 4 \\ 3 & 6 & 4 & 13 & 4 \\ 7 & 12 & 13 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$
- [60] $1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -60, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.40 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 3 \\ 3 & 4 & 6 & 9 & 4 \\ 3 & 6 & 4 & 9 & 4 \\ 6 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$
- [61] $1^{112}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 8 & 3 \\ 2 & 1 & 5 & 9 & 3 \\ 3 & 5 & 4 & 13 & 4 \\ 8 & 9 & 13 & 27 & 9 \\ 3 & 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$
- [62] $1^{11}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.81 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 4 & 6 & 9 & 4 \\ 3 & 6 & 4 & 9 & 4 \\ 5 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$
- [63] $21^{111}3^11^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -64, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 6 & 5 & 12 & 4 \\ 6 & 4 & 6 & 12 & 4 \\ 5 & 6 & 4 & 13 & 4 \\ 12 & 12 & 13 & 27 & 9 \\ 4 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$
- [64] $1^{12}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -66, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.12 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 5 & 4 & 0 & 11 & 5 \\ 2 & 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$
- [65] $1^{11}3^11^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -71, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.92 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 3 \\ 2 & 1 & 5 & 9 & 3 \\ 3 & 5 & 4 & 13 & 4 \\ 7 & 9 & 13 & 27 & 9 \\ 3 & 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$
- [68] $1^{12}3^11^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 7 & 8 & 0 & 35 & 9 \\ 2 & 2 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[66] $1^{14} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.44 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 9 & 6 & 0 & 39 & 9 \\ 2 & 2 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[67] $1^{113} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -72, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 3 \\ 2 & 1 & 5 & 9 & 3 \\ 3 & 5 & 4 & 12 & 4 \\ 7 & 9 & 12 & 24 & 8 \\ 3 & 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

[69] $1^{113} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -74, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 11 & 3 \\ 2 & 1 & 5 & 12 & 3 \\ 3 & 5 & 4 & 17 & 4 \\ 11 & 12 & 17 & 48 & 12 \\ 3 & 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

[71] $1^{12} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 4 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 5 & 4 & 0 & 11 & 5 \\ 2 & 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[70] $1^{11} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -76, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.26 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 4 & 4 & 5 & 0 & 11 & 5 \\ 2 & 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[72] $1^{12} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -77, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.74 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 8 & 10 & 0 & 53 & 11 \\ 2 & 2 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

[73] $1^{13} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -83, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 7 & 5 & 0 & 23 & 7 \\ 2 & 2 & 2 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

[75] $1^{11} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 4 & 2 \\ 1 & 2 & 0 & 1 & 5 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 4 & 5 & 0 & 11 & 5 \\ 2 & 2 & 2 & -1 & 5 & 1 \end{pmatrix}_{b_0}$$

[74] $1^{14} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.43 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 11 & 10 & 0 & 71 & 13 \\ 2 & 2 & 2 & -1 & 13 & 1 \end{pmatrix}_{b_0}$$

[76] $1^{11} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -84, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.86 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 6 & 9 & 0 & 39 & 9 \\ 2 & 2 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[77] $1^{11} 3^1 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -85, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 5 & 7 & 0 & 23 & 7 \\ 2 & 2 & 2 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

[78] $1^{13} 3^1 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -87, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 9 & 9 & 0 & 51 & 11 \\ 2 & 2 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

[80] $21^{12} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 4 & 6 & 12 & 4 \\ 4 & 6 & 4 & 13 & 4 \\ 9 & 12 & 13 & 27 & 9 \\ 3 & 4 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[79] $21^{11} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -91, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 3 & 8 & 6 & 4 \\ 3 & 1 & 5 & 3 & 3 \\ 8 & 5 & 9 & 8 & 5 \\ 6 & 3 & 8 & 3 & 3 \\ 4 & 3 & 5 & 3 & 1 \end{pmatrix}_{b_0}$$

[81] $1^{11} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -92, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.48 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 13 & 3 \\ 2 & 1 & 5 & 15 & 3 \\ 3 & 5 & 4 & 22 & 4 \\ 13 & 15 & 22 & 84 & 16 \\ 3 & 3 & 4 & 16 & 2 \end{pmatrix}_{b_0}$$

[82] $21^{11} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -96, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 3 & 8 & 9 & 4 \\ 3 & 1 & 5 & 6 & 3 \\ 8 & 5 & 9 & 12 & 5 \\ 9 & 6 & 12 & 12 & 6 \\ 4 & 3 & 5 & 6 & 2 \end{pmatrix}_{b_0}$$

[83] $21^{11} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.21 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 4 & 6 & 9 & 4 \\ 4 & 6 & 4 & 9 & 4 \\ 6 & 9 & 9 & 12 & 6 \\ 3 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[84] $1^{12} 3^1 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -99, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.78 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 12 & 3 \\ 2 & 1 & 5 & 15 & 3 \\ 3 & 5 & 4 & 22 & 4 \\ 12 & 15 & 22 & 84 & 16 \\ 3 & 3 & 4 & 16 & 2 \end{pmatrix}_{b_0}$$

[85] $21^{11} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -100, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.64 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 6 & 5 & 6 & 4 \\ 6 & 4 & 6 & 6 & 4 \\ 5 & 6 & 4 & 5 & 4 \\ 6 & 6 & 5 & 3 & 3 \\ 4 & 4 & 4 & 3 & 2 \end{pmatrix}_{b_0}$$

[86] $1^{11} 3^1 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -102, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.06 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 9 & 3 \\ 2 & 1 & 5 & 12 & 3 \\ 3 & 5 & 4 & 17 & 4 \\ 9 & 12 & 17 & 48 & 12 \\ 3 & 3 & 4 & 12 & 1 \end{pmatrix}_{b_0}$$

[87] $1^{12} 3^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -105, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.09 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 5 & 6 & 6 & 0 & 21 & 7 \\ 2 & 2 & 2 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

[88] $1^{11} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -107, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.50 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 5 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 5 & 7 & 0 & 23 & 7 \\ 2 & 2 & 2 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

[89] $1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.60 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 10 & 3 \\ 2 & 1 & 5 & 12 & 3 \\ 3 & 5 & 4 & 17 & 4 \\ 10 & 12 & 17 & 48 & 12 \\ 3 & 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

[90] $1^{14} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -109, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 10 & 8 & 0 & 53 & 11 \\ 2 & 2 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

[91] $1^{13} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -112, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.39 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 10 & 11 & 0 & 71 & 13 \\ 2 & 2 & 2 & -1 & 13 & 1 \end{pmatrix}_{b_0}$$

[92] $1^{13} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -116, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.07 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 8 & 7 & 0 & 35 & 9 \\ 2 & 2 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[94] $21^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 1 & 5 & 6 & 3 \\ 4 & 5 & 4 & 9 & 4 \\ 6 & 6 & 9 & 12 & 6 \\ 3 & 3 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$

[93] $1^{11} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -120, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.67 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 6 & 9 & 0 & 39 & 9 \\ 2 & 2 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

[95] $1^{12} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.13 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 6 & 2 \\ 1 & 2 & 0 & 1 & 6 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 6 & 6 & 0 & 21 & 7 \\ 2 & 2 & 2 & -1 & 7 & 1 \end{pmatrix}_{b_0}$$

[96] $1^{14} 3^1 1^{14} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -129, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.32 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 12 & 12 & 0 & 93 & 15 \\ 2 & 2 & 2 & -1 & 15 & 1 \end{pmatrix}_{b_0}$$

[97] $1^{12} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -134, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.15 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 6 & 7 & 8 & 0 & 35 & 9 \\ 2 & 2 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$

- [98] $21^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -135, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.16 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 6 & 5 & 9 & 4 \\ 6 & 4 & 6 & 9 & 4 \\ 5 & 6 & 4 & 9 & 4 \\ 9 & 9 & 9 & 12 & 6 \\ 4 & 4 & 4 & 6 & 2 \end{pmatrix}_{b_0}$$
- [99] $21^{111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -151, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.25 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 3 & 3 \\ 3 & 1 & 5 & 3 & 3 \\ 4 & 5 & 4 & 4 & 4 \\ 3 & 3 & 4 & 0 & 2 \\ 3 & 3 & 4 & 2 & 2 \end{pmatrix}_{b_0}$$
- [100] $1^{12}31^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -153, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.27 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 8 & 10 & 0 & 53 & 11 \\ 2 & 2 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$
- [101] $1^{13}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -156, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 8 & 2 \\ 1 & 2 & 0 & 1 & 7 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 8 & 7 & 0 & 35 & 9 \\ 2 & 2 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$
- [102] $21^{111}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -163, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 10.42 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 4 & 6 & 8 & 3 & 4 \\ 6 & 4 & 9 & 3 & 4 \\ 8 & 9 & 9 & 4 & 5 \\ 3 & 3 & 4 & 0 & 2 \\ 4 & 4 & 5 & 2 & 2 \end{pmatrix}_{b_0}$$
- [104] $1^{13}31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -165, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 9 & 9 & 0 & 51 & 11 \\ 2 & 2 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$
- [103] $1^{14}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -165, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.05 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 10 & 8 & 0 & 53 & 11 \\ 2 & 2 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$
- [105] $1^{12}31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -172, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.14 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 0 & 7 & 2 \\ 1 & 2 & 0 & 1 & 8 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 7 & 7 & 8 & 0 & 35 & 9 \\ 2 & 2 & 2 & -1 & 9 & 1 \end{pmatrix}_{b_0}$$
- [106] $1^{14}31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -178, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 11 & 10 & 0 & 71 & 13 \\ 2 & 2 & 2 & -1 & 13 & 1 \end{pmatrix}_{b_0}$$
- [107] $21^{112}31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -181, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.31 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 6 & 3 \\ 3 & 1 & 5 & 6 & 3 \\ 4 & 5 & 4 & 8 & 4 \\ 6 & 6 & 8 & 9 & 5 \\ 3 & 3 & 4 & 5 & 2 \end{pmatrix}_{b_0}$$
- [108] $1^{13}31^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -204, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.10 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 10 & 11 & 0 & 71 & 13 \\ 2 & 2 & 2 & -1 & 13 & 1 \end{pmatrix}_{b_0}$$

[109] $1^{12} 3 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -205, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.20 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 10 & 11 & 0 & 71 & 13 \\ 2 & 2 & 2 & -1 & 13 & 1 \end{pmatrix}_{b_0}$$

[110] $21^{112} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -214, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 10.06 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 1 & 5 & 9 & 3 \\ 4 & 5 & 4 & 13 & 4 \\ 9 & 9 & 13 & 27 & 9 \\ 3 & 3 & 4 & 9 & 2 \end{pmatrix}_{b_0}$$

[111] $1^{12} 3 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -219, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.02 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 0 & 9 & 2 \\ 1 & 2 & 0 & 1 & 9 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 8 & 9 & 9 & 0 & 51 & 11 \\ 2 & 2 & 2 & -1 & 11 & 1 \end{pmatrix}_{b_0}$$

[112] $21^{113} 31^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -231, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.33 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 9 & 3 \\ 3 & 1 & 5 & 9 & 3 \\ 4 & 5 & 4 & 12 & 4 \\ 9 & 9 & 12 & 24 & 8 \\ 3 & 3 & 4 & 8 & 2 \end{pmatrix}_{b_0}$$

[113] $21^{113} 31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -235, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 9.99 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 15 & 3 \\ 3 & 1 & 5 & 15 & 3 \\ 4 & 5 & 4 & 22 & 4 \\ 15 & 15 & 22 & 84 & 16 \\ 3 & 3 & 4 & 16 & 3 \end{pmatrix}_{b_0}$$

[114] $1^{14} 3 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -237, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 12 & 12 & 0 & 93 & 15 \\ 2 & 2 & 2 & -1 & 15 & 1 \end{pmatrix}_{b_0}$$

[115] $1^{14} 3 1^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -248, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 11 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 11 & 10 & 0 & 71 & 13 \\ 2 & 2 & 2 & -1 & 13 & 2 \end{pmatrix}_{b_0}$$

[116] $1^{12} 3 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -272, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 0 & 10 & 2 \\ 1 & 2 & 0 & 1 & 11 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 9 & 10 & 11 & 0 & 71 & 13 \\ 2 & 2 & 2 & -1 & 13 & 1 \end{pmatrix}_{b_0}$$

[117] $1^{14} 3 1^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -321, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.6$
 $T_{\min} = 9.98 \rightarrow 10, T_{\max} = 12, T \in [10, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 0 & 12 & 2 \\ 1 & 2 & 0 & 1 & 12 & 2 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 10 & 12 & 12 & 0 & 93 & 15 \\ 2 & 2 & 2 & -1 & 15 & 2 \end{pmatrix}_{b_0}$$

[118] $21^{113} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 7, T_{\max} = 12, \Delta = -93, \det = -328, (n_+, n_-, n_0) = (1, 7, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 10.04 \rightarrow 11, T_{\max} = 12, T \in [11, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 12 & 3 \\ 3 & 1 & 5 & 12 & 3 \\ 4 & 5 & 4 & 17 & 4 \\ 12 & 12 & 17 & 48 & 12 \\ 3 & 3 & 4 & 12 & 2 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
& T = 8 \quad [119] \, 21^{11} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 4, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 4 & 6 & 5 & 6 & 4 \\ 6 & 4 & 6 & 6 & 4 \\ 5 & 6 & 4 & 7 & 4 \\ 6 & 6 & 7 & 7 & 5 \\ 4 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0} \\
& [120] \, 21^{11} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 6, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 4 & 3 & 5 & 3 & 4 \\ 3 & 1 & 5 & 2 & 3 \\ 5 & 5 & 4 & 3 & 4 \\ 3 & 2 & 3 & 0 & 2 \\ 4 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
& [121] \, 1^{11} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
& [122] \, 21^{11} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 18, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.78 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 4 & 3 & 5 & 3 & 4 \\ 3 & 1 & 4 & 3 & 3 \\ 5 & 4 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 4 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
& [125] \, 21^{11} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 4 & 6 & 6 & 4 \\ 4 & 6 & 4 & 7 & 4 \\ 5 & 6 & 7 & 7 & 5 \\ 3 & 4 & 4 & 5 & 3 \end{pmatrix}_{b_0} \\
& [123] \, 1^{12} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 3 & 6 & 3 \\ 2 & 1 & 4 & 5 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 6 & 5 & 7 & 7 & 5 \\ 3 & 3 & 4 & 5 & 3 \end{pmatrix}_{b_0} \\
& [124] \, 21^{11} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 19, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.84 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 4 & 3 & 5 & 6 & 4 \\ 3 & 1 & 4 & 5 & 3 \\ 5 & 4 & 4 & 7 & 4 \\ 6 & 5 & 7 & 7 & 5 \\ 4 & 3 & 4 & 5 & 3 \end{pmatrix}_{b_0} \\
& [126] \, 1^{11} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 10.67 \rightarrow 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 1 & 4 & 0 & 3 \\ 3 & 4 & 4 & 0 & 4 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 4 & -1 & 3 \end{pmatrix}_{b_0} \\
& [129] \, 1^{11} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 1 & 2 & 3 & 2 & 3 \\ 2 & 1 & 4 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
& [128] \, 21^{11} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11] \\
& \begin{pmatrix} 4 & 3 & 5 & 3 & 4 \\ 3 & 1 & 4 & 2 & 3 \\ 5 & 4 & 4 & 4 & 4 \\ 3 & 2 & 4 & 0 & 2 \\ 4 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0} \\
& [127] \, 1^{11} 41^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 \quad T = 8, T_{\max} = 11, \Delta = -84, \det = 24, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 6.0 \\
& T_{\min} = 11, T_{\max} = 11, T \in [11, 11]
\end{aligned}$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 4 & 4 \\ 3 & 2 & 4 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[130] \ 21^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 27, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0 \\ T_{\min} = 10.89 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 1 & 4 & 3 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[131] \ 21^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 33, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0 \\ T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 4 & 3 & 3 \\ 4 & 4 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[132] \ 221^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 37, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0 \\ T_{\min} = 10.92 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 5 & 5 & 3 \\ 3 & 4 & 7 & 6 & 4 \\ 5 & 7 & 8 & 11 & 6 \\ 5 & 6 & 11 & 7 & 5 \\ 3 & 4 & 6 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[133] \ 1^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 41, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0 \\ T_{\min} = 10.83 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 3 \\ 2 & 1 & 4 & 5 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 5 & 5 & 7 & 7 & 5 \\ 3 & 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[134] \ 21^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 43, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0 \\ T_{\min} = 10.93 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 4 & 6 & 3 & 4 \\ 4 & 6 & 4 & 3 & 4 \\ 2 & 3 & 3 & 0 & 2 \\ 3 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[135] \ 221^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 45, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0 \\ T_{\min} = 10.76 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 6 & 5 & 3 \\ 3 & 4 & 9 & 6 & 4 \\ 6 & 9 & 13 & 12 & 7 \\ 5 & 6 & 12 & 7 & 5 \\ 3 & 4 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[136] \ 21^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 58, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0 \\ T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 3 \\ 3 & 1 & 4 & 5 & 3 \\ 3 & 4 & 4 & 7 & 4 \\ 5 & 5 & 7 & 7 & 5 \\ 3 & 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[137] \ 21^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 63, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0 \\ T_{\min} = 10.95 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 4 & 6 & 5 & 3 & 4 \\ 6 & 4 & 6 & 3 & 4 \\ 5 & 6 & 4 & 3 & 4 \\ 3 & 3 & 3 & 0 & 2 \\ 4 & 4 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[138] \ 21^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 73, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0 \\ T_{\min} = 10.79 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 2 & 4 & 5 & 3 \\ 2 & 1 & 4 & 5 & 3 \\ 4 & 4 & 4 & 7 & 4 \\ 5 & 5 & 7 & 7 & 5 \\ 3 & 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[139] \ 21^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 76, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0 \\ T_{\min} = 10.84 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 1 & 5 & 2 & 3 \\ 4 & 5 & 4 & 3 & 4 \\ 2 & 2 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[140] \ 221^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 85, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0 \\ T_{\min} = 10.78 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 5 & 3 & 5 & 3 \\ 5 & 8 & 7 & 10 & 6 \\ 3 & 7 & 4 & 7 & 4 \\ 5 & 10 & 7 & 7 & 5 \\ 3 & 6 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[141] \ 31^{11}31^{11}3 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -80, \ \det = 90, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.73 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 3 & 2 \\ 3 & 4 & 4 & 6 & 3 & 4 \\ 3 & 4 & 1 & 4 & 3 & 3 \\ 3 & 6 & 4 & 3 & 3 & 3 \\ 3 & 3 & 3 & 0 & 2 & 2 \\ 2 & 4 & 3 & 3 & 2 & 2 \end{pmatrix}_{b_0}$$

$$[142] \ 21^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 114, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.89 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 1 & 5 & 5 & 3 \\ 4 & 5 & 4 & 7 & 4 \\ 5 & 5 & 7 & 7 & 5 \\ 3 & 3 & 4 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[143] \ 31^{11}31^{11}3 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -80, \ \det = 198, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.70 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 0 & 3 & 3 & 3 & 3 & 2 \\ 3 & 4 & 7 & 3 & 6 & 4 \\ 3 & 7 & 4 & 4 & 6 & 4 \\ 3 & 3 & 4 & 0 & 3 & 2 \\ 3 & 6 & 6 & 3 & 3 & 3 \\ 2 & 4 & 4 & 2 & 3 & 2 \end{pmatrix}_{b_0}$$

$$[144] \ 221^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 8, \ T_{\max} = 11, \ \Delta = -84, \ \det = 333, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 10.77 \rightarrow 11, \ T_{\max} = 11, \ T \in [11, 11]$$

$$\begin{pmatrix} 1 & 5 & 6 & 5 & 3 \\ 5 & 8 & 13 & 10 & 6 \\ 6 & 13 & 13 & 12 & 7 \\ 5 & 10 & 12 & 7 & 5 \\ 3 & 6 & 7 & 5 & 3 \end{pmatrix}_{b_0}$$

$$T = 9 \quad [145] \ 1^{11}51^{11}2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -6, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 4 & 3 & 5 & 3 & 4 \\ 3 & 1 & 5 & 2 & 3 \\ 5 & 5 & 4 & 3 & 4 \\ 3 & 2 & 3 & 0 & 2 \\ 4 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[148] \ 21^{11}51^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -7, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 2 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[146] \ 21^{11}51^{11}2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -7, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 3 \\ 3 & 1 & 4 & 4 & 3 \\ 3 & 4 & 1 & 5 & 3 \\ 4 & 4 & 5 & 7 & 5 \\ 3 & 3 & 3 & 5 & 3 \end{pmatrix}_{b_0}$$

$$[147] \ 1^{11}51^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -7, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.57 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 2 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[150] \ 21^{11}51^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -21, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.67 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 4 & 3 & 5 & 3 & 4 \\ 3 & 1 & 4 & 2 & 3 \\ 5 & 4 & 4 & 3 & 4 \\ 3 & 2 & 3 & 0 & 2 \\ 4 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[149] \ 1^{11}51^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, \ T_{\max} = 12, \ \Delta = -79, \ \det = -21, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 11.67 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 3 \\ 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 2 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[151] $21^{\frac{1}{1}} 5^{\frac{1}{1}} 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -28, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 3 \\ 3 & 1 & 4 & 2 & 3 \\ 3 & 4 & 4 & 3 & 4 \\ 2 & 2 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[152] $21^{\frac{1}{1}} 5^{\frac{1}{1}} 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -37, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 3 \\ 2 & 1 & 4 & 2 & 3 \\ 4 & 4 & 4 & 3 & 4 \\ 2 & 2 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[153] $1^{\frac{1}{1}} 5^{\frac{1}{1}} 1^{11} 2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -76, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.84 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 1 & 5 & 2 & 3 \\ 4 & 5 & 4 & 3 & 4 \\ 2 & 2 & 3 & 0 & 2 \\ 3 & 3 & 4 & 2 & 3 \end{pmatrix}_{b_0}$$

[154] $221^{\frac{1}{1}} 5^{\frac{1}{1}} 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 9, T_{\max} = 12, \Delta = -79, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 6 & 2 & 3 \\ 5 & 8 & 13 & 4 & 6 \\ 6 & 13 & 13 & 5 & 7 \\ 2 & 4 & 5 & 0 & 2 \\ 3 & 6 & 7 & 2 & 3 \end{pmatrix}_{b_0}$$

$T = 10$ [155] $2^{\frac{1}{1}} 3^{\frac{1}{1}} 5^{\frac{1}{1}} 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 3 \\ 3 & 2 & 4 & 4 & 4 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[156] $21^{\frac{1}{1}} 5^{\frac{1}{1}} 131^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 3, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 3 \\ 2 & 2 & 3 & 4 & 4 \\ 3 & 3 & 1 & 3 & 3 \\ 2 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

[157] $1^{\frac{1}{1}} 5^{\frac{1}{1}} 131^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 2 & 4 & 4 & 4 \\ 3 & 4 & 4 & 3 & 4 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 4 & 3 & 4 \end{pmatrix}_{b_0}$$

[158] $1^{\frac{1}{1}} 5^{\frac{1}{1}} 1^1 231^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 12, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 6 & 3 & 3 & 3 \\ 6 & 19 & 11 & 7 & 9 \\ 3 & 11 & 5 & 5 & 5 \\ 3 & 7 & 5 & 1 & 3 \\ 3 & 9 & 5 & 3 & 4 \end{pmatrix}_{b_0}$$

[159] $1^{\frac{1}{1}} 5^{\frac{1}{1}} 131^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 1 & 4 & 0 & 3 \\ 3 & 4 & 5 & 0 & 5 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 3 & 5 & -1 & 4 \end{pmatrix}_{b_0}$$

[160] $1^{\frac{1}{1}} 5^{\frac{1}{1}} 131^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 13, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.85 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 \\ 1 & 2 & 0 & 2 & 0 \\ 1 & 1 & 2 & 0 & 0 \\ 1 & 0 & 0 & -3 & -1 \\ 2 & 2 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

[161] $1^{11} 5^{11} 131^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 16, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 2 & 3 & 3 & 3 \\ 2 & 1 & 4 & 2 & 3 \\ 3 & 4 & 5 & 5 & 5 \\ 3 & 2 & 5 & 1 & 3 \\ 3 & 3 & 5 & 3 & 4 \end{pmatrix}_{b_0}$

[162] $1^{11} 5^{11} 1231^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 36, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 0 & 3 & 1 & 1 & 2 \\ 3 & 10 & 4 & 4 & 0 \\ 1 & 4 & 0 & 2 & 0 \\ 1 & 4 & 2 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 \end{pmatrix}_{b_0}$

[163] $2^{11} 3^{11} 151^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 48, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 2 & 3 & 4 & 4 \\ 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$

[164] $2^{11} 3^{11} 151^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 53, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 2 & 3 \\ 3 & 2 & 6 & 2 & 4 \\ 4 & 6 & 5 & 3 & 5 \\ 2 & 2 & 3 & 0 & 2 \\ 3 & 4 & 5 & 2 & 4 \end{pmatrix}_{b_0}$

[165] $2^{11} 3^{11} 151^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 94, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 5 & 3 \\ 3 & 2 & 6 & 6 & 4 \\ 4 & 6 & 5 & 8 & 5 \\ 5 & 6 & 8 & 8 & 6 \\ 3 & 4 & 5 & 6 & 4 \end{pmatrix}_{b_0}$

[166] $2^{11} 3^{11} 151^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 149, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 3 & 4 & 8 & 3 \\ 3 & 2 & 6 & 10 & 4 \\ 4 & 6 & 5 & 13 & 5 \\ 8 & 10 & 13 & 24 & 10 \\ 3 & 4 & 5 & 10 & 4 \end{pmatrix}_{b_0}$

[167] $22^1 3^{11} 151^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 312, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 6 & 5 & 3 \\ 5 & 9 & 14 & 11 & 7 \\ 6 & 14 & 14 & 13 & 8 \\ 5 & 11 & 13 & 8 & 6 \\ 3 & 7 & 8 & 6 & 4 \end{pmatrix}_{b_0}$

[168] $22^1 3^{11} 151^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 371, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 5 & 5 & 3 \\ 5 & 9 & 12 & 11 & 7 \\ 5 & 12 & 9 & 12 & 7 \\ 5 & 11 & 12 & 8 & 6 \\ 3 & 7 & 7 & 6 & 4 \end{pmatrix}_{b_0}$

[169] $22^1 3^{11} 151^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 10, T_{\max} = 12, \Delta = -59, \det = 384, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 7.2$
 $T_{\min} = 11.92 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$
 $\begin{pmatrix} 1 & 5 & 6 & 10 & 3 \\ 5 & 9 & 14 & 22 & 7 \\ 6 & 14 & 14 & 26 & 8 \\ 10 & 22 & 26 & 41 & 13 \\ 3 & 7 & 8 & 13 & 4 \end{pmatrix}_{b_0}$

$T = 11$ [170] $1^1 41^1 321^1 51^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -36, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$
 $\begin{pmatrix} 1 & 5 & 6 & 1 & 3 \\ 5 & 8 & 8 & 0 & 6 \\ 6 & 8 & 6 & 0 & 6 \\ 1 & 0 & 0 & -2 & 0 \\ 3 & 6 & 6 & 0 & 4 \end{pmatrix}_{b_0}$

[171] $2^1 3^{11} 2161^1 + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 \quad T = 11, T_{\max} = 13, \Delta = -58, \det = -48, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 6.0$
 $T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 3 \\ 3 & 2 & 3 & 4 & 4 \\ 2 & 3 & 1 & 3 & 3 \\ 3 & 4 & 3 & 1 & 3 \\ 3 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[172] \ 1^{11} 2^3 2^1 1^1 6^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 11, \ T_{\max} = 13, \ \Delta = -58, \ \det = -58, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.97 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 7 & 2 & 3 & 3 \\ 7 & 19 & 7 & 7 & 9 \\ 2 & 7 & 1 & 3 & 3 \\ 3 & 7 & 3 & 1 & 3 \\ 3 & 9 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[173] \ 41^{11} 2^3 1^5 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 \quad T = 11, \ T_{\max} = 13, \ \Delta = -58, \ \det = -134, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 6.0$$

$$T_{\min} = 12.99 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 8 & 9 & 1 & 3 \\ 8 & 42 & 49 & 0 & 14 \\ 9 & 49 & 51 & 0 & 15 \\ 1 & 0 & 0 & -3 & -1 \\ 3 & 14 & 15 & -1 & 4 \end{pmatrix}_{b_0}$$

6.136 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6$

$$T = 8 \quad [0] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 64, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 \\ 3 & 5 & 4 & 5 & 0 \\ 3 & 3 & 5 & 3 & 0 \\ 1 & 0 & 0 & 0 & -6 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 79, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 \\ 4 & 5 & 9 & 4 & 0 \\ 2 & 3 & 4 & 0 & 0 \\ 1 & 0 & 0 & 0 & -6 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{12} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 79, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.94 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 2 & 4 & 7 & 1 & 3 \\ 2 & 1 & 5 & 9 & 0 & 3 \\ 4 & 5 & 9 & 16 & 0 & 5 \\ 7 & 9 & 16 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 3 & 5 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111} 3^1 1^{13} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 124, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.81 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 6 & 15 & 0 & 4 \\ 3 & 6 & 4 & 17 & 0 & 4 \\ 9 & 15 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 12 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 166, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 0 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 189, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 5 & 6 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 3 & 4 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{112} 3^1 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 189, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 98, \ T \in [13, 98]$$

$$\begin{pmatrix} 1 & 2 & 3 & 10 & 1 & 3 \\ 2 & 1 & 5 & 12 & 0 & 3 \\ 3 & 5 & 4 & 18 & 0 & 4 \\ 10 & 12 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 3 & 4 & 13 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 \quad T = 8, \ T_{\max} = 98, \ \Delta = -142, \ \det = 190, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 \\ 3 & 5 & 4 & 4 & 0 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 3 & 4 & 2 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{112}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 193, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.14 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 3 & 5 & 4 & 8 & 0 & 4 \\ 4 & 6 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 3 & 4 & 5 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 202, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 6 & 0 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 \\ 3 & 6 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 3 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{11}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 204, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 13.10 \rightarrow 14, T_{\max} = 98, T \in [14, 98]$$

$$\begin{pmatrix} 1 & 2 & 4 & 5 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 4 & 5 & 9 & 12 & 0 & 5 \\ 5 & 6 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 3 & 5 & 6 & -4 & -1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{111}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 274, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.72 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 6 & 12 & 0 & 4 \\ 3 & 6 & 4 & 13 & 0 & 4 \\ 7 & 12 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 300, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$$

$$T_{\min} = 12.81 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 9 & 0 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 4 & 4 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{112}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 316, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 1 & 3 \\ 2 & 1 & 5 & 9 & 0 & 3 \\ 3 & 5 & 4 & 13 & 0 & 4 \\ 7 & 9 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 3 & 4 & 9 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{113}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 442, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.65 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 2 & 3 & 12 & 1 & 3 \\ 2 & 1 & 5 & 15 & 0 & 3 \\ 3 & 5 & 4 & 22 & 0 & 4 \\ 12 & 15 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 3 & 4 & 16 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{113}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 \quad T = 8, T_{\max} = 98, \Delta = -142, \det = 445, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.74 \rightarrow 13, T_{\max} = 98, T \in [13, 98]$$

$$\begin{pmatrix} 1 & 2 & 3 & 9 & 1 & 3 \\ 2 & 1 & 5 & 12 & 0 & 3 \\ 3 & 5 & 4 & 17 & 0 & 4 \\ 9 & 12 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 3 & 3 & 4 & 12 & -4 & 0 \end{pmatrix}_{b_0}$$

6.137 $\text{NHC}(-2)(-2)(-3) + \mathbf{su}_3 + \mathbf{e}_6 + \text{NHC}(-2)(-3)$

$$T = 10 \quad [0] \ 1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 266, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 4 & 3 \\ 3 & 3 & 5 & 3 & 0 & 4 & 3 & 3 \\ 3 & 3 & 4 & 2 & -4 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 10, \ T_{\max} = 14, \ \Delta = -92, \ \det = 326, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.45 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 6 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 5 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 5 & 3 & 0 & 4 & 4 & 4 \\ 2 & 3 & 4 & 3 & 0 & 4 & 0 & 2 \\ 3 & 4 & 4 & 2 & -4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

6.138 $\text{NHC}(-2)(-2)(-3) + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{so}_8$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{so}_8 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -196, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.73 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 8 & 11 & 9 & 1 & 5 \\ 8 & 10 & 17 & 12 & 0 & 6 \\ 11 & 17 & 20 & 17 & 0 & 8 \\ 9 & 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 6 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{so}_8 \quad T = 9, \ T_{\max} = 99, \ \Delta = -141, \ \det = -559, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.74 \rightarrow 14, \ T_{\max} = 99, \ T \in [14, 99]$$

$$\begin{pmatrix} 5 & 9 & 11 & 9 & 1 & 5 \\ 9 & 13 & 18 & 15 & 0 & 7 \\ 11 & 18 & 20 & 17 & 0 & 8 \\ 9 & 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 7 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

6.139 $\text{NHC}(-2)(-2)(-3) + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2$

$$T = 9 \quad [0] \ 1^{\overline{112}} 3^{\overline{1}} 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -5, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.2$$

$$T_{\min} = 16.20 \rightarrow 17, \ T_{\max} = 23, \ T \in [17, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 1 & 3 & 3 \\ 2 & 1 & 5 & 9 & 0 & 3 & 3 \\ 3 & 5 & 4 & 13 & 0 & 5 & 4 \\ 7 & 9 & 13 & 27 & 0 & 12 & 9 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 3 & 5 & 12 & 0 & 4 & 4 \\ 3 & 3 & 4 & 9 & -4 & 4 & -4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{11}} 3^{\overline{1}} 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -44, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.55 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 3 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 \\ 4 & 5 & 9 & 4 & 0 & 8 & 5 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 3 & 8 & 3 & 0 & 4 & 4 \\ 3 & 3 & 5 & 2 & -4 & 4 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -65, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.55 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 5 & 3 \\ 2 & 1 & 5 & 3 & 0 & 6 & 3 \\ 3 & 5 & 4 & 5 & 0 & 9 & 4 \\ 3 & 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 6 & 9 & 9 & 0 & 13 & 7 \\ 3 & 3 & 4 & 3 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{1}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -68, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.24 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 \\ 3 & 5 & 4 & 5 & 0 & 4 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 3 & 4 & 3 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -78, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 14.87 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 5 & 1 & 2 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 & 3 \\ 4 & 5 & 9 & 12 & 0 & 4 & 5 \\ 5 & 6 & 12 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 6 & 0 & 1 & 3 \\ 3 & 3 & 5 & 6 & -4 & 3 & -2 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1112}} 3 1^{\overline{12}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -128, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.94 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 10 & 3 \\ 2 & 1 & 5 & 3 & 0 & 12 & 3 \\ 3 & 5 & 4 & 4 & 0 & 18 & 4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 12 & 18 & 12 & 0 & 58 & 14 \\ 3 & 3 & 4 & 2 & -4 & 14 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{1112}} 3 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -138, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.81 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 4 & 3 \\ 2 & 1 & 5 & 6 & 0 & 6 & 3 \\ 3 & 5 & 4 & 9 & 0 & 8 & 4 \\ 5 & 6 & 9 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 6 & 8 & 12 & 0 & 10 & 6 \\ 3 & 3 & 4 & 6 & -4 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3 1^{\overline{113}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -143, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.46 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 11 & 3 \\ 2 & 1 & 5 & 3 & 0 & 12 & 3 \\ 4 & 5 & 9 & 4 & 0 & 28 & 5 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 12 & 28 & 12 & 0 & 73 & 15 \\ 3 & 3 & 5 & 2 & -4 & 15 & -1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{11}} 3 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -146, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 2 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}} 3 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -155, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.92 \rightarrow 14, \ T_{\max} = 23, \ T \in [14, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 2 & 3 \\ 3 & 4 & 6 & 6 & 0 & 3 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 & 4 \\ 3 & 6 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 3 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{111}} 3 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -191, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 6 & 3 & 0 & 6 & 4 \\ 3 & 6 & 4 & 4 & 0 & 5 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 6 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 4 & 2 & -4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{111}} 3 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -203, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.95 \rightarrow 13, \ T_{\max} = 23, \ T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 5 & 3 \\ 2 & 1 & 5 & 3 & 0 & 6 & 3 \\ 4 & 5 & 9 & 4 & 0 & 12 & 5 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 6 & 12 & 6 & 0 & 13 & 7 \\ 3 & 3 & 5 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{112}} 3 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 23, \ \Delta = -112, \ \det = -257, \ (n_+, n_-, n_0) =$$

$$(1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.02 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 7 & 3 \\ 2 & 1 & 5 & 3 & 0 & 9 & 3 \\ 3 & 5 & 4 & 5 & 0 & 13 & 4 \\ 3 & 3 & 5 & 3 & 0 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 9 & 13 & 12 & 0 & 28 & 10 \\ 3 & 3 & 4 & 3 & -4 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{114}31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -281, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.13 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 11 & 3 \\ 2 & 1 & 5 & 3 & 0 & 15 & 3 \\ 3 & 5 & 4 & 5 & 0 & 21 & 4 \\ 3 & 3 & 5 & 3 & 0 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 15 & 21 & 18 & 0 & 76 & 16 \\ 3 & 3 & 4 & 3 & -4 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{111}31^{112} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -287, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 8 & 3 \\ 2 & 1 & 5 & 3 & 0 & 9 & 3 \\ 4 & 5 & 9 & 4 & 0 & 20 & 5 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 9 & 20 & 9 & 0 & 37 & 11 \\ 3 & 3 & 5 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{111}31^{121} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -299, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.73 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 6 & 12 & 0 & 3 & 4 \\ 3 & 6 & 4 & 13 & 0 & 4 & 4 \\ 7 & 12 & 13 & 27 & 0 & 9 & 9 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 4 & 9 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{113}31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -317, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 6 & 3 & 0 & 21 & 4 \\ 3 & 6 & 4 & 4 & 0 & 19 & 4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 21 & 19 & 12 & 0 & 73 & 15 \\ 3 & 4 & 4 & 2 & -4 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{111}31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -317, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.04 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 5 & 3 \\ 2 & 1 & 5 & 3 & 0 & 6 & 3 \\ 3 & 5 & 4 & 4 & 0 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 6 & 9 & 6 & 0 & 13 & 7 \\ 3 & 3 & 4 & 2 & -4 & 7 & -1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{111}31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -318, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 2 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 & 3 \\ 3 & 5 & 4 & 9 & 0 & 4 & 4 \\ 5 & 6 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 6 & 0 & 1 & 3 \\ 3 & 3 & 4 & 6 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{113}31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -329, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.01 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 9 & 3 \\ 2 & 1 & 5 & 3 & 0 & 12 & 3 \\ 3 & 5 & 4 & 5 & 0 & 17 & 4 \\ 3 & 3 & 5 & 3 & 0 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 17 & 15 & 0 & 49 & 13 \\ 3 & 3 & 4 & 3 & -4 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{112}31^{112} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -338, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.97 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 10 & 3 \\ 2 & 1 & 5 & 3 & 0 & 12 & 3 \\ 4 & 5 & 9 & 4 & 0 & 24 & 5 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 12 & 24 & 12 & 0 & 58 & 14 \\ 3 & 3 & 5 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[21] \quad 1^{1111} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -351, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.88 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 2 & 3 \\ 3 & 4 & 6 & 9 & 0 & 3 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 6 & 0 & 1 & 3 \\ 3 & 4 & 4 & 6 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[22] \quad 1^{1111} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -378, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 5 & 3 \\ 2 & 1 & 5 & 6 & 0 & 6 & 3 \\ 3 & 5 & 4 & 9 & 0 & 9 & 4 \\ 5 & 6 & 9 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 6 & 9 & 15 & 0 & 13 & 7 \\ 3 & 3 & 4 & 6 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[23] \quad 1^{1111} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -383, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 \\ 3 & 5 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 3 & 4 & 2 & -4 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[24] \quad 1^{1112} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.04 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 4 & 3 \\ 2 & 1 & 5 & 3 & 0 & 6 & 3 \\ 3 & 5 & 4 & 4 & 0 & 8 & 4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 4 & 6 & 8 & 6 & 0 & 10 & 6 \\ 3 & 3 & 4 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[26] \quad 1^{1111} 31^{112} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -407, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 6 & 3 & 0 & 12 & 4 \\ 3 & 6 & 4 & 4 & 0 & 13 & 4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 12 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 4 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[25] \quad 1^{1113} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -407, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 6 & 3 \\ 2 & 1 & 5 & 3 & 0 & 9 & 3 \\ 3 & 5 & 4 & 4 & 0 & 12 & 4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 6 & 9 & 12 & 9 & 0 & 25 & 9 \\ 3 & 3 & 4 & 2 & -4 & 9 & -1 \end{pmatrix}_{b_0}$$

$$[27] \quad 1^{1114} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -428, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.32 \rightarrow 14, T_{\max} = 23, T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 8 & 3 \\ 2 & 1 & 5 & 3 & 0 & 12 & 3 \\ 3 & 5 & 4 & 4 & 0 & 16 & 4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 12 & 16 & 12 & 0 & 46 & 12 \\ 3 & 3 & 4 & 2 & -4 & 12 & -1 \end{pmatrix}_{b_0}$$

$$[28] \quad 1^{1111} 31^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 23, \Delta = -112, \det = -485, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 23, T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[29] \quad 1^{112} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -542, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 6 & 3 & 0 & 15 & 4 \\ 3 & 6 & 4 & 4 & 0 & 14 & 4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 8 & 15 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 4 & 2 & -4 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[30] \quad 1^{113} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -563, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.98 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 15 & 3 \\ 2 & 1 & 5 & 3 & 0 & 18 & 3 \\ 4 & 5 & 9 & 4 & 0 & 36 & 5 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 18 & 36 & 18 & 0 & 133 & 21 \\ 3 & 3 & 5 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[31] \quad 1^{112} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -620, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 13 & 3 \\ 2 & 1 & 5 & 3 & 0 & 15 & 3 \\ 4 & 5 & 9 & 4 & 0 & 32 & 5 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 15 & 32 & 15 & 0 & 100 & 18 \\ 3 & 3 & 5 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[32] \quad 1^{112} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -632, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 6 & 3 & 0 & 24 & 4 \\ 3 & 6 & 4 & 4 & 0 & 26 & 4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 24 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 4 & 2 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[33] \quad 1^{112} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -674, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 13.01 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 16 & 3 \\ 2 & 1 & 5 & 3 & 0 & 18 & 3 \\ 4 & 5 & 9 & 4 & 0 & 40 & 5 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 16 & 18 & 40 & 18 & 0 & 154 & 22 \\ 3 & 3 & 5 & 2 & -4 & 22 & -1 \end{pmatrix}_{b_0}$$

$$[34] \quad 1^{112} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -696, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 7 & 3 \\ 2 & 1 & 5 & 6 & 0 & 9 & 3 \\ 3 & 5 & 4 & 9 & 0 & 13 & 4 \\ 5 & 6 & 9 & 12 & 0 & 21 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 9 & 13 & 21 & 0 & 28 & 10 \\ 3 & 3 & 4 & 6 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[35] \quad 1^{112} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -698, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 7 & 3 \\ 2 & 1 & 5 & 3 & 0 & 9 & 3 \\ 3 & 5 & 4 & 4 & 0 & 13 & 4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 7 & 9 & 13 & 9 & 0 & 28 & 10 \\ 3 & 3 & 4 & 2 & -4 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[36] \quad 1^{114} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -878, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.99 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 20 & 3 \\ 2 & 1 & 5 & 3 & 0 & 24 & 3 \\ 4 & 5 & 9 & 4 & 0 & 48 & 5 \\ 2 & 3 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 20 & 24 & 48 & 24 & 0 & 238 & 28 \\ 3 & 3 & 5 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

$$[37] \quad 1^{1113} 31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -887, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 13.01 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 12 & 3 \\ 2 & 1 & 5 & 3 & 0 & 15 & 3 \\ 3 & 5 & 4 & 4 & 0 & 22 & 4 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 15 & 22 & 15 & 0 & 85 & 17 \\ 3 & 3 & 4 & 2 & -4 & 17 & -1 \end{pmatrix}_{b_0}$$

$$[38] \quad 1^{1113} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -894, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 9 & 3 \\ 2 & 1 & 5 & 6 & 0 & 12 & 3 \\ 3 & 5 & 4 & 9 & 0 & 17 & 4 \\ 5 & 6 & 9 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 17 & 27 & 0 & 49 & 13 \\ 3 & 3 & 4 & 6 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[39] \quad 1^{112} 31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -944, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 6 & 3 & 0 & 18 & 4 \\ 3 & 6 & 4 & 4 & 0 & 18 & 4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 10 & 18 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 4 & 2 & -4 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[40] \quad 1^{1114} 31^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -950, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.14 \rightarrow 14, \quad T_{\max} = 23, \quad T \in [14, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 17 & 3 \\ 2 & 1 & 5 & 3 & 0 & 21 & 3 \\ 3 & 5 & 4 & 4 & 0 & 31 & 4 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 17 & 21 & 31 & 21 & 0 & 172 & 24 \\ 3 & 3 & 4 & 2 & -4 & 24 & -1 \end{pmatrix}_{b_0}$$

$$[41] \quad 1^{1114} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -972, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 11 & 3 \\ 2 & 1 & 5 & 6 & 0 & 15 & 3 \\ 3 & 5 & 4 & 9 & 0 & 21 & 4 \\ 5 & 6 & 9 & 12 & 0 & 33 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 15 & 21 & 33 & 0 & 76 & 16 \\ 3 & 3 & 4 & 6 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[42] \quad 1^{112} 31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -974, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.5$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 6 & 3 & 0 & 21 & 4 \\ 3 & 6 & 4 & 4 & 0 & 22 & 4 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 12 & 21 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 4 & 2 & -4 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[43] \quad 1^{113} 31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1043, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 18 & 3 \\ 2 & 1 & 5 & 3 & 0 & 21 & 3 \\ 4 & 5 & 9 & 4 & 0 & 44 & 5 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 21 & 44 & 21 & 0 & 193 & 25 \\ 3 & 3 & 5 & 2 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[44] \quad 1^{1113} 31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1085, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.94 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 9 & 3 \\ 2 & 1 & 5 & 3 & 0 & 12 & 3 \\ 3 & 5 & 4 & 4 & 0 & 17 & 4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 9 & 12 & 17 & 12 & 0 & 49 & 13 \\ 3 & 3 & 4 & 2 & -4 & 13 & 0 \end{pmatrix}_{b_0}$$

$$[45] \quad 1^{114}31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1172, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 6 & 3 & 0 & 30 & 4 \\ 3 & 6 & 4 & 4 & 0 & 28 & 4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 16 & 30 & 28 & 18 & 0 & 154 & 22 \\ 3 & 4 & 4 & 2 & -4 & 22 & 0 \end{pmatrix}_{b_0}$$

$$[46] \quad 1^{113}31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1199, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 6 & 3 & 0 & 24 & 4 \\ 3 & 6 & 4 & 4 & 0 & 23 & 4 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 13 & 24 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 4 & 2 & -4 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[47] \quad 1^{114}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1478, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 11 & 3 \\ 2 & 1 & 5 & 3 & 0 & 15 & 3 \\ 3 & 5 & 4 & 4 & 0 & 21 & 4 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 11 & 15 & 21 & 15 & 0 & 76 & 16 \\ 3 & 3 & 4 & 2 & -4 & 16 & 0 \end{pmatrix}_{b_0}$$

$$[48] \quad 1^{114}31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1652, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.95 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 14 & 3 \\ 2 & 1 & 5 & 3 & 0 & 18 & 3 \\ 3 & 5 & 4 & 4 & 0 & 26 & 4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 14 & 18 & 26 & 18 & 0 & 118 & 20 \\ 3 & 3 & 4 & 2 & -4 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[49] \quad 1^{113}31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1709, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 6 & 3 & 0 & 27 & 4 \\ 3 & 6 & 4 & 4 & 0 & 27 & 4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 15 & 27 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 4 & 2 & -4 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[50] \quad 1^{113}31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -1847, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 13.8$$

$$T_{\min} = 12.77 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 6 & 3 & 0 & 30 & 4 \\ 3 & 6 & 4 & 4 & 0 & 31 & 4 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 17 & 30 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 4 & 2 & -4 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[51] \quad 1^{114}31^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2162, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 6 & 3 & 0 & 33 & 4 \\ 3 & 6 & 4 & 4 & 0 & 32 & 4 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 18 & 33 & 32 & 21 & 0 & 193 & 25 \\ 3 & 4 & 4 & 2 & -4 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[52] \quad 1^{114}31^{114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 \quad T = 9, \quad T_{\max} = 23, \quad \Delta = -112, \quad \det = -2780, \quad (n_+, n_-, n_0) = (1, 9, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \quad T_{\max} = 23, \quad T \in [13, 23]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 6 & 3 & 0 & 36 & 4 \\ 3 & 6 & 4 & 4 & 0 & 36 & 4 \\ 2 & 3 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 20 & 36 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 4 & 2 & -4 & 28 & 0 \end{pmatrix}_{b_0}$$

6.140 NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 10 \quad [0] \quad 1^{\overline{1111}} 31^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 100, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 & 3 \\ 3 & 4 & 6 & 3 & 0 & 6 & 6 & 4 \\ 3 & 6 & 4 & 4 & 0 & 5 & 5 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 5 & 3 & 0 & 4 & 6 & 4 \\ 3 & 6 & 5 & 3 & 0 & 6 & 4 & 4 \\ 3 & 4 & 4 & 2 & -4 & 4 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{112}} 31^{\overline{1114}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 136, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.5$$

$$T_{\min} = 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 14 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 24 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 26 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 18 & 3 \\ 14 & 24 & 26 & 18 & 0 & 18 & 118 & 20 \\ 3 & 4 & 4 & 2 & -4 & 3 & 20 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{111}} 31^{\overline{1112}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 253, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.80 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 13 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 9 & 3 \\ 7 & 12 & 13 & 9 & 0 & 9 & 28 & 10 \\ 3 & 4 & 4 & 2 & -4 & 3 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{113}} 31^{\overline{1112}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 325, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 13 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 24 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 23 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 15 & 3 \\ 13 & 24 & 23 & 15 & 0 & 15 & 100 & 18 \\ 3 & 4 & 4 & 2 & -4 & 3 & 18 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{\overline{111}} 31^{\overline{1111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 409, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{\overline{1111}} 31^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 447, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 2 & 5 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 & 6 & 3 \\ 3 & 5 & 4 & 9 & 0 & 4 & 9 & 4 \\ 5 & 6 & 9 & 12 & 0 & 6 & 15 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 6 & 0 & 1 & 6 & 3 \\ 5 & 6 & 9 & 15 & 0 & 6 & 13 & 7 \\ 3 & 3 & 4 & 6 & -4 & 3 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{\overline{1111}} 31^{\overline{1111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 496, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.96 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 5 & 5 & 3 \\ 2 & 1 & 5 & 3 & 0 & 6 & 6 & 3 \\ 4 & 5 & 9 & 4 & 0 & 12 & 12 & 5 \\ 2 & 3 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 12 & 6 & 0 & 13 & 15 & 7 \\ 5 & 6 & 12 & 6 & 0 & 15 & 13 & 7 \\ 3 & 3 & 5 & 2 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{1111}} 3 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 499, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 6 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 5 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 6 & 5 & 3 & 0 & 4 & 9 & 4 \\ 5 & 9 & 9 & 6 & 0 & 9 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 4 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{1111}} 3 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 516, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 9 & 0 & 4 & 4 & 4 \\ 5 & 6 & 9 & 12 & 0 & 6 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 6 & 0 & 1 & 3 & 3 \\ 2 & 3 & 4 & 6 & 0 & 3 & 1 & 3 \\ 3 & 3 & 4 & 6 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{112}} 3 1^{\overline{1112}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 760, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 10 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 18 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 18 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 12 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 12 & 3 \\ 10 & 18 & 18 & 12 & 0 & 12 & 58 & 14 \\ 3 & 4 & 4 & 2 & -4 & 3 & 14 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{1111}} 3 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 772, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.99 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 4 & 0 & 4 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 3 & 4 & 2 & -4 & 3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{112}} 3 1^{\overline{1113}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 820, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.5$$

$$T_{\min} = 12.78 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 12 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 21 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 22 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 15 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 15 & 3 \\ 12 & 21 & 22 & 15 & 0 & 15 & 85 & 17 \\ 3 & 4 & 4 & 2 & -4 & 3 & 17 & 0 \end{pmatrix}_{b_0}$$

$$[12] \ 1^{\overline{114}} 3 1^{\overline{1113}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 928, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 18 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 33 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 32 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 21 & 3 \\ 18 & 33 & 32 & 21 & 0 & 21 & 193 & 25 \\ 3 & 4 & 4 & 2 & -4 & 3 & 25 & 0 \end{pmatrix}_{b_0}$$

$$[13] \ 1^{\overline{1111}} 3 1^{\overline{121}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 931, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.77 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 12 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 13 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 12 & 13 & 9 & 0 & 28 & 21 & 10 \\ 5 & 9 & 9 & 6 & 0 & 21 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 10 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[14] \ 1^{\overline{1111}} 3 1^{\overline{122}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 13, \ \Delta = -82, \ \det = 964, \ (n_+, n_-, n_0) =$$

$$(1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 7 & 3 \\ 3 & 4 & 6 & 3 & 0 & 12 & 12 & 4 \\ 3 & 6 & 4 & 4 & 0 & 13 & 13 & 4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 12 & 13 & 9 & 0 & 28 & 30 & 10 \\ 7 & 12 & 13 & 9 & 0 & 30 & 28 & 10 \\ 3 & 4 & 4 & 2 & -4 & 10 & 10 & 0 \end{pmatrix}_{b_0}$$

$$[15] \ 1^{1111} 3^1 1^{1111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1276, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.76 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 9 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 15 & 7 \\ 5 & 9 & 9 & 6 & 0 & 15 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 7 & 7 & 0 \end{pmatrix}_{b_0}$$

$$[16] \ 1^{113} 3^1 1^{1113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1345, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 15 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 27 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 27 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 18 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 18 & 3 \\ 15 & 27 & 27 & 18 & 0 & 18 & 133 & 21 \\ 3 & 4 & 4 & 2 & -4 & 3 & 21 & 0 \end{pmatrix}_{b_0}$$

$$[17] \ 1^{113} 3^1 1^{1114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 1621, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.8$$

$$T_{\min} = 12.78 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 17 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 30 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 31 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 21 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 21 & 3 \\ 17 & 30 & 31 & 21 & 0 & 21 & 172 & 24 \\ 3 & 4 & 4 & 2 & -4 & 3 & 24 & 0 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{114} 3^1 1^{1114} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -82, \det = 2164, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 12.77 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 20 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 36 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 36 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 24 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 24 & 3 \\ 20 & 36 & 36 & 24 & 0 & 24 & 238 & 28 \\ 3 & 4 & 4 & 2 & -4 & 3 & 28 & 0 \end{pmatrix}_{b_0}$$

6.141 NHC(-2)(-2)(-3) + $\mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3$

$$T = 10 \quad [0] \ 1^{1111} 3^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 84, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.24 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & 3 & -4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{1121} 3^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 462, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.12 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 7 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 9 & 3 & 3 \\ 3 & 5 & 4 & 5 & 0 & 13 & 4 & 4 \\ 3 & 3 & 5 & 3 & 0 & 12 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 7 & 9 & 13 & 12 & 0 & 28 & 9 & 10 \\ 2 & 3 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 3 & 4 & 3 & -4 & 10 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{112} 3^1 1^{1111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, T_{\max} = 14, \Delta = -91, \det = 498, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.96 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 8 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 15 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 14 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 8 & 15 & 14 & 9 & 0 & 37 & 9 & 11 \\ 2 & 3 & 4 & 3 & 0 & 9 & 0 & 2 \\ 3 & 4 & 4 & 2 & -4 & 11 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{1141}} 3^{\overline{1111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 534, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.19 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 11 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 15 & 3 & 3 \\ 3 & 5 & 4 & 5 & 0 & 21 & 4 & 4 \\ 3 & 3 & 5 & 3 & 0 & 18 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 11 & 15 & 21 & 18 & 0 & 76 & 15 & 16 \\ 2 & 3 & 4 & 3 & 0 & 15 & 0 & 2 \\ 3 & 3 & 4 & 3 & -4 & 16 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{1111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 555, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.05 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 1 & 6 & 3 \\ 5 & 9 & 9 & 6 & 0 & 6 & 12 & 6 \\ 3 & 4 & 4 & 2 & -4 & 3 & 6 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{1131}} 3^{\overline{1111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 678, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.14 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 9 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 12 & 3 & 3 \\ 3 & 5 & 4 & 5 & 0 & 17 & 4 & 4 \\ 3 & 3 & 5 & 3 & 0 & 15 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 9 & 12 & 17 & 15 & 0 & 49 & 12 & 13 \\ 2 & 3 & 4 & 3 & 0 & 12 & 0 & 2 \\ 3 & 3 & 4 & 3 & -4 & 13 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{1111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_2 + \mathbf{su}_3 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 894, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 9 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 13 & 6 & 7 \\ 2 & 3 & 4 & 3 & 0 & 6 & 0 & 2 \\ 3 & 4 & 4 & 2 & -4 & 7 & 2 & -1 \end{pmatrix}_{b_0}$$

6.142 NHC(-2)(-2)(-3) + su₃ + e₆ + su₃

$$T = 9 \quad [0] \ 1^{\overline{11}} 3^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -123, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.54 \rightarrow 15, \ T_{\max} = 68, \ T \in [15, 68]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 3 & 3 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 & 3 \\ 4 & 5 & 9 & 4 & 0 & 8 & 5 & 5 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 3 & 3 & 8 & 3 & 0 & 3 & 3 & 3 \\ 3 & 3 & 5 & 2 & -4 & 3 & -2 & 2 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -132, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 0 & 2 & 2 \\ 3 & 3 & 4 & 3 & -4 & 2 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -282, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 5 & 3 & 3 \\ 2 & 1 & 5 & 3 & 0 & 6 & 3 & 3 \\ 4 & 5 & 9 & 4 & 0 & 12 & 5 & 5 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 6 & 12 & 6 & 0 & 12 & 6 & 6 \\ 3 & 3 & 5 & 2 & -4 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -312, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.38 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -357, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.21 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 6 & 3 & 0 & 6 & 4 \\ 3 & 6 & 4 & 4 & 0 & 5 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 3 & 6 & 5 & 3 & 0 & 3 & 3 \\ 3 & 4 & 4 & 2 & -4 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -507, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.07 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 2 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 & 3 \\ 3 & 5 & 4 & 9 & 0 & 4 & 4 \\ 5 & 6 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 6 & 0 & 0 & 2 \\ 3 & 3 & 4 & 6 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -573, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.44 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 \\ 3 & 5 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 0 & 2 \\ 3 & 3 & 4 & 2 & -4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 68, \ \Delta = -121, \ \det = -651, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.05 \rightarrow 14, \ T_{\max} = 68, \ T \in [14, 68]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 4 & 2 & -4 & 6 & -1 \end{pmatrix}_{b_0}$$

6.143 NHC(-2)(-2)(-3) + $\mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 84, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.24 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 3 & 0 & 3 & 1 & 3 \\ 3 & 3 & 4 & 3 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 462, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.12 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 2 & 7 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 9 & 3 \\ 3 & 5 & 4 & 5 & 0 & 4 & 13 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 & 12 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 0 & 9 & 2 \\ 7 & 9 & 13 & 12 & 0 & 9 & 28 & 10 \\ 3 & 3 & 4 & 3 & -4 & 2 & 10 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{112}} 3 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{e}_6 + \mathbf{su}_3 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 498, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.5$$

$$T_{\min} = 13.96 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 8 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 15 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 14 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 9 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 0 & 9 & 2 \\ 8 & 15 & 14 & 9 & 0 & 9 & 37 & 11 \\ 3 & 4 & 4 & 2 & -4 & 2 & 11 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}1} 3^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 534, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.19 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 2 & 11 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 15 & 3 \\ 3 & 5 & 4 & 5 & 0 & 4 & 21 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 & 18 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 0 & 15 & 2 \\ 11 & 15 & 21 & 18 & 0 & 15 & 76 & 16 \\ 3 & 3 & 4 & 3 & -4 & 2 & 16 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}1} 3^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 555, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.05 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 9 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 5 & 9 & 9 & 6 & 0 & 12 & 6 & 6 \\ 2 & 3 & 4 & 3 & 0 & 6 & 1 & 3 \\ 3 & 4 & 4 & 2 & -4 & 6 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}1} 3^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 678, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.8$$

$$T_{\min} = 13.14 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 2 & 9 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 12 & 3 \\ 3 & 5 & 4 & 5 & 0 & 4 & 17 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 & 15 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 0 & 12 & 2 \\ 9 & 12 & 17 & 15 & 0 & 12 & 49 & 13 \\ 3 & 3 & 4 & 3 & -4 & 2 & 13 & -1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}1} 3^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 10, \ T_{\max} = 14, \ \Delta = -91, \ \det = 894, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.39 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 6 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 0 & 6 & 2 \\ 5 & 9 & 9 & 6 & 0 & 6 & 13 & 7 \\ 3 & 4 & 4 & 2 & -4 & 2 & 7 & -1 \end{pmatrix}_{b_0}$$

6.144 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{111}1} 3^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 216, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.44 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 & 3 \\ 3 & 5 & 4 & 5 & 0 & 4 & 4 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 0 & 3 & 2 \\ 2 & 3 & 4 & 3 & 0 & 3 & 0 & 2 \\ 3 & 3 & 4 & 3 & -4 & 2 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}1} 3^{\overline{111}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathfrak{su}_3 \quad T = 10, \ T_{\max} = 22, \ \Delta = -100, \ \det = 324, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 16.0$$

$$T_{\min} = 13.41 \rightarrow 14, \ T_{\max} = 22, \ T \in [14, 22]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 5 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 3 & 2 \\ 1 & 0 & 0 & 0 & -6 & 0 & 0 & -4 \\ 2 & 3 & 4 & 3 & 0 & 0 & 3 & 2 \\ 3 & 6 & 5 & 3 & 0 & 3 & 3 & 3 \\ 3 & 4 & 4 & 2 & -4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

6.145 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{f}_4$

$$T = 8 \quad [0] \ 1^{\overline{11}1} 3^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 \quad T = 8, \ T_{\max} = 28, \ \Delta = -116, \ \det = 49, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 11.2$$

$$T_{\min} = 12.08 \rightarrow 13, \ T_{\max} = 28, \ T \in [13, 28]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 \\ 3 & 5 & 4 & 5 & 0 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 4 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[1] $1^{11}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 \\ 4 & 5 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 5 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[2] $1^{12}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 62, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.10 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 2 & 4 & 7 & 1 & 3 \\ 2 & 1 & 5 & 9 & 0 & 3 \\ 4 & 5 & 9 & 16 & 0 & 5 \\ 7 & 9 & 16 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 5 & 9 & -3 & 0 \end{pmatrix}_{b_0}$$

[3] $1^{111}3^11^3 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 6 & 15 & 0 & 4 \\ 3 & 6 & 4 & 17 & 0 & 4 \\ 9 & 15 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$

[4] $1^{11}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 131, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 0 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 2 & -3 & 1 \end{pmatrix}_{b_0}$$

[6] $1^{112}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 12.39 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 3 & 5 & 4 & 8 & 0 & 4 \\ 4 & 6 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 4 & 5 & -3 & 0 \end{pmatrix}_{b_0}$$

[8] $1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 5 & 6 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 4 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[7] $1^{112}3^11^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.4$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 28, T \in [12, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 10 & 1 & 3 \\ 2 & 1 & 5 & 12 & 0 & 3 \\ 3 & 5 & 4 & 18 & 0 & 4 \\ 10 & 12 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 4 & 13 & -3 & 1 \end{pmatrix}_{b_0}$$

[5] $1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 147, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.12 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 \\ 3 & 5 & 4 & 4 & 0 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 4 & 2 & -3 & 0 \end{pmatrix}_{b_0}$$

[9] $1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 159, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.34 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 6 & 0 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 \\ 3 & 6 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 3 & -3 & 0 \end{pmatrix}_{b_0}$$

[10] $1^{11}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 \quad T = 8, T_{\max} = 28, \Delta = -116, \det = 160, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 12.27 \rightarrow 13, T_{\max} = 28, T \in [13, 28]$

$$\begin{pmatrix} 1 & 2 & 4 & 5 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 4 & 5 & 9 & 12 & 0 & 5 \\ 5 & 6 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 5 & 6 & -3 & 0 \end{pmatrix}_{b_0}$$

[11] $1^{111}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 215$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.88 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 6 & 12 & 0 & 4 \\ 3 & 6 & 4 & 13 & 0 & 4 \\ 7 & 12 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 236$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 11.2$
 $T_{\min} = 11.97 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 9 & 0 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 4 & 4 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{112}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 245$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.4$
 $T_{\min} = 11.90 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 1 & 3 \\ 2 & 1 & 5 & 9 & 0 & 3 \\ 3 & 5 & 4 & 13 & 0 & 4 \\ 7 & 9 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 4 & 9 & -3 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{113}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 343$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 9 & 1 & 3 \\ 2 & 1 & 5 & 12 & 0 & 3 \\ 3 & 5 & 4 & 17 & 0 & 4 \\ 9 & 12 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 4 & 12 & -3 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{113}31^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4$ $T = 8$, $T_{\max} = 28$, $\Delta = -116$, $\det = 343$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 13.2$
 $T_{\min} = 11.81 \rightarrow 12$, $T_{\max} = 28$, $T \in [12, 28]$

$$\begin{pmatrix} 1 & 2 & 3 & 12 & 1 & 3 \\ 2 & 1 & 5 & 15 & 0 & 3 \\ 3 & 5 & 4 & 22 & 0 & 4 \\ 12 & 15 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 3 & 3 & 4 & 16 & -3 & 1 \end{pmatrix}_{b_0}$$

6.146 $\text{NHC}(-2)(-2)(-3) + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{so}_8$

$T = 9$ [0] $1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -147$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 5 & 8 & 11 & 9 & 1 & 5 \\ 8 & 10 & 17 & 12 & 0 & 6 \\ 11 & 17 & 20 & 17 & 0 & 8 \\ 9 & 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 6 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

[1] $1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{so}_8$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -440$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 5 & 9 & 11 & 9 & 1 & 5 \\ 9 & 13 & 18 & 15 & 0 & 7 \\ 11 & 18 & 20 & 17 & 0 & 8 \\ 9 & 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 7 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

6.147 $\text{NHC}(-2)(-2)(-3) + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$

$T = 9$ [0] $1^{112}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9$, $T_{\max} = 12$, $\Delta = -86$, $\det = -98$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 12.7$

$T_{\min} = 11.92 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 4 & 3 \\ 2 & 1 & 5 & 6 & 0 & 6 & 3 \\ 3 & 5 & 4 & 9 & 0 & 8 & 4 \\ 5 & 6 & 9 & 12 & 0 & 12 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 8 & 12 & 0 & 10 & 6 \\ 3 & 3 & 4 & 6 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

- [1] $1^{11}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -115, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 3 & 0 & 1 & 3 \\ 3 & 4 & 4 & 2 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$
- [2] $1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -150, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.89 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 6 & 3 & 0 & 6 & 4 \\ 3 & 6 & 4 & 4 & 0 & 5 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 5 & 3 & 0 & 4 & 4 \\ 3 & 4 & 4 & 2 & -3 & 4 & 1 \end{pmatrix}_{b_0}$$
- [3] $1^{111}31^{121} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 2 & 3 \\ 3 & 4 & 6 & 12 & 0 & 3 & 4 \\ 3 & 6 & 4 & 13 & 0 & 4 & 4 \\ 7 & 12 & 13 & 27 & 0 & 9 & 9 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 9 & 0 & 1 & 3 \\ 3 & 4 & 4 & 9 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$
- [4] $1^{113}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -241, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 6 & 3 & 0 & 21 & 4 \\ 3 & 6 & 4 & 4 & 0 & 19 & 4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 21 & 19 & 12 & 0 & 73 & 15 \\ 3 & 4 & 4 & 2 & -3 & 15 & 1 \end{pmatrix}_{b_0}$$
- [5] $1^{111}31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -245, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 2 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 & 3 \\ 3 & 5 & 4 & 9 & 0 & 4 & 4 \\ 5 & 6 & 9 & 12 & 0 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 6 & 0 & 1 & 3 \\ 3 & 3 & 4 & 6 & -3 & 3 & 1 \end{pmatrix}_{b_0}$$
- [6] $1^{111}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -294, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 5 & 3 \\ 2 & 1 & 5 & 6 & 0 & 6 & 3 \\ 3 & 5 & 4 & 9 & 0 & 9 & 4 \\ 5 & 6 & 9 & 12 & 0 & 15 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 6 & 9 & 15 & 0 & 13 & 7 \\ 3 & 3 & 4 & 6 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$
- [7] $1^{111}31^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -318, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 6 & 3 & 0 & 12 & 4 \\ 3 & 6 & 4 & 4 & 0 & 13 & 4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 13 & 9 & 0 & 28 & 10 \\ 3 & 4 & 4 & 2 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$
- [8] $1^{111}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -381, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$
 $T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 6 & 0 & 13 & 7 \\ 3 & 4 & 4 & 2 & -3 & 7 & 1 \end{pmatrix}_{b_0}$$
- [9] $1^{112}31^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2$ $T = 9, T_{\max} = 12, \Delta = -86, \det = -423, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} =$

12.7

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 8 & 3 \\ 3 & 4 & 6 & 3 & 0 & 15 & 4 \\ 3 & 6 & 4 & 4 & 0 & 14 & 4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 8 & 15 & 14 & 9 & 0 & 37 & 11 \\ 3 & 4 & 4 & 2 & -3 & 11 & 1 \end{pmatrix}_{b_0}$$

[10] $1^{112}3^11^{14} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -486, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 11.94 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 14 & 3 \\ 3 & 4 & 6 & 3 & 0 & 24 & 4 \\ 3 & 6 & 4 & 4 & 0 & 26 & 4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 14 & 24 & 26 & 18 & 0 & 118 & 20 \\ 3 & 4 & 4 & 2 & -3 & 20 & 1 \end{pmatrix}_{b_0}$$

[11] $1^{112}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -539, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 7 & 3 \\ 2 & 1 & 5 & 6 & 0 & 9 & 3 \\ 3 & 5 & 4 & 9 & 0 & 13 & 4 \\ 5 & 6 & 9 & 12 & 0 & 21 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 9 & 13 & 21 & 0 & 28 & 10 \\ 3 & 3 & 4 & 6 & -3 & 10 & 1 \end{pmatrix}_{b_0}$$

[12] $1^{113}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -686, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 9 & 3 \\ 2 & 1 & 5 & 6 & 0 & 12 & 3 \\ 3 & 5 & 4 & 9 & 0 & 17 & 4 \\ 5 & 6 & 9 & 12 & 0 & 27 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 12 & 17 & 27 & 0 & 49 & 13 \\ 3 & 3 & 4 & 6 & -3 & 13 & 1 \end{pmatrix}_{b_0}$$

[13] $1^{114}3^11^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -735, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 11 & 3 \\ 2 & 1 & 5 & 6 & 0 & 15 & 3 \\ 3 & 5 & 4 & 9 & 0 & 21 & 4 \\ 5 & 6 & 9 & 12 & 0 & 33 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 15 & 21 & 33 & 0 & 76 & 16 \\ 3 & 3 & 4 & 6 & -3 & 16 & 1 \end{pmatrix}_{b_0}$$

[14] $1^{112}3^11^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -738, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 10 & 3 \\ 3 & 4 & 6 & 3 & 0 & 18 & 4 \\ 3 & 6 & 4 & 4 & 0 & 18 & 4 \\ 2 & 3 & 4 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 10 & 18 & 18 & 12 & 0 & 58 & 14 \\ 3 & 4 & 4 & 2 & -3 & 14 & 1 \end{pmatrix}_{b_0}$$

[15] $1^{112}3^11^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -759, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.7$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 12 & 3 \\ 3 & 4 & 6 & 3 & 0 & 21 & 4 \\ 3 & 6 & 4 & 4 & 0 & 22 & 4 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 12 & 21 & 22 & 15 & 0 & 85 & 17 \\ 3 & 4 & 4 & 2 & -3 & 17 & 1 \end{pmatrix}_{b_0}$$

[16] $1^{114}3^11^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -906, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 16 & 3 \\ 3 & 4 & 6 & 3 & 0 & 30 & 4 \\ 3 & 6 & 4 & 4 & 0 & 28 & 4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 16 & 30 & 28 & 18 & 0 & 154 & 22 \\ 3 & 4 & 4 & 2 & -3 & 22 & 1 \end{pmatrix}_{b_0}$$

[17] $1^{113}3^11^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{f}_4 + \text{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -934, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 13 & 3 \\ 3 & 4 & 6 & 3 & 0 & 24 & 4 \\ 3 & 6 & 4 & 4 & 0 & 23 & 4 \\ 2 & 3 & 4 & 0 & 0 & 15 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 13 & 24 & 23 & 15 & 0 & 100 & 18 \\ 3 & 4 & 4 & 2 & -3 & 18 & 1 \end{pmatrix}_{b_0}$$

$$[18] \ 1^{113}3^11^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1333, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 15 & 3 \\ 3 & 4 & 6 & 3 & 0 & 27 & 4 \\ 3 & 6 & 4 & 4 & 0 & 27 & 4 \\ 2 & 3 & 4 & 0 & 0 & 18 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 15 & 27 & 27 & 18 & 0 & 133 & 21 \\ 3 & 4 & 4 & 2 & -3 & 21 & 1 \end{pmatrix}_{b_0}$$

$$[19] \ 1^{113}3^11^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1438, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 17 & 3 \\ 3 & 4 & 6 & 3 & 0 & 30 & 4 \\ 3 & 6 & 4 & 4 & 0 & 31 & 4 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 17 & 30 & 31 & 21 & 0 & 172 & 24 \\ 3 & 4 & 4 & 2 & -3 & 24 & 1 \end{pmatrix}_{b_0}$$

$$[20] \ 1^{114}3^11^{13} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -1683, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 18 & 3 \\ 3 & 4 & 6 & 3 & 0 & 33 & 4 \\ 3 & 6 & 4 & 4 & 0 & 32 & 4 \\ 2 & 3 & 4 & 0 & 0 & 21 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 18 & 33 & 32 & 21 & 0 & 193 & 25 \\ 3 & 4 & 4 & 2 & -3 & 25 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{114}3^11^{14} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, T_{\max} = 12, \Delta = -86, \det = -2166, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 20 & 3 \\ 3 & 4 & 6 & 3 & 0 & 36 & 4 \\ 3 & 6 & 4 & 4 & 0 & 36 & 4 \\ 2 & 3 & 4 & 0 & 0 & 24 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 20 & 36 & 36 & 24 & 0 & 238 & 28 \\ 3 & 4 & 4 & 2 & -3 & 28 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{6.148} \quad \mathbf{NHC}(-2)(-2)(-3) + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3)$$

$$T = 10 \quad [0] \ 1^{113}3^11^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 194, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 22 & 3 \\ 3 & 4 & 6 & 3 & 0 & 33 & 4 \\ 3 & 6 & 4 & 4 & 0 & 43 & 4 \\ 2 & 3 & 4 & 0 & 0 & 32 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 23 & 43 & 43 & 0 & 12 & 3 \\ 2 & 3 & 3 & 2 & 0 & 20 & 2 \\ 3 & 4 & 4 & 2 & -3 & 32 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 244, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 32 & 3 \\ 3 & 4 & 6 & 3 & 0 & 63 & 4 \\ 3 & 6 & 4 & 4 & 0 & 53 & 4 \\ 2 & 3 & 4 & 0 & 0 & 32 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 5 & 3 & 0 & 43 & 4 \\ 2 & 3 & 3 & 2 & 0 & 30 & 2 \\ 3 & 4 & 4 & 2 & -3 & 42 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^11^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, T_{\max} = 12, \Delta = -58, \det = 302, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 22 & 3 \\ 2 & 1 & 5 & 6 & 0 & 32 & 3 \\ 3 & 5 & 4 & 9 & 0 & 43 & 4 \\ 5 & 6 & 9 & 12 & 0 & 65 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 6 & 0 & 12 & 3 \\ 2 & 2 & 3 & 5 & 0 & 20 & 2 \\ 3 & 3 & 4 & 6 & -3 & 32 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{111}} 1^{\overline{112}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 484, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 12 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 13 & 3 & 4 \\ 2 & 3 & 4 & 0 & 0 & 9 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 13 & 9 & 0 & 28 & 7 & 10 \\ 2 & 3 & 3 & 2 & 0 & 7 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 10 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{111}} 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \rightarrow (-3) \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 622, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 9 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 3 & 4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 5 & 9 & 9 & 6 & 0 & 13 & 5 & 7 \\ 2 & 3 & 3 & 2 & 0 & 5 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 7 & 2 & 1 \end{pmatrix}_{b_0}$$

6.149 NHC(-2)(-2)(-3) + $\mathbf{su}_3$ + $\mathbf{f}_4$ + $\mathbf{su}_2$ + $\mathbf{su}_2\mathbf{n}_3\mathbf{mix}$

$$T = 10 \quad [0] \ 1^{\overline{111}} 3^{\overline{111}} 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 389, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 12 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 4 & 3 & 0 & 1 & 8 & 3 \\ 7 & 12 & 12 & 8 & 0 & 8 & 25 & 9 \\ 3 & 4 & 4 & 2 & -3 & 3 & 9 & 1 \end{pmatrix}_{b_0}$$

6.150 NHC(-2)(-2)(-3) + $\mathbf{su}_3$ + $\mathbf{f}_4$ + $\mathbf{su}_2 \rightarrow (-3)$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{111}} 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -54, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.52 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 3 & 0 & 2 & 3 & 3 \\ 3 & 5 & 4 & 5 & 0 & 3 & 4 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 3 & 3 & 4 & 3 & -3 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{111}} 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -100, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 3 & 0 & 2 & 3 & 3 \\ 4 & 5 & 9 & 4 & 0 & 4 & 5 & 5 \\ 2 & 3 & 4 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 4 & 2 & 0 & 0 & 2 & 2 \\ 3 & 3 & 5 & 2 & -3 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{112}} 3^{\overline{111}} 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -146, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.4$$

$$T_{\min} = 12.49 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 6 & 0 & 2 & 3 & 3 \\ 3 & 5 & 4 & 8 & 0 & 3 & 4 & 4 \\ 4 & 6 & 8 & 9 & 0 & 4 & 5 & 5 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 4 & 0 & 0 & 2 & 2 \\ 3 & 3 & 4 & 5 & -3 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{111}} 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -194, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.12 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 2 & 3 & 3 \\ 2 & 1 & 5 & 3 & 0 & 2 & 3 & 3 \\ 3 & 5 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 2 & 0 & 0 & 2 & 2 \\ 3 & 3 & 4 & 2 & -3 & 2 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -210, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 2 & 3 \\ 2 & 1 & 5 & 6 & 0 & 2 & 3 \\ 3 & 5 & 4 & 9 & 0 & 3 & 4 \\ 5 & 6 & 9 & 12 & 0 & 5 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 2 & 3 & 5 & 0 & 0 & 2 \\ 3 & 3 & 4 & 6 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111}3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -226, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 3 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{111}3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -334, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.4$$

$$T_{\min} = 11.92 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 1 & 2 & 3 \\ 2 & 1 & 5 & 9 & 0 & 2 & 3 \\ 3 & 5 & 4 & 13 & 0 & 3 & 4 \\ 7 & 9 & 13 & 27 & 0 & 7 & 9 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 2 & 3 & 7 & 0 & 0 & 2 \\ 3 & 3 & 4 & 9 & -3 & 2 & 1 \end{pmatrix}_{b_0}$$

6.151 NHC(-2)(-2)(-3) + $\mathbf{su}_3$ + $\mathbf{f}_4$ + $\mathbf{su}_2 \rightarrow (-3)$ + $\mathbf{su}_2$

$$T = 10 \quad [0] \ 1^{111}3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 194, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 3 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 2 & 2 \\ 2 & 3 & 4 & 3 & 0 & 2 & 1 & 3 \\ 3 & 4 & 4 & 2 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111}3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 244, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 6 & 4 \\ 3 & 6 & 4 & 4 & 0 & 3 & 5 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 3 & 2 \\ 3 & 6 & 5 & 3 & 0 & 3 & 4 & 4 \\ 3 & 4 & 4 & 2 & -3 & 2 & 4 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111}3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 302, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 2 & 2 & 3 \\ 2 & 1 & 5 & 6 & 0 & 2 & 3 & 3 \\ 3 & 5 & 4 & 9 & 0 & 3 & 4 & 4 \\ 5 & 6 & 9 & 12 & 0 & 5 & 6 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 2 & 3 & 5 & 0 & 0 & 2 & 2 \\ 2 & 3 & 4 & 6 & 0 & 2 & 1 & 3 \\ 3 & 3 & 4 & 6 & -3 & 2 & 3 & 1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{111}3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 484, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.97 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 7 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 12 & 4 \\ 3 & 6 & 4 & 4 & 0 & 3 & 13 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 & 9 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 7 & 2 \\ 7 & 12 & 13 & 9 & 0 & 7 & 28 & 10 \\ 3 & 4 & 4 & 2 & -3 & 2 & 10 & 1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111}3^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 622, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.96 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 3 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 2 & 3 & 3 & 2 & 0 & 0 & 5 & 2 \\ 5 & 9 & 9 & 6 & 0 & 5 & 13 & 7 \\ 3 & 4 & 4 & 2 & -3 & 2 & 7 & 1 \end{pmatrix}_{b_0}$$

6.152 NHC(-2)(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{f}_4$ + su2n3mix

$$T = 9 \quad [0] \quad 1^{112} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -82, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 6 & 3 \\ 2 & 1 & 5 & 3 & 0 & 8 & 3 \\ 3 & 5 & 4 & 5 & 0 & 11 & 4 \\ 3 & 3 & 5 & 3 & 0 & 9 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 6 & 8 & 11 & 9 & 0 & 20 & 8 \\ 3 & 3 & 4 & 3 & -3 & 8 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -89, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 7 & 3 \\ 2 & 1 & 5 & 3 & 0 & 8 & 3 \\ 3 & 5 & 4 & 4 & 0 & 12 & 4 \\ 2 & 3 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 8 & 12 & 8 & 0 & 25 & 9 \\ 3 & 3 & 4 & 2 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -97, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.55 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 5 & 3 \\ 2 & 1 & 5 & 3 & 0 & 5 & 3 \\ 4 & 5 & 9 & 4 & 0 & 12 & 5 \\ 2 & 3 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 5 & 12 & 5 & 0 & 12 & 6 \\ 3 & 3 & 5 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -117, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.11 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 4 & 3 \\ 2 & 1 & 5 & 3 & 0 & 5 & 3 \\ 3 & 5 & 4 & 5 & 0 & 7 & 4 \\ 3 & 3 & 5 & 3 & 0 & 6 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 5 & 7 & 6 & 0 & 7 & 5 \\ 3 & 3 & 4 & 3 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -127, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 4 & 3 \\ 3 & 4 & 6 & 3 & 0 & 6 & 4 \\ 3 & 6 & 4 & 4 & 0 & 7 & 4 \\ 2 & 3 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 6 & 7 & 5 & 0 & 7 & 5 \\ 3 & 4 & 4 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{111} 3^1 1^{113} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -157, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.26 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 11 & 3 \\ 3 & 4 & 6 & 3 & 0 & 18 & 4 \\ 3 & 6 & 4 & 4 & 0 & 20 & 4 \\ 2 & 3 & 4 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 11 & 18 & 20 & 14 & 0 & 69 & 15 \\ 3 & 4 & 4 & 2 & -3 & 15 & 0 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{f}_4 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 13, \Delta = -88, \det = -175, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.10 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 7 & 3 \\ 2 & 1 & 5 & 3 & 0 & 8 & 3 \\ 4 & 5 & 9 & 4 & 0 & 16 & 5 \\ 2 & 3 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 8 & 16 & 8 & 0 & 25 & 9 \\ 3 & 3 & 5 & 2 & -3 & 9 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}3^1}1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -202, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 8 & 4 \\ 2 & 3 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 8 & 5 & 0 & 12 & 6 \\ 3 & 4 & 4 & 2 & -3 & 6 & 1 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{\overline{111}3^1}1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -329, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.93 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 4 & 3 \\ 2 & 1 & 5 & 6 & 0 & 5 & 3 \\ 3 & 5 & 4 & 9 & 0 & 7 & 4 \\ 5 & 6 & 9 & 12 & 0 & 11 & 6 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 5 & 7 & 11 & 0 & 7 & 5 \\ 3 & 3 & 4 & 6 & -3 & 5 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{\overline{111}3^1}1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -362, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 12.17 \rightarrow 13, \ T_{\max} = 13, \ T \in [13, 13]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 4 & 3 \\ 2 & 1 & 5 & 3 & 0 & 5 & 3 \\ 3 & 5 & 4 & 4 & 0 & 7 & 4 \\ 2 & 3 & 4 & 0 & 0 & 5 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 4 & 5 & 7 & 5 & 0 & 7 & 5 \\ 3 & 3 & 4 & 2 & -3 & 5 & 0 \end{pmatrix}_{b_0}$$

$$[10] \ 1^{\overline{111}3^1}1^{\overline{112}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -466, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 9 & 3 \\ 3 & 4 & 6 & 3 & 0 & 15 & 4 \\ 3 & 6 & 4 & 4 & 0 & 16 & 4 \\ 2 & 3 & 4 & 0 & 0 & 11 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 9 & 15 & 16 & 11 & 0 & 44 & 12 \\ 3 & 4 & 4 & 2 & -3 & 12 & 1 \end{pmatrix}_{b_0}$$

$$[11] \ 1^{\overline{111}3^1}1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 \quad T = 9, \ T_{\max} = 13, \ \Delta = -88, \ \det = -481, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 13, \ T \in [12, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 3 \\ 3 & 4 & 6 & 3 & 0 & 12 & 4 \\ 3 & 6 & 4 & 4 & 0 & 12 & 4 \\ 2 & 3 & 4 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 7 & 12 & 12 & 8 & 0 & 25 & 9 \\ 3 & 4 & 4 & 2 & -3 & 9 & 1 \end{pmatrix}_{b_0}$$

6.153 $\text{NHC}(-2)(-2)(-3) + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_{2n3\text{mix}} + \mathbf{su}_2$

$$T = 10 \quad [0] \ 1^{\overline{111}3^1}1^{\overline{1111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_2 + \mathbf{su}_2 \quad T = 10, \ T_{\max} = 12, \ \Delta = -58, \ \det = 389, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 11.95 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 7 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 12 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 12 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 8 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & 0 & -3 \\ 7 & 12 & 12 & 8 & 0 & 25 & 8 & 9 \\ 2 & 3 & 4 & 3 & 0 & 8 & 1 & 3 \\ 3 & 4 & 4 & 2 & -3 & 9 & 3 & 1 \end{pmatrix}_{b_0}$$

6.154 $\text{NHC}(-2)(-2)(-3) + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3$

$$T = 9 \quad [0] \ 1^{\overline{11}3^1}1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -95, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 13.86 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 3 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 \\ 4 & 5 & 9 & 4 & 0 & 8 & 5 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 3 & 8 & 3 & 0 & 3 & 3 \\ 3 & 3 & 5 & 2 & -3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}3^1}1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, \ T_{\max} = 14, \ \Delta = -95, \ \det = -98, \ (n_+, n_-, n_0) = (1, 9, 0), \ c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 \\ 3 & 5 & 4 & 5 & 0 & 4 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 & 3 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 3 & 0 & 0 & 2 \\ 3 & 3 & 4 & 3 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{111}} 1^{\overline{111}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -221, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 5 & 3 \\ 2 & 1 & 5 & 3 & 0 & 6 & 3 \\ 4 & 5 & 9 & 4 & 0 & 12 & 5 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 6 & 12 & 6 & 0 & 12 & 6 \\ 3 & 3 & 5 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -246, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.54 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 6 & 3 & 0 & 3 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 3 & 0 & 0 & 2 \\ 3 & 4 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -281, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.37 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 6 & 3 & 0 & 6 & 4 \\ 3 & 6 & 4 & 4 & 0 & 5 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 3 & 6 & 5 & 3 & 0 & 3 & 3 \\ 3 & 4 & 4 & 2 & -3 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -392, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 5 & 3 \\ 2 & 1 & 5 & 3 & 0 & 6 & 3 \\ 3 & 5 & 4 & 4 & 0 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 6 & 9 & 6 & 0 & 12 & 6 \\ 3 & 3 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -441, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.67 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 2 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 & 3 \\ 3 & 5 & 4 & 4 & 0 & 4 & 4 \\ 2 & 3 & 4 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 2 & 3 & 4 & 3 & 0 & 0 & 2 \\ 3 & 3 & 4 & 2 & -3 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{111}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{f}_4 + \mathbf{su}_3 \quad T = 9, T_{\max} = 14, \Delta = -95, \det = -512, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.2$$

$$T_{\min} = 12.21 \rightarrow 13, T_{\max} = 14, T \in [13, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 5 & 3 \\ 3 & 4 & 6 & 3 & 0 & 9 & 4 \\ 3 & 6 & 4 & 4 & 0 & 9 & 4 \\ 2 & 3 & 4 & 0 & 0 & 6 & 2 \\ 1 & 0 & 0 & 0 & -5 & 0 & -3 \\ 5 & 9 & 9 & 6 & 0 & 12 & 6 \\ 3 & 4 & 4 & 2 & -3 & 6 & 0 \end{pmatrix}_{b_0}$$

6.155 NHC(-2)(-2)(-3) + su₃ + NHC(-2)(-3)

$$T = 11 \quad [0] \ 1^{\overline{11}} 5^{\overline{1}} 1^{\overline{1}} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + [\mathbf{su}_2 \cdot \mathbf{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 2 & 1 & 3 \\ 2 & 1 & 4 & 2 & 3 & 0 & 3 \\ 3 & 4 & 4 & 3 & 3 & 0 & 4 \\ 2 & 2 & 3 & 0 & 2 & 0 & 2 \\ 2 & 3 & 3 & 2 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 3 & 4 & 2 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

6.156 NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{so}_8$

$$T = 8 \quad [0] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 1 & 3 \\ 2 & 1 & 5 & 0 & 0 & 3 \\ 3 & 5 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 6 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 6 & 6 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 11 & 16 & 13 & 6 \\ 11 & 10 & 17 & 12 & 6 \\ 16 & 17 & 25 & 20 & 9 \\ 13 & 12 & 20 & 12 & 6 \\ 6 & 6 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{11} 3^{11} 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 3 & 1 & 2 \\ 1 & 0 & 2 & 0 & 2 & 2 & 2 \\ 1 & 2 & 0 & 1 & 4 & 1 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 3 & 2 & 4 & 0 & 5 & 2 & 3 \\ 1 & 2 & 1 & 0 & 2 & -2 & 0 \\ 2 & 2 & 2 & -1 & 3 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{12} 3^{11} 1^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 21, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$$

$$T_{\min} = 11.43 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 11 & 1 & 3 \\ 2 & 1 & 5 & 12 & 0 & 3 \\ 3 & 5 & 4 & 18 & 0 & 4 \\ 11 & 12 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 1 & 1 & 3 \\ 2 & 1 & 5 & 0 & 0 & 3 \\ 4 & 5 & 9 & 0 & 0 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 5 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 32, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 3 & 0 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 \\ 3 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \ 21^{11} 31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 16 & 12 & 6 \\ 12 & 13 & 18 & 15 & 7 \\ 16 & 18 & 20 & 17 & 8 \\ 12 & 15 & 17 & 12 & 6 \\ 6 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[8] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 34, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.24 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 \\ 3 & 5 & 4 & 5 & 0 & 4 \\ 3 & 3 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \ 1^{11} 3^{11} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 41, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.41 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 15 & 1 & 6 \\ 12 & 13 & 18 & 0 & 7 \\ 15 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 7 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

- [10] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 4 & 2 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 \\ 4 & 5 & 9 & 4 & 0 & 5 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 5 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$
- [11] $1^{12}31^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 45, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 4 & 7 & 1 & 3 \\ 2 & 1 & 5 & 9 & 0 & 3 \\ 4 & 5 & 9 & 16 & 0 & 5 \\ 7 & 9 & 16 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 5 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$
- [12] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 48, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.42 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 5 \\ 8 & 10 & 17 & 0 & 6 \\ 12 & 17 & 25 & 0 & 9 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 6 & 9 & -1 & 2 \end{pmatrix}_{b_0}$$
- [13] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 51, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.18 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 11 & 15 & 12 & 6 \\ 11 & 10 & 17 & 12 & 6 \\ 15 & 17 & 20 & 17 & 8 \\ 12 & 12 & 17 & 12 & 6 \\ 6 & 6 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$
- [14] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 52, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 1 & 3 \\ 3 & 4 & 6 & 0 & 0 & 4 \\ 3 & 6 & 4 & 0 & 0 & 4 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$
- [15] $1^{11}31^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 56, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.57 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 8 & 12 & 10 & 5 \\ 8 & 10 & 17 & 12 & 6 \\ 12 & 17 & 25 & 20 & 9 \\ 10 & 12 & 20 & 12 & 6 \\ 5 & 6 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$
- [18] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 \\ 1 & 2 & 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & -3 & 0 & 0 \\ 1 & 0 & 0 & 0 & -3 & 0 \\ 1 & 2 & 1 & 0 & 0 & -2 \\ 2 & 2 & 2 & -1 & -1 & 0 \end{pmatrix}_{b_0}$$
- [17] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 15 & 12 & 6 \\ 12 & 13 & 18 & 15 & 7 \\ 15 & 18 & 20 & 17 & 8 \\ 12 & 15 & 17 & 12 & 6 \\ 6 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$
- [16] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 61, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 11 & 15 & 1 & 6 \\ 11 & 10 & 17 & 0 & 6 \\ 15 & 17 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 6 & 6 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$
- [19] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 66, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 12 & 14 & 1 & 7 \\ 12 & 10 & 13 & 0 & 6 \\ 14 & 13 & 13 & 1 & 7 \\ 1 & 0 & 1 & -3 & -1 \\ 7 & 6 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$
- [20] $1^{111}31^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 68, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.35 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 9 & 1 & 3 \\ 3 & 4 & 6 & 15 & 0 & 4 \\ 3 & 6 & 4 & 17 & 0 & 4 \\ 9 & 15 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[21] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 70, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.63 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 2 & 0 & 0 & -2 & 0 \\ 2 & 2 & 2 & -1 & -1 & 0 & 0 \end{pmatrix}_{b_0}$$

$$[22] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 73, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.23 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 13 & 14 & 12 & 1 & 7 \\ 14 & 13 & 13 & 0 & 7 \\ 12 & 13 & 10 & 1 & 6 \\ 1 & 0 & 1 & -3 & -1 \\ 7 & 7 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[24] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.12 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 0 & 0 & 3 & 2 \\ 1 & 2 & 0 & 1 & 0 & 3 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 3 & 0 & 0 & 2 & 2 \\ 2 & 2 & 2 & -1 & -1 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[23] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 75, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.07 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 10 & 12 & 15 & 13 & 6 \\ 12 & 13 & 18 & 15 & 7 \\ 15 & 18 & 20 & 17 & 8 \\ 13 & 15 & 17 & 12 & 6 \\ 6 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

$$[25] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 76, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.26 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 \\ 3 & 5 & 4 & 4 & 0 & 4 \\ 3 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[26] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 76, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.26 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 9 & 8 & 1 & 5 \\ 9 & 13 & 13 & 0 & 7 \\ 8 & 13 & 10 & 1 & 6 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 7 & 6 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[27] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 80, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.15 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 5 & 8 & 9 & 1 & 5 \\ 8 & 10 & 13 & 0 & 6 \\ 9 & 13 & 13 & 1 & 7 \\ 1 & 0 & 1 & -3 & -1 \\ 5 & 6 & 7 & -1 & 2 \end{pmatrix}_{b_0}$$

$$[28] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 84, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.43 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 4 & 6 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 4 & 5 & 9 & 12 & 0 & 5 \\ 6 & 6 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 5 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[29] \ 1^{11} 3^1 1^{11} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 84, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.67 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 4 & 1 & 3 \\ 3 & 4 & 6 & 6 & 0 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 \\ 4 & 6 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[30] \ 1^{11} 3^1 1^{13} + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{green}{g}_2] + \textcolor{brown}{su}_3 + \textcolor{red}{so}_8 \quad T = 8, \ T_{\max} = 12, \ \Delta = -92, \ \det = 84, \ (n_+, n_-, n_0) = (1, 8, 0), \ c_{\text{ext}} = 10.0$$

$$T_{\min} = 11.10 \rightarrow 12, \ T_{\max} = 12, \ T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 10 & 1 & 3 \\ 3 & 4 & 6 & 15 & 0 & 4 \\ 3 & 6 & 4 & 17 & 0 & 4 \\ 10 & 15 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[32] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 11 & 10 & 5 \\ 9 & 13 & 18 & 15 & 7 \\ 11 & 18 & 20 & 17 & 8 \\ 10 & 15 & 17 & 12 & 6 \\ 5 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[31] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 89, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.37 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 5 \\ 9 & 13 & 18 & 0 & 7 \\ 11 & 18 & 20 & 0 & 8 \\ 1 & 0 & 0 & -3 & -1 \\ 5 & 7 & 8 & -1 & 2 \end{pmatrix}_{b_0}$$

[33] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 96, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.12 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 4 & 6 & 3 & 0 & 4 \\ 3 & 6 & 4 & 4 & 0 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[34] $1^{112}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 101, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.88 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 3 & 5 & 4 & 8 & 0 & 4 \\ 4 & 6 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$

[35] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 104, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.46 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 3 \\ 2 & 1 & 5 & 3 & 0 & 3 \\ 3 & 5 & 4 & 4 & 0 & 4 \\ 2 & 3 & 4 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 2 & -2 & 1 \end{pmatrix}_{b_0}$$

[37] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 3 & 5 & 4 & 9 & 0 & 4 \\ 5 & 6 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[36] $1^{112}31^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 105, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.16 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 10 & 1 & 3 \\ 2 & 1 & 5 & 12 & 0 & 3 \\ 3 & 5 & 4 & 18 & 0 & 4 \\ 10 & 12 & 18 & 57 & 0 & 13 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 13 & -2 & 1 \end{pmatrix}_{b_0}$$

[38] $21^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 109, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.38 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 12 & 20 & 15 & 7 \\ 12 & 10 & 17 & 12 & 6 \\ 20 & 17 & 25 & 20 & 9 \\ 15 & 12 & 20 & 12 & 6 \\ 7 & 6 & 9 & 6 & 2 \end{pmatrix}_{b_0}$$

[40] $1^{11}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.59 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 4 & 5 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 4 & 5 & 9 & 12 & 0 & 5 \\ 5 & 6 & 12 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 5 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

[39] $1^{111}31^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{so}_8 \quad T = 8, T_{\max} = 12, \Delta = -92, \det = 116, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.62 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 6 & 6 & 0 & 4 \\ 3 & 6 & 4 & 5 & 0 & 4 \\ 3 & 6 & 5 & 3 & 0 & 3 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 3 & -2 & 1 \end{pmatrix}_{b_0}$$

- [41] $1^{11} 1^3 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 119, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.78 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 5 & 9 & 11 & 9 & 5 \\ 9 & 13 & 18 & 15 & 7 \\ 11 & 18 & 20 & 17 & 8 \\ 9 & 15 & 17 & 12 & 6 \\ 5 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$
- [42] $1^{112} 1^{11} 3^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 129, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.32 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 5 & 1 & 3 \\ 2 & 1 & 5 & 6 & 0 & 3 \\ 3 & 5 & 4 & 8 & 0 & 4 \\ 5 & 6 & 8 & 9 & 0 & 5 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 5 & -2 & 1 \end{pmatrix}_{b_0}$$
- [43] $1^{112} 1^{11} 3^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 146, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 8 & 1 & 3 \\ 2 & 1 & 5 & 9 & 0 & 3 \\ 3 & 5 & 4 & 13 & 0 & 4 \\ 8 & 9 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$
- [44] $1^{111} 1^1 3^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 7 & 1 & 3 \\ 3 & 4 & 6 & 12 & 0 & 4 \\ 3 & 6 & 4 & 13 & 0 & 4 \\ 7 & 12 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$
- [45] $1^{111} 1^{11} 3^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.05 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 8 & 1 & 3 \\ 3 & 4 & 6 & 12 & 0 & 4 \\ 3 & 6 & 4 & 13 & 0 & 4 \\ 8 & 12 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$
- [46] $1^{111} 1^{11} 3^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 156, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.15 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 6 & 1 & 3 \\ 3 & 4 & 6 & 9 & 0 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 \\ 6 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [47] $21^{11} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 10 & 12 & 16 & 12 & 6 \\ 12 & 10 & 17 & 12 & 6 \\ 16 & 17 & 20 & 17 & 8 \\ 12 & 12 & 17 & 12 & 6 \\ 6 & 6 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$
- [48] $1^{111} 1^1 3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 172, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.23 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 5 & 1 & 3 \\ 3 & 4 & 6 & 9 & 0 & 4 \\ 3 & 6 & 4 & 9 & 0 & 4 \\ 5 & 9 & 9 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 4 & 4 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$
- [49] $1^{112} 1^1 3^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 174, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 11.2$
 $T_{\min} = 11.21 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 1 & 3 \\ 2 & 1 & 5 & 9 & 0 & 3 \\ 3 & 5 & 4 & 13 & 0 & 4 \\ 7 & 9 & 13 & 27 & 0 & 9 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 9 & -2 & 1 \end{pmatrix}_{b_0}$$
- [50] $1^{113} 1^{11} 3^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 180, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 12.0$
 $T_{\min} = 11.40 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 7 & 1 & 3 \\ 2 & 1 & 5 & 9 & 0 & 3 \\ 3 & 5 & 4 & 12 & 0 & 4 \\ 7 & 9 & 12 & 24 & 0 & 8 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 8 & -2 & 1 \end{pmatrix}_{b_0}$$
- [51] $21^{11} 31^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{so}_8$ $T = 8, T_{\max} = 12, \Delta = -92, \det = 199, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 10.0$
 $T_{\min} = 11.07 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 13 & 15 & 17 & 15 & 7 \\ 15 & 13 & 18 & 15 & 7 \\ 17 & 18 & 20 & 17 & 8 \\ 15 & 15 & 17 & 12 & 6 \\ 7 & 7 & 8 & 6 & 2 \end{pmatrix}_{b_0}$$

[52] $21^{113}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 205$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 10.0$
 $T_{\min} = 11.28 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 13 & 15 & 20 & 12 & 7 \\ 15 & 13 & 21 & 12 & 7 \\ 20 & 21 & 25 & 16 & 9 \\ 12 & 12 & 16 & 9 & 5 \\ 7 & 7 & 9 & 5 & 2 \end{pmatrix}_{b_0}$$

[53] $1^{113}31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 216$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 10.98 \rightarrow 11$, $T_{\max} = 12$, $T \in [11, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 13 & 1 & 3 \\ 2 & 1 & 5 & 15 & 0 & 3 \\ 3 & 5 & 4 & 22 & 0 & 4 \\ 13 & 15 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 16 & -2 & 2 \end{pmatrix}_{b_0}$$

[54] $1^{113}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 241$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.27 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 9 & 1 & 3 \\ 2 & 1 & 5 & 12 & 0 & 3 \\ 3 & 5 & 4 & 17 & 0 & 4 \\ 9 & 12 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

[55] $1^{113}31^{12} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 244$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.10 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 12 & 1 & 3 \\ 2 & 1 & 5 & 15 & 0 & 3 \\ 3 & 5 & 4 & 22 & 0 & 4 \\ 12 & 15 & 22 & 84 & 0 & 16 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 16 & -2 & 1 \end{pmatrix}_{b_0}$$

[56] $1^{113}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 8$, $T_{\max} = 12$, $\Delta = -92$, $\det = 269$, $(n_+, n_-, n_0) = (1, 8, 0)$, $c_{\text{ext}} = 12.0$
 $T_{\min} = 11.06 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 10 & 1 & 3 \\ 2 & 1 & 5 & 12 & 0 & 3 \\ 3 & 5 & 4 & 17 & 0 & 4 \\ 10 & 12 & 17 & 48 & 0 & 12 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 12 & -2 & 1 \end{pmatrix}_{b_0}$$

$T = 9$ [57] $1^{11}41^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8$ $T = 9$, $T_{\max} = 12$, $\Delta = -83$, $\det = -69$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 10.0$

$T_{\min} = 11.78 \rightarrow 12$, $T_{\max} = 12$, $T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 3 \\ 2 & 1 & 4 & 3 & 0 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 3 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 3 & 3 & 4 & 2 & -2 & 2 \end{pmatrix}_{b_0}$$

6.157 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{e}_6$

$T = 9$ [0] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{e}_6$ $T = 9$, $T_{\max} = 99$, $\Delta = -141$, $\det = -559$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 16.0$

$T_{\min} = 13.74 \rightarrow 14$, $T_{\max} = 99$, $T \in [14, 99]$

$$\begin{pmatrix} 5 & 9 & 11 & 9 & 1 & 5 \\ 9 & 13 & 18 & 15 & 0 & 7 \\ 11 & 18 & 20 & 17 & 0 & 8 \\ 9 & 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -6 & -4 \\ 5 & 7 & 8 & 6 & -4 & 0 \end{pmatrix}_{b_0}$$

6.158 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4$

$T = 9$ [0] $1^{111}31^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{f}_4$ $T = 9$, $T_{\max} = 26$, $\Delta = -115$, $\det = -440$, $(n_+, n_-, n_0) = (1, 9, 0)$, $c_{\text{ext}} = 15.2$

$T_{\min} = 12.90 \rightarrow 13$, $T_{\max} = 26$, $T \in [13, 26]$

$$\begin{pmatrix} 5 & 9 & 11 & 9 & 1 & 5 \\ 9 & 13 & 18 & 15 & 0 & 7 \\ 11 & 18 & 20 & 17 & 0 & 8 \\ 9 & 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -5 & -3 \\ 5 & 7 & 8 & 6 & -3 & 1 \end{pmatrix}_{b_0}$$

6.159 NHC(-2)(-2)(-3) + $\mathfrak{su}_3$ + $\mathfrak{so}_8$ + $\mathfrak{so}_8$

$$T = 9 \quad [0] \quad 1^{111} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -61, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.28 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 8 & 11 & 1 & 1 & 5 \\ 8 & 10 & 17 & 0 & 0 & 6 \\ 11 & 17 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 6 & 8 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{11} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -75, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.23 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 8 & 12 & 1 & 1 & 5 \\ 8 & 10 & 17 & 0 & 0 & 6 \\ 12 & 17 & 25 & 0 & 0 & 9 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 6 & 9 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3^1 1^1 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -98, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.24 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 8 & 11 & 9 & 1 & 5 \\ 8 & 10 & 17 & 12 & 0 & 6 \\ 11 & 17 & 20 & 17 & 0 & 8 \\ 9 & 12 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 6 & 8 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{11} 3^1 1^{111} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -132, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.27 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 8 & 12 & 10 & 1 & 5 \\ 8 & 10 & 17 & 12 & 0 & 6 \\ 12 & 17 & 25 & 20 & 0 & 9 \\ 10 & 12 & 20 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 6 & 9 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -164, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.29 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 11 & 1 & 1 & 5 \\ 9 & 13 & 18 & 0 & 0 & 7 \\ 11 & 18 & 20 & 0 & 0 & 8 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 7 & 8 & -1 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -194, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.12 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 0 & 0 & 2 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 2 & 1 & 0 & 0 & -2 & 1 & 0 \\ 1 & 1 & 2 & 0 & 0 & 1 & -2 & 0 \\ 2 & 2 & 2 & -1 & -1 & 0 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -201, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.08 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 11 & 10 & 1 & 5 \\ 9 & 13 & 18 & 15 & 0 & 7 \\ 11 & 18 & 20 & 17 & 0 & 8 \\ 10 & 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 7 & 8 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{111} 3^1 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{so}_8 + \mathfrak{so}_8 \quad T = 9, T_{\max} = 13, \Delta = -91, \det = -321, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.18 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 5 & 9 & 11 & 9 & 1 & 5 \\ 9 & 13 & 18 & 15 & 0 & 7 \\ 11 & 18 & 20 & 17 & 0 & 8 \\ 9 & 15 & 17 & 12 & 0 & 6 \\ 1 & 0 & 0 & 0 & -4 & -2 \\ 5 & 7 & 8 & 6 & -2 & 1 \end{pmatrix}_{b_0}$$

6.160 NHC(-2)(-2)(-3) + $\mathfrak{su}_3 + \mathfrak{su}_2$

$$T = 8 \quad [0] \ 1^{11} 3 \ 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 12, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 3 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 16, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 3 \\ 3 & 4 & 4 & 0 & 0 & 4 \\ 2 & 4 & 1 & 1 & 0 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 \\ 3 & 4 & 3 & -1 & 0 & 3 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 \ 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 28, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 4 \\ 2 & 4 & 1 & 1 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & -1 \\ 2 & 3 & 3 & 0 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{14} 3 \ 1^{131} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 8, T_{\max} = 10, \Delta = -63, \det = 245, (n_+, n_-, n_0) = (1, 8, 0), c_{\text{ext}} = 9.6$$

$$T_{\min} = 9.99 \rightarrow 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 9 & 1 & 2 \\ 1 & 0 & 2 & 0 & 11 & 1 & 2 \\ 1 & 2 & 0 & 1 & 10 & 2 & 2 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 9 & 11 & 10 & 0 & 71 & 10 & 13 \\ 1 & 1 & 2 & 0 & 10 & 0 & 2 \\ 2 & 2 & 2 & -1 & 13 & 2 & 2 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [4] \ 2 \ 3 \ 1^{111} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 5 & 3 \\ 3 & 2 & 4 & 4 & 6 & 4 \\ 3 & 4 & 4 & 3 & 7 & 4 \\ 3 & 4 & 3 & 1 & 6 & 3 \\ 5 & 6 & 7 & 6 & 9 & 7 \\ 3 & 4 & 4 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[5] \ 2 \ 3 \ 1^{112} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -14, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 10 & 3 \\ 3 & 2 & 4 & 4 & 12 & 4 \\ 3 & 4 & 4 & 3 & 14 & 4 \\ 3 & 4 & 3 & 1 & 12 & 3 \\ 10 & 12 & 14 & 12 & 42 & 14 \\ 3 & 4 & 4 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{11} 5 \ 1^{1231} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 3 & 1 & 1 & 1 & 1 & 2 \\ 3 & 10 & 4 & 4 & 0 & 3 & 6 \\ 1 & 4 & 0 & 2 & 0 & 2 & 2 \\ 1 & 4 & 2 & 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 3 & 2 & 1 & 0 & 0 & 2 \\ 2 & 6 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 2 \ 3 \ 1^{113} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -24, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 3 & 3 & 15 & 3 \\ 3 & 2 & 4 & 4 & 18 & 4 \\ 3 & 4 & 4 & 3 & 21 & 4 \\ 3 & 4 & 3 & 1 & 18 & 3 \\ 15 & 18 & 21 & 18 & 97 & 21 \\ 3 & 4 & 4 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[8] \ 2 \ 3 \ 1^{111} 1^{11} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -108, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 3 & 3 \\ 3 & 2 & 6 & 2 & 4 & 4 \\ 4 & 6 & 5 & 3 & 5 & 5 \\ 2 & 2 & 3 & 0 & 2 & 2 \\ 3 & 4 & 5 & 2 & 2 & 4 \\ 3 & 4 & 5 & 2 & 4 & 4 \end{pmatrix}_{b_0}$$

[9] $2^{\overset{1}{1}\overset{112}{3}} 3^{\overset{1}{1}} 15 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -114, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.5$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 6 & 3 \\ 3 & 2 & 6 & 2 & 8 & 4 \\ 4 & 6 & 5 & 3 & 10 & 5 \\ 2 & 2 & 3 & 0 & 4 & 2 \\ 6 & 8 & 10 & 4 & 14 & 8 \\ 3 & 4 & 5 & 2 & 8 & 4 \end{pmatrix}_{b_0}$$

[10] $2^{\overset{1}{1}\overset{113}{3}} 3^{\overset{1}{1}} 15 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -124, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.8$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 9 & 3 \\ 3 & 2 & 6 & 2 & 12 & 4 \\ 4 & 6 & 5 & 3 & 15 & 5 \\ 2 & 2 & 3 & 0 & 6 & 2 \\ 9 & 12 & 15 & 6 & 34 & 12 \\ 3 & 4 & 5 & 2 & 12 & 4 \end{pmatrix}_{b_0}$$

[11] $2^{\overset{1}{1}\overset{111}{3}} 3^{\overset{1}{1}} 15 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -128, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 2 & 3 & 5 & 3 \\ 3 & 2 & 3 & 4 & 6 & 4 \\ 2 & 3 & 1 & 3 & 5 & 3 \\ 3 & 4 & 3 & 1 & 6 & 3 \\ 5 & 6 & 5 & 6 & 9 & 7 \\ 3 & 4 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

[12] $2^{\overset{1}{1}\overset{114}{3}} 3^{\overset{1}{1}} 15 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -138, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 12 & 3 \\ 3 & 2 & 6 & 2 & 16 & 4 \\ 4 & 6 & 5 & 3 & 20 & 5 \\ 2 & 2 & 3 & 0 & 8 & 2 \\ 12 & 16 & 20 & 8 & 62 & 16 \\ 3 & 4 & 5 & 2 & 16 & 4 \end{pmatrix}_{b_0}$$

[13] $2^{\overset{1}{1}\overset{115}{3}} 3^{\overset{1}{1}} 15 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -156, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.1$
 $T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 15 & 3 \\ 3 & 2 & 6 & 2 & 20 & 4 \\ 4 & 6 & 5 & 3 & 25 & 5 \\ 2 & 2 & 3 & 0 & 10 & 2 \\ 15 & 20 & 25 & 10 & 98 & 20 \\ 3 & 4 & 5 & 2 & 20 & 4 \end{pmatrix}_{b_0}$$

[14] $2^{\overset{1}{1}\overset{116}{3}} 3^{\overset{1}{1}} 15 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -178, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.2$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 18 & 3 \\ 3 & 2 & 6 & 2 & 24 & 4 \\ 4 & 6 & 5 & 3 & 30 & 5 \\ 2 & 2 & 3 & 0 & 12 & 2 \\ 18 & 24 & 30 & 12 & 142 & 24 \\ 3 & 4 & 5 & 2 & 24 & 4 \end{pmatrix}_{b_0}$$

[15] $2^{\overset{1}{1}\overset{1111}{3}} 3^{\overset{1}{1}} 15 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -190, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 5 & 3 & 3 \\ 3 & 2 & 6 & 6 & 4 & 4 \\ 4 & 6 & 5 & 8 & 5 & 5 \\ 5 & 6 & 8 & 8 & 6 & 6 \\ 3 & 4 & 5 & 6 & 2 & 4 \\ 3 & 4 & 5 & 6 & 4 & 4 \end{pmatrix}_{b_0}$$

[16] $2^{\overset{1}{1}\overset{117}{3}} 3^{\overset{1}{1}} 15 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -204, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 8.3$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 21 & 3 \\ 3 & 2 & 6 & 2 & 28 & 4 \\ 4 & 6 & 5 & 3 & 35 & 5 \\ 2 & 2 & 3 & 0 & 14 & 2 \\ 21 & 28 & 35 & 14 & 194 & 28 \\ 3 & 4 & 5 & 2 & 28 & 4 \end{pmatrix}_{b_0}$$

[17] $22^{\overset{1}{1}\overset{1111}{3}} 3^{\overset{1}{1}} 15 1^{\overset{1}{1}} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -696, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 6 & 5 & 5 & 3 \\ 5 & 9 & 14 & 11 & 11 & 7 \\ 6 & 14 & 14 & 13 & 13 & 8 \\ 5 & 11 & 13 & 8 & 11 & 6 \\ 5 & 11 & 13 & 11 & 9 & 7 \\ 3 & 7 & 8 & 6 & 7 & 4 \end{pmatrix}_{b_0}$$

6.161 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2$

$$T = 9 \quad [0] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -8, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 3 & 4 \\ 2 & 4 & 1 & 1 & 3 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 \\ 3 & 4 & 4 & 0 & 3 & 3 & 4 \\ 2 & 4 & 1 & 1 & 3 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 3 & 0 & 1 & 3 & 3 \\ 2 & 3 & 3 & 0 & 3 & 1 & 3 \\ 3 & 4 & 3 & -1 & 3 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 3^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 + \mathfrak{su}_2 \quad T = 9, T_{\max} = 10, \Delta = -33, \det = -16, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 10, T_{\max} = 10, T \in [10, 10]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 1 & 2 & 3 \\ 3 & 4 & 4 & 0 & 0 & 3 & 4 \\ 2 & 4 & 1 & 1 & 0 & 3 & 3 \\ 1 & 0 & 1 & -3 & 0 & 0 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 2 & 3 & 3 & 0 & 0 & 1 & 3 \\ 3 & 4 & 3 & -1 & 0 & 3 & 3 \end{pmatrix}_{b_0}$$

6.162 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3)$

$$T = 11 \quad [0] \quad 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 2 & 3 \\ 2 & 2 & 4 & 4 & 3 & 4 \\ 3 & 4 & 4 & 3 & 3 & 4 \\ 3 & 4 & 3 & 1 & 3 & 3 \\ 2 & 3 & 3 & 3 & 1 & 3 \\ 3 & 4 & 4 & 3 & 3 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -11, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.91 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 3 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{11}} 5^{\overline{11}} 13^{\overline{11}} 1^{\overline{11}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -19, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 3 \\ 2 & 1 & 4 & 0 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 4 & 0 & 1 & 3 \\ 3 & 3 & 5 & -1 & 3 & 4 \end{pmatrix}_{b_0}$$

6.163 $\text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2$

$$T = 12 \quad [7] \quad 1^{\overline{13}} 5^{\overline{13}} 13^{\overline{14}} 1^{\overline{14}} + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{su}_2 \rightarrow (-3) + \mathfrak{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 14 & 3 \\ 2 & 1 & 4 & 0 & 2 & 17 & 5 \\ 3 & 4 & 5 & 0 & 4 & 29 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 2 & 4 & 0 & 1 & 18 & 3 \\ 14 & 17 & 29 & 0 & 18 & 137 & 25 \\ 3 & 3 & 5 & -1 & 3 & 25 & 4 \end{pmatrix}_{b_0}$$

$$\begin{aligned}
[4] \quad & 1^{11}_{1^1} 5^{11} 13^1 1^{111} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 3 & 2 & 4 & 3 \\ 2 & 2 & 4 & 4 & 3 & 6 & 4 \\ 3 & 4 & 4 & 3 & 3 & 7 & 4 \\ 3 & 4 & 3 & 1 & 3 & 6 & 3 \\ 2 & 3 & 3 & 3 & 1 & 6 & 3 \\ 4 & 6 & 7 & 6 & 5 & 9 & 7 \\ 3 & 4 & 4 & 3 & 3 & 7 & 4 \end{pmatrix}_{b_0} \\
[5] \quad & 1^{12}_{1^1} 5^{13} 13^1 1^{112} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 3 & 2 & 8 & 3 \\ 2 & 2 & 4 & 4 & 3 & 12 & 4 \\ 3 & 4 & 4 & 3 & 3 & 14 & 4 \\ 3 & 4 & 3 & 1 & 3 & 12 & 3 \\ 2 & 3 & 3 & 3 & 1 & 10 & 3 \\ 8 & 12 & 14 & 12 & 10 & 42 & 14 \\ 3 & 4 & 4 & 3 & 3 & 14 & 4 \end{pmatrix}_{b_0} \\
[6] \quad & 1^{13}_{1^1} 5^{13} 13^1 1^{113} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 3 & 2 & 12 & 3 \\ 2 & 2 & 4 & 4 & 3 & 18 & 4 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \\ 3 & 4 & 3 & 1 & 3 & 18 & 3 \\ 2 & 3 & 3 & 3 & 1 & 15 & 3 \\ 12 & 18 & 21 & 18 & 15 & 97 & 21 \\ 3 & 4 & 4 & 3 & 3 & 21 & 4 \end{pmatrix}_{b_0} \\
[8] \quad & 1^{11}_{1^1} 5^{13} 13^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 4, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 2 & 3 \\ 2 & 1 & 4 & 0 & 2 & 3 & 3 \\ 3 & 4 & 5 & 0 & 4 & 3 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 2 & 4 & 0 & 1 & 2 & 3 \\ 2 & 3 & 3 & 0 & 2 & 1 & 3 \\ 3 & 3 & 5 & -1 & 3 & 3 & 4 \end{pmatrix}_{b_0} \\
[9] \quad & 1^{12}_{1^1} 5^{13} 13^1 1^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 6 & 3 \\ 2 & 1 & 4 & 0 & 2 & 8 & 3 \\ 3 & 4 & 5 & 0 & 4 & 11 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 2 & 4 & 0 & 1 & 7 & 3 \\ 6 & 8 & 11 & 0 & 7 & 22 & 10 \\ 3 & 3 & 5 & -1 & 3 & 10 & 4 \end{pmatrix}_{b_0} \\
[10] \quad & 1^{11}_{1^{11}} 5^{13} 13^1 1^1 + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 9, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.0 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 1 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 1 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 2 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 1 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 1 & 2 \\ 1 & 1 & 2 & 1 & 0 & 1 & 0 & 2 \\ 2 & 2 & 2 & 2 & -1 & 2 & 2 & 3 \end{pmatrix}_{b_0} \\
[11] \quad & 1^{13}_{1^1} 5^{13} 13^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 16, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.8 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12] \\
& \begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 10 & 3 \\ 2 & 1 & 4 & 0 & 2 & 13 & 3 \\ 3 & 4 & 5 & 0 & 4 & 19 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 2 & 4 & 0 & 1 & 12 & 3 \\ 10 & 13 & 19 & 0 & 12 & 65 & 17 \\ 3 & 3 & 5 & -1 & 3 & 17 & 4 \end{pmatrix}_{b_0} \\
[12] \quad & 1^{12}_{1^{11}} 5^{13} 13^1 1^{12} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 25, (n_+, n_-, n_0) = \\
& (1, 12, 0), c_{\text{ext}} = 8.5 \\
& T_{\min} = 12, T_{\max} = 12, T \in [12, 12]
\end{aligned}$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 7 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 9 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 7 & 2 \\ 5 & 7 & 6 & 9 & 0 & 7 & 36 & 12 \\ 2 & 2 & 2 & 2 & -1 & 2 & 12 & 3 \end{pmatrix}_{b_0}$$

$$[14] \quad 1^{11} \overset{1}{5} 13^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 3 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 4 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 4 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 5 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 4 & 2 \\ 3 & 4 & 4 & 5 & 0 & 4 & 11 & 7 \\ 2 & 2 & 2 & 2 & -1 & 2 & 7 & 3 \end{pmatrix}_{b_0}$$

$$[13] \quad 1^{12} \overset{1}{5} 13^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 38, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 6 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 5 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 5 & 2 \\ 4 & 6 & 5 & 6 & 0 & 5 & 21 & 9 \\ 2 & 2 & 2 & 2 & -1 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

$$[15] \quad 1^{11} \overset{1}{5} 13^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 39, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 4 & 3 \\ 2 & 1 & 4 & 0 & 2 & 5 & 3 \\ 3 & 4 & 5 & 0 & 4 & 8 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 2 & 4 & 0 & 1 & 5 & 3 \\ 4 & 5 & 8 & 0 & 5 & 9 & 7 \\ 3 & 3 & 5 & -1 & 3 & 7 & 4 \end{pmatrix}_{b_0}$$

$$[16] \quad 1^{12} \overset{1}{5} 13^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 42, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 11.95 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 8 & 3 \\ 2 & 1 & 4 & 0 & 2 & 10 & 3 \\ 3 & 4 & 5 & 0 & 4 & 16 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 2 & 4 & 0 & 1 & 10 & 3 \\ 8 & 10 & 16 & 0 & 10 & 42 & 14 \\ 3 & 3 & 5 & -1 & 3 & 14 & 4 \end{pmatrix}_{b_0}$$

$$[17] \quad 1^{13} \overset{1}{5} 13^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 2 & 12 & 3 \\ 2 & 1 & 4 & 0 & 2 & 15 & 3 \\ 3 & 4 & 5 & 0 & 4 & 24 & 5 \\ 1 & 0 & 0 & -3 & 0 & 0 & -1 \\ 2 & 2 & 4 & 0 & 1 & 15 & 3 \\ 12 & 15 & 24 & 0 & 15 & 97 & 21 \\ 3 & 3 & 5 & -1 & 3 & 21 & 4 \end{pmatrix}_{b_0}$$

$$[18] \quad 1^{13} \overset{1}{5} 13^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 47, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$$

$$T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 9 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 10 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 8 & 2 \\ 6 & 9 & 7 & 10 & 0 & 8 & 52 & 14 \\ 2 & 2 & 2 & 2 & -1 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

$$[19] \quad 1^{12} \overset{1}{5} 13^{11} + [(-2) \cdot \mathbf{sp}_1 \cdot \mathbf{g}_2] + \mathbf{su}_3 + \mathbf{su}_2 \rightarrow (-3) + \mathbf{su}_2 \quad T = 12, T_{\max} = 12, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 7 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 7 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 7 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 6 & 2 \\ 5 & 7 & 7 & 7 & 0 & 6 & 33 & 11 \\ 2 & 2 & 2 & 2 & -1 & 2 & 11 & 3 \end{pmatrix}_{b_0}$$

[20] $1^{11}_{112} \overset{1}{5} 13^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 49, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.0$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 5 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 6 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 6 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 5 & 2 \\ 4 & 5 & 6 & 6 & 0 & 5 & 21 & 9 \\ 2 & 2 & 2 & 2 & -1 & 2 & 9 & 3 \end{pmatrix}_{b_0}$$

[21] $1^{13}_{112} \overset{1}{5} 13^1 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 81, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 11 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 14 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 11 & 2 \\ 8 & 11 & 10 & 14 & 0 & 11 & 93 & 19 \\ 2 & 2 & 2 & 2 & -1 & 2 & 19 & 3 \end{pmatrix}_{b_0}$$

[22] $1^{12}_{112} \overset{1}{5} 13^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 86, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 6 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 8 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 8 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 10 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 8 & 2 \\ 6 & 8 & 8 & 10 & 0 & 8 & 50 & 14 \\ 2 & 2 & 2 & 2 & -1 & 2 & 14 & 3 \end{pmatrix}_{b_0}$$

[23] $1^{13}_{112} \overset{1}{5} 13^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 108, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.96 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 10 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 9 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 11 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 9 & 2 \\ 7 & 10 & 9 & 11 & 0 & 9 & 68 & 16 \\ 2 & 2 & 2 & 2 & -1 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[24] $1^{12}_{113} \overset{1}{5} 13^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 121, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.5$
 $T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 9 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 10 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 11 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 9 & 2 \\ 7 & 9 & 10 & 11 & 0 & 9 & 68 & 16 \\ 2 & 2 & 2 & 2 & -1 & 2 & 16 & 3 \end{pmatrix}_{b_0}$$

[25] $1^{13}_{113} \overset{1}{5} 13^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 143, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 11 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 11 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 12 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 10 & 2 \\ 8 & 11 & 11 & 12 & 0 & 10 & 88 & 18 \\ 2 & 2 & 2 & 2 & -1 & 2 & 18 & 3 \end{pmatrix}_{b_0}$$

[26] $1^{13}_{113} \overset{1}{5} 13^1 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \rightarrow (-3) + \text{su}_2$ $T = 12, T_{\max} = 12, \Delta = 0, \det = 166, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 8.8$
 $T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 1 & 12 & 2 \\ 1 & 2 & 0 & 2 & 0 & 1 & 12 & 2 \\ 1 & 1 & 2 & 0 & 0 & 2 & 15 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & 0 & -1 \\ 1 & 1 & 1 & 2 & 0 & 0 & 12 & 2 \\ 9 & 12 & 12 & 15 & 0 & 12 & 115 & 21 \\ 2 & 2 & 2 & 2 & -1 & 2 & 21 & 3 \end{pmatrix}_{b_0}$$

6.164 NHC(-2)(-2)(-3) + su₃ + su₂n₃mix

$$T = 11 \quad [0] \quad 1^{12} \overset{1}{5} 13 \overset{1}{1} 1^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 10 & 3 \\ 2 & 2 & 4 & 4 & 15 & 4 \\ 3 & 4 & 4 & 3 & 17 & 4 \\ 3 & 4 & 3 & 1 & 15 & 3 \\ 10 & 15 & 17 & 15 & 65 & 17 \\ 3 & 4 & 4 & 3 & 17 & 4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{11} \overset{1}{5} 13 \overset{1}{1} 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 6 & 3 \\ 2 & 2 & 4 & 4 & 9 & 4 \\ 3 & 4 & 4 & 3 & 10 & 4 \\ 3 & 4 & 3 & 1 & 9 & 3 \\ 6 & 9 & 10 & 9 & 22 & 10 \\ 3 & 4 & 4 & 3 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{13} \overset{1}{5} 13 \overset{1}{1} 1^{113} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -2, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 14 & 3 \\ 2 & 2 & 4 & 4 & 21 & 4 \\ 3 & 4 & 4 & 3 & 24 & 4 \\ 3 & 4 & 3 & 1 & 21 & 3 \\ 14 & 21 & 24 & 21 & 130 & 24 \\ 3 & 4 & 4 & 3 & 24 & 4 \end{pmatrix}_{b_0}$$

$$[5] \quad 1^{11} \overset{1}{5} 13 \overset{1}{1} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 4 & 3 \\ 2 & 1 & 4 & 2 & 5 & 3 \\ 3 & 4 & 5 & 5 & 7 & 5 \\ 3 & 2 & 5 & 1 & 5 & 3 \\ 4 & 5 & 7 & 5 & 8 & 6 \\ 3 & 3 & 5 & 3 & 6 & 4 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{12} \overset{1}{5} 13 \overset{1}{1} 1^{111} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 8 & 3 \\ 2 & 1 & 4 & 2 & 10 & 3 \\ 3 & 4 & 5 & 5 & 15 & 5 \\ 3 & 2 & 5 & 1 & 11 & 3 \\ 8 & 10 & 15 & 11 & 39 & 13 \\ 3 & 3 & 5 & 3 & 13 & 4 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{13} \overset{1}{5} 13 \overset{1}{1} 1^{112} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 3 & 12 & 3 \\ 2 & 1 & 4 & 2 & 15 & 3 \\ 3 & 4 & 5 & 5 & 23 & 5 \\ 3 & 2 & 5 & 1 & 17 & 3 \\ 12 & 15 & 23 & 17 & 92 & 20 \\ 3 & 3 & 5 & 3 & 20 & 4 \end{pmatrix}_{b_0}$$

$$[6] \quad 1^{11} \overset{1}{5} 13 \overset{1}{1} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -18, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 2 & 1 & 0 & 3 & 2 \\ 1 & 2 & 0 & 2 & 0 & 2 & 2 \\ 1 & 1 & 2 & 0 & 0 & 3 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 3 & 2 & 3 & 0 & 4 & 4 \\ 2 & 2 & 2 & 2 & -1 & 4 & 3 \end{pmatrix}_{b_0}$$

$$[7] \quad 1^{11} \overset{1}{5} 13 \overset{1}{1} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -31, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 6 & 3 \\ 2 & 1 & 4 & 0 & 7 & 3 \\ 3 & 4 & 5 & 0 & 12 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 6 & 7 & 12 & 0 & 22 & 10 \\ 3 & 3 & 5 & -1 & 10 & 4 \end{pmatrix}_{b_0}$$

$$[8] \quad 1^{11} \overset{1}{5} 13 \overset{1}{1} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2 \quad T = 11, T_{\max} = 12, \Delta = -30, \det = -32, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 4 & 2 \\ 1 & 0 & 2 & 1 & 0 & 5 & 2 \\ 1 & 2 & 0 & 2 & 0 & 5 & 2 \\ 1 & 1 & 2 & 0 & 0 & 7 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 4 & 5 & 5 & 7 & 0 & 21 & 9 \\ 2 & 2 & 2 & 2 & -1 & 9 & 3 \end{pmatrix}_{b_0}$$

[9] $1^{12}_{1^1} 5^{11} 13^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -47$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 10 & 3 \\ 2 & 1 & 4 & 0 & 12 & 3 \\ 3 & 4 & 5 & 0 & 20 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 10 & 12 & 20 & 0 & 65 & 17 \\ 3 & 3 & 5 & -1 & 17 & 4 \end{pmatrix}_{b_0}$$

[10] $1^{13}_{1^{11}} 5^{11} 13^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -50$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 10 & 2 \\ 1 & 2 & 0 & 2 & 0 & 8 & 2 \\ 1 & 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 10 & 8 & 12 & 0 & 70 & 16 \\ 2 & 2 & 2 & 2 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[12] $1^{12}_{1^{11}} 5^{11} 13^1 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -67$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 5 & 2 \\ 1 & 0 & 2 & 1 & 0 & 7 & 2 \\ 1 & 2 & 0 & 2 & 0 & 6 & 2 \\ 1 & 1 & 2 & 0 & 0 & 8 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 5 & 7 & 6 & 8 & 0 & 33 & 11 \\ 2 & 2 & 2 & 2 & -1 & 11 & 3 \end{pmatrix}_{b_0}$$

[11] $1^{13}_{1^1} 5^{11} 13^1 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -67$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 2 & 3 & 1 & 14 & 3 \\ 2 & 1 & 4 & 0 & 17 & 3 \\ 3 & 4 & 5 & 0 & 28 & 5 \\ 1 & 0 & 0 & -3 & 0 & -1 \\ 14 & 17 & 28 & 0 & 130 & 24 \\ 3 & 3 & 5 & -1 & 24 & 4 \end{pmatrix}_{b_0}$$

[13] $1^{13}_{1^{12}} 5^{11} 13^1 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -72$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 12 & 2 \\ 1 & 2 & 0 & 2 & 0 & 11 & 2 \\ 1 & 1 & 2 & 0 & 0 & 16 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 9 & 12 & 11 & 16 & 0 & 117 & 21 \\ 2 & 2 & 2 & 2 & -1 & 21 & 3 \end{pmatrix}_{b_0}$$

[14] $1^{12}_{1^{12}} 5^{11} 13^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -91$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 11.99 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 7 & 2 \\ 1 & 0 & 2 & 1 & 0 & 9 & 2 \\ 1 & 2 & 0 & 2 & 0 & 9 & 2 \\ 1 & 1 & 2 & 0 & 0 & 12 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 7 & 9 & 9 & 12 & 0 & 68 & 16 \\ 2 & 2 & 2 & 2 & -1 & 16 & 3 \end{pmatrix}_{b_0}$$

[15] $1^{13}_{1^{12}} 5^{11} 13^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -154$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 8 & 2 \\ 1 & 0 & 2 & 1 & 0 & 11 & 2 \\ 1 & 2 & 0 & 2 & 0 & 10 & 2 \\ 1 & 1 & 2 & 0 & 0 & 13 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 8 & 11 & 10 & 13 & 0 & 88 & 18 \\ 2 & 2 & 2 & 2 & -1 & 18 & 3 \end{pmatrix}_{b_0}$$

[16] $1^{13}_{1^{13}} 5^{11} 13^1 1^{13} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11$, $T_{\max} = 12$, $\Delta = -30$, $\det = -188$, $(n_+, n_-, n_0) = (1, 11, 0)$, $c_{\text{ext}} = 7.0$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 10 & 2 \\ 1 & 0 & 2 & 1 & 0 & 13 & 2 \\ 1 & 2 & 0 & 2 & 0 & 13 & 2 \\ 1 & 1 & 2 & 0 & 0 & 17 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 10 & 13 & 13 & 17 & 0 & 141 & 23 \\ 2 & 2 & 2 & 2 & -1 & 23 & 3 \end{pmatrix}_{b_0}$$

[17] $1^{13} \overset{1}{5} 13^1 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -30, \det = -242, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 9 & 2 \\ 1 & 0 & 2 & 1 & 0 & 12 & 2 \\ 1 & 2 & 0 & 2 & 0 & 12 & 2 \\ 1 & 1 & 2 & 0 & 0 & 14 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 9 & 12 & 12 & 14 & 0 & 110 & 20 \\ 2 & 2 & 2 & 2 & -1 & 20 & 3 \end{pmatrix}_{b_0}$$

6.165 NHC(-2)(-2)(-3) + su₃ + su₃

$T = 10$ [0] $21^1 \overset{1}{5} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 4, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 4 & 3 & 5 & 3 & 3 & 4 \\ 3 & 1 & 5 & 2 & 2 & 3 \\ 5 & 5 & 4 & 3 & 3 & 4 \\ 3 & 2 & 3 & 0 & 2 & 2 \\ 3 & 2 & 3 & 2 & 0 & 2 \\ 4 & 3 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

[1] $1^1 \overset{1}{5} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 40, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 11.90 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 2 & 3 & 2 & 1 & 3 \\ 2 & 1 & 4 & 2 & 0 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 3 & 4 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[2] $21^1 \overset{1}{5} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + \text{su}_3$ $T = 10, T_{\max} = 12, \Delta = -58, \det = 116, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 11.93 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 4 & 2 & 2 & 3 \\ 3 & 1 & 5 & 2 & 2 & 3 \\ 4 & 5 & 4 & 3 & 3 & 4 \\ 2 & 2 & 3 & 0 & 2 & 2 \\ 2 & 2 & 3 & 2 & 0 & 2 \\ 3 & 3 & 4 & 2 & 2 & 3 \end{pmatrix}_{b_0}$$

6.166 NHC(-2)(-2)(-3) + su₃ + (-3)→(-2)

$T = 10$ [0] $1^1 \overset{1}{5} 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 61, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 3 \\ 3 & 1 & 4 & 2 & 0 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 \\ 2 & 2 & 3 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 3 & 4 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

[1] $1^1 \overset{2^1 2^1}{5} 1^{11} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2)$ $T = 10, T_{\max} = 12, \Delta = -59, \det = 413, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 8.0$

$T_{\min} = 12.00 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 5 & 6 & 2 & 1 & 3 \\ 5 & 8 & 13 & 4 & 0 & 6 \\ 6 & 13 & 13 & 5 & 0 & 7 \\ 2 & 4 & 5 & 0 & 0 & 2 \\ 1 & 0 & 0 & 0 & -3 & -1 \\ 3 & 6 & 7 & 2 & -1 & 3 \end{pmatrix}_{b_0}$$

6.167 NHC(-2)(-2)(-3) + su₃ + (-3)→(-2) + su₂

$T = 11$ [0] $1^1 \overset{1}{5} 1^{12} + [(-2) \cdot \text{sp}_1 \cdot \text{g}_2] + \text{su}_3 + (-3) \rightarrow (-2) + \text{su}_2$ $T = 11, T_{\max} = 12, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$

$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$

$$\begin{pmatrix} 1 & 3 & 3 & 2 & 1 & 2 & 3 \\ 3 & 1 & 4 & 2 & 0 & 2 & 3 \\ 3 & 4 & 4 & 3 & 0 & 4 & 1 \\ 2 & 2 & 3 & 0 & 0 & 2 & 2 \\ 1 & 0 & 0 & 0 & -3 & 0 & -1 \\ 2 & 2 & 4 & 2 & 0 & 1 & 3 \\ 3 & 3 & 4 & 2 & -1 & 3 & 3 \end{pmatrix}_{b_0}$$

6.168 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-2)(-3)$

$$T = 9 \quad [0] \ 1^{\overline{111}} 3^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -3, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 16.33 \rightarrow 17, T_{\max} = 24, T \in [17, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 7 & 3 & 4 \\ 3 & 5 & 0 & 7 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 4 & 4 & 2 & -4 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -129, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.03 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 4 & 1 & 4 & 3 \\ 3 & 5 & 0 & 3 & 4 & 3 & 3 \\ 3 & 4 & -4 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -130, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.20 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 4 & 1 & 4 & 3 \\ 2 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 4 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[3] \ 1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -234, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.40 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 4 & 1 & 4 & 3 \\ 2 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -275, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 24, T \in [13, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 24, \Delta = -113, \det = -322, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.18 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 1 & 4 & 2 & 3 & 3 & 1 & 3 \\ 4 & 9 & 4 & 8 & 8 & 0 & 5 \\ 2 & 4 & -3 & 3 & 3 & 0 & -1 \\ 3 & 8 & 3 & 4 & 7 & 0 & 4 \\ 3 & 8 & 3 & 7 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & -1 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$T = 10 \quad [6] \ 1^{\overline{111}} 3^{\overline{21}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -85, \det = 216, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 12.98 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 21 & 27 & 1 & 3 & 3 & 9 \\ 27 & 33 & 1 & 4 & 2 & 11 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 3 & 3 & 4 & 2 & 0 & 13 \\ 9 & 11 & -1 & 1 & 3 & 3 \end{pmatrix}_{b_0}$$

$$[7] \ 1^{\overline{11}} 3^{\overline{111}} 3^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_6 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 428, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.16 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 4 & 0 & 6 & 6 & 3 & 4 \\ 1 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 6 & 3 & 0 & 4 & 7 & 3 & 4 \\ 2 & 6 & 3 & 0 & 7 & 4 & 4 & 4 \\ 1 & 3 & 3 & 0 & 3 & 4 & 0 & 2 \\ 2 & 4 & 2 & -4 & 4 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [8] \quad 1^{11} 1^{11} 3^{11} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -41, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 3 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 3 & 4 \\ 3 & 3 & 0 & 4 & 1 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[9] \quad 1^{11} 1^{11} 3^{11} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -45, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.84 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 & 2 & 3 \\ 4 & 4 & 0 & 4 & 5 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 4 & 0 & 1 & 4 & 2 & 3 \\ 3 & 5 & 0 & 4 & 4 & 4 & 4 \\ 2 & 3 & 0 & 2 & 4 & 0 & 2 \\ 3 & 4 & -3 & 3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$[10] \quad 1^{11} 1^{11} 3^{11} 4^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -215, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 13.0$$

$$T_{\min} = 13.97 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 4 & 1 & 3 & 3 & 2 & 3 \\ 4 & 4 & 0 & 5 & 4 & 3 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 3 & 5 & 0 & 4 & 4 & 3 & 4 \\ 3 & 4 & 0 & 4 & 1 & 3 & 3 \\ 2 & 3 & 0 & 3 & 3 & 0 & 2 \\ 3 & 4 & -3 & 4 & 3 & 2 & 1 \end{pmatrix}_{b_0}$$

6.169 $\text{NHC}(-2)(-3) + \textcolor{green}{e}_6 + \text{NHC}(-2)(-2)(-3) + \textcolor{blue}{su}_2$

$$T = 10 \quad [0] \quad 1^{111} 1^{11} 3^{111} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 3 \\ 3 & 5 & 0 & 3 & 6 & 3 & 4 \\ 3 & 4 & -4 & 3 & 4 & 2 & 4 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 1^{11} 3^{111} 1^{112} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 8 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 14 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 9 & 3 \\ 3 & 5 & 0 & 4 & 4 & 4 & 15 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 9 & 2 \\ 8 & 14 & 0 & 9 & 15 & 9 & 37 & 11 \\ 3 & 4 & -4 & 3 & 4 & 2 & 11 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 1^{11} 3^{111} 1^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 294, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 & 3 \\ 3 & 5 & 0 & 4 & 4 & 4 & 3 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 3 & 2 \\ 2 & 4 & 0 & 3 & 3 & 1 & 3 & 3 \\ 3 & 4 & -4 & 3 & 4 & 2 & 3 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{111} 1^{11} 3^{111} 1^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] + \textcolor{blue}{su}_2 \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 721, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 5 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 9 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 6 & 3 \\ 3 & 5 & 0 & 4 & 4 & 4 & 9 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 6 & 2 \\ 5 & 9 & 0 & 6 & 9 & 6 & 13 & 7 \\ 3 & 4 & -4 & 3 & 4 & 2 & 7 & 0 \end{pmatrix}_{b_0}$$

6.170 $\text{NHC}(-2)(-3) + \mathfrak{e}_6 + \text{NHC}(-2)(-2)(-3) + \mathfrak{su}_3$

$$T = 10 \quad [0] \ 1^{\overline{1111}} 3^1 1^{\overline{11}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{c}_6 + [(-2) \cdot \textcolor{red}{s}\textcolor{blue}{p}_1 \cdot \textcolor{blue}{g}_2] + \textcolor{red}{s}\textcolor{blue}{u}_3 \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 266, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 & -4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 & 3 \\ 2 & 4 & 0 & 4 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 3 & 4 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 3 & 3 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 3 & 2 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3 1^{111} + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + [(-2) \cdot \textcolor{red}{s} \textcolor{blue}{p}_1 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{s} \textcolor{violet}{u}_3 \quad T = 10, \quad T_{\max} = 14, \quad \Delta = -92, \quad \det = 326, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 4 & 4 & 4 & 6 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 3 & 2 \\ 3 & 5 & 0 & 3 & 6 & 3 & 3 & 3 \\ 3 & 4 & -4 & 3 & 4 & 2 & 3 & -1 \end{pmatrix}_{b_0}$$

6.171 $\mathbf{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_2 + \mathbf{NHC}(-2)(-2)(-3)$

$$T = 10 \quad [0] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{blue}{1}} 3^{\textcolor{blue}{1}} 1^{\textcolor{blue}{1}\textcolor{blue}{1}} + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{brown}{s}\textcolor{blue}{u}_2 + [(-2) \cdot \textcolor{blue}{s}\textcolor{blue}{p}_1 \cdot \textcolor{blue}{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 4 & 3 & 6 & 3 & 4 \\ 2 & 4 & 0 & 3 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 6 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 4 & 3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{1111} 3^1 1^{21} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{c}_6 + \textcolor{brown}{s}u_2 + [(-2) \cdot \textcolor{blue}{s}p_1 \cdot \textcolor{blue}{g}_2] \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -83, \quad \det = 150, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 8 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 14 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 8 & 14 & 0 & 37 & 9 & 15 & 9 & 11 \\ 2 & 4 & 0 & 9 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 15 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 9 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 11 & 3 & 4 & 2 & 0 \end{pmatrix} b_0$$

$$[2] \quad 1^{1111} 3 1^{11} + [\textcolor{red}{s} \textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{s} \textcolor{blue}{u}_2 + [(-2) \cdot \textcolor{blue}{s} \textcolor{red}{p}_1 \cdot \textcolor{green}{g}_2] \quad T = 10, \, T_{\max} = 13, \, \Delta = -83, \, \det = 294, \, (n_+, n_-, n_0) = (1, 10, 0), \, c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 1 & 3 & 3 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 3 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 4 & 2 & 0 \end{pmatrix} b_0$$

$$[3] \quad 1^{1111} 3 1^{111} + [\textcolor{red}{s}u_2 \cdot \textcolor{blue}{g}_2] + \textcolor{violet}{c}_6 + \textcolor{brown}{s}u_2 + [(-2) \cdot \textcolor{blue}{s}p_1 \cdot \textcolor{violet}{g}_2] \quad T = 10, \quad T_{\max} = 13, \quad \Delta = -83, \quad \det = 721, \quad (n_+, n_-, n_0) = (1, 10, 0), \quad c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 1 & 5 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 9 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 5 & 9 & 0 & 13 & 6 & 9 & 6 & 7 \\ 2 & 4 & 0 & 6 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 9 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 6 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 7 & 3 & 4 & 2 & 0 \end{pmatrix} b_0$$

6.172 $\mathbf{NHC}(-2)(-3) + \mathfrak{e}_6 + \mathfrak{su}_3 + \mathbf{NHC}(-2)(-2)(-3)$

$$T = 10 \quad [0] \ 1^{\textcolor{red}{1}\textcolor{blue}{1}\textcolor{teal}{1}\textcolor{brown}{1}} 3^{\textcolor{red}{1}} 1 + [\textcolor{red}{s}\textcolor{blue}{u}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{teal}{c}_6 + \textcolor{brown}{s}\textcolor{blue}{u}_3 + [(-2) \cdot \textcolor{red}{s}\textcolor{blue}{p}_1 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 14, \ \Delta = -92, \ \det = 266, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 2 & 2 & 3 & 3 \\ 3 & 4 & 0 & 4 & 4 & 4 & 5 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 2 & 4 & 0 & 3 & 1 & 4 & 3 & 3 \\ 2 & 4 & 0 & 3 & 4 & 1 & 4 & 3 \\ 3 & 5 & 0 & 3 & 3 & 4 & 3 & 3 \\ 3 & 4 & -4 & 2 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^{111} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_6 + \textcolor{orange}{su}_3 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 14, \ \Delta = -92, \ \det = 326, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.45 \rightarrow 14, \ T_{\max} = 14, \ T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 1 & 3 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 5 & 4 & 5 & 4 & 4 \\ 1 & 0 & -6 & 0 & 0 & 0 & 0 & -4 \\ 3 & 5 & 0 & 3 & 3 & 6 & 3 & 3 \\ 2 & 4 & 0 & 3 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 6 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 3 & 4 & 0 & 2 \\ 3 & 4 & -4 & 3 & 3 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

6.173 NHC(-2)(-3) + $\textcolor{green}{e}_7$ + NHC(-2)(-2)(-3)

$$T = 10 \quad [0] \ 1^{11} 3^{111} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 100, \ \Delta = -140, \ \det = 6, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 16.33 \rightarrow 17, \ T_{\max} = 100, \ T \in [17, 100]$$

$$\begin{pmatrix} 14 & 13 & 1 & 7 & 8 \\ 13 & 12 & 1 & 7 & 8 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 7 & 0 & -2 & 0 \\ 8 & 8 & -1 & 0 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{111} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 100, \ \Delta = -140, \ \det = 159, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.07 \rightarrow 16, \ T_{\max} = 100, \ T \in [16, 100]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -4 & 0 & 4 & -2 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 4 & 8 & 0 & 4 & 1 & 9 & 5 \\ 3 & 4 & -1 & -2 & 0 & 5 & -2 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{111} 3^{11} 21^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] \quad T = 10, \ T_{\max} = 100, \ \Delta = -140, \ \det = 334, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.74 \rightarrow 15, \ T_{\max} = 100, \ T \in [15, 100]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 9 & 5 \\ 7 & 8 & 1 & 5 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -3 & 6 & -1 \\ 9 & 12 & 0 & 6 & 13 & 7 \\ 5 & 6 & -1 & -1 & 7 & 0 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [3] \ 1^{11} 3^{111} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 24, \ \Delta = -111, \ \det = -3, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 18.33 \rightarrow 19, \ T_{\max} = 24, \ T \in [19, 24]$$

$$\begin{pmatrix} 63 & 62 & 1 & 15 \\ 62 & 61 & 1 & 15 \\ 1 & 1 & -3 & -1 \\ 15 & 15 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[4] \ 1^{111} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 24, \ \Delta = -111, \ \det = -130, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.20 \rightarrow 16, \ T_{\max} = 24, \ T \in [16, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 4 & -1 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 4 & 8 & 0 & 4 & 1 & 9 & 5 \\ 3 & 4 & -1 & -1 & 0 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[5] \ 1^{111} 3^{11} 221^1 + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 24, \ \Delta = -111, \ \det = -275, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.83 \rightarrow 15, \ T_{\max} = 24, \ T \in [15, 24]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 9 & 5 \\ 7 & 8 & 1 & 5 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 6 & 0 \\ 9 & 12 & 0 & 6 & 13 & 7 \\ 5 & 6 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

$$[6] \ 1^{1231} 1^{111} 3^{11} + [\textcolor{blue}{su}_2 \cdot \textcolor{blue}{g}_2] + \textcolor{green}{e}_7 + [(-2) \cdot \textcolor{blue}{sp}_1 \cdot \textcolor{blue}{g}_2] \quad T = 11, \ T_{\max} = 48, \ \Delta = -119, \ \det = -525, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 15.05 \rightarrow 16, \ T_{\max} = 48, \ T \in [16, 48]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -4 & 3 & 3 & 0 & -2 \\ 2 & 6 & 0 & 3 & 4 & 7 & 0 & 4 \\ 2 & 6 & 0 & 3 & 7 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 0 & 1 & -3 & -1 \\ 2 & 4 & -1 & -2 & 4 & 4 & -1 & -2 \end{pmatrix}_{b_0}$$

6.174 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \text{NHC}(-2)(-2)(-3) + \mathfrak{su}_2$

$$T = 11 \quad [0] \ 1^{111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 11, \ T_{\max} = 23, \ \Delta = -110, \ \det = -62, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.84 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 9 & 1 & 5 \\ 7 & 8 & 1 & 5 & 12 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & -3 & 6 & 0 & -1 \\ 9 & 12 & 0 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 5 & 6 & -1 & -1 & 7 & 0 & 0 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, \ T_{\max} = 15, \ \Delta = -81, \ \det = 51, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.94 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 9 & 1 & 5 \\ 7 & 8 & 1 & 5 & 12 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 6 & 0 & 0 \\ 9 & 12 & 0 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 5 & 6 & -1 & 0 & 7 & 0 & 1 \end{pmatrix}_{b_0}$$

6.175 $\text{NHC}(-2)(-3) + \mathfrak{e}_7 + \mathfrak{su}_2 + \text{NHC}(-2)(-2)(-3)$

$$T = 11 \quad [0] \ 1^{111} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 11, \ T_{\max} = 23, \ \Delta = -110, \ \det = -62, \ (n_+, n_-, n_0) = (1, 11, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.84 \rightarrow 15, \ T_{\max} = 23, \ T \in [15, 23]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 1 & 9 & 5 \\ 7 & 8 & 1 & 5 & 0 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & -3 & 0 & 6 & -1 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 12 & 0 & 6 & 0 & 13 & 7 \\ 5 & 6 & -1 & -1 & 0 & 7 & 0 \end{pmatrix}_{b_0}$$

$$T = 12 \quad [1] \ 1^{111} 3 \ 221^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_7 + \mathfrak{su}_2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 12, \ T_{\max} = 15, \ \Delta = -81, \ \det = 51, \ (n_+, n_-, n_0) = (1, 12, 0), \ c_{\text{ext}} = 15.0$$

$$T_{\min} = 14.94 \rightarrow 15, \ T_{\max} = 15, \ T \in [15, 15]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 1 & 9 & 5 \\ 7 & 8 & 1 & 5 & 0 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 12 & 0 & 6 & 0 & 13 & 7 \\ 5 & 6 & -1 & 0 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

6.176 $\text{NHC}(-2)(-3) + \mathfrak{e}'_7 + \text{NHC}(-2)(-2)(-3)$

$$T = 10 \quad [0] \ 1^{11} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 3, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 17.33 \rightarrow 18, \ T_{\max} = 24, \ T \in [18, 24]$$

$$\begin{pmatrix} 63 & 62 & 1 & 15 \\ 62 & 61 & 1 & 15 \\ 1 & 1 & -3 & -1 \\ 15 & 15 & -1 & -1 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{11} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 130, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.20 \rightarrow 15, \ T_{\max} = 24, \ T \in [15, 24]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -3 & 0 & 4 & -1 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 4 & 8 & 0 & 4 & 1 & 9 & 5 \\ 3 & 4 & -1 & -1 & 0 & 5 & -1 \end{pmatrix}_{b_0}$$

$$[2] \ 1^{11} 3 \ 21^1 + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}'_7 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, \ T_{\max} = 24, \ \Delta = -112, \ \det = 275, \ (n_+, n_-, n_0) = (1, 10, 0), \ c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.83 \rightarrow 14, T_{\max} = 24, T \in [14, 24]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 9 & 5 \\ 7 & 8 & 1 & 5 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 6 & 0 \\ 9 & 12 & 0 & 6 & 13 & 7 \\ 5 & 6 & -1 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

$$T = 11 \quad [3] \quad 1^{111} 3 \quad 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + [(-2) \cdot \textcolor{red}{\mathfrak{sp}}_1 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 14, \Delta = -83, \det = -216, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 13.98 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 21 & 27 & 1 & 33 & 9 \\ 27 & 33 & 1 & 42 & 11 \\ 1 & 1 & -3 & 0 & -1 \\ 33 & 42 & 0 & 49 & 13 \\ 9 & 11 & -1 & 13 & 3 \end{pmatrix}_{b_0}$$

$$[4] \quad 1^{1231} 1^{111} 31^{11} + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + [(-2) \cdot \textcolor{red}{\mathfrak{sp}}_1 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 15, \Delta = -91, \det = -428, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 14.16 \rightarrow 15, T_{\max} = 15, T \in [15, 15]$$

$$\begin{pmatrix} 0 & 2 & 1 & 1 & 2 & 2 & 1 & 2 \\ 2 & 4 & 1 & 3 & 6 & 6 & 0 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & 0 & -1 \\ 1 & 3 & 0 & -3 & 3 & 3 & 0 & -1 \\ 2 & 6 & 0 & 3 & 4 & 7 & 0 & 4 \\ 2 & 6 & 0 & 3 & 7 & 4 & 1 & 4 \\ 1 & 0 & 0 & 0 & 0 & 1 & -3 & -1 \\ 2 & 4 & -1 & -1 & 4 & 4 & -1 & -1 \end{pmatrix}_{b_0}$$

$$\mathbf{6.177} \quad \text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{e}}'_7 + \text{NHC}(-2)(-2)(-3) + \textcolor{red}{\mathfrak{su}}_2$$

$$T = 11 \quad [0] \quad 1^{111} 3 \quad 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + [(-2) \cdot \textcolor{red}{\mathfrak{sp}}_1 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{su}}_2 \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -51, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 9 & 1 & 5 \\ 7 & 8 & 1 & 5 & 12 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 6 & 0 & 0 \\ 9 & 12 & 0 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 5 & 6 & -1 & 0 & 7 & 0 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{6.178} \quad \text{NHC}(-2)(-3) + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{red}{\mathfrak{su}}_2 + \text{NHC}(-2)(-2)(-3)$$

$$T = 11 \quad [0] \quad 1^{111} 3 \quad 21^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{green}{\mathfrak{e}}'_7 + \textcolor{red}{\mathfrak{su}}_2 + [(-2) \cdot \textcolor{red}{\mathfrak{sp}}_1 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 14, \Delta = -82, \det = -51, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.94 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 1 & 9 & 5 \\ 7 & 8 & 1 & 5 & 0 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & -2 & 0 & 6 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 12 & 0 & 6 & 0 & 13 & 7 \\ 5 & 6 & -1 & 0 & 0 & 7 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{6.179} \quad \text{NHC}(-2)(-3) + \textcolor{red}{\mathfrak{e}}_8 + \text{NHC}(-2)(-2)(-3)$$

$$T = 11 \quad [0] \quad 1^{111} 3 \quad 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_8 + [(-2) \cdot \textcolor{red}{\mathfrak{sp}}_1 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 106, \Delta = -226, \det = -15, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 19.13 \rightarrow 20, T_{\max} = 106, T \in [20, 106]$$

$$\begin{pmatrix} 14 & 13 & 1 & 7 & 8 \\ 13 & 12 & 1 & 7 & 8 \\ 1 & 1 & -3 & 0 & -1 \\ 7 & 7 & 0 & -5 & -3 \\ 8 & 8 & -1 & -3 & -3 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{111} 3 \quad 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_8 + [(-2) \cdot \textcolor{red}{\mathfrak{sp}}_1 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 106, \Delta = -226, \det = -246, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.86 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 1 & 4 & 3 \\ 3 & 4 & 1 & 3 & 0 & 8 & 4 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 2 & 3 & 0 & -7 & 0 & 4 & -5 \\ 1 & 0 & 0 & 0 & -2 & 1 & 0 \\ 4 & 8 & 0 & 4 & 1 & 9 & 5 \\ 3 & 4 & -1 & -5 & 0 & 5 & -4 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{111} 3 \quad 221^1 + [\textcolor{red}{\mathfrak{su}}_2 \cdot \textcolor{blue}{\mathfrak{g}}_2] + \textcolor{red}{\mathfrak{e}}_8 + [(-2) \cdot \textcolor{red}{\mathfrak{sp}}_1 \cdot \textcolor{blue}{\mathfrak{g}}_2] \quad T = 11, T_{\max} = 106, \Delta = -226, \det = -511, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 18.58 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 9 & 5 \\ 7 & 8 & 1 & 5 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & -1 \\ 4 & 5 & 0 & -6 & 6 & -4 \\ 9 & 12 & 0 & 6 & 13 & 7 \\ 5 & 6 & -1 & -4 & 7 & -3 \end{pmatrix}_{b_0}$$

$$\mathbf{6.180} \quad \mathbf{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathbf{NHC}(-2)(-2)(-3) + \mathfrak{su}_2$$

$$T = 12 \quad [0] \ 1^{\overline{111}} \ 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 \quad T = 12, T_{\max} = 106, \Delta = -196, \det = 95, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.67 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 9 & 1 & 5 \\ 7 & 8 & 1 & 5 & 12 & 0 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & -6 & 6 & 0 & -4 \\ 9 & 12 & 0 & 6 & 13 & 0 & 7 \\ 1 & 0 & 0 & 0 & 0 & -2 & 0 \\ 5 & 6 & -1 & -4 & 7 & 0 & -3 \end{pmatrix}_{b_0}$$

$$\mathbf{6.181} \quad \mathbf{NHC}(-2)(-3) + \mathfrak{e}_8 + \mathfrak{su}_2 + \mathbf{NHC}(-2)(-2)(-3)$$

$$T = 12 \quad [0] \ 1^{\overline{111}} \ 3 \ 221^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{e}_8 + \mathfrak{su}_2 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 12, T_{\max} = 106, \Delta = -196, \det = 95, (n_+, n_-, n_0) = (1, 12, 0), c_{\text{ext}} = 16.0$$

$$T_{\min} = 18.67 \rightarrow 19, T_{\max} = 106, T \in [19, 106]$$

$$\begin{pmatrix} 5 & 7 & 1 & 4 & 1 & 9 & 5 \\ 7 & 8 & 1 & 5 & 0 & 12 & 6 \\ 1 & 1 & -3 & 0 & 0 & 0 & -1 \\ 4 & 5 & 0 & -6 & 0 & 6 & -4 \\ 1 & 0 & 0 & 0 & -2 & 0 & 0 \\ 9 & 12 & 0 & 6 & 0 & 13 & 7 \\ 5 & 6 & -1 & -4 & 0 & 7 & -3 \end{pmatrix}_{b_0}$$

$$\mathbf{6.182} \quad \mathbf{NHC}(-2)(-3) + \mathfrak{f}_4 + \mathbf{NHC}(-2)(-2)(-3)$$

$$T = 9 \quad [0] \ 1^{\overline{111}} \ 3 \ 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{f}_4 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 9, T_{\max} = 12, \Delta = -87, \det = -216, (n_+, n_-, n_0) = (1, 9, 0), c_{\text{ext}} = 12.2$$

$$T_{\min} = 11.98 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 0 & 4 & 5 & 4 & 4 \\ 1 & 0 & -5 & 0 & 0 & 0 & -3 \\ 2 & 4 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 0 & 4 & 4 & 4 & 4 \\ 2 & 4 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & -3 & 3 & 4 & 2 & 1 \end{pmatrix}_{b_0}$$

$$\mathbf{6.183} \quad \mathbf{NHC}(-2)(-3) + \mathbf{NHC}(-2)(-2)(-3)$$

$$T = 11 \quad [0] \ 1^{\overline{1}} \ 5 \ 131^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -8, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 7.0$$

$$T_{\min} = 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 1 & 3 \\ 3 & 5 & 3 & 5 & 0 & 5 \\ 2 & 3 & 1 & 3 & 0 & 3 \\ 2 & 5 & 3 & 2 & 1 & 4 \\ 1 & 0 & 0 & 1 & -3 & -1 \\ 3 & 5 & 3 & 4 & -1 & 4 \end{pmatrix}_{b_0}$$

$$\mathbf{6.184} \quad \mathbf{NHC}(-2)(-3) + \mathfrak{su}_2 + \mathfrak{e}_6 + \mathbf{NHC}(-2)(-2)(-3)$$

$$T = 10 \quad [0] \ 1^{\overline{111}} \ 3 \ 1^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 51, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.94 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 & 4 \\ 3 & 5 & 4 & 0 & 3 & 6 & 3 & 4 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 6 & 0 & 4 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & 4 & -4 & 3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[1] \ 1^{\overline{111}} \ 3 \ 1^{\overline{121}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 150, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.92 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 8 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 14 & 0 & 4 & 5 & 4 & 4 \\ 8 & 14 & 37 & 0 & 9 & 15 & 9 & 11 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 9 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 15 & 0 & 4 & 4 & 4 & 4 \\ 2 & 4 & 9 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & 11 & -4 & 3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[2] \quad 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 294, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.84 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 4 & 0 & 4 & 5 & 4 & 4 \\ 2 & 4 & 1 & 0 & 3 & 3 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 3 & 0 & 4 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

$$[3] \quad 1^{\overline{1111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_2 + \mathfrak{e}_6 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 13, \Delta = -83, \det = 721, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 14.0$$

$$T_{\min} = 12.83 \rightarrow 13, T_{\max} = 13, T \in [13, 13]$$

$$\begin{pmatrix} 1 & 3 & 5 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 9 & 0 & 4 & 5 & 4 & 4 \\ 5 & 9 & 13 & 0 & 6 & 9 & 6 & 7 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 6 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 9 & 0 & 4 & 4 & 4 & 4 \\ 2 & 4 & 6 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & 7 & -4 & 3 & 4 & 2 & 0 \end{pmatrix}_{b_0}$$

6.185 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \mathfrak{e}_6 + \text{NHC}(-2)(-2)(-3)$

$$T = 10 \quad [0] \quad 1^{\overline{1111}} 3 1^{\overline{11}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 266, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.15 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 2 & 1 & 2 & 2 & 3 & 3 \\ 3 & 4 & 4 & 0 & 4 & 4 & 5 & 4 \\ 2 & 4 & 0 & 0 & 3 & 3 & 3 & 2 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 4 & 3 & 3 \\ 2 & 4 & 3 & 0 & 4 & 1 & 4 & 3 \\ 3 & 5 & 3 & 0 & 3 & 4 & 3 & 3 \\ 3 & 4 & 2 & -4 & 3 & 3 & 3 & -1 \end{pmatrix}_{b_0}$$

$$[1] \quad 1^{\overline{111}} 3 1^{\overline{111}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + \mathfrak{e}_6 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 10, T_{\max} = 14, \Delta = -92, \det = 326, (n_+, n_-, n_0) = (1, 10, 0), c_{\text{ext}} = 15.0$$

$$T_{\min} = 13.45 \rightarrow 14, T_{\max} = 14, T \in [14, 14]$$

$$\begin{pmatrix} 1 & 3 & 3 & 1 & 2 & 3 & 2 & 3 \\ 3 & 4 & 5 & 0 & 4 & 5 & 4 & 4 \\ 3 & 5 & 3 & 0 & 3 & 6 & 3 & 3 \\ 1 & 0 & 0 & -6 & 0 & 0 & 0 & -4 \\ 2 & 4 & 3 & 0 & 1 & 4 & 3 & 3 \\ 3 & 5 & 6 & 0 & 4 & 4 & 4 & 4 \\ 2 & 4 & 3 & 0 & 3 & 4 & 0 & 2 \\ 3 & 4 & 3 & -4 & 3 & 4 & 2 & -1 \end{pmatrix}_{b_0}$$

6.186 $\text{NHC}(-2)(-3) + \mathfrak{su}_3 + \text{NHC}(-2)(-2)(-3)$

$$T = 11 \quad [0] \quad 1^{\overline{11}} 5^{\overline{1}} 1^{\overline{1}} + [\mathfrak{su}_2 \cdot \mathfrak{g}_2] + \mathfrak{su}_3 + [(-2) \cdot \mathfrak{sp}_1 \cdot \mathfrak{g}_2] \quad T = 11, T_{\max} = 12, \Delta = -29, \det = -35, (n_+, n_-, n_0) = (1, 11, 0), c_{\text{ext}} = 9.0$$

$$T_{\min} = 11.97 \rightarrow 12, T_{\max} = 12, T \in [12, 12]$$

$$\begin{pmatrix} 1 & 3 & 2 & 2 & 2 & 1 & 3 \\ 3 & 4 & 3 & 4 & 3 & 0 & 4 \\ 2 & 3 & 0 & 2 & 2 & 0 & 2 \\ 2 & 4 & 2 & 1 & 3 & 0 & 3 \\ 2 & 3 & 2 & 3 & 1 & 1 & 3 \\ 1 & 0 & 0 & 0 & 1 & -3 & -1 \\ 3 & 4 & 2 & 3 & 3 & -1 & 3 \end{pmatrix}_{b_0}$$

7
NHC
 $(-2)(-4)$
External
(1 combos, 1 bases)

7.1
nhc₂₄

$$\begin{aligned}
 T = 9 \quad [0] \quad 1^{\textcolor{blue}{1}}2231321^{\textcolor{red}{1}} + [\textcolor{red}{s}\textcolor{red}{u}\textcolor{red}{8} \cdot \textcolor{blue}{s}\textcolor{blue}{o}\textcolor{blue}{16}] \quad T = 9, \, T_{\max} = 12, \, \Delta = -85, \, \det = -4, \, (n_+, n_-, n_0) = (1, 9, 0), \, c_{\text{ext}} = 5.0 \\
 T_{\min} = 12, \, T_{\max} = 12, \, T \in [12, 12] \\
 \begin{pmatrix} 0 & 1 & 1 & 2 \\ 1 & -2 & 2 & 0 \\ 1 & 2 & 2 & 4 \\ 2 & 0 & 4 & 4 \end{pmatrix}_{b_0}
 \end{aligned}$$